

Mathematics

A Complete Undergraduate Course

Adam Kelly

University of Cambridge

Preface

These notes began as an act of self-defence.

A Cambridge lecture moves at about the speed of the lecturer's handwriting, which is much faster than the speed of understanding. You arrive, you copy down a definition, a theorem and a proof, and an hour later you leave with a notebook full of statements that are correct on the page and only approximately correct in your head. My response, in my first term, was to go home in the evening and write the lecture again from scratch: in my own words, in my own order, with the missing steps put back in wherever I had lost the thread. It was much slower than reading. It was also the only thing that reliably worked, because you cannot write a proof out honestly without discovering exactly which line you did not follow.

I did this for one course, and then for another, and then it became simply what I did. Three years later there were forty-four of these documents sitting in a folder, covering every course I took for Parts IA, IB and II of the Mathematical Tripos, somewhere close to three thousand pages in all. This book is that folder, bound together and given a spine.

What this book is

It is a single volume containing the complete set of my Cambridge undergraduate lecture notes: eight courses from Part IA, fifteen from Part IB, and twenty-one from Part II. Each chapter is one course, more or less exactly as I wrote it at the time. The assembling is new; the mathematics is not.

That is worth being precise about, because it gives the book a texture you would never choose if you sat down to write a textbook. The Tripos is a collection of courses, not a single narrative, and courses overlap. Cauchy's theorem is proved carefully in Complex Analysis and then used briskly in Complex Methods. The Gauss–Bonnet theorem appears once in IB Geometry and again, from a greater height, in II Differential Geometry. The Fourier transform turns up in four separate chapters wearing four separate sets of hypotheses, and the Baire category theorem is proved twice. Notation drifts between chapters, and a symbol defined in one chapter means nothing in the next.

I have deliberately left all of this alone. Smoothing it out would have meant rewriting forty-four courses into one, which is a different book and not one I have written. What you have instead is an honest record of what the subject actually looks like from inside an undergraduate degree – overlapping, repetitive in places, and occasionally inconsistent with itself. The repetitions are, I think, among the more instructive things here. Seeing the same theorem proved twice at two levels of generality tells you something about the theorem that seeing it once does not.

What this book is not

It is not a substitute for a textbook, and it is certainly not a substitute for going to the lectures. Where a textbook would motivate at length, cite the literature and give three

proofs, these notes give one proof and move on. They were written under time pressure by somebody meeting the material for the first time, and they show it.

Nor is this a light tour. The book that inspired the shape of this front matter – Evan Chen’s *An Infinitely Large Napkin* – sets out to give you the feeling of a field in forty pages, and cheerfully omits any proof that does not teach something. This book does close to the opposite, not because that approach is wrong but because these notes were written for a different purpose. They are working notes: rigorous, complete against the schedules, and in places genuinely terse. There are stretches that are close to unreadable unless you already half know what is going on, which is what happens when you write for the version of yourself who sat in the lecture that morning.

Finally, these are not transcriptions. Each chapter says so at its start, and each one means it. I reordered material, replaced proofs I did not like, and added examples that were never lectured. All lectured content should be covered, but the route through it is frequently my own, and any error you find in a proof is far more likely to be mine than the lecturer’s.

Who it is for

Mostly, honestly, for me – somewhere to look things up, and a record of what three years of work actually amounted to. But there are two other readers I had in mind while putting it together.

The first is a Cambridge student sitting one of these courses now, who wants a second account of the lectures to set beside their own notes, and would rather have all of them in one file than in forty-four. The second is anyone learning this material from outside Cambridge who wants to see what a complete and rigorous undergraduate path through pure and applied mathematics looks like when it is written down end to end. The Tripod is unusually broad – there are not many degrees in which the same student proves the Riemann–Roch theorem and computes the flow of a viscous fluid down a slope – and that breadth is most of what makes a document like this worth assembling at all.

Debts

The largest debt is to the lecturers. Almost every good idea in this book was explained to me first by somebody standing at a blackboard, and the choice of material, the order of the arguments and very often the proofs themselves are theirs. What I have added is the writing-out, and occasionally the impertinence of thinking I could arrange it better.

I owe a second debt to the other note-takers. Cambridge has a generous and slightly anarchic tradition of students typing up their courses and putting them online for everybody else, and I leaned on it constantly – Dexter Chua’s notes in particular were a fixture of my degree, and there were plenty of others whose PDFs I went to when my own record of a lecture had gone hopelessly wrong. The style file all of this is set in descends, in its turn, from Evan Chen’s and Tony Zhang’s. Very little here was invented from nothing, and that is rather the point.

Errors

There are certainly some. If you find one – a broken proof, a missing hypothesis, a figure that says the opposite of the text – I would genuinely like to hear about it, at ak2316@cam.ac.uk. The notes are kept in a public repository as well, so corrections can go there directly.

Adam Kelly

How to Use This Book

The shape of the book

The book is in three parts, matching the three years of the Cambridge Mathematical Tripos, and each part is a year's worth of lectures.

- **Part IA** (eight chapters) is the first year. Everybody takes everything; there is no choice at all.
- **Part IB** (fifteen chapters) is the second year, where the syllabus opens up and a student picks perhaps ten of the courses on offer. What is collected here is what I took.
- **Part II** (twenty-one chapters) is the third year, where the courses become genuinely specialised and largely independent of one another.

Each of the three parts opens with a short essay about what that year is doing and how its courses hang together. Those essays are the closest thing this book has to a narrative, and they are worth reading before the mathematics.

Every chapter is a course

This is the single most important thing to know about the book. Each chapter is one lecture course, written as a standalone document and then dropped in here essentially untouched. In particular:

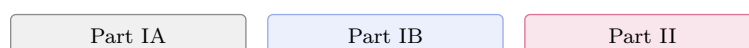
- **Chapters do not continue from one another.** Chapter 17 does not pick up where Chapter 16 left off; it starts again from its own beginning, with its own introduction saying what the course is about.
- **Notation is local.** A macro or a piece of notation established in one chapter is dissolved at the end of it. Two chapters may use the same letter for completely different things, and two chapters covering overlapping material may well use different conventions for it.
- **Numbering is by chapter and section.** A reference to Theorem 4.2.1 means the first numbered result in the second section of the fourth chapter. Definitions, theorems, lemmas and examples share a single counter within each section, so the numbers run in reading order.
- **Occasional references point outwards.** Because these were written as separate documents, you will now and then find a phrase like “see my Numbers and Sets notes”. That means the corresponding chapter of this book.

The upside of all this is that you can open the book anywhere. The downside is that reading it front to back is not obviously the right thing to do, which brings us to the next section.

Reading order, and the dependency graphs

Only Part IA is really meant to be read from a standing start. After that the courses fan out, and by Part II several of them share no content at all – the reader of Statistical Physics and the reader of Logic and Set Theory can happily ignore each other.

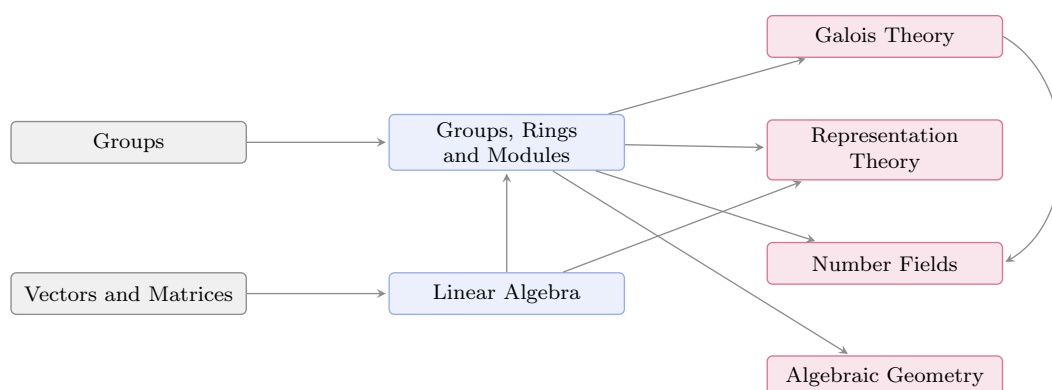
My honest advice is not to read this book linearly. Pick a chapter you want, look up what it stands on, and read backwards until you hit something you already know. The diagrams below are there to make that possible. They show the main lines of dependency running through the three years: an arrow from A to B means that B genuinely leans on A , not merely that the two are thematically related. Position from left to right indicates the direction of that flow, and the colour of a box tells you which part the course belongs to.



The graphs are deliberately incomplete. Every course assumes the whole of Part IA in the background, and drawing that in would have turned each picture into a scribble, so the arrows record only the specific and load-bearing dependencies.

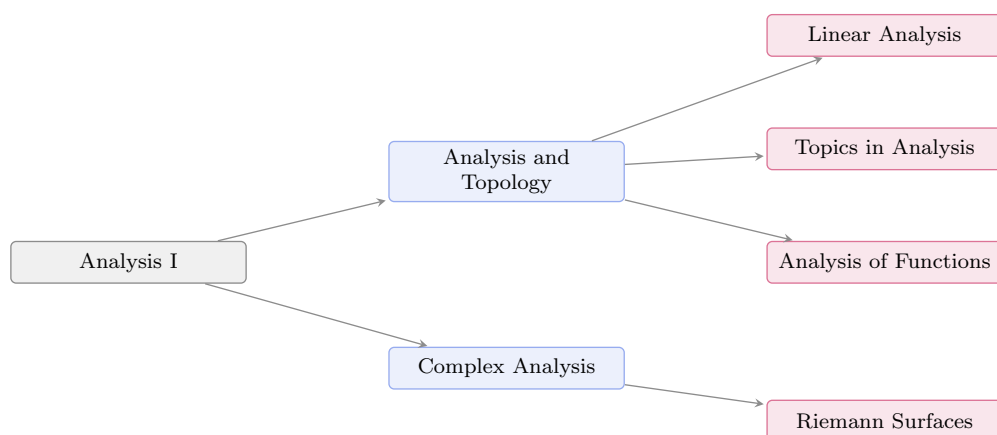
Algebra

The longest and cleanest chain in the book. Groups gives the objects, Linear Algebra gives the technique, Groups, Rings and Modules assembles both into a general theory of algebraic structure, and Part II then spends four courses cashing it in. Number Fields in addition wants a little Galois theory, and quite a lot of the Number Theory chapter.



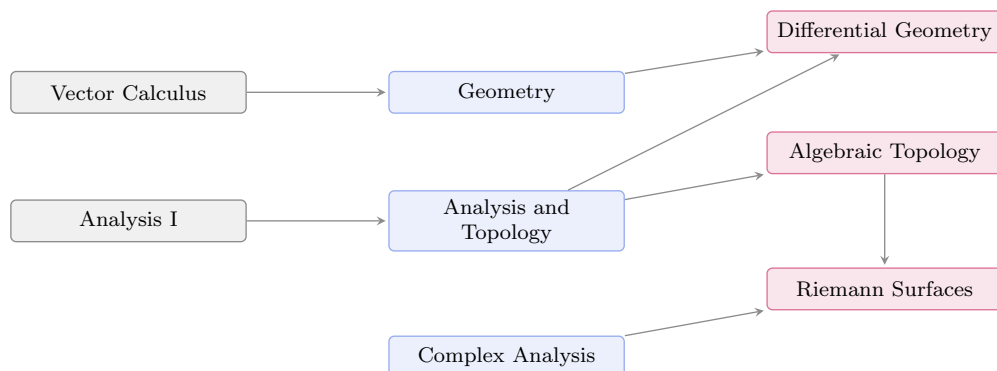
Analysis

Everything here descends from the ϵ - N definition of a limit in Analysis I. Analysis and Topology is the pivot: it replaces $\mathbb{R}$ by a metric or topological space, and once that is done the three Part II analysis courses become possible. Analysis of Functions also assumes the measure theory built in Probability and Measure, and repays the Hahn–Banach and Baire theorems of Linear Analysis.



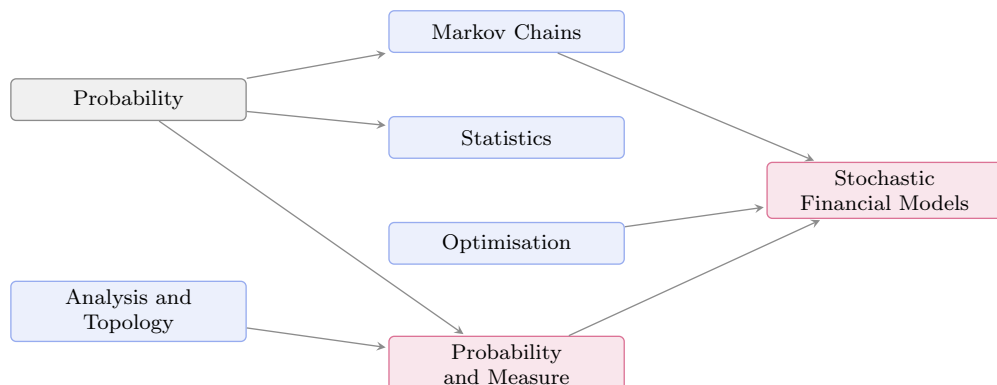
Geometry and topology

Two strands that meet in Part II. One is metric and computational, running from the curves and surfaces of Vector Calculus through Geometry to Differential Geometry; the other is the topology introduced in Analysis and Topology, which Algebraic Topology turns into a machine for telling spaces apart. Riemann Surfaces sits at the junction, needing complex analysis for its local theory and covering spaces for its global one.



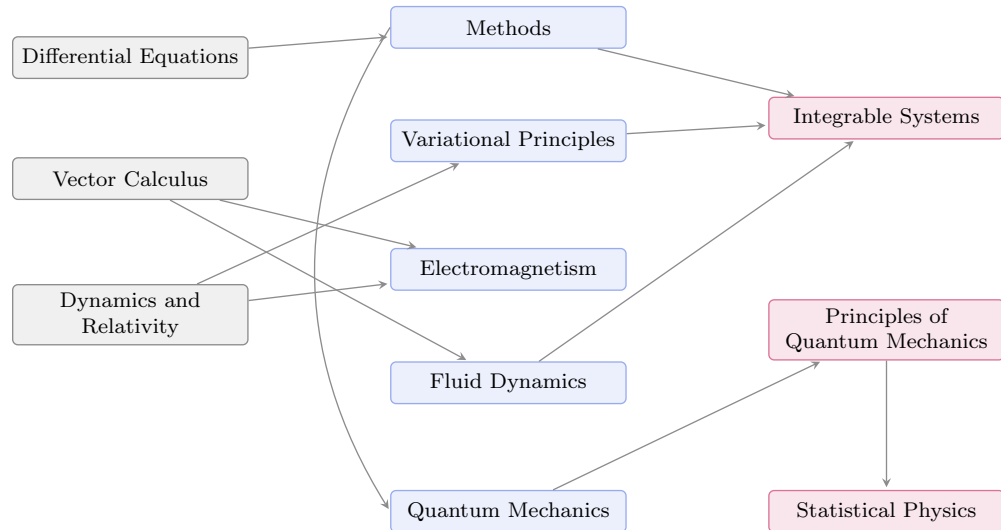
Probability and statistics

Part IA Probability is combinatorial and concrete; Probability and Measure goes back and rebuilds it on the measure theory of Analysis and Topology, at which point the limit theorems become theorems rather than slogans. Stochastic Financial Models is where the whole strand, optimisation included, is put to work.



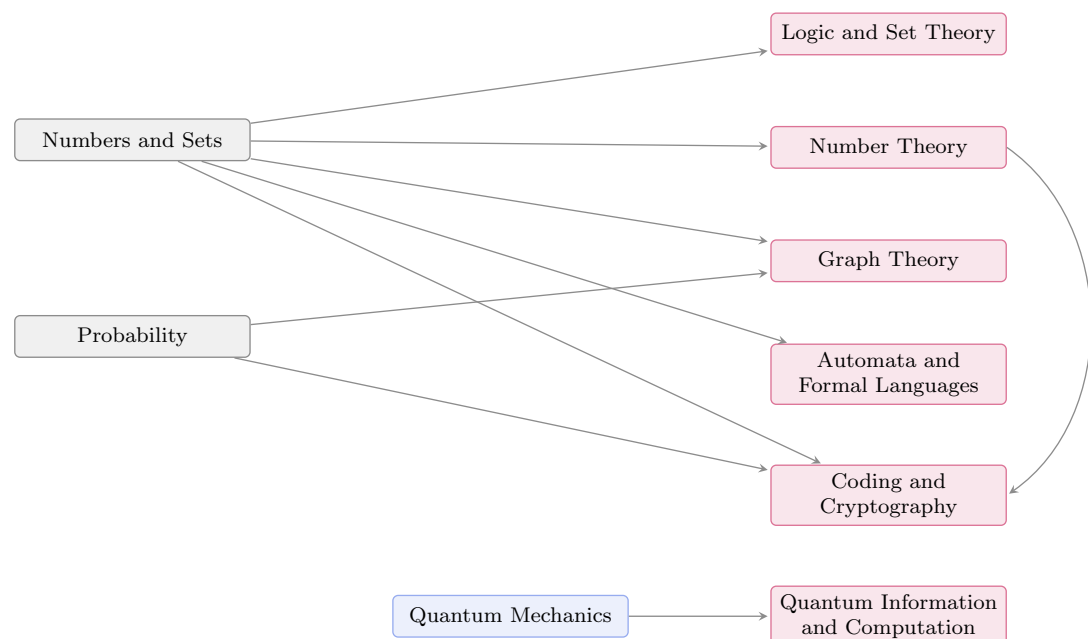
Methods and mathematical physics

The applied side of the book. Differential Equations and Vector Calculus supply the machinery, Dynamics and Relativity supplies the physics, and the Part IB courses are where the two are finally allowed to meet. Note how much converges on Methods: Fourier series, Sturm–Liouville theory and Green’s functions are the common language of almost everything applied that follows, and it is Methods that supplies Quantum Mechanics with the separation of variables and the special functions it needs in three dimensions.



Discrete mathematics, logic and computation

The most self-contained group in the book, and the natural place to start if you want to read a Part II course without first reading two years of anything. Numbers and Sets is very nearly the only prerequisite, and most of what it supplies is the habit of proof rather than any particular theorem.



The colours

The notes lean fairly heavily on coloured panels, and once you know the scheme you can navigate a chapter by scanning it. Everything in a coloured box is a formal statement; everything between the boxes is me talking.

- **Pale green, with a teal heading** is a *definition*. Axioms and physical laws share this colour, on the grounds that they too are things being stipulated rather than deduced.
- **Pale lilac, with a dark blue heading** is a *result*: theorems, propositions, lemmas, corollaries, facts and claims all live here. There is no significance to the choice between those words beyond the usual one of importance.
- **Pale pink, with a rust-coloured heading** is an *example* or a question. These are usually worked, and they are usually the fastest way into a section you find opaque.
- **Grey, with a rule down the left-hand edge**, is a *proof*. The rule runs the height of the argument, so you can see at a glance where a long proof ends and the discussion starts again.
- **Bold blue** in running text marks a term at the point where it is defined.

A word of warning about the pale green boxes in particular. Because definitions are so visually distinctive it is tempting to read only those, which is exactly the wrong way round: the prose between the boxes is where any motivation lives, and in these notes there is not so much of it that you can afford to skip it.

A closing caution

These are lecture notes, and they behave like lecture notes. Sometimes a chapter develops an idea patiently over three pages; sometimes it disposes of an entire lecture in half a column because that was all the reminder I needed. If you hit a passage that seems to assume you already understand it, that is usually because it was written by somebody who had just been told it, for somebody who was about to be examined on it. Read around it, come back, and it generally opens up.

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I

Part IA

About Part IA

The first year is the only part of the Tripos with no choices in it. Everybody sits the same eight courses, and the reason is that the year has a job to do: it has to take a room of people who arrived knowing how to differentiate and turn them into people who can write a proof. Almost everything in this part is either a foundation that later years will build on without comment, or a piece of technique that later years will assume you can execute in your sleep.

The year splits fairly cleanly in two. Numbers and Sets, Groups, Analysis I and Vectors and Matrices are the pure half, and between them they introduce essentially every kind of argument the rest of the book uses. Numbers and Sets comes first in more than just the ordering – it is where induction, the Euclidean algorithm, equivalence relations, countability and the least upper bound axiom appear, and it is the course whose real content is the idea of proof itself. Analysis I then takes the least upper bound axiom and does something startling with it, building limits, continuity, differentiation and the Riemann integral out of nothing but ϵ and N . Groups and Vectors and Matrices run in parallel as the first taste of abstraction: Groups is abstract from its first line and stays concrete in its examples, working through the dihedral, symmetric and Möbius groups by hand, while Vectors and Matrices spends most of its time on $\mathbb{R}^3$ and $\mathbb{C}^n$ before quietly showing you that a matrix is a linear map in disguise.

The applied half – Differential Equations, Vector Calculus, Dynamics and Relativity, and Probability – is deliberately less careful, and it is worth being clear that this is a choice rather than an oversight. Differential Equations solves equations without ever proving that solutions exist. Vector Calculus differentiates under integral signs and exchanges limits with abandon. Probability defines a random variable as a function on a sample space and then never worries about whether the sample space is big enough. All three of these debts are called in later, in Analysis and Topology, in Methods, and in Probability and Measure respectively; the first year buys fluency now and pays for rigour later.

There are through-lines worth watching for. Vector Calculus is the most quietly important course of the year, because the divergence theorem and Stokes' theorem are the entire grammar of Electromagnetism and Fluid Dynamics, and its chapter on curves and surfaces is the first draft of everything in Geometry and Differential Geometry. Groups is the seed of the longest chain in the book, running through Groups, Rings and Modules to Galois Theory and Representation Theory. Vectors and Matrices is the same material as Linear Algebra with the abstraction removed, which makes reading the two in succession an unusually good way to see what abstraction actually buys you. And Dynamics and Relativity, which looks at first like a physics course that wandered into the wrong degree, turns out to be where the variational and Hamiltonian ideas of the next two years get their motivation.

If you are reading this book rather than sitting the exams, Part IA is the one part that genuinely rewards being read in order.

1 Analysis

At its heart, analysis is the study of ideas that depend on the notion of *limits*. The main concepts of analysis (such as convergence, continuity, differentiation and integration) will all depend quite fundamentally on a limiting process.

This article constitutes my notes for the ‘Analysis I’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered¹.

§1.1 Sequences and Convergence

One of the fundamental objects of study in analysis is *sequences*. In particular, in almost all areas of this course we will be concerned (either implicitly or explicitly) with the notion of convergence.

§1.1.1 Limits & The Reals

We will define convergence as follows.

Definition 1.1.1 (Convergence)

A sequence $a_1, a_2, \dots \in \mathbb{R}$ is said to **converge** to the limit $a \in \mathbb{R}$ if given any $\epsilon > 0$, we can find an integer N such that $|a_n - a| < \epsilon$ for all $n \geq N$. We write $\lim_{n \rightarrow \infty} a_n = a$, or $a_n \rightarrow a$ as $n \rightarrow \infty$.

This definition is (notably) purely algebraic. We can sensibly define this notion of convergence for any ordered field (for example, $\mathbb{Q}$). What takes us from algebra to analysis is the fundamental property of the real numbers.

Axiom 1.1.2 (Fundamental Axiom of Analysis)

If $a_1, a_2, \dots \in \mathbb{R}$ is an increasing sequence and there exists $A \in \mathbb{R}$ such that $a_i \leq A$ for all $i \in \mathbb{N}$, then there exists $a \in \mathbb{R}$ such that $a_n \rightarrow a$ as $n \rightarrow \infty$.

In other words – an increasing sequence of real numbers that is bounded above converges. Also this clearly implies that a decreasing sequence of reals bounded below converges.

Limits obey the properties that you would naturally expect.

Proposition 1.1.3 (Uniqueness of Limits)

If $a_n \rightarrow a$ and $a_n \rightarrow b$ as $n \rightarrow \infty$, then $a = b$.

¹A tiny bit of analysis is assumed, namely the content covered in the ‘Numbers and Sets’ course at Cambridge. Specifically the reader should be aware of least upper bounds/suprema along with the least upper bound axiom. If the reader is unfamiliar with this content, they are referred to Chapter 2 of my Numbers and Sets notes

Proof. Assume that $a \neq b$. Given given any $\epsilon > 0$, we can find integers N_1 and N_2 such that

$$\begin{aligned} |a_n - a| &\leq \epsilon, & \text{for all } n \geq N_1 \\ |a_n - b| &\leq \epsilon, & \text{for all } n \geq N_2 \end{aligned}$$

Then letting $\epsilon = |a - b|/3$ and taking $N = \max\{N_1, N_2\}$, we have by the triangle inequality

$$|a - b| \leq |a_n - a| + |a_n - b| \leq 2\epsilon = \frac{2}{3}|a - b|$$

for all $n \geq N$. Thus we must have $|a - b| = 0$, and $a = b$. $\square$

Proposition 1.1.4 (Convergence of Subsequences)

If $a_n \rightarrow a$ as $n \rightarrow \infty$ and $n(1) < n(2) < \dots$, then $a_{n(j)} \rightarrow a$ as $j \rightarrow \infty$.

Proof. We note that $n(j) \geq j$. Now $a_n \rightarrow a$ implies that given some ϵ , we can find an N such that $|a_j - a| < \epsilon$ for all $j \geq N$. But then this implies that $|a_{n(j)} - a| < \epsilon$ for all $j \geq N$, since $j \geq N$ implies $n(j) \geq N$. So $a_{n(j)} \rightarrow a$ also. $\square$

Proposition 1.1.5 (Manipulating Limits)

Let a_n and b_n be sequences. Then the following hold.

- (i) If $a_n \rightarrow a$ and $b_n \rightarrow b$ then $a_n + b_n \rightarrow a + b$.
- (ii) If $a_n \rightarrow a$ then $ca_n \rightarrow ac$ for a constant c .
- (iii) If $a_n \rightarrow a$ and $b_n \rightarrow b$ then $a_nb_n \rightarrow ab$.
- (iv) If $a_n \rightarrow a$ and $a, a_n \neq 0$ for all n , then $1/a_n \rightarrow 1/a$.

Proof. We prove each individually.

- (i) Given some $\epsilon > 0$, we can find integers N_a and N_b such that $|a_n - a| \leq \epsilon/2$ for all $n \geq N_a$ and $|b_n - b| \leq \epsilon/2$ for all $n \geq N_b$.

Then letting $N = \max\{N_a, N_b\}$, by the triangle inequality we have

$$|(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all $n \geq N$. Thus $a_n + b_n \rightarrow a + b$.

- (ii) Given $\epsilon > 0$, we can find some N such that $|a_n - a| \leq \epsilon/|c|$ for all $n \geq N$. Then $|ca_n - ca| \leq \epsilon$ for all $n \geq N$. So $ca_n \rightarrow ca$.
- (iii) Given some $\epsilon > 0$, we can find integers N_a and N_b such that $|a_n - a| \leq \sqrt{\epsilon}$ for all $n \geq N_a$ and $|b_n - b| \leq \sqrt{\epsilon}$ for all $n \geq N_b$. Then again letting $N = \max\{N_a, N_b\}$, we have

$$|(a_n - a)(b_n - b)| \leq \epsilon,$$

for all $n \geq N$. Hence $(a_n - a)(b_n - b) \rightarrow 0$, so $a_nb_n - ab_n - ba_n + ab \rightarrow 0$, and using the previous two properties we have $a_nb_n \rightarrow ab$.

- (iv) Since the sequence converges to $a \neq 0$, there must be some $r > 0$ such that $|a_n| > r$, for all n . Then given some $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a| < |a|\epsilon r$ for all $n \geq N$. That is,

$$\left| \frac{1}{a_n} - \frac{1}{a} \right| = \left| \frac{a - a_n}{aa_n} \right| < \frac{|a|\epsilon r}{|a|r} = \epsilon,$$

for all $n \geq N$. Hence $1/a_n \rightarrow 1/a$. $\square$

Proposition 1.1.6 (Squeeze Theorem)

Let a_n, b_n and x_n be sequences such that $a_n \leq x_n \leq b_n$ for all n . Then if $a_n \rightarrow \ell$ and $b_n \rightarrow \ell$ as $n \rightarrow \infty$, we also have $x_n \rightarrow \ell$.

Proof. Given some $\epsilon > 0$, we can find integers N_a and N_b such that $|a_n - \ell| < \epsilon$ for all $n \geq N_a$ and $|b_n - \ell| < \epsilon$ for all $n \geq N_b$. Then letting $N = \max\{N_a, N_b\}$, we have $\ell - \epsilon < a_n \leq x_n \leq b_n < \ell + \epsilon$ for all $n \geq N$. That is, $|x_n - \ell| < \epsilon$ for all $n \geq N$. Hence $x_n \rightarrow \ell$. $\square$

With these results in the our toolbox, we can then prove our first actual analysis result.

Lemma 1.1.7 (Axiom of Archimedes)

$\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$.

Proof. The sequence $\frac{1}{n}$ is a decreasing sequence bounded below, and thus has a limit a . Considering the sequence $\frac{1}{2n} = \frac{1}{2} \cdot \frac{1}{n}$, this tends to a limit $\frac{a}{2}$, but since $\frac{1}{2n}$ is a subsequence of $\frac{1}{n}$, it also tends to a limit a . Thus $\frac{a}{2} = a$, so $a = 0$ as required. $\square$

Now while this is an article about real analysis, we will frequently employ the complex numbers. Of course, to do that we need to be able to do some analysis with $\mathbb{C}$, and indeed the definition of a limit still makes sense in $\mathbb{C}$. All of the above properties also hold, apart from the squeeze theorem (we need to be careful because $\mathbb{C}$ cannot be ordered like $\mathbb{R}$!)

§1.1.2 Bolzano–Weierstrass

An equivalent but important and quite useful form of the fundamental axiom is the ‘Bolzano–Weierstrass theorem’, that every bounded sequence has a convergent subsequence.

Theorem 1.1.8 (Bolzano–Weierstrass)

If $a_n \in \mathbb{R}$ and there exists K such that $|a_n| \leq K$ for all n , then we can find $n_1 < n_2 < n_3 < \dots$ and $a \in \mathbb{R}$ such that $a_{n_j} \rightarrow a$ as $j \rightarrow \infty$.

Proof. Consider a real sequence a_n . We say that an element a_i of the sequence is *good* if $a_j \geq a_i$ for all $j \geq i$. If the sequence has an infinite number of good elements, then taking them as a subsequence will be increasing. If there is a finite number of good elements, then letting N be the index of the last good element, we have that for any a_j with $j \geq N + 1$, there exists some a_k with $k > j$ such that $a_k < a_j$. Then repeating this, we can get a collection of decreasing elements. Thus every real sequence has a monotonic subsequence. Since the sequence is bounded, we then have that this subsequence converges. $\square$

Remark. This theorem says nothing about the uniqueness of the subsequence's limit. For example, consider the sequence $x_n = (-1)^n$. Then $x_{2n+1} \rightarrow -1$ and $x_{2n} \rightarrow 1$.

The proof given above is quite clean but is not the only standard proof of the theorem. Another common method which involves the method of 'repeated bisection', which is colloquially known as 'lion hunting'. This method will be discussed shortly.

The Bolzano-Weierstrass theorem also holds in $\mathbb{C}$ (and the same proof method will show with a little induction that it holds in $\mathbb{R}^n$, but we won't dwell on that in this article).

Theorem 1.1.9 (Bolzano-Weierstrass in $\mathbb{C}$)

If $a_n \in \mathbb{C}$ is a bounded sequence, then it has a convergent subsequence.

Proof. Begin by writing $a_n = x_n + iy_n$. Then since a_n is bounded, the sequences x_n and y_n are also bounded. We can apply Bolzano-Weierstrass to x_n to get a subsequence $x_{n(1)}, x_{n(2)}, \dots$ with $x_{n(j)} \rightarrow x$ as $j \rightarrow \infty$, for some $x \in \mathbb{R}$. Then we can apply Bolzano-Weierstrass again to $y_{n(j)}$ to get a subsequence $y_{n(k(1))}, y_{n(k(2))}, \dots$ with $y_{n(k(j))} \rightarrow y$ as $j \rightarrow \infty$ for some $y \in \mathbb{R}$.

Thus $x_{n(k(j))} \rightarrow x$ and $y_{n(k(j))} \rightarrow y$ and thus $a_{n(k(j))} \rightarrow x + iy$ as $j \rightarrow \infty$, so we have a convergent subsequence. $\square$

Before we end our discussion of the Bolzano-Weierstrass theorem, we will take a short digression on Lion Hunting (you need to know this proof though!)

Aside: How To Hunt Lions

The term 'lion hunting' comes most likely from the satirical paper 'A Contribution to the Mathematical Theory of Big Game Hunting', by a group of mathematicians under the pseudonym 'H. Pétard'. They present the following method of hunting a lion (that we know exists) in the Sahara Desert.

Bisect the desert by a line running N-S. The lion is either in the E portion or in the W portion; let us suppose him to be in the W portion. Bisect this portion by a line running E-W. The lion is either in the N portion or in the S portion; let us suppose him to be in the N portion. We continue this process indefinitely, constructing a sufficiently strong fence about the chosen portion at each step. The diameter of the chosen portions approaches zero, so that the lion is ultimately surrounded by a fence of arbitrarily small perimeter.

While this is tongue and cheek, it does give us a method of finding things by 'bisecting intervals', which we can apply to quite a few different theorems in Analysis.

Now it's time to prove Bolzano-Weierstrass by hunting lions.

Proof (Bolzano-Weierstrass, Lion Hunting Style). We are going to define two sequences a_n and b_n inductively as follows. Begin by setting $[a_1, b_1] = [-K, K]$, and let $c_1 = (a_1 + b_1)/2$ be the midpoint of this interval.

Then there is two possibilities:

1. $x_n \in [a_1, c_1]$ for infinitely many values of n .
2. $x_n \in [c_1, b_1]$ for infinitely many values of n .

Of course both of these can hold at the same time, but if the first one holds we set $a_2 = a_1$ and $b_2 = c_1$, and if it doesn't then we set $a_2 = c_1$ and $b_2 = b_1$.

Repeating this process we construct a_n and b_n such that $x_m \in [a_n, b_n]$ for infinitely many values of m ^a. Then we have $a_{n-1} \leq a_n \leq b_n \leq b_{n-1}$, and also $b_n - a_n = (b_{n-1} - a_{n-1})/2$.

Now a_n is increasing and bounded above, and b_n is decreasing and bounded below and thus $a_n \rightarrow a \in [a_1, b_1]$ and $b_n \rightarrow b \in [a_1, b_1]$. Then we have $b - a = (b - a)/2$ using the above result, and thus $a = b$.

Since $x_m \in [a_n, b_n]$ for infinitely many values of m , we can construct a sequence n_j such that $n_{j+1} > n_j$ and $x_{n_{j+1}} \in [a_{j+1}, b_{j+1}]$. Then $a_j \leq x_{n_j} \leq b_j$, and thus $x_{n_j} \rightarrow a$. $\square$

^aThis is lion hunting! You can kind of imagine that we are hunting for a number that a subsequence converges to, using the fact that there must be infinitely many terms near that number

§1.1.3 Cauchy Sequences & The General Principle of Convergence

So far we have defined the notion of a sequence converging to some explicit limit. However, it is possible to determine if a sequence converges without considering this limit explicitly. This is done by considering how 'close' the terms in the sequence eventually get, as we shall see.

We begin by describing what it means for a sequence to be *Cauchy*.

Definition 1.1.10 (Cauchy Sequence)

A sequence $a_1, a_2, \dots \in \mathbb{C}$ is said to be **Cauchy** if for every $\epsilon > 0$ there exists an integer N such that for all $n, m \geq N$ we have $|a_n - a_m| \leq \epsilon$.

We can almost immediately write down our first lemma.

Lemma 1.1.11 (Convergence Implies Cauchy)

If a sequence converges then it is Cauchy.

Proof. Consider a convergent sequence $a_n \rightarrow a$. Given $\epsilon > 0$, we can find N such

that $|a_n - a| \leq \epsilon/2$ for all $n \geq N$. Then by the triangle inequality we have

$$|a_n - a_m| \leq |a_n - a| + |a - a_m| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon,$$

for all $m, n \geq N$. Thus the sequence is Cauchy. $\square$

The converse of this result is also true! This gives us a powerful result about convergence, which is quite widely applicable (particularly because we can avoid talking about the limit explicitly, as we said before).

Lemma 1.1.12 (Completeness)

If a sequence is Cauchy then it converges.

Proof. Let $a_1, a_2, \dots$ be a Cauchy sequence. We will first show that the sequence is bounded. Because the sequence is Cauchy, can find an integer $N \in \mathbb{N}$ such that $n, m \geq N$ implies that $|a_n - a_m| \leq 1$. This implies that if $n \geq N$ we have

$$|x_n| \leq |x_n - x_N| + |x_N| \leq 1 + |x_N|.$$

Thus we have $|x_n| \leq \max_{1 \leq r \leq N} |x_r| + 1$ for all n , so the sequence is bounded.

Applying the Bolzano–Weierstrass theorem, we then have a subsequence $a_{n_1}, a_{n_2}, \dots$ with $a_{n_j} \rightarrow a$ as $j \rightarrow \infty$.

So given $\epsilon > 0$, we can find M such that $|a_{n_j} - a| \leq \epsilon/2$ for all $n_j \geq M$. We can also find M' such that $n, m \geq M'$ implies $|a_n - a_m| < \epsilon/2$.

Then for all $n \geq M'$, choosing n_j such that $n_j \geq \max\{M, M'\}$, we have

$$|a_n - a| \leq |a_n - a_{n_j}| + |a_{n_j} - a| \leq \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus the sequence converges (namely to a). $\square$

Stripping away the detail, the reason this result holds is since Cauchy sequences are bounded, we can get a convergent subsequence. Then since all of the terms get arbitrarily close, the whole sequence must converge along with the subsequence.

Combining these two lemmas gives us the ‘general principle of convergence’, a result also known as Cauchy’s criterion.

Theorem 1.1.13 (General Principle of Convergence/Cauchy’s Criterion)

A sequence converges if and only if it is a Cauchy sequence.

§1.1.4 Limits of Functions

So far we have developed the notion of a limit of a sequence, but we can sensibly come up with a definition for what limits are in the context of functions. We will need to do a little bit of setup first, however.

We first need to be able to distinguish points (element of $\mathbb{C}$) where its possible to converge to those points with a sequence of elements that are all in some set A .

Definition 1.1.14 (Limit Point)

Let $A \subseteq \mathbb{C}$ and $a \in \mathbb{C}$. We say that a is a **limit point** of A if for any $\delta > 0$ there is some $z \in A$ such that $0 < |z - a| < \delta$.

We can then define a limit of a function by its behaviour as we approach a given limit point.

Definition 1.1.15 (Limit of a Function)

Let $A \subseteq \mathbb{C}$ and let $a \in \mathbb{C}$ be a limit point of A . Then for $f : A \rightarrow \mathbb{C}$, we say that the **limit of f** as z approaches a is ℓ , written $\lim_{z \rightarrow a} f(z) = \ell$, if given $\epsilon > 0$ there is some $\delta > 0$ such that whenever $0 < |z - a| < \delta$ and $z \in A$ we have $|f(z) - \ell| < \epsilon$.

These definitions should match the informal notion of a limits that would have been given in a calculus course. The reader should note that there is no requirement that $f(a) = \ell$, or even that f be defined at a at all.

There is a natural relation between this notion of a limit point and a limit and our previous definition of limits of sequences.

Proposition 1.1.16 (Sequence Definition of Limit Points)

Let $A \subseteq \mathbb{C}$ and $a \in \mathbb{C}$. Then a is a limit point of A if and only if there is a sequence $z_n \in A$ such that $z_n \rightarrow a$ as $n \rightarrow \infty$ and $z_n \neq a$ for all n .

Proof. If a is a limit point then for every positive integer n we can take $\delta = 1/n$ and obtain a $z_n \in A$ such that $0 < |z_n - a| < 1/n$. Then by the squeeze theorem $z_n \rightarrow a$ as $n \rightarrow \infty$, and also $z_n \neq a$ for all n .

Conversely, if there is such a sequence $z_n \in A$, then given $\delta > 0$ there is an N such that $z_N \in A$ and $0 < |z_N - a| < \delta$, so a is a limit point of A . $\square$

Proposition 1.1.17 (Sequence Definition of Limits)

Let $A \subseteq \mathbb{C}$ and let $a \in \mathbb{C}$ be a limit point of A . Then for $f : A \rightarrow \mathbb{C}$ we have $\lim_{z \rightarrow a} f(z) = \ell$ if and only if $\lim_{n \rightarrow \infty} f(z_n) = \ell$ for every sequence $z_n \in A$ satisfying $z_n \neq a$ and $z_n \rightarrow a$ has $f(z_n) \rightarrow \ell$.

Proof. If $\lim_{z \rightarrow a} f(z) = \ell$, consider a sequence $z_n \in A$ where $z_n \neq a$ and $z_n \rightarrow a$. Given $\epsilon > 0$, there exists $\delta > 0$ such that $0 < |z - a| < \delta$ and $z \in A$ implies $|f(z) - \ell| < \epsilon$.

Also there exists N such that $n \geq N$ implies $0 < |z_n - a| < \delta$. Thus for $n \geq N$ we have $|f(z_n) - \ell| < \epsilon$, so $\lim_{n \rightarrow \infty} f(z_n) = \ell$.

Conversely, if $\lim_{z \rightarrow a} f(z)$ was not ℓ , then there would be some ϵ such that for every δ there is a $z \in A$ such that $0 < |z - a| < \delta$ but $|f(z) - \ell| \geq \epsilon$. Then taking $\delta = 1/n$ for each positive integer n we could construct a sequence $z_n \in A$ with $z_n \neq a$ where $0 < |z_n - a| < 1/n$ but $|f(z_n) - \ell| \geq \epsilon$ for all n . But then $z_n \rightarrow a$ and $f(z_n) \not\rightarrow \ell$, which is a contradiction. $\square$

In practice, which of these definitions is more useful will depend on context, however the sequences definition does allow us to use the results we developed at the start of this section.

Proposition 1.1.18 (Properties of Limits of Functions)

Let $A \subseteq \mathbb{C}$ and let a be a limit point of A . Then if $f, g : A \rightarrow \mathbb{C}$ are functions the following hold.

- (i) If $\lim_{z \rightarrow a} f(z) = \ell$ and $\lim_{z \rightarrow a} g(z) = m$, then $\lim_{z \rightarrow a} f(z) + g(z) = \ell + m$.
- (ii) If $\lim_{z \rightarrow a} f(z) = \ell$ then $\lim_{z \rightarrow a} cf(z) = c\ell$.
- (iii) If $\lim_{z \rightarrow a} f(z) = \ell$ and $\lim_{z \rightarrow a} g(z) = m$ then $\lim_{z \rightarrow a} f(z)g(z) = \ell m$.
- (iv) If $\lim_{z \rightarrow a} f(z) = \ell$ and $\ell \neq 0$ then $\lim_{z \rightarrow a} 1/f(z) = 1/\ell$.

Proof. These all follow directly from the sequence definition of limits along with [Proposition 1.1.5](#). □

§1.2 Infinite Series

The notion of convergence lets us talk quite sensibly about what it would mean to add up an infinite number of things, which is quite exciting.

§1.2.1 Convergent & Divergent Series

The definition of convergence for an infinite series is rather natural, and comes from considering the sequence of partial sums.

Definition 1.2.1 (Convergence of an Infinite Series)

For a sequence $a_j \in \mathbb{C}$, we say that the series $\sum_{j=1}^{\infty} a_j$ **converges** to S if the sequence of partial sums converges to S . That is, $\sum_{j=1}^n a_j \rightarrow S$ as $n \rightarrow \infty$. Otherwise we say that it **diverges**.

Remark (Notation). If a series converges, then we will typically write $\sum_{j=1}^{\infty} a_j = S$, but some care is needed as if the series does not converge then this is nonsense. We will also write S_n to denote the partial sum of a series, if it is clear from context what series is being referred to.

In this chapter we will be primarily concerned with finding ways to show whether a given series converges. But first, a few preliminaries.

Proposition 1.2.2 (Adding Series)

If two series converge, say $\sum_{j=1}^{\infty} a_j = a$ and $\sum_{j=1}^{\infty} b_j = b$, then $\sum_{j=1}^{\infty} (\lambda a_j + \mu b_j)$ also converges, and we have

$$\sum_{j=1}^{\infty} (\lambda a_j + \mu b_j) = \lambda a + \mu b.$$

Proof. Considering the sequence of partial sums we have

$$\sum_{j=1}^n (\lambda a_j + \mu b_j) = \lambda \sum_{j=1}^n a_j + \mu \sum_{j=1}^n b_j \longrightarrow \lambda a + \mu b$$

as $n \rightarrow \infty$, as required. $\square$

It should also be intuitively clear that the first few terms of an infinite series will not affect its convergence.

Proposition 1.2.3 (Ignoring Initial Terms)

Suppose there exists N such that $a_j = b_j$ for all $j \geq N$. Then either $\sum_{j=1}^{\infty} a_j$ and $\sum_{j=1}^{\infty} b_j$ both converge or they both diverge.

Proof. For $n \geq N$, we have $\sum_{j=1}^n a_n - \sum_{j=1}^n b_n = \sum_{j=1}^N a_n - \sum_{j=1}^N b_n$, which is a constant. Thus the sequences of partial sums either both converge or diverge. $\square$

We can also apply the general principle of convergence to infinite series. This gives us

another useful way of proving that a series converges, particularly when we don't know what the series converges to which is often the case.

Theorem 1.2.4 (General Principle of Convergence/Cauchy's Criterion for Series)

The infinite series $\sum_{j=1}^{\infty} a_j$ converges if and only if for every $\epsilon > 0$ there is an integer N such that for all $n, m \geq N$ we have $|\sum_{j=n}^m a_j| \leq \epsilon$.

Proof. This follows directly from the definition of the sequence of partial sums being Cauchy. $\square$

§1.2.2 Convergence Tests

In this section we will develop some 'tests' of convergence, so that given some infinite series we will have some tools to try and determine relatively quickly if it converges and diverges.

We begin with the observation that if the terms of the series do not tend to 0, then the series must diverge.

Proposition 1.2.5 (Limit of Terms in a Convergent Series)

Suppose that $\sum_{j=1}^{\infty} a_j$ converges. Then $a_j \rightarrow 0$ as $j \rightarrow \infty$.

Proof. Let $S_n = \sum_{j=1}^n a_j$. Then $S_n \rightarrow S$ for some S , and thus we must have $S_{n+1} \rightarrow S$. Adding these sequences we get $S_{n+1} - S_n \rightarrow 0$, that is, $a_{n+1} \rightarrow 0$ as $n \rightarrow \infty$ as required. $\square$

Somewhat unfortunately, while necessary this is not a sufficient condition for convergence. The most commonly cited counterexample to this is the Harmonic series.

Example 1.2.6 (Divergence of the Harmonic Series)

Consider the infinite series with $a_k = 1/k$. Then $a_k \rightarrow 0$, but $\sum_{k=1}^{\infty} a_k$ diverges.

To see this, let $H_n = \sum_{k=1}^n a_k$ be the partial sum of the series. Then we have

$$\begin{aligned} H_{2n} &= H_n + \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \\ &\geq H_n + \underbrace{\frac{1}{2n} + \frac{1}{2n} + \cdots + \frac{1}{2n}}_{n \text{ times}} = H_n + \frac{1}{2}. \end{aligned}$$

So if the series converged, we would have $H_n \rightarrow a$ and $H_{2n} \rightarrow a$, giving $a \geq a + 1/2$, which is a contradiction.

The result about the limit of terms in a series can be helpful though! For example, let's consider a series you are likely familiar with.

Example 1.2.7 (Convergence/Divergence of the Geometric Series)

Consider the infinite series $\sum_{j=1}^{\infty} x^{j-1}$. Writing it's partial sum as $S_n = \sum_{j=1}^n x^{j-1}$, A little bit of algebra gives us that

$$S_n = \begin{cases} \frac{1-x^n}{1-x} & \text{if } x \neq 1, \\ n & \text{if } x = 1. \end{cases}$$

So if $|x| < 1$, then $x^n \rightarrow 0$ and $S_n \rightarrow \frac{1}{1-x}$. Otherwise, if $|x| \geq 1$ then the series cannot converge since the terms do not tend to 0.

It's best to think of [Proposition 1.2.5](#) as giving the sort of a 'bare minimum' property that a series has to have in order to converge. Of course, it's also quite helpful for sanity checks!

Series of Non-Negative Terms

We just saw a necessary condition for convergence, now let's have a look at some *sufficient* conditions for convergence. For the time being we will restrict ourselves to considering infinite series where all of the terms are *real and non-negative*.

The first result we have is a direct consequence of our fundamental axiom.

Theorem 1.2.8 (Bounded Partial Sums)

If a_n is a non-negative sequence and the partial sums S_n are bounded above, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof. Since all of the terms are non-negative, S_n is a bounded and increasing sequence, and thus converges by our fundamental axiom. $\square$

In a similar vein is the comparison test, which allows us to show convergence by comparing a series with another series whose convergence we know.

Theorem 1.2.9 (The Comparison Test)

Suppose that $0 \leq a_j \leq b_j$ for all j . Then if $\sum_{j=1}^{\infty} b_j$ converges, so does $\sum_{j=1}^{\infty} a_j$.

Proof. Since $B_n = \sum_{j=1}^n b_j$ is an increasing sequence whose limit is say B , we have $B_n \leq B$ for all n . Thus $\sum_{j=1}^n a_j \leq B_n \leq B$, so we have a series of non-negative terms whose partial sums are bounded. Thus $\sum_{j=1}^{\infty} a_n$ converges. $\square$

Example 1.2.10 (Using the Comparison Test)

We will prove that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges using the comparison test.

Since $\frac{1}{n^2} < \frac{1}{n(n-1)}$ for $n \geq 2$, and $\sum_{n=2}^{\infty} \frac{1}{n(n-1)}$ converges as $\sum_{n=2}^N \frac{1}{n(n-1)} = 1 - \frac{1}{N} \rightarrow 1$ as $N \rightarrow \infty$, we get that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by comparison.

Using the comparison test we can derive two more useful tests, both of which come from comparing a series with the geometric series. You'll notice that the proofs for both results are relatively similar!

Theorem 1.2.11 (The Root Test)

Let a_n be a non-negative sequence and suppose $\sqrt[n]{a_n} \rightarrow a$ as $n \rightarrow \infty$. Then if $a < 1$, $\sum_{n=1}^{\infty} a_n$ converges, and if $a > 1$, $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. If $a < 1$, we can choose b such that $a < b < 1$, and there exists an integer N such that for all $n \geq N$ we have $\sqrt[n]{a_n} < b$, that is, $a_n < b^n$. But then $\sum_{n=N}^{\infty} b^n$ converges since $b < 1$. Thus by comparison $\sum_{n=1}^{\infty} a_n$ converges too.

If $a > 1$, then for $n \geq N$ $\sqrt[n]{a_n} > 1$ implies that $a_n > 1$, but then the terms in the series do not tend to zero and thus $\sum_{n=1}^{\infty} a_n$ diverges. $\square$

Theorem 1.2.12 (The Ratio Test)

Let a_n be a non-negative sequence and suppose that $\frac{a_{n+1}}{a_n} \rightarrow \ell$ as $n \rightarrow \infty$. Then if $\ell < 1$, $\sum_{n=1}^{\infty} a_n$ converges, and if $\ell > 1$, $\sum_{n=1}^{\infty} a_n$ diverges.

Proof. If $\ell < 1$, we can choose b such that $\ell < b < 1$, and there exists an integer N such that $\frac{a_{n+1}}{a_n} < b$ for all $n \geq N$. Therefore

$$\begin{aligned} a_n &= \frac{a_n}{a_{n-1}} \cdot \frac{a_{n-1}}{a_{n-2}} \cdots \frac{a_{N+1}}{a_N} \cdot a_N \\ &< (a_N b^{-N}) b^n, \end{aligned}$$

for $n > N$. But then $\sum_{n=N+1}^{\infty} b^n$ converges as $b < 1$, and since $a_N b^{-N}$ is a constant, $\sum_{n=1}^{\infty} a_n$ converges by comparison.

If $\ell > 1$, then $a_{n+1} > a_n$ and the terms in the series do not tend to zero and thus $\sum_{n=1}^{\infty} a_n$ diverges. $\square$

Remark (A Deadly Sin). In both the root and ratio test, if we find that either $a = 1$ or $\ell = 1$, we *cannot draw any conclusions about the convergence of the series*. To see this, consider $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$.

Some examples of using the root and ratio tests are shown below, but we won't do too many since there's a most likely a few already on your example sheets (and there's not really much to the basic technique).

Example 1.2.13 (Using the Ratio Test)

We will prove that $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges using the ratio test.

Let $a_n = n/2^n$. Then we have $\frac{a_{n+1}}{a_n} = \frac{n+1}{2^{n+1}} \cdot \frac{2^n}{n} = \frac{n+1}{2n}$, and thus $\frac{a_{n+1}}{a_n} \rightarrow \frac{1}{2} < 1$ as $n \rightarrow \infty$. Then by the ratio test $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

Example 1.2.14 (Using the Root Test)

We will prove that $\sum_{n=1}^{\infty} \left(\frac{n+1}{3n+5}\right)^n$ converges using the root test.

Let $a_n = \left(\frac{n+1}{3n+5}\right)^n$. Then $\sqrt[n]{a_n} = \frac{n+1}{3n+5}$, and thus $\sqrt[n]{a_n} \rightarrow \frac{1}{3}$ as $n \rightarrow \infty$. Then by the

root test $\sum_{n=1}^{\infty} \left(\frac{n+1}{3n+5}\right)^n$ converges.

In the case that we have our sequence of non-negative terms is decreasing, we can determine the convergence by looking at a related series involving a relatively small subsequence of terms.

Theorem 1.2.15 (Cauchy's Condensation Test)

If a_n is a decreasing sequence of positive numbers, then $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=0}^{\infty} 2^n a_{2^n}$ either both converge or both diverge.

Proof. It suffices to show that the partial sums of both series are either both bounded or both unbounded. Let $S_n = \sum_{j=1}^n a_j$ and $T_n = \sum_{j=0}^n 2^j a_{2^j}$ be the partial sums of each series.

If $n < 2^k$ we have

$$\begin{aligned} S_n &\leq a_1 + (a_2 + a_3) + \cdots + (a_{2^k} + \cdots + a_{2^{k+1}-1}) \\ &\leq a_1 + 2a_2 + \cdots + 2^k a_{2^k} = T_k, \end{aligned}$$

thus $S_n \leq T_k$. So if T_n is bounded so is S_n . Also if $n > 2^k$, we have

$$\begin{aligned} S_n &\geq a_1 + a_2 + (a_3 + a_4) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + \cdots + 2^{k-1}a_{2^k} = \frac{1}{2}T_k, \end{aligned}$$

so $2S_n \geq T_k$. So if S_n is bounded so is T_k . Thus S_n and T_n are either both bounded or both unbounded. $\square$

You may notice that this proof somewhat resembles our proof that the harmonic series diverges, and indeed you can show the following stronger result.

Theorem 1.2.16 (Convergence of $\sum n^{-\alpha}$)

$\sum_{n=1}^{\infty} \frac{1}{n^\alpha}$ converges if and only if $\alpha > 1$.

Proof. If $\alpha \leq 0$ then the terms in the series do not tend to zero, so the series diverges. So for $\alpha > 0$ we apply Cauchy's condensation test. This series converges if and only if the series

$$\sum_{k=0}^{\infty} 2^k \cdot \frac{1}{2^{\alpha k}} = \sum_{k=0}^{\infty} 2^{(1-\alpha)k}$$

converges. By comparison with the geometric series, this will converge if and only if $2^{1-\alpha} < 1$, that is, $\alpha > 1$. $\square$

In a way Cauchy's condensation test is slightly different to the others convergence tests since it just gives us a different series that may be easier to show convergence or divergence for.

Still, it is natural to reach for Cauchy's condensation test when the root or ratio tests fail. For example, all of the series below cannot be shown to converge or diverge using

the root or ratio tests:

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}, \quad \sum_{n=2}^{\infty} \frac{1}{n \log n}, \quad \sum_{n=2}^{\infty} \frac{1}{(\log n)^2}, \quad \sum_{n=2}^{\infty} \frac{1}{n(\log n)^2}, \quad \text{and} \quad \sum_{n=2}^{\infty} \frac{\log n}{n^2}.$$

If you try, you will find that these tests are inconclusive. However, all can be shown to converge/diverge using Cauchy's condensation test.²

Alternating Series

We will now look at a convergence test for series whose terms alternate from positive to negative.

Theorem 1.2.17 (Alternating Series Test)

If a_n is a decreasing sequence of positive numbers with $a_n \rightarrow 0$ as $n \rightarrow \infty$ then $\sum_{j=1}^{\infty} (-1)^{j+1} a_j$ converges.

Proof. Let $S_n = \sum_{j=1}^n (-1)^{j+1} a_j$ denote the partial sum of the series. Then $S_{2n} = S_{2n-2} + (a_{2n-1} - a_{2n}) \geq S_{2n-2}$. Also

$$S_{2n} = a_1 - (a_2 - a_3) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n} \leq a_1.$$

Therefore S_{2n} is increasing and bounded above, and thus converges. Now let $S_{2n} \rightarrow S$ as $n \rightarrow \infty$. Then $S_{2n+1} = S_{2n} + a_{2n+1} \rightarrow S + 0 = S$ as $n \rightarrow \infty$. Thus $S_n \rightarrow S$, and $\sum_{j=1}^{\infty} (-1)^{j+1} a_j$ converges. $\square$

Example 1.2.18

We will prove that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges using the alternating series test.^a

We note that $\frac{1}{n}$ is a decreasing sequence of positive numbers with $\frac{1}{n} \rightarrow 0$. Thus by the alternating series test $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges.

^aLater on we will see that this is equal to $\log 2$.

§1.2.3 Absolute Convergence

To finish our discussion on infinite series (for now!) we will introduce the stronger notion of *absolute convergence*. The main motivation for introducing this will become clear as we develop some of the properties of absolute convergence, but informally knowing that a series converges absolutely will allow us to be a little less careful when working with the series.

Definition 1.2.19 (Absolute Convergence)

For some sequence $a_n \in \mathbb{C}$, we say that the series $\sum_{n=1}^{\infty} a_n$ converges **absolutely** if the series $\sum_{n=1}^{\infty} |a_n|$ converges.

²You could also use the integration test (which we have not discussed), but that's most likely going to be slower as you have to integrate things.

We can see that absolute convergence is a strictly stronger property than ‘regular’ convergence. Indeed absolute convergence implies convergence, but not the other way round.

Theorem 1.2.20 (Absolute Convergence Implies Convergence)

If $\sum_{j=1}^{\infty} a_j$ converges absolutely then it converges.

Proof. By the triangle inequality we have

$$\left| \sum_{j=n}^m a_j \right| \leq \sum_{j=n}^m |a_j|,$$

and thus by Cauchy’s criterion $\sum_{j=1}^{\infty} a_j$ converges. $\square$

To see that the converse isn’t true (and that indeed it is a strictly stronger notion), consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$, which converges but does not converge absolutely. We sometimes say such series converge **conditionally**.

When trying to determine whether a series converges absolutely, since $|a_n| \geq 0$ we are free to apply all of the results that we developed in the previous subsection. We will see this in the next example.

Example 1.2.21 (Showing Absolute Convergence)

Consider the infinite series $\sum_{n=1}^{\infty} z^n/2^n$. We will determine for which values of z this infinite series converges absolutely.

By the root test, we know that $\sum_{n=1}^{\infty} |z^n/2^n|$ converges if $|z| < 2$ and diverges if $|z| > 2$. Also for $|z| = 2$ we can see that the terms of the series do not go to zero, so the series diverges.

Thus $\sum_{n=1}^{\infty} z^n/2^n$ converges absolutely for $|z| < 2$ and diverges for $|z| \geq 2$.

Now it was claimed at the start of this subsection that absolute convergence allows us to be a ‘little less careful’. Allow me to elaborate on this.

In general, *you must be very careful when manipulating infinite series*. To see why, consider this informal example.

$$\begin{aligned} 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots &= \log 2 \\ 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \cdots &= \frac{3}{2} \log 2 \end{aligned}$$

The two series above both converge, and they both have all of the same terms. The only difference is that the terms are in a different order, and this change has completely altered the value of the series.

What’s nice about absolutely convergent series is that *we don’t have to worry about this*.

Definition 1.2.22 (Rearrangement)

A **rearrangement** of the series $\sum_{n=1}^{\infty} a_n$ is another series $\sum_{n=1}^{\infty} a_{\sigma(n)}$, where $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ is a bijection.

Theorem 1.2.23 (Rearranging Absolutely Convergent Series)

If $\sum_{j=1}^{\infty} a_j$ is absolutely convergent then any rearrangement of the series will converge to the same value.

Proof. Let $\sum_{j=1}^{\infty} a_{\sigma(j)}$ be a rearrangement of the series. Given $\epsilon > 0$, we wish to show that there exists an integer N such that $|\sum_{j=1}^{\infty} a_j - \sum_{j=1}^n a_{\sigma(j)}| < \epsilon$ for all $n \geq N$.

By Cauchy's criterion there exists an integer m such that $\sum_{j=m+1}^{\infty} |a_j| < \epsilon$. We then choose N such that $\{a_1, a_2, \dots, a_m\} \subseteq \{a_{\sigma(1)}, a_{\sigma(2)}, \dots, a_{\sigma(N)}\}$. This can be done by setting $N = \max_{1 \leq i \leq m} \sigma^{-1}(i)$. Then if $n \geq N$ we have

$$\sum_{j=1}^{\infty} a_j - \sum_{j=1}^n a_{\sigma(j)} = \sum_{j \in A_n} a_j,$$

where $A_n \subseteq \{m+1, m+2, \dots\}$. Then applying the triangle inequality we have

$$\left| \sum_{j=1}^{\infty} a_j - \sum_{j=1}^n a_{\sigma(j)} \right| = \left| \sum_{j \in A_n} a_j \right| \leq \left| \sum_{j=m+1}^{\infty} a_j \right| \leq \sum_{j=m+1}^{\infty} |a_j| < \epsilon,$$

as required. □

To see just how not-true this is for series that do not converge absolutely, we just need to read the statement of Riemann's Rearrangement Theorem.

Theorem 1.2.24 (Riemann's Rearrangement Theorem)

Let $a_j \in \mathbb{R}$. Suppose that $\sum_{j=1}^{\infty} a_j$ is convergent but not absolutely convergent. Then given any $x \in \mathbb{R}$ there is a rearrangement of this series such that $\sum_{j=1}^{\infty} a_{\sigma(j)} = x$.

Proof. Left as an exercise – it's on the example sheet! □

§1.3 Continuity

The next idea we will develop is that of ‘continuity’.

§1.3.1 Continuity of Functions

Continuity is all about the behaviour of a function around a point. For a function to be continuous at some x , we need $f(y)$ to be close to $f(x)$ whenever y is sufficiently close to x .

Definition 1.3.1 (Continuity)

Let $A \subseteq \mathbb{C}$ and $f : A \rightarrow \mathbb{C}$. We say that f is **continuous at** $a \in A$ if given any $\epsilon > 0$ we can find a $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ for all $y \in A$ such that $|x - y| < \delta$.

We say that f is **continuous** if it is continuous at every $a \in A$.

You may notice that this definition uses exactly the definition of the limit of a function mentioned in [subsection 1.1.4](#). This immediately gives us two more equivalent definitions of continuity.

Proposition 1.3.2 (Limit Definition of Continuity)

Let $f : A \rightarrow \mathbb{C}$. Then f is continuous at $a \in A$ if and only if $\lim_{z \rightarrow a} f(z) = f(a)$.

Proposition 1.3.3 (Sequence Definition of Continuity)

Let $f : A \rightarrow \mathbb{C}$. Then f is continuous at $a \in A$ if and only if for every sequence $a_n \in A$ with $a_n \rightarrow a$ we have $f(a_n) \rightarrow f(a)$.

Each of the definitions have various advantages or disadvantages. Still it is worth noting that the sequential/limit continuity definitions can be used to quite easily show certain properties of continuity that would otherwise be quite fiddly with the ϵ - δ definition. Using sequential continuity is also a natural way to show that a function *is not* continuous at a point. Let’s take a look at some examples.

Example 1.3.4 (The Constant Function is Continuous)

If $f(x) = c$ then f is continuous. To see this, we can take any value of δ in the definition.

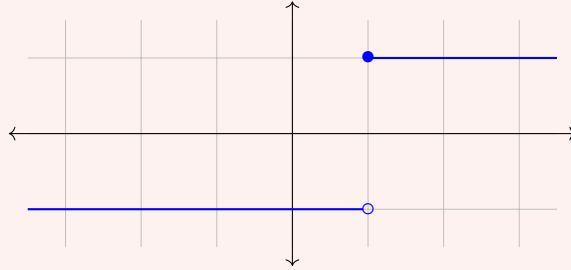
Example 1.3.5 ($f(x) = x$ is Continuous)

If $f(x) = x$ then f is continuous. To see this, we can take $\delta = \epsilon$ in the definition.

Example 1.3.6 (Step Functions are Discontinuous)

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with

$$f(x) = \begin{cases} -1 & \text{if } x < 1, \\ 1 & \text{if } x \geq 1. \end{cases}$$



This function is not continuous at 1. To see this, note that $1 - 1/n \rightarrow 1$ but $f(1 - 1/n) \rightarrow -1 \neq f(1)$ as $n \rightarrow \infty$.

Example 1.3.7 (The Domain Matters)

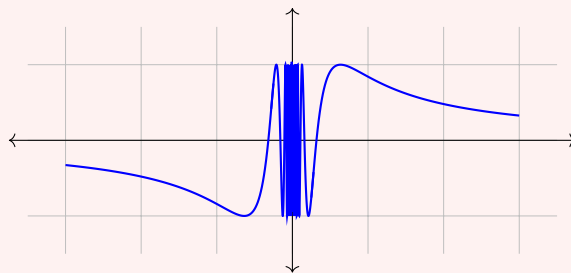
Consider the function $f : \mathbb{Q} \rightarrow \mathbb{R}$ with

$$\begin{cases} 1 & \text{if } x^2 > 2, \\ 0 & \text{if } x^2 < 2. \end{cases}$$

This function is continuous, since for every $a \in \mathbb{Q}$ there is an interval about a for which f is constant, so f is continuous at a . If f was defined on $\mathbb{R}$ instead of $\mathbb{Q}$, then it would be discontinuous only at $\pm\sqrt{2}$. But these points are not in our domain, so we don't need to worry about them.

Example 1.3.8 (Continuity of $\sin 1/x$)

Consider the function^a $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \sin(1/x)$ if $x \neq 0$ and $f(0) = 0$.



This function is not continuous at $x = 0$. To see this, consider the sequence $a_n = \frac{1}{2\pi n + \pi/2}$. Then $a_n \rightarrow 0$, but $f(a_n) = 1 \rightarrow 1 \neq f(0)$ as $n \rightarrow \infty$.

^aI know that we haven't defined what $\sin$ is, but we will use it in examples because they are instructive. If this bothers you, feel free to come back after our discussion of power series.

Example 1.3.9 (A Functional Equation with Continuity)

We wish to find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) + f(2x) = 0$.

Letting $x = 0$ we get $f(0) = 0$. Then rearranging we have $f(x) = -f(x/2)$, and repeatedly using this gives

$$f(x) = -f\left(\frac{x}{2}\right) = f\left(\frac{x}{4}\right) = -f\left(\frac{x}{8}\right) = \cdots = (-1)^n f\left(\frac{x}{2^n}\right).$$

Then since f is continuous we have $f(x) = \lim_{n \rightarrow \infty} (-1)^n f(x/2^n) = f(0) = 0$. Thus $f(x) = 0$ is the only such function, which clearly works.

When attempting to determine if a function is continuous, one should keep in mind the following properties of continuity (all of which follow directly from our basic properties of limits).

Proposition 1.3.10 (Sums, Products and Reciprocals of Continuous Functions)

Let $f, g : A \rightarrow \mathbb{C}$ be functions. Then the following hold.

- (i) If f and g are continuous at a , then so is their sum $f + g$.
- (ii) If f and g are continuous at a , then so is their product fg .
- (iii) If f is continuous at a and $f(a) \neq 0$ then so is $1/f$.

Proof. We prove each individually using the limit definition of continuity.

- (i) If $\lim_{z \rightarrow a} f(z) = f(a)$ and $\lim_{z \rightarrow a} g(z) = g(a)$ then $\lim_{z \rightarrow a} (f + g)(z) = (f + g)(a)$.
- (ii) If $\lim_{z \rightarrow a} f(z) = f(a)$ and $\lim_{z \rightarrow a} g(z) = g(a)$ then $\lim_{z \rightarrow a} (fg)(z) = (fg)(a)$.
- (iii) If $\lim_{z \rightarrow a} f(z) = f(a)$ and $f(a) \neq 0$, then $\lim_{z \rightarrow a} 1/f(z) = 1/f(a)$. □

Along with the fact that $f(x) = x$ and $f(x) = c$ are continuous, these properties imply that all polynomials are continuous.

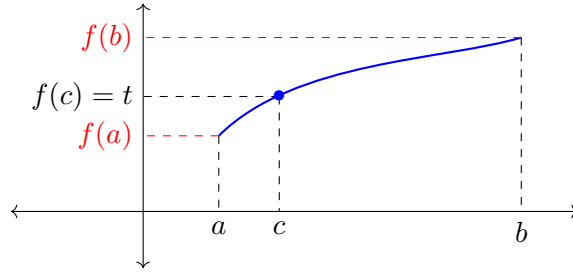
Proposition 1.3.11 (The Composition of Continuous Functions is Continuous)

Let $f : A \rightarrow B$ and $g : B \rightarrow \mathbb{C}$ with $A, B \subseteq \mathbb{C}$. Then if f is continuous at $a \in A$ and g is continuous at $f(a)$, then $g \circ f$ is also continuous at a .

Proof. If $a_n \rightarrow a$ then by the continuity of f we have $f(a_n) \rightarrow f(a)$, and by the continuity of g we have $g(f(a_n)) \rightarrow g(f(a))$. So $g \circ f$ is continuous at a . □

§1.3.2 The Intermediate Value Theorem

Continuous functions are nice because they have a few nice properties. The first of these that we will prove is the *intermediate value theorem*, a central theorem in elementary analysis. The idea of the intermediate value theorem is that if a continuous function starts at one value and ends at a different value, then it must take on all of the values in between.



In the proof below we will employ suprema³, but we will give another proof afterwards that does not use this (at the expense of being slightly longer).

Theorem 1.3.12 (The Intermediate Value Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then for every t between $f(a)$ and $f(b)$ there is an $c \in [a, b]$ with $f(c) = t$.

Proof. Suppose without loss of generality^a that $f(a) < t < f(b)$. Consider the set $S = \{x \in [a, b] : f(x) < t\}$. This set is bounded and since $a \in S$ it is non-empty. So we can let $c = \sup S$, and note that $a \leq c \leq b$.

Let $n \geq 1$ be an integer. Then since c is the supremum there must exist some $x_n \in S$ such that $c - \frac{1}{n} < x_n \leq c$. By the squeeze theorem we have $x_n \rightarrow c$ as $n \rightarrow \infty$. Also f is continuous so $f(x_n) \rightarrow f(c)$. We also constructed x_n to be in S , giving $f(x_n) < t$ for all n . This implies that $f(c) \leq t$.

We know that $c \in [a, b]$ but $c \neq b$, so there is some integer N such that for $n \geq N$ we have $c + 1/n \leq b$. Using that c is the supremum, we have $c + 1/n \notin S$ for all $n \geq N$, that is, $f(c + 1/n) \geq t$ for all $n \geq N$. Then by continuity we have $f(c) \geq t$.

But then $f(c) \geq t$ and $f(c) \leq t$, so we must have $f(c) = t$. $\square$

^aIf $t = f(a)$ or $t = f(b)$ then we are done. Also the case $f(a) > t > f(b)$ follows similarly.

Before we look at some examples of using the intermediate value theorem, there's a few things worth noting about the intermediate value theorem.

First of all, we say absolutely nothing about uniqueness in the theorem. It is very much possible for a function to take on an intermediate value multiple times. Second of all, when applying the intermediate value theorem (commonly abbreviated to IVT) a good general problem solving technique is trying to apply IVT to other related functions such as $g(x) = f(x) - x$ or $g(x) = f(x) - t$ and things like that. You will see this 'trick' show up in the examples below, and also again when we get to Rolle's theorem and the mean value theorem.

Example 1.3.13 (Fixed Points of Continuous Maps from $[0, 1]$ to Itself)

We will prove that if $f : [0, 1] \rightarrow [0, 1]$ is a continuous function then there is some $c \in [0, 1]$ such that $f(c) = c$.

Consider the function $g(x) = f(x) - x$. Then g is continuous with $g(0) = f(0)$ and $g(1) = f(1) - 1$. Since $0 \leq f(x) \leq 1$, we must have $g(0) \geq 0$ and $g(1) \leq 0$. Thus by

³If you are unfamiliar with this term and/or the least upper bound principle, feel free to have a look at Chapter 2 of my 'Numbers and Sets' course notes.

the intermediate value theorem there exists some $c \in [0, 1]$ such that $g(c) = 0$. But then we have $f(c) - c = 0$, that is, $f(c) = c$ as required.

Example 1.3.14 (Existence n th Roots)

We will show that for any positive integer n and $y > 0$ there exists a real number x such that $x^n = y$.

Consider the function $f(x) = x^n$. This function is continuous on $[0, 1 + y]$, and since $f(0) = 0$ and $f(1 + y) = (1 + y)^n$, we have $f(0) < y < f(1 + y)$. Thus by the intermediate value theorem there is some $x \in [0, 1 + y]$ such that $f(x) = y$, that is, $x^n = y$.

Now before we look on let's have a look at an alternative proof for the intermediate value theorem. This proof uses the same idea as the previous proof (try to see why) but avoids the use of suprema.

Aside: Hunting For Lions Intermediate Values

It was mentioned before that the ‘lion hunting’ technique that we used to prove the Bolzano–Weierstrass theorem could also be used to prove other theorems in Analysis. One such example is the intermediate value theorem.

The steps in the proof are quite similar to that of the proof of the Bolzano–Weierstrass theorem, the only difference is what we are ‘hunting for’, which in this case is our intermediate value, and how we finish off the proof.

Proof (Intermediate Value Theorem, Lion Hunting Style). Again suppose without loss of generality that $f(a) < t < f(b)$. We are going to define two sequences a_n and b_n inductively as follows. Begin by setting $a_1 = a$ and $b_1 = b$, and let $c = (a_1 + b_1)/2$ be the midpoint a_1 and b_1 .

Then there is two possibilities:

1. $f(c_1) \geq t$.
2. $f(c_1) < t$.

If the first one holds, we set $a_2 = a_1$ and $b_2 = c_1$, and if it doesn't then we set $a_2 = c_1$ and $b_2 = b_1$. Repeating this process, we construct a_n and b_n such that $f(a_n) \leq t \leq f(b_n)$, $a_{n-1} \leq a_n \leq b_n \leq b_{n-1}$, and also $b_n - a_n = (b_{n-1} - a_{n-1})/2$.

Now a_n is increasing and bounded above, and b_n is decreasing and bounded below and thus $a_n \rightarrow c \in [a, b]$, and $b_n \rightarrow c' \in [a, b]$. Then we have $c - c' = (c - c')/2$ using the above result, and thus $c = c'$.

Since f is continuous at c and $a_n \rightarrow c$, we know that $f(a_n) \rightarrow f(c)$ as $n \rightarrow \infty$. Since $f(a_n) \leq t$, we also know that $f(c) \leq t$. Similarly we also have $f(b_n) \geq t$, so $f(c) \geq t$. But then $t \leq f(c) \leq t$, and thus $f(c) = t$, and we are done. $\square$

To get a feel for the construction, try drawing a rough sketch of the process. If you consider the case of a function that takes on the desired intermediate value multiple times, you should also see that this proof does not give the same value for c as the

previous proof.

A natural question upon learning of the intermediate value theorem is if it is equivalent to definition of continuity. The answer to this question is no – there are functions that have the ‘intermediate value property’, without being continuous. One example is $\sin(1/x)$ which we showed is discontinuous at $x = 0$ in [Example 1.3.8](#). Another more elaborate example is ‘Conway’s Base-13 Function’, which has the intermediate value property but is discontinuous everywhere. You can read more about this function on [Wikipedia](#).

§1.3.3 The Boundedness Theorem

The last result about continuous functions that we will prove is the *boundedness theorem*, another central theorem in elementary analysis. The statement of this result is straightforward: a continuous function on a closed bounded interval is bounded and attains its bounds.

First we will prove that a continuous function defined on some closed interval is indeed bounded.

Lemma 1.3.15 (Continuous Function on Closed Interval is Bounded)

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous then there exists K such that $|f(x)| \leq K$ for all $x \in [a, b]$.

Proof. If f was unbounded, then given any integer $n \geq 1$ there exists $x_n \in [a, b]$ such that $|f(x_n)| > n$. By Bolzano–Weierstrass, x_n has a convergent subsequence $x_{n_j} \rightarrow x$. Also since $a \leq x \leq b$, we must have $x \in [a, b]$. By continuity of f , $f(x_{n_j}) \rightarrow f(x)$, but $|f(x_{n_j})| > n_j$, and thus does not tend to a limit. Thus we have a contradiction. $\square$

Now we can prove that such a continuous function attains its bounds, giving us the boundedness theorem.

Theorem 1.3.16 (Boundedness Theorem)

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous then there exists $x_1, x_2 \in [a, b]$ such that $f(x_1) \leq f(x) \leq f(x_2)$ for all $x \in [a, b]$.

Proof. By our previous lemma, we know that f is bounded. Now we define the set

$$A = \{f(x) : x \in [a, b]\}.$$

This set is bounded and non-empty and thus has a supremum $M = \sup A$. Then for each positive integer n , $M - 1/n$ cannot be an upper bound for A . This implies that there is some $x_n \in [a, b]$ such that $M - 1/n < f(x_n) \leq M$.

By Bolzano–Weierstrass, x_n has a convergent subsequence $x_{n_j} \rightarrow x$. Since $a \leq x_n \leq b$, we know $a \leq x \leq b$. By the continuity of f , $f(x_{n_j}) \rightarrow f(x)$, but $f(x_{n_j}) \rightarrow M$ by construction. So $f(x) = M$. So $x_2 = x$. The minimum follows analogously. $\square$

Alternate Proof. As before, let $M = \sup A$ and suppose that there was no x_2 such

that $f(x_2) = M$. Then let $g(x) = \frac{1}{M-f(x)}$ for $x \in [a, b]$. This function is well defined and continuous. Now g must be bounded by the previous lemma, so $g(x) \leq K$ for all $x \in [a, b]$, for some K . This means that $f(x) \leq M - 1/k$ on $[a, b]$. This is contradiction since we set M as the supremum. $\square$

§1.4 Differentiability

You are likely familiar with differentiability (and particularly the computation of derivatives) from calculus. While this knowledge should certainly not be disregarded, we are going to go from the beginning, doing everything with a little more care than it got in calculus. Throughout this section we will deal with functions $f : A \subseteq \mathbb{R} \rightarrow \mathbb{R}$, though the basic definitions will apply to $\mathbb{C}$.⁴

Here's our basic definition:

Definition 1.4.1 (Differentiability)

Let $A \subseteq \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. Then if x is a limit point of A , we say that f is **differentiable at x** with derivative $f'(x)$ if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x).$$

We say that f is **differentiable** on A if f is differentiable at every $x \in A$.

Let's pause for a moment. The core idea of differentiation is we want to *approximate a function around some point using a linear map*⁵. Indeed, our definition of the derivative is directly equivalent to the following: f is differentiable at x if

$$f(x+h) = f(x) + f'(x)h + \varepsilon(h),$$

where $\lim_{h \rightarrow 0} \varepsilon(h)/h = 0$. That is, it is differentiable if we can approximate the function with some linear map where the error decreases faster than linearly.

§1.4.1 Properties of Differentiable Functions

With that spiel over, we can move on to some properties of differentiable functions. The first is that differentiability implies continuity.

Proposition 1.4.2 (Differentiability Implies Continuity)

If f is differentiable at x , then f is continuous at x .

Proof. Using the limit definition of continuity we have

$$\lim_{h \rightarrow 0} [f(x+h) - f(x)] = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot (h) = f'(x) \cdot 0 = 0,$$

thus $\lim_{h \rightarrow 0} f(x+h) = f(x)$, so f is continuous at x . □

Now we can prove some basic rules for computing derivatives.

⁴We will use differentiation over $\mathbb{C}$ in our discussion of power series though. One should also note that the other basic rules such as the sum, product and chain rules will also hold over $\mathbb{C}$, but being differentiable over $\mathbb{C}$ is a stronger condition than being differentiable over $\mathbb{R}$ or $\mathbb{R}^2$. For example, consider $f(z) = \bar{z}$, which is not complex differentiable.

⁵Indeed it is which this idea, not really that of 'the tangent to a curve' that is used to generalise the derivative to functions of multiple variables. Of course they are closely related.

Proposition 1.4.3 (Sum, Product & Quotient Rules)

Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable at x with derivatives $f'(x)$ and $g'(x)$. Then the following hold.

- (i) $f + g$ is differentiable at x , with $(f + g)'(x) = f'(x) + g'(x)$,
- (ii) fg is differentiable at x , with $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$,
- (iii) If $g'(x) \neq 0$ then f/g is differentiable at x , with $(f/g)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$.

Proof. We prove each individually.

(i) This follows from the sum of limits.

(ii) We have by the continuity of f

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[f(x+h) \frac{g(x+h) - g(x)}{h} + g(x) \frac{f(x+h) - f(x)}{h} \right], \end{aligned}$$

and by the continuity of f and g at x we have $(fg)'(x) = f(x)g'(x) + f'(x)g(x)$.

(iii) Similarly we have

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{f(x+h)/g(x+h) - f(x)/g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{g(x)g(x+h)} \left[g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h} \right], \end{aligned}$$

and by the continuity of f and g we have $(f/g)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}$. $\square$

We will now prove the chain rule, which tells us how to compute the derivative of the composition of functions. Unfortunately this proof is quite ‘tricky’, and trying to do something like we did in the proof above will not work. Instead, we need to return to our equivalent definition of differentiability, with $f(x+h) = f(x) + f'(x)h + \varepsilon_f(h)$, where $\lim_{h \rightarrow 0} \varepsilon_f(h)/h = 0$.

Proposition 1.4.4 (Chain Rule)

Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at x and $g : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at $f(x)$. Then $g \circ f$ is differentiable at x with $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$.

Proof. We have

$$\begin{aligned} f(x+h) &= f(x) + f'(x)h + \varepsilon_f(h) \\ g(f(x) + h) &= g(f(x)) + g'(f(x))h + \varepsilon_g(h), \end{aligned}$$

where $\varepsilon_f(h)/h, \varepsilon_g(h)/h \rightarrow 0$ as $h \rightarrow 0$. From this we obtain

$$g(f(x+h)) = g(f(x) + f'(x)h + \varepsilon_f(h))$$

$$= g(f(x)) + g'(f(x))(f'(x)h + \varepsilon_f(h)) + \varepsilon_g(f'(x)h + \varepsilon_f(h)).$$

Rearranging we get

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{g(f(x+h)) - g(f(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{g'(f(x))(f'(x)h + \varepsilon_f(h)) + \varepsilon_g(f'(x)h + \varepsilon_f(h))}{h} \\ &= g'(f(x))f'(x) + \lim_{h \rightarrow 0} g'(f(x)) \frac{\varepsilon_f(h)}{h} + \lim_{h \rightarrow 0} \frac{\varepsilon_g(f'(x)h + \varepsilon_f(h))}{h}. \end{aligned}$$

Then as $h \rightarrow 0$ we have $\varepsilon_f(h)/h \rightarrow 0$ and $f'(x)h + \varepsilon_f(h) \rightarrow 0$ so $\varepsilon_g(f'(x)h + \varepsilon_f(h))/h \rightarrow 0$, and thus we have $(g \circ f)'(x) = g'(f(x))f'(x)$. $\square$

Example 1.4.5 (Using the Chain Rule)

Let $f(x) = \sin(x^2)$. Then since $(\sin x)' = \cos x$ we have $f'(x) = 2x \cos(x^2)$.

Example 1.4.6 (A Continuous Everywhere But Non-Differentiable Function)

Consider the function $f(x) = x \sin(1/x)$ when $x \neq 0$, and $f(0) = 0$. This function is continuous everywhere^a, but we will show that it is not differentiable at $x = 0$.

At $x \neq 0$ this is differentiable by the chain rule, but at $x = 0$ we would need the following limit to exist:

$$\lim_{h \rightarrow 0} \frac{(0+h) \sin\left(\frac{1}{x+h}\right) - 0 \sin\left(\frac{1}{x}\right)}{h} = \lim_{h \rightarrow 0} \sin\left(\frac{1}{h}\right),$$

but this limit does not exist (a similar construction to [Example 1.3.8](#) will show this), and thus f is not differentiable at $x = 0$.

^aFeel free to check this!

Now in the example above we have a continuous everywhere function that's not differentiable at 0. A related example is one of a function that is differentiable everywhere (and hence continuous), but whose derivative is discontinuous.

Example 1.4.7 (A Function with Discontinuous Derivative)

Consider the function $f(x) = x^2 \sin(1/x)$ when $x \neq 0$ and $f(0) = 0$.

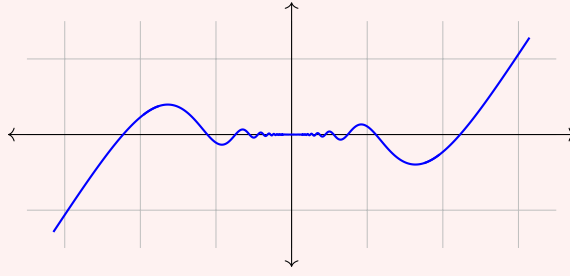
For $x \neq 0$ we have that f is differentiable by the chain rule, and the derivative is given by

$$f'(x) = x \sin\left(\frac{1}{x}\right) + \cos\left(\frac{1}{x}\right).$$

For $x = 0$, we compute the derivative directly. We have

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin\left(\frac{1}{h}\right)}{h} = 0.$$

Thus we can see that f' is not continuous, since $f'(x) \not\rightarrow 0$ as $x \rightarrow 0$ (as $\cos 1/x$ is not continuous).



There is however a limit as to how discontinuous a derivative can be. In particular the derivative of a differentiable function must have the intermediate value property.

Theorem 1.4.8 (Darboux's Theorem)

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is differentiable then f' has the intermediate value property. That is to say, if $a < b$ and $f'(a) < z < f'(b)$ then there exists $c \in (a, b)$ with $f'(c) = z$.

Proof. Given $a < b$ and $z \in \mathbb{R}$ such that $f'(a) < z < f'(b)$, we wish to show there exists $c \in (a, b)$ such that $f'(c) = z$. We can rewrite the condition as $f'(a) - z < 0 < f'(b) - z$. Now define $g(x) = f(x) - zx$, and note that we have $g'(a) < 0 < g'(b)$.

We want to find $c \in (a, b)$ such that $g'(c) = 0$. Now, g is continuous (since f is continuous), and thus is bounded on $[a, b]$, and the minimum is attained say at some point k . We can't have $k = a$, as that would imply that $g'(a) \geq 0$, and we also can't have $k = b$ as that would imply $g'(b) \leq 0$. Thus $k \in (a, b)$, and we must then have $g'(k) = 0$, and we are done. $\square$

§1.4.2 Rolle's Theorem & The Mean Value Theorem

The sign of a function's derivative at a point can tell us quite a bit about the behaviour of that function near that point. In particular, knowing that the derivative at a point vanishes can be quite useful;

Definition 1.4.9 (Local Maxima & Minima)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. If there exists δ such that $|x - x_0| < \delta$ implies that $f(x_0) \geq f(x)$, then we call x a **local maximum**. Similarly if this implies that $f(x_0) \leq f(x)$, we call x a **local minimum**.

Our important result is as follows:

Lemma 1.4.10 (Derivative of Maxima)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. If x is a local maximum or minimum of f and f is differentiable at x then $f'(x) = 0$.

Proof. Without loss of generality, assume that x is a local maximum. Then there exists δ such that $|h| < \delta_1$ implies that $f(x + h) \leq f(x)$.

Then if $0 < h < \delta$ we have

$$\frac{f(x+h) - f(x)}{h} \geq 0,$$

and thus we have $f'(x) \geq 0$.

Similarly if $-\delta < h < 0$ we have

$$\frac{f(x+h) - f(x)}{h} \leq 0,$$

and thus we have $f'(x) \leq 0$. Hence $f'(x) = 0$. $\square$

When we combine this result with the boundedness theorem from our discussion about continuity, we end up with *Rolle's theorem*.

Theorem 1.4.11 (Rolle's Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function which is differentiable on (a, b) . Then if $f(a) = f(b)$ then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Proof. Since f is differentiable, it is also continuous and thus by the boundedness theorem f is bounded on $[a, b]$ and these bounds are achieved. Thus we can let $M = \max_{x \in [a, b]} f(x)$ and $m = \min_{x \in [a, b]} f(x)$.

If $M = m$, then f must be constant and $f'(x) = 0$ for all $x \in (a, b)$, so we are done. Otherwise $M > f(a)$ or $m < f(a)$. If $M > f(a)$ then since the bounds are attained there exists $c \in (a, b)$ such that $f(c) = M$. But then M is a maximum, so by our previous lemma we have $f'(c) = 0$. A similar argument works if $m < f(a)$. $\square$

A direct consequence of Rolle's theorem⁶ is another classic theorem of analysis, the *mean value theorem*, which is frequently abbreviated to MVT.

Theorem 1.4.12 (The Mean Value Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function which is differentiable on (a, b) . Then there exists $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$.

Proof. Let $k = \frac{f(a) - f(b)}{a - b}$, and define $\phi(x) = f(x) - kx$. Note that $\phi(a) = \phi(b)$. Then by Rolle's theorem there exists $c \in (a, b)$ such that $\phi'(c) = 0$, that is $f'(c) = k$. $\square$

The mean value theorem says something notable: the size of the derivative controls the size of the function, or (in rougher terms) it puts a restriction on how 'badly behaved' the function can be. Also, if we appeal to geometric intuition (as we have tried not to, it's an easy way to go wrong) we can see that the mean value theorem says, as R. P. Burn wrote in *Numbers and functions*, "for the graph of a differentiable function, there is always a tangent parallel to the chord". Of course, we will quickly move on from geometrical thinking.⁷

⁶It's possible to establish this theorem without Rolle's theorem, and then Rolle's theorem pops out as a special case. The proof is (more or less) just what we did for Rolle's theorem laid out explicitly in the proof of this theorem.

⁷I guess here is a good place to mention how we can reason geometrically in analysis. Drawing pictures

Now let's think about applying the mean value theorem to two different functions. Suppose that $f, g : [a, b] \rightarrow \mathbb{R}$ is continuous and differentiable on (a, b) , and $g(a) \neq g(b)$. Then the mean value theorem gives us $s, t \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{(b-a)f'(s)}{(b-a)g'(t)} = \frac{f'(s)}{g'(t)}.$$

A stronger version of the mean value theorem says that we can take $s = t$.

Theorem 1.4.13 (Cauchy's Mean Value Theorem)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions which are differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$(f(b) - f(a))g'(c) = f'(c)(g(b) - g(a)).$$

Proof. We define the function

$$\begin{aligned} \phi(x) &= \begin{vmatrix} 1 & 1 & 1 \\ f(a) & f(x) & f(b) \\ g(a) & g(x) & g(b) \end{vmatrix} \\ &= [f(x)g(b) - f(b)g(x)] - [f(a)g(b) - f(b)g(a)] + [f(a)g(x) - f(x)g(a)]. \end{aligned}$$

Then ϕ is continuous on $[a, b]$ and differentiable on (a, b) ^a. Also $\phi(a) = \phi(b) = 0$. Then by Rolle's theorem, there exists $c \in (a, b)$ such that $\phi'(c) = 0$.

Differentiating ϕ we then have

$$\phi'(x) = f'(x)[g(b) - g(a)] + g'(x)[f(a) - f(b)],$$

and thus $\phi'(c) = 0$ gives the desired result. $\square$

^aIf the use of determinants bothers you, then feel free to just look at the expansion. If also remember how integrals work, you can obtain ϕ from working backwards from what we need to arrive at.

Cauchy's mean value theorem has many applications. For example, we can use the mean value theorem to establish L'Hôpital's rule for evaluating limits.

Example 1.4.14 (L'Hôpital's Rule)

We wish to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$.

By Cauchy's mean value theorem, there exist $c \in (0, x)$ such that $\frac{e^x - e^0}{\sin x - \sin 0} = \frac{e^c}{\cos c}$. Then $c \rightarrow 0$ as $x \rightarrow 0$, and thus we get

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x} = \lim_{c \rightarrow 0} \frac{e^c}{\cos c} = 1.$$

In more generality:

and thinking geometrically is a great way to understand how things work, to come up with counterexamples and much more. Still it is important to remember that *basically nothing in this course is provable by appealing to geometric intuition*. Instead this type of thinking should just inform the 'analysis' side of us how to approach things.

Theorem 1.4.15 (L'Hôpital's Rule)

Suppose $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous and differentiable on (a, b) . Suppose that $f(a) = g(a) = 0$, that $g'(x)$ does not vanish near a and $f'(x)/g'(x) \rightarrow \ell$ as $x \rightarrow a$. Then $f(x)/g(x) \rightarrow \ell$ as $x \rightarrow a$.

Proof. Since $g'(x)$ does not vanish near a , we can suppose that $g'(x) \neq 0$ for $x \in (a, b)$, as otherwise we could just consider the subinterval (a, b') , defined so that this is the case.

By Cauchy's mean value theorem we have

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)}{g'(c)}$$

where $c \in (a, x)$. Then $c \rightarrow a$ as $x \rightarrow a$, and hence $f(x)/g(x) \rightarrow \ell$ as $x \rightarrow a$. $\square$

The mean value theorem can also help us extend [Lemma 1.4.10](#). Knowing the sign of a derivative over some interval can tell us immediately if it is constant, increasing or decreasing.

Proposition 1.4.16 (Sign of the Derivative)

Let $f : [a, b] \rightarrow \mathbb{R}$ be differentiable on (a, b) . Then the following hold.

- (i) If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing.
- (ii) If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant.
- (iii) If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing.

Proof. By the mean value theorem we have $f(x_2) - f(x_1) = (x_2 - x_1)f'(x)$ where $a < x_1 < x < x_2 < b$. Then all of the results follow by considering the sign of $f'(x)$. $\square$

§1.4.3 Inverses of Functions

Before we move to slightly more exciting material, there's a few things about inverses of functions that we need to get out of the way. The proofs below are straightforward but require some care (by which I mean you have to avoid messing up which letter means what).

First of all, if we know that a continuous function is strictly increasing⁸ then we can deduce that the function has an inverse that is continuous and strictly increasing also. A similar result holds for strictly decreasing functions.

Theorem 1.4.17 (Continuous Inverse Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous and strictly increasing function. Then if $c = f(a)$ and $d = f(b)$, $f : [a, b] \rightarrow [c, d]$ is bijective and the inverse $f^{-1} : [c, d] \rightarrow [a, b]$ is

⁸If a continuous function was not strictly increasing or strictly decreasing then we couldn't have a unique inverse.

continuous and strictly increasing.

Proof. As f is strictly increasing, it is an injection. It is also a surjection since if $c < t < d$ then there exists $x \in (a, b)$ such that $f(x) = t$. Thus f is a bijection, and has an inverse f^{-1} .

Since f is strictly increasing, we can write this as $x_1 < x_2 \iff f(x_1) < f(x_2)$, so letting $f(x_1) = y_1$ and $f(x_2) = y_2$ we have $f^{-1}(y_1) < f^{-1}(y_2) \iff y_1 < y_2$. Thus f^{-1} is also strictly increasing.

Now we will show f^{-1} is continuous at some $y \in [c, d]$. Given $\epsilon > 0$, let $x = f^{-1}(y)$. If $x \neq a, b$, we can find δ such that

$$f(x - \epsilon) \leq f(x) - \delta < f(x) < f(x) + \delta \leq f(x + \epsilon).$$

But then $|z - y| < \delta$ implies that $x - \epsilon < f^{-1}(z) < x + \epsilon$, so $|f^{-1}(z) - f^{-1}(y)| < \epsilon$. Thus f^{-1} is continuous on (a, b)

Otherwise, if $x = a$ we have $f(x) < f(x + \epsilon)$. Then $|z - y| < f(x + \epsilon) - f(x)$ implies that $|f^{-1}(z) - f^{-1}(y)| < \epsilon$, so f^{-1} is continuous at c . A similar argument shows that it is continuous at d . $\square$

Now this *is* the differentiability section, so we can also describe what the derivative of a differentiable function's inverse is. This result is known as the one variable *inverse function theorem*.

Theorem 1.4.18 (Inverse Function Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, strictly increasing, and differentiable on (a, b) . Then let $f(a) = c$ and $f(b) = d$. Then $f : [a, b] \rightarrow [c, d]$ is a bijection, and f^{-1} is differentiable on (c, d) , with

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}.$$

Proof. By our previous theorem, f is a bijection and f^{-1} is a continuous and strictly increasing function. Let $y = f(x)$, with $x \in (a, b)$. We wish to show that $g'(y) = 1/f'(x)$.

Given $k \neq 0$, let h be given by $y + k = f(x + h)$, that is, $f^{-1}(y + k) = x + h$, where $h \neq 0$. Then

$$\frac{f^{-1}(y + k) - f^{-1}(y)}{k} = \frac{h}{f(x + h) - f(x)}.$$

Then since $f'(x) > 0$ and f^{-1} is continuous (so $h \rightarrow 0$ as $k \rightarrow 0$) we have $k \rightarrow 0$

$$\lim_{k \rightarrow 0} \frac{f^{-1}(y + k) - f^{-1}(y)}{k} = \lim_{h \rightarrow 0} \frac{x + h - x}{f(x + h) - f(x)} = \frac{1}{f'(x)},$$

as required. $\square$

§1.4.4 Taylor's Theorem

We are now going to think about functions where we can take derivatives a number of times (getting higher-order derivatives). We will begin by looking at how we can get an analogue of Rolle's theorem for higher-order derivatives. I will apologise in advance that this discussion is a bit long, so feel free to just skip to Taylor's theorem.

The following generalisation will come directly from Rolle's theorem.

Theorem 1.4.19 (Higher-Order Rolle's Theorem)

Let f be continuous and $n - 1$ -times differentiable on $[a, b]$ and also n -times differentiable on (a, b) . Then if $f(a) = f'(a) = f''(a) = \dots = f^{(n-1)}(a) = 0$ and $f(b) = 0$ then there is a $c \in (a, b)$ such that $f^{(n)}(c) = 0$.

Proof. Since $f(a) = f(b)$, by Rolle's theorem there exists $c_1 \in (a, b)$ such that $f'(c_1) = 0$. Then since $f'(a) = f'(c_1) = 0$, we can use Rolle's theorem again to obtain $c_2 \in (a, c_1)$ such that $f''(c_2) = 0$. Continuing on like this we obtain a $c_n \in (a, b)$ such that $f^{(n)}(c_n) = 0$, as required. $\square$

We can also attempt to get some sort of 'higher-order' mean value theorem. Recall that our proof of the mean value theorem was more or less as follows:

Define the function $\phi(x) = f(x) - kx$, where we choose k such that the conditions of Rolle's theorem are satisfied. Then apply Rolle's theorem to ϕ .

We can try and do the same with a more elaborate construction. We will do this in three steps.

1. Construct a polynomial $P(x)$ such that $P(a) = f(a)$, $P'(a) = f'(a)$, and so on until $P^{(n-1)}(a) = f^{(n-1)}(a)$.
2. Construct another polynomial⁹ $E(x)$ such that $E^{(r)}(a) = 0$ for $r = 0, 1, \dots, n - 1$, but with $E(b) = f(b) - P(b)$.
3. Take $\phi(x) = f(x) - P(x) - E(x)$.

If we construct $\phi(x)$ in this way, then we will have $\phi(a) = \phi'(a) = \dots = \phi^{(n-1)}(a) = 0$, and also $\phi(b) = 0$, so it will satisfy our higher-order Rolle's theorem.

Aside: Constructing Our Polynomials

While it is likely you have seen a construction for P before, I'd like to take a minute to spell it out in detail because it has some nice ideas.

To construct this polynomial, what we really need is some construction of a term which affects the value of the polynomial at a point a if and only if we are considering a certain derivative. We can do this as follows:

$$Q_k(x) = \frac{(x - a)^k}{k!}.$$

⁹We can think of this as the 'error correcting term', because it fixes the discrepancy between $f(b)$ and $P(b)$ while not ruining the previous construction.

We can then (by taking derivatives) find that the value of this term is

$$Q_k^{(j)}(a) = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k \end{cases}$$

With this we can immediately write down an explicit construction for $P(x)$.

$$\begin{aligned} P(x) &= \sum_{k=0}^{n-1} f^{(k)}(a) Q_k(x) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (x-a)^k \\ &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!} (x-a)^{n-1}. \end{aligned}$$

This polynomial is known as the **Taylor polynomial** of degree $n-1$ for f about a .

We can use the exact same method to construct $E(x)$. We need to not affect the first $n-1$ derivatives at $x=a$, so we will need to add some multiple of $Q_n(x)$. However, this time we care about the value of the undifferentiated term at $x=b$:

$$Q_n(b) = \frac{(b-a)^n}{n!}.$$

Since we want $E(b) = f(b) - P(b)$, if we are going to have $E(x) = \lambda Q_n(x)$ we are going to need

$$\begin{aligned} \lambda Q_n(b) &= \lambda \frac{(b-a)^n}{n!} = f(b) - P(b) \\ \implies \lambda &= \frac{[f(b) - P(b)] \cdot n!}{(b-a)^n}. \end{aligned}$$

And thus we get

$$\begin{aligned} E(x) &= \frac{[f(b) - P(b)] \cdot n!}{(b-a)^n} Q_n(x) \\ &= [f(b) - P(b)] \left(\frac{x-a}{b-a} \right)^n. \end{aligned}$$

Now with the constructions as above, we can write down $\phi(x) = f(x) - P(x) - E(x)$. Then since $\phi(x)$ satisfies all of the conditions of our higher-order Rolle's theorem, there exists $c \in (a, b)$ such that $\phi^{(n)}(c) = 0$.

Taking the n th derivative, since P is a degree $n-1$ polynomial and thus has a zero derivative, we get

$$\begin{aligned} \phi^{(n)}(c) &= f^{(n)}(c) - [f(b) - P(b)] \cdot \frac{n!}{(b-a)^n} = 0 \\ \implies f(b) &= P(b) + \frac{f^{(n)}(c)}{n!} (b-a)^n. \end{aligned}$$

for some $c \in (a, b)$.

This result is Taylor's theorem, specially Taylor's theorem with the Lagrange form of the remainder. For completeness, this proof is given in a standalone form below.

Theorem 1.4.20 (Taylor's Theorem with Lagrange Remainder)

Let f be continuous and $(n-1)$ -times differentiable on $[a, b]$ and n -times differentiable on (a, b) . Then we have

$$f(b) = f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \frac{f^{(n)}(c)}{n!}(b-a)^n,$$

for some $c \in (a, b)$.

Proof. We define

$$\phi(x) = f(x) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(x-a)^k - M \left(\frac{x-a}{b-a} \right)^n,$$

where M is chosen such that $\phi(a) = \phi(b) = 0$. Then differentiating we have $\phi(a) = \phi'(a) = \cdots = \phi^{(n-1)}(a) = 0$.

Then since $\phi(a) = \phi(b)$, there exists $c_1 \in (a, b)$ such that $\phi'(c_1) = 0$. Similarly $\phi'(a) = \phi'(c_1) = 0$, and thus there exists $c_2 \in (a, c_1)$ such that $\phi''(c_2) = 0$. Continuing on in this way, we find a c_n such that $\phi^{(n)}(c_n) = 0$, where $c_n \in (a, b)$.

Then differentiating again we have

$$\phi^{(n)}(c) = f^{(n)}(c) - \frac{M \cdot n!}{(b-a)^n} = 0.$$

Now we know that M is such that $\phi(b) = 0$, so we get

$$M = f(b) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(b-a)^k.$$

Thus we have

$$\begin{aligned} f^{(n)}(c) &= \frac{M \cdot n!}{(b-a)^n} = \left(f(b) - \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!}(b-a)^k \right) \cdot \frac{n!}{(b-a)^n} \\ \implies f(b) &= f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \frac{f^{(n)}(c)}{n!}(b-a)^n, \end{aligned}$$

where $c \in (a, b)$ as required. □

So we have Taylor's theorem (which in this case can be thought of as a higher-order mean value theorem), let's try and do something with it. One of the main uses of Taylor's theorem is in giving us an infinite series which converges to the value of our function. This involves considering the terms of the degree n Taylor polynomial for increasing values of n , as we shall see.

Doing this type of problem has two main steps.

1. Write down Taylor's theorem for the case of the function being considered.
2. Show that the error term tends to zero as n goes to infinity.

We see how this is done using a Tripos question from 2010 (so eh spoilers I guess).

Example 1.4.21 (Using Taylor's Theorem – Part IA, 2010 Paper 1, Q10)

Problem. Suppose that $e : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $e(0) = 1$ and $e'(x) = e(x)$ for all $x \in \mathbb{R}$. Use Taylor's theorem with the remainder in Lagrange's form to prove that

$$e(x) = \sum_{n \geq 0} \frac{x^n}{n!} \quad \text{for all } x \in \mathbb{R}$$

[No property of the exponential function may be assumed.]

Solution. We begin by noting that $e(x)$ is infinitely differentiable, with $e^{(n)}(x) = e(x)$. Indeed this is true for $n = 1$, and then it follows by induction for all n . Thus Taylor's theorem with Lagrange Remainder holds for all $n \in \mathbb{N}$.

Thus for all $n \in \mathbb{N}$ we have (using $e^{(n)}(0) = 1$ and $e^{(n)}(c) = e(c)$)

$$e(x) = \sum_{k=0}^{n-1} \frac{x^k}{k!} + \frac{e(c)x^n}{n!}$$

for some $c \in (0, x)$. And thus it suffices to show that $\frac{e(c)x^n}{n!} \rightarrow 0$ as $n \rightarrow \infty$, since that would imply that $e(x) - \sum_{k=0}^{n-1} \frac{x^k}{k!} \rightarrow 0$ as $n \rightarrow \infty$. Now $e(t)$ is differentiable and hence continuous on $[0, x]$, and thus is bounded. So it suffices to show that $x^n/n! \rightarrow 0$, which holds.

Thus

$$e(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

as required.

Now it is *not* the case that the remainder term always goes to zero. A common counterexample is $f(x) = \exp(-1/x^2)$ for $x \neq 0$ with $f(0) = 0$.

§1.5 Power Series

Following on from our discussion of infinite series in Chapter 2, we are going to discuss a particular type of infinite series known as a *power series*. Throughout this section, we are going to work mostly in $\mathbb{C}$.

Definition 1.5.1 (Complex Power Series)

A **complex power series** is an infinite series of the form

$$\sum_{n=0}^{\infty} a_n z^n,$$

where $z \in \mathbb{C}$ and $a_n \in \mathbb{C}$.

In this section we will look at some properties of power series, and we will use them to (finally) define some functions that we have alluded to quite a bit.

§1.5.1 Radius of Convergence

Before we develop the concept of the radius of convergence, we first need to develop a lemma about the convergence of power series.

Lemma 1.5.2 (Absolute Convergence of Power Series)

Suppose that $\sum_{n=0}^{\infty} a_n z_0^n$ converges for some $z_0 \in \mathbb{C}$. Then $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely for all $z \in \mathbb{C}$ with $|z| < |z_0|$.

Proof. Since $\sum_{n=0}^{\infty} a_n z_1^n$ converges, we have $a_n z_1^n \rightarrow 0$, implying this sequence is bounded. Then there is some $K > 0$ such that $|a_n z_1^n| < K$ for all n .

So if $|z| < |z_0|$, we have $|a_n z^n| \leq K |z/z_1|^n$. Since the geometric series $\sum_{n=0}^{\infty} |z/z_1|^n$ converges, our result follows by comparison. $\square$

With this lemma we can prove one of our key results in the study of power series, the existence of the *radius of convergence*.

Theorem 1.5.3 (Radius of Convergence)

Suppose that $a_n \in \mathbb{C}$. Then either the power series $\sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in \mathbb{C}$, or there exists a real number R with $R \geq 0$ such that

- (i) $\sum_{n=0}^{\infty} a_n z^n$ converges absolutely if $|z| < R$,
- (ii) $\sum_{n=0}^{\infty} a_n z^n$ diverges if $|z| > R$.

We say that R is the **radius of convergence** of the power series.

Proof. If $\sum_{n=0}^{\infty} a_n z^n$ converges for all $z \in \mathbb{C}$, then we are done. Otherwise, there must exist $z_1 \in \mathbb{C}$ such that $\sum_{n=0}^{\infty} a_n z_1^n$ diverges. Then by our previous lemma the power series must diverge for all $z \in \mathbb{C}$ where $|z| > |z_1|$.

Now define the set $S = \{|z| : \sum_{n=0}^{\infty} a_n z^n \text{ converges}\}$. This set is bounded above by $|z_1|$, and is non-empty since the power series converges for $z = 0$. Thus it has a supremum, say $R = \sup S$.

Then by definition $\sum_{n=0}^{\infty} a_n z^n$ diverges if $|z| > R$, and then suppose $|z| < R$. Then there must be some z_0 with $|z| < |z_0| < R$ such that the power series converges for z_0 (by the definition of the supremum again). Then by our previous lemma the power series converges absolutely for z . $\square$

What's quite lovely is that when we are inside the radius of convergence, the power series converges absolutely! This means that we can avoid all of those issues with rearranging series that show up when things don't converge absolutely. This can be quite helpful when actually working with power series.

What's also quite nice is that we can employ all of those lovely results we developed in [section 1.2](#) to find what the radius of convergence of a power series is. Let's look at two lemmas that come from the ratio and root tests, which can give the radius of convergence.

Lemma 1.5.4 (The Ratio Test for Radius of Convergence)

Let $\sum_{n=0}^{\infty} a_n z^n$ be a power series with radius of convergence R . Then if $|a_{n+1}/a_n| \rightarrow \ell$ as $n \rightarrow \infty$, we have $R = 1/\ell$ if $\ell \neq 0$, and convergence everywhere if $\ell = 0$.

Proof. By the ratio test, we have absolute convergence if

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \cdot \frac{z^{n+1}}{z^n} \right| < 1.$$

So if $\ell = 0$, then we have convergence everywhere. Alternatively, if $|z| < 1/\ell$ we have absolute convergence, and if $|z| > 1/\ell$ the series diverges, again by the ratio test. $\square$

Lemma 1.5.5 (The Root Test for Radius of Convergence)

Let $\sum_{n=0}^{\infty} a_n z^n$ be a power series with radius of convergence R . Then if $|a_n|^{1/n} \rightarrow \ell$ as $n \rightarrow \infty$, we have $R = 1/\ell$.

Proof. By the root test, we have absolute convergence if

$$\lim_{n \rightarrow \infty} |a_n z^n|^{1/n} < 1.$$

So if $\ell = 0$, then we have convergence everywhere. Alternatively, if $|z| < 1/\ell$ we have absolute convergence, and if $|z| > 1/\ell$ the series diverges, again by the root test. $\square$

Now it is time for an important conceptual point: if we have a power series with radius of convergence R , then this only specifies the convergence/divergence inside or outside the circle $|z| = R$ (the *circle of convergence*). Specifically **it says nothing about the behaviour on the circle**. To see this, we will look at a few examples¹⁰.

¹⁰This should remind you of the deadly sin mentioned in [section 1.2](#), where the root or ratio test said nothing if the limit was 1.

Example 1.5.6 (Convergence on $|z| = R$)

Consider the following power series:

$$(i) \sum_{n=0}^{\infty} z^n, \quad (ii) \sum_{n=1}^{\infty} \frac{z^n}{n} \quad (iii) \sum_{n=1}^{\infty} \frac{z^n}{n^2}.$$

All of these have a radius of convergence $R = 1$, however for $|z| = 1$ for each power series we have:

1. Divergence if $|z| = 1$.
2. Convergence^a if $|z| = 1$ with $z \neq 1$, and divergence if $z = 1$.
3. Convergence if $|z| = 1$.

Thus we can't say anything about the case $|z| = R$ and each case really has to be treated separately.

^aHave a look at Example Sheet 1 for this result

§1.5.2 Differentiating Power Series

So things are generally quite nice inside the circle of convergence. One of the nice things we can do is differentiate! Even better, we can differentiate power series just like we do polynomials! The following results you need to know (including the lemmas), but the proofs are not examinable (feel free to skip them).

We are going to show that the power series $\sum_{n=0}^{\infty} a_n z^n$ with radius of convergence R has the derivative $\sum_{n=0}^{\infty} n a_n z^{n-1}$, whenever $|z| < R$. Before showing this, we will need a lemma which gives us that this second power series converges for $|z| < R$.

Lemma 1.5.7

Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be a power series whose radius of convergence is R . Then if $0 < r < R$, the power series $\sum_{n=1}^{\infty} n |a_n| r^{n-1}$ converges.

Proof. Pick w such that $r < |w| < R$. Then the power series $\sum_{n=0}^{\infty} a_n w^n$ converges, so the terms $|a_n w^n|$ are bounded above by say M . Then we have

$$n |a_n| r^n = n |a_n w^n| \frac{r^n}{|w|^n} \leq M n \frac{r^n}{|w|^n}.$$

But then the series $\sum_{n=1}^{\infty} M n (r/|w|)^n$ converges by the ratio test, and thus by the comparison test $\sum_{n=1}^{\infty} n |a_n| r^{n-1}$ converges. $\square$

Applying this lemma twice also gives us that the power series $\sum_{n=2}^{\infty} n(n-1) |a_n| r^{n-2}$ converges. We are now ready to prove our theorem about differentiating power series.

Theorem 1.5.8 (Differentiating Power Series)

Let $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be a power series whose radius of convergence is R . Then f is differentiable at all points with $|z| < R$, and $f'(z) = \sum_{n=0}^{\infty} n a_n z^{n-1}$.

Proof. Using the difference of powers factorisation, we have

$$\begin{aligned}\frac{f(z+h) - f(z)}{h} &= \sum_{n=0}^{\infty} a_n \frac{(z+h)^n - z^n}{h} \\ &= \sum_{n=1}^{\infty} a_n [(z+h)^{n-1} + (z+h)^{n-2}z + \cdots + z^{n-1}].\end{aligned}$$

Then, since $\sum_{n=1}^{\infty} na_n z^{n-1}$ converges by our lemma, we have

$$\begin{aligned}\frac{f(z+h) - f(z)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} &= \sum_{n=1}^{\infty} a_n \left(\sum_{j=0}^{n-1} z^j (z+h)^{n-1-j} - z^{n-1} \right) \\ &= \sum_{n=1}^{\infty} a_n \left(\sum_{j=0}^{n-1} z^j [(z+h)^{n-1-j} - z^{n-1-j}] \right).\end{aligned}$$

Now we want to bound the terms of this inside sum. Choose r such that $|z| < r < R$ and h such that $|z| + |h| < r$. Then employing the difference of powers factorisation again we get

$$\begin{aligned}|(z+h)^{n-1-j} - z^{n-1-j}| &= |h((z+h)^{n-2-j} + z(z+h)^{n-3-j} + \cdots + z^{n-2-j})| \\ &\leq |h|(n-1-j)r^{n-2-j}.\end{aligned}$$

Applying this bound, assuming the right hand sum converges we have

$$\begin{aligned}\left| \frac{f(z+h) - f(z)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} \right| &\leq \sum_{n=1}^{\infty} |a_n| \left(\sum_{j=0}^{n-1} r^j |h|(n-1-j)r^{n-2-j} \right) \\ &= |h| \sum_{n=1}^{\infty} |a_n| r^{n-2} \left(\sum_{j=0}^{n-1} j \right) \\ &= \frac{1}{2} |h| \sum_{n=2}^{\infty} |a_n| n(n-1) r^{n-2}.\end{aligned}$$

Using our previous lemma twice shows that the right hand sum converges, and thus

$$\frac{f(z+h) - f(z)}{h} - \sum_{n=1}^{\infty} na_n z^{n-1} \rightarrow 0 \quad \text{as } h \rightarrow 0,$$

that is, $f'(z) = \sum_{n=0}^{\infty} na_n z^{n-1}$. □

§1.5.3 Exponential, Trigonometric & Hyperbolic Functions

With all that out of the way, let's define some functions! You'll hopefully be familiar with all of these anyway, but we will now be able to define them properly.

We are going to need one little lemma before we proceed.

Lemma 1.5.9 (Constant Value Theorem Over $\mathbb{C}$)

If $F : \mathbb{C} \rightarrow \mathbb{C}$ has $F'(z) = 0$ for all $z \in \mathbb{C}$, then $F(z)$ is constant for all z .

Proof. For some $z \in \mathbb{C}$, we define the functions $u_z, v_z : \mathbb{R} \rightarrow \mathbb{R}$ such that $F(tz) = u_z(t) + iv_z(t)$. By the chain rule $F'(tz)z = u'_z(t) + iv'_z(t) = 0$ for any $t \in \mathbb{R}$. Then comparing real and imaginary parts, this implies $u'_z(t) = 0$ and $v'_z(t) = 0$, for all $t \in \mathbb{R}$. So u_z and v_z are constant.

Then for any $z \in \mathbb{C}$ we have $F(z) = u_z(1) + iv_z(1) = u_z(0) + iv_z(0) = F(0)$, so F is constant too. $\square$

The Exponential & Logarithm Function

We will begin with one of the most important functions, the exponential function.

Definition 1.5.10 (The Exponential Function)

We define the **exponential function** $\exp : \mathbb{C} \rightarrow \mathbb{C}$ by the power series

$$\exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

When we state a definition using power series, we pretty much always have to check convergence straight away so that we do not end up writing complete nonsense. For this power series, we can see by the ratio test that it converges for all $z \in \mathbb{C}$. By the result in the previous section, we can also see that it's differentiable.

Now we can prove all of the properties that you already are aware of.

Proposition 1.5.11 (Properties of the Exponential Function)

Let $z, w \in \mathbb{C}$. Then the following hold.

- (i) $\exp'(z) = \exp(z)$.
- (ii) $\exp(z) \exp(w) = \exp(z + w)$.
- (iii) $\exp(x) > 0$ for $x \in \mathbb{R}$.
- (iv) $\exp(x)$ is strictly increasing on $\mathbb{R}$.
- (v) $\exp(x) \rightarrow \infty$ as $x \rightarrow \infty$, $\exp(x) \rightarrow 0$ as $x \rightarrow -\infty$.
- (vi) $\exp : \mathbb{R} \rightarrow (0, \infty)$ is a bijection.

Proof. We prove each individually.

- (i) This follows from differentiating the power series.
- (ii) Let $a, b \in \mathbb{C}$ and define $F(z) = \exp(a + b - z) \exp(z)$. Then $F'(z) = -\exp(a + b - z) \exp(z) + \exp(a + b - z) \exp(z) = 0$, and thus by the constant value theorem, F is constant. So $\exp(a + b - z) \exp(z) = F(0) = \exp(a + b)$, and setting $z = b$ gives $\exp(a) \exp(b) = \exp(a + b)$.

- (iii) Clearly $\exp(x) > 0$ for all $x \geq 0$, and $\exp(0) = 1$. Then $\exp(0) = \exp(x - x) = \exp(x) \exp(-x) = 1$, and thus $\exp(-x) > 0$ for $x > 0$.
- (iv) Differentiating, $\exp'(x) = \exp(x) > 0$, and thus $\exp$ is strictly increasing on $\mathbb{R}$.
- (v) Truncating the power series, $\exp(x) > 1 + x$ for $x > 0$. Thus if $x \rightarrow \infty$, $\exp(x) \rightarrow \infty$. Then $\exp(-x) = 1/\exp(x)$, so $\exp(x) \rightarrow 0$ as $x \rightarrow -\infty$.
- (vi) Injectivity follows directly from $\exp$ being strictly increasing. For surjectivity, take $y > 0$. Then by (v) there exists $a, b \in \mathbb{R}$ such that $\exp(a) < y < \exp(b)$, and by the intermediate value theorem there is $c \in (a, b)$ such that $\exp(c) = y$. $\square$

Since the exponential function is a bijection $\mathbb{R} \rightarrow \mathbb{R}^+$, it also has a well defined inverse.

Definition 1.5.12 (The Logarithm Function)

We define the **logarithm function** $\log : (0, \infty) \rightarrow \mathbb{R}$ to be the inverse of $\exp$.

By the inverse function theorem, this is a differentiable function, and its derivative is given by $\log'(t) = 1/t$ for $t > 0$.

Remark. As defined here, the logarithm function is an inverse of $\exp : \mathbb{R} \rightarrow (0, \infty)$, not over the whole of $\mathbb{C}$. In general, $\exp$ is *not* a bijection, and thus it having an inverse doesn't make sense (with the tools we have developed, this is remedied in later courses).

Using both the exponential and logarithm functions, we can define powers in the general case. For $x > 0$ and $\alpha \in \mathbb{R}$, we define $x^\alpha = \exp(\alpha \log x)$. With this we get the normal 'rules of indices' being properties of the exponential function, and it is not hard to see that this definition matches how powers would be previously defined for $\alpha \in \mathbb{Q}$. This also gives us a new shorthand for the exponential function: $\exp(z) = e^z$.

Aside: Multiplying Infinite Series

In [Proposition 1.5.11](#) we proved a result involving multiplying two exponential functions together. We also managed to do this in a way that avoided using the power series definition of the function. Despite this, I think it's worth taking a moment to look properly at the multiplication of infinite series¹¹.

When we talk about trying multiplying infinite series, we are really talking about taking two infinite series so that they can be combined into a new infinite series¹², which hopefully converges to the product of the two original series.

So how can we do this? A natural approach would be to just 'expand everything', but unfortunately that's not really how infinite series work, so we can't do that. Still, we can use this idea as inspiration for a more mathematically sound approach.

Being quite informal and hand-wavy, let's consider two power series $\sum_{n=0}^{\infty} a_n z^n$ and $\sum_{n=0}^{\infty} b_n z^n$, multiply them term by term and then group by powers of z . Doing this, we

¹¹This section is completely not-examinable and the proofs are relatively long, so I suggest that you skip it if you are using these notes for revision.

¹²If we don't care about obtaining a new infinite series, this discussion can be completely ignored.

get something like

$$(a_0 + a_1z + a_2z^2 + a_3z^3 + \cdots) \cdot (b_0 + b_1z + b_2z^2 + b_3z^3 + \cdots) \\ = a_0b_0 + (a_0b_1 + a_1b_0)z + (a_0b_2 + a_1b_1 + a_2b_0)z^2 + (a_0b_3 + a_1b_2 + a_2b_1 + a_3b_0)z^3 + \cdots.$$

Taking inspiration from this, for two sequence a_n and b_n we can define their **convolution** to be the sequence c_n where

$$c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0.$$

It is this construction that will form the basis of our product of infinite series.

Definition (Cauchy Product)

Given two infinite series $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$, their **Cauchy product** is the infinite series

$$\sum_{n=0}^{\infty} c_n,$$

where $c_n = a_0b_n + a_1b_{n-1} + \cdots + a_nb_0$.

The Cauchy product does give us a way to obtain a ‘product’ of infinite series, but before we go any further some important caveats need to be pointed out. The most notable is the following: the infinite series obtained with the Cauchy product *does not necessarily converge*¹³.

Example (Convergent Series Whose Cauchy Product Diverges)

Consider the convergent^a series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n+1}}$. Then the Cauchy product of this series with itself is given by

$$\sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(-1)^k}{\sqrt{k+1}} \frac{(-1)^{n-k}}{\sqrt{n-k+1}}.$$

But then since

$$(k+1)(n-k+1) \leq \left(\frac{n}{2} + 1\right)^2 - \left(\frac{n}{2} - k\right)^2 \leq \left(\frac{n}{2} + 1\right)^2,$$

the terms of the Cauchy product satisfy

$$\left| \sum_{k=0}^n \frac{(-1)^k}{\sqrt{k+1}} \frac{(-1)^{n-k}}{\sqrt{n-k+1}} \right| \geq \sum_{k=0}^n \frac{1}{n/2 + 1} = \frac{n+1}{n/2 + 1},$$

and thus they don’t tend to 0 so the series cannot converge. This implies that we have two convergent series whose Cauchy product does not converge.

^aExercise!

This is obviously not a very good thing to happen, and we will spend the rest of this aside trying to deal with this possible issue. Luckily enough, this issue can be completely avoided if we know that one of the series being ‘multiplied’ converges absolutely¹⁴.

¹³It’s also possible for two divergent series to have a convergent Cauchy product. Try coming up with an example as an exercise.

¹⁴This is another reason why absolute convergence is a great property to have.

Theorem (Mertens' Theorem)

Suppose that $\sum_{n=0}^{\infty} a_n = A$ converges absolutely, and $\sum_{n=0}^{\infty} b_n = B$ converges. Then their Cauchy product $\sum_{n=0}^{\infty} c_n$ converges, and

$$\sum_{n=0}^{\infty} c_n = AB.$$

Proof. Let the partial sums of the series be

$$A_n = \sum_{k=0}^n a_k, \quad B_n = \sum_{k=0}^n b_k, \quad C_n = \sum_{k=0}^n c_k.$$

Then we have

$$\begin{aligned} C_n &= a_0 b_0 + (a_0 b_1 + a_1 b_0) + (a_0 b_2 + a_1 b_1 + a_2 b_0) + \cdots + (a_0 b_n + \cdots + a_{n-1} b_n) \\ &= a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= (a_0 B + \cdots + a_n B) - (a_0 B + \cdots + a_n B) + a_0 B_n + a_1 B_{n-1} + \cdots + a_n B_0 \\ &= A_n B + a_0 (B_n - B) + a_1 (B_{n-1} - B) + \cdots + a_n (B_0 - B). \end{aligned}$$

We wish to show $C_n \rightarrow AB$, and we will do so using three bounds. Given $\epsilon > 0$, since $A_n \rightarrow A$, there is an integer L such that $n \geq L$ implies

$$|A_n - A| \leq \frac{\epsilon/3}{|B|}.$$

Then since $\sum_{k=0}^{\infty} |a_k|$ converges and $B_n \rightarrow B$, there is an integer M such that $n \geq M$ implies

$$|B_n - B| \leq \frac{\epsilon/3}{\sum_{k=0}^{\infty} |a_k|}.$$

Also, since $\sum_{k=0}^{\infty} a_k$ converges, $a_n \rightarrow 0$, and thus there is an integer N such that $n \geq N$ implies that

$$|a_n| \leq \frac{\epsilon/3}{\sum_{k=0}^{M-1} |B_k - B|}.$$

Combining these^a, for $N \geq \max\{L, M + N\}$ we obtain

$$\begin{aligned} |C_n - AB| &= \left| (A_n - A)B + \sum_{k=0}^n a_{n-k}(B_k - B) \right| \\ &\leq |A_n - A||B| + \sum_{k=0}^{M-1} |a_{n-k}||B_k - B| + \sum_{k=M}^n |a_{n-k}||B_k - B| \leq \epsilon \end{aligned}$$

Thus $\sum_{n=0}^{\infty} c_n = AB$. □

^aWe use $M + N$ because we will need $n - (M - 1) \geq N$ to make the bounds work.

We can apply this result to power series. Recalling that power series converge absolutely inside their circle of convergence, we can safely multiply power series using the Cauchy product.

Corollary (Multiplying Power Series)

Let $\sum_{n=0}^{\infty} a_n z^n$ and $\sum_{n=0}^{\infty} b_n z^n$ be power series with radius of convergence R_1 and R_2 respectively. Then if $|z| < \min\{R_1, R_2\}$ we have

$$\left(\sum_{n=0}^{\infty} a_n z^n\right) \cdot \left(\sum_{n=0}^{\infty} b_n z^n\right) = \left(\sum_{n=0}^{\infty} c_n z^n\right),$$

where $c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0$.

So to summarise: when inside the circle of convergence, we can multiply power series in the natural way, and the result will be as we expect. Using this corollary, we could prove $\exp(a+b) = \exp(a)\exp(b)$ directly (though admittedly the other proof is much faster).

The Trigonometric Functions

We can use the exponential function to define the familiar trigonometric functions.

Definition 1.5.13 (Sine & Cosine)

We define the **sine** and **cosine functions** by the power series

$$\begin{aligned}\sin z &= \frac{e^{iz} - e^{-iz}}{2i} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots \\ \cos z &= \frac{e^{iz} + e^{-iz}}{2} = 1 - \frac{z^2}{2} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots\end{aligned}$$

where $z \in \mathbb{R}$.

Stating these definitions in terms of the exponential function make lots of identities quite easy to derive.

Proposition 1.5.14 (Properties of sin and cos)

Let $z, w \in \mathbb{R}$. Then the following hold.

- (i) $\sin' z = \cos z$, $\cos' z = -\sin z$.
- (ii) $\sin^2 z + \cos^2 z = 1$.
- (iii) $|\sin z|, |\cos z| \leq 1$.
- (iv) $\cos(z+w) = \cos z \cos w - \sin z \sin w$.
- (v) $\sin(z+w) = \sin z \cos w + \cos z \sin w$.

Proof. We prove each individually, mainly by computation.

- (i) For sin we have $\sin' z = [(e^{iz} - e^{-iz})/2i]' = (e^{iz} + e^{-iz})/2 = \cos z$, and for cos we have $\cos' z = [(e^{iz} + e^{-iz})/2]' = (e^{-iz} - e^{iz})/2i = -\sin z$.
- (ii) $\sin^2 z + \cos^2 z = -(e^{2iz} + e^{-2iz} + 2)/4 + (e^{2iz} + e^{-2iz} - 2)/4 = 1$, as required.
- (iii) This follows from the above result.

(iv) Expanding the right hand expression gives

$$\begin{aligned}
 & \cos z \cos w - \sin z \sin w \\
 &= \frac{(e^{iz} + e^{-iz})(e^{iw} + e^{-iw})}{4} + \frac{(e^{iz} - e^{-iz})(e^{iw} - e^{-iw})}{4} \\
 &= \frac{e^{i(z+w)} + e^{-i(z+w)}}{2} \\
 &= \cos(z + w).
 \end{aligned}$$

(v) Differentiating the above result with respect to z , we have $-\sin z \cos w - \cos z \sin w = -\sin(z + w)$, giving us our desired result. $\square$

One of the more notable properties of $\sin$ and $\cos$ is that they are periodic, which is what we are going to show next. To establish this, we can try evaluating one of them (say $\cos$) at some values near where we expect a root to be (guided by our previous knowledge of these functions).

Lemma 1.5.15 (Smallest Root of $\cos$)

There is a smallest positive x such that $\cos(x) = 0$.

Proof. We begin by computing the sign of the derivative over $(0, 2)$. Since $\cos'(x) = -\sin(x)$, we can use the inequality $\frac{x^{2n-1}}{(2n-1)!} > \frac{x^{2n+1}}{(2n+1)!}$ to bound

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots > 0,$$

so $\cos$ has a negative derivative and must be decreasing on $(0, 2)$.

Evaluating the power series for $\cos$ at $\sqrt{2}$ and $\sqrt{3}$, we can see that the function changes sign

$$\cos \sqrt{2} = 1 - \frac{2}{2} + \frac{2^2}{4!} - \frac{2^3}{6!} + \cdots > 0, \cos \sqrt{3} = 1 - \frac{3}{2} + \frac{3^2}{4!} - \frac{3^3}{6!} + \cdots < 0.$$

Then by the intermediate value theorem, there must be a root in the interval $(\sqrt{2}, \sqrt{3})$. Also since $\cos$ is decreasing on $(0, 2)$, this must also be the smallest positive root. $\square$

The existence of this root gives us a way to define π .

Definition 1.5.16 (π)

π is the smallest positive real number such that $\cos \pi/2 = 0$.

We can easily determine the value of $\sin \pi/2$ too. We have $\sin^2 \pi/2 + \cos^2 \pi/2 = 1$, so $\sin^2 \pi/2 = 1$, and knowing from the proof of [Lemma 1.5.15](#) that $\sin(x) > 0$ on $(0, 2)$, we can deduce that $\sin \pi/2 = 1$. These results are all we need to show that $\sin$ and $\cos$ are periodic, and also that the two functions are just translated versions of each-other.

Proposition 1.5.17 (Periodicity of sin and cos)

The functions sin and cos are periodic with a period of 2π , and are related as follows.

$$\begin{aligned}\sin(z + \pi/2) &= \cos z, & \cos(z + \pi/2) &= -\sin z, \\ \sin(z + \pi) &= -\sin z, & \cos(z + \pi) &= -\cos z, \\ \sin(z + 2\pi) &= \sin z, & \cos(z + 2\pi) &= \cos z.\end{aligned}$$

Proof. This follows directly from $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$ and the angle addition formulas. $\square$

Remark (Periodicity of exp). If we use the power series for sin and cos, we can write $\exp(iz) = \cos z + i \sin z$, from which the above result implies that exp is periodic with a period of 2π .

With sin and cos have been defined properly, we can then define tan, sec, csc and all of those fun functions in the normal way.

The Hyperbolic Functions

The last set of functions we will touch on is the (probably) slightly less familiar hyperbolic functions. As with the standard trigonometric functions, we can state the definitions using the exponential function.

Definition 1.5.18 (Hyperbolic Sine & Cosine)

We define the **hyperbolic sine** and **hyperbolic cosine functions** by the power series

$$\begin{aligned}\sinh z &= \frac{e^z - e^{-z}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots \\ \cosh z &= \frac{e^z + e^{-z}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots\end{aligned}$$

where $z \in \mathbb{R}$.

Remark. The basic hyperbolic and trigonometric functions are related by $\cosh z = \cos(iz)$ and $\sinh z = -i \sin(iz)$. Remembering this can allow you to adapt facts about sin and cos to facts about sinh and cosh.

We will state a few properties for convenience.

Proposition 1.5.19 (Properties of sinh & cosh)

Let $z, w \in \mathbb{R}$. Then the following hold.

- (i) $\sinh' z = \cosh z$, $\cosh' z = \sinh z$.
- (ii) $\cosh^2 z - \sinh^2 z = 1$.
- (iii) $\sinh(z + w) = \sinh z \cosh w + \cosh z \sinh w$.
- (iv) $\cosh(z + w) = \cosh z \cosh w + \sinh z \sinh w$.

Proof. We will use the observation made in the above remark.

- (i) For $\sinh$ we have $\sinh' z = [-i \sin(iz)]' = \cos(iz) = \cosh z$, and for $\cosh$ we have $\cosh' z = [\cos(iz)]' = -i \sin(iz) = \sinh z$.
- (ii) $\cosh^2 z - \sinh^2 z = \cos^2(iz) - (-i \sin(iz))^2 = \cos^2(iz) + \sin^2(iz) = 1$.
- (iii) $\sinh(z + w) = -i \sin(iz + iw) = \sinh(z) \cosh(w) + \cosh(z) \sinh(w)$.
- (iv) $\cosh(z + w) = \cos(iz + iw) = \cosh(z) \cosh(w) + \sinh(z) \sinh(w)$. □

§1.6 Integration

We are now prepared, both technically and emotionally, to talk about *integrals*. The idea behind integration is to assign some sort of ‘area’ to sets. The first issue we run into is that it is not quite clear what ‘area’ actually means (in mathematical terms), so to avoid this difficulty, we will instead define the notion of an integral (specifically the *Riemann integral*, there are other types) and we will use this as our definition of area (not vice versa).

Throughout this section, we will deal with the integration of bounded functions defined on bounded intervals, $f : [a, b] \rightarrow \mathbb{R}$. We will discuss a slight generalisation later on.

§1.6.1 Dissections, Upper & Lower Sums, and Riemann Integrals

As with most ideas in analysis, integration depends fundamentally on a ‘limiting process’. For Riemann integrals, this is conducted by considering *dissections* of the interval in which we will be integrating over.

Definition 1.6.1 (Dissection)

A **dissection** $\mathcal{D}$ of a finite interval $[a, b]$ is a finite subset of $[a, b]$ containing the endpoints. We typically write $\mathcal{D} = \{x_0, x_1, x_2, \dots, x_n\}$ with $a = x_0 \leq x_1 \leq \dots \leq x_n = b$.

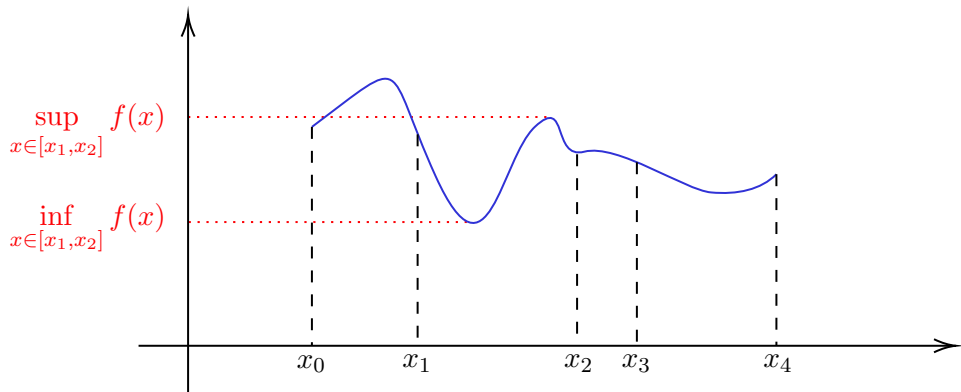
For a given dissection, because we are dealing with bounded functions we can sensibly define the lower and upper sums of a function as follows.

Definition 1.6.2 (Upper & Lower Sums)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function, and let $\mathcal{D}$ be a dissection of $[a, b]$. We define the **upper sum** $S_{\mathcal{D}}(f)$ and **lower sum** $s_{\mathcal{D}}(f)$ by

$$S_{\mathcal{D}}(f) = \sum_{j=1}^n (x_j - x_{j-1}) \sup_{x \in [x_{j-1}, x_j]} f(x),$$

$$s_{\mathcal{D}}(f) = \sum_{j=1}^n (x_j - x_{j-1}) \inf_{x \in [x_{j-1}, x_j]} f(x).$$



Let's make some observations. By construction we have $S_{\mathcal{D}}(f) \geq s_{\mathcal{D}}(f)$, and we can also see intuitively that replacing a dissection by a more elaborate one will decrease the upper

sum and increase the lower sum. Indeed, if we say that $\mathcal{D}'$ **refines** $\mathcal{D}$ when $\mathcal{D}' \supseteq \mathcal{D}$, then we can formalise this intuition as follows.

Lemma 1.6.3 (Refinement Lemma)

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and $\mathcal{D}, \mathcal{D}'$ are dissections of $[a, b]$ such that $\mathcal{D} \subseteq \mathcal{D}'$, then

$$s_{\mathcal{D}}(f) \leq s_{\mathcal{D}'}(f) \leq S_{\mathcal{D}'}(f) \leq S_{\mathcal{D}}(f).$$

Proof. It suffices to show that this holds when $\mathcal{D}'$ and $\mathcal{D}$ differ by a single element (since then we can just repeatedly apply this argument to obtain the general case). Suppose that $\mathcal{D}' = \{x_0, \dots, x_n\}$, and that $\mathcal{D} = \mathcal{D}' \setminus \{x_i\}$ where $i \neq 0, n$. Then let

$$A = \sup_{x \in [x_{i-1}, x_i]} f(x), \quad \text{and} \quad B = \sup_{x \in [x_i, x_{i+1}]} f(x).$$

We can then establish the right hand side of the inequality with

$$\begin{aligned} S_{\mathcal{D}}(f) - S_{\mathcal{D}'}(f) &= (x_{i+1} - x_{i-1}) \max\{A, B\} - (x_i - x_{i-1})A - (x_{i+1} - x_i)B \\ &\geq \max\{A, B\}[(x_{i+1} - x_{i-1}) - (x_i - x_{i-1}) - (x_{i+1} - x_i)] = 0. \end{aligned}$$

Then the left hand side of the inequality follows similarly (taking infima). $\square$

A direct consequence of this lemma is that lower sums can *never* exceed upper sums.

Lemma 1.6.4 (Key Integration Property)

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and $\mathcal{D}_1, \mathcal{D}_2$ are dissections of $[a, b]$ then

$$s_{\mathcal{D}_1}(f) \leq S_{\mathcal{D}_2}(f).$$

Proof. This follows from the previous lemma by noting that $s_{\mathcal{D}_1}(f) \leq s_{\mathcal{D}_1 \cup \mathcal{D}_2}(f)$, and $S_{\mathcal{D}_1 \cup \mathcal{D}_2}(f) \leq S_{\mathcal{D}_2}(f)$. $\square$

We are now (finally) ready to say what it means for a function to be Riemann integrable.

Definition 1.6.5 (Upper & Lower Integrals)

For a bounded function $f : [a, b] \rightarrow \mathbb{R}$ we define the **upper integral** as $I^*(f) = \inf_{\mathcal{D}} S_{\mathcal{D}}(f)$ and the **lower integral** as $I_*(f) = \sup_{\mathcal{D}} s_{\mathcal{D}}(f)$.

Note that [Lemma 1.6.4](#), along with the boundedness of the function, guarantees that the upper and lower integrals always exist.

Definition 1.6.6 (Riemann Integrable)

If $f : [a, b] \rightarrow \mathbb{R}$ is a bounded function and $I^*(f) = I_*(f)$, then we say f is **Riemann integrable** and write

$$\int_a^b f(x) \, dx = I^*(f).$$

So this is what it means for a function to be Riemann integrable¹⁵. In general using this definition directly on a function to show that it's integrable is quite tricky. As an example, we can attempt to show that $f(x) = x$ is integrable.

Example 1.6.7 ($f(x) = x$ is Integrable)

Consider the function $f(x) = x$ defined on the interval $[a, b]$. Then for any dissection $\mathcal{D} = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ we have the upper sum given by

$$\begin{aligned} S_{\mathcal{D}}(f) &= \sum_{j=1}^n (x_j - x_{j-1})x_j \\ &= \sum_{j=1}^n \left(\frac{x_j^2}{2} - \frac{x_{j-1}^2}{2} + \frac{x_j^2}{2} + \frac{x_{j-1}^2}{2} - x_{j-1}x_j \right) \\ &= \sum_{j=1}^n \left(\frac{x_j^2}{2} - \frac{x_{j-1}^2}{2} \right) + \frac{1}{2} \sum_{j=1}^n (x_j - x_{j-1})^2 \\ &= \left(\frac{b^2}{2} - \frac{a^2}{2} \right) + \frac{1}{2} \sum_{j=1}^n (x_j - x_{j-1})^2. \end{aligned}$$

If $x_j - x_{j-1} < \delta$ for all j , then we can obtain the upper bound

$$S_{\mathcal{D}}(f) \leq \left(\frac{b^2}{2} - \frac{a^2}{2} \right) + \frac{1}{2} \sum_{j=1}^n \delta (x_j - x_{j-1}) = \left(\frac{b^2}{2} - \frac{a^2}{2} \right) + \frac{\delta}{2} (b - a).$$

So given $\epsilon > 0$, there is some dissection $\mathcal{D}$ such that $S_{\mathcal{D}}(f) \leq (b^2 - a^2)/2 + \epsilon$, and a similar argument also gives $s_{\mathcal{D}}(f) \geq (b^2 - a^2)/2 - \epsilon$. So by our key integration property, we have

$$\frac{1}{2}(b^2 - a^2) - \epsilon \leq I^*(f), I_*(f) \leq \frac{1}{2}(b^2 - a^2) + \epsilon,$$

for any $\epsilon > 0$, and thus $I^*(f) = I_*(f) = (b^2 - a^2)/2$. This shows that $f(x) = x$ is integrable over $[a, b]$, with

$$\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}.$$

There are a few ways that we can streamline this argument. Later on we will develop more general integrability results, but the following criterion is also quite useful for showing particular functions are integrable.

Theorem 1.6.8 (Riemann's Integrability Criterion)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then f is Riemann integrable if and only if given any $\epsilon > 0$, we can find a dissection $\mathcal{D}$ with

$$S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon.$$

¹⁵We will regularly drop the 'Riemann', and when referring to 'integrability' we will always be talking only about Riemann integrability.

Proof. If f is Riemann integrable, then given $\epsilon > 0$ there must exist dissections $\mathcal{D}_1, \mathcal{D}_2$ such that

$$S_{\mathcal{D}_1}(f) < \int_a^b f(x) \, dx + \frac{\epsilon}{2} \quad \text{and} \quad s_{\mathcal{D}_2}(f) > \int_a^b f(x) \, dx - \frac{\epsilon}{2}.$$

Then taking $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$ we have $S_{\mathcal{D}}(f) \leq S_{\mathcal{D}_1}(f)$, and $s_{\mathcal{D}}(f) \geq s_{\mathcal{D}_2}(f)$, and thus

$$S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) \leq S_{\mathcal{D}_1}(f) - s_{\mathcal{D}_2}(f) < \epsilon.$$

Otherwise if for any $\epsilon > 0$ we can find a dissection $\mathcal{D}$ such that $S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon$, then we would have

$$\inf_{\mathcal{D}'} S_{\mathcal{D}'}(f) - \sup_{\mathcal{D}'} s_{\mathcal{D}'}(f) \leq S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon,$$

that is, $I^*(f) - I_*(f) < \epsilon$ for all $\epsilon > 0$. So we must have $I^*(f) = I_*(f)$, implying f is Riemann integrable. $\square$

§1.6.2 Properties of Integration

We now need to prove many of the properties of integrals that you are (most likely) already familiar with. The first result we would like to prove is that an arbitrary linear combination of integrable functions is integrable, but we will need to prove some smaller results first.

The proofs below are a little bit fiddly, but they all follow the same basic idea: simplify the problem as much as possible, think a bit about what's going on with the suprema/infima, and use our key property of integration, Riemann's criterion and so on.

Proposition 1.6.9 (Adding Integrable Functions)

If $f, g : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable, then so is $f + g$ and

$$\int_a^b f(x) + g(x) \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx.$$

Proof. We begin by noting the general fact

$$\sup_{x \in [x_{j-1}, x_j]} (f(x) + g(x)) \leq \sup_{x \in [x_{j-1}, x_j]} f(x) + \sup_{x \in [x_{j-1}, x_j]} g(x),$$

which directly implies that $S_{\mathcal{D}}(f + g) \leq S_{\mathcal{D}}(f) + S_{\mathcal{D}}(g)$. A similar fact about infima implies $s_{\mathcal{D}}(f + g) \geq s_{\mathcal{D}}(f) + s_{\mathcal{D}}(g)$.

Now let $\mathcal{D}_1$ and $\mathcal{D}_2$ be dissections, and let $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$. Then we have

$$S_{\mathcal{D}}(f + g) \leq S_{\mathcal{D}}(f) + S_{\mathcal{D}}(g) \leq S_{\mathcal{D}_1}(f) + S_{\mathcal{D}_2}(g),$$

which implies that $I^*(f + g) \leq I^*(f) + I^*(g)$. Similarly we have $I_*(f + g) \geq I_*(f) + I_*(g)$.

Since f and g are integrable, their upper and lower integrals are equal, so combining

the two previous inequalities gives

$$\int_a^b f(x) \, dx + \int_a^b g(x) \, dx \leq I_*(f+g) \leq I^*(f+g) \leq \int_a^b f(x) \, dx + \int_a^b g(x) \, dx,$$

so $I_*(f+g) = I^*(f+g)$, showing $f+g$ is integrable, with the integral given by $\int_a^b f(x) + g(x) \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$. $\square$

Proposition 1.6.10 (Negating Integrable Functions)

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then so is $-f$, and

$$\int_a^b (-f(x)) \, dx = - \int_a^b f(x) \, dx.$$

Proof. Let $\mathcal{D}$ be a dissection of $[a, b]$. Then since $\sup_x -f(x) = -\inf_x f(x)$, we have

$$\begin{aligned} S_{\mathcal{D}}(-f) &= \sum_{j=1}^n (x_j - x_{j-1}) \sup_{x \in [x_{j-1}, x_j]} (-f(x)) \\ &= - \sum_{j=1}^n (x_j - x_{j-1}) \inf_{x \in [x_{j-1}, x_j]} f(x) \\ &= -s_{\mathcal{D}}(f). \end{aligned}$$

Similarly $s_{\mathcal{D}}(-f) = -S_{\mathcal{D}}(f)$. This then implies that $I^*(-f) = -I_*(f)$ and $I_*(-f) = -I^*(f)$. But then f is integrable, so $I^*(-f) = I_*(-f) = -I^*(f)$. Thus $-f$ is also integrable, and $\int_a^b (-f(x)) \, dx = - \int_a^b f(x) \, dx$. $\square$

Proposition 1.6.11 (Scaling Integrable Functions)

If $\lambda \in \mathbb{R}$ and $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then so is λf and

$$\int_a^b \lambda f(x) \, dx = \lambda \int_a^b f(x) \, dx.$$

Proof. Without loss of generality^a, we may assume that $\lambda \geq 0$. Then if $\mathcal{D}$ is a dissection of $[a, b]$ we have

$$S_{\mathcal{D}}(\lambda f) = \sum_{j=1}^n (x_j - x_{j-1}) \sup_{x \in [x_{j-1}, x_j]} \lambda f(x) = \lambda S_{\mathcal{D}}(f),$$

and similarly $s_{\mathcal{D}}(\lambda f) = \lambda s_{\mathcal{D}}(f)$. Then $I^*(\lambda f) = \lambda I^*(f)$ and $I_*(\lambda f) = \lambda I_*(f)$. Since f is integrable, we then have $I^*(\lambda f) = I_*(\lambda f) = \lambda I^*(f)$, so λf is integrable and $\int_a^b \lambda f(x) \, dx = \lambda \int_a^b f(x) \, dx$. $\square$

^aIf $\lambda < 0$, we can consider $|\lambda|f$, then by the previous lemma $-|\lambda|f = \lambda f$ will be integrable.

To summarise these results, if $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable and $\lambda, \mu \in \mathbb{R}$ then $\lambda f + \mu g$

is integrable and

$$\int_a^b \lambda f(x) + \mu g(x) \, dx = \lambda \int_a^b f(x) \, dx + \mu \int_a^b g(x) \, dx.$$

Let's now turn our attention¹⁶ to another set of integral properties which are still natural but probably ever-so-slightly less familiar: integral inequalities. The first result is one you'd expect to be true.

Proposition 1.6.12 (Basic Integral Inequality)

If $f, g : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable and $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x) \, dx \leq \int_a^b g(x) \, dx.$$

Proof. If $f \leq g$ then for all dissections $\mathcal{D}$ we have $S_{\mathcal{D}}(f) \leq S_{\mathcal{D}}(g)$, and hence $\int_a^b f(x) \, dx = \inf_{\mathcal{D}} S_{\mathcal{D}}(f) \leq \inf_{\mathcal{D}} S_{\mathcal{D}}(g) = \int_a^b g(x) \, dx$. $\square$

The next result shows that $|f|$ is integrable if f is, and also something like a triangle inequality for integrals.

Proposition 1.6.13 (Integral Triangle Inequality)

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable, then so is $|f|$ and

$$\int_a^b |f(x)| \, dx \geq \left| \int_a^b f(x) \, dx \right|.$$

Proof. We define $f_+(x) = \max\{f(x), 0\}$. We will show that $f_+(x)$ is integrable. For some interval I we have

$$\sup_{x \in I} f_+(x) = \max \left\{ 0, \sup_{x \in I} f(x) \right\}, \quad \inf_{x \in I} f_+(x) = \max \left\{ 0, \inf_{x \in I} f(x) \right\},$$

and then considering the cases $\sup_{x \in I} f(x) \geq 0$ and $\sup_{x \in I} f(x) \leq 0$, we find that the following inequality holds:

$$\begin{aligned} \sup_{x \in I} f_+(x) - \inf_{x \in I} f_+(x) &= \max \left\{ 0, \sup_{x \in I} f(x) \right\} - \max \left\{ 0, \inf_{x \in I} f(x) \right\} \\ &\leq \sup_{x \in I} f(x) - \inf_{x \in I} f(x). \end{aligned}$$

Then since f is integrable, given $\epsilon > 0$ there is a dissection $\mathcal{D}$ such that

$$S_{\mathcal{D}}(f_+) - s_{\mathcal{D}}(f_+) \leq S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon,$$

and thus f_+ is integrable. We can write $|f| = 2f_+ - f$, and thus $|f|$ is also integrable.

¹⁶Don't get too sad – we still have to show that the product of integrable functions is integrable, but we will come back to that once we have shown $|f|$ is integrable

We also have $-|f| \leq f \leq |f|$, so $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$. $\square$

We can use this property to prove that the product of two integrable functions fg is integrable. To do this, the expansion $2fg = (f+g)^2 - f^2 - g^2$ can be used, so all we need to show is that f^2 is integrable if f is. Also since $f^2 = |f| \cdot |f|$, we can work only with the case that f is positive.

Unlike some of the previous results, we will not write down what the value of this integral will be (there's not really a nice rule). We will only be proving integrability.

Proposition 1.6.14 (Multiplying Integrable Functions)

If $f, g : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable, then so is fg .

Proof. We will first show that if $f : [a, b] \rightarrow \mathbb{R}$ is an integrable function such that $f(x) \geq 0$ for all $x \in [a, b]$, then f^2 is integrable.

Since f is integrable (and hence bounded), we can choose some $K > 0$ such that $|f(x)| \leq K$ for all $x \in [a, b]$. Then given any $\epsilon > 0$, there exists some dissection $\mathcal{D}$ where $S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon/2K$.

Using $\mathcal{D}$, we define $M_j = \sup_{x \in [x_{j-1}, x_j]} f(x)$ and $m_j = \inf_{x \in [x_{j-1}, x_j]} f(x)$. Then we have

$$\begin{aligned} S_{\mathcal{D}}(f^2) - s_{\mathcal{D}}(f^2) &= \sum_{j=1}^n (x_j - x_{j-1})(M_j^2 - m_j^2) \\ &= \sum_{j=1}^n (x_j - x_{j-1})(M_j + m_j)(M_j - m_j) \\ &\leq 2K(S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f)) < \epsilon, \end{aligned}$$

showing that f^2 is integrable.

Since $f^2 = |f| \cdot |f|$, and $|f| \geq 0$, this argument shows that f^2 is integrable for any integrable function f . Using this and previous properties then shows that $[(f+g)^2 - f^2 - g^2]/2 = fg$ is integrable. $\square$

§1.6.3 Integrable Functions

At this point we have our definition of integrability, along with some basic properties of integrals. However, we have so far only really seen that $f(x) = x$ and $f(x) = |x|$ are integrable. In this section we will prove some results that will expand this list quite widely.

We have so only looked at bounded functions, and a natural question is whether bounded is a sufficient criterion for integrability. This sadly is not the case, and there is a standard counterexample due to Dirichlet.

Example 1.6.15 (A Bounded Non-Integrable Function)

The function $f : [0, 1] \rightarrow \mathbb{R}$ given by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

is not Riemann integrable.

To see this, note that for any dissection $\mathcal{D}$, since $\mathbb{Q}$ is dense in $\mathbb{R}$, we have $s_{\mathcal{D}}(f) = 0$ and $S_{\mathcal{D}}(f) = 1$.

So let's look at some types of integrable functions. The first is that monotonic functions are integrable, which can be shown by taking an evenly spaced dissection causing the sum to telescope.

Proposition 1.6.16 (Monotonic Functions are Integrable)

Let $f : [a, b] \rightarrow \mathbb{R}$ be monotonic. Then f is Riemann integrable.

Proof. Without loss of generality, assume that f is increasing. Then given $\epsilon > 0$, for any integer $n > [(b-a)(f(b)-f(a))]/\epsilon$ we can define the dissection $\mathcal{D} = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$, where $x_0 = a$, and $x_j - x_{j-1} = (b-a)/n$. Then we have

$$\begin{aligned} S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) &= \sum_{j=1}^n (x_j - x_{j-1}) \left[\sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \right] \\ &= \sum_{j=1}^n \frac{b-a}{n} [f(x_j) - f(x_{j-1})] \\ &= \frac{(b-a)(f(b) - f(a))}{n} < \epsilon, \end{aligned}$$

and thus f is integrable. □

An interesting consequence of this is that if we can write $f(x) = f_1(x) - f_2(x)$ with f_1, f_2 are increasing, then f is integrable. Such functions are known as ‘functions of bounded variation’.

The next result we would like to show is that every continuous function is Riemann integrable on some bounded interval. Before jumping into the proof, let's take a moment to reflect on what might be needed¹⁷. For some continuous function, we want to get $S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) < \epsilon$ with some dissection. What this comes down to is finding some way to bound

$$\sup_{x \in I} f(x) - \inf_{x \in I} f(x),$$

under some sufficiently small interval I . Now we know that f is continuous, and continuity tells us that there if the interval is of size at most some δ (which can depend on where the interval is) then we can get $|f(x) - f(y)| < \epsilon$. This works for one interval, but we need to construct a dissection, so we would need to make this type of argument work for intervals which cover the whole of $[a, b]$.

What would be really nice is if we could find some δ that would work no matter where

¹⁷This is not the only possible proof! You can prove things without uniform continuity, but this is a more natural (and more slick) argument.

the sub-interval in $[a, b]$ is. If this was the case, then the dissection could just be made up of points which are about a distance of δ apart. It turns out that this is always possible, and gives us the notion of *uniform continuity*.¹⁸

Definition 1.6.17 (Uniform Continuity)

Let $A \subset \mathbb{R}$ and $f : A \rightarrow \mathbb{R}$. We say that f is **uniformly continuous** on A if given any $\epsilon > 0$ we can find a $\delta > 0$ such that for $x, y \in A$, whenever $|x - y| < \delta$ we have $|f(x) - f(y)| < \epsilon$.

Note that unlike the previous definition, we have no knowledge of where in the set A the points x and y are, just how far apart they are. It turns out that on closed bounded intervals (like what we have when we talk about integration, at least in this section), continuous functions are always uniformly continuous. The proof of this is quite natural – we use Bolzano-Weierstrass to show that a contradiction to uniform continuity is a contradiction to continuity.

Theorem 1.6.18 (Continuous Functions are Uniformly Continuous)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then f is uniformly continuous.

Proof. Suppose f was not uniformly continuous, that is, there exists some $\epsilon > 0$ such that for all $\delta > 0$ there is some x, y with $|x - y| < \delta$ such that $|f(x) - f(y)| \geq \epsilon$.

Taking $\delta = 1/n$, we can find some sequences $x_n, y_n \in [a, b]$ such that $|x_n - y_n| < 1/n$, and $|f(x_n) - f(y_n)| \geq \epsilon$. Then by Bolzano-Weierstrass, we can find some convergent subsequence $x_{n_j} \rightarrow x$ for some $x \in [a, b]$. But then $|x_{n_j} - y_{n_j}| < 1/n_j$ for all j , so we must have $y_{n_j} \rightarrow x$ also.

Then $|f(x_{n_j}) - f(y_{n_j})| \geq \epsilon$ for every j , which implies that $f(x_{n_j})$ and $f(y_{n_j})$ cannot converge to the same value. But then by continuity we have $f(x_{n_j}) \rightarrow f(x)$ and $f(y_{n_j}) \rightarrow f(x)$, which is a contradiction. Thus f must be uniformly continuous. $\square$

With this result, we can prove that continuous functions are integrable in the way we discussed previously.

Theorem 1.6.19 (Continuous Functions are Integrable)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then f is Riemann integrable.

Proof. Given $\epsilon > 0$, since f is uniformly continuous, there is some $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/(b - a)$ whenever $|x - y| < \delta$ and $x, y \in [a, b]$.

Now choose some integer $n \geq (b - a)/\delta$, and define the dissection $\mathcal{D} = \{x_0, x_1, \dots, x_n\}$ with $x_j = a + j(b - a)/n$. Then we have

$$\sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \leq \frac{\epsilon}{(b - a)},$$

¹⁸The ‘uniform’ part of the definition is part of a larger class of definitions that you will come across in later analysis courses.

for all $1 \leq j \leq n$, and thus

$$\begin{aligned} S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) &= \sum_{j=1}^n (x_j - x_{j-1}) \left[\sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \right] \\ &\leq \sum_{j=1}^n \frac{b-a}{n} \cdot \frac{\epsilon}{b-a} = \epsilon, \end{aligned}$$

and thus f is Riemann integrable. $\square$

This theorem shows that most of the functions we deal with regularly (and all of the ones discussed in the previous chapter) are integrable as well as continuous.

Also, we have so far spent relatively little time on the ‘evaluating integrals’ side of things, but knowing a function is integrable can help in determining the actual value of the integral. A straightforward example is shown below.

Example 1.6.20 (Integral of $f(x) = 1$)

Consider the function $f : [a, b] \rightarrow \mathbb{R}$ defined by $f(x) = 1$. This function is continuous and hence integrable. Then taking the dissection $\mathcal{D} = \{a, b\}$, we have $S_{\mathcal{D}}(f) = s_{\mathcal{D}}(f) = b - a$, so the value of the integral is

$$\int_a^b 1 \, dx = b - a.$$

§1.6.4 The Fundamental Theorem of Calculus

With all of that out of the way, we can now prove a central result in the theory of integration: the fundamental theorem of calculus. The fundamental theorem of calculus connects the theory that we built up relating to differentiation and integration, showing that (in some sense) they are ‘inverses’.

Theorem 1.6.21 (Fundamental Theorem of Calculus)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. For $x \in [a, b]$, we define

$$F(x) = \int_a^x f(t) \, dt.$$

Then F is differentiable, with derivative $F'(x) = f(x)$ for every x .

Proof. For $|h| > 0$, we note that

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

Now given $\epsilon > 0$, since f is continuous at x there is some $\delta > 0$ such that when

$|x - y| < \delta$ we have $|f(x) - f(y)| < \epsilon$. Hence if $0 < |h| < \delta$ we have

$$\begin{aligned} \left| \frac{1}{h} \int_x^{x+h} f(t) \, dt - f(x) \right| &= \left| \frac{1}{h} \int_x^{x+h} f(t) - f(x) \, dt \right| \\ &\leq \frac{1}{|h|} \left| \int_x^{x+h} |f(t) - f(x)| \, dt \right| \\ &\leq \frac{1}{|h|} \cdot \epsilon |h| = \epsilon, \end{aligned}$$

hence $\lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} = f(x)$, as required. $\square$

Another way to view this theorem (which is quite useful in evaluating integrals) is the corollary below.

Corollary 1.6.22 (Fundamental Theorem of Calculus – Again)

Suppose that $f : [a, b] \rightarrow \mathbb{R}$ has a continuous derivative. Then

$$\int_a^b f'(t) \, dt = f(b) - f(a).$$

Proof. Define $g(x) = \int_a^x f'(t) \, dt$. Then by the fundamental theorem of calculus, $g'(x) = f'(x)$. Also $g(a) = 0$.

Let $h(x) = g(x) - f(x)$. Then $h'(x) = 0$, so by the mean value theorem, $h(x)$ is constant. Taking the value at a , we then have $h(a) = -f(a) = g(a) - f(a) = -f(a)$ for all x . So if $x = b$, this becomes $g(b) = f(b) - f(a)$, which is our desired result. $\square$

One of the most useful applications of the fundamental theorem of calculus is that we can translate results about derivatives to results about integrals. This works in both coming up with the integrals of functions (by thinking about which functions result in a given function when differentiated), but also in the development of general methods for evaluating integrals. Two results we will look at are integration by parts, which comes from the product rule of differentiation, and integration by substitution, which comes from the chain rule.

Corollary 1.6.23 (Integration By Parts)

Suppose that the derivatives f' and g' exist and are continuous on $[a, b]$. Then

$$\int_a^b f'(x)g(x) \, dx = f(b)g(b) - f(a)g(a) - \int_a^b f(x)g'(x) \, dx.$$

Proof. By the product rule $(fg)' = f'g + fg'$. Then by the fundamental theorem of calculus we have $f(b)g(b) - f(a)g(a) = \int_a^b f'(x)g(x) \, dx + \int_a^b f(x)g'(x) \, dx$. $\square$

Corollary 1.6.24 (Integration By Substitution)

Let $g : [\alpha, \beta] \rightarrow [a, b]$ with $g(\alpha) = a$ and $g(\beta) = b$, let g' exist and be continuous on

$[\alpha, \beta]$. Also let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then

$$\int_a^b f(x) \, dx = \int_\alpha^\beta f(g(t))g'(t) \, dt.$$

Proof. Set $F(x) = \int_a^x f(t) \, dt$, and let $h(t) = F(g(t))$, which is defined since g takes values in $[a, b]$. Then by the fundamental theorem of calculus we have

$$\int_\alpha^\beta f(g(t))g'(t) \, dt = \int_\alpha^\beta F'(g(t))g'(t) \, dt$$

Then by the chain rule, we have

$$\int_\alpha^\beta F'(g(t))g'(t) \, dt = \int_\alpha^\beta h'(t) \, dt = h(\beta) - h(\alpha) = F(b) - F(a) = \int_a^b f(x) \, dx.$$

□

§1.6.5 Taylor's Theorem Again

With some results now built up around integration, we are going to spend the next two sections discussing two applications of integration to topics we have discussed earlier. The first will be revisiting Taylor's theorem, and after that we will look back at the convergence of infinite series.

A nice application of integration by parts (and by extension the fundamental theorem of calculus) is in proving Taylor's theorem. You may recall from our discussion of differentiation that the proof of Taylor's theorem with Lagrange remainder was somewhat cumbersome. Using integration, we can derive another form of Taylor's theorem (this time with the 'integral remainder'). This other form can then even be used to obtain Lagrange's form of the remainder, under some slightly stronger assumptions about our function¹⁹.

Theorem 1.6.25 (Taylor's Theorem with Integral Remainder)

Let f be n -times continuously differentiable on $[a, b]$. Then we have

$$f(b) = f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \int_a^b \frac{f^{(n)}(t)}{(n-1)!}(b-t)^{n-1} \, dt$$

Proof. We do induction on n . For $n = 1$, the result becomes $f(b) - f(a) = \int_a^b f'(t) \, dt$, which is true by the fundamental theorem of calculus.

Now if f is $(n+1)$ -times continuously differentiable, integrating by parts we have

$$\int_a^b \frac{f^{(n)}(t)}{(n-1)!}(b-t)^{n-1} \, dt = \left[-\frac{f^{(n)}(t)}{n!}(b-t)^n \right]_a^b + \int_a^b \frac{f^{(n+1)}(t)}{n!}(b-t)^n \, dt$$

¹⁹This is why we don't just throw out the old proof! We need a stronger continuity condition than before - specifically $f^{(n)}$ being continuous, not just existing.

$$= \frac{f^{(n)}(a)}{n!}(b-a)^n + \int_a^b \frac{f^{(n+1)}(t)}{n!}(b-t)^n dt.$$

Thus if the result is true for n , it is also true for $n+1$, so we are done by induction. $\square$

With this result in hand, we can try and re-obtain the Lagrange form of the remainder. Looking back at the form of this, we really want to some way take the $f^{(n)}(t)$ term ‘out of the integral’, turning it into some scale factor $f^{(n)}(c)$, for some $c \in (a, b)$.

The way we worked with this type of idea before was using the mean value theorem, so we are going to prove an analogue of the mean value theorem for integrals. To do this we will (naturally) use the mean value theorem along with the fundamental theorem of calculus.

Lemma 1.6.26 (Integral Mean Value Theorem)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous with $g(x) \neq 0$ for all $x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\int_a^b f(x)g(x) dx = f(c) \int_a^b g(x) dx.$$

Proof. Let $F(x) = \int_a^x f(t)g(t) dt$ and $G(x) = \int_a^x g(t) dt$. By Cauchy’s mean value theorem, there exists some $c \in (a, b)$ such that

$$(F(b) - F(a))G'(c) = F'(c)(G(b) - G(a)),$$

which by the fundamental theorem of calculus implies

$$\left(\int_a^b f(t)g(t) dt \right) g(c) = f(c)g(c) \int_a^b g(t) dt.$$

Then since $g(c) \neq 0$, the result follows. $\square$

And now we can write down a weaker form of Taylor’s theorem with Lagrange remainder.

Theorem 1.6.27 (Taylor’s Theorem with Lagrange Remainder – Weaker)

Let f be n -times continuously differentiable on $[a, b]$. Then we have

$$f(b) = f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \frac{f^{(n)}(c)}{n!}(b-a)^n,$$

for some $c \in (a, b)$.

Proof. By Taylor’s theorem with integral remainder, we have

$$f(b) = f(a) + f'(a)(b-a) + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(b-a)^{n-1} + \int_a^b \frac{f^{(n)}(t)}{(n-1)!}(b-t)^{n-1} dt.$$

Then the integral mean value theorem gives us that

$$\int_a^b \frac{f^{(n)}(t)}{(n-1)!} (b-t)^{n-1} dt = f^{(n)}(c) \int_a^b \frac{(b-t)^{n-1}}{(n-1)!} dt = \frac{f^{(n)}(c)}{n!} (b-a)^n,$$

for some $c \in (a, b)$, giving our result. $\square$

Now, to just emphasize one last time: this version of the result imposes *an extra continuity condition*, and *does not replace the result and proof we had earlier*. That said, it's still a perfectly good result on its own.

§1.6.6 Improper Integrals & The Integral Test

So far our definition of Riemann integration applies only to functions that are bounded over some closed interval. In this subsection we are going to try and extend this definition in a natural way to discuss the integration of functions that are not necessarily bounded and integration over an interval that may not be closed (and may be infinite).

Definition 1.6.28 (Improper Integrals)

Suppose $f : [a, b) \rightarrow \mathbb{R}$ is integrable (and bounded) on every interval $[a, R]$, where $b \in \mathbb{R} \cup \{\infty\}$, and

$$\int_a^R f(x) dx \rightarrow \ell \quad \text{as } R \rightarrow b.$$

Then we say that the **improper integral** $\int_a^b f(x) dx$ converges, and that its value is ℓ . Otherwise, we say that it diverges.

Let's have a look at some examples.

Example 1.6.29 (Integral of $1/x^k$ from 1 to ∞)

We will show that the improper integral $\int_1^\infty \frac{1}{x^k} dx$ converges if and only if $k > 1$.

If $k \neq 1$, then considering the integral of $1/x^k$ over $[1, R]$, we have

$$\begin{aligned} \int_1^R \frac{1}{x^k} dx &= \left. \frac{x^{1-k}}{1-k} \right|_1^R \\ &= \frac{R^{1-k} - 1}{1-k}, \end{aligned}$$

which tends to a finite limit as $R \rightarrow \infty$, namely $-1/(1-k)$, if and only if $k > 1$.

Otherwise, if $k = 1$, then over $[1, R]$ we have

$$\int_1^R \frac{1}{x} dx = \log R,$$

which does not tend to a finite limit as $R \rightarrow \infty$. Thus the integral converges if and only if $k > 1$.

Example 1.6.30 (Integral of $1/\sqrt{x}$ from 0 to 1)

We will show that the improper integral $\int_0^1 \frac{1}{\sqrt{x}} dx$ converges.

For any $\delta > 0$, we have

$$\int_{\delta}^1 \frac{1}{\sqrt{x}} dx = 2\sqrt{x} \Big|_{\delta}^1 = 2 - 2\sqrt{\delta} \rightarrow 2 \quad \text{as } \delta \rightarrow 0,$$

and thus the improper integral converges to 2.

Note the example above shows that it is possible to have an improper integral which converges when the function is not bounded on the interval being integrated over.²⁰

The definition of improper integrals also gives us a sensible way to say if $\int_{-\infty}^{\infty} f(x) dx$ converges. If for some a we have $\int_{-\infty}^a f(x) dx = \ell_1$ and $\int_a^{\infty} f(x) dx = \ell_2$, then we say that $\int_{-\infty}^{\infty} f(x) dx = \ell_1 + \ell_2$.

Remark (Warning). This is a strictly stronger notion than saying that $\int_{-R}^R f(x) dx$ converges to some limit as $R \rightarrow \infty$.

It is with improper integrals that we get the last application of integration to our previously discussed topics: infinite series.

Theorem 1.6.31 (The Integral Comparison Test)

If $f : [1, \infty) \rightarrow \mathbb{R}$ is a non-negative decreasing function with $f(x) \rightarrow 0$ as $x \rightarrow \infty$ then

$$\sum_1^{\infty} f(n) \quad \text{and} \quad \int_1^{\infty} f(x) dx$$

either both diverge or both converge.

Proof. Since f is decreasing and is non-negative, we have for $n \geq 2$ that

$$\int_n^{n+1} f(x) dx \leq f(n) \leq \int_{n-1}^n f(x) dx.$$

That is, since $\int_1^2 f(x) dx \leq f(1)$, we have

$$\int_1^{N+1} f(x) dx \leq \sum_{n=1}^N f(n) \leq \int_1^N f(x) dx + f(1),$$

and then our result follows as $N \rightarrow \infty$. □

The integral comparison test is useful because in many cases it is easier to evaluate an integral than it is to evaluate a sum (of course this isn't always true, and there's many examples in both directions). This test also gives us a way to determine the convergence of sums that previously required tools such as Cauchy's condensation test²¹. An example

²⁰It is also possible to have a function f that is unbounded on the interval $[1, \infty)$, and yet $\int_1^{\infty} f(x) dx$ converges. If you haven't seen this before, try coming up with an example (as a hint, one way to do it is by drawing triangles in a somewhat clever way).

²¹Though it can be rather cumbersome to evaluate the integrals in certain cases, in which case reaching

is shown below.

Example 1.6.32 (Using the Integral Test)

We will prove that $\sum_{n=2}^{\infty} \frac{\log n}{n^2}$ converges using the integral test.

We compute using integration by parts that

$$\begin{aligned}\int_2^R \frac{\log x}{x^2} dx &= -\frac{\log x}{x} \Big|_2^R + \int_2^R \frac{1}{x^2} dx \\ &= -\frac{\log R}{R} + \frac{\log 2}{2} - \frac{1}{R} + \frac{1}{2}.\end{aligned}$$

This converges to $(1+\log 2)/2$ as $R \rightarrow \infty$, and thus $\int_2^{\infty} \log x/x^2 dx$ and $\sum_{n=2}^{\infty} \log n/n^2$ both converge by the integral test.

for Cauchy's condensation test first might be still helpful. Using the integral test also is a straightforward way to derive Cauchy's condensation test

2 Differential Equations

A differential equation is an equation that relates a function to its derivatives. Such equations are the natural language in which much of science is written – whenever the rate at which something changes depends on its current state, a differential equation is lurking. This course is about developing a collection of techniques for solving (and, when solving is too hard, understanding the behaviour of) differential equations and their discrete cousins, difference equations.

This article constitutes my notes for the ‘Differential Equations’ course, held in Michaelmas 2020 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§2.1 A Revision of Calculus

This course leans heavily on the machinery of calculus, so we will begin with a quick (and informal) review of differentiation, along with some notation for describing the sizes of functions that will be used constantly in what follows. This section is not meant to be a first introduction to calculus¹, and the reader should feel free to skim it.

§2.1.1 Differentiation

We recall the basic definition of the derivative of a function.

Definition 2.1.1 (Derivative)

The **derivative** of a function $f(x)$ with respect to x is

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

when this limit exists. We say f is **differentiable** at x if the limit exists, which requires the limits from the left and from the right to exist and agree.

The derivative measures the instantaneous rate of change of f , and geometrically it is the slope of the graph of f . We will use the notations

$$\frac{df}{dx} = f'(x) = \frac{d}{dx}f(x)$$

interchangeably, and write f'' , f''' , $f^{(n)}$ and so on for higher derivatives. In this course we will also frequently use Newton’s notation $\dot{y}$, $\ddot{y}$ for derivatives with respect to time.

One thing worth being slightly careful about: the notation f' always denotes the derivative of f *with respect to its argument*. So $f'(2x)$ means the derivative of f evaluated at $2x$, not the derivative of the function $x \mapsto f(2x)$.

¹The reader who wants to see calculus done *properly*, with all of the analytic details filled in, should consult the ‘Analysis I’ course. In this course we take a rather more relaxed attitude towards questions of convergence and rigour.

Example 2.1.2 (A Function That is Not Differentiable)

The function $f(x) = |x|$ is not differentiable at $x = 0$, since

$$\lim_{h \rightarrow 0^+} \frac{|h| - |0|}{h} = 1, \quad \lim_{h \rightarrow 0^-} \frac{|h| - |0|}{h} = -1,$$

and these two one-sided limits disagree.

§2.1.2 Big- O and Little- o Notation

Throughout this course (and indeed throughout mathematics), it is useful to have a compact notation for comparing the sizes of functions near a point.

Definition 2.1.3 (Little- o)

We write $f(x) = o(g(x))$ as $x \rightarrow x_0$ if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

Informally, ‘ f is much smaller than g near x_0 ’.

Definition 2.1.4 (Big- O)

We write $f(x) = O(g(x))$ as $x \rightarrow x_0$ if $f(x)/g(x)$ is bounded as $x \rightarrow x_0$. Informally, ‘ f is at most comparable in size to g near x_0 ’.

Usually x_0 will be 0 or ∞ , and often we will not say which explicitly when it is clear from context. Note that $f = o(g)$ implies $f = O(g)$, but not conversely. Also, for $f = O(g)$ we do *not* require that $\lim f/g$ exists – boundedness is enough.

There is a standard abuse of notation here that is worth pointing out. When we write $f(x) = o(g(x))$, we are not saying that $f(x)$ literally equals some function called ‘ $o(g(x))$ ’; rather, $o(g(x))$ stands for the *class* of functions small compared to g , and we are asserting that f belongs to it. In particular such ‘equations’ should only be read left to right.

Example 2.1.5

As illustrations of the notation:

- (i) $x = o(\sqrt{x})$ as $x \rightarrow 0$, while $\sqrt{x} = o(x)$ as $x \rightarrow \infty$.
- (ii) $\sin 2x = O(x)$ as $x \rightarrow 0$, since $\sin \theta \approx \theta$ for small θ .
- (iii) $\sin 2x = O(1)$ as $x \rightarrow \infty$, even though $\lim_{x \rightarrow \infty} \sin 2x$ does not exist.

The real utility of this notation is that it lets us do calculus ‘to leading order’, keeping track of the error terms without needing to write them down explicitly. As a first example of this, we can rephrase differentiability.

Proposition 2.1.6 (Derivative as Linear Approximation)

If f is differentiable at x_0 then

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h) \quad \text{as } h \rightarrow 0.$$

Proof. By the definition of the derivative,

$$\frac{f(x_0 + h) - f(x_0)}{h} - f'(x_0) \rightarrow 0 \quad \text{as } h \rightarrow 0,$$

so the left-hand side is $o(h)/h$, that is, $f(x_0 + h) - f(x_0) - f'(x_0)h = o(h)$. $\square$

In other words – near x_0 , a differentiable function looks like a straight line, up to an error which is small compared to h . This point of view (functions being locally linear) is the correct way to think about differentiation, and it is the form of the definition which generalises to higher dimensions.

§2.1.3 Rules for Differentiation

The basic rules of differentiation can all be derived quite cleanly using the o notation. We will do the chain rule as an example.

Theorem 2.1.7 (Chain Rule)

If $f(x) = F(g(x))$, then

$$\frac{df}{dx} = \frac{dF}{dg} \frac{dg}{dx} = F'(g(x)) g'(x).$$

Proof. Assuming g is differentiable at x and F is differentiable at $g(x)$, we compute

$$\begin{aligned} \frac{df}{dx} &= \lim_{h \rightarrow 0} \frac{F(g(x+h)) - F(g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{F(g(x) + hg'(x) + o(h)) - F(g(x))}{h} \\ &= \lim_{h \rightarrow 0} \frac{F(g(x)) + (hg'(x) + o(h))F'(g(x)) + o(hg'(x) + o(h)) - F(g(x))}{h} \\ &= \lim_{h \rightarrow 0} \left[g'(x)F'(g(x)) + \frac{o(h)}{h} \right] = g'(x)F'(g(x)). \square \end{aligned}$$

Theorem 2.1.8 (Product Rule)

If $f(x) = u(x)v(x)$, then $f'(x) = u'(x)v(x) + u(x)v'(x)$.

The product rule extends to higher derivatives in a way that looks suspiciously like the binomial theorem (and indeed the proof is an induction of exactly the same shape).

Theorem 2.1.9 (Leibniz's Rule)

If $f = uv$ then the n th derivative of f is

$$f^{(n)}(x) = \sum_{r=0}^n \binom{n}{r} u^{(r)}(x) v^{(n-r)}(x).$$

Example 2.1.10

Let's find the n th derivative of $f(x) = x^2 e^x$. Take $u = x^2$ and $v = e^x$. All derivatives of v are e^x , while $u' = 2x$, $u'' = 2$, and all higher derivatives of u vanish. So only three terms of the sum survive:

$$f^{(n)}(x) = \binom{n}{0} x^2 e^x + \binom{n}{1} \cdot 2x \cdot e^x + \binom{n}{2} \cdot 2 \cdot e^x = (x^2 + 2nx + n(n-1)) e^x.$$

Leibniz's rule is at its best in situations like this, where one factor is eventually annihilated by differentiation.

§2.1.4 Taylor Series

Linear approximation says a differentiable function looks like a line up to $o(h)$. If the function has more derivatives, we can do better.

Theorem 2.1.11 (Taylor's Theorem)

If f is n -times differentiable at x , then

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \cdots + \frac{h^n}{n!} f^{(n)}(x) + E_n,$$

where $E_n = o(h^n)$ as $h \rightarrow 0$. If additionally $f^{(n+1)}$ exists, then $E_n = O(h^{n+1})$.

It is sometimes convenient to relabel and write this as

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + E_n,$$

an expansion of f *about the point* x_0 . Letting $n \rightarrow \infty$ (and not worrying too much about when this is valid) gives the **Taylor series** of f about x_0 .

We should stress that Taylor's theorem is a purely *local* statement: it describes the behaviour of f arbitrarily close to x_0 , and by itself says nothing about what f does far away.

Example 2.1.12 (Standard Taylor Series)

Some expansions about $x = 0$ that we will use freely throughout the course:

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \\ \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \quad (|x| < 1), \end{aligned}$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots \quad (|x| < 1).$$

Each follows by computing derivatives at 0 and applying the theorem. The most common use in this course is truncation: statements like $e^\varepsilon = 1 + \varepsilon + O(\varepsilon^2)$ or $\sin \varepsilon = \varepsilon + O(\varepsilon^3)$, valid for small ε , are the engine behind every ‘linearisation’ we will perform.

§2.1.5 L'Hôpital's Rule

A pleasant application of Taylor's theorem is a systematic way of evaluating limits of the form $0/0$.

Theorem 2.1.13 (L'Hôpital's Rule)

Let f and g be differentiable at x_0 , with $\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} g(x) = 0$. Then

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)},$$

provided the right-hand limit exists.

Proof. By Taylor's theorem, $f(x) = f(x_0) + (x - x_0)f'(x_0) + o(x - x_0)$, and similarly for g . Since $f(x_0) = g(x_0) = 0$,

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{(x - x_0)f'(x_0) + o(x - x_0)}{(x - x_0)g'(x_0) + o(x - x_0)} = \lim_{x \rightarrow x_0} \frac{f'(x_0) + \frac{o(x-x_0)}{x-x_0}}{g'(x_0) + \frac{o(x-x_0)}{x-x_0}} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)}.$$

□

Example 2.1.14

Let's compute $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$. Both numerator and denominator vanish at 0, so

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{1}{2},$$

applying the rule twice. (Alternatively, just expand $\cos x = 1 - x^2/2 + O(x^4)$ and read off the answer – Taylor series are frequently faster than repeated applications of L'Hôpital.)

Example 2.1.15

As another quick illustration,

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x}{1} = 1.$$

In fact, whenever $g(x) = x$ and $f(0) = 0$, L'Hôpital's rule simply restates the definition of $f'(0)$ as a limiting difference quotient – the rule is a generalisation of the derivative itself.

§2.1.6 Integration

We will also need integration, which we think of informally as computing the area under a graph: for a partition $a = x_0 < x_1 < \cdots < x_N = b$ with spacing Δx ,

$$\int_a^b f(x) \, dx = \lim_{\Delta x \rightarrow 0} \sum_n f(x_n) \Delta x.$$

The connection between integration and differentiation is the fundamental theorem of calculus.

Theorem 2.1.16 (Fundamental Theorem of Calculus)

Let $F(x) = \int_a^x f(t) \, dt$. Then $F'(x) = f(x)$.

Proof. We compute directly, using the fact that the integral from x to $x+h$ is an area of width h and height $f(x) + o(1)$:

$$F'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} f(t) \, dt - \int_a^x f(t) \, dt \right] = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) \, dt = \lim_{h \rightarrow 0} \frac{f(x)h + o(h)}{h} = f(x). \quad \square$$

Some direct consequences worth recording: $\frac{d}{dx} \int_x^b f(t) \, dt = -f(x)$, and by the chain rule

$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x)) g'(x).$$

From the product rule we obtain the standard technique of **integration by parts**,

$$\int uv' \, dx = uv - \int u'v \, dx,$$

and from the chain rule we get **integration by substitution**. We will assume fluency with both.

Example 2.1.17 (Integration by Substitution)

Consider $\int \sqrt{2x - x^2} \, dx$. Completing the square, $2x - x^2 = 1 - (x - 1)^2$, which suggests the trigonometric substitution $x - 1 = \sin \theta$, $dx = \cos \theta \, d\theta$. The integral becomes

$$\begin{aligned} \int \cos^2 \theta \, d\theta &= \int \frac{1 + \cos 2\theta}{2} \, d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + C \\ &= \frac{1}{2} \sin^{-1}(x - 1) + \frac{1}{2}(x - 1)\sqrt{2x - x^2} + C, \end{aligned}$$

undoing the substitution with $\sin 2\theta = 2 \sin \theta \cos \theta$ and $\cos \theta = \sqrt{2x - x^2}$.

Example 2.1.18 (Integration by Parts)

Consider $\int \log x \, dx$. Taking $u = \log x$ and $v' = 1$, we have $u' = 1/x$ and $v = x$, so

$$\int \log x \, dx = x \log x - \int dx = x \log x - x + C.$$

As another quick example, $\int_0^\infty x e^{-x} dx$: taking $u = x$ and $v' = e^{-x}$,

$$\int_0^\infty x e^{-x} dx = [-x e^{-x}]_0^\infty + \int_0^\infty e^{-x} dx = 0 + 1 = 1.$$

§2.2 Functions of Several Variables

Many of the differential equations we will meet involve functions of more than one variable, so before going any further we should establish the basics of multivariate calculus. We will return to this subject in more depth later in the course; for now we need just enough to get going.

§2.2.1 Partial Derivatives

If $f(x, y)$ is a function of two variables, we can differentiate it with respect to either one, holding the other fixed.

Definition 2.2.1 (Partial Derivative)

The **partial derivative** of $f(x, y)$ with respect to x is the rate of change of f as x varies with y held constant:

$$\left. \frac{\partial f}{\partial x} \right|_y = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}.$$

In practice, computing a partial derivative is no harder than ordinary differentiation – we just treat the other variables as constants. When it is clear which variables are being held constant we drop the subscript and write $\partial f / \partial x$, and we will also use the compact notation

$$f_x = \frac{\partial f}{\partial x}, \quad f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y \partial x}.$$

Example 2.2.2

Let $f(x, y) = x^2 + y^3 + e^{xy^2}$. Then

$$f_x = 2x + y^2 e^{xy^2}, \quad f_y = 3y^2 + 2xy e^{xy^2},$$

and differentiating again,

$$f_{xx} = 2 + y^4 e^{xy^2}, \quad f_{xy} = 2y e^{xy^2} + 2xy^3 e^{xy^2}.$$

The reader is invited to check that computing f_{yx} gives the same answer as f_{xy} .

That last observation is no coincidence.

Theorem 2.2.3 (Symmetry of Mixed Partial Derivatives)

If f has continuous second partial derivatives, then $f_{xy} = f_{yx}$.

We will not prove this (the proof belongs to an analysis course), but we will use it freely.

§2.2.2 The Multivariate Chain Rule and Differentials

Suppose we perturb both arguments of $f(x, y)$ at once, sending $(x, y) \mapsto (x + \delta x, y + \delta y)$. The change in f can be computed by moving one coordinate at a time:

$$\begin{aligned}\delta f &= f(x + \delta x, y + \delta y) - f(x, y) \\ &= [f(x + \delta x, y + \delta y) - f(x + \delta x, y)] + [f(x + \delta x, y) - f(x, y)] \\ &= f_y(x + \delta x, y) \delta y + o(\delta y) + f_x(x, y) \delta x + o(\delta x) \\ &= \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + o(\delta x, \delta y).\end{aligned}$$

Taking the limit of small displacements, we write this relation between infinitesimals as follows.

Theorem 2.2.4 (Chain Rule for Partial Derivatives)

For $f = f(x, y)$,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

The expression on the right is called the **differential** of f . This one formula packages up every version of the chain rule we could want. For instance, if x and y are both functions of a parameter t , dividing by dt gives the derivative of f along the path $(x(t), y(t))$:

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Similarly, if we move along a path given as a graph $y = y(x)$, then

$$\frac{df}{dx} = \frac{\partial f}{\partial x} \Big|_y + \frac{\partial f}{\partial y} \frac{dy}{dx},$$

and if we change to new independent variables, say polar coordinates $x = x(r, \theta)$, $y = y(r, \theta)$, then

$$\frac{\partial f}{\partial \theta} \Big|_r = \frac{\partial f}{\partial x} \Big|_y \frac{\partial x}{\partial \theta} \Big|_r + \frac{\partial f}{\partial y} \Big|_x \frac{\partial y}{\partial \theta} \Big|_r.$$

Example 2.2.5

Let $f(x, y) = x^2 + y^2$, and let's differentiate f along the circular path $x = \cos t$, $y = \sin t$:

$$\frac{df}{dt} = 2x \cdot (-\sin t) + 2y \cdot \cos t = -2 \cos t \sin t + 2 \sin t \cos t = 0,$$

as it should be – the path runs along a contour of f , so f doesn't change along it.

§2.2.3 Exact Differentials

A general expression of the form $P(x, y) dx + Q(x, y) dy$ is called a *differential form*. Sometimes such a form is the differential of some function, and sometimes it is not.

Definition 2.2.6 (Exact Differential)

The differential form $P(x, y) dx + Q(x, y) dy$ is **exact** if there is a function $f(x, y)$ such that

$$df = P(x, y) dx + Q(x, y) dy.$$

If the form is exact, then comparing with $df = f_x dx + f_y dy$ (and noting the displacements dx, dy are arbitrary) we must have

$$\frac{\partial f}{\partial x} = P, \quad \frac{\partial f}{\partial y} = Q.$$

Differentiating the first with respect to y and the second with respect to x , symmetry of mixed partials forces a necessary condition:

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

This condition is not quite sufficient in general, but it is if the domain we are working on has no ‘holes’.

Definition 2.2.7 (Simply Connected Domain)

A domain $\mathcal{D}$ is **simply connected** if it is connected and every closed curve in $\mathcal{D}$ can be continuously shrunk to a point without leaving $\mathcal{D}$.

For example, a disc is simply connected, but a disc with a point removed is not (a loop around the missing point cannot be shrunk down).

Theorem 2.2.8

If $\partial P/\partial y = \partial Q/\partial x$ everywhere on a simply connected domain $\mathcal{D}$, then $P dx + Q dy$ is the exact differential of some single-valued function on $\mathcal{D}$.

We won’t prove this, but we will make good use of it when we come to solve ‘exact’ differential equations shortly.

§2.2.4 Implicit Differentiation and Differentiating Integrals

Suppose a surface is given implicitly by $F(x, y, z) = c$ for a constant c . This implicitly defines z as a function $z(x, y)$, even when we cannot solve for z explicitly. We can still compute its partial derivatives: taking the differential of both sides,

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial z} dz = 0.$$

Holding y constant ($dy = 0$) and dividing by dx gives the following.

Proposition 2.2.9 (Implicit Differentiation)

If $F(x, y, z) = c$ defines $z = z(x, y)$, then

$$\left. \frac{\partial z}{\partial x} \right|_y = -\frac{\partial F / \partial x}{\partial F / \partial z}.$$

Example 2.2.10

Let $xy^2 + yz^2 + z^5x = 5$. Solving for $z(x, y)$ explicitly would require solving a quintic, which is hopeless, but differentiating with respect to x (holding y fixed) gives

$$y^2 + 2yz \frac{\partial z}{\partial x} + z^5 + 5z^4x \frac{\partial z}{\partial x} = 0 \implies \frac{\partial z}{\partial x} = -\frac{y^2 + z^5}{2yz + 5z^4x}.$$

One more tool in a similar spirit: differentiating an integral with respect to a parameter. If $I(b, c) = \int_0^b f(x, c) \, dx$, then $\partial I / \partial b = f(b, c)$ by the fundamental theorem of calculus, and (interchanging the limit and the integral, which we will do without comment)

$$\frac{\partial I}{\partial c} = \lim_{\delta c \rightarrow 0} \int_0^b \frac{f(x, c + \delta c) - f(x, c)}{\delta c} \, dx = \int_0^b \frac{\partial f}{\partial c} \, dx.$$

Combining these with the chain rule gives the general formula for differentiating an integral whose limit and integrand both depend on x .

Theorem 2.2.11 (Differentiation Under the Integral Sign)

$$\frac{d}{dx} \int_0^{b(x)} f(y, c(x)) \, dy = f(b, c) b'(x) + c'(x) \int_0^b \frac{\partial f}{\partial c} \, dy.$$

§2.3 First-Order Linear Differential Equations

We now begin the course proper. A *first-order* differential equation is one involving only first derivatives of the unknown function. In this section we deal with the linear ones, which we can solve completely.

Let's first fix some terminology, which applies to differential equations of any order.

Definition 2.3.1 (Linearity)

A differential equation for $y(x)$ is **linear** if the dependent variable y and its derivatives appear only linearly.

Definition 2.3.2 (Homogeneous Equation)

A linear differential equation is **homogeneous** if $y = 0$ is a solution.

Definition 2.3.3 (Constant Coefficients)

A linear differential equation has **constant coefficients** if the coefficients multiplying y and its derivatives do not depend on the independent variable.

So, for example, $5y' - 3y = 0$ is linear, homogeneous, and has constant coefficients; $y' + xy = x^2$ is linear with non-constant coefficients and is not homogeneous; and $yy' = 1$ is not linear.

§2.3.1 The Exponential Function

The solutions of linear differential equations are built almost entirely out of exponentials, so it is worth pausing to recall why the exponential function is so special.

Consider $f(x) = a^x$ for some constant $a > 0$. Then

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} = \lambda a^x,$$

where $\lambda = f'(0)$ is a constant depending only on a . So the derivative of an exponential is a *multiple of itself*. The natural choice of base is the one for which this multiple is exactly 1.

Definition 2.3.4 (Exponential Function)

The function $\exp(x) = e^x$ is the unique function f with $f'(x) = f(x)$ and $f(0) = 1$. Its inverse is written $\ln x$ or $\log x$.

With this in hand, if $y = a^x = e^{x \ln a}$ then $y' = a^x \ln a$, so the constant λ above is $\ln a$.

The reason exponentials matter so much for us is the following language.

Definition 2.3.5 (Eigenfunction)

A function f is an **eigenfunction** of the differential operator $\frac{d}{dx}$ if

$$\frac{df}{dx} = \lambda f$$

for some constant λ , the **eigenvalue**.

The functions e^{mx} are eigenfunctions of $\frac{d}{dx}$ with eigenvalue m , since $\frac{d}{dx}e^{mx} = me^{mx}$. Differentiation doesn't change the *shape* of an exponential, only its magnitude – and this is precisely why substituting exponentials into constant coefficient equations works so well, as we are about to see.

§2.3.2 Homogeneous Equations with Constant Coefficients

Everything in this section rests on the following observation about exponentials.

Proposition 2.3.6

Every linear, homogeneous ordinary differential equation with constant coefficients has solutions of the form e^{mx} .

Rather than proving this in general (the general statement will fall out of our later work), let's see the idea in action.

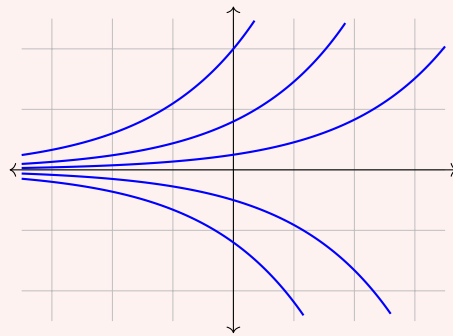
Example 2.3.7

Consider $5y' - 3y = 0$. Trying $y = e^{mx}$, we get

$$5me^{mx} - 3e^{mx} = 0.$$

Since e^{mx} is never zero we may divide by it, giving $5m - 3 = 0$, so $m = 3/5$ and $y = e^{3x/5}$ is a solution.

Because the equation is linear and homogeneous, any constant multiple of a solution is again a solution, so $y = Ae^{3x/5}$ solves the equation for any constant A . In fact this is the *general* solution: a first-order linear equation has exactly a one-parameter family of solutions^a, and we have found it. The constant A is then determined by an initial or boundary condition, such as the value $y(0)$. Here is the family of solutions $y = Ae^{3x/5}$, one curve for each value of A :



^aMore generally, an n th-order linear equation has an n -parameter family of solutions – the solution space of the homogeneous equation is a vector space of dimension n . We will see why when we discuss phase space in §2.6.4.

Comparison with a Discrete Equation

Before moving on, it is illuminating to see how the exponential arises as the limit of a *discrete* process. Suppose we wanted to solve $5y' - 3y = 0$ with $y(0) = y_0$ numerically. We could discretise: take small steps of size h , write y_n for our approximation to $y(x_n)$ where $x_n = nh$, and replace the derivative by a difference quotient:

$$5 \frac{y_{n+1} - y_n}{h} - 3y_n \approx 0.$$

Rearranging, $y_{n+1} = (1 + \frac{3}{5}h) y_n$, and applying this repeatedly,

$$y_n = \left(1 + \frac{3}{5}h\right)^n y_0.$$

Now fix a point x and let the step size shrink, taking $h = x/n$ with $n \rightarrow \infty$:

$$y(x) = \lim_{n \rightarrow \infty} y_0 \left(1 + \frac{3x/5}{n}\right)^n = y_0 e^{3x/5},$$

recovering exactly the solution of the differential equation, using the standard limit $e^t = \lim_{n \rightarrow \infty} (1 + t/n)^n$. The exponential is what compound growth becomes when the compounding is continuous.

Series Solution

There is yet a third way to solve $5y' - 3y = 0$, which foreshadows a technique we will use seriously later in the course: assume the solution has a power series expansion

$$y = \sum_{n=0}^{\infty} a_n x^n,$$

substitute it in, and see what the coefficients must be. We have $y' = \sum_n n a_n x^{n-1}$, so

$$5y' - 3y = \sum_n (5(n+1)a_{n+1} - 3a_n) x^n = 0.$$

For this to vanish identically, the coefficient of each power of x must vanish, so $a_{n+1} = \frac{3}{5(n+1)}a_n$ for all $n \geq 0$. Iterating this recurrence,

$$a_n = \frac{3}{5n} a_{n-1} = \frac{3^2}{5^2 n(n-1)} a_{n-2} = \cdots = \left(\frac{3}{5}\right)^n \frac{1}{n!} a_0,$$

with a_0 arbitrary (it is our constant of integration). Hence

$$y = a_0 \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{3x}{5}\right)^n = a_0 e^{3x/5},$$

as expected.

§2.3.3 Forced Equations

An equation like $5y' - 3y = 10$ or $5y' - 3y = x^2$, with ‘extra’ terms not involving y , is called **forced** (or inhomogeneous), and the extra terms are **forcing terms**. The name comes from physics: such terms typically represent an external force being applied to a system.

The general strategy for solving forced linear equations is one we will use over and over again:

- (1) Find *any* single solution y_p of the forced equation, called a **particular integral**. This is often done by an educated guess.
- (2) Solve the associated homogeneous equation to find the general **complementary function** y_c .
- (3) The general solution is $y = y_p + y_c$.

Why does this work? If y is any solution of the forced equation, then $y - y_p$ solves the homogeneous equation by linearity, so $y - y_p$ is a complementary function. It is worth internalising this structure: the solutions of a forced linear equation form the translate of a vector space (the homogeneous solutions) by a particular solution.

Example 2.3.8 (Constant Forcing)

Consider $5y' - 3y = 10$. We spot the constant (equilibrium) solution $y_p = -10/3$. The complementary function satisfies $5y'_c - 3y_c = 0$, which we solved above. So the general solution is

$$y = -\frac{10}{3} + A e^{3x/5}.$$

A word of warning: any boundary condition must be applied to the *full* solution y ,

not just to the complementary function.

Eigenfunction Forcing

A particularly clean situation is when the forcing term is itself an eigenfunction of the differential operator. The natural guess for the particular integral is then a multiple of the forcing.

Example 2.3.9 (Radioactive Decay Chains)

In a radioactive rock, isotope A decays into isotope B at a rate proportional to the number a of remaining A nuclei, and B in turn decays at a rate proportional to the number b of B nuclei. We would like to determine $b(t)$.

The setup gives us the pair of equations

$$\frac{da}{dt} = -k_a a, \quad \frac{db}{dt} = k_a a - k_b b.$$

The first equation is homogeneous, with solution $a = a_0 e^{-k_a t}$. Substituting into the second,

$$\frac{db}{dt} + k_b b = k_a a_0 e^{-k_a t}.$$

The forcing term is an eigenfunction of d/dt , so we guess a particular integral $b_p = C e^{-k_a t}$. Substituting,

$$-k_a C + k_b C = k_a a_0 \implies C = \frac{k_a}{k_b - k_a} a_0.$$

The complementary function satisfies $b'_c + k_b b_c = 0$, so $b_c = D e^{-k_b t}$, and altogether

$$b = \frac{k_a}{k_b - k_a} a_0 e^{-k_a t} + D e^{-k_b t}.$$

With the natural initial condition $b(0) = 0$ we get $D = -\frac{k_a}{k_b - k_a} a_0$, so

$$b = \frac{k_a}{k_b - k_a} a_0 (e^{-k_a t} - e^{-k_b t}).$$

This little computation has a genuinely useful application: the isotope ratio is

$$\frac{b}{a} = \frac{k_a}{k_b - k_a} [1 - e^{(k_a - k_b)t}],$$

so measuring b/a in a rock sample today, with the decay rates k_a, k_b known from laboratory measurement, determines t – the age of the rock. This is (a simplified version of) how radiometric dating works.

Example 2.3.10 (Newton's Law of Cooling)

A cup of tea at temperature $T(t)$ sits in a room at constant ambient temperature T_a . Newton's law of cooling says the tea loses heat at a rate proportional to the

temperature difference:

$$\frac{dT}{dt} = -k(T - T_a), \quad k > 0.$$

This is a forced linear equation, $\dot{T} + kT = kT_a$, with constant forcing. The equilibrium (particular) solution is $T_p = T_a$, the complementary function is Ce^{-kt} , and with initial temperature $T(0) = T_0$,

$$T(t) = T_a + (T_0 - T_a)e^{-kt}.$$

The temperature difference decays exponentially with rate k ; after a few multiples of the timescale $1/k$, the tea is at room temperature. Note how the structure ‘particular + complementary’ matches the physics: the equilibrium is where the system ends up, and the complementary function describes the decaying transient.

§2.3.4 Non-Constant Coefficients: The Integrating Factor

Now we drop the assumption of constant coefficients and consider the general first-order linear equation

$$a(x)y' + b(x)y = c(x).$$

Dividing through by $a(x)$, we may as well consider the *standard form*

$$y' + p(x)y = f(x).$$

The trick to solving this is a clever piece of reverse-engineering. Multiply the whole equation by a yet-to-be-chosen function $\mu(x)$:

$$\mu y' + \mu p y = \mu f.$$

The left-hand side would be exactly $(\mu y)'$ – a perfect derivative – if only $\mu' = \mu p$. But we can simply *choose* μ to make that happen! Solving $\mu'/\mu = p$ by integrating,

$$\ln \mu = \int p \, dx \implies \mu(x) = \exp\left(\int p(x) \, dx\right).$$

This μ is called the **integrating factor**. With this choice, the equation reads $(\mu y)' = \mu f$, and integrating once,

$$y = \frac{1}{\mu} \int \mu f \, dx, \quad \mu = \exp\left(\int p \, dx\right),$$

where the constant of integration hiding in $\int \mu f \, dx$ supplies the expected one-parameter family.

Example 2.3.11

Let's solve $xy' + (1-x)y = 1$. In standard form this is

$$y' + \frac{1-x}{x}y = \frac{1}{x},$$

so the integrating factor is

$$\mu = \exp\left(\int \left(\frac{1}{x} - 1\right) dx\right) = e^{\ln x - x} = xe^{-x}.$$

Then $(xe^{-x}y)' = xe^{-x} \cdot \frac{1}{x} = e^{-x}$, so

$$xe^{-x}y = -e^{-x} + C \implies y = -\frac{1}{x} + \frac{C}{x}e^x.$$

Suppose we are told that y remains finite as $x \rightarrow 0$. Since $y = (Ce^x - 1)/x$, we need $Ce^x - 1 \rightarrow 0$ as $x \rightarrow 0$, forcing $C = 1$. Then by L'Hôpital's rule, $y \rightarrow 1$ as $x \rightarrow 0$ – so the boundary condition picks out the unique well-behaved solution.

§2.4 Nonlinear First-Order Equations

Life gets harder (and more interesting) when we allow nonlinearity. A reasonably general first-order equation has the form

$$Q(x, y) \frac{dy}{dx} + P(x, y) = 0.$$

There is no method that solves all such equations, but there are two important classes we *can* solve exactly, and for the rest we will develop graphical and qualitative techniques that are often more informative than an explicit formula anyway.

§2.4.1 Separable Equations

The simplest nonlinear equations we can hope to solve are those where the variables can be pried apart.

Definition 2.4.1 (Separable Equation)

A first-order equation is **separable** if it can be rearranged into the form

$$q(y) dy = p(x) dx,$$

with all the y dependence on one side and all the x dependence on the other. It can then be solved directly by integrating both sides:

$$\int q(y) dy = \int p(x) dx.$$

This is the easy kind, and there isn't much to say beyond doing an example.

Example 2.4.2

Consider $(x^2y - 3y) \frac{dy}{dx} - 2xy^2 = 4x$. Rearranging,

$$\frac{dy}{dx} = \frac{4x + 2xy^2}{x^2y - 3y} = \frac{2x(2 + y^2)}{y(x^2 - 3)},$$

and now the variables separate:

$$\int \frac{y}{2 + y^2} dy = \int \frac{2x}{x^2 - 3} dx.$$

Both sides are of the form $\int u'/u$, giving

$$\frac{1}{2} \ln(2 + y^2) = \ln(x^2 - 3) + C \implies \sqrt{y^2 + 2} = A(x^2 - 3),$$

where $A = e^C$. This is a perfectly good (implicit) description of the family of solution curves.

§2.4.2 Exact Equations

The second solvable class makes direct use of the exact differentials we studied in the previous section.

Definition 2.4.3 (Exact Equation)

The equation $Q(x, y) \frac{dy}{dx} + P(x, y) = 0$ is **exact** if the differential form $P dx + Q dy$ is exact, i.e. if there is a function $f(x, y)$ with

$$df = P dx + Q dy.$$

If the equation is exact, then along any solution curve we have $df = 0$, so the solutions are simply the level sets

$$f(x, y) = \text{constant}.$$

And we know from before how to detect exactness: check whether $\partial P/\partial y = \partial Q/\partial x$ (on a simply connected domain). To find f , integrate $f_x = P$ and $f_y = Q$.

Example 2.4.4

Consider

$$6y(y - x) \frac{dy}{dx} + (2x - 3y^2) = 0.$$

Here $P = 2x - 3y^2$ and $Q = 6y(y - x) = 6y^2 - 6xy$. We check:

$$\frac{\partial P}{\partial y} = -6y = \frac{\partial Q}{\partial x},$$

so the equation is exact. Now we hunt for f . Integrating $f_x = P$ with respect to x ,

$$f = x^2 - 3xy^2 + h(y),$$

where the ‘constant of integration’ h may depend on y (we were holding x fixed). Differentiating with respect to y and comparing with $f_y = Q$:

$$-6xy + h'(y) = 6y^2 - 6xy \implies h'(y) = 6y^2 \implies h(y) = 2y^3 + C.$$

So the solutions are the curves

$$x^2 - 3xy^2 + 2y^3 = \text{constant}.$$

Remark. If an equation fails the exactness test, all is not necessarily lost: multiplying through by a suitable function can sometimes *make* it exact. Indeed, the integrating factor μ for first-order linear equations is exactly this idea – multiplying $y' + py - f = 0$

by μ turned the left-hand side into the exact derivative $(\mu y)'$. Finding integrating factors for general nonlinear equations, however, is an art rather than an algorithm.

§2.4.3 Sketching Solution Curves

For equations we cannot solve explicitly – and even for ones we can – it is extremely useful to be able to *sketch* the family of solutions of

$$\frac{dy}{dt} = f(t, y)$$

directly from the equation. The equation tells us the slope of the solution curve through every point of the (t, y) plane, and we can piece together the qualitative picture from this slope field. The main tools are:

- *Constant solutions*: values $y = c$ with $f(t, c) = 0$ for all t . Solution curves cannot cross these horizontal lines, so they partition the plane into invariant strips.
- *Isoclines*: curves along which the slope $f(t, y)$ takes a constant value D . Marking a few isoclines (in particular the $D = 0$ isocline, where solutions have horizontal tangents) pins down the flow.
- *Signs*: in each region cut out by the isocline $f = 0$, the sign of f tells us whether solutions rise or fall.

Example 2.4.5

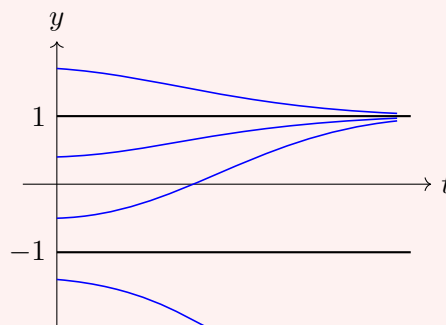
Consider

$$\frac{dy}{dt} = t(1 - y^2).$$

This is actually separable, and solving it (using partial fractions) gives

$$\int \frac{dy}{1 - y^2} = \int t \, dt \implies \frac{1}{2} \ln \frac{1 + y}{1 - y} = \frac{1}{2} t^2 + C \implies y = \frac{A - e^{-t^2}}{A + e^{-t^2}}.$$

But let's instead see how much we can say without solving. First, $y = 1$ and $y = -1$ are constant solutions. Second, $y' = 0$ whenever $t = 0$, so every solution crosses the y -axis horizontally. Third, for $t > 0$ we have $y' > 0$ when $-1 < y < 1$ and $y' < 0$ otherwise. So for $t > 0$, solutions in the strip between -1 and 1 climb towards $y = 1$; solutions above $y = 1$ descend towards it; and solutions below $y = -1$ plunge downwards. The isoclines $t(1 - y^2) = D$ can be sketched to refine the picture. All of this matches the exact solution above: as $t \rightarrow \infty$, every solution with $y(0) > -1$ tends to 1 .

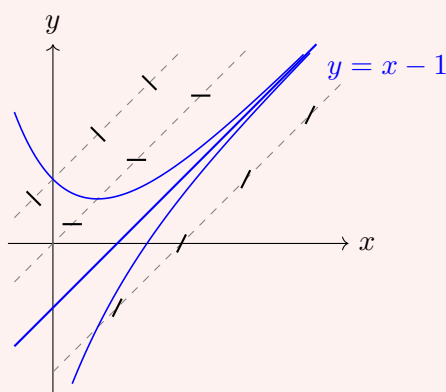


Note the qualitative structure: $y = 1$ attracts nearby solutions, while $y = -1$ repels them.

Example 2.4.6 (Straight-Line Isoclines)

Consider $\frac{dy}{dx} = x - y$. The isoclines $x - y = D$ are parallel straight lines of unit slope, along which all solution curves have slope D . Something special happens on the isocline $x - y = 1$: the prescribed slope there is 1, which is exactly the slope of the isocline itself – so that line is *itself a solution*, $y = x - 1$.

Above this line ($y > x - 1$), slopes are less than 1; below it, greater than 1; and one can convince oneself from the slope field that all solutions converge towards the line $y = x - 1$ as $x \rightarrow \infty$. In the sketch below, the dashed lines are the isoclines $D = -1, 0, 2$ (with short strokes showing the prescribed slope on each), and the blue curves are solutions.



Since the equation is actually linear, we can confirm this: the integrating factor is e^x , giving $(e^x y)' = x e^x$, so $e^x y = (x - 1)e^x + C$ and

$$y = x - 1 + C e^{-x}.$$

Every solution is the special straight line plus an exponentially decaying correction – precisely the convergence the isoclines predicted.

§2.4.4 Equilibrium Points and Stability

The behaviour in the last example – solutions being attracted to or repelled from constant solutions – is so important that it deserves proper terminology.

Definition 2.4.7 (Equilibrium Point)

An **equilibrium point** (or **fixed point**) of a differential equation is a constant solution $y = c$, i.e. one with $\frac{dy}{dt} = 0$ for all t .

Equilibria are usually the easiest solutions to find: just set the right-hand side of the equation to zero and solve.

Definition 2.4.8 (Stability)

An equilibrium $y = c$ is **stable** if whenever y is perturbed slightly from c , we have $y \rightarrow c$ as $t \rightarrow \infty$. It is **unstable** if small perturbations grow.

Perturbation Analysis

To determine stability we use *perturbation analysis*, which is really just a Taylor expansion. Suppose $y = a$ is a fixed point of $\dot{y} = f(y, t)$, so $f(a, t) = 0$ for all t . Set

$$y(t) = a + \varepsilon(t),$$

where ε is a small perturbation. Substituting into the equation and Taylor expanding f in its first argument about a :

$$\dot{\varepsilon} = f(a + \varepsilon, t) = \underbrace{f(a, t)}_{=0} + \varepsilon \frac{\partial f}{\partial y}(a, t) + O(\varepsilon^2).$$

While ε remains small the $O(\varepsilon^2)$ term is negligible, and we are left with the **linearised equation**

$$\dot{\varepsilon} \simeq \varepsilon \frac{\partial f}{\partial y}(a, t),$$

a linear equation which we know how to solve. If the resulting ε decays, the equilibrium is stable; if it grows, unstable².

Example 2.4.9

Return to $\dot{y} = t(1 - y^2)$, with fixed points $y = \pm 1$. We have

$$\frac{\partial f}{\partial y} = -2yt = \begin{cases} -2t & \text{at } y = 1, \\ 2t & \text{at } y = -1. \end{cases}$$

At $y = 1$: $\dot{\varepsilon} = -2t\varepsilon$, so $\varepsilon = \varepsilon_0 e^{-t^2} \rightarrow 0$, and $y = 1$ is stable. At $y = -1$: $\dot{\varepsilon} = 2t\varepsilon$, so $\varepsilon = \varepsilon_0 e^{t^2}$ grows, and $y = -1$ is unstable – exactly as our sketch suggested.

Autonomous Equations and the Phase Line

Very often the physics of a system doesn't change with time, so that $\dot{y}$ depends only on y itself.

Definition 2.4.10 (Autonomous Equation)

An equation of the form $\dot{y} = f(y)$, with no explicit t dependence, is called **autonomous**.

For autonomous equations the perturbation analysis is particularly simple. Near a fixed point $y = a$ with $f(a) = 0$, the linearisation is

$$\dot{\varepsilon} = f'(a) \varepsilon = k\varepsilon \implies \varepsilon = \varepsilon_0 e^{kt},$$

with $k = f'(a)$ constant. So the rule is simply:

$$f'(a) < 0 \implies \text{stable}, \quad f'(a) > 0 \implies \text{unstable}.$$

²Strictly, once ε has grown large the linearisation is no longer valid, so we cannot conclude that $\varepsilon \rightarrow \infty$ – only that the perturbation initially grows and the solution moves away from the equilibrium. That is exactly what instability means.

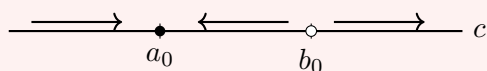
Graphically, this is obvious: plot $f(y)$ against y . Where $f > 0$, y increases; where $f < 0$, y decreases. So y flows to the right where the graph is above the axis and to the left where it is below, and the fixed points where f crosses the axis *downwards* are the stable ones. Recording just the fixed points and flow directions on a copy of the y -axis gives the **phase portrait** (or phase line) of the equation, a diagram which captures the entire qualitative behaviour at a glance.

Example 2.4.11 (Chemical Kinetics)

Consider the reaction $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$, with a molecules of NaOH and b of HCl present, starting from a_0 and b_0 respectively, and c molecules of each product formed. In dilute solution the reaction rate is proportional to ab , so

$$\frac{dc}{dt} = \lambda ab = \lambda(a_0 - c)(b_0 - c) = f(c).$$

Suppose (without loss of generality) that $a_0 < b_0$. The graph of $f(c)$ is an upward parabola with roots at $c = a_0$ and $c = b_0$; between the roots $f > 0$, so c increases towards a_0 from below. Checking stability: $f'(c) = \lambda(2c - a_0 - b_0)$, so $f'(a_0) = \lambda(a_0 - b_0) < 0$ and $f'(b_0) = \lambda(b_0 - a_0) > 0$. Hence $c = a_0$ is stable and $c = b_0$ is unstable.



Physically, of course, only $0 \leq c \leq a_0$ makes sense (we cannot have a negative amount of NaOH!), and every physical solution tends to $c = a_0$: the reaction runs until the scarcer reagent is used up.

For completeness, the equation is separable and can be solved exactly:

$$c = \frac{a_0 b_0 [1 - e^{-(b_0 - a_0)\lambda t}]}{b_0 - a_0 e^{-(b_0 - a_0)\lambda t}},$$

which indeed tends to a_0 as $t \rightarrow \infty$ – but notice how much less work the phase portrait was.

§2.4.5 The Logistic Equation

The logistic equation is the classic model of a population limited by finite resources, and it makes a nice showcase for everything in this section.

Consider a population of size y with birth rate αy and death rate βy . Then $\dot{y} = (\alpha - \beta)y$, giving exponential growth or decay – fine for small populations, but unsustainable forever. To model competition for limited resources, note that the rate of pairwise encounters between individuals is proportional to y^2 , so fighting over scarce food contributes an extra death rate γy^2 . This gives

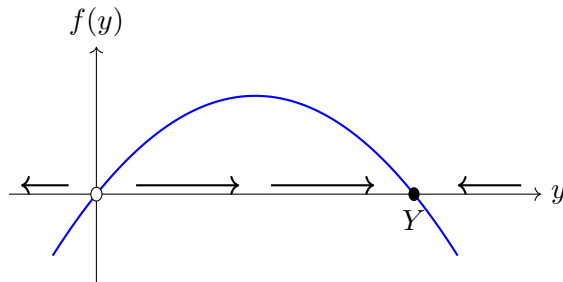
$$\frac{dy}{dt} = (\alpha - \beta)y - \gamma y^2 = ry \left(1 - \frac{y}{Y}\right),$$

where $r = \alpha - \beta$ and $Y = r/\gamma$. This is the **logistic equation**.

The equation is separable and can be solved explicitly, but the phase portrait tells the whole story with far less effort. Take $r > 0$. The graph of $f(y) = ry(1 - y/Y)$ is a

downward parabola with zeros at $y = 0$ and $y = Y$:

- $f'(0) = r > 0$, so $y = 0$ is unstable;
- $f'(Y) = -r < 0$, so $y = Y$ is stable.



For small populations $\dot{y} \simeq ry$ and growth is exponential; as y approaches the *carrying capacity* Y the growth saturates, and every positive solution tends to Y . The characteristic S-shaped (‘sigmoid’) solution curves interpolate between exponential growth at the bottom and saturation at the top.

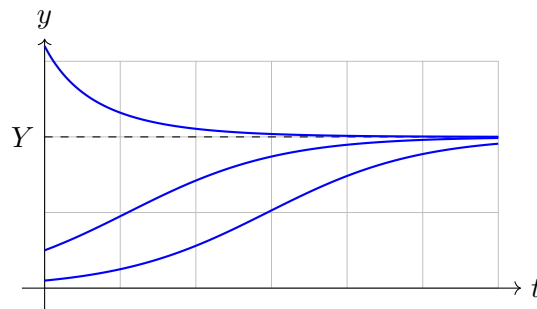
For completeness, let’s also solve the equation exactly. Separating variables and using partial fractions,

$$\int \frac{dy}{y(1 - y/Y)} = \int \left(\frac{1}{y} + \frac{1/Y}{1 - y/Y} \right) dy = \ln \frac{y}{1 - y/Y} = rt + C,$$

so $\frac{y}{1 - y/Y} = Ae^{rt}$, and solving for y with $y(0) = y_0$:

$$y(t) = \frac{Y y_0 e^{rt}}{Y + y_0(e^{rt} - 1)}.$$

As a sanity check: for small t (while $y \ll Y$) this is approximately $y_0 e^{rt}$, and as $t \rightarrow \infty$ it tends to Y , from below if $y_0 < Y$ and from above if $y_0 > Y$ – exactly the picture the phase portrait predicted, with rather more algebra.



§2.5 First-Order Difference Equations

Differential equations describe quantities evolving continuously; but many systems evolve in discrete steps – populations with annual breeding seasons, and (importantly) any differential equation being solved numerically on a computer. The discrete analogue of a first-order ODE is a **difference equation** (or *recurrence*, or *map*)

$$x_{n+1} = f(x_n),$$

which generates a sequence from a starting value x_0 . The theory runs parallel to the continuous case, but with some genuine surprises.

§2.5.1 Fixed Points and Stability

As with differential equations, the natural starting point is to look for solutions that don't change at all.

Definition 2.5.1 (Fixed Point of a Map)

A **fixed point** of $x_{n+1} = f(x_n)$ is a value x^* with $f(x^*) = x^*$, so that the constant sequence $x_n = x^*$ is a solution.

To analyse stability we perturb, just as before. Write $x_n = x^* + \varepsilon_n$ and expand:

$$x^* + \varepsilon_{n+1} = f(x^* + \varepsilon_n) = f(x^*) + \varepsilon_n f'(x^*) + O(\varepsilon_n^2),$$

so to leading order $\varepsilon_{n+1} \simeq f'(x^*) \varepsilon_n$, giving $\varepsilon_n \simeq \varepsilon_0 [f'(x^*)]^n$. The perturbation shrinks precisely when the multiplier has modulus less than one:

$$|f'(x^*)| < 1 \implies \text{stable}, \quad |f'(x^*)| > 1 \implies \text{unstable}.$$

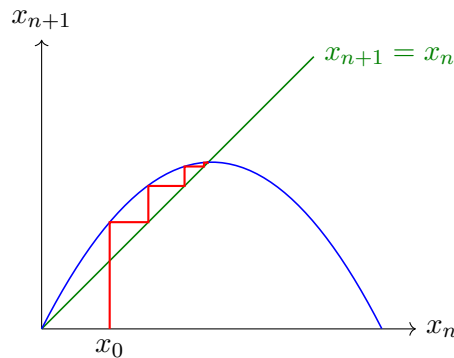
Note the contrast with the continuous case, where the criterion was on the *sign* of f' ; here it is on the *magnitude*. Note also that $f'(x^*)$ negative and small gives *oscillatory* convergence, with the iterates alternating sides of the fixed point.

Remark (Reconciling the Two Criteria). The discrete and continuous stability criteria fit together neatly. The Euler discretisation of $\dot{y} = f(y)$ with step h is the map $y_{n+1} = y_n + hf(y_n)$, whose fixed points coincide with the equilibria of the ODE, and whose multiplier there is

$$1 + hf'(a).$$

If $f'(a) < 0$ (stable for the ODE), then for small enough h we have $|1 + hf'(a)| < 1$ and the map is stable too, consistent with the ODE. But if $h > 2/|f'(a)|$, the multiplier drops below -1 and the discretisation is unstable even though the ODE is perfectly stable – a numerical instability caused purely by too large a time step. Numerical analysts spend a great deal of effort designing schemes that avoid this.

A useful way to visualise iteration is a **cobweb diagram**: draw the graphs of $x_{n+1} = f(x_n)$ and the diagonal $x_{n+1} = x_n$, then trace vertically to the curve and horizontally to the diagonal, repeatedly. Fixed points are where curve meets diagonal, and the stability criterion $|f'| < 1$ says the curve crosses the diagonal less steeply than 45° .



(The cobweb above shows convergence towards a stable fixed point where the blue curve cuts the diagonal.)

§2.5.2 The Logistic Map

Discretising the logistic equation with the Euler scheme (time step Δt):

$$\frac{y_{n+1} - y_n}{\Delta t} = r y_n \left(1 - \frac{y_n}{Y}\right),$$

and rescaling via $\lambda = 1 + r\Delta t$ and $x_n = \frac{r\Delta t}{1+r\Delta t} \cdot \frac{y_n}{Y}$, one arrives (after a little algebra) at the famous **logistic map**

$$x_{n+1} = \lambda x_n (1 - x_n).$$

Its fixed points satisfy $\lambda x(1 - x) = x$, giving

$$x^* = 0 \quad \text{or} \quad x^* = 1 - \frac{1}{\lambda},$$

and $f'(x) = \lambda(1 - 2x)$, so $f'(0) = \lambda$ and $f'(1 - 1/\lambda) = 2 - \lambda$. The stability analysis then predicts strikingly rich behaviour as λ increases:

- (i) For $0 < \lambda < 1$: only $x^* = 0$ is relevant, $|f'(0)| = \lambda < 1$, and the population dies out.
- (ii) For $1 < \lambda < 2$: $x^* = 0$ becomes unstable, and $x^* = 1 - 1/\lambda$ is stable with $0 < f' < 1$, so iterates converge to it monotonically.
- (iii) For $2 < \lambda < 3$: still stable, but now $-1 < f' < 0$, so the convergence is oscillatory, with iterates spiralling into the fixed point on the cobweb diagram.
- (iv) For $\lambda > 3$: $|f'(x^*)| > 1$, so *both* fixed points are unstable. The iterates settle instead into a *2-cycle*, oscillating between two values (a solution of $x_{n+2} = x_n$). As λ increases further to $1 + \sqrt{6} \approx 3.449$, the 2-cycle destabilises in favour of a 4-cycle, then an 8-cycle, and so on – a *period-doubling cascade* which accumulates at $\lambda \approx 3.57$, beyond which the dynamics become chaotic.

The moral is worth remembering: a discretisation can have wildly different behaviour from the differential equation it approximates. The continuous logistic equation is completely tame – every positive solution glides monotonically to the carrying capacity – while its discrete counterpart, for large enough λ (i.e. large enough time step), is chaotic.

§2.6 Second-Order Linear Differential Equations

We now move up an order. Second-order linear equations are probably the most important differential equations in all of physics – Newton's second law is one – and they will occupy us for a good while. Almost everything we do in this section extends routinely to higher-order linear equations, but second order is where all the ideas already appear.

§2.6.1 Constant Coefficients and the Characteristic Equation

The general second-order linear equation with constant coefficients is

$$ay'' + by' + cy = f(x).$$

Following the strategy from the first-order case, we solve this in two steps: find the general solution of the homogeneous equation $ay'' + by' + cy = 0$ (the complementary function), then find any one particular integral of the full equation, and add them.

For the complementary function, we again exploit the fact that $e^{\lambda x}$ is an eigenfunction of $\frac{d}{dx}$, and hence also of $\frac{d^2}{dx^2}$. Trying $y = e^{\lambda x}$ gives $a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$, and dividing by the (never-zero) exponential:

Definition 2.6.1 (Characteristic Equation)

The **characteristic equation** of $ay'' + by' + cy = 0$ is

$$a\lambda^2 + b\lambda + c = 0.$$

This is a quadratic with two roots λ_1, λ_2 . If they are distinct, we get two genuinely different solutions $e^{\lambda_1 x}$ and $e^{\lambda_2 x}$, and (as we will justify carefully in §2.6.4) the general complementary function is

$$y_c = Ae^{\lambda_1 x} + Be^{\lambda_2 x}.$$

Example 2.6.2

Consider $y'' - 5y' + 6y = 0$. The characteristic equation is $\lambda^2 - 5\lambda + 6 = 0$, with roots $\lambda = 2, 3$. So the general solution is

$$y = Ae^{2x} + Be^{3x}.$$

Example 2.6.3 (Simple Harmonic Motion)

Consider $y'' + 4y = 0$. The characteristic equation is $\lambda^2 + 4 = 0$, so $\lambda = \pm 2i$ and

$$y = Ae^{2ix} + Be^{-2ix}.$$

The constants A, B are allowed to be complex, but if this is modelling a physical oscillation we would like the answer to at least *look* real. Expanding with Euler's formula,

$$y = A(\cos 2x + i \sin 2x) + B(\cos 2x - i \sin 2x) = \alpha \cos 2x + \beta \sin 2x,$$

where $\alpha = A + B$ and $\beta = i(A - B)$. All we have done is change basis for the solution space, from $\{e^{2ix}, e^{-2ix}\}$ to $\{\cos 2x, \sin 2x\}$; whichever basis is more convenient may be used. More generally, complex roots $\lambda = \sigma \pm i\omega$ give solutions $e^{\sigma x}(\alpha \cos \omega x + \beta \sin \omega x)$ – oscillations with exponentially growing or decaying amplitude.

Remark (Higher-Order Equations). Everything here generalises immediately: an n th-order linear equation with constant coefficients has characteristic polynomial of degree n , and each distinct root λ contributes $e^{\lambda x}$ to the complementary function. For example, radioactive decay *sequences* $A \rightarrow B \rightarrow C \rightarrow \dots$, which we treated one step at a time in the previous section, can be packaged as a single higher-order equation whose characteristic roots are $-k_a, -k_b, -k_c, \dots$ – each isotope contributing its own decaying exponential to the mix.

§2.6.2 Degeneracy and Detuning

Something goes wrong when the characteristic equation has a repeated root: our method produces only *one* solution, but a second-order equation should have a two-parameter

family.

Example 2.6.4 (A Repeated Root)

Consider $y'' - 4y' + 4y = 0$. The characteristic equation is $(\lambda - 2)^2 = 0$, so $\lambda = 2$ twice, and we find only the single solution family Ae^{2x} . The solution space is two-dimensional, so we are missing a second, independent solution.

Here is a rather beautiful way of finding it, called **detuning**. The problem is that our two roots collided, so let's separate them slightly: perturb the equation to

$$y'' - 4y' + (4 - \varepsilon^2)y = 0,$$

which has roots $\lambda = 2 \pm \varepsilon$, and recovers our equation as $\varepsilon \rightarrow 0$. For $\varepsilon \neq 0$ the general solution is

$$y = Ae^{(2+\varepsilon)x} + Be^{(2-\varepsilon)x} = e^{2x} [Ae^{\varepsilon x} + Be^{-\varepsilon x}].$$

Now Taylor expand the small exponentials:

$$y = e^{2x} [(A + B) + \varepsilon x(A - B) + O(\varepsilon^2 A, \varepsilon^2 B)].$$

Fix constants α and β , and for each small ε choose $A = \frac{1}{2}(\alpha + \beta/\varepsilon)$ and $B = \frac{1}{2}(\alpha - \beta/\varepsilon)$, so that $A + B = \alpha$ and $\varepsilon(A - B) = \beta$. Since $A, B = O(1/\varepsilon)$, the error terms are $O(\varepsilon)$, and letting $\varepsilon \rightarrow 0$,

$$y \rightarrow e^{2x}(\alpha + \beta x).$$

So the missing second solution is xe^{2x} .

The pattern in this example is completely general, and worth committing to memory: *if the characteristic equation has a repeated root λ , the complementary function is $(A + Bx)e^{\lambda x}$.*

§2.6.3 Reduction of Order

Detuning relied on constant coefficients. There is a more robust technique for manufacturing a second solution out of a known one, which works for any linear second-order equation: look for a solution of the form

$$y_2(x) = v(x) y_1(x),$$

where y_1 is the solution we already have and v is to be found. The point is that substituting this in always produces a *first-order* equation for v' (the v terms cancel because y_1 is a solution) – hence the name **reduction of order**.

Example 2.6.5

Take $y'' - 4y' + 4y = 0$ again, with known solution $y_1 = e^{2x}$. Try $y_2 = ve^{2x}$. Then

$$y_2' = (v' + 2v)e^{2x}, \quad y_2'' = (v'' + 4v' + 4v)e^{2x},$$

and substituting in,

$$[(v'' + 4v' + 4v) - 4(v' + 2v) + 4v] e^{2x} = 0 \implies v'' = 0.$$

So $v = Ax + B$, and $y_2 = (Ax + B)e^{2x}$ – recovering the answer from detuning, this time with essentially no cleverness required.

§2.6.4 Phase Space, Linear Independence and the Wronskian

We have been asserting that a second-order linear equation has a two-parameter family of solutions, and that certain pairs of solutions are ‘independent’. Let’s now make this precise.

Consider a general n th-order linear equation. Its solution is not determined by the value $y(x_0)$ alone: we also need $y'(x_0)$, and so on up to $y^{(n-1)}(x_0)$. Once these n numbers are known, the equation itself determines $y^{(n)}(x_0)$, and by differentiating the equation, all higher derivatives too – so the entire Taylor series of the solution at x_0 is pinned down. This strongly suggests (and it is true, though a real proof needs the machinery of IB Analysis) that a solution is uniquely determined by the n initial conditions $y(x_0), y'(x_0), \dots, y^{(n-1)}(x_0)$.

This motivates packaging those numbers into a single object.

Definition 2.6.6 (Solution Vector and Phase Space)

The **solution vector** of a solution y of an n th-order differential equation is

$$\mathbf{Y}(x) = (y(x), y'(x), \dots, y^{(n-1)}(x)).$$

The n -dimensional space in which $\mathbf{Y}$ lives is called **phase space**. As x varies, $\mathbf{Y}(x)$ traces out a **trajectory** in phase space, and each point of phase space (viewed as an initial condition) has a unique trajectory through it.

Example 2.6.7

Consider $y'' + 4y = 0$. The initial condition $y(0) = 1, y'(0) = 0$ gives the solution $y_1 = \cos 2x$, with solution vector $\mathbf{Y}_1(x) = (\cos 2x, -2 \sin 2x)$ – which traces out an ellipse in phase space, traversed clockwise. The initial condition $y(0) = 0, y'(0) = 2$ gives $y_2 = \sin 2x$, with $\mathbf{Y}_2(x) = (\sin 2x, 2 \cos 2x)$.

Notice that the two initial vectors $(1, 0)$ and $(0, 2)$ are linearly independent, and one can check that $\mathbf{Y}_1(x)$ and $\mathbf{Y}_2(x)$ remain linearly independent for *every* x . This is not a coincidence, as we will now see.

For a second-order equation, phase space is a plane, and a natural question is whether two solutions y_1, y_2 give solution vectors forming a basis of that plane. The determinant testing this has a name.

Definition 2.6.8 (Wronskian)

The **Wronskian** of two solutions y_1, y_2 of a second-order differential equation is the determinant

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

Definition 2.6.9 (Independent Solutions)

Solutions y_1 and y_2 are **independent** if their solution vectors $\mathbf{Y}_1(x), \mathbf{Y}_2(x)$ are linearly independent for some x – that is, if $W(x) \neq 0$ for some x .

Example 2.6.10

For $y'' + 4y = 0$ with $y_1 = \cos 2x$, $y_2 = \sin 2x$:

$$W = \cos 2x \cdot 2 \cos 2x - \sin 2x \cdot (-2 \sin 2x) = 2 \neq 0.$$

For $y'' - 4y' + 4y = 0$ with $y_1 = e^{2x}$, $y_2 = xe^{2x}$:

$$W = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & (1+2x)e^{2x} \end{vmatrix} = e^{4x}(1+2x-2x) = e^{4x} \neq 0.$$

In both examples the Wronskian was non-vanishing *everywhere*. Could the Wronskian vanish at some points but not others? Remarkably, no.

Theorem 2.6.11 (Abel's Theorem)

Let y_1, y_2 be solutions of $y'' + p(x)y' + q(x)y = 0$. Then either $W \equiv 0$, or $W(x) \neq 0$ for all x . In particular, if two solutions are independent at one point, they are independent everywhere.

Proof. Since y_1 and y_2 are both solutions,

$$\begin{aligned} y_2(y_1'' + py_1' + qy_1) &= 0, \\ y_1(y_2'' + py_2' + qy_2) &= 0. \end{aligned}$$

Subtracting the first from the second,

$$(y_1y_2'' - y_2y_1'') + p(y_1y_2' - y_2y_1') = 0.$$

Now observe that $W = y_1y_2' - y_2y_1'$ and, differentiating, $W' = y_1y_2'' - y_2y_1''$ (the cross terms cancel). So the equation above says precisely

$$W' + p(x)W = 0,$$

a first-order linear equation for W , with solution

$$W(x) = W_0 e^{-\int p \, dx}.$$

Since the exponential never vanishes, either $W_0 = 0$ and $W \equiv 0$, or W is nowhere zero. $\square$

Remark. Abel's theorem generalises: writing an n th-order homogeneous linear equation as a first-order system $\mathbf{Y}' = A\mathbf{Y}$ (as we will do in §2.11), one can show $W' + \text{tr}(A)W = 0$, so $W = W_0 e^{-\int \text{tr } A \, dx}$ and the same dichotomy holds.

§2.6.5 Particular Integrals

With the complementary function understood, we return to the forced equation

$$ay'' + by' + cy = f(x)$$

and the problem of finding a particular integral. The first method, as ever, is educated guessing: choose a trial form matching the forcing, with unknown coefficients, and substitute. The standard guesses are:

$f(x)$	trial $y_p(x)$
e^{mx}	Ae^{mx}
$\sin kx$ or $\cos kx$	$A \sin kx + B \cos kx$
polynomial of degree n	$a_n x^n + \cdots + a_1 x + a_0$

Note that for trigonometric forcing we must include *both* $\sin$ and $\cos$ in the guess (differentiation mixes them), and for polynomial forcing all lower-order terms. Since the equation is linear, a sum of forcing terms can be handled term by term and the particular integrals superposed.

Example 2.6.12

Consider $y'' - 5y' + 6y = 2x + e^{4x}$. We guess $y_p = ax + b + ce^{4x}$. Then $y'_p = a + 4ce^{4x}$ and $y''_p = 16ce^{4x}$, and substituting in:

$$16ce^{4x} - 5(a + 4ce^{4x}) + 6(ax + b + ce^{4x}) = 2x + e^{4x}.$$

Comparing coefficients of e^{4x} , x and 1 respectively:

$$\begin{aligned} 16c - 20c + 6c &= 1 \implies c = \frac{1}{2}, \\ 6a &= 2 \implies a = \frac{1}{3}, \\ -5a + 6b &= 0 \implies b = \frac{5}{18}. \end{aligned}$$

The complementary function was found earlier, so the general solution is

$$y = Ae^{3x} + Be^{2x} + \frac{1}{2}e^{4x} + \frac{1}{3}x + \frac{5}{18}.$$

§2.6.6 Resonance

The guessing procedure breaks down in one important situation: when the forcing is itself a complementary function.

Consider a simple harmonic oscillator forced at its own natural frequency:

$$\ddot{y} + \omega_0^2 y = \sin \omega_0 t.$$

The complementary function is $y_c = A \sin \omega_0 t + B \cos \omega_0 t$, and if we try $y_p = C \sin \omega_0 t + D \cos \omega_0 t$, we just get $\ddot{y}_p + \omega_0^2 y_p = 0$ – the guess is annihilated by the operator, and can never balance the forcing. This phenomenon is called **resonance**.

To understand what actually happens, we detune again: force at a nearby frequency $\omega \neq \omega_0$,

$$\ddot{y} + \omega_0^2 y = \sin \omega t,$$

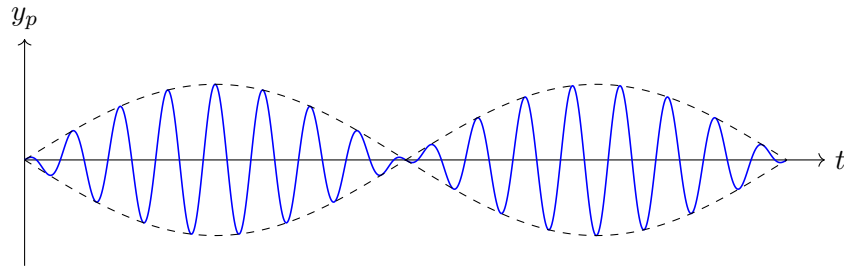
and take the limit $\omega \rightarrow \omega_0$ at the end. Guessing the antisymmetric combination $y_p = C(\sin \omega t - \sin \omega_0 t)$ (chosen so that $y_p \rightarrow 0$ as $\omega \rightarrow \omega_0$, keeping the limit finite), we have

$$\ddot{y}_p = C(-\omega^2 \sin \omega t + \omega_0^2 \sin \omega_0 t),$$

and substituting in gives $C(\omega_0^2 - \omega^2) = 1$. So

$$y_p = \frac{\sin \omega t - \sin \omega_0 t}{\omega_0^2 - \omega^2} = \frac{2}{\omega_0^2 - \omega^2} \cos\left(\frac{\omega_0 + \omega}{2}t\right) \sin\left(\frac{\omega - \omega_0}{2}t\right),$$

using the sum-to-product formula. Writing $\Delta\omega = \omega_0 - \omega$, this is a fast oscillation at roughly the natural frequency, modulated by a slow envelope $\sin(\Delta\omega t/2)$ of amplitude $O(1/\Delta\omega)$ and period $O(1/\Delta\omega)$. This slow waxing and waning of amplitude, familiar from tuning musical instruments, is called **beating**: it occurs whenever a system is forced close to (but not exactly at) its natural frequency.



Now let $\Delta\omega \rightarrow 0$. The envelope's wavelength stretches to infinity, and using $\sin \theta \approx \theta$ for the slow factor,

$$y_p \rightarrow -\frac{t}{2\omega_0} \cos \omega_0 t.$$

So at exact resonance, the response is an oscillation whose amplitude *grows linearly in time* – the initial stretch of linear growth is all that remains of the beat envelope. This also tells us the correct guess to make at resonance, and the rule is general: *if the forcing is a complementary function, multiply the usual guess by the independent variable*. (If that is somehow still a complementary function, multiply by it again.)

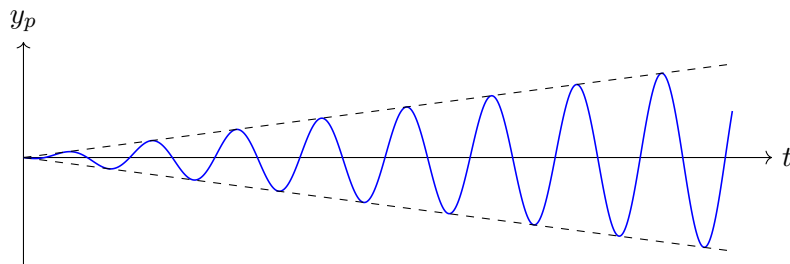
Example 2.6.13

For $\ddot{y} + \omega_0^2 y = \sin \omega_0 t$, try $y_p = t(C \sin \omega_0 t + D \cos \omega_0 t)$. Substituting in, the t -proportional terms cancel (as they must – they are complementary functions) leaving

$$2C\omega_0 \cos \omega_0 t - 2D\omega_0 \sin \omega_0 t = \sin \omega_0 t,$$

so $C = 0$, $D = -1/(2\omega_0)$, and $y_p = -\frac{t}{2\omega_0} \cos \omega_0 t$, in agreement with the limit above.

The resonant response, with its linearly growing envelope, looks like this:



§2.6.7 Variation of Parameters

Guessing works when the forcing is simple. **Variation of parameters** is a systematic method that works for *any* forcing, provided we know the complementary functions. It is more work than guessing, so it should be a last resort – but it always succeeds.

Consider

$$y'' + p(x)y' + q(x)y = f(x),$$

with independent complementary functions y_1, y_2 . The idea comes from phase space: for each fixed x , the solution vectors $\mathbf{Y}_1(x)$ and $\mathbf{Y}_2(x)$ form a basis of the plane, so the particular solution vector we seek can be expanded in that basis, with coefficients that vary with x :

$$\mathbf{Y}_p(x) = u(x)\mathbf{Y}_1(x) + v(x)\mathbf{Y}_2(x).$$

Reading off the two components,

$$y_p = uy_1 + vy_2, \quad (\text{a})$$

$$y'_p = uy'_1 + vy'_2. \quad (\text{b})$$

These two equations carry a hidden constraint: differentiating (a) gives $y'_p = uy'_1 + vy'_2 + u'y_1 + v'y_2$, which is consistent with (b) only if

$$u'y_1 + v'y_2 = 0. \quad (\text{c})$$

Now differentiate (b):

$$y''_p = uy''_1 + vy''_2 + u'y'_1 + v'y'_2,$$

and substitute everything into the differential equation. The terms multiplying u and v vanish (since y_1, y_2 are complementary functions), leaving

$$u'y'_1 + v'y'_2 = f. \quad (\text{d})$$

Equations (c) and (d) are two linear equations for u' and v' :

$$\begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix} \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 \\ f \end{pmatrix},$$

and the matrix is invertible precisely because its determinant is the Wronskian $W \neq 0$. Inverting,

$$u' = -\frac{y_2 f}{W}, \quad v' = \frac{y_1 f}{W},$$

and integrating gives u and v , hence y_p . (It is far better to remember the derivation than the formula.)

Example 2.6.14

Let's redo resonance by this method: $y'' + 4y = \sin 2x$, with $y_1 = \sin 2x$, $y_2 = \cos 2x$ and $W = y_1 y'_2 - y_2 y'_1 = -2$. Then

$$u' = \frac{\cos 2x \sin 2x}{2} = \frac{\sin 4x}{4}, \quad v' = -\frac{\sin^2 2x}{2} = \frac{\cos 4x - 1}{4},$$

so

$$u = -\frac{\cos 4x}{16}, \quad v = \frac{\sin 4x}{16} - \frac{x}{4}.$$

Therefore

$$y_p = \frac{1}{16} (-\cos 4x \sin 2x + \sin 4x \cos 2x) - \frac{x}{4} \cos 2x = \frac{1}{16} \sin 2x - \frac{x}{4} \cos 2x.$$

The term $-\frac{x}{4} \cos 2x$ is the resonant response we found by detuning (here $\omega_0 = 2$), and $\frac{1}{16} \sin 2x$ is merely a complementary function, so the answers agree.

Example 2.6.15 (When Guessing is Hopeless)

Consider $y'' + y = \tan x$. No finite combination of elementary functions suggests itself as a guess – differentiating $\tan x$ generates ever more complicated expressions – so variation of parameters is the way forward. With $y_1 = \sin x$, $y_2 = \cos x$ and $W = \sin x \cdot (-\sin x) - \cos x \cdot \cos x = -1$:

$$u' = -\frac{y_2 f}{W} = \cos x \tan x = \sin x, \quad v' = \frac{y_1 f}{W} = -\sin x \tan x = \cos x - \sec x,$$

where we wrote $-\sin^2 x / \cos x = (\cos^2 x - 1) / \cos x$. Integrating,

$$u = -\cos x, \quad v = \sin x - \ln |\sec x + \tan x|,$$

so

$$y_p = uy_1 + vy_2 = -\cos x \sin x + \sin x \cos x - \cos x \ln |\sec x + \tan x| = -\cos x \ln |\sec x + \tan x|.$$

The general solution is $y = A \sin x + B \cos x - \cos x \ln |\sec x + \tan x|$ – involving a logarithm that no reasonable person would have guessed.

§2.6.8 Equidimensional Equations

There is one important class of *non*-constant coefficient equations that we can solve just as easily.

Definition 2.6.16 (Equidimensional Equation)

An equation is **equidimensional** if it has the form

$$ax^2y'' + bxy' + cy = f(x),$$

with a, b, c constants.

The name comes from dimensional analysis: if x carries some physical dimension, then each of x^2y'' , xy' and y has the same dimensions, so the equation is invariant under rescaling $x \mapsto \lambda x$. (These are often called ‘homogeneous’ equations, but that clashes horribly with our earlier usage, so we avoid it.)

Just as $e^{\lambda x}$ is the eigenfunction of $\frac{d}{dx}$, the powers x^k are eigenfunctions of the operator $x \frac{d}{dx}$, and this suggests trying $y = x^k$. Since $xy' = kx^k$ and $x^2y'' = k(k-1)x^k$, substituting into the homogeneous equation gives

$$ak(k-1) + bk + c = 0,$$

with roots k_1, k_2 , and complementary function $y_c = Ax^{k_1} + Bx^{k_2}$ when the roots are distinct.

Alternatively, substituting $z = \ln x$ transforms the equation into one with constant coefficients:

$$a \frac{d^2 y}{dz^2} + (b - a) \frac{dy}{dz} + cy = f(e^z),$$

whose characteristic equation $a\lambda(\lambda - 1) + b\lambda + c = 0$ is precisely the one above. This substitution is the easiest way to deal with the degenerate cases: a repeated root k gives $y_c = (A + B \ln x)x^k$ (from $(A + Bz)e^{kz}$), and resonant forcing proportional to x^{k_1} gives a particular integral proportional to $x^{k_1} \ln x$.

Example 2.6.17

Let's solve $x^2 y'' - 3xy' + 4y = x^2$ for $x > 0$. Trying $y = x^k$ in the homogeneous equation:

$$k(k - 1) - 3k + 4 = k^2 - 4k + 4 = (k - 2)^2 = 0,$$

a repeated root $k = 2$. So the complementary function is

$$y_c = (A + B \ln x) x^2.$$

The forcing x^2 is resonant (it is a complementary function!), so following the rule above we try $y_p = Cx^2(\ln x)^2$ – the usual resonant guess $x^2 \ln x$ is *also* a complementary function here, so we multiply by $\ln x$ once more. Substituting (a slightly tedious computation, best done via $z = \ln x$, where the equation becomes $\frac{d^2 y}{dz^2} - 4\frac{dy}{dz} + 4y = e^{2z}$ with double-resonant forcing and trial $Cz^2 e^{2z}$):

$$C(2 + 8z + 4z^2 - 8z - 8z^2 + 4z^2) e^{2z} = e^{2z} \implies 2C = 1,$$

so $C = \frac{1}{2}$ and the general solution is

$$y = (A + B \ln x + \frac{1}{2}(\ln x)^2) x^2.$$

§2.6.9 Transients and Damping

Let's use what we have built to study a genuinely physical system: a damped, forced oscillator. Consider a car of mass M riding on suspension; the springs supply a restoring force kx and the shock absorbers a friction force $l\dot{x}$ opposing motion, while $F(t)$ is an external vertical force. Newton's second law gives

$$M\ddot{x} = F(t) - kx - l\dot{x} \implies \ddot{x} + \frac{l}{M}\dot{x} + \frac{k}{M}x = \frac{F(t)}{M}.$$

Without damping and forcing this would be simple harmonic motion with angular frequency $\sqrt{k/M}$. It is illuminating to non-dimensionalise: measure time in units of $\sqrt{M/k}$ by setting $t = \tau \sqrt{M/k}$. The equation becomes

$$\ddot{x} + 2\kappa\dot{x} + x = f(\tau),$$

where now dots denote $d/d\tau$, and

$$\kappa = \frac{l}{2\sqrt{kM}}, \quad f = \frac{F}{k}.$$

The original three parameters M, l, k have collapsed into the single dimensionless damping parameter κ – this sort of reduction is one of the main payoffs of non-dimensionalisation.

Consider first the free response, $f = 0$. The characteristic equation $\lambda^2 + 2\kappa\lambda + 1 = 0$ has roots

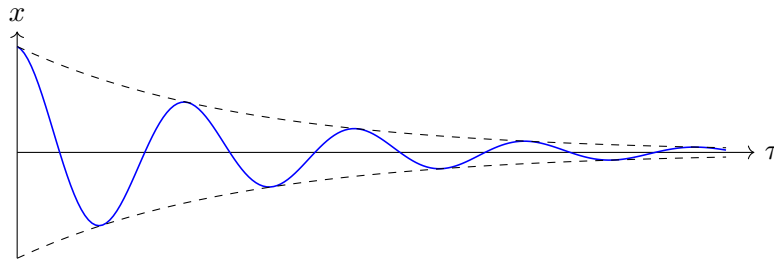
$$\lambda = -\kappa \pm \sqrt{\kappa^2 - 1},$$

and the character of the motion depends entirely on where κ sits relative to 1.

- (i) *Underdamped* ($\kappa < 1$). The roots are complex, $\lambda = -\kappa \pm i\sqrt{1 - \kappa^2}$, and

$$x = e^{-\kappa\tau} \left(A \sin \sqrt{1 - \kappa^2} \tau + B \cos \sqrt{1 - \kappa^2} \tau \right).$$

The system oscillates with period $2\pi/\sqrt{1 - \kappa^2}$ – note that damping *lengthens* the period – inside an envelope decaying over a timescale $O(1/\kappa)$. As $\kappa \rightarrow 1$ the period tends to infinity.



- (ii) *Critically damped* ($\kappa = 1$). Repeated root $\lambda = -1$, so

$$x = (A + B\tau)e^{-\tau}.$$

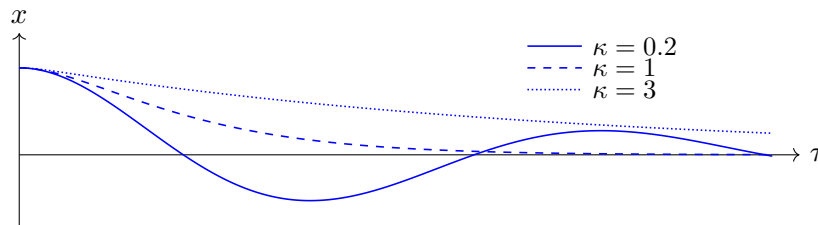
The solution can rise at most once before decaying; both the rise and decay happen on the $O(1)$ timescale (i.e. $O(\sqrt{M/k})$ in dimensional terms).

- (iii) *Overdamped* ($\kappa > 1$). Two real negative roots $-\lambda_1 > -\lambda_2$, and

$$x = Ae^{-\lambda_1\tau} + Be^{-\lambda_2\tau}, \quad \lambda_{1,2} = \kappa \mp \sqrt{\kappa^2 - 1}.$$

The fast mode dies on a timescale $O(1/\lambda_2)$, and the solution then decays slowly, on the long timescale $O(1/\lambda_1)$. Heavy damping makes the system *slower* to settle.

The three regimes are compared below, each released from rest at $x = 1$: the underdamped solution overshoots and oscillates, the critically damped one settles fastest, and the overdamped one creeps home.



For car suspension we would tune κ close to critical damping: strong enough to kill oscillations, not so strong that the car takes forever to level out.

Example 2.6.18 (An Underdamped Oscillator)

Let's solve a concrete initial value problem:

$$\ddot{x} + 2\dot{x} + 5x = 0, \quad x(0) = 1, \quad \dot{x}(0) = 0.$$

The characteristic equation $\lambda^2 + 2\lambda + 5 = 0$ gives $\lambda = -1 \pm 2i$, so

$$x = e^{-t}(A \cos 2t + B \sin 2t).$$

The condition $x(0) = 1$ gives $A = 1$; then $\dot{x} = e^{-t}[(-A + 2B) \cos 2t + (-B - 2A) \sin 2t]$, and $\dot{x}(0) = 0$ gives $B = \frac{1}{2}$. So

$$x = e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right),$$

a decaying oscillation: the mass overshoots the origin, swings back, and each successive swing is smaller by a factor $e^{-\pi} \approx 0.04$ (one half-period is $\pi/2$, so a full period costs $e^{-\pi}$ in amplitude). Within a few swings the motion is imperceptible.

Now switch the forcing on, say $f(\tau) = \sin \tau$. In all three cases the roots of the characteristic equation have negative real part, so the complementary function decays exponentially: it is a **transient**, describing the system's short-time adjustment to its initial conditions. What survives at large times is the particular integral, the *steady-state response*. Guessing $x_p = C \sin \tau + D \cos \tau$ and substituting into $\ddot{x} + 2\kappa\dot{x} + x = \sin \tau$:

$$(-C - 2\kappa D + C) \sin \tau + (-D + 2\kappa C + D) \cos \tau = \sin \tau,$$

so $C = 0$ and $D = -1/(2\kappa)$, giving

$$x \sim -\frac{1}{2\kappa} \cos \tau \quad \text{as } \tau \rightarrow \infty.$$

Notice that the long-time response is 90° out of phase with the forcing (and its amplitude blows up as $\kappa \rightarrow 0$ – our old friend resonance, since the forcing here is at the natural frequency).

§2.7 Second-Order Difference Equations

Just as first-order ODEs had discrete analogues, so do second-order equations. Consider

$$a y_{n+2} + b y_{n+1} + c y_n = f_n.$$

The whole solution theory transfers almost verbatim from the continuous setting, because the same structural facts hold: the equation is linear, and we know an eigenfunction of the basic operator. Here the basic operator is the *shift* $D[y_n] = y_{n+1}$, and its eigenfunctions are the geometric sequences $y_n = k^n$, since $D[k^n] = k \cdot k^n$.

So, to solve the homogeneous equation $a y_{n+2} + b y_{n+1} + c y_n = 0$, we try $y_n = k^n$:

$$a k^{n+2} + b k^{n+1} + c k^n = 0 \implies a k^2 + b k + c = 0.$$

If the roots k_1, k_2 are distinct, the general complementary function is

$$y_n^{(c)} = A k_1^n + B k_2^n,$$

and if they coincide, $y_n^{(c)} = (A + Bn)k^n$ – multiplication by n playing exactly the role that multiplication by x played before.

For particular integrals we guess, following the same principles:

f_n	trial $y_n^{(p)}$
k^n ($k \neq k_1, k_2$)	Ak^n
k_1^n	Ank_1^n
n^p	$An^p + Bn^{p-1} + \dots + C$

with the resonant case again handled by an extra factor of n .

Example 2.7.1 (The Fibonacci Sequence)

The Fibonacci numbers satisfy

$$y_{n+2} - y_{n+1} - y_n = 0, \quad y_0 = y_1 = 1.$$

Trying $y_n = k^n$ gives $k^2 - k - 1 = 0$, so

$$k = \frac{1 \pm \sqrt{5}}{2}.$$

Write $\varphi_1 = \frac{1+\sqrt{5}}{2}$ (the golden ratio) and $\varphi_2 = \frac{1-\sqrt{5}}{2}$; note $\varphi_1\varphi_2 = -1$. The general solution is $y_n = A\varphi_1^n + B\varphi_2^n$, and the initial conditions give

$$A + B = 1, \quad A\varphi_1 + B\varphi_2 = 1,$$

from which $A = \varphi_1/\sqrt{5}$ and $B = -\varphi_2/\sqrt{5}$. Hence

$$y_n = \frac{\varphi_1^{n+1} - \varphi_2^{n+1}}{\sqrt{5}}.$$

Since $|\varphi_2| < 1$, the second term rapidly becomes negligible: the Fibonacci numbers grow like $\varphi_1^{n+1}/\sqrt{5}$, and the ratio of successive terms tends to the golden ratio. (It is mildly miraculous that this expression, stuffed full of $\sqrt{5}$'s, is an integer for every n – but of course it must be.)

Example 2.7.2 (A Forced Difference Equation)

Consider

$$y_{n+2} - 5y_{n+1} + 6y_n = 2^n.$$

The characteristic equation $k^2 - 5k + 6 = 0$ has roots $k = 2, 3$, so the complementary function is $A \cdot 2^n + B \cdot 3^n$. But the forcing 2^n is resonant with the root $k = 2$, so we try $y_n^{(p)} = Cn2^n$. Substituting:

$$C(n+2)2^{n+2} - 5C(n+1)2^{n+1} + 6Cn2^n = 2^n [4C(n+2) - 10C(n+1) + 6Cn] = -2C \cdot 2^n,$$

so $C = -\frac{1}{2}$, and the general solution is

$$y_n = A \cdot 2^n + B \cdot 3^n - \frac{n}{2} \cdot 2^n.$$

Everything about resonance transfers from the continuous world: the resonant response grows like n times the forcing, just as it grew like t times the forcing before.

§2.8 Impulses and Discontinuous Forcing

So far our forcing functions have been smooth. But many physical forces are not: a hammer blow, a ball bouncing, a switch being flicked. In this section we develop the (initially rather alarming) mathematical objects used to model instantaneous and switched-on forces.

§2.8.1 Impulses and the Dirac Delta Function

Consider a ball bouncing on the ground. When the ball hits the ground at time T , the ground exerts a large force $F(t)$ on it, but only during a very short interval $(T-\varepsilon, T+\varepsilon)$; at all other times $F(t) = 0$. Typically we neither know nor care about the detailed profile of F during the impact. What actually matters?

Newton's second law during the bounce reads $m\ddot{x} = F(t) - mg$. Integrating over the impact,

$$\int_{T-\varepsilon}^{T+\varepsilon} m\ddot{x} \, dt = \int_{T-\varepsilon}^{T+\varepsilon} F(t) \, dt - \int_{T-\varepsilon}^{T+\varepsilon} mg \, dt,$$

which is to say

$$\Delta p = I - O(\varepsilon), \quad I = \int_{T-\varepsilon}^{T+\varepsilon} F(t) \, dt,$$

where Δp is the change in momentum. The quantity I – the area under the force curve, called the **impulse** – is the *only* feature of F that affects the motion on macroscopic timescales. So for a short impact we may as well idealise: let $\varepsilon \rightarrow 0$ and imagine the force acting instantaneously, with the impulse held fixed.

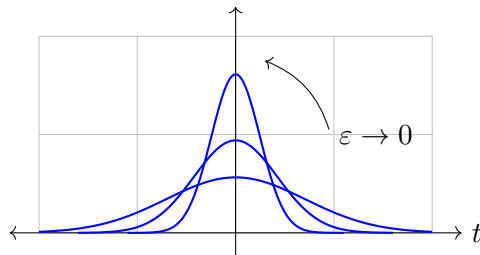
To make this idealisation usable inside differential equations, consider (taking $T = 0$ for convenience) a family of functions $D(t; \varepsilon)$ that become taller and narrower as $\varepsilon \rightarrow 0$ while keeping unit area:

$$\lim_{\varepsilon \rightarrow 0} D(t; \varepsilon) = 0 \text{ for } t \neq 0, \quad \lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} D(t; \varepsilon) \, dt = 1.$$

A standard choice is the Gaussian family

$$D(t; \varepsilon) = \frac{1}{\varepsilon\sqrt{\pi}} e^{-t^2/\varepsilon^2},$$

of height $O(1/\varepsilon)$ and width $O(\varepsilon)$. An idealised unit impulse at $t = 0$ is then modelled by the ‘limit’ of this family.



Definition 2.8.1 (Dirac Delta Function)

The **Dirac delta function** is defined by

$$\delta(x) = \lim_{\varepsilon \rightarrow 0} D(x; \varepsilon),$$

on the strict understanding that only its *integral* properties may ever be used: an expression $\int_{-\infty}^{\infty} g(x) \delta(x) dx$ means, by definition,

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^{\infty} g(x) D(x; \varepsilon) dx.$$

The delta function is not a function in any honest sense – the pointwise limit at 0 doesn't even exist – but the integrals it appears in are perfectly meaningful³. The key rule, obtained by noting that all the mass of $D(\cdot; \varepsilon)$ concentrates at the origin, is the **sampling property**:

$$\int_a^b g(x) \delta(x - c) dx = \begin{cases} g(c) & c \in (a, b), \\ 0 & c \notin [a, b], \end{cases}$$

for g continuous at c . Integrating against $\delta(x - c)$ simply evaluates at c .

With this notation, our bouncing ball obeys the single equation

$$m\ddot{x} = -mg + I\delta(t - T),$$

valid for all time, with the impact incorporated as a delta-function forcing.

Example 2.8.2 (Manipulating Delta Functions)

All computations with δ reduce to the sampling property, usually after a substitution. For instance, for a constant $a \neq 0$, substituting $u = ax$:

$$\int_{-\infty}^{\infty} g(x) \delta(ax) dx = \frac{1}{|a|} \int_{-\infty}^{\infty} g(u/a) \delta(u) du = \frac{g(0)}{|a|},$$

(the modulus appearing because $a < 0$ also reverses the limits). So, as distributions, $\delta(ax) = \delta(x)/|a|$. Similarly,

$$\int_0^5 (x^2 + 1) \delta(x - 2) dx = 5, \quad \int_3^5 (x^2 + 1) \delta(x - 2) dx = 0,$$

the second because the spike at $x = 2$ lies outside the range of integration.

§2.8.2 Delta-Function Forcing of Second-Order Equations

What does a delta-forced solution actually look like? The guiding principle is:

Differentiation makes functions less smooth, so the discontinuity in a delta-forced equation is always absorbed by the highest derivative present.

³The proper home for objects like δ is the theory of *distributions*, but the operational rules stated here are all we need.

For a second-order equation $y'' + py' + qy = \delta(x - c)$: the second derivative behaves like a delta function at $x = c$; hence y' has a finite jump there; and y itself is continuous. (If y itself jumped, y' would contain a delta and y'' something worse – which could not be balanced by the right-hand side.) The size of the jump in y' is found by integrating the equation across the singularity.

Example 2.8.3

Let's solve

$$y'' - y = 3\delta\left(x - \frac{\pi}{2}\right), \quad y(0) = y(\pi) = 0.$$

Away from $x = \pi/2$ the equation is just $y'' - y = 0$, with general solution built from $\sinh$ and $\cosh$. It is convenient to write the solutions in a form adapted to each boundary:

- On $0 \leq x < \pi/2$: $y = A \sinh x + B \cosh x$, and $y(0) = 0$ forces $B = 0$.
- On $\pi/2 < x \leq \pi$: $y = C \sinh(\pi - x) + D \cosh(\pi - x)$, and $y(\pi) = 0$ forces $D = 0$.

Now we join the pieces at $x = \pi/2$. First, continuity of y :

$$A \sinh \frac{\pi}{2} = C \sinh \frac{\pi}{2} \implies A = C.$$

Second, the jump condition. Integrating the equation from $\frac{\pi}{2}^-$ to $\frac{\pi}{2}^+$:

$$[y']_{\pi/2^-}^{\pi/2^+} - \int_{\pi/2^-}^{\pi/2^+} y \, dx = 3.$$

The integral of the (bounded, continuous) function y over a vanishing interval is zero, so the jump in y' is exactly 3:

$$-C \cosh \frac{\pi}{2} - A \cosh \frac{\pi}{2} = 3 \implies A = C = \frac{-3}{2 \cosh \frac{\pi}{2}}.$$

So the solution is

$$y = \begin{cases} \frac{-3 \sinh x}{2 \cosh \frac{\pi}{2}} & 0 \leq x \leq \frac{\pi}{2}, \\ \frac{-3 \sinh(\pi - x)}{2 \cosh \frac{\pi}{2}} & \frac{\pi}{2} \leq x \leq \pi, \end{cases}$$

a tent-like curve, continuous everywhere but with a corner (jump in y') at the point of impact.

§2.8.3 The Heaviside Step Function

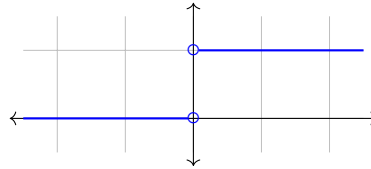
Closely related to the delta function is the function that models a switch being flicked on.

Definition 2.8.4 (Heaviside Step Function)

The **Heaviside step function** is

$$H(x) = \int_{-\infty}^x \delta(t) dt = \begin{cases} 0 & x < 0, \\ 1 & x > 0, \end{cases}$$

with $H(0)$ left undefined.



By the fundamental theorem of calculus (applied, as always with these objects, in the sense of integrals),

$$\frac{dH}{dx} = \delta(x).$$

This gives a tidy hierarchy of smoothness: H has a jump, its derivative δ is an impulse, and going the other way, the integral of H is the continuous ‘ramp’ function $xH(x)$. Solving an equation with step-function forcing works just like the delta case, but one degree smoother: for $y'' + py' + qy = H(x - c)$, both y and y' are continuous at $x = c$, and it is y'' that jumps.

Example 2.8.5 (Switching On an Oscillator)

Consider an undamped oscillator, at rest, which at $t = 0$ has a constant force switched on:

$$\ddot{y} + y = H(t), \quad y = \dot{y} = 0 \text{ for } t < 0.$$

For $t < 0$ the solution is $y = 0$. For $t > 0$ the equation is $\ddot{y} + y = 1$, with particular integral $y_p = 1$ and general solution

$$y = 1 + A \cos t + B \sin t.$$

At $t = 0$ we require both y and $\dot{y}$ to be continuous (only $\ddot{y}$ may jump, matching the jump in H). So $y(0) = 0$ gives $A = -1$, and $\dot{y}(0) = 0$ gives $B = 0$:

$$y = \begin{cases} 0 & t < 0, \\ 1 - \cos t & t > 0. \end{cases}$$

The oscillator swings between 0 and 2 forever: switching on a constant force doesn’t move an undamped oscillator to its new equilibrium $y = 1$, but sets it oscillating about it – with, note, twice the amplitude one might naively expect. In a real system, damping would cause these oscillations to decay and the solution to settle at $y = 1$; the oscillation is the transient.

§2.9 Series Solutions

Many important second-order equations – Legendre’s, Bessel’s, and others that arise when PDEs are solved in special geometries – have non-constant coefficients and no closed-form solution in elementary functions. The method of this section constructs their solutions directly as power series.

We consider homogeneous equations of the form

$$p(x)y'' + q(x)y' + r(x)y = 0.$$

Prominent members of this family, which the reader will meet properly in IB, include *Legendre's equation* $(1 - x^2)y'' - 2xy' + \ell(\ell + 1)y = 0$ and *Bessel's equation* $x^2y'' + xy' + (x^2 - \nu^2)y = 0$; their series solutions (Legendre polynomials, Bessel functions) are among the standard 'special functions' of mathematical physics.

§2.9.1 Ordinary and Singular Points

The behaviour of series solutions about a point x_0 is governed by whether the leading coefficient vanishes there.

Definition 2.9.1 (Ordinary and Singular Points)

The point $x = x_0$ is an **ordinary point** of the equation if q/p and r/p have Taylor series about x_0 (i.e. are analytic there). Otherwise x_0 is a **singular point**.

A singular point x_0 is a **regular singular point** if the equation can be written as

$$P(x)(x - x_0)^2y'' + Q(x)(x - x_0)y' + R(x)y = 0$$

with Q/P and R/P analytic at x_0 .

Loosely: at a regular singular point the coefficients are allowed to blow up, but only as fast as the equidimensional equation's do.

Example 2.9.2

Some quick classifications:

- (i) $(1 - x^2)y'' - 2xy' + 2y = 0$ (Legendre's equation): $x = 0$ is ordinary; $x = \pm 1$ are regular singular points.
- (ii) $(\sin x)y'' + (\cos x)y' + 2y = 0$: the points $x = n\pi$ are regular singular points; all others are ordinary.
- (iii) $(1 + \sqrt{x})y'' - 2xy' + 2y = 0$: $x = 0$ is an *irregular* singular point, since $\sqrt{x}$ has no Taylor series at 0.

The fundamental facts – which we will use but not prove – are:

- (1) If x_0 is an *ordinary point*, the equation has two linearly independent solutions of Taylor form

$$y = \sum_{n=0}^{\infty} a_n(x - x_0)^n,$$

convergent in some neighbourhood of x_0 .

- (2) If x_0 is a *regular singular point*, there is at least one solution of the form

$$y = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+\sigma}, \quad a_0 \neq 0,$$

a **Frobenius series**, where the **index** σ need not be an integer (nor even real). Requiring $a_0 \neq 0$ makes σ well defined. Equivalently, $y = (x - x_0)^\sigma f(x)$ with f analytic and $f(x_0) \neq 0$.

§2.9.2 Solutions About an Ordinary Point

The method is best introduced by simply watching it work on an important equation.

Example 2.9.3 (Legendre's Equation)

Let's find a series solution of

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

about the ordinary point $x = 0$. Try $y = \sum_{n \geq 0} a_n x^n$. A small trick makes the bookkeeping cleaner: multiply the equation through by x^2 so that every term is a combination of the equidimensional building blocks $x^2 y''$, xy' and y (with polynomial coefficients), each of which acts diagonally on powers:

$$x^2 y'' = \sum_n n(n-1) a_n x^n, \quad xy' = \sum_n n a_n x^n.$$

We get

$$(1-x^2)(x^2 y'') - 2x^2(xy') + 2x^2 y = 0 \implies \sum_n a_n [n(n-1) + (-n^2 - n + 2)x^2] x^n = 0.$$

Setting the coefficient of x^n to zero gives the recurrence

$$n(n-1)a_n = [(n-2)^2 + (n-2) - 2] a_{n-2} = (n^2 - 3n) a_{n-2}.$$

We resist the urge to divide, because the prefactors may vanish – and indeed the cases $n = 0$ and $n = 1$ read $0 \cdot a_0 = 0$ and $0 \cdot a_1 = 0$, so a_0 and a_1 are both arbitrary: they are our two constants of integration. For $n \geq 2$ we may divide:

$$a_n = \frac{n-3}{n-1} a_{n-2}.$$

For odd n , the chain passes through $a_3 = 0 \cdot a_1 = 0$, so all odd coefficients beyond a_1 vanish. For even $n = 2k$, the product telescopes:

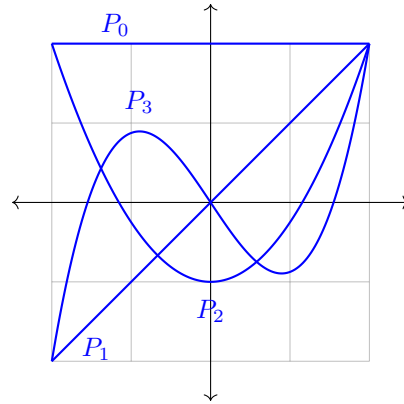
$$a_{2k} = \frac{2k-3}{2k-1} \cdot \frac{2k-5}{2k-3} \cdots \frac{-1}{1} a_0 = \frac{-1}{2k-1} a_0.$$

So the general solution is

$$y = a_0 \left[1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \cdots \right] + a_1 x = a_0 \left[1 - \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) \right] + a_1 x,$$

where the closed form of the even series can be checked by expanding the logarithm. Two things are worth noticing: the polynomial solution $y = x$ (Legendre equations always have one polynomial solution for the right parameter values), and the logarithmic divergence of the other solution at $x = \pm 1$ – precisely at the singular points of the equation. The series knows about the singularities.

The same happens for every parameter value $\ell(\ell+1)$ with ℓ a non-negative integer: one of the two series terminates, leaving a polynomial solution of degree ℓ . Normalised so that $y(1) = 1$, these are the **Legendre polynomials** $P_\ell(x)$; the first few are $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$ and $P_3 = \frac{1}{2}(5x^3 - 3x)$, sketched below on $[-1, 1]$.



§2.9.3 Solutions About a Regular Singular Point: Frobenius

To see why singular points demand something new, let's try our hand at an example.

Example 2.9.4

Consider

$$4xy'' + 2(1 - x^2)y' - xy = 0.$$

The point $x = 0$ is singular (the coefficient of y'' vanishes), but multiplying by x puts the equation in the form

$$4(x^2y'') + 2(1 - x^2)(xy') - x^2y = 0,$$

and comparing with the definition, $Q/P = (1 - x^2)/2$ and $R/P = -x^2/4$ are analytic, so $x = 0$ is a *regular* singular point. We therefore try a Frobenius series

$$y = \sum_{n=0}^{\infty} a_n x^{n+\sigma}, \quad a_0 \neq 0.$$

Using the diagonal action $x^2y'' = \sum a_n(n+\sigma)(n+\sigma-1)x^{n+\sigma}$ and $xy' = \sum a_n(n+\sigma)x^{n+\sigma}$:

$$\sum_n a_n x^{n+\sigma} [4(n+\sigma)(n+\sigma-1) + 2(n+\sigma)] - \sum_n a_n x^{n+\sigma+2} [2(n+\sigma) + 1] = 0.$$

The coefficient of $x^{n+\sigma}$ gives the recurrence

$$2(n+\sigma)(2n+2\sigma-1)a_n = (2n+2\sigma-3)a_{n-2}.$$

Now the crucial step: setting $n = 0$ (where $a_{-2} = 0$) gives

$$2\sigma(2\sigma-1)a_0 = 0,$$

and since $a_0 \neq 0$ by fiat, we obtain the **indicial equation** $2\sigma(2\sigma - 1) = 0$, so

$$\sigma = 0 \quad \text{or} \quad \sigma = \frac{1}{2}.$$

The index $\sigma = 0$ will give an analytic solution; $\sigma = \frac{1}{2}$ a genuinely non-analytic one (a series in x multiplied by $\sqrt{x}$).

Case $\sigma = 0$. The recurrence is $2n(2n - 1)a_n = (2n - 3)a_{n-2}$. Here a_0 is arbitrary, $n = 1$ gives $a_1 = 0$ (and hence all odd terms vanish), and for even n ,

$$a_n = \frac{2n - 3}{2n(2n - 1)} a_{n-2} \implies y_1 = a_0 \left[1 + \frac{1}{4 \cdot 3} x^2 + \frac{5}{8 \cdot 7 \cdot 4 \cdot 3} x^4 + \cdots \right].$$

Case $\sigma = \frac{1}{2}$. The recurrence is $(2n + 1)(2n)a_n = (2n - 2)a_{n-2}$. Again a_0 (call it b_0 to avoid confusion) is arbitrary and odd terms vanish, and for even n ,

$$a_n = \frac{n - 1}{n(2n + 1)} a_{n-2} \implies y_2 = b_0 x^{1/2} \left[1 + \frac{1}{2 \cdot 5} x^2 + \frac{3}{2 \cdot 5 \cdot 4 \cdot 9} x^4 + \cdots \right].$$

Together, $y = y_1 + y_2$ gives the general solution.

§2.9.4 When the Indices Differ by an Integer

The indicial equation is a quadratic with roots σ_1, σ_2 (say $\operatorname{Re} \sigma_1 \leq \operatorname{Re} \sigma_2$), and the structure of the solutions depends on the difference $\sigma_2 - \sigma_1$.

- (1) If $\sigma_2 - \sigma_1$ is *not* an integer, everything is pleasant: there are two independent Frobenius solutions, one for each index, and near x_0 the general solution behaves like $(x - x_0)^{\sigma_1}$.
- (2) If $\sigma_2 - \sigma_1$ *is* an integer (including the repeated case $\sigma_1 = \sigma_2$), the larger index σ_2 always yields a Frobenius solution y_1 , but the smaller one generally does not: the recurrence for $\sigma = \sigma_1$ breaks down when it reaches the offending coefficient. The second solution instead (usually) takes the logarithmic form

$$y_2 = \ln(x - x_0) y_1 + \sum_{n=0}^{\infty} b_n (x - x_0)^{n+\sigma_1},$$

with the coefficients b_n found by substituting this form into the equation. The appearance of the logarithm is a resonance phenomenon between the two solutions – entirely analogous to the resonant forcing we met earlier, where the natural guess had to be corrected by a factor breaking the degeneracy. (Occasionally the resonance ‘accidentally’ avoids itself and two ordinary Frobenius solutions exist after all.)

Example 2.9.5

Consider $x^2 y'' - xy = 0$, already in equidimensional form. Trying $y = \sum a_n x^{n+\sigma}$ with $a_0 \neq 0$:

$$\sum_n a_n x^{n+\sigma} (n + \sigma)(n + \sigma - 1) - \sum_n a_n x^{n+\sigma+1} = 0,$$

giving the recurrence

$$(n + \sigma)(n + \sigma - 1)a_n = a_{n-1}.$$

The indicial equation is $\sigma(\sigma - 1) = 0$, so $\sigma = 0, 1$ – differing by an integer, so we should brace ourselves.

Index $\sigma = 1$ (the larger). The recurrence is $(n + 1)n a_n = a_{n-1}$, so a_0 is arbitrary and for $n \geq 1$,

$$a_n = \frac{a_{n-1}}{n(n+1)} = \frac{a_0}{(n+1)(n!)^2},$$

giving

$$y_1 = a_0 x \left(1 + \frac{x}{2} + \frac{x^2}{12} + \frac{x^3}{144} + \cdots \right).$$

Index $\sigma = 0$. The recurrence is $n(n-1)a_n = a_{n-1}$. At $n = 1$ this reads $0 \cdot a_1 = a_0$, which is impossible since $a_0 \neq 0$. So there is *no* Frobenius solution with index 0, exactly as advertised, and the second solution has the logarithmic form

$$y_2 = y_1 \ln x + \sum_{n=0}^{\infty} b_n x^n,$$

whose coefficients could be ground out by substitution.

Remark. It is no accident that we saw logarithms appearing at singular points before – in the Legendre example, the second solution behaved like $\ln\left(\frac{1+x}{1-x}\right)$ near the regular singular points $x = \pm 1$. Singular points are exactly where solutions are allowed to misbehave, and the possible misbehaviours at a *regular* singular point are tightly controlled: a branch-point power $(x - x_0)^\sigma$, possibly decorated with a logarithm. Irregular singular points can do far wilder things (consider $e^{1/x}$ near 0), which is why the theory stops there.

§2.10 The Gradient and Stationary Points

We now return to functions of several variables, and develop the tools needed to do calculus in genuinely multi-dimensional settings: the gradient vector, Taylor series in $\mathbb{R}^n$, and the classification of stationary points. Beyond their intrinsic interest, these are exactly the tools we will need to analyse systems of differential equations in the next section.

§2.10.1 The Directional Derivative and the Gradient

Consider $f(x, y)$ and a small displacement $ds = (dx, dy)$ in the plane. The chain rule gives the change in f :

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = ds \cdot \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right).$$

The vector appearing on the right is important enough to name. Writing $ds = \hat{s} ds$ with $|\hat{s}| = 1$:

Definition 2.10.1 (Directional Derivative and Gradient)

The **gradient** of f , written ∇f , is the vector such that for every unit vector $\hat{\mathbf{s}}$, the **directional derivative** of f in the direction $\hat{\mathbf{s}}$ is

$$\frac{df}{ds} = \hat{\mathbf{s}} \cdot \nabla f.$$

In Cartesian coordinates, $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$.

(One can take either statement as the definition; taking the coordinate-free one, the Cartesian formula becomes a small theorem, following from the chain rule computation above.)

Writing $\frac{df}{ds} = |\nabla f| \cos \theta$, where θ is the angle between $\hat{\mathbf{s}}$ and ∇f , we can read off the geometry of the gradient:

- (i) ∇f points in the direction in which f increases fastest;
- (ii) $|\nabla f|$ is that maximal rate of increase;
- (iii) if $\hat{\mathbf{s}}$ is tangent to a contour of f (a curve of constant f), then $\frac{df}{ds} = 0$, so ∇f is perpendicular to the contours.

These three facts are worth having thoroughly absorbed – they are used constantly, in this course and in Vector Calculus.

Example 2.10.2

Let $f(x, y) = x^2y$, and let's find the rate of change of f at the point $(1, 2)$ in the direction of the unit vector $\hat{\mathbf{s}} = \frac{1}{5}(3, 4)$. We have

$$\nabla f = (2xy, x^2), \quad \nabla f(1, 2) = (4, 1),$$

so

$$\frac{df}{ds} = \hat{\mathbf{s}} \cdot \nabla f = \frac{1}{5}(3 \cdot 4 + 4 \cdot 1) = \frac{16}{5}.$$

The maximum possible rate of change at this point is $|\nabla f| = \sqrt{17}$, attained in the direction $\frac{1}{\sqrt{17}}(4, 1)$, and the level curve of f through $(1, 2)$ has tangent perpendicular to this, i.e. along $(1, -4)$.

§2.10.2 Taylor Series in Several Variables

To classify stationary points we need the second-order Taylor expansion of a multivariate function. Let $\mathbf{x}_0$ be a point and $\delta \mathbf{x} = \mathbf{x} - \mathbf{x}_0$ a small displacement, along a straight line in the direction $\hat{\mathbf{s}}$, so $\delta \mathbf{x} = \hat{\mathbf{s}} \delta s$. Along that line, f is a function of the single variable s with $\frac{d}{ds} = \hat{\mathbf{s}} \cdot \nabla$, and the ordinary Taylor expansion gives

$$f = f(\mathbf{x}_0) + \delta s (\hat{\mathbf{s}} \cdot \nabla) f + \frac{1}{2} (\delta s)^2 (\hat{\mathbf{s}} \cdot \nabla)(\hat{\mathbf{s}} \cdot \nabla) f + \dots$$

The first-order term is $\delta \mathbf{x} \cdot \nabla f$. For the second-order term, expanding in coordinates,

$$(\delta \mathbf{x} \cdot \nabla)(\delta \mathbf{x} \cdot \nabla) f = (\delta x)^2 f_{xx} + 2 \delta x \delta y f_{xy} + (\delta y)^2 f_{yy} = \delta \mathbf{x}^T H \delta \mathbf{x},$$

where H is the following matrix of second derivatives.

Definition 2.10.3 (Hessian Matrix)

The **Hessian** of f is the matrix of second partial derivatives

$$H = \nabla \nabla f = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix},$$

(and the analogous $n \times n$ matrix in n variables). By symmetry of mixed partials, H is a symmetric matrix.

So we have the coordinate-free second-order Taylor expansion

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \delta \mathbf{x} \cdot \nabla f(\mathbf{x}_0) + \frac{1}{2} \delta \mathbf{x}^T H(\mathbf{x}_0) \delta \mathbf{x} + o(|\delta \mathbf{x}|^2),$$

which is the multivariate analogue of ‘constant + linear + quadratic’.

Example 2.10.4

Let’s expand $f(x, y) = e^{xy}$ to second order about $(0, 0)$. We have

$$f_x = ye^{xy}, \quad f_y = xe^{xy}, \quad f_{xx} = y^2 e^{xy}, \quad f_{xy} = (1 + xy)e^{xy}, \quad f_{yy} = x^2 e^{xy},$$

so at the origin $f = 1$, $\nabla f = (0, 0)$ and

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Hence

$$e^{xy} = 1 + \frac{1}{2} \begin{pmatrix} x & y \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \cdots = 1 + xy + \cdots,$$

which agrees, reassuringly, with substituting $u = xy$ into $e^u = 1 + u + \cdots$. Note in passing that the origin is a stationary point of e^{xy} , and the Hessian has eigenvalues ± 1 (eigenvectors $(1, \pm 1)$): a saddle, as a sketch of the function quickly confirms.

§2.10.3 Stationary Points and Their Classification

At any point there is always at least one direction with $df/ds = 0$, namely along the contour. A *stationary point* is where this happens in every direction.

Definition 2.10.5 (Stationary Point)

A **stationary point** of f is a point where $\nabla f = \mathbf{0}$, i.e. where all partial derivatives vanish.

In one variable, stationary points are maxima, minima or (rarely) inflections. In two or more variables there is a genuinely new type: the **saddle point**, where f increases in some directions and decreases in others (like the middle of a mountain pass – or a horse’s saddle). We define a saddle to be any stationary point that is neither a local maximum nor a local minimum.

Near a stationary point the Taylor expansion reads

$$f(\mathbf{x}) \approx f(\mathbf{x}_0) + \frac{1}{2} \delta \mathbf{x}^T H \delta \mathbf{x},$$

so the local behaviour is governed entirely by the sign of the quadratic form $\delta \mathbf{x}^T H \delta \mathbf{x}$:

- minimum: $\delta \mathbf{x}^T H \delta \mathbf{x} > 0$ for all $\delta \mathbf{x} \neq \mathbf{0}$ (H is *positive definite*);
- maximum: $\delta \mathbf{x}^T H \delta \mathbf{x} < 0$ for all $\delta \mathbf{x} \neq \mathbf{0}$ (H is *negative definite*);
- saddle: the form takes both signs (H is *indefinite*).

To determine which case we are in, we use the fact that H is symmetric, and hence diagonalisable with respect to orthogonal axes (its principal axes). In those axes,

$$\delta \mathbf{x}^T H \delta \mathbf{x} = \lambda_1(\delta x_1)^2 + \lambda_2(\delta x_2)^2 + \cdots + \lambda_n(\delta x_n)^2,$$

where λ_i are the eigenvalues of H . Reading off the signs:

Proposition 2.10.6 (Classification by Eigenvalues)

At a stationary point with Hessian H :

- (i) if all eigenvalues of H are positive, the point is a local minimum;
- (ii) if all are negative, a local maximum;
- (iii) if there are eigenvalues of both signs, a saddle;
- (iv) if some eigenvalue is zero (and the rest are of one sign), the quadratic term is degenerate and higher-order terms must be examined.

Computing eigenvalues can be tedious, and there is a shortcut using determinants.

Definition 2.10.7 (Signature)

The **signature** of H is the sequence of signs of the leading principal minors

$$|H_1| = f_{x_1 x_1}, \quad |H_2| = \begin{vmatrix} f_{x_1 x_1} & f_{x_1 x_2} \\ f_{x_2 x_1} & f_{x_2 x_2} \end{vmatrix}, \quad \dots, \quad |H_n| = |H|.$$

Proposition 2.10.8 (Sylvester's Criterion)

H is positive definite if and only if its signature is $+, +, \dots, +$, and negative definite if and only if its signature is $-, +, -, \dots, (-1)^n$ (alternating, starting negative).

In two dimensions this boils down to something very memorable: compute $\det H = f_{xx}f_{yy} - f_{xy}^2$. If $\det H < 0$, saddle. If $\det H > 0$, it is a max or min according to the sign of f_{xx} .

Example 2.10.9

Let's find and classify the stationary points of

$$f(x, y) = 4x^3 - 12xy + y^2 + 10y + 6.$$

The first derivatives are

$$f_x = 12x^2 - 12y, \quad f_y = -12x + 2y + 10.$$

Setting $f_x = 0$ gives $y = x^2$; substituting into $f_y = 0$ gives $-12x + 2x^2 + 10 = 0$, i.e. $x^2 - 6x + 5 = (x - 1)(x - 5) = 0$. So the stationary points are $(1, 1)$ and $(5, 25)$.

The second derivatives are $f_{xx} = 24x$, $f_{xy} = -12$, $f_{yy} = 2$.

- At $(1, 1)$: $H = \begin{pmatrix} 24 & -12 \\ -12 & 2 \end{pmatrix}$, with $|H_1| = 24 > 0$ and $|H_2| = 48 - 144 = -96 < 0$. Signature $+, -$: indefinite, so a *saddle point*.
- At $(5, 25)$: $H = \begin{pmatrix} 120 & -12 \\ -12 & 2 \end{pmatrix}$, with $|H_1| = 120 > 0$ and $|H_2| = 240 - 144 = 96 > 0$. Signature $+, +$: positive definite, so a *local minimum*.

A final remark on the geometry of contours. Near a maximum or minimum, in principal axes the contours $\lambda_1 X^2 + \lambda_2 Y^2 = \text{const}$ (with $\lambda_1 \lambda_2 > 0$) are locally *ellipses*; near a saddle ($\lambda_1 \lambda_2 < 0$) they are locally *hyperbolae*, and the two branches of the contour through the saddle itself cross there. Indeed, contour lines of a function can only cross at saddle points – a fact which makes sketching contour maps much easier: find and classify the stationary points, draw the local pictures, and join everything up consistently.

§2.11 Systems of Differential Equations

So far we have studied single equations for a single unknown function. But interesting systems usually involve several interacting quantities – predators and prey, coupled circuits, positions and momenta – each with its own differential equation, coupled together. In this section we solve linear systems using matrices and eigenvalues, and then use *linearisation* to understand nonlinear systems near their equilibria.

§2.11.1 Linear Systems and Matrix Methods

Consider two functions $y_1(t)$, $y_2(t)$ satisfying the coupled linear equations

$$\dot{y}_1 = ay_1 + by_2 + f_1(t), \quad \dot{y}_2 = cy_1 + dy_2 + f_2(t),$$

which we write in matrix form as

$$\dot{\mathbf{Y}} = M\mathbf{Y} + \mathbf{F}, \quad \mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}.$$

First, let us note that first-order systems and higher-order scalar equations are two descriptions of the same thing. In one direction, differentiating the first equation and eliminating y_2 turns the system into a single second-order equation for y_1 :

$$\ddot{y}_1 - (a + d)\dot{y}_1 + (ad - bc)y_1 = bf_2 - df_1 + \dot{f}_1.$$

(Notice the trace and determinant of M appearing – this is no accident, as we will see in a moment.) In the other direction, given a second-order equation $\ddot{y} + a\dot{y} + by = f$, setting $y_1 = y$ and $y_2 = \dot{y}$ gives the first-order system

$$\dot{\mathbf{Y}} = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix} \mathbf{Y} + \begin{pmatrix} 0 \\ f \end{pmatrix}.$$

This second direction is the more useful one: it says every ODE can be viewed as a first-order system, i.e. as a flow in phase space, and lets us bring linear algebra to bear.

To solve the homogeneous system $\dot{\mathbf{Y}} = M\mathbf{Y}$, we make the by-now-familiar exponential ansatz, but with a constant *vector* attached:

$$\mathbf{Y} = \mathbf{v} e^{\lambda t}.$$

Substituting in, $\lambda \mathbf{v} e^{\lambda t} = M\mathbf{v} e^{\lambda t}$, so

$$M\mathbf{v} = \lambda \mathbf{v}$$

– an eigenvalue problem! The exponents are the eigenvalues of M (roots of $\det(M - \lambda I) = 0$), and the directions are the corresponding eigenvectors. With two distinct eigenvalues, the general solution is

$$\mathbf{Y} = A\mathbf{v}_1 e^{\lambda_1 t} + B\mathbf{v}_2 e^{\lambda_2 t}.$$

For forced systems, we find a particular integral by guessing in the same spirit as before, now with vector-valued coefficients.

Example 2.11.1

Let's solve

$$\dot{\mathbf{Y}} = \begin{pmatrix} -4 & 24 \\ 1 & -2 \end{pmatrix} \mathbf{Y} + \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^t.$$

Complementary function. The characteristic equation is

$$\begin{vmatrix} -4 - \lambda & 24 \\ 1 & -2 - \lambda \end{vmatrix} = \lambda^2 + 6\lambda - 16 = (\lambda + 8)(\lambda - 2) = 0,$$

so $\lambda = 2, -8$. For $\lambda = 2$, solving $(M - 2I)\mathbf{v} = \mathbf{0}$:

$$\begin{pmatrix} -6 & 24 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \implies \mathbf{v}_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

For $\lambda = -8$:

$$\begin{pmatrix} 4 & 24 \\ 1 & 6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0} \implies \mathbf{v}_2 = \begin{pmatrix} -6 \\ 1 \end{pmatrix}.$$

So the complementary function is

$$\mathbf{Y}_c = A \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{2t} + B \begin{pmatrix} -6 \\ 1 \end{pmatrix} e^{-8t}.$$

Particular integral. The forcing is e^t , not resonant with either eigenvalue, so try $\mathbf{Y}_p = \mathbf{u} e^t$. Substituting,

$$\mathbf{u} - M\mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \implies \begin{pmatrix} 5 & -24 \\ -1 & 3 \end{pmatrix} \mathbf{u} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \implies \mathbf{u} = \frac{1}{-9} \begin{pmatrix} 3 & 24 \\ 1 & 5 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}.$$

So the general solution is

$$\mathbf{Y} = A \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{2t} + B \begin{pmatrix} -6 \\ 1 \end{pmatrix} e^{-8t} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^t.$$

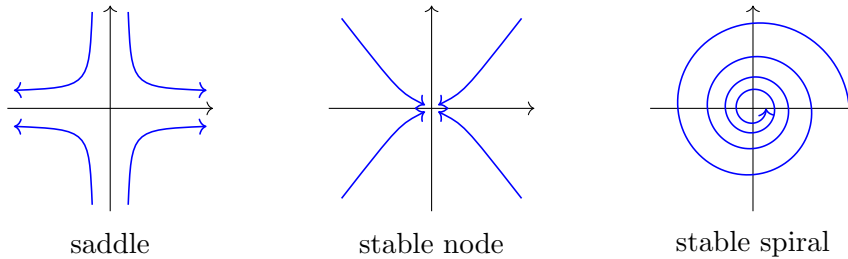
§2.11.2 Phase Portraits in the Plane

For an autonomous planar system $\dot{\mathbf{Y}} = M\mathbf{Y}$, we can draw the trajectories $(y_1(t), y_2(t))$ in the (y_1, y_2) plane – the **phase portrait** of the system. The eigenvector directions

organise the entire picture: along an eigenvector $\mathbf{v}_i$, the trajectory stays on the line through the origin spanned by $\mathbf{v}_i$, moving outwards if $\lambda_i > 0$ and inwards if $\lambda_i < 0$. Every other trajectory is a combination of the two eigen-motions, dominated for large t by whichever mode grows fastest (and for large negative t by the other).

There are three main cases, according to the eigenvalues of M .

- (1) *Real eigenvalues of opposite sign* ($\lambda_1 > 0 > \lambda_2$): a **saddle point**. Trajectories come in along the stable (λ_2) direction and are flung out along the unstable (λ_1) direction, sweeping past the origin on hyperbola-like paths. (This is the shape of the example above, with $\lambda = 2, -8$.)
- (2) *Real eigenvalues of the same sign*: a **node**. All trajectories head into the origin (if both $\lambda_i < 0$: a *stable node*) or away from it (both positive: an *unstable node*). Trajectories are tangent to the ‘slow’ eigenvector direction (smaller $|\lambda|$) near the origin, and asymptotically parallel to the ‘fast’ one far away.
- (3) *Complex conjugate eigenvalues* $\lambda = \sigma \pm i\omega$: a **spiral** (or, if $\sigma = 0$, a family of concentric ellipses – a *centre*). The solutions are of the form $e^{\sigma t} \times (\text{oscillation})$, so trajectories spiral inwards if $\sigma < 0$ (*stable spiral*) and outwards if $\sigma > 0$ (*unstable spiral*). The sense of rotation can be found by evaluating $\dot{\mathbf{Y}} = M\mathbf{Y}$ at one convenient point, say $(1, 0)$, and seeing which way the flow points.



It is worth noting that everything here can be read off from just the trace and determinant of M , since $\lambda_1\lambda_2 = \det M$ and $\lambda_1 + \lambda_2 = \text{tr } M$: negative determinant means a saddle; positive determinant and negative trace a stable node or spiral (spiral if $(\text{tr } M)^2 < 4\det M$); and so on.

Example 2.11.2 (The Damped Oscillator, Revisited)

The damped oscillator $\ddot{x} + 2\kappa\dot{x} + x = 0$ from earlier becomes, upon setting $\mathbf{Y} = (x, \dot{x})$, the planar system

$$\dot{\mathbf{Y}} = \begin{pmatrix} 0 & 1 \\ -1 & -2\kappa \end{pmatrix} \mathbf{Y},$$

with $\text{tr } M = -2\kappa$ and $\det M = 1$, so the eigenvalues are our old friends $\lambda = -\kappa \pm \sqrt{\kappa^2 - 1}$. Reading off the classification:

- $\kappa = 0$ (undamped): purely imaginary eigenvalues, a *centre* – the closed ellipses of simple harmonic motion in phase space, energy conserved.
- $0 < \kappa < 1$ (underdamped): complex eigenvalues with negative real part, a *stable spiral* – the trajectory winds around the origin as the oscillator swings back and forth, spiralling inwards as the amplitude decays.
- $\kappa > 1$ (overdamped): two negative real eigenvalues, a *stable node* – no more

winding; trajectories slide into the origin, tangent to the slow eigenvector.

The transition through critical damping $\kappa = 1$ is precisely the moment the spiral degenerates into a node, i.e. the eigenvalues collide on the real axis. Phase-portrait language and the earlier time-domain analysis are two views of the same underlying picture.

Example 2.11.3 (An Initial Value Problem)

Continuing the earlier example, let's impose the initial condition $\mathbf{Y}(0) = (0, 0)$ on

$$\mathbf{Y} = A \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{2t} + B \begin{pmatrix} -6 \\ 1 \end{pmatrix} e^{-8t} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^t.$$

At $t = 0$:

$$4A - 6B - 4 = 0, \quad A + B - 1 = 0,$$

so $B = 1 - A$ and $4A - 6 + 6A = 4$, giving $A = 1$, $B = 0$. The solution through the origin is

$$\mathbf{Y} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} (e^{2t} - e^t) :$$

as it happens the e^{-8t} mode is not excited at all, and this particular trajectory runs along the fixed direction $(4, 1)$ for all time.

§2.11.3 Nonlinear Systems and Linearisation

Finally we come to nonlinear autonomous systems,

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y),$$

where f and g have no explicit time-dependence. Solving such systems exactly is usually hopeless, but – just as for single autonomous equations – we can understand the phase portrait by finding the equilibria and analysing the flow near each.

Definition 2.11.4 (Equilibrium Point of a System)

An **equilibrium point** of the system is a point (x_0, y_0) where $\dot{x} = \dot{y} = 0$, i.e. where $f(x_0, y_0) = g(x_0, y_0) = 0$ simultaneously.

Near an equilibrium, write $x = x_0 + \xi$, $y = y_0 + \eta$ for small perturbations ξ, η , and Taylor expand:

$$\dot{\xi} = f(x_0 + \xi, y_0 + \eta) = \underbrace{f(x_0, y_0)}_{=0} + \xi f_x + \eta f_y + O(\xi^2, \xi\eta, \eta^2),$$

and similarly for $\dot{\eta}$. Dropping the quadratic terms, we obtain the **linearised system**

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix},$$

where the matrix of partial derivatives (the *Jacobian*) is evaluated at the equilibrium. This is a linear system of exactly the type we have just classified, so near each equilibrium the phase portrait looks (in the non-degenerate cases) like a saddle, node or spiral, determined by the Jacobian's eigenvalues. The global phase portrait is then assembled by patching together these local pictures.

Example 2.11.5 (A Predator–Prey System)

Suppose $x(t)$ counts prey and $y(t)$ predators. A reasonable model:

$$\dot{x} = \underbrace{\alpha x}_{\text{births}} - \underbrace{\beta x^2}_{\text{competition}} - \underbrace{\gamma xy}_{\text{predation}}, \quad \dot{y} = \underbrace{\varepsilon xy}_{\text{food supply}} - \underbrace{\delta y}_{\text{deaths}}.$$

Take, concretely,

$$\dot{x} = 8x - 2x^2 - 2xy, \quad \dot{y} = xy - y.$$

Equilibria. We need $x(8 - 2x - 2y) = 0$ and $y(x - 1) = 0$. The first gives $x = 0$ or $y = 4 - x$; the second gives $y = 0$ or $x = 1$. Combining, the equilibria are

$$(0, 0), \quad (4, 0), \quad (1, 3).$$

Near (0, 0). The linearisation is

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} 8 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix},$$

with eigenvalues $8, -1$ along the coordinate axes: a *saddle*, unstable along the prey axis (a few prey with no predators multiply) and stable along the predator axis (predators with no prey starve).

Near (4, 0). Set $x = 4 + \xi$, $y = \eta$. Rather than computing the Jacobian abstractly, we can substitute directly and discard quadratic terms:

$$\begin{aligned} \dot{\xi} &= (4 + \xi)(8 - 8 - 2\xi - 2\eta) = -8\xi - 8\eta + O(2), \\ \dot{\eta} &= \eta(4 + \xi - 1) = 3\eta + O(2). \end{aligned}$$

The matrix $\begin{pmatrix} -8 & -8 \\ 0 & 3 \end{pmatrix}$ has eigenvalues -8 and 3 (it is triangular), with eigenvectors $(1, 0)$ and $(8, -11)$: another *saddle*. A prey-only population at carrying capacity is unstable to the introduction of predators.

Near (1, 3). Set $x = 1 + \xi$, $y = 3 + \eta$:

$$\begin{aligned} \dot{\xi} &= (1 + \xi)(8 - 2 - 2\xi - 6 - 2\eta) \approx -2\xi - 2\eta, \\ \dot{\eta} &= (3 + \eta)\xi \approx 3\xi. \end{aligned}$$

The matrix $\begin{pmatrix} -2 & -2 \\ 3 & 0 \end{pmatrix}$ has characteristic equation $\lambda^2 + 2\lambda + 6 = 0$, giving

$$\lambda = -1 \pm i\sqrt{5}.$$

Complex with negative real part: a *stable spiral*. To find the sense of rotation, perturb to the right, $(\xi, \eta) = (\xi, 0)$ with $\xi > 0$: then $(\dot{\xi}, \dot{\eta}) = (-2\xi, 3\xi)$, pointing up and to the left, so the spiral is anticlockwise.

The global picture. Patching the three local portraits together: trajectories leave the two saddles along their unstable directions and are drawn into the spiral. Almost every solution with $x, y > 0$ spirals in towards the coexistence state $(1, 3)$ – predators and prey settle down, through decaying oscillations, to stable coexistence.

§2.12 Partial Differential Equations

We end the course with a first look at *partial* differential equations – equations involving partial derivatives of a function of several variables. The subject is vast (much of IB and beyond is devoted to it), but even with the tools we already have, we can solve some of the most important examples: the wave equation and the diffusion equation.

§2.12.1 The First-Order Wave Equation

Consider, for a function $y(x, t)$ and a constant c , the equation

$$\frac{\partial y}{\partial t} = c \frac{\partial y}{\partial x},$$

known as the **first-order wave equation**. The key idea for solving it – the **method of characteristics** – is to stop thinking of y as a function on the whole (x, t) plane, and instead watch how y varies along well-chosen *paths* $x = x(t)$.

Along a path $x(t)$, the multivariate chain rule gives

$$\frac{d}{dt}y(x(t), t) = \frac{\partial y}{\partial t} + \frac{dx}{dt} \frac{\partial y}{\partial x} = \left(c + \frac{dx}{dt}\right) \frac{\partial y}{\partial x},$$

where we used the PDE in the last step. Now *choose* the paths to kill the right-hand side: along paths with

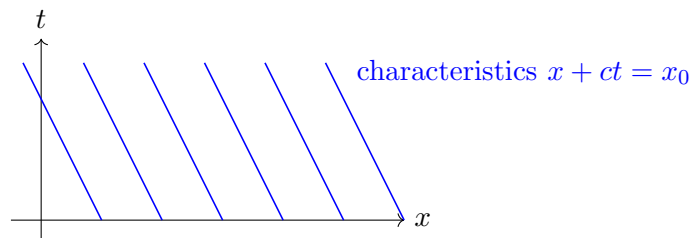
$$\frac{dx}{dt} = -c \quad \text{we have} \quad \frac{dy}{dt} = 0.$$

We have traded one PDE for a pair of ODEs. The chosen paths, called the **characteristics** of the equation, are the straight lines $x = x_0 - ct$, one for each constant $x_0 = x + ct$; and along each characteristic, y is constant. So the value of y at (x, t) depends only on which characteristic the point lies on – that is, only on $x + ct$:

$$y = f(x + ct)$$

for an arbitrary function f (one can check directly that any such y solves the equation). This is a *wave*: the profile f translates rigidly to the left with speed c , unchanged in shape. Note that in place of the arbitrary *constants* in the general solutions of ODEs, the general solution of a PDE contains an arbitrary *function*.

It is helpful to picture the characteristics in the (x, t) plane: a family of parallel lines of slope $-1/c$, each carrying a single constant value of y , transported up from its intersection with the x -axis.



The arbitrary function is pinned down by initial data. Given $y(x, 0)$, we have $f(x) = y(x, 0)$ and hence the full solution: to find y at (x, t) , follow the characteristic through that point back down to the x -axis and read off the initial value there.

Example 2.12.1

Solve $y_t = cy_x$ with $y(x, 0) = x^2 - 3$. Since $y = f(x + ct)$ with $f(x) = x^2 - 3$,

$$y = (x + ct)^2 - 3,$$

the initial parabola sliding left at speed c .

Forced versions succumb to the same method – along the characteristics, y is no longer constant, but satisfies an ODE we can solve.

Example 2.12.2

Solve

$$\frac{\partial y}{\partial t} + 5 \frac{\partial y}{\partial x} = e^{-t}, \quad y(x, 0) = e^{-x^2}.$$

The characteristics are now $\frac{dx}{dt} = 5$, i.e. the lines $x_0 = x - 5t$, and along them

$$\frac{dy}{dt} = e^{-t} \implies y = F(x_0) - e^{-t},$$

with a ‘constant of integration’ $F(x_0)$ that may differ from characteristic to characteristic. At $t = 0$ we have $x = x_0$ and $y = F(x_0) - 1 = e^{-x_0^2}$, so $F(x_0) = 1 + e^{-x_0^2}$, and

$$y = 1 + e^{-(x-5t)^2} - e^{-t}.$$

A rightward-travelling Gaussian pulse, riding on a background that relaxes from 0 up to 1.

§2.12.2 The Second-Order Wave Equation

The equation usually called *the* wave equation is second order:

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}.$$

This models, for instance, small transverse vibrations of a taut string, with $y(x, t)$ the displacement at position x and time t : the acceleration of each element of string is proportional to the local *curvature* y_{xx} , since a straight (even sloped) string feels no net sideways force, while a curved one does.

To solve it, factorise the differential operator:

$$\frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2} = \left(\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right) \left(\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right),$$

where the two first-order factors commute. Anything annihilated by either factor is annihilated by the product, and each factor is a first-order wave equation, solved above. So $y = f(x + ct)$ and $y = g(x - ct)$ are both solutions, and by linearity so is their sum. In fact this is everything:

Proposition 2.12.3 (d’Alembert’s Solution)

The general solution of $y_{tt} = c^2 y_{xx}$ is

$$y = f(x + ct) + g(x - ct)$$

for arbitrary (twice-differentiable) functions f and g : a superposition of a leftward- and a rightward-travelling wave, each moving at speed c with fixed profile.

Proof. That these are solutions is a direct check. For the converse, change variables to $\xi = x + ct$ and $\eta = x - ct$; the chain rule turns the equation into

$$y_{tt} - c^2 y_{xx} = -4c^2 \frac{\partial^2 y}{\partial \xi \partial \eta} = 0,$$

and integrating once in each variable gives $y = f(\xi) + g(\eta)$. $\square$

Since the equation is second order in t (and in x), we expect to need two initial conditions and two boundary conditions to pin down a unique solution – one pair for each of the two arbitrary functions.

Example 2.12.4

Solve $y_{tt} = c^2 y_{xx}$ subject to

$$y(x, 0) = \frac{1}{1 + x^2}, \quad y_t(x, 0) = 0, \quad y \rightarrow 0 \text{ as } x \rightarrow \pm\infty.$$

Write $y = f(x + ct) + g(x - ct)$. The initial conditions give

$$f(x) + g(x) = \frac{1}{1 + x^2}, \quad cf'(x) - cg'(x) = 0.$$

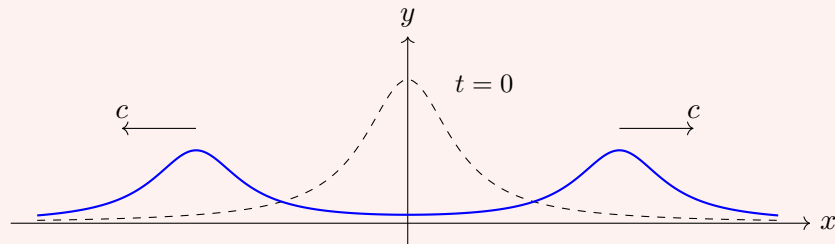
The second says $f' = g'$, so $f = g + A$ for a constant A ; since only $f + g$ enters the solution, we may absorb the constant and take $f = g$, whence

$$f(x) = g(x) = \frac{1}{2(1 + x^2)},$$

and

$$y = \frac{1}{2} \left[\frac{1}{1 + (x + ct)^2} + \frac{1}{1 + (x - ct)^2} \right].$$

The initial hump splits into two half-height copies, travelling apart at speeds $\pm c$.



§2.12.3 The Diffusion Equation

Our final equation models heat conduction in one dimension. If $T(x, t)$ is the temperature in a solid bar, then

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2},$$

where the constant $\kappa > 0$ is the **diffusivity**. This is the **diffusion equation** (or heat equation). Structurally it looks like the wave equation with one time derivative removed, and this changes the character of solutions completely: for the wave equation, curvature drives *acceleration* and disturbances propagate forever; here, curvature drives *velocity*, so bumps in the temperature profile simply flatten themselves out. Diffusion is dissipative, not oscillatory.

Remark (Nomenclature). Second-order linear PDEs are traditionally classified by loose analogy with conic sections: the wave equation $y_{tt} - c^2 y_{xx} = 0$ is called *hyperbolic* (compare $t^2 - c^2 x^2 = \text{const}$), the diffusion equation is *parabolic* ($t = x^2$ -like, one time derivative against two space derivatives), and Laplace's equation $y_{xx} + y_{yy} = 0$, which the reader will meet in Vector Calculus, is *elliptic*. The analogy is purely formal, but the classification is genuinely meaningful: hyperbolic equations propagate signals along characteristics, parabolic ones smooth and dissipate, and elliptic ones describe equilibrium states.

We will solve one classic problem, which also introduces the useful idea of a *similarity solution*.

Example 2.12.5 (Heating a Semi-Infinite Bar)

Consider an infinitely long bar occupying $x \geq 0$, initially cold, whose end $x = 0$ is abruptly heated to unit temperature at $t = 0$ and held there:

$$T(x, 0) = 0, \quad T(0, t) = H(t), \quad T(x, t) \rightarrow 0 \text{ as } x \rightarrow \infty,$$

where H is the Heaviside function.

The key observation is dimensional. The only parameter in the problem is κ , with dimensions $(\text{length})^2/(\text{time})$, and neither the initial nor boundary data introduces any length scale. So the only dimensionless combination of x and t available is

$$\eta = \frac{x}{2\sqrt{\kappa t}}$$

(the factor 2 is for later convenience), and it is natural to seek a **similarity solution** depending on x and t only through η :

$$T(x, t) = \theta(\eta).$$

The temperature profiles at different times should all be the same shape, stretched horizontally by $\sqrt{\kappa t}$. Let's compute the derivatives by the chain rule:

$$\begin{aligned} \frac{\partial T}{\partial x} &= \theta'(\eta) \frac{\partial \eta}{\partial x} = \frac{1}{2\sqrt{\kappa t}} \theta'(\eta), & \frac{\partial^2 T}{\partial x^2} &= \frac{1}{4\kappa t} \theta''(\eta), \\ \frac{\partial T}{\partial t} &= \theta'(\eta) \frac{\partial \eta}{\partial t} = -\frac{1}{2} \cdot \frac{x}{2\sqrt{\kappa} t^{3/2}} \theta'(\eta) = -\frac{\eta}{2t} \theta'(\eta). \end{aligned}$$

Substituting into the PDE:

$$-\frac{\eta}{2t}\theta' = \frac{1}{4t}\theta'' \implies \theta'' + 2\eta\theta' = 0.$$

All the x 's and t 's have vanished, leaving an ODE for $\theta(\eta)$ – the sign that our similarity guess was consistent. Moreover it is an ODE we can solve: it is first order in θ' , with integrating factor $\mu = \exp(\int 2\eta d\eta) = e^{\eta^2}$, so

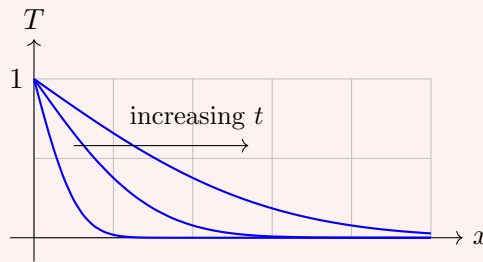
$$(e^{\eta^2}\theta')' = 0 \implies \theta' = Ae^{-\eta^2} \implies \theta = A \int_0^\eta e^{-u^2} du + B.$$

This integral has a standard name: the *error function* $\text{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-u^2} du$, normalised so that $\text{erf}(\eta) \rightarrow 1$ as $\eta \rightarrow \infty$. So $\theta = \alpha \text{erf}(\eta) + B$.

Now the boundary conditions, translated into η : the heated end $x = 0$ (with $t > 0$) corresponds to $\eta = 0$, so $\theta(0) = 1$, giving $B = 1$. The far field $x \rightarrow \infty$ (or the initial instant $t \rightarrow 0^+$) corresponds to $\eta \rightarrow \infty$, so $\theta(\infty) = 0$, giving $\alpha = -1$. Hence $\theta = 1 - \text{erf}(\eta)$, and

$$T(x, t) = \text{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right),$$

where $\text{erfc} = 1 - \text{erf}$ is the complementary error function. The temperature profiles at successive times are stretched copies of one another:



The physical picture: at each time t , the temperature falls smoothly from 1 at the hot end towards 0, over a penetration depth of order $\sqrt{\kappa t}$. Heat creeps into the bar with the characteristic square-root-of-time spreading of diffusion – to heat twice as deep takes four times as long. (This also tells us when a *finite* bar of length L can be treated as infinite: precisely when $\sqrt{\kappa t} \ll L$, i.e. $t \ll L^2/\kappa$.)

And that is where we stop. We have seen equations solved exactly, approximated, discretised, linearised and sketched; the recurring themes – eigenfunctions, linearity, superposition, equilibria, stability – will reappear throughout the rest of the Tripos, in Methods and Analysis and far beyond. Differential equations are never really finished; one simply stops.

3 Dynamics and Relativity

Dynamics and relativity sets the scene for studying the problems of Newtonian dynamics and, towards the end, Einstein's theory of special relativity.

The first part of this course develops *Newtonian mechanics* – a mathematical framework built around the notion of a force, which turns out to be capable of describing an astonishing range of physical phenomena: projectiles, oscillations, planetary orbits, spinning tops and much more. Along the way we will meet the great conserved quantities of physics (energy, momentum and angular momentum), and we will see how symmetry principles constrain what a physical law is allowed to look like. The second part of the course takes one such symmetry principle seriously enough to break Newtonian mechanics entirely: insisting that the speed of light is the same for all observers leads to *special relativity*, where space and time mix together, moving clocks run slow, and mass is just another form of energy.

This article constitutes my notes for the 'Dynamics and Relativity' course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§3.1 Newton's Laws and Forces

In Newtonian mechanics, we use the idea of a force to build up a mathematical framework in which we can describe the motion of objects, typically particles.

§3.1.1 Particles & Inertial Frames

Newtonian mechanics is at its cleanest when describing the motion of *particles*.

Definition 3.1.1 (Particle)

A **particle** is an idealised object of negligible size and no internal structure, to which we can assign physical properties such as mass $m > 0$ and electric charge q .

Real objects are, of course, not points. This idealisation is still tremendously useful for two reasons. First, an extended body can be treated as a particle whenever its size is irrelevant to the problem at hand – the Earth, viewed on the scale of the solar system, is quite point-like. Secondly, as we will see in Section 3.5, an extended body can be modelled as a large collection of particles, and its overall motion is then governed by the very same laws.

To describe the positions of particles over time, we use a *reference frame*, which consists of a choice of origin and some coordinate system which can be used to consistently describe position.

For a particle with position vector $\mathbf{x}(t)$, a displacement vector from the origin of our reference frame, we then define the following quantities.

- *Velocity.* The velocity $\mathbf{v}$ of the particle is $\mathbf{v} = \dot{\mathbf{x}} = d\mathbf{x}(t)/dt$. This vector is always tangent to the particle's trajectory. The speed is then $v = |\mathbf{v}|$.
- *Acceleration.* The acceleration $\mathbf{a}$ of a particle is $\mathbf{a} = \ddot{\mathbf{x}} = d^2\mathbf{x}(t)/dt^2$.
- *Momentum.* The momentum $\mathbf{p}$ is then $\mathbf{p} = m\mathbf{v} = m\dot{\mathbf{x}}$.

Throughout these notes, an overdot will always denote a derivative with respect to time.

§3.1.2 Newton's Laws

The whole of Newtonian mechanics rests on three laws, which we will treat as axioms of our theory.

- Law 3.1.2 (Newton's Laws of Motion)**
1. *Galileo's Law of Inertia.* There exist **inertial frames**, reference frames in which particles travel at constant velocity when the force acting on them vanishes.
 2. *Newton's Second Law.* If a force $\mathbf{F}$ acts on a particle with momentum $\mathbf{p}$, then $d\mathbf{p}/dt = \mathbf{F}$.
 3. *Newton's Third Law.* Every action has an equal and opposite reaction: if a particle A exerts a force $\mathbf{F}$ on a particle B , then B exerts a force $-\mathbf{F}$ on A .

All of these are tied together with the *principle of Galilean relativity*, which is that the laws of physics are the same in every inertial reference frame.

A few comments are in order. The first law looks at a glance like it is just a special case of the second (if $\mathbf{F} = \mathbf{0}$ then $\mathbf{p}$ is constant) – its actual content is the *existence* of inertial frames, which is what makes the second law meaningful at all. A frame fixed to a spinning roundabout is not inertial: a particle with no force on it will appear to accelerate. Throughout the Newtonian part of this course we will work in inertial frames unless stated otherwise (and we will study what happens in rotating, non-inertial frames in Section 3.4).

For a particle whose mass is constant, the second law reads

$$m\ddot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \dot{\mathbf{x}}, t),$$

which is a second order ordinary differential equation for the trajectory $\mathbf{x}(t)$. Assuming the force is well behaved, such an equation has a unique solution once we specify the initial position $\mathbf{x}(t_0)$ and initial velocity $\dot{\mathbf{x}}(t_0)$. This is worth dwelling on – it means Newtonian mechanics is *deterministic*: the position and velocity of a particle right now determine its entire future (and past) motion. Solving dynamics problems is therefore really the art of solving differential equations.

§3.1.3 Galilean Transformations

A natural question is how one can transform between different inertial frames. That is, under what transformations do the laws of physics remain unchanged. It is straightforward to see that if we have some $(\mathbf{x}, t)$ in an inertial frame $\mathcal{F}$, then we can construct another inertial frame $\mathcal{F}'$ by combining:

- *Translations.* $\mathbf{x}' = \mathbf{x} + \mathbf{a}$ for a constant vector $\mathbf{a}$, and $t' = t + t_0$ for a constant t_0 .

- *Rotations and Reflections.* $\mathbf{x}' = R\mathbf{x}$, where R is some orthogonal matrix.
- *Boosts.* $\mathbf{x}' = \mathbf{x} + \mathbf{v}t$, for a constant velocity vector $\mathbf{v}$.

This set of transformations forms the *Galilean group*, the group of symmetries of Newton's laws. Indeed, each of these transformations maps straight-line motion to straight-line motion, so each maps inertial frames to inertial frames, and the general element of the group is a composition of the above types.

The principle of Galilean relativity can then be restated as: *the laws of physics are invariant under Galilean transformations*. Note that any Galilean transformation has $\ddot{\mathbf{x}}' = R\ddot{\mathbf{x}}$, so a particle's acceleration is the same in all inertial frames (up to the fixed rotation of the axes). Since Newton's second law involves only acceleration, and not absolute position or absolute velocity, it is compatible with this principle – there is no mechanical experiment that can detect a state of 'absolute rest'. A passenger in a smoothly moving train can pour their tea exactly as if they were at rest.

When working with Newton's laws, we assume that there is some 'absolute time', with the times t, t' in two inertial frames differing by at most a constant¹. All inertial observers therefore agree on the time interval between two events. This seems so obvious as to not be worth stating, but it is really a physical assumption – and we will see in Section 3.7 that it is *false* when speeds involved approach the speed of light.

Example 3.1.3 (Projectiles on a Train)

Let's check covariance concretely. In the frame $\mathcal{F}$ of the ground, a projectile obeys $m\ddot{\mathbf{x}} = m\mathbf{g}$, giving parabolic trajectories. Now watch from the frame $\mathcal{F}'$ of a train moving at constant velocity $\mathbf{v}$, so $\mathbf{x}' = \mathbf{x} - \mathbf{v}t$. Then

$$\ddot{\mathbf{x}}' = \ddot{\mathbf{x}} = \mathbf{g},$$

so the equation of motion takes *exactly the same form* in the train frame, and the trajectory is again a parabola – just one with different initial conditions (the initial velocity is shifted by $-\mathbf{v}$). A ball tossed straight up on a smoothly moving train lands back in your hand, not behind you; no experiment with projectiles can reveal the train's motion. Contrast an *accelerating* train: then $\ddot{\mathbf{x}}' = \mathbf{g} - \mathbf{a}$, the form of the law changes, and the tossed ball drifts backwards – accelerated frames are detectably not inertial.

§3.1.4 Examples of Forces

Newton's laws tell us how a particle responds to a force, but they are silent on what forces actually arise in nature – that information must be supplied separately, by *force laws*. At the fundamental level, remarkably few are needed: gravity and electromagnetism account for essentially everything we will meet in this course (the strong and weak nuclear forces operate only at subatomic ranges). The everyday 'contact' forces – tension in strings, normal reactions from surfaces, friction – are all electromagnetic in origin, arising from the interactions of vast numbers of atoms, and we model them phenomenologically rather than from first principles.

The *Lorentz force* describes the force on a particle moving in an electromagnetic field. If a particle has position $\mathbf{x}$ and charge q , and it moves in an electric field $\mathbf{E}$ and magnetic

¹Once we fix the units, of course.

field $\mathbf{B}$, then the force on the particle is given by

$$\mathbf{F} = q(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}).$$

The *gravitational force* describes the force due to gravity on a particle. If two particles have masses m_1 and m_2 and positions $\mathbf{x}_1$ and $\mathbf{x}_2$, then the force of gravity between them is given by

$$\mathbf{F}_1 = -\frac{Gm_1m_2}{|\mathbf{x}_1 - \mathbf{x}_2|^3}(\mathbf{x}_1 - \mathbf{x}_2) = -\mathbf{F}_2,$$

where G is the **gravitational constant**.

§3.1.5 Potentials in One Dimension

Roughly, the framework of Newtonian mechanics takes in particles and forces, and from that we get the dynamics at play. However, in many cases we think of forces as coming from a more primitive idea, one of *potentials*. We will work in one dimension for the time being.

Definition 3.1.4 (Potential Energy)

For a force field $F(x)$, depending only on position, we define the **potential energy** to be a function $V(x)$ such that

$$F = -\frac{dV}{dx},$$

so $V = -\int F dx$, defined up to a constant.

With this, Newton's second law becomes $\dot{p} = -dV/dx$. The arbitrary constant is harmless: only differences in potential energy will ever matter physically, so we may fix the constant however is convenient.

With forces in one dimension depending on position only, we have a naturally conserved quantity. We define the *kinetic energy* of a particle to be $T = \frac{1}{2}m\dot{x}^2$.

Proposition 3.1.5 (Conservation of Energy)

Consider a particle of mass m moving under a force $F(x) = -dV/dx$. Then the **total energy**

$$E = T + V = \frac{1}{2}m\dot{x}^2 + V(x)$$

is conserved, that is, $\dot{E} = 0$.

Proof. Differentiating with respect to time, using the chain rule and then Newton's second law,

$$\frac{dE}{dt} = m\dot{x}\ddot{x} + \frac{dV}{dx}\dot{x} = \dot{x}(m\ddot{x} - F(x)) = 0,$$

as required. □

It is important in this proposition that F depends only on position – if F depends on, say, velocity (as friction does), the argument breaks down and energy is typically not conserved.

Remark. A theme worth watching throughout this course: conserved quantities are systematically tied to *symmetries*. Energy conservation will hold when the physics is unchanged by translation in time; momentum conservation when it is unchanged by translation in space; angular momentum conservation when it is unchanged by rotation. This correspondence is made precise by *Noether's theorem*, which appears properly in next year's 'Variational Principles' course – but you can already spot the pattern in every conservation law we derive.

One nice thing about having such a conserved quantity is that

$$E = \frac{1}{2}m\dot{x}^2 + V(x) \implies \dot{x} = \pm\sqrt{\frac{2}{m}(E - V(x))},$$

which gives us a *first order* differential equation for position, which we can (at least in principle) solve by separating variables. The conservation of energy has done one integration of Newton's second law for us. Let's see this in action.

Example 3.1.6 (The Harmonic Oscillator)

The **harmonic oscillator** is the system with quadratic potential

$$V(x) = \frac{1}{2}kx^2,$$

which describes, for example, a particle attached to a spring of stiffness k : the force $F = -dV/dx = -kx$ is a restoring force proportional to the displacement (this is 'Hooke's law'). The equation of motion

$$m\ddot{x} = -kx$$

is linear, and has general solution

$$x(t) = A \cos \omega t + B \sin \omega t, \quad \text{where } \omega = \sqrt{k/m}.$$

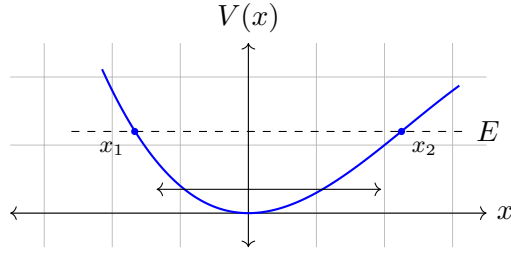
The motion is periodic with *angular frequency* ω , that is, with period $\tau = 2\pi/\omega$. Note that the period is independent of the amplitude of the oscillation. Substituting the solution back in, the energy is

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = \frac{1}{2}k(A^2 + B^2),$$

which is indeed constant. Energy sloshes back and forth between kinetic and potential, twice per period.

The reason the harmonic oscillator is *the* single most important system in physics is that it approximates almost everything, as we are about to see.

We can also get a lot of qualitative information about the motion of a particle by sketching the graph of the potential V against position. Since $T = E - V(x) \geq 0$, the particle can only ever be found in regions where $V(x) \leq E$, and it is instantaneously at rest where $V(x) = E$. Thinking of the graph of V as a landscape in which a ball rolls with total energy E gives entirely correct intuition: a particle with energy E trapped between two points where $V = E$ oscillates back and forth between them, while a particle that never meets a point with $V = E$ runs away to infinity.



Points where the force vanishes are of particular interest.

Definition 3.1.7 (Equilibrium Point)

A point x_0 is an **equilibrium point** if $V'(x_0) = 0$, so that a particle placed at rest at x_0 remains there for all time.

To understand the behaviour *near* an equilibrium point, we Taylor expand the potential about x_0 . Using $V'(x_0) = 0$,

$$V(x) \approx V(x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \cdots,$$

and so the equation of motion is approximately

$$m\ddot{x} \approx -V''(x_0)(x - x_0).$$

There are two cases worth distinguishing.

- (i) If $V''(x_0) > 0$, so x_0 is a local minimum of the potential, then this is exactly the harmonic oscillator equation: the particle oscillates about x_0 with angular frequency

$$\omega = \sqrt{\frac{V''(x_0)}{m}}.$$

We call such an equilibrium point *stable*, and these oscillations *small oscillations* about equilibrium.

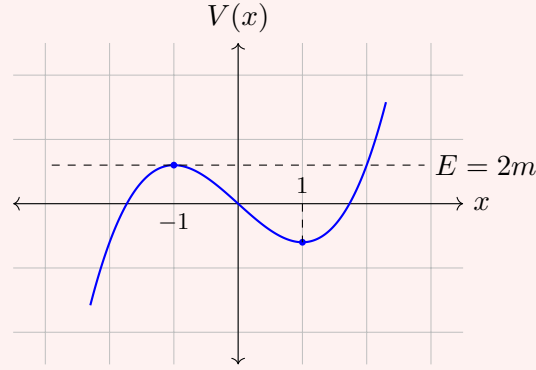
- (ii) If $V''(x_0) < 0$, so x_0 is a local maximum, then the solutions are $e^{\pm\gamma t}$ with $\gamma = \sqrt{-V''(x_0)/m}$, and a generic small displacement grows exponentially. The equilibrium is *unstable* – think of a ball balanced on the top of a hill.

Example 3.1.8 (Motion in a Cubic Potential)

Let's carry out this programme for a particle of mass m in the potential

$$V(x) = m(x^3 - 3x),$$

sketching-first. We have $V'(x) = 3m(x^2 - 1)$, so the equilibria are at $x = \pm 1$, and $V''(x) = 6mx$ tells us that $x = 1$ (where $V = -2m$) is a stable local minimum while $x = -1$ (where $V = 2m$) is an unstable local maximum. The graph falls away to $-\infty$ on the far left.



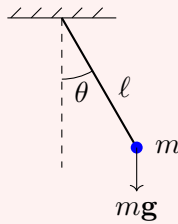
Now release the particle from rest at various starting points and read off the motion from the graph:

- (i) Released near $x = 1$, the particle oscillates in the potential well about the stable equilibrium; for small displacements the angular frequency is $\omega = \sqrt{V''(1)/m} = \sqrt{6}$.
- (ii) Released anywhere with total energy $E < 2m$ and to the right of the potential hill, the particle remains trapped in the well, oscillating between the two solutions of $V(x) = E$ flanking $x = 1$.
- (iii) Released with $E > 2m$, the particle clears the hill at $x = -1$ and escapes to $x \rightarrow -\infty$, never to return.
- (iv) The critical case $E = 2m$ is the interesting one: a particle heading leftward towards the hilltop slows down more and more, *approaching* the unstable equilibrium $x = -1$ but never reaching it in finite time (near the top, $\dot{x} \propto (x+1)$, giving exponential approach). The hilltop is reached only ‘after infinite time’.

All of this qualitative information came from a sketch and no integration whatsoever.

Example 3.1.9 (The Pendulum)

A pendulum consists of a mass m attached to a light rod of length ℓ , swinging from a fixed pivot.



If θ is the angle from the downward vertical, the equation of motion is (as can be derived by resolving forces, or by using the energy below)

$$m\ell\ddot{\theta} = -mg \sin \theta.$$

This is of our potential form with generalised coordinate $x = \ell\theta$, with energy

$$E = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell \cos \theta.$$

The equilibria are $\theta = 0$ (stable) and $\theta = \pi$ (unstable, the pendulum balanced upside-down). Near $\theta = 0$ we have $\sin \theta \approx \theta$, and we recover a harmonic oscillator with angular frequency $\omega = \sqrt{g/\ell}$, so the period of small oscillations is

$$\tau = 2\pi\sqrt{\frac{\ell}{g}},$$

independent of the amplitude and of the mass. For larger swings the period does depend on the amplitude: using conservation of energy, a pendulum released from rest at angle θ_0 has

$$\dot{\theta} = \pm \sqrt{\frac{2g}{\ell}(\cos \theta - \cos \theta_0)},$$

and separating variables gives the period as an integral,

$$\tau = 4\sqrt{\frac{\ell}{2g}} \int_0^{\theta_0} \frac{d\theta}{\sqrt{\cos \theta - \cos \theta_0}},$$

which cannot be evaluated in elementary terms (it is an *elliptic integral*), but which can happily be studied asymptotically or numerically.

§3.1.6 Potentials in Three Dimensions

The concept of potentials generalises into three dimensions, but there is some groundwork needed to be done first. We first will need to redefine kinetic energy to make sense in multiple dimensions. This is done naturally with

$$T = \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}}.$$

Differentiating and using Newton's second law, we get $\dot{T} = m\ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} = \mathbf{F} \cdot \dot{\mathbf{x}}$, so the kinetic energy changes at a rate given by the component of force along the direction of motion. In particular, a force which is always perpendicular to the velocity does not change the particle's speed.

Now consider a force field $\mathbf{F}(\mathbf{x})$, depending only on position. If the force $\mathbf{F}$ acts on a particle, moving it from $\mathbf{x}(t_1)$ to $\mathbf{x}(t_2)$ along a trajectory $\mathcal{C}$, we define the *work done* W to be

$$W = \int_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{x} = \int_{t_1}^{t_2} \mathbf{F} \cdot \frac{d\mathbf{x}}{dt} dt.$$

Note that if $\mathbf{F} = m\ddot{\mathbf{x}}$ (as in there is no change in mass), then integrating the expression for $\dot{T}$ above shows that the work done is

$$W = m \int_{t_1}^{t_2} \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} dt = T(t_2) - T(t_1),$$

so the work done by the force is the change in kinetic energy of the particle.

In one dimension, every force $F(x)$ had a potential – we just integrated. In three dimensions this is no longer true: the work done by a general force field can depend on the *path* taken between the endpoints, in which case there is no way to consistently assign a potential energy to positions. The forces for which this does work get a special name.

Definition 3.1.10 (Conservative Force)

A force field $\mathbf{F}(\mathbf{x})$ is **conservative** if it can be written as

$$\mathbf{F} = -\nabla V$$

for some scalar potential $V(\mathbf{x})$, or equivalently^a if the work done by $\mathbf{F}$ along a trajectory depends only on the endpoints of the trajectory.

^aThe equivalence of these conditions (on suitable domains), along with the equivalent statement $\nabla \times \mathbf{F} = \mathbf{0}$, is established properly in the ‘Vector Calculus’ course.

Proposition 3.1.11

If $\mathbf{F} = -\nabla V$, then the total energy

$$E = T + V(\mathbf{x}) = \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + V(\mathbf{x})$$

is conserved.

Proof. We differentiate, using the chain rule $\frac{d}{dt}V(\mathbf{x}(t)) = \nabla V \cdot \dot{\mathbf{x}}$:

$$\frac{dE}{dt} = m\ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} + \nabla V \cdot \dot{\mathbf{x}} = (m\ddot{\mathbf{x}} - \mathbf{F}) \cdot \dot{\mathbf{x}} = 0. \quad \square$$

Conversely, if a force depending only on position conserves energy for every trajectory, it must be conservative in the above sense. Happily, the fundamental forces of nature that we will meet – gravity and (static) electromagnetism – are conservative.

§3.1.7 Gravity

Gravity is an example of a conserved force. For a particle of mass m , under the influence of gravity from a particle of mass M at the origin, we have the potential energy

$$V(r) = -\frac{GMm}{r}.$$

The force of gravity is then given by

$$\mathbf{F} = -\nabla V = -\frac{GMm}{r^2}\hat{\mathbf{r}},$$

with $\hat{\mathbf{r}}$ in the direction of the particle. Note that this force points towards the origin. It is sometimes convenient to remove the dependence on the mass of the particle m , which we do by defining the *gravitational potential* $\phi(r) = -GM/r$, with $V = \phi m$.

If there are many particles say of masses M_i and positions $\mathbf{r}_i$, the gravitational potential just adds, with

$$\begin{aligned} \phi(\mathbf{r}) &= -G \sum_i \frac{M_i}{|\mathbf{r} - \mathbf{r}_i|} \\ \Rightarrow \mathbf{F} &= -Gm \sum_i \frac{M_i}{|\mathbf{r} - \mathbf{r}_i|^3} (\mathbf{r} - \mathbf{r}_i). \end{aligned}$$

A remarkable fact (which we will take on trust – it is a pleasant exercise in vector calculus) is that the external gravitational potential of a spherically symmetric body of total mass M is exactly that of a point mass M at its centre. This is what lets us treat planets and stars as particles when studying orbits.

Example 3.1.12 (Gravity Near the Earth's Surface)

Consider a particle of mass m at height $z \ll R$ above the surface of the Earth, which has mass M and radius R . The potential energy is

$$V = -\frac{GMm}{R+z} = -\frac{GMm}{R} \left(1 - \frac{z}{R} + \dots\right) \approx \text{const} + mgz,$$

where we define

$$g = \frac{GM}{R^2} \approx 9.8 \text{ m s}^{-2}.$$

So near the surface of the Earth we may use the familiar uniform-gravity potential $V = mgz$, and every particle falls with the same acceleration g .

There is a genuinely deep assumption hiding in all of this. The mass appearing in Newton's second law (the *inertial mass*, measuring reluctance to accelerate) and the mass appearing in the law of gravitation (the *gravitational mass*, playing the role of a 'gravitational charge') have no a priori reason to be equal – yet experimentally they agree to extraordinary precision, which is why all bodies fall at the same rate. This equality, the *equivalence principle*, later became the starting point for Einstein's theory of general relativity.

§3.1.8 Electromagnetism

If we have time-independent electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$, then $\mathbf{E} = -\nabla\phi$ for some $\phi(\mathbf{x})$, the *electric potential*. We have conservation of energy for time-independent fields, with the total energy²

$$E = \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + q\phi(\mathbf{x}).$$

Just as objects with mass both feel the effect of and produce gravitational force, the same is true of objects with charge. At the origin, a particle of charge Q creates an electric field as

$$\mathbf{E} = -\nabla \left(\frac{Q}{4\pi\epsilon_0 r} \right) = \frac{Q}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2},$$

where $r^2 = \mathbf{x} \cdot \mathbf{x}$. The force between objects of charge Q and q is $\mathbf{F} = q\mathbf{E}$, or

$$\mathbf{F} = \frac{Qq}{4\pi\epsilon_0} \frac{\hat{\mathbf{r}}}{r^2}.$$

This is the *Coulomb force*, and has the same form as gravity – an inverse square law – with one important difference: charges come in both signs, so the Coulomb force can be attractive (unlike charges) or repulsive (like charges).

The magnetic part of the Lorentz force is stranger, since it depends on velocity. Note however that the magnetic force $q\dot{\mathbf{x}} \times \mathbf{B}$ is always perpendicular to the velocity, so it does no work – a magnetic field can bend a particle's path, but never change its speed.

²Again, this can be checked by differentiating.

Example 3.1.13 (Motion in a Uniform Magnetic Field)

Consider a particle of charge q and mass m in a uniform magnetic field $\mathbf{B} = B\mathbf{e}_z$, with no electric field. The equation of motion $m\ddot{\mathbf{x}} = q\dot{\mathbf{x}} \times \mathbf{B}$ has components

$$m\ddot{x} = qB\dot{y}, \quad m\ddot{y} = -qB\dot{x}, \quad m\ddot{z} = 0.$$

The z motion is uniform. For the rest, write $\omega = qB/m$ and combine the first two equations using the complex variable $\zeta = x + iy$:

$$\ddot{\zeta} = \ddot{x} + i\ddot{y} = \omega\dot{y} - i\omega\dot{x} = -i\omega\dot{\zeta},$$

so $\dot{\zeta} = \dot{\zeta}_0 e^{-i\omega t}$ and hence $\zeta = \zeta_0 + \frac{i\dot{\zeta}_0}{\omega} e^{-i\omega t}$. In the (x, y) plane the particle therefore moves in a circle, traversed with constant angular speed

$$\omega = \frac{qB}{m},$$

called the **gyrofrequency** (or cyclotron frequency). Combined with the uniform drift in z , the general trajectory is a helix winding around the magnetic field lines. Note that, as promised, the speed of the particle never changes.

§3.1.9 Friction

In many physical scenarios we deal with forces that do not come from potentials. Friction is one such force, and it is a force that is used to approximate the complex relationship between objects in contact.

One common form of friction is *dry friction*, which acts when solid bodies are in contact. If N is the magnitude of the normal reaction force between the surfaces, then the frictional force acts tangentially to the surfaces, opposing relative motion, with magnitude

$$|\mathbf{F}| = \mu N$$

when the surfaces slide over each other, where μ is the *coefficient of friction* (which depends on the materials involved). When the surfaces are not sliding, the friction takes whatever magnitude is required to prevent slipping, up to a maximum of roughly μN . Note that friction opposes relative motion – it never causes it.

Another common type of friction is *fluid drag*, the resistance felt by a body moving through a fluid. Fluid drag is typically modelled as either

- *linear* drag, with $\mathbf{F} = -\gamma\mathbf{v}$, suitable for small, slow-moving objects in viscous fluids (for a sphere of radius R , Stokes' law gives $\gamma = 6\pi\eta R$ where η is the fluid's viscosity); or
- *quadratic* drag, with $\mathbf{F} = -\gamma|\mathbf{v}|\mathbf{v}$, suitable for larger, fast-moving objects in less viscous fluids such as air, where the resistance comes mostly from the momentum transferred to the fluid pushed out of the way.

In either case, the drag force does negative work $\mathbf{F} \cdot \mathbf{v} < 0$, so the particle continually loses energy to the fluid (where it ends up as heat).

Let's now put several of our forces together in a classic worked example – a projectile moving under gravity, experiencing linear drag.

Example 3.1.14 (Projectile with Linear Drag)

A projectile of mass m is launched from the origin with initial velocity $\mathbf{u}$, moving under uniform gravity and linear air resistance. The equation of motion is

$$m \frac{d\mathbf{v}}{dt} = m\mathbf{g} - \gamma\mathbf{v}.$$

This is a linear first order equation for $\mathbf{v}$, which we can solve with an integrating factor $e^{\gamma t/m}$:

$$\frac{d}{dt} (e^{\gamma t/m} \mathbf{v}) = e^{\gamma t/m} \mathbf{g} \implies \mathbf{v}(t) = \frac{m\mathbf{g}}{\gamma} + \left(\mathbf{u} - \frac{m\mathbf{g}}{\gamma} \right) e^{-\gamma t/m}.$$

As $t \rightarrow \infty$, the velocity tends to the **terminal velocity** $\mathbf{v}_\infty = m\mathbf{g}/\gamma$, in which the drag exactly balances gravity – and it does so exponentially quickly, on the timescale m/γ , regardless of the initial conditions.

Integrating once more gives the trajectory:

$$\mathbf{x}(t) = \frac{m\mathbf{g}}{\gamma} t + \frac{m}{\gamma} \left(\mathbf{u} - \frac{m\mathbf{g}}{\gamma} \right) (1 - e^{-\gamma t/m}).$$

Taking $\mathbf{g} = -g\mathbf{e}_z$ and $\mathbf{u} = (u \cos \theta, 0, u \sin \theta)$, the components read

$$x(t) = \frac{mu \cos \theta}{\gamma} (1 - e^{-\gamma t/m}), \quad z(t) = -\frac{mg}{\gamma} t + \frac{m}{\gamma} \left(u \sin \theta + \frac{mg}{\gamma} \right) (1 - e^{-\gamma t/m}).$$

Two sanity checks are worth performing. First, as $\gamma \rightarrow 0$ (expanding the exponentials to second order) we recover the familiar drag-free parabola $x = ut \cos \theta$, $z = ut \sin \theta - \frac{1}{2}gt^2$. Secondly, note that x is bounded: $x < mu \cos \theta/\gamma$ always, so unlike the parabola, the trajectory has a vertical asymptote – the projectile ends up falling essentially straight down at terminal velocity.

Example 3.1.15 (Falling Raindrop with Quadratic Drag)

For a raindrop falling vertically under quadratic drag, taking v to be the downward speed,

$$m \frac{dv}{dt} = mg - \gamma v^2 = \gamma (v_\infty^2 - v^2), \quad \text{where } v_\infty = \sqrt{\frac{mg}{\gamma}}.$$

Separating variables and using partial fractions,

$$\int \frac{dv}{v_\infty^2 - v^2} = \frac{\gamma}{m} t + \text{const} \implies v(t) = v_\infty \tanh \left(\frac{gt}{v_\infty} \right)$$

for a drop released from rest. Again the speed approaches the terminal velocity v_∞ , this time with the characteristic tanh profile.

§3.2 Dimensional Analysis

Before going any further with dynamics proper, we pause for a very useful tool. Physical quantities are not pure numbers – each carries a *dimension*, which is roughly the kind of thing that its units measure. Keeping track of dimensions is a powerful (and almost

free) source of information about physical problems.

§3.2.1 Dimensions of Physical Quantities

For the problems of dynamics, three basic dimensions suffice:

- length, L ;
- mass, M ;
- time, T .

The dimensions of any other physical quantity X in mechanics, written $[X]$, can then be expressed uniquely as a product of powers of L , M and T . For example, working directly from the definitions and from equations we know to be true,

$$\begin{aligned} [\text{area}] &= L^2, & [\text{velocity}] &= LT^{-1}, & [\text{acceleration}] &= LT^{-2}, \\ [\text{force}] &= MLT^{-2}, & [\text{energy}] &= ML^2T^{-2}, & [\text{angular momentum}] &= ML^2T^{-1}, \end{aligned}$$

where, say, the dimensions of force are read off from $F = ma$ and the dimensions of energy from $E = \frac{1}{2}mv^2$. Dimensionful physical constants are treated the same way: from $F = GMm/r^2$ we find $[G] = M^{-1}L^3T^{-2}$.

The key principle is that of *dimensional consistency*: in any meaningful physical equation, both sides must have the same dimensions, and we may only add or subtract terms of equal dimensions. Adding a length to an area is as meaningless as adding a vector to a matrix. Moreover, arguments of functions like $\exp$, $\sin$ or $\log$ must be *dimensionless*: if x had a dimension, the series $e^x = 1 + x + \frac{1}{2}x^2 + \dots$ would be a sum of terms of different dimensions. (It is a good check on our projectile-with-drag example above that $\gamma t/m$ is dimensionless, since $[\gamma] = [F]/[v] = MT^{-1}$.)

Units are conventional reference amounts of dimensional quantities – in the SI system, the metre, kilogram and second for L , M and T respectively. The choice is arbitrary and historical, and no law of physics can depend on it; dimensional consistency is exactly what guarantees that our equations hold in any consistent system of units.

§3.2.2 Scaling

These restrictions are not a hindrance – they are a gift. If we believe that some quantity Y depends only on the quantities X_1, X_2, X_3 , and the dependence is a power law

$$Y = CX_1^{p_1}X_2^{p_2}X_3^{p_3}$$

with C a dimensionless constant, then taking dimensions of both sides gives three linear equations for the three exponents p_1, p_2, p_3 (one for each of L, M, T). Provided the dimensions of X_1, X_2, X_3 are independent, these determine the exponents uniquely – so we learn the entire form of the physical law, up to the constant C , without solving any dynamics at all.

If there are more than three quantities involved, their dimensions can no longer be independent, and we can form dimensionless combinations $\lambda_1, \lambda_2, \dots$ of the variables. The most general relationship then takes the form

$$Y = f(\lambda_1, \lambda_2, \dots) X_1^{p_1} X_2^{p_2} X_3^{p_3},$$

where f is an undetermined dimensionless function³. Dimensional analysis cannot pin down f – but knowing the scaling structure is often most of the battle.

Example 3.2.1 (The Simple Pendulum, by Dimensional Analysis)

Consider again the pendulum: a mass m on a light rod of length ℓ , released from rest a horizontal distance d from the vertical through the pivot, swinging under gravity g . What can the period τ depend on? Our candidate variables have dimensions

$$[m] = M, \quad [\ell] = L, \quad [g] = LT^{-2}, \quad [d] = L,$$

and $[\tau] = T$. The first three have independent dimensions, and from the fourth we can form the dimensionless group d/ℓ . So the relationship must be of the form

$$\tau = f\left(\frac{d}{\ell}\right) m^{p_1} \ell^{p_2} g^{p_3},$$

and balancing dimensions,

$$T = M^{p_1} L^{p_2+p_3} T^{-2p_3} \implies p_1 = 0, \quad p_3 = -\frac{1}{2}, \quad p_2 = \frac{1}{2}.$$

Hence

$$\tau = f\left(\frac{d}{\ell}\right) \sqrt{\frac{\ell}{g}}.$$

Without solving any differential equation, we have learnt that the period is independent of the mass, and scales as $\sqrt{\ell/g}$: if ℓ and d are both quadrupled, the period doubles. Comparing with our earlier small-oscillation result $\tau = 2\pi\sqrt{\ell/g}$, we see that $f(x) \rightarrow 2\pi$ as $x \rightarrow 0$.

Example 3.2.2 (The Speed of Water Waves)

How fast do waves travel on deep water? Plausibly the wave speed v depends on the wavelength Λ , the water density ρ , and gravity g (which provides the restoring force). These three have independent dimensions, so we look for

$$v = C \Lambda^{p_1} \rho^{p_2} g^{p_3},$$

and balancing dimensions,

$$LT^{-1} = L^{p_1} (ML^{-3})^{p_2} (LT^{-2})^{p_3} \implies p_2 = 0, \quad p_3 = \frac{1}{2}, \quad p_1 = \frac{1}{2}.$$

So

$$v = C \sqrt{g\Lambda}.$$

The density drops out entirely, and long waves travel faster than short ones – which is why the long, slow swell from a distant storm arrives at the beach before the choppy short waves do. (A full fluid-dynamical calculation gives $C = 1/\sqrt{2\pi}$.)

³This is a special case of what is formally known as the *Buckingham Pi theorem*.

Example 3.2.3 (Taylor's Estimate of a Blast Energy)

A famous (and true) story: in 1950, G. I. Taylor estimated the (then classified) energy yield of the first atomic bomb from declassified photographs of the fireball, using nothing but dimensional analysis. Suppose the radius R of the blast wave at time t after the explosion depends only on t , the energy released E , and the density ρ of the surrounding air:

$$R = CE^{p_1} \rho^{p_2} t^{p_3}.$$

Balancing dimensions with $[E] = ML^2T^{-2}$ and $[\rho] = ML^{-3}$,

$$L = M^{p_1+p_2} L^{2p_1-3p_2} T^{-2p_1+p_3} \implies p_1 = \frac{1}{5}, \quad p_2 = -\frac{1}{5}, \quad p_3 = \frac{2}{5}.$$

So

$$R = C \left(\frac{Et^2}{\rho} \right)^{1/5} \implies E \sim \frac{\rho R^5}{t^2},$$

taking the undetermined constant C to be of order 1. Reading off R at known times t from the photographs gave Taylor an estimate of about 20 kilotons of TNT – embarrassingly close to the classified value.

Dimensional analysis is not magic, and it is worth being honest about its limits: it cannot produce dimensionless constants (the 2π in the pendulum period), it cannot help if we have misidentified which quantities are relevant, and when there are many dimensionless groups it constrains the answer only loosely. But as a first line of attack on any physical problem – and as an ever-present check on the sanity of our formulae – it is hard to beat.

§3.3 Orbits

We now turn to one of the triumphs of Newtonian mechanics – the description of the motion of planets, comets, moons and satellites. The key structural feature of all of these problems is that the force always points towards a fixed centre, and we will see that this alone constrains the motion enormously.

§3.3.1 Central Forces

If a force is given by a potential V , depending only on the distance from the origin, then

$$\mathbf{F}(\mathbf{x}) = -\nabla V(|\mathbf{x}|) = -\frac{dV}{dr} \hat{\mathbf{r}},$$

where $\hat{\mathbf{r}} = \mathbf{x}/|\mathbf{x}|$. Such forces are known as **central forces** – the force is always parallel to the position vector. Gravity due to a mass fixed at the origin and the Coulomb force due to a fixed charge are the standard examples.

§3.3.2 Angular Momentum

An important result when dealing with central forces is the existence of another conserved quantity, *angular momentum*.

Definition 3.3.1 (Angular Momentum)

The **angular momentum** of a particle about the origin is

$$\mathbf{L} = \mathbf{x} \times \mathbf{p} = m\mathbf{x} \times \dot{\mathbf{x}}.$$

Note that, unlike momentum, angular momentum depends on the choice of origin – it measures something like ‘the amount of rotation about the origin’. Let’s see how it changes in time. With a general force $\mathbf{F}$,

$$\frac{d\mathbf{L}}{dt} = m\dot{\mathbf{x}} \times \dot{\mathbf{x}} + m\mathbf{x} \times \ddot{\mathbf{x}} = \mathbf{x} \times \mathbf{F} = \boldsymbol{\tau},$$

a quantity named *torque* (about the origin). This is the rotational analogue of Newton’s second law: torque is the rate of change of angular momentum.

Proposition 3.3.2

For motion in a central force, angular momentum is conserved.

Proof. If the force is central, then $\mathbf{F} \parallel \mathbf{x}$, so $\boldsymbol{\tau} = \mathbf{x} \times \mathbf{F} = \mathbf{0}$, and thus $\dot{\mathbf{L}} = \mathbf{0}$. $\square$

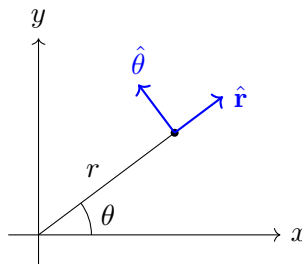
This conservation law has an immediate geometric consequence. Since $\mathbf{L} = m\mathbf{x} \times \dot{\mathbf{x}}$ is a constant vector and $\mathbf{x} \cdot \mathbf{L} = 0$ always, the position vector is confined to the plane through the origin perpendicular to $\mathbf{L}$.⁴ So *motion in a central force is planar*, and a problem that started in three dimensions is now a problem in two. We will shortly reduce it to one.

§3.3.3 Polar Coordinates in the Plane

Since the motion is planar, it makes sense to use coordinates adapted to the geometry: plane polar coordinates (r, θ) , with $x = r \cos \theta$ and $y = r \sin \theta$. We use the orthonormal basis vectors

$$\hat{\mathbf{r}} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \hat{\boldsymbol{\theta}} = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix},$$

pointing in the directions of increasing r and θ respectively.



The subtlety, and the entire content of what follows, is that these basis vectors *depend on the position of the particle*, so they change as the particle moves. Differentiating the expressions above with respect to θ ,

$$\frac{d\hat{\mathbf{r}}}{d\theta} = \hat{\boldsymbol{\theta}}, \quad \frac{d\hat{\boldsymbol{\theta}}}{d\theta} = -\hat{\mathbf{r}}.$$

⁴If $\mathbf{L} = \mathbf{0}$, then $\mathbf{x}$ and $\dot{\mathbf{x}}$ are always parallel, and the motion takes place on a straight line through the origin – which is certainly also planar.

Proposition 3.3.3 (Velocity and Acceleration in Polar Coordinates)

For a particle with position $\mathbf{x} = r\hat{\mathbf{r}}$,

$$\begin{aligned}\dot{\mathbf{x}} &= \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta}, \\ \ddot{\mathbf{x}} &= (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}.\end{aligned}$$

Proof. By the chain rule, $\dot{\hat{\mathbf{r}}} = \dot{\theta}\hat{\theta}$ and $\dot{\hat{\theta}} = -\dot{\theta}\hat{\mathbf{r}}$. So differentiating $\mathbf{x} = r\hat{\mathbf{r}}$,

$$\dot{\mathbf{x}} = \dot{r}\hat{\mathbf{r}} + r\dot{\hat{\mathbf{r}}} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\theta},$$

and differentiating again,

$$\ddot{\mathbf{x}} = \ddot{r}\hat{\mathbf{r}} + \dot{r}\dot{\hat{\mathbf{r}}} + \dot{r}\dot{\hat{\theta}} + r\ddot{\hat{\theta}} - r\dot{\theta}^2\hat{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}. \quad \square$$

The term $-r\dot{\theta}^2\hat{\mathbf{r}}$ should be familiar: for a particle moving on a *circle* of constant radius r with angular speed $\omega = \dot{\theta}$, the acceleration is $-r\omega^2\hat{\mathbf{r}}$, of magnitude $r\omega^2 = v^2/r$ directed towards the centre – the *centripetal acceleration*. Some force must supply this if a particle is to move in a circle.

Example 3.3.4 (Circular Orbit Speed)

For a satellite in a circular orbit of radius r about the Earth, gravity supplies the centripetal force:

$$\frac{mv^2}{r} = \frac{GMm}{r^2} \implies v = \sqrt{\frac{GM}{r}}.$$

The period is then $\tau = 2\pi r/v = 2\pi r^{3/2}/\sqrt{GM}$ – a first glimpse of Kepler's third law, to which we will return.

§3.3.4 Reduction to One Dimension

Suppose we had a potential $V(r)$, with $\mathbf{F} = -\nabla V = -\frac{dV}{dr}\hat{\mathbf{r}}$. Newton's second law in the plane with plane polar coordinates (r, θ) is (for constant mass)

$$\begin{aligned}m\ddot{\mathbf{x}} &= -\frac{dV}{dr}\hat{\mathbf{r}} \\ \implies m(\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta} &= -\frac{dV}{dr}\hat{\mathbf{r}}.\end{aligned}$$

Then since the components are orthogonal, we can then obtain two separate equations. The first (taking the $\hat{\theta}$ component) is

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0 \implies \frac{1}{r} \frac{d}{dt}(r^2\dot{\theta}) = 0.$$

This is not a surprise but a consistency check: the angular momentum of the system is $\mathbf{L} = \mathbf{x} \times m\dot{\mathbf{x}} = mr^2\dot{\theta}(\hat{\mathbf{r}} \times \hat{\theta})$, and thus $|\mathbf{L}| = mr^2\dot{\theta} = m\ell$, where $\ell = r^2\dot{\theta}$ is the angular momentum per unit mass. So this equation just says that angular momentum is conserved, which we already knew.

The other component then gives us the equation

$$m(\ddot{r} - r\dot{\theta}^2) = -\frac{dV}{dr} \implies m\ddot{r} = -\frac{dV}{dr} + mr\dot{\theta}^2 = -\frac{dV}{dr} + \frac{m\ell^2}{r^3},$$

where in the last step we eliminated $\dot{\theta}$ using $\dot{\theta} = \ell/r^2$. Thus we have reduced our three dimensional problem of a central force into a one dimensional problem involving the radial component only – the price being an extra, angular-momentum-dependent term in the force.

§3.3.5 The Effective Potential

Using our reduced problem, we can ‘forget’ about the θ component, and consider a one dimensional system, under the influence of the *effective potential*

$$V_{\text{eff}}(r) = V(r) + \frac{m\ell^2}{2r^2}.$$

The rightmost term is the *angular momentum barrier*, and it stops the particle from getting too close to the origin: as $r \rightarrow 0$ it blows up like $1/r^2$, which (for potentials less singular than this, such as gravity) dominates and repels the particle.

We can even use this viewpoint in energy conservation, as

$$E = \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} + V(r) = \frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r),$$

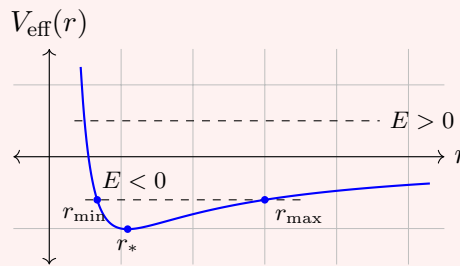
using $|\dot{\mathbf{x}}|^2 = \dot{r}^2 + r^2\dot{\theta}^2$ and $\dot{\theta} = \ell/r^2$. So all our one dimensional technology – sketching the potential, classifying allowed regions by energy, finding equilibria – applies directly to the radial motion.

Example 3.3.5 (Orbits in an Inverse Square Law Potential)

For gravity, $V(r) = -GMm/r$, the effective potential

$$V_{\text{eff}}(r) = -\frac{GMm}{r} + \frac{m\ell^2}{2r^2}$$

tends to $+\infty$ as $r \rightarrow 0$ (the angular momentum barrier wins), and tends to 0^- as $r \rightarrow \infty$, with a single minimum in between at $r_* = \ell^2/GM$.



Reading off the qualitative motion from the graph:

- (i) If $E = V_{\text{eff}}(r_*)$, then $r = r_*$ is constant, and the orbit is a *circle*.
- (ii) If $V_{\text{eff}}(r_*) < E < 0$, the radius oscillates between two values $r_{\min} \leq r \leq r_{\max}$ (the *perihelion* and *aphelion*) – the orbit is bounded.
- (iii) If $E \geq 0$, the particle can escape to $r \rightarrow \infty$ – the orbit is unbounded, and the particle approaches, swings past, and leaves, never to return.

We will see shortly that the bounded orbits are precisely ellipses, and the unbounded ones hyperbolae (or, in the marginal case $E = 0$, parabolae).

Example 3.3.6 (Finding the Turning Points)

The perihelion and aphelion distances can be computed directly from the conserved quantities. At a turning point of the radial motion, $\dot{r} = 0$, so $E = V_{\text{eff}}(r)$:

$$E = -\frac{GMm}{r} + \frac{m\ell^2}{2r^2} \implies Er^2 + GMmr - \frac{m\ell^2}{2} = 0,$$

a quadratic in r . For bound orbits ($E < 0$) it has two positive roots, $r_{\min}$ and $r_{\max}$, and by the standard formulae for the sum and product of roots,

$$r_{\min} + r_{\max} = -\frac{GMm}{E} \quad (= 2a), \quad r_{\min} r_{\max} = -\frac{m\ell^2}{2E} \quad (= b^2),$$

which the reader may enjoy checking against the geometry of the ellipse. For $E \geq 0$ there is a single positive root – the distance of closest approach of an unbound trajectory.

§3.3.6 Stability of Circular Orbits

A circular orbit at radius r_* exists whenever $V'_{\text{eff}}(r_*) = 0$, and just as for one dimensional potentials, the orbit is *stable* (small perturbations of the radius oscillate rather than grow) precisely when r_* is a minimum of the effective potential, that is when $V''_{\text{eff}}(r_*) > 0$. In that case a slightly perturbed orbit has radius oscillating about r_* with angular frequency ω_r given by $m\omega_r^2 = V''_{\text{eff}}(r_*)$.

Example 3.3.7 (Stability for Power Law Forces)

Suppose the force is attractive with $F(r) = -mk/r^n$ for some n , so $V(r) = -\frac{mk}{(n-1)r^{n-1}}$ for $n \neq 1$. Then

$$V_{\text{eff}}(r) = -\frac{mk}{(n-1)r^{n-1}} + \frac{m\ell^2}{2r^2},$$

and circular orbits satisfy $V'_{\text{eff}}(r_*) = 0$, giving $\ell^2 = kr_*^{3-n}$. Computing the second derivative and substituting this in,

$$V''_{\text{eff}}(r_*) = -\frac{nmk}{r_*^{n+1}} + \frac{3m\ell^2}{r_*^4} = \frac{mk}{r_*^{n+1}} (3-n).$$

So circular orbits are stable if and only if $n < 3$. In particular the inverse square law $n = 2$ gives stable circular orbits (reassuring, given that we live on a planet in a nearly circular orbit), while an inverse cube force $n = 3$ or stronger cannot support stable circles – a slight nudge sends the particle spiralling inwards or outwards.

§3.3.7 Escape Velocity

Consider a projectile launched from the surface of a planet of mass M and radius R . Ignoring air resistance, when can it escape the planet's gravity entirely? By conservation

of energy,

$$E = \frac{1}{2}mv^2 - \frac{GMm}{r},$$

and (as we saw from the effective potential) the projectile can reach $r \rightarrow \infty$ precisely when $E \geq 0$. Launching from $r = R$, this requires

$$v \geq v_{\text{esc}} = \sqrt{\frac{2GM}{R}},$$

the **escape velocity** of the planet. For the Earth, $v_{\text{esc}} \approx 11.2 \text{ km s}^{-1}$. Note that v_{esc} is independent of the mass of the projectile, and (perhaps surprisingly) of the direction of launch: energy conservation cares only about speed. A particle launched at exactly v_{esc} reaches infinity with nothing to spare, arriving with zero speed.

§3.3.8 The $u(\theta)$ Equation

We now want the *shape* of a general orbit, that is, r as a function of θ rather than of t . It turns out that the substitution $u = 1/r$ just seems to work really well when dealing with the system we have set up. Using the chain rule with $\dot{\theta} = \ell u^2$, we have

$$\begin{aligned}\dot{r} &= \frac{dr}{d\theta} \dot{\theta} = \frac{dr}{d\theta} \ell = -\ell \frac{du}{d\theta}, \\ \ddot{r} &= \frac{d}{dt} \left(-\ell \frac{du}{d\theta} \right) = -\ell \frac{d^2u}{d\theta^2} \dot{\theta} = -\ell^2 u^2 \frac{d^2u}{d\theta^2}.\end{aligned}$$

Then our equation of motion goes from $m\ddot{r} - m\ell^2/r^3 = F(r)$ to *Binet's equation*:

$$-m\ell^2 u^2 \left(\frac{d^2u}{d\theta^2} + u \right) = F\left(\frac{1}{u}\right).$$

This is generally still nonlinear – but for inverse square laws (such as gravity), where $F(1/u) \propto u^2$, the factor of u^2 on the left cancels, and it becomes a *linear* equation, one we know exactly how to solve. After it is solved, we can get back the time dependence with $\dot{\theta} = \ell u^2$.

§3.3.9 The Kepler Problem

Binet's equation can be used to find the shapes of orbits under an inverse square law. Let

$$V(r) = -\frac{mk}{r}, \quad F(r) = -\frac{mk}{r^2},$$

where $k = GM$ for gravity due to a mass M fixed at the origin. Then Binet's equation is

$$\frac{d^2u}{d\theta^2} + u = \frac{k}{\ell^2}.$$

This is the harmonic oscillator equation (in the variable θ !) with a constant forcing term, so the general solution is

$$u = \frac{k}{\ell^2} + A \cos(\theta - \theta_0).$$

Taking $\theta_0 = 0$ (which we can arrange by rotating our coordinates) and $A \geq 0$, we conclude that for a planet in orbit about the origin, the shape of the orbit is given by

$$r = \frac{p}{1 + e \cos \theta},$$

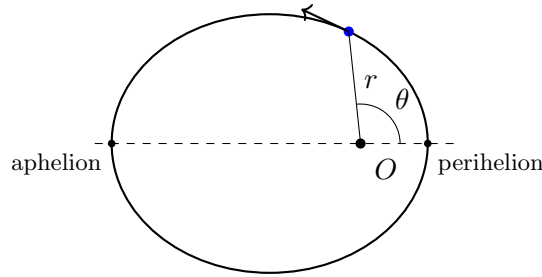
with $p = \ell^2/k$ and $e = A\ell^2/k \geq 0$. This is the polar-coordinate equation of a *conic section* with a focus at the origin, where e is the **eccentricity**. Notice that r is minimised at $\theta = 0$ (the closest approach, or *perihelion* for orbits about the Sun) and, when the orbit is bounded, maximised at $\theta = \pi$ (*aphelion*). The nature of the conic depends on the eccentricity:

- (i) If $e = 0$ then $r = p$ is constant, and the orbit is a *circle*.
- (ii) If $0 < e < 1$, the denominator is always positive, so r is bounded with $\frac{p}{1+e} \leq r \leq \frac{p}{1-e}$, and the orbit is an *ellipse* with the origin at one focus. In Cartesian coordinates centred appropriately, it has semi-major and semi-minor axes

$$a = \frac{p}{1-e^2}, \quad b = \frac{p}{\sqrt{1-e^2}}.$$

- (iii) If $e = 1$, then $r \rightarrow \infty$ as $\theta \rightarrow \pm\pi$: the orbit is an unbounded *parabola*.
- (iv) If $e > 1$, then $r \rightarrow \infty$ as $\theta \rightarrow \pm\alpha$ where $\alpha = \cos^{-1}(1/e) \in (\pi/2, \pi)$: the orbit is one branch of a *hyperbola*, with asymptotes at angles $\pm\alpha$.

The bounded case is the one Kepler cared about – an ellipse with the attracting body at a focus:



Example 3.3.8 (Determining an Orbit from Initial Conditions)

A satellite is at distance r_0 from the Earth's centre, moving with speed v_0 *perpendicular* to its position vector. What orbit does it follow? Since $\dot{\mathbf{x}} \perp \mathbf{x}$, the angular momentum per unit mass is $\ell = r_0 v_0$, and this point is an apse of the orbit (a maximum or minimum of r), so it lies at $\theta = 0$ or $\theta = \pi$ in our standard form. Comparing r_0 with the orbit equation at $\theta = 0$,

$$r_0 = \frac{p}{1+e} = \frac{r_0^2 v_0^2}{GM(1+e)} \implies e = \frac{r_0 v_0^2}{GM} - 1.$$

Reading off the cases:

- (i) If $v_0^2 = GM/r_0$, then $e = 0$ – the circular orbit speed we found before.
- (ii) If $GM/r_0 < v_0^2 < 2GM/r_0$, then $0 < e < 1$: an ellipse with the launch point as perihelion.
- (iii) If $v_0^2 < GM/r_0$, the formula gives $e < 0$, which just means the launch point is the *aphelion* instead (replace $\theta \mapsto \theta + \pi$): the satellite falls inwards on an ellipse.
- (iv) If $v_0^2 \geq 2GM/r_0$, then $e \geq 1$ and the satellite escapes on a parabola or hyper-

bola – and indeed $v_0 = \sqrt{2GM/r_0}$ is precisely the escape velocity from radius r_0 that we found earlier.

§3.3.10 Energy and Eccentricity

Which orbit we get is determined by the energy. Evaluating the conserved energy at perihelion (where $\dot{r} = 0$, $r = p/(1+e)$ and so $\dot{\mathbf{x}} = r\dot{\theta}\hat{\theta}$ has magnitude ℓ/r),

$$E = \frac{1}{2} \frac{m\ell^2}{r^2} - \frac{mk}{r} \Big|_{r=p/(1+e)} = \frac{mk^2}{2\ell^2} (e^2 - 1).$$

So the classification by eccentricity is exactly the classification by energy that we sketched using the effective potential:

$$\begin{aligned} e < 1 &\iff E < 0 && \text{(ellipse),} \\ e = 1 &\iff E = 0 && \text{(parabola),} \\ e > 1 &\iff E > 0 && \text{(hyperbola).} \end{aligned}$$

Bound orbits are precisely those with negative total energy – the particle does not have enough kinetic energy to climb out of the potential well.

For an elliptical orbit there is a particularly clean way to package the energy. Using $\ell^2 = pk$ and $p = a(1 - e^2)$,

$$E = \frac{mk^2(e^2 - 1)}{2\ell^2} = \frac{mk^2(e^2 - 1)}{2a(1 - e^2)k} = -\frac{mk}{2a},$$

so the energy of an orbit depends only on its *semi-major axis*, not on its eccentricity – a circle of radius a and a wildly elongated ellipse of semi-major axis a have exactly the same energy. Writing out $E = \frac{1}{2}mv^2 - mk/r$ and rearranging gives the handy *vis-viva equation*,

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right),$$

which gives the speed at any point of the orbit. Sanity checks: for a circle ($r = a$) it reduces to $v^2 = GM/r$, and letting $a \rightarrow \infty$ (the parabolic, marginally-escaping orbit) recovers $v^2 = 2GM/r$, the escape velocity.

§3.3.11 Kepler's Laws

Long before Newton, Kepler had extracted three empirical laws of planetary motion from Tycho Brahe's astronomical observations.

Law 3.3.9 (Kepler's Laws of Planetary Motion)

1. The orbit of a planet is an ellipse, with the Sun at one focus.
2. The line between the planet and the Sun sweeps out equal areas in equal times.
3. The square of the orbital period is proportional to the cube of the semi-major axis, $\tau^2 \propto a^3$.

Newton's great achievement was to show that all three follow from his laws of motion together with the inverse square law of gravitation – and we have now assembled everything needed to check this ourselves.

The first law follows from our discussion of the Kepler problem: bounded orbits under an inverse square law are ellipses with a focus at the origin.

The second law is just conservation of angular momentum in geometric clothing, and in fact holds for *any* central force. When the position vector sweeps through a small angle $d\theta$, the area swept out is that of a thin triangle, $dA = \frac{1}{2}r \cdot r d\theta$, so

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{\ell}{2},$$

which is constant. A planet therefore moves fastest at perihelion and slowest at aphelion.

Example 3.3.10

Conservation of angular momentum relates the extreme speeds directly: at perihelion and aphelion the velocity is purely tangential, so $\ell = r_{\min}v_{\max} = r_{\max}v_{\min}$, giving

$$\frac{v_{\max}}{v_{\min}} = \frac{r_{\max}}{r_{\min}} = \frac{1+e}{1-e}.$$

The Earth's orbital eccentricity is a modest $e \approx 0.017$, so its speed varies by only about 3.4% over the year (roughly 30.3 versus 29.3 km s⁻¹). Halley's comet, with $e \approx 0.97$, moves about 66 times faster at perihelion than at aphelion – it spends decades dawdling in the outer solar system and mere months whipping around the Sun.

For the third law, integrate the second law over a full orbit: the area of an ellipse is $A = \pi ab$, so

$$\pi ab = \frac{\ell}{2}\tau.$$

Now using $b^2 = a^2(1 - e^2)$ and $\ell^2 = pk = a(1 - e^2)k$,

$$\tau^2 = \frac{4\pi^2 a^2 b^2}{\ell^2} = \frac{4\pi^2 a^4 (1 - e^2)}{a(1 - e^2)k} = \frac{4\pi^2}{GM} a^3,$$

recalling that $k = GM$. So $\tau^2 \propto a^3$, with a constant of proportionality depending only on the mass of the Sun – which is why the same law holds for every planet in the solar system.

Example 3.3.11 (Geostationary Orbit)

At what radius does a satellite orbit the Earth once per day, so as to hover over a fixed point on the equator? Rearranging the third law (for a circular orbit, $a = r$),

$$r = \left(\frac{GM\tau^2}{4\pi^2} \right)^{1/3},$$

and inserting $GM \approx 4.0 \times 10^{14} \text{ m}^3 \text{ s}^{-2}$ for the Earth and $\tau = 86\,400 \text{ s}$ gives $r \approx 4.2 \times 10^7 \text{ m}$, i.e. about 42 000 km from the Earth's centre, or some 36 000 km above the surface. This is where communications and weather satellites live.

Example 3.3.12 (Halley's Comet)

The third law works just as well in reverse. Halley's comet returns every 76 years; measuring in years and astronomical units (in which the Earth's orbit has $\tau = 1$ and $a = 1$, conveniently normalising the constant of proportionality), its semi-major axis is

$$a = \tau^{2/3} = 76^{2/3} \approx 17.9 \text{ AU.}$$

Since the comet's orbit is highly eccentric ($e \approx 0.97$), its distances of closest and furthest approach are $a(1 - e) \approx 0.6 \text{ AU}$ (inside Venus's orbit) and $a(1 + e) \approx 35 \text{ AU}$ (out past Neptune) – all deduced from its period and one observation of its perihelion.

§3.3.12 Rutherford Scattering

Finally, let's consider what happens when the inverse square force is *repulsive*, so

$$V(r) = +\frac{mk}{r}, \quad F(r) = +\frac{mk}{r^2}, \quad k > 0.$$

The classic physical setting is Rutherford's famous experiment: alpha particles (charge q) fired at gold atoms are repelled by the nucleus (charge Q), with $k = Qq/4\pi\epsilon_0 m > 0$ by the Coulomb force law.

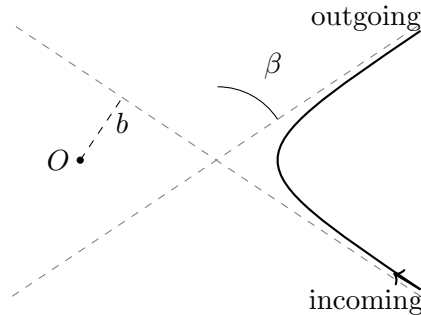
Binet's equation now reads $u'' + u = -k/\ell^2$, with general solution

$$u = -\frac{k}{\ell^2} + A \cos(\theta - \theta_0),$$

and taking $\theta_0 = 0$, $A \geq 0$ as usual we get

$$r = \frac{p}{e \cos \theta - 1}, \quad p = \frac{\ell^2}{k}, \quad e = \frac{A\ell^2}{k}.$$

For r to be positive we need $e > 1$: *every* orbit in a repulsive inverse square force is a hyperbola (which makes sense – there can be no bound orbits when the force pushes you away). The particle comes in from infinity, reaches a closest approach at $\theta = 0$, and escapes to infinity again, with $r \rightarrow \infty$ as $\theta \rightarrow \pm\alpha$ where $\alpha = \cos^{-1}(1/e)$. This time the trajectory is the branch of the hyperbola on the *far side* from the focus, curving away from the origin.



The interesting physical question is: by what angle is the particle deflected? Characterise the incoming trajectory by its speed v at infinity, and its **impact parameter** b – the perpendicular distance between the incoming asymptote and the origin. Evaluating the conserved quantities far away, the angular momentum per unit mass is $\ell = bv$.

The two asymptotes make angles $\pm\alpha$ with the axis of the hyperbola, so the total *scattering angle* (between the incoming and outgoing directions) is $\beta = \pi - 2\alpha$. Then

$$\frac{1}{e} = \cos \alpha = \cos \left(\frac{\pi}{2} - \frac{\beta}{2} \right) = \sin \frac{\beta}{2}.$$

A little algebra with the geometry of the hyperbola shows that the impact parameter is $b = p/\sqrt{e^2 - 1}$, so

$$b = \frac{\ell^2}{k} \frac{1}{\sqrt{e^2 - 1}} = \frac{(bv)^2}{k} \tan \frac{\beta}{2},$$

and rearranging,

$$\beta = 2 \tan^{-1} \left(\frac{k}{bv^2} \right).$$

Small impact parameters $b \ll k/v^2$ (nearly head-on collisions) give scattering angles approaching π – the particle bounces almost straight back. It was precisely the observation of such large-angle scattering of alpha particles that led Rutherford to conclude that an atom's positive charge is concentrated in a tiny nucleus, rather than being smeared throughout the atom: as he put it, “it was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you.”

§3.4 Rotating Frames

Newton's laws hold in inertial frames – but many frames we care about are not inertial. The most important example is right beneath our feet: the Earth rotates once a day, so a laboratory fixed to its surface is a rotating frame. In this section we work out what Newton's second law looks like to an observer in a rotating frame, and find that they can continue to use it, provided they are willing to admit some additional ‘fictitious’ forces into their bookkeeping.

§3.4.1 Motion in Rotating Frames

Suppose that $\mathcal{F}$ is an inertial frame, and $\mathcal{F}'$ is rotating about the z axis with angular velocity $\omega = \omega \mathbf{e}_z$ with respect to $\mathcal{F}$.

Suppose we have basis vectors $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$ in $\mathcal{F}$ and $\mathcal{F}'$ respectively. If a particle is at rest in $\mathcal{F}'$, then in $\mathcal{F}$ its velocity is given by

$$\left(\frac{d\mathbf{r}}{dt} \right)_{\mathcal{F}} = \omega \times \mathbf{r}.$$

Of course, this also applies to the basis vectors in $\mathcal{F}'$, with

$$\left(\frac{d\mathbf{e}'_i}{dt} \right)_{\mathcal{F}} = \omega \times \mathbf{e}'_i.$$

Now for some vector $\mathbf{a}$, we can write it in the $\{\mathbf{e}'_i\}$ basis as

$$\mathbf{a} = \sum_i a'_i(t) \mathbf{e}'_i.$$

In the frame $\mathcal{F}'$, the basis vectors $\mathbf{e}'_i$ are constant, and thus the derivative of $\mathbf{a}$ is given by

$$\left(\frac{d\mathbf{a}}{dt}\right)_{\mathcal{F}'} = \sum_i \frac{da'_i(t)}{dt} \mathbf{e}'_i.$$

In the frame $\mathcal{F}$ however, the basis vectors $\{\mathbf{e}'_i\}$ are not constant, and we have

$$\left(\frac{d\mathbf{a}}{dt}\right)_{\mathcal{F}} = \sum_i \frac{da'_i(t)}{dt} \mathbf{e}'_i + \sum_i a'_i(t) \boldsymbol{\omega} \times \mathbf{e}'_i = \left(\frac{d\mathbf{a}}{dt}\right)_{\mathcal{F}'} + \boldsymbol{\omega} \times \mathbf{a}$$

§3.4.2 Change of Frame Operator

Let $\mathcal{F}$ be an inertial frame, and $\mathcal{F}'$ be rotating relative to $\mathcal{F}$ with angular velocity $\boldsymbol{\omega}$. Then we have

$$\left(\frac{d}{dt}\right)_{\mathcal{F}} = \left(\frac{d}{dt}\right)_{\mathcal{F}'} + \boldsymbol{\omega} \times$$

§3.4.3 Velocity and Acceleration

Using the change of frame operator, we can see that

$$\left(\frac{d\mathbf{r}}{dt}\right)_{\mathcal{F}} = \left(\frac{d\mathbf{r}}{dt}\right)_{\mathcal{F}'} + \boldsymbol{\omega} \times \mathbf{r},$$

and applying the operator again (and noting that $\dot{\boldsymbol{\omega}}$ is the same in both frames), we have

$$\left(\frac{d^2\mathbf{r}}{dt^2}\right)_{\mathcal{F}} = \left(\frac{d^2\mathbf{r}}{dt^2}\right)_{\mathcal{F}'} + 2\boldsymbol{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_{\mathcal{F}'} + \dot{\boldsymbol{\omega}} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}).$$

§3.4.4 Force in the Rotating Frame

Since $\mathcal{F}$ is an inertial frame, we have $m \left(\frac{d^2\mathbf{r}}{dt^2} \right)_{\mathcal{F}} = \mathbf{F}$ by Newton's laws. This allows us to write down what the force appears to be in the frame $\mathcal{F}'$ (as if the observer in the frame was trying to apply Newton's laws).

$$m \left(\frac{d^2\mathbf{r}}{dt^2}\right)_{\mathcal{F}'} = \mathbf{F} - 2m\boldsymbol{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_{\mathcal{F}'} - m\dot{\boldsymbol{\omega}} \times \mathbf{r} - m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}).$$

The additional terms on the right are known as *fictitious forces*, each with a different name.

1. *Coriolis force*. $-2m\boldsymbol{\omega} \times \left(\frac{d\mathbf{r}}{dt}\right)_{\mathcal{F}'}$.
2. *Euler force*. $-m\dot{\boldsymbol{\omega}} \times \mathbf{r}$.
3. *Centrifugal force*. $-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$.

These are not forces in the sense of Newton's laws – nothing exerts them, and they have no third-law reaction partners. They are artefacts of insisting on doing mechanics in a non-inertial frame. Still, to an observer in the rotating frame they feel perfectly real, as anyone who has stood on a spinning roundabout can attest. For a frame rotating at constant angular velocity (such as the Earth, to good approximation) the Euler force vanishes, and we are left with the centrifugal and Coriolis forces, which we now examine in turn.

§3.4.5 The Centrifugal Force

Let's look more closely at the centrifugal force, $-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$. Using the vector triple product identity $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$,

$$-m\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) = -m((\boldsymbol{\omega} \cdot \mathbf{r})\boldsymbol{\omega} - |\boldsymbol{\omega}|^2\mathbf{r}) = m\omega^2\mathbf{r}_\perp,$$

where $\mathbf{r}_\perp = \mathbf{r} - (\hat{\boldsymbol{\omega}} \cdot \mathbf{r})\hat{\boldsymbol{\omega}}$ is the component of $\mathbf{r}$ perpendicular to the rotation axis. So the centrifugal force has magnitude $m\omega^2 d$, where $d = |\mathbf{r}_\perp|$ is the distance from the axis, and points *directly away from the axis of rotation*. It is also conservative: we can write it as

$$-\nabla \left(-\frac{1}{2}m\omega^2|\mathbf{r}_\perp|^2 \right),$$

so it can be absorbed into an effective potential.

Example 3.4.1 (Apparent Gravity on the Earth)

For a particle at rest on the surface of the rotating Earth at latitude λ , the effective gravitational acceleration measured in the rotating frame is

$$\mathbf{g}_{\text{eff}} = \mathbf{g} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}),$$

the sum of true gravity (pointing to the Earth's centre) and the centrifugal term (pointing away from the rotation axis, with magnitude $\omega^2 R \cos \lambda$). With $\omega = 2\pi/86400 \text{ s} \approx 7.3 \times 10^{-5} \text{ s}^{-1}$ and $R \approx 6400 \text{ km}$, the centrifugal term has magnitude at most $\omega^2 R \approx 0.03 \text{ m s}^{-2}$ – about 0.3% of g , largest at the equator and vanishing at the poles. Two consequences: the measured g is slightly smaller at the equator than at the poles, and a plumb line does not point exactly towards the centre of the Earth except at the poles and equator. (For the same reason the Earth itself is not quite spherical, but bulges at the equator.)

Example 3.4.2 (The Rotating Bucket)

Spin a bucket of water about its vertical axis at constant angular speed ω , and wait for the water to rotate with it. What shape is the surface? In the rotating frame the water is at rest, subject to gravity, the centrifugal force, and pressure forces. Since the centrifugal force is conservative, each fluid element sits in the effective potential (per unit mass)

$$\Phi_{\text{eff}} = gz - \frac{1}{2}\omega^2 r^2,$$

where r is the distance from the axis. The free surface of a fluid at rest is a surface of constant potential, so on the surface

$$gz - \frac{1}{2}\omega^2 r^2 = \text{const} \implies z = z_0 + \frac{\omega^2 r^2}{2g}.$$

The surface is a *paraboloid*, climbing up the sides of the bucket – as you can verify by stirring your tea. (Liquid mirror telescopes exploit exactly this: a slowly rotating dish of mercury forms a perfect parabolic mirror.)

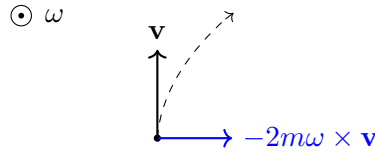
§3.4.6 The Coriolis Force

The Coriolis force

$$\mathbf{F}_{\text{cor}} = -2m\boldsymbol{\omega} \times \mathbf{v}, \quad \text{where } \mathbf{v} = \left(\frac{d\mathbf{r}}{dt} \right)_{\mathcal{F}'},$$

is stranger: it depends on the particle's velocity *in the rotating frame*, and vanishes for objects at rest in that frame. Since $\mathbf{F}_{\text{cor}}$ is always perpendicular to $\mathbf{v}$, it does no work – like a magnetic force, it deflects motion sideways without changing speed.

To get oriented, consider the Northern hemisphere and motion in a horizontal plane. The locally vertical component of $\boldsymbol{\omega}$ points up out of the ground, and one can check from the cross product that the Coriolis force then deflects a moving object *to the right* of its velocity (in the Southern hemisphere, to the left). This is the origin of the rotation of weather systems: air being drawn towards a low pressure region is deflected rightwards as it flows, and the result is that cyclones in the Northern hemisphere rotate anticlockwise. On everyday scales the effect is tiny – it does *not* determine which way your bathtub drains, despite the folklore – but over the scale of an atmosphere it dominates.



In the diagram (drawn for the Northern hemisphere, viewed from above, with $\boldsymbol{\omega}$ pointing out of the page), a particle moving with velocity $\mathbf{v}$ feels a Coriolis force at right angles to its motion, and its trajectory (dashed) is continually deflected to the right.

Let's do a genuine calculation.

Example 3.4.3 (Dropping a Particle from a Tower)

Stand at the equator, and drop a particle from a tower of height h . Where does it land? In the rotating frame of the Earth, working to first order in the small quantity ω (so ignoring the centrifugal force, which is order ω^2),

$$\ddot{\mathbf{r}} = \mathbf{g} - 2\boldsymbol{\omega} \times \dot{\mathbf{r}}.$$

We solve perturbatively. At zeroth order, $\dot{\mathbf{r}}_0 = \mathbf{g}t$, and substituting this into the Coriolis term,

$$\ddot{\mathbf{r}} \approx \mathbf{g} - 2\boldsymbol{\omega} \times \mathbf{g}t \implies \mathbf{r}(t) \approx \mathbf{r}_0 + \frac{1}{2}\mathbf{g}t^2 - \frac{1}{3}\boldsymbol{\omega} \times \mathbf{g}t^3.$$

At the equator, $\boldsymbol{\omega}$ points north and $\mathbf{g}$ points down, so $-\boldsymbol{\omega} \times \mathbf{g}$ points *east*, with magnitude ωg . The particle hits the ground at time $t = \sqrt{2h/g}$, displaced eastward by

$$d = \frac{1}{3}\omega g \left(\frac{2h}{g} \right)^{3/2}.$$

For $h = 100$ m, this is about 2 cm – small, but measurable, and it was one of the early direct demonstrations of the Earth's rotation. Intuitively, the top of the tower moves faster eastward than its base, and the particle keeps its excess eastward velocity as it falls.

Example 3.4.4 (The Foucault Pendulum)

A related and more famous demonstration: suspend a long pendulum at latitude λ and let it swing freely for hours. Set up local axes with x pointing east, y north and z up. At latitude λ , the Earth's angular velocity in these axes is

$$\boldsymbol{\omega} = \omega(0, \cos \lambda, \sin \lambda).$$

For small swings of a pendulum of length ℓ , the bob moves essentially horizontally, with displacement (x, y) and a restoring force $-(g/\ell)(x, y)$ per unit mass, plus the Coriolis force. Computing $-2\boldsymbol{\omega} \times \mathbf{v}$ with $\mathbf{v} \approx (\dot{x}, \dot{y}, 0)$ and keeping the horizontal components, the equations of motion are

$$\ddot{x} = -\frac{g}{\ell}x + 2\Omega\dot{y}, \quad \ddot{y} = -\frac{g}{\ell}y - 2\Omega\dot{x}, \quad \text{where } \Omega = \omega \sin \lambda$$

is the locally vertical component of the Earth's rotation. The two equations beg to be combined into one complex variable $\zeta = x + iy$:

$$\ddot{\zeta} + 2i\Omega\dot{\zeta} + \omega_0^2\zeta = 0, \quad \omega_0^2 = \frac{g}{\ell}.$$

Seeking solutions $e^{i\alpha t}$ gives $\alpha = -\Omega \pm \sqrt{\omega_0^2 + \Omega^2} \approx -\Omega \pm \omega_0$, since $\Omega \ll \omega_0$. So

$$\zeta(t) \approx e^{-i\Omega t} (Ae^{i\omega_0 t} + Be^{-i\omega_0 t}).$$

The bracket is an ordinary pendulum oscillation; the prefactor $e^{-i\Omega t}$ slowly *rotates* the whole pattern. The plane of swing therefore precesses (clockwise viewed from above, in the Northern hemisphere) at angular rate $\Omega = \omega \sin \lambda$, completing a full turn in

$$T = \frac{24 \text{ h}}{\sin \lambda}.$$

At the poles this is exactly one day – the pendulum swings in a fixed inertial plane while the Earth turns beneath it, which is a helpful way to see the result must be true. At the latitude of Cambridge the period is around 30 hours, and at the equator there is no precession at all. Foucault's original 1851 demonstration in the Panthéon in Paris was the first direct mechanical evidence of the Earth's rotation.

§3.5 Systems of Particles

So far we have described the dynamics of single particles. Now let's have a look at systems involving multiple particles, and see how much of the single-particle story – momentum, angular momentum, energy – survives. The punchline is that a composite system behaves, in aggregate, remarkably like a single particle located at its centre of mass; this is precisely why we could get away with treating planets as points for so long.

§3.5.1 Centre of Mass & Conservation of Momentum

Suppose we have N particles, where particle i has position $\mathbf{x}_i$, mass m_i , and momentum $\mathbf{p}_i = m_i \dot{\mathbf{x}}_i$. Newton's second law for each particle becomes

$$\dot{\mathbf{p}}_i = \mathbf{F}_i,$$

and we can helpfully split the force on particle i as

$$\mathbf{F}_i = \mathbf{F}_i^{\text{ext}} + \sum_{j \neq i} \mathbf{F}_{ij},$$

where $\mathbf{F}_i^{\text{ext}}$ is the external force on particle i (from outside the system) and $\mathbf{F}_{ij}$ is the internal force on particle i from particle j . By Newton's third law, we have $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$.

Definition 3.5.1 (Centre of Mass)

The **centre of mass** of the system is the mass-weighted average position

$$\mathbf{R} = \frac{1}{M} \sum_i m_i \mathbf{x}_i, \quad \text{where } M = \sum_i m_i \text{ is the total mass.}$$

The total momentum of the system is then $\mathbf{P} = \sum_i \mathbf{p}_i = M\dot{\mathbf{R}}$ – exactly the momentum of a single particle of mass M moving with the centre of mass.

Proposition 3.5.2 (Motion of the Centre of Mass)

$$M\ddot{\mathbf{R}} = \dot{\mathbf{P}} = \sum_i \mathbf{F}_i^{\text{ext}}.$$

Proof. Summing Newton's second law over the particles,

$$\dot{\mathbf{P}} = \sum_i \dot{\mathbf{p}}_i = \sum_i \mathbf{F}_i^{\text{ext}} + \sum_i \sum_{j \neq i} \mathbf{F}_{ij} = \sum_i \mathbf{F}_i^{\text{ext}} + \frac{1}{2} \sum_{i \neq j} (\mathbf{F}_{ij} + \mathbf{F}_{ji}) = \sum_i \mathbf{F}_i^{\text{ext}},$$

where the internal forces cancel in pairs by Newton's third law. $\square$

Thus the motion of the centre of mass is not influenced by internal forces at all – when a firework explodes, its centre of mass continues serenely along a parabola. This proposition is also what justifies treating extended bodies as particles: as far as its total momentum is concerned, a body *is* a particle of mass M at $\mathbf{R}$. If there is no external force, we then have *conservation of momentum*, $\dot{\mathbf{P}} = \mathbf{0}$. In this case the centre of mass moves uniformly, and it is often convenient to change to an inertial frame in which $\mathbf{R}$ is at rest at the origin – the **centre of mass frame**.

Example 3.5.3 (An Exploding Shell)

A shell is launched from the origin and, at the top of its parabolic flight, moving horizontally with speed v , explodes into two equal fragments. One fragment is observed to be momentarily at rest just after the explosion. Where does the other one go? Conservation of momentum across the explosion (the explosive forces are internal, and gravity's impulse during the brief explosion is negligible) gives

$$mv = \frac{m}{2} \cdot 0 + \frac{m}{2} v_2 \implies v_2 = 2v.$$

The second fragment moves off horizontally at double the speed. Both fragments then fall freely, and – as the proposition guarantees – the centre of mass of the pair continues along the *original* parabola exactly as if nothing had happened, with the

stationary fragment dropping straight down and the fast one landing correspondingly further along.

§3.5.2 Angular Momentum of a System

We define the total angular momentum of the system about the origin as

$$\mathbf{L} = \sum_i \mathbf{x}_i \times \mathbf{p}_i.$$

Differentiating, and using Newton's second and third laws as before,

$$\dot{\mathbf{L}} = \sum_i \mathbf{x}_i \times \dot{\mathbf{p}}_i = \sum_i \mathbf{x}_i \times \mathbf{F}_i^{\text{ext}} + \frac{1}{2} \sum_{i \neq j} (\mathbf{x}_i - \mathbf{x}_j) \times \mathbf{F}_{ij}.$$

To make the final sum vanish we need slightly more than Newton's third law: we need the internal forces to be *central*, that is $\mathbf{F}_{ij} \parallel (\mathbf{x}_i - \mathbf{x}_j)$, so that each term of the sum is zero. This is true of gravity and of electrostatic forces, so we will assume it. Then

$$\dot{\mathbf{L}} = \sum_i \mathbf{x}_i \times \mathbf{F}_i^{\text{ext}} = \boldsymbol{\tau}^{\text{ext}},$$

the total *external* torque. So, just as internal forces cannot change the total momentum, (central) internal forces cannot change the total angular momentum, and if the total external torque vanishes then $\mathbf{L}$ is conserved.

§3.5.3 Motion Relative to the Centre of Mass

Externally, then, a system of particles behaves like a point mass at $\mathbf{R}$. What about the internal motion? Write

$$\mathbf{x}_i = \mathbf{R} + \mathbf{y}_i,$$

where $\mathbf{y}_i$ is the position of particle i *relative to the centre of mass*. Directly from the definition of $\mathbf{R}$ we get the two useful identities

$$\sum_i m_i \mathbf{y}_i = \mathbf{0}, \quad \sum_i m_i \dot{\mathbf{y}}_i = \mathbf{0}.$$

Substituting into the angular momentum and expanding, the cross terms vanish by these identities, leaving

$$\mathbf{L} = M\mathbf{R} \times \dot{\mathbf{R}} + \sum_i m_i \mathbf{y}_i \times \dot{\mathbf{y}}_i.$$

Similarly for the kinetic energy:

$$T = \frac{1}{2} \sum_i m_i |\dot{\mathbf{x}}_i|^2 = \underbrace{\frac{1}{2} M |\dot{\mathbf{R}}|^2}_{\text{motion of the CoM}} + \underbrace{\frac{1}{2} \sum_i m_i |\dot{\mathbf{y}}_i|^2}_{\text{motion about the CoM}}.$$

Both quantities split cleanly into a 'centre of mass part' plus an 'internal part', with no interference between the two. These decompositions will be the key to understanding rigid bodies in Section 3.6.

Finally, energy. If the external forces are conservative with $\mathbf{F}_i^{\text{ext}} = -\nabla_i V_i(\mathbf{x}_i)$, and the internal forces are conservative with potentials depending on separations only, $\mathbf{F}_{ij} = -\nabla_i V_{ij}(\mathbf{x}_i - \mathbf{x}_j)$, then the total energy

$$E = T + \sum_i V_i(\mathbf{x}_i) + \frac{1}{2} \sum_{i \neq j} V_{ij}(\mathbf{x}_i - \mathbf{x}_j)$$

is conserved – the proof is the same differentiate-and-cancel argument as ever.

§3.5.4 The Two-Body Problem

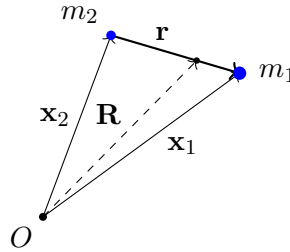
The simplest multi-particle system has just two particles, interacting through a force obeying Newton's third law – say two bodies moving under mutual gravity. This is the *two-body problem*, and it can be solved completely by a change of variables.

Let the particles have masses m_1, m_2 and positions $\mathbf{x}_1, \mathbf{x}_2$. The clever move is to trade $\mathbf{x}_1, \mathbf{x}_2$ for the centre of mass and the *separation vector*:

$$\mathbf{R} = \frac{m_1 \mathbf{x}_1 + m_2 \mathbf{x}_2}{M}, \quad \mathbf{r} = \mathbf{x}_1 - \mathbf{x}_2,$$

where $M = m_1 + m_2$, so that

$$\mathbf{x}_1 = \mathbf{R} + \frac{m_2}{M} \mathbf{r}, \quad \mathbf{x}_2 = \mathbf{R} - \frac{m_1}{M} \mathbf{r}.$$



With no external forces, we know $\ddot{\mathbf{R}} = \mathbf{0}$, so the centre of mass moves uniformly and can be taken as the origin of an inertial frame. Meanwhile,

$$\ddot{\mathbf{r}} = \ddot{\mathbf{x}}_1 - \ddot{\mathbf{x}}_2 = \frac{\mathbf{F}_{12}}{m_1} - \frac{\mathbf{F}_{21}}{m_2} = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \mathbf{F}_{12},$$

which we can rewrite as

$$\mu \ddot{\mathbf{r}} = \mathbf{F}_{12}(\mathbf{r}), \quad \text{where } \mu = \frac{m_1 m_2}{m_1 + m_2}$$

is called the **reduced mass**. This is exactly the equation for a *single* particle of mass μ at position $\mathbf{r}$ moving about a fixed centre of force – which is the problem we solved in the orbits section! The two-body problem thus reduces entirely to the one-body Kepler problem, and one can check that the angular momentum and kinetic energy also split accordingly:

$$\mathbf{L} = M \mathbf{R} \times \dot{\mathbf{R}} + \mu \mathbf{r} \times \dot{\mathbf{r}}, \quad T = \frac{1}{2} M |\dot{\mathbf{R}}|^2 + \frac{1}{2} \mu |\dot{\mathbf{r}}|^2.$$

Example 3.5.4 (Binary Orbits)

For two bodies under mutual gravity, $\mathbf{F}_{12} = -Gm_1m_2\hat{\mathbf{r}}/|\mathbf{r}|^2$, so

$$\mu\ddot{\mathbf{r}} = -\frac{Gm_1m_2}{|\mathbf{r}|^2}\hat{\mathbf{r}} \implies \ddot{\mathbf{r}} = -\frac{GM}{|\mathbf{r}|^2}\hat{\mathbf{r}}.$$

So the *separation* traces out a conic exactly as before, but with the total mass M appearing in the force law. In the centre of mass frame, each body individually moves in an ellipse about the (fixed) centre of mass, with

$$\mathbf{x}_1 = \frac{m_2}{M}\mathbf{r}, \quad \mathbf{x}_2 = -\frac{m_1}{M}\mathbf{r},$$

so the two ellipses are similar, scaled by the mass ratio, with the bodies always on opposite sides of the centre of mass. When $m_1 \ll m_2$ (a planet and the Sun), $\mu \approx m_1$ and the centre of mass sits essentially at the heavier body – recovering our fixed-Sun approximation, with a small correction: the Sun itself executes a miniature ellipse. It is exactly this tiny stellar ‘wobble’ that provided some of the first detections of planets around other stars.

§3.5.5 Collisions

A *collision* is an interaction between two bodies which is brief and localised: before and after, the bodies move freely. Whatever complicated internal forces act during the collision, they are internal, so the *total momentum is always conserved*:

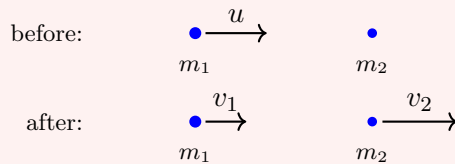
$$m_1\mathbf{u}_1 + m_2\mathbf{u}_2 = m_1\mathbf{v}_1 + m_2\mathbf{v}_2,$$

where $\mathbf{u}_i$ are velocities before and $\mathbf{v}_i$ after. Kinetic energy is a different matter – it may be transferred into heat, sound or deformation. We distinguish:

- *Elastic collisions*, in which the total kinetic energy is also conserved;
- *Inelastic collisions*, in which some kinetic energy is lost (a *totally* inelastic collision being one in which the bodies coalesce).

Example 3.5.5 (Elastic Collision in One Dimension)

Let a particle of mass m_1 and velocity u strike a stationary particle of mass m_2 , elastically, in one dimension.



Conservation of momentum and energy give

$$m_1u = m_1v_1 + m_2v_2, \quad \frac{1}{2}m_1u^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2.$$

Rearranging the first as $m_1(u - v_1) = m_2v_2$ and the second as $m_1(u - v_1)(u + v_1) = m_2v_2^2$, then dividing (assuming a genuine collision, $v_2 \neq 0$), we get $u + v_1 = v_2$, and

hence

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2}u, \quad v_2 = \frac{2m_1}{m_1 + m_2}u.$$

The limiting cases are instructive. If $m_1 = m_2$, then $v_1 = 0$ and $v_2 = u$: the particles simply exchange velocities, as anyone who has played snooker knows. If $m_2 \gg m_1$ (a ball bouncing off a wall), $v_1 \approx -u$: the light particle bounces straight back. If $m_1 \gg m_2$, the heavy particle barrels on essentially unaffected, and the light one is flung ahead at $v_2 \approx 2u$.

Example 3.5.6 (Totally Inelastic Collision)

If instead the two particles coalesce, momentum conservation alone determines the outcome:

$$v = \frac{m_1 u}{m_1 + m_2}.$$

The kinetic energy lost is

$$\Delta T = \frac{1}{2}m_1 u^2 - \frac{1}{2}(m_1 + m_2)v^2 = \frac{1}{2}\mu u^2,$$

where μ is the reduced mass – precisely the kinetic energy of the *relative* motion. This makes sense: the centre of mass motion cannot be destroyed (momentum conservation), so the most energy a collision can possibly dissipate is the internal part, and a totally inelastic collision dissipates exactly that.

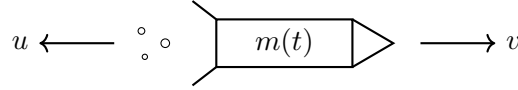
In higher dimensions, the same conservation laws apply, but they no longer determine the outcome uniquely – the scattering angle remains free. A classic result: for an elastic collision between *equal* masses, one initially at rest, conservation of momentum gives $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{v}_2$, and conservation of energy gives $|\mathbf{u}_1|^2 = |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2$. Squaring the first equation and comparing, $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$: the particles move off *at right angles* (unless one of them is at rest). We will revisit this in special relativity, where the right angle closes up.

Remark. Real collisions sit between the elastic and totally inelastic extremes. A common phenomenological model (‘Newton’s experimental law’) assigns a collision a *coefficient of restitution* $e \in [0, 1]$: the relative velocity of separation is e times the relative velocity of approach. Then $e = 1$ is the elastic case, $e = 0$ the totally inelastic one, and one can check that the kinetic energy lost is $\frac{1}{2}\mu(1 - e^2)|\mathbf{u}_1 - \mathbf{u}_2|^2$, interpolating between our two examples.

§3.5.6 Variable Mass Problems & The Rocket Equation

Newton’s second law reads $\dot{\mathbf{p}} = \mathbf{F}$ with $\mathbf{p} = m\mathbf{v}$ – and nothing in it requires m to be constant. Systems that gain or lose mass are everywhere: rockets burning fuel, raindrops accreting from a cloud, snowballs rolling downhill. The subtlety is that the mass that leaves (or joins) carries momentum with it, so we must be careful to apply the second law to a fixed collection of matter.

Consider the classic case: a rocket of mass $m(t)$ moving in a straight line with velocity $v(t)$, propelling itself by ejecting exhaust backwards with speed u *relative to the rocket*.



Consider the total system of rocket plus exhaust between times t and $t + \delta t$. At time t , the momentum is $m(t)v(t)$. At time $t + \delta t$, the rocket has mass $m(t + \delta t)$ and velocity $v(t + \delta t)$, and the mass $m(t) - m(t + \delta t)$ of ejected exhaust travels with velocity $v(t) - u + O(\delta t)$. So the change in total momentum is

$$\begin{aligned}\delta p &= m(t + \delta t)v(t + \delta t) + [m(t) - m(t + \delta t)][v(t) - u] - m(t)v(t) + O(\delta t^2) \\ &= (m + \dot{m}\delta t)(v + \dot{v}\delta t) - \dot{m}\delta t(v - u) - mv + O(\delta t^2) \\ &= (m\dot{v} + \dot{m}u)\delta t + O(\delta t^2).\end{aligned}$$

Newton's second law applied to the whole system (whose total mass *is* constant) says $\delta p / \delta t \rightarrow F$, the external force on the rocket. We obtain:

Proposition 3.5.7 (The Rocket Equation)

$$m \frac{dv}{dt} + u \frac{dm}{dt} = F.$$

Note $\dot{m} < 0$ for a rocket, so the second term acts as a forward thrust of magnitude $u|\dot{m}|$.

Example 3.5.8 (Rocket in Free Space)

In deep space, $F = 0$, and taking the exhaust speed u constant,

$$m \frac{dv}{dt} = -u \frac{dm}{dt} \implies \frac{dv}{dm} = -\frac{u}{m} \implies v = v_0 + u \log \left(\frac{m_0}{m} \right),$$

the *Tsiolkovsky rocket equation*. Two features deserve comment. First, the final velocity depends only on the exhaust speed and the fraction of mass burnt – not on *how fast* the fuel is burnt. Second, the dependence on the mass ratio is only logarithmic, which is the tyranny of rocketry: to gain speed $u \log 10 \approx 2.3u$, ninety percent of your initial mass must be fuel. This is why real rockets are almost entirely fuel tank, and why they shed stages as they climb – dropping an empty tank of mass m_{tank} improves every subsequent burn's mass ratio, and multi-stage rockets squeeze substantially more final velocity out of the same fuel.

Example 3.5.9 (A Falling Raindrop)

For a raindrop falling through a stationary cloud and accreting mass, the accreted mass is at rest, so relative to the drop it arrives with speed $u = v$ (opposing the motion). With gravity as the external force, the rocket equation gives

$$m \frac{dv}{dt} + v \frac{dm}{dt} = \frac{d}{dt}(mv) = mg,$$

that is, the momentum mv grows at rate mg . To go further we need a model for the accretion rate; for instance, if $\dot{m} = kmv$ (sweeping up mass in proportion to the volume swept out), one can show the drop approaches a constant terminal *acceleration* of $g/7$ – a fun exercise.

§3.6 Rigid Bodies

In many cases, we want to study a system of particles such that the particles are constrained such that their relative positions are fixed. Such systems correspond to solid objects that cannot deform, break or bend. Such systems are called *rigid bodies*. Everything we established for general systems of particles – the motion of the centre of mass, $\dot{\mathbf{P}} = \mathbf{F}^{\text{ext}}$ and $\dot{\mathbf{L}} = \boldsymbol{\tau}^{\text{ext}}$, the decompositions of energy and angular momentum – applies verbatim; what the rigidity constraint buys us is that the internal motion is always a *rotation*, so the whole configuration is captured by just a position and an orientation.

§3.6.1 Angular Velocity

Suppose a rigid body $\mathcal{B}$ is rotating about a fixed axis with angular speed ω . If $\hat{\mathbf{n}}$ is the unit vector parallel to the rotation axis, we define the *vector angular velocity* of $\mathcal{B}$ to be

$$\boldsymbol{\omega} = \omega \hat{\mathbf{n}},$$

where we choose the sign such that a positive ω corresponds to a right-handed rotation about the axis in the direction $\hat{\mathbf{n}}$.

If P is a particle in the rigid body $\mathcal{B}$, then its velocity due to the rotation of the body is given by

$$\mathbf{v} = \boldsymbol{\omega} \times (\mathbf{r} - \mathbf{b}),$$

where $\mathbf{b}$ is the position vector corresponding to any fixed point B on the rotation axis.

§3.6.2 Moment of Inertia

Now suppose we have a rigid body made up of N particles, rotating with angular velocity $\boldsymbol{\omega}$. Then we could write down the kinetic energy for the rigid body as

$$T = \frac{1}{2} \sum_i m_i \dot{\mathbf{x}}_i \cdot \dot{\mathbf{x}}_i = \frac{1}{2} I \omega^2,$$

where

$$I = \sum_{i=1}^N m_i d_i^2$$

is the **moment of inertia**, and d_i is the perpendicular distance from particle i to the axis of rotation.

The intuition behind moments of inertia is that they are to rotations what mass is to translations: the larger I is, the more energy you need to supply to the body to make it spin.

§3.6.3 Angular Momentum and Moments of Inertia

Recall that the angular momentum of a body is

$$\mathbf{L} = \sum_i m_i \mathbf{x}_i \times \dot{\mathbf{x}}_i.$$

Suppose that the body is rotating with angular velocity $\omega \hat{\mathbf{n}}$ about an axis through the origin. Then the component of angular momentum in the $\hat{\mathbf{n}}$ direction is given by

$$\begin{aligned}\mathbf{L} \cdot \hat{\mathbf{n}} &= \sum_i m_i \mathbf{x}_i \times (\omega \times \mathbf{x}_i) \cdot \hat{\mathbf{n}} \\ &= \omega \sum_i m_i (\mathbf{x}_i \times \hat{\mathbf{n}}) \cdot (\mathbf{x}_i \times \hat{\mathbf{n}}) \\ &= I\omega,\end{aligned}$$

since $|\mathbf{x}_i \times \hat{\mathbf{n}}| = d_i$, the perpendicular distance of particle i from the axis. (The components of $\mathbf{L}$ perpendicular to $\hat{\mathbf{n}}$ need not vanish, and in general $\mathbf{L}$ is *not* parallel to ω – but we will not need this complication in this course.) If there's a torque τ acting on the rigid body in the same direction as the angular velocity, so $\tau = \tau \hat{\mathbf{n}}$, then $I\dot{\omega} = \tau$.

§3.6.4 Calculating Moments of Inertia

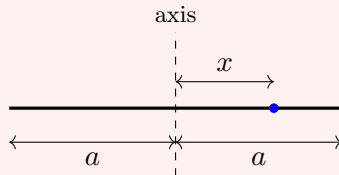
If we consider the rigid body as a continuous object with density distribution $\rho(\mathbf{x})$, then the sums of the previous section become integrals, and the moment of inertia is given by

$$I = \int \rho(\mathbf{x}) x_{\perp}^2 dV,$$

where $x_{\perp}$ is the perpendicular distance from $\mathbf{x}$ to the axis of rotation. Let's compute a few of the standard examples properly.

Example 3.6.1 (Thin Rod)

Take a uniform thin rod of mass M and length $2a$, rotating about an axis through its centre, perpendicular to the rod.



The mass per unit length is $\rho = M/2a$, and a point at position $x \in [-a, a]$ along the rod is at distance $|x|$ from the axis. So

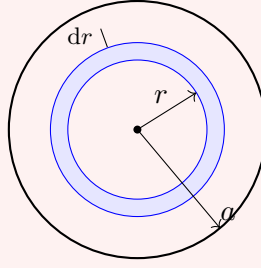
$$I = \int_{-a}^a \frac{M}{2a} x^2 dx = \frac{M}{2a} \cdot \frac{2a^3}{3} = \frac{1}{3}Ma^2.$$

Example 3.6.2 (Circular Hoop and Disc)

For a hoop of mass M and radius a , about the axis through its centre perpendicular to its plane, every point is at distance exactly a from the axis, so immediately $I = Ma^2$.

For a uniform disc of mass M and radius a about the same axis, the surface density is $\rho = M/\pi a^2$, and working in plane polar coordinates with area element $r dr d\theta$,

$$I = \int_0^{2\pi} \int_0^a \frac{M}{\pi a^2} r^2 \cdot r dr d\theta = \frac{M}{\pi a^2} \cdot 2\pi \cdot \frac{a^4}{4} = \frac{1}{2}Ma^2.$$



Here we have summed over rings of radius r and thickness dr (shaded), each of which contributes mass at a common distance r from the axis.

Example 3.6.3 (Solid Sphere)

For a uniform solid sphere of mass M and radius a about an axis through its centre, take the axis to be the z -axis and use the density $\rho = 3M/4\pi a^3$. Slicing the sphere into discs of thickness dz at height z , each of radius $\sqrt{a^2 - z^2}$, and using the disc result,

$$I = \int_{-a}^a \frac{1}{2} (\rho\pi(a^2 - z^2)) (a^2 - z^2) dz = \frac{\rho\pi}{2} \int_{-a}^a (a^2 - z^2)^2 dz = \frac{\rho\pi}{2} \cdot \frac{16a^5}{15} = \frac{2}{5}Ma^2.$$

Collecting these (all worth remembering):

1. *Thin Rod.* With mass M and length $2a$, about an axis perpendicular to the rod through its centre: $I = \frac{1}{3}Ma^2$.
2. *Circular Hoop.* With mass M and radius a , about an axis perpendicular to the plane of the hoop through its centre: $I = Ma^2$.
3. *Circular Disc.* With mass M and radius a , about an axis perpendicular to the plane of the disc through its centre: $I = \frac{1}{2}Ma^2$.
4. *Solid Sphere.* With mass M and radius a , about an axis through its centre: $I = \frac{2}{5}Ma^2$.

Two general theorems let us transport these results to other axes without redoing any integrals.

Theorem 3.6.4 (Parallel Axis Theorem)

If a rigid body of mass M has moment of inertia I_G about an axis through its centre of mass, then its moment of inertia about a parallel axis a distance d away is

$$I = I_G + Md^2.$$

Proof. Set up coordinates with origin at the centre of mass and the first axis along $\mathbf{e}_z$, and let the parallel axis pass through the point $d\mathbf{e}_x$. Writing $\mathbf{x}_i = (x_i, y_i, z_i)$ for the particles of the body, the perpendicular distance from particle i to the new axis satisfies $d_i'^2 = (x_i - d)^2 + y_i^2$. So

$$I = \sum_i m_i ((x_i - d)^2 + y_i^2) = \sum_i m_i (x_i^2 + y_i^2) - 2d \sum_i m_i x_i + d^2 \sum_i m_i = I_G + Md^2,$$

where the middle sum vanishes because the centre of mass is at the origin. $\square$

Note in particular that $I \geq I_G$: of all axes in a given direction, the one through the centre of mass gives the smallest moment of inertia.

Theorem 3.6.5 (Perpendicular Axis Theorem)

For a two dimensional body (a *lamina*) lying in the (x, y) plane, the moments of inertia about the coordinate axes satisfy

$$I_z = I_x + I_y.$$

Proof. Since $z_i = 0$ for every particle, we have $I_x = \sum m_i y_i^2$ and $I_y = \sum m_i x_i^2$, while $I_z = \sum m_i (x_i^2 + y_i^2) = I_x + I_y$. $\square$

Example 3.6.6

For the uniform disc, by symmetry $I_x = I_y$, and $I_z = \frac{1}{2}Ma^2$ from before. So the moment of inertia of a disc about one of its *diameters* is $I_x = \frac{1}{4}Ma^2$ – no integration required. Similarly, using the parallel axis theorem, a rod of length $2a$ spun about one of its *ends* has $I = \frac{1}{3}Ma^2 + Ma^2 = \frac{4}{3}Ma^2$.

Example 3.6.7 (Rectangular Lamina)

For a uniform rectangular plate of mass M with sides $2a$ and $2b$, the moment of inertia about the in-plane axis through the centre parallel to the side of length $2b$ only ‘sees’ the x -extent of the mass, so it equals that of a rod: $I_y = \frac{1}{3}Ma^2$, and likewise $I_x = \frac{1}{3}Mb^2$. The perpendicular axis theorem then gives, for the axis through the centre perpendicular to the plate,

$$I_z = I_x + I_y = \frac{1}{3}M(a^2 + b^2),$$

with all three answers obtained from a single one dimensional integral.

§3.6.5 Motion of Rigid Bodies

The general motion of a rigid body is a translation of its centre of mass along some trajectory $\mathbf{R}(t)$, together with a rotation about an axis through the centre of mass. Writing $\mathbf{x}_i = \mathbf{R} + \mathbf{y}_i$ as in Section 3.5, the rigidity constraint means the relative positions $\mathbf{y}_i$ evolve by pure rotation, so

$$\dot{\mathbf{x}}_i = \dot{\mathbf{R}} + \boldsymbol{\omega} \times \mathbf{y}_i$$

for some angular velocity $\boldsymbol{\omega}(t)$. Our kinetic energy decomposition then becomes

$$T = \frac{1}{2}M|\dot{\mathbf{R}}|^2 + \frac{1}{2}\sum_i m_i |\boldsymbol{\omega} \times \mathbf{y}_i|^2 = \underbrace{\frac{1}{2}M|\dot{\mathbf{R}}|^2}_{\text{translational}} + \underbrace{\frac{1}{2}I_G \omega^2}_{\text{rotational}},$$

where I_G is the moment of inertia about the axis through the centre of mass in the direction of $\boldsymbol{\omega}$. The motion is governed by our two equations for systems of particles,

$$\dot{\mathbf{P}} = \mathbf{F}^{\text{ext}}, \quad \dot{\mathbf{L}} = \boldsymbol{\tau}^{\text{ext}},$$

where the angular momentum and torque may be taken about any fixed point of an inertial frame, *or* about the centre of mass – the latter works even when the centre of mass is accelerating, since the fictitious forces this introduces have zero net torque about the centre of mass.⁵

In a *uniform* gravitational field, the external force and torque take a particularly nice form:

$$\mathbf{F} = \sum_i m_i \mathbf{g} = M \mathbf{g}, \quad \boldsymbol{\tau} = \sum_i \mathbf{x}_i \times m_i \mathbf{g} = M \mathbf{R} \times \mathbf{g}.$$

Both are exactly what would act on a single particle of mass M at the centre of mass. In particular, the gravitational torque *about* the centre of mass vanishes.

Example 3.6.8 (A Thrown Stick)

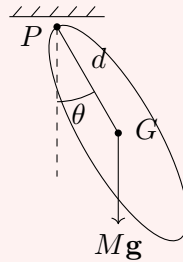
Throw a uniform stick spinning through the air (ignoring air resistance). The centre of mass traces a parabola, exactly as a point particle would. Meanwhile, since gravity exerts no torque about the centre of mass, $\dot{\mathbf{L}}^{\text{CoM}} = \mathbf{0}$, and the stick rotates with constant angular velocity about its centre as it flies.

§3.6.6 The Compound Pendulum

Our equation for rotation about a fixed axis is already enough to handle a classic example – a pendulum made from an arbitrary rigid body.

Example 3.6.9 (The Compound Pendulum)

A rigid body is free to swing under gravity in a vertical plane about a fixed horizontal axis through a point P , which is a distance d from the centre of mass G . Let θ be the angle of PG from the downward vertical, and let I be the moment of inertia of the body about the axis through P (computable via the parallel axis theorem: $I = I_G + Md^2$).



The body rotates about the fixed point P , so $L = I\dot{\theta}$, and the torque about P is due to gravity acting at the centre of mass:

$$I\ddot{\theta} = -Mgd \sin \theta.$$

This is exactly the simple pendulum equation, so all our pendulum analysis applies:

⁵In the accelerating centre of mass frame, each particle feels a fictitious force $-m_i \ddot{\mathbf{R}}$, and the total torque of these about the centre of mass is $-(\sum_i m_i \mathbf{y}_i) \times \ddot{\mathbf{R}} = \mathbf{0}$.

small oscillations have angular frequency

$$\omega = \sqrt{\frac{Mgd}{I}},$$

i.e. the body behaves as a simple pendulum of *effective length* $\ell_{\text{eff}} = I/Md$.

For instance, for a uniform rod of length ℓ pivoted at one end, we have $d = \ell/2$ and (from our earlier example, with $2a = \ell$) $I = \frac{1}{3}M\ell^2$, so

$$\ddot{\theta} = -\frac{3g}{2\ell} \sin \theta,$$

and the rod swings like a simple pendulum of length $2\ell/3$. The same equation drops out of energy conservation: differentiating $E = \frac{1}{2}I\dot{\theta}^2 - Mgd\cos\theta$ with respect to time and cancelling $\dot{\theta}$ recovers the equation of motion.

Example 3.6.10 (The Centre of Percussion)

Here is a question every cricketer cares about: where should the ball strike the bat so that your hands feel no jarring? Model the bat as a uniform rod of mass M and length ℓ , held (pivoted) at one end, and suppose the ball delivers a sharp horizontal impulse J at a distance b from the pivot. During the blow, the pivot exerts some impulsive reaction R on the rod.

Take angular impulse about the pivot (about which $I = \frac{1}{3}M\ell^2$): the reaction has no moment there, so

$$I \Delta\omega = Jb \implies \Delta\omega = \frac{Jb}{I}.$$

Meanwhile the linear impulse determines the change in velocity of the centre of mass, which sits at $d = \ell/2$ from the pivot and moves with velocity $d \Delta\omega$ after the blow:

$$J + R = Md \Delta\omega = \frac{MdJb}{I} \implies R = J \left(\frac{Mdb}{I} - 1 \right).$$

The reaction at the hands vanishes exactly when

$$b = \frac{I}{Md} = \frac{\frac{1}{3}M\ell^2}{M\ell/2} = \frac{2\ell}{3}.$$

This point is called the **centre of percussion** – strike there and the bat pivots cleanly about your hands with no sting; strike elsewhere and $R \neq 0$, which your palms report immediately. Note it is the same point $2\ell/3$ that appeared as the ‘effective length’ of the swinging rod, which is no coincidence: both are the point where the rod behaves like an equivalent simple pendulum.

§3.6.7 Rolling Without Slipping

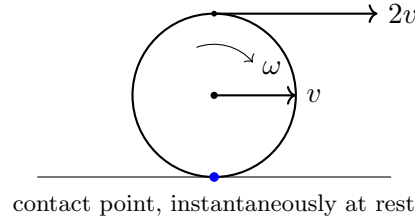
Consider a circular object – a cylinder, hoop or sphere – of radius a , moving along a surface, with its centre moving at speed v and the body rotating at angular speed ω about its centre. The velocity of the material point of the body instantaneously in contact with the surface is

$$v_{\text{slip}} = v - a\omega.$$

There are two qualitatively different regimes:

- *Sliding*: $v_{\text{slip}} \neq 0$. The contact point scrapes along the surface, kinetic friction μN acts to oppose the slipping, and mechanical energy is dissipated.
- *Rolling*: $v_{\text{slip}} = 0$, i.e. $v = a\omega$. This is the **rolling without slipping** condition. The contact point is instantaneously stationary – so although friction may act (and is often what *enforces* rolling), it acts at a point moving with zero velocity, and hence *does no work*. Energy is conserved for rolling bodies.

A rolling body can equivalently be viewed as rotating with angular speed ω about the (instantaneously stationary) contact point.



Example 3.6.11 (Rolling Down a Slope)

A uniform cylinder or sphere of mass M , radius a and moment of inertia I (about its axis through the centre of mass) rolls without slipping down a plane inclined at angle α . Let x be the distance travelled down the slope, so the rolling condition gives $\omega = \dot{x}/a$. The energy

$$E = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}I\omega^2 - Mgx \sin \alpha = \frac{1}{2} \left(M + \frac{I}{a^2} \right) \dot{x}^2 - Mgx \sin \alpha$$

is conserved (friction does no work while rolling), so differentiating and cancelling $\dot{x}$,

$$\ddot{x} = \frac{g \sin \alpha}{1 + I/Ma^2}.$$

The body accelerates more slowly than the sliding value $g \sin \alpha$, because some of the released potential energy must be spent spinning it up. The acceleration depends only on the *shape* through the ratio I/Ma^2 : a solid sphere ($I = \frac{2}{5}Ma^2$) has $\ddot{x} = \frac{5}{7}g \sin \alpha$, a solid cylinder ($\frac{1}{2}Ma^2$) has $\frac{2}{3}g \sin \alpha$, and a hoop (Ma^2) has $\frac{1}{2}g \sin \alpha$. So in a race down a hill, the sphere beats the cylinder, which beats the hoop, with the masses and radii entirely irrelevant. The same result follows from the force/torque equations $M\dot{v} = Mg \sin \alpha - F$ and $I\dot{\omega} = aF$, where F is the friction force enforcing the rolling.

Example 3.6.12 (The Struck Snooker Ball)

A snooker ball (a uniform sphere, $I = \frac{2}{5}Ma^2$) is struck centrally, so it initially translates with speed v_0 but does not spin: $v = v_0$, $\omega = 0$, and so $v_{\text{slip}} = v_0 > 0$. The ball is sliding, so kinetic friction $F = \mu_k Mg$ acts backwards at the contact point, and

$$M\dot{v} = -\mu_k Mg, \quad I\dot{\omega} = a\mu_k Mg.$$

So the ball slows down while spinning up:

$$v(t) = v_0 - \mu_k g t, \quad a\omega(t) = \frac{5}{2} \mu_k g t,$$

and the slip velocity $v_{\text{slip}} = v_0 - \frac{7}{2} \mu_k g t$ decreases to zero at time $t_{\text{roll}} = \frac{2v_0}{7\mu_k g}$. At that moment $v = a\omega = \frac{5}{7}v_0$, rolling begins, friction ceases to act (no applied force, no tendency to slip), and the ball rolls on at constant speed $\frac{5}{7}v_0$. The kinetic energy at this point is

$$\frac{1}{2} \left(M + \frac{I}{a^2} \right) v^2 = \frac{7}{10} M \cdot \frac{25}{49} v_0^2 = \frac{5}{14} M v_0^2 < \frac{1}{2} M v_0^2,$$

the difference having been dissipated by friction during the sliding phase.

Example 3.6.13 (When Does Rolling Fail?)

Return to the body on the inclined plane, and ask: how steep can the slope be before rolling becomes impossible? While rolling, the friction force required to sustain the motion is, from the equations $M\dot{v} = Mg \sin \alpha - F$ and $I\dot{\omega} = aF$ with $\dot{v} = a\dot{\omega}$,

$$F = \frac{I/a^2}{M + I/a^2} Mg \sin \alpha.$$

But static friction can supply at most $\mu N = \mu Mg \cos \alpha$. So rolling is sustainable only if

$$\tan \alpha \leq \mu \left(1 + \frac{Ma^2}{I} \right).$$

On steeper slopes the contact point must slip, and the body slides as it spins. Note the shape-dependence again: a hoop ($I = Ma^2$) demands the most friction and starts slipping first, while a solid sphere holds on to steeper slopes.

§3.7 Special Relativity

We now leave Newtonian mechanics behind. The trouble began with electromagnetism: Maxwell's equations predict that light travels at a definite speed $c \approx 3 \times 10^8 \text{ m s}^{-1}$ – but a speed relative to *what*? Under Galilean transformations, velocities simply add, so different inertial observers should measure different speeds of light, and experiments (most famously that of Michelson and Morley) resolutely failed to detect any such difference. Einstein's 1905 resolution was radical: keep the principle of relativity, keep the constancy of the speed of light, and let our assumptions about space and time – including Galilean transformations and absolute time itself – fall where they may.

§3.7.1 Axioms of Special Relativity

The whole of special relativity is built upon two deceptively simple postulates.

Axiom 3.7.1 (Postulates of Special Relativity) 1. *Principle of relativity.* The laws of physics are the same in all inertial reference frames.

2. *Constancy of the speed of light.* The speed of light in a vacuum is the same in all inertial reference frames.

The first postulate is just Galilean relativity as before. The second sounds innocuous but is flatly incompatible with the Galilean transformations: if a light ray passes you at speed c , and I chase after it at speed $c/2$, the second postulate insists that I *also* measure the light receding at speed c , not $c/2$. Clearly velocity addition – and with it, our notions of time and length – must be rebuilt from scratch.

Aside: The Michelson–Morley Experiment

Nineteenth-century physics held that light waves, like all other waves, must propagate in *some* medium – the ‘luminiferous ether’ – and that Maxwell’s c was the speed of light relative to it. The Earth, orbiting the Sun at about 30 km s^{-1} , ought then to be moving through the ether, and light should travel at measurably different speeds in different directions.

In 1887, Michelson and Morley set out to detect this using an interferometer: a light beam is split, sent down two perpendicular arms, reflected back, and recombined, with any difference in travel times showing up as a shift in the interference fringes as the apparatus is rotated. The expected effect was well within their sensitivity. They found *nothing* – no fringe shift, in any season, in any orientation. It is one of the most famous null results in the history of science.

Various rescues were attempted, most notably by FitzGerald and Lorentz, who observed that the null result could be explained if objects moving through the ether contracted by a factor $\sqrt{1 - v^2/c^2}$ in their direction of motion – the right formula, proposed as a mysterious dynamical accident. Einstein’s 1905 insight was that no ether and no accident are needed: if the speed of light is simply the same in every inertial frame, the contraction (and much else) follows from the geometry of space and time itself. That is the theory we now develop.

§3.7.2 Lorentz Transformations

In special relativity we think in terms of **events**: instantaneous, point-like occurrences. These are specified by four coordinates, one of time and three of position, like (t, x, y, z) . These coordinates will be measured differently in different inertial frames, and to make our axioms hold we need to use a new set of transformation laws.

Let $\mathcal{F}$ and $\mathcal{F}'$ be inertial frames, with $\mathcal{F}'$ moving at speed v relative to $\mathcal{F}$ in the x direction, in a *standard configuration*: axes aligned, origins coinciding at $t = t' = 0$. What could the transformation between them look like? We will sketch how the axioms pin it down. First, the transformation must map unaccelerated motion to unaccelerated motion (both frames are inertial), and this forces it to be *linear* in (t, x) . Motion transverse to the boost cannot be affected (by symmetry of the setup) so $y' = y$ and $z' = z$, and the origin of $\mathcal{F}'$ (i.e. $x' = 0$) must move along $x = vt$, so

$$x' = \gamma(x - vt)$$

for some factor $\gamma(v)$. By the principle of relativity, the inverse transformation must take the same form with $v \mapsto -v$ (and the same γ , which can depend only on $|v|$, again by symmetry):

$$x = \gamma(x' + vt').$$

Now impose the second postulate: a light ray through the origin has $x = ct$ in $\mathcal{F}$, and must also have $x' = ct'$ in $\mathcal{F}'$. Substituting both into the two equations above,

$$ct' = \gamma t(c - v), \quad ct = \gamma t'(c + v),$$

and multiplying these together gives $c^2 = \gamma^2(c - v)(c + v)$, so

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

the famous **Lorentz factor**. Eliminating x' between the two boxed-style relations then yields the transformation of time. The result is the **Lorentz transformation**:

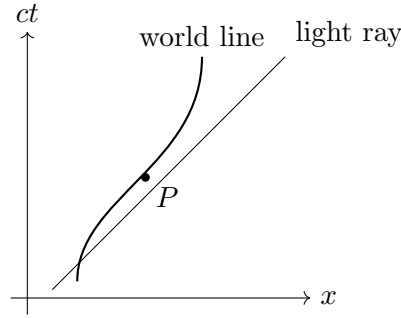
$$\begin{aligned} ct' &= \gamma \left(ct - \frac{v}{c}x \right) \\ x' &= \gamma \left(x - \frac{v}{c}ct \right) \\ y' &= y \\ z' &= z. \end{aligned}$$

A few observations to orient ourselves.

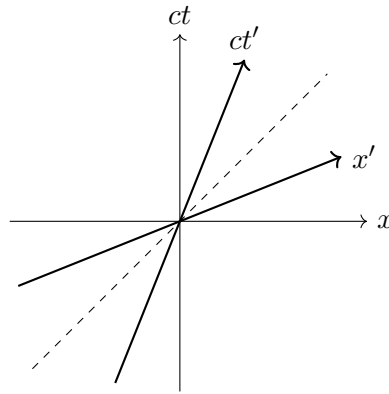
- (i) *Time transforms.* The first line is the revolution: t' depends on x as well as t . Different frames genuinely disagree about time, and the disagreement grows with separation in space. Absolute time is gone.
- (ii) *The Newtonian limit.* When v/c is very small, $\gamma \approx 1$ and the transformations reduce to $x' = x - vt$, $t' = t$: the Galilean boost. All of Newtonian mechanics survives as the low-speed approximation, which is why it took humanity so long to notice anything was wrong.
- (iii) *The speed limit.* The factor γ is real only for $|v| < c$, and $\gamma \rightarrow \infty$ as $|v| \rightarrow c$. The speed of light is built into the geometry as a limiting speed for frames (and, as we will see, for particles).
- (iv) *Symmetry.* The map treats ct and x almost symmetrically, mixing them rather like a rotation mixes coordinates – a hint we will make precise when we study rapidity.

§3.7.3 Spacetime Diagrams

Relativity problems reward drawing pictures. A **spacetime diagram** plots position x horizontally against time *scaled as* ct vertically (so that both axes carry dimensions of length). A point on the diagram is an event; the history of a particle traces out a curve called its **world line**. A particle at rest has a vertical world line; a particle in uniform motion has a straight world line tilted from the vertical; and a light ray has a world line at exactly 45° , since it satisfies $x = \pm ct$.



Now the crucial construction: how do the axes of a moving frame $\mathcal{F}'$ appear on the diagram of $\mathcal{F}$? The ct' axis is the set of events with $x' = 0$, which is the line $x = vt$ – a line through the origin with gradient $c/v > 1$, i.e. tilted from the vertical. The x' axis is the set of events with $t' = 0$, which by the Lorentz transformation is the line $ct = \frac{v}{c}x$ – tilted *up* from the horizontal by the same angle. So the primed axes close in symmetrically, like a pair of scissors, about the 45° light ray:



The primed axes are *not* orthogonal on the page – but both frames agree on the world lines of light rays, which bisect the axes in each frame. Almost every counterintuitive result of the subject (the relativity of simultaneity, time dilation, length contraction) becomes nearly obvious once drawn on such a diagram, so we will use them liberally in what follows.

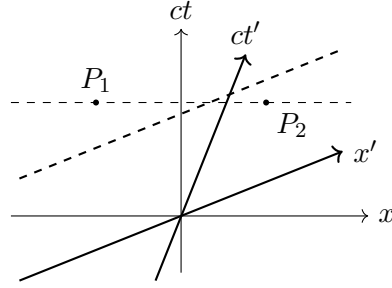
§3.7.4 Simultaneity

We say two events P_1 and P_2 are *simultaneous* in the frame $\mathcal{F}$ if they have the same time coordinate t . On a spacetime diagram, the events simultaneous with a given event form a horizontal line – a *line of simultaneity* of $\mathcal{F}$.

But the lines of simultaneity of $\mathcal{F}'$ are the lines of constant t' , which by the Lorentz transformation are the lines

$$ct - \frac{v}{c}x = \text{const},$$

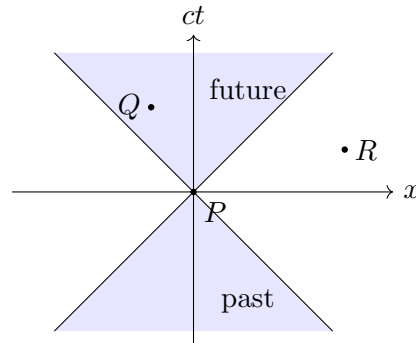
which are *tilted* lines parallel to the x' axis. Events with the same t no longer correspond to events with equal t' , so what is simultaneous in one frame is not necessarily simultaneous in another.



In the diagram, P_1 and P_2 are simultaneous in $\mathcal{F}$ (they lie on a horizontal dashed line), but the tilted line of simultaneity of $\mathcal{F}'$ through P_1 passes *above* P_2 – so in $\mathcal{F}'$, the event P_2 happens first. This is a genuine disagreement, not an artefact of light taking time to reach anyone's eyes: the coordinates already account for all such delays.

§3.7.5 Causality & Light Cones

If observers can disagree about the order of events, should we worry about effects preceding their causes? Consider an event P , which we may as well place at the origin. The light rays through P divide spacetime into regions, and this division is *frame independent*, since all observers agree on the speed of light:



The region $|x| \leq ct$ (shaded) is the **light cone** of P : the *future light cone* consists of all events reachable from P by influences travelling at up to the speed of light, and the *past light cone* of all events that can influence P . For an event Q inside the future light cone of P , one can check from the Lorentz transformation that *every* inertial frame agrees that Q occurs after P – so cause and effect are never inverted. For an event R *outside* the light cone, frames can disagree about the order of P and R (and there is even a frame in which they are simultaneous) – but no signal travelling at speed at most c can connect them, so neither can influence the other, and no causal paradox arises. Causality survives, provided nothing travels faster than light.

§3.7.6 Time Dilation

Consider a clock sitting stationary at the origin of the frame $\mathcal{F}'$, ticking at intervals of T' . The tick events in frame $\mathcal{F}'$ occur at $(ct'_1, 0), (c(t'_1 + T'), 0), \dots$, all with $x' = 0$. Using the inverse Lorentz transformation $t = \gamma(t' + vx'/c^2)$ with $x' = 0$, the interval between ticks observed in $\mathcal{F}$ is

$$T = \gamma T' > T'.$$

So the interval between ticks is longer in the frame in which the clock moves: *moving clocks run slowly*. The effect is mutual – a clock at rest in $\mathcal{F}$ runs slowly as measured

in $\mathcal{F}'$ – which sounds paradoxical, but the two statements compare different pairs of measurements, made along different lines of simultaneity, and no contradiction can be manufactured (as the twin ‘paradox’ below illustrates).

Example 3.7.2 (Muons in the Atmosphere)

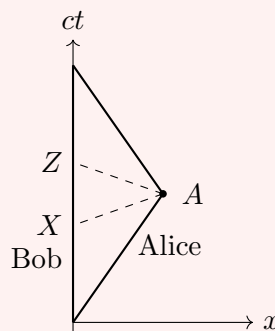
Muons are unstable particles with a lifetime of about $2.2\ \mu\text{s}$, created by cosmic rays striking the upper atmosphere, around 10 km up. Even travelling at essentially the speed of light, a muon should manage only about $c \times 2.2\ \mu\text{s} \approx 660\text{ m}$ before decaying – yet muons are detected in abundance at sea level. The resolution is time dilation: for a muon with $\gamma \approx 20$, its internal clock ticks 20 times slower as observed from the ground, giving it ample time to complete the journey. (From the muon’s own point of view, its clock runs normally – but, as we will see next, the atmosphere is length-contracted to a twentieth of its height.)

Nor is time dilation exotic: the clocks on GPS satellites, orbiting at about 3.9 km s^{-1} , run slow relative to the ground by around 7 microseconds per day due to time dilation (partially offset by a general-relativistic effect of the weaker gravity up there). Uncorrected, the resulting positioning error would accumulate at kilometres per day – your phone does relativity every time it finds you on a map.

Example 3.7.3 (The Twin Paradox)

Consider twins Alice and Bob. Bob stays at home (an inertial frame). Alice travels at constant speed v to a distant star, a distance d away in Bob’s frame, then immediately turns around and returns at the same speed. In Bob’s frame the journey takes $T = d/v$ each way, so Bob ages $2T$. Alice’s clock, moving throughout, runs slow by a factor γ , so she ages only $2T/\gamma$: Alice returns *genuinely younger* than her twin.

The apparent paradox: from Alice’s point of view, isn’t it Bob who moves, so shouldn’t Bob be younger? The asymmetry is that Bob remains inertial throughout while Alice does not – she turns around, and the situation is not symmetric. The spacetime diagram makes it precise:



On the outward leg, Alice’s line of simultaneity through the turnaround event A meets Bob’s world line at X – at that moment (by her reckoning) Bob has aged only T/γ^2 , and indeed each twin considers the other’s clock slow, symmetrically. But the moment Alice turns around, her frame changes, and her new line of simultaneity through A meets Bob’s world line at Z . From Alice’s perspective, Bob ages abruptly from X to Z during her turnaround. Adding up the pieces, both twins agree at the

reunion: Bob has aged $2T$, and Alice $2T/\gamma$. There is no paradox – only the (false) assumption that Alice’s viewpoint is that of a single inertial frame.

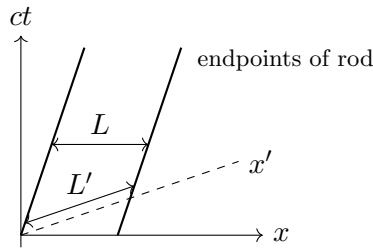
§3.7.7 Length Contraction

Consider a rod of length L' , stationary in the frame $\mathcal{F}'$. What is its length measured in $\mathcal{F}$? Some care is needed: the length of a moving object means the distance between its two ends *measured at the same time* – and we now know that ‘at the same time’ is a frame-dependent notion, which is exactly where the interesting physics hides.

The world lines of the endpoints of the rod are given by $x' = 0$ and $x' = L'$, which map under the Lorentz transformation into $x = vt$ and $x = vt + L'/\gamma$. So in $\mathcal{F}$, measuring at a fixed time t , the length of the rod is

$$L = \frac{L'}{\gamma} < L',$$

and thus lengths of moving objects are *contracted* in the direction of motion. (Lengths perpendicular to the motion are unaffected.) To deal with this, we define the **proper length** to be the length measured in an object’s rest frame; it is the longest length any inertial observer measures.



On the spacetime diagram, $\mathcal{F}$ measures the rod along a horizontal line of simultaneity, while $\mathcal{F}'$ measures it along the tilted x' axis – they are simply measuring *different* pairs of events, which is how both can be right.

Example 3.7.4 (The Train in the Station)

A train of proper length $2L$ travels at speed v , with $\gamma = 2$, through a station whose platform has length L . For observers on the platform, the train is contracted to length $2L/\gamma = L$, so for one instant it fits perfectly alongside the platform. But for observers on the train, it is the *platform* that is contracted, to $L/2$, and the train (length $2L$) never comes close to fitting!

Who is right? Both. ‘The train fits’ means ‘both ends of the train are alongside the platform *simultaneously*’ – and simultaneity is frame-dependent. The two events ‘front of train reaches far end of platform’ and ‘back of train passes near end of platform’ are simultaneous in the platform frame, but occur in a different order in the train frame. Once stated carefully, there is no contradiction, and the spacetime diagram of the two pairs of world lines shows exactly how the frames disagree.

§3.7.8 Composition of Velocities

Suppose a particle moves with constant velocity u' in frame $\mathcal{F}'$, which moves with velocity v relative to $\mathcal{F}$. We want to find its velocity u in the frame $\mathcal{F}$.

In $\mathcal{F}'$, the world line of the particle is $x' = u't'$. Substituting this into the (inverse) Lorentz transformation laws, we have

$$u = \frac{x}{t} = \frac{\gamma(x' + vt')}{\gamma(t' + vx'/c^2)} = \frac{u' + v}{1 + u'v/c^2}.$$

This is the relativistic law for the **composition of velocities**, replacing the Galilean $u = u' + v$. Some features worth noting:

- (i) When $u'v \ll c^2$, the denominator is close to 1 and we recover the Galilean answer – velocities approximately add at everyday speeds.
- (ii) If $u' = c$ then $u = (c+v)/(1+v/c) = c$, for *any* v : composing any velocity with the speed of light gives the speed of light, exactly as the second postulate demands.
- (iii) If $|u'| < c$ and $|v| < c$, then $|u| < c$. Try as we might, we cannot boost a particle past the speed of light by chaining together boosts – the speed of light is unreachable, not merely unreachd.

Example 3.7.5

A ship recedes from Earth at $\frac{1}{2}c$ and fires a probe forwards at $\frac{1}{2}c$ relative to itself. Galileo would have the probe moving at c ; instead,

$$u = \frac{\frac{1}{2}c + \frac{1}{2}c}{1 + \frac{1}{4}} = \frac{4}{5}c.$$

If the probe in turn launches its own sub-probe at $\frac{1}{2}c$, the Earth measures $\frac{(4/5+1/2)c}{1+2/5} = \frac{13}{14}c$ – each stage claws closer to c , with ever diminishing returns.

§3.7.9 Geometry of Minkowski Space

Consider two events P_1 and P_2 have coordinates (t_1, x_1) and (t_2, x_2) in the frame $\mathcal{F}$. These events are separated by $\Delta t = t_1 - t_2$ in time and $\Delta x = x_1 - x_2$.

We define the *invariant interval* between P_1 and P_2 to be

$$\Delta s^2 = c^2 \Delta t^2 - \Delta x^2.$$

We say it is invariant because it is the same in all inertial reference frames, that is, it is invariant under Lorentz transformations. This is a direct computation:

$$\begin{aligned} c^2 \Delta t'^2 - \Delta x'^2 &= \gamma^2 \left(c \Delta t - \frac{v}{c} \Delta x \right)^2 - \gamma^2 \left(\Delta x - \frac{v}{c} c \Delta t \right)^2 \\ &= \gamma^2 \left(1 - \frac{v^2}{c^2} \right) (c^2 \Delta t^2 - \Delta x^2) = c^2 \Delta t^2 - \Delta x^2, \end{aligned}$$

where the cross terms cancel. (In more than one spatial dimension, the interval is $\Delta s^2 = c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$, and the transverse coordinates are untouched by the boost.) The invariant interval plays the role in spacetime that squared distance plays in Euclidean geometry: it is the quantity that all observers agree on, even as they disagree about times and lengths separately.

Example 3.7.6

In the frame $\mathcal{F}$, two events are separated by $\Delta x = 3$ light-years and $c\Delta t = 5$ light-years. What does a ship travelling at $v = \frac{3}{5}c$ (so $\gamma = \frac{5}{4}$) between the two events make of them? Applying the Lorentz transformation to the separations,

$$\Delta x' = \gamma \left(\Delta x - \frac{v}{c} c\Delta t \right) = \frac{5}{4}(3-3) = 0, \quad c\Delta t' = \gamma \left(c\Delta t - \frac{v}{c} \Delta x \right) = \frac{5}{4} \left(5 - \frac{9}{5} \right) = 4.$$

In the ship's frame the two events happen at the *same place* (of course – the ship is present at both), separated by 4 years of ship time. And indeed the invariant interval agrees in both frames:

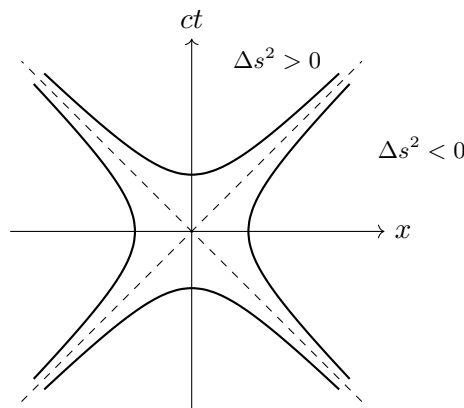
$$\Delta s^2 = 5^2 - 3^2 = 16 = 4^2 - 0^2,$$

with $\Delta s/c = 4$ years being precisely the time experienced aboard – the *proper time* between the events, computed without ever leaving the frame $\mathcal{F}$.

It is possible for Δs^2 to be either positive or negative. If it is positive, we say the events are *timelike* separated, and if it is negative we say they are *spacelike* separated, and if it is zero, we say they are *lightlike* (or *null*) separated. This classification is exactly the light cone structure from before: timelike separated events are in each other's light cones and can influence one another, spacelike separated events cannot, and lightlike separated events can be joined only by a light ray.

Each case has a natural interpretation. For timelike separated events, there is a frame in which the two events happen at the same place, and $\Delta s/c$ is the time between them in that frame – the *proper time*, of which much more shortly. For spacelike separated events, there is a frame in which they are simultaneous, and $\sqrt{-\Delta s^2}$ is their separation in that frame (the *proper distance*).

There is a nice picture of all this. On a spacetime diagram, the events at fixed invariant interval from the origin form *hyperbolae* $c^2t^2 - x^2 = \text{const}$, with the light cone $x = \pm ct$ as their asymptotes:



Lorentz transformations preserve the interval, so they slide events *along* these hyperbolae – just as rotations of the plane slide points along circles. This is the precise sense in which Lorentz transformations are the ‘rotations’ of spacetime geometry, and we will shortly make the analogy exact using rapidities.

§3.7.10 4-Vectors

We can view Minkowski space as a vector space equipped with the *Minkowski inner product*⁶. The coordinates of some event P in the frame $\mathcal{F}$ can be written as a *4-vector*

$$X = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix},$$

and then the Minkowski inner product is given by

$$X \cdot Y = X^T \eta Y, \quad \text{where} \quad \eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Taking $X \cdot X$ gives the invariant interval between the origin and P , and is known as the *Minkowski metric*⁷.

More generally, a **4-vector** is any four-component quantity that transforms under a change of inertial frame in the same way that X does (via the Lorentz transformation matrices of the next subsection). The point of the definition is that the Minkowski inner product of any two 4-vectors is then automatically frame independent – so building physics out of 4-vectors and their inner products is a machine for generating statements that all inertial observers agree on. We will exploit this repeatedly in the final section.

A quick note on notation, for compatibility with the wider world: the components of a 4-vector are commonly written with a Greek superscript, X^μ for $\mu = 0, 1, 2, 3$, with $X^0 = ct$ the time component. We will not need the index machinery in this course (it comes into its own in the Part IB ‘Electromagnetism’ and Part II relativity courses), but it is useful to recognise it.

§3.7.11 The Lorentz Group

Using 4-vectors, we can view Lorentz transformations as linear transformations from the coordinates of one inertial frame $\mathcal{F}$ to another $\mathcal{F}'$. This would be represented by a 4×4 matrix Λ , with

$$X' = \Lambda X.$$

Since such transformations must preserve the invariant interval, they must preserve the Minkowski inner product⁸ and hence satisfy

$$\Lambda^T \eta \Lambda = \eta.$$

The group of all such matrices is the *Lorentz group*, and is generated by

$$\left(\begin{array}{c|ccc} 1 & 0 & 0 & 0 \\ \hline 0 & & & \\ 0 & & R & \\ 0 & & & \end{array} \right) \quad \text{and} \quad \left(\begin{array}{cc|cc} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right),$$

where the first corresponds to a rotation of space, leaving time intact (so we need R to be orthogonal), and the second corresponds to a Lorentz boost along the x direction.

⁶This is not an inner product in the normal sense since it's not positive definite

⁷Again, this isn't actually a metric in the normal sense since it's not positive definite

⁸These are the analogue of orthogonal matrices for Minkowski space

Remark. Taking determinants of $\Lambda^T \eta \Lambda = \eta$ gives $(\det \Lambda)^2 = 1$, so $\det \Lambda = \pm 1$, and one can also show $|\Lambda_0^0| \geq 1$. The physically relevant transformations – those continuously connected to the identity – have $\det \Lambda = +1$ and $\Lambda_0^0 \geq 1$: they preserve both spatial orientation and the direction of time. The remaining components of the group involve spatial reflections or time reversal.

§3.7.12 Rapidities

Consider a 2×2 matrix corresponding to a Lorentz boost by v in the x direction. Then if $\beta = v/c$, we have

$$\Lambda[\beta] = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$$

and the composition law $\Lambda[\beta_1]\Lambda[\beta_2] = \Lambda\left[\frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}\right]$, which is exactly our composition law from before.

A slightly nicer way to deal with this is with *rapidities*. We define the *rapidity* of a Lorentz boost ϕ such that

$$\beta = \tanh \phi, \quad \gamma = \cosh \phi, \quad \gamma\beta = \sinh \phi.$$

Then

$$\Lambda[\beta] = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix} = \Lambda(\phi),$$

and also

$$\Lambda(\phi_1)\Lambda(\phi_2) = \Lambda(\phi_1 + \phi_2),$$

which is much nicer. This also shows that Lorentz boosts correspond to hyperbolic rotations in spacetime, just as the rotation matrices (with $\cos$ and $\sin$) represent ordinary rotations of space. The trigonometry of Minkowski space is hyperbolic trigonometry.

Rapidities also give the slickest possible derivation of the composition of velocities: composing boosts of rapidities ϕ_1 and ϕ_2 gives a boost of rapidity $\phi_1 + \phi_2$, whose velocity is

$$\frac{u}{c} = \tanh(\phi_1 + \phi_2) = \frac{\tanh \phi_1 + \tanh \phi_2}{1 + \tanh \phi_1 \tanh \phi_2} = \frac{v_1/c + v_2/c}{1 + v_1 v_2 / c^2},$$

by the addition formula for $\tanh$ – our composition law falls straight out. And since $\tanh$ maps $\mathbb{R}$ to $(-1, 1)$, no amount of adding rapidities ever reaches the speed of light: on the rapidity scale, velocities add plainly, and c sits at infinity.

§3.8 Relativistic Physics

Having rebuilt the geometry of space and time, we now rebuild *dynamics*: velocity, momentum, energy and force, all in a form that respects Lorentz transformations. The guiding principle is to phrase everything in terms of 4-vectors and Lorentz invariants, so that the physics manifestly takes the same form in every inertial frame. The payoff comes at the end of the section, where conservation of 4-momentum will let us analyse particle collisions, decays and creation processes with remarkable ease.

§3.8.1 Proper Time

To do kinematics, it helps to have a notion of time that all observers can agree on. We do this by defining **proper time** τ , which is the time experienced by a particle in its own reference frame. It is given by

$$\Delta\tau = \frac{\Delta s}{c}$$

along the particle's world line, in all reference frames – this is well defined precisely because Δs is Lorentz invariant. (For a massive particle, successive events on the world line are timelike separated, so $\Delta s^2 > 0$ and the proper time is real.)

We can relate proper time to the time measured in any particular frame by considering small changes. If $\mathbf{u}$ is the velocity of the particle in the frame $\mathcal{F}$, we get

$$d\tau = \frac{1}{c} ds = \frac{1}{c} \sqrt{c^2 dt^2 - |\mathbf{dx}|^2} = \sqrt{1 - u^2/c^2} dt,$$

so

$$\frac{dt}{d\tau} = \gamma_u, \quad \text{where } \gamma_u = \frac{1}{\sqrt{1 - u^2/c^2}}.$$

Note that γ_u here is built from the *particle's* speed in $\mathcal{F}$, which need not be constant. We can then get the total time experienced by a particle along a segment of its world line as

$$T = \int d\tau = \int \frac{1}{\gamma_u} dt \leq \int dt.$$

So of all observers passing through the same two events, the one who experiences the most time is the one moving inertially – a restatement of the twin paradox.

Example 3.8.1

An astronaut spends six months on the International Space Station, orbiting at about 7.7 km s^{-1} . Over the mission, the proper time deficit relative to mission control is

$$\Delta t - \Delta\tau = \left(1 - \frac{1}{\gamma}\right) \Delta t \approx \frac{v^2}{2c^2} \Delta t \approx \frac{(7.7 \times 10^3)^2}{2(3 \times 10^8)^2} \times 1.6 \times 10^7 \text{ s} \approx 5 \text{ ms}.$$

The astronaut returns five milliseconds younger than if they had stayed home – a small effect, but one routinely confirmed by flying atomic clocks around the world.

§3.8.2 4-Velocity

Using proper time, we can parametrise the trajectory of a particle using a 4-vector

$$X(\tau) = \begin{pmatrix} ct(\tau) \\ \mathbf{x}(\tau) \end{pmatrix},$$

the *4-position*, and then define the **4-velocity** as its derivative with respect to proper time:

$$U = \frac{dX}{d\tau} = \begin{pmatrix} c dt/d\tau \\ d\mathbf{x}/d\tau \end{pmatrix} = \frac{dt}{d\tau} \begin{pmatrix} c \\ \mathbf{u} \end{pmatrix} = \gamma_u \begin{pmatrix} c \\ \mathbf{u} \end{pmatrix},$$

where $\mathbf{u} = d\mathbf{x}/dt$ is the ordinary velocity in the frame. The reason to differentiate with respect to τ rather than t is that τ is an invariant: since $X' = \Lambda X$ and τ is frame

independent, the 4-velocity genuinely transforms as a 4-vector, $U' = \Lambda U$. (The naive candidate dX/dt transforms in no nice way at all.)

Like any 4-vector, U has an invariant inner product with itself. Easiest computed in the particle's rest frame, where $U = (c, \mathbf{0})$:

$$U \cdot U = c^2,$$

and indeed in a general frame $U \cdot U = \gamma_u^2(c^2 - |\mathbf{u}|^2) = c^2$. Every particle traverses spacetime with '4-speed' c .

§3.8.3 Transforming Velocities & Aberration

Since $U' = \Lambda U$, transforming velocities between frames is now just linear algebra – which lets us generalise our composition law to velocities in arbitrary directions. Suppose in frame $\mathcal{F}$ a particle moves with speed u at angle θ to the x -axis, in the (x, y) plane, and let $\mathcal{F}'$ move at speed v in the x direction as usual. Applying the boost matrix to $U = \gamma_u(c, u \cos \theta, u \sin \theta, 0)^T$:

$$\begin{pmatrix} \gamma_{u'} c \\ \gamma_{u'} u' \cos \theta' \\ \gamma_{u'} u' \sin \theta' \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma_v & -\gamma_v v/c & 0 & 0 \\ -\gamma_v v/c & \gamma_v & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \gamma_u c \\ \gamma_u u \cos \theta \\ \gamma_u u \sin \theta \\ 0 \end{pmatrix}.$$

Dividing the second row by the first eliminates the γ factors and gives

$$u' \cos \theta' = \frac{u \cos \theta - v}{1 - \frac{uv}{c^2} \cos \theta},$$

the familiar composition law applied to the parallel component. Dividing the third row by the second gives

$$\tan \theta' = \frac{u \sin \theta}{\gamma_v (u \cos \theta - v)},$$

which describes **aberration**: the direction of motion of a particle depends on the motion of the observer. There is a Newtonian part to this effect – walk through vertical rain and it appears to slant towards you – but the γ_v factor is genuinely relativistic. For light ($u = c$), aberration causes the apparent positions of stars to shift annually as the Earth swings around its orbit, an effect measured long before relativity explained it correctly.

§3.8.4 4-Momentum & Relativistic Energy

With the 4-velocity in hand, we can define the relativistic analogue of momentum.

Definition 3.8.2 (4-Momentum)

For a particle of mass m , the **4-momentum** is

$$P = mU = \begin{pmatrix} mc\gamma_u \\ m\gamma_u \mathbf{u} \end{pmatrix}.$$

The spatial components of P give us the *relativistic 3-momentum*, $\mathbf{p} = m\gamma_u \mathbf{u}$, which is the Newtonian momentum corrected by a factor of γ_u – note that $|\mathbf{p}| \rightarrow \infty$ as $|\mathbf{u}| \rightarrow c$.

The fundamental law of relativistic dynamics is that *4-momentum is conserved*: for a system of particles the total 4-momentum (the sum of the individual 4-momenta) is unchanged by collisions and interactions, in the absence of external forces. This single conservation law packages together the conservation of momentum *and* of energy, as we now see by interpreting the time component.

Expanding P^0 for speeds small compared to c ,

$$P^0 = m\gamma_u c = \frac{1}{c} \left(mc^2 + \frac{1}{2}m|\mathbf{u}|^2 + \dots \right).$$

The second term is the Newtonian kinetic energy, so we recognise $P^0 c$ as an energy.

Definition 3.8.3 (Relativistic Energy)

The **relativistic energy** of a particle is $E = P^0 c = m\gamma_u c^2$, so that

$$P = \begin{pmatrix} E/c \\ \mathbf{p} \end{pmatrix}, \quad \text{and} \quad E = mc^2 + \underbrace{(\gamma_u - 1)mc^2}_{\text{kinetic energy}}.$$

For a stationary particle we have the most famous equation in physics:

$$E = mc^2.$$

Mass is a form of energy – and an enormously concentrated one, with $c^2 \approx 9 \times 10^{16} \text{ J kg}^{-1}$ as the exchange rate. In relativity, mass is *not* separately conserved: it can be converted to kinetic energy and back, as happens in nuclear reactions and particle colliders, and only the total energy (including mass energy) is conserved.

Since we can calculate the Lorentz invariant quantity $P \cdot P$ in the particle's rest frame, we have

$$P \cdot P = \frac{E^2}{c^2} - |\mathbf{p}|^2 = m^2 c^2,$$

so generally we have the **energy-momentum relation**

$$E^2 = |\mathbf{p}|^2 c^2 + m^2 c^4.$$

This invariant is the workhorse of collision problems: given any 4-momentum (perhaps a sum over several particles), dotting it with itself produces frame-independent information.

Massless particles. For particles that have zero mass (like photons), we cannot define proper time or 4-velocity – but we can still perfectly well assign them a 4-momentum, with $P \cdot P = 0$. Such particles travel at the speed of light, along lightlike trajectories, and can still carry momentum and energy, related by $E = |\mathbf{p}|c$. Thus

$$P = \frac{E}{c} \begin{pmatrix} 1 \\ \mathbf{n} \end{pmatrix},$$

where $\mathbf{n}$ is a unit vector in the direction of propagation. Quantum mechanics supplies the energy in terms of the light's frequency f (or wavelength $\lambda = c/f$) via the *Planck relation* $E = hf = hc/\lambda$, where h is Planck's constant.

§3.8.5 The Relativistic Doppler Effect

Suppose a source of light recedes from an observer at speed v , emitting light of frequency f' (as measured in the source's own rest frame). What frequency does the observer measure? Two effects stack: successive wave crests are emitted at intervals lengthened by time dilation, *and* each crest is emitted from further away than the last, adding travel time.

Quantitatively: in the observer's frame, the source's clock ticks slowly, so crests are emitted at intervals $\gamma T'$, where $T' = 1/f'$. Between successive emissions the source recedes a further distance $v\gamma T'$, delaying the later crest's arrival by $v\gamma T'/c$. So crests arrive separated by

$$T = \gamma T' + \frac{v\gamma T'}{c} = T' \gamma \left(1 + \frac{v}{c}\right) = T' \sqrt{\frac{1 + v/c}{1 - v/c}},$$

and hence the observed frequency is

$$f = f' \sqrt{\frac{1 - v/c}{1 + v/c}}.$$

For a receding source, $f < f'$: the light is shifted towards the red end of the spectrum (*redshift*). For an approaching source, replace v with $-v$ to get a *blueshift*. An elegant alternative derivation: apply a Lorentz boost to the photon's 4-momentum, $E = \gamma(E' - vE'/c \cdot 1)$ for a photon emitted backwards, and use $E = hf$ – the same factor drops out.

Note that even the *transverse* Doppler effect (source moving perpendicular to the line of sight) is nontrivial in relativity, giving $f = f'/\gamma$ purely from time dilation, with no Newtonian counterpart.

The Doppler effect is a workhorse of astronomy: the redshifts of spectral lines tell us how fast stars and galaxies move along our line of sight, and the systematic redshift of distant galaxies – Hubble's observation that essentially everything is receding from us – is the observational cornerstone of the expanding universe.

It also gives a satisfying second look at the twin paradox. Suppose the travelling twin Alice sends Bob a light pulse every year of *her* time. On the outward leg, Bob receives the pulses at the slowed rate $\sqrt{(1 - \beta)/(1 + \beta)}$ per year; after her turnaround, at the sped-up rate $\sqrt{(1 + \beta)/(1 - \beta)}$. But here is the asymmetry: Alice's turnaround changes what she *sends* at her half-way point, while Bob only starts *receiving* the sped-up pulses late in the trip, when the last outbound pulse finally crawls to him. Counting the pulses each twin receives over the whole journey (a pleasant exercise) reproduces exactly the asymmetric ageing – no frames or lines of simultaneity required, just counting flashes of light.

§3.8.6 4-Force

We can now write down the relativistic version of Newton's second law. The natural formulation uses proper time and 4-vectors throughout.

Definition 3.8.4 (4-Force)

The **4-force** on a particle with 4-momentum P is

$$F = \frac{dP}{d\tau}.$$

Since P is a 4-vector and τ is invariant, F is a genuine 4-vector, transforming as $F' = \Lambda F$ – so if this law holds in one inertial frame, it holds in all of them, exactly as the principle of relativity requires. Writing out the components in a frame where the particle has velocity $\mathbf{u}$, and comparing with $P = (E/c, \mathbf{p})$, we can express the 4-force in terms of the ordinary 3-force $\mathbf{F} = d\mathbf{p}/dt$:

$$F = \gamma_u \begin{pmatrix} \mathbf{F} \cdot \mathbf{u}/c \\ \mathbf{F} \end{pmatrix},$$

where the factors of γ_u come from $dt/d\tau = \gamma_u$. The two components carry familiar content:

$$\frac{dE}{dt} = \mathbf{F} \cdot \mathbf{u}, \quad \frac{d\mathbf{p}}{dt} = \mathbf{F},$$

that is, the rate of working of the force equals the rate of change of (relativistic) energy, and force is the rate of change of (relativistic) momentum – Newtonian mechanics upgraded, with γ factors hiding inside E and $\mathbf{p}$.

For a particle of constant mass we can also write $F = mA$, where

$$A = \frac{dU}{d\tau}$$

is the *4-acceleration*. Differentiating the identity $U \cdot U = c^2$ with respect to τ gives

$$U \cdot A = 0,$$

so the 4-acceleration is always Minkowski-orthogonal to the 4-velocity. In the instantaneous rest frame, $U = (c, \mathbf{0})$ and $A = (0, \mathbf{a})$, making the orthogonality manifest.

Example 3.8.5 (Motion Under a Constant Force)

Push a particle of mass m with a constant 3-force F (in one dimension) forever. Newtonian mechanics says $v = Ft/m$ grows without bound; what does relativity say? We have $dp/dt = F$, so starting from rest,

$$p = m\gamma_u u = Ft \implies u = \frac{Ft/m}{\sqrt{1 + (Ft/mc)^2}},$$

solving for u . At early times ($Ft \ll mc$) this is the Newtonian $u \approx Ft/m$; at late times $u \rightarrow c$ but never reaches it – the momentum grows linearly forever, but ever more of the ‘push’ goes into the γ factor rather than into speed. Constant proper acceleration traces out a *hyperbola* on the spacetime diagram, asymptotic to a light ray: one more instance of hyperbolae being relativity’s circles.

§3.8.7 Particle Collisions & Decays

We finish the course with relativistic particle physics – the study of collisions, decays and creation of particles, where conservation of total 4-momentum

$$P = \begin{pmatrix} E/c \\ \mathbf{p} \end{pmatrix}$$

does essentially all of the work. A frame that makes calculations dramatically easier:

Definition 3.8.6 (Centre of Momentum Frame)

The **centre of momentum frame** of a system of particles is an inertial frame in which the total 3-momentum vanishes, $\sum_i \mathbf{p}_i = \mathbf{0}$.

Such a frame exists for any system except one consisting entirely of massless particles moving in a single direction. It is the relativistic analogue of the centre of mass frame from Newtonian collisions.

Remark. A closely related tool: for any system with total 4-momentum P_{tot} , the quantity

$$M_{\text{inv}}^2 c^2 = P_{\text{tot}} \cdot P_{\text{tot}}$$

defines the *invariant mass* of the system – it is frame independent, equal to the total energy in the centre of momentum frame divided by c^2 , and (because 4-momentum is conserved) unchanged by any decay or collision the system undergoes. Note that the invariant mass of a system is generally *greater* than the sum of the rest masses of its constituents, the excess being internal kinetic energy. Reconstructing invariant masses from measured decay products is how new particles are actually discovered.

Particle Decay

Suppose a particle of mass m_1 decays into two particles of masses m_2 and m_3 . Conservation of 4-momentum reads

$$P_1 = P_2 + P_3,$$

which contains both $E_1 = E_2 + E_3$ and $\mathbf{p}_1 = \mathbf{p}_2 + \mathbf{p}_3$. Work in the rest frame of the decaying particle (its own centre of momentum frame). There, $E_1 = m_1 c^2$, and using the energy-momentum relation for the products,

$$m_1 c^2 = \sqrt{|\mathbf{p}_2|^2 c^2 + m_2^2 c^4} + \sqrt{|\mathbf{p}_3|^2 c^2 + m_3^2 c^4} \geq m_2 c^2 + m_3 c^2.$$

So decay is possible only if

$$m_1 \geq m_2 + m_3$$

– a particle can only decay into products whose total mass does not exceed its own, with the excess mass converted into the kinetic energy of the products. (Mass is not conserved!)

Example 3.8.7 (Higgs Decay to Photons)

One of the decay channels in which the Higgs boson was discovered at the LHC is a decay into two photons, $h \rightarrow \gamma\gamma$ (allowed, since $m_\gamma = 0$). In the Higgs rest frame,

$$P_h = \begin{pmatrix} m_h c \\ \mathbf{0} \end{pmatrix} = P_{\gamma_1} + P_{\gamma_2}.$$

The spatial components force $\mathbf{p}_{\gamma_1} = -\mathbf{p}_{\gamma_2}$: the photons fly off back-to-back. Since $E_\gamma = |\mathbf{p}_\gamma|c$ for each photon, the two photons have equal energies, and the time component then gives

$$E_{\gamma_1} = E_{\gamma_2} = \frac{1}{2} m_h c^2.$$

All of the Higgs mass is converted into electromagnetic energy. Experimentally one runs this backwards: measuring the photon pair and reconstructing the invariant

$(P_{\gamma_1} + P_{\gamma_2}) \cdot (P_{\gamma_1} + P_{\gamma_2}) = m_h^2 c^2$ reveals the mass of the particle that decayed.

Example 3.8.8 (Photon Emission & Recoil)

An atom of mass m , at rest, drops to a lower energy state of mass $m' < m$ (the internal energy released being $\Delta E = (m - m')c^2$), emitting a single photon. Naively the photon should carry energy ΔE – but the atom must recoil to conserve momentum, and the recoil steals some energy. Conservation of 4-momentum gives $P_{\text{atom}} = P_\gamma + P'_{\text{atom}}$, so isolating and squaring the final atom's 4-momentum,

$$m'^2 c^2 = (P_{\text{atom}} - P_\gamma) \cdot (P_{\text{atom}} - P_\gamma) = m^2 c^2 - 2P_{\text{atom}} \cdot P_\gamma = m^2 c^2 - 2mE_\gamma,$$

using $P_\gamma \cdot P_\gamma = 0$ and evaluating $P_{\text{atom}} \cdot P_\gamma = mE_\gamma$ in the rest frame. Hence

$$E_\gamma = \frac{(m^2 - m'^2)c^2}{2m} = \Delta E \left(1 - \frac{\Delta E}{2mc^2} \right).$$

The photon carries slightly *less* than the transition energy, the deficit being carried off as the atom's recoil kinetic energy. For optical transitions the correction is minuscule ($\Delta E/mc^2 \sim 10^{-10}$), but for gamma-ray emission from nuclei it matters enormously – and clever ways around the recoil (the Mössbauer effect) enabled some of the most precise tests of relativity ever performed.

Particle Scattering

Next, consider an *elastic* collision: two identical particles of mass m collide and retain their identities, so

$$P_1 + P_2 = P_3 + P_4.$$

Set up the standard experiment in the lab frame $\mathcal{F}$: particle 1 moves with speed u along the x -axis and strikes particle 2 at rest, after which the particles move off at angles θ and ϕ to the axis.

The trick is to analyse the collision in the centre of momentum frame $\mathcal{F}'$, which moves at some speed v along the x -axis relative to the lab. In $\mathcal{F}'$, the two particles approach head-on with equal and opposite momenta, and (comparing the time components of 4-momentum before and after) they must also *recede* with equal and opposite momenta of the same magnitude, each with the same speed v as before. The only freedom is the scattering angle θ' :

$$P'_3 = \begin{pmatrix} m\gamma_v c \\ m\gamma_v v \cos \theta' \\ m\gamma_v v \sin \theta' \\ 0 \end{pmatrix}, \quad P'_4 = \begin{pmatrix} m\gamma_v c \\ -m\gamma_v v \cos \theta' \\ -m\gamma_v v \sin \theta' \\ 0 \end{pmatrix}.$$

Now boost back to the lab frame with the inverse Lorentz transformation. The incoming particle's speed matches the collision setup provided

$$u = \frac{2v}{1 + v^2/c^2}$$

(the composition of v with v , naturally, since the target is at rest in the lab and moves at v in $\mathcal{F}'$). Transforming P'_3 and P'_4 and taking ratios of components to extract the

angles, the half-angle formulae give

$$\tan \theta = \frac{\sin \theta'}{\gamma_v(1 + \cos \theta')} = \frac{1}{\gamma_v} \tan \frac{\theta'}{2}, \quad \tan \phi = \frac{\sin \theta'}{\gamma_v(1 - \cos \theta')} = \frac{1}{\gamma_v} \cot \frac{\theta'}{2}.$$

The individual angles depend on θ' , which is not determined by conservation laws alone – but their product is universal:

$$\tan \theta \tan \phi = \frac{1}{\gamma_v^2}.$$

In the Newtonian limit $\gamma_v \rightarrow 1$ we recover $\tan \theta \tan \phi = 1$, i.e. $\theta + \phi = \pi/2$: the equal-mass particles separate at right angles, exactly as we found in Section 3.5. Relativistically, $\tan \theta \tan \phi < 1$, so the opening angle between the outgoing particles is *less* than a right angle – fast collisions throw both products forward, and measuring this closing-up of the angle is a direct test of special relativity.

Example 3.8.9 (Compton Scattering)

A lovely application of 4-momentum bookkeeping: scatter a photon off an electron (mass m) at rest, and ask how the photon's wavelength changes. Let the incoming photon have energy E , the outgoing photon energy E' at angle θ to the incident direction, and let the electron recoil with 4-momentum P'_e . Conservation says

$$P_\gamma + P_e = P'_\gamma + P'_e.$$

We don't care about the electron's recoil, so isolate it and take the invariant square:

$$P'_e \cdot P'_e = (P_\gamma + P_e - P'_\gamma) \cdot (P_\gamma + P_e - P'_\gamma).$$

The left side is $m^2 c^2$. Expanding the right side using $P_\gamma \cdot P_\gamma = P'_\gamma \cdot P'_\gamma = 0$ and $P_e \cdot P_e = m^2 c^2$,

$$m^2 c^2 = m^2 c^2 + 2P_\gamma \cdot P_e - 2P'_\gamma \cdot P_e - 2P_\gamma \cdot P'_\gamma,$$

and computing each inner product in the lab frame (where $P_e = (mc, \mathbf{0})$):

$$P_\gamma \cdot P_e = mE, \quad P'_\gamma \cdot P_e = mE', \quad P_\gamma \cdot P'_\gamma = \frac{EE'}{c^2}(1 - \cos \theta).$$

Substituting and tidying up,

$$\frac{1}{E'} - \frac{1}{E} = \frac{1 - \cos \theta}{mc^2},$$

or, in terms of wavelengths via the Planck relation $E = hc/\lambda$,

$$\lambda' - \lambda = \frac{h}{mc}(1 - \cos \theta).$$

The scattered light is shifted to longer wavelengths, by an amount depending only on the scattering angle and the universal length h/mc (the *Compton wavelength* of the electron). Compton's measurement of exactly this shift in 1923 was decisive evidence that light really does carry momentum in particle-like quanta.

Particle Creation & Threshold Energies

Finally, creation of new particles: collide two particles of mass m fast enough, and the excess kinetic energy can be converted into the mass of a new particle:

$$P_1 + P_2 = P_3 + P_4 + P_5,$$

where particles 3 and 4 have mass m and the newly created particle 5 has mass M .

Work in the centre of momentum frame, where the two initial particles approach with equal speeds v . The total 4-momentum is

$$P_1 + P_2 = \begin{pmatrix} 2m\gamma_v c \\ \mathbf{0} \end{pmatrix},$$

and after the collision, the total energy is $E_3 + E_4 + E_5 \geq (2m + M)c^2$, with equality when all three products are at rest. So creation is possible only if

$$2m\gamma_v c^2 \geq (2m + M)c^2, \quad \text{i.e.} \quad \gamma_v \geq 1 + \frac{M}{2m},$$

or in terms of kinetic energy: the initial kinetic energy in the centre of momentum frame must be at least Mc^2 . This is the **threshold energy** for the process – entirely reasonable: every joule of kinetic energy in this frame is available for making mass.

The situation is dramatically different in a *fixed-target* experiment, where one initial particle is at rest in the lab. The lab frame moves at speed v relative to the centre of momentum frame, so the moving particle has speed $u = 2v/(1 + v^2/c^2)$, for which one can check the useful identity

$$\gamma_u = 2\gamma_v^2 - 1.$$

At threshold, $\gamma_v = 1 + M/2m$, so we require

$$\gamma_u \geq 2 \left(1 + \frac{M}{2m}\right)^2 - 1 = 1 + \frac{2M}{m} + \frac{M^2}{2m^2},$$

that is, an initial kinetic energy of

$$m(\gamma_u - 1)c^2 \geq \left(2 + \frac{M}{2m}\right) Mc^2.$$

This is far more than Mc^2 when $M \gg m$: most of the beam's energy is wasted on the kinetic energy that the products are forced to carry (the total momentum is nonzero and must be conserved). This is precisely why modern particle physics is done with *colliders*, smashing two beams together head-on – the lab *is* the centre of momentum frame, and every unit of beam energy is available for creating new particles. To create a Higgs boson ($M \approx 130$ proton masses) with a fixed-target proton experiment would require a beam energy hundreds of times larger than the LHC's colliding beams need.

And that – from Newton's laws to the design principles of the Large Hadron Collider – is where we end the course.

4 Groups

Informally, group theory is the study of certain structures that occur when we think about symmetries in a mathematical way. In this course, we will look at what groups are, their basic properties – with lots of examples.

This article constitutes my notes for the ‘Groups’ course, held in Michaelmas 2020 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas.

§4.1 Groups

‘Groups’ is a course which introduces you to the subject of *Abstract Algebra*. Indeed, while groups are one of the simplest and most basic of all the algebraic structures, they are immensely useful and appear in almost every area of mathematics.

§4.1.1 Definition of a Group

We will begin our study of the subject by defining formally what a group is.

Definition 4.1.1 (Group)

A **group** is a set G with a binary operation^a $*$ which satisfies the axioms:

- *Identity*. There is an element $e \in G$ such that $g * e = e * g = g$ for every $g \in G$.
- *Inverses*. For every element $g \in G$, there is an element $g^{-1} \in G$ such that $g * g^{-1} = g^{-1} * g = e$.
- *Associativity*. The operation $*$ is associative.

^aSome texts include an additional *closure* axiom, but this is implied by $*$ being a binary operation on G .

We typically refer to a group as defined above by $(G, *)$, which explicitly states that $*$ is the group operation. When the operation being used is clear, we can refer to the group by just G . We will also be omitting the group’s operation symbol quite often, for example writing $gh = g * h$. If G is finite, we call $|G|$ the **order** of the group.

In a later section, we will look at some non-trivial examples of groups.

Elementary Properties of Groups

With the notion of a group now defined, we can now consider some basic facts that follow directly from the definition of a group. We will first address whether it is possible for a group to have multiple identity elements, or for an element to have multiple inverses (no).

Proposition 4.1.2 (Uniqueness of the Identity and Inverse)

Let $(G, *)$ be a group. Then there is a unique identity element, and for every $g \in G$, g^{-1} is unique.

Proof. To prove that the identity element is unique, let e and e' be identity elements of G . Then $e * e' = e$ and $e * e' = e'$ by definition, giving $e = e'$.

To prove that the inverses are unique, suppose that for some $g, h, k \in G$ we have $g * h = g * k = e$. Then $g^{-1} * g * h = g^{-1} * g * k$, implying $h = k$. The case of $h * g = k * g = e$ follows analogously. $\square$

The next useful fact is the *cancellation law*, whose proof bears a large resemblance to the proof that inverses are unique.

Proposition 4.1.3 (Cancellation Law)

If $(G, *)$ is a group, and $a, b, c \in G$, then $a * b = a * c$ and $b * a = c * a$ both imply $b = c$.

Proof. Taking $a * b = a * c$ and left-multiplying by a^{-1} we have $a^{-1} * a * b = a^{-1} * a * c$, that is, $b = c$. The other case follows analogously. $\square$

The last proposition we will prove in this section gives us a useful result about computing inverses.

Proposition 4.1.4 (Computing Inverses)

Let $(G, *)$ be a group, and let $g, h \in G$. Then the following hold:

- (i) $(g * h)^{-1} = h^{-1} * g^{-1}$.
- (ii) $(g^{-1})^{-1} = g$.

Proof.

- (i) We have $(g * h) * (h^{-1} * g^{-1}) = g * (h * h^{-1}) * g^{-1} = g * g^{-1} = e$, so $(g * h)^{-1} = h^{-1} * g^{-1}$.
- (ii) Similarly, $g^{-1} * g = e$, so $(g^{-1})^{-1} = g$. $\square$

Examples of Groups

It's probably of some use to have concrete examples of groups in your head, so you can get a feel for what they are. In this section we will present some non-trivial examples of groups (and some examples of non-groups).

It should be recognized that commutativity is *not* a group axiom, and the majority of groups are not commutative. We do have a name for groups where the binary operation is commutative though.

Definition 4.1.5 (Abelian Groups)

We say a group $(G, *)$ is **abelian** if $*$ is commutative, that is, if for any $g, h \in G$, $g * h = h * g$.

In this section, we will consider examples of both abelian and non-abelian groups¹. In the first few cases, the reasons why they are a group are stated. For the others, you should consider how they satisfy the group axioms yourself.

Example 4.1.6 (The Trivial Group)

The **trivial group** is a group whose only element is the identity, $\{e\}$.

Example 4.1.7 (Additive Group of Integers)

$(\mathbb{Z}, +)$ is an group. We have

- The identity element $0 \in \mathbb{Z}$, as $a + 0 = 0 + a = a$ for any $a \in \mathbb{Z}$
- The inverse of $a \in \mathbb{Z}$ being $-a$, as $a + (-a) = (-a) + a = 0$.
- The operation $+$ is associative and commutative.

We also have the additive group of rationals $(\mathbb{Q}, +)$, of reals $(\mathbb{R}, +)$, and of complex numbers $(\mathbb{C}, +)$ for the same reasons.

Example 4.1.8 (Addition Modulo n)

Let $n \in \mathbb{N}$, and let $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$ denote the set of residues modulo n . Then $(\mathbb{Z}/n\mathbb{Z}, +)$ is a group (where addition is done modulo n). We have

- The identity element is $0 \pmod{n}$, as $a + 0 \equiv 0 + a \equiv a \pmod{n}$.
- The inverse of $a \in \mathbb{Z}/n\mathbb{Z}$ is $-a$, as $a + (-a) \equiv 0 \pmod{n}$.
- Addition modulo n is associative.

Example 4.1.9 (Non-Zero Rationals)

Let $\mathbb{Q}^\times$ denote the set of non-zero rationals. Then $(\mathbb{Q}^\times, \times)$ is a group.

Similarly, we also have the groups $(\mathbb{R}^\times, \times)$ and $(\mathbb{C}^\times, \times)$.

Example 4.1.10 (Multiplication Modulo p)

Let p be a prime, and let $(\mathbb{Z}/p\mathbb{Z})^\times$ denote the set of non-zero residues modulo p . Then $((\mathbb{Z}/p\mathbb{Z})^\times, \times)$ is a group (where multiplication is done modulo p).

Example 4.1.11 (General Linear Group)

Let $\text{GL}_n(\mathbb{R})$ be the set of $n \times n$ matrices with non-zero determinant. Then $(\text{GL}_n(\mathbb{R}), \times)$ is the **general linear group**^a.

^aUsing matrix multiplication

¹If you are not familiar with some of the concepts used, such as matrices or modular arithmetic, feel free to ignore those examples.

Example 4.1.12 (Special Linear Group)

Let $\text{SL}_n(\mathbb{R})$ be the set of $n \times n$ matrices with determinant 1. Then $(\text{SL}_n(\mathbb{R}), \times)$ is the **special linear group**.

Non-Examples of Groups

We will now give some examples of sets with operations that are not groups. It should be useful to think about why each example does not satisfy the group axioms.

Example 4.1.13 (Non-Examples of Groups)

The following are all *not* groups.

- $(\mathbb{Z}, \times)$
- $(\mathbb{Q}, \times)$
- The set of 2×2 matrices with matrix multiplication.
- $(\mathbb{R}, *)$ where $r * s = r \times r \times s$
- $(\mathbb{N}, *)$ where $n * m = |n - m|$.

§4.1.2 Subgroups

Given any mathematical structure, it can be useful to know about its *substructure*. In the case of a group $(G, *)$, one might ask the question is there some subset $H \subseteq G$ that still acts like a group? This motivates the introduction of *subgroups*.

Definition 4.1.14 (Subgroups)

Let $(G, *)$ be a group. A subset $H \subseteq G$ is a **subgroup** of G if $(H, *)$ is also a group. If H is a subgroup of G , we will write $H \leq G$.

Example 4.1.15 (Examples of Subgroups)

The following are subgroups.

- For any group G , we have the **trivial subgroups** $\{e\} \leq G$ and $G \leq G$.
- $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$ with addition.
- $\{0, 2, 4, \dots\} \leq \mathbb{Z}$ with addition.
- $\text{SL}_n(\mathbb{R}) \leq \text{GL}_n(\mathbb{R})$ with matrix multiplication.

Checking whether something is a subgroup is easier than checking if something is a group, since we already know about the structure of the group. To check whether H is a subgroup of $(G, *)$, we can just check the following hold:

- *Closure.* $*$ is closed in H .
- *Identity.* $e \in H$.
- *Inverses.* For $h \in H$, we also have $h^{-1} \in H$.

These can all be combined into a single test, that is sometimes known as the ‘subgroup checking lemma’.

Lemma 4.1.16 (Subgroup Criterion)

A subset H of a set G is a subgroup of $(G, *)$ if and only if H is non-empty and $x * y^{-1} \in H$ for all $x, y \in H$.

Proof Sketch. First check that the conditions of H being non-empty and $x * y^{-1} \in H$ imply that it’s a subgroup. Then, show that if H is not a subgroup, then either H is empty or $x * y^{-1} \notin H$ for some $x, y \in H$. $\square$

As an example of using subgroups, let’s try to characterize all of the subgroups of $(\mathbb{Z}, +)$.

Theorem 4.1.17 (Subgroups of $\mathbb{Z}$)

The subgroups of $(\mathbb{Z}, +)$ are precisely the subsets of the form $n\mathbb{Z}$ for $n \in \mathbb{N}$, where $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$.

Proof. First, we prove that $n\mathbb{Z}$ is a subgroup. Fix $n \in \mathbb{N}$.

- *Closure.* Given $nk_1, nk_2 \in n\mathbb{Z}$, then $nk_1 + nk_2 = n(k_1 + k_2) \in n\mathbb{Z}$.
- *Identity.* $0 = n \cdot 0 \in n\mathbb{Z}$.
- *Inverses.* The inverse of nk is $-nk = n(-k) \in n\mathbb{Z}$.

Thus each is subgroup. Now we prove that there is no other subgroups.

Let $H \leq \mathbb{Z}$. If $H = \{0\}$, then $H \equiv 0\mathbb{Z}$. If not, then take the smallest positive element in H (namely n). Since H is a subgroup, it’s closed and contains inverses, so $n + n + \cdots + n \in H$ and $-n - n - n - \cdots - n \in H$, so $n\mathbb{Z} \subseteq H$.

Suppose, for a contradiction, there is some $k \in H$ such that $k \notin n\mathbb{Z}$. So, there is some integer m such that $nm < k < n(m + 1)$. But then $0 \leq k - nm < n$, and $k - nm \in H$ which is a contradiction, so $H = n\mathbb{Z}$. $\square$

We can use the definition of a subgroup to prove some elementary facts.

Proposition 4.1.18 (Elementary Properties of Subgroups)

Let G be a group.

- (i) Let H and K be subgroups of G . Then $H \cap K \leq G$.
- (ii) If $K \leq H$ and $H \leq G$ then $K \leq G$ (being a subgroup is transitive).
- (iii) If $K \subset H$, $H \leq G$ and $K \leq G$, then $K \leq H$.

Proof. There is multiple ways to prove these, but we will use the subgroup criterion as an example of it being used.

- (i) Note that $H \cap K$ is not empty as $e \in H$ and $e \in K$. Then, for any $x, y \in H \cap K$, it suffices to show that $x * y^{-1} \in H$. By the subgroup criterion, we have

$x * y^{-1} \in H$ and $x * y^{-1} \in K$, thus $x * y^{-1} \in H \cap K$, and we are done.

- (ii) If $K \leq H$, then for any $x, y \in K$, we have $x * y^{-1} \in K$. Then as $K \subset H \subset G$, we must have $x * y^{-1} \in G$, and thus $K \leq G$.
- (iii) As $K \leq G$, we know K is non-empty. Thus it suffices to show that $x * y^{-1} \in K$ for any $x, y \in H$. But this is implied by $K \leq G$ and the subgroup criterion, and thus as $K \subset H$, $K \leq H$. $\square$

Generators

We will now consider a certain kind of subgroup, which is specified by some of the elements it contains.

Definition 4.1.19 (Subgroup Generated By A Subset)

For some set $X \subseteq G$, we define the **subgroup generated by X** , $\langle X \rangle$, to be the smallest subgroup of G which contains X .

From this definition, we can see that we must have $e \in \langle X \rangle$ and $X \subseteq \langle X \rangle$. Also, $\langle X \rangle$ must contain all products of elements in X and their inverses. We can put this in a more useful form with the following proposition.

Proposition 4.1.20

Let X be a non-empty subset of G . Then $\langle X \rangle$ is the set of elements of G of the form $x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_k^{\alpha_k}$ where $x_i \in X$ (not necessarily distinct), $\alpha_i = \pm 1$ and $k \geq 0$ (For $k = 0$, we say the element is e).

Proof. Let T be the set of such elements. Clearly $T \subseteq \langle X \rangle$, and also clearly T is a subgroup of G . We also have that $X \subseteq T$ so $\langle X \rangle \subseteq T$. Thus $T = \langle X \rangle$. $\square$

Example 4.1.21

We have $(\mathbb{Z}, +) = \langle 1 \rangle = \langle 2, 3 \rangle^a$, and $\mathbb{Z}/5\mathbb{Z} = \langle 1 \rangle = \langle 3 \rangle$.

^aNote that we write $\langle 2, 3 \rangle$ instead of $\langle \{2, 3\} \rangle$.

In the above examples, we found that there was some subset of the elements in each of the group where if we considered the subgroup generated by those elements, we get the entire group. There is a special name for such subsets.

Definition 4.1.22 (Generators)

If X is a subset of G such that $\langle X \rangle = G$, then we call X a **generating set** of G .

Notably, these generators are not necessarily unique, as can be seen in the example above.

§4.1.3 Homomorphisms

Imagine you had two groups, G and H and you wanted to think of a function from H to G that preserved some of the structure of the group. Let's say the function was

$\phi : H \rightarrow G$. We could take any two elements $h_1, h_2 \in H$, and we could find $h_1 h_2$, and then apply ϕ to get $\phi(h_1 h_2)$. Alternatively, we could try and find $\phi(h_1)$ and $\phi(h_2)$, and then get $\phi(h_1)\phi(h_2)$. If these were the same, then the function ϕ would indeed preserve some of the structure of the group. This motivates the introduction of *homomorphisms*.

Definition 4.1.23 (Homomorphism)

Let $(G, *_G)$ and $(H, *_H)$ be groups. A function $\phi : H \rightarrow G$ is a **group homomorphism** if for all $a, b \in H$,

$$\phi(a *_H b) = \phi(a) *_G \phi(b).$$

Example 4.1.24 (Inclusion Function)

If $H \leq G$, then the function $\iota : H \rightarrow G$ that has $\iota(h) = h$ for $h \in H$ is a homomorphism. It is also injective.

Example 4.1.25

The function $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ with $\phi(k) = k \pmod{n}$ is a homomorphism, since for $k, l \in \mathbb{Z}$,

$$\phi(k + l) = (k + l) \pmod{n} = (k \pmod{n}) + (l \pmod{n}) = \phi(k) + \phi(l).$$

ϕ is also surjective, since $\{0, 1, \dots, n-1\}$ are all the possible residues modulo n .

Example 4.1.26

The function $\phi : (\mathbb{R}, +) \rightarrow (\mathbb{R}^*, \cdot)$ where $x \mapsto e^x$ is a homomorphism. We have

$$\phi(x + y) = e^{x+y} = e^x \cdot e^y = \phi(x) \cdot \phi(y).$$

It is injective, as $e^x = e^y$ implies $x = y$ using logarithms, and surjective, as given $a \in \mathbb{R}^*$, $\phi(\log a) = e^{\log a} = a$.

We can see some natural consequences of this definition of a homomorphism, which shows how well it preserves the group's structure.

Proposition 4.1.27 (Properties of Homomorphisms)

Let $\phi : H \rightarrow G$ be a homomorphism.

- (i) $\phi(e_H) = e_G$.
- (ii) $\phi(h^{-1}) = \phi(h)^{-1}$ for all $h \in H$.
- (iii) If $\psi : G \rightarrow K$ is another homomorphism, then $\psi \circ \phi : H \rightarrow K$ is also a homomorphism.

Proof.

- (i) We have $e_H *_H e_H = e_H$, so $\phi(e_H *_H e_H) = \phi(e_H) *_G \phi(e_H) = \phi(e_H)$, so by the

cancellation law, $\phi(e_H) = e_G$.

(ii) Consider $\phi(h) *_{\mathcal{G}} \phi(h^{-1}) = \phi(h *_{\mathcal{H}} h^{-1}) = \phi(e_H) = e_G$, by (i). So $\phi(h) *_{\mathcal{G}} \phi(h^{-1}) = e_G$ which is the defining property of an inverse, so $\phi(h^{-1}) = \phi(h)^{-1}$.

(iii) We have

$$\begin{aligned} (\psi \circ \phi)(a *_{\mathcal{H}} b) &= \psi(\phi(a *_{\mathcal{H}} b)) \\ &= \psi(\phi(a) *_{\mathcal{G}} \phi(b)) \\ &= \psi(\phi(a)) *_{\mathcal{K}} \psi(\phi(b)) \\ &= (\psi \circ \phi)(a) *_{\mathcal{K}} (\psi \circ \phi)(b), \end{aligned}$$

so $\psi \circ \phi$ is a homomorphism from $H \rightarrow K$. □

There is a special case of homomorphism, which we can use to define when two groups ‘are the same’.

Definition 4.1.28 (Isomorphism)

If a function $\phi : H \rightarrow G$ is bijection, and ϕ is also a homomorphism from $H \rightarrow G$, then we say it is an **isomorphism**. We say two groups H, G are **isomorphic**, written $H \cong G$ if there is an isomorphism from $H \rightarrow G$.

Having an isomorphism between two groups can be thought of in a few ways. Because we have a bijection function between the two groups, the groups must have the same order. But also, because a homomorphism preserves the structure of the group, we must also have the same group-structure within each group. Thus, when we have two isomorphic groups, we can think of them as two different descriptions of the same group.

For example, we might claim that ‘there is exactly one group of order 2’, and what we mean is that for any group of order 2, we can find an isomorphism to any other group of order 2.

Example 4.1.29

Consider the group $G = \{1, i, -1, -i\}$ with complex multiplication. Then $G \cong \mathbb{Z}/4\mathbb{Z}$. This is isomorphic with the isomorphism $\phi : G \rightarrow \mathbb{Z}/4\mathbb{Z}$, where

$$\begin{aligned} \phi(1) &= 0, \\ \phi(i) &= 1, \\ \phi(-1) &= 2, \\ \phi(-i) &= 3 \end{aligned}$$

The general case is true too, where the group $H = \{e^{2\pi i k/n} : 0 \leq k \leq n-1\}$ with complex multiplication is isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Example 4.1.30 ($\mathbb{Z}$ ’s subgroups are isomorphic)

$\mathbb{Z} \cong n\mathbb{Z}$ for $n \in \mathbb{Z}$, as defined in [Theorem 4.1.17](#).

It’s worth noting that because isomorphisms are bijective, we have the following result.

Proposition 4.1.31 (Inverses of isomorphisms are isomorphisms)

Let $\phi : H \rightarrow G$ be an isomorphism. Then $\phi^{-1} : G \rightarrow H$ is also an isomorphism.

Proof Sketch. Check that ϕ^{-1} is a homomorphism. □

Kernels

When dealing with homomorphisms, say $\phi : H \rightarrow G$, it is useful to be able to think about what elements in H our homomorphism ‘reaches’. Another useful idea is thinking about what elements in H get mapped to the identity of G . To think about these questions, we use concepts of a homomorphism’s *image* and *kernel*.

Definition 4.1.32 (Image)

Let $\phi : H \rightarrow G$ be a homomorphism. We define the **image** of ϕ to be the set

$$\text{img}(\phi) = \{g \in G : g = \phi(h) \text{ for some } h \in H\}.$$

Definition 4.1.33 (Kernel)

Let $\phi : H \rightarrow G$ be a homomorphism. We define the **kernel** of ϕ to be the set

$$\ker(\phi) = \{h \in H : \phi(h) = e_G\}.$$

Indeed, while both of these are subsets of G and G respectively, they are also subgroups.

Proposition 4.1.34 (The Image and Kernel are Subgroups)

Let H and G be groups and let $\phi : H \rightarrow G$ be a homomorphism. Then $\text{img}(\phi)$ is a subgroup of G , and $\ker(\phi)$ is a subgroup of H .

Proof. We consider the two sets separately.

1. We will show $\text{img}(\phi) \leq G$. For any $x, y \in \text{img}(\phi)$, let $x = \phi(x')$ and $y = \phi(y')$ for $x', y' \in H$. Then

$$\phi(x'y'^{-1}) = \phi(x')\phi(y')^{-1} = xy^{-1} \in \text{img}(\phi),$$

thus by the subgroup criterion $\text{img}(\phi) \leq G$.

2. Now we show $\ker(\phi) \leq H$. For $x, y \in \ker(\phi)$, we have $xy^{-1} \in \ker(\phi)$, as

$$\phi(xy^{-1}) = \phi(x)\phi(y)^{-1} = \phi(x)\phi(y)^{-1} = e_G,$$

so again using the subgroup criterion, $\ker(\phi) \leq H$. □

Example 4.1.35

$\phi : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, where $\phi(k) = k \pmod{n}$ has $\text{img}(\phi) = \mathbb{Z}/n\mathbb{Z}$ and $\ker(\phi) = n\mathbb{Z}$.

One of the beauties of introducing the kernel and image is that it allows us to easily see whether a homomorphism is surjective or injective.

Proposition 4.1.36 (Surjectivity and Injectivity Criterion)

Let $\phi : H \rightarrow G$ be a homomorphism.

- (i) ϕ is surjective iff $\text{img}(\phi) = G$.
- (ii) ϕ is injective iff $\ker(\phi) = \{e\}$.

Proof. The first is true by definition, so we prove (ii). Suppose ϕ is injective, then as we have $\phi(e_H) = e_G$, so e_H must be the only element sent to e_G (by the definition of injectivity), which implies that $\ker(\phi) = \{e_H\}$. Now suppose that $\ker(\phi) = \{e_H\}$. Then if $\phi(a) = \phi(b)$ for some $a, b \in H$, we have $\phi(ab^{-1}) = \phi(a)\phi(b)^{-1} = \phi(b)\phi(b)^{-1} = e_G$. However, this implies $ab^{-1} = e_H$, so $a = b$, and ϕ is injective. $\square$

Direct Products

How can we easily find a group that will have two given groups G, H as subgroups? With the aim of getting the simplest construction possible, we can ‘stick them together’: by defining a group operation on the product $G \times H = \{(g, h) : g \in G, h \in H\}$ (a set of ordered pairs).

Definition 4.1.37 (Direct Product)

The **direct product** of two groups G, H is the set $G \times H$ with the operation of component-wise composition,

$$(g_1, h_1) * (g_2, h_2) = (g_1 *_G g_2, h_1 *_H h_2).$$

Proposition 4.1.38

The direct product of two groups G and H is a group.

Proof Sketch. Check everything component-wise. $\square$

This group contains subgroups isomorphic to G and H , taking $G \times \{e_H\}$ and $\{e_G\} \times H$.

A useful idea might be to try and recognize when a group is a direct product of two groups. This can be done with the following theorem.

Theorem 4.1.39 (Direct Product Theorem)

Let $H, K \leq G$ such that

- (i) $H \cap K = \{e\}$
- (ii) $\forall h \in H$ and $k \in K$, we have $hk = kh$
- (iii) $\forall g \in G$, there exists $h \in H, k \in K$ such that $g = hk$

then $G \cong H \times K$.

Proof. Consider the function $\phi : H \times K \rightarrow G$, where $\phi(h, k) = hk$. ϕ is a homomorphism, as

$$\phi((h_1, k_1) \cdot (h_2, k_2)) = \phi(h_1 h_2, k_1 k_2) = h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \phi(h_1, k_1) \phi(h_2, k_2).$$

ϕ is surjective by (ii), and now we will show ϕ is injective. Suppose that $(h, k) \in \ker \phi$. Then $h = k^{-1}$, which implies that $h, k \in H \cap K$ by (i), and thus $(h, k) = (e_H, e_K)$. Thus $\ker \phi = \{(e_H, e_K)\}$, so ϕ is injective by the injectivity criterion. $\square$

We now have two ways to think about the direct product.

- If we have two groups H, K , we can form their direct product $H \times K$, and view H and K as subgroups, namely $H \times \{e_K\}$ and $\{e_H\} \times K$.
- Given a group with subgroups H and K , which satisfy the conditions of the direct product theorem, then we know that we are really dealing with $H \times K$.

Indeed these are just two descriptions of the same thing. The convention is often to refer to $H \times \{e_K\}$ and $\{e_H\} \times K$ as just H and K respectively.

§4.2 Important Groups

Now that we have seen some properties of groups, we will now consider some important examples of groups.

§4.2.1 Cyclic Groups

Recall the notion of a generator from [Definition 4.1.22](#).

Definition 4.2.1 (Cyclic)

If G is a group and there is some $a \in G$ such that $\langle a \rangle = G$, then we say G is **cyclic**.

Notably, if this is the case, for all $b \in G$, there exists $k \in \mathbb{Z}$ such that $b = a^k$.

Example 4.2.2 (Examples of Cyclic Groups)

The following groups are all cyclic.

- $(\mathbb{Z}, +)$, which is generated by $\langle 1 \rangle$ or $\langle -1 \rangle$.
- $(\mathbb{Z}/n\mathbb{Z}, +)$, generated by $\langle 1 \rangle$. Indeed, any k coprime to n will satisfy $\langle k \rangle = \mathbb{Z}/n\mathbb{Z}$.
- Let $G = \{e^{2\pi i k/n} : 0 \leq k \leq n-1\}$. Then $(G, \cdot)$ is generated by $\langle e^{2\pi i/n} \rangle$ where k is coprime to n .

These groups all have the same ‘feel’ to them, and indeed they are all isomorphic to the following group.

Definition 4.2.3 (Cyclic Group C_n)

Let C_n be the group of elements $\{e = a^0, a, a^2, \dots, a^{n-1}\}$, where $a^k * a^j = a^{k+j \pmod n}$.

Then $(C_n, *)$ is the **cyclic group of order n** .

Theorem 4.2.4 (Cyclic Groups are Isomorphic)

A cyclic group G is isomorphic to $\mathbb{Z}$ or to C_n for some $n \in \mathbb{N}$.

Proof. As G is cyclic, we have $\langle b \rangle = G$, for some $b \in G$. Now let's suppose that there's some n such that $b^n = e$. Then define $\phi : C_n \rightarrow G$ by $\phi(a^k) = b^k$ for $0 \leq k \leq n-1$. Then for any a^j and $a^k \in C_n$, we trivially have that $\phi(a^j a^k) = \phi(a^{j+k}) = b^{j+k} = b^j b^k = \phi(a^j) \phi(a^k)$. Thus ϕ is a homomorphism. ϕ is also surjective as all elements in G can be written as b^k , $0 \leq k < n$. It is also injective, since $\phi(a^k) = e \implies b^k = e$ and so $k = 0$ (otherwise it contradicts the minimality of n). So ϕ is an isomorphism, and $G \cong C_n$.

If there is no such n , then we define $\phi : \mathbb{Z} \rightarrow G$ by $\phi(k) = b^k$. Then $\phi(k+m) = b^{k+m} = b^k b^m = \phi(k) \phi(m)$, so ϕ is a homomorphism. It is also clearly surjective. Now suppose $m \in \ker(\phi)$. Then $\phi(m) = b^m = e$, and $\phi(-m) = b^{-m} = e$, so if $m \neq 0$, we would get a contradiction to the fact that there is no $n > 0$ with $b^n = e$. So $m = 0$, $\ker(\phi) = \{0\}$ and ϕ must be an isomorphism. Thus $G \cong \mathbb{Z}$. $\square$

Because of this theorem, we will often just write C_n or $\mathbb{Z}$ for a cyclic group, regardless of its description.

Proposition 4.2.5

Cyclic groups are abelian.

Proof Sketch. Check definitions. $\square$

The idea of there being some k such that $g^k = e$ for some g is a frequently occurring concept.

Definition 4.2.6 (Order of an Element)

The **order of an element** $g \in G$ is the smallest $n \in \mathbb{N}$ such that $g^n = e$. This is sometimes written $\text{ord}_G(g) = n$. If there is no such n , we say g has **infinite order**.

Theorem 4.2.7 (Fundamental Theorem of Orders)

Let G be a group, and let $g \in G$ have finite order n . Then if $g^k = 1$, we have $n \mid k$.

Proof. By the division algorithm, we can write $k = qn + r$ uniquely with $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then we have

$$g^k = g^{qn+r} = g^{qn} g^r = (g^n)^q g^r = g^r = e.$$

But we defined n to be the smallest positive power for which $g^n = e$, and as $r < n$, we must have $r = 0$, otherwise we contradict the minimality of n . Thus $k = qn$, that is, $n \mid k$. $\square$

§4.2.2 Dihedral Group

Group theory is frequently thought of as the ‘algebraic study of symmetry’. With this rather vague claim in mind, we will now look at some groups related to geometry – the symmetries of a regular n -gon. Let’s define what we mean by a ‘symmetry’ of a regular polygon.

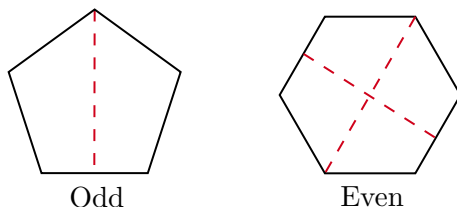
Definition 4.2.8

A **symmetry** of a regular n -gon is a transformation of the n -gon, so that when the transformed n -gon is placed on the original n -gon, it exactly covers it.

Definition 4.2.9

The **dihedral group** D_{2n} is the group of symmetries of a regular n -gon, where the group operation is the composition of symmetries.

Clearly in this group, we will have n rotations (clockwise) of the angle $\frac{2\pi k}{n}$, $0 \leq k < n$ ($k = 0$ gives the identity or ‘do nothing’ symmetry). There is also n reflections.



When n is odd, the n reflections are in axis through the center and each of the vertices. For even n , we have $n/2$ reflections in axis through pairs of opposite vertices, and $n/2$ reflections in axes through pairs of opposite midpoints of edges.

From this you should count $2n$ elements, and we will now see that there is no other elements.

Proposition 4.2.10

A regular n -gon has $2n$ symmetries.

Proof. Let $g \in D_{2n}$. Since g is a symmetry of our n -gon, it must send vertices to vertices and edges to edges. So if v_1 is a vertex who’s adjacent vertices are v_2 and v_n and we have $g(v_1) = v_i$, then we must know $g(v_2)$ and $g(v_n)$, so we must know exactly what g . Since there is n possibilities for where v_1 is sent, and 2 possibilities for where v_2 is sent, there must be $2n$ elements in total. $\square$

Proposition 4.2.11

D_{2n} is a group.

Proof. We have closure by ‘composition of symmetries are also symmetries’, identity with the ‘do nothing’ symmetry and also inverses, as a rotation by $\frac{2\pi k}{n}$ has an inverse of a $\frac{2\pi(n-k)}{n}$ rotation, and reflections are self inverse. We also have associativity, as

the composition of functions is associative. Thus D_{2n} is a group. $\square$

It's possible to generate every element in the group with just a single rotation and a reflection. Let r be the rotation by $\frac{2\pi}{n}$, and let s be the reflection about the axis through v_1 and the center. Then r^k gives the rotation by $\frac{2\pi k}{n}$ and we can perform any reflection by first rotating the n -gon, then applying the reflection, and then rotating back.

D_{2n} is also not abelian, and indeed we have $rs = sr^{-1}$.

Aside: Group Presentations

One way to write groups is with a **presentation**. This is an expression of the form

$$\langle \text{generators} \mid \text{relations between generators} \rangle.$$

As an example, we can express the cyclic and dihedral groups using generators as follows

$$\begin{aligned} C_n &= \langle a \mid a^n = e \rangle \\ D_{2n} &= \langle r, s \mid r^n = e, s^2 = e, rs = sr^{-1} \rangle \end{aligned}$$

You should be able to deduce all things that are true in the group from the relations in the presentation. However, you should be aware that there are some 'caveats', for example if we wrote down

$$\langle r, s \mid r^n = e, s^2 = e \rangle \neq D_{2n}.$$

It is, in general, quite hard to write down a presentation for a given group, or even to determine the group from a given presentation. In this course, we will not look at the 'mathematical tools' which allow us to discuss presentations in a rigorous way.

Example 4.2.12

The group

$$\langle a, b, c \mid aba^{-1}b^{-1} = b, bcb^{-1}c^{-1} = c, cac^{-1}a^{-1} = a \rangle = \{e\},$$

but the group

$$\langle a, b, c, d \mid aba^{-1}b^{-1} = b, bcb^{-1}c^{-1} = c, cdc^{-1}d^{-1} = d, dad^{-1}a^{-1} = a \rangle$$

is the **Higman group**, and it is infinite. It should be clear from this example that it is quite hard to determine a group from just its presentation.

§4.2.3 Permutation Groups

We are now going to discuss groups made up of *permutations*.

Definition 4.2.13 (Permutations)

Given a set X , a **permutation** of X is a bijective function $\sigma : X \rightarrow X$. The set of

all permutations of X is denoted $\text{Sym } X$.

Theorem 4.2.14

For any set X , $\text{Sym } X$ is a group with respect to composition.

Proof. We check the group axioms individually.

- *Closure.* The composition of two bijective functions from $X \rightarrow X$ is a bijective function from $X \rightarrow X$.
- *Associativity.* Composition of functions is associative.
- *Identity.* The identity function $\text{id}(x) = x$ is bijective.
- *Inverses.* Every bijective function has a bijective inverse.

Thus $\text{Sym } X$ is a group. □

Definition 4.2.15 (Symmetric Group)

If $|X| = n$, we write S_n for (the isomorphism class of) $\text{Sym } X$. S_n is the **symmetric group** on n elements.

It should be reasonably clear that $|S_n| = n(n-1) \cdots 1 = n!$. We will also normally use $X = \{1, 2, 3, \dots, n\}$ when we study S_n . When dealing with permutation groups, it's helpful to have some notation to express permutations. For a general $\sigma \in S_n$, we write

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & \cdots & n \\ \sigma(1) & \sigma(2) & \sigma(3) & \cdots & \sigma(n) \end{pmatrix}.$$

Example 4.2.16

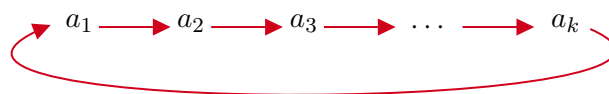
If we had some $\sigma \in S_3$ such that $\sigma(1) = 2$, $\sigma(2) = 3$, and $\sigma(3) = 1$, we would write

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

A slightly better notation for when we have a permutation that ‘cycles’ some elements $a_1, \dots, a_k \in \{1, 2, \dots, n\}$ and leaves the other elements unchanged, we can write

$$\sigma = (a_1 \ a_2 \ \cdots \ a_k)$$

which denotes the permutation mapping the elements as follows



The cyclic nature of this notation also implies that the two permutations $(a_1 \ a_2 \ \cdots \ a_k) =$

$(a_2 \ a_3 \ \cdots \ a_k \ a_1)$. To define this notation slightly more formally, we have

$$(a_1 \ a_2 \ \cdots \ a_k)(x) = \begin{cases} a_{i+1} & \text{if } x = a_i, (i < k) \\ a_1 & \text{if } x = a_k \\ x & \text{if } x \notin \{a_1, a_2, \dots, a_k\}. \end{cases}$$

We distinguish between permutations that can be written directly in this form in the following way.

Definition 4.2.17 (Cycles and Transpositions)

A permutation of the form $\sigma = (a_1 \ a_2 \ \cdots \ a_k)$ is a **k -cycle**. If $k = 2$ then we call it a **transposition**.

As cycles are permutations, we can compose them.

Example 4.2.18 (Composing Cycles)

If we consider the composition of two cycles $(1 \ 2 \ 3 \ 4)(3 \ 2 \ 4)$, this should be a permutation in S_4 . Indeed we have

$$\begin{aligned} 1 &\mapsto 1 \mapsto 2 \\ 2 &\mapsto 4 \mapsto 1 \\ 3 &\mapsto 2 \mapsto 3 \\ 4 &\mapsto 3 \mapsto 4 \end{aligned}$$

So we actually have that the composition of these cycles is also a cycle^a, namely $(1 \ 2 \ 3 \ 4)(3 \ 2 \ 4) = (1 \ 2)$.

^aThis is, in general, not the case

In the example above, the two cycles involved elements that were in both cycles. We have a specific term for when this is not the case.

Definition 4.2.19 (Disjoint Cycles)

We say that two cycles are **disjoint** if no number appears in both cycles.

Lemma 4.2.20

Disjoint cycles commute.

Proof. Let $\sigma, \tau \in S_n$ be two disjoint cycles. We want to show that $\sigma\tau = \tau\sigma$, that is, for any $x \in \{1, 2, \dots, n\}$, we have $\sigma(\tau(x)) = \tau(\sigma(x))$. We have two cases.

If x is in neither σ or τ , then $\sigma(x) = \tau(x) = x$, and thus $\sigma(\tau(x)) = \tau(\sigma(x)) = x$.

Otherwise x is in exactly one of σ or τ . WLOG let it be in σ . Then $\sigma(x)$ is also in σ (and hence not τ), so $\tau(x) = x$ and $\tau(\sigma(x)) = \sigma(x)$. Thus $\sigma(\tau(x)) = \sigma(x)$, so they commute. $\square$

Slightly more surprising is the following theorem

Theorem 4.2.21 (Writing Permutations with Cycles)

Any $\sigma \in S_n$ can be written uniquely^a as the composition of disjoint cycles.

^aUp to the order of the cycles in the composition

Proof. First we show that any permutation can be written as the composition of cycles. Take $\sigma \in S_n$, and consider $1, \sigma(1), \sigma^2(1), \dots$. Since $\{1, 2, \dots, n\}$ is finite, there must exist $a > b$ such that $\sigma^a(1) = \sigma^b(1)$. So $\sigma^{a-b}(1) = 1$. Now let $k > 0$ be the smallest integer such that $\sigma^k(1) = 1$, which must exist by the previous argument. Then for $0 \leq l < m < k$, if $\sigma^m(1) = \sigma^l(1)$, then $\sigma^{m-l}(1) = 1$, which contradicts the minimality of k . So all of $1, \sigma(1), \sigma^2(1), \dots, \sigma^{k-1}(1)$ are distinct. This gives us our first cycle $(1 \ \sigma(1) \ \sigma^2(1) \ \dots \ \sigma^{k-1}(1))$. We can repeat this process for the next number in $\{1, 2, \dots, n\}$ that has not already appeared, until eventually every element has appeared. As σ is a bijection, no element can reappear.

We now show that this composition of cycles is unique up to the order of composition. Suppose we have two such decompositions

$$\begin{aligned} \sigma &= (a_1 \ \dots \ a_{k_1})(a_{k_1+1} \ \dots \ a_{k_2}) \dots (a_{k_{m-1}+1} \ \dots \ a_{k_m}) \\ &= (b_1 \ \dots \ b_{k_1})(b_{k_1+1} \ \dots \ b_{k_2}) \dots (b_{k_{m-1}+1} \ \dots \ b_{k_m}) \end{aligned}$$

and each $j \in \{1, 2, \dots, n\}$ appears exactly once in both. Then we have $a_1 = b_t$ for some t , and the other numbers in the cycle are uniquely determined by $\sigma(a_1), \sigma^2(a_1), \dots$. So we have

$$(a_1 \ \dots \ a_{k_1})(\dots) = (b_t \ \dots)(\dots),$$

since disjoint cycles commute and we can ‘cycle’ the elements in cycles. If we continue this, we will find that all other cycles match too. $\square$

Now let’s consider an element $\sigma \in S_n$, and specifically we will look at the order of σ .

Definition 4.2.22

The set of cycle lengths of the disjoint cycle decomposition of a permutation σ is its **cycle type**.

Example 4.2.23

$(1 \ 2 \ 3)(5 \ 6)$ has a cycle type of 3, 2 (or 2, 3).

Theorem 4.2.24

The order of $\sigma \in S_n$ is the least common multiple of the cycle lengths in its cycle type.

Proof. First note that the order of a k -cycle is k . Suppose that $\sigma = \tau_1 \tau_2 \dots \tau_r$, where τ_i is a cycle disjoint to the others. Then we have $\sigma^m = \tau_1^m \tau_2^m \dots \tau_r^m$, since disjoint cycles commute. Let each τ_i be a k_i -cycle, then if $\sigma^m = e$, we have $\tau_1^m \tau_2^m \dots \tau_r^m = e$,

and thus $\tau_i^m = e$ for all i as they are disjoint. By the fundamental theorem of orders, we must have $k_i \mid m$, and thus $m = \text{lcm}(k_1, k_2, \dots, k_r)$ by minimality. $\square$

This theorem gives us an easy way to find the order of the elements in S_n : write them in cycle notation.

Disjoint cycle notation is a useful way to express elements of S_n . Another useful notation is writing elements as the product of transpositions.

Theorem 4.2.25 (Writing Permutations with Transpositions)

Let $\sigma \in S_n$. Then σ is a product of transpositions.

Proof. It suffices to show that we can write any cycle as a product of transpositions. We observe that

$$(a_1 \ a_2 \ \cdots \ a_k) = (a_1 \ a_2)(a_2 \ a_3) \cdots (a_{k-1} \ a_k).$$

$\square$

Unlike the disjoint cycle decomposition, this isn't unique. For example, $(1 \ 2 \ 3 \ 4) = (1 \ 2)(2 \ 3)(3 \ 4) = (1 \ 2)(2 \ 3)(1 \ 2)(3 \ 4)(1 \ 2)$. However, the parity of the number of transpositions *is* invariant among decompositions.

Theorem 4.2.26 (Parity of Transpositions)

Writing $\sigma \in S_n$ as a product of transpositions in different ways, the number of transpositions used is always either even or odd, that is, the parity is invariant with respect to σ .

Proof. Let's write $\chi(\sigma)$ for the number of cycles in σ in its disjoint cycle decomposition, including any 1-cycles. We will consider what happens to $\chi(\sigma)$ when we multiply σ by a transposition $\tau = (c \ d)$.

- If a cycle does not contain c or d , it will not be affected.
- If c and d are in the same cycle, say $(c \ a_2 \ a_3 \ \cdots \ a_{k-1} \ d \ a_{k+1} \ \cdots \ a_l)$, then composing with $(c \ d)$ gives $(c \ a_{k+1} \ \cdots \ a_l)(d \ a_2 \ \cdots \ a_{k-1})$. So $\chi(\sigma\tau) = \chi(\sigma) + 1$.
- If c and d are in different cycles, we have

$$(c \ a_2 \ \cdots \ a_k)(d \ b_2 \ \cdots \ b_l)(c \ d) = (c \ b_2 \ \cdots \ b_l \ d \ a_2 \ \cdots \ a_k).$$

$$\text{So } \chi(\sigma\tau) = \chi(\sigma) - 1.$$

Thus for any σ and any transposition τ , $\chi(\sigma) \equiv \chi(\sigma\tau) + 1 \pmod{2}$. We know that $\chi(\sigma)$ is uniquely determined by σ , and if we write

$$\sigma = e\tau_1 \cdots \tau_k = e\tau'_1 \cdots \tau'_l,$$

we can use our result to get

$$\begin{aligned}\chi(\sigma) &\equiv \chi(e) + k \equiv n + k \pmod{2} \\ \chi(\sigma) &\equiv \chi(e) + l \equiv n + l \pmod{2},\end{aligned}$$

and thus $k \equiv l \pmod{2}$. □

Because of this invariance, we can distinguish between odd and even permutations.

Definition 4.2.27 (Sign of a Permutation)

Writing $\sigma \in S_n$ as a product of transpositions $\sigma = \tau_1 \tau_2 \cdots \tau_k$, the **sign** of σ is defined as $\text{sign}(\sigma) = (-1)^k$. If k is even, we say that σ is **even**, and if k is odd, we say that σ is **odd**.

Proposition 4.2.28

For $n \geq 2$, $\text{sign} : S_n \rightarrow \{\pm 1\}$ is a surjective homomorphism.

Proof. We already know that sign is well defined, and if $\chi(\sigma) = k$ and $\chi(\sigma') = l$ for $\sigma, \sigma' \in S_n$, then $\sigma\sigma'$ can be written with $k + l$ transpositions, so $\text{sign}(\sigma\sigma') = (-1)^{k+l} = (-1)^k(-1)^l = \text{sign}(\sigma)\text{sign}(\sigma')$, so sign is a homomorphism. It is also surjective since $\text{sign}(e) = 1$ and $\text{sign}(1\ 2) = -1$. □

There is an important group that comes from sign being a homomorphism.

Definition 4.2.29 (Alternating Group)

The **alternating group** A_n is the kernel of the homomorphism $\text{sign} : S_n \rightarrow \{\pm 1\}$, that is, it's the group of even permutations.

§4.2.4 Möbius Groups

In the previous section we discussed many of the properties of permutations of a finite set. In this section, we will look at some permutations of an infinite set. We will be looking at functions $\mathbb{C} \rightarrow \mathbb{C}$ - but since $\mathbb{C}$ has some intrinsic geometric properties (lines, circles, etc) unlike $\{1, 2, \dots, n\}$, we will restrict ourselves to functions that interact well with its geometry.

More precisely, we will be looking at functions of the form $f : \mathbb{C} \rightarrow \mathbb{C}$, where

$$f(z) = \frac{az + b}{cz + d},$$

with $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$. We have this condition because

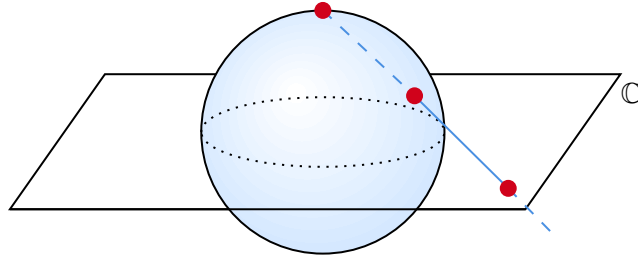
$$f(z) - f(w) = \frac{az + b}{cz + d} - \frac{aw + b}{cw + d} = \frac{(ad - bc)(z - w)}{(cw + d)(cz + d)},$$

so if $ad - bc = 0$, we would have $f(z) = f(w)$ and f would be constant. We want to avoid this case as we wish to somehow form a group from these functions.

We also have another point which could cause some trouble, namely when $z = -d/c$, and the denominator of f is 0. To fix this, we are going to introduce a new point ' ∞ '

to $\mathbb{C}$ to form the **extended complex plane**, $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. This can be visualized through ‘stereographic projection’.

Consider a sphere with the complex plane cutting it at the equator.



We get a correspondence between points on the sphere and points in $\mathbb{C}$ by drawing a line from the north pole to a point on the sphere, and seeing where it intersects the plane as shown. The north pole corresponds to ∞ in $\hat{\mathbb{C}}$.

Definition 4.2.30 (Möbius Maps)

A **Möbius map** is a function $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ of the form

$$f(z) = \frac{az + b}{cz + d},$$

with $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$, and with $f(-d/c) = \infty$ and

$$f(\infty) = \begin{cases} \frac{a}{c} & \text{if } c \neq 0 \\ \infty & \text{if } c = 0 \end{cases}.$$

Let's look at some properties of Möbius maps.

Proposition 4.2.31

Möbius maps are bijections $\hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$.

Proof Sketch. Define $f^{-1}(z) = \frac{dz-b}{-cz+a}$, and check that $f^{-1}(f(z)) = z$. □

We now have the main result of this section.

Theorem 4.2.32 (Möbius Group)

The set of Möbius maps forms a group $\mathcal{M}$ under composition.

Proof.

- *Closure.* Let $f_1(z) = \frac{a_1z+b_1}{c_1z+d_1}$ and $f_2(z) = \frac{a_2z+b_2}{c_2z+d_2}$. Then

$$f_2(f_1(z)) = \frac{(a_1a_2 + b_2c_1)z + (a_2b_1 + b_2d_1)}{(c_2a_1 + d_2c_1)z + (c_2b_1 + d_2d_1)} = \frac{a'z + b'}{c'z + d'},$$

and $a'd' - b'c' \neq 0$.

- *Identity.* Letting $a = 1, b = 0, c = 0$ and $d = 1$ gives $f(z) = z$.

- *Inverses.* For $f(z) = \frac{az+b}{cz+d}$, we have $f^{-1}(z) = \frac{dz-b}{-cz+a}$.

Thus the set of Möbius maps form a group. $\square$

Remark. When working with Möbius maps in $\hat{\mathbb{C}}$, we will (somewhat improperly) use the notation ' $\frac{1}{\infty} = 0$ ', ' $\frac{1}{0} = \infty$ ' and ' $\frac{a\infty}{c\infty} = \frac{a}{c}$ ' – but take care not to use this notation accidentally in other circumstances.

This group has an interesting set of generators, which provide some insight as to what Möbius maps do to the extended complex plane.

Theorem 4.2.33 (Generators of $\mathcal{M}$)

Every Möbius map can be written as a composition of maps of the following forms:

1. $f(z) = az$, $a \neq 0$ – dilation/rotation;
2. $f(z) = z + b$ – translation;
3. $f(z) = \frac{1}{z}$ – inversion;

Proof. Let $f(z) = \frac{az+b}{cz+d}$. Then if $c \neq 0$, $f(z)$ is the composition

$$\begin{aligned} z &\mapsto z + \frac{d}{c} \mapsto \frac{1}{z + \frac{d}{c}} \mapsto \frac{(-ad + bc)c^{-2}}{z + \frac{d}{c}} \\ &\mapsto \frac{a}{c} + \frac{(-ad + bc)c^{-2}}{z + \frac{d}{c}} = \frac{az + b}{cz + d}. \end{aligned}$$

If $c = 0$, then $z \mapsto \frac{a}{d}z \mapsto \frac{a}{d}z + \frac{b}{d}$. $\square$

§4.3 Lagrange's Theorem

We will now begin to add to our algebraic toolkit by developing some results that will shed some light on the internal structure of a group with respect to a subgroup. We will begin by defining the notion of a *coset*, which will be used frequently throughout the rest of the course.

Definition 4.3.1 (Coset)

Let G be a group and H a subgroup of G . For any element $g \in G$, the symbol

$$gH = \{gh : h \in H\}$$

is the set of elements gh , where h ranges over elements of H . gH is a **left coset** of H in G . We can also define $Hg = \{hg : h \in H\}$ which is the **right coset** of H in G .

The cosets of H in G are subsets of G , but they aren't (typically) subgroups. You should think of a coset as being a 'translated copy' of H that has the same number of elements as H , but may or may not be a subgroup.

Example 4.3.2 (Cosets of $2\mathbb{Z}$)

Let $H = 2\mathbb{Z} \leq \mathbb{Z}$. Then (using additive notation) the coset $0 + 2\mathbb{Z} = \{0 + k : k \in 2\mathbb{Z}\}$. The coset $1 + 2\mathbb{Z} = \{1 + k : k \in 2\mathbb{Z}\}$ is the set of all odd integers.

These are the only cosets we can obtain with this subgroup, as if we fix some $n \in \mathbb{Z}$, then $n + 2\mathbb{Z}$ will be in $2\mathbb{Z}$ if n is even, and in $1 + 2\mathbb{Z}$ if n is odd.

Example 4.3.3

Let $H = \{e, (1\ 2)\} \leq S_3 = \{e, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$.

Considering the cosets of H , we have

$$\begin{aligned} eH &= \{e, (1\ 2)\} = H \\ (1\ 2)H &= \{(1\ 2), e\} = H \\ (1\ 3)H &= \{(1\ 3), (1\ 3)(1\ 2\ 3)\} \\ (2\ 3)H &= \{(2\ 3), (1\ 3\ 2)\} \\ (1\ 2\ 3)H &= \{(1\ 2\ 3), (1\ 3)\} = (1\ 3)H \\ (1\ 3\ 2)H &= \{(1\ 3\ 2), (2\ 3)\} = (2\ 3)H, \end{aligned}$$

and so all together we have 3 distinct cosets.

From the last example, there are some notable details:

- Whenever we choose the identity element, we get the subgroup back: $eH = H$.
- Also, whenever we choose an element of H , we get H back: $hH = H$ for $h \in H$.
- The cosets of a subgroup are always the same size as that subgroup.
- Every element in G appears in at least one coset, that is, $\bigcup_{g \in G} gH = G$.

This leads up to *Lagrange's Theorem*.

Theorem 4.3.4 (Lagrange's Theorem)

Let $H \leq G$ be a subset of a finite group G . Then

- (i) $|H| = |gH|$ for all $g \in G$;
- (ii) For $g_1, g_2 \in G$, either $g_1H = g_2H$ or $g_1H \cap g_2H = \emptyset$;
- (iii) $G = \bigcup_{g \in G} gH$

In particular, defining the **index of H in G** as $|G : H|$ to be the number of distinct cosets of H in G , then $|G| = |G : H| \cdot |H|$.

Proof.

- (i) The function $H \rightarrow gH, h \rightarrow gh$ defined a bijection between H and gH .
- (ii) Suppose $g_1H \cap g_2H \neq \emptyset$. Then there exists $g \in g_1H \cap g_2H$. So $g = g_1h_1 = g_2h_2$

for some $h_1, h_2 \in H$. Then $g_1 = g_2 h_2 h_1^{-1}$, so for any $h \in H$, we have $g_1 h = g_2 h_2 h_1^{-1} h \in g_2 H$, so $g_1 H \subseteq g_2 H$. Similarity, $g_2 H \subseteq g_1 H$, thus $g_2 H = g_1 H$.

- (iii) Given some $g \in G$, then $g \in gH$ (since $e \in H$). Thus $G \subseteq \bigcup_{g \in G} gH$, and certainly $\bigcup gH \subseteq G$, thus $G = \bigcup_{g \in G} gH$.

Finally, $|G|$ is partitioned into the number of distinct cosets of H , thus $|G| = |G : H| \cdot |H|$. $\square$

What this theorem is saying is ‘cosets pave the group’, as all the paving stones (the cosets) are the same size, they don’t overlap, and they cover the whole group.

Here, we used left cosets but we could have used right cosets and would have gotten an analogous result. In general, the left and right cosets are not equal: $gH \neq Hg$. When this is true, it is an interesting property of a subgroup, that we will look at later in this chapter.

So left and right cosets are not generally, the same but what about two left cosets?

Proposition 4.3.5

If H is a subgroup of a group G , and $g_1, g_2 \in G$, then

$$g_1 H = g_2 H \iff g_1^{-1} g_2 \in H.$$

Proof Sketch. Follows from definitions. $\square$

In particular, taking $g' \in gH$ gives $g'H = gH$.

Let us take an element from each of the distinct cosets of a subgroup

$$g_1, g_2, \dots, g_{|G:H|},$$

then

$$G = \bigcup_{i=1}^{|G:H|} g_i H,$$

where the union is the union of disjoint sets. The elements g_i are called **coset representatives** of H in G .

We can use Lagrange’s theorem to immediately derive some useful results.

Corollary 4.3.6 (Order of an Element Divides Order of a Group)

Let G be a finite group, and let $g \in G$. Then $\text{ord}_G(g) \mid |G|$.

Proof. Notice that the subgroup $\langle g \rangle$ is a cyclic group of order $\text{ord}_G(g)$. Then $|\langle g \rangle| \mid |G|$ by Lagrange’s theorem. $\square$

Corollary 4.3.7

Let G be a finite group, and $g \in G$. Then $g^{|G|} = e$.

Proof. The order of g divides $|G|$, and thus $|G| = \text{ord}_G(g) \cdot k$ for some k . Thus $g^{|G|} = g^{\text{ord}_G(g) \cdot k} = e^k = e$. $\square$

Corollary 4.3.8

Groups of prime order are cyclic, and are generated by any non-identity element.

Proof. Let's suppose we have a group G with $|G| = p$, for a prime p . Take $g \in G$. Then $|\langle g \rangle| \mid |G|$ by Lagrange. So $|\langle g \rangle| = 1$ or p . If $g \neq e$, then $|\langle g \rangle| \neq 1$, so it must be p . So $\langle g \rangle = G$. $\square$

We will now see how a theorem from number theory can be proved using Lagrange's theorem. First, recall that $(\mathbb{Z}/n\mathbb{Z})^\times$ is a group made up of the elements of $\{1, 2, \dots, n-1\}$ that have an inverse. We also have $|(\mathbb{Z}/n\mathbb{Z})^\times| = \phi(n)$, which you should recall from 'Numbers and Sets'. We will now prove the following theorem.

Theorem 4.3.9 (Fermat-Euler Theorem)

Let $n \geq 1$, and $a \in \mathbb{Z}$ coprime to n . Then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof Sketch. Note that $a \in (\mathbb{Z}/n\mathbb{Z})^\times$ which is a group, then $a^{\phi(n)} = a^{|\mathbb{Z}/n\mathbb{Z}^\times|} = e$, by the corollary we proved earlier. $\square$

§4.3.1 Exploring Group Using Lagrange

Recall that we have $|G| = |G : H| \cdot |H|$, so based on the order of G , we can deduce the possible orders of a subgroup H .

Remark. Just because some $k \mid |G|$, that doesn't imply that there is some subgroup of order k .

Example 4.3.10 (Subgroups of D_{10})

Consider the group D_{10} . The order of this group is 10, so its subgroups must be of order 1, 2, 5 and 10 by Lagrange's theorem. 1 does occur with the subgroup $\{e\}$. Also 10 occurs with the subgroup D_{10} .

If we want a subgroup of order 2, we need to have the identity e and some other element of order 2. There are 5 such elements in D_{10} (reflections), which gives us 5 subgroups of order 2.

If we want a subgroup of order 5, it must be cyclic by [Corollary 4.3.8](#). We have 4 elements of order 5 in D_{10} .

These are all of the subgroups of D_{10} .

We will study groups of small order in this course, and we can already classify groups of order ≤ 5 using Lagrange.

- If $|G| = 1$, then $G = \{e\}$.

- If $|G| = 2, 3$, or 5 , then G is the cyclic group C_2, C_3 , and C_5 respectively, by Corollary 4.3.8.
- If $|G| = 4$, we have two possibilities, as we will see in the following proposition.

Proposition 4.3.11 (Groups of Order 4)

If $|G| = 4$, then $G \cong C_4$ or $G \cong C_2 \times C_2$.

Proof. By Lagrange, the possible orders of elements of G are 1, 2 or 4. If there is an element of order 4, say g , then $\langle g \rangle = G \cong C_4$. If there is no element of order 4, then all non-identity elements have order 2. We proved in Example Sheet 1 (apologies if you have not done this question) that G is abelian. We can take two distinct elements of order 2, say b, c . Then $\langle b \rangle \cap \langle c \rangle = \{e, b\} \cap \{e, c\} = \{e\}$. Also everything commutes as G is abelian, and if we have bc , then $bc \neq b$ and $bc \neq c$. Also $bc \neq e$ as otherwise $b = c^{-1} = c$. Thus bc is the fourth element of the group. Thus by the direct product theorem, $G = \langle b \rangle \times \langle c \rangle \cong C_2 \times C_2$. $\square$

To study groups of order 6 and greater, we are going to need more algebraic machinery.

§4.4 Quotients of Groups

Using the direct product, we have a notion of ‘multiplying’ groups together. We will now think about how and when it makes sense to ‘divide’ one group by another.

§4.4.1 Normal Subgroups

We will also be interested in subgroups for which the left and right cosets coincide.

Definition 4.4.1 (Normal Subgroups)

A subgroup $N \leq L$ is **normal** if $\forall g \in G, gN = Ng$. We write $N \trianglelefteq G$.

Proposition 4.4.2 (Equivalent Conditions for Normal Subgroups)

The following are all equivalent conditions for a subgroup being normal.

- (i) $\forall g \in G, gN = Ng$.
- (ii) $\forall g \in G$, and $\forall n \in N, g^{-1}ng \in N$ (or $gng^{-1} \in N$).
- (iii) $\forall g \in G, g^{-1}Ng = N$.

Proof Sketch. Follows from definitions. $\square$

We will use all of these conditions interchangeably.

Example 4.4.3

The following are all normal subgroups

- (i) $n\mathbb{Z} \trianglelefteq \mathbb{Z}$, as for all $a \in \mathbb{Z}$, we have $a + n\mathbb{Z} = \{a + nk : k \in \mathbb{Z}\} = \{nk + a : k \in \mathbb{Z}\} = n\mathbb{Z} + a$.
- (ii) $A_3 \trianglelefteq S_3$, where $A_3 = \{e, (1\ 2\ 3), (1\ 3\ 2)\}$. Clearly $eA_3 = A_3e$, and $(1\ 2\ 3)A_3 = A_3(1\ 2\ 3)$, $(1\ 3\ 2)A_3 = A_3(1\ 3\ 2)$. Also $(1\ 2)A_3 = \{(1\ 2), (2\ 3), (1\ 3)\} = A_3(1\ 2)$, and similarly for $(1\ 3)$ and $(2\ 3)$.

Proposition 4.4.4

Any subgroup of an abelian group is normal.

Proof. For any subgroup N , if G is abelian, we have $g^{-1}ng = n \in N$, so N is normal. $\square$

Proposition 4.4.5

Any subgroup of index 2 is normal.

Proof. If $H \leq G$ is a subgroup with $|G : H| = 2$, then there are only two cosets. $H = eH = He$ is one of the cosets, and since cosets are disjoint (by Lagrange), the other coset has to be $G \setminus H$ both for left and right cosets. So H is normal. $\square$

These propositions give us two rich sources of normal subgroups, and we will now see something that gives all possible normal subgroups.

Proposition 4.4.6

If $\phi : G \rightarrow H$ is a homomorphism, then $\ker \phi \trianglelefteq G$.

Proof. First we note that the kernel of a homomorphism is a subgroup. Then given some $k \in \ker \phi$, $g \in G$, then

$$\phi(g^{-1}kg) = \phi(g)^{-1}\phi(k)\phi(g) = \phi(g)^{-1}e\phi(g) = e,$$

so $g^{-1}kg \in \ker \phi$, for all $g \in G$ and $k \in \ker \phi$. $\square$

Later on, we will see that normal subgroups are *exactly* kernels of homomorphisms.

Example 4.4.7

The following are all normal subgroups.

1. $\mathrm{SL}_2(\mathbb{R}) \trianglelefteq \mathrm{GL}_2(\mathbb{R})$, as $\mathrm{SL}_2(\mathbb{R}) = \ker(\det)$.
2. $A_n \trianglelefteq S_n$ as A_n is the kernel of the sign homomorphism. Also $|S_n : A_n| = 2$.

With the notion of normal subgroups, we can continue to classify groups of small order.

Proposition 4.4.8 (Groups of Order 6)

If $|G| = 6$, then $G \cong C_6$ or D_6 .

Proof. By Lagrange, possible element orders are 1 (for e), 2, 3 and 6. If there is an element of order 6, then $G \cong C_6$. If there is no such element, then by one of the questions on the example sheet (again sorry), then there is an element of order 3, say r , as otherwise our group would have an order that is a power of 2. So $|\langle r \rangle| = 3$ and by Lagrange, $|G| = 6 = |G : \langle r \rangle| \cdot |\langle r \rangle| = 3|G : \langle r \rangle|$. So $\langle r \rangle \trianglelefteq G$.

There must also be an element s of order 2, since $|G|$ is even (again see the example sheet). We can now consider what $s^{-1}rs \in \langle r \rangle$ can be.

If $s^{-1}rs = e$, then $r = e$, which is not the case. If $s^{-1}rs = r$, then $sr = rs$, which implies that sr has order 6, which we assumed there wasn't. Thus we must have $s^{-1}rs = r^2$, so $G = \langle r, s \rangle$ with $r^3 = s^2 = e$, and $sr = r^2s = r^{-1}s$, which is how we define D_6 . $\square$

§4.4.2 Quotients

We said that the goal of this chapter was to develop the notion of ‘dividing’ groups, but what does that actually mean? As in, why would we attempt to make sense of such an idea? In a previous chapter, we looked at how to recognize when there was some surjective homomorphism from a group G to another group H . We are now going to take this idea further, to see how we can construct *all* groups H such that there is a surjective homomorphism from a given group G to H . Using this, we will be able to select exactly the properties of G that we wish to preserve in H , and which we want to ignore. In the previous section we developed the idea of *normal* subgroups. This will be a key aspect in developing this idea.

We are first going to define how we can ‘multiply’ cosets.

Definition 4.4.9 (Coset Multiplication)

Let G be a group with a subgroup H . We define the multiplication of the coset of a and of b to be the coset of ab , that is,

$$aH \cdot bH = (ab)H.$$

There are some inherent problems with this definition. For one, it's not immediately clear that such a product is well defined. If we had some cosets $aH = a'H$ and $bH = b'H$, it is not necessarily the case that their product will be equal, that is

$$aH = a'H, \quad bH = b'H \quad \text{does not imply} \quad (ab)H = (a'b')H.$$

So when is such a product well defined? It turns out that the required condition is that H must be normal.

Proposition 4.4.10 (Coset Multiplication for Normal Subgroups)

Let H be a normal subgroup of G . Then if $aH = a'H$ and $bH = b'H$, we have $(ab)H = (a'b')H$.

Proof. If $aH = a'H$, then $a \in a'H$, so $a = a'h_1$ for some $h_1 \in H$. Also if $bH = b'H$, we have $b = b'h_2$ for $h_2 \in H$. Thus $ab = a'h_1b'h_2$. Then as H is normal, $h_1b' = b'h_3$ for some $h_3 \in H$. Then $ab = a'b'h_3h_2$, and as $h_3h_2 \in H$, this must be in $(a'b')H$. $\square$

We now get to the central result of this section.

Proposition 4.4.11

Let $N \trianglelefteq G$. The set of cosets of N in G forms a group under coset multiplication.

Proof. This operation is well defined by the proposition above. We now check the group axioms.

- *Closure.* If g_1N and g_2N are cosets, so is g_1g_2N .
- *Identity.* $eN = N$.
- *Associativity.* This follows from the associativity of G : $(g_1N \cdot g_2N) \cdot g_3N = (g_1g_2N) \cdot g_3N = ((g_1g_2)g_3)N = (g_1(g_2g_3))N = g_1N \cdot (g_2N \cdot g_3N)$.

□

Definition 4.4.12 (Quotient Group)

If $N \trianglelefteq G$, then G/N is the group of cosets of N in G is called the **quotient group** of G by N .

Example 4.4.13 (Quotient Groups)

- (i) The group of integers modulo n is the quotient group of $n\mathbb{Z}$ in $\mathbb{Z}$ is a group, $\mathbb{Z}/n\mathbb{Z}$.
- (ii) As $A_3 \trianglelefteq S_3$, we have the quotient group S_3/A_3 , which has 2 elements since $|S_3 : A_3| = 2$. Thus $S_3/A_3 \cong C_2$.
- (iii) If $G = H \times K$, then both H and K are normal in G , and $G/H \cong K$ and $G/K \cong H$.
- (iv) If $N = \langle r^2 \rangle \leq D_8$, then it is a normal subgroup, and $D_8/N \cong C_2 \times C_2$.

Remark. In general, quotients are *not* subgroups. In general, they may not even be isomorphic to a subgroup in a group.

Theorem 4.4.14

Given $N \trianglelefteq G$, the function $\pi : G \rightarrow G/N$ where $\pi(g) = gN$ is a surjective homomorphism called the **quotient map**, and $\ker \pi = N$.

Proof. First we prove that π is a homomorphism. We have $\pi(g)\pi(h) = gN \cdot hN = ghN = \pi(gh)$. It's clearly injective, and also $\pi(g) = gN = N \iff g \in N$. Thus $\ker \pi = N$. □

This is a key result: *normal subgroups are exactly kernels of homomorphisms.*

§4.4.3 The Isomorphism Theorems

Using the idea of quotient groups, we can prove the following theorem.

Theorem 4.4.15 (First Isomorphism Theorem)

Let $\phi : G \rightarrow H$ be a homomorphism. Then $G/\ker \phi \cong \text{img } \phi$.

Proof. Define $\bar{\phi} : G/\ker \phi \rightarrow \text{img } \phi$ via $g\ker \phi \rightarrow \phi(g)$. This is well defined, as if $g_1\ker \phi = g_2\ker \phi$, then $g_1 = g_2k$ for some $k \in \ker \phi$. So $\bar{\phi}(g_1\ker \phi) = \phi(g_1) = \phi(g_2k) = \phi(g_2)\phi(k) = \phi(g_2) = \bar{\phi}(g_2\ker \phi)$.

We can show that $\bar{\phi}$ is a homomorphism, as

$$\bar{\phi}(g\ker \phi \cdot g'\ker \phi) = \bar{\phi}(gg'\ker \phi) = \phi(gg') = \phi(g)\phi(g') = \bar{\phi}(g\ker \phi) \cdot \bar{\phi}(g'\ker \phi).$$

We now show that $\bar{\phi}$ is also an isomorphism, by showing it is a bijection. First $\bar{\phi}$ is surjective as all elements in $\text{img } \phi$ are of the form $\phi(g)$ for some $g \in G$. It is also injective, as if $\bar{\phi}(g\ker \phi) = e = \phi(g)$ in $\text{img } \phi$, then $g \in \ker \phi$, so $g\ker \phi = \ker \phi$. $\square$

Example 4.4.16

- (i) $\det : \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R}^\times$ has $\text{img } \det = \mathbb{R}^\times$, $\ker \det = \text{SL}_2(\mathbb{R})$, so $\text{GL}_2(\mathbb{R})/\text{SL}_2(\mathbb{R}) \cong \mathbb{R}^\times$.
- (ii) Consider the map $\phi : \mathbb{R} \rightarrow \mathbb{C}^\times$, where $\phi(r) = e^{2\pi ir}$. This is a homomorphism, and its image is the circle of radius 1 in $\mathbb{C}$ (which is S^1), and the kernel is $\mathbb{Z}$. Then by the first isomorphism theorem, we have $\mathbb{R}/\mathbb{Z} \cong S^1$.

Quotient groups also have an interesting subgroup structure.

Theorem 4.4.17 (Correspondence Theorem)

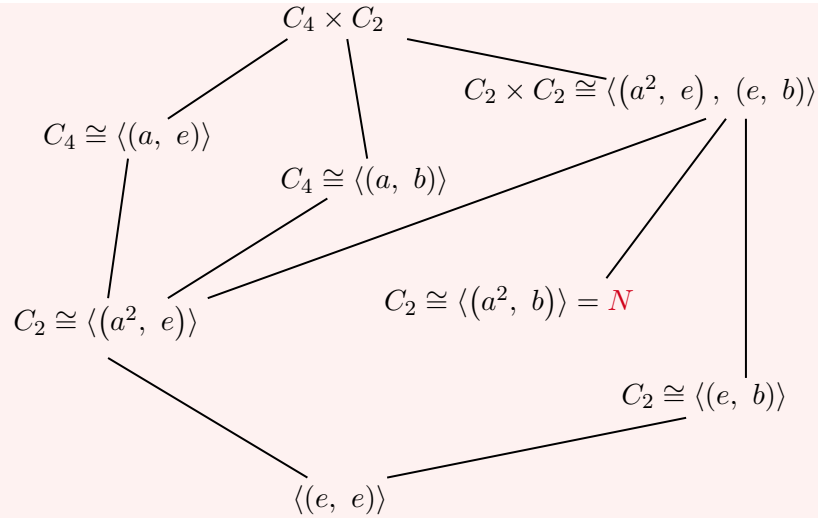
Let $N \trianglelefteq G$, then the subgroups of G/N are in bijective correspondence with subgroups of G containing N .

Proof. If we have some $N \leq M \leq G$, with $N \trianglelefteq G$, then $N \trianglelefteq M$, and clearly $M/N \leq G/N$. Conversely, for every subgroup $H \leq G/N$, we can take the preimage of H under the quotient map $\phi : G \rightarrow G/N$, that is, $\phi^{-1}(H) = \{g \in G : gN \in H\}$. We can check that this is a subgroup of G by checking the group axioms. We have closure by as $g_1, g_2 \in \phi^{-1}(H)$, then $g_1g_2N = g_1N \cdot g_2N$, so $g_1g_2N \in H$, and we inherit identity, inverses and associativity from the parent group. Now $\phi^{-1}(H)$ contains N , since $\forall n \in N$, $nN = N \in H$ (it's the identity coset). We can check that for $N \leq M \leq G$, $\phi^{-1}(M/N) = M$, and for $H \leq G/N$, $\phi^{-1}(H)/N = H$. So this correspondence is bijective. $\square$

This correspondence preserves lots of structure, for example indices, normality, containment.

Example 4.4.18

Suppose $C_4 = \{e, a, a^2, a^3\}$ and $C_2 = \{e, b\}$. Consider the group $\mathbb{C}_4 \times C_2$. Its subgroups are as follows.



We will define $N = \langle (a^2, b) \rangle$, which is normal because our group is abelian. Then if we look at $C_4 \times C_2/N$, then the correspondence theorem tells us that its subgroups are the intermediate subgroups between $C_4 \times C_2$ and N , and thus it has the following subgroup lattice.

$$C_4 \times C_2/N \quad \text{————} \quad C_2 \cong \{N, (e, b)N\} \quad \text{————} \quad \{N\}$$

This also shows us that because $C_4 \times C_2/N$ has order 4 by Lagrange, it must be isomorphic to C_4 (as otherwise it would be isomorphic to $C_2 \times C_2$, which would have a different subgroup lattice).

Now, if we had some subgroup $H \leq G$ that didn't contain $N \trianglelefteq G$, we can still make a normal subgroup of H by intersecting. This is the second isomorphism theorem.

Theorem 4.4.19 (Second Isomorphism Theorem)

Let $H \leq G$ and $N \trianglelefteq G$. Then $H \cap N \trianglelefteq H$ and $H/(H \cap N) \cong HN/N$.

Proof Sketch. When $N \trianglelefteq G$ and $H \leq G$, then $HN = \{hn : h \in H, n \in N\}$ is a subgroup of H , and $HN = \langle H, N \rangle$ (the smallest subgroups of G containing H and N). Then consider the function $\phi : H \rightarrow HN/N$, with $\phi(h) = hN$. We can check this is a well defined, surjective homomorphism, and its kernel is $N \cap H$, and we can apply the first isomorphism to get $H/(N \cap H) \cong HN/N \leq G/N$. $\square$

We mentioned previously that normality is preserved. We can even say something about quotients.

Theorem 4.4.20 (Third Isomorphism Theorem)

Let $N \leq M \leq G$ such that $N \trianglelefteq G$ and $M \trianglelefteq G$. Then $M/N \trianglelefteq G/N$ and $(G/N)/(M/N) \cong G/M$.

Proof. We define $\phi : G/N \rightarrow G/M$ by $\phi(gN) = gM$. ϕ is well-defined since $N \leq M$, and it's a surjective homomorphism. Its kernel is M/N , and thus by the first isomorphism theorem, $(G/N)/(M/N) \cong G/M$. $\square$

Let's have a look at some examples of the isomorphism theorems.

Example 4.4.21

Consider the group $\mathbb{Z}$, and define $H = 3\mathbb{Z}$ and $N = 5\mathbb{Z}$. Note that Then by the second isomorphism theorem as , $H/(H \cap N) \cong HN/N \cong \mathbb{Z}/5\mathbb{Z}$, as $\langle H, N \rangle = \mathbb{Z}$ by Bezout's lemma.

§4.4.4 Simple Groups

We will finish this chapter with the introduction of *simple groups*.

Definition 4.4.22 (Simple Groups)

A group G is **simple** if its only normal subgroups are $\{e\}$ and G .

Example 4.4.23 (Examples of Simple Groups)

The following are simple groups:

- C_p for a prime p .
- A_5 (we will prove this later on).

Finite simple groups are important as they can be thought of as the 'building blocks' of all finite groups. If we have some finite group with a normal subgroup, we can take the quotient by that normal subgroup, and you would obtain two objects (the quotient and the normal subgroup) that are smaller than the original group. These objects can help you understand the groups more, but simple groups can't be decomposed further.

Recently, the classification of all finite simple groups was completed, and you can find a large book with all finite simple groups in it!

§4.5 Group Actions

The next aspect of group theory we shall look at is *group actions*, where we study how groups interact with other objects. This isn't such a foreign concept – if you consider the examples of groups that we have looked at previously, we have in many cases been able to identify elements by their effect on some set. For example, we determined the elements of S_n by how it permuted elements of the set $\{1, 2, \dots, n\}$, and we defined D_{2n} based on the symmetries of an n -gon.

Of course, using the group axioms we can somewhat forget that these were ever groups that acted on certain objects, but by introducing these objects that groups can act upon adds a certain richness to the subject.

Definition 4.5.1 (Group Action)

Let G be a group, and let X be a set. An **action** of G on X is a function $\alpha : G \times X \rightarrow X$ written $\alpha(g, x) = \alpha_g(x)$, satisfying

- $\alpha_g(x) \in X$ for all $g \in G$ and $x \in X$.

- $\alpha_e(x) = x$ for all $x \in X$.
- $\alpha_g \circ \alpha_h(x) = \alpha_{gh}(x)$ for any $g, h \in G$ and $x \in X$.

Example 4.5.2

The following are all examples of groups that act on various sets.

- (i) The group S_n acts on X by permutation.
- (ii) The group D_{2n} acts on the vertices of a regular n -gon, and if we label these vertices $1, 2, \dots, n$, we get an action on the set $\{1, 2, \dots, n\}$.
- (iii) The symmetries of a cube act on a set of vertices, a set of edges, a set of faces, and a set of pairs of opposite faces.

Remark. These examples show us that more than one group can act on a given set, for example in (i) and (ii). A group can also act on many sets.

Lemma 4.5.3

For all $g \in G$ and an action $\alpha_g : X \rightarrow X$, the map $x \mapsto \alpha_g(x)$ is a bijection.

Proof. We have $\alpha_g(\alpha_{g^{-1}}(x)) = \alpha_{gg^{-1}}(x) = \alpha_e(x) = x$ for all $x \in X$, and similarly, $\alpha_{g^{-1}}(\alpha_g(x)) = x$ for all $x \in X$. Hence $\alpha_g \circ \alpha_{g^{-1}}$ and $\alpha_{g^{-1}} \circ \alpha_g$ are the identity functions, so α_g is a bijection. $\square$

We can also define actions by linking G to $\text{Sym}(X)$ in the following way.

Proposition 4.5.4

Let G be a group, and X be a set. Then $\alpha : G \times X \rightarrow X$ is an action if and only if the function $\rho : G \rightarrow \text{Sym}(X)$ where $\rho(g) = \alpha(g)$ is a homomorphism.

Proof. If α is an action, then by [Lemma 4.5.3](#), α_g is a bijection from $X \rightarrow X$, so $\alpha_g \in \text{Sym}(X)$. Now $\rho(gh) = \alpha_{gh}$ and for all $x \in X$, $\alpha_{gh}(x) = \alpha_g \circ \alpha_h(x)$, so $\rho(gh) = \alpha_{gh} = \alpha_g \circ \alpha_h = \rho(g)\rho(h)$, so ρ is a homomorphism.

Otherwise, if we have some homomorphism $\rho : G \rightarrow \text{Sym}(X)$, we can define $\alpha : G \times X \rightarrow X$ by $\alpha(g, x) = \alpha_g(x) = \rho(g)(x)$. Then α is an action. This is true because $\rho(g) \in \text{Sym}(X)$, so $\rho(g)(x) = \alpha_g(x) \in X$, and $\rho(e)$ is the identity in $\text{Sym}(X)$ which implies $\rho(e)(x) = x$, and finally, $\rho(gh) = \rho(g)\rho(h)$ implies that $\alpha_{gh}(x) = \alpha_g \circ \alpha_h(x)$ for all $x \in X$. $\square$

We can make the notation slightly easier. When we write $\alpha_g : X \rightarrow X$, we can really think of g as being a function on x , and we can write $g(x)$ instead.

With the above proposition in mind, we can define the *kernel* of an action.

Definition 4.5.5

The **kernel of an action** $\alpha : G \times X \rightarrow X$ is the kernel of the homomorphism $\rho : G \rightarrow \text{Sym}(X)$ (as above).

These are all of the elements of G that act as the identity of $\text{Sym}(X)$, that is, they do nothing to every $x \in X$. Note that by the first isomorphism theorem, this also implies that $G/\ker \rho \cong \text{img } \rho \leq \text{Sym}(X)$, and thus if $\ker \rho = \{e\}$, then $G \leq \text{Sym}(X)$.

Example 4.5.6

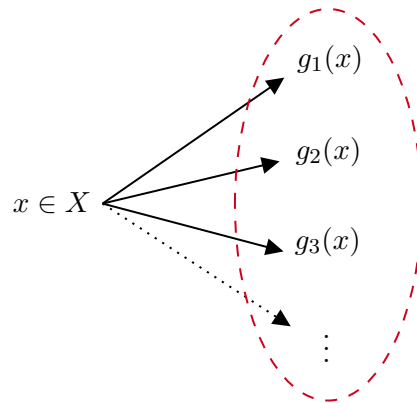
- (i) D_{2n} acting on $\{1, \dots, n\}$ (the labelled vertices on an n -gon) has $\ker \rho = \{e\}$, as every non-trivial element of D_{2n} moves at least one vertex. Thus $D_{2n} \leq S_n$.
- (ii) Let G be the symmetries of a cube, and consider X to be the set of unordered pairs of opposite faces. Then $|X| = 3$. So we get a homomorphism $\rho : G \rightarrow S_3$. Clearly there are symmetries of the cube that realize all of the permutation of X , so ρ is surjective. So, $G/\ker \rho \cong S_3$.

Definition 4.5.7 (Faithful Actions)

An action of G on X is called **faithful** if $\ker \rho = \{e\}$.

§4.5.1 Orbits and Stabilisers

Consider the following two questions: for a group G acting on a set X , what elements of X we can ‘get to’ from a certain $x \in X$ using the action of G ? Also, which group elements leave a given $x \in X$ unchanged?



Definition 4.5.8 (Orbit)

Let G act on X , and let $x \in X$. The **orbit** of x is

$$\text{Orb}(x) = \{g(x) : g \in G\} \subseteq X.$$

Definition 4.5.9 (Stabilizer)

The **stabilizer** of x is

$$\text{Stab}(x) = \{g \in G : g(x) = x\} \subseteq G.$$

We say that an action of **transitive** if $\text{Orb}(x) = X$, that is, if we can get to any element from any other element.

Example 4.5.10

For $G = S_3 = \{e, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_4$. G acts on $\{1, 2, 3, 4\}$, and we have $\text{Orb}(1) = \text{Orb}(2) = \text{Orb}(3) = \{1, 2, 3\}$, but $\text{Orb}(4) = \{4\}$. Also $\text{Stab}(1) = \{e, (2\ 3)\}$, $\text{Stab}(2) = \{e, (1\ 3)\}$, $\text{Stab}(3) = \{e, (1\ 2)\}$ and $\text{Stab}(4) = G$.

Lemma 4.5.11

For any $x \in X$, $\text{Stab}(x)$ is a subgroup of G .

Proof. We check the group axioms.

- *Closure.* If $g, h \in \text{Stab}(x)$, then $(gh)(x) = g(h(x)) = g(x) = x$, so $gh \in \text{Stab}(x)$.
- *Identity.* $e(x) = x$, so $e \in \text{Stab}(x)$.
- *Inverses.* If $g \in \text{Stab}(x)$ then $g(x) = x$ so $x = g^{-1}(x)$, and $g^{-1} \in \text{Stab}(x)$.

□

Now recall that a partition of a set X is a set of subsets of X such that each $x \in X$ belongs to exactly one subset in the partition.

Lemma 4.5.12

Let G act on X . Then the orbits partition X .

Proof. Firstly, for any $x \in X$, $x \in \text{Orb}(x)$. Secondly, if $z \in \text{Orb}(x) \cap \text{Orb}(y)$, then there exists $g_1 \in G$ such that $g_1(x) = z$, and there exists $g_2 \in G$ such that $g_2(y) = z$, that is, $y = g_2^{-1}(z)$. So $y = g_2^{-1}g_1(x)$, and thus for any $g \in G$, $g(y) = (gg_2^{-1}g_1)(x) \in \text{Orb}(x)$. So $\text{Orb}(y) \subseteq \text{Orb}(x)$. Similarly, $\text{Orb}(x) \subseteq \text{Orb}(y)$. Thus orbits are either disjoint or equal. □

Recall the proof that any permutation could be written as the product of disjoint cycles. What we were really doing was finding the orbits in $\{1, 2, \dots, n\}$ under $\langle \sigma \rangle$, which are disjoint.

Remark. Note that unlike cosets, the sizes of orbits can be different.

When we have an action, we can say even more about the structure of our group.

Theorem 4.5.13 (Orbit-Stabiliser Theorem)

Let a finite group G act on X . Then for any $x \in X$,

$$|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|.$$

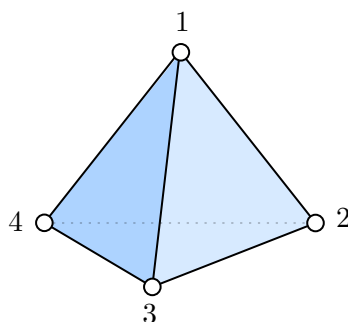
Proof. Note that $g(x) = h(x)$ occurs if and only if $h^{-1}g(x) = x$, that is, when $h^{-1}g \in \text{Stab}(x)$. Then this is true iff the cosets $g\text{Stab}(x) = h\text{Stab}(x)$. Rephrasing, distinct points in the orbit of x are in bijection with distinct cosets of $\text{Stab}(x)$. So $|\text{Orb}(x)| = |G : \text{Stab}(x)| = |G|/|\text{Stab}(x)|$ by Lagrange, and the result follows. □

In particular, notice that all elements in a given coset $g \text{Stab}(x)$ do the same thing to x as g , as an element of this coset $g \text{Stab}(x)$ has the form gh , $h \in \text{Stab}(x)$, so $gh(x) = g(h(x)) = g(x)$.

We can use the orbit-stabiliser theorem to investigate groups further. For example, we already know $|D_{2n}| = 2n$, but we can also show this by the orbit-stabiliser theorem. D_{2n} acts transitively on $\{1, \dots, n\}$, so $|\text{Orb}(1)| = n$, and $\text{Stab}(1) = \{e, s\}$, as the first vertex is fixed by one reflection and the identity. Thus $|D_{2n}| = |\text{Orb}(1)| \cdot |\text{Stab}(1)| = n \cdot 2 = 2n$.

Symmetries of the Tetrahedron

To see the power of using Orbits and Stabilisers, we are going to consider two more involved examples, beginning with the symmetries of a tetrahedron.



The tetrahedron has

- 4 faces (all regular triangles),
- 4 vertices, labelled $\{1, 2, 3, 4\}$,
- 6 edges.

So let G be the group of symmetries of the tetrahedron. Clearly G acts transitively on the vertices, and there's no non-identity symmetry that fixes all of the vertices. So we have an injective homomorphism $\rho : G \rightarrow S_4$.

We can consider the orbit and stabiliser for a vertex of the tetrahedron, say 1. For orbits, there is a symmetry that allows us to get from 1 to any other vertex, thus $\text{Orb}(1) = \{1, 2, 3, 4\}$. Then, the symmetries that leave the vertex 1 fixed are going to be the symmetries of the bottom face (the 2, 3, 4 triangle). Thus $\text{Stab}(1) \cong D_6$, the symmetries of a triangle.

With this information, then orbit stabiliser tells us that

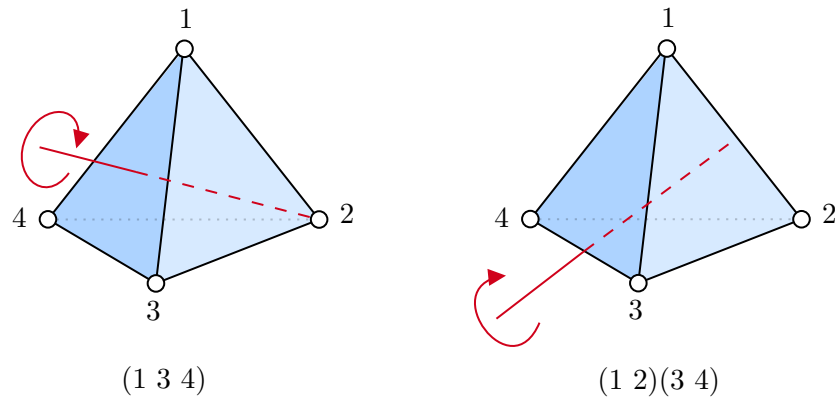
$$|G| = |\text{Orb}(1)| \cdot |\text{Stab}(1)| = 4 \cdot 6 = 24,$$

and as $|S_4| = 4! = 24$, and as $G \leq S_4$, we must have $G = S_4$. So just by considering the symmetries relating to one vertex, we were able to deduce the order of the group G , and also find a group that it's isomorphic to!

We can also consider a group G^+ of the rotations in G . Then in G^+ , $\text{Orb}(1) = \{1, 2, 3, 4\}$ as we can only use rotations, and then $\text{Stab}(1)$ is going to be the rotations of the bottom triangle, thus $\text{Stab}(1) \cong C_3$. Then again by the orbit stabiliser theorem,

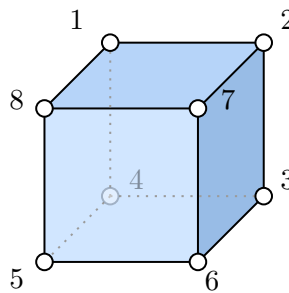
$$|G^+| = |\text{Orb}(1)| \cdot |\text{Stab}(1)| = 4 \cdot 3 = 12,$$

and as $G^+ \leq G \cong S_4$, we must have $G^+ = A_4$! Indeed, we can check that G^+ is made up of all of the even permutations of the four vertices, that is, all of the 3-cycles and the elements of the form $(1\ 2)(3\ 4)$.



Symmetries of the Cube

We are now going to look at the symmetries of a cube in a similar way.

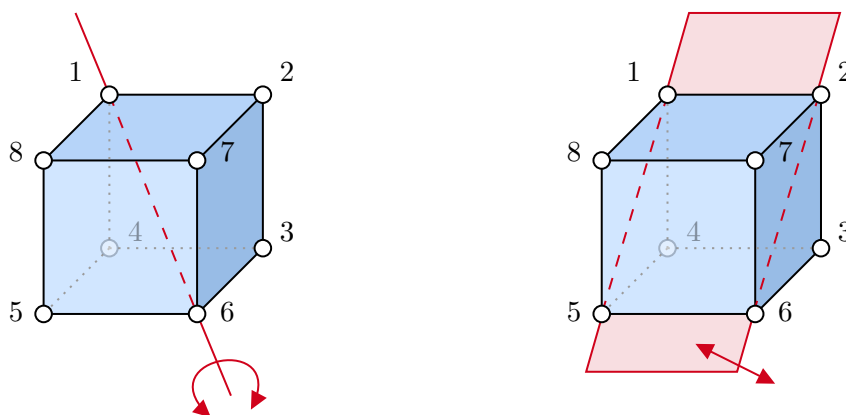


We have

- 6 faces (all square),
- 8 vertices, labelled $\{1, 2, 3, 4, 5, 6, 7, 8\}$,
- 12 edges.

Let G be the group of symmetries of the cube. In this example, we will find that it is interesting to consider either acting on the vertices or on the faces.

When G acts on the vertices of the cube, we can find the size of the orbit and stabiliser of a vertex geometrically. G is transitive, hence $|\text{Orb}(1)| = 8$. Then the elements that stabilize the vertex 1 are going to be the identity, the rotations about an axis through 1, and the reflections about planes through 1 and an outgoing edge. These are shown below.



So $|\text{Stab}(1)| = 6$. Then by the orbit-stabiliser theorem,

$$|G| = |\text{Orb}(1)| \cdot |\text{Stab}(1)| = 8 \cdot 6 = 48.$$

We will determine this group completely later.

Now let's look at the group of rotations, G^+ , acting on the 8 vertices. We have $|\text{Orb}(1)| = 8$ as before, but the stabilisers of 1 are only going to be the rotations, so $|\text{Stab}(1)| = 3$. Thus

$$|G^+| = |\text{Orb}(1)| \cdot |\text{Stab}(1)| = 8 \cdot 3 = 24.$$

Now if we let G^+ act on the four diagonals of the cube, giving $\rho : G^+ \rightarrow S_4$. We will have all 4-cycles in $\text{img } \rho$, and also all transpositions². So we have $(1\ 2)$ and $(1\ 2\ 3\ 4)$, but we found previously (in Example Sheet 2) that this generates S_4 , so ρ is surjective. Then as $|G^+| = 24 = |S_4|$, so we must have $G^+ \cong S_4$.

Platonic Solids

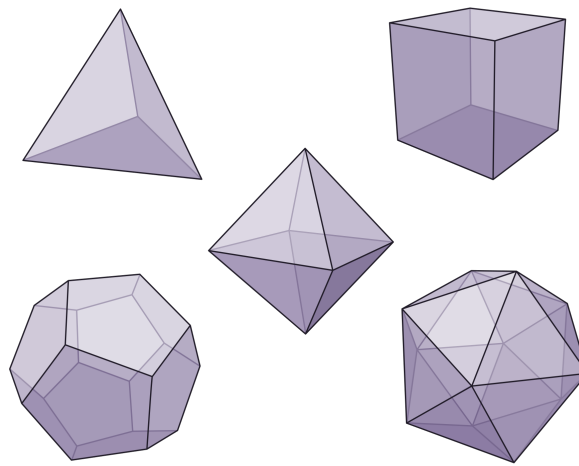
The last two shapes were two of the five *platonic solids*.

Definition 4.5.14 (Platonic Solid)

A **platonic solid** is a solid shape in $\mathbb{R}^3$ that has polygonal faces, straight edges and vertices such that the group of the shape's symmetries acts transitively on triples of the form (vertex, incident edge, incident face).

So these solids are particularly symmetric. The other platonic solids are the *octohedron*, *dodecahedron* and *icosahedron*.

²If you don't see why, I encourage you to draw a cube with a diagonal elongated, and see what the rotations do to it.



There are also links between the symmetries of these shapes - the cube and octahedron are **dual**, that is, they can be inscribed in each other with vertices in the middle of faces. The icosahedron and dodecahedron are also dual. In fact, this dual property implies that they have the same symmetry groups. So there is only 3 distinct symmetry groups of platonic solids.

§4.5.2 Cauchy's Theorem

We are now going to see how group actions can be used to prove a beautiful theorem. Back in Example Sheet 1, we had the following problem.

Problem. Let G be a group of even order. Show that G contains an element of order two.

It is *not true* in general that if $k \mid |G|$, then there is an element of order k – but it is true for primes. This result is *Cauchy's theorem*.

Theorem 4.5.15 (Cauchy's Theorem)

Let G be a finite group, and let p be a prime such that $p \mid |G|$. Then G has an element of order p .

Proof. Consider the group $G^p = \underbrace{G \times G \times \cdots \times G}_{p \text{ times}}$, that is, the group formed of p -tuples of elements of G , with component wise composition. Consider the subset $X \subseteq G^p$,

$$X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\},$$

Informally, ' p -tuples multiplying to e '. Note that $g \in G$ has order p if and only if $(g, \dots, g) \in X$ where g is not the identity.

Now take a cyclic group $C_p = \langle a \rangle$, and let C_p act on X by 'cycling', where $a(g_1, g_2, \dots, g_p) = (g_2, \dots, g_p, g_1)$. It is easily checked that this is an action. Since orbits partition X , the sum of the sizes of distinct orbits must be $|X|$, but $|X| = |G|^{p-1}$, since we can choose anything to put in the $p-1$ entries, and set the last entry to the inverse of their product. So since $p \mid |G|$, $p \mid |X|$. We have by the orbit-stabiliser

theorem that

$$|\text{Orb}(g_1, \dots, g_p)| \cdot |\text{Stab}(g_1, \dots, g_p)| = |C_p| = p.$$

So any orbit has size 1 or p , and they sum to $|X|$, so

$$|X| = \sum_{\text{orbits of size 1}} 1 + \sum_{\text{orbits of size } p} p.$$

Clearly $|\text{Orb}(e, e, \dots, e)| = 1$, and thus as $p \mid |X|$, there must be some other orbits of size 1. But orbits of size 1 must be of the form $(g, g, \dots, g)$, so there exists some $g \neq e \in G$ such that $g^p = e$. $\square$

§4.5.3 Important Actions

Once you have recovered from how nice the previous theorem was, we will look at some important examples of group actions. We will begin with left multiplication actions.

Lemma 4.5.16 (Left Multiplication is an Action)

Let G be a group. Then G acts on itself by left multiplication. This action is faithful and transitive.

Proof. For any $g \in G$ and $x \in G$, then $gx \in G$. Also, $ex = x$ for any $x \in G$, and lastly, $(gh)x = g(hx)$ is true by the associativity of G , so it is an action. It is faithful as if $gx = x$ then $g = e$ by the uniqueness of the identity. It's also transitive as given $x, y \in G$, set $g = yx^{-1}$, then $g(x) = gx = yx^{-1}x = y$. $\square$

Definition 4.5.17 (Left Regular Action)

The left multiplication action of a group on itself is called the **left regular action**.

With this group action, we can finally prove a result that we have been vaguely alluding to (and which should make intuitive sense).

Theorem 4.5.18 (Cayley's Theorem)

Every group is isomorphic to a subgroup of a symmetric group.

Proof. Let G act on itself by the left regular action. This gives a homomorphism $\rho : G \rightarrow \text{Sym}(G)$, with $\ker \rho = \{e\}$ since the action is faithful. Hence by the first isomorphism theorem, $G/\ker \rho \cong \text{img } \rho \leq \text{Sym}(G)$. $\square$

Proposition 4.5.19 (Left-Coset Action)

Let $H \leq G$. Then G acts on the set of left-cosets by left multiplication, and the action is transitive. This is the **left-coset action**.

Proof. We have $g(g_1H) = gg_1H$, so $g(g_1H)$ is a left coset. Also $e(g_1H) = eg_1H = g_1H$. Finally, $(gg')(g_1H) = gg'g_1H = g(g'(g_1H))$, so this is an action. Also given

g_1H and g_2H , we have $(g_1g_2^{-1})(g_2H) = g_1g_2^{-1}g_2H = g_1H$, so this action is transitive. $\square$

Remark. For $H = \{e\}$, then this is the left-regular action, as the cosets are just the elements of G .

This action gives us a way to induce actions of G onto its quotient groups G/N .

Conjugation

The next idea we will discuss is *conjugation*.

Definition 4.5.20 (Conjugation)

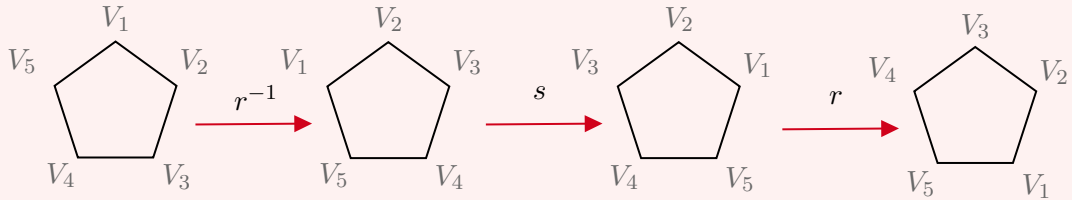
Given $g, h \in G$, the element $ghg^{-1} \in G$ is the conjugate of g by h .

Conjugate elements should be thought of as doing the ‘same thing’ but from ‘different points of view’. The exact meaning of this will be more clear as we look at some examples later on.

Example 4.5.21

Consider the group D_{10} , and consider the conjugate s and rsr^{-1} , where s is reflection through the vertex v_1 , and r is a clockwise rotation sending v_1 to v_2 and so on.

What rsr^{-1} does is rotating so that the reflection can be done through a different axis.



So in this example, we are still performing an action, just from a different point of view.

Example 4.5.22

In the group $GL_n \mathbb{R}$, the conjugate of a matrix represents the same transformation but written with respect to a different basis.

So it's natural to expect that conjugate elements have similar properties. Let's prove some facts about conjugation.

Proposition 4.5.23 (Conjugation is an Action)

A group G acts on itself by conjugation.

Proof. We have $g(x) = gxg^{-1} \in G$ for all $g, x \in G$. Also, $e(x) = exe^{-1} = x$ for all x , and finally, $g(h(x)) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = (gh)(x)$, for all $g, h, x \in G$. So this is an action. $\square$

This action's kernel, orbits and stabilisers even have their own names.

Definition 4.5.24 (Center)

The kernel of the conjugation action of G on itself is the **center** $Z(G)$ of G :

$$Z(G) = \{g \in G : ghg^{-1} = h, \forall h \in G\}.$$

Equivalently, the center of a group is the set of all elements that commute with every element in the group.

Definition 4.5.25 (Conjugacy Class)

An orbit of the conjugation action is a **conjugacy class**:

$$\text{ccl}(h) = \{ghg^{-1} : g \in G\}.$$

Definition 4.5.26 (centralizers)

The stabilisers of the conjugation action are the **centralizers**:

$$C(h) = \{g \in G : ghg^{-1} = h\}.$$

Equivalently, $C(h)$ is the set of elements that commute with h .

There is a connection between the center and the centralizers, namely

$$Z(G) = \bigcap_{h \in G} C_G(h).$$

Definition 4.5.27 (Subgroup Conjugation)

If $H \leq G$, $g \in G$, then the **conjugate of H by g** is

$$gHg^{-1} = \{ghg^{-1} : h \in H\}.$$

Proposition 4.5.28

Let $H \leq G$ and $g \in G$. Then gHg^{-1} is also a subgroup of G .

Proof. Clearly $e \in gHg^{-1}$, so it is nonempty. Thus we can apply the subgroup criterion. Let $g_1g_2^{-1}$ and $g_3g_4^{-1} \in gHg^{-1}$. Then

$$(g_1g_2^{-1})(g_3g_4^{-1})^{-1} = (g_1g_2^{-1}g_3^{-1}g_4) = g(xy^{-1})g^{-1},$$

and as $H \leq G$, we have $xy^{-1} \in H$ so $g(xy^{-1})g^{-1} \in gHg^{-1}$, so it is a subgroup. $\square$

In fact, $gHg^{-1} \cong H$, which should match our intuition about how conjugate elements act.

Proposition 4.5.29 (Subgroup Conjugation is an Action)

A group G acts by conjugation on the set of its subgroups. The singleton orbits are the normal subgroups.

Proof. Let G act on the set of subsets by conjugation, that is, for $H \leq G$, we define $g(H) = gHg^{-1}$. We will check that this is a group action.

We have closure by the argument above. We have identity, as $eHe^{-1} = H$ for any subgroup H . Also for $g, k \in G$, we have $g \circ k(H) = g(kHk^{-1}) = gkHk^{-1}g^{-1} = (gk)H(gk)^{-1} = (gk)(H)$, so this is indeed an action.

The singleton orbits are the normal subgroups, as a subgroup N is normal if $gNg^{-1} = N$ for any $g \in G$ (which corresponds to every element in G stabilising N). $\square$

Another way to look at normal subgroup is using conjugation. We already established that if and only if N is normal in G , we have $gNg^{-1} = N$, for any $g \in G$.

Proposition 4.5.30

Normal subgroups are unions of conjugacy classes.

Proof. Let $N \trianglelefteq G$. Then if $h \in N$, then $ghg^{-1} \in N$ for any $g \in G$. Hence $\text{ccl}(h) \subseteq N$. So N is a union of conjugacy classes of its elements, that is, $N = \bigcup_{h \in N} \text{ccl}(h)$. Conversely, suppose that we have a subgroup H that is the union of conjugacy classes. Then taking any $g \in G$ and $h \in H$, then $ghg^{-1} \in H$, so $H \trianglelefteq G$. $\square$

Example 4.5.31

Consider the group $A_3 = \{e, (1\ 2\ 3), (1\ 3\ 2)\} \trianglelefteq S_3$. We can write this as $A_3 = \{e\} \cup \{(1\ 2\ 3), (1\ 3\ 2)\}$, which are both conjugacy classes.

Conjugation in S_n and A_n is rather interesting. Let's begin with a lemma.

Lemma 4.5.32 (Conjugation Permutes Elements in a Cycle)

Given k -cycle $(a_1 \ \dots \ a_k)$ and $\sigma \in S_n$, we have

$$\sigma(a_1 \ \dots \ a_k)\sigma^{-1} = (\sigma(a_1) \ \sigma(a_2) \ \dots \ \sigma(a_k)),$$

which is also a k -cycle.

Proof. Note that if we consider the action of $\sigma(a_1 \ \dots \ a_k)\sigma^{-1}$ on a_i , we have

$$\sigma(a_i) \mapsto a_i \mapsto a_{i+1} \mapsto \sigma(a_{i+1}).$$

So $\sigma(a_1 \ \dots \ a_k)\sigma^{-1}$ does the same thing as $(\sigma(a_1) \ \sigma(a_2) \ \dots \ \sigma(a_k))$ on the set $\{\sigma(a_1), \dots, \sigma(a_k)\}$. Then for any $a \notin \{\sigma(a_1), \dots, \sigma(a_k)\}$, then $\sigma(a_1 \ \dots \ a_k)\sigma^{-1}$ and $(\sigma(a_1) \ \sigma(a_2) \ \dots \ \sigma(a_k))$ both leave a unchanged. $\square$

Proposition 4.5.33

Two elements of S_n are conjugate in S_n if and only if they have the same cycle type.

Proof. Two elements that are conjugate will certainly have the same cycle type, as given $\sigma \in S_n$, we can write σ as the product of disjoint cycles $\sigma = \sigma_1 \sigma_2 \cdots \sigma_m$. Then if $\rho \in S_n$, $\rho \sigma \rho^{-1} = (\rho \sigma_1 \rho^{-1})(\rho \sigma_2 \rho^{-1}) \cdots (\rho \sigma_m \rho^{-1})$, and by our previous lemma, $\rho \sigma_i \rho^{-1}$ is a cycle of the same length as σ_i , and $\rho \sigma_i \rho^{-1}$ are all disjoint (as ρ is a bijection). Conversely, if we have σ and τ that are the same cycle type, then we can write

$$\begin{aligned}\sigma &= (a_1 \ a_2 \ \cdots \ a_{k_1})(a_{k_1+1} \ a_{k_1+2} \ \cdots \ a_{k_2}) \cdots \\ \tau &= (b_1 \ b_2 \ \cdots \ b_{k_1})(b_{k_1+1} \ b_{k_1+2} \ \cdots \ b_{k_2}) \cdots\end{aligned}$$

in disjoint cycle notation, including any singletons (those fixed by σ and τ) so that all of $\{1, \dots, n\}$ appears in both σ and τ . Then setting ρ to be $\rho(a_i) = b_i$, we will have $\rho \sigma \rho^{-1} = \tau$. $\square$

Let's have a look at an example: the conjugacy classes of S_4 .

Example 4.5.34 (Normal subgroups of S_4)

Consider the following table of conjugacy classes of S_4 .

Cycle Type	Example Element	Size of ccl	Size of C_{S_4}	Sign
1, 1, 1, 1	e	1	24	+1
2, 1, 1	$(1 \ 2)$	$4 \cdot 3/2 = 6$	4	-1
2, 2	$(1 \ 2)(3 \ 4)$	$4 \cdot 3/(2 \cdot 2) = 3$	8	+1
3, 1	$(1 \ 2 \ 3)$	$4 \cdot 3 \cdot 2/3 = 8$	3	+1
4	$(1 \ 2 \ 3 \ 4)$	$4 \cdot 3 \cdot 2/4 = 6$	4	-1

From this table, we can work out the normal subgroups of S_4 . Each normal subgroup must contain e , and must be the union of conjugacy classes, and we must have the order of the subgroup dividing the order of the group. We can check what combinations of conjugacy classes satisfies these.

We can consider these possibilities by considering the divisors of 24, the order of the group: 1, 2, 3, 4, 6, 8, 12, 24. From these, we have

- Order 1: $\{e\}$
- Order 4: $\{e, (1 \ 2)(3 \ 4), (1 \ 3)(2 \ 4), (1 \ 4)(2 \ 3)\}$, which is $C_2 \times C_2$, also known as V_4 , the Klein four group.
- Order 12: A_4 .
- Order 24: S_4 .

Now because we have all of the normal subgroups, we can also get all of the possible quotients of S_4 , as

- $S_4/\{e\} \cong S_4$.
- $S_4/V_4 \cong S_3$.

- $S_4/A_4 \cong C_2$.
- $S_4/S_4 \cong \{e\}$.

Now let's look at conjugation in the alternating group. Note that $\text{ccl}_{S_n}(\sigma) = \{\tau\sigma\tau^{-1} : \tau \in S_n\}$ and $\text{ccl}_{A_n}(\sigma) = \{\tau\sigma\tau^{-1} : \tau \in A_n\}$, so $\text{ccl}_{A_n}(\sigma) \subseteq \text{ccl}_{S_n}(\sigma)$. But elements that are conjugate in S_n may not be conjugate in A_n . For example, $(2\ 3)(1\ 2\ 3)(2\ 3) = (1\ 3\ 2) \in S_3$, but $(2\ 3) \notin A_3$, and there are no elements $\tau \in A_3$ such that $\tau(1\ 2\ 3)\tau^{-1} = (1\ 3\ 2)$.

Some conjugacy classes of S_n will split into smaller conjugacy classes in A_n . By orbit-stabiliser, $|S_n| = |\text{ccl}_{S_n}(\sigma)| \cdot |C_{S_n}(\sigma)|$, and $|A_n| = |\text{ccl}_{A_n}(\sigma)| \cdot |C_{A_n}(\sigma)|$. But $|S_n| = 2 \cdot |A_n|$ and $|\text{ccl}_{A_n}(\sigma)| \leq |\text{ccl}_{S_n}(\sigma)|$, so either $\text{ccl}_{A_n}(\sigma) = \text{ccl}_{S_n}(\sigma)$ and $|C_{A_n}(\sigma)| = \frac{1}{2}|C_{S_n}(\sigma)|$, or $|\text{ccl}_{A_n}(\sigma)| = \frac{1}{2}|\text{ccl}_{S_n}(\sigma)|$ and $|C_{A_n}(\sigma)| = |C_{S_n}(\sigma)|$.

To consider this further, we will introduce a natural definition.

Definition 4.5.35 (Splitting Conjugacy Classes)

When $|\text{ccl}_{A_n}(\sigma)| = \frac{1}{2}|\text{ccl}_{S_n}(\sigma)|$, we say that the conjugacy class of σ **splits** in A_n .

So when does this happen?

Proposition 4.5.36

The conjugacy class of $\sigma \in A_n$ splits in A_n if and only if no odd permutations commute with σ .

Proof. $|\text{ccl}_{A_n}(\sigma)| = \frac{1}{2}|\text{ccl}_{S_n}(\sigma)| \iff C_{A_n}(\sigma) = C_{S_n}(\sigma)$. Now $C_{A_n}(\sigma) = A_n \cap C_{S_n}(\sigma)$, and $A_n \cap C_{S_n}(\sigma) = C_{S_n}(\sigma)$ if and only if $C_{S_n}(\sigma)$ doesn't contain any odd elements, that is, when no odd permutations commute with σ . $\square$

Example 4.5.37 (Conjugacy classes in A_4)

We will consider the conjugacy classes in A_4 .

Cycle Type	Example Element	Odd Element in C_{S_4} ?	Size of ccl_{S_4}	Size of ccl_{A_4}
1, 1, 1, 1	e	yes, eg. $(1\ 2)$	1	1
2, 2	$(1\ 2)(3\ 4)$	yes, eg. $(1\ 2)$	3	3
3, 1	$(1\ 2\ 3)$	no	8	4

Example 4.5.38 (Conjugacy classes in A_5)

We will consider the conjugacy classes in A_5 .

Cycle Type	Example Element	Odd Element in C_{S_5} ?	Size of ccl_{S_5}	Size of ccl_{A_5}
1, 1, 1, 1	e	yes, eg. $(1\ 2)$	1	1
2, 2, 1	$(1\ 2)(3\ 4)$	yes, eg. $(1\ 2)$	15	15
3, 1, 1	$(1\ 2\ 3)$	yes, eg. $(4\ 5)$	20	20
5	$(1\ 2\ 3\ 4\ 5)$	no	24	12

We can use this result to prove something that was mentioned earlier.

Theorem 4.5.39

A_5 is simple.

Proof. Normal subgroups must be unions of conjugacy classes, must contain e and must divide $|A_5| = 60$. The size of conjugacy classes in A_5 are 1, 15, 20, 12 and 12. To sum these to get a divisor of 60, we can only have 1, and $1 + 15 + 20 + 12 + 12 = 60$, so the only normal subgroups are $\{e\}$ and A_5 . $\square$

Remark. All A_n for $n \geq 5$ are simple.

§4.6 The Möbius Group, Revisited

With group actions, we now have more tools to study the Möbius group M . Recall that we defined the Möbius group as the set of Möbius maps, under composition.

Definition (Möbius Maps)

A **Möbius map** is a function $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ of the form

$$f(z) = \frac{az + b}{cz + d},$$

with $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$, and with $f(-d/c) = \infty$ and

$$f(\infty) = \begin{cases} \frac{a}{c} & \text{if } c \neq 0 \\ \infty & \text{if } c = 0 \end{cases}.$$

This definition defined an action of the Möbius group on the extended complex plane. This action is faithful.

Proposition 4.6.1

The action of the Möbius group $\mathcal{M}$ on the extended complex plane is faithful, and so $\mathcal{M} \leq \text{Sym}(\hat{\mathbb{C}})$.

Proof. Consider $\rho : M \rightarrow \text{Sym}(\hat{\mathbb{C}})$ given by $\rho(f)(z) = f(z)$. Then if $\rho(f)$ is the identity permutation, then f is the identity in M . So ρ is injective and the action is faithful. $\square$

We can think about points that are invariant under Möbius map.

Definition 4.6.2 (Fixed Point)

A **fixed point** of a Möbius map $f : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ is a point $z \in \hat{\mathbb{C}}$ such that $f(z) = z$.

Theorem 4.6.3

A Möbius map with at least three fixed points is the identity.

Proof. Let $f(z) = \frac{az+b}{cz+d}$ have at least 3 fixed points. If ∞ is not a fixed point of f , then $\frac{az+b}{cz+d} = z$ for more than three complex numbers z , that is, $cz^2 + (d-a)z - b = 0$ has more than three roots in $\mathbb{C}$. But this is quadratic, so we can't have more than two roots, unless $c = b = 0$, $d = a$ for all z . But then $f(z) = z$. If ∞ is a fixed point of f , then we have $a/c = \infty \implies c = 0$, and thus $z = \frac{az+b}{d}$ for more than two complex numbers z , which can only occur when $a = d$, $b = 0$, so $f(z) = z$. $\square$

What this theorem really tells us is that knowing what a Möbius map does on 3 points in $\hat{\mathbb{C}}$ uniquely determines it.

Corollary 4.6.4

If two Möbius maps coincide on three distinct points in $\hat{\mathbb{C}}$, then they are equal.

Proof. Let $f, g \in \mathcal{M}$ be such that $f(z_1) = g(z_1)$, $f(z_2) = g(z_2)$ and $f(z_3) = g(z_3)$ for three distinct points z_1, z_2 and $z_3 \in \hat{\mathbb{C}}$. Then $g^{-1}f(z_i) = g^{-1}g(z_i) = z_i$ for $i = 1, 2, 3$, and thus $g^{-1}f$ fixed more than 3 points, and it must be the identity. So $f = g$. $\square$

Theorem 4.6.5 (Existence of Unique Möbius Maps)

There is a unique Möbius map sending any three distinct points of $\hat{\mathbb{C}}$ to any three distinct points of $\hat{\mathbb{C}}$. That is, given $z_1, z_2, z_3 \in \hat{\mathbb{C}}$ and $w_1, w_2, w_3 \in \hat{\mathbb{C}}$ all distinct then there's a unique $f \in \mathcal{M}$ such that $f(z_i) = w_i$ for $i = 1, 2, 3$.

Proof. Note that uniqueness follows from existence by the previous corollary. We will construct the map by supposing that $w_1 = 0$, $w_2 = 1$ and $w_3 = \infty$, and then we will use the group structure. Define $f(z) = \frac{(z_2 - z_3)(z - z_1)}{(z_2 - z_1)(z - z_3)}$. This satisfies $f(z_i) = w_i$ for all i . There are some special cases. If $z_1 = \infty$ use $f(z) = \frac{z_2 - z_3}{z - z_3}$, if $z_2 = \infty$ use $f(z) = \frac{z - z_1}{z - z_3}$, and if $z_3 = \infty$ use $f(z) = \frac{z - z_1}{z_2 - z_1}$.

Thus we can find some f_1 sending (z_1, z_2, z_3) to $(0, 1, \infty)$. But then in the same way we can find some f_2 sending (w_1, w_2, w_3) to $(0, 1, \infty)$. But then we can use the group structure, considering $f = f_2^{-1} \circ f_1$, this will send (z_1, z_2, z_3) to (w_1, w_2, w_3) , as required. $\square$

§4.6.1 Conjugation

For two Möbius maps f and h , we can consider what happens when we conjugate hfh^{-1} . This acts in much the same way that any conjugate in a group structure does. For example, $\text{ord}(hfh^{-1}) = \text{ord}(f)$, since $(hfh^{-1})^n = hf^n h^{-1}$. Also, If f fixes z , then

hfh^{-1} fixes z . In particular, the number of fixed points of a conjugate is the same as that of the original Möbius map. This observation has a partial converse.

Theorem 4.6.6

Every non-identity $f \in \mathcal{M}$ has either 1 or 2 fixed points. If f has 1 fixed point, then it will be conjugate to the map $z \mapsto z + 1$, and if f has 2 fixed points, it will be conjugate to a map of the form $z \mapsto az$ for some non-zero $a \in \mathbb{C}$.

Proof. We know that a non-identity Möbius map has at most 2 fixed points, but we show that there can't be zero fixed points. If $f(z) = \frac{az+b}{cz+d}$, then considering the quadratic $cz^2 + (d-a)z - b = 0$, formed by considering $f(z) = z$, this must have at least one solution.

Now we consider the two cases. If f has exactly 1 fixed point, say z_0 , then choose some $z_1 \in \mathbb{C}$ which is not fixed by f . Then consider the triple $(z_1, f(z_1), z_0)$, which are all distinct points. So there is some $g \in \mathcal{M}$ so that $(z_1, f(z_1), z_0) \mapsto (0, 1, \infty)$. Then we can consider the conjugate of f by g . Then gfg^{-1} sends $0 \mapsto z_1 \mapsto f(z_1) \mapsto 1$, and $\infty \mapsto z_0 \mapsto z_0 \mapsto \infty$, so $gfg^{-1}(0) = 1$, $gfg^{-1}(\infty) = \infty$, so gfg^{-1} must be equal to $z \mapsto az + 1$ for $a \in \mathbb{C}$. If $a \neq 1$, then there is a non-infinity fixed point, $1/(1-a)$, but this is a contradiction, so $a = 1$. Thus f is conjugate via g to the map $z \mapsto z + 1$, as required.

If f has exactly 2 fixed points, z_0 and z_1 , then let g be any Möbius map sending $(z_0, z_1) \mapsto (0, \infty)$. So gfg^{-1} sends $0 \mapsto z_0 \mapsto z_0 \mapsto 0$, and $\infty \mapsto z_1 \mapsto z_1 \mapsto \infty$. So this conjugate fixes 0 and ∞ . We can deduce that gfg^{-1} must have the form $z \mapsto az$, where $a = gfg^{-1}(1)$. $\square$

We can use this to efficiently compute powers of Möbius maps, f^n for all $f \in \mathcal{M}$. We can see that $(gfg^{-1})^m = gf^mg^{-1}$, but this will be easy to compute, because the conjugate can have a nice form. Then we can conjugate back to get f^m .

§4.6.2 Circles and Lines – Geometric Properties of Möbius Maps

We've seen that the image of 3 points in $\mathbb{C}$ under a given Möbius map uniquely determines that map. We also know (from geometry) that 3 points determine lines or circles.

In the complex plane, the a circle with center $b \in \mathbb{C}$ and radius $r > 0 \in \mathbb{C}$ is the locus of points satisfying

$$|z - b| = r.$$

We can then write this as $|z - b|^2 - r^2 = 0$, or $(z - b)(\overline{z - b}) - r^2 = 0$, which we can write as

$$z\bar{z} - \bar{b}z - b\bar{z} + b\bar{b} - r^2 = 0. \quad (*)$$

Also, the equation of a straight line in the complex plane (for $a, b, c \in \mathbb{R}$) is

$$a \cdot \operatorname{re}(z) + bi \cdot \operatorname{im}(z) = c,$$

that is

$$\frac{\overline{a + ib}}{2}z + \frac{a + ib}{2}\bar{z} - c = 0. \quad (\dagger)$$

The form of these equations is slightly unusual but they are put in this form to emphasize the following definition. While we require three points in $\mathbb{C}$ to determine a circle, we

only need two points to determine a line. But if we work in $\hat{\mathbb{C}}$, we consider the point at infinity, ∞ to be on *every* line. In this way, a line is determined by exactly 3 points in $\hat{\mathbb{C}}$, namely two points on the complex plane and a point at infinity.

Thinking in this way, we can unify the concepts of circles and lines in the extended complex plane, by introducing *clines*, a neologism covering both circles and lines.

Definition 4.6.7 (Clines in $\hat{\mathbb{C}}$)

A **cline** in $\hat{\mathbb{C}}$ is the set of points satisfying the equation

$$Az\bar{z} + \bar{B}z + B\bar{z} + C = 0,$$

with $A, C \in \mathbb{R}$, $B \in \mathbb{C}$, and $|B|^2 > AC$. We consider $\infty \in \hat{\mathbb{C}}$ to be a solution if and only if $A = 0$.

Any equation of the form in the definition will match with exactly one of equation $(*)$ or $(\dagger)$.

Looking back at Möbius maps, we get the rather interesting result.

Theorem 4.6.8 (Möbius Maps Preserve Clines)

Möbius send clines in $\hat{\mathbb{C}}$ to clines in $\hat{\mathbb{C}}$.

Proof. Recall that the Möbius group $\mathcal{M}$ is generated by a composition of dilations/rotations, translations and inversions. That is, $z \mapsto az$, $z \mapsto z + b$ and $z \mapsto 1/z$.

Writing $S(A, B, C)$ for the circle satisfying $Az\bar{z} + \bar{B}z + B\bar{z} + C = 0$, we can check what happens under the various maps.

Under $z \mapsto az$, we have $S(A, B, C) \mapsto S\left(\frac{A}{aa}, \frac{B}{a}, C\right)$. Under $z \mapsto z + b$, $S(A, B, C) \mapsto S(A, B - Ab, C + Ab\bar{b} - B\bar{b} - \bar{B}b)$. Finally, under $z \mapsto 1/z$, we have $S(A, B, C) \mapsto S(C, \bar{B}, A)$. $\square$

In practice, because both clines and Möbius maps are determined by 3 points (or where they send 3 points), it is quite straightforward to find a Möbius map which sends a given cline to another cline.

Example 4.6.9

Let's say we wanted to find $f \in \mathcal{M}$ which send the unit circle to the real line.

If we pick three points on the unit circle, say $\{-1, i, 1\}$, and three points on $\mathbb{R}$, say $\{-1, 0, 1\}$, then we could find a map f such that -1 and 1 are fixed, but so that i is sent to 0 . For example,

$$f(z) = \frac{z - i}{1 - iz}$$

works.

§4.6.3 Cross Ratios

Recall that given three points $z_1, z_2, z_3 \in \hat{\mathbb{C}}$, we have a unique Möbius map f such that $f(z_1) = 0$, $f(z_2) = 1$ and $f(z_3) = \infty$.

Definition 4.6.10 (Cross Ratio)

If we have four distinct points $z_1, z_2, z_3, z_4 \in \hat{\mathbb{C}}$, then their **cross-ratio** (z_1, z_2, z_3, z_4) is defined to be $f(z_4)$, where f is the unique Möbius map such that $f(z_1) = 0$, $f(z_2) = 1$ and $f(z_3) = \infty$.

In particular, the cross ratio of $(0, 1, \infty, w) = w$, for all $w \in \hat{\mathbb{C}}$. This is because f is forced to be the identity. There is a way to compute the cross ratio.

Theorem 4.6.11 (Computing Cross Ratios)

For points z_1, z_2, z_3 and $z_4 \in \hat{\mathbb{C}}$, we have

$$(z_1, z_2, z_3, z_4) = \frac{(z_4 - z_1)(z_2 - z_3)}{(z_2 - z_1)(z_4 - z_3)}.$$

Proof Sketch. This follows from the construction of the unique Möbius map. □

Remark. There are many conventions for the cross-ratio, depending on the order of 0, 1 and ∞ . Some of these will be the same.

Proposition 4.6.12

Double transpositions of the z_i fix the cross-ratio, that is,

$$(z_1, z_2, z_3, z_4) = (z_2, z_1, z_4, z_3) = (z_3, z_4, z_1, z_2) = (z_4, z_3, z_2, z_1).$$

Proof Sketch. Note that this follows from the previous theorem. □

There is a connection between cross ratios and Möbius maps.

Theorem 4.6.13

Möbius maps preserve cross-ratio.

Proof. Let $f \in \mathcal{M}$ be the unique Möbius map such that $f : (z_1, z_2, z_3) \mapsto (0, 1, \infty)$, so that $f(z_4) = (z_1, z_2, z_3, z_4)$. Then for any Möbius map $g \in \mathcal{M}$, we have $f \circ g^{-1}$ sends $g(z_1) \mapsto 0$, $g(z_2) \mapsto 1$, and $g(z_3) \mapsto \infty$. Also $f \circ g^{-1}$ is unique, as Möbius maps are uniquely determined by how they transform three points. So, if we have $(g(z_1), g(z_2), g(z_3), g(z_4)) = (f \circ g^{-1})(g(z_4)) = f(z_4) = (z_1, z_2, z_3, z_4)$. □

This has an interesting geometric corollary.

Theorem 4.6.14

Four distinct points $z_1, z_2, z_3, z_4 \in \hat{\mathbb{C}}$ lie on a cline if and only if $(z_1, z_2, z_3, z_4) \in \mathbb{R}$.

Proof. Let f be the unique Möbius map sending $(z_1, z_2, z_3) \rightarrow (0, 1, \infty)$, so that $f(z_4) = (z_1, z_2, z_3, z_4)$. The cline C passing through z_1, z_2, z_3 is sent to the cline through $0, 1, \infty$ (the real axis), and so z_4 lies on C if and only if $f(z_4)$ lies on the real axis, that is, $f(z_4) \in \mathbb{R} \cup \{\infty\}$. But $f(z_3) = \infty$, so $f(z_4) \neq \infty$. Thus z_4 lies on C if and only if $z_4 \in \mathbb{R}$. $\square$

§4.7 Matrix Groups

In this chapter we will look at various groups of matrices. We will look at the actions of matrix groups on related actions, and we will study distance preserving maps (or *isometries*) on $\mathbb{R}^2$ and $\mathbb{R}^3$.

§4.7.1 Examples of Matrix Groups

We are going to write $M_{n \times n}(\mathbb{F})$ to denote the set of $n \times n$ matrices over the field $\mathbb{F}$, which will typically be either $\mathbb{R}$ or $\mathbb{C}$.

Definition 4.7.1 (General Linear Group)

$\text{GL}_n(\mathbb{F}) = \{A \in M_{n \times n}(\mathbb{F}) \mid A \text{ is invertible}\}$ is the **general linear group** over $\mathbb{F}$.

This is indeed a group, and the determinant is a surjective homomorphism, namely $\det : \text{GL}_n(\mathbb{F}) \rightarrow \mathbb{F}^*$, the set of non-zero elements in $\mathbb{F}$.

Definition 4.7.2 (Special Linear Group)

The **special linear group** $\text{SL}_n(\mathbb{F}) \leq \text{GL}_n(\mathbb{F})$ is the kernel of the $\det$ homomorphism, that is, the set of elements of $\text{GL}_n(\mathbb{F})$ with determinant 1.

Recall the following facts about matrices. Given $A, B \in \text{GL}_n(\mathbb{R})$,

- $(AB)^T = B^T A^T$,
- $(A^{-1})^T = (A^T)^{-1}$,
- $AA^T = I \iff A^T A = I \iff A^T = A^{-1}$,
- $(A^T)^T = A$,
- $\det(A^T) = \det(A)$.

With these we can define some groups.

Definition 4.7.3 (Orthogonal Group)

The **orthogonal group** is the group $O_n = \{A \in \text{GL}_n(\mathbb{R}) \mid A^T A = I\}$.

Proposition 4.7.4

$\det : O_n \rightarrow \{\pm 1\}$ is a surjective homomorphism.

Proof. This is a homomorphism as the determinant is a homomorphism in $\mathrm{GL}_n(\mathbb{R})$, and for $A \in \mathrm{O}_n$, then $A^T A = I$, and $1 = \det(AA^T) = \det(A) \det(A^T) = \det(A)^2$, so $\det(A) = \pm 1$. It is surjective as $\det I = 1$, and $\det(\mathrm{diag}(-1, 1, 1, \dots, 1)) = -1$. $\square$

Definition 4.7.5 (Special Orthogonal Group)

The **special orthogonal group** SO_n is the kernel of the determinant homomorphism, that is, $\mathrm{SO}_n = \{A \in \mathrm{O}_n \mid \det A = 1\}$.

These groups of matrices will be the main focus of this chapter. All of the matrix groups defined above act on the corresponding vector spaces. For example, $\mathrm{GL}_n(\mathbb{F})$ and $\mathrm{SL}_n(\mathbb{F})$ act on $\mathbb{F}^n$, and O_n , SO_n acts on $\mathbb{R}^n$.

Example 4.7.6

Let $G \leq \mathrm{GL}_2(\mathbb{R})$ act on $\mathbb{R}^2$. We will find the orbits. Note that $\{\mathbf{0}\}$ is always a singleton orbit, since we are acting by linear maps.

- If $G = \mathrm{GL}_2(\mathbb{R})$, then G acts transitively on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$ since we can complete any $\mathbf{v} \neq \mathbf{0}$ to a basis, and we have an invertible change of basis matrix sending any basis to an other basis.
- If $G = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{R}) \right\} = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \mid a, d \neq 0 \right\}$, then $\mathrm{Orb}(\mathbf{0}) = \{\mathbf{0}\}$, $\mathrm{Orb} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \left\{ \begin{pmatrix} a \\ 0 \end{pmatrix} \mid a \neq 0 \right\}$, and $\mathrm{Orb} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \left\{ \begin{pmatrix} b \\ d \end{pmatrix} \mid d \neq 0 \right\}$. This is all of the orbits, since the union gives all of $\mathbb{R}^2$.

§4.7.2 Möbius Maps as Matrices

Let's first examine the connection between Möbius maps and matrices.

Proposition 4.7.7

The function $\phi : \mathrm{SL}_2(\mathbb{C}) \rightarrow \mathcal{M}$ where

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto f, \quad \text{where } f(z) = \frac{az + b}{cz + d}$$

is a surjective homomorphism, with kernel $\{\pm I\}$.

Proof. First we show that ϕ is a homomorphism. If $f_1(z) = \frac{a_1 z + b_1}{c_1 z + d_1}$ and $f_2(z) = \frac{a_2 z + b_2}{c_2 z + d_2}$, then we have $f_2 \circ f_1(z) = \frac{az+b}{cz+d}$, where

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}.$$

It follows directly that ϕ is a homomorphism.

To see that it is surjective, we note that if $\frac{az+b}{cz+d}$ is a Möbius map, then the matrix

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{C}),$$

since $ad - bc \neq 0$. But the determinant may not be zero, so letting $D^2 = \det M$, consider the matrix

$$\frac{M}{D^2} = \begin{pmatrix} \frac{a}{D} & \frac{b}{D} \\ \frac{c}{D} & \frac{d}{D} \end{pmatrix},$$

this has determinant one (and thus is in $\mathrm{SL}_2(\mathbb{C})$) and $\frac{\frac{a}{D}z + \frac{b}{D}}{\frac{c}{D}z + \frac{d}{D}} = \frac{az+b}{cz+d}$, so this homomorphism is surjective.

As for the kernel of ϕ , If $\phi \begin{pmatrix} a & b \\ c & d \end{pmatrix} = Id \in \mathcal{M}$, then $\frac{az+b}{cz+d} = z$, for all $z \in \hat{\mathbb{C}}$. That is, $c = b = 0$ and $a = d$, and thus $\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$. Since this has determinant 1, we also need $a = \pm 1$. Thus $\ker \phi = \{I, -I\}$. $\square$

Corollary 4.7.8

$\mathcal{M} \cong \mathrm{SL}_2(\mathbb{C})/\{\pm I\}$.

Proof. Follows from the first isomorphism theorem. $\square$

This quotient $\mathrm{SL}_2(\mathbb{C})/\{\pm I\}$ is the **projective special linear group**, $\mathrm{PSL}_2(\mathbb{C})$.

§4.7.3 Conjugation and Changes of Basis

We can look at how the conjugation action of $\mathrm{GL}_n(\mathbb{F})$ on $M_{n \times n}(\mathbb{F})$ using the idea of changes of basis.

Recall that if we have some linear map $\alpha : \mathbb{F}^n \rightarrow \mathbb{F}^n$, we can represent α as a matrix A with respect to a basis $\{e_1, e_2, \dots, e_n\}$ of $\mathbb{F}^n$. se a different basis $\{f_1, f_2, \dots, f_n\}$, then α will be represented with respect to this basis by the matrix $P^{-1}AP$, where P is the **change of basis matrix**, defined by

$$f_j = \sum_{i=1}^n P_{ij}e_i.$$

This is an example of conjugation.

Proposition 4.7.9

$\mathrm{GL}_n(\mathbb{F})$ acts on $M_{n \times n}$ by conjugation. The orbit of a matrix $A \in M_{n \times n}(\mathbb{F})$ is the set of matrices representing the same linear map as A with respect to different bases.

Proof. To see that this is an action, if we have a matrix $A \in M_{n \times n}(\mathbb{F})$, then $P(A) = PAP^{-1} \in M_{n \times n}(\mathbb{F})$ for any $P \in \mathrm{GL}_n(\mathbb{F})$. Also, $I(A) = IAI^{-1} = A$, for any matrix A , and finally, if we have $Q, P \in \mathrm{GL}_n(\mathbb{F})$, then $(QP)(A) = QPA(QP)^{-1} = Q(PAP^{-1})Q^{-1} = Q(P(A))$, for any matrix A . So $\mathrm{GL}_n(\mathbb{F})$ acts on $M_{n \times n}(\mathbb{F})$.

By the argument above, we have that A and B are in the same orbit if and only if $A = PBP^{-1}$ for some $P \in \text{GL}_n(\mathbb{F})$, which is true if and only if $B = P^{-1}AP$. That is, B represents the same linear map as A with respect to a different basis. Precisely, this is the basis obtained by the change of basis corresponding to P . $\square$

Let's look at an example of this action.

Example 4.7.10 (Jordan Normal Form)

Recall that any matrix in $M_{2 \times 2}(\mathbb{C})$ is conjugate to a matrix in Jordan Normal Form (JNF). That is, it's conjugate to one of the following types of matrices:

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ (where } \lambda_1 \neq \lambda_2 \text{), } \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}, \quad \text{or} \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

In the case of the first type of matrix, the values λ_1 and λ_2 are uniquely determined by the matrix (look at the eigenvalues), but the order is not determined uniquely. Other than that, no two matrices on this list are conjugate.

This gives us a complete description of the orbits of the action of $\text{GL}_2(\mathbb{C})$ on $M_{2 \times 2}(\mathbb{C})$ via conjugation.

§4.7.4 Geometry of Orthogonal Groups

We are now going to look more closely at the orthogonal and special orthogonal group, and also the symmetries of $\mathbb{R}^2$ and $\mathbb{R}^3$.

Consider the standard inner product in $\mathbb{R}^n$: $x \cdot y = \sum_{i=1}^n x_i y_i$. Thinking of these as column vectors, we can write this as $x^T y$. If we consider the columns $p_1, \dots, p_n$ of $P \in O_n$, we have $(P^T P)_{ij} = p_i^T p_j = p_i \cdot p_j$. So since $P \in O_n$ if and only if $P^T P = I$, we have that

$$P \in O_n \iff p_i \cdot p_j = \delta_{ij},$$

using the kronecker delta. This proves the following proposition.

Proposition 4.7.11

$P \in O_n$ if and only if the columns of P form an orthonormal basis.

Thinking of $P \in O_n$ as a change of basis matrix, we get the following result.

Proposition 4.7.12

Consider O_n acting on $M_{n \times n}(\mathbb{R})$ by conjugation^a. Then two matrices will be in the same orbit if and only if they represent the same linear map with respect to two orthonormal bases.

^aThis is an action as O_n is a subgroup of GL_n

Another characterization of O_n is that it is inner-product preserving.

Proposition 4.7.13 (O_n preserves inner products)

$P \in O_n$ if and only if $(Px) \cdot (Py) = x \cdot y$ for any $x, y \in \mathbb{R}^n$.

Proof. If $P \in O_n$, then

$$(Px) \cdot (Py) = (Px)^T(Py) = x^T P^T Py = x^T I y = x^T y = x \cdot y.$$

If $(Px) \cdot (Py) = x \cdot y$ for all $x, y \in \mathbb{R}^n$, then taking the basis vectors e_i and e_j , we have

$$Pe_i \cdot Pe_j = e_i \cdot e_j = \delta_{ij}.$$

so the vectors $Pe_1, \dots, Pe_n$ are orthonormal. Then the columns of P are orthonormal, so $P \in O_n$ by our previous proposition. $\square$

Corollary 4.7.14

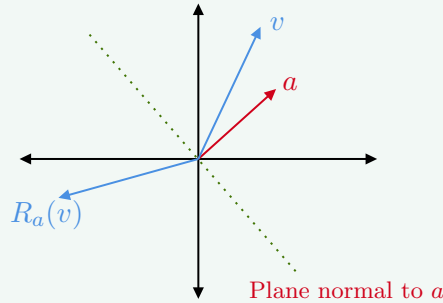
For $P \in O_n$, P preserves the length of vectors and the angle between vectors.

Proof. This follows directly from the preservation of inner product. $\square$

Let's investigate what the elements of O_n and SO_n look like. First, a bit of background.

Definition 4.7.15

If $a \in \mathbb{R}^n$ with $|a| = 1$, then the **reflection** in the plane normal to a is the linear map $R_a : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $x \rightarrow x - 2(x \cdot a)a$.



Lemma 4.7.16

R_a lies in O_n .

Proof. Let $x, y \in \mathbb{R}^n$. Then

$$\begin{aligned} R_a(x) \cdot R_a(y) &= (x - 2(x \cdot a)a) \cdot (y - 2(y \cdot a)a) \\ &= x \cdot y - 2(x \cdot a)(a \cdot y) - 2(y \cdot a)(x \cdot a) + 4(x \cdot a)(y \cdot a)(a \cdot a) \\ &= x \cdot y, \end{aligned}$$

so $R_a \in O_n$, as it preserves inner products. $\square$

As we might expect, conjugates of reflections are also reflections.

Lemma 4.7.17 (Conjugates of Reflections are Reflections)

Given $P \in O_n$, $PR_aP^{-1} = R_{Pa}$.

Proof. We have

$$\begin{aligned} PR_a P^{-1}(x) &= P(P^{-1}(x) - 2(P^{-1}(x) \cdot a)a) \\ &= x - 2(P^{-1}(x) \cdot a)(Pa), \end{aligned}$$

but $P^{-1} = P^T$, so $P^{-1}(x) \cdot a = P^T x \cdot a = x^T Pa$, which is just $x \cdot Pa$. So $PR_a P^{-1}(x) = x - 2(x \cdot Pa)(Pa) = R_{Pa}(x)$. $\square$

So reflections lie in O_n , but do they lie in SO_n ? We need to know the determinant of a reflection R_a . We know that the determinant of a matrix is the product of its eigenvalues, so eigenvalues of R_a might be a helpful thing to find.

We can spot some straightforward eigenvectors:

$$R_a(a) = a - 2(a \cdot a)a = -a,$$

so a is an eigenvector with eigenvalue -1 . And, for x in the plane normal to a , we get

$$R_a(x) = x - 2(x \cdot a)a = x.$$

so x is an eigenvector with eigenvalue 1 . This is all of the eigenvalues, as the eigenvalue 1 has multiplicity $n - 1$, as we can find $n - 1$ linearly independent eigenvectors in the plane normal to a . Thus we have

$$\det(R_a) = (-1) \cdot (1)^{n-1} = -1.$$

This proves the following proposition.

Proposition 4.7.18 (Reflections are not in SO_n)

$R_a \in O_n \setminus SO_n$.

To think about the elements that *are* in SO_n , we will begin by thinking about SO_2 .

Theorem 4.7.19 (Elements of SO_2)

Every element of SO_2 is of the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$$

for some $\theta \in [0, 2\pi)$.^a Conversely every such element lies in SO_2 .

^aThis is an anticlockwise rotation of $\mathbb{R}^2$ about the origin by angle θ .

Proof. Consider the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SO_2.$$

We have $A^T A = I$, and $\det A = 1$. Thus $A^T = A^{-1}$, so

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

and $a = d$, $b = -c$. Also since $ad - bc = 1$, so $a^2 + c^2 = 1$. So we can write $a = \cos \theta$ and $c = \sin \theta$ for a unique $\theta \in [0, 2\pi)$. Hence every matrix in SO_2 has the form as claimed. Conversely, the determinant of the matrix given is always one, and is in O_2 , and hence lies in SO_2 . $\square$

Theorem 4.7.20

The elements of $\text{O}_2 \setminus \text{SO}_2$ are the reflections in lines through the origin.

Proof. Consider the matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{O}_2 \setminus \text{SO}_2.$$

Then $A^T A = I$, and $\det A = -1$. Then we have

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix} = - \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = A^{-1},$$

and so $a = -d$, $b = c$. Since $ad - bc = -1$, we have $a^2 + c^2 = 1$, so $a = \cos \theta$, $c = \sin \theta$ for unique $\theta \in [0, 2\pi)$, and

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

We can check that

$$A \begin{pmatrix} \sin \theta/2 \\ -\cos \theta/2 \end{pmatrix} = - \begin{pmatrix} \sin \theta/2 \\ -\cos \theta/2 \end{pmatrix}, \quad \text{and} \quad A \begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 \end{pmatrix} = - \begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 \end{pmatrix},$$

so A is the reflection in the plane orthogonal the unit vector $(\sin \theta/2 \quad -\cos \theta/2)$.

Conversely any reflection in a line through the origin will have this form, and so is in $\text{O}_2 \setminus \text{SO}_2$. $\square$

Corollary 4.7.21

Every element of O_2 is the composition of at most two reflections.

Proof. Every element in $\text{O}_2 \setminus \text{SO}_2$ is a reflection, so if $A \in \text{SO}_2$, then

$$A = \left[A \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right] \left[\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \right],$$

and each of the matrices on the right is in $\text{O}_2 \setminus \text{SO}_2$, and is thus a reflection. $\square$

Moving from two to three dimensions, we have the following theorem.

Theorem 4.7.22

If $A \in \text{SO}_3$, then there exists a vector $v \in \mathbb{R}^3$ such that $|v| = 1$ and $Av = v$.

Proof. Such a v is an eigenvector for the eigenvalue 1, so it suffices to show 1 is an eigenvalue of A . This is equivalent to showing the determinant $\det(A - I) = 0$. We have

$$\begin{aligned}
 \det(A - I) &= \det(A - AA^T) \\
 &= \det(A(I - A^T)) \\
 &= \det A \cdot \det(I - A^T) \\
 &= \det(I - A^T) \\
 &= \det((I - A)^T) \\
 &= \det(I - A) \\
 &= \det(-I) \cdot \det(A - I) = -\det(A - I),
 \end{aligned}$$

so $\det(A - I) = 0$, as required. $\square$

Corollary 4.7.23

Every $A \in \text{SO}_3$ is conjugate in SO_3 to a matrix of the form

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}.$$

Proof. By the previous theorem, there exists $v_1 \in \mathbb{R}^3$ such that $|v_1| = 1$, and $Av_1 = v_1$. We can extend this to an orthonormal basis $\{v_1, v_2, v_3\}$ of $\mathbb{R}^3$. Then for $i = 2, 3$, we have

$$Av_i \cdot v_1 = Av_i \cdot Av_1 = v_i \cdot v_1 = 0,$$

so Av_2 and Av_3 lie in $\langle v_2, v_3 \rangle$. So A maps $\langle v_2, v_3 \rangle$ to itself. Now consider the restriction of A to $\langle v_2, v_3 \rangle$. This still has determinant 1, so A is an element of SO_2 , and its matrix must therefore be of the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

So A has the matrix

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix},$$

with respect to the basis $\{v_1, v_2, v_3\}$. The change of basis matrix P lies in O_3 since $\{v_1, v_2, v_3\}$ is an orthonormal basis, and if P does not lie in SO_3 , then instead we can use the basis $\{-v_1, v_2, v_3\}$. $\square$

Geometrically, this tells us in particular that every element in SO_3 is a rotation about some axis.

Corollary 4.7.24

Every element of O_3 is the composition of at most 3 reflections.

Proof. If $A \in \text{SO}_3$, then there exists $P \in \text{SO}_3$ such that $PAP^{-1} = B$, with

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}.$$

Since this matrix is the composition of at most 2 reflections by our result in two dimensions, so $B = B_1B_2$. Thus A also is, as the conjugate of a reflection is a reflection.

If $A \in \text{O}_3 \setminus \text{SO}_3$, then $\det A = -1$, then we can write

$$A = \left[A \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] \left[\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right],$$

and the first lies in SO_3 and is thus the product of at most 2 reflections, and second matrix is a reflection. $\square$

§4.7.5 Symmetries of the Cube Revisited

We will finish our discussion of matrix groups by again considering the symmetries of the cube. We can think of the symmetry groups of any of the platonic solids as subgroups of O_3 . We can do this by placing our solid at the origin, and then any symmetry of the solid will be an element of O_3 .

One fact about O_3 is that we can write it as a direct product.

Lemma 4.7.25

$\text{O}_3 \cong \text{SO}_3 \times C_2$.

Proof Sketch. Apply the direct product theorem. $\square$

Here, the subgroup C_2 is generated by the map $v \mapsto -v$. So if $v \mapsto -v$ is a symmetry of our platonic solid, then its group of symmetries will also split as the direct product $G^+ \times C_2$. Now this is a symmetry of the cube, so we have that the symmetry group of the cube is the group of its rotational symmetries with C_2 , so

$$G^+ \times C_2 \cong S_4 \times C_2,$$

which matches our result from Chapter 5.

§4.8 Groups of Small Order

We are going to finish off by characterizing all groups up to order 8. We have already found some of these, which we will review.

§4.8.1 Groups of Order Up To 7

So far we have established the following classifications.

1. *Prime Order.* Then we only have the cyclic group.
2. *Order 4.* We have C_4 or $C_2 \times C_2$.
3. *Order 6.* We have C_6 and D_6 .

§4.8.2 Groups of Order 8

To study groups of order 8, we'll need to introduce a new group.

Definition 4.8.1 (Quaternions)

Consider the subset of matrices in $\text{GL}_2(\mathbb{C})$ given by

$$\mathbf{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{i} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \mathbf{j} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \mathbf{k} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The set $\{\pm \mathbf{1}, \pm \mathbf{i}, \pm \mathbf{j}, \pm \mathbf{k}\}$ forms the **quaternion group**, Q_8 .

The elements of this group satisfy the following relations:

- (i) $g^4 = \mathbf{1}$ for any $g \in Q_8$.
- (ii) $(-\mathbf{1})^2 = \mathbf{1}$.
- (iii) $\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -\mathbf{1}$.

With that defined, let's get on to classifying groups of order 8. First, a little lemma.

Lemma 4.8.2

If a finite group has all non-identity elements of order 2, then it is isomorphic to $C_2 \times C_2 \times \cdots \times C_2$.

Proof. Recall from example sheet 1 that G must be abelian, and $|G| = 2^n$. If $|G| = 2$, then $G \cong C_2$, and if $|G| > 2$, then choose a_1 of order 2 in G . Then there exists $a_2 \notin \langle a_1 \rangle$, and by the direct product theorem, $\langle a_1, a_2 \rangle \cong \langle a_1 \rangle \times \langle a_2 \rangle \cong C_2 \times C_2$. If this is isomorphic to G , then we are done. Otherwise, pick $a_3 \notin \langle a_1, a_2 \rangle$, and we get $\langle a_1, a_2, a_3 \rangle = \langle a_1 \rangle \times \langle a_2 \rangle \times \langle a_3 \rangle \cong C_2 \times C_2 \times C_2$. Continuing in this way, we get $G \cong C_2 \times C_2 \times \cdots \times C_2$. $\square$

And now we can get to the heart of the matter.

Theorem 4.8.3 (Groups of order 8)

A group of order 8 is isomorphic to exactly one of C_8 , $C_4 \times C_2$, $C_2 \times C_2 \times C_2$, D_8 , or Q_8 .

Proof. Note that C_8 , $C_4 \times C_2$ and $C_2 \times C_2 \times C_2$ are all abelian and are distinguished by the maximal order of an element in the group. Then D_8 and Q_8 are non-abelian, and are distinguished from each other by the number of elements of order 2.

Not let G be a group such that $|G| = 8$. Then every element has order 1, 2, 4 or 8 by Lagrange. If there is an element of order 8, then $G \cong C_8$. If every element has

order 2, then $G \cong C_2 \times C_2 \times C_2$ by our lemma. Now let's assume that G has at least one element g of order 4, and no element of order 8.

Now $|G : \langle g \rangle| = 2$, and thus if $h \notin \langle g \rangle$, $G = \langle g \rangle \cup h\langle g \rangle$. If $h^2 \in h\langle g \rangle$, we would have $h \in \langle g \rangle$, which is a contradiction. Thus $h^2 \in \langle g \rangle$.

Now if $h^2 = g$ or g^3 , then g would have order 8, which is a contradiction. So $h^2 = e$ or g^2 . We have $\langle g \rangle \trianglelefteq G$, thus $hgh^{-1} = g^k$, for some k . We also have $g = h^2gh^{-2} = hg^kh^{-1} = (hgh^{-1})^k = g^{k^2}$, as g and h^2 commute. This implies $k^2 \equiv 1 \pmod{4}$, giving two cases.

If $k \equiv 1 \pmod{4}$, then $hgh^{-1} = g$, implying G is abelian. If $h^2 = e$, then $G = \langle g \rangle \times \langle h \rangle \cong C_4 \times C_2$ by the direct product theorem. If $h^2 = g^2$, then $(hg^{-1})^2 = e$, so $G = \langle g, hg^{-1} \rangle \cong C_4 \times C_2$.

If $k \equiv -1 \pmod{4}$, then $hgh^{-1} = g^{-1}$. Then if $h^2 = e$, we have $G = \langle g, h \mid g^4 = h^2 = e, gh = hg^{-1} \rangle \cong D_8$. Otherwise, $h^2 = g^2$, and $G \cong Q_8$. $\square$

Remark. We know that in an abelian group, every subgroup is normal. The converse is not true, and Q_8 is a counterexample.

5 Numbers and Sets

‘Numbers and Sets’ is one of the first course in pure mathematics that you will take as an undergraduate at Cambridge. In a sense, it is the ‘starting course’, in that it will introduce you to the ‘pure maths’ way of thinking about things. This introduction will happen through the lense of thinking about objects, beginning with the natural and real numbers. You will be introduced to the ‘thoughtful way’ of thinking about such objects, that you can carry through to almost every other course in pure mathematics.

This article constitutes my notes for the ‘Numbers and Sets’ course, held in Michaelmas 2020 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas.

§5.1 Elementary Number Theory

Number theory is the branch of mathematics that studies the properties of *numbers*, with a particular emphasis on the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$ and occasionally the rationals $\mathbb{Q}$. In this section, we will study some of the *additive* and *multiplicative* structure of the integers, looking at divisors, primes and tools such as modular arithmetic.

One of the aims of this course (and this section in particular) is to study numbers ‘from the ground up’, being quite careful about what we assume. This goal immediately presents us with a question: what exactly is a ‘number’?

§5.1.1 The Peano Axioms

What are the natural numbers? Intuitively, we might say that they are a set¹ $\mathbb{N} = \{1, 2, 3, \dots\}$, created by starting at 1 and counting forward indefinitely, each time obtaining an object distinct from all of the previous ones. This does answer our question (a natural number is any element of $\mathbb{N}$), but has created a series of other questions. For example, what does it mean to ‘count forward’, and how can it be done ‘indefinitely’? How are we allowed to use these natural numbers, in regard to defining things like addition and multiplication?

Instead of attempting to answer these questions using our informal, intuitive definition of the natural numbers, we will instead use a definition that is more precise. Namely, we will define the natural numbers using the Peano axioms. We will state the rules or *axioms* that natural numbers satisfy, which will define the natural numbers in terms of *how they work*, rather than *what they are*. After clearly setting out this definition, we will be in a much stronger position to write concrete mathematical proofs about the natural numbers.

Definition 5.1.1 (Peano Axioms)

The **natural numbers** are a set $\mathbb{N}$, along with a function $S : \mathbb{N} \rightarrow \mathbb{N}$ and an object ‘1’ satisfying the following axioms:

¹We will look at sets later in the course, but an informal familiarity will be assumed from the beginning

1. $1 \in \mathbb{N}$.
2. If $n \in \mathbb{N}$, then $S(n) \in \mathbb{N}$.
3. $S(n) \neq 1$ for every $n \in \mathbb{N}$.
4. If $n, m \in \mathbb{N}$ and $n \neq m$, then $S(n) \neq S(m)$.
5. *Induction.* Let $P(n)$ be any property about a natural number n . Suppose that $P(0)$ is true, and suppose that whenever $P(n)$ is true, $P(S(n))$ is also true. Then $P(n)$ is true for every natural number n .

This should match with our original, informal definition of the natural numbers. We have formalized the ‘counting forward’ process with the *successor function* $S(n)$.

Remark. We are going to assume various things about the way we write down natural numbers using the decimal system. You can assume that when we write something like $n = 3$, we really mean $n = S(S(1))$ etc.

Addition

Now we have defined natural numbers, but as of yet we can do nothing more look upon them fondly and increment them using the function $S(n)$. We will now begin to remedy that by defining addition and multiplication.

Definition 5.1.2 (Addition)

We define **addition** to be an operation $+$ such that for $m, n \in \mathbb{N}$, we have $m + 1 = S(m)$, and $m + (n + 1) = (m + n) + 1$.

This definition defines addition for all natural numbers by induction. We are now able to state and prove various properties of addition.

Proposition 5.1.3 (Properties of Addition)

For all $a, b, c \in \mathbb{N}$. Then

- (i) *Addition is commutative.* $a + b = b + a$.
- (ii) *Addition is associative.* $(a + b) + c = a + (b + c)$.
- (iii) *Cancellation law.* If $a + b = a + c$ then $b = c$.

Proof Sketch. Use induction.^a □

^aThe proofs for these sorts of statements tend to be slightly dull and laborious, and for this reason they have been excluded. If you wish to read them, I encourage you to consult a textbook/some other reference material.

Order

We can now use addition to define an ordering on the natural numbers.

Definition 5.1.4 (Ordering of the Natural Numbers)

Let $n, m \in \mathbb{N}$. We say that n is **greater than or equal to** m , written $n \geq m$ if and

only if $n = m$ or $n = m + a$ for some $a \in \mathbb{N}$. We say n is **strictly greater than** m if $n \geq m$ and $n \neq m$.

Proposition 5.1.5 (Properties of Ordering)

Let $a, b, c \in \mathbb{N}$. Then

- (i) *Order is reflective.* $a \geq a$.
- (ii) *Order is transitive.* If $a \geq b$ and $b \geq c$, then $a \geq c$.
- (iii) *Order is anti-symmetric.* If $a \geq b$ and $b \geq a$, then $a = b$.
- (iv) *Addition preserves order.* $a \geq b$ if and only if $a + c \geq b + c$.
- (v) $a > b$ if and only if $a \geq b + 1$.

Proposition 5.1.6 (Trichotomy)

Let a and b be natural numbers. Then exactly one of the following is true: $a < b$, $a = b$ or $a > b$.

Multiplication

We can also define another familiar operation, multiplication, in the same inductive/recursive fashion that we defined addition.

Definition 5.1.7 (Multiplication)

We define **multiplication** to be an operation $\times$ such that for $m, n \in \mathbb{N}$, $m \times 1 = m$, and $m \times (n + 1) = (m \times n) + m$.

As before, induction implies that this is defined for all natural numbers. It also guarantees that multiplying two natural numbers is a natural number.

Notation. We will use $a \times b = a \cdot b = ab$ when referring to multiplication.

Proposition 5.1.8 (Properties of Multiplication)

For all $a, b, c \in \mathbb{N}$. Then

- (i) *Multiplication is commutative.* $a \times b = b \times a$.
- (ii) *Multiplication is associative.* $(a \times b) \times c = a \times (b \times c)$.
- (iii) *Distributive law.* $a \times (b + c) = a \times b + a \times c$.
- (iv) *Cancellation law.* If $a \times b = a \times c$ then $b = c$.
- (v) *Multiplication preserves order.* If $a < b$, then $a \times c < b \times c$.

Remark. The final two statements in the proposition above, the cancellation law and that multiplication preserves order, only hold because we are dealing with natural numbers. These properties do not hold in general if we allow a, b or c to be integers.

We could go further and define other common operations such as exponentiation, factorials and so on, all of which can be defined in the same fashion. However, in the interest

of space, these definitions have been omitted.

§5.1.2 Strong Induction

There is a more useful form of induction that can be used, now that we have defined an ordering on the natural numbers.

Proposition 5.1.9 (Strong Induction)

Suppose that we have some property $P(n)$ about a natural number n . If we have $P(1)$, and for all $n \in \mathbb{N}$ we have that $P(m)$ for $m \leq n$ implies $P(n+1)$, then $P(n)$ holds for all $n \in \mathbb{N}$.

Proof. This follows from ordinary induction using the property “ $P(n)$ for all $m \leq n$ ”. $\square$

Informally, the principle of strong induction means that whenever we are proving some property $P(n)$ holds for all $n \in \mathbb{N}$ by induction, we can feel free to assume that $P(m)$ holds for $m \in \mathbb{N}$ with $m < n$.

Remark (For Pedants). Technically, we don’t need to check the case $P(1)$ separately, as it is implied by the condition if suitably interpreted. Still, it’s safer to just check $P(1)$.

Some other equivalent but useful² forms of induction are listed below. Let $P(n)$ be a property, then

- *Existence of a Minimal Counterexample*

If $P(n)$ is false for some $n \in \mathbb{N}$, then there exists $n_0 \in \mathbb{N}$ such that $P(n_0)$ is false but $P(m)$ is true for all $m < n_0$.

- *The Well-Ordering Principle*

If $P(n)$ is true for some $n \in \mathbb{N}$, then there exists a minimal n_0 such that $P(n_0)$ is true.

§5.1.3 The Integers and Rationals

The previous section defined the natural numbers, along with some ways that we can use them. We will now go one step further and define the *integers*, and the *rational*s.

Definition 5.1.10 (Integers)

The **integers** are a set $\mathbb{Z}$ consisting of all symbols n , $-n$ and 0 , where $n \in \mathbb{N}$.

We can then define addition, multiplication and subtraction in (and subtraction) on the integers by extending our definition on the natural numbers in the obvious way. We can also check that the properties of addition we had before still hold. There are some additional properties that the integers have.

²Some texts will claim that we can replace the induction axiom in Peano axioms with one of these other forms. This is incorrect, and typically one will need to add additional axioms alongside to keep the set of axioms equivalent.

Proposition 5.1.11 (Algebraic Properties of the Integers)

Let $a, b \in \mathbb{Z}$. Then^a

- (i) *Identity.* $a + 0 = a$.
- (ii) *Existence of an Additive Inverse.* For all $a \in \mathbb{Z}$, there exists $b \in \mathbb{Z}$ such that $a + b = 0$.

^aThese properties imply that the integers are a *group*.

Proof. $a + 0 = a$ holds automatically due to our definition of addition on the integers. Then, we always have a $b \in \mathbb{Z}$ such that $a + b = 0$, as we can let $b = -a$. $\square$

We also obtain another interesting property of multiplication.

Proposition 5.1.12 (Zero Product Law)

Let a and b be integers such that $a \times b = 0$. Then either $a = 0$, $b = 0$ or both.

Proof. Assume for the purposes of contradiction that both $a \neq 0$ and $b \neq 0$. Then we must have either $a > 0$ or $a < 0$ by trichotomy. If $a > 0$, then $a \times b > 0$ if $b > 0$, or $a \times b < 0$ if $b < 0$. Otherwise, if $a < 0$, then $a \times b < 0$ if $b > 0$, or $a \times b > 0$ if $b < 0$. These are all possible cases, and thus we never have that $a \times b = 0$. Thus at least one of a and b must be zero. $\square$

Remark (Caveats). We noted earlier that there was some properties of the natural numbers that don't hold over the integers. Specifically, if we have $a < b$ for $a, b \in \mathbb{Z}$, and we multiply by a negative number, then the order is no longer preserved (it is switched). Also, the cancellation law only applies when we are cancelling a non-zero integer.

We can now use the integers to define the *rationals*.

Definition 5.1.13 (Rationals)

The **rationals** are a set $\mathbb{Q}$ of expressions a/b for some $a, b \in \mathbb{Z}$, with $b \neq 0$. We will define equality between rationals such that $a/b = c/d \iff ad = bc$.

This definition implicitly defines $\mathbb{Q}$ using *equivalence classes*, which will be discussed later. We need to be slightly more careful in defining addition on $\mathbb{Q}$, as we will need to ensure that it respects the equality relation between rationals.

Definition 5.1.14 (Addition on $\mathbb{Q}$)

For a/b and $c/d \in \mathbb{Q}$, we define addition such that

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}.$$

We will now need to explicitly check that this definition is valid.

Proposition 5.1.15

Addition is well defined on $\mathbb{Q}$.

Proof. Let $a/b = a'/b'$ and $c/d = c'/d'$ be rationals. We show that

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{a'}{b'} + \frac{c'}{d'} \\ \iff \frac{ad + bc}{bd} &= \frac{a'd' + b'c'}{b'd'} \\ \iff (ad + bc)(b'd') &= (a'd' + b'c')(bd) \\ \iff (ab')dd' + (cd')bb' &= (a'b)d'd + (c'd)bb', \end{aligned}$$

which follows from $ab' = a'b$ and $cd' = c'd$. $\square$

To see why such a check was needed, consider the following example.

Example 5.1.16

We *cannot* define an operation on $\mathbb{Q}$ sending $a/b \rightarrow a^2/b^3$, as we would have $1/2 \rightarrow 1/8$ and $2/4 \rightarrow 4/64 = 1/16$, which is inconsistent.

We can then define multiplication in the same way as it was defined for integers, and we can check that all of the usual properties still hold. We can also define ordering, where $a/b < c/d$ if $ab < bc$. The rules from ordering $\mathbb{Z}$ still hold.

With all of these definitions in place, we can now start looking at some of the more interesting properties of numbers.

§5.1.4 Primes and Divisibility

We will now begin to discuss some actual number theory, beginning with the incredibly important concept of *divisibility*. You are likely to be familiar with this already, so we will jump into some definitions.

Definition 5.1.17 (Divisibility)

For $a, b \in \mathbb{N}$, we say that a **divides** b , written $a \mid b$ if there exists some $c \in \mathbb{N}$ such that $b = ac$.

If this is the case, we say that b is a **multiple** of a , and that a is a **divisor** of b . We can now define one of the most fundamental objects in number theory, the *primes*.

Definition 5.1.18 (Primes)

A natural number $n \geq 2$ is a **prime** if its only divisors are 1 and n . If a natural number is not prime, then it is **composite**.

With this definition, we can begin to state and prove some interesting results about the primes, which should hint at their importance in number theory.

Theorem 5.1.19

Every natural number is expressible as a product of primes.

Proof. We use induction on n . For $n = 2$ this is true, as 2 is prime. Now given $n > 2$, if n is prime, then we are done. If not, then n is composite so $n = ab$ for some $1 < a, b < n$. By our induction hypothesis, we have $a = p_1 p_2 \cdots p_k$ and $b = q_1 q_2 \cdots q_l$ for some (not necessarily distinct) primes $p_1, \dots, p_k, q_1, \dots, q_l$, hence $ab = p_1 \cdots p_k q_1 \cdots q_l$, which is the product of primes. Thus we are done by induction. $\square$

A nice consequence of this theorem is that the primes go on forever.

Theorem 5.1.20 (Euclid)

There are infinitely many primes.

Proof. Suppose there was finitely many primes, say $p_1, \dots, p_k$. Then consider the number $N = p_1 p_2 \cdots p_k + 1$. Then N has no prime factors, contradicting the fact that n can be written as the product of primes^a. $\square$

^aThis theorem has an amusingly large number of proofs. A short discussion can be found in the book ‘Proofs from the Book’.

Remark. There is no ‘pattern’ to the primes, in the sense that there is no algebraic formula for the n th prime.

So we know that a number can be written as the product of primes, but is this unique (up to some reordering)? This would seem to be true from experience, so why can’t we write a number as the product of primes in two ways. For example, why can’t we have $41 \times 101 = 67 \times 73$? Informally, we might think that this is impossible because we can’t have 41 dividing ‘a bit of’ 67 and ‘a bit of 73’. What we really need to show that prime factorization is unique is the following: For a prime p , if $p \mid ab$ then $p \mid a$ or $p \mid b$.

Now should this lemma be easy to prove, or hard to prove? That is, should this follow straight from definitions, or not? The answer is that it *cannot* follow straight from definitions. This is because it’s about ‘primes dividing things’, but we define the primes with ‘things dividing it’. This is the wrong way round! It will take some amount of work to be able to prove this.

Greatest Common Divisors

This section will begin to build some machinery that will allow us to tackle the problem of showing that prime factorizations are unique. We will begin by defining the notion of *greatest common divisor*, also known as *highest common factor*³.

Definition 5.1.21 (Greatest Common Divisor)

For $a, b \in \mathbb{N}$, a natural number d is the **greatest common divisor** of a and b , written $d = \gcd(a, b)$ if

- (i) $d \mid a$ and $d \mid b$;
- (ii) If $c \mid a$ and $c \mid b$, then $c \mid d$.

We want to show that the greatest common divisor always exists.

³This was the term used in the lectures, but I prefer greatest common divisor, so I will use that instead.

Proposition 5.1.22 (Division Algorithm)

For $n, k \in \mathbb{N}$, we can write $n = qk + r$, where $q, r \in \mathbb{N}$ and $0 \leq r < k$.

Proof. We use induction on N . For $n = 1$, this is clearly true. Otherwise, given $n > 1$, we can write $n - 1 = qk + r$ by our inductive hypothesis. Then, if $r < k - 1$, we can write $n = qk + (r + 1)$, and if $r = k - 1$, we can write $n = q(k + 1)$. $\square$

The division algorithm is related to the greatest common divisor using the following lemma.

Lemma 5.1.23 (Euclid)

For $a, b \in \mathbb{N}$, we can write $a = bq + r$ using the division algorithm. Then $\gcd(a, b) = \gcd(b, r)$.

Proof. If $d \mid a$ and $d \mid b$, then $d \mid a - bq = r$. Otherwise, if $d \mid b$ and $d \mid r$, then $d \mid bq + r = a$, hence the set of common divisors between a and b is the same as the set of divisors between b and r . Thus $\gcd(a, b) = \gcd(b, r)$. $\square$

Theorem 5.1.24 (Euclidean Algorithm)

For $a, b \in \mathbb{N}$, if we repeatedly apply the division algorithm to get

$$\begin{aligned} a &= q_1b + r_1 \\ b &= q_2r_1 + r_2 \\ r_1 &= q_3r_2 + r_3 \\ &\vdots \\ r_{n-2} &= q_nr_{n-1}, \end{aligned}$$

then $\gcd(a, b) = r_{n-1}$.

Proof. We repeatedly apply [Lemma 5.1.23](#) until we have $\gcd(a, b) = \gcd(r_{n-1}, 0) = r_{n-1}$. $\square$

The Euclidean Algorithm gives us both a proof that $\gcd(a, b)$ exists for any natural numbers a and b , and it also gives us an efficient way to find it.

Example 5.1.25

To find $\gcd(87, 52)$, we perform the following series of steps.

$$\begin{aligned} 87 &= 1 \times 52 + 35 \\ 52 &= 1 \times 35 + 17 \\ 35 &= 2 \times 17 + 1 \\ 17 &= 17 \times 1, \end{aligned}$$

thus $\gcd(87, 52) = 1$.

Notation. We will sometimes write $\gcd(87, 52) = \text{hcf}(87, 52) = (87, 52) = 1$. And when $\gcd(a, b) = 1$, we say that a and b are **coprime** or **relatively prime**.

The example above showed that $\gcd(87, 52) = 1$, so is it possible to write $1 = 87x + 52y$ for some $x, y \in \mathbb{Z}$? Looking at the computation we performed to find the greatest common divisor, we can work backwards to get

$$\begin{aligned} 1 &= 1 \times 35 - 2 \times 17 \\ &= 1 \times 35 - 2 \times (52 - 35) \\ &= -2 \times 52 + 3 \times 35 \\ &= -2 \times 52 + 3 \times (87 - 52) \\ &= 3 \times 87 - 5 \times 52, \end{aligned}$$

so it is indeed possible. In fact, there was nothing special about 87 and 52 here, we just used the sequence of steps performed in the Euclidean algorithm. We can formalize this.

Theorem 5.1.26

For all $a, b \in \mathbb{N}$, there exists $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$.

Proof. Run the Euclidean algorithm on a and b , say with some output r_n . Then we can write r_n as some integer linear combination of r_{n-1} and r_{n-2} . We can continue this to write any r_i as an integer linear combination of r_{i-1} and r_{i-2} . Thus eventually, we be able to write $r_n = ax + by$ for some $x, y \in \mathbb{Z}$, and as $r_n = \gcd(a, b)$, we are done. $\square$

Alternate Proof. Let h be the least positive integer linear combination of a and b . We claim that $h = \gcd(a, b)$. If $d \mid a$ and $d \mid b$, then $d \mid ax + by$ for any $x, y \in \mathbb{Z}$, thus $d \mid h$. Now suppose that $h \nmid a$. Then $a = qh + r$ for some $q, r \in \mathbb{Z}$, with $0 < r < h$. So $r - ah = a - q(ax + by)$, which is also a linear combination of a and b . But this is impossible, as it contradicts the minimality of h . Thus $h \mid a$, and by the same argument $h \mid b$. Thus $h = \gcd(a, b)$. $\square$

Remark. The second proof of the theorem above tells us that the greatest common divisor exists, but offers no way to find it, nor a way to find x or y .

This theorem has an interesting application: solving linear diophantine⁴ equations.

Corollary 5.1.27 (Bezout's Lemma)

Let $a, b, c \in \mathbb{N}$. Then the equation

$$ax + by = c$$

has an integer solution if and only if $\gcd(a, b) = c$.

Proof. Let $h = \gcd(a, b)$. If we have a solution $ax + by = c$, then $h \mid a$ and $h \mid b$, thus $h \mid c$. Now, if $h \mid c$, then we have $h = ax + by$ for $a, b \in \mathbb{Z}$ by our previous theorem, and thus we can write $c = \frac{c}{h}ax + \frac{c}{h}by$. Note that $\frac{c}{h}$ is guaranteed to be an integer by our previous argument. $\square$

⁴A diophantine equation is one where you seek only integer solutions.

The Fundamental Theorem of Arithmetic

We are now in a position to prove the lemma that was discussed quite a few pages ago, which will lead us directly to the uniqueness of prime factorizations.

Theorem 5.1.28 (Euclid's Lemma)

Let p be a prime, and $a, b \in \mathbb{N}$. Then $p \mid ab$ implies $p \mid a$ or $p \mid b$.

Proof. Suppose that $p \nmid a$. We wish to show that $p \mid b$. Then we have $\gcd(a, p) = 1$, which implies that we can write $ax + py = 1$, for some integers x and y . Then we can multiply by b to get $abx + pby = b$. But then $p \mid ab$ by our hypothesis, and thus $p \mid abx + pby = b$, and so we are done. $\square$

Remark. This theorem implies that if $p \mid a_1 a_2 \cdots a_k$, then p must divide at least one of a_i , for $1 \leq i \leq k$.

Theorem 5.1.29 (Fundamental Theorem of Arithmetic)

Every natural number $n \geq 2$ is uniquely expressible as a product of primes, up to re-ordering.

Proof. We already proved that such a product exists, so we will now prove that such a product is unique. We use induction on n . Suppose that $n = p_1 \cdots p_k = q_1 \cdots q_l$, where p_i, q_i are all primes. We wish to show that $k = l$, and after reordering, $p_i = q_i$ for all i . We have $p_1 \mid q_1 \cdots q_l$, so $p_1 \mid q_i$ for some i . We can (without loss of generality) label that prime q_1 . Hence $p_1 = q_1$. So $n/p_1 = p_2 \cdots p_k = q_2 \cdots q_l$, thus by induction $k = l$ and $p_2 = q_2, p_3 = q_3$, etc. $\square$

Aside: Why Unique Factorization Isn't Obvious

In the fundamental theorem of arithmetic, we took the ‘things that can't be broken up’ (the primes) and we broke up every number as a product of these, uniquely. The same concept makes sense in other places.

Consider instead $\mathbb{Z}[\sqrt{-3}]$, the set of complex numbers of the form $x + y\sqrt{-3}$, where $x, y \in \mathbb{Z}$. For example, $2 + 7\sqrt{-3}$. With these objects, we can both addition and multiplication two elements in $\mathbb{Z}[\sqrt{-3}]$, and always get back an element in $\mathbb{Z}[\sqrt{-3}]$. This allows us to talk about the notions of ‘divides’ and ‘multiple of’ etc in $\mathbb{Z}[\sqrt{-3}]$.

You might think that you can take the ‘things you can't break up’ in $\mathbb{Z}[\sqrt{-3}]$, and take any element and break it up into those. This is obviously correct (by definition). You may then want to say that this is unique, but that would not be true. For example,

$$4 = 2 \times 2 = (1 + \sqrt{-3})(1 - \sqrt{-3})$$

and all of the terms in this product are ‘things you can't break up’ in $\mathbb{Z}[\sqrt{-3}]$. This allows us to conclude that unique factorization *fails* in $\mathbb{Z}[\sqrt{-3}]$.

The fundamental theorem of arithmetic has some useful applications.

- *Factors*

Let's consider what are the factors of $2 \cdot 3^7 \cdot 11$ are. Certainly any $2^a \cdot 3^b \cdot 11^c$ where $0 \leq a \leq 3$, $0 \leq b \leq 7$ and $0 \leq c \leq 1$ is a factor. Also, there can't be any others. For example, if $7 \mid 2 \cdot 3^7 \cdot 11$, then we'd get a prime factorization involving a 7, which would contradict the uniqueness of prime factorization.

In general, the factors of $n = p_1^{a_1} \cdots p_k^{a_k}$ are precisely numbers of the form $p_1^{b_1} \cdots p_k^{b_k}$, where $0 \leq b_i \leq a_i$, for all $1 \leq i \leq k$.

- *Greatest Common Divisors*

If we wanted to find the common factors of $2^3 \cdot 3^7 \cdot 5 \cdot 11^3$ and $2^4 \cdot 3^2 \cdot 11 \cdot 13$, we could note that the common factors are $2^a \cdot 3^b \cdot 11^c$, where $0 \leq a \leq 3$, $0 \leq b \leq 2$ and $0 \leq c \leq 1$. So the greatest common divisor is $2^3 \cdot 3^2 \cdot 11$.

In general the greatest common divisor of $p_1^{a_1} \cdots p_k^{a_k}$ and $p_1^{b_1} \cdots p_k^{b_k}$ (with $a_i, b_i \geq 0$) is $p_1^{\min(a_1, b_1)} \cdots p_k^{\min(a_k, b_k)}$.

- *Lowest Common Multiples*

The common multiples of $2^3 \cdot 3^7 \cdot 5 \cdot 11^3$ and $2^4 \cdot 3^2 \cdot 11 \cdot 13$ are all of the form $2^a \cdot 3^b \cdot 5^c \cdot 11^d \cdot 13^e \cdot n$ (for a positive integer n) where $a \geq 4$, $b \geq 7$, $c \geq 1$, $d \geq 3$ and $e \geq 1$. We can then get the lowest common multiple as $2^4 \cdot 3^7 \cdot 5^1 \cdot 11^3 \cdot 13^1$.

Then in general, the lowest common multiple of $p_1^{a_1} \cdots p_k^{a_k}$ and $p_1^{b_1} \cdots p_k^{b_k}$ (with $a_i, b_i \geq 0$) is $p_1^{\max(a_1, b_1)} \cdots p_k^{\max(a_k, b_k)}$.

§5.1.5 Modular Arithmetic

It is a common occurrence in number theory that we will consider numbers that differ by a multiple of some fixed number to be equivalent. An example of this is the value of $(-1)^n$, which depends only on whether n is odd or even – that is, the values of n that differ by a multiple of 2 will give the same result. To give another example, the last digit of two numbers will be the same when the numbers differ by some multiple of 10. Modular arithmetic (or congruence notation) is a way of expressing such equivalences, where we have two integers a and b that differ by some fixed number m .

Definition 5.1.30 (Integers Modulo n)

Let $n \in \mathbb{N}$. The **integers modulo n** , written $\mathbb{Z}/n\mathbb{Z}$ consist of the integers where we regard two to be the same if they differ by a multiple of n .

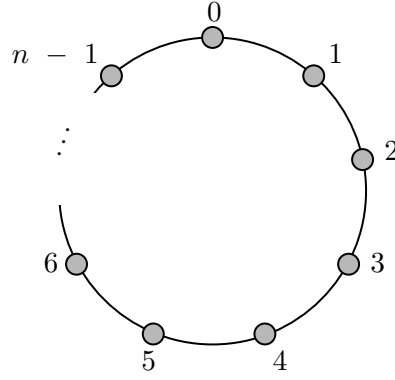
Definition 5.1.31 (Congruence Notation)

If x and y are integers that are the same modulo n , then we write

$$x \equiv y \pmod{n}.$$

This notation implies that $x \equiv y \pmod{n}$ if and only if $n \mid x - y$, that is, if $x = y + kn$ for some $k \in \mathbb{Z}$. Note that no two of $0, 1, \dots, n-1$ are congruent $\pmod{n}$, and every $x \in \mathbb{Z}$ is congruent to exactly one of these modulo n (this follows from the division algorithm).

We can view $\mathbb{Z}/n\mathbb{Z}$ as the following picture (indeed this is the 'correct picture' of $\mathbb{Z}/n\mathbb{Z}$).



With this mental picture, we can begin to build up some properties of modular arithmetic.

Proposition 5.1.32 (Arithmetic Modulo n)

If $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, then $a + b \equiv a' + b' \pmod{n}$ and $ab \equiv a'b' \pmod{n}$.

Proof. We have $a' = a + kn$ and $b' = b + jn$, for some $k, j \in \mathbb{Z}$. So $a' + b' = a + b + (k + j)n \equiv a + b \pmod{n}$, and $a'b' = (a + kn)(b + jn) = ab + (bk + aj + kjn)n \equiv ab \pmod{n}$. $\square$

Many of the other rules of arithmetic are inherited from $\mathbb{Z}$, for example we have $a + b \equiv b + a \pmod{n}$ as $a + b = b + a$ in $\mathbb{Z}$. Also some of the number-theoretic facts we have previously established can be expressed in modular arithmetic.

Example 5.1.33 (Euclid's Lemma in Modular Arithmetic)

The statement that for a prime p , ' $p \mid ab \implies p \mid a$ or $p \mid b$ ' is equivalent saying that $ab \equiv 0 \pmod{p}$ implies that $a \equiv 0 \pmod{p}$ or $b \equiv 0 \pmod{p}$.

Modular Inverses

We will now begin to look at the multiplicative structure of $\mathbb{Z}/n\mathbb{Z}$.

Definition 5.1.34 (Modular Inverses)

For $a, b \in \mathbb{Z}$, we say that b is an **inverse** of a if $ab \equiv 1 \pmod{n}$.^a

^aThis should be reminiscent of the notion of inverses from group theory.

Example 5.1.35

In $\mathbb{Z}/10\mathbb{Z}$, the inverse of 3 is 7, as $3 \times 7 = 21 \equiv 1 \pmod{10}$. The inverse of 4 does not exist, as for all $x \in \mathbb{Z}$, $4x \not\equiv 1 \pmod{10}$, as $4x$ is even.

This example shows that inverses do not always exist for an arbitrary modulus n .

Notation. We write a^{-1} to mean the inverse of a (modulo some n).

Proposition 5.1.36 (Properties of Modular Inverses)

In $\mathbb{Z}/n\mathbb{Z}$,

- (i) If a modular inverse exists, then it is unique modulo n .
- (ii) If a has an inverse, and $ab \equiv ac \pmod{n}$, then $b \equiv c \pmod{n}$. That is, we can cancel invertible elements.^a

^aWe cannot cancel elements in general.

Proof. To show (i), suppose there exists $b, c \in \mathbb{Z}$ such that for some a , $ab \equiv ac \equiv n \pmod{n}$. Then $b(ac) \equiv bab \implies c \equiv b \pmod{n}$. Then to show (ii), we can just multiply both sides by a^{-1} . $\square$

When we are working modulo a prime, things tend to be ‘nicer’.

Proposition 5.1.37

Let p be a prime, then every $a \not\equiv 0 \pmod{p}$ has an inverse modulo p .

Proof. We have $\gcd(a, p) = 1$, so we can write $ax + py = 1$ for some $x, y \in \mathbb{Z}$. But then $ax = 1 - py$, and thus $ax \equiv 1 \pmod{p}$. $\square$

Alternate Proof. In $\mathbb{Z}/n\mathbb{Z}$, consider the multiplies of a : $0 \times a, 1 \times a, 2 \times a, \dots, (p-1) \times a$. We wish to show that one of them is 1. We have p items written down, and I claim these are all *distinct* in $\mathbb{Z}/p\mathbb{Z}$. If $ia = ja \implies (i-j)a \equiv 0$, which implies $a \equiv 0$ or $i-j \equiv 0 \implies i \equiv j \pmod{p}$. Hence, there must be $0, 1, \dots, p-1$ in this list, in some order. Thus $xa \equiv 1 \pmod{p}$, for some $x \in \mathbb{Z}/n\mathbb{Z}$. $\square$

We can generalize this proposition by considering which properties of the primes we used in the previous proof.

Proposition 5.1.38

Let $n \in \mathbb{N}$. Then a has an inverse modulo n if and only if $\gcd(a, n) = 1$.

Proof. We have $\gcd(a, n) = 1 \iff ax + ny = 1$, for some $x, y \in \mathbb{Z}$. Then $ax = 1 - ny \equiv 1 \pmod{n}$. $\square$

There is a commonly used function to count how many elements have an inverse modulo some n .

Definition 5.1.39 (Euler's Totient Function)

Euler's totient function $\phi(n)$ is the number of numbers $1, 2, \dots, n$ that are relatively prime to n . This function counts the number of invertibles or **units** in $\mathbb{Z}/n\mathbb{Z}$.

Example 5.1.40

If p is a prime, then $\phi(p) = p - 1$, $\phi(p^2) = p^2 - p$ (we cannot have $p, 2p, \dots, pp$). If p and q are distinct primes, then $\phi(pq) = pq - p - q + 1$.

Exponentiation

Consider the powers of 2 in $\mathbb{Z}/7\mathbb{Z}$. We have the sequence

$$\begin{aligned} 2^1 &\equiv 2 \pmod{7} \\ 2^2 &\equiv 4 \pmod{7} \\ 2^3 &\equiv 1 \pmod{7} \\ 2^4 &\equiv 2 \pmod{7} \\ &\vdots \end{aligned}$$

This sequence will repeat periodically, with a period of 3. It turns out that in general, this will always occur. This can be seen in the following theorem, which turns out to be an incredibly important result which is used all the time. This theorem has many proofs, but we present one particularly nice one, which emphasizes the theory that we have developed so far.

Theorem 5.1.41 (Fermat's Little Theorem)

Let p be a prime, then for $a \not\equiv 0 \pmod{p}$, we have

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the numbers $1 \cdot a, 2 \cdot a, \dots, (p-1) \cdot a$ modulo p . They are distinct, as if $ai \equiv aj$ then $i \equiv j \pmod{p}$, because a is invertible. They are also non-zero. So we must have the numbers $1, 2, \dots, p-1 \pmod{p}$ in some order. Then, if we multiply together we have

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p},$$

so we can cancel $(p-1)!$ as it's the product of invertibles, to obtain $a^{p-1} \equiv 1 \pmod{p}$. $\square$

As before, we can consider what happens in the case of a non-prime modulus.

Theorem 5.1.42 (Fermat-Euler Theorem)

Let $n \in \mathbb{N}$. Then for $a \not\equiv 0 \pmod{n}$, we have

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

Proof. We copy the proof of Fermat's little theorem. Let the units in $\mathbb{Z}/n\mathbb{Z}$ be $x_1, \dots, x_{\phi(n)}$. Consider $a \cdot x_1, a \cdot x_2, \dots, a \cdot x_{\phi(n)}$. They are distinct and invertible as before, so they must be $x_1, \dots, x_{\phi(n)}$ in some order. Multiplying them all together,

$$a^{\phi(n)} x_1 x_2 \cdots x_{\phi(n)} \equiv x_1 x_2 \cdots x_{\phi(n)} \pmod{n},$$

and then we can cancel each x_i to obtain $a^{\phi(n)} \equiv 1 \pmod{n}$. $\square$

We can pause for a moment and notice that in the proof of Fermat's little theorem, we could cancel $(p-1)! \pmod{p}$ because it was non-zero. So what exactly was it? We can try an example of $p = 5$, then we have $4! = 1 \equiv 24 \equiv -1 \pmod{5}$. We can also try

$p = 7$ to get $6! = 720 \equiv -1 \pmod{7}$. It can be proved that this is always the case.

First, we prove a small lemma.

Lemma 5.1.43

Let p be prime, then $x^2 \equiv 1 \pmod{p}$ implies $x \equiv 1$ or $x \equiv -1 \pmod{p}$.^a

^aNote that this is only true because p is prime. For example, $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$.

Proof. $x^2 \equiv 1 \implies x^2 - 1 \equiv 0 \pmod{p}$, which we can factor as $(x+1)(x-1) \equiv 0 \pmod{p}$ and thus $x \equiv 1$ or $x \equiv -1 \pmod{p}$. $\square$

Remark. It turns out that for a prime p , a non-zero polynomial in $\mathbb{Z}/p\mathbb{Z}$ of degree k always has at most k roots.

We can now prove our result about the value of $(p-1)! \pmod{p}$.

Theorem 5.1.44 (Wilson's Theorem)

Let p be a prime, then

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. Note that this is true for $p = 2$, so we may assume that $p \geq 3$. Consider the numbers $1, 2, \dots, p-1$ modulo p . We can pair up each a with its inverse a^{-1} for $a \neq a^{-1}$. But $a = a^{-1} \iff a^2 = 1$, so the only elements that are their own inverse are 1 and -1 . Thus $1, 2, \dots, p-1$ consists of some pairs a, a^{-1} and 1 and -1 . Thus when we multiply, $(p-1)! \equiv 1^{\frac{p-3}{2}} \cdot 1 \cdot -1 \equiv -1 \pmod{p}$. $\square$

To see how we can use both Wilson's theorem and Fermat's little theorem together, we can consider another question that follows naturally from [Lemma 5.1.43](#).

Lemma 5.1.45 ($x^2 + 1$ Lemma)

Let p be an odd prime. Then $x^2 \equiv -1 \pmod{p}$ has a solution if and only if $p \equiv 1 \pmod{4}$.

Proof. If $p = 4k + 3$, suppose that $x^2 \equiv -1 \pmod{p}$. Then we have from Fermat's little theorem that $x^{4k+2} \equiv 1 \pmod{p}$, but $x^{4k+2} \equiv (x^2)^{2k+1} \equiv (-1)^{2k+1} \equiv -1$, which is a contradiction.

We now construct an example to show that $p \equiv 1 \pmod{4}$ works. By Wilson's theorem, $(4k!) \equiv -1 \pmod{p}$. Then, noting that $4k-j \equiv -j-1 \pmod{p}$, $(2k)!^2 \equiv (2k!)^2 \cdot (-1)^{2k} \equiv (4k!) \equiv -1 \pmod{p}$, so a solution exists. $\square$

Congruence Equations and the Chinese Remainder Theorem

We now return back to the topic of linear diophantine equations, this time looking specifically at linear congruence equations. Let's look at an example.

Example 5.1.46 (Solving a Linear Congruence Equation)

Solve $7x \equiv 4 \pmod{30}$.

Solution. First we *find a solution*.

We have $\gcd(7, 30) = 1$, so we can run the Euclidean algorithm on 7 and 30 to obtain $7 \times 13 - 30 \times 3 = 1$. So $7 \times 13 \equiv 1 \pmod{30}$, hence $7 \times 52 \equiv 4 \pmod{30}$. So $x \equiv 52 \pmod{30}$ is a solution.

Now we show there is *no more solutions*.

If x' is a solution, then we have $7x \equiv 4 \pmod{30}$ and $7x' \equiv 4 \pmod{30}$. Thus $7x \equiv 7x' \pmod{30}$, which implies that $x \equiv x' \pmod{30}$ as 7 is invertible. So $x \equiv 52 \pmod{30}$ is the only solution. $\square$

There is a shorter method that can be used too, if we can quickly spot the inverse of the coefficient of our variable.

Example 5.1.47

Solve $3x \equiv 9 \pmod{28}$.

Solution. $3x \equiv 9 \pmod{28} \iff 19 \cdot 3x \equiv 19 \cdot 9 \pmod{28}$, as 19 is invertible. Thus $x \equiv 171 \equiv 3 \pmod{28}$. $\square$

If we don't have that the coefficient and the modulus is coprime, it's typically easier to write the equation as a standard linear diophantine equation.

Example 5.1.48

Solve $10x \equiv 12 \pmod{34}$.

Solution. $10x \equiv 12 \pmod{34} \iff 10x = 12 + 34y$ for some $y \in \mathbb{Z}$. Then $10x = 12 + 34y \iff 5x = 6 + 17y$, which we can write as $5x \equiv 6 \pmod{17}$. We can then solve as before. $\square$

Now consider a simultaneous congruence equation. Specifically, consider the equations

$$x \equiv 6 \pmod{17}$$

$$x \equiv 2 \pmod{19}.$$

Do we expect a solution to this? We might guess yes, as 17 and 19 are coprime, so intuitively 'modulo 17 and modulo 19 should be independent of each other'. What about these equations:

$$x \equiv 6 \pmod{34}$$

$$x \equiv 11 \pmod{36}.$$

We shouldn't expect solutions to these equations. Even if you try and figure out if x is even or odd, you'll run into issues. This intuition can be formalized into a theorem.

Theorem 5.1.49 (Chinese Remainder Theorem)

Let m and n be relatively prime positive integers. Then for any integers a and b ,

there is solution to

$$\begin{aligned}x &\equiv a \pmod{m} \\x &\equiv b \pmod{n},\end{aligned}$$

and this solution is unique modulo mn .

Proof. First we prove existence. We have $ms + nt = 1$ for some $s, t \in \mathbb{Z}$ by Bezout's lemma. Then $ms \equiv 0 \pmod{m}$ and $1 \pmod{n}$ and $nt \equiv 0 \pmod{n}$ and $1 \pmod{m}$. Hence $x = a(nt) + b(ms)$ has $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$.

To show uniqueness, certainly any $x' \equiv x \pmod{mn}$ is also a solution. Conversely, suppose x' has $x' \equiv a \pmod{m}$ and $b \pmod{n}$. Then $x' \equiv x \pmod{m}$ and $x' \equiv x \pmod{n}$, that is, $m \mid x' - x$ and $n \mid x' - x$. Thus $mn \mid x' - x$, as $\gcd(m, n) = 1$, and thus $x' \equiv x \pmod{mn}$. $\square$

This theorem generalizes directly.

Theorem 5.1.50 (General Chinese Remainder Theorem)

Let $m_1, \dots, m_k$ be pairwise relatively prime positive integers, and let

$$M = m_1 m_2 \cdots m_k.$$

Then for all integers $x_1, \dots, x_k$, the equations

$$\begin{aligned}x &\equiv x_1 \pmod{m_1} \\x &\equiv x_2 \pmod{m_2} \\&\vdots \\x &\equiv x_k \pmod{m_k}\end{aligned}$$

have a solution x that is unique modulo M .

Proof Sketch. Reduce the number of equations by applying the two variable Chinese remainder theorem to the last two equations. $\square$

An Application of the Fermat-Euler Theorem: RSA Encryption

RSA is an algorithm that underpins a huge amount of modern cryptography and security.

Normally, to send a coded message to someone, you would encode the message, which you could then send to the intended receiver who would decode it. This decoding would be performing the encoding operation in reverse – they are inverse operations. Because of this, it would seem obvious in this way that knowing how to encode is the same as knowing how to decode. For example, if your encoding is ‘add 1 to each letter’, a *Caesar cipher*, then you immediately know how to decode: ‘take 1 from each letter’. For another example, if your encoding is ‘add 1 to each letter then reverse each word’, then to decode you ‘reverse each word and subtract 1 to each letter’. This belief, that knowing how to encrypt implies knowing how to decrypt, was held for centuries, but it isn't quite correct. We will see this in the RSA encryption system.

Begin with two large distinct primes p and q , and let $n = pq$. Let's say we want to

encrypt some element $x \in \mathbb{Z}/n\mathbb{Z}$ – this may involve taking the message we want to send and splitting it up in some way. We will do that in the following way.

Fix a ‘coding exponent’ e . To encode a message $x \in \mathbb{Z}/n\mathbb{Z}$, raise it to the power e .

$$x \mapsto x^e.$$

Now let’s figure out how to decode this message. We want some $d \in \mathbb{Z}/n\mathbb{Z}$ so that $(x^e)^d \equiv x \pmod{n}$. First, assume that x and n are coprime (ensure that the message x is constructed to have this condition). We know from the Fermat-Euler theorem that $x^{\phi(n)} \equiv 1 \pmod{n}$, and also that $x^{k\phi(n)} \equiv 1 \pmod{n}$, for all $k \in \mathbb{Z}$. So we have

$$x^{k\phi(n)+1} \equiv x \pmod{n}.$$

So it suffices to find a d such that $de = k\phi(n) + 1$, for some $k \in \mathbb{Z}$. That is, we want an inverse of e modulo $\phi(n)$. We can do this by running the Euclidean algorithm on the numbers e and $\phi(n)$. This implies we needed to have picked e coprime with $\phi(n)$.

With the encryption and decryption methods defined, we can now consider what information is needed to perform each step.

To encode, we take $x \mapsto x^e \pmod{n}$ ⁵. This means we need to know that will need to know n (so you know what modulus to use) and also the power e .

To decode, we take $y \mapsto y^d \pmod{n}$, so we need to know n and d . But we worked out d as the inverse of e modulo $\phi(n)$, so we really need n , e and $\phi(n)$. Recall that

$$\phi(n) = n - p - q + 1.$$

So to compute $\phi(n)$, we need to know the factors of n . However, the problem of factoring n is thought to be incredibly hard, and the best known methods of today would take longer than the age of the universe to factor n into two 100 digit long primes. This means that if e and n were published, then anyone could encode a message, but only you (or someone who knows the factors of n) would be able to decode. So, with RSA encryption, anyone can encrypt a message and send it to you, but only you can decrypt it.

§5.2 The Reals

Moving on from number theory, this chapter will center on the questions of *what is a real number* and *what can we assume like them*. Unlike when we defined the natural numbers, in this chapter we will not just define the reals formally and then proceed as normal; instead, we will try and understand some of the more subtle aspects of the real numbers, that will eventually be carried through in ‘Analysis’.

§5.2.1 Why We Need Real Numbers

Recall that in [section 5.1](#), we began by defining the natural numbers $\mathbb{N}$, extended them to the integers $\mathbb{Z}$, and extended them to the rationals $\mathbb{Q}$. So why would we not stop there? In the following example, we will show that for some (many) purposes, the rationals are not adequate.

⁵This process can be done quite quickly by squaring the number whenever possible.

Proposition 5.2.1

There is no $x \in \mathbb{Q}$ such that $x^2 = 2$.

Proof. Assume that $x \geq 0$. If there was such a rational, we could write

$$x = \frac{a}{b},$$

where $a, b \in \mathbb{Z}$. We can square this to get

$$x^2 = \frac{a^2}{b^2} = 2 \implies a^2 = 2b^2,$$

but this is a contradiction of the fundamental theorem of arithmetic, as the power of 2 in a^2 would have to be odd^a. $\square$

Alternate Proof. Suppose that for some $x = a/b \in \mathbb{Q}$ where $a, b \in \mathbb{N}$, we have $x^2 = 2$. Note that for any integers c and d , the number $cx + d$ is of the form e/b , for some $e \in \mathbb{Z}$. So, if $cx + d > 0$, then $cd + d > \frac{1}{b}$. But $x^2 = 2 \implies 0 < x - 1 < 1$, we can say $0 < (x - 1)^k < 1/b$ for large enough k . But this is a contradiction, because $(x - 1)^k$ is of the form $cx + d$. $\square$

^aThe same proof shows that if there exists $x \in \mathbb{Q}$ so that $x^2 = n$, for $n \in \mathbb{N}$, then n must be a square number. This is because each exponent in the prime factorisation of n must be even.

Informally, what this result shows is that $\mathbb{Q}$ has a ‘gap’ at $\sqrt{2}$, and in constructing the reals, we will try formally define a number system that has no such ‘gaps’. This immediately raises the question of how we can define such a system, formalizing our rather vague notion of a ‘gap’. This must be done while only making statements about the rationals.

Consider again our example that there is no rational x such that $x^2 = 2$. With this in mind, define the set S as follows:

$$S = \{x \in \mathbb{Q} : x^2 < 2\}.$$

We will now think about bounds on the elements in this set.

Definition 5.2.2

A set S is **bounded above** if there exists $x \in \mathbb{R}$ with $x \geq y$ for all $y \in S$. Such an x is a **least upper bound** of S if x is an upper bound for S , and every upper bound x' of S satisfies $x < x'$.

It should be clear that 1.5 is an upper bound on the elements in this set. 1.42 is also an upper bound. By setting such upper bounds, we can get arbitrarily close to a number x such that $x^2 = 1$, that is there cannot be a *least* upper bound. If there was, it would have to be a number so that $x^2 = 2$, but we know there isn’t such a rational. It’s using this idea of ‘least upper bounds’ that we can define the reals.

Definition 5.2.3 (The Reals)

The **reals** are a set $\mathbb{R}$ with elements 0 and 1 (where $0 \neq 1$), along with operations $+$ and $\times$ and an ordering $<$ such that:

1. $+$ is commutative, associative, has an identity 0, and every $x \in \mathbb{R}$ has an inverse.
2. $\times$ is commutative, associative, has an identity 1, and every $x \in \mathbb{R}$ where $x \neq 0$ has an inverse.
3. $\times$ is distributive over $+$.
4. $\forall a, b$, exactly one of $a < b$, $a = b$ and $b < a$ holds. Also if $a < b$ and $b < c \implies a < c$.
5. $\forall a, b, c \in \mathbb{R}$, $a < b \implies a + c < b + c$ and $a < b \implies ac < bc$ if $c > 0$.
6. *Least Upper Bound.* Every non-empty set of reals that is bounded above has a least upper bound.

It's worth noting that the conditions in axiom 6 of the set of reals being 'non-empty' and 'bounded above' is needed because if there was no upper bound, then there is certainly no *least* one. Also, if the set was empty then every $x \in \mathbb{R}$ would be an upper bound, and there would be no least one.

From these axioms, it's straightforward to check some very basic facts, for example that $0 < 1$.

Example 5.2.4

We have $0 < 1$.

Proof. If that were not the case, as we have $0 \neq 1$ as an axiom, we would need $1 < 0$. This would imply $0 < -1$ and thus it is 'positive', so we can multiply by the 'positive' -1 to get $0 < 1$, which contradicts our assumption. $\square$

We can also (as you may expect) view $\mathbb{Q}$ as contained in $\mathbb{R}$, by identifying $a/b \in \mathbb{Q}$ with $a, b^{-1} \in \mathbb{R}$. In this system, axioms 1 to 5 hold, but axiom 6 does not hold. We have defined the reals here by specifying the axioms that they satisfy, but that is not the only way. It is possible to construct $\mathbb{R}$ from $\mathbb{Q}$, in a way that is (vaguely) similar to the way we constructed $\mathbb{Q}$ from $\mathbb{Z}$. This is, surprisingly, not as helpful as one might imagine, so we will skip over this topic for now.

Least Upper Bounds

The notion of least upper bounds is so central to the real numbers that we will consider some examples of different sets and their least upper bounds, in order for you to get see some of the properties. We will use the following notation for the least upper bound.

Definition 5.2.5 (Supremum)

The least upper bound of a set S is also known as the **supremum** of S , written $\sup S$.

Example 5.2.6 (Sets and Least Upper Bounds)

1. The set $S = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$ has a least upper bound $x = 1$. This is

because 1 is an upper bound, and also every upper bound y has $y \geq 1$, as $1 \in S$.

2. Consider the set $S = \{x \in \mathbb{R} : 0 < x < 1\}$. We have 2 as an upper bound, as $x \in S$ has $x < 1 < 2$. Also $3/4$ is not an upper bound, as $7/8 \in S$, and $3/4 < 7/8$. Now the least upper bound of S is 1, because 1 is an upper bound, and no upper bound c has $c < 1$, as otherwise, $c \in S$, and we can take $c + (1 - c)/2 \in S$.
3. $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\}$. Clearly 1 is an upper bound, and we will return to this example to show that it is the least upper bound.

Theorem 5.2.7 (Axiom of Archimedes)

$\mathbb{N}$ is not bounded above in $\mathbb{R}$.

Proof. Let's suppose that it was bounded above, in which case we would have $c = \sup \mathbb{N}$, for some $c \in \mathbb{R}$. So $c - 1$ is *not* an upper bound for $\mathbb{N}$. That implies that there is a natural number n with $n > c - 1$. But then $n + 1 \in \mathbb{N}$, and $n + 1 > c$, which is a contradiction. $\square$

Corollary 5.2.8

For each $t > 0$, there exists $n \in \mathbb{N}$ with $\frac{1}{n} < t$.

Proof. We have some $n \in \mathbb{N}$ with $n > \frac{1}{t}$ by the previous result, so $\frac{1}{n} < t$. $\square$

Remark. Informally, these two results are telling us that $\mathbb{R}$ does not contain any ‘infinitely big’ or ‘infinitesimally small’ elements.

We can return to the third example above, and we can now show that $\sup S = 1$, as if there was some $c < 1$ that was an upper bound, then $1 - \frac{1}{n} < c$ for every $n \in \mathbb{N}$, so $1 - c < \frac{1}{n}$, which is impossible (by the corollary we just proved).

Remark (Warning). If a set S has a greatest element (like $[0, 1]$), then that greatest element is $\sup S$, so $\sup S \in S$. But if S has no greatest element (like $(0, 1)$), then $\sup S \notin S$.

We have been discussing upper bounds of sets, but the least upper bound axiom also implies some facts about lower bounds.

Definition 5.2.9

A set S is **bounded below** if there exists $x \in \mathbb{R}$ with $x \leq y$ for all $y \in S$. Such an x is a **greatest lower bound** of S if x is a lower bound for S , and every lower bound $x' \neq x$ of S satisfies $x < x'$. We write $x = \inf S$.

Proposition 5.2.10

If a set S is non-empty and has a lower bound, then it has a greatest lower bound.

Proof. Let the set $-S = \{-y : y \in S\}$. It is non-empty and bounded above, so it has a least upper bound c . So $-c$ is the greatest lower bound of S , $\inf S$. $\square$

Rationals and Irrationals

Let's return to our original question which prompted our discussion of the real numbers: that there is no $x \in \mathbb{Q}$ with $x^2 = 2$. If we have in any way succeeded, we should be able to do somewhat better than that in the reals.

Proposition 5.2.11

There is some $x \in \mathbb{R}$ such that $x^2 = 2$.

Proof. Let $S = \{x \in \mathbb{R}, x^2 < 2\}$. We have S is nonempty, and it is bounded above, so $c = \sup S$ for some $c \in \mathbb{R}$. We claim that $c^2 = 2$. If this wasn't the case, then either $c^2 < 2$ or $c^2 > 2$. If $c^2 < 2$, then for $t < \frac{2-c^2}{4}$ we have $(c+t)^2 < 2$, which contradicts c being an upper bound for S . If $c^2 > 2$, then we can let $t < \frac{c^2-2}{4}$ and then $(c-t)^2 > 2$, which contradicts that c is the *least* upper bound for S . So we must have $c^2 = 2$. $\square$

This same proof also shows that $\sqrt[n]{x}$ exists for all $n \in \mathbb{N}$ and $x \in \mathbb{R}$. We regularly distinguish between a real number that is also a rational number, and a real number that is not also a rational.

Definition 5.2.12

A real that is not rational is called an **irrational**.

Example 5.2.13

$\sqrt{2}$, $\sqrt{3}$, $\sqrt{6}$ and $\sqrt{15}$ are all irrational, as is $2 + 3\sqrt{5}$.

It's worth noting that the rationals are **dense** in the reals.

Theorem 5.2.14 ($\mathbb{Q}$ is dense in $\mathbb{R}$)

The rationals are **dense** in $\mathbb{R}$, that is, for $a, b \in \mathbb{R}$, there exists $q \in \mathbb{Q}$ such that $a \leq q \leq b$.

Proof. Without loss of generality, assume that $a, b \geq 0$. Then choose some $n \in \mathbb{N}$ such that $\frac{1}{n} < b - a$. Then among $\frac{0}{n}, \frac{1}{n}, \dots$, there is a *final* one that is less than or equal to a , say $\frac{k}{n}$. So $a < \frac{k+1}{n} < b$. $\square$

Slightly more interesting is the following result.

Theorem 5.2.15

The irrationals are dense, that is, for $a, b \in \mathbb{R}$ with $a < b$, there exists an irrational c such that $a < c < b$.

Proof. This follows from the rational case, as we can find a rational c between $\sqrt{2}a$ and $\sqrt{2}b$, then $c/\sqrt{2}$ is irrational, and lies between a and b . $\square$

§5.2.2 Sequences

We are now going to discuss one of the major uses of the least upper bound statement – sequences. We are going to consider some questions like

What should $1 + \frac{1}{2} + \frac{1}{4} + \dots = 2$ mean?

and

What should $0.3333\dots = \frac{1}{3}$ mean?

For the first question, presumably the first sequence should ‘tend to’ 2, and the second should ‘tend to’ $1/3$. So what does it mean for a sequence $x_1, x_2, \dots$ to ‘tend to’ some c ? It should not mean that x_n is getting closer to c , for example, we would not want $1/2, 2/3, 3/4, \dots$ to tend to 17. We also don’t mean that x_n gets arbitrarily close to c , as for example, the sequence $1/2, 10, 2/3, 10, \dots$ should not tend to 1. The correct way to think of it is that we will get arbitrarily close *and stay* within some value of c .

Definition 5.2.16

We say that a sequence of reals $x_1, x_2, \dots$ **tends to** $c \in \mathbb{R}$ if $\forall \varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|x_n - c| < \varepsilon.$$

We will also say that c is the **limit** of $x_1, x_2, \dots$.

Remark (Informal). We can think of this say saying ‘ $\forall \varepsilon > 0$, x_n eventually is within ε of c ’.

There is quite a lot of subtlety to this definition, so we will spend some time considering its consequences.

Notation. If x_n tends to c , we will write $x_n \rightarrow c$, or $x_n \rightarrow c$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = c$.

Example 5.2.17

Consider the sequence

$$\frac{1}{2}, \frac{1}{2} + \frac{1}{4}, \frac{1}{2} + \frac{1}{4} + \frac{1}{8}, \dots,$$

this is the sequence $x_1, x_2, \dots$ where $x_n = 1 - \frac{1}{2^n}$ (inductively). We claim $x_n \rightarrow 1$.

Proof. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$. Then $\forall n \geq N$, $|x_n - 1| = \frac{1}{2^n} \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. $\square$

Example 5.2.18

The constant sequence $c, c, c, \dots$ (where $x_n = c$ for all n), then $x_n \rightarrow c$.

Proof. Given $\epsilon > 0$, we have $|x_n - c| < \epsilon$ for all n . $\square$

Example 5.2.19

The sequence $-1, 1, -1, 1, \dots$ where $x_n = (-1)^n$ has no $c \in \mathbb{R}$ with $x_n \rightarrow c$.

Proof. Suppose that $x_n \rightarrow c$, then choose $\epsilon = 1$. So there exists $N \in \mathbb{N}$ such that for $n \geq N$, we have $|x_n - c| < 1$, in particular, $|1 - c|$ and $|(-1) - c| < 1$, so $|1 - (-1)| < 2$ by the triangle inequality, which is a contradiction. $\square$

Example 5.2.20

The sequence $1, 0, 1/3, 0, 1/5, \dots$ with x_n given by

$$x_n = \begin{cases} 1 & \text{if } n \equiv 0 \pmod{5} \\ 5 & \text{if } n \equiv 1 \pmod{5} \\ \frac{1}{n} & \text{if } n \text{ is odd} \\ 0 & \text{if } n \text{ is even} \end{cases},$$

then $x_n \rightarrow 0$.

Proof. Given $\epsilon > 0$, choose $N \in \mathbb{N}$ with $\frac{1}{N} < \epsilon$. Then $\forall n \geq N$, $x_n = \frac{1}{n}$ or 0 , so $|x_n - 0| \leq \frac{1}{n} \leq \frac{1}{N} < \epsilon$. $\square$

Definition 5.2.21 (Convergent and Divergent Sequences)

If $x_n \rightarrow c$ for some c , we say that the sequence $x_1, x_2, x_3, \dots$ is **convergent**. If a sequence is not convergent, we say it is **divergent**.

We now show that if a sequence converges, then it converges to a unique value.

Proposition 5.2.22 (Limits Are Unique)

If a sequence $(x_n) \rightarrow c$, and $(x_n) \rightarrow d$, then $c = d$.

Proof. Suppose $c \neq d$, and choose $\epsilon = \frac{1}{2}|c - d|$. then $\exists N \in \mathbb{N}$ with $|x_n - c| < \epsilon$ for all $n \geq N$, and some $M \in \mathbb{N}$ with $|x_n - d| < \epsilon$ for all $n \geq M$. But now for any $n \geq \max(M, N)$, we have $|x_n - c|, |x_n - d| < \epsilon$, so $|c - d| < 2\epsilon$, which is a contradiction. $\square$

Some special notation can be used if a sequence is written in a particular form.

Definition 5.2.23 (Series)

A sequence given in the form $x_1, x_1 + x_2, x_1 + x_2 + x_3 + \dots$ is called a **series**, and can be written as

$$\sum_{n=1}^{\infty} x_n.$$

The k th term of this sequence is $\sum_{n=1}^k x_n$, the k th **partial sum** of this series.

If the series $x_1, x_1 + x_2, x_1 + x_2 + x_3, \dots$ is convergent say to c , then we will write

$$\sum_{n=1}^{\infty} x_n = c.$$

Remark (Warning). We *cannot* write $\sum_{n=1}^{\infty} x_n$ unless we know that the limit of the partial sums exist.

Limits do generally behave as we would expect.

Proposition 5.2.24

If $x_n \rightarrow c$ and $y_n \rightarrow d$, then $x_n + y_n \rightarrow c + d$.

Proof. Given $\epsilon > 0$, then as $x_n \rightarrow c$, so there exists $N \in \mathbb{N}$ such that $|x_n - c| < \epsilon/2 \forall n \geq N$. Also, as $y_n \rightarrow d$, so there exists $M \in \mathbb{N}$ such that $|x_m - d| < \epsilon/2$ for $m \geq N$. Then we can take $m \geq \max(M, N)$, and $|(x_n + y_n) - (c + d)| \leq |x_n - c| + |y_n - d| < \epsilon/2 + \epsilon/2 = \epsilon$. $\square$

In the proofs above, we have only been claiming what the limit is, and then proving that it converges to that. This isn't always necessary, and sometimes you can prove that a sequence converges without knowing the limit that it converges to.

Definition 5.2.25 (Increasing Sequence)

A sequence $x_1, x_2, \dots$ is **increasing** if $x_{n+1} \geq x_n$ for all n .

Theorem 5.2.26 (Monotone Convergence Theorem)

If an increasing sequence is bounded above, then it converges.^a

^aThis also holds for a decreasing sequence, bounded below.

Proof. Let the sequence be $x_1, x_2, \dots$, and let $c = \sup\{x_1, x_2, \dots\}$. I claim that this sequence converges to c . Then there exists N such that $x_N > c - \epsilon$ (as otherwise $c - \epsilon$ would be an upper bound, which is a contradiction). So for all $n \geq N$, we have $c - \epsilon < x_n \leq x_N < c$, hence $|x_n - c| < \epsilon$. $\square$

To see how this can be used to prove that a series converges (or diverges) without finding the limit, consider the following two examples.

Proposition 5.2.27

- (i) $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.
- (ii) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

Proof.

- (i) We have $1/3 + 1/4 \geq 1/2$, and $1/5 + 1/6 + 1/7 + 1/8 \geq 1/2$. In general,

$$\frac{1}{2^n + 1} + \frac{1}{2^n + 2} + \cdots + \frac{1}{2^{n+1} - 1} \geq \frac{2^n}{2^{n+1}} = \frac{1}{2}, \quad \forall n.$$

Hence the partial sum of $\sum_{n=1}^{\infty} \frac{1}{n}$ is unbounded, and thus the series is not convergent.

- (ii) We have $1/2^2 + 1/3^2 \leq 2/2^2 = 1/2$, and $1/4^2 + 1/5^2 + 1/6^2 + 1/7^2 \leq 4/4^2 = 1/4$, and in general,

$$\frac{1}{(2^n)^2} + \frac{1}{(2^n + 1)^2} + \cdots + \frac{1}{(2^{n+1} - 1)^2} \leq \frac{2^n}{(2^n)^2} = \frac{1}{2^n}, \quad \forall n.$$

Hence the partial sums of $\sum_{n=1}^{\infty} \frac{1}{n^2}$ are bounded by $1 + 1/2 + 1/4 + \cdots = 2$, and thus this series converges by the monotone convergence theorem.

□

The value of the second series (the value that it converges to) is $\pi^2/6$. Also, the first series is quite important, and it has its own name.

Definition 5.2.28 (Harmonic Series)

The series

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

is the **harmonic series**.

The monotone convergence theorem also tells us something about decimal expansions. Let the decimal expansion

$$0.a_1a_2a_3\ldots$$

with $0 \leq a_n < 9$ for each n to be

$$\sum_{n=1}^{\infty} \frac{a_n}{10^n}$$

This must converge by the monotone convergence theorem, as all of the terms are at least zero and the partial sums are bounded by (for example 1). The converse of this is true, that given some $x \in \mathbb{R}$ with $0 < x < 1$, we can write it as $0.a_1a_2\ldots$ for $a_i \in \{0, \dots, 9\}$.

Begin by choosing the greatest $a_1 \in \{0, 1, \dots, 9\}$ such that $\frac{a_1}{10} \leq x$ thus $0 \leq x - \frac{a_1}{10} < \frac{1}{10}$. Then we can take the greatest $a_2 \in \{0, 1, \dots, 9\}$ such that $\frac{a_1}{10} + \frac{a_2}{100} \leq x$, so $0 \leq x - \frac{a_1}{10} - \frac{a_2}{100} < \frac{1}{100}$. We can continue this inductively to obtain $a_1, a_2, \dots \in \{0, 1, \dots, 9\}$ such that

$$0 \leq x - \sum_{n=1}^k \frac{a_n}{10^n} < \frac{1}{10^k},$$

for all k . Thus

$$\sum_{n=1}^{\infty} \frac{a_n}{10^n} = x.$$

Remark. We call a decimal expansion $0.a_1a_2\dots$ **recurrent** if there exists some $a_{n+k} = a_n$ for all $n \geq N$, for some N and k . For example, $0.3178426426426\dots$ is recurrent. It can be checked that a number has a recurrent decimal expansion if and only if that number is rational.

Remark (Uniqueness of Expansions). Decimal expansions need not be unique. For example, $0.370000\dots = 0.36999\dots$. However, this is the only case of non-uniqueness (where we have a ‘terminating’ decimal or a string of 9s).

Theorem 5.2.29 (e Exists)

We define

$$e = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots = \sum_{n=0}^{\infty} \frac{1}{n!},$$

and we claim that it exists.

Proof. The sequence of partial sums converge as all terms are positive, and the partial sums are bounded by $1 + 1 + \frac{1}{2} + \frac{1}{4} + \dots = 3$, since $1/n! \leq \frac{1}{2^{n-1}}$ (which is true by induction). $\square$

We can also show that this number e is irrational.

Theorem 5.2.30 (Irrationality of e)

e is not rational.

Proof. Suppose that e was rational, then write $e = p/q$ where $p, q \in \mathbb{N}$, and also suppose that $q > 1$. So, multiplying by $q!$ we have

$$\sum_{n=0}^{\infty} \frac{q!}{n!} = p(q-1)!,$$

which is an integer. This works, as if $x_n \rightarrow c$, then $kx_n \rightarrow kc$ for $k \in \mathbb{R}$. So

$$p(q-1)! = \underbrace{q! + \frac{q!}{1!} + \frac{q!}{2!} + \dots + \frac{q!}{q!}}_{\in \mathbb{Z}} + \frac{q!}{(q+1)!} + \dots$$

But in general,

$$\frac{q!}{(q+n)!} \leq \frac{1}{(q+1)^n}.$$

so we can bound part of the series with

$$\sum_{n=q+1}^{\infty} \frac{q!}{(q+n)!} \leq \frac{1}{q+1} + \frac{1}{(q+1)^2} + \dots = \frac{1}{q} < 1,$$

and thus we have a contradiction, as it implies $p(q-1)!$ is not an integer. $\square$

§5.2.3 Algebraic Numbers

Now we know that numbers can be rational or irrational, but we can also divide up numbers based on if they are the root of some integer polynomial.

Definition 5.2.31 (Algebraic)

We say that a real number x is **algebraic** if it is a root of a (non-zero) polynomial with integer coefficients. A number that is not algebraic is **transcendental**.

When we talk about algebraic numbers, we typically care about *irrational* algebraic numbers, because of the following proposition.

Proposition 5.2.32

Every rational is algebraic.

Proof. The number p/q for $p, q \in \mathbb{Z}$ with $q \neq 0$ is a root of $qx - p = 0$. $\square$

Let's have a look at some irrational algebraic numbers.

Example 5.2.33

$\sqrt{2}$ is algebraic as it satisfies $x^2 - 2 = 0$.

A natural question to ask is whether there are any non-algebraic, that is, transcendental numbers. We are going to construct one.

Theorem 5.2.34 (Example of a Transcendental Number)

The following number c is transcendental^a.

$$c = \sum_{n=1}^{\infty} \frac{1}{10^{n!}} = 0.110001000000000000000000010000\dots$$

^aThis is going to be the hardest proof on the course.

To prove that this number is indeed transcendental, we are going to need recall some facts about polynomials.

Proposition 5.2.35 (Properties of Polynomials)

- (i) For any polynomial p , there exists a constant k such that $|p(x) - p(y)| \leq k|x - y|$ for all $0 \leq x, y \leq 1$.
- (ii) A non-zero polynomial of degree d has at most d roots.

Proof.

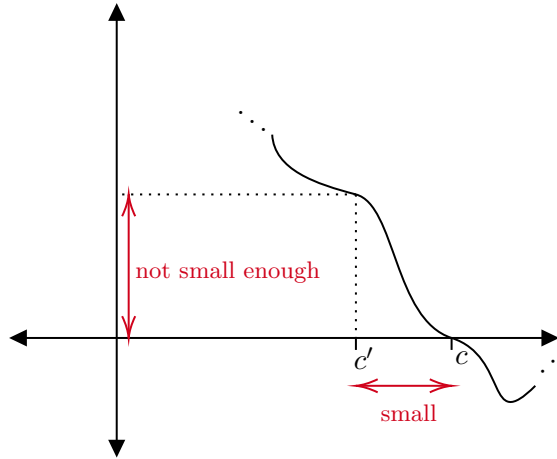
- (i) Say $p(x) = a_d x^d + \dots + a_0$. Then $p(x) - p(y) = a_d(x^d - y^d) + \dots + a_1(x - y) =$

$(x - y) [a_d(x^{d-1} + x^{d-2}y + \cdots + y^{d-1}) + \cdots + a_1]$. so $|p(x) - p(y)| \leq |x - y| (|a_d|d + |a_{d-1}|d + \cdots + |a_1|d)$, because $0 \leq x, y \leq 1$.

- (ii) Given a polynomial p of degree d , if p has no roots, then we are done. Otherwise, it has a root, say a , then $p(x) = (x - a)q(x)$ for some polynomial q of degree $d - 1$. So each root of p is either a or a root of q , but q has at most $d - 1$ roots by induction.

□

We are now ready to prove that our number c is transcendental. The idea of the proof is summarized in the image below



Proof of Theorem 5.2.34. Write $c_n = \sum_{k=0}^n \frac{1}{10^{k!}}$, so $c_n \rightarrow c$. Suppose the polynomial $p = a_dx^d + \cdots + a_0$ of degree d has c as a root. Then there exists k such that $|p(x) - p(y)| \leq k|x - y|$, for all $0 \leq x, y \leq 1$. Now

$$|c - c_n| = \sum_{k=1}^{\infty} \frac{1}{10^{k!}} \leq \frac{2}{10^{(n+1)!}}.$$

Also, $c_n = \frac{1}{10^{n!}}$ for some $a \in \mathbb{Z}$, so $p(c_n) = \frac{b}{10^{dn!}}$, for some $b \in \mathbb{Z}$ (since $p(\frac{s}{t}) = \frac{q}{t^d}$ for some integer q whenever $s, t \in \mathbb{Z}$). But, for n large enough, c_n is *not* a root of p (as it has finitely many roots). So $|p(c_n)| \geq \frac{1}{10^{dn!}}$.

This implies that $|p(c_n) - p(c)| \geq \frac{1}{10^{dn!}}$. Thus $\frac{1}{10^{dn!}} \leq k \frac{2}{10^{(n+1)!}}$, which is a contradiction for n sufficiently large. □

Remark. The same proof shows that any real x such that $\forall n$, there exists $\frac{p}{q} \in \mathbb{Q}$ with $0 < |x - \frac{p}{q}| < \frac{1}{q^n}$ is transcendental. Informally, ‘ x has very good rational approximations’. Such x are often called **Liouville numbers**, and this says that ‘every Liouville number’ is transcendental.

§5.3 Sets and Functions

In a sense, this is a chapter of notation. We really just get some terminology, notation and straightforward concepts and make sure we agree with them. It’s got some theorems, but there isn’t any amazing theorems or proofs that are really hard, and the theorems are going to be rather unsurprising.

However, the concepts will be important, as is the way of thinking that we will develop.

§5.3.1 Sets

Let's begin with a definition, that will have a few caveats.

Definition 5.3.1 (Set)

A **set** is any^a collection of (mathematical) objects.

^awe are going to change this later

Example 5.3.2

$\mathbb{R}$, $\mathbb{N}$, $\mathbb{Q}$ are all sets.

We say that two sets with the same members are the same (that is, a set is determined by its members). Symbolically, if A and B are sets, then if $\forall x, x \in A \iff x \in B$, then $A = B$. For example, the sets $\{1, 3, 7\}$ and $\{1, 7, 3\}$ equal, as are $\{3, 4, 4, 6\}$ and $\{3, 4, 6\}$.

So that's what a set is, but what's more important is how we can build sets. We are going to enter a series of sections which informally can be grouped as 'creating new sets from old sets'.

Subsets

When we have a set, we are able to choose certain elements of that set with some condition, to create another set.

Axiom 5.3.3 (Subset Selection)

Given a set A and a property $p(x)$, we can form $\{x \in A : p(x)\}$, the set of all members of A with property p .

Definition 5.3.4 (Subsets)

We say that B is a **subset** of A if $\forall x$ where $x \in B$ implies $x \in A$. This is written $B \subseteq A$.

Note that this implies that $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Unions and Intersections

We have two main ways of joining two sets.

Definition 5.3.5 (Union and Intersection)

Given sets A and B , we can form their **union** $A \cup B = \{x : x \in A \text{ or } x \in B\}$, and their **intersection** $A \cap B = \{x : x \in A \text{ and } x \in B\}$.

Definition 5.3.6 (Disjoint)

If the intersection of two sets A and B is the empty set, $A \cap B = \emptyset$, then we say that A and B are **disjoint**.

Note that union and intersection are commutative and associative. Also, union is distributive over intersection, that is, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. This can be proven by checking definitions.

We can also take a bigger union.

Example 5.3.7 (Infinite Union)

If $A_n = \{n^2, n^3\}$ for each $n \in \mathbb{N}$, then $A_1 \cup A_2 \cup \dots$ is going to be the set of all squares or cubes. We can write this as

$$A_1 \cup A_2 \cup \dots = \bigcup_{n=1}^{\infty} A_n = \bigcup_{n \in \mathbb{Z}} A_n.$$

Of course, unlike our previous use of similar notation, this is not a limiting process but an *indexing process*.

In general, given a set I and sets A_i where $i \in I$, then we can form

$$\bigcup_{i \in I} A_i = \{x : x \in A_i \text{ for some } i \in I\},$$

and similarly,

$$\bigcap_{i \in I} A_i = \{x : x \in A_i \text{ } \forall i \in I\}.$$

However, this section set can only be defined for a *non-empty index set*, for reasons that we will see later.

Ordered Pairs

Definition 5.3.8 (Ordered Pair)

For any a, b , we can form the **ordered pair** (a, b) , such that $(a, b) = (c, d) \iff a = c \text{ and } b = d$.

Using the idea of an ordered pair, we can form the *product* of sets

Definition 5.3.9 (Cartesian Product)

For sets A and B , their **cartesian product** $A \times B = \{(a, b) : a \in A, b \in B\}$.

Remark. If we wished, we could define ordered pairs using sets, $(a, b) = \{\{a\}, \{a, b\}\}$, and then we can check that the equality case is the same.

Power Set

Definition 5.3.10

For any set X , we can form the **power set** $\mathcal{P}(X)$ consisting of all subsets of X .

$$\mathcal{P}(X) = \{Y : Y \subseteq X\}.$$

Example 5.3.11

If $X = \{1, 2\}$, then $\mathcal{P}(X) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Defining Sets

For a set A , we can form $\{x \in A : p(x)\}$. However, we *cannot* form $\{x, p(x)\}$. Indeed, if we could, then consider the set

$$X = \{x : x \notin X\}.$$

So is $X \in X$? If yes, then $X \notin X$, but this is a contradiction. But if no, then $x \notin X$, but that would imply that it is in X ! So this doesn't make any sense. This is Russell's paradox.

Similarity, there is no 'universal' set Y , such that $\forall x, x \in Y$. This is because by subset selection, we could create the group $X = \{x : x \notin X\}$, which would be a contradiction.

Thus to guarantee that a given set exists, we must obtain it in some way (for example with the rules given above) from known sets.

§5.3.2 Finite Sets

Now let's have a look at the sizes of sets. We are going to write $\mathbb{N}_0 = \mathbb{N} \cup \{0\} = \{0, 1, \dots\}$.

Definition 5.3.12 (Size)

For $n \in \mathbb{N}_0$, a set A has **size** n if it can be written $A = \{a_1, a_2, \dots, a_n\}$ with the a_i all distinct.

Definition 5.3.13 (Finite)

We will say that a set A is **finite** if there exists some n such that the size of A is n . Otherwise, we say that the size of the set is **infinite**.

Proposition 5.3.14 (Counting Subsets)

A set of size n has exactly 2^n subsets.

Proof. Assume that our set is $\{1, \dots, n\}$. To specify a subset S , we must say if $1 \in S$, or $1 \notin S$, then if $2 \in S$ or if $2 \notin S$, and so on. So the number of independent choices needed to form S is $\underbrace{2 \times 2 \times \dots \times 2}_{n \text{ times}} = 2^n$.

Alternate Proof. We do induction on n . This is true for $n = 0$, and given some $n > 0$, for $T \subseteq \{1, 2, \dots, n-1\}$, then we can count how many $S \subseteq \{1, 2, \dots, n\}$ have $S \cap \{1, \dots, n-1\} = T$. There is exactly 2, namely T and $T \cup \{n\}$. Hence

the number of subsets of $\{1, \dots, n\}$ is twice the number of subsets of $\{1, \dots, n-1\}$. Then we are done by induction. $\square$

Remark. We could view the second proof as a ‘more formal version’ of the first proof.

As a bit of notation, if A has size n , then we write $|A| = n$. Using this notation, our counting subsets proposition says that

$$|A| = n \implies |\mathcal{P}(A)| = 2^n.$$

§5.3.3 Finite Sets and their Sizes

Given a set A , we are now going to think of how many subsets of A there is of size k .

Definition 5.3.15 (Binomial Coefficients)

For $n \in \mathbb{N}_0$ and $0 \leq k \leq n$, we write $\binom{n}{k}$ for the number of subsets of an n -element set that are of size k :

$$\binom{n}{k} = |\{S \subset \{1, 2, \dots, n\} : |S| = k\}|.$$

Example 5.3.16

The 2-element sets in a 4-element set, say $\{1, 2, 3, 4\}$ are $\{1, 2\}$, $\{1, 3\}$, $\{1, 4\}$, $\{2, 3\}$, $\{2, 4\}$, $\{3, 4\}$. So $\binom{4}{2} = 6$.

We will also write $\binom{n}{0} = 1$, $\binom{n}{n} = 1$, and note that $\binom{n}{1} = n$.

With this definition of binomial coefficients, we can deduce some nice properties.

Proposition 5.3.17 (Properties of Binomial Coefficients)

- (i) $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$.
- (ii) $\binom{n}{k} = \binom{n}{n-k}$, for all $n \in \mathbb{N}_0$ with $0 \leq k \leq n$.
- (iii) $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, for all $n \geq 1$, and $0 < k < n$.^a

^aThis property can be used to compute small binomial coefficients with ‘Pascal’s triangle’.

Proof. (i) Each side counts the number of subsets of $\{1, \dots, n\}$.

(ii) Picking a k element subset from an n element set, is the same as choosing $n - k$ elements to not pick.

(iii) The number of subsets of $\{1, 2, \dots, n\}$ with k elements is made up of the k -element subsets of $\{1, 2, \dots, n-1\}$ and also the $k-1$ -element subsets of $\{1, 2, \dots, n-1\}$, along with n .

$\square$

Sometimes we will have to compute binomial coefficients, and indeed there is a formula for binomial coefficients. However, what’s more important is where the formula comes from.

Proposition 5.3.18 (Binomial Coefficient Formula)

We have for integers n and k ,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}.$$

Proof. The number of ways to have a k -set is $n(n-1)\cdots(n-k+1)$, as there is n ways to name one element, $n-1$ ways to name a different element, and so on. This does grossly overcount though. The number of times a given k -set was named is $k(k-1)\cdots 1$. Hence the number of k -sets is

$$\frac{n(n-1)\cdots(n-k+1)}{k!}.$$

□

Example 5.3.19

We have

$$\binom{n}{2} = \frac{n(n+1)}{2}.$$

Remark. You could, if you wanted, write

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

but this is rather odd, as we have just multiplied my loads of terms that will cancel.

The formula we derived is also good for approximating the size of binomial coefficients. For example,

$$\binom{n}{3} = \frac{n(n-1)(n-2)}{6} \approx \frac{n^3}{6}$$

for large n .

An important application of binomial coefficients (and why we call them binomial coefficients) is the binomial theorem.

Theorem 5.3.20 (Binomial Theorem)

For a positive integer n ,

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{n}b^n = \sum_{k=0}^n \binom{n}{k}a^k b^{n-k}.$$

Proof. When we expand $(a+b)^n = (a+b)(a+b)\cdots(a+b)$, we obtain terms of the form $a^k b^{n-k}$, where $0 \leq k \leq n$. The number of such terms $a^k b^{n-k}$ is the number of ways of choosing k brackets with an a , out of n . That is, the number of times the

term $a^k b^{n-k}$ occurs is $\binom{n}{k}$. So

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

□

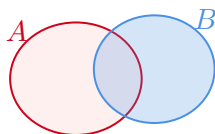
Example 5.3.21

We can expand $(1+x)^n$ using the binomial theorem as

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \cdots + nx^{n-1} + x^n,$$

so for small x , a good approximation for $(1+x)^n$ is $1+nx$. For example, $1.00001^8 \approx 1.0008$. A better approximation could be obtained by taking an extra term.

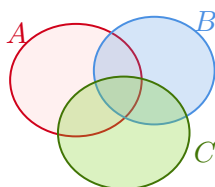
So this describes sizes of a single finite set, but what about the sizes of unions and intersections of finite sets? For example, how big is $|A \cup B|$?



We could try and write down $|A \cup B| = |A| + |B|$, but then we would count every element that's in both sets twice. That is, we have double counted $A \cap B$. So we really have

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

For three sets, we could try and write down $|A \cup B \cup C| = |A| + |B| + |C|$, but then we have over counted many elements.



Specifically, we would have overcounted every element that's in A and B , every element in A and C , and every element in B and C . So we need to subtract and get $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C|$. But we have then undercounted elements in all 3 sets! We really have

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

In general, we have the following theorem.

Theorem 5.3.22 (Inclusion-Exclusion Theorem)

Let $S_1, \dots, S_n$ be finite sets. Then

$$|S_1 \cup \dots \cup S_n| = \sum_{|A|=1} |S_A| - \sum_{|A|=2} |S_A| + \sum_{|A|=3} |S_A| - \dots + (-1)^{n+1} \sum_{|A|=n} |S_A|,$$

where S_A is $\bigcap_{i \in A} S_i$, and $\sum_{|A|=r}$ is taken over all $A \subset \{1, \dots, n\}$ of size r .

Proof. Let x be some element in one of the sets. We wish to show that it has been counted exactly once. Suppose that x is contained in exactly k of the sets. Then the number of A such that $|A| = r$ and $x \in S_A$ is $\binom{k}{r}$, as we must choose r of the sets of which x is an element. So the number of times x is counted on the right hand side is

$$k - \binom{k}{2} + \binom{k}{3} - \dots + (-1)^{k+1} \binom{k}{k},$$

for $r \leq k$. But $(1 + (-1))^k$ is exactly this by the binomial theorem, so the number of times x is counted is exactly $1 - (1 + (-1))^k = 1$. $\square$

§5.3.4 Functions

For some sets A and B , we would intuitively think of a function as some rule that each $a \in A$ is assigned to a point $f(a) \in B$. We can define this notion more precisely.

Definition 5.3.23 (Function)

A **function** from A to B is a set $f \subset A \times B$ such that for each $x \in A$ there exists a unique $y \in B$ with $(x, y) \in f$.

If $(x, y) \in f$, we write $f(x) = y$.

Example 5.3.24 (Examples of Functions)

$f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$ is a function.

$f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = 1$ if x is rational, and $f(x) = 0$ otherwise is a function.

The second example given is slightly subtle. Using the intuitive notion of a function, one might wonder whether the ‘rule’ of $f(x) = 1$ if x is rational and $f(x) = 0$ otherwise is valid. For example, we know that $f(2) = 1$ and $f(e) = 0$, but what about $f(e + \pi)$? Well the definition of a function tells us that we only need to make sure that this is a ‘well defined’ thing, and indeed as every real number is either rational or irrational, this is a valid function. Let’s state some not-so valid functions.

Example 5.3.25 (Non-Examples of Functions)

$f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \frac{1}{x}$ is *not* a function because it doesn’t specify the value of $f(0)$.

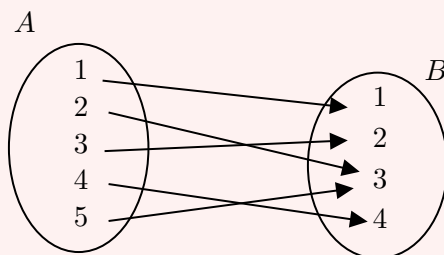
$f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = \pm\sqrt{|x|}$ is *not* a function because it assigns multiple values to every value of $|x|$.

Let’s continue looking at examples of valid functions.

Example 5.3.26

The following are all functions.

- (i) Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 4, 5\}$. Then $f : A \rightarrow B$ given by:



is a function.

- (ii) f from $\{1, 2, 3\}$ to $\{1, 2, 3\}$ with $1 \mapsto 1$, $2 \mapsto 3$, and $3 \mapsto 2$ is a function.

We frequently deal with functions that have certain interesting properties, which we will describe in the following definitions.

Definition 5.3.27 (Injectivity)

A function $f : A \rightarrow B$ is **injective** if for any $a, a' \in A$, then $a \neq a' \implies f(a) \neq f(a')$.

Definition 5.3.28 (Surjectivity)

A function $f : A \rightarrow B$ is **surjective** if for all $b \in B$, there exists $a \in A$ such that $f(a) = b$.

Definition 5.3.29 (Bijectivity)

A function $f : A \rightarrow B$ is **bijective** if it is injective and surjective.

We also name certain parts of a function.

Definition 5.3.30

For $f : A \rightarrow B$, we say A is the **domain**, and B is the **range**. We also say the **image** of f is the set $\{f(a) \mid a \in A\}$.

Also, when specifying a function, you must say what the domain and range are. For example, is the function $f(x) = x^2$ injective? The answer is that this is meaningless, as we haven't defined what the domain or range are. If $f : \mathbb{N} \rightarrow \mathbb{N}$, then it is injective, but it is not injective if $f : \mathbb{R} \rightarrow \mathbb{R}$.

When we are dealing with functions on finite sets, some obvious facts are true.

Proposition 5.3.31

For A and B finite, the following are true:

- (i) There is no surjection from $A \rightarrow B$ if $|B| > |A|$.

- (ii) There is no injection from $A \rightarrow B$ if $|A| > |B|$.
- (iii) For $f : A \rightarrow A$, if f is injective then f is also surjective (and vice versa).
- (iv) There is no bijection from A to any proper subset of A .

Proof Sketch. Check definitions. □

However, this *does not* hold when A and/or B is not finite. For example, consider the function $f_0 : \mathbb{N} \rightarrow \mathbb{N}$ with $f(x) = x+1$. Then f_0 is injective but not surjective. Similarly, define $f_0 : \mathbb{N} \rightarrow \mathbb{N}$, so that $f(1) = 1$ and $f(x) = x-1$ if $x > 1$. Then this function is surjective but not injective. Lastly, define $g : \mathbb{N} \rightarrow \mathbb{N} \setminus \{0\}$ with $x \mapsto x+1$. Then this is a bijection between a proper subset of $\mathbb{N}$ to $\mathbb{N}$.

Let's look at more functions!

Example 5.3.32

The following are functions.

- (i) For any set X , the **identity function** $1_X : X \rightarrow X$ such that $1_X(x) = x$ for all $x \in X$.
- (ii) For any set X and $A \subset X$ the **indicator function** or **characteristic function** $\chi_A : X \rightarrow \{0, 1\}$ with $\chi_A(x) = 1$ if $x \in A$ and 0 otherwise.

This function is important because it carries all of the information about what is in the set A .

- (iii) A sequence $x_1, x_2, \dots$ of reals is a function from $\mathbb{N} \rightarrow \mathbb{R}$ with $n \mapsto x_n$.
- (iv) The operation $+$ on $\mathbb{N}$ is a function from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$.
- (v) A set X has size $|n|$ if there is a bijective function from $\{1, 2, \dots, n\} \rightarrow X$.

One thing that we regularly do with functions is *compose* them, that is, apply one after another.

Definition 5.3.33

Given $f : A \rightarrow B$ and $g : B \rightarrow C$, the **composition** $g \circ f : A \rightarrow C$ is defined by

$$(g \circ f)(a) = g(f(a)).$$

Example 5.3.34

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $f(x) = 2x$, and $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = x+1$, then

$$(g \circ f)(x) = 2x + 1, \quad \text{and} \quad (f \circ g)(x) = 2x + 2.$$

In general (and in the example above), composition isn't commutative. However, composition is associative.

Composition is useful in talking about the idea of inverting functions.

Definition 5.3.35

We say a function $f : A \rightarrow B$ is **invertible** if there exists a function $g : B \rightarrow A$ such that $g \circ f = 1_A$ and $f \circ g = 1_B$.

Remark. It is not sufficient to show just that $g \circ f = 1_A$ or $f \circ g = 1_B$. Both must be checked.

Proposition 5.3.36

A function $f : A \rightarrow B$ is invertible if and only if it is bijective.

Proof. Let's say that g is inverse to f . Then for any $b \in B$, we have $f(g(b)) = b$, so f is surjective. Also, $f(a) = f(a')$ implies $g(f(a)) = g(f(a'))$, and $a = a'$, so f is injective and thus bijective.

Now let's suppose that f is bijective, and let $g(b)$ be the unique $a \in A$ with $f(a) = b$. Then g is the inverse of f . $\square$

§5.3.5 Equivalence Relations

The last item we will discuss in this chapter is equivalence relations. These appear throughout mathematics, and they induce phenomena that quite commonly appear when we define something.

First, we must define what a relation is.

Definition 5.3.37 (Relation)

A **relation** on a set X is a subset R of $X \times X$.

We normally write aRb for $(a, b) \in R$.

Example 5.3.38

The following are all relations.

- (i) On $\mathbb{N}$, aRb if $a \cong b \pmod{5}$. For example, $2R12$ but not $2R11$.
- (ii) On $\mathbb{N}$, aRb if $a \mid b$.
- (iii) On $\mathbb{N}$, aRb if $a = b \pm 1$.

Some properties a relation might satisfy are the following.

Definition 5.3.39 (Reflexive)

A relation R is **reflexive** if for all $x \in X$, xRx .

Definition 5.3.40 (Symmetric)

A relation R is **symmetric** if for all $x, y \in X$, then $xRy \implies yRx$.

Definition 5.3.41 (Transitive)

A relation R is **transitive** if for all $x, y, z \in X$, then xRy and yRz implies xRz .

When all of these properties are satisfied, we have an equivalence relation.

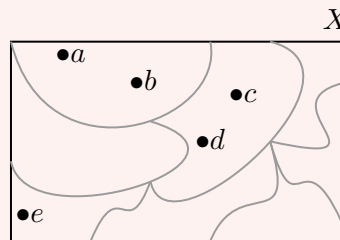
Definition 5.3.42 (Equivalence Relation)

An equivalence relation is a relation that is reflexive, symmetric and transitive.

To get a feel for what equivalence relations actually are, consider the following slightly natural yet slightly contrived example.

Example 5.3.43

Imagine we had a map of the world X , divided up into countries.



There are people in the world, and let's suppose each person is in just one country and there are no border disputes.

The relation 'is in the same country as' is an equivalence relation.

This relation is reflexive (you're in the same country as yourself), symmetric (if a is in the same country as b , then surely b is in the same country as a), and it's also transitive (if x and y are in the same country, and y and z are in the same country, then x and z are in the same country).

We can phrase this example in a more sensible way.

Example 5.3.44

Let X be a set, and let $\{C_i \mid i \in I\}$ be a **partition** of X , such that each C_i is non-empty, they are disjoint, and $\bigcup_{i \in I} C_i = X$.

Then aRb if there exists i such that $a, b \in C_i$ is an equivalence relation on X .

It actually turns out that all examples look like this! We will need a definition first.

Definition 5.3.45 (Equivalence Class)

For an equivalence relation R on a set X , and $x \in X$, the **equivalence class**, $[x]$, is the set $\{y \in X \mid yRx\}$. That is, the set of everything in X that is related to x .

Proposition 5.3.46 (Equivalence Classes Partition a Set)

Let R be an equivalence relation on a set X . Then the equivalence classes of R partition X .

Proof. Each equivalence class $[x]$ is non-empty (as $x \in [x]$), and $\cup_{x \in X} [x] = X$, as $x \in [x]$ for all $x \in X$. Thus it suffices to show that all equivalence classes are either disjoint or equal.

Given x, y with $[x] \cap [y] \neq \emptyset$. We wish to show $[x] = [y]$. Take $z \in [x] \cap [y]$. Then zRx and zRy so xRy . Thus $[x] = [y]$. $\square$

Definition 5.3.47 (Quotient and Quotient Map)

Given an equivalence relation R on a set X , the **quotient** of x by R is $X/R = \{[x] \mid x \in X\}$.

The map $q : X \rightarrow X/R$ is the **quotient map**, $x \mapsto [x]$. It is also known as the **projection map**.

§5.4 Countability

In this chapter we will discuss something we completely ignored in the previous discussion of sets – the sizes of infinite sets. Of course, we are going to have to introduce some ways of talking about the sizes of infinite sets...

Definition 5.4.1 (Countability)

We say that a set X is **countable** if X is finite or if there is a bijection with $\mathbb{N}$. Equivalently, X is countable if and only if we can list X as $a_1, a_2, \dots$ which might terminate.

§5.4.1 Countable Sets

Let's use this definition to look at some countable sets.

Example 5.4.2 (Trivially Countable Sets)

Any finite set is countable, and $\mathbb{N}$ is countable, by definition.

Example 5.4.3 ($\mathbb{Z}$ is Countable)

$\mathbb{Z}$ is countable, as we can list it as $0, 1, -1, 2, -2, \dots$.

It's sometimes easier to work with the following characterization of countability.

Proposition 5.4.4

A set X is countable if and only if there exists an injection $f : X \rightarrow \mathbb{N}$.

Proof. If X is finite or bijects with $\mathbb{N}$, then of course X injects in $\mathbb{N}$.

Now assume that X is not finite (if it was finite, then it's already countable), and has an injection $f : X \rightarrow \mathbb{N}$. Then we have a bijection from X with its image $f(X) = \{f(x) \mid x \in X\}$.

Set a_1 to be $\min f(X)$, a_2 to be $\min f(X) \setminus \{a_1\}$, and so on. Then $f(X) = \{a_1, a_2, \dots\}$ – each $a \in X$ is a_n for some n , because $a = a_n$ for some $n \leq a$. So $f(x)$ is countable. $\square$

Remark (A Warning). In $\mathbb{R}$, let $X = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\} \cup \{1\}$. Then X is countable, as we can list it as $1, 1/2, 2/3, \dots$. But if we counted by ‘least element’, 1 would not be on the list. So the counting method used in the proof above does not always work.

Let's look at some more countable sets.

Theorem 5.4.5

$\mathbb{N} \times \mathbb{N}$ is countable.

Proof. Define $a_1 = (1, 1)$, and inductively letting $a_{n-1} = (p, q)$,

$$a_n = \begin{cases} (p-1, q+1) & \text{if } p \neq 1, \\ (1, p+1) & \text{if } p = 1. \end{cases}$$

Then a_n hits every point $(x, y) \in \mathbb{N} \times \mathbb{N}$, by induction on $x + y$, so we have listed $\mathbb{N} \times \mathbb{N}$, and thus it is countable. $\square$

Alternate Proof. We will look for an injection $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Indeed, $f(x, y) = 2^x 3^y$ works, and thus $\mathbb{N} \times \mathbb{N}$ is countable. $\square$

The same proof shows the following theorem.

Theorem 5.4.6

Let $A_1, A_2, \dots$ be countable sets. Then $A_1 \cup A_2 \cup \dots$ is countable.

Proof. For each i , we have A_i countable, so we can list A_i as $a_{i,1}, a_{i,2}, \dots$, which might terminate. Then define $f : \bigcup_{n \in \mathbb{N}} A_n \rightarrow \mathbb{N}$, so that $x \rightarrow 2^i 3^j$, where $x = a_{ij}$ for the least such i . This is an injection. $\square$

This is a quite powerful way of showing that sets are countable, as we shall see.

Example 5.4.7

$\mathbb{Q}$ is countable, as $\mathbb{Q} = \mathbb{Z} \cup \frac{1}{2}\mathbb{Z} \cup \frac{1}{3}\mathbb{Z} \cup \dots$, where $\frac{1}{k}\mathbb{Z} = \{\frac{x}{k} \mid x \in \mathbb{Z}\}$. So $\mathbb{Q}$ is countable.

Example 5.4.8

The set $\mathbb{A}$ of all algebraic numbers is countable.

Indeed, it's enough to show that the set of all integer polynomials is countable, as each has finitely many roots, so $\mathbb{A}$ would be a countable union of finite sets.

So, to show that the collection of integer polynomials is countable, it's enough to

show that for each d , the set of all integer polynomials of degree d is countable. But this set injects into $\mathbb{Z}^{d+1}$ with its coefficients, and $\mathbb{Z}^n$ is countable for any n , by our previous theorem.

§5.4.2 Uncountable Sets

All of the examples we have given so far have been countable sets, but don't start thinking that all sets are countable!

Definition 5.4.9 (Uncountability)

A set X is **uncountable** if it is not countable.

Theorem 5.4.10

$\mathbb{R}$ is uncountable.

Proof. It suffices to show that $(0, 1)$ is uncountable. Given a sequence $r_1, r_2, \dots$ in $(0, 1)$, we wish to find some $s \in (0, 1)$ that's not in this list.

For each r_n , we have a decimal expansion $r_n = 0.r_{n,1}r_{n,2}\dots$. Define $s = 0.s_1s_2\dots$ by

$$s_i = \begin{cases} 5 & \text{if } r_{i,i} \neq 5, \\ 6 & \text{if } r_{i,i} = 5. \end{cases}$$

Then s cannot be on our list (as $s \neq r_n$ as they differ at decimal digit n). $\square$

The proof is known as a **diagonal argument**, or **cantor's diagonal argument**.

Interestingly, now that we know $\mathbb{R}$ is uncountable, but $\mathbb{A}$ is countable, there must be a transcendental number! Indeed, 'most' numbers are transcendental, in that $\mathbb{R} \setminus \mathbb{A}$ is uncountable.

Let's look at another uncountable set, which we will show is uncountable using a diagonal argument.

Theorem 5.4.11

The power set of $\mathbb{N}$, $\mathcal{P}(\mathbb{N})$ is uncountable.

Proof. Suppose $\mathcal{P}(\mathbb{N})$ is listed as $S_1, S_2, \dots$. We wish to find some $S \subset \mathbb{N}$ that's not on this list. For every $i \in \mathbb{N}$, we take $i \in S$ iff $i \notin S_i$. Then S cannot be on this list, since $S \neq S_n$ for all n , as they differ at element n . Hence $\mathcal{P}(\mathbb{N})$ is uncountable. $\square$

In fact, our result about the power set of $\mathbb{N}$ shows the following theorem.

Theorem 5.4.12

For any set X , there is no bijection from X to $\mathcal{P}(X)$.

Proof. Given any function $f : X \rightarrow \mathcal{P}(X)$, we will show f is not surjective.

Let $S = \{x \in X \mid x \notin f(x)\}$. Then S does not belong to the image of f , since for every x , $S \neq f(x)$, as S and $f(x)$ differ at element x . $\square$

Remark. This proof is vaguely reminiscent of Russell's paradox. In fact, this theorem proves that there is no universal set, just as Russell's paradox did. This is because otherwise, we would have $\mathcal{P}(V) \subset V$, where certain that V would surject to $\mathcal{P}(V)$.

Now we're going to look at an example of how we can think about countability.

Example 5.4.13

Let A_i , $i \in I$ be a family of open intervals that are pairwise disjoint. Must the family be countable?

The answer is yes. Each A_i contains a rational, and $\mathbb{Q}$ is countable. So this family is countable.

Another proof is that we can note the set $\{i \in I \mid A_i \text{ has length } \geq \frac{1}{k}\}$ is countable (it injects into $\mathbb{Z}$). So the family be countable.

In general, to show a set X is uncountable, we had two approaches.

1. Run the diagonal argument on X .
2. Inject your favorite uncountable set into X .

To show a set X is countable, we could do something like

1. List it (which can be ugly)
2. Inject it into $\mathbb{N}$
3. Use 'countable union of countable sets is countable'. This is usually the best one.
4. If in or near $\mathbb{R}$, sometimes using the countability of $\mathbb{Q}$ helps.

§5.4.3 Bijections

Intuitively, we can think of ' A bijects with B ' as saying that A and B are the same size. Similarly, we can think of ' A injects into B ' as sort of saying that A is at most as large as B . Similarly we can think of ' A surjects to B ' as saying A is at least as large as B .

For these to make sense, we want for A and B nonempty that there exists an injection $f : A \rightarrow B$ if and only if there exists a surjection $g : B \rightarrow A$.

This is easy to show. Fix some $a_0 \in A$, and define $g : B \rightarrow A$ so that b maps to the unique $a \in A$ with $f(a) = b$, and a_0 if that does not exist. Also, if g is surjective, then for each $a \in A$ we have $a' \in B$ with $g(a') = a$. Then let $f(a) = a'$. Then f is injective.

Theorem 5.4.14 (Schröder-Bernstein Theorem)

If $f : A \rightarrow B$ and $g : B \rightarrow A$ are injections, then there exists a bijection $h : A \rightarrow B$.

Proof. For $a \in A$, we write $g^{-1}(a)$ for the $b \in B$ such that $g(b) = a$, if it exists. We write $f^{-1}(b)$ for $b \in B$ similarly.

The *ancestor sequence* of $a \in A$ is $g^{-1}(a), f^{-1}(g^{-1}(a)), \dots$, which may terminate.

Let A_0 be the set of $a \in A$ such that the ancestor sequence stops in A , and A_1 the set of $a \in A$ such that the sequence stops in B . Lastly define A_∞ for the $a \in A$ whose ancestor sequence does not stop. Define B_0, B_1 and B_∞ similarly.

Then f bijects A_0 with B_1 (noting that every $b \in B_1$ is $f(a)$ for some $a \in A_0$). Similarly g bijects A_1 with B_0 . Finally, f or g bijects A_∞ with B_∞ .

Thus the function $h : A \rightarrow B$ with

$$h(a) = \begin{cases} f(a) & \text{if } a \in A_0, \\ g^{-1}(a) & \text{if } a \in A_1, \\ f(a) & \text{if } a \in A_\infty \end{cases}$$

is a bijection. □

Example 5.4.15

Do $[0, 1]$ and $[0, 1] \cup [2, 3]$ biject?

By Schröder-Bernstein this is trivial. Consider the identity function $f : [0, 1] \rightarrow [0, 1] \cup [2, 3]$, and the function $g : [0, 1] \cup [2, 3] \rightarrow [0, 1]$ with $g(x) = x/3$ are both injections, so there is a bijection between these sets.

In the same vein, a nice result to have would be ‘for any sets A and B , either A injects into B or B injects into A ’. This is true, but is not proved in this course.

§5.4.4 Sizes of Sets

Going back to the sizes of sets, we have seen that we can have finite sets, countable sets and uncountable sets. We also know that in the realm of uncountable sets, taking power sets makes you bigger. So the sequence $\mathbb{N}, \mathcal{P}(\mathbb{N}), \mathcal{P}(\mathcal{P}(\mathbb{N})), \dots$ may make us wonder does every set X inject into one of these?

This is not true. For example, take the set $X = \mathbb{N} \cup \mathcal{P}(\mathbb{N}) \cup \mathcal{P}(\mathcal{P}(\mathbb{N})) \cup \dots$. This isn’t the biggest set either, as we can take $X' = X \cup \mathcal{P}(X) \cup \dots$. And this isn’t biggest either, as we could just do the same thing all over again, and again, and again.

6 Probability

Probability is the art of reasoning precisely about uncertainty. Starting from a small collection of axioms, we will build up a theory that lets us make (often surprisingly strong) quantitative statements about randomness – from coin tosses and card shuffles all the way up to random walks, branching processes and the celebrated central limit theorem. One of the joys of the subject is how far a few simple ideas (independence, expectation, indicators, generating functions) can be pushed, and we will try to emphasise these recurring techniques throughout.

This article constitutes my notes for the ‘Probability’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§6.1 Setting Up Probability

We are going to build up probability from the very basic axioms. This will give us a firm mathematical basis to work from, but also will match the intuitive notions of probability that you have hopefully developed before reading this handout.

§6.1.1 Probability Spaces

The basic object of study in probability theory is that of *probability spaces*. We begin by defining how we can talk about the idea of ‘events’ formally. What we want from our definition will be some way to have a set of possible ‘outcomes’ (a *sample space*), from which we can combine different outcomes to form various ‘events’. This leads almost directly to the definition of an *event space*.

Definition 6.1.1 (Event Space)

The collection $\mathcal{F}$ of subsets of the sample space Ω is an **event space** or **σ -algebra** if

1. $\mathcal{F}$ is non-empty,
2. if $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$,
3. if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Within this system, $A \cup B$ corresponds to either A or B occurring, $A \cap B$ corresponds to both A and B occurring, and $\Omega \setminus A$ corresponds with anything but A occurring. You get the idea.

Our main goal in the study of probability is to assign some sort of ‘likelihood’ to events in the event space. This is done through the introduction of a *probability measure*.

Definition 6.1.2 (Probability Measure)

We say that a function $\mathbb{P} : \mathcal{F} \rightarrow \mathbb{R}$ is a **probability measure** on $(\Omega, \mathcal{F})$ if

1. $\mathbb{P}(A) \geq 0$ for all $A \in \mathcal{F}$,
2. $\mathbb{P}(\Omega) = 1$,
3. if $A_1, A_2, \dots$ are disjoint events in $\mathcal{F}$, then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

These properties of the probability measure should match your own intuition about the way that probability works. Putting all the pieces together, we get our main object of study: the *probability space*.

Definition 6.1.3 (Probability Space)

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ such that Ω is non empty, $\mathcal{F}$ is an event space on Ω , and $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$.

If we only have to deal with say finite (or even countable) Ω , then it makes sense to usually take $\mathcal{F}$ to just be the power set of Ω . In this case, if $\Omega = \{\omega_1, \omega_2, \dots\}$, then we can construct any probability measure $\mathbb{P}$ by just specifying $\mathbb{P}(\{\omega_i\})$ for all i , as then for $A \in \mathcal{F}$, we can get $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$.

Before going any further, we should note some basic properties of probability measures, all of which follow pretty quickly from the definition.

Proposition 6.1.4 (Properties of Probability Measures)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $A, B \in \mathcal{F}$. Then

- (i) $\mathbb{P}(\Omega \setminus A) = 1 - \mathbb{P}(A)$,
- (ii) $\mathbb{P}(\emptyset) = 0$,
- (iii) if $A \subseteq B$ then $\mathbb{P}(A) \leq \mathbb{P}(B)$,
- (iv) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Proof. For (i), the events A and $\Omega \setminus A$ are disjoint with union Ω , so $\mathbb{P}(A) + \mathbb{P}(\Omega \setminus A) = \mathbb{P}(\Omega) = 1$. Then (ii) follows by taking $A = \Omega$. For (iii), we can write B as the disjoint union $B = A \cup (B \setminus A)$, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. Finally for (iv), we write $A \cup B$ as the disjoint union $A \cup (B \setminus (A \cap B))$, giving

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus (A \cap B)) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

where the last step uses that $A \cap B \subseteq B$. □

§6.1.2 Classical Probability and Counting

The setting in which probability historically began (think gambling problems in the 17th century) is what's now called *classical probability*: a finite sample space in which every outcome is equally likely. That is, we take $\Omega = \{\omega_1, \dots, \omega_n\}$ and

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|} \quad \text{for } A \subseteq \Omega.$$

Computing probabilities in this setting amounts to *counting*, so it is worth pausing to build up a small toolbox of combinatorial facts. We will state them briefly, since they

should be mostly familiar.

Fact 6.1.5 (Counting Toolbox)

Let Ω be a set of n elements. Then

- (i) the number of orderings (permutations) of Ω is $n! = n(n-1) \cdots 1$,
- (ii) the number of subsets of Ω of size k is $\binom{n}{k} = \frac{n!}{k!(n-k)!}$,
- (iii) the number of ways to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $n_1 + \cdots + n_k = n$ is the **multinomial coefficient**

$$\binom{n}{n_1, n_2, \dots, n_k} = \binom{n}{n_1} \binom{n-n_1}{n_2} \cdots \binom{n-n_1-\cdots-n_{k-1}}{n_k} = \frac{n!}{n_1! n_2! \cdots n_k!}.$$

Armed with these, let's do some classical probability. The following examples are, well, classics.

Example 6.1.6 (Aces on Top)

A standard deck of 52 cards is shuffled so that all $52!$ orderings are equally likely. What is the probability that the top two cards are both aces?

We take Ω to be the set of all $52!$ orderings. The number of orderings with aces in the top two positions is $4 \times 3 \times 50!$ (choose which aces go first and second, then order the rest freely). So the probability is

$$\frac{4 \cdot 3 \cdot 50!}{52!} = \frac{4 \cdot 3}{52 \cdot 51} = \frac{1}{221}.$$

Example 6.1.7 (The Birthday Problem)

There are n people in a room. What is the probability that at least two of them share a birthday^a?

Take $\Omega = \{1, \dots, 365\}^n$, with all outcomes equally likely. Let A be the event that at least two birthdays coincide. It's much easier to count the complement, the event that all n birthdays are distinct. There are $365 \cdot 364 \cdots (365 - n + 1)$ such outcomes, so

$$\mathbb{P}(A) = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

Plugging in numbers gives a somewhat surprising result: with $n = 22$ we get $\mathbb{P}(A) \approx 0.476$, and with $n = 23$ we get $\mathbb{P}(A) \approx 0.507$. So in a room of just 23 people, it's more likely than not that two people share a birthday.

^aWe'll assume that there's 365 days in a year, that no one is born on the 29th of February, and that all birthdays are equally likely.

The trick of 'counting the complement' used above is worth remembering – it is frequently much easier to count the outcomes avoiding some property than those with it.

Since factorials appear all over the place in these calculations, it is quite helpful to know how fast $n!$ actually grows. The definitive answer is given by *Stirling's formula*. We write $a_n \sim b_n$ to mean $a_n/b_n \rightarrow 1$ as $n \rightarrow \infty$.

Theorem 6.1.8 (Stirling's Formula)

As $n \rightarrow \infty$, we have

$$n! \sim n^n e^{-n} \sqrt{2\pi n}.$$

We will prove the following weaker statement, which is enough for many applications (it pins down $\log n!$ to first order, but not the constant $\sqrt{2\pi}$).

Proposition 6.1.9 (Weak Stirling)

As $n \rightarrow \infty$, $\log n! \sim n \log n$.

Proof. Let $l_n = \log n! = \log 2 + \log 3 + \cdots + \log n$. Since $\log$ is increasing, for any integer k we have $\log k \leq \log x \leq \log(k+1)$ for $x \in [k, k+1]$, and integrating over such intervals gives

$$\sum_{k=1}^{n-1} \log k \leq \int_1^n \log x \, dx \leq \sum_{k=2}^n \log k,$$

that is, $l_{n-1} \leq n \log n - n + 1 \leq l_n$. Using both bounds, for all n we get

$$n \log n - n + 1 \leq l_n \leq (n+1) \log(n+1) - n,$$

and dividing through by $n \log n$ and letting $n \rightarrow \infty$ gives $l_n/(n \log n) \rightarrow 1$, as required. $\square$

The proof of the full result is a little involved (the constant $\sqrt{2\pi}$ is usually extracted from the asymptotics of the central binomial coefficient $\binom{2n}{n}$, using Wallis' product), and we will not give it here. A fact worth keeping from that analysis is

$$2^{-2n} \binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}},$$

which says that the probability of getting exactly n heads in $2n$ tosses of a fair coin decays like $1/\sqrt{\pi n}$ – slowly!

§6.1.3 Conditional Probability and Independence

In our day-to-day experience, we know that knowledge about one event can influence how likely we determine another event is to occur. This type of thinking motivates the introduction of *conditional probability*.

Definition 6.1.10 (Conditional Probability)

If $A, B \in \mathcal{F}$ and $\mathbb{P}(B) > 0$, the **probability of A given B** is denoted $\mathbb{P}(A \mid B)$, and is given by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

It's not hard to check that $\mathbb{P}(\cdot \mid B)$ is a probability measure, as we would expect – it's just assigning probabilities to events in $\mathcal{F}$ in a different way!

A natural thing to care about is if knowledge about an event *doesn't* influence how likely another event is. We call such events independent.

Definition 6.1.11 (Independent Events)

We say that events A, B in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

and **dependent** otherwise.

We can see immediately that A, B independent implies that

$$\mathbb{P}(A \mid B) = \mathbb{P}(A) \quad \text{and} \quad \mathbb{P}(B \mid A) = \mathbb{P}(B),$$

if $\mathbb{P}(A), \mathbb{P}(B) \neq 0$. However, this definition is slightly more general than this property since it handles events with zero probability. We can generalise this notion of independence to multiple events, and we get a criterion that's stronger than pairwise independence¹.

Definition 6.1.12 (Mutually Independent Events)

We say that the events $A_1, A_2, A_3, \dots$ are **mutually independent** if

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_r}) = \mathbb{P}(A_{i_1})\mathbb{P}(A_{i_2}) \cdots \mathbb{P}(A_{i_r}),$$

for any $i_1, i_2, \dots, i_r$ and $r \geq 2$.

So if conditional probability tells us about the likelihood of a certain event in one particular scenario, then it's easy to imagine that knowing about the likelihood in every scenario would then just give you the likelihood in general. Such thinking is formalised in the *law of total probability*, also known as the *partition theorem*.

Theorem 6.1.13 (Partition Theorem)

Let $\{B_1, B_2, \dots\}$ be a collection of disjoint events such that $\mathbb{P}(B_i) > 0$ for all i and $\bigcup_{i=1}^{\infty} B_i = \Omega$. Then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i),$$

for any $A \in \mathcal{F}$.

Proof. We can compute that

$$\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i) = \sum_i \mathbb{P}(A \cap B_i) = \mathbb{P}\left(\bigcup_i (A \cap B_i)\right) = \mathbb{P}(A),$$

as required. □

Using the partition theorem, we can establish the famous **Bayes' theorem**, which allows us to use knowledge about $\mathbb{P}(A \mid B_j)$ to give information about $\mathbb{P}(B_j \mid A)$:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

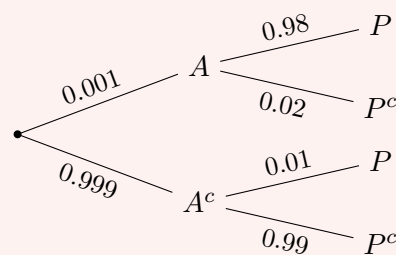
¹It's not hard to construct an example of events that are pairwise independent but wouldn't be independent in any reasonable sense.

Bayes' theorem is the basis of essentially all of Bayesian statistics – it lets us 'reverse the direction' of conditioning, updating our beliefs about which scenario B_j we are in, given that we observed A . It also has a habit of producing answers that clash violently with intuition, as the next example shows.

Example 6.1.14 (False Positives)

Suppose that 0.1% of the population suffer from a certain rare condition, and that we have a test for the condition which is positive for 98% of sufferers, but also (falsely) positive for 1% of non-sufferers. A person is chosen at random and tests positive. What is the probability that they actually have the condition?

Let A be the event that the person has the condition, and P the event that they test positive. The situation is summarised by the following tree, with the two stages of conditioning shown left to right.



Then by Bayes' theorem,

$$\begin{aligned}\mathbb{P}(A | P) &= \frac{\mathbb{P}(P | A)\mathbb{P}(A)}{\mathbb{P}(P | A)\mathbb{P}(A) + \mathbb{P}(P | A^c)\mathbb{P}(A^c)} \\ &= \frac{0.98 \times 0.001}{0.98 \times 0.001 + 0.01 \times 0.999} \approx 0.09.\end{aligned}$$

So despite the test looking quite accurate, a positive result only means a 9% chance of having the condition! The point is that the false positive rate (1%) utterly dwarfs the prevalence of the condition (0.1%): in a population of 1000 people, we expect about 1 true positive but about 10 false positives, so only about 1 in 11 positive tests is genuine.

Another recurring subtlety is that the answer to a conditional probability question depends delicately on *exactly* what information you condition on.

Example 6.1.15 (Two Children)

I have two children. Consider the following three statements:

- (a) at least one of them is a boy;
- (b) the eldest is a boy;
- (c) at least one of them is a boy born on a Thursday.

Given each statement in turn, what is the probability that I have two boys? (Assume boys and girls, and days of the week, are equally likely and independent.)

Writing the sexes in age order, the equally likely possibilities are BB, BG, GB, GG .

Given (a), we condition on $\{BB, BG, GB\}$, so

$$\mathbb{P}(BB \mid BB \cup BG \cup GB) = \frac{1/4}{3/4} = \frac{1}{3}.$$

Given (b), we condition on $\{BB, BG\}$, and the answer is $\frac{1}{2}$ – so (a) and (b) genuinely differ, even though both ‘tell you there’s a boy’.

For (c), splitting each boy into ‘born on Thursday’ (T , probability $\frac{1}{14}$ per child) or ‘boy not born on Thursday’ (N , probability $\frac{6}{14}$), a short computation with the equally likely fine-grained outcomes gives

$$\mathbb{P}(\text{two boys} \mid \text{at least one } T) = \frac{\frac{1}{14} \cdot \frac{1}{14} + 2 \cdot \frac{1}{14} \cdot \frac{6}{14}}{\frac{1}{14} \cdot \frac{1}{14} + 2 \cdot \frac{1}{14} \cdot \frac{6}{14} + 2 \cdot \frac{1}{14} \cdot \frac{7}{14}} = \frac{13}{27} \approx 0.48.$$

Astonishingly, the (apparently irrelevant!) day of birth moves the answer from $\frac{1}{3}$ towards $\frac{1}{2}$. Intuitively, ‘a boy born on a Thursday’ is more likely to be reportable in a two-boy family, since there are two chances for it to occur.

Conditioning can be genuinely subtle, and it’s worth seeing one famous trap: aggregating data across dissimilar populations can *reverse* the direction of an inequality between conditional probabilities.

Example 6.1.16 (Simpson’s Paradox)

A university admits students from two types of school, and it’s observed that applicants from state schools are admitted at a higher rate than applicants from independent schools *both* among men and among women – yet when the data is aggregated, independent school applicants are admitted at a higher rate overall.

This can genuinely happen with real numbers, and there is no contradiction. Writing A for admission, C for coming from a state school and B for being a man, the ‘paradoxical’ data amounts to

$$\mathbb{P}(A \mid B \cap C) > \mathbb{P}(A \mid B \cap C^c), \quad \mathbb{P}(A \mid B^c \cap C) > \mathbb{P}(A \mid B^c \cap C^c),$$

but $\mathbb{P}(A \mid C) < \mathbb{P}(A \mid C^c)$. If we try to ‘average’ the first two inequalities using the partition theorem, we find

$$\mathbb{P}(A \mid C) = \mathbb{P}(A \mid B \cap C)\mathbb{P}(B \mid C) + \mathbb{P}(A \mid B^c \cap C)\mathbb{P}(B^c \mid C),$$

and similarly for C^c – but the weights $\mathbb{P}(B \mid C)$ and $\mathbb{P}(B \mid C^c)$ need not be equal! It is only when the populations are mixed in the same proportions ($\mathbb{P}(B \mid C) = \mathbb{P}(B \mid C^c)$) that the intuitive conclusion $\mathbb{P}(A \mid C) > \mathbb{P}(A \mid C^c)$ actually follows. The moral: be very careful about pooling conditional probabilities.

§6.1.4 Continuity of Probability Measures

In the definitions for event spaces (σ -algebras) and probability measures, you may notice that we specify rules for countably infinite sequences of events. This is for good reason – there’s plenty of natural scenarios where such a setup is useful. For example, consider the experiment of tossing a coin until you get heads, in which it is possible to get an arbitrarily long run of tails before you stop tossing. To make working with such scenarios

easier, we can establish a result which allows us study events as limits of a sequence of events – that the probability measure is continuous.

Definition 6.1.17 (Limits of Events)

We say that a sequence of events $A_1, A_2, \dots$ is *increasing* if $A_1 \subseteq A_2 \subseteq \dots$. We define the **limit** as

$$\lim_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} A_n.$$

Theorem 6.1.18 (Continuity of Probability Measures)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. If $A_1, A_2, \dots$ is an increasing sequence of events in $\mathcal{F}$ with limit A , then $\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n)$.

Proof. Define $B_i = A_i \setminus A_{i-1}$, with $B_1 = A_1$. Then we have

$$A = B_1 \cup B_2 \cup B_3 \cup \dots$$

which is the union of disjoint events in $\mathcal{F}$. Thus we have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}\left(\bigcup_{n=1}^{\infty} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} B_n\right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \mathbb{P}(B_1 \cup \dots \cup B_n) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}(A_n), \end{aligned}$$

as required. □

As a side note, the type of construction used to define B_i can be quite helpful when we want to deal with a disjoint union, rather than some non-disjoint sequence of events (since it works nicely with the definition of the probability space!)

Another rather intuitive notion is that of probabilities being ‘subadditive’ when events are *not* disjoint, which we can see in the following result.

Theorem 6.1.19 (Boole’s Inequality – The Union Bound)

For any sequence of events $A_1, A_2, \dots$ we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Proof. We use a similar construction to the previous proof. Let $B_1 = A_1$, and define $B_i = A_i \setminus \bigcup_{k=1}^{i-1} A_k$. Then we have

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(B_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i),$$

where the inequality occurs since $B_i \subseteq A_i$ implies $\mathbb{P}(B_i) \leq \mathbb{P}(A_i)$. $\square$

§6.1.5 Inclusion-Exclusion

We saw earlier that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$: if we just add up the two probabilities, we've double counted the overlap, so we subtract it off once. If we try the same thing with three events, adding up the pairwise corrections over-corrects for the triple overlap, and we need to add that back on:

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(A \cap C) + \mathbb{P}(A \cap B \cap C).$$

Continuing this alternating pattern of corrections gives the general *inclusion-exclusion formula*.

Theorem 6.1.20 (Inclusion-Exclusion)

Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}).$$

Proof. We induct on n , with the case $n = 2$ already established. Assuming the result for $n - 1$ events, we have

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right).$$

Now we apply the inductive hypothesis to both unions on the right hand side, the second with $B_i = A_i \cap A_n$. The first gives all of the terms not involving A_n , the term $\mathbb{P}(A_n)$ gives the term involving only A_n , and expanding the second union gives (after noting $B_{i_1} \cap \dots \cap B_{i_k} = A_{i_1} \cap \dots \cap A_{i_k} \cap A_n$, and with a sign flip from the minus sign) exactly the terms involving A_n and at least one other event. Collecting everything up gives the required formula. $\square$

Remark. In the classical setting where $\mathbb{P}(A) = |A|/|\Omega|$, multiplying through by $|\Omega|$ gives the corresponding counting formula

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^n (-1)^{k+1} \sum_{i_1 < \dots < i_k} |A_{i_1} \cap \dots \cap A_{i_k}|,$$

which you may well have seen in ‘Numbers and Sets’.

If we don't need an exact answer, we can truncate the inclusion-exclusion sum, and it's useful to know which way the truncation errs. The answer alternates: stopping after an odd number of terms overestimates, and after an even number underestimates. These are the *Bonferroni inequalities*.

Theorem 6.1.21 (Bonferroni Inequalities)

Let $A_1, \dots, A_n$ be events and $1 \leq r \leq n$. If r is odd, then

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{k=1}^r (-1)^{k+1} \sum_{i_1 < \dots < i_k} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}),$$

and if r is even the reverse inequality holds.

Proof. We induct on n , exactly as in the proof of inclusion-exclusion (the case $n = 2$ is easy to check directly). Suppose r is odd, and write

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right),$$

where $B_i = A_i \cap A_n$. Applying the inductive hypothesis with r (odd) to the first union, and with $r - 1$ (even) to the second union – noting the minus sign flips the direction of that inequality – and collecting terms as before gives the required overestimate. The case of r even is identical with the inequalities reversed. $\square$

Note that the case $r = 1$ of the Bonferroni inequalities is just Boole's inequality (for finitely many events).

Let's now see inclusion-exclusion doing some real work.

Example 6.1.22 (Derangements)

A **derangement** is a permutation with no fixed points. If we pick a permutation σ of $\{1, \dots, n\}$ uniformly at random, what is the probability that σ is a derangement?

Let A_i be the event that $\sigma(i) = i$. Then we want to compute $1 - \mathbb{P}(A_1 \cup \dots \cup A_n)$. For any distinct indices $i_1 < \dots < i_k$, the permutations fixing all of $i_1, \dots, i_k$ number $(n - k)!$, so

$$\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \frac{(n - k)!}{n!}.$$

There are $\binom{n}{k}$ such sets of indices, so by inclusion-exclusion,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n - k)!}{n!} = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!},$$

and thus

$$\mathbb{P}(\sigma \text{ is a derangement}) = 1 - \sum_{k=1}^n \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^n \frac{(-1)^k}{k!} \rightarrow \frac{1}{e}$$

as $n \rightarrow \infty$. So for large n , a bit over a third of permutations are derangements – and remarkably, the answer barely depends on n .

Example 6.1.23 (Counting Surjections)

How many functions $f : \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ are surjective? Let A_i be the set of functions that miss the value i . Then the surjections are exactly the functions

in none of the A_i , and $|A_{i_1} \cap \dots \cap A_{i_k}| = (m - k)^n$ since such functions map into a set of size $m - k$. So by the counting form of inclusion-exclusion, the number of surjections is

$$m^n - \sum_{k=1}^m (-1)^{k+1} \binom{m}{k} (m - k)^n = \sum_{k=0}^m (-1)^k \binom{m}{k} (m - k)^n.$$

§6.2 Discrete Random Variables

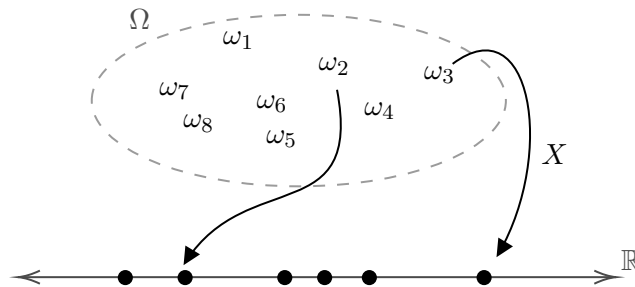
Imagine the following scenario. You are in a lecture hall, and you decide to perform an experiment where choose a student at random and ask them how they performed in the admissions test. It would be natural to model this scenario with a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\Omega = \{\omega_1, \dots, \omega_n\}$ where each ω_i corresponds to a student in the lecture hall, $\mathcal{F}$ is all possible subsets of Ω , and $\mathbb{P}$ is the probability measure describing the likelihoods of given students being chosen in the experiment.

So how does the ‘asking for the student’s test score’ fit into this setup? A natural way to deal with this is by defining a function $X : \Omega \rightarrow \mathbb{R}$ where $X(\omega_i)$ is the admissions test score of the student ω_i . We call such a function a *discrete random variable*.

Definition 6.2.1 (Discrete Random Variables)

A **discrete random variable** X on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a function $X : \Omega \rightarrow \mathbb{R}$ such that the image $X(\Omega)$ is a countable subset of $\mathbb{R}$ and $\{\omega \in \Omega : X(\omega) = x\} \in \mathcal{F}$ for $x \in \mathbb{R}$.

A natural mental picture to go along with this definition (which roughly follows our motivating example) is shown below.



§6.2.1 Probability Mass Functions

With the introduction of discrete random variables, we can change our view from probabilities of events occurring to the probabilities that a discrete random variable takes on specific values. A nice way to do this is by introducing the *probability mass function*.

Definition 6.2.2 (Probability Mass Function)

The **probability mass function** or **pmf** of a discrete random variable X is a function $p_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$p_X(x) = \mathbb{P}(X = x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}).$$

It turns out that the probability mass function encodes pretty much all information about the discrete random variable X , so much so that if we have some function p_X that acts like a probability mass function, we can say things like ‘let X be a random variable taking value x_i with probability $p_X(x_i)$ ’ and have it be completely well defined.

Theorem 6.2.3

Let $p_X : \mathbb{R} \rightarrow [0, 1]$ be a function such that $p_X(x) \neq 0$ for countably many x and $\sum_{x \in \mathbb{R}} p_X(x) = 1$. Then there is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X on this space where X has probability mass function p_X .

Proof. Take $\Omega = \{x \in \mathbb{R} : p_X(x) \neq 0\}$, $\mathcal{F}$ to be all subsets of Ω , and

$$\mathbb{P}(A) = \sum_{\omega \in A} p_X(\omega) \quad \text{for } A \in \mathcal{F},$$

and define $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$. □

§6.2.2 Expected Value

A common question when dealing with discrete random variables is ‘what is the mean of this discrete random variable?’. Thinking back to our lecture hall example, this would correspond with asking what the mean test score in the lecture hall was. We answer such questions by defining the *expectation* of a random variable.

Definition 6.2.4 (Expected Value)

If X is a discrete random variable, we define the **expected value** of X to be

$$\mathbb{E}[X] = \sum_{x \in \text{img } X} x \mathbb{P}(X = x)$$

whenever this sum converges absolutely^a.

^aWe require this because we rearranging the sum shouldn’t change the expected value – see Analysis I.

There are a few ‘tricks’ which make the computation of expected value somewhat easier. For example, if we have a discrete random variable defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then we can sum over the elements of Ω rather than the elements of $\text{img } X$. Also, since random variables are really just functions, we can obtain new random variables by composing them with other functions, for example $X \mapsto X^2$. In such scenarios, we can use a seemingly obvious result.

Theorem 6.2.5 (Law of The Subconscious Statistician)

If X is a discrete random variable and $g : \mathbb{R} \rightarrow \mathbb{R}$, then

$$\mathbb{E}[g(X)] = \sum_{x \in \text{img } X} g(x) \mathbb{P}(X = x),$$

whenever this sum converges absolutely.

Proof. Let $Y = g(X)$ be a discrete random variable. Then

$$\begin{aligned}\mathbb{E}[Y] &= \sum_{y \in \text{img } Y} y \mathbb{P}(Y = y) = \sum_{y \in \text{img } Y} y \sum_{x \in \text{img } X: g(x)=y} \mathbb{P}(X = x) \\ &= \sum_{x \in \text{img } X} g(x) \mathbb{P}(X = x),\end{aligned}$$

provided this converges absolutely. $\square$

Another thing we can do very easily is find the expected value of a sum of random variables, using *linearity of expectation*.

Theorem 6.2.6 (Linearity of Expectation)

For any random variables $X_1, X_2, \dots, X_n$ for which the following expectations exist,

$$\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].$$

Proof. We sum over the elements of the sample space Ω to get

$$\begin{aligned}\mathbb{E}[X] &= \sum_{\omega \in \Omega} p(\omega)[X_1(\omega) + \dots + X_n(\omega)] \\ &= \sum_{\omega \in \Omega} p(\omega)X_1(\omega) + \dots + \sum_{\omega \in \Omega} p(\omega)X_n(\omega) \\ &= \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].\end{aligned}$$

$\square$

It's important to note that there's absolutely no requirement that the random variables be independent for this to hold – one of the reasons why it's an incredibly useful result!

So if expected value measures something like the ‘mean’ of a random variable, it's natural to think about how we can measure how we can expect the random variable deviates from this mean. This is done through the notion of *variance*.

Definition 6.2.7 (Variance)

The **variance** $\text{Var}(X)$ of a discrete random variable X is defined to be

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2].$$

By just working through the definitions we can see that $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, which is often an easy way to calculate this quantity.

§6.2.3 Indicator Functions

There's one type of random variable which is so useful that it needs to be brought to your attention immediately – indicator functions.

Definition 6.2.8 (Indicator Functions)

The **indicator function** of an event A is the random variable $\mathbb{1}_A$ where

$$\mathbb{1}_A(\omega) = \begin{cases} 1 & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

Indicator functions have nice algebraic facts which reflect common ways to manipulate events:

$$\begin{aligned} \mathbb{1}_{A^c} &= 1 - \mathbb{1}_A, \\ \mathbb{1}_{A \cap B} &= \mathbb{1}_A \mathbb{1}_B, \\ \mathbb{1}_{A \cup B} &= \mathbb{1}_A + \mathbb{1}_B - \mathbb{1}_A \mathbb{1}_B. \end{aligned}$$

Still one of the best things about indicator functions is that they allow us to talk about probabilities of events using expectations, since

$$\mathbb{E}[\mathbb{1}_A] = \mathbb{P}(A).$$

All of these properties are straightforward and easy to check (just do casework), but they can make proving things and solving problems much easier. Nontrivial example time!

Example 6.2.9 (Solving Problems With Indicator Functions)

A total of n bar magnets are placed end to end in a line with random independent orientations. Adjacent like poles repel, ends with opposite polarities join to form blocks. Let X be the number of blocks of joined magnets. Find $\mathbb{E}[X]$ and $\text{Var}(X)$.

Solution. The situation looks something like this:



Let $\mathbb{1}_i$ be the indicator function for magnets i and $i+1$ repelling, for $1 \leq i \leq n-1$. We can then rewrite X as

$$X = \mathbb{1}_1 + \mathbb{1}_2 + \cdots + \mathbb{1}_{n-1} + 1.$$

Then since $\mathbb{E}[\mathbb{1}_i] = \mathbb{P}(\text{two magnets of random orientation repel}) = 1/2$, we have

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[\mathbb{1}_1] + \cdots + \mathbb{E}[\mathbb{1}_{n-1}] + 1 \\ &= \frac{1}{2} + \cdots + \frac{1}{2} + 1 = \frac{n+1}{2}. \end{aligned}$$

To compute the variance, we need to calculate $\mathbb{E}[X^2]$. Writing $S = \mathbb{1}_1 + \cdots + \mathbb{1}_{n-1}$, so that $X = 1 + S$, we can expand

$$X^2 = 1 + 2S + S^2 = 1 + 2S + (\mathbb{1}_1^2 + \cdots + \mathbb{1}_{n-1}^2) + \sum_{i \neq j} \mathbb{1}_i \mathbb{1}_j = 1 + 3S + \sum_{i \neq j} \mathbb{1}_i \mathbb{1}_j,$$

using that $\mathbb{1}_i^2 = \mathbb{1}_i$ for indicators. Now $\mathbb{E}[\mathbb{1}_i \mathbb{1}_j] = \mathbb{P}(\text{both pairs repel}) = 1/4$ for $i \neq j$ (the two events involve at least one distinct magnet orientation each, and one

can check they are independent), and there are $(n-1)(n-2)$ ordered pairs $i \neq j$. So

$$\mathbb{E}[X^2] = 1 + 3 \cdot \frac{n-1}{2} + (n-1)(n-2) \cdot \frac{1}{4} = \frac{3n-1}{2} + \frac{(n-1)(n-2)}{4},$$

and thus

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{3n-1}{2} + \frac{(n-1)(n-2)}{4} - \frac{(n+1)^2}{4} \\ &= \frac{2(3n-1) + (n^2 - 3n + 2) - (n^2 + 2n + 1)}{4} = \frac{n-1}{4}. \end{aligned}$$

As a sanity check: the indicators $\mathbb{1}_1, \dots, \mathbb{1}_{n-1}$ are actually mutually independent (each depends on whether consecutive orientations differ), so $X-1$ is a sum of $n-1$ independent ‘coin flips’ and we should expect variance $(n-1) \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{n-1}{4}$, which matches. $\square$

One more nice trick with indicators: if X is a random variable taking values in the non-negative integers, then writing $X = \sum_{k \geq 1} \mathbb{1}_{\{X \geq k\}}$ (an integer m is the sum of m ones!) and taking expectations gives the occasionally very convenient formula

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$

§6.2.4 A Zoo of Standard Distributions

There are a handful of distributions that show up over and over again, both in applications and throughout the rest of this course, so we will now go through them and compute their expectations and variances. Recall that the distribution of a discrete random variable is specified entirely by its probability mass function.

The Bernoulli Distribution

The simplest random experiment imaginable has two outcomes: success (with probability p) or failure. We say X has the **Bernoulli distribution** with parameter $p \in [0, 1]$, written $X \sim \text{Bernoulli}(p)$, if

$$\mathbb{P}(X = 1) = p, \quad \mathbb{P}(X = 0) = 1 - p.$$

Immediately we get $\mathbb{E}[X] = p$, and since $X^2 = X$,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1 - p).$$

Of course, an indicator function $\mathbb{1}_A$ is exactly a Bernoulli random variable with parameter $\mathbb{P}(A)$.

The Binomial Distribution

Now toss the same coin n times independently, and count the number of successes. This gives the **binomial distribution** $X \sim \text{Bin}(n, p)$, with

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \{0, 1, \dots, n\},$$

the binomial coefficient counting which of the n trials were the successes. The binomial theorem shows these probabilities sum to 1 (hence the name).

Writing $X = \mathbb{1}_1 + \cdots + \mathbb{1}_n$ as a sum of indicators of success in each trial, linearity of expectation instantly gives

$$\mathbb{E}[X] = np.$$

For the variance, the slickest elementary route is via $\mathbb{E}[X(X-1)]$ (a *factorial moment* – these interact very well with binomial coefficients):

$$\begin{aligned}\mathbb{E}[X(X-1)] &= \sum_{k=2}^n k(k-1) \binom{n}{k} p^k (1-p)^{n-k} \\ &= n(n-1)p^2 \sum_{k=2}^n \binom{n-2}{k-2} p^{k-2} (1-p)^{n-k} = n(n-1)p^2,\end{aligned}$$

using the identity $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$ and then the binomial theorem. Hence

$$\text{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p).$$

Remark (The Multinomial Distribution). If each trial has not two but k possible outcomes, with probabilities $p_1, \dots, p_k$ summing to 1, then the vector $(N_1, \dots, N_k)$ of counts of each outcome over n independent trials has the **multinomial distribution**:

$$\mathbb{P}(N_1 = n_1, \dots, N_k = n_k) = \binom{n}{n_1, \dots, n_k} p_1^{n_1} \cdots p_k^{n_k},$$

for $n_1 + \cdots + n_k = n$, with the multinomial coefficient counting the arrangements of outcomes among the trials. Each marginal N_i is $\text{Bin}(n, p_i)$.

The Geometric Distribution

Instead of a fixed number of tosses, toss the coin repeatedly until the first success, and let X be the number of tosses used. This is the **geometric distribution** $X \sim \text{Geom}(p)$ (with $p > 0$), where

$$\mathbb{P}(X = k) = (1-p)^{k-1}p, \quad k \in \{1, 2, 3, \dots\},$$

since we need $k-1$ failures followed by a success². Summing a geometric series confirms the total probability is 1 – in particular, the first success comes eventually with probability 1.

For the expectation, note $\mathbb{P}(X \geq k) = (1-p)^{k-1}$ (the first $k-1$ tosses all fail), so using our tail-sum formula,

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=1}^{\infty} (1-p)^{k-1} = \frac{1}{p},$$

which matches intuition: an event of probability p takes about $1/p$ attempts to see. For the variance we again use factorial moments, this time with the second derivative of the geometric series $\sum_k q^k = (1-q)^{-1}$:

$$\mathbb{E}[X(X-1)] = \sum_{k=2}^{\infty} k(k-1)(1-p)^{k-1}p = p(1-p) \sum_{k=2}^{\infty} k(k-1)(1-p)^{k-2} = \frac{2(1-p)}{p^2},$$

²Some authors instead count the number of *failures* before the first success, giving pmf $(1-p)^k p$ on $\{0, 1, 2, \dots\}$ – watch out for the convention in use.

and thus

$$\text{Var}(X) = \frac{2(1-p)}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1-p}{p^2}.$$

The geometric distribution has a property worth flagging now, as we will meet its continuous cousin later: it is *memoryless*. If the first s tosses were all failures, the number of *further* tosses needed is distributed just like a fresh geometric:

$$\mathbb{P}(X > s+t \mid X > s) = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t = \mathbb{P}(X > t).$$

The coin does not remember its past – which is also why the ‘gambler’s fallacy’ (a run of losses making a win ‘due’) is a fallacy.

The Hypergeometric Distribution

Suppose an urn contains N balls, K of which are red, and we draw n balls *without replacement*. The number X of red balls drawn has the **hypergeometric distribution**, with

$$\mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}},$$

for the possible values of k . Indicators make the expectation painless: letting $\mathbb{1}_i$ indicate that the i -th draw is red, each draw is (by symmetry) equally likely to be any of the N balls, so $\mathbb{E}[\mathbb{1}_i] = K/N$ and

$$\mathbb{E}[X] = \frac{nK}{N}.$$

Note this matches the binomial mean with $p = K/N$ – drawing without replacement doesn’t change the expectation, though it does shrink the variance.

The Poisson Distribution

Our final distribution models the number of events occurring in a fixed window of time, when events occur ‘at a constant rate, independently’: the number of buses passing in an hour, misprints on a page, radioactive decays in a second, and so on. We say X has the **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \in \{0, 1, 2, \dots\}.$$

The total probability is $e^{-\lambda} \sum_k \lambda^k / k! = e^{-\lambda} e^\lambda = 1$, as needed. The expectation is

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda,$$

so the parameter λ is exactly the average number of events. Similarly $\mathbb{E}[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k \geq 2} \lambda^{k-2} / (k-2)! = \lambda^2$, giving

$$\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

The mean and the variance are both λ – a distinctive fingerprint of the Poisson distribution.

Where does the Poisson distribution come from? It arises naturally as a limit of binomials with many trials, each individually unlikely – this is sometimes called the *law of rare events*.

Theorem 6.2.10 (Poisson Approximation to the Binomial)

Fix $\lambda > 0$, and let $X_n \sim \text{Bin}(n, \lambda/n)$. Then for each fixed $k \geq 0$,

$$\mathbb{P}(X_n = k) \longrightarrow e^{-\lambda} \frac{\lambda^k}{k!} \quad \text{as } n \rightarrow \infty.$$

Proof. We compute directly:

$$\begin{aligned} \mathbb{P}(X_n = k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \cdot \frac{n(n-1) \cdots (n-k+1)}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}. \end{aligned}$$

As $n \rightarrow \infty$ with k fixed, the middle factor tends to 1, the last factor tends to 1, and $(1 - \lambda/n)^n \rightarrow e^{-\lambda}$. So $\mathbb{P}(X_n = k) \rightarrow e^{-\lambda} \lambda^k / k!$ as claimed. $\square$

To interpret this: divide an hour into n tiny intervals, and suppose a bus arrives in each interval with probability λ/n , independently. The number of buses in the hour is $\text{Bin}(n, \lambda/n)$, and letting the intervals shrink gives exactly $\text{Poisson}(\lambda)$.

§6.3 Multiple Random Variables

Most interesting probabilistic questions involve the interaction of *several* random variables – and this is where notions like independence, covariance and conditional expectation come into play.

§6.3.1 Joint Distributions and Independence

So far we have studied random variables one at a time. To capture how several random variables interact, we record all of their values at once.

Definition 6.3.1 (Joint Distribution)

Let $X_1, \dots, X_n$ be discrete random variables on a probability space. Their **joint distribution** is the collection of probabilities

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n),$$

over all possible values $x_1, \dots, x_n$.

The joint distribution knows everything about the individual random variables: summing out all but one variable recovers what is called the **marginal distribution**,

$$\mathbb{P}(X_1 = x_1) = \sum_{x_2, \dots, x_n} \mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n),$$

which is just the partition theorem in disguise. The converse fails badly – the marginals alone don't determine the joint distribution, as they say nothing about how the variables *depend* on each other. The simplest possible dependence structure is no dependence at all.

Definition 6.3.2 (Independent Random Variables)

Discrete random variables $X_1, \dots, X_n$ are **independent** if for all values $x_1, \dots, x_n$,

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \cdots \mathbb{P}(X_n = x_n).$$

Example 6.3.3 (Coin Tossing, Formally)

Let's model tossing a biased coin n times, formally. Take $\Omega = \{0, 1\}^n$, and for $\omega = (\omega_1, \dots, \omega_n)$ set

$$\mathbb{P}(\{\omega\}) = \prod_{k=1}^n p^{\omega_k} (1-p)^{1-\omega_k},$$

and let $X_k(\omega) = \omega_k$ be the result of the k th toss. Then each $X_k \sim \text{Bernoulli}(p)$, and multiplying out shows that $\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_k \mathbb{P}(X_k = x_k)$, so the tosses are genuinely independent in our formal sense. The number of heads $S_n = X_1 + \dots + X_n$ then has the $\text{Bin}(n, p)$ distribution.

A very useful consequence of independence is that expectations of products factorise.

Theorem 6.3.4 (Expectation of a Product)

If X and Y are independent discrete random variables and $f, g : \mathbb{R} \rightarrow \mathbb{R}$, then

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)] \mathbb{E}[g(Y)],$$

whenever the expectations on the right exist.

Proof. Summing over the possible pairs of values,

$$\begin{aligned} \mathbb{E}[f(X)g(Y)] &= \sum_{x,y} f(x)g(y)\mathbb{P}(X=x, Y=y) \\ &= \sum_{x,y} f(x)g(y)\mathbb{P}(X=x)\mathbb{P}(Y=y) \\ &= \left(\sum_x f(x)\mathbb{P}(X=x) \right) \left(\sum_y g(y)\mathbb{P}(Y=y) \right) = \mathbb{E}[f(X)] \mathbb{E}[g(Y)], \end{aligned}$$

where the rearrangement is justified by absolute convergence. $\square$

Contrast this with linearity of expectation, which required no independence whatsoever: for products, independence is really doing work.

We can also ask about the distribution of a *sum* of independent random variables. Conditioning on the value of one of them gives

$$\mathbb{P}(X + Y = z) = \sum_y \mathbb{P}(X = z - y, Y = y) = \sum_y \mathbb{P}(X = z - y)\mathbb{P}(Y = y),$$

a formula known as the **convolution** of the two distributions.

Example 6.3.5 (Sum of Independent Poissons)

Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent. Then

$$\begin{aligned}\mathbb{P}(X + Y = n) &= \sum_{r=0}^n e^{-\lambda} \frac{\lambda^r}{r!} \cdot e^{-\mu} \frac{\mu^{n-r}}{(n-r)!} \\ &= \frac{e^{-(\lambda+\mu)}}{n!} \sum_{r=0}^n \binom{n}{r} \lambda^r \mu^{n-r} = e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!},\end{aligned}$$

using the binomial theorem. So $X + Y \sim \text{Poisson}(\lambda + \mu)$ – the Poisson family is closed under adding independent members, which is exactly what you’d hope for from a distribution modelling counts of events (two independent streams of buses form one stream with the combined rate).

§6.3.2 Covariance

If independence is the absence of dependence, we’d also like a way to *measure* dependence. The first-order tool for this is covariance.

Definition 6.3.6 (Covariance)

The **covariance** of random variables X and Y is

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])],$$

whenever this exists.

Roughly, $\text{Cov}(X, Y) > 0$ means X and Y tend to be large together, and $\text{Cov}(X, Y) < 0$ means one tends to be large when the other is small. Covariance has a pile of easily-verified properties, which we collect here.

Proposition 6.3.7 (Properties of Covariance)

For random variables X, Y, Z and a constant c :

- (i) $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, and $\text{Cov}(X, X) = \text{Var}(X)$,
- (ii) $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$,
- (iii) $\text{Cov}(cX, Y) = c \text{Cov}(X, Y)$ and $\text{Cov}(X + c, Y) = \text{Cov}(X, Y)$,
- (iv) $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$,
- (v) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y)$.

Proof. Part (i) is immediate from the definition. For (ii), expand

$$(X - \mathbb{E}[X])(Y - \mathbb{E}[Y]) = XY - X\mathbb{E}[Y] - Y\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y]$$

and take expectations. Parts (iii) and (iv) then follow from linearity of expectation applied to (ii). For (v), write $\tilde{X} = X - \mathbb{E}[X]$ and $\tilde{Y} = Y - \mathbb{E}[Y]$; then

$$\text{Var}(X + Y) = \mathbb{E}[(\tilde{X} + \tilde{Y})^2] = \mathbb{E}[\tilde{X}^2] + \mathbb{E}[\tilde{Y}^2] + 2\mathbb{E}[\tilde{X}\tilde{Y}],$$

which is exactly the claim. $\square$

Iterating (iv) and (v) gives the general and very useful formula

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j).$$

If X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ by the product theorem, so property (ii) gives $\text{Cov}(X, Y) = 0$, and hence variance becomes additive:

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{for independent } X, Y.$$

(This gives another derivation of the binomial variance $np(1-p)$, summing the variances of n independent Bernoulli trials.)

It's very important to remember that the converse is *false*: zero covariance does not imply independence.

Example 6.3.8 (Uncorrelated but Dependent)

Let X_1, X_2, X_3 be independent Bernoulli($1/2$) random variables, and set $Y_i = 2X_i - 1$ (so the Y_i are random signs), and consider $Z_1 = X_3Y_1$ and $Z_2 = X_3Y_2$. Then $\mathbb{E}[Z_1] = \mathbb{E}[Z_2] = 0$ and

$$\text{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1Z_2] = \mathbb{E}[X_3^2] \mathbb{E}[Y_1] \mathbb{E}[Y_2] = 0,$$

using independence of X_3, Y_1, Y_2 . But Z_1, Z_2 are certainly not independent: since Y_1, Y_2 are never zero, $Z_1 = 0$ happens exactly when $X_3 = 0$, and likewise for Z_2 , so

$$\mathbb{P}(Z_1 = 0, Z_2 = 0) = \mathbb{P}(X_3 = 0) = \frac{1}{2} \neq \frac{1}{4} = \mathbb{P}(Z_1 = 0)\mathbb{P}(Z_2 = 0).$$

Since covariance scales with the variables (property (iii)), it is often more meaningful to normalise it, defining the **correlation**

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}},$$

when the variances are finite and nonzero. We will see later (from the Cauchy-Schwarz inequality) that correlation always lies in $[-1, 1]$.

§6.3.3 Conditional Expectation

We now bring conditioning into the world of random variables. If B is an event with $\mathbb{P}(B) > 0$, we define the **conditional expectation** of X given B to be

$$\mathbb{E}[X \mid B] = \frac{\mathbb{E}[X \mathbb{1}_B]}{\mathbb{P}(B)} = \sum_x x \mathbb{P}(X = x \mid B),$$

the average of X over only those outcomes lying in B . Just as knowing conditional probabilities in every scenario gives the total probability, conditional expectations recombine to give the full expectation.

Theorem 6.3.9 (Law of Total Expectation)

Let $\{B_1, B_2, \dots\}$ be disjoint events with $\bigcup_i B_i = \Omega$ and $\mathbb{P}(B_i) > 0$. Then for any random variable X for which $\mathbb{E}[X]$ exists,

$$\mathbb{E}[X] = \sum_i \mathbb{E}[X \mid B_i] \mathbb{P}(B_i).$$

Proof. Since $\mathbb{1}_{B_1} + \mathbb{1}_{B_2} + \dots = 1$, we can write $X = \sum_i X \mathbb{1}_{B_i}$, and taking expectations (using linearity, or countable additivity of expectation in the infinite case, applied first to non-negative parts) gives

$$\mathbb{E}[X] = \sum_i \mathbb{E}[X \mathbb{1}_{B_i}] = \sum_i \mathbb{E}[X \mid B_i] \mathbb{P}(B_i). \quad \square$$

Now for a genuinely important idea. Given two random variables X and Y , for each possible value y of Y we can compute the number $g(y) = \mathbb{E}[X \mid Y = y]$. This defines a function of y , and plugging the random variable Y back into it yields a *random variable*

$$\mathbb{E}[X \mid Y] = g(Y),$$

called the **conditional expectation of X given Y** . You should think of $\mathbb{E}[X \mid Y]$ as the best guess of X made by someone who knows the value of Y : it is random precisely because Y is.

Example 6.3.10

Toss a biased coin n times, let X_i be the indicator that toss i is a head, and let $Y_n = X_1 + \dots + X_n$ be the total number of heads. What is $\mathbb{E}[X_1 \mid Y_n]$?

We compute, for $y \geq 1$,

$$\begin{aligned} \mathbb{E}[X_1 \mid Y_n = y] &= \mathbb{P}(X_1 = 1 \mid Y_n = y) = \frac{\mathbb{P}(X_1 = 1, X_2 + \dots + X_n = y - 1)}{\mathbb{P}(Y_n = y)} \\ &= \frac{p \binom{n-1}{y-1} p^{y-1} (1-p)^{n-y}}{\binom{n}{y} p^y (1-p)^{n-y}} = \frac{\binom{n-1}{y-1}}{\binom{n}{y}} = \frac{y}{n}, \end{aligned}$$

and this also holds (trivially) for $y = 0$. So $\mathbb{E}[X_1 \mid Y_n] = Y_n/n$. Note the answer doesn't even depend on p : given the total number of heads, each toss is equally likely to have contributed, by symmetry.

The key properties of conditional expectation are as follows.

Proposition 6.3.11 (Properties of Conditional Expectation)

Let X, Y, Z be discrete random variables, and $c \in \mathbb{R}$. Then, whenever the relevant expectations exist:

- (i) (Linearity) $\mathbb{E}[cX + Z \mid Y] = c\mathbb{E}[X \mid Y] + \mathbb{E}[Z \mid Y]$.
- (ii) (Tower property) $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$.
- (iii) (Taking out what is known) $\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$ for any function h .

(iv) (Independence) If X and Y are independent, then $\mathbb{E}[X | Y] = \mathbb{E}[X]$.

Proof. Part (i) holds since $\mathbb{E}[\cdot | Y = y]$ is an expectation with respect to the probability measure $\mathbb{P}(\cdot | Y = y)$, and expectations are linear.

For (ii), we use the law of total expectation with the partition $\{Y = y\}$:

$$\mathbb{E}[\mathbb{E}[X | Y]] = \sum_y \mathbb{E}[X | Y = y] \mathbb{P}(Y = y) = \mathbb{E}[X].$$

For (iii), conditionally on $Y = y$ the random variable $h(Y)$ is just the constant $h(y)$, so

$$\mathbb{E}[h(Y)X | Y = y] = h(y)\mathbb{E}[X | Y = y],$$

and substituting Y for y gives the claim. Finally for (iv), if X, Y are independent then $\mathbb{P}(X = x | Y = y) = \mathbb{P}(X = x)$ for every y , so $\mathbb{E}[X | Y = y] = \mathbb{E}[X]$ for all y . $\square$

The tower property is used *constantly* – it lets us compute a difficult expectation by first conditioning on some helpful piece of information. We will lean on it heavily when we study branching processes.

§6.4 Inequalities and the Law of Large Numbers

Exact computations are all well and good, but much of the power of probability theory comes from *inequalities* – results that let us bound probabilities and expectations with only partial information about a distribution. We will build up the standard toolkit, and then use it to prove our first genuine limit theorem.

Everything in this section works for arbitrary random variables (discrete or otherwise); the proofs we give need nothing beyond basic properties of expectation.

§6.4.1 Markov and Chebyshev

Our first inequality says: a non-negative random variable with a small mean is unlikely to be large. Crude, but astonishingly useful.

Theorem 6.4.1 (Markov's Inequality)

Let $X \geq 0$ be a random variable. Then for any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Proof. The whole proof is the pointwise inequality $X \geq a\mathbb{1}_{\{X \geq a\}}$, which is easily checked in both the cases $X \geq a$ and $X < a$ (using $X \geq 0$ for the latter). Taking expectations,

$$\mathbb{E}[X] \geq a \mathbb{E}[\mathbb{1}_{\{X \geq a\}}] = a \mathbb{P}(X \geq a),$$

and dividing by a gives the result. $\square$

Applying Markov's inequality to the squared deviation of X from its mean gives a bound

in terms of the variance – a much sharper tool, since it captures fluctuations rather than just size.

Theorem 6.4.2 (Chebyshev's Inequality)

Let X be a random variable with finite mean. Then for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}.$$

Proof. The event $\{|X - \mathbb{E}[X]| \geq a\}$ is the same as the event $\{(X - \mathbb{E}[X])^2 \geq a^2\}$, and applying Markov's inequality to the non-negative random variable $(X - \mathbb{E}[X])^2$ gives

$$\mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} = \frac{\text{Var}(X)}{a^2}. \quad \square$$

In words: a random variable is unlikely to be many standard deviations away from its mean – the probability of being at least k standard deviations away is at most $1/k^2$, whatever the distribution. This universality is what makes Chebyshev so valuable.

§6.4.2 Cauchy-Schwarz

The next inequality should look familiar from any setting with an inner product; expectation of a product behaves exactly like one.

Theorem 6.4.3 (Cauchy-Schwarz Inequality)

For random variables X and Y ,

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2] \mathbb{E}[Y^2]}.$$

Proof. If either of $\mathbb{E}[X^2]$ or $\mathbb{E}[Y^2]$ is infinite the inequality is trivial, so assume both are finite (note $|XY| \leq \frac{1}{2}(X^2 + Y^2)$ ensures $\mathbb{E}[XY]$ exists). Further, if $\mathbb{E}[Y^2] = 0$ then $Y = 0$ with probability 1 and both sides vanish, so assume $\mathbb{E}[Y^2] > 0$.

Now for any $t \in \mathbb{R}$, we have $(X - tY)^2 \geq 0$, so taking expectations,

$$f(t) = \mathbb{E}[(X - tY)^2] = \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2] \geq 0.$$

This is a quadratic in t which never goes negative, so evaluating at its minimum $t^* = \mathbb{E}[XY]/\mathbb{E}[Y^2]$,

$$0 \leq f(t^*) = \mathbb{E}[X^2] - \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]},$$

which rearranges to $\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2]$, as required. $\square$

Remark. Chasing through the proof shows when equality holds: we need $\mathbb{E}[(X - t^*Y)^2] = 0$, which forces $X = t^*Y$ with probability 1. So equality in Cauchy-Schwarz means the variables are (almost surely) proportional.

Applying Cauchy-Schwarz to the centred variables $X - \mathbb{E}[X]$ and $Y - \mathbb{E}[Y]$ gives

$$|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}(X) \text{Var}(Y)},$$

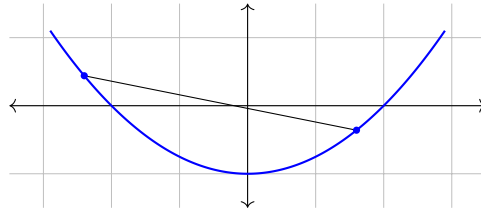
which is exactly the promised fact that correlations lie in $[-1, 1]$, with ± 1 attained precisely for (almost surely) linear relationships.

§6.4.3 Jensen's Inequality

Our final inequality concerns *convex* functions. Recall that $f : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if for all $x, y \in \mathbb{R}$ and $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y),$$

that is, chords of the graph of f lie above the graph. We say f is **concave** if $-f$ is convex.



The definition of convexity says that f of an average of *two* points is at most the average of the values. Jensen's inequality upgrades this from two-point averages to arbitrary expectations.

Theorem 6.4.4 (Jensen's Inequality)

Let X be a random variable with finite mean and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex. Then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]).$$

A helpful mnemonic for the direction: $f(x) = x^2$ is convex, and $\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$ is just the statement $\text{Var}(X) \geq 0$.

The proof rests on a fundamental geometric property of convex functions: at every point, there is a line through the graph lying entirely below it (a *supporting line*).

Lemma 6.4.5 (Supporting Lines)

Let f be convex and $m \in \mathbb{R}$. Then there exist $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \geq ax + b$ for all x .

Proof. Fix $x < m < y$. Writing m as a convex combination of x and y and rearranging the convexity inequality, one obtains

$$\frac{f(m) - f(x)}{m - x} \leq \frac{f(y) - f(m)}{y - m},$$

i.e. slopes of chords to the left of m are at most slopes of chords to the right. So if we let

$$a = \sup_{x < m} \frac{f(m) - f(x)}{m - x},$$

then a is finite, and for all $x < m < y$ we have

$$\frac{f(m) - f(x)}{m - x} \leq a \leq \frac{f(y) - f(m)}{y - m}.$$

Rearranging both inequalities gives $f(x) \geq a(x - m) + f(m)$ for every x (on both sides of m , and trivially at $x = m$). Taking $b = f(m) - am$ finishes the proof. $\square$

Proof of Jensen's Inequality. Let $m = \mathbb{E}[X]$, and take a supporting line $f(m) = am + b$ with $f(x) \geq ax + b$ for all x . Then the random variable $f(X)$ satisfies $f(X) \geq aX + b$ pointwise, so taking expectations,

$$\mathbb{E}[f(X)] \geq a\mathbb{E}[X] + b = am + b = f(\mathbb{E}[X]). \quad \square$$

Remark. If f is *strictly* convex (the supporting line touches only at m), then chasing the equality case shows $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$ forces $X = \mathbb{E}[X]$ with probability 1, i.e. X is essentially constant.

As a first application, we can obtain a classical inequality about means, seemingly for free.

Corollary 6.4.6 (AM-GM Inequality)

For any positive reals $x_1, \dots, x_n$,

$$\left(\prod_{k=1}^n x_k \right)^{1/n} \leq \frac{1}{n} \sum_{k=1}^n x_k.$$

Proof. Let X be a random variable taking each of the values $x_1, \dots, x_n$ with probability $1/n$, and let $f(x) = -\log x$, which is convex on the positive reals (its second derivative $1/x^2$ is positive, and the supporting line argument goes through verbatim on $(0, \infty)$). Jensen's inequality gives $\mathbb{E}[-\log X] \geq -\log \mathbb{E}[X]$, that is,

$$-\frac{1}{n} \sum_{k=1}^n \log x_k \geq -\log \left(\frac{1}{n} \sum_{k=1}^n x_k \right).$$

Negating and exponentiating (both order-reversing and order-preserving respectively) gives exactly AM-GM. $\square$

Remark. For *concave* f the inequality reverses: $\mathbb{E}[f(X)] \leq f(\mathbb{E}[X])$, simply by applying Jensen to $-f$. Thus for instance $\mathbb{E}[\log X] \leq \log \mathbb{E}[X]$ for positive X – which is really what powered the AM-GM proof above. Weighted versions of AM-GM come for free too: giving the value x_k probability w_k instead of $1/n$ proves $\prod_k x_k^{w_k} \leq \sum_k w_k x_k$ for any weights $w_k \geq 0$ summing to 1.

§6.4.4 The Weak Law of Large Numbers

We now have the tools to prove something that underpins the whole interpretation of probability: if you repeat an experiment independently many times, the empirical average converges to the expectation. This is the mathematical justification for the

frequentist intuition that ‘probability is long-run frequency’.

Theorem 6.4.7 (Weak Law of Large Numbers)

Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and finite variance σ^2 , and let $S_n = X_1 + \dots + X_n$. Then for every fixed $\epsilon > 0$,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof. By linearity, $\mathbb{E}[S_n/n] = \mu$, and by independence the variances add:

$$\text{Var}\left(\frac{S_n}{n}\right) = \frac{1}{n^2} \text{Var}(S_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

So by Chebyshev’s inequality,

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \epsilon\right) \leq \frac{\sigma^2}{n\epsilon^2} \longrightarrow 0. \quad \square$$

Look at how little was needed: Chebyshev plus the fact that variances of independent variables add. The theorem remains true assuming only that the mean exists (no variance needed), but the proof is then harder.

Example 6.4.8 (Frequencies Converge to Probabilities)

Fix an event A in some repeatable experiment, and let $X_i = \mathbb{1}_A$ for the i th independent repetition, so that S_n/n is the proportion of the first n trials in which A occurred. The X_i are independent Bernoulli(p) variables with $p = \mathbb{P}(A)$, so the weak law gives

$$\mathbb{P}\left(\left|\frac{S_n}{n} - \mathbb{P}(A)\right| > \epsilon\right) \leq \frac{p(1-p)}{n\epsilon^2} \longrightarrow 0.$$

So observed frequencies converge (in probability) to the abstract probability assigned by the measure – the theory validates the intuition that inspired its axioms. Note also the explicit bound: since $p(1-p) \leq \frac{1}{4}$, roughly $n \approx 1/(4\delta\epsilon^2)$ trials suffice to know $\mathbb{P}(A)$ to within ϵ with confidence $1 - \delta$, whatever the experiment.

Remark (The Strong Law). The type of convergence appearing in the weak law is called **convergence in probability**. There is a genuinely stronger statement, the *strong law of large numbers*: under the same hypotheses,

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1,$$

i.e. the sample averages converge to μ **almost surely**. We won’t prove this here (a proof assuming finite fourth moments is a nice exercise using Markov’s inequality on S_n^4), but it is worth knowing the distinction between the two modes of convergence: the strong law says that with probability one the entire sequence of averages settles down to μ , not merely that large deviations at each fixed time become unlikely.

§6.5 Random Walks

We now look at our first *random process* – a sequence of random variables $X_0, X_1, X_2, \dots$ evolving in time. The simplest and most fundamental example is the random walk.

Definition 6.5.1 (Simple Random Walk)

Let $Y_1, Y_2, \dots$ be independent identically distributed random variables with

$$\mathbb{P}(Y_i = 1) = p, \quad \mathbb{P}(Y_i = -1) = q = 1 - p.$$

The process

$$X_n = x + Y_1 + \dots + Y_n$$

is called a **simple random walk** started from $x \in \mathbb{Z}$. If $p = q = \frac{1}{2}$ the walk is called **symmetric**.

You should picture a particle on the integers, taking a step up with probability p or down with probability q at each tick of a clock – or a gambler making a sequence of one pound bets. A crucial feature of the walk is that it has no memory: where it goes next depends only on where it is now, not on how it got there.

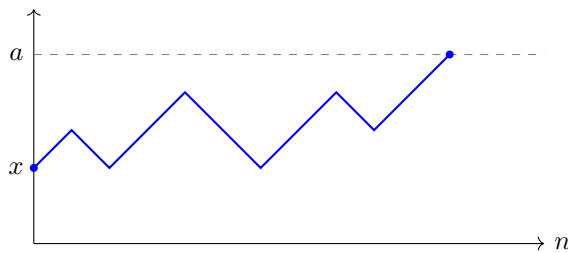
§6.5.1 Gambler's Ruin

Here is the classical question. A gambler starts with $\mathcal{L}x$ and repeatedly bets $\mathcal{L}1$, winning each bet with probability p ; they stop when they either hit their target of $\mathcal{L}a$ or go bankrupt. What is the probability they reach the target?

In terms of the walk, we start at x with $0 \leq x \leq a$, and we want the probability that the walk hits a before it hits 0. We write $\mathbb{P}_x$ for probabilities of the walk started at x , and define

$$h(x) = \mathbb{P}_x(X_n \text{ hits } a \text{ before hitting } 0).$$

A typical path of the walk, started from x and absorbed at the barrier a , looks like this:



The technique – worth internalising, since it applies to a huge family of such problems – is to *condition on the first step*. By the partition theorem, for $0 < x < a$,

$$\begin{aligned} h(x) &= \mathbb{P}_x(\text{hit } a \text{ first} \mid Y_1 = 1)p + \mathbb{P}_x(\text{hit } a \text{ first} \mid Y_1 = -1)q \\ &= ph(x+1) + qh(x-1), \end{aligned}$$

since after the first step the walk is just a fresh random walk started from $x \pm 1$. Together with the boundary conditions $h(0) = 0$ and $h(a) = 1$, this recurrence determines h completely.

Theorem 6.5.2 (Gambler's Ruin)

For the simple random walk with $0 \leq x \leq a$,

$$h(x) = \begin{cases} \frac{(q/p)^x - 1}{(q/p)^a - 1} & \text{if } p \neq q, \\ \frac{x}{a} & \text{if } p = q = \frac{1}{2}. \end{cases}$$

Proof. We solve the linear recurrence $ph(x+1) - h(x) + qh(x-1) = 0$. Trying $h(x) = \lambda^x$ gives the auxiliary equation $p\lambda^2 - \lambda + q = 0$, which factors as $(\lambda - 1)(p\lambda - q) = 0$, so $\lambda = 1$ or $\lambda = q/p$.

If $p \neq q$ these roots are distinct, so the general solution is $h(x) = A + B(q/p)^x$. The boundary conditions $h(0) = 0$, $h(a) = 1$ give $A + B = 0$ and $A + B(q/p)^a = 1$, so

$$h(x) = \frac{(q/p)^x - 1}{(q/p)^a - 1}.$$

If $p = q = \frac{1}{2}$, the root $\lambda = 1$ is repeated and the general solution is $h(x) = A + Bx$; the boundary conditions immediately give $h(x) = x/a$. $\square$

There is a sobering message here for gamblers. Suppose $p < \frac{1}{2}$ (as in every casino), so $q/p > 1$. Letting the target $a \rightarrow \infty$, the probability of ever reaching it is

$$h(x) = \frac{(q/p)^x - 1}{(q/p)^a - 1} \rightarrow 0,$$

and moreover the probability of *avoiding bankruptcy forever* tends to 0 too: the walk with downward drift hits 0 with probability

$$1 - \lim_{a \rightarrow \infty} h(x) = 1.$$

Even in a fair game ($p = 1/2$), the probability x/a of reaching a large target before ruin is small – the moral being that with a finite bankroll against an effectively infinite house, ruin is essentially certain.

§6.5.2 Expected Duration

We can play the same first-step game with expectations. Let T be the first time the walk hits 0 or a , and let $\tau_x = \mathbb{E}_x[T]$ denote its expected value for the walk started at x . Conditioning on the first step, using the law of total expectation, gives for $0 < x < a$

$$\tau_x = p(1 + \tau_{x+1}) + q(1 + \tau_{x-1}) = 1 + p\tau_{x+1} + q\tau_{x-1},$$

with boundary conditions $\tau_0 = \tau_a = 0$. (The ‘1+’ accounts for the step just taken.)

Proposition 6.5.3 (Expected Time to Absorption)

For the symmetric walk ($p = q = \frac{1}{2}$),

$$\tau_x = x(a - x),$$

and for $p \neq q$,

$$\tau_x = \frac{x}{q-p} - \frac{a}{q-p} \cdot \frac{(q/p)^x - 1}{(q/p)^a - 1}.$$

Proof. This is now an inhomogeneous linear recurrence, so we need a particular solution plus the general solution of the homogeneous equation found above.

If $p = q = \frac{1}{2}$: trying $\tau_x = Ax^2$ as a particular solution gives

$$Ax^2 = 1 + \frac{A}{2} ((x+1)^2 + (x-1)^2) = 1 + Ax^2 + A,$$

so $A = -1$. The general solution is thus $\tau_x = -x^2 + Bx + C$, and the boundary conditions give $C = 0$ and $B = a$, so $\tau_x = x(a - x)$.

If $p \neq q$: trying $\tau_x = Cx$ gives $Cx = 1 + pC(x+1) + qC(x-1) = 1 + Cx + C(p-q)$, so $C = 1/(q-p)$. The general solution is

$$\tau_x = \frac{x}{q-p} + A + B \left(\frac{q}{p}\right)^x,$$

and imposing $\tau_0 = \tau_a = 0$ gives $A = -B$ and $B((q/p)^a - 1) = -a/(q-p)$, yielding the stated formula. $\square$

It's worth pausing at the symmetric case: starting in the middle, $\tau_{a/2} = a^2/4$, so the duration of a fair game grows *quadratically* in the stakes – fair games are long games.

§6.6 Probability Generating Functions

We now introduce one of the most powerful bookkeeping devices in discrete probability. The idea is to encode the entire distribution of a random variable into a single function, whereupon many operations on random variables (particularly adding independent copies) become simple algebraic operations on functions.

Definition 6.6.1 (Probability Generating Function)

Let X be a random variable taking values in $\{0, 1, 2, \dots\}$, with $p_r = \mathbb{P}(X = r)$. The **probability generating function** (or **pgf**) of X is

$$p(z) = \mathbb{E}[z^X] = \sum_{r=0}^{\infty} p_r z^r, \quad \text{for } |z| \leq 1.$$

Since $\sum_r p_r = 1$, the series converges absolutely for $|z| \leq 1$, so the pgf is always well defined there, and $p(1) = 1$. The first thing to establish is that no information is lost in passing to the pgf.

Theorem 6.6.2 (The pgf Determines the Distribution)

The probability generating function of X uniquely determines the distribution of X .

Explicitly,

$$p_n = \frac{1}{n!} p^{(n)}(0).$$

Proof. Suppose (p_r) and (q_r) are pmfs whose generating functions agree for all $|z| \leq 1$. Setting $z = 0$ gives $p_0 = q_0$. Inductively, if $p_r = q_r$ for all $r \leq n$, then

$$\sum_{r=n+1}^{\infty} p_r z^r = \sum_{r=n+1}^{\infty} q_r z^r$$

for $0 < z < 1$; dividing by z^{n+1} and letting $z \rightarrow 0^+$ gives $p_{n+1} = q_{n+1}$ (the tails of both series tend to 0, being bounded by $z \cdot \sum_r p_r$ and so on). So the distributions agree. The explicit formula is just the standard formula for coefficients of a power series in terms of derivatives at 0. $\square$

§6.6.1 Extracting Moments

Differentiating the series $\sum_r p_r z^r$ term by term (valid inside the radius of convergence) gives $p'(z) = \sum_r r p_r z^{r-1}$, and evaluating at $z = 1$ *formally* gives $\sum_r r p_r = \mathbb{E}[X]$. Making this precise requires a small limiting argument, since the radius of convergence may be exactly 1.

Theorem 6.6.3 (Moments From the pgf)

Let X have pgf p . Then

$$\lim_{z \rightarrow 1^-} p'(z) = \mathbb{E}[X],$$

where both sides may be $+\infty$, and we write $p'(1^-)$ for this limit. More generally,

$$p^{(k)}(1^-) = \mathbb{E}[X(X-1)\cdots(X-k+1)].$$

Proof. For $0 < z < 1$ we have $p'(z) = \sum_r r p_r z^{r-1}$, which is an increasing function of z , bounded above by $\sum_r r p_r = \mathbb{E}[X]$. So the limit $p'(1^-)$ exists (possibly infinite) and is at most $\mathbb{E}[X]$.

For the reverse bound, fix $\epsilon > 0$ (supposing first that $\mathbb{E}[X] < \infty$) and pick N with $\sum_{r=0}^N r p_r \geq \mathbb{E}[X] - \epsilon$. Then for $0 < z < 1$,

$$p'(z) \geq \sum_{r=1}^N r p_r z^{r-1} \longrightarrow \sum_{r=1}^N r p_r \geq \mathbb{E}[X] - \epsilon \quad \text{as } z \rightarrow 1^-,$$

so $p'(1^-) \geq \mathbb{E}[X] - \epsilon$ for every ϵ , giving the result. If $\mathbb{E}[X] = \infty$, the same argument with $\sum_{r=0}^N r p_r \geq M$ for arbitrary M shows $p'(1^-) = \infty$. The claim for higher derivatives follows in exactly the same way. $\square$

In particular, we get a formula for the variance which is often the fastest route in computations:

$$\text{Var}(X) = p''(1^-) + p'(1^-) - p'(1^-)^2.$$

Example 6.6.4 (Geometric Moments, the Painless Way)

Let $X \sim \text{Geom}(p)$, with pgf $p(z) = pz/(1 - qz)$ where $q = 1 - p$ (computed below, but easy to check directly). Differentiating with the quotient rule,

$$p'(z) = \frac{p}{(1 - qz)^2}, \quad p''(z) = \frac{2pq}{(1 - qz)^3},$$

so $\mathbb{E}[X] = p'(1) = p/p^2 = 1/p$, and

$$\text{Var}(X) = \frac{2pq}{p^3} + \frac{1}{p} - \frac{1}{p^2} = \frac{2q + p - 1}{p^2} = \frac{q}{p^2} = \frac{1 - p}{p^2},$$

recovering our earlier answers with strictly less summation.

§6.6.2 Sums of Independent Random Variables

Here is where pgfs really earn their keep.

Theorem 6.6.5 (pgfs of Sums)

Let $X_1, \dots, X_n$ be independent random variables with pgfs $p_1, \dots, p_n$. Then the pgf of $X_1 + \dots + X_n$ is

$$\mathbb{E}[z^{X_1 + \dots + X_n}] = p_1(z) p_2(z) \cdots p_n(z).$$

Proof. Since the X_i are independent, so are the random variables $z^{X_1}, \dots, z^{X_n}$ (they are functions of the X_i), and so the expectation of their product factorises:

$$\mathbb{E}[z^{X_1} z^{X_2} \cdots z^{X_n}] = \mathbb{E}[z^{X_1}] \cdots \mathbb{E}[z^{X_n}]. \quad \square$$

So convolution of distributions – a messy sum – becomes multiplication of functions. Let's compute the pgfs of our standard distributions and put this to use.

Example 6.6.6 (Standard pgfs) (i) If $X \sim \text{Bernoulli}(p)$, then $\mathbb{E}[z^X] = (1 - p) + pz$.

(ii) If $X \sim \text{Bin}(n, p)$, then by the binomial theorem (or by (i) and independence!)

$$\mathbb{E}[z^X] = \sum_{r=0}^n \binom{n}{r} (pz)^r (1 - p)^{n-r} = (1 - p + pz)^n.$$

(iii) If $X \sim \text{Geom}(p)$ (support $\{1, 2, \dots\}$), then

$$\mathbb{E}[z^X] = \sum_{r=1}^{\infty} (1 - p)^{r-1} p z^r = \frac{pz}{1 - (1 - p)z}.$$

(iv) If $X \sim \text{Poisson}(\lambda)$, then

$$\mathbb{E}[z^X] = \sum_{r=0}^{\infty} e^{-\lambda} \frac{(\lambda z)^r}{r!} = e^{\lambda(z-1)}.$$

Example 6.6.7 (Sums Revisited)

If $X \sim \text{Bin}(n, p)$ and $Y \sim \text{Bin}(m, p)$ are independent, then $X + Y$ has pgf

$$(1 - p + pz)^n (1 - p + pz)^m = (1 - p + pz)^{n+m},$$

so $X + Y \sim \text{Bin}(n + m, p)$ by uniqueness – obvious probabilistically (concatenate the trials), but now completely mechanical. Similarly for independent Poissons, $e^{\lambda(z-1)} e^{\mu(z-1)} = e^{(\lambda+\mu)(z-1)}$ recovers $X + Y \sim \text{Poisson}(\lambda + \mu)$ with none of the earlier convolution work.

§6.6.3 Random Sums

Pgfs handle beautifully a situation that would otherwise be quite awkward: adding up a *random number* of random variables. Think of the total claim against an insurer in a year (a random number of claims, each of random size), or – as we'll see in the next section – the number of children in a generation of a branching process.

Theorem 6.6.8 (Random Sums – Compounding)

Let $X_1, X_2, \dots$ be independent identically distributed random variables with pgf $p(z)$, and let N be a random variable with values in $\{0, 1, 2, \dots\}$, independent of the X_i , with pgf $q(z)$. Then the pgf of the random sum

$$S_N = X_1 + \dots + X_N$$

is $r(z) = q(p(z))$.

Proof. We condition on N using the tower property:

$$r(z) = \mathbb{E}[z^{S_N}] = \mathbb{E}[\mathbb{E}[z^{S_N} | N]].$$

Given $N = n$, the sum S_N is just $X_1 + \dots + X_n$, which is independent of N , so

$$\mathbb{E}[z^{S_N} | N = n] = \mathbb{E}[z^{X_1 + \dots + X_n}] = p(z)^n.$$

Hence $\mathbb{E}[z^{S_N} | N] = p(z)^N$, and

$$r(z) = \mathbb{E}[p(z)^N] = q(p(z)).$$

□

Differentiating $r = q \circ p$ and evaluating at 1^- (using $p(1^-) = 1$) gives the pleasingly interpretable formulas

$$\mathbb{E}[S_N] = q'(1^-)p'(1^-) = \mathbb{E}[N] \mathbb{E}[X_1],$$

and, after a short computation with second derivatives,

$$\text{Var}(S_N) = \mathbb{E}[N] \text{Var}(X_1) + \text{Var}(N) \mathbb{E}[X_1]^2.$$

The first says the expected total is the expected count times the expected size – entirely reasonable – while the second shows both sources of randomness contribute to the variance.

§6.7 Branching Processes

Armed with generating functions, we can analyse a rather beautiful random process. Imagine an organism that produces a random number of offspring, each of which independently produces its own offspring in the same way, and so on – a model equally at home describing family surnames, nuclear chain reactions, or the early spread of an epidemic. The central question: does the population survive forever, or die out?

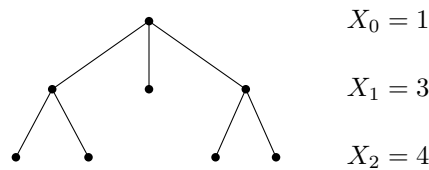
Definition 6.7.1 (Branching Process)

Let $(Y_{k,n})_{k,n \geq 1}$ be independent identically distributed random variables with values in $\{0, 1, 2, \dots\}$, where $Y_{k,n}$ represents the number of children of the k th individual in generation n . The **branching process** (X_n) is defined by $X_0 = 1$ and

$$X_{n+1} = \begin{cases} Y_{1,n} + \dots + Y_{X_n,n} & \text{if } X_n \geq 1, \\ 0 & \text{if } X_n = 0. \end{cases}$$

So X_n is the size of the n th generation. The common distribution of the $Y_{k,n}$ is called the **offspring distribution**.

Note that each generation is exactly a *random sum* of the type we just studied: X_{n+1} is a sum of X_n independent copies of the offspring distribution. This observation makes the theory almost write itself. A typical family tree looks something like the following.



§6.7.1 Expected Generation Size

The first natural question to ask about a branching process is how big we expect the n th generation to be.

Theorem 6.7.2 (Mean Growth)

Let $\mu = \mathbb{E}[X_1]$ be the mean of the offspring distribution. Then

$$\mathbb{E}[X_n] = \mu^n.$$

Proof. By our random sums formula (or directly by conditioning on X_n with the tower property),

$$\mathbb{E}[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1} \mid X_n]] = \mathbb{E}[X_n \mu] = \mu \mathbb{E}[X_n],$$

and since $\mathbb{E}[X_0] = 1$, induction gives $\mathbb{E}[X_n] = \mu^n$. $\square$

So in expectation the population grows or decays geometrically, with the critical threshold at $\mu = 1$. This suggests – correctly, as we will see – that the fate of the process hinges on whether each individual has more or less than one child on average.

§6.7.2 Generating Functions of Generations

Let $G(z) = \mathbb{E}[z^{X_1}]$ be the pgf of the offspring distribution, and let $G_n(z) = \mathbb{E}[z^{X_n}]$ be the pgf of the n th generation.

Theorem 6.7.3 (Iterating the pgf)

For all $n \geq 0$,

$$G_{n+1}(z) = G_n(G(z)) = \underbrace{G(G(\cdots G(z)\cdots))}_{n+1 \text{ times}}.$$

Proof. Since X_{n+1} is a random sum of X_n independent copies of the offspring distribution, the compounding theorem gives $G_{n+1}(z) = G_n(G(z))$ directly. Unwinding the recursion (with $G_1 = G$) shows G_n is the n -fold iterate of G , and in particular the composition can be taken in either order. $\square$

So the entire distributional structure of the branching process is encoded in iterating a single function – a lovely fact, even if iterates are rarely explicitly computable.

§6.7.3 Extinction

We say the process goes **extinct** if $X_n = 0$ for some n . Let

$$q = \mathbb{P}(\text{extinction}), \quad q_n = \mathbb{P}(X_n = 0).$$

Since $X_n = 0$ forces $X_{n+1} = 0$, the events $\{X_n = 0\}$ are increasing, and their union is the event of extinction. So by continuity of the probability measure (finally paying off!),

$$q_n \longrightarrow q \quad \text{as } n \rightarrow \infty.$$

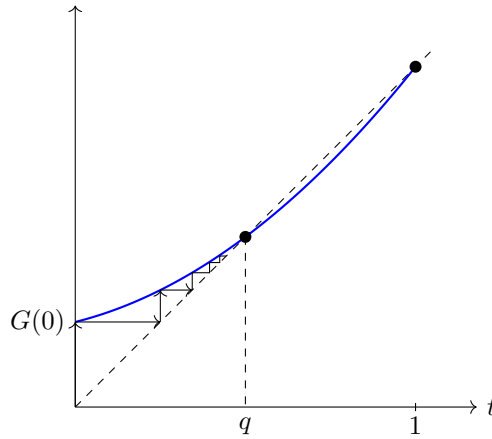
Moreover, $q_n = \mathbb{P}(X_n = 0) = G_n(0)$, so the extinction probabilities are obtained by iterating G starting from 0:

$$q_{n+1} = G(q_n),$$

and letting $n \rightarrow \infty$, continuity of G on $[0, 1]$ gives the key equation

$$q = G(q).$$

So the extinction probability is a fixed point of the offspring pgf. But G may have several fixed points in $[0, 1]$ (note $G(1) = 1$ always!), so which one is q ? The picture below, showing the iteration $q_{n+1} = G(q_n)$ climbing the ‘staircase’ between the graph of G and the diagonal, should be rather suggestive: starting from $q_0 = 0$, the iterates converge to the *first* crossing of the curve and the diagonal.

**Theorem 6.7.4** (Extinction Probability)

The extinction probability q is the *smallest* non-negative solution of $G(t) = t$. Moreover, if $\mathbb{P}(X_1 = 1) < 1$, then

$$q < 1 \iff \mu > 1,$$

where $\mu = \mathbb{E}[X_1]$.

Proof. First, minimality. Let $t \geq 0$ satisfy $G(t) = t$; we show $q_n \leq t$ for all n by induction, whence $q = \lim q_n \leq t$. Indeed $q_0 = 0 \leq t$, and if $q_n \leq t$ then since G has non-negative power series coefficients it is increasing on $[0, 1]$, so

$$q_{n+1} = G(q_n) \leq G(t) = t.$$

Since q itself solves $G(q) = q$, it must be the smallest such solution.

Now for the criterion. Write $g_r = \mathbb{P}(X_1 = r)$. If $g_0 + g_1 = 1$ (so no individual ever has two or more children) then $\mu \leq 1$; and unless $g_1 = 1$ (excluded), extinction is certain since the single line of descent dies at each step with probability $g_0 > 0$ – consistent with the claim. So suppose $g_0 + g_1 < 1$, which makes

$$G''(z) = \sum_{r \geq 2} r(r-1)g_r z^{r-2} > 0 \quad \text{for } z \in (0, 1),$$

i.e. G is strictly convex on $(0, 1)$. Consider $H(z) = G(z) - z$, which satisfies $H(1) = 0$ and $H(0) = g_0 \geq 0$, and is strictly convex. A strictly convex function can have at most two roots in $[0, 1]$, so besides $z = 1$ there is at most one other root of H .

Suppose $\mu = G'(1^-) > 1$. Then $H'(1^-) = G'(1^-) - 1 > 0$, so $H(z) < 0$ for z slightly less than 1. Since $H(0) \geq 0$ and H is continuous, H has a root $r \in [0, 1)$, and by minimality $q \leq r < 1$. So the process survives with positive probability.

Conversely suppose $\mu \leq 1$. Then for $z \in (0, 1)$, by strict convexity $G'(z) < G'(1^-) = \mu \leq 1$, so $H'(z) = G'(z) - 1 < 0$ on $(0, 1)$: H is strictly decreasing there. Since $H(1) = 0$, we get $H(z) > 0$ for all $z \in [0, 1)$, i.e. there is no fixed point below 1, and hence $q = 1$: extinction is certain. $\square$

This is a genuinely striking result. If $\mu \leq 1$ the population dies out with probability 1 – even in the critical case $\mu = 1$ where the *expected* population size stays constant forever!

And if $\mu > 1$, survival is possible but by no means guaranteed: the survival probability is $1 - q$ where q is the smaller fixed point.

Example 6.7.5

Suppose each individual has 0, 1 or 2 children with probability $\frac{1}{4}, \frac{1}{4}, \frac{1}{2}$ respectively. Then $\mu = \frac{5}{4} > 1$, so we expect $q < 1$. The pgf is $G(z) = \frac{1}{4} + \frac{1}{4}z + \frac{1}{2}z^2$, and solving $G(t) = t$:

$$\frac{1}{2}t^2 - \frac{3}{4}t + \frac{1}{4} = 0 \iff 2t^2 - 3t + 1 = 0 \iff (2t - 1)(t - 1) = 0,$$

so $t = \frac{1}{2}$ or $t = 1$. The extinction probability is the smaller root, $q = \frac{1}{2}$.

§6.8 Continuous Random Variables

So far all of our random variables have taken countably many values. But plenty of natural quantities – waiting times, positions, measurement errors – vary over a continuum, and for these a new framework is needed: no pmf can describe a random variable for which every individual value has probability zero.

§6.8.1 Distribution Functions

The right object to focus on turns out to be the following.

Definition 6.8.1 (Distribution Function)

Let X be a random variable, i.e. a function $X : \Omega \rightarrow \mathbb{R}$ with $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for all x^a . The **distribution function** of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

^aNote this condition is a slight refinement of the one we used for discrete random variables – for those, the two conditions agree.

The distribution function makes sense for *any* random variable, discrete or not, and it completely determines the distribution. Its basic properties are as follows.

Proposition 6.8.2 (Properties of Distribution Functions)

Let $F = F_X$ be a distribution function. Then

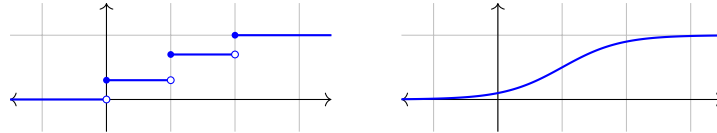
- (i) F is non-decreasing,
- (ii) $\mathbb{P}(a < X \leq b) = F(b) - F(a)$ for $a < b$,
- (iii) F is right continuous, and has left limits with $F(x^-) = \mathbb{P}(X < x)$,
- (iv) $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow \infty$.

Proof. (i) follows since $\{X \leq x\} \subseteq \{X \leq y\}$ for $x \leq y$, and (ii) since $\{X \leq b\}$ is the disjoint union of $\{X \leq a\}$ and $\{a < X \leq b\}$.

For (iii), the events $A_n = \{x < X \leq x + \frac{1}{n}\}$ are decreasing with empty intersection,

so their complements are increasing with union $\Omega \setminus \bigcap_n A_n$, and continuity of $\mathbb{P}$ gives $\mathbb{P}(A_n) \rightarrow 0$; but $\mathbb{P}(A_n) = F(x + \frac{1}{n}) - F(x)$, which together with monotonicity gives right continuity. Similarly, $B_n = \{X \leq x - \frac{1}{n}\}$ is an increasing sequence of events with union $\{X < x\}$, so $F(x - \frac{1}{n}) \rightarrow \mathbb{P}(X < x)$, and monotonicity upgrades this to the full left limit. Part (iv) is another application of continuity of $\mathbb{P}$, with the events $\{X \leq -n\}$ (decreasing to $\emptyset$) and $\{X \leq n\}$ (increasing to Ω). $\square$

For a discrete random variable, F is a step function, jumping by $\mathbb{P}(X = x)$ at each x in the image (shown below on the left). The random variables of interest in this section sit at the other extreme, with no jumps at all (below right).



Definition 6.8.3 (Continuous Random Variable)

A random variable X is **continuous** if its distribution function F is continuous. In this course we work with the (slightly stronger, but much more usable) assumption that F is differentiable^a, and we call

$$f(x) = F'(x)$$

the **probability density function** (or **density**) of X .

^aStrictly, such random variables are called *absolutely continuous*; we will not fuss over the distinction, and ‘continuous random variable’ will always mean this.

Note that for a continuous random variable, $\mathbb{P}(X = x) = F(x) - F(x^-) = 0$ for every single x – and yet X certainly takes *some* value. Individual values are all negligible; only ranges of values carry probability. By the fundamental theorem of calculus,

$$\mathbb{P}(a < X \leq b) = F(b) - F(a) = \int_a^b f(x) \, dx,$$

and more generally $\mathbb{P}(X \in S) = \int_S f(x) \, dx$ for reasonable sets S . So the density is exactly what its name suggests: probability per unit length. For a small interval,

$$\mathbb{P}(x < X \leq x + \delta) \approx f(x) \delta,$$

which is the right intuition to have – $f(x)$ is the continuous analogue of $\mathbb{P}(X = x)$, but it is not itself a probability (indeed, densities can exceed 1!). Any density satisfies $f \geq 0$ and $\int_{-\infty}^{\infty} f(x) \, dx = 1$, and conversely any such function defines a distribution.

§6.8.2 Expectation and Variance

The definition of expectation transfers from the discrete world by replacing sums with integrals (and pmfs with densities).

Definition 6.8.4 (Expectation)

Let X be a continuous random variable with density f . If $X \geq 0$, we define

$$\mathbb{E}[X] = \int_0^\infty x f(x) \, dx,$$

which is either finite or $+\infty$. In general, writing $X = X^+ - X^-$ with $X^+ = \max(X, 0)$ and $X^- = \max(-X, 0)$, we set $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-] = \int_{-\infty}^\infty x f(x) \, dx$, provided at least one of the two parts is finite.

The expectation retains all of its old properties: it is linear, monotone, and satisfies the law of the subconscious statistician

$$\mathbb{E}[g(X)] = \int_{-\infty}^\infty g(x) f(x) \, dx,$$

along with the same definitions and identities for variance and covariance as before, e.g. $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$. The tail-sum formula also survives, now in integral form.

Proposition 6.8.5 (Tail Formula)

If $X \geq 0$ is a continuous random variable, then

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq x) \, dx.$$

Proof. We write $x = \int_0^x dy$ inside the definition and swap the order of integration^a:

$$\mathbb{E}[X] = \int_0^\infty \int_0^x dy f(x) \, dx = \int_0^\infty \int_y^\infty f(x) \, dx \, dy = \int_0^\infty \mathbb{P}(X \geq y) \, dy. \quad \square$$

^aJustified since the integrand is non-negative (this is Fubini's theorem, or can be checked directly here).

Likewise, our inequalities (Markov, Chebyshev, Cauchy-Schwarz, Jensen) were proved using only properties of expectation, so they hold verbatim for continuous random variables.

§6.8.3 The Uniform and Exponential Distributions

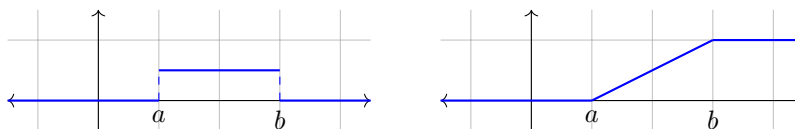
Let's meet the continuous analogues of the 'standard' discrete distributions.

The Uniform Distribution

The **uniform distribution** on $[a, b]$, written $X \sim U[a, b]$, models a point chosen 'completely at random' from the interval:

$$f(x) = \begin{cases} \frac{1}{b-a} & x \in [a, b], \\ 0 & \text{otherwise,} \end{cases} \quad F(x) = \frac{x-a}{b-a} \text{ for } x \in [a, b].$$

The density (left) and distribution function (right) look as follows.



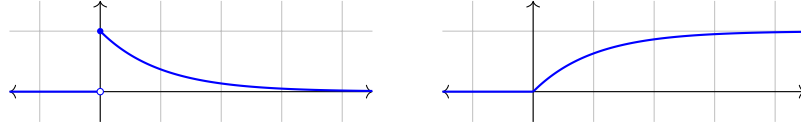
A quick integration gives $\mathbb{E}[X] = \frac{a+b}{2}$ (no surprises), and computing $\mathbb{E}[X^2] = \frac{a^2+ab+b^2}{3}$ leads to $\text{Var}(X) = \frac{(b-a)^2}{12}$.

The Exponential Distribution

The **exponential distribution** with rate $\lambda > 0$, written $X \sim \text{Exp}(\lambda)$, has

$$f(x) = \lambda e^{-\lambda x} \quad (x > 0), \quad F(x) = 1 - e^{-\lambda x}.$$

It is the continuous model of a waiting time for an event occurring ‘at rate λ ’. The density decays exponentially (left), and the distribution function climbs towards 1 (right); here $\lambda = 1$.



Integrating by parts (or by the tail formula, $\int_0^\infty e^{-\lambda x} dx$) gives

$$\mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

The exponential distribution should be thought of as a continuous limit of the geometric distribution: if $T \sim \text{Exp}(\lambda)$ then $\lfloor nT \rfloor$ satisfies

$$\mathbb{P}(\lfloor nT \rfloor \geq k) = \mathbb{P}(T \geq k/n) = (e^{-\lambda/n})^k,$$

which is a geometric tail with success probability $1 - e^{-\lambda/n} \approx \lambda/n$ – checking a coin with tiny success chance many times a second.

The property that really characterises the exponential distribution is that it is *memoryless*: knowing you have already waited s minutes tells you nothing about how much longer you’ll wait.

Theorem 6.8.6 (Memorylessness)

Let T be a positive continuous random variable, not identically 0 or ∞ . Then

$$\mathbb{P}(T > t + s \mid T > s) = \mathbb{P}(T > t) \text{ for all } s, t \geq 0$$

if and only if $T \sim \text{Exp}(\lambda)$ for some $\lambda > 0$.

Proof. If $T \sim \text{Exp}(\lambda)$, then

$$\mathbb{P}(T > t + s \mid T > s) = \frac{\mathbb{P}(T > t + s)}{\mathbb{P}(T > s)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} = \mathbb{P}(T > t).$$

Conversely, suppose T is memoryless, and let $g(t) = \mathbb{P}(T > t)$, so that g satisfies the functional equation $g(t + s) = g(t)g(s)$. Then for positive integers m, n ,

$$g(m) = g(1)^m, \quad g(m/n)^n = g(m) \implies g(m/n) = g(1)^{m/n},$$

so $g(q) = g(1)^q$ for all rationals $q > 0$. Now $g(1) \in (0, 1)$: if $g(1) = 0$ then $g(q) = 0$ for all rational $q > 0$ and $T = 0$ almost surely; if $g(1) = 1$ then similarly $T = \infty$ almost surely – both excluded. So we may define $\lambda = -\log g(1) > 0$, giving

$g(q) = e^{-\lambda q}$ on the rationals. Finally, g is decreasing, so for real t , squeezing t between rationals $r \leq t \leq s$ with $s - r$ small gives $e^{-\lambda s} \leq g(t) \leq e^{-\lambda r}$, and hence $g(t) = e^{-\lambda t}$ everywhere. $\square$

§6.8.4 The Normal Distribution

We now come to the most important distribution in all of probability – the reason for its ubiquity being the central limit theorem, which we will meet in due course.

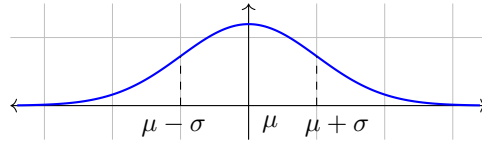
Definition 6.8.7 (Normal Distribution)

The **normal** (or **Gaussian**) **distribution** with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, written $N(\mu, \sigma^2)$, has density

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

The case $N(0, 1)$ is called the **standard normal distribution**, and its distribution function is denoted Φ , with density $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$.

The density is the famous ‘bell curve’, symmetric about μ , with the standard deviation σ setting the width of the bell ($\mu \pm \sigma$ are precisely its points of inflection):



We should first check that this is actually a density – the source of the mysterious $\sqrt{2\pi}$.

Proposition 6.8.8

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = 1.$$

Proof. Substituting $u = (x - \mu)/\sigma$ reduces to showing $I = \int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{2\pi}$. We use the famous squaring trick: writing I^2 as a double integral and changing to polar coordinates $(u, v) = (r \cos \theta, r \sin \theta)$,

$$I^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(u^2+v^2)/2} du dv = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r dr d\theta = 2\pi \left[-e^{-r^2/2}\right]_0^{\infty} = 2\pi,$$

and since $I > 0$ we get $I = \sqrt{2\pi}$. $\square$

The parameters deserve their names: if $X \sim N(\mu, \sigma^2)$ then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x-\mu}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2} dx + \mu = 0 + \mu = \mu,$$

the first integral vanishing since the integrand is odd about μ ; and substituting $u =$

$(x - \mu)/\sigma$ and integrating by parts,

$$\text{Var}(X) = \sigma^2 \int_{-\infty}^{\infty} \frac{u^2}{\sqrt{2\pi}} e^{-u^2/2} du = \sigma^2.$$

Also note the symmetry $\phi(x) = \phi(-x)$, which gives the frequently-used relation $\Phi(-x) = 1 - \Phi(x)$.

§6.8.5 Transformations of Random Variables

Given a random variable X with known density and a function g , what is the density of $g(X)$? For monotone g there is a clean answer, and the proof is simply ‘differentiate the distribution function’.

Theorem 6.8.9 (Densities Under Monotone Maps)

Let X be a continuous random variable with density f , and let g be strictly monotone with differentiable inverse. Then $Y = g(X)$ is a continuous random variable with density

$$f_Y(y) = f(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Proof. Suppose first that g is strictly increasing. Then $g(X) \leq y \iff X \leq g^{-1}(y)$, so

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = F(g^{-1}(y)),$$

and differentiating with the chain rule gives $f_Y(y) = f(g^{-1}(y)) \cdot (g^{-1})'(y)$, where the derivative is positive. If instead g is strictly decreasing, then $g(X) \leq y \iff X \geq g^{-1}(y)$, so $F_Y(y) = 1 - F(g^{-1}(y))$, and differentiating gives $f_Y(y) = f(g^{-1}(y)) \cdot (-(g^{-1})'(y))$, where now $-(g^{-1})' = |(g^{-1})'|$. Both cases match the claimed formula. $\square$

Rather than memorising this, it is usually safer to redo the two-line calculation with distribution functions – but the formula gives an important structural fact about the normal family.

Corollary 6.8.10 (Linear Maps of Normals)

If $X \sim N(\mu, \sigma^2)$ and $a \neq 0, b \in \mathbb{R}$, then

$$aX + b \sim N(a\mu + b, a^2\sigma^2).$$

In particular, $(X - \mu)/\sigma \sim N(0, 1)$.

Proof. Applying the theorem with $g(x) = ax + b$, so $g^{-1}(y) = (y - b)/a$,

$$f_Y(y) = \frac{1}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{(\frac{y-b}{a} - \mu)^2}{2\sigma^2}\right) \cdot \frac{1}{|a|} = \frac{1}{\sqrt{2\pi}a^2\sigma^2} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right),$$

which is the $N(a\mu + b, a^2\sigma^2)$ density. $\square$

The process of passing from X to $(X - \mu)/\sigma$ is called *standardisation*, and is how one

uses tables of Φ in practice: every question about a normal random variable reduces to one about the standard normal.

§6.9 Joint Densities

Just as in the discrete setting, the interesting questions about continuous random variables involve several of them at once, and for this we need joint densities.

§6.9.1 Joint Densities and Independence

Just as in the discrete setting, understanding several continuous random variables at once means studying their joint behaviour.

Definition 6.9.1 (Joint Density)

Random variables $X_1, \dots, X_n$ have **joint density** $f : \mathbb{R}^n \rightarrow [0, \infty)$ if

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(y_1, \dots, y_n) \, dy_1 \cdots dy_n$$

for all $x_1, \dots, x_n \in \mathbb{R}$.

As before, this extends to $\mathbb{P}((X_1, \dots, X_n) \in B) = \int_B f$ for reasonable sets $B \subseteq \mathbb{R}^n$, and the density can be recovered by (partial) differentiation of the joint distribution function. The density of a single coordinate, called its **marginal density**, is obtained by integrating out the others:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, y_2, \dots, y_n) \, dy_2 \cdots dy_n.$$

For independence, we can no longer compare with products of point probabilities (they're all zero!), so we phrase everything in terms of distribution functions: $X_1, \dots, X_n$ are **independent** if

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \mathbb{P}(X_1 \leq x_1) \cdots \mathbb{P}(X_n \leq x_n) \quad \text{for all } x_i.$$

The key practical criterion is that independence corresponds exactly to the joint density factorising.

Theorem 6.9.2 (Factorisation Criterion)

Suppose $(X_1, \dots, X_n)$ has joint density f .

- (i) If $X_1, \dots, X_n$ are independent with marginal densities $f_1, \dots, f_n$, then

$$f(x_1, \dots, x_n) = f_1(x_1) \cdots f_n(x_n).$$

- (ii) Conversely, if $f(x_1, \dots, x_n) = g_1(x_1) \cdots g_n(x_n)$ for some non-negative functions g_i , then the X_i are independent, with densities proportional to the g_i .

Proof. For (i), independence lets us factor the joint distribution function into $\prod_i \int_{-\infty}^{x_i} f_i$, which is precisely the statement that $\prod_i f_i$ serves as a joint density.

For (ii), let $c_i = \int_{-\infty}^{\infty} g_i(y) dy$. Since f integrates to 1 over $\mathbb{R}^n$, we get $c_1 c_2 \cdots c_n = 1$. Now integrating out all variables except the i th shows the marginal density of X_i is $g_i \cdot \prod_{j \neq i} c_j = g_i/c_i$. Then

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_i \int_{-\infty}^{x_i} g_i(y_i) dy_i = \prod_i \int_{-\infty}^{x_i} \frac{g_i(y_i)}{c_i} dy_i = \prod_i \mathbb{P}(X_i \leq x_i),$$

using $\prod_i c_i = 1$, so the X_i are independent with the claimed densities. $\square$

§6.9.2 Sums of Continuous Random Variables

The distribution of a sum of independent continuous random variables is again given by a convolution, with the sum over values replaced by an integral.

Proposition 6.9.3 (Convolution)

Let X and Y be independent with densities f_X and f_Y . Then $X + Y$ has density

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

Proof. We compute the distribution function of the sum by integrating the joint density $f_X(x)f_Y(y)$ over the half plane $\{x + y \leq z\}$:

$$\mathbb{P}(X+Y \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_X(x) f_Y(y) dy dx = \int_{-\infty}^z \left(\int_{-\infty}^{\infty} f_X(x) f_Y(u - x) dx \right) du,$$

substituting $u = y + x$ in the inner integral and then swapping the order of integration. The result is the integral up to z of the claimed density, which proves the claim. $\square$

§6.9.3 Conditional Densities

Conditioning on the value of a continuous random variable looks dangerous – the conditioning event $\{Y = y\}$ has probability zero – but at the level of densities everything goes through smoothly.

Definition 6.9.4 (Conditional Density)

Let X, Y have joint density $f_{X,Y}$, and let y be such that $f_Y(y) > 0$. The **conditional density of X given $Y = y$** is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

This is the exact analogue of $\mathbb{P}(X = x | Y = y)$, and one can justify it by conditioning on the event $\{y < Y \leq y + \delta\}$ and letting $\delta \rightarrow 0$. Integrating out the conditioning value

gives the continuous law of total probability,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x | y) f_Y(y) \, dy,$$

and we can define conditional expectation just as before: $\mathbb{E}[X | Y] = g(Y)$ where

$$g(y) = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) \, dx.$$

All the properties from the discrete case (linearity, the tower property, taking out what is known) continue to hold.

§6.9.4 Transformations in Higher Dimensions

The change of variables formula for densities generalises to $\mathbb{R}^d$, with the derivative replaced by a Jacobian determinant – exactly mirroring the change of variables formula for multiple integrals.

Theorem 6.9.5 (Multidimensional Change of Variables)

Let X be a random variable with values in $D \subseteq \mathbb{R}^d$ and density f_X , and let $g : D \rightarrow g(D)$ be a bijection with continuous derivative and $\det g'(x) \neq 0$ on D . Then $Y = g(X)$ has density

$$f_Y(y) = f_X(g^{-1}(y)) |J|, \quad \text{where } J = \det \left(\frac{\partial x_i}{\partial y_j} \right)_{i,j}$$

is the Jacobian of the inverse map $x = g^{-1}(y)$.

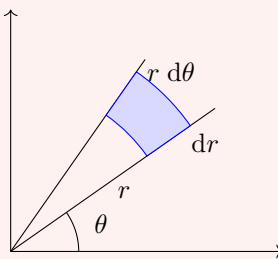
We won't prove this (it is essentially the substitution rule for multiple integrals, which we take on trust from Vector Calculus), but let's see it in action on a genuinely useful example.

Example 6.9.6 (Polar Coordinates of a Gaussian Pair)

Let X, Y be independent standard normals, and let (R, Θ) be the polar coordinates of the point (X, Y) , so $X = R \cos \Theta$ and $Y = R \sin \Theta$. The Jacobian of the map $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is

$$J = \det \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix} = r,$$

which is geometrically just the statement that a small 'polar rectangle' with sides dr and $r \, d\theta$ has area $r \, dr \, d\theta$:



So for $r > 0$ and $\theta \in [0, 2\pi)$,

$$f_{R,\Theta}(r, \theta) = f_{X,Y}(r \cos \theta, r \sin \theta) \cdot r = \frac{1}{2\pi} r e^{-r^2/2} = \underbrace{\frac{1}{2\pi}}_{f_{\Theta}(\theta)} \cdot \underbrace{r e^{-r^2/2}}_{f_R(r)}.$$

The joint density factorises, so by the factorisation criterion R and Θ are *independent*, with $\Theta \sim U[0, 2\pi)$ and R having density $r e^{-r^2/2}$. (Run backwards, this observation is a practical recipe for generating normal random variables – the ‘Box-Muller’ method – which we will return to at the end of these notes.)

§6.9.5 Order Statistics

Given independent identically distributed random variables $X_1, \dots, X_n$, we often care about their sorted values – the smallest, the largest, the median. Sorting them into increasing order gives the **order statistics**

$$Y_1 = X_{(1)} \leq Y_2 = X_{(2)} \leq \dots \leq Y_n = X_{(n)}.$$

For instance, with $n = 5$ the picture might be as follows – the samples X_i land in some scattered order on the line, and the order statistics simply relabel them from left to right.



Suppose the X_i have distribution function F and density f . The extremes are easy to handle directly:

$$\begin{aligned} \mathbb{P}(Y_n \leq x) &= \mathbb{P}(X_1 \leq x, \dots, X_n \leq x) = F(x)^n, \\ \mathbb{P}(Y_1 \leq x) &= 1 - \mathbb{P}(X_1 > x, \dots, X_n > x) = 1 - (1 - F(x))^n, \end{aligned}$$

using independence, and differentiating gives their densities

$$f_{Y_n}(x) = nF(x)^{n-1}f(x), \quad f_{Y_1}(x) = n(1 - F(x))^{n-1}f(x).$$

For the full joint behaviour, we have the following.

Proposition 6.9.7 (Joint Density of Order Statistics)

The order statistics $(Y_1, \dots, Y_n)$ have joint density

$$f_{Y_1, \dots, Y_n}(x_1, \dots, x_n) = \begin{cases} n! f(x_1) \cdots f(x_n) & \text{if } x_1 < x_2 < \dots < x_n, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. For $x_1 < \dots < x_n$, the event $\{Y_1 \leq x_1, \dots, Y_n \leq x_n\}$ splits according to which of the X_i lands in which position of the ordering; since the X_i are independent and identically distributed (and ties have probability zero), each of the $n!$ orderings contributes equally, and so

$$\mathbb{P}(Y_1 \leq x_1, \dots, Y_n \leq x_n) = n! \mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n, X_1 < X_2 < \dots < X_n).$$

Writing the right hand side as an integral of $f(u_1) \cdots f(u_n)$ over the region $\{u_1 \leq$

$x_1, \dots, u_n \leq x_n, u_1 < \dots < u_n\}$ and differentiating in each variable gives the claimed density. $\square$

Note the factor $n!$ and the ordering constraint: the joint density does *not* factorise, so the order statistics are certainly not independent (knowing the minimum is large forces the maximum to be larger still). For the exponential distribution, though, something special happens with the *gaps*.

Example 6.9.8 (Exponential Spacings)

Let $X_1, \dots, X_n$ be independent $\text{Exp}(\lambda)$, with order statistics $Y_1 \leq \dots \leq Y_n$, and consider the spacings $Z_1 = Y_1$ and $Z_i = Y_i - Y_{i-1}$ for $i \geq 2$. The map from $(y_1, \dots, y_n)$ to $(z_1, \dots, z_n)$ is linear with determinant 1, and $y_i = z_1 + \dots + z_i$, so

$$\begin{aligned} f_{Z_1, \dots, Z_n}(z_1, \dots, z_n) &= n! \lambda^n e^{-\lambda(y_1 + \dots + y_n)} = n! \lambda^n e^{-\lambda(nz_1 + (n-1)z_2 + \dots + z_n)} \\ &= \prod_{i=1}^n (n-i+1) \lambda e^{-(n-i+1)\lambda z_i}. \end{aligned}$$

The joint density factorises! So the spacings are independent, with $Z_i \sim \text{Exp}((n-i+1)\lambda)$. This is really memorylessness at work: when $i-1$ of the clocks have rung, the remaining $n-i+1$ have (conditionally) fresh exponential lifetimes, and the minimum of k independent $\text{Exp}(\lambda)$ variables is $\text{Exp}(k\lambda)$ – a fact worth checking directly:

$$\mathbb{P}(\min(X_1, \dots, X_k) > z) = \prod_{i=1}^k \mathbb{P}(X_i > z) = e^{-k\lambda z}.$$

§6.10 Moment Generating Functions

Probability generating functions served us wonderfully for random variables with values in $\{0, 1, 2, \dots\}$, but they make no sense for continuous random variables. The right generalisation is to replace z^X with $e^{\theta X}$.

Definition 6.10.1 (Moment Generating Function)

The **moment generating function** (or **mgf**) of a random variable X is

$$m(\theta) = \mathbb{E} \left[e^{\theta X} \right],$$

defined for those $\theta \in \mathbb{R}$ where the expectation is finite. For a continuous random variable with density f , this is $\int_{-\infty}^{\infty} e^{\theta x} f(x) dx$.

Note $m(0) = 1$ always, but the mgf may well be infinite for every other θ , in which case it's useless. The good situation is when m is finite on an open interval around 0; then the mgf shares the two crucial properties of the pgf. The first we must take on trust (its proof needs tools beyond this course).

Theorem 6.10.2 (Uniqueness)

If the moment generating function of X is finite on some open interval containing

0, then it uniquely determines the distribution of X .

Theorem 6.10.3 (Moments From the mgf)

If m is finite on an open interval around 0, then for each $r \geq 0$,

$$m^{(r)}(0) = \mathbb{E}[X^r].$$

The idea behind the second result is simply to expand $e^{\theta X} = \sum_k \theta^k X^k / k!$ and differentiate term by term (the finiteness assumption is what justifies the exchanges of limits) – so the mgf is precisely the exponential generating function of the moments, whence the name. And as with pgfs, independence turns sums into products:

Theorem 6.10.4 (mgfs of Sums)

If $X_1, \dots, X_n$ are independent, then

$$\mathbb{E} \left[e^{\theta(X_1 + \dots + X_n)} \right] = \prod_{i=1}^n \mathbb{E} \left[e^{\theta X_i} \right].$$

Proof. The random variables $e^{\theta X_i}$ are independent, so the expectation of their product factorises. $\square$

§6.10.1 Examples: Gamma, Normal, and a Warning

Let's put moment generating functions to work on some concrete distributions.

Example 6.10.5 (The Gamma Distribution)

The **gamma distribution** $\Gamma(n, \lambda)$, for $n \in \mathbb{N}$ and $\lambda > 0$, has density

$$f(x) = e^{-\lambda x} \frac{\lambda^n x^{n-1}}{(n-1)!}, \quad x > 0.$$

(One checks this integrates to 1 by repeated integration by parts, reducing to the exponential case $n = 1$: indeed $\Gamma(1, \lambda) = \text{Exp}(\lambda)$.) For $\theta < \lambda$, the mgf is

$$\begin{aligned} m(\theta) &= \int_0^\infty e^{\theta x} e^{-\lambda x} \frac{\lambda^n x^{n-1}}{(n-1)!} dx = \left(\frac{\lambda}{\lambda - \theta} \right)^n \int_0^\infty e^{-(\lambda - \theta)x} \frac{(\lambda - \theta)^n x^{n-1}}{(n-1)!} dx \\ &= \left(\frac{\lambda}{\lambda - \theta} \right)^n, \end{aligned}$$

since the final integrand is the $\Gamma(n, \lambda - \theta)$ density. (For $\theta \geq \lambda$ the integral diverges, but finiteness on $(-\infty, \lambda)$ is plenty.) Multiplying mgfs, we conclude that if $X \sim \Gamma(n, \lambda)$ and $Y \sim \Gamma(m, \lambda)$ are independent then $X + Y \sim \Gamma(n + m, \lambda)$; in particular, a sum of n independent $\text{Exp}(\lambda)$ waiting times has the $\Gamma(n, \lambda)$ distribution.

Example 6.10.6 (The Normal Distribution)

Let $X \sim N(\mu, \sigma^2)$. Completing the square in the exponent,

$$\theta x - \frac{(x - \mu)^2}{2\sigma^2} = \theta\mu + \frac{\theta^2\sigma^2}{2} - \frac{(x - (\mu + \theta\sigma^2))^2}{2\sigma^2},$$

so

$$m(\theta) = \exp\left(\theta\mu + \frac{\theta^2\sigma^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - (\mu + \theta\sigma^2))^2}{2\sigma^2}\right) dx = e^{\theta\mu + \theta^2\sigma^2/2},$$

the integral being that of the $N(\mu + \theta\sigma^2, \sigma^2)$ density. This mgf is finite for *all* θ – the normal distribution has very thin tails. As an immediate payoff, if $X \sim N(\mu, \sigma^2)$ and $Y \sim N(\nu, \tau^2)$ are independent then

$$\mathbb{E}[e^{\theta(X+Y)}] = e^{\theta\mu + \theta^2\sigma^2/2} e^{\theta\nu + \theta^2\tau^2/2} = e^{\theta(\mu+\nu) + \theta^2(\sigma^2+\tau^2)/2},$$

so $X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2)$: sums of independent normals are normal.

Example 6.10.7 (The Cauchy Distribution – A Warning)

The **Cauchy distribution** has density

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}.$$

Its tails decay so slowly that $\mathbb{E}[e^{\theta X}] = \infty$ for every $\theta \neq 0$ – indeed even $\mathbb{E}[X]$ fails to exist. So mgfs tell us nothing here: X and $2X$ have identical mgfs but different distributions. The moral is that all the mgf machinery genuinely requires finiteness on an open interval.

§6.11 The Central Limit Theorem

We have arrived at the crown jewel of the course. The weak law of large numbers says the sample mean S_n/n concentrates around μ ; the central limit theorem describes, with remarkable precision, the *shape* of the fluctuations around μ – and the answer is universal: whatever distribution you start with, the fluctuations look normal.

First, we need to say what it means for distributions to converge.

Definition 6.11.1 (Convergence in Distribution)

Random variables X_n **converge in distribution** to X , written $X_n \rightarrow X$ in distribution, if

$$F_{X_n}(x) \longrightarrow F_X(x) \quad \text{as } n \rightarrow \infty$$

for every $x \in \mathbb{R}$ at which F_X is continuous.

(The restriction to continuity points is a technicality that avoids silly counterexamples with point masses; for a continuous limit like the normal distribution, it is no restriction at all.) The bridge between mgfs and convergence in distribution is the following theorem, which we state without proof.

Theorem 6.11.2 (Continuity Theorem)

Let X have mgf m , finite on an open interval around 0, and let X_n have mgfs m_n . If $m_n(\theta) \rightarrow m(\theta)$ for all θ , then $X_n \rightarrow X$ in distribution.

With this in hand, the central limit theorem is within reach.

Theorem 6.11.3 (Central Limit Theorem)

Let $X_1, X_2, \dots$ be independent identically distributed random variables with mean μ and variance $\sigma^2 \in (0, \infty)$, and let $S_n = X_1 + \dots + X_n$. Then for every $x \in \mathbb{R}$,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \rightarrow \Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \quad \text{as } n \rightarrow \infty.$$

That is, $(S_n - n\mu)/\sigma\sqrt{n} \rightarrow N(0, 1)$ in distribution.

Note the choice of scaling: $S_n - n\mu$ has variance $n\sigma^2$, so dividing by $\sigma\sqrt{n}$ is exactly what's needed to stabilise the fluctuations at unit size. Informally, the theorem says

$$S_n \approx n\mu + \sigma\sqrt{n}Z, \quad Z \sim N(0, 1),$$

for large n : the sum is its mean, plus a normal fluctuation of order $\sqrt{n}$.

Proof (assuming an mgf exists). We give the proof under the additional assumption that the mgf of X_1 is finite on an open interval around 0 – this is how the result is proved in this course, and covers all the standard distributions; the general case requires characteristic functions.

By replacing X_i with $(X_i - \mu)/\sigma$, we may assume $\mu = 0$ and $\sigma = 1$, and must show $S_n/\sqrt{n} \rightarrow N(0, 1)$ in distribution. Let $m(\theta) = \mathbb{E}[e^{\theta X_1}]$. By independence,

$$\mathbb{E}\left[e^{\theta S_n/\sqrt{n}}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{(\theta/\sqrt{n})X_i}\right] = m\left(\frac{\theta}{\sqrt{n}}\right)^n,$$

so by the continuity theorem it suffices to show that for each fixed θ ,

$$m\left(\frac{\theta}{\sqrt{n}}\right)^n \rightarrow e^{\theta^2/2} = \mathbb{E}[e^{\theta Z}].$$

Since m is finite near 0, it has derivatives of all orders at 0 given by the moments, and Taylor's theorem gives, as $t \rightarrow 0$,

$$m(t) = 1 + t \mathbb{E}[X_1] + \frac{t^2}{2} \mathbb{E}[X_1^2] + o(t^2) = 1 + \frac{t^2}{2} + o(t^2),$$

using $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = \text{Var}(X_1) = 1$. Applying this with $t = \theta/\sqrt{n}$,

$$m\left(\frac{\theta}{\sqrt{n}}\right)^n = \left(1 + \frac{\theta^2}{2n} + o\left(\frac{1}{n}\right)\right)^n,$$

and taking logarithms, $n \log\left(1 + \frac{\theta^2}{2n} + o\left(\frac{1}{n}\right)\right) = \frac{\theta^2}{2} + o(1) \rightarrow \frac{\theta^2}{2}$, using $\log(1+u) = u + o(u)$. So $m(\theta/\sqrt{n})^n \rightarrow e^{\theta^2/2}$, completing the proof. $\square$

The really remarkable feature is *universality*: the limit doesn't remember anything about the distribution of the X_i except its mean and variance. This is why the normal distribution is everywhere – any quantity that is a sum of many small independent contributions (measurement errors, say, or heights in a population) should be approximately normally distributed.

§6.11.1 Applications

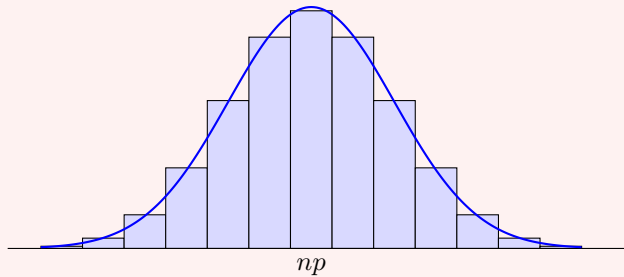
The real value of the central limit theorem is as an approximation tool, so let's see it in action.

Example 6.11.4 (Normal Approximation to the Binomial)

If $S_n \sim \text{Bin}(n, p)$, then S_n is a sum of n independent Bernoulli variables, so by the CLT

$$\frac{S_n - np}{\sqrt{np(1-p)}} \approx N(0, 1), \quad \text{i.e.} \quad S_n \approx N(np, np(1-p))$$

for large n . The picture below shows the pmf of $\text{Bin}(16, \frac{1}{2})$ with the approximating $N(8, 4)$ density drawn over it – the fit is already rather good.



Note the contrast with the Poisson approximation: there we let $p = \lambda/n$ shrink as n grew (so the CLT does not apply – the summands change with n), whereas here p is fixed. The rough rule: rare events give Poisson, common events give normal. Similarly, $\text{Poisson}(n)$ is a sum of n independent $\text{Poisson}(1)$ variables, so $\text{Poisson}(n) \approx N(n, n)$ for large n .

Example 6.11.5 (Opinion Polling)

Suppose each member of a large population independently supports a proposition with unknown probability p , and we poll N random individuals, estimating p by $\hat{p}_N = S_N/N$ where $S_N \sim \text{Bin}(N, p)$. How large must N be so that $|\hat{p}_N - p| \leq 0.04$ with probability at least 0.99?

By the CLT, $\hat{p}_N \approx p + \sqrt{p(1-p)/N} Z$ with $Z \sim N(0, 1)$, so we need

$$\mathbb{P} \left(\sqrt{\frac{p(1-p)}{N}} |Z| \leq 0.04 \right) \geq 0.99.$$

From tables, $\mathbb{P}(|Z| \leq 2.58) = 0.99$, so we require $0.04\sqrt{N/p(1-p)} \geq 2.58$. The

worst case is $p = \frac{1}{2}$ (maximising $p(1-p) = \frac{1}{4}$), giving

$$N \geq \frac{2.58^2}{4 \times 0.04^2} \approx 1040.$$

Only about a thousand people – regardless of the size of the population! The $1/\sqrt{N}$ decay of sampling error is entirely responsible for the feasibility of opinion polling.

§6.12 The Multivariate Normal Distribution

Given how central the normal distribution is, we should understand its multidimensional version – both because it arises in its own right (via a multidimensional CLT), and because it has some genuinely special structure not shared by other distributions.

In this section vectors are columns, and we write $u^\top$ for the transpose.

§6.12.1 Gaussian Vectors

We begin by pinning down exactly what it means for a whole vector of random variables to be normally distributed.

Definition 6.12.1 (Gaussian Vector)

A random vector $X = (X_1, \dots, X_n)^\top$ is a **Gaussian vector** (or **multivariate normal**) if for every $u \in \mathbb{R}^n$, the scalar random variable

$$u^\top X = u_1 X_1 + \dots + u_n X_n$$

is normally distributed (where we allow degenerate ‘normals’ of zero variance, i.e. constants).

Note this asks for far more than each X_i being normal individually – *every* linear combination must be normal. An immediate consequence of the definition is stability under affine maps: if X is Gaussian in $\mathbb{R}^n$, A is an $m \times n$ matrix and $b \in \mathbb{R}^m$, then $AX + b$ is Gaussian in $\mathbb{R}^m$, since $u^\top (AX + b) = (A^\top u)^\top X + u^\top b$ is a normal random variable plus a constant.

The distribution of a Gaussian vector turns out to be pinned down by two pieces of data: the **mean vector** and **covariance matrix**

$$\mu = \mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])^\top, \quad V = \mathbb{E}[(X - \mu)(X - \mu)^\top],$$

so that $V_{ij} = \text{Cov}(X_i, X_j)$. The matrix V is symmetric, and for any $u \in \mathbb{R}^n$, bilinearity of covariance gives

$$\mathbb{E}[u^\top X] = u^\top \mu, \quad \text{Var}(u^\top X) = \sum_{i,j} u_i u_j \text{Cov}(X_i, X_j) = u^\top V u.$$

In particular $u^\top V u = \text{Var}(u^\top X) \geq 0$, so V is non-negative definite, and

$$u^\top X \sim N(u^\top \mu, u^\top V u).$$

This last observation computes the mgf of X for free: taking the definition $m(\lambda) = \mathbb{E}[e^{\lambda^\top X}]$ for $\lambda \in \mathbb{R}^n$, the exponent $\lambda^\top X$ is $N(\lambda^\top \mu, \lambda^\top V \lambda)$, so evaluating the one-dimensional normal mgf at $\theta = 1$,

$$m(\lambda) = \exp\left(\lambda^\top \mu + \frac{\lambda^\top V \lambda}{2}\right).$$

By uniqueness of mgfs, we conclude that a Gaussian vector is completely determined by (μ, V) , and we write $X \sim N(\mu, V)$.

§6.12.2 Construction and Density

Do Gaussian vectors with arbitrary prescribed μ and (non-negative definite) V actually exist? Yes – and the construction is a direct generalisation of writing $X = \mu + \sigma Z$ in one dimension.

Start with $Z = (Z_1, \dots, Z_n)^\top$ where the Z_i are independent standard normals. Then Z is a Gaussian vector: for any u , the mgf of $u^\top Z$ factorises over the independent coordinates as

$$\mathbb{E}[e^{\theta u^\top Z}] = \prod_{i=1}^n e^{\theta^2 u_i^2 / 2} = e^{\theta^2 |u|^2 / 2},$$

which is the mgf of $N(0, |u|^2)$. So $Z \sim N(0, I)$.

Now given μ and non-negative definite V , diagonalise $V = U^\top D U$ with U orthogonal and D diagonal with non-negative entries, and let $\sigma = U^\top \sqrt{D} U$, where $\sqrt{D}$ takes square roots of the diagonal. Then σ is symmetric with $\sigma^2 = V$, and setting

$$X = \mu + \sigma Z$$

gives a Gaussian vector (affine image of a Gaussian vector) with $\mathbb{E}[X] = \mu$ and

$$\text{Var}(X) = \mathbb{E}[\sigma Z Z^\top \sigma^\top] = \sigma \mathbb{E}[Z Z^\top] \sigma^\top = \sigma I \sigma = V,$$

so $X \sim N(\mu, V)$ as desired.

When V is actually positive definite (no zero eigenvalues), σ is invertible and we can compute the density of X from that of Z by our change of variables formula, with $z = \sigma^{-1}(x - \mu)$ and Jacobian $|\det \sigma^{-1}| = 1/\sqrt{\det V}$:

$$f_X(x) = \frac{1}{(2\pi)^{n/2}} e^{-|z|^2/2} \cdot \frac{1}{\sqrt{\det V}} = \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x - \mu)^\top V^{-1}(x - \mu)}{2}\right),$$

using $z^\top z = (x - \mu)^\top \sigma^{-2}(x - \mu) = (x - \mu)^\top V^{-1}(x - \mu)$. (If V is singular, X lives on a lower-dimensional affine subspace and has no density on $\mathbb{R}^n$; restricting to that subspace recovers the formula above.)

§6.12.3 Independence and Covariance for Gaussians

We saw earlier that zero covariance does not imply independence in general. For Gaussian vectors, wonderfully, it does.

Theorem 6.12.2 (Independence Equals Zero Covariance)

Let $X = (X_1, \dots, X_n)^\top$ be a Gaussian vector. Then $X_1, \dots, X_n$ are independent if and only if $V = \text{Var}(X)$ is diagonal, i.e. $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$.

Proof. Independence always forces the off-diagonal covariances to vanish, so only the converse needs proof. Suppose V is diagonal with entries λ_i . Then the mgf factorises:

$$m(\theta) = \exp\left(\theta^\top \mu + \frac{\theta^\top V \theta}{2}\right) = \prod_{i=1}^n \exp\left(\theta_i \mu_i + \frac{\theta_i^2 \lambda_i}{2}\right),$$

which is the product of the mgfs of independent $N(\mu_i, \lambda_i)$ random variables. By uniqueness of (multivariate) mgfs, X has the same distribution as a vector of independent normals, so the X_i are independent. (When V is positive definite one can equally see this from the density, which factorises as a product of one-dimensional normal densities.) $\square$

§6.12.4 The Bivariate Normal

The case $n = 2$ deserves special attention. Let (X_1, X_2) be a Gaussian vector, with means μ_k , variances $\sigma_k^2 > 0$, and correlation

$$\rho = \text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2} \in [-1, 1].$$

The covariance matrix is then

$$V = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix},$$

and any such matrix with $\rho \in [-1, 1]$ is a legitimate covariance matrix (one can check $u^\top V u \geq 0$ directly by writing it as $(1 - \rho)(\sigma_1^2 u_1^2 + \sigma_2^2 u_2^2) + \rho(\sigma_1 u_1 + \sigma_2 u_2)^2$ and the analogous expression with $1 + \rho$).

By the previous theorem, X_1 and X_2 are independent exactly when $\rho = 0$. Even when they're correlated, we can compute conditional expectations cleanly with a decorrelation trick that is worth remembering.

Proposition 6.12.3 (Conditioning in the Bivariate Normal)

Let (X_1, X_2) be bivariate normal as above, and let $a = \rho \sigma_2 / \sigma_1$. Then

$$\mathbb{E}[X_2 \mid X_1] = \mu_2 + a(X_1 - \mu_1),$$

and conditionally on X_1 , X_2 has the $N(\mu_2 + a(X_1 - \mu_1), \sigma_2^2(1 - \rho^2))$ distribution.

Proof. Consider $Y = X_2 - aX_1$, and choose a so that Y is uncorrelated with X_1 :

$$\text{Cov}(Y, X_1) = \text{Cov}(X_2, X_1) - a \text{Var}(X_1) = \rho \sigma_1 \sigma_2 - a \sigma_1^2 = 0 \iff a = \frac{\rho \sigma_2}{\sigma_1}.$$

Now (X_1, Y) is a Gaussian vector (an affine image of (X_1, X_2)), so uncorrelated

implies *independent*. Writing $X_2 = Y + aX_1$,

$$\mathbb{E}[X_2 | X_1] = \mathbb{E}[Y | X_1] + a\mathbb{E}[X_1 | X_1] = \mathbb{E}[Y] + aX_1 = \mu_2 - a\mu_1 + aX_1,$$

using independence for the first term and ‘taking out what is known’ for the second. Moreover, given X_1 , the only remaining randomness in $X_2 = Y + aX_1$ is the independent normal Y , so conditionally X_2 is normal with the stated mean and variance

$$\text{Var}(Y) = \text{Var}(X_2) + a^2 \text{Var}(X_1) - 2a \text{Cov}(X_1, X_2) = \sigma_2^2 + \rho^2 \sigma_2^2 - 2\rho^2 \sigma_2^2 = \sigma_2^2(1 - \rho^2). \quad \square$$

Notice that the conditional expectation is an affine function of X_1 – the ‘regression line’ – and the conditional variance $\sigma_2^2(1 - \rho^2)$ doesn’t depend on the observed value of X_1 at all. Both are special features of the Gaussian world.

§6.13 Geometric Probability and Simulation

We close with two lighter topics that showcase continuous probability in action: problems about ‘random’ geometric objects, and the question of how to actually *generate* random variables with a computer.

§6.13.1 Bertrand's Paradox

Geometric probability has a famous health warning attached, and it is best delivered immediately: the phrase ‘at random’ has no meaning until you specify the distribution.

Example 6.13.1 (Bertrand's Paradox)

Draw a chord ‘at random’ in a circle of radius r . What is the probability that its length C is less than r ?

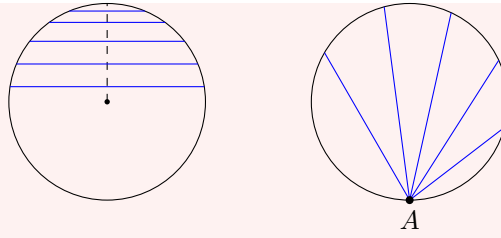
Interpretation 1. Choose the chord by picking a distance $X \sim \text{U}[0, r]$ and drawing the chord perpendicular to a fixed radius at distance X from the centre. By Pythagoras, $C = 2\sqrt{r^2 - X^2}$, so

$$\mathbb{P}(C \leq r) = \mathbb{P}(4(r^2 - X^2) \leq r^2) = \mathbb{P}\left(X \geq \frac{\sqrt{3}}{2}r\right) = 1 - \frac{\sqrt{3}}{2} \approx 0.134.$$

Interpretation 2. Choose the chord by fixing one endpoint A on the circle and picking the other endpoint at a uniformly random angle: let $\Theta \sim \text{U}[0, 2\pi]$ be the angle subtended at the centre. Then $C = 2r \sin(\Theta/2)$, so

$$\mathbb{P}(C \leq r) = \mathbb{P}\left(\sin \frac{\Theta}{2} \leq \frac{1}{2}\right) = \mathbb{P}\left(\Theta \leq \frac{\pi}{3}\right) + \mathbb{P}\left(\Theta \geq \frac{5\pi}{3}\right) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}.$$

The two procedures are illustrated below: chords perpendicular to a fixed radius at a uniform distance (left), and chords from a fixed point A at a uniform angle (right).



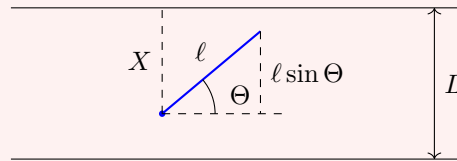
Two perfectly reasonable readings of ‘a random chord’; two different answers. There is no paradox once we accept that the two procedures generate genuinely different distributions on chords – the lesson is that in continuous settings, the model must be specified, not assumed.

§6.13.2 Buffon’s Needle

Our second classic has a happier punchline: a probabilistic experiment that computes π .

Example 6.13.2 (Buffon’s Needle)

A floor is ruled with parallel lines a distance L apart, and a needle of length $\ell \leq L$ is dropped on it at random. What is the probability that the needle crosses a line?



We model the drop by two independent random variables: the distance $X \sim U[0, L]$ from the needle’s lower end^a to the nearest line above it, and the angle $\Theta \sim U[0, \pi]$ the needle makes with the lines. The needle crosses a line precisely when $\ell \sin \Theta \geq X$, so

$$p = \mathbb{P}(X \leq \ell \sin \Theta) = \int_0^\pi \int_0^L \frac{1}{\pi L} \mathbb{1}_{\{x \leq \ell \sin \theta\}} dx d\theta = \frac{1}{\pi L} \int_0^\pi \ell \sin \theta d\theta = \frac{2\ell}{\pi L}.$$

^aLower in the direction perpendicular to the lines, say.

Since $\pi = 2\ell/pL$, we can *estimate* π by dropping n needles, letting $\hat{p}_n$ be the observed proportion of crossings, and computing $\hat{\pi}_n = 2\ell/\hat{p}_n L$. How good is this? This is exactly the sampling problem from the previous section: by the CLT,

$$\hat{p}_n \approx p + \sqrt{\frac{p(1-p)}{n}} Z, \quad Z \sim N(0, 1),$$

and applying the first-order Taylor expansion $f(\hat{p}_n) \approx f(p) + (\hat{p}_n - p)f'(p)$ to $f(x) = 2\ell/xL$ (which has $f'(p) = -\pi/p$),

$$\hat{\pi}_n - \pi \approx -\pi \sqrt{\frac{1-p}{pn}} Z.$$

Asking for $|\hat{\pi}_n - \pi| \leq 0.001$ with probability 0.99 and consulting tables ($\mathbb{P}(|Z| \leq 2.58) = 0.99$) leads to n of order 10^7 drops. As numerical integration schemes go, this one converges *very* slowly – the $1/\sqrt{n}$ error decay of averages is no match for even basic quadrature – but as a party trick it remains unbeatable.

§6.13.3 Simulating Random Variables

Suppose we have a source of uniform randomness – a computer’s random number generator handing us independent $U[0, 1]$ variables – and we want to manufacture a random variable with some other prescribed distribution F . There is a beautifully simple recipe: feed the uniform through the inverse of F .

Theorem 6.13.3 (Inverse Transform Sampling)

Let F be a continuous, strictly increasing distribution function, and let $U \sim U[0, 1]$. Then

$$X = F^{-1}(U) \quad \text{has distribution function } F.$$

Proof. Since F is increasing, $F^{-1}(U) \leq x \iff U \leq F(x)$, and so

$$\mathbb{P}(X \leq x) = \mathbb{P}(U \leq F(x)) = F(x),$$

as $F(x) \in [0, 1]$ and U is uniform. $\square$

The intuition: $F^{-1}(p)$ is the value that the distribution places probability p below, so sampling a uniform ‘probability level’ and inverting produces values with exactly the right frequencies.

Example 6.13.4

Taking $F(x) = 1 - e^{-x}$, we get that if $U \sim U[0, 1]$ then $F^{-1}(U) = -\log(1 - U)$ is $\text{Exp}(1)$ – and since $1 - U$ is also uniform, more simply $-\log U \sim \text{Exp}(1)$. Similarly, the Box-Muller observation from our polar coordinates example gives a normal sampler: with independent $U_1, U_2 \sim U[0, 1]$, the variables

$$X = \sqrt{-2 \log U_1} \cos(2\pi U_2), \quad Y = \sqrt{-2 \log U_1} \sin(2\pi U_2)$$

are independent standard normals, since $\sqrt{-2 \log U_1}$ has the density $re^{-r^2/2}$ of the Gaussian radius and $2\pi U_2$ is a uniform angle.

Inverse transform sampling needs a computable F^{-1} , which is often unavailable (already for the normal distribution one has to work around it, as above). Our final method needs only the density – and the willingness to throw samples away.

Theorem 6.13.5 (Rejection Sampling)

Let $A \subseteq [0, 1]^d$ have volume $|A| > 0$, and let $U_1, U_2, \dots$ be independent uniform points in $[0, 1]^d$. Let $N = \min\{n \geq 1 : U_n \in A\}$. Then U_N is uniformly distributed on A .

Proof. Let $B \subseteq [0, 1]^d$. Splitting according to the value of N and using independence,

$$\mathbb{P}(U_N \in B) = \sum_{n=1}^{\infty} \mathbb{P}(U_n \in A \cap B, U_{n-1} \notin A, \dots, U_1 \notin A)$$

$$= \sum_{n=1}^{\infty} |A \cap B| (1 - |A|)^{n-1} = \frac{|A \cap B|}{|A|},$$

summing the geometric series. This is exactly the uniform distribution on A . $\square$

To simulate a random variable on $[0, 1]^{d-1}$ with a bounded density $f \leq \lambda$, we apply this with the region *under the (scaled) graph of f* :

$$A = \left\{ (x_1, \dots, x_d) \in [0, 1]^d : x_d \leq \frac{f(x_1, \dots, x_{d-1})}{\lambda} \right\}.$$

Generate a uniform point of A by rejection, and keep only its first $d - 1$ coordinates X . For any $B \subseteq [0, 1]^{d-1}$, integrating out the last coordinate gives

$$\mathbb{P}(X \in B) = \frac{|(B \times [0, 1]) \cap A|}{|A|} = \frac{\frac{1}{\lambda} \int_B f(x) \, dx}{\frac{1}{\lambda} \int f(x) \, dx} = \int_B f(x) \, dx,$$

so X has density f , as desired. In words: throw uniform darts at the box, keep the ones landing under the graph of the density, and read off their horizontal coordinates. It is a fittingly concrete note on which to end – the abstract machinery of densities, built up from three axioms about measures, cashing out as instructions you could implement in ten lines of code.

7 Vector Calculus

Vector calculus takes the differentiation and integration that we know and love from single-variable calculus and asks what they look like in two, three (and more) dimensions. The objects of study become scalar and vector *fields*, and the ways of differentiating them – the gradient, divergence and curl – turn out to be linked to integration by a family of remarkable ‘integral theorems’, all of which generalise the fundamental theorem of calculus. This language is the natural setting for much of mathematical physics, and we will see it at work in electromagnetism through Maxwell’s equations.

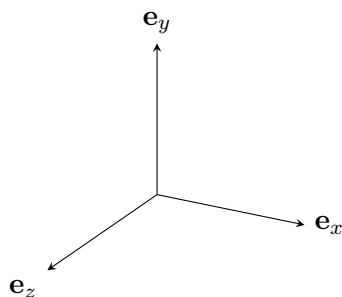
This article constitutes my notes for the ‘Vector Calculus’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. If you spot any errors or have any feedback, I can be contacted at ak2316@cam.ac.uk.

§7.1 Notation

Throughout this course, a column vector

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

is to be interpreted as the vector $\mathbf{x} = a\mathbf{e}_x + b\mathbf{e}_y + c\mathbf{e}_z$, where $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ are basis vectors aligned with the fixed Cartesian x , y and z axes in $\mathbb{R}^3$.



We may also use the notation $\mathbf{e}_1 = \mathbf{e}_x$, $\mathbf{e}_2 = \mathbf{e}_y$ and $\mathbf{e}_3 = \mathbf{e}_z$, and then we can write $\mathbf{x} = x_i \mathbf{e}_i$.

We will also be using the summation convention frequently, which you can read more about in the ‘Vectors and Matrices’ course notes.

§7.2 Differential Geometry of Curves

We will begin by studying the differential geometry of curves in $\mathbb{R}^3$, which sounds exciting but isn’t.

§7.2.1 Parameterized Curves & Arc Length

The first object we shall study is *parameterized curves*.

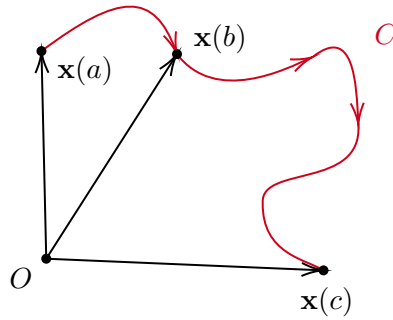
Definition 7.2.1 (Parameterized Curve)

A **parameterized curve** C in $\mathbb{R}^3$ is the image of a continuous map $\mathbf{x} : [a, b] \rightarrow \mathbb{R}^3$ in which $t \mapsto \mathbf{x}(t)$.

In Cartesian coordinates, we can write

$$\mathbf{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix} = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}.$$

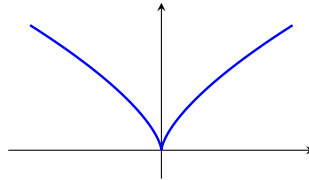
We also give the curve an orientation, based on the map going from the image of A to the image of B .



Definition 7.2.2 (Differentiability and Smoothness)

We say that a curve C is **differentiable** if each of the components $\{x_1(t), x_2(t), x_3(t)\}$ are differentiable, and say C is **regular** if $|\mathbf{x}'(t)| \neq 0$. If C is differentiable and regular then we say C is **smooth**.

Remark. The ‘regular’ condition exists to avoid ‘spikes’ in the curve. For example, consider the curve $\mathbf{x}(t) = (t^3, t^2)$. Clearly this is differentiable, but $\mathbf{x}(t)$ has a cusp at $t = 0$. At $t = 0$, we have $|\mathbf{x}'(0)| = 0$. If this was not the case, there would be no cusp.

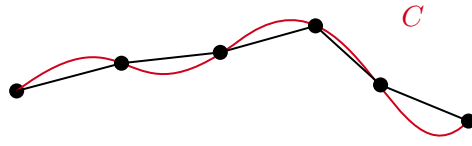


Recall that the function $x_i(t)$ is differentiable at t if $x_i(t+h) = x_i(t) + x'_i(t)h + o(h)$, where $\frac{o(h)}{h} \rightarrow 0$ as $h \rightarrow 0$. We can write this in terms of vectors, where

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \mathbf{x}'(t)h + o(h),$$

where $o(h)$ is a vector for which $\frac{|o(h)|}{h} \rightarrow 0$ as $h \rightarrow 0$.

Now given some curve C , we may want to find the length of the curve. We can try to do this by approximating the curve using straight lines.



For $C : [a, b] \rightarrow \mathbb{R}^3$ with $t \mapsto \mathbf{x}(t)$, we introduce a partition P of $[a, b]$ with $t_0 = a$, $t_N = b$ and $t_0 < t_1 < \dots < t_N$, and we set $\Delta t_i = t_{i+1} - t_i$ and $\Delta t = \max_i \Delta t_i$.

Definition 7.2.3 (Length Relative to a Partition)

For some curve C and partition P as above, we define the **length** of C relative to P by

$$\ell(C, P) = \sum_{i=0}^{N-1} |\mathbf{x}(t_{i+1}) - \mathbf{x}(t_i)|.$$

We would expect that this length would get closer to the true length of C as $\Delta t \rightarrow 0$. Indeed, we will define the length of C in this way.

Definition 7.2.4 (Length of a Curve)

We define the length of a curve C by

$$\ell(C) = \lim_{\Delta t \rightarrow 0} \sum_{i=0}^{N-1} |\mathbf{x}(t_{i+1}) - \mathbf{x}(t_i)| = \lim_{\Delta t \rightarrow 0} \ell(C, P).$$

If this limit does not exist, we say the curve is **non-rectifiable**.

In this course, we will not worry too much about non-rectifiable curves, so we will assume that this notion is always well defined.

Now suppose C is a differentiable curve. Then we have that

$$\begin{aligned} \mathbf{x}(t_{i+1}) &= \mathbf{x}(t_i + t_{i+1} - t_i) \\ &= \mathbf{x}(t_i + \Delta t_i) \\ &= \mathbf{x}(t_i) + \mathbf{x}'(t_i) \Delta t_i + o(\Delta t_i). \end{aligned}$$

It follows that

$$|\mathbf{x}(t_{i+1}) - \mathbf{x}(t_i)| = |\mathbf{x}'(t_i)| \Delta t_i + o(\Delta t_i).$$

So if C is differentiable, we get the expression

$$\ell(C, P) = \sum_{i=0}^{N-1} (|\mathbf{x}'(t_i)| \Delta t_i + o(\Delta t_i)).$$

Recall that $o(\Delta t_i)$ represents a function for which $\frac{o(\Delta t_i)}{\Delta t_i} \rightarrow 0$ as $\Delta t_i \rightarrow 0$. So for any $\epsilon > 0$, if $\Delta t = \max_i \Delta t_i$ is sufficiently small, then we have

$$|o(\Delta t_i)| < \left(\frac{\epsilon}{b-a} \right) \Delta t_i,$$

for $i = \{0, \dots, N-1\}$. So

$$\begin{aligned} \left| \ell(C, P) - \sum_{i=0}^{N-1} |\mathbf{x}'(t_i)| \Delta t_i \right| &= \left| \sum_{i=0}^{N-1} o(\Delta t_i) \right| \\ &< \frac{\epsilon}{b-a} \sum_{i=0}^{N-1} \Delta t_i = \epsilon. \end{aligned}$$

Thus the LHS tends to zero as $\Delta t \rightarrow 0$. So we get that

$$\begin{aligned} \ell(C) &= \lim_{\Delta t \rightarrow 0} \ell(C, P) = \lim_{\Delta t \rightarrow 0} \sum_{i=0}^{N-1} |\mathbf{x}'(t_i)| \Delta t_i \\ &= \int_a^b |\mathbf{x}'(t)| \, dt, \end{aligned}$$

by the definition of the Riemann integral. So, we can restate our definition in this other form.

Definition 7.2.5 (Length of a Curve)

If $C : [a, b] \rightarrow \mathbb{R}^3$ is a differentiable curve with $t \mapsto \mathbf{x}(t)$, then its **length** is

$$\ell(C) = \int_a^b |\mathbf{x}'(t)| \, dt = \int_C ds,$$

where ds is the **arc-length element**, $ds = |\mathbf{x}'(t)| \, dt$.

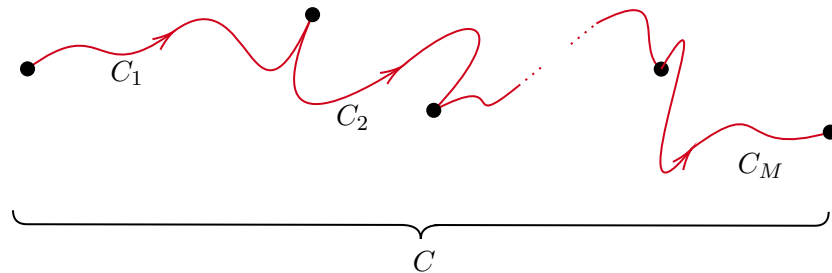
With this defined, we can define the integral of a function along a curve.

Definition 7.2.6 (Integral on a Curve)

For a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and a differentiable curve $C : [a, b] \rightarrow \mathbb{R}^3$ with $t \mapsto \mathbf{x}(t)$, we define the **integral of f along C** to be

$$\int_C f(\mathbf{x}) \, ds = \int_a^b f(\mathbf{x}(t)) |\mathbf{x}'(t)| \, dt.$$

Now consider a curve C made up of M smooth curves $C_1, C_2, \dots, C_M$.



Then writing $C = C_1 + C_2 + \dots + C_M$, we can define

$$\int_C f(\mathbf{x}) \, ds = \sum_{i=1}^M \int_{C_i} f(\mathbf{x}) \, ds,$$

so we can integrate over piecewise smooth curves.

Note that informally, we have

$$ds = |\mathbf{x}'(t)| \, dt = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt.$$

So being somewhat sacrilegious, we have

$$ds^2 = dx^2 + dy^2 + dz^2.$$

Example 7.2.7 (Circumference of a Circle)

Let C be a circle of radius $r > 0$ in $\mathbb{R}^3$,

$$\mathbf{x}(t) = \begin{pmatrix} r \cos t \\ r \sin t \\ 0 \end{pmatrix}, \quad t \in [0, 2\pi].$$

We can differentiate this to get

$$\mathbf{x}'(t) = \begin{pmatrix} -r \sin t \\ r \cos t \\ 0 \end{pmatrix}, \quad t \in [0, 2\pi].$$

Then integrating over C we have

$$\begin{aligned} \int_C ds &= \int_0^{2\pi} \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} \, dt \\ &= \int_0^{2\pi} r \, dt \\ &= 2\pi r, \end{aligned}$$

as we would expect.

Example 7.2.8 (Integrating over a Circle)

With C being a circle as before, we can integrate over the curve. For example

$$\begin{aligned} \int_C x^2 y \, ds &= \int_0^{2\pi} (r \cos t)^2 (r \sin t) \underbrace{r \, dt}_{|\mathbf{x}'(t)| \, dt} \\ &= 0 \end{aligned}$$

There is one subtlety that we have looked over - does $\ell(C)$ depend on the parameterization used?

For example, if we had $\mathbf{x}(t) = (r \cos t, r \sin t, 0)$ with $t \in [0, 2\pi]$ and $\tilde{\mathbf{x}}(t) = (r \cos(2t), r \sin(2t), 0)$ with $t \in [0, \pi]$, these represent the same circle and they should have the same length.

We will clear up this possible ambiguity now.

Proposition 7.2.9

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a function and C is a differentiable curve, then $\int_C f(\mathbf{x}) \, ds$ is independent of the parameterization of C used.

Proof. Suppose C has two different parameterizations,

$$\begin{aligned}\mathbf{x} &= \mathbf{x}_1(t), & a \leq t \leq b \\ \mathbf{x} &= \mathbf{x}_2(\tau), & \alpha \leq \tau \leq \beta\end{aligned}$$

We must then have $\mathbf{x}_2(\tau) = \mathbf{x}_1(t(\tau))$ for some function $t(\tau)$. We can assume that $\frac{dt}{d\tau} \neq 0$ so the map between t and τ is invertible and differentiable (this is the inverse function theorem, covered in Analysis and Topology in Part IB). Note then that

$$\begin{aligned}\mathbf{x}'_2(\tau) &= \frac{d}{d\tau} \mathbf{x}_2(t) \\ &= \frac{d}{d\tau} \mathbf{x}_1(t(\tau)) \\ &= \frac{dt}{d\tau} \mathbf{x}'_1(t(\tau)).\end{aligned}$$

Then from our definition,

$$\int_C f(\mathbf{x}) \, ds = \int_a^b f(\mathbf{x}_1(t)) |\mathbf{x}'_1(t)| \, dt.$$

We can make the substitution $t = t(\tau)$, and assuming $dt/d\tau > 0$, the latter integral becomes

$$\int_\alpha^\beta f(\mathbf{x}_2(\tau)) \underbrace{|\mathbf{x}'_1(t(\tau))| \frac{dt}{d\tau}}_{|\mathbf{x}'_2(\tau)|} \, d\tau,$$

which is precisely the same as $\int_C f(\mathbf{x}) \, ds$ using the $\mathbf{x}_2(t)$ parameterization.

We assumed here that $\frac{dt}{d\tau} > 0$, but the same holds if it is negative. Thus our definition of integrating over a curve C does not depend on the choice of parameterization of C . $\square$

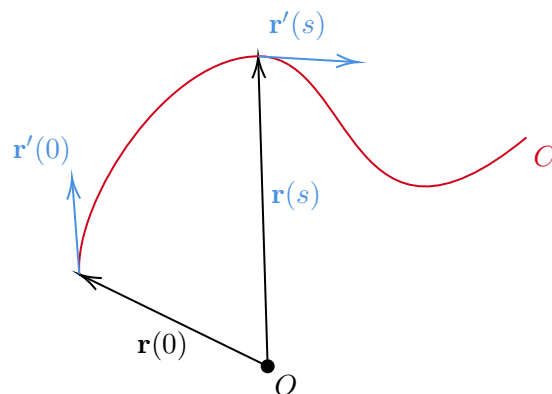
We can parameterize curves in many different ways. One useful way to parameterise regular curves is with respect to arc-length. If we write $\mathbf{r}(s) = \mathbf{x}(t(s))$ where $0 \leq s \leq \ell(C)$, then by the chain rule

$$\frac{dt}{ds} = \frac{1}{ds/dt} = \frac{1}{|\mathbf{x}'(t(s))|},$$

so

$$\begin{aligned}\mathbf{r}'(s) &= \frac{d}{ds} \mathbf{x}(t(s)) \\ &= \frac{dt}{ds} \mathbf{x}'(t(s)) \\ &= \frac{\mathbf{x}'(t(s))}{|\mathbf{x}'(t(s))|},\end{aligned}$$

that is, $|\mathbf{r}'(s)| = 1$. This (consistently) gives $\ell(C) = \int_0^{\ell(C)} |\mathbf{r}'(s)| \, ds = \int_0^{\ell(C)} 1 \, ds$. So the way to picture $\mathbf{r}'(s)$ is as the unit tangent vector to the curve.



§7.2.2 Curvature and Torsion

Throughout this section, we will talk about a generic regular curve C , parameterized by arc-length, and we will write $s \mapsto \mathbf{r}(s)$.

Definition 7.2.10 (Tangent Vector)

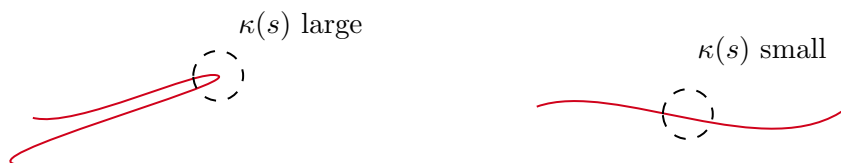
For a regular curve C parameterized by arc length, we define the **tangent vector** to be $\mathbf{t}(s) = \mathbf{r}'(s)$.

We already established that $|\mathbf{t}(s)| = 1$, and thus as the second derivative $\mathbf{r}''(s) = \mathbf{t}'(s)$ only measures the change in *direction*.

Intuitively, if $|\mathbf{r}''(s)|$ is large, then the curve rapidly changes in direction, whereas if $|\mathbf{r}''(s)|$ is small, we would expect the curve to be approximately flat.

Definition 7.2.11 (Curvature)

We define the **curvature** of C to be $\kappa(s) = |\mathbf{r}''(s)| = |\mathbf{t}'(s)|$.



Now since $\mathbf{t} = \mathbf{r}'(s)$ is a unit vector, if we differentiate $\mathbf{t} \cdot \mathbf{t} = 1$ we get $\mathbf{t} \cdot \mathbf{t}' = 0$. Because of this we can introduce the following definition.

Definition 7.2.12 (Principal Normal)

The **principal normal** $\mathbf{n}$ is the vector given by $\mathbf{t}' = \kappa \mathbf{n}$.

The principal normal $\mathbf{n}$ is everywhere normal to the curve C , since it's always perpendicular to the tangent vector ($\mathbf{t} \cdot \mathbf{n} = 0$). Then we have two perpendicular vectors, so we can extend $\{\mathbf{t}, \mathbf{n}\}$ to an orthonormal basis by defining the following.

Definition 7.2.13 (Binormal)

The **binormal** $\mathbf{b}$ is the unit vector perpendicular to $\mathbf{t}$ and $\mathbf{n}$, $\mathbf{b} = \mathbf{t} \times \mathbf{n}$.

Since $|\mathbf{b}| = 1$, we have $\mathbf{b}' \cdot \mathbf{b} = 0$. Also since $\mathbf{t} \cdot \mathbf{b} = 0$ and $\mathbf{n} \cdot \mathbf{b} = 0$,

$$\begin{aligned} 0 &= (\mathbf{t} \cdot \mathbf{b})' = \mathbf{t}' \cdot \mathbf{b} + \mathbf{t} \cdot \mathbf{b}' \\ &= \cancel{\kappa \mathbf{n} \cdot \mathbf{b}} + \mathbf{t} \cdot \mathbf{b}', \end{aligned}$$

thus $\mathbf{b}'$ is orthogonal to $\mathbf{t}$ and $\mathbf{b}$, and thus it is parallel to $\mathbf{n}$. We define the following.

Definition 7.2.14 (Torsion)

We define the **torsion** τ by $\mathbf{b}' = -\tau \mathbf{n}$.

Informally, torsion measures a twisting motion along a curve. So we have two equations.

$$\mathbf{t}' = \kappa \mathbf{n}, \quad \mathbf{b}' = -\tau \mathbf{n}.$$

Proposition 7.2.15 (Fundamental Theorem of Curves)

The curvature $\kappa(s)$ and torsion $\tau(s)$ define a curve up to translation and orientation.

Proof Sketch. Since $\mathbf{n} = \mathbf{b} \times \mathbf{t}$, we have

$$\mathbf{t}' = \kappa(\mathbf{b} \times \mathbf{t}), \quad \mathbf{b}' = -\tau(\mathbf{b} \times \mathbf{t}).$$

This gives six equations for six unknowns. Given $\kappa(s)$, $\tau(s)$, $\mathbf{t}(0)$ and $\mathbf{b}(0)$, we can construct the functions $\mathbf{t}(s)$, $\mathbf{b}(s)$, and hence $\mathbf{n} = \mathbf{b} \times \mathbf{t}$. Hence the result. $\square$

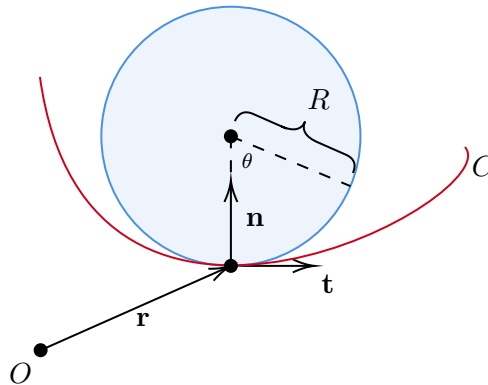
§7.2.3 Radius of Curvature

Consider some generic curve $s \mapsto \mathbf{r}(s)$, and consider the Taylor expansion about $s = 0$. Write $\mathbf{t} = \mathbf{t}(0)$, $\mathbf{n} = \mathbf{n}(0)$, etc. Then we get

$$\begin{aligned} \mathbf{r}(s) &= \mathbf{r}(0) + s\mathbf{r}'(0) + \frac{1}{2}s^2\mathbf{r}''(0) + o(s^2) \\ &= \mathbf{r} + s\mathbf{t} + \frac{1}{2}s^2\kappa\mathbf{n} + o(s^2). \end{aligned}$$

So this gives us a good approximation of the curve around $s = 0$, and now we want to try and draw a circle through the point $\mathbf{r}(0)$ such that the circle is tangent to the curve at that point. Specifically, what would the radius of that circle be?

Suppose, without loss of generality, that $\mathbf{t}$ is horizontal.



A point on this circle can be described by

$$\mathbf{x}(\theta) = \mathbf{r} + R(1 - \cos \theta)\mathbf{n} + R \sin \theta \mathbf{t}.$$

Then expanding for $|\theta|$ small, we get

$$\mathbf{x}(\theta) = \mathbf{r} + R\theta\mathbf{t} + \frac{1}{2}R\theta^2\mathbf{n} + o(\theta^2).$$

On this circle, the arc-length is $s = R\theta$. Thus if we express the above in terms of arc-length, we get

$$\mathbf{x}(\theta) = \mathbf{r} + s\mathbf{t} + \frac{1}{2R}s^2\mathbf{n} + o(s^2).$$

As we want this to match the curve, we compare it to the equation for the curve near $\mathbf{r}$, and to match quadratic terms we need $R = 1/\kappa$.

Definition 7.2.16 (Radius of Curvature)

We define $R(s) = 1/\kappa(s)$ to be the **radius of curvature** of the curve $s \mapsto \mathbf{r}(s)$.

§7.3 Coordinates, Differentials and Gradients

§7.3.1 Differentials and First Order Changes

Recall that for $f = f(u_1, \dots, u_n)$, we define the **differential** of f , written df , by

$$df = \frac{\partial f}{\partial u_i} du_i,$$

using the summation convention. We call $\{du_i\}$ **differential forms**. We will treat these differential forms as formal objects that are the elements of some abstract vector space, and treat them as linear independent if $\{u_1, \dots, u_n\}$ are independent, that is, if $\alpha_i du_i = 0$ implies $\alpha_i = 0$ for $i = 1, \dots, n$.

Similarly, if we had $\mathbf{x} = \mathbf{x}(u_1, \dots, u_n)$, we define

$$d\mathbf{x} = \frac{\partial \mathbf{x}}{\partial u_i} du_i.$$

Example 7.3.1 (Using Differentials)

If $f(u, v, w) = u^2 + w \sin(v)$, then

$$df = 2u du + w \cos(v) dv + \sin(v) dw.$$

Similarly, if we had $\mathbf{x}(u, v, w) = \begin{pmatrix} u^2 - v^2 \\ w \\ e^v \end{pmatrix}$, then

$$d\mathbf{x} = \begin{pmatrix} 2u \\ 0 \\ 0 \end{pmatrix} du + \begin{pmatrix} -2v \\ 0 \\ e^v \end{pmatrix} dv + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} dw.$$

Differentials encode information about how a function/field changes when we ‘wobble’ our coordinates. Indeed, by calculus we know

$$f(u_1 + \delta u_1, \dots, u_n + \delta u_n) - f(u_1, \dots, u_n) = \frac{\partial f}{\partial u_i} \delta u_i + o(\delta \mathbf{u}),$$

where $\delta \mathbf{u} = (\delta u_1, \dots, \delta u_n)$. So differentials allow to discuss first order changes in f .

So if δf denotes a change in $f(u_1, \dots, u_n)$ under perturbation of coordinates $(u_1, \dots, u_n) \mapsto (u_1 + \delta u_1, \dots, u_n + \delta u_n)$, we have, to first order,

$$\delta f \approx \frac{\partial f}{\partial u_i} \delta u_i.$$

Similarly for vector fields,

$$\delta \mathbf{x} \approx \frac{\partial \mathbf{x}}{\partial u_i} \delta u_i.$$

§7.3.2 Coordinates and Line Elements

You have already seen at least two sets of coordinates before taking this course for $\mathbb{R}^2$: Cartesian coordinates (x, y) , and polar coordinates (r, θ) . They have an invertible (except at the origin) relationship,

$$x = r \cos \theta, y = r \sin \theta.$$

A general set of coordinates (u, v) on $\mathbb{R}^2$ can be specified by their relationship to (x, y) , the Cartesian coordinates. That is, we can specify smooth functions $x = x(u, v)$, $y = y(u, v)$ that are a bijection. Of course we have the same thing for $\mathbb{R}^3$.

Let’s look at some examples.

Example 7.3.2 (Cartesian Coordinates)

In the standard Cartesian coordinate system we have $\mathbf{x}(x, y) = \begin{pmatrix} x \\ y \end{pmatrix} = x\mathbf{e}_x + y\mathbf{e}_y$, where $\{\mathbf{e}_x, \mathbf{e}_y\}$ are orthogonal vectors. We have $\mathbf{e}_x$ is in the direction of changing x with y fixed.

Said differently, we have

$$\mathbf{e}_x = \frac{\frac{\partial}{\partial x} \mathbf{x}(x, y)}{\left| \frac{\partial}{\partial x} \mathbf{x}(x, y) \right|}, \quad \text{and} \quad \mathbf{e}_y = \frac{\frac{\partial}{\partial y} \mathbf{x}(x, y)}{\left| \frac{\partial}{\partial y} \mathbf{x}(x, y) \right|}.$$

A feature of Cartesian coordinates is that

$$d\mathbf{x} = \frac{\partial \mathbf{x}}{\partial x} dx + \frac{\partial \mathbf{x}}{\partial y} dy = dx\mathbf{e}_x + dy\mathbf{e}_y,$$

that is, a change in coordinate $x \mapsto x + \delta x$ cause a change to the vector (to the first order) by $\mathbf{x} \mapsto \mathbf{x} + \delta x\mathbf{e}_x$. We call $d\mathbf{x}$ the **line element**, which tells us how small changes in coordinates produce changes in position vectors.

That was slightly tame, we’re going to look at something slightly more interesting.

Example 7.3.3 (Polar Coordinates)

For polar coordinates (r, θ) we have

$$\mathbf{x}(r, \theta) = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix} = r \mathbf{e}_r.$$

We have used the basis vectors $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ where

$$\mathbf{e}_r = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}, \quad \mathbf{e}_\theta = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}.$$

Note that $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are orthonormal at each (r, θ) , but not the same for each (r, θ) . Note that as before,

$$\mathbf{e}_r = \frac{\frac{\partial}{\partial r} \mathbf{x}(r, \theta)}{\left| \frac{\partial}{\partial r} \mathbf{x}(r, \theta) \right|}, \quad \text{and} \quad \mathbf{e}_\theta = \frac{\frac{\partial}{\partial \theta} \mathbf{x}(r, \theta)}{\left| \frac{\partial}{\partial \theta} \mathbf{x}(r, \theta) \right|}.$$

Since $\{\mathbf{e}_r, \mathbf{e}_\theta\}$ are orthogonal, it makes sense to call (r, θ) **orthogonal, curvilinear coordinates**.

We also have the line element

$$d\mathbf{x} = \frac{\partial \mathbf{x}}{\partial r} dr + \frac{\partial \mathbf{x}}{\partial \theta} d\theta = \mathbf{e}_r dr + r d\theta \mathbf{e}_\theta,$$

and so we can see that a change $\theta \mapsto \theta + \delta\theta$ produces a (first order) change

$$\mathbf{x} \mapsto \mathbf{x} + r\delta\theta \mathbf{e}_\theta,$$

which is notably *not* $\mathbf{x} \mapsto \mathbf{x} + \delta\theta \mathbf{e}_\theta$.

Orthogonal Curvilinear Coordinates

We can state properly what it means for a coordinate system to be orthogonal curvilinear.

Definition 7.3.4 (Orthogonal Curvilinear)

We say that (u, v, w) are a set of **orthogonal curvilinear coordinates** if the vectors

$$\mathbf{e}_u = \frac{\partial \mathbf{x} / \partial u}{\left| \partial \mathbf{x} / \partial u \right|}, \quad \mathbf{e}_v = \frac{\partial \mathbf{x} / \partial v}{\left| \partial \mathbf{x} / \partial v \right|}, \quad \mathbf{e}_w = \frac{\partial \mathbf{x} / \partial w}{\left| \partial \mathbf{x} / \partial w \right|}$$

form a right-handed^a, orthonormal basis for each (u, v, w) .

^a $\mathbf{e}_u \times \mathbf{e}_v = \mathbf{e}_w$.

Just as with polar coordinates, we have these vectors $\{\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w\}$ form an orthonormal basis for $\mathbb{R}^3$ with each choice of (u, v, w) , but *not* necessarily the same basis at each point.

It is standard to write

$$h_u = \left| \frac{\partial \mathbf{x}}{\partial u} \right|, \quad h_v = \left| \frac{\partial \mathbf{x}}{\partial v} \right|, \quad h_w = \left| \frac{\partial \mathbf{x}}{\partial w} \right|,$$

which we call **scale factors**. Note that the line element is

$$\begin{aligned} d\mathbf{x} &= \frac{\partial \mathbf{x}}{\partial u} du + \frac{\partial \mathbf{x}}{\partial v} dv + \frac{\partial \mathbf{x}}{\partial w} dw \\ &= h_u \mathbf{e}_u du + h_v \mathbf{e}_v dv + h_w \mathbf{e}_w dw. \end{aligned}$$

Scale factors tell us how small changes in coordinates ‘scale-up’ to changes in position $\mathbf{x}$.

Cylindrical Polar Coordinates

We define the coordinate system (ρ, ϕ, z) by

$$\mathbf{x}(\rho, \phi, z) = \begin{pmatrix} \rho \cos \phi \\ \rho \sin \phi \\ z \end{pmatrix},$$

with $0 \leq \rho < \infty$, $0 \leq \phi < 2\pi$ and $-\infty < z < \infty$.

We find that

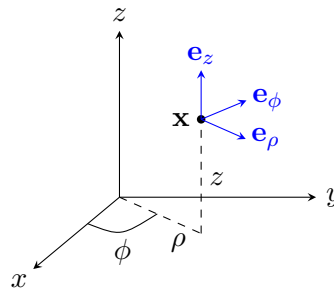
$$\mathbf{e}_\rho = \begin{pmatrix} \cos \phi \\ \sin \phi \\ 0 \end{pmatrix}, \quad \mathbf{e}_\phi = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix}, \quad \mathbf{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

and $h_\rho = 1$, $h_\phi = \rho$, $h_z = 1$. Thus the line element is

$$d\mathbf{x} = d\rho \mathbf{e}_\rho + \rho d\phi \mathbf{e}_\phi + dz \mathbf{e}_z.$$

Note also that we can write

$$\mathbf{x} = \begin{pmatrix} \rho \cos \phi \\ \rho \sin \phi \\ z \end{pmatrix} = \rho \begin{pmatrix} \cos \phi \\ \sin \phi \\ 0 \end{pmatrix} + z \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \rho \mathbf{e}_\rho + z \mathbf{e}_z.$$



Spherical Polar Coordinates

We can define the spherical polar coordinates (r, θ, ϕ) by

$$\mathbf{x}(r, \theta, \phi) = \begin{pmatrix} r \cos \phi \sin \theta \\ r \sin \phi \sin \theta \\ r \cos \theta \end{pmatrix},$$

with $0 \leq r < \infty$, $0 \leq \theta \leq \pi$ and $0 \leq \phi < 2\pi$.

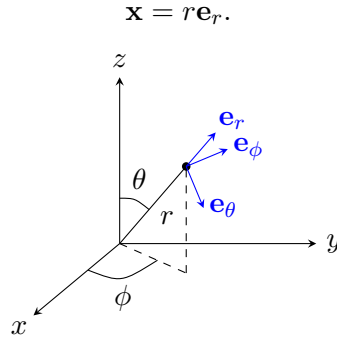
We have the corresponding basis vectors

$$\mathbf{e}_r = \begin{pmatrix} \cos \phi \sin \theta \\ \sin \phi \sin \theta \\ \cos \theta \end{pmatrix}, \quad \mathbf{e}_\theta = \begin{pmatrix} \cos \phi \cos \theta \\ \sin \phi \cos \theta \\ -\sin \theta \end{pmatrix}, \quad \mathbf{e}_\phi = \begin{pmatrix} -\sin \phi \\ \cos \phi \\ 0 \end{pmatrix}.$$

We also have $h_r = 1$, $h_\theta = r$ and $h_\phi = r \sin \theta$, so

$$d\mathbf{x} = dr\mathbf{e}_r + r d\theta\mathbf{e}_\theta + r \sin \theta d\phi\mathbf{e}_\phi,$$

and we can write



§7.3.3 The Gradient Operator

We will now introduce an ‘upgraded’ version of differentiation.

Definition 7.3.5 (Gradient)

For $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, we define the **gradient** of f written ∇f by

$$f(\mathbf{x} + \mathbf{h}) = f(\mathbf{x}) + \nabla f(\mathbf{x}) \cdot \mathbf{h} + o(\mathbf{h}),$$

as $\mathbf{h} \rightarrow 0$.

The gradient doesn’t exist for every function (just like the derivative), but in this course we will deal with functions for which it does.

We can interpret the gradient using directional derivatives.

Definition 7.3.6 (Directional Derivative)

The **directional derivative** of f in the direction $\mathbf{v}$, denoted $D_{\mathbf{v}}f$ or $\frac{\partial f}{\partial \mathbf{v}}$, is defined by

$$D_{\mathbf{v}}f(\mathbf{x}) = \lim_{t \rightarrow 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t},$$

that is, $f(\mathbf{x} + t\mathbf{v}) = f(\mathbf{x}) + tD_{\mathbf{v}}f(\mathbf{x}) + o(t)$, as $t \rightarrow 0$.

This expression is the same as if we substituted $\mathbf{h} = t\mathbf{v}$ into the definition of the gradient operator. So

$$D_{\mathbf{v}}f = \mathbf{v} \cdot \nabla f.$$

By Cauchy-Schwartz, we know that $\mathbf{a} \cdot \mathbf{b}$ is maximized when $\mathbf{a}$ and $\mathbf{b}$ are parallel. Thus to maximize the directional derivative, we would choose $\mathbf{v}$ to be in the same direction as ∇f .

Thus ∇f points in the direction of the greatest increase of f . Similarly, $-\nabla f$ points in the direction of greatest decrease of f .

Example 7.3.7

Suppose $f(\mathbf{x}) = \frac{1}{2}|\mathbf{x}|^2$. Then

$$\begin{aligned} f(\mathbf{x} + \mathbf{h}) &= \frac{1}{2}(\mathbf{x} + \mathbf{h}) \cdot (\mathbf{x} + \mathbf{h}) \\ &= \frac{1}{2}|\mathbf{x}|^2 + \frac{1}{2}(2\mathbf{x} \cdot \mathbf{h}) + \frac{1}{2}|\mathbf{h}|^2 \\ &= f(\mathbf{x}) + \mathbf{x} \cdot \mathbf{h} + o(\mathbf{h}), \end{aligned}$$

as $\mathbf{h} \rightarrow 0$. Thus $\nabla f(\mathbf{x}) = \mathbf{x}$.

Now suppose we have a curve $t \mapsto \mathbf{x}(t)$. We might wonder how f changes as we evaluate it along the curve. We write $F(t) = f(\mathbf{x}(t))$, and we will write $\delta\mathbf{x} = \mathbf{x}(t + \delta t) - \mathbf{x}(t)$. We want to consider

$$\begin{aligned} F(t + \delta t) &= f(\mathbf{x}(t + \delta t)) \\ &= f(\mathbf{x}(t) + \delta\mathbf{x}) \\ &= f(\mathbf{x}(t)) + \nabla f(\mathbf{x}(t)) \cdot \delta\mathbf{x} + o(\delta\mathbf{x}). \end{aligned}$$

Since $\delta\mathbf{x} = \mathbf{x}'(t)\delta t + o(\delta t)$, we get

$$F(t + \delta t) = F(t) + \mathbf{x}'(t) \cdot \nabla f(\mathbf{x}(t))\delta t + o(\delta t).$$

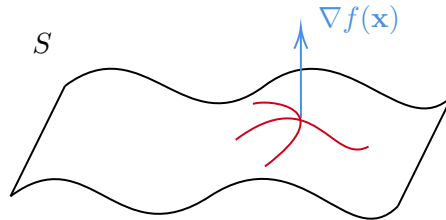
Thus

$$\frac{dF}{dt} = \frac{d}{dt}f(\mathbf{x}(t)) = \frac{d\mathbf{x}}{dt} \cdot \nabla f(\mathbf{x}(t)).$$

Another use of the gradient is with surfaces. Suppose a surface S is defined implicitly by $S = \{\mathbf{x} \in \mathbb{R}^3 \mid f(\mathbf{x}) = 0\}$. If $t \mapsto \mathbf{x}(t)$ is *any* curve in S , then $f(\mathbf{x}(t)) = 0$ identically. So if we differentiated, we would get

$$0 = \frac{d}{dt}f(\mathbf{x}(t)) = \nabla f(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt}.$$

So ∇f is orthogonal to the tangent vector of *any* curve in S . That is, $\nabla f(\mathbf{x})$ is *normal* to the surface at the point $\mathbf{x}$.



§7.3.4 Computing Gradients

In the example in the previous section, computing the gradient was somewhat difficult, requiring various expansions. We might wonder if it is possible to compute it more efficiently, in some orthogonal curvilinear coordinates.

In Cartesian coordinates, things are straightforward. To get the change $\mathbf{x} \mapsto \mathbf{x} + \mathbf{h}$, we could just get the change $x \mapsto x + h_1$, $y \mapsto y + h_2$, $z \mapsto z + h_3$, then

$$\begin{aligned} f(\mathbf{x} + \mathbf{h}) &= f(x + h_1, y + h_2, z + h_3) \\ &= f(\mathbf{x}) + \frac{\partial f}{\partial x} h_1 + \frac{\partial f}{\partial y} h_2 + \frac{\partial f}{\partial z} h_3 + o(\mathbf{h}) \\ &= f(\mathbf{x}) + \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{pmatrix} \cdot \mathbf{h} + o(\mathbf{h}), \end{aligned}$$

so

$$\nabla f = \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{pmatrix}.$$

Or, using suffix notation,

$$\nabla f = \mathbf{e}_i \frac{\partial f}{\partial x_i}, \quad \text{or} \quad [\nabla f]_i = \frac{\partial f}{\partial x_i}.$$

We can see that ∇ is a kind of vector differential operator.

So let's try and see how to apply this to other coordinate systems. Recall that in Cartesian coordinates, the line element was

$$d\mathbf{x} = dx\mathbf{e}_x + dy\mathbf{e}_y + dz\mathbf{e}_z = dx_i\mathbf{e}_i.$$

Also a function $f = f(x, y, z)$ has differential

$$df = \frac{\partial f}{\partial x_i} dx_i.$$

Then

$$\begin{aligned} \nabla f \cdot d\mathbf{x} &= \left(\mathbf{e}_i \frac{\partial f}{\partial x_i} \right) \cdot (\mathbf{e}_j dx_j) \\ &= \frac{\partial f}{\partial x_i} \underbrace{\mathbf{e}_i \cdot \mathbf{e}_j}_{\delta_{ij}} dx_j \\ &= \frac{\partial f}{\partial x_i} dx_i \\ &= df. \end{aligned}$$

So we have the following important result:

$$\boxed{\nabla f \cdot d\mathbf{x} = df}$$

This result is important because it is *coordinate independent*.

Remark (Abuse of Notation). So far we have jumped from writing $f(\mathbf{x}) \rightarrow f(x, y, z)$. Really, what we should have written is $F(x, y, z) = f(\mathbf{x}(x, y, z))$. This seems over the top in Cartesian coordinates, but would be far more proper to write $F(u, v, w) = f(\mathbf{x}(u, v, w))$.

Proposition 7.3.8

If (u, v, w) are orthogonal curvilinear coordinates and $f = f(u, v, w)$, then

$$\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w.$$

Proof. If $f = f(u, v, w)$ and $\mathbf{x} = \mathbf{x}(u, v, w)$, then

$$df = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv + \frac{\partial f}{\partial w} dw,$$

and

$$d\mathbf{x} = h_u du \mathbf{e}_u + h_v dv \mathbf{e}_v + h_w dw \mathbf{e}_w.$$

Using $df = \nabla f \cdot d\mathbf{x}$, and writing

$$\nabla f = [\nabla f]_u \mathbf{e}_u + [\nabla f]_v \mathbf{e}_v + [\nabla f]_w \mathbf{e}_w,$$

we find that

$$\frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv + \frac{\partial f}{\partial w} dw = h_u [\nabla f]_u du + h_v [\nabla f]_v dv + h_w [\nabla f]_w dw.$$

Since $\{du, dv, dw\}$ are linearly independent, we can compare components, and we get

$$[\nabla f]_u = \frac{1}{h_u} \frac{\partial f}{\partial u}, \quad [\nabla f]_v = \frac{1}{h_v} \frac{\partial f}{\partial v}, \quad [\nabla f]_w = \frac{1}{h_w} \frac{\partial f}{\partial w},$$

as required. $\square$

Example 7.3.9 (Gradient in Cylindrical/Spherical Coordinates)

In cylindrical polar coordinates (ρ, ϕ, z) , we have $h_\rho = 1$, $h_\phi = \rho$, and $h_z = 1$. So

$$\nabla f = \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi + \frac{\partial f}{\partial z} \mathbf{e}_z.$$

In spherical polar coordinates (r, θ, ϕ) , we have $h_r = 1$, $h_\theta = r$ and $h_\phi = r \sin \theta$. Then

$$\nabla f = \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \mathbf{e}_\phi.$$

§7.4 Integration over Lines, Surfaces and Volumes

We are now going to move on to the other (possibly more interesting) side of Calculus – integration! We are going to study how integration can be performed over various objects, beginning with lines (a rather simple generalization of ‘normal’ integration), followed by surfaces and volumes.

§7.4.1 Line Integrals

We will begin by defining the line integral.

Definition 7.4.1 (Line Integral)

For a vector field $\mathbf{F} = \mathbf{F}(\mathbf{x})$ and a piecewise smooth parameterized curve $t \mapsto \mathbf{x}(t)$ with $t \in [a, b]$, we define the **line integral**

$$\int_C \mathbf{F} \cdot d\mathbf{x} = \int_a^b \mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} dt.$$

Note that the direction is important (we integrate over ‘increasing t ’). If we wanted to integrate in the opposite direction over the curve, we would consider $\int_{-C}$.

Example 7.4.2

Consider the vector field

$$\mathbf{F} = \begin{pmatrix} x^2y \\ yz \\ 2zx \end{pmatrix},$$

and consider two curves connecting the origin to $(1, 1, 1)$,

$$C_1 : t \mapsto \begin{pmatrix} t \\ t \\ t \end{pmatrix}, \quad t \in [0, 1], \quad C_2 : t \mapsto \begin{pmatrix} t \\ t \\ t^2 \end{pmatrix}, \quad t \in [0, 1].$$

So

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \begin{pmatrix} t^3 \\ t^2 \\ 2t^2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} dt = \frac{5}{4}.$$

and

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{x} = \int_0^1 \begin{pmatrix} t^3 \\ t^2 \\ 2t^2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2t \end{pmatrix} dt = \frac{13}{10}.$$

We can see from the above example that the line integral between two points depends on the path taken.

Example 7.4.3

A particle at $\mathbf{x}$ experiences a force in cylindrical polars of

$$\mathbf{F}(\mathbf{x}) = z\rho\mathbf{e}_\phi.$$

We will calculate the work done by travelling along

$$C : t \mapsto \begin{pmatrix} a \cos t \\ a \sin t \\ t \end{pmatrix}, \quad t \in [0, 2\pi].$$

Recall the line element in cylindrical polars is

$$d\mathbf{x} = d\rho\mathbf{e}_\rho + \rho d\phi\mathbf{e}_\phi + dz\mathbf{e}_z.$$

So

$$\mathbf{F} \cdot d\mathbf{x} = z\rho^2 d\phi.$$

Also on the path $(\rho, \phi, z) = (a, t, t)$, so $(d\rho, d\phi, dz) = (0, dt, dt)$. So $\mathbf{F} \cdot d\mathbf{x} = a^2 t dt$. Finally,

$$\int_C \mathbf{F} \cdot d\mathbf{x} = a^2 \int_0^{2\pi} t dt = 2\pi^2 a^2.$$

A curve $\mathbf{x}(t)$ with $t \in [a, b]$ might be such that $\mathbf{x}(a) = \mathbf{x}(b)$, that is, the curve is *closed*. In this case, we write

$$\oint_C \mathbf{F} \cdot d\mathbf{x}.$$

Sometimes we call integrals of this form the **circulation** of $\mathbf{F}$ about C .

Example 7.4.4

Using C_1 and C_2 from [Example 7.4.2](#), we define the curve $C = C_1 - C_2$. Then

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = \int_{C_1} \mathbf{F} \cdot d\mathbf{x} - \int_{C_2} \mathbf{F} \cdot d\mathbf{x} = -\frac{2}{15}.$$

§7.4.2 Conservative Forces and Exact Differentials

We have seen how to interpret things like $\mathbf{F} \cdot d\mathbf{x}$ when they're inside an integral. This is another differential form, that is, in coordinates (u, v, w) ,

$$\mathbf{F} \cdot d\mathbf{x} = () du + () dv + () dw,$$

where there's *something* inside the brackets.

Definition 7.4.5 (Exactness)

We say that $\mathbf{F} \cdot d\mathbf{x}$ is **exact** if $\mathbf{F} \cdot d\mathbf{x} = df$, for some scalar function f .

Recall that $df = \nabla f \cdot d\mathbf{x}$, so $\mathbf{F} \cdot d\mathbf{x}$ is exact if and only if $\mathbf{F} = \nabla f$ for some scalar function f . We call such vector fields **conservative**.

For a vector field $\mathbf{F}$, we have $\mathbf{F} \cdot d\mathbf{x}$ is exact if and only if $\mathbf{F}$ is conservative.

Using properties such as $d(\alpha f + \beta g) = \alpha df + \beta dg$ for constant α, β and $d(fg) = g df + f dg$, usually it is easy to see if a differential form $\mathbf{F} \cdot d\mathbf{x}$ is exact.

Proposition 7.4.6

If θ is an exact differential form, then

$$\oint_C \theta = 0,$$

for any closed curve C .

Proof. If θ is exact, then $\theta = \nabla f \cdot d\mathbf{x}$ for some scalar f . If $C : t \mapsto \mathbf{x}(t)$ with

$t \in [a, b]$,

$$\begin{aligned}\oint_C \theta &= \oint_C \nabla f \cdot d\mathbf{x} \\ &= \int_a^b \nabla f(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt} dt \\ &= \int_a^b \frac{d}{dt} [f(\mathbf{x}(t))] dt \\ &= f(\mathbf{x}(b)) - f(\mathbf{x}(a)) \\ &= 0,\end{aligned}$$

if $\mathbf{x}(a) = \mathbf{x}(b)$. □

Equivalently, if $\mathbf{F}$ is conservative, then the circulation of $\mathbf{F}$ around any closed loop curve C vanishes, $\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$.

Also if $\mathbf{F}$ is conservative (and $\mathbf{F} \cdot d\mathbf{x}$ is exact), then the line integral between points $A = \mathbf{x}(a)$ and $B = \mathbf{x}(b)$ is *independent* of path.

Now let (u, v, w) be an arbitrary set of orthonormal curvilinear coordinates. Let

$$\mathbf{F} \cdot d\mathbf{x} = \theta = \underbrace{A(u, v, w)}_{\theta_1} du + \underbrace{B(u, v, w)}_{\theta_2} dv + \underbrace{C(u, v, w)}_{\theta_3} dw = \theta_i du_i.$$

A *necessary* condition for θ to be exact is

$$\frac{\partial \theta_i}{\partial u_j} = \frac{\partial \theta_j}{\partial u_i}, \quad \text{for each } i, j.$$

Indeed, if θ is exact, then $\theta = df$ for some f . Then

$$\theta = \frac{\partial f}{\partial u_i} du_i \iff \theta_i = \frac{\partial f}{\partial u_i}, \quad (*)$$

and so

$$\frac{\partial \theta_i}{\partial u_j} = \frac{\partial^2 f}{\partial u_j \partial u_i} = \frac{\partial^2 f}{\partial u_i \partial u_j} = \frac{\partial \theta_j}{\partial u_i}.$$

If a differential form satisfies $(*)$, we call it **closed**. So if θ is exact, then θ is closed.

The reverse implication is true if the domain $\Omega \subset \mathbb{R}^3$ on which θ is defined is simply-connected¹. We will be able to prove this later on, using Stokes' theorem.

Example 7.4.7

Consider $\theta = y dx - x dy$. We will check if it is closed. If so, we need to have

$$\frac{\partial y}{\partial y} = \frac{\partial (-x)}{\partial x},$$

which is clearly not the case, as $1 \neq -1$. Thus θ is not exact.

¹All closed loops in Ω can be 'continuously shrunk' to any point inside Ω without leaving it.

Example 7.4.8 (Silly Example)

We will compute the line integral

$$\oint_C 3x^2y \, dx + x^3 \, dy,$$

over the curve

$$C : t \mapsto \begin{pmatrix} \cos(\operatorname{Im}(\zeta(1/2 + it))) \\ \sin(\operatorname{Re}(\zeta(1/2 + it))) \\ 0 \end{pmatrix}, \quad t \in [\alpha_1, \alpha_{100}],$$

where α_1 and α_{100} are the 1st and 100th zero of $\zeta(1/2 + it)$, so that $\zeta(1/2 + i\alpha_1) = \zeta(1/2 + i\alpha_{100}) = 0$.

Now this differential form is exact, as $3x^2y \, dx + x^3 \, dy = d(x^3y)$, and C is a closed curve (both endpoints are $(1, 0, 0)$), thus $\oint_C 3x^2y \, dx + x^3 \, dy = 0$.

§7.4.3 Integration in $\mathbb{R}^2$

We want to integrate over some bounded region say $D \subset \mathbb{R}^2$. To do this, we will cover D with small, disjoint sets A_{ij} , each with area δA_{ij} , so that each of these are contained in a disk of radius $\epsilon > 0$. Let (x_i, y_j) be points contained in each A_{ij} .

Now define

$$\int_D f(\mathbf{x}) \, dA = \lim_{\epsilon \rightarrow 0} \sum_{i,j} f(x_i, y_j) \delta A_{ij}$$

We will say that the integral exists if it is independent of our choice of A_{ij} and choice of (x_i, y_j) .

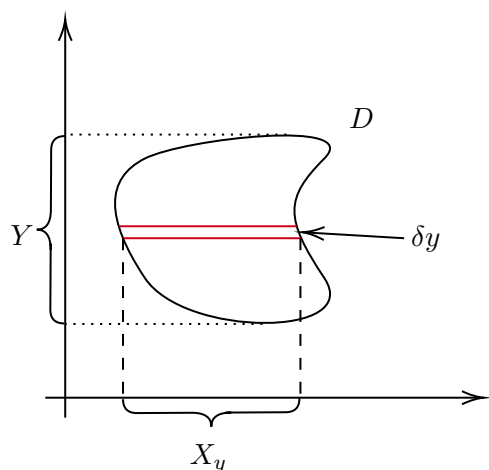
An obvious choice of partition would be to use rectangles with $\delta A_{ij} = \delta x \delta y$. We could sum over horizontal strips of width δy , then take the limit as $\delta x \rightarrow 0$. This gives

$$\delta y \int_{X_y} f(x, y) \, dx,$$

where $X_y = \{x \mid (x, y) \in D\}$. Summing over each such strip, we'd end up with

$$\int_D f(x, y) \, dA = \int_Y \left(\int_{X_y} f(x, y) \, dx \right) dy,$$

where Y is as in the diagram below.



If we instead sum over vertical strips, we get

$$\int_D f(x, y) \, dA = \int_X \left(\int_{Y_x} f(x, y) \, dy \right) dx.$$

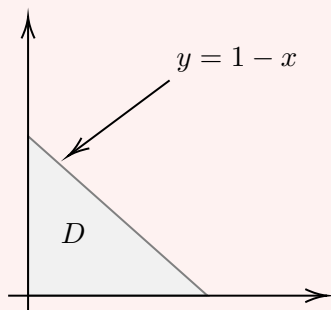
More concisely, we have

$$dA = dx \, dy = dy \, dx.$$

This is known as Fubini's theorem (with a few caveats that we won't worry about), which you may meet properly in a later course.

Example 7.4.9

Let D be a triangle with vertices at $(0, 0)$, $(1, 0)$ and $(0, 1)$.



If $f(x, y) = xy^2$, then if we integrate over horizontal strips,

$$\begin{aligned} \int_D f \, dA &= \int_0^1 \left(\int_0^{1-y} xy^2 \, dx \right) dy \\ &= \int_0^1 y^2 \left[\frac{1}{2}x^2 \right]_0^{1-y} dy \\ &= \int_0^1 \frac{1}{2}y^2(1-y)^2 dy = \frac{1}{60}. \end{aligned}$$

If we did the same with vertical strips, we could get

$$\int_D f \, dA = \int_0^1 \left(\int_0^{1-x} xy^2 \, dy \right) dx$$

$$= \int_0^1 x \left[\frac{1}{3} y^3 \right]_0^{1-x} dx = \frac{1}{60},$$

and they are the same.

A special case is when $f(x, y) = g(x)h(y)$ and D is a rectangle, $D = \{(x, y) \mid a \leq x \leq b, c \leq y \leq d\}$, then

$$\int_D f(x, y) \, dA = \left(\int_a^b g(x) \, dx \right) \left(\int_c^d h(y) \, dy \right).$$

It is often useful to introduce a change of variables to compute $\int_a^b f(x) \, dx$. If we introduce $x = x(u)$ with $x(\alpha) = a$ and $x(\beta) = b$, then

$$\int_a^b f(x) \, dx = \begin{cases} + \int_\alpha^\beta f(x(u)) \frac{dx}{du} \, du & \text{if } \beta > \alpha, \\ - \int_\beta^\alpha f(x(u)) \frac{dx}{du} \, du & \text{if } \alpha > \beta. \end{cases}$$

So if $I = [a, b]$ is the image under x of an interval I' , then

$$\int_I f(x) \, dx = \int_{I'} f(x(u)) \left| \frac{dx}{du} \right| \, du.$$

There is a similar formula in two dimensions.

Proposition 7.4.10

Let $x = x(u, v)$ and $y = y(u, v)$ be a smooth, invertible transformation with smooth inverse that maps the region D' in the (u, v) plane to the region D in the (x, y) -plane. Then

$$\int_D f(x, y) \, dx \, dy = \int_{D'} f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| \, du \, dv,$$

where

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{pmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{pmatrix}$$

is the *Jacobian*, often denoted by J .

The short version is that $dx \, dy = |J| \, du \, dv$.

Example 7.4.11

Using polar coordinates (ρ, ϕ) ,

$$x(\rho, \phi) = \rho \cos \phi, \quad y(\rho, \phi) = \rho \sin \phi.$$

Hence

$$J = \det \begin{pmatrix} \cos \phi & -\rho \sin \phi \\ \sin \phi & \rho \cos \phi \end{pmatrix} = \rho,$$

So if $D = \{(x, y) \mid x > 0, y > 0, x^2 + y^2 < R^2\}$, then $D' = \{(\rho, \phi) \mid 0 < \rho < R, 0 < \phi < \pi/2\}$, and

$$\int_D f(x, y) \, dx \, dy = \int_{D'} f(\rho \cos \phi, \rho \sin \phi) \rho \, d\rho \, d\phi,$$

so $dx dy = \rho d\rho d\phi$.

Taking $R \rightarrow \infty$, we get

$$\int_{x=0}^{\infty} \int_{y=0}^{\infty} f(x, y) dy dx = \int_{\phi=0}^{\pi/2} \int_{\rho=0}^{\infty} f(\rho \cos \phi, \rho \sin \phi) \rho d\rho d\phi.$$

Now consider $I = \int_0^{\infty} e^{-x^2} dx$. Using the previous result, we have

$$\begin{aligned} I^2 &= \int_0^{\infty} e^{-x^2} dx \cdot \int_0^{\infty} e^{-y^2} dy \\ &= \int_{x=0}^{\infty} \int_{y=0}^{\infty} e^{-x^2-y^2} dx dy \\ &= \int_{\phi=0}^{\pi/2} \left(\int_{\rho=0}^{\infty} e^{-\rho^2} \rho d\rho \right) d\phi \\ &= \frac{\pi}{2} \int_0^{\infty} \frac{d}{d\rho} \left(-\frac{1}{2} e^{-\rho^2} \right) d\rho = \frac{\pi}{4}. \end{aligned}$$

Thus $I = \frac{\sqrt{\pi}}{2}$.

§7.4.4 Integration in $\mathbb{R}^3$

To integrate over a region V in $\mathbb{R}^3$, we use similar ideas to that of the previous section. Let

$$\int_V f(\mathbf{x}) dV = \lim_{\epsilon \rightarrow 0} \sum_{i,j,k} f(x_i, y_j, z_k) \delta V_{ijk},$$

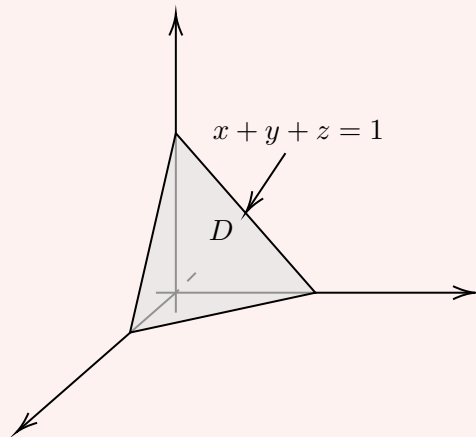
where the *volume element* satisfies

$$dV = dx dy dz.$$

We can do the integrals in any order (again Fubini's theorem).

Example 7.4.12

Consider the region V bound by planes $x + y + z = 1$ and $x = 0, y = 0$ and $z = 0$. We will find the volume (integrating 1 over the region).



We have

$$\begin{aligned}\int_V dV &= \int_{x=0}^1 \left(\int_{y=0}^{1-x} \left(\int_{z=0}^{1-x-y} dz \right) dy \right) dx \\ &= \int_{x=0}^1 \int_{y=0}^{1-x} (1-x-y) dy dx \\ &= \frac{1}{6}.\end{aligned}$$

If we wanted to compute say the center of mass, then assuming it had constant density say $\rho = 1$, the centre of mass $\mathbf{X}$ is

$$\mathbf{X} = \frac{1}{M} \int_V \rho \mathbf{x} dV = \frac{1}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

We can write down the change of variables for three dimensional integrals.

Proposition 7.4.13

Let $x = x(u, v, w)$, $y = y(u, v, w)$ and $z = z(u, v, w)$ be a continuously differentiable bijection with continuously differentiable inverse that maps the volume V' to V . Then

$$\int_V f(x, y, z) dx dy dz = \int_{V'} f(x(u, v, w), y(u, v, w), z(u, v, w)) |J| du dv dw,$$

where

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)}$$

Again the short version is $dx dy dz = |J| du dv dw$.

Example 7.4.14

In cylindrical polars $(u, v, w) = (\rho, \phi, z)$,

$$dV = \rho d\rho d\phi dz,$$

and $|J| = \rho$.

In spherical polars $(u, v, w) = (r, \theta, \phi)$,

$$dV = r^2 \sin \theta dr d\theta d\phi,$$

and $|J| = r^2 \sin \theta$.

Example 7.4.15

Suppose we wanted to calculate the volume of the ball of radius R , $V = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq R^2\}$.

We could try a ‘bare hands’ approach in Cartesian coordinates.

$$\begin{aligned}
 \int_V dV &= \int_{z=-R}^R \left(\int_{y=-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} \left(\int_{x=-\sqrt{R^2-z^2-y^2}}^{\sqrt{R^2-z^2-y^2}} dx \right) dy \right) dz \\
 &= \int_{z=-R}^R \left[\int_{y=-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} 2\sqrt{R^2-z^2-y^2} dy \right] dz \\
 &= \int_{z=-R}^R \left[y\sqrt{R^2-z^2-y^2} + (R^2-z^2) \tan^{-1} \left(\frac{y}{\sqrt{R^2-z^2-y^2}} \right) \right]_{-\sqrt{R^2-z^2}}^{\sqrt{R^2-z^2}} dz \\
 &= \int_{-R}^R \pi(R^2-z^2) dz = \frac{4\pi R^3}{3}.
 \end{aligned}$$

Alternatively, since we have even a limited amount of self respect, we can use spherical polars. We have $V' = \{(r, \theta, \phi) \mid 0 \leq r \leq R, 0 \leq \theta \leq \pi, 0 \leq \phi < 2\pi\}$. Then

$$\begin{aligned}
 \int_V dV &= \int_{\phi=0}^{2\pi} \left[\int_{\theta=0}^{\pi} \left[\int_{r=0}^R r^2 \sin \theta dr \right] d\theta \right] d\phi \\
 &= \int_{\theta=0}^{\pi} \frac{2\pi R^3}{3} \sin \theta d\theta = \frac{4\pi R^3}{3}.
 \end{aligned}$$

And this is clearly much nicer.

Example 7.4.16

Consider a ball of radius a with a cylinder of radius $b < a$ removed. The symmetry suggests that we use cylindrical polar coordinates. In Cartesian coordinates, we would have

$$V = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq a^2, x^2 + y^2 \geq b^2\},$$

but in cylindrical polars we would have

$$V' = \{(\rho, \phi, z) \mid b \leq \rho \leq a, 0 \leq z^2 + \rho^2 \leq a^2, 0 \leq \phi < 2\pi\}.$$

So we have

$$\begin{aligned}
 \int_V dV &= \int_{\rho=b}^a \left[\int_{\phi=0}^{2\pi} \left[\int_{z=-\sqrt{a^2-\rho^2}}^{\sqrt{a^2-\rho^2}} dz \right] d\phi \right] \rho d\rho \\
 &= 2\pi \int_b^a 2\rho \sqrt{a^2-\rho^2} d\rho \\
 &= \frac{4\pi}{3} (a^2 - b^2)^{3/2}.
 \end{aligned}$$

§7.4.5 Integration Over Surfaces

We have now integrated over curves, over regions of the plane, and over volumes. The remaining object to integrate over is a *surface* in $\mathbb{R}^3$, and this is where things get in-

teresting – surface integrals are the key ingredient in the integral theorems that we are building towards.

Surfaces and Their Boundaries

A two dimensional surface in $\mathbb{R}^3$ can be defined *implicitly* using a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, by setting

$$S = \{\mathbf{x} \in \mathbb{R}^3 \mid f(\mathbf{x}) = 0\}.$$

We saw before that ∇f is normal to the level sets of f , so the normal to S at $\mathbf{x}$ is parallel to $\nabla f(\mathbf{x})$. In analogy with what we did for curves, we say the surface is **regular** if $\nabla f(\mathbf{x}) \neq \mathbf{0}$ on S , which rules out any ‘spikes’ or ‘creases’.

Example 7.4.17 (The Sphere)

Take $f(\mathbf{x}) = x^2 + y^2 + z^2 - 1$, so that

$$S = \{(x, y, z) \mid x^2 + y^2 + z^2 - 1 = 0\}$$

is the unit sphere. Then $\nabla f = 2\mathbf{x}$, which is never zero on S , and is indeed normal to the sphere at $\mathbf{x}$.

Some surfaces have a **boundary**, which we write as ∂S . For example, the hemisphere

$$S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1, z \geq 0\}$$

has boundary the circle

$$\partial S = \{(x, y, z) \mid x^2 + y^2 = 1, z = 0\}.$$

In this course, a surface will either have no boundary at all, or its boundary will be made up of piecewise smooth curves. If S has no boundary (like the sphere), we say that S is a **closed** surface.

Just as it was useful to parameterize curves, it is useful to parameterize surfaces. We do this using *two* coordinates (u, v) , writing

$$S = \{\mathbf{x} = \mathbf{x}(u, v) \mid (u, v) \in D\},$$

where D is some region in the (u, v) -plane.

Example 7.4.18 (Parameterizing the Hemisphere)

For the upper unit hemisphere we can use the angles from spherical polar coordinates as our parameters, writing

$$S = \left\{ \mathbf{x}(\theta, \phi) = \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} \mid 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi < 2\pi \right\}.$$

Holding v fixed and varying u traces out a curve in S with tangent vector $\partial \mathbf{x} / \partial u$, and similarly for v . So $\partial \mathbf{x} / \partial u$ and $\partial \mathbf{x} / \partial v$ are both tangent to S , and (provided they are linearly independent) their cross product is *normal* to S . With this in mind, we say the parameterization is **regular** if

$$\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \neq \mathbf{0}$$

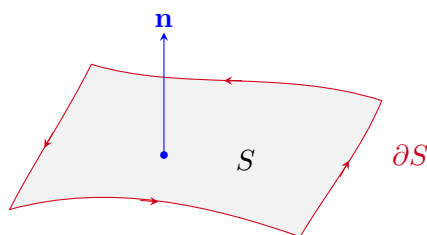
for all $(u, v) \in D$, and in that case we obtain a unit normal

$$\mathbf{n} = \frac{\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}}{\left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right|},$$

which varies smoothly over the (smooth parts of the) surface.

Of course, $-\mathbf{n}$ is also a perfectly good unit normal, so there is a choice to be made here. A surface for which a consistent choice of unit normal can be made over all of S is called **orientable**, and the choice of normal is an **orientation**².

An orientation of S also induces an orientation on the boundary curve ∂S : we make the convention that ∂S is traversed so that the normal vectors near the boundary point out of your *left-hand side* as you walk along it. Equivalently, this is the ‘right-hand rule’: if the thumb of your right hand points along $\mathbf{n}$, your fingers curl in the direction of ∂S .



The Area Element

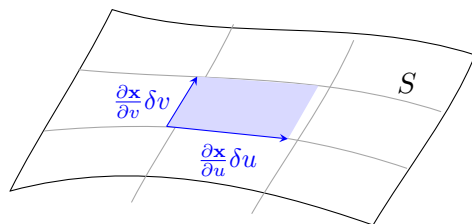
Now let's integrate. Take a regular parameterized surface $S = \{\mathbf{x}(u, v) \mid (u, v) \in D\}$. We might naively try to define the integral of f over S as the integral of $f(\mathbf{x}(u, v))$ over the region D , but this can't be right – a small patch of area $\delta u \delta v$ in D does not generally correspond to a patch of the same area on S , and different parameterizations would give different answers.

Instead, let's see what a small patch actually looks like. Perturbing the coordinates one at a time,

$$\begin{aligned}\mathbf{x}(u + \delta u, v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial u} \delta u + o(\delta u), \\ \mathbf{x}(u, v + \delta v) - \mathbf{x}(u, v) &= \frac{\partial \mathbf{x}}{\partial v} \delta v + o(\delta v).\end{aligned}$$

So to first order, the small rectangle $[u, u + \delta u] \times [v, v + \delta v]$ in D is mapped to a small *parallelogram* on S spanned by the vectors $\frac{\partial \mathbf{x}}{\partial u} \delta u$ and $\frac{\partial \mathbf{x}}{\partial v} \delta v$. Recalling from ‘Vectors & Matrices’ that the area of a parallelogram is the magnitude of the cross product of its sides, this patch has area

$$\left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| \delta u \delta v.$$



²Not every surface is orientable – the Möbius band famously has only one side, and if you slide a normal vector all the way around the band it comes back pointing the other way. We will only work with orientable surfaces in this course.

This tells us exactly what the correct ‘scaling factor’ is, and we make the following definition.

Definition 7.4.19 (Surface Integral)

Let $S = \{\mathbf{x}(u, v) \mid (u, v) \in D\}$ be a regular parameterized surface. The **scalar area element** on S is

$$dS = \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv,$$

and for $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ we define

$$\int_S f(\mathbf{x}) dS = \int_D f(\mathbf{x}(u, v)) \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv.$$

In particular, taking $f = 1$ gives the *area* of the surface S .

Example 7.4.20 (Area of a Hemisphere)

Let S be the hemisphere of radius R , parameterized by

$$\mathbf{x}(\theta, \phi) = \begin{pmatrix} R \sin \theta \cos \phi \\ R \sin \theta \sin \phi \\ R \cos \theta \end{pmatrix} = R \mathbf{e}_r, \quad 0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi < 2\pi.$$

Differentiating (or just recalling the scale factors for spherical polars),

$$\frac{\partial \mathbf{x}}{\partial \theta} = R \mathbf{e}_\theta, \quad \frac{\partial \mathbf{x}}{\partial \phi} = R \sin \theta \mathbf{e}_\phi,$$

and since $\mathbf{e}_\theta \times \mathbf{e}_\phi = \mathbf{e}_r$ is a unit vector,

$$dS = R^2 \sin \theta d\theta d\phi.$$

The area of the hemisphere is then

$$\int_S dS = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} R^2 \sin \theta d\theta d\phi = 2\pi R^2,$$

as expected.

As with curves, we should pause and check that our definition doesn’t depend on the parameterization we happened to pick.

Proposition 7.4.21

The integral $\int_S f(\mathbf{x}) dS$ is independent of the (regular) parameterization of S used.

Proof. Suppose $\mathbf{x} = \mathbf{x}(u, v)$ for $(u, v) \in D$ and $\mathbf{x} = \tilde{\mathbf{x}}(\tilde{u}, \tilde{v})$ for $(\tilde{u}, \tilde{v}) \in \tilde{D}$ are two parameterizations of S . Then there is a relationship between the coordinates,

$\mathbf{x}(u, v) = \tilde{\mathbf{x}}(\tilde{u}(u, v), \tilde{v}(u, v))$, and by the chain rule

$$\begin{aligned} \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} &= \left(\frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \frac{\partial \tilde{u}}{\partial u} + \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \frac{\partial \tilde{v}}{\partial u} \right) \times \left(\frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \frac{\partial \tilde{u}}{\partial v} + \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \frac{\partial \tilde{v}}{\partial v} \right) \\ &= \left(\frac{\partial \tilde{u}}{\partial u} \frac{\partial \tilde{v}}{\partial v} - \frac{\partial \tilde{u}}{\partial v} \frac{\partial \tilde{v}}{\partial u} \right) \left(\frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \times \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \right) \\ &= \frac{\partial(\tilde{u}, \tilde{v})}{\partial(u, v)} \left(\frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \times \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \right), \end{aligned}$$

where the cross terms vanished since $\mathbf{a} \times \mathbf{a} = \mathbf{0}$. So the two normal vectors differ exactly by a factor of the Jacobian of the change of coordinates $(u, v) \mapsto (\tilde{u}, \tilde{v})$.

Then using the change of variables formula for double integrals,

$$\begin{aligned} \int_{\tilde{D}} f(\tilde{\mathbf{x}}(\tilde{u}, \tilde{v})) \left| \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \times \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \right| d\tilde{u} d\tilde{v} &= \int_D f(\mathbf{x}(u, v)) \left| \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{u}} \times \frac{\partial \tilde{\mathbf{x}}}{\partial \tilde{v}} \right| \left| \frac{\partial(\tilde{u}, \tilde{v})}{\partial(u, v)} \right| du dv \\ &= \int_D f(\mathbf{x}(u, v)) \left| \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right| du dv, \end{aligned}$$

so both parameterizations give the same integral. $\square$

Flux Integrals

For vector fields, the natural thing to integrate over a surface involves the *normal* direction.

Definition 7.4.22 (Vector Area Element and Flux)

On a regular parameterized surface S with unit normal $\mathbf{n}$, the **vector area element** is

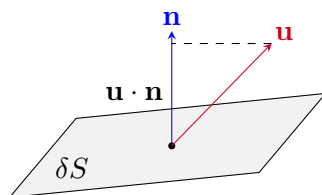
$$d\mathbf{S} = \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} du dv = \mathbf{n} dS.$$

For a vector field $\mathbf{F}$, the **flux** of $\mathbf{F}$ through S is

$$\int_S \mathbf{F} \cdot d\mathbf{S} = \int_S \mathbf{F} \cdot \mathbf{n} dS.$$

Note that the flux depends on the orientation: swapping $\mathbf{n}$ for $-\mathbf{n}$ flips the sign. When S is a closed surface, the convention is to take $\mathbf{n}$ to be the *outward* pointing normal.

To see why this is a sensible quantity to consider, suppose that a fluid has velocity $\mathbf{u}(\mathbf{x})$, and we want to know how much fluid crosses a surface S per unit time. Over a small patch δS of the surface, only the component of $\mathbf{u}$ *normal* to the patch actually carries fluid across it – flow tangent to the surface just slides along it. So in time δt the volume of fluid crossing the patch is $(\mathbf{u} \cdot \mathbf{n})\delta S \delta t$, and the total volume crossing S per unit time is exactly the flux $\int_S \mathbf{u} \cdot d\mathbf{S}$.



Example 7.4.23 (Flux Through a Sphere)

Let S be the sphere of radius R centred at the origin with outward normal, and let $\mathbf{F}(\mathbf{x}) = \mathbf{x}$. Using spherical polars as before,

$$d\mathbf{S} = \mathbf{e}_r R^2 \sin \theta \, d\theta \, d\phi,$$

and on the sphere $\mathbf{F} = R\mathbf{e}_r$, so $\mathbf{F} \cdot d\mathbf{S} = R^3 \sin \theta \, d\theta \, d\phi$. Then

$$\int_S \mathbf{F} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} R^3 \sin \theta \, d\theta \, d\phi = 4\pi R^3,$$

which is notably three times the volume of the ball – a fact that will stop looking like a coincidence once we meet the divergence theorem.

Example 7.4.24 (Flux Through a Hemisphere)

Let S be the upper hemisphere of radius R with upward-pointing normal, and take $\mathbf{F} = z\mathbf{e}_z$. On S we have $z = R \cos \theta$ and $\mathbf{e}_z \cdot \mathbf{e}_r = \cos \theta$, so

$$\begin{aligned} \int_S \mathbf{F} \cdot d\mathbf{S} &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi/2} (R \cos \theta)(\cos \theta) R^2 \sin \theta \, d\theta \, d\phi \\ &= 2\pi R^3 \left[-\frac{\cos^3 \theta}{3} \right]_0^{\pi/2} = \frac{2\pi R^3}{3}. \end{aligned}$$

§7.5 Div, Curl and the Laplacian

So far the only way we have of differentiating fields is the gradient, which takes a scalar field and produces a vector field. In this section we complete the toolkit: we will meet two ways of differentiating a *vector* field – the divergence and the curl – along with a natural second-order operator, the Laplacian. These operators are the stars of the rest of the course.

§7.5.1 The Divergence and Curl

Recall that in Cartesian coordinates the gradient operator can be written as

$$\nabla = \mathbf{e}_i \frac{\partial}{\partial x_i},$$

a ‘vector of derivatives’. Treating ∇ formally as a vector, there are two natural products we can form with a vector field $\mathbf{F}$: a dot product and a cross product.

Definition 7.5.1 (Divergence)

For a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, we define the **divergence** of $\mathbf{F}$ to be $\nabla \cdot \mathbf{F}$, which in Cartesian coordinates is

$$\nabla \cdot \mathbf{F} = \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \frac{\partial F_3}{\partial x_3}.$$

Note that the divergence of a vector field is a *scalar* field. To see where the Cartesian expression comes from, note that in Cartesian coordinates the basis vectors are constant, so

$$\nabla \cdot \mathbf{F} = \left(\mathbf{e}_i \frac{\partial}{\partial x_i} \right) \cdot (F_j \mathbf{e}_j) = \underbrace{(\mathbf{e}_i \cdot \mathbf{e}_j)}_{\delta_{ij}} \frac{\partial F_j}{\partial x_i} = \frac{\partial F_i}{\partial x_i}.$$

This warning matters: in curvilinear coordinates the basis vectors themselves depend on position and must also be differentiated, so the divergence is *not* just ‘ $\partial F_u / \partial u + \dots$ ’. We will sort out what happens in general coordinates at the end of this section.

Definition 7.5.2 (Curl)

For a vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, we define the **curl** of $\mathbf{F}$ to be $\nabla \times \mathbf{F}$, which in Cartesian coordinates has components

$$[\nabla \times \mathbf{F}]_i = \epsilon_{ijk} \frac{\partial F_k}{\partial x_j}.$$

The curl of a vector field is another vector field. Again the Cartesian expression follows from a short computation:

$$\nabla \times \mathbf{F} = \left(\mathbf{e}_j \frac{\partial}{\partial x_j} \right) \times (F_k \mathbf{e}_k) = (\mathbf{e}_j \times \mathbf{e}_k) \frac{\partial F_k}{\partial x_j} = \epsilon_{ijk} \frac{\partial F_k}{\partial x_j} \mathbf{e}_i.$$

It is often convenient to remember this as a ‘formal determinant’,

$$\nabla \times \mathbf{F} = \det \begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ F_1 & F_2 & F_3 \end{pmatrix}.$$

Unlike the gradient and divergence, the curl is genuinely a three-dimensional construction – it relies on the cross product, which doesn’t generalise naively to other dimensions.

Let’s compute some simple examples, which will also suggest how to *think* about these operators.

Example 7.5.3 (A Radial Field)

Let $\mathbf{F}(\mathbf{x}) = \mathbf{x}$. Then in Cartesian coordinates,

$$\nabla \cdot \mathbf{F} = \frac{\partial x_i}{\partial x_i} = \delta_{ii} = 3, \quad [\nabla \times \mathbf{F}]_i = \epsilon_{ijk} \frac{\partial x_k}{\partial x_j} = \epsilon_{ijk} \delta_{kj} = 0.$$

So a field pointing radially outwards has positive divergence and no curl.

Example 7.5.4 (A Rotating Field)

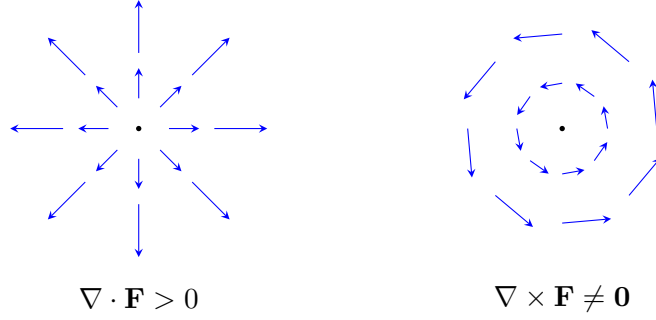
Let $\mathbf{F}(\mathbf{x}) = (-y, x, 0)$, the velocity field of a rigid rotation about the z -axis. Then

$$\nabla \cdot \mathbf{F} = 0, \quad \nabla \times \mathbf{F} = \det \begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \partial_x & \partial_y & \partial_z \\ -y & x & 0 \end{pmatrix} = 2\mathbf{e}_3.$$

So a rotating field has no divergence, but its curl points along the axis of rotation

(with magnitude twice the angular speed).

These examples are entirely representative. If we think of $\mathbf{F}$ as the velocity field of a fluid, then a positive divergence at a point means the fluid is (locally) flowing *away* from that point, so the point acts like a *source*; negative divergence gives a *sink*. A field with $\nabla \cdot \mathbf{F} = 0$ everywhere models an incompressible fluid. Meanwhile the curl measures the *local rotation* of the fluid: if you dropped a tiny paddle wheel into the flow at $\mathbf{x}$ with its axis along $\hat{\mathbf{k}}$, it would spin at a rate proportional to $\hat{\mathbf{k}} \cdot (\nabla \times \mathbf{F})$. We will be able to make both of these statements precise once we have the integral theorems.



Finally, we can compose our operators to get a second-order operator.

Definition 7.5.5 (Laplacian)

For a scalar field $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, the **Laplacian** of f is

$$\nabla^2 f = \nabla \cdot (\nabla f),$$

which in Cartesian coordinates is

$$\nabla^2 f = \frac{\partial^2 f}{\partial x_i \partial x_i} = \frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \frac{\partial^2 f}{\partial x_3^2}.$$

The Laplacian shows up all over mathematical physics – in the heat equation, the wave equation, Laplace’s equation and Poisson’s equation, some of which we will meet later in these notes.

We can also apply the Laplacian to a *vector* field. In Cartesian coordinates we simply apply it to each component,

$$\nabla^2 \mathbf{F} = (\nabla^2 F_i) \mathbf{e}_i,$$

but this recipe fails in curvilinear coordinates (where the basis vectors move). The coordinate-independent statement is the following.

Proposition 7.5.6

In Cartesian coordinates, $\nabla^2 \mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$.

Proof. We compute the curl of the curl in index notation:

$$[\nabla \times (\nabla \times \mathbf{F})]_i = \epsilon_{ijk} \frac{\partial}{\partial x_j} \left(\epsilon_{klm} \frac{\partial F_m}{\partial x_l} \right)$$

$$\begin{aligned}
&= \epsilon_{kij}\epsilon_{klm} \frac{\partial^2 F_m}{\partial x_j \partial x_l} \\
&= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}) \frac{\partial^2 F_m}{\partial x_j \partial x_l} \\
&= \frac{\partial}{\partial x_i} \left(\frac{\partial F_j}{\partial x_j} \right) - \frac{\partial^2 F_i}{\partial x_j \partial x_j} \\
&= [\nabla(\nabla \cdot \mathbf{F})]_i - \nabla^2 F_i,
\end{aligned}$$

using the identity $\epsilon_{kij}\epsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$ from ‘Vectors & Matrices’. Rearranging gives the result. $\square$

Since the right-hand side $\nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F})$ makes sense in any coordinate system, we take it as the *definition* of $\nabla^2 \mathbf{F}$ in general.

§7.5.2 Identities in Index Notation

Just as ordinary differentiation has a product rule, our new operators satisfy a family of ‘Leibniz identities’. These are used constantly, and proving a few of them is excellent practice with the index notation.

Proposition 7.5.7 (Leibniz Identities)

For scalar fields f, g and vector fields $\mathbf{F}, \mathbf{G}$, we have

- (i) $\nabla(fg) = f\nabla g + g\nabla f$,
- (ii) $\nabla \cdot (f\mathbf{F}) = (\nabla f) \cdot \mathbf{F} + f(\nabla \cdot \mathbf{F})$,
- (iii) $\nabla \times (f\mathbf{F}) = (\nabla f) \times \mathbf{F} + f(\nabla \times \mathbf{F})$,
- (iv) $\nabla \cdot (\mathbf{F} \times \mathbf{G}) = (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G})$,
- (v) $\nabla \times (\mathbf{F} \times \mathbf{G}) = \mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}$,
- (vi) $\nabla(\mathbf{F} \cdot \mathbf{G}) = \mathbf{F} \times (\nabla \times \mathbf{G}) + \mathbf{G} \times (\nabla \times \mathbf{F}) + (\mathbf{F} \cdot \nabla)\mathbf{G} + (\mathbf{G} \cdot \nabla)\mathbf{F}$.

Here $\mathbf{F} \cdot \nabla$ denotes the differential operator $F_j \frac{\partial}{\partial x_j}$, so that

$$[(\mathbf{F} \cdot \nabla)\mathbf{G}]_i = F_j \frac{\partial G_i}{\partial x_j}.$$

Be careful to distinguish the operator $\mathbf{F} \cdot \nabla$ from the scalar field $\nabla \cdot \mathbf{F}$ – they are certainly not the same thing!

Proof. All of these identities are statements about fields, so they can be checked in any coordinate system we like – and we will naturally use Cartesian coordinates, where the basis vectors are constant. We prove (ii), (iv) and (v); the others are entirely similar (and are good exercises).

For (ii), by the ordinary product rule,

$$\nabla \cdot (f\mathbf{F}) = \frac{\partial}{\partial x_i} (fF_i) = \frac{\partial f}{\partial x_i} F_i + f \frac{\partial F_i}{\partial x_i} = (\nabla f) \cdot \mathbf{F} + f(\nabla \cdot \mathbf{F}).$$

For (iv),

$$\begin{aligned}
 \nabla \cdot (\mathbf{F} \times \mathbf{G}) &= \frac{\partial}{\partial x_i} (\epsilon_{ijk} F_j G_k) \\
 &= \epsilon_{ijk} \frac{\partial F_j}{\partial x_i} G_k + \epsilon_{ijk} F_j \frac{\partial G_k}{\partial x_i} \\
 &= \left(\epsilon_{kij} \frac{\partial F_j}{\partial x_i} \right) G_k - F_j \left(\epsilon_{jik} \frac{\partial G_k}{\partial x_i} \right) \\
 &= (\nabla \times \mathbf{F}) \cdot \mathbf{G} - \mathbf{F} \cdot (\nabla \times \mathbf{G}),
 \end{aligned}$$

where we just cycled and swapped indices of ϵ , picking up a sign in the second term.

For (v), we use the ϵ - δ identity:

$$\begin{aligned}
 [\nabla \times (\mathbf{F} \times \mathbf{G})]_i &= \epsilon_{ijk} \frac{\partial}{\partial x_j} (\epsilon_{klm} F_l G_m) \\
 &= \epsilon_{kij} \epsilon_{klm} \left(\frac{\partial F_l}{\partial x_j} G_m + F_l \frac{\partial G_m}{\partial x_j} \right) \\
 &= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \left(\frac{\partial F_l}{\partial x_j} G_m + F_l \frac{\partial G_m}{\partial x_j} \right) \\
 &= \frac{\partial F_i}{\partial x_j} G_j + F_i \frac{\partial G_j}{\partial x_j} - \frac{\partial F_j}{\partial x_j} G_i - F_j \frac{\partial G_i}{\partial x_j} \\
 &= [(\mathbf{G} \cdot \nabla) \mathbf{F}]_i + [\mathbf{F}(\nabla \cdot \mathbf{G})]_i - [\mathbf{G}(\nabla \cdot \mathbf{F})]_i - [(\mathbf{F} \cdot \nabla) \mathbf{G}]_i,
 \end{aligned}$$

as required. $\square$

There are also two identities involving *second* derivatives which are so important that they deserve their own proposition. They say, slightly cryptically, that ‘curl grad is zero’ and ‘div curl is zero’.

Proposition 7.5.8

For any (twice continuously differentiable) scalar field f and vector field $\mathbf{F}$,

$$\nabla \times (\nabla f) = \mathbf{0} \quad \text{and} \quad \nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

Proof. Both proofs use the same trick: a symmetric object contracted with an antisymmetric one gives zero. For the first,

$$[\nabla \times (\nabla f)]_i = \epsilon_{ijk} \frac{\partial}{\partial x_j} \frac{\partial f}{\partial x_k} = \epsilon_{ijk} \frac{\partial^2 f}{\partial x_j \partial x_k}.$$

Now ϵ_{ijk} is antisymmetric under swapping $j \leftrightarrow k$, while $\frac{\partial^2 f}{\partial x_j \partial x_k}$ is symmetric (partial derivatives commute). Relabelling the dummy indices,

$$\epsilon_{ijk} \frac{\partial^2 f}{\partial x_j \partial x_k} = \epsilon_{ikj} \frac{\partial^2 f}{\partial x_k \partial x_j} = -\epsilon_{ijk} \frac{\partial^2 f}{\partial x_j \partial x_k},$$

so the expression equals its own negative and must vanish. Identically,

$$\nabla \cdot (\nabla \times \mathbf{F}) = \epsilon_{ijk} \frac{\partial^2 F_k}{\partial x_i \partial x_j} = 0$$

by antisymmetry of ϵ_{ijk} in (i, j) . □

This proposition gives us a useful vocabulary. We say $\mathbf{F}$ is **irrotational** if $\nabla \times \mathbf{F} = \mathbf{0}$, and **solenoidal** if $\nabla \cdot \mathbf{F} = 0$. The proposition then says:

- (i) every conservative field $\mathbf{F} = \nabla f$ is irrotational;
- (ii) every field of the form $\mathbf{F} = \nabla \times \mathbf{A}$ is solenoidal. We call such an $\mathbf{A}$ a **vector potential** for $\mathbf{F}$.

Remarkably, both statements have converses if the domain of $\mathbf{F}$ is topologically nice enough: an irrotational field on a simply connected domain is conservative (we will prove this using Stokes' theorem), and a solenoidal field on a suitable domain (one where every closed surface can be shrunk to a point, such as all of $\mathbb{R}^3$) has a vector potential.

In practice, checking $\nabla \times \mathbf{F} = \mathbf{0}$ is by far the easiest way to show a field is conservative, so this is a result worth internalising.

§7.5.3 Div and Curl in Orthogonal Curvilinear Coordinates

We now settle the question of what the divergence and curl look like in a general orthogonal curvilinear coordinate system (u, v, w) , with scale factors h_u, h_v, h_w and right-handed orthonormal basis $\{\mathbf{e}_u, \mathbf{e}_v, \mathbf{e}_w\}$. The subtlety, as mentioned before, is that the basis vectors vary from point to point, and naive componentwise differentiation is wrong.

Rather than grinding through derivatives of basis vectors, there is a rather slick route using two small observations. First, recall that

$$\nabla u = \frac{1}{h_u} \mathbf{e}_u,$$

which is just the formula for the gradient in curvilinear coordinates applied to the function $f = u$. Second, since the basis is right-handed and orthonormal, $\mathbf{e}_u = \mathbf{e}_v \times \mathbf{e}_w$, so

$$\mathbf{e}_u = h_v h_w \nabla v \times \nabla w,$$

and similarly for cyclic permutations of (u, v, w) .

Proposition 7.5.9 (Divergence in Curvilinear Coordinates)

If $\mathbf{F} = F_u \mathbf{e}_u + F_v \mathbf{e}_v + F_w \mathbf{e}_w$, then

$$\nabla \cdot \mathbf{F} = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} (h_v h_w F_u) + \frac{\partial}{\partial v} (h_u h_w F_v) + \frac{\partial}{\partial w} (h_u h_v F_w) \right].$$

Proof. Consider the first term $F_u \mathbf{e}_u = (F_u h_v h_w) \nabla v \times \nabla w$. Using the Leibniz identity for $\nabla \cdot (f \mathbf{G})$,

$$\nabla \cdot (F_u \mathbf{e}_u) = \nabla (F_u h_v h_w) \cdot (\nabla v \times \nabla w) + F_u h_v h_w \nabla \cdot (\nabla v \times \nabla w).$$

But by the identity for the divergence of a cross product, along with ‘curl grad is zero’,

$$\nabla \cdot (\nabla v \times \nabla w) = (\nabla \times \nabla v) \cdot \nabla w - \nabla v \cdot (\nabla \times \nabla w) = 0,$$

so the second term dies. For the first term, $\nabla v \times \nabla w = \mathbf{e}_u / (h_v h_w)$, and taking the $\mathbf{e}_u$ component of the gradient,

$$\nabla(F_u h_v h_w) \cdot \frac{\mathbf{e}_u}{h_v h_w} = \frac{1}{h_u h_v h_w} \frac{\partial}{\partial u} (h_v h_w F_u).$$

Adding the corresponding expressions for the $\mathbf{e}_v$ and $\mathbf{e}_w$ terms (which follow by cyclic symmetry) gives the result. $\square$

Proposition 7.5.10 (Curl in Curvilinear Coordinates)

With $\mathbf{F}$ as above,

$$\nabla \times \mathbf{F} = \frac{1}{h_u h_v h_w} \det \begin{pmatrix} h_u \mathbf{e}_u & h_v \mathbf{e}_v & h_w \mathbf{e}_w \\ \frac{\partial}{\partial u} & \frac{\partial}{\partial v} & \frac{\partial}{\partial w} \\ h_u F_u & h_v F_v & h_w F_w \end{pmatrix}.$$

Proof. This time write $F_u \mathbf{e}_u = (h_u F_u) \nabla u$. Then by the Leibniz identity for the curl of a product, and ‘curl grad is zero’ once more,

$$\nabla \times (F_u \mathbf{e}_u) = \nabla(h_u F_u) \times \nabla u + (h_u F_u) \underbrace{\nabla \times \nabla u}_{=0} = \nabla(h_u F_u) \times \frac{\mathbf{e}_u}{h_u}.$$

Expanding the gradient in the curvilinear basis,

$$\begin{aligned} \nabla(h_u F_u) \times \frac{\mathbf{e}_u}{h_u} &= \left(\frac{1}{h_u} \frac{\partial(h_u F_u)}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial(h_u F_u)}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial(h_u F_u)}{\partial w} \mathbf{e}_w \right) \times \frac{\mathbf{e}_u}{h_u} \\ &= \frac{1}{h_w h_u} \frac{\partial(h_u F_u)}{\partial w} \mathbf{e}_v - \frac{1}{h_u h_v} \frac{\partial(h_u F_u)}{\partial v} \mathbf{e}_w, \end{aligned}$$

using $\mathbf{e}_v \times \mathbf{e}_u = -\mathbf{e}_w$ and $\mathbf{e}_w \times \mathbf{e}_u = \mathbf{e}_v$. These are precisely the terms involving $h_u F_u$ in the expansion of the determinant, and the other columns follow by cyclic symmetry. $\square$

Corollary 7.5.11 (Laplacian in Curvilinear Coordinates)

For a scalar field f ,

$$\nabla^2 f = \frac{1}{h_u h_v h_w} \left[\frac{\partial}{\partial u} \left(\frac{h_v h_w}{h_u} \frac{\partial f}{\partial u} \right) + \frac{\partial}{\partial v} \left(\frac{h_u h_w}{h_v} \frac{\partial f}{\partial v} \right) + \frac{\partial}{\partial w} \left(\frac{h_u h_v}{h_w} \frac{\partial f}{\partial w} \right) \right].$$

Proof. Apply the divergence formula to $\nabla f = \frac{1}{h_u} \frac{\partial f}{\partial u} \mathbf{e}_u + \frac{1}{h_v} \frac{\partial f}{\partial v} \mathbf{e}_v + \frac{1}{h_w} \frac{\partial f}{\partial w} \mathbf{e}_w$. $\square$

It is worth recording the two special cases that get used over and over.

Example 7.5.12 (Cylindrical Polar Coordinates)

In cylindrical polars (ρ, ϕ, z) , where $(h_\rho, h_\phi, h_z) = (1, \rho, 1)$,

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho F_\rho) + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}, \\ \nabla \times \mathbf{F} &= \frac{1}{\rho} \det \begin{pmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_\rho & \rho F_\phi & F_z \end{pmatrix}, \\ \nabla^2 f &= \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.\end{aligned}$$

Example 7.5.13 (Spherical Polar Coordinates)

In spherical polars (r, θ, ϕ) , where $(h_r, h_\theta, h_\phi) = (1, r, r \sin \theta)$,

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}, \\ \nabla \times \mathbf{F} &= \frac{1}{r^2 \sin \theta} \det \begin{pmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & r F_\theta & r \sin \theta F_\phi \end{pmatrix}, \\ \nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}.\end{aligned}$$

Example 7.5.14

Let's put these formulas to work.

- (i) Consider $f(\mathbf{x}) = 1/r$ in spherical polars, away from the origin. Then

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \left(-\frac{1}{r^2} \right) \right) = 0,$$

so $1/r$ is a solution of Laplace's equation on $\mathbb{R}^3 \setminus \{0\}$ – a fact which underpins much of the theory of gravity and electrostatics, as we will see.

- (ii) Consider the rotating field from before, which in cylindrical polars is $\mathbf{F} = \rho \mathbf{e}_\phi$. Then

$$\nabla \times \mathbf{F} = \frac{1}{\rho} \det \begin{pmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & \rho^2 & 0 \end{pmatrix} = \frac{1}{\rho} \cdot 2\rho \mathbf{e}_z = 2\mathbf{e}_z,$$

matching our earlier Cartesian computation.

§7.6 The Integral Theorems

We have arrived at the heart of the course. The fundamental theorem of calculus says that

$$\int_a^b f'(x) \, dx = f(b) - f(a),$$

that is, integrating the derivative of a function over an interval only requires knowing the function on the *boundary* of the interval. The integral theorems of vector calculus

are all statements of exactly this shape: the integral of some derivative of a field over a region equals an integral of the field itself over the *boundary* of the region. There is one theorem for each type of region – planar regions, surfaces, and volumes.

§7.6.1 Green's Theorem

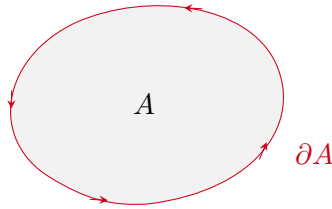
We begin in the plane.

Theorem 7.6.1 (Green's Theorem)

Let $P = P(x, y)$ and $Q = Q(x, y)$ be continuously differentiable on a planar region $A \cup \partial A$, where the boundary ∂A is piecewise smooth. Then

$$\oint_{\partial A} P \, dx + Q \, dy = \int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA,$$

where ∂A is traversed so that A lies to the *left* (anticlockwise, for a simple region).



We will prove Green's theorem (along with the other integral theorems) in §7.6.4, but we can quickly convince ourselves it's plausible by checking it on a rectangle $A = [a, b] \times [c, d]$. There,

$$\begin{aligned} \int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, dA &= \int_c^d \left(\int_a^b \frac{\partial Q}{\partial x} \, dx \right) \, dy - \int_a^b \left(\int_c^d \frac{\partial P}{\partial y} \, dy \right) \, dx \\ &= \int_c^d [Q(b, y) - Q(a, y)] \, dy + \int_a^b [P(x, c) - P(x, d)] \, dx, \end{aligned}$$

using the fundamental theorem of calculus on each inner integral. But this is exactly $\oint_{\partial A} P \, dx + Q \, dy$: the four terms are the contributions of the four sides of the rectangle, traversed anticlockwise (say, the P integral along the top edge runs from $x = b$ back to $x = a$, giving the $-P(x, d)$ term). One can then imagine building up an arbitrary region out of many small rectangles – the boundary contributions of adjacent rectangles cancel along shared edges, leaving only the outer boundary.

A cute application of Green's theorem is computing areas using only line integrals.

Example 7.6.2 (Area of an Ellipse)

Take $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$, so Green's theorem says

$$\text{area}(A) = \int_A \, dA = \frac{1}{2} \oint_{\partial A} x \, dy - y \, dx.$$

Now let A be the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1$, with ∂A parameterized by $t \mapsto (a \cos t, b \sin t)$

for $t \in [0, 2\pi]$. Then

$$\text{area}(A) = \frac{1}{2} \int_0^{2\pi} (ab \cos^2 t + ab \sin^2 t) \, dt = \pi ab.$$

§7.6.2 Stokes' Theorem

Green's theorem relates a double integral over a flat region to an integral over its boundary. Stokes' theorem is the version for a *curved* surface in $\mathbb{R}^3$, and it tells us that the derivative being integrated is precisely the curl.

Theorem 7.6.3 (Stokes' Theorem)

Let $\mathbf{F}$ be a continuously differentiable vector field, and let S be an orientable, piecewise regular surface with piecewise smooth boundary ∂S . Then

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{x},$$

where the orientation of ∂S is compatible with the choice of normal on S , as described previously (the normal is on your left as you traverse ∂S ; equivalently, use the right-hand rule).

Again, this is the fundamental theorem of calculus in disguise: the integral of the 'derivative' $\nabla \times \mathbf{F}$ over S is determined by the values of $\mathbf{F}$ on the boundary ∂S . The only difference from the one dimensional case is that the boundary is now a curve rather than a pair of points, so 'evaluating on the boundary' means integrating around it.

An immediate consequence: if S is a *closed* surface then $\partial S = \emptyset$, so

$$\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$$

for any closed surface S . Notice also that the theorem implies the (perhaps surprising) fact that $\int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ depends only on the boundary curve ∂S – any two surfaces sharing the same boundary give the same integral.

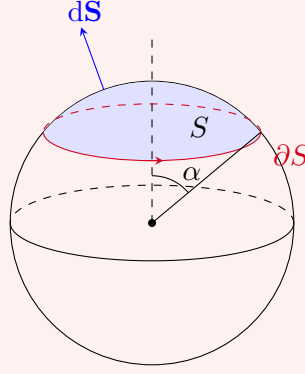
Example 7.6.4 (Verifying Stokes' Theorem on a Spherical Cap)

Let S be the cap of the unit sphere given in spherical polars by

$$S = \{\mathbf{x}(\theta, \phi) = \mathbf{e}_r \mid 0 \leq \theta \leq \alpha, 0 \leq \phi < 2\pi\},$$

and consider

$$\mathbf{F}(\mathbf{x}) = \begin{pmatrix} -x^2 y \\ 0 \\ 0 \end{pmatrix}, \quad \text{so that} \quad \nabla \times \mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ x^2 \end{pmatrix}.$$



On S we have $d\mathbf{S} = \mathbf{e}_r \sin \theta \, d\theta \, d\phi$ and $x = \sin \theta \cos \phi$, and since $\mathbf{e}_z \cdot \mathbf{e}_r = \cos \theta$,

$$\begin{aligned} \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} &= \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\alpha} (\sin \theta \cos \phi)^2 \cos \theta \sin \theta \, d\theta \, d\phi \\ &= \pi \int_0^{\alpha} \sin^3 \theta \cos \theta \, d\theta = \frac{\pi}{4} \sin^4 \alpha. \end{aligned}$$

Now the boundary: ∂S is the circle $t \mapsto (\sin \alpha \cos t, \sin \alpha \sin t, \cos \alpha)$ for $t \in [0, 2\pi]$, so

$$d\mathbf{x} = \sin \alpha \begin{pmatrix} -\sin t \\ \cos t \\ 0 \end{pmatrix} dt,$$

and on this circle $-x^2 y = -\sin^3 \alpha \cos^2 t \sin t$. Then

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{x} = \sin^4 \alpha \int_0^{2\pi} (-\cos^2 t \sin t)(-\sin t) \, dt = \frac{\pi}{4} \sin^4 \alpha,$$

in agreement with Stokes' theorem.

Stokes' theorem lets us tie up a loose end from our discussion of conservative fields: it links the vanishing of circulations to the vanishing of the curl.

Proposition 7.6.5

Let $\mathbf{F}$ be a continuously differentiable vector field. Then $\oint_C \mathbf{F} \cdot d\mathbf{x} = 0$ for every closed curve C if and only if $\nabla \times \mathbf{F} = \mathbf{0}$.

Proof. ($\Leftarrow$) Given a closed curve C , take any surface S with $\partial S = C$ (for the curves we work with this can always be done). Then by Stokes' theorem,

$$\oint_C \mathbf{F} \cdot d\mathbf{x} = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0.$$

($\Rightarrow$) Suppose for contradiction that $\nabla \times \mathbf{F} \neq \mathbf{0}$, so there is a point $\mathbf{x}_0$ and a unit vector $\hat{\mathbf{k}}$ with

$$\hat{\mathbf{k}} \cdot (\nabla \times \mathbf{F})(\mathbf{x}_0) = \lambda > 0.$$

Since $\nabla \times \mathbf{F}$ is continuous, there is $\delta > 0$ such that

$$\hat{\mathbf{k}} \cdot (\nabla \times \mathbf{F})(\mathbf{x}) > \frac{\lambda}{2} \quad \text{whenever } |\mathbf{x} - \mathbf{x}_0| < \delta.$$

Now let S be a small flat disc centred at $\mathbf{x}_0$, contained in this ball, lying in the plane through $\mathbf{x}_0$ with normal $\hat{\mathbf{k}}$. Applying Stokes' theorem to its boundary circle,

$$0 = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{x} = \int_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{k}} dS > \frac{\lambda}{2} \int_S dS > 0,$$

a contradiction. □

The same sort of argument gives us a coordinate-free way of *seeing* the curl, making precise the ‘paddle wheel’ picture from the last section. Let S_ϵ be a flat disc of radius ϵ , centred at $\mathbf{x}_0$, with unit normal $\hat{\mathbf{k}}$. Then

$$\int_{S_\epsilon} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = (\nabla \times \mathbf{F})(\mathbf{x}_0) \cdot \hat{\mathbf{k}} \text{area}(S_\epsilon) + \underbrace{\int_{S_\epsilon} [(\nabla \times \mathbf{F})(\mathbf{x}) - (\nabla \times \mathbf{F})(\mathbf{x}_0)] \cdot d\mathbf{S}}_{o(\text{area}(S_\epsilon)) \text{ by continuity}},$$

so dividing by the area and using Stokes' theorem,

$$(\nabla \times \mathbf{F})(\mathbf{x}_0) \cdot \hat{\mathbf{k}} = \lim_{\epsilon \rightarrow 0} \frac{1}{\text{area}(S_\epsilon)} \oint_{\partial S_\epsilon} \mathbf{F} \cdot d\mathbf{x}.$$

In other words, the component of the curl in the direction $\hat{\mathbf{k}}$ is the *infinitesimal circulation per unit area* about the axis $\hat{\mathbf{k}}$.

§7.6.3 The Divergence Theorem

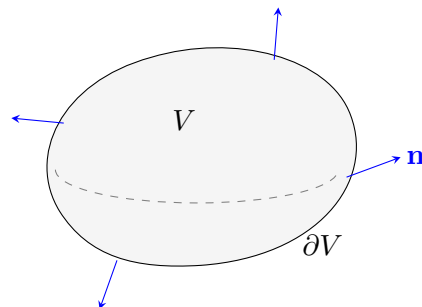
The final theorem of the trio concerns volumes, and the relevant derivative is the divergence.

Theorem 7.6.6 (Divergence Theorem)

Let $\mathbf{F}$ be a continuously differentiable vector field, and let V be a volume whose boundary ∂V is a piecewise regular surface, with normal pointing *out* of V . Then

$$\int_V \nabla \cdot \mathbf{F} dV = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S}.$$

That is, the total flux of $\mathbf{F}$ out of a region equals the integral of the divergence over that region – the divergence really does measure ‘net outflow’.



There is also a two dimensional version: if D is a planar region with piecewise smooth boundary ∂D , then

$$\int_D \nabla \cdot \mathbf{F} \, dA = \oint_{\partial D} \mathbf{F} \cdot \mathbf{n} \, ds,$$

where $\mathbf{n}$ is the outward unit normal to the boundary curve and ds is the arc length element.

Example 7.6.7 (Verifying the Divergence Theorem on a Cylinder)

Let V be the solid cylinder given in cylindrical polars by $0 \leq \rho \leq R$, $-h \leq z \leq h$, and take $\mathbf{F}(\mathbf{x}) = \mathbf{x}$, so $\nabla \cdot \mathbf{F} = 3$. The left-hand side is

$$\int_V \nabla \cdot \mathbf{F} \, dV = 3 \operatorname{vol}(V) = 6\pi R^2 h.$$

The boundary has three pieces: the curved side $S_R = \{\rho = R\}$, and the top and bottom discs $S_{\pm} = \{z = \pm h\}$. On S_R we have $d\mathbf{S} = \mathbf{e}_\rho R \, d\phi \, dz$ and $\mathbf{x} \cdot \mathbf{e}_\rho = \rho = R$, so

$$\int_{S_R} \mathbf{F} \cdot d\mathbf{S} = \int_{z=-h}^h \int_{\phi=0}^{2\pi} R^2 \, d\phi \, dz = 4\pi R^2 h.$$

On $S_{\pm}$ we have $d\mathbf{S} = \pm \mathbf{e}_z \rho \, d\rho \, d\phi$ and $\mathbf{x} \cdot (\pm \mathbf{e}_z) = \pm z = h$ on each disc, so each contributes

$$\int_{S_{\pm}} \mathbf{F} \cdot d\mathbf{S} = \int_{\phi=0}^{2\pi} \int_{\rho=0}^R h\rho \, d\rho \, d\phi = \pi R^2 h.$$

The total flux is $4\pi R^2 h + 2\pi R^2 h = 6\pi R^2 h$, as expected.

Exactly as with the curl, the divergence theorem characterises solenoidal fields, and gives a coordinate-free description of the divergence.

Proposition 7.6.8

Let $\mathbf{F}$ be continuously differentiable. Then $\int_S \mathbf{F} \cdot d\mathbf{S} = 0$ for every closed surface S if and only if $\nabla \cdot \mathbf{F} = 0$.

Proof. ($\Leftarrow$) is immediate from the divergence theorem applied to the volume enclosed by S . For ($\Rightarrow$), suppose $(\nabla \cdot \mathbf{F})(\mathbf{x}_0) = \lambda > 0$ at some point (if negative, the argument is identical). By continuity there is $\delta > 0$ with $\nabla \cdot \mathbf{F} > \lambda/2$ on the ball $|\mathbf{x} - \mathbf{x}_0| < \delta$. Taking V to be this ball,

$$0 = \int_{\partial V} \mathbf{F} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{F} \, dV > \frac{\lambda}{2} \operatorname{vol}(V) > 0,$$

a contradiction. □

Similarly, if V_ϵ is a ball of radius ϵ centred at $\mathbf{x}_0$, then by continuity

$$\int_{V_\epsilon} \nabla \cdot \mathbf{F} \, dV = (\nabla \cdot \mathbf{F})(\mathbf{x}_0) \operatorname{vol}(V_\epsilon) + o(\operatorname{vol}(V_\epsilon)),$$

so by the divergence theorem

$$(\nabla \cdot \mathbf{F})(\mathbf{x}_0) = \lim_{\epsilon \rightarrow 0} \frac{1}{\operatorname{vol}(V_\epsilon)} \int_{\partial V_\epsilon} \mathbf{F} \cdot d\mathbf{S},$$

that is, the divergence is the *infinitesimal flux per unit volume*. A point of positive divergence really is a ‘source’ of the field.

§7.6.4 Proving the Integral Theorems

We now pay our debts. The strategy is: prove the divergence theorem directly, deduce Green’s theorem from its two dimensional version, and then deduce Stokes’ theorem from Green’s theorem.

Proof of the Divergence Theorem. By linearity it suffices to prove the theorem separately for fields pointing along each coordinate axis, so first suppose $\mathbf{F} = F_z(x, y, z)\mathbf{e}_z$. Then the claim is that

$$\int_V \frac{\partial F_z}{\partial z} dV = \int_{\partial V} F_z \mathbf{e}_z \cdot d\mathbf{S}. \quad (\dagger)$$

Suppose first that V is a ‘nice’ volume, by which we mean that we can split the boundary as $\partial V = S_+ \cup S_-$, where S_+ collects the parts of the boundary whose outward normal has positive $\mathbf{e}_z$ component and S_- those with negative $\mathbf{e}_z$ component, and each is a graph over the projection A of V onto the (x, y) -plane^a. That is,

$$S_{\pm} = \left\{ \begin{pmatrix} x \\ y \\ g_{\pm}(x, y) \end{pmatrix} \mid (x, y) \in A \right\},$$

for some functions $g_- \leq g_+$. Then integrating in z first and using the fundamental theorem of calculus,

$$\begin{aligned} \int_V \frac{\partial F_z}{\partial z} dV &= \int_A \left(\int_{z=g_-(x,y)}^{g_+(x,y)} \frac{\partial F_z}{\partial z} dz \right) dx dy \\ &= \int_A [F_z(x, y, g_+(x, y)) - F_z(x, y, g_-(x, y))] dx dy. \end{aligned}$$

Now for the right-hand side of $(\dagger)$. Using (x, y) as parameters on $S_{\pm}$,

$$\frac{\partial \mathbf{x}}{\partial x} \times \frac{\partial \mathbf{x}}{\partial y} = \begin{pmatrix} 1 \\ 0 \\ \partial g_{\pm}/\partial x \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ \partial g_{\pm}/\partial y \end{pmatrix} = \begin{pmatrix} -\partial g_{\pm}/\partial x \\ -\partial g_{\pm}/\partial y \\ 1 \end{pmatrix},$$

which has positive $\mathbf{e}_z$ component. Since we need the normal to point *out* of V – upwards on S_+ but downwards on S_- – we get

$$d\mathbf{S} = \pm \begin{pmatrix} -\partial g_{\pm}/\partial x \\ -\partial g_{\pm}/\partial y \\ 1 \end{pmatrix} dx dy \quad \text{on } S_{\pm}.$$

Dotting with $F_z \mathbf{e}_z$, only the third component survives:

$$\int_{\partial V} F_z \mathbf{e}_z \cdot d\mathbf{S} = \int_A F_z(x, y, g_+(x, y)) dx dy - \int_A F_z(x, y, g_-(x, y)) dx dy,$$

which matches the left-hand side, establishing $(\dagger)$ for nice volumes. A general volume can be cut up into finitely many nice pieces; summing $(\dagger)$ over the pieces, the flux

integrals over the internal cut surfaces cancel in pairs (each internal surface is shared by two pieces with opposite outward normals), leaving exactly the integral over ∂V .

The same argument with the roles of the axes exchanged gives

$$\int_V \frac{\partial F_x}{\partial x} dV = \int_{\partial V} F_x \mathbf{e}_x \cdot d\mathbf{S}, \quad \int_V \frac{\partial F_y}{\partial y} dV = \int_{\partial V} F_y \mathbf{e}_y \cdot d\mathbf{S},$$

and adding the three identities gives the divergence theorem. $\square$

^aSome boundaries also have vertical portions where the outward normal is horizontal; these contribute nothing to either side of ($\dagger$), so we can safely ignore them.

The two dimensional divergence theorem is proved in exactly the same way (splitting ∂D into upper and lower graphs over an interval), and we will take it as established. From it, Green's theorem follows almost immediately.

Proof of Green's Theorem. Given P and Q , consider the planar vector field

$$\mathbf{F} = \begin{pmatrix} Q(x, y) \\ -P(x, y) \end{pmatrix}, \quad \text{so} \quad \nabla \cdot \mathbf{F} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

By the two dimensional divergence theorem,

$$\int_A \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \oint_{\partial A} \mathbf{F} \cdot \mathbf{n} ds.$$

Parameterize ∂A by arc length, traversed with A on the left. The unit tangent is $\mathbf{t} = (x'(s), y'(s))$, and the *outward* unit normal is $\mathbf{t}$ rotated clockwise by $\pi/2$,

$$\mathbf{n} = \begin{pmatrix} y'(s) \\ -x'(s) \end{pmatrix}.$$

Then

$$\oint_{\partial A} \mathbf{F} \cdot \mathbf{n} ds = \oint_{\partial A} \left(Q \frac{dy}{ds} + P \frac{dx}{ds} \right) ds = \oint_{\partial A} P dx + Q dy,$$

as required. $\square$

Proof of Stokes' Theorem. Let $S = \{\mathbf{x}(u, v) \mid (u, v) \in A\}$ be a regular parameterized surface, so that ∂S is the image of ∂A . The idea is to 'pull back' everything to the (u, v) -plane and apply Green's theorem there. Define

$$P(u, v) = \mathbf{F}(\mathbf{x}(u, v)) \cdot \frac{\partial \mathbf{x}}{\partial u}, \quad Q(u, v) = \mathbf{F}(\mathbf{x}(u, v)) \cdot \frac{\partial \mathbf{x}}{\partial v}.$$

Then along any curve in A and its image in S ,

$$P du + Q dv = \mathbf{F} \cdot \left(\frac{\partial \mathbf{x}}{\partial u} du + \frac{\partial \mathbf{x}}{\partial v} dv \right) = \mathbf{F} \cdot d\mathbf{x},$$

so

$$\oint_{\partial A} P du + Q dv = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{x}.$$

Now we compute the integrand of Green's theorem. By the chain and product rules,

$$\frac{\partial Q}{\partial u} = \frac{\partial x_j}{\partial u} \frac{\partial F_i}{\partial x_j} \frac{\partial x_i}{\partial v} + F_i \frac{\partial^2 x_i}{\partial u \partial v}, \quad \frac{\partial P}{\partial v} = \frac{\partial x_j}{\partial v} \frac{\partial F_i}{\partial x_j} \frac{\partial x_i}{\partial u} + F_i \frac{\partial^2 x_i}{\partial v \partial u}.$$

The second derivative terms cancel when we subtract, leaving

$$\begin{aligned} \frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v} &= \frac{\partial F_i}{\partial x_j} \left(\frac{\partial x_j}{\partial u} \frac{\partial x_i}{\partial v} - \frac{\partial x_j}{\partial v} \frac{\partial x_i}{\partial u} \right) \\ &= (\delta_{jp} \delta_{iq} - \delta_{jq} \delta_{ip}) \frac{\partial F_i}{\partial x_j} \frac{\partial x_p}{\partial u} \frac{\partial x_q}{\partial v} \\ &= \epsilon_{kji} \epsilon_{kpq} \frac{\partial F_i}{\partial x_j} \frac{\partial x_p}{\partial u} \frac{\partial x_q}{\partial v} \\ &= \left(\epsilon_{kji} \frac{\partial F_i}{\partial x_j} \right) \left(\epsilon_{kpq} \frac{\partial x_p}{\partial u} \frac{\partial x_q}{\partial v} \right) \\ &= [\nabla \times \mathbf{F}]_k \left[\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right]_k \\ &= (\nabla \times \mathbf{F}) \cdot \left(\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right). \end{aligned}$$

So by Green's theorem,

$$\begin{aligned} \oint_{\partial S} \mathbf{F} \cdot d\mathbf{x} &= \int_A \left(\frac{\partial Q}{\partial u} - \frac{\partial P}{\partial v} \right) du dv \\ &= \int_A (\nabla \times \mathbf{F}) \cdot \left(\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \right) du dv = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}, \end{aligned}$$

which is Stokes' theorem. (One can check that the orientation conventions match up: traversing ∂A anticlockwise corresponds exactly to traversing ∂S compatibly with the normal $\frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v}$.) $\square$

§7.6.5 Conservative Fields Revisited

Way back when we discussed line integrals, we showed that exact differentials are closed, and claimed that the converse holds on simply connected domains. With Stokes' theorem in hand we can finally prove this, in the language of vector fields: recall that $\mathbf{F} \cdot d\mathbf{x}$ being closed corresponds precisely to $\nabla \times \mathbf{F} = \mathbf{0}$, and being exact corresponds to $\mathbf{F} = \nabla f$.

Theorem 7.6.9

Let $\Omega \subseteq \mathbb{R}^3$ be open and simply connected, and let $\mathbf{F}$ be a continuously differentiable vector field on Ω with $\nabla \times \mathbf{F} = \mathbf{0}$. Then $\mathbf{F}$ is conservative: there exists a scalar field f with $\mathbf{F} = \nabla f$ on Ω .

Proof. Fix a base point $\mathbf{x}_0 \in \Omega$, and define

$$f(\mathbf{x}) = \int_{C(\mathbf{x})} \mathbf{F} \cdot d\mathbf{x}',$$

where $C(\mathbf{x})$ is any piecewise smooth curve in Ω from $\mathbf{x}_0$ to $\mathbf{x}$.

First we check this is well defined. If C_1 and C_2 are two such curves, then $C_1 - C_2$ is a closed loop in Ω . Since Ω is simply connected, this loop can be continuously shrunk to a point within Ω , and the shrinking process sweeps out a surface $S \subseteq \Omega$ with $\partial S = C_1 - C_2$. By Stokes' theorem,

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{x}' - \int_{C_2} \mathbf{F} \cdot d\mathbf{x}' = \oint_{C_1 - C_2} \mathbf{F} \cdot d\mathbf{x}' = \int_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0,$$

so the definition of f does not depend on the choice of curve.

Now we check $\nabla f = \mathbf{F}$. Fix $\mathbf{x} \in \Omega$ and a small $\mathbf{h}$, and choose the path to $\mathbf{x} + \mathbf{h}$ to be a path to $\mathbf{x}$ followed by the straight line segment from $\mathbf{x}$ to $\mathbf{x} + \mathbf{h}$ (which lies in Ω for $|\mathbf{h}|$ small, as Ω is open). Then

$$f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) = \int_0^1 \mathbf{F}(\mathbf{x} + t\mathbf{h}) \cdot \mathbf{h} dt = \mathbf{F}(\mathbf{x}) \cdot \mathbf{h} + o(\mathbf{h}),$$

using the continuity of $\mathbf{F}$. Comparing with the definition of the gradient, $\nabla f = \mathbf{F}$. $\square$

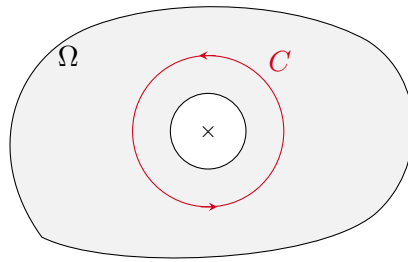
So on a simply connected domain,

$$\mathbf{F} \text{ conservative} \iff \nabla \times \mathbf{F} = \mathbf{0} \iff \oint_C \mathbf{F} \cdot d\mathbf{x} = 0 \text{ for all closed } C.$$

The hypothesis on the domain really is needed: the field

$$\mathbf{F} = \frac{1}{x^2 + y^2} \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}$$

is irrotational on $\mathbb{R}^3$ minus the z -axis (a domain which is *not* simply connected), but its circulation around the unit circle in the (x, y) -plane is $2\pi \neq 0$, so it cannot be conservative there. Looking down the z -axis, the domain has a 'hole', and the circle C below cannot be shrunk to a point without crossing it.



There is an analogous result for the divergence, which we state without proof: if $\nabla \cdot \mathbf{F} = 0$ on a domain in which every closed surface can be shrunk to a point (such as $\mathbb{R}^3$ itself, or $\mathbb{R}^3$ minus the z -axis – but *not* $\mathbb{R}^3$ minus the origin), then $\mathbf{F} = \nabla \times \mathbf{A}$ for some vector potential $\mathbf{A}$.

§7.7 Maxwell's Equations

Having assembled all this machinery, it would be a shame not to use it. In this section we take a brief tour of electromagnetism, which is governed by a system of partial differential

equations – Maxwell’s equations – written entirely in the language of vector calculus. Almost everything we have developed will make an appearance.

§7.7.1 Conservation Laws

Before writing down Maxwell’s equations, we make a general observation. Many equations of mathematical physics take the form

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

where $\rho(\mathbf{x}, t)$ is the *density* of some quantity (mass, electric charge, energy, ...) and $\mathbf{J}(\mathbf{x}, t)$ is its *current density*, describing how the quantity flows. An equation of this form is called a **conservation law**, and the name is justified by the following computation. Suppose that ρ and $|\mathbf{J}|$ decay rapidly as $|\mathbf{x}| \rightarrow \infty$, and define the total charge

$$Q = \int_{\mathbb{R}^3} \rho(\mathbf{x}, t) \, dV.$$

Then, differentiating under the integral and using the conservation law and then the divergence theorem,

$$\begin{aligned} \frac{dQ}{dt} &= \int_{\mathbb{R}^3} \frac{\partial \rho}{\partial t} \, dV = - \int_{\mathbb{R}^3} \nabla \cdot \mathbf{J} \, dV \\ &= - \lim_{R \rightarrow \infty} \int_{|\mathbf{x}| \leq R} \nabla \cdot \mathbf{J} \, dV = - \lim_{R \rightarrow \infty} \int_{|\mathbf{x}|=R} \mathbf{J} \cdot d\mathbf{S} = 0, \end{aligned}$$

since $\mathbf{J}$ decays rapidly. So the total charge Q is constant in time – charge is *conserved*. The same argument on a finite volume V says that the rate of change of the charge inside V equals minus the flux of $\mathbf{J}$ out through ∂V : charge only changes by flowing through the boundary.

§7.7.2 The Equations

Electromagnetism concerns two vector fields: the electric field $\mathbf{E}(\mathbf{x}, t)$ and the magnetic field $\mathbf{B}(\mathbf{x}, t)$. These are produced by (and act upon) electric charges, described by the charge density $\rho(\mathbf{x}, t)$ and current density $\mathbf{J}(\mathbf{x}, t)$. The dynamics are governed by **Maxwell’s equations**:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \tag{M1}$$

$$\nabla \cdot \mathbf{B} = 0, \tag{M2}$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \tag{M3}$$

$$\nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \tag{M4}$$

Here ϵ_0 and μ_0 are constants (the permittivity and permeability of free space), which satisfy the rather suggestive relation

$$\frac{1}{\mu_0 \epsilon_0} = c^2,$$

where $c \approx 3 \times 10^8 \, \text{m s}^{-1}$ is the speed of light. We will see why shortly.

As a first check of consistency, take the divergence of (M4). Since ‘div curl is zero’,

$$0 = \nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) + \mu_0 \nabla \cdot \mathbf{J} = \mu_0 \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} \right),$$

using (M1). So Maxwell’s equations *force* the conservation law

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

that is, electric charge is automatically conserved.

§7.7.3 Integral Forms

Each of Maxwell’s equations has an equivalent integral formulation, obtained using our integral theorems, and each integral form has a satisfying physical reading.

Integrating (M1) over a volume V and applying the divergence theorem gives **Gauss’ law**,

$$\int_{\partial V} \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0},$$

where Q is the total charge inside V : the flux of the electric field out of any closed surface measures the charge enclosed.

Doing the same to (M2),

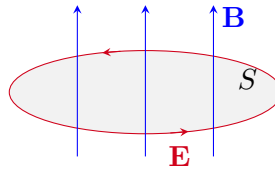
$$\int_{\partial V} \mathbf{B} \cdot d\mathbf{S} = 0,$$

so there is no net magnetic flux through any closed surface. In other words, there are no ‘magnetic charges’ – you cannot have a magnet with only a north pole.

Integrating (M3) over a (fixed) surface S and applying Stokes’ theorem gives **Faraday’s law**,

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{x} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}.$$

A changing magnetic flux through S induces a circulation of the electric field around the boundary – this is the principle behind electrical generators, where turning a magnet drives a current around a wire.



Finally, integrating (M4) over a surface and applying Stokes’ theorem gives **Ampère’s law** (with Maxwell’s correction),

$$\oint_{\partial S} \mathbf{B} \cdot d\mathbf{x} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{S}.$$

So a current flowing through a wire generates a magnetic field circulating around the wire (and so does a changing electric flux).

§7.7.4 Electromagnetic Waves

Here is the payoff of writing everything with div and curl. Consider empty space, where $\rho = 0$ and $\mathbf{J} = \mathbf{0}$. Recall that the vector Laplacian satisfies

$$\nabla^2 \mathbf{F} = \nabla(\nabla \cdot \mathbf{F}) - \nabla \times (\nabla \times \mathbf{F}).$$

Applying this to $\mathbf{E}$ and using Maxwell's equations repeatedly,

$$\begin{aligned} \nabla^2 \mathbf{E} &= \nabla \underbrace{(\nabla \cdot \mathbf{E})}_{=0 \text{ by (M1)}} - \nabla \times (\nabla \times \mathbf{E}) \\ &= \nabla \times \frac{\partial \mathbf{B}}{\partial t} \quad \text{by (M3)} \\ &= \frac{\partial}{\partial t}(\nabla \times \mathbf{B}) \\ &= \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \text{by (M4)}. \end{aligned}$$

That is,

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \mathbf{0},$$

which is the *wave equation* for waves travelling at speed c . An identical computation (using (M2) and swapping the roles of (M3) and (M4)) gives

$$\nabla^2 \mathbf{B} - \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} = \mathbf{0}.$$

So in a vacuum, disturbances of the electric and magnetic fields propagate as waves at the speed of light. This is no coincidence: light *is* an electromagnetic wave, and this computation – essentially Maxwell's own – was the first theoretical evidence for that fact.

§7.7.5 Electrostatics and Potentials

Now suppose instead that all fields and sources are independent of time. Then Maxwell's equations decouple into a pair of systems,

$$\begin{cases} \nabla \cdot \mathbf{E} = \rho/\epsilon_0 \\ \nabla \times \mathbf{E} = \mathbf{0} \end{cases} \quad \begin{cases} \nabla \cdot \mathbf{B} = 0 \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J}, \end{cases}$$

one governing *electrostatics* and one *magnetostatics*.

Look at the electric system. Since $\mathbf{E}$ is irrotational on the simply connected domain $\mathbb{R}^3$, our theory of conservative fields says $\mathbf{E} = -\nabla\phi$ for some scalar field ϕ , called the **electric potential** (the sign is a physics convention). Substituting into the first equation,

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.$$

Similarly, since $\mathbf{B}$ is solenoidal on $\mathbb{R}^3$, it has a vector potential: $\mathbf{B} = \nabla \times \mathbf{A}$.

So electrostatics boils down to solving the equation $\nabla^2 \phi = -\rho/\epsilon_0$ – an instance of *Poisson's equation*, the subject of the next section.

§7.8 Poisson's Equation and Laplace's Equation

As we've just seen, a great many problems in mathematical physics reduce to solving

$$\nabla^2 \phi = F$$

on some domain $\Omega \subseteq \mathbb{R}^n$ (for us, $n = 2$ or 3), for a given 'source' function F . This is called **Poisson's equation**, and in the special case $F \equiv 0$ it is called **Laplace's equation**. The same equation describes electric and gravitational potentials, steady-state temperature distributions, and much else besides.

§7.8.1 Boundary Value Problems

On its own, Poisson's equation has far too many solutions (we can add to ϕ any solution of Laplace's equation), so a well-posed problem must come with *boundary conditions*. Two kinds are standard. The **Dirichlet problem** prescribes the values of ϕ on the boundary:

$$\nabla^2 \phi = F \text{ in } \Omega, \quad \phi = f \text{ on } \partial\Omega.$$

The **Neumann problem** prescribes the normal derivative:

$$\nabla^2 \phi = F \text{ in } \Omega, \quad \frac{\partial \phi}{\partial n} = g \text{ on } \partial\Omega,$$

where $\frac{\partial \phi}{\partial n} = \mathbf{n} \cdot \nabla \phi$ and $\mathbf{n}$ is the outward normal. If Ω is unbounded (say $\Omega = \mathbb{R}^3$), the boundary condition is replaced by a decay condition as $|\mathbf{x}| \rightarrow \infty$. In both cases we require ϕ and $\nabla \phi$ to extend continuously to the boundary.

Example 7.8.1

Let's solve

$$\nabla^2 \phi = r \text{ in } \{r < a\}, \quad \phi = 1 \text{ on } \{r = a\},$$

where r is the spherical radial coordinate. Since the problem is spherically symmetric, we guess a solution of the form $\phi = \phi(r)$, and use the radial part of the spherical polar Laplacian:

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = r \implies \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = r^3.$$

Integrating twice, the general spherically symmetric solution is

$$\phi(r) = \frac{r^3}{12} + A - \frac{B}{r}.$$

But a term B/r blows up at the origin, which lies inside our domain, so we must take $B = 0$ for ϕ to be a genuine solution on all of $\{r < a\}$. The boundary condition $\phi(a) = 1$ then fixes $A = 1 - a^3/12$, giving

$$\phi(r) = 1 + \frac{1}{12} (r^3 - a^3).$$

There is something slightly dishonest about 'guessing' that ϕ depends only on r – how do we know we haven't missed other, non-symmetric solutions? The answer is the following uniqueness theorem, which is itself a lovely application of the divergence theorem.

§7.8.2 Uniqueness

The divergence theorem gives us a surprisingly quick proof that the boundary conditions above pin down the solution essentially uniquely.

Theorem 7.8.2 (Uniqueness for Poisson's Equation)

The solution of the Dirichlet problem for Poisson's equation, if it exists, is unique. The solution of the Neumann problem is unique up to adding a constant.

Proof. Suppose ϕ_1 and ϕ_2 both solve the problem, and let $\psi = \phi_1 - \phi_2$. By linearity, ψ satisfies the corresponding *homogeneous* problem:

$$\nabla^2 \psi = 0 \text{ in } \Omega, \quad \psi = 0 \text{ (resp. } \frac{\partial \psi}{\partial n} = 0) \text{ on } \partial\Omega.$$

Now consider the quantity

$$I[\psi] = \int_{\Omega} |\nabla \psi|^2 \, dV \geq 0.$$

Using the Leibniz identity $\nabla \cdot (\psi \nabla \psi) = \nabla \psi \cdot \nabla \psi + \psi \nabla^2 \psi$ and $\nabla^2 \psi = 0$,

$$\begin{aligned} I[\psi] &= \int_{\Omega} \nabla \cdot (\psi \nabla \psi) \, dV - \int_{\Omega} \psi \nabla^2 \psi \, dV \\ &= \int_{\partial\Omega} \psi \nabla \psi \cdot \mathbf{dS} = \int_{\partial\Omega} \psi \frac{\partial \psi}{\partial n} \, dS, \end{aligned}$$

by the divergence theorem. But on $\partial\Omega$ either $\psi = 0$ (Dirichlet) or $\partial\psi/\partial n = 0$ (Neumann), so in both cases $I[\psi] = 0$. Since the integrand $|\nabla \psi|^2$ is non-negative and continuous, this forces $\nabla \psi = \mathbf{0}$ throughout Ω , so ψ is constant.

In the Dirichlet case $\psi = 0$ on the boundary, so the constant is zero and $\phi_1 = \phi_2$. In the Neumann case no such conclusion is available, and indeed adding a constant to any solution gives another. $\square$

So in the previous example, the spherically symmetric solution we found is *the* solution. This ‘guess cleverly, then invoke uniqueness’ strategy is extremely common, and we will use it repeatedly below.

Example 7.8.3 (Inside a Charged Shell)

Suppose a charge density is zero for $r < a$ (and arbitrary but spherically symmetric outside). What is the electric field in the cavity $r < a$? The potential satisfies

$$\nabla^2 \phi = 0 \text{ in } \{r < a\}, \quad \phi = \text{const on } \{r = a\}$$

(the boundary value is constant by symmetry). But a constant function solves this Dirichlet problem, so by uniqueness it is the only solution. Hence $\mathbf{E} = -\nabla \phi = \mathbf{0}$: there is no electric field anywhere inside the shell. This is a version of Newton's shell theorem, here obtained with essentially no computation.

§7.8.3 Gauss' Flux Method

For source terms with spherical or cylindrical symmetry, the divergence theorem gives a wonderfully efficient way of solving Poisson's equation, known as **Gauss' flux method**.

Suppose we want to solve $\nabla^2 \phi = F$ on $\mathbb{R}^3$ where $F = F(r)$ is spherically symmetric. By symmetry (and uniqueness!) we look for $\phi = \phi(r)$, for which

$$\nabla \phi = \phi'(r) \mathbf{e}_r.$$

Integrate $\nabla \cdot (\nabla \phi) = F$ over the ball $|\mathbf{x}| < R$ and apply the divergence theorem:

$$\int_{|\mathbf{x}|=R} \nabla \phi \cdot d\mathbf{S} = \int_{|\mathbf{x}|<R} F dV =: Q(R),$$

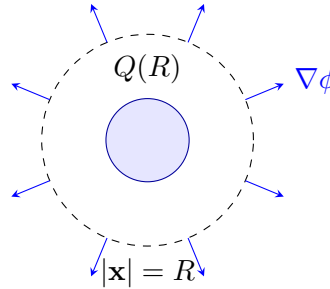
where we think of $Q(R)$ as the 'total charge' (or mass, etc.) inside radius R . On the sphere $|\mathbf{x}| = R$ we have $d\mathbf{S} = \mathbf{e}_r R^2 \sin \theta d\theta d\varphi$, so $\nabla \phi \cdot d\mathbf{S} = \phi'(R) dS$, and $\phi'(R)$ is constant over the sphere. Hence

$$Q(R) = \phi'(R) \int_{|\mathbf{x}|=R} dS = 4\pi R^2 \phi'(R),$$

which rearranges to the key formula

$$\phi'(R) = \frac{Q(R)}{4\pi R^2}, \quad \text{i.e.} \quad \nabla \phi = \frac{Q(r)}{4\pi r^2} \mathbf{e}_r.$$

The flux through a sphere only sees the charge *inside* it.



Example 7.8.4 (A Uniformly Charged Ball)

Let's find the electric field of a ball of uniform charge density ρ_0 and radius a . The potential satisfies $\nabla^2 \phi = -\rho(r)/\epsilon_0$ where $\rho = \rho_0$ for $r \leq a$ and 0 beyond. The enclosed charge is

$$Q(r) = \begin{cases} \frac{4}{3}\pi r^3 \rho_0 & r \leq a, \\ \frac{4}{3}\pi a^3 \rho_0 = Q & r > a, \end{cases}$$

where Q is the total charge. By Gauss' flux method (applied to $-\epsilon_0 \phi$),

$$\mathbf{E} = -\nabla \phi = \frac{1}{4\pi \epsilon_0} \frac{Q(r)}{r^2} \mathbf{e}_r = \begin{cases} \frac{\rho_0 r}{3\epsilon_0} \mathbf{e}_r & r \leq a, \\ \frac{Q}{4\pi \epsilon_0 r^2} \mathbf{e}_r & r > a. \end{cases}$$

Two features are worth noting. Outside the ball, the field is exactly that of a *point charge* Q at the origin – the details of the charge distribution are invisible. And

letting $a \rightarrow 0$ with Q fixed, we obtain the field of a point charge,

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{e}_r,$$

which is the inverse square law. The same mathematics with ϕ the gravitational potential and $\nabla^2 \phi = 4\pi G\rho$ gives Newton's inverse square law of gravitation, and shows that a spherical planet attracts as if all its mass were concentrated at its centre.

The corresponding computation for cylindrically symmetric sources, integrating over a cylinder of radius R and height h (the flat ends contribute nothing since $\nabla\phi = \phi'(\rho)\mathbf{e}_\rho$ is horizontal), gives

$$2\pi R h \phi'(R) = \int_V F \, dV \implies \phi'(\rho) = \frac{1}{\rho} \int_0^\rho s F(s) \, ds.$$

For example, an infinite wire with charge λ per unit length produces a field decaying like $1/\rho$, not $1/\rho^2$.

§7.8.4 Harmonic Functions

Solutions of Laplace's equation $\nabla^2 \phi = 0$ are important enough to have their own name: they are called **harmonic** functions. Harmonic functions have some rather striking rigidity properties, and the divergence theorem is the key to all of them.

Theorem 7.8.5 (Mean Value Property)

If ϕ is harmonic on a domain $\Omega \subseteq \mathbb{R}^3$, then for any $\mathbf{a} \in \Omega$ and any $r > 0$ small enough that the closed ball $|\mathbf{x} - \mathbf{a}| \leq r$ lies in Ω ,

$$\phi(\mathbf{a}) = \frac{1}{4\pi r^2} \int_{|\mathbf{x}-\mathbf{a}|=r} \phi(\mathbf{x}) \, dS.$$

In words: the value of a harmonic function at a point equals its *average* over any sphere centred at that point.

Proof. Let $F(r)$ denote the average on the right-hand side. Parameterizing the sphere with spherical polar angles, $\mathbf{x} = \mathbf{a} + r\mathbf{e}_r$ and $dS = r^2 \sin \theta \, d\theta \, d\varphi$, so the factors of r^2 cancel:

$$F(r) = \frac{1}{4\pi} \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi} \phi(\mathbf{a} + r\mathbf{e}_r) \sin \theta \, d\theta \, d\varphi.$$

Note that as $r \rightarrow 0$, $F(r) \rightarrow \phi(\mathbf{a})$ by continuity. Now differentiate with respect to r , using $\frac{d}{dr} \phi(\mathbf{a} + r\mathbf{e}_r) = \mathbf{e}_r \cdot \nabla \phi(\mathbf{a} + r\mathbf{e}_r)$:

$$\begin{aligned} F'(r) &= \frac{1}{4\pi} \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi} \mathbf{e}_r \cdot \nabla \phi(\mathbf{a} + r\mathbf{e}_r) \sin \theta \, d\theta \, d\varphi \\ &= \frac{1}{4\pi r^2} \int_{|\mathbf{x}-\mathbf{a}|=r} \nabla \phi \cdot d\mathbf{S} \end{aligned}$$

$$= \frac{1}{4\pi r^2} \int_{|\mathbf{x}-\mathbf{a}|<r} \nabla^2 \phi \, dV = 0,$$

where we recognised the integral as a flux integral, applied the divergence theorem, and used that ϕ is harmonic. So F is constant, and since $F(r) \rightarrow \phi(\mathbf{a})$ as $r \rightarrow 0$, we get $F(r) = \phi(\mathbf{a})$ for all valid r . $\square$

Theorem 7.8.6 (Maximum Principle)

If ϕ is harmonic on a (connected) domain Ω , then ϕ has no strict local maximum in the interior of Ω . Moreover, if ϕ attains its maximum value at an interior point, then ϕ is constant.

Proof. Suppose ϕ attains a maximum at an interior point $\mathbf{a}$, so $\phi(\mathbf{a}) \geq \phi(\mathbf{x})$ for all $\mathbf{x} \in \Omega$. Pick $\epsilon > 0$ small enough that the ball $|\mathbf{x} - \mathbf{a}| \leq \epsilon$ lies inside Ω . By the mean value property,

$$0 = \frac{1}{4\pi\epsilon^2} \int_{|\mathbf{x}-\mathbf{a}|=\epsilon} (\phi(\mathbf{a}) - \phi(\mathbf{x})) \, dS.$$

The integrand is continuous and non-negative, so it must vanish identically: $\phi = \phi(\mathbf{a})$ on the whole sphere. Since ϵ was arbitrary (below some threshold), ϕ is constant on a ball around $\mathbf{a}$.

Now take any other point $\mathbf{y} \in \Omega$. Connect $\mathbf{a}$ to $\mathbf{y}$ by a chain of overlapping balls inside Ω , each centred at a point of the previous ball. Applying the argument above repeatedly along the chain, ϕ is constant on each ball, so $\phi(\mathbf{y}) = \phi(\mathbf{a})$. Hence ϕ is constant on Ω . $\square$

Corollary 7.8.7

If ϕ is harmonic on a bounded domain Ω and continuous up to the boundary, then

$$\max_{\Omega \cup \partial\Omega} \phi = \max_{\partial\Omega} \phi,$$

that is, ϕ attains its maximum on the boundary. (Applying this to $-\phi$, the same holds for the minimum.)

This gives another proof of uniqueness for the Dirichlet problem: the difference of two solutions is harmonic and zero on the boundary, so its max and min are both zero.

§7.9 Cartesian Tensors

To finish the course, we are going to revisit a foundational question: what actually *is* a vector? We have been writing vectors as columns of components, but the components depend on a choice of basis, while the vector itself – an arrow in space – does not. Making this precise leads to the notion of a *tensor*, which simultaneously generalises scalars, vectors and matrices, and which gives the right language for many physical laws. Throughout this section we work exclusively with *Cartesian* coordinates, that is, right-handed orthonormal bases of $\mathbb{R}^3$, and we use the summation convention relentlessly.

§7.9.1 Changing Coordinates

Let $\{\mathbf{e}_i\}$ and $\{\mathbf{e}'_i\}$ be two right-handed orthonormal bases of $\mathbb{R}^3$. A vector $\mathbf{x}$ can be expanded in either basis,

$$\mathbf{x} = x_j \mathbf{e}_j = x'_j \mathbf{e}'_j,$$

and we would like to relate the two sets of components. Since the bases are orthonormal, $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$ and $\mathbf{e}'_i \cdot \mathbf{e}'_j = \delta_{ij}$, so dotting the expansion with $\mathbf{e}'_i$,

$$x'_i = \mathbf{e}'_i \cdot \mathbf{x} = (\mathbf{e}'_i \cdot \mathbf{e}_j) x_j.$$

So if we define the 3×3 array

$$R_{ij} = \mathbf{e}'_i \cdot \mathbf{e}_j,$$

then the components transform according to

$$x'_i = R_{ij} x_j.$$

Dotting instead with $\mathbf{e}_i$ gives the inverse relationship $x_i = R_{ji} x'_j$.

What kind of matrix is R ? Substituting one relation into the other,

$$x_i = R_{ji} x'_j = R_{ji} R_{jk} x_k,$$

and since this holds for every vector $\mathbf{x}$, we must have $R_{ji} R_{jk} = \delta_{ik}$, that is,

$$R^\top R = R R^\top = I,$$

so R is an *orthogonal* matrix. Moreover, using $\mathbf{e}_j = R_{ij} \mathbf{e}'_i$ and the right-handedness of both bases,

$$1 = \mathbf{e}_1 \cdot (\mathbf{e}_2 \times \mathbf{e}_3) = R_{i1} R_{j2} R_{k3} \mathbf{e}'_i \cdot (\mathbf{e}'_j \times \mathbf{e}'_k) = R_{i1} R_{j2} R_{k3} \epsilon_{ijk} = \det R,$$

so $\det R = 1$ and R is a *rotation* matrix. To summarise: under a change of right-handed orthonormal basis, the components of a vector transform as $v'_i = R_{ij} v_j$ for a rotation matrix R .

Contrast this with a *scalar* quantity, like the dot product of two vectors. If $\sigma = \mathbf{a} \cdot \mathbf{b} = a_i b_i$, then in the new basis

$$\sigma' = a'_i b'_i = R_{ip} a_p R_{iq} b_q = \delta_{pq} a_p b_q = a_p b_p = \sigma,$$

using orthogonality of R . So scalars don't transform at all – as had better be the case, since $\mathbf{a} \cdot \mathbf{b}$ has a geometric meaning independent of coordinates.

Now consider a *linear map*. Let $\mathbf{n}$ be a fixed unit vector and let T be the orthogonal projection onto the plane normal to $\mathbf{n}$,

$$T(\mathbf{x}) = \mathbf{x} - (\mathbf{x} \cdot \mathbf{n}) \mathbf{n}.$$

Writing $\mathbf{y} = T(\mathbf{x})$ in components, $y_i = (\delta_{ij} - n_i n_j) x_j$, so with respect to the basis $\{\mathbf{e}_i\}$ the map is described by the array

$$T_{ij} = \delta_{ij} - n_i n_j.$$

In a second basis, the same formula holds with primes, and using $n'_i = R_{ip} n_p$ and $\delta_{ij} = R_{ip} R_{jq} \delta_{pq}$,

$$T'_{ij} = \delta_{ij} - n'_i n'_j = R_{ip} R_{jq} (\delta_{pq} - n_p n_q) = R_{ip} R_{jq} T_{pq}.$$

So the components of a linear map transform with *two* copies of R . There is a clear pattern emerging, and it leads us to the central definition of the section.

§7.9.2 Tensors and Their Rank

Guided by how vectors themselves transform, we make the following definition.

Definition 7.9.1 (Cartesian Tensor)

A **(Cartesian) tensor of rank n** is an object with components $T_{ij\dots k}$ (n indices) with respect to each right-handed orthonormal basis, which transform under a change of basis according to

$$T'_{ij\dots k} = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r},$$

where $R_{ij} = \mathbf{e}'_i \cdot \mathbf{e}_j$.

So a tensor of rank 0 is a scalar, a tensor of rank 1 is a vector, and our discussion above shows that linear maps are tensors of rank 2. The point of the definition is that a tensor is a genuinely basis-independent object: its components in any one basis determine its components in every other, by a fixed rule.

Let's build up a stock of examples.

Example 7.9.2 (Outer Products)

If $\mathbf{u}, \mathbf{v}, \dots, \mathbf{w}$ are n vectors, then

$$T_{ij\dots k} = u_i v_j \cdots w_k$$

defines a tensor of rank n , since each factor transforms with its own copy of R :

$$T'_{ij\dots k} = u'_i v'_j \cdots w'_k = R_{ip}u_p R_{jq}v_q \cdots R_{kr}w_r = R_{ip}R_{jq}\cdots R_{kr}T_{pq\dots r}.$$

(Not every tensor is of this special form, but sums of such tensors give everything.)

Example 7.9.3 (The Kronecker Delta)

The symbol δ_{ij} is defined by the same rule in every basis, so $\delta'_{ij} = \delta_{ij}$. Is this consistent with the tensor transformation law? We need

$$R_{ip}R_{jq}\delta_{pq} = R_{ip}R_{jp} = \delta_{ij} = \delta'_{ij},$$

which holds by orthogonality. So δ is a rank 2 tensor whose components are the same in every basis.

Example 7.9.4 (The Levi-Civita Symbol)

Similarly, ϵ_{ijk} is defined identically in every basis, and

$$R_{ip}R_{jq}R_{kr}\epsilon_{pqr} = (\det R)\epsilon_{ijk} = \epsilon_{ijk},$$

using the standard determinant identity and $\det R = 1$. So ϵ is a rank 3 tensor. (Note the role of right-handedness here: if we allowed reflections, $\det R = -1$ and ϵ would pick up a sign.)

Example 7.9.5 (The Conductivity Tensor)

Tensors arise naturally in physics whenever one vector quantity depends linearly on another. For instance, in many materials an applied electric field $\mathbf{E}$ produces a current density $\mathbf{J}$ depending linearly on $\mathbf{E}$:

$$J_i = \sigma_{ij} E_j,$$

where σ_{ij} is the **electrical conductivity tensor** of the material. (For an isotropic material $\sigma_{ij} = \sigma \delta_{ij}$ and $\mathbf{J}$ is parallel to $\mathbf{E}$; in a crystal, current may flow more easily in some directions than others, and $\mathbf{J}$ need not be parallel to $\mathbf{E}$.)

The array σ_{ij} really is a tensor. Both $\mathbf{J}$ and $\mathbf{E}$ are vectors, so

$$\sigma'_{ij} E'_j = J'_i = R_{ip} J_p = R_{ip} \sigma_{pq} E_q = R_{ip} \sigma_{pq} R_{jq} E'_j,$$

where in the last step we used $E_q = R_{jq} E'_j$. Since $\mathbf{E}$ is arbitrary, $\sigma'_{ij} = R_{ip} R_{jq} \sigma_{pq}$.

Remark. Having indices does not make something a tensor! We could define an array A_{ij} of numbers in one particular basis and declare $A'_{ij} = 0$ in every other basis – this is a perfectly well-defined indexed object, but it does not obey the transformation law. The transformation law is the entire content of being a tensor.

§7.9.3 Operations on Tensors

Tensors of the same rank can be added and multiplied by scalars componentwise, and the results are again tensors (the transformation law is linear in the components). More interestingly, there are two operations that change rank.

Definition 7.9.6 (Tensor Product)

If U has rank m and V has rank n , their **tensor product** $U \otimes V$ is the rank $m + n$ tensor with components

$$(U \otimes V)_{ij \dots k pq \dots r} = U_{ij \dots k} V_{pq \dots r}.$$

That this is a tensor is checked exactly as for outer products of vectors – each factor brings its own copies of R .

Definition 7.9.7 (Contraction)

If T has rank $n \geq 2$, the **contraction** of T on a pair of indices (say the first two) is the rank $n - 2$ tensor with components

$$\delta_{ij} T_{ijk \dots l} = T_{iik \dots l}.$$

To see that contraction lowers the rank by exactly two (rather than breaking the tensor property altogether), we transform:

$$T'_{iik \dots l} = R_{ip} R_{iq} R_{kr} \dots R_{ls} T_{pqr \dots s} = \delta_{pq} R_{kr} \dots R_{ls} T_{pqr \dots s} = R_{kr} \dots R_{ls} T_{ppr \dots s},$$

using orthogonality to kill the two contracted copies of R . The surviving indices transform correctly.

Contraction is everywhere once you look for it: the trace T_{ii} of a rank 2 tensor is a scalar, the dot product $a_i b_i$ is a contraction of the tensor product $a_i b_j$, and matrix multiplication $(ST)_{ik} = S_{ij} T_{jk}$ is a contraction of $S \otimes T$.

There is also a very useful criterion for *recognising* tensors, which saves us from checking the transformation law by hand.

Proposition 7.9.8

A rank 2 array T_{ij} (defined in each basis) is a tensor if and only if the map

$$(\mathbf{a}, \mathbf{b}) \mapsto T_{ij} a_i b_j$$

defines a bilinear map $\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$ independent of the choice of basis (that is, a bilinear *scalar*). The analogous statement holds for higher ranks with multilinear maps.

Proof. If T is a tensor, then $T_{ij} a_i b_j$ is a full contraction of the rank 4 tensor $T_{ij} a_k b_l$, hence a scalar. Conversely, suppose $t(\mathbf{a}, \mathbf{b}) = T_{ij} a_i b_j$ is a basis-independent bilinear map. Then $T_{ij} = t(\mathbf{e}_i, \mathbf{e}_j)$, and changing basis with $\mathbf{e}'_i = R_{ip} \mathbf{e}_p$ (which follows from $R_{ip} = \mathbf{e}'_i \cdot \mathbf{e}_p$),

$$T'_{ij} = t(\mathbf{e}'_i, \mathbf{e}'_j) = t(R_{ip} \mathbf{e}_p, R_{jq} \mathbf{e}_q) = R_{ip} R_{jq} t(\mathbf{e}_p, \mathbf{e}_q) = R_{ip} R_{jq} T_{pq},$$

using bilinearity. □

Theorem 7.9.9 (Quotient Theorem)

Suppose $T_{i \dots j p \dots q}$ is an array of numbers defined in each Cartesian coordinate system, such that whenever $u_{p \dots q}$ is a tensor (of rank m), the array

$$v_{i \dots j} = T_{i \dots j p \dots q} u_{p \dots q}$$

is also a tensor (of rank n). Then T is a tensor of rank $n + m$.

The name comes from thinking of T as the ‘quotient’ v/u . This is exactly the argument we made for the conductivity tensor, done once and for all.

Proof. It suffices to consider tensors of the special form $u_{p \dots q} = c_p \dots d_q$ for vectors $\mathbf{c}, \dots, \mathbf{d}$. By hypothesis

$$v_{i \dots j} = T_{i \dots j p \dots q} c_p \dots d_q$$

is a tensor, so contracting fully with further arbitrary vectors $\mathbf{a}, \dots, \mathbf{b}$,

$$T_{i \dots j p \dots q} a_i \dots b_j c_p \dots d_q$$

is a scalar, independent of basis. But this expression is multilinear in $(\mathbf{a}, \dots, \mathbf{b}, \mathbf{c}, \dots, \mathbf{d})$, so by the multilinear map criterion above, T is a tensor. □

§7.9.4 Symmetric, Antisymmetric and Isotropic Tensors

We say a tensor T is **symmetric** in a pair of indices if swapping them leaves the components unchanged, e.g. $T_{ij \dots} = T_{ji \dots}$, and **antisymmetric** if swapping them flips the

sign. Crucially, these are properties of the *tensor*, not of the basis used: if $T_{ij} = T_{ji}$, then

$$T'_{ij} = R_{ip}R_{jq}T_{pq} = R_{ip}R_{jq}T_{qp} = T'_{ji},$$

by relabelling dummy indices, and similarly for antisymmetry. A tensor is **totally symmetric** (or antisymmetric) if it is so in every pair of indices; δ_{ij} and $a_i a_j a_k$ are totally symmetric, while ϵ_{ijk} is totally antisymmetric.

Any rank 2 tensor splits into symmetric and antisymmetric parts,

$$T_{ij} = \underbrace{\frac{1}{2}(T_{ij} + T_{ji})}_{S_{ij}} + \underbrace{\frac{1}{2}(T_{ij} - T_{ji})}_{A_{ij}}.$$

Counting components: S_{ij} has 6 independent components and A_{ij} has 3 (the entries above the diagonal), which adds up to the 9 components of a general rank 2 tensor. The count of 3 is suspicious – it suggests the antisymmetric part is secretly a *vector*, and indeed it is.

Proposition 7.9.10

Every rank 2 tensor T_{ij} can be written uniquely as

$$T_{ij} = S_{ij} + \epsilon_{ijk}\omega_k,$$

where S_{ij} is symmetric and $\omega_k = \frac{1}{2}\epsilon_{kpq}T_{pq}$.

Proof. Taking $S_{ij} = \frac{1}{2}(T_{ij} + T_{ji})$, we need only show $\epsilon_{ijk}\omega_k$ recovers the antisymmetric part. Indeed,

$$\epsilon_{ijk}\omega_k = \frac{1}{2}\epsilon_{ijk}\epsilon_{kpq}T_{pq} = \frac{1}{2}(\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})T_{pq} = \frac{1}{2}(T_{ij} - T_{ji}) = A_{ij}.$$

Uniqueness follows since the symmetric and antisymmetric parts of a tensor are determined by T (symmetrise or antisymmetrise both sides), and ω is determined by A . $\square$

Example 7.9.11 (Strain and Rotation in an Elastic Body)

Here is a physical situation where performing this decomposition is genuinely informative. Suppose an elastic body deforms, with the point $\mathbf{x}$ moving to $\mathbf{x} + \mathbf{u}(\mathbf{x})$ for a small displacement field $\mathbf{u}$. Two nearby points originally separated by $\delta\mathbf{x}$ end up separated by

$$\delta\mathbf{x} + \mathbf{u}(\mathbf{x} + \delta\mathbf{x}) - \mathbf{u}(\mathbf{x}), \quad \text{where} \quad u_i(\mathbf{x} + \delta\mathbf{x}) - u_i(\mathbf{x}) = \frac{\partial u_i}{\partial x_j}\delta x_j + o(\delta\mathbf{x}).$$

Decompose the tensor $\partial u_i/\partial x_j$ as above:

$$\frac{\partial u_i}{\partial x_j} = e_{ij} + \epsilon_{ijk}\omega_k, \quad e_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \quad \omega_k = -\frac{1}{2}[\nabla \times \mathbf{u}]_k.$$

Then

$$u_i(\mathbf{x} + \delta\mathbf{x}) - u_i(\mathbf{x}) = e_{ij}\delta x_j + [\delta\mathbf{x} \times \boldsymbol{\omega}]_i + o(\delta\mathbf{x}).$$

The antisymmetric part contributes a small *rotation* of the body about the axis ω , which doesn't deform the body at all. The actual stretching and shearing is captured entirely by the symmetric part e_{ij} , which is called the **linear strain tensor**.

For symmetric rank 2 tensors we can say more, by borrowing a theorem from linear algebra.

Proposition 7.9.12 (Principal Axes)

If T_{ij} is a symmetric rank 2 tensor, then there is a basis in which its components are diagonal. The corresponding coordinate axes are called the **principal axes** of the tensor.

Proof. A real symmetric matrix can be diagonalised by an orthogonal matrix, which we may take to have determinant 1 (flip the sign of a column if necessary). The change of basis formula for matrices, $T' = RTR^\top$, is componentwise exactly the transformation law $T'_{ij} = R_{ip}R_{jq}T_{pq}$ for rank 2 tensors, so applying that rotation as our change of basis diagonalises the components. $\square$

Finally, we single out the tensors that look the same in *every* basis.

Definition 7.9.13 (Isotropic Tensor)

A tensor is **isotropic** if its components are the same in every (right-handed, orthonormal) basis, that is,

$$R_{ip}R_{jq} \cdots R_{kr}T_{pq \cdots r} = T_{ij \cdots k} \quad \text{for every rotation } R.$$

We have already met the key examples: every scalar is isotropic, and so are δ_{ij} and ϵ_{ijk} . Remarkably, in three dimensions this is essentially a complete list.

Theorem 7.9.14 (Classification of Isotropic Tensors)

In $\mathbb{R}^3$, up to scalar multiples:

- (i) every rank 0 tensor is isotropic;
- (ii) the only isotropic tensor of rank 1 is $\mathbf{0}$;
- (iii) the isotropic tensors of rank 2 are $\alpha\delta_{ij}$;
- (iv) the isotropic tensors of rank 3 are $\beta\epsilon_{ijk}$;
- (v) the isotropic tensors of rank 4 are

$$\alpha\delta_{ij}\delta_{kl} + \beta\delta_{ik}\delta_{jl} + \gamma\delta_{il}\delta_{jk};$$

and every isotropic tensor of higher rank is a linear combination of products of δ 's and ϵ 's.

Proof (Sketch). The idea is to test the isotropy condition against enough specific rotations. We illustrate ranks 1 and 2.

For rank 1: suppose $v_i = R_{ij}v_j$ for all rotations R . Taking R to be the rotation by π about the x_3 -axis, which sends $(v_1, v_2, v_3) \mapsto (-v_1, -v_2, v_3)$, we get $v_1 = v_2 = 0$. Rotating by π about the x_1 -axis then forces $v_3 = 0$.

For rank 2: let R be the rotation by $\pi/2$ about the x_3 -axis,

$$R = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Then isotropy gives $T_{13} = R_{1p}R_{3q}T_{pq} = T_{23}$ but also $T_{23} = R_{2p}R_{3q}T_{pq} = -T_{13}$, so $T_{13} = T_{23} = 0$, and similarly all off-diagonal entries vanish. Moreover $T_{11} = R_{1p}R_{1q}T_{pq} = T_{22}$, and using rotations about the other axes, $T_{11} = T_{22} = T_{33}$. Hence $T_{ij} = \alpha\delta_{ij}$.

Higher ranks proceed with the same technique, only with more indices to chase. $\square$

Isotropic tensors give a beautiful trick for evaluating certain integrals without doing any real work.

Proposition 7.9.15

Let f be a function of $r = |\mathbf{x}|$ only. Then

$$T_{ij\dots k} = \int_{|\mathbf{x}| < R} f(r) x_i x_j \cdots x_k \, dV$$

is an isotropic tensor (as is the corresponding integral over all of $\mathbb{R}^3$, when it converges).

Proof. T is a tensor (each basis gives an array, and one can check the multilinear criterion). For isotropy, transform and substitute $y_i = R_{ip}x_p$ in the integral: a rotation has $|\det| = 1$, so dV is unchanged, and both $f(r)$ and the ball $|\mathbf{x}| < R$ are rotation-invariant. Hence

$$T'_{ij\dots k} = \int_{|\mathbf{x}| < R} f(r) R_{ip}x_p R_{jq}x_q \cdots R_{kr}x_r \, dV = \int_{|\mathbf{y}| < R} f(r) y_i y_j \cdots y_k \, dV = T_{ij\dots k},$$

as required. $\square$

Example 7.9.16

Let's compute

$$T_{ij} = \int_{\mathbb{R}^3} e^{-r^5} x_i x_j \, dV.$$

By the proposition, T_{ij} is isotropic of rank 2, so $T_{ij} = \alpha\delta_{ij}$ for some scalar α – and to find α we need only compute *one* number, which we do by contracting:

$$3\alpha = T_{ii} = \int_{\mathbb{R}^3} e^{-r^5} r^2 \, dV = 4\pi \int_0^\infty e^{-r^5} r^4 \, dr = 4\pi \left[-\frac{e^{-r^5}}{5} \right]_0^\infty = \frac{4\pi}{5}.$$

Hence $T_{ij} = \frac{4\pi}{15}\delta_{ij}$, and we have computed nine integrals for the price of one.

§7.9.5 Tensor Calculus

Just as scalar and vector fields assign a scalar or vector to each point of space, a **tensor field** of rank n assigns a rank n tensor $T_{ij\dots k}(\mathbf{x})$ to each point. Our differential operators interact with tensor fields in a very clean way, and the starting observation is that *differentiation itself transforms like a tensor index*.

Since $x'_i = R_{ij}x_j$ and equivalently $x_j = R_{ij}x'_i$, we have $\partial x_j / \partial x'_i = R_{ij}$, so by the chain rule

$$\frac{\partial}{\partial x'_i} = \frac{\partial x_j}{\partial x'_i} \frac{\partial}{\partial x_j} = R_{ij} \frac{\partial}{\partial x_j}.$$

Proposition 7.9.17

If $T_{i\dots j}$ is a rank n tensor field, then

$$\frac{\partial}{\partial x_p} \cdots \frac{\partial}{\partial x_q} T_{i\dots j}(\mathbf{x}) \quad (m \text{ derivatives})$$

is a rank $n + m$ tensor field.

Proof. Under a change of basis, using the transformation law for T and for each derivative,

$$\begin{aligned} \frac{\partial}{\partial x'_p} \cdots \frac{\partial}{\partial x'_q} T'_{i\dots j} &= R_{pa} \frac{\partial}{\partial x_a} \cdots R_{qb} \frac{\partial}{\partial x_b} (R_{ic} \cdots R_{jd} T_{c\dots d}) \\ &= R_{pa} \cdots R_{qb} R_{ic} \cdots R_{jd} \frac{\partial}{\partial x_a} \cdots \frac{\partial}{\partial x_b} T_{c\dots d}, \end{aligned}$$

where we could pull the (constant!) matrices R out of the derivatives. $\square$

Note the parenthetical remark: this argument works precisely because in Cartesian coordinates the change of basis matrix is constant. (This is why this topic is called *Cartesian* tensors – in general curvilinear coordinates, differentiation of tensor fields is much more subtle.)

This proposition packages up several facts we already knew: ∇f is a vector field (rank $0 + 1$), $\nabla \cdot \mathbf{F} = \partial F_i / \partial x_i$ is a scalar field (a contraction of the rank 2 field $\partial F_j / \partial x_i$), and $[\nabla \times \mathbf{F}]_i = \epsilon_{ijk} \partial F_k / \partial x_j$ is a vector field (contract ϵ with $\partial F_k / \partial x_j$ twice).

The divergence theorem also upgrades to tensor fields, and the proof is a pleasant application of what we know.

Theorem 7.9.18 (Divergence Theorem for Tensor Fields)

For a tensor field $T_{ij\dots k\dots l}$ and a volume V with piecewise regular boundary,

$$\int_V \frac{\partial}{\partial x_k} T_{ij\dots k\dots l} \, dV = \int_{\partial V} T_{ij\dots k\dots l} n_k \, dS,$$

where $\mathbf{n}$ is the outward unit normal, and the contraction may be on any chosen index.

Proof. Fix constant vectors $\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}$ and consider the vector field with components

$$v_k = a_i b_j \cdots c_l T_{ij \dots k \dots l},$$

contracting every index of T except the k . Applying the ordinary divergence theorem to $\mathbf{v}$,

$$\begin{aligned} a_i b_j \cdots c_l \int_V \frac{\partial}{\partial x_k} T_{ij \dots k \dots l} dV &= \int_V \frac{\partial v_k}{\partial x_k} dV \\ &= \int_{\partial V} v_k n_k dS = a_i b_j \cdots c_l \int_{\partial V} T_{ij \dots k \dots l} n_k dS. \end{aligned}$$

Since $\mathbf{a}, \mathbf{b}, \dots, \mathbf{c}$ were arbitrary, the two multilinear forms agree, so the tensors themselves are equal. $\square$

§7.9.6 The Inertia Tensor

We end with the classic example of a rank 2 tensor in physics. Consider a rigid body occupying a volume V with mass density $\rho(\mathbf{x})$, rotating with angular velocity $\boldsymbol{\omega}$ about an axis through the origin. Each point of the body moves with velocity $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{x}$, so the total angular momentum about the origin is

$$\mathbf{L} = \int_V \rho(\mathbf{x}) \mathbf{x} \times \mathbf{v} dV = \int_V \rho(\mathbf{x}) \mathbf{x} \times (\boldsymbol{\omega} \times \mathbf{x}) dV.$$

Expanding the triple product in index notation,

$$[\mathbf{x} \times (\boldsymbol{\omega} \times \mathbf{x})]_i = \epsilon_{ijk} x_j \epsilon_{klm} \omega_l x_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) x_j \omega_l x_m = (x_k x_k) \omega_i - x_i x_j \omega_j,$$

so

$$L_i = \left[\int_V \rho(\mathbf{x}) (x_k x_k \delta_{ij} - x_i x_j) dV \right] \omega_j = I_{ij} \omega_j,$$

where we define the **inertia tensor** of the body,

$$I_{ij} = \int_V \rho(\mathbf{x}) (x_k x_k \delta_{ij} - x_i x_j) dV.$$

So angular momentum depends *linearly* on angular velocity, but – unlike its familiar one dimensional counterpart $L = I\omega$ – the angular momentum need not be parallel to the rotation axis. That I_{ij} really is a tensor follows from the quotient theorem (or directly, since $x_k x_k \delta_{ij} - x_i x_j$ is built from tensors and the integral transforms as in our isotropy proposition).

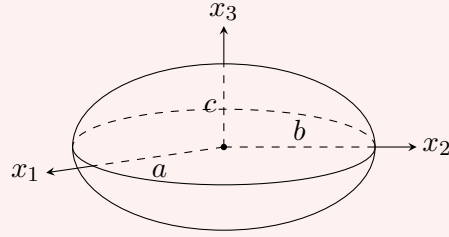
Since I_{ij} is manifestly symmetric, the principal axes theorem applies: every rigid body has three perpendicular *principal axes* for which the inertia tensor is diagonal, and rotation about a principal axis gives $\mathbf{L}$ parallel to $\boldsymbol{\omega}$.

Example 7.9.19 (Inertia Tensor of an Ellipsoid)

Let V be the solid ellipsoid

$$V = \left\{ \mathbf{x} \mid \frac{x_1^2}{a^2} + \frac{x_2^2}{b^2} + \frac{x_3^2}{c^2} \leq 1 \right\}$$

with uniform density ρ_0 and mass $M = \frac{4}{3}\pi\rho_0 abc$.



To evaluate the integrals, use ‘scaled’ spherical polar coordinates

$$x_1 = ar \sin \theta \cos \varphi, \quad x_2 = br \sin \theta \sin \varphi, \quad x_3 = cr \cos \theta,$$

with $r \in [0, 1]$, which have Jacobian $|J| = abc r^2 \sin \theta$.

First, the off-diagonal terms. For $i \neq j$, the integrand $x_i x_j$ is odd under flipping the sign of one coordinate while the region and density are symmetric, so

$$I_{ij} = - \int_V \rho_0 x_i x_j \, dV = 0, \quad i \neq j.$$

So the coordinate axes are principal axes – as we’d expect, they are the axes of symmetry of the ellipsoid. For the diagonal terms,

$$\begin{aligned} I_{33} &= \int_V \rho_0 (x_1^2 + x_2^2) \, dV \\ &= \rho_0 abc \int_{\varphi=0}^{2\pi} \int_{\theta=0}^{\pi} \int_{r=0}^1 r^2 \sin^2 \theta (a^2 \cos^2 \varphi + b^2 \sin^2 \varphi) r^2 \sin \theta \, dr \, d\theta \, d\varphi \\ &= \frac{\rho_0 abc}{5} \cdot \pi (a^2 + b^2) \int_{\theta=0}^{\pi} \sin^3 \theta \, d\theta \\ &= \frac{\rho_0 abc}{5} \cdot \pi (a^2 + b^2) \cdot \frac{4}{3} = \frac{M}{5} (a^2 + b^2), \end{aligned}$$

using $\int_0^{2\pi} \cos^2 \varphi \, d\varphi = \int_0^{2\pi} \sin^2 \varphi \, d\varphi = \pi$ and $\int_0^{\pi} \sin^3 \theta \, d\theta = \frac{4}{3}$. By symmetry,

$$I_{11} = \frac{M}{5} (b^2 + c^2), \quad I_{22} = \frac{M}{5} (a^2 + c^2),$$

so

$$I = \frac{M}{5} \begin{pmatrix} b^2 + c^2 & 0 & 0 \\ 0 & a^2 + c^2 & 0 \\ 0 & 0 & a^2 + b^2 \end{pmatrix}.$$

Example 7.9.20 (Inertia Tensor of a Ball, the Slick Way)

For a uniform ball of radius R and mass $M = \frac{4}{3}\pi\rho_0 R^3$, we could set $a = b = c = R$ above, but isotropy gives a cleaner route. The region and density are rotation-invariant, so I_{ij} is an isotropic rank 2 tensor: $I_{ij} = \alpha \delta_{ij}$. Contracting,

$$3\alpha = I_{ii} = \int_{|\mathbf{x}| \leq R} \rho_0 (3r^2 - r^2) \, dV = 2\rho_0 \int_{|\mathbf{x}| \leq R} r^2 \, dV = 2\rho_0 \cdot 4\pi \int_0^R r^4 \, dr = \frac{8\pi\rho_0 R^5}{5},$$

so $\alpha = \frac{8\pi\rho_0 R^5}{15} = \frac{2}{5}MR^2$, giving

$$I_{ij} = \frac{2}{5}MR^2 \delta_{ij}.$$

Every axis through the centre of a ball is a principal axis, with the familiar moment of inertia $\frac{2}{5}MR^2$.

And with that, we have reached the end of the course. From arc lengths to inertia tensors, the recurring themes have been the same: differential operators built from ∇ , integrals over curves, surfaces and volumes, the integral theorems binding the two together, and index notation quietly doing most of the heavy lifting.

8 Vectors and Matrices

Vectors and Matrices covers topics in both algebra and geometry, and the ways in which they relate to one another. Starting from the complex numbers, we will work our way up through vectors in three dimensions, vectors in general, and then to linear maps and the matrices that represent them – along the way meeting determinants, eigenvalues, and the various canonical forms a matrix can be put into. The approaches used throughout are quite varied in their nature (they can be abstract or concrete, conceptual or computational, and so on), and being able to switch fluently between these viewpoints is one of the main skills this course develops.

This article constitutes my notes for the ‘Vectors and Matrices’ course, held in Michaelmas 2020 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§8.1 Complex Numbers

The complex numbers arose through the study of polynomials, but there is hardly an area of modern mathematics where they are not of use, with applications stretching from number theory to geometry to quantum mechanics.

§8.1.1 Defining the Complex Numbers

We are going to start right at the beginning, though some familiarity with complex numbers is assumed. We will construct the set of complex numbers, denoted by $\mathbb{C}$, from the real numbers $\mathbb{R}$ by adjoining an element i with the property that $i^2 = -1$.

Definition 8.1.1 (Complex Numbers)

A **complex number** is a number $z \in \mathbb{C}$ of the form $z = x + yi$ with $x, y \in \mathbb{R}$ such that $i^2 = -1$.

We call $x = \operatorname{Re}(z)$ the **real part**, and $y = \operatorname{Im}(z)$ the **imaginary part** of z .

The reals are a subset of the complex numbers, as for $x \in \mathbb{R}$, we have $x + 0i \in \mathbb{C}$.

Basic Operations

With the complex numbers defined, we have to come up with some things to do with them. We can add, subtract, multiply and divide complex numbers in a sensible way, and indeed they form a *field* (we will elaborate on this later).

Definition 8.1.2 (Basic Operations in $\mathbb{C}$)

Let $z_1 = x_1 + y_1i$ and $z_2 = x_2 + y_2i$ be two complex numbers. Then we define **addition** and **multiplication** as

$$z_1 + z_2 = (x_1 + x_2) + (y_1 + y_2)i,$$

$$z_1 \times z_2 = (x_1x_2 - y_1y_2) + (x_1y_2 + x_2y_1)i.$$

We can see from this definition that the operation of multiplication respects our original requirement that $i^2 = -1$. From these definitions we can also immediately define **subtraction** and **division** as the inverse operations of these, with

$$\begin{aligned} z_1 - z_2 &= (x_1 - x_2) + (y_1 - y_2)i \\ \frac{z_1}{z_2} &= \frac{x_1 + y_1i}{x_2 + y_2i} \cdot \frac{x_2 - y_2i}{x_2 - y_2i} = \left(\frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} \right) + \left(\frac{x_2y_1 - x_1y_2}{x_2^2 + y_2^2} \right)i. \end{aligned}$$

From the definitions of addition and multiplication, it is clear that both operations are commutative and associative. Indeed, as we also have inverse operations, the complex numbers form groups.

Proposition 8.1.3 ($\mathbb{C}$ is a Group)

$\mathbb{C}$ with the operation $+$ is an abelian group with identity 0, and $\mathbb{C} \setminus \{0\}$ with the operation $\times$ is an abelian group with identity 1.

Proof Sketch. Check definitions and verify that $+$ and $\times$ are indeed associative. $\square$

Another useful operation is the *complex conjugate*.

Definition 8.1.4 (Complex Conjugation)

For a complex number $z = x + iy$, we define its **complex conjugate** to be $\bar{z} = z^* = x - iy$.

Complex conjugation allows us to extract the real and imaginary parts of a complex number, with

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \text{and} \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}.$$

For complex numbers z_1 and z_2 , we have $\overline{\bar{z}} = z$, $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$ and $\overline{z_1 z_2} = \bar{z}_1 \cdot \bar{z}_2$.

Aside: The History of Complex Numbers

The complex numbers arose through the study of polynomials, and their serious study really began in the study of the *cubic* polynomials of the form¹

$$x^3 - 3px - 2q = 0.$$

The sixteenth century mathematicians del Ferro, Tartaglia and Cardano discovered that this equation could be solved explicitly, with the result being published in Cardano's 1545 work *Ars Magna* (where he mentioned del Ferro as a first author, and noted that Tartaglia had independently found the same result). The solution they had obtained was

$$x = \sqrt[3]{q + \sqrt{q^2 - p^3}} + \sqrt[3]{q - \sqrt{q^2 - p^3}}.$$

For example, for the cubic $x^3 - 12x - 20$ we obtain the solution $x = \sqrt[3]{16} + \sqrt[3]{2}$.

¹This form of a cubic equation is a 'depressed', and any general cubic can be expressed in this form.

Some 30 years later, Rafael Bombelli discussed one particular cubic in his work *l'Algebra*. He looked at $x^3 - 15x - 4 = 0$. One might observe (through mere trial and error) that $x = 4$ is a solution, but using Cardano's formula, we obtain the solution

$$x = \sqrt[3]{2 + \sqrt{-121}} + \sqrt[3]{2 - \sqrt{-121}}.$$

Indeed, Bombelli noticed this and gave an explanation for how these two solutions were essentially the same, even with the presence of the rather puzzling term $\sqrt{-121}$. His explanation essentially began by noting that if we wrote $\sqrt{-121} = 11i$, then we could find that

$$\begin{aligned}\sqrt[3]{2 + \sqrt{-121}} &= 2 + ni \\ \sqrt[3]{2 - \sqrt{-121}} &= 2 - ni,\end{aligned}$$

and adding would then result in $2 + ni + 2 - ni = 4$, our other solution.

Inherent in this explanation was an expectation that this i would obey many of the same rules of arithmetic that we would expect of, say, a real number. After this work, contributions from mathematicians such as Descartes, Wallis, de Moivre, Euler, Wessel, Argand, Hamilton, Gauss, Cauchy and many others began to build up a rich theory of the complex numbers, later developing the field of complex analysis, all centered on this construction of some i with the property that $i^2 = -1$.

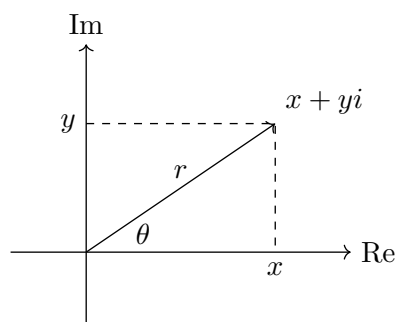
Looking again at the solutions to polynomials it turns out that the complex numbers are sufficient to solve *every* polynomial equation. This result is known as the 'Fundamental Theorem of Algebra', and was proved in the 18th century. We will use it frequently in this course, but you will have to wait until a later course to see a proof.

Theorem 8.1.5 (Fundamental Theorem of Algebra)

Let $p(z)$ be a polynomial of degree $n \geq 1$ with complex coefficients. Then $p(z) = 0$ has precisely n (not necessarily distinct) complex roots, counted with multiplicity.

The Argand Diagram

There is a geometric way to interpret complex numbers and the way that they operate. We can plot a complex number $z = x + yi$ on the **Argand diagram** or **complex plane** by associating it with a point (x, y) . We call the x axis the 'real' axis, and the y axis the 'imaginary' axis.



From the complex number written $z = x + yi$, we can plot it in rectangular coordinates. However, it can also be useful to plot it using polar coordinates. With this in mind, we introduce some useful definitions.

Definition 8.1.6 (Modulus)

For a complex number $z = x + yi$, we define its **modulus** to be $r = |z|$ such that $r \geq 0$ and $r^2 = |z|^2 = x^2 + y^2$.

Indeed, this definition matches the distance from the origin of the Argand diagram to the point (x, y) .

Definition 8.1.7 (Argument)

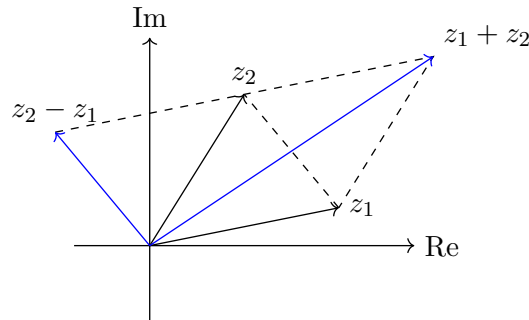
For a non-zero complex number z , we define its **argument** to be $\theta = \arg(z)$, a real number such that $z = r(\cos \theta + i \sin \theta)$. We say this is the **polar form** of z .

From this, we get that for $z = x + yi$,

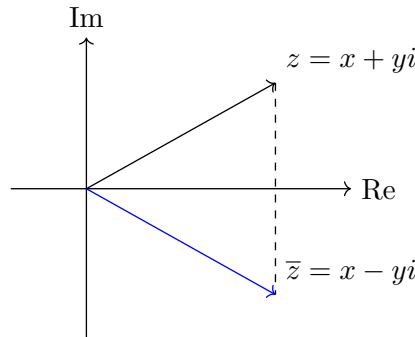
$$\cos \theta = \frac{x}{\sqrt{x^2 + y^2}}, \quad \sin \theta = \frac{y}{\sqrt{x^2 + y^2}}, \quad \text{and} \quad \tan \theta = \frac{y}{x}.$$

An important point to note is that $\arg(z)$ is only defined up to multiples of 2π , as mapping $\theta \mapsto \theta + 2\pi$ leaves z unchanged. So, to make θ unique (which is often useful), we typically restrict its range. For example, if we restrict $-\pi < \theta \leq \pi$, we get the **principal value** of θ .

Using the Argand diagram, there is a geometric way to define addition, multiplication and complex conjugation. Addition and subtraction correspond to parallelograms constructed as shown.



Complex conjugation corresponds to reflection in the real axis.

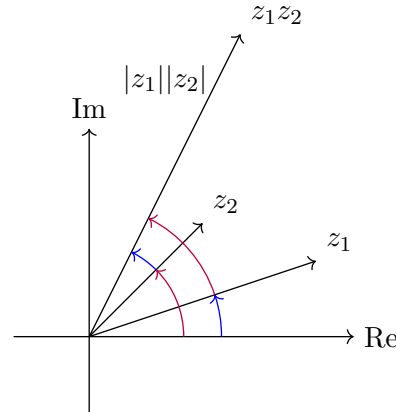


Multiplication is slightly more complex. If we have two complex numbers in polar form, $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then expanding and applying the angle sum identities gives $z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$. This gives the following.

Proposition 8.1.8 (Moduli Multiply and Arguments Add)

For complex numbers z_1 and z_2 , we have $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ and $|z_1 z_2| = |z_1| \cdot |z_2|$.

This is shown in the diagram below.



To see how useful a geometric perspective can be, consider the following result.

Proposition 8.1.9 (Triangle Inequality)

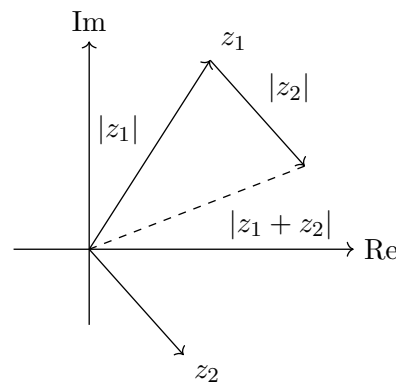
For complex numbers z_1 and z_2 , $|z_1 + z_2| \leq |z_1| + |z_2|$.

Algebraic Proof. We have $|z_1 + z_2|^2 = (z_1 + z_2)\overline{(z_1 + z_2)} = |z_1|^2 + z_1\overline{z_2} + \overline{z_1}z_2 + |z_2|^2$, so it suffices to show that $z_1\overline{z_2} + \overline{z_1}z_2 \leq 2|z_1||z_2|$. To show this, we have

$$\begin{aligned} z_1\overline{z_2} + \overline{z_1}z_2 &\leq 2|z_1||z_2| \\ \iff \frac{z_1\overline{z_2} + \overline{z_1}z_2}{2} &\leq |z_1||\overline{z_2}| \\ \iff \operatorname{Re}(z_1\overline{z_2}) &\leq |z_1\overline{z_2}|, \end{aligned}$$

which is true. □

Now put everything on an Argand diagram and stare at it for a minute.



Geometric Proof. The shortest distance between 0 and $z_1 + z_2$ is $|z_1 + z_2|$. This is a side length of the triangle at 0, z_1 and $z_1 + z_2$, and this must be shorter than the

other two sides of the triangle, which has length $|z_1| + |z_2|$. This gives the triangle inequality. $\square$

Alternate forms of the triangle inequality are

$$|z_2 - z_1| \geq |z_2| - |z_1|, \quad \text{and} \quad |z_2 - z_1| \geq |z_1| - |z_2|.$$

One of these tends to be trivially true, however, due to one side being negative. Still, we can combine these into a single statement

$$|z_2 - z_1| \geq ||z_2| - |z_1||.$$

We will end our discussion with the following theorem which will repeatedly appear.

Theorem 8.1.10 (De Moivre's Theorem)

For $n \in \mathbb{Z}$ and $\theta \in \mathbb{R}$, we have

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta,$$

where $z^0 = 1$ for non-zero z .

Proof Sketch. Use induction and angle sum identities. A different proof will be given after exponential functions are defined. $\square$

§8.1.2 Exponential and Trigonometric Functions

We define the functions $\exp$, $\cos$ and $\sin$ on the complex numbers by

$$\begin{aligned} \exp(z) &= \sum_{n=0}^{\infty} \frac{1}{n!} z^n = e^z \\ \cos(z) &= \frac{1}{2}(e^{iz} + e^{-iz}) = 1 - \frac{1}{2!}z^2 + \frac{1}{4!}z^4 - \dots \\ \sin(z) &= \frac{1}{2i}(e^{iz} - e^{-iz}) = z - \frac{1}{3!}z^3 + \frac{1}{5!}z^5 - \dots \end{aligned}$$

These functions are defined by **power series**, which will be explored more in Analysis. They converge for all z , and can be manipulated and rearranged as you may expect.

Aside: Power Series (Preview of Analysis I)

A **power series** about a point x_0 is a sum of the form

$$\sum_{n=0}^{\infty} a_n(x - x_0)^n,$$

where $a_n \in \mathbb{C}$. A power series is said to **converge** at a point x if the finite sums $\sum_{n=0}^N a_n(x - x_0)^n$ tend to a limit as $N \rightarrow \infty$. With each power series there is a **radius of convergence** ρ such that if $|x - x_0| < \rho$, then the power series converges absolutely. If $|x - x_0| > \rho$, then the series does not converge. If $|x - x_0| = \rho$, then it may or may not converge. In the functions defined above, the radius of convergence is infinity.

We can manipulate power series as you might expect. Notably, we can differentiate term by term (and the radius of convergence will stay the same), and we can add and multiply power series if we are in the radius of convergence of both.

Power series will be discussed in detail in Analysis I.

These definitions reduce to familiar definitions when $z \in \mathbb{R}$. Indeed, if we differentiate the series, we get

$$\frac{d}{dx} \exp(x) = \exp(x), \quad \frac{d}{dx} \cos(x) = -\sin(x), \quad \frac{d}{dx} \sin(x) = \cos(x),$$

and $\exp(0) = 1$, $\cos(0) = 1$, $\sin(0) = 0$, which together uniquely defines these functions over the reals. However, the behavior over $\mathbb{C}$ is somewhat richer.

Proposition 8.1.11 (Properties of the Complex Exponential)

For $z = x + yi$,

- (i) $e^z = e^x(\cos y + i \sin y)$.
- (ii) e^z can take on all values in $\mathbb{C}$ except 0.
- (iii) $e^z = 1$ if and only if $z = 2n\pi i$ for $n \in \mathbb{Z}$.

Proof. (i) $e^{x+iy} = e^x e^{iy}$, and $e^{iy} = \cos y + i \sin y$.

(ii) $|e^z| = |e^x|$, which takes on all positive real values, and $\arg(e^z) = y$, which takes on all real values.

(iii) $e^z = 1 \iff e^x = 1$ and $\cos y = 1$, $\sin y = 0$ which gives $x = 0$ and $y = 2n\pi$ as required. $\square$

Returning to polar form, we can write

$$z = r(\cos \theta + i \sin \theta) = re^{i\theta},$$

where $r = |z|$ and $\theta = \arg(z)$. Then De Moivre's theorem follows immediately from $(e^{i\theta})^n = e^{in\theta}$.

§8.1.3 Roots of Unity

An important application of these results are *roots of unity*, which answers the question 'find all z such that $z^n = 1$ '.

Definition 8.1.12 (Roots of Unity)

An n th root of unity is a complex number z such that $z^n = 1$.

Proposition 8.1.13 (Finding Roots of Unity)

The n th roots of unity are

$$z = e^{2k\pi i/n} = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n},$$

where $k \in \{0, 1, \dots, n-1\}$.

Proof. Writing z in polar form, these are easy to find. We have $z = re^{i\theta}$, then

$$z^n = r^n e^{in\theta} = 1,$$

so $r = 1$, and $\theta = \frac{2k\pi}{n}$ for some $k \in \mathbb{Z}$. This gives n distinct solutions. $\square$

To get a feel for this, we can plot the cases for $n = 2, 3, 4$ and 5 on an Argand diagram. This hints at a geometric interpretation of the roots of unity.

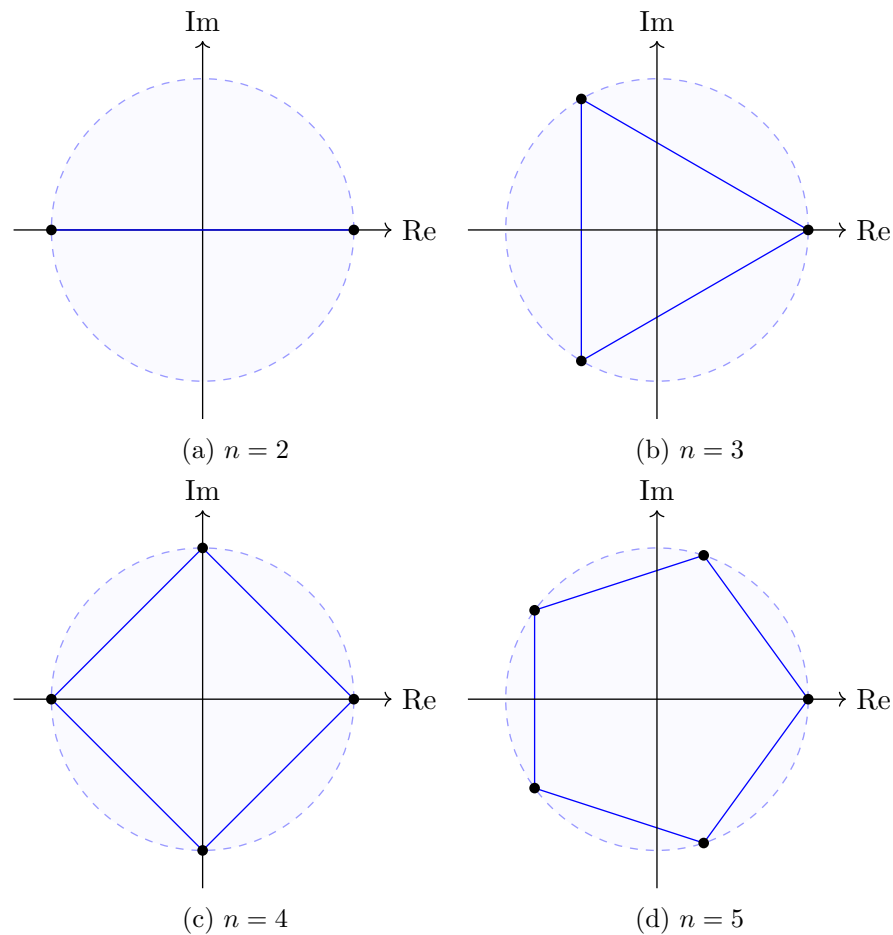


Figure 8.1: n th roots of unity on an Argand diagram

Proposition 8.1.14 (Geometric Interpretation of Roots of Unity)

The n th roots of unity correspond to the vertices of a regular n -gon, inscribed within the unit circle with a vertex at 1.

Proof. Follows directly from roots of unity written in polar form. $\square$

§8.1.4 Logarithms and Complex Powers

Over $\mathbb{R}$, the exponential function is one to one. This means that we can quite easily define the inverse function, the logarithm, on the positive reals (the image of the exponential over $\mathbb{R}$).

This is not true over $\mathbb{C}$, as the complex exponential function is many to one. For example, $e^{3\pi i/2} = e^{7\pi i/2} = -i$. However, we do know that if we have $e^{i\theta} = e^{i\gamma}$, then θ and γ differ by a multiple of 2π ². Thus we can define a *multi-valued* logarithm function.

The Complex Logarithm

To see how to define the logarithm over $\mathbb{C}$, let z be any non-zero complex number. Then if we let $\theta = \arg(z)$ with $-\pi < \theta \leq \pi$, then we can write

$$z = re^{i\theta} = e^{\log r} e^{i\theta + 2n\pi i} = e^{\log r + i(\theta + 2n\pi)},$$

for any $n \in \mathbb{Z}$.

Definition 8.1.15 (Complex Logarithm)

For a complex number z , we define the **complex logarithm** $\log z$ to be the multi-valued function

$$\log z = \log r + i(\theta + 2n\pi),$$

where $r = |z|$, $\theta = \arg(z)$ with $-\pi < \theta \leq \pi$ and $n \in \mathbb{Z}$.

Example 8.1.16 (Calculating Complex Logarithms)

Let $z = \frac{5}{2} + \frac{5\sqrt{3}i}{2}$. We will find $\log z$. We have $|z| = 5$, and the principal value is $\arg(z) = \pi/3$. Thus we have

$$\log z = \log 5 + i(\pi/3 + 2n\pi), \quad n \in \mathbb{Z}.$$

Sometimes we will want single-valued behavior from the complex logarithm, and we can define the **principal logarithm** so that we fix $n = 0$. A warning though: if this is done, not all of the familiar properties of the logarithm still hold.³

Complex Powers

The multi-valued nature of the logarithm carries into the notion of taking powers of complex numbers. For real numbers, we can define $x^a = e^{a \ln x}$ for $x > 0$. We can do the same for complex numbers.

Definition 8.1.17 (Complex Powers)

We define **complex powers** for $z, \alpha \in \mathbb{C}$ with $z \neq 0$ to be the multi-valued

$$z^\alpha = e^{\alpha \log z}.$$

²This comes from the argument of a complex number only being defined up to a multiple of 2π

³When taking the principal logarithm, we are really taking a *branch* of the multi-valued logarithm function. This corresponds with taking a slice of the Riemann surface that $\log z$ defines. This gives some weird behavior as we move past certain sets of points (singular points) which breaks many of the identities that we use frequently for the real valued logarithm.

This is not *always* multi valued. For example, if $\alpha \in \mathbb{Z}$, then z^α is unique. If $\alpha \in \mathbb{Q}$, then there will be only finitely many values. However, in most cases there will be infinitely many values.

Example 8.1.18 (Calculating i^i)

We have

$$\begin{aligned} i^i &= e^{i \log i} \\ &= e^{i \log 1 + i(\pi/2 + 2n\pi)} \\ &= e^{-(2n+1/2)\pi}, \end{aligned}$$

for any $n \in \mathbb{Z}$. Notably $i^i \in \mathbb{R}$.

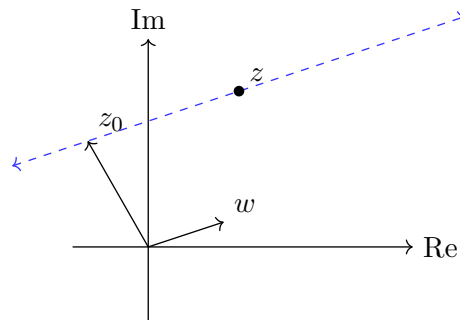
§8.1.5 Transformations, Lines and Circles

Consider the following transformations on $\mathbb{C}$.

- $z \mapsto z + \alpha$, translation by $\alpha \in \mathbb{C}$.
- $z \mapsto \lambda z$, scaling by $\lambda \in \mathbb{R}$,
- $z \mapsto e^{i\theta} z$, rotation by $\theta \in \mathbb{R}$.
- $z \mapsto \bar{z}$, reflection in the real axis.
- $z \mapsto \frac{1}{z}$, inversion.

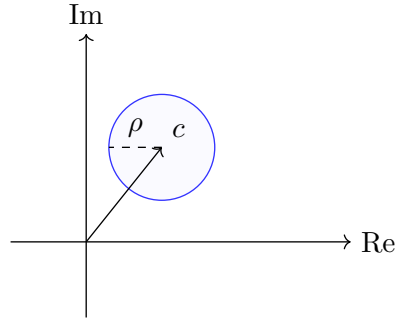
Using these transformations, we can think geometrically to do things like find equations of various geometric objects.

First we will find the general point on a line in $\mathbb{C}$ through z_0 , parallel to $w \in \mathbb{C}$ where $w \neq 0$.



We can write $z = z_0 + \lambda w$ for $\lambda \in \mathbb{R}$. To eliminate λ , we can take the conjugate $\bar{z} = \bar{z}_0 + \lambda \bar{w}$, and then multiplying by w and subtracting we obtain $\bar{w}z - w\bar{z} = \bar{w}z_0 - w\bar{z}_0$.

Similarly, we can find the general point on a circle with a center $c \in \mathbb{C}$ and positive real radius ρ .



A general point on this circle can be written $z = c + \rho e^{i\alpha}$, $\alpha \in \mathbb{R}$. To get rid of α , note that we could instead write $|z - c| = \rho$.

We can unify these two forms in the following way.

Proposition 8.1.19 (Circles and Lines in $\mathbb{C}$)

The equation of any line, and of any circle, may be written respectively as

$$\bar{B}z + B\bar{z} + C = 0 \quad \text{and} \quad z\bar{z} + \bar{B}z + B\bar{z} + C = 0$$

for some complex B and real C .

In particular, the first is a line in the direction iB , and the second (if $|B|^2 > C$) is a circle with centre $-B$ and radius $\sqrt{|B|^2 - C}$.

Proof. For the line, we found above that a general line through z_0 parallel to w satisfies $\bar{w}z - w\bar{z} = \bar{w}z_0 - w\bar{z}_0$. The right-hand side is a constant, and both sides are purely imaginary (each is equal to the negative of its own conjugate). Multiplying through by i , we can write this as $\bar{B}z + B\bar{z} + C = 0$ where $B = -iw$ and $C \in \mathbb{R}$, and then the direction of the line is $w = iB$ (rescaling to absorb factors). This process is reversible, so every equation of this form with $B \neq 0$ gives a line.

For the circle, expanding $|z - c|^2 = \rho^2$ gives

$$z\bar{z} - \bar{c}z - c\bar{z} + |c|^2 - \rho^2 = 0,$$

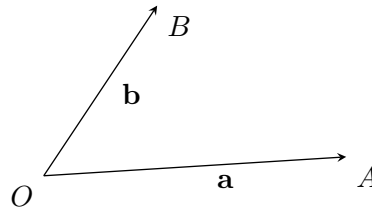
which is of the stated form with $B = -c$ and $C = |c|^2 - \rho^2 \in \mathbb{R}$. Conversely, given such an equation we can complete the square to get $|z + B|^2 = |B|^2 - C$, a circle with centre $-B$ and radius $\sqrt{|B|^2 - C}$, provided $|B|^2 > C$. $\square$

§8.2 Vectors in Three Dimensions

Fundamentally, vectors are mathematical objects that we can add together and take multiples of, in both cases obtaining another vector. We will consider vectors in a general setting later on, but in this chapter we will consider vectors in $\mathbb{R}^3$, three dimensional space. Specifically, we will take a geometric approach, thinking of vectors as position vectors and using Euclidean notions of points, lines, planes, etc.

We begin by choosing a point O as the origin. Then points A and B have position vectors

$$\mathbf{a} = \overrightarrow{OA}, \quad \mathbf{b} = \overrightarrow{OB}.$$



The vectors have length $|\mathbf{a}| = |\overrightarrow{OA}|$, the distance between O and A . We also let $\mathbf{0}$ denote the position vector of O .

§8.2.1 Vector Addition and Scalar Multiplication

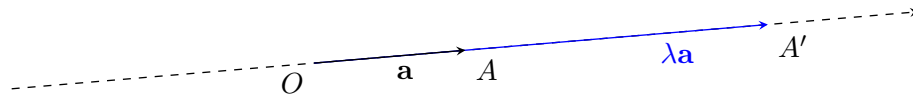
The most important operations on vectors (in $\mathbb{R}^3$ and in general) are vector addition and scalar multiplication. We define these geometrically as follows.

Definition 8.2.1 (Scalar Multiplication in $\mathbb{R}^3$)

Given $\mathbf{a}$, the position vector for a point A , and a scalar $\lambda \in \mathbb{R}$, we define **scalar multiplication** $\lambda\mathbf{a}$ to be the position vector for the point A' on the line OA such that

$$|\lambda\mathbf{a}| = |OA'| = |\lambda||\mathbf{a}|,$$

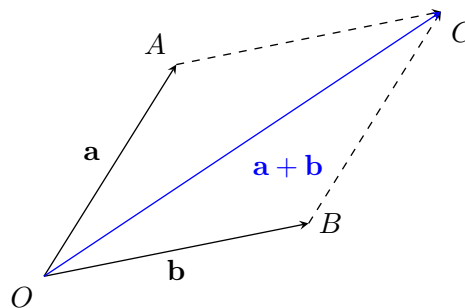
with the vector being in the direction $\overrightarrow{OA}$ if λ is positive, and in the opposite direction otherwise.



If two vectors are scalar multiples of each other, then we say that they are **parallel**. Specifically, we write $\mathbf{a} \parallel \mathbf{b}$ if and only if $\mathbf{a} = \lambda\mathbf{b}$ or $\mathbf{b} = \lambda\mathbf{a}$ for some $\lambda \in \mathbb{R}$. Notably, $\mathbf{a} \parallel \mathbf{0}$ for all vectors $\mathbf{a}$.

Definition 8.2.2 (Vector Addition in $\mathbb{R}^3$)

Given $\mathbf{a}$ and $\mathbf{b}$, position vectors of points A and B , if $\mathbf{a} \parallel \mathbf{b}$, we construct the parallelogram $OACB$, and define **vector addition** $\mathbf{a} + \mathbf{b} = \mathbf{c}$.



If $\mathbf{a} \parallel \mathbf{b}$, then writing $\mathbf{a} = \alpha\mathbf{u}$ and $\mathbf{b} = \beta\mathbf{u}$ where $\mathbf{u}$ is a unit vector, then $\mathbf{a} + \mathbf{b} = (\alpha + \beta)\mathbf{u}$.

Given some set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$, we can form a **linear combination**

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \dots + \lambda_k\mathbf{v}_k,$$

where $\lambda_i \in \mathbb{R}$. With this in mind, we can consider the set of all vectors that can be formed as a linear combination of some vectors.

Definition 8.2.3 (Span)

For vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$, we define their **span** to be the set

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} = \{\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k \mid \lambda_i \in \mathbb{R}\}.$$

If two vectors $\mathbf{a} \nparallel \mathbf{b}$, then $\text{span}\{\mathbf{a}, \mathbf{b}\}$ is a plane through OAB .

Vector addition and scalar multiplication obey some basic properties that you should keep in mind, as they will be used again when we define vectors in a general sense.

Proposition 8.2.4 (Basic Properties of Vector Operations)

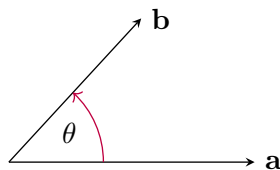
For any vectors $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$,

- (i) Vectors along with vector addition form an abelian group.
- (ii) $\lambda(\mathbf{a} + \mathbf{b}) = \lambda\mathbf{a} + \lambda\mathbf{b}$.
- (iii) $(\lambda + \mu)\mathbf{a} = \lambda\mathbf{a} + \mu\mathbf{a}$.
- (iv) $\lambda(\mu\mathbf{a}) = (\lambda\mu)\mathbf{a}$.

Proof Sketch. Check geometric definitions. Checking associativity of vector addition will require the construction of a parallelepiped. $\square$

§8.2.2 The Dot Product

We saw in the previous section how to add vectors, and how to multiply a vector by a scalar. In the next two sections, we will consider how to multiply a vector and a vector. We will first define the *scalar* or *dot product*. Consider two vectors $\mathbf{a}$ and $\mathbf{b}$, with the angle between them being θ as shown.



Definition 8.2.5 (Dot Product)

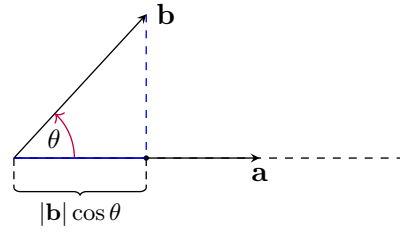
For two vectors $\mathbf{a}$ and $\mathbf{b}$, and θ the angle between them as shown, we define the **scalar** or **dot product** as

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta.$$

[Note that θ is defined unless $\mathbf{a}$ or $\mathbf{b}$ is zero, but then $|\mathbf{a}| = 0$ or $|\mathbf{b}| = 0$ and $\mathbf{a} \cdot \mathbf{b} = 0$.]

The scalar product encodes the angle between vectors, and in particular it gives us a test for perpendicularity. We say vectors $\mathbf{a}$ and $\mathbf{b}$ are **orthogonal** or **perpendicular** if and only if $\mathbf{a} \cdot \mathbf{b} = 0$. This corresponds with $\theta = \pm\pi/2$, or with either $|\mathbf{a}| = 0$ or $|\mathbf{b}| = 0$.

There is a geometric interpretation of the dot product. Fundamentally, the dot product is a *projection*. For $\mathbf{a} \neq \mathbf{0}$, the quantity $|\mathbf{b}| \cos \theta$ is the component of $\mathbf{b}$ when projected along the direction of $\mathbf{a}$.



Proposition 8.2.6 (Basic Properties of the Dot Product)

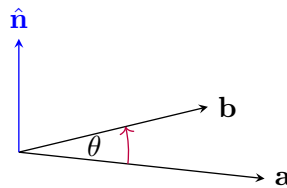
For any vectors $\mathbf{a}$ and $\mathbf{b}$,

- (i) $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.
- (ii) $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2 \geq 0$, and $\mathbf{a} \cdot \mathbf{a} = 0$ if and only if $\mathbf{a} = \mathbf{0}$.
- (iii) $(\lambda \mathbf{a}) \cdot \mathbf{b} = \lambda(\mathbf{a} \cdot \mathbf{b}) = \mathbf{a} \cdot (\lambda \mathbf{b})$.
- (iv) $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$.

Proof. Properties (i) to (iii) follow directly from the definition of the dot product. Property (iv) follows from the geometric interpretation of the dot product shown above. $\square$

§8.2.3 The Cross Product

The next operation only works with vectors in three dimensions, and is the *vector* or *cross product*. As before, consider two vectors $\mathbf{a}$ and $\mathbf{b}$, and let θ be measured as shown with respect to $\hat{\mathbf{n}}$, a unit normal to the plane spanned by $\mathbf{a}$ and $\mathbf{b}$.



Definition 8.2.7 (Cross Product)

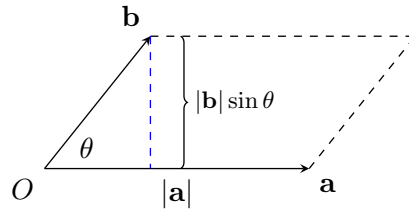
For two vectors $\mathbf{a}$ and $\mathbf{b}$ with angle θ measured as shown, we define the **vector** or **cross product** as

$$\mathbf{a} \times \mathbf{b} = |\mathbf{a}||\mathbf{b}| \sin \theta \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane spanned by $\mathbf{a}$ and $\mathbf{b}$.

[$\hat{\mathbf{n}}$ is defined up to a sign if $\mathbf{a} \nparallel \mathbf{b}$, but changing the sign changes θ to $2\pi - \theta$, and thus leaves $\mathbf{a} \times \mathbf{b}$ unchanged. When $\hat{\mathbf{n}}$ is not defined ($\mathbf{a} \parallel \mathbf{b}$ or θ not defined), $\mathbf{a} \times \mathbf{b} = \mathbf{0}$.]

The geometric interpretation of the cross product is that $|\mathbf{a} \times \mathbf{b}|$ is the area of the parallelogram as shown.



The direction of $\mathbf{a} \times \mathbf{b}$ gives the orientation of this parallelogram in space.

For an alternate geometric interpretation, fix a vector $\mathbf{a}$ and consider some vector $\mathbf{x}$ with $\mathbf{x} \perp \mathbf{a}$. Then the transformation $\mathbf{x} \mapsto \mathbf{a} \times \mathbf{x}$ corresponds with scaling $\mathbf{x}$ by $|\mathbf{a}|$ and rotating by $\pi/2$ in the plane perpendicular to $\mathbf{a}$.

Proposition 8.2.8 (Basic Properties of the Cross Product)

For vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$,

- (i) $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$.
- (ii) $(\lambda \mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b})$.
- (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.
- (iv) $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ if and only if $\mathbf{a} \parallel \mathbf{b}$.
- (v) $\mathbf{a} \times \mathbf{b} \perp \mathbf{a}, \mathbf{b}$, so $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

Proof Sketch. Check definitions. □

§8.2.4 Orthonormal Bases and Components

So far we have considered vectors in a purely geometric sense. But just as a coordinate system can be used to reason about geometry, basis (and by extension components) can be used to reason about vectors. We will look at bases again in a more general setting later on, but for now we will focus on vectors in three dimensions.

When we use a cartesian coordinate system, we pick a set of axes and using them, we can describe the position of any point. In a similar way, a *basis* is a set of vectors that span the space. That is, any vector can be written as a linear combination of vectors in the set. However we also have an additional requirement: this linear combination must be unique.

Definition 8.2.9 (Basis – Informal)

We say that a set of vectors is a **basis** if any vector can be uniquely written as a linear combination of vectors in the set.

Before we continue, it's helpful when working with vectors in three dimensions (and in other settings) to pick basis vectors in a certain way. We introduce the following definition.

Definition 8.2.10 (Orthonormal)

We say that a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is **orthonormal** if they are all unit

vectors and are orthogonal to each other. That is, if $|\mathbf{v}_i| = 1$ and

$$\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

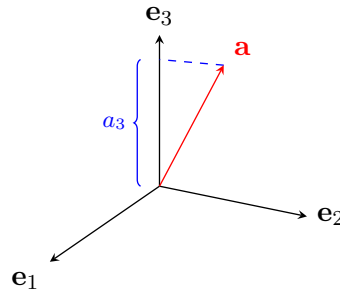
for all $1 \leq i, j \leq n$.

Now choose vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ that are **orthonormal**. Then the set $\{\mathbf{e}_i\}$ is a basis, so for any vector $\mathbf{a}$, we can write

$$\mathbf{a} = \sum_i a_i \mathbf{e}_i = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3,$$

and each coefficient is uniquely defined by $a_i = \mathbf{e}_i \cdot \mathbf{a}$. Note that this only works because of our orthonormal condition.

Recalling that dot products correspond to projections, we can see that this really is analogous to a cartesian coordinate system.



Once we have specified the basis vectors $\{\mathbf{e}_i\}$, we can then identify the vector $\mathbf{a}$ by its **components**, writing

$$\mathbf{a} = (a_1, a_2, a_3), \quad \text{or} \quad \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix},$$

which is a **row** or **column** vector respectively⁴.

An advantage of writing vectors in this form is that we can work directly with the components of vectors, rather than using a purely geometric approach.

Dot Product in Components

For two vectors $\mathbf{a} = (a_1, a_2, a_3)$ and $\mathbf{b} = (b_1, b_2, b_3)$, their dot product is

$$\mathbf{a} \cdot \mathbf{b} = (a_1, a_2, a_3) \cdot (b_1, b_2, b_3) = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

This comes directly from writing $\mathbf{a}$ and $\mathbf{b}$ in terms of $\mathbf{e}_i$, and using the orthonormality condition.

We can use this to directly calculate the length of the vector, with

$$\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2 = a_1^2 + a_2^2 + a_3^2.$$

⁴The difference will be important when we move onto matrices.

Cross Product in Components

The cross product is orientation dependent, so it helps to choose the basis in a specific way. Usually, we pick a **right-handed** bases, which is one that satisfies

$$\begin{aligned}\mathbf{e}_1 \times \mathbf{e}_2 &= \mathbf{e}_3 \\ \mathbf{e}_2 \times \mathbf{e}_3 &= \mathbf{e}_1 \\ \mathbf{e}_3 \times \mathbf{e}_1 &= \mathbf{e}_2,\end{aligned}$$

and note that just one of these implies all of the others. Then we can calculate the cross product of two vectors in component form.

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= (a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3) \times (b_1\mathbf{e}_1 + b_2\mathbf{e}_2 + b_3\mathbf{e}_3) \\ &= (a_2b_3 - a_3b_2)\mathbf{e}_1 + (a_3b_1 - a_1b_3)\mathbf{e}_2 + (a_1b_2 - a_2b_1)\mathbf{e}_3.\end{aligned}$$

Example 8.2.11 (Computing a Cross Product)

The following is an example of computing a cross product using vector components.

$$\begin{aligned}\mathbf{a} \times \mathbf{b} &= \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \times \begin{pmatrix} 7 \\ -3 \\ 5 \end{pmatrix} \\ &= \begin{pmatrix} 0 \times 5 - (-3) \times (-1) \\ 7 \times (-1) - 2 \times 5 \\ 2 \times (-3) - 7 \times 0 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ -17 \\ -6 \end{pmatrix}.\end{aligned}$$

Later on in this chapter, we will develop some more tools for working efficiently with components.

§8.2.5 Triple Products

Scalar Triple Product

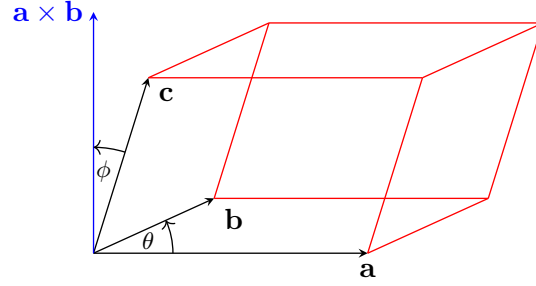
Given three vectors, we define the *scalar triple product* as follows.

Definition 8.2.12 (Scalar Triple Product)

Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be vectors. The **scalar triple product** of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is

$$\begin{aligned}[\mathbf{a}, \mathbf{b}, \mathbf{c}] &= \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) \\ &= -\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b}) = -\mathbf{b} \cdot (\mathbf{a} \times \mathbf{c}) = -\mathbf{c} \cdot (\mathbf{b} \times \mathbf{a}).\end{aligned}$$

The most natural interpretation of the scalar triple product is as the ‘signed’ volume of the parallelepiped formed by $\mathbf{a}, \mathbf{b}, \mathbf{c}$.



In the sketch above (where we assume $\theta, \phi < \pi/2$), we have

$$|\mathbf{c}| |\mathbf{a} \times \mathbf{b}| \cos \phi = \underbrace{|\mathbf{a}| |\mathbf{b}| \sin \theta}_{\text{base area}} \cdot \underbrace{|\mathbf{c}| \cos \phi}_{\text{height}},$$

which is the volume as shown.

We do have a sign, and indeed we have that $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) > 0$ implies that $\mathbf{a}, \mathbf{b}, \mathbf{c}$ is a right handed set. Also $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = 0$ if and only if $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are coplanar (corresponding with the parallelepiped having zero volume).

We can compute the scalar triple product using components. For $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{b} = (b_1, b_2, b_3)$, $\mathbf{c} = (c_1, c_2, c_3)$, then we have

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= a_1 b_2 c_3 - a_1 b_3 c_2 \\ &\quad + a_2 b_3 c_1 - a_2 b_1 c_3 \\ &\quad + a_3 b_1 c_2 - a_3 b_2 c_1 \\ &= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}, \end{aligned}$$

which is a determinant (we shall study these in Section 5).

Vector Triple Product

Another slightly less used product on three vectors is the *vector triple product*.

Definition 8.2.13 (Vector Triple Product)

For vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, we define the **vector triple product** to be

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

We will prove this identity later on, but note that this product is not associative. As the cross product is anticommutative, we do have

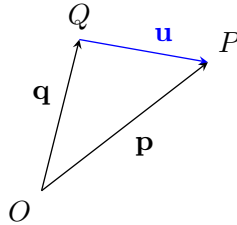
$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = -(\mathbf{b} \times \mathbf{c}) \times \mathbf{a}.$$

There's not really a sensible geometric interpretation of the vector triple product, but it is used in other vector formulas as it can shorten certain expressions.

§8.2.6 Lines, Planes & Vector Equations

We defined vectors as position vectors from an origin O , but with the introduction of vector addition, we can also use vectors to describe the displacement between two points,

$$\mathbf{u} = \overrightarrow{QP} = \mathbf{p} - \mathbf{q}.$$



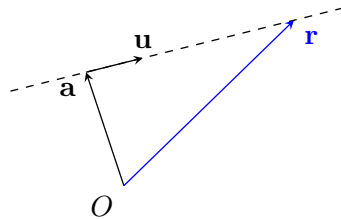
With this description of displacement vectors, we can define geometric objects such as lines and planes using equations involving vectors.

Lines

A general point on a line through $\mathbf{a}$ in the direction of $\mathbf{u} \neq \mathbf{0}$ can be written as

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{u}, \quad \lambda \in \mathbb{R}.$$

This is known as the **parametric form** of the line. Note the similarity between this equation and the one for a line in the complex plane.



To eliminate the parameter from this equation, we can take the cross product with $\mathbf{u}$

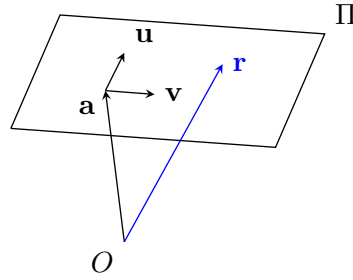
$$\begin{aligned} \mathbf{u} \times \mathbf{r} &= \mathbf{u} \times (\mathbf{a} + \lambda \mathbf{u}) \\ &= \mathbf{u} \times \mathbf{a} \\ \implies \mathbf{u} \times (\mathbf{r} - \mathbf{a}) &= \mathbf{0}. \end{aligned}$$

With this in mind, consider also the related equation $\mathbf{u} \times \mathbf{r} = \mathbf{c}$ with $\mathbf{u}, \mathbf{c} \neq \mathbf{0}$. From this, $\mathbf{u} \cdot \mathbf{c} = \mathbf{u} \cdot (\mathbf{u} \times \mathbf{r}) = 0$, so if $\mathbf{u} \cdot \mathbf{c} \neq 0$, this is inconsistent and there are no solutions. Otherwise, if $\mathbf{u} \cdot \mathbf{c} = 0$, then $\mathbf{u} \times (\mathbf{u} \times \mathbf{c}) = -|\mathbf{u}|^2 \mathbf{c}$, so $\mathbf{a} = \frac{-1}{|\mathbf{u}|^2} (\mathbf{u} \times \mathbf{c})$ is one solution. Then $\mathbf{r} = \mathbf{a} + \lambda \mathbf{u}$ is the general solution.

Planes

A general point on a plane Π through $\mathbf{a}$ with directions $\mathbf{u}, \mathbf{v}$ in the plane ($\mathbf{u} \nparallel \mathbf{v}$) is

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{u} + \mu \mathbf{v}, \quad \lambda, \mu \in \mathbb{R}.$$



An alternative form comes from considering a normal to the plane, $\mathbf{n} = \mathbf{u} \times \mathbf{v} (\neq \mathbf{0})$. With this, we get

$$\begin{aligned} \mathbf{n} \cdot \mathbf{r} &= \mathbf{n} \cdot \mathbf{a} \\ \implies \mathbf{n} \cdot (\mathbf{r} - \mathbf{a}) &= 0. \end{aligned}$$

For a geometric interpretation of this, note that $\mathbf{r} - \mathbf{a}$ is the displacement in the direction of the plane, and thus should be perpendicular to $\mathbf{n}$, a normal to the plane.

Other Vector Equations

To handle other vector equations, a general strategy is to manipulate equations and dot/cross with constant vectors.

Example 8.2.14 (Vector Equation of a Sphere)

We will attempt to find a geometric interpretation of the vector equation

$$|\mathbf{r}|^2 + \mathbf{r} \cdot \mathbf{a} = k,$$

with $\mathbf{a}$, k constant.

Here, we can ‘complete the square’ to get

$$\left| \mathbf{r} + \frac{1}{2}\mathbf{a} \right|^2 = k + \frac{1}{4}|\mathbf{a}|^2,$$

which is a sphere with center $-\frac{1}{2}\mathbf{a}$ and radius $(k + \frac{1}{4}|\mathbf{a}|^2)^{1/2}$, if $k > -\frac{1}{4}|\mathbf{a}|^2$. If this is not the case, then there are no solutions.

Example 8.2.15 (Arbitrary Vector Equation)

We will solve the vector equation

$$\mathbf{r} + \mathbf{a} \times (\mathbf{b} \times \mathbf{r}) = \mathbf{c},$$

where $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are given.

We will use the ‘dot and cross stuff’ approach. We use the vector triple product identity to rewrite this as

$$\mathbf{r} + (\mathbf{a} \cdot \mathbf{r})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{r} = \mathbf{c}. \quad (\dagger)$$

Then dotting with $\mathbf{a}$ we get

$$\begin{aligned}\mathbf{a} \cdot \mathbf{r} + (\mathbf{a} \cdot \mathbf{r})(\mathbf{b} \cdot \mathbf{a}) - (\mathbf{a} \cdot \mathbf{b})(\mathbf{a} \cdot \mathbf{r}) &= \mathbf{c} \cdot \mathbf{a} \\ \implies \mathbf{a} \cdot \mathbf{r} &= \mathbf{c} \cdot \mathbf{a}. \quad (\dagger\dagger)\end{aligned}$$

Taking a dot product has ‘lost information’, so we substitute $(\dagger\dagger)$ into $(\dagger)$ to get

$$(1 - \mathbf{a} \cdot \mathbf{b})\mathbf{r} = \mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}.$$

If $\mathbf{a} \cdot \mathbf{b} \neq 1$, then we can rearrange to get our solution

$$\mathbf{r} = \frac{1}{1 - \mathbf{a} \cdot \mathbf{b}}(\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b}),$$

which is a point.

If $\mathbf{a} \cdot \mathbf{b} = 1$, then $\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b} \neq \mathbf{0}$ would cause the equation to be inconsistent. So if $\mathbf{c} - (\mathbf{a} \cdot \mathbf{c})\mathbf{b} = \mathbf{0}$, then $(\dagger)$ can be rewritten

$$(\mathbf{a} \cdot \mathbf{r} - \mathbf{a} \cdot \mathbf{c})\mathbf{b} = \mathbf{0},$$

of which the set of solutions is given by $(\dagger\dagger)$, a plane.

§8.2.7 Index Notation & The Summation Convention

Returning to the component approach that we began to develop earlier, we will now introduce some machinery for working with components algebraically in a more streamlined way. The notation may look like mere bookkeeping at first, but it makes otherwise painful vector identities almost mechanical to prove.

Components, δ and ε

Begin by writing vectors $\mathbf{a}$, $\mathbf{b}$, ... in terms of components a_i, b_i with respect to an orthonormal, right-handed basis $\{\mathbf{e}_i\}$. In the following discussion, the indices $i, j, k, l, p, q, \dots$ will take on values $1, 2, 3$ ⁵.

Using indices, we can take some vector $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$ and express the components as $c_i = [\alpha\mathbf{a} + \beta\mathbf{b}]_i = \alpha a_i + \beta b_i$, for $i = 1, 2, 3$ (a **free variable**). Using the same notation, we can write things like

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i,$$

and

$$\begin{aligned}\mathbf{x} &= \mathbf{a} + (\mathbf{b} \cdot \mathbf{c})\mathbf{d} \\ \iff x_j &= a_j + \left(\sum_k b_k c_k \right) d_j, \quad \text{for } j = 1, 2, 3.\end{aligned}$$

⁵This corresponds to the number of basis vectors, and the approach used will naturally generalize to indices taking on more values.

Definition 8.2.16 (Kronecker Delta Symbol)

We define the **Kronecker delta** symbol to be

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

With this definition, we get $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$, which encodes our orthonormality condition from before. Then

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \left(\sum_i a_i \mathbf{e}_i \right) \cdot \left(\sum_j b_j \mathbf{e}_j \right) \\ &= \sum_{ij} a_i b_j \mathbf{e}_i \cdot \mathbf{e}_j \\ &= \sum_{ij} a_i b_j \delta_{ij} \\ &= \sum_i a_i b_i. \end{aligned}$$

Definition 8.2.17 (Levi-Civita Symbol)

We define the **Levi-Civita symbol** such that

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i \ j \ k) \text{ is an even permutation of } (1 \ 2 \ 3), \\ -1 & \text{if } (i \ j \ k) \text{ is an odd permutation of } (1 \ 2 \ 3), \\ 0 & \text{otherwise} \end{cases}$$

For example,

$$\begin{aligned} \varepsilon_{123} &= \varepsilon_{231} = \varepsilon_{312} = +1, \\ \varepsilon_{321} &= \varepsilon_{213} = \varepsilon_{132} = -1, \\ \varepsilon_{ijk} &= 0 \text{ if any of } i, j, k \text{ repeat.} \end{aligned}$$

The Levi-Civita symbol is **totally antisymmetric**, in that exchanging a pair of indices produces a change in sign.

With this definition, in our right-handed orthonormal basis we have

$$\mathbf{e}_i \times \mathbf{e}_j = \sum_k \varepsilon_{ijk} \mathbf{e}_k,$$

for example

$$\mathbf{e}_2 \times \mathbf{e}_1 = \sum_k \varepsilon_{21k} \mathbf{e}_k = \varepsilon_{213} \mathbf{e}_3 = -\mathbf{e}_3.$$

So we can compute the cross product of two vectors as

$$\mathbf{a} \times \mathbf{b} = \left(\sum_i a_i \mathbf{e}_i \right) \times \left(\sum_j b_j \mathbf{e}_j \right)$$

$$\begin{aligned}
&= \sum_{ij} a_i b_j \mathbf{e}_i \times \mathbf{e}_j \\
&= \sum_{ijk} a_i b_j \varepsilon_{ijk} \mathbf{e}_k \\
&\iff (\mathbf{a} \times \mathbf{b})_k = \sum_{ij} \varepsilon_{ijk} a_i b_j.
\end{aligned}$$

For example, $(\mathbf{a} \times \mathbf{b})_3 = \sum_{ij} \varepsilon_{ij3} a_i b_j = \varepsilon_{123} a_1 b_2 + \varepsilon_{213} a_2 b_1 = a_1 b_2 - a_2 b_1$, as we had before.

Summation Convention

With components and index notation, indices that appear twice in a given term are usually summed over. In the **summation convention**, we omit the sigma ($\sum$) signs for repeated indices, and the sums are implicitly understood.

Example 8.2.18 (Examples of the Summation Convention)

The following all use the summation convention.

- (i) $a_i \delta_{ij} = a_1 \delta_{1j} + a_2 \delta_{2j} + a_3 \delta_{3j} = a_j$.
- (ii) $\mathbf{a} \cdot \mathbf{b} = a_i b_i$.
- (iii) $(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k$.
- (iv) $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \varepsilon_{ijk} a_i b_j c_k$.
- (v) $\delta_{ii} = \delta_{11} + \delta_{22} + \delta_{33} = 3$.
- (vi) $[(\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}]_i = (\mathbf{a} \cdot \mathbf{c})b_i - (\mathbf{a} \cdot \mathbf{b})c_i = a_j c_j b_i - a_k b_k c_i$.

We can explicitly define the rules of the summation convention as follows.

Definition 8.2.19 (Summation Convention Rules)

When using the summation convention, we have the following rules.

- (i) An index occurring exactly once in any given term must appear once in every term of an equation, and it can take any value – a **free index**.
- (ii) Any index occurring exactly *twice* in a given term is summed over – a **repeated**, **contracted** or **dummy index**.
- (iii) No index can occur more than twice in a given term.

Let's have a look at applying the summation convention to prove the vector triple product identity we stated earlier.

Proposition 8.2.20 (Vector Triple Product Identity)

For vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$, we have

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

Proof. We have

$$\begin{aligned} [\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_i &= \varepsilon_{ijk} a_j (\mathbf{b} \times \mathbf{c})_k \\ &= \varepsilon_{ijk} a_j \varepsilon_{kpq} b_p c_q \\ &= (\varepsilon_{ijk} \varepsilon_{pqk}) a_j b_p c_q, \end{aligned}$$

now $\varepsilon_{ijk} \varepsilon_{pqk} = \delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp}$ (we will prove this later on), hence

$$\begin{aligned} [\mathbf{a} \times (\mathbf{b} \times \mathbf{c})]_i &= (\delta_{ip} \delta_{jq}) a_j b_p c_q - (\delta_{iq} \delta_{jp}) a_j b_p c_q \\ &= [(\mathbf{a} \cdot \mathbf{c}) \mathbf{b} - (\mathbf{a} \cdot \mathbf{b}) \mathbf{c}]_i, \end{aligned}$$

as required. □

εε Identities

We will finish this section by looking at εε identities, which can be used in various computations involving δ and ε symbols.

- $\varepsilon_{ijk} \varepsilon_{pqk} = \delta_{ip} \delta_{jq} - \delta_{iq} \delta_{jp} = \varepsilon_{kij} \varepsilon_{kpq}$.

To check this, note that RHS and LHS are antisymmetric under $i \leftrightarrow j$, $p \leftrightarrow q$, so both vanish if i & j or p & q take the same value. Now it suffices to check examples $i = p = 1, j = q = 2$, where LHS and RHS are both +1, or $i = q = 1, j = p = 2$, where LHS and RHS are both −1. All other index choices giving non-zero results work similarly.

- $\varepsilon_{ijk} \varepsilon_{pjk} = 2\delta_{ip}$.

This comes from the above identity, where the LHS becomes $\delta_{ip} \delta_{jj} - \delta_{ij} \delta_{jp} = 3\delta_{ip} - \delta_{ip} = 2\delta_{ip}$.

- $\varepsilon_{ijk} \varepsilon_{ijk} = 6$.

This follows directly from the previous result, with $i = p$. Then $2\delta_{ii} = 6$.

§8.3 Vectors in a General Setting, $\mathbb{R}^n$ and $\mathbb{C}^n$

In the previous chapter, we studied vectors as they occur when considering position vectors in three dimensions. However, vectors are more general objects, which we will see in this chapter. First, we will consider how to immediately generalise vectors from $\mathbb{R}^3$ (as in the previous chapter) to $\mathbb{R}^n$.

§8.3.1 Vectors in $\mathbb{R}^n$

By regarding vectors as sets of components, it's easy to generalize from 3 to n dimensions.

Definition 8.3.1 ($\mathbb{R}^n$ Space)

We define $\mathbb{R}^n$ to be the set of n -tuples of real numbers,

$$\mathbb{R}^n = \{(x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R}\}.$$

We also define **vector addition** and **scalar multiplication** as

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n),$$

$$\text{and } \lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n).$$

As before, we can take arbitrary linear combinations and we also get a notion of parallelism, where $\mathbf{x} \parallel \mathbf{y}$ if and only if $\mathbf{x} = \lambda \mathbf{y}$ or $\mathbf{y} = \lambda \mathbf{x}$ for $\lambda \in \mathbb{R}$.

We also still have a dot product, but in more general settings, such operations are normally referred to as *inner products*.

Definition 8.3.2 (Inner Product)

The inner product on $\mathbb{R}^n$ is defined by

$$\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i = x_1 y_1 + x_2 y_2 + \dots + x_n y_n.$$

Proposition 8.3.3 (Properties of the Inner Product)

In $\mathbb{R}^n$, the inner product satisfies

- (i) *Symmetric.* $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.
- (ii) *Bilinear.* $(\lambda \mathbf{x} + \lambda' \mathbf{x}') \cdot \mathbf{y} = \lambda \mathbf{x} \cdot \mathbf{y} + \lambda' \mathbf{x}' \cdot \mathbf{y}$.
- (iii) *Positive Definite.* $\mathbf{x} \cdot \mathbf{x} = \sum_i x_i^2 \geq 0$, and is equal 0 if and only if $\mathbf{x} = \mathbf{0}$.

Also, if we return back to vectors in three-dimensions, we had an equivalent definition of the dot product as

$$\mathbf{x} \cdot \mathbf{y} = |\mathbf{x}| |\mathbf{y}| \cos \theta,$$

where θ is the angle between the vectors. Using this as a definition of the inner product in $\mathbb{R}^n$, we get a definition for the angle between vectors.

When we look at vector spaces in general, we will define the inner product in general to be any operation that satisfies these properties.

We also have the notions of length and orthogonality, much the same as in three-dimensions.

Definition 8.3.4 (Length/Norm)

For $\mathbf{x} \in \mathbb{R}^n$, we define its **length** or **norm** to be $|\mathbf{x}| \geq 0$ with $|\mathbf{x}|^2 = \mathbf{x} \cdot \mathbf{x}$.

Definition 8.3.5 (Orthogonality)

We say two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

When working in $\mathbb{R}^n$, we typically use a set of basis vectors known as the **standard basis**. They are

$$\mathbf{e}_1 = (1, 0, 0, \dots, 0),$$

$$\mathbf{e}_2 = (0, 1, 0, \dots, 0),$$

$$\begin{aligned} & \vdots \\ \mathbf{e}_n &= (0, 0, 0, \dots, 1). \end{aligned}$$

Then we have

$$\mathbf{x} = \sum_i x_i \mathbf{e}_i,$$

and $\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij}$, so this basis is orthonormal.

Cauchy-Schwarz and the Triangle Inequality

We will now state an inequality about the inner product of two vectors in $\mathbb{R}^n$.

Proposition 8.3.6 (Cauchy-Schwarz Inequality)

For all vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, we have

$$|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| \cdot |\mathbf{y}|,$$

with equality if and only if $\mathbf{x} \parallel \mathbf{y}$.

Proof. If $\mathbf{y} = \mathbf{0}$, then we are done. Otherwise, consider

$$\begin{aligned} |\mathbf{x} - \lambda \mathbf{y}|^2 &= (\mathbf{x} - \lambda \mathbf{y}) \cdot (\mathbf{x} - \lambda \mathbf{y}) \\ &= |\mathbf{x}|^2 - 2\lambda \mathbf{x} \cdot \mathbf{y} + \lambda^2 |\mathbf{y}|^2 \geq 0 \end{aligned}$$

for any $\lambda \in \mathbb{R}$. This is a real quadratic in λ with at most one real root, so its discriminant satisfies

$$(-2\mathbf{x} \cdot \mathbf{y})^2 - 4|\mathbf{x}|^2 |\mathbf{y}|^2 \leq 0,$$

which rearranges to give $|\mathbf{x} \cdot \mathbf{y}| \leq |\mathbf{x}| |\mathbf{y}|$. Equality holds if and only if the discriminant is zero, that is, if $\mathbf{x} = \lambda \mathbf{y}$ for some λ , as required. $\square$

As vectors in $\mathbb{R}^n$ are just sets of components, we can state this inequality in terms of real numbers. For sequences of real numbers (a_i) and (b_i) , we have

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right),$$

where we have equality if and only if $\frac{a_i}{b_i}$ is a constant for all i where $a_i b_i \neq 0$.

We also have the triangle inequality, similar to the one stated in Section 1.

Proposition 8.3.7 (Triangle Inequality)

For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\mathbf{x} + \mathbf{y}| \leq |\mathbf{x}| + |\mathbf{y}|$$

Proof. We have

$$\begin{aligned} |\mathbf{x} + \mathbf{y}|^2 &= |\mathbf{x}|^2 + 2\mathbf{x} \cdot \mathbf{y} + |\mathbf{y}|^2 \\ &\leq |\mathbf{x}|^2 + 2|\mathbf{x}| \cdot |\mathbf{y}| + |\mathbf{y}|^2 \\ &= (|\mathbf{x}| + |\mathbf{y}|)^2, \end{aligned}$$

as required. □

A comment about $\mathbb{R}^n$ and $\mathbb{R}^3$

The definition we gave for the inner product on $\mathbb{R}^n$ can be written as

$$\mathbf{a} \cdot \mathbf{b} = \delta_{ij} a_i b_j,$$

for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ where we use the summation convention. For $n = 3$, this matches with the compact definition of the dot product we gave previously.

However, in three dimensions we also had a compact definition of the cross product with

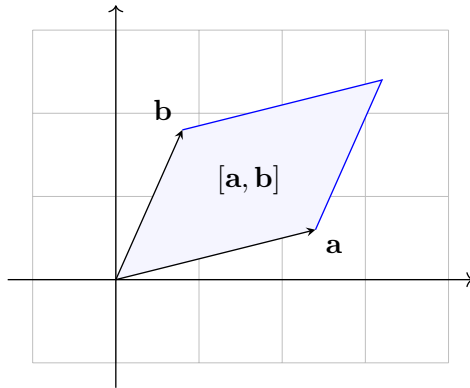
$$(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k.$$

In $\mathbb{R}^n$ however, we have $\varepsilon_{ij\dots k}$ where there are n indices (we will look at this again in Section 5). This is totally antisymmetric, so we cannot use ε to define a vector-valued product for any n except 3.

In $\mathbb{R}^2$ however, we have ε_{ij} where $\varepsilon_{12} = -\varepsilon_{21} = 1$, and we can define an additional scalar product

$$[\mathbf{a}, \mathbf{b}] = \varepsilon_{ij} a_i b_j = a_1 b_2 - a_2 b_1.$$

This corresponds with the signed area of a parallelogram, and $|[\mathbf{a}, \mathbf{b}]| = |\mathbf{a}||\mathbf{b}|\sin\theta$. Compare this with $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = \varepsilon_{ijk} a_i b_j c_k$, the signed volume of the parallelepiped in three dimensions we defined earlier.



§8.3.2 Vector Spaces

We will now go one step further and define what a vector is in a much more general sense. This will be done by defining the properties that vectors have, rather than what they are themselves.

Definition 8.3.8 (Vector Space)

Let V be a set of objects called **vectors** with operations $\mathbf{v} + \mathbf{w} \in V$ defined for all $\mathbf{v}, \mathbf{w} \in V$, and $\lambda \mathbf{v} \in V$ for all $\mathbf{v} \in V$ and $\lambda \in F$, where F is some field.

Then V is a **vector space** if

- (i) V with $+$ is an abelian group.
- (ii) $\lambda(\mathbf{v} + \mathbf{w}) = \lambda\mathbf{v} + \lambda\mathbf{w}$.
- (iii) $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$.

$$(iv) \lambda(\mu \mathbf{v}) = (\lambda\mu)\mathbf{v}.$$

$$(v) 1\mathbf{v} = \mathbf{v}.$$

In this course, we will deal only with the fields $\mathbb{R}$ and $\mathbb{C}$. If $F = \mathbb{R}$, then we call V a **real vector space**, and if $F = \mathbb{C}$, then we call it a **complex vector space**.

Example 8.3.9 (Example of a Real Vector Space)

Let V be the set of all functions $f : [0, 1] \rightarrow \mathbb{R}$ where $f(0) = f(1) = 0$ and f is smooth. Then V is a real vector space.

Indeed we have

$$\begin{aligned} (f + g)(x) &= f(x) + g(x), \\ (\lambda f)(x) &= \lambda(f(x)). \end{aligned}$$

The other properties can be checked as required.

We can also define a *subspace* in the natural way.

Definition 8.3.10 (Subspace)

A **subspace** of a vector space V is a subset $U \subseteq V$ that is also a vector space.

Note that a non-empty subset is a subspace if and only if $\mathbf{v}, \mathbf{w} \in U$ implies $\lambda\mathbf{v} + \mu\mathbf{w} \in U$ where λ, μ are some arbitrary scalars.

For any vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$, their span

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_r\} = \{\lambda_1\mathbf{v}_1 + \dots + \lambda_r\mathbf{v}_r \mid \lambda_i \in \mathbb{R} \text{ or } \mathbb{C}\}.$$

is a subspace. Also note that V and $\{\mathbf{0}\}$ are both subspaces of a vector space V .

Linear Dependence & Independence

We will now think about whether sets of vectors are independent.

Definition 8.3.11 (Linear Dependence/Independence)

For $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in V$, some vector space, consider the **linear relation**

$$\lambda_1\mathbf{v}_1 + \lambda_2\mathbf{v}_2 + \dots + \lambda_r\mathbf{v}_r = \mathbf{0} \quad (\dagger)$$

If $(\dagger)$ implies that $\lambda_i = 0$ for all i , then these vectors form a **linearly independent** set. If $(\dagger)$ holds with some $\lambda_i \neq 0$, then they form a **linearly dependent** set.

Example 8.3.12 (Example of a Linearly Dependent Set)

In $\mathbb{R}^2$, the set $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$ is linearly dependent, since

$$0 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which is a non-trivial linear relation.

Any set containing $\mathbf{0}$ is linearly dependent, since $\lambda\mathbf{0} = \mathbf{0}$ for any $\lambda \neq 0$.

In $\mathbb{R}^3$, the set of vectors $\{\mathbf{a}, \mathbf{b}, \mathbf{c}\}$ are linearly independent if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \neq 0$, since if $\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c} = \mathbf{0}$, then dotting with $\mathbf{b} \times \mathbf{c}$ gives $\alpha\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0 \implies \alpha = 0$, and β, γ similarly.

Inner Products

A real vector space can have an additional structure that we can define by axioms.

Definition 8.3.13 (Inner Product)

For a real vector space V and $\mathbf{v}, \mathbf{w} \in V$, an **inner product** denoted by

$$\mathbf{v} \cdot \mathbf{w} \quad \text{or} \quad (\mathbf{v}, \mathbf{w}) \in \mathbb{R}$$

is a map that is symmetric, bilinear, and positive definite.

If these conditions hold, then we get things like the Cauchy-Schwarz inequality and a norm as by the previous argument.

Example 8.3.14 (Inner Product on a Real Vector Space)

Consider the real vector space of functions $V = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ smooth, and } f(0) = f(1) = 0\}$. We can define the inner product as

$$(f, g) = \int_0^1 f(x)g(x) \, dx,$$

which satisfies the conditions specified above. Then we have

$$|(f, g)| \leq \|f\| \cdot \|g\|,$$

that is,

$$\left| \int_0^1 f(x)g(x) \, dx \right| \leq \left(\int_0^1 f(x)^2 \, dx \right)^{1/2} \left(\int_0^1 g(x)^2 \, dx \right)^{1/2}.$$

We can also say something about linear independence from an inner product.

Lemma 8.3.15

In any real vector space V with an inner product, if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r$ are non-zero and orthogonal vectors, then they are linearly independent.

Proof. If $\sum_i \alpha_i \mathbf{v}_i = \mathbf{0}$, then $(\mathbf{v}_j, \sum_i \alpha_i \mathbf{v}_i) = 0$ for a fixed j , so $\alpha_j(\mathbf{v}_j, \mathbf{v}_j) = 0$, and thus $\alpha_j = 0$. $\square$

§8.3.3 Basis and Dimension

We can generalize the notion of a basis to an arbitrary vector space.

Definition 8.3.16 (Basis)

For a vector space V , a **basis** is a set $B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ such that

- (i) B spans V , that is, any $\mathbf{v} \in V$ can be written $\mathbf{v} = \sum_{i=1}^n v_i \mathbf{e}_i$.
- (ii) B is linearly independent.

The linear independence implies that the coefficients v_i in (i) are unique. We say that v_i are the components of $\mathbf{v}$ with respect to the basis B .

Example 8.3.17 (Standard Basis)

The standard basis for $\mathbb{R}^n$ consists of the vectors

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

This is a basis since it satisfies the definition as $\mathbf{x} \in \mathbb{R}^n$, we have

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n,$$

and $\mathbf{x} = \mathbf{0}$ if and only if $x_1 = x_2 = \dots = x_n = 0$.

Of course in the example above we just used one particular base. For example in $\mathbb{R}^2$ another basis consists of $\{(1, 0), (1, 1)\}$.

Theorem 8.3.18 (The Basis Theorem)

If $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ are both bases of a vector space V , then $n = m$.

Proof. We can find coefficients A_{ai} and B_{ia} such that $\mathbf{f}_a = \sum_i A_{ai} \mathbf{e}_i$ and $\mathbf{e}_i = \sum_a B_{ia} \mathbf{f}_a$. So we have that

$$\begin{aligned} \mathbf{f}_a &= \sum_i A_{ai} \left(\sum_b B_{ib} \mathbf{f}_b \right) \\ &= \sum_b \left(\sum_i A_{ai} B_{ib} \right) \mathbf{f}_b, \end{aligned}$$

but coefficients with respect to a basis are unique, hence

$$\sum_i A_{ai} B_{ib} = \delta_{ab}.$$

Similarly, $\sum_a B_{ia}A_{aj} = \delta_{ij}$. Finally,

$$\begin{aligned}\sum_{ia} A_{ai}B_{ia} &= \sum_a \delta_{aa} = m \\ &= \sum_i \delta_{ii} = n,\end{aligned}$$

hence $n = m$. □

This theorem allows us to state the following definition.

Definition 8.3.19 (Dimension)

The **dimension** of a vector space V , written $\dim V$, is the size of its basis.

Note that $\mathbb{R}^n$ has dimension n .

The steps in the proof of the basis theorem are within the scope of the course, but you will not be expected to prove this theorem without prompts (it is non-examinable in the schedules). The same applies with the following.

Proposition 8.3.20

Let V be a vector space with finite subsets $Y = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ that spans V , and $X = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ that is linearly independent. Then $k \leq n \leq m$ where $n = \dim V$. Also

- (i) A basis can be formed from subsets of Y ,
- (ii) A basis can be obtained from X by adding additional vectors (from Y if necessary).

Proof. (i) If Y is linearly independent, then it's a basis and $m = n = \dim V$. If Y is not linearly independent, then

$$\sum_{i=1}^m \lambda_i \mathbf{w}_i = \mathbf{0}, \quad \text{with some } \lambda_i \neq 0.$$

Take $\lambda_m \neq 0$ without loss of generality, then $\mathbf{w}_m = -\frac{1}{\lambda_m} \sum_{i=1}^{m-1} \lambda_i \mathbf{w}_i$, so $\text{span } Y = \text{span } Y'$, with $Y' = \{\mathbf{w}_1, \dots, \mathbf{w}_{m-1}\}$. This can be repeated until a basis is obtained.

- (ii) If X spans V then it's a basis and $k = n = \dim V$. If not, then there exists $\mathbf{u}_{k+1} \in V$ not in $\text{span } X$. But then $\sum_{i=1}^{k+1} \mu_i \mathbf{u}_i = \mathbf{0}$ implies $\mu_{k+1} = 0$, thus $X' = \{\mathbf{u}_1, \dots, \mathbf{u}_{k+1}\}$ is linearly independent. Furthermore, we can choose $\mathbf{u}_{k+1} \in Y$ as otherwise $Y \subseteq \text{span } X \implies \text{span } Y \subseteq \text{span } X \implies \text{span } X = V$. We can repeat this process until we obtain a basis (and this must stop since Y is a finite set). □

Not every vector space has a finite dimension, but in this course we will deal only with vector spaces that do, save for the occasional example.

Example 8.3.21 (Example of an Infinite Dimensional Vector Space)

Consider the vector space

$$V = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ smooth}, f(0) = f(1) = 0\}.$$

Then consider $S_n(x) = \sqrt{2} \sin(n\pi x)$, $n = 1, 2, \dots$. These belong to V and

$$(S_n, S_m) = 2 \int_0^1 \sin(n\pi x) \sin(m\pi x) dx = \delta_{mn},$$

so they are all linearly independent. But we have infinitely many of them, so V is infinite dimensional.

§8.3.4 Vectors in $\mathbb{C}^n$

So far we have looked at real vector spaces, and particularly at $\mathbb{R}^n$. We can define a similar vector space over the complex numbers.

Definition 8.3.22 ($\mathbb{C}^n$ Space)

We define $\mathbb{C}^n$ to be the set of n -tuples of complex numbers,

$$\mathbb{C}^n = \{(z_1, z_2, \dots, z_n) \mid z_j \in \mathbb{C}\}.$$

We also define **addition** and scalar multiplication with

$$(z_1, \dots, z_n) + (w_1, \dots, w_n) = (z_1 + w_1, \dots, z_n + w_n) \\ \text{and } \lambda(z_1, \dots, z_n) = (\lambda z_1, \dots, \lambda z_n)$$

Note that we could have used real scalars and have gotten a real vector space, but having a complex vector space is more natural.

The distinction between real and complex vector spaces is important. For example, if we have $\mathbf{z} = (z_1, \dots, z_n) \in \mathbb{C}^n$, with $z_j = x_j + iy_j$ then

$$\mathbf{z} = \sum_j x_j \mathbf{e}_j + \sum_j y_j \mathbf{f}_j, \quad \text{where } \mathbf{f}_j = i\mathbf{e}_j,$$

is a real linear combination, and thus $\{\mathbf{e}_1, \dots, \mathbf{e}_n, \mathbf{f}_1, \dots, \mathbf{f}_n\}$ is a basis for $\mathbb{C}^n$ as a real vector space, and it has dimension $2n$. However, taking $\mathbb{C}^n$ as defined before, it is of dimension n which is clearly different.

From now on we will view $\mathbb{C}^n$ as a complex vector space unless we say otherwise.

Complex Inner Product

We can define an inner product on $\mathbb{C}^n$.

Definition 8.3.23 (Complex Inner Product)

For $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$, we define the inner product to be

$$(\mathbf{z}, \mathbf{w}) = \sum_j \bar{z}_j w_j = \bar{z}_1 w_1 + \cdots + \bar{z}_n w_n.$$

Proposition 8.3.24 (Properties of the Complex Inner Product)

The complex inner product satisfies

- (i) *Hermitian*. $(\mathbf{w}, \mathbf{z}) = \overline{(\mathbf{z}, \mathbf{w})}$.
- (ii) *Linear/anti-linear*. $(\mathbf{z}, \lambda \mathbf{w} + \lambda' \mathbf{w}') = \lambda(\mathbf{z}, \mathbf{w}) + \lambda'(\mathbf{z}, \mathbf{w}')$ and $(\lambda \mathbf{z} + \lambda' \mathbf{z}', \mathbf{w}) = \bar{\lambda}(\mathbf{z}, \mathbf{w}) + \bar{\lambda}'(\mathbf{z}', \mathbf{w})$.
- (iii) *Positive definite*. $(\mathbf{z}, \mathbf{z}) = \sum_j |z_j|^2$, which is real, non-negative and zero if and only if $\mathbf{z} = \mathbf{0}$.

Proof Sketch. Check definitions. □

We also define the *length* or *norm* of $\mathbf{z}$ to be $|\mathbf{z}| \geq 0$ with $|\mathbf{z}|^2 = (\mathbf{z}, \mathbf{z})$.

We can also say that $\mathbf{z}, \mathbf{w} \in \mathbb{C}^n$ are orthogonal if $(\mathbf{z}, \mathbf{w}) = 0$. With this, note that the standard basis for $\mathbb{C}^n$ is orthonormal.

§8.4 Matrices and Linear Maps

Vector spaces have two basic operations: addition and scalar multiplication. The prominence of these operations makes thinking about functions that preserve this structure quite useful. Such functions are called linear maps, and in the setting of finite dimensional vector spaces we have a useful way of studying them: matrices. In this chapter we will study both ideas and how they link together.

§8.4.1 Linear Maps

Let's begin by defining what we mean exactly by a linear map.

Definition 8.4.1 (Linear Map)

A **linear map** or **linear transformation** is a function $T : V \rightarrow W$ between two vector spaces V and W such that

$$T(\lambda \mathbf{x} + \mu \mathbf{y}) = \lambda T(\mathbf{x}) + \mu T(\mathbf{y}),$$

for all $\mathbf{x}, \mathbf{y} \in V$ and $\lambda, \mu \in \mathbb{R}$ or $\mathbb{C}$ ^a.

^aDepending on whether we are working with real or complex vector spaces.

From this definition, it is immediately clear that a linear map is completely determined by its action on a set of basis vectors.

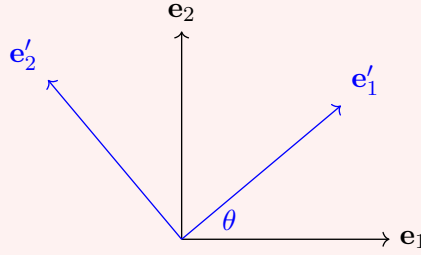
Let's have a look at some examples of linear maps that appear in geometry.

Example 8.4.2 (Geometric Linear Maps)

The following are all examples of linear maps.

- (i) A rotation is an example of a linear map. In $\mathbb{R}^2$, a rotation about $\mathbf{0}$ through an angle θ is defined by

$$\begin{aligned}\mathbf{e}_1 &\mapsto \mathbf{e}'_1 = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2 \\ \mathbf{e}_2 &\mapsto \mathbf{e}'_2 = -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2\end{aligned}$$



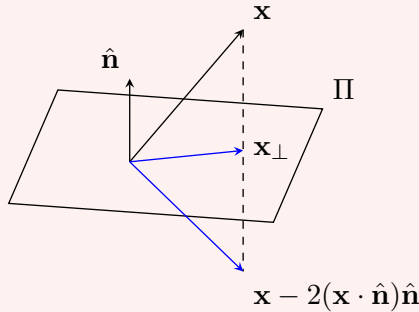
In $\mathbb{R}^3$, a rotation about an axis given by $\mathbf{e}_3$ is defined as above, but with $\mathbf{e}_3 \mapsto \mathbf{e}'_3 = \mathbf{e}_3$.

- (ii) A projection onto a plane through the origin with unit normal $\hat{\mathbf{n}}$ is a linear map, defined by

$$\mathbf{x} \mapsto \mathbf{x}' = \mathbf{x}_{\perp} = \mathbf{x} - (\mathbf{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}.$$

Similarly, the reflection about that plane is a linear map, defined by

$$\mathbf{x} \mapsto \mathbf{x}' = \mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$$



- (iii) A dilation along the basis vectors with scale factors α, β, γ is a linear map, defined by

$$\mathbf{e}_1 \mapsto \mathbf{e}'_1 = \alpha \mathbf{e}_1, \quad \mathbf{e}_2 \mapsto \mathbf{e}'_2 = \beta \mathbf{e}_2, \quad \mathbf{e}_3 \mapsto \mathbf{e}'_3 = \gamma \mathbf{e}_3.$$

- (iv) For orthogonal unit vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$, we can define a *shear* to be the linear map

$$\mathbf{x} \mapsto \mathbf{x}' = \mathbf{x} + \lambda \mathbf{a}(\mathbf{x} \cdot \mathbf{b}).$$

This displaces each point in the direction of $\mathbf{a}$, by an amount proportional to how far along $\mathbf{b}$ the point lies.

When we have a linear map $T : V \rightarrow W$, it can be helpful to think about what vectors in the vector space W are in the image of this function, and also what in V will just be

sent to $\mathbf{0}$. Because we deal with these so commonly, we introduce standard terms.

Definition 8.4.3 (Image and Kernel)

If $T : V \rightarrow W$ is a linear map, we say that its **image** is the set

$$\text{img}(T) = \{\mathbf{x}' \in W \mid \mathbf{x}' = T(\mathbf{x}) \text{ for some } \mathbf{x} \in V\},$$

and its **kernel** is the set

$$\ker(T) = \{\mathbf{x} \in V \mid T(\mathbf{x}) = \mathbf{0}\}.$$

Proposition 8.4.4 (Image and Kernel are Subspaces)

If $T : V \rightarrow W$ is a linear map, then $\ker(T)$ is a subspace of V and $\text{img}(T)$ is a subspace of W .

Proof Sketch. Use the properties of T as a linear map to check definitions. $\square$

These subspaces have a special property relating to their dimension. We call the dimension of the image the **rank** and dimension of the kernel as the **nullity** of a linear map. Then we have the following result (the proof of which is non-examinable).

Theorem 8.4.5 (Rank-Nullity Theorem)

For a linear map $T : V \rightarrow W$, we have $\text{rank}(T) + \text{null}(T) = \dim V$.

Proof. Let $\mathbf{e}_1, \dots, \mathbf{e}_k$ be a basis for $\ker(T)$. Extend this by adding in $\mathbf{e}_{k+1}, \dots, \mathbf{e}_n$ to get a basis for V . We claim that $B = \{T(\mathbf{e}_{k+1}), \dots, T(\mathbf{e}_n)\}$ is a basis for $\text{img}(T)$ (from which the result would follow).

First we have that B spans $\text{img}(T)$, since for any $\mathbf{x} = \sum_{i=1}^n x_i \mathbf{e}_i \in V$, $T(\mathbf{x}) = \sum_{i=k+1}^n x_i T(\mathbf{e}_i)$. Also, B is linearly independent since $\sum_{i=k+1}^n \lambda_i T(\mathbf{e}_i) = \mathbf{0}$ implies that $T(\sum_{i=k+1}^n \lambda_i \mathbf{e}_i) = \mathbf{0}$. But then that linear combination must be in the kernel, so we must have $\lambda_i = 0$, as $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ are linearly independent. Thus B is a basis, as required. $\square$

§8.4.2 Matrices

In the examples above, we described linear maps by giving their action on basis vectors, or by giving an explicit formula for the image of a general vector $\mathbf{x}$. We are now going to introduce *matrices*, which give us a uniform (and highly computable) way of doing this.

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map, and write $\mathbf{x}' = T(\mathbf{x})$. Taking the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of $\mathbb{R}^n$, we can write any $\mathbf{x} \in \mathbb{R}^n$ as $\mathbf{x} = \sum_j x_j \mathbf{e}_j$. Then by linearity,

$$\mathbf{x}' = T\left(\sum_j x_j \mathbf{e}_j\right) = \sum_j x_j T(\mathbf{e}_j),$$

so taking components in $\mathbb{R}^m$,

$$x'_i = \sum_j [T(\mathbf{e}_j)]_i x_j.$$

This motivates the following definition.

Definition 8.4.6 (Matrix of a Linear Map)

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map. The **matrix** of T (with respect to the standard bases) is the $m \times n$ array of numbers $A = \{A_{ij}\}$ with entries

$$A_{ij} = [T(\mathbf{e}_j)]_i,$$

that is, the j th column of A is the image of the j th basis vector. We write

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1n} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{mn} \end{pmatrix},$$

where A_{ij} is the entry in row i and column j , and we define the action of A on a vector by $\mathbf{x}' = A\mathbf{x}$ where $x'_i = \sum_j A_{ij}x_j$, or with the summation convention just $x'_i = A_{ij}x_j$.

The fact that *the columns of the matrix are the images of the basis vectors* is worth internalising – it is how one actually writes down matrices in practice, and we will use it constantly. The same definitions work with $\mathbb{C}^n$ and $\mathbb{C}^m$ (or indeed between any vector spaces with a choice of basis, as we will see when changing bases in Section 7).

Let's use this to write down matrices for the geometric linear maps we met before, working in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Example 8.4.7 (Matrices of Geometric Maps) (i) A rotation about $\mathbf{0}$ through angle θ in $\mathbb{R}^2$ sends $\mathbf{e}_1 \mapsto \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2$ and $\mathbf{e}_2 \mapsto -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2$. Putting these images in as columns, the matrix of the rotation is

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

In $\mathbb{R}^3$, a rotation by θ about the $\mathbf{e}_3$ axis fixes $\mathbf{e}_3$, giving

$$R = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(ii) A reflection in the plane through the origin with unit normal $\hat{\mathbf{n}}$ is $\mathbf{x} \mapsto \mathbf{x} - 2(\mathbf{x} \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$, which in components reads $x'_i = x_i - 2x_j n_j n_i$. So the matrix has entries

$$H_{ij} = \delta_{ij} - 2n_i n_j.$$

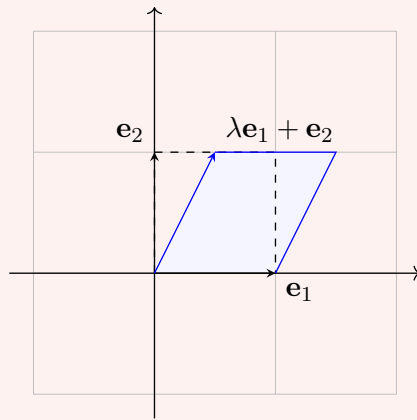
(iii) A dilation with scale factors α, β, γ along the coordinate axes has matrix

$$\begin{pmatrix} \alpha & 0 & 0 \\ 0 & \beta & 0 \\ 0 & 0 & \gamma \end{pmatrix}.$$

(iv) The shear $\mathbf{x} \mapsto \mathbf{x} + \lambda \mathbf{a}(\mathbf{x} \cdot \mathbf{b})$ with $\mathbf{a} = \mathbf{e}_1$, $\mathbf{b} = \mathbf{e}_2$ has matrix

$$S_\lambda = \begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

which displaces points in the $\mathbf{e}_1$ direction by an amount proportional to their $\mathbf{e}_2$ component. The picture to have in mind (in the plane, with $\lambda = 1/2$) is the following, with the unit square being tilted into a parallelogram.



It's also worth knowing what a rotation about a *general* axis looks like, and index notation makes the derivation quite pleasant.

Proposition 8.4.8 (Rotation About a General Axis)

Let R be the rotation through angle θ about an axis through the origin in the direction of the unit vector $\hat{\mathbf{n}}$ (in the positive sense given by the right-hand rule). Then

$$\mathbf{x}' = \mathbf{x} \cos \theta + (1 - \cos \theta)(\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}} + \sin \theta (\hat{\mathbf{n}} \times \mathbf{x}),$$

or in components, $x'_i = R_{ij}x_j$ where

$$R_{ij} = \cos \theta \delta_{ij} + (1 - \cos \theta)n_i n_j - \sin \theta \varepsilon_{ijk} n_k.$$

Proof. We resolve $\mathbf{x}$ into a part parallel to the axis, $(\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}}$, and a part perpendicular to it, $\mathbf{x}_\perp = \mathbf{x} - (\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}}$. The parallel part is unchanged by the rotation. The perpendicular part rotates by θ in the plane perpendicular to $\hat{\mathbf{n}}$, which is spanned by the orthogonal pair $\mathbf{x}_\perp$ and $\hat{\mathbf{n}} \times \mathbf{x}$ (which have the same length). So $\mathbf{x}_\perp \mapsto \cos \theta \mathbf{x}_\perp + \sin \theta (\hat{\mathbf{n}} \times \mathbf{x})$, and adding the parallel part back gives

$$\mathbf{x}' = (\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}} + \cos \theta (\mathbf{x} - (\hat{\mathbf{n}} \cdot \mathbf{x})\hat{\mathbf{n}}) + \sin \theta (\hat{\mathbf{n}} \times \mathbf{x}),$$

which rearranges to the stated result. The component form follows by writing

$$(\hat{\mathbf{n}} \times \mathbf{x})_i = \varepsilon_{ijk} n_j x_k = -\varepsilon_{ijk} x_j n_k.$$

□

§8.4.3 Matrix Algebra

Since matrices represent linear maps, the natural operations on linear maps (adding them, scaling them, and composing them) give us corresponding operations on matrices.

Definition 8.4.9 (Matrix Addition and Scalar Multiplication)

For $m \times n$ matrices A, B and a scalar λ , we define $A + B$ and λA to be the $m \times n$ matrices with entries

$$(A + B)_{ij} = A_{ij} + B_{ij}, \quad (\lambda A)_{ij} = \lambda A_{ij}.$$

These represent the maps $\mathbf{x} \mapsto A\mathbf{x} + B\mathbf{x}$ and $\mathbf{x} \mapsto \lambda(A\mathbf{x})$ respectively.

Composition is more interesting, and gives us *matrix multiplication*. Suppose A is an $n \times \ell$ matrix representing $S : \mathbb{R}^\ell \rightarrow \mathbb{R}^n$, and B is an $m \times n$ matrix representing $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Then the composite $T \circ S : \mathbb{R}^\ell \rightarrow \mathbb{R}^m$ sends

$$x''_i = B_{ik} x'_k = B_{ik} (A_{kj} x_j) = (B_{ik} A_{kj}) x_j,$$

using the summation convention. So the composite is represented by the $m \times \ell$ matrix BA with entries as follows.

Definition 8.4.10 (Matrix Multiplication)

For an $m \times n$ matrix B and an $n \times \ell$ matrix A , the **product** BA is the $m \times \ell$ matrix with entries

$$(BA)_{ij} = \sum_{k=1}^n B_{ik} A_{kj},$$

that is, the (i, j) entry of BA is the i th row of B dotted with the j th column of A .

Note that the product is only defined when the number of columns of B matches the number of rows of A – exactly the condition needed for the maps to compose. Matrix multiplication inherits its properties from composition of functions: it is associative, $A(BC) = (AB)C$, and distributes over addition. What it is *not* is commutative.

Example 8.4.11 (Matrix Multiplication is Not Commutative)

Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}. \quad \text{Then} \quad AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \neq \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = BA.$$

Geometrically this should not be a surprise – performing two shears in different directions, the order genuinely matters.

We write I for the **identity matrix**, with entries $I_{ij} = \delta_{ij}$, which represents the identity map and satisfies $AI = IA = A$.

There are a few more operations on matrices which don't come from composing linear maps, but which we will use constantly.

Definition 8.4.12 (Transpose)

If A is an $m \times n$ matrix, its **transpose** A^T is the $n \times m$ matrix with entries

$$(A^T)_{ij} = A_{ji}.$$

Proposition 8.4.13 (Properties of the Transpose)

For matrices A and B of appropriate sizes,

- (i) $(A^T)^T = A$,
- (ii) $(AB)^T = B^T A^T$,
- (iii) if $\mathbf{x}$ is a column vector, then $\mathbf{x}^T$ is a row vector, and $\mathbf{x}^T \mathbf{y} = \mathbf{x} \cdot \mathbf{y}$ for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Proof. (i) and (iii) are immediate from the definitions. For (ii), we compute using the summation convention:

$$((AB)^T)_{ij} = (AB)_{ji} = A_{jk} B_{ki} = (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij},$$

where we are free to reorder the (scalar!) factors in the middle. $\square$

Over the complex numbers, the natural analogue of the transpose also conjugates the entries.

Definition 8.4.14 (Hermitian Conjugate)

For a complex matrix A , its **Hermitian conjugate** is

$$A^\dagger = \overline{(A^T)} = \overline{A}^T,$$

that is, $(A^\dagger)_{ij} = \overline{A_{ji}}$. As with the transpose, $(AB)^\dagger = B^\dagger A^\dagger$.

The Hermitian conjugate interacts with the complex inner product on $\mathbb{C}^n$ in the same way the transpose interacts with the dot product: $(\mathbf{z}, \mathbf{w}) = \mathbf{z}^\dagger \mathbf{w}$.

These operations pick out some particularly important classes of square matrices.

Definition 8.4.15 (Symmetric and Hermitian Matrices)

A square matrix A is **symmetric** if $A^T = A$, and **antisymmetric** if $A^T = -A$. A complex square matrix is **Hermitian** if $A^\dagger = A$, and **skew-Hermitian** if $A^\dagger = -A$.

Note the diagonal entries of an antisymmetric matrix are all zero, the diagonal entries of a Hermitian matrix are real, and those of a skew-Hermitian matrix are purely imaginary. Hermitian matrices will play a starring role when we study eigenvalues in Section 6.

Definition 8.4.16 (Trace)

The **trace** of an $n \times n$ matrix A is the sum of its diagonal entries,

$$\text{tr}(A) = A_{ii} = A_{11} + A_{22} + \cdots + A_{nn}.$$

Proposition 8.4.17 (Cyclic Property of Trace)

For $n \times n$ matrices A and B , $\text{tr}(AB) = \text{tr}(BA)$.

Proof. $\text{tr}(AB) = (AB)_{ii} = A_{ik}B_{ki} = B_{ki}A_{ik} = (BA)_{kk} = \text{tr}(BA)$. $\square$

Example 8.4.18 (Trace of a Reflection)

For the reflection matrix $H_{ij} = \delta_{ij} - 2n_i n_j$, we get

$$\text{tr}(H) = H_{ii} = \delta_{ii} - 2n_i n_i = 3 - 2|\hat{\mathbf{n}}|^2 = 1,$$

independent of the choice of plane – a first hint that the trace captures something basis-independent about the underlying map, a fact we will return to.

There is a useful way of decomposing an arbitrary square matrix using these notions. Any $n \times n$ matrix B can be split into symmetric and antisymmetric parts,

$$B_{ij} = \underbrace{\frac{1}{2}(B_{ij} + B_{ji})}_{S_{ij}} + \underbrace{\frac{1}{2}(B_{ij} - B_{ji})}_{A_{ij}},$$

and the symmetric part can be further split off from its trace, writing $S_{ij} = \frac{1}{n} \text{tr}(S) \delta_{ij} + T_{ij}$ where $\text{tr}(T) = 0$. Altogether,

$$B = \underbrace{\frac{1}{n} \text{tr}(B) I}_{\text{isotropic}} + \underbrace{\left(\frac{1}{2}(B + B^T) - \frac{1}{n} \text{tr}(B) I \right)}_{\text{symmetric, traceless}} + \underbrace{\frac{1}{2}(B - B^T)}_{\text{antisymmetric}}.$$

In three dimensions, the antisymmetric part has only three independent entries, and can be packaged as a vector: setting $A_{ij} = \varepsilon_{ijk} \omega_k$ gives

$$A = \begin{pmatrix} 0 & \omega_3 & -\omega_2 \\ -\omega_3 & 0 & \omega_1 \\ \omega_2 & -\omega_1 & 0 \end{pmatrix},$$

and then $A\mathbf{x} = -\boldsymbol{\omega} \times \mathbf{x}$ where $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)$. This decomposition appears throughout applied mathematics (for instance in describing the local behaviour of fluid flows, where the three parts correspond to expansion, strain, and rotation).

§8.4.4 Inverses, Orthogonal and Unitary Matrices

We now consider when the linear map represented by a matrix can be undone.

Definition 8.4.19 (Inverse)

Let A be an $m \times n$ matrix. An $n \times m$ matrix B is a **left inverse** of A if $BA = I$, and a **right inverse** if $AB = I$. If A is square and has both, then they coincide, since

$$B = B(AC) = (BA)C = C,$$

and we write A^{-1} for this **inverse** of A , satisfying $AA^{-1} = A^{-1}A = I$. A square

matrix with an inverse is called **invertible** or **non-singular**.

Not every square matrix is invertible – the zero matrix clearly isn't, and neither is the matrix of a projection (having squashed the component along $\hat{\mathbf{n}}$, there is no way to recover it). We will find a clean criterion for invertibility in the next section, using determinants. For now, we note the basic properties.

Proposition 8.4.20

If A and B are invertible $n \times n$ matrices, then

- (i) $(A^{-1})^{-1} = A$,
- (ii) $(AB)^{-1} = B^{-1}A^{-1}$,
- (iii) $(A^T)^{-1} = (A^{-1})^T$.

Proof. (i) is immediate. For (ii), $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$. For (iii), transposing $AA^{-1} = I$ gives $(A^{-1})^T A^T = I$. $\square$

In two dimensions we can just write the inverse down: if $ad - bc \neq 0$ then

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix},$$

as can be checked by multiplying out. The quantity $ad - bc$ appearing here is the *determinant*, which is the subject of the next section.

We finish with two classes of matrices whose inverses are essentially free.

Definition 8.4.21 (Orthogonal and Unitary Matrices)

A real $n \times n$ matrix R is **orthogonal** if

$$R^T R = R R^T = I, \quad \text{i.e.} \quad R^{-1} = R^T.$$

A complex $n \times n$ matrix U is **unitary** if

$$U^\dagger U = U U^\dagger = I, \quad \text{i.e.} \quad U^{-1} = U^\dagger.$$

The condition $R^T R = I$ says, in components, $R_{ki} R_{kj} = \delta_{ij}$ – that is, *the columns of R are orthonormal* (and by considering $R R^T = I$, so are the rows). Similarly the columns of a unitary matrix are orthonormal with respect to the complex inner product. This gives orthogonal matrices their geometric significance.

Proposition 8.4.22 (Orthogonal Matrices Preserve the Dot Product)

A real $n \times n$ matrix R is orthogonal if and only if

$$(R\mathbf{x}) \cdot (R\mathbf{y}) = \mathbf{x} \cdot \mathbf{y} \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^n.$$

In particular, orthogonal matrices preserve lengths and angles.

Proof. If R is orthogonal, then $(R\mathbf{x}) \cdot (R\mathbf{y}) = (R\mathbf{x})^T (R\mathbf{y}) = \mathbf{x}^T R^T R \mathbf{y} = \mathbf{x}^T \mathbf{y} = \mathbf{x} \cdot \mathbf{y}$. Conversely, if the dot product is preserved then taking $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ gives $(R^T R)_{ij} = (R\mathbf{e}_i) \cdot (R\mathbf{e}_j) = \delta_{ij}$, so $R^T R = I$. □

In fact, preserving lengths alone is enough: if $|R\mathbf{x}| = |\mathbf{x}|$ for all $\mathbf{x}$, then the identity $\mathbf{x} \cdot \mathbf{y} = \frac{1}{2}(|\mathbf{x} + \mathbf{y}|^2 - |\mathbf{x}|^2 - |\mathbf{y}|^2)$ shows the dot product is preserved as well, so the length-preserving matrices are precisely the orthogonal ones.

Example 8.4.23 (Rotations and Reflections are Orthogonal) (i) The reflection matrix $H_{ij} = \delta_{ij} - 2n_i n_j$ satisfies

$$\begin{aligned} (HH^T)_{ij} &= H_{ik} H_{jk} = (\delta_{ik} - 2n_i n_k)(\delta_{jk} - 2n_j n_k) \\ &= \delta_{ij} - 2n_i n_j - 2n_j n_i + 4n_i n_j (n_k n_k) = \delta_{ij}, \end{aligned}$$

using $n_k n_k = 1$. So H is orthogonal (as it must be – reflections certainly preserve lengths and angles). Note also $H^T = H$, so $H^{-1} = H$: reflecting twice does nothing.

(ii) For the rotation $R(\theta, \hat{\mathbf{n}})$ about a general axis, the component formula from before gives $R_{ji}(\theta, \hat{\mathbf{n}}) = R_{ij}(-\theta, \hat{\mathbf{n}})$, i.e. R^T is the rotation by $-\theta$. So $R^T R = I$, and rotations are orthogonal too.

We will return to orthogonal and unitary matrices repeatedly: in the next section we will see what determinants say about them, in Section 6 they will be the key to diagonalising symmetric and Hermitian matrices, and in Section 7 we will study them as groups of transformations in their own right.

§8.5 Determinants and Inverses

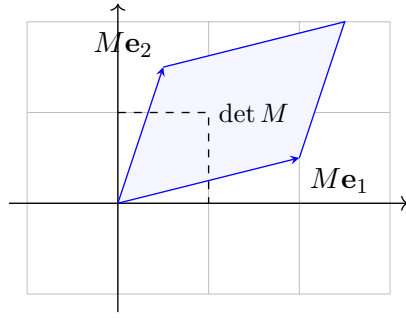
We saw at the end of the last section that a 2×2 matrix is invertible exactly when the quantity $ad - bc$ is non-zero. In this section we will develop the analogous quantity for $n \times n$ matrices – the *determinant*. As promised in the introduction, we will take two points of view: we need to be able to *compute* determinants, but we also want to understand what they *mean*. The meaning comes first: the determinant measures how the linear map scales volume.

§8.5.1 Motivation: Scaling Areas and Volumes

Consider a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with matrix M . The unit square spanned by $\mathbf{e}_1, \mathbf{e}_2$ is mapped to the parallelogram spanned by $M\mathbf{e}_1$ and $M\mathbf{e}_2$, which has signed area

$$[M\mathbf{e}_1, M\mathbf{e}_2] = \varepsilon_{ij} (M\mathbf{e}_1)_i (M\mathbf{e}_2)_j = \varepsilon_{ij} M_{i1} M_{j2} = M_{11} M_{22} - M_{21} M_{12},$$

recalling the two dimensional alternating symbol ε_{ij} from Section 3. By linearity, *every* region has its area scaled by this same factor.



Similarly, a linear map of $\mathbb{R}^3$ sends the unit cube to the parallelepiped spanned by the images of the basis vectors, which has signed volume given by the scalar triple product

$$[Me_1, Me_2, Me_3] = \varepsilon_{ijk} M_{i1} M_{j2} M_{k3}.$$

These are the determinants in two and three dimensions, and the general definition is the natural extension of this pattern to n indices. To write it down we need the Levi-Civita symbol with n indices, and for that we need a brief word about permutations.

§8.5.2 Permutations and the ε Symbol in n Dimensions

Recall (from Groups) that a **permutation** of $\{1, 2, \dots, n\}$ is a bijection from the set to itself, and that the permutations form a group S_n of order $n!$ under composition. A **transposition** is a permutation swapping two elements and fixing the rest. Every permutation σ can be written as a product of transpositions, and while such an expression is far from unique, the parity of the number of transpositions used is an invariant of σ . We define the **sign** $\varepsilon(\sigma) = (-1)^k$ where σ is a product of k transpositions, and call σ **even** or **odd** according to whether $\varepsilon(\sigma)$ is $+1$ or -1 . The sign is multiplicative: $\varepsilon(\sigma\tau) = \varepsilon(\sigma)\varepsilon(\tau)$, and $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$.

Definition 8.5.1 (Levi-Civita Symbol in n Dimensions)

The **Levi-Civita** or **alternating symbol** with n indices is defined by

$$\varepsilon_{ij\dots l} = \begin{cases} +1 & \text{if } i, j, \dots, l \text{ is an even permutation of } 1, 2, \dots, n, \\ -1 & \text{if } i, j, \dots, l \text{ is an odd permutation of } 1, 2, \dots, n, \\ 0 & \text{otherwise (if any index repeats).} \end{cases}$$

In particular, $\varepsilon_{\sigma(1)\sigma(2)\dots\sigma(n)} = \varepsilon(\sigma)$ for any $\sigma \in S_n$, and exchanging any pair of indices changes the sign.

Just as in three dimensions, we can combine n vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$ into a scalar.

Definition 8.5.2 (Alternating Form)

Given vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in \mathbb{R}^n$ or $\mathbb{C}^n$, their **alternating form** is

$$[\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = \varepsilon_{ij\dots l} (\mathbf{v}_1)_i (\mathbf{v}_2)_j \cdots (\mathbf{v}_n)_l = \sum_{\sigma \in S_n} \varepsilon(\sigma) (\mathbf{v}_1)_{\sigma(1)} (\mathbf{v}_2)_{\sigma(2)} \cdots (\mathbf{v}_n)_{\sigma(n)}.$$

For $n = 2$ this is the signed area $[\mathbf{a}, \mathbf{b}] = a_1 b_2 - a_2 b_1$, and for $n = 3$ it is the scalar triple product. The key properties, which follow directly from the corresponding properties of ε , are the following.

Proposition 8.5.3 (Properties of the Alternating Form) (i) *Multilinearity.* The form is linear in each argument separately, e.g.

$$[\alpha \mathbf{u} + \beta \mathbf{w}, \mathbf{v}_2, \dots, \mathbf{v}_n] = \alpha [\mathbf{u}, \mathbf{v}_2, \dots, \mathbf{v}_n] + \beta [\mathbf{w}, \mathbf{v}_2, \dots, \mathbf{v}_n].$$

(ii) *Total antisymmetry.* $[\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(n)}] = \varepsilon(\sigma)[\mathbf{v}_1, \dots, \mathbf{v}_n]$; in particular exchanging two arguments changes the sign, and the form vanishes if two arguments are equal.

(iii) *Normalisation.* $[\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n] = 1$.

Proof Sketch. (i) and (iii) are immediate from the definition. For (ii), it suffices to check a single transposition (as these generate S_n and the sign is multiplicative), and exchanging two arguments amounts to relabelling the summation variable $\sigma \mapsto \sigma\tau$, which flips the sign of each term. $\square$

A consequence of multilinearity and antisymmetry which will do a lot of work for us: if one of the vectors is a linear combination of the others, the form vanishes. In fact, we can say more.

Proposition 8.5.4

$[\mathbf{v}_1, \dots, \mathbf{v}_n] \neq 0$ if and only if $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly independent.

Proof. If the vectors are linearly dependent, then one of them, say $\mathbf{v}_p$, is a linear combination of the others. Substituting this combination in and expanding by multilinearity, each resulting term has a repeated argument, so vanishes by antisymmetry.

Conversely, if the vectors are linearly independent then they form a basis of $\mathbb{R}^n$ (being n of them), so each standard basis vector can be expanded as $\mathbf{e}_i = U_{ai}\mathbf{v}_a$ for some coefficients U_{ai} . Then by multilinearity and antisymmetry,

$$1 = [\mathbf{e}_1, \dots, \mathbf{e}_n] = U_{a1}U_{b2} \cdots U_{cn}[\mathbf{v}_a, \mathbf{v}_b, \dots, \mathbf{v}_c] = U_{a1}U_{b2} \cdots U_{cn} \varepsilon_{ab \cdots c} [\mathbf{v}_1, \dots, \mathbf{v}_n],$$

so $[\mathbf{v}_1, \dots, \mathbf{v}_n]$ cannot be zero. $\square$

§8.5.3 The Determinant

We can now make our main definition.

Definition 8.5.5 (Determinant)

The **determinant** of an $n \times n$ matrix M is the alternating form of its columns $\mathbf{C}_a = M\mathbf{e}_a$,

$$\begin{aligned} \det M &= [M\mathbf{e}_1, M\mathbf{e}_2, \dots, M\mathbf{e}_n] = \varepsilon_{ij \cdots l} M_{i1} M_{j2} \cdots M_{ln} \\ &= \sum_{\sigma \in S_n} \varepsilon(\sigma) M_{\sigma(1)1} M_{\sigma(2)2} \cdots M_{\sigma(n)n}. \end{aligned}$$

We also write $\det M = |M|$, displaying the array between vertical bars.

By our earlier discussion, $\det M$ is precisely the factor by which the map $\mathbf{x} \mapsto M\mathbf{x}$ scales signed volume. The proposition above then immediately gives us the link to invertibility that we were looking for:

$$\det M \neq 0 \iff \text{the columns of } M \text{ are linearly independent} \iff \text{img } M = \mathbb{R}^n,$$

and by rank-nullity this is also equivalent to $\ker M = \{\mathbf{0}\}$. (We will construct the inverse explicitly below.)

Example 8.5.6 (Small Determinants)

For $n = 2$ we recover

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc,$$

and for $n = 3$, expanding the sum over the six permutations of S_3 ,

$$\begin{vmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{vmatrix} = M_{11}M_{22}M_{33} + M_{12}M_{23}M_{31} + M_{13}M_{21}M_{32} \\ - M_{11}M_{23}M_{32} - M_{12}M_{21}M_{33} - M_{13}M_{22}M_{31}.$$

Also, if M is diagonal (or indeed upper or lower triangular), the only permutation contributing a non-zero term is the identity, so $\det M = M_{11}M_{22} \cdots M_{nn}$.

The determinant doesn't care whether we work with rows or columns.

Proposition 8.5.7 ($\det M^T = \det M$)

For any $n \times n$ matrix M , we have

$$\det M = \sum_{\sigma \in S_n} \varepsilon(\sigma) M_{1\sigma(1)} M_{2\sigma(2)} \cdots M_{n\sigma(n)} = \det(M^T).$$

Proof. Take a term $M_{\sigma(1)1} M_{\sigma(2)2} \cdots M_{\sigma(n)n}$ of the defining sum. Reordering the factors so that the *row* indices are in increasing order, it becomes $M_{1\rho(1)} M_{2\rho(2)} \cdots M_{n\rho(n)}$ where $\rho = \sigma^{-1}$. Since $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$, and summing over σ is the same as summing over $\rho = \sigma^{-1}$, the two sums coincide. $\square$

Row and Column Operations

The properties of the alternating form translate directly into how the determinant behaves under *row and column operations*, and this gives an efficient way of computing determinants. Writing $\mathbf{C}_a$ for the columns (everything below applies equally to rows, using $\det M = \det M^T$):

- *Scaling*: if $\mathbf{C}_a \mapsto \lambda \mathbf{C}_a$ for a single fixed a , then $\det M \mapsto \lambda \det M$, by multilinearity. Note in particular that $\det(\lambda M) = \lambda^n \det M$ – the determinant is very much not linear in M .
- *Exchanges*: if two columns are exchanged, then $\det M \mapsto -\det M$, by antisymmetry.

- *Operations*: if $\mathbf{C}_a \mapsto \mathbf{C}_a + \lambda \mathbf{C}_b$ with $b \neq a$, then $\det M$ is *unchanged*, since the extra term is a determinant with a repeated column.

Example 8.5.8 (Computing a Determinant with Row and Column Operations)

Let's compute, for $a \in \mathbb{C}$,

$$\det A = \begin{vmatrix} 1 & 1 & a \\ a & 1 & 1 \\ 1 & a & 1 \end{vmatrix}.$$

Subtracting the third column from the first ($\mathbf{C}_1 \mapsto \mathbf{C}_1 - \mathbf{C}_3$), and then taking out the common factor,

$$\det A = \begin{vmatrix} 1-a & 1 & a \\ a-1 & 1 & 1 \\ 0 & a & 1 \end{vmatrix} = (1-a) \begin{vmatrix} 1 & 1 & a \\ -1 & 1 & 1 \\ 0 & a & 1 \end{vmatrix}.$$

Now $\mathbf{C}_2 \mapsto \mathbf{C}_2 - \mathbf{C}_3$ followed by taking out $(1-a)$ again gives

$$\det A = (1-a)^2 \begin{vmatrix} 1 & 1 & a \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix},$$

and finally adding the second and third rows to the first ($\mathbf{R}_1 \mapsto \mathbf{R}_1 + \mathbf{R}_2 + \mathbf{R}_3$),

$$\det A = (1-a)^2 \begin{vmatrix} 0 & 0 & a+2 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = (1-a)^2(a+2),$$

where the last determinant is found by expanding the sum directly (only one permutation contributes). We will use this particular matrix again shortly when solving linear systems.

The Determinant of a Product

The single most important algebraic property of the determinant is that it is *multiplicative*. This is geometrically obvious – composing two maps composes their volume-scaling factors – but it deserves an honest proof. We first record a small generalisation of the definition.

Lemma 8.5.9

For an $n \times n$ matrix M ,

$$\varepsilon_{i_1 i_2 \dots i_n} M_{i_1 a_1} M_{i_2 a_2} \dots M_{i_n a_n} = (\det M) \varepsilon_{a_1 a_2 \dots a_n}.$$

Proof. The left hand side is totally antisymmetric under exchanging any pair of the free indices $a_1, \dots, a_n$ (exchanging a_p and a_q can be undone by relabelling the summed indices $i_p \leftrightarrow i_q$, at the cost of a sign from ε). Any totally antisymmetric n -index object is a multiple of $\varepsilon_{a_1 \dots a_n}$, and setting $a_j = j$ shows that the constant of proportionality is $\det M$. $\square$

Theorem 8.5.10 (Determinants are Multiplicative)

For $n \times n$ matrices M and N ,

$$\det(MN) = \det M \det N.$$

Proof. Directly from the definition and the lemma,

$$\begin{aligned} \det(MN) &= \varepsilon_{i_1 \dots i_n} (MN)_{i_1 1} \cdots (MN)_{i_n n} \\ &= \varepsilon_{i_1 \dots i_n} M_{i_1 k_1} N_{k_1 1} \cdots M_{i_n k_n} N_{k_n n} \\ &= (\varepsilon_{i_1 \dots i_n} M_{i_1 k_1} \cdots M_{i_n k_n}) N_{k_1 1} \cdots N_{k_n n} \\ &= (\det M) \varepsilon_{k_1 \dots k_n} N_{k_1 1} \cdots N_{k_n n} = \det M \det N. \square \end{aligned}$$

Corollary 8.5.11 (i) If M is invertible, then $\det(M^{-1}) = (\det M)^{-1}$.

(ii) If R is orthogonal, then $\det R = \pm 1$.

(iii) If U is unitary, then $|\det U| = 1$.

Proof. (i) $\det(M^{-1}) \det(M) = \det(M^{-1}M) = \det I = 1$.

(ii) $R^T R = I$ gives $(\det R)^2 = \det(R^T) \det(R) = 1$.

(iii) $U^\dagger U = I$ gives $\overline{\det U} \det U = 1$, since $\det(U^\dagger) = \det(\overline{U}^T) = \overline{\det U}$. $\square$

In $\mathbb{R}^3$ this splits the orthogonal matrices into two families: the rotations, with $\det R = 1$ (preserving orientation), and the reflections, with $\det R = -1$ (reversing it). We will pick this thread up again in Section 7.

§8.5.4 Minors, Cofactors and the Laplace Expansion

The next tool both organises hand computation of determinants and leads directly to the formula for the inverse.

Definition 8.5.12 (Minors and Cofactors)

Let M be an $n \times n$ matrix. The **minor** M^{ia} is the determinant of the $(n-1) \times (n-1)$ matrix obtained from M by deleting row i and column a . The **cofactor** of the entry M_{ia} is

$$\Delta_{ia} = (-1)^{i+a} M^{ia}.$$

Theorem 8.5.13 (Laplace Expansion)

For an $n \times n$ matrix M and any fixed column a (or fixed row i),

$$\det M = \sum_{i=1}^n M_{ia} \Delta_{ia} = \sum_{a=1}^n M_{ia} \Delta_{ia}.$$

Proof. Expanding column a as $\mathbf{C}_a = \sum_i M_{ia} \mathbf{e}_i$ and using multilinearity,

$$\det M = [\mathbf{C}_1, \dots, \mathbf{C}_a, \dots, \mathbf{C}_n] = \sum_i M_{ia} [\mathbf{C}_1, \dots, \mathbf{e}_i, \dots, \mathbf{C}_n],$$

so it remains to identify $[\mathbf{C}_1, \dots, \mathbf{e}_i, \dots, \mathbf{C}_n]$ with the cofactor Δ_{ia} . In this determinant, column a is $\mathbf{e}_i$; adding suitable multiples of column a to the others, we may assume every other entry of row i is zero without changing the value. Now move column a to the last position ($n - a$ adjacent exchanges) and row i to the last position ($n - i$ adjacent exchanges), picking up a sign $(-1)^{(n-a)+(n-i)} = (-1)^{i+a}$. The result is a block matrix

$$(-1)^{i+a} \begin{vmatrix} \tilde{M} & \mathbf{0} \\ * & 1 \end{vmatrix} = (-1)^{i+a} \det \tilde{M},$$

where $\tilde{M}$ is M with row i and column a deleted – here the block determinant reduces to $\det \tilde{M}$ since any permutation not fixing the last index contributes zero. This is exactly $(-1)^{i+a} M^{ia} = \Delta_{ia}$. The row version follows from $\det M = \det M^T$. $\square$

Example 8.5.14 (Laplace Expansion in Practice)

Expanding along columns or rows with many zeros keeps the computation short. For instance, expanding along the second column,

$$\begin{vmatrix} 2 & 0 & 2 \\ 3 & 2 & 1 \\ 2 & 0 & 1 \end{vmatrix} = -0 + 2 \begin{vmatrix} 2 & 2 \\ 2 & 1 \end{vmatrix} - 0 = 2(2 - 4) = -4.$$

In practice one uses row and column operations to create zeros, and then expands.

Example 8.5.15 (A Vandermonde-type Determinant)

As a slightly more substantial example, let us show

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c)(c - a).$$

Subtracting row 2 from row 1, and row 3 from row 2,

$$\begin{vmatrix} 0 & a - b & a^2 - b^2 \\ 0 & b - c & b^2 - c^2 \\ 1 & c & c^2 \end{vmatrix} = (a - b)(b - c) \begin{vmatrix} 0 & 1 & a + b \\ 0 & 1 & b + c \\ 1 & c & c^2 \end{vmatrix},$$

and expanding along the first column gives

$$(a - b)(b - c) \cdot 1 \cdot \begin{vmatrix} 1 & a + b \\ 1 & b + c \end{vmatrix} = (a - b)(b - c)(c - a).$$

§8.5.5 The Adjugate and the Inverse

Here is the key observation: in the Laplace expansion $\det M = \sum_i M_{ia}\Delta_{ia}$, what happens if we expand the entries of one column against the cofactors of a *different* column?

Lemma 8.5.16

For an $n \times n$ matrix M ,

$$\sum_i M_{ib}\Delta_{ia} = (\det M) \delta_{ab}.$$

Proof. If $a = b$ this is the Laplace expansion. If $a \neq b$, then $\sum_i M_{ib}\Delta_{ia} = [\mathbf{C}_1, \dots, \mathbf{C}_b, \dots, \mathbf{C}_n]$ with $\mathbf{C}_b$ appearing in position a – but $\mathbf{C}_b$ also appears in its own position b , so this is a determinant with a repeated column, which vanishes. $\square$

Definition 8.5.17 (Adjugate)

The **adjugate** of M is the transpose of the matrix of cofactors,

$$(\operatorname{adj} M)_{ai} = \Delta_{ia}.$$

With this notation, the lemma says precisely that $(\operatorname{adj} M)M = (\det M)I$, giving us the general formula for the inverse.

Theorem 8.5.18 (Inverse via the Adjugate)

If $\det M \neq 0$, then M is invertible with

$$M^{-1} = \frac{1}{\det M} \operatorname{adj} M, \quad \text{that is,} \quad (M^{-1})_{ai} = \frac{\Delta_{ia}}{\det M}.$$

Conversely, if $\det M = 0$ then M has no inverse.

Proof. The formula follows from Lemma 8.5.16 (and the analogous row version gives $M \operatorname{adj} M = (\det M)I$ as well). If $\det M = 0$, then for any B we have $\det(BM) = \det B \det M = 0 \neq \det I$, so no inverse can exist. $\square$

Example 8.5.19 (Inverting a 3×3 Matrix)

Consider the shear matrix from the previous section,

$$S_\lambda = \begin{pmatrix} 1 & \lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This is triangular, so $\det S_\lambda = 1$. Computing the cofactors, the only non-trivial one is $\Delta_{21} = -\lambda$, and we find

$$S_\lambda^{-1} = \begin{pmatrix} 1 & -\lambda & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = S_{-\lambda},$$

which is geometrically obvious: to undo a shear, shear back the other way.

While the adjugate formula is theoretically clean, as a computational method it is dreadful: computing an $n \times n$ determinant from cofactors takes on the order of $n!$ operations. For actually solving equations, the method of choice is *Gaussian elimination*, discussed below, which takes on the order of n^3 operations.

§8.5.6 Systems of Linear Equations

Much of the theory we have built up comes together in the problem of solving simultaneous linear equations. A system of n equations in n unknowns $x_1, \dots, x_n$ can be packaged as

$$A\mathbf{x} = \mathbf{b},$$

where A is an $n \times n$ matrix of coefficients and $\mathbf{b} \in \mathbb{R}^n$ is given. We say the system is **homogeneous** if $\mathbf{b} = \mathbf{0}$, and **inhomogeneous** otherwise.

Thinking of A as a linear map makes the structure of the solutions transparent. There are three possibilities.

- (i) If $\det A \neq 0$, then A^{-1} exists and there is a *unique* solution

$$\mathbf{x} = A^{-1}\mathbf{b}$$

for any $\mathbf{b}$ (which is $\mathbf{x} = \mathbf{0}$ in the homogeneous case).

- (ii) If $\det A = 0$ and $\mathbf{b} \notin \text{img } A$, then there are *no* solutions – the system is inconsistent.
 (iii) If $\det A = 0$ and $\mathbf{b} \in \text{img } A$, then there are *infinitely many* solutions, of the form

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{u},$$

where $\mathbf{x}_0$ is any *particular solution* (so $A\mathbf{x}_0 = \mathbf{b}$) and $\mathbf{u}$ ranges over $\ker A$. Writing $\mathbf{u}_1, \dots, \mathbf{u}_k$ for a basis of the kernel (where $k = \text{null } A > 0$), the general solution is $\mathbf{x} = \mathbf{x}_0 + \sum_j \mu_j \mathbf{u}_j$ for scalars μ_j .

To see that these are the only possibilities: if $\mathbf{x}$ and $\mathbf{x}_0$ are both solutions, then $A(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, so any two solutions differ by an element of the kernel. And $\det A \neq 0$ if and only if $\text{rank } A = n$, if and only if (by rank-nullity) $\ker A = \{\mathbf{0}\}$.

You may notice the structure of case (iii) – particular solution plus general solution of the homogeneous problem – is precisely the ‘particular integral plus complementary function’ pattern familiar from linear differential equations. This is no coincidence: both are instances of solving a linear equation.

Example 8.5.20 (A Family of Linear Systems)

Let us solve $A\mathbf{x} = \mathbf{b}$ where

$$A = \begin{pmatrix} 1 & 1 & a \\ a & 1 & 1 \\ 1 & a & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ c \\ 1 \end{pmatrix}, \quad a, c \in \mathbb{R}.$$

We computed earlier that $\det A = (1 - a)^2(a + 2)$.

Case $a \neq 1, -2$. Here $\det A \neq 0$, so for every c there is the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. Geometrically, the three planes described by the rows meet in a single point.

Case $a = 1$. All three rows become $(1, 1, 1)$, so

$$\text{img } A = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}, \quad \ker A = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

If $c \neq 1$ then $\mathbf{b} \notin \text{img } A$ and there are no solutions. If $c = 1$, then $\mathbf{x}_0 = (1, 0, 0)$ is a particular solution, and the general solution is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

which is a plane (all three planes coincide).

Case $a = -2$. Solving $A\mathbf{u} = \mathbf{0}$ directly gives $\ker A = \text{span}\{(1, 1, 1)\}$, and $\text{img } A$ is spanned by $(1, -2, 1)$ and $(1, 1, -2)$. One can check $\mathbf{b} \in \text{img } A$ if and only if $c = -2$, in which case $\mathbf{x}_0 = (1, 0, 0)$ is a particular solution^a and the general solution is the line $\mathbf{x} = \mathbf{x}_0 + \lambda(1, 1, 1)$. For $c \neq -2$ there are no solutions.

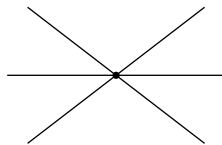
^aFound by inspection: any single solution will do.

Geometric Interpretation

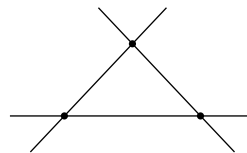
In $\mathbb{R}^3$, each row of the system $A\mathbf{x} = \mathbf{b}$ is the equation of a plane, $\mathbf{R}_i \cdot \mathbf{x} = b_i$, with normal $\mathbf{R}_i$ (assuming $\mathbf{R}_i \neq \mathbf{0}$). Solving the system means intersecting the three planes, and the possibilities line up with the rank of A :

- If $\text{rank } A = 3$, the three normals are linearly independent and the planes meet in exactly one point.
- If $\text{rank } A = 2$, the normals span a plane. For the homogeneous problem the solution set is a line. In the inhomogeneous case, the planes may intersect in a common line, or may form a ‘sheaf’ – intersecting pairwise in three parallel lines, like a triangular prism – with no common point (no solutions).
- If $\text{rank } A = 1$, the normals are all parallel. The planes are parallel, and there are solutions only if they all coincide, in which case the solution set is that plane.

The two rank 2 configurations are worth visualising. Viewing the planes edge-on (looking along the direction of the would-be common line), they appear as three lines: either all passing through one point, or forming a triangle.



a common line (end-on)



a sheaf (end-on)

§8.5.7 Gaussian Elimination

Finally, let’s discuss how solving linear systems actually gets done. The method is **Gaussian elimination**: use one equation to eliminate a variable from the others, and

repeat. It applies to any system of m equations in n unknowns,

$$A_{11}x_1 + A_{12}x_2 + \cdots + A_{1n}x_n = d_1, \quad \dots, \quad A_{m1}x_1 + \cdots + A_{mn}x_n = d_m.$$

Assuming $A_{11} \neq 0$ (reordering the equations if necessary), we can subtract multiples of the first equation from the others to eliminate x_1 from all of them. Then we use the (new) second equation to eliminate x_2 from those below it, and so on. Whenever a leading coefficient vanishes we reorder to fix it, and the process halts when no equations with non-zero coefficients remain. We are left with a system in *echelon form*:

$$\begin{aligned} A_{11}x_1 + A_{12}x_2 + \cdots + \cdots + A_{1n}x_n &= d_1 \\ A_{22}^{(2)}x_2 + \cdots + \cdots + A_{2n}^{(2)}x_n &= d_2^{(2)} \\ &\vdots \\ A_{rr}^{(r)}x_r + \cdots + A_{rn}^{(r)}x_n &= d_r^{(r)} \\ 0 &= d_{r+1}^{(r)} \\ &\vdots \\ 0 &= d_m^{(r)}, \end{aligned}$$

where the superscripts track how many times a coefficient has been updated, and the leading coefficients $A_{ii}^{(i)}$ are non-zero. The number r of non-trivial equations remaining is exactly $\text{rank } A$, and reading off the possibilities:

- (i) If some $d_j^{(r)} \neq 0$ for $j > r$, the system is inconsistent – *no solutions* (it is *overdetermined*).
- (ii) If $r = n$ and the leftover equations all read $0 = 0$, we can solve the n th equation for x_n , substitute back into the $(n-1)$ th to find x_{n-1} , and so on – a *unique solution*, found by *back substitution*.
- (iii) If $r < n$ and the leftover equations read $0 = 0$, then $x_{r+1}, \dots, x_n$ can be chosen freely and the remaining variables are determined – *infinitely many solutions* (the system is *underdetermined*).

Example 8.5.21 (Gaussian Elimination in Action)

Consider the system

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ 6x_1 + 3x_2 + 3x_3 &= 0 \\ 6x_1 + 2x_2 + 4x_3 &= 6. \end{aligned}$$

Subtracting twice the first equation from the second and third,

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ -x_2 + x_3 &= -6 \\ -2x_2 + 2x_3 &= 0, \end{aligned}$$

and then subtracting twice the second from the third leaves $0 = 12$. The system is inconsistent, and there are no solutions.

Gaussian elimination also gives an efficient way to compute determinants: for a square matrix, the elimination steps are row operations which leave the determinant unchanged, apart from row exchanges which flip its sign. The end result is upper triangular, so

$$\det A = (-1)^k A_{11} A_{22}^{(2)} \cdots A_{nn}^{(n)},$$

where k is the number of row exchanges used. In particular $\det A \neq 0$ if and only if the elimination reaches $r = n$, tying together our earlier characterisations.

§8.5.8 Matrix Rank

We defined the rank of a linear map as the dimension of its image. For a matrix, there are two natural candidate notions, and it is a small theorem that they agree.

Definition 8.5.22 (Column and Row Rank)

The **column rank** of a matrix A is the maximum number of linearly independent columns of A , i.e. the dimension of the span of the columns. The **row rank** is defined analogously using rows.

Since the columns of A are the images $A\mathbf{e}_j$ of the basis vectors, the column rank is precisely $\text{rank } A = \dim \text{img } A$.

Theorem 8.5.23 (Row Rank Equals Column Rank)

For any $m \times n$ matrix A , the row rank and column rank are equal.

Proof. Let r be the row rank of A , and let $\mathbf{v}_1^T, \dots, \mathbf{v}_r^T$ be a maximal linearly independent set of rows. Then every row of A is a linear combination of these, say

$$\mathbf{r}_i^T = \sum_{k=1}^r C_{ik} \mathbf{v}_k^T$$

for some coefficients C_{ik} . Looking at the j th component of this equation,

$$A_{ij} = \sum_{k=1}^r C_{ik} (\mathbf{v}_k)_j,$$

which says that the j th column of A is the linear combination $\sum_k (\mathbf{v}_k)_j \mathbf{c}_k$ of the r fixed vectors $\mathbf{c}_k = (C_{1k}, \dots, C_{mk})^T$. Hence the column span has dimension at most r , i.e. column rank $\leq$ row rank. Applying the same argument to A^T gives the reverse inequality. $\square$

§8.6 Eigenvalues and Eigenvectors

This section is about the core question raised in the introduction: given a linear map, what does it act on in a really simple way? For most vectors $\mathbf{x}$, applying a matrix

A moves them somewhere with no obvious relation to where they started. But some special vectors are simply *scaled* – the map doesn't turn them at all. Understanding these special directions turns out to essentially unlock the structure of the whole map.

§8.6.1 Definitions and the Characteristic Polynomial

Let's make precise the notion of a direction which a linear map simply scales.

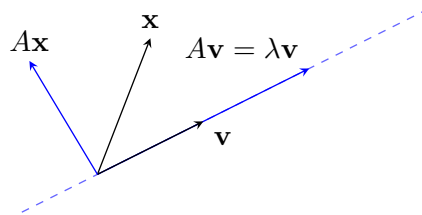
Definition 8.6.1 (Eigenvectors and Eigenvalues)

Let A be an $n \times n$ matrix acting on $V = \mathbb{R}^n$ or $\mathbb{C}^n$. A vector $\mathbf{v} \in V$ with $\mathbf{v} \neq \mathbf{0}$ is an **eigenvector** of A with **eigenvalue** λ if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Note that we insist $\mathbf{v} \neq \mathbf{0}$ (every scalar would be an eigenvalue for $\mathbf{0}$, which would tell us nothing), but $\lambda = 0$ is allowed – an eigenvector with eigenvalue 0 is just a non-zero vector of the kernel.

Geometrically, an eigenvector picks out an *invariant line* through the origin: a typical vector $\mathbf{x}$ gets moved off its own direction by A , but an eigenvector $\mathbf{v}$ stays on the dashed line below, merely stretched by a factor of λ .



How do we find eigenvalues? The equation $A\mathbf{v} = \lambda\mathbf{v}$ rearranges to $(A - \lambda I)\mathbf{v} = \mathbf{0}$, which asks for a non-zero vector in the kernel of $A - \lambda I$. By the theory of the last section, such a vector exists precisely when $\det(A - \lambda I) = 0$. We have proven:

Proposition 8.6.2

λ is an eigenvalue of A if and only if

$$\det(A - \lambda I) = 0.$$

This is called the **characteristic equation** of A .

Definition 8.6.3 (Characteristic Polynomial)

The **characteristic polynomial** of an $n \times n$ matrix A is

$$\chi_A(t) = \det(A - tI).$$

The eigenvalues of A are exactly the roots of χ_A .

Expanding the determinant, $\chi_A(t) = c_0 + c_1t + \cdots + c_nt^n$ is a polynomial of degree n in t . This simple observation, combined with the fundamental theorem of algebra from Section 1, already tells us a surprising amount.

Proposition 8.6.4 (Properties of the Characteristic Polynomial)

Let A be an $n \times n$ matrix with characteristic polynomial $\chi_A(t) = c_0 + c_1t + \cdots + c_nt^n$, and let $\lambda_1, \dots, \lambda_n$ be the roots of χ_A counted with multiplicity. Then

- (i) A has n eigenvalues counted with multiplicity (and in particular, working over $\mathbb{C}$, at least one);
- (ii) $\text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n$;
- (iii) $\det(A) = \lambda_1\lambda_2 \cdots \lambda_n$;
- (iv) if A is real, the non-real eigenvalues come in complex conjugate pairs.

Proof. Since $c_n = (-1)^n \neq 0$, χ_A has degree exactly n , and (i) follows from the fundamental theorem of algebra. Writing

$$\chi_A(t) = (-1)^n(t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n),$$

we can compare coefficients with the determinant expansion. The terms of degree n and $n - 1$ in $\det(A - tI)$ can only arise from the product of diagonal entries $(A_{11} - t) \cdots (A_{nn} - t)$ (any other permutation omits at least two diagonal entries), so

$$c_{n-1} = (-1)^{n-1}(A_{11} + \cdots + A_{nn}) = (-1)^{n-1} \text{tr}(A),$$

while expanding the factorised form gives $c_{n-1} = (-1)^{n-1}(\lambda_1 + \cdots + \lambda_n)$. This proves (ii). For (iii), set $t = 0$: $\chi_A(0) = \det A$, and the factorised form gives $\chi_A(0) = \lambda_1 \cdots \lambda_n$. Finally, if A is real then the c_i are real, so $\chi_A(\lambda) = 0$ implies $\chi_A(\bar{\lambda}) = \overline{\chi_A(\lambda)} = 0$, giving (iv). $\square$

Let's compute some examples, which will also serve to illustrate the various things that can happen.

Example 8.6.5 (Finding Eigenvalues and Eigenvectors) (i) Take $A = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$ acting on $\mathbb{C}^2$. The characteristic equation is

$$\det \begin{pmatrix} 2 - \lambda & i \\ -i & 2 - \lambda \end{pmatrix} = (2 - \lambda)^2 - 1 = 0,$$

so $\lambda = 1$ or 3 . For $\lambda = 1$, solving $(A - I)\mathbf{v} = \mathbf{0}$ gives $\mathbf{v} = \alpha \begin{pmatrix} 1 \\ i \end{pmatrix}$, and for $\lambda = 3$

we get $\mathbf{v} = \beta \begin{pmatrix} 1 \\ -i \end{pmatrix}$, for any non-zero α, β .

(ii) Take the shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Then $\chi_A(t) = (1 - t)^2$, so $\lambda = 1$ is the only eigenvalue, with multiplicity two. But solving $(A - I)\mathbf{v} = \mathbf{0}$ forces $v_2 = 0$, so the only eigenvectors are $\mathbf{v} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ – there is only *one* linearly independent eigenvector. (Geometrically: a shear fixes its horizontal axis and genuinely tilts every other direction.)

(iii) Take the rotation $U = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$. Then

$$\chi_U(t) = t^2 - 2t \cos \theta + 1,$$

which has roots $\lambda = e^{\pm i\theta}$ – for θ not a multiple of π there are *no real* eigenvalues (as expected: a genuine rotation of the plane preserves no direction), but over $\mathbb{C}$ there are two, with eigenvectors $\begin{pmatrix} 1 \\ \mp i \end{pmatrix}$.

§8.6.2 Eigenspaces and Multiplicities

For a given eigenvalue, the set of all associated eigenvectors (together with $\mathbf{0}$) forms a subspace, since it is a kernel.

Definition 8.6.6 (Eigenspace)

For an eigenvalue λ of A , the **eigenspace** is

$$E_\lambda = \{\mathbf{v} : A\mathbf{v} = \lambda\mathbf{v}\} = \ker(A - \lambda I).$$

There are now two different ways to count ‘how many times’ λ occurs as an eigenvalue, and their interplay is the crux of this section.

Definition 8.6.7 (Algebraic and Geometric Multiplicity)

Let λ be an eigenvalue of A .

- The **algebraic multiplicity** M_λ is the multiplicity of λ as a root of $\chi_A(t)$, i.e. the largest k such that $(t - \lambda)^k$ divides $\chi_A(t)$.
- The **geometric multiplicity** m_λ is $\dim E_\lambda = \text{null}(A - \lambda I)$, the maximum number of linearly independent eigenvectors with eigenvalue λ .

If $M_\lambda > 1$ we say the eigenvalue is **degenerate** or **repeated**.

It is a fact (which we will be able to prove once we can change bases, in Section 7) that

$$1 \leq m_\lambda \leq M_\lambda,$$

that is, the geometric multiplicity never exceeds the algebraic one. The shear above shows the inequality can be strict: there $M_1 = 2$ but $m_1 = 1$.

Example 8.6.8 (Multiplicities) (i) Let

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}.$$

Computing, $\chi_A(t) = -(t - 5)(t + 3)^2$, so the eigenvalues are 5 and -3 with

$M_5 = 1$, $M_{-3} = 2$. Solving $(A - 5I)\mathbf{v} = \mathbf{0}$ gives

$$E_5 = \text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right\},$$

while for $\lambda = -3$ the system $(A + 3I)\mathbf{v} = \mathbf{0}$ reduces to the single equation $v_1 + 2v_2 - 3v_3 = 0$, so

$$E_{-3} = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Hence $m_5 = M_5 = 1$ and $m_{-3} = M_{-3} = 2$: three linearly independent eigenvectors in total.

(ii) Let

$$B = \begin{pmatrix} -3 & -1 & 1 \\ -1 & -3 & 1 \\ -2 & -2 & 0 \end{pmatrix}.$$

Here $\chi_B(t) = -(t+2)^3$, so $\lambda = -2$ with $M_{-2} = 3$. But $(B + 2I)\mathbf{v} = \mathbf{0}$ reduces to the single equation $v_1 + v_2 - v_3 = 0$, giving

$$E_{-2} = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\},$$

so $m_{-2} = 2 < 3 = M_{-2}$. There are only two independent eigenvectors to be found.

- (iii) For a reflection H in a plane through the origin with unit normal $\hat{\mathbf{n}}$: clearly $H\hat{\mathbf{n}} = -\hat{\mathbf{n}}$, and $H\mathbf{u} = \mathbf{u}$ for any $\mathbf{u}$ in the plane. So the eigenvalues are -1 , with $E_{-1} = \text{span}\{\hat{\mathbf{n}}\}$, and 1 , with E_1 the plane itself: $M_{-1} = m_{-1} = 1$ and $M_1 = m_1 = 2$.
- (iv) For a rotation R through θ about the axis $\hat{\mathbf{n}}$ in $\mathbb{R}^3$: the axis satisfies $R\hat{\mathbf{n}} = \hat{\mathbf{n}}$, giving eigenvalue 1 , and restricting to the plane perpendicular to the axis we get a two dimensional rotation, contributing complex eigenvalues $e^{\pm i\theta}$ as computed above.

§8.6.3 Linear Independence and Diagonalisation

The importance of having ‘enough’ eigenvectors is the following: if the eigenvectors of A form a *basis*, then in that basis the map just scales each basis direction, and its matrix is diagonal. Working with diagonal matrices is a delight – powers, inverses and polynomials of them can be simply read off – so we would very much like to know when this happens.

The first main result is that eigenvectors for different eigenvalues never interfere with one another.

Proposition 8.6.9 (Independence of Eigenvectors)

Let $\mathbf{v}_1, \dots, \mathbf{v}_r$ be eigenvectors of A with eigenvalues $\lambda_1, \dots, \lambda_r$. If the eigenvalues are distinct, then the eigenvectors are linearly independent.

Proof. Suppose we have a linear relation $\sum_{j=1}^r \alpha_j \mathbf{v}_j = \mathbf{0}$. Fix any k , and apply the matrix

$$\prod_{j \neq k} (A - \lambda_j I)$$

to both sides. Each eigenvector $\mathbf{v}_i$ with $i \neq k$ is annihilated by its own factor $(A - \lambda_i I)$, while $\mathbf{v}_k$ just picks up scalars, leaving

$$\alpha_k \prod_{j \neq k} (\lambda_k - \lambda_j) \mathbf{v}_k = \mathbf{0}.$$

The product of the (non-zero!) factors $\lambda_k - \lambda_j$ is non-zero, and $\mathbf{v}_k \neq \mathbf{0}$, so $\alpha_k = 0$. As k was arbitrary, the relation was trivial. $\square$

Corollary 8.6.10

Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of A , and let B_i be a basis of the eigenspace E_{λ_i} . Then the union $B = B_1 \cup B_2 \cup \dots \cup B_r$ is linearly independent.

Proof. A general linear combination of the elements of B can be grouped as $\mathbf{w}_1 + \dots + \mathbf{w}_r$ with $\mathbf{w}_i \in E_{\lambda_i}$. If this sum is $\mathbf{0}$, then applying $\prod_{j \neq k} (A - \lambda_j I)$ exactly as in the proposition^a shows each $\mathbf{w}_k = \mathbf{0}$. But $\mathbf{w}_k$ is a linear combination of the basis B_k , so all its coefficients vanish. $\square$

^aThe argument only used that $\mathbf{w}_i$ is an eigenvector or zero.

Definition 8.6.11 (Diagonalisable)

An $n \times n$ matrix A is **diagonalisable** if there is an invertible matrix P with

$$P^{-1}AP = D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n),$$

a diagonal matrix. (We will see in Section 7 that this says exactly that A represents the same map as D , but with respect to a different basis.)

Proposition 8.6.12 (Diagonalisability Criterion)

For an $n \times n$ matrix A , the following are equivalent:

- (i) there is a basis of $V = \mathbb{C}^n$ consisting of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ of A ;
- (ii) A is diagonalisable, and indeed $P^{-1}AP = D$ where the columns of P are the eigenvectors $\mathbf{v}_i$ and the diagonal entries of D are the corresponding eigenvalues.

Proof. The key computation: for any matrix P with columns $\mathbf{v}_i$, the columns of AP are $A\mathbf{v}_i$, while the columns of PD are $\lambda_i \mathbf{v}_i$. So

$$AP = PD \iff A\mathbf{v}_i = \lambda_i \mathbf{v}_i \text{ for each } i.$$

If we have a basis of eigenvectors, take these as the columns of P ; then P is invertible (its columns are linearly independent) and $P^{-1}AP = D$. Conversely if $P^{-1}AP = D$ with P invertible, the columns of P are n linearly independent eigenvectors, hence a basis. $\square$

Combining this with Corollary 8.6.10, we get workable criteria.

Theorem 8.6.13 (Tests for Diagonalisability)

Let A be an $n \times n$ matrix.

- (i) If A has n distinct eigenvalues, then A is diagonalisable.
- (ii) In general, A is diagonalisable over $\mathbb{C}$ if and only if $m_\lambda = M_\lambda$ for every eigenvalue λ .

Proof. For (i), the n corresponding eigenvectors are linearly independent by the proposition above, and n linearly independent vectors in $\mathbb{C}^n$ form a basis.

For (ii), take bases of all the eigenspaces; by Corollary 8.6.10 their union is linearly independent, and it has $\sum_\lambda m_\lambda$ elements. If $m_\lambda = M_\lambda$ for all λ , then since $\sum_\lambda M_\lambda = n$ (the characteristic polynomial has n roots with multiplicity over $\mathbb{C}$), this union is a basis of eigenvectors. Conversely, if $m_\mu < M_\mu$ for some μ , then the maximum possible number of linearly independent eigenvectors is $\sum_\lambda m_\lambda < n$, so no eigenvector basis exists. $\square$

Note carefully that (i) is sufficient but not necessary – the identity matrix has just one eigenvalue but is already diagonal!

Example 8.6.14 (Diagonalising a Matrix)

Returning to the matrix

$$A = \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix}$$

from before, we found eigenvalues $5, -3, -3$ with three independent eigenvectors. Assembling them into the columns of

$$P = \begin{pmatrix} 1 & -2 & 3 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad \text{with} \quad P^{-1} = \frac{1}{8} \begin{pmatrix} 1 & 2 & -3 \\ -2 & 4 & 6 \\ 1 & 2 & 5 \end{pmatrix},$$

we get $P^{-1}AP = \text{diag}(5, -3, -3)$: the matrix is diagonalisable even though its eigenvalues are not distinct.

On the other hand, the matrix B from the same example has $m_{-2} = 2 < 3 = M_{-2}$, so B is *not* diagonalisable. (There is also a slicker way to see this: if B were diagonalisable, then $P^{-1}BP = -2I$ would give $B = P(-2I)P^{-1} = -2I$, which is false.) Similarly the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is not diagonalisable.

§8.6.4 Hermitian and Symmetric Matrices

We now specialise to Hermitian matrices ($A^\dagger = A$), which include the real symmetric matrices ($A^T = A$). These matrices turn out to have astonishingly good eigenvalue behaviour, and they appear everywhere – notably as the matrices of quadratic forms below, and (though beyond this course) as the observables of quantum mechanics.

The key observation is that for a Hermitian matrix, using the complex inner product $(\mathbf{v}, \mathbf{w}) = \mathbf{v}^\dagger \mathbf{w}$ on $\mathbb{C}^n$,

$$(A\mathbf{v})^\dagger \mathbf{w} = \mathbf{v}^\dagger A^\dagger \mathbf{w} = \mathbf{v}^\dagger (A\mathbf{w}),$$

i.e. A can be moved from one side of the inner product to the other.

Theorem 8.6.15 (Eigenvalues and Eigenvectors of Hermitian Matrices)

Let A be an $n \times n$ Hermitian matrix. Then

- (i) every eigenvalue of A is real;
- (ii) eigenvectors with distinct eigenvalues are orthogonal;
- (iii) if A is real (and hence symmetric), the eigenvectors in (ii) can be chosen to be real.

Proof. (i) Let $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Then

$$\lambda \mathbf{v}^\dagger \mathbf{v} = \mathbf{v}^\dagger (A\mathbf{v}) = (A\mathbf{v})^\dagger \mathbf{v} = \bar{\lambda} \mathbf{v}^\dagger \mathbf{v},$$

and since $\mathbf{v}^\dagger \mathbf{v} = |\mathbf{v}|^2 \neq 0$, we get $\lambda = \bar{\lambda}$.

(ii) Let $A\mathbf{v} = \lambda\mathbf{v}$ and $A\mathbf{w} = \mu\mathbf{w}$ with $\lambda \neq \mu$ (both real by (i)). Then

$$\mu \mathbf{v}^\dagger \mathbf{w} = \mathbf{v}^\dagger (A\mathbf{w}) = (A\mathbf{v})^\dagger \mathbf{w} = \lambda \mathbf{v}^\dagger \mathbf{w},$$

so $(\lambda - \mu)\mathbf{v}^\dagger \mathbf{w} = 0$, forcing $\mathbf{v}^\dagger \mathbf{w} = 0$.

(iii) Given $A\mathbf{v} = \lambda\mathbf{v}$ with A and λ real, write $\mathbf{v} = \mathbf{u} + i\mathbf{u}'$ with $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^n$. Taking real and imaginary parts of the (linear) eigenvector equation, $A\mathbf{u} = \lambda\mathbf{u}$ and $A\mathbf{u}' = \lambda\mathbf{u}'$, and at least one of $\mathbf{u}, \mathbf{u}'$ is non-zero. $\square$

If a Hermitian matrix has distinct eigenvalues, this hands us an *orthogonal* basis of eigenvectors, which we can normalise to be orthonormal. The remarkable fact – and the main theorem about Hermitian matrices – is that this works even when there are repeated eigenvalues.

To handle the repeated case we want to turn bases of eigenspaces into *orthonormal* bases, and there is a standard procedure for doing exactly that.

Proposition 8.6.16 (Gram-Schmidt Orthogonalisation)

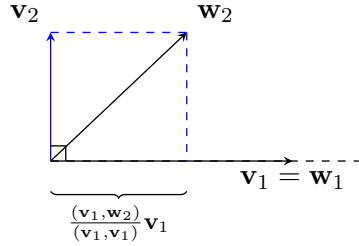
Let $\mathbf{w}_1, \dots, \mathbf{w}_r$ be linearly independent vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$. Define, iteratively,

$$\mathbf{v}_1 = \mathbf{w}_1, \quad \mathbf{v}_k = \mathbf{w}_k - \sum_{j=1}^{k-1} \frac{(\mathbf{v}_j, \mathbf{w}_k)}{(\mathbf{v}_j, \mathbf{v}_j)} \mathbf{v}_j.$$

Then the $\mathbf{v}_i$ are non-zero, pairwise orthogonal, and span the same subspace as the $\mathbf{w}_i$. Normalising each $\mathbf{v}_i$ then yields an orthonormal basis of that subspace.

Proof Sketch. At each step we subtract from $\mathbf{w}_k$ its components along the previously constructed directions, so a direct computation gives $(\mathbf{v}_j, \mathbf{v}_k) = 0$ for $j < k$. Each $\mathbf{v}_k$ is non-zero since $\mathbf{w}_k$ is not in the span of $\mathbf{w}_1, \dots, \mathbf{w}_{k-1}$. $\square$

The geometry of a single step is exactly the projection picture from Section 2: we remove from $\mathbf{w}_2$ its projection onto $\mathbf{v}_1$, and what is left over is orthogonal to $\mathbf{v}_1$.



Theorem 8.6.17 (Unitary Diagonalisation of Hermitian Matrices)

Every $n \times n$ Hermitian matrix A is diagonalisable, and moreover the basis of eigenvectors $\mathbf{u}_1, \dots, \mathbf{u}_n$ can be chosen orthonormal. Equivalently, there is a *unitary* matrix U (whose columns are the $\mathbf{u}_i$) with

$$U^\dagger A U = D = \text{diag}(\lambda_1, \dots, \lambda_n),$$

with all λ_i real. If A is real and symmetric, U can be taken real, i.e. *orthogonal*: $Q^T A Q = D$ with $Q^T Q = I$.

Proof (Non-Examinable). We induct on n , the case $n = 1$ being trivial. Let λ_1 be an eigenvalue of A (one exists, by the fundamental theorem of algebra) with normalised eigenvector $\mathbf{u}_1$. Extend $\mathbf{u}_1$ to a basis of $\mathbb{C}^n$ and apply Gram-Schmidt to obtain an orthonormal basis $\mathbf{u}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$; let P be the unitary matrix with these as columns. Then the first column of $P^\dagger A P$ is $P^\dagger A \mathbf{u}_1 = \lambda_1 P^\dagger \mathbf{u}_1 = \lambda_1 \mathbf{e}_1$, and moreover $P^\dagger A P$ is still Hermitian, so its first row is also $\lambda_1 \mathbf{e}_1^T$. That is,

$$P^\dagger A P = \begin{pmatrix} \lambda_1 & \mathbf{0}^T \\ \mathbf{0} & C \end{pmatrix},$$

where C is an $(n-1) \times (n-1)$ Hermitian matrix. By induction there is a unitary $\tilde{Q}$ with $\tilde{Q}^\dagger C \tilde{Q}$ diagonal, and bordering $\tilde{Q}$ with a 1 in the top-left corner gives a unitary Q for which $(PQ)^\dagger A (PQ)$ is diagonal. Since products of unitary matrices are unitary, we are done; the columns of PQ are the required orthonormal eigenvectors. In the real symmetric case, all eigenvalues are real and every step can be carried out with real vectors and orthogonal matrices. $\square$

Example 8.6.18 (Orthogonal Diagonalisation with a Repeated Eigenvalue)

Let

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix},$$

which is real and symmetric. The characteristic polynomial is $\chi_A(t) = -(t+1)^2(t-2)$, so the eigenvalues are $-1, -1, 2$. For $\lambda = 2$ we find the eigenvector $(1, 1, 1)$, and for $\lambda = -1$ the eigenspace is the plane $v_1 + v_2 + v_3 = 0$, e.g. spanned by

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

These are not orthogonal to each other, so we apply Gram-Schmidt within the eigenspace: keep $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, -1, 0)$, and replace $\mathbf{w}_2$ by

$$\mathbf{w}'_2 = \mathbf{w}_2 - (\mathbf{u}_1 \cdot \mathbf{w}_2)\mathbf{u}_1 = \begin{pmatrix} 1/2 \\ 1/2 \\ -1 \end{pmatrix}, \quad \text{normalised to} \quad \mathbf{u}_2 = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

Together with $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(1, 1, 1)$, we have an orthonormal basis of eigenvectors (note $\mathbf{u}_1, \mathbf{u}_2 \perp \mathbf{u}_3$ automatically, being in eigenspaces of different eigenvalues). So with

$$Q = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ -1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \\ 0 & -2/\sqrt{6} & 1/\sqrt{3} \end{pmatrix},$$

an orthogonal matrix, we get $Q^T A Q = \text{diag}(-1, -1, 2)$.

Remark. Hermitian matrices are not the end of the story: a complex matrix can be unitarily diagonalised if and only if it is **normal**, meaning $AA^\dagger = A^\dagger A$. This class includes Hermitian, skew-Hermitian and unitary matrices (and their real cousins) all at once, but we won't pursue this further here.

§8.6.5 Quadratic Forms

One of the payoffs of orthogonal diagonalisation is a complete understanding of *quadratic forms* – expressions like $2x_1^2 - 4x_1x_2 + 5x_2^2$ in which every term has degree two. These arise throughout mathematics and physics: as conics and quadric surfaces, as second-order terms in Taylor expansions, as moments of inertia, and so on.

Definition 8.6.19 (Quadratic Form)

A **quadratic form** on $\mathbb{R}^n$ is a function $\mathcal{F} : \mathbb{R}^n \rightarrow \mathbb{R}$ of the form

$$\mathcal{F}(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} = A_{ij} x_i x_j,$$

where A is a real symmetric $n \times n$ matrix.

There is no loss of generality in taking A symmetric: an antisymmetric matrix B satisfies $\mathbf{x}^T B \mathbf{x} = 0$ (as $\mathbf{x}^T B \mathbf{x}$ is a scalar equal to its own transpose $-\mathbf{x}^T B \mathbf{x}$), so only the

symmetric part of a general matrix contributes to the form.⁶

Since A is real symmetric, Theorem 8.6.17 provides an orthogonal Q with $Q^T A Q = D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Substituting $\mathbf{x} = Q\mathbf{x}'$, i.e. $\mathbf{x}' = Q^T \mathbf{x}$,

$$\mathcal{F}(\mathbf{x}) = (Q\mathbf{x}')^T A (Q\mathbf{x}') = (\mathbf{x}')^T (Q^T A Q) \mathbf{x}' = \lambda_1 x_1'^2 + \lambda_2 x_2'^2 + \dots + \lambda_n x_n'^2.$$

In other words: *in the right orthonormal coordinates, every quadratic form is a weighted sum of squares*, with the eigenvalues of A as weights. The new coordinate axes – the directions of the orthonormal eigenvectors $\mathbf{u}_i$ – are called the **principal axes** of the form.

Example 8.6.20 (Diagonalising a Quadratic Form)

Consider $\mathcal{F}(\mathbf{x}) = 2x_1^2 - 4x_1x_2 + 5x_2^2$. We have $\mathcal{F}(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}$$

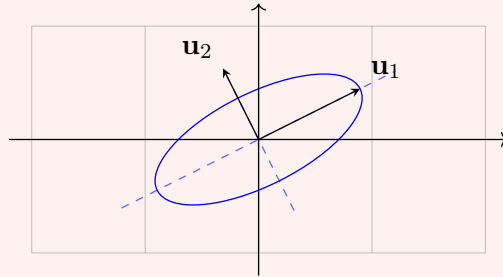
(note the off-diagonal entries are each *half* the cross-term coefficient). The eigenvalues of A are 1 and 6, with orthonormal eigenvectors

$$\mathbf{u}_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

So setting $x_1' = \frac{1}{\sqrt{5}}(2x_1 + x_2)$ and $x_2' = \frac{1}{\sqrt{5}}(-x_1 + 2x_2)$, the form becomes

$$\mathcal{F} = x_1'^2 + 6x_2'^2.$$

In particular the level set $\mathcal{F} = 1$ is an ellipse, with principal axes along $\mathbf{u}_1$ and $\mathbf{u}_2$ and semi-axes 1 and $1/\sqrt{6}$.



The signs of the eigenvalues determine the character of the form. If all $\lambda_i > 0$ the form is **positive definite** (like $|\mathbf{x}|^2$); if the signs are mixed, the level sets change character entirely. In $\mathbb{R}^2$, the level set $\mathcal{F} = k$ ($k > 0$) is an ellipse if $\lambda_1, \lambda_2 > 0$ and a hyperbola if $\lambda_1 \lambda_2 < 0$; in $\mathbb{R}^3$, positive eigenvalues give ellipsoids, while mixed signs give hyperboloids (of one or two sheets, according to the sign pattern and k).

Example 8.6.21 (A Hyperboloid)

Take the matrix A with entries $A_{ij} = 1 - \delta_{ij}$ from the previous subsection, so

$$\mathcal{F}(\mathbf{x}) = 2x_1x_2 + 2x_2x_3 + 2x_3x_1 = -x_1'^2 - x_2'^2 + 2x_3'^2$$

⁶Over $\mathbb{C}$, the analogous object is a *Hermitian form* $F(\mathbf{z}) = \mathbf{z}^\dagger A \mathbf{z}$ with A Hermitian; the computation $\overline{F(\mathbf{z})} = (\mathbf{z}^\dagger A \mathbf{z})^\dagger = \mathbf{z}^\dagger A \mathbf{z}$ shows that Hermitian forms take real values.

in principal axes coordinates. The surface $\mathcal{F} = 1$, i.e. $2x_3'^2 = 1 + x_1'^2 + x_2'^2$, is a hyperboloid of two sheets (rotationally symmetric about the x_3' axis), while $\mathcal{F} = -1$ gives $x_1'^2 + x_2'^2 = 1 + 2x_3'^2$, a hyperboloid of one sheet.

§8.6.6 Quadrics and Conics

Allowing linear terms as well as quadratic ones, we arrive at the general notion of a quadric.

Definition 8.6.22 (Quadric)

A **quadric** in $\mathbb{R}^n$ is the solution set of

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + c = 0,$$

where $A \neq 0$ is a real symmetric $n \times n$ matrix, $\mathbf{b} \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

If A is invertible, we can absorb the linear term by completing the square: setting $\mathbf{y} = \mathbf{x} + \frac{1}{2}A^{-1}\mathbf{b}$ (a translation of the origin), a short computation gives

$$\mathbf{y}^T A \mathbf{y} = \mathbf{x}^T A \mathbf{x} + \mathbf{b}^T \mathbf{x} + \frac{1}{4}\mathbf{b}^T A^{-1} \mathbf{b},$$

so $Q(\mathbf{x}) = 0$ becomes $\mathbf{y}^T A \mathbf{y} = k$ where $k = \frac{1}{4}\mathbf{b}^T A^{-1} \mathbf{b} - c$. Diagonalising as before, every quadric with $\det A \neq 0$ is – after a translation and a rotation of coordinates, which change nothing geometrically – a level set of a sum of squares:

$$\lambda_1 x_1^2 + \lambda_2 x_2^2 + \cdots + \lambda_n x_n^2 = k.$$

Conics

In $\mathbb{R}^2$ quadrics are curves, called **conics**, and we can classify them completely. Suppose first $\det A \neq 0$, so as above the conic is $\lambda_1 x_1^2 + \lambda_2 x_2^2 = k$ in suitable coordinates.

- If $\lambda_1, \lambda_2 > 0$: an *ellipse* for $k > 0$, a single point for $k = 0$, and empty for $k < 0$.
- If $\lambda_1 > 0 > \lambda_2$: a *hyperbola* for $k \neq 0$ (with the two orientations for the two signs of k), and for $k = 0$ a *pair of lines* through the origin, since $\lambda_1 x_1^2 - |\lambda_2| x_2^2$ factorises.

If $\det A = 0$, then exactly one eigenvalue vanishes (as $A \neq 0$), say $\lambda_2 = 0$, and the translation trick above is unavailable. In principal axis coordinates the equation reads

$$\lambda_1 x_1'^2 + b_1' x_1' + b_2' x_2' + c = 0.$$

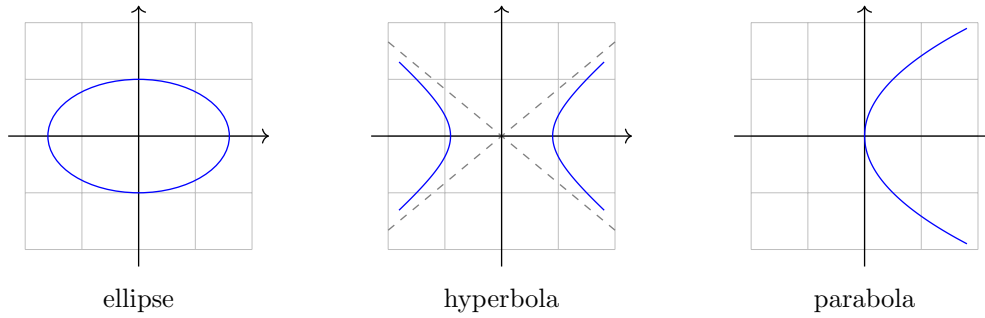
Completing the square in x_1' removes the b_1' term. If $b_2' \neq 0$, absorbing the constant into x_2' then leaves

$$\lambda_1 x_1''^2 + b_2' x_2'' = 0,$$

a *parabola*. If $b_2' = 0$, we instead get $\lambda_1 x_1''^2 + c' = 0$, which is a pair of parallel lines, a single line, or empty, according to the sign of c'/λ_1 .

So, up to translations and rotations, a conic is an ellipse, a hyperbola, a parabola, or a degenerate case (points and lines). The three non-degenerate conics have standard forms

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (\text{ellipse}), \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad (\text{hyperbola}), \quad y^2 = 4ax \quad (\text{parabola}).$$

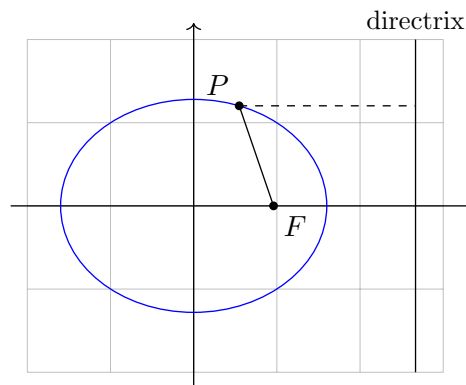


The Focus-Directrix Property

The name ‘conic’ comes from the fact that these curves are the sections of a cone by planes – and there is also a beautiful unified description in terms of a focus and directrix, which is the form in which conics appear in Dynamics and Relativity when studying planetary orbits.

Fix a constant $e > 0$ (the **eccentricity**), a point F (the **focus**), and a line (the **directrix**). Consider the locus of points P whose distance from F is e times the distance from the directrix. Taking $F = (ae, 0)$ and the directrix $x = a/e$ (with $a > 0$), the condition is

$$\sqrt{(x - ae)^2 + y^2} = e \left| \frac{a}{e} - x \right|.$$



Squaring and simplifying,

$$\frac{x^2}{a^2} + \frac{y^2}{a^2(1 - e^2)} = 1.$$

For $e < 1$ this is an ellipse with semi-minor axis $b = a\sqrt{1 - e^2}$ (with $e \rightarrow 0$ approaching a circle); for $e > 1$ the second denominator is negative and we get the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{a^2(e^2 - 1)} = 1$; and a similar computation with $e = 1$, focus $(a, 0)$ and directrix $x = -a$ gives the parabola $y^2 = 4ax$. So the eccentricity interpolates between all three non-degenerate conics.

In polar coordinates centred on the focus, all three cases combine into the single equation

$$r = \frac{\ell}{1 + e \cos \theta},$$

where $\ell = a|1 - e^2|$ (for the parabola, $\ell = 2a$). Note that r stays bounded when $e < 1$, while for $e \geq 1$ the denominator can vanish and the curve escapes to infinity – exactly the distinction between bound (elliptic) and unbound (parabolic, hyperbolic) orbits.

§8.7 Changing Basis, Canonical Forms and Symmetries

In this final section, we tie together several loose threads. We have repeatedly used expressions of the form $P^{-1}AP$, promising that they mean ‘the same map, written with respect to a different basis’ – it is time to make this precise. Once we can change basis freely, the natural question becomes: given a linear map, how simple can we make its matrix look? The answers (diagonal matrices where possible, and *Jordan blocks* where not) are some of the most useful results in linear algebra. We will finish with the Cayley-Hamilton theorem, and a look at matrix groups of symmetries.

§8.7.1 Change of Basis

Let’s first set up the matrix of a linear map with respect to *arbitrary* bases, generalising what we did in Section 4 for standard bases.

Let $T : V \rightarrow W$ be a linear map between real or complex vector spaces with $\dim V = n$ and $\dim W = m$, and choose bases $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of V and $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ of W . The **matrix of T with respect to these bases** is the $m \times n$ matrix A defined by

$$T(\mathbf{e}_i) = \sum_a \mathbf{f}_a A_{ai},$$

i.e. column i of A holds the components of $T(\mathbf{e}_i)$ with respect to the basis $\{\mathbf{f}_a\}$. This is set up exactly so that the abstract statement $\mathbf{y} = T(\mathbf{x})$ becomes the matrix statement $y_a = A_{ai}x_i$ on components: indeed

$$T\left(\sum_i x_i \mathbf{e}_i\right) = \sum_i x_i T(\mathbf{e}_i) = \sum_a \left(\underbrace{\sum_i A_{ai}x_i}_{y_a}\right) \mathbf{f}_a.$$

Now suppose we pick *new* bases $\{\mathbf{e}'_i\}$ of V and $\{\mathbf{f}'_a\}$ of W . The relationship between old and new is captured by **change of basis matrices** P ($n \times n$) and Q ($m \times m$), defined by

$$\mathbf{e}'_i = \sum_j \mathbf{e}_j P_{ji}, \quad \mathbf{f}'_a = \sum_b \mathbf{f}_b Q_{ba},$$

so the columns of P are the new basis vectors written in terms of the old (and similarly for Q). Both are invertible, since they map a basis to a basis.

Theorem 8.7.1 (Change of Basis Formula)

Let A be the matrix of T with respect to $\{\mathbf{e}_i\}$, $\{\mathbf{f}_a\}$, and A' its matrix with respect to $\{\mathbf{e}'_i\}$, $\{\mathbf{f}'_a\}$. Then

$$A' = Q^{-1}AP.$$

Proof. We compute $T(\mathbf{e}'_i)$ in two ways. Directly from the definitions,

$$T(\mathbf{e}'_i) = T\left(\sum_j \mathbf{e}_j P_{ji}\right) = \sum_j T(\mathbf{e}_j) P_{ji} = \sum_{j,a} \mathbf{f}_a A_{aj} P_{ji},$$

while using the matrix with respect to the new bases,

$$T(\mathbf{e}'_i) = \sum_b \mathbf{f}'_b A'_{bi} = \sum_{a,b} \mathbf{f}_a Q_{ab} A'_{bi}.$$

Comparing coefficients of the basis vectors $\mathbf{f}_a$ (whose coefficients are unique),

$$(AP)_{ai} = (QA')_{ai},$$

so $AP = QA'$ and $A' = Q^{-1}AP$. $\square$

It is also instructive to see how *components of vectors* transform. If $\mathbf{x} = \sum_i x_i \mathbf{e}_i = \sum_j x'_j \mathbf{e}'_j$, then substituting the definition of $\mathbf{e}'_j$ and comparing coefficients gives

$$x_i = \sum_j P_{ij} x'_j, \quad \text{i.e.} \quad X = PX' \text{ and } X' = P^{-1}X,$$

writing X, X' for the column vectors of components. (Note the slight inversion: the *new* components are obtained using P^{-1} .) Then from $Y = AX$ we get $QY' = APX'$, so $Y' = (Q^{-1}AP)X'$ – the change of basis formula again.

Example 8.7.2 (Changing Bases)

Define $T : V \rightarrow W$ with $\dim V = 2$, $\dim W = 3$ by

$$T(\mathbf{e}_1) = \mathbf{f}_1 + 2\mathbf{f}_2 - \mathbf{f}_3, \quad T(\mathbf{e}_2) = -\mathbf{f}_1 + 2\mathbf{f}_2 + \mathbf{f}_3, \quad \text{so} \quad A = \begin{pmatrix} 1 & -1 \\ 2 & 2 \\ -1 & 1 \end{pmatrix}.$$

Now take new bases $\mathbf{e}'_1 = \mathbf{e}_1 - \mathbf{e}_2$, $\mathbf{e}'_2 = \mathbf{e}_1 + \mathbf{e}_2$ and $\mathbf{f}'_1 = \mathbf{f}_1 - \mathbf{f}_3$, $\mathbf{f}'_2 = \mathbf{f}_2$, $\mathbf{f}'_3 = \mathbf{f}_1 + \mathbf{f}_3$, so

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}, \quad Q^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}.$$

The change of basis formula gives

$$A' = Q^{-1}AP = \begin{pmatrix} 2 & 0 \\ 0 & 4 \\ 0 & 0 \end{pmatrix},$$

and indeed one can check directly that $T(\mathbf{e}'_1) = 2\mathbf{f}_1 - 2\mathbf{f}_3 = 2\mathbf{f}'_1$ and $T(\mathbf{e}'_2) = 4\mathbf{f}_2 = 4\mathbf{f}'_2$. A good choice of basis has made the map look about as simple as possible.

§8.7.2 Similar Matrices

The most important case of the change of basis formula is $V = W$ with $\{\mathbf{e}_i\} = \{\mathbf{f}_i\}$ and $\{\mathbf{e}'_i\} = \{\mathbf{f}'_i\}$, so that $Q = P$ and

$$A' = P^{-1}AP.$$

Definition 8.7.3 (Similar Matrices)

Two $n \times n$ matrices A and B are **similar** if $B = P^{-1}AP$ for some invertible matrix

P .

Similarity is an equivalence relation, and by the discussion above, two matrices are similar precisely when they represent *the same linear map with respect to different bases*. Anything intrinsic to the underlying map must therefore be shared by all matrices similar to it. Here are the key examples, which we have already met.

Proposition 8.7.4 (Invariants Under Similarity)

If A and B are similar, then

- (i) $\det B = \det A$,
- (ii) $\operatorname{tr} B = \operatorname{tr} A$,
- (iii) $\chi_B(t) = \chi_A(t)$, so A and B have the same eigenvalues with the same algebraic multiplicities.

Proof. Let $B = P^{-1}AP$.

- (i) $\det B = \det(P^{-1}) \det A \det P = \det A$, using multiplicativity.
- (ii) $\operatorname{tr} B = \operatorname{tr}(P^{-1}(AP)) = \operatorname{tr}((AP)P^{-1}) = \operatorname{tr} A$, using the cyclic property.
- (iii) Since $tI = P^{-1}(tI)P$, we have

$$\chi_B(t) = \det(P^{-1}AP - tI) = \det(P^{-1}(A - tI)P) = \det(A - tI) = \chi_A(t). \quad \square$$

So the determinant, trace and characteristic polynomial are genuinely properties of the linear map, not of the particular matrix chosen to represent it. (The converse fails, incidentally: matrices with the same characteristic polynomial need not be similar – compare I and the shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.)

As promised in Section 6, we can now prove the relationship between the two multiplicities.

Proposition 8.7.5 (Geometric vs Algebraic Multiplicity)

For any eigenvalue λ of an $n \times n$ matrix A , we have $1 \leq m_\lambda \leq M_\lambda$.

Proof. Let $m = m_\lambda$, and choose a basis $\mathbf{v}_1, \dots, \mathbf{v}_m$ of the eigenspace E_λ , extended to a basis $\mathbf{v}_1, \dots, \mathbf{v}_n$ of $\mathbb{C}^n$. Let P be the (invertible) matrix with these as columns. Since $A\mathbf{v}_i = \lambda\mathbf{v}_i$ for $i \leq m$, the first m columns of $\tilde{A} = P^{-1}AP$ are $\lambda\mathbf{e}_1, \dots, \lambda\mathbf{e}_m$, so $\tilde{A}$ has the block form

$$\tilde{A} = \begin{pmatrix} \lambda I_m & B \\ 0 & C \end{pmatrix}$$

for some blocks B and C . Then, expanding the determinant repeatedly along the first columns,

$$\chi_A(t) = \chi_{\tilde{A}}(t) = \det \begin{pmatrix} (\lambda - t)I_m & B \\ 0 & C - tI \end{pmatrix} = (\lambda - t)^m \chi_C(t),$$

so $(t - \lambda)^m$ divides $\chi_A(t)$, i.e. $M_\lambda \geq m$. $\square$

§8.7.3 Canonical Forms

We saw that a diagonalisable matrix is exactly one that is similar to a diagonal matrix. What is the simplest form we can reach when a matrix is *not* diagonalisable? For 2×2 matrices we can answer this completely.

Theorem 8.7.6 (Canonical Forms for 2×2 Matrices)

Any 2×2 complex matrix A is similar to exactly one of the following:

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \ (\lambda_1 \neq \lambda_2), \quad \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}, \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

Proof. Over $\mathbb{C}$, the characteristic polynomial χ_A has two roots, counted with multiplicity. We consider the possible cases.

- (i) If the roots $\lambda_1 \neq \lambda_2$ are distinct, then the corresponding eigenvectors are linearly independent, and taking P with the eigenvectors as columns diagonalises A , giving the first form.
- (ii) If there is a repeated root λ with $m_\lambda = 2$, then $E_\lambda = \mathbb{C}^2$, and any basis diagonalises: A is similar to λI . (In fact then $A = P(\lambda I)P^{-1} = \lambda I$ already.)
- (iii) If there is a repeated root λ with $m_\lambda = 1$, then A is not diagonalisable, and we claim it is similar to the third matrix. Since $A \neq \lambda I$, we can pick a vector $\mathbf{w}$ with

$$\mathbf{u} = (A - \lambda I)\mathbf{w} \neq \mathbf{0}.$$

Now, here is the trick: since $\chi_A(t) = (t - \lambda)^2$, the Cayley-Hamilton theorem (proven independently below – or verified directly in the 2×2 case) gives $(A - \lambda I)^2 = \chi_A(A) = 0$, so

$$(A - \lambda I)\mathbf{u} = (A - \lambda I)^2\mathbf{w} = \mathbf{0},$$

i.e. $\mathbf{u}$ is an eigenvector. Then $\{\mathbf{u}, \mathbf{w}\}$ is linearly independent (as $\mathbf{w}$ is not an eigenvector but $\mathbf{u}$ is), and on this basis

$$A\mathbf{u} = \lambda\mathbf{u}, \quad A\mathbf{w} = \mathbf{u} + \lambda\mathbf{w},$$

which is precisely the third matrix, with respect to the basis $\{\mathbf{u}, \mathbf{w}\}$.

Finally, no two of the three forms are similar to each other (compare characteristic polynomials, and diagonalisability), so the case is uniquely determined by A . $\square$

Example 8.7.7 (Finding a Canonical Form)

Let

$$A = \begin{pmatrix} 1 & 4 \\ -1 & 5 \end{pmatrix}, \quad \text{so} \quad \chi_A(t) = t^2 - 6t + 9 = (t - 3)^2.$$

The eigenvalue 3 is repeated, and $A \neq 3I$, so we are in the third case. Take $\mathbf{w} = (1, 0)^T$; then

$$\mathbf{u} = (A - 3I)\mathbf{w} = \begin{pmatrix} -2 & 4 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix},$$

which is indeed an eigenvector of A . With

$$P = \begin{pmatrix} -2 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{we get} \quad P^{-1}AP = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}.$$

What happens for larger matrices? The building blocks are the $k \times k$ **Jordan blocks**

$$J_k(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} = \lambda I + N,$$

where N has 1s just above the diagonal and 0s elsewhere. The matrix N acts on the standard basis by $\mathbf{e}_k \mapsto \mathbf{e}_{k-1} \mapsto \cdots \mapsto \mathbf{e}_1 \mapsto \mathbf{0}$ (so $N^k = 0$), and one checks that $J_k(\lambda)$ has characteristic polynomial $(\lambda - t)^k$ with $M_\lambda = k$ but $m_\lambda = 1$: it is the ‘most non-diagonalisable’ matrix with a single eigenvalue.

Theorem 8.7.8 (Jordan Normal Form)

Any $n \times n$ complex matrix is similar to a block diagonal matrix

$$\begin{pmatrix} J_{n_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{n_r}(\lambda_r) \end{pmatrix},$$

where $n_1 + \cdots + n_r = n$, the λ_i are eigenvalues (possibly with repeats among different blocks), and the collection of blocks is unique up to reordering. A matrix is diagonalisable if and only if all its blocks are 1×1 .

The proof is well beyond this course (it belongs to Linear Algebra, and Groups, Rings and Modules, in Part IB), but the statement is worth knowing: it completely classifies complex matrices up to similarity, and the 2×2 theorem above is exactly its two dimensional case.

§8.7.4 The Cayley-Hamilton Theorem

If $p(t) = c_0 + c_1t + \cdots + c_k t^k$ is a polynomial and A a square matrix, we can form the matrix

$$p(A) = c_0I + c_1A + \cdots + c_k A^k.$$

Matrix polynomials – and indeed power series, such as the matrix exponential $\exp A = \sum_{r=0}^{\infty} \frac{1}{r!} A^r$ – behave pleasantly with respect to similarity and diagonalisation. Since $(P^{-1}AP)^r = P^{-1}A^rP$, we have

$$p(P^{-1}AP) = P^{-1}p(A)P,$$

and if $D = \text{diag}(\lambda_1, \dots, \lambda_n)$ then $p(D) = \text{diag}(p(\lambda_1), \dots, p(\lambda_n))$.

Now for one of the most striking results of the course: what happens if we substitute a matrix into its *own* characteristic polynomial?

Theorem 8.7.9 (Cayley-Hamilton Theorem)

Every $n \times n$ complex matrix satisfies its own characteristic equation:

$$\chi_A(A) = c_0 I + c_1 A + \cdots + c_n A^n = 0.$$

Before the proof, a warning about a tempting non-proof: it is *not* valid to argue ' $\chi_A(A) = \det(A - AI) = \det(0) = 0$ '. The expression $\chi_A(A)$ means substituting A into the polynomial $\chi_A(t)$ after expanding it out, which is not the same as plugging a matrix into the determinant.

Proof (for diagonalisable matrices). Suppose $P^{-1}AP = D = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then, using the observations above,

$$\chi_A(D) = \text{diag}(\chi_A(\lambda_1), \dots, \chi_A(\lambda_n)) = 0,$$

since the λ_i are eigenvalues of A , i.e. roots of χ_A . Hence

$$\chi_A(A) = \chi_A(PDP^{-1}) = P\chi_A(D)P^{-1} = 0. \quad \square$$

Proof (general, non-examinable). Let $M = A - tI$, so that $\det M = \chi_A(t) = \sum_{r=0}^n c_r t^r$. Each entry of the adjugate $\text{adj } M$ is (up to sign) an $(n-1) \times (n-1)$ minor of M , hence a polynomial in t of degree at most $n-1$, so we can write

$$\text{adj } M = \sum_{r=0}^{n-1} B_r t^r$$

for some matrices $B_0, \dots, B_{n-1}$ independent of t . The fundamental adjugate identity $(\text{adj } M)M = (\det M)I$ then reads

$$\left(\sum_{r=0}^{n-1} B_r t^r \right) (A - tI) = \sum_{r=0}^n c_r t^r I.$$

Both sides are polynomials in t with matrix coefficients, so we may compare coefficients of each power of t :

$$c_0 I = B_0 A, \quad c_1 I = B_1 A - B_0, \quad \dots, \quad c_{n-1} I = B_{n-1} A - B_{n-2}, \quad c_n I = -B_{n-1}.$$

Now multiply the equation for $c_r I$ on the right by A^r and sum over r : the right hand sides telescope,

$$\chi_A(A) = \sum_{r=0}^n c_r A^r = B_0 A + (B_1 A^2 - B_0 A) + \cdots + (B_{n-1} A^n - B_{n-2} A^{n-1}) - B_{n-1} A^n = 0. \quad \square$$

A nice consequence: Cayley-Hamilton gives a formula for the inverse in terms of *positive* powers of A . Rearranging $\chi_A(A) = 0$,

$$-c_0 I = A(c_1 I + c_2 A + \cdots + c_n A^{n-1}),$$

and since $c_0 = \chi_A(0) = \det A$, if A is invertible then

$$A^{-1} = \frac{-1}{\det A} (c_1 I + c_2 A + \cdots + c_n A^{n-1}).$$

Example 8.7.10 (Inverting with Cayley-Hamilton)

For $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, we have $\chi_A(t) = t^2 - 5t - 2$, so $A^2 - 5A - 2I = 0$, giving

$$A^{-1} = \frac{1}{2}(A - 5I) = \frac{1}{2} \begin{pmatrix} -4 & 2 \\ 3 & -1 \end{pmatrix} = \frac{1}{-2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix},$$

agreeing with the 2×2 inverse formula (here $\det A = -2$).

§8.7.5 Symmetries and Transformation Groups

We finish the course by returning to orthogonal matrices, now viewed as *groups of symmetries* of space. Recall from Section 4 that the orthogonal matrices are exactly the length-preserving ones.

Proposition 8.7.11 (The Orthogonal Group)

The set of $n \times n$ orthogonal matrices forms a group under matrix multiplication, the **orthogonal group** $O(n)$. The subset with determinant 1 forms a subgroup, the **special orthogonal group** $SO(n)$.

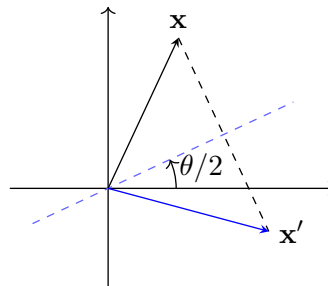
Proof. For closure, if $R^T R = S^T S = I$ then $(RS)^T(RS) = S^T(R^T R)S = S^T S = I$. The identity is orthogonal; the inverse of an orthogonal matrix is R^T , which is orthogonal since $(R^T)^T R^T = R R^T = I$; and matrix multiplication is associative. For $SO(n)$, closure follows from multiplicativity of the determinant. $\square$

Since $\det R = \pm 1$ for orthogonal R , the group $O(n)$ splits into two halves: $SO(n)$, the *rotations*, which preserve orientation (and hence signed volumes), and the remainder, the *reflections*, with determinant -1 . Any element of the second half can be written as (a fixed reflection) $\times$ (a rotation).

In two dimensions we can describe everything explicitly: requiring the columns to be orthonormal forces any element of $O(2)$ to have the form

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix},$$

the first being the rotations (determinant 1) and the second the reflections (determinant -1 , being reflections in the line at angle $\theta/2$ to the x -axis).



There is one conceptual point worth making about transformations $x'_i = R_{ij}x_j$: they admit two readings.

- *Active*: R moves the vectors of space, and x'_i are the components of the *new* vector in the old basis.
- *Passive*: nothing moves; x'_i are the components of the *same* vector with respect to a new (rotated) orthonormal basis. This is the change-of-basis reading, with $P = R^{-1} = R^T$.

Both readings are useful, and translating between them carelessly is a well-known source of sign errors.

Minkowski Space and Lorentz Transformations

The definition of ‘orthogonal’ depended on the dot product, which measures length. If we change the way we measure length, we change which matrices count as symmetries – and there is one non-standard choice of enormous physical importance.

Definition 8.7.12 (Minkowski Inner Product)

The **Minkowski inner product** on $\mathbb{R}^2$ is

$$(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T J \mathbf{y}, \quad \text{where} \quad J = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

i.e. $(\mathbf{x}, \mathbf{y}) = x_0 y_0 - x_1 y_1$, indexing coordinates from 0. With this inner product, $\mathbb{R}^2$ is called **Minkowski space**: the coordinate x_0 plays the role of time and x_1 of space.

This is symmetric and bilinear, but *not* positive definite: $(\mathbf{x}, \mathbf{x}) = x_0^2 - x_1^2$ can take either sign (it is a quadratic form with eigenvalues ± 1). The analogue of the circle $|\mathbf{x}|^2 = k$ is the hyperbola $x_0^2 - x_1^2 = k$.

Which matrices preserve this inner product? We need $(M\mathbf{x}, M\mathbf{y}) = (\mathbf{x}, \mathbf{y})$ for all $\mathbf{x}, \mathbf{y}$, i.e. $\mathbf{x}^T M^T J M \mathbf{y} = \mathbf{x}^T J \mathbf{y}$ for all vectors, which forces

$$M^T J M = J,$$

in analogy with $R^T I R = I$ for orthogonal matrices. Taking determinants gives $\det M = \pm 1$ as before, and looking at the $(0, 0)$ entry of the equation gives $M_{00}^2 = 1 + M_{10}^2 \geq 1$.

Proposition 8.7.13 (Lorentz Transformations)

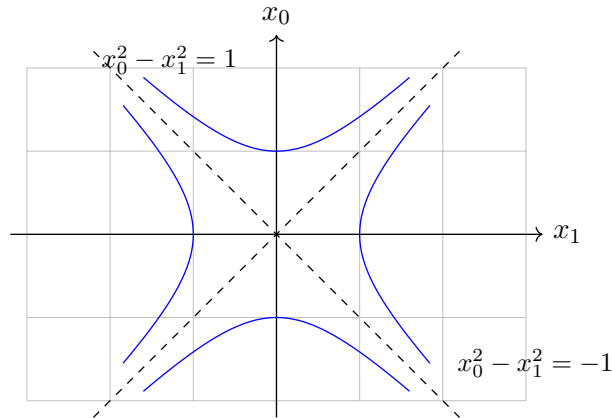
The matrices with $M^T J M = J$, $\det M = 1$ and $M_{00} \geq 1$ are exactly those of the form

$$M(\theta) = \begin{pmatrix} \cosh \theta & \sinh \theta \\ \sinh \theta & \cosh \theta \end{pmatrix}, \quad \theta \in \mathbb{R},$$

and they form a group – the **Lorentz group** – with $M(\theta_1)M(\theta_2) = M(\theta_1 + \theta_2)$.

Proof Sketch. The condition $M^T J M = J$ says the columns $M\mathbf{e}_0, M\mathbf{e}_1$ are ‘orthonormal’ with respect to the Minkowski product. From $(M\mathbf{e}_0, M\mathbf{e}_0) = M_{00}^2 - M_{10}^2 = 1$ with $M_{00} \geq 1$, we can write $(M_{00}, M_{10}) = (\cosh \theta, \sinh \theta)$ for a unique $\theta \in \mathbb{R}$. Then $(M\mathbf{e}_0, M\mathbf{e}_1) = 0$ and $(M\mathbf{e}_1, M\mathbf{e}_1) = -1$ force the second column to be $\pm(\sinh \theta, \cosh \theta)$, and $\det M = 1$ fixes the sign $+$. The group law follows from the addition formulae for $\cosh$ and $\sinh$, which mirror those for $\cos$ and $\sin$. $\square$

These **hyperbolic rotations** preserve each branch of the hyperbolas $x_0^2 - x_1^2 = k$, just as ordinary rotations preserve circles.



To connect with physics, substitute $v = \tanh \theta$ (so $-1 < v < 1$) and write $\gamma = (1 - v^2)^{-1/2} = \cosh \theta$. Renaming $(x_0, x_1) = (t, x)$,

$$M = \gamma \begin{pmatrix} 1 & v \\ v & 1 \end{pmatrix}, \quad \text{i.e.} \quad t' = \gamma(t + vx), \quad x' = \gamma(x + vt).$$

This is the **Lorentz boost** of special relativity, relating the time and space coordinates of observers in relative motion with velocity v , in units where the speed of light is 1 – the factor γ is responsible for time dilation and length contraction. The group law becomes the relativistic velocity addition formula

$$v_3 = \tanh(\theta_1 + \theta_2) = \frac{v_1 + v_2}{1 + v_1 v_2},$$

which in particular never exceeds the speed of light. The Lorentz transformations will be picked up in earnest in Dynamics and Relativity; for us, they are a fitting final example of how the framework of this course – vector spaces, inner products, and the matrix groups preserving them – extends beyond Euclidean geometry.

II

Part IB

About Part IB

If Part IA is about technique, Part IB is about abstraction. This is the year in which the objects you have been computing with are replaced by the axioms they satisfy, and it is by some distance the largest single jump in the degree. It is also the year with the most courses in it – fifteen chapters here, out of a longer list on offer – because the second year is where a mathematician’s tastes are supposed to start declaring themselves.

The clearest example of the shift is Linear Algebra. Vectors and Matrices spent a year in $\mathbb{R}^3$ and $\mathbb{C}^n$; Linear Algebra deletes the coordinates and starts from a vector space over an arbitrary field, and suddenly the dual space, the trace, the determinant and the Jordan normal form all become statements about linear maps rather than statements about arrays of numbers. The same thing happens to analysis. Analysis and Topology takes the ϵ - N arguments of Analysis I and asks what they actually needed; the answer is a metric, and then not even that, and by the end of the chapter you are proving compactness results in a topological space with no numbers in sight. Groups, Rings and Modules performs the same trick on Groups, generalising the integers and the polynomials into a single theory of factorisation, and then generalising vector spaces into modules so thoroughly that the classification of finitely generated modules over a Euclidean domain hands you back both the structure of finite abelian groups and the Jordan normal form as corollaries.

That triple – Linear Algebra, Analysis and Topology, Groups, Rings and Modules – is the spine of the year, and almost every Part II course leans on at least one of them.

Alongside the abstraction there is theory-building of a different kind. Complex Analysis is the outstanding course of the second year, and possibly of the degree: a short list of hypotheses about a function of a complex variable turns out to force an extraordinary amount of structure, and Cauchy’s theorem propagates through the chapter until residues, the argument principle and Rouché’s theorem all fall out of it. Complex Methods covers overlapping ground from the other direction, caring about contour integrals you can actually evaluate and conformal maps you can actually draw; reading the two chapters side by side is instructive precisely because they disagree about what a proof is for. Geometry, meanwhile, does for surfaces what Analysis and Topology does for spaces, working up through the two fundamental forms to the Theorema Egregium and the Gauss–Bonnet theorem, which is the first result in the book where a local computation and a global topological invariant are forced to agree.

The applied courses inherit Part IA’s machinery and finally put it to work. Vector Calculus’s integral theorems become Maxwell’s equations in Electromagnetism and the transport theorems in Fluid Dynamics. Methods is the workhorse of the year, and worth singling out: Fourier series, Sturm–Liouville theory, distributions and Green’s functions form a toolkit that reappears in nearly every applied course afterwards. Variational Principles reframes mechanics in terms of stationary actions, which is the language Principles of Quantum Mechanics and Statistical Physics will later use, and Quantum Mechanics itself is in large part an exercise in solving the equations that Methods taught you to solve.

Four courses here sit slightly apart. Statistics and Markov Chains take Part IA Probability seriously enough to build real theory on it – estimators, confidence intervals and hypothesis tests in one case, recurrence, invariant distributions and convergence to equilibrium in the other. Optimisation, with its convexity, duality and simplex method, and Numerical Analysis, with its interpolation, quadrature and stability theory, are the two

most computational chapters in the book. None of them is a prerequisite for a great deal, but Optimisation and Markov Chains both feed into Stochastic Financial Models, and Numerical Analysis shares more than you would expect with Topics in Analysis.

The character of the year, in short, is that objects stop being defined by what they are and start being defined by what they do. It is uncomfortable for a while, and then it becomes the only way you want to think.

9 Analysis and Topology

This course is a second course in Analysis and a first course in Topology. We will study both concrete results over $\mathbb{R}$ and $\mathbb{C}$ concerning uniform convergence and continuity, and we will also move to more abstract settings to discuss metric and topological spaces, completeness, connectedness and compactness.

This article constitutes my notes for the ‘Analysis and Topology’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§9.1 Uniform Convergence and Uniform Continuity

§9.1.1 Uniform Convergence

Recall the notion of convergence for a sequence in $\mathbb{R}$ or $\mathbb{C}$:

Definition 9.1.1 (Convergence of a Sequence)

A sequence $a_1, a_2, \dots \in \mathbb{R}$ is said to **converge** to the limit a if given any $\epsilon > 0$, we can find an integer N such that $|a_n - a| < \epsilon$ for all $n \geq N$. We write $a_n \rightarrow a$ as $n \rightarrow \infty$.

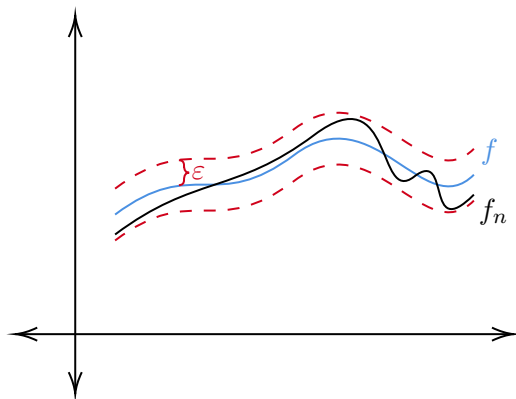
That is, given any ϵ , there is some point in the sequence after which the terms of the sequence are ϵ close to a .

Our aim is going to define a similar notion to make sense of $f_n \rightarrow f$, where f_n is a sequence of functions.

Definition 9.1.2 (Uniform Convergence)

A sequence of functions $f_1, f_2, \dots$ with $f_i : S \rightarrow \mathbb{R}$ is said to **converge uniformly** on S to a function $f : S \rightarrow \mathbb{R}$ if given any $\epsilon > 0$, we can find an integer N such that $|f_n(x) - f(x)| < \epsilon$ for all $n \geq N$ and any $x \in S$.

Remark. In the above definition, our N can depend only on ϵ , and must be independent of the particular choice of $x \in S$. This is why we call this *uniform* convergence – because the property has to hold uniformly across the domain.



Equivalently, we could say that for all $\varepsilon > 0$ there's some $N \in \mathbb{N}$ such that for all $n \geq N$ we have $\sup_{x \in S} |f_n(x) - f(x)| < \varepsilon$.

The above definition implies that if we fix some value of x that $f_1(x), f_2(x), \dots$ converges to $f(x)$. This implies that the function f is unique, due to the uniqueness of limits in $\mathbb{R}$ and $\mathbb{C}$. We sometimes call f the **uniform limit**.

Definition 9.1.3 (Pointwise Convergence)

We say that f_n **converges pointwise** on S to f if $f_n(x) \rightarrow f(x)$ for every $x \in S$.

Remark. In this definition our ' N ' can depend on ε and x ! This makes it a much weaker notion than uniform convergence, and clearly uniform convergence implies pointwise convergence.

Example 9.1.4 (Checking Uniform Convergence)

Consider the sequence of functions $f_n(x) = x^2 \cdot e^{-nx}$ for $n \in \mathbb{N}$ and $x \in \mathbb{R}^+$. We want to know if this sequence of functions converges uniformly on this domain.

Since pointwise convergence is implied by uniform convergence, we can first check the pointwise limit exists and use that to specify f in our definition of uniform convergence.

Fix $x \geq 0$. Then $x^2 e^{-nx} \rightarrow 0$ as $n \rightarrow \infty$. So f_n tends to 0 (the zero function) pointwise on $\mathbb{R}^+$. We can now check if f_n converges uniformly to the zero function.

We can attempt to compute the quantity

$$\sup_{x \geq 0} |f_n(x) - 0| = \sup_{x \geq 0} f_n(x).$$

One approach would be to differentiate it, which would need some care. A better way is to find an upper bound on $|f_n(x) - f(x)|$ which does not depend on x . In this case we can bound

$$0 \leq x^2 e^{-nx} = \frac{x^2}{1 + nx + n^2 x^2 / 2 + \dots} \leq \frac{2}{n^2},$$

for all $x \geq 0$. Thus $\sup_{x \geq 0} f_n(x) \leq 2/n^2 \rightarrow 0$ as $n \rightarrow \infty$. So indeed $f_n \rightarrow 0$ uniformly on $\mathbb{R}^+$.

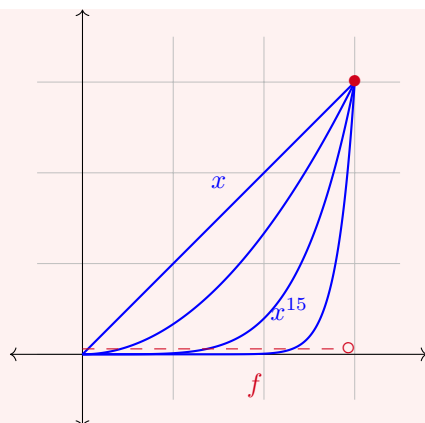
Example 9.1.5 (Showing Uniform Convergence Doesn't Hold)

Consider the sequence of functions $f_n(x) = x^n$ for $n \in \mathbb{N}$, over the domain $[0, 1]$.

We can compute the limit as

$$x^n \rightarrow f(x) = \begin{cases} 0 & \text{if } 0 \leq x < 1, \\ 1 & \text{if } x = 1. \end{cases}$$

This implies that $\sup_{x \in [0, 1]} |f_n(x) - f(x)| = 1$. So this doesn't tend to zero, and thus the sequence of functions f_n converges pointwise but not uniformly.



The picture above shows what is going wrong – as n grows, the graph of x^n sags towards zero, but it always has to climb back up to 1 at the right endpoint. The convergence near $x = 1$ is genuinely slower, and no single N works across the whole interval.

Alternatively, we could compute $\sup_{x \in [0,1]} f_n(x) \geq f_n((1/2)^{1/n}) = 1/2$, which shows that f_n does not converge uniformly.

Remark. The statement “ $f_n \not\rightarrow f$ uniformly on the domain S ” means there exists some ε such that for all $N \in \mathbb{N}$, there’s some $n \geq N$ and $x \in S$ such that $|f_n(x) - f(x)| \geq \varepsilon$. In many cases when thinking about this, it’s almost easier just to negate the statement symbolically.

We will now see that the uniform limit function retains certain properties from the original sequence. For example, the uniform limit of continuous functions is continuous.

Theorem 9.1.6 (Continuity of the Uniform Limit)

Let $S \subseteq \mathbb{R}$ or $\mathbb{C}$. Given a sequence of functions $f_n : S \rightarrow \mathbb{R}$ (or $\mathbb{C}$), if f_n is continuous for all n , and $f_n \rightarrow f$ uniformly on S , then f is continuous.

Proof. We will show that f is continuous at some arbitrary $a \in S$. Given $\varepsilon > 0$, there exists some $n \in \mathbb{N}$ such that for $x \in S$ we have $|f_n(x) - f(x)| < \varepsilon/3$. Then since f_n is continuous, there exists $\delta > 0$ such that $|x - a| < \delta$ implies that $|f_n(x) - f_n(a)| < \varepsilon/3$. Then if $x \in S$ and $|x - a| < \delta$, we have

$$|f(x) - f(a)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(a)| + |f_n(a) - f(a)| \leq \varepsilon,$$

as required^a. □

^aThis type of proof is called a ‘ $\varepsilon/3$ ’ proof, or a 3ε proof depending on your outlook

Remark. This result *does not* hold for just pointwise convergence. For example, consider $f_n(x) = x^n$ on the interval $[0, 1]$. However, the set of points at which it can be discontinuous is relatively small. Also this result does not hold for differentiability.

Another way of viewing this result is that it gives us a case where swapping limits is okay¹, that is

$$\lim_{x \rightarrow a} \lim_{n \rightarrow \infty} f_n(x) = \lim_{x \rightarrow a} f(x) = f(a),$$

¹Generally, swapping limits is bad.

if f_n converges uniformly to f .

We can also prove that boundedness is a property preserved by the uniform limit.

Lemma 9.1.7 (Uniform Limit of Bounded Functions is Bounded)

Assume that $f_n \rightarrow f$ uniformly on some set S . If f_n is bounded for every n , then so is f .

Proof. Fix some $n \in \mathbb{N}$ such that for all $x \in S$ we have $|f_n(x) - f(x)| < 1$. Then since f_n is bounded, there is an $M \in \mathbb{R}$ such that $|f_n(x)| \leq M$ for all $x \in S$. Thus

$$|f(x)| \leq |f(x) - f_n(x)| + |f_n(x)| \leq 1 + M,$$

so f is bounded. □

Another natural property that's preserved by the uniform limit is integrability. Since we have f bounded, we can sensibly talk about the upper and lower sums, and just need to check that Riemann's criterion is satisfied.

Theorem 9.1.8 (Uniform Limit of Integrable Functions is Integrable)

Let $f_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of integrable functions. If $f_n \rightarrow f$ uniformly on $[a, b]$ then f is integrable and

$$\int_a^b f_n \rightarrow \int_a^b f,$$

as $n \rightarrow \infty$.

Proof. Since every f_n is integrable, they must all be bounded and hence f is also bounded. Thus it suffices to check that Riemann's criterion is satisfied.

Given $\varepsilon > 0$, we can fix an $n \in \mathbb{N}$ such that $x \in [a, b]$ implies $|f_n(x) - f(x)| < \varepsilon$. Then since f_n is integrable, there exists a dissection $\mathcal{D} = \{x_0, x_1, \dots, x_m\}$ of $[a, b]$ such that $S_{\mathcal{D}}(f_n) - s_{\mathcal{D}}(f_n) < \varepsilon$.

Now, for each $k \in \{1, \dots, m\}$ and any $x, y \in [x_{k-1}, x_k]$, we have

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(y)| + |f_n(y) - f(y)| \\ &\leq |f_n(x) - f_n(y)| + 2\varepsilon. \end{aligned}$$

Then noting that we can write the difference between the supremum and infimum of f in the interval $[x_{k-1}, x_k]$ as

$$\sup_{x, y \in [x_{k-1}, x_k]} |f(x) - f(y)| \leq \sup_{x, y \in [x_{k-1}, x_k]} |f_n(x) - f_n(y)| + 2\varepsilon,$$

we can multiply by $(x_k - x_{k-1})$ and sum over k to get

$$S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) \leq S_{\mathcal{D}}(f_n) - s_{\mathcal{D}}(f_n) + 2\varepsilon(b - a) \leq \varepsilon(2(b - a) + 1),$$

so f is integrable.

Finally, we can write

$$\left| \int_a^b f_n - \int_a^b f \right| \leq \int_a^b |f_n - f| \leq (b-a) \cdot \sup_{[a,b]} |f_n - f| \rightarrow 0$$

as $n \rightarrow \infty$. □

Similar to the previous theorem, this result can be viewed as giving us a case where swapping integrals (which are some form of limit) and limits is okay, so that

$$\int_a^b \lim_{n \rightarrow \infty} f_n(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx,$$

whenever $f_n \rightarrow f$ uniformly.

From these results², we obtain some results on how we can both integrate and differentiate sufficiently nice functions term by term.

Theorem 9.1.9 (Term-wise Integration)

Let $g_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of integrable functions. Then if $\sum_{n=1}^{\infty} g_n(x)$ converges uniformly on $[a, b]$, the function $x \mapsto \sum_{n=1}^{\infty} g_n(x)$ is integrable and

$$\int_a^b \sum_{n=1}^{\infty} g_n(x) dx = \sum_{n=1}^{\infty} \int_a^b g_n(x) dx.$$

Proof Sketch. Let f_n be the sequence of partial sums, and notice that f_n converging uniformly implies that we can apply the previous theorem. □

Theorem 9.1.10 (Term-wise Differentiation)

Let $g_n : [a, b] \rightarrow \mathbb{R}$ be a sequence of continuously differentiable functions. If $\sum_{j=1}^n g_j(c)$ converges for *some* $c \in [a, b]$ and $\sum_{j=1}^n g'_j$ converges uniformly as $n \rightarrow \infty$, then $\sum_{j=1}^{\infty} g_j$ converges uniformly to a continuously differentiable function g , and

$$g'(x) = \frac{d}{dx} \left(\sum_{j=1}^{\infty} g_j(x) \right) = \sum_{j=1}^{\infty} g'_j(x).$$

Proof. We are going to try and solve $g' = h$, where we let $h(x) = \sum_{k=1}^{\infty} g'_k(x)$, subject to the initial condition $g(c) = \sum_{n=1}^{\infty} g_n(c)$.

Since $\sum_{n=1}^{\infty} g'_n(x) \rightarrow h$ uniformly, and since g'_n is continuous, we know that h is continuous and thus integrable.

Now let $\lambda = \sum_{n=1}^{\infty} g_n(c)$, and define

$$g(x) = \lambda + \int_c^x h(t) dt,$$

for $x \in [a, b]$. Since h is continuous, by the fundamental theorem of calculus, g is

²The derivation is straightforward, just consider the sequence of partial sums!

differentiable with $g' = h$, and moreover, $g(c) = \lambda$.

Again by the fundamental theorem of calculus, we also have

$$g_k(x) = g_k(c) + \int_c^x g'_k(t) dt,$$

for $x \in [a, b]$ and $k \in \mathbb{N}$.

Now given $\varepsilon > 0$, there exists a positive integer N such that for $n \geq N$ we have

$$\left| \lambda - \sum_{k=1}^n g_k(c) \right| < \frac{\varepsilon}{2}, \quad \text{and} \quad \left| h(t) - \sum_{k=1}^n g'_k(t) \right| < \frac{\varepsilon}{2(b-a)},$$

for $t \in [a, b]$. Then for $x \in [a, b]$, we have

$$\begin{aligned} \left| g(x) - \sum_{k=1}^n g_k(x) \right| &= \left| \lambda + \int_c^x h(t) dt - \sum_{k=1}^n \left(g_k(c) + \int_c^x g'_k(t) dt \right) \right| \\ &\leq \left| \lambda - \sum_{k=1}^n g_k(c) \right| + \left| \int_c^x \left(h(t) - \sum_{k=1}^n g'_k(t) \right) dt \right| \\ &\leq \frac{\varepsilon}{2} + |x - c| \cdot \frac{\varepsilon}{2(b-a)} \\ &\leq \varepsilon. \end{aligned}$$

This shows that $\sum_{k=1}^n g_k(x) \rightarrow g(x)$ uniformly on $[a, b]$, and we also know that g is differentiable and $g' = h$ is continuous. $\square$

§9.1.2 The General Principle of Uniform Convergence

If a sequence $x_n \in \mathbb{R}$ is Cauchy, that is, if for all $\varepsilon > 0$ there exists a positive integer N such that $m, n \geq N$ implies $|x_m - x_n| < \varepsilon$, then the *general principle of convergence* says that the sequence converges. Said differently, terms in a sequence getting arbitrarily close together is necessary and sufficient for convergence.

A similar concept can be applied to sequences of functions.

Definition 9.1.11 (Uniformly Cauchy)

A sequence f_n of functions on a set S is **uniformly Cauchy** if for all $\varepsilon > 0$ there exists a positive integer N such that $n, m \geq N$ implies that for any $x \in S$ we have $|f_m(x) - f_n(x)| < \varepsilon$.

From this, we get an analogue of the general principle of convergence for uniform convergence.

Theorem 9.1.12 (General Principle of Uniform Convergence)

If f_n is a uniformly Cauchy sequence of functions on a set S , then it converges uniformly on S to some function f .

Proof. For any $x \in S$, the sequence of functions being uniformly Cauchy implies that the sequence of reals $f_n(x)$ is Cauchy. Thus $f_n(x)$ converges for all $x \in S$, and f_n converges pointwise to some function f .

Given $\varepsilon > 0$, there is some positive integer N such that for all $m, n \geq N$ and all $x \in S$, $|f_m(x) - f_n(x)| < \varepsilon/2$. We first fix $x \in S$, and fix $n \geq N$. Since $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$, we can choose $m \in \mathbb{N}$ such that $|f_m(x) - f(x)| < \varepsilon/2$, and $m \geq N$. Then

$$|f_n(x) - f(x)| \leq |f_n(x) - f_m(x)| + |f_m(x) - f(x)| < \varepsilon,$$

as required, and thus f_n converges uniformly to f . $\square$

This is an incredibly important theorem, and has lots of applications. One example is proving the uniform convergence of certain power series, which we will see below.

Theorem 9.1.13 (Weierstrass M -test)

Let f_n be a sequence of functions on a set S . Assume that for every $n \in \mathbb{N}$ there is an $M_n \in \mathbb{R}^+$ such that $|f_n(x)| \leq M_n$ for all $x \in S$. If $\sum_{n=1}^{\infty} M_n$ converges, then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on S .

Proof. Let $F_n(x) = \sum_{k=1}^n f_k(x)$ where $x \in S$ and $n \in \mathbb{N}$. For $x \in S$ and $n \geq m$ we have

$$|F_n(x) - F_m(x)| \leq \sum_{k=m+1}^n |f_k(x)| \leq \sum_{k=m+1}^n M_k.$$

Now given $\varepsilon > 0$, we can choose $N \in \mathbb{N}$ such that $\sum_{k=N+1}^{\infty} M_k < \varepsilon$. Then for every $x \in S$ and $n \geq m \geq N$, we have

$$|F_n(x) - F_m(x)| \leq \sum_{k=m+1}^{\infty} M_k < \varepsilon,$$

so F_n is uniformly Cauchy on S , and hence converges uniformly on S by the general principle of uniform convergence. $\square$

Definition 9.1.14 (Open Disk)

We define the **complex open disk** $D(a, R)$ to be the set

$$D(a, R) = \{z \in \mathbb{C} \mid |z - a| < R\}.$$

Example 9.1.15 (Uniform Convergence of $\sum_{n=1}^{\infty} z^n/n^2$)

We want to know if the power series $\sum_{n=1}^{\infty} z^n/n^2$ converges uniformly.

It's easy to check that the radius of convergence is 1, so now we consider the terms in the sequence $f_n(z) = z^n/n^2$ where $z \in D(0, 1)$.

For all such z , $|f_n(z)| \leq 1/n^2$, and since $\sum_{n=1}^{\infty} 1/n^2$ is convergent, by the M -test, this power series converges uniformly on $D(0, 1)$.

Of course, not every power series converges uniformly.

For example, the power series $\sum_{n=0}^{\infty} z^n$ has radius of convergence 1, and has bounded partial sums for $z \in D(0, 1)$. However, it converges to $1/(1 - z)$, which is not bounded (and hence the convergence can't be uniform, as otherwise boundedness would be preserved).

Despite this example, we can see that uniform convergence basically failed near the radius of convergence, and we can still recover a nice result if we just move *slightly* away, we can still get uniform convergence.

Theorem 9.1.16 (Uniform Convergence Near the Radius of Convergence)

Assume the power series $\sum_{n=0}^{\infty} c_n(z - a)^n$ has radius of convergence R . Then for any r with $0 < r < R$, the power series converges uniformly on $D(a, r)$.

Proof. Fix some $w \in \mathbb{C}$ such that $r < |w - a| < R$. We set $\rho = \frac{r}{|w - a|}$, and we note that $\rho \in (0, 1)$.

Since $\sum_{n=0}^{\infty} c_n(w - a)^n$ converges, we have $c_n(w - a)^n \rightarrow 0$ as $n \rightarrow \infty$. Thus there exists some $M \in \mathbb{R}^+$ such that $|c_n(w - a)^n| \leq M$ for all $n \in \mathbb{N}$.

Now if we take $z \in D(a, r)$ and $n \in \mathbb{N}$, we have

$$\begin{aligned} |c_n(z - a)^n| &= |c_n(w - a)^n| \cdot \left(\frac{|z - a|}{|w - a|} \right)^n \\ &\leq M \left(\frac{r}{|w - a|} \right)^n \leq M \rho^n. \end{aligned}$$

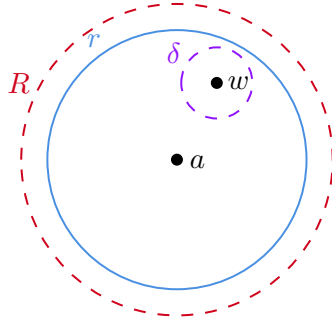
Then since $\sum_{n=0}^{\infty} M \rho^n$ is convergent, by the M -test we have $\sum_{n=0}^{\infty} c_n(z - a)^n$ converging uniformly for $z \in D(a, r)$. $\square$

This result allows us to use our results about uniform convergence to say things about power series. For example:

1. Since a power series is the uniform limit of polynomials inside the radius of convergence, and polynomials are continuous, we get that power series are also continuous inside their radius of convergence.
2. Using our term-wise differentiability result, we get that power series can be differentiated term-wise inside their radius of convergence, and that the power series $\sum_{n=0}^{\infty} c_n(z - a)^n$ has derivative $\sum_{n=1}^{\infty} c_n \cdot n(z - a)^{n-1}$.

We saw previously that it's possible for a power series to not converge uniformly on its whole open disk of convergence. However, the previous theorem showed that we could get arbitrarily close to the radius of convergence, and we'd still have *some* amount of uniform convergence.

To be exact, consider a power series $\sum_{n=0}^{\infty} c_n(z - a)^n$ with radius of convergence R . Then if we take some $w \in D(a, R)$, and fix some r such that $|w - a| < r < R$, we can find some $\delta > 0$ such that $|w - a| + \delta < r$.



Then if $|z - w| < \delta$, $|z - a| \leq |z - w| + |w - a| < \delta + |w - a| < r$, so $D(w, \delta) \subset D(a, r)$, and thus $\sum_{n=0}^{\infty} c_n(z - a)^n$ converges uniformly on $D(w, \delta)$.

This inspires a helpful definition.

Definition 9.1.17 (Open)

A subset U of $\mathbb{C}$ is **open** if for all $w \in U$ there exists $\delta > 0$ such that $D(w, \delta) \subset U$.

Definition 9.1.18 (Local Uniform Convergence)

Let U be an open subset of $\mathbb{C}$ and f_n be a sequence of functions on U . We say that f_n **converges locally uniformly** on U if for all $w \in U$ there exists some $\delta > 0$ such that f_n converges uniformly on $D(w, \delta) \subset U$.

With this the result we discussed above can be stated as follows.

Theorem 9.1.19 (Local Uniform Convergence of Power Series)

A power series centered at a with radius of convergence R converges locally uniformly on $D(a, R)$.

We will return to this when we discuss compactness later on.

§9.1.3 Uniform Continuity

Recall the standard notion of continuity.

Definition 9.1.20 (Continuity)

Let $A \subseteq \mathbb{C}$ and $f : A \rightarrow \mathbb{C}$. We say that f is **continuous at** $a \in A$ if given any $\epsilon > 0$ we can find a $\delta > 0$ such that $|f(a) - f(x)| < \epsilon$ for all $x \in A$ such that $|x - a| < \delta$.

We say that f is **continuous** if it is continuous at every $a \in A$.

In the above definition, our δ is allowed to depend on both ϵ and x .

Definition 9.1.21 (Uniform Continuity)

Let $A \subset \mathbb{C}$ and $f : A \rightarrow \mathbb{C}$. We say that f is **uniformly continuous** on A if given any $\epsilon > 0$ we can find a $\delta > 0$ such that for all $x, y \in A$ we have $|x - y| < \delta$ implying $|f(x) - f(y)| < \epsilon$.

Here, we again impose this *uniformity* constraint, that is, δ is allowed to depend on ε only.

It's easy to see that uniform continuity implies continuity, but the converse doesn't hold in general.

Example 9.1.22 (Uniformly Continuous Function)

We will show that $f(x) = 2x + 17$ is uniformly continuous over $\mathbb{R}$.

Given $\varepsilon > 0$, let $\delta = \varepsilon/2$. Then for all $x, y \in \mathbb{R}$ with $|x - y| < \delta$ we have $|f(x) - f(y)| = 2|x - y| < \varepsilon$, as required.

Example 9.1.23 (A Non-Uniformly Continuous Function)

The function $f(x) = x^2$ over $\mathbb{R}$ is continuous, but not uniformly continuous.

Consider $\varepsilon = 1$. Then given some δ , let $x = 1/\delta$ and $y = 1/\delta + \delta/2$. Then we have $|x - y| < \delta$ but $|f(x) - f(y)| = 1 + \delta^2/4 > 1 = \varepsilon$, so f is not uniformly continuous.

Here, our uniform continuity was shown to not hold by taking larger and larger x , which forced the slope of the function to get too large for a given value of δ to be sufficient. It turns out that this is the only failure point of uniform continuity, and indeed continuity implies uniform continuity when we are working over a bounded interval.

The proof of this is quite natural – we use Bolzano-Weierstrass to show that a contradiction to uniform continuity is a contradiction to continuity.³

Theorem 9.1.24 (Continuous Functions are Uniformly Continuous)

Let $f : [a, b] \rightarrow \mathbb{C}$ be continuous. Then f is uniformly continuous.

Proof. Suppose f was not uniformly continuous, that is, there exists some $\epsilon > 0$ such that for all $\delta > 0$ there is some x, y with $|x - y| < \delta$ such that $|f(x) - f(y)| \geq \epsilon$.

Taking $\delta = 1/n$, we can find some sequences $x_n, y_n \in [a, b]$ such that $|x_n - y_n| < 1/n$, and $|f(x_n) - f(y_n)| \geq \epsilon$. Then by Bolzano-Weierstrass, we can find some convergent subsequence $x_{n_j} \rightarrow x$ for some $x \in [a, b]$. But then $|x_{n_j} - y_{n_j}| < 1/n_j$ for all j , so we must have $y_{n_j} \rightarrow x$ also.

Then $|f(x_{n_j}) - f(y_{n_j})| \geq \epsilon$ for every j , which implies that $f(x_{n_j})$ and $f(y_{n_j})$ cannot converge to the same value. But then by continuity we have $f(x_{n_j}) \rightarrow f(x)$ and $f(y_{n_j}) \rightarrow f(x)$, which is a contradiction. Thus f must be uniformly continuous. $\square$

With this result, we can prove the familiar result that continuous functions are integrable.

Theorem 9.1.25 (Continuous Functions are Integrable)

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then f is Riemann integrable.

³Another argument which is more direct can be made using compactness, but we will look past this for now.

Proof. Given $\epsilon > 0$, since f is uniformly continuous, there is some $\delta > 0$ such that $|f(x) - f(y)| < \epsilon/(b-a)$ whenever $|x - y| < \delta$ and $x, y \in [a, b]$.

Now choose some integer n large enough that $(b-a)/n < \delta$, and define the dissection $\mathcal{D} = \{x_0, x_1, \dots, x_n\}$ with $x_j = a + j(b-a)/n$. Then we have

$$\sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \leq \frac{\epsilon}{(b-a)},$$

for all $1 \leq j \leq n$, and thus

$$\begin{aligned} S_{\mathcal{D}}(f) - s_{\mathcal{D}}(f) &= \sum_{j=1}^n (x_j - x_{j-1}) \left[\sup_{x \in [x_{j-1}, x_j]} f(x) - \inf_{x \in [x_{j-1}, x_j]} f(x) \right] \\ &\leq \sum_{j=1}^n \frac{b-a}{n} \cdot \frac{\epsilon}{b-a} = \epsilon, \end{aligned}$$

and thus f is Riemann integrable. $\square$

§9.2 Metric Spaces

§9.2.1 Defining Metric Spaces

In $\mathbb{R}$ and $\mathbb{C}$, we measured the ‘closeness’ of two points x, y using $|x - y|$. We used this throughout our study of analysis both in this course and previously, and possibly the most important property of this was the triangle inequality,

$$|x - z| \leq |x - y| + |y - z|.$$

The triangle inequality acts, more or less, as saying that being close is *transitive*. That is, if x is close to y , and y is close to z , then x is close to z .

In fact, we can abstract away quite naturally from the absolute value into a more general setting in which a measure of distance has this property.

Definition 9.2.1 (Metric)

Let M be a set. A **metric** on M is a function $d : M \times M \rightarrow \mathbb{R}$ such that

- *Positivity.* For all $x, y \in M$, $d(x, y) \geq 0$, with equality if and only if $x = y$.
- *Symmetry.* For all $x, y \in M$, $d(x, y) = d(y, x)$.
- *Triangle inequality.* For all $x, y, z \in M$,

$$d(x, z) \leq d(x, y) + d(y, z).$$

Definition 9.2.2 (Metric Space)

A **metric space** is a pair (M, d) where M is a set and d is a metric on M .

Example 9.2.3 (Metric Spaces on $\mathbb{R}$)

The real line $\mathbb{R}$ is a metric space under the metric $d(x, y) = |x - y|$. In fact, any subset of $\mathbb{R}$ is a metric space with the same metric.

Example 9.2.4 (Euclidean Space $\mathbb{R}^n$)

In $\mathbb{R}^n$, we can define the **Euclidean norm** of $x \in \mathbb{R}^n$ by

$$\|x\| = \|x\|_2 = \left(\sum_{k=1}^n |x_k|^2 \right)^{\frac{1}{2}}.$$

This satisfies the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. This ‘norm’ then induces a metric on $\mathbb{R}^n$

$$d(x, y) = d_2(x, y) = \|x - y\|_2 = \left(\sum_{k=1}^n |x_k - y_k|^2 \right)^{\frac{1}{2}}.$$

This metric is the **Euclidean distance**, and with it we get a metric space $(\mathbb{R}^n, d_2)$ called **Euclidean space**. This is the standard metric space on $\mathbb{R}^n$.

Remark. We sometimes denote the n -dimensional real or complex Euclidean space by ℓ_2^n . The Euclidean norm is called the ℓ_2 -norm, and the corresponding Euclidean distance is called the ℓ_2 -metric.

Example 9.2.5 (Taxicab Metric)

It’s also possible to use the **taxicab metric** on $\mathbb{R}^n$, given by

$$d(x, y) = \sum_{k=1}^n |x_k - y_k|.$$

This is also known as the ℓ_1 -metric.

Example 9.2.6 (ℓ_∞ Metric)

In $\mathbb{R}^n$ or $\mathbb{C}^n$, we can define the ℓ_∞ norm of x by

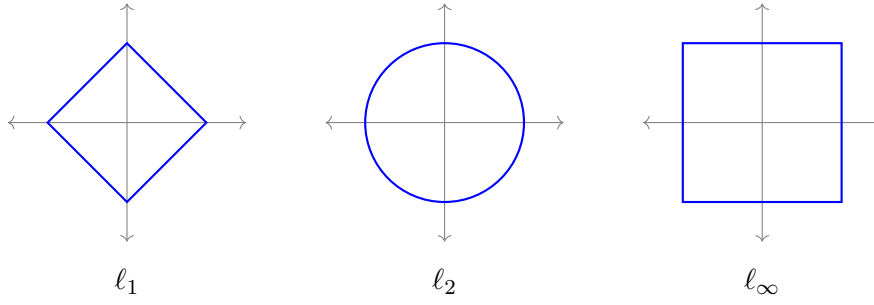
$$\|x\|_\infty = \max_{1 \leq k \leq n} |x_k|,$$

and then we can define the **ℓ_∞ -metric** by

$$d(x, y) = \|x - y\|_\infty = \max_{1 \leq k \leq n} |x_k - y_k|.$$

We then get a metric space which we denote by ℓ_∞^n .

To get some feeling for how these three metrics on $\mathbb{R}^n$ differ, it’s instructive to look at the ‘unit circle’ $\{x \mid d(x, 0) = 1\}$ in each of them. In the plane, we get a diamond, a circle and a square respectively.



In particular, distances measured in these metrics differ, but not by much – each of these unit circles can be sandwiched between scaled copies of the others. We will return to this observation when we discuss *equivalence* of metrics.

Example 9.2.7 (Uniform Metric)

Let S be a set, and let $\ell_\infty(S)$ be the set of all bounded scalar functions on S . We define the ℓ_∞ -norm of $f \in \ell_\infty(S)$ by

$$\|f\| = \|f\|_\infty = \sup_{x \in S} |f(x)|,$$

also called the sup norm or the uniform norm. From this we can define $d(f, g) = \|f - g\|_\infty$ called the **uniform metric** on $\ell_\infty(S)$.

Example 9.2.8 (L_p -Norm)

Let $\mathcal{C}[a, b]$ be the set of all continuous functions defined on the closed and bounded interval $[a, b]$. Then for $p \geq 1$ we can define the **L_p -norm** of $f \in \mathcal{C}[a, b]$ by

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx \right)^{1/p}.$$

This defines^a the **L_p -metric**, $d_p(f, g) = \|f - g\|_p$.

^aIf you want to check that d_p satisfies the metric space axioms, you do need the *Minkowski inequality*, which we will not discuss.

Example 9.2.9 (Discrete Metric)

Let M be any set. Then

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

defines the metric called the **discrete metric**, and (M, d) is called a **discrete metric space**.

Example 9.2.10 (Word Metric)

Let G be a group generated by $S \subset G$, where $e \notin S$ and $x \in S$ implies $x^{-1} \in S$.

Then

$$d(x, y) = \min\{n \geq 0 \mid \text{there exists } s_1, \dots, s_n \in S \text{ such that } y = xs_1 \cdots s_n\}$$

defines the **word metric**.

Example 9.2.11 (p -adic Metric)

Fix a prime $p \in \mathbb{N}$. We define

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ p^{-n} & \text{if } x \neq y \text{ where } x - y = p^n m \text{ with } p \nmid m. \end{cases}$$

defines a metric on $\mathbb{Z}$ called the **p -adic metric**.

§9.2.2 Subspaces

As with many mathematical objects, it can be useful to discuss the substructure to a metric space.

Definition 9.2.12 (Subspace)

Let (M, d) be a metric space and let $N \subset M$. Then d is a metric on N , and (N, d) is a **subspace** of M .

Example 9.2.13 ($\mathbb{Q}$ is a subspace of $\mathbb{R}$)

$\mathbb{Q}$ with the metric $d(x, y) = |x - y|$ is a subspace of $\mathbb{R}$.

Example 9.2.14 ($\mathcal{C}[a, b]$ is a subspace of $\ell_\infty([a, b])$)

Since every continuous function on a closed, bounded interval is bounded, it follows that $\mathcal{C}[a, b]$ is a subset of $\ell_\infty([a, b])$, so $\mathcal{C}[a, b]$ with the uniform metric is a subspace of $\ell_\infty([a, b])$.

Given two metric spaces, there is a number of natural construction of another metric space which has those two metric spaces as subspaces.

Definition 9.2.15 (Product Space)

Let (M, d) and (M', d') be metric spaces. Then each of the following defines a metric on $M \times M'$:

$$\begin{aligned} d_1((x, x'), (y, y')) &= d(x, y) + d'(x', y'), \\ d_2((x, x'), (y, y')) &= (d(x, y)^2 + d'(x', y')^2)^{1/2}, \\ d_\infty((x, x'), (y, y')) &= \max\{d(x, y), d'(x', y')\}. \end{aligned}$$

We denote the metric space $(M \times M', d_p)$ by $M \oplus_p M'$.

Remark. There is no canonical choice of which of $\{d_1, d_2, d_\infty\}$ we take as the product

metric. Indeed, it's straightforward to show that

$$d_\infty \leq d_2 \leq d_1 \leq 2d_\infty,$$

and later on when we discuss concepts such as convergence we will see that because of this fact, the exact choice of metric doesn't matter *that* much.

Of course, this construction extends naturally to any finite number of metric spaces.

§9.2.3 Convergence

Metric spaces give us a measure of closeness, and we know that in $\mathbb{R}$ and $\mathbb{C}$ that convergence is really just a notion of a sequence getting close to some 'limiting value'. Indeed, convergence as we defined it before directly translates into the language of metric spaces.

Definition 9.2.16 (Convergence)

Let (M, d) be a metric space. A sequence $a_1, a_2, \dots \in M$ is said to **converge** to the limit $a \in M$ if given any $\varepsilon > 0$, we can find an integer N such that $d(a_n, a) < \varepsilon$ for all $n \geq N$. We write $a_n \rightarrow a$ as $n \rightarrow \infty$.

A straightforward but useful thing to note is that this definition implies $a_n \rightarrow a$ if and only if $d(a_n, a) \rightarrow 0$ in $\mathbb{R}$.

Now we need to go back to all of the theorems about convergence that were proved in an earlier Analysis course and check if they still hold for a general Metric space (that is, if they don't really use the structure of $\mathbb{R}$, apart from closeness).

Lemma 9.2.17 (Uniqueness of Limits)

If $a_n \rightarrow a$ and $a_n \rightarrow b$ in a metric space M , then $a = b$.

Proof. Assume that $a \neq b$. Given any $\epsilon > 0$, we can find integers N_1 and N_2 such that

$$\begin{aligned} d(a_n, a) &\leq \epsilon, & \text{for all } n \geq N_1 \\ d(a_n, b) &\leq \epsilon, & \text{for all } n \geq N_2 \end{aligned}$$

Then letting $\epsilon = d(a, b)/3$ and taking $N = \max\{N_1, N_2\}$, we have by the triangle inequality

$$d(a, b) \leq d(a, a_n) + d(a_n, b) \leq 2\epsilon = \frac{2}{3}d(a, b)$$

for all $n \geq N$. Thus we must have $d(a, b) = 0$, and $a = b$, contradicting our assumption. $\square$

Lemma 9.2.18 (Convergence of Subsequences)

If $a_n \rightarrow a$ as $n \rightarrow \infty$ in a metric space M , and $n(1) < n(2) < \dots$, then $a_{n(j)} \rightarrow a$ as $j \rightarrow \infty$.

Proof Sketch. Take the old proof from $\mathbb{R}$ and $\mathbb{C}$ and replace the absolute values with the metric! $\square$

§9.2.4 Continuity

As with convergence, we also get a direct generalization of the notion of continuity.

Definition 9.2.19 (Continuity)

Let $f : M \rightarrow M'$ be a function between metric spaces (M, d) and (M', d') . Then for $a \in M$, we say that f is **continuous** at a if given any $\varepsilon > 0$ we can find a $\delta > 0$ such that $d'(f(x), f(a)) < \varepsilon$ for all $x \in M$ such that $d(x, a) < \delta$.

We say that f is **continuous** if f is continuous at a for all $a \in M$.

Proposition 9.2.20 (Sequence Definition of Continuity)

Let $f : M \rightarrow M'$ be a function between metric spaces, and let $a \in M$. Then f is continuous at a if and only if $x_n \rightarrow a$ in M implies that $f(x_n) \rightarrow f(a)$ in M' .

We also keep the same standard properties of continuity as before.

Proposition 9.2.21 (Sums, Products, and Reciprocals of Continuous Functions)

Let $f, g : M \rightarrow \mathbb{R}$ be scalar functions on a metric space M . Then the following hold.

- (i) If f and g are continuous at a , then so is their sum $f + g$.
- (ii) If f and g are continuous at a , then so is their product fg .
- (iii) If $N = \{x \in M \mid g(x) \neq 0\}$, then if f and g are continuous at $a \in N$, then so is f/g .

Proposition 9.2.22 (The Composition of Continuous Functions is Continuous)

Let $f : M \rightarrow M'$ and $g : M' \rightarrow M''$ with M, M', M'' all metric spaces. Then if f is continuous at $a \in M$ and g is continuous at $f(a)$, then $g \circ f$ is also continuous at a .

Example 9.2.23 (Constant Function)

Consider the constant function $f : M \rightarrow M'$ with $f(x) = b$ for some $b \in M'$. Then f is continuous.

Example 9.2.24 (Identity Function)

Consider the identity function $\text{id} : M \rightarrow M$. This is continuous at all $a \in M$, as given $\varepsilon > 0$ we can take $\delta = \varepsilon$, and if $d(x, a) < \delta$ then $d(\text{id}(x), \text{id}(a)) = d(x, a) < \delta = \varepsilon$.

Example 9.2.25 (Polynomials)

Using the examples above, we get that polynomials are always continuous.

Example 9.2.26 (Metrics are Continuous)

Let (M, d) be a metric space. Then d is a function between metric spaces, $d : M \oplus_p M \rightarrow \mathbb{R}$ (where we take some $p \in \{1, 2, \infty\}$).

Given $v = (x, x')$ and $w = (y, y') \in M \oplus_p M$, then

$$|d(v) - d(w)| = |d(x, x') - d(y, y')| \leq d(x, y) + d(x', y') = d_1(v, w) \leq 2d_p(v, w).$$

So $\delta = \varepsilon/2$ will do.

Definition 9.2.27 (Isometric and Lipschitz)

Let $f : M \rightarrow M'$ be a function between metric spaces. Then f is **isometric** if for all $x, y \in M$ we have $d'(f(x), f(y)) = d(x, y)$.

We say that f is **Lipschitz** if there exists some $c \in \mathbb{R}^+$ such that $d'(f(x), f(y)) \leq cd(x, y)$.

Definition 9.2.28 (Uniform Continuity)

Let $f : M \rightarrow M'$ be a function between metric spaces. We say that f is **uniformly continuous** if given $\varepsilon > 0$, there exists some $\delta > 0$ such that for all $x, y \in M$ with $d(x, y) < \delta$ we have $d'(f(x), f(y)) < \varepsilon$.

We can easily see that f isometric implies that it's Lipschitz, and f Lipschitz implies that it's uniformly continuous. Also, if f is isometric then it must be injective (but not necessarily surjective). If f is also surjective then we call it a **isometry**.

If there exists an isometry between two metric spaces M and M' , we say that they are **isometric**.

Example 9.2.29

Let (M, d) and (M', d') be metric spaces, and fix some $y \in M'$. Then if we define $f : M \rightarrow M \oplus_p M'$ by $f(x) = (x, y)$, we have $d_p(f(x), f(z)) = d_p((x, y), (z, y)) = d(x, z)$. So f is an isometry, and $M \times \{y\}$ is an isometric copy of M in the space $M \oplus_p M'$.

§9.2.5 The Topology of a Metric Space

While we will discuss topology and topological spaces later, it's worth looking at a few of the core ideas in the context of metric spaces.

It seems that there are some concepts in metric spaces that don't *really* rely on the metric, just about the closeness of points. The point of this section is looking to see exactly what type of relations are these, so that we can generalize them to a setting where we don't use a metric.

We begin with the familiar concept of an open ball.

Definition 9.2.30 (Open Ball)

Let M be a metric space, $x \in M$ and $r > 0$. The **open ball** in M of center x and radius r is the set $D_r(x)$ given by

$$D_r(x) = \{y \in M \mid d(x, y) < r\}.$$

The idea of open balls is a natural one, and it even fits naturally into some of our previous definitions.

For example, we say that $x_n \rightarrow x$ in M if and only if for all $\varepsilon > 0$, there is some positive integer N such that $n \geq N$ implies that $x_n \in D_\varepsilon(x)$.

Similarly, for a function $f : M \rightarrow M'$ we say that f is continuous at $x \in M$ if and only if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that $f(D_\delta(x)) \subset D_\varepsilon(f(x))$.

Definition 9.2.31 (Closed Ball)

Let M be a metric space, $x \in M$ and $r > 0$. The **closed ball** in M of center x and radius r is the set $B_r(x)$ given by

$$B_r(x) = \{y \in M \mid d(x, y) \leq r\}.$$

Example 9.2.32 (Open and Closed Ball in $\mathbb{R}$)

In $\mathbb{R}$, we have $D_r(x) = (x - r, x + r)$, and $B_r(x) = [x - r, x + r]$.

We now move to define some concepts that will be a little less straightforward, but should come together later on as we discuss more topology.

Definition 9.2.33 (Neighborhood)

Let M be a metric space, and let $U \subset M$. For $x \in M$, we say that U is a **neighborhood** of x in M if there exists some $r > 0$ such that $D_r(x) \subset U$.

Definition 9.2.34 (Open)

We say that U is **open** in M if for all x in U there is some $r > 0$ such that $D_r(x) \subset U$. That is, if U is a neighborhood for all of its points.

Example 9.2.35 (Upper Half of the Complex Plane is Not Open)

Consider the set $H = \{z \in \mathbb{C} \mid \operatorname{Im} z \geq 0\}$. Then let $w \in H$, and let $\delta = \operatorname{Im} w$. If $\delta > 0$, then $D_\delta(w) \subset H$, and if $\delta = 0$, then for all $r > 0$ $D_r(w) \not\subset H$. Thus H is not open.

We commonly use the seeming tautological fact that open balls are open.

Lemma 9.2.36

Open balls are open.

Proof. Consider some open ball $D_r(x)$ in a metric space M . Then let $y \in D_r(x)$, and set $\delta = r - d(x, y)$. Then $\delta > 0$, and if $z \in D_\delta(y)$ then

$$d(z, x) \leq d(z, y) + d(y, x) < \delta + d(x, y) = r,$$

so $z \in D_r(x)$. This shows that $D_\delta(y) \subset D_r(x)$. $\square$

Corollary 9.2.37

Let M be a metric space, $U \subset M$ and $x \in M$. Then U is a neighborhood of x if and only if there exists an open subset V of M such that $x \in V \subset U$.

Proof Sketch. Check definitions. $\square$

This notion of neighborhoods and open sets is sufficient to describe both convergence and continuity in a metric space on its own.

Proposition 9.2.38 (Convergence using Open Sets & Neighborhoods)

In a metric space M , the following are equivalent.

- (i) $x_n \rightarrow x$;
- (ii) For all neighborhoods U of x in M , there exists some positive integer N such that $n \geq N$ implies $x_n \in U$;
- (iii) For all open subsets U of M with $x \in U$, there is a positive integer N such that for all $n \geq N$ we have $x_n \in U$.

Proof. (i) implies (ii). Let U be a neighborhood of x in M . Then by definition there exists some $\varepsilon > 0$ such that $D_\varepsilon(x) \subset U$. Then since $x_n \rightarrow x$, there's an $N \in \mathbb{N}$ such that $n \geq N$ implies $d(x_n, x) < \varepsilon$, that is, $x_n \in D_\varepsilon(x)$.

(ii) implies (iii). This is clear since any open set U with $x \in U$ is a neighborhood of x .

(iii) implies (i). Given some $\varepsilon > 0$, $U = D_\varepsilon(x)$ is open and $x \in U$. Then by (iii) there's an $N \in \mathbb{N}$ such that $n \geq N$ implies $x_n \in U$, that is, $d(x_n, x) < \varepsilon$. $\square$

Proposition 9.2.39 (Local Continuity using Open Sets & Neighborhoods)

Let $f : M \rightarrow M'$ be a function between metric spaces, and let $x \in M$. Then the following are equivalent:

- (i) f is continuous at x ;
- (ii) For all neighborhoods V of $f(x)$ in M' , there exists a neighborhood U of x in M such that $f(U) \subset V$;
- (iii) For all neighborhoods V of $f(x)$ in M' , $f^{-1}(V)$ is a neighborhood of x in M .

Proof. (i) implies (ii). Let V be a neighborhood of $f(x)$ in M' . Then by definition

there exists some $\varepsilon > 0$ such that $D_\varepsilon(f(x)) \subset V$. Since f is continuous at x , there exists $\delta > 0$ such that $f(D_\delta(x)) \subset D_\varepsilon(f(x))$. Then $U = D_\delta(x)$ is a neighborhood of x in M , and $f(U) \subset V$.

(ii) *implies* (iii). Let V be a neighborhood of $f(x)$ in M' . Then by (ii) there is a neighborhood U of x in M such that $f(U) \subset V$. Then $U \subset f^{-1}(V)$, and since U is a neighborhood of x in M , there is some $r > 0$ with $D_r(x) \subset U \subset f^{-1}(V)$. Thus $f^{-1}(V)$ is a neighborhood of x in M .

(iii) *implies* (i). Given $\varepsilon > 0$, $V = D_\varepsilon(f(x))$ is a neighborhood of $f(x)$ in M' . By (iii), $f^{-1}(V)$ is a neighborhood of x in M . So there exists $\delta > 0$ with $D_\delta(x) \subset f^{-1}(V)$. Then $f(D_\delta(x)) \subset V = D_\varepsilon(f(x))$. $\square$

Proposition 9.2.40 (Global Continuity using Open Sets & Neighborhoods)

Let $f : M \rightarrow M'$ be a function between metric spaces. Then the following are equivalent:

- (i) f is continuous;
- (ii) $f^{-1}(V)$ is open in M for all open subsets V of M' .

Proof. (i) *implies* (ii). Let V be open in M' . Let $x \in f^{-1}(V)$. Then $f(x) \in V$. Since V is open, there's some $\varepsilon > 0$ such that $D_\varepsilon(f(x)) \subset V$. Since f is continuous at x , there's a $\delta > 0$ such that $f(D_\delta(x)) \subset D_\varepsilon(f(x))$. Therefore $D_\delta(x) \subset f^{-1}(D_\varepsilon(f(x))) \subset f^{-1}(V)$. Thus $f^{-1}(V)$ is open in M .

(ii) *implies* (i). Let $x \in M$, and let $\varepsilon > 0$. Then $V = D_\varepsilon(f(x))$ is open in M' . By (ii) $f^{-1}(V)$ is open in M , also $x \in f^{-1}(V)$ and $f(x) \in V$. Then by definition there's a $\delta > 0$ such that $D_\delta(x) \subset f^{-1}(V)$, so $f(D_\delta(x)) \subset V = D_\varepsilon(f(x))$. $\square$

We can now write down what the *topology* of a metric space is. We are going to study topology in general later on, but we can use some properties that occur here to inform how we define things more generally afterwards.

Definition 9.2.41 (Topology of a Metric Space)

The **topology** of a metric space M is the family of all open subsets of M .

Proposition 9.2.42 (Basic Properties)

The topology of a metric space satisfies the following.

- (i) $\emptyset$ and M are open.
- (ii) If U_i is open in M for all i in some index set I , then $\bigcup_{i \in I} U_i$ is open in M .
- (iii) U, V open in M implies $U \cap V$ open in M .

Proof. Part (i). This is clear.

Part (ii). Given $x \in \bigcup_{i \in I} U_i$, then there's some $i_0 \in I$ with $x \in U_{i_0}$. Then U_{i_0} is open, so by definition there's an $r > 0$ with $D_r(x) \subset U_{i_0} \subset \bigcup_{i \in I} U_i$.

Part (iii). Given $x \in U \cap V$, since U is open and $x \in U$, there's an $r > 0$ with $D_r(x) \subset U$, and since V is open and $x \in V$, there's an $s > 0$ with $D_s(x) \subset V$. Then letting $t = \min\{r, s\}$, we have $t > 0$ and $D_t(x) = D_r(x) \cap D_s(x) \subset U \cap V$. $\square$

Definition 9.2.43 (Closed)

A subset A of a metric space M is **closed** in M if for every sequence $x_n \in A$ that is convergent in M , we have $\lim_{n \rightarrow \infty} x_n \in A$.

Lemma 9.2.44

Closed balls are closed.

Proof. Consider $B_r(x) = \{y \in M \mid d(y, x) \leq r\}$ in a metric space M , and a sequence $x_n \in B_r(x)$ such that $x_n \rightarrow z$. We need to show that $z \in B_r(x)$. We have

$$\begin{aligned} d(z, x) &\leq d(z, x_n) + d(x_n, x) \\ &\leq d(z, x_n) + r \rightarrow r \end{aligned}$$

as $n \rightarrow \infty$. Thus $d(z, x) \leq r$, and hence $z \in B_r(x)$. $\square$

Example 9.2.45 (Closed but Not Open, and Open but Not Closed Sets)

The set $[0, 1] = B_{\frac{1}{2}}(1/2)$ is closed in $\mathbb{R}$. $[0, 1]$ is not open, for example $D_r(0) \not\subset [0, 1]$ for any $r > 0$.

The set $(0, 1) = D_{\frac{1}{2}}(1/2)$ is open, but it is not closed. Take $1/(n+1) \in (0, 1)$ for all $n \in \mathbb{N}$. Then $1/(n+1) \rightarrow 0$ in $\mathbb{R}$ but $0 \notin (0, 1)$.

Example 9.2.46 (An Open and Closed Set)

$\mathbb{R}$ is open and closed in $\mathbb{R}$.

Example 9.2.47 (A Neither Open or Closed Set)

The set $(0, 1]$ in $\mathbb{R}$ is neither open nor closed.

Lemma 9.2.48 (Relating Closed and Open)

Let A be a subset of a metric space M . Then A is closed in M if and only if $M \setminus A$ is open in M .

Proof. Assume A is closed, and that $M \setminus A$ is not open. Then there exists $x \in M \setminus A$ such that for all $r > 0$ we have $D_r(x) \not\subset M \setminus A$. Hence for all n there exists $x_n \in D_{1/n}(x) \cap A$. Then $d(x_n, x) < 1/n \rightarrow 0$, so $x_n \rightarrow x$ in M and $x_n \in A$ for all n , which contradicts that A is closed.

Now suppose that $M \setminus A$ is open but A is not closed. Then there exists some sequence

x_n in A such that $x_n \rightarrow x$ in M but $x \notin A$. Since $x \in M \setminus A$ and $M \setminus A$ is open, there exists $\varepsilon > 0$ with $D_\varepsilon(x) \subset M \setminus A$. Then since $x_n \rightarrow x$, there is some $N \in \mathbb{N}$ such that for all $n \geq N$ we have $x_n \in D_\varepsilon(x)$, and hence $x_n \in M \setminus A$, which is a contradiction. $\square$

We previously met isometries, which were maps between metric spaces that preserved the distance function. Now that we are trying to look slightly away from metrics and more towards open sets and neighborhoods, it's natural to think of a new definition of a function between metric spaces that preserves open sets.

Definition 9.2.49 (Homeomorphism)

A map $f : M \rightarrow M'$ between metric spaces is a **homeomorphism** if f is a bijection, and f and f^{-1} are both continuous.

Equivalently, f is a homeomorphism if for all open sets V in M' , $f^{-1}(V)$ is open in M , and for all open sets U in M , $f(U)$ is open in M' .

If there is a homeomorphism between M and M' , we say that M and M' are **homeomorphic**.

Example 9.2.50

The metric spaces $(0, \infty)$ and $(0, 1)$ are homeomorphic, with the homeomorphism $x \mapsto 1/(x+1)$ and inverse $x \mapsto 1/x - 1$.

While it is true that every isometry is a homeomorphism, the converse is absolutely false.

We will say that two metrics on a set are equivalent if they induce the same topology.

Definition 9.2.51 (Equivalent Metrics)

Let d and d' be metrics on a set M . We say that d and d' are **equivalent**, denoted $d \sim d'$, if they define the same topology on M . That is, for $U \subset M$, U is open in (M, d) if and only if U is open in (M, d') .

So $d \sim d'$ if and only if $\text{id} : (M, d) \rightarrow (M, d')$ is a homeomorphism.

Note that if $d \sim d'$, then (M, d) and (M, d') have the same convergent sequences and the same continuous maps.

Definition 9.2.52 (Uniformly and Lipschitz Equivalent)

Let d and d' be metrics on a set M . We say that d and d' are **uniformly equivalent** if $\text{id} : (M, d) \rightarrow (M, d')$ and $\text{id} : (M, d') \rightarrow (M, d)$ are uniformly continuous, and write $d \sim_u d'$.

We will say d and d' are **Lipschitz equivalent** if these functions are Lipschitz, and write $d \sim_{Lip} d'$.

Note that $d \sim_{Lip} d'$ if and only if there exists $a > 0$ and $b > 0$ such that

$$ad(x, y) \leq d'(x, y) \leq bd(x, y),$$

and also note that $d \sim_{Lip} d'$ implies $d \sim_u d'$ implies $d \sim d'$.

Example 9.2.53

Given a metric space (M, d) , $d'(x, y) = \min\{1, d(x, y)\}$ defines a metric on M , and $d' \sim_u d$.

Example 9.2.54

On a product space $M \times M'$, the metrics d_1 , d_2 and d_∞ are pairwise Lipschitz equivalent.

Example 9.2.55

On $C[0, 1]$, the L_1 -metric and the uniform metric are *not* equivalent.

We will return to these topological ideas later on.

§9.3 Completeness and the Contraction Mapping Theorem

§9.3.1 Completeness

In $\mathbb{R}$ and $\mathbb{C}$, every Cauchy sequence is convergent. In what other spaces is this true?

Definition 9.3.1 (Cauchy Sequence)

A sequence x_n in a metric space M is **Cauchy** if given $\varepsilon > 0$, there's some positive integer N such that $n, m \geq N$ implies that $d(x_n, x_m) < \varepsilon$.

Definition 9.3.2 (Bounded)

A sequence x_n in a metric space M is **bounded** if there exists some $z \in M$ and $r > 0$ such that for all n we have $x_n \in B_r(z)$.

Lemma 9.3.3 (Convergent Implies Cauchy Implies Bounded)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. Let x_n be a sequence in metric space M .

Suppose x_n is convergent in M , and let $x = \lim_{n \rightarrow \infty} x_n$. Given $\varepsilon > 0$, there is some positive integer N such that $n \geq N$ implies $d(x_n, x) < \varepsilon/2$. Then for all $m, n \geq N$, $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon$. Thus if a sequence converges, then it's Cauchy.

Now assume that x_n is Cauchy in M . Then there's a positive integer N such that for all $m, n \geq N$ we have $d(x_m, x_n) < 1$. In particular, $d(x_n, x_N) < 1$ for $n \geq N$, that is, $x_n \in B_1(x_N)$ for $n \geq N$. So let $r = \max\{d(x_1, x_N), \dots, d(x_{N-1}, x_N), 1\}$. Then $x_n \in B_r(x_N)$ for all $n \in \mathbb{N}$, and so x_n is bounded. $\square$

In general, being bounded does not imply that a sequence is Cauchy (for example, take the sequence $0, 1, 0, 1, \dots$).

Slightly more subtle is that being Cauchy doesn't imply convergence, and we can construct a somewhat silly example. For example, $x_n = 1/n$ is Cauchy but doesn't converge in the metric space $(0, \infty)$. It seems that this example works because there's something 'missing' from the metric space – the limit of a sequence. Indeed, in this section we will care about metric spaces where such an implication *does hold*, in 'complete' metric spaces.

Definition 9.3.4 (Completeness)

A metric space M is **complete** if every Cauchy sequence in M converges in M .

Example 9.3.5

$\mathbb{R}$ and $\mathbb{C}$ are complete.

Proposition 9.3.6

If M and M' are complete metric spaces, then so is $M \oplus_p M'$.

Proof. Let $z_n = (x_n, x'_n)$ be a Cauchy sequence in $M \oplus_p M'$. Since $d(x_m, x_n) \leq d_p(z_m, z_n)$ and $d'(x'_m, x'_n) \leq d_p(z_m, z_n)$, both x_n and x'_n are Cauchy in M and M' respectively. By completeness, $x_n \rightarrow x$ in M and $x'_n \rightarrow x'$ in M' , and then letting $z = (x, x')$ we have

$$d_p(z_n, z) \leq d_1(z_n, z) = d(x_n, x) + d'(x'_n, x') \rightarrow 0,$$

so $z_n \rightarrow z$ in $M \oplus_p M'$. □

Example 9.3.7

$\mathbb{R}^n$ and $\mathbb{C}^n$ are complete in the ℓ_p -metric.

Theorem 9.3.8 (Completeness of a Function Space)

Let S be any set. Then $\ell_\infty(S)$ is complete in the uniform metric D .

Proof. Let f_n be a Cauchy sequence in $\ell_\infty(S)$. Then given $\varepsilon > 0$, there is some positive integer N such that for all $n, m \geq N$ we have

$$D(f_m, f_n) = \sup_{x \in S} |f_m(x) - f_n(x)| < \varepsilon.$$

So f_n is uniformly Cauchy. By the general principle of uniform convergence, f_n converges uniformly to some function f on S . We also know that f is bounded, so $f \in \ell_\infty(S)$. Now given $\varepsilon > 0$, there's some $N \in \mathbb{N}$ such that for all $n \geq N$ and $x \in S$ we have

$$|f_n(x) - f(x)| < \varepsilon,$$

and thus for $n \geq N$ we have $\sup_{x \in S} |f_n(x) - f(x)| = D(f_n, f) \leq \varepsilon$, so $f_n \rightarrow f$ in $(\ell_\infty(S), D)$. □

Proposition 9.3.9 (Completeness of a Subspace)

Let N be a subspace of a metric space M .

- (i) If N is complete, then N is closed in M .
- (ii) If M is complete and N is closed in M , then N is complete.

Proof. Part (i). Let x_n be a sequence in N and assume that $x_n \rightarrow x$ in M . We then need $x \in N$. Since x_n is convergent in M , x_n must also be Cauchy in M , and thus it is Cauchy in N . Then since N is complete, we must have $x_n \rightarrow y \in N$. But then $x_n \rightarrow y \in M$, and by uniqueness of limits, $y = x$ and $x \in N$.

Part (ii). Let x_n be a Cauchy sequence in N . Then x_n is Cauchy in M . Since M is complete, $x_n \rightarrow x$ in M for some $x \in M$. Since N is closed in M , we have $x \in N$. So $x_n \rightarrow x$ in N . $\square$

Theorem 9.3.10

Let (M, d) be a metric space, and define

$$C_b(M) = \{f \in \ell_\infty(M) \mid f \text{ is continuous}\}.$$

This is a subspace of $\ell_\infty(M)$ in the uniform metric D .

Then $C_b(M)$ is complete in the uniform metric.

Proof. It suffices to show that $C_b(M)$ is closed in $\ell_\infty(M)$. Let f_n be a sequence in $C_b(M)$, and assume that $f_n \rightarrow f$ in $\ell_\infty(M)$. We need to show that f is continuous.

Given some $a \in M$ and $\varepsilon > 0$, since $f_n \rightarrow f$ in $\ell_\infty(M)$, we can fix an $n \in \mathbb{N}$ such that $D(f_n, f) < \varepsilon$. Since f_n is continuous at a , there's a $\delta > 0$ such that for all $x \in M$ we have $d(x, a) < \delta$ implying $|f_n(x) - f_n(a)| < \varepsilon$. Hence for all $x \in M$ if $d(x, a) < \delta$ then

$$\begin{aligned} |f(x) - f(a)| &\leq |f(x) - f_n(x)| + |f_n(x) - f_n(a)| + |f_n(a) - f(a)| \\ &\leq 2D(f_n, f) + |f_n(x) - f_n(a)| \\ &\leq 3\varepsilon. \end{aligned}$$

$\square$

Corollary 9.3.11

$C[a, b]$, the space of continuous functions on the closed bounded interval $[a, b]$ is complete in the uniform metric.

This last corollary is going to be surprisingly useful. Completeness of function spaces is what will let us *construct* functions as limits – and in particular, it is the key to proving the existence of solutions to differential equations, as we are about to see.

§9.3.2 The Contraction Mapping Theorem

We now come to the highlight of this section, which is a genuinely useful theorem about complete metric spaces. It concerns *fixed points* of certain well-behaved maps from a metric space to itself.

Definition 9.3.12 (Contraction)

Let (M, d) be a metric space. A map $f : M \rightarrow M$ is a **contraction** if there exists some λ with $0 \leq \lambda < 1$ such that

$$d(f(x), f(y)) \leq \lambda d(x, y),$$

for all $x, y \in M$.

That is, a contraction is a map which is Lipschitz with constant strictly less than 1 – it shrinks distances by a definite factor. In particular every contraction is (uniformly) continuous.

Theorem 9.3.13 (Contraction Mapping Theorem)

Let M be a non-empty complete metric space, and let $f : M \rightarrow M$ be a contraction. Then f has a unique fixed point, that is, there is a unique $z \in M$ such that $f(z) = z$.

Proof. Let $\lambda \in [0, 1)$ be such that $d(f(x), f(y)) \leq \lambda d(x, y)$ for all $x, y \in M$.

We first show uniqueness. If $f(z) = z$ and $f(w) = w$, then

$$d(z, w) = d(f(z), f(w)) \leq \lambda d(z, w),$$

and since $\lambda < 1$ this forces $d(z, w) = 0$, so $z = w$.

Now for existence, the idea is to just pick a point and repeatedly apply f , and show that the sequence we get is Cauchy. So fix any $x_0 \in M$ (using non-emptiness), and define the sequence $x_n = f(x_{n-1})$ for $n \geq 1$. Applying the contraction property repeatedly,

$$d(x_n, x_{n+1}) = d(f(x_{n-1}), f(x_n)) \leq \lambda d(x_{n-1}, x_n) \leq \cdots \leq \lambda^n d(x_0, x_1).$$

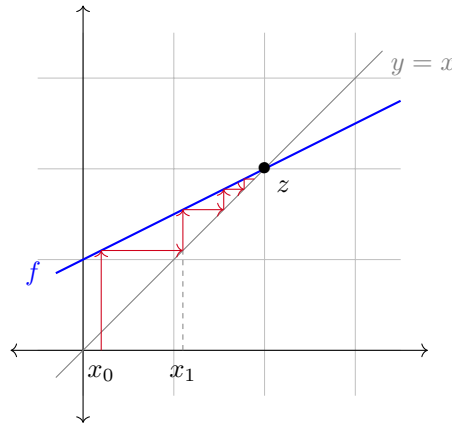
Then for $m \geq n$, the triangle inequality gives

$$d(x_n, x_m) \leq \sum_{k=n}^{m-1} d(x_k, x_{k+1}) \leq \sum_{k=n}^{m-1} \lambda^k d(x_0, x_1) \leq \frac{\lambda^n}{1-\lambda} d(x_0, x_1).$$

Since $\lambda^n/(1-\lambda) \rightarrow 0$ as $n \rightarrow \infty$, given $\varepsilon > 0$ we can find N such that $d(x_n, x_m) < \varepsilon$ for all $m \geq n \geq N$. So the sequence x_n is Cauchy, and since M is complete, $x_n \rightarrow z$ for some $z \in M$.

Finally, f is continuous, so $x_{n+1} = f(x_n) \rightarrow f(z)$, but also $x_{n+1} \rightarrow z$, so by uniqueness of limits $f(z) = z$. $\square$

There is a nice picture of what is going on here for a contraction on $\mathbb{R}$. Starting anywhere and ‘cobwebbing’ between the graph of f and the line $y = x$, the iterates spiral into the fixed point, which is where the graphs cross.



Remark. Letting $m \rightarrow \infty$ in the bound for $d(x_n, x_m)$ in the proof above shows that

$$d(x_n, z) \leq \frac{\lambda^n}{1 - \lambda} d(x_0, x_1),$$

so the iterates converge to the fixed point *exponentially fast*. This makes the contraction mapping theorem practical as a numerical method, not just a theoretical tool.

Both hypotheses in the theorem – completeness of the space, and the Lipschitz constant being strictly less than 1 – are genuinely needed, as the following examples show.

Example 9.3.14 (Completeness is Necessary)

Consider $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ given by $f(x) = x/2$. This is a contraction (with $\lambda = 1/2$), but it has no fixed point – the ‘missing’ fixed point is 0, and the problem is that $\mathbb{R} \setminus \{0\}$ is not complete.

Example 9.3.15 (Shrinking Distances is Not Enough)

Consider $f : [1, \infty) \rightarrow [1, \infty)$ given by $f(x) = x + 1/x$. The domain is closed in $\mathbb{R}$ and hence complete, and it’s easy to check that $|f(x) - f(y)| < |x - y|$ for all $x \neq y$. But f has no fixed point, since $f(x) - x = 1/x > 0$ everywhere.

The issue is that there is no single $\lambda < 1$ that works for *all* pairs of points – the map shrinks distances, but not uniformly so. Similarly, $f(x) = x + 1$ on $\mathbb{R}$ is an isometry ($\lambda = 1$) with no fixed point.

§9.3.3 Application: Solving Differential Equations

At first glance, the contraction mapping theorem might look like a curiosity. Its real power comes from applying it to *function spaces*: if we can express ‘being a solution to a problem’ as ‘being a fixed point of a map on a complete space of functions’, then we get existence and uniqueness of solutions in one stroke. The classic application is to differential equations.

Let’s warm up with a specific example before proving the general theorem.

Example 9.3.16 (An Unusual Differential Equation)

Fix $y_0 \in \mathbb{R}$. We will show that there is a unique differentiable function $f : [0, \frac{1}{2}] \rightarrow \mathbb{R}$

such that $f(0) = y_0$ and

$$f'(t) = f(t^2) \quad \text{for all } t \in [0, \tfrac{1}{2}].$$

First note that if f is such a solution, then f is differentiable and hence continuous, so $f'(t) = f(t^2)$ is also continuous. Then by the fundamental theorem of calculus,

$$f(t) = y_0 + \int_0^t f(s^2) ds.$$

Conversely, if $f \in C[0, \frac{1}{2}]$ satisfies this integral equation, then by the fundamental theorem of calculus (the integrand being continuous) f is differentiable with $f'(t) = f(t^2)$, and $f(0) = y_0$. So solving the differential equation is *equivalent* to solving the integral equation in $C[0, \frac{1}{2}]$.

Now let $M = C[0, \frac{1}{2}]$ with the uniform metric D , which we know is non-empty and complete, and define $T : M \rightarrow M$ by

$$(Tg)(t) = y_0 + \int_0^t g(s^2) ds.$$

By the reasoning above, f solves our problem if and only if $Tf = f$. But for $g, h \in M$ and $t \in [0, \frac{1}{2}]$,

$$|(Tg)(t) - (Th)(t)| = \left| \int_0^t [g(s^2) - h(s^2)] ds \right| \leq t \sup_s |g(s^2) - h(s^2)| \leq \frac{1}{2} D(g, h).$$

Taking the supremum over t , we get $D(Tg, Th) \leq \frac{1}{2} D(g, h)$, so T is a contraction. By the contraction mapping theorem, T has a unique fixed point, which is the unique solution to our differential equation.

Remark. The choice of the interval $[0, \frac{1}{2}]$ mattered only in making the map a contraction – the same argument gives a unique solution f_δ on $[0, \delta]$ for any $\delta < 1$. By uniqueness, these solutions agree on common domains, so they patch together into a unique solution on $[0, 1)$.

The same circle of ideas gives the fundamental existence and uniqueness theorem for ordinary differential equations. Before we state it, we need to make sense of integrating a vector-valued function: for continuous $f = (f_1, \dots, f_n) : [c, d] \rightarrow \mathbb{R}^n$, we integrate coordinate-wise,

$$\int_c^d f(t) dt = \left(\int_c^d f_1(t) dt, \dots, \int_c^d f_n(t) dt \right),$$

and differentiability of such an f also means differentiability of each coordinate, with $f' = (f'_1, \dots, f'_n)$. A useful fact, which follows from the Cauchy-Schwarz inequality⁴, is that

$$\left\| \int_c^d f(t) dt \right\| \leq \int_c^d \|f(t)\| dt \leq (d - c) \sup_{t \in [c, d]} \|f(t)\|.$$

⁴Write $v = \int_c^d f(t) dt$. Then $\|v\|^2 = \sum_k v_k \int f_k = \int \langle v, f(t) \rangle dt \leq \int \|v\| \|f(t)\| dt$, and divide by $\|v\|$.

Theorem 9.3.17 (Picard-Lindelöf)

Let $y_0 \in \mathbb{R}^n$, let $a < b$ be reals, $R > 0$, and let

$$\phi : [a, b] \times B_R(y_0) \rightarrow \mathbb{R}^n$$

be a continuous function. Suppose ϕ is Lipschitz in the second variable, that is, there is some $K > 0$ such that

$$\|\phi(t, x) - \phi(t, y)\| \leq K\|x - y\|$$

for all $t \in [a, b]$ and $x, y \in B_R(y_0)$. Then there exists $\varepsilon > 0$ such that for any $t_0 \in [a, b]$, the initial value problem

$$f'(t) = \phi(t, f(t)), \quad f(t_0) = y_0$$

has a unique solution on $[c, d] = [t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]$.

Proof. Since closed balls are closed, $B_R(y_0)$ is closed and bounded in $\mathbb{R}^n$, and one can check (say, coordinate-wise using Bolzano-Weierstrass, as we did for uniform continuity) that a continuous function on a closed bounded subset of $\mathbb{R}^{n+1}$ is bounded^a. So we may set

$$C = \sup\{\|\phi(t, x)\| \mid t \in [a, b], x \in B_R(y_0)\} < \infty,$$

and define $\varepsilon = \min\{R/C, 1/(2K)\}$. Fix $t_0 \in [a, b]$ and let $[c, d] = [t_0 - \varepsilon, t_0 + \varepsilon] \cap [a, b]$.

As before, we convert the problem into an integral equation. By the fundamental theorem of calculus applied coordinate-wise, a function $f : [c, d] \rightarrow B_R(y_0)$ solves the initial value problem if and only if f is continuous and

$$f(t) = y_0 + \int_{t_0}^t \phi(s, f(s)) ds \quad \text{for all } t \in [c, d].$$

Now we set up our function space. Since $B_R(y_0)$ is closed in the complete space $\mathbb{R}^n$, it is complete, and hence the space $M = C([c, d], B_R(y_0))$ of continuous functions $[c, d] \rightarrow B_R(y_0)$ is complete in the uniform metric D (as before, being uniformly Cauchy implies uniform convergence, and the uniform limit of continuous functions into a closed set is continuous into that set). It's also non-empty, as it contains the constant function $t \mapsto y_0$.

Define $T : M \rightarrow M$ by

$$(Tg)(t) = y_0 + \int_{t_0}^t \phi(s, g(s)) ds.$$

We should check this is well defined. If $g \in M$ then $s \mapsto \phi(s, g(s))$ is continuous, so the integral makes sense and Tg is continuous (indeed differentiable). Moreover for $t \in [c, d]$,

$$\|(Tg)(t) - y_0\| = \left\| \int_{t_0}^t \phi(s, g(s)) ds \right\| \leq |t - t_0| \sup_{s \in [c, d]} \|\phi(s, g(s))\| \leq \varepsilon C \leq R,$$

so Tg does map into $B_R(y_0)$, and $Tg \in M$. By construction, f solves the initial value problem if and only if $Tf = f$.

Finally, we check that T is a contraction. For $g, h \in M$ and $t \in [c, d]$,

$$\begin{aligned} \|(Tg)(t) - (Th)(t)\| &= \left\| \int_{t_0}^t [\phi(s, g(s)) - \phi(s, h(s))] ds \right\| \\ &\leq |t - t_0| \sup_{s \in [c, d]} \|\phi(s, g(s)) - \phi(s, h(s))\| \\ &\leq \varepsilon K \sup_{s \in [c, d]} \|g(s) - h(s)\| \leq \frac{1}{2} D(g, h), \end{aligned}$$

using the Lipschitz hypothesis and $\varepsilon \leq 1/(2K)$. Taking the supremum over t gives $D(Tg, Th) \leq \frac{1}{2} D(g, h)$, so T is a contraction on the non-empty complete metric space M , and by the contraction mapping theorem it has a unique fixed point. This fixed point is the unique solution to the initial value problem. $\square$

^aWe will be able to see this much more cleanly once we have discussed *compactness*.

Remark. A few comments on this theorem.

- (i) The theorem is *local* – it only guarantees a solution close to t_0 . In general there really is no solution on all of $[a, b]$; consider $f' = f^2$, $f(0) = 1$, whose solution $f(t) = 1/(1 - t)$ blows up in finite time.
- (ii) Any n -th order ODE $y^{(n)} = \psi(t, y, y', \dots, y^{(n-1)})$ can be handled by this theorem, by the standard trick of introducing the vector $(y, y', \dots, y^{(n-1)}) \in \mathbb{R}^n$ and rewriting the equation as a first-order system.

§9.4 Topological Spaces

When we studied metric spaces, we found that many important concepts – convergence, continuity, open and closed sets – could be phrased entirely in terms of *open sets*, with no direct mention of the metric. This suggests a further abstraction: throw away the metric entirely, and take the open sets themselves as the fundamental structure. This is the starting point of *topology*.

§9.4.1 Topologies

Recall the basic properties satisfied by the open subsets of a metric space: the empty set and the whole space are open, arbitrary unions of open sets are open, and finite intersections of open sets are open. We take precisely these properties as our axioms.

Definition 9.4.1 (Topology)

Let X be a set. A **topology** on X is a family τ of subsets of X such that

- (i) $\emptyset \in \tau$ and $X \in \tau$;
- (ii) if $U_i \in \tau$ for all i in some index set I , then $\bigcup_{i \in I} U_i \in \tau$;
- (iii) if $U, V \in \tau$, then $U \cap V \in \tau$.

A **topological space** is a pair (X, τ) where X is a set and τ is a topology on X . The members of τ are called the **open sets** of X .

By induction, (iii) implies that any *finite* intersection of open sets is open. As with metric spaces, we usually just refer to ‘the topological space X ’, leaving the topology implicit.

Example 9.4.2 (Metric Topology)

Let (M, d) be a metric space. Then the family of all d -open subsets of M is a topology on M , called the **metric topology** induced by d – this is exactly what we proved at the end of our discussion of the topology of metric spaces.

Definition 9.4.3 (Metrizable)

A topological space (X, τ) is **metrizable** if there exists a metric d on X which induces the topology τ .

Note that equivalent metrics induce the same topology, so a metrizable space never remembers *which* metric it came from. Not every topological space is metrizable, as we shall see shortly. First, some examples of topologies that don’t obviously come from metrics.

Example 9.4.4 (Discrete Topology)

For any set X , the family $\tau = \mathcal{P}(X)$ of *all* subsets of X is a topology, called the **discrete topology**. It is metrizable – it is induced by the discrete metric, since in that metric $D_{1/2}(x) = \{x\}$, so every singleton (and hence every set) is open.

Example 9.4.5 (Indiscrete Topology)

For any set X , the family $\tau = \{\emptyset, X\}$ is a topology, called the **indiscrete topology**. If $|X| \geq 2$, this is *not* metrizable. Indeed, suppose d were a metric inducing it, and take distinct $x, y \in X$. Then with $r = d(x, y) > 0$, the ball $D_r(x)$ is d -open, contains x but not y , and so is neither $\emptyset$ nor X – contradiction.

Example 9.4.6 (Cofinite Topology)

For any set X , the family

$$\tau = \{\emptyset\} \cup \{U \subset X \mid X \setminus U \text{ is finite}\}$$

is a topology on X , called the **cofinite topology**. If X is finite this is just the discrete topology, but if X is infinite then any two non-empty open sets intersect (their complements are finite, and X is not).

Definition 9.4.7 (Coarser and Finer)

If $\tau_1 \subset \tau_2$ are topologies on the same set X , we say τ_1 is **coarser** than τ_2 , and τ_2 is

finer than τ_1 .

So the indiscrete topology is the coarsest topology on X , and the discrete topology is the finest.

Often it is convenient to specify a topology by giving only some of its open sets, and generating the rest by taking unions. This gives us the notion of a *basis*.

Definition 9.4.8 (Basis)

Let (X, τ) be a topological space. A family $\mathcal{B} \subset \tau$ is a **basis** for τ if every member of τ is a union of members of $\mathcal{B}$. Equivalently, $\mathcal{B}$ is a basis if for every open U and every $x \in U$, there is some $B \in \mathcal{B}$ with $x \in B \subset U$.

Example 9.4.9 (Open Balls Form a Basis)

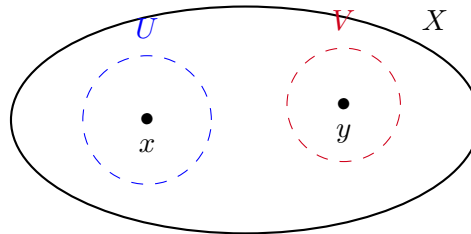
In a metric space M , the open balls $\{D_r(x) \mid x \in M, r > 0\}$ form a basis for the metric topology: if U is open then by definition every $x \in U$ has some $D_{r_x}(x) \subset U$, and $U = \bigcup_{x \in U} D_{r_x}(x)$. In fact the balls of radius $1/n$ for $n \in \mathbb{N}$ already suffice.

The next definition captures the most important property that distinguishes ‘metric-like’ spaces from stranger topologies: the ability to separate points.

Definition 9.4.10 (Hausdorff)

A topological space X is **Hausdorff** if for all $x \neq y$ in X , there exist open sets U, V in X such that $x \in U$, $y \in V$ and $U \cap V = \emptyset$.

In other words, distinct points can be ‘housed off’ from each other by disjoint open sets.



Proposition 9.4.11 (Metric Spaces are Hausdorff)

Every metrizable topological space is Hausdorff.

Proof. Let d be a metric inducing the topology, and let $x \neq y$. Set $r = d(x, y)/2 > 0$. Then $U = D_r(x)$ and $V = D_r(y)$ are open sets containing x and y respectively, and they are disjoint: if $z \in U \cap V$ then $d(x, y) \leq d(x, z) + d(z, y) < 2r = d(x, y)$, a contradiction. $\square$

Example 9.4.12 (Non-Metrizable Topologies)

The indiscrete topology on a set with at least two points is not Hausdorff (there aren’t enough open sets), which gives another proof that it isn’t metrizable. Sim-

ilarly, the cofinite topology on an infinite set is not Hausdorff, since any two non-empty open sets intersect – so it is not metrizable either.

§9.4.2 Closed Sets, Neighborhoods and Convergence

In metric spaces we could define closed sets via convergent sequences, and we then proved that a set is closed if and only if its complement is open. In a general topological space, sequences are no longer expressive enough to detect the topology, so we take the complement property as the *definition*.

Definition 9.4.13 (Closed Set)

A subset A of a topological space X is **closed** in X if $X \setminus A$ is open in X .

Taking complements in the axioms for a topology (using De Morgan's laws) immediately gives the following.

Proposition 9.4.14 (Properties of Closed Sets)

In a topological space X :

- (i) $\emptyset$ and X are closed;
- (ii) if A_i is closed for all i in a non-empty index set I , then $\bigcap_{i \in I} A_i$ is closed;
- (iii) if A, B are closed, then $A \cup B$ is closed.

Note that sets are not doors: a set can be open, closed, both (like $\emptyset$), or neither. For example, in a discrete space every set is both open and closed, while in the cofinite topology on an infinite set, the closed sets are exactly the finite sets and the whole space.

Definition 9.4.15 (Neighborhood)

Let X be a topological space, $x \in X$ and $U \subset X$. We say U is a **neighborhood** of x in X if there is an open set V with $x \in V \subset U$.

This matches our characterisation of neighborhoods in metric spaces. Just as before, open sets are precisely the sets that are neighborhoods of all of their points.

Proposition 9.4.16

A subset U of a topological space X is open if and only if U is a neighborhood of every $x \in U$.

Proof. If U is open and $x \in U$, then $V = U$ witnesses that U is a neighborhood of x . Conversely, if for each $x \in U$ there's an open V_x with $x \in V_x \subset U$, then $U = \bigcup_{x \in U} V_x$ is a union of open sets, hence open. $\square$

We can also define convergence, mimicking the neighborhood description of convergence that we proved for metric spaces.

Definition 9.4.17 (Convergence)

A sequence x_n in a topological space X **converges** to $x \in X$ if for every neighborhood U of x , there is some $N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$. We write $x_n \rightarrow x$.

It is equivalent to only require this for every *open* U containing x (any neighborhood contains such an open set). In metric spaces, this recovers exactly our old definition. In general topological spaces, however, convergence can behave very oddly.

Example 9.4.18 (Convergence Can Be Strange)

In an indiscrete topological space X , the only neighborhood of any point is X itself, so *every* sequence converges to *every* point.

In the cofinite topology on an infinite set X , a sequence x_n converges to x if and only if for each $y \neq x$, the set $\{n \mid x_n = y\}$ is finite. In particular, a sequence of *distinct* points converges to every point of X at once.

The property needed to rule out this behaviour and make limits unique is precisely the Hausdorff property.

Proposition 9.4.19 (Uniqueness of Limits in Hausdorff Spaces)

Let X be a Hausdorff topological space. If $x_n \rightarrow x$ and $x_n \rightarrow y$ in X , then $x = y$.

Proof. Suppose $x \neq y$. Then there are disjoint open sets $U \ni x$ and $V \ni y$. Since $x_n \rightarrow x$ there is N_1 with $x_n \in U$ for $n \geq N_1$, and since $x_n \rightarrow y$ there is N_2 with $x_n \in V$ for $n \geq N_2$. But then for $n \geq \max\{N_1, N_2\}$ we have $x_n \in U \cap V = \emptyset$, a contradiction. $\square$

Remark. In a metric space, we characterised closed sets as those closed under taking limits of sequences. In a general topological space only one direction survives: if A is closed, $x_n \in A$ and $x_n \rightarrow x$, then $x \in A$ (otherwise $X \setminus A$ would be an open neighborhood of x containing some x_n). But there are non-closed sets that are closed under limits – sequences are simply too short-sighted to see all of a general topology.

§9.4.3 Interiors, Closures and Density

Given an arbitrary subset A of a topological space, there is a canonical way to ‘round it down’ to an open set, and to ‘round it up’ to a closed one.

Definition 9.4.20 (Interior and Closure)

Let X be a topological space and $A \subset X$. The **interior** of A is

$$A^\circ = \bigcup \{U \subset X \mid U \text{ open in } X, U \subset A\},$$

and the **closure** of A is

$$\overline{A} = \bigcap \{F \subset X \mid F \text{ closed in } X, F \supset A\}.$$

Since unions of open sets are open, A° is open, and it is clearly the *largest* open subset of A . Similarly $\overline{A}$ is the smallest closed set containing A (the intersection is over a non-empty family, as X itself is closed). In particular, A is open if and only if $A = A^\circ$, and closed if and only if $A = \overline{A}$.

There is a more hands-on description of both.

Proposition 9.4.21 (Pointwise Descriptions)

Let X be a topological space and $A \subset X$. Then:

- (i) $A^\circ = \{x \in X \mid A \text{ is a neighborhood of } x\}$;
- (ii) $\overline{A} = \{x \in X \mid \text{every neighborhood of } x \text{ meets } A\}$.

Proof. Part (i). If A is a neighborhood of x , there is an open U with $x \in U \subset A$, so $x \in A^\circ$. Conversely if $x \in A^\circ$, then A° itself is an open set with $x \in A^\circ \subset A$.

Part (ii). Suppose $x \notin \overline{A}$. Then there is a closed $F \supset A$ with $x \notin F$, and $U = X \setminus F$ is a neighborhood of x with $U \cap A = \emptyset$. Conversely, suppose some neighborhood U of x misses A . Shrinking, we may take U open. Then $F = X \setminus U$ is closed and contains A , so $\overline{A} \subset F$, and since $x \notin F$ we get $x \notin \overline{A}$. $\square$

Example 9.4.22

In $\mathbb{R}$, take $A = [0, 1) \cup \{2\}$. Then $A^\circ = (0, 1)$ and $\overline{A} = [0, 1] \cup \{2\}$. For the rationals, $\mathbb{Q}^\circ = \emptyset$ (every interval contains irrationals) and $\overline{\mathbb{Q}} = \mathbb{R}$ (every interval contains rationals). For the integers, $\mathbb{Z}^\circ = \emptyset$ and $\overline{\mathbb{Z}} = \mathbb{Z}$.

Remark. In a *metric* space, $x \in \overline{A}$ if and only if there is a sequence in A converging to x (take $x_n \in D_{1/n}(x) \cap A$). In a general topological space, having such a sequence still implies $x \in \overline{A}$, but the converse can fail.

Definition 9.4.23 (Dense and Separable)

A subset A of a topological space X is **dense** in X if $\overline{A} = X$. We say X is **separable** if it has a countable dense subset.

Example 9.4.24

$\mathbb{R}^n$ is separable, since $\mathbb{Q}^n$ is countable and dense. An uncountable set with the discrete topology is *not* separable, since every subset is closed and so the only dense subset is the whole space.

§9.4.4 Subspaces

If Y is a subset of a topological space X , then Y inherits a topology from X in a natural way.

Definition 9.4.25 (Subspace Topology)

Let (X, τ) be a topological space and $Y \subset X$. The **subspace topology** on Y is

$$\tau|_Y = \{V \cap Y \mid V \in \tau\}.$$

It's easy to check that this is a topology (intersecting with Y commutes with unions and intersections). Note that a set open in Y need not be open in X .

Example 9.4.26

Let $X = \mathbb{R}$, $Y = [0, 2]$ and $U = (1, 2]$. Then $U = (1, 3) \cap Y$ is open in Y , but U is certainly not open in $\mathbb{R}$ – no interval around 2 fits inside U .

Remark. Reassuringly, the constructions we already have are all consistent with each other:

- (i) If $Z \subset Y \subset X$, the subspace topology that Z inherits from Y (with its subspace topology) is the same as the one it inherits directly from X .
- (ii) If (M, d) is a metric space and $N \subset M$, then the metric topology of the metric subspace (N, d) coincides with the subspace topology induced by the metric topology of M . This is because balls in N are exactly balls of M intersected with N .

Closed sets in a subspace have exactly the description you'd hope for.

Proposition 9.4.27 (Closed Sets and Closures in Subspaces)

Let X be a topological space, $A \subset Y \subset X$. Then A is closed in Y if and only if $A = B \cap Y$ for some closed B in X . Moreover, writing cl_Y for closure inside the space Y ,

$$\text{cl}_Y(A) = \overline{A} \cap Y,$$

where $\overline{A}$ denotes closure in X .

Proof. For the first claim, A is closed in Y if and only if $Y \setminus A$ is open in Y , that is, $Y \setminus A = V \cap Y$ for some open V in X . Taking $B = X \setminus V$, this says exactly $A = B \cap Y$ with B closed in X .

For the second, $\overline{A} \cap Y$ is closed in Y by the first part and contains A , so $\text{cl}_Y(A) \subset \overline{A} \cap Y$. Conversely, write $\text{cl}_Y(A) = B \cap Y$ with B closed in X . Then $A \subset B$, so $\overline{A} \subset B$, and hence $\overline{A} \cap Y \subset B \cap Y = \text{cl}_Y(A)$. $\square$

The corresponding statement for interiors is *false* in general – for example, with $A = Y = \{0\}$ inside $X = \mathbb{R}$, we have $\text{int}_Y(A) = \{0\}$ but $A^\circ = \emptyset$ in $\mathbb{R}$.

§9.4.5 Continuity and Homeomorphisms

Our characterisation of global continuity in metric spaces – preimages of open sets are open – becomes the *definition* of continuity for topological spaces.

Definition 9.4.28 (Continuity)

A function $f : X \rightarrow Y$ between topological spaces is **continuous** if for every open set V in Y , the preimage $f^{-1}(V)$ is open in X .

For metric spaces this agrees with the ε - δ definition, by our earlier work. Let's collect the basic properties.

Proposition 9.4.29 (Basic Properties of Continuity)

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be functions between topological spaces.

- (i) Constant functions are continuous, and the identity $\text{id} : X \rightarrow X$ is continuous.
- (ii) f is continuous if and only if $f^{-1}(B)$ is closed in X for every closed B in Y .
- (iii) If f and g are continuous, then so is $g \circ f$.
- (iv) If $Y \subset X$ has the subspace topology, the inclusion $i : Y \rightarrow X$ is continuous, and hence if $f : X \rightarrow Z$ is continuous, so is the restriction $f|_Y = f \circ i$.

Proof. Part (i). If $f(x) = y_0$ for all x , then $f^{-1}(V)$ is X or $\emptyset$ depending on whether $y_0 \in V$, and both are open. For the identity, $\text{id}^{-1}(V) = V$.

Part (ii). This follows from $f^{-1}(Y \setminus B) = X \setminus f^{-1}(B)$, together with the fact that a set is open if and only if its complement is closed.

Part (iii). For open $W \subset Z$, we have $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$, which is open since $g^{-1}(W)$ is open by the continuity of g , and preimages of open sets under f are open.

Part (iv). For open V in X , $i^{-1}(V) = V \cap Y$, which is open in Y by the definition of the subspace topology. $\square$

Definition 9.4.30 (Homeomorphism)

A function $f : X \rightarrow Y$ between topological spaces is a **homeomorphism** if f is a bijection and both f and f^{-1} are continuous. If such an f exists we say X and Y are **homeomorphic**, written $X \cong Y$.

A homeomorphism induces a bijection between the topologies of X and Y , so homeomorphic spaces are indistinguishable as topological spaces. This makes the following notion the central ‘sameness’ concept of the subject.

Definition 9.4.31 (Topological Property)

A property $\mathcal{P}$ of topological spaces is a **topological property** (or **topological invariant**) if whenever $X \cong Y$, the space X has property $\mathcal{P}$ if and only if Y does.

Example 9.4.32

Being Hausdorff, being metrizable and being separable are all topological properties. On the other hand, being *complete* with respect to a specific metric is not: $\mathbb{R} \cong (0, 1)$ as topological spaces, but the usual metric on $\mathbb{R}$ is complete while the usual metric on $(0, 1)$ is not. Completeness is a property of a metric, not of a topology.

Much of the power of topological invariants is in proving that spaces are *not* homeomorphic – if we can find a topological property that one space has and another lacks,

no homeomorphism can exist. We'll see genuinely useful invariants (connectedness and compactness) in the coming sections.

Definition 9.4.33 (Open Map)

A function $f : X \rightarrow Y$ is an **open map** if $f(U)$ is open in Y for every open U in X .

If f is a bijection, then f^{-1} is continuous exactly when f is an open map (since $(f^{-1})^{-1}(U) = f(U)$). So the homeomorphisms are exactly the continuous open bijections.

§9.4.6 Products

Given topological spaces X and Y , we want a sensible topology on the product $X \times Y$. Certainly if U is open in X and V is open in Y , we would like the 'open box' $U \times V$ to be open in $X \times Y$. Boxes are closed under finite intersections, since

$$(U \times V) \cap (U' \times V') = (U \cap U') \times (V \cap V'),$$

but a union of boxes need not be a box, so we must throw the unions in too. That is, we take the boxes as a *basis*.

Definition 9.4.34 (Product Topology)

Let X and Y be topological spaces. The **product topology** on $X \times Y$ is the topology whose open sets are the sets of the form

$$W = \bigcup_{i \in I} U_i \times V_i,$$

where each U_i is open in X and each V_i is open in Y .

Equivalently, $W \subset X \times Y$ is open if and only if for every $(x, y) \in W$ there are open sets $U \ni x$ in X and $V \ni y$ in Y with $U \times V \subset W$.

We should immediately check this is consistent with what we already do for metric spaces.

Example 9.4.35 (Products of Metric Spaces)

Let (M, d) and (M', d') be metric spaces, and consider the product metric d_∞ on $M \times M'$ (recall all the product metrics d_1, d_2, d_∞ are Lipschitz equivalent, so induce the same topology). For $z = (x, x')$ and $r > 0$,

$$D_r(z) = \{(y, y') \mid d(x, y) < r \text{ and } d'(x', y') < r\} = D_r(x) \times D_r(x'),$$

so open balls in d_∞ are precisely 'square' boxes. It follows easily that a subset of $M \times M'$ is d_∞ -open if and only if it is open in the product topology – the two constructions agree. In particular, the product topology on $\mathbb{R} \times \mathbb{R}$ is the usual Euclidean topology on $\mathbb{R}^2$.

The product topology is 'designed' to make the coordinate projections behave well, in the following precise sense.

Proposition 9.4.36 (Continuity and Products)

Let X, Y be topological spaces and give $X \times Y$ the product topology. Write $q_X : X \times Y \rightarrow X$ and $q_Y : X \times Y \rightarrow Y$ for the coordinate projections. Then:

- (i) q_X and q_Y are continuous;
- (ii) for any topological space Z , a function $g : Z \rightarrow X \times Y$ is continuous if and only if both $q_X \circ g$ and $q_Y \circ g$ are continuous.

Proof. Part (i). For open U in X , we have $q_X^{-1}(U) = U \times Y$, a box, hence open. Similarly for q_Y .

Part (ii). If g is continuous then $q_X \circ g$ and $q_Y \circ g$ are compositions of continuous maps. Conversely, suppose $h = q_X \circ g$ and $k = q_Y \circ g$ are continuous, so that $g(z) = (h(z), k(z))$. For open sets U in X and V in Y ,

$$g^{-1}(U \times V) = h^{-1}(U) \cap k^{-1}(V),$$

which is open in Z . A general open set W in $X \times Y$ is a union of boxes $\bigcup_i U_i \times V_i$, so $g^{-1}(W) = \bigcup_i g^{-1}(U_i \times V_i)$ is a union of open sets, hence open. $\square$

In other words, a map *into* a product is continuous exactly when its components are. This all extends to finite products $X_1 \times \cdots \times X_n$ in the obvious way, with basis the boxes $U_1 \times \cdots \times U_n$. If each X_i is metrizable, so is the product, using for example the d_∞ metric.

§9.4.7 Quotients

Products let us build bigger spaces from smaller ones; *quotients* let us build new spaces by *gluing*. The setup is a topological space X together with an equivalence relation R on X (we write $x \sim y$ for $(x, y) \in R$). We write X/R for the set of equivalence classes, and $q : X \rightarrow X/R$ for the **quotient map** sending each point to its equivalence class. The idea is that in X/R , equivalent points of X have been glued into a single point.

Definition 9.4.37 (Quotient Topology)

Let X be a topological space and R an equivalence relation on X . The **quotient topology** on X/R is

$$\tau = \{V \subset X/R \mid q^{-1}(V) \text{ is open in } X\}.$$

This is a topology because taking preimages commutes with unions and intersections: $q^{-1}(\emptyset) = \emptyset$, $q^{-1}(X/R) = X$, $q^{-1}(\bigcup V_i) = \bigcup q^{-1}(V_i)$ and $q^{-1}(U \cap V) = q^{-1}(U) \cap q^{-1}(V)$. Note that q is continuous by construction – indeed the quotient topology is the *finest* topology on X/R making q continuous.

Example 9.4.38 (The Circle as a Quotient)

Consider $\mathbb{R}$ with the equivalence relation $x \sim y$ if and only if $x - y \in \mathbb{Z}$, giving the quotient $\mathbb{R}/\mathbb{Z}$. Every real number is equivalent to one in $[0, 1]$, and the only identification within $[0, 1]$ is that $0 \sim 1$. So $\mathbb{R}/\mathbb{Z}$ is the interval $[0, 1]$ with its two

endpoints glued together – which should be a circle! We will prove that $\mathbb{R}/\mathbb{Z} \cong S^1$ once we have compactness available, which makes the proof essentially effortless.

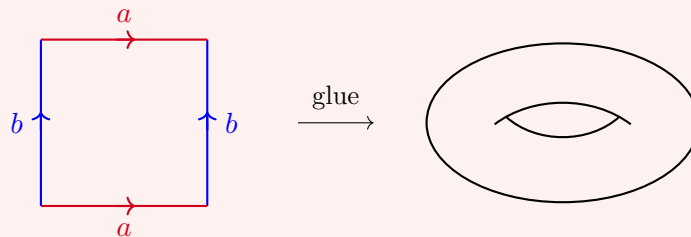
Example 9.4.39 (Quotients Can Be Horrible)

Let $x \sim y$ if and only if $x - y \in \mathbb{Q}$, and consider $\mathbb{R}/\mathbb{Q}$. If $V \subset \mathbb{R}/\mathbb{Q}$ is open and non-empty, then $q^{-1}(V)$ is open and non-empty, so contains an interval (a, b) . But then given *any* $x \in \mathbb{R}$, we can pick a rational $r \in (a - x, b - x)$, and then $x + r \in (a, b)$, so $q(x) = q(x + r) \in V$. Hence $V = \mathbb{R}/\mathbb{Q}$, and the quotient topology on $\mathbb{R}/\mathbb{Q}$ is indiscrete!

In particular, quotients of metrizable spaces (indeed, of $\mathbb{R}$ itself) can fail to be Hausdorff or metrizable. Quotients require some care.

Example 9.4.40 (Gluing a Square)

Let $Q = [0, 1] \times [0, 1] \subset \mathbb{R}^2$. Identifying the left and right edges of Q via $(0, y) \sim (1, y)$ rolls the square up into a *cylinder*. If we additionally identify the top and bottom edges via $(x, 0) \sim (x, 1)$ (and consequently all four corners become one point), we get the **torus** – the surface of a doughnut. The picture to have in mind is:



Formally, the resulting quotient Q/R is homeomorphic to $\mathbb{R}^2/\mathbb{Z}^2$ (quotienting by the relation $x \sim y \iff x - y \in \mathbb{Z}^2$), and also to the familiar torus surface embedded in $\mathbb{R}^3$.

To actually *work* with quotients, the key tool is the way that maps out of X descend to maps out of X/R .

Proposition 9.4.41 (Factoring Maps Through Quotients)

Let X be a set, R an equivalence relation on X , and $q : X \rightarrow X/R$ the quotient map. Let $f : X \rightarrow Y$ be a function which **respects** R , meaning $x \sim y$ implies $f(x) = f(y)$. Then there is a unique map $\tilde{f} : X/R \rightarrow Y$ such that $f = \tilde{f} \circ q$.

Moreover $\tilde{f}$ has the same image as f , and if f **fully respects** R , meaning $x \sim y \iff f(x) = f(y)$, then $\tilde{f}$ is injective.

Proof. Any class $z \in X/R$ is $q(x)$ for some x , and the requirement $\tilde{f}(q(x)) = f(x)$ pins down $\tilde{f}(z)$ – this is well defined precisely because f respects R . Uniqueness and the image claim follow since q is surjective. If f fully respects R and $\tilde{f}(q(x)) = \tilde{f}(q(y))$, then $f(x) = f(y)$, so $x \sim y$ and $q(x) = q(y)$, giving injectivity. $\square$

Proposition 9.4.42 (Continuity of Factored Maps)

In the setting above, suppose additionally that X and Y are topological spaces and that X/R carries the quotient topology. Then:

- (i) if f is continuous, then so is $\tilde{f}$;
- (ii) if f is an open map, then so is $\tilde{f}$.

In particular, if f is a continuous surjection that fully respects R , then $\tilde{f}$ is a continuous bijection, and if f is additionally an open map, then $\tilde{f}$ is a homeomorphism $X/R \cong Y$.

Proof. Part (i). For open V in Y ,

$$q^{-1}(\tilde{f}^{-1}(V)) = (\tilde{f} \circ q)^{-1}(V) = f^{-1}(V),$$

which is open in X , so by the definition of the quotient topology, $\tilde{f}^{-1}(V)$ is open in X/R .

Part (ii). Let V be open in X/R , and set $U = q^{-1}(V)$, which is open in X . Since q is surjective, $q(U) = V$, so

$$\tilde{f}(V) = \tilde{f}(q(U)) = f(U),$$

which is open in Y since f is an open map. □

Finally, let us record when a quotient has a chance of being Hausdorff. Thinking of R as a subset of $X \times X$ (with the product topology), we have the following.

Proposition 9.4.43 (Hausdorff Quotients)

Let X be a topological space and R an equivalence relation on X .

- (i) If X/R is Hausdorff, then R is closed in $X \times X$.
- (ii) If R is closed in $X \times X$ and q is an open map, then X/R is Hausdorff.

Proof. Part (i). Let $W = (X \times X) \setminus R$; we show W is open. Given $(x, y) \in W$ we have $q(x) \neq q(y)$, so there are disjoint open sets $S \ni q(x)$ and $T \ni q(y)$ in X/R . Then $U = q^{-1}(S)$ and $V = q^{-1}(T)$ are open in X with $x \in U$, $y \in V$, and for any $(a, b) \in U \times V$ we have $q(a) \in S$ and $q(b) \in T$, so $q(a) \neq q(b)$, that is, $a \not\sim b$. Hence $(x, y) \in U \times V \subset W$, and W is open.

Part (ii). Let $z \neq w$ in X/R , and pick $x, y \in X$ with $q(x) = z$, $q(y) = w$. Then $x \not\sim y$, so (x, y) lies in the open set $W = (X \times X) \setminus R$, and there are open sets $U \ni x$, $V \ni y$ with $U \times V \subset W$. Since q is an open map, $q(U)$ and $q(V)$ are open, and they contain z and w respectively. They are disjoint: if $q(a) = q(b)$ with $a \in U$, $b \in V$, then $(a, b) \in R \cap (U \times V)$, contradicting $U \times V \subset W$. So X/R is Hausdorff. □

§9.5 Connectedness

We now come to our first substantial topological property. Recall the intermediate value theorem from Analysis I: a continuous function on an interval takes all values between any two of its values. Which feature of intervals makes this work? Intuitively, an interval is ‘all in one piece’ – and this is the idea we now make precise, for arbitrary topological spaces.

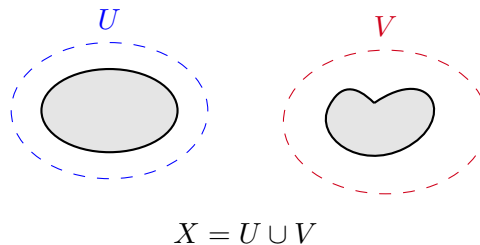
§9.5.1 Connected Spaces

Definition 9.5.1 (Connected)

A topological space X is **disconnected** if there exist open sets U, V in X such that

$$U \cap V = \emptyset, \quad U \cup V = X, \quad U, V \neq \emptyset.$$

In this case we say U and V **disconnect** X . We say X is **connected** if it is not disconnected.



Equivalently (taking complements), X is disconnected if and only if it has a subset which is open, closed, non-empty and not the whole space.

There is a slick reformulation of connectedness in terms of continuous functions, which is frequently the easiest version to use in proofs. Here $\mathbb{Z}$ carries the subspace topology from $\mathbb{R}$, which is its discrete topology.

Theorem 9.5.2 (Characterisations of Connectedness)

Let X be a topological space. The following are equivalent:

- (i) X is connected;
- (ii) if $f : X \rightarrow \mathbb{R}$ is continuous, then $f(X)$ is an interval;
- (iii) every continuous function $f : X \rightarrow \mathbb{Z}$ is constant.

Proof. (i) implies (ii). Suppose $f : X \rightarrow \mathbb{R}$ is continuous but $f(X)$ is not an interval. Then there are $a < b < c$ with $a, c \in f(X)$ but $b \notin f(X)$. Let $U = f^{-1}((-\infty, b))$ and $V = f^{-1}((b, \infty))$. These are open (preimages of open sets), disjoint, non-empty (they contain points mapping to a and c), and since b is not a value of f their union is all of X . So U, V disconnect X , a contradiction.

(ii) implies (iii). If $f : X \rightarrow \mathbb{Z}$ is continuous, then viewed as a map into $\mathbb{R}$ it is continuous, so $f(X)$ is an interval consisting of integers – which must be a single point.

(iii) implies (i). Suppose U, V disconnect X . Define $f : X \rightarrow \mathbb{Z}$ by $f = 0$ on U and $f = 1$ on V . For any $W \subset \mathbb{Z}$, the preimage $f^{-1}(W)$ is one of $\emptyset, U, V, X$, all of which are open, so f is continuous. But f is not constant, contradicting (iii). $\square$

We can now identify the connected subsets of the real line. Here, and always, a subset of a topological space is called connected if it is connected in its subspace topology. Recall that $I \subset \mathbb{R}$ is an **interval** if whenever $x < y < z$ with $x, z \in I$, we have $y \in I$.

Theorem 9.5.3 (Connected Subsets of $\mathbb{R}$)

A subset $X \subset \mathbb{R}$ is connected if and only if X is an interval.

Proof. If X is connected, then the inclusion $i : X \rightarrow \mathbb{R}$ is continuous, so $X = i(X)$ is an interval by the theorem above.

Conversely, suppose X is an interval and $f : X \rightarrow \mathbb{R}$ is continuous. If $x < y$ are points of X then $[x, y] \subset X$, and by the intermediate value theorem from Analysis I, f takes every value between $f(x)$ and $f(y)$ on $[x, y]$. Hence $f(X)$ is an interval, and by the theorem above, X is connected. $\square$

Note the pleasing exchange that has taken place here: the intermediate value theorem told us that intervals are connected, and conversely, once we know a space is connected, part (ii) of Theorem 9.5.2 is the intermediate value theorem for that space.

Example 9.5.4

Any indiscrete space is connected (there aren't enough open sets to disconnect it). The cofinite topology on an infinite set is connected (any two non-empty open sets meet). A discrete space with at least two points is disconnected.

When working with subspaces, it's convenient to phrase disconnection using open sets of the ambient space.

Lemma 9.5.5 (Disconnecting Subspaces)

Let Y be a subspace of a topological space X . Then Y is disconnected if and only if there are open sets U, V in X such that

$$U \cap V \cap Y = \emptyset, \quad Y \subset U \cup V, \quad U \cap Y \neq \emptyset, \quad V \cap Y \neq \emptyset.$$

In this situation we say that U and V disconnect Y in X .

Proof. Open subsets of Y are exactly the sets $U \cap Y$ for U open in X , so this is just unwinding the definition of the subspace topology. $\square$

Proposition 9.5.6 (Closures of Connected Sets)

Let Y be a connected subspace of a topological space X . Then $\overline{Y}$ is connected. More generally, any Z with $Y \subset Z \subset \overline{Y}$ is connected.

Proof. Suppose U, V are open in X and disconnect Z . Since $Y \subset Z$, we have $U \cap V \cap Y = \emptyset$ and $Y \subset U \cup V$. If both $U \cap Y$ and $V \cap Y$ were non-empty, then U, V would disconnect Y , which is connected – so without loss of generality $V \cap Y = \emptyset$. Then $Y \subset X \setminus V$, which is closed, so $\overline{Y} \subset X \setminus V$, and hence $V \cap Z \subset V \cap \overline{Y} = \emptyset$. This contradicts U, V disconnecting Z . $\square$

Theorem 9.5.7 (Continuous Images of Connected Spaces)

Let $f : X \rightarrow Y$ be a continuous function between topological spaces. If X is connected, then $f(X)$ is connected.

Proof. Suppose U, V are open in Y and disconnect $f(X)$. Then $f^{-1}(U)$ and $f^{-1}(V)$ are open in X , they are non-empty (as U and V meet $f(X)$), their union is X (as $f(X) \subset U \cup V$), and they are disjoint (if x was in both, $f(x) \in U \cap V \cap f(X) = \emptyset$). So they disconnect X , a contradiction. $\square$

In particular, connectedness is a topological property, and any quotient of a connected space is connected (the quotient map is a continuous surjection).

Lemma 9.5.8 (Unions of Overlapping Connected Sets)

Let $\mathcal{A}$ be a family of connected subsets of a topological space X such that $A \cap B \neq \emptyset$ for all $A, B \in \mathcal{A}$. Then $\bigcup_{A \in \mathcal{A}} A$ is connected.

Proof. Let $Y = \bigcup_{A \in \mathcal{A}} A$ and let $f : Y \rightarrow \mathbb{Z}$ be continuous; by Theorem 9.5.2 it's enough to show f is constant. For each $A \in \mathcal{A}$, the restriction $f|_A$ is continuous, so it is constant since A is connected. For any $A, B \in \mathcal{A}$, picking a point of $A \cap B$ shows these constants agree. Hence f is constant on Y . $\square$

Theorem 9.5.9 (Products of Connected Spaces)

If X and Y are connected topological spaces, then $X \times Y$ is connected.

Proof. We may assume both are non-empty. Fix $x_0 \in X$. The map $y \mapsto (x_0, y)$ is continuous (its components are constant and the identity), so its image $\{x_0\} \times Y$ is connected; similarly each $X \times \{y\}$ is connected. Now for $y \in Y$, let

$$A_y = (\{x_0\} \times Y) \cup (X \times \{y\}).$$

The two pieces meet in the point (x_0, y) , so each A_y is connected by Lemma 9.5.8. Furthermore, any two A_y, A_z contain $\{x_0\} \times Y$, so they pairwise intersect, and hence

$$\bigcup_{y \in Y} A_y = X \times Y$$

is connected, again by Lemma 9.5.8. $\square$

Example 9.5.10

$\mathbb{R}^n$ is connected for every n , being a product of copies of the interval $\mathbb{R}$.

§9.5.2 Path-Connectedness

There is a second, more intuitive notion of being ‘all in one piece’: you can walk from any point to any other without leaving the space.

Definition 9.5.11 (Path)

Let X be a topological space and $x, y \in X$. A **path** from x to y in X is a continuous function $\gamma : [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. We say X is **path-connected** if for all $x, y \in X$ there is a path from x to y in X .

Example 9.5.12 (Convex Sets are Path-Connected)

Any convex subset $C \subset \mathbb{R}^n$ (one containing the straight line segment between any two of its points) is path-connected: for $y, z \in C$ take the straight-line path $\gamma(t) = (1-t)y + tz$, which is continuous with image in C . In particular open balls $D_r(x)$ are path-connected, since for $y, z \in D_r(x)$,

$$\|\gamma(t) - x\| = \|(1-t)(y-x) + t(z-x)\| \leq (1-t)\|y-x\| + t\|z-x\| < r.$$

Theorem 9.5.13 (Path-Connected Implies Connected)

If a topological space X is path-connected, then X is connected.

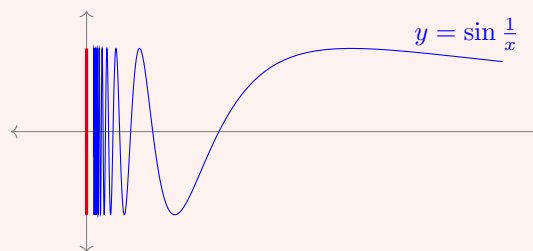
Proof. Suppose U, V disconnect X , and take $x \in U, y \in V$. Let γ be a path from x to y . Then $\gamma^{-1}(U)$ and $\gamma^{-1}(V)$ are open, non-empty (they contain 0 and 1), disjoint, and cover $[0, 1]$ – so they disconnect $[0, 1]$. But $[0, 1]$ is an interval, hence connected, a contradiction. $\square$

The converse is false, and the standard counterexample is important enough to have a name: the *topologist’s sine curve*.

Example 9.5.14 (The Topologist’s Sine Curve)

Consider the set

$$X = \underbrace{\left\{ \left(x, \sin \frac{1}{x} \right) \mid x > 0 \right\}}_Y \cup \underbrace{\left\{ (0, y) \mid -1 \leq y \leq 1 \right\}}_Z \subset \mathbb{R}^2.$$



We claim that X is connected but not path-connected.

X is connected. The curve Y is the image of the connected set $(0, \infty)$ under the continuous map $x \mapsto (x, \sin \frac{1}{x})$, so Y is connected. We claim $X = \bar{Y}$; then X is connected by Proposition 9.5.6. To see $X \subset \bar{Y}$, fix $y \in [-1, 1]$. For each n , the map $x \mapsto 1/x$ sends $(0, \frac{1}{n})$ onto (n, ∞) , and $\sin$ takes the value y somewhere in any interval of length 2π , so there is $x_n \in (0, \frac{1}{n})$ with $\sin \frac{1}{x_n} = y$. Then $(x_n, y) \in Y$ and $(x_n, y) \rightarrow (0, y)$, so $(0, y) \in \bar{Y}$. Conversely X is closed: if $(x_n, y_n) \in X$ converges to (x, y) , then $y \in [-1, 1]$, and either $x = 0$, in which case $(x, y) \in Z \subset X$, or $x > 0$, in which case $x_n > 0$ eventually, so $y_n = \sin \frac{1}{x_n} \rightarrow \sin \frac{1}{x}$ by continuity and $(x, y) \in Y$. Hence $\bar{Y} \subset X$, so $X = \bar{Y}$.

X is not path-connected. Suppose $\gamma = (\gamma_1, \gamma_2) : [0, 1] \rightarrow X$ were a path from $(0, 0)$ to $(1, \sin 1)$. The idea is that any path leaving the vertical segment must traverse infinitely many oscillations in finite time. Suppose $\gamma_1(t) > 0$ for some t . By the intermediate value theorem, $\gamma_1([0, t]) \supset [0, \gamma_1(t)]$, so for any large n we can find $s < t$ with $\gamma_1(s) = \frac{1}{2\pi n}$, whence $\gamma_2(s) = \sin(2\pi n) = 0$; and similarly some $s' < t$ with $\gamma_1(s') = \frac{1}{2\pi n + \pi/2}$, whence $\gamma_2(s') = 1$. Iterating this (each time applying the argument on the interval up to the previously found time, with larger n), we obtain a decreasing sequence of times $1 \geq t_1 > t_2 > \cdots > 0$ with $\gamma_2(t_k)$ alternating between 0 and 1. The sequence t_k is decreasing and bounded below, so it converges to some $t^* \in [0, 1]$; but then $\gamma_2(t_k) \rightarrow \gamma_2(t^*)$ by continuity, which is impossible for a sequence alternating between 0 and 1. So no such path exists.

Despite this counterexample, for *open* subsets of Euclidean space the two notions do coincide. First, a lemma that is endlessly useful for building paths (and other functions) piece by piece.

Lemma 9.5.15 (Gluing Lemma)

Let $X = A \cup B$ be a topological space with A, B closed in X , and let $g : A \rightarrow Y$ and $h : B \rightarrow Y$ be continuous functions which agree on $A \cap B$. Then the function $f : X \rightarrow Y$ defined by

$$f(x) = \begin{cases} g(x) & \text{if } x \in A, \\ h(x) & \text{if } x \in B \end{cases}$$

is well defined and continuous.

Proof. Well-definedness is exactly the agreement on $A \cap B$. Now let V be closed in Y . Then

$$f^{-1}(V) = g^{-1}(V) \cup h^{-1}(V),$$

where $g^{-1}(V)$ is closed in A , and hence closed in X (it equals $A \cap G$ for some closed G in X , and an intersection of closed sets is closed); similarly $h^{-1}(V)$ is closed in X . So $f^{-1}(V)$ is closed in X for every closed V , and f is continuous. $\square$

With this, being joined by a path is an equivalence relation on any topological space: constant paths give reflexivity, reversing a path ($t \mapsto \gamma(1 - t)$) gives symmetry, and for transitivity we concatenate paths, defining

$$\eta(t) = \begin{cases} \gamma(2t) & \text{if } 0 \leq t \leq \frac{1}{2}, \\ \delta(2t - 1) & \text{if } \frac{1}{2} \leq t \leq 1, \end{cases}$$

which is continuous by the gluing lemma applied to $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. The equivalence classes are called the **path-connected components** of X .

Theorem 9.5.16 (Open Subsets of $\mathbb{R}^n$)

Let U be an open subset of $\mathbb{R}^n$. Then U is connected if and only if U is path-connected.

Proof. Path-connected implies connected always, so suppose U is connected and non-empty, and fix $x_0 \in U$. Let

$$P = \{x \in U \mid \text{there is a path from } x_0 \text{ to } x \text{ in } U\}.$$

We will show P is both open and closed in U . Since U is connected and $x_0 \in P \neq \emptyset$, it will follow that $P = U$, which is what we want.

P is open. Let $x \in P$. Since U is open there is $r > 0$ with $D_r(x) \subset U$. Any $y \in D_r(x)$ is joined to x by a straight-line path in $D_r(x)$, and concatenating with a path from x_0 to x shows $y \in P$. So $D_r(x) \subset P$.

$U \setminus P$ is open. Let $x \in U \setminus P$, and again take $D_r(x) \subset U$. If some $y \in D_r(x)$ had a path from x_0 , then concatenating with the segment from y to x would put x in P – contradiction. So $D_r(x) \subset U \setminus P$.

Hence P is open and closed in U , and $P = U$ as claimed. $\square$

§9.5.3 Connected Components

A disconnected space breaks up into natural maximal connected pieces.

Definition 9.5.17 (Connected Component)

Let X be a topological space. Define a relation on X by $x \sim y$ if and only if there is some connected $A \subset X$ with $x, y \in A$. This is an equivalence relation^a, and its equivalence classes are called the **connected components** of X .

^aReflexivity: $\{x\}$ is connected. Symmetry is clear. Transitivity: if $x, y \in A$ and $y, z \in B$ with A, B connected, then $A \cup B$ is connected by Lemma 9.5.8 since $y \in A \cap B$, and it contains x and z .

Proposition 9.5.18 (Properties of Components)

The connected components of a topological space X are precisely the maximal connected subsets of X ; they are closed, and they partition X .

Proof. That they partition X is immediate as they are equivalence classes. Let $C = C_x$ be the component of x .

C is connected. For each $y \in C$ there is a connected $A_y \ni x, y$. Note $A_y \subset C$, since every point of A_y is related to x . Then $C = \bigcup_{y \in C} A_y$, the sets A_y pairwise intersect (all contain x), so C is connected by Lemma 9.5.8.

C is maximal. If $C \subset A$ with A connected, then every $y \in A$ satisfies $y \sim x$, so

$A \subset C$.

C is closed. $\overline{C}$ is connected by Proposition 9.5.6, so by maximality $\overline{C} = C$. $\square$

Example 9.5.19

The components of $\mathbb{R} \setminus \{0\}$ are $(-\infty, 0)$ and $(0, \infty)$. The components of $\mathbb{Q}$ (as a subspace of $\mathbb{R}$) are the singletons: any subset of $\mathbb{Q}$ with two points $p < q$ is disconnected by cutting at an irrational between them. Note this shows components need *not* be open in general.

Since connectedness is a topological invariant, so is the number of connected components, and this gives a genuinely useful way of telling spaces apart. Here is the classic application.

Theorem 9.5.20 ($\mathbb{R} \not\cong \mathbb{R}^n$ for $n \geq 2$)

For $n \geq 2$, the spaces $\mathbb{R}$ and $\mathbb{R}^n$ are not homeomorphic.

Proof. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}^n$ were a homeomorphism. Removing a point, f restricts to a homeomorphism from $\mathbb{R} \setminus \{0\}$ to $\mathbb{R}^n \setminus \{f(0)\}$. But $\mathbb{R} \setminus \{0\}$ is disconnected, while $\mathbb{R}^n \setminus \{\text{point}\}$ for $n \geq 2$ is path-connected (any two points can be joined by a path of at most two straight segments avoiding the removed point) and hence connected. This contradicts connectedness being a topological property. $\square$

Remark. It is true in general that $\mathbb{R}^m \not\cong \mathbb{R}^n$ for $m \neq n$, but this is substantially harder – the standard proofs need tools from algebraic topology.

§9.6 Compactness

Our second major topological property is *compactness*. It is best thought of as the topological version of being finite: compact spaces are the ones on which ‘local’ information can always be patched into ‘global’ information using only finitely many patches.

§9.6.1 Open Covers and Compactness

To motivate the definition, recall from Analysis I that a continuous function on a closed bounded interval is bounded. Let’s think about what makes this work for a continuous $f : X \rightarrow \mathbb{R}$ on a general space. Continuity gives us *local* boundedness for free: every $x \in X$ lies in the open set

$$U_x = f^{-1}((f(x) - 1, f(x) + 1)),$$

and f is bounded on U_x . The sets U_x cover X . If we could always pass to *finitely many* of them, say $U_{x_1}, \dots, U_{x_k}$, still covering X , then f would be bounded on all of X by $\max_j |f(x_j)| + 1$. This finite-subcover property is exactly compactness.

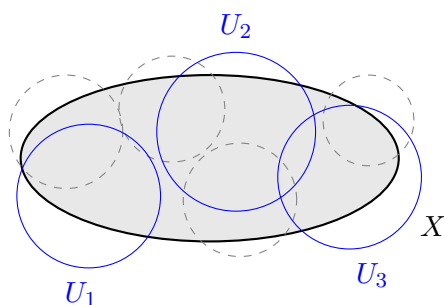
Definition 9.6.1 (Open Cover and Compactness)

Let X be a topological space. An **open cover** of X is a family $\mathcal{U}$ of open subsets

of X with $\bigcup_{U \in \mathcal{U}} U = X$. A **subcover** of $\mathcal{U}$ is a subfamily $\mathcal{V} \subset \mathcal{U}$ which still covers X , and it is a **finite subcover** if $\mathcal{V}$ is finite.

We say X is **compact** if every open cover of X has a finite subcover.

For a subspace $Y \subset X$, it is easy to check (using the definition of the subspace topology) that Y is compact if and only if every family $\mathcal{U}$ of open sets of X whose union contains Y admits a finite subfamily whose union still contains Y . We use both formulations interchangeably.



The picture: out of a possibly enormous open cover (dashed), we can always select finitely many sets (solid) which still cover a compact space.

The motivating discussion above proves most of the following theorem.

Theorem 9.6.2 (Boundedness and the Extreme Value Theorem)

Let X be a compact topological space and $f : X \rightarrow \mathbb{R}$ continuous. Then f is bounded, and if $X \neq \emptyset$ then f attains its bounds.

Proof. For boundedness, cover X by the open sets $U_n = \{x \mid |f(x)| < n\}$ for $n \in \mathbb{N}$ (open as preimages of open intervals). By compactness finitely many suffice, and since the U_n are nested, $X = U_N$ for some N , so $|f| < N$ on X .

Now suppose $X \neq \emptyset$, and let $\alpha = \inf_X f$, which exists as f is bounded. Suppose for contradiction that $f(x) > \alpha$ for all $x \in X$. Then the open sets

$$V_n = f^{-1}\left(\left(\alpha + \frac{1}{n}, \infty\right)\right), \quad n \in \mathbb{N}$$

cover X , so again by compactness and nestedness, $X = V_N$ for some N . But then $f(x) > \alpha + \frac{1}{N}$ for all x , so $\inf_X f \geq \alpha + \frac{1}{N} > \alpha$, a contradiction. The supremum is attained by applying this to $-f$. $\square$

Of course, all this is only useful if we have a good supply of compact spaces. The fundamental example is the closed bounded interval.

Theorem 9.6.3 ($[0, 1]$ is Compact)

The closed interval $[0, 1]$ is compact.

Proof. Let $\mathcal{U}$ be a family of open sets in $\mathbb{R}$ covering $[0, 1]$. Say that $\mathcal{U}$ *finitely covers* a set $A \subset [0, 1]$ if finitely many members of $\mathcal{U}$ cover A ; note that if $A = B \cup C$ and

$\mathcal{U}$ finitely covers B and C , then it finitely covers A .

Suppose $\mathcal{U}$ does not finitely cover $[0, 1]$. Bisecting, at least one of $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$ is not finitely covered; call it $[a_1, b_1]$. Bisecting again and repeating, we get nested intervals

$$[0, 1] \supset [a_1, b_1] \supset [a_2, b_2] \supset \cdots, \quad b_n - a_n = 2^{-n},$$

none of which is finitely covered by $\mathcal{U}$. The sequence a_n is increasing and bounded, so $a_n \rightarrow x$ for some $x \in [0, 1]$, and $b_n = a_n + 2^{-n} \rightarrow x$ too.

Since $\mathcal{U}$ covers $[0, 1]$, there's some $U \in \mathcal{U}$ with $x \in U$, and since U is open, $(x - \varepsilon, x + \varepsilon) \subset U$ for some $\varepsilon > 0$. But for large enough n , both a_n and b_n lie in $(x - \varepsilon, x + \varepsilon)$, so $[a_n, b_n] \subset U$ is covered by the single set U – contradicting that $[a_n, b_n]$ is not finitely covered. $\square$

Example 9.6.4 (Compact and Non-Compact Spaces) (i) Any finite topological space is compact, as is any set with the indiscrete or cofinite topology^a.

(ii) If $x_n \rightarrow x$ in a topological space, then $\{x_n \mid n \in \mathbb{N}\} \cup \{x\}$ is compact: any cover contains some $U \ni x$, which contains all but finitely many of the x_n , and finitely many more sets mop up the rest.

(iii) An infinite discrete space is not compact: the cover by singletons has no finite subcover.

(iv) $\mathbb{R}$ is not compact: the cover by the intervals $(-n, n)$ for $n \in \mathbb{N}$ has no finite subcover. Similarly $(0, 1]$ is not compact, using the cover $(1/n, 2)$ for $n \in \mathbb{N}$ – note how this cover exploits the ‘missing’ endpoint 0.

^aFor the cofinite topology: pick any non-empty U in the cover; its complement is finite, and each missing point needs only one more set.

§9.6.2 Building New Compact Spaces

The next two results tell us how compactness interacts with subspaces, and they display a pleasing duality between ‘compact’ and ‘closed’.

Theorem 9.6.5 (Compactness and Closed Subspaces)

Let Y be a subspace of a topological space X .

- (i) If X is compact and Y is closed in X , then Y is compact.
- (ii) If X is Hausdorff and Y is compact, then Y is closed in X .

Proof. Part (i). Let $\mathcal{U}$ be a family of open sets of X covering Y . Since Y is closed, $\mathcal{U} \cup \{X \setminus Y\}$ is an open cover of X , which by compactness has a finite subcover. Discarding $X \setminus Y$ from it if present, we are left with finitely many members of $\mathcal{U}$ covering Y .

Part (ii). We show $X \setminus Y$ is open. Let $x \in X \setminus Y$. For each $y \in Y$, using the Hausdorff property pick disjoint open sets $U_y \ni x$ and $V_y \ni y$. The V_y cover Y , so

by compactness there is a finite $F \subset Y$ with $Y \subset \bigcup_{y \in F} V_y$. Then

$$U = \bigcap_{y \in F} U_y$$

is a finite intersection of open sets, so open, contains x , and misses every V_y for $y \in F$, hence misses Y . So $X \setminus Y$ is a neighborhood of each of its points, and Y is closed. $\square$

Note that part (ii) genuinely needs the Hausdorff hypothesis: in an infinite indiscrete space, every subset is compact, but only $\emptyset$ and the whole space are closed.

Theorem 9.6.6 (Continuous Images of Compact Spaces)

Let $f : X \rightarrow Y$ be a continuous function between topological spaces. If X is compact, then $f(X)$ is compact.

Proof. Let $\mathcal{U}$ be a family of open sets in Y covering $f(X)$. Then $\{f^{-1}(U) \mid U \in \mathcal{U}\}$ is an open cover of X , so finitely many $f^{-1}(U_1), \dots, f^{-1}(U_k)$ cover X , and then $U_1, \dots, U_k$ cover $f(X)$. $\square$

So compactness is a topological property, and any quotient of a compact space is compact. Also, combining with Theorem 9.6.3: any closed interval $[a, b] \cong [0, 1]$ is compact, and Theorem 9.6.2 recovers the familiar extreme value theorem of Analysis I.

Here is a first genuinely topological payoff. Recall that a continuous bijection need not have a continuous inverse – but compactness fixes this.

Theorem 9.6.7 (Topological Inverse Function Theorem)

Let $f : X \rightarrow Y$ be a continuous bijection from a compact space X to a Hausdorff space Y . Then f^{-1} is continuous, and hence f is a homeomorphism.

Proof. It's enough to show f is an open map, or equivalently (taking complements, f being a bijection) that f maps closed sets to closed sets. So let $K \subset X$ be closed. Since X is compact, K is compact; hence $f(K)$ is compact as a continuous image; hence $f(K)$ is closed in Y , since Y is Hausdorff. Done – notice how the previous two theorems slotted together perfectly. $\square$

Example 9.6.8 ($\mathbb{R}/\mathbb{Z} \cong S^1$)

We can now deliver on our promise from the section on quotients: $\mathbb{R}/\mathbb{Z}$ is homeomorphic to the circle

$$S^1 = \{v \in \mathbb{R}^2 \mid \|v\| = 1\}.$$

Define $f : \mathbb{R} \rightarrow S^1$ by $f(t) = (\cos 2\pi t, \sin 2\pi t)$. Then f is continuous (its components are), surjective, and $f(s) = f(t)$ if and only if $s - t \in \mathbb{Z}$, so f fully respects the relation defining $\mathbb{R}/\mathbb{Z}$. By our results on quotients, f descends to a continuous bijection $\tilde{f} : \mathbb{R}/\mathbb{Z} \rightarrow S^1$.

Now, $\mathbb{R}/\mathbb{Z} = q([0, 1])$ is the image of a compact space under the (continuous) quo-

tion map, so it is compact. Also S^1 is a metric space, hence Hausdorff. So $\tilde{f}$ is a continuous bijection from a compact space to a Hausdorff space, and by the topological inverse function theorem, it is a homeomorphism.

Next, products. The general form of this theorem – for infinite products – is a famous and much deeper result of Tychonov, but for finite products the proof is a pleasant exercise in cover manipulation.

Theorem 9.6.9 (Tychonov's Theorem, Finite Case)

If X and Y are compact topological spaces, then $X \times Y$ is compact.

Proof. Let $\mathcal{U}$ be an open cover of $X \times Y$. We first reduce to covers by boxes: each $z \in X \times Y$ lies in some $W_z \in \mathcal{U}$, and by the definition of the product topology there are open U_z, V_z with $z \in U_z \times V_z \subset W_z$. If finitely many of the boxes $U_z \times V_z$ cover $X \times Y$, then the corresponding W_z form a finite subcover of $\mathcal{U}$. So we may assume every member of $\mathcal{U}$ is a box $U \times V$.

Fix $x \in X$. The slice $\{x\} \times Y$ is the continuous image of Y (under $y \mapsto (x, y)$), so it is compact, and it is covered by $\mathcal{U}$. Hence there are finitely many boxes

$$U_{x,1} \times V_{x,1}, \dots, U_{x,n_x} \times V_{x,n_x} \in \mathcal{U}$$

covering $\{x\} \times Y$, and we may discard any that don't actually meet the slice, so that $x \in U_{x,j}$ for each j . Now let

$$U_x = \bigcap_{j=1}^{n_x} U_{x,j},$$

an open set containing x . The key point is that the same finite family covers the whole 'tube' over U_x :

$$U_x \times Y \subset \bigcup_{j=1}^{n_x} U_{x,j} \times V_{x,j},$$

since for $(u, y) \in U_x \times Y$, the point (x, y) lies in some $U_{x,j} \times V_{x,j}$, and then (u, y) does too as $u \in U_x \subset U_{x,j}$.

Finally, $\{U_x \mid x \in X\}$ is an open cover of the compact space X , so $X = \bigcup_{x \in F} U_x$ for some finite $F \subset X$. Then

$$X \times Y = \bigcup_{x \in F} U_x \times Y \subset \bigcup_{x \in F} \bigcup_{j=1}^{n_x} U_{x,j} \times V_{x,j},$$

a finite subcover. □

By induction, any finite product of compact spaces is compact. Combining everything, we arrive at the definitive description of the compact subsets of Euclidean space.

Theorem 9.6.10 (Heine-Borel)

A subset $K \subset \mathbb{R}^n$ is compact if and only if K is closed and bounded.

Proof. Suppose K is compact. $\mathbb{R}^n$ is a metric space, hence Hausdorff, so K is closed by Theorem 9.6.5. And $x \mapsto \|x\|$ is continuous, so it is bounded on K by Theorem 9.6.2, which says exactly that K is bounded.

Conversely, suppose K is closed and bounded, say $\|x\| \leq M$ for all $x \in K$. Then $K \subset [-M, M]^n$. Each $[-M, M]$ is compact (being homeomorphic to $[0, 1]$), so $[-M, M]^n$ is compact by Tychonov, and K is a closed subset of it, hence compact. $\square$

Example 9.6.11

Closed balls $B_r(x)$ in $\mathbb{R}^n$ are compact, as are spheres. And we can finally tie up a loose end from the proof of the Picard-Lindelöf theorem: a continuous function on $[a, b] \times B_R(y_0) \subset \mathbb{R}^{n+1}$ is bounded, since its domain is closed and bounded, hence compact.

§9.6.3 Sequential Compactness

In metric spaces we made heavy use of the Bolzano-Weierstrass property – every bounded sequence has a convergent subsequence. There is a corresponding abstract notion.

Definition 9.6.12 (Sequentially Compact)

A topological space X is **sequentially compact** if every sequence in X has a subsequence converging to a point of X .

It will be convenient to have some notation: given a sequence x_n and an infinite subset $M = \{m_1 < m_2 < \dots\} \subset \mathbb{N}$, we write $(x_m)_{m \in M}$ for the subsequence $x_{m_1}, x_{m_2}, \dots$. Note that if $L \subset M$ are infinite, then $(x_m)_{m \in L}$ is a subsequence of $(x_m)_{m \in M}$.

Example 9.6.13 (Bolzano-Weierstrass in $\mathbb{R}^n$)

Any closed bounded subset $K \subset \mathbb{R}^n$ is sequentially compact. Indeed, let $x_m = (x_{m,1}, \dots, x_{m,n})$ be a sequence in K ; each coordinate sequence is bounded. By Bolzano-Weierstrass applied to the first coordinate, there's an infinite $M_1 \subset \mathbb{N}$ such that $(x_{m,1})_{m \in M_1}$ converges. The second coordinate restricted to M_1 is still bounded, so there's an infinite $M_2 \subset M_1$ with $(x_{m,2})_{m \in M_2}$ convergent – and note the first coordinate still converges along M_2 . Continuing, we get $M_n \subset \dots \subset M_1$ such that all coordinates converge along M_n , so $(x_m)_{m \in M_n}$ converges in $\mathbb{R}^n$, with limit in K since K is closed.

So for subsets of $\mathbb{R}^n$, sequential compactness and compactness agree (both are ‘closed and bounded’). This is no accident: we will now show that they agree in *every* metric space. (For general topological spaces the two notions genuinely diverge – there are compact spaces that are not sequentially compact and vice versa – but the examples are exotic and beyond this course.)

We will actually prove a three-way equivalence, bringing in completeness as well. The third notion we need is a strengthening of boundedness.

Definition 9.6.14 (Totally Bounded)

Let (M, d) be a metric space. For $\varepsilon > 0$, a subset $F \subset M$ is an **ε -net** for M if

$$M = \bigcup_{y \in F} B_\varepsilon(y),$$

that is, every point of M is within ε of some point of F . We say M is **totally bounded** if for every $\varepsilon > 0$ there is a *finite* ε -net for M .

Totally bounded implies bounded, but not conversely: an infinite set with the discrete metric is bounded but has no finite $\frac{1}{2}$ -net. Similarly the closed unit ball of $\ell_\infty(\mathbb{N})$ is bounded but not totally bounded.

Definition 9.6.15 (Diameter)

For a non-empty subset A of a metric space, the **diameter** of A is $\text{diam } A = \sup\{d(x, y) \mid x, y \in A\}$.

Lemma 9.6.16 (Chopping Up a Totally Bounded Space)

Let M be a totally bounded metric space, let $A \subset M$ be non-empty and closed, and let $\varepsilon > 0$. Then we can write $A = B_1 \cup \cdots \cup B_K$ where each B_k is non-empty and closed with $\text{diam } B_k \leq \varepsilon$.

Proof. Let F be a finite $\frac{\varepsilon}{2}$ -net for M , so that $A = \bigcup_{x \in F} (A \cap B_{\varepsilon/2}(x))$. Throw away the x for which the intersection is empty, and for the rest let $B_x = A \cap B_{\varepsilon/2}(x)$. Each B_x is non-empty, closed (an intersection of closed sets), has $\text{diam } B_x \leq \text{diam } B_{\varepsilon/2}(x) \leq \varepsilon$, and together they cover A . $\square$

Theorem 9.6.17 (The Big Equivalence)

For a metric space (M, d) , the following are equivalent:

- (i) M is compact;
- (ii) M is sequentially compact;
- (iii) M is complete and totally bounded.

Proof. (i) implies (ii). Let x_n be a sequence in M , and for each n let $T_n = \{x_k \mid k > n\}$ be its n -th tail. We first claim $\bigcap_n \overline{T_n} \neq \emptyset$. If not, then the open sets $M \setminus \overline{T_n}$ cover M , so by compactness finitely many do; since the T_n are decreasing, this means $M = M \setminus \overline{T_N}$ for some N , that is, $\overline{T_N} = \emptyset$ – absurd, since $T_N \neq \emptyset$.

So pick $x \in \bigcap_n \overline{T_n}$. Since $x \in \overline{T_1}$, the ball $D_1(x)$ meets T_1 , so there is $k_1 > 1$ with $d(x_{k_1}, x) < 1$. Since $x \in \overline{T_{k_1}}$, the ball $D_{1/2}(x)$ meets T_{k_1} , giving $k_2 > k_1$ with $d(x_{k_2}, x) < \frac{1}{2}$. Continuing inductively, we get $k_1 < k_2 < \cdots$ with $d(x_{k_n}, x) < \frac{1}{n}$, so $x_{k_n} \rightarrow x$.

(ii) implies (iii). For completeness, let x_n be Cauchy, and use sequential compactness to extract a subsequence $x_{k_n} \rightarrow x$. Given $\varepsilon > 0$, take N so that $d(x_m, x_n) < \varepsilon$

for $m, n \geq N$. For fixed $n \geq N$, since $k_m \geq m$,

$$d(x_n, x) \leq d(x_n, x_{k_m}) + d(x_{k_m}, x) \leq \varepsilon + d(x_{k_m}, x) \rightarrow \varepsilon$$

as $m \rightarrow \infty$. So $d(x_n, x) \leq \varepsilon$ for all $n \geq N$, and $x_n \rightarrow x$.

For total boundedness, suppose it fails for some $\varepsilon > 0$: no finite set is an ε -net. Then we can inductively pick $x_1 \in M$ and

$$x_n \in M \setminus \bigcup_{j=1}^{n-1} B_\varepsilon(x_j),$$

since the union on the right never covers M . The resulting sequence has $d(x_m, x_n) > \varepsilon$ for all $m \neq n$, so it has no Cauchy subsequence, and hence no convergent subsequence – contradicting sequential compactness.

(iii) implies (i). Let $\mathcal{U}$ be an open cover of M , and suppose for contradiction that no finite subfamily covers M . We construct a nested sequence of ‘bad’ sets: let $A_0 = M$, and suppose we have found a non-empty closed A_{n-1} which is not finitely covered by $\mathcal{U}$. By Lemma 9.6.16 we can write A_{n-1} as a finite union of non-empty closed sets of diameter at most $\frac{1}{n}$; if each piece were finitely covered, so would A_{n-1} be, so some piece is not finitely covered – call it A_n . This gives non-empty closed sets

$$M = A_0 \supset A_1 \supset A_2 \supset \cdots, \quad \text{diam } A_n \leq \frac{1}{n},$$

with no A_n finitely covered by $\mathcal{U}$.

Now pick $x_n \in A_n$ for each n . For $m, n \geq N$ both x_m, x_n lie in A_N , so $d(x_m, x_n) \leq \frac{1}{N}$, and the sequence is Cauchy. By completeness $x_n \rightarrow x$ for some $x \in M$; since each A_N is closed and contains the tail of the sequence, $x \in A_N$ for every N . Take $U \in \mathcal{U}$ with $x \in U$ and $r > 0$ with $D_r(x) \subset U$. Choosing n with $d(x_n, x) < \frac{r}{2}$ and $\text{diam } A_n < \frac{r}{2}$, every $y \in A_n$ satisfies

$$d(y, x) \leq d(y, x_n) + d(x_n, x) < \frac{r}{2} + \frac{r}{2} = r,$$

so $A_n \subset D_r(x) \subset U$. But then A_n is covered by the single set $U \in \mathcal{U}$, contradicting that it is not finitely covered. $\square$

Remark. Note that (iii) makes the slogan ‘compact = complete + totally bounded’ precise: compactness is completeness plus an approximate finiteness. It also explains why compactness is such a strong hypothesis in analysis. As sanity checks: combining Example 9.6.13 with the theorem re-proves the hard direction of Heine-Borel from Bolzano-Weierstrass, and one can also check directly that a finite product of sequentially compact spaces is sequentially compact, giving an alternative proof of Tychonov’s theorem for metric spaces.

As a final application, we can deliver on a promise made back in our study of uniform continuity: the Bolzano-Weierstrass argument we used there works verbatim on any compact metric space.

Theorem 9.6.18 (Heine-Cantor)

Let $f : M \rightarrow M'$ be a continuous map between metric spaces, with M compact.

Then f is uniformly continuous.

Proof. If not, there is some $\varepsilon > 0$ such that for every n we can find $x_n, y_n \in M$ with $d(x_n, y_n) < \frac{1}{n}$ but $d'(f(x_n), f(y_n)) \geq \varepsilon$. By sequential compactness, passing to a subsequence we may assume $x_n \rightarrow x \in M$, and then $y_n \rightarrow x$ too since $d(x_n, y_n) \rightarrow 0$. By continuity, $f(x_n) \rightarrow f(x)$ and $f(y_n) \rightarrow f(x)$, so

$$d'(f(x_n), f(y_n)) \leq d'(f(x_n), f(x)) + d'(f(x), f(y_n)) \rightarrow 0,$$

contradicting $d'(f(x_n), f(y_n)) \geq \varepsilon$. □

§9.7 Differentiation from $\mathbb{R}^m$ to $\mathbb{R}^n$

For the final part of the course, we return from the abstract world of topological spaces to concrete multivariable analysis, and develop the theory of differentiation for functions $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. The metric and topological machinery we've built – open sets, completeness, connectedness, the contraction mapping theorem – will all earn its keep here.

§9.7.1 The Derivative as a Linear Map

What should it mean for $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ to be differentiable at a point? The one-dimensional definition,

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h},$$

doesn't directly generalize – we can't divide by a vector $h \in \mathbb{R}^m$. The right move is to reformulate. In one dimension, f is differentiable at a with derivative λ if and only if

$$f(a+h) = f(a) + \lambda h + h\varepsilon(h),$$

where $\varepsilon(h) \rightarrow 0$ as $h \rightarrow 0$. Read this way, the derivative is not really a number – it's the *linear map* $h \mapsto \lambda h$ which best approximates the change in f near a . Linear maps make perfect sense between $\mathbb{R}^m$ and $\mathbb{R}^n$, and this is the definition that generalizes.

Before making this precise, we need a tiny bit of setup about the space of linear maps itself. Write $L(\mathbb{R}^m, \mathbb{R}^n)$ for the vector space of linear maps $T : \mathbb{R}^m \rightarrow \mathbb{R}^n$. Fixing standard bases $e_1, \dots, e_m$ of $\mathbb{R}^m$ and $e'_1, \dots, e'_n$ of $\mathbb{R}^n$, each T is represented by the $n \times m$ matrix with entries $T_{ji} = \langle Te_i, e'_j \rangle$, so $L(\mathbb{R}^m, \mathbb{R}^n)$ can be identified with $\mathbb{R}^{nm}$, and we give it the Euclidean norm

$$\|T\| = \left(\sum_{i=1}^m \sum_{j=1}^n T_{ji}^2 \right)^{1/2} = \left(\sum_{i=1}^m \|Te_i\|^2 \right)^{1/2}.$$

In particular $L(\mathbb{R}^m, \mathbb{R}^n)$ is a metric space with $d(S, T) = \|S - T\|$, and everything we know about $\mathbb{R}^{nm}$ (completeness, Heine-Borel, and so on) applies to it.

Lemma 9.7.1 (Linear Maps are Lipschitz)

For $T \in L(\mathbb{R}^m, \mathbb{R}^n)$ and $x \in \mathbb{R}^m$ we have $\|Tx\| \leq \|T\|\|x\|$. Consequently every linear map is Lipschitz, hence continuous. Moreover, for $S \in L(\mathbb{R}^n, \mathbb{R}^p)$ we have

$$\|ST\| \leq \|S\|\|T\|.$$

Proof. Writing $x = \sum_i x_i e_i$, we have $Tx = \sum_i x_i Te_i$, so by the triangle inequality and Cauchy-Schwarz,

$$\|Tx\| \leq \sum_{i=1}^m |x_i| \|Te_i\| \leq \left(\sum_{i=1}^m x_i^2 \right)^{1/2} \left(\sum_{i=1}^m \|Te_i\|^2 \right)^{1/2} = \|x\| \|T\|.$$

Then $\|Tx - Ty\| = \|T(x - y)\| \leq \|T\| \|x - y\|$, so T is Lipschitz. Finally,

$$\|ST\| = \left(\sum_{i=1}^m \|S(Te_i)\|^2 \right)^{1/2} \leq \left(\sum_{i=1}^m \|S\|^2 \|Te_i\|^2 \right)^{1/2} = \|S\| \|T\|. \quad \square$$

Now we can make our main definition. We work on open subsets, since differentiability at a should only depend on the behaviour of f near a .

Definition 9.7.2 (Differentiability)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$ and $a \in U$. We say f is **differentiable at a** if there exists a linear map $T \in L(\mathbb{R}^m, \mathbb{R}^n)$ such that

$$\frac{f(a+h) - f(a) - T(h)}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Equivalently, $f(a+h) = f(a) + T(h) + \|h\| \varepsilon(h)$, where $\varepsilon(h) \rightarrow 0 = \varepsilon(0)$ as $h \rightarrow 0$.

If f is differentiable at every point of U , we say f is **differentiable on U** .

Proposition 9.7.3 (Uniqueness of the Derivative)

The linear map T in the definition above is unique.

Proof. Suppose S and T both work. Subtracting the two conditions gives

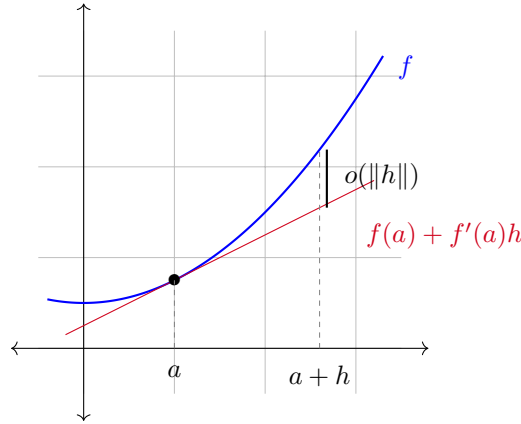
$$\frac{S(h) - T(h)}{\|h\|} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Fix $x \neq 0$ and take $h = x/k$ for $k \in \mathbb{N}$, so $h \rightarrow 0$ as $k \rightarrow \infty$. By linearity, $(S(h) - T(h))/\|h\|$ doesn't depend on k at all – it always equals $(S(x) - T(x))/\|x\|$. So this quantity is 0, and $S(x) = T(x)$ for all x . $\square$

We call this unique T the **derivative** of f at a , and write $f'(a) = Df(a) = T$. So $f'(a)$ is a *linear map* $\mathbb{R}^m \rightarrow \mathbb{R}^n$, and the defining property is

$$f(a+h) = f(a) + f'(a)(h) + o(\|h\|),$$

where as usual $o(\|h\|)$ denotes a quantity which, divided by $\|h\|$, tends to 0 as $h \rightarrow 0$. The right picture is that the affine function $x \mapsto f(a) + f'(a)(x - a)$ is the *best linear approximation* to f near a – its graph is the tangent line (or plane) to the graph of f , and the error dies off faster than linearly.



Remark (Recovering the One-Dimensional Derivative). When $m = 1$, a linear map $\mathbb{R} \rightarrow \mathbb{R}^n$ is just multiplication by a fixed vector $v \in \mathbb{R}^n$, and the condition becomes $(f(a+h) - f(a))/h \rightarrow v$: the familiar limit definition, now taking vector values. So for curves, the derivative-as-linear-map and the derivative-as-vector are the same information.

Let's compute some derivatives from the definition.

Example 9.7.4 (Constant and Linear Maps)

If $f(x) = b$ is constant, then $f(a+h) = f(a) + 0(h) + 0$, so f is differentiable with $f'(a) = 0$ for all a .

If $f(x) = Tx$ for a fixed $T \in L(\mathbb{R}^m, \mathbb{R}^n)$, then $f(a+h) = f(a) + T(h) + 0$, so $f'(a) = T$ for every a – a linear map is its own best linear approximation, everywhere.

Example 9.7.5 (The Norm-Squared Function)

Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be $f(x) = \|x\|^2$. Then

$$f(a+h) = \|a\|^2 + 2\langle a, h \rangle + \|h\|^2 = f(a) + 2\langle a, h \rangle + o(\|h\|),$$

since $\|h\|^2/\|h\| = \|h\| \rightarrow 0$. The map $h \mapsto 2\langle a, h \rangle$ is linear, so f is differentiable with $f'(a)(h) = 2\langle a, h \rangle$.

Example 9.7.6 (Squaring a Matrix)

Let M_n denote the space of $n \times n$ real matrices, identified with $\mathbb{R}^{n^2}$, and let $f : M_n \rightarrow M_n$ be $f(A) = A^2$. Then

$$f(A+H) = A^2 + AH + HA + H^2,$$

and $H \mapsto AH + HA$ is linear, while $\|H^2\| \leq \|H\|^2 = o(\|H\|)$ using Lemma 9.7.1. So $f'(A)(H) = AH + HA$. (For $n = 1$ this is the statement that the derivative of x^2 is multiplication by $2x$ – but note that for $n > 1$ we cannot collapse $AH + HA$ to $2AH$, as matrices don't commute.)

As in one dimension, differentiability is stronger than continuity.

Proposition 9.7.7 (Differentiable Implies Continuous)

If $f : U \rightarrow \mathbb{R}^n$ is differentiable at $a \in U$, then f is continuous at a .

Proof. Write $f(x) = f(a) + f'(a)(x - a) + \|x - a\| \varepsilon(x - a)$. As $x \rightarrow a$: the middle term tends to 0 since linear maps are continuous, and the error term tends to 0. So $f(x) \rightarrow f(a)$. $\square$

The chain rule survives intact, and in fact its statement becomes cleaner than in one dimension: the derivative of a composition is the *composition* of the derivatives.

Theorem 9.7.8 (Chain Rule)

Let $U \subset \mathbb{R}^m$ and $V \subset \mathbb{R}^n$ be open, and let $f : U \rightarrow \mathbb{R}^n$ and $g : V \rightarrow \mathbb{R}^p$ with $f(U) \subset V$. If f is differentiable at a and g is differentiable at $b = f(a)$, then $g \circ f$ is differentiable at a , with

$$(g \circ f)'(a) = g'(b) \circ f'(a).$$

Proof. Write $S = f'(a)$ and $T = g'(b)$, so that

$$f(a + h) = f(a) + S(h) + \|h\| \varepsilon(h), \quad g(b + k) = g(b) + T(k) + \|k\| \zeta(k),$$

with $\varepsilon(h) \rightarrow 0$ and $\zeta(k) \rightarrow 0$. Substituting $k = k(h) = S(h) + \|h\| \varepsilon(h)$,

$$\begin{aligned} (g \circ f)(a + h) &= g(b + k) = g(b) + T(S(h) + \|h\| \varepsilon(h)) + \|k\| \zeta(k) \\ &= (g \circ f)(a) + (T \circ S)(h) + \underbrace{\|h\| T(\varepsilon(h)) + \|k\| \zeta(k)}_{\eta(h)}. \end{aligned}$$

It remains to show $\eta(h) = o(\|h\|)$. For the first piece, $\|T(\varepsilon(h))\| \leq \|T\| \|\varepsilon(h)\| \rightarrow 0$. For the second, note

$$\frac{\|k\|}{\|h\|} \leq \frac{\|S(h)\| + \|h\| \|\varepsilon(h)\|}{\|h\|} \leq \|S\| + \|\varepsilon(h)\|,$$

which is bounded for small h ; and $k \rightarrow 0$ as $h \rightarrow 0$, so $\zeta(k) \rightarrow 0$. Hence

$$\frac{\|\eta(h)\|}{\|h\|} \leq \|T(\varepsilon(h))\| + \frac{\|k\|}{\|h\|} \|\zeta(k)\| \rightarrow 0,$$

as required. $\square$

Just as for continuity, differentiability can be checked component by component.

Proposition 9.7.9 (Components)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$, and write $f_j = \pi_j \circ f$ for the components of f , where $\pi_j : \mathbb{R}^n \rightarrow \mathbb{R}$ is the j -th coordinate projection. Then f is differentiable at a if and only if every f_j is, and in that case

$$f'(a)(h) = \sum_{j=1}^n f'_j(a)(h) e'_j.$$

Proof. If f is differentiable at a , then since π_j is linear (so differentiable, and equal to its own derivative), the chain rule gives $f'_j(a) = \pi_j \circ f'(a)$, which is the claim. Conversely, if each f_j is differentiable at a , write $f_j(a+h) = f_j(a) + f'_j(a)(h) + \|h\|\varepsilon_j(h)$; summing against e'_j ,

$$f(a+h) = f(a) + \sum_{j=1}^n f'_j(a)(h)e'_j + \|h\| \sum_{j=1}^n \varepsilon_j(h)e'_j,$$

and the final error term is $o(\|h\|)$ since each $\varepsilon_j(h) \rightarrow 0$. $\square$

This is a very useful reduction – it means that in many proofs we may assume $n = 1$ without loss of generality. Finally, the usual algebraic rules.

Proposition 9.7.10 (Sums and Products)

Let $U \subset \mathbb{R}^m$ be open and let $f, g : U \rightarrow \mathbb{R}^n$ and $\phi : U \rightarrow \mathbb{R}$ all be differentiable at $a \in U$. Then $f + g$ and ϕf are differentiable at a , with

$$(f + g)'(a) = f'(a) + g'(a), \quad (\phi f)'(a)(h) = \phi(a)f'(a)(h) + [\phi'(a)(h)]f(a).$$

Proof. The sum rule follows immediately by adding the two defining expansions. For the product, expanding $\phi(a+h)f(a+h)$ using both expansions gives

$$(\phi f)(a) + \phi(a)f'(a)(h) + \phi'(a)(h)f(a) + \rho(h),$$

where $\rho(h)$ collects the remaining terms: $\phi'(a)(h) \cdot f'(a)(h)$, and various products involving an error term. The first satisfies $\|\phi'(a)(h)f'(a)(h)\| \leq \|\phi'(a)\|\|f'(a)\|\|h\|^2 = o(\|h\|)$, and the rest are also $o(\|h\|)$ since each contains a factor $\|h\|$ (or a bounded multiple of it) alongside a factor tending to zero. The map $h \mapsto \phi(a)f'(a)(h) + \phi'(a)(h)f(a)$ is linear, so the result follows. Taking $m = n = 1$ recovers the familiar product rule. $\square$

§9.7.2 Partial Derivatives

The definition of the derivative is clean, but to *compute* with it we want to reduce to one-dimensional derivatives. This is what directional and partial derivatives do.

Definition 9.7.11 (Directional and Partial Derivatives)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$, $a \in U$, and let $u \in \mathbb{R}^m$ be non-zero. If the limit

$$D_u f(a) = \lim_{t \rightarrow 0} \frac{f(a + tu) - f(a)}{t}$$

exists (in $\mathbb{R}^n$), we call it the **directional derivative** of f at a in the direction u .

When $u = e_i$ is a standard basis vector, we write $D_i f(a) = D_{e_i} f(a)$ and call it the **i -th partial derivative** of f at a . For scalar components we use the familiar notation $D_i f_j(a) = \frac{\partial f_j}{\partial x_i}(a)$.

Note that $D_u f(a)$ is exactly the derivative at 0 of the one-variable function $t \mapsto f(a + tu)$

– we restrict f to a line through a and differentiate along it.

Proposition 9.7.12 (The Derivative Determines All Directional Derivatives)

If f is differentiable at a , then $D_u f(a)$ exists for every direction $u \neq 0$, and

$$D_u f(a) = f'(a)(u).$$

In particular, writing $h = \sum_i h_i e_i$,

$$f'(a)(h) = \sum_{i=1}^m h_i D_i f(a),$$

so the derivative is determined by the m partial derivatives.

Proof. Setting $h = tu$ in the definition of differentiability,

$$\frac{f(a + tu) - f(a)}{t} = f'(a)(u) + \frac{|t|}{t} \|u\| \varepsilon(tu),$$

and the error term tends to 0 as $t \rightarrow 0$ (the factor $|t|/t$ is bounded). The second claim follows from linearity of $f'(a)$. $\square$

Definition 9.7.13 (Jacobian Matrix)

If f is differentiable at a , the matrix of $f'(a)$ with respect to the standard bases is called the **Jacobian matrix** $J_f(a)$. By the proposition, its i -th column is $D_i f(a)$, so its entries are

$$(J_f(a))_{ji} = D_i f_j(a) = \frac{\partial f_j}{\partial x_i}(a).$$

It is important that the converse of Proposition 9.7.12 fails: a function can have all directional derivatives at a point without being differentiable there.

Example 9.7.14 (Directional Derivatives Are Not Enough)

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \frac{x^3}{x^2 + y^2} \text{ for } (x, y) \neq (0, 0), \quad f(0, 0) = 0.$$

For any direction $u = (u_1, u_2)$, we have

$$\frac{f(tu) - f(0)}{t} = \frac{1}{t} \cdot \frac{t^3 u_1^3}{t^2 u_1^2 + t^2 u_2^2} = \frac{u_1^3}{\|u\|^2},$$

so all directional derivatives at the origin exist (and equal this value). But $u \mapsto u_1^3/\|u\|^2$ is *not linear* in u , so by Proposition 9.7.12, f cannot be differentiable at the origin.

However, if the partial derivatives exist *near* a and are continuous *at* a , then we do get differentiability. This is the workhorse criterion for checking differentiability in practice.

Theorem 9.7.15 (Continuous Partial Derivatives Imply Differentiability)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$, and $a \in U$. Suppose there is an open neighborhood $V \subset U$ of a on which all partial derivatives $D_i f$ exist, and that each $D_i f : V \rightarrow \mathbb{R}^n$ is continuous at a . Then f is differentiable at a .

Proof. By Proposition 9.7.9 we may assume $n = 1$. We also present the proof for $m = 2$ – the general case works in exactly the same way, one coordinate at a time, at the cost of heavier notation. Write $a = (p, q)$, and let

$$\psi(h, k) = f(p + h, q + k) - f(p, q) - hD_1f(p, q) - kD_2f(p, q).$$

We must show $\psi(h, k) = o(\|(h, k)\|)$; then f is differentiable at a with derivative $(h, k) \mapsto hD_1f(a) + kD_2f(a)$. The trick is to move from (p, q) to $(p + h, q + k)$ one coordinate at a time:

$$\psi(h, k) = \underbrace{[f(p + h, q + k) - f(p + h, q) - kD_2f(p, q)]}_{(1)} + \underbrace{[f(p + h, q) - f(p, q) - hD_1f(p, q)]}_{(2)}.$$

Term (2) is $o(|h|)$, hence $o(\|(h, k)\|)$, directly by the definition of $D_1f(p, q)$.

For term (1), fix h, k small and consider $\phi(t) = f(p + h, q + tk)$ for $t \in [0, 1]$. Then ϕ is differentiable with $\phi'(t) = kD_2f(p + h, q + tk)$, so by the one-dimensional mean value theorem there's some $\theta = \theta(h, k) \in (0, 1)$ with

$$f(p + h, q + k) - f(p + h, q) = \phi(1) - \phi(0) = kD_2f(p + h, q + \theta k).$$

Hence term (1) equals $k[D_2f(p + h, q + \theta k) - D_2f(p, q)]$. As $(h, k) \rightarrow 0$, the point $(p + h, q + \theta k) \rightarrow (p, q)$, so by continuity of D_2f at a , the bracket tends to 0, and term (1) is $o(|k|) = o(\|(h, k)\|)$. $\square$

Definition 9.7.16 (C^1 Function)

Let $U \subset \mathbb{R}^m$ be open. A function $f : U \rightarrow \mathbb{R}^n$ is **C^1 on U** (or **continuously differentiable**) if f is differentiable on U and the derivative map $f' : U \rightarrow L(\mathbb{R}^m, \mathbb{R}^n)$ is continuous.

Since the entries of the Jacobian are the partial derivatives, and convergence in $L(\mathbb{R}^m, \mathbb{R}^n) \cong \mathbb{R}^{nm}$ is entry-wise, f is C^1 on U if and only if all partial derivatives $D_i f_j$ exist and are continuous on U – combining Theorem 9.7.15 with Proposition 9.7.12.

§9.7.3 The Mean Value Inequality

The mean value *theorem* – that $f(b) - f(a) = f'(c)(b - a)$ for some intermediate c – fails for vector-valued functions; different components want different intermediate points⁵. But in analysis we almost always used the mean value theorem to *bound* increments, and the bound survives beautifully.

⁵Consider $f(t) = (\cos t, \sin t)$: we have $f(2\pi) - f(0) = 0$, but f' never vanishes.

Theorem 9.7.17 (Mean Value Inequality)

Let $U \subset \mathbb{R}^m$ be open and $f : U \rightarrow \mathbb{R}^n$ differentiable on U . Suppose $a, b \in U$ are such that the segment $[a, b] = \{(1-t)a + tb \mid t \in [0, 1]\}$ lies in U , and that $\|f'(z)\| \leq M$ for all $z \in [a, b]$. Then

$$\|f(b) - f(a)\| \leq M\|b - a\|.$$

Proof. Let $u = b - a$ (we may assume $u \neq 0$) and let $v = f(b) - f(a)$. The trick is to convert to a scalar problem by projecting onto v : define

$$\phi(t) = \langle f(a + tu), v \rangle, \quad t \in [0, 1].$$

By the chain rule (composing $t \mapsto a + tu$ with f , then with the linear map $\langle \cdot, v \rangle$), ϕ is differentiable with

$$\phi'(t) = \langle f'(a + tu)(u), v \rangle.$$

By the one-dimensional mean value theorem, there's some $\theta \in (0, 1)$ with $\phi(1) - \phi(0) = \phi'(\theta)$. But $\phi(1) - \phi(0) = \langle f(b) - f(a), v \rangle = \|v\|^2$, so by Cauchy-Schwarz and Lemma 9.7.1,

$$\|v\|^2 = \langle f'(a + \theta u)(u), v \rangle \leq \|f'(a + \theta u)\| \|u\| \|v\| \leq M\|b - a\| \|v\|.$$

If $v = 0$ there is nothing to prove; otherwise divide by $\|v\|$. □

Here is the promised payoff from connectedness.

Corollary 9.7.18 (Zero Derivative Implies Constant)

Let $U \subset \mathbb{R}^m$ be open and *connected*, and let $f : U \rightarrow \mathbb{R}^n$ be differentiable with $f'(a) = 0$ for all $a \in U$. Then f is constant.

Proof. For any $x \in U$, take $r > 0$ with $D_r(x) \subset U$. The ball is convex, so for $y \in D_r(x)$ the segment $[x, y]$ lies in U , and the mean value inequality with $M = 0$ gives $f(y) = f(x)$. So f is *locally constant*.

Now fix $x_0 \in U$ and let $P = \{x \in U \mid f(x) = f(x_0)\}$. By local constancy, both P and $U \setminus P$ are open. Since U is connected and $P \ni x_0$ is non-empty, we must have $P = U$. □

Note that connectedness is genuinely needed – on a disconnected open set, a function can be constant on each piece with different values.

§9.7.4 Second Derivatives

If $f : U \rightarrow \mathbb{R}^n$ is differentiable on U , its derivative is itself a function $f' : U \rightarrow L(\mathbb{R}^m, \mathbb{R}^n)$ into the (finite dimensional, normed) space of linear maps, and we can ask for *its* derivative.

Definition 9.7.19 (Second Derivative)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$, and $a \in U$. Suppose f is differentiable on some open neighborhood $V \subset U$ of a . We say f is **twice differentiable at a** if

$f' : V \rightarrow L(\mathbb{R}^m, \mathbb{R}^n)$ is differentiable at a , and write $f''(a)$ for its derivative, so

$$f''(a) \in L(\mathbb{R}^m, L(\mathbb{R}^m, \mathbb{R}^n)).$$

An element of $L(\mathbb{R}^m, L(\mathbb{R}^m, \mathbb{R}^n))$ is awkward to think about directly, but there is a standard way out: such a gadget is the same thing as a *bilinear map* $\mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^n$. Explicitly, if $T \in L(\mathbb{R}^m, L(\mathbb{R}^m, \mathbb{R}^n))$, we identify it with the bilinear map $(h, k) \mapsto T(h)(k)$, and we will freely write $f''(a)(h, k)$ for $(f''(a)(h))(k)$.

Proposition 9.7.20 (Second Derivative as a Bilinear Approximation)

In the setting above, f is twice differentiable at a if and only if there is a bilinear map $T : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ such that for every fixed $k \in \mathbb{R}^m$,

$$f'(a+h)(k) = f'(a)(k) + T(h, k) + o(\|h\|) \quad \text{as } h \rightarrow 0,$$

and in that case $T = f''(a)$.

Proof. If f is twice differentiable at a , then $f'(a+h) = f'(a) + f''(a)(h) + \|h\|\varepsilon(h)$ where $\varepsilon(h) \rightarrow 0$ in $L(\mathbb{R}^m, \mathbb{R}^n)$. Evaluating at k and using $\|\varepsilon(h)(k)\| \leq \|\varepsilon(h)\|\|k\| \rightarrow 0$ gives the displayed formula.

Conversely, suppose the formula holds for each fixed k , and let

$$\varepsilon(h) = \frac{f'(a+h) - f'(a) - T(h)}{\|h\|} \in L(\mathbb{R}^m, \mathbb{R}^n).$$

By hypothesis $\varepsilon(h)(k) \rightarrow 0$ for each fixed k ; taking $k = e_1, \dots, e_m$ and using

$$\|\varepsilon(h)\| = \left(\sum_{i=1}^m \|\varepsilon(h)(e_i)\|^2 \right)^{1/2} \rightarrow 0,$$

we get $\varepsilon(h) \rightarrow 0$ in $L(\mathbb{R}^m, \mathbb{R}^n)$, which is twice differentiability with $f''(a) = T$. $\square$

Example 9.7.21

If f is linear then f' is the constant function $a \mapsto f$, so $f''(a) = 0$ everywhere. If $f : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^p$ is *bilinear*, then a similar computation to the matrix-squaring example shows $f'(a, b)(h, k) = f(a, k) + f(h, b)$, which is *linear* as a function of (a, b) – so f'' is constant, equal to the bilinear map $((h, k), (h', k')) \mapsto f(h, k') + f(h', k)$.

Example 9.7.22 (Cubing a Matrix)

Let $f : M_n \rightarrow M_n$ be $f(A) = A^3$. Expanding $(A+H)^3$ and collecting terms of degree 1 in H ,

$$f'(A)(H) = A^2H + AHA + HA^2,$$

with the degree- ≥ 2 terms being $o(\|H\|)$. Then expanding $f'(A+H)(K) - f'(A)(K)$ and collecting the terms linear in H ,

$$f''(A)(H, K) = AHK + HAK + AKH + HKA + KAH + KHA,$$

which is indeed bilinear in (H, K) , and symmetric under swapping H and K . In one dimension these formulas collapse to the familiar $f'(a) = 3a^2$ and $f''(a) = 6a$.

The symmetry visible in that example is no coincidence. To see what it means concretely, evaluate the second derivative on directions: if f is twice differentiable at a , then for fixed v , the map $x \mapsto D_v f(x) = f'(x)(v)$ is differentiable at a , and by the proposition above,

$$D_u D_v f(a) = (D_v f)'(a)(u) = f''(a)(u, v).$$

In particular $D_i D_j f(a) = f''(a)(e_i, e_j)$: second partial derivatives are the entries of the second derivative.

Theorem 9.7.23 (Symmetry of Second Derivatives)

Let $U \subset \mathbb{R}^m$ be open, $f : U \rightarrow \mathbb{R}^n$, and $a \in U$. Suppose f is twice differentiable on an open V with $a \in V \subset U$, and that $f'' : V \rightarrow \text{Bil}(\mathbb{R}^m \times \mathbb{R}^m, \mathbb{R}^n)$ is continuous at a . Then for all u, v ,

$$f''(a)(u, v) = f''(a)(v, u),$$

that is, $f''(a)$ is a symmetric bilinear map. Equivalently, $D_u D_v f(a) = D_v D_u f(a)$.

Proof. Considering components (which commute with everything in sight), we may assume $n = 1$. The key object is the *second difference*

$$\phi(s, t) = f(a + su + tv) - f(a + tv) - f(a + su) + f(a),$$

for small $s, t > 0$ – the alternating sum of f over the vertices of the parallelogram spanned by su and tv . It is manifestly symmetric between the roles of (s, u) and (t, v) , and we will compute its size in two ways.

Fix s, t and let $\psi(y) = f(a + yu + tv) - f(a + yu)$, so that $\phi(s, t) = \psi(s) - \psi(0)$. By the mean value theorem, there's some $\alpha \in (0, 1)$ with

$$\phi(s, t) = s\psi'(\alpha s) = s[D_u f(a + \alpha su + tv) - D_u f(a + \alpha su)].$$

Applying the mean value theorem again, to $y \mapsto D_u f(a + \alpha su + yv)$, gives $\beta \in (0, 1)$ with

$$\phi(s, t) = st D_v D_u f(a + \alpha su + \beta tv) = st f''(a + \alpha su + \beta tv)(v, u).$$

Since f'' is continuous at a , letting $s, t \rightarrow 0$ gives

$$\frac{\phi(s, t)}{st} \rightarrow f''(a)(v, u).$$

But by symmetry of ϕ , running the same argument with $\psi(y) = f(a + su + yv) - f(a + yv)$ instead gives

$$\frac{\phi(s, t)}{st} \rightarrow f''(a)(u, v).$$

Limits are unique, so $f''(a)(u, v) = f''(a)(v, u)$. □

Remark. Taking $u = e_i$, $v = e_j$, this is the familiar statement that mixed partial derivatives commute,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(a) = \frac{\partial^2 f}{\partial x_j \partial x_i}(a),$$

for (say) twice continuously differentiable f – so for scalar f , the *Hessian matrix* $(D_i D_j f(a))_{ij}$ is symmetric. Some continuity hypothesis is genuinely needed; without it there are standard counterexamples where the two mixed partials at a point disagree.

§9.7.5 The Inverse Function Theorem

We finish the course with a genuine highlight – a theorem that combines nearly everything we have done: completeness and the contraction mapping theorem, open sets and continuity, the mean value inequality, and the derivative as a linear map.

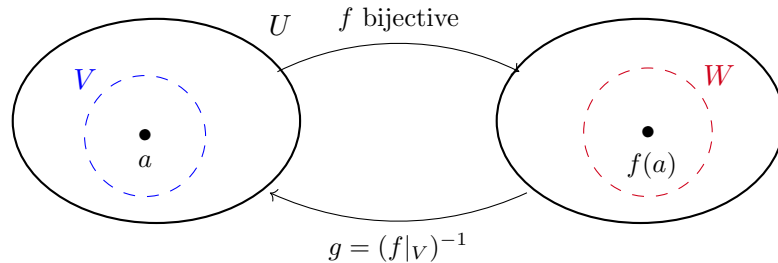
The question is: when does a function f have a differentiable inverse? There is an obvious necessary condition. If $f : V \rightarrow W$ is a bijection between open sets, differentiable at a with differentiable inverse g at $f(a)$, then differentiating $g \circ f = \text{id}$ and $f \circ g = \text{id}$ with the chain rule shows that $g'(f(a)) \circ f'(a)$ and $f'(a) \circ g'(f(a))$ are both identity maps – so $f'(a)$ must be an *invertible* linear map⁶.

The inverse function theorem says that, for C^1 functions, this necessary condition is *locally sufficient*: wherever the derivative is invertible, the function itself is invertible near that point, with C^1 inverse.

Theorem 9.7.24 (Inverse Function Theorem)

Let $U \subset \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^n$ a C^1 function. Let $a \in U$ and suppose $f'(a)$ is invertible. Then there exist open sets $V \ni a$ and $W \ni f(a)$, with $V \subset U$, such that $f|_V : V \rightarrow W$ is a bijection, and its inverse $g : W \rightarrow V$ is C^1 with

$$g'(y) = [f'(g(y))]^{-1} \quad \text{for all } y \in W.$$



Proof. We proceed in several steps.

Step 0: normalisation. We first argue that we may assume $a = 0$, $f(a) = 0$ and $f'(a) = I$, the identity map. Indeed, let $T = f'(a)$ and set

$$h(x) = T^{-1}(f(x+a) - f(a)),$$

defined on the open set $U - a \ni 0$. By the chain rule (linear maps being their own derivatives), h is differentiable with $h'(x) = T^{-1} \circ f'(x+a)$, so $h(0) = 0$ and $h'(0) = I$; moreover

$$\|h'(x) - h'(y)\| \leq \|T^{-1}\| \|f'(x+a) - f'(y+a)\|,$$

⁶Taking traces of the two identities also shows $m = n$: differentially-invertible maps only exist between spaces of equal dimension.

so h is also C^1 . If the theorem holds for h , then it holds for f , since $f(x) = T(h(x - a)) + f(a)$ and composing with the fixed homeomorphisms $x \mapsto x - a$, $y \mapsto T(y) + f(a)$ (whose derivatives are invertible) transports all the conclusions back. So assume $f(0) = 0$ and $f'(0) = I$.

Step 1: f doesn't collapse distances near 0. Since f' is continuous, there is $r > 0$ such that $B_r(0) \subset U$ and

$$\|f'(x) - I\| \leq \frac{1}{2} \quad \text{for all } x \in B_r(0).$$

Consider $p(x) = f(x) - x$, so $p'(x) = f'(x) - I$ has $\|p'(x)\| \leq \frac{1}{2}$ on $B_r(0)$. The ball is convex, so the mean value inequality gives, for $x, y \in B_r(0)$,

$$\|p(x) - p(y)\| \leq \frac{1}{2}\|x - y\|,$$

and hence

$$\|f(x) - f(y)\| \geq \|x - y\| - \|p(x) - p(y)\| \geq \frac{1}{2}\|x - y\|.$$

In particular f is injective on $B_r(0)$.

Step 2: f hits everything near 0. Let $s = r/2$. We claim that for every $w \in D_s(0)$ there is a unique $x \in D_r(0)$ with $f(x) = w$. Fix such a w and define

$$q(x) = w - p(x) = w - f(x) + x,$$

so that $f(x) = w$ if and only if $q(x) = x$: solving our equation means finding a fixed point of q . Now for $x \in B_r(0)$, using $p(0) = 0$,

$$\|q(x)\| \leq \|w\| + \|p(x) - p(0)\| < s + \frac{1}{2}\|x\| \leq \frac{r}{2} + \frac{r}{2} = r,$$

so q maps $B_r(0)$ into $D_r(0) \subset B_r(0)$. Moreover, for $x, y \in B_r(0)$,

$$\|q(x) - q(y)\| = \|p(x) - p(y)\| \leq \frac{1}{2}\|x - y\|,$$

so q is a contraction of the non-empty *complete* metric space $B_r(0)$ (closed and bounded in $\mathbb{R}^n$, hence complete). By the contraction mapping theorem, q has a unique fixed point $x \in B_r(0)$, and since q maps into the open ball, in fact $x \in D_r(0)$. This proves the claim.

Step 3: the local inverse, and its continuity. Let

$$W = D_s(0), \quad V = D_r(0) \cap f^{-1}(W).$$

Then W is open, and V is open as an intersection of open sets (f being continuous), with $0 \in V$. By Step 2, $f|_V : V \rightarrow W$ is a bijection; call its inverse $g : W \rightarrow V$. For $u, v \in W$, writing $x = g(u)$, $y = g(v)$, Step 1 gives

$$\|g(u) - g(v)\| = \|x - y\| \leq 2\|f(x) - f(y)\| = 2\|u - v\|,$$

so g is 2-Lipschitz, and in particular continuous.

Step 4: differentiability of the inverse. First note that for $x \in B_r(0)$, the linear map $f'(x)$ is invertible: if $f'(x)v = 0$ then

$$\|v\| = \|(I - f'(x))v\| \leq \|I - f'(x)\|\|v\| \leq \frac{1}{2}\|v\|,$$

so $v = 0$, and an injective linear map $\mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible.

Now fix $y \in W$ and let $x = g(y) \in V$, $S = f'(x)$. Let k be small enough that $y + k \in W$, and set $h = g(y + k) - g(y)$, so that $f(x + h) = y + k$ and, by the Lipschitz bound on g , $\|h\| \leq 2\|k\|$. Differentiability of f at x gives

$$k = f(x + h) - f(x) = S(h) + \|h\|\varepsilon(h), \quad \varepsilon(h) \rightarrow 0 \text{ as } h \rightarrow 0.$$

Applying S^{-1} ,

$$g(y + k) - g(y) - S^{-1}(k) = h - S^{-1}(S(h) + \|h\|\varepsilon(h)) = -\|h\|S^{-1}(\varepsilon(h)),$$

and hence

$$\frac{\|g(y + k) - g(y) - S^{-1}(k)\|}{\|k\|} \leq \frac{\|h\|}{\|k\|} \|S^{-1}\| \|\varepsilon(h)\| \leq 2\|S^{-1}\| \|\varepsilon(h)\|.$$

As $k \rightarrow 0$ we have $h \rightarrow 0$ (continuity of g), so the right hand side tends to 0. This shows g is differentiable at y with

$$g'(y) = S^{-1} = [f'(g(y))]^{-1}.$$

Finally, g' is continuous: it is the composition of the continuous maps g (Step 3), f' (as f is C^1), and matrix inversion, which is continuous on the set of invertible matrices since the entries of A^{-1} are rational functions (with non-vanishing denominator $\det A$) of the entries of A . So g is C^1 , completing the proof. $\square$

Remark. A couple of closing remarks on this theorem.

- (i) The conclusion is genuinely *local*. The map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given in complex notation by $z \mapsto e^z$ has everywhere-invertible derivative, and is locally invertible near every point, but it is not injective globally (as $e^{z+2\pi i} = e^z$).
- (ii) The inverse function theorem is the gateway to the *implicit function theorem* and to the theory of manifolds and submanifolds – it is the tool that lets us straighten out nonlinear maps locally, and it will reappear throughout courses in differential geometry and analysis.

10 Complex Analysis

Complex analysis is the study of differentiable functions of a complex variable. Asking for a function $\mathbb{C} \rightarrow \mathbb{C}$ to be differentiable turns out to be a shockingly strong requirement – such functions are automatically infinitely differentiable, are determined on huge regions by their values on tiny ones, and their integrals along closed curves detect delicate information about singularities. The theory that develops from this single definition is remarkably rigid, remarkably beautiful, and (as we will see towards the end of these notes) genuinely useful for down-to-earth tasks like computing real integrals.

This article constitutes my notes for the ‘Complex Analysis’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§10.1 Basic Notions

§10.1.1 Complex Differentiation

This course is about complex valued functions of a (single) complex variable, that is, functions

$$f : A \rightarrow \mathbb{C}, \quad \text{where } A \subseteq \mathbb{C}.$$

In this course we will focus on differentiable functions f (the definition of which we will give soon), but let’s start by recalling the notion of continuity.

Definition 10.1.1 (Continuity)

A function $f : A \rightarrow \mathbb{C}$ is **continuous** at a point $w \in A$ if for all $\varepsilon > 0$ there exists some $\delta > 0$ such that if $z \in A$ with $|z - w| < \delta$ then $|f(z) - f(w)| < \varepsilon$.

Note that by identifying $\mathbb{C}$ with $\mathbb{R}^2$ in the usual way, we can write $f(z) = u(x, y) + iv(x, y)$ for $z = x + iy \in A$ with $u, v : A \rightarrow \mathbb{R}$. With this, the triangle inequality implies that f is continuous at $w = c + id$ if and only if u, v are continuous at (c, d) .

We can now define the notion of differentiation for complex functions.

Definition 10.1.2 (Complex Differentiation)

Let $f : U \rightarrow \mathbb{C}$ where $U \subseteq \mathbb{C}$ is open. We say that f is **differentiable** at $w \in U$ if the limit

$$f'(w) = \lim_{z \rightarrow w} \frac{f(z) - f(w)}{z - w}$$

exists as a complex number.

Definition 10.1.3 (Holomorphic)

We say that f is **holomorphic at** $w \in U$ if there is $\varepsilon > 0$ such that $D(w, \varepsilon) \subset U$ where f is differentiable at every point in $D(w, \varepsilon)$. We say that f is **holomorphic**

if it is holomorphic at every point in U .

Remark. Sometime authors use the word ‘analytic’ instead of the word ‘holomorphic’.

Definition 10.1.4 (Entire)

If $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic on all of $\mathbb{C}$, we say f is **entire**.

It’s straightforward to check that the same rules of differentiation for real variables hold for complex functions (product rule, quotient rule, chain rule, etc), which makes computing complex derivatives relatively straightforward.

Example 10.1.5 (Complex Differentiable Functions)

- (i) Polynomials $p(z) = \sum_{k=0}^n a_k z^k$, $a_0, \dots, a_n \in \mathbb{C}$ are holomorphic on all of $\mathbb{C}$.
- (ii) If p and q are polynomials, then p/q is holomorphic on $\mathbb{C} \setminus \{z \mid q(z) = 0\}$.

We saw above that the notion of continuity is basically the same for both real and complex variables. A natural question is then if the same is true for complex differentiability. Indeed, the answer is no! Complex differentiability is a much stronger condition than asserting real differentiability on the real and imaginary components of a function.

We do however have a criterion for establishing complex differentiability from the real and imaginary parts.

Theorem 10.1.6 (Cauchy-Riemann Equations)

A function $f = u + iv : U \rightarrow \mathbb{C}$ is differentiable at $w = c + id \in U$ if and only if $u, v : U \rightarrow \mathbb{R}$ are differentiable at $(c, d) \in U$ and u, v satisfy the **Cauchy-Riemann** equations at (c, d) :

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \quad \text{at } (c, d). \end{aligned}$$

If f is differentiable at $w = c + id$, then $f'(w) = \frac{\partial u}{\partial x}(c, d) + i \frac{\partial v}{\partial x}(c, d)$.

Proof. The function f is differentiable at w with derivative $f'(w) = p + iq$ if and only if

$$\lim_{z \rightarrow w} \frac{f(z) - f(w)}{z - w} = p + iq \iff \lim_{z \rightarrow w} \frac{f(z) - f(w) - (z - w)(p + iq)}{|z - w|} = 0.$$

Separating real and imaginary parts and writing $f = u + iv$, the above holds if and only if

$$\lim_{(x,y) \rightarrow (c,d)} \frac{u(x, y) - u(c, d) - p(x - c) + q(y - d)}{\sqrt{(x - c)^2 + (y - d)^2}} = 0$$

and

$$\lim_{(x,y) \rightarrow (c,d)} \frac{v(x, y) - v(c, d) - q(x - c) - p(y - d)}{\sqrt{(x - c)^2 + (y - d)^2}} = 0.$$

That is, if and only if u is differentiable at (c, d) with $Du(c, d)(x, y) = px - qy$, and v is differentiable at (c, d) with $Dv(c, d)(x, y) = qx + py$.

Taking partial derivatives, this is true if and only if u, v are differentiable at (c, d) and $u_x(c, d) = p = v_y(c, d)$ and $u_y(c, d) = -q = -v_x(c, d)$, that is, the Cauchy-Riemann equations hold at (c, d) .

We also get from the above that if f is differentiable at w , then $f'(w) = p + iq = u_x(c, d) + iv_x(c, d)$. $\square$

Remark (Warning). The real and imaginary parts u, v satisfying the Cauchy-Riemann equations at a point does *not* guarantee the differentiability of f , and a counterexample is given on the first example sheet.

We can easily use this criterion to show that many functions are not complex differentiable.

Example 10.1.7 (Conjugation isn't Differentiable)

Consider the function $f(z) = \bar{z} = x - iy$. For this, $u(x, y) = x$ and $v(x, y) = -y$, so $u_x = 1$ and $u_y = 0$, $v_x = 0$ and $v_y = -1$. So the Cauchy-Riemann equations do not hold anywhere, and f is not differentiable at any point.

Now, if we just wanted to show that the differentiability of $f = u + iv$ at $w = c + id$ implies that the partial derivatives u_x, u_y, v_x and v_y exist at (c, d) and satisfy the Cauchy-Riemann equations, then we can proceed more simply as follows.

Begin with the definition of differentiability

$$f'(w) = \lim_{h \rightarrow 0} \frac{f(w+h) - f(w)}{h}.$$

Then taking $h = t \in \mathbb{R}$, we get

$$f'(w) = \lim_{t \rightarrow 0} \left(\frac{u(c+t, d) - u(c, d)}{t} + i \frac{v(c+t, d) - v(c, d)}{t} \right),$$

from which the real and imaginary parts imply that the limits

$$\lim_{t \rightarrow 0} \frac{u(c+t, d) - u(c, d)}{t} \quad \text{and} \quad \lim_{t \rightarrow 0} \frac{v(c+t, d) - v(c, d)}{t}$$

both exist, that is, $u_x(c, d)$ and $v_x(c, d)$ exists, and $f'(w) = u_x(c, d) + iv_x(c, d)$. Similarly, taking $h = it$ with $t \in \mathbb{R}$ gives $f'(w) = v_y(c, d) - iu_y(c, d)$, so $u_x = v_y$ and $u_y = -v_x$ at (c, d) .

Because partial derivatives are a little easier to work with than the standard derivative of a function on $\mathbb{R}^2$, we can use a straightforward corollary of the Cauchy-Riemann equations for checking complex differentiability¹.

Corollary 10.1.8 (Cauchy-Riemann and Continuous Partial derivatives implies Holomorphic)

Let $f = u + iv : U \rightarrow \mathbb{C}$. If u, v have continuous partial derivatives at a point $(c, d) \in U$ and satisfy the Cauchy-Riemann equations at (c, d) , then f is differentiable

¹Remarkably, the *Looman-Menchoff* theorem says that it suffices for u and v to be continuous and satisfy the Cauchy-Riemann equations to achieve complex differentiability, but this is quite non-trivial to prove.

at $w = c + id$.

Proof. Continuity of partial derivatives of u implies that u is differentiable, and similarly for v . So this follows from our original theorem. $\square$

Complex differentiability is a *much* stronger notion than the differentiability of real and imaginary parts. This leads to some surprising theorems compared to the real case. For example:

- (a) *Liouville's Theorem.* If $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic and bounded then f is constant.
- (b) If $f : U \rightarrow \mathbb{C}$ is holomorphic, then f is automatically infinitely differentiable on U .

We will prove these and much more later on.

Remark (Harmonic Functions). Note that the second theorem implies that the partial derivatives of u, v exist to all orders, so we can differentiate the Cauchy-Riemann equations to get

$$u_{xx} = v_{yx}, \quad \text{and} \quad u_{yy} = -v_{xy},$$

and since $v_{yx} = v_{xy}$, we get

$$\nabla^2 u = u_{xx} + u_{yy} = 0 \quad \text{in } U,$$

and similarly $\nabla^2 v = 0$ in U . So *real and imaginary parts of a holomorphic function are harmonic*.

We will now consider curves in the complex plane.

Definition 10.1.9 (Curve)

A **curve** is a continuous map $\gamma : [a, b] \rightarrow \mathbb{C}$, where $[a, b] \subset \mathbb{R}$ is a closed interval. We say γ is a **C^1 curve** if γ' exists and is continuous on $[a, b]$.

Definition 10.1.10 (Path Connected)

An open set $U \subset \mathbb{C}$ is **path connected** if for any two points $z, w \in U$ there is a curve $\gamma : [0, 1] \rightarrow U$ such that $\gamma(0) = z$ and $\gamma(1) = w$.

Definition 10.1.11 (Domain)

A **domain** is a non-empty, open, path connected subset of $\mathbb{C}$.

Corollary 10.1.12 (Zero Derivative implies Constant)

If $U \subset \mathbb{C}$ is a domain and $f : U \rightarrow \mathbb{C}$ is holomorphic with $f'(z) = 0$ for every $z \in U$, then f is constant.

Proof. Write $f = u + iv$. By the Cauchy-Riemann equations, $f' = 0$ implies that $Du = 0$ and $Dv = 0$ in U . Then since U is a domain, we get that u and v are constant. $\square$

§10.1.2 Power Series

So far we've only seen very few explicit holomorphic functions (namely polynomials on $\mathbb{C}$ and rational functions on their domains). We'd like to generate more, which we can do by looking at *power series*.

Recall the notion of *radius of convergence*.

Theorem 10.1.13 (Radius of Convergence)

If $c_n \in \mathbb{C}$ is a sequence of complex numbers, then there is a unique $R \in [0, \infty]$, the **radius of convergence**, such that the power series

$$\sum_{n=0}^{\infty} c_n (z - a)^n, \quad z, a \in \mathbb{C}$$

converges absolutely if $|z - a| < R$ and diverges if $|z - a| > R$. If $0 < r < R$, then the series converges uniformly (with respect to z) on the compact disk $\overline{D}(a, r) = \{z \in \mathbb{C} : |z - a| \leq r\}$.

Note that there are no claims about convergence when $|z - a| = R$ for $R > 0$. There are also various expressions for R , for example

$$R = \sup\{r \geq 0 \mid |c_n| r^n \rightarrow 0 \text{ as } n \rightarrow \infty\}$$

and

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n}.$$

We get quite natural behavior of power series with respect to complex differentiability.

Theorem 10.1.14 (Derivatives of Power Series)

Let $\sum_{n=0}^{\infty} c_n (z - a)^n$ be a power series with radius of convergence $R > 0$. Fix some $a \in \mathbb{C}$, and define $f : D(a, R) \rightarrow \mathbb{C}$ by $f(z) = \sum_{n=0}^{\infty} c_n (z - a)^n$. Then

- (i) f is holomorphic on $D(a, R)$;
- (ii) the derived series $\sum_{n=1}^{\infty} n c_n (z - a)^{n-1}$ also has radius convergence R , and is equal to $f'(z)$ for $z \in D(a, R)$.
- (iii) f has derivatives of all orders on $D(a, R)$, and $c_n = \frac{f^{(n)}(a)}{n!}$.
- (iv) if f vanishes on $D(a, \varepsilon)$ for some $\varepsilon > 0$, then $f \equiv 0$ on $D(a, R)$.

Proof. Parts (i) and (ii). Without loss of generality, take $a = 0$. Now we have $f(z) = \sum_{n=0}^{\infty} c_n z^n$ for $z \in D(0, R)$, where R is the radius of convergence.

The derived series $\sum_{n=1}^{\infty} n c_n z^{n-1}$ will have some radius of convergence $R_1 \in [0, \infty]$. To see that $R_1 \geq R$, let $z \in D(0, R)$, and choose ρ such that $|z| < \rho < R$. Then

$$n |c_n| |z|^{n-1} = \left(n \left| \frac{z}{\rho} \right|^{n-1} \right) |c_n| \rho^{n-1} \leq |c_n| \rho^{n-1}$$

for sufficiently large n (as $n |z/\rho|^{n-1} \rightarrow 0$ as $n \rightarrow \infty$). Since $\sum |c_n| \rho^n$ converges, it

follows that $\sum_{n=1}^{\infty} n|c_n||z|^{n-1}$ converges. Thus $D(0, R) \subset D(0, R_1)$ that is, $R_1 \geq R$.

Also, since $|c_n||z|^n \leq n|c_n||z|^{n-1} = |z|(n|c_n||z|^{n-1})$, if $\sum n|c_n||z|^{n-1}$ converges then so does $\sum |c_n||z|^n$, so $R_1 \leq R$. This implies that we must have $R_1 = R$.

To prove that f is differentiable with $f'(z) = \sum_{n=1}^{\infty} nc_n z^{n-1}$, fix some $z \in D(0, R)$. Then our key idea is that this assertion is equivalent to the continuity at z of the function $g : D(0, R) \rightarrow \mathbb{C}$ with

$$g(w) = \begin{cases} \frac{f(w)-f(z)}{w-z} & \text{if } w \neq z, \\ \sum_{n=1}^{\infty} nc_n z^{n-1} & \text{if } w = z. \end{cases}$$

By substituting for f , we get that $g(w) = \sum_{n=1}^{\infty} h_n(w)$ for $w \in D(0, R)$ where h_n is a sequence of functions given by

$$h_n(w) = \begin{cases} \frac{c_n(w^n - z^n)}{w - z} & \text{if } w \neq z, \\ nc_n z^{n-1} & \text{if } w = z. \end{cases}$$

Now h_n is continuous on $D(0, R)$ for all n , then using that $\frac{w^n - z^n}{w - z} = \sum_{j=0}^{n-1} z^j w^{n-1-j}$, we get that for any r with $|z| < r < R$ and any $w \in D(0, r)$ we have $|h_n(w)| \leq n|c_n|r^{n-1}$. Defining $M_n = n|c_n|r^{n-1}$, we can note that $\sum M_n < \infty$, so by the Weierstrass M-test, $\sum h_n$ converges uniformly on $D(0, r)$. But a uniform limit of continuous functions is continuous, so $g = \sum h_n$ is continuous in $D(0, r)$ and in particular at z , as required.

Part (iii). Repeatedly apply (ii), and the formula $c_n = f^{(n)}(a)/n!$ follows by differentiating the series n times and setting $z = a$.

Part (iv). If $f = 0$ on $D(a, \varepsilon)$, then $f^{(n)}(a) = 0$ for all n , so $c_n = 0$ for all n . Hence $f = 0$ on $D(a, R)$. $\square$

Remark. This theorem provides a way to generate a large class of holomorphic functions on a disk. Later we will show that every holomorphic function is *locally* given by a power series. Once we have that, (iii) above gives that holomorphic functions are automatically infinitely differentiable in their domain.

§10.1.3 The Exponential and the Logarithm

Let's now consider a specific example of a power series, the *complex exponential* function, which you are likely familiar with.

Definition 10.1.15 (Complex Exponential)

The **complex exponential** function is defined by the power series

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Some basic properties of this function will also be familiar.

Proposition 10.1.16 (Properties of the Exponential)

- (i) e^z is entire, with $(e^z)' = e^z$.
- (ii) $e^z \neq 0$ and $e^{z+w} = e^z \cdot e^w$ for all $z, w \in \mathbb{C}$.
- (iii) $e^{x+iy} = e^x(\cos y + i \sin y)$ for $x, y \in \mathbb{R}$.
- (iv) $e^z = 1$ if and only if $z = 2\pi ni$ for some integer n .
- (v) Let $z \in \mathbb{C}$. Then there exists $w \in \mathbb{C}$ such that $e^w = z$ if and only if $z \neq 0$.

Proof. Part (i). The radius of convergence of the series is ∞ , and to see that $(e^z)' = e^z$, differentiate the series term-by-term.

Part (ii). Fix any $w \in \mathbb{C}$ and set $F(z) = e^{z+w} \cdot e^{-z}$. Then $F'(z) = -e^{z+w} \cdot e^{-z} + e^{z+w} \cdot e^{-z} = 0$, so $F(z)$ is constant, and equal to e^w . Thus $e^{z+w} \cdot e^{-z} = e^w$ for all $z, w \in \mathbb{C}$, and the result follows noting that $e^0 = 1$.

Part (iii). By the previous part, $e^{x+iy} = e^x \cdot e^{iy}$. Now use the definition of e^{iy} and the series for $\sin y$ and $\cos y$ with $y \in \mathbb{R}$.

Part (iv) and (v). Both follow directly from (iii). □

We can now consider the ‘inverse’ of this function, the *logarithm*.

Definition 10.1.17 (Logarithm)

Given $z \in \mathbb{C}$, we say that $w \in \mathbb{C}$ is a **logarithm** of z if $e^w = z$.

By our above theorem, we note that z has a logarithm if and only if $z \neq 0$, and also if z *does* have a logarithm then there are infinitely many such logarithms, each differing from each other by $2\pi ni$ for some integer n .

If w is a logarithm of z , then $e^{\operatorname{Re}(w)} = |z|$, so $\operatorname{Re}(w) = \ln |z|$ (the real logarithm of the positive number $|z|$). In particular, $\operatorname{Re}(w)$ is uniquely determined by z .

Definition 10.1.18 (Branches of the Logarithm)

Let $U \subset \mathbb{C} \setminus \{0\}$ be open. Then a **branch** of the logarithm on U is a continuous function $\lambda : U \rightarrow \mathbb{C}$ such that $e^{\lambda(z)} = z$ for each $z \in U$.

Remark. If λ is a branch of $\log$ on U , then λ is automatically holomorphic in U , with $\lambda'(z) = 1/z$, with the proof following by direct computation of limits.

Definition 10.1.19 (Principal Branch of the Logarithm)

The **principal branch** of the logarithm is the function

$$\operatorname{Log} : U_1 = \mathbb{C} \setminus \{x \in \mathbb{R} : x \leq 0\} \rightarrow \mathbb{C}$$

defined by

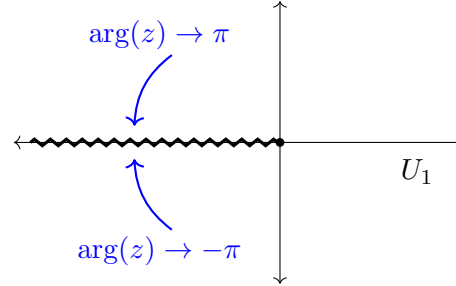
$$\operatorname{Log}(z) = \ln |z| + i \arg(z),$$

where $\arg(z)$ is the unique argument of $z \in U_1$ in $(-\pi, \pi)$.

Remark. Log does not have a continuous extension to $\mathbb{C} \setminus \{0\}$, since $\arg(z) \rightarrow \pi$ as $z \rightarrow -1$ with $\operatorname{Im}(z) > 0$, and $\arg(z) \rightarrow -\pi$ as $z \rightarrow -1$ with $\operatorname{Im}(z) < 0$. We will later see

that it is impossible to define a branch of $\log$ on $\mathbb{C} \setminus \{0\}$.

The negative real axis, which we removed from the domain of Log , is called a **branch cut**. The picture to have in mind is the following: as z crosses the cut, the argument (and hence the imaginary part of any would-be logarithm) jumps by 2π .



There was nothing special about our choice to cut along the negative real axis – removing any ray from the origin (indeed, any reasonable curve joining 0 to infinity) leaves a domain on which a branch of the logarithm can be defined. The principal branch is merely the standard convention.

Proposition 10.1.20 (Power Series for Log)

For $|z| < 1$, we have

$$\text{Log}(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n}.$$

Proof. Note that the radius of convergence of the series is 1 and $|z| < 1$ implies that $1+z \in U_1$, so both sides are defined on $|z| < 1$. Then let $F(z) = \text{Log}(1+z) - \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n}$, for $|z| < 1$. Then $F'(z) = \frac{1}{1+z} - \sum_{n=1}^{\infty} (-z)^{n-1} = 0$. So $F(z)$ is constant, and $F(z) = F(0) = 0$. $\square$

Using $\exp$ and Log we can define further useful functions.

- (i) For any constant $\alpha \in \mathbb{C}$, define

$$z^\alpha = e^{\alpha \text{Log}(z)}, \quad z \in U_1.$$

This is the principal branch of z^α . We have that z^α is holomorphic on U_1 with $(z^\alpha)' = \alpha z^{\alpha-1}$.

- (ii) We can define

$$\begin{aligned} \cos(z) &= \frac{e^{iz} + e^{-iz}}{2} \\ \sin(z) &= \frac{e^{iz} - e^{-iz}}{2i} \\ \cosh(z) &= \frac{e^z + e^{-z}}{2} \\ \sinh(z) &= \frac{e^z - e^{-z}}{2}. \end{aligned}$$

All of these functions are entire and their derivatives are given by the familiar expressions from real variables.

§10.1.4 Conformal Maps

We now look at a geometric property of holomorphic functions called *conformality*. Let $f : U \rightarrow \mathbb{C}$ be holomorphic, where $U \subset \mathbb{C}$ is open, as usual. Let $w \in U$, and suppose that $f'(w) \neq 0$.

Take two C^1 curves $\gamma_1, \gamma_2 : [-1, 1] \rightarrow U$, such that $\gamma_i(0) = w$ and $\gamma'_i(0) \neq 0$. Then $f \circ \gamma_i$ are C^1 curves passing through $f(w)$. Moreover, $(f \circ \gamma_i)'(0) = f'(w)\gamma'_i(0) \neq 0$.

This implies that

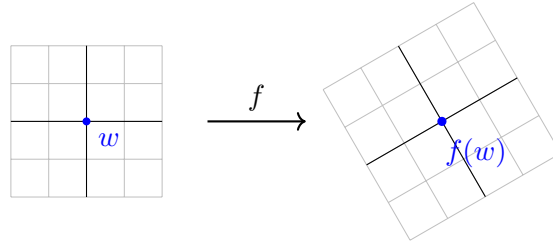
$$\frac{(f \circ \gamma_1)'(0)}{(f \circ \gamma_2)'(0)} = \frac{\gamma'_1(0)}{\gamma'_2(0)},$$

and hence

$$\arg(f \circ \gamma_1)'(0) - \arg(f \circ \gamma_2)'(0) = \arg \gamma'_1(0) - \arg \gamma'_2(0).$$

This means that the angle that the curves γ_1, γ_2 make at w is the same as the angle their images $f \circ \gamma_1$ and $f \circ \gamma_2$ make at $f(w)$, in size as well as in orientation. That is, f is ‘angle-preserving at w ’ whenever $f'(w) \neq 0$.

Geometrically, this is because multiplication by the complex number $f'(w) = re^{i\theta} \neq 0$ is a rotation by θ composed with a scaling by r . To first order, f acts near w by exactly this rotation and scaling: a small grid of curves through w is carried to a rotated and scaled grid through $f(w)$, with all angles intact.



Remark. If f is a C^1 map on U , the converse of the above also holds. That is, if $w \in U$ is such that $(f \circ \gamma)'(0) \neq 0$ for every C^1 curve γ with $\gamma(0) = w$ and $\gamma'(0) \neq 0$, and if f is angle preserving at w in the above sense, then $f'(w)$ exists and $f'(w) \neq 0$.

Definition 10.1.21 (Conformal)

A holomorphic function $f : U \rightarrow \mathbb{C}$ on an open set U is said to be **conformal** at the point $w \in U$ if $f'(w) \neq 0$.

Definition 10.1.22 (Conformal Equivalence)

Let $U, \tilde{U}$ be domains in $\mathbb{C}$. A map $f : U \rightarrow \tilde{U}$ is said to be a **conformal equivalence** between U and $\tilde{U}$ if f is a bijective holomorphic map with $f'(z) \neq 0$ for every $z \in U$.

Remark (Redundancy). If f is holomorphic and injective, then $f'(z) \neq 0$ for each z (we will see this later in the course). So, in the above definition, the requirement $f'(z) \neq 0$ is redundant.

Remark (Inverse Functions). It is automatic that the inverse $f^{-1} : \tilde{U} \rightarrow U$ is holomorphic. This follows from the **holomorphic inverse function theorem**, which can be proved using the real inverse function theorem.

We will now consider some examples of conformal equivalences.

Example 10.1.23 (Möbius Maps Give a Conformal Equivalence)

Consider the Möbius maps

$$f(z) = \frac{az + b}{cz + d}$$

where $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$. This function f is conformal on $\mathbb{C} \setminus \{-d/c\}$ if $c \neq 0$, and conformal on $\mathbb{C}$ if $c = 0$.

Möbius maps sometimes serve as explicit conformal equivalences between subdomains of $\mathbb{C}$.

For example, let $\mathbb{H}$ be the open upper-half plane, $\mathbb{H} = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$. Then

$$\begin{aligned} z \in \mathbb{H} &\iff z \text{ is closer to } i \text{ than to } -i \\ &\iff |z - i| < |z + i| \\ &\iff \left| \frac{z - i}{z + i} \right| < 1. \end{aligned}$$

Thus $g = \frac{z-i}{z+i}$ maps $\mathbb{H}$ onto $D(0, 1)$, so $g : \mathbb{H} \rightarrow D(0, 1)$ is a conformal equivalence. The inverse is also a Möbius map, namely $w \mapsto i\frac{1+w}{1-w}$, so the disc and the half-plane are two interchangeable models of the ‘same’ domain.

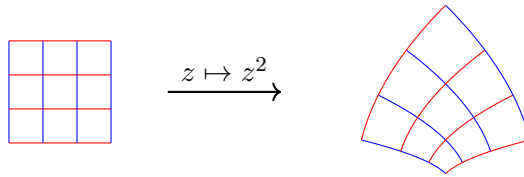
Example 10.1.24 (Powers Give a Conformal Equivalence)

Let $f(z) = z^n$ where $n \in \mathbb{N}$, with

$$f : \{z \in \mathbb{C} \setminus \{0\} \mid 0 < \arg(z) < \pi/n\} \rightarrow \mathbb{H}.$$

This is a conformal equivalence, with inverse $f^{-1}(z) = z^{1/n}$.

It is worth pausing to actually *look* at what a map like $z \mapsto z^2$ does. Away from the origin (where the derivative vanishes) it is conformal, so a grid of horizontal and vertical lines is carried to two families of curves (in fact parabolas) which still meet each other at right angles.



Example 10.1.25 (Exponentials Give a Conformal Equivalence)

The restricted exponential function given by

$$\exp : \{z \in \mathbb{C} \mid -\pi < \text{Im}(z) < \pi\} \rightarrow \mathbb{C} \setminus \{x \in \mathbb{R} \mid x \leq 0\}$$

is a conformal equivalence, whose inverse is $\text{Log}(z)$.

Aside: The Riemann Mapping Theorem

There is an interesting result about many domains being conformally equivalent.

Theorem 10.1.26 (Riemann Mapping Theorem)

Any simply connected domain $U \subset \mathbb{C}$ with $U \neq \mathbb{C}$ is conformally equivalent to $D(0, 1)$.

Here, $U \subset \mathbb{C}$ is simply connected if every continuous map $\gamma : S^1 = \partial D(0, 1) \rightarrow U$ extends to a continuous map $\Gamma : \overline{D(0, 1)} \rightarrow U$ with $\Gamma|_{\partial D(0, 1)} = \gamma$. We will discuss simply connected domains in detail later on, but intuitively one should think of simply connected domains as the ones ‘without holes’.

A proof of this result can be found in many textbooks on complex analysis.

§10.2 Complex Integration

§10.2.1 Extending Riemann Integration

As we did with differentiation, we now want to extend our real (Riemann) notion of integration to the integration of complex functions $f : U \rightarrow \mathbb{C}$ with $U \subset \mathbb{C}$ along curves in U .

First we will look at complex functions of a single real variable.

Definition 10.2.1 (Integral on a Real Interval)

If $f : [a, b] \subset \mathbb{R} \rightarrow \mathbb{C}$ is a complex function and if f is continuous^a then we define the **integral** of f over $[a, b]$ to be

$$\int_a^b f(t) \, dt = \int_a^b \operatorname{Re}(f(t)) \, dt + i \int_a^b \operatorname{Im}(f(t)) \, dt.$$

^aOr, more generally, f is Riemann integrable in its real and imaginary parts.

In particular,

$$\int_a^b i g(t) \, dt = i \int_a^b g(t) \, dt$$

for a real function $g : [a, b] \rightarrow \mathbb{R}$. Hence, by direct calculation,

$$\int_a^b w f(t) \, dt = w \int_a^b f(t) \, dt$$

for any $w \in \mathbb{C}$.

Proposition 10.2.2 (Basic Estimate)

If $f : [a, b] \rightarrow \mathbb{C}$ is continuous then

$$\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt \leq (b-a) \sup_{t \in [a, b]} |f(t)|$$

with equality if and only if f is constant.

Proof. If $\int_a^b f(t) dt = 0$ then we are done. Otherwise, write $\int_a^b f(t) dt = re^{i\theta}$, with $\theta \in [0, 2\pi)$. Let $M = \sup_{t \in [a, b]} |f(t)|$. Then

$$\begin{aligned} \left| \int_a^b f(t) dt \right| &= r = e^{-i\theta} \int_a^b f(t) dt = \int_a^b e^{-i\theta} f(t) dt \\ &= \int_a^b \operatorname{Re}(e^{-i\theta} f(t)) dt + i \int_a^b \operatorname{Im}(e^{-i\theta} f(t)) dt. \end{aligned}$$

Since the left-hand-side is real,

$$\begin{aligned} \left| \int_a^b f(t) dt \right| &= \int_a^b \operatorname{Re}(e^{-i\theta} f(t)) dt \\ &\leq \int_a^b |e^{-i\theta} f(t)| dt \\ &= \int_a^b |f(t)| dt \\ &\leq (b-a)M. \end{aligned}$$

Equality holds if and only if $|f(t)| = M$ and $\operatorname{Re}(e^{-i\theta} f(t)) = M$ for all $t \in [a, b]$, that is, if and only if $|f(t)| = M$ and $\arg(f(t)) = \theta$ for all $t \in [a, b]$, that is, f is constant. $\square$

We can now define a more general notion of an integral along a curve in the complex plane (not just somewhere on the real line).

Definition 10.2.3 (Integral on a Complex Curve)

Let $U \subset \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ be continuous. Let $\gamma : [a, b] \rightarrow U$ be a C^1 curve. Then the **integral of f along γ** is

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

This integral has lots of nice basic properties that one might expect.

Proposition 10.2.4 (Invariance under Reparameterisation)

Let $\varphi : [a_1, b_1] \rightarrow [a, b]$ be C^1 and injective with $\varphi(a_1) = a$ and $\varphi(b_1) = b$. Let $\delta = \gamma \circ \varphi : [a_1, b_1] \rightarrow U$. Then we have

$$\int_{\delta} f(z) dz = \int_{\gamma} f(z) dz$$

Proof. Letting $s = \varphi(t)$, we have

$$\begin{aligned} \int_{\delta} f(z) \, dz &= \int_{a_1}^{b_1} f(\gamma \circ \varphi(t)) \gamma'(\varphi(t)) \varphi'(t) \, dt \\ &= \int_a^b f(\gamma(s)) \gamma'(s) \, ds \\ &= \int_{\gamma} f(z) \, dz. \end{aligned}$$

The second equality follows from the definition of the integral of a function of a real variable and the change of variables formula for integrals of real functions of a real variable. $\square$

Proposition 10.2.5 (Linearity)

We have

$$\int_{\gamma} (c_1 f_1(z) + c_2 f_2(z)) \, dz = c_1 \int_{\gamma} f_1(z) \, dz + c_2 \int_{\gamma} f_2(z) \, dz,$$

for constants $c_1, c_2 \in \mathbb{C}$.

Proof Sketch. Check definitions. $\square$

Proposition 10.2.6 (Additivity)

If $\gamma : [a, b] \rightarrow U$ is a C^1 curve and $a < c < b$, then

$$\int_{\gamma} f(z) \, dz = \int_{\gamma|_{[a, c]}} f(z) \, dz + \int_{\gamma|_{[c, b]}} f(z) \, dz.$$

Proof Sketch. Check definitions. $\square$

Proposition 10.2.7 (Inverse Path)

Define the inverse path $(-\gamma) : [-b, -a] \rightarrow U$ by $(-\gamma)(t) = \gamma(-t)$ for $-b \leq t \leq -a$. Then

$$\int_{(-\gamma)} f(z) \, dz = - \int_{\gamma} f(z) \, dz.$$

Proof Sketch. Check definitions. $\square$

We can get the euclidean length of a curve in the complex plane by the following (which we take as a definition).

Definition 10.2.8 (Length of a Curve)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a C^1 curve. The **length** of γ is defined by

$$\text{length}(\gamma) = \int_a^b |\gamma'(t)| \, dt$$

Definition 10.2.9 (Piecewise C^1 Curve)

A **piecewise C^1 curve** is a continuous map $\gamma : [a, b] \rightarrow \mathbb{C}$ such that there exists a finite subdivision $a = a_0 < a_1 < \cdots < a_n = b$ with the property that $\gamma_j = \gamma|_{[a_{j-1}, a_j]} : [a_{j-1}, a_j] \rightarrow \mathbb{C}$ is C^1 for $1 \leq j \leq n$.

If γ is a piecewise C^1 curve as above, we *define*

$$\int_{\gamma} f(z) \, dz = \sum_{j=1}^n \int_{\gamma_j} f(z) \, dz$$

and

$$\text{length}(\gamma) = \sum_{j=1}^n \text{length}(\gamma_j) = \sum_{j=1}^n \int_{a_{j-1}}^{a_j} |\gamma'(t)| \, dt.$$

Remark. Note that both definitions are independent of the subdivisions by the additivity property above.

With this, from now on a ‘curve’ will mean a piecewise C^1 curve.

Definition 10.2.10 (Sum of Curves)

If $\gamma_1 : [a, b] \rightarrow \mathbb{C}$ and $\gamma_2 : [c, d] \rightarrow \mathbb{C}$ are curves with $\gamma_1(b) = \gamma_2(c)$, we define the **sum** of γ_1 and γ_2 to be the curve $(\gamma_1 + \gamma_2) : [a, b + d - c] \rightarrow \mathbb{C}$, with

$$(\gamma_1 + \gamma_2)(t) = \begin{cases} \gamma_1(t) & \text{if } a \leq t \leq b, \\ \gamma_2(t - b + c) & \text{if } b \leq t \leq b + d - c. \end{cases}$$

We can then give another basic estimate for integrals over curves in the complex plane.

Proposition 10.2.11 (Another Basic Estimate)

For any continuous function $f : U \rightarrow \mathbb{C}$ and any curve $\gamma : [a, b] \rightarrow \mathbb{C}$, we have that

$$\left| \int_{\gamma} f(z) \, dz \right| \leq \text{length}(\gamma) \sup_{\gamma} |f|$$

where $\sup_{\gamma} |f| = \sup_{t \in [a, b]} |f(\gamma(t))|$.

Proof. If γ is C^1 , then

$$\left| \int_{\gamma} f(z) \, dz \right| = \left| \int_a^b f(\gamma(t)) \gamma'(t) \, dt \right|$$

$$\begin{aligned} &\leq \int_a^b |f(\gamma(t))| |\gamma'(t)| \, dt \\ &\leq \sup_{t \in [a,b]} |f(\gamma(t))| \text{length}(\gamma). \end{aligned}$$

If γ is piecewise C^1 then the result follows from the definition

$$\int_{\gamma} f(z) \, dz = \sum_{j=1}^n \int_{\gamma_j} f(z) \, dz$$

where γ_j is C^1 , and the triangle inequality. $\square$

§10.2.2 The Fundamental Theorem of Calculus

In real analysis, the fundamental theorem of calculus links differentiation and integration. The same is true for integrals along curves in the complex plane, and (as in the real case) it is the main computational tool we have for evaluating integrals directly.

Definition 10.2.12 (Antiderivative)

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ be continuous. An **antiderivative** of f is a holomorphic function $F : U \rightarrow \mathbb{C}$ such that $F'(z) = f(z)$ for all $z \in U$.

Theorem 10.2.13 (Fundamental Theorem of Calculus)

Let $U \subseteq \mathbb{C}$ be open and let $f : U \rightarrow \mathbb{C}$ be continuous with antiderivative F . Then for any curve $\gamma : [a, b] \rightarrow U$ we have

$$\int_{\gamma} f(z) \, dz = F(\gamma(b)) - F(\gamma(a)).$$

In particular, if γ is a *closed* curve (that is, $\gamma(a) = \gamma(b)$), then $\int_{\gamma} f(z) \, dz = 0$.

Proof. Suppose first that γ is C^1 . Then by the chain rule,

$$\int_{\gamma} f(z) \, dz = \int_a^b f(\gamma(t)) \gamma'(t) \, dt = \int_a^b \frac{d}{dt} [F(\gamma(t))] \, dt = F(\gamma(b)) - F(\gamma(a)),$$

where the last equality is the real fundamental theorem of calculus applied to the real and imaginary parts. If γ is only piecewise C^1 , apply the above on each piece and note that the sum telescopes. $\square$

Note that the theorem *assumes* that an antiderivative F exists – it says nothing about when this happens. Whether or not every holomorphic function on U has an antiderivative turns out to depend delicately on the geometry of U , and this question is really the engine behind the next few sections. Let's do the fundamental example.

Example 10.2.14 (Integrating z^n around a Circle)

Fix $R > 0$ and an integer n , and let $\gamma(t) = Re^{2\pi it}$ for $t \in [0, 1]$, the circle of radius R traversed once anticlockwise. We compute $\int_{\gamma} z^n dz$.

If $n \neq -1$, then $z^{n+1}/(n+1)$ is an antiderivative of z^n on $\mathbb{C} \setminus \{0\}$ (or on $\mathbb{C}$ if $n \geq 0$), so since γ is closed,

$$\int_{\gamma} z^n dz = 0.$$

If $n = -1$, we instead compute directly from the definition:

$$\int_{\gamma} \frac{dz}{z} = \int_0^1 \frac{2\pi i R e^{2\pi it}}{R e^{2\pi it}} dt = 2\pi i.$$

This example is small but its consequences are large. Since $2\pi i \neq 0$, the function $1/z$ has *no* antiderivative on any open set containing the circle $\{|z| = R\}$, for any $R > 0$. But if λ were a branch of the logarithm on $\mathbb{C} \setminus \{0\}$, then λ would be exactly such an antiderivative. So, as promised earlier, *there is no branch of the logarithm on $\mathbb{C} \setminus \{0\}$.*

Now, the fundamental theorem of calculus has a converse: if all integrals of f around closed curves vanish, then an antiderivative can be built.

Theorem 10.2.15 (Converse to the Fundamental Theorem of Calculus)

Let $U \subseteq \mathbb{C}$ be a domain and $f : U \rightarrow \mathbb{C}$ be continuous. If

$$\int_{\gamma} f(z) dz = 0$$

for every closed curve γ in U , then f has an antiderivative on U .

Proof. Fix a base point $a_0 \in U$. Since U is path connected, for each $w \in U$ we can choose^a a curve $\gamma_w : [0, 1] \rightarrow U$ from a_0 to w , and we define

$$F(w) = \int_{\gamma_w} f(z) dz.$$

This is well defined, i.e. independent of the choice of curve: if γ_w and $\tilde{\gamma}_w$ are two such curves, then $\gamma_w + (-\tilde{\gamma}_w)$ is a closed curve in U , so by hypothesis

$$0 = \int_{\gamma_w + (-\tilde{\gamma}_w)} f(z) dz = \int_{\gamma_w} f(z) dz - \int_{\tilde{\gamma}_w} f(z) dz.$$

We now check that F is differentiable with $F' = f$. Fix $w \in U$ and take $r > 0$ with $D(w, r) \subset U$. For $h \in \mathbb{C}$ with $0 < |h| < r$, let $\delta_h(t) = w + th$, $t \in [0, 1]$, be the radial segment from w to $w + h$. Then $\gamma_w + \delta_h + (-\gamma_{w+h})$ is a closed curve in U , so its integral vanishes, giving

$$F(w + h) = F(w) + \int_{\delta_h} f(z) dz = F(w) + hf(w) + \int_{\delta_h} (f(z) - f(w)) dz,$$

where we used $\int_{\delta_h} dz = h$. Hence by our basic estimate,

$$\left| \frac{F(w+h) - F(w)}{h} - f(w) \right| \leq \frac{1}{|h|} \cdot \text{length}(\delta_h) \sup_{z \in \delta_h} |f(z) - f(w)| = \sup_{z \in \delta_h} |f(z) - f(w)|,$$

and this tends to 0 as $h \rightarrow 0$ by the continuity of f at w . So $F'(w) = f(w)$. $\square$

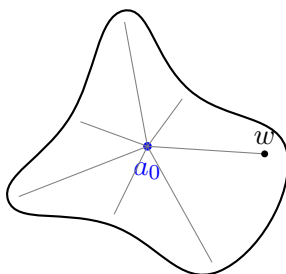
^aStrictly, path connectedness gives a continuous curve, but for open sets one can always upgrade this to a piecewise C^1 curve, say by covering the curve with finitely many disks and taking line segments within them.

§10.2.3 Cauchy's Theorem for Star-Shaped Domains

We have just seen that ‘all closed curve integrals vanish’ is equivalent to having an antiderivative. The central theorem of this course, *Cauchy's theorem*, says that for a *holomorphic* function on a domain with reasonable geometry, all closed curve integrals really do vanish. In this section we prove this for *star-shaped* domains, which covers a surprising number of applications.

Definition 10.2.16 (Star-Shaped Domain)

A domain $U \subseteq \mathbb{C}$ is **star-shaped** if there is a point $a_0 \in U$ (a **centre**) such that for every $w \in U$, the straight line segment $[a_0, w]$ from a_0 to w lies entirely in U .



A star-shaped domain need not be convex (the domain above is not!), but every point can ‘see’ the centre along a straight line. Some quick observations, each of which is straightforward to check:

- (i) every disk $D(a, r)$ is convex;
- (ii) every convex domain is star-shaped (any point works as a centre);
- (iii) every star-shaped domain is path connected.

None of the reverse implications hold in general. For instance, $\mathbb{C} \setminus \{x \in \mathbb{R} : x \leq 0\}$ is star-shaped (take $a_0 = 1$, say) but certainly not convex, and an annulus $\{1 < |z| < 2\}$ is path connected but not star-shaped.

The strategy for proving Cauchy's theorem on a star-shaped domain is to first establish it for the simplest possible closed curves – boundaries of triangles – and then to leverage the converse of the fundamental theorem of calculus.

Definition 10.2.17 (Triangle)

A **triangle** T in $\mathbb{C}$ is the convex hull of three points $z_1, z_2, z_3 \in \mathbb{C}$, that is

$$T = \{az_1 + bz_2 + cz_3 \mid a, b, c \geq 0, a + b + c = 1\}.$$

We write ∂T for the boundary of T , viewed as the closed curve which traverses the three sides of T once.

Corollary 10.2.18 (Antiderivatives on Star-Shaped Domains)

Let U be a star-shaped domain and $f : U \rightarrow \mathbb{C}$ be continuous with

$$\int_{\partial T} f(z) \, dz = 0$$

for every triangle $T \subseteq U$. Then f has an antiderivative on U .

Proof. We repeat the proof of the converse to the fundamental theorem of calculus, being economical about which curves we use. Let a_0 be a centre of U , and for $w \in U$ define

$$F(w) = \int_{[a_0, w]} f(z) \, dz,$$

integrating along the straight segment, which lies in U as U is star-shaped. Given $w \in U$, choose $r > 0$ with $D(w, r) \subset U$, and let $h \in \mathbb{C}$ with $0 < |h| < r$. The key point is that the closed curve $[a_0, w] + [w, w+h] + [w+h, a_0]$ is precisely the boundary of the triangle T with vertices $a_0, w, w+h$, and this triangle lies in U : its sides do, and every point of the segment $[w, w+h]$ lies in $D(w, r) \subseteq U$, so every segment from a_0 to a point of T lies in U by the star-shape property^a. Hence by hypothesis

$$F(w+h) - F(w) = \int_{[w, w+h]} f(z) \, dz,$$

and exactly as before, continuity of f gives $F'(w) = f(w)$. □

^aTo see $T \subset U$ carefully: every point of T lies on a segment from a_0 to a point of $[w, w+h]$, and $[w, w+h] \subset D(w, r) \subset U$.

So everything now rests on showing that integrals of holomorphic functions around triangles vanish. This is the technical heart of Cauchy's theorem, and the proof – due to Goursat – is a beautiful ‘divide and conquer’ argument.

Theorem 10.2.19 (Cauchy's Theorem for Triangles)

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ be holomorphic. Then

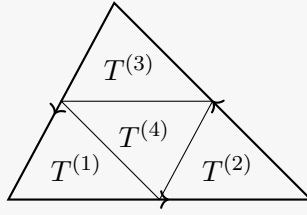
$$\int_{\partial T} f(z) \, dz = 0$$

for every triangle $T \subseteq U$.

Proof. For a triangle T , write $\eta(T) = \int_{\partial T} f(z) \, dz$; our goal is $\eta(T) = 0$.

Subdivide T into four smaller triangles $T^{(1)}, \dots, T^{(4)}$ by joining the midpoints of its

sides, as in the picture below.



Traversing each $\partial T^{(j)}$ anticlockwise, the integrals over the three interior sides cancel in pairs (each is traversed once in each direction), so

$$\eta(T) = \sum_{j=1}^4 \eta(T^{(j)}).$$

By the triangle inequality, there is some j with $|\eta(T^{(j)})| \geq |\eta(T)|/4$. Set $T_0 = T$ and $T_1 = T^{(j)}$, and repeat this construction inductively to obtain nested triangles $T_0 \supset T_1 \supset T_2 \supset \dots$ with

$$|\eta(T_n)| \geq \frac{|\eta(T_0)|}{4^n}, \quad \text{and} \quad \text{length}(\partial T_n) = \frac{\text{length}(\partial T_0)}{2^n},$$

the latter since each subdivision halves all side lengths.

The T_n are nested non-empty compact sets with diameters tending to zero, so their intersection is a single point: $\bigcap_{n \geq 0} T_n = \{z_0\}$, with $z_0 \in T \subseteq U$.

Now we finally use that f is differentiable at z_0 . Given $\varepsilon > 0$, there is $\delta > 0$ such that $D(z_0, \delta) \subseteq U$ and

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0| \quad \text{for all } z \in D(z_0, \delta).$$

Also, since the affine function $z \mapsto f(z_0) + f'(z_0)(z - z_0)$ has an antiderivative, its integral around any closed curve vanishes, so for every n ,

$$\eta(T_n) = \int_{\partial T_n} (f(z) - f(z_0) - f'(z_0)(z - z_0)) \, dz.$$

Choose n large enough that $T_n \subseteq D(z_0, \delta)$. Then for $z \in \partial T_n$ we have $|z - z_0| \leq \text{length}(\partial T_n)$, so estimating the integral,

$$\frac{|\eta(T_0)|}{4^n} \leq |\eta(T_n)| \leq \varepsilon \text{length}(\partial T_n)^2 = \frac{\varepsilon \text{length}(\partial T_0)^2}{4^n}.$$

Hence $|\eta(T_0)| \leq \varepsilon \text{length}(\partial T_0)^2$ for every $\varepsilon > 0$, forcing $\eta(T_0) = 0$. $\square$

For later applications (in particular when we prove the Cauchy integral formula) it is extremely useful to relax the hypothesis slightly, allowing f to fail to be differentiable at finitely many points, so long as it remains continuous there.

Theorem 10.2.20 (Cauchy's Theorem for Triangles, with Exceptional Points)

Let $U \subseteq \mathbb{C}$ be open, $S \subset U$ be finite, and let $f : U \rightarrow \mathbb{C}$ be continuous on U and

holomorphic on $U \setminus S$. Then $\int_{\partial T} f(z) dz = 0$ for every triangle $T \subseteq U$.

Proof. Subdivide T as before, but n times over, keeping *all* of the 4^n small triangles $T_1, \dots, T_N$ ($N = 4^n$). Again the interior sides cancel, so

$$\int_{\partial T} f(z) dz = \sum_{j=1}^N \int_{\partial T_j} f(z) dz.$$

By the previous theorem, $\int_{\partial T_j} f(z) dz = 0$ for every T_j that avoids S . Now, each point of S can lie in at most 6 of the small triangles (a point can be a vertex of at most 6 of them), and each small triangle has $\text{length}(\partial T_j) = 2^{-n} \text{length}(\partial T)$. Also f , being continuous on the compact set T , is bounded there, say $|f| \leq M$ on T . Hence

$$\left| \int_{\partial T} f(z) dz \right| \leq 6|S|M \cdot \frac{\text{length}(\partial T)}{2^n},$$

and letting $n \rightarrow \infty$ gives the result. $\square$

Combining our work, we obtain the form of Cauchy's theorem that we will use constantly.

Theorem 10.2.21 (Cauchy's Theorem for Star-Shaped Domains)

Let U be a star-shaped domain, and let $f : U \rightarrow \mathbb{C}$ be continuous on U and holomorphic on $U \setminus S$ for some finite set S . Then

$$\int_{\gamma} f(z) dz = 0$$

for every closed curve γ in U . In particular this holds when U is convex, and when f is holomorphic on all of U .

Proof. By the previous theorem, $\int_{\partial T} f(z) dz = 0$ for every triangle $T \subseteq U$. Since U is star-shaped, our corollary gives that f has an antiderivative on U , and then the fundamental theorem of calculus gives $\int_{\gamma} f(z) dz = 0$ for every closed γ . $\square$

Remark (Deforming Curves). The reader should think of Cauchy's theorem as a statement about the *geometry* of U as much as about f . The correct general statement, which we will prove later in the course, is that $\int_{\gamma} f dz = 0$ whenever the closed curve γ can be continuously shrunk to a point within U (that is, γ is *null-homotopic*), and more generally that $\int_{\gamma} f dz$ is unchanged when γ is continuously deformed within U . Star-shaped domains are exactly domains where every closed curve can be shrunk to the centre, sliding along straight lines. What can go wrong in general is visible already for $f(z) = 1/z$ on $\mathbb{C} \setminus \{0\}$: a circle around the origin cannot be shrunk to a point without crossing 0, and correspondingly its integral is $2\pi i \neq 0$.

§10.3 Cauchy's Integral Formula and its Applications

§10.3.1 The Cauchy Integral Formula

We now come to a truly remarkable result: the value of a holomorphic function at a point is completely determined by its values on a circle around that point, via an explicit integral formula. Almost all of the ‘magic’ properties of holomorphic functions flow from this.

Throughout, for a disk $D(a, \rho)$ we write $\int_{\partial D(a, \rho)} f(z) \, dz$ for $\int_{\gamma} f(z) \, dz$ where $\gamma(t) = a + \rho e^{2\pi i t}$, $t \in [0, 1]$, the boundary circle traversed once anticlockwise.

We will need a simple lemma about swapping limits and integrals.

Lemma 10.3.1 (Uniform Convergence and Integrals)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a curve and let f_n be a sequence of continuous functions on the image of γ , converging uniformly on the image of γ to f . Then

$$\int_{\gamma} f_n(z) \, dz \rightarrow \int_{\gamma} f(z) \, dz.$$

Proof. By our basic estimate,

$$\left| \int_{\gamma} f_n(z) \, dz - \int_{\gamma} f(z) \, dz \right| \leq \text{length}(\gamma) \sup_{\gamma} |f_n - f| \rightarrow 0.$$

□

Theorem 10.3.2 (Cauchy Integral Formula for a Disk)

Let $D = D(a, r)$ and let $f : D \rightarrow \mathbb{C}$ be holomorphic. Then for any $0 < \rho < r$ and any $w \in D(a, \rho)$,

$$f(w) = \frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{z - w} \, dz.$$

Proof. Fix $w \in D(a, \rho)$ and define $h : D \rightarrow \mathbb{C}$ by

$$h(z) = \begin{cases} \frac{f(z) - f(w)}{z - w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w. \end{cases}$$

Then h is continuous on D and holomorphic on $D \setminus \{w\}$. Since D is convex, Cauchy's theorem (with one exceptional point) gives

$$0 = \int_{\partial D(a, \rho)} h(z) \, dz = \int_{\partial D(a, \rho)} \frac{f(z)}{z - w} \, dz - f(w) \int_{\partial D(a, \rho)} \frac{dz}{z - w}.$$

So it suffices to prove that

$$\int_{\partial D(a, \rho)} \frac{dz}{z - w} = 2\pi i.$$

For $z \in \partial D(a, \rho)$ we have $|w - a| < \rho = |z - a|$, so expanding as a geometric series,

$$\frac{1}{z - w} = \frac{1}{(z - a) \left(1 - \frac{w - a}{z - a}\right)} = \sum_{j=0}^{\infty} \frac{(w - a)^j}{(z - a)^{j+1}},$$

with the convergence uniform in $z \in \partial D(a, \rho)$ by the Weierstrass M-test (the terms are bounded by $(|w - a|/\rho)^j/\rho$). Hence by our lemma we may integrate term by term:

$$\int_{\partial D(a, \rho)} \frac{dz}{z - w} = \sum_{j=0}^{\infty} (w - a)^j \int_{\partial D(a, \rho)} \frac{dz}{(z - a)^{j+1}}.$$

For $j \geq 1$ the integrand $(z - a)^{-j-1}$ has an antiderivative near the circle, so those terms vanish; and for $j = 0$ we computed the integral directly before, obtaining $2\pi i$. This completes the proof. $\square$

Taking $w = a$ and parameterising the circle explicitly, we obtain a particularly pleasant special case.

Corollary 10.3.3 (Mean Value Property)

If $f : D(a, r) \rightarrow \mathbb{C}$ is holomorphic and $0 < \rho < r$, then

$$f(a) = \int_0^1 f(a + \rho e^{2\pi i t}) dt.$$

In other words – the value of a holomorphic function at a point is the *average* of its values on any circle centred at that point.

§10.3.2 Liouville's Theorem and the Fundamental Theorem of Algebra

Here is the first of the surprising rigidity theorems that we promised back in the first section.

Theorem 10.3.4 (Liouville's Theorem)

Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be entire and bounded. Then f is constant.

Proof. Say $|f| \leq M$ on $\mathbb{C}$. Fix $w \in \mathbb{C}$, and let $\rho > 2|w|$. By the Cauchy integral formula applied on a large disk $D(0, \rho')$ with $\rho < \rho'$,

$$f(w) - f(0) = \frac{1}{2\pi i} \int_{\partial D(0, \rho)} f(z) \left(\frac{1}{z - w} - \frac{1}{z} \right) dz = \frac{1}{2\pi i} \int_{\partial D(0, \rho)} \frac{wf(z)}{z(z - w)} dz.$$

For z on the circle $|z| = \rho$ we have $|z - w| \geq \rho - |w| \geq \rho/2$, so estimating,

$$|f(w) - f(0)| \leq \frac{1}{2\pi} \cdot 2\pi\rho \cdot \frac{|w|M}{\rho \cdot \rho/2} = \frac{2|w|M}{\rho}.$$

Letting $\rho \rightarrow \infty$ gives $f(w) = f(0)$. Since w was arbitrary, f is constant. $\square$

Notice how strong this is: there is no entire analogue of $\sin$ on $\mathbb{R}$, bounded and wiggling forever – indeed $\sin$ itself is wildly unbounded on $\mathbb{C}$ (consider $\sin(iy)$). One spectacular

application is a genuinely short proof of the fundamental theorem of algebra.

Theorem 10.3.5 (Fundamental Theorem of Algebra)

Every non-constant polynomial with complex coefficients has a root in $\mathbb{C}$.

Proof. Let $p(z) = a_n z^n + \cdots + a_1 z + a_0$ with $n \geq 1$ and $a_n \neq 0$. For $z \neq 0$,

$$|p(z)| \geq |z|^n \left(|a_n| - \frac{|a_{n-1}|}{|z|} - \cdots - \frac{|a_0|}{|z|^n} \right),$$

and the bracket tends to $|a_n| > 0$ as $|z| \rightarrow \infty$. So there is $R > 0$ such that $|p(z)| \geq 1$ for all $|z| > R$.

Now suppose for contradiction that p has no root. Then $g = 1/p$ is entire. By the above, $|g| \leq 1$ outside $D(0, R)$; and $|g|$ is continuous on the compact set $\overline{D(0, R)}$, hence bounded there too. So g is a bounded entire function, and by Liouville's theorem g is constant – but then p is constant, a contradiction. $\square$

Remark. It follows by induction (dividing out roots) that every polynomial of degree $n \geq 1$ factors completely as $p(z) = a_n(z - z_1) \cdots (z - z_n)$. So $\mathbb{C}$ is algebraically closed – the analytic input of Liouville's theorem proves a purely algebraic statement.

§10.3.3 Taylor Expansions

We now show that holomorphic functions are *analytic*: locally, they are given by convergent power series. Together with our earlier work on power series, this instantly upgrades ‘differentiable once’ to ‘differentiable infinitely often’.

Theorem 10.3.6 (Taylor Expansion)

Let $f : D(a, R) \rightarrow \mathbb{C}$ be holomorphic. Then f has a convergent power series representation on $D(a, R)$:

$$f(w) = \sum_{n=0}^{\infty} c_n (w - a)^n, \quad \text{where} \quad c_n = \frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z - a)^{n+1}} dz$$

for any $0 < \rho < R$.

Proof. Let $0 < \rho < R$ and $w \in D(a, \rho)$. By the Cauchy integral formula and the geometric series expansion from its proof (which converges uniformly for $z \in \partial D(a, \rho)$),

$$\begin{aligned} f(w) &= \frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{z - w} dz = \frac{1}{2\pi i} \int_{\partial D(a, \rho)} f(z) \sum_{n=0}^{\infty} \frac{(w - a)^n}{(z - a)^{n+1}} dz \\ &= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z - a)^{n+1}} dz \right) (w - a)^n, \end{aligned}$$

where the interchange of sum and integral is justified by our uniform convergence lemma (f is bounded on the circle, so the partial sums converge uniformly there).

Write $c_n(\rho)$ for the coefficient above. We have shown $f(w) = \sum c_n(\rho)(w-a)^n$ on $D(a, \rho)$, so by our theorem on derivatives of power series, $c_n(\rho) = f^{(n)}(a)/n!$. In particular $c_n(\rho)$ does not depend on ρ , so we may write c_n unambiguously. $\square$

Corollary 10.3.7 (Holomorphic Functions are Infinitely Differentiable)

If f is holomorphic on an open set $U \subseteq \mathbb{C}$, then f has derivatives of all orders on U , and they are all holomorphic on U .

Proof. Around any point of U there is a disk on which f is a convergent power series, and power series have derivatives of all orders inside their disk of convergence. $\square$

This settles the promise made at the start of the course, and it also completes the picture for the Cauchy-Riemann equations: if $f = u + iv$ is holomorphic then u, v are automatically C^∞ , and in particular the harmonic function remark from Section 1 applies in earnest.

Remark (Holomorphic = Analytic). We say f is **analytic** at a if it is given by a convergent power series in some neighbourhood of a . Combining the Taylor expansion theorem with our theorem on differentiating power series, the following are equivalent for a function on an open set U : f is holomorphic on U ; f is analytic at every point of U ; f has complex derivatives of all orders on U . Nothing like this is true on $\mathbb{R}$: the function $x \mapsto e^{-1/x^2}$ (extended by 0) is C^∞ but is not given by its Taylor series at 0.

From the Taylor coefficients we can also read off integral formulas for all derivatives, along with very useful bounds.

Theorem 10.3.8 (Cauchy Integral Formula for Derivatives & Cauchy Estimates)

Let $f : D(a, R) \rightarrow \mathbb{C}$ be holomorphic. Then for any $0 < \rho < R$ and $w \in D(a, \rho)$,

$$f^{(k)}(w) = \frac{k!}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z-w)^{k+1}} dz.$$

Moreover, we have the estimate

$$\sup_{w \in D(a, R/2)} |f^{(k)}(w)| \leq \frac{k! 2^{k+1}}{R^k} \sup_{z \in D(a, R)} |f(z)|.$$

Proof. The case $k = 0$ is the Cauchy integral formula. For $k = 1$, consider $g(z) = f(z)/(z-w)$, holomorphic on $D(a, R) \setminus \{w\}$, with

$$g'(z) = \frac{f'(z)}{z-w} - \frac{f(z)}{(z-w)^2}.$$

Now g is an antiderivative of g' on $D(a, R) \setminus \{w\}$, which contains the circle $\partial D(a, \rho)$, so $\int_{\partial D(a, \rho)} g'(z) dz = 0$ by the fundamental theorem of calculus. Hence

$$\int_{\partial D(a, \rho)} \frac{f'(z)}{z-w} dz = \int_{\partial D(a, \rho)} \frac{f(z)}{(z-w)^2} dz,$$

and applying the Cauchy integral formula to the holomorphic function f' gives the $k = 1$ case. For general k we induct: assuming the formula for k (for all holomorphic functions on $D(a, R)$), the same trick with $g(z) = f(z)/(z - w)^{k+1}$ gives

$$\int_{\partial D(a, \rho)} \frac{f'(z)}{(z - w)^{k+1}} dz = (k + 1) \int_{\partial D(a, \rho)} \frac{f(z)}{(z - w)^{k+2}} dz,$$

and applying the inductive hypothesis to f' we get

$$f^{(k+1)}(w) = \frac{k!}{2\pi i} \int_{\partial D(a, \rho)} \frac{f'(z)}{(z - w)^{k+1}} dz = \frac{(k + 1)!}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z - w)^{k+2}} dz.$$

For the estimate, we may assume $M = \sup_{D(a, R)} |f| < \infty$. Let $\rho \in (R/2, R)$ and $w \in D(a, R/2)$. Then $|z - w| \geq \rho - R/2$ for $z \in \partial D(a, \rho)$, so

$$|f^{(k)}(w)| \leq \frac{k!}{2\pi} \cdot 2\pi\rho \cdot \frac{M}{(\rho - R/2)^{k+1}} = \frac{k!\rho M}{(\rho - R/2)^{k+1}},$$

and letting $\rho \rightarrow R$ gives $|f^{(k)}(w)| \leq k!RM/(R/2)^{k+1} = k!2^{k+1}M/R^k$. $\square$

We can also now prove a converse to Cauchy's theorem, which gives a criterion for holomorphy requiring no differentiability assumptions whatsoever.

Theorem 10.3.9 (Morera's Theorem)

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ be continuous such that

$$\int_{\gamma} f(z) dz = 0$$

for every closed curve γ in U . Then f is holomorphic on U .

Proof. The statement is local, so we may assume U is a disk (restricting to a disk around each point, where the hypothesis still holds). Then by the converse to the fundamental theorem of calculus, f has an antiderivative F on U , i.e. $F' = f$. But F is holomorphic, so it is infinitely differentiable, and in particular $f = F'$ is holomorphic. $\square$

Corollary 10.3.10 (Removing Exceptional Points)

Let $U \subseteq \mathbb{C}$ be open, $S \subset U$ finite, and let $f : U \rightarrow \mathbb{C}$ be continuous on U and holomorphic on $U \setminus S$. Then f is holomorphic on U .

Proof. Around each point $a \in U$ take a disk $D(a, r) \subseteq U$. By Cauchy's theorem for star-shaped domains (with exceptional points), every closed curve integral of f in $D(a, r)$ vanishes, so by Morera's theorem f is holomorphic on $D(a, r)$. $\square$

§10.3.4 Zeros and the Identity Theorem

Power series give us very tight control over the zeros of holomorphic functions. Suppose $f : D(a, R) \rightarrow \mathbb{C}$ is holomorphic and not identically zero, and write its Taylor expansion

$f(z) = \sum_{n \geq 0} c_n(z-a)^n$. Since $f \not\equiv 0$, some coefficient is non-zero (by our power series theorem, if all $c_n = 0$ then $f \equiv 0$ on the disk), so we may let

$$m = \min\{n \geq 0 \mid c_n \neq 0\}.$$

Then we can factor

$$f(z) = (z-a)^m g(z), \quad \text{where} \quad g(z) = \sum_{n=m}^{\infty} c_n(z-a)^{n-m}$$

is holomorphic on $D(a, R)$ with $g(a) = c_m \neq 0$.

Definition 10.3.11 (Order of a Zero)

In the above setting, if $f(a) = 0$ (that is, $m \geq 1$), we say f has a **zero of order m** at a . Equivalently, m is the least n with $f^{(n)}(a) \neq 0$. A zero of order 1 is called a **simple zero**.

Theorem 10.3.12 (Principle of Isolated Zeros)

Let $f : D(a, R) \rightarrow \mathbb{C}$ be holomorphic and not identically zero. Then there is some $0 < r \leq R$ such that $f(z) \neq 0$ for all z with $0 < |z-a| < r$.

Proof. If $f(a) \neq 0$, this is immediate from continuity. If $f(a) = 0$, write $f(z) = (z-a)^m g(z)$ as above with $g(a) \neq 0$. By continuity of g there is $r > 0$ with $g \neq 0$ on $D(a, r)$, and then $f(z) \neq 0$ for $0 < |z-a| < r$. $\square$

In other words – the zeros of a non-trivial holomorphic function are *isolated*: each zero has a punctured neighbourhood free of other zeros. There is no holomorphic function vanishing on a little segment, or on a little disk, other than the zero function. This has a striking global consequence.

Theorem 10.3.13 (Identity Theorem)

Let $U \subseteq \mathbb{C}$ be a domain and $f, g : U \rightarrow \mathbb{C}$ be holomorphic. If the set $S = \{z \in U \mid f(z) = g(z)\}$ contains a non-isolated point^a, then $f = g$ on all of U .

^aA point $w \in S$ is *non-isolated* if every punctured disk $D(w, r) \setminus \{w\}$ meets S .

Proof. Let $h = f - g$, so $S = \{z \in U \mid h(z) = 0\}$, and suppose $w \in S$ is a non-isolated point of S . Define

$$\begin{aligned} U_0 &= \{z \in U \mid h \equiv 0 \text{ on some } D(z, r)\}, \\ U_1 &= \{z \in U \mid h^{(n)}(z) \neq 0 \text{ for some } n \geq 0\}. \end{aligned}$$

These are disjoint: if h vanishes identically near z then so do all its derivatives at z . They also cover U : if $z \notin U_1$ then every Taylor coefficient of h at z vanishes, and so $h \equiv 0$ on any disk $D(z, r) \subseteq U$, meaning $z \in U_0$.

Moreover both sets are open: U_0 is open essentially by definition, and U_1 is open since each $h^{(n)}$ is continuous. Since U is a domain (hence connected), one of U_0, U_1 is empty.

Finally, the non-isolated point w lies in U_0 : if it did not, then $h \not\equiv 0$ near w , so by the principle of isolated zeros h would be non-vanishing on some punctured disk around w , contradicting non-isolation. So $U_0 \neq \emptyset$, hence $U_1 = \emptyset$ and $U = U_0$, that is, $h \equiv 0$ on U . $\square$

Remark (Analytic Continuation). The identity theorem says that a holomorphic function on a domain is completely determined by its values on any set with a limit point – say, a tiny disk, or a convergent sequence. If $U \subseteq V$ are domains and $f : U \rightarrow \mathbb{C}$ is holomorphic, then there is *at most one* holomorphic $g : V \rightarrow \mathbb{C}$ with $g|_U = f$; such a g is called the **analytic continuation** of f to V . It may or may not exist! For instance $f(z) = \sum_{n \geq 0} z^n$ is holomorphic on $D(0, 1)$, and admits the analytic continuation $1/(1 - z)$ to $\mathbb{C} \setminus \{1\}$, far beyond the disk of convergence. In contrast, one can show that $f(z) = \sum_{n \geq 0} z^{n!}$ (also holomorphic on $D(0, 1)$) admits no analytic continuation past the unit disk whatsoever.

Example 10.3.14 (Real Identities Propagate)

Since $\sin^2 x + \cos^2 x = 1$ holds for all real x , the holomorphic function $\sin^2 z + \cos^2 z - 1$ vanishes on $\mathbb{R}$, a set with (many!) non-isolated points. By the identity theorem it vanishes identically, so $\sin^2 z + \cos^2 z = 1$ for *all* $z \in \mathbb{C}$. The same argument transfers essentially any polynomial identity from $\mathbb{R}$ to $\mathbb{C}$.

§10.3.5 The Maximum Modulus Principle

Another consequence of the mean value property is that a holomorphic function can never have an interior maximum of modulus, unless it is constant.

Theorem 10.3.15 (Local Maximum Modulus Principle)

Let $f : D(a, R) \rightarrow \mathbb{C}$ be holomorphic. If $|f(z)| \leq |f(a)|$ for all $z \in D(a, R)$, then f is constant.

Proof. Let $0 < \rho < R$. By the mean value property and the basic estimate,

$$|f(a)| = \left| \int_0^1 f(a + \rho e^{2\pi i t}) dt \right| \leq \sup_{t \in [0, 1]} |f(a + \rho e^{2\pi i t})| \leq |f(a)|,$$

using the hypothesis for the final inequality. So both inequalities are equalities. Equality in the basic estimate forces $t \mapsto f(a + \rho e^{2\pi i t})$ to be constant, with modulus $|f(a)|$ (by the first equality). Since $\rho \in (0, R)$ was arbitrary, $|f| \equiv |f(a)|$ is constant on $D(a, R)$.

Finally, a holomorphic function of constant modulus is constant. If $|f|^2 = u^2 + v^2$ is a non-zero constant, differentiating gives $uu_x + vv_x = 0$ and $uu_y + vv_y = 0$; using the Cauchy-Riemann equations to eliminate the y -derivatives, $uu_x - vv_x = 0$ and $uu_y + vv_x = 0$, and solving this linear system (whose determinant is $u^2 + v^2 \neq 0$) gives $u_x = u_y = 0$, and similarly for v . So $f' = 0$ and f is constant on the (connected) disk. If instead $|f| \equiv 0$ then $f \equiv 0$. $\square$

Corollary 10.3.16 (Maximum Modulus Principle)

Let U be a bounded open set and let $f : \overline{U} \rightarrow \mathbb{C}$ be continuous on $\overline{U}$ and holomorphic on U . Then $|f|$ attains its maximum over $\overline{U}$ on the boundary ∂U .

Proof. Since $\overline{U}$ is compact and $|f|$ continuous, the maximum is attained at some $w \in \overline{U}$. If $w \in \partial U$ we are done, so suppose $w \in U$, and take $D(w, r) \subseteq U$. By the local maximum modulus principle, f is constant on $D(w, r)$, and hence by the identity theorem f is constant on the connected component U' of U containing w . By continuity f is constant on $\overline{U'}$, so the value $|f(w)|$ is also attained at points of $\partial U' \subseteq \partial U$.^a $\square$

^aTo see $\partial U' \subseteq \partial U$: a boundary point z of U' cannot lie in U , for otherwise a small disk around z in U would be connected to points of U' , forcing $z \in U'$, which is impossible as U' is open. So $z \in \overline{U} \setminus U = \partial U$.

§10.4 Uniform Limits of Holomorphic Functions

We now look at sequences of holomorphic functions that converge uniformly – or rather, uniformly on the parts of the domain that matter.

Definition 10.4.1 (Locally Uniform Convergence)

Let $U \subset \mathbb{C}$ be open, and let $f_n : U \rightarrow \mathbb{C}$ be a sequence of functions. We say that f_n converges **locally uniformly** on U if for each $a \in U$ there is some $r > 0$ such that f_n converges uniformly on $D(a, r)$.

Example 10.4.2 (Sequence of Functions Converging Locally Uniformly)

Consider the sequence of functions $f_n(z) = z^n$. Then $f_n \rightarrow 0$ locally uniformly on $D(0, 1)$, but not uniformly on $D(0, 1)$. This is because $f_n \rightarrow 0$ uniformly on any disk $D(0, r)$ if $r < 1$, but $\sup_{z \in D(0, 1)} |f_n(z)| = 1$ for each n .

Proposition 10.4.3

The sequence of functions f_n converges locally uniformly on U if and only if f_n converges uniformly on each compact subset $K \subset U$.

Proof. For the forward direction, suppose f_n converges locally uniformly, and let $K \subset U$ be compact. Each $a \in K$ has a disk $D(a, r_a)$ on which f_n converges uniformly, and these disks cover K , so by compactness finitely many of them, say $D(a_1, r_{a_1}), \dots, D(a_m, r_{a_m})$, cover K . Given $\varepsilon > 0$, for each i there is N_i such that $|f_n(z) - f(z)| < \varepsilon$ for all $n \geq N_i$ and $z \in D(a_i, r_{a_i})$; taking $N = \max_i N_i$ handles all $z \in K$ at once. So f_n converges uniformly on K .

Conversely, for each $a \in U$ there is $r > 0$ with the compact disk $\overline{D(a, r)} \subset U$, and uniform convergence on $\overline{D(a, r)}$ implies uniform convergence on $D(a, r)$. $\square$

Theorem 10.4.4 (Uniform Limits of Holomorphic Functions)

Let $U \subset \mathbb{C}$ be open, and $f_n : U \rightarrow \mathbb{C}$ be a sequence of holomorphic functions. If f_n converges locally uniformly on U to some function $f : U \rightarrow \mathbb{C}$, then f is holomorphic. Moreover, $f'_n \rightarrow f'$ locally uniformly on U ^a.

^aBy applying this result iteratively, we also get $f_n^{(k)} \rightarrow f^{(k)}$ locally uniformly on U , for every k .

Proof. Let $a \in U$, and choose $r > 0$ such that $\overline{D(a, r)} \subset U$ and $f_n \rightarrow f$ uniformly on $\overline{D(a, r)}$. A uniform limit of continuous functions is continuous, so f is continuous on $\overline{D(a, r)}$.

Now let γ be any closed curve in $D(a, r)$. The disk $D(a, r)$ is convex, so Cauchy's theorem gives $\int_{\gamma} f_n(z) dz = 0$ for every n . By uniform convergence on the image of γ (a subset of $\overline{D(a, r)}$), our lemma on swapping limits and integrals gives

$$\int_{\gamma} f(z) dz = \lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = 0.$$

Then Morera's theorem shows that f is holomorphic on $D(a, r)$, and since $a \in U$ was arbitrary, f is holomorphic on U .

For the convergence of derivatives, again take $a \in U$ and r as above. Applying the Cauchy estimate (with $k = 1$, $R = r$) to the holomorphic function $f_n - f$,

$$\sup_{z \in D(a, r/2)} |f'_n(z) - f'(z)| \leq \frac{4}{r} \sup_{z \in D(a, r)} |f_n(z) - f(z)| \rightarrow 0$$

as $n \rightarrow \infty$. So $f'_n \rightarrow f'$ uniformly on $D(a, r/2)$, giving locally uniform convergence. $\square$

This result spectacularly fails for real functions, as seen by the following theorem (which we will not prove).

Theorem 10.4.5 (Weierstrass Approximation Theorem)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on a compact interval $[a, b] \subset \mathbb{R}$. Then there is a sequence of polynomials converging to f uniformly on $[a, b]$.

Since there are continuous nowhere-differentiable functions on $[a, b]$, a uniform limit of real polynomials – the nicest functions imaginable – need not be differentiable at even a single point. For holomorphic functions, by contrast, local uniform limits preserve *everything*.

§10.5 Winding Numbers and the General Cauchy Theorem

So far, Cauchy's theorem needs the domain to be star-shaped. To go further we need a way of measuring how a closed curve sits inside a general domain – in particular, how many times it wraps around any given point. This is the *winding number*, and with it we can state and prove Cauchy's theorem in essentially complete generality.

§10.5.1 Winding Numbers

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed curve and let w be a point not on the curve. For each t we can certainly write

$$\gamma(t) = w + r(t)e^{i\theta(t)},$$

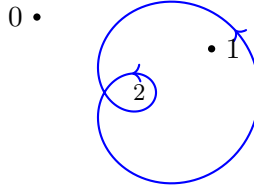
where $r(t) = |\gamma(t) - w| > 0$ is uniquely determined and continuous. The angle $\theta(t)$ is only determined up to adding multiples of 2π – but as we will show below, it is possible to make a *continuous* choice of θ , and this is what makes the following definition sensible.

Definition 10.5.1 (Winding Number)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed curve, $w \notin \text{image}(\gamma)$, and let $\theta : [a, b] \rightarrow \mathbb{R}$ be a continuous function with $\gamma(t) = w + r(t)e^{i\theta(t)}$. The **winding number** (or **index**) of γ about w is

$$I(\gamma; w) = \frac{\theta(b) - \theta(a)}{2\pi}.$$

Two things need checking. First, since $\gamma(a) = \gamma(b)$, we have $e^{i(\theta(b) - \theta(a))} = 1$, so $I(\gamma; w)$ is an *integer*. Second, it is well defined: if θ_1 is another continuous choice, then $(\theta_1 - \theta)/2\pi$ is a continuous integer-valued function on an interval, hence constant, so both choices give the same value of $I(\gamma; w)$.



The picture shows a closed curve along with the winding number of the curve about points of the various regions of the complement – points it wraps around once, twice, and not at all.

For the curves we care about (piecewise C^1), a continuous angle function always exists, and better still, the winding number is computed by an integral.

Lemma 10.5.2 (Existence of Continuous Argument)

Let $\gamma : [a, b] \rightarrow \mathbb{C} \setminus \{w\}$ be a (piecewise C^1) curve. Then there is a piecewise C^1 function $\theta : [a, b] \rightarrow \mathbb{R}$ with $\gamma(t) = w + r(t)e^{i\theta(t)}$, where $r(t) = |\gamma(t) - w|$. Moreover, if γ is closed, then

$$I(\gamma; w) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z - w}.$$

Proof. Define

$$h(t) = \int_a^t \frac{\gamma'(s)}{\gamma(s) - w} ds.$$

The integrand is bounded and continuous away from the finitely many points where γ' jumps, so h is continuous and piecewise C^1 , with $h'(t) = \gamma'(t)/(\gamma(t) - w)$ away

from those points. Then

$$\frac{d}{dt} [(\gamma(t) - w)e^{-h(t)}] = \gamma'(t)e^{-h(t)} - (\gamma(t) - w)h'(t)e^{-h(t)} = 0$$

except at finitely many points, so by continuity $(\gamma(t) - w)e^{-h(t)}$ is constant, equal to $\gamma(a) - w$. Writing $\gamma(a) - w = r(a)e^{i\alpha}$, we get

$$\gamma(t) - w = r(a)e^{\operatorname{Re} h(t)}e^{i(\alpha + \operatorname{Im} h(t))},$$

so $\theta(t) = \alpha + \operatorname{Im} h(t)$ is a continuous (piecewise C^1) argument function, and comparing moduli, $r(t) = r(a)e^{\operatorname{Re} h(t)}$.

Now suppose γ is closed. Then $r(b) = r(a)$, so $\operatorname{Re} h(b) = 0$, and

$$I(\gamma; w) = \frac{\theta(b) - \theta(a)}{2\pi} = \frac{\operatorname{Im} h(b)}{2\pi} = \frac{h(b)}{2\pi i} = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(s)}{\gamma(s) - w} ds = \frac{1}{2\pi i} \int_\gamma \frac{dz}{z - w}.$$

□

Proposition 10.5.3 (Winding Numbers are Locally Constant)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed curve. Then $w \mapsto I(\gamma; w)$ is locally constant on $\mathbb{C} \setminus \operatorname{image}(\gamma)$; in particular, it is constant on each connected component of this set.

Proof. The function $w \mapsto \frac{1}{2\pi i} \int_\gamma \frac{dz}{z - w}$ is continuous on the open set $\mathbb{C} \setminus \operatorname{image}(\gamma)$ (the integrand depends continuously on w , uniformly for z on the curve, since $|z - w|$ is bounded below near any fixed w_0). A continuous integer-valued function is locally constant, and a locally constant function is constant on connected components. □

Proposition 10.5.4 (Winding Number Far Away)

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a closed curve. Then $\mathbb{C} \setminus \operatorname{image}(\gamma)$ has exactly one unbounded connected component Ω , and $I(\gamma; w) = 0$ for all $w \in \Omega$.

Proof. The image of γ is compact, so it lies in some disk $D(0, R)$. The connected set $\mathbb{C} \setminus D(0, R)$ lies in a single component Ω of the complement of the curve, which is then unbounded; every other component avoids $\mathbb{C} \setminus D(0, R)$ and hence is contained in $D(0, R)$, i.e. is bounded. Finally, if $w \in \mathbb{C} \setminus D(0, R)$, then $z \mapsto 1/(z - w)$ is holomorphic on the convex set $D(0, R)$ which contains γ , so $I(\gamma; w) = 0$ by Cauchy's theorem for convex domains. By local constancy, $I(\gamma; \cdot) = 0$ on all of Ω . □

§10.5.2 The General Cauchy Theorem

We can now state the theorem. The correct condition on the curve turns out to be the following.

Definition 10.5.5 (Homologous to Zero)

Let $U \subseteq \mathbb{C}$ be open. A closed curve γ in U is **homologous to zero** in U if

$I(\gamma; w) = 0$ for every $w \in \mathbb{C} \setminus U$.

Intuitively $-\gamma$ does not wind around any of the ‘holes’ of U (points outside U). Note this is automatic if $U = \mathbb{C}$.

Before the main theorem, we need a somewhat technical lemma on difference quotients.

Lemma 10.5.6 (Difference Quotients are Jointly Continuous)

Let $U \subseteq \mathbb{C}$ be open and $f : U \rightarrow \mathbb{C}$ be holomorphic. Define $g : U \times U \rightarrow \mathbb{C}$ by

$$g(z, w) = \begin{cases} \frac{f(z) - f(w)}{z - w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w. \end{cases}$$

Then g is continuous. Moreover, for any closed curve γ in U , the function $h(w) = \int_{\gamma} g(z, w) dz$ is holomorphic on U .

Proof. Continuity of g is clear at points (z, w) with $z \neq w$. For a point (a, a) , let $\varepsilon > 0$, and use continuity of f' (recall f' is holomorphic, in particular continuous) to pick $\delta > 0$ with $D(a, \delta) \subseteq U$ and $|f'(\zeta) - f'(a)| < \varepsilon$ for $\zeta \in D(a, \delta)$. For $z, w \in D(a, \delta)$ with $z \neq w$, the segment from w to z lies in $D(a, \delta)$, and

$$f(z) - f(w) = \int_0^1 \frac{d}{dt} f(tz + (1-t)w) dt = (z - w) \int_0^1 f'(tz + (1-t)w) dt,$$

so that

$$\begin{aligned} |g(z, w) - g(a, a)| &= \left| \int_0^1 [f'(tz + (1-t)w) - f'(a)] dt \right| \\ &\leq \sup_{t \in [0,1]} |f'(tz + (1-t)w) - f'(a)| < \varepsilon. \end{aligned}$$

The case $z = w$ is simply $|f'(z) - f'(a)| < \varepsilon$. Hence g is continuous at (a, a) .

Now let γ be a closed curve in U and $h(w) = \int_{\gamma} g(z, w) dz$. First, h is continuous: if $w_n \rightarrow w_0$ in U , choose $\delta > 0$ with $\overline{D(w_0, \delta)} \subset U$; then g is uniformly continuous on the compact set $\text{image}(\gamma) \times \overline{D(w_0, \delta)}$, so $g(\cdot, w_n) \rightarrow g(\cdot, w_0)$ uniformly on $\text{image}(\gamma)$, and hence $h(w_n) \rightarrow h(w_0)$ by our uniform convergence lemma.

To see h is holomorphic, we use Morera. Fix $w_0 \in U$ and a disk $D(w_0, \delta) \subseteq U$, and let β be any closed curve in $D(w_0, \delta)$. Writing out both integrals as integrals over parameter intervals, Fubini’s theorem^a lets us swap the order of integration:

$$\int_{\beta} h(w) dw = \int_{\beta} \left(\int_{\gamma} g(z, w) dz \right) dw = \int_{\gamma} \left(\int_{\beta} g(z, w) dw \right) dz.$$

But for each fixed z , the function $w \mapsto g(z, w)$ is continuous on U and holomorphic on $U \setminus \{z\}$, hence holomorphic on U by our corollary on removing exceptional points. Since $D(w_0, \delta)$ is convex, $\int_{\beta} g(z, w) dw = 0$ by Cauchy’s theorem. Hence

$\int_{\beta} h(w) dw = 0$ for all closed β in $D(w_0, \delta)$, so h is holomorphic on $D(w_0, \delta)$ by Morera's theorem, and w_0 was arbitrary. $\square$

^aFor continuous integrands on products of compact intervals, Fubini's theorem – that the two iterated integrals agree – follows by uniform continuity: approximate the integrand uniformly by step functions, for which the interchange is immediate.

Theorem 10.5.7 (General Cauchy Theorem & Integral Formula)

Let $U \subseteq \mathbb{C}$ be a non-empty open set and γ a closed curve in U , homologous to zero in U . Then for every holomorphic $f : U \rightarrow \mathbb{C}$:

- (i) $I(\gamma; w)f(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - w} dz$ for every $w \in U \setminus \text{image}(\gamma)$;
- (ii) $\int_{\gamma} f(z) dz = 0$.

Proof. Let us first note that (i) implies (ii): apply (i) to the holomorphic function $F(z) = (z - w)f(z)$ for any fixed $w \in U \setminus \text{image}(\gamma)$; since $F(w) = 0$, we get $\int_{\gamma} f(z) dz = 0$.

So we prove (i), which (multiplying up) is precisely the statement that $h(w) = \int_{\gamma} g(z, w) dz = 0$ for all $w \in U \setminus \text{image}(\gamma)$, where g is the difference quotient function of the previous lemma. By that lemma, h is holomorphic on U . The plan is to extend h to an entire function which vanishes at infinity, and then invoke Liouville.

Let

$$V = \{w \in \mathbb{C} \setminus \text{image}(\gamma) \mid I(\gamma; w) = 0\}.$$

Since γ is homologous to zero in U , we have $\mathbb{C} \setminus U \subseteq V$, i.e. $U \cup V = \mathbb{C}$. Also V is open, since $I(\gamma; \cdot)$ is locally constant. On the overlap $U \cap V$, for $w \in U \cap V$,

$$h(w) = \int_{\gamma} \frac{f(z) - f(w)}{z - w} dz = \int_{\gamma} \frac{f(z)}{z - w} dz - 2\pi i I(\gamma; w)f(w) = \int_{\gamma} \frac{f(z)}{z - w} dz,$$

as $I(\gamma; w) = 0$ there. So h agrees on $U \cap V$ with the function

$$h_1(w) = \int_{\gamma} \frac{f(z)}{z - w} dz, \quad w \in V,$$

which is holomorphic on V (by the standard argument: it is continuous, and holomorphic under the integral sign – or simply repeat the Morera argument of the lemma). Hence

$$H(w) = \begin{cases} h(w) & \text{if } w \in U, \\ h_1(w) & \text{if } w \in V \end{cases}$$

is a well-defined entire function.

Now pick R with $\text{image}(\gamma) \subset D(0, R)$. Then $\mathbb{C} \setminus D(0, R)$ lies in the unbounded component of the complement of γ , where the winding number vanishes, so $\mathbb{C} \setminus D(0, R) \subseteq$

V. For $|w| > R$,

$$|H(w)| = |h_1(w)| \leq \frac{\text{length}(\gamma)}{|w| - R} \sup_{z \in \text{image}(\gamma)} |f(z)| \rightarrow 0 \quad \text{as } |w| \rightarrow \infty.$$

In particular H is bounded (continuous on a compact disk, small outside it), so by Liouville's theorem H is constant – and since $H(w) \rightarrow 0$, that constant is 0. Hence $h \equiv 0$ on U , as required. $\square$

This machine has some immediate and very useful corollaries about systems of curves.

Corollary 10.5.8 (Cauchy's Theorem for Several Curves)

Let $U \subseteq \mathbb{C}$ be open and let $\gamma_1, \dots, \gamma_n$ be closed curves in U such that

$$\sum_{j=1}^n I(\gamma_j; w) = 0 \quad \text{for all } w \in \mathbb{C} \setminus U.$$

Then for every holomorphic $f : U \rightarrow \mathbb{C}$,

$$\sum_{j=1}^n \int_{\gamma_j} f(z) \, dz = 0, \quad \text{and} \quad f(w) \sum_{j=1}^n I(\gamma_j; w) = \sum_{j=1}^n \frac{1}{2\pi i} \int_{\gamma_j} \frac{f(z)}{z - w} \, dz$$

for every $w \in U$ not on any of the curves.

Proof. Run the proof of the theorem again, defining $h(w) = \sum_j \int_{\gamma_j} g(z, w) \, dz$ and $V = \{w \mid \sum_j I(\gamma_j; w) = 0\}$; every step goes through verbatim. $\square$

Corollary 10.5.9 (Curves with Equal Winding Numbers)

Let $U \subseteq \mathbb{C}$ be open and β_1, β_2 be closed curves in U with $I(\beta_1; w) = I(\beta_2; w)$ for every $w \in \mathbb{C} \setminus U$. Then

$$\int_{\beta_1} f(z) \, dz = \int_{\beta_2} f(z) \, dz$$

for every holomorphic $f : U \rightarrow \mathbb{C}$.

Proof. Apply the previous corollary to the curves $\gamma_1 = \beta_1$ and $\gamma_2 = -\beta_2$, noting $I(-\beta_2; w) = -I(\beta_2; w)$. $\square$

§10.5.3 Homotopy and Simply Connected Domains

The condition ‘homologous to zero’ is exactly the right one, but it can be awkward to verify directly. A more intuitive (and slightly stronger) condition comes from the notion of *continuously deforming* curves.

Definition 10.5.10 (Homotopy)

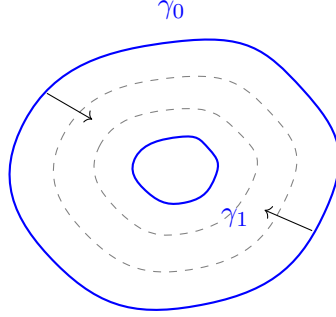
Let $U \subseteq \mathbb{C}$ be a domain and $\gamma_0, \gamma_1 : [a, b] \rightarrow U$ be closed curves. We say γ_0 and γ_1

are **homotopic** in U if there is a continuous map $H : [0, 1] \times [a, b] \rightarrow U$ such that

$$H(0, t) = \gamma_0(t), \quad H(1, t) = \gamma_1(t), \quad H(s, a) = H(s, b)$$

for all $s \in [0, 1]$ and $t \in [a, b]$.

Writing $\gamma_s(t) = H(s, t)$, the definition says exactly that $\{\gamma_s\}_{s \in [0, 1]}$ is a family of closed curves in U deforming γ_0 continuously into γ_1 , without ever leaving U .



Definition 10.5.11 (Null-Homotopic)

A closed curve in a domain U is **null-homotopic** if it is homotopic in U to a constant curve.

The key fact is that deforming a curve cannot change any of the winding numbers that matter.

Theorem 10.5.12 (Homotopic Curves have Equal Winding Numbers)

If $\gamma_0, \gamma_1 : [a, b] \rightarrow U$ are homotopic closed curves in a domain U , then $I(\gamma_0; w) = I(\gamma_1; w)$ for every $w \in \mathbb{C} \setminus U$. In particular, a null-homotopic closed curve is homologous to zero in U .

Proof. We will use the following fact, which is an exercise on the example sheets: if $\gamma, \tilde{\gamma}$ are closed curves with

$$|\gamma(t) - \tilde{\gamma}(t)| < |w - \gamma(t)| \quad \text{for all } t,$$

then $I(\gamma; w) = I(\tilde{\gamma}; w)$. (Intuitively – if $\tilde{\gamma}$ stays closer to γ than γ is to w , then in walking along $\tilde{\gamma}$ instead of γ we can never gain or lose a loop around w .)

Let H be a homotopy from γ_0 to γ_1 , and fix $w \in \mathbb{C} \setminus U$. The image $K = H([0, 1] \times [a, b])$ is compact and contained in U , so there is $\varepsilon > 0$ with $|w - H(s, t)| > 2\varepsilon$ for all (s, t) . By uniform continuity of H , there is $n \in \mathbb{N}$ such that

$$|s - s'| + |t - t'| \leq \frac{\max\{1, b - a\}}{n} \implies |H(s, t) - H(s', t')| < \varepsilon.$$

For $k = 0, 1, \dots, n$ let $\Gamma_k(t) = H(k/n, t)$, closed curves interpolating from γ_0 to γ_1 with $|\Gamma_{k-1}(t) - \Gamma_k(t)| < \varepsilon < |w - \Gamma_{k-1}(t)|$ for all t .

If each Γ_k were piecewise C^1 we would be done by the example sheet fact. In general they are merely continuous, so we approximate: let $a_j = a + j(b - a)/n$ for $0 \leq j \leq n$,

and let $\tilde{\Gamma}_k$ be the closed polygonal curve agreeing with Γ_k at each a_j and affine on each $[a_{j-1}, a_j]$. On $[a_{j-1}, a_j]$, the value $\tilde{\Gamma}_k(t)$ is a convex combination of $\Gamma_k(a_{j-1})$ and $\Gamma_k(a_j)$, both within ε of $\Gamma_k(t)$ by uniform continuity; so $|\tilde{\Gamma}_k(t) - \Gamma_k(t)| < \varepsilon$ for all t . Next we estimate the distance between consecutive polygonal curves: $\tilde{\Gamma}_{k-1}(t) - \tilde{\Gamma}_k(t)$ is a convex combination of $\Gamma_{k-1}(a_{j-1}) - \Gamma_k(a_{j-1})$ and $\Gamma_{k-1}(a_j) - \Gamma_k(a_j)$, each of modulus less than ε , so $|\tilde{\Gamma}_{k-1}(t) - \tilde{\Gamma}_k(t)| < \varepsilon$. Also

$$|w - \tilde{\Gamma}_{k-1}(t)| \geq |w - \Gamma_{k-1}(t)| - |\Gamma_{k-1}(t) - \tilde{\Gamma}_{k-1}(t)| > 2\varepsilon - \varepsilon = \varepsilon.$$

So the example sheet fact applies to each consecutive pair, giving $I(\tilde{\Gamma}_0; w) = I(\tilde{\Gamma}_1; w) = \cdots = I(\tilde{\Gamma}_n; w)$. Finally, $|\tilde{\Gamma}_0(t) - \gamma_0(t)| < \varepsilon < |w - \gamma_0(t)|$ gives $I(\tilde{\Gamma}_0; w) = I(\gamma_0; w)$, and similarly at the other end, $I(\tilde{\Gamma}_n; w) = I(\gamma_1; w)$. Hence $I(\gamma_0; w) = I(\gamma_1; w)$.

For the last statement: a constant curve has all winding numbers zero, so a null-homotopic curve has $I(\gamma; w) = 0$ for every $w \notin U$. $\square$

Corollary 10.5.13 (Deformation Invariance of Integrals)

If γ_0, γ_1 are homotopic closed curves in a domain U , then

$$\int_{\gamma_0} f(z) \, dz = \int_{\gamma_1} f(z) \, dz$$

for every holomorphic $f : U \rightarrow \mathbb{C}$.

Proof. Combine the theorem with our corollary on curves with equal winding numbers. $\square$

Definition 10.5.14 (Simply Connected)

A domain U is **simply connected** if every closed curve in U is null-homotopic.

Intuitively, simply connected domains are the domains ‘without holes’. For example, every star-shaped domain U with centre a_0 is simply connected: given a closed curve γ , the map $H(s, t) = (1 - s)\gamma(t) + sa_0$ is a homotopy in U (each point $H(s, t)$ lies on the segment from $\gamma(t)$ to the centre) from γ to the constant curve at a_0 .

Theorem 10.5.15 (Cauchy's Theorem for Simply Connected Domains)

Let U be a simply connected domain and $f : U \rightarrow \mathbb{C}$ be holomorphic. Then

$$\int_{\gamma} f(z) \, dz = 0$$

for every closed curve γ in U .

Proof. Every closed curve in U is null-homotopic, hence homologous to zero, so the general Cauchy theorem applies. $\square$

Remark. The converse is also true (though harder to prove): if $\int_{\gamma} f \, dz = 0$ for all

holomorphic f and closed curves γ in U , then U is simply connected. However, at the level of individual curves the two notions genuinely differ: in $U = \mathbb{C} \setminus \{w_1, w_2\}$ (two punctures), there are closed curves that are homologous to zero but *not* null-homotopic – the classic example traces around both punctures in a clever ‘figure-of-eight-like’ pattern so that both winding numbers cancel to zero, yet the curve cannot be shrunk to a point. Homology counts winding; homotopy remembers more.

§10.6 Isolated Singularities and Laurent Series

The Cauchy integral formula lets us integrate functions of the form $f(z)/(z - a)$ where f is holomorphic at a . But many natural functions – say $e^{1/z}$ near $z = 0$ – fail to be holomorphic at a point in a more serious way, and we would still like to integrate them around that point. This leads us to classify how a holomorphic function can behave near an isolated ‘bad point’, and to a generalisation of Taylor series adapted to such points.

§10.6.1 Removable Singularities and Poles

Definition 10.6.1 (Isolated Singularity)

Let $U \subseteq \mathbb{C}$ be open and $a \in U$. If $f : U \setminus \{a\} \rightarrow \mathbb{C}$ is holomorphic, we say f has an **isolated singularity** at a .

The mildest possibility is that the singularity is illusory.

Definition 10.6.2 (Removable Singularity)

An isolated singularity a of f is a **removable singularity** if f can be defined at a in such a way that the extended function is holomorphic on U .

Proposition 10.6.3 (Characterising Removable Singularities)

Let U be open, $a \in U$, and $f : U \setminus \{a\} \rightarrow \mathbb{C}$ be holomorphic. The following are equivalent:

- (i) f has a removable singularity at a ;
- (ii) $\lim_{z \rightarrow a} f(z)$ exists in $\mathbb{C}$;
- (iii) $|f|$ is bounded on $D(a, \varepsilon) \setminus \{a\}$ for some $\varepsilon > 0$;
- (iv) $\lim_{z \rightarrow a} (z - a)f(z) = 0$.

Proof. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) are each immediate. The content is (iv) $\Rightarrow$ (i). Define $h : U \rightarrow \mathbb{C}$ by

$$h(z) = \begin{cases} (z - a)^2 f(z) & \text{if } z \neq a, \\ 0 & \text{if } z = a. \end{cases}$$

Then h is holomorphic on $U \setminus \{a\}$, and at a ,

$$\lim_{z \rightarrow a} \frac{h(z) - h(a)}{z - a} = \lim_{z \rightarrow a} (z - a)f(z) = 0,$$

using (iv). So h is differentiable at a too, with $h(a) = h'(a) = 0$, i.e. h is holomorphic on U . Its Taylor expansion at a therefore starts at the quadratic term: on some $D(a, r)$ we have $h(z) = (z - a)^2 g(z)$ with g holomorphic on $D(a, r)$. Comparing with the definition of h , we get $f = g$ on $D(a, r) \setminus \{a\}$, so defining $f(a) = g(a)$ extends f holomorphically. $\square$

Example 10.6.4

The function $f(z) = \frac{e^z - 1}{z}$ is holomorphic on $\mathbb{C} \setminus \{0\}$, and $zf(z) = e^z - 1 \rightarrow 0$ as $z \rightarrow 0$, so the singularity at 0 is removable. Indeed, dividing the Taylor series of $e^z - 1$ by z exhibits the extension $\sum_{k \geq 1} z^{k-1}/k!$ directly, an entire function.

If a singularity is not removable, then by (iii) above f is unbounded near a . The tame way this can happen is if f blows up cleanly.

Definition 10.6.5 (Pole)

An isolated singularity a of f is a **pole** if $|f(z)| \rightarrow \infty$ as $z \rightarrow a$.

Definition 10.6.6 (Essential Singularity)

An isolated singularity which is neither removable nor a pole is called an **essential singularity**.

Example 10.6.7

- (i) $f(z) = (z - a)^{-k}$ for $k \in \mathbb{N}$ has a pole at a .
- (ii) $f(z) = e^{1/z}$ has an essential singularity at 0. Indeed, $|f(iy)| = 1$ for real $y \neq 0$, while $f(x) \rightarrow \infty$ as $x \rightarrow 0^+$ along the reals – so $|f|$ neither stays bounded nor tends to infinity.

Poles are completely understood in terms of zeros of holomorphic functions.

Proposition 10.6.8 (Characterising Poles)

Let $f : U \setminus \{a\} \rightarrow \mathbb{C}$ be holomorphic. The following are equivalent:

- (i) f has a pole at a ;
- (ii) there is $\varepsilon > 0$ and a holomorphic $h : D(a, \varepsilon) \rightarrow \mathbb{C}$ with $h(a) = 0$, $h \neq 0$ on $D(a, \varepsilon) \setminus \{a\}$, and $f = 1/h$ on $D(a, \varepsilon) \setminus \{a\}$;
- (iii) there is a unique integer $k \geq 1$ and a unique holomorphic $g : U \rightarrow \mathbb{C}$ with $g(a) \neq 0$ such that

$$f(z) = (z - a)^{-k} g(z) \quad \text{on } U \setminus \{a\}.$$

Proof. (i) $\Rightarrow$ (ii). Since $|f(z)| \rightarrow \infty$, there is $\varepsilon > 0$ with $|f| \geq 1$ on $D(a, \varepsilon) \setminus \{a\}$, so $1/f$ is holomorphic and bounded (by 1) there. By the removable singularity criterion, $1/f$ extends to a holomorphic h on $D(a, \varepsilon)$, and since $|f| \rightarrow \infty$ we must have $h(a) = 0$; clearly $h \neq 0$ away from a and $f = 1/h$.

(ii) $\Rightarrow$ (iii). By our factorisation of zeros, $h(z) = (z - a)^k h_1(z)$ for some $k \geq 1$ and holomorphic h_1 with $h_1 \neq 0$ on $D(a, \varepsilon)$ (shrinking ε if needed). Then $g_1 = 1/h_1$ is holomorphic on $D(a, \varepsilon)$ and $f(z) = (z - a)^{-k} g_1(z)$ there. Now define $g : U \rightarrow \mathbb{C}$ by $g = g_1$ on $D(a, \varepsilon)$ and $g(z) = (z - a)^k f(z)$ on $U \setminus \{a\}$; the two definitions agree on the overlap, so g is well-defined, holomorphic, and $g(a) = g_1(a) \neq 0$.

For uniqueness, if $(z - a)^{-k} g(z) = (z - a)^{-\tilde{k}} \tilde{g}(z)$ with $g(a), \tilde{g}(a) \neq 0$, then $g(z) = (z - a)^{k-\tilde{k}} \tilde{g}(z)$ away from a ; if $k \neq \tilde{k}$ this forces one of $g(a), \tilde{g}(a)$ to vanish (letting $z \rightarrow a$), a contradiction. So $k = \tilde{k}$, and then $g = \tilde{g}$ off a , hence everywhere by continuity.

(iii) $\Rightarrow$ (i). Immediate: $|f(z)| = |z - a|^{-k} |g(z)| \rightarrow \infty$ as $z \rightarrow a$ since $g(a) \neq 0$. $\square$

Definition 10.6.9 (Order of a Pole)

If f has a pole at a , the unique integer $k \geq 1$ above is the **order** of the pole. A pole of order 1 is called a **simple pole**.

Definition 10.6.10 (Meromorphic)

Let U be open and $S \subset U$ be a discrete subset (all of its points are isolated). If $f : U \setminus S \rightarrow \mathbb{C}$ is holomorphic and every point of S is a removable singularity or a pole of f , we say f is **meromorphic** on U .

Remark (The Riemann Sphere). A pole is not really a ‘bad point’ at all, if one is willing to enlarge the target. Adding a point at infinity to $\mathbb{C}$ yields the *Riemann sphere* $\mathbb{C} \cup \{\infty\}$, and a meromorphic function on U becomes a genuinely continuous (indeed holomorphic, suitably interpreted) map $U \rightarrow \mathbb{C} \cup \{\infty\}$ by sending poles to ∞ . From this point of view the only true singularities are essential ones. This perspective is developed properly in the Part II course *Riemann Surfaces*.

Near an essential singularity, on the other hand, the behaviour is truly wild.

Theorem 10.6.11 (Casorati-Weierstrass)

Let $f : U \setminus \{a\} \rightarrow \mathbb{C}$ be holomorphic with an essential singularity at a . Then for every $\varepsilon > 0$, the image $f(D(a, \varepsilon) \setminus \{a\})$ is *dense* in $\mathbb{C}$: for any $b \in \mathbb{C}$ and any $\delta > 0$, there are points z arbitrarily close to a with $|f(z) - b| < \delta$.

Proof. Suppose not: there are $b \in \mathbb{C}$, $\delta > 0$ and $\varepsilon > 0$ such that $|f(z) - b| \geq \delta$ for all $z \in D(a, \varepsilon) \setminus \{a\}$. Then

$$g(z) = \frac{1}{f(z) - b}$$

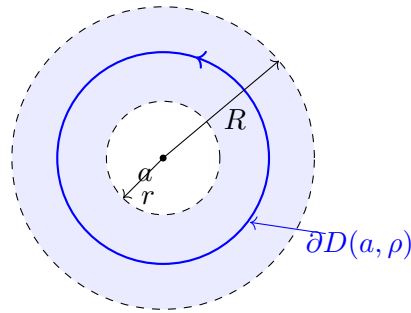
is holomorphic and bounded (by $1/\delta$) on $D(a, \varepsilon) \setminus \{a\}$, so g extends holomorphically over a . If $g(a) \neq 0$ then $f = b + 1/g$ is bounded near a , so a is removable; if $g(a) = 0$

then $|f(z)| = |b + 1/g(z)| \rightarrow \infty$ as $z \rightarrow a$, so a is a pole. Either way we contradict the singularity being essential. $\square$

Remark (Picard's Theorem). Much more is true: *Picard's theorem* states that near an essential singularity, f attains every complex value, with at most a single exception, infinitely often. (For $e^{1/z}$ the exceptional value is 0.) The proof is well beyond this course.

§10.6.2 Laurent Series

We can now generalise Taylor series to functions holomorphic on an *annulus* – in particular on a punctured disk, the natural home of an isolated singularity. The price is that negative powers of $(z - a)$ appear; the reward is that they tell us everything about the singularity.



Theorem 10.6.12 (Laurent Expansion)

Let $0 \leq r < R \leq \infty$ and let f be holomorphic on the annulus

$$A = \{z \in \mathbb{C} \mid r < |z - a| < R\}.$$

Then:

- (i) f has a unique convergent series expansion on A ,

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n = \sum_{n=1}^{\infty} c_{-n} (z - a)^{-n} + \sum_{n=0}^{\infty} c_n (z - a)^n;$$

- (ii) for any $\rho \in (r, R)$, the coefficients are given by

$$c_n = \frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z - a)^{n+1}} dz;$$

- (iii) if $r < \rho' \leq \rho < R$, the two series in (i) converge uniformly on $\{\rho' \leq |z - a| \leq \rho\}$.

Proof. Fix $w \in A$, and choose ρ_1, ρ_2 with $r < \rho_1 < |w - a| < \rho_2 < R$. As in the proof of the Cauchy integral formula, the function

$$g(z) = \begin{cases} \frac{f(z) - f(w)}{z - w} & \text{if } z \neq w, \\ f'(w) & \text{if } z = w \end{cases}$$

is continuous on A and holomorphic on $A \setminus \{w\}$, hence holomorphic on A (removable singularity). The circles $\partial D(a, \rho_1)$ and $\partial D(a, \rho_2)$ are homotopic in A (deform the radius continuously), so

$$\int_{\partial D(a, \rho_1)} g(z) \, dz = \int_{\partial D(a, \rho_2)} g(z) \, dz.$$

Substituting the definition of g and using $I(\partial D(a, \rho_1); w) = 0$ (as $|w - a| > \rho_1$) and $I(\partial D(a, \rho_2); w) = 1$, we obtain the annular Cauchy integral formula

$$f(w) = \frac{1}{2\pi i} \int_{\partial D(a, \rho_2)} \frac{f(z)}{z - w} \, dz - \frac{1}{2\pi i} \int_{\partial D(a, \rho_1)} \frac{f(z)}{z - w} \, dz.$$

Now expand each integrand in a geometric series, exactly as for Taylor series. On $\partial D(a, \rho_2)$ we have $|w - a| < |z - a|$, so

$$\frac{1}{z - w} = \sum_{n=0}^{\infty} \frac{(w - a)^n}{(z - a)^{n+1}},$$

uniformly in z ; while on $\partial D(a, \rho_1)$ we have $|z - a| < |w - a|$, so

$$\frac{1}{z - w} = - \sum_{n=0}^{\infty} \frac{(z - a)^n}{(w - a)^{n+1}},$$

again uniformly. Integrating term by term (using our uniform convergence lemma) gives

$$f(w) = \sum_{n=0}^{\infty} c_n (w - a)^n + \sum_{n=1}^{\infty} c_{-n} (w - a)^{-n},$$

where c_n (for $n \geq 0$) is given by the integral over $\partial D(a, \rho_2)$ as in (ii), and c_{-n} (for $n \geq 1$) by the integral over $\partial D(a, \rho_1)$. Since any two circles $\partial D(a, \rho)$, $\rho \in (r, R)$, are homotopic in A and the integrands are holomorphic on A , the integrals do not depend on the radius used, giving (ii) as stated.

For (iii) and uniqueness, suppose $f(z) = \sum_{n=-\infty}^{\infty} c_n (z - a)^n$ on A for some coefficients c_n . The non-negative part $\sum_{n \geq 0} c_n (z - a)^n$ converges on A , so its radius of convergence is at least R , giving uniform convergence on $|z - a| \leq \rho < R$. For the negative part, substitute $\zeta = (z - a)^{-1}$: the series $\sum_{n \geq 1} c_{-n} \zeta^n$ converges for $1/R < |\zeta| < 1/r$, so has radius of convergence at least $1/r$, and converges uniformly on $|\zeta| \leq 1/\rho'$, i.e. on $|z - a| \geq \rho'$. This proves (iii). Then for any $m \in \mathbb{Z}$ we may integrate the uniformly convergent series term by term against $(z - a)^{-m-1}$:

$$\int_{\partial D(a, \rho)} \frac{f(z)}{(z - a)^{m+1}} \, dz = \sum_{n=-\infty}^{\infty} c_n \int_{\partial D(a, \rho)} (z - a)^{n-m-1} \, dz = 2\pi i c_m,$$

since $\int_{\partial D(a, \rho)} (z - a)^j \, dz = 0$ for $j \neq -1$ (antiderivative!) and $= 2\pi i$ for $j = -1$. So the coefficients are forced to be those of (ii), giving uniqueness. $\square$

Remark. If f happens to extend holomorphically to the full disk $D(a, R)$, then each coefficient c_{-m} with $m \geq 1$ vanishes by Cauchy's theorem (the integrand is then holomorphic on the disk), and the Laurent series is just the Taylor series. The new content

is precisely for functions with no such extension.

Applying this on a punctured disk ($r = 0$), the shape of the Laurent series reads off the type of singularity.

Proposition 10.6.13 (Classification of Singularities via Laurent Series)

Let f be holomorphic on the punctured disk $D(a, R) \setminus \{a\}$, with Laurent expansion $f(z) = \sum_{n \in \mathbb{Z}} c_n(z - a)^n$. Then:

- (i) a is a removable singularity if and only if $c_n = 0$ for all $n < 0$;
- (ii) a is a pole of order k if and only if $c_{-k} \neq 0$ and $c_n = 0$ for all $n < -k$;
- (iii) a is an essential singularity if and only if $c_n \neq 0$ for infinitely many $n < 0$.

Proof. If $c_n = 0$ for $n < 0$, the series defines a holomorphic function on $D(a, R)$ agreeing with f away from a , so the singularity is removable; conversely if the singularity is removable the Laurent series is the Taylor series of the extension.

If $c_{-k} \neq 0$ and lower coefficients vanish, then $f(z) = (z - a)^{-k}g(z)$ where $g(z) = \sum_{n \geq 0} c_{n-k}(z - a)^n$ is holomorphic on $D(a, R)$ with $g(a) = c_{-k} \neq 0$; by our characterisation, a is a pole of order k . Conversely a pole of order k has this form, and expanding g in a Taylor series produces such a Laurent series (unique!).

Finally (iii) holds since (i) and (ii) are equivalences and the three cases are exhaustive and mutually exclusive. $\square$

Example 10.6.14 (Laurent Series of $e^{1/z}$)

Substituting $1/z$ into the exponential series,

$$e^{1/z} = \sum_{n=0}^{\infty} \frac{z^{-n}}{n!} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \cdots, \quad z \neq 0.$$

Infinitely many negative coefficients are non-zero, confirming that 0 is an essential singularity.

Example 10.6.15 (Different Annuli, Different Series)

A function can have different Laurent expansions on different annuli. Take

$$f(z) = \frac{1}{z(1-z)} = \frac{1}{z} + \frac{1}{1-z}.$$

On the punctured disk $0 < |z| < 1$ we expand $\frac{1}{1-z}$ as a geometric series:

$$f(z) = \frac{1}{z} + 1 + z + z^2 + \cdots,$$

exhibiting a simple pole at 0. But on the annulus $1 < |z| < \infty$ we must instead

expand in powers of $1/z$:

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-1/z} = -\frac{1}{z} - \frac{1}{z^2} - \dots,$$

so there $f(z) = -z^{-2} - z^{-3} - \dots$. The moral: a Laurent expansion is always relative to a choice of annulus.

§10.7 The Residue Theorem

We now assemble everything into the single most useful computational theorem of the course. It turns out that when integrating around an isolated singularity, only *one* Laurent coefficient survives.

§10.7.1 Residues

Definition 10.7.1 (Residue and Principal Part)

Let $f : D(a, R) \setminus \{a\} \rightarrow \mathbb{C}$ be holomorphic, with Laurent expansion $\sum_{n=-\infty}^{\infty} c_n(z-a)^n$. The **residue** of f at a is

$$\operatorname{Res}_{z=a} f = c_{-1},$$

and the **principal part** of f at a is the series

$$f_P(z) = \sum_{n=1}^{\infty} c_{-n}(z-a)^{-n}.$$

Why is c_{-1} singled out? Because $(z-a)^n$ has an antiderivative for every n *except* $n = -1$. So when we integrate the Laurent series term by term around a closed curve, only the c_{-1} term can contribute – and it contributes exactly $2\pi i c_{-1}$ times the winding number. Let us make this precise, in full generality.

Note first that the series defining f_P converges for all $z \neq a$ (its convergence in $0 < |z-a| < R$ forces, via the substitution $\zeta = (z-a)^{-1}$, a radius of convergence of at least $1/r$ for every $r > 0$), and the convergence is uniform on compact subsets of $\mathbb{C} \setminus \{a\}$. In particular f_P is holomorphic on $\mathbb{C} \setminus \{a\}$, and $f - f_P$ has a removable singularity at a .

Theorem 10.7.2 (Residue Theorem)

Let $U \subseteq \mathbb{C}$ be open, let $a_1, \dots, a_k \in U$ be distinct points, and let

$$f : U \setminus \{a_1, \dots, a_k\} \rightarrow \mathbb{C}$$

be holomorphic. If γ is a closed curve in U , homologous to zero in U , with no a_j on the curve, then

$$\frac{1}{2\pi i} \int_{\gamma} f(z) \, dz = \sum_{j=1}^k I(\gamma; a_j) \operatorname{Res}_{z=a_j} f.$$

Proof. Let $f_P^{(j)}$ be the principal part of f at a_j , a holomorphic function on $\mathbb{C} \setminus \{a_j\}$, and set

$$h = f - (f_P^{(1)} + \cdots + f_P^{(k)}),$$

holomorphic on $U \setminus \{a_1, \dots, a_k\}$. Fix j : near a_j , the difference $f - f_P^{(j)}$ has a removable singularity (its Laurent series has no negative terms), and each $f_P^{(\ell)}$ with $\ell \neq j$ is holomorphic at a_j . So h has a removable singularity at each a_j , and extends to a holomorphic function on all of U . By the general Cauchy theorem, $\int_\gamma h(z) dz = 0$, so

$$\frac{1}{2\pi i} \int_\gamma f(z) dz = \sum_{j=1}^k \frac{1}{2\pi i} \int_\gamma f_P^{(j)}(z) dz.$$

Finally, the series for $f_P^{(j)}$ converges uniformly on the (compact) image of γ , so we may integrate term by term; every term $(z - a_j)^{-n}$ with $n \geq 2$ has an antiderivative on $\mathbb{C} \setminus \{a_j\}$ and integrates to zero, leaving

$$\frac{1}{2\pi i} \int_\gamma f_P^{(j)}(z) dz = \frac{c_{-1}^{(j)}}{2\pi i} \int_\gamma \frac{dz}{z - a_j} = I(\gamma; a_j) \operatorname{Res}_{z=a_j} f.$$

□

In practice γ will wind once around each singularity we care about and zero times around everything else, and then the theorem reads: *the integral is $2\pi i$ times the sum of the residues inside.*

§10.7.2 Computing Residues

To use the residue theorem we need to actually compute residues. Thankfully, for poles this rarely requires finding the whole Laurent series.

Proposition 10.7.3 (Residue Computation Rules)

- (i) If f has a simple pole at a , then

$$\operatorname{Res}_{z=a} f = \lim_{z \rightarrow a} (z - a)f(z).$$

- (ii) If $f(z) = (z - a)^{-k}g(z)$ with g holomorphic at a and $g(a) \neq 0$ (a pole of order k), then

$$\operatorname{Res}_{z=a} f = \frac{g^{(k-1)}(a)}{(k-1)!}.$$

- (iii) If $f = g/h$ with g, h holomorphic at a , $g(a) \neq 0$, and h having a simple zero at a , then

$$\operatorname{Res}_{z=a} f = \frac{g(a)}{h'(a)}.$$

Proof. For (ii) (which contains (i) as the case $k = 1$): expanding g in its Taylor series at a , the coefficient of $(z - a)^{-1}$ in the Laurent series of $(z - a)^{-k}g(z)$ is the

coefficient of $(z - a)^{k-1}$ in the Taylor series of g , namely $g^{(k-1)}(a)/(k-1)!$.

For (iii): $h(z) = (z - a)h_1(z)$ with h_1 holomorphic, $h_1(a) = h'(a) \neq 0$, so f has a simple pole at a and by (i)

$$\operatorname{Res}_{z=a} f = \lim_{z \rightarrow a} \frac{(z - a)g(z)}{h(z)} = \lim_{z \rightarrow a} \frac{g(z)}{\left(\frac{h(z)-h(a)}{z-a}\right)} = \frac{g(a)}{h'(a)}.$$

□

Example 10.7.4

Let $f(z) = \frac{e^{iz}}{1+z^2}$. The denominator factors as $(z - i)(z + i)$, so f has simple poles at $\pm i$, and by rule (iii),

$$\operatorname{Res}_{z=i} f = \frac{e^{i \cdot i}}{2i} = \frac{e^{-1}}{2i}.$$

§10.7.3 Evaluating Real Integrals

We now come to the payoff: the residue theorem is a machine for computing real integrals that resist elementary methods. The general strategy is always the same:

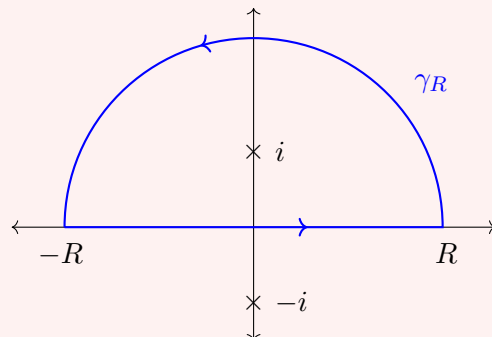
- (1) pick a holomorphic (or meromorphic) function related to the real integrand;
- (2) pick a closed contour, part of which runs along (a piece of) the real axis;
- (3) compute the contour integral by residues;
- (4) show the contribution of the auxiliary parts of the contour vanishes (or converges to something known) in a limit.

Example 10.7.5 (A Semicircular Contour)

We compute

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} \quad (= \pi, \text{ as we know from } \arctan).$$

Let $f(z) = \frac{1}{1+z^2}$, which has simple poles at $\pm i$. For $R > 1$, let γ be the closed contour consisting of the segment $[-R, R]$ along the real axis, followed by the semicircular arc $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$, in the upper half-plane.



The contour is a closed curve in the star-shaped domain $\mathbb{C}$ (so certainly homologous to zero), winding once around i and zero times around $-i$. By the residue theorem

and rule (iii),

$$\int_{\gamma} f(z) \, dz = 2\pi i \operatorname{Res}_{z=i} \frac{1}{1+z^2} = 2\pi i \cdot \frac{1}{2i} = \pi.$$

On the arc, $|1+z^2| \geq R^2 - 1$, so

$$\left| \int_{\gamma_R} f(z) \, dz \right| \leq \pi R \cdot \frac{1}{R^2 - 1} \rightarrow 0 \quad \text{as } R \rightarrow \infty.$$

Hence

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \lim_{R \rightarrow \infty} \int_{-R}^R \frac{dx}{1+x^2} = \pi.$$

That example could have been done in first year – but the same contour computes many integrals that certainly could not. When the integrand involves oscillating factors like e^{ix} , the arc estimate needs more care, and the following lemma is the standard tool.

Lemma 10.7.6 (Jordan's Lemma)

Let f be a continuous complex-valued function on the semicircle $C_R^+ = \{Re^{it} \mid t \in [0, \pi]\}$, and let $\gamma_R^+(t) = Re^{it}$, $t \in [0, \pi]$. Then for $\alpha > 0$,

$$\left| \int_{\gamma_R^+} f(z) e^{i\alpha z} \, dz \right| \leq \frac{\pi}{\alpha} \sup_{z \in C_R^+} |f(z)|.$$

In particular, if $\sup_{z \in C_R^+} |f(z)| \rightarrow 0$ as $R \rightarrow \infty$, then $\int_{\gamma_R^+} f(z) e^{i\alpha z} \, dz \rightarrow 0$.

Proof. Let $M_R = \sup_{C_R^+} |f|$. On the arc, $|e^{i\alpha z}| = e^{-\alpha R \sin t}$, so

$$\left| \int_{\gamma_R^+} f(z) e^{i\alpha z} \, dz \right| \leq \int_0^\pi M_R e^{-\alpha R \sin t} R \, dt = 2RM_R \int_0^{\pi/2} e^{-\alpha R \sin t} \, dt,$$

using the symmetry $\sin(\pi - t) = \sin t$. Now on $(0, \pi/2]$ we have the classical bound $\sin t \geq 2t/\pi$ (the function $\sin t/t$ decreases on this interval, with value $2/\pi$ at $\pi/2$), so

$$2RM_R \int_0^{\pi/2} e^{-\alpha R \sin t} \, dt \leq 2RM_R \int_0^{\pi/2} e^{-2\alpha R t/\pi} \, dt = \frac{\pi M_R}{\alpha} (1 - e^{-\alpha R}) \leq \frac{\pi M_R}{\alpha}.$$

□

The point of Jordan's lemma is that decay of f alone (however slow) suffices – the oscillation of $e^{i\alpha z}$ does the rest. A companion lemma handles small arcs around simple poles that sit *on* our contour.

Lemma 10.7.7 (Small Arcs around Simple Poles)

Let f be holomorphic on $D(a, R) \setminus \{a\}$ with a simple pole at a , and for $0 < \varepsilon < R$

let $\gamma_\varepsilon : [\alpha, \beta] \rightarrow \mathbb{C}$ be the arc $\gamma_\varepsilon(t) = a + \varepsilon e^{it}$. Then

$$\lim_{\varepsilon \rightarrow 0^+} \int_{\gamma_\varepsilon} f(z) \, dz = (\beta - \alpha) i \operatorname{Res}_{z=a} f.$$

Proof. Write $f(z) = \frac{c}{z-a} + g(z)$ with $c = \operatorname{Res}_{z=a} f$ and g holomorphic on $D(a, R)$ (this is exactly the Laurent expansion of a simple pole). Then g is bounded near a , say $|g| \leq M$ on $D(a, R/2)$, so

$$\left| \int_{\gamma_\varepsilon} g(z) \, dz \right| \leq (\beta - \alpha) \varepsilon M \rightarrow 0,$$

while direct computation gives

$$\int_{\gamma_\varepsilon} \frac{c}{z-a} \, dz = c \int_\alpha^\beta \frac{i\varepsilon e^{it}}{\varepsilon e^{it}} \, dt = (\beta - \alpha) ic.$$

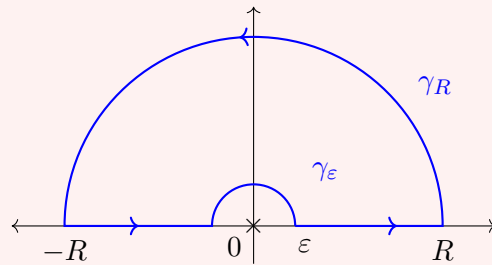
□

Example 10.7.8 (The Dirichlet Integral)

We compute the famous integral

$$\int_0^\infty \frac{\sin x}{x} \, dx = \frac{\pi}{2}.$$

The natural function to use is $f(z) = e^{iz}/z$, which has a simple pole at 0 – right on the contour we would like to use! So we *indent*: for $0 < \varepsilon < R$, consider the closed contour γ consisting of the segment $[\varepsilon, R]$, the large semicircular arc $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$, the segment $[-R, -\varepsilon]$, and the small *clockwise* arc γ_ε from $-\varepsilon$ to ε over the top of the origin.



Now f is holomorphic on $\mathbb{C} \setminus \{0\}$, and γ winds zero times around 0 – the indentation was designed exactly to avoid it – so γ is homologous to zero in $\mathbb{C} \setminus \{0\}$, and the general Cauchy theorem gives $\int_\gamma f(z) \, dz = 0$. That is,

$$\int_{[\varepsilon, R]} \frac{e^{ix}}{x} \, dx + \int_{\gamma_R} f \, dz + \int_{[-R, -\varepsilon]} \frac{e^{ix}}{x} \, dx + \int_{\gamma_\varepsilon} f \, dz = 0.$$

Substituting $x \mapsto -x$ in the third integral and combining with the first,

$$\int_\varepsilon^R \frac{e^{ix} - e^{-ix}}{x} \, dx = 2i \int_\varepsilon^R \frac{\sin x}{x} \, dx.$$

By Jordan's lemma (with $f(z) = 1/z$, $\sup_{C_R^+} |1/z| = 1/R \rightarrow 0$), the integral over γ_R tends to 0 as $R \rightarrow \infty$. The small arc γ_ε is traversed clockwise from angle π to 0, so by the small arcs lemma its integral tends to $-\pi i \operatorname{Res}_{z=0} f = -\pi i$ as $\varepsilon \rightarrow 0^+$. Putting everything together,

$$2i \int_0^\infty \frac{\sin x}{x} dx - \pi i = 0, \quad \text{that is} \quad \int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

Next, an integral involving a branch cut, where the multi-valuedness of z^α is not an obstacle but the very thing that makes the method work.

Example 10.7.9 (A Keyhole Contour)

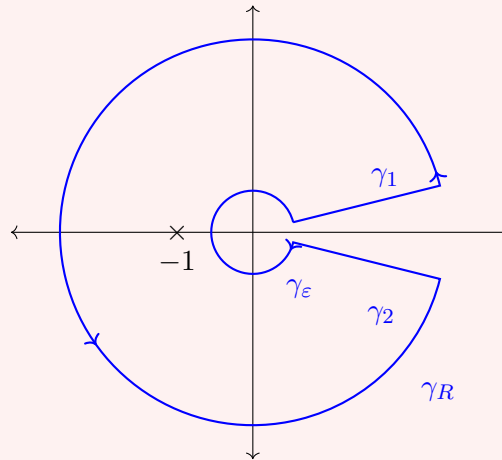
For $0 < \alpha < 1$, we show

$$\int_0^\infty \frac{x^{-\alpha}}{1+x} dx = \frac{\pi}{\sin \pi \alpha}.$$

Work on the cut plane $U = \mathbb{C} \setminus \{x \in \mathbb{R} \mid x \geq 0\}$, on which we choose the branch $g(z) = z^{-\alpha} = e^{-\alpha \ell(z)}$, where $\ell(z) = \ln |z| + i \arg(z)$ with $\arg(z) \in (0, 2\pi)$. Let $f(z) = g(z)/(1+z)$, holomorphic on $U \setminus \{-1\}$ with a simple pole at -1 and

$$\operatorname{Res}_{z=-1} f = g(-1) = e^{-\alpha(\ln 1 + i\pi)} = e^{-i\pi\alpha}.$$

For $0 < \varepsilon < 1 < R$ and small $\theta > 0$, let γ be the 'keyhole contour' below: out along the ray $\arg z = \theta$ from radius ε to R , anticlockwise around the big circle to the ray $\arg z = 2\pi - \theta$, back along that ray, and clockwise around the small circle.



The domain U is not star-shaped, but the winding number of γ about every point of the positive real axis (and about 0) is zero, so γ is homologous to zero in U ; and $I(\gamma; -1) = 1$. By the residue theorem,

$$\int_\gamma f(z) dz = 2\pi i e^{-i\pi\alpha}.$$

Consider the pieces. On the outgoing ray $\gamma_1(t) = te^{i\theta}$, $t \in [\varepsilon, R]$,

$$\int_{\gamma_1} f(z) dz = \int_\varepsilon^R \frac{t^{-\alpha} e^{-i\alpha\theta} e^{i\theta}}{1 + te^{i\theta}} dt \xrightarrow{\theta \rightarrow 0^+} \int_\varepsilon^R \frac{t^{-\alpha}}{1+t} dt,$$

with uniform convergence of the integrands on $[\varepsilon, R]$. On the returning ray γ_2 ($\arg z = 2\pi - \theta$, traversed inwards from radius R down to radius ε), the branch of $z^{-\alpha}$ picks up an extra factor tending to $e^{-2\pi i\alpha}$ as $\theta \rightarrow 0^+$, so in that limit this piece contributes

$$-e^{-2\pi i\alpha} \int_{\varepsilon}^R \frac{t^{-\alpha}}{1+t} dt,$$

the minus sign coming from the direction of traversal. Meanwhile, on the circular arcs we estimate

$$\left| \int_{\gamma_R} f dz \right| \leq 2\pi R \cdot \frac{R^{-\alpha}}{R-1} = \frac{2\pi R^{1-\alpha}}{R-1}, \quad \left| \int_{\gamma_{\varepsilon}} f dz \right| \leq \frac{2\pi \varepsilon^{1-\alpha}}{1-\varepsilon},$$

both independent of θ . Since $0 < \alpha < 1$, both bounds vanish as $R \rightarrow \infty$, $\varepsilon \rightarrow 0^+$ respectively. Taking $\theta \rightarrow 0^+$, then $\varepsilon \rightarrow 0^+$ and $R \rightarrow \infty$,

$$(1 - e^{-2\pi i\alpha}) \int_0^{\infty} \frac{t^{-\alpha}}{1+t} dt = 2\pi i e^{-i\pi\alpha},$$

and dividing through,

$$\int_0^{\infty} \frac{t^{-\alpha}}{1+t} dt = \frac{2\pi i}{e^{i\pi\alpha} - e^{-i\pi\alpha}} = \frac{\pi}{\sin \pi\alpha}.$$

§10.7.4 Summing Series

The residue theorem can also sum infinite series. The trick is to find a meromorphic function with poles at all the integers with known residues – and $\pi \cot(\pi z)$, which has a simple pole of residue 1 at every integer, is precisely such a function.

Example 10.7.10 (The Basel Problem)

We prove

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Consider

$$f(z) = \frac{\pi \cot(\pi z)}{z^2} = \frac{\pi \cos(\pi z)}{z^2 \sin(\pi z)}.$$

The zeros of $\sin(\pi z)$ are exactly the integers, and are simple. So f has simple poles at each $n \in \mathbb{Z} \setminus \{0\}$ with, by our rules,

$$\operatorname{Res}_{z=n} f = \frac{\pi \cos(\pi n)/n^2}{\pi \cos(\pi n)} = \frac{1}{n^2},$$

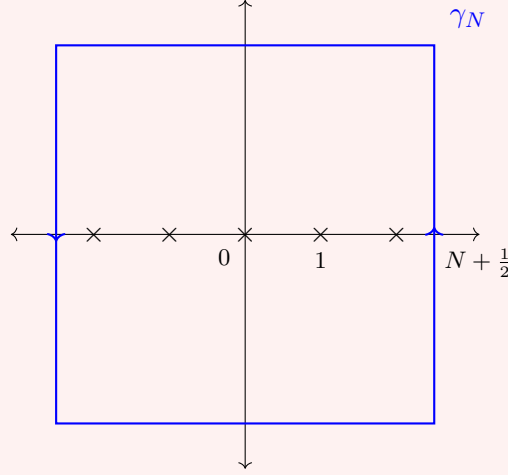
and a pole of order 3 at 0. There, from the Taylor expansions of $\cos$ and $\sin$,

$$\begin{aligned} \cot z &= \frac{1 - z^2/2 + O(z^4)}{z(1 - z^2/6 + O(z^4))} = \frac{1}{z} \left(1 - \frac{z^2}{2} + O(z^4) \right) \left(1 + \frac{z^2}{6} + O(z^4) \right) \\ &= \frac{1}{z} - \frac{z}{3} + O(z^3), \end{aligned}$$

so that

$$f(z) = \frac{\pi \cot(\pi z)}{z^2} = \frac{1}{z^3} - \frac{\pi^2}{3z} + O(z), \quad \text{giving} \quad \text{Res}_{z=0} f = -\frac{\pi^2}{3}.$$

Now, for $N \in \mathbb{N}$, let γ_N be the positively oriented boundary of the square with vertices $(\pm(N + \frac{1}{2}), \pm(N + \frac{1}{2}))$.



The square passes half-way between consecutive integers, so it encloses the poles $0, \pm 1, \dots, \pm N$, each with winding number 1. By the residue theorem,

$$\int_{\gamma_N} f(z) \, dz = 2\pi i \left(2 \sum_{n=1}^N \frac{1}{n^2} - \frac{\pi^2}{3} \right).$$

On the other hand, one can check that $|\cot(\pi z)|$ is bounded on γ_N by a constant C independent of N (on the vertical sides $\cot(\pi z)$ is purely imaginary with modulus at most 1; on the horizontal sides $|\cot(\pi z)| \leq \coth(\pi/2)$). Since $|z| \geq N + \frac{1}{2}$ on the square and $\text{length}(\gamma_N) = 4(2N + 1)$,

$$\left| \int_{\gamma_N} f(z) \, dz \right| \leq \frac{\pi C}{(N + \frac{1}{2})^2} \cdot 4(2N + 1) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Letting $N \rightarrow \infty$ in the previous display therefore gives

$$2 \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{3}, \quad \text{that is} \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The same method, with $\pi \cot(\pi z)$ replaced by $\pi \operatorname{cosec}(\pi z)$ and other kernels, evaluates a whole family of classical series.

§10.8 The Argument Principle and Rouché's Theorem

Our final circle of ideas uses the residue theorem to *count*. The quantity $\frac{f'}{f}$ – the ‘logarithmic derivative’ of f – turns out to have poles exactly at the zeros and poles of f , with residues recording their orders, and integrating it counts zeros and poles inside a contour.

§10.8.1 The Argument Principle

Proposition 10.8.1 (Logarithmic Derivatives)

If f has a zero (respectively, a pole) of order k at $z = a$, then f'/f has a simple pole at a with residue k (respectively $-k$).

Proof. If a is a zero of order k , write $f(z) = (z - a)^k g(z)$ on some disk $D(a, r)$, with g holomorphic and non-vanishing there. Then

$$\frac{f'(z)}{f(z)} = \frac{k(z - a)^{k-1}g(z) + (z - a)^k g'(z)}{(z - a)^k g(z)} = \frac{k}{z - a} + \frac{g'(z)}{g(z)},$$

and g'/g is holomorphic on $D(a, r)$, so f'/f has a simple pole at a with residue k . For a pole of order k , write $f(z) = (z - a)^{-k} g(z)$ and the same computation gives residue $-k$. $\square$

Definition 10.8.2

For a zero or pole a of f , we write $\text{ord}_f(a)$ for its order.

Theorem 10.8.3 (Argument Principle)

Let f be a meromorphic function on a domain U , with finitely many zeros $a_1, \dots, a_k$ and poles $b_1, \dots, b_\ell$. Let γ be a closed curve in U , homologous to zero in U , with no zero or pole of f on the curve. Then

$$\frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = \sum_{i=1}^k I(\gamma; a_i) \text{ord}_f(a_i) - \sum_{j=1}^{\ell} I(\gamma; b_j) \text{ord}_f(b_j).$$

Proof. The function $g = f'/f$ is holomorphic away from the zeros and poles of f (at a removable singularity of f , take the holomorphic extension; if that extension is non-vanishing there, g is holomorphic there too, and if it vanishes we have simply found another zero). So the singularities of g in U are precisely $a_1, \dots, a_k, b_1, \dots, b_\ell$, all simple poles with residues $\text{ord}_f(a_i)$ and $-\text{ord}_f(b_j)$ by the proposition. Now apply the residue theorem to g . $\square$

Why is this called the *argument* principle? Let $\Gamma = f \circ \gamma$, a closed curve avoiding 0 (since f has no zeros or poles on γ). Then, writing $[a, b]$ for the parameter interval of γ ,

$$I(\Gamma; 0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{dw}{w} = \frac{1}{2\pi i} \int_a^b \frac{f'(\gamma(t))\gamma'(t)}{f(\gamma(t))} dt = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz.$$

So the left-hand side of the argument principle is precisely *the number of times $f \circ \gamma$ winds around the origin* – the total change in $\arg f$ along γ , divided by 2π . Zeros inside make the image wind anticlockwise; poles, clockwise.

To turn the argument principle into clean counting statements, we want curves for which all the winding numbers are 0 or 1.

Definition 10.8.4 (Bounding a Domain)

Let Ω be a domain and γ a closed curve. We say γ **bounds** Ω if

$$I(\gamma; w) = \begin{cases} 1 & \text{for } w \in \Omega, \\ 0 & \text{for } w \notin \Omega \cup \text{image}(\gamma). \end{cases}$$

For example, $\partial D(0, 1)$ bounds $D(0, 1)$. If γ bounds Ω then Ω is bounded (all winding numbers vanish far away), and one can check $\partial\Omega \subseteq \text{image}(\gamma)$. The fundamental supply of such curves comes from the *Jordan curve theorem*: every simple closed curve γ (one with no self-intersections) splits the plane into exactly two components, one bounded and one unbounded, and γ or $-\gamma$ bounds the bounded component. This is intuitively obvious and surprisingly hard to prove; we will take it as known, and in any concrete example the winding numbers can be checked by hand anyway.

Corollary 10.8.5 (Counting Zeros and Poles)

Let γ be a closed curve bounding a domain Ω , and let f be meromorphic on an open set $U \supseteq \Omega \cup \text{image}(\gamma)$, with no zeros or poles on $\text{image}(\gamma)$. Then f has finitely many zeros and poles in Ω , and if N and P denote the number of zeros and poles respectively, counted with multiplicity,

$$N - P = \frac{1}{2\pi i} \int_{\gamma} \frac{f'(z)}{f(z)} dz = I(f \circ \gamma; 0).$$

Proof. First, finiteness. The closure $\overline{\Omega}$ is compact (as Ω is bounded) and contained in $\Omega \cup \text{image}(\gamma) \subseteq U$. If f had infinitely many poles in $\overline{\Omega}$, they would accumulate at some $w \in \overline{\Omega} \subseteq U$; but the singular set of a meromorphic function is discrete, and a non-singular point has a neighbourhood free of singularities – either way, a contradiction. If f had infinitely many zeros in $\overline{\Omega}$, they would accumulate at some $w \in \overline{\Omega}$, which is either a regular point or a removable singularity of f (it cannot be a pole, since $f \rightarrow \infty$ at a pole but $f = 0$ along the accumulating zeros); then by the identity theorem $f \equiv 0$ near w , and hence (applying the identity theorem on the domain Ω minus the singular set, which is still connected) f vanishes throughout Ω , contradicting $f \neq 0$ on the boundary curve.

Since γ bounds Ω , we have $I(\gamma; w) = 0$ for every $w \notin \Omega \cup \text{image}(\gamma)$; in particular γ is homologous to zero in U . Every zero or pole in Ω has winding number 1, and those outside contribute nothing, so the argument principle gives exactly $N - P$; and we identified the integral with $I(f \circ \gamma; 0)$ above. $\square$

§10.8.2 The Local Degree and the Open Mapping Theorem

The argument principle also tells us what a holomorphic map looks like *locally*.

Definition 10.8.6 (Local Degree)

Let f be holomorphic and non-constant on a disk $D(a, R)$. The **local degree** of f at a , written $\deg_f(a)$, is the order of the zero of $f(z) - f(a)$ at $z = a$.

Note this is a well-defined positive integer: $f(z) - f(a)$ is not identically zero (else f is constant), so has a zero of some finite order $d \geq 1$ at a . For instance, $f(z) = (z-1)^4 + 1$ has $\deg_f(1) = 4$.

Theorem 10.8.7 (Local Degree Theorem)

Let $f : D(a, R) \rightarrow \mathbb{C}$ be holomorphic and non-constant with $\deg_f(a) = d$. Then there is $r_0 > 0$ such that for every $r \in (0, r_0]$ there is $\varepsilon > 0$ with the property: for every w with $0 < |w - f(a)| < \varepsilon$, the equation $f(z) = w$ has exactly d distinct roots in $D(a, r) \setminus \{a\}$.

Proof. Let $g(z) = f(z) - f(a)$. Since g is non-constant, $g \not\equiv 0$ and $g' \not\equiv 0$ on $D(a, R)$, so by the principle of isolated zeros we may choose $r_0 \in (0, R)$ such that g and g' have no zeros on $D(a, r_0) \setminus \{a\}$.

Fix $r \in (0, r_0]$, let $\gamma(t) = a + re^{2\pi it}$, and let $\Gamma = g \circ \gamma$. The image of Γ is compact and misses 0 (as $g \not\equiv 0$ on the circle), so there is $\varepsilon > 0$ with $D(0, \varepsilon) \cap \text{image}(\Gamma) = \emptyset$.

Now let $0 < |w - f(a)| < \varepsilon$. Since winding numbers are constant on the connected set $D(0, \varepsilon)$,

$$I(\Gamma; w - f(a)) = I(\Gamma; 0).$$

By our counting corollary applied to g on the disk (the circle γ bounds $D(a, r)$), $I(\Gamma; 0)$ is the number of zeros of g in $D(a, r)$, which is d (the single zero at a , counted with multiplicity d). On the other hand, $I(\Gamma; w - f(a))$ counts the zeros of $g(z) - (w - f(a)) = f(z) - w$ in $D(a, r)$, with multiplicity. So $f(z) = w$ has d roots in $D(a, r)$ counted with multiplicity.

Finally, these roots are distinct and none equals a : since $w \neq f(a)$, $z = a$ is not a root; and at any root $z \neq a$ we have $f'(z) = g'(z) \neq 0$ by our choice of r_0 , so each root is simple. $\square$

So near a point of local degree d , a holomorphic map is d -to-one onto a neighbourhood of the image point (think of $z \mapsto z^d$ near 0, which is the model case). One consequence deserves to be a headline theorem.

Theorem 10.8.8 (Open Mapping Theorem)

A non-constant holomorphic function on a domain maps open sets to open sets.

Proof. Let $f : U \rightarrow \mathbb{C}$ be holomorphic and non-constant on the domain U , let $V \subseteq U$ be open, and let $b = f(a) \in f(V)$ with $a \in V$. Note f is non-constant on every disk in U (by the identity theorem, constancy on a disk forces constancy on U). Choose $r > 0$ small enough that $D(a, r) \subseteq V$ and the local degree theorem applies with some $\varepsilon > 0$; then every $w \in D(b, \varepsilon)$ with $w \neq b$ is attained by f on $D(a, r)$, and b itself is attained at a . Hence $D(b, \varepsilon) \subseteq f(D(a, r)) \subseteq f(V)$, so $f(V)$ is open. $\square$

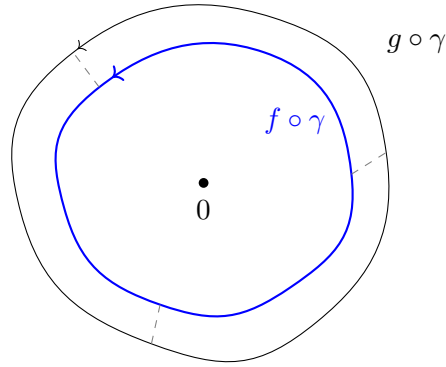
Remark. Compare with the real situation: $x \mapsto x^2$ maps the open interval $(-1, 1)$ onto $[0, 1)$, which is not open. Once again holomorphic functions are far more rigid than smooth real ones.

The local degree theorem also settles a promise from Section 1: *if f is holomorphic and injective, then $f' \neq 0$ everywhere.* Indeed, if $f'(a) = 0$ then $f(z) - f(a)$ has a zero of

order $d \geq 2$ at a , and the local degree theorem produces points near a where f takes the same value $d \geq 2$ times – contradicting injectivity. So in our definition of conformal equivalence, the non-vanishing derivative condition was automatic.

§10.8.3 Rouché's Theorem

Our last theorem is a wonderfully practical tool for locating zeros. The idea is often called ‘walking the dog’. Suppose you walk a closed loop around a lamppost (the origin), holding a dog on a lead. If the lead always stays shorter than your distance to the lamppost, then the dog is dragged around the lamppost exactly as many times as you walk around it – the dog can never sneak an extra loop, nor miss one. Here you are $g \circ \gamma$, the dog is $f \circ \gamma$, and the lead has length $|f - g|$.



Theorem 10.8.9 (Rouché's Theorem)

Let γ be a closed curve bounding a domain Ω , and let f, g be holomorphic on an open set $U \supseteq \Omega \cup \text{image}(\gamma)$. If

$$|f(z) - g(z)| < |g(z)| \quad \text{for all } z \in \text{image}(\gamma),$$

then f and g have the same number of zeros in Ω , counted with multiplicity.

Proof. The strict inequality forces $g \neq 0$ on the curve (else $0 \leq |f - g| < 0$) and then also $f \neq 0$ on the curve ($|f| \geq |g| - |f - g| > 0$). Let $h = f/g$, which is meromorphic on U , holomorphic and non-vanishing on an open set containing $\text{image}(\gamma)$. Both f and g have finitely many zeros in Ω by our counting corollary (neither vanishes identically, as neither vanishes on the curve).

The hypothesis says exactly that

$$|h(z) - 1| < 1 \quad \text{on } \text{image}(\gamma),$$

so the closed curve $h \circ \gamma$ lies in the disk $D(1, 1)$, which does not contain 0. Since the winding number about 0 vanishes on the unbounded component of the complement of $h \circ \gamma$, and 0 can be joined to infinity without crossing $D(1, 1)$, we get $I(h \circ \gamma; 0) = 0$.

By the argument principle (counting corollary) applied to h ,

$$0 = I(h \circ \gamma; 0) = N_h - P_h,$$

where N_h, P_h are the numbers of zeros and poles of h in Ω with multiplicity. Now at any point $w \in \Omega$, writing ord_w for the order of vanishing at w (negative at a pole), we have $\text{ord}_w(h) = \text{ord}_w(f) - \text{ord}_w(g)$, and summing over all relevant points,

$$N_h - P_h = \sum_{w \in \Omega} \text{ord}_w(h) = \sum_{w \in \Omega} \text{ord}_w(f) - \sum_{w \in \Omega} \text{ord}_w(g) = N_f - N_g,$$

where N_f, N_g count the zeros of f and g in Ω with multiplicity (the sums are finite, over the zeros of f and g). Hence $N_f = N_g$. $\square$

Example 10.8.10 (Locating Roots of $z^4 + 6z + 3$)

Let $f(z) = z^4 + 6z + 3$. We claim f has exactly three roots (with multiplicity) in the annulus $\{1 < |z| < 2\}$.

On the circle $|z| = 2$: taking $g(z) = z^4$, we have

$$|f(z) - g(z)| = |6z + 3| \leq 15 < 16 = |z^4| = |g(z)|,$$

so by Rouché's theorem f has the same number of zeros in $D(0, 2)$ as z^4 , namely 4. So all four roots of f lie in $|z| < 2$ (and none on $|z| = 2$, by the strict inequality).

On the circle $|z| = 1$: taking $g(z) = 6z$ instead,

$$|f(z) - g(z)| = |z^4 + 3| \leq 4 < 6 = |6z| = |g(z)|,$$

so f has as many zeros in $D(0, 1)$ as $6z$, namely one (and none on $|z| = 1$).

Subtracting, exactly $4 - 1 = 3$ roots lie in $\{1 < |z| < 2\}$.

Example 10.8.11 (A Perturbation Result)

Rouché's theorem explains why roots of polynomials move continuously with the coefficients. For instance, suppose $p(z) = z^n + c_{n-1}z^{n-1} + \cdots + c_0$ and z_0 is a simple root. For any sufficiently small $\delta > 0$, we have $m = \min_{|z-z_0|=\delta} |p(z)| > 0$, and then any polynomial q with $|q(z) - p(z)| < m$ on the circle $|z - z_0| = \delta$ has, by Rouché's theorem (with $g = p$, $f = q$), exactly one root in $D(z_0, \delta)$. Sufficiently small changes of the coefficients therefore move the root only slightly.

And that – from a single innocuous-looking limit defining $f'(w)$, all the way to counting the roots of polynomials by walking dogs around lampposts – is the IB Complex Analysis course.

11 Complex Methods

Complex analysis has a strange quality to it – results that are hard-won over the reals (and often false there!) come almost for free once we work over $\mathbb{C}$. A function that is complex differentiable once is differentiable infinitely often, is given locally by its Taylor series, and is essentially determined everywhere by its behaviour on a tiny region. This course is about what you can *do* with all of that structure. We will use complex functions to map complicated regions onto simple ones (and solve Laplace’s equation while doing so), evaluate real integrals that resist every technique of real calculus, and invert Fourier and Laplace transforms to solve differential equations.

This article constitutes my notes for the ‘Complex Methods’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. These notes are also designed to complement my notes for the ‘Complex Analysis’ course, which covers the same core theory but with an emphasis on rigour¹. In this article we take the applied point of view: proofs are given when they are short or genuinely illuminating, and are otherwise sketched or omitted with a reference.

§11.1 Analytic Functions

The star of this course is the *analytic function* – a complex function which is complex differentiable. This first section develops what that means, how to recognise such functions, and the surprisingly rigid structure that differentiability imposes. Already in this section we will meet the connection with Laplace’s equation, which drives many of the physical applications later on.

§11.1.1 The Extended Complex Plane

We will assume basic familiarity with the complex numbers themselves: every $z \in \mathbb{C}$ can be written as $z = x + iy$ with $x, y \in \mathbb{R}$, or in polar form as

$$z = re^{i\theta} = r(\cos \theta + i \sin \theta),$$

where $r = |z|$ and $\theta = \arg z$. The argument is only defined up to adding multiples of 2π , a seemingly innocent remark that will cause us a surprising amount of grief in Section 2.

An ‘open set’ will mean the same as it does in analysis: $D \subseteq \mathbb{C}$ is **open** if for every $z \in D$ there is some $\epsilon > 0$ such that the disc $\{w : |w - z| < \epsilon\}$ lies in D . A **domain** is a non-empty open set in which any two points can be joined by a path lying in the set. Domains are the natural habitat of the functions in this course.

It is frequently convenient to add a single ‘point at infinity’ to the complex plane.

¹The two courses share their central theorems – Cauchy’s theorem, Taylor and Laurent series, the residue theorem – and lead to the same examination questions on that material. Where a careful proof exists in my Complex Analysis notes I will generally cite it rather than repeat it, and spend the space here on computations and applications instead.

Definition 11.1.1 (Extended Complex Plane)

The **extended complex plane** is $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$, the complex plane together with a single extra point ∞ .

One can visualise $\mathbb{C}^*$ as a sphere (the *Riemann sphere*): place a sphere on the plane, and join each point $z \in \mathbb{C}$ to the north pole by a straight line. That line meets the sphere at one other point, and this gives a bijection between $\mathbb{C}$ and the sphere minus the north pole. The north pole itself then corresponds to ∞ . Points far from the origin in any direction are all close to the north pole, which is why a single point at infinity (rather than one for each direction, as one does with $\pm\infty$ in $\mathbb{R}$) is the natural choice.

To study the behaviour of a function f ‘at infinity’, we simply study $f(1/z)$ at $z = 0$. We will use this idea repeatedly, and it is worth internalising: *infinity is not a special place, it just looks far away from where we are standing.*

§11.1.2 Complex Differentiation and the Cauchy–Riemann Equations

We define differentiability exactly as we do over the reals.

Definition 11.1.2 (Differentiability)

A function $f : \mathbb{C} \rightarrow \mathbb{C}$ is **differentiable** at z if the limit

$$f'(z) = \lim_{\delta z \rightarrow 0} \frac{f(z + \delta z) - f(z)}{\delta z}$$

exists.

The definition looks identical to the real one, but it is far more demanding. The increment δz is a complex number, and may approach 0 from *any direction* in the plane – along the real axis, along the imaginary axis, along a spiral – and the limit must be the same however it does so. This is a strong constraint, and it is what gives complex differentiable functions their remarkable properties.

Definition 11.1.3 (Analytic Function)

A function f is **analytic**^a at a point z if it is differentiable throughout some open set containing z . A function analytic on all of $\mathbb{C}$ is called **entire**.

^aThe words *holomorphic* and *regular* are also used, particularly in the Complex Analysis course. In this course ‘analytic’ is traditional.

Note the subtlety: analyticity at z requires differentiability *near* z , not just at z . This makes analyticity a much more robust notion, and it is the one we care about.

Let’s extract something concrete from the definition. Write $z = x + iy$ and

$$f(z) = u(x, y) + iv(x, y),$$

where u and v are real-valued functions of two real variables. Suppose f is differentiable at z . Then we may compute $f'(z)$ by letting $\delta z \rightarrow 0$ along whichever direction we please, and comparing the answers.

Taking $\delta z = \delta x$ real, we get

$$f'(z) = \lim_{\delta x \rightarrow 0} \frac{f(z + \delta x) - f(z)}{\delta x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}.$$

Taking $\delta z = i \delta y$ purely imaginary instead,

$$f'(z) = \lim_{\delta y \rightarrow 0} \frac{f(z + i \delta y) - f(z)}{i \delta y} = \frac{1}{i} \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}.$$

These must be equal, and comparing real and imaginary parts gives the fundamental equations of the subject.

Theorem 11.1.4 (Cauchy–Riemann Equations)

If $f = u + iv$ is differentiable at $z = x + iy$, then at that point

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

The converse is almost true: it can fail if the partial derivatives exist but are badly discontinuous.

Theorem 11.1.5 (Converse to Cauchy–Riemann)

If u and v have continuous first-order partial derivatives satisfying the Cauchy–Riemann equations throughout an open set D , then $f = u + iv$ is analytic on D .

Proof (Sketch). With continuous partials we can expand to first order:

$$\begin{aligned} \delta f &= u(x + \delta x, y + \delta y) - u(x, y) + i(v(x + \delta x, y + \delta y) - v(x, y)) \\ &= \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) \delta x + \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) \delta y + o(|\delta z|). \end{aligned}$$

Applying the Cauchy–Riemann equations to the second bracket turns the right hand side into $\left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) (\delta x + i \delta y) + o(|\delta z|)$, so $\delta f / \delta z$ tends to a limit independent of the direction of δz . $\square$

So, up to a regularity condition that never bites in practice, *analytic* = *Cauchy–Riemann*. This gives us a practical test, and it's worth seeing it in action both on functions that pass and functions that fail.

Example 11.1.6 (Analytic Functions) (i) Any polynomial in z is entire – sums, products and compositions of differentiable functions are differentiable, with the same proofs as in real analysis (the familiar product, quotient and chain rules all hold).

(ii) The exponential

$$e^z = e^x (\cos y + i \sin y)$$

is entire, as one can check directly from the Cauchy–Riemann equations: $u = e^x \cos y$ and $v = e^x \sin y$ give $u_x = e^x \cos y = v_y$ and $u_y = -e^x \sin y = -v_x$. Differentiating, $\frac{d}{dz} e^z = u_x + i v_x = e^z$, as we'd hope.

(iii) Any rational function $P(z)/Q(z)$ is analytic away from the zeros of Q . These

‘almost analytic’ functions will be of great interest when we come to residue calculus – the places where analyticity fails are exactly where the action is.

Example 11.1.7 (Non-Analytic Functions)

Consider $f(z) = \bar{z} = x - iy$, so $u = x$ and $v = -y$. Then $u_x = 1$ but $v_y = -1$, so the Cauchy–Riemann equations fail *everywhere*, and $\bar{z}$ is nowhere differentiable, despite being a perfectly smooth function of (x, y) . Similarly $f(z) = |z|^2 = x^2 + y^2$ has $u_x = 2x$, $v_y = 0$, $u_y = 2y$, $v_x = 0$, so the equations hold only at $z = 0$. Thus $|z|^2$ is differentiable at the origin only, and analytic *nowhere* – there is no open set of differentiability.

The moral of these examples: a function is analytic when it is genuinely a function ‘of z alone’, and not of $\bar{z}$. This slogan can be made precise – writing $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$, one can treat z and $\bar{z}$ as independent variables, and then the Cauchy–Riemann equations become the single equation

$$\frac{\partial f}{\partial \bar{z}} = 0.$$

Before moving on, let’s assemble the standard library of analytic functions, since we will compute with them constantly. Each is defined by the natural extension of its real Taylor series (or in terms of e^z), and each is entire:

$$\begin{aligned} \sin z &= \frac{e^{iz} - e^{-iz}}{2i}, & \cos z &= \frac{e^{iz} + e^{-iz}}{2}, \\ \sinh z &= \frac{e^z - e^{-z}}{2}, & \cosh z &= \frac{e^z + e^{-z}}{2}. \end{aligned}$$

Over $\mathbb{C}$, the trigonometric and hyperbolic families merge into one: from the definitions,

$$\sin(iz) = i \sinh z, \quad \cos(iz) = \cosh z,$$

and all the usual addition formulas hold verbatim (they are identities between analytic functions that agree on $\mathbb{R}$ – and, as is shown in Complex Analysis, analytic functions agreeing on a line agree everywhere). Some facts worth internalising now, as they shape all the contour arguments to come:

- (i) e^z is periodic with period $2\pi i$, and is *never zero*: $|e^z| = e^x > 0$.
- (ii) $\sin z$ and $\cos z$ are *unbounded* off the real axis – e.g. $\cos(iy) = \cosh y \rightarrow \infty$. Boundedness of $\sin$ and $\cos$ is a purely real-variable accident.
- (iii) The zeros of $\sin z$ and $\cos z$ are exactly the familiar real ones, $n\pi$ and $(n + \frac{1}{2})\pi$. Writing $\sin(x+iy) = \sin x \cosh y + i \cos x \sinh y$, both terms vanish only when $y = 0$ and $\sin x = 0$.
- (iv) Correspondingly $\sinh z$ has zeros exactly at $z = n\pi i$, and $\cosh z$ at $z = (n + \frac{1}{2})\pi i$ – on the *imaginary* axis. These lattices of zeros will reappear as lattices of poles of $1/\cosh$ and friends in Section 6.

For working in polar coordinates – which we will do constantly once branch cuts appear – it is useful to have the polar form of the Cauchy–Riemann equations.

Proposition 11.1.8 (Cauchy–Riemann in Polar Form)

Let $f(z) = u(r, \theta) + iv(r, \theta)$ where $z = re^{i\theta}$. Then the Cauchy–Riemann equations take the form

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta},$$

and then $f'(z) = e^{-i\theta}(u_r + iv_r)$.

Proof. This is a chain rule computation. From $x = r \cos \theta$, $y = r \sin \theta$ we get

$$\frac{\partial u}{\partial r} = u_x \cos \theta + u_y \sin \theta, \quad \frac{\partial u}{\partial \theta} = -u_x r \sin \theta + u_y r \cos \theta,$$

and similarly for v . Substituting the Cartesian Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ into these expressions gives

$$\frac{1}{r} \frac{\partial v}{\partial \theta} = -v_x \sin \theta + v_y \cos \theta = u_y \sin \theta + u_x \cos \theta = \frac{\partial u}{\partial r},$$

and the second equation follows in the same way. For the derivative, approach along a ray of constant θ , so $\delta z = e^{i\theta} \delta r$, giving $f'(z) = e^{-i\theta}(u_r + iv_r)$. $\square$

Example 11.1.9 (Powers and Logarithms)

Take $f(z) = z^c$ for a real constant c , defined (for now, for $r > 0$ and some range of θ) by $f = r^c e^{ic\theta}$, so $u = r^c \cos c\theta$ and $v = r^c \sin c\theta$. Then

$$u_r = cr^{c-1} \cos c\theta = \frac{1}{r} v_\theta, \quad v_r = cr^{c-1} \sin c\theta = -\frac{1}{r} u_\theta,$$

so z^c is analytic where defined, with

$$f'(z) = e^{-i\theta} \cdot cr^{c-1} e^{ic\theta} = cz^{c-1}.$$

An identical computation shows $\log z = \log r + i\theta$ satisfies the polar Cauchy–Riemann equations with $\frac{d}{dz} \log z = e^{-i\theta} \cdot \frac{1}{r} = \frac{1}{z}$. The phrase ‘where defined’ is hiding real content here – pinning down where these functions can be defined is the subject of Section 2.

§11.1.3 Harmonic Functions

We now come to the first connection with applied mathematics, and it is a spectacular one. Recall Laplace’s equation for a function $\phi(x, y)$:

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.$$

This is arguably the most important partial differential equation there is: it governs steady-state temperature distributions, electrostatic potentials, gravitational potentials, and the flow of ideal fluids.

Definition 11.1.10 (Harmonic Function)

A function $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfying $\nabla^2 \phi = 0$ is **harmonic**.

Here is the connection.

Proposition 11.1.11

If $f = u + iv$ is analytic on a domain D , then u and v are both harmonic on D .

Proof. We use the fact (immediate from results in Complex Analysis, where it is shown that analytic functions are infinitely differentiable) that u and v have continuous second derivatives. Then, using the Cauchy–Riemann equations twice,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \frac{\partial v}{\partial x} = \frac{\partial}{\partial y} \left(-\frac{\partial u}{\partial y} \right) = -\frac{\partial^2 u}{\partial y^2},$$

so u is harmonic, and likewise for v . □

So the real and imaginary parts of *any* analytic function hand us solutions of Laplace’s equation, entirely for free. The functions u and v obtained in this way are not unrelated, and we give the relationship a name.

Definition 11.1.12 (Harmonic Conjugate)

Two harmonic functions u and v satisfying the Cauchy–Riemann equations are **harmonic conjugates**, and then $u + iv$ is analytic.

Given a harmonic function u , we can reconstruct its harmonic conjugate (up to a constant) directly from the Cauchy–Riemann equations, by integrating one equation and differentiating the result. This is best seen by example.

Example 11.1.13 (Finding a Harmonic Conjugate)

Let $u(x, y) = x^2 - y^2$, which is easily checked to be harmonic. We seek v with

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 2x \quad \implies \quad v = 2xy + g(x)$$

for some function g of x alone. The other Cauchy–Riemann equation demands

$$\frac{\partial v}{\partial x} = 2y + g'(x) = -\frac{\partial u}{\partial y} = 2y,$$

so g is a constant, and $v = 2xy + c$. Sure enough,

$$u + iv = x^2 - y^2 + 2ixy + ic = z^2 + ic,$$

and we have recognised our harmonic function as the real part of an analytic one.

One caution about existence. The construction above is local – it works on any disc – but on domains with holes, a harmonic function need not have a *single-valued* conjugate globally. The standard example is $u = \log |z|$ on the punctured plane: its conjugate wants to be $\theta = \arg z$, which (as Section 2 will make much of) cannot be defined continuously all the way around the origin. On simply connected domains this problem never arises,

and every harmonic function is globally the real part of an analytic one.

Example 11.1.14 (A Conjugate in Polar Coordinates)

Let $u = r^2 \cos 2\theta$ (which is $\operatorname{Re} z^2$ in disguise, but let's pretend we don't know that). The polar Cauchy–Riemann equations give

$$\frac{\partial v}{\partial \theta} = r \frac{\partial u}{\partial r} = 2r^2 \cos 2\theta \quad \implies \quad v = r^2 \sin 2\theta + g(r),$$

and then $\frac{\partial v}{\partial r} = 2r \sin 2\theta + g'(r)$ must equal $-\frac{1}{r}u_\theta = 2r \sin 2\theta$, so g is constant. Thus $v = r^2 \sin 2\theta + c$ and $u + iv = r^2 e^{2i\theta} + ic = z^2 + ic$, as expected.

The level curves of u and v carry geometric information too. If $f = u + iv$ is analytic, then

$$\nabla u \cdot \nabla v = u_x v_x + u_y v_y = -u_x u_y + u_y u_x = 0,$$

using the Cauchy–Riemann equations. So *the level curves of u and v are orthogonal* wherever $f' \neq 0$. If u is an electrostatic potential, the curves $u = \text{const}$ are equipotentials and the curves $v = \text{const}$ are field lines; if v is the stream function of a fluid flow, the curves $v = \text{const}$ are streamlines and u is the velocity potential. One analytic function packages the whole picture. We will exploit this systematically in Section 3.

Let's record the fluid dynamics dictionary now, as we will want it later.

Remark (Complex Potentials for Fluid Flow). For steady, incompressible, irrotational flow in the plane with velocity (u_1, u_2) , there is a velocity potential ϕ (with $\nabla \phi = (u_1, u_2)$, existing since the flow is irrotational) and a stream function ψ (existing since the flow is incompressible), and both are harmonic. The **complex potential** is the analytic function

$$w(z) = \phi + i\psi,$$

and the velocity is recovered from $w'(z) = u_1 - iu_2$. The flow is along the level curves of ψ . For example, $w(z) = Uz$ is uniform flow with speed U in the x -direction, and $w(z) = U(z + a^2/z)$ will turn out to describe flow past a circular cylinder of radius a .

§11.2 Multivalued Functions and Branch Cuts

We have already brushed against the problem twice: both $\log z$ and z^c were only defined once we chose a range for $\theta = \arg z$. This section confronts the issue properly. It is a genuinely subtle business – more subtle than it first appears – and it matters a great deal in practice, since the placement of branch cuts is exactly what makes certain real integrals computable in Section 6.

§11.2.1 The Problem with the Logarithm

For $z = re^{i\theta} \neq 0$ we would like to define

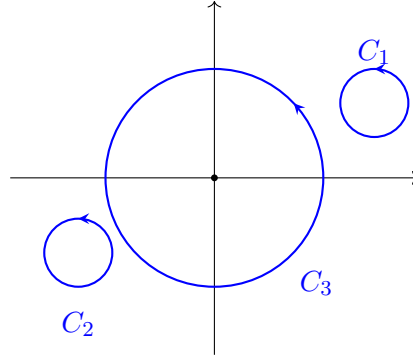
$$\log z = \log r + i\theta.$$

But θ is only defined up to multiples of 2π , so this definition assigns infinitely many values to each point:

$$\log i = \frac{\pi i}{2} \quad \text{or} \quad \frac{5\pi i}{2} \quad \text{or} \quad -\frac{3\pi i}{2} \quad \text{or} \quad \dots$$

We say $\log$ is a **multivalued function**. One might hope to fix this by just picking a value at each point – say, always take $\theta \in (0, 2\pi]$. But an arbitrary choice like this creates discontinuities, and where those discontinuities can and cannot go is the heart of the matter.

Consider tracking a value of $\log z$ continuously along each of the following three curves.



On C_1 , which stays in the right half-plane, we can fix $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ throughout, and $\log z$ is continuous and single-valued all the way round. On C_2 we can similarly fix a range of θ and all is well. But on C_3 , which encircles the origin, there is no possible choice: following the curve anticlockwise once, θ must increase continuously by 2π , so when we return to our starting point the value of $\log z$ has increased by $2\pi i$. The function must either jump somewhere on C_3 , or fail to be single-valued. The origin is the source of the problem, and we give such points a name.

Definition 11.2.1 (Branch Point)

A **branch point** of a function is a point which it is impossible to encircle with a closed curve on which the function is both continuous and single-valued. The function is said to have a **branch point singularity** there.

Example 11.2.2 (Spotting Branch Points) (i) $\log(z - a)$ has a branch point at $z = a$.

(ii) $z^c = r^c e^{ic\theta}$ has a branch point at the origin whenever $c \notin \mathbb{Z}$: going once anticlockwise around a circle of radius r_0 , continuity forces the value to move from r_0^c to $r_0^c e^{2\pi ic}$, and these differ unless c is an integer.

(iii) $\log z$ also has a branch point *at infinity*: setting $\zeta = 1/z$, we get $\log z = -\log \zeta$, which has a branch point at $\zeta = 0$. Similarly z^c has a branch point at infinity for $c \notin \mathbb{Z}$.

(iv) $f(z) = \log\left(\frac{z-1}{z+1}\right) = \log(z-1) - \log(z+1)$ has branch points at ± 1 , but *not* at infinity: with $\zeta = 1/z$ we get $\log\left(\frac{1-\zeta}{1+\zeta}\right)$, and for ζ near 0 the quantity $\frac{1-\zeta}{1+\zeta}$ stays near 1, well away from the troublesome points of $\log$. So small circles about $\zeta = 0$ cause no jump. (Equivalently: a giant circle in the z -plane winds around *both* branch points, and the two jumps of $\pm 2\pi i$ cancel.)

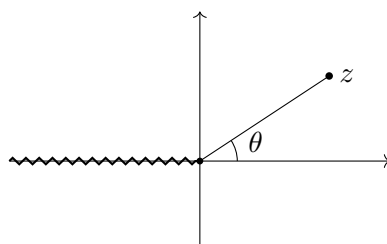
§11.2.2 Branch Cuts

Since the trouble comes from curves that wind around branch points, the remedy is blunt but effective: forbid them.

Definition 11.2.3 (Branch Cut)

A **branch cut** is a curve in the plane, joining branch points (or running from a branch point to infinity), which curves are forbidden from crossing. On the cut plane we can then make a consistent choice of value of the function; such a choice is called a **branch**, and it is single-valued and continuous on any curve avoiding the cut.

For $\log z$, the standard choice is to cut along the negative real axis, and take $\theta \in (-\pi, \pi]$:



This is called the **principal branch** of the logarithm, and its value the **principal value**. The principal branch is analytic on the cut plane, with $\frac{d}{dz} \log z = \frac{1}{z}$, but on the cut itself it is not even continuous: just above the cut at $z = x + i0^+$ (with $x < 0$) we have $\log z = \log |x| + i\pi$, while just below at $z = x + i0^-$ we have $\log z = \log |x| - i\pi$. The function jumps by $2\pi i$ across the cut. This jump is not a defect – in Section 6 it will be precisely the tool that lets us evaluate integrals.

Some remarks on the choices being made here:

- (i) The cut itself is a choice. Any curve from 0 to ∞ works – the non-negative imaginary axis (with, say, $\theta \in (-\frac{3\pi}{2}, \frac{\pi}{2}]$), or even a wiggly curve. All that matters is that curves can no longer wind around the branch point.
- (ii) Even after fixing the cut, the branch is a further choice: with the cut on the negative real axis we could take $\theta \in (\pi, 3\pi]$ instead, which changes $\log z$ by $2\pi i$ everywhere.
- (iii) In any problem where a branch matters, you must specify it – either by giving the range of θ explicitly, or by giving the location of the cut together with the value of the branch at one point (the values everywhere else are then pinned down by continuity).

The same considerations apply verbatim to z^c for $c \notin \mathbb{Z}$, whose branches on the cut plane are $r^c e^{ic\theta}$ with θ in a chosen range of length 2π . Note the different branches of z^c differ by constant factors $e^{2\pi i ck}$, and for $z^{1/2}$ there are exactly two branches, differing by a sign – just as we would expect for a square root.

Remark (Riemann Surfaces). There is a more elegant point of view, due to Riemann: rather than forbidding curves from crossing the cut, imagine infinitely many copies of the cut plane stacked above one another, with the top edge of each cut glued to the bottom edge of the cut on the next sheet up. Crossing the cut moves you to the adjacent sheet, where $\log z$ takes its next branch's values. On this spiral staircase – the *Riemann surface*

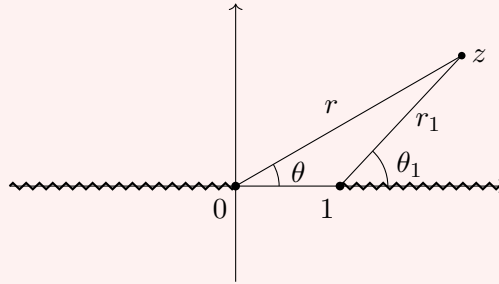
of $\log$ – the logarithm is a perfectly ordinary single-valued analytic function. Riemann surfaces get an entire Part II course, and we say no more about them here.

§11.2.3 Multiple Branch Cuts

When a function has several branch points, we may need several cuts, and there is some genuine artistry in choosing them well.

Example 11.2.4 (Two Cuts to Infinity)

Consider $f(z) = (z(z-1))^{1/3}$, with branch points at 0 and 1 (and at ∞). One valid choice is to cut from 0 out to $-\infty$ along the real axis, and from 1 out to $+\infty$:



Writing $z = re^{i\theta}$ with $\theta \in (-\pi, \pi]$ and $z - 1 = r_1 e^{i\theta_1}$ with $\theta_1 \in [0, 2\pi)$, we define

$$f(z) = \sqrt[3]{rr_1} e^{i(\theta+\theta_1)/3}.$$

Crossing either cut makes one of the angles jump, so f jumps – but away from the cuts, f is continuous and single-valued, as no curve can now encircle either branch point alone.

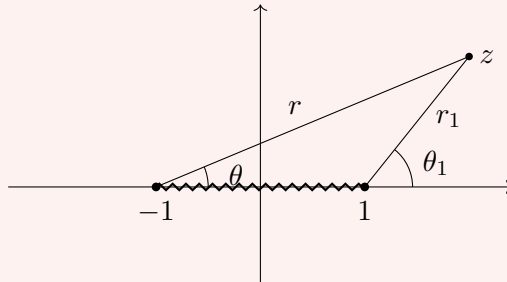
However, sometimes we can get away with less. If a function has *no* branch point at infinity, then large curves winding around all the finite branch points are harmless – so we shouldn't waste a cut ruling them out.

Example 11.2.5 (A Finite Cut)

Take again

$$f(z) = \log \left(\frac{z-1}{z+1} \right),$$

with branch points at ± 1 but not at ∞ . Here we may take a single finite cut along the segment $[-1, 1]$:



Write $z + 1 = re^{i\theta}$ and $z - 1 = r_1 e^{i\theta_1}$, and – this is the part requiring thought – take

both $\theta, \theta_1 \in [0, 2\pi)$. Then define

$$f(z) = \log(r_1/r) + i(\theta_1 - \theta).$$

Let's check this is continuous away from the cut. Crossing the real axis to the *left* of the cut ($x < -1$): both θ and θ_1 pass smoothly through π , so nothing happens. Crossing to the *right* of the cut ($x > 1$): both θ and θ_1 jump by 2π – but f depends only on the difference $\theta_1 - \theta$, so it doesn't notice. Crossing the cut itself ($-1 < x < 1$): θ_1 jumps by 2π while θ varies smoothly, so f jumps by $2\pi i$, exactly as a branch cut should.

This example is worth understanding well – finite branch cuts of this type appear frequently on example sheets and in applications, and choosing the angle ranges correctly is the whole game. Let's do one more of the same flavour, since the square-root version is the one that appears most often.

Example 11.2.6 (Branches of $\sqrt{z^2 - 1}$)

The function $f(z) = (z^2 - 1)^{1/2} = ((z - 1)(z + 1))^{1/2}$ has branch points at ± 1 . Is there a branch point at infinity? With $\zeta = 1/z$,

$$f = \frac{1}{\zeta}(1 - \zeta^2)^{1/2},$$

and $(1 - \zeta^2)^{1/2}$ is single-valued near $\zeta = 0$ (the argument stays near 1) – so no: for large $|z|$, $f(z) \approx \pm z$ consistently. We may therefore use a single finite cut along $[-1, 1]$.

Write $z - 1 = r_1 e^{i\theta_1}$ and $z + 1 = r_2 e^{i\theta_2}$ with $\theta_1, \theta_2 \in [0, 2\pi)$, and set

$$f(z) = \sqrt{r_1 r_2} e^{i(\theta_1 + \theta_2)/2}.$$

Crossing the real axis to the right of the cut, both angles jump by 2π , so $(\theta_1 + \theta_2)/2$ jumps by 2π – invisible in the exponential. Crossing to the left of the cut, both angles pass smoothly through π . Crossing the cut itself, only θ_2 jumps, so f changes sign: a genuine discontinuity, as expected. On the top of the cut ($z = x$ with $-1 < x < 1$, approached from above) we have $\theta_1 = \pi$, $\theta_2 = 0$, so $f = i\sqrt{1 - x^2}$; on the bottom, $f = -i\sqrt{1 - x^2}$. For large real $z = x > 1$ we get $f = \sqrt{x^2 - 1} > 0$, so this branch behaves like $+z$ at infinity.

Functions of exactly this type appear when one inverts Joukowski-type maps ($w = \frac{1}{2}(z + \frac{1}{z})$ gives $z = w + \sqrt{w^2 - 1}$), and the cut along $[-1, 1]$ is the image of the unit circle – the two branches correspond to the inside and outside of the circle.

A final practical point: a branch is often specified not by angle ranges but by its *value at one point*, and one must then be able to evaluate it elsewhere by continuity. With the branch of $\sqrt{z^2 - 1}$ above (cut $[-1, 1]$, $f(2) = +\sqrt{3}$), what is $f(-2)$? Reading off the angles at $z = -2$: $\theta_1 = \theta_2 = \pi$, so $f(-2) = \sqrt{1 \cdot 3} e^{i(\pi + \pi)/2} = -\sqrt{3}$. Note this is *not* $+\sqrt{(-2)^2 - 1}$ – a single consistent branch of $\sqrt{z^2 - 1}$ must behave like $+z$ at one end of the real axis and hence like z (i.e. negatively) at the other. Blindly taking positive square roots on both ends describes *two different branches*, and is the single most common branch-cut error.

§11.3 Conformal Maps

Here is the plan for this section. Laplace's equation is easy to solve on nice domains (half-planes, discs, strips) and hard to solve on nasty ones. Analytic functions, viewed as maps from one region of the plane to another, turn out to preserve exactly the structure Laplace's equation cares about. So if we can map our nasty domain onto a nice one by an analytic bijection, we can solve our problem on the nice domain and pull the solution back. This section builds up a toolkit of such maps, and then puts them to work.

§11.3.1 Möbius Maps

We begin with a class of maps you have met before, in IA Groups.

Definition 11.3.1 (Möbius Map)

A **Möbius map** is a map of the form

$$z \mapsto w = \frac{az + b}{cz + d},$$

where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$.

The condition $ad - bc \neq 0$ rules out the degenerate case where the map is constant. A Möbius map is analytic everywhere except at $z = -d/c$, where the denominator vanishes. The right way to think about this is as a map of the *extended* complex plane: setting

$$-\frac{d}{c} \mapsto \infty, \quad \infty \mapsto \frac{a}{c},$$

a Möbius map becomes a bijection $\mathbb{C}^* \rightarrow \mathbb{C}^*$, with inverse

$$w \mapsto \frac{-dw + b}{cw - a},$$

which is another Möbius map. Composites of Möbius maps are also Möbius maps – indeed they form a group under composition.

For our purposes their key property concerns circles and lines. It is convenient to treat these uniformly: a line is simply a circle through ∞ .

Definition 11.3.2 (Circline)

A **circline** is a circle or a straight line.

Proposition 11.3.3

Möbius maps send circlines to circlines.

Proof. Every circline can be written as a circle of Apollonius

$$|z - z_1| = \lambda |z - z_2|$$

for some $z_1, z_2 \in \mathbb{C}$ and $\lambda > 0$, with $\lambda = 1$ giving lines and $\lambda \neq 1$ circles (this was on the first IA Vectors and Matrices example sheet). Substituting $z = \frac{-dw+b}{cw-a}$ and

clearing denominators, this becomes

$$|(cz_1 + d)w - (az_1 + b)| = \lambda |(cz_2 + d)w - (az_2 + b)|.$$

If $cz_1 + d$ and $cz_2 + d$ are both non-zero, dividing through puts this in Apollonius form again, so the image is a circline; if one of them vanishes the equation directly describes a circle (or line). $\square$

Note carefully what is *not* claimed: circles may map to lines and vice versa. A circle passing through the point sent to ∞ must map to a line.

Proposition 11.3.4 (Three-Point Determination)

Given distinct points $\alpha, \beta, \gamma \in \mathbb{C}^*$ and distinct $\alpha', \beta', \gamma' \in \mathbb{C}^*$, there is a Möbius map sending $\alpha \mapsto \alpha', \beta \mapsto \beta', \gamma \mapsto \gamma'$.

Proof. The map

$$f_1(z) = \frac{\beta - \gamma}{\beta - \alpha} \cdot \frac{z - \alpha}{z - \gamma}$$

sends $\alpha \mapsto 0, \beta \mapsto 1, \gamma \mapsto \infty$. Building f_2 in the same way from the primed points, the composition $f_2^{-1} \circ f_1$ does the job (and is Möbius, since Möbius maps form a group). $\square$

Since three points determine a circline, this means we can find a Möbius map taking any circline to any other – and hence, with a little care about which side is which, taking any disc or half-plane to any other disc or half-plane. This is exactly the flexibility we will want.

§11.3.2 Conformal Maps

Definition 11.3.5 (Conformal Map)

A **conformal map** $f : U \rightarrow V$, where U, V are open subsets of $\mathbb{C}$, is an analytic map with $f'(z) \neq 0$ for all $z \in U$. If additionally f is a bijection $U \rightarrow V$, we call it a **conformal equivalence**.

The name comes from the following geometric property.

Proposition 11.3.6 (Conformal Maps Preserve Angles)

A conformal map preserves the angle – in both magnitude and orientation – between intersecting curves.

Proof. Let $z_1(t)$ be a curve passing through $z_0 = z_1(t_1)$ with non-zero tangent $z_1'(t_1)$, making angle $\phi = \arg z_1'(t_1)$ with the real axis. The image curve $Z_1(t) = f(z_1(t))$ has tangent

$$Z_1'(t_1) = f'(z_0) z_1'(t_1),$$

which makes angle $\arg f'(z_0) + \phi$ with the real axis (using $f'(z_0) \neq 0$ so its argument makes sense). So *every* curve through z_0 has its tangent direction rotated by the same angle $\arg f'(z_0)$, and hence angles between pairs of curves are preserved. $\square$

In fact the converse holds too (an angle-preserving map is conformal), though we won't prove it. Do note where the proof uses $f' \neq 0$: at a point where f' vanishes, angles are genuinely *not* preserved – we will see in a moment that $z \mapsto z^2$ doubles angles at the origin.

When computing with conformal maps, the following procedure for finding the image $f(U)$ is invaluable: *find the image of the boundary ∂U , which will be the boundary of V ; then take any single convenient point of U and see which side of that boundary its image lands on.*

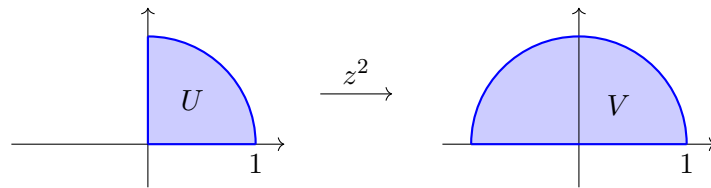
§11.3.3 A Catalogue of Conformal Maps

In practice one builds complicated conformal maps by composing simple standard ones (compositions of conformal maps are conformal, by the chain rule). Here is the standard toolkit.

(i) Linear maps. $z \mapsto az + b$ with $a \neq 0$ rotates by $\arg a$, scales by $|a|$, and translates by b . Conformal everywhere.

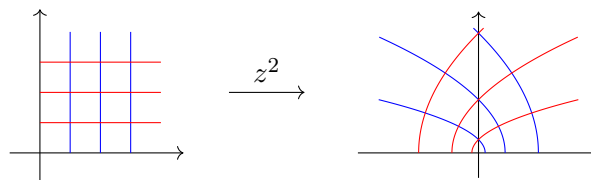
(ii) Powers. $z \mapsto z^\alpha$ (with a suitable branch when $\alpha \notin \mathbb{Z}$) is conformal away from the origin, and maps the sector $0 < \arg z < \phi$ to the sector $0 < \arg w < \alpha\phi$: it *multiplies angles at the origin by α* . For instance, $z \mapsto z^2$ maps the quarter-disc to the half-disc:

$$U = \{z : |z| < 1, 0 < \arg z < \frac{\pi}{2}\} \longrightarrow V = \{w : |w| < 1, 0 < \arg w < \pi\}.$$



The right angles of ∂U at $z = 1$ and $z = i$ are preserved (as f is conformal there), but the right angle at the origin is doubled to a straight line – which is fine, because $0 \notin U$ (U is open, and $f'(0) = 0$ means f was never claimed to be conformal there).

It is illuminating to watch $w = z^2$ act on a coordinate grid. Writing $w = u + iv$, the vertical line $x = c$ maps to $u = c^2 - y^2$, $v = 2cy$ – a leftward-opening parabola – and the horizontal line $y = c$ to a rightward-opening one. The two families of parabolas intersect *at right angles*, as the images of an orthogonal grid under a conformal map must:



Note also how the first quadrant (the region shown) opens out to fill the whole upper half-plane, in accordance with the doubling of angles at the origin.

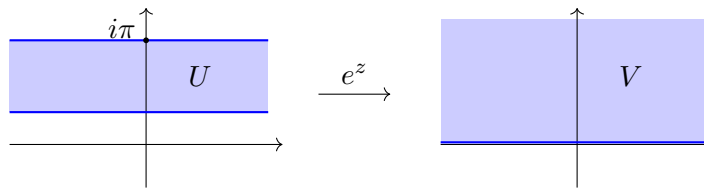
Going the other way, fractional powers *open out* sectors. To map the left half-plane $U = \{\operatorname{Re} z < 0\}$ to the wedge $\{-\frac{\pi}{4} < \arg w < \frac{\pi}{4}\}$, we want to halve angles, so we try $z^{1/2}$ – but we must place the branch cut *outside* U , since $z^{1/2}$ is not even continuous, let alone analytic, on its cut. Cutting along the negative imaginary axis and taking

$\theta \in (-\frac{\pi}{2}, \frac{3\pi}{2}]$, the map $re^{i\theta} \mapsto \sqrt{r}e^{i\theta/2}$ sends U to the wedge $\frac{\pi}{4} < \arg w < \frac{3\pi}{4}$, and following with a rotation we arrive at

$$f(z) = -iz^{1/2}.$$

(iii) The exponential. $w = e^z = e^x e^{iy}$ sends horizontal lines $y = c$ to rays $\arg w = c$, and vertical lines $x = c$ to circles $|w| = e^c$. Consequently it maps a rectangle $x_1 < x < x_2$, $y_1 < y < y_2$ conformally onto a sector of an annulus, and in the degenerate cases:

- the horizontal strip $0 < \operatorname{Im} z < \pi$ maps to the upper half-plane;
- the half-strip $\operatorname{Re} z < 0$, $0 < \operatorname{Im} z < \pi$ maps to the upper half-disc.

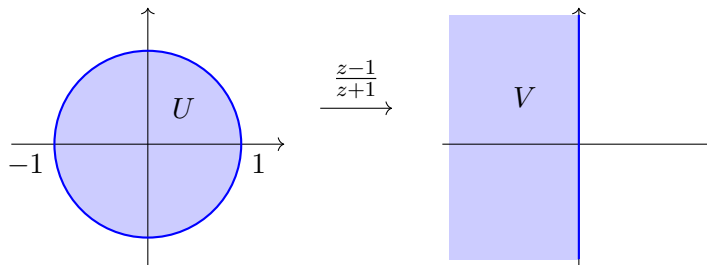


With an appropriate branch, $\log z$ performs the inverse operations, flattening sectors and annuli into strips and rectangles. This pair of maps is the standard tool for ‘unrolling’ angular regions.

(iv) Möbius maps. As discussed, these shuttle between discs and half-planes. The single most useful example is

$$f(z) = \frac{z-1}{z+1},$$

which maps the unit disc to the left half-plane. To see this, note the boundary circle passes through $-1 \mapsto \infty$, $i \mapsto i$ and $1 \mapsto 0$, so its image is the circline through ∞ , i , 0 – the imaginary axis. Checking an interior point, $f(0) = -1$, so the disc maps to the left half-plane.



A useful alternative derivation: $w = \frac{z-1}{z+1}$ rearranges to $z = -\frac{w+1}{w-1}$, and then

$$|z| < 1 \iff |w+1| < |w-1|,$$

which says exactly that w is closer to -1 than to 1 , i.e. $\operatorname{Re} w < 0$.

The map $z \mapsto 1/z$ is also just a Möbius map, and is particularly handy for dealing with regions bounded by lines and circles through the origin. For instance, the vertical line $\operatorname{Re} z = c$ (with $c > 0$) is a circline through ∞ avoiding 0 , so its image under $1/z$ is a circle through the origin; tracking the points $c \mapsto \frac{1}{c}$ and $c \pm i\infty \mapsto 0$, it is the circle of radius $\frac{1}{2c}$ centred at $\frac{1}{2c}$. Consequently $1/z$ maps the half-plane $\operatorname{Re} z > c$ to the open disc bounded by that circle – a useful trick for problems involving two nested or tangent circles.

Let's see the toolkit in action on a composite problem.

Example 11.3.7 (Half-Disc to Disc)

Let's find a conformal equivalence from the upper half-disc $U = \{|z| < 1, \operatorname{Im} z > 0\}$ to the unit disc.

A tempting first guess is $z \mapsto z^2$, since it maps the upper half-disc onto the disc minus the segment $[0, 1)$ – but that slit is missing, so this is not onto, and no good. Instead we go via half-planes:

1. $f_1(z) = \frac{z-1}{z+1}$ maps U to the second quadrant. (Check: the diameter $(-1, 1)$ maps to the negative real axis, the semicircular arc to the positive imaginary axis, and the interior point $i/2$ maps to $\frac{i/2-1}{i/2+1} = \frac{-3+4i}{5}$, indeed in the second quadrant.)
2. $f_2(z) = iz^2$: squaring maps the second quadrant to the lower half-plane, and multiplying by i rotates it to the right half-plane.
3. $f_3(z) = \frac{z-1}{z+1}$ maps the right half-plane to the unit disc: the imaginary axis maps to the unit circle (it is the set of points equidistant from 1 and -1 , which maps to $|w| = 1$ by the displayed equivalence above), and $f_3(1) = 0$.

The required map is $f_3 \circ f_2 \circ f_1$. One could expand this into a single formula, but there is rarely any reason to.

Example 11.3.8 (Strip to Disc)

Similarly, let's map the strip $S = \{0 < \operatorname{Im} z < 1\}$ conformally onto the unit disc. First scale to the standard strip: $z \mapsto \pi z$ gives $0 < \operatorname{Im} z < \pi$. Then exponentiate: e^z maps this strip to the upper half-plane (from the catalogue). Finally, a Möbius map takes the upper half-plane to the disc: $w \mapsto \frac{w-i}{w+i}$ sends the real axis to the unit circle (each real w is equidistant from i and $-i$) and the interior point i to 0. Composing,

$$f(z) = \frac{e^{\pi z} - i}{e^{\pi z} + i}.$$

Notice what happened to the boundary: the two edges of the strip, each a copy of $\mathbb{R}$, are wrapped onto complementary arcs of the circle, meeting at the images of the two 'ends' of the strip, $f(-\infty) = -1$ and $f(+\infty) = 1$. Keeping track of boundary correspondences like this is exactly what transported the boundary conditions in the Laplace problems below.

Remark (The Riemann Mapping Theorem). One might worry about how far this game can be pushed – which regions *can* be conformally mapped onto the disc? The answer is remarkably generous: *every* simply connected domain other than $\mathbb{C}$ itself is conformally equivalent to the unit disc (the *Riemann mapping theorem*). The proof is thoroughly non-constructive and belongs to a later course; for us its value is moral support. The map we need always exists – the examples in this section are about actually finding it.

§11.3.4 The Joukowski Map

One more map deserves special mention, both for its historical importance and because it shows conformal maps doing real engineering work.

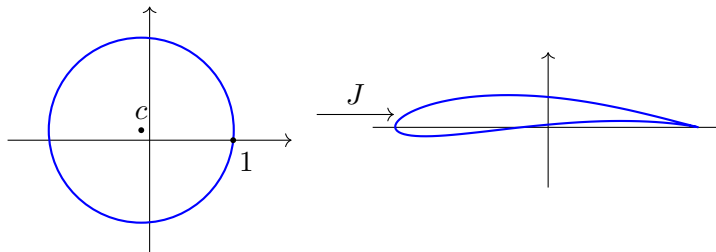
Definition 11.3.9 (Joukowski Map)

The **Joukowski map** is

$$w = J(z) = \frac{1}{2} \left(z + \frac{1}{z} \right).$$

This is conformal except at $z = \pm 1$, where $J'(z) = \frac{1}{2}(1 - 1/z^2)$ vanishes. Since $J(z) = J(1/z)$, the map is two-to-one on $\mathbb{C} \setminus \{0\}$; it is a conformal equivalence from the *exterior* of the unit circle to the complement of the segment $[-1, 1]$. Indeed, on the unit circle itself $z = e^{i\theta}$ we get $w = \cos \theta$: the circle is flattened onto the segment $[-1, 1]$, traversed twice.

Circles of radius $r > 1$ map to ellipses with semi-axes $\frac{1}{2}(r + \frac{1}{r})$ and $\frac{1}{2}(r - \frac{1}{r})$. The magic happens when we apply J to a circle which passes through $z = 1$ but encloses $z = -1$ slightly off-centre: because J fails to be conformal at $z = 1$, the smooth circle acquires a sharp cusp there, and the image is an *aerofoil* – a wing cross-section with a rounded leading edge and sharp trailing edge.



The point of this construction: we understand ideal fluid flow past a circular cylinder completely (we are about to write it down), and the Joukowski map transports that flow conformally to flow past the aerofoil. This was how early aerodynamicists first computed the lift on a wing.

§11.3.5 Solving Laplace's Equation with Conformal Maps

Now we put the machinery to work. The key fact is the one from Section 1: harmonic functions are (locally) real parts of analytic functions, and compositions of analytic functions are analytic. This gives:

Proposition 11.3.10

Let $f : U \rightarrow V$ be conformal and let Φ be harmonic on V . Then

$$\phi(x, y) = \Phi(\operatorname{Re} f(x + iy), \operatorname{Im} f(x + iy))$$

is harmonic on U .

Proof. Since Φ is harmonic, we can write $\Phi = \operatorname{Re} F$ for some analytic F on V (its analytic completion, $F = \Phi + i\Psi$ with Ψ the harmonic conjugate^a). Then $F \circ f$ is analytic on U , being a composition of analytic functions, and $\phi = \operatorname{Re}(F \circ f)$ is harmonic. $\square$

^aOn a simply connected domain this always exists; the nice target domains we choose will always be simply connected.

In other words: *harmonicity survives conformal changes of variable*. This justifies the following algorithm for solving $\nabla^2\phi = 0$ on a difficult domain U with given (Dirichlet, say) boundary conditions:

1. Find a conformal equivalence $f : U \rightarrow V$ with V nice (a half-plane, strip or disc).
2. Transport the boundary conditions: each boundary point $z \in \partial U$ carries its boundary value over to $f(z) \in \partial V$.
3. Solve $\nabla^2\Phi = 0$ on V with the transported boundary conditions – ideally by inspection.
4. Then $\phi = \Phi \circ f$ (in the sense of the proposition) solves the original problem.

Example 11.3.11 (Laplace's Equation on a Quadrant)

Let's find a bounded solution of $\nabla^2\phi = 0$ on the first quadrant $x, y > 0$, with $\phi(x, 0) = 0$ and $\phi(0, y) = 1$.

The quadrant is a sector of angle $\frac{\pi}{2}$, so we unroll it with a logarithm: $f(z) = \log z$ (principal branch) maps it to the strip

$$V = \left\{ w : 0 < \operatorname{Im} w < \frac{\pi}{2} \right\},$$

with the positive real axis (where $\phi = 0$) going to the boundary line $\operatorname{Im} w = 0$, and the positive imaginary axis (where $\phi = 1$) to the line $\operatorname{Im} w = \frac{\pi}{2}$.

On the strip, the boundary conditions do not depend on $\operatorname{Re} w$, so we look for a solution depending only on $\operatorname{Im} w = v$, and $\nabla^2\Phi = 0$ forces Φ to be linear in v :

$$\Phi(u, v) = \frac{2}{\pi}v.$$

Pulling back,

$$\phi(x, y) = \frac{2}{\pi} \operatorname{Im} \log z = \frac{2}{\pi} \arg z = \frac{2}{\pi} \tan^{-1} \left(\frac{y}{x} \right).$$

In hindsight the answer is obvious – the level sets are rays from the origin, interpolating linearly in angle between the two boundary rays – but the method required no cleverness, and works just as well when the answer is not guessable.

Example 11.3.12 (Temperature in a Half-Disc)

Let's solve $\nabla^2 T = 0$ in the upper half-disc $|z| < 1$, $\operatorname{Im} z > 0$, with $T = 1$ on the curved boundary and $T = 0$ on the diameter.

From our catalogue, $f_1(z) = \frac{z-1}{z+1}$ maps the half-disc to the second quadrant, sending the diameter to the negative real axis and the arc to the positive imaginary axis. Then $w = \log \left(\frac{z-1}{z+1} \right)$, taking the branch with $\arg \in [\frac{\pi}{2}, \pi]$ on the closed quadrant, maps the region to the strip $\frac{\pi}{2} < \operatorname{Im} w < \pi$, with the arc's image at $\operatorname{Im} w = \frac{\pi}{2}$ and the diameter's at $\operatorname{Im} w = \pi$. The solution linear in $\operatorname{Im} w$ with the right boundary values is $\Phi = 2 - \frac{2}{\pi} \operatorname{Im} w$, so

$$T(x, y) = 2 - \frac{2}{\pi} \arg \left(\frac{z-1}{z+1} \right).$$

(As always, one should pause and sanity-check: on the arc, $\arg \frac{z-1}{z+1} = \frac{\pi}{2}$ – the two boundary points ± 1 subtend a right angle from any point of the semicircle – giving $T = 1$, and on the diameter the argument is π , giving $T = 0$.)

Example 11.3.13 (Uniform Flow Past a Cylinder)

As a final showpiece, consider ideal fluid flow, uniform at infinity with speed U parallel to the x -axis, past a solid cylinder $|z| \leq a$. We claim the complex potential is

$$w(z) = U \left(z + \frac{a^2}{z} \right).$$

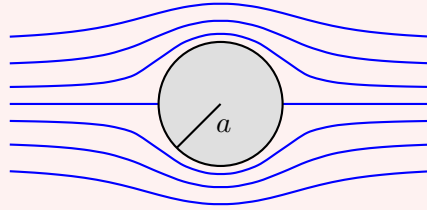
Why: this is analytic in the flow region $|z| > a$; as $z \rightarrow \infty$, $w'(z) = U(1 - a^2/z^2) \rightarrow U$, giving the right behaviour at infinity; and on the boundary circle $z = ae^{i\theta}$,

$$w = U(ae^{i\theta} + ae^{-i\theta}) = 2Ua \cos \theta,$$

which is real – so the stream function $\psi = \text{Im } w$ vanishes on the cylinder, making it a streamline, as a solid boundary must be. (Notice this is just a scaled Joukowski map!) The stream function is

$$\psi(x, y) = Uy \left(1 - \frac{a^2}{x^2 + y^2} \right),$$

and the streamlines $\psi = \text{const}$ look as follows.



The velocity on the boundary is $|w'(ae^{i\theta})| = 2U|\sin \theta|$: the flow is stationary at the front and rear *stagnation points* $\theta = 0, \pi$, and fastest at the top and bottom. Composing with a Joukowski-type map now transports this entire flow field to the flow past an aerofoil, streamline by streamline – conformality guarantees the boundary remains a streamline.

Example 11.3.14 (Flow in a Corner)

Conformal maps also generate flows directly: if $w(z)$ is a complex potential for a flow in a region V , and $g : U \rightarrow V$ is conformal, then $w \circ g$ is a complex potential in U (it is analytic, and boundary streamlines pull back to boundary streamlines).

Uniform flow in the upper half-plane has potential $w(z) = Uz$. The map $g(z) = z^2$ takes the quarter-plane $x, y > 0$ to the upper half-plane. So

$$w(z) = Uz^2 = U(x^2 - y^2) + 2iUxy$$

is the potential for flow in a right-angled corner: the stream function is $\psi = 2Uxy$, whose level curves are the hyperbolas $xy = \text{const}$, sliding along the two walls (which

form the streamline $\psi = 0$). The velocity $w'(z) = 2Uz$ vanishes at the corner itself – a stagnation point, as the geometry demands. More generally $w = Uz^{\pi/\phi}$ describes flow in a corner of opening angle ϕ , and the exponent controls the (in)famous behaviour of the velocity at the vertex: for $\phi > \pi$ (flow around a protruding edge) the velocity is infinite there.

The same technique solves electrostatics problems: the potential between conductors held at fixed voltages is harmonic with Dirichlet boundary conditions, so we conformally map the gap between the conductors to a strip or annulus where the potential is obvious, and pull back. The half-disc example above, reread with ‘temperature’ replaced by ‘potential’ and the boundaries by conductors, is already such an example. Here is one more, with cylindrical rather than flat geometry.

Example 11.3.15 (A Coaxial Cable)

Two conducting cylinders, radii $r_1 < r_2$, are held at potentials 0 and V ; we want the potential ϕ in the annulus between them. The map $w = \log z$ (any branch, locally) sends the annulus to the vertical strip $\log r_1 < \operatorname{Re} w < \log r_2$, and the boundary circles to the strip’s edges. In the strip the solution linear in $\operatorname{Re} w$ is immediate, and pulling back,

$$\phi = V \frac{\log(r/r_1)}{\log(r_2/r_1)}, \quad r = |z|.$$

Strictly the annulus is not simply connected and $\log z$ is not single-valued on it – but its *real* part $\log |z|$ is, and that is all the solution uses. The field lines are the level curves of the conjugate $\operatorname{Im} \log z = \theta$: radial lines, orthogonal to the equipotential circles, as they must be.

§11.4 Contour Integration and Cauchy’s Theorem

We now change gears entirely, and spend the rest of the course on the theory of *integration* of complex functions. If differentiating complex functions felt like a modest generalisation of the real theory, integrating them will not: complex integration has a completely different character, dominated by one theorem (Cauchy’s), and it leads to computational techniques that look like magic the first few times you see them.

Throughout this part of the course, our treatment runs parallel to the Complex Analysis course. The definitions below are the same as the ones given there; the difference is that we will move quickly to computation, citing my Complex Analysis notes for the careful proofs.

§11.4.1 Contours and Integrals

Definition 11.4.1 (Contour)

A **curve** γ is a continuous map $\gamma : [a, b] \rightarrow \mathbb{C}$. A **contour** is a piecewise smooth curve; it is **closed** if $\gamma(a) = \gamma(b)$, and **simple** if it does not intersect itself (except that a closed contour returns to its start).

Definition 11.4.2 (Contour Integral)

The **contour integral** of f along a contour γ is

$$\int_{\gamma} f(z) \, dz = \int_a^b f(\gamma(t)) \gamma'(t) \, dt.$$

For a closed contour we sometimes write $\oint_{\gamma} f(z) \, dz$, and unless stated otherwise closed contours are traversed *anticlockwise*.

One can check (by the chain rule) that the integral is independent of the parametrisation chosen, so it really is an integral ‘along the curve’. Equivalently, the integral is a limit of Riemann sums $\sum f(z_k) \delta z_k$ over subdivisions of the contour, which is the picture to keep in mind: we sum up values of f weighted by *complex* displacements along the curve.

The elementary properties are what you would expect: the integral is linear in f , additive over concatenation of contours, and changes sign if the contour is traversed backwards. The workhorse inequality is the following, which we will use dozens of times.

Lemma 11.4.3 (Estimation Lemma)

If $|f(z)| \leq M$ on a contour γ of length L , then

$$\left| \int_{\gamma} f(z) \, dz \right| \leq ML.$$

Proof. Bounding the Riemann sums, $|\sum f(z_k) \delta z_k| \leq M \sum |\delta z_k| \leq ML$. (A careful proof is in my Complex Analysis notes.) $\square$

In practice this appears as ‘length times sup’, often in asymptotic form: a semicircle of radius R has length πR , so if $|f| = O(R^{-2})$ on it then the integral is $O(R^{-1}) \rightarrow 0$. Estimates of exactly this shape will kill off the ‘unwanted’ parts of our contours in Section 6.

An important question: does a contour integral depend on the contour, or only on its endpoints? For general functions, it certainly depends on the contour.

Example 11.4.4 (Path Dependence)

Let’s integrate $f(z) = \bar{z}$ from 0 to $1+i$ along two different contours: γ_1 , the straight line $\gamma_1(t) = (1+i)t$ for $t \in [0, 1]$; and γ_2 , going along the real axis to 1 and then vertically up to $1+i$.

Along γ_1 :

$$\int_{\gamma_1} \bar{z} \, dz = \int_0^1 \overline{(1+i)t} \cdot (1+i) \, dt = |1+i|^2 \int_0^1 t \, dt = 1.$$

Along γ_2 : on the horizontal leg $z = t$, $\bar{z} = t$, contributing $\int_0^1 t \, dt = \frac{1}{2}$; on the vertical leg $z = 1 + it$, $dz = i \, dt$, contributing $\int_0^1 (1 - it)i \, dt = i + \frac{1}{2}$. Total: $1 + i$.

So the two routes disagree – unsurprisingly, since $\bar{z}$ is not analytic. Repeating the

computation with the analytic function $f(z) = z$ gives $\frac{(1+i)^2}{2} = i$ along *both* routes (as the reader should check). Path-independence for analytic functions is not an accident; it is Cauchy's theorem, to which we are heading.

For functions with an antiderivative, the fundamental theorem of calculus works exactly as over $\mathbb{R}$:

Proposition 11.4.5 (Fundamental Theorem of Calculus)

If F is analytic on a domain containing the contour γ from a to b , and $F' = f$, then

$$\int_{\gamma} f(z) \, dz = F(b) - F(a).$$

In particular the integral depends only on the endpoints, and vanishes if γ is closed.

Proof. Parametrise γ over $[t_0, t_1]$ and apply the chain rule: $\int_{t_0}^{t_1} f(\gamma(t))\gamma'(t) \, dt = \int_{t_0}^{t_1} \frac{d}{dt} F(\gamma(t)) \, dt = F(b) - F(a)$. $\square$

Example 11.4.6 (A Fundamental Computation)

Let γ be the circle of radius r about the origin, traversed anticlockwise, and let $n \in \mathbb{Z}$. Parametrising $\gamma(\theta) = re^{i\theta}$,

$$\oint_{\gamma} z^n \, dz = \int_0^{2\pi} r^n e^{in\theta} \cdot ire^{i\theta} \, d\theta = ir^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} \, d\theta = \begin{cases} 2\pi i & n = -1, \\ 0 & n \neq -1. \end{cases}$$

This tiny computation is secretly the entire subject. Every power of z integrates to zero around a closed loop – except $\frac{1}{z}$, which detects the loop and contributes $2\pi i$. The residue theorem in Section 6 is little more than this example dressed up.

Note why $n = -1$ is special: z^n has antiderivative $\frac{z^{n+1}}{n+1}$ for $n \neq -1$, and integrals of functions with (single-valued) antiderivatives around closed loops vanish, by the fundamental theorem of calculus. But the antiderivative of $\frac{1}{z}$ is $\log z$, which is multivalued – going once around the circle, $\log z$ climbs by $2\pi i$, and that is exactly the value of the integral. Branch cuts and contour integrals are two faces of the same phenomenon.

The same computation viewed through open contours is also instructive. Integrating $\frac{1}{z}$ from -1 to 1 along the semicircle $e^{i\theta}$, θ from π down to 0 , through the *upper* half-plane:

$$\int_{\pi}^0 \frac{1}{e^{i\theta}} ie^{i\theta} \, d\theta = -i\pi,$$

while through the *lower* half-plane (θ from $-\pi$ up to 0) the answer is $+i\pi$. The two routes differ by $2\pi i$ – the closed-loop value – and each individually is ‘ $\log 1 - \log(-1)$ ’ for a different branch of $\log$. Everything is consistent, provided one keeps track of which side of the singularity the contour passes.

§11.4.2 Cauchy's Theorem

We need one topological notion.

Definition 11.4.7 (Simply Connected)

A domain D is **simply connected** if every closed curve in D encloses only points of D – informally, D has no holes.

Theorem 11.4.8 (Cauchy's Theorem)

If f is analytic on a simply connected domain D , then for every closed contour γ in D ,

$$\oint_{\gamma} f(z) \, dz = 0.$$

Proof (Sketch). This is *the* theorem of the course, and it deserves a real proof – which is given in my Complex Analysis notes. The quick heuristic, assuming f' continuous, comes from Green's theorem: writing $f = u + iv$ and $dz = dx + i \, dy$,

$$\oint_{\gamma} f \, dz = \oint_{\gamma} (u \, dx - v \, dy) + i \oint_{\gamma} (v \, dx + u \, dy),$$

and applying Green's theorem to each part turns the integrands into $(-v_x - u_y)$ and $(u_x - v_y)$ respectively over the enclosed region – both of which vanish by the Cauchy–Riemann equations. The honest proof avoids assuming f' is continuous, and is genuinely harder. $\square$

The hypothesis of simple connectedness is doing real work: it guarantees that f is analytic *everywhere inside* γ , not just on it. Integrating $\frac{1}{z}$ around the unit circle – which gave $2\pi i$, not 0 – does not contradict the theorem, because no simply connected domain contains the circle while avoiding the singularity at 0.

Cauchy's theorem has two immediate consequences that we use constantly, usually without comment.

Proposition 11.4.9 (Deformation of Contours) (i) If γ_1 and γ_2 are two contours from a to b , and f is analytic on a region containing both and the area between them, then $\int_{\gamma_1} f \, dz = \int_{\gamma_2} f \, dz$.

(ii) If γ_1 and γ_2 are closed contours, one inside the other, and f is analytic on the region between them, then $\oint_{\gamma_1} f \, dz = \oint_{\gamma_2} f \, dz$.

Proof. For (i), the concatenation of γ_1 with γ_2 reversed is a closed contour to which Cauchy's theorem applies. For (ii), join the two contours by a thin 'cross-cut': the resulting single closed contour encloses only the region between γ_1 and γ_2 , where f is analytic, so the integral around it vanishes; the two passes along the cross-cut cancel in the limit as it narrows, leaving $\oint_{\gamma_1} f \, dz - \oint_{\gamma_2} f \, dz = 0$. $\square$

The slogan for (ii): *contours may be freely deformed, so long as they never cross a singularity*. This is the practical form in which Cauchy's theorem enters computations – we shrink complicated contours onto small circles around the singularities they enclose.

§11.4.3 Cauchy's Integral Formula

Theorem 11.4.10 (Cauchy's Integral Formula)

Let f be analytic on and inside a simple closed contour γ , and let z_0 lie inside γ . Then

$$f(z_0) = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{z - z_0} dz.$$

Proof. The integrand is analytic except at z_0 , so by deformation we may replace γ by the circle $z = z_0 + \epsilon e^{i\theta}$ of small radius ϵ . On this circle,

$$\oint \frac{f(z)}{z - z_0} dz = \int_0^{2\pi} \frac{f(z_0 + \epsilon e^{i\theta})}{\epsilon e^{i\theta}} i\epsilon e^{i\theta} d\theta = i \int_0^{2\pi} f(z_0 + \epsilon e^{i\theta}) d\theta \rightarrow 2\pi i f(z_0)$$

as $\epsilon \rightarrow 0$, by continuity of f . Since the left hand side is independent of ϵ , it equals $2\pi i f(z_0)$. $\square$

Pause on how strange this is: the values of an analytic function *everywhere inside* a contour are completely determined by its values *on* the contour. Nothing remotely like this is true for real differentiable functions. It reflects the rigidity of analytic functions – being differentiable in the complex sense ties all the values together.

Differentiating under the integral sign (which can be justified) gives formulas for all derivatives:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz.$$

In particular *an analytic function is infinitely differentiable* – the fact we borrowed back in Section 1 to prove harmonicity. The same formulas power the theory of Taylor series, to which we now turn shortly; first, two quick consequences that show off how much information the integral formula packs.

Proposition 11.4.11 (Mean Value Property)

If f is analytic on a disc about z_0 , then $f(z_0)$ is the average of f over any circle $|z - z_0| = r$ in the disc:

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta.$$

Taking real parts, harmonic functions also have the mean value property.

Proof. Apply Cauchy's integral formula on the circle and parametrise, exactly as in its proof (this time with no limit needed). $\square$

Theorem 11.4.12 (Liouville's Theorem)

A bounded entire function is constant.

Proof. Say $|f| \leq M$ everywhere. For any z_0 , apply the derivative formula on a circle

of radius R about z_0 and estimate:

$$|f'(z_0)| = \left| \frac{1}{2\pi i} \oint \frac{f(z)}{(z - z_0)^2} dz \right| \leq \frac{1}{2\pi} \cdot 2\pi R \cdot \frac{M}{R^2} = \frac{M}{R} \rightarrow 0$$

as $R \rightarrow \infty$. So $f' \equiv 0$, and f is constant. $\square$

Liouville's theorem is the standard route to the fundamental theorem of algebra (if a polynomial P had no root, $1/P$ would be a bounded entire function), and for us it also explains a fact we keep using implicitly: interesting entire functions – e^z , $\sin z$ – *must* be unbounded somewhere. Both results, along with much more in this vein (Morera's theorem, the identity theorem, the maximum modulus principle), are treated properly in my Complex Analysis notes.

§11.5 Laurent Series and Singularities

Analytic functions are exactly the functions given locally by power series – and near their singularities, they are given by power series allowing negative powers. This section sets up that machinery and uses it to classify singularities, which is precisely what we need to compute residues efficiently.

§11.5.1 Taylor and Laurent Series

Theorem 11.5.1 (Taylor Series)

If f is analytic on the disc $|z - z_0| < R$, then on that disc

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz,$$

where γ is any simple closed contour in the disc encircling z_0 . The series converges (uniformly on compact subsets) in the disc.

The proof (expand $\frac{1}{z-w}$ as a geometric series inside Cauchy's integral formula and integrate term by term) is in my Complex Analysis notes. Note the remarkable strength of the statement compared with the real case: mere differentiability guarantees an everywhere-convergent local power series. There is no complex analogue of the smooth-but-not-analytic function e^{-1/x^2} .

Example 11.5.2

The classics, valid for all $z \in \mathbb{C}$:

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \quad \sin z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}, \quad \cos z = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!},$$

and for $|z| < 1$,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n, \quad \log(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{n}$$

(principal branch). The radius of convergence of a Taylor series is always the distance from z_0 to the nearest singularity – which finally explains a real-analysis mystery, namely why $\frac{1}{1+x^2}$, a perfectly smooth function on all of $\mathbb{R}$, has Taylor series with radius of convergence only 1: the function has (complex!) singularities at $\pm i$, at distance 1 from the origin.

What if f has a singularity at z_0 itself, or we are working in an annulus around a hole? We then allow negative powers.

Theorem 11.5.3 (Laurent Series)

If f is analytic in an annulus $R_1 < |z - z_0| < R_2$, then on that annulus

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz,$$

where γ is any simple closed contour in the annulus encircling z_0 . The expansion is unique, and converges uniformly on compact subsets of the annulus.

Uniqueness is worth emphasising for practical reasons: *any* valid method of producing such an expansion – long division, substituting into known series, partial fractions – must give *the* Laurent series. We never compute the coefficient integrals directly.

Example 11.5.4 (Computing Laurent Series) (i) Around $z_0 = 0$,

$$\frac{e^z}{z^3} = \frac{1}{z^3} \left(1 + z + \frac{z^2}{2} + \cdots \right) = \frac{1}{z^3} + \frac{1}{z^2} + \frac{1}{2z} + \frac{1}{6} + \frac{z}{24} + \cdots$$

(ii) $e^{1/z}$ has Laurent series $\sum_{n=0}^{\infty} \frac{z^{-n}}{n!}$ around 0, with *infinitely many* negative powers.

(iii) The function $\frac{1}{z(z-1)}$ has *different* Laurent expansions in the annuli $0 < |z| < 1$ and $|z| > 1$:

$$\frac{1}{z(z-1)} = -\frac{1}{z} \cdot \frac{1}{1-z} = -\frac{1}{z} - 1 - z - z^2 - \cdots \quad (0 < |z| < 1),$$

while for $|z| > 1$ we expand in powers of $\frac{1}{z}$ instead:

$$\frac{1}{z(z-1)} = \frac{1}{z^2} \cdot \frac{1}{1-1/z} = \frac{1}{z^2} + \frac{1}{z^3} + \frac{1}{z^4} + \cdots$$

Same function, same centre, different annuli, different series – uniqueness is per-annulus.

(iv) Expansions about points other than the singularity work the same way. For $\frac{1}{z(z-2)}$ about $z_0 = 1$: writing $\zeta = z - 1$,

$$\frac{1}{z(z-2)} = \frac{1}{(\zeta+1)(\zeta-1)} = \frac{1}{\zeta^2-1} = -(1 + \zeta^2 + \zeta^4 + \cdots) \quad (|\zeta| < 1)$$

– a Taylor series, no negative powers, since the function is analytic at $z = 1$; the nearest singularities (at 0 and 2) set the radius of convergence to 1. In the outer region $|\zeta| > 1$ we would instead expand in $\frac{1}{\zeta}$ and find only negative powers.

§11.5.2 Zeros and the Classification of Singularities

Definition 11.5.5 (Order of a Zero)

A **zero** of an analytic function f is a point with $f(z_0) = 0$. Its **order** is the index of the first non-vanishing Taylor coefficient at z_0 ; a zero of order 1 is called **simple**.

So $z^3 + iz = z(z^2 + i)$ has a simple zero at the origin (and at the two square roots of $-i$), and $\sinh^3 z$ has a zero of order 3 at $z = n\pi i$ for each $n \in \mathbb{Z}$. Zeros of order k look locally like $c(z - z_0)^k$.

Definition 11.5.6 (Isolated Singularity)

If f is analytic on a punctured neighbourhood $0 < |z - z_0| < r$ of z_0 but not at z_0 itself, then z_0 is an **isolated singularity** of f .

Branch points are the standard example of *non-isolated* singularities in this course – there is no punctured neighbourhood of a branch point on which the function is even single-valued, so no Laurent expansion exists there. At an isolated singularity, however, we have a Laurent series, and we classify the singularity by how badly the series fails to be a Taylor series. Suppose $f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$ on $0 < |z - z_0| < r$.

- (i) If $a_n = 0$ for all $n < 0$, then z_0 is a **removable singularity**: defining $f(z_0) = a_0$ extends f analytically across z_0 . Example: $\frac{\sin z}{z}$ at 0.
- (ii) If $a_n = 0$ for $n < -N$ but $a_{-N} \neq 0$ (some $N \geq 1$), then z_0 is a **pole of order N** . Poles of order 1, 2, 3 are called *simple*, *double*, *triple*. Example: $\frac{e^z}{z^3}$ has a triple pole at 0, as we computed. Near a pole, $|f(z)| \rightarrow \infty$.
- (iii) Otherwise – infinitely many negative coefficients – z_0 is an **essential singularity**. Example: $e^{1/z}$ at 0.

Poles are the well-behaved singularities, and essentially all singularities met in this course are poles. Note f has a pole of order N at z_0 exactly when $\frac{1}{f}$ has a zero of order N there (poles of f are zeros of $1/f$ and vice versa), which is usually the fastest way to find the order of a pole: for instance $\frac{1}{\sinh^3 z}$ has a triple pole at each $z = n\pi i$.

Essential singularities are genuinely wild: in any neighbourhood of an essential singularity, f takes values arbitrarily close to *every* complex number (Casorati–Weierstrass; Picard proved the stronger statement that it takes every value with at most one exception, infinitely often). We will treat this as a good reason to stay away from them.

Let's practice the classification – this is the sort of thing one must be able to do at sight.

Example 11.5.7 (Classifying Singularities) (i) $\frac{z}{\sin z}$: at $z = 0$, both numerator and denominator have simple zeros, so the singularity is *removable* (with value 1). At $z = n\pi$ for $n \neq 0$, the denominator has a simple zero and the numerator

doesn't vanish: *simple poles*.

- (ii) $\tan z = \frac{\sin z}{\cos z}$: simple poles at the zeros $(n + \frac{1}{2})\pi$ of $\cos$, each with residue $\frac{\sin z}{-\sin z} = -1$ by the $\frac{g}{h'}$ rule (coming shortly).
- (iii) $\frac{\cos z - 1}{z^4}$: the numerator has a *double* zero at 0 (its series starts at $-\frac{z^2}{2}$), so this is a double pole, with Laurent series $-\frac{1}{2z^2} + \frac{1}{24} - \dots$. Note the residue is 0 – poles can have zero residue.
- (iv) $\sin \frac{1}{z}$: essential singularity at 0 (the Laurent series $\frac{1}{z} - \frac{1}{3!z^3} + \dots$ has infinitely many negative terms).
- (v) $\frac{1}{\sin(1/z)}$: this has simple poles at $z = \frac{1}{n\pi}$, which *accumulate* at 0. So $z = 0$ is a singularity but not an isolated one – there is no punctured disc around 0 free of other singularities – and it is therefore none of the three types. No Laurent series about 0 exists.

Remark (Behaviour at Infinity). As always, we classify the point at infinity by looking at $f(1/z)$ at 0. A polynomial of degree n has a pole of order n at ∞ ; a rational function is analytic at ∞ iff its denominator degree is at least its numerator degree; and e^z has an *essential* singularity at ∞ (as $e^{1/z}$ does at 0) – which is one way to understand why e^z attains every non-zero value in every half-plane $\operatorname{Re} z > R$.

§11.5.3 Residues

For the applications, one Laurent coefficient towers above the rest in importance.

Definition 11.5.8 (Residue)

The **residue** of f at an isolated singularity z_0 is the coefficient a_{-1} of its Laurent expansion about z_0 , written $\operatorname{Res}_{z=z_0} f(z)$.

Why single out a_{-1} ? Integrate the Laurent series term by term around a small circle about z_0 : by our fundamental computation, every term dies except the $\frac{1}{z-z_0}$ term, and

$$\oint f(z) \, dz = 2\pi i a_{-1}.$$

So the residue is exactly what survives integration. This observation *is* the residue theorem, up to bookkeeping; but before stating it, let's build the toolkit for computing residues without ever writing down a full Laurent expansion.

Proposition 11.5.9 (Computing Residues)

Let z_0 be a pole of f .

- (i) If the Laurent series is available or easy, read off a_{-1} .
- (ii) If z_0 is a simple pole,

$$\operatorname{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} (z - z_0) f(z).$$

(iii) If $f = \frac{g}{h}$ with $g(z_0) \neq 0$ and h having a simple zero at z_0 , then

$$\operatorname{Res}_{z=z_0} f(z) = \frac{g(z_0)}{h'(z_0)}.$$

(iv) If z_0 is a pole of order N ,

$$\operatorname{Res}_{z=z_0} f(z) = \lim_{z \rightarrow z_0} \frac{1}{(N-1)!} \frac{d^{N-1}}{dz^{N-1}} \left((z - z_0)^N f(z) \right).$$

Proof. For (ii) and (iv), write out the Laurent series: $(z - z_0)^N f(z) = a_{-N} + a_{-N+1}(z - z_0) + \dots$ is analytic at z_0 , and differentiating $N - 1$ times and evaluating at z_0 isolates $(N - 1)! a_{-1}$. For (iii), note

$$\lim_{z \rightarrow z_0} (z - z_0) \frac{g(z)}{h(z)} = g(z_0) \lim_{z \rightarrow z_0} \frac{z - z_0}{h(z) - h(z_0)} = \frac{g(z_0)}{h'(z_0)}. \quad \square$$

Method (iii) is used more than all the others combined. A quick gallery:

Example 11.5.10 (Computing Residues) (i) $\frac{e^z}{z^3}$ at $z = 0$: from the Laurent series above, the residue is $\frac{1}{2}$.

(ii) $\frac{1}{1+z^4}$ at $z_0 = e^{i\pi/4}$: by (iii), the residue is $\frac{1}{4z_0^3} = \frac{z_0}{4z_0^4} = -\frac{1}{4}e^{i\pi/4}$. (Multiplying top and bottom by z_0 to use $z_0^4 = -1$ is a trick worth remembering.)

(iii) $\frac{e^{iz}}{\sinh z}$ at $z = \pi i$: $\sinh$ has a simple zero there, so by (iii) the residue is $\frac{e^{i\pi i}}{\cosh \pi i} = \frac{e^{-\pi}}{\cos \pi} = -e^{-\pi}$.

(iv) $\frac{1}{(z^2 + a^2)^2}$ at $z = ia$ (a double pole, $a > 0$): by (iv),

$$\operatorname{Res} = \lim_{z \rightarrow ia} \frac{d}{dz} \frac{1}{(z + ia)^2} = \frac{-2}{(2ia)^3} = -\frac{i}{4a^3}.$$

(v) Essential singularities have residues too – the definition only needs the Laurent coefficient – but the pole formulas above do not apply, and one must go back to the series. For instance $e^{1/z}$ has residue 1 at the origin (the $n = 1$ term of $\sum \frac{z^{-n}}{n!}$), and $ze^{1/z}$, whose expansion $z + 1 + \frac{1}{2z} + \dots$ has residue $\frac{1}{2}$.

§11.6 The Calculus of Residues

This section contains the computational heart of the course. The residue theorem turns the evaluation of contour integrals into the algebra of computing residues – and with the right contours, an enormous variety of *real* integrals fall to it. The section is essentially a sequence of worked examples, each illustrating a standard contour or trick; between them they cover everything that appears on example sheets and exams.

§11.6.1 The Residue Theorem

Theorem 11.6.1 (Residue Theorem)

Let f be analytic on and inside a simple closed contour γ (traversed anticlockwise), except at finitely many isolated singularities $z_1, \dots, z_n$ inside γ . Then

$$\oint_{\gamma} f(z) \, dz = 2\pi i \sum_{k=1}^n \operatorname{Res}_{z=z_k} f(z).$$

Proof. Deform γ , without crossing any singularity, into small circles γ_k around each z_k joined by cross-cuts (the cross-cut contributions cancel in pairs as they narrow, exactly as in the proof of contour deformation). This gives

$$\oint_{\gamma} f(z) \, dz = \sum_{k=1}^n \oint_{\gamma_k} f(z) \, dz.$$

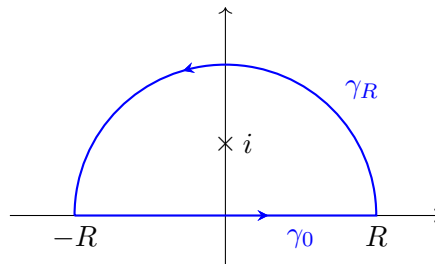
On each small circle, integrate the Laurent expansion of f about z_k term by term: every power of $(z - z_k)$ integrates to zero except the (-1) -th, which gives $2\pi i a_{-1} = 2\pi i \operatorname{Res}_{z=z_k} f$. $\square$

A fully rigorous version of this argument (with winding numbers, and without the appeal to pictures) is in my Complex Analysis notes. For this course, the picture *is* the point: the integral around γ only sees the singularities inside, each contributing $2\pi i$ times its residue.

Everything else in this section is technique for *choosing* γ .

§11.6.2 Integrals Along the Real Axis

The prototype problem: evaluate $\int_{-\infty}^{\infty} f(x) \, dx$ for a rational function f decaying at infinity. The strategy is to ‘close the contour’: integrate $f(z)$ around the contour consisting of the real segment $[-R, R]$ together with the semicircle γ_R of radius R in the upper half-plane, apply the residue theorem, and let $R \rightarrow \infty$, arguing that the semicircle’s contribution dies.



Example 11.6.2 (The Prototype)

Let's evaluate

$$I = \int_0^{\infty} \frac{dx}{1+x^2}.$$

Of course we know the answer is $\arctan \infty = \frac{\pi}{2}$; the point is to establish the method

on familiar ground. Consider

$$\oint_{\gamma} \frac{dz}{1+z^2}$$

around the contour above. The integrand factors as $\frac{1}{(z-i)(z+i)}$, so the only enclosed singularity is the simple pole at $z = i$, with residue $\frac{1}{2i}$. The residue theorem gives, for all $R > 1$,

$$\int_{\gamma_0} \frac{dz}{1+z^2} + \int_{\gamma_R} \frac{dz}{1+z^2} = 2\pi i \cdot \frac{1}{2i} = \pi.$$

As $R \rightarrow \infty$, the first integral tends to $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = 2I$ by evenness. For the second, on γ_R we have $|1+z^2| \geq R^2 - 1$, so by the estimation lemma

$$\left| \int_{\gamma_R} \frac{dz}{1+z^2} \right| \leq \frac{\pi R}{R^2 - 1} \rightarrow 0.$$

Hence $2I = \pi$, so $I = \frac{\pi}{2}$ as expected.

Some remarks before the next examples, all of them important for practice:

- (i) The informal version of the tail estimate, universally used: on γ_R , the integrand is $O(R^{-2})$ and the contour has length πR , so the integral is $O(R^{-1}) \rightarrow 0$. Any rational function with denominator degree exceeding numerator degree by at least 2 dies this way.
- (ii) We could equally have closed in the *lower* half-plane. The enclosed pole is then $-i$ with residue $-\frac{1}{2i}$, but the contour is traversed clockwise, contributing a compensating sign: same answer. Close wherever the estimate works; when both work, pick whichever contains nicer poles.
- (iii) The method's value is its generality – replace $1+x^2$ by $1+x^8$ and the trigonometric substitution collapses, but the contour argument goes through verbatim.

Example 11.6.3 (A Double Pole)

For $a > 0$,

$$\int_0^{\infty} \frac{dx}{(x^2 + a^2)^2}.$$

Same contour. The only enclosed singularity is the *double* pole at $z = ia$, whose residue we computed in Section 5: $-\frac{i}{4a^3}$. The semicircle integrand is $O(R^{-4})$, so its integral is $O(R^{-3}) \rightarrow 0$, and

$$2I = 2\pi i \left(-\frac{i}{4a^3} \right) = \frac{\pi}{2a^3}, \quad I = \frac{\pi}{4a^3}.$$

Example 11.6.4 (Choosing a Cleverer Contour)

Consider

$$I = \int_0^{\infty} \frac{dx}{1+x^4}.$$

The semicircular contour works: the enclosed poles are $e^{i\pi/4}$ and $e^{3i\pi/4}$, with residues

$-\frac{1}{4}e^{i\pi/4}$ and $-\frac{1}{4}e^{3i\pi/4}$ (using the $\frac{g}{h}$ formula and the $z_0^4 = -1$ trick), giving after a little algebra $I = \frac{\pi}{2\sqrt{2}}$.

But since the integrand has the symmetry $f(iz) = f(z)$, there is a slicker choice: the quarter-circle contour $0 \rightarrow R \rightarrow iR \rightarrow 0$ (out along the real axis, around the arc, back down the imaginary axis). The return leg contributes, substituting $z = iy$,

$$\int_R^0 \frac{i \, dy}{1 + y^4} \rightarrow -iI,$$

the arc dies as usual, and only the single pole $e^{i\pi/4}$ is enclosed. So

$$(1 - i)I = 2\pi i \left(-\frac{1}{4}e^{i\pi/4}\right), \quad I = \frac{-\frac{\pi i}{2}e^{i\pi/4}}{1 - i} = \frac{\pi}{2\sqrt{2}},$$

with less residue arithmetic. Matching the contour to the symmetry of the integrand is a recurring theme.

Example 11.6.5 (A Wedge Contour)

The quarter-circle idea generalises to any integrand with a rotational symmetry. Consider

$$I = \int_0^\infty \frac{dx}{1 + x^3}.$$

(Note we *cannot* close a semicircle here: the integrand is not even, so $\int_{-\infty}^\infty$ is not $2I$ – indeed the integrand has a pole at $x = -1$ on the negative real axis.) The symmetry is $z \mapsto e^{2\pi i/3}z$, which permutes the poles and fixes the integrand. So take the wedge contour: out along the real axis to R , around the arc of angle $\frac{2\pi}{3}$, and back along the ray $\arg z = \frac{2\pi}{3}$. On the returning ray, $z = e^{2\pi i/3}t$ gives $1 + z^3 = 1 + t^3$ and $dz = e^{2\pi i/3} dt$, so that leg contributes $-e^{2\pi i/3}I$; the arc dies as $O(R^{-2})$.

The wedge encloses exactly one pole, $z_0 = e^{i\pi/3}$, with residue $\frac{1}{3z_0^2} = \frac{z_0}{3z_0^3} = -\frac{z_0}{3}$. Hence

$$(1 - e^{2\pi i/3})I = -\frac{2\pi i}{3}e^{i\pi/3},$$

and factoring $1 - e^{2\pi i/3} = -e^{i\pi/3}(e^{i\pi/3} - e^{-i\pi/3}) = -2ie^{i\pi/3} \sin \frac{\pi}{3}$,

$$I = \frac{\pi/3}{\sin(\pi/3)} = \frac{2\pi}{3\sqrt{3}}.$$

The same wedge of angle $\frac{2\pi}{n}$ handles $\int_0^\infty \frac{dx}{1+x^n}$ for any $n \geq 2$, giving $\frac{\pi/n}{\sin(\pi/n)}$.

Example 11.6.6 (The Fresnel Integrals)

A wedge can rotate an integral onto an axis where it becomes elementary, even with *no poles at all*. The Fresnel integrals

$$\int_0^\infty \cos(x^2) \, dx \quad \text{and} \quad \int_0^\infty \sin(x^2) \, dx$$

(which converge, delicately, by the increasingly rapid oscillation) are the real and imaginary parts of $\int_0^\infty e^{ix^2} dx$. Integrate the *entire* function e^{iz^2} around the wedge of angle $\frac{\pi}{4}$: out to R , around the arc, and back along the ray $z = e^{i\pi/4}t$. Cauchy's theorem says the total is zero.

The point of the angle $\frac{\pi}{4}$: on the returning ray, $z^2 = it^2$, so

$$e^{iz^2} = e^{i \cdot it^2} = e^{-t^2},$$

and the wild oscillatory integrand becomes the Gaussian. The ray leg contributes $-e^{i\pi/4} \int_0^R e^{-t^2} dt$. On the arc, $|e^{iz^2}| = e^{-R^2 \sin 2\theta}$, and the Jordan-style bound $\sin 2\theta \geq \frac{4\theta}{\pi}$ on $[0, \frac{\pi}{4}]$ gives

$$\left| \int_{\text{arc}} \right| \leq R \int_0^{\pi/4} e^{-4R^2\theta/\pi} d\theta \leq \frac{\pi}{4R} \rightarrow 0.$$

Hence, letting $R \rightarrow \infty$ and using $\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$,

$$\int_0^\infty e^{ix^2} dx = e^{i\pi/4} \cdot \frac{\sqrt{\pi}}{2},$$

and separating real and imaginary parts,

$$\int_0^\infty \cos(x^2) dx = \int_0^\infty \sin(x^2) dx = \frac{1}{2} \sqrt{\frac{\pi}{2}}.$$

No residues were harmed in this example – the entire content of the calculation was the freedom to rotate the contour.

§11.6.3 Trigonometric Integrals

For integrals of a rational function of $\sin \theta$ and $\cos \theta$ over a full period, no limiting process is needed at all: the unit circle *is* the contour.

Substitute $z = e^{i\theta}$, so that as θ runs from 0 to 2π , z traverses the unit circle γ anticlockwise, and

$$\cos \theta = \frac{1}{2} \left(z + \frac{1}{z} \right), \quad \sin \theta = \frac{1}{2i} \left(z - \frac{1}{z} \right), \quad d\theta = \frac{dz}{iz}.$$

Example 11.6.7

For $a > 1$, let's compute

$$I = \int_0^{2\pi} \frac{d\theta}{a + \cos \theta}.$$

Substituting,

$$I = \oint_\gamma \frac{1}{a + \frac{1}{2}(z + z^{-1})} \frac{dz}{iz} = \frac{2}{i} \oint_\gamma \frac{dz}{z^2 + 2az + 1}.$$

The denominator has roots $z_\pm = -a \pm \sqrt{a^2 - 1}$, real, with $z_+ z_- = 1$; since $a > 1$ we

have $|z_-| > 1 > |z_+|$, so only z_+ is enclosed. Its residue is $\frac{1}{z_+ - z_-} = \frac{1}{2\sqrt{a^2 - 1}}$, and

$$I = \frac{2}{i} \cdot 2\pi i \cdot \frac{1}{2\sqrt{a^2 - 1}} = \frac{2\pi}{\sqrt{a^2 - 1}}.$$

Example 11.6.8 (The Poisson Kernel Integral)

For $|a| < 1$, let's compute

$$I = \int_0^{2\pi} \frac{d\theta}{1 - 2a \cos \theta + a^2},$$

which the reader may recognise as the normalisation of the Poisson kernel from IB Methods. The denominator factors beautifully after the substitution: since $1 - 2a \cos \theta + a^2 = (1 - ae^{i\theta})(1 - ae^{-i\theta})$,

$$I = \oint_{\gamma} \frac{1}{(1 - az)(1 - a/z)} \frac{dz}{iz} = \frac{1}{i} \oint_{\gamma} \frac{dz}{(1 - az)(z - a)}.$$

The integrand has simple poles at $z = a$ (inside the unit circle) and $z = \frac{1}{a}$ (outside) – note how the two poles are inversions of each other, mirroring the symmetry of the original expression. Only $z = a$ contributes:

$$I = \frac{1}{i} \cdot 2\pi i \cdot \frac{1}{1 - a^2} = \frac{2\pi}{1 - a^2}.$$

§11.6.4 Jordan's Lemma

Integrals like $\int \frac{\cos x}{1+x^2} dx$ bring a new difficulty. To use contour methods we must extend the integrand off the real axis, and the natural-looking extension $\frac{\cos z}{1+z^2}$ is a trap: $\cos z = \frac{e^{iz} + e^{-iz}}{2}$ grows exponentially as $\text{Im } z \rightarrow \pm\infty$ (in *both* half-planes!), so no closing semicircle can die. The correct move is to integrate $\frac{e^{iz}}{1+z^2}$, which is bounded in the upper half-plane since $|e^{iz}| = e^{-\text{Im } z}$, and take real parts at the end.

For decay of the semicircle we then need a slightly finer estimate than ‘length times sup’, because $|e^{iz}|$ is only small on *most* of the semicircle – near the real axis it is of size 1. The right tool:

Lemma 11.6.9 (Jordan's Lemma)

Let f be analytic in the upper half-plane apart from finitely many singularities, with $f(z) \rightarrow 0$ as $|z| \rightarrow \infty$. Then for any constant $\lambda > 0$,

$$\int_{\gamma_R} f(z) e^{i\lambda z} dz \rightarrow 0 \quad \text{as } R \rightarrow \infty,$$

where γ_R is the semicircle of radius R in the upper half-plane. For $\lambda < 0$ the same holds with the semicircle in the lower half-plane.

Proof. Parametrise $z = Re^{i\theta}$. The whole point is the bound $|e^{i\lambda z}| = e^{-\lambda R \sin \theta}$,

together with the elementary inequality $\sin \theta \geq \frac{2\theta}{\pi}$ on $[0, \frac{\pi}{2}]$ (the sine curve lies above its chord). Splitting the semicircle symmetrically and using this:

$$\begin{aligned} \left| \int_{\gamma_R} f(z) e^{i\lambda z} dz \right| &\leq R \sup_{z \in \gamma_R} |f(z)| \int_0^\pi e^{-\lambda R \sin \theta} d\theta \leq 2R \sup_{z \in \gamma_R} |f(z)| \int_0^{\pi/2} e^{-2\lambda R \theta / \pi} d\theta \\ &= \frac{\pi}{\lambda} (1 - e^{-\lambda R}) \sup_{z \in \gamma_R} |f(z)| \rightarrow 0, \end{aligned}$$

since $\sup_{\gamma_R} |f| \rightarrow 0$ by hypothesis. $\square$

The lesson to remember alongside the lemma: *the factor $e^{i\lambda z}$ decides which half-plane to close in* – upper for $\lambda > 0$, lower for $\lambda < 0$. If f decays like R^{-2} we don't need the lemma (crude estimation suffices), but when $f \sim R^{-1}$ – as for Fourier transforms of rational functions with degree gap 1 – Jordan's lemma is essential.

Example 11.6.10

For $\alpha > 0$, let's find

$$I = \int_0^\infty \frac{\cos \alpha x}{1+x^2} dx.$$

Close $\frac{e^{i\alpha z}}{1+z^2}$ in the upper half-plane (as $\alpha > 0$). The enclosed pole is $z = i$ with residue $\frac{e^{i\alpha \cdot i}}{2i} = \frac{e^{-\alpha}}{2i}$, so

$$\int_{-\infty}^\infty \frac{e^{i\alpha x}}{1+x^2} dx = 2\pi i \cdot \frac{e^{-\alpha}}{2i} = \pi e^{-\alpha}.$$

Taking real parts and halving (the integrand $\frac{\cos \alpha x}{1+x^2}$ is even; its companion $\frac{\sin \alpha x}{1+x^2}$ is odd, so the answer was real all along),

$$I = \frac{\pi}{2} e^{-\alpha}.$$

It is worth noticing the answer is *not analytic in α* at 0 – it is $\frac{\pi}{2} e^{-|\alpha|}$ for real α – a hint that something interesting (closing in opposite half-planes) happens as α changes sign.

Example 11.6.11 (Where Jordan is Genuinely Needed)

For $a > 0$, let's compute

$$I = \int_0^\infty \frac{x \sin x}{x^2 + a^2} dx.$$

Close $f(z)e^{iz}$ with $f(z) = \frac{z}{z^2+a^2}$ in the upper half-plane. This time $f \sim \frac{1}{R}$ on the arc, so the crude bound gives $\pi R \cdot O(R^{-1}) = O(1)$ – useless – and Jordan's lemma is doing real work. The enclosed pole is $z = ia$:

$$\int_{-\infty}^\infty \frac{x e^{ix}}{x^2 + a^2} dx = 2\pi i \operatorname{Res}_{z=ia} \frac{z e^{iz}}{z^2 + a^2} = 2\pi i \cdot \frac{ia e^{-a}}{2ia} = \pi i e^{-a}.$$

Taking imaginary parts and halving (the integrand is even),

$$I = \frac{\pi}{2} e^{-a}.$$

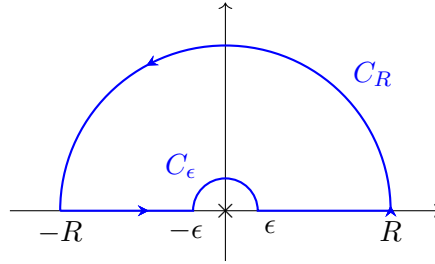
(The real part of the identity reads $\text{PV} \int \frac{x \cos x}{x^2 + a^2} dx = 0$, correct by oddness.)

§11.6.5 Indented Contours and Principal Values

Next complication: a pole *on* the contour. The classic example is

$$I = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx.$$

The real integrand is perfectly finite at $x = 0$ (removable singularity), but the function we want to close with, $\frac{e^{iz}}{z}$, has a genuine pole at the origin, sitting exactly on our contour. The fix is to *indent*: walk along the real axis from $-R$ to $-\epsilon$, hop over the pole via a small semicircle C_ϵ in the upper half-plane, continue from ϵ to R , and close with the big semicircle C_R .



The small detour contributes something computable in the limit:

Lemma 11.6.12 (Indentation Lemma)

If f has a *simple* pole at z_0 , and C_ϵ is an arc of angular width ϕ on the circle $|z - z_0| = \epsilon$, traversed anticlockwise, then

$$\int_{C_\epsilon} f(z) dz \rightarrow i\phi \operatorname{Res}_{z=z_0} f(z) \quad \text{as } \epsilon \rightarrow 0.$$

Proof. Write $f(z) = \frac{a_{-1}}{z-z_0} + g(z)$ with g analytic near z_0 . The g part is bounded, so its integral is $O(\epsilon)$. Parametrising the principal part over the arc gives $a_{-1} \int i d\theta = i\phi a_{-1}$. $\square$

So a *semicircular* indentation picks up exactly *half* a residue, $\pm i\pi \operatorname{Res} f$ (sign according to direction). Note this is special to simple poles – for higher-order poles the indentation integral diverges, matching the fact that the corresponding principal value integrals genuinely diverge.

Example 11.6.13 (The Dirichlet Integral)

Continuing with $\int_{-\infty}^{\infty} \frac{\sin x}{x} dx$: the indented contour encloses *no* singularities, so by

Cauchy's theorem

$$\left(\int_{-R}^{-\epsilon} + \int_{\epsilon}^R \right) \frac{e^{ix}}{x} dx + \int_{C_\epsilon} \frac{e^{iz}}{z} dz + \int_{C_R} \frac{e^{iz}}{z} dz = 0.$$

The big arc dies by Jordan's lemma ($f(z) = \frac{1}{z} \rightarrow 0$, and note we genuinely need Jordan here – crude estimation gives $\pi R \cdot \frac{1}{R} \not\rightarrow 0$). The indentation is traversed *clockwise* over angle π , contributing $-i\pi \operatorname{Res}_{z=0} \frac{e^{iz}}{z} = -i\pi$ in the limit. Hence

$$\operatorname{PV} \int_{-\infty}^{\infty} \frac{e^{ix}}{x} dx = i\pi,$$

and taking imaginary parts,

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \pi.$$

(The real part says $\operatorname{PV} \int \frac{\cos x}{x} dx = 0$, which is true by oddness – but only as a principal value, the symmetric limit $\lim_{\epsilon \rightarrow 0} (\int_{-\infty}^{-\epsilon} + \int_{\epsilon}^{\infty})$. For $\frac{\sin x}{x}$ the singularity is removable, so the principal value is an honest integral and no caveat is needed.)

The general bookkeeping is worth extracting from the example. Closing an indented contour and sending $\epsilon \rightarrow 0$ always yields, for simple poles *on* the (say, real-axis) contour,

$$\operatorname{PV} \int_{-\infty}^{\infty} f(x) dx = 2\pi i \sum_{\text{inside}} \operatorname{Res} f + \pi i \sum_{\text{on the contour}} \operatorname{Res} f$$

(when closing in the upper half-plane, with the arc dying) – poles sitting on the contour count with weight $\frac{1}{2}$, because the indentation walks half-way around them.

Example 11.6.14 (A Double Pole on the Axis, Almost)

Let's compute

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx.$$

Writing $\sin^2 x = \frac{1 - \cos 2x}{2}$, the natural complex extension is

$$f(z) = \frac{1 - e^{2iz}}{2z^2},$$

whose *real-axis* real part is our integrand. Although $\frac{1}{2z^2}$ suggests a double pole, the numerator vanishes at 0 too: expanding, $1 - e^{2iz} = -2iz + 2z^2 + \dots$, so

$$f(z) = -\frac{i}{z} + 1 + O(z),$$

a *simple* pole with residue $-i$. Close the indented contour in the upper half-plane; it encloses nothing, and the two pieces of f both die on the big arc ($\frac{1}{2z^2}$ crudely, $\frac{e^{2iz}}{2z^2}$ by Jordan or crudely too). The half-residue rule then gives

$$\operatorname{PV} \int_{-\infty}^{\infty} \frac{1 - e^{2ix}}{2x^2} dx = \pi i \cdot (-i) = \pi,$$

and taking real parts (the principal value is again an honest integral, the singularity of $\frac{\sin^2 x}{x^2}$ being removable),

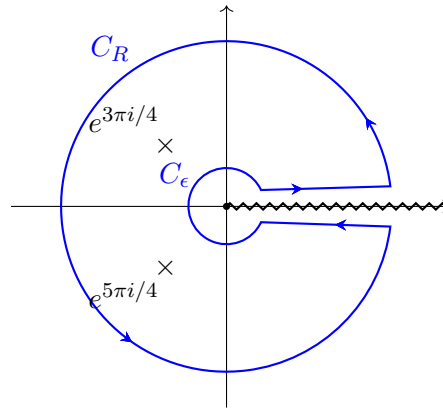
$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2} dx = \pi.$$

Amusingly, this is the same value as $\int \frac{\sin x}{x} dx$ – a coincidence the reader may enjoy explaining via integration by parts.

§11.6.6 Integrals with Branch Cuts: the Keyhole Contour

Now for the technique promised in Section 2, where the discontinuity across a branch cut is turned from nuisance into tool. The standard problem: integrals of the form $\int_0^\infty x^\alpha R(x) dx$ with $\alpha \notin \mathbb{Z}$ and R rational.

The trick: integrate $z^\alpha R(z)$ with the branch cut along the *positive real axis* – right where we want to integrate! – using the branch $z^\alpha = r^\alpha e^{i\alpha\theta}$ with $\theta \in [0, 2\pi)$. Just above the cut, $z^\alpha \approx x^\alpha$; just below, $z^\alpha \approx x^\alpha e^{2\pi i\alpha}$. Because $\alpha \notin \mathbb{Z}$ these differ, so the two sides of the cut do *not* cancel – they combine to a non-zero multiple of the integral we want. The contour is the **keyhole**: a large circle C_R , a small circle C_ϵ around the branch point, and two straight legs hugging the cut from above and below.



Example 11.6.15 (A Keyhole Integral)

For $-1 < \alpha < 1$ (so that the integral converges at both ends), let's evaluate

$$I = \int_0^\infty \frac{x^\alpha}{1 + \sqrt{2}x + x^2} dx.$$

Integrate $f(z) = \frac{z^\alpha}{1 + \sqrt{2}z + z^2}$, with the branch of z^α fixed above, around the keyhole. Four pieces:

- On C_R : the integrand is $O(R^{\alpha-2})$ and the length is $O(R)$, so the integral is $O(R^{\alpha-1}) \rightarrow 0$, using $\alpha < 1$.
- On C_ϵ : integrand $O(\epsilon^\alpha)$, length $O(\epsilon)$, total $O(\epsilon^{\alpha+1}) \rightarrow 0$ since $\alpha > -1$.
- Just above the cut ($\theta = 0$): $\int_\epsilon^R \frac{x^\alpha dx}{1 + \sqrt{2}x + x^2} \rightarrow I$.
- Just below the cut ($\theta = 2\pi$, traversed right to left): $\int_R^\epsilon \frac{x^\alpha e^{2\pi i\alpha} dx}{1 + \sqrt{2}x + x^2} \rightarrow$

$$-e^{2\pi i\alpha} I.$$

So the whole contour integral tends to $(1 - e^{2\pi i\alpha})I$. The poles are the roots of $z^2 + \sqrt{2}z + 1$, namely $e^{3\pi i/4}$ and $e^{5\pi i/4}$ – and here the branch choice matters: the lower pole must be labelled $e^{5\pi i/4}$, *not* $e^{-3\pi i/4}$, since our θ lives in $[0, 2\pi)$, and using the wrong label produces the wrong value of z^α . By the $\frac{g}{h'}$ formula, with $h' = 2z + \sqrt{2}$ giving $h'(e^{3\pi i/4}) = \sqrt{2}i$ and $h'(e^{5\pi i/4}) = -\sqrt{2}i$, the residues are

$$\frac{e^{3\pi i\alpha/4}}{\sqrt{2}i}, \quad \frac{e^{5\pi i\alpha/4}}{-\sqrt{2}i}.$$

The residue theorem then gives

$$(1 - e^{2\pi i\alpha})I = \frac{2\pi i}{\sqrt{2}i} (e^{3\pi i\alpha/4} - e^{5\pi i\alpha/4}) = \sqrt{2}\pi e^{\pi i\alpha} (e^{-\pi i\alpha/4} - e^{\pi i\alpha/4}).$$

Writing $1 - e^{2\pi i\alpha} = -e^{\pi i\alpha}(e^{\pi i\alpha} - e^{-\pi i\alpha}) = -2ie^{\pi i\alpha} \sin \pi\alpha$ and the bracket as $-2i \sin \frac{\pi\alpha}{4}$,

$$I = \frac{\sqrt{2}\pi \sin(\pi\alpha/4)}{\sin \pi\alpha}.$$

As a sanity check, $\alpha \rightarrow 0$ gives $I \rightarrow \frac{\sqrt{2}\pi}{4} = \frac{\pi}{2\sqrt{2}}$, agreeing with $\int_0^\infty \frac{dx}{1+\sqrt{2}x+x^2}$ computed by completing the square.

Not every branch-cut integral needs the full keyhole. If the integral is over the whole real line, it is often enough to place the cut in the half-plane we are *not* closing in (e.g. cut along the negative imaginary axis and close upwards), so the closed contour never meets the cut at all. And for integrals against $\log x$, a useful alternative to the keyhole is a semicircular contour with the branch cut pointing downwards, comparing the contributions of $\log$ on the two halves of the real axis. Let's see this in action.

Example 11.6.16 (A Logarithmic Integral)

Let's evaluate

$$I = \int_0^\infty \frac{\log x}{1+x^2} dx.$$

Integrate $f(z) = \frac{\log z}{1+z^2}$ around the *indented semicircular* contour of the Dirichlet-integral example – along the real axis from $-R$ to R , hopping over the origin by a small semicircle, closed by the big upper semicircle – with the branch cut of $\log$ along the negative *imaginary* axis, $\theta \in (-\frac{\pi}{2}, \frac{3\pi}{2})$, so the cut stays clear of the contour.

The small arc contributes $O(\epsilon \log \frac{1}{\epsilon}) \rightarrow 0$ (the logarithmic singularity is too weak to matter – note this is different from the simple-pole indentations of before), and the big arc contributes $O(\frac{\log R}{R}) \rightarrow 0$. On the positive real axis, $\log z = \log x$; on the negative real axis, $z = x = |x|e^{i\pi}$ with our branch, so $\log z = \log |x| + i\pi$. Hence in the limit the contour integral equals

$$\int_0^\infty \frac{\log x}{1+x^2} dx + \int_0^\infty \frac{\log x + i\pi}{1+x^2} dx = 2I + i\pi \cdot \frac{\pi}{2}.$$

The enclosed pole is at $z = i = e^{i\pi/2}$, with residue

$$\frac{\log i}{2i} = \frac{i\pi/2}{2i} = \frac{\pi}{4},$$

so the residue theorem gives $2I + \frac{i\pi^2}{2} = 2\pi i \cdot \frac{\pi}{4} = \frac{i\pi^2}{2}$. The imaginary parts balance identically – a free consistency check on the branch bookkeeping – and

$$I = 0.$$

A surprising answer, but a correct one: substituting $x \mapsto \frac{1}{x}$ maps the integral to its own negative. The same contour with $\frac{\log z}{(1+z^2)^2}$ or $\frac{(\log z)^2}{1+z^2}$ gives genuinely non-zero values (e.g. $-\frac{\pi}{4}$ for the first), and the method is identical.

§11.6.7 Rectangular Contours

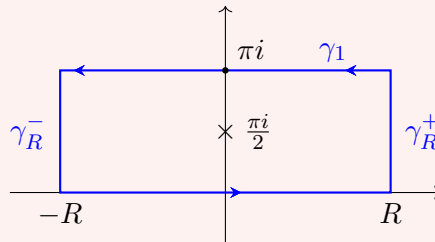
Semicircular contours fail for integrands built from cosh, sinh and friends: these are periodic in the *imaginary* direction, so a growing semicircle swallows more and more poles, and the arc never dies. The remedy is to respect the periodicity: use a *rectangle* of fixed height, chosen so that the top edge reproduces the bottom edge up to a constant factor.

Example 11.6.17

For $-1 < \alpha < 1$, let's compute

$$I = \int_{-\infty}^{\infty} \frac{e^{\alpha x}}{\cosh x} dx.$$

The poles of $\frac{1}{\cosh}$ sit at $z = (n + \frac{1}{2})\pi i$, an infinite ladder up the imaginary axis. Since $\cosh(z + \pi i) = -\cosh z$, we take the rectangle with corners $\pm R$, $\pm R + \pi i$, which encloses exactly one pole, at $\frac{\pi i}{2}$.



The bottom edge tends to I . The top edge, parametrised as $z = x + \pi i$ with x from R to $-R$, gives

$$\int_R^{-R} \frac{e^{\alpha x} e^{\alpha \pi i}}{-\cosh x} dx \rightarrow e^{\alpha \pi i} I,$$

the two sign changes (reversed direction, cosh flipping sign) cancelling. On the right-hand edge $z = R + iy$ we have the pleasant identity $|\cosh z|^2 = \sinh^2 R + \cos^2 y \geq \sinh^2 R$, so the integrand is $O(e^{(\alpha-1)R}) \rightarrow 0$ (here $\alpha < 1$ matters), and its integral over the fixed height π dies; similarly the left edge dies using $\alpha > -1$.

The enclosed residue, by the $\frac{g}{h}$ rule, is

$$\operatorname{Res}_{z=\pi i/2} \frac{e^{\alpha z}}{\cosh z} = \frac{e^{\alpha \pi i/2}}{\sinh \frac{\pi i}{2}} = \frac{e^{\alpha \pi i/2}}{i \sin \frac{\pi}{2}} = -ie^{\alpha \pi i/2}.$$

Assembling with the residue theorem:

$$(1 + e^{\alpha \pi i})I = 2\pi i \left(-ie^{\alpha \pi i/2}\right) = 2\pi e^{\alpha \pi i/2},$$

and dividing through by $e^{\alpha \pi i/2}$,

$$I(e^{-\alpha \pi i/2} + e^{\alpha \pi i/2}) = 2\pi \implies I = \frac{\pi}{\cos(\alpha \pi/2)}.$$

(Check: at $\alpha = 0$ this says $\int \frac{dx}{\cosh x} = \pi$, which is the classic $[2 \tan^{-1}(\tanh \frac{x}{2}) \cdot 2]$ computation – or just believe the contour.)

Example 11.6.18 (Summing a Series by Residues)

Rectangular (here square) contours can even sum series. Consider

$$\oint_{\gamma_N} \frac{\cot \pi z}{z^2} dz$$

where γ_N is the square with corners $(N + \frac{1}{2})(\pm 1 \pm i)$ – positioned so the sides pass between the poles of $\cot \pi z$, which lie at the integers. One can show $|\cot \pi z|$ is bounded on these squares uniformly in N (by $\coth \frac{\pi}{2}$, say: on the vertical sides $\cot \pi z$ reduces to $\mp \tanh \pi y$, and on the horizontal sides it is within $\coth \pi(N + \frac{1}{2})$ of $\mp i$), so the integral is $O(N^{-1}) \rightarrow 0$.

Meanwhile the residue theorem counts the poles inside: at each non-zero integer n , $\cot \pi z$ has a simple pole with residue $\frac{1}{\pi}$, contributing $\frac{1}{\pi n^2}$; and at $z = 0$ there is a triple pole with residue $-\frac{\pi}{3}$ (from $\cot \pi z = \frac{1}{\pi z} - \frac{\pi z}{3} - \dots$). Letting $N \rightarrow \infty$,

$$0 = 2\pi i \left(2 \sum_{n=1}^{\infty} \frac{1}{\pi n^2} - \frac{\pi}{3} \right) \implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

This is, admittedly, an extravagant derivation of Euler's sum – but unlike most derivations it generalises instantly: the same contour applied to $\frac{\cot \pi z}{z^{2k}}$ evaluates $\sum n^{-2k}$ for every k .

Let's close the section by collecting the decision procedure, since in an exam the choice of contour *is* the question:

Integral looks like	Contour to reach for
$\int_{-\infty}^{\infty}$ (rational, decay ≥ 2)	semicircle, crude estimate
$\int_{-\infty}^{\infty}$ (rational) $\times e^{i\lambda x}$, $\cos$, $\sin$	semicircle ($e^{i\lambda z}$ version!), Jordan
$\int_0^{2\pi}$ (rational in $\cos \theta, \sin \theta$)	unit circle, $z = e^{i\theta}$
pole on the contour / PV	indent by a small semicircle: half-residues
$\int_0^{\infty} x^{\alpha} \times$ (rational), $\alpha \notin \mathbb{Z}$	keyhole around the positive real axis
$\int_0^{\infty} \log x \times$ (rational)	indented semicircle, or keyhole with $(\log z)^2$
$\cosh$, $\sinh$, $e^{\alpha x}$ in the integrand	rectangle of height matching the period
series $\sum f(n)$	square through half-integers, with $\cot \pi z$

In every case the workflow is the same four steps: choose the complex extension of the integrand (and branch, if needed); choose the contour; kill the unwanted pieces by estimation, Jordan or indentation lemmas; and count residues.

§11.7 Fourier Transforms

The last two sections of the course concern *transform theory*: the Fourier transform, which you met in IB Methods, and its cousin the Laplace transform, which you did not (for reasons that will become clear – its inversion formula is a contour integral). The new weapon of residue calculus lets us actually *compute* transforms and inverse transforms that were previously out of reach.

§11.7.1 Definition and Basic Properties

Definition 11.7.1 (Fourier Transform)

The **Fourier transform** of a function $f : \mathbb{R} \rightarrow \mathbb{C}$ (decaying suitably at $\pm\infty$) is

$$\tilde{f}(k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} dx,$$

and the function is recovered by the **inverse transform**

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk.$$

(Conventions differ – the signs in the exponent and the placement of the 2π get shuffled between authors. Any statement you prove should survive a change of convention.) We write $\tilde{f} = \mathcal{F}(f)$, and when f is a function of time t the transform variable is traditionally called ω .

From IB Methods we recall the key properties, all following from elementary manipulations of the integral: the transform is linear, and

$$\mathcal{F}(f(x-a)) = e^{-ika} \tilde{f}(k), \quad \mathcal{F}(f(ax)) = \frac{1}{|a|} \tilde{f}\left(\frac{k}{a}\right), \quad \mathcal{F}(f'(x)) = ik \tilde{f}(k),$$

together with the convolution theorem $\mathcal{F}(f * g) = \tilde{f} \tilde{g}$. The derivative rule is the engine of applications: differential equations become algebraic equations in transform space.

There is one refinement of the inversion formula worth knowing. The correct general statement is

$$\frac{1}{2} (f(x^+) + f(x^-)) = \frac{1}{2\pi} \text{PV} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk :$$

at a jump discontinuity the inverse transform returns the *average* of the two one-sided values, and the integral needs only exist as a principal value, $\lim_{R \rightarrow \infty} \int_{-R}^R$. This is excellent news for us, because principal values at infinity are exactly what closing contours computes.

§11.7.2 Transforms via Contour Integration

Example 11.7.2 (The Gaussian)

Let $f(x) = e^{-x^2/2}$. Completing the square,

$$\tilde{f}(k) = \int_{-\infty}^{\infty} e^{-x^2/2 - ikx} dx = e^{-k^2/2} \int_{-\infty}^{\infty} e^{-(x+ik)^2/2} dx,$$

and the remaining integral is $\int e^{-z^2/2} dz$ along the horizontal line $\text{Im } z = k$. Since $e^{-z^2/2}$ is entire, we may shift this line down to the real axis: integrate around the rectangle with corners $\pm R$, $\pm R + ik$, note the vertical ends have integrand bounded by $e^{-(R^2 - k^2)/2} \rightarrow 0$, and apply Cauchy's theorem. So the integral equals $\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$, giving

$$\tilde{f}(k) = \sqrt{2\pi} e^{-k^2/2}.$$

The Gaussian is (up to scaling) its own Fourier transform.

Example 11.7.3 (Inversion by Jordan's Lemma)

Let

$$f(x) = \begin{cases} e^{-ax} & x > 0 \\ 0 & x < 0 \end{cases}$$

for a constant $a > 0$. A direct computation gives $\tilde{f}(k) = \frac{1}{a + ik}$. Let's now run the inversion integral through our contour machinery and check we recover f :

$$f(x) = \frac{1}{2\pi} \text{PV} \int_{-\infty}^{\infty} \frac{e^{ikx}}{a + ik} dk.$$

The integrand, as a function of complex k , has a single simple pole at $k = ia$ (in the upper half-plane), and decays like $|k|^{-1}$ – too slowly for crude estimates, so Jordan's lemma it is. The factor e^{ikx} tells us where to close: upwards for $x > 0$, downwards for $x < 0$.

For $x > 0$, closing upwards encircles the pole:

$$f(x) = \frac{1}{2\pi} \cdot 2\pi i \text{Res}_{k=ia} \frac{e^{ikx}}{i(k - ia)} = e^{-ax}.$$

For $x < 0$, closing downwards encircles nothing, so $f(x) = 0$. And at the jump $x = 0$: the principal value gives $\frac{1}{2\pi} \text{PV} \int \frac{dk}{a + ik} = \frac{1}{2}$ (the $\frac{a}{a^2 + k^2}$ part integrates to π ; the odd part cancels) – precisely the average $\frac{f(0^+) + f(0^-)}{2}$, as the refined inversion formula promised.

This half-plane dichotomy – close up for positive ‘time’, down for negative – is the

mechanism behind causality in physical applications, and we are about to see it again as the reason Laplace transforms describe systems that start at $t = 0$.

Example 11.7.4 (The Lorentzian Pair)

For $a > 0$ consider $f(x) = e^{-a|x|}$. Directly,

$$\tilde{f}(k) = \int_{-\infty}^0 e^{(a-ik)x} dx + \int_0^{\infty} e^{-(a+ik)x} dx = \frac{1}{a-ik} + \frac{1}{a+ik} = \frac{2a}{a^2 + k^2},$$

the *Lorentzian*. Inverting by residues confirms the round trip: the inversion integrand $2a e^{ikx}/(k-ia)(k+ia)$ has poles at $\pm ia$, and for $x > 0$ we close upwards around $k = ia$,

$$f(x) = \frac{1}{2\pi} \cdot 2\pi i \cdot \frac{2a e^{-ax}}{2ia} = e^{-ax},$$

and for $x < 0$ close downwards around $k = -ia$ (collecting a sign for the clockwise direction) to get e^{ax} . So $f(x) = e^{-a|x|}$, as it should be. This pair is worth remembering: exponential decay in one domain corresponds to algebraic decay with poles at $\pm ia$ in the other, and the distance a of the pole from the real axis sets the decay rate.

Remark (Parseval's Identity). We recall from IB Methods, without proof, that

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk.$$

Combined with residue-computed transforms this evaluates yet more real integrals: applying it to the pair above, for instance, gives $\int \frac{4a^2 dk}{(a^2+k^2)^2} = 2\pi \int e^{-2a|x|} dx = \frac{2\pi}{a}$, recovering our earlier result $\int_{-\infty}^{\infty} \frac{dk}{(k^2+a^2)^2} = \frac{\pi}{2a^3}$ with no contour at all.

§11.7.3 Solving Differential Equations

A quick reminder of the standard workflow from IB Methods, which transfers wholesale to this course. To solve a linear constant-coefficient equation, transform it, solve algebraically, and invert – with the inversion now done by residues where a table fails.

Example 11.7.5 (An ODE on the Real Line)

Let's find the decaying solution of

$$y'' - y = -f(x)$$

for given decaying f . Transforming, $-k^2 \tilde{y} - \tilde{y} = -\tilde{f}$, so

$$\tilde{y} = \frac{\tilde{f}(k)}{1+k^2},$$

and by the convolution theorem $y = f * g$ where g has transform $\frac{1}{1+k^2}$. We can find g by the same inversion technique as above: for $x > 0$ close upwards around the pole $k = i$,

$$g(x) = \frac{1}{2\pi} \cdot 2\pi i \cdot \frac{e^{i(i)x}}{2i} = \frac{1}{2} e^{-x},$$

and by symmetry $g(x) = \frac{1}{2}e^{-|x|}$ in general. So

$$y(x) = \frac{1}{2} \int_{-\infty}^{\infty} e^{-|x-x'|} f(x') \, dx',$$

which the reader may recognise as the Green's function solution from IB Methods – with the Green's function now produced by a residue rather than by patching boundary conditions.

The same scheme solves PDEs, transforming in one variable to leave an ODE in the others.

Example 11.7.6 (The Heat Equation on the Line)

Let's solve the diffusion equation on the whole line,

$$\frac{\partial u}{\partial t} = \kappa \frac{\partial^2 u}{\partial x^2}, \quad u(x, 0) = u_0(x),$$

by transforming in x . Writing $\tilde{u}(k, t)$ for the transform, the PDE becomes an ODE in t for each mode:

$$\frac{\partial \tilde{u}}{\partial t} = -\kappa k^2 \tilde{u} \quad \implies \quad \tilde{u}(k, t) = \tilde{u}_0(k) e^{-\kappa k^2 t}.$$

Each Fourier mode simply decays, at a rate growing like k^2 – diffusion is a low-pass filter. To return to real space, recognise the product as a convolution: we need the function whose transform is $e^{-\kappa k^2 t}$, which by our Gaussian computation (run backwards, with the scaling property) is

$$\frac{1}{\sqrt{4\pi\kappa t}} e^{-x^2/4\kappa t}.$$

Hence

$$u(x, t) = \frac{1}{\sqrt{4\pi\kappa t}} \int_{-\infty}^{\infty} e^{-(x-x')^2/4\kappa t} u_0(x') \, dx' :$$

the initial data smeared by a spreading Gaussian (the *heat kernel*), recovering the IB Methods result with the Gaussian transform now honestly derived by contour shifting.

For *initial value* problems in time – where the equation starts at $t = 0$ with prescribed data, rather than living on the whole line – the Laplace transform of the next section is the natural tool, and we will return to PDEs at the end of it.

§11.8 Laplace Transforms

The Fourier transform has two blind spots as a tool for differential equations. First, it requires decay: functions like e^t , or even the constant function 1, have no Fourier transform, so growing solutions are out of reach. Second, it has no natural slot for *initial conditions* – transforming an ODE on the whole line yields a particular solution with no arbitrary constants. The Laplace transform fixes both, at the modest price of only handling functions defined on $t \geq 0$.

Throughout this section, every function is implicitly zero for negative time: when we write $f(t) = e^t$ we mean $e^t H(t)$, where H is the Heaviside step function.

§11.8.1 Definition and First Examples

Definition 11.8.1 (Laplace Transform)

The **Laplace transform** of $f(t)$ (with $f(t) = 0$ for $t < 0$) is

$$\hat{f}(p) = \mathcal{L}(f)(p) = \int_0^\infty f(t)e^{-pt} dt,$$

defined for those p for which the integral converges – which includes all $\operatorname{Re} p$ sufficiently large provided f grows no faster than exponentially.

(The transform variable is called s at least as often as p .) Note the close kinship with the Fourier transform: formally $\hat{f}(p) = \tilde{f}(-ip)$. The exponential weight e^{-pt} with $\operatorname{Re} p$ large is what tames growing functions.

Example 11.8.2 (Basic Transforms)

All of the following are direct computations with the definition:

$$(i) \quad \mathcal{L}(1) = \int_0^\infty e^{-pt} dt = \frac{1}{p}.$$

$$(ii) \quad \mathcal{L}(t) = \frac{1}{p^2}, \text{ integrating by parts; more generally } \mathcal{L}(t^n) = \frac{n!}{p^{n+1}}.$$

$$(iii) \quad \mathcal{L}(e^{\lambda t}) = \int_0^\infty e^{(\lambda-p)t} dt = \frac{1}{p-\lambda}, \text{ valid for } \operatorname{Re} p > \operatorname{Re} \lambda.$$

$$(iv) \quad \text{Writing } \sin t = \frac{1}{2i}(e^{it} - e^{-it}) \text{ and using (iii) twice,}$$

$$\mathcal{L}(\sin t) = \frac{1}{2i} \left(\frac{1}{p-i} - \frac{1}{p+i} \right) = \frac{1}{p^2 + 1},$$

$$\text{and similarly } \mathcal{L}(\cos t) = \frac{p}{p^2 + 1}.$$

For reference (and revision), here is the resulting table, which together with the properties of the next subsection covers nearly everything met in practice:

$f(t)$	1	t^n	$e^{\lambda t}$	$\sin \omega t$	$\cos \omega t$	$t^{-1/2}$
$\hat{f}(p)$	$\frac{1}{p}$	$\frac{n!}{p^{n+1}}$	$\frac{1}{p-\lambda}$	$\frac{\omega}{p^2 + \omega^2}$	$\frac{p}{p^2 + \omega^2}$	$\sqrt{\frac{\pi}{p}}$

(The last entry, whose derivation is a Gaussian integral, will make a curious appearance when we meet branch cuts in the inversion formula.)

Each computation is initially valid only in some half-plane $\operatorname{Re} p > c$, but the resulting formulas – $\frac{1}{p}$, $\frac{1}{p-\lambda}$ and so on – are analytic functions on the whole plane minus their singularities. We henceforth regard $\hat{f}$ as defined everywhere by **analytic continuation**: same function, bigger domain. This matters because the inversion formula will want to roam over the whole p -plane and take residues at the singularities.

§11.8.2 Properties of the Laplace Transform

Proposition 11.8.3 (Properties of the Laplace Transform)

Let f have transform $\hat{f}(p)$. Then:

- (i) (Linearity) $\mathcal{L}(\alpha f + \beta g) = \alpha \hat{f} + \beta \hat{g}$.
- (ii) (Translation) $\mathcal{L}(f(t - t_0)H(t - t_0)) = e^{-pt_0} \hat{f}(p)$ for $t_0 > 0$.
- (iii) (Scaling) $\mathcal{L}(f(\lambda t)) = \frac{1}{\lambda} \hat{f}(p/\lambda)$ for $\lambda > 0$.
- (iv) (Shifting) $\mathcal{L}(e^{p_0 t} f(t)) = \hat{f}(p - p_0)$.
- (v) (Derivative of function) $\mathcal{L}(f'(t)) = p\hat{f}(p) - f(0)$, and iterating,

$$\mathcal{L}(f''(t)) = p^2 \hat{f}(p) - pf(0) - f'(0).$$
- (vi) (Derivative of transform) $\hat{f}'(p) = \mathcal{L}(-tf(t))$, and more generally we have $\hat{f}^{(n)}(p) = \mathcal{L}((-t)^n f(t))$.
- (vii) (Asymptotic limits) $p\hat{f}(p) \rightarrow f(0)$ as $p \rightarrow \infty$, and $p\hat{f}(p) \rightarrow f(\infty)$ as $p \rightarrow 0$ (when the latter limit exists).

Proof. Properties (i)–(iv) are immediate substitutions in the integral. For (v), integrate by parts:

$$\int_0^\infty f'(t)e^{-pt} dt = [f(t)e^{-pt}]_0^\infty + p \int_0^\infty f(t)e^{-pt} dt = p\hat{f}(p) - f(0).$$

For (vi), differentiate under the integral sign. For (vii), take $p \rightarrow \infty$ and $p \rightarrow 0$ respectively in the identity $p\hat{f}(p) = f(0) + \int_0^\infty f'(t)e^{-pt} dt$ from (v). $\square$

Property (v) is the whole reason the Laplace transform exists as a tool: differentiation becomes multiplication by p , and the initial conditions enter automatically. Compare the Fourier rule $\mathcal{F}(f') = ik\hat{f}$, which has no memory of $f(0)$.

Property (vi) is also worth appreciating as a computational device, read from right to left:

$$\mathcal{L}(t \sin t) = -\frac{d}{dp} \frac{1}{p^2 + 1} = \frac{2p}{(p^2 + 1)^2}.$$

Example 11.8.4 (A Timely Warning)

Property (ii) inserts a Heaviside function, and forgetting it is the classic error. For example

$$\mathcal{L}^{-1}\left(\frac{e^{-p}}{p^2}\right) = (t - 1)H(t - 1),$$

the ramp that switches on at $t = 1$ – not $(t - 1)$, which would be negative before then.

§11.8.3 The Inverse Laplace Transform

Now for the part that belongs to this course rather than to Methods. Inverting a Laplace transform is a contour integral along a vertical line.

Theorem 11.8.5 (Bromwich Inversion Formula)

Let $\hat{f}$ be the Laplace transform of f . Then

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \hat{f}(p) e^{pt} dp,$$

where the **Bromwich contour** $\operatorname{Re} p = c$ is a vertical line lying to the *right* of all singularities of $\hat{f}$.

Proof. This is Fourier inversion in disguise. Since f grows at most exponentially, choose c so large that $g(t) = f(t)e^{-ct}$ decays; then g has a Fourier transform, and

$$\tilde{g}(\omega) = \int_{-\infty}^{\infty} f(t)e^{-ct}e^{-i\omega t} dt = \hat{f}(c + i\omega),$$

using $f = 0$ for $t < 0$. Fourier inversion gives

$$f(t)e^{-ct} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(c + i\omega) e^{i\omega t} d\omega,$$

and substituting $p = c + i\omega$ (so $dp = i d\omega$) and multiplying by e^{ct} yields the formula. The condition on c ensures the half-plane $\operatorname{Re} p \geq c$ is free of singularities, which is what makes g decay – and, as we now see, what forces $f(t) = 0$ for $t < 0$. $\square$

In practice we do not integrate along the line directly; we close the contour and use residues.

Proposition 11.8.6 (Inversion by Residues)

Suppose $\hat{f}(p)$ has finitely many isolated singularities $p_1, \dots, p_n$ and $\hat{f}(p) \rightarrow 0$ as $|p| \rightarrow \infty$. Then for $t > 0$,

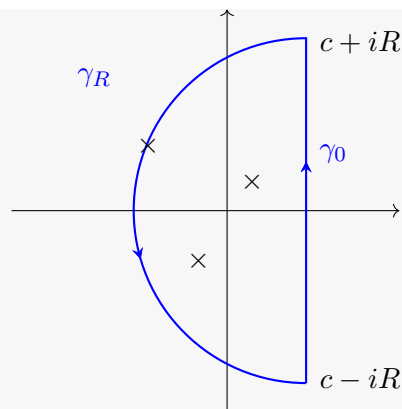
$$f(t) = \sum_{k=1}^n \operatorname{Res}_{p=p_k} (\hat{f}(p) e^{pt}),$$

and $f(t) = 0$ for $t < 0$.

Proof. The factor e^{pt} plays the role of $e^{i\lambda z}$ in Jordan's lemma, rotated a quarter turn: $|e^{pt}| = e^{t \operatorname{Re} p}$ is small where $t \operatorname{Re} p$ is very negative.

For $t < 0$, close the Bromwich contour with a large semicircle to the *right*, where $\operatorname{Re} p \geq c$ and so $|e^{pt}| \leq e^{ct}$. The closed contour encloses no singularities – they all lie to the left of the line – so the integral vanishes by Cauchy's theorem, and (the arc dying by the Jordan-type estimate) $f(t) = 0$. This is the promised consistency check: functions with Laplace transforms vanish for negative t , and it is the placement of the Bromwich line to the right of all singularities that encodes it.

For $t > 0$, close to the *left* instead. For R large enough the semicircle engulfs every singularity:



The arc integral tends to zero (crudely if $\hat{f} = o(|p|^{-1})$; by the Jordan-type estimate in general, using only $\hat{f} \rightarrow 0$), so the residue theorem applied to the closed contour gives

$$f(t) = \frac{1}{2\pi i} \cdot 2\pi i \sum_{k=1}^n \operatorname{Res}_{p=p_k} (\hat{f}(p)e^{pt}),$$

as claimed. $\square$

Example 11.8.7 (Recovering Old Friends) (i) $\hat{f}(p) = \frac{1}{p-1}$: a single simple pole at $p = 1$ (so take $c > 1$). For $t > 0$,

$$f(t) = \operatorname{Res}_{p=1} \frac{e^{pt}}{p-1} = e^t. \quad \checkmark$$

(ii) $\hat{f}(p) = p^{-n}$: a pole of order n at the origin. For $t > 0$,

$$f(t) = \operatorname{Res}_{p=0} \frac{e^{pt}}{p^n} = \lim_{p \rightarrow 0} \frac{1}{(n-1)!} \frac{d^{n-1}}{dp^{n-1}} e^{pt} = \frac{t^{n-1}}{(n-1)!}. \quad \checkmark$$

(iii) $\hat{f}(p) = \frac{1}{p^2 + \omega^2}$: simple poles at $\pm i\omega$, on the imaginary axis (so any $c > 0$ works). For $t > 0$,

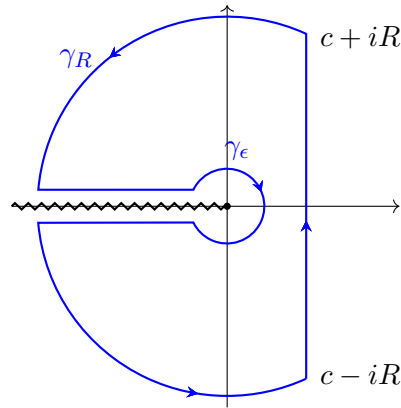
$$f(t) = \frac{e^{i\omega t}}{2i\omega} + \frac{e^{-i\omega t}}{-2i\omega} = \frac{\sin \omega t}{\omega}. \quad \checkmark$$

Poles on the imaginary axis give pure oscillation – no decay, no growth – consistent with the pole-diagram picture we will draw below.

(iv) $\hat{f}(p) = \frac{e^{-p}}{p}$: here $\hat{f} \not\rightarrow 0$ in the left half-plane (the numerator e^{-p} blows up), so we must not blindly close and take residues. Instead absorb the exponential into the time variable: the Bromwich integrand is $\frac{e^{p(t-1)}}{p}$, so writing $t' = t - 1$ we close right for $t' < 0$ (getting 0) and left for $t' > 0$ (getting $\operatorname{Res}_{p=0} \frac{e^{pt'}}{p} = 1$). Thus $f(t) = H(t - 1)$, in accordance with the translation property.

What if $\hat{f}$ has a *branch point*? Then there is no annulus of analyticity, no Laurent series, no residue – and the contour must be reshaped so that it never crosses the cut. The standard geometry, for a branch point at the origin with cut along the negative real axis, is the *Bromwich keyhole*: close to the left, but detour along both sides of the cut and

around the branch point.



Example 11.8.8 (Inversion Through a Branch Cut)

Let's invert $\hat{f}(p) = p^{-1/2}$ (principal branch, cut along the negative real axis). Around the Bromwich keyhole the integrand $e^{pt}p^{-1/2}$ is analytic, so the integral over the whole closed contour vanishes; the large arcs die as before, and the little circle around the branch point contributes $O(\epsilon^{1/2}) \rightarrow 0$. So the Bromwich integral equals minus the contribution of the two legs along the cut.

Above the cut, $p = xe^{i\pi}$ with x running from ∞ to 0: there $p^{-1/2} = x^{-1/2}e^{-i\pi/2} = -ix^{-1/2}$ and $dp = -dx$. Below the cut, $p = xe^{-i\pi}$ with x from 0 to ∞ : there $p^{-1/2} = ix^{-1/2}$ and $dp = -dx$. Adding,

$$\begin{aligned} f(t) &= -\frac{1}{2\pi i} \left(\int_{\infty}^0 e^{-xt}(-ix^{-1/2})(-dx) + \int_0^{\infty} e^{-xt}(ix^{-1/2})(-dx) \right) \\ &= \frac{1}{\pi} \int_0^{\infty} e^{-xt} x^{-1/2} dx. \end{aligned}$$

Substituting $u = xt$, the integral is $t^{-1/2}\Gamma(\frac{1}{2}) = \sqrt{\pi/t}$, and so

$$f(t) = \frac{1}{\sqrt{\pi t}}.$$

This matches the known transform $\mathcal{L}(t^{-1/2}) = \sqrt{\pi/p}$. Notice the qualitative moral: poles of $\hat{f}$ produce exponentials $e^{p_k t}$ in f , while a branch cut produces a *continuum* of decaying exponentials – here giving algebraic decay $t^{-1/2}$, a behaviour no finite collection of poles could create.

§11.8.4 Solving Differential Equations

The Laplace transform converts ODEs (in t) into algebraic equations, and initial value problems are its natural habitat: the data $y(0), y'(0)$ are built into the derivative rule. The workflow is: transform, solve for $\hat{y}$, invert by residues (or partial fractions and the table).

Example 11.8.9 (A Damped Oscillator)

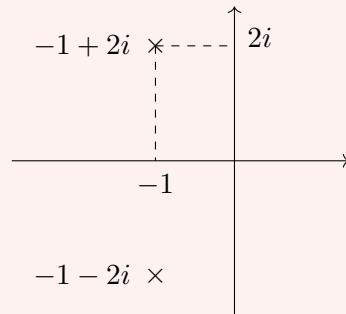
Let's solve

$$\ddot{y} + 2\dot{y} + 5y = 0, \quad y(0) = 1, \quad \dot{y}(0) = 1.$$

Transforming, using $\mathcal{L}(\ddot{y}) = p^2\hat{y} - py(0) - \dot{y}(0)$ and $\mathcal{L}(\dot{y}) = p\hat{y} - y(0)$:

$$(p^2 + 2p + 5)\hat{y} - p - 1 - 2 = 0 \implies \hat{y} = \frac{p+3}{p^2 + 2p + 5}.$$

The poles sit at the roots of $p^2 + 2p + 5$, namely $p_{\pm} = -1 \pm 2i$, and it is worth drawing them, because the pole diagram of $\hat{y}$ is a portrait of the solution: the real part of the poles (-1) is the decay rate, the imaginary part (± 2) the oscillation frequency.



Inverting by residues, for $t > 0$:

$$y(t) = \sum_{p_{\pm}} \text{Res} \frac{(p+3)e^{pt}}{(p-p_+)(p-p_-)} = \frac{(2+2i)e^{(-1+2i)t}}{4i} + \frac{(2-2i)e^{(-1-2i)t}}{-4i},$$

and combining the conjugate terms,

$$y(t) = e^{-t} (\cos 2t + \sin 2t).$$

A quick check: $y(0) = 1$, and $\dot{y}(0) = -1 + 2 = 1$. The pole diagram told us the form $e^{-t}(\dots \cos 2t + \dots \sin 2t)$ before we computed anything.

Example 11.8.10 (Switched-On Forcing)

The translation property makes forcing that switches on at a later time completely painless – a situation where classical ‘particular integral’ methods get awkward. Let's solve

$$\ddot{y} + 3\dot{y} + 2y = H(t-1), \quad y(0) = \dot{y}(0) = 0:$$

a system at rest, kicked by a constant force starting at $t = 1$. Transforming (the zero initial conditions keep things clean, and $\mathcal{L}(H(t-1)) = \frac{e^{-p}}{p}$ by translation):

$$(p^2 + 3p + 2)\hat{y} = \frac{e^{-p}}{p} \implies \hat{y} = \frac{e^{-p}}{p(p+1)(p+2)}.$$

First invert the rational part by partial fractions (or equally by residues at $0, -1, -2$):

$$\frac{1}{p(p+1)(p+2)} = \frac{1/2}{p} - \frac{1}{p+1} + \frac{1/2}{p+2} \implies g(t) = \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}.$$

Then the factor e^{-p} translates in time:

$$y(t) = g(t-1)H(t-1) = \left(\frac{1}{2} - e^{-(t-1)} + \frac{1}{2}e^{-2(t-1)}\right)H(t-1).$$

The solution is identically zero until the forcing arrives (causality again, automatic), rises smoothly, and settles to the static value $\frac{1}{2}$ forced by the equation as $t \rightarrow \infty$ – which we could have predicted from the transform alone via the asymptotic property: $p\hat{y}(p) \rightarrow \frac{1}{1.2} = \frac{1}{2}$ as $p \rightarrow 0$.

Example 11.8.11 (A Variable-Coefficient Equation)

The transform can even help when coefficients are polynomial in t , via the derivative-of-transform rule – the price is that multiplication by t becomes $-\frac{d}{dp}$, so we trade our ODE in t for an ODE in p ; the trade is worthwhile when the new equation is first order. Consider Laguerre's equation (with parameter 1),

$$t\ddot{y} + (1-t)\dot{y} + y = 0, \quad y(0) = 1.$$

Using $\mathcal{L}(tg(t)) = -\frac{d}{dp}\hat{g}(p)$ with $g = \ddot{y}$ and $g = \dot{y}$:

$$\begin{aligned}\mathcal{L}(t\dot{y}) &= -\frac{d}{dp}(p\hat{y} - y(0)) = -p\hat{y}' - \hat{y}, \\ \mathcal{L}(t\ddot{y}) &= -\frac{d}{dp}(p^2\hat{y} - py(0) - \dot{y}(0)) = -p^2\hat{y}' - 2p\hat{y} + y(0).\end{aligned}$$

Substituting into the equation, the initial-condition terms cancel and everything tidies down to

$$p(p-1)\hat{y}' = (2-p)\hat{y} \quad \implies \quad \frac{\hat{y}'}{\hat{y}} = \frac{2-p}{p(p-1)} = -\frac{2}{p} + \frac{1}{p-1},$$

by partial fractions. Integrating, $\log \hat{y} = -2 \log p + \log(p-1) + \text{const}$, i.e.

$$\hat{y} = C \frac{p-1}{p^2} = C \left(\frac{1}{p} - \frac{1}{p^2} \right)$$

for a constant of integration C . We can fix C from the asymptotic property (vii): $p\hat{y} = C(1 - \frac{1}{p}) \rightarrow C$ as $p \rightarrow \infty$, and this must equal $y(0) = 1$, so $C = 1$. Inverting term by term,

$$y(t) = 1 - t,$$

which indeed satisfies the equation: $t \cdot 0 + (1-t)(-1) + (1-t) = 0$. (This is the Laguerre polynomial L_1 ; the second, non-polynomial solution of the ODE grows too fast in a non-smooth way at the origin to be captured with these initial data.)

Example 11.8.12 (Systems of ODEs)

A linear system $\dot{\mathbf{x}} = M\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$ transforms to $p\hat{\mathbf{x}} - \mathbf{x}_0 = M\hat{\mathbf{x}}$, so

$$\hat{\mathbf{x}}(p) = (pI - M)^{-1}\mathbf{x}_0.$$

The entries of $(pI - M)^{-1}$ are rational with denominator $\det(pI - M)$: the poles of the transform are the *eigenvalues* of M , and inverting by residues recovers the usual expansion in modes $e^{\lambda_i t}$. Once again the pole locations are the physics.

§11.8.5 The Convolution Theorem

Definition 11.8.13 (Convolution)

The **convolution** of f and g is

$$(f * g)(t) = \int_{-\infty}^{\infty} f(t - t')g(t') \, dt',$$

which for functions vanishing on $t < 0$ reduces to $(f * g)(t) = \int_0^t f(t - t')g(t') \, dt'$.

Theorem 11.8.14 (Convolution Theorem)

$$\mathcal{L}(f * g) = \hat{f}(p) \hat{g}(p).$$

Proof. Write out the double integral and swap the order (both orders describe the wedge $0 \leq t' \leq t < \infty$ of the (t, t') plane):

$$\mathcal{L}(f * g)(p) = \int_0^\infty \int_0^t f(t - t')g(t')e^{-pt} \, dt' \, dt = \int_0^\infty g(t') \left(\int_{t'}^\infty f(t - t')e^{-pt} \, dt \right) dt'.$$

Substituting $u = t - t'$ in the inner integral factors it as $e^{-pt'} \hat{f}(p)$, and the remaining integral over t' gives $\hat{g}(p)$. $\square$

This is how forced systems are solved once and for all. If a system has *transfer function* $\frac{1}{Q(p)}$ – meaning the transform of the equation reads $Q(p)\hat{y} = \hat{f}$ for forcing f – then

$$y = (\mathcal{L}^{-1}(1/Q)) * f,$$

a single convolution against a kernel computed once by residues.

Remark (Impulses and Transfer Functions). The kernel $\mathcal{L}^{-1}(1/Q)$ has a physical name: the **impulse response**. Allowing the (IB Methods) delta function into the formalism, $\mathcal{L}(\delta(t - t_0)) = e^{-pt_0}$ for $t_0 \geq 0$, so in particular $\mathcal{L}(\delta) = 1$: an impulsive kick at $t = 0$ has transform 1, and the response it provokes – the Green's function $g = \mathcal{L}^{-1}(1/Q)$ – is what the convolution then smears over the actual forcing history. The pole diagram of the transfer function $\frac{1}{Q(p)}$ is precisely the engineer's stability picture: all poles in the left half-plane means every impulse response decays.

Example 11.8.15 (Forced Oscillator via Convolution)

Solve $\ddot{y} + y = f(t)$ with $y(0) = \dot{y}(0) = 0$, for arbitrary forcing f . Transforming, $(p^2 + 1)\hat{y} = \hat{f}$, so $\hat{y} = \frac{\hat{f}}{p^2 + 1}$. Since $\mathcal{L}^{-1}\left(\frac{1}{p^2 + 1}\right) = \sin t$,

$$y(t) = \int_0^t \sin(t - t')f(t') \, dt'.$$

This is exactly the Green's function solution of IB Methods, derived in three lines.

The convolution theorem also works as an *inversion* device, splitting a complicated transform into a convolution of simple ones.

Example 11.8.16

Let's find $\mathcal{L}^{-1}\left(\frac{1}{p^2(p^2+1)}\right)$. Rather than partial fractions or a fourth-order residue computation, write it as $\frac{1}{p^2} \cdot \frac{1}{p^2+1}$, the product of the transforms of t and $\sin t$. So the inverse is their convolution:

$$\int_0^t (t-t') \sin t' dt' = \left[-(t-t') \cos t' \right]_0^t - \int_0^t \cos t' dt' = t - \sin t.$$

(Partial fractions, $\frac{1}{p^2(p^2+1)} = \frac{1}{p^2} - \frac{1}{p^2+1}$, confirms this instantly – as usual there are several routes, and it pays to pick the shortest.)

§11.8.6 Solving PDEs with Transforms

Finally, let's see a transform dispose of a partial differential equation. The strategy: transform in *time*, reducing the PDE to an ODE in the spatial variable with p as a parameter; solve it; invert.

Example 11.8.17 (Waves on a Half-Line)

A semi-infinite string lies along $x \geq 0$, initially at rest ($u = u_t = 0$ at $t = 0$), and from time zero its end is waggled: $u(0, t) = f(t)$. The displacement obeys the wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

Transform in t , writing $\hat{u}(x, p) = \mathcal{L}(u(x, \cdot))(p)$. The time derivatives become multiplications, and the zero initial conditions kill the boundary terms:

$$p^2 \hat{u} = c^2 \frac{\partial^2 \hat{u}}{\partial x^2}.$$

For each fixed p this is a constant-coefficient ODE in x , with general solution

$$\hat{u}(x, p) = A(p)e^{-px/c} + B(p)e^{px/c}.$$

For $\text{Re } p > 0$ the second solution grows as $x \rightarrow \infty$, so boundedness forces $B = 0$; the boundary condition gives $A(p) = \hat{f}(p)$. Hence

$$\hat{u}(x, p) = \hat{f}(p) e^{-px/c},$$

and we recognise the translation property from our table – no contour needed:

$$u(x, t) = f\left(t - \frac{x}{c}\right) H\left(t - \frac{x}{c}\right).$$

The physics falls out of the formula: the end's motion propagates rightward at speed c , and a point at distance x sits perfectly still until the signal arrives at time x/c .

Causality, which we built into the Laplace transform by fiat ($f = 0$ for $t < 0$), emerges in the solution. Had we transformed the heat equation instead, the same procedure would produce a $\sqrt{p}$ – and its branch point would send us back to the Bromwich keyhole. Since this is the last computation of the course, let's see it through.

Example 11.8.18 (Heat Conduction into a Half-Line)

A semi-infinite rod along $x \geq 0$ starts at temperature 0, and from $t = 0$ its end is held at constant temperature u_0 . The temperature obeys the heat equation $u_t = \kappa u_{xx}$. Transforming in t (zero initial data, so $\mathcal{L}(u_t) = p\hat{u}$):

$$p\hat{u} = \kappa \frac{\partial^2 \hat{u}}{\partial x^2} \implies \hat{u}(x, p) = \frac{u_0}{p} e^{-x\sqrt{p/\kappa}},$$

taking the branch of $\sqrt{p}$ positive on the positive real axis (cut along the negative real axis), discarding the growing solution $e^{+x\sqrt{p/\kappa}}$, and fitting the boundary condition $\hat{u}(0, p) = \frac{u_0}{p}$.

Now invert with the Bromwich keyhole. The pole at $p = 0$ sits at the branch point's doorstep; shrinking the little circle around it contributes its residue-like term $u_0 \cdot e^0 = u_0$ (on the small circle $e^{-x\sqrt{p/\kappa}} \rightarrow 1$, and $\frac{1}{2\pi i} \oint \frac{dp}{p} = 1$ counting the full loop). The two legs of the cut give, with $p = se^{\pm i\pi}$ so that $\sqrt{p} = \pm i\sqrt{s}$,

$$-\frac{u_0}{2\pi i} \int_0^\infty \frac{e^{-st}}{s} \left(e^{-ix\sqrt{s/\kappa}} - e^{ix\sqrt{s/\kappa}} \right) ds = -\frac{u_0}{\pi} \int_0^\infty \frac{e^{-st}}{s} \sin(x\sqrt{s/\kappa}) ds.$$

Substituting $s = \kappa v^2$ (so $\frac{ds}{s} = \frac{2dv}{v}$), the solution assembles as

$$u(x, t) = u_0 \left(1 - \frac{2}{\pi} \int_0^\infty e^{-\kappa v^2 t} \frac{\sin xv}{v} dv \right).$$

The remaining integral is a Gaussian classic. Writing $F(x) = \int_0^\infty e^{-\kappa v^2 t} \frac{\sin xv}{v} dv$ and differentiating under the integral sign,

$$F'(x) = \int_0^\infty e^{-\kappa v^2 t} \cos(xv) dv = \frac{1}{2} \sqrt{\frac{\pi}{\kappa t}} e^{-x^2/4\kappa t};$$

integrating back up from $F(0) = 0$ then yields $F(x) = \frac{\pi}{2} \operatorname{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right)$, where $\operatorname{erf}(y) = \frac{2}{\sqrt{\pi}} \int_0^y e^{-w^2} dw$ is the error function. Hence

$$u(x, t) = u_0 \left(1 - \operatorname{erf}\left(\frac{x}{2\sqrt{\kappa t}}\right) \right) = u_0 \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right).$$

Contrast this with the string: there is no wavefront – for any $t > 0$ the temperature is (slightly) positive at *every* x , and the solution depends on x and t only through the diffusive similarity variable $x/\sqrt{\kappa t}$. All of that physics was encoded in the branch cut: a continuum of decaying modes rather than a clean travelling pole.

And that is the course. Looking back, the through-line is a single idea: analyticity is rigidity, and rigidity is exploitable. The Cauchy–Riemann equations turned complex

functions into conformal maps and harmonic solvers; Cauchy's theorem turned integration into a game of counting singularities; and residues turned that game into a machine for real integrals, transforms, and differential equations. The Complex Analysis course makes all of this bulletproof; here, we have made it *work*.

12 Electromagnetism

Electromagnetism is the theory of two vector fields – the electric field and the magnetic field – and of the charges which produce them and are pushed around by them. Between them, these fields are responsible for very nearly every phenomenon we encounter that is not gravity: the binding of electrons to nuclei and of atoms into molecules, the rigidity of solid objects, chemistry, electricity, magnets, and light itself.

What makes the subject beautiful, rather than merely useful, is that all of this is encoded in *four* equations. Maxwell’s equations are a system of linear partial differential equations relating the fields to their sources, and essentially the whole of this course consists of finding and interpreting their solutions. We will start with the static solutions – stationary charges (electrostatics) and steady currents (magnetostatics) – where the equations decouple and a great deal can be computed by hand. We will then let things depend on time, at which point the electric and magnetic fields become inextricably linked: changing magnetic fields drive currents, and the equations acquire wave solutions which travel at precisely the speed of light. Finally we will discover that the whole structure is really a relativistic one. Written in the language of special relativity, Maxwell’s four equations collapse into two, and the magnetic field turns out to be nothing but the electric field seen by a moving observer.

This article constitutes my notes for the ‘Electromagnetism’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. The reader is assumed to be comfortable with the vector calculus of the IA course – gradients, divergences, curls, and the integral theorems of Green, Stokes and Gauss¹. If you spot any errors or have any feedback, I can be contacted at ak2316@cam.ac.uk.

§12.1 Charges, Fields and Maxwell’s Equations

We begin by setting up the objects the theory is about. There are two kinds: the *sources*, which are electric charges and their motion, and the *fields*, which the sources produce and which in turn exert forces on the sources. The whole theory then amounts to two statements – a force law telling us how fields push charges, and Maxwell’s equations telling us how charges make fields.

§12.1.1 Charge and Charge Density

Every particle carries a number, its **electric charge**, which measures how strongly it couples to the electromagnetic force. Charge is measured in Coulombs, and in this course we will happily allow it to be any real number. This is a lie, but a harmless one: at the fundamental level charge is quantised, and every particle carries $q = ne$ for some $n \in \mathbb{Z}$, where

$$e \approx 1.6 \times 10^{-19} \text{ C}$$

¹The final section also uses special relativity at the level of the IA ‘Dynamics and Relativity’ course, though we will review everything we need.

is the charge on a proton. The electron has $n = -1$, the proton $n = +1$, the neutron $n = 0$. Since e is so tiny compared to the charges we deal with in practice, treating charge as a continuum is an excellent approximation, in exactly the same way that treating water as a continuous fluid is.

Rather than track individual particles, we therefore describe charge by a density.

Definition 12.1.1 (Charge density)

The **charge density** $\rho(\mathbf{x}, t)$ is the charge per unit volume, so that the total charge contained in a region $V \subseteq \mathbb{R}^3$ is

$$Q = \int_V \rho(\mathbf{x}, t) \, dV.$$

We will also want to consider idealised charge distributions which are concentrated on a surface, on a curve, or at a single point. In those cases the appropriate object is a charge per unit area σ , a charge per unit length η , or simply a charge Q together with a delta function. Which one is meant will always be clear from context.

§12.1.2 Current and Current Density

Charge that moves is a *current*. Again the natural object is a density.

Definition 12.1.2 (Current density)

The **current density** $\mathbf{J}(\mathbf{x}, t)$ is a vector field with the property that, for any surface S ,

$$I = \int_S \mathbf{J} \cdot d\mathbf{S}$$

is the charge crossing S per unit time. The number I is the **current** through S .

So $\mathbf{J}$ points in the direction that charge is flowing, and its magnitude is the current per unit area. If the charge density ρ is moving with velocity $\mathbf{v}(\mathbf{x}, t)$, then in time δt the charge crossing a small patch of area δA with unit normal $\hat{\mathbf{n}}$ is the charge in a slanted cylinder of volume $(\mathbf{v} \cdot \hat{\mathbf{n}})\delta A \delta t$, and so

$$\mathbf{J} = \rho \mathbf{v}.$$

This is worth remembering; it is the bridge between the field description and the particle description.

Example 12.1.3 (Current in a wire)

Consider a wire: a cylinder of cross-sectional area A containing n mobile electrons per unit volume, each of charge $q = -e$, all drifting along the wire with speed v . Then

$$\rho = nq = -ne, \quad \mathbf{J} = nq\mathbf{v}, \quad I = |nqvA| = nevA.$$

It is worth noting how slow this drift is in practice. For a copper wire of a square millimetre carrying an amp, v works out at less than a tenth of a millimetre per sec-

ond. The *signal* travels essentially at the speed of light, but the electrons themselves shuffle along at a snail's pace.

§12.1.3 Conservation of Charge

It is a fact of nature that charge cannot be created or destroyed. But this statement, as usually phrased, is much weaker than the truth. ‘The total charge in the universe is constant’ would permit a charge to vanish from Cambridge and simultaneously appear on the Moon. What actually happens is that charge is conserved *locally*: if the charge in some region decreases, it is because charge flowed out through the boundary.

The mathematical expression of this is a continuity equation, exactly as in the IA Vector Calculus course.

Law 12.1.4 (Continuity equation)

The charge density and current density satisfy

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0.$$

To see that this says what we claimed, fix a region V with boundary $S = \partial V$ and let $Q(t) = \int_V \rho \, dV$ be the charge inside. Then differentiating under the integral sign and applying the divergence theorem,

$$\frac{dQ}{dt} = \int_V \frac{\partial \rho}{\partial t} \, dV = - \int_V \nabla \cdot \mathbf{J} \, dV = - \int_S \mathbf{J} \cdot d\mathbf{S} = -I,$$

where I is the total current flowing out through S . In other words – the charge in V decreases at exactly the rate at which charge leaves through the boundary. Nothing teleports.

Taking $V = \mathbb{R}^3$ and assuming, as we always shall, that $\mathbf{J}$ decays at infinity, the surface term vanishes and we recover the global statement $dQ/dt = 0$. So local conservation implies global conservation, but not conversely.

§12.1.4 Fields and the Lorentz Force Law

In modern physics we do not believe in action at a distance. Instead, forces are mediated by *fields*: dynamical objects which assign a value to every point of space and time, and which carry energy and momentum in their own right. Electromagnetism has two of them,

$\mathbf{E}(\mathbf{x}, t)$ the **electric field**, $\mathbf{B}(\mathbf{x}, t)$ the **magnetic field**,

both vector fields on $\mathbb{R}^3$.

The fields interact with charges in two directions, and it is worth keeping the two firmly separate in one's head.

- Fields push charges. This is governed by the Lorentz force law.
- Charges make fields. This is governed by Maxwell's equations.

The first is quickly stated.

Law 12.1.5 (Lorentz force law)

A particle of charge q moving with velocity $\mathbf{v}$ in fields $\mathbf{E}$ and $\mathbf{B}$ experiences a force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

Two features deserve immediate comment. The electric part $q\mathbf{E}$ acts on a charge whether or not it is moving, and points along $\mathbf{E}$. The magnetic part $q\mathbf{v} \times \mathbf{B}$ acts only on moving charges, and is perpendicular to the velocity. In particular

$$\mathbf{F} \cdot \mathbf{v} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} = q\mathbf{E} \cdot \mathbf{v},$$

so *magnetic fields do no work*. They can bend a particle's trajectory but never change its speed. Every bit of energy a charged particle gains comes from the electric field.

For a continuous distribution of charge rather than a single particle, we should integrate. A volume element dV carries charge ρdV moving with velocity $\mathbf{v}$, so it feels a force $(\rho\mathbf{E} + \rho\mathbf{v} \times \mathbf{B}) dV$, giving

$$\mathbf{F} = \int_V (\rho\mathbf{E} + \mathbf{J} \times \mathbf{B}) dV$$

using $\mathbf{J} = \rho\mathbf{v}$. We will use the magnetic half of this repeatedly when we compute forces between wires.

§12.1.5 Maxwell's Equations

Now for the other half of the story: how charges produce fields. Remarkably, the answer fits on one line each.

Law 12.1.6 (Maxwell's equations)

The electromagnetic fields produced by a charge density ρ and current density $\mathbf{J}$ obey

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad (\text{M1})$$

$$\nabla \cdot \mathbf{B} = 0, \quad (\text{M2})$$

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}, \quad (\text{M3})$$

$$\nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J}. \quad (\text{M4})$$

Here ϵ_0 and μ_0 are constants of nature, the **electric constant** and the **magnetic constant**^a, with numerical values

$$\epsilon_0 \approx 8.85 \times 10^{-12} \text{ m}^{-3} \text{ kg}^{-1} \text{ s}^2 \text{ C}^2, \quad \mu_0 = 4\pi \times 10^{-7} \text{ m kg C}^{-2}.$$

^aThese are also called the permittivity and permeability of free space, which sound more impressive but mean the same thing.

That is the entire content of classical electromagnetism. Everything that follows in this course – Coulomb's law, capacitors, solenoids, induction, light – is a consequence of these four equations together with the Lorentz force law. It is genuinely astonishing that so much should follow from so little, and the reader is encouraged to keep the equations somewhere visible while reading the rest of these notes.

A few structural remarks before we start solving them.

They are linear. If $(\mathbf{E}_1, \mathbf{B}_1)$ solves the equations with sources $(\rho_1, \mathbf{J}_1)$ and $(\mathbf{E}_2, \mathbf{B}_2)$ solves them with sources $(\rho_2, \mathbf{J}_2)$, then the sum solves them with the summed sources. This is the **principle of superposition**, and it is why electromagnetism is tractable at all. Of the four fundamental forces, it is the only one with this property.

They are inconsistent without the last term. Take the divergence of (M4). Since $\nabla \cdot (\nabla \times \mathbf{B}) = 0$ always,

$$0 = \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) + \mu_0 \nabla \cdot \mathbf{J} = \mu_0 \left(\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} \right),$$

where we used (M1) in the last step. So Maxwell's equations do not merely permit charge conservation – they *imply* it. We will see in Section 12.4.4 that had we written (M4) without the $\partial \mathbf{E} / \partial t$ term (which is what Ampère originally wrote), the same computation would have forced $\partial \rho / \partial t = 0$, which is plainly false. The extra term is Maxwell's, and it is also, as we shall see, the term responsible for light.

They come in pairs. Equations (M2) and (M3) contain no sources at all: they are constraints on the fields themselves, and they are exactly the conditions which allow us to introduce potentials. Equations (M1) and (M4) are the ones that say where the fields come from.

They have an integral form. Applying the integral theorems to each gives a statement with a direct physical reading; we will derive each in turn as we meet it, but for reference the four integral forms are

$$\begin{aligned} \int_{\partial V} \mathbf{E} \cdot d\mathbf{S} &= \frac{Q}{\epsilon_0}, & (\text{Gauss' law}) \\ \int_{\partial V} \mathbf{B} \cdot d\mathbf{S} &= 0, & (\text{no magnetic monopoles}) \\ \oint_{\partial S} \mathbf{E} \cdot d\mathbf{x} &= -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S}, & (\text{Faraday's law}) \\ \oint_{\partial S} \mathbf{B} \cdot d\mathbf{x} &= \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{S}. & (\text{Ampère's law}) \end{aligned}$$

The reader who has done the IA Vector Calculus course will have seen a version of this already; we now do the whole thing properly.

§12.1.6 The Static Limit

The most important simplification is to look for solutions which do not depend on time. Setting all time derivatives to zero, Maxwell's equations fall apart into two entirely independent systems:

$$\begin{cases} \nabla \cdot \mathbf{E} = \rho / \epsilon_0, \\ \nabla \times \mathbf{E} = \mathbf{0}, \end{cases} \quad \begin{cases} \nabla \cdot \mathbf{B} = 0, \\ \nabla \times \mathbf{B} = \mu_0 \mathbf{J}. \end{cases}$$

The first pair involves only $\mathbf{E}$ and ρ ; the second only $\mathbf{B}$ and $\mathbf{J}$. This is the sense in which electricity and magnetism are separate phenomena in the static world – they only become entangled once things start changing in time. Accordingly we spend the next two sections on **electrostatics** (stationary charges, $\rho = \rho(\mathbf{x})$, $\mathbf{J} = \mathbf{0}$) and **magnetostatics** (steady currents, $\rho = 0$, $\mathbf{J} = \mathbf{J}(\mathbf{x})$) respectively, and only afterwards let time back in.

§12.2 Electrostatics

In electrostatics nothing moves. We take $\mathbf{J} = \mathbf{0}$, we take $\rho = \rho(\mathbf{x})$ independent of time, and we look for a time-independent electric field with $\mathbf{B} = \mathbf{0}$. As we saw, the equations we have to solve are

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{E} = \mathbf{0},$$

and (M2) and (M4) are satisfied trivially. The goal of this section is to determine $\mathbf{E}$ from ρ : first in highly symmetric situations, where a single application of the divergence theorem does everything, and then in general.

§12.2.1 Gauss' Law

The integral form of (M1) is so useful that it gets its own name. Let V be a region with boundary $S = \partial V$. Integrating (M1) over V ,

$$\int_V \nabla \cdot \mathbf{E} \, dV = \frac{1}{\epsilon_0} \int_V \rho \, dV,$$

and applying the divergence theorem on the left gives the following.

Law 12.2.1 (Gauss' law)

For any closed surface S bounding a region V ,

$$\int_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q}{\epsilon_0},$$

where $Q = \int_V \rho \, dV$ is the total charge enclosed by S .

Definition 12.2.2 (Flux)

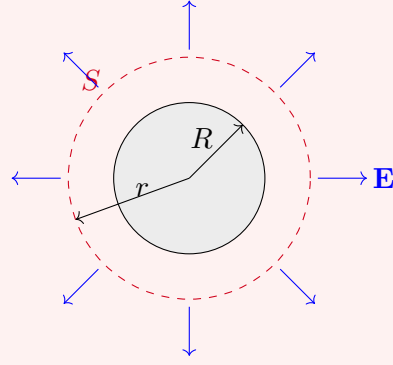
The **flux** of $\mathbf{E}$ through a surface S is $\int_S \mathbf{E} \cdot d\mathbf{S}$.

Gauss' law says something slightly surprising. The flux out of S depends only on the charge *inside* S , and not at all on the charge outside. This is not to say that external charges make no field on S – of course they do – but their field lines enter through one part of the surface and leave through another, and the two contributions cancel exactly.

The reason Gauss' law is so powerful is that, when the charge distribution has enough symmetry, symmetry alone determines the direction of $\mathbf{E}$ and the surfaces on which $|\mathbf{E}|$ is constant. Gauss' law then determines the one remaining unknown function. Let's see this in action.

Example 12.2.3 (Coulomb's law)

Take a spherically symmetric charge distribution $\rho = \rho(r)$ with $\rho(r) = 0$ for $r > R$, so all the charge lies inside a ball of radius R , and let Q be the total charge. We take for S a sphere of radius $r > R$ centred at the origin.



By symmetry the field can only point radially and can only depend on r , so

$$\mathbf{E} = E(r)\hat{\mathbf{r}}.$$

(Note this automatically has $\nabla \times \mathbf{E} = \mathbf{0}$, so we need only worry about (M1).) The flux is then

$$\int_S \mathbf{E} \cdot d\mathbf{S} = \int_S E(r) \hat{\mathbf{r}} \cdot d\mathbf{S} = E(r) \cdot 4\pi r^2,$$

since $\hat{\mathbf{r}}$ is precisely the outward normal to S and $E(r)$ is constant on S . Gauss' law now gives $4\pi r^2 E(r) = Q/\epsilon_0$, so

$$\mathbf{E}(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, \quad r > R.$$

Combining with the Lorentz force law, a second charge q placed at $\mathbf{r}$ feels a force

$$\mathbf{F} = \frac{Qq}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}},$$

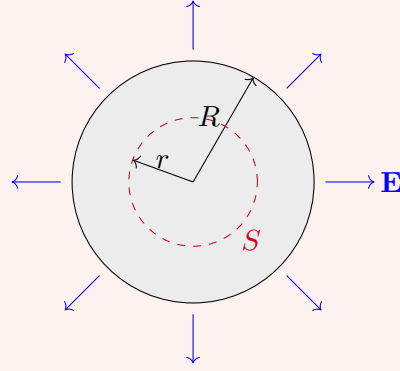
which is **Coulomb's law**: an inverse square law, repulsive for like charges and attractive for unlike ones. It is worth pausing over the fact that we never needed to know $\rho(r)$, only the total charge Q . From outside, any spherically symmetric lump of charge is indistinguishable from a point charge at its centre.

Example 12.2.4 (Inside a uniformly charged ball)

Now take the charge to be uniformly spread through the ball,

$$\rho(r) = \begin{cases} \rho_0 & r < R, \\ 0 & r > R, \end{cases}$$

with total charge $Q = \frac{4}{3}\pi R^3 \rho_0$, and ask what happens *inside*. Take S to be a sphere of radius $r < R$.



The charge enclosed is now only Qr^3/R^3 , so Gauss' law reads

$$4\pi r^2 E(r) = \frac{Q}{\epsilon_0} \frac{r^3}{R^3},$$

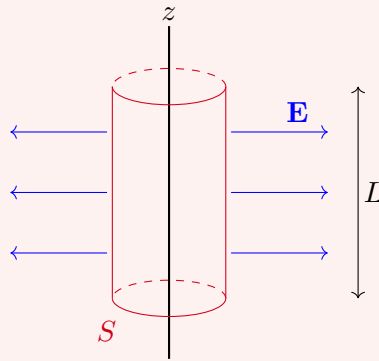
giving

$$\mathbf{E}(\mathbf{r}) = \frac{Qr}{4\pi\epsilon_0 R^3} \hat{\mathbf{r}}, \quad r < R.$$

So inside the ball the field grows *linearly* with r , reaching its maximum on the surface $r = R$, where it matches continuously onto the exterior solution $Q/4\pi\epsilon_0 r^2$ found above. It vanishes at the centre, as it must by symmetry.

Example 12.2.5 (Line charge)

Consider an infinite straight line of charge along the z -axis, with uniform charge per unit length η . We use cylindrical polars (r, φ, z) .



By symmetry (translations along z , rotations about z , and reflection in any plane containing the axis) the field must be radial and depend only on r :

$$\mathbf{E} = E(r) \hat{\mathbf{r}}.$$

Take S to be a cylinder of radius r and length L about the axis. The end caps contribute nothing, since $\mathbf{E}$ is tangent to them, and the curved part has area $2\pi rL$ with outward normal $\hat{\mathbf{r}}$. So

$$\int_S \mathbf{E} \cdot d\mathbf{S} = 2\pi rL E(r) = \frac{\eta L}{\epsilon_0},$$

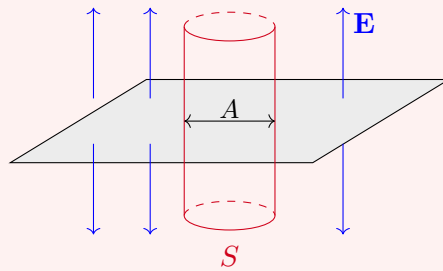
giving

$$\mathbf{E} = \frac{\eta}{2\pi\epsilon_0 r} \hat{\mathbf{r}}.$$

Note that the field falls off like $1/r$ rather than $1/r^2$. This is exactly what one should expect: we have one extra dimension's worth of charge contributing, so the field dies away more slowly.

Example 12.2.6 (Plane of charge)

Finally, let all the charge sit on the plane $z = 0$ with uniform charge per unit area σ .



By symmetry the field is vertical and depends only on z , and reflection in the plane forces it to be odd:

$$\mathbf{E} = E(z)\hat{\mathbf{z}}, \quad E(-z) = -E(z).$$

Take for S a cylinder (a ‘Gaussian pillbox’) of cross-sectional area A with its flat faces at heights $\pm z$. Now it is the curved side that contributes nothing, and the two caps that do:

$$\int_S \mathbf{E} \cdot d\mathbf{S} = E(z)A - E(-z)A = 2E(z)A = \frac{\sigma A}{\epsilon_0}.$$

Hence

$$E(z) = \frac{\sigma}{2\epsilon_0} \quad \text{for } z > 0,$$

which is *constant* – the field of an infinite charged plane does not decay at all. (Each successive dimension of charge costs us one power of r , and we have run out of powers.)

The last example exhibits an important general phenomenon: $\mathbf{E}$ jumps across a layer of surface charge. Here

$$E(0^+) - E(0^-) = \frac{\sigma}{\epsilon_0},$$

and this is not special to the plane.

Proposition 12.2.7 (Discontinuity across a surface charge)

Let S be a surface carrying surface charge density σ , with unit normal $\hat{\mathbf{n}}$ pointing

from the $-$ side to the $+$ side. Then

$$\hat{\mathbf{n}} \cdot \mathbf{E}_+ - \hat{\mathbf{n}} \cdot \mathbf{E}_- = \frac{\sigma}{\epsilon_0},$$

while the components of $\mathbf{E}$ tangential to S are continuous.

Proof. For the normal component, apply Gauss' law to a small pillbox straddling S , with flat faces of area δA parallel to S on either side, and then shrink the height of the pillbox to zero. The contribution of the curved side vanishes in the limit, the enclosed charge tends to $\sigma \delta A$, and the two faces contribute $(\hat{\mathbf{n}} \cdot \mathbf{E}_+ - \hat{\mathbf{n}} \cdot \mathbf{E}_-) \delta A$. Dividing by δA gives the result.

For the tangential component, use $\nabla \times \mathbf{E} = \mathbf{0}$ instead. Take a small rectangular loop C straddling S , with two sides of length $\delta \ell$ parallel to S (one either side) and two short sides crossing it, and let $\hat{\mathbf{t}}$ be the unit tangent along the long sides. By Stokes' theorem $\oint_C \mathbf{E} \cdot d\mathbf{x} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = 0$. Shrinking the short sides to zero length leaves $(\hat{\mathbf{t}} \cdot \mathbf{E}_+ - \hat{\mathbf{t}} \cdot \mathbf{E}_-) \delta \ell = 0$, and since $\hat{\mathbf{t}}$ was an arbitrary tangent direction, the tangential components agree. $\square$

§12.2.2 The Electrostatic Potential

Symmetry will only take us so far. To handle a general ρ we must actually solve

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}, \quad \nabla \times \mathbf{E} = \mathbf{0}.$$

The trick, familiar from vector calculus, is that the second equation solves itself. A curl-free field on $\mathbb{R}^3$ (which is simply connected) is a gradient.

Definition 12.2.8 (Electrostatic potential)

If $\mathbf{E} = -\nabla\phi$, we call the scalar field ϕ the **electrostatic potential**.

The minus sign is a convention, and a sensible one: it makes ϕ agree with potential energy per unit charge, so that positive charges roll downhill. Substituting into (M1) turns the whole of electrostatics into a single scalar equation.

Proposition 12.2.9 (Poisson's equation)

The electrostatic potential satisfies

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.$$

In a region free of charge this reduces to **Laplace's equation** $\nabla^2 \phi = 0$.

Two remarks. First, ϕ is only determined up to an additive constant, since ∇ kills constants. We fix this in the usual way by demanding $\phi(\mathbf{r}) \rightarrow 0$ as $r \rightarrow \infty$, which is possible whenever the charge is confined to a bounded region. Secondly, Poisson's equation is linear, which is superposition again: the potential of a collection of charges is the sum of the individual potentials.

Poisson's equation was studied at length in the IA Vector Calculus course, and we will

lean on it heavily. The single most useful fact is the uniqueness theorem, which is worth recalling in full since we will use it to justify some outrageous-looking guesses later on.

Theorem 12.2.10 (Uniqueness for Poisson's equation)

Let V be a bounded region with boundary ∂V , and consider the problem

$$\nabla^2 \phi = -\frac{\rho}{\epsilon_0} \quad \text{in } V,$$

subject to either ϕ specified on ∂V (Dirichlet conditions) or $\partial\phi/\partial n = \hat{\mathbf{n}} \cdot \nabla \phi$ specified on ∂V (Neumann conditions). Then the solution is unique – in the Dirichlet case outright, and in the Neumann case up to an additive constant.

Proof. Suppose ϕ_1 and ϕ_2 are both solutions, and set $\psi = \phi_1 - \phi_2$. Then $\nabla^2 \psi = 0$ in V , and on ∂V we have either $\psi = 0$ (Dirichlet) or $\partial\psi/\partial n = 0$ (Neumann). Now consider

$$\int_V \nabla \cdot (\psi \nabla \psi) \, dV = \int_V (|\nabla \psi|^2 + \psi \nabla^2 \psi) \, dV = \int_V |\nabla \psi|^2 \, dV,$$

using harmonicity of ψ . On the other hand, by the divergence theorem the left hand side is

$$\int_{\partial V} \psi \nabla \psi \cdot d\mathbf{S} = \int_{\partial V} \psi \frac{\partial \psi}{\partial n} \, dS = 0,$$

since one factor or the other vanishes on ∂V in each case. Hence $\int_V |\nabla \psi|^2 \, dV = 0$, and as the integrand is non-negative and continuous, $\nabla \psi = \mathbf{0}$ throughout V . So ψ is constant, and in the Dirichlet case that constant is zero. $\square$

The theorem also holds on unbounded regions provided we impose a decay condition at infinity, which for us will always be $\phi \rightarrow 0$ as $r \rightarrow \infty$. The moral is a practical one: if by any means whatsoever – inspired guesswork included – we produce a function satisfying the equation and the boundary conditions, we have found *the* answer, and are not obliged to explain how we found it.

Example 12.2.11 (The point charge, again)

A point charge Q at the origin has

$$\rho(\mathbf{r}) = Q \delta^3(\mathbf{r}),$$

where δ^3 is the three-dimensional delta function. Away from the origin we must solve Laplace's equation with spherical symmetry, and the general such solution is $\phi = \alpha/r + \text{const}$; the constant is zero by our boundary condition at infinity.

To fix α , integrate Poisson's equation over a ball V of radius r . The left hand side is

$$\int_V \nabla^2 \phi \, dV = \int_S \nabla \phi \cdot d\mathbf{S} = \int_S \left(-\frac{\alpha}{r^2}\right) \hat{\mathbf{r}} \cdot d\mathbf{S} = -4\pi\alpha,$$

while the right hand side is $-Q/\epsilon_0$. Hence $\alpha = Q/4\pi\epsilon_0$, and

$$\phi = \frac{Q}{4\pi\epsilon_0 r}, \quad \mathbf{E} = -\nabla \phi = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}},$$

which is Coulomb's law once more, now derived by a completely different route.

§12.2.3 General Charge Distributions

The point charge solution is the key to everything else, because it is (up to a constant) the Green's function for the Laplacian. Recall that this is the function G solving

$$\nabla^2 G(\mathbf{r}; \mathbf{r}') = \delta^3(\mathbf{r} - \mathbf{r}'),$$

and the calculation we just did says

$$G(\mathbf{r}; \mathbf{r}') = -\frac{1}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{r}'|}.$$

Proposition 12.2.12 (Potential of a general distribution)

Let ρ be a charge distribution confined to a bounded region V . Then

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV',$$

and correspondingly

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\mathbf{r}') \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} dV'.$$

Proof. Since $\nabla^2 \phi = -\rho/\epsilon_0$ and G is the Green's function,

$$\phi(\mathbf{r}) = -\frac{1}{\epsilon_0} \int_V \rho(\mathbf{r}') G(\mathbf{r}; \mathbf{r}') dV' = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV'.$$

For the field we differentiate under the integral sign, noting that ∇ acts on $\mathbf{r}$ only, and

$$\nabla \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) = -\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}.$$

Then $\mathbf{E} = -\nabla \phi$ gives the stated formula. $\square$

This is a complete answer, but for a complicated ρ it is a complete answer of the least useful kind: an integral one cannot do. What saves us is that we are usually far away from the charge, and then we can expand.

§12.2.4 The Multipole Expansion and Dipoles

Suppose all the charge lies within some bounded region near the origin, and we look at the potential at $|\mathbf{r}| \gg |\mathbf{r}'|$. Taylor expanding the Green's function in $\mathbf{r}'$,

$$\frac{1}{|\mathbf{r} - \mathbf{r}'|} = \frac{1}{r} + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^3} + O\left(\frac{r'^2}{r^3}\right),$$

and substituting into Proposition 12.2.12 gives

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r} + \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^2} + \cdots \right),$$

where

$$Q = \int_V \rho(\mathbf{r}') \, dV', \quad \mathbf{p} = \int_V \mathbf{r}' \rho(\mathbf{r}') \, dV'.$$

This is the beginning of the **multipole expansion**. The leading term says that from far enough away any lump of charge looks like a point charge with the same total charge. The correction $\mathbf{p}$ is the first thing you see when the total charge vanishes.

Definition 12.2.13 (Electric dipole moment)

The **electric dipole moment** of a charge distribution is

$$\mathbf{p} = \int \mathbf{r}' \rho(\mathbf{r}') \, dV'.$$

Note that $\mathbf{p}$ depends on the choice of origin in general – but if $Q = 0$, shifting the origin by $\mathbf{a}$ changes $\mathbf{p}$ by $-\mathbf{a}Q = \mathbf{0}$, so for a neutral distribution the dipole moment is well defined. This is the case of interest.

The cleanest example is a pair of equal and opposite point charges.

Example 12.2.14 (The dipole)

Put a charge $+Q$ at the origin and $-Q$ at $-\mathbf{d}$. By superposition,

$$\phi(\mathbf{r}) = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{1}{|\mathbf{r} + \mathbf{d}|} \right).$$

This is exact but unenlightening; we expand for $r \gg d$. Using the general Taylor expansion $f(\mathbf{r} + \mathbf{d}) = f(\mathbf{r}) + \mathbf{d} \cdot \nabla f + \frac{1}{2}(\mathbf{d} \cdot \nabla)^2 f + \dots$ with $f = 1/r$,

$$\frac{1}{|\mathbf{r} + \mathbf{d}|} = \frac{1}{r} - \frac{\mathbf{d} \cdot \mathbf{r}}{r^3} + O\left(\frac{d^2}{r^3}\right),$$

so the $1/r$ terms cancel and

$$\phi(\mathbf{r}) \approx \frac{Q}{4\pi\epsilon_0} \frac{\mathbf{d} \cdot \mathbf{r}}{r^3} = \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{4\pi\epsilon_0 r^2}, \quad \mathbf{p} = Q\mathbf{d}.$$

This agrees with our general formula, as it must. Note the convention: $\mathbf{p}$ points from the negative charge to the positive one.

Differentiating, and using $\nabla(\mathbf{p} \cdot \mathbf{r}) = \mathbf{p}$ and $\nabla(r^{-3}) = -3\hat{\mathbf{r}}/r^4$,

$$\mathbf{E} = -\nabla\phi = \frac{1}{4\pi\epsilon_0} \left(\frac{3(\mathbf{p} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}}{r^3} \right).$$

So the dipole field falls off like $1/r^3$, faster than a point charge, which is unsurprising: at large distances the two charges nearly cancel.

Dipoles are not only things that produce fields; they are also things that respond to them, and the way they do so explains a good deal of everyday physics.

Proposition 12.2.15 (Dipole in an external field)

A dipole $\mathbf{p}$ sitting in an external field $\mathbf{E}$ has potential energy

$$U = -\mathbf{p} \cdot \mathbf{E},$$

experiences a torque

$$\boldsymbol{\tau} = \mathbf{p} \times \mathbf{E}$$

about its centre, and (if the field is non-uniform) a net force

$$\mathbf{F} = \nabla(\mathbf{p} \cdot \mathbf{E}).$$

Proof. Model the dipole as charges $\pm Q$ at $\mathbf{r} \pm \frac{1}{2}\mathbf{d}$, with $\mathbf{p} = Q\mathbf{d}$. The total energy in the external potential ϕ is

$$U = Q\phi(\mathbf{r} + \frac{1}{2}\mathbf{d}) - Q\phi(\mathbf{r} - \frac{1}{2}\mathbf{d}) = Q\mathbf{d} \cdot \nabla\phi(\mathbf{r}) + O(d^2) = -\mathbf{p} \cdot \mathbf{E}(\mathbf{r}),$$

Taylor expanding and using $\mathbf{E} = -\nabla\phi$. The force is minus the gradient of the energy, $\mathbf{F} = -\nabla U = \nabla(\mathbf{p} \cdot \mathbf{E})$, which indeed vanishes for a uniform field. For the torque, the two charges feel forces $\pm Q\mathbf{E}$, so about the centre

$$\boldsymbol{\tau} = \frac{1}{2}\mathbf{d} \times (Q\mathbf{E}) + (-\frac{1}{2}\mathbf{d}) \times (-Q\mathbf{E}) = Q\mathbf{d} \times \mathbf{E} = \mathbf{p} \times \mathbf{E}. \quad \square$$

The energy $-\mathbf{p} \cdot \mathbf{E}$ is minimised when $\mathbf{p}$ is aligned with $\mathbf{E}$, and the torque is exactly what rotates it into alignment. This is the mechanism behind a great deal of chemistry – a water molecule is a permanent dipole, and it is why microwave ovens work and why water is such a good solvent. The force formula also explains why a charged comb picks up small neutral pieces of paper: the field of the comb *induces* a dipole in the paper, and being non-uniform, then pulls on it.

§12.2.5 Field Lines and Equipotentials

Drawing a vector field is awkward – we cannot draw an arrow at every point. The standard device is to draw *field lines* instead.

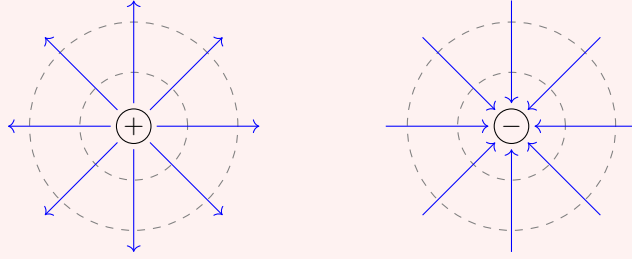
Definition 12.2.16 (Field lines and equipotentials)

A **field line** is a curve everywhere tangent to $\mathbf{E}$, drawn with the convention that the density of lines is proportional to $|\mathbf{E}|$. An **equipotential** is a surface on which ϕ is constant.

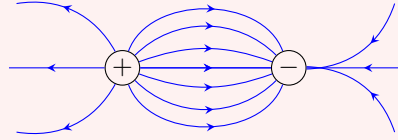
Field lines begin on positive charges and end on negative charges (or run off to infinity), and they never cross, since $\mathbf{E}$ has a well-defined direction at each point. Equipotentials always meet field lines at right angles, since $\mathbf{E} = -\nabla\phi$ is normal to the level sets of ϕ .

Example 12.2.17 (Point charges and dipoles)

For a single positive charge the lines run radially outward and the equipotentials are concentric spheres; for a negative charge everything reverses.



For a dipole the lines leave the positive charge and curl round to the negative one.



§12.2.6 Electrostatic Energy

How much energy does it take to assemble a configuration of charges? Recall from IA Dynamics and Relativity that a charge q sitting at $\mathbf{r}$ in a potential ϕ has potential energy $U = q\phi(\mathbf{r})$; this is the work done in bringing it in from infinity, since

$$W = - \int_{\infty}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{x} = -q \int_{\infty}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{x} = q \int_{\infty}^{\mathbf{r}} \nabla \phi \cdot d\mathbf{x} = q [\phi(\mathbf{r}) - \phi(\infty)] = q\phi(\mathbf{r}).$$

Now consider N point charges q_i at positions $\mathbf{r}_i$, and build the configuration up one charge at a time. Bringing in the first costs nothing. The second must be dragged in against the field of the first, costing

$$W_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\mathbf{r}_1 - \mathbf{r}_2|},$$

the third must fight both of its predecessors, and so on. Summing,

$$U = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|} = \frac{1}{8\pi\epsilon_0} \sum_{i \neq j} \frac{q_i q_j}{|\mathbf{r}_i - \mathbf{r}_j|},$$

the second form counting each pair twice and compensating with a factor of $\frac{1}{2}$. Since the potential at $\mathbf{r}_i$ due to all the *other* charges is $\phi(\mathbf{r}_i) = \frac{1}{4\pi\epsilon_0} \sum_{j \neq i} q_j / |\mathbf{r}_i - \mathbf{r}_j|$, we can write this more compactly as

$$U = \frac{1}{2} \sum_{i=1}^N q_i \phi(\mathbf{r}_i),$$

which generalises to a continuous distribution as

$$U = \frac{1}{2} \int \rho(\mathbf{r}) \phi(\mathbf{r}) dV.$$

This is already a nice formula, but we can do better and remove the charges entirely.

Proposition 12.2.18 (Energy in the electric field)

The energy of an electrostatic configuration is

$$U = \frac{\epsilon_0}{2} \int_{\mathbb{R}^3} |\mathbf{E}|^2 dV.$$

Proof. Using $\rho = \epsilon_0 \nabla \cdot \mathbf{E}$ and the product rule $\nabla \cdot (\phi \mathbf{E}) = \phi \nabla \cdot \mathbf{E} + \mathbf{E} \cdot \nabla \phi$,

$$U = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) \phi dV = \frac{\epsilon_0}{2} \int [\nabla \cdot (\phi \mathbf{E}) - \mathbf{E} \cdot \nabla \phi] dV.$$

The first term is a total divergence, so by the divergence theorem it becomes a surface integral over a sphere at infinity. For a bounded charge distribution $\phi \sim 1/r$ and $|\mathbf{E}| \sim 1/r^2$ while the area grows like r^2 , so the integrand decays like $1/r$ and the term vanishes. In the second, $\nabla \phi = -\mathbf{E}$, leaving

$$U = \frac{\epsilon_0}{2} \int \mathbf{E} \cdot \mathbf{E} dV. \quad \square$$

This is a genuinely different picture of where the energy lives. The formula $U = \frac{1}{2} \int \rho \phi$ suggests the energy is stored ‘in the charges’, while $U = \frac{\epsilon_0}{2} \int |\mathbf{E}|^2$ says it is stored in the field, spread out through all of space with energy density $\frac{\epsilon_0}{2} |\mathbf{E}|^2$. The second picture is the correct one, though our derivation of it from the first is not a proof of that.

Remark. The two formulae are not quite equivalent, and it is instructive to see how they differ. For a single point charge the discrete formula gives $U = 0$ (there are no pairs), but the field is non-zero, so $\frac{\epsilon_0}{2} \int |\mathbf{E}|^2$ certainly is not – indeed it diverges, since $|\mathbf{E}|^2 \sim r^{-4}$ near the origin. The discrepancy is the (infinite) *self-energy* of the point charge, which the discrete sum quietly discards. This is the first hint of a genuine difficulty with point particles in classical field theory, and we will say no more about it here.

§12.2.7 Conductors

Definition 12.2.19 (Conductor)

A **conductor** is a region of space containing charges which are free to move.

The single fact that governs everything about conductors in electrostatics is this: inside a conductor, $\mathbf{E} = \mathbf{0}$. The argument is a one-liner. If $\mathbf{E}$ were non-zero anywhere inside, the free charges there would feel a force and move; so we would not be in a static situation. In equilibrium the charges have rearranged themselves precisely so as to cancel the field inside. That is the whole story, and a surprising number of consequences follow from it.

Proposition 12.2.20 (Properties of conductors)

In electrostatic equilibrium, a conductor satisfies:

- (i) ϕ is constant throughout the conductor;
- (ii) $\rho = 0$ inside the conductor, so any net charge lives on the surface;
- (iii) the surface of the conductor is an equipotential, and $\mathbf{E}$ just outside it is perpendicular to the surface;

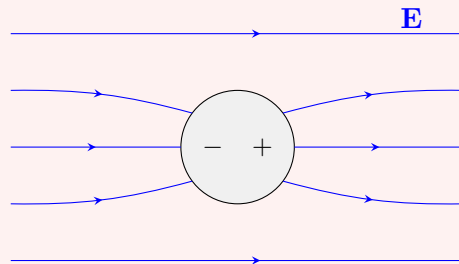
(iv) just outside the surface, $\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$, where σ is the surface charge density and $\hat{\mathbf{n}}$ the outward normal.

Proof. (i) is immediate from $\mathbf{E} = -\nabla\phi = \mathbf{0}$. For (ii), $\rho = \epsilon_0 \nabla \cdot \mathbf{E} = 0$. Statement (iii) follows from (i), since the surface is part of the conductor and hence has ϕ constant on it, and $\mathbf{E}$ is normal to equipotentials. Physically, a tangential component of $\mathbf{E}$ at the surface would push the surface charges around. Finally (iv) is Proposition 12.2.7 with $\mathbf{E}_{\text{inside}} = \mathbf{0}$. $\square$

Part (iv) is extremely convenient in practice: it lets us read off the induced surface charge from the field, or vice versa.

Example 12.2.21 (Induced surface charge)

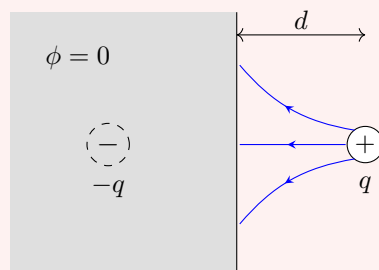
Take a neutral spherical conductor and place it in a uniform external field, say produced by a positively charged plate on the left and a negatively charged one on the right. Far away, the field lines are straight. Near the conductor they must bend so as to meet the surface perpendicularly, and since field lines terminate on charges, there must be negative charge on the left of the sphere and positive charge on the right.



The total induced charge is of course zero – the conductor was neutral and stays so – but it has been redistributed. This is why a conductor screens its interior: a hollow conductor (a ‘Faraday cage’) has $\mathbf{E} = \mathbf{0}$ inside no matter what is going on outside.

Example 12.2.22 (The method of images)

Let a conductor fill the half-space $x < 0$, grounded so that $\phi = 0$ throughout it, and place a point charge q at $\mathbf{d} = (d, 0, 0)$ with $d > 0$.



We must solve $\nabla^2\phi = -\rho/\epsilon_0$ in $x > 0$ with the boundary condition $\phi = 0$ on $x = 0$. By the uniqueness theorem, *any* solution we can find is *the* solution, so we are entitled to guess.

Here is the guess. Forget the conductor and instead put an *image charge* $-q$ at $(-d, 0, 0)$. By symmetry the potential of this pair vanishes on the plane $x = 0$, which is exactly the boundary condition we want. So we set

$$\phi = \begin{cases} \frac{q}{4\pi\epsilon_0} \left[\frac{1}{|\mathbf{r} - \mathbf{d}|} - \frac{1}{|\mathbf{r} + \mathbf{d}|} \right] & x > 0, \\ 0 & x \leq 0. \end{cases}$$

In $x > 0$ this satisfies Poisson's equation with the single source at $\mathbf{d}$ (the image charge lies outside the region, so contributes no source there), it satisfies Laplace's equation in $x < 0$, and it is continuous across $x = 0$. Uniqueness does the rest.

It remains to find the surface charge that makes this consistent. We need E_x at $x = 0^+$:

$$E_x = -\frac{\partial\phi}{\partial x} = \frac{q}{4\pi\epsilon_0} \left(\frac{x-d}{|\mathbf{r} - \mathbf{d}|^3} - \frac{x+d}{|\mathbf{r} + \mathbf{d}|^3} \right),$$

and setting $x = 0$ gives $E_x|_{x=0} = -\frac{q}{2\pi\epsilon_0} \frac{d}{(d^2 + y^2 + z^2)^{3/2}}$. The outward normal of the conductor is $\hat{\mathbf{n}} = \hat{\mathbf{x}}$, so

$$\sigma = \epsilon_0 E_x|_{x=0} = -\frac{q}{2\pi} \frac{d}{(d^2 + y^2 + z^2)^{3/2}}.$$

Integrating over the plane (most easily in polar coordinates $s^2 = y^2 + z^2$),

$$\int \sigma \, dy \, dz = -qd \int_0^\infty \frac{s \, ds}{(d^2 + s^2)^{3/2}} = -q,$$

so the induced charge is exactly $-q$: every field line leaving the charge terminates on the conductor. As a byproduct, the force on the charge is the force exerted by the image, namely $q^2/16\pi\epsilon_0 d^2$ towards the plane – a charge is always attracted to a nearby grounded conductor.

Example 12.2.23 (A charge outside an earthed sphere)

The method is not limited to planes. Take an earthed conducting sphere of radius a centred at the origin, and a charge q at $\mathbf{d}$ with $d > a$. We look for a potential vanishing on $|\mathbf{r}| = a$ and on the sphere's interior.

Try a single image charge q' placed at $\mathbf{d}' = \lambda\mathbf{d}$ for some λ to be determined, so that

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{|\mathbf{r} - \mathbf{d}|} + \frac{q'}{|\mathbf{r} - \lambda\mathbf{d}|} \right).$$

On the sphere $|\mathbf{r}| = a$, writing θ for the angle between $\mathbf{r}$ and $\mathbf{d}$,

$$|\mathbf{r} - \mathbf{d}|^2 = a^2 + d^2 - 2ad \cos \theta, \quad |\mathbf{r} - \lambda\mathbf{d}|^2 = a^2 + \lambda^2 d^2 - 2\lambda ad \cos \theta.$$

For ϕ to vanish for *all* θ we need the ratio of these to be a constant, and inspection shows that taking $\lambda = a^2/d^2$ works: then

$$|\mathbf{r} - \lambda\mathbf{d}|^2 = a^2 + \frac{a^4}{d^2} - \frac{2a^3 \cos \theta}{d} = \frac{a^2}{d^2} (d^2 + a^2 - 2ad \cos \theta) = \frac{a^2}{d^2} |\mathbf{r} - \mathbf{d}|^2.$$

So $|\mathbf{r} - \lambda \mathbf{d}| = (a/d)|\mathbf{r} - \mathbf{d}|$ on the sphere, and $\phi = 0$ there provided

$$q' = -\frac{a}{d}q.$$

The image charge therefore sits at distance a^2/d from the centre – inside the sphere, hence outside the region where we are solving, as required – and carries charge $-qa/d$.

By uniqueness this is the answer. Two consequences are worth noting: the total induced charge on the sphere is $-qa/d$ (not $-q$, since the sphere is earthed and can exchange charge with the ground), and the force on q is the attraction towards the image,

$$F = \frac{1}{4\pi\epsilon_0} \frac{q|q'|}{(d - a^2/d)^2} = \frac{q^2 ad}{4\pi\epsilon_0(d^2 - a^2)^2},$$

directed towards the sphere.

§12.2.8 Capacitance

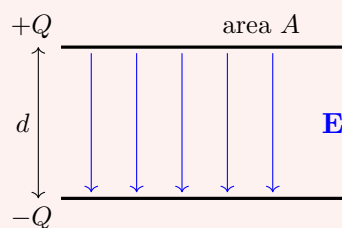
A pair of conductors carrying equal and opposite charges $\pm Q$ makes a **capacitor**. Since Poisson's equation is linear, doubling all the charges doubles all the potentials, so the potential difference V between the two conductors is proportional to Q .

Definition 12.2.24 (Capacitance)

The **capacitance** of a capacitor is $C = Q/V$, where $\pm Q$ are the charges on the two conductors and V is the potential difference between them. It depends only on the geometry.

Example 12.2.25 (Parallel plate capacitor)

Take two parallel plates of area A , a distance d apart, carrying charges $\pm Q$, and suppose $d \ll \sqrt{A}$ so we may pretend the plates are infinite and ignore what happens at the edges.



Each plate carries surface charge $\sigma = \pm Q/A$, and by our earlier calculation each produces a field of magnitude $\sigma/2\epsilon_0$ pointing away from the positive plate and towards the negative one. Between the plates these add; outside, they cancel. So

$$\mathbf{E} = \frac{Q}{\epsilon_0 A} \hat{\mathbf{n}}$$

between the plates, pointing from $+$ to $-$, and $\mathbf{E} = \mathbf{0}$ outside. The potential

difference is then

$$V = \int \mathbf{E} \cdot d\mathbf{x} = \frac{Qd}{\epsilon_0 A},$$

so

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}.$$

To make a large capacitance one wants large plates very close together, which is exactly what a real capacitor looks like.

We can also compute the energy stored, and it is a good check on Proposition 12.2.18. The field occupies a volume Ad and has magnitude $Q/\epsilon_0 A$, so

$$U = \frac{\epsilon_0}{2} |\mathbf{E}|^2 \cdot Ad = \frac{\epsilon_0}{2} \frac{Q^2}{\epsilon_0^2 A^2} Ad = \frac{Q^2 d}{2\epsilon_0 A} = \frac{Q^2}{2C} = \frac{1}{2} CV^2,$$

which is the formula every physicist knows. Alternatively one can get it directly: to move a further charge dq from one plate to the other against the existing potential difference q/C costs $dW = (q/C) dq$, and integrating from 0 to Q gives $Q^2/2C$ again.

Example 12.2.26 (Spherical capacitor)

For a slightly less trivial geometry, take two concentric spherical conducting shells of radii $a < b$, the inner carrying charge Q and the outer $-Q$.

Between the shells, Gauss' law on a sphere of radius $a < r < b$ encloses charge Q , so

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}},$$

while inside the inner shell and outside the outer shell the enclosed charge is zero and $\mathbf{E} = \mathbf{0}$. The potential difference is

$$V = \int_a^b \frac{Q}{4\pi\epsilon_0 r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right),$$

so

$$C = 4\pi\epsilon_0 \frac{ab}{b-a}.$$

Two limits are worth noting. If $b = a + d$ with $d \ll a$, then $C \approx 4\pi\epsilon_0 a^2/d = \epsilon_0 A/d$ where $A = 4\pi a^2$: we recover the parallel plate answer, as we should, since a thin shell looks locally flat. And if we let $b \rightarrow \infty$, we get $C = 4\pi\epsilon_0 a$, the 'self-capacitance' of an isolated sphere.

§12.3 Magnetostatics

We now turn on currents and turn off time dependence. In magnetostatics we take $\rho = 0$ and $\mathbf{J} = \mathbf{J}(\mathbf{x})$ a steady current, and look for a time-independent magnetic field. As we noted, Maxwell's equations then reduce to

$$\nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$

with $\mathbf{E} = \mathbf{0}$.

It is worth being clear about what $\rho = 0$ means here. It does *not* mean there are no charges around – we certainly need charges, or there would be no current. It means

the positive and negative charges balance everywhere, and continue to do so. In a copper wire, the mobile electrons drift along while the positive ions stay put, and the two densities cancel; charge does not pile up anywhere. Mathematically, the continuity equation with $\partial\rho/\partial t = 0$ says

$$\nabla \cdot \mathbf{J} = 0,$$

which is the precise statement that current does not accumulate: whatever flows into a region flows out again. This condition is essential – indeed, taking the divergence of $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ forces it.

§12.3.1 Ampère's Law

Just as Gauss' law is the integral form of (M1), the integral form of $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ is Ampère's law. Take a surface S with boundary $C = \partial S$, integrate over S and apply Stokes' theorem:

$$\oint_C \mathbf{B} \cdot d\mathbf{x} = \int_S (\nabla \times \mathbf{B}) \cdot d\mathbf{S} = \mu_0 \int_S \mathbf{J} \cdot d\mathbf{S}.$$

Law 12.3.1 (Ampère's law)

For any closed curve C bounding a surface S ,

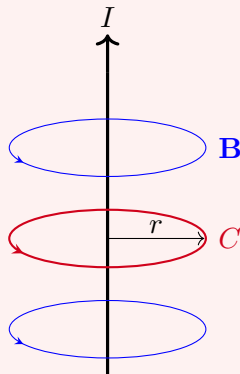
$$\oint_C \mathbf{B} \cdot d\mathbf{x} = \mu_0 I,$$

where I is the total current passing through S , with orientations related by the right-hand rule.

Note that, because $\nabla \cdot \mathbf{J} = 0$, the current through S depends only on the boundary curve C and not on which surface we span it with – a fact we will have cause to revisit when we come to the displacement current. As with Gauss' law, the utility of Ampère's law lies in exploiting symmetry.

Example 12.3.2 (The straight wire)

Take an infinite straight wire along the z -axis carrying current I , and use cylindrical polars (r, φ, z) .



Symmetry under translations in z and rotations about the axis says $\mathbf{B}$ can depend only on r . It can have no z component (reflect in a plane perpendicular to the wire), and no radial component, since $\nabla \cdot \mathbf{B} = 0$ applied to a cylinder about the axis would

then be violated. So

$$\mathbf{B} = B(r)\hat{\phi}.$$

Take C to be a circle of radius r about the axis. Then $d\mathbf{x} = r d\phi \hat{\phi}$, so

$$\oint_C \mathbf{B} \cdot d\mathbf{x} = 2\pi r B(r) = \mu_0 I,$$

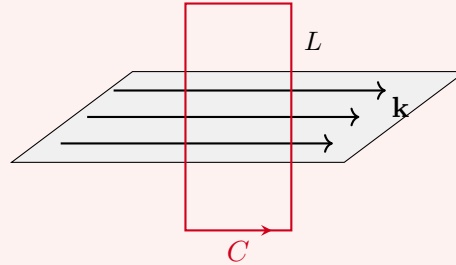
and hence

$$\mathbf{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi}.$$

The field circles the wire, falling off as $1/r$, with a direction given by the right-hand rule: point your right thumb along the current and your fingers curl the way $\mathbf{B}$ does.

Example 12.3.3 (Surface current)

Take the plane $z = 0$ to carry a **surface current density** $\mathbf{k}$ – current per unit length, measured across a line in the plane perpendicular to the flow – with $\mathbf{k} = k\hat{\mathbf{x}}$.



Think of this as infinitely many parallel wires laid side by side. Each produces a field circling it, and adding them up, the components perpendicular to the sheet cancel in pairs while those parallel to it survive. So $\mathbf{B}$ points in the $\pm\hat{\mathbf{y}}$ direction, depends only on z , and is odd in z :

$$\mathbf{B} = -B(z)\hat{\mathbf{y}}, \quad B(-z) = -B(z).$$

Take a rectangular loop C of height $2z$ and width L , lying in the y - z plane and straddling the sheet. Only the two horizontal sides contribute, and the current through the loop is kL , so

$$\oint_C \mathbf{B} \cdot d\mathbf{x} = LB(z) - LB(-z) = 2LB(z) = \mu_0 kL,$$

giving

$$B(z) = \frac{\mu_0 k}{2} \quad \text{for } z > 0.$$

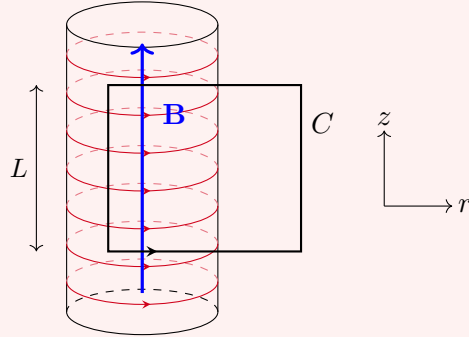
Once again the field is constant, and once again there is a discontinuity across the sheet – this time in the *tangential* component of $\mathbf{B}$. The general statement, proved by shrinking an Ampèrian loop across an arbitrary surface exactly as in Proposition 12.2.7, is

$$\hat{\mathbf{n}} \times \mathbf{B}_+ - \hat{\mathbf{n}} \times \mathbf{B}_- = \mu_0 \mathbf{k},$$

with the normal component of $\mathbf{B}$ continuous.

Example 12.3.4 (The solenoid)

A **solenoid** is a wire wound tightly into a cylindrical helix, so that it approximates a cylindrical surface current. Let there be N turns per unit length, each carrying current I .



Symmetry (translation in z , rotation about the axis) together with $\nabla \cdot \mathbf{B} = 0$ gives $\mathbf{B} = B(r)\hat{\mathbf{z}}$. Outside the solenoid there is no current, so $\nabla \times \mathbf{B} = \mathbf{0}$, and for a field of this form that says $\partial B/\partial r = 0$. So B is constant outside; since it must vanish at infinity, it vanishes everywhere outside.

Now apply Ampère's law to the rectangle C shown, with one vertical side of length L inside the solenoid and the other outside. The outside side contributes nothing (the field is zero there) and the horizontal sides contribute nothing (they are perpendicular to $\mathbf{B}$, or lie where $\mathbf{B} = \mathbf{0}$). The number of wires threading the rectangle is NL , so

$$BL = \mu_0 INL \implies B = \mu_0 NI.$$

So the field inside a solenoid is uniform, parallel to the axis, and independent of the radius. This is how one makes a uniform magnetic field in practice.

§12.3.2 The Vector Potential and Gauge

For a general current distribution we need to deal with both equations at once, and as in electrostatics we begin by solving one of them identically. This time it is $\nabla \cdot \mathbf{B} = 0$, and a divergence-free field on $\mathbb{R}^3$ is a curl.

Definition 12.3.5 (Vector potential)

If $\mathbf{B} = \nabla \times \mathbf{A}$, then the vector field $\mathbf{A}$ is a **vector potential** for $\mathbf{B}$.

Before going on it is worth appreciating what (M2) is telling us. In electrostatics the analogous equation was $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$, with a source on the right. The magnetic equation has a zero there instead, and the physical content of that zero is that there are no *magnetic charges*: no isolated north pole has ever been found. Break a bar magnet in half and you get two smaller magnets, never a lone north and a lone south.

Substituting $\mathbf{B} = \nabla \times \mathbf{A}$ into the remaining equation and using the identity $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ gives

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}. \quad (*)$$

This looks unpleasant. What rescues us is that $\mathbf{A}$ is far from unique.

Definition 12.3.6 (Gauge transformation)

Given a vector potential $\mathbf{A}$ and any scalar function $\chi(\mathbf{x})$, the field

$$\mathbf{A}' = \mathbf{A} + \nabla\chi$$

is another vector potential for the same $\mathbf{B}$, since $\nabla \times \nabla\chi = \mathbf{0}$. The map $\mathbf{A} \mapsto \mathbf{A} + \nabla\chi$ is a **gauge transformation**, and a choice of $\mathbf{A}$ is a choice of **gauge**.

So $\mathbf{A}$ carries redundant information: only the gauge-invariant combination $\nabla \times \mathbf{A}$ is physical. We may exploit this freedom to make (*) as simple as possible, and the obvious thing to try is to kill the first term.

Definition 12.3.7 (Coulomb gauge)

A vector potential with $\nabla \cdot \mathbf{A} = 0$ is said to be in **Coulomb gauge**.

Proposition 12.3.8

Any vector potential can be brought into Coulomb gauge.

Proof. Suppose $\nabla \cdot \mathbf{A} = \psi(\mathbf{x})$. Under a gauge transformation,

$$\nabla \cdot \mathbf{A}' = \nabla \cdot \mathbf{A} + \nabla^2\chi = \psi + \nabla^2\chi,$$

so we need χ with $\nabla^2\chi = -\psi$. This is Poisson's equation, which we know always has a solution (write it down with the Green's function if you like). So such a χ exists. $\square$

In Coulomb gauge, (*) collapses to

$$\nabla^2\mathbf{A} = -\mu_0\mathbf{J},$$

or in Cartesian components, $\nabla^2 A_i = -\mu_0 J_i$: three independent copies of Poisson's equation, one for each component. We solved that in electrostatics, so we can simply write down the answer.

Proposition 12.3.9

In Coulomb gauge, the vector potential of a localised steady current $\mathbf{J}$ is

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} dV'.$$

Proof. Applying the Green's function solution of Poisson's equation componentwise gives this formula, so all that remains is to check the result really is in Coulomb gauge – otherwise it would not solve the original equation (*). Taking the divergence (with ∇ acting on $\mathbf{r}$),

$$\nabla \cdot \mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \cdot \nabla \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) dV' = -\frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \cdot \nabla' \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) dV',$$

using the fact that differentiating $|\mathbf{r} - \mathbf{r}'|^{-1}$ with respect to $\mathbf{r}'$ is minus differentiating with respect to $\mathbf{r}$. Now that the derivative acts on the primed variable we may integrate by parts:

$$\nabla \cdot \mathbf{A} = -\frac{\mu_0}{4\pi} \int \left[\nabla' \cdot \left(\frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right) - \frac{\nabla' \cdot \mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right] dV'.$$

The first term is a total divergence, and becomes a surface integral at infinity which vanishes because the current is localised. The second vanishes because the current is steady, $\nabla' \cdot \mathbf{J} = 0$. So $\nabla \cdot \mathbf{A} = 0$. $\square$

Taking the curl gives the magnetostatic analogue of the Coulomb integral.

Law 12.3.10 (Biot–Savart law)

The magnetic field of a localised steady current is

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} dV'.$$

If the current is confined to a thin wire along a curve C , then $\mathbf{J} dV' \rightarrow I d\mathbf{r}'$ and this becomes

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C d\mathbf{r}' \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}.$$

Proof. Take the curl of Proposition 12.3.9. Since ∇ acts on $\mathbf{r}$ and $\mathbf{J}$ depends on $\mathbf{r}'$,

$$\nabla \times \left(\frac{\mathbf{J}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \right) = \nabla \left(\frac{1}{|\mathbf{r} - \mathbf{r}'|} \right) \times \mathbf{J}(\mathbf{r}') = -\frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3} \times \mathbf{J}(\mathbf{r}') = \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3},$$

and the result follows. $\square$

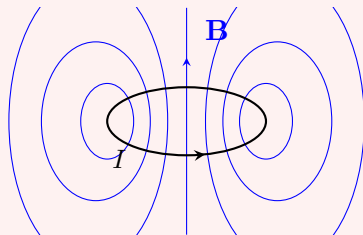
The Biot–Savart law is the exact analogue, for magnetostatics, of the formula $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \rho(\mathbf{r} - \mathbf{r}')/|\mathbf{r} - \mathbf{r}'|^3$ we found in electrostatics – with the cross product accounting for the fact that magnetic fields circulate rather than radiate.

§12.3.3 Magnetic Dipoles

There are no magnetic monopoles, so the leading behaviour of a magnetic field far from its source is a dipole. Let us compute it.

Example 12.3.11 (The current loop)

Take a planar loop of wire C carrying current I , and look at the field far away.



By Proposition 12.3.9 specialised to a wire,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C \frac{d\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|},$$

and expanding as before,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \oint_C \left(\frac{1}{r} + \frac{\mathbf{r} \cdot \mathbf{r}'}{r^3} + \dots \right) d\mathbf{r}'.$$

The leading term is a constant times $\oint_C d\mathbf{r}' = \mathbf{0}$, so it drops out – exactly as we should expect, since a $1/r$ term in $\mathbf{A}$ would be a magnetic monopole.

For the next term we use a corollary of Stokes' theorem: for any scalar f ,

$$\oint_C f d\mathbf{r}' = \int_S d\mathbf{S}' \times \nabla' f.$$

(To see this, dot both sides with a constant vector $\mathbf{c}$; the left becomes $\oint_C (f\mathbf{c}) \cdot d\mathbf{r}'$, which by Stokes' theorem is $\int_S \nabla' \times (f\mathbf{c}) \cdot d\mathbf{S}' = \int_S (\nabla' f \times \mathbf{c}) \cdot d\mathbf{S}'$, and rearranging the triple product gives $\mathbf{c} \cdot \int_S d\mathbf{S}' \times \nabla' f$.)

Taking $f(\mathbf{r}') = \mathbf{g} \cdot \mathbf{r}'$ for a constant vector $\mathbf{g}$ gives $\nabla' f = \mathbf{g}$, so

$$\oint_C (\mathbf{g} \cdot \mathbf{r}') d\mathbf{r}' = \int_S d\mathbf{S}' \times \mathbf{g} = \mathbf{S} \times \mathbf{g}, \quad \text{where } \mathbf{S} = \int_S d\mathbf{S}'$$

is the vector area of the loop. Applying this with $\mathbf{g} = \mathbf{r}/r^3$,

$$\mathbf{A}(\mathbf{r}) \approx \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3},$$

where we have defined the following.

Definition 12.3.12 (Magnetic dipole moment)

The **magnetic dipole moment** of a current loop is $\mathbf{m} = I\mathbf{S}$, with $\mathbf{S}$ the vector area of the loop.

Taking the curl (a slightly tedious but routine index computation) gives

$$\mathbf{B} = \nabla \times \mathbf{A} = \frac{\mu_0}{4\pi} \left(\frac{3(\mathbf{m} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}}{r^3} \right),$$

which is precisely the same shape as the electric field of an electric dipole, with $\mathbf{p}/\epsilon_0$ replaced by $\mu_0 \mathbf{m}$. The difference is that the magnetic expansion has no $1/r^2$ monopole term at all.

The same conclusion holds for an arbitrary localised current, though the computation takes a little more care.

Proposition 12.3.13 (Dipole moment of a general current)

For a localised steady current $\mathbf{J}$,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \frac{\mathbf{m} \times \mathbf{r}}{r^3} + O(r^{-3}), \quad \mathbf{m} = \frac{1}{2} \int \mathbf{r}' \times \mathbf{J}(\mathbf{r}') dV'.$$

Proof. Expanding Proposition 12.3.9 in components,

$$A_i(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \left(\frac{J_i(\mathbf{r}')}{r} + \frac{J_i(\mathbf{r}')(\mathbf{r} \cdot \mathbf{r}')}{r^3} + \dots \right) dV'.$$

For the first term, note that since $\partial'_j J_j = \nabla' \cdot \mathbf{J} = 0$ and $\partial'_j r'_i = \delta_{ij}$,

$$\partial'_j (J_j r'_i) = (\partial'_j J_j) r'_i + J_j \delta_{ij} = J_i,$$

so J_i is a total derivative and its integral vanishes.

For the second, apply the same trick to $J_j r'_i r'_k$:

$$\partial'_j (J_j r'_i r'_k) = J_i r'_k + J_k r'_i,$$

so the symmetric part of $J_i r'_k$ integrates to zero, and only the antisymmetric part survives:

$$\int J_i(\mathbf{r} \cdot \mathbf{r}') dV' = \int \frac{r_k}{2} (J_i r'_k - J_k r'_i) dV' = \frac{1}{2} \int [\mathbf{r} \times (\mathbf{J} \times \mathbf{r}')]_i dV'.$$

Rearranging the triple product and comparing with $(\mathbf{m} \times \mathbf{r})_i$ gives the result with the stated $\mathbf{m}$. $\square$

For a planar loop the two definitions agree: $\frac{1}{2} \oint \mathbf{r}' \times I d\mathbf{r}'$ is I times the vector area, which is $I\mathbf{S}$.

§12.3.4 Magnetic Forces

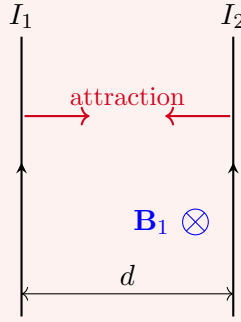
Currents make magnetic fields, and magnetic fields push currents. Hence currents push each other. The relevant formula is the continuum Lorentz force law from Section 1,

$$\mathbf{F} = \int \mathbf{J} \times \mathbf{B} dV,$$

which for a thin wire becomes $\mathbf{F} = I \oint_C d\mathbf{r} \times \mathbf{B}$.

Example 12.3.14 (Force between parallel wires)

Take two infinite parallel wires a distance d apart, carrying currents I_1 and I_2 in the same direction, say along $\hat{\mathbf{z}}$, and separated along $\hat{\mathbf{x}}$.



The first wire produces at the location of the second a field of magnitude

$$B_1 = \frac{\mu_0 I_1}{2\pi d},$$

pointing in the $\hat{\phi}$ direction, which at the second wire is $-\hat{\mathbf{y}}$ (taking the wires along $\hat{\mathbf{z}}$ separated in $\hat{\mathbf{x}}$). The force per unit length on the second wire is then

$$\frac{\mathbf{F}}{L} = I_2 \hat{\mathbf{z}} \times \mathbf{B}_1 = -\frac{\mu_0 I_1 I_2}{2\pi d} \hat{\mathbf{z}} \times \hat{\mathbf{y}} = \frac{\mu_0 I_1 I_2}{2\pi d} \hat{\mathbf{x}},$$

which points from the second wire towards the first. So parallel currents *attract*, and (reversing I_2) antiparallel currents repel. This is exactly backwards from the electrostatic case, where like charges repel, and it is the reason the definition of the Ampere used to be phrased in terms of the force between two wires.

Example 12.3.15 (General force between current loops)

More generally, a current I_1 on a loop C_1 produces, by Biot–Savart,

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} d\mathbf{r}_1 \times \frac{\mathbf{r} - \mathbf{r}_1}{|\mathbf{r} - \mathbf{r}_1|^3},$$

and a second loop C_2 carrying I_2 then feels

$$\mathbf{F} = \frac{\mu_0 I_1 I_2}{4\pi} \oint_{C_2} \oint_{C_1} d\mathbf{r}_2 \times \left(d\mathbf{r}_1 \times \frac{\mathbf{r}_2 - \mathbf{r}_1}{|\mathbf{r}_2 - \mathbf{r}_1|^3} \right).$$

This double integral is not something one wants to evaluate. If the loops are far apart, though, each looks like a dipole, and the force can be shown to be

$$\mathbf{F} = \frac{\mu_0}{4\pi} \nabla \left(\frac{3(\mathbf{m}_1 \cdot \hat{\mathbf{r}})(\mathbf{m}_2 \cdot \hat{\mathbf{r}}) - \mathbf{m}_1 \cdot \mathbf{m}_2}{r^3} \right),$$

the gradient of an interaction energy of exactly the same form as for two electric dipoles. We omit the (long) proof.

There is one more result worth recording, on a small current loop placed in an *external* field. It is the exact magnetic analogue of Proposition 12.2.15.

Proposition 12.3.16 (Magnetic dipole in an external field)

A small current loop of dipole moment $\mathbf{m}$ in an external field $\mathbf{B}$ experiences a torque

$$\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B},$$

and has interaction energy $U = -\mathbf{m} \cdot \mathbf{B}$, so that in a non-uniform field it feels a force $\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$.

Proof. If $\mathbf{B}$ is essentially constant over the loop, the net force is

$$\mathbf{F} = I \oint_C d\mathbf{r} \times \mathbf{B} = I \left(\oint_C d\mathbf{r} \right) \times \mathbf{B} = \mathbf{0},$$

since the integral of $d\mathbf{r}$ round a closed curve vanishes. The torque about the centre of the loop is

$$\boldsymbol{\tau} = I \oint_C \mathbf{r} \times (d\mathbf{r} \times \mathbf{B}) = I \oint_C [d\mathbf{r} (\mathbf{r} \cdot \mathbf{B}) - \mathbf{B} (\mathbf{r} \cdot d\mathbf{r})],$$

by the vector triple product. The second term vanishes, since $\oint_C \mathbf{r} \cdot d\mathbf{r} = \oint_C d(\frac{1}{2}r^2) = 0$. For the first, apply the identity $\oint_C (\mathbf{g} \cdot \mathbf{r}) d\mathbf{r} = \mathbf{S} \times \mathbf{g}$ established in the current loop example, with $\mathbf{g} = \mathbf{B}$:

$$\boldsymbol{\tau} = I \mathbf{S} \times \mathbf{B} = \mathbf{m} \times \mathbf{B},$$

as claimed. Finally, $U = -\mathbf{m} \cdot \mathbf{B}$ is the energy function whose angular derivative reproduces this torque, and whose spatial gradient gives the force. $\square$

This is why a compass needle turns to point along the field, and why an electric motor turns: a current loop in a magnetic field twists until its moment aligns, and if you flip the current at the right moment it keeps twisting. A bar magnet is, at bottom, an enormous number of microscopic current loops – electron spins, really – whose dipole moments have lined up, and the force $\nabla(\mathbf{m} \cdot \mathbf{B})$ is what makes it stick to a fridge.

We have now spent a whole section on ‘magnets’ which look nothing at all like the ones in a kitchen drawer. The bridge between the two is the observation just made, and the details belong to a course on the electromagnetism of media.

§12.4 Electrodynamics

We now allow the fields to depend on time, and the two halves of the theory become inseparable. A changing magnetic field produces an electric field, and a changing electric field produces a magnetic field; put the two together and you get a self-sustaining disturbance which propagates through empty space at the speed of light.

§12.4.1 Induction and Faraday’s Law

We begin with the Maxwell equation we have so far ignored,

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = \mathbf{0}.$$

Integrate it over a fixed surface S with boundary C , and apply Stokes' theorem:

$$\oint_C \mathbf{E} \cdot d\mathbf{x} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = - \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S},$$

the last step being legitimate because S is not moving. Both sides deserve names.

Definition 12.4.1 (Electromotive force and magnetic flux)

The **electromotive force** (emf) around a closed curve C is

$$\mathcal{E} = \oint_C \mathbf{E} \cdot d\mathbf{x},$$

and the **magnetic flux** through a surface S is

$$\Phi = \int_S \mathbf{B} \cdot d\mathbf{S}.$$

The name 'electromotive force' is a historical embarrassment: $\mathcal{E}$ is not a force but a work done per unit charge, and it is measured in volts. If C is a loop of wire, $\mathcal{E}$ is the voltage that the loop sees.

Law 12.4.2 (Faraday's law of induction)

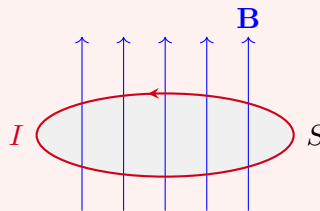
$$\mathcal{E} = - \frac{d\Phi}{dt}.$$

So changing the flux through a circuit drives a current around it. There are many ways to change the flux – move a magnet towards the loop, switch an electromagnet on or off, rotate the loop – and all of them work. Essentially every electrical generator in the world runs on this principle.

The minus sign is worth dwelling on. It says that the induced current flows in the direction whose own magnetic field *opposes* the change in flux. This is **Lenz's law**, and it is a statement about stability: were the sign the other way, an increase in flux would induce a current reinforcing the increase, which would induce more current, and so on. The world would rapidly become unpleasant.

Example 12.4.3 (A loop in a decreasing field)

Consider a circular loop of wire with a magnetic field passing up through it, and suppose we decrease $|\mathbf{B}|$.



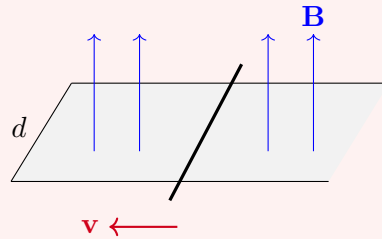
Then $d\Phi/dt < 0$, so $\mathcal{E} > 0$ and the current flows anticlockwise when viewed from above. By the right-hand rule this current produces a magnetic field pointing *up* through the loop, which is to say it tries to maintain the flux that we are removing.

Lenz's law in action.

There is a second, superficially quite different, way to change the flux: keep $\mathbf{B}$ fixed and move the circuit. Remarkably, the same formula holds.

Example 12.4.4 (The sliding bar)

Take a conducting bar of length d sliding with speed v along a pair of rails, in a uniform field $\mathbf{B}$ perpendicular to the plane of the circuit.



Here nothing is changing in time at any fixed point of space, so the Maxwell equation we used above has nothing to say. Instead, use the Lorentz force law directly: each charge q in the bar is being carried along at velocity $\mathbf{v}$, so it feels a force $q\mathbf{v} \times \mathbf{B}$ of magnitude qvB directed along the bar. The work done per unit charge in going once round the circuit is therefore

$$\mathcal{E} = vBd,$$

since only the charges in the moving bar have any force on them. Meanwhile the area of the circuit is shrinking at rate vd , so

$$\frac{d\Phi}{dt} = -vBd,$$

and once again $\mathcal{E} = -d\Phi/dt$. The same law, but for a completely different reason – one is Maxwell's equation, the other is the Lorentz force law. That these two conspire to give the same answer is a strong hint that something deeper is going on, and in Section 12.5 we will see what.

Let us do the general case, in which both effects occur at once.

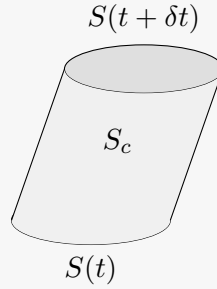
Theorem 12.4.5 (The flux rule)

Let $C(t)$ be a moving closed curve bounding a moving surface $S(t)$, with the curve moving with local velocity $\mathbf{v}$. Define the emf as

$$\mathcal{E} = \oint_{C(t)} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x}.$$

Then $\mathcal{E} = -d\Phi/dt$.

Proof. As the curve moves from time t to $t + \delta t$, it sweeps out a cylindrical surface S_c , and $S(t)$, $S(t + \delta t)$ and S_c together bound a volume.



Now compute the change in flux, expanding $\mathbf{B}(t + \delta t) = \mathbf{B}(t) + \delta t \partial \mathbf{B} / \partial t + O(\delta t^2)$:

$$\begin{aligned} \Phi(t + \delta t) - \Phi(t) &= \int_{S(t+\delta t)} \mathbf{B}(t + \delta t) \cdot d\mathbf{S} - \int_{S(t)} \mathbf{B}(t) \cdot d\mathbf{S} \\ &= \delta t \int_{S(t)} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \left[\int_{S(t+\delta t)} - \int_{S(t)} \right] \mathbf{B}(t) \cdot d\mathbf{S} + O(\delta t^2). \end{aligned}$$

Since $\nabla \cdot \mathbf{B} = 0$, the flux of $\mathbf{B}(t)$ out of the closed surface formed by $S(t + \delta t)$, $S(t)$ and S_c is zero, which says exactly

$$\left[\int_{S(t+\delta t)} - \int_{S(t)} \right] \mathbf{B}(t) \cdot d\mathbf{S} = - \int_{S_c} \mathbf{B}(t) \cdot d\mathbf{S}.$$

On S_c , an element of the curve $d\mathbf{x}$ sweeps out area $d\mathbf{S} = d\mathbf{x} \times \mathbf{v} \delta t$, so $\mathbf{B} \cdot d\mathbf{S} = \delta t (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x}$. Dividing by δt and letting $\delta t \rightarrow 0$,

$$\frac{d\Phi}{dt} = \int_{S(t)} \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} - \oint_{C(t)} (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x}.$$

Finally substitute $\partial \mathbf{B} / \partial t = -\nabla \times \mathbf{E}$ from (M3) and apply Stokes' theorem to the first term:

$$\frac{d\Phi}{dt} = - \oint_{C(t)} \mathbf{E} \cdot d\mathbf{x} - \oint_{C(t)} (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{x} = -\mathcal{E}. \quad \square$$

So the two mechanisms are two terms of one formula, and the flux rule is completely general.

§12.4.2 Inductance and Magnetic Energy

Consider a loop of wire C carrying current I . This produces a magnetic field, and hence a flux through the loop itself. By linearity, the flux is proportional to the current.

Definition 12.4.6 (Inductance)

The **inductance** of a circuit C is $L = \Phi/I$, the flux through C per unit current flowing in C . It depends only on the geometry of C .

Example 12.4.7 (Inductance of a solenoid)

Take a solenoid of length ℓ and cross-sectional area A , with N turns per unit length, and suppose $\ell \gg \sqrt{A}$ so that we may use $B = \mu_0 N I$ throughout. The flux through

a single turn is $\Phi_0 = \mu_0 N I A$, and there are $N\ell$ turns in total, so

$$\Phi = \mu_0 N^2 I A \ell = \mu_0 N^2 I V,$$

where $V = A\ell$ is the volume. Hence

$$L = \mu_0 N^2 V.$$

There is also a version of this for two circuits, which is what makes transformers possible.

Definition 12.4.8 (Mutual inductance)

Given two circuits C_1 and C_2 , the **mutual inductance** M is defined by $\Phi_2 = M I_1$, where Φ_2 is the flux through C_2 produced by a current I_1 in C_1 .

It is a small miracle – and a very useful one – that M is symmetric in the two circuits.

Proposition 12.4.9

The flux through C_2 due to unit current in C_1 equals the flux through C_1 due to unit current in C_2 .

Proof. Using $\mathbf{B} = \nabla \times \mathbf{A}$, Stokes' theorem, and the expression for the vector potential of a wire,

$$\Phi_2 = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{S} = \oint_{C_2} \mathbf{A}_1 \cdot d\mathbf{r}_2 = \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\mathbf{r}_1 \cdot d\mathbf{r}_2}{|\mathbf{r}_1 - \mathbf{r}_2|}.$$

The right hand side is manifestly symmetric under exchanging the roles of C_1 and C_2 , so M is too. (This is the **Neumann formula**.) $\square$

Inductance is the key to computing the energy stored in a magnetic field, by the following argument. To build up a current we must change it, and by Faraday's law a changing current induces an emf

$$\mathcal{E} = -\frac{d\Phi}{dt} = -L \frac{dI}{dt}$$

which, by Lenz's law, opposes what we are trying to do. So we must do work against it. In a time δt a charge $I\delta t$ goes round the loop, and the work we must do against the induced emf is

$$\delta W = -\mathcal{E} I \delta t = L I \frac{dI}{dt} \delta t = \frac{L}{2} \frac{d(I^2)}{dt} \delta t.$$

Integrating from zero current up to I ,

$$W = \frac{1}{2} L I^2 = \frac{1}{2} I \Phi.$$

Proposition 12.4.10 (Energy in the magnetic field)

The energy stored in a magnetic field is

$$U = \frac{1}{2\mu_0} \int_{\mathbb{R}^3} |\mathbf{B}|^2 dV.$$

Proof. Start from $U = \frac{1}{2}I\Phi$, write $\mathbf{B} = \nabla \times \mathbf{A}$ and use Stokes' theorem:

$$U = \frac{1}{2}I \int_S \mathbf{B} \cdot d\mathbf{S} = \frac{1}{2}I \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \frac{1}{2}I \oint_C \mathbf{A} \cdot d\mathbf{x} = \frac{1}{2} \int_{\mathbb{R}^3} \mathbf{J} \cdot \mathbf{A} dV,$$

the last step because the current is confined to the wire, where $\mathbf{J} dV = I d\mathbf{x}$. Now use $\mathbf{J} = (\nabla \times \mathbf{B})/\mu_0$ and the identity $\nabla \cdot (\mathbf{B} \times \mathbf{A}) = \mathbf{A} \cdot (\nabla \times \mathbf{B}) - \mathbf{B} \cdot (\nabla \times \mathbf{A})$:

$$U = \frac{1}{2\mu_0} \int (\nabla \times \mathbf{B}) \cdot \mathbf{A} dV = \frac{1}{2\mu_0} \int [\nabla \cdot (\mathbf{B} \times \mathbf{A}) + \mathbf{B} \cdot (\nabla \times \mathbf{A})] dV.$$

The first term is a total divergence and integrates to a surface term at infinity, which vanishes provided $\mathbf{B} \times \mathbf{A}$ decays fast enough. The second is $|\mathbf{B}|^2$, giving the result. $\square$

Comparing with Proposition 12.2.18, the total energy stored in the electromagnetic field is

$$U = \int_{\mathbb{R}^3} \left(\frac{\epsilon_0}{2} |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \right) dV.$$

It is worth flagging that this does not follow logically from the two separate results: in principle, when both fields are present, they might interact and contribute cross terms $\mathbf{E} \cdot \mathbf{B}$ to the energy. They do not, but seeing why requires the Poynting theorem, which we prove below.

§12.4.3 Resistance and Circuits

So far our story is: change the flux, get an emf, accelerate the charges. But charges that accelerate forever would give infinite currents, and in reality they do not. What stops them is a form of friction: the charges collide with the atoms of the wire, losing energy. The upshot is that the emf ends up proportional to the charges' *velocity* rather than their acceleration.

Law 12.4.11 (Ohm's law)

For most materials, the current is proportional to the applied emf:

$$\mathcal{E} = IR,$$

where R is the **resistance** of the circuit. In terms of fields, this reads $\mathbf{J} = \sigma \mathbf{E}$, where σ is the **conductivity** of the material.

Since $\mathcal{E} = \oint \mathbf{E} \cdot d\mathbf{x}$ is the potential difference V in the static case, this is the familiar $V = IR$. For a wire of length ℓ and cross-section A made of a material of **resistivity** $\varrho = 1/\sigma$, one finds $R = \varrho \ell / A$; the resistivity is a property of the substance alone, while the resistance depends on the shape.

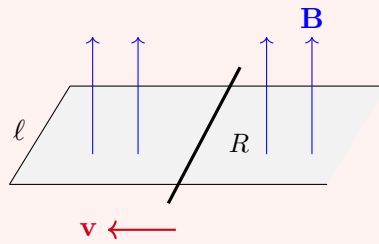
Resistance costs energy. In time δt a charge $I \delta t$ moves round the circuit through a potential difference $\mathcal{E}$, so

$$\frac{dW}{dt} = \mathcal{E}I = I^2 R.$$

This is **Joule heating**: the energy lost to the collisions, which reappears as heat. It is how an electric kettle works, and why a computer needs a fan.

Example 12.4.12 (The decelerating bar)

Return to the sliding bar, and now give the bar resistance R (taking the rails to be perfect conductors).



There are two unknowns: the position $x(t)$ of the bar and the current $I(t)$. A current I in the bar feels a force $I\ell B$ along the rails, so

$$m\ddot{x} = I\ell B.$$

Meanwhile the emf is $\mathcal{E} = -d\Phi/dt = -B\ell\dot{x}$, so Ohm's law gives $IR = -B\ell\dot{x}$. Eliminating I ,

$$m\ddot{x} = -\frac{B^2\ell^2}{R}\dot{x},$$

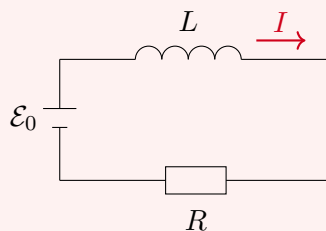
which integrates to

$$\dot{x}(t) = -v e^{-B^2\ell^2 t/mR}.$$

The bar decelerates exponentially. This is Lenz's law again – the induced current opposes the motion that produced it – and its kinetic energy is dissipated as Joule heating in the bar. It is exactly the mechanism behind magnetic braking.

Example 12.4.13 (The LR circuit)

Put an inductor L , a resistor R and a battery of emf $\mathcal{E}_0$ in series, and close the switch at $t = 0$.



Going round the loop, the battery supplies $\mathcal{E}_0$, the inductor opposes with $-L dI/dt$, and the resistor drops IR :

$$\mathcal{E}_0 - L \frac{dI}{dt} = IR.$$

This is a first order linear ODE with solution

$$I(t) = \frac{\mathcal{E}_0}{R} \left(1 - e^{-Rt/L}\right),$$

so the current does not jump to its final value $\mathcal{E}_0/R$ instantly but rises on a timescale L/R . Inductors resist changes in current, in exactly the way that masses resist

changes in velocity.

§12.4.4 The Displacement Current

We have now used every one of Maxwell's equations except for the final term in (M4),

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right),$$

in which the quantity $\epsilon_0 \partial \mathbf{E} / \partial t$ is called, for historical reasons, the **displacement current**. It is the smallest looking term in Maxwell's equations and by far the most consequential.

Its necessity is a matter of pure mathematics, and was noticed as such. Suppose we drop it, as Ampère did, and write $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. Taking the divergence, and using that the divergence of a curl vanishes,

$$\mu_0 \nabla \cdot \mathbf{J} = \nabla \cdot (\nabla \times \mathbf{B}) = 0.$$

But charge conservation says $\dot{\rho} + \nabla \cdot \mathbf{J} = 0$, so this forces $\dot{\rho} = 0$ everywhere and always. That is absurd – we can perfectly well pick up a charge and carry it somewhere else, changing ρ as we go. So the equations as they stood were inconsistent with the conservation of charge.

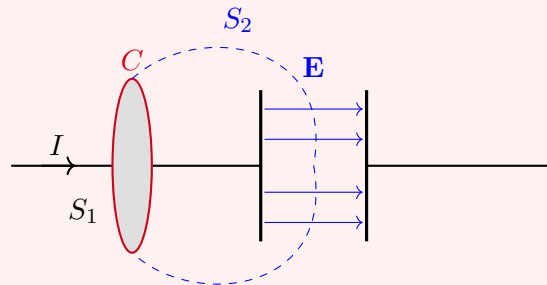
With the extra term the computation instead reads

$$0 = \nabla \cdot \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = \nabla \cdot \mathbf{J} + \epsilon_0 \frac{\partial}{\partial t} (\nabla \cdot \mathbf{E}) = \nabla \cdot \mathbf{J} + \frac{\partial \rho}{\partial t},$$

using (M1). So not only is the theory now consistent with charge conservation, it *implies* it. This is a beautiful piece of physics: Maxwell found the term by demanding mathematical consistency, and it turned out to predict light.

Example 12.4.14 (The charging capacitor)

Here is the standard illustration of why the term is needed. Take a wire carrying current I into one plate of a capacitor, and consider Ampère's law on a circle C around the wire. We may span C with a flat disc S_1 cut by the wire, or with a bulging surface S_2 that passes between the plates.



Through S_1 the current is I ; through S_2 it is zero, since no charge crosses the gap. Without the displacement current Ampère's law would give two different answers, which is not acceptable. With it, we note that the charge Q on the plate gives a

field $E = Q/\epsilon_0 A$ in the gap, so the displacement current through S_2 is

$$\epsilon_0 \frac{d}{dt} \int_{S_2} \mathbf{E} \cdot d\mathbf{S} = \epsilon_0 A \frac{d}{dt} \left(\frac{Q}{\epsilon_0 A} \right) = \frac{dQ}{dt} = I,$$

exactly making up the difference. The two surfaces agree, as they must.

§12.4.5 Electromagnetic Waves

Now for the payoff. We look for solutions of Maxwell's equations in vacuum, so $\rho = 0$ and $\mathbf{J} = \mathbf{0}$:

$$\nabla \cdot \mathbf{E} = 0, \quad \nabla \cdot \mathbf{B} = 0, \quad \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Notice that this system is not empty: the source-free equations couple $\mathbf{E}$ and $\mathbf{B}$ to each other, so we can have fields with no charges anywhere at all.

Theorem 12.4.15 (The wave equation)

In vacuum, each of $\mathbf{E}$ and $\mathbf{B}$ satisfies the wave equation

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} - \nabla^2 \mathbf{E} = \mathbf{0}, \quad \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} - \nabla^2 \mathbf{B} = \mathbf{0},$$

with wave speed

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

Proof. Differentiate (M4) with respect to t and commute the derivatives:

$$\begin{aligned} \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} &= \frac{\partial}{\partial t} (\nabla \times \mathbf{B}) = \nabla \times \frac{\partial \mathbf{B}}{\partial t} = -\nabla \times (\nabla \times \mathbf{E}) \\ &= \nabla^2 \mathbf{E} - \nabla (\nabla \cdot \mathbf{E}) = \nabla^2 \mathbf{E}, \end{aligned}$$

using (M3), then the vector identity, then (M1) with $\rho = 0$. The computation for $\mathbf{B}$ is identical with the roles of (M3) and (M4) exchanged. $\square$

Now put in the numbers. With $\epsilon_0 \approx 8.85 \times 10^{-12}$ and $\mu_0 = 4\pi \times 10^{-7}$ in SI units,

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \approx 3 \times 10^8 \text{ m s}^{-1},$$

which is the speed of light. This is not a coincidence, and it was not regarded as one at the time: two constants measured in table-top experiments with wires and capacitors, combined in this particular way, reproduce the speed at which light travels. Maxwell's conclusion was the correct one – light *is* an electromagnetic wave.

Let us find the solutions explicitly. Look for a wave travelling in the x direction and independent of y and z , so $\mathbf{E} = \mathbf{E}(x, t)$. Then $\nabla \cdot \mathbf{E} = \partial E_x / \partial x = 0$, so E_x is constant, and we may take it to be zero. Choosing axes so that the field oscillates along $\hat{\mathbf{y}}$,

$$\mathbf{E} = (0, E(x, t), 0), \quad \frac{1}{c^2} \frac{\partial^2 E}{\partial t^2} - \frac{\partial^2 E}{\partial x^2} = 0,$$

whose general solution, as we know, is

$$E(x, t) = f(x - ct) + g(x + ct)$$

for arbitrary functions f, g : a wave travelling right and a wave travelling left. The most important special case is a **monochromatic** wave

$$E = E_0 \sin(kx - \omega t).$$

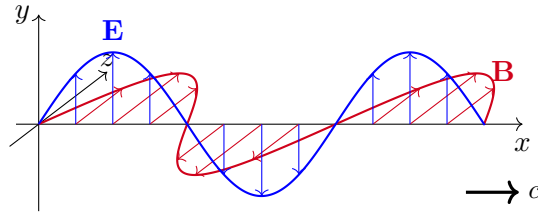
Definition 12.4.16 (Amplitude, wavenumber, frequency)

In the above, E_0 is the **amplitude**, k the **wavenumber** and ω the (angular) **frequency**. The wavelength is $\lambda = 2\pi/k$, and the wave equation forces the **dispersion relation**

$$\omega^2 = c^2 k^2.$$

To find the accompanying magnetic field, use (M3). With $\mathbf{E}$ as above, $\nabla \times \mathbf{E} = (\partial E / \partial x) \hat{\mathbf{z}}$, so $\mathbf{B} = (0, 0, B)$ with

$$\frac{\partial B}{\partial t} = -\frac{\partial E}{\partial x} \implies B = \frac{E_0}{c} \sin(kx - \omega t).$$



So $\mathbf{E}$ and $\mathbf{B}$ oscillate in phase, perpendicular to one another and to the direction of travel, with $|\mathbf{B}| = |\mathbf{E}|/c$. The magnetic field is not free to be chosen: it is determined entirely by the electric one.

For general directions of propagation it is cleaner to use complex exponentials, and take real parts at the end. Write

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}, \quad \mathbf{B} = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)},$$

where $\mathbf{k}$ is the (real) **wavevector** with $\omega = c|\mathbf{k}|$. Maxwell's equations then impose

$$\nabla \cdot \mathbf{E} = 0 \implies \mathbf{k} \cdot \mathbf{E}_0 = 0,$$

$$\nabla \cdot \mathbf{B} = 0 \implies \mathbf{k} \cdot \mathbf{B}_0 = 0,$$

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B} \implies \mathbf{k} \times \mathbf{E}_0 = \omega \mathbf{B}_0.$$

So both fields are transverse – perpendicular to the direction of propagation – and if $\mathbf{E}_0, \mathbf{B}_0$ are real then $\mathbf{k}, \mathbf{E}_0$ and $\mathbf{B}_0$ form a right-handed orthogonal triad.

§12.4.6 Polarisation

The vector $\mathbf{E}_0$ has two remaining degrees of freedom (it must be perpendicular to $\mathbf{k}$), and how it uses them is the **polarisation** of the wave.

Definition 12.4.17 (Linear polarisation)

A wave with $\mathbf{E}_0$, $\mathbf{B}_0$ and $\mathbf{k}$ all real is **linearly polarised**.

Such a wave oscillates back and forth in a fixed plane containing $\mathbf{k}$, and this is what polarising sunglasses filter for.

If instead $\mathbf{E}_0$ is complex, we may write $\mathbf{E}_0 = \alpha + i\beta$ with $\alpha, \beta \in \mathbb{R}^3$, and taking the real part,

$$\text{Re } \mathbf{E} = \alpha \cos(\mathbf{k} \cdot \mathbf{x} - \omega t) - \beta \sin(\mathbf{k} \cdot \mathbf{x} - \omega t).$$

At a fixed point in space, as t increases the tip of $\mathbf{E}$ traces out an ellipse in the plane spanned by α and β (both of which must be perpendicular to $\mathbf{k}$).

Definition 12.4.18 (Elliptic and circular polarisation)

A wave with complex $\mathbf{E}_0$ is **elliptically polarised**. In the special case $|\alpha| = |\beta|$ and $\alpha \cdot \beta = 0$, the ellipse is a circle and the wave is **circularly polarised**.

Example 12.4.19 (Why metals are shiny)

Let a conductor fill the region $x > 0$, and shine a linearly polarised light wave at it:

$$\mathbf{E}_{\text{inc}} = E_0 \hat{\mathbf{y}} e^{i(kx - \omega t)}, \quad \omega = ck.$$

Inside a conductor $\mathbf{E} = \mathbf{0}$, and the tangential component of $\mathbf{E}$ is continuous, so we need $\mathbf{E} \cdot \hat{\mathbf{y}} = 0$ at $x = 0$. The incident wave alone certainly does not satisfy this.

The fix is to add a reflected wave travelling the other way,

$$\mathbf{E}_{\text{ref}} = -E_0 \hat{\mathbf{y}} e^{i(-kx - \omega t)},$$

so that the total field $\mathbf{E} = \mathbf{E}_{\text{inc}} + \mathbf{E}_{\text{ref}}$ solves Maxwell's equations (by linearity) and vanishes on $x = 0$ (by construction). So a wave *must* be reflected – there is no consistent solution without one, and that is why metals are shiny.

Working out the corresponding magnetic fields from (M3) gives

$$\mathbf{B}_{\text{inc}} = \frac{E_0}{c} \hat{\mathbf{z}} e^{i(kx - \omega t)}, \quad \mathbf{B}_{\text{ref}} = \frac{E_0}{c} \hat{\mathbf{z}} e^{i(-kx - \omega t)},$$

so at the surface $\mathbf{B} \cdot \hat{\mathbf{z}}|_{x=0^-} = (2E_0/c)e^{-i\omega t}$, while $\mathbf{B} = \mathbf{0}$ inside. That is a discontinuity in the tangential magnetic field, which we know signals a surface current: from the jump condition,

$$\mathbf{k}_{\text{surf}} = \frac{2E_0}{\mu_0 c} \hat{\mathbf{y}} e^{-i\omega t}.$$

So the physical picture is that the incoming light drives an oscillating current in the surface of the metal, and it is that oscillating current which radiates the reflected wave.

§12.4.7 Energy Transport and the Poynting Vector

Electromagnetic waves carry energy – this is how sunlight warms the Earth. Let us make that precise by asking how the field energy in a region changes with time.

Theorem 12.4.20 (Poynting's theorem)

Let $U = \int_V \left(\frac{\epsilon_0}{2} |\mathbf{E}|^2 + \frac{1}{2\mu_0} |\mathbf{B}|^2 \right) dV$ be the field energy in a region V with boundary S . Then

$$\underbrace{\frac{dU}{dt} + \int_V \mathbf{J} \cdot \mathbf{E} dV}_{\text{rate of change of energy in } V} = \underbrace{- \int_S \mathbf{S}_P \cdot d\mathbf{S}}_{\text{energy flowing out through } \partial V}, \quad \mathbf{S}_P = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}.$$

Proof. Differentiate under the integral sign and substitute Maxwell's equations for the time derivatives:

$$\begin{aligned} \frac{dU}{dt} &= \int_V \left(\epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} + \frac{1}{\mu_0} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} \right) dV \\ &= \int_V \left(\frac{1}{\mu_0} \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \mathbf{E} \cdot \mathbf{J} - \frac{1}{\mu_0} \mathbf{B} \cdot (\nabla \times \mathbf{E}) \right) dV, \end{aligned}$$

using (M4) for $\partial \mathbf{E} / \partial t$ and (M3) for $\partial \mathbf{B} / \partial t$. Now the vector identity

$$\nabla \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{B})$$

turns the first and third terms into a total divergence, and the divergence theorem gives

$$\frac{dU}{dt} = - \int_V \mathbf{J} \cdot \mathbf{E} dV - \frac{1}{\mu_0} \int_S (\mathbf{E} \times \mathbf{B}) \cdot d\mathbf{S},$$

which rearranges to the stated form. $\square$

The interpretation is exactly a conservation law for energy, of the same shape as the continuity equation for charge. The term $\int_V \mathbf{J} \cdot \mathbf{E}$ is the rate at which the field does work on the charges in V (recall the work done on a charge q is $q\mathbf{v} \cdot \mathbf{E} \delta t$), so the left hand side is the total rate of change of energy in V , fields and particles together. The right hand side must therefore be the energy flowing in through the boundary.

Definition 12.4.21 (Poynting vector)

The **Poynting vector** is

$$\mathbf{S}_P = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B},$$

the flux of electromagnetic energy: energy per unit time per unit area.

Example 12.4.22 (Energy carried by a wave)

For a linearly polarised plane wave,

$$\mathbf{E} = \mathbf{E}_0 \sin(\mathbf{k} \cdot \mathbf{x} - \omega t), \quad \mathbf{B} = \frac{1}{c} (\hat{\mathbf{k}} \times \mathbf{E}_0) \sin(\mathbf{k} \cdot \mathbf{x} - \omega t),$$

so, using $\mathbf{E}_0 \times (\hat{\mathbf{k}} \times \mathbf{E}_0) = |\mathbf{E}_0|^2 \hat{\mathbf{k}}$ since $\mathbf{E}_0 \perp \hat{\mathbf{k}}$,

$$\mathbf{S}_P = \frac{E_0^2}{c\mu_0} \hat{\mathbf{k}} \sin^2(\mathbf{k} \cdot \mathbf{x} - \omega t).$$

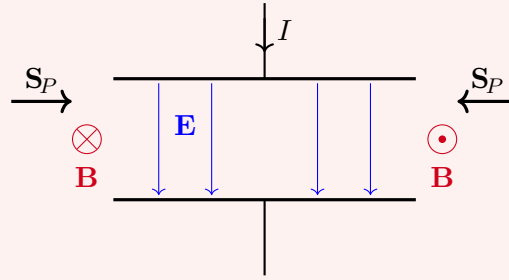
Averaging over one period $T = 2\pi/\omega$, and using $\langle \sin^2 \rangle = \frac{1}{2}$,

$$\langle \mathbf{S}_P \rangle = \frac{E_0^2}{2c\mu_0} \hat{\mathbf{k}}.$$

The energy flows in the direction of propagation, as it should, at a rate proportional to the square of the amplitude.

Example 12.4.23 (Charging a capacitor)

Here is a more surprising application. Consider a parallel plate capacitor being charged at a steady rate, so the charge is $Q(t)$ and the current into the plates is $I = \dot{Q}$.



Between the plates $\mathbf{E}$ points from the positive plate to the negative one and has magnitude $Q/\epsilon_0 A$. The displacement current generates a magnetic field circulating around the axis; by Ampère's law applied to a circle of radius r between the plates,

$$2\pi r B = \mu_0 \epsilon_0 \frac{d}{dt} \left(\frac{Q}{\epsilon_0 A} \pi r^2 \right) \implies B = \frac{\mu_0 I r}{2\pi R^2},$$

where R is the radius of the plates. So at the rim $r = R$ we find $\mathbf{S}_P = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}$ pointing radially *inwards*, with magnitude

$$|\mathbf{S}_P| = \frac{1}{\mu_0} \frac{Q}{\epsilon_0 \pi R^2} \cdot \frac{\mu_0 I}{2\pi R} = \frac{QI}{2\pi^2 \epsilon_0 R^3}.$$

Multiplying by the area of the rim, $2\pi R d$, the total power flowing in is

$$P = \frac{QId}{\pi \epsilon_0 R^2} = \frac{d}{dt} \left(\frac{Q^2 d}{2\pi \epsilon_0 R^2} \right) = \frac{d}{dt} \left(\frac{Q^2}{2C} \right),$$

which is exactly the rate at which the stored energy increases. So the energy does not flow *down the wires* into the capacitor at all – it flows in sideways through the gap, carried by the fields. This is a striking demonstration that the field picture is the right one.

Remark (Radiation pressure). Electromagnetic fields carry momentum as well as energy, with momentum density $\epsilon_0 \mathbf{E} \times \mathbf{B} = \mathbf{S}_P/c^2$. A wave absorbed by a surface therefore delivers momentum at a rate $\langle |\mathbf{S}_P| \rangle / c$ per unit area, exerting a **radiation pressure**

$$P = \frac{\langle |\mathbf{S}_P| \rangle}{c} = \frac{E_0^2}{2c^2 \mu_0}$$

(and twice that if the surface is a perfect reflector, since the momentum is reversed rather than absorbed). Sunlight at the Earth delivers roughly 1.4 kW m^{-2} , so the pressure is

about 5×10^{-6} Pa – utterly negligible for everyday purposes, but quite enough to push a solar sail, and quite enough to blow the tail of a comet away from the Sun.

§12.5 Electromagnetism and Relativity

We have already seen two hints that special relativity is lurking. The first is that Maxwell's equations have wave solutions travelling at a definite speed c , and the natural question is: with respect to which frame? The answer, and Einstein's starting point, is: all of them. The second is that the flux rule came out the same whether we moved the magnet or moved the loop – as though the split between 'electric' and 'magnetic' effects were frame-dependent.

In this final section we make this precise. We will rewrite Maxwell's equations in relativistic notation, at which point four equations become two and the theory becomes almost embarrassingly simple. Along the way we will discover that the magnetic field is not really an independent object at all: it is what the electric field looks like to a moving observer.

§12.5.1 A Review of Special Relativity

Recall from the IA Dynamics and Relativity course that in special relativity we treat space and time as a single four-dimensional object, and events are labelled by a **4-vector**

$$X^\mu = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}, \quad \mu = 0, 1, 2, 3.$$

We use Greek indices μ, ν, ρ, σ for the range $0, \dots, 3$ and Roman indices i, j, k when we mean only the spatial components $1, 2, 3$. The summation convention is in force throughout.

The geometry of spacetime is encoded in the **Minkowski metric**

$$\eta_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

which plays the role of the dot product, so that

$$X \cdot X = \eta_{\mu\nu} X^\mu X^\nu = (ct)^2 - (x^2 + y^2 + z^2)$$

is the spacetime interval. This is not positive definite – vectors can have zero or negative 'length squared' – but it is invertible, with inverse $\eta^{\mu\nu}$ numerically the same matrix.

The failure of positive-definiteness means we must be more careful than in Euclidean space about the position of indices, and it is worth understanding why. A vector, informally, is a direction; a *covector* is a linear map eating a vector and returning a number. In Euclidean space with an orthonormal basis, the metric is the identity and one can identify the two without noticing. In Minkowski space one cannot, because the metric has minus signs, and a vector's components and its associated covector's components genuinely differ. We therefore write vector components with upper indices V^μ and covector components with lower indices V_μ , and adopt the following rule.

Definition 12.5.1 (Raising and lowering)

The metric converts between the two:

$$V_\mu = \eta_{\mu\nu} V^\nu, \quad V^\mu = \eta^{\mu\nu} V_\nu.$$

Concretely, lowering an index flips the sign of the spatial components and leaves the time component alone.

The rule of the summation convention now becomes sharper: *an upper index may only be contracted with a lower index*. Contracting two upper indices is meaningless – it corresponds to trying to feed a vector to a vector.

Definition 12.5.2 (Lorentz transformation)

A **Lorentz transformation** is a change of basis Λ preserving the metric,

$$\Lambda^\mu{}_\rho \eta_{\mu\nu} \Lambda^\nu{}_\sigma = \eta_{\rho\sigma},$$

equivalently $\Lambda^T \eta \Lambda = \eta$.

The Lorentz transformations are exactly the boosts, rotations, reflections and time reversals. For instance the standard boost with velocity v along the x -axis,

$$ct' = \gamma \left(ct - \frac{v}{c} x \right), \quad x' = \gamma(x - vt), \quad y' = y, \quad z' = z, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}},$$

corresponds to

$$\Lambda^\mu{}_\nu = \begin{pmatrix} \gamma & -\gamma v/c & 0 & 0 \\ -\gamma v/c & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

while a spatial rotation R corresponds to $\Lambda = \text{diag}(1, R)$ in block form.

We can now give the working definition of these objects, which is the one we will actually use: an object is a vector, covector, or tensor according to how its components transform.

Definition 12.5.3 (Vectors, covectors, tensors)

A **4-vector** is an assignment of four numbers V^μ to each inertial frame, transforming as $V^\mu \mapsto \Lambda^\mu{}_\nu V^\nu$. A **covector** transforms as $V_\mu \mapsto \Lambda_\mu{}^\nu V_\nu$. More generally, a **tensor** of type (m, n) is an object $T^{\mu_1 \dots \mu_m}{}_{\nu_1 \dots \nu_n}$ transforming with one factor of Λ for each index, in the appropriate position.

By assumption X^μ is a vector, and by differentiating along a worldline so is dX^μ/ds for any invariant parameter s . Two derived objects we will need:

Definition 12.5.4 (4-velocity and 4-momentum)

For a particle with worldline $X^\mu(\tau)$ parameterised by proper time τ (so $dt/d\tau =$

$\gamma(\mathbf{u})$ where $\mathbf{u}$ is the ordinary velocity), the **4-velocity** and **4-momentum** are

$$U^\mu = \frac{dX^\mu}{d\tau} = \gamma \begin{pmatrix} c \\ \mathbf{u} \end{pmatrix}, \quad P^\mu = mU^\mu = \begin{pmatrix} E/c \\ \mathbf{p} \end{pmatrix},$$

where E is the relativistic energy and $\mathbf{p} = m\gamma\mathbf{u}$.

Definition 12.5.5 (4-derivative)

The **4-derivative** is

$$\partial_\mu = \frac{\partial}{\partial X^\mu} = \left(\frac{1}{c} \frac{\partial}{\partial t}, \nabla \right).$$

The index on ∂_μ is naturally a *lower* one, i.e. the derivative is a covector, which we can check by the chain rule: under $X \mapsto X' = \Lambda X$,

$$\frac{\partial}{\partial X'^\mu} = \frac{\partial X^\nu}{\partial X'^\mu} \frac{\partial}{\partial X^\nu} = (\Lambda^{-1})^\nu{}_\mu \partial_\nu = \Lambda_\mu{}^\nu \partial_\nu,$$

which is exactly the covector transformation rule. This matches the intuition from the IA courses that the derivative of a function eats a direction and returns a rate of change.

§12.5.2 The 4-Current and Charge Conservation

Our first job is to package the sources relativistically. The natural guess is to combine the charge density and current density into a single object.

Definition 12.5.6 (4-current)

The **4-current** is

$$J^\mu = \begin{pmatrix} \rho c \\ \mathbf{J} \end{pmatrix}.$$

Of course, writing four numbers in a column does not make them a 4-vector; we should check that they transform correctly. The full verification is a good exercise, but the following special case makes it entirely believable. Suppose in frame S we have a static charge density $\rho_0(\mathbf{x})$ with no current, so $J^\mu = (\rho_0 c, \mathbf{0})$. In a frame S' boosted with velocity $\mathbf{v}$, the transformation rule would give

$$J'^\mu = \Lambda^\mu{}_\nu J^\nu = \begin{pmatrix} \gamma \rho_0 c \\ -\gamma \rho_0 \mathbf{v} \end{pmatrix}.$$

Both entries are exactly what we would expect physically. The charge density has gone up by a factor of γ , which is right: charge is invariant, but lengths in the direction of motion are Lorentz contracted, so the same charge occupies less volume. And the current density is that charge density times the velocity $-\mathbf{v}$ at which the charge now appears to move.

With this notation, local charge conservation takes a startlingly compact form.

Proposition 12.5.7

The continuity equation is

$$\partial_\mu J^\mu = 0,$$

and this equation is Lorentz invariant.

Proof. Expanding, $\partial_\mu J^\mu = \frac{1}{c} \frac{\partial(\rho c)}{\partial t} + \nabla \cdot \mathbf{J} = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J}$, which is the continuity equation. Invariance is immediate from the index structure: every index is contracted, one up with one down, so the object is a scalar. $\square$

This is the first indication of how much the relativistic notation buys us. The statement ‘charge is conserved locally’ and the statement ‘all observers agree that charge is conserved locally’ are now the same statement, and both are one symbol long.

§12.5.3 Gauge Potentials and the Field Tensor

In the static case we introduced potentials ϕ with $\mathbf{E} = -\nabla\phi$ and $\mathbf{A}$ with $\mathbf{B} = \nabla \times \mathbf{A}$. In the time-dependent case the first of these fails, since $\nabla \times \mathbf{E} \neq \mathbf{0}$ in general. But (M3) says $\nabla \times \mathbf{E} = -\partial_t(\nabla \times \mathbf{A})$, i.e. $\nabla \times (\mathbf{E} + \partial_t \mathbf{A}) = \mathbf{0}$, so it is $\mathbf{E} + \partial_t \mathbf{A}$ which is a gradient. Hence we should define the potentials by

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}.$$

With this definition, (M2) and (M3) hold automatically, and the whole content of Maxwell’s equations sits in (M1) and (M4).

The potentials are still not unique. For any function $\chi(\mathbf{x}, t)$, the transformation

$$\mathbf{A} \mapsto \mathbf{A} + \nabla\chi, \quad \phi \mapsto \phi - \frac{\partial\chi}{\partial t}$$

leaves both $\mathbf{E}$ and $\mathbf{B}$ unchanged, as one checks directly. This is the time-dependent version of the gauge transformations we met in magnetostatics.

Now, we have one scalar and one 3-vector, which is four functions. The obvious thing to do is the right thing.

Definition 12.5.8 (4-potential)

The **gauge potential** is the 4-vector

$$A^\mu = \begin{pmatrix} \phi/c \\ \mathbf{A} \end{pmatrix}.$$

Lowering the index gives $A_\mu = (\phi/c, -\mathbf{A})$, and the gauge transformation becomes simply

$$A_\mu \mapsto A_\mu - \partial_\mu \chi.$$

This one line contains both of the transformations above; the reader is encouraged to check it.

Since $\mathbf{E}$ and $\mathbf{B}$ are built from derivatives of A_μ , and since we want a gauge-invariant object, the following definition suggests itself.

Definition 12.5.9 (Electromagnetic field tensor)

The **electromagnetic field tensor** is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

This is antisymmetric, $F_{\mu\nu} = -F_{\nu\mu}$, so its diagonal vanishes and it has $(16 - 4)/2 = 6$ independent components – precisely the right number to hold the three components of $\mathbf{E}$ and the three of $\mathbf{B}$. It is gauge invariant, since under $A_\mu \mapsto A_\mu - \partial_\mu \chi$,

$$F_{\mu\nu} \mapsto F_{\mu\nu} - \partial_\mu \partial_\nu \chi + \partial_\nu \partial_\mu \chi = F_{\mu\nu}$$

by the symmetry of mixed partials. So F is a genuinely physical object, while A is not.

Let us identify the components. Taking $\mu = 0, \nu = 1$ and remembering that the spatial part of A_μ carries a minus sign,

$$F_{01} = \frac{1}{c} \frac{\partial}{\partial t} (-A_x) - \frac{\partial}{\partial x} \left(\frac{\phi}{c} \right) = \frac{1}{c} \left(-\frac{\partial A_x}{\partial t} - \frac{\partial \phi}{\partial x} \right) = \frac{E_x}{c},$$

and similarly $F_{02} = E_y/c$, $F_{03} = E_z/c$. For the purely spatial components,

$$F_{12} = \frac{\partial}{\partial x} (-A_y) - \frac{\partial}{\partial y} (-A_x) = -(\nabla \times \mathbf{A})_z = -B_z,$$

and cyclically. Assembling,

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix}.$$

Raising both indices with $F^{\mu\nu} = \eta^{\mu\rho} \eta^{\nu\sigma} F_{\rho\sigma}$ flips the sign of any component with exactly one spatial index, so

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}.$$

The electric entries change sign (one spatial index) and the magnetic ones do not (two spatial indices, so two sign flips).

§12.5.4 Transformation of the Fields

The whole point of assembling $F^{\mu\nu}$ is that we now know how $\mathbf{E}$ and $\mathbf{B}$ transform: as a tensor,

$$F'^{\mu\nu} = \Lambda^\mu_\rho \Lambda^\nu_\sigma F^{\rho\sigma}.$$

Let us extract what this means in familiar language.

For a rotation $\Lambda = \text{diag}(1, R)$ one gets, unsurprisingly, $\mathbf{E}' = R\mathbf{E}$ and $\mathbf{B}' = R\mathbf{B}$: both fields are ordinary 3-vectors as far as rotations are concerned. For a boost with speed v along the x -axis, grinding through the matrix multiplication gives the following.

Proposition 12.5.10 (Transformation of $\mathbf{E}$ and $\mathbf{B}$)

Under a boost with velocity v along $\hat{\mathbf{x}}$,

$$\begin{aligned} E'_x &= E_x, & B'_x &= B_x, \\ E'_y &= \gamma(E_y - vB_z), & B'_y &= \gamma\left(B_y + \frac{v}{c^2}E_z\right), \\ E'_z &= \gamma(E_z + vB_y), & B'_z &= \gamma\left(B_z - \frac{v}{c^2}E_y\right). \end{aligned}$$

The components along the boost are unchanged, and the transverse components mix. Crucially, they mix $\mathbf{E}$ with $\mathbf{B}$: a purely electric field in one frame has a magnetic part in another. This is the sense in which the two fields are not separate things.

Example 12.5.11 (The boosted line charge)

Consider an infinite line of charge along the x -axis with charge per unit length η , at rest in frame S . We computed its field long ago:

$$\mathbf{E} = \frac{\eta}{2\pi\epsilon_0(y^2 + z^2)} \begin{pmatrix} 0 \\ y \\ z \end{pmatrix}, \quad \mathbf{B} = \mathbf{0}.$$

Now boost with velocity v along the wire. Applying the transformation rules, and noting that $y' = y$ and $z' = z$ since the boost is along x ,

$$\mathbf{E}' = \frac{\gamma\eta}{2\pi\epsilon_0(y'^2 + z'^2)} \begin{pmatrix} 0 \\ y' \\ z' \end{pmatrix}, \quad \mathbf{B}' = \frac{\gamma\eta v}{2\pi\epsilon_0 c^2(y'^2 + z'^2)} \begin{pmatrix} 0 \\ z' \\ -y' \end{pmatrix}.$$

Look at what these say. The electric field is that of a line charge with density $\gamma\eta$ – which is right, because in S' the charge is Lorentz contracted. And writing $s' = \sqrt{y'^2 + z'^2}$ and $\hat{\phi}' = (0, -z', y')/s'$, and using $\epsilon_0 c^2 = 1/\mu_0$, the magnetic field is

$$\mathbf{B}' = \frac{\mu_0 I'}{2\pi s'} \hat{\phi}', \quad I' = -\gamma\eta v,$$

which is precisely the field of a straight wire carrying current I' , exactly as we computed from Ampère's law.

It is worth being clear about how remarkable this is. We did not use Ampère's law, or the Biot–Savart law, or anything magnetic at all. We used Gauss' law and a Lorentz boost. Magnetism, in other words, is what you get when you look at electrostatics from a moving frame. Relativity is not only a theory of things going very fast; it is there every time you stick a magnet on the fridge.

§12.5.5 The Field of a Moving Charge

Let us do the same for a point charge, which is a more delicate calculation but gives a memorable answer.

Example 12.5.12 (Boosted point charge)

A charge Q at rest at the origin of S has

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0(x^2 + y^2 + z^2)^{3/2}} \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad \mathbf{B} = \mathbf{0}.$$

Boosting with velocity v along $\hat{\mathbf{x}}$, the transformation rules give $E'_x = E_x$ and $E'_{y,z} = \gamma E_{y,z}$, so

$$\mathbf{E}' = \frac{Q}{4\pi\epsilon_0(x^2 + y^2 + z^2)^{3/2}} \begin{pmatrix} x \\ \gamma y \\ \gamma z \end{pmatrix},$$

but this is expressed in the wrong coordinates. Substituting $x = \gamma(x' + vt')$, $y = y'$, $z = z'$,

$$\mathbf{E}' = \frac{\gamma Q}{4\pi\epsilon_0(\gamma^2(x' + vt')^2 + y'^2 + z'^2)^{3/2}} \begin{pmatrix} x' + vt' \\ y' \\ z' \end{pmatrix}.$$

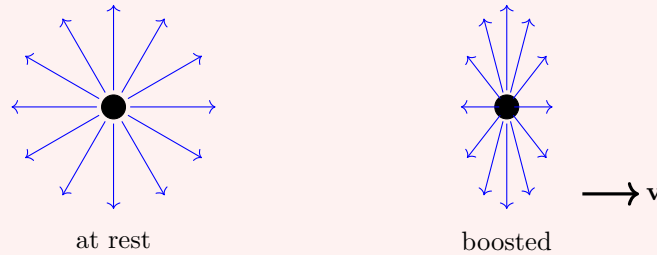
In S' the particle sits at $(-vt', 0, 0)$, so this is a radial field centred on the particle, as it should be. Evaluate at $t' = 0$, where the particle is at the origin of S' and $\mathbf{r}' = (x', y', z')$. Writing θ for the angle between $\mathbf{r}'$ and the x' -axis, the denominator simplifies:

$$\begin{aligned} \gamma^2 x'^2 + y'^2 + z'^2 &= (\gamma^2 - 1)x'^2 + r'^2 = \frac{v^2\gamma^2}{c^2}x'^2 + r'^2 \\ &= \left(\frac{v^2\gamma^2}{c^2} \cos^2 \theta + 1\right) r'^2 = \gamma^2 \left(1 - \frac{v^2}{c^2} \sin^2 \theta\right) r'^2, \end{aligned}$$

using $\gamma^2 - 1 = \gamma^2 v^2/c^2$ in the second step and $\gamma^{-2} = 1 - v^2/c^2$ in the last. Hence

$$\mathbf{E}' = \frac{1}{\gamma^2 \left(1 - \frac{v^2}{c^2} \sin^2 \theta\right)^{3/2}} \frac{Q}{4\pi\epsilon_0 r'^2} \hat{\mathbf{r}}'.$$

The field is still radial, but no longer isotropic. The prefactor is smallest at $\theta = 0$ (ahead of and behind the charge) and largest at $\theta = \pi/2$ (to the sides), so the field is squashed in the direction of motion and enhanced perpendicular to it. Drawing field lines, the effect is a ‘pancaking’ of the field:



There is also a magnetic field, which one obtains from the same transformation rules:

$$\mathbf{B}' = \frac{\mu_0 \gamma Q v}{4\pi(\gamma^2(x' + vt')^2 + y'^2 + z'^2)^{3/2}} \begin{pmatrix} 0 \\ z' \\ -y' \end{pmatrix},$$

circulating around the direction of motion, as one would expect of a moving charge.

As a historical aside: this squashing of the field of a moving charge was found by Lorentz by solving Maxwell's equations directly, before relativity existed, and it is precisely what led him to write down the transformations that bear his name.

§12.5.6 Lorentz Invariants

Since $\mathbf{E}$ and $\mathbf{B}$ mix under boosts, it is natural to ask whether any combination of them is agreed upon by all observers. With index notation this is easy to answer: we need to contract all indices away.

The first candidate is the obvious one.

Proposition 12.5.13

The quantity

$$\frac{1}{2}F_{\mu\nu}F^{\mu\nu} = |\mathbf{B}|^2 - \frac{|\mathbf{E}|^2}{c^2}$$

is Lorentz invariant.

Proof. Invariance is automatic, since all indices are contracted. The evaluation is a direct computation from the matrices above: the electric entries appear as (something) $\times$ (−something), contributing negatively, and the magnetic ones contribute positively. $\square$

For the second invariant we need a new object. Define the totally antisymmetric symbol in four dimensions,

$$\varepsilon^{\mu\nu\rho\sigma} = \begin{cases} +1 & \mu\nu\rho\sigma \text{ an even permutation of } 0123, \\ -1 & \text{an odd permutation,} \\ 0 & \text{otherwise,} \end{cases}$$

the spacetime analogue of ϵ_{ijk} . Is it a tensor? Transforming it gives another totally antisymmetric object, and (exactly as in $\mathbb{R}^3$) the space of such objects is one-dimensional, so $\varepsilon'^{\mu\nu\rho\sigma} = a \varepsilon^{\mu\nu\rho\sigma}$ for some constant a . Testing on a single component,

$$\varepsilon'^{0123} = \Lambda^0_{\kappa} \Lambda^1_{\lambda} \Lambda^2_{\alpha} \Lambda^3_{\beta} \varepsilon^{\kappa\lambda\alpha\beta} = \det \Lambda,$$

by the definition of the determinant. Lorentz transformations have $\det \Lambda = \pm 1$, and restricting to the ones continuously connected to the identity (rotations and boosts, but not reflections or time reversal) we have $\det \Lambda = 1$ and ε is indeed invariant.

Definition 12.5.14 (Dual field tensor)

The **dual electromagnetic tensor** is

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} F_{\rho\sigma} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z/c & -E_y/c \\ B_y & -E_z/c & 0 & E_x/c \\ B_z & E_y/c & -E_x/c & 0 \end{pmatrix}.$$

The factor of $\frac{1}{2}$ is there because each pair $\{\rho, \sigma\}$ is counted twice in the sum. Comparing with $F^{\mu\nu}$, the dual is obtained by the substitution

$$\mathbf{E} \mapsto c\mathbf{B}, \quad \mathbf{B} \mapsto -\mathbf{E}/c,$$

and the reader should note the minus sign: the duality is like a rotation by ninety degrees, and doing it twice gives minus the identity. In particular $\tilde{F}_{\mu\nu}\tilde{F}^{\mu\nu} = -F_{\mu\nu}F^{\mu\nu}$, so contracting the dual with itself gives nothing new. Contracting it with F , however, does.

Proposition 12.5.15

The quantity

$$\frac{1}{4}\tilde{F}^{\mu\nu}F_{\mu\nu} = \frac{\mathbf{E} \cdot \mathbf{B}}{c}$$

is Lorentz invariant.

So all observers agree on $|\mathbf{B}|^2 - |\mathbf{E}|^2/c^2$ and on $\mathbf{E} \cdot \mathbf{B}$. In particular, if the fields are perpendicular in one frame they are perpendicular in every frame, and if $|\mathbf{E}| = c|\mathbf{B}|$ in one frame – as for a plane wave – then the same holds in all frames.

§12.5.7 Maxwell's Equations in Covariant Form

We now have everything we need to write down the theory in its final form.

Theorem 12.5.16 (Maxwell's equations, relativistic form)

Maxwell's four equations are equivalent to the pair

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu, \quad \partial_\mu \tilde{F}^{\mu\nu} = 0.$$

Proof. We check each equation component by component. Take the first, with $\nu = 0$. Since $F^{00} = 0$, only spatial μ contribute, and $F^{i0} = E_i/c$, so

$$\partial_i F^{i0} = \nabla \cdot \left(\frac{\mathbf{E}}{c} \right) = \mu_0 J^0 = \mu_0 \rho c,$$

i.e. $\nabla \cdot \mathbf{E} = \mu_0 c^2 \rho = \rho/\epsilon_0$, which is (M1). Taking instead $\nu = i$ spatial, the $\mu = 0$ term gives $\frac{1}{c}\partial_t(-E_i/c)$ and the spatial terms assemble into $(\nabla \times \mathbf{B})_i$, so

$$-\frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} + \nabla \times \mathbf{B} = \mu_0 \mathbf{J},$$

which is (M4).

Now the second equation. With $\nu = 0$ we get $\partial_i \tilde{F}^{i0} = \nabla \cdot \mathbf{B} = 0$, which is (M2); with $\nu = i$ we get $\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E} = \mathbf{0}$, which is (M3). $\square$

Two equations, each one line long, replacing four. It is hard to overstate how much cleaner this is: the constants c , ϵ_0 , μ_0 have been almost entirely absorbed into the definitions, and the equations are manifestly Lorentz covariant, since both sides carry a single free upper index and so transform the same way.

Several facts we worked for now become immediate.

First, charge conservation. Since $F^{\mu\nu}$ is antisymmetric and $\partial_\nu\partial_\mu$ is symmetric, $\partial_\nu\partial_\mu F^{\mu\nu} = 0$ identically, and hence

$$\partial_\nu J^\nu = 0.$$

Compare this with the fiddly computation we did in Section 12.4.4.

Secondly, the second equation is automatic once we have potentials. If $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ then

$$\partial_\mu \tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} \partial_\mu F_{\rho\sigma} = \frac{1}{2} \varepsilon^{\mu\nu\rho\sigma} \partial_\mu (\partial_\rho A_\sigma - \partial_\sigma A_\rho) = 0,$$

since ε is antisymmetric in $\mu\rho$ while $\partial_\mu\partial_\rho$ is symmetric. So the entire theory reduces to

$$\partial_\mu F^{\mu\nu} = \mu_0 J^\nu \quad \text{where} \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Thirdly, the asymmetry between the two equations – a source on one, zero on the other – is exactly the statement that there are electric charges but no magnetic ones.

§12.5.8 The Lorenz Gauge

In magnetostatics we exploited the freedom in $\mathbf{A}$ by choosing Coulomb gauge, $\nabla \cdot \mathbf{A} = 0$, which turned the field equation into Poisson's equation. There is a relativistic version of the same trick, and it is prettier.

Definition 12.5.17 (Lorenz gauge)

A gauge potential satisfying

$$\partial_\mu A^\mu = 0$$

is said to be in **Lorenz gauge**^a. In three-dimensional language this reads $\nabla \cdot \mathbf{A} + \frac{1}{c^2} \frac{\partial \phi}{\partial t} = 0$.

^aAfter Ludvig Lorenz, who is not Hendrik Lorentz of the Lorentz transformations. The near-coincidence of names is one of the crueller accidents in the history of physics.

Unlike Coulomb gauge, this condition is Lorentz invariant: it involves only a contracted pair of indices, so an observer in any frame will agree that it holds.

Proposition 12.5.18

Any gauge potential can be brought into Lorenz gauge.

Proof. Suppose $\partial_\mu A^\mu = \psi$. Under a gauge transformation $A_\mu \mapsto A_\mu - \partial_\mu \chi$ we get

$$\partial_\mu A^\mu \mapsto \psi - \partial_\mu \partial^\mu \chi,$$

so we need to solve $\square \chi = \psi$, where $\square = \partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2$ is the **d'Alembertian**. This is the inhomogeneous wave equation, which always has solutions. $\square$

The payoff is immediate. Substituting $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ into the Maxwell equation,

$$\mu_0 J^\nu = \partial_\mu F^{\mu\nu} = \partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu) = \square A^\nu,$$

the second term vanishing precisely because we are in Lorenz gauge. So the whole of electromagnetism reduces to

Theorem 12.5.19

In Lorenz gauge, Maxwell's equations are

$$\square A^\nu = \mu_0 J^\nu.$$

Four decoupled wave equations, one for each component of the potential, sourced by the corresponding component of the current. In empty space $J^\nu = \mathbf{0}$ and this is $\square A^\nu = 0$: the potential propagates as a wave at speed c , which is our electromagnetic waves reappearing in a much more economical form. And the fact that the equation is a *wave* equation rather than Poisson's equation says something physically important – changing a charge here does not instantaneously change the field there. Influence propagates at the speed of light.

Remark. The Lorenz gauge does not fix A^μ completely. We may still perform a further gauge transformation with any χ satisfying $\square\chi = 0$, and remain in Lorenz gauge. This residual freedom is what allows one to argue that an electromagnetic wave has only two physical polarisation states, rather than the four one might naively count from A^μ – exactly the two we found in Section 4.

§12.5.9 The Lorentz Force in Covariant Form

One thing remains: the force law. In non-relativistic notation it is

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}),$$

and we should like a version which is manifestly covariant. The right hand side involves $\mathbf{E}$, $\mathbf{B}$ and $\mathbf{u}$, which we now know how to package as $F^{\mu\nu}$ and U^μ ; the left is a rate of change of momentum, which becomes $dP^\mu/d\tau$. There is essentially only one way to combine these.

Law 12.5.20 (Lorentz force law, relativistic form)

A particle of charge q obeys

$$\frac{dP^\mu}{d\tau} = q F^{\mu\nu} U_\nu.$$

Proposition 12.5.21

This is equivalent to the Lorentz force law together with the work-energy relation.

Proof. Take $\mu = i$ spatial. Using $U_\nu = \gamma(c, -\mathbf{u})$ and reading off $F^{i\nu}$ from the matrix,

$$\frac{d\mathbf{p}}{d\tau} = q\gamma(\mathbf{E} + \mathbf{u} \times \mathbf{B}),$$

and since $dt/d\tau = \gamma$, the chain rule gives

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B}),$$

the familiar law – with the understanding that $\mathbf{p} = m\gamma\mathbf{u}$ is the relativistic momentum.

Now take $\mu = 0$. Then $F^{0\nu}U_\nu = -\frac{\mathbf{E}}{c} \cdot (-\gamma\mathbf{u}) = \frac{\gamma}{c}\mathbf{E} \cdot \mathbf{u}$, so

$$\frac{dP^0}{d\tau} = \frac{1}{c} \frac{dE}{d\tau} = \frac{q\gamma}{c} \mathbf{E} \cdot \mathbf{u} \implies \frac{dE}{dt} = q\mathbf{E} \cdot \mathbf{u},$$

which is the statement that the electric field does work at a rate $q\mathbf{E} \cdot \mathbf{u}$ – and, incidentally, that the magnetic field does none. $\square$

So the time component of the covariant force law, which has no non-relativistic analogue as a separate statement, is the energy equation. This is a recurring theme in relativity: equations that look independent in three dimensions turn out to be components of a single four-dimensional statement.

We close with two standard applications.

Example 12.5.22 (Motion in a constant electric field)

Take $\mathbf{E} = (E, 0, 0)$ and start the particle from rest, so the motion stays along $\hat{\mathbf{x}}$. The equation of motion is

$$m \frac{d(\gamma u)}{dt} = qE \implies m\gamma u = qEt,$$

taking $u = 0$ at $t = 0$. Solving for u ,

$$u = \frac{dx}{dt} = \frac{qEt}{\sqrt{m^2 + q^2 E^2 t^2 / c^2}}.$$

Note that $u \rightarrow c$ as $t \rightarrow \infty$, but never reaches it: no amount of pushing gets a massive particle to the speed of light. Integrating,

$$x = \frac{mc^2}{qE} \left(\sqrt{1 + \frac{q^2 E^2 t^2}{m^2 c^2}} - 1 \right),$$

and for small t this expands as $x \approx \frac{qE}{2m} t^2$, which is the Newtonian answer, as it must be.

Example 12.5.23 (Motion in a constant magnetic field)

Take $\mathbf{B} = (0, 0, B)$ with $\mathbf{E} = \mathbf{0}$. The $\mu = 0$ equation gives

$$\frac{dP^0}{d\tau} = 0 \implies E = m\gamma c^2 \text{ is constant,}$$

so γ , and hence $|\mathbf{u}|$, is constant – as we knew, since magnetic fields do no work. The spatial equation is then

$$m\gamma \frac{d\mathbf{u}}{dt} = q\mathbf{u} \times \mathbf{B},$$

the γ coming outside the derivative because it is constant. This is the equation we solved in Dynamics and Relativity, except for the factor of γ : the particle moves in

a circle (or a helix, if it also has velocity along $\mathbf{B}$) with angular frequency

$$\omega = \frac{qB}{m\gamma}.$$

The relativistic correction is what makes a cyclotron stop working at high energies, and is why one builds synchrotrons instead.

That is the end of the course. It is worth taking stock of where we have got to. We began with four experimental laws about wires and magnets and charges, wrote them as differential equations, and found that they contained – as inevitable consequences – the existence of light, the impossibility of magnetic monopoles, the conservation of charge, and finally the structure of spacetime itself. Very little in mathematics repays this kind of attention so generously.

13 Fluid Dynamics

Fluids are everywhere. Air, water, blood, magma, the interstellar medium and the honey on your toast are all fluids, and the same handful of equations governs all of them. What makes something a fluid rather than a solid is a matter of how it responds to being pushed sideways: lean on a wall and it deforms a little and then stops, but lean on air or water and it simply keeps on moving. A solid resists shear with a finite deformation; a fluid cannot resist shear at all, and so we stop asking how far a fluid has moved and start asking how fast it is moving. Velocity, not displacement, is the central object of this subject.

This course is an introduction to that story. We will begin with the simplest flows imaginable – fluid dragged between two flat plates – and derive their equations of motion from scratch. We will then develop the general kinematic language of fluid motion, state the Navier–Stokes equations, and spend most of our time exploiting the two great simplifications available to us: that many flows of interest are effectively *inviscid* (viscosity is negligible), and that many inviscid flows are *irrotational* (they carry no local spin), in which case the whole problem collapses to Laplace’s equation and the machinery of vector calculus takes over. Along the way we will find out why aeroplanes fly, why a spinning ball swerves, how fast a tsunami travels, and why cyclones go around the wrong way in Australia.

This article constitutes my notes for the ‘Fluid Dynamics’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered¹. If you spot any errors or have any feedback, I can be contacted at ak2316@cam.ac.uk.

§13.1 What is a Fluid?

Before we can do any mathematics we need to decide what it is that we are modelling, and the honest answer is that we are going to model something that does not quite exist.

§13.1.1 The Continuum Hypothesis

A real fluid is made of molecules. A cubic millimetre of air at room temperature contains something like 2.5×10^{16} of them, each careering about at several hundred metres per second and colliding with its neighbours every nanosecond or so. Nobody wants to solve 10^{16} coupled equations of motion, and in any case we are almost never interested in the answer: what we want to know is how the air as a whole moves, on scales of centimetres and seconds.

So we make an idealisation.

¹We assume throughout a working knowledge of the material in the ‘Vector Calculus’ course – in particular the divergence theorem, Stokes’ theorem, the vector identities for $\nabla \times (\mathbf{a} \times \mathbf{b})$ and friends, and the expressions for ∇ , $\nabla \cdot$ and ∇^2 in cylindrical and spherical coordinates. Cross-references to my notes for that course appear throughout.

Definition 13.1.1 (Continuum Hypothesis)

The **continuum hypothesis** is the assumption that a fluid may be treated as a continuous medium, so that at every point $\mathbf{x}$ of the region occupied by the fluid and at every time t we may speak of a density $\rho(\mathbf{x}, t)$, a pressure $p(\mathbf{x}, t)$ and a velocity $\mathbf{u}(\mathbf{x}, t)$, all varying smoothly with $\mathbf{x}$ and t .

What justifies this? The idea is that we choose a length scale ℓ which is enormous compared with the spacing between molecules, but tiny compared with the scale L on which the flow varies. A blob of side ℓ then contains so many molecules that its average density is a perfectly well-defined and steady number – the statistical fluctuations are of relative size $N^{-1/2}$, which for $N \sim 10^{10}$ is utterly negligible – and yet the blob is so small compared with L that we may pretend it is a point. The ‘fluid particle’ at $\mathbf{x}$ is really such a blob, and $\mathbf{u}(\mathbf{x}, t)$ is the mean velocity of the molecules within it.

For everyday flows this works beautifully. It begins to fail in very rarefied gases (the upper atmosphere, where the mean free path is metres) and at very small scales, and we will say no more about that.

Two properties of the fluid enter our equations, and we should say what they are. The **density** ρ is the mass per unit volume; for water $\rho \approx 1000 \text{ kg m}^{-3}$ and for air $\rho \approx 1 \text{ kg m}^{-3}$. The other is the viscosity, which we come to in a moment.

§13.1.2 Stress, Strain and Newtonian Fluids

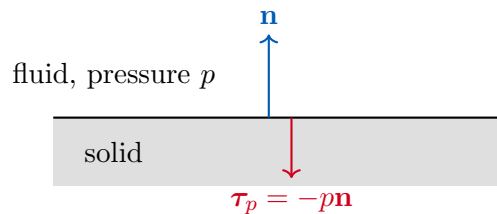
Forces in a fluid come in two flavours. A **body force** acts on the bulk of the fluid, and is specified as a force per unit volume $\mathbf{f}$; gravity, with $\mathbf{f} = \rho\mathbf{g}$, is the only one we shall meet. A **surface force** acts across a surface – either a genuine physical boundary, or an imaginary surface that we have drawn inside the fluid for the purposes of calculation – and is specified as a force per unit area.

Definition 13.1.2 (Stress)

A **stress** is a force per unit area exerted across a surface. If $\mathbf{n}$ is the unit normal to the surface, the stress is resolved into a **normal stress** parallel to $\mathbf{n}$ and a **tangential** (or **shear**) **stress** perpendicular to it.

The normal stress is one we already know about. If a fluid of pressure p lies on one side of a surface with unit normal $\mathbf{n}$ pointing *into* the fluid, then the fluid pushes on the surface with a force per unit area

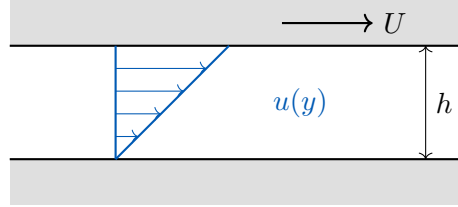
$$\boldsymbol{\tau}_p = -p\mathbf{n}.$$



Pressure by itself does nothing very interesting: it pushes equally in all directions and the net effect on a small blob cancels. What drives a flow is a pressure *gradient*. Integrating the pressure force over the surface of a small volume and applying the divergence

theorem, the net force per unit volume is $-\nabla p$, so fluid is pushed from high pressure towards low pressure.

Tangential stresses are the new ingredient, and they are what distinguishes a fluid from an ideal ‘no friction’ medium. Consider the classic experiment: two large parallel plates a distance h apart with fluid in between, the lower held fixed and the upper dragged along at speed U .



To keep the upper plate moving we must keep pushing it, so the fluid exerts a retarding tangential stress on it. Experiment tells us that for air, water, oil and most other simple fluids this stress is directly proportional to U/h .

Definition 13.1.3 (Newtonian Fluid, Viscosity)

A **Newtonian fluid** is one in which the tangential stress is proportional to the rate of strain, that is, to the velocity gradient. In the configuration above the tangential stress is

$$\tau_s = \mu \frac{U}{h},$$

and the constant of proportionality μ is the **dynamic viscosity** of the fluid.

The word ‘strain’ here means extension per unit length, and the *rate* of strain is its time derivative, which is exactly a velocity gradient. Air, water and oil are Newtonian. Paint, toothpaste, blood and shampoo are **non-Newtonian**: their long chain molecules give them stress-strain relationships that are nonlinear, or that depend on the history of the flow, and they are far harder to model. Sand and rice are **granular** and are not really fluids at all, though they sometimes behave like them. We will only ever consider Newtonian fluids.

Dimensionally, $[\tau_s] = ML^{-1}T^{-2}$ and $[U/h] = T^{-1}$, so

$$[\mu] = ML^{-1}T^{-1},$$

measured in $\text{kg m}^{-1} \text{s}^{-1}$. It is a property of the fluid alone. For water $\mu \approx 10^{-3} \text{ kg m}^{-1} \text{s}^{-1}$ and for air $\mu \approx 10^{-5} \text{ kg m}^{-1} \text{s}^{-1}$.

In general, for a flow with velocity u_T tangential to some surface with inward normal $\mathbf{n}$, the tangential stress exerted by the fluid on the surface is

$$\tau_s = \mu \frac{\partial u_T}{\partial n},$$

directed along the tangential component of the velocity. This is the definition we shall actually use.

Remark. In full generality the stress is a rank-2 tensor field σ_{ij} , and the Newtonian constitutive relation reads $\sigma_{ij} = -p\delta_{ij} + 2\mu e_{ij}$ where e_{ij} is the rate-of-strain tensor. We will not need this machinery in this course – we will only ever consider situations simple enough that we can identify ‘the’ shear stress by inspection – but it is what makes the arguments below rigorous, and it is the starting point of the Part II course.

§13.2 Parallel Viscous Flow

We are going to begin with the simplest flows there are, and derive their equations of motion honestly from a force balance. This has the great virtue that when we later write down the Navier–Stokes equations without deriving them, we will at least recognise them as a generalisation of something we understand.

Definition 13.2.1 (Steady, Parallel Flow)

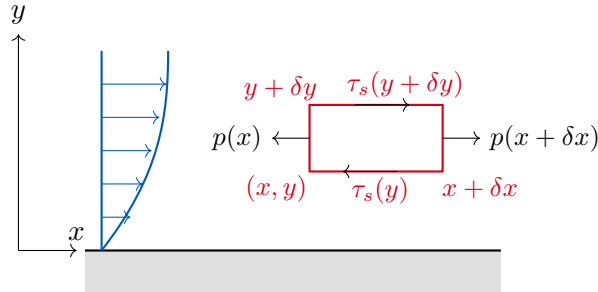
A flow is **steady** if the velocity at each fixed point does not change in time, so that $\partial \mathbf{u} / \partial t = \mathbf{0}$. A flow is **parallel** if the fluid moves in one direction only, so that (choosing axes appropriately)

$$\mathbf{u} = (u(y, t), 0, 0).$$

Note the claim buried in the definition: the velocity of a parallel flow does not depend on x . This is forced on us by incompressibility. If u increased with x then fluid would be leaving any given slab faster than it arrives, and fluid would have to be created or destroyed to compensate. We will prove this properly in Section 13.3; for now we take it as read.

§13.2.1 The Equations of Parallel Flow

Consider a small rectangular box of fluid, of side δx by δy and unit width in the z direction, sitting inside a steady parallel flow.



The box moves in the x direction without accelerating, so the forces on it must balance. In the x direction there are normal stresses (pressures) on the two vertical faces, and tangential stresses on the two horizontal faces. Being careful with signs, the net force per unit width is

$$p(x) \delta y - p(x + \delta x) \delta y + \tau_s(y + \delta y) \delta x + \tau_s(y) \delta x = 0.$$

Now the normal to the top face points upwards *into* the fluid above, and the normal to the bottom face points downwards, so by our sign convention

$$\tau_s(y + \delta y) = \mu \frac{\partial u}{\partial y}(y + \delta y), \quad \tau_s(y) = -\mu \frac{\partial u}{\partial y}(y).$$

Dividing through by $\delta x \delta y$ we get

$$-\frac{p(x + \delta x) - p(x)}{\delta x} + \mu \frac{1}{\delta y} \left(\frac{\partial u}{\partial y}(y + \delta y) - \frac{\partial u}{\partial y}(y) \right) = 0,$$

and letting $\delta x, \delta y \rightarrow 0$,

$$-\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} = 0.$$

The same argument in the y direction, where nothing at all is happening, gives $\partial p / \partial y = 0$.

There is no reason to restrict ourselves to steady flow with no body forces. If the flow is unsteady, the left hand side of the force balance becomes mass times acceleration, $\rho \delta x \delta y \partial u / \partial t$; and if there is a body force per unit volume $(f_x, f_y, 0)$ we should add $f_x \delta x \delta y$ and $f_y \delta x \delta y$ to the two equations. Doing so gives the following.

Proposition 13.2.2 (Equations of Parallel Flow)

For a parallel flow $\mathbf{u} = (u(y, t), 0, 0)$ of a Newtonian fluid of constant density ρ and viscosity μ , subject to a body force per unit volume $(f_x, f_y, 0)$,

$$\begin{aligned} \rho \frac{\partial u}{\partial t} &= -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} + f_x, \\ 0 &= -\frac{\partial p}{\partial y} + f_y. \end{aligned}$$

Notice what the second equation says: across the flow, the pressure is purely hydrostatic. All the dynamics is in the first equation, which is a diffusion equation for u forced by the pressure gradient. Notice too that these equations are *linear* in u , so we may superpose solutions – a fact we will use immediately.

Finally, we need boundary conditions. There are two possibilities.

Definition 13.2.3 (Boundary Conditions for Viscous Flow)

At a boundary, a viscous fluid satisfies one of:

- (i) the **no-slip condition**, that the tangential velocity of the fluid equals the tangential velocity of the boundary. In particular $u_T = 0$ at a stationary rigid boundary.
- (ii) the **stress condition**, that a prescribed tangential stress τ is applied, so that $-\mu \partial u_T / \partial n = \tau$.

No-slip is the appropriate condition at a solid boundary: fluid really does stick to solids, which is why dust does not blow off a fan blade. The stress condition is appropriate at a boundary between two fluids, where the stresses must match; in the important special case of a liquid with air above it, the air is so much less viscous that we may take $\tau = 0$ and speak of a **free surface**.

§13.2.2 Couette and Poiseuille Flow

The two canonical parallel flows are driven, respectively, by a moving boundary and by a pressure gradient.

Example 13.2.4 (Plane Couette Flow)

Fluid occupies $0 < y < h$ between two plates; the lower is at rest, the upper moves with speed U . There is no pressure gradient and no body force, and the flow is steady, so our equation reduces to

$$\frac{\partial^2 u}{\partial y^2} = 0.$$

Hence u is linear in y . The no-slip conditions are $u(0) = 0$ and $u(h) = U$, so

$$u = \frac{Uy}{h},$$

a uniform shear. The velocity gradient, and hence the stress, is the same everywhere in the fluid.

Example 13.2.5 (Plane Poiseuille Flow)

Now hold both plates at rest and instead drive the flow with a pressure difference: the pressure is P_1 at $x = 0$ and $P_0 < P_1$ at $x = L$. We will include gravity, taking $\mathbf{f} = (0, -g\rho, 0)$. Our equations read

$$\begin{aligned} -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial y^2} &= 0, \\ -\frac{\partial p}{\partial y} - g\rho &= 0. \end{aligned}$$

Integrating the second, $p = -g\rho y + F(x)$ for some function F . Substituting into the first,

$$\mu \frac{\partial^2 u}{\partial y^2} = F'(x).$$

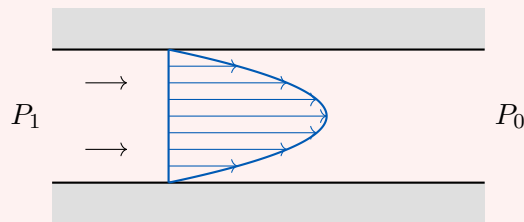
The left hand side is a function of y alone, the right of x alone, so both are equal to some constant G . Since $F(L) - F(0) = P_0 - P_1$ we identify

$$G = -\frac{P_1 - P_0}{L},$$

and integrating twice with $u(0) = u(h) = 0$ gives

$$u = \frac{P_1 - P_0}{2\mu L} y(h - y).$$

This is the famous parabolic profile, fastest in the middle at $y = h/2$ where $u = (P_1 - P_0)h^2/8\mu L$.



Since the equations are linear, a flow driven by *both* a moving boundary and a pressure gradient is simply the sum of the two solutions above.

It is worth extracting a couple of derived quantities from these solutions, since they are the things one actually measures.

Definition 13.2.6 (Volume Flux)

The **volume flux** q across a cross-section is the volume of fluid crossing it per unit time. For a parallel flow, per unit transverse width,

$$q = \int_0^h u(y) \, dy.$$

Example 13.2.7

For the Couette flow,

$$q = \int_0^h \frac{Uy}{h} \, dy = \frac{Uh}{2},$$

which is just what we would get by moving all the fluid at the average of the two plate speeds. For the Poiseuille flow, writing $G = (P_1 - P_0)/L$,

$$q = \int_0^h \frac{G}{2\mu} y(h-y) \, dy = \frac{Gh^3}{12\mu}.$$

The h^3 here is worth remembering: doubling the width of a pipe increases its throughput eightfold at fixed pressure gradient, which is why arterial narrowing is so serious.

We can also ask how much local rotation there is in these flows.

Definition 13.2.8 (Vorticity)

The **vorticity** of a flow is

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}.$$

We will have a great deal to say about vorticity later. For a parallel flow $\mathbf{u} = (u(y, t), 0, 0)$ we simply have

$$\boldsymbol{\omega} = \left(0, 0, -\frac{\partial u}{\partial y}\right).$$

Example 13.2.9

For the Couette flow, $\boldsymbol{\omega} = (0, 0, -U/h)$ is uniform: every fluid particle is spinning at the same rate, clockwise. This is a little counterintuitive, since the flow is perfectly straight, but a small paddle-wheel placed in the fluid would indeed turn, because the fluid is faster above it than below.

For the Poiseuille flow,

$$\boldsymbol{\omega} = \left(0, 0, \frac{G}{\mu} \left(y - \frac{h}{2}\right)\right),$$

which vanishes on the centre-line and has opposite signs in the two halves of the channel – the flow spins one way below the centre-line and the other way above it.

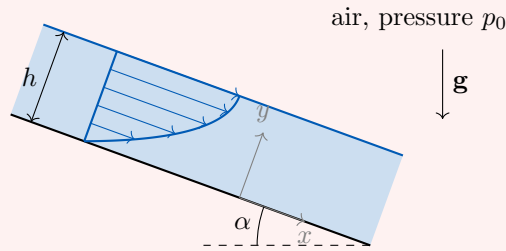
Lastly, the tangential stress exerted *on the boundaries*, which is what we would feel if we

tried to hold the plates in place, is $\tau_s = \mu \partial u / \partial n$ with $\mathbf{n}$ into the fluid. For Couette flow this is $\mu U/h$ at $y = 0$ and $-\mu U/h$ at $y = h$: the fluid drags the stationary lower plate forwards and drags backwards on the moving upper plate, as it must. For Poiseuille flow it is $Gh/2$ at both walls, pulling both forwards, and – strikingly – independent of μ . That makes sense: the total force on the walls must balance the pressure force driving the flow, which knows nothing about the fluid.

813 2 3 Two More Examples

Example 13.2.10 (Gravity-Driven Flow Down a Slope)

Suppose a layer of liquid of depth h flows steadily down a plane inclined at angle α to the horizontal, with air above.



We take x down the slope and y normal to it, so that the body force is

$$\rho \mathbf{g} = (\rho g \sin \alpha, -\rho g \cos \alpha, 0).$$

The y equation then gives

$$\frac{\partial p}{\partial y} = -\rho g \cos \alpha \implies p = p_0 - \rho g \cos \alpha (y - h),$$

using $p = p_0$ at the free surface $y = h$ (air is so light that we may treat its pressure as uniform). Note that p is independent of x , so there is no pressure gradient driving the flow – gravity does it all. The x equation is

$$\mu \frac{\partial^2 u}{\partial y^2} = -\rho g \sin \alpha.$$

The boundary conditions are no-slip at the bottom, $u(0) = 0$, and *no stress* at the free surface, $\partial u / \partial y = 0$ at $y = h$, since air is so much less viscous that it cannot exert any appreciable drag. Integrating,

$$u = \frac{\rho g \sin \alpha}{2\mu} y(2h - y).$$

This is exactly half of a Poiseuille profile: instead of returning to zero at $y = h$ it turns over there and would return to zero at $y = 2h$. That is precisely what we should expect, since a free surface is a plane of symmetry for the shear stress.

The volume flux is

$$q = \int_0^h u \, dy = \frac{\rho g \sin \alpha}{2\mu} \left(h^3 - \frac{h^3}{3} \right) = \frac{\rho g h^3 \sin \alpha}{3\mu}.$$

Everything so far has been steady. Let us do one unsteady problem, which will teach us something important about how viscosity actually operates.

Example 13.2.11 (Impulsively Started Plate)

Fluid is initially at rest in $y > 0$, above a flat plate at $y = 0$. At $t = 0$ the plate is suddenly set moving with constant speed U in the x direction. There is no pressure gradient and no body force, so our equation is

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}, \quad \nu := \frac{\mu}{\rho}.$$

This is the diffusion equation!

Definition 13.2.12 (Kinematic Viscosity)

The **kinematic viscosity** is $\nu = \mu/\rho$, with dimensions L^2T^{-1} .

So ν is a diffusivity, and what it diffuses is momentum – or, equivalently, vorticity. This is the single most useful way to think about viscosity.

Before solving, let us extract the answer by dimensional analysis. The only quantities in the problem are U , ν , y and t ; there is no imposed length scale at all, since the fluid is infinitely deep. Suppose that after a time T the motion has penetrated a distance δ into the fluid. Balancing the two sides of the equation on those scales,

$$\frac{U}{T} \sim \nu \frac{U}{\delta^2} \implies \delta \sim \sqrt{\nu T}.$$

The U has cancelled, as it must, since the equation is linear.

This is a very small number. For water $\nu \approx 10^{-6} \text{ m}^2 \text{ s}^{-1}$ and for air $\nu \approx 10^{-5} \text{ m}^2 \text{ s}^{-1}$, so after a full minute the disturbance has spread only a few millimetres. Viscous effects are confined to thin layers near boundaries – a fact which will dominate the rest of the course.

Now let us solve it properly. Since there is no extrinsic length scale, we look for a **similarity solution**: we guess that the profile always has the same shape, merely stretched by the growing thickness $\sqrt{\nu t}$. Write

$$u(y, t) = U f(\eta), \quad \eta = \frac{y}{\sqrt{\nu t}}.$$

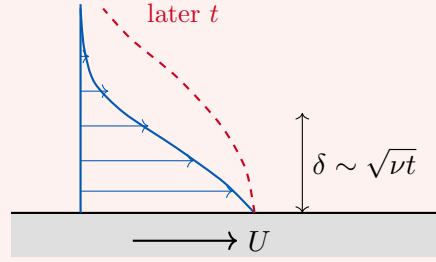
Then $\partial u / \partial t = -U f'(\eta) \eta / (2t)$ and $\partial^2 u / \partial y^2 = U f''(\eta) / (\nu t)$, so the equation becomes an ordinary differential equation,

$$f'' + \frac{1}{2} \eta f' = 0,$$

with $f(0) = 1$ and $f \rightarrow 0$ as $\eta \rightarrow \infty$. This integrates at once: $f' = C e^{-\eta^2/4}$, and imposing the boundary conditions gives

$$u = U \operatorname{erfc}\left(\frac{y}{2\sqrt{\nu t}}\right),$$

where $\operatorname{erfc}(s) = 1 - \frac{2}{\sqrt{\pi}} \int_0^s e^{-r^2} dr$ is the complementary error function.



The stress on the plate is

$$\tau_s = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = -\frac{\mu U}{\sqrt{\pi \nu t}} = -U \sqrt{\frac{\rho \mu}{\pi t}},$$

which is infinite at $t = 0$ (we did, after all, ask for an infinite acceleration) and decays like $t^{-1/2}$ thereafter.

The combination $\sqrt{\rho \mu}$ appearing here is worth a comment. It is about 1 for water and about $10^{-2.5}$ for air, so for a given surface velocity, water transmits far more stress than air does. This is why the wind blowing across the ocean generates a surface current a hundred times slower than itself, rather than dragging the sea along with it.

§13.3 Kinematics

So far we have looked at one very special family of flows. We now build up the general language needed to describe any flow at all. This section contains no dynamics whatsoever – no forces appear – it is purely about how to describe motion and what conservation of mass has to say about it.

§13.3.1 The Material Derivative

Suppose we want to measure how some quantity f – pressure, temperature, the concentration of a dye – is changing. There are two natural things we might mean, and they are not the same.

We could fix a probe at a point in space and watch the reading change. This gives us $\partial f / \partial t$, the rate of change at fixed $\mathbf{x}$, and it is called the **Eulerian** time derivative. Alternatively we could climb into a boat, let ourselves drift with the current, and watch the reading change as we go. This is the **Lagrangian** derivative, and it is the one which physics actually cares about: Newton's second law is about the acceleration of a *particle*, not about the rate of change of velocity at a fixed point in the room.

Relating them is an application of the chain rule. If $\mathbf{x}(t)$ is any path through the flow, then

$$\frac{d}{dt} f(\mathbf{x}(t), t) = \frac{\partial f}{\partial t} + \dot{\mathbf{x}} \cdot \nabla f.$$

If the path is that of a fluid particle, then by definition $\dot{\mathbf{x}} = \mathbf{u}$.

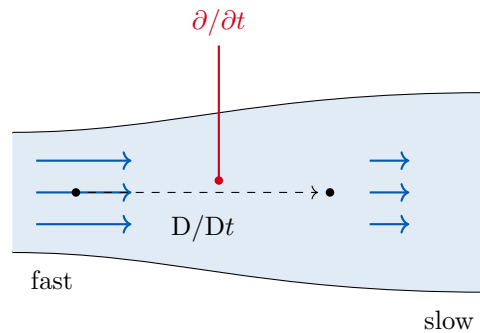
Definition 13.3.1 (Material Derivative)

The **material derivative** (or convective, or Lagrangian derivative) of a field f is

$$\frac{Df}{Dt} = \frac{\partial f}{\partial t} + \mathbf{u} \cdot \nabla f.$$

It is the rate of change of f experienced by a fluid particle moving with the flow.

The two terms have names. The first, $\partial f / \partial t$, is the local or Eulerian rate of change: how fast things are changing at a fixed point. The second, $\mathbf{u} \cdot \nabla f$, is the **advective** term: the change we experience because we have been carried somewhere where f is different.



The picture to have in mind is a river which widens as it flows. The flow is steady, so a probe fixed to the bank sees nothing changing at all and $\partial \mathbf{u} / \partial t = \mathbf{0}$. But a leaf drifting downstream visibly *decelerates* as the river widens, and this deceleration is entirely due to the advective term $\mathbf{u} \cdot \nabla \mathbf{u}$. Steady does not mean unaccelerated.

The material derivative is where all the difficulty of fluid dynamics comes from, incidentally: applied to $\mathbf{u}$ itself, the term $\mathbf{u} \cdot \nabla \mathbf{u}$ is quadratic in the unknown, and it is this nonlinearity which makes the equations of fluid dynamics so intractable (and which is responsible for turbulence).

§13.3.2 Streamlines, Pathlines and Streaklines

There are three natural families of curves associated with a flow, and they coincide only for steady flows – confusing them is a classic source of error.

Definition 13.3.2 (Streamlines, Pathlines, Streaklines)

Let $\mathbf{u}(\mathbf{x}, t)$ be a velocity field.

- (i) A **streamline** at time t_0 is a curve which is everywhere parallel to $\mathbf{u}(\cdot, t_0)$; that is, a solution of

$$\frac{d\mathbf{x}}{ds} = \mathbf{u}(\mathbf{x}, t_0)$$

with t_0 held *fixed*. Streamlines give an instantaneous snapshot of the flow direction.

- (ii) A **pathline** (or particle path) is the trajectory actually followed by a given fluid particle, a solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}, t).$$

- (iii) A **streakline** is the locus at a given time of all particles which have previously

passed through some fixed point – the curve you see when you release dye continuously from a nozzle.

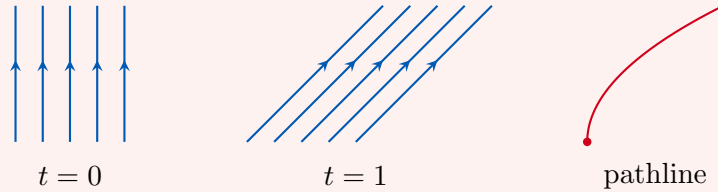
If the flow is steady then $\mathbf{u}$ does not depend on t and the two differential equations above are identical, so streamlines, pathlines and streaklines all coincide. If the flow is unsteady they can be wildly different.

Example 13.3.3 (Streamlines Need Not Be Pathlines)

Consider the (rather artificial) unsteady flow

$$\mathbf{u} = (t, 1, 0).$$

At any instant the velocity is uniform in space, so the streamlines are parallel straight lines: vertical at $t = 0$, at 45° at $t = 1$, and steadily flattening thereafter.



Yet no fluid particle follows any of these lines. A particle released at (x_0, y_0) obeys $\dot{x} = t$, $\dot{y} = 1$, so

$$x = \frac{1}{2}t^2 + x_0, \quad y = t + y_0,$$

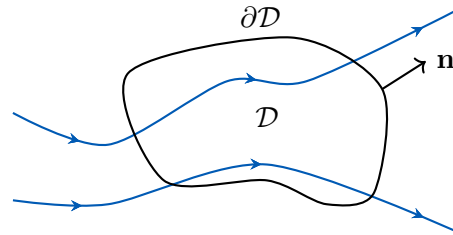
and eliminating t , its pathline is the parabola

$$x - x_0 = \frac{1}{2}(y - y_0)^2.$$

Not a straight line in sight.

§13.3.3 Conservation of Mass

Now for our first real physical law. Fix an arbitrary region of space $\mathcal{D}$, with boundary $\partial\mathcal{D}$ and outward unit normal $\mathbf{n}$, and let fluid flow through it.



The mass inside $\mathcal{D}$ can change only because fluid flows in or out across $\partial\mathcal{D}$. In a time δt , the fluid crossing a patch δS of the boundary occupies a slanted cylinder of volume $(\mathbf{u} \cdot \mathbf{n}) \delta S \delta t$, carrying mass $\rho(\mathbf{u} \cdot \mathbf{n}) \delta S \delta t$. Hence

$$\frac{d}{dt} \int_{\mathcal{D}} \rho \, dV = - \int_{\partial\mathcal{D}} \rho \mathbf{u} \cdot \mathbf{n} \, dS,$$

the minus sign because $\mathbf{n}$ points out. Since $\mathcal{D}$ is a *fixed* region we may take the derivative inside the integral, and by the divergence theorem we may convert the right hand side

to a volume integral:

$$\int_{\mathcal{D}} \left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right) dV = 0.$$

This holds for every region $\mathcal{D}$, and the integrand is continuous, so the integrand must vanish identically.

Proposition 13.3.4 (Conservation of Mass)

At every point of the fluid,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

This is the prototype of a conservation law: the rate of change of the density of some quantity, plus the divergence of its flux, is zero. Exactly the same structure appears for charge in electromagnetism, and for probability in quantum mechanics.

Expanding the divergence, $\nabla \cdot (\rho \mathbf{u}) = \mathbf{u} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{u}$, and recognising a material derivative, we may write this as

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0.$$

This form makes the physical content much clearer. A fluid particle's density changes exactly when the flow is locally expanding or contracting.

§13.3.4 Incompressibility

We are now in a position to say precisely what we mean by an incompressible fluid.

Definition 13.3.5 (Incompressible Fluid)

A fluid is **incompressible** if the density of each fluid particle is constant, i.e. $D\rho/Dt = 0$. By conservation of mass this is equivalent to the **continuity equation**

$$\nabla \cdot \mathbf{u} = 0.$$

Notice this is weaker than saying the density is uniform: a stably stratified ocean, denser at the bottom, can perfectly well be incompressible in this sense. In this course we will almost always take ρ to be a constant as well, and we will say so when we do not.

Two immediate remarks. First, this justifies our claim about parallel flow: if $\mathbf{u} = (u, 0, 0)$ then $\nabla \cdot \mathbf{u} = \partial u / \partial x$, so incompressibility forces u to be independent of x .

Secondly, incompressibility is an approximation, and it is worth knowing when it is a good one. Nothing is truly incompressible – sound waves are compression waves, and if air were incompressible we could not hear. A cleaner way to say it is that incompressibility amounts to assuming that the speed of sound is infinite, so that pressure changes are communicated instantaneously. That is a good approximation precisely when the flow speed is small compared to the sound speed, which is 340 m s^{-1} in air and 1500 m s^{-1} in water. Almost everything we will do is safely in this regime.

§13.3.5 Kinematic Boundary Conditions

Whatever the dynamics, there is one thing fluid may never do: cross a boundary.

Suppose a boundary moves with velocity $\mathbf{U}$ (which may vary from place to place along it) and has unit normal $\mathbf{n}$. In a frame moving with the boundary, the fluid velocity is $\mathbf{u} - \mathbf{U}$, and the requirement that no fluid crosses the boundary is that this relative velocity has no normal component. Thus:

Definition 13.3.6 (Kinematic Boundary Condition)

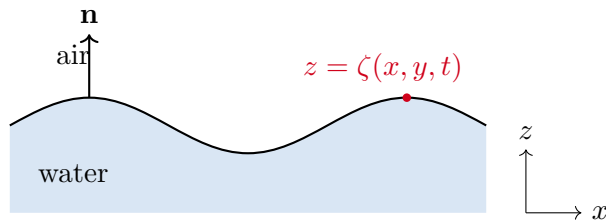
At a boundary moving with velocity $\mathbf{U}$ and unit normal $\mathbf{n}$,

$$\mathbf{u} \cdot \mathbf{n} = \mathbf{U} \cdot \mathbf{n}.$$

At a stationary rigid boundary this reduces to $\mathbf{u} \cdot \mathbf{n} = 0$.

This is a purely kinematic statement – no forces, no viscosity – and it holds whether or not we are making the inviscid approximation. It should not be confused with the no-slip condition, which is a statement about the *tangential* component and only holds for a viscous fluid.

Free surfaces are more interesting, since we do not know in advance where they are. Suppose the surface of the water is given by $z = \zeta(x, y, t)$; think of an oil-water interface, or of a water wave.



Define $F(x, y, z, t) = z - \zeta(x, y, t)$, so that the surface is the level set $F = 0$. Since F is a material surface – the same fluid particles remain on it forever – we must have

$$\frac{DF}{Dt} = 0 \quad \text{on } F = 0.$$

Writing this out with $\mathbf{u} = (u, v, w)$,

$$\frac{\partial}{\partial t}(z - \zeta) + \mathbf{u} \cdot \nabla(z - \zeta) = -\frac{\partial \zeta}{\partial t} + w - u \frac{\partial \zeta}{\partial x} - v \frac{\partial \zeta}{\partial y} = 0,$$

that is,

$$w = \frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} = \frac{D\zeta}{Dt} \quad \text{on } z = \zeta.$$

In words: the vertical velocity of the fluid at the surface equals the rate at which the surface height changes, following the fluid. We will lean heavily on this when we do water waves.

§13.3.6 The Streamfunction

Incompressibility, $\nabla \cdot \mathbf{u} = 0$, is a constraint on three unknown functions. It would be nice to solve it once and for all, and vector calculus tells us how: a divergence-free field on a suitable domain is a curl.

Definition 13.3.7 (Vector Potential)

A **vector potential** for an incompressible flow is a field $\mathbf{A}$ with $\mathbf{u} = \nabla \times \mathbf{A}$.

In three dimensions this trades three unknowns for three unknowns and is not obviously progress. In two dimensions it is a genuine simplification. Suppose the flow is two-dimensional,

$$\mathbf{u} = (u(x, y, t), v(x, y, t), 0).$$

Then we may take $\mathbf{A} = (0, 0, \psi(x, y, t))$, and computing the curl,

$$\mathbf{u} = \left(\frac{\partial \psi}{\partial y}, -\frac{\partial \psi}{\partial x}, 0 \right).$$

Definition 13.3.8 (Streamfunction)

For a two-dimensional incompressible flow, the **streamfunction** is the scalar $\psi(x, y, t)$ with

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}.$$

So a two-dimensional incompressible flow is described by a *single* scalar function, and any ψ whatsoever gives an incompressible flow, since $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ automatically. That is a genuine reduction in the difficulty of the problem. The streamfunction has three lovely properties, which we now establish.

Proposition 13.3.9 (Properties of the Streamfunction)

Let ψ be the streamfunction of a two-dimensional incompressible flow. Then

- (i) The streamlines of the flow are exactly the contours of ψ .
- (ii) The volume flux (per unit length in z) across any curve from $\mathbf{x}_0$ to $\mathbf{x}_1$ is $\psi(\mathbf{x}_1) - \psi(\mathbf{x}_0)$, and in particular depends only on the endpoints.
- (iii) ψ is constant on any stationary rigid boundary.

Proof. For (i), a contour of ψ has normal $\mathbf{n} \parallel \nabla \psi = (\psi_x, \psi_y, 0)$, and

$$\mathbf{u} \cdot \nabla \psi = \frac{\partial \psi}{\partial y} \frac{\partial \psi}{\partial x} - \frac{\partial \psi}{\partial x} \frac{\partial \psi}{\partial y} = 0.$$

So $\mathbf{u}$ is everywhere tangent to the contours, which is exactly the definition of a streamline.

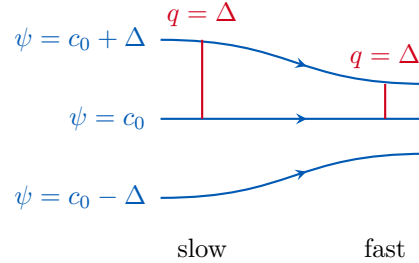
For (ii), parameterise the curve by arclength. A line element $d\boldsymbol{\ell} = (dx, dy)$ has (rightward) normal $\mathbf{n} d\ell = (dy, -dx)$, so the flux across the curve is

$$q = \int_{\mathbf{x}_0}^{\mathbf{x}_1} \mathbf{u} \cdot \mathbf{n} d\ell = \int_{\mathbf{x}_0}^{\mathbf{x}_1} (u dy - v dx) = \int_{\mathbf{x}_0}^{\mathbf{x}_1} \left(\frac{\partial \psi}{\partial y} dy + \frac{\partial \psi}{\partial x} dx \right) = \int_{\mathbf{x}_0}^{\mathbf{x}_1} d\psi,$$

which is $\psi(\mathbf{x}_1) - \psi(\mathbf{x}_0)$.

For (iii), a stationary rigid boundary satisfies $\mathbf{u} \cdot \mathbf{n} = 0$, so no flux crosses it, so by (ii) ψ takes the same value at any two of its points. $\square$

Part (ii) is the one to remember, and it has a very useful visual consequence. Suppose we draw streamlines at equally spaced values $\psi = c_0, c_0 + \Delta, c_0 + 2\Delta, \dots$. Then the volume flux between any two adjacent streamlines is the same, so where the streamlines are close together, the fluid must be moving faster.



By property (iii) we can always choose the constant so that $\psi = 0$ on a boundary, and it is conventional to do so.

It is often convenient to work in plane polar coordinates (r, θ) , viewed as cylindrical polars with the z direction ignored. Then $\mathbf{A} = (0, 0, \psi)$ and

$$\mathbf{u} = \nabla \times \mathbf{A} = \frac{1}{r} \begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ 0 & 0 & \psi \end{vmatrix} = \left(\frac{1}{r} \frac{\partial \psi}{\partial \theta}, -\frac{\partial \psi}{\partial r}, 0 \right).$$

One checks directly that this is divergence free using the polar form

$$\nabla \cdot \mathbf{u} = \frac{1}{r} \frac{\partial}{\partial r}(ru_r) + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta}.$$

There is a third case which comes up often enough to be worth recording, namely axisymmetric flow. If in spherical polars (r, θ, φ) the flow has no φ component and no φ dependence, we may take $\mathbf{A} = (0, 0, \Psi/(r \sin \theta))$, giving

$$u_r = \frac{1}{r^2 \sin \theta} \frac{\partial \Psi}{\partial \theta}, \quad u_\theta = -\frac{1}{r \sin \theta} \frac{\partial \Psi}{\partial r}.$$

Definition 13.3.10 (Stokes Streamfunction)

The function $\Psi(r, \theta)$ above is the **Stokes streamfunction**. Its contours are again streamlines, and $2\pi(\Psi(\mathbf{x}_1) - \Psi(\mathbf{x}_0))$ is the volume flux through the annular surface swept out by rotating a curve from $\mathbf{x}_0$ to $\mathbf{x}_1$ about the axis.

The factor of $r \sin \theta$ is there precisely to make the flux interpretation work: it is the distance from the axis, so that $2\pi r \sin \theta$ is the circumference of the circle traced out by a point.

§13.4 Dynamics

We have spent long enough describing motion; it is time to explain it. In this section we write down the general equation of motion for a fluid, work out when we are allowed to throw parts of it away, and then extract two of the most useful results in the subject: the momentum integral and Bernoulli's equation.

§13.4.1 The Navier–Stokes Equations

The general equation of motion for a Newtonian fluid is the following. Deriving it properly requires the stress tensor, and is a job for the Part II course; what we can say is that it is exactly Newton’s second law, with the same three forces we met in the parallel case, and that it reduces to our parallel-flow equations in that special case.

Law 13.4.1 (Navier–Stokes Equation)

For an incompressible Newtonian fluid of constant density ρ and viscosity μ ,

$$\rho \frac{D\mathbf{u}}{Dt} = \rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f},$$

together with the continuity equation $\nabla \cdot \mathbf{u} = 0$.

Let us read this off term by term. On the left is mass per unit volume times acceleration – and note carefully that the acceleration of a fluid *particle* is the material derivative of $\mathbf{u}$, not the partial derivative. On the right are the three forces per unit volume: the pressure gradient, the viscous force, and any body force $\mathbf{f}$ (for us, always gravity, $\mathbf{f} = \rho \mathbf{g}$).

A few remarks are in order.

- (i) The equation is *nonlinear*, because of the advective term $\mathbf{u} \cdot \nabla \mathbf{u}$. This is the source of essentially all the difficulty in fluid dynamics, and the reason that the existence and smoothness of solutions in three dimensions is an open problem with a million dollar prize attached.
- (ii) Together with $\nabla \cdot \mathbf{u} = 0$ we have four scalar equations for four unknowns (u, v, w and p), which is at least the right count. Note that there is no separate equation for the pressure: the pressure is whatever it has to be in order to keep the flow incompressible.
- (iii) In Cartesian coordinates the viscous term acts componentwise, so that $\nabla^2 \mathbf{u}$ has components $(\nabla^2 u_x, \nabla^2 u_y, \nabla^2 u_z)$. In other coordinate systems it emphatically does not, and it is then safest to use the identity

$$\nabla^2 \mathbf{u} = \nabla(\nabla \cdot \mathbf{u}) - \nabla \times (\nabla \times \mathbf{u}) = -\nabla \times \boldsymbol{\omega}$$

for an incompressible flow. This exhibits the viscous force as being entirely due to gradients of vorticity, a fact we will return to.

- (iv) Setting $\mathbf{u} = (u(y, t), 0, 0)$ recovers exactly the parallel flow equations of Section 2, as the reader should check.

§13.4.2 Pressure, Gravity and Reduced Pressure

Before doing anything hard, let us dispose of gravity.

Suppose the fluid is at rest, $\mathbf{u} = \mathbf{0}$. Then Navier–Stokes reduces to $\nabla p_H = \rho \mathbf{g}$, where we write p_H for the pressure in this static case. Taking $\mathbf{g} = (0, 0, -g)$ and ρ constant, this integrates to

Definition 13.4.2 (Hydrostatic Pressure)

The **hydrostatic pressure** in a fluid of constant density at rest is

$$p_H = p_0 - g\rho z,$$

where p_0 is the pressure at $z = 0$.

This says exactly what you would expect: the pressure at a point is the weight of fluid above it. It has one famous consequence.

Proposition 13.4.3 (Archimedes' Principle)

A body $\mathcal{D}$ immersed in a fluid at rest experiences a net pressure force equal to minus the weight of the fluid it displaces.

Proof. The force is the integral of $-p_H \mathbf{n}$ over the surface of the body, and by the divergence theorem

$$\mathbf{F} = - \int_{\partial \mathcal{D}} p_H \mathbf{n} \, dS = - \int_{\mathcal{D}} \nabla p_H \, dV = - \int_{\mathcal{D}} \rho \mathbf{g} \, dV = -M \mathbf{g},$$

where $M = \rho \, \text{vol}(\mathcal{D})$ is the mass of fluid displaced. $\square$

So a body denser than the fluid sinks, one less dense floats, and one of exactly the same density is neutrally buoyant and just sits there.

Now suppose the fluid is moving. The pressure will not be hydrostatic, but we may always split it up,

$$p = p_H + p',$$

where p' is the deviation from hydrostatic caused by (or causing) the motion. Substituting into Navier–Stokes, the term $-\nabla p_H$ cancels precisely against $\rho \mathbf{g}$, leaving

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p' + \mu \nabla^2 \mathbf{u}.$$

Definition 13.4.4 (Reduced Pressure)

The **reduced** (or **modified**) **pressure** is $p' = p - p_H = p + g\rho z$.

This is a genuinely useful trick, and it deserves to be stated loudly: *if the density is constant, gravity has no dynamical effect at all*. Every fluid particle is neutrally buoyant, and we may drop gravity from the equations entirely provided we agree to work with the reduced pressure. From now on we will usually drop the prime and simply write p , understanding that we mean the reduced pressure.

The one place where we cannot do this is when there is a density difference in the problem – most importantly, at a free surface, where water meets air. There, gravity matters enormously, and it is the restoring force for water waves. We will restore it when we get there.

§13.4.3 The Reynolds Number

The Navier–Stokes equations are hard, so we would like to know when we can throw a term away. The systematic way to do this is to estimate the size of each term.

Suppose the flow has a characteristic speed U and a characteristic length scale L , both externally imposed by the geometry – for flow in a pipe, U might be the mean speed and L the diameter. The natural time scale is then $T = L/U$, the time taken to traverse the region. Dividing Navier–Stokes by ρ and writing $\nu = \mu/\rho$,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u},$$

and the four terms have magnitudes of order

$$\frac{U}{L/U}, \quad \frac{U^2}{L}, \quad \frac{P}{\rho L}, \quad \frac{\nu U}{L^2},$$

where P is a characteristic pressure *difference* (only differences drive flow). Dividing throughout by U^2/L ,

$$1, \quad 1, \quad \frac{P}{\rho U^2}, \quad \frac{\nu}{UL}.$$

Definition 13.4.5 (Reynolds Number)

The **Reynolds number** of a flow is the dimensionless quantity

$$Re = \frac{UL}{\nu}.$$

The two inertial terms are both of relative size 1, and the viscous term of relative size $1/Re$. So the Reynolds number measures the importance of inertia relative to viscosity, and it is the single most important number in the subject. We need not worry about the pressure term: the pressure will always scale so as to balance whatever the dominant terms are, since its job is to enforce incompressibility.

Definition 13.4.6 (Dynamic Similarity)

Two flows with geometrically similar boundaries and the same Reynolds number are **dynamically similar**: they are described by the same dimensionless solution.

This is why wind tunnels work. It is also why lava, despite being fabulously viscous, flows down a mountain looking very much like water flowing down a river – the length scales are so much larger that the Reynolds numbers are comparable. It is the Reynolds number, not U , L or ν separately, that determines the character of a flow.

Example 13.4.7

An aircraft has $U \sim 250 \text{ m s}^{-1}$, $L \sim 10 \text{ m}$ and (in air) $\nu \sim 10^{-5} \text{ m}^2 \text{ s}^{-1}$, so $Re \sim 10^8$: viscosity is utterly negligible on the scale of the wing. A bacterium swimming in water has $U \sim 10^{-5} \text{ m s}^{-1}$, $L \sim 10^{-6} \text{ m}$, $\nu \sim 10^{-6} \text{ m}^2 \text{ s}^{-1}$, so $Re \sim 10^{-5}$: for the bacterium, inertia is meaningless and water feels like treacle.

The two limiting cases are worth naming. When $Re \ll 1$ the inertial terms are negligible, the pressure scales as $P \sim \mu U/L$ to balance the viscous stress, and we obtain the linear **Stokes equations**

$$\mathbf{0} = -\nabla p + \mu \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0.$$

When $Re \gg 1$ the viscous term is negligible, the pressure scales as $P \sim \rho U^2$ to balance inertia, and we obtain the **Euler equations**

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p, \quad \nabla \cdot \mathbf{u} = 0.$$

The rest of this course is essentially about the second of these.

There is however a serious catch, and we should be honest about it. Dropping the viscous term reduces the order of the equation from two to one, so we can no longer impose as many boundary conditions – and the one we must give up is no-slip. An inviscid fluid is allowed to slide along a solid wall. Since real fluids do not, our approximation must be failing somewhere near the boundary.

Where? At the scale δ on which the viscous and inertial terms are comparable:

$$\frac{U^2}{\delta} \sim \frac{\nu U}{\delta^2} \implies \delta \sim \frac{\nu}{U}, \quad \text{equivalently} \quad \frac{\delta}{L} \sim \frac{1}{Re}.$$

So when Re is large there is a thin **boundary layer** of thickness $\delta \ll L$ hugging the wall, inside which viscosity matters and the velocity adjusts rapidly from zero at the wall to the inviscid ‘slip’ value just outside. Away from that layer the inviscid solution is excellent. This is the picture to keep in mind whenever we set $\nu = 0$ in what follows.

§13.4.4 A Case Study: Stagnation Point Flow

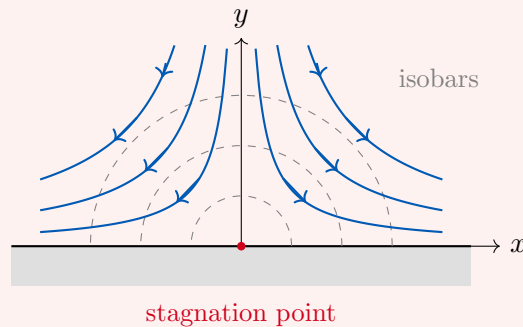
Let us make all of that concrete with a single carefully worked example, which is well worth the effort because it displays the boundary layer explicitly.

Example 13.4.8 (Stagnation Point Flow)

We look for a solution of the Navier–Stokes equations in the half-plane $y \geq 0$ with $\mathbf{u} = \mathbf{0}$ on $y = 0$ and

$$\mathbf{u} \rightarrow (Ex, -Ey, 0) \quad \text{as } y \rightarrow \infty,$$

for a constant $E > 0$. The far field here is fluid falling onto a plate and spreading sideways, with the streamlines being hyperbolae $xy = \text{const}$.



Notice there is no imposed length scale in this problem: $[E] = T^{-1}$ and $[\nu] = L^2 T^{-1}$,

so the only length we can build is

$$\delta = \sqrt{\frac{\nu}{E}},$$

and we look for a similarity solution in $\eta = y/\delta$. Incompressibility suggests we posit a streamfunction of the form

$$\psi = \sqrt{\nu E} x g(\eta),$$

which gives

$$u = \frac{\partial \psi}{\partial y} = E x g'(\eta), \quad v = -\frac{\partial \psi}{\partial x} = -\sqrt{\nu E} g(\eta) = -E \delta g(\eta).$$

Substituting into the two components of Navier–Stokes gives

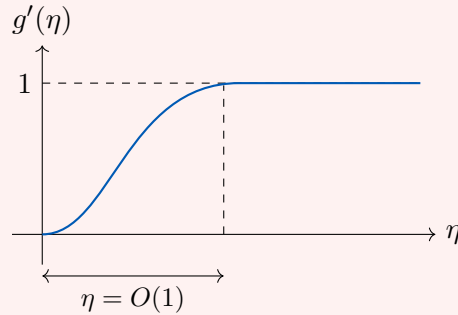
$$\begin{aligned} E^2 x (g'^2 - g g'') &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + E^2 x g''', \\ E \sqrt{\nu E} g g' &= -\frac{1}{\rho} \frac{\partial p}{\partial y} - E \sqrt{\nu E} g''. \end{aligned}$$

To eliminate the pressure we cross-differentiate: take $\partial/\partial y$ of the first and $\partial/\partial x$ of the second and subtract. The second equation has no x dependence at all, so we simply get

$$g' g'' - g g''' = g^{(4)},$$

a single fourth order ordinary differential equation. The boundary conditions are $g(0) = g'(0) = 0$ from no-slip, and $g'(\eta) \rightarrow 1$, $g(\eta) \sim \eta$ as $\eta \rightarrow \infty$ to match the far field.

We do not need to solve this (it has no closed form), and that is rather the point: all the dimensional parameters have been absorbed into g and η , so one numerical solution serves for every E and ν . Its shape is easy to guess.



The horizontal velocity rises from zero at the wall to its inviscid value Ex over a distance $y \sim \delta$, and is essentially inviscid beyond that. So δ is the boundary layer thickness, and at that scale $Re_\delta = U\delta/\nu = O(1)$, exactly as advertised.

Outside the boundary layer we may drop viscosity, and one checks directly that $\mathbf{u} = (Ex, -Ey, 0)$ solves the Euler equations exactly, with

$$p = p_0 - \frac{1}{2} \rho E^2 (x^2 + y^2).$$

The isobars are the semicircles drawn in the figure. As fluid falls in from above it moves into regions of higher pressure, so the pressure gradient is **adverse** and it decelerates; as it turns and moves outwards it moves into regions of lower pressure, so the gradient is **favourable** and it accelerates. At the origin the velocity vanishes and the pressure attains its maximum p_0 . This is a **stagnation point**, and the association it displays – high pressure with low velocity and vice versa – is a foretaste of Bernoulli's equation.

§13.4.5 The Euler Momentum Equation

We stated Navier–Stokes without proof, but in the inviscid case we can do better and derive it honestly, by considering the momentum budget of a fixed region of space. This is worth doing because the intermediate result – the momentum integral – is a powerful tool in its own right.

Fix a region $\mathcal{D}$ with boundary $\partial\mathcal{D}$ and outward normal $\mathbf{n}$. The momentum inside $\mathcal{D}$ changes for three reasons (ignoring viscous surface forces, which is our inviscid approximation): momentum is carried across the boundary by the flow, pressure acts on the boundary, and body forces act throughout. So

$$\frac{d}{dt} \int_{\mathcal{D}} \rho \mathbf{u} \, dV = - \int_{\partial\mathcal{D}} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) \, dS - \int_{\partial\mathcal{D}} p \mathbf{n} \, dS + \int_{\mathcal{D}} \mathbf{f} \, dV.$$

The first term on the right is the momentum flux: fluid crossing the boundary at rate $(\mathbf{u} \cdot \mathbf{n}) \, dS$ per unit volume carries momentum $\rho \mathbf{u}$ with it.

To make progress, switch to suffix notation, since the divergence theorem applies to each component separately:

$$\frac{d}{dt} \int_{\mathcal{D}} \rho u_i \, dV = - \int_{\partial\mathcal{D}} \rho u_i u_j n_j \, dS - \int_{\partial\mathcal{D}} p n_i \, dS + \int_{\mathcal{D}} f_i \, dV.$$

Applying the divergence theorem to the surface integrals and collecting everything on one side,

$$\int_{\mathcal{D}} \left(\rho \frac{\partial u_i}{\partial t} + \rho \frac{\partial}{\partial x_j} (u_i u_j) + \frac{\partial p}{\partial x_i} - f_i \right) dV = 0.$$

Since $\mathcal{D}$ is arbitrary the integrand vanishes. Finally

$$\frac{\partial}{\partial x_j} (u_i u_j) = u_j \frac{\partial u_i}{\partial x_j} + u_i \frac{\partial u_j}{\partial x_j} = u_j \frac{\partial u_i}{\partial x_j},$$

the last term vanishing by incompressibility, and we recognise the material derivative.

Proposition 13.4.9 (Euler Momentum Equation)

For an inviscid incompressible fluid of constant density,

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{f}.$$

This is of course exactly Navier–Stokes with $\mu = 0$, but it is reassuring to have got it from momentum conservation directly.

Very often the body force is conservative, $\mathbf{f} = -\nabla\chi$. For gravity with constant ρ we have $\mathbf{f} = \rho\mathbf{g} = -\nabla(-\rho\mathbf{g} \cdot \mathbf{x})$, so

$$\chi = -\rho\mathbf{g} \cdot \mathbf{x} = g\rho z$$

when $\mathbf{g} = (0, 0, -g)$. (Of course, by the discussion of reduced pressure we could equally absorb χ into p and forget about it.)

Specialising the integral form to a *steady* flow, the left hand side vanishes, and converting the body force integral back to a surface integral gives a result which is extremely useful for practical problems.

Proposition 13.4.10 (Momentum Integral for Steady Flow)

For a steady inviscid flow with conservative body force $\mathbf{f} = -\nabla\chi$, and any fixed region $\mathcal{D}$,

$$\int_{\partial\mathcal{D}} (\rho\mathbf{u}(\mathbf{u} \cdot \mathbf{n}) + p\mathbf{n} + \chi\mathbf{n}) \, dS = \mathbf{0}.$$

The power of this is that it involves only surface integrals: if we can choose a control surface on which we know the flow, we can compute forces without knowing anything about the messy interior.

§13.4.6 Bernoulli's Equation

The other classical consequence of the Euler equation comes from the vector identity

$$\mathbf{u} \cdot \nabla\mathbf{u} = \nabla\left(\frac{1}{2}|\mathbf{u}|^2\right) - \mathbf{u} \times (\nabla \times \mathbf{u}) = \nabla\left(\frac{1}{2}|\mathbf{u}|^2\right) - \mathbf{u} \times \boldsymbol{\omega}.$$

Substituting this into the Euler equation with $\mathbf{f} = -\nabla\chi$ and rearranging,

$$\rho \frac{\partial\mathbf{u}}{\partial t} - \rho \mathbf{u} \times \boldsymbol{\omega} = -\nabla\left(\frac{1}{2}\rho|\mathbf{u}|^2 + p + \chi\right).$$

This is worth a name.

Definition 13.4.11 (Bernoulli Quantity)

The **Bernoulli quantity** of a flow is

$$H = \frac{1}{2}\rho|\mathbf{u}|^2 + p + \chi.$$

Now dot the rearranged equation with $\mathbf{u}$. The term $\mathbf{u} \cdot (\mathbf{u} \times \boldsymbol{\omega})$ vanishes identically, and we are left with the following.

Theorem 13.4.12 (Bernoulli's Theorem)

For an inviscid incompressible flow of constant density with conservative body force,

$$\frac{1}{2}\rho \frac{\partial}{\partial t}|\mathbf{u}|^2 = -\mathbf{u} \cdot \nabla H.$$

In particular:

- (i) If the flow is *steady*, then $\mathbf{u} \cdot \nabla H = 0$, so H is constant along streamlines.
- (ii) If the flow is *irrotational* ($\boldsymbol{\omega} = \mathbf{0}$) and steady, then $\nabla H = \mathbf{0}$, so H is constant

everywhere.

Proof. The displayed equation is what we derived. For (i), a steady flow has $\partial \mathbf{u} / \partial t = \mathbf{0}$, so $\mathbf{u} \cdot \nabla H = 0$, which says exactly that H does not change as we move along a streamline. For (ii), if $\boldsymbol{\omega} = \mathbf{0}$ then the term $\rho \mathbf{u} \times \boldsymbol{\omega}$ was absent to begin with, so the rearranged Euler equation reads $\rho \partial \mathbf{u} / \partial t = -\nabla H$, and if additionally the flow is steady then $\nabla H = \mathbf{0}$. $\square$

The distinction between (i) and (ii) is important and frequently muddled. In a general steady flow, H is constant on each streamline but may take different values on different streamlines. Only if the flow is irrotational may we compare H at two points which are not joined by a streamline. Since (as we will see in Section 13.5) flows which start from rest are irrotational, we will usually be in the happy case (ii).

The content of Bernoulli's theorem is the trade-off between speed and pressure: where a steady flow speeds up, the pressure must fall. This is the single most-used fact in fluid dynamics, and it is worth pausing to see *why* it is true, since it often gets mystified. It is nothing but energy conservation. H is (up to the constant χ) the sum of kinetic energy density and pressure, and a fluid particle can only speed up if something pushes it, i.e. if it is moving from high pressure to low.

Remark. Even when the flow is unsteady, H retains a meaning. Integrating Bernoulli's theorem over a fixed region $\mathcal{D}$ and using incompressibility,

$$\frac{d}{dt} \int_{\mathcal{D}} \frac{1}{2} \rho |\mathbf{u}|^2 dV = - \int_{\partial \mathcal{D}} H \mathbf{u} \cdot \mathbf{n} dS,$$

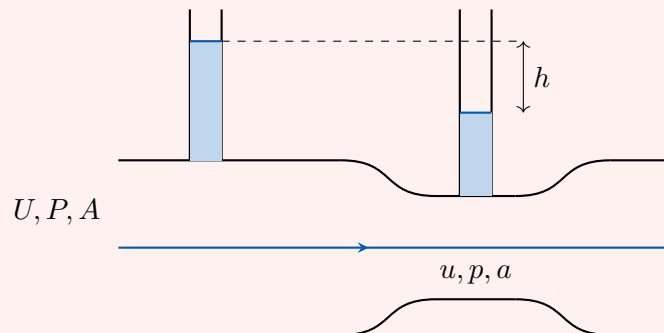
so H is the density of *transportable energy*: the rate at which the kinetic energy in a region changes is the flux of H across its boundary.

§13.4.7 Applications of Bernoulli and the Momentum Integral

Let us put these two results to work. The examples below are all worked in the same style: identify the streamline, apply Bernoulli at two points on it, and use conservation of mass to relate the speeds.

Example 13.4.13 (The Venturi Meter)

Consider a horizontal pipe of cross-sectional area A with a constriction of area $a < A$ in the middle, and two vertical tubes let into it so that we may read off the pressure.



Conservation of mass gives the volume flux

$$q = UA = ua.$$

Now apply Bernoulli along the central streamline. The pipe is horizontal so the χ terms are equal and cancel, giving

$$\frac{1}{2}\rho U^2 + P = \frac{1}{2}\rho u^2 + p.$$

Substituting $U = q/A$ and $u = q/a$,

$$\left(\frac{1}{A^2} - \frac{1}{a^2}\right) q^2 = \frac{2(p - P)}{\rho}.$$

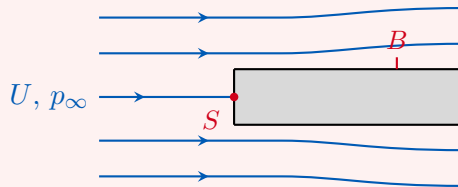
Since $a < A$ the left hand side is negative, so $p < P$: the fluid is faster and hence at lower pressure in the constriction. The vertical tubes are hydrostatic, so $p - P = -\rho gh$, and rearranging,

$$q = \sqrt{2gh} \frac{Aa}{\sqrt{A^2 - a^2}}.$$

So by measuring a height difference with a ruler we can measure the flow rate through the pipe. This is a Venturi meter, and it is how the fuel flow in a carburettor is metered.

Example 13.4.14 (The Pitot Tube)

How does an aircraft measure its airspeed? Point a tube directly into the oncoming flow and seal the far end. The fluid at the mouth of the tube is brought to rest – it is a stagnation point – while fluid passing a second, side-facing opening is still moving at the free stream speed U .



Bernoulli along the central streamline, from far upstream to the stagnation point S , gives

$$p_S = p_\infty + \frac{1}{2}\rho U^2.$$

The pressure p_S is called the **stagnation** or **total** pressure, and $\frac{1}{2}\rho U^2$ the **dynamic** pressure. At the side opening B the fluid is still moving at essentially U , so it reads $p_B \approx p_\infty$. Hence

$$U = \sqrt{\frac{2(p_S - p_B)}{\rho}},$$

and a differential pressure gauge between the two openings is an airspeed indicator.

Example 13.4.15 (Torricelli's Law)

A large tank of water has a small hole of area a in its side, a depth H below the free surface. How fast does the water come out?

Apply Bernoulli along the streamline running from the free surface down to the hole, and now we must *not* drop gravity, since we are comparing two points at different heights on a free surface. At the surface the pressure is atmospheric and (the tank being large) the velocity is negligible. At the hole the pressure is again atmospheric, since the emerging jet is surrounded by air. So with $\chi = \rho g z$,

$$p_{\text{atm}} + \rho g H = p_{\text{atm}} + \frac{1}{2} \rho u^2,$$

giving

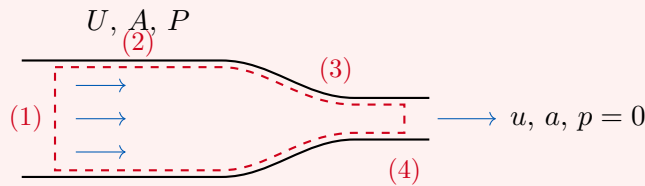
$$u = \sqrt{2gH}.$$

This is Torricelli's law, and it is precisely the speed a particle would reach falling freely through a height H – which is no coincidence, Bernoulli being energy conservation in disguise.

The next example uses the momentum integral instead, and shows off its real advantage: we compute a force on a body whose shape we never specify.

Example 13.4.16 (Force on a Fire Hose Nozzle)

Water enters a nozzle with speed U , area A and gauge pressure P , and leaves as a jet with speed u , area a and gauge pressure 0 (the jet is open to the atmosphere). With what force does the water try to tear the nozzle off the hose?



Apply the steady momentum integral in the x direction over the dashed control surface, which consists of the inlet (1), the walls (2) and (3), and the outlet (4). We may ignore gravity (χ contributes nothing net to a horizontal force).

- On the inlet (1), $\mathbf{n} = -\hat{\mathbf{x}}$, so the momentum flux contributes $\rho U(-U)A = -\rho U^2 A$ and the pressure contributes $-PA$.
- On the cylindrical wall (2), the velocity is parallel to the surface so there is no momentum flux, and the pressure force is purely radial: nothing in the x direction.
- On the tapering wall (3), again no momentum flux, but the pressure force does have an x component. Its total is $\int_{\text{nozzle}} p \mathbf{n} \cdot \hat{\mathbf{x}} \, dS$, which is exactly (minus) the force F that we want.
- On the outlet (4), the momentum flux is $\rho u \cdot ua = \rho u^2 a$ and the pressure is zero.

Summing to zero, the force exerted by the water on the nozzle in the x direction is

$$F = \rho A U^2 - \rho a u^2 + P A.$$

We still have P in there, but Bernoulli along the central streamline gives $\frac{1}{2} \rho U^2 + P =$

$\frac{1}{2}\rho u^2$, and with $q = UA = ua$ this simplifies, after a little algebra, to

$$F = \frac{\rho q^2}{2a} \left(1 - \frac{a}{A}\right)^2.$$

Let us put numbers in. A hose of radius 10 cm with a nozzle of radius 5 cm has $A/a = 4$, and a typical flow rate is $q = 0.01 \text{ m}^3 \text{ s}^{-1}$. Then

$$F = \frac{1000 \times 10^{-4}}{2 \times (0.05)^2 \pi} \left(\frac{3}{4}\right)^2 \approx 14 \text{ N},$$

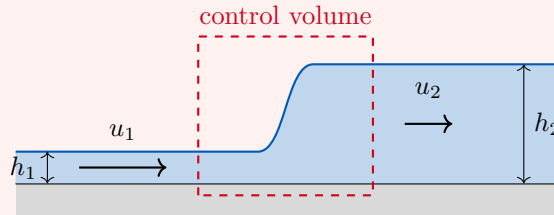
a couple of kilograms' worth. That is why fire hoses have handles.

Remark. It is instructive to see what would have happened if we had tried to get this force by integrating the pressure over the nozzle directly. We would have needed to know the shape of the nozzle, and the flow inside it, in complete detail. The momentum integral lets us put a box around our ignorance.

The last example in this vein is a favourite, because it is one where Bernoulli's equation is *wrong* and only the momentum integral will do.

Example 13.4.17 (The Hydraulic Jump)

Turn on a kitchen tap onto a flat sink and look at the bottom. There is a disc of very fast, very thin water around the point of impact, bounded by an abrupt circular step beyond which the water is slower and deeper. That step is a **hydraulic jump**, and something similar happens at the foot of a spillway or a weir.



Take a control volume straddling the jump, with the flow uniform and parallel on each side. Conservation of mass (per unit width) gives

$$q = u_1 h_1 = u_2 h_2.$$

For the momentum, the pressure on each face is hydrostatic, $p = p_{\text{atm}} + \rho g(h_i - z)$, and integrating over the depth gives a total pressure force per unit width of $\frac{1}{2}\rho g h_i^2$ (above atmospheric). Balancing horizontal momentum flux plus pressure force,

$$\rho u_1^2 h_1 + \frac{1}{2}\rho g h_1^2 = \rho u_2^2 h_2 + \frac{1}{2}\rho g h_2^2.$$

Substituting $u_i = q/h_i$ and dividing by ρ ,

$$\frac{q^2}{h_1} - \frac{q^2}{h_2} = \frac{g}{2} (h_2^2 - h_1^2) \implies \frac{q^2(h_2 - h_1)}{h_1 h_2} = \frac{g}{2} (h_2 - h_1)(h_2 + h_1).$$

The trivial root $h_2 = h_1$ is 'no jump'; cancelling it gives

$$q^2 = \frac{1}{2} g h_1 h_2 (h_1 + h_2),$$

a quadratic determining h_2 from h_1 and q . Writing this in terms of the upstream Froude number $Fr_1 = u_1/\sqrt{gh_1}$ and solving,

$$\frac{h_2}{h_1} = \frac{1}{2} \left(\sqrt{1 + 8Fr_1^2} - 1 \right).$$

So a jump ($h_2 > h_1$) is possible only if $Fr_1 > 1$: the upstream flow must be *super-critical*, moving faster than shallow-water waves can propagate, so that it cannot be warned of the obstruction ahead. Downstream one checks that $Fr_2 < 1$, so the flow is subcritical.

Where does Bernoulli go? Computing $H = \frac{1}{2}\rho u^2 + \rho gh$ on the two sides, one finds it *decreases* across the jump, by

$$\Delta H = \frac{\rho g(h_2 - h_1)^3}{4h_1 h_2} > 0.$$

Energy has been lost, to the turbulent churning that is so visible at a real jump. This is precisely why we had to use momentum: momentum is conserved across the jump but energy is not, and applying Bernoulli here would give a flatly wrong answer. It is a useful reminder that Bernoulli's theorem was derived for a smooth inviscid flow, and a hydraulic jump is neither.

§13.5 Vorticity

We met the vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$ briefly in Section 2. It is time to take it seriously, because it turns out to be the quantity which really controls the character of a flow. In particular, we are going to prove that a flow which starts out with no vorticity keeps none – which is the licence we need to spend the second half of the course on potential flow.

§13.5.1 What Vorticity Measures

A good way to see the meaning of $\boldsymbol{\omega}$ is to look at a general flow very close to a chosen point $\mathbf{x}_0$. Taylor expanding,

$$\mathbf{u}(\mathbf{x}) = \mathbf{u}_0 + \mathbf{r} \cdot \nabla \mathbf{u}_0 + O(|\mathbf{r}|^2), \quad \mathbf{r} = \mathbf{x} - \mathbf{x}_0,$$

so to first order the flow is described by the matrix $\partial u_i / \partial x_j$. Any matrix splits uniquely into symmetric and antisymmetric parts,

$$\frac{\partial u_i}{\partial x_j} = E_{ij} + \Omega_{ij}, \quad E_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad \Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right),$$

and the antisymmetric part is precisely the vorticity in disguise. Indeed, using $\varepsilon_{ijk}\varepsilon_{klm} = \delta_{il}\delta_{jm} - \delta_{im}\delta_{jl}$,

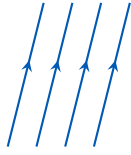
$$(\boldsymbol{\omega} \times \mathbf{r})_i = \varepsilon_{ijk}\varepsilon_{jlm} \frac{\partial u_m}{\partial x_l} r_k = \left(\frac{\partial u_i}{\partial x_k} - \frac{\partial u_k}{\partial x_i} \right) r_k = 2\Omega_{ik}r_k.$$

Hence the local structure of any flow is

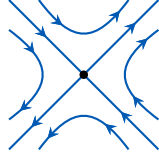
Proposition 13.5.1 (Decomposition of a Linear Flow)

Near any point,

$$\mathbf{u} = \underbrace{\mathbf{u}_0}_{\text{uniform flow}} + \underbrace{E\mathbf{r}}_{\text{pure strain}} + \underbrace{\frac{1}{2}\boldsymbol{\omega} \times \mathbf{r}}_{\text{rigid rotation}} + O(|\mathbf{r}|^2).$$



uniform flow

pure strain, $E\mathbf{r}$ rotation, $\frac{1}{2}\boldsymbol{\omega} \times \mathbf{r}$

So the vorticity at a point is *twice the local angular velocity* of the fluid there. A tiny paddle-wheel dropped into the flow spins at angular velocity $\frac{1}{2}\boldsymbol{\omega}$. Meanwhile E , the rate-of-strain tensor, describes how a small blob is stretched and squashed; incompressibility says $\text{tr } E = \nabla \cdot \mathbf{u} = 0$, so in principal axes $E = \text{diag}(E_1, E_2, E_3)$ with $E_1 + E_2 + E_3 = 0$, and any stretching in one direction must be paid for by compression in another.

Definition 13.5.2 (Irrotational Flow, Circulation)

A flow is **irrotational** if $\boldsymbol{\omega} = \mathbf{0}$ everywhere. The **circulation** around a closed curve C is

$$\Gamma = \oint_C \mathbf{u} \cdot d\boldsymbol{\ell} \stackrel{\text{Stokes}}{=} \int_S \boldsymbol{\omega} \cdot \mathbf{n} \, dS$$

for any surface S spanning C .

Circulation is the integrated form of vorticity, and Stokes' theorem relating them is the reason both are useful. Notice that an irrotational flow has zero circulation around every curve which bounds a surface lying in the fluid – but *not* necessarily around every curve, if the region is not simply connected. That loophole is exactly where lift comes from, as we shall see.

§13.5.2 The Vorticity Equation

We know how $\mathbf{u}$ evolves; let us find out how $\boldsymbol{\omega}$ evolves. Start from Navier–Stokes with a conservative body force, in the form

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times \boldsymbol{\omega} = -\frac{1}{\rho} \nabla p - \frac{1}{\rho} \nabla \chi + \nu \nabla^2 \mathbf{u},$$

and take the curl. Every gradient dies, leaving

$$\frac{\partial \boldsymbol{\omega}}{\partial t} - \nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = \nu \nabla^2 \boldsymbol{\omega}.$$

Now expand the remaining curl using the standard identity

$$\nabla \times (\mathbf{u} \times \boldsymbol{\omega}) = (\nabla \cdot \boldsymbol{\omega}) \mathbf{u} + \boldsymbol{\omega} \cdot \nabla \mathbf{u} - (\nabla \cdot \mathbf{u}) \boldsymbol{\omega} - \mathbf{u} \cdot \nabla \boldsymbol{\omega}.$$

The first term vanishes because a curl is divergence free, and the third because the flow is incompressible. Rearranging and recognising the material derivative gives the following.

Proposition 13.5.3 (Vorticity Equation)

For an incompressible Newtonian fluid of constant density,

$$\frac{D\boldsymbol{\omega}}{Dt} = \boldsymbol{\omega} \cdot \nabla \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}.$$

Two things are remarkable about this equation. First, the pressure has vanished completely: vorticity does not care about pressure. Secondly, in the inviscid case the equation is *homogeneous* in $\boldsymbol{\omega}$ – there is no source term. Vorticity cannot be created out of nothing in the interior of an inviscid flow; it can only be moved around and stretched.

Let us read the two surviving terms.

The term $\nu \nabla^2 \boldsymbol{\omega}$ is diffusion: viscosity spreads vorticity out, exactly as we saw in the impulsively started plate problem, and eventually dissipates it. It is also, importantly, the only mechanism by which vorticity can be *generated*, and this happens at boundaries: the no-slip condition forces a steep velocity gradient at a wall, which is to say a sheet of vorticity, and viscosity then diffuses it into the fluid. Every bit of vorticity in the world was born at a boundary.

The term $\boldsymbol{\omega} \cdot \nabla \mathbf{u}$ is more interesting and is called **vortex stretching**. To see what it does, take the inviscid case and dot with $\boldsymbol{\omega}$:

$$\frac{D}{Dt} \left(\frac{1}{2} |\boldsymbol{\omega}|^2 \right) = \omega_i (E_{ij} + \Omega_{ij}) \omega_j = \omega_i E_{ij} \omega_j,$$

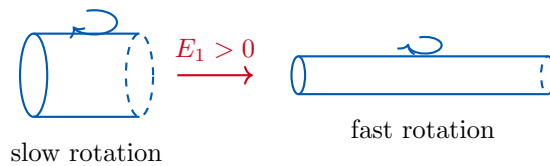
the antisymmetric part dropping out against the symmetric $\omega_i \omega_j$. In principal axes,

$$\frac{D}{Dt} \left(\frac{1}{2} |\boldsymbol{\omega}|^2 \right) = E_1 \omega_1^2 + E_2 \omega_2^2 + E_3 \omega_3^2.$$

Suppose the flow is being stretched along $\mathbf{e}_1$, so $E_1 > 0$ and (since the E_i sum to zero) $E_2, E_3 < 0$. A vortex aligned with the stretching, $\boldsymbol{\omega} = (\omega_1, 0, 0)$, then obeys

$$\frac{D}{Dt} \left(\frac{1}{2} \omega_1^2 \right) = E_1 \omega_1^2,$$

so $|\boldsymbol{\omega}|$ grows exponentially. Stretching a vortex tube spins it up.



This is nothing but conservation of angular momentum: squeezing a rotating tube of fluid radially inwards brings its mass closer to the axis, and it must spin faster to compensate. It is the same physics as an ice skater pulling in their arms, and it is why bath plugholes, tornadoes and dust devils are all so vigorous.

In fact, this is a special case of a beautiful general statement. Let $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ be two neighbouring fluid particles and let $\delta \boldsymbol{\ell} = \mathbf{x}_2 - \mathbf{x}_1$ be the material line element joining them. Then

$$\frac{D\delta \boldsymbol{\ell}}{Dt} = \mathbf{u}(\mathbf{x}_2) - \mathbf{u}(\mathbf{x}_1) = \delta \boldsymbol{\ell} \cdot \nabla \mathbf{u}$$

to first order. But this is *exactly* the same equation as the inviscid vorticity equation. So in an inviscid flow, vorticity behaves exactly like a material line element: vortex lines are frozen into the fluid and are stretched and rotated along with it.

Remark. If the density is not constant then the derivation above picks up an extra term, and one finds

$$\frac{D\boldsymbol{\omega}}{Dt} = \boldsymbol{\omega} \cdot \nabla \mathbf{u} + \frac{1}{\rho^2} \nabla \rho \times \nabla p + \nu \nabla^2 \boldsymbol{\omega}.$$

The new term is called the **baroclinic** term, and it *does* generate vorticity from nothing whenever the surfaces of constant density and constant pressure are not parallel. It is what drives convection: a heater at one end of a room produces a horizontal density gradient which, crossed with the vertical (hydrostatic) pressure gradient, spins the air up into a circulation.

§13.5.3 Kelvin's Circulation Theorem

Circulation obeys a conservation law of its own, which is arguably the most elegant result in the subject.

Theorem 13.5.4 (Kelvin's Circulation Theorem)

Let $C(t)$ be a closed *material* curve – one which moves with the fluid – in an inviscid, incompressible flow of constant density with conservative body forces. Then the circulation

$$\Gamma = \oint_{C(t)} \mathbf{u} \cdot d\boldsymbol{\ell}$$

is independent of t .

Proof. Differentiate following the fluid. Both the velocity and the line element change, so

$$\frac{d\Gamma}{dt} = \oint_{C(t)} \frac{D\mathbf{u}}{Dt} \cdot d\boldsymbol{\ell} + \oint_{C(t)} \mathbf{u} \cdot \frac{D(d\boldsymbol{\ell})}{Dt}.$$

For the first integral, the Euler equation gives $D\mathbf{u}/Dt = -\nabla(p/\rho + \chi/\rho)$, and the integral of a gradient around a closed curve vanishes.

For the second, we showed above that a material line element obeys $D(d\boldsymbol{\ell})/Dt = d\boldsymbol{\ell} \cdot \nabla \mathbf{u}$, so

$$\mathbf{u} \cdot \frac{D(d\boldsymbol{\ell})}{Dt} = u_i d\ell_j \frac{\partial u_i}{\partial x_j} = d\boldsymbol{\ell} \cdot \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right),$$

which is again the integral of a gradient around a closed curve, and vanishes. Hence $d\Gamma/dt = 0$. $\square$

By Stokes' theorem this says that the flux of vorticity through any material surface is constant, which is a precise version of the statement that vortex lines are frozen into an inviscid fluid. Viscosity is what allows vortex lines to slip through the fluid and reconnect, and its absence is why inviscid flows have such long memories.

§13.5.4 Irrotational Flow Stays Irrotational

Here is the payoff, and it is a corollary of either of the last two results.

Corollary 13.5.5 (Persistence of Irrotational Flow)

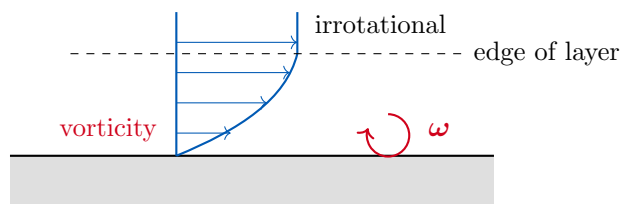
Suppose an inviscid, incompressible flow of constant density with conservative body force is irrotational at some initial time. Then it is irrotational for all time.

Proof. In the inviscid case the vorticity equation reads $D\boldsymbol{\omega}/Dt = \boldsymbol{\omega} \cdot \nabla \mathbf{u}$. This is a linear ordinary differential equation for $\boldsymbol{\omega}$ along each particle path, homogeneous in $\boldsymbol{\omega}$, so if $\boldsymbol{\omega} = \mathbf{0}$ at $t = 0$ on a given particle path then $\boldsymbol{\omega} = \mathbf{0}$ on it for all time.

Alternatively: by Kelvin's theorem the circulation around every material curve is preserved, and initially every such circulation is zero. $\square$

This is enormously useful, because the great majority of the flows we care about *do* start from rest. Water in a tank before we disturb it; air far upstream of an aeroplane; the ocean before a storm. All of these are at rest, hence trivially irrotational, hence – to the extent that we may neglect viscosity – irrotational forever.

We should be honest about the limits of this. Real fluids are viscous, and viscosity generates vorticity at boundaries. What actually happens at high Reynolds number is that vorticity is confined to thin boundary layers on solid surfaces and to the wakes shed behind bodies, while the bulk of the fluid remains beautifully irrotational. So the irrotational model is excellent away from surfaces and wakes, and hopeless inside them – an assessment we will have cause to remember when we compute the drag on a sphere and get zero.



§13.6 Inviscid Irrotational Flow

We now specialise, for the rest of the course, to flows which are incompressible, inviscid *and* irrotational. We have just seen that the last assumption is self-consistent, so let us see what it buys us.

If $\boldsymbol{\omega} = \nabla \times \mathbf{u} = \mathbf{0}$ on a simply connected region, then $\mathbf{u}$ is a gradient.

Definition 13.6.1 (Velocity Potential)

A **velocity potential** for an irrotational flow is a scalar ϕ with

$$\mathbf{u} = \nabla \phi.$$

(Note the sign: unlike gravity or electrostatics, we do not put a minus sign in front. Fluid dynamicists are lazy.) Now impose incompressibility:

$$\nabla \cdot \mathbf{u} = \nabla^2 \phi = 0.$$

Definition 13.6.2 (Potential Flow)

A **potential flow** is an incompressible irrotational flow, described by a velocity

potential satisfying Laplace's equation $\nabla^2\phi = 0$.

This is a spectacular simplification. We have replaced a nonlinear system of four coupled partial differential equations by a single *linear* equation, and moreover the best-understood linear equation in all of mathematics. In particular we may superpose solutions, so we can build up complicated flows by adding simple ones. The velocity comes from $\mathbf{u} = \nabla\phi$ and the pressure afterwards from Bernoulli, which (the flow being irrotational) holds globally, not just along streamlines.

Boundary conditions are as follows: at a stationary rigid boundary, $\mathbf{u} \cdot \mathbf{n} = \partial\phi/\partial n = 0$, a Neumann condition. We have already given up no-slip.

It is worth pausing on the relationship between ϕ and the streamfunction ψ , since students often confuse them. In two dimensions both exist, and

$$u = \frac{\partial\phi}{\partial x} = \frac{\partial\psi}{\partial y}, \quad v = \frac{\partial\phi}{\partial y} = -\frac{\partial\psi}{\partial x}.$$

These are the Cauchy–Riemann equations! We will exploit that fact in Section 13.6.5. But note that ψ exists whenever the flow is incompressible (whether or not it is irrotational), while ϕ exists whenever it is irrotational (whether or not it is incompressible). It is only when we have both that they come as a pair. Their contours are the streamlines and the equipotentials respectively, and they meet at right angles.

§13.6.1 Three-Dimensional Potential Flows

We solve Laplace's equation by separation of variables, exactly as in the Methods course. In spherical polars (r, θ, φ) ,

$$\nabla^2\phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\phi}{\partial r} \right) + \frac{1}{r^2 \sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial\phi}{\partial\theta} \right) + \frac{1}{r^2 \sin^2\theta} \frac{\partial^2\phi}{\partial\varphi^2},$$

and the velocity is

$$\mathbf{u} = \nabla\phi = \left(\frac{\partial\phi}{\partial r}, \frac{1}{r} \frac{\partial\phi}{\partial\theta}, \frac{1}{r \sin\theta} \frac{\partial\phi}{\partial\varphi} \right).$$

Start with the simplest possible case, $\phi = \phi(r)$. Then Laplace's equation is $(r^2\phi')' = 0$, so $\phi' = A/r^2$ and

$$\phi = -\frac{A}{r}$$

(discarding an irrelevant additive constant). The velocity is purely radial and falls off like r^{-2} .

What is A ? Compute the volume flux out of the sphere $r = a$:

$$q = \int_S \mathbf{u} \cdot \mathbf{n} \, dS = \int_S \frac{\partial\phi}{\partial r} \, dS = \frac{A}{a^2} \cdot 4\pi a^2 = 4\pi A.$$

Definition 13.6.3 (Point Source)

A **point source** of strength q at the origin has potential

$$\phi = -\frac{q}{4\pi r}, \quad u_r = \frac{q}{4\pi r^2}.$$

If $q < 0$ we call it a **sink**.

We could have found this without any calculus: incompressibility says the flux through every sphere about the origin is the same, and the area grows like r^2 , so the velocity must fall like r^{-2} . Note that $\nabla^2 \phi = q \delta(\mathbf{x})$, so up to a constant ϕ is the Green's function for the Laplacian – the source is really a singularity that we have inserted by hand, and the flow is genuinely a potential flow only away from the origin.

Allowing θ dependence but keeping axisymmetry, separation of variables gives the general solution

$$\phi = \sum_{n=0}^{\infty} (A_n r^n + B_n r^{-n-1}) P_n(\cos \theta),$$

where P_n are the Legendre polynomials, with $P_0 = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$. The $n = 1$ decaying solution, $\phi \propto \cos \theta / r^2$, is a **dipole**, and it is the workhorse of what follows.

Example 13.6.4 (Uniform Flow Past a Sphere)

Let us find the flow past a fixed sphere of radius a , with uniform stream $\mathbf{u} \rightarrow U \hat{\mathbf{x}}$ at infinity. Since $\phi = Ux = Ur \cos \theta$ generates the uniform stream, we must solve

$$\begin{aligned} \nabla^2 \phi &= 0 & r > a, \\ \phi &\rightarrow Ur \cos \theta & r \rightarrow \infty, \\ \frac{\partial \phi}{\partial r} &= 0 & r = a, \end{aligned}$$

the last being the condition that no fluid enters the sphere.

The far-field condition is proportional to $P_1(\cos \theta)$, and the P_n are orthogonal, so only the $n = 1$ terms can appear:

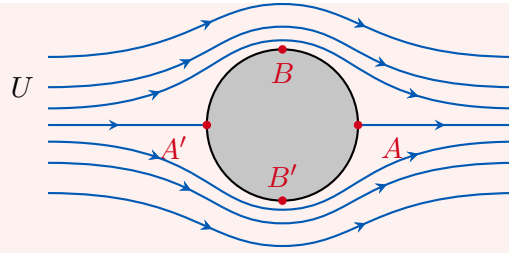
$$\phi = \left(Ar + \frac{B}{r^2} \right) \cos \theta.$$

Matching at infinity gives $A = U$, and the boundary condition at $r = a$ gives $U - 2B/a^3 = 0$, so $B = Ua^3/2$. Thus

$$\phi = U \left(r + \frac{a^3}{2r^2} \right) \cos \theta.$$

The flow is a uniform stream plus the dipole response of the sphere. Differentiating,

$$\begin{aligned} u_r &= U \left(1 - \frac{a^3}{r^3} \right) \cos \theta, \\ u_\theta &= -U \left(1 + \frac{a^3}{2r^3} \right) \sin \theta. \end{aligned}$$



At $\theta = 0$ and $\theta = \pi$ on the surface both components vanish: these are the front and rear stagnation points A and A' . At the ‘poles’ $\theta = \pm\pi/2$ we have $u_r = 0$ and $|u_\theta| = \frac{3}{2}U$, so the flow over the top of the sphere is fifty percent faster than the free stream. (This is why it is windier at the top of a hill than in the valley, and why the streamlines in the figure bunch together there.)

Now for the pressure. Bernoulli applies globally, so comparing the surface with infinity,

$$p_\infty + \frac{1}{2}\rho U^2 = p + \frac{1}{2}\rho \cdot \frac{9}{4}U^2 \sin^2 \theta,$$

giving the surface pressure

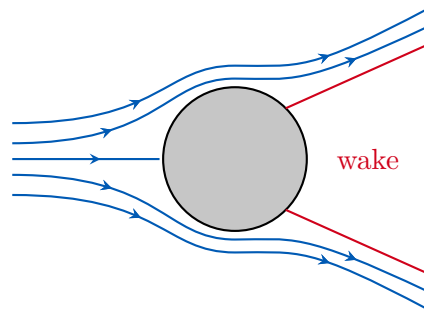
$$p = p_\infty + \frac{1}{2}\rho U^2 \left(1 - \frac{9}{4} \sin^2 \theta\right).$$

At the stagnation points A and A' this is $p_\infty + \frac{1}{2}\rho U^2$, the maximum; at the poles it is $p_\infty - \frac{5}{8}\rho U^2$, the minimum.

And now something has gone wrong. The pressure depends only on $\sin^2 \theta$, so it is exactly the same at the front and the back of the sphere. Integrating $-p\mathbf{n}$ over the surface gives *zero net force*. The fluid exerts no drag whatsoever.

Remark (d’Alembert’s Paradox). The conclusion of the last example is completely general: a body moving steadily through an inviscid, irrotational, incompressible fluid experiences no drag. This is **d’Alembert’s paradox**, and it is obviously false – if it were true, cycling would be effortless.

The resolution is that we threw away viscosity. What happens in reality at high Reynolds number is that the flow does look very much like our solution over the front of the sphere, but the adverse pressure gradient at the rear (the pressure is rising as the fluid moves from the poles towards A') causes the boundary layer to *separate*. Behind the sphere is a turbulent wake in which the pressure is roughly p_∞ rather than recovering to $p_\infty + \frac{1}{2}\rho U^2$, and this fore-aft asymmetry is the drag.



Empirically one writes the drag force as

$$F = C_D \cdot \frac{1}{2}\rho U^2 \pi a^2,$$

defining a dimensionless **drag coefficient** C_D which is a function of Re alone (by dynamic similarity), and which must be measured.

Not everything about the sphere solution is wrong, though. The kinetic energy it carries is a real and measurable thing.

Example 13.6.5 (Added Mass and the Rising Bubble)

Change frame so that the sphere moves at speed U through fluid at rest; we simply subtract $U\hat{\mathbf{x}}$ from the velocity field of Example 13.6.4, leaving only the dipole part

$$\phi = \frac{Ua^3}{2r^2} \cos \theta.$$

The kinetic energy of the fluid is then

$$\int_{r>a} \frac{1}{2}\rho|\mathbf{u}|^2 dV = \frac{\pi}{3}a^3\rho U^2 = \frac{1}{2}M_A U^2,$$

where

$$M_A = \frac{2}{3}\pi a^3\rho = \frac{1}{2}M_D$$

is the **added mass** and $M_D = \frac{4}{3}\pi a^3\rho$ is the mass of fluid displaced. In other words, to move a sphere through a fluid you must also accelerate the fluid around it, and the sphere behaves as if it had an extra $\frac{1}{2}M_D$ of inertia.

Here is a cute application. A small spherical bubble rises through water. It is essentially massless, so the only energy in the problem is the kinetic energy of the water and the potential energy released as the bubble rises a distance h :

$$\frac{1}{2}M_A U^2 - M_D gh = \text{const.}$$

Differentiating with respect to time and using $\dot{h} = U$,

$$M_A U \dot{U} = M_D g U \implies \dot{U} = \frac{M_D}{M_A} g = 2g.$$

So an inviscid bubble accelerates upwards at *twice* the acceleration due to gravity. (In practice viscosity and deformation intervene long before the bubble gets very fast, but the factor of two is real and has been measured.)

§13.6.2 Potential Flows in Two Dimensions

Two-dimensional flows are where most of the classical theory lives, partly because they are easier, and partly because a long thin wing is nearly two-dimensional. In plane polars

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2}, \quad \mathbf{u} = \left(\frac{\partial \phi}{\partial r}, \frac{1}{r} \frac{\partial \phi}{\partial \theta} \right),$$

and separation of variables gives the general solution

$$\phi = A \log r + B\theta + \sum_{n=1}^{\infty} (A_n r^n + B_n r^{-n}) \begin{Bmatrix} \cos n\theta \\ \sin n\theta \end{Bmatrix}.$$

Let us go through the building blocks one at a time.

Example 13.6.6 (Line Source)

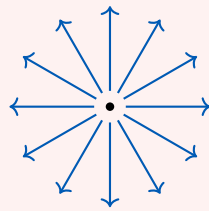
Take $\phi = A \log r$. By conservation of mass, if q is the volume flux per unit length in the z direction then $2\pi r u_r = q$, so $u_r = q/2\pi r$ and

$$\phi = \frac{q}{2\pi} \log r, \quad \psi = \frac{q}{2\pi} \theta.$$

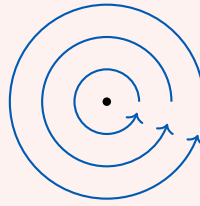
The streamlines are radial lines and the equipotentials are circles, meeting at right angles as promised.

Example 13.6.7 (Line Vortex)

Take $\phi = B\theta$. Then $u_r = 0$ and $u_\theta = B/r$: the flow goes round in circles, faster near the centre.



source



line vortex

To fix B , compute the circulation around a circle of radius a :

$$\kappa = \oint_{r=a} \mathbf{u} \cdot d\boldsymbol{\ell} = \int_0^{2\pi} \frac{B}{a} a d\theta = 2\pi B.$$

So

$$\phi = \frac{\kappa}{2\pi} \theta, \quad u_\theta = \frac{\kappa}{2\pi r}, \quad \psi = -\frac{\kappa}{2\pi} \log r.$$

Now, this flow *looks* rotational – everything is going round in circles – but it is not. One checks directly that $\nabla \times \mathbf{u} = \mathbf{0}$ for $r \neq 0$: a paddle-wheel placed off-centre translates around the vortex without spinning, because the shear on its inner side exactly cancels the shear on its outer side. All the vorticity is concentrated at the singular point $r = 0$.

This has a consequence which we cannot stress enough. For any simple closed curve C ,

$$\oint_C \mathbf{u} \cdot d\boldsymbol{\ell} = \begin{cases} \kappa & \text{if } C \text{ encloses the origin,} \\ 0 & \text{otherwise,} \end{cases}$$

so the circulation is not zero, even though the vorticity vanishes everywhere in the fluid. There is no contradiction with Stokes' theorem, because the region occupied by the fluid ($r > 0$) is not simply connected and no surface spanning C lies entirely in the fluid. Note also that $\phi = \kappa\theta/2\pi$ is not single-valued: it increases by κ each time we go round.

Example 13.6.8 (Line Dipole)

Place a source of strength q at $(\varepsilon, 0)$ and a sink of strength $-q$ at $(-\varepsilon, 0)$, and let

$\varepsilon \rightarrow 0$ with $q\varepsilon$ held fixed. Then

$$\phi = \frac{q}{2\pi} (\log |\mathbf{x} - \varepsilon \hat{\mathbf{x}}| - \log |\mathbf{x} + \varepsilon \hat{\mathbf{x}}|) \rightarrow -\frac{q\varepsilon}{\pi} \cdot \frac{\cos \theta}{r},$$

so up to normalisation a two-dimensional dipole has $\phi \propto \cos \theta / r$. This is precisely the $n = 1$ decaying term in our general solution, and it is what a cylinder in a stream will generate.

§13.6.3 Flow Past a Cylinder, and Lift

We now assemble the pieces into the most important example in the course.

Example 13.6.9 (Uniform Flow Past a Cylinder with Circulation)

We solve

$$\begin{aligned} \nabla^2 \phi &= 0 & r > a, \\ \phi &\rightarrow Ur \cos \theta & r \rightarrow \infty, \\ \frac{\partial \phi}{\partial r} &= 0 & r = a. \end{aligned}$$

As with the sphere, the far field picks out the $\cos \theta$ terms; but now we are also allowed to add a line vortex, since the region $r > a$ is not simply connected and a circulation κ round the cylinder is perfectly consistent with irrotational flow. So

$$\phi = U \left(r + \frac{a^2}{r} \right) \cos \theta + \frac{\kappa}{2\pi} \theta,$$

with

$$\begin{aligned} u_r &= U \left(1 - \frac{a^2}{r^2} \right) \cos \theta, \\ u_\theta &= -U \left(1 + \frac{a^2}{r^2} \right) \sin \theta + \frac{\kappa}{2\pi r}, \end{aligned}$$

and streamfunction

$$\psi = Ur \sin \theta \left(1 - \frac{a^2}{r^2} \right) - \frac{\kappa}{2\pi} \log r.$$

Physically, the circulation κ is a stand-in for the vorticity trapped in the viscous boundary layer on the cylinder – for instance because the cylinder is spinning.

Stagnation points. We need $u_r = u_\theta = 0$. The first gives $r = a$ or $\theta = \pm\pi/2$. On $r = a$ the second gives

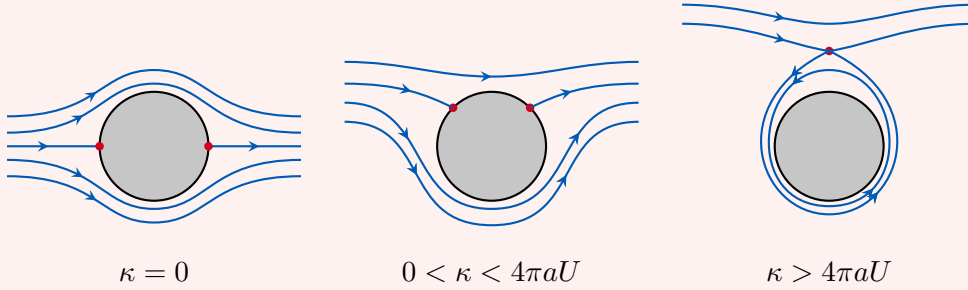
$$\sin \theta = \frac{\kappa}{4\pi a U}.$$

So provided $|\kappa| \leq 4\pi a U$ there are two stagnation points on the surface of the cylinder, symmetrically placed about the vertical axis. As κ increases from zero they start at $\theta = 0, \pi$ and migrate upwards (for $\kappa > 0$) until they merge at the top when

$\kappa = 4\pi aU$. Beyond that they leave the cylinder altogether: the remaining solution is on $\theta = \pi/2$ with

$$\frac{\kappa}{2\pi r} = U \left(1 + \frac{a^2}{r^2} \right),$$

a single stagnation point above the cylinder, with a region of fluid trapped circulating around it.



The force. On the surface $r = a$ the speed is $|u_\theta| = |-2U \sin \theta + \kappa/2\pi a|$, so Bernoulli (globally valid, the flow being irrotational and steady) gives

$$p_\infty + \frac{1}{2}\rho U^2 = p + \frac{1}{2}\rho \left(\frac{\kappa}{2\pi a} - 2U \sin \theta \right)^2,$$

that is,

$$p = p_\infty + \frac{1}{2}\rho U^2 - \frac{\rho \kappa^2}{8\pi^2 a^2} + \frac{\rho \kappa U \sin \theta}{\pi a} - 2\rho U^2 \sin^2 \theta.$$

This is still symmetric under $\theta \mapsto \pi - \theta$, so there is still no drag – d’Alembert has not gone away. But it is *not* symmetric under $\theta \mapsto -\theta$, and computing the transverse force per unit length,

$$F_y = - \int_0^{2\pi} p \sin \theta \, a \, d\theta = - \int_0^{2\pi} \frac{\rho \kappa U}{\pi a} \sin^2 \theta \, a \, d\theta = -\rho U \kappa,$$

every other term integrating to zero by parity. So there is a sideways force, at right angles to the oncoming stream, directly proportional to the circulation.

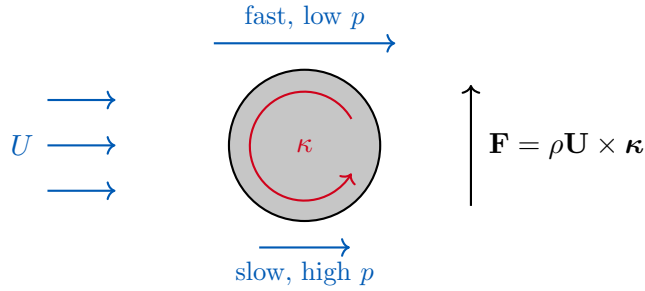
This is the central fact of aerodynamics, and it deserves to be stated on its own.

Theorem 13.6.10 (Magnus Force / Kutta–Joukowski Lift)

A two-dimensional body in a uniform stream $\mathbf{U}$ with circulation κ about it experiences a force per unit length

$$\mathbf{F} = \rho \mathbf{U} \times \boldsymbol{\kappa},$$

where $\boldsymbol{\kappa} = \kappa \hat{\mathbf{z}}$. There is no drag.



The mechanism is now transparent: the circulation adds to the free stream on one side of the body and subtracts on the other, so the flow is faster above than below, so by Bernoulli the pressure is lower above than below, so there is a net upward force. This is why a sliced tennis ball swerves, why a spinning cricket ball dips, and – with a little more work – why an aerofoil generates lift. (For an aerofoil the circulation is not imposed by spinning it; it is generated automatically by the sharp trailing edge, via the Kutta condition, and comes with an equal and opposite ‘starting vortex’ shed downstream, in accordance with Kelvin’s theorem.)

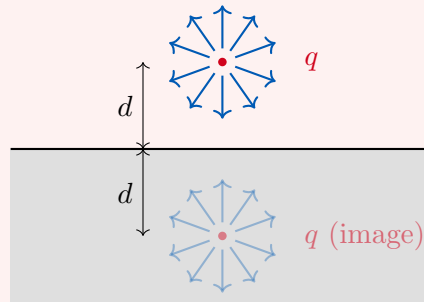
§13.6.4 The Method of Images

Laplace’s equation is linear, and boundaries are just conditions we have to satisfy. Sometimes we can satisfy them by adding an extra singularity outside the fluid, exactly as in electrostatics.

Example 13.6.11 (Source Next to a Plane Wall)

Suppose there is a rigid wall along $y = 0$ and a line source of strength q at $(0, d)$, with fluid in $y > 0$. The wall requires $v = 0$ on $y = 0$.

Place an image source, also of strength q , at $(0, -d)$ – outside the fluid, so it does not disturb Laplace’s equation where we need it. By symmetry the two vertical velocities on $y = 0$ are equal and opposite, so $v = 0$ there automatically.



Hence in $y > 0$,

$$\phi = \frac{q}{4\pi} \log(x^2 + (y - d)^2) + \frac{q}{4\pi} \log(x^2 + (y + d)^2),$$

and the flow in the physical region $y > 0$ is exactly that of two equal sources. Notice we have solved a problem with a boundary by solving a problem without one.

The same idea handles a vortex next to a wall (image vortex of *opposite* sign, so that the wall is a streamline; the pair then translates parallel to the wall, which is why smoke rings bounce along ceilings), and a source outside a circular cylinder (the Milne–Thomson

circle theorem, which puts an image at the inverse point a^2/d .

§13.6.5 Complex Potentials

We noted earlier that in two dimensions,

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \quad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x},$$

which are exactly the Cauchy–Riemann equations for $\phi + i\psi$ as a function of $z = x + iy$. This is not a coincidence, and it is the doorway to an enormously powerful set of techniques.

Definition 13.6.12 (Complex Potential)

The **complex potential** of a two-dimensional potential flow is the holomorphic function

$$w(z) = \phi(x, y) + i\psi(x, y),$$

and its derivative is the **complex velocity**

$$\frac{dw}{dz} = u - iv.$$

Every one of our building blocks is now a one-line formula: a uniform stream is $w = Uz$; a source of strength q at the origin is $w = (q/2\pi) \log z$; a line vortex of circulation κ is $w = -(i\kappa/2\pi) \log z$; a dipole is $w = m/z$; and flow past a cylinder with circulation is

$$w(z) = U \left(z + \frac{a^2}{z} \right) - \frac{i\kappa}{2\pi} \log z,$$

whose real and imaginary parts the reader may check reproduce the ϕ and ψ of Example 13.6.9. Better still, if $\zeta = f(z)$ is a conformal map then $w(\zeta(z))$ is again a complex potential, so we can transplant the solution for a cylinder onto any shape a conformal map can reach – including, via the Joukowski map, an aerofoil.

We will not develop this here, since it is properly the business of the ‘Complex Methods’ course, and the reader is referred to my notes for that course ([../part-ib/complex-methods/](#)), where conformal mapping and its fluid applications are treated at length. The only point to take away now is that potential flow in two dimensions and complex analysis are the same subject.

§13.6.6 Unsteady Potential Flows

Everything so far has been steady. If the flow is unsteady but still irrotational, Bernoulli takes a slightly different and very useful form. Starting from Euler in the form

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times \boldsymbol{\omega} \right) = -\nabla p - \nabla \chi$$

and setting $\boldsymbol{\omega} = \mathbf{0}$, $\mathbf{u} = \nabla \phi$, everything becomes a gradient:

$$\nabla \left(\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho |\nabla \phi|^2 + p + \chi \right) = \mathbf{0}.$$

Proposition 13.6.13 (Unsteady Bernoulli)

For an unsteady irrotational flow of an inviscid incompressible fluid of constant density,

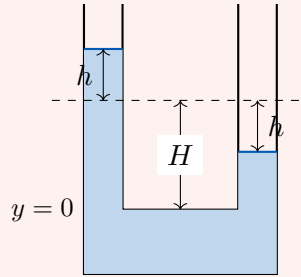
$$\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho |\nabla \phi|^2 + p + \chi = f(t)$$

for some function f of time alone.

The point is that f is the same at every point in space, so we may equate the left hand side at two different places, however far apart, and thereby couple the dynamics of distant parts of the flow. Two classic examples follow.

Example 13.6.14 (Oscillations in a Manometer)

A U-tube of uniform cross section contains liquid, with the equilibrium level a height H above the bottom bend, and we displace the level in the left arm up by $h(t)$.



Dimensional analysis already tells us a lot: the only parameters are g and H , so the frequency must be a multiple of $\sqrt{g/H}$. Let us get the constant.

Take the reservoir at the bottom to be wide, so the fluid there is essentially at rest and $\phi = 0$ there. In the left arm the fluid moves vertically at $\dot{h}$, so $\phi = \dot{h}y$ and $\partial\phi/\partial t = \ddot{h}y$; in the right arm the fluid moves down at $\dot{h}$, so $\phi = -\dot{h}y$ and $\partial\phi/\partial t = -\ddot{h}y$. The free surfaces are at $y = H + h$ and $y = H - h$, both at atmospheric pressure.

Applying unsteady Bernoulli at each free surface and equating (both equal $f(t)$),

$$\rho \ddot{h}(H + h) + \frac{1}{2} \rho \dot{h}^2 + p_{\text{atm}} + g\rho(H + h) = -\rho \ddot{h}(H - h) + \frac{1}{2} \rho \dot{h}^2 + p_{\text{atm}} + g\rho(H - h).$$

A great deal cancels, leaving $2\rho H\ddot{h} + 2g\rho h = 0$, that is,

$$\ddot{h} + \frac{g}{H}h = 0.$$

Simple harmonic motion with frequency exactly $\sqrt{g/H}$ – the dimensionless constant is 1.

Example 13.6.15 (Oscillations of a Bubble)

A spherical bubble of radius $a(t)$ sits in an infinite fluid and pulses radially. The induced flow is spherically symmetric, so

$$\nabla^2 \phi = 0 \quad (r > a), \quad \phi \rightarrow 0 \quad (r \rightarrow \infty), \quad \frac{\partial \phi}{\partial r} = \dot{a} \quad (r = a).$$

The decaying spherically symmetric solution is $\phi = A(t)/r$, so $u_r = -A/r^2$, and the boundary condition gives $-A/a^2 = \dot{a}$. Hence

$$\phi = -\frac{a^2\dot{a}}{r}, \quad \left. \frac{\partial\phi}{\partial t} \right|_{r=a} = -(a\ddot{a} + 2\dot{a}^2).$$

Neglect gravity and apply unsteady Bernoulli at $r = a$ and at infinity:

$$-\rho(a\ddot{a} + 2\dot{a}^2) + \frac{1}{2}\rho\dot{a}^2 + p(a, t) = p_\infty,$$

that is, the **Rayleigh–Plesset equation**

$$\rho(a\ddot{a} + \frac{3}{2}\dot{a}^2) = p(a, t) - p_\infty.$$

This is nonlinear, so linearise: put $a = a_0 + \eta$ with $\eta \ll a_0$, and to first order

$$\rho a_0 \ddot{\eta} = p(a, t) - p_\infty = \delta p.$$

To relate δp to η we need thermodynamics. If the oscillation is fast there is no time for heat exchange, so it is adiabatic and pV^γ is constant, where γ is the ratio of specific heats. Taking logarithms and varying,

$$\frac{\delta p}{p_\infty} = -\gamma \frac{\delta V}{V_0} = -3\gamma \frac{\eta}{a_0},$$

using $V = \frac{4}{3}\pi a^3$. Therefore

$$\rho a_0 \ddot{\eta} = -\frac{3\gamma p_\infty}{a_0} \eta,$$

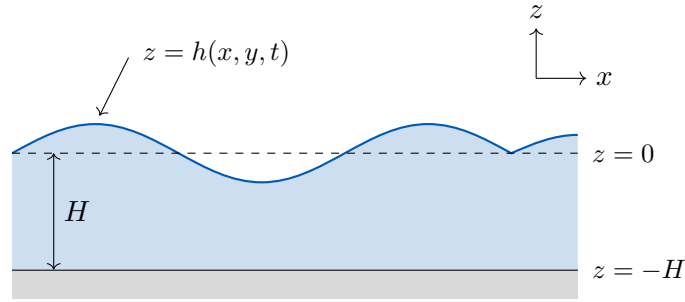
simple harmonic motion with frequency

$$\omega = \left(\frac{3\gamma p_\infty}{\rho a_0^2} \right)^{1/2}.$$

Putting numbers in for a bubble of radius 1 mm in water gives $\omega \sim 2 \times 10^4 \text{ s}^{-1}$, comfortably inside the human audible range of 20 to 20 000 s^{-1} . This is the sound of a babbling brook, or of a glass of fizzy water: we are literally hearing individual bubbles ring.

§13.7 Water Waves

We now apply the machinery of potential flow to what is probably the most familiar fluid phenomenon of all. The set-up is a layer of water of undisturbed depth H over a flat bottom, with air above, and a free surface at $z = h(x, y, t)$.



We will start with dimensional analysis, which gets us surprisingly far, and then solve the problem properly.

§13.7.1 What Dimensional Analysis Tells Us

Consider waves of wavenumber $k = 2\pi/\lambda$ on water of depth H , and suppose the fluid is inviscid. The wave speed c can depend only on k , g and H , so dimensionally

$$c = \sqrt{gH} f(kH)$$

for some dimensionless function f .

Now consider the two extreme regimes.

Deep water, $kH \gg 1$: the wavelength is short compared with the depth, so the wave has no way of knowing how deep the water is, and c cannot depend on H . The only way to arrange that is $f(s) \propto s^{-1/2}$, giving

$$c = \alpha \sqrt{\frac{g}{k}}$$

for some constant α . Long waves travel faster than short ones.

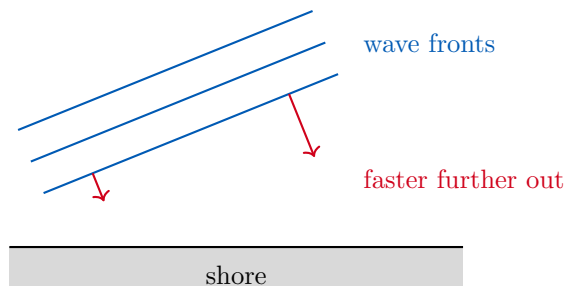
Shallow water, $kH \ll 1$: the wavelength is enormous compared with the depth, and the wave cannot resolve individual wavelengths, so c cannot depend on k . Hence f is a constant β and

$$c = \beta \sqrt{gH}.$$

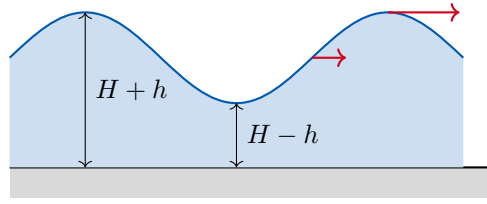
All waves travel at the same speed, set by the depth.

We cannot get α and β this way, nor can we interpolate between the two regimes; for that we need to solve the equations, and we will find that $\alpha = \beta = 1$. But two well-known phenomena already follow.

First, *why wave crests are always parallel to the beach*. Approaching a shore the water gets shallower, so waves slow down. If a wavefront comes in at an angle, the end that is further out is in deeper water and travels faster, so the front swings round until it is parallel to the shore. It is refraction, exactly as in optics.



Secondly, *why waves break*. In shallow water $c = \sqrt{gH}$, so where the water is deeper the wave moves faster. But at a crest the water *is* deeper, by the wave amplitude. When the amplitude becomes comparable with the depth, the crest overtakes the trough in front of it, the front face steepens, and the wave breaks.



§13.7.2 The Linearised Problem

Now let us set the problem up properly. The water starts at rest, so by the persistence of irrotationality it is irrotational forever, and being incompressible it has a potential satisfying

$$\nabla^2 \phi = 0, \quad -H < z < h.$$

There are three boundary conditions.

- (i) *No flow through the bottom*: $\frac{\partial \phi}{\partial z} = 0$ on $z = -H$.
- (ii) *Kinematic condition at the free surface*: by Section 3.5, the surface is a material surface, so

$$\frac{\partial \phi}{\partial z} = \frac{Dh}{Dt} = \frac{\partial h}{\partial t} + u \frac{\partial h}{\partial x} + v \frac{\partial h}{\partial y} \quad \text{on } z = h.$$

- (iii) *Dynamic condition at the free surface*: the pressure there must equal the (constant) atmospheric pressure p_0 . By unsteady Bernoulli, with $\chi = g\rho z$,

$$\rho \frac{\partial \phi}{\partial t} + \frac{1}{2} \rho |\nabla \phi|^2 + g\rho h + p_0 = f(t) \quad \text{on } z = h.$$

Note that gravity has come back, and had to: at a free surface the density is not constant across the boundary, so the reduced-pressure trick is unavailable. Gravity is the restoring force for these waves.

The equation is easy; the boundary conditions are horrible. They are nonlinear, and worse, they are imposed on a surface $z = h$ whose location we do not know until we have solved the problem. So we linearise. Assume:

- small amplitude, $h \ll H$;
- small slope, $|\partial h / \partial x|, |\partial h / \partial y| \ll 1$;

and throw away everything quadratic in small quantities. The velocities are themselves small (the water would be at rest if h were), so $u \partial h / \partial x$ goes, and so does all of $|\nabla \phi|^2$. Finally, Taylor expanding about $z = 0$,

$$\left. \frac{\partial \phi}{\partial z} \right|_{z=h} = \left. \frac{\partial \phi}{\partial z} \right|_{z=0} + h \left. \frac{\partial^2 \phi}{\partial z^2} \right|_{z=0} + \cdots \approx \left. \frac{\partial \phi}{\partial z} \right|_{z=0},$$

so we may apply the surface conditions on the fixed plane $z = 0$ instead. We are left with a completely linear problem.

Proposition 13.7.1 (Linear Water Wave Equations)

$$\begin{aligned}
\nabla^2 \phi &= 0 & -H < z \leq 0, \\
\frac{\partial \phi}{\partial z} &= 0 & z = -H, \\
\frac{\partial \phi}{\partial z} &= \frac{\partial h}{\partial t} & z = 0, \\
\frac{\partial \phi}{\partial t} + gh &= 0 & z = 0,
\end{aligned}$$

the last after absorbing constants into ϕ .

§13.7.3 The Dispersion Relation

Consider straight-crested waves travelling in the x direction, so nothing depends on y . Try a travelling wave

$$h = h_0 e^{i(kx - \omega t)},$$

(with the convention that we take real parts at the end), and correspondingly

$$\phi = \hat{\phi}(z) e^{i(kx - \omega t)}.$$

Laplace's equation gives $\hat{\phi}'' - k^2 \hat{\phi} = 0$, so $\hat{\phi}$ is a combination of $e^{\pm kz}$; the bottom condition $\hat{\phi}'(-H) = 0$ picks out

$$\hat{\phi} = \phi_0 \cosh k(z + H).$$

There are now three unknowns h_0, ϕ_0, ω and two conditions left. The kinematic condition at $z = 0$ gives

$$k\phi_0 \sinh kH = -i\omega h_0,$$

and the dynamic condition gives

$$-i\omega\phi_0 \cosh kH + gh_0 = 0.$$

Eliminating ϕ_0 and h_0 (the alternative being the trivial solution) gives $\omega^2 \cosh kH = gk \sinh kH$.

Theorem 13.7.2 (Dispersion Relation for Water Waves)

Linear surface gravity waves on water of depth H satisfy

$$\omega^2 = gk \tanh kH,$$

and hence travel with phase speed

$$c = \frac{\omega}{k} = \sqrt{\frac{g}{k} \tanh kH}.$$

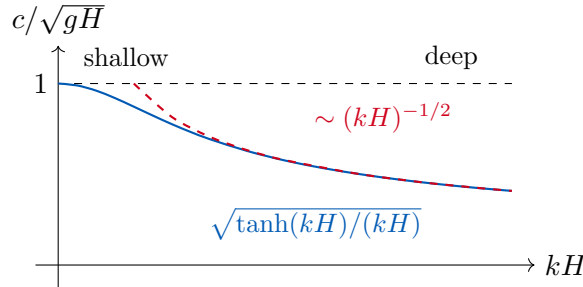
Let us check our limits. As $kH \rightarrow \infty$, $\tanh kH \rightarrow 1$, so in deep water

$$\omega^2 = gk, \quad c = \sqrt{\frac{g}{k}} = \sqrt{\frac{g\lambda}{2\pi}}.$$

As $kH \rightarrow 0$, $\tanh kH \rightarrow kH$, so in shallow water

$$\omega^2 = gHk^2, \quad c = \sqrt{gH}.$$

Exactly as dimensional analysis predicted, with $\alpha = \beta = 1$.



Several things follow from the shape of this curve.

The wave speed decreases monotonically with k : long waves are fast, short waves are slow. Hence a disturbance of a general shape – a square wave, say, or the splash of a stone – disperses, its long components racing ahead and its short ones lagging behind. That is what the word ‘dispersion’ means.

There is a *maximum* wave speed $\sqrt{gH}$, attained in the shallow-water limit. Nothing on the surface can travel faster than this. If a boat travels faster than $\sqrt{gH}$ then all the waves it produces are left behind, and it produces a wake with a sharp bow shock, exactly like a supersonic aircraft. The relevant dimensionless number is

Definition 13.7.3 (Froude Number)

The **Froude number** of a flow of speed U over depth H is

$$Fr = \frac{U}{\sqrt{gH}},$$

the analogue of the Mach number for surface waves.

Example 13.7.4 (Tsunami)

A tsunami has wavelength $\lambda \sim 400$ km in an ocean of depth $H \sim 4$ km, so $kH \sim 0.06 \ll 1$ and we are firmly in the shallow water regime. Hence

$$c = \sqrt{gH} = \sqrt{10 \times 4 \times 10^3} = 200 \text{ m s}^{-1},$$

roughly the speed of a jet aircraft, crossing the Pacific in under a day. Note that the speed depends only on the depth, so the topography of the ocean floor refracts and focusses tsunamis – knowing the bathymetry lets one predict where a tsunami will strike hardest, which saves lives.

Note also that in deep water a tsunami has an amplitude of perhaps half a metre spread over 400 km: a ship at sea would not notice it passing. It is only as H decreases near the coast, so that c decreases and (by conservation of energy flux) the amplitude must grow, that it becomes destructive.

§13.7.4 Particle Paths

What do individual fluid particles do as a wave passes? They certainly do not travel with the wave – a cork bobs, it does not surf.

Take the deep water case for cleanliness. With $h = h_0 \cos(kx - \omega t)$ (real part taken), the relations above give

$$\phi = \frac{h_0 g}{\omega} e^{kz} \sin(kx - \omega t),$$

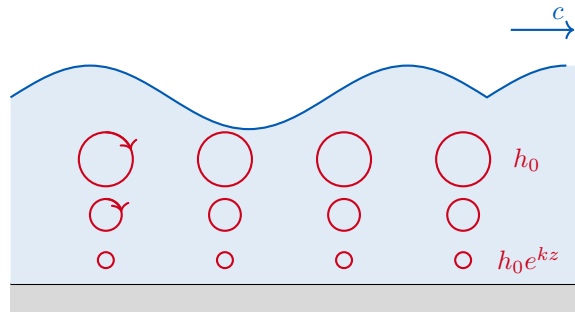
so

$$u = \frac{\partial \phi}{\partial x} = h_0 \omega e^{kz} \cos(kx - \omega t), \quad w = \frac{\partial \phi}{\partial z} = h_0 \omega e^{kz} \sin(kx - \omega t),$$

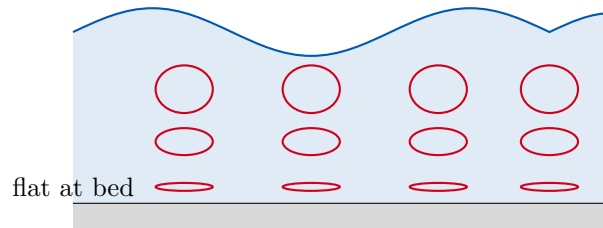
using $\omega^2 = gk$. To leading order a particle near (x_0, z_0) therefore has

$$x - x_0 = -h_0 e^{kz_0} \sin(kx_0 - \omega t), \quad z - z_0 = h_0 e^{kz_0} \cos(kx_0 - \omega t).$$

So each particle moves in a *circle* of radius $h_0 e^{kz_0}$, traversed once per wave period. The radius decays exponentially with depth, with e-folding scale $1/k = \lambda/2\pi$: at one wavelength down the motion is only $e^{-2\pi} \approx 0.2\%$ of that at the surface, which is why submarines are calm in a storm.



In shallow water the story is different: the exponentials become cosh and sinh, the horizontal motion is nearly independent of depth and the vertical motion vanishes at the bottom, so the circles flatten into ellipses which become more and more elongated as we descend, degenerating into pure horizontal to-and-fro at the bed.

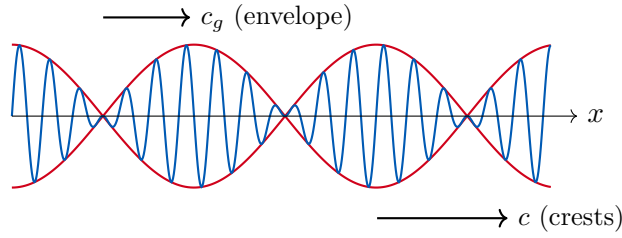


§13.7.5 Group Velocity

Suppose we superpose two waves of nearly equal wavenumber. Using a trigonometric identity,

$$\cos k_1 x + \cos k_2 x = 2 \cos \left(\frac{k_1 + k_2}{2} x \right) \cos \left(\frac{k_1 - k_2}{2} x \right).$$

The first factor oscillates rapidly at the mean wavenumber; the second oscillates slowly and acts as an envelope. The result is a train of wave ‘packets’.



The crests move at the phase speed $c = \omega/k$, but the envelope – and hence the energy, and hence the information – moves at a different speed.

Definition 13.7.5 (Group Velocity)

The **group velocity** of a wave with dispersion relation $\omega(k)$ is

$$c_g = \frac{d\omega}{dk}.$$

To see why, write the envelope factor above with $\Delta k = (k_1 - k_2)/2$ and restore the time dependence: the envelope is $\cos(\Delta k x - \Delta \omega t)$, which moves at speed $\Delta \omega / \Delta k \rightarrow d\omega / dk$.

For deep water waves, $\omega = \sqrt{gk}$, so

$$c_g = \frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} c.$$

The group travels at half the speed of the individual crests. If you watch a group of swell in the open sea carefully you can see this: crests appear at the back of the group, race forward through it, and vanish at the front. For shallow water waves, on the other hand, $\omega = k\sqrt{gH}$ is linear in k , so $c_g = c = \sqrt{gH}$ and the waves are *non-dispersive*: a shallow water wave keeps its shape.

§13.7.6 Standing Waves and Instability

Example 13.7.6 (Standing Waves in a Container)

Take a rectangular tank $0 < x < L$ with vertical end walls, where the condition is $u = \partial\phi/\partial x = 0$. Superposing left- and right-moving waves of the same wavenumber gives a standing wave

$$\phi = \phi_0 \cosh k(z + H) \cos kx \cos \omega t,$$

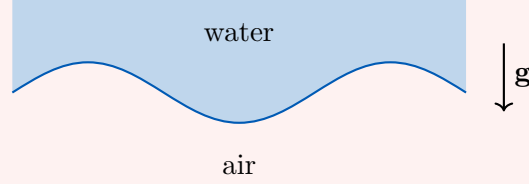
which satisfies the wall conditions provided $\sin kL = 0$, that is,

$$k = \frac{n\pi}{L}, \quad n = 1, 2, \dots$$

So the tank has a discrete set of normal modes, with frequencies $\omega_n^2 = gk_n \tanh k_n H$. This is exactly the phenomenon of *sloshing* in a bath or a fuel tank, and the $n = 1$ mode is the one with the longest period and the largest amplitude. (In a harbour the same modes are called *seiches*, and they can be excited resonantly by tides or by tsunamis, sometimes with dramatic results.)

Example 13.7.7 (Rayleigh–Taylor Instability)

Here is a delightful use of the dispersion relation. Suppose we turn the problem upside down and put the water *above* the air. Every equation is the same, except that gravity now points from the water into the air, so we replace g by $-g$.



In the deep water limit this turns $\omega^2 = gk$ into

$$\omega^2 = -gk \implies \omega = \pm i\sqrt{gk},$$

so that

$$h \propto Ae^{\sqrt{gk}t} + Be^{-\sqrt{gk}t}.$$

One of these solutions grows exponentially: the configuration is **unstable**, and the water falls. This is the **Rayleigh–Taylor instability**, and note that the growth rate $\sqrt{gk}$ increases with k , so the shortest wavelengths grow fastest – which is why the interface breaks up into a fine-scale mess of fingers rather than tipping over in one piece. (In reality surface tension, which we have neglected throughout, cuts the instability off at short wavelengths, which is why you can hold water in an upside-down glass covered with a fine mesh.)

§13.8 Flow in a Rotating Frame

The Earth rotates, so anyone who wants to describe the ocean or the atmosphere must work in a rotating frame. This turns out to change everything: rotating fluids behave in ways that are, at first sight, deeply counterintuitive.

§13.8.1 Equations of Motion in a Rotating Frame

From the Dynamics and Relativity course, the acceleration of a particle in a frame rotating with angular velocity $\boldsymbol{\Omega}$ is

$$\frac{D\mathbf{u}}{Dt} + 2\boldsymbol{\Omega} \times \mathbf{u} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}),$$

the extra terms being the Coriolis and centrifugal accelerations. So the Euler equation becomes

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} + 2\boldsymbol{\Omega} \times \mathbf{u} \right) = -\nabla p - \rho \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}) + \rho \mathbf{g}.$$

This is worse than what we started with, so let us simplify it.

First, the centrifugal term. For the Earth $|\boldsymbol{\Omega}| = 2\pi/(1 \text{ day}) \approx 7 \times 10^{-5} \text{ s}^{-1}$, and taking the largest relevant scale $|\mathbf{x}| \sim 10^7 \text{ m}$,

$$\frac{|\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x})|}{|\mathbf{g}|} \lesssim \frac{(7 \times 10^{-5})^2 \times 10^7}{10} \approx 5 \times 10^{-3}.$$

Utterly negligible compared with gravity. (In any case it is a gradient, $\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{x}) = -\nabla \left(\frac{1}{2} |\boldsymbol{\Omega} \times \mathbf{x}|^2 \right)$, so we could absorb it into the pressure. It is, in effect, already inside what we call ‘gravity’ – it is why the Earth is an oblate spheroid.)

Secondly, the nonlinear term. Comparing it with the Coriolis term, their scales are U^2/L and ΩU , so we may drop the nonlinearity provided

Definition 13.8.1 (Rossby Number)

The **Rossby number** is

$$Ro = \frac{U}{\Omega L},$$

measuring inertia relative to Coriolis effects.

For the atmosphere, $U \sim 10 \text{ m s}^{-1}$ and $L \sim 10^6 \text{ m}$, giving $Ro \approx 0.1$: small, so rotation dominates. (For your bath, $U \sim 0.1$ and $L \sim 0.3$, giving $Ro \sim 5000$; rotation is irrelevant, and the direction your bath drains has nothing whatsoever to do with the Coriolis force.)

Proposition 13.8.2 (Euler’s Equation in a Rotating Frame)

For small Rossby number,

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} = -\frac{1}{\rho} \nabla p + \mathbf{g}, \quad \mathbf{f} := 2\boldsymbol{\Omega}.$$

Definition 13.8.3 (Coriolis Parameter)

The vector $\mathbf{f} = 2\boldsymbol{\Omega}$ is the **planetary vorticity**. Since only the component of $\boldsymbol{\Omega}$ perpendicular to the flow matters, and geophysical flows are nearly horizontal, the relevant quantity is the local vertical component

$$f = 2\Omega \sin \theta,$$

where θ is the latitude. This is the **Coriolis parameter**.

Note that $f > 0$ in the northern hemisphere, $f < 0$ in the southern, and $f = 0$ on the equator. This sign change is the reason cyclones spin the other way in Australia.

§13.8.2 The Shallow Water Equations

Geophysical flows are shallow: the Atlantic is 4 km deep and 5000 km across, giving an aspect ratio like a sheet of paper. Let us exploit this.

Consider a layer of fluid with a free surface at $z = h(x, y, t)$ and $p = p_0$ there. Write $\mathbf{u} = (u, v, w)$ and suppose horizontal scales L vastly exceed the vertical scale H . Incompressibility gives

$$\frac{\partial w}{\partial z} = -\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y},$$

whose terms scale as W/H , U/L and V/L . Since $H \ll L$ we find $W \ll U, V$: the motion is overwhelmingly horizontal, which is intuitively obvious – there is nowhere to go vertically.

So take $\mathbf{u} = (u, v, 0)$ and $\mathbf{f} = (0, 0, f)$. The three components of the rotating Euler equation are then

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -\frac{1}{\rho} \frac{\partial p}{\partial x}, \\ \frac{\partial v}{\partial t} + fu &= -\frac{1}{\rho} \frac{\partial p}{\partial y}, \\ 0 &= -\frac{1}{\rho} \frac{\partial p}{\partial z} - g.\end{aligned}$$

The last is the **hydrostatic approximation**: the vertical momentum equation has collapsed to a statement of hydrostatic balance, because vertical accelerations are so tiny. Integrating it with $p = p_0$ at $z = h$,

$$p = p_0 + g\rho(h - z).$$

Substituting into the horizontal equations, the z dependence drops out entirely and we obtain

Proposition 13.8.4 (Shallow Water Momentum Equations)

$$\begin{aligned}\frac{\partial u}{\partial t} - fv &= -g \frac{\partial h}{\partial x}, \\ \frac{\partial v}{\partial t} + fu &= -g \frac{\partial h}{\partial y}.\end{aligned}$$

The right hand sides are independent of z , so if the velocities are initially independent of z they remain so. Pressure gradients have been replaced entirely by *gradients of the free surface height*: the weather is driven by the shape of the surface.

We also need mass conservation. Taking a fixed horizontal region $\mathcal{D}$ and noting that the mass per unit area is ρh ,

$$\frac{d}{dt} \int_{\mathcal{D}} \rho h \, dA = - \int_{\partial \mathcal{D}} \rho h \mathbf{u} \cdot \mathbf{n} \, d\ell,$$

and applying the (two-dimensional) divergence theorem and taking $\mathcal{D}$ arbitrary gives

Proposition 13.8.5 (Shallow Water Continuity Equation)

$$\frac{\partial h}{\partial t} + \nabla \cdot (h\mathbf{u}) = 0.$$

If fluid converges horizontally at a point, the surface there rises. That is the whole content.

§13.8.3 Geostrophic Balance

Now specialise to steady flow, so the time derivatives vanish:

$$-fv = -g \frac{\partial h}{\partial x}, \quad fu = -g \frac{\partial h}{\partial y},$$

which we may rearrange as

$$u = \frac{\partial}{\partial y} \left(-\frac{gh}{f} \right), \quad v = -\frac{\partial}{\partial x} \left(-\frac{gh}{f} \right).$$

But these are exactly the equations defining a streamfunction!

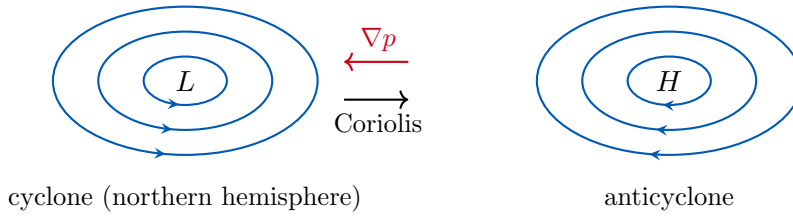
Definition 13.8.6 (Geostrophic Balance)

A flow in which the Coriolis force exactly balances the horizontal pressure gradient is in **geostrophic balance**. The **shallow water streamfunction** is

$$\psi = -\frac{gh}{f} = -\frac{p}{\rho f},$$

so that the streamlines are the contours of constant surface height, equivalently the isobars.

This is a genuinely strange conclusion, and it is worth stating in plain words: *in a rotating fluid, the flow runs along the isobars, not across them*. Naively one expects fluid to flow from high pressure to low, and in a non-rotating fluid it does. Here, instead, fluid accelerating down the pressure gradient is deflected by the Coriolis force until it is moving at right angles to that gradient, at which point the two forces balance and the motion is steady.



This is why a weather map is useful: the isobars *are* the streamlines, so you can read the wind straight off the chart. Around a low pressure system in the northern hemisphere the flow is anticlockwise (a cyclone); around a high it is clockwise. In the southern hemisphere $f < 0$ and both are reversed.

§13.8.4 Potential Vorticity and the Rossby Radius

Finally, let us find the conserved quantity that governs how a rotating fluid adjusts. Linearise about a state of rest with uniform depth, writing

$$h = h_0 + \eta, \quad \eta \ll h_0,$$

so that the equations become

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{f} \times \mathbf{u} = -g \nabla \eta \quad (*)$$

and, dropping products of small quantities,

$$\frac{\partial \eta}{\partial t} + h_0 \nabla \cdot \mathbf{u} = 0. \quad (\dagger)$$

Take the curl of (*), writing $\boldsymbol{\zeta} = \nabla \times \mathbf{u}$ for the (vertical) relative vorticity. Using $\nabla \times (\mathbf{f} \times \mathbf{u}) = \mathbf{f}(\nabla \cdot \mathbf{u})$ for constant $\mathbf{f}$ and horizontal $\mathbf{u}$,

$$\frac{\partial \boldsymbol{\zeta}}{\partial t} + \mathbf{f} \nabla \cdot \mathbf{u} = 0,$$

and substituting $\nabla \cdot \mathbf{u} = -\eta_t/h_0$ from (†) gives

$$\frac{\partial}{\partial t} \left(\zeta - \frac{\eta}{h_0} f \right) = 0.$$

Definition 13.8.7 (Potential Vorticity)

The **potential vorticity** is

$$Q = \zeta - \frac{\eta}{h_0} f,$$

and it is conserved: $\partial Q / \partial t = 0$.

The interpretation is that a column of fluid which is stretched (larger η) must spin up (larger ζ) to compensate – exactly the vortex stretching of Section 13.5, with the planetary vorticity f supplying the ambient rotation. This single conservation law explains an enormous amount of atmospheric and oceanic dynamics.

To see it in action, take the divergence of (*) and use (†) again:

$$-\frac{1}{h_0} \frac{\partial^2 \eta}{\partial t^2} - f \zeta = -g \nabla^2 \eta,$$

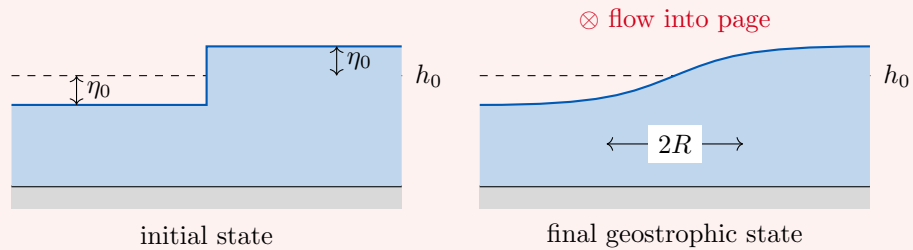
and eliminating ζ using $\zeta = Q_0 + f\eta/h_0$,

$$\frac{\partial^2 \eta}{\partial t^2} - gh_0 \nabla^2 \eta + f^2 \eta = -h_0 f Q_0,$$

where Q_0 is the (time-independent, hence known from the initial data) potential vorticity.

Example 13.8.8 (Rossby Adjustment)

Suppose we start with a step in the surface height: $\eta = -\eta_0$ for $x < 0$ and $\eta = +\eta_0$ for $x > 0$, with the fluid at rest. What is the final state?



Without rotation, the answer is obvious: the water sloshes about and eventually settles flat, with no flow. With rotation, that answer is *impossible*, because a flat surface with no flow has $Q = 0$ everywhere, and the initial data has (since $\zeta = 0$ initially)

$$Q_0 = \mp \frac{\eta_0}{h_0} f \quad \text{for } x \gtrless 0,$$

which is not zero. Conservation of Q forbids the naive answer.

Look for a steady final state independent of y . Then our equation reduces to

$$\frac{d^2 \eta}{dx^2} - \frac{f^2}{gh_0} \eta = \frac{f}{g} Q_0 = \mp \frac{f^2}{gh_0} \eta_0.$$

Definition 13.8.9 (Rossby Radius of Deformation)

The length scale

$$R = \frac{\sqrt{gh_0}}{f}$$

is the **Rossby radius of deformation** – the distance a shallow water wave travels in one rotation time.

In terms of R the equation is $\eta'' - \eta/R^2 = \mp\eta_0/R^2$, and requiring $\eta \rightarrow \pm\eta_0$ as $x \rightarrow \pm\infty$ with η, η' continuous at the origin gives

$$\eta = \eta_0 \begin{cases} 1 - e^{-x/R} & x > 0, \\ -\left(1 - e^{x/R}\right) & x < 0. \end{cases}$$

So the step does not flatten out; it slumps into a smooth front of width $2R$, and it stops there. Computing the velocity from geostrophic balance,

$$u = -\frac{g}{f} \frac{\partial \eta}{\partial y} = 0, \quad v = \frac{g}{f} \frac{\partial \eta}{\partial x} = \eta_0 \sqrt{\frac{g}{h_0}} e^{-|x|/R},$$

there is a jet flowing along the front, whose Coriolis force holds up the remaining pressure gradient. The final state is not rest; it is geostrophic balance.

Putting numbers in, the Rossby radius in the atmosphere is

$$R = \frac{\sqrt{10 \times 10^4}}{10^{-4}} \approx 10^6 \text{ m} = 1000 \text{ km},$$

which is precisely the scale of the weather systems on a synoptic chart – that is why cyclones are the size they are. In the ocean, h_0 is effectively much smaller (one should use the depth of the thermocline), and one finds $R \sim 100 \text{ km}$: ocean eddies are ten times smaller than atmospheric ones, and correspondingly much harder to observe. The Rossby radius is the fundamental length scale of any rotating system in which gravity also matters, and it is a good note on which to end.

14 Geometry

Geometry is the study of shapes – but shapes viewed with an eye towards *measurement*. In this course we will study surfaces: spaces which locally look like the plane, but which globally can be curved, twisted or glued together in interesting ways. We will begin by pinning down what a surface actually is, and then gradually add structure: smoothness, ways of measuring lengths, angles and areas, and finally curvature. Along the way we will meet the three classical two-dimensional geometries – Euclidean, spherical and hyperbolic – and the course will culminate in the Gauss–Bonnet theorem, a genuinely remarkable result linking the curvature of a surface to its topology.

This article constitutes my notes for the ‘Geometry’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§14.1 Topological and Smooth Surfaces

§14.1.1 Topological Surfaces

We will begin immediately with a definition that will occupy us for some time.

Definition 14.1.1 (Topological Surface)

A topological surface is a topological space Σ such that

- (i) Each $p \in \Sigma$ has an open neighborhood U with $p \in U$ such that U is homeomorphic to $\mathbb{R}^2$, with its usual Euclidean topology.
- (ii) Σ is Hausdorff and second countable.

Recall that a space X is Hausdorff if for $p \neq q$ in X there exist disjoint open sets $p \in U$ and $q \in V$ in X , and that a space is second countable if its topology has a countable basis. In some ways, the real nature of topological surfaces comes from the condition (i), and the condition (ii) is really there for technical honesty.

§14.1.2 Examples of Topological Surfaces

Let’s now take some time to consider some examples of topological surfaces.

Example 14.1.2 ($\mathbb{R}^2$)

The plane $\mathbb{R}^2$ is a topological surface.

Example 14.1.3 (Subsets of $\mathbb{R}^2$)

Any open subset of $\mathbb{R}^2$ is a topological surface. For example

- (i) $\mathbb{R}^2 \setminus \{0\}$ is a topological surface;

(ii) Let $Z = \{(0, 0)\} \cup \{(1, 1/n) \mid n \in \mathbb{N}\}$, then $\mathbb{R}^2 \setminus Z$ is a topological surface.

Example 14.1.4 (Graphs)

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. Then the **graph**

$$\Gamma_f = \{(x, y, f(x, y)) \mid (x, y) \in \mathbb{R}^2\} \subseteq \mathbb{R}^3,$$

taken with the subspace topology, is a topological surface. Let's see why.

Recall that if X and Y are topological spaces, the product topology on $X \times Y$ has basis open sets $U \times V$ with $U \subseteq X$ and $V \subseteq Y$ both open sets.

It has the feature that $f : Z \rightarrow X \times Y$ is continuous if and only if $\pi_x \circ f : Z \rightarrow X$ and $\pi_y \circ f : Z \rightarrow Y$ are continuous.

So if $\Gamma_f \subseteq X \times Y$ and $f : X \rightarrow Y$ is continuous then Γ_f is homeomorphic to X , with the map $s : x \mapsto (x, f(x))$, so that $\pi|_{\Gamma_f}$ and s are inverse homeomorphisms.

So $\Gamma_f \cong \mathbb{R}^2$ for *any* continuous $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, and Γ_f is a topological surface.

As a note, the topological surface Γ_f is independent of f . Later on as we develop more tools in geometry we will be able to better reflect the structure of the function f .

Example 14.1.5 (Stereographic Projection)

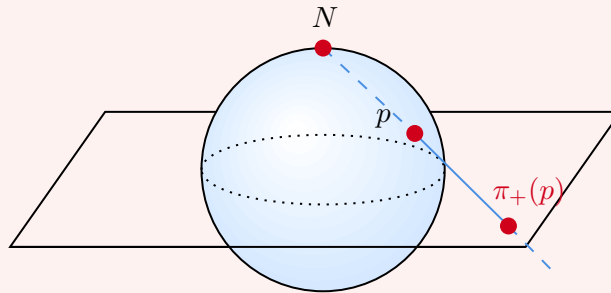
Consider the sphere

$$S^2 = \{(x, y, t) \in \mathbb{R}^3 \mid x^2 + y^2 + t^2 = 1\}.$$

We can consider the stereographic projection

$$\begin{aligned} \pi_+ : S^2 \setminus \{(0, 0, 1)\} &\rightarrow \mathbb{R}^2 (z = 0) \subseteq \mathbb{R}^3 \\ (x, y, t) &\mapsto \left(\frac{x}{1-t}, \frac{y}{1-t} \right). \end{aligned}$$

Such a projection is shown below.



Note that π_+ is continuous and has an inverse

$$(u, v) \mapsto \left(\frac{2u}{u^2 + v^2 + 1}, \frac{2v}{u^2 + v^2 + 1}, \frac{u^2 + v^2 - 1}{u^2 + v^2 + 1} \right).$$

So π_+ is a continuous bijection with continuous inverse and hence a homeomorphism.

Of course we could also have projected from the south pole, to get a homeomorphism π_- from $S^2 \setminus \{(0, 0, -1)\}$ to $\mathbb{R}^2$, so indeed every point lies in an open set which is homeomorphic through either π_+ or π_- to $\mathbb{R}^2$. So S^2 is a topological surface.

Remark. S^2 is compact as a topological space, since it is a closed bounded set in $\mathbb{R}^3$.

Example 14.1.6 (Real Projective Plane)

The group $\mathbb{Z}/2\mathbb{Z}$ acts on S^2 by homeomorphisms via the **antipodal map** $a : S^2 \rightarrow S^2$ with

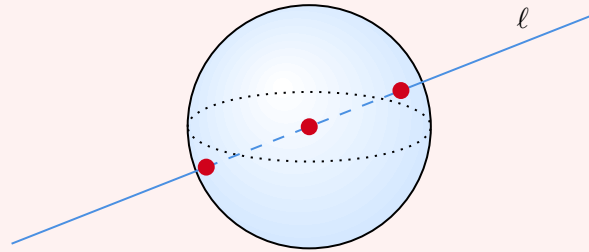
$$a(x, y, t) = (-x, -y, -t).$$

That is, there exists a homomorphism $\mathbb{Z}/2\mathbb{Z} \rightarrow \text{Homeo}(S^2)$ sending the non-identity element to a .

The **real projective plane** is the quotient space of S^2 given by identifying every point with its antipodal image: $\mathbb{RP}^2 = S^2/(\mathbb{Z}/2\mathbb{Z}) = S^2/\sim$ with $x \sim a(x)$.

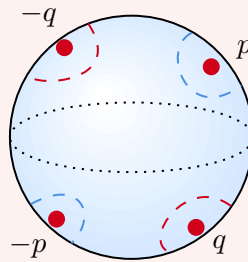
Note that $\sim$ is the equivalence relation of belonging to the same orbit under the given action.

As a set, $\mathbb{RP}^2$ is naturally in bijection with the set of straight lines in $\mathbb{R}^3$ through the origin, with the bijection given by mapping lines with the identified points on the sphere that they pass through.



We can also check that $\mathbb{RP}^2$ is a topological surface.

We must first check that it is Hausdorff. Recall that if X is a space and $q : X \rightarrow Y$ is a quotient map, then $V \subseteq Y$ is open if and only if $q^{-1}(V) \subseteq X$ is open.

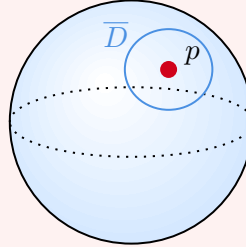


If $[p] \neq [q] \in \mathbb{RP}^2$ then $\pm p$ and $\pm q \in S^2$ are distinct antipodal points. We can then take small open discs^a about these in S^2 as in the picture, which give us disjoint open neighborhoods of $[p]$ or $[q]$ in $\mathbb{RP}^2$. That is, it's Hausdorff.

We can also check that $\mathbb{RP}^2$ is second countable. Let U be a countable basis for the topology on S^2 , and without loss of generality for all $u \in U$, the antipodal image is in U . Let $\overline{U}$ be the family of open sets in $\mathbb{RP}^2$ of the form $q(u) \cup q(-u)$, $u \in U$. Now if $V \subseteq \mathbb{RP}^2$ is open, by definition $q^{-1}(V)$ is open in S^2 , and so $q^{-1}(V)$ contains

some $u \in U$ and hence contains $u \cup (-u)$. So $\overline{U}$ is a countable base for the quotient topology on $\mathbb{RP}^2$.

Finally, let $p \in S^2$ and let $[p] \in \mathbb{RP}^2$ be its image. Let $\overline{D}$ be a small^b closed disc neighborhood of $p \in S^2$.



Then the quotient map $q|_{\overline{D}} : \overline{D} \rightarrow q(\overline{D}) \subseteq \mathbb{RP}^2$ is a continuous map from a compact space to a Hausdorff space. Also, on $\overline{D}$ the map q is injective. So by the topological inverse function theorem, this map $q|_{\overline{D}}$ is a homeomorphism. This induces (by taking interiors) a homeomorphism $q|_D : D \rightarrow q(D) \subseteq \mathbb{RP}^2$. So $[p] \in q(D)$ has an open neighborhood in $\mathbb{RP}^2$ homeomorphic to an open disk, and we are done.

^aWe could take small balls $B_{\pm p}(\delta)$ and $B_{\pm q}(\delta)$ ($\delta \ll 1$ small), which meet S^2 in open sets. If $q : S^2 \rightarrow \mathbb{RP}^2$ is the quotient map, then $q(B_p(\delta))$ is open since $q^{-1}(q(B_p(\delta))) = B_p(\delta) \cup (-B_p(\delta))$, the union with the antipodal image.

^bContained in an open hemisphere

Example 14.1.7 (Torus)

Let $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$, and define the **torus** $S^1 \times S^1$ with the subspace topology from $\mathbb{C}^2$ (which is the product topology). We will show that the torus is a topological surface.

Consider the map

$$\begin{aligned} \mathbb{R}^2 &\xrightarrow{e} S^1 \times S^1 \subseteq \mathbb{C} \times \mathbb{C} \\ (s, t) &\mapsto (e^{2\pi i s}, e^{2\pi i t}). \end{aligned}$$

Note that this induces a map (in a set theoretic sense)

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{e} & S^1 \times S^1 \\ \downarrow & \nearrow \hat{e} & \\ \mathbb{R}^2/\mathbb{Z}^2 & & \end{array}$$

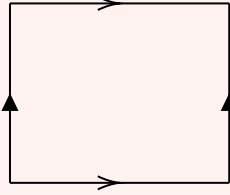
that is, on the equivalence relation on $\mathbb{R}^2$ given by translating by $\mathbb{Z}^2$, e is constant on equivalence classes and so induces a map of sets $\mathbb{R}^2/\mathbb{Z}^2 \rightarrow S^1 \times S^1$. Note we are viewing $\mathbb{R}^2/\mathbb{Z}^2$ as the quotient space for $q : \mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$.

So the map $[0, 1]^2 \hookrightarrow \mathbb{R}^2 \xrightarrow{q} \mathbb{R}^2/\mathbb{Z}^2$ is onto, so $\mathbb{R}^2/\mathbb{Z}^2$ is compact. So $\hat{e}$ is a continuous map from a compact space to a Hausdorff space and is a bijection, and is thus a homeomorphism by the topological inverse function theorem.

Note we already know that $S^1 \times S^1$ is compact and Hausdorff (closed and bounded

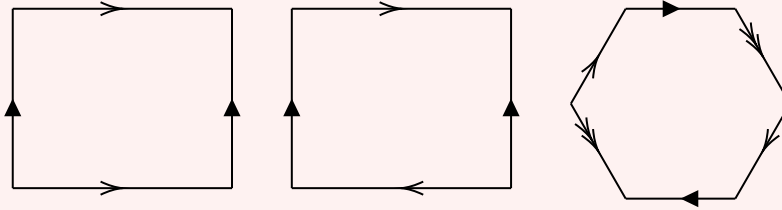
in $\mathbb{R}^4$). As for $S^2 \rightarrow \mathbb{RP}^2$, pick $[p] = q(p)$ for $p \in \mathbb{R}^2$ and some small closed disc $\overline{D}(p) \subseteq \mathbb{R}^2$ such that for all integers $(n, m) \neq (0, 0)$ we have $\overline{D}(p) \cap (\overline{D}(p) + (n, m)) = \emptyset$. Then $e|_{\overline{D}(p)}$ is injective and $q|_{\overline{D}(p)}$ is injective. Now restricting to the open disc as before, we get an open disc neighborhood of $[p] \in S^1 \times S^1$. Since $[p]$ is arbitrary, $S^1 \times S^1$ is a topological surface.

Another viewpoint: $\mathbb{R}^2/\mathbb{Z}^2$ is also given by imposing on $[0, 1]^2$ the equivalence relation generated by $(x, 0) \sim (x, 1)$, $(0, y) \sim (1, y)$.



Example 14.1.8 (Gluing Edges)

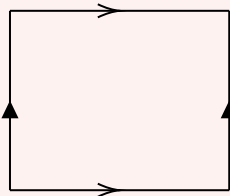
Let P be a planar Euclidean polygon. We will assume the edges are *oriented* and paired. For simplicity, we can suppose that e and e' have the same Euclidean length whenever $\{e, e'\}$ are paired.



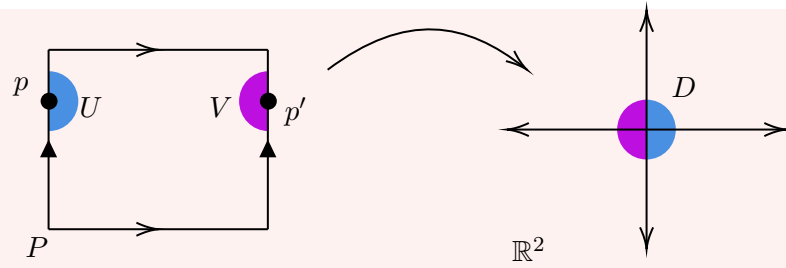
If $\{e, e'\}$ are paired edges, there is a unique isometry from e to e' respecting their orientations, say $f_{ee'} : e \rightarrow e'$. These maps generate an equivalence relation on P , where I identify $x \in P$ with $f_{ee'}(x)$ whenever $x \in e$.

Lemma. $P/\sim$ (with the quotient topology) is a topological surface.

Before we prove this, we will consider the specific example of the torus as $[0, 1]^2/\sim$.



If $P = [0, 1]^2$ and p is in the interior of P , then we can pick $\delta > 0$ sufficiently small so that $B_p(\delta)$ and $\overline{B_p(\delta)}$ in $\mathbb{R}^2$ lie in the interior of P . Now we argue as before: the quotient map is injective on $\overline{B_p(\delta)}$ and a homeomorphism on its interior. If p is on an edge of P , then we take two half disks of sufficiently small radius δ so that they don't meet a vertex of P .

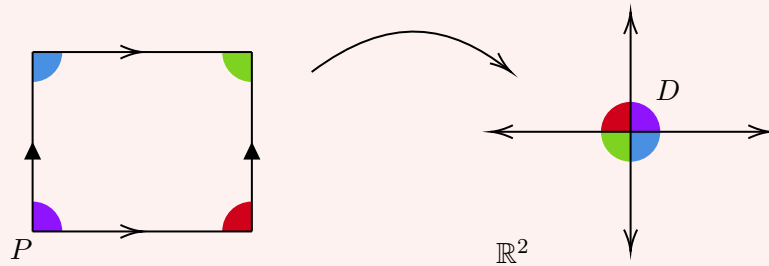


Then we can define a map f from the union of these half disks to the disk of the same radius at the origin of $\mathbb{R}^2$ as above.

Explicitly, we define f_U and f_V such that they are continuous on the half disks $U, V \in [0, 1]^2$. These induce a continuous map on $q(U)$ and $q(V) \subseteq T^2$, ($q : [0, 1]^2 \rightarrow [0, 1]^2 / \sim = T^2$). In T^2 , the half disks $q(U)$ and $q(V)$ overlap *but* overlaps agree on the closed intersection locus (as f_U and f_V are compatible with the equivalence relation). So f_U and f_V glue to define a continuous map f on an open neighborhood $[p] \in T^2$ to $B_0(\delta) \subseteq \mathbb{R}^2$.

Now we can apply our ‘usual argument’ (pass to a closed disk, apply the topological inverse function theorem, pass back to the interior) to show that if $[q] \in T^2$ lies on the image of an edge of $[0, 1]^2$, it has an open neighborhood homeomorphic to a disk.

Analogously, at a vertex of $[0, 1]^2$, the same argument with a slight modification works.



This shows that $[0, 1]^2 / \sim$ is a topological surface.

For a general planar polygon P . We will now see how to address the general case. We again are trying to show that each point has an open neighborhood that is homeomorphic to a disk. For interior points, a suitably sized disk (chosen so that it does not intersect any edges or vertices) works exactly as before.

Now we have our equivalence relation on points in the polygon $x \sim f_{ee'}(x)$ where $x \in e \in \text{Edge}(P)$ and e, e' are paired with $f : e \rightarrow e'$ compatible with orientation. This relation induces an equivalence relation on $\text{Vert}(P)$.

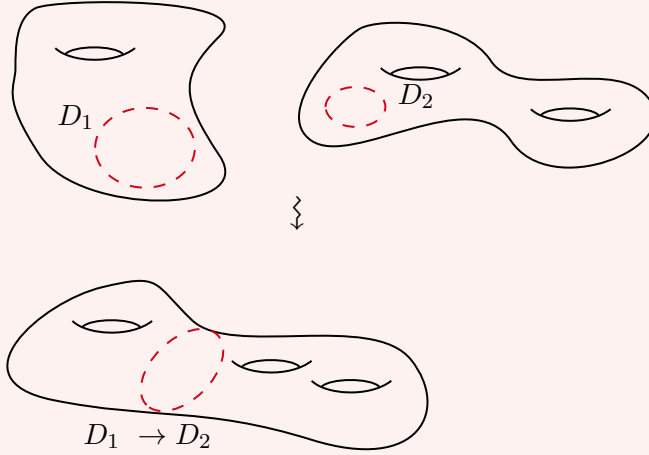
If $v \in \text{Vert}(P)$ has r vertices in its equivalence class, then these contribute r sectors in P , of total angle α_v say. Rescaling angles by a factor of $2\pi/\alpha_v$, these sectors can be glued around the image of v to give a full sector $\{(\rho, \theta) \mid 0 \leq \rho < \delta, \theta \in [0, 2\pi]\}$, which is homeomorphic to a disk.

To see that $P / \sim$ is Hausdorff, we can take disks on the interior with a sufficiently small radius and distinct points will have disjoint disks. For second countable, consider disks on the interior of P with rational radii and centre, and if $e \in \text{Edge}(P)$ then $e \mapsto [0, \text{length}(e)]$ is an isometry, so take only half disks on e which are centred

at rational points, with rational radii. And at vertices, allow rational radius sectors. This gives a countable basis.

Example 14.1.9 (Connect Sum)

Given topological surfaces Σ_1 and Σ_2 , we can remove an open disk from each and glue the resulting boundary circles.



Explicitly, I take $\Sigma_1 \setminus D_1$ and $\Sigma_2 \setminus D_2$ and impose an equivalence relation

$$\theta \in \partial D_1 \sim \theta \in \partial D_2,$$

where θ parameterises the boundary circles $S^1 \cong \partial D_i$.

The result $\Sigma_1 \# \Sigma_2$ is the **connect sum** of Σ_1 and Σ_2 . Note that in principle this connect sum depends on a number of choices, but we suppress this from the notation.

Indeed, the connect sum of topological surfaces is a topological surface. This is proved as before.

§14.1.3 Subdivisions and Triangulations

Before discussing smooth surfaces, we want to talk a bit about subdivisions and triangulations of compact surfaces.

Definition 14.1.10 (Subdivision)

A **subdivision** of a compact topological surface Σ consists of

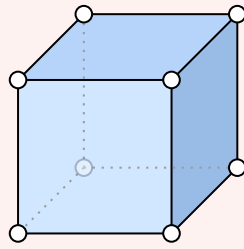
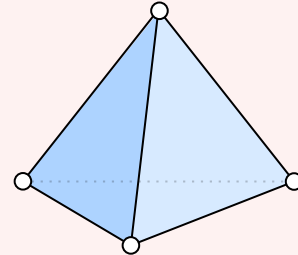
- (i) A finite set $V \subseteq \Sigma$ of **vertices**;
- (ii) A finite collection of continuous embeddings $\{e_i : [0, 1] \rightarrow \Sigma\}$ called **edges**, each of which has endpoints in V , and any two of which meet at endpoints;
- (iii) The connected components of $\Sigma \setminus [V \cup \bigcup_i e_i([0, 1])]$ are each homeomorphic to an open disk called a **face**.

Definition 14.1.11 (Triangulation)

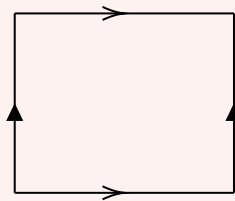
A subdivision is a **triangulation** if each *closed* face (the closure of a face) contains exactly 3 edges and two closed faces are either disjoint or meet in exactly one edge.

Example 14.1.12 (Subdivisions and Triangulations of S^2 and T^2)

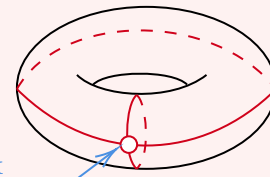
The cube displays a subdivision of the sphere S^2 , and the tetrahedron displays a triangulation of the sphere S^2 .

Subdivision of S^2 Triangulation of S^2

One convenient way of displaying a subdivision is with the diagrams we had for specifying surfaces by gluing edges. For example, the below shows a subdivision of T^2 with 1 vertex, 2 edges and 1 face.

Subdivision of T^2

2 Edges, 1 Vertex



As a more degenerate example, a single vertex of S^2 is indeed a subdivision, leaving 1 vertex, 0 edges and 1 face.

Indeed, the notion of subdivisions gives us an interesting topological invariant for compact topological surfaces.

Definition 14.1.13 (Euler Characteristic)

The **Euler characteristic** of a subdivision is the number

$$V - E + F$$

where V is number of vertices, E is the number of edges and F is the number of faces.

Theorem 14.1.14 (Euler's Relation)

- (i) Every compact topological surface admits subdivisions, and indeed triangulations.

- (ii) The Euler characteristic, denoted $\chi(\Sigma)$, does not depend on the choice of subdivision, and thus defines a topological invariant of the surface.

We aren't going to prove this theorem because it's relatively hard and won't really say anything about the true nature of what is going on. A nice proof will be given in a course on Algebraic Topology.

Example 14.1.15 (Euler's Relation for Various Surfaces)

Considering some of our previous examples, we can see that

- (i) $\chi(S^2) = 2$;
- (ii) $\chi(T^2) = 0$;
- (iii) If Σ_1 and Σ_2 are compact topological surfaces, we can form $\Sigma_1 \# \Sigma_2$ by removing an open disk which is the face of a triangulation, and gluing the boundary circles gives us that

$$\chi(\Sigma_1 \# \Sigma_2) = \chi(\Sigma_1) + \chi(\Sigma_2) - 2.$$

In particular, if a surface has g holes, then considering it as homeomorphic to connect sum of g copies of T^2 , we get that $\chi(\Sigma_g) = 2 - 2g$. The number g is called the **genus** of Σ .

§14.1.4 Charts and Atlases

Recall that if Σ is a topological surface, each $p \in \Sigma$ lies in an open neighbourhood $p \in U \subseteq \Sigma$ with U homeomorphic to an open disk.

Definition 14.1.16 (Chart)

A pair (U, ϕ) where $U \subseteq \Sigma$ and ϕ is a homeomorphism $\phi : U \rightarrow V$ where V is open in $\mathbb{R}^2$ is called a **chart** for Σ .

Definition 14.1.17 (Atlas)

A collection of charts $\{(U_i, \phi_i) \mid i \in I\}$ such that $\bigcup_{i \in I} U_i = \Sigma$ is called an **atlas** for Σ .

Example 14.1.18 (Trivial Example of an Atlas)

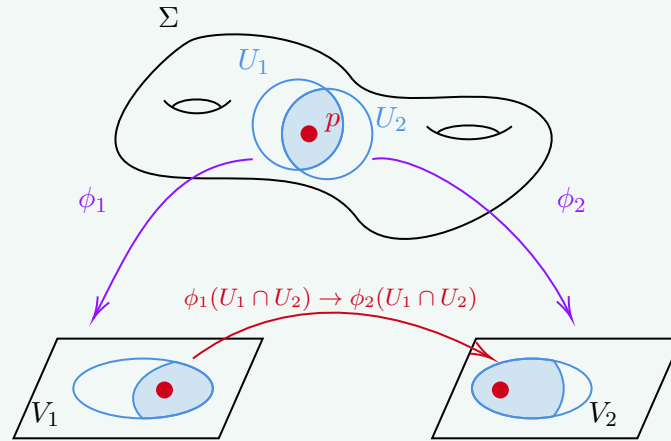
Recall that any open subset U of $\mathbb{R}^2$ is a topological surface. Then taking (U, Id) gives us an atlas with one chart.

Example 14.1.19 (Atlas for S^2)

For S^2 , we have an atlas with 2 charts, the 2 stereographic projections.

Definition 14.1.20 (Transition Maps)

Let (U_1, ϕ_1) and (U_2, ϕ_2) be charts containing a point p on a topological surface Σ .



Considering the intersection, we get a map $\phi_1(U_1 \cap U_2) \rightarrow \phi_2(U_1 \cap U_2)$. Such a map is called a **transition map** between the charts, and this is a homeomorphism of open sets in $\mathbb{R}^2$.

§14.1.5 Smooth Surfaces

Recall that if V and V' are open subsets of $\mathbb{R}^n$, then a map $f : V \rightarrow V'$ is **smooth** if it is infinitely differentiable. If f is a homeomorphism between V and V' , then we call it a **diffeomorphism** if it's smooth and has a smooth inverse.

Definition 14.1.21 (Abstract Smooth Surface)

An **abstract smooth surface** Σ is a topological surface with an atlas of charts $\{(U_i, \phi_i) \mid i \in I\}$ with $\bigcup_{i \in I} U_i = \Sigma$ such that all of the transition maps $\phi_j(U_i \cap U_j) \rightarrow \phi_i(U_i \cap U_j)$ are diffeomorphisms of open sets in $\mathbb{R}^2$.

Remark. It would *not* make sense to ask for the maps ϕ_i themselves to be smooth, as Σ is just a topological space.

Example 14.1.22 (S^2 with Stereographic Projection)

The atlas of 2 charts with stereographic projection gives S^2 the structure of an abstract smooth surface.

Remark (Philosophical). Being a topological surface is *structure*, one can ask if a topological space X is a topological surface or not. But being an abstract smooth surface is *data*, in that one has to be given an atlas of charts with smooth transition maps with smooth inverses, and there could be many.

Definition 14.1.23 (Smooth at a Point)

Let Σ be an abstract smooth surface, and if $f : \Sigma \rightarrow \mathbb{R}^n$ is a continuous map then we say that f is **smooth** at $p \in \Sigma$ if whenever (U, ϕ) is a chart at p belonging to the smooth atlas for Σ , the map $f \circ \phi^{-1} : \phi(U) \rightarrow \mathbb{R}^n$ is smooth.

Note that smoothness of f at p is *independent* of the choice of chart since the transition maps between two such are diffeomorphisms.

Definition 14.1.24 (Smoothness of Maps Between Surfaces)

If Σ_1 and Σ_2 are abstract smooth surfaces, a map $f : \Sigma_1 \rightarrow \Sigma_2$ is smooth if “it is smooth in the local charts”. That is, given charts (U, ϕ) at p and (U', ϕ') at $f(p)$ in our chosen smooth atlases, we want $\phi' \circ f \circ \phi^{-1}$ to be smooth at $\phi(p)$.

Again, smoothness of f does not depend on our choice of charts at p and $f(p)$ provided we take charts from our smooth atlas.

We can now say what it means for abstract smooth surfaces to be diffeomorphic.

Definition 14.1.25 (Diffeomorphic)

Two abstract smooth surfaces Σ_1 and Σ_2 are **diffeomorphic** if there exists a homeomorphism $f : \Sigma_1 \rightarrow \Sigma_2$ which is smooth and has a smooth inverse.

Remark. We often pass from a given smooth atlas for an abstract smooth surfaces Σ to the maximal ‘compatible’ such atlas, that is, I add to the atlas $\{(U_i, \phi_i) \mid i \in I\}$ for Σ all charts (V, ψ) with the property that the transition maps are still all diffeomorphisms. Such a process technically involves Zorn’s lemma.

§14.2 Smooth Surfaces in $\mathbb{R}^3$

So far, this has all been rather abstract, but in this course what we really care about is smooth surfaces in $\mathbb{R}^3$, where we can do honest calculus. Recall if $V \subseteq \mathbb{R}^n$ and $V' \subseteq \mathbb{R}^m$ are open subsets then $f : V \rightarrow V'$ is smooth if it is infinitely differentiable.

Definition 14.2.1 (Smoothness of Maps on Subsets of $\mathbb{R}^n$)

If $Z \subseteq \mathbb{R}^n$ is an arbitrary subset, we say that a continuous map $f : Z \rightarrow \mathbb{R}^m$ is **smooth** at a point $p \in Z$ if there exists an open set $B \subseteq \mathbb{R}^n$ with $p \in B$ and a smooth map $F : B \rightarrow \mathbb{R}^m$ such that

$$F|_{B \cap Z} = f|_{B \cap Z}.$$

That is, f is *locally* the restriction of a smooth map defined on an open set.

With this notion in hand, it makes sense to talk about subsets of Euclidean spaces being diffeomorphic.

Definition 14.2.2 (Diffeomorphic Subsets)

If $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$ are arbitrary subsets, we say that X and Y are **diffeomorphic** if there is a homeomorphism $f : X \rightarrow Y$ such that both f and f^{-1} are smooth.

§14.2.1 Definitions and Equivalent Characterisations

We can now say what it means for a subset of $\mathbb{R}^3$ to be a smooth surface.

Definition 14.2.3 (Smooth Surface in $\mathbb{R}^3$)

A **smooth surface in $\mathbb{R}^3$** is a subset $\Sigma \subseteq \mathbb{R}^3$ such that each point $p \in \Sigma$ has an open neighbourhood $U \subseteq \Sigma$ which is diffeomorphic to an open set in $\mathbb{R}^2$.

Unpacking the definitions, this says: for each $p \in \Sigma$ there is an open ball $p \in B \subseteq \mathbb{R}^3$ and an open set $V \subseteq \mathbb{R}^2$ such that, setting $U = B \cap \Sigma$, there is a homeomorphism $\phi : U \rightarrow V$ which is smooth (in the sense of the previous definition), and whose inverse $V \rightarrow U \subseteq \mathbb{R}^3$ is also smooth.

There are several other ways one might have tried to define a smooth surface in $\mathbb{R}^3$ – as a level set of a smooth function, as a graph, or via nice parametrisations – and thankfully they all agree. Before stating this, we single out the parametrisations that will do most of the work in this course.

Definition 14.2.4 (Allowable Parametrisation)

Let $V \subseteq \mathbb{R}^2$ be open and let $U \subseteq \Sigma \subseteq \mathbb{R}^3$ be open in Σ . A map $\sigma : V \rightarrow U$ is an **allowable parametrisation** if σ is a homeomorphism, is smooth as a map $V \rightarrow \mathbb{R}^3$, and $D\sigma|_x$ has rank 2 for every $x \in V$. If moreover $\sigma(0) = p$, we say σ is an allowable parametrisation *near* p .

The rank condition is what stops the parametrisation from degenerating – it guarantees that the two partial derivatives σ_u and σ_v are linearly independent, so the image genuinely looks two-dimensional.

Theorem 14.2.5 (Characterisations of Smooth Surfaces)

For a subset $\Sigma \subseteq \mathbb{R}^3$, the following are equivalent.

- (i) Σ is a smooth surface in $\mathbb{R}^3$.
- (ii) Σ is *locally a graph* over one of the three coordinate planes: for each $p \in \Sigma$ there is an open ball $p \in B \subseteq \mathbb{R}^3$ and an open set $V \subseteq \mathbb{R}^2$ such that

$$\Sigma \cap B = \{(x, y, g(x, y)) \mid (x, y) \in V\}$$

for some smooth $g : V \rightarrow \mathbb{R}$, or the same with the roles of the coordinates permuted.

- (iii) Σ is *locally cut out* by a smooth function: for each $p \in \Sigma$ there is an open ball $p \in B \subseteq \mathbb{R}^3$ and a smooth $f : B \rightarrow \mathbb{R}$ such that

$$\Sigma \cap B = f^{-1}(0), \quad \text{with } Df|_x \neq 0 \text{ for all } x \in B.$$

- (iv) Σ is locally the image of an allowable parametrisation near each of its points.

Remark. Characterisation (ii) implies that every $p \in \Sigma$ lies in a chart (U, ϕ) where ϕ is the restriction of one of the coordinate projections $\pi_{xy}, \pi_{yz}, \pi_{xz} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$. The transition map between two graphical charts over the *same* plane is the identity, and between charts over different planes it looks like

$$(x, y) \mapsto (x, y, g(x, y)) \mapsto (y, g(x, y)),$$

which is smooth with smooth inverse of the same shape. So a smooth surface in $\mathbb{R}^3$ carries a natural atlas whose transition maps are diffeomorphisms – that is, it is an abstract smooth surface, and the two uses of the word ‘smooth’ are consistent.

Some of the implications in this theorem are easy, but the substantial ones rest on the inverse function theorem, which we now recall (a proof is given in Analysis and Topology).

§14.2.2 The Inverse and Implicit Function Theorems

Theorem 14.2.6 (Inverse Function Theorem)

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}^n$ be continuously differentiable. Let $p \in U$, $q = f(p)$, and suppose the derivative $Df|_p$ is invertible. Then there exists an open neighbourhood V of q and a continuously differentiable map $g : V \rightarrow \mathbb{R}^n$ with $g(q) = p$, whose image is an open neighbourhood $U' \subseteq U$ of p , such that $f \circ g = \text{Id}_V$. Moreover if f is smooth then so is g .

By the chain rule, the inverse satisfies $Dg|_q = (Df|_p)^{-1}$. The inverse function theorem deals with maps $\mathbb{R}^n \rightarrow \mathbb{R}^n$ whose derivative is an isomorphism. If instead we have $f : \mathbb{R}^{k+\ell} \rightarrow \mathbb{R}^\ell$ whose derivative is *surjective*, we can say something about the level sets of f – this is the implicit function theorem.

Theorem 14.2.7 (Implicit Function Theorem)

Let $p = (x_0, y_0) \in U$ where $U \subseteq \mathbb{R}^k \times \mathbb{R}^\ell$ is open, and let $f : U \rightarrow \mathbb{R}^\ell$ be continuously differentiable with $f(p) = 0$, such that the $\ell \times \ell$ matrix of partial derivatives $(\partial f_i / \partial y_j)$ is invertible at p . Then there is an open neighbourhood V of x_0 in $\mathbb{R}^k$ and a continuously differentiable map $g : V \rightarrow \mathbb{R}^\ell$ with $g(x_0) = y_0$ such that, for $(x, y) \in U$ sufficiently close to p ,

$$f(x, y) = 0 \iff y = g(x).$$

Moreover if f is smooth then so is g .

Proof. Define $F : U \rightarrow \mathbb{R}^k \times \mathbb{R}^\ell$ by $F(x, y) = (x, f(x, y))$. Then in block form

$$DF = \begin{pmatrix} I & 0 \\ * & \partial f_i / \partial y_j \end{pmatrix},$$

which is invertible at (x_0, y_0) . So by the inverse function theorem, F is locally invertible near $F(x_0, y_0) = (x_0, 0)$; let $G : V \times V' \rightarrow U' \subseteq U$ be a continuously differentiable local inverse, defined on a product of open sets. Writing $G(x, y) = (\varphi(x, y), \psi(x, y))$, the relation $F \circ G = \text{Id}$ gives

$$(\varphi(x, y), f(\varphi(x, y), \psi(x, y))) = (x, y),$$

so $\varphi(x, y) = x$ and $f(x, \psi(x, y)) = y$ for $(x, y) \in V \times V'$. In particular $f(x, y) = 0$ exactly when $y = \psi(x, 0)$, so we may take $g(x) = \psi(x, 0)$. $\square$

Example 14.2.8 (Level Sets in $\mathbb{R}^3$)

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be smooth with $f(x_0, y_0, z_0) = 0$, and consider the level set $\Sigma = f^{-1}(0)$. Suppose $Df \neq 0$ at (x_0, y_0, z_0) ; permuting coordinates, we may assume $\partial f / \partial z \neq 0$ there. The implicit function theorem (with $k = 2$, $\ell = 1$) then provides an open neighbourhood V of (x_0, y_0) and a smooth $g : V \rightarrow \mathbb{R}$ such that near (x_0, y_0, z_0) , the level set Σ is exactly the graph $\{(x, y, g(x, y)) \mid (x, y) \in V\}$. In other words – near a point where its defining function has non-vanishing derivative, a level set is locally a graph.

§14.2.3 Proof of the Characterisation Theorem

We can now assemble the proof of Theorem 14.2.5.

Proof of Theorem 14.2.5. We first show that (ii) implies all of the other conditions. Suppose Σ is locally the graph $\{(x, y, g(x, y))\}$ over (say) the xy -plane. Then the coordinate projection π_{xy} restricts to a homeomorphism from this graph to V , it is defined and smooth on an open neighbourhood in $\mathbb{R}^3$, and its inverse $\sigma(x, y) = (x, y, g(x, y))$ is smooth. So Σ is a smooth surface in $\mathbb{R}^3$, giving (i). Also, the graph is cut out by the function $f(x, y, z) = z - g(x, y)$, which has $\partial f / \partial z = 1 \neq 0$, giving (iii). Finally $\sigma(x, y) = (x, y, g(x, y))$ is an allowable parametrisation: it is a smooth homeomorphism, and its derivative contains the 2×2 identity block in the first two rows, so has rank 2. This gives (iv).

Next, (i) implies (iv): if $\phi : U \rightarrow V$ is a diffeomorphism as in the definition of a smooth surface, then near a given point, ϕ^{-1} extends implicitly to a smooth map on an open subset of $\mathbb{R}^2$, and (shrinking if necessary) $\sigma = \phi^{-1}$ is a smooth homeomorphism. Its derivative has rank 2: writing F for a local smooth extension of ϕ to an open set $B \subseteq \mathbb{R}^3$, we have $F \circ \sigma = \text{Id}$ near a point of V , so by the chain rule $DF|_p \circ D\sigma|_x = I_2$, which forces $D\sigma|_x$ to be injective, i.e. of rank 2. So σ is allowable.

That (iii) implies (ii) is exactly the level set example of the implicit function theorem above.

Finally we show (iv) implies (ii) and (i), which completes the cycle. Let $\sigma : V \rightarrow U$ be an allowable parametrisation with $\sigma(0) = p$, and write $\sigma = (\sigma_1, \sigma_2, \sigma_3)$, so

$$D\sigma = \begin{pmatrix} \partial\sigma_1/\partial u & \partial\sigma_1/\partial v \\ \partial\sigma_2/\partial u & \partial\sigma_2/\partial v \\ \partial\sigma_3/\partial u & \partial\sigma_3/\partial v \end{pmatrix}.$$

This matrix has rank 2, so some two rows form an invertible 2×2 matrix – without loss of generality the first two rows. Let $\text{pr} = \pi_{xy}$, and consider $\text{pr} \circ \sigma : V \rightarrow \mathbb{R}^2$. Its derivative at 0 is an isomorphism, so by the inverse function theorem it has a smooth local inverse, and hence Σ is locally the graph of the smooth function $\sigma_3 \circ (\text{pr} \circ \sigma)^{-1}$ over the xy -plane. This gives (ii). For (i), the map $(x, y, z) \mapsto (\text{pr} \circ \sigma)^{-1}(x, y)$ is defined and smooth on an open ball about p in $\mathbb{R}^3$, and restricts on Σ to a local inverse of σ . So σ^{-1} is smooth, and Σ is a smooth surface in $\mathbb{R}^3$. $\square$

Let's see some examples.

Example 14.2.9 (The Sphere)

The unit sphere $S^2 \subseteq \mathbb{R}^3$ is $f^{-1}(0)$ for $f(x, y, z) = x^2 + y^2 + z^2 - 1$. At any point of S^2 we have $Df = (2x, 2y, 2z) \neq 0$, so S^2 is a smooth surface in $\mathbb{R}^3$ by criterion (iii).

Example 14.2.10 (Surfaces of Revolution)

Let $\gamma : (a, b) \rightarrow \mathbb{R}^3$ be a smooth injective curve in the xz -plane, say

$$\gamma(t) = (f(t), 0, g(t)), \quad \text{with } f > 0 \text{ and } \gamma' \neq 0 \text{ everywhere.}$$

The **surface of revolution** obtained by rotating γ about the z -axis is covered by parametrisations

$$\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u)),$$

with $(u, v) \in (a, b) \times (\theta, \theta + 2\pi)$ for fixed θ . Here

$$\sigma_u = (f' \cos v, f' \sin v, g'), \quad \sigma_v = (-f \sin v, f \cos v, 0),$$

and one computes $\|\sigma_u \times \sigma_v\|^2 = f^2((f')^2 + (g')^2) \neq 0$, so these are allowable parametrisations and the surface of revolution is a smooth surface in $\mathbb{R}^3$.

Example 14.2.11 ($O(3)$ Acting on S^2)

The orthogonal group $O(3)$ acts on S^2 by diffeomorphisms: any $A \in O(3)$ defines a linear (hence smooth) map $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ preserving S^2 , whose restriction to S^2 is a homeomorphism that is smooth with smooth inverse in the sense above. This is analogous to the action of the Möbius group on $\mathbb{C} \cup \{\infty\} \cong S^2$, which we shall meet again when we study hyperbolic geometry.

§14.2.4 Orientability

There is one more piece of global structure we should discuss before we start measuring things: *orientation*. Intuitively, an orientation of a surface is a consistent choice of what ‘anticlockwise’ means at each point, and the Möbius band shows this is not always possible.

Definition 14.2.12 (Orientation-Preserving)

Let V, V' be open sets in $\mathbb{R}^2$ and $f : V \rightarrow V'$ a diffeomorphism, so $Df|_x \in \text{GL}(2, \mathbb{R})$ for every $x \in V$. We say f is **orientation-preserving** if $\det Df|_x > 0$ for all $x \in V$.

Definition 14.2.13 (Orientable)

An abstract smooth surface Σ is **orientable** if it admits an atlas in which all transition maps are orientation-preserving. A choice of such an atlas is an **orientation** of Σ , and Σ equipped with an orientation is **oriented**.

Orientability behaves well under diffeomorphism, as one would hope.

Lemma 14.2.14

If Σ_1 and Σ_2 are diffeomorphic abstract smooth surfaces, then Σ_1 is orientable if and only if Σ_2 is orientable.

Proof. Let $f : \Sigma_1 \rightarrow \Sigma_2$ be a diffeomorphism and suppose Σ_2 is equipped with an oriented atlas. Consider the atlas on Σ_1 whose charts are $(f^{-1}(U), \psi \circ f)$ for charts (U, ψ) in the oriented atlas of Σ_2 . The transition map between two such charts,

$$(\psi_1 \circ f) \circ (\psi_2 \circ f)^{-1} = \psi_1 \circ \psi_2^{-1},$$

is exactly a transition map of the oriented atlas on Σ_2 , so is orientation-preserving. Hence Σ_1 is orientable, and the claim follows by symmetry. $\square$

Example 14.2.15 (The Möbius Band)

The **Möbius band** is the surface obtained from the square by gluing a pair of opposite open edges with a flip:



where the dashed edges are omitted (so this is a surface without boundary). One can check that the Möbius band is *not* orientable, and in fact a surface is orientable if and only if it contains no open subsurface homeomorphic to the Möbius band. This gives a way of extending the notion of orientability to topological surfaces, where the definition through determinants of derivatives makes no sense.

Example 14.2.16 (S^2 and T^2 are Orientable)

For S^2 with the atlas of two stereographic projections, the transition map works out to be

$$(u, v) \mapsto \left(\frac{u}{u^2 + v^2}, \frac{v}{u^2 + v^2} \right)$$

on $\mathbb{R}^2 \setminus \{0\}$, which does not have positive determinant everywhere (it is a reflection composed with an inversion). But replacing π_- by its composite with the reflection $(u, v) \mapsto (u, -v)$ fixes this, and the resulting atlas is oriented. So S^2 is orientable. For the torus $T^2 = \mathbb{R}^2 / \mathbb{Z}^2$, we found an atlas whose transition maps are restrictions of translations of $\mathbb{R}^2$, and translations certainly have positive determinant. So T^2 is orientable.

Remark. One can define other classes of surfaces by restricting which subgroup of $\text{GL}(2, \mathbb{R})$ the derivatives of transition maps may lie in. Asking for derivatives in $\text{GL}^+(2, \mathbb{R})$ gives oriented surfaces; asking for transition maps that are restrictions of translations gives *Euclidean* surfaces (like the torus above); asking for derivatives in $\text{GL}(1, \mathbb{C}) \subseteq \text{GL}(2, \mathbb{R})$ gives *Riemann surfaces*.

§14.2.5 Tangent Planes

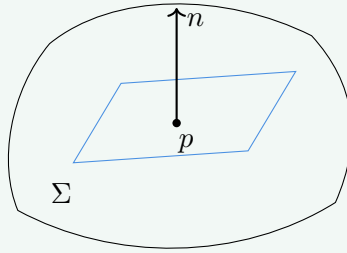
If a smooth surface is the collection of points that locally looks like a plane, its tangent plane at p is the best flat approximation at p .

Definition 14.2.17 (Tangent Plane)

Let Σ be a smooth surface in $\mathbb{R}^3$, $p \in \Sigma$, and let $\sigma : V \rightarrow U$ be an allowable parametrisation near p (so $\sigma(0) = p$). The **tangent plane** is

$$T_p\Sigma = \text{Image}(D\sigma|_0) = \langle \sigma_u, \sigma_v \rangle \subseteq \mathbb{R}^3,$$

a two-dimensional vector subspace of $\mathbb{R}^3$. The **affine tangent plane** at p is the translate $p + T_p\Sigma$.



Of course, we should check that this does not depend on which allowable parametrisation we picked.

Lemma 14.2.18

$T_p\Sigma$ is independent of the choice of allowable parametrisation.

Proof. Suppose $\sigma : V \rightarrow U$ and $\tilde{\sigma} : \tilde{V} \rightarrow \tilde{U}$ are allowable parametrisations with $\sigma(0) = \tilde{\sigma}(0) = p$. The transition map $\sigma^{-1} \circ \tilde{\sigma}$ is a diffeomorphism of open sets of $\mathbb{R}^2$ (shrinking domains so both parametrise the same neighbourhood), and

$$\tilde{\sigma} = \sigma \circ (\sigma^{-1} \circ \tilde{\sigma}).$$

Applying the chain rule at 0, $D\tilde{\sigma}|_0 = D\sigma|_0 \circ D(\sigma^{-1} \circ \tilde{\sigma})|_0$, and the second factor is an isomorphism of $\mathbb{R}^2$. So the images of $D\tilde{\sigma}|_0$ and $D\sigma|_0$ agree. $\square$

There is a more geometric way to think about the tangent plane. If $\gamma : (-\epsilon, \epsilon) \rightarrow \mathbb{R}^3$ is any smooth curve with image in Σ and $\gamma(0) = p$, then writing $\gamma(t) = \sigma(u(t), v(t))$ for smooth u, v we get

$$\gamma'(0) = \sigma_u u'(0) + \sigma_v v'(0) \in T_p\Sigma,$$

and conversely every vector of $T_p\Sigma$ arises this way. So $T_p\Sigma$ is precisely the set of velocity vectors at p of smooth curves on Σ through p – which makes it visibly independent of any parametrisation.

Definition 14.2.19 (Normal Direction)

If Σ is a smooth surface in $\mathbb{R}^3$ and $p \in \Sigma$, the **normal direction** at p is $(T_p\Sigma)^\perp$, the Euclidean orthogonal complement of the tangent plane. There are exactly two

unit vectors in the normal direction.

Definition 14.2.20 (Two-Sided)

A smooth surface in $\mathbb{R}^3$ is **two-sided** if it admits a continuous global choice of unit normal vector $p \mapsto n(p)$.

Two-sidedness is how orientability manifests itself for surfaces sitting in $\mathbb{R}^3$.

Lemma 14.2.21

A smooth surface Σ in $\mathbb{R}^3$ is orientable if and only if it is two-sided.

Proof. Given an allowable parametrisation $\sigma : V \rightarrow U$ near p , define the *positive unit normal with respect to σ*

$$n_\sigma(p) = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|},$$

which is the unit normal making $\{\sigma_u, \sigma_v, n_\sigma\}$ a positively oriented basis of $\mathbb{R}^3$. Now suppose $\tilde{\sigma}$ is another allowable parametrisation near p and $\sigma = \tilde{\sigma} \circ \varphi$ for the transition map φ . Writing $D\varphi|_0 = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, the chain rule gives $\sigma_u = \alpha\tilde{\sigma}_u + \gamma\tilde{\sigma}_v$ and $\sigma_v = \beta\tilde{\sigma}_u + \delta\tilde{\sigma}_v$, and hence

$$\sigma_u \times \sigma_v = \det(D\varphi|_0) \tilde{\sigma}_u \times \tilde{\sigma}_v. \quad (\dagger)$$

So if Σ is oriented and we only use parametrisations compatible with the orientation, the determinant above is positive and the positive unit normal is independent of the parametrisation. Since n_σ is manifestly continuous on each U , these local normals patch to a global continuous unit normal, and Σ is two-sided.

Conversely, suppose Σ has a global continuous unit normal n . Consider the collection of all allowable parametrisations σ such that $\{\sigma_u, \sigma_v, n\}$ is a positively oriented basis of $\mathbb{R}^3$ at every point (any allowable parametrisation can be corrected to have this property by precomposing with $(u, v) \mapsto (v, u)$ if needed, so these charts cover Σ). By $(\dagger)$, the transition map φ between two such parametrisations satisfies $\det D\varphi > 0$. So this atlas is oriented and Σ is orientable. $\square$

The following little lemma gets used constantly: symmetries of a surface act on its tangent planes.

Lemma 14.2.22

Let Σ be a smooth surface in $\mathbb{R}^3$, and let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a smooth map with $A(\Sigma) \subseteq \Sigma$. Then for $p \in \Sigma$, the derivative $DA|_p$ maps $T_p\Sigma$ into $T_{A(p)}\Sigma$.

Proof. Recall $T_p\Sigma$ is spanned by vectors $\gamma'(0)$ for smooth curves $\gamma : (-\epsilon, \epsilon) \rightarrow \Sigma$ with $\gamma(0) = p$. For such a curve, $A \circ \gamma$ is a smooth curve on Σ through $A(p)$, and by the chain rule

$$DA|_p(\gamma'(0)) = (A \circ \gamma)'(0) \in T_{A(p)}\Sigma. \quad \square$$

Example 14.2.23 (The Sphere is Two-Sided)

For the unit sphere S^2 , the normal direction at p is spanned by p itself: the rotation group $\text{SO}(3)$ acts transitively on S^2 and preserves tangent planes by Lemma 14.2.22, so it suffices to check this at the north pole, where it is clear. So $n(p) = p$ (the outward normal) is a global continuous choice of unit normal, and S^2 is two-sided, hence orientable – in agreement with our computation with stereographic charts.

Example 14.2.24 (The Möbius Band is Not Two-Sided)

An explicit embedding of the Möbius band in $\mathbb{R}^3$ is

$$\sigma(t, \theta) = \left((1 - t \sin \frac{\theta}{2}) \cos \theta, (1 - t \sin \frac{\theta}{2}) \sin \theta, t \cos \frac{\theta}{2} \right),$$

where $t \in (-1/2, 1/2)$ and θ ranges over $(0, 2\pi)$ or $(-\pi, \pi)$ for the two charts. Geometrically we take a unit interval centred on each point of the unit circle, and let it rotate by half a turn as θ goes around the circle once. A direct computation along the central circle $t = 0$ gives

$$\sigma_t \times \sigma_\theta = \left(-\cos \theta \cos \frac{\theta}{2}, -\sin \theta \cos \frac{\theta}{2}, -\sin \frac{\theta}{2} \right),$$

which is already a unit vector. As $\theta \rightarrow 0^+$ this tends to $(-1, 0, 0)$, while as $\theta \rightarrow 2\pi^-$ it tends to $(1, 0, 0)$ – the normal comes back *flipped* after a loop around the band. So there is no continuous global choice of unit normal, and the Möbius band is not two-sided (hence not orientable).

Remark. There is no sensible classification of all smooth surfaces – already the surfaces $\mathbb{R}^2 \setminus Z$ for closed sets Z form an uncountable zoo. But for *compact* smooth surfaces the situation is remarkably clean: a compact smooth surface is determined up to diffeomorphism by its orientability and its Euler characteristic. This classification is stated here for culture; it will be proved in the Part II course *Algebraic Topology*.

§14.3 The First Fundamental Form

Up until now everything has been topology in disguise – we have said when surfaces are the same as topological or smooth objects, but we have not *measured* anything. In this section we start doing geometry properly: we will measure lengths of curves on a surface, angles between curves, and areas of regions. All of this information is captured by a single gadget called the first fundamental form.

§14.3.1 Isometries of Euclidean Space

It is worth first pausing to understand distance-preserving maps of flat space itself, since these are the ‘symmetries’ against which all our later notions are calibrated.

Definition 14.3.1 (Isometry)

A map $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an **isometry** if it preserves distances, that is

$$\|f(x) - f(y)\| = \|x - y\| \quad \text{for all } x, y \in \mathbb{R}^n.$$

The isometries of $\mathbb{R}^n$ form a group $\text{Isom}(\mathbb{R}^n)$ under composition.

Obvious examples are translations $x \mapsto x + b$, and the linear maps preserving the inner product – the **orthogonal group**

$$\text{O}(n) = \{A \in \text{GL}(n, \mathbb{R}) \mid A^T A = I\}.$$

It turns out these examples generate everything.

Theorem 14.3.2

Every isometry of $\mathbb{R}^n$ has the form $f(x) = Ax + b$ for some $A \in \text{O}(n)$ and $b \in \mathbb{R}^n$.

Proof. Let $b = f(0)$, and set $g(x) = f(x) - b$, an isometry fixing the origin. Since g fixes 0 and preserves distances, it preserves norms, and then by the polarisation identity

$$\langle x, y \rangle = \frac{1}{2} (\|x\|^2 + \|y\|^2 - \|x - y\|^2)$$

it preserves inner products: $\langle g(x), g(y) \rangle = \langle x, y \rangle$ for all x, y . In particular, if $e_1, \dots, e_n$ is the standard orthonormal basis then $g(e_1), \dots, g(e_n)$ is again orthonormal, and for any x ,

$$\left\langle g(x) - \sum_i \langle x, e_i \rangle g(e_i), g(e_j) \right\rangle = \langle x, e_j \rangle - \langle x, e_j \rangle = 0$$

for each j . As the $g(e_i)$ form a basis, the vector on the left is orthogonal to everything, so is zero, i.e.

$$g(x) = \sum_i \langle x, e_i \rangle g(e_i).$$

So g is linear, and being a linear map preserving the inner product, its matrix A satisfies $A^T A = I$, that is $A \in \text{O}(n)$. $\square$

Recall that $\det A = \pm 1$ for $A \in \text{O}(n)$. Isometries with $\det A = 1$ are **orientation-preserving** (or *rigid motions*); those with $\det A = -1$ are orientation-reversing. The simplest orientation-reversing isometries are **reflections**: given an affine hyperplane $H = \{x \mid \langle x, u \rangle = c\}$ with $\|u\| = 1$, the reflection in H is

$$R_H(x) = x - 2(\langle x, u \rangle - c)u,$$

which fixes H pointwise and swaps the two sides. Remarkably, these building blocks suffice.

Theorem 14.3.3 (Reflections Generate)

Every isometry of $\mathbb{R}^n$ is a composite of at most $n+1$ reflections in affine hyperplanes.

Proof. First observe that if an isometry f fixes points $p_0, \dots, p_k$, then it fixes every point of their affine span: indeed a point x is determined by its distances to $p_0, \dots, p_k$ together with the affine subspace it lies in, and one can check directly that f fixes the affine span setwise and preserves those distances. In particular an isometry fixing $n+1$ affinely independent points (whose span is all of $\mathbb{R}^n$) is the identity.

Now take any isometry f , and choose affinely independent points $p_0, \dots, p_n$. We correct f one point at a time. If $f(p_0) \neq p_0$, let R_1 be the reflection in the perpendicular bisector hyperplane of p_0 and $f(p_0)$, so $R_1 f$ fixes p_0 . Inductively, suppose $g = R_k \cdots R_1 f$ fixes $p_0, \dots, p_{k-1}$. If $g(p_k) \neq p_k$, let R_{k+1} be the reflection in the perpendicular bisector of p_k and $g(p_k)$; since

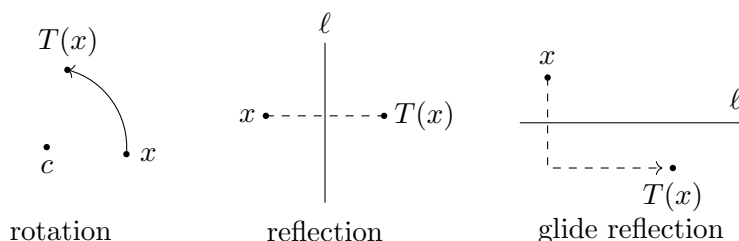
$$\|p_i - p_k\| = \|g(p_i) - g(p_k)\| = \|p_i - g(p_k)\| \quad \text{for } i < k,$$

the points $p_0, \dots, p_{k-1}$ lie on this bisector, so are still fixed by $R_{k+1}g$, which now also fixes p_k . After at most $n+1$ steps we obtain a composite of reflections and f which fixes all of $p_0, \dots, p_n$, hence equals the identity. Rearranging, f is a composite of at most $n+1$ reflections. $\square$

In low dimensions we can be completely explicit. In $\mathbb{R}^2$, an isometry is a composite of at most 3 reflections, and working through the cases gives the classical classification: every isometry of the plane is one of

- (i) the identity;
- (ii) a *translation* $x \mapsto x + b$;
- (iii) a *rotation* about some point c ;
- (iv) a *reflection* in a line ℓ ;
- (v) a *glide reflection* – a reflection in a line ℓ followed by a translation along ℓ .

The first three are orientation-preserving (composites of two reflections), the last two orientation-reversing (one or three reflections).



The moral for the rest of the course is this: the plane $\mathbb{R}^2$ has a very large group of isometries – enough to take any point to any other point, and any direction at a point to any other direction at any other point. We will meet exactly two other surfaces with this much symmetry: the round sphere, and the hyperbolic plane.

§14.3.2 Lengths of Curves

Let $\gamma : (a, b) \rightarrow \mathbb{R}^3$ be a smooth curve. We define its **length** to be

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt,$$

whenever this integral is finite. The length does not depend on the parametrisation of the curve: if $s : (A, B) \rightarrow (a, b)$ is monotonic increasing and smooth, and $\tau = \gamma \circ s$, then by the chain rule and the substitution rule

$$L(\tau) = \int_A^B \|\gamma'(s(t))\| |s'(t)| \, dt = \int_a^b \|\gamma'(t')\| \, dt' = L(\gamma).$$

Lemma 14.3.4

If $\gamma : (a, b) \rightarrow \mathbb{R}^3$ is continuously differentiable with $\gamma'(t) \neq 0$ everywhere, then γ can be reparametrised by **arc length**, so that $\|\gamma'\| \equiv 1$.

Proof. Fix $t_0 \in (a, b)$ and let $s(t) = \int_{t_0}^t \|\gamma'(w)\| dw$. Then s is continuously differentiable with $s'(t) = \|\gamma'(t)\| > 0$, so s is a monotonic bijection onto its image with differentiable inverse. Setting $\tau = \gamma \circ s^{-1}$, the chain rule gives $\|\tau'\| = \|\gamma'\|/s' = 1$. $\square$

§14.3.3 The First Fundamental Form

Now let Σ be a smooth surface in $\mathbb{R}^3$, and $\sigma : V \rightarrow U \subseteq \Sigma$ an allowable parametrisation. If γ is a smooth curve with image in U , we can write $\gamma(t) = \sigma(u(t), v(t))$ for smooth functions $(u, v) : (a, b) \rightarrow V$, and then

$$\gamma'(t) = \sigma_u u'(t) + \sigma_v v'(t).$$

Taking the squared norm,

$$\|\gamma'(t)\|^2 = E u'(t)^2 + 2F u'(t)v'(t) + G v'(t)^2,$$

where

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle$$

are smooth functions on V depending only on σ , not on the curve.

Definition 14.3.5 (First Fundamental Form)

The **first fundamental form** of Σ in the allowable parametrisation σ is the expression

$$E du^2 + 2F du dv + G dv^2,$$

with E, F, G as above. The notation records the fact that if $\gamma(t) = \sigma(u(t), v(t))$ then

$$L(\gamma) = \int_a^b \sqrt{E(u')^2 + 2F u' v' + G(v')^2} dt.$$

Remark. There is a cleaner way to think about this. The Euclidean inner product of $\mathbb{R}^3$ restricts to an inner product on the subspace $T_p \Sigma$, and *that* restriction is the real object of study: a symmetric positive definite bilinear form on tangent planes, varying smoothly with p . The functions E, F, G are just its matrix

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

in the basis $\{\sigma_u, \sigma_v\}$ of $T_p \Sigma$ supplied by a parametrisation. Indeed, since the columns of $D\sigma$ are σ_u and σ_v , we can write the matrix of the first fundamental form compactly as $(D\sigma)^T(D\sigma)$.

Let's compute some examples.

Example 14.3.6 (The Plane)

The plane $\mathbb{R}_{xy}^2 \subseteq \mathbb{R}^3$ has parametrisation $\sigma(u, v) = (u, v, 0)$, giving $\sigma_u = e_1$, $\sigma_v = e_2$ and first fundamental form

$$du^2 + dv^2.$$

We could instead use polar coordinates $\sigma(r, \theta) = (r \cos \theta, r \sin \theta, 0)$ on the plane minus a half-line, giving $\sigma_r = (\cos \theta, \sin \theta, 0)$ and $\sigma_\theta = (-r \sin \theta, r \cos \theta, 0)$, so the first fundamental form is

$$dr^2 + r^2 d\theta^2.$$

Note the same surface can have quite different-looking fundamental forms in different parametrisations.

Example 14.3.7 (The Sphere)

The sphere of radius a has an allowable parametrisation

$$\sigma(u, v) = (a \cos u \cos v, a \cos u \sin v, a \sin u), \quad u \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad v \in (0, 2\pi),$$

covering the complement of a half great circle (u is latitude and v longitude). Here

$$\sigma_u = (-a \sin u \cos v, -a \sin u \sin v, a \cos u), \quad \sigma_v = (-a \cos u \sin v, a \cos u \cos v, 0),$$

so $E = a^2$, $F = 0$, $G = a^2 \cos^2 u$, and the first fundamental form is

$$a^2 du^2 + a^2 \cos^2 u dv^2.$$

Example 14.3.8 (Surfaces of Revolution)

For the surface of revolution of $\eta(t) = (f(t), 0, g(t))$ about the z -axis, with parametrisation $\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u))$, we compute

$$\sigma_u = (f' \cos v, f' \sin v, g'), \quad \sigma_v = (-f \sin v, f \cos v, 0),$$

giving first fundamental form

$$((f')^2 + (g')^2) du^2 + f^2 dv^2.$$

If the profile curve η is parametrised by arc length, this simplifies to $du^2 + f^2 dv^2$.

§14.3.4 Isometric Surfaces

The first fundamental form is exactly the data needed to measure lengths, so it is the right thing to preserve if we want maps of surfaces that preserve geometry.

Definition 14.3.9 (Isometric Surfaces)

Smooth surfaces Σ, Σ' in $\mathbb{R}^3$ are **isometric** if there is a diffeomorphism $f : \Sigma \rightarrow \Sigma'$ which preserves lengths of curves: for every smooth curve γ on Σ , $L(f \circ \gamma) = L(\gamma)$. Such an f is an **isometry**.

Example 14.3.10

If $f(x) = Ax + b$ is a global isometry of $\mathbb{R}^3$ (with $A \in O(3)$), then for any smooth surface Σ , the surfaces Σ and $f(\Sigma)$ are isometric – rigid motions preserve the lengths of all curves in $\mathbb{R}^3$. But note the definition asks for much less: f need only be defined on Σ , and isometric surfaces certainly need not be related by a rigid motion of the ambient space, as we will see with the cone below.

Definition 14.3.11 (Locally Isometric)

Σ and Σ' are **locally isometric near $p \in \Sigma$ and $q \in \Sigma'$** if there are open neighbourhoods $U \ni p$, $U' \ni q$ which are isometric. We say Σ and Σ' are locally isometric if each point of either surface has a neighbourhood isometric to an open set of the other.

The first fundamental form detects local isometry completely.

Lemma 14.3.12

Smooth surfaces $\Sigma, \Sigma' \subseteq \mathbb{R}^3$ are locally isometric near $p \in \Sigma$ and $q \in \Sigma'$ if and only if there are allowable parametrisations $\sigma : V \rightarrow U \subseteq \Sigma$ and $\sigma' : V \rightarrow U' \subseteq \Sigma'$, with the *same* domain V , near p and q respectively, whose first fundamental forms are equal as functions on V .

Proof. ($\Leftarrow$) If such parametrisations exist, then $f = \sigma' \circ \sigma^{-1} : U \rightarrow U'$ is a diffeomorphism, and given a curve $\gamma = \sigma(u(t), v(t))$ in U , the image curve is $f \circ \gamma = \sigma'(u(t), v(t))$. The two length integrals are computed from the respective first fundamental forms with the identical functions $(u(t), v(t))$, so they agree. Hence f is an isometry.

($\Rightarrow$) Suppose $f : U \rightarrow U'$ is an isometry with $f(p) = q$, and let $\sigma : V \rightarrow U$ be an allowable parametrisation near p ; then $\sigma' = f \circ \sigma$ is an allowable parametrisation of U' near q with the same domain. We must show lengths of curves determine the first fundamental form – then, since σ and σ' assign equal lengths to corresponding curves, their fundamental forms must agree. Without loss of generality let $V = B(0, \delta)$ and $\sigma(0) = p$. For $\epsilon < \delta$, consider the curves

$$\gamma_\epsilon(t) = \sigma(t, 0), \quad t \in [0, \epsilon].$$

Then

$$L(\gamma_\epsilon) = \int_0^\epsilon \sqrt{E(t, 0)} \, dt \implies \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} L(\gamma_\epsilon) = \sqrt{E(0, 0)},$$

so lengths of curves near p determine E at p (and similarly at every point of V). Using instead $t \mapsto \sigma(0, t)$ determines G , and $t \mapsto \sigma(t, t)$ determines $E + 2F + G$, hence F . So the first fundamental form is determined by lengths, as required. $\square$

Example 14.3.13 (The Cone)

Consider the cone with angle $\arctan(a)$ to the vertical (minus its apex and a ruling

line), parametrised for $u > 0, v \in (0, 2\pi)$ by

$$\sigma(u, v) = (au \cos v, au \sin v, u),$$

with first fundamental form

$$(1 + a^2) du^2 + a^2 u^2 dv^2.$$

Intuitively, we can cut the cone along a ruling and flatten it onto the plane – rolling a paper cone flat – to obtain a plane sector of radius $\sqrt{1 + a^2} u$ and total angle $2\pi a/\sqrt{1 + a^2}$. Concretely, the map into the plane

$$\sigma'(u, v) = \left(\sqrt{1 + a^2} u \cos \left(\frac{av}{\sqrt{1 + a^2}} \right), \sqrt{1 + a^2} u \sin \left(\frac{av}{\sqrt{1 + a^2}} \right), 0 \right)$$

with the same domain has first fundamental form

$$(1 + a^2) du^2 + a^2 u^2 dv^2$$

as well. So by Lemma 14.3.12 the cone is *locally isometric to the plane* – despite looking curved, it is flat from the point of view of any measurements made along the surface. The cone and the plane are not *globally* isometric, since they are not even homeomorphic.

Finally, let's record how the first fundamental form transforms when we change parametrisation.

Lemma 14.3.14

Suppose $\sigma : V \rightarrow U$ and $\sigma' : V' \rightarrow U$ are allowable parametrisations of the same open set $U \subseteq \Sigma$, related by the transition map $\varphi = (\sigma')^{-1} \circ \sigma : V \rightarrow V'$. Then their first fundamental forms are related by

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} = (D\varphi)^T \begin{pmatrix} E' & F' \\ F' & G' \end{pmatrix} (D\varphi).$$

Proof. Since the matrix of the first fundamental form is $(D\sigma)^T(D\sigma)$ and $\sigma = \sigma' \circ \varphi$, the chain rule $D\sigma = (D\sigma')(D\varphi)$ gives

$$(D\sigma)^T(D\sigma) = (D\varphi)^T(D\sigma')^T(D\sigma')(D\varphi),$$

which is the claim. □

§14.3.5 Angles and Conformality

For vectors $v, w \in \mathbb{R}^3$ we have $\langle v, w \rangle = \|v\| \|w\| \cos \theta$, so the angle between them is determined by inner products. If $v, w \in T_p \Sigma$ are tangent vectors, say $v = D\sigma|_0(\hat{v})$ and $w = D\sigma|_0(\hat{w})$, then writing $I(\cdot, \cdot)$ for the first fundamental form as a bilinear form,

$$\cos \theta = \frac{I(\hat{v}, \hat{w})}{\sqrt{I(\hat{v}, \hat{v})} \sqrt{I(\hat{w}, \hat{w})}}.$$

So the first fundamental form measures angles as well as lengths. In particular, if two curves $\gamma(t) = \sigma(u(t), v(t))$ and $\tilde{\gamma}(t) = \sigma(\tilde{u}(t), \tilde{v}(t))$ meet at p , the angle θ at which they

meet satisfies

$$\cos \theta = \frac{E\dot{u}\dot{u} + F(\dot{u}\dot{v} + \dot{v}\dot{u}) + G\dot{v}\dot{v}}{\sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}\sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}}.$$

Definition 14.3.15 (Conformal)

An allowable parametrisation $\sigma : V \rightarrow U$ is **conformal** if it preserves angles: whenever two curves in V meet at some angle, their images under σ meet at the same angle.

Lemma 14.3.16

σ is conformal if and only if $E = G$ and $F = 0$.

Proof. Suppose σ is conformal. The curves $t \mapsto (t, 0)$ and $t \mapsto (0, t)$ meet at angle $\pi/2$ in V , so their images meet at angle $\pi/2$; by the angle formula this gives $F = 0$. Likewise the curves $t \mapsto (t, t)$ and $t \mapsto (t, -t)$ meet at right angles, and the angle formula (using $F = 0$) gives $E - G = 0$.

Conversely if $E = G$ and $F = 0$, then the first fundamental form is $\rho(du^2 + dv^2)$ where $\rho = E > 0$. So at each point it is a pointwise rescaling of the Euclidean inner product, and rescaling an inner product does not change the angles it defines. Hence σ preserves angles. $\square$

Remark. Conformal parametrisations are historically important in cartography – a conformal map of the Earth renders compass bearings faithfully, even though (as we will see) no map can render distances faithfully. The existence of conformal charts on any surface is a hard theorem, closely tied to the theory of *Riemann surfaces*, where surfaces are modelled on $\mathbb{C}$ rather than $\mathbb{R}^2$.

§14.3.6 Area

Recall that the parallelogram spanned by $v, w \in \mathbb{R}^3$ has area

$$\|v \times w\| = \sqrt{\|v\|^2\|w\|^2 - \langle v, w \rangle^2}.$$

Applying this to the ‘infinitesimal parallelogram’ spanned by σ_u and σ_v , whose sides span the image of a small coordinate square under σ , we find its area is

$$\|\sigma_u \times \sigma_v\| = \sqrt{EG - F^2}.$$

This motivates the following definition.

Definition 14.3.17 (Area)

Let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parametrisation. The **area** of U is

$$\text{area}(U) = \int_V \sqrt{EG - F^2} \, du \, dv.$$

Remark. This is independent of the parametrisation. If $\tilde{\sigma} : \tilde{V} \rightarrow U$ is another allowable parametrisation, with transition map $\varphi : \tilde{V} \rightarrow V$, then from the transformation law for the first fundamental form,

$$\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = |\det D\varphi| \sqrt{EG - F^2},$$

and the change of variables formula for double integrals gives

$$\int_{\tilde{V}} \sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} \, du \, dv = \int_V \sqrt{EG - F^2} \, du \, dv.$$

For an open set (or a compact surface) not contained in a single chart, we compute the area by breaking the set into finitely many pieces each lying in a chart, and adding up.

Example 14.3.18 (Graphs Stretch Area)

Let Σ be the graph of a smooth function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, with its global parametrisation $\sigma(u, v) = (u, v, f(u, v))$. Then $\sigma_u = (1, 0, f_u)$ and $\sigma_v = (0, 1, f_v)$, so

$$\sqrt{EG - F^2} = \sqrt{1 + f_u^2 + f_v^2}.$$

If U_R is the part of the graph over the disc $B(0, R)$, then

$$\text{area}(U_R) = \int_{B(0, R)} \sqrt{1 + f_u^2 + f_v^2} \, du \, dv \geq \pi R^2,$$

with equality if and only if $f_u \equiv f_v \equiv 0$ on the disc, i.e. the graph is horizontal. In other words – projecting a graph to the plane always shrinks area, unless the graph was flat to begin with.

Example 14.3.19 (Archimedes)

Let S^2 be the unit sphere, enclosed in the cylinder $\{x^2 + y^2 = 1\}$ of unit radius, and consider the radial projection taking a point of the sphere (other than the poles) horizontally outward to the cylinder. A pleasant computation (on the example sheets) shows this map is *area-preserving*: whatever area a region gains by being pushed outwards horizontally, it loses by being compressed vertically. This is Archimedes' theorem, and gives $\text{area}(S^2) = 4\pi$ immediately – the cylinder unrolls to a $2\pi \times 2$ rectangle. It also underlies the Lambert equal-area projection used in cartography.

§14.4 The Second Fundamental Form and Gaussian Curvature

The first fundamental form records the *intrinsic* geometry of a surface – everything a small ant living on the surface could measure. We now introduce a second quadratic form which records how the surface *bends* inside $\mathbb{R}^3$, and from it we will extract the single most important function in this course: the Gaussian curvature.

§14.4.1 The Second Fundamental Form

Let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parametrisation. By Taylor's theorem, for small (h, ℓ) ,

$$\sigma(u + h, v + \ell) = \sigma(u, v) + h \sigma_u + \ell \sigma_v$$

$$+ \frac{1}{2} (h^2 \sigma_{uu} + 2h\ell \sigma_{uv} + \ell^2 \sigma_{vv}) + O(h^3, \ell^3).$$

Write $p = \sigma(u, v)$ and let n be the unit normal at p . The first order terms lie in $T_p \Sigma$, so the perpendicular distance from $\sigma(u + h, v + \ell)$ to the affine tangent plane $p + T_p \Sigma$ is

$$\langle \sigma(u + h, v + \ell) - \sigma(u, v), n \rangle = \frac{1}{2} (Lh^2 + 2Mh\ell + N\ell^2) + O(h^3, \ell^3),$$

where we define

$$L = \langle n, \sigma_{uu} \rangle, \quad M = \langle n, \sigma_{uv} \rangle, \quad N = \langle n, \sigma_{vv} \rangle.$$

So the following quadratic form measures, to leading order, how the surface pulls away from its tangent plane.

Definition 14.4.1 (Second Fundamental Form)

The **second fundamental form** of Σ in the allowable parametrisation σ is the quadratic form

$$L du^2 + 2M du dv + N dv^2, \quad \text{where } n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|},$$

with L, M, N as above, thought of as the matrix $\begin{pmatrix} L & M \\ M & N \end{pmatrix}$ with respect to the basis $\{\sigma_u, \sigma_v\}$ of $T_p \Sigma$.

Note that unlike the first fundamental form, which is always positive definite, the second fundamental form can be degenerate, indefinite, or zero. Indeed, vanishing identically means exactly what you would guess.

Lemma 14.4.2

Let V be connected and $\sigma : V \rightarrow U \subseteq \Sigma$ an allowable parametrisation whose second fundamental form vanishes identically. Then U is contained in an affine plane.

Proof. Differentiating the identities $\langle n, \sigma_u \rangle = 0 = \langle n, \sigma_v \rangle$ with respect to u and v gives

$$\begin{aligned} L &= \langle n, \sigma_{uu} \rangle = -\langle n_u, \sigma_u \rangle, & M &= \langle n, \sigma_{uv} \rangle = -\langle n_v, \sigma_u \rangle = -\langle n_u, \sigma_v \rangle, \\ N &= \langle n, \sigma_{vv} \rangle = -\langle n_v, \sigma_v \rangle. \end{aligned}$$

If $L = M = N \equiv 0$, then n_u is orthogonal to σ_u and σ_v ; but also $\langle n, n \rangle = 1$ gives $\langle n_u, n \rangle = 0$. Since $\{\sigma_u, \sigma_v, n\}$ is a basis of $\mathbb{R}^3$, we get $n_u \equiv 0$, and similarly $n_v \equiv 0$. As V is connected, n is a constant vector, and then

$$\frac{\partial}{\partial u} \langle \sigma, n \rangle = \langle \sigma_u, n \rangle = 0, \quad \frac{\partial}{\partial v} \langle \sigma, n \rangle = 0,$$

so $\langle \sigma, n \rangle$ is constant – which says exactly that $U = \sigma(V)$ lies in a plane normal to n . $\square$

The identities in the proof above deserve to be displayed on their own: the second fundamental form can be written as

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = - \begin{pmatrix} \langle n_u, \sigma_u \rangle & \langle n_u, \sigma_v \rangle \\ \langle n_v, \sigma_u \rangle & \langle n_v, \sigma_v \rangle \end{pmatrix} = -(Dn)^T (D\sigma),$$

in perfect analogy with $(D\sigma)^T(D\sigma)$ for the first fundamental form. From this one can also derive the transformation rule under a transition map φ :

$$\begin{pmatrix} \tilde{L} & \tilde{M} \\ \tilde{M} & \tilde{N} \end{pmatrix} = \pm (D\varphi)^T \begin{pmatrix} L & M \\ M & N \end{pmatrix} (D\varphi),$$

where the sign is $+$ if φ is orientation-preserving (so the two parametrisations induce the same normal) and $-$ otherwise.

Example 14.4.3 (The Cylinder)

Consider the cylinder of radius a , parametrised by $\sigma(u, v) = (a \cos u, a \sin u, v)$. Then $\sigma_{uv} = \sigma_{vu} = 0$, so $M = N = 0$, while $\sigma_{uu} = (-a \cos u, -a \sin u, 0)$ and $n = (\cos u, \sin u, 0)$ (the outward normal), giving $L = -a$. So the second fundamental form is

$$-a \, du^2.$$

The cylinder bends in one direction and is straight in the other – exactly what the degenerate matrix $\begin{pmatrix} -a & 0 \\ 0 & 0 \end{pmatrix}$ records.

§14.4.2 The Gauss Map

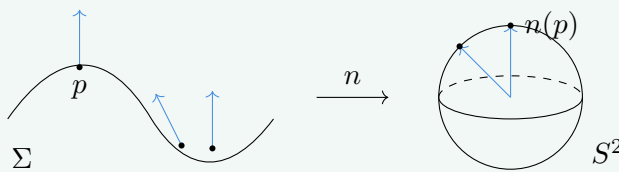
The appearance of Dn above suggests that we study the normal vector as a map in its own right.

Definition 14.4.4 (Gauss Map)

Let Σ be a smooth oriented surface in $\mathbb{R}^3$. The **Gauss map** is

$$n : \Sigma \rightarrow S^2, \quad p \mapsto n(p),$$

sending each point to its (positively oriented) unit normal, viewed as a point of the unit sphere.



Lemma 14.4.5

The Gauss map is smooth.

Proof. Smoothness is a local property, so we can check it in an allowable parametrisation σ compatible with the orientation, where

$$n \circ \sigma = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}.$$

The numerator is smooth and the denominator is smooth and non-vanishing (since

σ is allowable), so $n \circ \sigma$ is smooth, which is what smoothness of n means. $\square$

Remark. If $\Sigma = f^{-1}(0)$ is cut out by $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ with non-vanishing derivative along Σ , the Gauss map is simply

$$n(p) = \frac{\nabla f}{\|\nabla f\|}.$$

The key observation about the Gauss map is that its derivative at p is an endomorphism of the tangent plane. Indeed, the tangent plane to S^2 at $n(p)$ is $(n(p))^\perp = T_p\Sigma$ – the two planes are orthogonal to the same vector! So the derivative of the Gauss map is a linear map

$$Dn|_p : T_p\Sigma \rightarrow T_{n(p)}S^2 = T_p\Sigma.$$

Concretely, if $v = \gamma'(0)$ for a curve γ on Σ through p , then $Dn|_p(v) = (n \circ \gamma)'(0)$.

We can now give the slick, parametrisation-free description of both fundamental forms. For an oriented smooth surface Σ in $\mathbb{R}^3$ and $p \in \Sigma$:

- (i) The *first* fundamental form is the symmetric bilinear form on $T_p\Sigma$ obtained by restricting the Euclidean inner product:

$$I_p(v, w) = \langle v, w \rangle.$$

- (ii) The *second* fundamental form is the symmetric bilinear form

$$II_p(v, w) = I_p(-Dn|_p(v), w).$$

In an allowable (correctly oriented) parametrisation these have matrices $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$ and $\begin{pmatrix} L & M \\ M & N \end{pmatrix}$ in the basis $\{\sigma_u, \sigma_v\}$, recovering our earlier definitions – this is exactly the identity $-(Dn)^T(D\sigma) = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$ from before. The forms themselves do not depend on σ .

Lemma 14.4.6

The derivative of the Gauss map is self-adjoint with respect to the first fundamental form:

$$I_p(Dn|_p(v), w) = I_p(v, Dn|_p(w)) \quad \text{for all } v, w \in T_p\Sigma.$$

Proof. In a parametrisation, both fundamental forms are represented by *symmetric* matrices ($F = F$ trivially, and $M = \langle n, \sigma_{uv} \rangle = \langle n, \sigma_{vu} \rangle$ by symmetry of mixed partials). Hence

$$I_p(Dn(v), w) = -II_p(v, w) = -II_p(w, v) = I_p(Dn(w), v) = I_p(v, Dn(w)). \quad \square$$

Remark. A theorem worth knowing (though beyond this course) is the *fundamental theorem of surfaces in $\mathbb{R}^3$* : a smooth oriented connected surface in $\mathbb{R}^3$ is determined, up to rigid motion, by its two fundamental forms. So I and II really are a complete set of invariants for local surface theory.

§14.4.3 Gaussian Curvature

Definition 14.4.7 (Gaussian Curvature)

The **Gaussian curvature** of a smooth surface Σ in $\mathbb{R}^3$ is the function

$$\kappa : \Sigma \rightarrow \mathbb{R}, \quad \kappa(p) = \det(Dn|_p).$$

Remark. This is well-defined even when Σ is not orientable: locally there are two choices of normal, differing by a sign, but in two dimensions $\det(-A) = \det(A)$, so the determinant is unaffected.

Let's express κ in terms of the fundamental forms. Fix a parametrisation, and let $A = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $B = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, and let S be the matrix of $Dn|_p$ in the basis $\{\sigma_u, \sigma_v\}$. The relation $\Pi_p(v, w) = \mathbb{I}_p(-Dn(v), w)$ reads $B = -S^T A$, so $S = -(A^{-1}B)^T$ and

$$\kappa = \det S = \frac{\det B}{\det A} = \frac{LN - M^2}{EG - F^2}.$$

Under a transition map φ , A and B transform by $(D\varphi)^T(\cdot)(D\varphi)$ up to the same sign, so this ratio is genuinely independent of the parametrisation.

Example 14.4.8 (The Cylinder is Flat)

For the cylinder, the image of the Gauss map lies in the equator of S^2 : the normal $(\cos u, \sin u, 0)$ is always horizontal. So for any curve γ on the cylinder, $n \circ \gamma$ stays in a fixed circle, and $Dn|_p$ has image of dimension at most 1. Hence $\kappa \equiv 0$. Alternatively, $LN - M^2 = (-a)(0) - 0 = 0$. A surface with $\kappa \equiv 0$ is called **flat**.

Example 14.4.9 (Graphs, and the Sphere)

If Σ is the graph of a smooth function f , a computation from the example sheets gives

$$\kappa = \frac{f_{uu}f_{vv} - f_{uv}^2}{(1 + f_u^2 + f_v^2)^2}.$$

For instance, $f(u, v) = \sqrt{r^2 - u^2 - v^2}$ describes a piece of the sphere of radius r near its north pole, and at the origin $f_u = f_v = 0$, $f_{uu} = f_{vv} = -1/r$, $f_{uv} = 0$, giving

$$\kappa(0, 0, r) = \frac{1}{r^2}.$$

Since $O(3)$ acts transitively on the sphere by rigid motions, which preserve both fundamental forms, $\kappa \equiv 1/r^2$ on the whole sphere of radius r . Notice that a big sphere is gently curved and a small sphere is sharply curved, and that a dilation by a scales κ by $1/a^2$.

Example 14.4.10 (The Paraboloid)

Let $\Sigma = \{z = x^2 + y^2\}$. By the graph formula, at the origin $\kappa = 4 > 0$, and

everywhere

$$\kappa = \frac{4}{(1 + 4x^2 + 4y^2)^2} > 0.$$

One can also check that (for the inward orientation) the image of the Gauss map is precisely the open northern hemisphere: Σ is rotationally symmetric, the Gauss map commutes with rotations about the z -axis, and along $\{y = 0\}$ we have $n(x, 0, x^2) = (-2x, 0, 1)/\sqrt{1 + 4x^2}$, which sweeps out an open arc from pole towards (but never reaching) the equator.

§14.4.4 Elliptic, Hyperbolic and Parabolic Points

Definition 14.4.11

A point p of a smooth surface $\Sigma \subseteq \mathbb{R}^3$ is **elliptic** if $\kappa(p) > 0$, **hyperbolic** if $\kappa(p) < 0$, and **parabolic** if $\kappa(p) = 0$.

The sign of the curvature has a directly visible meaning.

Lemma 14.4.12

Near an elliptic point p , the surface lies entirely on one side of the affine tangent plane $p + T_p\Sigma$. Near a hyperbolic point, the surface meets both sides of the tangent plane in every neighbourhood of p .

Proof. Since the first fundamental form is positive definite, $EG - F^2 > 0$, and the sign of κ is the sign of $LN - M^2 = \det \Pi_p$. Recall the Taylor computation: for $w = h\sigma_u + \ell\sigma_v \in T_p\Sigma$, the signed distance from $\sigma(h, \ell)$ to the affine tangent plane is

$$\frac{1}{2} \Pi_p(w, w) + O(h^3, \ell^3).$$

If p is elliptic, Π_p has two eigenvalues of the same sign, so it is positive or negative definite; on the unit circle of directions w the form is bounded away from 0, so for small (h, ℓ) the quadratic term dominates the error and the signed distance has constant sign. If p is hyperbolic, Π_p has eigenvalues of opposite signs, so along the two eigendirections the signed distance takes opposite signs arbitrarily close to p . $\square$

Remark. At parabolic points nothing can be concluded. The cylinder has $\kappa \equiv 0$ and lies on one side of each tangent plane. On the other hand the *monkey saddle* $\sigma(u, v) = (u, v, u^3 - 3v^2u)$ has a parabolic point at the origin, but crosses its tangent plane there (it has three ‘valleys’ – two for the legs and one for the tail).

Proposition 14.4.13

Every compact smooth surface in $\mathbb{R}^3$ has an elliptic point.

Proof. Being compact, Σ is closed and bounded, so is contained in some closed ball; let R be the smallest radius such that $\Sigma \subseteq \overline{B(0, R)}$ (after translating so that the minimal enclosing ball is centred at the origin). Then Σ touches the sphere $S^2(R)$ at

some point, and applying a rotation we may assume $p = (0, 0, R) \in \Sigma$. The tangent planes of Σ and $S^2(R)$ at p agree (both are characterised by minimality/tangency; more carefully, if they differed then points of Σ near p would leave the ball). So near p we can write Σ as a graph $\{(u, v, f(u, v))\}$ with

$$f(u, v) \leq \sqrt{R^2 - u^2 - v^2},$$

expressing that Σ stays inside the ball, with equality at $(0, 0)$. Since $(0, 0)$ is thus a maximum of $f(u, v) - \sqrt{R^2 - u^2 - v^2}$, second-order Taylor expansion of both sides gives, for small (u, v) ,

$$\frac{1}{2} (f_{uu}u^2 + 2f_{uv}uv + f_{vv}v^2) \leq -\frac{1}{2R}(u^2 + v^2) + O((u^2 + v^2)^{3/2}).$$

So the Hessian of f at 0 is negative definite, and by the graph formula of Example 14.4.9 (using $f_u = f_v = 0$ at the origin),

$$\kappa(p) = f_{uu}f_{vv} - f_{uv}^2 \geq \frac{1}{R^2} > 0. \quad \square$$

This innocuous-looking proposition has a striking consequence: the flat torus $\mathbb{R}^2/\mathbb{Z}^2$, which we will construct properly in Section 14.6, admits *no* isometric embedding as a smooth surface in $\mathbb{R}^3$, since any compact surface in $\mathbb{R}^3$ has a point of positive curvature.

Here is one more interpretation of κ : it is the infinitesimal amount by which the Gauss map distorts area.

Theorem 14.4.14

Let $p \in \Sigma$ with $\kappa(p) \neq 0$, and let (A_i) be a sequence of open neighbourhoods of p which shrink to p , in the sense that for every $\epsilon > 0$ we have $A_i \subseteq B(p, \epsilon)$ for all large i . Then

$$|\kappa(p)| = \lim_{i \rightarrow \infty} \frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)}.$$

Proof. Fix an allowable parametrisation σ near p and let $V_i = \sigma^{-1}(A_i)$, so $\bigcap_i V_i = \{0\}$. First,

$$\text{area}_{\Sigma}(A_i) = \int_{V_i} \|\sigma_u \times \sigma_v\| \, du \, dv.$$

Next, by the chain rule, $Dn|_{\sigma(u,v)}(\sigma_u) = (n \circ \sigma)_u$ and similarly for v ; since $\kappa(p) \neq 0$, the map $n \circ \sigma$ has derivative of rank 2 at 0, so (by the inverse function theorem, shrinking V) it is an allowable parametrisation of a neighbourhood of $n(p)$ in S^2 . Hence for large i ,

$$\begin{aligned} \text{area}_{S^2}(n(A_i)) &= \int_{V_i} \|n_u \times n_v\| \, du \, dv = \int_{V_i} |\det Dn| \|\sigma_u \times \sigma_v\| \, du \, dv \\ &= \int_{V_i} |\kappa| \|\sigma_u \times \sigma_v\| \, du \, dv, \end{aligned}$$

using $n_u \times n_v = Dn(\sigma_u) \times Dn(\sigma_v) = \det(Dn) \sigma_u \times \sigma_v$. Finally, κ is continuous, so

given $\epsilon > 0$, for all large i we have $|\kappa| \in (|\kappa(p)| - \epsilon, |\kappa(p)| + \epsilon)$ on V_i , and hence

$$|\kappa(p)| - \epsilon \leq \frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)} \leq |\kappa(p)| + \epsilon.$$

Letting $i \rightarrow \infty$ and then $\epsilon \rightarrow 0$ gives the result. $\square$

§14.4.5 The Theorema Egregium

We defined κ using the Gauss map – that is, using the way Σ sits inside $\mathbb{R}^3$. The astonishing discovery of Gauss is that κ doesn't actually depend on the embedding at all.

Theorem 14.4.15 (Theorema Egregium)

The Gaussian curvature is invariant under isometries: if $f : \Sigma_1 \rightarrow \Sigma_2$ is an isometry of smooth surfaces in $\mathbb{R}^3$, then

$$\kappa_{\Sigma_2}(f(p)) = \kappa_{\Sigma_1}(p) \quad \text{for all } p \in \Sigma_1.$$

In other words, the Gaussian curvature is determined by the first fundamental form alone.

The 'remarkable theorem' is remarkable because κ was defined through the second fundamental form, which is definitely *not* an isometry invariant: the plane and the cylinder are locally isometric (unroll the cylinder!) but have different second fundamental forms. Somehow the particular combination $(LN - M^2)/(EG - F^2)$ only depends on E, F, G .

One can prove this by brute force – differentiate everything in sight and grind until κ is expressed in terms of E, F, G and their derivatives – but that proof teaches you nothing. We will instead take a more geometric route: in the next section we will find, on any surface, special parametrisations ('geodesic normal coordinates') in which the first fundamental form looks like

$$du^2 + G(u, v) dv^2,$$

and in such coordinates one can show (example sheet) that

$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}}.$$

Since geodesics – and hence these coordinates – will be defined using lengths of curves only, this formula exhibits κ as an invariant of the first fundamental form, proving the Theorema Egregium.

An immediate and famous consequence: there is no map of the Earth (or any part of it) which preserves distances to scale, since the sphere has $\kappa = 1/r^2 > 0$ while the plane has $\kappa \equiv 0$. Every atlas of the world necessarily distorts.

§14.5 Geodesics

On the plane, the shortest path between two points is a straight line. What is the analogue of a straight line on a curved surface? The right notion turns out to be that of

a *geodesic*, which we will define variationally: a geodesic is a critical point of an energy functional. This section develops the theory and computes geodesics on our favourite examples.

§14.5.1 The Energy Functional

Let Σ be a smooth surface in $\mathbb{R}^3$ and $\gamma : [a, b] \rightarrow \Sigma$ a smooth curve. Alongside the length $L(\gamma) = \int_a^b \|\gamma'\| dt$ we define the **energy**

$$E(\gamma) = \int_a^b \|\gamma'(t)\|^2 dt.$$

Energy is slightly less geometric than length (it is not invariant under reparametrisation), but it is analytically much better behaved – no square roots – and we will see that its critical points are exactly the constant-speed shortest paths.

Definition 14.5.1 (Variation)

A **one-parameter variation with fixed endpoints** of γ is a smooth map

$$\Gamma : (-\epsilon, \epsilon) \times [a, b] \rightarrow \Sigma, \quad \gamma_s(t) = \Gamma(s, t),$$

such that $\gamma_0 = \gamma$, and $\gamma_s(a) = \gamma(a)$, $\gamma_s(b) = \gamma(b)$ for all s .

Definition 14.5.2 (Geodesic)

A smooth curve $\gamma : [a, b] \rightarrow \Sigma$ is a **geodesic** if for every one-parameter variation (γ_s) of γ with fixed endpoints,

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = 0.$$

That is, γ is a critical point of the energy functional on paths from $\gamma(a)$ to $\gamma(b)$.

§14.5.2 The Geodesic Equations

Let's turn the variational condition into differential equations. Suppose γ has image in the image of an allowable parametrisation $\sigma : V \rightarrow U$ with first fundamental form $E du^2 + 2F du dv + G dv^2$ (context will always distinguish the function E from the energy). For small s write $\gamma_s(t) = \sigma(u(s, t), v(s, t))$ and let

$$R = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2, \quad E(\gamma_s) = \int_a^b R dt,$$

where dots denote t -derivatives. Differentiating under the integral,

$$\begin{aligned} \frac{\partial R}{\partial s} &= (E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2) \frac{\partial u}{\partial s} + (E_v \dot{u}^2 + 2F_v \dot{u}\dot{v} + G_v \dot{v}^2) \frac{\partial v}{\partial s} \\ &\quad + 2(E\dot{u} + F\dot{v}) \frac{\partial \dot{u}}{\partial s} + 2(F\dot{u} + G\dot{v}) \frac{\partial \dot{v}}{\partial s}. \end{aligned}$$

Integrating the last two terms by parts in t – the boundary terms vanish since $\partial u/\partial s = \partial v/\partial s = 0$ at $t = a, b$ (fixed endpoints!) – we obtain

$$\left. \frac{d}{ds} \right|_{s=0} E(\gamma_s) = \int_a^b \left(A \frac{\partial u}{\partial s} + B \frac{\partial v}{\partial s} \right) dt,$$

where

$$\begin{aligned} A &= E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2 - 2 \frac{d}{dt} (E\dot{u} + F\dot{v}), \\ B &= E_v \dot{u}^2 + 2F_v \dot{u}\dot{v} + G_v \dot{v}^2 - 2 \frac{d}{dt} (F\dot{u} + G\dot{v}). \end{aligned}$$

Since $\partial u/\partial s$ and $\partial v/\partial s$ can be arbitrary smooth functions vanishing at the endpoints, the integral vanishes for every variation exactly when $A \equiv B \equiv 0$.

Corollary 14.5.3 (Geodesic Equations)

A smooth curve $\gamma(t) = \sigma(u(t), v(t))$ is a geodesic if and only if

$$\begin{aligned} \frac{d}{dt} (E\dot{u} + F\dot{v}) &= \frac{1}{2} (E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2), \\ \frac{d}{dt} (F\dot{u} + G\dot{v}) &= \frac{1}{2} (E_v \dot{u}^2 + 2F_v \dot{u}\dot{v} + G_v \dot{v}^2). \end{aligned}$$

Remark. Being a geodesic looks like a global condition (it quantifies over all variations) but is actually local: any subarc of a geodesic is a geodesic, since a variation of the subarc extends by the constant variation, and conversely the equations above are differential equations, checked pointwise. This is important – it means we may verify the geodesic condition chart by chart.

What is the relationship between minimising energy and minimising length? The Cauchy–Schwarz inequality applied to $f = \|\gamma'\|$ and $g = 1$ gives

$$L(\gamma)^2 = \left(\int_a^b \|\gamma'\| \cdot 1 \, dt \right)^2 \leq (b-a) \int_a^b \|\gamma'\|^2 \, dt = (b-a)E(\gamma),$$

with equality if and only if $\|\gamma'\|$ is constant, i.e. γ is parametrised proportionally to arc length. From this we can deduce:

Corollary 14.5.4

If γ has constant speed and locally minimises length, then γ is a geodesic. Conversely, a curve which globally minimises energy among paths with the given endpoints also globally minimises length, and has constant speed.

We stress that geodesics need not globally minimise length – an arc of a great circle going the ‘long way round’ the sphere between two points is still a geodesic, as we are about to see, but is certainly not shortest. Geodesics are only *critical points* of the length/energy functionals: straightest, not necessarily shortest.

§14.5.3 Geodesics on the Plane and the Sphere

Example 14.5.5 (The Plane)

For $\sigma(u, v) = (u, v, 0)$ we have $E = G = 1$, $F = 0$, and all derivatives of E, F, G vanish. The geodesic equations reduce to

$$\ddot{u} = 0, \quad \ddot{v} = 0,$$

so $u(t) = \alpha t + \beta$ and $v(t) = \gamma t + \delta$: geodesics in the plane are straight lines traversed at constant speed, as they should be.

Example 14.5.6 (The Sphere)

Consider the unit sphere with the latitude–longitude parametrisation of Example 14.3.7, with first fundamental form $du^2 + \cos^2 u \, dv^2$, so $E = 1$, $F = 0$, $G = \cos^2 u$. The geodesic equations become

$$\ddot{u} + \sin u \cos u \dot{v}^2 = 0, \quad \frac{d}{dt} (\cos^2 u \dot{v}) = 0.$$

Geodesics have constant speed, and rescaling time we may assume unit speed:

$$\dot{u}^2 + \cos^2 u \dot{v}^2 = 1.$$

The second geodesic equation gives $\dot{v} = C / \cos^2 u$ for a constant C , and substituting into the unit speed condition,

$$\dot{u}^2 = 1 - \frac{C^2}{\cos^2 u} = \frac{\cos^2 u - C^2}{\cos^2 u}.$$

Assuming we are not on a parallel, we can divide to obtain

$$\frac{dv}{du} = \frac{\dot{v}}{\dot{u}} = \frac{C}{\cos u \sqrt{\cos^2 u - C^2}} = \frac{C \sec^2 u}{\sqrt{1 - C^2 \sec^2 u} \sqrt{1 - C^2} / \sqrt{1 - C^2}},$$

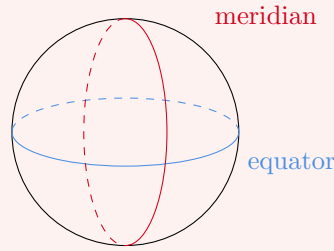
and the substitution $w = C \tan u / \sqrt{1 - C^2}$ turns the integral for v into $\int dw / \sqrt{1 - w^2}$, so

$$v = \arcsin \left(\frac{C \tan u}{\sqrt{1 - C^2}} \right) + \delta, \quad \text{i.e.} \quad \sin(v - \delta) = \lambda \tan u$$

for constants λ, δ . Expanding $\sin(v - \delta)$ and multiplying through by $\cos u$,

$$\underbrace{(\cos u \sin v)}_y \cos \delta - \underbrace{(\cos u \cos v)}_x \sin \delta - \lambda \underbrace{\sin u}_z = 0.$$

This is a linear equation in (x, y, z) – so the geodesic lies on a plane through the origin, and hence on a **great circle**, the intersection of such a plane with the sphere.



Conversely all constant-speed parametrisations of arcs of great circles are geodesics – this will drop out of the symmetry argument below with no computation at all.

Example 14.5.7 (The Torus of Revolution)

Take the torus obtained by revolving the circle of radius 1 centred at $(a, 0, 0)$ (with $a > 1$) in the xz -plane about the z -axis:

$$\sigma(u, v) = ((a + \cos u) \cos v, (a + \cos u) \sin v, \sin u),$$

with first fundamental form $du^2 + (a + \cos u)^2 dv^2$. Formally this is the sphere computation with $\cos u$ replaced by $a + \cos u$, and the same manipulations give

$$\frac{dv}{du} = \frac{C}{(a + \cos u) \sqrt{(a + \cos u)^2 - C^2}}.$$

But this integral cannot be evaluated in elementary functions – it is an *elliptic integral*, and the study of such integrals is precisely where elliptic functions historically came from. The moral: even on very symmetric surfaces, geodesics can be genuinely transcendental objects.

§14.5.4 A Second Characterisation of Geodesics

In the plane, straight lines are not just shortest – they are *straightest*: their tangent vector never turns. On a surface in $\mathbb{R}^3$, a curve's tangent vector may be forced to turn simply because the surface is curved, but a geodesic should turn no more than the surface forces it to. This is captured by the following characterisation.

Proposition 14.5.8

Let Σ be a smooth surface in $\mathbb{R}^3$. A smooth curve $\gamma : [a, b] \rightarrow \Sigma$ is a geodesic if and only if $\ddot{\gamma}(t)$ is normal to Σ (that is, orthogonal to $T_{\gamma(t)}\Sigma$) for every t .

Remark. Before the proof, note a pleasant consequence: if $\ddot{\gamma} \perp T_{\gamma}\Sigma$ then

$$\frac{d}{dt} \langle \dot{\gamma}, \dot{\gamma} \rangle = 2 \langle \ddot{\gamma}, \dot{\gamma} \rangle = 0,$$

since $\dot{\gamma}$ is tangent. So geodesics are automatically parametrised at constant speed.

Proof. Both conditions are local, so we may work in an allowable parametrisation and write $\gamma(t) = \sigma(u(t), v(t))$, with $\dot{\gamma} = \sigma_u \dot{u} + \sigma_v \dot{v}$. Now $\ddot{\gamma}$ is normal to Σ if and

only if it is orthogonal to both σ_u and σ_v . Let's expand the first condition:

$$0 = \left\langle \frac{d}{dt}(\sigma_u \dot{u} + \sigma_v \dot{v}), \sigma_u \right\rangle = \frac{d}{dt} \langle \sigma_u \dot{u} + \sigma_v \dot{v}, \sigma_u \rangle - \left\langle \sigma_u \dot{u} + \sigma_v \dot{v}, \frac{d}{dt} \sigma_u \right\rangle.$$

The first term is $\frac{d}{dt}(E\dot{u} + F\dot{v})$. For the second, $\frac{d}{dt}\sigma_u = \sigma_{uu}\dot{u} + \sigma_{uv}\dot{v}$, so it expands as

$$\dot{u}^2 \langle \sigma_u, \sigma_{uu} \rangle + \dot{u}\dot{v} (\langle \sigma_u, \sigma_{uv} \rangle + \langle \sigma_v, \sigma_{uu} \rangle) + \dot{v}^2 \langle \sigma_v, \sigma_{uv} \rangle.$$

Differentiating $E = \langle \sigma_u, \sigma_u \rangle$, $F = \langle \sigma_u, \sigma_v \rangle$ and $G = \langle \sigma_v, \sigma_v \rangle$ with respect to u gives

$$E_u = 2\langle \sigma_u, \sigma_{uu} \rangle, \quad F_u = \langle \sigma_u, \sigma_{uv} \rangle + \langle \sigma_v, \sigma_{uu} \rangle, \quad G_u = 2\langle \sigma_v, \sigma_{uv} \rangle,$$

so the condition $\langle \ddot{\gamma}, \sigma_u \rangle = 0$ becomes precisely the first geodesic equation

$$\frac{d}{dt}(E\dot{u} + F\dot{v}) = \frac{1}{2} (E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2),$$

and symmetrically $\langle \ddot{\gamma}, \sigma_v \rangle = 0$ is the second geodesic equation. $\square$

§14.5.5 Planes of Symmetry

Here is a beautifully economical way to find geodesics.

Proposition 14.5.9

Let Σ be a smooth surface in $\mathbb{R}^3$ and $\Pi \subseteq \mathbb{R}^3$ a plane such that Σ is setwise preserved by reflection in Π , and $C = \Pi \cap \Sigma$ is a smooth embedded curve. Then C , parametrised at constant speed, is a geodesic.

Proof. Let $p \in C$ and let R denote reflection in Π . Since R preserves Σ and fixes p , its derivative (which is R itself, suitably translated) preserves $T_p\Sigma$ by Lemma 14.2.22. The tangent plane therefore splits into the (± 1) -eigenspaces of the reflection restricted to it; since $T_p\Sigma \not\subseteq \Pi$ would force... let us argue directly: R restricted to $T_p\Sigma$ is a linear isometry of order 2, so $T_p\Sigma$ has an orthogonal basis of eigenvectors with eigenvalues ± 1 . The $+1$ -eigenspace is $T_p\Sigma \cap \Pi$, which is the tangent line to C ; it is one-dimensional (as C is an embedded curve), so the -1 -eigenspace is also one-dimensional. Now the normal line $\mathbb{R}n_p = (T_p\Sigma)^\perp$ is preserved by R ; if R acted by -1 on it, then R would act by -1 on a two-dimensional subspace spanned by n_p and the -1 -eigenvector, and by $+1$ on the tangent line to C – but a reflection acts by -1 only on the one-dimensional $\Pi^\perp$. So $Rn_p = n_p$, that is $n_p \in \Pi$.

Now parametrise C near p at constant speed by $\gamma(t)$. Since γ lies in the plane Π , both $\dot{\gamma}$ and $\ddot{\gamma}$ lie in (the direction space of) Π . Constant speed gives $\langle \dot{\gamma}, \ddot{\gamma} \rangle = 0$, so within the plane Π , $\ddot{\gamma}$ is orthogonal to the tangent line of C . But Π 's direction space is spanned by the tangent line of C together with n_p (both lie in Π , and they are orthogonal). Hence $\ddot{\gamma} \in \mathbb{R}n_p$, i.e. $\ddot{\gamma}$ is normal to Σ , and γ is a geodesic by Proposition 14.5.8. $\square$

Example 14.5.10

Any great circle on S^2 is the intersection of the sphere with a plane through the

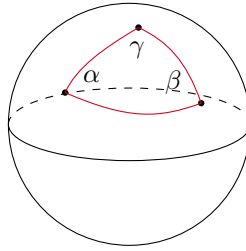
origin, which is a plane of reflective symmetry of the sphere. So all great circles, parametrised at constant speed, are geodesics – no computation needed. Together with the computation of Example 14.5.6 (and uniqueness of geodesics, coming shortly), the geodesics of the sphere are *precisely* the constant-speed arcs of great circles.

§14.5.6 Spherical Triangles

Since great circles are the ‘straight lines’ of the sphere, it is natural to do some classical geometry with them – this will also give us a first taste of the Gauss–Bonnet phenomenon. Note the sharp contrast with the plane: any two distinct great circles meet (in two antipodal points), so *there are no parallel lines in spherical geometry*.

Definition 14.5.11 (Spherical Triangle)

A **spherical triangle** on S^2 is a region bounded by three arcs of great circles, each of length less than π , meeting pairwise in three vertices. The **angle** at a vertex is the angle between the tangent vectors of the two arcs meeting there.



Theorem 14.5.12 (Gauss–Bonnet for Spherical Triangles)

Let T be a spherical triangle on the unit sphere with internal angles α, β, γ . Then

$$\text{area}(T) = \alpha + \beta + \gamma - \pi.$$

In particular the angles of a spherical triangle always sum to *more* than π .

Proof. The key player is the *lune*: the region between two great semicircles meeting at a pair of antipodal points with angle θ . Since the area of a lune is proportional to its angle and the lune of angle π is a hemisphere (area 2π), a lune of angle θ has area 2θ .

Now let T have vertices A, B, C . Each pair of the three great circles through the sides of T divides the sphere; consider the three lunes with angles α, β, γ containing T , together with their antipodal images. These six lunes cover the whole sphere; moreover every point of the sphere outside $T \cup (-T)$ is covered exactly once, while the points of T are covered by three lunes and points of the antipodal triangle $-T$ by the three antipodal lunes. Adding up areas,

$$2(2\alpha) + 2(2\beta) + 2(2\gamma) = \text{area}(S^2) + 4 \text{area}(T) = 4\pi + 4 \text{area}(T),$$

using $\text{area}(-T) = \text{area}(T)$. Rearranging gives $\text{area}(T) = \alpha + \beta + \gamma - \pi$. $\square$

This is a first instance of curvature controlling angle sums: on the unit sphere $\kappa \equiv 1$,

and the theorem says

$$\int_T \kappa \, dA = \text{area}(T) = (\alpha + \beta + \gamma) - \pi,$$

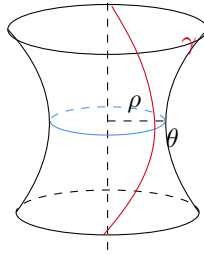
the *angle excess* of the triangle. We will later prove the same statement for geodesic triangles on an arbitrary surface – the local Gauss–Bonnet theorem – and the hyperbolic plane will exhibit the opposite behaviour, with angle sums falling short of π .

§14.5.7 Surfaces of Revolution and Clairaut's Relation

Let Σ be the surface of revolution of the curve $\eta(u) = (f(u), 0, g(u))$, with $f > 0$, which we may as well parametrise by arc length so the first fundamental form is $du^2 + f(u)^2 dv^2$ (Example 14.3.8).

Definition 14.5.13 (Parallels and Meridians)

The circles obtained by rotating a fixed point of η about the axis are called **parallels**; the copies of η obtained by rotating through a fixed angle are called **meridians**.



Every plane containing the axis of revolution is a plane of reflective symmetry of Σ , so by Proposition 14.5.9:

Corollary 14.5.14

All meridians of a surface of revolution are geodesics.

Parallels, on the other hand, are usually *not* geodesics – think of a circle of latitude on the sphere near the north pole, which is visibly not straight.

Lemma 14.5.15

The parallel $u = u_0$, parametrised at constant speed, is a geodesic if and only if $f'(u_0) = 0$, i.e. the parallel sits at a critical radius.

Proof. With $E = 1$, $F = 0$, $G = f^2$, the geodesic equations read

$$\ddot{u} = f f' \dot{v}^2, \quad \frac{d}{dt} (f^2 \dot{v}) = 0.$$

On the parallel $u \equiv u_0$ with constant speed, $\dot{v}$ is a non-zero constant, so the second equation holds automatically and the first reduces to $0 = f(u_0) f'(u_0) \dot{v}^2$. Since $f > 0$ and $\dot{v} \neq 0$, this holds precisely when $f'(u_0) = 0$. $\square$

For general geodesics on a surface of revolution there is a striking conservation law. Let

γ be a unit speed curve on Σ , let $\rho = f(u)$ be the radius of the parallel through a point, and let θ be the angle at which γ crosses that parallel.

Proposition 14.5.16 (Clairaut's Relation)

If γ is a geodesic, then $\rho \cos \theta$ is constant along γ .

Proof. Write $\gamma(t) = \sigma(u(t), v(t))$, so $\dot{\gamma} = \sigma_u \dot{u} + \sigma_v \dot{v}$, and the tangent to the parallel is σ_v , of length $\|\sigma_v\| = \sqrt{G} = f = \rho$. By the angle formula from the first fundamental form ($F = 0$),

$$\cos \theta = \frac{\langle \sigma_v, \sigma_u \dot{u} + \sigma_v \dot{v} \rangle}{\|\sigma_v\| \|\dot{\gamma}\|} = \frac{G \dot{v}}{\rho} = \rho \dot{v},$$

using $\|\dot{\gamma}\| = 1$. So $\rho \cos \theta = \rho^2 \dot{v} = f^2 \dot{v}$, and the second geodesic equation says precisely that this is constant. $\square$

Example 14.5.17 (Geodesics Trapped Between Parallels)

Consider an ellipsoid of revolution, and a geodesic γ which crosses some parallel of radius ρ_0 at angle θ_0 , and is not a meridian, so $c = \rho_0 \cos \theta_0 > 0$. Along γ we always have $\rho \cos \theta = c$, and since $|\cos \theta| \leq 1$ this forces $\rho \geq c$: the geodesic can never enter the polar caps where the parallels are narrower than c . Such a geodesic is ‘trapped’ in the band between two parallels of radius c , bouncing between them indefinitely. In particular, any geodesic through a pole must have $c = 0$, i.e. must be a meridian.

§14.5.8 Local Existence of Geodesics

Solving the geodesic equations globally is generally hopeless (we already met elliptic integrals on the torus), but locally the general theory of differential equations does everything for us. Recall *Picard's theorem* from Analysis and Topology: an ODE $\dot{x} = h(t, x)$ with h continuous and Lipschitz in x has a unique local solution through each initial condition, and if h is smooth, the solution is smooth and depends smoothly on the initial conditions.

The geodesic equations can be put in this form. Writing them as

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \ddot{u} \\ \ddot{v} \end{pmatrix} = R(u, v, \dot{u}, \dot{v})$$

with R built smoothly from E, F, G and their derivatives, we can invert the (positive definite, hence invertible) matrix – and matrix inversion is smooth – to get

$$\ddot{u} = A(u, v, \dot{u}, \dot{v}), \quad \ddot{v} = B(u, v, \dot{u}, \dot{v})$$

with A, B smooth. Introducing $p = \dot{u}$, $q = \dot{v}$ converts this to a first-order system in (u, v, p, q) to which Picard's theorem applies.

Corollary 14.5.18

Let Σ be a smooth surface in $\mathbb{R}^3$, $p \in \Sigma$ and $0 \neq v \in T_p \Sigma$. Then there is some $\epsilon > 0$ and a geodesic $\gamma : [0, \epsilon) \rightarrow \Sigma$ with

$$\gamma(0) = p, \quad \dot{\gamma}(0) = v,$$

and γ is unique and depends smoothly on (p, v) .

This local existence gives us the promised ‘best possible’ parametrisations. Fix $p \in \Sigma$ and a unit speed geodesic arc $\gamma(v)$ through $p = \gamma(0)$. For each small v , let γ_v be the unit speed geodesic through $\gamma(v)$ orthogonal to γ , and define

$$\sigma(u, v) = \gamma_v(u).$$

Lemma 14.5.19

For sufficiently small $\epsilon, \delta > 0$, the map $\sigma : (u, v) \mapsto \gamma_v(u)$ on $(-\epsilon, \epsilon) \times (-\delta, \delta)$ is an allowable parametrisation of an open set in Σ .

Proof. Smoothness of σ is the smooth dependence on initial conditions in Picard’s theorem. At the origin, $\sigma_u = \dot{\gamma}_0(0)$ and $\sigma_v = \dot{\gamma}(0)$ are orthogonal unit vectors by construction, so $D\sigma$ has rank 2 at the origin, and by continuity in a neighbourhood; shrinking further makes σ injective. So σ is allowable. $\square$

Corollary 14.5.20 (Geodesic Normal Coordinates)

Every smooth surface in $\mathbb{R}^3$ admits, near each point, an allowable parametrisation with first fundamental form

$$du^2 + G(u, v) dv^2, \quad \text{with} \quad G(0, v) = 1, \quad G_u(0, v) = 0.$$

Proof. Take $\sigma(u, v) = \gamma_v(u)$ as above. For fixed v_0 , the curve $u \mapsto \sigma(u, v_0)$ is a unit speed geodesic, so $E = \|\sigma_u\|^2 = 1$ everywhere, and in particular $E_v = 0$. Substituting the curve $u(t) = t, v(t) = v_0$ (so $\dot{u} = 1, \dot{v} = 0$) into the second geodesic equation,

$$\frac{d}{dt}(F\dot{u} + G\dot{v}) = \frac{1}{2}(E_v\dot{u}^2 + 2F_v\dot{u}\dot{v} + G_v\dot{v}^2) \implies \frac{d}{dt}F = 0 \implies F_u = 0,$$

so F is independent of u . At $u = 0$ we have $F = \langle \dot{\gamma}_v(0), \dot{\gamma}(v) \rangle = 0$ by the orthogonality in the construction; hence $F \equiv 0$. Next, $\sigma_v(0, v) = \dot{\gamma}(v)$ has unit length, so $G(0, v) = 1$. Finally $u = 0$ gives the unit speed geodesic γ itself (with $\dot{u} = 0, \dot{v} = 1$), and the first geodesic equation applied to it gives

$$\frac{d}{dt}(E\dot{u} + F\dot{v}) = \frac{1}{2}(E_u\dot{u}^2 + 2F_u\dot{u}\dot{v} + G_u\dot{v}^2) \implies 0 = \frac{1}{2}G_u(0, v). \quad \square$$

In geodesic normal coordinates one shows (on the example sheets, by direct computation with the formula $\kappa = (LN - M^2)/(EG - F^2)$ and repeated differentiation of the relations among σ_u, σ_v, n) that

$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}}.$$

Since geodesics, and hence these coordinates, are defined purely in terms of lengths of curves, this expresses κ in terms of the first fundamental form alone – completing the proof of the Theorema Egregium (Theorem 14.4.15), as promised.

§14.5.9 Surfaces of Constant Curvature

The formula $\kappa = -(\sqrt{G})_{uu}/\sqrt{G}$ lets us completely determine the local geometry of surfaces of constant curvature. Note first that under a dilation $x \mapsto ax$ of $\mathbb{R}^3$, the functions E, F, G scale by a^2 while L, M, N scale by a , so κ scales by $1/a^2$; thus it suffices to understand $\kappa \equiv 0, \pm 1$.

Proposition 14.5.21

Let Σ be a smooth surface in $\mathbb{R}^3$.

- (i) If $\kappa \equiv 0$, then Σ is locally isometric to the plane $(\mathbb{R}^2, du^2 + dv^2)$.
- (ii) If $\kappa \equiv 1$, then Σ is locally isometric to the unit sphere $(S^2, du^2 + \cos^2 u dv^2)$.

Proof. Take geodesic normal coordinates about a point, so that the first fundamental form is $du^2 + G dv^2$ with $G(0, v) = 1$, $G_u(0, v) = 0$, and

$$(\sqrt{G})_{uu} + \kappa\sqrt{G} = 0.$$

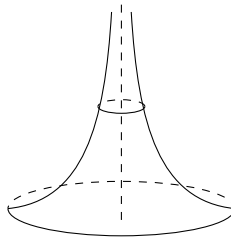
If $\kappa \equiv 0$: then $\sqrt{G} = A(v)u + B(v)$, and the boundary conditions force $A \equiv 0$, $B \equiv 1$, so $G \equiv 1$ and the first fundamental form is $du^2 + dv^2$ – that of the plane. By Lemma 14.3.12, Σ is locally isometric to the plane.

If $\kappa \equiv 1$: then $\sqrt{G} = A(v)\sin u + B(v)\cos u$, and the boundary conditions force $A \equiv 0$, $B \equiv 1$, giving first fundamental form $du^2 + \cos^2 u dv^2$ – that of the unit sphere in latitude–longitude coordinates (Example 14.3.7). $\square$

What about $\kappa \equiv -1$? The same argument gives $\sqrt{G} = \cosh u$, so the first fundamental form is

$$du^2 + \cosh^2 u dv^2.$$

There is a surface of revolution with this geometry: the **pseudosphere** (or tractoid), obtained by revolving the *tractrix* – the curve traced by a weight dragged on the end of a taut string whose other end moves along a line – about its asymptotic axis.



But the pseudosphere is geometrically unsatisfying: it has an edge (the rim), and geodesics run off it in finite time. A theorem of Hilbert says this is unavoidable – *no* complete surface of constant curvature -1 embeds smoothly in $\mathbb{R}^3$. If we want a good home for constant negative curvature, we must leave $\mathbb{R}^3$ behind and work with *abstract* Riemannian metrics. That is the subject of the next section, and the resulting geometry – hyperbolic geometry – will occupy us for most of the rest of the course.

Before moving on, we record a change of variables we will meet again: substituting $V = e^v \tanh u$, $W = e^v \operatorname{sech} u$ transforms $du^2 + \cosh^2 u dv^2$ into

$$\frac{dV^2 + dW^2}{W^2},$$

which is the ‘standard’ form of the hyperbolic metric on a half-plane.

§14.6 Abstract Riemannian Metrics

The first fundamental form of a surface in $\mathbb{R}^3$ was induced from the ambient Euclidean inner product. But nothing stops us writing down expressions like $E du^2 + 2F du dv + G dv^2$ on an abstract smooth surface directly, without any embedding – and the Theorema Egregium tells us that curvature will still make sense. This idea unlocks geometries, like the hyperbolic plane, which genuinely do not exist inside $\mathbb{R}^3$.

§14.6.1 Riemannian Metrics

Definition 14.6.1 (Riemannian Metric on an Open Set)

Let $V \subseteq \mathbb{R}^2$ be open. A **Riemannian metric** on V is a smooth map from V to the positive definite symmetric bilinear forms on $\mathbb{R}^2$,

$$x \mapsto \begin{pmatrix} E(x) & F(x) \\ F(x) & G(x) \end{pmatrix}, \quad \text{with } E > 0, \ G > 0, \ EG - F^2 > 0,$$

where $E, F, G : V \rightarrow \mathbb{R}$ are smooth. We write it as $E du^2 + 2F du dv + G dv^2$.

Given such a metric, we measure exactly as before: a tangent vector w at $x \in V$ has length $\|w\|^2 = w^T \begin{pmatrix} E & F \\ F & G \end{pmatrix} w$, and a smooth curve $\gamma(t) = (u(t), v(t))$ in V has length

$$L(\gamma) = \int_a^b (E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2)^{1/2} dt.$$

Angles and areas are defined by the same formulae as for surfaces in $\mathbb{R}^3$; and geodesics are again critical points of the energy, characterised by the same geodesic equations (our derivation only ever used E, F, G).

To put a metric on an abstract surface, we put compatible metrics on all its charts.

Definition 14.6.2 (Riemannian Metric on an Abstract Surface)

Let Σ be an abstract smooth surface with atlas $\{(U_i, \phi_i)\}$, $\phi_i : U_i \rightarrow V_i \subseteq \mathbb{R}^2$. A **Riemannian metric** g on Σ is a choice of Riemannian metric on each V_i such that the transition maps are isometries: writing f for the transition map between charts i and j ,

$$(Df)^T \begin{pmatrix} E_j & F_j \\ F_j & G_j \end{pmatrix} (Df) = \begin{pmatrix} E_i & F_i \\ F_i & G_i \end{pmatrix}$$

wherever this makes sense.

The compatibility condition is precisely the transformation law that first fundamental forms of surfaces in $\mathbb{R}^3$ obey under transition maps – so a smooth surface in $\mathbb{R}^3$ with its first fundamental form is an example. The point is that there are now many more examples.

Example 14.6.3 (The Flat Torus)

Recall the torus $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, which has an atlas of charts inverse to $q|_D$ for small

discs $D \subseteq \mathbb{R}^2$, whose transition maps are restrictions of translations of $\mathbb{R}^2$. Equip every chart with the Euclidean metric $du^2 + dv^2$. If f is a translation then $Df = I$, so the compatibility condition

$$(Df)^T I (Df) = I$$

holds trivially, and we obtain a global Riemannian metric on T^2 : the **flat torus**. Every point has a chart which is an isometry onto a Euclidean disc, so this metric has $\kappa \equiv 0$.

Contrast this with the torus of revolution in $\mathbb{R}^3$: by Proposition 14.4.13, any compact surface in $\mathbb{R}^3$ has an elliptic point, so *no* embedding of the torus in $\mathbb{R}^3$ can induce the flat metric. The flat torus is a genuinely abstract object.

Example 14.6.4 ($\mathbb{RP}^2$ and the Klein Bottle)

The real projective plane $\mathbb{RP}^2 = S^2 / \pm 1$ has an atlas of charts lifted from charts of S^2 on open sets contained in hemispheres, with transition maps locally equal to the identity or the antipodal map. The antipodal map is an isometry of the round sphere, so the round metric descends to a Riemannian metric on $\mathbb{RP}^2$ of constant curvature $+1$.

Similarly, the Klein bottle (the square with one pair of sides glued directly and the other with a flip) has an atlas whose transition maps are translations and translations composed with a reflection – all Euclidean isometries – so it inherits a flat Riemannian metric. Neither $\mathbb{RP}^2$ nor the Klein bottle embeds in $\mathbb{R}^3$ at all, so abstract metrics are the only ones available.

Definition 14.6.5 (Isometry of Riemannian Surfaces)

Let (Σ_1, g_1) and (Σ_2, g_2) be abstract smooth surfaces with Riemannian metrics. A diffeomorphism $f : \Sigma_1 \rightarrow \Sigma_2$ is an **isometry** if it preserves the lengths of all curves.

Example 14.6.6 (Pullback Metrics)

If (Σ_2, g_2) is given and $f : \Sigma_1 \rightarrow \Sigma_2$ is any diffeomorphism of smooth surfaces, there is a unique metric on Σ_1 making f an isometry: define the inner product of two tangent vectors to be the g_2 -inner product of their images under Df . This is the **pullback metric** f^*g_2 .

§14.6.2 The Length Metric

A Riemannian metric on Σ measures lengths of curves; from this we get a candidate distance function between points, in the usual ‘infimum over paths’ way.

Definition 14.6.7 (Length Metric)

Let (Σ, g) be a connected abstract smooth surface with a Riemannian metric. For

$p, q \in \Sigma$ define

$$d_g(p, q) = \inf\{L(\gamma) \mid \gamma \text{ a piecewise smooth path from } p \text{ to } q\}.$$

We should check this is an honest metric – the worry being that two distinct points might be joined by paths of arbitrarily small length.

Proposition 14.6.8

d_g is a metric on Σ , and it induces the original topology of Σ .

Proof. First, d_g is well-defined and finite: a connected surface is path-connected, and given a continuous path from p to q we can cover its (compact) image by finitely many charts $U_1, \dots, U_N$ and pick intermediate points $p = x_0, x_1 \in U_1 \cap U_2, \dots, x_N = q$; within each chart, the straight line segment between the images of consecutive points gives a smooth path, and concatenating produces a piecewise smooth path from p to q . Symmetry of d_g follows by reversing paths, and the triangle inequality by concatenating them.

The only issue is showing $d_g(p, q) = 0 \implies p = q$; equivalently, that distinct points are a positive distance apart. Let $p \neq q$, fix a chart (U, ϕ) at p with $\phi(p) = 0$ and $\phi(U) = B(0, 1)$, and choose $\epsilon \in (0, 1)$ small enough that $q \notin \phi^{-1}(\overline{B(0, \epsilon)})$. Any path γ from p to q must cross the circle $\phi^{-1}(\{|z| = \epsilon\})$, so it suffices to bound below the length of the part of γ inside $\phi^{-1}(\overline{B(0, \epsilon)})$.

On the compact set $\overline{B(0, \epsilon)}$, the metric $\begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is a continuous family of positive definite forms, so there is $\delta > 0$ such that

$$\begin{pmatrix} E - \delta & F \\ F & G - \delta \end{pmatrix}$$

is still positive semi-definite everywhere on $\overline{B(0, \epsilon)}$ – for instance because the smallest eigenvalue of the metric is a continuous positive function on a compact set, hence bounded below by some $\delta > 0$. Then for any tangent vector w at a point of $\overline{B(0, \epsilon)}$,

$$\|w\|_g^2 \geq \delta \|w\|_{\text{Euc}}^2,$$

so g -lengths of curves inside the ball dominate $\sqrt{\delta}$ times their Euclidean lengths. The part of $\phi \circ \gamma$ from 0 to the circle of radius ϵ has Euclidean length at least ϵ , whence

$$d_g(p, q) \geq \sqrt{\delta} \epsilon > 0.$$

Finally, the two-sided comparison between g and (a scaled) Euclidean metric on compact sets, argued exactly as above, shows that small d_g -balls contain and are contained in small chart-balls, so d_g induces the given topology. $\square$

We now have all the language we need. The plane $\mathbb{R}^2$ with its flat metric and the sphere S^2 with its round metric are homogeneous, highly symmetric geometries of curvature 0 and +1. The missing member of the family – constant curvature -1 – is the subject of the next section.

§14.7 Hyperbolic Geometry

We now construct the hyperbolic plane: the homogeneous geometry of constant curvature -1 . Historically this was the geometry that resolved the two-thousand-year controversy over Euclid's parallel postulate – in the hyperbolic plane, through any point not on a line ℓ there are *infinitely many* lines not meeting ℓ . We will build it concretely, as a disc (or half-plane) with a rescaled Euclidean metric, and its symmetries will come from the Möbius group.

§14.7.1 The Hyperbolic Disc

Definition 14.7.1 (Hyperbolic Disc)

Let $D = \{z \in \mathbb{C} \mid |z| < 1\}$ be the open unit disc. The **hyperbolic metric** on D is the Riemannian metric

$$g_{\text{hyp}} = \frac{4(du^2 + dv^2)}{(1 - u^2 - v^2)^2} = \frac{4|dz|^2}{(1 - |z|^2)^2}.$$

The disc D with this metric is the **disc model** of the hyperbolic plane.

Since $E = G$ and $F = 0$, the hyperbolic metric is a *conformal* rescaling of the Euclidean metric – hyperbolic angles are Euclidean angles. But lengths are distorted, and dramatically so near the boundary: a smooth curve $\gamma : [0, 1] \rightarrow D$ has hyperbolic length

$$L_{\text{hyp}}(\gamma) = \int_0^1 \frac{2|\dot{\gamma}(t)|}{1 - |\gamma(t)|^2} dt,$$

and the factor $2/(1 - |z|^2)$ blows up as $|z| \rightarrow 1$. The boundary circle is ‘infinitely far away’ – as we will verify shortly – and it is *not* part of the hyperbolic plane.

The flat plane and the round sphere both have very large isometry groups. The same is true here, and the isometries come from a group we know well from IA Groups: the Möbius group

$$\text{Möb} = \left\{ z \mapsto \frac{az + b}{cz + d} \mid \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbb{C}) \right\}$$

acting on $\mathbb{C} \cup \{\infty\}$.

Lemma 14.7.2

The subgroup of Möbius maps preserving the disc, $\text{Möb}(D) = \{T \in \text{Möb} \mid T(D) = D\}$, is

$$\text{Möb}(D) = \left\{ z \mapsto e^{i\theta} \frac{z - a}{1 - \bar{a}z} \mid a \in D, \theta \in \mathbb{R} \right\}.$$

Proof. First we check the given maps preserve D . For $|a| < 1$,

$$\begin{aligned} \left| \frac{z - a}{1 - \bar{a}z} \right| = 1 &\iff (z - a)(\bar{z} - \bar{a}) = (1 - \bar{a}z)(1 - a\bar{z}) \\ &\iff |z|^2 - a\bar{z} - \bar{a}z + |a|^2 = 1 - \bar{a}z - a\bar{z} + |a|^2|z|^2 \\ &\iff (|z|^2 - 1)(1 - |a|^2) = 0 \iff |z| = 1. \end{aligned}$$

So the map preserves the unit circle; being a homeomorphism of $\mathbb{C} \cup \{\infty\}$, it maps D to either the inside or the outside, and since $a \mapsto 0 \in D$, it preserves D . The rotation factor $e^{i\theta}$ clearly preserves D too.

Conversely, suppose $T \in \text{Möb}$ preserves D , and let $a = T^{-1}(0) \in D$. Then composing T with the inverse of $z \mapsto (z - a)/(1 - \bar{a}z)$, we get a Möbius map preserving D and fixing 0; it also preserves the unit circle, and one checks (using preservation of inverse points, or a direct computation) that a Möbius map fixing 0 and preserving the unit circle is a rotation $z \mapsto e^{i\theta}z$. So T has the stated form. $\square$

Lemma 14.7.3

$\text{Möb}(D)$ acts on (D, g_{hyp}) by isometries.

Proof. The group $\text{Möb}(D)$ is generated by rotations $z \mapsto e^{i\theta}z$ and the maps $T_a : z \mapsto \frac{z-a}{1-\bar{a}z}$. Rotations are Euclidean isometries fixing $|z|$, so preserve g_{hyp} , which only depends on $|z|$ and $|dz|$. For T_a , write $w = \frac{z-a}{1-\bar{a}z}$. Differentiating,

$$dw = \left(\frac{1}{1-\bar{a}z} + \frac{(z-a)\bar{a}}{(1-\bar{a}z)^2} \right) dz = \frac{1-|a|^2}{(1-\bar{a}z)^2} dz.$$

Also

$$1-|w|^2 = \frac{|1-\bar{a}z|^2 - |z-a|^2}{|1-\bar{a}z|^2} = \frac{(1-|a|^2)(1-|z|^2)}{|1-\bar{a}z|^2},$$

where the numerator is the same computation as in Lemma 14.7.2. Dividing,

$$\frac{|dw|}{1-|w|^2} = \frac{(1-|a|^2)|dz|}{|1-\bar{a}z|^2} \cdot \frac{|1-\bar{a}z|^2}{(1-|a|^2)(1-|z|^2)} = \frac{|dz|}{1-|z|^2}.$$

So T_a preserves the expression $2|dz|/(1-|z|^2)$, i.e. pulls back g_{hyp} to itself. Hence every element of $\text{Möb}(D)$ is an isometry. $\square$

Note that $\text{Möb}(D)$ acts *transitively* on D (the map T_a sends a to 0), so the hyperbolic plane looks the same at every point – just like the flat plane and the round sphere.

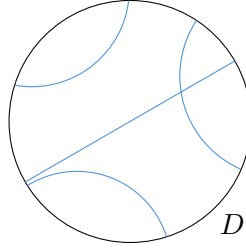
Now we can find the geodesics.

Proposition 14.7.4

In (D, g_{hyp}) :

- (i) any two points are joined by a unique geodesic, up to reparametrisation;
- (ii) the geodesics are the diameters of D and the arcs of circles meeting the boundary circle ∂D at right angles.

The complete geodesics (defined on all of $\mathbb{R}$) are called **hyperbolic lines**.



Proof. Consider first two points on the real diameter, say 0 and $a \in (0, 1)$. Let $\gamma(t) = (u(t), v(t))$ be any smooth path from 0 to a in D . Then

$$L_{\text{hyp}}(\gamma) = \int_0^1 \frac{2\sqrt{\dot{u}^2 + \dot{v}^2}}{1 - u^2 - v^2} dt \geq \int_0^1 \frac{2|\dot{u}|}{1 - u^2} dt \geq \int_0^1 \frac{2\dot{u}}{1 - u^2} dt = 2 \operatorname{artanh} a,$$

where the first inequality is an equality if and only if $\dot{v} \equiv 0$ and $v \equiv 0$ (dropping $\dot{v}^2$ and enlarging the denominator), and the second iff u is monotonic. So the straight segment $[0, a]$, monotonically parametrised, is the unique shortest path, of hyperbolic length $2 \operatorname{artanh} a$; being length-minimising with constant speed (after reparametrising) it is a geodesic, and any geodesic between the two points must equal it.

Now let $p, q \in D$ be arbitrary. Using the transitive isometric action of $\operatorname{Möb}(D)$, choose $T \in \operatorname{Möb}(D)$ with $T(p) = 0$, and then a rotation moves $T(q)$ to some $a \in (0, 1)$. Since isometries carry geodesics to geodesics, p and q are joined by a unique geodesic: the preimage of the segment $[0, a]$. Finally, Möbius maps send circles-and-lines to circles-and-lines and preserve angles, and diameters are exactly the circles-and-lines through 0 meeting ∂D at right angles; so the geodesics through arbitrary points are exactly diameters and circular arcs orthogonal to ∂D . $\square$

Corollary 14.7.5

The hyperbolic distance on D is

$$d_{\text{hyp}}(p, q) = 2 \operatorname{artanh} \left| \frac{p - q}{1 - \bar{p}q} \right|.$$

Proof. The map $z \mapsto \frac{z-p}{1-\bar{p}z}$ is an isometry taking p to 0 and q to $\frac{q-p}{1-\bar{p}q}$, and a rotation about 0 takes this to the positive real point $a = \left| \frac{q-p}{1-\bar{p}q} \right|$, at distance $2 \operatorname{artanh} a$ from 0. $\square$

Notice that $d_{\text{hyp}}(0, a) = 2 \operatorname{artanh} a \rightarrow \infty$ as $a \rightarrow 1$: the boundary really is infinitely far away, so the hyperbolic plane is ‘complete’ in a way the pseudosphere was not.

Remark. A pleasant uniqueness statement: up to a constant scale, g_{hyp} is the *only* Riemannian metric on D invariant under $\operatorname{Möb}(D)$ for which the diameters are length-minimising – essentially because invariance forces the distance function $p(a) = d(0, a)$ to satisfy the functional equation $p(b) - p(a) = p\left(\frac{b-a}{1-\bar{a}b}\right)$, whose differentiable solutions are multiples of artanh . The factor 4 in the metric is chosen precisely so that the curvature is -1 , as we will see.

§14.7.2 The Upper Half-Plane

There is a second standard model of the same geometry, often more convenient for computation.

Definition 14.7.6 (Upper Half-Plane)

The **hyperbolic upper half-plane** is

$$\mathfrak{h} = \{z \in \mathbb{C} \mid \operatorname{Im} z > 0\}, \quad \text{with the metric} \quad g_{\text{hyp}} = \frac{dx^2 + dy^2}{y^2} = \frac{|dz|^2}{(\operatorname{Im} z)^2}.$$

Lemma 14.7.7

(D, g_{hyp}) and $(\mathfrak{h}, g_{\text{hyp}})$ are isometric.

Proof. Consider the mutually inverse Möbius maps

$$T : \mathfrak{h} \rightarrow D, \quad T(w) = \frac{w - i}{w + i}, \quad T^{-1}(z) = i \frac{1 + z}{1 - z}.$$

(T sends $\mathbb{R} \cup \{\infty\}$ to the unit circle and $i \mapsto 0$, so maps $\mathfrak{h}$ onto D .) We compute

$$T'(w) = \frac{2i}{(w + i)^2},$$

and, with $z = T(w)$,

$$1 - |z|^2 = \frac{|w + i|^2 - |w - i|^2}{|w + i|^2} = \frac{4 \operatorname{Im} w}{|w + i|^2}.$$

Hence

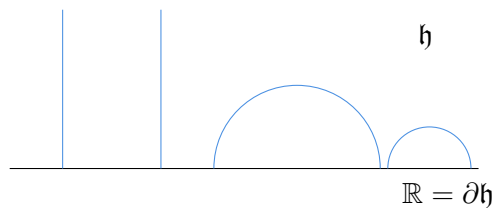
$$\frac{2|dz|}{1 - |z|^2} = \frac{2|T'(w)| |dw|}{1 - |T(w)|^2} = \frac{4|dw|}{|w + i|^2} \cdot \frac{|w + i|^2}{4 \operatorname{Im} w} = \frac{|dw|}{\operatorname{Im} w}.$$

So T pulls the disc metric back to the half-plane metric, i.e. T is an isometry. $\square$

Corollary 14.7.8

Any two points of $\mathfrak{h}$ are joined by a unique geodesic, and the hyperbolic lines in $\mathfrak{h}$ are the vertical half-lines and the semicircles centred on the real axis.

Proof. The isometry $T^{-1} : D \rightarrow \mathfrak{h}$ is a Möbius map, so takes circles-and-lines to circles-and-lines, preserves angles, and takes ∂D to $\mathbb{R} \cup \{\infty\}$. So the images of the hyperbolic lines of D (circles-and-lines orthogonal to ∂D) are precisely the circles-and-lines orthogonal to the real axis: vertical lines and semicircles centred on $\mathbb{R}$. $\square$



In the half-plane model the orientation-preserving symmetries take a particularly clean form: one can check that the Möbius maps preserving $\mathfrak{h}$ are exactly those with *real* coefficients and positive determinant,

$$\text{Möb}(\mathfrak{h}) = \left\{ z \mapsto \frac{az+b}{cz+d} \mid a, b, c, d \in \mathbb{R}, ad - bc > 0 \right\} \cong \text{PSL}(2, \mathbb{R}),$$

where $\text{PSL}(2, \mathbb{R}) = \text{SL}(2, \mathbb{R})/\{\pm I\}$, since scaling the matrix doesn't change the map. For example $z \mapsto z + b$ ($b \in \mathbb{R}$) and $z \mapsto \lambda z$ ($\lambda > 0$) are hyperbolic isometries – the second showing that all vertical lines are ‘translates’ of one another.

Remark (The Hyperboloid Model). There is a third model worth knowing about, which makes the analogy with the sphere vivid. Consider $\mathbb{R}^3$ with the *Lorentzian* inner product $\langle x, y \rangle_L = x_1y_1 + x_2y_2 - x_3y_3$, and the upper sheet of the hyperboloid

$$\{x \in \mathbb{R}^3 \mid \langle x, x \rangle_L = -1, x_3 > 0\},$$

which is the ‘sphere of radius $\sqrt{-1}$ ’ for this inner product. The restriction of $\langle \cdot, \cdot \rangle_L$ to its tangent planes is positive definite, and gives a metric of constant curvature -1 isometric to (D, g_{hyp}) – indeed, radial projection from $(0, 0, -1)$ maps the hyperboloid onto the disc $D \times \{0\}$ and carries this metric to g_{hyp} , in perfect analogy with stereographic projection of the sphere. The isometry group becomes the group $\text{O}^+(2, 1)$ of Lorentz transformations preserving the sheet, mirroring $\text{O}(3)$ acting on S^2 .

§14.7.3 Isometries of the Hyperbolic Plane

We have exhibited many isometries of $\mathbb{H}$ (we write $\mathbb{H}$ for the hyperbolic plane in either model). Are there others? Certainly one: complex conjugation $z \mapsto \bar{z}$ on D visibly preserves g_{hyp} but is orientation-reversing. It is the hyperbolic analogue of a reflection, and generalises as follows.

Definition 14.7.9 (Inversion)

Let $\Gamma \subseteq \mathbb{C} \cup \{\infty\}$ be a circle or line. Points z, z' are **inverse with respect to Γ** if every circle-or-line through z orthogonal to Γ also passes through z' . Given z , the inverse point exists and is unique, and the map $J_\Gamma : z \mapsto z'$ is called **inversion** in Γ .

To see existence and uniqueness: for $\Gamma = \mathbb{R} \cup \{\infty\}$, the circles through z orthogonal to $\mathbb{R}$ are precisely those centred on $\mathbb{R}$, and they all pass through $\bar{z}$ and through no other common point; so $J_{\mathbb{R} \cup \{\infty\}}(z) = \bar{z}$. For general Γ , choose $T \in \text{Möb}$ with $T(\mathbb{R} \cup \{\infty\}) = \Gamma$; since Möbius maps preserve circles and angles, they preserve the inversion relation, so

$$J_\Gamma = T \circ (z \mapsto \bar{z}) \circ T^{-1}.$$

Inversion in a straight line is ordinary reflection; inversion in the unit circle is $z \mapsto 1/\bar{z}$. Inversions fix Γ pointwise and swap its two sides, and the composite of two inversions is a Möbius map (conjugating twice removes the conjugations: if $T(z) = \frac{az+b}{cz+d}$, then $z \mapsto \overline{T(\bar{z})} = \frac{\bar{a}z+\bar{b}}{\bar{c}z+\bar{d}}$ is again Möbius).

Theorem 14.7.10

The isometry group of the hyperbolic plane is generated by inversions in hyperbolic

lines; the orientation-preserving isometries are exactly $\text{Möb}(\mathbb{H})$.

Proof. We work in the disc model. First, inversions in hyperbolic lines are isometries: inversion in the diameter $\mathbb{R} \cap D$ is $z \mapsto \bar{z}$, which preserves g_{hyp} ; and a general hyperbolic line is the image of $\mathbb{R} \cap D$ under some $T \in \text{Möb}(D)$, so inversion in it is $T \circ (z \mapsto \bar{z}) \circ T^{-1}$, a composite of isometries.

Now let α be any isometry of (D, g_{hyp}) . Choose $T \in \text{Möb}(D)$ with $T(\alpha(0)) = 0$, so $T \circ \alpha$ is an isometry fixing 0. An isometry fixing 0 permutes the geodesics through 0, i.e. the diameters, and preserves angles up to sign; composing with a rotation R we may assume $R \circ T \circ \alpha$ fixes 0 and maps the positive real radius $D \cap \mathbb{R}_+$ to itself (using that isometries preserve $d_{\text{hyp}}(0, \cdot)$, which is a strictly increasing function of $|z|$, so points of $D \cap \mathbb{R}_+$ go to points at the same radius). Composing if necessary with conjugation $C : z \mapsto \bar{z}$, we get an isometry β fixing 0, fixing $D \cap \mathbb{R}_+$ pointwise (again since distances from 0 are preserved), and preserving the upper half of the disc. We claim $\beta = \text{Id}$: any point $w \in D$ is determined by its distances to three non-collinear points of $D \cap \mathbb{R}$ together with the side of $\mathbb{R}$ it lies on – distances to two points of the fixed diameter determine w up to reflection. So $\alpha = (C) \circ (R \circ T)^{-1}$ or $(R \circ T)^{-1}$, a composite of Möbius maps in $\text{Möb}(D)$ and possibly one conjugation. Since elements of $\text{Möb}(D)$ are themselves composites of (two or four) inversions – this is an exercise on the example sheets, mirroring the Euclidean fact that rotations and translations are composites of two reflections – the full isometry group is generated by inversions. And an orientation-preserving isometry must involve an even number of inversions, hence lies in $\text{Möb}(D)$. $\square$

It is worth recording the trichotomy of non-identity orientation-preserving isometries, in analogy with rotations and translations of the Euclidean plane. Any $\alpha \in \text{Möb}(\mathbb{H})$, $\alpha \neq \text{Id}$, extends to the closed disc (or closed half-plane) and has a fixed point there, and is exactly one of:

- (i) **elliptic**: α fixes a point $p \in \mathbb{H}$. Conjugating so $p = 0 \in D$, α is a rotation $z \mapsto e^{i\theta}z$ – hyperbolic rotations about p .
- (ii) **parabolic**: α fixes exactly one point of the boundary $\partial\mathbb{H}$ and none of $\mathbb{H}$. Conjugating so the fixed point is $\infty \in \partial\mathbb{H}$, α is $z \mapsto z + b$.
- (iii) **hyperbolic**: α fixes exactly two points of $\partial\mathbb{H}$. It then preserves the hyperbolic line joining them (its *axis*) and translates along it; conjugating so the fixed points are $0, \infty \in \partial\mathbb{H}$, α is $z \mapsto \lambda z$ with $\lambda > 0$.

There is a corresponding trichotomy for pairs of hyperbolic lines ℓ, ℓ' : they are **intersecting** if they meet in $\mathbb{H}$; **parallel** if they meet only ‘at infinity’, i.e. their closures share a boundary point; and **ultraparallel** if their closures are disjoint. Given a hyperbolic line ℓ and $p \notin \ell$, there are exactly two lines through p parallel to ℓ (one for each end of ℓ), and a continuum of lines through p ultraparallel to ℓ – so the parallel postulate fails as extravagantly as possible.

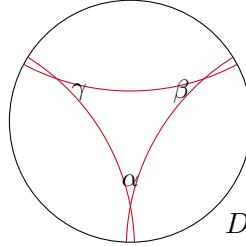
§14.7.4 Hyperbolic Triangles

Definition 14.7.11 (Hyperbolic Triangle)

A **hyperbolic triangle** is a region of $\mathbb{H}$ bounded by three hyperbolic line segments,

meeting pairwise in three vertices. We allow vertices to lie on the boundary $\partial\mathbb{H}$ (where the two sides meet at angle 0); such vertices are called **ideal**, and a triangle with all three vertices ideal is an **ideal triangle**.

We adopt the classical labelling: side lengths A, B, C , with opposite angles α, β, γ at vertices a, b, c . Since the hyperbolic metric is conformal, hyperbolic angles are just Euclidean angles in either model. Here is a hyperbolic triangle in the disc model – note how the sides bow inwards, pinching the angles:



As the picture suggests, the angles of a hyperbolic triangle sum to *less* than π ; we will prove this precisely with the area formula below. First, the hyperbolic law of cosines, which controls the metric side of the story.

Proposition 14.7.12 (Hyperbolic Cosine Rule)

For a hyperbolic triangle with the labelling above,

$$\cosh C = \cosh A \cosh B - \sinh A \sinh B \cos \gamma.$$

Proof. Applying an isometry, we may place the vertex c at 0 in the disc model and the vertex b on the positive real axis; then the sides through c are Euclidean radii, meeting at (Euclidean, hence hyperbolic) angle γ at the origin. From $d_{\text{hyp}}(0, z) = 2 \operatorname{artanh} |z|$, the vertices are

$$b = \tanh \frac{A}{2} \in \mathbb{R}_+, \quad a = e^{i\gamma} \tanh \frac{B}{2},$$

(the side ca has length B since it is opposite b , and so on), and the distance formula gives

$$\tanh \frac{C}{2} = \left| \frac{a - b}{1 - \bar{a}b} \right|.$$

Now use the identities, for $t = \tanh \frac{\lambda}{2}$,

$$\cosh \lambda = \frac{1 + t^2}{1 - t^2}, \quad \sinh \lambda = \frac{2t}{1 - t^2}.$$

So

$$\cosh C = \frac{|1 - \bar{a}b|^2 + |a - b|^2}{|1 - \bar{a}b|^2 - |a - b|^2}.$$

Expanding the numerator and denominator,

$$|1 - \bar{a}b|^2 \pm |a - b|^2 = 1 - \bar{a}b - a\bar{b} + |a|^2|b|^2 \pm (|a|^2 - \bar{a}b - a\bar{b} + |b|^2)$$

$$= \begin{cases} (1 + |a|^2)(1 + |b|^2) - 2(\bar{a}b + a\bar{b}) & (+) \\ (1 - |a|^2)(1 - |b|^2) & (-) \end{cases}$$

Since $b \in \mathbb{R}$ and $\bar{a}b + a\bar{b} = 2b \operatorname{Re} a = 2|a|b \cos \gamma$, we get

$$\begin{aligned} \cosh C &= \frac{(1 + |a|^2)(1 + |b|^2)}{(1 - |a|^2)(1 - |b|^2)} - \frac{2|a|}{1 - |a|^2} \cdot \frac{2b}{1 - |b|^2} \cos \gamma \\ &= \cosh A \cosh B - \sinh A \sinh B \cos \gamma, \end{aligned}$$

reading off the hyperbolic functions of A and B via the identities above (note $|b| = \tanh \frac{A}{2}$, $|a| = \tanh \frac{B}{2}$). $\square$

Remark. For small A, B, C , using $\cosh \lambda \approx 1 + \frac{\lambda^2}{2}$ and $\sinh \lambda \approx \lambda$ and keeping second order terms, the formula degenerates to

$$C^2 \approx A^2 + B^2 - 2AB \cos \gamma,$$

the Euclidean cosine rule – at small scales, hyperbolic geometry looks Euclidean, as it must since any Riemannian metric is approximately flat at small scales. Also, since $\cos \gamma \geq -1$,

$$\cosh C \leq \cosh A \cosh B + \sinh A \sinh B = \cosh(A + B),$$

and $\cosh$ is increasing on $\mathbb{R}_{\geq 0}$, so $C \leq A + B$: the triangle inequality, with a sharper quantitative form.

§14.7.5 The Area of a Hyperbolic Triangle

We now come to the hyperbolic counterpart of the spherical triangle theorem (Theorem 14.5.12) – the Gauss–Bonnet theorem for hyperbolic triangles.

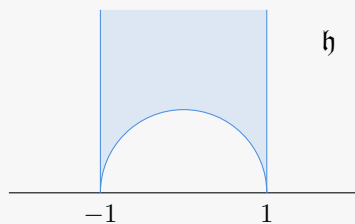
Theorem 14.7.13 (Gauss–Bonnet for Hyperbolic Triangles)

Let $T \subseteq \mathbb{H}$ be a hyperbolic triangle with internal angles α, β, γ (with angle 0 at any ideal vertex). Then

$$\operatorname{area}_{\text{hyp}}(T) = \pi - \alpha - \beta - \gamma.$$

In particular the angle sum of a hyperbolic triangle is always strictly less than π , and no triangle has area exceeding π .

Proof. Step 1: ideal triangles. $\operatorname{Möb}(\mathbb{H})$ (together with an inversion if needed) acts transitively on triples of distinct boundary points, so all ideal triangles are isometric and we may compute with the ideal triangle in $\mathfrak{h}$ with vertices $-1, 1, \infty$ – the region above the unit semicircle between the vertical lines $x = \pm 1$:



The hyperbolic area element is $\sqrt{EG - F^2} \, dx \, dy = \frac{1}{y^2} \, dx \, dy$, so

$$\text{area}(T) = \int_{-1}^1 \int_{\sqrt{1-x^2}}^{\infty} \frac{dy}{y^2} \, dx = \int_{-1}^1 \frac{dx}{\sqrt{1-x^2}} = \pi = \pi - 0 - 0 - 0,$$

as claimed for ideal triangles.

Step 2: triangles with two ideal vertices. Let $A(\alpha)$ denote the area of a triangle with angles $\alpha, 0, 0$; this is well defined, as such triangles are determined up to isometry by α (put the vertex at $0 \in D$ with one side along $\mathbb{R}_+$; the ideal vertices are then determined). Sliding the finite vertex continuously shows A is continuous, and A is decreasing in α since a triangle of larger angle sits inside one of smaller angle with the same ideal vertices. Now consider the configuration of a triangle with angle $\alpha + \beta$ split by a hyperbolic line into triangles of angles α and β , with all other vertices ideal: comparing areas of the four triangles formed by the lines in that figure yields the identity

$$A(\alpha) + A(\beta) = \pi + A(\alpha + \beta).$$

(One can see this cleanly in $\mathfrak{h}$: the triangle $\{-1, 1, \infty\}$ is divided by the vertical line through a point $e^{i\theta}$ of the unit semicircle into two triangles with two ideal vertices each.) Setting $F(\alpha) = \pi - A(\alpha)$, this reads $F(\alpha) + F(\beta) = F(\alpha + \beta)$; F is continuous and monotonic, so $F(\alpha) = \lambda\alpha$ for some constant $\lambda > 0$. Finally the two triangles with angles α and $\pi - \alpha$ obtained by cutting a ‘bigon’ with both vertices ideal along a line satisfy $A(\alpha) + A(\pi - \alpha) = \pi$; substituting gives $\lambda = 1$. So

$$A(\alpha) = \pi - \alpha.$$

Step 3: one ideal vertex. Next consider a triangle T_1 with one ideal vertex w and finite vertices a, b with angles α, β . Extend the side from a through b to an ideal point u . The triangle auw has two ideal vertices and angle α at a , so has area $A(\alpha) = \pi - \alpha$, and it decomposes as T_1 together with the triangle buw , which has two ideal vertices and angle $\pi - \beta$ at b . Hence

$$\pi - \alpha = \text{area}(T_1) + A(\pi - \beta) = \text{area}(T_1) + \beta \implies \text{area}(T_1) = \pi - \alpha - \beta.$$

Step 4: general triangles. Finally let T have finite vertices a, b, c with angles α, β, γ . Extend the side from a through b to an ideal point u , and consider the triangle auc : it has one ideal vertex, angle α at a , and angle $\gamma + \delta$ at c , where δ is the angle of the triangle buc at c . By Step 3,

$$\text{area}(auc) = \pi - \alpha - (\gamma + \delta),$$

while the triangle buc has one ideal vertex, angle $\pi - \beta$ at b and angle δ at c , so

$$\text{area}(buc) = \pi - (\pi - \beta) - \delta = \beta - \delta.$$

Since auc decomposes as $T \cup buc$,

$$\text{area}(T) = (\pi - \alpha - \gamma - \delta) - (\beta - \delta) = \pi - \alpha - \beta - \gamma. \quad \square$$

Corollary 14.7.14 (Hyperbolic Polygons)

A **geodesic n -gon** in $\mathbb{H}$ – a disc bounded by n hyperbolic segments – with internal angles $\alpha_1, \dots, \alpha_n$ has area

$$(n-2)\pi - \sum_{i=1}^n \alpha_i.$$

Proof. Cut the polygon into $n-2$ triangles by geodesic diagonals (possible since any two points of $\mathbb{H}$ are joined by a unique geodesic) and sum the triangle formula. $\square$

Compare our three geometries: a triangle with angles α, β, γ has

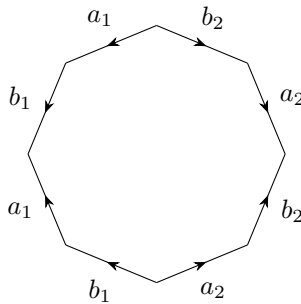
$$\text{area} = (\alpha + \beta + \gamma) - \pi \text{ on } S^2, \quad \alpha + \beta + \gamma = \pi \text{ on } \mathbb{R}^2, \quad \text{area} = \pi - (\alpha + \beta + \gamma) \text{ on } \mathbb{H},$$

which is exactly the statement $\int_T \kappa \, dA = (\alpha + \beta + \gamma) - \pi$ with $\kappa \equiv 1, 0, -1$ respectively. In particular the hyperbolic plane has constant curvature -1 , as promised (and this can be checked directly: the half-plane metric is $\frac{1}{y^2}(dx^2 + dy^2)$, and the formula for the curvature of a conformal metric gives $\kappa \equiv -1$).

§14.7.6 Hyperbolic Surfaces

We built a flat metric on the torus by gluing the sides of a Euclidean square. The same idea, run in the hyperbolic plane, produces metrics of constant curvature -1 on all surfaces of genus at least 2 – so every compact orientable surface carries one of our three geometries: S^2 carries spherical geometry, T^2 flat geometry, and everything else hyperbolic geometry.

Recall that the surface Σ_g of genus g is obtained from a $4g$ -gon by gluing its edges in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$; for $g = 2$:



(Each label is traversed once in each direction; all eight vertices are identified to a single point of Σ_2 .)

To copy the flat torus construction we need the polygon to be a *hyperbolic* polygon, and for a smooth structure at the glued vertex we need the interior angles to add up to exactly 2π .

Lemma 14.7.15

For each $g \geq 2$ there is a regular hyperbolic $4g$ -gon with all internal angles equal to $\frac{2\pi}{4g} = \frac{\pi}{2g}$.

Proof. Place the vertices at the points $r\omega^k$, $k = 0, \dots, 4g - 1$, in the disc model, where $\omega = e^{2\pi i/4g}$ and $r \in (0, 1)$, joined by hyperbolic segments; by symmetry this is a regular hyperbolic $4g$ -gon with some internal angle $\beta(r)$. As $r \rightarrow 1$ the polygon converges to the *ideal* regular $4g$ -gon, with all angles 0; as $r \rightarrow 0$, small hyperbolic polygons are nearly Euclidean, so $\beta(r)$ tends to the Euclidean regular $4g$ -gon angle $\frac{(4g-2)\pi}{4g}$. The angle $\beta(r)$ varies continuously, so by the intermediate value theorem it takes the value $\frac{\pi}{2g}$ for some r , since

$$0 < \frac{\pi}{2g} < \frac{(4g-2)\pi}{4g} \iff 2 < 4g - 2,$$

which holds exactly when $g \geq 2$ (this is where genus ≥ 2 is needed – for the torus, $g = 1$, the required angle $\pi/2$ is not available for hyperbolic squares, whose angles are strictly less than $\pi/2$... consistent with the torus being flat rather than hyperbolic). $\square$

Theorem 14.7.16

For every $g \geq 2$, the compact orientable surface Σ_g of genus g admits a Riemannian metric locally isometric to $\mathbb{H}$; in particular it has constant curvature $\kappa \equiv -1$.

Proof (sketch). Take the regular hyperbolic $4g$ -gon P of Lemma 14.7.15 with angles $\frac{\pi}{2g}$, with edges labelled and oriented in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} \dots$. For each pair of edges to be glued there is a hyperbolic isometry carrying one to the other, respecting the orientations and exchanging the sides of the edges (isometries act transitively on ‘flags’ – a point on an oriented line together with a choice of side). Now define $\Sigma_g = P / \sim$ as a topological surface (as in Section 1), and give it charts as follows:

- (i) at the image of an interior point of P , take a small hyperbolic disc inside P , included into $\mathbb{H}$;
- (ii) at the image of an edge point, combine a half-disc at the point with the half-disc at the paired point, transported by the gluing isometry; the two half-discs assemble to an honest disc in $\mathbb{H}$;
- (iii) at the single vertex of Σ_g : the $4g$ corners of P have angles summing to $4g \cdot \frac{\pi}{2g} = 2\pi$, so transporting them by gluing isometries, they fit together exactly around a point of $\mathbb{H}$, forming a disc.

All transition maps between these charts are restrictions of hyperbolic isometries, which are smooth and preserve g_{hyp} . So the locally-defined hyperbolic metrics agree on overlaps and define a Riemannian metric on Σ_g , locally isometric to $\mathbb{H}$ by construction. $\square$

Remark. The quotient description of the flat torus as $\mathbb{R}^2/\mathbb{Z}^2$ has a hyperbolic analogue: the side-pairing isometries generate a subgroup $\Gamma \leq \text{Möb}(\mathbb{H})$, and $\Sigma_g = \mathbb{H}/\Gamma$. This point of view is developed properly in Part II *Algebraic Topology*. There is also a more flexible cut-and-paste construction, assembling Σ_g from $2g - 2$ hyperbolic ‘pairs of pants’ – three-holed spheres with geodesic boundary, themselves glued from two right-angled hyperbolic hexagons – which exhibits many essentially different hyperbolic metrics on the same surface. The space of such metrics, the *moduli space*, is among the most-studied

objects in geometry.

§14.8 The Gauss–Bonnet Theorem

We have now seen the same phenomenon three times: on the sphere, the plane and the hyperbolic plane, the failure of a triangle's angles to sum to π is exactly the integral of the curvature over the triangle. This is no coincidence, and its vast generalisation – the Gauss–Bonnet theorem – is the crowning result of the course: it says that curvature, a purely local quantity, integrates up to a purely *topological* one.

Throughout this section, curvature on an abstract surface with a Riemannian metric means the expression from the Theorema Egregium discussion: in geodesic normal coordinates ($E = 1$, $F = 0$),

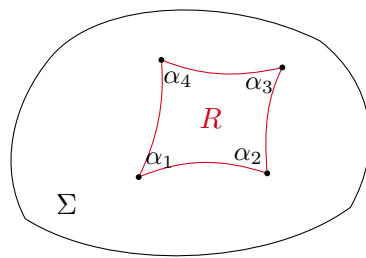
$$\kappa = -\frac{(\sqrt{G})_{uu}}{\sqrt{G}},$$

which makes sense for any Riemannian metric, embedded or not, and agrees with $\det(Dn)$ for surfaces in $\mathbb{R}^3$. (Geodesic normal coordinates exist in the abstract setting by the same Picard argument as before – the geodesic equations never mentioned the embedding.)

§14.8.1 The Local Gauss–Bonnet Theorem

Definition 14.8.1 (Geodesic Polygon)

A **geodesic n -gon** on a surface Σ with a Riemannian metric is an open region $R \subseteq \Sigma$ homeomorphic to a disc, whose boundary is a piecewise smooth closed curve made up of n geodesic arcs, meeting at n vertices with internal angles $\alpha_1, \dots, \alpha_n \in (0, 2\pi)$.



Theorem 14.8.2 (Local Gauss–Bonnet Theorem)

Let Σ be an abstract smooth surface with Riemannian metric g , and let $R \subseteq \Sigma$ be a geodesic n -gon with internal angles $\alpha_1, \dots, \alpha_n$. Then

$$\int_R \kappa \, dA = \sum_{i=1}^n \alpha_i - (n-2)\pi.$$

For $n = 3$ this is exactly ‘ $\int_T \kappa \, dA$ equals the angle excess’, generalising Theorems 14.5.12 and 14.7.13 simultaneously. It is essential that R is a disc: the theorem fails for regions with holes (as we will see below, this failure is precisely what makes the global theorem interesting).

We will not give a complete proof, but the mechanism deserves to be seen, because it is exactly Green's theorem from Vector Calculus. This discussion is non-examinable. Suppose for simplicity that R lies in the image of a single parametrisation in geodesic normal coordinates, so $E = 1$, $F = 0$, and let

$$e = \sigma_u, \quad f = \frac{\sigma_v}{\sqrt{G}}$$

be the resulting orthonormal frame of the tangent planes. Parametrise ∂R by arc length and consider

$$I = \int_{\partial R} \langle e, \dot{f} \rangle dt.$$

On one hand, writing $\dot{f} = f_u \dot{u} + f_v \dot{v}$ and setting $P = \langle e, f_u \rangle$, $Q = \langle e, f_v \rangle$, Green's theorem gives

$$I = \int_{\partial R} P du + Q dv = \int_R (Q_u - P_v) du dv,$$

and a direct computation using the orthonormality relations shows

$$Q_u - P_v = \langle e_u, f_v \rangle - \langle e_v, f_u \rangle = -(\sqrt{G})_{uu} = \kappa \sqrt{G},$$

where $\sqrt{G} = \sqrt{EG - F^2}$ is exactly the area element – so $I = \int_R \kappa dA$. On the other hand, let $\theta(t)$ be the angle from e to the tangent $\dot{\gamma}$ of the boundary, so $\dot{\gamma} = \cos \theta e + \sin \theta f$. Along a geodesic arc, differentiating and using $\langle \ddot{\gamma}, -\sin \theta e + \cos \theta f \rangle = 0$ (the acceleration has no tangential component) together with orthonormality yields

$$\dot{\theta} = \langle e, \dot{f} \rangle.$$

So I also computes the total rotation of the boundary direction relative to the frame, which for a piecewise-geodesic boundary of a disc is 2π minus the sum of the exterior angles:

$$I = 2\pi - \sum_{i=1}^n (\pi - \alpha_i) = \sum_i \alpha_i - (n-2)\pi.$$

Equating the two computations of I gives the theorem.

§14.8.2 The Global Gauss–Bonnet Theorem

Theorem 14.8.3 (Global Gauss–Bonnet Theorem)

Let Σ be a compact abstract smooth surface with Riemannian metric g . Then

$$\int_{\Sigma} \kappa dA = 2\pi \chi(\Sigma),$$

where $\chi(\Sigma)$ is the Euler characteristic.

Pause on what this says. The left-hand side depends on the metric – deform the surface, and κ changes pointwise, drastically. The right-hand side is a topological invariant, a difference of counts of vertices, edges and faces. The theorem says all the local changes in curvature miraculously cancel: *total* curvature cannot be changed by any deformation whatsoever. For instance, however you dent a sphere, extra negative curvature in the dent is exactly balanced by positive curvature around its rim.

The global theorem follows from the local one, given one plausible (but nontrivial) fact whose proof, via the exponential map, belongs to Part II:

Lemma 14.8.4

A compact surface with a Riemannian metric admits subdivisions all of whose faces are geodesic polygons.

Proof of Theorem 14.8.3. Take such a subdivision, with V vertices, E edges and F faces. Summing the local Gauss–Bonnet theorem over all faces,

$$\int_{\Sigma} \kappa \, dA = \sum_{\text{faces } P} \int_P \kappa \, dA = \sum_{\text{faces } P} \left(\sum_{\text{angles of } P} \alpha - (n_P - 2)\pi \right),$$

where n_P is the number of edges of the face P . Now:

- (i) The angles around each vertex sum to 2π , so summing all angles over all faces gives $2\pi V$.
- (ii) Each edge is an edge of exactly two faces, so $\sum_P n_P = 2E$.
- (iii) $\sum_P 2\pi = 2\pi F$.

Putting these together,

$$\int_{\Sigma} \kappa \, dA = 2\pi V - 2\pi E + 2\pi F = 2\pi \chi(\Sigma). \quad \square$$

Remark. Note the pleasing corollary in the other direction: since $\int_{\Sigma} \kappa \, dA$ manifestly does not depend on any choice of subdivision, the Gauss–Bonnet theorem re-proves that the Euler characteristic is independent of the subdivision – for surfaces admitting a Riemannian metric, which is all of them.

Let’s sanity-check the theorem against everything we have built.

Example 14.8.5 (The Sphere)

The round sphere of radius r has $\kappa \equiv 1/r^2$ and area $4\pi r^2$, so

$$\int_{S^2} \kappa \, dA = \frac{4\pi r^2}{r^2} = 4\pi = 2\pi \cdot 2 = 2\pi \chi(S^2).$$

Note that Gauss–Bonnet forbids a metric of $\kappa \leq 0$ on the sphere: any compact surface with non-positive curvature has $\chi \leq 0$.

Example 14.8.6 (The Flat Torus)

The flat torus of Example 14.6.3 has $\kappa \equiv 0$, so $\int \kappa \, dA = 0 = 2\pi \chi(T^2)$ – consistent with $\chi(T^2) = 0$. Conversely, Gauss–Bonnet shows any metric on T^2 must have total curvature zero; combined with Proposition 14.4.13, this gives another proof that a torus embedded in $\mathbb{R}^3$ must have regions of both positive and negative curvature.

Example 14.8.7 (Hyperbolic Surfaces)

The hyperbolic surface Σ_g ($g \geq 2$) built from the regular $4g$ -gon with angles $\frac{\pi}{2g}$ has

$\kappa \equiv -1$, and its total area is the area of the polygon,

$$\text{area}(\Sigma_g) = (4g - 2)\pi - 4g \cdot \frac{\pi}{2g} = (4g - 4)\pi,$$

by the hyperbolic polygon area formula. So

$$\int_{\Sigma_g} \kappa \, dA = -(4g - 4)\pi = 2\pi(2 - 2g) = 2\pi\chi(\Sigma_g),$$

in perfect agreement. In fact Gauss–Bonnet shows that *every* metric of curvature -1 on Σ_g has the same total area $(4g - 4)\pi$ – a first hint of the rigidity of hyperbolic geometry.

Example 14.8.8 (Closed Geodesics Bounding Discs)

Let Σ be a surface with $\kappa \equiv 0$ (say the flat torus) and suppose γ is a closed geodesic bounding a smooth disc R . Marking two points of γ as vertices makes R a geodesic 2-gon with both internal angles equal to π , and the local Gauss–Bonnet theorem would give

$$0 = \int_R \kappa \, dA = (\pi + \pi) - (2 - 2)\pi = 2\pi,$$

a contradiction. So on a flat surface, no closed geodesic bounds a disc – every closed geodesic on the flat torus must wrap around the torus. Contrast the sphere, where great circles bound hemispheres: there positive curvature supplies the missing 2π .

§14.8.3 Flat Metrics on the Torus

We close with a brief look at a question the course has equipped us to pose: *how many* geometries can a given surface carry? Take the simplest interesting case, flat metrics on T^2 .

The construction of Example 14.6.3 works for any parallelogram, not just the unit square: if $Q \subseteq \mathbb{R}^2$ is the parallelogram spanned by linearly independent v_1, v_2 , then $T^2 = \mathbb{R}^2 / (\mathbb{Z}v_1 \oplus \mathbb{Z}v_2)$ inherits a flat metric g_Q , of total area equal to the Euclidean area of Q . Different parallelograms can give genuinely different geometries:

Lemma 14.8.9

The flat tori built from $Q = [0, 1]^2$ and $\hat{Q} = [0, 10] \times [0, \frac{1}{10}]$ have equal area but are not isometric.

Proof. Geodesics on a flat torus are exactly the images of straight lines in $\mathbb{R}^2$ (the charts are local isometries with $\mathbb{R}^2$, and Picard uniqueness means there are no others). A line through a lift $\hat{p}$ of p descends to a *closed* geodesic exactly when it passes through another lift of p , i.e. has rational slope with respect to the lattice. The shortest closed geodesic therefore has length equal to the shortest nonzero lattice vector: 1 for $\mathbb{Z}^2$, but $\frac{1}{10}$ for $10\mathbb{Z} \oplus \frac{1}{10}\mathbb{Z}$. Isometries preserve the set of lengths of closed geodesics, so the tori are not isometric. $\square$

To organise all flat tori, normalise: rotating, reflecting and dilating Q (which changes

the metric only by a global isometry and scale), we can take $v_1 = 1 \in \mathbb{C}$ and $v_2 = \tau$ with $\text{Im } \tau > 0$. So the upper half-plane $\mathfrak{h}$ – of all places! – parametrises flat metrics on T^2 up to scale. Two parameters τ, τ' give isometric tori (by an orientation-preserving diffeomorphism) exactly when the lattices agree after rotation, which happens exactly when

$$\tau' = \frac{a\tau + b}{c\tau + d} \quad \text{for some} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z}),$$

the subgroup $\text{SL}(2, \mathbb{Z}) \leq \text{SL}(2, \mathbb{R})$ acting on $\mathfrak{h}$ by Möbius maps as in the previous section. One obtains:

Theorem 14.8.10

The set of flat metrics on T^2 , up to dilation and orientation-preserving diffeomorphism, is in natural bijection with the quotient $\mathfrak{h}/\text{SL}(2, \mathbb{Z})$.

The quotient $\mathfrak{h}/\text{SL}(2, \mathbb{Z})$ is called the **moduli space** of flat tori, and it is a remarkable object: a question purely about *flat* geometry is answered by a quotient of the *hyperbolic* plane. The analogous moduli spaces of hyperbolic metrics on Σ_g for $g \geq 2$ are central objects of modern geometry.

Where Next

Several Part II courses pick up where these notes leave off. *Algebraic Topology* proves the classification of surfaces and the topological invariance of χ , and constructs $\Sigma_g = \mathbb{H}/\Gamma$ properly. *Differential Geometry* develops the second fundamental form further (the trace of Dn , the mean curvature, governs soap films as the determinant governs Gauss–Bonnet). *Riemann Surfaces* studies surfaces through complex analysis, where $\text{Möb}(D)$ reappears as the automorphism group of the disc. And *General Relativity* takes the idea that geometry is encoded in a metric and its geodesics, and applies it to spacetime itself.

15 Groups, Rings and Modules

In this course we will continue our study of groups, focusing on simple groups, p -groups and p -subgroups along with the Sylow theorems. We will then move on to discuss rings and fields, finishing up by discussing modules where the scalars belong to a ring instead of a field.

This article constitutes my notes for the ‘Groups, Rings and Modules’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§15.1 The Theory of Groups

§15.1.1 Definition of a Group

We will begin by recalling the basic definition of a group.

Definition 15.1.1 (Group)

A **group** is a pair $(G, *)$ consisting of a set G and a binary operation $* : G \times G \rightarrow G$ satisfying the axioms:

- *Identity.* There is an element $e \in G$ such that $e * g = g * e = g$ for all $g \in G$,
- *Inverses.* For every element $g \in G$, there is an element $g^{-1} \in G$ such that $g * g^{-1} = g^{-1} * g = e$.
- *Associativity.* The operation $*$ is associative.

Remark. We will usually either use additive or multiplicative notation for groups, and in these cases we will often write 0 or 1 for the identity respectively.

Definition 15.1.2 (Subgroup)

A subset $H \subseteq G$ is a **subgroup** of G , written $H \leq G$, if it is a group with respect to the operation $*$ defined on $H \times H$.

There is a way to test the conditions needed for a subset to be a subgroup in just a few lines, but it does have limited utility.

Lemma 15.1.3 (Fast Subgroup Checking)

A nonempty subset $H \subseteq G$ is a subgroup if $a, b \in H$ implies $a * b^{-1} \in H$.

Proof Sketch. Check that this implies the definition. □

Example 15.1.4 (Examples of Groups)

The following are all examples of groups.

- (i) The additive groups $(\mathbb{Z}, +) \leq (\mathbb{Q}, +) \leq (\mathbb{R}, +)$.
- (ii) The cyclic group of order n , C_n .
- (iii) The dihedral group D_{2n} of the symmetries of a regular n -gon.
- (iv) The symmetric group S_n and alternating group A_n , where S_n is the group of permutations of $\{1, 2, \dots, n\}$ and $A_n \leq S_n$ is the group of even permutations.
- (v) The quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$.
- (vi) The matrix groups over some field F , $\text{GL}_n(F)$ of all $n \times n$ matrices over F with non-zero determinant, and $\text{SL}_n(F) \leq \text{GL}_n(F)$, the subgroup of matrices with determinant 1.

Definition 15.1.5 (Direct Product)

The **direct product** of groups G and H is $G \times H$ with operation $(g_1, h_1) * (g_2, h_2) = (g_1 g_2, h_1 h_2)$.

§15.1.2 Lagrange's Theorem

For a subgroup $H \leq G$, the **left cosets** of H in G are the sets $gH = \{gh \mid h \in H\}$ where $g \in G$. The left cosets partition G , and it is straightforward to show that each has the same cardinality as H . These facts immediately give us *Lagrange's theorem*.

Theorem 15.1.6 (Lagrange's Theorem)

If G is a finite group and $H \leq G$, then

$$|G| = |H| \cdot |G : H|,$$

where $|G : H|$ is the **index** of H in G , the number of left cosets of H in G .

Remark. The proof of Lagrange's theorem is equally valid when using right cosets, and as a consequence there is always the same number of left and right cosets for a given subgroup.

Definition 15.1.7 (Order of an Element)

Let G be a group and $g \in G$. The least n such that $g^n = 1$ is the **order** of g . If no such n exists, we say that g has **infinite order**.

Remark. If g has order d , then the division algorithm implies $g^n = 1 \iff d \mid n$. Also by taking powers we can generate a cyclic subgroup of G with $\{1, g, g^2, \dots, g^{d-1}\} \leq G$, so if G is finite then by Lagrange, $d \mid |G|$.

§15.1.3 Homomorphisms and Isomorphisms

As with many mathematical objects, there are many instances in which we are interested in the structure preserving maps between groups. Such maps are called *homomorphisms*.

Definition 15.1.8 (Homomorphism)

If G and H are groups, a function $\phi : G \rightarrow H$ is a **group homomorphism** if $\phi(g_1g_2) = \phi(g_1)\phi(g_2)$ for all $g_1, g_2 \in G$.

We frequently care about what elements in the group H can be obtained through this map, and also what elements in G just get sent to the identity.

Definition 15.1.9 (Kernel)

The **kernel** of ϕ is $\ker(\phi) = \{g \in G \mid \phi(g) = e\}$.

Definition 15.1.10 (Image)

The **image** of ϕ is $\text{img}(\phi) = \{\phi(g) \mid g \in G\}$.

It is straightforward to see that $\ker(\phi) \leq G$ and $\text{img}(\phi) \leq H$.

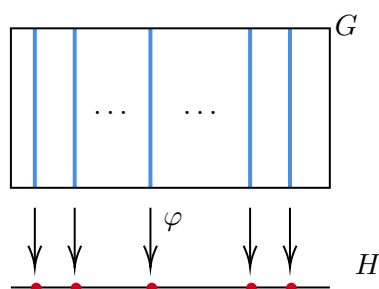
When we have a homomorphism that gives a bijection between two groups, the groups must have the same underlying algebraic structure. For this reason, we take this to be a notion of two groups being the same.

Definition 15.1.11 (Isomorphism)

We say that two groups G and H are **isomorphic** written $G \cong H$ if there is an *isomorphism* between them. That is, if there is a bijective homomorphism $\phi : G \rightarrow H$ between the groups.

§15.1.4 Quotient Groups and the Isomorphism Theorems

Quotient groups arise naturally in the study of homomorphisms. When we have two groups G and H , and some homomorphism $\varphi : G \rightarrow H$, we can see that the elements of G mapping to certain elements of H naturally arrange themselves together.



Indeed, if two elements of G map to the same element of H under φ , we would find that the two elements must differ by an element of the kernel of φ . So, going back to how the elements of G arrange themselves, it would seem that they are arranged into cosets of the kernel of φ . It is this observation that leads us to the definition of quotient groups and the isomorphism theorems.

From the discussion above, it is natural to see when we can make the cosets of a given subgroup into another group. It turns out that this only works when we have a specific type of subgroup.

Definition 15.1.12 (Normal Subgroup)

We say that a subgroup N of a group G is **normal** if the left and right cosets coincide, that is, $gN = Ng$ for all $g \in G$. If N is normal in G we write $N \trianglelefteq G$.

Now we can show that the set of cosets of N in G , written G/N , forms a group.

Theorem 15.1.13 (Quotient Groups)

If $N \trianglelefteq G$, then the set of cosets of N in G forms the **quotient group** G/N with the operation of coset multiplication, where $gN \cdot g'N = (gg')N$.

Proof. We first need to check that coset multiplication is well defined. If we had $a_1N = a_2N$ and $b_1N = b_2N$, then $a_1 = a_2n_a$ and $b_1 = b_2n_b$ for some $n_a, n_b \in N$. Then $(a_1b_1)N = (a_2n_ab_2n_b)N = (a_2b_2)N$, since N is normal in G . So our group operation is well defined, and we are just left to check the group axioms.

We have closure since g_1N and g_2N being cosets implies $(g_1g_2)N$ is too. Then we have the identity element N , and inverses with $gN \cdot g^{-1}N = N$. Finally we get associativity inherited from G , thus we do indeed have a group. $\square$

Now we initially tried to motivate thinking about groups of cosets by noticing that elements in the same coset of the kernel of ϕ had the same image. We then used this ‘normal’ property of a subgroup to make this actually work. It turns out that these express the same idea – normal subgroups are *exactly* the kernels of homomorphisms.

Theorem 15.1.14 (Kernels are Normal Subgroups)

If $\phi : G \rightarrow H$ is a homomorphism, then $\ker \phi \trianglelefteq G$.

Proof. We already know that the kernel of a homomorphism is a subgroup, so we just need to check that it is normal. Let $n \in \ker \phi$, and $g \in G$. Then $\phi(gng^{-1}) = \phi(g)\phi(n)\phi(g^{-1}) = \phi(g)\phi(g^{-1}) = e$, thus $gng^{-1} \in \ker \phi$, so $\ker \phi \trianglelefteq G$ as required. $\square$

Theorem 15.1.15 (Normal Subgroups are Kernels)

Given $N \trianglelefteq G$, the map $\pi : G \rightarrow G/N$ with $\pi(g) = gN$ is a surjective homomorphism called the **quotient map**, and $\ker \pi = N$.

Proof. We first check that π is a homomorphism. For $g, h \in G$, we have $\pi(gh) = (gh)N = gN \cdot hN = \pi(g)\pi(h)$ as required. This map is clearly surjective, and also $\pi(g) = gN = N \iff g \in N$, so $\ker \pi = N$, as required. $\square$

Looking back again to our original motivation, we can see that the cosets of the kernel seem to form the group that is the image of the homomorphism. This is the exact idea encompassed in quotient groups, and is formalised in the *first isomorphism theorem*.

Theorem 15.1.16 (First Isomorphism Theorem)

Let $\phi : G \rightarrow H$ be a group homomorphism. Then $\ker(\phi) \trianglelefteq G$, and $G/\ker(\phi) \cong$

$\text{img}(\phi)$.

Proof. Let $K = \ker(\phi)$. We already checked that $K \trianglelefteq G$. Now define $\Phi : G/K \rightarrow \text{img}(\phi)$ by $gK \mapsto \phi(g)$.

We first check Φ is well defined and injective. We have

$$\begin{aligned} g_1K = g_2K &\iff g_2^{-1}g_1 \in K \\ &\iff \phi(g_2^{-1}g_1) = e \\ &\iff \phi(g_2)^{-1}\phi(g_1) = e \\ &\iff \phi(g_1) = \phi(g_2). \end{aligned}$$

Then we check that Φ is a group homomorphism, with

$$\begin{aligned} \Phi(g_1K g_2K) &= \Phi(g_1g_2K) \\ &= \phi(g_1g_2) \\ &= \phi(g_1)\phi(g_2) \\ &= \Phi(g_1K)\Phi(g_2K). \end{aligned}$$

Lastly we check that it is surjective. Let $x \in \text{img}(\phi)$, say $x = \phi(g)$ for some $g \in G$. Then $x = \Phi(gK) \in \text{img}(\Phi)$. $\square$

In a sense, this is the way that you should think about normal subgroups and quotient groups – as the kernels and images of homomorphisms.

Example 15.1.17 (Using the First Isomorphism Theorem)

Let $\phi : \mathbb{C} \rightarrow \mathbb{C}^*$ with $z \mapsto e^z$. As $e^{z+w} = e^z e^w$, this is a group homomorphism from $(\mathbb{C}, +)$ to $(\mathbb{C}^*, \times)$.

We find that $\ker(\phi) = \{z \in \mathbb{C} \mid e^z = 1\} = 2\pi i\mathbb{Z}$, and $\text{img}(\phi) = \mathbb{C}^*$. Thus $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$.

With the first isomorphism theorem, it is not enough to know the statement and proof – you have to know when to employ it. For example, if asked to prove $\mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$, you should be able to think of a strategy similar to the one used above.

The first isomorphism theorem is sometimes just called the ‘isomorphism theorem’, and it tends to be more important than the corollaries that we will state.

Corollary 15.1.18 (Second Isomorphism Theorem)

Let $H \leq G$ and $K \trianglelefteq G$. Then $HK = \{hk \mid h \in H, k \in K\} \leq G$ and $H \cap K \trianglelefteq H$, moreover $HK/K \cong H/H \cap K$.

Proof. Let $h_1k_1, h_2k_2 \in HK$. We have

$$\begin{aligned} h_1k_1(h_2k_2)^{-1} &= h_1k_1k_2^{-1}h_2^{-1} \\ &= h_1h_2^{-1}h_2k_1k_2^{-1}h_2^{-1} \\ &= (h_1h_2^{-1})(h_2k_1k_2^{-1}h_2^{-1}) \in HK, \end{aligned}$$

thus $HK \leq G$, as required.

Now let $\phi : H \mapsto G/K$ with $h \mapsto hK$. This is the composite of the inclusion $H \hookrightarrow G$ and the quotient map $G \rightarrow G/K$, thus ϕ is a group homomorphism.

We note $\ker(\phi) = \{h \in H \mid hK = K\} = H \cap K \trianglelefteq H$, and $\text{img}(\phi) = \{hK \mid h \in H\} = HK/K$. Thus by the first isomorphism theorem, $H/H \cap K \cong HK/K$. $\square$

Before we state the third isomorphism theorem, consider the following motivation. Suppose $K \trianglelefteq G$. There is a bijection

$$\{\text{subgroups of } G/K\} \longleftrightarrow \{\text{subgroups of } G \text{ containing } K\},$$

obtained by considering the maps $X \mapsto \{g \in G \mid gK \in X\}$ and $H \mapsto H/K$. This restricts to a bijection between

$$\{\text{normal subgroups of } G/K\} \longleftrightarrow \{\text{normal subgroups of } G \text{ containing } K\}.$$

In pictures, the portion of the subgroup lattice of G sitting above K matches the entire subgroup lattice of G/K .

$$\begin{array}{ccc} G & \longleftrightarrow & G/K \\ | & & | \\ H & \longleftrightarrow & H/K \\ | & & | \\ K & \longleftrightarrow & \{e\} \\ | & & \\ \{e\} & & \end{array}$$

Corollary 15.1.19 (Third Isomorphism Theorem)

Let $K \leq H \leq G$ be normal subgroups of G . Then

$$(G/K)/(H/K) \cong G/H.$$

Proof. Let $\phi : G/K \rightarrow G/H$ with $gK \mapsto gH$. If $g_1K = g_2K$, then $g_2^{-1}g_1 \in K \leq H$, so $g_1H = g_2H$, and thus ϕ is well defined. Also ϕ is a surjective group homomorphism with kernel $\ker(\phi) = H/K$. Then apply the first isomorphism. $\square$

§15.1.5 Simple Groups

If $K \trianglelefteq G$, then studying the groups K and the quotient group G/K gives some information about G . However, this approach is not always available.

Definition 15.1.20 (Simple Group)

A group G is **simple** if $\{e\}$ and G are its only normal subgroups.

Lemma 15.1.21 (Abelian Simple Groups)

An abelian group is simple if and only if it is isomorphic to C_p for some prime p .

Proof. By Lagrange's theorem, a subgroup $H \leq C_p$ has order dividing $|C_p| = p$, which is a prime. Hence H has order 1 or p , and H is either $\{e\}$ or $H = G$.

Now let G be an abelian simple group, and $g \in G$ with $g \neq e$. Note that any subgroup of an abelian group is normal, and thus G must have no subgroups other than G and $\{e\}$. But then G has subgroup $\langle g \rangle = \{\dots, g^{-2}, g^{-1}, e, g, g^2, \dots\}$. Since G is simple, this must be the whole group, that is, G is cyclic.

If G is the infinite cyclic group, then $G \cong (\mathbb{Z}, +)$, which is not simple (as $2\mathbb{Z} \leq \mathbb{Z}$). Thus $G \cong C_n$ for some n . Let g be a generator for C_n . If $m \mid n$, then $\langle g^{n/m} \rangle$ is a subgroup of order m , but G is simple thus $m = 1$ or n , hence n must be prime. $\square$

Lemma 15.1.22 (Composition Series of Finite Groups)

If G is a finite group then G has a composition series $\{e\} = G_0 \triangleleft G_1 \triangleleft G_2 \triangleleft \dots \triangleleft G_{m-1} \triangleleft G_m = G$, with each quotient G_i/G_{i-1} is simple.

[Note that G_i need not be normal in G .]

Proof. We induct on $|G|$. If $|G| = 1$ we are done. If $|G| > 1$, then let G_{m-1} be a normal subgroup of largest possible order (not $|G|$). Then G/G_{m-1} is simple, and by induction on G_{m-1} , we are done. $\square$

§15.2 Group Actions

A useful way to study a group is by studying how it 'acts' on some set. The way we look at this mathematically is through the lens of group actions.

§15.2.1 Definitions

We will begin by looking at groups of permutations of a set.

Definition 15.2.1 ($\text{Sym}(X)$)

For a set X , let $\text{Sym}(X)$ be the group of all bijections $X \rightarrow X$ under composition. We let the identity of this group be id .

Definition 15.2.2 (Permutation Group)

A group G is a **permutation group** (of degree n) if $G \leq \text{Sym}(X)$ (where $|X| = n$).

Example 15.2.3 (Examples of Permutation Groups)

The group $S_n = \text{Sym}(\{1, 2, \dots, n\})$ is a permutation group of degree n , as is the alternating group $A_n \leq S_n$.

The group D_n , the symmetries of a regular n -gon, is a permutation group as it is a subgroup of $\text{Sym}(\{\text{vertices of an } n\text{-gon}\})$.

We can now generalize the notion of a permutation group to the idea mentioned before

– slightly more useful (and general) than a permutation group is the notion of a group *acting* on a set.

Definition 15.2.4 (Group Action)

An action of a group G on a set X is a function $*$: $G \times X \rightarrow X$ satisfying

- (i) $e * x = x$ for all $x \in X$.
- (ii) $(g_1 g_2) * x = g_1 * (g_2 * x)$ for all $g_1, g_2 \in G$ and $x \in X$.

We can also think about group actions in the following way, and switching between the two points of view can be quite helpful.

Proposition 15.2.5

An action of a group G on a set X is equivalent to specifying a group homomorphism $\phi : G \rightarrow \text{Sym}(X)$.

Proof. For each $g \in G$, there is a function $\phi_g : X \rightarrow X$ given by $x \mapsto g * x$. We have $\phi_{g_1 g_2}(x) = (g_1 g_2) * x = g_1 * (g_2 * x) = \phi_{g_1}(\phi_{g_2}(x))$. Thus $\phi_{g_1 g_2} = \phi_{g_1} \circ \phi_{g_2}$.

In particular, $\phi_g \circ \phi_{g^{-1}} = \phi_{g^{-1}} \circ \phi_g = \phi_e = \text{id}$. Thus ϕ_g is a bijection, and $\phi_g \in \text{Sym}(X)$. We define $\phi : G \rightarrow \text{Sym}(X)$ with $g \mapsto \phi_g$. Then this is a group homomorphism by the above.

Conversely, let $\phi : G \rightarrow \text{Sym}(X)$ be a group homomorphism. Then defining $*$: $G \times X \rightarrow X$ with $(g, x) \mapsto \phi(g)(x)$, we have that this is a group action since $e * x = \phi(e)(x) = \text{id}(x) = x$ and $(g_1 g_2) * x = \phi(g_1 g_2)(x) = \phi(g_1)(\phi(g_2)(x)) = g_1 * (g_2 * x)$. $\square$

Definition 15.2.6 (Permutation Representation)

We say $\phi : G \rightarrow \text{Sym}(X)$ is a **permutation representation** of G .

§15.2.2 Orbits and Stabilisers

A useful notion is that of *orbits* and *stabilisers*. Informally, the orbit of an element x in a set S acted on by a group G is all of the elements that can be reached by applying elements of G . The stabiliser of x is the set of elements in G so that when we apply them, we still have the element x .

Definition 15.2.7 (Orbits and Stabilisers)

Let G act on a set X . The **orbit** of an element $x \in X$ is $\text{Orb}_G(x) = \{g * x \mid g \in G\}$, which is a subset of X . The **stabiliser** of $x \in X$ is $\text{Stab}_G(x) = \{g \in G \mid g * x = x\}$, which is a subgroup of G .

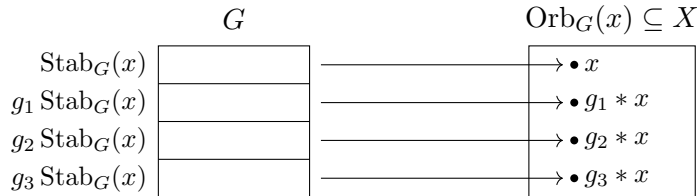
The orbits partition the set X . If there is only one orbit then we say that the group action is **transitive**. We recall the Orbit-Stabiliser theorem.

Theorem 15.2.8 (Orbit-Stabiliser Theorem)

There is a bijection between $\text{Orb}_G(x)$ and the set of left cosets of $\text{Stab}_G(x)$ in G . In particular, if G is finite, then

$$|G| = |\text{Orb}_G(x)| \cdot |\text{Stab}_G(x)|.$$

The picture to have in mind is the following: the cosets of $\text{Stab}_G(x)$ partition G , and all of the elements of a given coset $g \text{Stab}_G(x)$ move x to the *same* point $g * x$ of the orbit, with distinct cosets giving distinct points.



Remark. The kernel of ϕ , which can be written as $\ker \phi = \bigcap_{x \in X} \text{Stab}_G(x)$, is called the **kernel of the group action**. Also $\text{Stab}(g * x) = g \text{Stab}(x) g^{-1}$, so if $x, y \in X$ belong to the same orbit then their stabilisers are conjugate subgroups of G .

§15.2.3 Using Group Actions

In many cases, choosing the right group action can help make progress on a problem. With this in mind, it's helpful to have a list of standard group actions that you can apply.

Example 15.2.9 (Examples of Group Actions)

The following are all group actions.

- (i) Let G act on itself by left multiplication, with $g * x = gx$. The kernel of this action is $\{g \in G \mid gx = x\} = \{e\}$, and thus G injects into $\text{Sym}(G)$. This proves Cayley's theorem, that any finite group G is isomorphic to a subgroup of S_n for some n (say $n = |G|$).
- (ii) Let $H \leq G$. Then G acts on the left cosets of H in G by left multiplication. This group action is transitive, with $\text{Stab}_G(xH) = \{g \in G \mid gxH = xH\} = xHx^{-1}$. We have the kernel $\ker(\phi) = \bigcap_{x \in G} xHx^{-1}$, which is the largest normal subgroup of G contained in H .
- (iii) Let G act on itself by conjugation, so $g * x = gxg^{-1}$. Here, the orbits and stabilisers are $\text{Orb}_G(x) = \{gxg^{-1} \mid g \in G\} = \text{ccl}_G(x)$, the **conjugacy class** of x in G . The stabilisers are $\text{Stab}_G(x) = \{g \in G \mid gx = xg\} = C_G(x) \leq G$, the **centraliser** of x . The kernel of this action is the **center**, $Z(G) = \{g \in G \mid gx = xg \forall x \in G\}$.

G also acts by conjugation on any normal subgroup.

- (iv) Let X be the set of all subgroups of G . Then G acts on X by conjugation. That is, $g * H = gHg^{-1} \leq G$. The stabiliser of H is $\{g \in G \mid gHg^{-1} = H\} = N_G(H)$ is the **normaliser** of H in G . This is the largest subgroup of G to contain H as a normal subgroup. In particular $H \trianglelefteq G \iff N_G(H) = G$.

Of this last, the frequently useful actions (that you should readily reach for) are the left

multiplication of cosets, conjugation of elements and subgroups.

To see how we can apply group actions to proving a theorem, consider the following result (and mainly it's proof). Here we will use the action of G on the left cosets of a subgroup H .

Theorem 15.2.10

Let G be a non-abelian simple group, and $H \leq G$ a subgroup of index $n > 1$ in G . Then $n \geq 5$ and G is isomorphic to a subgroup of A_n .

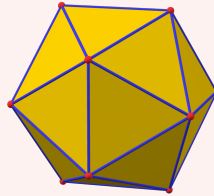
Proof. Let G act on X , the set of left cosets of H in G , by left multiplication, and let $\phi : G \rightarrow \text{Sym}(X)$ be the associated permutation representation. Then $\text{Sym}(X) = S_n$. As G is simple, we know that $\ker \phi = \{e\}$ or G . If $\ker \phi = G$, then $\text{img } \phi = \{e\}$, contradicting that G acts transitively on X (since $n > 1$). Thus $\ker \phi$ is trivial, and $G \cong \text{img } \phi \leq S_n$.

Since $G \leq S_n$ and $A_n \trianglelefteq S_n$, the second isomorphism theorem gives $G \cap A_n \trianglelefteq G$ and $G/G \cap A_n \leq S_n/A_n \cong C_2$. But G is simple so $G \cap A_n = \{e\}$ or G . If $G \cap A_n = \{e\}$, then $G \hookrightarrow C_2$, which contradicts that G is non-abelian. Otherwise, $G \cap A_n = G$, and $G \leq A_n$. Finally if $n \leq 4$, then A_n has no non-abelian simple subgroups (by listing them), so $n \geq 5$. $\square$

Another example, looking at the rotational symmetries of an icosahedron, is given below.

Example 15.2.11 (Rotational Symmetries of an Icosahedron)

Let G be the group of rotations of an icosahedron, as pictured.



In this shape, there is 20 faces, 12 vertices and 30 edges. We want to know the possible orders of elements in this group, and the number of elements of each order.

Order	Number of Elements in G
1	1
2	15
3	20
5	24

This gives us a total of 60 elements. We can verify this with the orbit-stabiliser theorem. For G acting on the vertices, and some vertex v , we have $|G| = |\text{Orb}(v)| \cdot |\text{Stab}(v)| = 12 \cdot 5 = 60$.

To consider the conjugacy classes, we note that two elements are conjugate if they 'look the same up to relabelling the icosahedron'.

The elements of order 2 are all conjugate, as are those by order 3. The elements of order 5 split into two conjugacy classes of equal size (12), as the rotations by $\pm \frac{2\pi}{5}$ are all conjugate and rotations by $\pm \frac{4\pi}{5}$ are all conjugate, but not to each-other.

We can use this to deduce G is simple. If $H \triangleleft G$, then $|H| = 1 + 15a + 20b + 12c$ for $a, b \in \{0, 1\}$ and $c \in \{0, 1, 2\}$, but as $|H| \mid 60$ by Lagrange, we get $|H| = 1$ or 60 . Thus G is simple.

In fact, G is isomorphic to A_5 . We will show that the sets $H \setminus \{1\}$ for $H \leq G$ a subgroup of order 4 partition the 15 elements of order 2 into 5 sets of 3.

1. If $|H| = 4$, then $H \cong C_2 \times C_2$ or $H \cong C_4$. But there is no elements of order 4 in G , so $H \cong C_2 \times C_2$. Note that this has three elements of order 2.
2. If $g \in G$ has order 2, then $g \in C_G(g)$ and $|C_G(g)| = \frac{|G|}{|\text{ccl}_G(g)|} = \frac{60}{15} = 4$.
3. Suppose $e \neq g \in H \cap K$ where H and K are distinct subgroups of order 4. Then $|C_G(g)| \geq |H \cup K|$ since H and K are abelian, and $|H \cup K| > 4$.

This proves the claim. Now let G act on the set X of subgroups of order 4 by conjugation. We obtain a group homomorphism $\phi : G \rightarrow \text{Sym}(X) \cong S_5$. Then G simple implies either $\ker \phi = \{e\}$ or G . If it was G , then we would get that there is a normal subgroup of order 4, which is a contradiction. Thus the kernel is trivial, and $G \hookrightarrow S_5$. Then exactly as before, either $G \hookrightarrow C_2$ or $G \leq A_5$, and thus $|G| = |A_5| = 60$, so $G \cong A_5$.

§15.3 Alternating Groups

We know from the ‘Groups’ course that two permutations in S_n are conjugate if and only if they have the same cycle type.

Example 15.3.1 (Conjugacy Classes in S_5)

In S_5 , we have the following:

Cycle Type	Number of Elements	Sign
1, 1, 1, 1, 1	1	+
2, 1, 1, 1	10	−
2, 2, 1	15	+
3, 1, 1	20	+
3, 2	20	−
4, 1	30	−
5	24	+

Thus we know the sizes of the conjugacy classes in S_5 . The same method works in general for S_n .

We want to think about the conjugacy classes of A_n .

§15.3.1 Conjugacy Classes & Simplicity of A_n

One way to think about conjugacy classes is to say that the group acts on itself by conjugation. Then the conjugacy classes correspond to orbits and centralizers correspond

to stabilisers.

So it's going to be useful to think about the centralisers of elements in A_n . Let $g \in A_n$. Then $C_{A_n}(g) = C_{S_n}(g) \cap A_n$. We have then got two cases.

- If there exists an odd permutation commuting with g , then $|C_{A_n}(g)| = \frac{1}{2}|C_{S_n}(g)|$, and $|\text{ccl}_{A_n}(g)| = |\text{ccl}_{S_n}(g)|$.
- Otherwise, $|C_{A_n}(g)| = |C_{S_n}(g)|$ and $|\text{ccl}_{A_n}(g)| = \frac{1}{2}|\text{ccl}_{S_n}(g)|$.

Example 15.3.2 (Conjugacy Classes in A_5)

We can now investigate the conjugacy classes in A_5 . Note that $(1\ 2)(3\ 4)$ commutes with the odd permutation $(1\ 2)$ and $(1\ 2\ 3)$ commutes with $(4\ 5)$. But if $h \in C_{S_5}(g)$ where $g = (1\ 2\ 3\ 4\ 5)$, then

$$(1\ 2\ 3\ 4\ 5) = h(1\ 2\ 3\ 4\ 5)h^{-1} = (h(1)\ h(2)\ h(3)\ h(4)\ h(5)).$$

Thus $h \in \langle g \rangle$, and $|\text{ccl}_{A_5}(g)| = \frac{1}{2}|\text{ccl}_{S_5}(g)| = 12$. Thus A_5 has conjugacy classes of size 1, 15, 20, 12, 12. By the same argument as before, this implies that A_5 is simple.

Lemma 15.3.3 (Generators of A_n)

A_n is generated by 3-cycles.

Proof. We know that each $\sigma \in A_n$ is the product of an even number of transpositions. So it suffices to write the product of any two transpositions as the product of 3-cycles.

For a, b, c, d distinct, we have $(a\ b)(b\ c) = (a\ b\ c)$, and $(a\ b)(c\ d) = (a\ c\ b)(a\ c\ d)$. $\square$

Lemma 15.3.4 (3-Cycles are Conjugate in A_n)

If $n \geq 5$, then all 3-cycles in A_n are conjugate.

Proof. We claim that every 3-cycle is conjugate to $(1\ 2\ 3)$. Indeed, if $(a\ b\ c)$ is a 3-cycle, then $(a\ b\ c) = \sigma(1\ 2\ 3)\sigma^{-1}$ for some $\sigma \in S_n$. If $\sigma \notin A_n$, then replace σ with $\sigma(4\ 5)$. $\square$

Theorem 15.3.5 (A_n is Simple)

The alternating group A_n is simple for all $n \geq 5$.

Proof. Let $N \trianglelefteq A_n$ be a non-trivial subgroup of A_n . It suffices to show that N contains a 3-cycle.

We take some $\sigma \in N$ with $\sigma \neq e$, and write σ as the product of disjoint cycles.

- *Case 1.* σ contains a cycle of length $r \geq 4$.

Without loss of generality, let $\sigma = (1\ 2\ 3 \cdots r)\tau$. Let $\delta = (1\ 2\ 3)$, then $\delta^{-1}\sigma\delta \in N$, and $\sigma^{-1}\delta^{-1}\sigma\delta \in N$. This can be written $(r \cdots 2\ 1)(1\ 3\ 2)(1\ 2 \cdots r)(1\ 2\ 3) =$

$(2\ 3\ r)$, which is a three cycle. Then N contains a three cycle.

- *Case 2.* σ contains two 3-cycles.

Again, without loss of generality say $\sigma = (1\ 2\ 3)(4\ 5\ 6)\tau$. Then let $\delta = (1\ 2\ 4)$, and $\sigma^{-1}\delta^{-1}\sigma\delta = (1\ 3\ 2)(4\ 6\ 5)(1\ 4\ 2)(1\ 2\ 3)(4\ 6\ 5)(1\ 2\ 3) = (1\ 2\ 4\ 3\ 6)$. Then we can apply case 1, and N contains a three cycle.

- *Case 3.* σ contains two 2-cycles.

Without loss of generality, say $\sigma = (1\ 2)(3\ 4)\tau$. Then let $\delta = (1\ 2\ 3)$. Then $\sigma^{-1}\delta^{-1}\sigma\delta = (1\ 2)(3\ 4)(1\ 3\ 2)(1\ 2)(3\ 4)(1\ 2\ 3) = (1\ 4)(2\ 3)$. Calling this π , and letting $\epsilon = (2\ 3\ 5)$, then $\pi^{-1}\epsilon^{-1}\pi\epsilon = (1\ 4)(2\ 3)(2\ 5\ 3)(1\ 4)(2\ 3)(2\ 3\ 5) = (2\ 5\ 3)$, so N contains a 3-cycle. Note that we use $n \geq 5$ here.

It remains to consider σ where it is a three-cycle. But then N contains a 3-cycle. $\square$

Definition 15.3.6 (Automorphism)

An **automorphism** of a group G is an isomorphism $G \rightarrow G$.

The automorphisms of a group form a subgroup $\text{Aut}(G) \leq \text{Sym}(G)$.

§15.4 p -Groups and p -Subgroups

§15.4.1 Basic Properties of p -Groups

Throughout this section, p will denote a fixed prime number.

Definition 15.4.1 (p -group)

A finite group G is a p -group if $|G| = p^n$ for $n \in \mathbb{N}$ where p is a prime.

We will now state some basic properties of p -groups that will be used in the following sections.

Theorem 15.4.2 (p -groups Have Non-trivial Centers)

If G is a p -group, then $Z(G) \neq \{e\}$.

Proof. For $g \in G$, we have $|\text{ccl}_G(g)| \cdot |C_G(g)| = |G| = p^n$. So each conjugacy class has size that is a power of p . Since G is a union of its conjugacy classes,

$$\begin{aligned} |G| &\equiv \#(\text{conj. classes of size 1}) \pmod{p} \\ \implies 0 &\equiv |Z(G)| \pmod{p}. \end{aligned}$$

In particular, $|Z(G)| > 1$. $\square$

Corollary 15.4.3 (Simple p -groups)

The only simple p -group is C_p .

Proof. Let G be a simple p -group. Since $Z(G) \trianglelefteq G$, we have either $Z(G) = \{e\}$ or $Z(G) = G$. The center cannot be trivial by the previous theorem, thus $Z(G) = G$, so G is abelian, and the only abelian simple groups are C_p for a prime p . $\square$

Corollary 15.4.4

Let G be a p -group of order p^n . Then G has a subgroup of order p^r for all $0 \leq r \leq n$.

Proof. We know that any finite group G has a composition series $\{e\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_{m-1} \triangleleft G_m = G$, with each quotient G_i/G_{i-1} simple. Then G is a p -group implies G_i/G_{i-1} is a p -group, and thus $G_i/G_{i-1} \cong C_p$. Then $|G_i| = p^i$ for all $0 \leq i \leq m$, and $m = n$. $\square$

Lemma 15.4.5

For G a group, if $G/Z(G)$ is cyclic, then G is abelian, and $G/Z(G)$ is trivial.

Proof. Let $gZ(G)$ be a generator for $G/Z(G)$. Then each coset is of the form $g^r Z(G)$ for some $r \in \mathbb{Z}$. Thus $G = \{g^r z \mid r \in \mathbb{Z}, z \in Z(G)\}$. Then $(g^{r_1} z_1)(g^{r_2} z_2) = g^{r_1+r_2} z_1 z_2$ since z_1 is in the center. Since z_2 is also in the center, we can also write this as $g^{r_1+r_2} z_2 z_1 = (g^{r_2} z_2)(g^{r_1} z_1)$. Thus G is abelian. $\square$

We can now start to consider what various p -groups actually look like. We know that a group of order p is cyclic, so we will begin with groups of order p^2 .

Corollary 15.4.6 (Groups of Order p^2)

If $|G| = p^2$, then G is abelian.

Proof. The center of G is non-trivial, and by thus by Lagrange, $|Z(G)|$ is either p or p^2 . If the order was p^2 , then we would be done. So consider the case $|Z(G)| = p$. Then $|G/Z(G)|$ must have order p , but then it is cyclic and contradicts our previous lemma. $\square$

We will leave considering groups of order p^3 as an exercise.

§15.4.2 Sylow Theorems

Let G be a finite group, and write $|G| = p^a m$ where p is a prime with $p \nmid m$.

Theorem 15.4.7 (First Sylow Theorem)

The set $\text{Syl}_p(G) = \{P \leq G \mid |P| = p^a\}$ of **Sylow p -subgroups** is non-empty.

Proof. We wish to show that G has a subgroup of order p^a . Let Ω be the set of all

subsets of G of size p^a . We have

$$|\Omega| = \binom{p^a m}{p^a} = \frac{p^a m}{p^a} \cdot \frac{p^a m - 1}{p^a - 1} \cdots \frac{p^a m - p^a + 1}{1}.$$

For $0 \leq k < p^a$, the numbers $p^a m - k$ and $p^a - k$ are divisible by the same power of p . Thus $|\Omega|$ is coprime to p .

Let G act on Ω by left multiplication, where for $g \in G$ and $X \in \Omega$, we put $g * X = \{gx \mid x \in X\} \in \Omega$. For any $X \in \Omega$, we have $|\text{Stab}_G(X)| \cdot |\text{Orb}_G(X)| = |G| = p^a m$. But by the coprimality described above, we can pick X such that $|\text{Orb}_G(X)|$ is coprime to p . Therefore $p^a \mid |\text{Stab}_G(X)|$.

On the other hand, if $g \in G$ and $x \in X$, then $g \in (gx^{-1}) * X$. Therefore $G = \bigcup_{g \in G} g * X$. Thus $|G| \leq |\text{Orb}_G(X)| \cdot |X|$, and thus $|\text{Stab}_G(X)| = \frac{|G|}{|\text{Orb}_G(X)|} \leq |X| = p^a$. So $|\text{Stab}_G(X)| = p^a$, and is thus a Sylow p -subgroup. $\square$

Theorem 15.4.8 (Second Sylow Theorem)

All elements of $\text{Syl}_p(G)$ are conjugate.

Proof. We claim that if $P \in \text{Syl}_p(G)$ and $Q \leq G$ is a p -subgroup then $Q \leq gPg^{-1}$ for some $g \in G$.

Let Q act on the set of left cosets of G by left multiplication, where $q * gP = qgP$. The orbit stabiliser theorem implies each orbit has size dividing $|Q|$, so it is either 1 or a multiple of p .

Since $|G : P| = m$ is coprime to p , there exists an orbit of size 1. Thus there exists $g \in G$ such that $qgP = gP$ for all $q \in Q$, and we get $g^{-1}qg \in P$, so $Q \subseteq gPg^{-1}$. $\square$

Theorem 15.4.9 (Third Sylow Theorem)

The number $n_p = |\text{Syl}_p(G)|$ of Sylow p -subgroups satisfies $n_p \equiv 1 \pmod{p}$, and $n_p \mid |G|$.

Proof. Let G act on $\text{Syl}_p(G)$ by conjugation. The second Sylow theorem implies that this action is transitive. So by the orbit stabiliser theorem, $n_p = |\text{Syl}_p(G)|$ divides $|G|$.

Now let $P \in \text{Syl}_p(G)$. P acts on $\text{Syl}_p(G)$ by conjugation. Then the orbits have size dividing $|P|$, so they are either 1 or a multiple of p .

To show $n_p \equiv 1 \pmod{p}$, it suffices to show that $\{P\}$ is the unique orbit of size 1. If $\{Q\}$ is an orbit of size 1, then P normalizes Q , so $P \leq N_G(Q)$. Now P and Q are Sylow p -subgroups of $N_G(Q)$. Hence by the second Sylow theorem, they are conjugate in $N_G(Q)$, and equal since $Q \trianglelefteq N_G(Q)$. Thus $\{P\}$ is the unique orbit of size 1. $\square$

§15.4.3 Applications of the Sylow Theorems

We will now look at how the Sylow theorems can be used to prove some facts about certain groups.

Corollary 15.4.10

If $n_p = 1$, then the unique Sylow p -subgroup is normal.

Proof. Let $g \in G$ and $P \in \text{Syl}_p(G)$. Then $gPg^{-1} \leq G$ is another Sylow p -subgroup, so we must have $gPg^{-1} = P$, for all $g \in G$. Then $P \trianglelefteq G$. $\square$

Example 15.4.11 (No Simple Groups of Order 1000)

Let $|G| = 1000 = 2^3 \cdot 5^3$. Then $n_5 \equiv 1 \pmod{5}$, and $n_5 \mid 8$. Checking, we find that $n_5 = 1$, and the unique Sylow 5-subgroup is normal. Then G is not simple.

Example 15.4.12 (No Simple Groups of Order 132)

Let $|G| = 132 = 2^2 \cdot 3 \cdot 11$. Then $n_{11} \equiv 1 \pmod{11}$, and $n_{11} \mid 12$. Then $n_{11} = 1$ or 12.

Suppose G was simple. Then $n_{11} \neq 1$, so $n_{11} = 12$. Now $n_3 \equiv 1 \pmod{3}$, and $n_3 \mid 44$, so $n_3 = 1, 4$ or 22. But G is simple, and either $n_3 = 4$ or 22. Supposing $n_3 = 4$, then letting G act on $\text{Syl}_3(G)$ by conjugation, we get a group homomorphism $\phi : G \rightarrow S_4$. Then $\ker(\phi) \trianglelefteq G$ implies $\ker(\phi) = \{e\}$ or G . It cannot be G by the second Sylow theorem, thus ϕ is injective. But then $G \hookrightarrow S_4$, and $132 \leq 24$, which is a contradiction.

Thus $n_3 = 22$, and $n_{11} = 12$. But then counting, G has $22 \cdot (3 - 1) = 44$ elements of order 3, and $12 \cdot (11 - 1) = 120$ elements of order 11. But then $120 + 44 = 164 > 132$, which is a contradiction. Thus there does not exist a simple group of order 132.

§15.5 Some Groups

§15.5.1 Matrix Groups

We are going to finish our study of groups by looking at some matrix groups, along with finite abelian groups.

Let F be a field (for example $\mathbb{C}$ or $\mathbb{Z}/p\mathbb{Z}$). Recall the groups $\text{GL}_n(F)$ of $n \times n$ invertible matrices over F , and $\text{SL}_n(F) = \ker(\text{GL}_n \xrightarrow{\det} F^\times) \trianglelefteq \text{GL}_n(F)$. Also let $Z \trianglelefteq \text{GL}_n(F)$ be the group of scalar matrices (scalar multiples of the identity matrices).

We define the *projective general/special linear groups* as follows.

Definition 15.5.1 (Projective General Linear Group)

The **projective general linear group** $\text{PGL}_n(F)$ is defined to be $\text{GL}_n(F)/Z$.

Definition 15.5.2 (Projective Special Linear Group)

The **projective special linear group** $\mathrm{PSL}_n(F)$ is the group $\mathrm{SL}_n(F)/(Z \cap \mathrm{SL}_n(F))$, which is isomorphic to $Z \cdot \mathrm{SL}_n(F)/Z$, by the second isomorphism theorem.

Example 15.5.3

Let $G = \mathrm{GL}_n(\mathbb{Z}/p\mathbb{Z})$. A list of n vectors in $(\mathbb{Z}/p\mathbb{Z})^n$ are the columns of some $A \in G$ if and only if they are linearly independent. Thus

$$\begin{aligned} |G| &= (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1}) \\ &= p^{1+2+\cdots+(n-1)}(p^n - 1)(p^{n-1} - 1) \cdots (p - 1) \\ &= p^{\binom{n}{2}} \prod_{i=1}^n (p^i - 1). \end{aligned}$$

We see that the Sylow p -subgroups have order $p^{\binom{n}{2}}$. One such is the subgroup of upper triangular matrices with 1's on the diagonal.

$$U = \left\{ \begin{pmatrix} 1 & * & * & \cdots & * \\ 0 & 1 & * & \cdots & * \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} \right\} \leq G.$$

Indeed there are $\binom{n}{2}$ entries $*$, each which can take p values, so the order is $p^{\binom{n}{2}}$.

Just as $\mathrm{PGL}_2(\mathbb{C})$ acts on $\mathbb{C} \cup \{\infty\}$ via Möbius maps, $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z})$ acts on $\mathbb{Z}/p\mathbb{Z} \cup \{\infty\}$. Indeed, $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ acts as

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} : z \mapsto \frac{az + b}{cz + d},$$

and since scalar matrices act trivially, this is an action of $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z})$.

Lemma 15.5.4

The permutation representation of $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z}) \rightarrow S_{p+1}$ is injective (and is an isomorphism if $p = 2$ or 3).

Proof. Suppose $\frac{az+b}{cz+d} = z$ for all $z \in \mathbb{Z}/p\mathbb{Z} \cup \{\infty\}$. Putting $z = 0$ shows $b = 0$. Putting $z = \infty$ shows $c = 0$. Putting $z = 1$ shows $a = d$, and thus this is a scalar matrix. $\square$

Lemma 15.5.5

If p is an odd prime, then $|\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{p(p-1)(p+1)}{2}$.

Proof. By the previous example, $|\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})| = p(p-1)(p^2-1)$. Then consider the group homomorphism $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) \xrightarrow{\det} (\mathbb{Z}/p\mathbb{Z})^\times$. This is surjective, since we

have

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \mapsto a,$$

so $|\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{|\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})|}{p-1} = p(p-1)(p+1)$.

Then counting the number of scalar matrices, if

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}),$$

then $\lambda^2 \equiv 1 \pmod{p}$ implies $p \mid (\lambda-1)(\lambda+1)$, so $\lambda \equiv \pm 1 \pmod{p}$. So there is two such scalar matrices, and $|\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{1}{2}|\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})| = \frac{p(p-1)(p+1)}{2}$. $\square$

Example 15.5.6

Let $G = \mathrm{PSL}_2(\mathbb{Z}/5\mathbb{Z})$. Then $|G| = 60 = 2^2 \cdot 3 \cdot 5$.

Let G act on $\mathbb{Z}/5\mathbb{Z} \cup \{\infty\}$ via Möbius maps. By our previous lemma, there is an injective group homomorphism $\phi : G \rightarrow \mathrm{Sym}(\{0, 1, \dots, 4, \infty\}) \cong S_6$.

Claim. $\mathrm{img}(\phi) \leq A_6$, that is, $\psi : G \rightarrow S_6 \xrightarrow{\mathrm{sgn}} \{\pm 1\}$ is trivial.

If m is odd, then $\psi(g) = e \iff \psi(g)^m = e \iff \psi(g^m) = e$. So it suffices to consider $g \in G$ with order a power of 2. We showed in the proof of the Sylow theorems that every such element belongs to a Sylow 2-subgroup. So it suffices to check that $\psi(H) = e$, where H is a Sylow 2-subgroup (using that we have any two Sylow 2-subgroups are conjugate, and ψ maps to an abelian group).

We take $H = \left\{ \pm \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \pm \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\} \leq G$. We compute:

$$\begin{aligned} \phi\left(\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\right) &= (1\ 4)(2\ 3) \\ \phi\left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right) &= (0\ \infty)(1\ 4). \end{aligned}$$

These are even permutations, so $\psi(H)$ is trivial, and our claim is true.

§15.5.2 Finite Abelian Groups

Later in this course, we will prove the following theorem.

Theorem 15.5.7 (The Fundamental Theorem of Finite Abelian Groups)

Every finite abelian group is isomorphic to the direct product of cyclic groups.

Remark (Non-Uniqueness). While this is possible, such a decomposition is not unique. For example, you may recall from IA groups that if m and n are coprime, then $C_m \times C_n \cong C_{mn}$.

Lemma 15.5.8

If m and n are coprime, then $C_{mn} = C_m \times C_n$.

Proof. Let g and h be generators of C_m and C_n . Then we have $(g, h) \in C_m \times C_n$, and $(g, h)^r = (g^r, h^r)$. In particular, $(g, h)^r = e$ if and only if $m \mid r$ and $n \mid r$, that is, if $mn \mid r$ since they are coprime. Thus (g, h) has order mn , which is $|C_m \times C_n|$. $\square$

Corollary 15.5.9

Let G be a finite abelian group. Then $G \cong C_{n_1} \times C_{n_2} \times \cdots \times C_{n_k}$ where each n_i is a prime power.

Proof. If $n = p_1^{a_1} \cdots p_r^{a_r}$, with p_i prime, then we can (by the previous lemma) write $C_n \cong C_{p_1^{a_1}} \times \cdots \times C_{p_r^{a_r}}$. Writing each of the cyclic groups in the fundamental theorem of abelian groups then gives the result. $\square$

In fact, we will prove the following refinement of the the fundamental theorem of finite abelian groups.

Theorem 15.5.10

Let G be a finite abelian group. Then

$$G \cong C_{d_1} \times C_{d_2} \times \cdots \times C_{d_t},$$

for some $d_1 \mid d_2 \mid \cdots \mid d_t$.

Remark. The integers $n_1, \dots, n_k$ in the previous corollary and $d_1, \dots, d_t$ in the previous theorem are uniquely determined by G . The proof (which we omit) works by counting the number of elements of each prime power order.

Example 15.5.11 (Abelian Groups of Order 8)

The abelian groups of order 8 are C_8 , $C_2 \times C_4$ and $C_2 \times C_2 \times C_2$.

Example 15.5.12 (Abelian Groups of Order 12)

The abelian groups of order 12 are $C_2 \times C_6 \cong C_2 \times C_2 \times C_3$, $C_4 \times C_3 \cong C_{12}$.

Definition 15.5.13 (Exponent)

The **exponent** of a group G is the least integer $n \geq 1$ such that $g^n = e$ for all $g \in G$. That is, it is the lowest common multiple of all of the orders of elements in G .

Example 15.5.14

A_4 has exponent 6.

Corollary 15.5.15

Every finite abelian group contains an element whose order is the exponent of the group.

Proof. If $G \cong C_{d_1} \times \cdots \times C_{d_t}$ with $d_1 \mid d_2 \mid \cdots \mid d_t$, then every $g \in G$ has order dividing d_t , and if $h \in C_{d_t}$ is a generator, then $(1, 1, 1, \dots, h) \in G$ has order d_t . So G has exponent d_t . $\square$

§15.6 Rings**§15.6.1 Basic Definitions**

We will now move on to another central algebraic object in this course – *rings*. Let's start with a definition.

Definition 15.6.1 (Ring)

A **ring** is a triple $(R, +, \cdot)$ consisting of a set R and two binary operations^a $+, \cdot : R \times R \rightarrow R$, referred to as *addition* and *multiplication*, that satisfy

- $(R, +)$ is an abelian group, with identity 0.
- Multiplication is associative and has an identity 1.
- *Distributivity.* For $x, y, z \in R$, we have $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(x + y) \cdot z = x \cdot z + y \cdot z$.

^aAgain this has an implicit *closure* axiom

Remark (Notation). For $x \in R$, we write $-x$ for its inverse under addition, and we abbreviate $x + (-y) = x - y$.

We note that in the definition we require $(R, +)$ to be an abelian group, but the fact that it is abelian is deducible from the other axioms. This means that if we talk about a ring being commutative, we must be referring to the multiplication.

Definition 15.6.2 (Commutative Ring)

A ring R is **commutative** if $x \cdot y = y \cdot x$ for all $x, y \in R$.

In this course, we will *only consider* commutative rings. So whenever we refer to an arbitrary ring, we will be referring to a commutative ring.¹

We can define subrings in the natural way.

Definition 15.6.3 (Subrings)

A subset $S \subset R$ is a **subring**, written $S \leq R$, if it is a ring under the same operations $+$ and $\cdot$, with the same identity elements 0 and 1.

¹There are some notable non-commutative rings such as ones made up of matrices, but they will not be considered in this course.

Example 15.6.4 (Examples of Rings and Subrings)

The following are examples of rings and subrings.

- (i) We have the subrings $\mathbb{Z} \leq \mathbb{Q} \leq \mathbb{R} \leq \mathbb{C}$, under addition and multiplication as normal.
- (ii) $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a ring, and $\mathbb{Z}[i] \leq \mathbb{C}$.
- (iii) $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\} \leq \mathbb{R}$.
- (iv) $\mathbb{Z}\left[\frac{1}{p}\right] = \left\{\frac{m}{p^n} \mid m \in \mathbb{Z}, n \geq 0\right\} \leq \mathbb{Q}$.
- (v) $\mathbb{Z}/n\mathbb{Z}$, the integers modulo n , is a ring of order n .

§15.6.2 Constructing Rings

Similar to the direct product of groups, there is a natural way to combine rings.

Definition 15.6.5 (Product Ring)

If R and S are rings, then their **product ring** $R \times S$ is a ring via $(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2)$ and $(r_1, s_1) \cdot (r_2, s_2) = (r_1 \cdot r_2, s_1 \cdot s_2)$.

We have $0_{R \times S} = (0_R, 0_S)$ and $1_{R \times S} = (1_R, 1_S)$.

Remark. Note that $R \times \{0\}$ is *not* a subring of $R \times S$ – this doesn't have the same identity elements.

There are some other interesting ways to construct rings from other rings. First of all, we could consider functions from some set onto R .

Definition 15.6.6

If R is a ring and X is a set, then the set of all functions $X \rightarrow R$ is a ring under pointwise operations, $(f + g)(x) = f(x) + g(x)$, and $(f \cdot g)(x) = f(x) \cdot g(x)$.

Other interesting examples appear as subrings. For example, the continuous functions $\mathbb{R} \rightarrow \mathbb{R}$.

We can also construct the *ring of polynomials*

Definition 15.6.7 (Ring of Polynomials)

Let R be a ring and define S to be the set of all sequences $(a_0, a_1, \dots)$ with $a_i \in R$ with $a_i = 0$ for all $i \geq N$ for some N . Then defining addition and multiplication as

$$\begin{aligned}(a_0, a_1, \dots) + (b_0, b_1, \dots) &= (a_0 + b_0, a_1 + b_1, \dots) \\ (a_0, a_1, \dots) \cdot (b_0, b_1, \dots) &= (c_0, c_1, \dots),\end{aligned}$$

where $c_n = \sum_{i=0}^n a_i b_{n-i}$ gives us $R[X]$, the **ring of polynomials in X with coefficients in R** .

Note that if we define $X = (0, 1, 0, \dots)$, then $X^n = (\underbrace{0, 0, \dots, 0}_{n \text{ times}}, 1, 0, \dots)$, and

$$(a_0, a_1, \dots, a_n, 0, \dots) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0.$$

This is why this is known as the polynomial ring.

You may wonder why we defined things in this way, and the answer is that we are not thinking about polynomials as functions, but as sequences of coefficients. To see why this matters, consider $R = \mathbb{Z}/p\mathbb{Z}$ with p prime and $f(x) = X^p - X$. Then the function $x \mapsto f(x)$ is identically zero, but the polynomial $X^p - X$ is non-zero.

So we can restate our definition in a clearer and more general way.

Definition 15.6.8 (Ring of Polynomials)

If R is a ring then $R[X_1, \dots, X_n]$ is the *ring of polynomials* in $X_1, \dots, X_n$ with coefficients in R .

Remark. We can define this inductively as $R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$ using our careful construction for $R[X]$ above.

Some extensions of this are the following rings.

Definition 15.6.9 (Power Series and Laurent Polynomials)

The **power series ring** is the ring $R[[X]] = \{a_0 + a_1 X + a_2 X^2 + \dots \mid a_i \in R\}$ and the **ring of Laurent polynomials** is $R[X, X^{-1}] = \{\sum_{i \in \mathbb{Z}} a_i X^i \mid a_i \in R, \text{ and only finitely many } a_i \neq 0\}$.

§15.6.3 Units and Fields

We now introduce an important definition. Recall that not every element of a ring necessarily has an inverse under multiplication. When an element *does* have one, we call it a *unit*.

Definition 15.6.10 (Unit)

An element $r \in R$ is a **unit** if it has an inverse under multiplication. That is, if there exists $s \in R$ such that $r \cdot s = 1$.

The units in a ring R form a group $(R^\times, \cdot)$ under multiplication. For example, $\mathbb{Z}^\times = \{\pm 1\}$, and $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$.

When every non-zero element in a ring is a unit, we call that ring a *field*.

Definition 15.6.11 (Field)

A **field** is a ring with $0 \neq 1$ such that every non-zero element is a unit.

Examples of fields are $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $\mathbb{Z}/p\mathbb{Z}$ where p is prime.

Remark. Note that we require $0 \neq 1$ in the definition of a field. If we did not have this, then if $0 = 1$, $x = 1 \cdot x = 0 \cdot x = 0$, for all $x \in R$. So if a ring has $0 = 1$, then it is the trivial ring $R = \{0\}$. And we don't want this to be a field.

The notion of a ring allows us to do some interesting things.

Lemma 15.6.12 (Polynomial Division Algorithm)

Let $f, g \in R[X]$. Suppose that the leading coefficient of g is a unit. Then there exists $q, r \in R[X]$ such that $f(X) = q(X)g(X) + r(X)$, where $\deg(r) < \deg(g)$.

Proof. We induct on the degree of f . If the degree is n , then we write $f(X) = a_nX^n + \cdots + a_1X + a_0$ with $a_n \neq 0$, and $g(X) = b_mX^m + \cdots + b_1X + b_0$, where $b_m \in R^\times$.

If $n < m$, put $q = 0$ and $r = f$, and we are done. Otherwise, we put $f_1(X) = f(X) - a_nb_m^{-1}X^{n-m}g(X)$. Then the coefficient of X^n is 0, and $\deg(f_1) < n$, and we apply the induction hypothesis to get $f_1(X) = q_1(X)g(X) + r(X)$, with $\deg(r) < \deg(g)$.

Then $f(X) = (q_1(X) + a_nb_m^{-1}X^{n-m})g(X) + r(X)$. $\square$

Remark. If R is a field, then we only need $g \neq 0$.

§15.6.4 Homomorphisms and Ideals

As with groups, we are interested in the structure preserving maps between rings. Since a ring has two operations, we ask that both are preserved, and (importantly) that the multiplicative identity is sent to the multiplicative identity.

Definition 15.6.13 (Ring Homomorphism)

Let R and S be rings. A function $\phi : R \rightarrow S$ is a **ring homomorphism** if

- (i) $\phi(r_1 + r_2) = \phi(r_1) + \phi(r_2)$ for all $r_1, r_2 \in R$,
- (ii) $\phi(r_1 \cdot r_2) = \phi(r_1) \cdot \phi(r_2)$ for all $r_1, r_2 \in R$,
- (iii) $\phi(1_R) = 1_S$.

A bijective ring homomorphism is a **ring isomorphism**.

Note that $\phi(0_R) = 0_S$ comes for free, since ϕ is in particular a group homomorphism $(R, +) \rightarrow (S, +)$. The same reasoning gives us the following.

Definition 15.6.14 (Kernel)

The **kernel** of a ring homomorphism $\phi : R \rightarrow S$ is $\ker(\phi) = \{r \in R \mid \phi(r) = 0\}$.

Lemma 15.6.15

A ring homomorphism ϕ is injective if and only if $\ker(\phi) = \{0\}$.

Proof. ϕ is, in particular, a homomorphism of additive groups, so this follows from the corresponding result for groups. $\square$

Now here is where the theory of rings begins to diverge slightly from the theory of groups. In group theory, the subsets that could appear as kernels of homomorphisms were the

normal subgroups. Let's think about what the kernel of a ring homomorphism looks like. It is certainly a subgroup under addition, but if $r \in R$ and $x \in \ker(\phi)$ then

$$\phi(rx) = \phi(r)\phi(x) = \phi(r) \cdot 0 = 0,$$

so the kernel 'absorbs' multiplication by arbitrary ring elements – something much stronger than being closed under multiplication. Subsets with this property are called *ideals*.

Definition 15.6.16 (Ideal)

A subset $I \subseteq R$ is an **ideal**, written $I \trianglelefteq R$, if

- (i) I is a subgroup of $(R, +)$,
- (ii) if $r \in R$ and $x \in I$, then $rx \in I$.

We say I is **proper** if $I \neq R$.

By the discussion above, if $\phi : R \rightarrow S$ is a ring homomorphism then $\ker(\phi) \trianglelefteq R$.

Remark. If an ideal I contains a unit u , then it contains $u^{-1}u = 1$, and then it contains $r \cdot 1 = r$ for every $r \in R$. So a proper ideal contains no units. In particular, note that a proper ideal is *not* a subring, since $1 \notin I$.

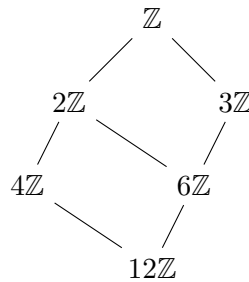
The most familiar example of an ideal comes from the integers.

Lemma 15.6.17 (Ideals of $\mathbb{Z}$)

The ideals of $\mathbb{Z}$ are exactly $n\mathbb{Z}$ for $n = 0, 1, 2, \dots$

Proof. Each $n\mathbb{Z}$ is easily checked to be an ideal. Conversely, let $I \trianglelefteq \mathbb{Z}$ be a non-zero ideal, and let n be its smallest positive element. Then $n\mathbb{Z} \subseteq I$. If $m \in I$, then writing $m = qn + r$ with $0 \leq r < n$, we get $r = m - qn \in I$, so by minimality $r = 0$ and $m \in n\mathbb{Z}$. Thus $I = n\mathbb{Z}$. $\square$

Since $n\mathbb{Z} \subseteq m\mathbb{Z}$ if and only if $m \mid n$, the ideals of $\mathbb{Z}$ ordered by inclusion form precisely the divisibility lattice, drawn below for the ideals containing $12\mathbb{Z}$.



This example generalises: in any ring we can consider the set of 'multiples' of a fixed collection of elements.

Definition 15.6.18 (Generated Ideals)

For $a \in R$, the **principal ideal** generated by a is $(a) = \{ra \mid r \in R\}$. More

generally, for $a_1, \dots, a_n \in R$ we write

$$(a_1, \dots, a_n) = \{r_1 a_1 + \dots + r_n a_n \mid r_i \in R\},$$

the **ideal generated by** $a_1, \dots, a_n$.

For example, every ideal of $\mathbb{Z}$ is principal by the lemma above. Notice also that $(a) = R$ if and only if a is a unit – a small observation that we will use constantly.

§15.6.5 Quotient Rings

Just as we formed quotient groups from normal subgroups, we can form quotient rings from ideals. The additive structure is exactly the quotient group $(R, +)/(I, +)$, and the ideal property is precisely what makes multiplication of cosets well defined.

Theorem 15.6.19 (Quotient Rings)

Let $I \trianglelefteq R$. Then the set R/I of cosets of I in $(R, +)$ is a ring, the **quotient ring**, with operations

$$\begin{aligned}(r_1 + I) + (r_2 + I) &= (r_1 + r_2) + I, \\ (r_1 + I) \cdot (r_2 + I) &= (r_1 r_2) + I,\end{aligned}$$

and identities $0_{R/I} = 0 + I$ and $1_{R/I} = 1 + I$. Moreover the **quotient map** $\pi : R \rightarrow R/I$ with $r \mapsto r + I$ is a surjective ring homomorphism with kernel I .

Proof. From the theory of groups, addition is well defined and gives an abelian group. For multiplication, suppose $r_1 + I = r'_1 + I$ and $r_2 + I = r'_2 + I$, so $r'_1 = r_1 + a_1$ and $r'_2 = r_2 + a_2$ with $a_1, a_2 \in I$. Then

$$r'_1 r'_2 = r_1 r_2 + \underbrace{a_1 r_2 + r_1 a_2 + a_1 a_2}_{\in I},$$

so $r'_1 r'_2 + I = r_1 r_2 + I$, and multiplication of cosets is well defined. The ring axioms are all inherited directly from R , as is easily checked. The quotient map is then a surjective homomorphism by construction, and $\pi(r) = 0 + I \iff r \in I$. $\square$

So exactly as in the group case, ideals are precisely the kernels of ring homomorphisms. Let's look at some examples of quotient rings.

Example 15.6.20 (Quotients of $\mathbb{Z}$)

The quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$ is the familiar ring $\mathbb{Z}/n\mathbb{Z}$ of integers modulo n .

Example 15.6.21 ($\mathbb{C}[X]/(X)$)

Consider the ideal $(X) \trianglelefteq \mathbb{C}[X]$, the set of polynomials with zero constant term. Given $f(X) = a_n X^n + \dots + a_1 X + a_0$, we have $f + (X) = a_0 + (X)$. So the map $\mathbb{C}[X]/(X) \rightarrow \mathbb{C}$ given by $f + (X) \mapsto f(0)$ is a well defined bijection, and it is easily checked to be a ring homomorphism. Thus $\mathbb{C}[X]/(X) \cong \mathbb{C}$.

Example 15.6.22 ($\mathbb{R}[X]/(X^2 + 1)$)

Consider the ideal $(X^2 + 1) \trianglelefteq \mathbb{R}[X]$. Using the polynomial division algorithm, any $f \in \mathbb{R}[X]$ can be written $f(X) = q(X)(X^2 + 1) + r(X)$ with $\deg r < 2$, so every coset has a representative $a + bX$ with $a, b \in \mathbb{R}$, and it's straightforward to check this representative is unique. The bijection

$$\mathbb{R}[X]/(X^2 + 1) \longrightarrow \mathbb{C}, \quad a + bX + (X^2 + 1) \longmapsto a + bi$$

respects addition, and for multiplication note that $(a + bX)(c + dX) = ac + (ad + bc)X + bdX^2 \equiv (ac - bd) + (ad + bc)X \pmod{X^2 + 1}$, matching $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. So $\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$ – quotienting by $(X^2 + 1)$ amounts to ‘adjoining a square root of -1 ’.

This last example illustrates an important idea – quotients of polynomial rings let us *build* rings in which some chosen polynomial has a root. We will return to this when we discuss fields and algebraic integers.

§15.6.6 The Isomorphism Theorems for Rings

Each of the isomorphism theorems for groups has an analogue for rings, with normal subgroups replaced by ideals. The proofs consist of applying the group versions to the additive groups, and then checking everything respects multiplication.

Theorem 15.6.23 (First Isomorphism Theorem)

Let $\phi : R \rightarrow S$ be a ring homomorphism. Then $\ker(\phi) \trianglelefteq R$, $\text{img}(\phi) \leq S$ is a subring, and

$$R/\ker(\phi) \cong \text{img}(\phi).$$

Proof. We have already seen $\ker(\phi) \trianglelefteq R$. The image is an additive subgroup, it's closed under multiplication as $\phi(r_1)\phi(r_2) = \phi(r_1r_2)$, and $1_S = \phi(1_R) \in \text{img}(\phi)$, so $\text{img}(\phi)$ is a subring of S .

Now let $K = \ker(\phi)$ and define $\Phi : R/K \rightarrow \text{img}(\phi)$ by $r + K \mapsto \phi(r)$. By the first isomorphism theorem for groups, this is a well defined bijective homomorphism of additive groups. Also

$$\Phi((r_1 + K)(r_2 + K)) = \Phi(r_1r_2 + K) = \phi(r_1r_2) = \phi(r_1)\phi(r_2) = \Phi(r_1 + K)\Phi(r_2 + K),$$

and $\Phi(1 + K) = \phi(1_R) = 1_S$, so Φ is a ring isomorphism. $\square$

Example 15.6.24

Consider the ‘evaluation at i ’ homomorphism $\phi : \mathbb{R}[X] \rightarrow \mathbb{C}$ given by $f \mapsto f(i)$. This is surjective, since $a + bX \mapsto a + bi$. Its kernel consists of real polynomials with i as a root, and dividing any such polynomial by $X^2 + 1$ shows $\ker(\phi) = (X^2 + 1)$. So the first isomorphism theorem gives $\mathbb{R}[X]/(X^2 + 1) \cong \mathbb{C}$ immediately – much more efficiently than our direct verification before.

Theorem 15.6.25 (Second Isomorphism Theorem)

Let $R \leq S$ be a subring and $J \trianglelefteq S$. Then $R \cap J \trianglelefteq R$, the set $R + J = \{r + a \mid r \in R, a \in J\}$ is a subring of S , and

$$R/(R \cap J) \cong (R + J)/J.$$

Proof. Let $\phi : R \rightarrow S/J$ be the composite of the inclusion $R \hookrightarrow S$ and the quotient map $S \rightarrow S/J$, so $\phi(r) = r + J$. This is a ring homomorphism, with

$$\ker(\phi) = \{r \in R \mid r + J = J\} = R \cap J \trianglelefteq R,$$

and $\text{img}(\phi) = \{r + J \mid r \in R\} = (R + J)/J$, which is then a subring of S/J . The result follows by the first isomorphism theorem. (One can also check directly that $R + J$ is a subring of S – it is an additive subgroup containing 1, and $(r_1 + a_1)(r_2 + a_2) = r_1r_2 + (r_1a_2 + r_2a_1 + a_1a_2) \in R + J$.) $\square$

For the third isomorphism theorem, we again start with a correspondence. If $I \trianglelefteq R$, then there is a bijection

$$\{\text{ideals of } R/I\} \longleftrightarrow \{\text{ideals of } R \text{ containing } I\},$$

given by $K \mapsto \{r \in R \mid r + I \in K\}$ and $J \mapsto J/I = \{r + I \mid r \in J\}$.

Theorem 15.6.26 (Third Isomorphism Theorem)

Let $I \subseteq J$ be ideals of R . Then $J/I \trianglelefteq R/I$ and

$$(R/I) / (J/I) \cong R/J.$$

Proof. Consider $\phi : R/I \rightarrow R/J$ with $r + I \mapsto r + J$. This is well defined (as $I \subseteq J$), and is a surjective ring homomorphism with kernel $\{r + I \mid r \in J\} = J/I$. Apply the first isomorphism theorem. $\square$

Let's see a genuinely useful consequence of this machinery.

Example 15.6.27 (The Characteristic of a Ring)

For any ring R , there is exactly one ring homomorphism $\iota : \mathbb{Z} \rightarrow R$. Indeed, we are forced to set $\iota(1) = 1_R$, and then $\iota(n) = 1_R + \cdots + 1_R$ (n times) for $n \geq 0$, and $\iota(-n) = -\iota(n)$; one checks this is a ring homomorphism.

Now $\ker(\iota) \trianglelefteq \mathbb{Z}$, so $\ker(\iota) = n\mathbb{Z}$ for a unique $n \geq 0$, called the **characteristic** of R . By the first isomorphism theorem, every ring contains a copy of $\mathbb{Z}/n\mathbb{Z}$ (interpreting $\mathbb{Z}/0\mathbb{Z}$ as $\mathbb{Z}$).

For example, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ have characteristic 0, while $\mathbb{Z}/p\mathbb{Z}$ and $(\mathbb{Z}/p\mathbb{Z})[X]$ have characteristic p .

§15.6.7 Integral Domains

Many familiar rings have the property that a product of non-zero elements is non-zero. This is such a useful property that rings with it get their own name.

Definition 15.6.28 (Integral Domain)

An **integral domain** is a ring R with $0 \neq 1$ such that for all $a, b \in R$,

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

A non-zero element $a \in R$ for which there is a non-zero b with $ab = 0$ is called a **zero divisor** – so an integral domain is a non-trivial ring with no zero divisors.

Example 15.6.29 (Examples of Integral Domains)

Every field is an integral domain: if $ab = 0$ and $a \neq 0$, then $b = a^{-1}(ab) = 0$. Also, any subring of an integral domain is an integral domain, so for example $\mathbb{Z} \leq \mathbb{Q}$ and $\mathbb{Z}[i] \leq \mathbb{C}$ are integral domains.

On the other hand, $\mathbb{Z} \times \mathbb{Z}$ is not an integral domain, since $(1, 0) \cdot (0, 1) = (0, 0)$, and $\mathbb{Z}/6\mathbb{Z}$ is not, since $2 \cdot 3 = 0$.

Working in an integral domain makes polynomials behave the way we would like them to.

Lemma 15.6.30

If R is an integral domain, then $R[X]$ is an integral domain, and $\deg(fg) = \deg f + \deg g$ for non-zero $f, g \in R[X]$.

Proof. Let $f, g \neq 0$ have leading coefficients $a_n \neq 0$ and $b_m \neq 0$. The coefficient of X^{n+m} in fg is a_nb_m , which is non-zero since R is an integral domain. So $fg \neq 0$ and $\deg(fg) = n + m$. $\square$

Lemma 15.6.31 (Roots of Polynomials)

Let R be an integral domain and $f \in R[X]$ be non-zero. Then f has at most $\deg(f)$ roots in R .

Proof. We induct on $\deg f$. If $\deg f = 0$ then f is a non-zero constant, with no roots. Now suppose a is a root of f . Dividing by the monic polynomial $X - a$, we can write $f(X) = q(X)(X - a) + r$ with r constant, and evaluating at a shows $r = 0$, so $f = q \cdot (X - a)$ with $\deg q = \deg f - 1$. Any other root $b \neq a$ of f satisfies $q(b)(b - a) = 0$, so $q(b) = 0$ since R is an integral domain. By induction q has at most $\deg q - 1$ roots, so f has at most $\deg f$ roots. $\square$

This innocent-looking fact has a rather striking consequence.

Theorem 15.6.32

Let F be a field. Then any finite subgroup $G \leq (F^\times, \cdot)$ is cyclic.

Proof. G is a finite abelian group. If G is not cyclic, then by the classification of finite abelian groups (using the invariant factor form $G \cong C_{d_1} \times \cdots \times C_{d_t}$ with

$d_1 \mid \cdots \mid d_t$ and $t \geq 2$), G contains a subgroup isomorphic to $C_d \times C_d$ for some $d \geq 2$. But then the polynomial $X^d - 1 \in F[X]$ has at least $d^2 > d$ roots in F , contradicting the previous lemma. $\square$

Corollary 15.6.33

$(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic for any prime p .

We record one more useful fact about integral domains.

Proposition 15.6.34 (Finite Integral Domains are Fields)

Any finite integral domain is a field.

Proof. Let R be a finite integral domain and $0 \neq a \in R$. Consider the map $\phi : R \rightarrow R$ given by $x \mapsto ax$. If $\phi(x) = \phi(y)$ then $a(x - y) = 0$, so $x = y$, that is, ϕ is injective. Since R is finite, ϕ is a bijection, so there is some $b \in R$ with $ab = 1$. Thus every non-zero element is a unit. $\square$

§15.6.8 The Field of Fractions

The integers are not a field, but they sit naturally inside one – the rationals, whose elements are ratios of integers. This construction works in any integral domain.

Theorem 15.6.35 (Field of Fractions)

Let R be an integral domain. Then there is a field F , the **field of fractions** of R , which contains R as a subring and in which every element can be written as ab^{-1} with $a, b \in R$, $b \neq 0$.

Proof. We mimic the construction of $\mathbb{Q}$ from $\mathbb{Z}$. Consider the set $S = \{(a, b) \mid a, b \in R, b \neq 0\}$ of ‘fractions’, with the equivalence relation

$$(a, b) \sim (c, d) \iff ad = bc.$$

This is clearly reflexive and symmetric. For transitivity, if $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$, then $adf = bcf = bde$, so $d(af - be) = 0$, and since $d \neq 0$ and R is an integral domain^a, $af = be$.

Write $\frac{a}{b}$ for the equivalence class of (a, b) , and let $F = S/\sim$ with operations

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}.$$

Note $bd \neq 0$, and one checks these operations are well defined and make F a ring, with $0_F = \frac{0}{1}$ and $1_F = \frac{1}{1}$. If $\frac{a}{b} \neq 0_F$ then $a \neq 0$, and $\frac{a}{b} \cdot \frac{b}{a} = 1_F$, so F is a field. Finally, identifying $r \in R$ with $\frac{r}{1} \in F$ embeds R as a subring of F , and $\frac{a}{b} = ab^{-1}$. $\square$

^aThis is where we need R to be an integral domain – without it, transitivity genuinely fails.

Example 15.6.36

The field of fractions of $\mathbb{Z}$ is $\mathbb{Q}$, and the field of fractions of $\mathbb{C}[X]$ is the field $\mathbb{C}(X)$ of **rational functions** $\frac{f(X)}{g(X)}$ with $f, g \in \mathbb{C}[X]$ and $g \neq 0$.

§15.6.9 Maximal and Prime Ideals

We now introduce two special classes of ideals, which correspond exactly to when the quotient ring is a field or an integral domain. First, a small but important observation.

Lemma 15.6.37 (Fields Have No Interesting Ideals)

A non-trivial ring R is a field if and only if its only ideals are $\{0\}$ and R .

Proof. Suppose R is a field and $I \trianglelefteq R$ is non-zero. Then I contains a non-zero element, which is a unit, so $I = R$.

Conversely, suppose the only ideals are $\{0\}$ and R , and let $0 \neq x \in R$. Then $(x) \neq \{0\}$, so $(x) = R$. In particular $1 \in (x)$, meaning $xy = 1$ for some y , so x is a unit. $\square$

Definition 15.6.38 (Maximal Ideal)

An ideal $I \trianglelefteq R$ is **maximal** if $I \neq R$ and the only ideals containing I are I and R .

Proposition 15.6.39

Let $I \trianglelefteq R$. Then I is maximal if and only if R/I is a field.

Proof. R/I is a field if and only if its only ideals are $\{0\} = I/I$ and R/I . By the correspondence between ideals of R/I and ideals of R containing I , this happens if and only if the only ideals of R containing I are I and R , that is, if and only if I is maximal. $\square$

Definition 15.6.40 (Prime Ideal)

An ideal $I \trianglelefteq R$ is **prime** if $I \neq R$ and whenever $ab \in I$, either $a \in I$ or $b \in I$.

Example 15.6.41 (Prime Ideals of $\mathbb{Z}$)

The ideal $n\mathbb{Z} \trianglelefteq \mathbb{Z}$ is prime if and only if $n = 0$ or n is a prime number. Indeed, for $n = p$ prime this is the classical fact that $p \mid ab$ implies $p \mid a$ or $p \mid b$, and (0) is prime since $\mathbb{Z}$ is an integral domain. Conversely if $n = uv$ with $u, v > 1$ then $uv \in n\mathbb{Z}$ but $u, v \notin n\mathbb{Z}$.

Proposition 15.6.42

Let $I \trianglelefteq R$. Then I is prime if and only if R/I is an integral domain.

Proof. Unwinding definitions: R/I is an integral domain if and only if $(a+I)(b+I) = 0 + I$ implies $a + I = 0 + I$ or $b + I = 0 + I$; that is, if and only if $ab \in I$ implies $a \in I$ or $b \in I$. (And $I \neq R$ corresponds to $0 \neq 1$ in R/I .) $\square$

Corollary 15.6.43

Every maximal ideal is prime.

Proof. If I is maximal then R/I is a field, hence an integral domain, hence I is prime. $\square$

The converse fails: $(0) \leq \mathbb{Z}$ is prime but not maximal. As an application of this circle of ideas, note that if R is an integral domain of characteristic n , then $\mathbb{Z}/n\mathbb{Z}$ embeds in R , so $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, and thus n is 0 or prime. So *the characteristic of an integral domain (in particular, of any field) is either 0 or a prime number.*

§15.7 Factorisation in Integral Domains

We now begin the study of factorisation, which will occupy us for some time. The story we are trying to generalise is the fundamental theorem of arithmetic: every integer factors into primes, essentially uniquely. It turns out that in a general integral domain this can fail in interesting ways, and understanding when it holds leads to a hierarchy of classes of rings.

Throughout this section, R is an integral domain.

§15.7.1 Prime and Irreducible Elements

We first need the right vocabulary for talking about divisibility.

Definition 15.7.1 (Divisibility & Associates)

For $a, b \in R$, we say a **divides** b , written $a \mid b$, if $b = ac$ for some $c \in R$. Equivalently, $(b) \subseteq (a)$.

Elements a, b are **associates** if $a = bu$ for a unit u . Equivalently, $(a) = (b)$.

Associate elements are ‘the same up to units’ – for factorisation purposes they are interchangeable, just as $6 = 2 \cdot 3 = (-2)(-3)$ in $\mathbb{Z}$.

Definition 15.7.2 (Irreducible & Prime Elements)

Let $r \in R$ be non-zero and not a unit. Then:

- (i) r is **irreducible** if $r = ab$ implies a or b is a unit;
- (ii) r is **prime** if $r \mid ab$ implies $r \mid a$ or $r \mid b$.

In $\mathbb{Z}$ these two notions coincide (both giving the usual prime numbers, up to sign), but in general they do not, and the distinction is the crux of this whole section. Note that

both notions depend on the ambient ring: 2 is prime in $\mathbb{Z}$ but a unit in $\mathbb{Q}$, and $2X$ is irreducible in $\mathbb{Q}[X]$ but not in $\mathbb{Z}[X]$ (where 2 is not a unit).

Let us relate these notions to ideals.

Lemma 15.7.3

$(r) \trianglelefteq R$ is a prime ideal if and only if $r = 0$ or r is prime.

Proof. Suppose (r) is prime and $r \neq 0$. Since prime ideals are proper, r is not a unit. If $r \mid ab$ then $ab \in (r)$, so $a \in (r)$ or $b \in (r)$, that is, $r \mid a$ or $r \mid b$. So r is prime.

Conversely, $(0) = \{0\}$ is prime as R is an integral domain. If r is prime then $(r) \neq R$, and $ab \in (r)$ means $r \mid ab$, so $r \mid a$ or $r \mid b$, giving $a \in (r)$ or $b \in (r)$. $\square$

Lemma 15.7.4 (Primes are Irreducible)

If $r \in R$ is prime, then r is irreducible.

Proof. Suppose $r = ab$. Then certainly $r \mid ab$, so without loss of generality $r \mid a$, say $a = rc$. Then $r = rc b$, so $r(1 - cb) = 0$, and since $r \neq 0$ and R is an integral domain, $cb = 1$. So b is a unit. $\square$

The converse is *false* in general, and it's worth seeing this fail concretely.

Example 15.7.5 (An Irreducible That Isn't Prime)

Consider the ring

$$R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\} \leq \mathbb{C},$$

an integral domain (being a subring of $\mathbb{C}$). Define the **norm** $N : R \rightarrow \mathbb{Z}_{\geq 0}$ by $N(a + b\sqrt{-5}) = a^2 + 5b^2$, which is multiplicative: $N(zw) = N(z)N(w)$.

First, the units of R are ± 1 : if $rs = 1$ then $N(r)N(s) = 1$, so $N(r) = 1$, forcing $r = \pm 1$.

Next, 2 is irreducible in R : if $2 = rs$ with neither a unit, then $N(r)N(s) = 4$ with $N(r), N(s) > 1$, so $N(r) = 2$ – but $a^2 + 5b^2 = 2$ has no integer solutions. Similarly 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are irreducible (there are no elements of norm 3 either).

Now observe

$$2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

So $2 \mid (1 + \sqrt{-5})(1 - \sqrt{-5})$, but 2 divides neither factor (taking norms: $N(2) = 4 \nmid 6 = N(1 \pm \sqrt{-5})$). Hence 2 is irreducible but *not prime*.

Notice what we have really found: two genuinely different factorisations of 6 into irreducibles, not related by reordering and units. Unique factorisation fails in $\mathbb{Z}[\sqrt{-5}]$.

§15.7.2 Principal Ideal Domains

Definition 15.7.6 (Principal Ideal Domain)

An integral domain R is a **principal ideal domain** (PID) if every ideal of R is principal, that is, of the form (a) for some $a \in R$.

We have already shown that $\mathbb{Z}$ is a PID. In a PID, the pathology of the previous example cannot occur.

Proposition 15.7.7 (Irreducible $\iff$ Prime in a PID)

Let R be a principal ideal domain. Then every irreducible element of R is prime.

Proof. Let r be irreducible and suppose $r \mid ab$ but $r \nmid a$; we must show $r \mid b$. Since R is a PID, $(a, r) = (d)$ for some $d \in R$. Then $d \mid r$, say $r = cd$. As r is irreducible, c or d is a unit.

If c were a unit, then $(r) = (d) = (a, r) \ni a$, so $r \mid a$, a contradiction. So d is a unit, and $(a, r) = R$. In particular $1 = sa + tr$ for some $s, t \in R$. Multiplying by b ,

$$b = sab + trb,$$

and r divides both terms on the right ($r \mid ab$ for the first). Hence $r \mid b$. $\square$

Lemma 15.7.8 (Irreducibles Generate Maximal Ideals in a PID)

Let R be a PID and $r \in R$ be non-zero. Then r is irreducible if and only if (r) is maximal.

Proof. Suppose r is irreducible, so $(r) \neq R$. If $(r) \subseteq J \subseteq R$, then $J = (a)$ for some a (as R is a PID), and $r = ab$ for some b . Since r is irreducible, either a is a unit, giving $J = R$, or b is a unit, making a and r associates, giving $J = (r)$. So (r) is maximal.

Conversely if (r) is maximal then r is not a unit, and if $r = ab$ then $(r) \subseteq (a) \subseteq R$, so $(a) = (r)$ (making b a unit) or $(a) = R$ (making a a unit). So r is irreducible. (This direction holds in any integral domain.) $\square$

Combining our results: in a PID, for non-zero r ,

$$(r) \text{ maximal} \iff r \text{ irreducible} \iff r \text{ prime} \iff (r) \text{ prime}.$$

So in a PID, the non-zero prime ideals are exactly the maximal ideals.

§15.7.3 Euclidean Domains

How do we actually *prove* that a ring is a PID? For $\mathbb{Z}$, the key was the division algorithm. Abstracting this gives the following definition.

Definition 15.7.9 (Euclidean Domain)

An integral domain R is a **Euclidean domain** (ED) if there is a function $\phi : R \setminus \{0\} \rightarrow \mathbb{Z}_{\geq 0}$, called a **Euclidean function**, such that

- (i) if $a \mid b$ (with $a, b \neq 0$) then $\phi(a) \leq \phi(b)$,
- (ii) if $a, b \in R$ with $b \neq 0$, then there are $q, r \in R$ with

$$a = bq + r, \quad \text{and either } r = 0 \text{ or } \phi(r) < \phi(b).$$

Example 15.7.10

$\mathbb{Z}$ is a Euclidean domain with $\phi(n) = |n|$, and for a field F , the polynomial ring $F[X]$ is a Euclidean domain with $\phi(f) = \deg f$ – property (ii) is exactly the polynomial division algorithm, which applies since every non-zero element of F is a unit.

Proposition 15.7.11 (ED implies PID)

Every Euclidean domain is a principal ideal domain.

Proof. Let R have Euclidean function ϕ , and let $I \trianglelefteq R$ be a non-zero ideal. Choose $b \in I \setminus \{0\}$ with $\phi(b)$ minimal. Certainly $(b) \subseteq I$. Now take any $a \in I$, and write $a = bq + r$ with $r = 0$ or $\phi(r) < \phi(b)$. But $r = a - bq \in I$, so by minimality of $\phi(b)$ we must have $r = 0$. Hence $a \in (b)$, and $I = (b)$. $\square$

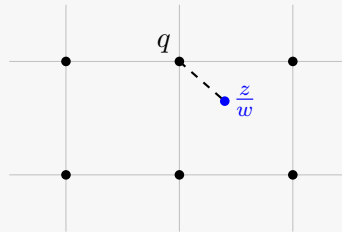
Remark. Only property (ii) was used here. The reason for including property (i) is that it lets us identify the units: they are exactly the non-zero elements with $\phi(u) = \phi(1)$, which is often handy in computations.

The most important new example is the ring of *Gaussian integers*.

Lemma 15.7.12 ($\mathbb{Z}[i]$ is a Euclidean Domain)

The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is a Euclidean domain with Euclidean function $N(a + bi) = a^2 + b^2$.

Proof. The norm is multiplicative, $N(zw) = N(z)N(w)$, which gives property (i) at once. For property (ii), let $z, w \in \mathbb{Z}[i]$ with $w \neq 0$, and consider the complex number $\frac{z}{w} \in \mathbb{C}$. The elements of $\mathbb{Z}[i]$ form a unit square lattice in $\mathbb{C}$, so there is some $q \in \mathbb{Z}[i]$ with $\left| \frac{z}{w} - q \right| < 1$ (any point of $\mathbb{C}$ has distance at most $\frac{\sqrt{2}}{2} < 1$ from the nearest lattice point).



Now set $r = z - wq \in \mathbb{Z}[i]$. Then $z = wq + r$, and if $r \neq 0$,

$$N(r) = |z - wq|^2 = |w|^2 \left| \frac{z}{w} - q \right|^2 < |w|^2 = N(w),$$

as required. □

So $\mathbb{Z}[i]$ is a PID, and $F[X]$ is a PID for any field F . We will use both facts extensively. Here are two quick illustrations of the power of the $F[X]$ case.

Example 15.7.13 (The Minimal Polynomial)

Let A be an $n \times n$ matrix over a field F , and let

$$I = \{f \in F[X] \mid f(A) = 0\}.$$

Then I is an ideal of $F[X]$ (differences and multiples of polynomials vanishing at A vanish at A). Since $F[X]$ is a PID, $I = (f)$ for a single polynomial f , which we may scale to be monic – this is precisely the *minimal polynomial* of A , and we get for free that it divides every polynomial killing A .

Example 15.7.14 (A Field of Order 8)

Let $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ and $f(X) = X^3 + X + 1 \in \mathbb{F}_2[X]$. If f factored non-trivially, it would have a linear factor and hence a root in $\mathbb{F}_2$, but $f(0) = f(1) = 1$. So f is irreducible, and since $\mathbb{F}_2[X]$ is a PID, the ideal (f) is maximal. Therefore

$$\mathbb{F}_2[X]/(X^3 + X + 1)$$

is a field. By the division algorithm each coset has a unique representative $aX^2 + bX + c$ with $a, b, c \in \mathbb{F}_2$, so this is a field with exactly 8 elements. This construction of finite fields is taken much further in Galois theory.

Not every PID-related fact extends to all polynomial rings, however.

Example 15.7.15 ($\mathbb{Z}[X]$ is Not a PID)

Consider the ideal $I = (2, X) \leq \mathbb{Z}[X]$, which consists of the polynomials with even constant term. Suppose $I = (f)$. Since $2 \in I$, we have $2 = fg$ for some g , so comparing degrees f is a constant, and $f = \pm 1$ or ± 2 . If $f = \pm 1$ then $I = \mathbb{Z}[X]$, but $1 \notin I$. If $f = \pm 2$ then $X \notin (f)$, but $X \in I$. Either way we have a contradiction, so I is not principal.

§15.7.4 Unique Factorisation Domains

We can now formalise ‘factorisation works like in $\mathbb{Z}$ ’.

Definition 15.7.16 (Unique Factorisation Domain)

An integral domain R is a **unique factorisation domain** (UFD) if

- (i) every non-zero, non-unit element is a product of irreducibles, and
- (ii) if $p_1 \cdots p_m = q_1 \cdots q_n$ with all p_i, q_j irreducible, then $m = n$ and, after reordering, p_i and q_i are associates for each i .

The uniqueness half turns out to be equivalent to the prime/irreducible distinction collapsing.

Proposition 15.7.17

Let R be an integral domain in which every non-zero, non-unit element is a product of irreducibles. Then R is a UFD if and only if every irreducible element of R is prime.

Proof. ($\Rightarrow$) Let p be irreducible with $p \mid ab$, say $ab = pc$. Writing a, b, c as products of irreducibles (or units), uniqueness of the factorisation of ab forces p to be an associate of one of the irreducible factors of a or of b ; hence $p \mid a$ or $p \mid b$.

($\Leftarrow$) Suppose every irreducible is prime, and $p_1 \cdots p_m = q_1 \cdots q_n$ with all factors irreducible, where without loss of generality $m \leq n$. Since p_1 is prime and divides $q_1 \cdots q_n$, it divides some q_i ; reorder so $p_1 \mid q_1$. Then $q_1 = p_1 u$, and u must be a unit as q_1 is irreducible and p_1 is not a unit. So p_1, q_1 are associates. Cancelling p_1 (valid in an integral domain) gives $p_2 \cdots p_m = u q_2 \cdots q_n$, and absorbing u into q_2 we may induct. If $m < n$ we would eventually find that a product of irreducibles equals a unit, which is impossible, so $m = n$ and all pairs are associates. $\square$

To show that PIDs are UFDs, the harder part is existence of factorisations, and for that we need a finiteness condition on chains of ideals – one which will reappear later when we study Noetherian rings.

Definition 15.7.18 (Ascending Chain Condition)

A ring R satisfies the **ascending chain condition** (ACC) if every increasing chain of ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ is eventually constant, that is, there is some N with $I_n = I_{n+1}$ for all $n \geq N$. Rings satisfying the ACC are called **Noetherian**.

Lemma 15.7.19 (PIDs Satisfy the ACC)

Every principal ideal domain satisfies the ascending chain condition.

Proof. Given a chain $I_1 \subseteq I_2 \subseteq \cdots$, the union $I = \bigcup_{i \geq 1} I_i$ is an ideal (a union of a chain of ideals is an ideal). Since R is a PID, $I = (a)$ for some a , and $a \in I_N$ for some N . Then for all $n \geq N$,

$$(a) \subseteq I_N \subseteq I_n \subseteq I = (a),$$

so all inclusions are equalities. $\square$

Theorem 15.7.20 (PID implies UFD)

Every principal ideal domain is a unique factorisation domain.

Proof. We first show every non-zero, non-unit x is a product of irreducibles. Suppose not. Then in particular x is not itself irreducible, so $x = x_1 y_1$ with neither factor a unit, and at least one of them – say x_1 – is not a product of irreducibles. Note $(x) \subset (x_1)$, and this inclusion is strict since y_1 is not a unit. Repeating with x_1 in

place of x , we build a strictly increasing chain

$$(x) \subset (x_1) \subset (x_2) \subset \cdots,$$

contradicting the ascending chain condition. So factorisations exist.

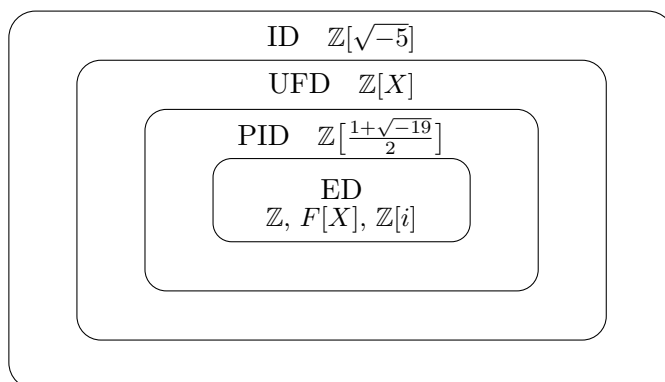
For uniqueness, by the previous proposition it suffices that every irreducible is prime – which we proved for PIDs earlier. $\square$

§15.7.5 The Hierarchy of Integral Domains

Assembling everything so far, we have a chain of implications

$$\text{ED} \implies \text{PID} \implies \text{UFD} \implies \text{ID},$$

and each implication is strict. The picture to have in mind is the following, with a witness sitting in each gap.



Example 15.7.21 (Separating Examples)

Let's justify the diagram.

- (i) $\mathbb{Z}$, $F[X]$ and $\mathbb{Z}[i]$ are Euclidean domains, as we have seen.
- (ii) $\mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ is a PID but not a Euclidean domain. (Both facts require some work, and are beyond this course – this example is treated in Part II Number Fields.)
- (iii) $\mathbb{Z}[X]$ is a UFD (as we will prove in the next section) but not a PID, since the ideal $(2, X)$ is not principal.
- (iv) $\mathbb{Z}[\sqrt{-5}]$ is an integral domain but not a UFD, since 6 factors into irreducibles in two genuinely different ways.

And of course $\mathbb{Z}/4\mathbb{Z}$ fails even to be an integral domain, as $2 \cdot 2 = 0$.

§15.7.6 Greatest Common Divisors

One pleasant consequence of unique factorisation is a good theory of greatest common divisors.

Definition 15.7.22 (GCD and LCM)

Let $a_1, \dots, a_n \in R$. An element $d \in R$ is a **greatest common divisor** of $a_1, \dots, a_n$ if $d \mid a_i$ for all i , and any d' with $d' \mid a_i$ for all i satisfies $d' \mid d$.

Similarly, m is a **least common multiple** if $a_i \mid m$ for all i , and every common multiple m' satisfies $m \mid m'$.

Greatest common divisors need not exist in a general integral domain, but when they do they are unique up to associates – so we allow ourselves to speak of ‘the’ gcd.

Proposition 15.7.23 (GCDs Exist in UFDs)

If R is a UFD, then greatest common divisors and least common multiples always exist.

Proof. Write each $a_i = u_i \prod_j p_j^{n_{ij}}$, where the p_j are pairwise non-associate irreducibles, u_i are units, and $n_{ij} \geq 0$. We claim

$$d = \prod_j p_j^{m_j}, \quad \text{where } m_j = \min_i n_{ij},$$

is a greatest common divisor. Certainly d divides each a_i . If $d' \mid a_i$ for all i , then writing d' as a product of irreducibles, uniqueness of factorisation shows that (up to associates) $d' = w \prod_j p_j^{t_j}$ with $t_j \leq n_{ij}$ for every i , so $t_j \leq m_j$ and $d' \mid d$. The lcm is analogous, taking $\max_i n_{ij}$ instead. $\square$

§15.8 Factorisation in Polynomial Rings

The goal of this section is the following theorem, together with some practical tools it generates for showing that specific polynomials are irreducible.

Theorem 15.8.1 (Polynomial Rings over UFDs)

If R is a unique factorisation domain, then $R[X]$ is a unique factorisation domain.

Note that this is exactly the sort of statement our hierarchy cannot give us directly: $\mathbb{Z}$ is a PID but $\mathbb{Z}[X]$ is not, so we cannot pass through principal ideals. Instead we will compare $R[X]$ with $F[X]$, where F is the field of fractions of R . Since $F[X]$ is a Euclidean domain, it is a UFD, and the strategy is to transfer factorisations back and forth between $R[X]$ and $F[X]$.

Throughout this section, R is a UFD with field of fractions F , so $R[X] \leq F[X]$.

§15.8.1 Content and Gauss’s Lemma

The bookkeeping device that makes the comparison work is the *content* of a polynomial.

Definition 15.8.2 (Content & Primitive Polynomials)

The **content** of $f = a_0 + a_1X + \cdots + a_nX^n \in R[X]$ is

$$c(f) = \gcd(a_0, a_1, \dots, a_n),$$

which is well defined up to associates (using that R is a UFD). We say f is **primitive** if $c(f)$ is a unit.

Any $f \in R[X]$ can be written $f = c(f)f_0$ with f_0 primitive – just pull out the gcd of the coefficients. The key technical fact is that primitivity is preserved by products.

Lemma 15.8.3 (Products of Primitive Polynomials)

If $f, g \in R[X]$ are primitive, then so is fg . Moreover, for any $f, g \in R[X]$, the elements $c(fg)$ and $c(f)c(g)$ are associates.

Proof. Let $f = \sum_i a_i X^i$ and $g = \sum_i b_i X^i$ be primitive, and suppose fg is not, so some irreducible (hence prime) $p \in R$ divides $c(fg)$. Since f and g are primitive, p does not divide all the a_i , nor all the b_i . Let k and ℓ be minimal with $p \nmid a_k$ and $p \nmid b_\ell$. The coefficient of $X^{k+\ell}$ in fg is

$$\sum_{i+j=k+\ell} a_i b_j = \underbrace{(\cdots + a_{k-1} b_{\ell+1})}_{\text{divisible by } p} + a_k b_\ell + \underbrace{(a_{k+1} b_{\ell-1} + \cdots)}_{\text{divisible by } p},$$

where the flanking terms are divisible by p by minimality of k and ℓ . Since p divides the whole coefficient, $p \mid a_k b_\ell$, so $p \mid a_k$ or $p \mid b_\ell$ as p is prime – a contradiction.

For the second part, write $f = c(f)f_0$ and $g = c(g)g_0$ with f_0, g_0 primitive. Then $fg = c(f)c(g)f_0g_0$ with f_0g_0 primitive, so taking contents gives $c(fg) = c(f)c(g)$ up to a unit. $\square$

Corollary 15.8.4

If $p \in R$ is prime in R , then p is prime as an element of $R[X]$.

Proof. Note $R[X]^\times = R^\times$ (degrees add in an integral domain), so p is a non-zero non-unit in $R[X]$. For $f \in R[X]$, we have $p \mid f$ in $R[X]$ if and only if $p \mid c(f)$ in R . So if $p \mid gh$, then $p \mid c(gh) = c(g)c(h)$ (up to units), hence $p \mid c(g)$ or $p \mid c(h)$, that is, $p \mid g$ or $p \mid h$ in $R[X]$. $\square$

The next lemma is the bridge between divisibility in $F[X]$ and in $R[X]$.

Lemma 15.8.5 (Descending Divisibility)

Let $f, g \in R[X]$ with g primitive. If $g \mid f$ in $F[X]$, then $g \mid f$ in $R[X]$.

Proof. Write $f = gh$ with $h \in F[X]$. Clearing denominators, there is a non-zero $a \in R$ with $ah \in R[X]$, and we can write $ah = c(ah)h_0$ with $h_0 \in R[X]$ primitive. Then

$$af = g \cdot ah = c(ah)gh_0,$$

and taking contents: $a \cdot c(f) = c(ah) \cdot c(gh_0) = c(ah)$ up to units, since gh_0 is a product of primitive polynomials, hence primitive. In particular $a \mid c(ah)$, so $h = \frac{c(ah)}{a}h_0 \in R[X]$, and $g \mid f$ in $R[X]$. $\square$

We can now prove the classical result relating irreducibility over R and over the field of fractions.

Lemma 15.8.6 (Gauss's Lemma)

Let $f \in R[X]$ be primitive. Then f is irreducible in $R[X]$ if and only if f is irreducible in $F[X]$.

Proof. ($\Rightarrow$) Since f is irreducible and primitive, $\deg f > 0$, so f is not a unit in $F[X]$. Suppose for contradiction that $f = gh$ with $g, h \in F[X]$ of positive degree. We claim we can arrange for g to lie in $R[X]$ and be primitive. Indeed, choose $b \in R$ non-zero with $bg \in R[X]$, and write $bg = c(bg)g_0$ with g_0 primitive; then setting $\lambda = c(bg)b^{-1} \in F^\times$, we have $g = \lambda g_0$, and replacing (g, h) with $(g_0, \lambda h)$ keeps $f = g_0(\lambda h)$. So assume $g \in R[X]$ is primitive. Then $g \mid f$ in $F[X]$, so by the previous lemma $g \mid f$ in $R[X]$, say $f = gh'$ with $h' \in R[X]$, where $\deg g, \deg h' > 0$. This contradicts irreducibility of f in $R[X]$.

($\Leftarrow$) If $f = gh$ in $R[X]$ with g, h non-units, then since f is primitive neither factor can be constant, so both have positive degree and this also factors f non-trivially in $F[X]$. $\square$

Lemma 15.8.7

Let $g \in R[X]$ be primitive. If g is prime in $F[X]$, then g is prime in $R[X]$.

Proof. g is not a unit in $R[X]$ (it has positive degree, as primitive constants are units). If $g \mid f_1f_2$ with $f_i \in R[X]$, then in $F[X]$ we get $g \mid f_1$ or $g \mid f_2$, and descending divisibility brings this back to $R[X]$. $\square$

With these tools, the main theorem falls out.

Proof that polynomial rings over UFDs are UFDs. Existence. Let $f \in R[X]$ be a non-zero non-unit, and write $f = c(f)f_0$ with f_0 primitive. Since R is a UFD, $c(f)$ factors into irreducibles of R , which remain irreducible in $R[X]$. For f_0 : if it is not irreducible, then $f_0 = gh$ with g, h non-units, and since f_0 is primitive both g and h have positive degree (and are primitive). By induction on degree, f_0 is a product of irreducibles.

Uniqueness. It suffices to show every irreducible $f \in R[X]$ is prime. Since f is irreducible, it is either a constant or primitive. If f is a constant, it is irreducible in R , hence prime in R (R being a UFD), hence prime in $R[X]$ by our corollary. If f is primitive, then by Gauss's lemma f is irreducible in $F[X]$; since $F[X]$ is a UFD, f is prime in $F[X]$, and by the previous lemma f is prime in $R[X]$. $\square$

Example 15.8.8

$\mathbb{Z}[X]$ is a UFD, even though it is not a PID. Iterating the theorem, $R[X_1, \dots, X_n] = R[X_1] \cdots [X_n]$ is a UFD whenever R is – for instance $\mathbb{Z}[X_1, \dots, X_n]$ and $F[X_1, \dots, X_n]$ for a field F .

§15.8.2 Eisenstein's Criterion

Gauss's lemma reduces irreducibility over $\mathbb{Q}$ to irreducibility over $\mathbb{Z}$, where we have more structure to exploit. The following criterion is the classic way to do so.

Theorem 15.8.9 (Eisenstein's Criterion)

Let R be a UFD and let $f(X) = a_n X^n + \cdots + a_1 X + a_0 \in R[X]$ be primitive with $\deg f = n \geq 1$. Suppose there is an irreducible $p \in R$ such that

- (i) $p \nmid a_n$,
- (ii) $p \mid a_i$ for all $i < n$,
- (iii) $p^2 \nmid a_0$.

Then f is irreducible in $R[X]$ (and hence in $F[X]$, by Gauss's lemma).

Proof. Suppose $f = gh$ with g, h non-units. Since f is primitive, g and h have positive degrees k and ℓ , with $k + \ell = n$. Write $g = \sum_{i=0}^k r_i X^i$ and $h = \sum_{i=0}^{\ell} s_i X^i$.

Since $p \nmid a_n = r_k s_\ell$, we have $p \nmid r_k$ and $p \nmid s_\ell$. Since $p \mid a_0 = r_0 s_0$ but $p^2 \nmid a_0$, exactly one of r_0, s_0 is divisible by p – without loss of generality $p \mid r_0$ and $p \nmid s_0$.

Let $j \leq k$ be minimal with $p \nmid r_j$ (this exists as $p \nmid r_k$), so $j \geq 1$. Consider

$$a_j = r_0 s_j + r_1 s_{j-1} + \cdots + r_{j-1} s_1 + r_j s_0.$$

We have $j \leq k < n$, so $p \mid a_j$, and p divides each of the first j terms by minimality of j . Hence $p \mid r_j s_0$, but $p \nmid r_j$ and $p \nmid s_0$ – a contradiction, as p is prime (R is a UFD). $\square$

Example 15.8.10

For a prime $p \in \mathbb{Z}$, the polynomial $X^n - p$ satisfies Eisenstein's criterion at p , so it is irreducible in $\mathbb{Z}[X]$, hence in $\mathbb{Q}[X]$. In particular $\sqrt[n]{p}$ is irrational for $n \geq 2$, and $\mathbb{Q}[X]/(X^n - p)$ is a field.

Sometimes the criterion doesn't apply directly, but does after a change of variable.

Example 15.8.11 (Cyclotomic Polynomials of Prime Order)

Let p be prime and consider

$$f(X) = X^{p-1} + X^{p-2} + \cdots + X + 1 = \frac{X^p - 1}{X - 1} \in \mathbb{Z}[X].$$

Eisenstein does not apply to f directly (all the coefficients are 1). But substituting

$$X = Y + 1,$$

$$f(Y + 1) = \frac{(Y + 1)^p - 1}{Y} = Y^{p-1} + \binom{p}{1}Y^{p-2} + \cdots + \binom{p}{p-2}Y + \binom{p}{p-1}.$$

Now $p \mid \binom{p}{i}$ for $1 \leq i \leq p-1$, and $\binom{p}{p-1} = p$ is not divisible by p^2 , so Eisenstein applies at p and $f(Y + 1)$ is irreducible. Since $X \mapsto Y + 1$ is an isomorphism $\mathbb{Z}[X] \rightarrow \mathbb{Z}[Y]$, the polynomial $f(X)$ is irreducible in $\mathbb{Z}[X]$, and thus in $\mathbb{Q}[X]$.

Example 15.8.12 (When Eisenstein Fails, Try Roots)

Consider $f(X) = X^3 + 2X + 5 \in \mathbb{Z}[X]$. No prime works for Eisenstein, but if f factored in $\mathbb{Z}[X]$, then (being cubic) it would have a linear factor $X + a$ with $a \mid 5$, so an integer root among $\pm 1, \pm 5$. Checking, none of these are roots, so f is irreducible in $\mathbb{Z}[X]$, and by Gauss's lemma also in $\mathbb{Q}[X]$. Hence $\mathbb{Q}[X]/(f)$ is a field.

§15.9 The Gaussian Integers

The ring $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ deserves a section of its own – it is a Euclidean domain (with the norm $N(a + bi) = a^2 + b^2$ as Euclidean function), hence a PID and a UFD, and this structure turns out to encode genuinely arithmetic information about $\mathbb{Z}$: which integers are sums of two squares.

Since N is multiplicative and $N(z) = 1 \iff z \in \{\pm 1, \pm i\}$, the units of $\mathbb{Z}[i]$ are exactly $\pm 1, \pm i$. Also note the handy fact that if $N(z)$ is prime in $\mathbb{Z}$, then z is irreducible in $\mathbb{Z}[i]$, since any factorisation splits the norm.

Example 15.9.1

The prime 2 is *not* irreducible in $\mathbb{Z}[i]$: we have $2 = (1 + i)(1 - i)$, and neither factor is a unit ($N(1 \pm i) = 2$). Similarly $5 = (2 + i)(2 - i)$ is not irreducible. But 3 *is* irreducible: $N(3) = 9$, so a non-trivial factorisation $3 = zw$ would need $N(z) = 3$, and $a^2 + b^2 = 3$ has no integer solutions. The same argument shows 7 is irreducible.

There is a clear pattern here, and pinning it down is the main theorem of this section.

Proposition 15.9.2

Let $p \in \mathbb{Z}$ be a prime number. The following are equivalent:

- (i) p is not prime in $\mathbb{Z}[i]$;
- (ii) $p = a^2 + b^2$ for some $a, b \in \mathbb{Z}$;
- (iii) $p = 2$ or $p \equiv 1 \pmod{4}$.

Proof. (i) $\Rightarrow$ (ii): Since $\mathbb{Z}[i]$ is a UFD, ‘not prime’ means ‘not irreducible’, so $p = zw$ with z, w non-units. Then $p^2 = N(p) = N(z)N(w)$ with $N(z), N(w) > 1$, forcing $N(z) = p$. Writing $z = a + bi$ gives $p = a^2 + b^2$.

(ii) $\Rightarrow$ (iii): The squares modulo 4 are 0 and 1, so $a^2 + b^2 \not\equiv 3 \pmod{4}$. Since p is

prime, either $p = 2$ or p is odd with $p \equiv 1 \pmod{4}$.

(iii) $\Rightarrow$ (i): We know $2 = (1+i)(1-i)$ is not prime, so suppose $p \equiv 1 \pmod{4}$. The group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic of order $p-1$, and $4 \mid p-1$, so there is an element x of order 4. Then x^2 has order 2, and the only element of order 2 in the cyclic group is -1 , so $x^2 \equiv -1 \pmod{p}$. Hence

$$p \mid x^2 + 1 = (x+i)(x-i) \quad \text{in } \mathbb{Z}[i].$$

But $p \nmid x \pm i$, since $\frac{x}{p} \pm \frac{1}{p}i \notin \mathbb{Z}[i]$. So p is not prime in $\mathbb{Z}[i]$. $\square$

It is worth pausing here: the statement ‘every prime $p \equiv 1 \pmod{4}$ is a sum of two squares’ is a theorem of Fermat, and makes no mention of complex numbers at all – yet the natural proof goes through the arithmetic of $\mathbb{Z}[i]$. Fermat’s original approach was his method of *descent*: from any solution of $a^2 + b^2 \equiv 0 \pmod{p}$ he constructed a smaller multiple of p expressible as a sum of two squares, descending until reaching p itself. The Euclidean algorithm in $\mathbb{Z}[i]$ is doing exactly this descent for us, packaged as the statement that $\mathbb{Z}[i]$ is a PID.

We can now describe all the primes of $\mathbb{Z}[i]$.

Theorem 15.9.3 (Primes in $\mathbb{Z}[i]$)

Up to associates, the primes of $\mathbb{Z}[i]$ are:

- (i) elements $a + bi$ with $N(a + bi) = a^2 + b^2 = p$ a prime number with $p = 2$ or $p \equiv 1 \pmod{4}$;
- (ii) prime numbers $p \in \mathbb{Z}$ with $p \equiv 3 \pmod{4}$.

Proof. Elements of type (i) are irreducible since their norm is prime, and elements of type (ii) are prime by the previous proposition. Conversely, let z be prime in $\mathbb{Z}[i]$, and note $\bar{z}$ is then also prime, with

$$N(z) = z\bar{z}$$

a factorisation into primes. Pick a prime number $p \in \mathbb{Z}$ dividing $N(z) \in \mathbb{Z}$.

If $p \equiv 3 \pmod{4}$, then p is prime in $\mathbb{Z}[i]$ and divides $z\bar{z}$, so $p \mid z$ or $p \mid \bar{z}$; in either case z is an associate of p (a prime dividing a prime), which is type (ii).

Otherwise $p = 2$ or $p \equiv 1 \pmod{4}$, so $p = a^2 + b^2 = (a + bi)(a - bi)$ with both factors of norm p , hence prime. Then $(a + bi)(a - bi) \mid z\bar{z}$, and unique factorisation in $\mathbb{Z}[i]$ shows z is an associate of $a + bi$ or $a - bi$, which is type (i). $\square$

Corollary 15.9.4 (Sums of Two Squares)

A positive integer n is a sum of two squares if and only if every prime $p \equiv 3 \pmod{4}$ appears in the factorisation of n to an even power.

Proof. Note that n is a sum of two squares if and only if $n = N(z)$ for some $z \in \mathbb{Z}[i]$. Factoring z into primes of $\mathbb{Z}[i]$ and taking norms, n is a product of norms of primes

in $\mathbb{Z}[i]$, which by the classification are the primes $p \not\equiv 3 \pmod{4}$ and the *squares* p^2 of primes $p \equiv 3 \pmod{4}$. The stated condition describes exactly the integers expressible as products of such numbers. Conversely if the condition holds, writing each $p \not\equiv 3 \pmod{4}$ as a norm and each p^{2k} as $N(p^k)$, multiplicativity of N expresses n as a norm. $\square$

Example 15.9.5 (Writing 65 as a Sum of Two Squares)

We have $65 = 5 \cdot 13$, and both primes are $1 \pmod{4}$, so 65 should be a sum of two squares. Factoring in $\mathbb{Z}[i]$:

$$5 = (2 + i)(2 - i), \quad 13 = (2 + 3i)(2 - 3i).$$

Grouping the four prime factors into conjugate pairs,

$$65 = N((2 + i)(2 + 3i)) = N(1 + 8i) = 1^2 + 8^2,$$

but the alternative grouping gives

$$65 = N((2 + i)(2 - 3i)) = N(7 - 4i) = 7^2 + 4^2.$$

So $65 = 1 + 64 = 49 + 16$ – and the two essentially different representations come precisely from the two ways of grouping the Gaussian primes.

§15.10 Algebraic Integers

We now apply our factorisation theory to a class of complex numbers of central importance in number theory.

Definition 15.10.1 (Algebraic Numbers and Algebraic Integers)

A number $\alpha \in \mathbb{C}$ is **algebraic** if it is a root of some non-zero polynomial in $\mathbb{Q}[X]$, and α is an **algebraic integer** if it is a root of some *monic* polynomial in $\mathbb{Z}[X]$.

For example, $\sqrt{2}$, i , and $\frac{-1+\sqrt{-3}}{2}$ (a root of $X^2 + X + 1$) are algebraic integers, while $\frac{1}{2}$ is algebraic but – as we shall see – not an algebraic integer.

For a subring $R \leq S$ and $\alpha \in S$, we write $R[\alpha]$ for the smallest subring of S containing both R and α . Concretely, $R[\alpha] = \text{img}(\phi)$, where $\phi : R[X] \rightarrow S$ is the **evaluation homomorphism** $g(X) \mapsto g(\alpha)$. This notation is consistent with our earlier usage of $\mathbb{Z}[i]$ and $\mathbb{Z}[\sqrt{-5}]$.

Definition 15.10.2 (Minimal Polynomial)

Let α be an algebraic number, and consider the evaluation homomorphism $\phi : \mathbb{Q}[X] \rightarrow \mathbb{C}$, $g \mapsto g(\alpha)$. Since $\mathbb{Q}[X]$ is a PID, $\ker(\phi) = (f)$ for some f , which is non-zero as α is algebraic, and which we scale to be monic. This f is the **minimal polynomial** of α .

Proposition 15.10.3

The minimal polynomial f of an algebraic number α is irreducible in $\mathbb{Q}[X]$, and $\mathbb{Q}[\alpha] \cong \mathbb{Q}[X]/(f)$ is a field.

Proof. By the first isomorphism theorem, $\mathbb{Q}[X]/(f) \cong \mathbb{Q}[\alpha] \leq \mathbb{C}$. A subring of a field is an integral domain, so (f) is a prime ideal, and hence f is prime, so irreducible. But then (f) is a *maximal* ideal ($\mathbb{Q}[X]$ is a PID), so $\mathbb{Q}[X]/(f) \cong \mathbb{Q}[\alpha]$ is a field. $\square$

The corresponding statement over $\mathbb{Z}$ requires more care, because $\mathbb{Z}[X]$ is not a PID – and this is exactly where Gauss’s lemma earns its keep.

Proposition 15.10.4

Let α be an algebraic *integer* with minimal polynomial $f \in \mathbb{Q}[X]$. Then $f \in \mathbb{Z}[X]$, and $\ker(\theta) = (f)$, where $\theta : \mathbb{Z}[X] \rightarrow \mathbb{C}$ is the evaluation homomorphism at α .

Proof. Choose $\lambda \in \mathbb{Q}^\times$ so that $\lambda f \in \mathbb{Z}[X]$ is primitive (clear denominators, then divide by the content). Then $\lambda f(\alpha) = 0$, so $\lambda f \in \ker(\theta)$.

We claim $\ker(\theta) = (\lambda f)$. Indeed, let $g \in \ker(\theta) \subseteq \mathbb{Z}[X]$. Then $g(\alpha) = 0$, so $f \mid g$ in $\mathbb{Q}[X]$, and hence $\lambda f \mid g$ in $\mathbb{Q}[X]$. Since λf is primitive, descending divisibility gives $\lambda f \mid g$ in $\mathbb{Z}[X]$, proving the claim.

Now, since α is an algebraic integer, there is a *monic* $g \in \mathbb{Z}[X]$ with $g(\alpha) = 0$, so $\lambda f \mid g$ in $\mathbb{Z}[X]$. Comparing leading coefficients, the leading coefficient of λf divides 1, so it is ± 1 ; and as f is monic, $\lambda = \pm 1$. Hence $f = \pm \lambda f \in \mathbb{Z}[X]$ and $\ker(\theta) = (f)$. $\square$

Applying the first isomorphism theorem to θ gives presentations of our favourite rings:

$$\mathbb{Z}[X]/(X^2+1) \cong \mathbb{Z}[i], \quad \mathbb{Z}[X]/(X^2-2) \cong \mathbb{Z}[\sqrt{2}], \quad \mathbb{Z}[X]/(X^2+X+1) \cong \mathbb{Z}\left[\frac{-1+\sqrt{-3}}{2}\right],$$

and $\mathbb{Z}[X]/(X^n - p) \cong \mathbb{Z}[\sqrt[n]{p}]$ for a prime p .

Corollary 15.10.5 (Rational Algebraic Integers)

If $\alpha \in \mathbb{Q}$ is an algebraic integer, then $\alpha \in \mathbb{Z}$.

Proof. If $\alpha \neq 0$ is rational, its minimal polynomial is $X - \alpha$, which by the previous proposition lies in $\mathbb{Z}[X]$. Hence $\alpha \in \mathbb{Z}$. (And $0 \in \mathbb{Z}$ trivially.) $\square$

This innocuous corollary is a genuinely useful tool – for instance, it is the cleanest way to see that $\frac{1}{2}$ is not an algebraic integer, and it underlies many irrationality proofs.

§15.11 Noetherian Rings

We finish our study of rings by returning to the ascending chain condition, which we met while proving that PIDs are UFDs. It defines a class of rings large enough to contain almost everything we care about, but restrictive enough to do serious work.

Definition 15.11.1 (Noetherian Ring)

A ring R is **Noetherian** if every ascending chain of ideals $I_1 \subseteq I_2 \subseteq \cdots$ is eventually constant: there is N such that $I_n = I_{n+1}$ for all $n \geq N$.

There is an equivalent characterisation which is often easier to verify and to use.

Lemma 15.11.2 (Noetherian $\iff$ Finitely Generated Ideals)

A ring R is Noetherian if and only if every ideal of R is finitely generated.

Proof. ($\Leftarrow$) Let $I_1 \subseteq I_2 \subseteq \cdots$ be a chain and $I = \bigcup_i I_i$, an ideal. By assumption $I = (a_1, \dots, a_k)$, and each a_j lies in some I_{n_j} ; taking $N = \max_j n_j$, all the generators lie in I_N , so for $n \geq N$,

$$I = (a_1, \dots, a_k) \subseteq I_N \subseteq I_n \subseteq I,$$

and the chain is constant from I_N onwards.

($\Rightarrow$) Suppose some ideal J is not finitely generated. Pick $a_1 \in J$; then $(a_1) \neq J$, so pick $a_2 \in J \setminus (a_1)$, and inductively $a_{n+1} \in J \setminus (a_1, \dots, a_n)$ – possible since $(a_1, \dots, a_n)$ is finitely generated but J is not. This produces a strictly increasing chain $(a_1) \subset (a_1, a_2) \subset \cdots$, contradicting the ACC. $\square$

In particular every PID is Noetherian – each ideal is generated by a single element. Fields are Noetherian for the same reason. The fundamental theorem about Noetherian rings is that the property passes to polynomial rings.

Theorem 15.11.3 (Hilbert Basis Theorem)

If R is Noetherian, then $R[X]$ is Noetherian.

Proof. Suppose for contradiction that $J \trianglelefteq R[X]$ is not finitely generated. Let $f_1 \in J$ have minimal degree; given $f_1, \dots, f_n$, let $f_{n+1} \in J \setminus (f_1, \dots, f_n)$ have minimal degree – such an element exists as J is not finitely generated. Note the degrees $\deg f_1 \leq \deg f_2 \leq \cdots$ are non-decreasing by construction.

Let $a_i \in R$ be the leading coefficient of f_i . In R we have the chain of ideals

$$(a_1) \subseteq (a_1, a_2) \subseteq (a_1, a_2, a_3) \subseteq \cdots,$$

which since R is Noetherian is eventually constant – so there is m with $a_{m+1} \in (a_1, \dots, a_m)$, say $a_{m+1} = \sum_{i=1}^m \lambda_i a_i$ with $\lambda_i \in R$. Now consider

$$g(X) = \sum_{i=1}^m \lambda_i X^{\deg f_{m+1} - \deg f_i} f_i \in (f_1, \dots, f_m),$$

which is engineered to have the same degree and the same leading coefficient as f_{m+1} . Then $f_{m+1} - g \in J$ with $\deg(f_{m+1} - g) < \deg f_{m+1}$, so by minimality of $\deg f_{m+1}$ we must have $f_{m+1} - g \in (f_1, \dots, f_m)$. But then $f_{m+1} = (f_{m+1} - g) + g \in (f_1, \dots, f_m)$, contradicting our choice of f_{m+1} . $\square$

Corollary 15.11.4

$\mathbb{Z}[X_1, \dots, X_n]$ and $F[X_1, \dots, X_n]$ (for a field F) are Noetherian.

Lemma 15.11.5 (Quotients of Noetherian Rings)

If R is Noetherian and $I \trianglelefteq R$, then R/I is Noetherian.

Proof. A chain of ideals $J'_1 \subseteq J'_2 \subseteq \dots$ in R/I corresponds, via the ideal correspondence, to a chain $J_1 \subseteq J_2 \subseteq \dots$ of ideals of R containing I , with $J'_i = J_i/I$. The latter chain stabilises since R is Noetherian, so the former does too. $\square$

So, for example, *any finitely generated ring* – any quotient of some $\mathbb{Z}[X_1, \dots, X_n]$ – is Noetherian. Let's see why one might care.

Example 15.11.6 (Algebraic Varieties)

Let $R = \mathbb{C}[X_1, \dots, X_n]$, and let $\mathcal{F} \subseteq R$ be any (possibly infinite) set of polynomials, with solution set

$$V = \{(a_1, \dots, a_n) \in \mathbb{C}^n \mid f(a_1, \dots, a_n) = 0 \text{ for all } f \in \mathcal{F}\},$$

an **algebraic variety**. Let $I \trianglelefteq R$ be the ideal generated by $\mathcal{F}$. Since R is Noetherian, $I = (g_1, \dots, g_r)$ is finitely generated, and V is exactly the common zero set of $g_1, \dots, g_r$. So *every* system of polynomial equations, however large, is equivalent to a finite system.

Example 15.11.7 (Non-Noetherian Rings)

The ring $\mathbb{Z}[X_1, X_2, \dots]$ of polynomials in countably many variables is not Noetherian: the chain $(X_1) \subset (X_1, X_2) \subset (X_1, X_2, X_3) \subset \dots$ never stabilises. Note that this ring is an integral domain, and in fact a UFD – so ‘UFD’ does not imply ‘Noetherian’.

More surprisingly, a subring of a Noetherian ring need not be Noetherian. Take

$$R = \{f \in \mathbb{Q}[X] \mid f(0) \in \mathbb{Z}\} \leq \mathbb{Q}[X].$$

Then $(X) \subset (\frac{X}{2}) \subset (\frac{X}{4}) \subset \dots$ is a strictly increasing chain in R (each inclusion is strict because 2 is not a unit in R), even though $\mathbb{Q}[X]$ itself is a Euclidean domain.

§15.12 Modules

We now come to the final act of the course. A *module* is what you get when you take the definition of a vector space and allow the scalars to come from a ring rather than a field. This modest-sounding change has dramatic consequences – most of the comfortable facts of linear algebra (every vector space has a basis, spanning sets contain bases, and so on) fail for general modules – but for modules over sufficiently nice rings we will prove a powerful structure theorem, and from it deduce, in one stroke, the classification of finite abelian groups *and* the theory of canonical forms of matrices (including Jordan normal

form).

§15.12.1 Definitions and Examples

Definition 15.12.1 (Module)

Let R be a ring. An R -module is a triple $(M, +, \cdot)$ consisting of a set M , an operation $+: M \times M \rightarrow M$, and a scalar multiplication $\cdot: R \times M \rightarrow M$, such that

- (i) $(M, +)$ is an abelian group with identity $0 = 0_M$,
- (ii) $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$,
- (iii) $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$,
- (iv) $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$,
- (v) $1_R \cdot m = m$,

for all $r, r_1, r_2 \in R$ and $m, m_1, m_2 \in M$.

These are exactly the vector space axioms, with the field replaced by a ring. The real content of the definition is in the examples.

Example 15.12.2 (Examples of Modules)

The following are all modules.

- (i) If $R = F$ is a field, an F -module is precisely a vector space over F .
- (ii) A $\mathbb{Z}$ -module is precisely an abelian group. Indeed, given an abelian group A , the only possible $\mathbb{Z}$ -action is

$$n \cdot a = \begin{cases} a + \cdots + a & (n \text{ times, if } n > 0), \\ 0 & (n = 0), \\ -(a + \cdots + a) & (|n| \text{ times, if } n < 0), \end{cases}$$

and this does define a module structure. So the theory of $\mathbb{Z}$ -modules *is* the theory of abelian groups.

- (iii) Let F be a field, V a vector space over F , and $\alpha: V \rightarrow V$ a linear map. Then V becomes an $F[X]$ -module, denoted V_α , via

$$f(X) \cdot v = f(\alpha)(v).$$

Note that the module structure remembers α , not just V – different choices of α give genuinely different $F[X]$ -modules on the same underlying space. This example will drive everything we do with canonical forms.

- (iv) Any ring R is an R -module over itself, and more generally R^n is an R -module with coordinatewise operations.
- (v) An ideal $I \trianglelefteq R$ is an R -module, as is the quotient ring R/I , with $r \cdot (s + I) = rs + I$.

- (vi) If $\phi : R \rightarrow S$ is a ring homomorphism, then any S -module M becomes an R -module via $r \cdot m = \phi(r) \cdot m$. In particular, a module over a ring is a module over any subring.

Definition 15.12.3 (Submodule)

Let M be an R -module. A subset $N \subseteq M$ is an **R -submodule**, written $N \leq M$, if N is a subgroup of $(M, +)$ and $rn \in N$ for all $r \in R, n \in N$.

Example 15.12.4

Viewing R as an R -module over itself, the submodules of R are exactly the ideals of R . If $R = F$ is a field, submodules are exactly vector subspaces.

It is worth noticing what has happened here: subrings and ideals, which looked like two different generalisations of ‘subgroup’, are unified by the module point of view – and it is the ideal, not the subring, that is the module-theoretic notion.

Definition 15.12.5 (Quotient Module)

Let $N \leq M$ be R -modules. The **quotient module** M/N is the quotient abelian group with scalar multiplication $r \cdot (m + N) = rm + N$.

This is well defined precisely because N is closed under scalar multiplication – note that, unlike for groups and rings, *every* submodule can be quotiented by; there is no analogue of normality to worry about.

§15.12.2 Homomorphisms and the Isomorphism Theorems

Definition 15.12.6 (Module Homomorphism)

Let M, N be R -modules. A function $f : M \rightarrow N$ is an **R -module homomorphism** if

$$f(m_1 + m_2) = f(m_1) + f(m_2), \quad \text{and} \quad f(r \cdot m) = r \cdot f(m),$$

for all $m, m_1, m_2 \in M$ and $r \in R$. A bijective module homomorphism is an **isomorphism**.

When R is a field, module homomorphisms are exactly linear maps. As always, we have kernels and images, and a family of isomorphism theorems, whose proofs at this point are routine: apply the group-theoretic versions to $(M, +)$, and check that all the maps involved respect scalar multiplication.

Theorem 15.12.7 (First Isomorphism Theorem)

Let $f : M \rightarrow N$ be an R -module homomorphism. Then

$$\ker(f) = \{m \in M \mid f(m) = 0\} \leq M, \quad \text{img}(f) = \{f(m) \mid m \in M\} \leq N,$$

and $M/\ker(f) \cong \text{img}(f)$, via the map $m + \ker(f) \mapsto f(m)$.

$$\begin{array}{ccc}
 M & \xrightarrow{f} & \text{img}(f) \\
 \pi \downarrow & \nearrow \cong & \\
 M/\ker(f) & &
 \end{array}$$

Theorem 15.12.8 (Second Isomorphism Theorem)

Let $A, B \leq M$ be R -submodules. Then $A + B = \{a + b \mid a \in A, b \in B\}$ and $A \cap B$ are submodules of M , and

$$A/(A \cap B) \cong (A + B)/B.$$

Theorem 15.12.9 (Third Isomorphism Theorem)

Let $N \leq L \leq M$ be R -submodules. Then $L/N \leq M/N$, and

$$(M/N)/(L/N) \cong M/L.$$

There is also the familiar correspondence: submodules of M/N biject with submodules of M containing N .

Remark. Since vector spaces are modules, these specialise to facts from linear algebra – for instance, applying the first isomorphism theorem to a linear map $f : V \rightarrow W$ between finite dimensional spaces and taking dimensions gives

$$\dim V - \dim \ker(f) = \dim \text{img}(f),$$

the rank-nullity theorem.

§15.12.3 Finitely Generated Modules

We need a notion of ‘finite dimensional’ for modules.

Definition 15.12.10 (Generation)

For $m \in M$, write $Rm = \{rm \mid r \in R\} \leq M$, the submodule **generated by** m . An R -module M is **finitely generated** if

$$M = Rm_1 + Rm_2 + \cdots + Rm_n$$

for some $m_1, \dots, m_n \in M$, that is, every element of M is an R -linear combination $r_1m_1 + \cdots + r_nm_n$.

The following reformulation is simple but surprisingly useful – it will be the opening move in the proof of the structure theorem.

Lemma 15.12.11

An R -module M is finitely generated if and only if there is a surjective R -module homomorphism $f : R^n \rightarrow M$ for some n .

Proof. If $M = Rm_1 + \cdots + Rm_n$, define $f : R^n \rightarrow M$ by $f(r_1, \dots, r_n) = r_1m_1 + \cdots + r_nm_n$. This is an R -module homomorphism, and it is surjective by assumption.

Conversely, given such an f , let $e_i \in R^n$ be the standard ‘basis’ element with 1 in position i , and put $m_i = f(e_i)$. Any $m \in M$ equals $f(r_1, \dots, r_n) = r_1m_1 + \cdots + r_nm_n$ for some r_i , so the m_i generate M . $\square$

Corollary 15.12.12

Any quotient of a finitely generated module is finitely generated.

Proof. If $f : R^n \rightarrow M$ is surjective and $N \leq M$, then composing with the quotient map gives a surjection $R^n \rightarrow M/N$. $\square$

Example 15.12.13 (Submodules Need Not Be Finitely Generated)

In contrast with vector spaces (where subspaces of finite dimensional spaces are finite dimensional), a submodule of a finitely generated module can fail to be finitely generated. Take a non-Noetherian ring R with an ideal I that is not finitely generated – say $R = \mathbb{Z}[X_1, X_2, \dots]$ and $I = (X_1, X_2, \dots)$. Then R is generated as an R -module by the single element 1, while its submodule I is not finitely generated.

If R is Noetherian, this cannot happen: submodules of finitely generated R -modules are finitely generated.

§15.12.4 Torsion and Direct Sums

Definition 15.12.14 (Torsion)

Let M be an R -module. An element $m \in M$ is a **torsion element** if $rm = 0$ for some $r \neq 0$ in R . We say M is a **torsion module** if every element is torsion, and **torsion-free** if 0 is the only torsion element.

Example 15.12.15

For $\mathbb{Z}$ -modules (abelian groups), the torsion elements are exactly the elements of finite order; so a finite abelian group is a torsion $\mathbb{Z}$ -module, and $\mathbb{Z}^n$ is torsion-free. Any module over a field is torsion-free. The $\mathbb{Z}$ -module $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is neither torsion nor torsion-free.

Definition 15.12.16 (Direct Sum)

Let $M_1, \dots, M_n$ be R -modules. The **direct sum** $M_1 \oplus \cdots \oplus M_n$ is the set $M_1 \times \cdots \times M_n$ with componentwise addition and scalar multiplication.

For example, $R^n = R \oplus \cdots \oplus R$. We record one convenient compatibility between direct sums and quotients.

Lemma 15.12.17

Let $M = M_1 \oplus \cdots \oplus M_n$ and $N_i \leq M_i$ for each i . Then $N = N_1 \oplus \cdots \oplus N_n \leq M$ and

$$\frac{M_1 \oplus \cdots \oplus M_n}{N_1 \oplus \cdots \oplus N_n} \cong \frac{M_1}{N_1} \oplus \cdots \oplus \frac{M_n}{N_n}.$$

Proof. Apply the first isomorphism theorem to the surjective homomorphism $M \rightarrow \bigoplus_i M_i/N_i$ given by $(m_1, \dots, m_n) \mapsto (m_1 + N_1, \dots, m_n + N_n)$, whose kernel is exactly N . $\square$

§15.12.5 Free Modules

We next isolate the modules that behave like vector spaces.

Definition 15.12.18 (Independence)

Elements $m_1, \dots, m_n \in M$ are **independent** if

$$r_1 m_1 + \cdots + r_n m_n = 0 \implies r_1 = \cdots = r_n = 0.$$

Definition 15.12.19 (Freely Generating Sets & Free Modules)

A subset $S \subseteq M$ **generates M freely** if S generates M , and every function $\psi : S \rightarrow N$ into an R -module N extends to an R -module homomorphism $\theta : M \rightarrow N$. (Such an extension is automatically unique, since S generates M .) A module with a freely generating set is called **free**, and S is a **free basis**.

Proposition 15.12.20

For a finite subset $S = \{m_1, \dots, m_n\} \subseteq M$, the following are equivalent:

- (i) S generates M freely;
- (ii) S generates M and is independent;
- (iii) every element of M is *uniquely* expressible as $r_1 m_1 + \cdots + r_n m_n$;
- (iv) the map $R^n \rightarrow M$, $(r_1, \dots, r_n) \mapsto r_1 m_1 + \cdots + r_n m_n$, is an isomorphism.

Proof. We show (i) $\implies$ (ii); the equivalence of (ii), (iii) and (iv) is exactly as in linear algebra. Suppose S generates M freely but is not independent, say $\sum_i r_i m_i = 0$ with $r_j \neq 0$. Consider the function $\psi : S \rightarrow R$ with $\psi(m_j) = 1$ and $\psi(m_i) = 0$ for $i \neq j$, and extend it to a homomorphism $\theta : M \rightarrow R$. Then

$$0 = \theta(0) = \theta\left(\sum_i r_i m_i\right) = \sum_i r_i \theta(m_i) = r_j \neq 0,$$

a contradiction. For (iii) $\implies$ (i), define $\theta(r_1 m_1 + \cdots + r_n m_n) = \sum r_i \psi(m_i)$; uniqueness of the expression makes this well defined. $\square$

So a finitely generated free module is just (isomorphic to) some R^n . But be warned – the intuition from vector spaces breaks down beyond this.

Example 15.12.21 (Modules Are Not Vector Spaces)

The $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$ is finitely generated, but not free: every element m satisfies $2m = 0$, so no non-empty subset is independent.

More subtly, $\{2, 3\}$ generates $\mathbb{Z}$ as a $\mathbb{Z}$ -module (as $3 - 2 = 1$), but is not independent ($3 \cdot 2 - 2 \cdot 3 = 0$), and *no subset* of $\{2, 3\}$ is a free basis – $\{2\}$ and $\{3\}$ generate proper submodules. So a generating set need not contain a basis, in stark contrast with linear algebra.

One fact from linear algebra does survive, though it now requires genuine proof.

Proposition 15.12.22 (Invariance of Rank)

Let R be a non-trivial ring. If $R^n \cong R^m$ as R -modules, then $n = m$.

Proof. The idea is to reduce to the vector space case by quotienting. For an ideal $I \trianglelefteq R$ and an R -module M , define

$$IM = \{\sum_i a_i m_i \mid a_i \in I, m_i \in M\} \leq M.$$

The quotient M/IM is then naturally an R/I -module, via $(r + I)(m + IM) = rm + IM$ – this is well defined precisely because IM absorbs multiplication by I .

Now let I be a *maximal* ideal of R (one exists, by a Zorn's lemma argument which we will not detail). Applying the above with $M = R^n$, note $IR^n = I^n$ (componentwise), so

$$(R/I)^n \cong R^n/IR^n \cong R^n/IR^n \cong (R/I)^m$$

as R/I -modules. But R/I is a field, so these are vector spaces, and comparing dimensions gives $n = m$. $\square$

§15.12.6 Matrices over Euclidean Domains

We now begin working towards the main structure theorem. The engine of the proof is a normal form for *matrices* over a Euclidean domain, obtained by row and column operations – essentially Gaussian elimination, done without division.

For the rest of this section, R is a Euclidean domain with Euclidean function ϕ , and we consider $m \times n$ matrices A with entries in R .

Definition 15.12.23 (Elementary Operations)

The **elementary row operations** on a matrix over R are:

- (i) add $\lambda \in R$ times row j to row i (where $i \neq j$),
- (ii) swap rows i and j ,
- (iii) multiply row i by a unit $u \in R^\times$.

Elementary column operations are defined analogously. Each is realised by multiplying on the left (respectively right) by an invertible matrix.

Definition 15.12.24 (Equivalence)

Matrices A and B are **equivalent** if one can be obtained from the other by a sequence of elementary row and column operations; equivalently, $B = QAP$ for invertible Q and P .

Notice what is *not* allowed: we cannot divide by non-units. The whole subtlety of the following algorithm is in working around this.

Theorem 15.12.25 (Smith Normal Form)

Any $m \times n$ matrix A over a Euclidean domain R is equivalent to a diagonal matrix

$$\begin{pmatrix} d_1 & & & & \\ & \ddots & & & \\ & & d_t & & \\ & & & 0 & \\ & & & & \ddots \end{pmatrix},$$

with $d_1 \mid d_2 \mid \cdots \mid d_t$. The d_i are called the **invariant factors** of A , and are unique up to associates.

Proof. If $A = 0$ we are done, so assume not, and by swapping rows and columns arrange $a_{11} \neq 0$. The plan is to repeatedly reduce $\phi(a_{11})$ until a_{11} divides everything in sight; since ϕ takes values in $\mathbb{Z}_{\geq 0}$, this cannot go on forever.

Step 1: clear the first row and column. If $a_{11} \nmid a_{1j}$ for some j , write $a_{1j} = qa_{11} + r$ with $\phi(r) < \phi(a_{11})$ (and $r \neq 0$, else $a_{11} \mid a_{1j}$). Subtracting q times column 1 from column j and then swapping those columns puts r in the top-left corner, strictly decreasing $\phi(a_{11})$. Do the same with rows if $a_{11} \nmid a_{i1}$. After finitely many iterations, a_{11} divides every entry of the first row and column, and subtracting suitable multiples of the first row and column makes all those entries zero:

$$A \sim \begin{pmatrix} d & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A' & \\ 0 & & & \end{pmatrix}, \quad d = a_{11}.$$

Step 2: make d divide everything. If $d \nmid a_{ij}$ for some entry of A' , add row i to row 1 – now the first row contains an entry not divisible by d , and we return to Step 1, which strictly decreases $\phi(a_{11})$ again. So after finitely many rounds, we reach the displayed form where moreover $d = d_1$ divides every entry of A' .

Step 3: recurse. Apply the algorithm to A' . The operations involved never touch the first row or column, and preserve the property that d_1 divides all entries (each entry of the new matrix is a combination of old ones). Inducting, A is equivalent to $\text{diag}(d_1, \dots, d_t, 0, \dots, 0)$ with $d_1 \mid d_2 \mid \cdots \mid d_t$.

Uniqueness of the d_i is proven below, using the machinery of Fitting ideals. □

Example 15.12.26

Let us put the integer matrix $A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$ into Smith normal form. Working over $\mathbb{Z}$:

$$\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} \xrightarrow{c_1 \mapsto c_1 + c_2} \begin{pmatrix} 1 & -1 \\ 3 & 2 \end{pmatrix} \xrightarrow{c_2 \mapsto c_2 + c_1} \begin{pmatrix} 1 & 0 \\ 3 & 5 \end{pmatrix} \xrightarrow{r_2 \mapsto r_2 - 3r_1} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix},$$

which is in Smith normal form since $1 \mid 5$: the invariant factors are 1, 5.

To prove uniqueness we introduce an invariant of the equivalence class of a matrix.

Definition 15.12.27 (Minors and Fitting Ideals)

A $k \times k$ **minor** of A is the determinant of a $k \times k$ submatrix of A (obtained by deleting all but k rows and k columns). The k -th **Fitting ideal** $\text{Fit}_k(A) \trianglelefteq R$ is the ideal generated by all $k \times k$ minors of A .

Lemma 15.12.28

Equivalent matrices have the same Fitting ideals.

Proof. By symmetry (transposing), it's enough to show row operations preserve Fit_k . Swapping rows and scaling a row by a unit change each minor by at most a unit factor, so preserve the ideal they generate. Consider then the operation adding λ times row j to row i , giving matrix A' , and let C' be a $k \times k$ submatrix of A' with corresponding submatrix C of A . If row i is not involved, $C' = C$. If both rows i and j are involved, then C' is obtained from C by a row operation, so $\det C' = \det C$. If row i is involved but row j is not, expanding the determinant along the modified row gives

$$\det C' = \det C \pm \lambda \det D,$$

where D is a $k \times k$ submatrix of A (using row j instead of row i , with rows reordered). In all cases $\det C' \in \text{Fit}_k(A)$, so $\text{Fit}_k(A') \subseteq \text{Fit}_k(A)$, and since the operation is invertible, equality holds. $\square$

Proof of uniqueness. Suppose the matrix A is equivalent to $\text{diag}(d_1, \dots, d_t, 0, \dots, 0)$ with $d_1 \mid \dots \mid d_t$. A $k \times k$ minor of this diagonal matrix is non-zero only when it uses rows and columns with matching indices, in which case it is a product $d_{i_1} \cdots d_{i_k}$ with $i_1 < \dots < i_k$. Each such product is divisible by $d_1 d_2 \cdots d_k$ (using the divisibility chain), and $d_1 \cdots d_k$ itself is a minor, so

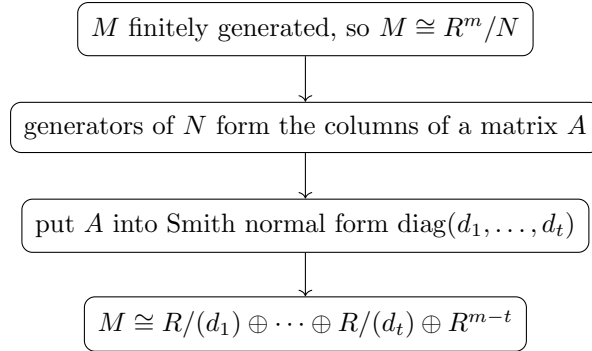
$$\text{Fit}_k(A) = (d_1 d_2 \cdots d_k).$$

Hence the products $d_1 \cdots d_k$ are determined by A up to associates for each k , and dividing successive products recovers each d_i up to associates. $\square$

Remark. Fitting ideals also give a way to *compute* invariant factors without running the algorithm. In the example above, $\text{Fit}_1(A) = (2, -1, 1, 2) = (1)$ gives $d_1 = \pm 1$, and $\text{Fit}_2(A) = (\det A) = (5)$ gives $d_1 d_2 = \pm 5$, so $d_2 = \pm 5$.

§15.12.7 The Structure Theorem

We now harvest the consequences of Smith normal form. The chain of reasoning runs as follows.



We need two preliminary results to make this rigorous. The first says submodules of R^m are manageable, and the second interprets row and column operations module-theoretically.

Lemma 15.12.29 (Submodules of Free Modules)

Let R be a Euclidean domain (or indeed any PID). Then any submodule $N \leq R^m$ is generated by at most m elements.

Proof. We induct on m . Consider the ideal

$$I = \{r \in R \mid (r, r_2, \dots, r_m) \in N \text{ for some } r_2, \dots, r_m \in R\},$$

the set of possible first coordinates of elements of N (an ideal, since N is a submodule). As R is a PID, $I = (a)$ for some a , and we may choose $n_1 = (a, a_2, \dots, a_m) \in N$.

Now take any $x = (r_1, \dots, r_m) \in N$. Then $r_1 = ra$ for some $r \in R$, and

$$x - rn_1 = (0, r_2 - ra_2, \dots, r_m - ra_m) \in N \cap (\{0\} \times R^{m-1}) =: N'.$$

So $N = Rn_1 + N'$, where N' is (identified with) a submodule of R^{m-1} , generated by at most $m-1$ elements $n_2, \dots, n_m$ by induction. Then $n_1, n_2, \dots, n_m$ generate N . $\square$

Theorem 15.12.30

Let R be a Euclidean domain and $N \leq R^m$ a submodule. Then there is a free basis $x_1, \dots, x_m$ of R^m such that N is generated by $d_1x_1, d_2x_2, \dots, d_tx_t$ for some $t \leq m$ and $d_1 \mid d_2 \mid \dots \mid d_t$.

Proof. By the lemma, N is generated by elements $y_1, \dots, y_n \in R^m$ with $n \leq m$. Form the $m \times n$ matrix A whose columns are the y_i , so that N is generated by the columns of A .

Now consider what the elementary operations do to this data. A column operation replaces the generators y_i by a different generating set for the *same* submodule N

(adding a multiple of one generator to another, swapping generators, or scaling by a unit). A row operation corresponds to changing the free basis of R^m : for instance, adding λ times row j to row i amounts to expressing everything in the new basis $e_1, \dots, e_j - \lambda e_i, \dots, e_m$, which is again a free basis.

Putting A into Smith normal form $\text{diag}(d_1, \dots, d_t, 0, \dots)$ therefore produces a free basis $x_1, \dots, x_m$ of R^m (the accumulated row operations) together with generators $d_1 x_1, \dots, d_t x_t$ of N (the non-zero columns). $\square$

Theorem 15.12.31 (Structure Theorem for Finitely Generated Modules over a Euclidean Domain)

Let R be a Euclidean domain and M a finitely generated R -module. Then

$$M \cong R/(d_1) \oplus R/(d_2) \oplus \cdots \oplus R/(d_t) \oplus R^k$$

for some $k \geq 0$ and non-zero $d_1 \mid d_2 \mid \cdots \mid d_t$. The d_i are called the **invariant factors** of M , and (after discarding those d_i which are units, for which $R/(d_i) = 0$) they are unique up to associates.

Proof. Since M is finitely generated, there is a surjective homomorphism $f : R^m \rightarrow M$, and by the first isomorphism theorem $M \cong R^m / \ker(f)$. By the previous theorem, there is a free basis $x_1, \dots, x_m$ of R^m with $\ker(f)$ generated by $d_1 x_1, \dots, d_t x_t$, where $d_1 \mid \cdots \mid d_t$. Using this basis to identify R^m with $R \oplus \cdots \oplus R$,

$$M \cong \frac{R \oplus \cdots \oplus R}{d_1 R \oplus \cdots \oplus d_t R \oplus 0 \oplus \cdots \oplus 0} \cong R/(d_1) \oplus \cdots \oplus R/(d_t) \oplus R^{m-t},$$

using our lemma on quotients of direct sums. We omit the proof of uniqueness (which can be carried out by counting, for each prime $p \in R$ and each j , the ‘sizes’ of the modules $p^j M / p^{j+1} M$). $\square$

Remark. The structure theorem holds more generally over any PID, with essentially the same statement. The proof runs along the same lines, but without a Euclidean function one needs a different induction to establish Smith normal form – one can measure the ‘size’ of a matrix entry by the number of irreducible factors, using unique factorisation.

Some immediate consequences are worth recording.

Corollary 15.12.32

Over a Euclidean domain, any finitely generated torsion-free module is free.

Proof. In the decomposition of the structure theorem, any factor $R/(d_i)$ with d_i a non-zero non-unit contains torsion (the class of 1 is killed by d_i). Torsion-freeness forces all d_i to be units, so $M \cong R^k$. $\square$

Example 15.12.33 (Computing with the Structure Theorem)

Let G be the abelian group generated by elements a and b subject to the relations

$$2a + b = 0, \quad -a + 2b = 0.$$

Formally, $G \cong \mathbb{Z}^2/N$, where $N \leq \mathbb{Z}^2$ is generated by $(2, 1)$ and $(-1, 2)$. These generators are the columns of

$$A = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix},$$

whose Smith normal form we computed: $\text{diag}(1, 5)$. So in a suitable basis of $\mathbb{Z}^2$, N is generated by x_1 and $5x_2$, and

$$G \cong \frac{\mathbb{Z} \oplus \mathbb{Z}}{\mathbb{Z} \oplus 5\mathbb{Z}} \cong \mathbb{Z}/5\mathbb{Z} \cong C_5.$$

§15.12.8 Finitely Generated Abelian Groups

Since abelian groups are exactly $\mathbb{Z}$ -modules and $\mathbb{Z}$ is a Euclidean domain, the structure theorem instantly classifies all finitely generated abelian groups.

Theorem 15.12.34 (Classification of Finitely Generated Abelian Groups)

Let G be a finitely generated abelian group. Then

$$G \cong C_{d_1} \times C_{d_2} \times \cdots \times C_{d_t} \times \mathbb{Z}^r$$

for some $r \geq 0$ and $d_1 \mid d_2 \mid \cdots \mid d_t$.

Proof. This is precisely the structure theorem for $R = \mathbb{Z}$, translated from module notation $(\mathbb{Z}/d\mathbb{Z}, \oplus)$ into group notation $(C_d, \times)$. $\square$

In particular, taking G finite (so $r = 0$), this proves the fundamental theorem of finite abelian groups in its invariant factor form – the statement we promised back when we studied finite abelian groups. The prime power decomposition also generalises. Recall the key lemma was that $C_{mn} \cong C_m \times C_n$ for coprime m, n ; here is the module version.

Lemma 15.12.35 (Chinese Remainder Theorem)

Let R be a PID and $a, b \in R$ with $\gcd(a, b)$ a unit. Then, as R -modules,

$$R/(ab) \cong R/(a) \oplus R/(b).$$

Proof. Since R is a PID, $(a, b) = (d)$ where d is a gcd of a and b , so $(a, b) = R$ and we can write $1 = ra + sb$ for some $r, s \in R$.

Define $\psi : R \rightarrow R/(a) \oplus R/(b)$ by $\psi(x) = (x + (a), x + (b))$, an R -module homomorphism. Note

$$\psi(sb) = (sb + (a), sb + (b)) = (1 - ra + (a), 0 + (b)) = (1 + (a), 0 + (b)),$$

and similarly $\psi(ra) = (0 + (a), 1 + (b))$. Hence $\psi(x \cdot sb + y \cdot ra) = (x + (a), y + (b))$,

so ψ is surjective.

For the kernel: clearly $(ab) \subseteq \ker(\psi)$. Conversely if $x \in \ker(\psi)$ then $a \mid x$ and $b \mid x$, and writing $x = x(ra + sb) = r(ax) + s(bx)$, both terms are divisible by ab (as $b \mid x$ in the first, $a \mid x$ in the second), so $x \in (ab)$. The first isomorphism theorem finishes the proof. $\square$

Theorem 15.12.36 (Primary Decomposition)

Let R be a Euclidean domain and M a finitely generated R -module. Then

$$M \cong R/(p_1^{n_1}) \oplus \cdots \oplus R/(p_k^{n_k}) \oplus R^r$$

where the p_i are primes of R (not necessarily distinct) and $r \geq 0$.

Proof. By the structure theorem, it suffices to decompose each factor $R/(d)$. As R is a UFD, write $d = up_1^{\alpha_1} \cdots p_s^{\alpha_s}$ with u a unit and the p_i pairwise non-associate primes. The prime powers $p_i^{\alpha_i}$ are pairwise coprime, so repeated application of the previous lemma gives

$$R/(d) \cong R/(p_1^{\alpha_1}) \oplus \cdots \oplus R/(p_s^{\alpha_s}).$$

$\square$

For $R = \mathbb{Z}$ this recovers the prime power form of the classification of finite abelian groups: every finite abelian group is a direct product of cyclic groups of prime power order.

§15.12.9 Rational Canonical Form

We now turn to the second great application of the structure theorem: canonical forms for matrices. The bridge is the $F[X]$ -module V_α from the start of this section – a vector space V over a field F together with a linear map $\alpha : V \rightarrow V$, with X acting as α .

Lemma 15.12.37

If V is finite dimensional, then V_α is a finitely generated $F[X]$ -module.

Proof. A basis $v_1, \dots, v_n$ of V as an F -vector space certainly generates V_α over the larger ring $F[X] \supseteq F$. $\square$

Since $F[X]$ is a Euclidean domain, the structure theorem applies to V_α . To interpret the outcome we must understand what the cyclic modules $F[X]/(f)$ look like as ‘vector space plus endomorphism’ data.

Example 15.12.38 (Companion Matrices)

Suppose $V_\alpha \cong F[X]/(f)$ where $f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_0$ is monic. Then $1, X, X^2, \dots, X^{n-1}$ is an F -vector space basis of $F[X]/(f)$ (by the division algorithm). The action of α is multiplication by X , which sends $X^i \mapsto X^{i+1}$ for $i < n-1$, and

$$X^{n-1} \mapsto X^n = -a_{n-1}X^{n-1} - \cdots - a_1X - a_0.$$

So with respect to this basis, α has matrix

$$C(f) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix},$$

the **companion matrix** of f .

Theorem 15.12.39 (Rational Canonical Form)

Let V be a finite dimensional vector space over a field F , and $\alpha : V \rightarrow V$ a linear map. Then

$$V_\alpha \cong F[X]/(f_1) \oplus F[X]/(f_2) \oplus \cdots \oplus F[X]/(f_t)$$

for monic polynomials $f_1 \mid f_2 \mid \cdots \mid f_t$, and with respect to a suitable basis of V , the map α has block diagonal matrix

$$\begin{pmatrix} C(f_1) & & & \\ & C(f_2) & & \\ & & \ddots & \\ & & & C(f_t) \end{pmatrix}.$$

Proof. V_α is a finitely generated module over the Euclidean domain $F[X]$, so the structure theorem gives

$$V_\alpha \cong F[X]/(f_1) \oplus \cdots \oplus F[X]/(f_t) \oplus F[X]^k$$

with $f_1 \mid \cdots \mid f_t$. Since $F[X]$ is infinite dimensional over F while V is finite dimensional, $k = 0$. Multiplying each f_i by a unit, we may take them monic (and discard unit f_i). Choosing the companion basis in each summand and concatenating gives the block matrix. $\square$

Remark. In matrix language: every square matrix over *any* field is similar to a block diagonal matrix of companion matrices with $f_1 \mid \cdots \mid f_t$. Reading off the diagonal: the *minimal polynomial* of α is f_t (it kills every summand, and nothing smaller kills the last), and the *characteristic polynomial* is $f_1 f_2 \cdots f_t$ (the characteristic polynomial of $C(f)$ is f). In particular the minimal polynomial divides the characteristic polynomial – we have proven the Cayley-Hamilton theorem along the way.

Example 15.12.40 (2×2 Matrices)

Let $\dim V = 2$. Then $\sum_i \deg f_i = 2$, so either $t = 1$ with $\deg f_1 = 2$, or $t = 2$ with both linear. In the second case $f_1 \mid f_2$ forces $f_1 = f_2 = X - \lambda$, so $V_\alpha \cong F[X]/(X - \lambda) \oplus F[X]/(X - \lambda)$ and $\alpha = \lambda \cdot \text{id}$ is scalar.

So: a non-scalar 2×2 matrix satisfies $V_\alpha \cong F[X]/(f)$ with f its characteristic polynomial, and hence *two non-scalar 2×2 matrices are similar if and only if they have the same characteristic polynomial* – both being similar to the same companion

matrix.

Definition 15.12.41 (Annihilator)

The **annihilator** of an R -module M is the ideal

$$\text{Ann}_R(M) = \{r \in R \mid rm = 0 \text{ for all } m \in M\} \trianglelefteq R.$$

Example 15.12.42

For an ideal $I \trianglelefteq R$ we have $\text{Ann}_R(R/I) = I$. For a finite abelian group A viewed as a $\mathbb{Z}$ -module, $\text{Ann}_{\mathbb{Z}}(A) = (e)$ where e is the exponent of A – this finally delivers the proof (promised earlier) that the exponent is realised: in the invariant factor decomposition $A \cong C_{d_1} \times \cdots \times C_{d_t}$, the exponent is d_t , the order of an actual element. And for V_α as above, $\text{Ann}_{F[X]}(V_\alpha) = (f_t)$ is generated by the minimal polynomial of α .

§15.12.10 Jordan Normal Form

Finally we specialise to $F = \mathbb{C}$, where the primes of the polynomial ring are as simple as possible.

Lemma 15.12.43

The irreducibles (equivalently, primes) in $\mathbb{C}[X]$ are exactly the linear polynomials $X - \lambda$ for $\lambda \in \mathbb{C}$, up to associates.

Proof. Linear polynomials are always irreducible. Conversely, by the fundamental theorem of algebra any non-constant $f \in \mathbb{C}[X]$ has a root $\lambda \in \mathbb{C}$, and then $X - \lambda$ divides f (divide and inspect the remainder), so any f of degree ≥ 2 is reducible. $\square$

So over $\mathbb{C}$, the primary decomposition of V_α is a direct sum of modules of the form $\mathbb{C}[X]/((X - \lambda)^n)$. Let's see what such a summand looks like.

Example 15.12.44 (A Single Jordan Block)

Suppose $V_\alpha \cong \mathbb{C}[X]/((X - \lambda)^n)$, and take the $\mathbb{C}$ -vector space basis

$$1, (X - \lambda), (X - \lambda)^2, \dots, (X - \lambda)^{n-1}.$$

The map $\alpha - \lambda \cdot \text{id}$ acts as multiplication by $X - \lambda$, sending each basis vector to the next and the last to $(X - \lambda)^n = 0$. So in this basis $\alpha - \lambda \text{id}$ is the shift matrix, and α has matrix

$$J_n(\lambda) = \begin{pmatrix} \lambda & & & \\ 1 & \lambda & & \\ & \ddots & \ddots & \\ & & 1 & \lambda \end{pmatrix},$$

an $n \times n$ **Jordan block**.^a

^aDepending on the choice of basis ordering, the 1's appear below or above the diagonal – both

conventions are in use.

Theorem 15.12.45 (Jordan Normal Form)

Let V be a finite dimensional vector space over $\mathbb{C}$ and $\alpha : V \rightarrow V$ a linear map. Then

$$V_\alpha \cong \mathbb{C}[X]/((X - \lambda_1)^{n_1}) \oplus \cdots \oplus \mathbb{C}[X]/((X - \lambda_t)^{n_t})$$

with $\lambda_i \in \mathbb{C}$ not necessarily distinct, and with respect to a suitable basis, α has block diagonal matrix

$$\begin{pmatrix} J_{n_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{n_t}(\lambda_t) \end{pmatrix}.$$

The Jordan blocks are unique up to reordering.

Proof. V_α is a finitely generated module over the Euclidean domain $\mathbb{C}[X]$, so the primary decomposition theorem applies; the primes of $\mathbb{C}[X]$ are the $X - \lambda$, and no free summand $\mathbb{C}[X]$ can appear by finite dimensionality. Choosing the basis of the previous example in each summand gives the block matrix. We will not prove uniqueness, but it follows by computing the dimensions of the generalised eigenspaces $\ker((\alpha - \lambda \text{id})^m)$ for each λ and m , which are determined by (and determine) the block sizes. $\square$

Remark (Reading Off Data From the JNF). Every square complex matrix is similar to one in Jordan normal form, and the form contains all similarity-invariant information:

- (i) the eigenvalues of α are the λ_i appearing in blocks;
- (ii) the characteristic polynomial is $\prod_\lambda (X - \lambda)^{a_\lambda}$, where a_λ is the sum of the sizes of the λ -blocks;
- (iii) the minimal polynomial is $\prod_\lambda (X - \lambda)^{c_\lambda}$, where c_λ is the size of the *largest* λ -block;
- (iv) the dimension of the λ -eigenspace is the *number* of λ -blocks.

In particular, α is diagonalisable if and only if all Jordan blocks have size 1, if and only if the minimal polynomial has no repeated roots.

This is a satisfying place to stop and look back. The single idea of doing linear algebra over a ring – keeping track of *which* scalars we are allowed to divide by – has unified the classification of finite abelian groups, the theory of canonical forms, and (through the rings $\mathbb{Z}[i]$ and $F[X]$) a good deal of number theory. Each of these threads is picked up again in Part II, in Number Fields, Galois Theory and Representation Theory.

16 Linear Algebra

Linear algebra is the area of mathematics concerned with *vector spaces* and the *linear maps* between them. On the face of it this sounds like quite a specialised topic, but the truth is quite the opposite – linearity is everywhere. Whether you are solving simultaneous equations, studying differential equations, working in geometry, or doing quantum mechanics, the underlying structure is frequently that of a vector space, and the machinery developed in this course (bases, matrices, determinants, eigenvalues, inner products and more) is the machinery used to understand it.

This article constitutes my notes for the ‘Linear Algebra’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. As a warning, this course has lots of results which involve just checking that a definition holds – such proofs have been intentionally omitted.

§16.1 Vector Spaces

§16.1.1 Vector Spaces and Subspaces

Linear algebra is, somewhat obviously, primarily about studying objects that are *linear* in nature. The objects we really care about are *vector spaces*, settings in which we can add elements and multiply by scalars. We are also going to consider *linear maps*, functions on vector spaces which preserve that linear structure – but more on that later.

Throughout the following discussion (and this course), $\mathbb{F}$ is going to denote an arbitrary field¹

Definition 16.1.1 ($\mathbb{F}$ -Vector Space)

An **$\mathbb{F}$ -vector space** is an abelian group $(V, +)$ together with a function $\mathbb{F} \times V \rightarrow V$, written $(\lambda, v) \mapsto \lambda v$ such that the following axioms hold:

- (i) *Distributivity in V* . $\lambda(v_1 + v_2) = \lambda v_1 + \lambda v_2$,
- (ii) *Distributivity in $\mathbb{F}$* . $(\lambda_1 + \lambda_2)v = \lambda_1 v + \lambda_2 v$,
- (iii) *Associativity*. $\lambda(\mu v) = (\lambda\mu)v$,
- (iv) *Identity*. $1v = v$.

We usually call elements of V **vectors** and elements of $\mathbb{F}$ **scalars**. The identity element in V is usually called the zero vector, and is written 0_V (or just 0 if the context is clear).

If $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, we use the terms ‘real vector space’ and ‘complex vector space’, since they’re so common.

¹A field $\mathbb{F}$ is a set $\mathbb{F}$ equipped with two operations $+$ (‘addition’) and $\cdot$ (‘multiplication’). We require $\mathbb{F}$ with addition to form an abelian group, and multiplication must be associative and have an identity element 1 . We also require every element except 0 to have an inverse with respect to multiplication, and multiplication must be distributive over addition.

Informally, you can think of a field as something you can do arithmetic in.

Example 16.1.2 (Examples of Vector Spaces)

- (i) The set of triples

$$\{(x, y, z) \mid x, y, z \in \mathbb{R}\}$$

forms a real vector space called $\mathbb{R}^3$, because you can add any two triples component wise.

- (ii) The set

$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$$

is a $\mathbb{Q}$ -vector space, where we add elements and scale by rational numbers in the obvious way.

- (iii) The set
- $\mathcal{C}[0, 1]$
- of all continuous functions
- $f : [0, 1] \rightarrow \mathbb{R}$
- forms a real vector space.

As with many new objects, it's helpful to be able to discuss its substructure. In the case of a vector space V , there's a pretty natural notion for what it means for a subset $U \subseteq V$ to still act like a vector space.

Definition 16.1.3 (Subspace)

Let V be a $\mathbb{F}$ -vector space. A subset $U \subseteq V$ is a **subspace** of V if U is also an $\mathbb{F}$ -vector space. If U is a subspace of V , we will write $U \leq V$.

Example 16.1.4 (Examples of Subspaces)

- (i) The set of vectors $\{(x, y, z) \mid x, y, z \in \mathbb{R}, x + y + z = 0\}$ is a subspace of $\mathbb{R}^3$.
- (ii) The set of polynomials with terms of even degree $\{a_0 + a_2x^2 + a_4x^4 + \cdots + a_{2k}x^{2k} \mid a_{2i} \in \mathbb{R}, k \in \mathbb{N}\}$ is a subspace of $\mathbb{R}[X]$, the vector space of polynomials with coefficients in $\mathbb{R}$.

As you would expect, checking that something is a subspace is usually easier than checking all of the axioms for a vector space. In particular, to check that U is a subspace of an $\mathbb{F}$ -vector space V , you can just check that the following hold:

- *Zero vector*². $0_V \in U$,
- *Closure under addition*. $u_1, u_2 \in U$ to imply $u_1 + u_2 \in U$,
- *Closure under scaling*. $\lambda \in \mathbb{F}$ and $u \in U$ to imply $\lambda u \in U$.

There are various ways in which we can manipulate subspaces, for example we can take the intersection of two subspaces, and we will get back another subspace.

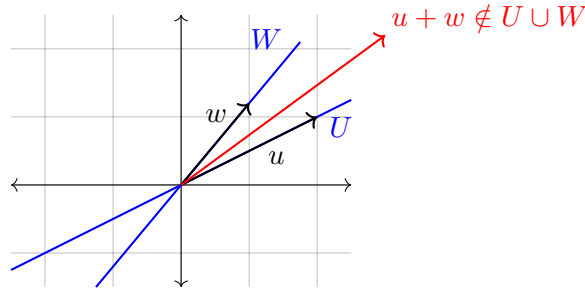
Proposition 16.1.5 (Intersecting Subspaces)

Let $U, W \leq V$. Then $U \cap W \leq V$.

²You may wonder why we need to check this when we already check that we are closed under scaling. To see why, notice that we still have to ensure U is non-empty!

Proof. Since U and W are both subspaces of V , we have $0_V \in U \cap W$, and also since they are both closed under addition and scaling, $u_1, u_2 \in U \cap W$ implies that $u_1 + u_2 \in U \cap W$, and $\lambda \in \mathbb{F}$ implies $\lambda u \in U \cap W$. Thus $U \cap W$ is a subspace of V . $\square$

However we can't manipulate subspaces however we want and expect magic. For example, the union of two subspaces is generally *not* a subspace, as it is typically not closed under addition. In fact, the union is only ever a subspace if one of the subspaces is contained in the other.³ You can see what goes wrong already in $\mathbb{R}^2$: if U and W are two distinct lines through the origin, then adding a vector from each takes you off both lines.



We can however try to ‘complete’ the union so that it becomes a subspace.

Definition 16.1.6 (Sum of Subspaces)

Let V be a vector space over $\mathbb{F}$, and let $U, W \leq V$. We define the **sum** of U and W to be the set

$$U + W = \{u + w \mid u \in U, w \in W\}.$$

This definition immediately forces $U + W \leq V$, and indeed it is the minimal such space (in that any subspace of V containing both U and W must also contain $U + W$).

§16.1.2 Quotient Spaces

Since a vector space V forms an abelian group $(V, +)$, we are able to take the quotient by any subspace $U \leq V$.

Definition 16.1.7 (Quotient Space)

Let V an $\mathbb{F}$ -vector space, and let $U \leq V$. The **quotient space** V/U is the abelian group V/U equipped with the scalar multiplication $F \times V/U \rightarrow V/U$ written $(\lambda, v + U) \mapsto \lambda v + U$.

With this definition, we need to check that this scalar multiplication operation is well defined. Indeed, if $v_1 + U = v_2 + U$ then

$$\begin{aligned} v_1 - v_2 &\in U \\ \implies \lambda(v_1 - v_2) &\in U \\ \implies \lambda v_1 + U &= \lambda v_2 + U \in V/U, \end{aligned}$$

so our operation is indeed well defined.

³There are some more exercises of this flavour on the example sheet.

As you would expect, taking a quotient gives you back a vector space.

Proposition 16.1.8 (Quotient Spaces are Vector Spaces)

V/U is an $\mathbb{F}$ -vector space.

Proof Sketch. Check definitions (most properties are inherited from V being a vector space). $\square$

§16.1.3 Basis and Dimension

You are likely informally familiar with the idea of *dimension*, a measure how much freedom exists in a system. Dimensionality is a rather natural concept with respect to vector spaces, but we will need to move through some technicalities to establish the results we want.

To discuss the amount of freedom, we first need a way to quantify what it means for a set of vectors to be independent from one another. This is the idea of *linear independence*.

Definition 16.1.9 (Linear Independence)

We say that $\{v_1, \dots, v_n\} \in V$ are **linearly independent** if

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n = 0$$

implies that $\lambda_1 = \dots = \lambda_n = 0$.

Remark. For an infinite subset $S \subseteq V$, we say it's linearly independent if every finite subset is linearly independent.

If a set of vectors is *not* linearly independent, then there's some vector in that set that can be written as a linear combination of the others – so it's not independent of them!

The next idea we need to pin down is being able to see if our set of vectors can ‘generate’ the rest of our vector space.

Definition 16.1.10 (Span)

Let V be a vector space over $\mathbb{F}$, and let $S \subset V$. We define the **span** of S , $\langle S \rangle$ to be the set of finite combinations of elements of S .

If $\langle S \rangle = V$, then we say S **spans** or **generates** V .

Remark. By convention, we also take $\langle \emptyset \rangle = \{0\}$. An equivalent definition is that $\langle S \rangle$ is the smallest subspace of V that contains S .

Example 16.1.11 (Quadratic Polynomials)

Let V be the vector space of quadratic polynomials over $\mathbb{R}$,

$$V = \{ax^2 + bx + c \mid a, b, c \in \mathbb{R}\}.$$

Then the subset $S \subseteq V$ with $S = \{1, x, x^2\}$ spans V .

Putting these two concepts together gives us the idea of *bases*, which are sets of linearly independent vectors that span a vector space.

Definition 16.1.12 (Basis)

A subset S of a vector space V is a **basis** if S is a set of linearly independent vectors that span V .

Example 16.1.13 (Basis for $\mathbb{R}^n$)

The **canonical basis** of $\mathbb{R}^n$ is the set of vectors

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\}.$$

Importantly, this is not the *only* basis of $\mathbb{R}^n$, just one that is quite convenient most of the time.

Remark. Note that in the definition of a basis there is no requirement for the set of basis vectors $S \subseteq V$ to be finite – only that any element in V must be representable using finitely many elements of S .

We'd intuitively want to say that the *dimension* of a vector space is the number of elements in its basis. However, we first need to check that this is a well defined notion. We can at this point distinguish between finite and infinite dimensional vector spaces at this point though⁴.

Definition 16.1.14 (Finite & Infinite Dimension)

We say a vector space V is **finite dimensional** if it has a finite basis, and we say it is **infinite dimensional** otherwise.

The next result about bases we will prove is that they induce *unique* representations of elements in the vector space.

Lemma 16.1.15 (Unique Representations with a Basis)

Let V be a vector space over $\mathbb{F}$. Then $S \subseteq V$ is a basis of V if and only if any vector $v \in V$ can be written uniquely as a linear combination of elements $v_1, \dots, v_n \in S$.

Proof. Suppose that S was a basis for V . Then if $v \in V$ can't be written as such a linear combination, then S wouldn't span V , contradicting it being a basis. Also, if v can be written as such a linear combination non-uniquely, then taking

$$v = \lambda_1 v_1 + \dots + \lambda_n v_n = \mu_1 v_1 + \dots + \mu_n v_n.$$

where $\lambda_i \neq \mu_i$ for at least one value of i , we'd have $0 = v - v = (\lambda_1 - \mu_1)v_1 + \dots + (\lambda_n - \mu_n)v_n$, and at least one of these coefficients must be non-zero, contradicting

⁴Can you see why this is well defined already?

S being linearly independent.

Alternatively, if any element in V can be written uniquely, then if $v_1, \dots, v_n \in S$ with $\lambda_1 v_1 + \dots + \lambda_n v_n = 0$ implies that $\lambda_1 = \dots = \lambda_n = 0$, giving that S must be linearly independent. Since S is also spanning by definition, we see that it therefore must be a basis of V . $\square$

With that out of the way, we can prove some results about finite dimensional vector spaces which will help us get towards our definition of dimension.

Lemma 16.1.16 (Spanning Sets Contain a Basis)

Let V be a finite dimensional vector space, and let $S = \{v_1, \dots, v_n\}$ be a set of vectors that spans V . Then there is some subset of S that is a basis of V .

Proof. If $\{v_1, \dots, v_n\}$ is linearly independent, then we are done. If it's not, then (up to reordering) we have $v_n \in \langle \{v_1, \dots, v_{n-1}\} \rangle$. But then $\langle \{v_1, \dots, v_n\} \rangle = \langle \{v_1, \dots, v_{n-1}\} \rangle$, so we can not include v_n in our subset. Not including elements in this way repeatedly, since there is finitely many elements in S , we must eventually get a linearly independent set that still spans V . $\square$

Theorem 16.1.17 (Steinitz Exchange Lemma)

Let V be a finite dimensional vector space. Then if $\{v_1, \dots, v_m\}$ is a set of linearly independent vectors, and $\{w_1, \dots, w_n\}$ spans V , then

- (i) $m \leq n$
- (ii) up to reordering, $\{v_1, \dots, v_m, w_{m+1}, \dots, w_n\}$ spans V .

Proof. We will prove this by induction. Suppose we have replaced $\ell \geq 0$ of the w_i , and that

$$\langle \{v_1, \dots, v_\ell, w_{\ell+1}, \dots, w_n\} \rangle = V.$$

If $m = \ell$, we are done, so assume that $\ell < m$. Then since this set is spanning, we can write $v_{\ell+1} \in V$ as

$$v_{\ell+1} = \alpha_1 v_1 + \dots + \alpha_\ell v_\ell + \beta_{\ell+1} w_{\ell+1} + \dots + \beta_n w_n.$$

Since having $\beta_i = 0$ for all $\ell + 1 \leq i \leq n$ would violate linear independence, we can suppose without loss of generality that $\beta_{\ell+1} \neq 0$. We also note that this implies that $\ell + 1 \leq n$, as otherwise this would not be possible.

Then $w_{\ell+1} \in \langle \{v_1, \dots, v_{\ell+1}, w_{\ell+2}, \dots, w_n\} \rangle$, and this set spans V .

Repeating this process, we will be done after m steps, and we have also shown (at the final step) that $m \leq n$. $\square$

Corollary 16.1.18 (Dimension)

Let V be a finite dimensional vector space over $\mathbb{F}$. Then any two bases of V have the same number of elements, called the **dimension** of V , denoted $\dim V$ or $\dim_{\mathbb{F}} V$.

Proof. Immediate by Steinitz exchange lemma. $\square$

So using Steinitz exchange lemma we have finally been able to pin down exactly what is meant by the dimension of a vector space – it’s the size of it’s basis. This should match up to the intuitive idea of ‘freedom’ that you had at the start of this section. Freedom in a vector space comes from varying coefficients, and in a basis we can both freely vary coefficients and also reach any element in a vector space uniquely, so the number of independent parameters really is the size of the basis.

Steinitz also gives us a few useful results for free.

Corollary 16.1.19

Let V be a vector space over $\mathbb{F}$ with finite dimension $n = \dim V$. Then

- (i) Any independent set of vectors has at most n elements, with equality if and only if it’s a basis.
- (ii) Any spanning set has at least n elements, with equality if and only if it’s a basis.

Proof. Immediate by Steinitz exchange lemma. $\square$

Working with basis and dimension can make the study of vector spaces much easier. For example, Steinitz allows us nicely extend bases. That is,

If we have a vector space V , and a subspace $U \leq V$,
then we can pick a nice basis $\{u_1, \dots, u_\ell\}$ of U , and
extend it to a basis $\{u_1, \dots, u_\ell, u_{\ell+1}, \dots, u_n\}$ of V .

Working with ideas like this can make many results easier to prove, as we will see in the following propositions.

Proposition 16.1.20 (Dimension of the Sum of Subspaces)

Let U, W be subspaces of a vector space V . If U and W are finite dimensional, then so is $U + W$, and $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

Proof. Pick a basis $\{v_1, \dots, v_a\}$ of $U \cap W$, and extend by Steinitz exchange lemma to a basis $\{v_1, \dots, v_a, u_1, \dots, u_b\}$ of U , and to a basis $\{v_1, \dots, v_a, w_1, \dots, w_c\}$ of W .

It suffices to prove that $\{v_1, \dots, v_a, u_1, \dots, u_b, w_1, \dots, w_c\}$ is a basis of $U + W$.

Clearly this set of vectors spans $U + W$, so we just need to check that they are linearly independent. Suppose that

$$\sum_{i=1}^a \alpha_i v_i + \sum_{i=1}^b \beta_i u_i + \sum_{i=1}^c \gamma_i w_i = 0.$$

Rewriting,

$$\sum_{i=1}^a \alpha_i v_i + \sum_{i=1}^b \beta_i u_i = - \sum_{i=1}^c \gamma_i w_i, \quad (\dagger)$$

where the LHS is in U and the RHS is in W . This implies that $\sum_{i=1}^c \gamma_i w_i \in U \cap W$, and can be written as $\sum_{i=1}^c \gamma_i w_i = \sum_{i=1}^a \mu_i v_i$, for some μ_i , and then substituting this back into (†),

$$\sum_{i=1}^a (\alpha_i + \mu_i) v_i + \sum_{i=1}^b \beta_i u_i = 0$$

which forces $\beta_i = 0$. A similar argument also gives $\gamma_i = 0$, which then finally forces $\alpha_i = 0$, since $\{v_1, \dots, v_a\}$ is a basis. $\square$

Proposition 16.1.21 (Dimension of the Quotient Space)

If V is a finite dimensional vector space over $\mathbb{F}$ and $U \leq V$, then U and V/U are also finite dimensional, and $\dim V = \dim U + \dim V/U$.

Proof. Let $\{u_1, \dots, u_\ell\}$ be a basis of U , and extend it via Steinitz exchange lemma to a basis $\{u_1, \dots, u_\ell, w_{\ell+1}, \dots, w_n\}$ of V .

It's easy to see that $\{w_{\ell+1} + U, \dots, w_n + U\}$ is a basis of V/U , as it clearly spans and linear independence is inherited from it being a basis of V . The result then follows. $\square$

§16.1.4 Direct Sums

Previously, we were able to look at the substructure of a vector space by looking at subspaces. Given two subspaces, we were then able to construct their sum, which is the set of all linear combinations of elements in each subspace.

When studying a vector space using its subspaces in this way (considering their sum), it can be useful to impose an *additional* constraint about the way in which the subspaces interact. In particular, it can be useful to impose a uniqueness constraint on the linear combinations that are created.

Definition 16.1.22 (Direct Sum)

Let V be a vector space over $\mathbb{F}$, and let $U, W \leq V$. We say that V is the **direct sum** of U and W , written $V = U \oplus W$ if and only if every element $v \in V$ can be decomposed as

$$v = u + w$$

with $u \in U$ and $w \in W$, with this decomposition being unique.

Of course, we can generalize the notion of a direct sum naturally to the case of multiple subspaces, in the way that you would expect.

Remark (Warning). We say that W is a direct complement of U in V . There is *no uniqueness* of such a complement!

Lemma 16.1.23

Let V be a vector space and let $U, W \leq V$. Then the following are equivalent.

- (i) $V = U \oplus W$

(ii) $V = U + W$ and $U \cap W = \{0\}$

(iii) For any basis B_1 of U and B_2 of W , the union $B = B_1 \cup B_2$ is a basis of V .

Proof. (ii) implies (i). Let $V = U + W$ with $U \cap W = \{0\}$. Then for all $v \in V$, we can write $v = u + w$ with $u \in U$ and $w \in W$. To see that this is unique, suppose that

$$v = u + w = u' + w'.$$

Then $(u - u') = -(w - w')$, and thus they are both in $U \cap W$, but the only element of this is 0, and thus $u = u'$ and $w = w'$, giving us uniqueness.

(i) implies (iii). Let B_1 be a basis of U and B_2 be a basis of W . Then $B = B_1 \cup B_2$ clearly spans V , so we need to check that it's linearly independent. Indeed, if

$$\sum_{u \in B_1} \lambda_u u + \sum_{w \in B_2} \lambda_w w = 0,$$

then $V = U \oplus W$ implies that since this is the sum of an element of U and an element of W , then each is zero, and linear independence follows from B_1, B_2 being sets of linearly independent vectors.

(iii) implies (ii). Clearly we have $V = U + W$, so we just need to check that $U \cap W = \{0\}$. Let $v \in U \cap W$. Then we can write

$$v = \sum_{u \in B_1} \lambda_u u = \sum_{w \in B_2} \lambda_w w \implies \sum_{u \in B_1} \lambda_u u - \sum_{w \in B_2} \lambda_w w = 0,$$

and since $B_1 \cup B_2$ is a basis for V , we must have $\lambda_u, \lambda_w = 0$ for all $u, w \in B_1, B_2$, implying that $v = 0$. $\square$

This result extends as you would expect for the case of direct products using multiple subspaces.

§16.2 Linear Maps

§16.2.1 Linear Maps and Isomorphisms

We are now going to turn our attention to *linear maps*, which are maps that preserve the underlying structure of the vector space that they act on.

Definition 16.2.1 (Linear Map)

Let V, W be vector spaces over $\mathbb{F}$. A map $\alpha : V \rightarrow W$ is a **linear map** if for all $\lambda_1, \lambda_2 \in \mathbb{F}$ and $v_1, v_2 \in V$ we have

$$\alpha(\lambda_1 v_1 + \lambda_2 v_2) = \lambda_1 \alpha(v_1) + \lambda_2 \alpha(v_2).$$

Example 16.2.2 (Examples of Linear Maps)

- (i) Consider the vector space $\mathcal{C}[0, 1]$ of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$. Then for any fixed point $x \in [0, 1]$, the map $\alpha : \mathcal{C}[0, 1] \rightarrow \mathbb{R}$ with $\alpha(f) = f(x)$ is a

linear map.

- (ii) The map $\mathbb{R}^3 \rightarrow \mathbb{R}$ given by $(a, b, c) \mapsto 4a + 2b + c$ is a linear map.
- (iii) The map $\alpha : Q[\sqrt{2}] \rightarrow Q[\sqrt{2}]$ of ‘multiplying by $\sqrt{2}$ ’ is a linear map, with $a + b\sqrt{2} \mapsto 2b + a\sqrt{2}$.

It’s easy to see that the identity map is a linear map, and also that linearity is preserved by composing linear maps.

One nice thing about linear maps is that they interact well with basis, because of linearity. Indeed, specifying a linear map on the basis is sufficient to define that map on all of a vector space.

Lemma 16.2.3 (Extension by Linearity)

Let V and W be vector spaces, and let B be a basis for V . Then if $\alpha_0 : B \rightarrow W$ is *any* map, then there exists a *unique* linear map $\alpha : V \rightarrow W$ extending α_0 , so that for all $v \in B$ we have $\alpha(v) = \alpha_0(v)$.

Proof. For $v \in V$, let $v = \sum_{i=1}^n \lambda_i v_i$. Then necessarily by linearity, we define

$$\alpha(v) = \alpha \left(\sum_{i=1}^n \lambda_i v_i \right) = \sum_{i=1}^n \lambda_i \alpha_0(v_i).$$

By construction this is a linear map and it extends α_0 . □

As with many other areas of pure mathematics, we have a special case for the structure preserving bijections on a given object.

Definition 16.2.4 (Isomorphism)

If V and W are vector spaces over $\mathbb{F}$, the linear map $\alpha : V \rightarrow W$ is an **isomorphism** if it also bijective. We say two vector spaces V and W are **isomorphic**, written $V \cong W$, if there is an isomorphism between them.

Remark. if $\alpha : V \rightarrow W$ is an isomorphism, then the inverse map α^{-1} exists and importantly it’s *also* linear.

Lemma 16.2.5 (Isomorphism is an Equivalence Relation)

$\cong$ is an equivalence relation on the class of all vector spaces over $\mathbb{F}$.

Proof. For reflexivity, we note that the identity map $id : V \rightarrow V$ is an isomorphism. For symmetry, we note that if $\alpha : V \rightarrow W$ is an isomorphism, then $\alpha^{-1} : W \rightarrow V$ is also an isomorphism. Lastly, we get transitivity because if $\alpha : U \rightarrow V$ and $\beta : V \rightarrow W$ are isomorphisms, then $\beta \circ \alpha : U \rightarrow W$ is also an isomorphism. □

From one perspective, isomorphism allows us to know when two objects are really just different representations of the same thing. With this in mind, the following result tells us something interesting (yet unsurprising) about the nature of finite dimensional vector spaces.

Theorem 16.2.6 (Isomorphism for Finite Dimensional Vector Spaces)

If V is an n dimensional vector space over $\mathbb{F}$, then $V \cong \mathbb{F}^n$, the vector space

$$\mathbb{F}^n = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \mid x_1, \dots, x_n \in \mathbb{F} \right\},$$

where addition is component-wise.

Proof. Let $B = \{v_1, \dots, v_n\}$ be a basis for V . Then we claim that the map $\alpha : V \rightarrow \mathbb{F}^n$ defined by

$$v = \sum_{i=1}^n \lambda_i v_i \mapsto \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$$

is an isomorphism. Indeed, it's easy to see that we have linearity inherited from $\mathbb{F}$, and also the map is a bijection, so indeed we do have $V \cong \mathbb{F}^n$. $\square$

In particular, this result tells us that when we choose a basis for a finite dimensional vector space V , it's really the same as picking some isomorphism from V to $\mathbb{F}^n$.

This theorem also directly implies (with isomorphism being an equivalence relation) a criterion for two finite dimensional vector spaces to be isomorphic.

Theorem 16.2.7 (Isomorphism Criterion)

Let V and W be finite dimensional vector spaces over $\mathbb{F}$. Then they are isomorphic if and only if they have the same dimension.

§16.2.2 Images and Kernels

Stepping back from isomorphisms, we need to define some frequently used concepts relating to linear maps.

Definition 16.2.8

Let V, W be vector spaces over F , and let $\alpha : V \rightarrow W$ be a linear map. We define the **kernel** of α as

$$\ker \alpha = \{v \in V \mid \alpha(v) = 0\},$$

and the **image** of α as

$$\text{img } \alpha = \{w \in W \mid w = \alpha(v), v \in V\}.$$

The structure of a linear map carries over to the image and the kernel, and naturally implies that they are both subspaces of their parent vector spaces.

Lemma 16.2.9 (Image and Kernel are Subspaces)

Let $\alpha : V \rightarrow W$ be a linear map. Then $\ker \alpha$ and $\text{img } \alpha$ are subspaces of V and W respectively.

Proof Sketch. Check definitions. $\square$

You may recognize the definitions of the kernel and image from a course in group theory, particularly in relation to the first isomorphism theorem⁵. Indeed, remembering that vector spaces with vector addition are groups, with we can use the kernel and image of a linear map to establish an isomorphism between vector spaces.

Theorem 16.2.10 (First Isomorphism Theorem For Vector Spaces)

Let V and W be $\mathbb{F}$ vector spaces, and let $\alpha : V \rightarrow W$ be a linear map. Then $V/\ker \alpha \cong \text{img } \alpha$.

Proof Sketch. Consider the isomorphism $\bar{\alpha} : V/\ker \alpha \rightarrow \text{img } \alpha$ defined by $\bar{\alpha}(v + \ker \alpha) = \alpha(v)$, and check definitions. $\square$

Now while the kernel and image being subspaces are rather straightforward, slightly more interesting is the *dimension* of these subspaces. This is described by the *rank-nullity theorem*.

Definition 16.2.11 (Rank and Nullity)

Let α be a linear map. Then the **rank** of α , $r(\alpha) = \dim \text{img } \alpha$, is the dimension of the image. The **nullity** of α , $n(\alpha) = \dim \ker \alpha$, is the dimension of the Kernel.

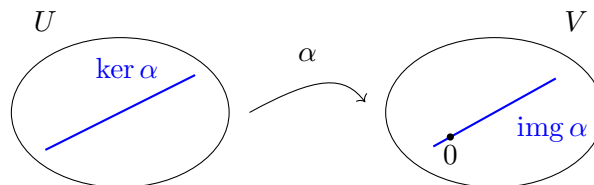
Theorem 16.2.12 (Rank-Nullity Theorem)

Let U and V be vector spaces such that U is finite dimensional. Let $\alpha : U \rightarrow V$ be a linear map. Then

$$\dim U = r(\alpha) + n(\alpha).$$

Proof. We know that $U/\ker \alpha \cong \text{img } \alpha$. Thus $\dim(U/\ker \alpha) = \dim(\text{img } \alpha)$, that is $\dim U - \dim \ker \alpha = \dim \text{img } \alpha$, or $\dim U = r(\alpha) + n(\alpha)$. $\square$

It is worth pausing to picture what this theorem is saying. A linear map $\alpha : U \rightarrow V$ crushes the kernel down to the single point 0, and maps what remains onto the image inside V . The theorem says that the dimensions ‘lost’ and ‘kept’ in this process must account for all of $\dim U$.



This result has an important consequence which is incredibly useful for quickly checking if a map is an isomorphism.

⁵Recall that if G and H are groups, and $\phi : G \rightarrow H$ is a homomorphism, then $G/\ker \phi \cong \text{img } \phi$.

Lemma 16.2.13 (Characterisation of Isomorphism)

Let V and W be vector spaces over $\mathbb{F}$ with *equal finite dimension*. Then the following are equivalent:

- (i) α is injective,
- (ii) α is surjective,
- (iii) α is an isomorphism.

Proof Sketch. Follows directly from the rank-nullity theorem. $\square$

§16.2.3 Matrices

When dealing with vector spaces, it's sometimes convenient to work with vector spaces in the abstract, and other times it's convenient to pick some nice basis which you can easily work with. When dealing with linear maps $\alpha : V \rightarrow W$, it can also be useful to work with bases. The way we do this is by introducing *matrices*.

It's important to say that matrices are not the essence of linear algebra, and in many cases viewing the subject in this way will just give you the impression that linear algebra is an opaque subject. Instead, you should think of matrices as a nice representation of linear maps for which it's easy to perform computations with. With that said, let's talk about linear maps.

In this section, we are going to discuss linear maps between two vector spaces V and W .

Definition 16.2.14 (Space of Linear Maps)

Let V and W be vector spaces over $\mathbb{F}$. We define

$$\mathcal{L}(V, W) = \{\alpha : V \rightarrow W \mid \alpha \text{ is a linear map}\}.$$

Proposition 16.2.15 (Space of Linear Maps is a Vector Space)

$\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$. Moreover, if V and W are finite dimensional, then

$$\dim \mathcal{L}(V, W) = \dim V \cdot \dim W.$$

Proof Sketch. To see that $\mathcal{L}(V, W)$ is a vector space, one just needs to check definitions. We will leave the proof of the dimension until later. $\square$

Now, given some linear map α , let's think about how we can represent this map with respect to a basis. Suppose that we had a finite dimensional vector space V with basis $\{v_1, \dots, v_m\}$, and a finite dimensional vector space W with basis $\{w_1, \dots, w_n\}$, and suppose that we had some linear map $\alpha : V \rightarrow W$.

Given some $v \in V$, if we wanted to work out $\alpha(v)$, we can write express v in the basis above, with

$$\alpha(v) = \alpha \left(\sum_{i=1}^m \lambda_i v_i \right) = \sum_{i=1}^m \lambda_i \alpha(v_i).$$

So, to compute what $\alpha(v)$ is for any $v \in V$, it suffices to know what $\alpha(v_1), \dots, \alpha(v_m)$ is. To specify what say $\alpha(v_1)$ is, we can use the above basis of W and write

$$\alpha(v_1) = a_{11}w_1 + a_{21}w_2 + \cdots + a_{n1}w_n,$$

for some coefficients a_{i1} . Repeating this for $v_2, \dots, v_n$ then totally specifies the linear map α . These collection of coefficients can be put into a *matrix*, which will then be a representation of α when coupled with the bases $\{v_i\}$ and $\{w_i\}$.

Definition 16.2.16 (Matrix)

An $n \times m$ **matrix** over $\mathbb{F}$ is an array with n rows and m columns, with entries in $\mathbb{F}$:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix}.$$

We denote the set of $n \times m$ matrices over $\mathbb{F}$ by $\mathcal{M}_{n,m}(\mathbb{F})$.

Proposition 16.2.17 (Vector Space of Matrices)

$\mathcal{M}_{n,m}(\mathbb{F})$ is an $\mathbb{F}$ -vector space.

Proof Sketch. Taking addition entry-wise we can just check definitions. $\square$

Proposition 16.2.18 (Dimension of $\mathcal{M}_{n,m}(\mathbb{F})$)

$\dim \mathcal{M}_{m,n}(\mathbb{F}) = m \times n$.

Proof. We define the *elementary matrix* $E_{i,j}$ such that every entry is 0 except for the i, j th entry, which is 1. Then the set $\{E_{i,j} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ is clearly a basis, and has $n \times m$ elements. $\square$

Following on from the discussion above, we can now write down how linear maps can be represented using matrices.

Definition 16.2.19 (Matrix of α in the B, C Basis)

Let $\alpha : V \rightarrow W$ be a linear map, and let $B = \{v_1, \dots, v_m\}$ be a basis of V and $C = \{w_1, \dots, w_n\}$ be a basis of W . Then we define the **matrix of α** to be

$$[\alpha]_{B,C} = \begin{pmatrix} \begin{array}{c} | \\ [\alpha(v_1)]_C \\ | \end{array} & \begin{array}{c} | \\ [\alpha(v_2)]_C \\ | \end{array} & \cdots & \begin{array}{c} | \\ [\alpha(v_m)]_C \\ | \end{array} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix},$$

where a_{ij} is the coefficient of w_i in $\alpha(v_j)$.

Remark (Notation). We use $[v]_C$ to denote the column vector representing the vector v in the basis C .

Proposition 16.2.20 (Matrix Representation of Linear Maps)

If V and W are vector spaces over $\mathbb{F}$ such that $\dim V = n$ and $\dim W = m$, then $\mathcal{L}(V, W) \cong \mathcal{M}_{m,n}(\mathbb{F})$.

Proof. Fix a basis B, C of V and W respectively. Then consider the linear map $\theta : \mathcal{L}(V, W) \rightarrow \mathcal{M}_{m,n}(\mathbb{F})$ given by $\alpha \mapsto [\alpha]_{B,C}$.

This map is clearly linear, and it's also clearly a bijection thus it's an isomorphism and our result follows. $\square$

Remark. This implies that $\dim \mathcal{L}(V, W) = m \times n$, filling in the gap left in the proposition at the start of this section.

What's nice about matrices is that they make working with linear maps *easy*. It's generally much easier to manipulate arbitrary matrices than it is to arbitrary linear maps (of course, this varies from case to case), and we will see this in the following set of computational results.

You should be familiar from previous courses with the notion of matrix multiplication.

Definition 16.2.21 (Matrix Multiplication)

Let $A = (a_{ij})$ be an $n \times p$ matrix, and $B = (b_{ij})$ be an $p \times m$ matrix. Then their **product** $C = A \cdot B$ is an $n \times m$ matrix $C = (c_{ij})$, with

$$c_{ij} = \sum_{k=1}^p a_{ik} b_{kj}.$$

We can use matrix multiplication in lots of nice ways.

Lemma 16.2.22 (Evaluating Linear Maps)

Let $\alpha : V \rightarrow W$ be a linear map, and let B, C be bases for V and W respectively. Then if $v \in V$, we have

$$[\alpha(v)]_C = [\alpha]_{B,C} \cdot [v]_B.$$

Proof. Let $v \in V$, and let $v = \sum_{j=1}^n \lambda_j v_j$. Then

$$\alpha(v) = \sum_{j=1}^n \lambda_j \alpha(v_j) = \sum_{j=1}^n \lambda_j \sum_{i=1}^m a_{ij} w_i = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} \lambda_j \right) w_i,$$

as required. $\square$

Lemma 16.2.23 (Composing Linear Maps)

Let $\alpha : V \rightarrow W$ and $\beta : U \rightarrow V$ be linear maps. Then if A, B, C are bases of U, V and W respectively, we have

$$[\alpha \circ \beta]_{A,C} = [\alpha]_{B,C} \cdot [\beta]_{A,B}.$$

Proof Sketch. Use the definition of matrix multiplication to check that this map corresponds to the correct thing. $\square$

§16.2.4 Change of Basis and Equivalent Matrices

Matrices represent a linear map with respect to a given basis, so a natural question is how do we rewrite a matrix in a different basis? That is, given two vector spaces V, W and a linear map $\alpha : V \rightarrow W$, we want to take the matrix $[\alpha]_{B,C}$ and express it with respect to the basis B', C' .

The way we do this is by introducing a *change of basis* matrix.

Definition 16.2.24 (Change of Basis Matrix)

The **change of basis** matrix from B' to B is the matrix $P = [I]_{B',B}$, where I is the identity map.

Remark. P is an invertible square matrix, and P^{-1} is the change of basis matrix from B to B' .

Given a vector represented in a basis B , the change of basis matrix allows that vector to be expressed in a different basis B' .

Lemma 16.2.25 (Change of Basis for a Vector)

Let B and B' be bases for a vector space V , and let $v \in V$. Then if P is the change of basis matrix from B' to B we have

$$[v]_B = P[v]_{B'}.$$

Proof. We know that $P = [I]_{B',B}$, and so $[v]_B = [I(v)]_B = [I]_{B',B}[v]_{B'} = P[v]_{B'}$. $\square$

Using this idea of change of basis matrices, we can express the matrix of a linear map with respect to another basis. The situation to keep in mind is the following square, where the vertical arrows are the identity map expressed in the relevant bases – the two ways of travelling from the top-left corner to the bottom-right corner must agree.

$$\begin{array}{ccc} V \text{ (basis } B') & \xrightarrow{A' = [\alpha]_{B',C'}} & W \text{ (basis } C') \\ \downarrow P & & \downarrow Q \\ V \text{ (basis } B) & \xrightarrow{A = [\alpha]_{B,C}} & W \text{ (basis } C) \end{array}$$

Proposition 16.2.26 (Change of Basis Formula)

Let $A = [\alpha]_{B,C}$ and $A' = [\alpha]_{B',C'}$, and let P be the change of basis from B' to B and Q be the change of basis matrix from C' to C . Then

$$A' = Q^{-1}AP.$$

Proof. For any $v \in V$ we have, using the change of basis lemma for vectors twice,

$$Q[\alpha(v)]_{C'} = [\alpha(v)]_C = A[v]_B = AP[v]_{B'},$$

and so $[\alpha(v)]_{C'} = Q^{-1}AP[v]_{B'}$. Since the matrix representing α with respect to B' and C' is unique, $A' = Q^{-1}AP$. $\square$

§16.2.5 Equivalent Matrices

A natural notion of equivalence comes from saying two matrices are equivalent if there is some choice of basis for which they represent the same linear map. Using the change of basis formula, we can pin down this notion exactly.

Definition 16.2.27 (Equivalent Matrices)

We say that two matrices $A, A' \in \mathcal{M}_{m,n}(\mathcal{F})$ are **equivalent** if $A' = Q^{-1}AP$ for some invertible matrices $Q \in \mathcal{M}_{m,m}(\mathbb{F})$ and $P \in \mathcal{M}_{n,n}$.

It's easy to check that this definition gives us an equivalence relation, and indeed it turns out that the equivalence classes of matrices are remarkably straightforward.

Proposition 16.2.28 (Natural Basis for Matrices)

Let V, W be vector spaces over $\mathbb{F}$, with $\dim V = n$ and $\dim W = m$. Let $\alpha : V \rightarrow W$ be a linear map. Then there exists a basis B of V and a basis C of W such that

$$[\alpha]_{B,C} = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right),$$

where $r = r(\alpha)$ and I_r is the $r \times r$ identity matrix.

Proof. We first choose a basis $\{v_{r+1}, \dots, v_n\}$ of $\ker \alpha$, and extend it to a basis $B = \{v_1, \dots, v_n\}$ of V . We will first show that $\{\alpha(v_1), \dots, \alpha(v_r)\}$ is a basis of $\text{img } \alpha$.

To see that it's spanning, given $v \in V$ we can write $\alpha(v) = \alpha(\sum_{i=1}^n \lambda_i v_i) = \sum_{i=1}^n \lambda_i \alpha(v_i) = \sum_{i=1}^r \lambda_i \alpha(v_i)$, since $\alpha(v_i) = 0$ for $i > r$.

To see that we have linear independence, suppose that $\sum_{i=1}^r \lambda_i \alpha(v_i) = 0$. Then we have $\alpha(\sum_{i=1}^r \lambda_i v_i) = 0$ implying that $\sum_{i=1}^r \lambda_i v_i \in \ker \alpha$, and since the kernel is spanned by $\{v_{r+1}, \dots, v_n\}$ and $\{v_1, \dots, v_n\}$ is a basis, we must have $\lambda_i = 0$, as required.

We can now extend this basis to a basis $C = \{\alpha(v_1), \dots, \alpha(v_r), w_{r+1}, \dots, w_m\}$ of W . Then we can express α as a matrix in the basis B and C as

$$[\alpha]_{B,C} = \left(\begin{array}{c|ccc|c} | & & & & | \\ \hline [\alpha(v_1)]_C & \cdots & [\alpha(v_r)]_C & [\alpha(v_{r+1})]_C & \cdots & [\alpha(v_n)]_C \\ \hline | & & | & | & & | \end{array} \right)$$

$$= \left(\begin{array}{c|c|c|c|c|c} | & & | & & | & \\ \hline [\alpha(v_1)]_C & \cdots & [\alpha(v_r)]_C & [0]_C & \cdots & [0]_C \\ \hline | & & | & & | & \end{array} \right),$$

and it can easily be seen that this matrix is of the form desired. $\square$

This result gives us a clear criterion for the equivalence of matrices, and also gives us a lovely result which is useful for choosing a basis which works nicely with the matrix representation of a linear map.

Corollary 16.2.29 (Matrix Equivalence Criterion)

Two $m \times n$ matrices are equivalent if the linear maps they represent have the same rank.

The equivalence criterion above is great, but doesn't give us a very easy way to see what the rank of the underlying linear map is, without doing a large amount of work. What we really want is a way to get this rank directly from the matrix. We can do this by considering the *column* and *row rank* of a matrix (which as we will see are really the same notion).

Definition 16.2.30 (Column and Row Rank)

Let $A \in \mathcal{M}_{m,n}(\mathbb{F})$. The **column rank** of A , $r(A)$, is the dimension of the subspace of $\mathbb{F}^m$ spanned by the column vectors of A .

Similarity, the **row rank** is the dimension of the subspace spanned by the row vectors of A (that is, it's the column rank of A^T).

Remark. If α is a linear map, represented by a matrix A with respect to some basis, then $r(A) = r(\alpha)$, so column rank is the same as the rank.

Proposition 16.2.31 (Equivalence from Row Rank)

Two matrices A, A' are equivalent if and only if $r(A) = r(A')$.

Proof. If they are equivalent, then they correspond to the same linear map α , expressed in two different bases. So the column rank is $r(A) = r(A') = r(\alpha)$, and we are done.

If $r(A) = r(A') = r$, then A and A' are both equivalent to

$$\left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right),$$

and by transitivity of equivalent, we have A and A' equivalent. $\square$

The distinction between column rank and row rank is really non-existent, as the following result shows. This also means that we can refer to the column and row rank as just the 'rank' of the matrix.

Theorem 16.2.32 (Column Rank Equals Row Rank)

$$r(A) = r(A^T)$$

Proof. Let $r = r(A)$. Then we know that we can write

$$Q^{-1}AP = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right),$$

then taking the transpose, we have $(Q^{-1}AP)^T = P^T A^T (Q^{-1})^T = P^T A^T (Q^T)^{-1}$ so $P^T A^T (Q^T)^{-1}$ is

$$\left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right)^T,$$

which is an $n \times m$ matrix. But then $r = r(A) = r(A^T)$. $\square$

Let's now consider the special case of **endomorphisms**, that is, linear maps $\alpha : V \rightarrow V$.

If B and B' are a basis for V , and P is the change of basis matrix from B' to B , then given α we have $[\alpha]_{B',B'} = P^{-1}[\alpha]_{B,B}P$, which is a special case of the change of basis formula. This idea of writing a matrix using the same basis for the domain and range gives us *similarity*, another notion of matrices being the same.

Definition 16.2.33 (Similar Matrices)

Let A, A' be two $n \times n$ square matrices. We say that A and A' are **similar** or **conjugate** if and only if $A' = P^{-1}AP$ for some invertible $n \times n$ matrix P .

This idea of similarity will be a central concept when we deal with the diagonalization of such matrices later on.

§16.2.6 Elementary Operations and Elementary Matrices

We are going to end this section by looking at the matrix operations that you will have used in previous courses to solve systems of linear equations, and seeing how they fit into the framework we have built up.

Definition 16.2.34 (Elementary Operations)

An **elementary column operation** on a matrix is one of the following:

- (i) swap two columns i and j ,
- (ii) multiply a column i by some $\lambda \in \mathbb{F}$ with $\lambda \neq 0$,
- (iii) add λ times column i to column j , where $i \neq j$.

Elementary row operations are defined analogously, acting on rows.

Each of these operations is invertible, and importantly each of them can be performed by matrix multiplication. That is, for each elementary operation there is a corresponding **elementary matrix**, obtained by performing that operation on the identity matrix.

For example, the elementary matrices

$$S_{ij} = \begin{pmatrix} 1 & & & \\ & 0 & \cdots & 1 \\ & \vdots & \ddots & \vdots \\ & 1 & \cdots & 0 \\ & & & & 1 \end{pmatrix}, \quad M_i(\lambda) = \begin{pmatrix} 1 & & & \\ & \lambda & & \\ & & \ddots & \\ & & & 1 \end{pmatrix}, \quad C_{ij}(\lambda) = I + \lambda E_{ij}$$

correspond to swapping columns i, j , scaling column i by λ , and adding λ times column i to column j respectively (here E_{ij} is the matrix which is zero everywhere except for a 1 in the (i, j) entry).

Lemma 16.2.35 (Elementary Operations as Matrix Multiplication)

Performing an elementary column operation on a matrix A is the same as multiplying A *on the right* by the corresponding elementary matrix. Similarly, performing an elementary row operation is the same as multiplying A *on the left* by the corresponding elementary matrix.

Proof Sketch. Check definitions, using the formula for matrix multiplication. $\square$

These operations give a *constructive* way to see the ‘natural basis for matrices’ proposition from earlier: given any $m \times n$ matrix A , if $A \neq 0$ we can pick some non-zero entry, swap rows and columns to move it to the top-left corner, scale it to be 1, and then use it to clear out the rest of the first row and column. Repeating this on the remaining $(m-1) \times (n-1)$ block, we eventually reach the matrix

$$\left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right),$$

having multiplied A on the left and right by invertible (elementary) matrices.

In the case of square invertible matrices, we can say a little more.

Proposition 16.2.36 (Invertible Matrices are Products of Elementary Matrices)

Let A be an invertible $n \times n$ matrix. Then A can be reduced to I_n using column operations alone (or row operations alone), and thus A is a product of elementary matrices.

Proof. We induct on the number of rows put into the correct form, supposing that using column operations we have transformed A into a matrix of the shape

$$\left(\begin{array}{c|c} I_k & 0 \\ \hline * & B \end{array} \right).$$

We claim there is some $j > k$ such that the $(k+1, j)$ entry is non-zero. If not, then row $k+1$ of our matrix would be a linear combination of the first k rows, which is impossible as our matrix is a product of A and invertible matrices, hence invertible. Swapping column j with column $k+1$ and scaling so that the $(k+1, k+1)$ entry is 1, we can then clear the rest of row $k+1$ with column operations, extending the identity block to size $k+1$.

Continuing, we obtain elementary matrices with $AE_1E_2\cdots E_m = I_n$. But then $A = E_m^{-1}\cdots E_1^{-1}$, and since the inverse of an elementary matrix is elementary, we are done. $\square$

This gives us a practical algorithm for inverting a matrix: perform column operations on A until you reach the identity, and apply the same operations to I_n as you go – what remains is A^{-1} .

§16.3 Dual Space

§16.3.1 Defining the Dual Space

We are now going to turn our attention to an interesting set of linear maps on a vector space.

Definition 16.3.1 (Dual Space)

Let V be a vector space over a field $\mathbb{F}$. The **dual space** V^* is the set of linear maps from V to $\mathbb{F}$,

$$V^* = \mathcal{L}(V, \mathbb{F}).$$

So how should we understand this space? As usual, things are easier for finite dimensional vector spaces, so we are going to start there. We know from the previous section that *any* linear map is determined by its action on basis vectors. So if we have some finite dimensional vector space V with basis $\{v_1, \dots, v_n\}$ and a linear map $\alpha : V \rightarrow \mathbb{F}$, we just need to know $\alpha(v_1), \dots, \alpha(v_n) \in \mathbb{F}$.

This tells us exactly what the nice way to think about the dual space is – through a basis that just picks out basis vectors so that we can deal with them individually. We call this the *dual basis*.⁶

Definition 16.3.2 (Dual Basis)

Let V be a finite dimensional vector space over $\mathbb{F}$ with a basis $B = \{v_1, \dots, v_n\}$. Then we define the **dual basis** for V^* to be $B^* = \{\varepsilon_1, \dots, \varepsilon_n\}$ where $\varepsilon_j(v_i) = \delta_{ij}$.

In particular, if $v \in V$ can be written as $v = \sum_{i=1}^n \lambda_i v_i$, then $\varepsilon_j(v) = \lambda_j$.

Since this is a basis of V^* , we can immediately see that $\dim V^* = \dim V$. Sometimes it's also convenient to think of V^* as the space of row vectors of length m over $\mathbb{F}$.⁷

§16.3.2 The Annihilator

When we have some subset U of a vector space V , we frequently care about the set of linear maps that are zero on everything in U . This construction ends up giving us a subspace of the dual space V^* .

⁶We just write it as a definition here, but we should really check that it is indeed a basis of the dual, but I think it's relatively clear so this is left to the reader.

⁷There's also a hidden inner product structure here, but we won't dwell on that.

Definition 16.3.3 (Annihilator)

If V is a vector space over $\mathbb{F}$, and $U \subseteq V$, then the **annihilator** of U is the set U^0 of linear maps $V \rightarrow \mathbb{F}$ that are zero on U ,

$$U^0 = \{\alpha \in V^* \mid \alpha(u) = 0 \text{ for all } u \in U\}.$$

Lemma 16.3.4 (Annihilator is a Subspace)

The annihilator is always a subspace of the dual, that is, $U^0 \leq V^*$.

Proof Sketch. Check definitions. □

By thinking about the annihilator, we also get a result that looks a little like rank-nullity.

Lemma 16.3.5 (Rank-Nullity for the Annihilator)

If V is a finite dimensional vector space and $U \leq V$, then $\dim V = \dim U + \dim U^0$.

Proof Sketch. Let $\{v_1, \dots, v_k\}$ be a basis of U , and extend it to a basis $B = \{v_1, \dots, v_n\}$ of V . Then the dual basis is $B' = \{\varepsilon_1, \dots, \varepsilon_n\}$. We then just need to check that $\{\varepsilon_{k+1}, \dots, \varepsilon_n\}$ is a basis of U^0 . □

§16.3.3 The Dual Map

We will now see that thinking about the dual will give us a basis-free interpretation of what corresponds with the transpose of a matrix.

Definition 16.3.6 (Dual Map)

Let V and W be vector spaces, and suppose that $\alpha : V \rightarrow W$ is a linear map. Then we define the **dual map** of α to be the linear map $\alpha^* : W^* \rightarrow V^*$ given by $\varepsilon \mapsto \varepsilon \circ \alpha$.

To see what this is all about, consider the following example.

Example 16.3.7 (Using the Dual Map)

Let V and W be $\mathbb{R}$ vector spaces, and let $B = \{v_1, v_2, v_3\}$ be a basis for V and $C = \{w_1, w_2\}$ be a basis for W .

We define the linear map $\alpha : V \rightarrow W$ by

$$\begin{aligned}\alpha(v_1) &= w_1 + w_2 \\ \alpha(v_2) &= 2w_1 - w_2 \\ \alpha(v_3) &= 6w_1 + 3w_2.\end{aligned}$$

Now we can see what the dual map $\alpha^* : W^* \rightarrow V^*$ looks like. Let $B^* = \{v_1^*, v_2^*, v_3^*\}$ be the corresponding dual basis of V , and let $C^* = \{w_1^*, w_2^*\}$ be the corresponding

dual basis of W . Then for any $v = av_1 + bv_2 + cv_3$ we have

$$\begin{aligned} [\alpha^*(w_1^*)](v) &= w_1^*(\alpha(av_1 + bv_2 + cv_3)) \\ &= w_1^*((a + 2b + 6c)w_1 + (a - b + 3c)w_2) \\ &= a + 2b + 6c, \end{aligned}$$

and in this way, we can really write

$$\alpha^*(w_1^*) = v_1^* + 2v_2^* + 6v_3^*, \quad \text{and} \quad \alpha^*(w_2^*) = v_1^* - v_2^* + 3v_3^*,$$

which totally specifies α^* .

A slightly more interesting way of viewing this example is through the lens of matrices.

Example 16.3.8 (Above with Matrices)

Continuing on from the example above, we can see that in the B, C basis we have

$$[\alpha]_{B,C} = \begin{pmatrix} 1 & 2 & 6 \\ 1 & -1 & 3 \end{pmatrix}.$$

Then α^* in the C^*, B^* basis is given by

$$[\alpha^*]_{C^*,B^*} = \begin{pmatrix} 1 & 1 \\ 2 & -1 \\ 6 & 3 \end{pmatrix}.$$

Of course, in finite dimensions this works in the general case.

Theorem 16.3.9 (Transpose Interpretation of the Dual Map)

Let V and W be finite dimensional vector spaces over $\mathbb{F}$. Let B be a basis for V and C be a basis for W , and let B^*, C^* be the corresponding dual spaces. Then if $\alpha : V \rightarrow W$ is a linear map, we have

$$[\alpha^*]_{C^*,B^*} = [\alpha]_{B,C}^T.$$

Proof. Let $B = \{v_1, \dots, v_n\}$ be a basis for V and $C = \{w_1, \dots, w_m\}$ be a basis for W , and let B^*, C^* be the corresponding dual spaces. Then

$$([\alpha]_{B,C})_{i,j} = ([\alpha(v_i)]_C)_j = ([w_j^* \circ \alpha]_{B^*})_i = ([\alpha^*]_{C^*,B^*})_{j,i},$$

as required. □

This result about viewing the matrix of α^* in terms of the matrix of α hints at a more general perspective where it can be easier to understand one of α and its dual α^* than the other, and indeed many properties of α^* can be derived from α .

For example, the result below is *immensely important*⁸ and is an example of the ‘duality approach’.

⁸But we won’t see in this course why that is.

Proposition 16.3.10 (Properties of the Dual Map)

Let V and W be vector spaces over $\mathbb{F}$, and let $\alpha : V \rightarrow W$ be a linear map. Then

- (i) $\ker \alpha^* = (\operatorname{img} \alpha)^0$, so α^* is injective if and only if α is surjective.
- (ii) $\operatorname{img} \alpha^* \leq (\ker \alpha)^0$, with equality if V and W are finite dimensional (in this case, α^* is surjective if and only if α is injective).

Proof. (i) Let $\varepsilon \in W^*$. Then $\varepsilon \in \ker \alpha^*$ if and only if $\alpha^*(\varepsilon) = 0$, that is, if $\varepsilon(\alpha(v)) = 0$ for all $v \in V$. But this occurs if and only if $\varepsilon \in (\operatorname{img} \alpha)^0$.

(ii) We will first show that $\operatorname{img} \alpha^* \leq (\ker \alpha)^0$. Indeed, let $\varepsilon \in \operatorname{img} \alpha^*$. Then $\varepsilon = \alpha^*(\phi)$ for some $\phi \in W^*$. But then for all $u \in \ker \alpha$ we have $\varepsilon(u) = (\alpha^*(\phi))(u) = \phi(\alpha(u)) = 0$. So then $\varepsilon \in (\ker \alpha)^0$, and thus $\operatorname{img} \alpha^* \leq (\ker \alpha)^0$ as required.

In finite dimensions, we can compute the dimensions of the two spaces. We have

$$\dim \operatorname{img} \alpha^* = r(\alpha^*) = r([\alpha^*]_{C^*, B^*}) = r([\alpha]_{B, C}^T) = r(\alpha).$$

So $\dim \operatorname{img} \alpha^* = r(\alpha)$, and by rank-nullity, this is $\dim V - \dim \ker \alpha = \dim (\ker \alpha)^0$. So $\operatorname{img} \alpha^* \leq (\ker \alpha)^0$ and also $\dim \operatorname{img} \alpha^* = \dim (\ker \alpha)^0$, giving us equality. $\square$

§16.3.4 Double Dual

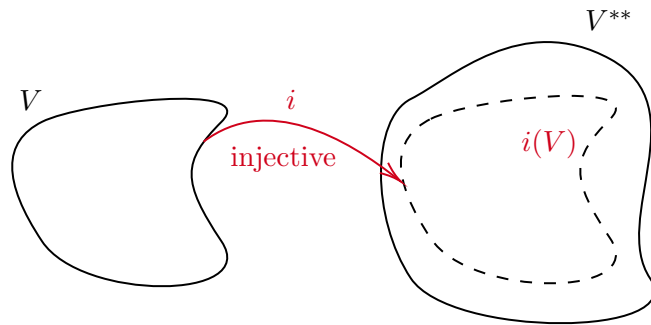
In general, there is no *obvious* relation between V and V^* , in that all of the relations we discussed in the previous section rely on choosing a basis – and choosing bases is an inherently arbitrary process. However, there *is* a canonical (that is, basis free) relation between V and $(V^*)^*$.

Definition 16.3.11 (Double Dual)

Let V be a vector space over $\mathbb{F}$, and $V^* = \mathcal{L}(V, \mathbb{F})$ be the dual of V . Then we define the **double dual** or **bidual** of V to be

$$V^{**} = \mathcal{L}(V^*, \mathbb{F}) = (V^*)^*.$$

Indeed, there is a *canonical* embedding from V to V^{**} , and there are non-trivial examples⁹ of infinite dimensional spaces where $V \cong V^{**}$.



This canonical embedding is described by ‘evaluating at v ’.

⁹Such as $L^p(\mathbb{R}^d) = \{f : \mathbb{R}^d \rightarrow \mathbb{R} \mid \int_{\mathbb{R}^d} |f(x)|^p dx < \infty\}$.

Theorem 16.3.12 (Canonical Embedding of V into V^{**})

There is a natural injective linear map from V to V^{**} , called the **evaluation map**.

Proof. Let V be an $\mathbb{F}$ vector space, and let $v \in V$. We define the evaluation at v map

$$\hat{v} : V^* \rightarrow \mathbb{F} \quad \text{by} \quad \hat{v}(\varepsilon) = \varepsilon(v).$$

Then $\hat{v}(\varepsilon_1 + \lambda\varepsilon_2) = (\varepsilon_1 + \lambda\varepsilon_2)(v) = \varepsilon_1(v) + \lambda\varepsilon_2(v)$, so $\hat{v} \in V^{**}$.

The proof that this is injective is intentionally omitted. $\square$

Theorem 16.3.13 (Canonical Embedding in Finite Dimensions)

If V is finite dimensional, then the evaluation map is an isomorphism.

Proof. We know that for $v \in V$ we have the evaluation map $\hat{v} \in V^{**}$. It's easy to see that the map $v \mapsto \hat{v}$ is linear in v , so it is left to check that this map is a bijection.

Let $e \in V \setminus \{0\}$. We can extend $\{e\}$ to a basis $B = \{e, e_2, \dots, e_n\}$ of V . Let B^* be the dual basis of V . Then

$$\hat{e}(e^*) = e^*(e) = 1,$$

and in particular, $\hat{e} \neq 0$, so the kernel of the map $v \mapsto \hat{v}$ contains only 0, and thus it is injective^a.

To see that this map is an isomorphism, we just need to compute dimensions. We know that $\dim V = \dim V^* = \dim(V^*)^* = \dim V^{**}$, and this along with injectivity implies that the map $v \mapsto \hat{v}$ is an isomorphism. $\square$

^aThis step of the proof has an analogue in infinite dimensions for a certain kind of vector space (Banach space), but requires the use of non-trivial results (the Hahn-Banach theorem).

This canonical embedding allows us to ‘identify’ V and V^{**} , which in turn can be used to prove various facts about say the annihilator.

Lemma 16.3.14 (Double Annihilator)

Let V be a finite dimensional vector space over $\mathbb{F}$, and $U \leq V$. Then $\hat{U} = U^{00}$, so after identification of V and V^{**} , we have $U = U^{00}$.

Proof. Let us first show that $U \leq U^{00}$. Indeed, let $u \in U$. Then for all $\varepsilon \in U^0$, $\varepsilon(u) = 0$. But then $\varepsilon(u) = \hat{u}(\varepsilon) = 0$, so $\hat{u} \in U^{00}$, and $\hat{U} \leq U^{00}$.

We can then compute dimensions, and $\dim U^{00} = \dim V - \dim U^0 = \dim U$, and since $v \mapsto \hat{v}$ is an isomorphism, $\dim \hat{U} = \dim U$. Thus $\dim \hat{U} = \dim U = \dim U^{00}$, and $\hat{U} = U^{00}$. $\square$

Remark. With the identification of V^{**} and V , for $T \leq V^*$ we can define $T^0 = \{v \in V \mid \alpha(v) = 0, \forall \alpha \in T\}$.

These results give us some unsurprising properties which are useful for doing computations.

Lemma 16.3.15 (Sums and Intersections of Annihilators)

Let V be a finite dimensional vector space over $\mathbb{F}$, and let $U_1, U_2 \leq V$. Then

- (i) $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$;
- (ii) $(U_1 \cap U_2)^0 = U_1^0 + U_2^0$.

Proof. (i) Let $\theta \in V^*$, then $\theta \in (U_1 + U_2)^0$ if and only if $\theta(u_1 + u_2) = 0$ for all $u_1 \in U_1$ and $u_2 \in U_2$. By linearity this is exactly when $\theta(u) = 0$ for $u \in U_1 \cup U_2$, which occurs if and only if $\theta \in U_1^0 \cap U_2^0$.

(ii) Take the annihilator of the result in (i), and use the fact that $U^{00} = U$. $\square$

§16.4 Bilinear Forms**§16.4.1 Introducing Bilinear Forms**

In the previous section we considered linear maps from a vector space to the scalar field that vector space is over. In this section we will look at a related but interesting set of maps which have *two* arguments and are linear in both. Such maps are *bilinear forms*.

Definition 16.4.1 (Bilinear Form)

Let U and V be vector spaces over $\mathbb{F}$. Then $\phi : U \times V \rightarrow \mathbb{F}$ is a **bilinear form** if it is linear in both components.

Example 16.4.2 (Examples of Bilinear Forms)

The following are all bilinear forms.

- (i) A canonical example – the scalar product on $U = V = \mathbb{R}^n$.
- (ii) The map $V \times V^* \rightarrow \mathbb{F}$ given by $(v, \alpha) \mapsto \alpha(v)$.
- (iii) For $U = V = \mathcal{C}([0, 1], \mathbb{R})$ the map $\phi(f, g) = \int_0^1 f(t)g(t) dt$.

There is a standard representation of bilinear forms using matrices.

Definition 16.4.3 (Matrix of a Bilinear Form)

Let $B = \{u_1, \dots, u_n\}$ be a basis of U and $C = \{v_1, \dots, v_m\}$ be a basis of V , and let $\phi : U \times V \rightarrow \mathbb{F}$ be a bilinear form.

We define the **matrix of ϕ** with respect to B and C by

$$([\phi]_{B,C})_{i,j} = \phi(u_i, v_j).$$

Lemma 16.4.4 (Evaluating a Bilinear Form)

Let $\phi : U \times V \rightarrow \mathbb{F}$ be a bilinear form, and let $u \in U$ and $v \in V$. Then if B is a

basis of U and C is a basis of V , we have

$$\phi(u, v) = [u]_B^T [\phi]_{B,C} [v]_C.$$

Proof Sketch. Check definitions. □

Remark. $[\phi]_{B,C}$ is the unique matrix such that this equation holds.

It's sometimes useful to study a bilinear form in just one of its arguments. This can be done by considering the induced maps.

Definition 16.4.5 (Induced Map of a Bilinear Form)

Let $\phi : U \times V \rightarrow \mathbb{F}$ be a bilinear form. Then ϕ determines two linear maps $\phi_L : U \rightarrow V^*$ and $\phi_R : V \rightarrow U^*$, with

$$(\phi_L(u))(v) = \phi(u, v) \quad \text{and} \quad (\phi_R(v))(u) = \phi(u, v).$$

These induced maps are easily obtained from the matrix of the bilinear form.

Lemma 16.4.6 (Matrices of the Induced Maps)

Let $\phi : U \times V \rightarrow \mathbb{F}$ be a bilinear map. Then if $B = \{u_1, \dots, u_n\}$ is a basis of U and $C = \{v_1, \dots, v_m\}$ is a basis of V , then if $A = [\phi]_{B,C}$ we have

$$[\phi_R]_{C,B^*} = A \quad \text{and} \quad [\phi_L]_{B,C^*} = A^T.$$

Proof Sketch. Check definitions. □

Just as with the linear maps that we studied previously, it's possible look at the kernel of a bilinear map by considering the left and right component.

Definition 16.4.7 (Left/Right Kernel)

Let $\phi : U \times V \rightarrow \mathbb{F}$ be a bilinear form. Then the **left kernel** of ϕ is $\ker \phi_L$, and the **right kernel** of ϕ is $\ker \phi_R$.

Definition 16.4.8 (Degeneracy)

We say that a bilinear form ϕ is **non-degenerate** if $\ker \phi_L = \{0\}$ and $\ker \phi_R = \{0\}$. Otherwise, we say that ϕ is **degenerate**.

Lemma 16.4.9 (Matrix Degeneracy Criterion)

Let U and V be finite dimensional vector spaces, and let B be a basis of U and C be a basis of V . Suppose $\phi : U \times V \rightarrow \mathbb{F}$ is a bilinear form with matrix $A = [\phi]_{B,C}$. Then ϕ is non-degenerate if and only if A is invertible.

Proof. We know that ϕ is non-degenerate if and only if $\ker \phi_L = \{0\}$ and $\ker \phi_R = \{0\}$, that is, if and only if $n(A^T) = 0$ and $n(A) = 0$. But then by rank-nullity, we

have this if and only if $r(A^T) = \dim U$ and $r(A) = \dim V$, that is, if and only if A is invertible. $\square$

Corollary 16.4.10

If $\phi : U \times V \rightarrow \mathbb{F}$ is a non-degenerate bilinear form, then $\dim U = \dim V$.

Corollary 16.4.11 (Choosing a Bilinear Form)

When U and V are finite dimensional vector spaces over $\mathbb{F}$, choosing a non-degenerate bilinear form $\phi : U \times V \rightarrow \mathbb{F}$ is equivalent to choosing an isomorphism $\phi_L : U \rightarrow V^*$.

As with matrices for normal linear maps $\alpha : V \rightarrow W$, we can change the basis that the matrix of a bilinear form is represented with.

Proposition 16.4.12 (Change of Basis for Bilinear Forms)

Let U and V be vector spaces over $\mathbb{F}$ and let B, B' be bases for U and C, C' be bases for V . Suppose that P is the change of basis matrix from B' to B and Q is the change of basis matrix from C' to C . Then if $\phi : U \times V \rightarrow \mathbb{F}$ is a bilinear form, then

$$[\phi]_{B',C'} = P^T [\phi]_{B,C} Q.$$

Proof. We have

$$\phi(u, v) = [u]_B^T [\phi]_{B,C} [v]_C = (P[u]_{B'})^T [\phi]_{B,C} (Q[v]_{C'}) = [u]_{B'}^T (P^T [\phi]_{B,C} Q) [v]_{C'},$$

and thus $[\phi]_{B',C'} = P^T [\phi]_{B,C} Q$. $\square$

This change of basis formula also allows us to define the rank of a bilinear map and have that definition be well defined.

Definition 16.4.13 (Rank of a Bilinear Map)

The rank of a bilinear map ϕ , $r(\phi)$, is the rank of any matrix representation of ϕ .

To see that this is well defined, note that $r(P^T A Q) = r(A)$ for any invertible P, Q .

§16.5 The Trace and Determinant

We will now look at the determinant and trace, two quantities you may be familiar with from the study of matrices. In this section, we will consider how these can be interpreted in a basis-free way – that is, how they can be seen as quantities attached to a linear map itself, not just to a particular matrix representation of it.

§16.5.1 The Trace

We begin with the trace, which is by far the simpler of the two.

Definition 16.5.1 (Trace)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$ be a square matrix. We define the trace of A by

$$\operatorname{tr} A = \sum_{i=1}^n A_{ii}.$$

In particular, we have

$$\operatorname{tr} \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} = a_{11} + \cdots + a_{nn}.$$

Remark. The trace is a linear form $\mathcal{M}_{n,n}(\mathbb{F}) \rightarrow \mathbb{F}$ given by $A \mapsto \operatorname{tr} A$.

The key computational fact about the trace is that it does not notice the order of a product.

Lemma 16.5.2 (Trace of a Product)

For any matrices $A, B \in \mathcal{M}_{n,n}(\mathbb{F})$, we have $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof. Directly computing, we have

$$\operatorname{tr}(AB) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^n b_{ji} a_{ij} = \operatorname{tr}(BA). \quad \square$$

Corollary 16.5.3 (Similar Matrices Have Equal Trace)

If A and A' are similar matrices, then $\operatorname{tr} A = \operatorname{tr} A'$.

Proof. If $A' = P^{-1}AP$, then $\operatorname{tr} A' = \operatorname{tr}(P^{-1}(AP)) = \operatorname{tr}((AP)P^{-1}) = \operatorname{tr} A. \quad \square$

This corollary is exactly what we need to make a basis-free definition. Recalling that two matrices represent the same endomorphism (in possibly different bases) precisely when they are similar, the following is well defined.

Definition 16.5.4 (Trace of an Endomorphism)

Let $\alpha : V \rightarrow V$ be an endomorphism of a finite dimensional vector space V . The **trace** of α is $\operatorname{tr} \alpha = \operatorname{tr}[\alpha]_{B,B}$, where B is any basis of V .

As a quick application of thinking this way, the trace interacts with duality in the way you might guess.

Lemma 16.5.5 (Trace of the Dual Map)

Let $\alpha : V \rightarrow V$ be an endomorphism of a finite dimensional vector space. Then $\operatorname{tr} \alpha^* = \operatorname{tr} \alpha$.

Proof. Picking a basis B of V , we have $\operatorname{tr} \alpha^* = \operatorname{tr}[\alpha^*]_{B^*, B^*} = \operatorname{tr}[\alpha]_{B, B}^T = \operatorname{tr}[\alpha]_{B, B} = \operatorname{tr} \alpha$, since transposing a matrix does not change its diagonal. $\square$

§16.5.2 Permutations

The determinant requires a bit more machinery than the trace, and the machinery it requires is that of *permutations*, which we briefly recall from IA Groups.

The **symmetric group** S_n is the group of permutations of $\{1, \dots, n\}$, that is, the group of bijections $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ under composition. A **transposition** $(k \ell)$ is a permutation which swaps k and ℓ and fixes everything else. Every permutation can be written as a product of transpositions, and while such a decomposition is not unique, the *parity* of the number of transpositions used is. This lets us define the **signature** $\varepsilon : S_n \rightarrow \{\pm 1\}$, with $\varepsilon(\sigma) = 1$ if σ is a product of an even number of transpositions and $\varepsilon(\sigma) = -1$ otherwise. Importantly, the signature is a group homomorphism.

§16.5.3 The Determinant

With that, we can write down the (somewhat intimidating looking) definition of the determinant.

Definition 16.5.6 (Determinant)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. The **determinant** of A is

$$\det A = \sum_{\sigma \in S_n} \varepsilon(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n}.$$

Each term in this sum picks exactly one entry from each column of A , with no two entries from the same row, and weights the resulting product by a sign. For small n , this recovers formulas you already know.

Example 16.5.7 (Determinants of Small Matrices)

For $n = 2$, the only permutations are the identity and the transposition $(1 \ 2)$, so

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

Similarly for $n = 3$, the six permutations of S_3 give the familiar six-term formula.

Directly from the definition we can compute the determinant of triangular matrices.

Lemma 16.5.8 (Determinant of a Triangular Matrix)

If A is an upper (or lower) triangular matrix with diagonal entries $a_{11}, \dots, a_{nn}$, then $\det A = a_{11}a_{22} \cdots a_{nn}$.

Proof. Suppose $a_{ij} = 0$ for $i > j$. For the term corresponding to σ in the sum to be non-zero, we need $\sigma(j) \leq j$ for all j . But then $\sigma(1) = 1$, which forces $\sigma(2) = 2$, and so on – so the only permutation contributing is the identity, and $\det A = a_{11} \cdots a_{nn}$.

The lower triangular case is analogous. $\square$

Lemma 16.5.9 (Determinant of the Transpose)

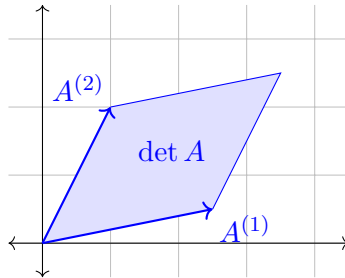
Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. Then $\det A^T = \det A$.

Proof. Reindexing the product inside the sum using σ^{-1} , and noting that $\sigma \mapsto \sigma^{-1}$ is a bijection $S_n \rightarrow S_n$ with $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$, we get

$$\begin{aligned} \det A &= \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n a_{\sigma(i)i} = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{j=1}^n a_{j\sigma^{-1}(j)} \\ &= \sum_{\tau \in S_n} \varepsilon(\tau) \prod_{j=1}^n a_{j\tau(j)} = \det A^T. \square \end{aligned}$$

§16.5.4 Volume Forms

The definition of the determinant above is useful for computing, but it's fair to say it doesn't exactly drip with meaning. The right way to understand the determinant is *geometric*: viewing the columns of A as n vectors in $\mathbb{F}^n$, the determinant is the *signed volume* of the parallelepiped they span.



To make this precise (and to prove the important properties of the determinant cleanly) we isolate the properties that any sensible notion of ‘signed volume’ should satisfy.

Definition 16.5.10 (Volume Form)

A **volume form** d on $\mathbb{F}^n$ is a function $d : \underbrace{\mathbb{F}^n \times \cdots \times \mathbb{F}^n}_{n \text{ times}} \rightarrow \mathbb{F}$ such that:

- (i) d is *multilinear*: for each i and each fixed choice of $v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n$, the map $v \mapsto d(v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n)$ is linear;
- (ii) d is *alternating*: if $v_i = v_j$ for some $i \neq j$, then $d(v_1, \dots, v_n) = 0$.

Both of these are natural if you think of d as measuring volume – scaling one edge of a parallelepiped scales its volume¹⁰, and a ‘parallelepiped’ with two equal edges is squashed flat, so should have zero volume.

¹⁰Additivity in each argument is perhaps slightly less obvious, but can be seen with a ‘shearing’ argument.

Lemma 16.5.11 (Determinant is a Volume Form)

The map $(\mathbb{F}^n)^n \rightarrow \mathbb{F}$ given by $(A^{(1)}, \dots, A^{(n)}) \mapsto \det A$, where A is the matrix with columns $A^{(1)}, \dots, A^{(n)}$, is a volume form.

Proof. For multilinearity, fix $\sigma \in S_n$ and consider the term $\prod_{i=1}^n a_{\sigma(i)i}$. This product contains exactly one entry from each column of A , so it is linear in each column separately. The determinant is a sum of such terms, so is multilinear.

For the alternating property, suppose columns $k \neq \ell$ are equal, and let $\tau = (k \ell)$. Then $a_{ij} = a_{i\tau(j)}$ for all i, j . Splitting $S_n = A_n \cup \tau A_n$ into even permutations and odd permutations, the terms for σ and $\tau\sigma$ have equal products but opposite signs, so cancel in pairs. Hence $\det A = 0$. $\square$

Lemma 16.5.12 (Swapping Entries)

Let d be a volume form. Then swapping two arguments of d changes its sign, that is,

$$d(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = -d(v_1, \dots, v_j, \dots, v_i, \dots, v_n).$$

Proof. Consider d applied with $v_i + v_j$ in both the i th and j th positions. By the alternating property this is zero, and expanding by multilinearity,

$$\begin{aligned} 0 &= \underbrace{d(\dots, v_i, \dots, v_i, \dots)}_{=0} + d(\dots, v_i, \dots, v_j, \dots) \\ &\quad + d(\dots, v_j, \dots, v_i, \dots) + \underbrace{d(\dots, v_j, \dots, v_j, \dots)}_{=0}, \end{aligned}$$

which rearranges to give the result. $\square$

Corollary 16.5.13 (Volume Forms and Permutations)

If d is a volume form and $\sigma \in S_n$, then $d(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \varepsilon(\sigma) d(v_1, \dots, v_n)$.

Proof. Write σ as a product of transpositions and apply the previous lemma once for each. $\square$

We can now prove the theorem that justifies this whole discussion: up to scaling, the determinant is the *only* volume form on $\mathbb{F}^n$.

Theorem 16.5.14 (Uniqueness of Volume Forms)

Let d be a volume form on $\mathbb{F}^n$, and let A be a matrix with columns $A^{(1)}, \dots, A^{(n)}$. Then

$$d(A^{(1)}, \dots, A^{(n)}) = \det A \cdot d(e_1, \dots, e_n).$$

In particular, the determinant is the unique volume form with $d(e_1, \dots, e_n) = 1$.

Proof. Writing $A^{(1)} = \sum_{i=1}^n a_{i1}e_i$ and using multilinearity in the first argument,

$$d(A^{(1)}, \dots, A^{(n)}) = \sum_{i=1}^n a_{i1} d(e_i, A^{(2)}, \dots, A^{(n)}).$$

Doing the same in every argument, we get

$$d(A^{(1)}, \dots, A^{(n)}) = \sum_{i_1, \dots, i_n} a_{i_1 1} a_{i_2 2} \cdots a_{i_n n} d(e_{i_1}, \dots, e_{i_n}).$$

Since d is alternating, the only non-zero terms are those where $i_1, \dots, i_n$ are distinct – that is, where $i_k = \sigma(k)$ for some permutation $\sigma \in S_n$. Applying the corollary above,

$$d(A^{(1)}, \dots, A^{(n)}) = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{k=1}^n a_{\sigma(k)k} \cdot d(e_1, \dots, e_n) = \det A \cdot d(e_1, \dots, e_n),$$

as required. $\square$

This uniqueness result is more than just aesthetically pleasing – it gives essentially free proofs of facts about the determinant that are painful to obtain from the formula. The prime example is multiplicativity.

Theorem 16.5.15 (Determinant is Multiplicative)

Let $A, B \in \mathcal{M}_{n,n}(\mathbb{F})$. Then $\det(AB) = \det A \det B$.

Proof. Fixing A , define $d_A : (\mathbb{F}^n)^n \rightarrow \mathbb{F}$ by

$$d_A(v_1, \dots, v_n) = \det(Av_1, \dots, Av_n),$$

where the right hand side denotes the determinant of the matrix with the given columns. Since $v \mapsto Av$ is linear and $\det$ is multilinear, d_A is multilinear, and if $v_i = v_j$ then $Av_i = Av_j$, so d_A is alternating. Thus d_A is a volume form, and by uniqueness $d_A = C_A \det$ for the constant

$$C_A = d_A(e_1, \dots, e_n) = \det(Ae_1, \dots, Ae_n) = \det A,$$

since Ae_i is exactly the i th column of A . Applying this with the columns of B ,

$$\det(AB) = d_A(B^{(1)}, \dots, B^{(n)}) = \det A \det B. \quad \square$$

§16.5.5 Singular and Non-Singular Matrices

The determinant gives us a clean test for invertibility.

Definition 16.5.16 (Singular Matrices)

A square matrix A is **singular** if $\det A = 0$, and **non-singular** otherwise.

Theorem 16.5.17 (Determinant Criterion for Invertibility)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. Then the following are equivalent:

- (i) A is invertible,
- (ii) A is non-singular,
- (iii) $r(A) = n$.

Proof. (i) implies (ii). If A is invertible then $\det A \det A^{-1} = \det(AA^{-1}) = \det I = 1$, so $\det A \neq 0$.

(i) is equivalent to (iii). This is the rank-nullity theorem: A is invertible if and only if the linear map it represents is bijective, which happens if and only if it is surjective, that is $r(A) = n$.

(ii) implies (iii). We show the contrapositive. If $r(A) < n$, then the columns $A^{(1)}, \dots, A^{(n)}$ are linearly dependent, so some $A^{(j)} = -\frac{1}{\lambda_j} \sum_{i \neq j} \lambda_i A^{(i)}$ for scalars λ_i with $\lambda_j \neq 0$. Expanding $\det A$ using multilinearity in the j th column expresses it as a sum of determinants of matrices with two equal columns, all of which vanish. So $\det A = 0$. $\square$

§16.5.6 The Determinant of a Linear Map

We can now play the same game that we played with the trace.

Lemma 16.5.18 (Similar Matrices Have Equal Determinant)

If $A' = P^{-1}AP$, then $\det A' = \det A$.

Proof. Using multiplicativity and $\det P^{-1} \det P = 1$, we have

$$\det(P^{-1}AP) = \det P^{-1} \det A \det P = \det A. \quad \square$$

Definition 16.5.19 (Determinant of an Endomorphism)

Let $\alpha : V \rightarrow V$ be an endomorphism of a finite dimensional vector space V . The **determinant** of α is $\det \alpha = \det[\alpha]_{B,B}$, where B is any basis of V .

The properties of the matrix determinant transfer directly: $\det(\text{id}) = 1$, $\det(\alpha\beta) = \det \alpha \det \beta$, and α is invertible if and only if $\det \alpha \neq 0$, in which case $\det \alpha^{-1} = (\det \alpha)^{-1}$.

There is one more computational result that will be repeatedly useful when we study eigenvalues.

Lemma 16.5.20 (Determinant of Block Triangular Matrices)

Let $A \in \mathcal{M}_{k,k}(\mathbb{F})$, $B \in \mathcal{M}_{\ell,\ell}(\mathbb{F})$ and $C \in \mathcal{M}_{k,\ell}(\mathbb{F})$, and consider the block matrix

$$M = \left(\begin{array}{c|c} A & C \\ \hline 0 & B \end{array} \right).$$

Then $\det M = \det A \det B$.

Proof. Let $n = k + \ell$ and write $M = (m_{ij})$. In the sum

$$\det M = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n m_{\sigma(i)i},$$

the term for σ vanishes unless $\sigma(i) \leq k$ for all $i \leq k$ (as the bottom-left block is zero). Such σ decompose uniquely as $\sigma = \sigma_1 \sigma_2$ where σ_1 permutes $\{1, \dots, k\}$ and σ_2 permutes $\{k+1, \dots, n\}$, with $\varepsilon(\sigma) = \varepsilon(\sigma_1)\varepsilon(\sigma_2)$. The sum then factors as

$$\det M = \left(\sum_{\sigma_1} \varepsilon(\sigma_1) \prod_{i=1}^k m_{\sigma_1(i)i} \right) \left(\sum_{\sigma_2} \varepsilon(\sigma_2) \prod_{i=k+1}^n m_{\sigma_2(i)i} \right) = \det A \det B. \quad \square$$

Applying this repeatedly, a block *upper* triangular matrix has determinant equal to the product of the determinants of its diagonal blocks.

§16.5.7 Expanding Determinants and the Adjugate

The formula defining the determinant has $n!$ terms, so is rarely how you want to compute one. A more practical approach reduces an $n \times n$ determinant to a combination of $(n-1) \times (n-1)$ determinants.

Definition 16.5.21 (Minor)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$ and $i, j \in \{1, \dots, n\}$. The **minor** $A_{\widehat{i}\widehat{j}} \in \mathcal{M}_{n-1,n-1}(\mathbb{F})$ is the matrix obtained from A by deleting the i th row and j th column.

Lemma 16.5.22 (Row and Column Expansions)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. Then for any fixed column j ,

$$\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det A_{\widehat{i}\widehat{j}},$$

and for any fixed row i , we similarly have $\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det A_{\widehat{i}\widehat{j}}$.

Proof. Write the j th column as $A^{(j)} = \sum_{i=1}^n a_{ij} e_i$. By multilinearity,

$$\det A = \sum_{i=1}^n a_{ij} \det(A^{(1)}, \dots, e_i, \dots, A^{(n)}),$$

with e_i in the j th position. Now move the j th column to the front ($j-1$ adjacent swaps) and the i th row to the top ($i-1$ adjacent swaps), picking up the sign $(-1)^{i+j-2} = (-1)^{i+j}$. The resulting matrix is block triangular with top-left block

(1) and bottom-right block $A_{\widehat{ij}}$, so by the block triangular lemma,

$$\det(A^{(1)}, \dots, e_i, \dots, A^{(n)}) = (-1)^{i+j} \det A_{\widehat{ij}},$$

giving the column expansion. The row expansion follows by transposing. $\square$

Collecting these signed minors into a matrix gives us an explicit (if computationally expensive) formula for the inverse of a matrix.

Definition 16.5.23 (Adjugate)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. The **adjugate** of A is the $n \times n$ matrix $\text{adj } A$ with

$$(\text{adj } A)_{ij} = (-1)^{i+j} \det A_{\widehat{ji}}.$$

Note the transposition of indices – the (i, j) entry of the adjugate involves deleting row j and column i . In the notation of the proof above, $(\text{adj } A)_{ji} = \det(A^{(1)}, \dots, e_i, \dots, A^{(n)})$ with e_i in position j .

Theorem 16.5.24 (Adjugate Formula)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$. Then

$$(\text{adj } A)A = (\det A)I.$$

In particular, if A is invertible then

$$A^{-1} = \frac{1}{\det A} \text{adj } A.$$

Proof. On the diagonal, the (j, j) entry of $(\text{adj } A)A$ is

$$\sum_{i=1}^n (\text{adj } A)_{ji} a_{ij} = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det A_{\widehat{ij}} = \det A,$$

by the column expansion. Off the diagonal, for $j \neq k$ the (j, k) entry is

$$\sum_{i=1}^n (\text{adj } A)_{ji} a_{ik} = \sum_{i=1}^n a_{ik} \det(A^{(1)}, \dots, \underbrace{e_i}_{j\text{th}}, \dots, A^{(n)}) = \det(A^{(1)}, \dots, \underbrace{A^{(k)}}_{j\text{th}}, \dots, A^{(n)})$$

after reassembling using multilinearity. But this last determinant has the column $A^{(k)}$ appearing twice (in positions j and k), so vanishes. $\square$

§16.5.8 Cramer's Rule

Finally, the adjugate machinery gives us an algorithm for computing the unique solution to $Ax = b$ without computing A^{-1} .

Proposition 16.5.25 (Cramer's Rule)

Let $A \in \mathcal{M}_{n,n}(\mathbb{F})$ be an invertible matrix, and let $b \in \mathbb{F}^n$. Then the unique solution

to $Ax = b$ is given by

$$x_i = \frac{1}{\det A} \det(A_{ib}),$$

where A_{ib} is the matrix obtained by replacing the i th column of A by b .

Proof. Since A is invertible, there exists a unique $x \in \mathbb{F}^n$ with $Ax = b$. Then computing we have

$$\begin{aligned} \det(A_{ib}) &= \det(A^{(1)}, \dots, A^{(i-1)}, b, A^{(i+1)}, \dots, A^{(n)}) \\ &= \det(A^{(1)}, \dots, A^{(i-1)}, Ax, A^{(i+1)}, \dots, A^{(n)}) \\ &= \det\left(A^{(1)}, \dots, A^{(i-1)}, \sum_{j=1}^n x_j A^{(j)}, A^{(i+1)}, \dots, A^{(n)}\right) \\ &= x_i \det(A^{(1)}, \dots, A^{(i-1)}, A^{(i)}, A^{(i+1)}, \dots, A^{(n)}) \\ &= x_i \det A, \end{aligned}$$

where in the second-to-last step every other term vanishes since it is the determinant of a matrix with a repeated column. $\square$

§16.6 Diagonalization, Eigenvectors and Eigenvalues

§16.6.1 Diagonal and Triangular Matrices

We are now going to take the first steps towards the diagonalisation of endomorphisms. Given a finite dimensional vector space V over $\mathbb{F}$ and some endomorphism $\alpha : V \rightarrow V$, we are going to be guided by the following general problem:

Can we find a basis B of V such that in this basis $[\alpha]_{B,B}$ has a ‘nice’ form?

With this in mind, we will first care about when this ‘nice’ form is a diagonal matrix, or more generally a triangular matrix.

Definition 16.6.1 (Diagonalizable)

Let V be a vector space and let $\alpha : V \rightarrow V$ be a linear map. We say that α is **diagonalizable** if there is a basis B of V such that

$$[\alpha]_{B,B} = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

where $\lambda_i \in \mathbb{F}$.

Definition 16.6.2 (Triangular)

Let V be a vector space and let $\alpha : V \rightarrow V$ be a linear map. We say that α is

triangulable if there is a basis B of V such that

$$[\alpha]_{B,B} = \begin{pmatrix} \lambda_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

where $\lambda_i \in \mathbb{F}$.

We will see that when thinking about such matrices, we will frequently turn to the concept of eigenvectors and eigenvalues.

Definition 16.6.3 (Eigenvectors and Eigenvalues)

If V is a vector space over $\mathbb{F}$ and $\alpha : V \rightarrow V$ is a linear map, we say that $\lambda \in \mathbb{F}$ is a **eigenvalue** if there exists some $v \in V \setminus \{0\}$ such that $\alpha(v) = \lambda v$. We say that v is a corresponding **eigenvector**.

Definition 16.6.4 (Eigenspace)

If λ is an eigenvalue of a linear map $\alpha : V \rightarrow V$, we define the corresponding **eigenspace** to be the subspace $V_\lambda = \{v \in V \mid \alpha(v) = \lambda v\}$.

When computing eigenvalues we frequently employ the following lemma.

Lemma 16.6.5 (Computing Eigenvalues)

Let V be a vector space over $\mathbb{F}$, and let $\alpha : V \rightarrow V$ be a linear map. Then $\lambda \in \mathbb{F}$ is an eigenvalue if and only if $\det(\alpha - \lambda \text{id}) = 0$.

Proof. There exists $v \in V \setminus \{0\}$ with $\alpha(v) = \lambda v$ if and only if $\ker(\alpha - \lambda \text{id}) \neq \{0\}$. This occurs if and only if $\alpha - \lambda \text{id}$ is not injective, and since α is an endomorphism, this occurs if and only if $\alpha - \lambda \text{id}$ is not invertible, that is, if $\det(\alpha - \lambda \text{id}) = 0$. $\square$

Applying this lemma gives us a polynomial in λ which one must find the roots of to obtain the eigenvalues of the linear map. We call this polynomial the *characteristic polynomial*.

Definition 16.6.6 (Characteristic Polynomial)

Let $\alpha : V \rightarrow V$ be an endomorphism. Then the **characteristic polynomial**^a of α is $\chi_\alpha(\lambda) = \det(\alpha - \lambda \text{id})$.

^aIt should be checked that this definition is basis free, but this is straightforward.

The heart of the matter is the following criterion of triangulability.

Theorem 16.6.7 (Triangulability Criterion)

Let V be a vector space over $\mathbb{F}$. A linear map $\alpha : V \rightarrow V$ is triangulable if and only

if

$$\chi_\alpha(t) = c \prod_{i=1}^n (t - \lambda_i),$$

for some $c, \lambda_i \in \mathbb{F}$.

Proof. Suppose α was triangulable. Then for some basis B of V we have

$$[\alpha]_{B,B} = \begin{pmatrix} \lambda_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix},$$

and so

$$\chi_\alpha(t) = \det \begin{pmatrix} \lambda_1 - t & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n - t \end{pmatrix} = \prod_{i=1}^n (\lambda_i - t),$$

which is of the required form.

Now suppose that this criterion holds. We will prove by induction on $n = \dim V$ that it implies triangulability. For $n = 1$, we are immediately done, so suppose that $n > 1$. Then let λ be a root of $\chi_\alpha(t)$ (which exists by assumption). Then λ is an eigenvalue of α . Let $U = V_\lambda$ be the corresponding eigenspace, and let $\{v_1, \dots, v_k\}$ be a basis of U . We can complete this to a basis $B = \{v_1, \dots, v_n\}$ of V . So then

$$[\alpha]_{B,B} = \left(\begin{array}{c|c} \lambda I & * \\ \hline 0 & C \end{array} \right).$$

Now α induces an endomorphism $\bar{\alpha} : V/U \rightarrow V/U$, and C is the matrix of $\bar{\alpha}$ with respect to the basis $\{v_{k+1} + U, \dots, v_n + U\}$. The characteristic polynomial of $\bar{\alpha}$ divides that of α , so it too is a product of linear factors, and by induction we can write $\bar{\alpha}$ in triangular form by choosing an appropriate basis $\{v'_{k+1} + U, \dots, v'_n + U\}$ of V/U . Taking the basis $B' = \{v_1, \dots, v_k, v'_{k+1}, \dots, v'_n\}$ then gives us $[\alpha]_{B',B'}$ in triangular form. $\square$

Remark. By the fundamental theorem of algebra, if $\mathbb{F} = \mathbb{C}$, then every matrix is triangulable.

§16.6.2 Polynomials

Before we can get a criterion for *diagonalisability*, we need to set up a small dictionary between endomorphisms and polynomials. Let's first recall some basic facts about polynomials over a field $\mathbb{F}$. We write $\mathbb{F}[t]$ for the vector space of polynomials

$$f(t) = a_n t^n + a_{n-1} t^{n-1} + \cdots + a_1 t + a_0$$

with coefficients in $\mathbb{F}$, and $\deg f$ for the degree of f . The facts we will need are the following.

- (i) If λ is a root of f , then $(t - \lambda)$ divides f . This can be seen by writing $f(t) = f(t) - f(\lambda)$ and factoring $(t - \lambda)$ out of each $t^k - \lambda^k$.
- (ii) We say λ is a root of **multiplicity** k if $(t - \lambda)^k$ divides f but $(t - \lambda)^{k+1}$ does not. A non-zero polynomial of degree n has at most n roots counted with multiplicity.

- (iii) Two polynomials of degree less than n agreeing at n distinct points are equal (their difference has too many roots).
- (iv) *The fundamental theorem of algebra*: every non-constant $f \in \mathbb{C}[t]$ has a complex root, and hence factors completely into linear factors.
- (v) Given polynomials $a, b \in \mathbb{F}[t]$ with $b \neq 0$, there are polynomials q, r with $a = qb + r$ and $\deg r < \deg b$ (polynomial division).

The dictionary is then this: given a polynomial $p(t) = a_n t^n + \cdots + a_0$ and an endomorphism α of V , we can form the endomorphism

$$p(\alpha) = a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha + a_0 \text{id},$$

where α^k denotes k -fold composition. Similarly for a square matrix A we define $p(A) = a_n A^n + \cdots + a_1 A + a_0 I$. Note that any two polynomials in the same α commute with each other – this simple observation gets used constantly below.

§16.6.3 A Criterion for Diagonalisability

We saw above that triangulability is governed by the characteristic polynomial. Diagonalisability turns out to be governed by polynomials too, in a rather clean way.

Theorem 16.6.8 (Diagonalisability Criterion)

Let V be a finite dimensional vector space over $\mathbb{F}$, and let $\alpha : V \rightarrow V$ be an endomorphism. Then α is diagonalisable if and only if there is a non-zero polynomial p that is a product of *distinct* linear factors such that $p(\alpha) = 0$.

Proof. Suppose first that α is diagonalisable, with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, and set $p(t) = \prod_{i=1}^k (t - \lambda_i)$. If v is a basis vector in the diagonalising basis, then $\alpha(v) = \lambda_i v$ for some i , and since the factors of $p(\alpha)$ commute we can apply $(\alpha - \lambda_i \text{id})$ to v first, giving $p(\alpha)(v) = 0$. Since $p(\alpha)$ kills a basis, $p(\alpha) = 0$.

The converse is the interesting direction. Suppose $p(\alpha) = 0$ where $p(t) = \prod_{i=1}^k (t - \lambda_i)$ with $\lambda_1, \dots, \lambda_k$ distinct, and let $V_{\lambda_i} = \ker(\alpha - \lambda_i \text{id})$. We claim that

$$V = \bigoplus_{i=1}^k V_{\lambda_i},$$

which suffices, since choosing a basis of each eigenspace and taking their union then gives a basis of V consisting of eigenvectors.

To prove the claim, consider the *Lagrange interpolation* polynomials

$$q_j(t) = \prod_{i \neq j} \frac{t - \lambda_i}{\lambda_j - \lambda_i},$$

which are engineered so that $q_j(\lambda_i) = \delta_{ij}$. The sum $q = q_1 + \cdots + q_k$ has degree at most $k - 1$ and takes the value 1 at the k distinct points $\lambda_1, \dots, \lambda_k$, so $q \equiv 1$.

Now define $\pi_j = q_j(\alpha)$. By the above, $\sum_{j=1}^k \pi_j = \text{id}$, so every $v \in V$ decomposes as

$$v = \sum_{j=1}^k \pi_j(v).$$

Moreover each $\pi_j(v)$ lands in the eigenspace V_{λ_j} : up to a scalar, $(\alpha - \lambda_j \text{id}) \circ q_j(\alpha)$ is exactly $p(\alpha) = 0$, so $(\alpha - \lambda_j \text{id})\pi_j(v) = 0$. Hence $V = \sum_j V_{\lambda_j}$.

It remains to check the sum is direct. Note that if $v \in V_{\lambda_j}$, then applying $q_j(\alpha)$ replaces each factor $(\alpha - \lambda_i \text{id})$ acting on v by the scalar $(\lambda_j - \lambda_i)$, so $\pi_j(v) = q_j(\lambda_j)v = v$; and if $v \in V_{\lambda_i}$ with $i \neq j$ then some factor of $q_j(\alpha)$ kills v , so $\pi_j(v) = 0$. So suppose

$$v_1 + v_2 + \cdots + v_k = 0, \quad v_i \in V_{\lambda_i}.$$

Applying π_j to both sides kills every term except the j th, leaving $v_j = 0$. This holds for each j , so the decomposition of 0 is unique and the sum is direct. $\square$

Remark. Along the way we proved something useful in its own right: for *any* endomorphism α , the sum of eigenspaces corresponding to distinct eigenvalues is always direct. Diagonalisability fails only when this direct sum is a proper subspace of V .

This criterion can feel abstract, so let's see it do something impressive with almost no work.

Example 16.6.9 (Matrices of Finite Order)

Let $A \in \mathcal{M}_{n,n}(\mathbb{C})$ with $A^m = I$ for some $m \geq 1$. Then A is diagonalisable, since the polynomial

$$p(t) = t^m - 1 = \prod_{j=1}^m (t - e^{2\pi i j/m})$$

is a product of distinct linear factors and satisfies $p(A) = 0$.

As another application, we can answer the natural question of when two diagonalisable maps can be diagonalised *at the same time*.

Theorem 16.6.10 (Simultaneous Diagonalisation)

Let α, β be diagonalisable endomorphisms of a finite dimensional vector space V . Then α and β are **simultaneously diagonalisable** (there is a single basis B with both $[\alpha]_{B,B}$ and $[\beta]_{B,B}$ diagonal) if and only if $\alpha\beta = \beta\alpha$.

Proof. One direction is immediate: diagonal matrices commute, so if $[\alpha]_{B,B}$ and $[\beta]_{B,B}$ are both diagonal then $\alpha\beta = \beta\alpha$.

Conversely, suppose α and β commute. Since α is diagonalisable, $V = \bigoplus_{i=1}^k V_{\lambda_i}$ where $\lambda_1, \dots, \lambda_k$ are its distinct eigenvalues. The key point is that β preserves each eigenspace of α : for $v \in V_{\lambda_j}$,

$$\alpha(\beta(v)) = \beta(\alpha(v)) = \beta(\lambda_j v) = \lambda_j \beta(v),$$

so $\beta(v) \in V_{\lambda_j}$. Now β is diagonalisable, so some polynomial p with distinct linear

factors has $p(\beta) = 0$, and hence certainly $p(\beta|_{V_{\lambda_j}}) = 0$. So each restriction $\beta|_{V_{\lambda_j}}$ is diagonalisable by our criterion, and we may pick a basis B_j of V_{λ_j} consisting of eigenvectors of β . Since each B_j consists of λ_j -eigenvectors of α as well, the union $B = B_1 \cup \dots \cup B_k$ diagonalises both maps simultaneously. $\square$

§16.6.4 The Minimal Polynomial

The diagonalisability criterion asks whether *some* polynomial with distinct linear factors kills α . It is natural to focus on the smallest polynomial killing α .

Definition 16.6.11 (Minimal Polynomial)

Let α be an endomorphism of a finite dimensional vector space V . The **minimal polynomial** m_α of α is the non-zero monic polynomial of least degree such that $m_\alpha(\alpha) = 0$.

Some polynomial killing α certainly exists: $\dim \mathcal{L}(V, V) = n^2$, so the $n^2 + 1$ endomorphisms $\text{id}, \alpha, \alpha^2, \dots, \alpha^{n^2}$ are linearly dependent, and any dependence gives a polynomial vanishing at α .

Lemma 16.6.12 (Divisibility Property of the Minimal Polynomial)

Let α be an endomorphism of V and $p \in \mathbb{F}[t]$. Then $p(\alpha) = 0$ if and only if m_α divides p . In particular, the minimal polynomial is unique.

Proof. If m_α divides p , say $p = m_\alpha q$, then $p(\alpha) = m_\alpha(\alpha)q(\alpha) = 0$. Conversely, suppose $p(\alpha) = 0$, and divide with remainder: $p = qm_\alpha + r$ with $\deg r < \deg m_\alpha$. Evaluating at α gives $r(\alpha) = 0$, and by minimality of $\deg m_\alpha$ this forces $r \equiv 0$, so m_α divides p .

For uniqueness, two monic minimal polynomials must divide each other and have the same degree, so are equal. $\square$

With this language, our diagonalisability criterion can be restated: α is diagonalisable if and only if m_α is a product of distinct linear factors. (If some product of distinct linear factors kills α , then m_α divides it, so m_α is itself such a product.)

Example 16.6.13 (Telling Matrices Apart)

Consider the matrices

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Both satisfy $p(t) = (t - 1)^2$, i.e. $p(A) = p(B) = 0$, so both minimal polynomials divide $(t - 1)^2$. For A we have $A - I = 0$, so $m_A(t) = t - 1$; but $B - I \neq 0$, so $m_B(t) = (t - 1)^2$. In particular B is *not* diagonalisable, since its minimal polynomial has a repeated root.

§16.6.5 The Cayley-Hamilton Theorem

We now come to one of the most famous theorems of linear algebra, relating the two polynomials we have attached to an endomorphism: *every endomorphism satisfies its own characteristic polynomial*.

Theorem 16.6.14 (Cayley-Hamilton)

Let α be an endomorphism of a finite dimensional vector space V . Then $\chi_\alpha(\alpha) = 0$. Equivalently, m_α divides χ_α .

Before the proof, one warning: it is tempting to argue ‘ $\chi_\alpha(t) = \det(\alpha - t \text{id})$, so $\chi_\alpha(\alpha) = \det(\alpha - \alpha) = 0$ ’. This is nonsense – $\chi_\alpha(\alpha)$ is an *endomorphism*, not a scalar, and substituting a matrix into the determinant formula in this way is not a valid operation.

We will give two proofs: a clean geometric one for $\mathbb{F} = \mathbb{C}$ (which also covers $\mathbb{F} = \mathbb{R}$, viewing a real matrix as a complex one), and an algebraic one that works over any field.

Proof (over $\mathbb{C}$). Since $\mathbb{F} = \mathbb{C}$, we can pick a basis $B = \{v_1, \dots, v_n\}$ in which $A = [\alpha]_{B,B}$ is upper triangular with diagonal entries $a_1, \dots, a_n$, so that

$$\chi_\alpha(t) = \prod_{i=1}^n (a_i - t).$$

Let $U_j = \langle v_1, \dots, v_j \rangle$, so that $U_0 = \{0\} \leq U_1 \leq \dots \leq U_n = V$. Triangularity says exactly that $\alpha(U_j) \leq U_j$, and moreover $(\alpha - a_j \text{id})(U_j) \leq U_{j-1}$, since the j th column of $A - a_j I$ has zero entries from row j downwards.

Now take any $v \in V = U_n$ and feed it through the factors of $\chi_\alpha(\alpha)$ from the right:

$$\chi_\alpha(\alpha)(v) = (\alpha - a_1 \text{id}) \cdots (\alpha - a_{n-1} \text{id}) \underbrace{(\alpha - a_n \text{id})(v)}_{\in U_{n-1}}.$$

Each successive factor pushes the vector down one step in the chain of subspaces, so after all n factors we land in $U_0 = \{0\}$. Hence $\chi_\alpha(\alpha) = 0$. $\square$

Proof (any field). Let A be the matrix of α in some basis, and write

$$\det(tI - A) = (-1)^n \chi_A(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0.$$

Recall the adjugate formula $(\text{adj } B)B = (\det B)I$, and apply it with $B = tI - A$. Every entry of $\text{adj}(tI - A)$ is (up to sign) the determinant of an $(n-1) \times (n-1)$ submatrix of $tI - A$, hence a polynomial in t of degree at most $n-1$. Collecting coefficients, we can write

$$\text{adj}(tI - A) = B_{n-1}t^{n-1} + \cdots + B_1t + B_0$$

for some matrices $B_i \in \mathcal{M}_{n,n}(\mathbb{F})$. The adjugate formula then reads

$$(B_{n-1}t^{n-1} + \cdots + B_0)(tI - A) = (t^n + a_{n-1}t^{n-1} + \cdots + a_0)I,$$

and since this holds for all t (as an identity of polynomials with matrix coefficients),

we may equate the coefficients of each power of t :

$$\begin{aligned} t^n : \quad & B_{n-1} = I \\ t^{n-1} : \quad & B_{n-2} - B_{n-1}A = a_{n-1}I \\ & \vdots \\ t^1 : \quad & B_0 - B_1A = a_1I \\ t^0 : \quad & -B_0A = a_0I. \end{aligned}$$

Multiplying the t^k relation on the right by A^k and summing all of the relations, the left hand side telescopes to zero, leaving

$$0 = A^n + a_{n-1}A^{n-1} + \cdots + a_1A + a_0I = (-1)^n \chi_A(A),$$

as required. $\square$

§16.6.6 Multiplicities of Eigenvalues

We now have three numbers we can attach to each eigenvalue of an endomorphism, and comparing them will give us a rather complete picture of diagonalisability.

Definition 16.6.15 (Algebraic and Geometric Multiplicity)

Let λ be an eigenvalue of an endomorphism α of V . We define:

- (i) the **algebraic multiplicity** a_λ , the multiplicity of λ as a root of χ_α ,
- (ii) the **geometric multiplicity** $g_\lambda = \dim V_\lambda$, the dimension of the corresponding eigenspace,
- (iii) the multiplicity c_λ of λ as a root of the minimal polynomial m_α .

Lemma 16.6.16 (Comparing Multiplicities)

If λ is an eigenvalue of α , then

$$1 \leq g_\lambda \leq a_\lambda \quad \text{and} \quad 1 \leq c_\lambda \leq a_\lambda.$$

Proof. Since λ is an eigenvalue there is a non-zero eigenvector, so $g_\lambda \geq 1$. For $g_\lambda \leq a_\lambda$, pick a basis $\{v_1, \dots, v_{g_\lambda}\}$ of V_λ and extend to a basis B of V . Then

$$[\alpha]_{B,B} = \left(\begin{array}{c|c} \lambda I_{g_\lambda} & * \\ \hline 0 & A_1 \end{array} \right),$$

and expanding using the block triangular lemma, $\chi_\alpha(t) = (\lambda - t)^{g_\lambda} \det(A_1 - tI)$. So $(t - \lambda)^{g_\lambda}$ divides $\chi_\alpha(t)$, that is, $g_\lambda \leq a_\lambda$.

For c_λ : by Cayley-Hamilton, m_α divides χ_α , so $c_\lambda \leq a_\lambda$. And if v is a λ -eigenvector, then $p(\alpha)(v) = p(\lambda)v$ for any polynomial p ; taking $p = m_\alpha$ gives $m_\alpha(\lambda)v = 0$ and so $m_\alpha(\lambda) = 0$, giving $c_\lambda \geq 1$. $\square$

Theorem 16.6.17 (Characterisations of Diagonalisability)

Let α be an endomorphism of a finite dimensional $\mathbb{C}$ -vector space V . Then the following are equivalent:

- (i) α is diagonalisable,
- (ii) $a_\lambda = g_\lambda$ for every eigenvalue λ of α ,
- (iii) $c_\lambda = 1$ for every eigenvalue λ of α .

Proof. Since $\mathbb{F} = \mathbb{C}$, the characteristic polynomial factors completely, so if $\lambda_1, \dots, \lambda_k$ are the distinct eigenvalues then $\sum_{i=1}^k a_{\lambda_i} = \deg \chi_\alpha = n$.

(i) is equivalent to (iii). Over $\mathbb{C}$, m_α factors as $\prod_i (t - \lambda_i)^{c_{\lambda_i}}$, and we know α is diagonalisable if and only if m_α has distinct linear factors, which happens exactly when every $c_{\lambda_i} = 1$.

(i) is equivalent to (ii). We know that the sum of the eigenspaces is always direct, so α is diagonalisable if and only if $\sum_{i=1}^k g_{\lambda_i} = \dim \bigoplus_i V_{\lambda_i}$ equals $n = \sum_{i=1}^k a_{\lambda_i}$. Since $g_{\lambda_i} \leq a_{\lambda_i}$ termwise, the sums are equal if and only if $g_{\lambda_i} = a_{\lambda_i}$ for every i . $\square$

Example 16.6.18 (Using the Minimal Polynomial)

Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Expanding along the bottom row, $\chi_A(t) = (1-t)^2(2-t)$, so the minimal polynomial is $(t-1)(t-2)$ or $(t-1)^2(t-2)$. Multiplying out,

$$(A - I)(A - 2I) = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & -2 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = 0,$$

so $m_A(t) = (t-1)(t-2)$, a product of distinct linear factors – hence A is diagonalisable.

§16.7 Jordan Normal Form

Not every endomorphism is diagonalisable, which raises the obvious question: how close can we get? Over $\mathbb{C}$, the answer is ‘remarkably close’, and the precise answer is the *Jordan normal form*. Throughout this section we work over $\mathbb{F} = \mathbb{C}$.

§16.7.1 Jordan Blocks

Definition 16.7.1 (Jordan Block and Jordan Normal Form)

For $\lambda \in \mathbb{C}$, the **Jordan block** of size m is the matrix

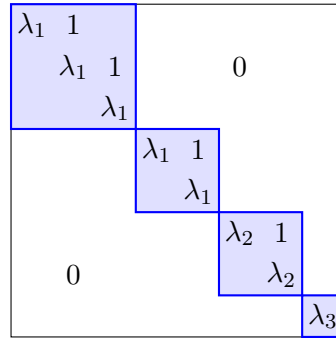
$$J_m(\lambda) = \begin{pmatrix} \lambda & 1 & & \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ & & & \lambda \end{pmatrix} \in \mathcal{M}_{m,m}(\mathbb{C}),$$

with λ on the diagonal, 1 just above it, and 0 everywhere else. A matrix is in **Jordan normal form** if it is block diagonal with each block a Jordan block,

$$A = \begin{pmatrix} J_{n_1}(\lambda_1) & & \\ & \ddots & \\ & & J_{n_k}(\lambda_k) \end{pmatrix},$$

where the λ_i need not be distinct.

So a matrix in Jordan normal form is ‘diagonal, plus some pesky 1s’. A diagonal matrix is the special case where every block has size 1. Schematically, the structure looks like this, with each shaded block a single Jordan block:



Note that repeated eigenvalues can appear in several different blocks (here λ_1 owns both a 3×3 and a 2×2 block).

The main theorem, which we will state but not prove¹¹, is that this form can always be reached.

Theorem 16.7.2 (Jordan Normal Form)

Every matrix $A \in \mathcal{M}_{n,n}(\mathbb{C})$ is similar to a matrix in Jordan normal form, which is unique up to reordering the Jordan blocks.

Example 16.7.3 (JNF in Two Dimensions)

Every 2×2 complex matrix is similar to exactly one of

$$\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}, \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix},$$

¹¹The proof is non-examinable, and follows from the structure theory of modules developed in IB Groups, Rings and Modules.

with respective minimal polynomials $(t - \lambda_1)(t - \lambda_2)$, $(t - \lambda)$ and $(t - \lambda)^2$.

§16.7.2 Generalised Eigenspaces

While we won't prove the Jordan normal form theorem in full, we can prove the decomposition of V that underlies it. The idea is to relax the eigenspace condition $(\alpha - \lambda \text{id})(v) = 0$, and instead only require that some *power* of $(\alpha - \lambda \text{id})$ kills v .

Theorem 16.7.4 (Generalised Eigenspace Decomposition)

Let V be a finite dimensional $\mathbb{C}$ -vector space and α an endomorphism of V with minimal polynomial

$$m_\alpha(t) = \prod_{j=1}^k (t - \lambda_j)^{c_j},$$

where the λ_j are distinct. Then

$$V = \bigoplus_{j=1}^k V_j, \quad \text{where } V_j = \ker((\alpha - \lambda_j \text{id})^{c_j})$$

is called the **generalised eigenspace** associated to λ_j .

Proof. The proof mirrors our diagonalisability criterion – we construct explicit projections onto the pieces. Let

$$p_j(t) = \prod_{i \neq j} (t - \lambda_i)^{c_i},$$

so that $p_j(t)(t - \lambda_j)^{c_j} = m_\alpha(t)$. The polynomials $p_1, \dots, p_k$ have no common factor, so by the Euclidean algorithm there are polynomials $q_1, \dots, q_k$ with

$$q_1 p_1 + \dots + q_k p_k = 1.$$

Define $\pi_j = (q_j p_j)(\alpha)$. Then immediately $\sum_j \pi_j = \text{id}$, so every $v \in V$ decomposes as $v = \sum_j \pi_j(v)$; and each piece lies where it should, since

$$(\alpha - \lambda_j \text{id})^{c_j} \pi_j(v) = q_j(\alpha) m_\alpha(\alpha)(v) = 0,$$

so $\pi_j(v) \in V_j$. Hence $V = \sum_j V_j$.

For directness, note that if $i \neq j$ then $p_i p_j$ is divisible by m_α , so $\pi_i \pi_j = 0$. Combining this with $\sum_j \pi_j = \text{id}$ gives $\pi_i = \pi_i \left(\sum_j \pi_j \right) = \pi_i^2$, so each π_i is a projection, and moreover π_i acts as the identity on V_i and as zero on V_j for $j \neq i$.^a Then given any relation $v_1 + \dots + v_k = 0$ with $v_j \in V_j$, applying π_j picks out $v_j = 0$ for each j , so the sum is direct. $\square$

^aTo see π_i acts as the identity on V_i : for $v \in V_i$, all the terms $\pi_j(v)$ with $j \neq i$ lie in V_j , and applying the identity $\text{id} = \sum_j \pi_j$ twice with $\pi_i \pi_j = 0$ collapses everything except $\pi_i(v)$.

Remark. Each V_j is stable under α , and the restriction of $(\alpha - \lambda_j \text{id})$ to V_j is *nilpotent* – its c_j th power is zero. So the Jordan normal form theorem is really a statement about how nilpotent maps can act, and the 1s above the diagonal record that nilpotent action.

It's worth seeing concretely how a single Jordan block acts. Writing $J_m(\lambda) = \lambda I + N$, the basis vectors form a *chain*: N sends each basis vector to the previous one, eventually falling off the end into 0.

$$v_m \xrightarrow{N} v_{m-1} \xrightarrow{N} \cdots \xrightarrow{N} v_2 \xrightarrow{N} v_1 \xrightarrow{N} 0$$

Here $N = J_m(\lambda) - \lambda I$, and N^k shifts the off-diagonal line of 1s one step further from the diagonal each time, so $N^m = 0$ but $N^{m-1} \neq 0$.

This picture lets us read off all three multiplicities of an eigenvalue λ directly from the Jordan normal form:

- a_λ is the *total size* of all blocks with eigenvalue λ (the number of times λ appears on the diagonal),
- g_λ is the *number* of blocks with eigenvalue λ (each block contributes exactly one independent eigenvector – the start of its chain),
- c_λ is the size of the *largest* block with eigenvalue λ (the longest chain takes the most steps to kill).

In particular, knowing $(a_\lambda, g_\lambda, c_\lambda)$ for each eigenvalue determines the Jordan normal form completely for small matrices, which makes computing it quite mechanical.

§16.7.3 Computing the Jordan Normal Form

Example 16.7.5 (Computing a JNF)

Let's find the Jordan normal form of

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix},$$

along with a basis realising it. The characteristic polynomial is $\chi_A(t) = t(2-t)+1 = (t-1)^2$, so $\lambda = 1$ is the only eigenvalue. Since $A - I \neq 0$, the minimal polynomial is $(t-1)^2$, so $c_1 = 2$ and the JNF has a single 2×2 block:

$$J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

To find the basis, we need a chain: a vector v_2 with $(A - I)v_2 = v_1 \neq 0$ and $(A - I)v_1 = 0$. We compute

$$A - I = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \text{so} \quad \ker(A - I) = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\rangle.$$

Taking $v_1 = (1, -1)^T$, we then solve $(A - I)v_2 = v_1$, which gives (for example) $v_2 = (-1, 0)^T$. In the basis $\{v_1, v_2\}$ the map acts by $v_1 \mapsto v_1$ and $v_2 \mapsto v_1 + v_2$, so with $P = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}$ we have $A = PJP^{-1}$.

§16.8 Symmetric, Quadratic and Hermitian Forms

We now return to bilinear forms, this time restricting our attention to forms $\phi : V \times V \rightarrow \mathbb{F}$ on a *single* vector space. Just as we asked when an endomorphism can be represented by a diagonal matrix, we will ask when a bilinear form can – and here the answer turns out to be much cleaner.

§16.8.1 Symmetric Bilinear Forms and Congruence

For a bilinear form $\phi : V \times V \rightarrow \mathbb{F}$ and a basis B of V , we write $[\phi]_B = [\phi]_{B,B}$. Specialising our change of basis result for bilinear forms, if B' is another basis and $P = [I]_{B',B}$ is the change of basis matrix, then

$$[\phi]_{B'} = P^T [\phi]_B P.$$

This motivates the following analogue of similarity.

Definition 16.8.1 (Congruent Matrices)

Two square matrices $A, A' \in \mathcal{M}_{n,n}(\mathbb{F})$ are **congruent** if $A' = P^T A P$ for some invertible matrix P .

Congruence is an equivalence relation, and two matrices are congruent exactly when they represent the same bilinear form in different bases. So the question ‘when can ϕ be represented by a diagonal matrix?’ is the question of classifying matrices up to congruence.

There is an immediate necessary condition. If $[\phi]_B = D$ is diagonal in some basis, then for any other basis, $[\phi]_{B'} = P^T D P$ satisfies $([\phi]_{B'})^T = P^T D^T P = [\phi]_{B'}$ – the matrix of ϕ is symmetric in *every* basis. This corresponds to a natural symmetry condition on the form itself.

Definition 16.8.2 (Symmetric Bilinear Form)

A bilinear form $\phi : V \times V \rightarrow \mathbb{F}$ is **symmetric** if $\phi(u, v) = \phi(v, u)$ for all $u, v \in V$.

It’s easy to check that ϕ is symmetric if and only if $[\phi]_B$ is a symmetric matrix for any (equivalently, every) basis B . Remarkably, this necessary condition for diagonalisability turns out to be sufficient, as we will prove shortly.

§16.8.2 Quadratic Forms

Before that, let’s meet the closely-related notion of a *quadratic form*, which is what you get when you feed a bilinear form the same vector twice.

Definition 16.8.3 (Quadratic Form)

A map $Q : V \rightarrow \mathbb{F}$ is a **quadratic form** if there is some bilinear form $\phi : V \times V \rightarrow \mathbb{F}$ with

$$Q(v) = \phi(v, v) \text{ for all } v \in V.$$

In coordinates, if $[\phi]_B = A$ and $x = [v]_B$, then $Q(v) = x^T A x$ is a homogeneous quadratic

polynomial in the coordinates of v – hence the name. Note that

$$x^T A x = \sum_{i,j} a_{ij} x_i x_j = \sum_{i,j} \frac{a_{ij} + a_{ji}}{2} x_i x_j = x^T \left(\frac{A + A^T}{2} \right) x,$$

so we lose nothing by assuming the underlying matrix is symmetric. This observation upgrades to a bijection between quadratic forms and symmetric bilinear forms.

Proposition 16.8.4 (Polarisation Identity)

Let $Q : V \rightarrow \mathbb{F}$ be a quadratic form (where the characteristic of $\mathbb{F}$ is not 2). Then there is a *unique* symmetric bilinear form ϕ with $Q(v) = \phi(v, v)$, given by

$$\phi(u, v) = \frac{1}{2} (Q(u + v) - Q(u) - Q(v)).$$

Proof. For existence, take any bilinear ψ with $Q(v) = \psi(v, v)$ and symmetrise: $\phi(u, v) = \frac{1}{2}(\psi(u, v) + \psi(v, u))$ is symmetric, bilinear, and has $\phi(v, v) = \psi(v, v) = Q(v)$.

For uniqueness, if ϕ is any symmetric bilinear form representing Q , then expanding gives

$$Q(u + v) = \phi(u + v, u + v) = Q(u) + Q(v) + 2\phi(u, v),$$

so ϕ is determined by Q via the stated formula. $\square$

So symmetric bilinear forms and quadratic forms are two views of the same data, and we can pass between them freely.

§16.8.3 Diagonalising Symmetric Bilinear Forms

Theorem 16.8.5 (Diagonalisability of Symmetric Bilinear Forms)

Let $\phi : V \times V \rightarrow \mathbb{F}$ be a symmetric bilinear form on a finite dimensional vector space V . Then there is a basis B of V such that $[\phi]_B$ is diagonal.

Proof. We induct on $n = \dim V$, the case $n \leq 1$ being trivial. If $\phi(v, v) = 0$ for every $v \in V$, then by the polarisation identity $\phi = 0$, which is certainly diagonal. Otherwise, pick e_1 with $\phi(e_1, e_1) \neq 0$, and let

$$U = \{v \in V \mid \phi(e_1, v) = 0\},$$

the kernel of the linear map $v \mapsto \phi(e_1, v)$. This map is a non-zero element of V^* (it doesn't vanish at e_1), so by rank-nullity $\dim U = n - 1$. Moreover $\langle e_1 \rangle \cap U = \{0\}$, since $\lambda e_1 \in U$ forces $\lambda \phi(e_1, e_1) = 0$, so $\lambda = 0$. Counting dimensions, $V = \langle e_1 \rangle \oplus U$.

By induction, the restriction $\phi|_{U \times U}$ is diagonal in some basis $\{e_2, \dots, e_n\}$ of U . Then in the basis $B = \{e_1, e_2, \dots, e_n\}$, using symmetry and $\phi(e_1, e_j) = 0$ for $j \geq 2$, the matrix $[\phi]_B$ is diagonal. $\square$

The proof is really an algorithm, but in practice there's often a faster route: *completing the square*.

Example 16.8.6 (Diagonalising by Completing the Square)

Consider the quadratic form on $\mathbb{R}^3$ given by

$$Q(x_1, x_2, x_3) = x_1^2 + x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3,$$

corresponding to the symmetric matrix

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 2 \end{pmatrix}$$

(note the off-diagonal entries are *half* the corresponding coefficients, since each cross term appears twice in $x^T Ax$). Gathering every appearance of x_1 into a single square, we compute

$$Q - (x_1 + x_2 + x_3)^2 = x_3^2 - 4x_2x_3,$$

and then completing the square again in x_3 ,

$$\begin{aligned} Q &= (x_1 + x_2 + x_3)^2 + x_3^2 - 4x_2x_3 \\ &= (x_1 + x_2 + x_3)^2 + (x_3 - 2x_2)^2 - 4x_2^2. \end{aligned}$$

So in the new coordinates $x'_1 = x_1 + x_2 + x_3$, $x'_2 = x_3 - 2x_2$, $x'_3 = 2x_2$, we have $Q = x'^2_1 + x'^2_2 - x'^2_3$, and the form is diagonal with diagonal entries 1, 1, -1.

§16.8.4 Sylvester's Law of Inertia

Over $\mathbb{C}$, we can go further than just 'diagonal': dividing each basis vector e_i by a square root of $\phi(e_i, e_i)$, every non-zero diagonal entry can be scaled to 1.

Corollary 16.8.7 (Canonical Form over $\mathbb{C}$)

Let ϕ be a symmetric bilinear form on a finite dimensional $\mathbb{C}$ -vector space V . Then there is a basis B with

$$[\phi]_B = \left(\begin{array}{c|c} I_r & 0 \\ \hline 0 & 0 \end{array} \right),$$

where $r = r(\phi)$. Equivalently, every complex symmetric matrix is congruent to exactly one matrix of this form.

Over $\mathbb{R}$ we cannot take square roots of negative numbers, so the best we can do is scale each non-zero diagonal entry to ± 1 .

Corollary 16.8.8 (Canonical Form over $\mathbb{R}$)

Let ϕ be a symmetric bilinear form on a finite dimensional $\mathbb{R}$ -vector space V . Then there is a basis B with

$$[\phi]_B = \begin{pmatrix} I_p & & \\ & -I_q & \\ & & 0 \end{pmatrix}$$

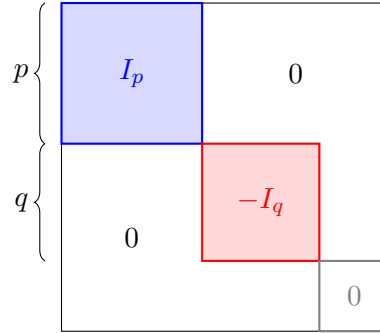
for some non-negative integers p, q with $p + q = r(\phi)$.

Proof. Diagonalise ϕ , reorder the basis so the positive diagonal entries $a_1, \dots, a_p$ come first, then the negative entries $a_{p+1}, \dots, a_{p+q}$, then the zeros, and replace e_i by $e_i/\sqrt{|a_i|}$ for $i \leq p+q$. $\square$

In terms of the associated quadratic form, this says that after a linear change of coordinates, any real quadratic form looks like

$$Q(x) = x_1^2 + \dots + x_p^2 - x_{p+1}^2 - \dots - x_{p+q}^2,$$

where p directions count positively, q count negatively, and the rest are invisible to Q .



The remarkable fact – *Sylvester’s law of inertia* – is that the numbers p and q do not depend on how you diagonalised.

Theorem 16.8.9 (Sylvester’s Law of Inertia)

Let ϕ be a symmetric bilinear form on a finite dimensional real vector space V . If ϕ is represented by the matrix above with parameters (p, q) in one basis and (p', q') in another, then $p = p'$ and $q = q'$.

Definition 16.8.10 (Signature)

The **signature** of a real symmetric bilinear form ϕ is $s(\phi) = p - q$.

To prove Sylvester’s law, we need a basis-independent characterisation of p , and the right notion is the following.

Definition 16.8.11 (Definiteness)

A symmetric bilinear form ϕ on a real vector space V is:

- (i) **positive definite** if $\phi(v, v) > 0$ for all $v \neq 0$,
- (ii) **positive semi-definite** if $\phi(v, v) \geq 0$ for all $v \in V$,

with **negative (semi-)definite** defined analogously with the inequalities reversed.

Proof of Sylvester’s Law. Suppose $B = \{v_1, \dots, v_n\}$ is a basis in which $[\phi]_B$ has the canonical form with parameters (p, q) . We claim

$$p = \max\{\dim X \mid X \leq V, \phi|_{X \times X} \text{ positive definite}\},$$

which is manifestly basis-independent, so proves that p (and by the same argument

applied to $-\phi$, also q) is well defined.

First, ϕ restricted to $X_0 = \langle v_1, \dots, v_p \rangle$ is positive definite, so the maximum is at least p . Conversely, let $Y = \langle v_{p+1}, \dots, v_n \rangle$, on which ϕ is negative semi-definite, and suppose X' is any subspace on which ϕ is positive definite. Then $X' \cap Y = \{0\}$, since any v in the intersection has $\phi(v, v) > 0$ and $\phi(v, v) \leq 0$ unless $v = 0$. Hence the sum $X' + Y$ is direct, and

$$\dim X' + (n - p) = \dim X' + \dim Y \leq \dim V = n,$$

giving $\dim X' \leq p$ as required. $\square$

Remark. Since $p + q = r(\phi)$ is the rank, and (p, q) determine the canonical form completely, we get a tidy classification: two real symmetric matrices are congruent if and only if they have the same rank and signature.

§16.8.5 Sesquilinear and Hermitian Forms

Over $\mathbb{C}$, symmetric bilinear forms are not quite the right notion for measuring ‘length’. The problem is visible even in one dimension: for the form $\phi(x, y) = xy$ on $\mathbb{C}$, we get $\phi(ix, ix) = -\phi(x, x)$, so no non-zero form of this kind takes only real positive values. The fix is to build in a complex conjugate.

Definition 16.8.12 (Sesquilinear Form)

Let V, W be $\mathbb{C}$ -vector spaces. A map $\phi : V \times W \rightarrow \mathbb{C}$ is **sesquilinear** if it is linear in the first argument and *antilinear* in the second, meaning

$$\phi(v, \lambda_1 w_1 + \lambda_2 w_2) = \overline{\lambda_1} \phi(v, w_1) + \overline{\lambda_2} \phi(v, w_2).$$

The matrix theory carries over with only cosmetic changes: with respect to bases B of V and C of W , we define $([\phi]_{B,C})_{ij} = \phi(v_i, w_j)$ as before, and then

$$\phi(v, w) = [v]_B^T [\phi]_{B,C} \overline{[w]_C}, \quad [\phi]_{B',C'} = P^T [\phi]_{B,C} \overline{Q}$$

for change of basis matrices P, Q .

The analogue of a symmetric form is the following.

Definition 16.8.13 (Hermitian Form)

A sesquilinear form $\phi : V \times V \rightarrow \mathbb{C}$ is **Hermitian** if

$$\phi(u, v) = \overline{\phi(v, u)} \quad \text{for all } u, v \in V.$$

Note that for a Hermitian form, $\phi(v, v) = \overline{\phi(v, v)}$ is always *real* – which is what makes definiteness meaningful over $\mathbb{C}$ – and $\phi(\lambda v, \lambda v) = |\lambda|^2 \phi(v, v)$, so scaling can’t change signs as it did above.

Lemma 16.8.14 (Matrix Criterion for Hermitian Forms)

A sesquilinear form ϕ on V is Hermitian if and only if for any basis B of V ,

$$[\phi]_B = \overline{[\phi]_B}^T =: [\phi]_B^\dagger.$$

Proof. If ϕ is Hermitian, then $a_{ji} = \phi(v_j, v_i) = \overline{\phi(v_i, v_j)} = \overline{a_{ij}}$. Conversely, if $A = A^\dagger$, then expanding $u = \sum_i \lambda_i v_i$ and $v = \sum_j \mu_j v_j$ gives

$$\phi(u, v) = \sum_{i,j} \lambda_i \overline{\mu_j} a_{ij}, \quad \overline{\phi(v, u)} = \overline{\sum_{i,j} \mu_j \overline{\lambda_i} a_{ji}} = \sum_{i,j} \lambda_i \overline{\mu_j} \overline{a_{ji}},$$

and these agree since $\overline{a_{ji}} = a_{ij}$. $\square$

There is also a Hermitian version of polarisation, this time requiring four terms:

$$\phi(u, v) = \frac{1}{4} (Q(u+v) - Q(u-v) + iQ(u+iv) - iQ(u-iv)),$$

where $Q(v) = \phi(v, v)$. So again, a Hermitian form is entirely determined by the real-valued function Q .

Sylvester's law then goes through essentially verbatim.

Theorem 16.8.15 (Sylvester's Law for Hermitian Forms)

Let ϕ be a Hermitian form on a finite dimensional $\mathbb{C}$ -vector space V . Then there is a basis B of V in which

$$[\phi]_B = \begin{pmatrix} I_p & & \\ & -I_q & \\ & & 0 \end{pmatrix},$$

and the integers p and q depend only on ϕ , not on the basis.

Proof Sketch. Existence follows the real symmetric case: if $\phi \neq 0$, polarisation gives some e_1 with $\phi(e_1, e_1) \neq 0$ (a real number), we normalise so that $\phi(v_1, v_1) = \pm 1$, split off $\langle v_1 \rangle^\perp$ using rank-nullity, and induct. Uniqueness follows Sylvester's law: p is again the maximal dimension of a subspace on which ϕ is positive definite, which makes sense as $\phi(v, v)$ is real. $\square$

§16.8.6 Skew-Symmetric Forms

For completeness, let us briefly mention the opposite symmetry class. A bilinear form ϕ on a real vector space is **skew-symmetric** if $\phi(u, v) = -\phi(v, u)$ for all u, v , or equivalently if its matrix satisfies $A = -A^T$ in any basis. Note that any bilinear form decomposes as a sum of a symmetric and a skew-symmetric form, via

$$\phi(u, v) = \underbrace{\frac{\phi(u, v) + \phi(v, u)}{2}}_{\text{symmetric}} + \underbrace{\frac{\phi(u, v) - \phi(v, u)}{2}}_{\text{skew-symmetric}}.$$

Theorem 16.8.16 (Canonical Form for Skew-Symmetric Forms)

Let ϕ be a skew-symmetric bilinear form on a finite dimensional real vector space V . Then there is a basis $B = \{v_1, w_1, \dots, v_m, w_m, v_{2m+1}, \dots, v_n\}$ in which $[\phi]_B$ is block diagonal, beginning with m blocks of the form

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

followed by zeros. In particular, a skew-symmetric matrix has even rank.

Proof Sketch. If $\phi \neq 0$, pick v_1, w_1 with $\phi(v_1, w_1) = 1$ (rescaling as needed); skew-symmetry gives $\phi(w_1, v_1) = -1$, and v_1, w_1 are independent since $\phi(v, v) = 0$ for every v . Set $U = \langle v_1, w_1 \rangle$ and let $W = \{v \in V \mid \phi(v_1, v) = \phi(w_1, v) = 0\}$. A dimension count shows $V = U \oplus W$, and we induct on W . $\square$

§16.9 Inner Product Spaces

In this final section we add the last piece of structure to our vector spaces: a notion of *length* and *angle*. All of the geometry that this makes possible comes from a single piece of data – an inner product – and the positive-definite forms of the previous section are exactly what we need. Throughout this section, V is a vector space over $\mathbb{R}$ or $\mathbb{C}$.

§16.9.1 Inner Products

Definition 16.9.1 (Inner Product)

An **inner product** on V is a positive definite symmetric bilinear form (if V is a real vector space) or positive definite Hermitian sesquilinear form (if V is complex). We write $\phi(u, v) = \langle u, v \rangle$, and call V equipped with an inner product an **inner product space**.

Example 16.9.2 (Examples of Inner Products)

- (i) On $\mathbb{R}^n$, the familiar dot product $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$; on $\mathbb{C}^n$, the standard inner product $\langle x, y \rangle = \sum_{i=1}^n x_i \overline{y_i}$.
- (ii) On $\mathcal{C}([0, 1], \mathbb{C})$, the L^2 inner product

$$\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} \, dt.$$

- (iii) More generally, given a continuous ‘weight’ function $w : [0, 1] \rightarrow \mathbb{R}_{>0}$, we can define $\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} w(t) \, dt$.

When checking that something is an inner product, linearity and symmetry are usually immediate, and the only real content is positive definiteness – often just the statement that $\langle v, v \rangle = 0$ implies $v = 0$.

Definition 16.9.3 (Norm)

Let V be an inner product space. The **norm** of $v \in V$ is

$$\|v\| = \sqrt{\langle v, v \rangle},$$

which is a non-negative real number, and is zero only for $v = 0$.

The single most useful inequality relating the norm and the inner product is the following.

Theorem 16.9.4 (Cauchy-Schwarz Inequality)

Let V be an inner product space, and $u, v \in V$. Then

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Proof. If $u = 0$ the result is trivial, so assume $u \neq 0$. For any scalar t , positive definiteness gives

$$0 \leq \|tu - v\|^2 = |t|^2 \|u\|^2 - 2 \operatorname{Re}(t \langle u, v \rangle) + \|v\|^2.$$

Choosing $t = \overline{\langle u, v \rangle} / \|u\|^2$ (the choice that makes the bound tightest), the right hand side becomes

$$\frac{|\langle u, v \rangle|^2}{\|u\|^2} - 2 \frac{|\langle u, v \rangle|^2}{\|u\|^2} + \|v\|^2 = \|v\|^2 - \frac{|\langle u, v \rangle|^2}{\|u\|^2} \geq 0,$$

which rearranges to the desired inequality. $\square$

Corollary 16.9.5 (Triangle Inequality)

In an inner product space, $\|u + v\| \leq \|u\| + \|v\|$.

Proof. Expanding and applying Cauchy-Schwarz,

$$\|u + v\|^2 = \|u\|^2 + 2 \operatorname{Re}\langle u, v \rangle + \|v\|^2 \leq \|u\|^2 + 2 \|u\| \|v\| + \|v\|^2 = (\|u\| + \|v\|)^2. \quad \square$$

§16.9.2 Orthogonality**Definition 16.9.6 (Orthogonal and Orthonormal Sets)**

A set $\{e_1, \dots, e_k\}$ of vectors in an inner product space is **orthogonal** if $\langle e_i, e_j \rangle = 0$ for all $i \neq j$, and **orthonormal** if additionally $\|e_i\| = 1$ for each i , so that $\langle e_i, e_j \rangle = \delta_{ij}$.

Orthogonality is a strong condition – strong enough to give linear independence for free, along with an explicit formula for coordinates.

Lemma 16.9.7 (Orthogonality Implies Independence)

Let $\{e_1, \dots, e_k\}$ be orthogonal non-zero vectors. Then they are linearly independent,

and if $v = \sum_{j=1}^k \lambda_j e_j$ then

$$\lambda_j = \frac{\langle v, e_j \rangle}{\|e_j\|^2}.$$

Proof. Taking the inner product of $v = \sum_i \lambda_i e_i$ with e_j , all terms with $i \neq j$ vanish, leaving $\langle v, e_j \rangle = \lambda_j \|e_j\|^2$, which gives the coordinate formula. In particular if $v = 0$ then every $\lambda_j = 0$, which is linear independence. $\square$

For orthonormal *bases* the coordinate formula becomes $\lambda_j = \langle v, e_j \rangle$, and the inner product itself takes a particularly clean coordinate form.

Corollary 16.9.8 (Parseval's Identity)

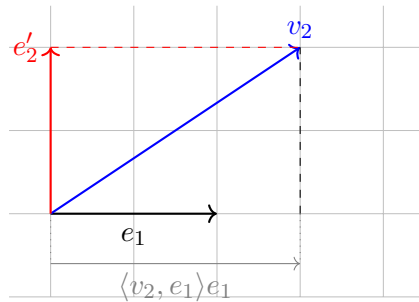
Let V be a finite dimensional inner product space with orthonormal basis $\{e_1, \dots, e_n\}$. Then for all $u, v \in V$,

$$\langle u, v \rangle = \sum_{i=1}^n \langle u, e_i \rangle \overline{\langle v, e_i \rangle}, \quad \text{and in particular} \quad \|u\|^2 = \sum_{i=1}^n |\langle u, e_i \rangle|^2.$$

Proof. Expand $u = \sum_i \langle u, e_i \rangle e_i$ and $v = \sum_j \langle v, e_j \rangle e_j$ and use sesquilinearity together with $\langle e_i, e_j \rangle = \delta_{ij}$. The second identity is the case $u = v$. $\square$

§16.9.3 The Gram-Schmidt Process

Orthonormal bases are clearly a pleasant thing to have, and happily there is an algorithm that manufactures them from arbitrary bases: the *Gram-Schmidt process*. The idea of each step is simple – subtract off the components of the next vector lying along the directions already constructed, then rescale what's left.



Theorem 16.9.9 (Gram-Schmidt Orthogonalisation)

Let V be an inner product space, and let $v_1, v_2, \dots$ be a (finite or countable) linearly independent family of vectors. Then there is an orthonormal family $e_1, e_2, \dots$ such that for every k ,

$$\langle v_1, \dots, v_k \rangle = \langle e_1, \dots, e_k \rangle.$$

Proof. We construct the e_i inductively. To start, take $e_1 = v_1 / \|v_1\|$. Now suppose

$e_1, \dots, e_k$ have been constructed, orthonormal and spanning $\langle v_1, \dots, v_k \rangle$. Define

$$e'_{k+1} = v_{k+1} - \sum_{i=1}^k \langle v_{k+1}, e_i \rangle e_i.$$

For each $j \leq k$ we then have

$$\langle e'_{k+1}, e_j \rangle = \langle v_{k+1}, e_j \rangle - \langle v_{k+1}, e_j \rangle = 0,$$

so e'_{k+1} is orthogonal to everything so far. It is also non-zero, since otherwise v_{k+1} would lie in $\langle e_1, \dots, e_k \rangle = \langle v_1, \dots, v_k \rangle$, contradicting linear independence. So we can set $e_{k+1} = e'_{k+1} / \|e'_{k+1}\|$, and the spans match up since e_{k+1} is a linear combination of v_{k+1} and $e_1, \dots, e_k$ (and vice versa). $\square$

Corollary 16.9.10 (Orthonormal Bases Exist)

Every finite dimensional inner product space has an orthonormal basis, and any orthonormal set of vectors can be extended to an orthonormal basis.

Proof. Apply Gram-Schmidt to any basis for the first claim. For the second, extend the orthonormal set to any basis and apply Gram-Schmidt – the process leaves the already-orthonormal initial vectors unchanged. $\square$

Interpreting Gram-Schmidt through matrices gives a classical matrix factorisation.

Definition 16.9.11 (Orthogonal and Unitary Matrices)

A matrix $A \in \mathcal{M}_{n,n}(\mathbb{R})$ is **orthogonal** if $A^T A = I$, that is $A^{-1} = A^T$. A matrix $A \in \mathcal{M}_{n,n}(\mathbb{C})$ is **unitary** if $A^\dagger A = I$, that is $A^{-1} = A^\dagger = \overline{A}^T$.

Note that A is orthogonal (or unitary) precisely when its columns form an orthonormal basis of $\mathbb{F}^n$ with the standard inner product.

Proposition 16.9.12 (QR Decomposition)

Every invertible matrix $A \in \mathcal{M}_{n,n}(\mathbb{R})$ (respectively $\mathbb{C}$) can be written as $A = QR$ with Q orthogonal (respectively unitary) and R upper triangular.

Proof Sketch. Apply Gram-Schmidt to the columns $v_1, \dots, v_n$ of A , producing an orthonormal basis $e_1, \dots, e_n$. By construction each v_k is a linear combination of $e_1, \dots, e_k$ only, so the matrix Q with columns e_i satisfies $A = QR$ with R upper triangular. $\square$

§16.9.4 Orthogonal Complements and Projections

Definition 16.9.13 (Orthogonal Complement)

Let V be an inner product space and $W \leq V$. The **orthogonal complement** of W is

$$W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

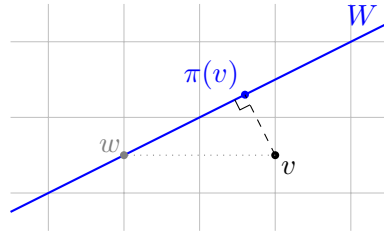
Lemma 16.9.14 (Orthogonal Decomposition)

Let V be a finite dimensional inner product space and $W \leq V$. Then $V = W \oplus W^\perp$.

Proof. If $v \in W \cap W^\perp$ then $\|v\|^2 = \langle v, v \rangle = 0$, so the intersection is trivial. For the sum, take an orthonormal basis $\{e_1, \dots, e_k\}$ of W and extend it to an orthonormal basis $\{e_1, \dots, e_n\}$ of V ; then $e_{k+1}, \dots, e_n \in W^\perp$, so $W + W^\perp$ contains a basis of V . $\square$

Compare this with the general situation for direct complements: earlier we emphasised that a subspace has *many* complements, with no way to prefer one. An inner product fixes this – $W^\perp$ is a *canonical* complement.

Whenever $V = U \oplus W$, we can define the **projection** $\pi : V \rightarrow V$ onto W by writing $v = u + w$ uniquely and setting $\pi(v) = w$; this is linear and satisfies $\pi^2 = \pi$. When the decomposition is the orthogonal one, π has a beautiful geometric property: it finds the *closest point* of W .

**Proposition 16.9.15** (Orthogonal Projection is the Closest Point)

Let V be an inner product space and $W \leq V$ a finite dimensional subspace with orthonormal basis $\{e_1, \dots, e_k\}$. Define

$$\pi(v) = \sum_{i=1}^k \langle v, e_i \rangle e_i.$$

Then π is the projection onto W along $W^\perp$, and for every $w \in W$,

$$\|v - \pi(v)\| \leq \|v - w\|,$$

with equality if and only if $w = \pi(v)$. That is, $\pi(v)$ is the unique closest point of W to v .

Proof. Certainly $\pi(v) \in W$. Moreover $v - \pi(v) \in W^\perp$, since for each basis vector e_j ,

$$\langle v - \pi(v), e_j \rangle = \langle v, e_j \rangle - \sum_{i=1}^k \langle v, e_i \rangle \langle e_i, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0.$$

So $v = (v - \pi(v)) + \pi(v)$ is exactly the decomposition of v with respect to $V = W^\perp \oplus W$, and π is the claimed projection.

For the closest point property, let $w \in W$ and split $v - w$ into orthogonal pieces:

$$\|v - w\|^2 = \left\| \underbrace{(v - \pi(v))}_{\in W^\perp} + \underbrace{(\pi(v) - w)}_{\in W} \right\|^2 = \|v - \pi(v)\|^2 + \|\pi(v) - w\|^2,$$

where the cross terms vanish by orthogonality (this is just Pythagoras). The right hand side is minimised exactly when $\pi(v) = w$. $\square$

§16.9.5 Adjoint Maps

We come now to the inner product space analogue of the dual map.

Theorem 16.9.16 (Existence of the Adjoint)

Let V, W be finite dimensional inner product spaces and let $\alpha : V \rightarrow W$ be a linear map. Then there is a unique linear map $\alpha^* : W \rightarrow V$, the **adjoint** of α , such that

$$\langle \alpha(v), w \rangle = \langle v, \alpha^*(w) \rangle \quad \text{for all } v \in V, w \in W.$$

Moreover, if B and C are *orthonormal* bases of V and W , then

$$[\alpha^*]_{C,B} = [\alpha]_{B,C}^\dagger,$$

the conjugate transpose of the matrix of α .

Proof. Let $B = \{v_1, \dots, v_n\}$, $C = \{w_1, \dots, w_m\}$ be orthonormal bases, and let $A = [\alpha]_{B,C}$. Define α^* to be the linear map with matrix $[\alpha^*]_{C,B} = A^\dagger$. Then for basis vectors, using orthonormality to compute inner products,

$$\langle \alpha(v_i), w_j \rangle = \left\langle \sum_k a_{ki} w_k, w_j \right\rangle = a_{ji}, \quad \langle v_i, \alpha^*(w_j) \rangle = \left\langle v_i, \sum_k \overline{a_{jk}} v_k \right\rangle = \overline{a_{ji}} = a_{ji},$$

and by (sesqui)linearity the two sides agree for all vectors. For uniqueness, if β also satisfies the defining property then $\langle v, (\alpha^* - \beta)(w) \rangle = 0$ for all v , and taking $v = (\alpha^* - \beta)(w)$ shows $(\alpha^* - \beta)(w) = 0$ for every w . $\square$

Remark. We are recycling the notation α^* from dual maps, and this is no accident. The inner product gives an (antilinear) identification $v \mapsto \langle \cdot, v \rangle$ of V with V^* , and under this identification the adjoint *is* the dual map. This is also why the matrix of the adjoint is a (conjugate) transpose, just as the matrix of the dual map was a transpose.

§16.9.6 Self-Adjoint Maps and Isometries

Two classes of endomorphisms of an inner product space deserve special attention – those that equal their own adjoint, and those whose adjoint is their inverse.

Definition 16.9.17 (Self-Adjoint and Isometric Maps)

Let V be a finite dimensional inner product space, and let $\alpha : V \rightarrow V$ be an

endomorphism.

- (i) α is **self-adjoint** if $\alpha^* = \alpha$, that is, $\langle \alpha(v), w \rangle = \langle v, \alpha(w) \rangle$ for all v, w . Over $\mathbb{R}$ such maps are called **symmetric**; over $\mathbb{C}$, **Hermitian**.
- (ii) α is an **isometry** if $\alpha^* = \alpha^{-1}$, or equivalently $\langle \alpha(v), \alpha(w) \rangle = \langle v, w \rangle$ for all v, w . Over $\mathbb{R}$ such maps are called **orthogonal**; over $\mathbb{C}$, **unitary**.

Let's check the claimed equivalence for isometries. If α preserves all inner products, then in particular $\|\alpha(v)\| = \|v\|$, so α is injective, hence invertible; and then

$$\langle v, \alpha^*(w) \rangle = \langle \alpha(v), w \rangle = \langle \alpha(v), \alpha(\alpha^{-1}(w)) \rangle = \langle v, \alpha^{-1}(w) \rangle$$

for all v, w , forcing $\alpha^* = \alpha^{-1}$. The converse is immediate: $\langle \alpha(v), \alpha(w) \rangle = \langle v, \alpha^* \alpha(w) \rangle = \langle v, w \rangle$. In fact, by polarisation, preserving norms alone is enough to make α an isometry.

In matrix terms: with respect to any *orthonormal* basis, self-adjoint maps are represented by symmetric (Hermitian) matrices, and isometries by orthogonal (unitary) matrices. The isometries of V form the **orthogonal group** $O(V)$ (respectively the **unitary group** $U(V)$) under composition.

§16.9.7 The Spectral Theorems

We now arrive at the crowning results of the course, the *spectral theorems*: self-adjoint maps (and over $\mathbb{C}$, isometries) are not just diagonalisable, but diagonalisable in an *orthonormal* basis. First, a lemma that makes everything work.

Lemma 16.9.18 (Eigenvalues of Self-Adjoint Maps)

Let α be a self-adjoint endomorphism of a finite dimensional inner product space V . Then:

- (i) every eigenvalue of α is real,
- (ii) eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof. (i) Suppose $\alpha(v) = \lambda v$ with $v \neq 0$. Then

$$\lambda \|v\|^2 = \langle \alpha(v), v \rangle = \langle v, \alpha(v) \rangle = \bar{\lambda} \|v\|^2,$$

so $\lambda = \bar{\lambda}$ is real.^a

(ii) If additionally $\alpha(w) = \mu w$ with $\mu \neq \lambda$, then

$$\lambda \langle v, w \rangle = \langle \alpha(v), w \rangle = \langle v, \alpha(w) \rangle = \bar{\mu} \langle v, w \rangle = \mu \langle v, w \rangle,$$

using that μ is real. Since $\lambda \neq \mu$, this forces $\langle v, w \rangle = 0$. □

^aIn the real case this argument should be read as applying to the complexification – concretely, a real symmetric matrix, viewed as a complex matrix, has real eigenvalues. This is exactly what lets the induction in the spectral theorem proceed over $\mathbb{R}$.

Theorem 16.9.19 (Spectral Theorem for Self-Adjoint Maps)

Let V be a finite dimensional (real or complex) inner product space, and let $\alpha : V \rightarrow$

V be self-adjoint. Then V has an orthonormal basis consisting of eigenvectors of α . In particular, α is diagonalisable in an orthonormal basis, and V is the orthogonal direct sum of the eigenspaces of α .

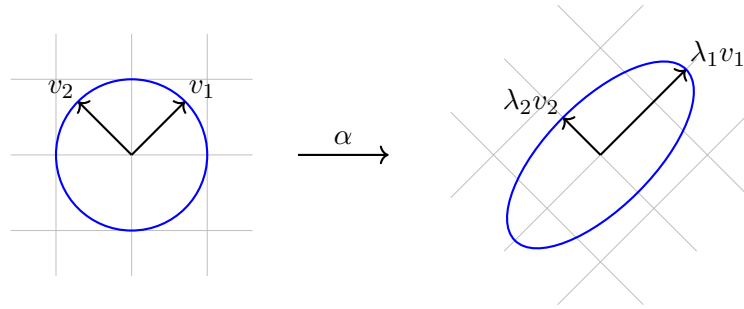
Proof. We induct on $\dim V$, with the case $\dim V = 1$ being trivial. The characteristic polynomial of α has a complex root λ by the fundamental theorem of algebra, and by the lemma λ is real – so in both the real and complex cases, λ is a genuine eigenvalue of α . Pick an eigenvector v_1 , scaled so that $\|v_1\| = 1$, and let $U = \langle v_1 \rangle^\perp$.

The key step is that U is α -stable. Indeed, for $u \in U$,

$$\langle \alpha(u), v_1 \rangle = \langle u, \alpha(v_1) \rangle = \lambda \langle u, v_1 \rangle = 0,$$

so $\alpha(u) \in U$. The restriction $\alpha|_U$ is a self-adjoint endomorphism of the inner product space U with $\dim U = \dim V - 1$, so by induction U has an orthonormal basis of eigenvectors $\{v_2, \dots, v_n\}$. Since $V = \langle v_1 \rangle \oplus U$ orthogonally, $\{v_1, v_2, \dots, v_n\}$ is an orthonormal basis of eigenvectors of V . $\square$

Geometrically, the spectral theorem says a self-adjoint map is just *scaling along some set of perpendicular axes*. Applied to the unit circle in the plane, a symmetric map produces an ellipse whose principal axes are the (orthogonal) eigendirections:



Over $\mathbb{C}$, the same conclusion holds for unitary maps, with almost the same proof.

Lemma 16.9.20 (Eigenvalues of Unitary Maps)

Let α be a unitary endomorphism of a finite dimensional complex inner product space. Then every eigenvalue of α satisfies $|\lambda| = 1$, and eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof. If $\alpha(v) = \lambda v$ with $v \neq 0$, then $\|v\| = \|\alpha(v)\| = |\lambda| \|v\|$, so $|\lambda| = 1$ (in particular $\lambda \neq 0$). If also $\alpha(w) = \mu w$ with $\mu \neq \lambda$, then

$$\lambda \langle v, w \rangle = \langle \alpha(v), w \rangle = \langle v, \alpha^{-1}(w) \rangle = \left\langle v, \frac{1}{\mu} w \right\rangle = \frac{1}{\mu} \langle v, w \rangle = \mu \langle v, w \rangle,$$

using $\mu \bar{\mu} = 1$, and since $\lambda \neq \mu$ we get $\langle v, w \rangle = 0$. $\square$

Theorem 16.9.21 (Spectral Theorem for Unitary Maps)

Let V be a finite dimensional complex inner product space, and let $\alpha : V \rightarrow V$ be unitary. Then V has an orthonormal basis of eigenvectors of α .

Proof. As before, we induct on the dimension. Pick an eigenvalue λ (which exists as $\mathbb{F} = \mathbb{C}$) with unit eigenvector v_1 , and let $U = \langle v_1 \rangle^\perp$. For $u \in U$,

$$\langle \alpha(u), v_1 \rangle = \langle u, \alpha^{-1}(v_1) \rangle = \langle u, \frac{1}{\lambda} v_1 \rangle = \frac{1}{\lambda} \langle u, v_1 \rangle = 0,$$

so U is α -stable, and $\alpha|_U$ is a unitary endomorphism of U . Induction and the orthogonal decomposition $V = \langle v_1 \rangle \oplus U$ finish the proof as before. $\square$

Remark. The restriction to $\mathbb{C}$ really is needed for isometries: the rotation matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is orthogonal, but has eigenvalues $e^{\pm i\theta}$, so is not diagonalisable over $\mathbb{R}$ for general θ – there is simply no real direction that a rotation preserves.

§16.9.8 Applications to Bilinear Forms

The spectral theorem lets us sharpen our earlier results on diagonalising symmetric bilinear forms: not only can we diagonalise, we can do so with an *orthonormal* change of basis.

Corollary 16.9.22 (Orthogonal Diagonalisation of Symmetric Matrices)

Let $A \in \mathcal{M}_{n,n}(\mathbb{R})$ be symmetric (or $A \in \mathcal{M}_{n,n}(\mathbb{C})$ Hermitian). Then there is an orthogonal (respectively unitary) matrix P such that $P^T A P$ (respectively $P^\dagger A P$) is diagonal with real entries.

Proof. View A as a self-adjoint endomorphism of $\mathbb{F}^n$ with the standard inner product. By the spectral theorem there is an orthonormal basis $\{v_1, \dots, v_n\}$ of eigenvectors; let P be the matrix with these as columns. Then P is orthogonal (unitary), and $P^{-1} A P = P^T A P$ (respectively $P^\dagger A P$) is diagonal, with entries the eigenvalues of A , which are real. $\square$

Note the happy coincidence that makes this work: for orthogonal matrices, the change of basis formula for linear maps ($P^{-1} A P$) and for bilinear forms ($P^T A P$) *agree*. So the diagonal entries produced are exactly the eigenvalues of A , and we can read off the signature of a real symmetric form: $s(\phi) = (\text{number of positive eigenvalues}) - (\text{number of negative eigenvalues})$ of any matrix representing ϕ .

We can also now prove a satisfying result about pairs of forms.

Corollary 16.9.23 (Simultaneous Diagonalisation of Forms)

Let V be a finite dimensional real (or complex) vector space, and let ϕ, ψ be symmetric (respectively Hermitian) forms on V with ϕ positive definite. Then there is

a single basis B of V in which $[\phi]_B$ and $[\psi]_B$ are both diagonal.

Proof. Since ϕ is positive definite, it makes V into an inner product space with $\langle u, v \rangle = \phi(u, v)$. By the previous corollary applied in this inner product space, there is a ϕ -orthonormal basis B of V in which $[\psi]_B$ is diagonal. But orthonormality of B with respect to ϕ says precisely that $[\phi]_B = I$, which is certainly diagonal. $\square$

In matrix language: if A, B are real symmetric (or Hermitian) matrices with A positive definite, then there is an invertible Q with $Q^T A Q$ and $Q^T B Q$ both diagonal. This result is the abstract heart of many applications – for instance, finding the normal modes of a mechanical system simultaneously diagonalises the kinetic and potential energy forms.

And with that, we have reached the end of the course. Starting from nothing but the axioms of a vector space, we have built up bases and dimension, classified linear maps up to change of basis, developed determinants, eigenvalues and canonical forms, and finally seen how an inner product turns linear algebra into geometry. These tools appear constantly throughout mathematics – and if the subject appeals to you, the story continues in courses on modules (IB Groups, Rings and Modules) and functional analysis, where infinite dimensional spaces take centre stage.

17 Markov Chains

A stochastic process is said to have the ‘Markov property’ if, conditional on its present value, the future is independent of the past. This is a *restrictive* assumption, but we do end up with a useful model with a rich mathematical theory, which we shall study in this course. Despite the restriction, Markov chains turn out to be everywhere: shuffling cards, random walks, population growth, queues, and the algorithms behind modern statistical sampling are all naturally described by them. The pay-off of the theory is that remarkably detailed questions – will the process return to where it started? how long does it take? what does it look like after a long time? – admit complete and often beautiful answers.

This article constitutes my notes for the ‘Markov Chains’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§17.1 The Markov Property

§17.1.1 What is a Markov Chain?

Let S be a countable set (the set of possible ‘states’), and let X_n be a sequence of random variables taking values in S .

Definition 17.1.1 (Markov Chain)

The sequence of random variables X_n is a **Markov chain** if it satisfies the **Markov property**

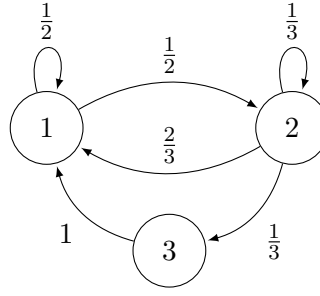
$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

The Markov chain is said to be **homogeneous** if for all $i, j \in S$ the conditional probability $\mathbb{P}(X_{n+1} = j \mid X_n = i)$ is independent of n .

In this course we are only going to study homogeneous Markov chains.

Markov chains are often best described by diagrams¹ which show the probability of moving from one state to another. For example, the Markov chain in the diagram below has three states which we label $\{1, 2, 3\}$, and the probability of moving from state 1 to state 2 is $1/2$, and the probability of moving from state 2 to state 3 is $1/3$, and so on.

¹You might notice that these diagrams are labelled directed graphs, and indeed you will see concepts from graph theory such as connectivity reoccur when we later talk about communicating classes.



In general, to calculate the probabilities associated with a Markov chain, we need to know two quantities.

- *The initial distribution.* We first need to know about the starting state of a Markov chain. This is described by the initial distribution $\lambda = (\lambda_i \mid i \in S)$, where $\lambda_i = \mathbb{P}(X_0 = i)$.
- *The transition probabilities.* We also need to know the probability of moving from a state $i \in S$ to a state $j \in S$. This is typically given by a *transition matrix* $P = (P_{i,j} \mid i, j \in S)$ with $p_{i,j} = \mathbb{P}(X_1 = j \mid X_0 = i)$.

These quantities are of course subject to some constraints, in that we require $\sum_{i \in S} \lambda_i = 1$ and the transition matrix P must be **stochastic**, in that $\sum_{j \in S} P_{i,j} = 1$ for all $i \in S$.

If a Markov chain X_n has initial distribution λ and transition matrix P , we say that it is **Markov**(λ, P).

Once we have these quantities, we can begin to actually establish various properties about the Markov chain, for example the probability that it goes through a given sequence of states.

Theorem 17.1.2 (Probability of a Sequence of States)

The sequence of random variables X_n is Markov(λ, P) if and only if

$$\mathbb{P}(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = \lambda_{i_0} P_{i_0, i_1} \cdots P_{i_{n-1}, i_n},$$

for all $n \geq 0$ and $i_0, \dots, i_n \in S$.

Proof. Let A_k denote the event $\{X_k = i_k\}$. First suppose that X_n is Markov(λ, P). We prove the result holds by induction. For $n = 0$ this is true by definition. Then if it holds up to n , we have

$$\begin{aligned} \mathbb{P}(A_0 \cap \cdots \cap A_n) &= \mathbb{P}(A_0 \cap \cdots \cap A_{n-1}) \mathbb{P}(A_n \mid A_0 \cap \cdots \cap A_{n-1}) \\ &= \mathbb{P}(A_0 \cap \cdots \cap A_{n-1}) \mathbb{P}(A_n \mid A_{n-1}) \\ &= (\lambda_{i_0} P_{i_0, i_1} \cdots P_{i_{n-2}, i_{n-1}}) P_{i_{n-1}, i_n}, \end{aligned}$$

which completes our induction.

Conversely, suppose that the result holds. Then with $n = 0$ we get that the initial distribution of X_n is λ . Then

$$\mathbb{P}(A_{n+1} \mid A_0 \cap \cdots \cap A_n) = \frac{\mathbb{P}(A_0 \cap \cdots \cap A_{n+1})}{\mathbb{P}(A_0 \cap \cdots \cap A_n)} = P_{i_n, i_{n+1}}.$$

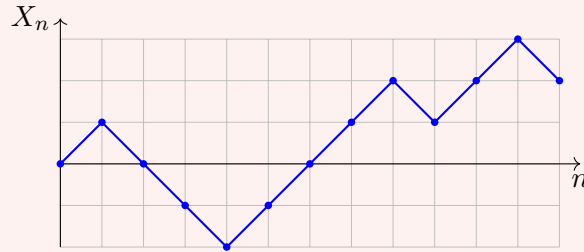
Since this does not depend on $i_0, \dots, i_{n-1}$, X_n is a homogeneous Markov chain with transition matrix P , as required. $\square$

Before moving on, let's get a feel for what is and isn't a Markov chain.

Example 17.1.3 (Some Markov Chains, and a Non-Example) (i) *An i.i.d. sequence.*

If $X_0, X_1, X_2, \dots$ are independent and identically distributed random variables taking values in S , then X_n is (rather boringly) a Markov chain: conditioning on the past changes nothing at all, and $P_{i,j} = \mathbb{P}(X_1 = j)$ doesn't even depend on i .

- (ii) *A random walk.* Let $\xi_1, \xi_2, \dots$ be i.i.d. integer-valued random variables and set $X_n = X_0 + \xi_1 + \dots + \xi_n$. Then X_n is a Markov chain on $\mathbb{Z}$, since given $X_n = x$, we have $X_{n+1} = x + \xi_{n+1}$, and ξ_{n+1} is independent of everything that came before. Random walks will be recurring characters throughout this course. A typical trajectory of the walk with steps $\xi_i = \pm 1$ looks like the following.



- (iii) *A non-example.* Let X_n be the random walk above with steps ± 1 , and let $M_n = \max_{0 \leq k \leq n} X_k$ be its running maximum. Then M_n is *not* a Markov chain in general: knowing that $M_n = M_{n-1} = m$ tells us the walk is currently strictly below its maximum, while knowing only $M_n = m$ leaves open the possibility $X_n = m$, and these lead to different probabilities that $M_{n+1} = m+1$. The past leaks information about the present position of the underlying walk, which the value of M_n alone doesn't capture.

Example 17.1.4 (Probability of a Path)

Let's also see the previous theorem in action on the three-state chain drawn at the start of this section, with $S = \{1, 2, 3\}$ and initial distribution $\lambda = \delta_1$ (i.e. we start at state 1 with certainty). The probability that the chain follows the path $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ is

$$\mathbb{P}(X_0 = 1, X_1 = 2, X_2 = 3, X_3 = 1) = \lambda_1 P_{1,2} P_{2,3} P_{3,1} = 1 \cdot \frac{1}{2} \cdot \frac{1}{3} \cdot 1 = \frac{1}{6}.$$

Notice how the answer is just the product of the probabilities written on the arrows of the diagram, along the path taken – this is a perfectly good way to think about the theorem.

§17.1.2 Simple Markov Property

An important aspect of Markov chains is that they are *memoryless*, in the future is independent of the past, conditional on the present. This is encapsulated in the **simple Markov property**.

Theorem 17.1.5 (Simple Markov Property)

Let X_n be a Markov chain. Then conditional on $X_m = i$, the sequence of random variables $(X_{m+n})_{n \geq 0}$ is^a Markov(δ_i, P), and is independent of $X_0, \dots, X_{m-1}$.

^aHere δ_{ij} is 1 if $i = j$ and 0 otherwise.

Proof. Given any event H determined by $X_0, \dots, X_{m-1}$, we want to show that for an event $F = \{X_m = i_m, \dots, X_{m+n} = i_{m+n}\}$ we have

$$\mathbb{P}(H \cap F \mid X_m = i) = \delta_{ii_m} P_{i_m, i_{m+1}} \cdots P_{i_{m+n-1}, i_{m+n}} \mathbb{P}(H \mid X_m = i).$$

Indeed, considering the case of $H = \{X_0 = i_0, \dots, X_m = i_m\}$ we have

$$\begin{aligned} \mathbb{P}(H \cap F \mid X_m = i) &= \frac{\lambda_{i_0} P_{i_0, i_1} \cdots P_{i_{m-1}, i_m} P_{i_m, i_{m+1}} \cdots P_{i_{m+n-1}, i_{m+n}}}{\mathbb{P}(X_m = i)} \\ &= \delta_{ii_m} P_{i_m, i_{m+1}} \cdots P_{i_{m+n-1}, i_{m+n}} \mathbb{P}(H \mid X_m = i), \end{aligned}$$

as required.

Then for a general H , we can write it as the disjoint union $H = \bigcup_{k=1}^{\infty} H_k$, and then the overall result follows by summing the above result for relevant H_k . $\square$

Remark. It is worth internalising what the simple Markov property really says: the chain, observed from time m onwards, is a fresh copy of the Markov chain started from wherever it happened to be at time m – and this fresh copy doesn't care at all about the route taken to get there. For instance, for any states i, j, k ,

$$\mathbb{P}(X_5 = j \mid X_3 = i, X_0 = k) = \mathbb{P}(X_5 = j \mid X_3 = i) = \mathbb{P}(X_2 = j \mid X_0 = i),$$

so conditional probabilities of the future can always be 're-rooted' to time zero. We will use manipulations of this sort constantly, and usually without comment.

§17.1.3 Transition Probabilities

We are now going to address how to find the probability that a Markov chain is in a given state after n steps. The core idea of this section is that we will be able to reduce such questions into questions about the transition matrix that we introduced earlier.

Recall that if X_n is a Markov chain with transition matrix P , then $P_{i,j}$ was the probability of moving from the state i to the state j . We call these the **one-step transition probabilities**. We can generalize this slightly.

Definition 17.1.6 (n -Step Transition Probabilities)

For a Markov chain X_n , the **n -step transition probabilities** are given by

$$p_{i,j}(n) = \mathbb{P}(X_n = j \mid X_0 = i).$$

These n -step transition probabilities naturally form the **n -step transition matrix** $P(n) = (p_{i,j}(n) \mid i, j \in S)$. The nice thing about writing these transition probabilities as a matrix is that they satisfy a lovely set of equations that relate extremely well to matrix algebra.

Theorem 17.1.7 (Chapman-Kolmogorov Equations)

We have that

$$p_{i,j}(n+m) = \sum_{k \in S} p_{i,k}(n)p_{k,j}(m),$$

where $i, j \in S$ and $m, n \geq 0$. In particular, $P(m+n) = P(m)P(n)$.

Proof. Using the partition theorem and simple Markov property,

$$\begin{aligned} p_{i,j}(n+m) &= \mathbb{P}(X_{n+m} = j \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j \mid X_n = k, X_0 = i) \mathbb{P}(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} \mathbb{P}(X_{m+n} = j \mid X_n = k) \mathbb{P}(X_n = k \mid X_0 = i) \\ &= \sum_{k \in S} p_{i,k}(n)p_{k,j}(m). \end{aligned}$$

□

So, if we have some n -step transition matrix $P(n)$, the above result shows us that it satisfies $P(n) = P^n$. This reduces our problem to just this:

To compute $p_{i,j}(n)$, we can compute powers of
the transition matrix, and take $(P^n)_{i,j}$.

In general, this makes our problem significantly easier, and if the state space is finite, we can use tools from linear algebra such as diagonalisation.

Example 17.1.8 (Predicting the Weather)

Suppose the weather in Cambridge is sunny, cloudy or rainy, and that tomorrow's weather depends only on today's^a, according to the transition matrix

$$P = \begin{pmatrix} 1/2 & 1/3 & 1/6 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/4 & 1/2 \end{pmatrix},$$

where the states are (sunny, cloudy, rainy) in that order. If it is sunny today, what is the probability that it rains two days from now?

By our discussion above, this is just the matrix entry

$$(P^2)_{1,3} = \sum_{k=1}^3 P_{1,k}P_{k,3} = \frac{1}{2} \cdot \frac{1}{6} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{1}{4},$$

where the three terms account for the three possible weathers tomorrow. There is nothing special about two days – the probability for any horizon n is read off from P^n , computable by repeated squaring or by diagonalisation as described above.

^aA bold modelling assumption, but this is a mathematics course.

While we are speaking the language of matrices, it is worth recording what happens when

the starting state is itself random. Treating the initial distribution λ as a *row vector*, the distribution of the chain at time n is obtained by simply multiplying by powers of P .

Proposition 17.1.9 (Distribution at Time n)

Let X_n be $\text{Markov}(\lambda, P)$. Then for all $n \geq 0$ and $j \in S$,

$$\mathbb{P}(X_n = j) = (\lambda P^n)_j.$$

Proof. Conditioning on the starting state and using the previous result,

$$\mathbb{P}(X_n = j) = \sum_{i \in S} \mathbb{P}(X_0 = i) \mathbb{P}(X_n = j \mid X_0 = i) = \sum_{i \in S} \lambda_i (P^n)_{i,j} = (\lambda P^n)_j. \quad \square$$

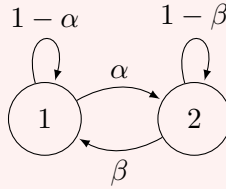
So the entire distributional evolution of a Markov chain is captured by the linear algebra of the pair (λ, P) – a fact we will lean on constantly, and which will become especially interesting when we ask what happens as $n \rightarrow \infty$ later in the course.

Example 17.1.10 (Computing Transition Probabilities)

Let $\alpha, \beta \in (0, 1)$. Consider the Markov chain X_n with states $S = \{1, 2\}$, with transition matrix

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix}.$$

A diagram of this Markov chain is shown below.



We want to find the n -step transition probabilities.

Method 1 (Difference Equations). We are first going to find $p_{1,1}(n)$ using a ‘bare-hands’ method. By the Chapman-Kolmogorov equations (that is, conditioning on X_n) we have

$$\begin{aligned} p_{1,1}(n+1) &= p_{1,1}(n)p_{1,1}(1) + p_{1,2}(n)p_{2,1}(1) \\ &= (1 - \alpha)p_{1,1}(n) + \beta p_{1,2}(n) \\ &= (1 - \alpha)p_{1,1}(n) + \beta(1 - p_{1,1}(n)) \\ &= p_{1,1}(n)(1 - \alpha - \beta) + \beta. \end{aligned}$$

This is a recurrence relation which we can then solve using the boundary condition $p_{1,1}(0) = 1$ to get

$$p_{1,1}(n) = \begin{cases} \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta} \cdot (1 - \alpha - \beta)^n & \text{if } \alpha + \beta > 0, \\ 1 & \text{if } \alpha + \beta = 0. \end{cases}$$

Method 2 (Diagonalisation). An alternative solution uses some tools from matrix algebra. To calculate the n -step transition matrix P^n , we are going to diagonalise P .

The eigenvalues of P are given by the solutions to $\det(P - \mu I) = 0$, which are $\{1, 1 - \alpha - \beta\}$. Thus for some invertible matrix U we have

$$P = U^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1 - \alpha - \beta \end{pmatrix} U \quad \text{and} \quad P^n = U^{-1} \begin{pmatrix} 1 & 0 \\ 0 & (1 - \alpha - \beta)^n \end{pmatrix} U.$$

Thus $p_{1,1}(n) = A + B(1 - \alpha - \beta)^n$, for constants A and B . We can find these by noting the boundary conditions $p_{1,1}(0) = 1$ and $p_{1,1}(1) = 1 - \alpha$, giving

$$p_{1,1}(n) = \begin{cases} \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta} \cdot (1 - \alpha - \beta)^n & \text{if } \alpha + \beta > 0, \\ 1 & \text{if } \alpha + \beta = 0. \end{cases}$$

In general, if the state space of a Markov chain is finite with $|S| = k$, then P is a $k \times k$ matrix with eigenvalues $\mu_1, \dots, \mu_k$.

If *all of the eigenvalues are distinct*, then P is diagonalisable, and we can write

$$P = U^{-1} \begin{pmatrix} \mu_1 & \cdots & 0 \\ 0 & \ddots & 0 \\ 0 & \cdots & \mu_k \end{pmatrix} U \quad \text{and} \quad P^n = U^{-1} \begin{pmatrix} \mu_1^n & \cdots & 0 \\ 0 & \ddots & 0 \\ 0 & \cdots & \mu_k^n \end{pmatrix} U,$$

and $p_{i,j} = a_1 \mu_1^n + \cdots + a_k \mu_k^n$, for some constants $a_1, \dots, a_k$, which are determined by the boundary conditions.

If some eigenvalue μ_k is complex, then its conjugate is also an eigenvalue, so if $\mu_k = re^{i\theta}$ we have also the eigenvalue $\overline{\mu_k} = re^{-i\theta}$, and we can write

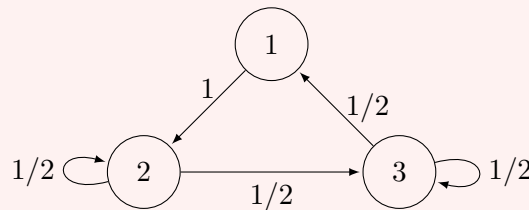
$$p_{i,j} = a_1 \mu_1^n + \cdots + a_{k-2} \mu_{k-2}^n + a_{k-1} r^n \cos(n\theta) + a_k r^n \sin(n\theta),$$

since $p_{i,j}$ is real and so all of the imaginary parts must cancel out.

If *some eigenvalues repeat*, then the situation is slightly more complicated. If an eigenvalue μ_k has multiplicity m , we can simply replace a_k by a degree m polynomial in n ².

Example 17.1.11 (Transition Matrix with Complex Eigenvalues)

Consider the Markov chain X_n with states $S = \{1, 2, 3\}$ as shown in the diagram below. We want to find the n -step transition probability $p_{i,i}(n)$.



²This comes from considering the Jordan Normal form of the transition matrix.

This Markov chain has the transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix},$$

which has distinct eigenvalues $\{1, i/2, -i/2\}$.

We can rewrite the complex eigenvalues using trigonometric functions as

$$\begin{aligned} \frac{i}{2} &= \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}, \\ -\frac{i}{2} &= \cos \frac{\pi}{2} - i \sin \frac{\pi}{2}. \end{aligned}$$

The general form for $p_{1,1}(n)$ is then given by

$$p_{1,1}(n) = A + B \cdot \left(\frac{1}{2}\right)^n \cos\left(\frac{n\pi}{2}\right) + C \cdot \left(\frac{1}{2}\right)^n \sin\left(\frac{n\pi}{2}\right).$$

The boundary conditions can be computed by hand as $p_{1,1}(0) = 1$, $p_{1,1}(1) = 0$ and $p_{1,1}(2) = 0$. This allows us to solve for A , B and C to get

$$p_{1,1}(n) = \frac{1}{5} + \left(\frac{1}{2}\right)^n \left(\frac{4}{5} \cos\left(\frac{n\pi}{2}\right) - \frac{2}{5} \sin\left(\frac{n\pi}{2}\right) \right).$$

§17.2 Class Structure

§17.2.1 Communicating Classes

In Graph Theory, one can often make a problem easier by looking at the *connected components* of the graph, effectively splitting it up into smaller disconnected chunks which are easier to understand on their own. This has a natural analogue in the study of Markov chains,³ where we consider which states can interact.

Definition 17.2.1 (Communication)

Let X_n be a Markov chain. For $i, j \in S$, we say that i **leads to** j , written $i \rightarrow j$, if $p_{i,j}(n) > 0$ for some $n \geq 0$.

If $i \rightarrow j$ and $j \rightarrow i$, we say that i and j **communicate**, and write $i \leftrightarrow j$.

We can use this idea of communication to break up our state space by noting that communication is an equivalence relation.

Theorem 17.2.2 (Communication is an Equivalence Relation)

The relation $\leftrightarrow$ is an equivalence relation on S .

³Indeed, one can view a Markov chain as a labelled directed graph, and then apply the same connected components argument to obtain the theory shown here.

Proof. For any $i \in S$, $p_{i,i}(0) = 1$ and thus $i \leftrightarrow i$, so the relation is reflexive. The relation is symmetric by definition, so we are just left to check transitivity.

Let $i, j, k \in S$, and suppose that $i \leftrightarrow j$ and $j \leftrightarrow k$. Then there exists $m, n \geq 0$ such that $p_{i,j}(n) > 0$ and $p_{j,k}(m) > 0$. Then by the Chapman-Kolmogorov equations,

$$p_{i,k}(m+n) = \sum_{l \in S} p_{i,l}(m)p_{l,k}(n) \geq p_{i,j}(n)p_{j,k}(m) > 0,$$

so $i \rightarrow k$. By the same argument $k \rightarrow i$, and thus $\leftrightarrow$ is transitive. $\square$

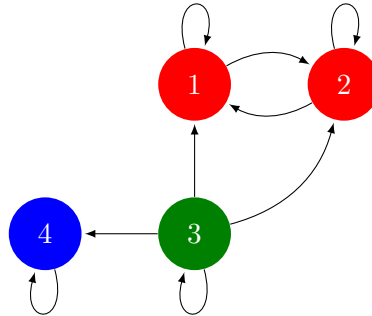
Because communication is an equivalence relation, it partitions the state space.

Definition 17.2.3 (Communicating Classes)

The equivalence classes induced by $\longleftrightarrow$ on S are called **communicating classes**.

Because communication is really a notion about directed graphs (and not *really* about individual probabilities), it's usually relatively easy to identify the communicating classes from the diagram of a Markov chain.

For example, the Markov chain below has its communicating classes coloured (and the transition probabilities are not labelled, as they are not needed to infer the communicating classes, aside from knowing they are non-zero along each edge).



In the Markov chain above, we can see that there are a few different ways that communicating classes can appear.

A communicating class C is **closed** if there is no way to leave it, in that if $i \in C$ and $i \rightarrow j$, then $j \in C$. For example, in the Markov chain above, the communicating class $\{1, 2\}$ is closed.

A state $i \in S$ is **absorbing** if there is no way to leave that state, that is, $\{i\}$ is a communicating class. In the Markov chain above, 4 is absorbing.

If a Markov chain has only one communicating class, we call it **irreducible**, as we cannot split it into smaller parts in this way.

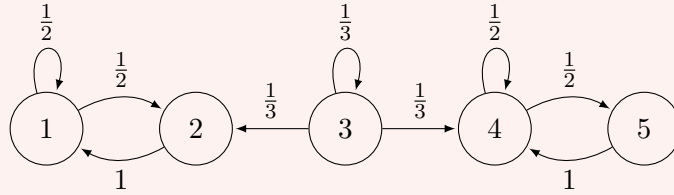
Let's put these definitions to use on a concrete example.

Example 17.2.4 (Finding the Communicating Classes)

Consider the Markov chain on $S = \{1, 2, 3, 4, 5\}$ with transition matrix

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1/3 & 0 & 1/3 & 1/3 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Drawing the diagram (which, as we said, is usually the right move) gives the following.



We can now just read off the class structure. States 1 and 2 communicate (each leads to the other in one step), and nothing outside $\{1, 2\}$ can be reached from them, so $\{1, 2\}$ is a *closed* communicating class. Similarly $\{4, 5\}$ is a closed class. State 3 leads to state 2 and state 4, but neither leads back to 3, so $\{3\}$ is a communicating class by itself, and it is *not* closed.

This chain is not irreducible – it has three communicating classes. Intuitively, the chain will linger at 3 for a (geometrically distributed) while, and then fall either left into $\{1, 2\}$ or right into $\{4, 5\}$, where it will remain trapped forever. Computing the probability of each of these outcomes is a matter of *hitting probabilities*, which is exactly the business of the next section.

§17.3 Hitting Times, Stopping Times and Absorption Probabilities

§17.3.1 Hitting Times

When studying Markov chains, a frequently occurring question is ‘*how long does it take to reach a given state $i \in S$?*’, a concept which we formulate as the *hitting time*.

Definition 17.3.1 (Hitting Time)

Let X_n be a Markov chain. The **hitting time** of some subset of states $A \subseteq S$ is the random variable T_A , which is the least n for which $X_n \in A$. In particular,

$$T_A = \inf\{n \geq 0 \mid X_n \in A\},$$

where this may be ∞ if $X_n \notin A$ for all $n \geq 0$.

In this section, we will also look some related quantities, such as the *hitting probability* and *mean hitting time*.

Definition 17.3.2 (Hitting Probability)

Let X_n be a Markov chain, and let $A \subseteq S$ be some subset of states. The **hitting probability** h_i^A of A is the probability that X_n eventually reaches a state in A , given $X_0 = i$. That is,

$$h_i^A = \mathbb{P}(T_A < \infty \mid X_0 = i),$$

where $i \in S$.

Definition 17.3.3 (Mean Hitting Time)

Let X_n be a Markov chain, and let $A \subseteq S$ be some subset of states. The **mean hitting time** k_i^A of A is the expected number of steps taken for X_n to reach a state in A , given $X_0 = i$. That is,

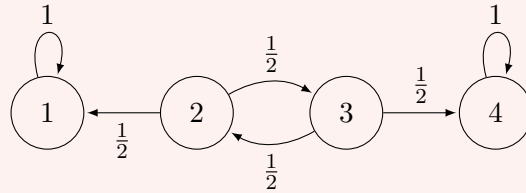
$$k_i^A = \mathbb{E}[T_A \mid X_0 = i],$$

where $i \in S$. We note that this may be ∞ .

So how do we go about calculating these quantities? An intuitive way is to try and consider what happens as we move from one state to another, as we will see in the example below.

Example 17.3.4 (Computing Hitting Probabilities and Mean Hitting Times)

Consider the Markov chain in the diagram below.



Suppose we took $A = \{4\}$ and wanted to find the hitting probability $h_2^A = \mathbb{P}(T_A < \infty \mid X_0 = 2)$. One might intuitively suppose that

$$h_2^A = P_{2,3}h_3^A = \frac{1}{2}h_3^A,$$

$$h_3^A = P_{3,2} \cdot h_2^A + P_{3,4} \cdot h_4^A = \frac{1}{2} \cdot h_2^A + \frac{1}{2}.$$

which when solved gives us $h_2^A = 1/3$.

Alternatively, if we took $B = \{1, 4\}$ and wanted to find the mean hitting time k_2^B , we could again try the intuitive approach of writing

$$k_2^B = 1 + \frac{1}{2}k_3^B,$$

and noting that $k_3^B = k_2^B$ by symmetry, giving $k_2^B = 2$.

In the computations above, we really should check that these are valid methods (though it *really is* quite intuitive).

Theorem 17.3.5 (Computing Hitting Probabilities)

Let $A \subseteq S$. Then the set of hitting probabilities $h^A = (h_i^A \mid i \in S)$ is the minimal^a non-negative solution to the equations

$$h_i^A = \begin{cases} 1 & \text{if } i \in A, \\ \sum_{j \in S} P_{i,j} h_j^A & \text{if } i \notin A. \end{cases}$$

^aMinimal in that if x is another non-negative solution, then $x_i \geq h_i^A$ for all $i \in S$.

Proof. We first check that h_i solves this system. Clearly if $i \in A$, then $h_i^A = 1$, so take $i \in S \setminus A$. Then conditioning on the first step in the chain,

$$\begin{aligned} h_i^A &= \sum_{j \in S} P_{i,j} \mathbb{P}(T_A < \infty \mid X_0 = i, X_1 = j) \\ &= \sum_{j \in S} P_{i,j} h_j^A, \end{aligned}$$

by the simple Markov property. So h_i does indeed solve this system.

We now check minimality. Let $x = (x_i \mid i \in S)$ be another non-negative solution. For $i \in A$, we have $x_i = h_i^A = 1$, so that holds. Then for $i \in S \setminus A$, since x satisfies the system of equations we can write

$$x_i = \sum_{j \in S} P_{i,j} x_j = \sum_{j \in A} P_{i,j} x_j + \sum_{j \in S \setminus A} P_{i,j} x_j. \quad (\dagger)$$

Then with $x_j = 1$ for $j \in A$ and $x \geq 0$, we have

$$x_i \geq \sum_{j \in A} P_{i,j} = \mathbb{P}(X_1 \in A \mid X_0 = i) = \mathbb{P}(T_A < 2 \mid X_0 = i).$$

Similarly, expanding $(\dagger)$, we have

$$\begin{aligned} x_i &= \mathbb{P}(X_1 \in A \mid X_0 = i) + \sum_{j \in S \setminus A} P_{i,j} \left(\sum_{k \in A} P_{j,k} x_k + \sum_{k \in S \setminus A} P_{j,k} x_k \right) \\ &\geq \mathbb{P}(X_1 \in A \mid X_0 = i) + \mathbb{P}(X_1 \notin A, X_2 \in A \mid X_0 = i) \\ &= \mathbb{P}(T_A < 3 \mid X_0 = i). \end{aligned}$$

We can repeat this argument to eventually establish that $x_i \geq \mathbb{P}(T_A < n \mid X_0 = i)$, for all $n \geq 0$. Then as $n \rightarrow \infty$, we have $x_i \geq \mathbb{P}(T_A < \infty \mid X_0 = i)$, as required. $\square$

Let's also settle a question we left dangling at the end of the previous section: for the five-state chain whose classes were $\{1, 2\}$ (closed), $\{3\}$ and $\{4, 5\}$ (closed), where does the chain end up when started from state 3?

Example 17.3.6 (Absorption Probabilities)

Take $A = \{1, 2\}$, and let $h_i = h_i^A$. We have $h_1 = h_2 = 1$, and since states 4 and 5 can never reach A , also $h_4 = h_5 = 0$. From state 3, the chain moves to each of

states 2, 3, 4 with probability $1/3$, so

$$h_3 = \frac{1}{3}h_2 + \frac{1}{3}h_3 + \frac{1}{3}h_4 = \frac{1}{3} + \frac{1}{3}h_3,$$

giving $h_3 = 1/2$. So the chain started from 3 is absorbed into $\{1, 2\}$ or into $\{4, 5\}$ with probability $1/2$ each – which makes sense, as the two escape routes from state 3 are equally likely, and dawdling at 3 doesn't favour either.

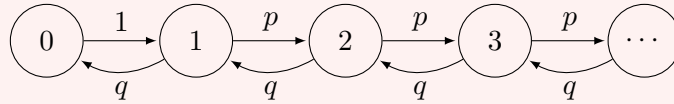
This method of computing hitting probabilities is perfectly valid on Markov chains with infinite state spaces.

Example 17.3.7 (Hitting Probabilities on an Infinite State Space)

Consider the Markov chain with states $S = \{0, 1, \dots\}$ such that $P_{0,1} = 1$, and

$$P_{i,i+1} = p, \quad P_{i,i-1} = q \quad \text{for all } i \geq 1,$$

where $p, q \in (0, 1)$ with $p + q = 1$.



Let $h_i = \mathbb{P}(T_0 < \infty \mid X_0 = i)$, and suppose that $p \neq q$. Then we clearly have $h_0 = 1$, and generally we can write down the recurrence

$$h_i = ph_{i+1} + qh_{i-1}.$$

This then implies $p(h_{i+1} - h_i) = q(h_i - h_{i-1})$, so that

$$h_{i+1} - h_i = \left(\frac{q}{p}\right)^i (h_1 - 1).$$

This inspires us to write h_i as a telescoping sum

$$\begin{aligned} h_i &= \sum_{j=1}^i (h_j - h_{j-1}) + 1 \\ &= (h_1 - 1) \sum_{j=1}^i \left(\frac{q}{p}\right)^j + 1, \end{aligned}$$

which gives us our general solution $h_i = a + b(q/p)^i$, which we can then solve.

In the case that $q > p$, then in order for h_i to be the minimal solution, we need to have $a = 1$ and $b = 0$, so $h_i = 1$.

We also need to check that our method for calculating the mean hitting time was valid.

Theorem 17.3.8 (Computing Mean Hitting Times)

Let $A \subseteq S$. Then the set of mean hitting times $k^A = (k_i^A \mid i \in S)$ is the minimal

non-negative solution to the equations

$$k_i^A = \begin{cases} 0 & \text{if } i \in A, \\ 1 + \sum_{j \in S \setminus A} P_{i,j} k_j^A & \text{if } i \notin A. \end{cases}$$

Proof. We first check that k_i solves this system. Clearly if $i \in A$, then $k_i^A = 0$, so take $i \in S \setminus A$. Then conditioning on the first step in the chain,

$$\begin{aligned} k_i^A &= \sum_{j \in S} P_{i,j} \mathbb{E}[T_A \mid X_0 = i, X_1 = j] \\ &= \sum_{j \in S} P_{i,j} (1 + \mathbb{E}[T_A \mid X_0 = j]) \\ &= 1 + \sum_{j \in S} P_{i,j} k_j^A, \end{aligned}$$

which is equivalent to our original sum.

We now show that the solution is minimal. Let $y = (y_i \mid i \in S)$ be another non-negative solution. Then for $i \in A$ we have $y_i = k_i^A = 0$, so consider $i \in S \setminus A$. We can then write

$$\begin{aligned} y_i &= 1 + \sum_{j \in S \setminus A} P_{i,j} y_j \\ &= 1 + \sum_{j \in S \setminus A} P_{i,j} \left(1 + \sum_{k \in S \setminus A} P_{j,k} y_k \right) \\ &\geq \mathbb{P}(T_A \geq 1 \mid X_0 = i) + \mathbb{P}(T_A \geq 2 \mid X_0 = i). \end{aligned}$$

By iterating this, we obtain

$$y_i \geq \sum_{m=1}^n \mathbb{P}(T_A \geq m \mid X_0 = i),$$

and taking the limit as $n \rightarrow \infty$, this gives

$$y_i \geq \sum_{m=1}^{\infty} \mathbb{P}(T_A \geq m \mid X_0 = i) = k_i^A,$$

where we note that for a random variable M taking non-negative integer values we have $\mathbb{E}[M] = \sum_{m=1}^{\infty} \mathbb{P}(M \geq m)$. $\square$

So it does work, how lovely.

§17.3.2 Gambler's Ruin

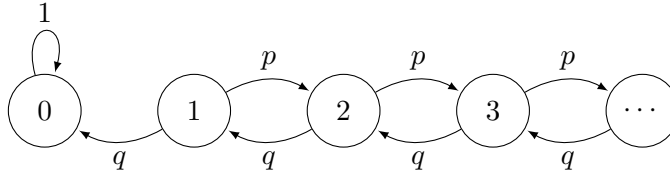
Let's now put these tools to work on one of the oldest problems in probability. A gambler walks into a casino with $\mathcal{L}i$ in their pocket, and repeatedly plays a game in which they win $\mathcal{L}1$ with probability p and lose $\mathcal{L}1$ with probability $q = 1 - p$. The gambler has no self-control, and will keep playing until they go bankrupt. We'd like to know the

probability that this happens⁴.

We can model this situation with a Markov chain on the states $S = \{0, 1, 2, \dots\}$, where the state tracks the gambler's current fortune. We set

$$P_{i,i+1} = p, \quad P_{i,i-1} = q \quad \text{for all } i \geq 1,$$

and $P_{0,0} = 1$, since a bankrupt gambler can no longer play. A diagram of this chain is shown below.



The probability of ruin starting from a fortune of i is exactly the hitting probability $h_i = h_i^{\{0\}}$, so we can use the machinery of the previous section. We know that $h = (h_i \mid i \in S)$ is the minimal non-negative solution to

$$h_0 = 1, \quad h_i = ph_{i+1} + qh_{i-1} \quad \text{for } i \geq 1.$$

This is a linear difference equation, which we know how to solve. Suppose first that $p \neq q$. Then the general solution is

$$h_i = A + B \left(\frac{q}{p} \right)^i,$$

and the boundary condition $h_0 = 1$ forces $A + B = 1$.

If $p < q$ (the game is stacked against the gambler, as it usually is), then $(q/p)^i \rightarrow \infty$ as $i \rightarrow \infty$, and since $h_i \in [0, 1]$ is a probability we must have $B = 0$. Thus $h_i = 1$ for all i – *the gambler is certain to go bankrupt*.

If instead $p > q$, then substituting $B = 1 - A$ gives

$$h_i = \left(\frac{q}{p} \right)^i + A \left(1 - \left(\frac{q}{p} \right)^i \right),$$

which is a genuine non-negative solution for any $A \geq 0$. Minimality then tells us to take $A = 0$, giving

$$h_i = \left(\frac{q}{p} \right)^i.$$

So with a favourable game the gambler has a positive chance $1 - (q/p)^i$ of playing forever.

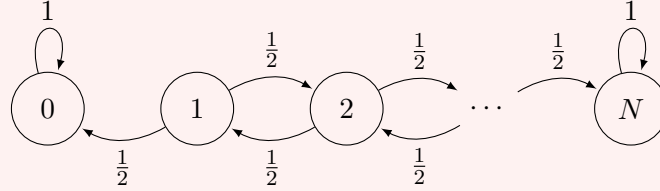
We are left with the fair case $p = q = 1/2$. Here the general solution to the difference equation is $h_i = A + Bi$, and for h_i to remain in $[0, 1]$ for all i we need $B = 0$, so $h_0 = 1$ forces $h_i = 1$ for all i . That is, *even in a perfectly fair game, ruin is certain*.

The same circle of ideas handles the case where the gambler resolves to walk away upon reaching some target fortune.

⁴Spoiler: unless the game is rigged in the gambler's favour, it happens with probability 1. The house always wins.

Example 17.3.9 (Gambling Against a Target)

Suppose the game is fair ($p = q = 1/2$), and the gambler starts with $\mathcal{L}i$ but will stop when they reach either $\mathcal{L}0$ or a target of $\mathcal{L}N$, where $0 \leq i \leq N$. Now both 0 and N are absorbing, and the interesting quantities are the ruin probability $h_i = h_i^{\{0\}}$ and the expected duration of the game $k_i = k_i^{\{0,N\}}$.



For the ruin probability, we have the same difference equation $h_i = \frac{1}{2}h_{i-1} + \frac{1}{2}h_{i+1}$ for $0 < i < N$, but now with *two* boundary conditions $h_0 = 1$ and $h_N = 0$, which pin down the solution uniquely (no appeal to minimality needed):

$$h_i = 1 - \frac{i}{N}.$$

Reassuringly, letting $N \rightarrow \infty$ recovers $h_i = 1$, matching the one-sided problem.

For the expected duration, the mean hitting time equations give

$$k_0 = k_N = 0, \quad k_i = 1 + \frac{1}{2}k_{i-1} + \frac{1}{2}k_{i+1} \text{ for } 0 < i < N.$$

The homogeneous solutions are $A + Bi$ as before, and since the inhomogeneous term is a constant, we try a particular solution of the form ci^2 ; substituting in gives $c = -1$. So $k_i = A + Bi - i^2$, and the boundary conditions force $A = 0$ and $B = N$, yielding the rather elegant

$$k_i = i(N - i).$$

So a fair game between a gambler with $\mathcal{L}i$ and an opponent with $\mathcal{L}(N - i)$ lasts $i(N - i)$ rounds on average – and note how the duration blows up quadratically as we let the opponent's wealth grow, consistent with the infinite expected duration we will find for the one-sided game.

Remark. There is something quite striking hiding in the fair case. Ruin occurs with probability 1, but we shall see shortly (using generating functions and the strong Markov property) that the *expected* time until ruin is infinite. Certain events can still take a rather long time to happen.

§17.3.3 Birth-Death Chains

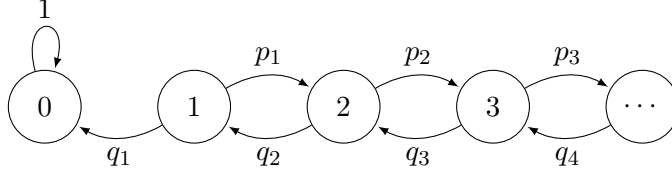
The gambler's ruin chain has the same transition probabilities at every state, which makes the associated difference equation easy to solve. We will now consider a generalisation where the transition probabilities vary from state to state, a model known as a *birth-death chain*. These chains model, for example, the size of a population, where the rates of births and deaths depend on the current population size.

Let $(p_i \mid i \geq 1)$ be a sequence with $p_i \in (0, 1)$, and let $q_i = 1 - p_i$. The **birth-death**

chain is the Markov chain on $S = \{0, 1, 2, \dots\}$ with transition probabilities

$$P_{i,i+1} = p_i, \quad P_{i,i-1} = q_i \quad \text{for } i \geq 1,$$

and $P_{0,0} = 1$ (once the population dies out, it stays extinct).



We would like to compute the probability of extinction $h_i = h_i^{\{0\}}$, starting from a population of size i . As always, h is the minimal non-negative solution to

$$h_0 = 1, \quad h_i = p_i h_{i+1} + q_i h_{i-1} \quad \text{for } i \geq 1.$$

This is *not* a constant-coefficient difference equation, so we need to be slightly cleverer. Using $p_i + q_i = 1$, we can rearrange the relation into

$$p_i(h_{i+1} - h_i) = q_i(h_i - h_{i-1}).$$

So setting $u_i = h_{i-1} - h_i$ (which will turn out to be non-negative), we get $u_{i+1} = (q_i/p_i)u_i$, and iterating this down to u_1 gives

$$u_{i+1} = \left(\frac{q_i}{p_i}\right) \left(\frac{q_{i-1}}{p_{i-1}}\right) \cdots \left(\frac{q_1}{p_1}\right) u_1 = \gamma_i u_1, \quad \text{where } \gamma_i = \prod_{j=1}^i \frac{q_j}{p_j},$$

and we take $\gamma_0 = 1$ for convenience. Summing $u_j = h_{j-1} - h_j$ over $j = 1, \dots, i$ telescopes to give

$$h_i = 1 - u_1(\gamma_0 + \gamma_1 + \cdots + \gamma_{i-1}).$$

All that remains is to identify the constant u_1 , and this is exactly what minimality is for. Everything hinges on whether the sum

$$\mathcal{S} = \sum_{i=0}^{\infty} \gamma_i$$

converges.

If $\mathcal{S} = \infty$, then any $u_1 > 0$ would make h_i negative for large i , so we must have $u_1 = 0$, and $h_i = 1$ for all i – extinction is certain.

If $\mathcal{S} < \infty$, then to minimise h_i we want u_1 as large as possible, subject to $h_i \geq 0$ for all i . Taking $i \rightarrow \infty$, the constraint becomes $1 - u_1 \mathcal{S} \geq 0$, so the minimal solution takes $u_1 = 1/\mathcal{S}$, giving

$$h_i = \frac{\sum_{j=i}^{\infty} \gamma_j}{\sum_{j=0}^{\infty} \gamma_j} < 1.$$

So the population survives forever with positive probability exactly when $\sum \gamma_i$ converges, that is, when the birth rates outweigh the death rates sufficiently strongly.

Example 17.3.10 (A Sanity Check)

Let's check this against the gambler's ruin, where $p_i = p$ and $q_i = q$ are constant. Then $\gamma_i = (q/p)^i$, a geometric series, and $\mathcal{S} = \sum_i (q/p)^i$ converges if and only if $q < p$. If $q \geq p$ then $h_i = 1$, and if $q < p$ then

$$h_i = \frac{\sum_{j=i}^{\infty} (q/p)^j}{\sum_{j=0}^{\infty} (q/p)^j} = \left(\frac{q}{p}\right)^i,$$

in agreement with what we found before. As a more interesting case, if the birth rate decays towards criticality like $p_i = \frac{1}{2} + \frac{c}{i}$ for a constant $c > 0$ (and large i), one can show γ_i behaves like i^{-4c} , so extinction is uncertain if $c > 1/4$ and certain if $c \leq 1/4$ – the boundary between the two behaviours is surprisingly delicate.

§17.3.4 Stopping Times and the Strong Markov Property

We previously proved the *simple Markov property*, that conditional on the value of a Markov chain at a given time m , the future is independent of the past. This property required that we take some fixed time m – but what if m itself is random? It's *not* true in general⁵ that the Markov property still holds for an arbitrary random time, but it is true for certain kinds of times.

Definition 17.3.11 (Stopping Times)

A random time $T : \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$ is a **stopping time** for a Markov chain X_n if for all $n \geq 0$ the event $\{T = n\}$ is given in terms of $X_0, X_1, \dots, X_n$ only.

In particular, random times that ‘look into the future’ are not stopping times. A straightforward example of a stopping time is the hitting time T^A that we defined in the previous section.

Example 17.3.12 (Stopping Times and Non-Stopping Times)

Let X_n track a gambler's fortune, and consider the following rules for when to leave the casino.

- (i) ‘Leave when your fortune first hits £10.’ This is the hitting time $T_{\{10\}}$: the event $\{T_{\{10\}} = n\}$ is exactly $\{X_0 \neq 10, \dots, X_{n-1} \neq 10, X_n = 10\}$, which depends only on $X_0, \dots, X_n$. So this is a stopping time – it is a rule you could actually follow.
- (ii) ‘Leave one round after your fortune first hits £10.’ This is $T_{\{10\}} + 1$, which is also a stopping time, as $\{T_{\{10\}} + 1 = n\}$ depends only on $X_0, \dots, X_{n-1}$.
- (iii) ‘Leave one round *before* your fortune first hits £10.’ This is $T_{\{10\}} - 1$, and it is *not* a stopping time: deciding whether to stop at time n requires knowing X_{n+1} . However appealing this rule sounds, there is no way to implement it.

With stopping times, we can indeed make a statement similar to the Markov property using random times.

⁵For example, let T be the first time that a Markov chain hits some value i . Then if we condition on time $T - 1$, the future is then determined since the next step will be to i ! You will see that this type of event is not allowed in our definition of stopping times.

Theorem 17.3.13 (Strong Markov Property)

Let X_n be a Markov chain with transition matrix P , and let T be a stopping time. Then conditional on $T < \infty$ and $X_T = i$, the sequence of random variables $(X_{T+n})_{n \geq 0}$ is Markov(δ_i, P) and is independent of $X_0, \dots, X_T$.

Proof. Let H be an event given in terms of $X_0, \dots, X_{T-1}$. Then it suffices to show that

$$\begin{aligned} & \mathbb{P}(X_{T+1} = i_1, \dots, X_{T+n} = i_n, H \mid T < \infty, X_T = i) \\ &= \mathbb{P}(X_1 = i_1, \dots, X_n = i_n \mid X_0 = i) \mathbb{P}(H \mid T < \infty, X_T = i). \end{aligned}$$

The event $H \cap \{T = m\}$ is given in terms of $X_1, X_2, \dots, X_m$ only. Furthermore, $X_T = X_m$ when $T = m$. We condition on the event $H \cap \{T = m\} \cap \{X_m = i\}$ and use the simple Markov property at time m to get

$$\begin{aligned} & \mathbb{P}(X_{T+1} = i_1, \dots, X_{T+n} = i_n, H \mid T < \infty, X_T = i) \\ &= \mathbb{P}(X_1 = i_1, \dots, X_n = i_n) \mathbb{P}(H, T = m, X_T = i). \end{aligned}$$

Summing this over $m = 0, 1, \dots$ and dividing by $\mathbb{P}(T < \infty, X_T = i)$ then gives the desired result. $\square$

The strong Markov property is one of those results whose usefulness is best appreciated by watching it in action. Let's return to our friend the gambler, and work out not just *whether* they go bankrupt, but *how long* it takes.

Example 17.3.14 (Duration of the Gambler's Ruin)

Consider again the gambler's ruin chain on $S = \{0, 1, 2, \dots\}$ with $P_{i,i+1} = p$ and $P_{i,i-1} = q = 1 - p$ for $i \geq 1$, and 0 absorbing. Let $H = T_{\{0\}}$ be the time of bankruptcy, and suppose the gambler starts with £1.

We are going to study the probability generating function

$$G(s) = \mathbb{E}[s^H \mid X_0 = 1] = \sum_{n=0}^{\infty} s^n \mathbb{P}(H = n \mid X_0 = 1), \quad \text{for } |s| < 1,$$

where we adopt the convention $s^\infty = 0$ (recall it is possible that $H = \infty$).

Conditioning on the first step, we have

$$G(s) = p\mathbb{E}[s^H \mid X_0 = 1, X_1 = 2] + q\mathbb{E}[s^H \mid X_0 = 1, X_1 = 0].$$

The second term is easy: if $X_1 = 0$ then $H = 1$, so $\mathbb{E}[s^H \mid X_0 = 1, X_1 = 0] = s$. For the first term, starting from 2 the chain must pass through 1 on its way to 0, so we can write $H = 1 + H' + H''$, where H' is the time taken to go from 2 to 1, and H'' is the time taken to go from 1 to 0 after that. The random time at which we first hit 1 is a stopping time, so by the strong Markov property H' and H'' are independent, and each is distributed as H . Hence

$$G(s) = psG(s)^2 + qs. \quad (*)$$

Solving this quadratic in $G(s)$ gives

$$G(s) = \frac{1 \pm \sqrt{1 - 4pqs^2}}{2ps}.$$

We should be a little careful here – a priori the sign might depend on s . But G is a power series and hence continuous on $(0, 1)$, so a single sign must be taken throughout, and since $G(s)$ remains bounded as $s \downarrow 0$ we are forced to take the negative root:

$$G(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}.$$

This function knows everything about the distribution of H . For example, the ruin probability is

$$\mathbb{P}(H < \infty \mid X_0 = 1) = \lim_{s \uparrow 1} G(s) = \frac{1 - \sqrt{1 - 4pq}}{2p} = \frac{1 - |p - q|}{2p} = \begin{cases} 1 & \text{if } p \leq q, \\ q/p & \text{if } p > q, \end{cases}$$

using $1 - 4pq = (p + q)^2 - 4pq = (p - q)^2$. This agrees (reassuringly!) with our hitting probability computation from before.

We can also extract the mean. If $p \leq q$ then differentiating (*) and rearranging gives

$$G'(s) = \frac{pG(s)^2 + q}{1 - 2psG(s)},$$

and letting $s \uparrow 1$ we obtain

$$\mathbb{E}[H \mid X_0 = 1] = \begin{cases} \frac{1}{q-p} & \text{if } p < q, \\ \infty & \text{if } p = q = \frac{1}{2}. \end{cases}$$

So in a fair game, ruin is certain but takes infinite expected time – exactly the curiosity we promised earlier.

§17.4 Recurrence and Transience

§17.4.1 Recurrent and Transient States

We now consider states that a Markov chain continues to come back to, or only visits finitely many times. It will turn out (perhaps surprisingly) that for each state, one of these two behaviours *must* occur, in quite a strong sense: either the chain returns to the state infinitely often with probability 1, or it returns only finitely many times with probability 1. We give these two behaviours names.

Definition 17.4.1 (Recurrence and Transience)

Let X_n be a Markov chain and let $i \in S$. We say the state i is **recurrent** if

$$\mathbb{P}(X_n = i \text{ for infinitely many } n \mid X_0 = i) = 1,$$

and we say i is **transient** if

$$\mathbb{P}(X_n = i \text{ for infinitely many } n \mid X_0 = i) = 0.$$

A priori, this probability could lie strictly between 0 and 1, in which case a state would be neither recurrent nor transient. The first goal of this section is to rule this out, and to obtain a usable criterion for telling the two cases apart. Doing this will be a lovely application of the strong Markov property.

We will need a little notation for ‘repeated visits’. Define the **first passage time** to a state $i \in S$ as

$$T_i = \inf\{n \geq 1 \mid X_n = i\},$$

which differs from the hitting time in that we do not count the chain as having returned to i at time 0. More generally, we let $T_i^{(0)} = 0$ and inductively define the **r -th passage time** by

$$T_i^{(r+1)} = \inf\{n \geq T_i^{(r)} + 1 \mid X_n = i\},$$

so that $T_i^{(1)} = T_i$. We also define the **return probability**

$$f_i = \mathbb{P}(T_i < \infty \mid X_0 = i),$$

and the total **number of visits**

$$V_i = \sum_{n=0}^{\infty} \mathbb{1}[X_n = i].$$

With this notation, i is recurrent exactly if $\mathbb{P}(V_i = \infty \mid X_0 = i) = 1$, and transient exactly if this probability is 0.

Intuitively, every time the chain returns to i , it ‘starts afresh’ by the strong Markov property, and so has probability f_i of returning again. So the number of returns should behave like a run of independent trials each with success probability f_i . The following lemma makes this precise.

Lemma 17.4.2 (Probability of Repeated Visits)

Let $i \in S$. Then $\mathbb{P}(V_i > r \mid X_0 = i) = f_i^r$, for all $r \in \mathbb{N}$.

Proof. We use induction on r . For $r = 0$ the claim is clear, since if $X_0 = i$ then $V_i \geq 1$. Now note that, given $X_0 = i$, the event $\{V_i > r\}$ is precisely the event $\{T_i^{(r)} < \infty\}$. So suppose the result is true for r . Then

$$\begin{aligned} \mathbb{P}(V_i > r + 1 \mid X_0 = i) &= \mathbb{P}(T_i^{(r+1)} < \infty \mid X_0 = i) \\ &= \mathbb{P}(T_i^{(r+1)} < \infty, T_i^{(r)} < \infty \mid X_0 = i) \\ &= \mathbb{P}(T_i^{(r+1)} < \infty \mid T_i^{(r)} < \infty, X_0 = i) \cdot \mathbb{P}(T_i^{(r)} < \infty \mid X_0 = i) \\ &= \mathbb{P}(T_i^{(r+1)} < \infty \mid T_i^{(r)} < \infty, X_0 = i) \cdot f_i^r. \end{aligned}$$

Now $T_i^{(r)}$ is a stopping time, and on the event $T_i^{(r)} < \infty$ we have $X_{T_i^{(r)}} = i$. So by the strong Markov property, conditional on $T_i^{(r)} < \infty$, the chain $(X_{T_i^{(r)}+n})_{n \geq 0}$ is

Markov(δ_i, P), and hence

$$\mathbb{P}(T_i^{(r+1)} < \infty \mid T_i^{(r)} < \infty, X_0 = i) = \mathbb{P}(T_i < \infty \mid X_0 = i) = f_i,$$

and thus $\mathbb{P}(V_i > r + 1 \mid X_0 = i) = f_i^{r+1}$, completing the induction. $\square$

So given $X_0 = i$, the number of visits V_i is a geometric random variable with parameter $1 - f_i$. In particular, if $f_i = 1$ then $V_i = \infty$ almost surely, and if $f_i < 1$ then V_i is almost surely finite with finite expectation. This immediately gives us the dichotomy we were after.

Theorem 17.4.3 (Recurrence and Transience Criterion)

Let X_n be a Markov chain with transition matrix P , and let $i \in S$. Then we have the following dichotomy:

- (1) If $\mathbb{P}(T_i < \infty \mid X_0 = i) = 1$, then i is recurrent and $\sum_{n=0}^{\infty} p_{i,i}(n)$ is infinite.
- (2) If $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$, then i is transient and $\sum_{n=0}^{\infty} p_{i,i}(n) < \infty$.

In particular, every state is either recurrent or transient.

Proof. Recall we can write V_i using indicator functions as $V_i = \sum_{n=0}^{\infty} \mathbb{1}[X_n = i]$, and then by linearity of expectation we can write

$$\mathbb{E}[V_i \mid X_0 = i] = \sum_{n=0}^{\infty} \mathbb{P}(X_n = i \mid X_0 = i) = \sum_{n=0}^{\infty} p_{i,i}(n).$$

Part (1). If $f_i = 1$, then for all r we have $\mathbb{P}(V_i > r \mid X_0 = i) = 1$, and thus $\mathbb{P}(V_i = \infty \mid X_0 = i) = 1$, implying that i is recurrent, and $\mathbb{E}[V_i \mid X_0 = i] = \infty$, implying that $\sum_{n=0}^{\infty} p_{i,i}(n) = \infty$.

Part (2). If $f_i < 1$, then by the previous lemma we have $\mathbb{E}[V_i \mid X_0 = i] = \sum_{r=0}^{\infty} \mathbb{P}(V_i > r \mid X_0 = i) = 1/(1 - f_i) < \infty$, implying that $\mathbb{P}(V_i < \infty \mid X_0 = i) = 1$, so i is transient, and $\sum_{n=0}^{\infty} p_{i,i}(n) < \infty$. $\square$

This criterion is genuinely useful – the quantity $\sum_n p_{i,i}(n)$ is frequently something that we can estimate, and we will do exactly this when we study random walks on $\mathbb{Z}^d$ in the next section. But first, let's see how recurrence and transience interact with the class structure that we developed earlier. The headline is that recurrence and transience are *class properties*: within a communicating class, all states behave in the same way.

Theorem 17.4.4 (Recurrence and Transience of Communicating States)

Let x and y be two states that communicate. Then either they are both recurrent or they are both transient.

Proof. Suppose that x is recurrent. We will show that y is also recurrent. Since $x \rightarrow y$, there is some $m, l \in \mathbb{N}$ such that $p_{x,y}(m) > 0$ and $p_{y,x}(l) > 0$. Then we know

that $\sum_{n=0}^{\infty} p_{x,x}(n) = \infty$, and then

$$\begin{aligned} \sum_{n=0}^{\infty} p_{y,y}(n) &\geq \sum_{n=0}^{\infty} p_{y,y}(n+m+l) \\ &\geq \sum_{n=0}^{\infty} p_{y,x}(l) p_{x,x}(n) p_{x,y}(m) \\ &\geq p_{y,x}(l) p_{x,y}(m) \sum_{n=0}^{\infty} p_{x,x}(n) = \infty, \end{aligned}$$

using the Chapman-Kolmogorov equations, and thus y is recurrent by our criterion. The transient case follows since if x was transient but y was recurrent, then swapping the roles of x and y in the above argument would make x recurrent, a contradiction. $\square$

This immediately can be applied to the entire class structure of a Markov chain.

Corollary 17.4.5 (Class Structure of Recurrence and Transience)

Either all of the states in a communicating class are recurrent or they are all transient.

Recurrent communicating classes have a certain flavour to them, which should be intuitive. Really, for a communicating class to be recurrent we have to avoid getting stuck outside of that given communicating class. This forces closure of the communicating class.

Theorem 17.4.6 (Closure of Recurrent Communicating Classes)

If C is a recurrent communicating class, then it must be closed.

Proof. Suppose that C was not closed. Then there exists $x \in C$ and $y \notin C$ such that $x \rightarrow y$. Let m be such that $p_{x,y}(m) > 0$. We first note that $y \not\rightarrow x$, since if we had $y \rightarrow x$ then x and y would communicate, and we would have $y \in C$. So if starting from x we hit y at some point, then we can never visit x again. Thus

$$\mathbb{P}(V_x < \infty \mid X_0 = x) \geq \mathbb{P}(X_m = y \mid X_0 = x) = p_{x,y}(m) > 0,$$

so x is not recurrent, which is a contradiction. $\square$

In the other direction, a closed class need not be recurrent in general (we will meet examples of this when we study random walks), but it *is* when the class is finite. This should feel right – a chain trapped in a finite set of states has to visit some state infinitely often.

Theorem 17.4.7 (Finite Closed Classes are Recurrent)

A finite closed class is recurrent.

Proof. Let C be a finite closed class, and pick any state $x \in C$. Starting from x , the chain remains in C forever (as C is closed), and since C is finite, there must be

some state $y \in C$ that is visited infinitely often with positive probability, that is^a,

$$\mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = x) > 0.$$

Now, for the chain to visit y infinitely often it must visit y at least once, so conditioning on the first visit and applying the strong Markov property,

$$\begin{aligned} 0 &< \mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = x) \\ &= \mathbb{P}(T_y < \infty \mid X_0 = x) \cdot \mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = y). \end{aligned}$$

In particular $\mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = y) > 0$. But by the dichotomy this probability is either 0 or 1, so it equals 1, and y is recurrent. Finally, since $x, y \in C$ they communicate, and recurrence is a class property, so all of C is recurrent. $\square$

^aThis is really the pigeonhole principle: infinitely many time steps must be distributed amongst finitely many states, so if each $y \in C$ was visited infinitely often with probability zero, then with probability one every state would be visited finitely often – which is impossible for a chain that stays in C forever.

Corollary 17.4.8 (Finite Irreducible Chains are Recurrent)

If S is finite and P is irreducible, then every state is recurrent.

Proof. The whole state space is a single communicating class, which is certainly closed, and it is finite. $\square$

So for *finite* chains the story is pleasantly complete: the transient states are precisely those in non-closed classes, and the recurrent states are precisely those in closed classes. Let's see this classification on our worked example from the section on class structure.

Example 17.4.9 (Classifying the States of a Finite Chain)

Recall the five-state chain whose communicating classes we found to be $\{1, 2\}$ (closed), $\{3\}$ (not closed) and $\{4, 5\}$ (closed). By our theorems:

- The classes $\{1, 2\}$ and $\{4, 5\}$ are finite and closed, hence recurrent – once the chain enters one of them, it revisits each of its states infinitely often.
- The class $\{3\}$ is not closed, hence not recurrent, so state 3 is transient. (Of course, we can also see this directly: on each visit to 3, the chain escapes forever with probability $2/3$, so the number of visits is geometric and finite.)

For an *infinite* chain the classification is genuinely more subtle – a closed class can be transient, as the asymmetric random walk on $\mathbb{Z}$ (which is irreducible, so its single class is closed) will shortly demonstrate.

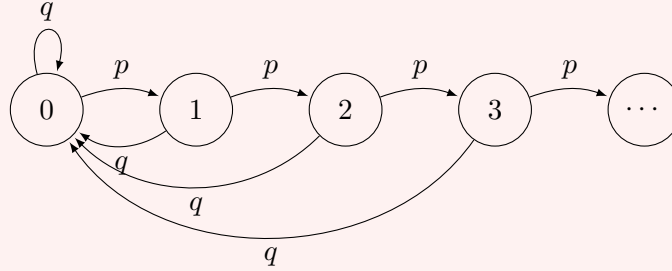
Sometimes we don't need any of this machinery, because the return probability f_i can simply be computed by hand. Here is an infinite chain where that happens.

Example 17.4.10 (The Success-Runs Chain)

Fix $p \in (0, 1)$ and $q = 1 - p$, and consider the chain on $S = \{0, 1, 2, \dots\}$ with

$$P_{i,i+1} = p, \quad P_{i,0} = q \quad \text{for all } i \geq 0.$$

This chain tracks the length of the current run of successes in a sequence of independent trials: each success extends the run, and each failure resets the counter to zero. The chain is irreducible (we can reach i from 0 via i successes, and return in one failure).



Let's compute the return probability of state 0 directly. Starting from 0, the chain fails to return to 0 for n consecutive steps precisely if the first n trials are all successes, so

$$\mathbb{P}(T_0 > n \mid X_0 = 0) = p^n \rightarrow 0,$$

giving $f_0 = \mathbb{P}(T_0 < \infty \mid X_0 = 0) = 1$, so 0 is recurrent – and hence, recurrence being a class property, the whole chain is recurrent. Moreover T_0 is geometrically distributed, with

$$m_0 = \mathbb{E}[T_0 \mid X_0 = 0] = \sum_{n=0}^{\infty} \mathbb{P}(T_0 > n \mid X_0 = 0) = \frac{1}{q} < \infty.$$

So state 0 takes on average $1/q$ steps to revisit, however long the runs of successes may occasionally be. (Once we have developed the theory of invariant distributions, this will tell us instantly that $\pi_0 = q$ – and it's a nice check that solving $\pi = \pi P$ gives the geometric distribution $\pi_i = qp^i$, whose value at 0 is indeed q .)

To close out this discussion, we prove one more result about irreducible, recurrent chains: not only does the chain return to where it started, it visits *every* state with certainty, no matter where it begins.

Theorem 17.4.11 (Recurrent Chains Visit Everything)

Let P be irreducible and recurrent. Then for all $x, y \in S$, $\mathbb{P}(T_y < \infty \mid X_0 = x) = 1$.

Proof. Fix $y \in S$, and let m be such that $p_{y,x}(m) > 0$, which exists by irreducibility. Since y is recurrent, using the simple Markov property at time m ,

$$\begin{aligned} 1 &= \mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = y) \\ &= \sum_{z \in S} \mathbb{P}(X_m = z \mid X_0 = y) \mathbb{P}(X_n = y \text{ for infinitely many } n \geq m \mid X_m = z) \\ &= \sum_{z \in S} p_{y,z}(m) \mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = z). \end{aligned}$$

Now each term $\mathbb{P}(X_n = y \text{ for infinitely many } n \mid X_0 = z) \leq \mathbb{P}(T_y < \infty \mid X_0 = z) \leq 1$, and the weights $p_{y,z}(m)$ sum to 1. So the only way the weighted average above can equal 1 is if

$$\mathbb{P}(T_y < \infty \mid X_0 = z) = 1 \quad \text{whenever } p_{y,z}(m) > 0.$$

In particular, taking $z = x$ (recalling $p_{y,x}(m) > 0$) gives the result. $\square$

Aside: A Generating Function Approach

There is a slick alternative route to the recurrence–transience dichotomy using generating functions, which is worth seeing. Fix a state i , write $f(n) = \mathbb{P}(T_i = n \mid X_0 = i)$ for the distribution of the first return time, and consider the two generating functions

$$F(s) = \sum_{n=1}^{\infty} f(n)s^n, \quad U(s) = \sum_{n=0}^{\infty} p_{i,i}(n)s^n, \quad |s| < 1.$$

Conditioning on the time of the *first* return to i , for $n \geq 1$ we have

$$p_{i,i}(n) = \sum_{k=1}^n f(k) p_{i,i}(n-k),$$

using the strong Markov property at the first return time. But this is precisely a convolution! So multiplying by s^n and summing over $n \geq 1$ (the $n = 0$ term of U being 1),

$$U(s) - 1 = F(s)U(s), \quad \text{that is, } U(s) = \frac{1}{1 - F(s)}.$$

Now we let $s \uparrow 1$. For power series with non-negative coefficients, we have $\lim_{s \uparrow 1} F(s) = \sum_n f(n) = f_i$ and $\lim_{s \uparrow 1} U(s) = \sum_n p_{i,i}(n)$, whether these are finite or infinite (this is *Abel's lemma*, which follows from monotone convergence). Hence

$$\sum_{n=0}^{\infty} p_{i,i}(n) = \frac{1}{1 - f_i},$$

with the convention $1/0 = \infty$. This identity contains the whole dichotomy at a glance: the left hand side is infinite if and only if $f_i = 1$. It also quantifies the relationship rather pleasingly – for a transient state, the expected total number of visits is exactly $1/(1 - f_i)$, the mean of the geometric distribution we found earlier.

§17.4.2 Random Walks on $\mathbb{Z}^d$ and Pólya's Theorem

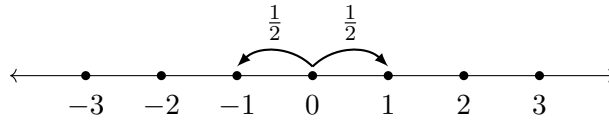
We now turn our attention to simple random walks in low and high dimensions, and in particular at an interesting result due to Pólya which informally says

A drunk man will find his way home, but a drunk bird may get lost forever.

Let's make this precise. The **simple random walk** on $\mathbb{Z}^d$ is the Markov chain with state space $\mathbb{Z}^d$ and transition probabilities

$$P_{x, x+e_i} = P_{x, x-e_i} = \frac{1}{2d} \quad \text{for } i = 1, \dots, d,$$

where $e_1, \dots, e_d$ are the standard basis vectors. That is, at each step the walker moves to one of its $2d$ neighbouring lattice points, chosen uniformly at random. For $d = 1$, this is a walk on the integers, moving left or right with probability $1/2$ each.



The chain is clearly irreducible, so all states are of the same type – either the walk is recurrent, and returns to its starting point over and over forever, or it is transient, and eventually escapes to infinity for good. Pólya's theorem tells us that which of these occurs depends only on the dimension.

Theorem 17.4.12 (Pólya's Recurrence Theorem)

The simple random walk on $\mathbb{Z}^d$ is recurrent for $d = 1$ and $d = 2$, and transient for $d \geq 3$.

The strategy for the proof is the one suggested by our recurrence and transience criterion: we will estimate the return probabilities $p_{0,0}(n)$, and determine whether $\sum_n p_{0,0}(n)$ converges or diverges. The main analytic input is Stirling's formula, which we will not prove here.

Fact 17.4.13 (Stirling's Formula)

As $n \rightarrow \infty$, we have $n! \sim n^n e^{-n} \sqrt{2\pi n}$, in the sense that the ratio of the two sides tends to 1.

We take the three cases $d = 1, 2, 3$ in turn.

Proof ($d = 1$ is recurrent). First note that the walk can only return to 0 at even times, since the parity of X_n changes at every step. So it suffices to study $p_{0,0}(2n)$.

For the walk to be back at 0 after $2n$ steps, it must have taken exactly n steps up and n steps down. There are $\binom{2n}{n}$ ways to choose which steps are the 'up' steps, and each arrangement occurs with probability $(1/2)^{2n}$. Hence

$$p_{0,0}(2n) = \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} = \frac{(2n)!}{(n!)^2} \cdot \frac{1}{2^{2n}}.$$

Applying Stirling's formula,

$$p_{0,0}(2n) \sim \frac{(2n)^{2n} e^{-2n} \sqrt{4\pi n}}{n^{2n} e^{-2n} (2\pi n)} \cdot \frac{1}{2^{2n}} = \frac{1}{\sqrt{\pi n}}.$$

In particular, there is a constant $A > 0$ and some n_0 such that $p_{0,0}(2n) \geq A/\sqrt{n}$ for all $n \geq n_0$. Thus

$$\sum_{n=0}^{\infty} p_{0,0}(n) \geq \sum_{n \geq n_0} \frac{A}{\sqrt{n}} = \infty,$$

and the walk is recurrent. □

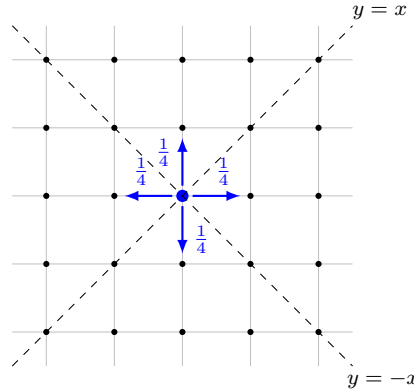
Remark (Asymmetric Walks are Transient). Suppose instead the walk moves up with

probability p and down with probability $q = 1 - p$, where $p \neq q$. The same counting gives

$$p_{0,0}(2n) = \binom{2n}{n} (pq)^n \sim \frac{(4pq)^n}{\sqrt{\pi n}},$$

and since $4pq < 1$ when $p \neq q$ (as $4pq = 1 - (p - q)^2$), the series $\sum_n p_{0,0}(n)$ converges by comparison with a geometric series. So *any* asymmetry makes the walk transient – the symmetric case is quite delicate!

Now let's move up to two dimensions. The key idea here is a rather beautiful trick: if we rotate the lattice by 45° , a simple random walk on $\mathbb{Z}^2$ splits into two *independent* simple random walks on (scaled copies of) $\mathbb{Z}$.



Proof ($d = 2$ is recurrent). Let X_n be the simple random walk on $\mathbb{Z}^2$, and write $X_n = \xi_1 + \cdots + \xi_n$, where the steps ξ_i are independent and uniform on the four unit vectors $\pm(1, 0)$ and $\pm(0, 1)$. We project onto the diagonal lines $y = x$ and $y = -x$, defining

$$X_n^+ = \sum_{i=1}^n \frac{\xi_i^{(1)} + \xi_i^{(2)}}{\sqrt{2}}, \quad X_n^- = \sum_{i=1}^n \frac{\xi_i^{(1)} - \xi_i^{(2)}}{\sqrt{2}},$$

where $\xi_i = (\xi_i^{(1)}, \xi_i^{(2)})$.

We claim that X_n^+ and X_n^- are independent simple random walks on $\frac{1}{\sqrt{2}}\mathbb{Z}$. Indeed, checking each of the four possible values of ξ_i , we see that $(\xi_i^{(1)} + \xi_i^{(2)})/\sqrt{2}$ and $(\xi_i^{(1)} - \xi_i^{(2)})/\sqrt{2}$ each take the values $\pm 1/\sqrt{2}$ with probability $1/2$, and moreover each of the four possible combinations of signs occurs with probability $1/4$ – so the two increments are independent. Since X_n^+ and X_n^- are built from independent increments in this way, they are independent simple random walks, as claimed.

Now, the walk X_n is at the origin if and only if both projections are at 0. So using independence and our one-dimensional estimate,

$$p_{0,0}(2n) = \mathbb{P}(X_{2n}^+ = 0) \mathbb{P}(X_{2n}^- = 0) \sim \frac{1}{\sqrt{\pi n}} \cdot \frac{1}{\sqrt{\pi n}} = \frac{1}{\pi n}.$$

Hence there is $A > 0$ with $p_{0,0}(2n) \geq A/n$ for all sufficiently large n , and $\sum_n p_{0,0}(n)$ diverges by comparison with the harmonic series. So the walk is recurrent. $\square$

Proof ($d = 3$ is transient). Again $p_{0,0}(n) = 0$ for odd n , so we consider $p_{0,0}(2n)$. For the walk to return to the origin after $2n$ steps, it must take i steps in each of

the $+x$ and $-x$ directions, j steps in each of the $\pm y$ directions and k steps in each of the $\pm z$ directions, for some $i + j + k = n$. Counting arrangements,

$$\begin{aligned} p_{0,0}(2n) &= \sum_{\substack{i,j,k \geq 0 \\ i+j+k=n}} \frac{(2n)!}{(i!)^2(j!)^2(k!)^2} \left(\frac{1}{6}\right)^{2n} \\ &= \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \sum_{\substack{i,j,k \geq 0 \\ i+j+k=n}} \left(\binom{n}{i,j,k} \left(\frac{1}{3}\right)^n \right)^2, \end{aligned}$$

where $\binom{n}{i,j,k} = \frac{n!}{i!j!k!}$ is the multinomial coefficient.

Now, the numbers $\binom{n}{i,j,k} (1/3)^n$ are exactly the probabilities of the possible outcomes when placing n balls into 3 boxes uniformly at random, so they sum to 1. Hence, bounding one factor in each square by the maximum,

$$p_{0,0}(2n) \leq \binom{2n}{n} \left(\frac{1}{2}\right)^{2n} \max_{i+j+k=n} \binom{n}{i,j,k} \left(\frac{1}{3}\right)^n.$$

The multinomial coefficient is maximised when i, j, k are as equal as possible^a, so writing $n = 3m$ we have

$$p_{0,0}(6m) \leq \binom{6m}{3m} \left(\frac{1}{2}\right)^{6m} \binom{3m}{m,m,m} \left(\frac{1}{3}\right)^{3m} \sim \frac{C}{m^{3/2}}$$

for some constant C , applying Stirling's formula to each factorial. In particular, the sum $\sum_m p_{0,0}(6m)$ converges.

We are not quite done, since we have only summed over times divisible by 6. But the walk can always waste time by stepping out and back, so

$$p_{0,0}(6m) \geq \left(\frac{1}{6}\right)^2 p_{0,0}(6m-2) \quad \text{and} \quad p_{0,0}(6m) \geq \left(\frac{1}{6}\right)^4 p_{0,0}(6m-4),$$

which gives $p_{0,0}(6m-2) \leq 36 p_{0,0}(6m)$ and $p_{0,0}(6m-4) \leq 6^4 p_{0,0}(6m)$. Hence

$$\sum_{n=0}^{\infty} p_{0,0}(2n) \leq (1 + 6^2 + 6^4) \sum_{m=0}^{\infty} p_{0,0}(6m) < \infty,$$

and the walk is transient. □

^aIf, say, $i \geq j + 2$, then swapping (i, j) for $(i-1, j+1)$ multiplies the coefficient by $i/(j+1) > 1$.

Remark. For $d > 3$, one can check that the first three coordinates of the simple random walk on $\mathbb{Z}^d$, observed at the times when one of them changes, form a simple random walk on $\mathbb{Z}^3$. If the walk on $\mathbb{Z}^d$ were recurrent, it would return to the origin infinitely often, forcing the embedded walk on $\mathbb{Z}^3$ to do the same – contradicting transience in three dimensions. So the walk is transient for all $d \geq 3$.

So the drunk man on the plane will stagger home eventually, but the drunk bird, with the extra dimension at its disposal, may drift away forever.

§17.5 Invariant Distributions

§17.5.1 What is an Invariant Distribution?

Having spent some time on the question of *whether* a Markov chain comes back, we now turn to the long-run behaviour of the chain: if we let a Markov chain run for a very long time, what does its distribution look like?

Back when we computed n -step transition probabilities for the two-state chain, we found

$$p_{1,1}(n) = \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta} \cdot (1 - \alpha - \beta)^n,$$

and if $\alpha + \beta < 2$ the second term dies away as $n \rightarrow \infty$, leaving $p_{1,1}(n) \rightarrow \beta/(\alpha + \beta)$. The chain forgets where it started, and settles down to some limiting distribution. We would like to understand this phenomenon in general: when does a Markov chain settle down, and to what?

If the distribution of X_n is to converge to something, a natural first step is to look for distributions that the chain leaves unchanged. Suppose X_0 has distribution π , and we would like X_1 to have distribution π as well. Conditioning on the first state,

$$\mathbb{P}(X_1 = j) = \sum_{i \in S} \mathbb{P}(X_0 = i) \mathbb{P}(X_1 = j \mid X_0 = i) = \sum_{i \in S} \pi_i P_{i,j},$$

so what we require is exactly that $\pi_j = \sum_i \pi_i P_{i,j}$ for all $j \in S$. Viewing π as a row vector, this says precisely $\pi = \pi P$.

Definition 17.5.1 (Invariant Distribution)

A distribution^a π on S is an **invariant distribution** (also called an *equilibrium* or *stationary* distribution) for the transition matrix P if

$$\pi = \pi P.$$

^aThat is, $\pi = (\pi_i \mid i \in S)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$.

In the language of linear algebra, π is a left eigenvector of P with eigenvalue 1, normalised to sum to 1. The name ‘invariant’ is justified by the following observation.

Proposition 17.5.2 (Invariance for All Time)

Let π be an invariant distribution for P , and let X_n be $\text{Markov}(\pi, P)$. Then X_n has distribution π for all $n \geq 0$.

Proof. The distribution of X_n is given by the row vector πP^n , and by induction $\pi P^n = \pi P^{n-1} = \dots = \pi$. $\square$

So a chain started in an invariant distribution is, in a distributional sense, in perfect equilibrium – at every time it looks statistically identical. Let’s compute an example.

Example 17.5.3 (Invariant Distribution of the Two-State Chain)

Consider our two-state chain with transition matrix

$$P = \begin{pmatrix} 1 - \alpha & \alpha \\ \beta & 1 - \beta \end{pmatrix},$$

with $\alpha, \beta \in (0, 1)$. To find an invariant distribution we solve $\pi = \pi P$, that is,

$$\begin{aligned}\pi_1 &= (1 - \alpha)\pi_1 + \beta\pi_2, \\ \pi_2 &= \alpha\pi_1 + (1 - \beta)\pi_2.\end{aligned}$$

Both equations reduce to $\alpha\pi_1 = \beta\pi_2$, and combining this with $\pi_1 + \pi_2 = 1$ we get

$$\pi = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right).$$

Notice that this is exactly the limit of $p_{1,1}(n)$ and $p_{2,1}(n)$ that we found by diagonalisation – the limiting distribution of the chain is the invariant distribution. This is not a coincidence, as we will see in the next section.

Example 17.5.4 (Invariant Distribution of a Three-State Chain)

Recall the three-state chain with complex eigenvalues from Section 1, with transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

Solving $\pi = \pi P$ coordinate by coordinate:

$$\begin{aligned}\pi_1 &= \frac{1}{2}\pi_3, \\ \pi_2 &= \pi_1 + \frac{1}{2}\pi_2 \implies \pi_2 = 2\pi_1, \\ \pi_3 &= \frac{1}{2}\pi_2 + \frac{1}{2}\pi_3 \implies \pi_3 = \pi_2.\end{aligned}$$

So $\pi \propto (1, 2, 2)$, and normalising gives $\pi = (1/5, 2/5, 2/5)$.

Back in Section 1 we computed, with some effort involving complex eigenvalues,

$$p_{1,1}(n) = \frac{1}{5} + \left(\frac{1}{2}\right)^n \left(\frac{4}{5} \cos\left(\frac{n\pi}{2}\right) - \frac{2}{5} \sin\left(\frac{n\pi}{2}\right) \right) \longrightarrow \frac{1}{5} = \pi_1,$$

so the mysterious constant $1/5$ appearing there was the invariant probability of state 1 all along. The oscillating terms, damped by the factor $(1/2)^n = |\mu|^n$, describe *how* the chain spirals in towards equilibrium.

The relationship between limits and invariant distributions goes both ways, at least when the state space is finite: if the n -step probabilities converge, then whatever they converge to must be invariant.

Theorem 17.5.5 (Limits are Invariant)

Let S be finite, and suppose that for some $i \in S$ we have $p_{i,j}(n) \rightarrow \pi_j$ as $n \rightarrow \infty$, for all $j \in S$. Then $\pi = (\pi_j \mid j \in S)$ is an invariant distribution.

Proof. We first check that π is a distribution. Certainly $\pi_j \geq 0$ for all j , and since S is finite we may interchange the (finite) sum and the limit to get

$$\sum_{j \in S} \pi_j = \sum_{j \in S} \lim_{n \rightarrow \infty} p_{i,j}(n) = \lim_{n \rightarrow \infty} \sum_{j \in S} p_{i,j}(n) = 1.$$

Next we check invariance. Using the Chapman-Kolmogorov equations, and again exchanging the limit with the finite sum,

$$\pi_j = \lim_{n \rightarrow \infty} p_{i,j}(n) = \lim_{n \rightarrow \infty} \sum_{k \in S} p_{i,k}(n-1)P_{k,j} = \sum_{k \in S} \pi_k P_{k,j},$$

so $\pi = \pi P$ as required. $\square$

Remark. Finiteness really is needed here, because exchanging limits with infinite sums is a dangerous business. For example, for the simple random walk on $\mathbb{Z}$ our estimates showed $p_{0,j}(n) \rightarrow 0$ for every j , and the zero vector is certainly not a probability distribution – the mass of the walk ‘escapes to infinity’ in the limit.

This remark shows that invariant distributions need not exist in general. The next portion of this section is devoted to pinning down exactly when they do exist, and when they are unique. The answer will turn out to be a lovely story involving recurrence.

§17.5.2 Existence of Invariant Measures

It will be convenient to temporarily drop the requirement that our distributions sum to 1. We say $\lambda = (\lambda_i \mid i \in S)$ is a **measure** on S if $\lambda_i \geq 0$ for all i , and an **invariant measure** if additionally $\lambda = \lambda P$. If an invariant measure has finite total mass $\sum_i \lambda_i < \infty$, we can normalise it into an invariant distribution.

So where might an invariant measure come from? Here is a probabilistic way to think about it: fix a state k , and consider an ‘excursion’ of the chain from k back to k . If the chain is in equilibrium, then the amount of time it spends in each state i per excursion ought to describe the equilibrium proportions. This motivates the following definition.

Definition 17.5.6 (Expected Time Spent Per Excursion)

Let $k \in S$. For each $i \in S$, define

$$\nu_k(i) = \mathbb{E} \left[\sum_{n=0}^{T_k-1} \mathbb{1}[X_n = i] \mid X_0 = k \right],$$

the expected number of visits to i during an excursion from k , before the chain first returns to k .

Note that $\nu_k(k) = 1$, since the chain starts the excursion at k and doesn’t count its return. The following theorem says that this excursion recipe really does produce an invariant measure, at least when the chain is recurrent (so that excursions actually finish).

Theorem 17.5.7 (Existence of Invariant Measures)

Let P be irreducible and recurrent. Then ν_k is an invariant measure, and moreover

$0 < \nu_k(i) < \infty$ for all $i \in S$.

Proof. Since P is irreducible and recurrent, we know $T_k < \infty$ almost surely, and $X_{T_k} = k$. Because the excursion begins and ends at k , we can shift our time window by one step: counting visits to i at times $0, 1, \dots, T_k - 1$ is the same as counting visits at times $1, 2, \dots, T_k$, since the only disagreement could be at the endpoints, where $X_0 = X_{T_k} = k$ contributes to both counts exactly when $i = k$. Hence, throughout taking probabilities conditional on $X_0 = k$,

$$\begin{aligned} \nu_k(i) &= \mathbb{E} \left[\sum_{n=1}^{T_k} \mathbb{1}[X_n = i] \right] = \mathbb{E} \left[\sum_{n=1}^{\infty} \mathbb{1}[X_n = i, T_k \geq n] \right] \\ &= \sum_{n=1}^{\infty} \mathbb{P}(X_n = i, T_k \geq n) \\ &= \sum_{n=1}^{\infty} \sum_{j \in S} \mathbb{P}(X_n = i \mid X_{n-1} = j, T_k \geq n) \mathbb{P}(X_{n-1} = j, T_k \geq n). \end{aligned}$$

Now here is the key point: the event $\{T_k \geq n\} = \{T_k \leq n-1\}^c$ is determined by $X_0, \dots, X_{n-1}$ only, so by the simple Markov property $\mathbb{P}(X_n = i \mid X_{n-1} = j, T_k \geq n) = P_{j,i}$. Continuing,

$$\begin{aligned} \nu_k(i) &= \sum_{j \in S} P_{j,i} \sum_{n=1}^{\infty} \mathbb{P}(X_{n-1} = j, T_k \geq n) \\ &= \sum_{j \in S} P_{j,i} \mathbb{E} \left[\sum_{n=0}^{T_k-1} \mathbb{1}[X_n = j] \right] = \sum_{j \in S} \nu_k(j) P_{j,i}, \end{aligned}$$

which is exactly the statement $\nu_k = \nu_k P$.

It remains to show that ν_k is finite and positive. Note that invariance gives $\nu_k = \nu_k P^n$ for all n . Given $i \in S$, by irreducibility we can pick n, m with $p_{k,i}(n) > 0$ and $p_{i,k}(m) > 0$. Then

$$\nu_k(i) = \sum_{j \in S} \nu_k(j) p_{j,i}(n) \geq \nu_k(k) p_{k,i}(n) = p_{k,i}(n) > 0,$$

and in the other direction

$$1 = \nu_k(k) = \sum_{j \in S} \nu_k(j) p_{j,k}(m) \geq \nu_k(i) p_{i,k}(m),$$

so $\nu_k(i) \leq 1/p_{i,k}(m) < \infty$. □

§17.5.3 Uniqueness of Invariant Measures

We now know that irreducible recurrent chains have at least one invariant measure. Conveniently, it turns out they have essentially *only* one – every invariant measure is a scalar multiple of ν_k .

Theorem 17.5.8 (Uniqueness of Invariant Measures)

Let P be irreducible, and let λ be an invariant measure with $\lambda_k = 1$. Then $\lambda \geq \nu_k$. If moreover P is recurrent, then $\lambda = \nu_k$.

Proof. For the first part, the idea is to repeatedly expand the invariance equation, each time splitting off the terms that pass through k . Indeed, for any i ,

$$\lambda_i = \sum_{j_1 \in S} \lambda_{j_1} P_{j_1, i} = P_{k, i} + \sum_{j_1 \neq k} \lambda_{j_1} P_{j_1, i},$$

using $\lambda_k = 1$. Substituting the invariance equation into λ_{j_1} again,

$$\lambda_i = P_{k, i} + \sum_{j_1 \neq k} P_{k, j_1} P_{j_1, i} + \sum_{j_1, j_2 \neq k} \lambda_{j_2} P_{j_2, j_1} P_{j_1, i},$$

and continuing in this way, after n expansions we obtain

$$\begin{aligned} \lambda_i &= P_{k, i} + \sum_{j_1 \neq k} P_{k, j_1} P_{j_1, i} + \cdots + \sum_{j_1, \dots, j_{n-1} \neq k} P_{k, j_{n-1}} \cdots P_{j_2, j_1} P_{j_1, i} \\ &\quad + \sum_{j_1, \dots, j_n \neq k} \lambda_{j_n} P_{j_n, j_{n-1}} \cdots P_{j_1, i}. \end{aligned}$$

The final term is non-negative, and the earlier terms have a probabilistic meaning: the m -th term is exactly the probability that the chain started at k is at i at time m without having returned to k in between. That is,

$$\lambda_i \geq \sum_{m=1}^n \mathbb{P}(X_m = i, T_k \geq m \mid X_0 = k),$$

and letting $n \rightarrow \infty$, the right hand side converges^a to $\nu_k(i)$. So $\lambda \geq \nu_k$.

Now suppose P is recurrent, so that ν_k is invariant by the previous theorem. Then $\mu = \lambda - \nu_k$ is also an invariant measure (it's non-negative by the first part), and $\mu_k = 1 - 1 = 0$. Given any $i \in S$, choose n with $p_{i, k}(n) > 0$ by irreducibility. Then

$$0 = \mu_k = \sum_{j \in S} \mu_j p_{j, k}(n) \geq \mu_i p_{i, k}(n),$$

forcing $\mu_i = 0$. Hence $\lambda = \nu_k$. □

^aHere we are using the shifted-window description of $\nu_k(i)$ from the proof of the existence theorem: $\nu_k(i) = \sum_{m \geq 1} \mathbb{P}(X_m = i, T_k \geq m \mid X_0 = k)$.

§17.5.4 Positive Recurrence and Existence of Invariant Distributions

Let's take stock. For an irreducible recurrent chain, we have a family of invariant measures ν_k (one for each $k \in S$), all proportional to one another. To produce an invariant *distribution* we need to normalise, which is possible exactly when ν_k has finite total

mass. And the total mass has a beautiful interpretation:

$$\sum_{i \in S} \nu_k(i) = \mathbb{E} \left[\sum_{n=0}^{T_k-1} \sum_{i \in S} \mathbb{1}[X_n = i] \mid X_0 = k \right] = \mathbb{E} \left[\sum_{n=0}^{T_k-1} 1 \mid X_0 = k \right] = \mathbb{E}[T_k \mid X_0 = k],$$

the expected length of an excursion from k . So everything comes down to whether the *mean return time* is finite. This suggests refining our notion of recurrence.

Definition 17.5.9 (Positive and Null Recurrence)

Let $i \in S$ be a recurrent state, and let $m_i = \mathbb{E}[T_i \mid X_0 = i]$ denote the **mean return time** of i . We say i is **positive recurrent** if $m_i < \infty$, and **null recurrent** if $m_i = \infty$.

Note that a recurrent state has $T_i < \infty$ almost surely, but this does *not* imply that $\mathbb{E}[T_i] < \infty$ – we saw exactly this phenomenon with the gambler’s ruin. The following theorem ties everything in this section together.

Theorem 17.5.10 (Existence and Uniqueness of Invariant Distributions)

Let P be irreducible. Then the following are equivalent:

- (i) every state is positive recurrent;
- (ii) some state is positive recurrent;
- (iii) P has an invariant distribution π .

Moreover, when these hold, the invariant distribution is unique and is given by

$$\pi_i = \frac{1}{m_i} \quad \text{for all } i \in S.$$

Proof. Clearly (i) implies (ii).

(ii) *implies* (iii). Let k be a positive recurrent state. Then k is in particular recurrent, and since recurrence is a class property and P is irreducible, P is recurrent. So by our existence theorem, ν_k is an invariant measure, with total mass $\sum_i \nu_k(i) = m_k < \infty$. Hence

$$\pi_i = \frac{\nu_k(i)}{m_k}$$

defines an invariant distribution.

(iii) *implies* (i). Suppose π is an invariant distribution, and let $k \in S$ be any state; we will show k is positive recurrent. First we claim $\pi_k > 0$: there is certainly some i with $\pi_i > 0$, and choosing n with $p_{i,k}(n) > 0$ by irreducibility, invariance gives

$$\pi_k = \sum_{j \in S} \pi_j p_{j,k}(n) \geq \pi_i p_{i,k}(n) > 0.$$

Before we can apply our uniqueness theorem, we need to know that P is recurrent. Suppose instead that P was transient. Then for all states i, j , we have $\sum_n p_{j,i}(n) < \infty$ and so certainly $p_{j,i}(n) \rightarrow 0$ as $n \rightarrow \infty$. But then, since $\pi_i = \sum_j \pi_j p_{j,i}(n)$ for

every n , and the summands are dominated^a by the summable sequence (π_j) , we may pass to the limit and conclude $\pi_i = 0$ for all i – contradicting $\sum_i \pi_i = 1$. So P is recurrent.

Now define $\lambda_i = \pi_i/\pi_k$. This is an invariant measure with $\lambda_k = 1$, so by the uniqueness theorem, $\lambda = \nu_k$. Hence

$$m_k = \sum_{i \in S} \nu_k(i) = \sum_{i \in S} \frac{\pi_i}{\pi_k} = \frac{1}{\pi_k} < \infty,$$

so k is positive recurrent, establishing (i). This computation also gives the final claim: $\pi_k = 1/m_k$ for every state k , and in particular the invariant distribution is unique, since its values are determined by the mean return times. $\square$

^aWe are using here the discrete version of the dominated convergence theorem: if $|f_n(j)| \leq g(j)$ with $\sum_j g(j) < \infty$ and $f_n(j) \rightarrow f(j)$ pointwise, then $\sum_j f_n(j) \rightarrow \sum_j f(j)$. In our case $\pi_j p_{j,i}(n) \leq \pi_j$, which is summable.

This is a genuinely satisfying result: for irreducible chains, the existence of an invariant distribution is a sharp dividing line, and when it exists, the equilibrium probability of a state is exactly the reciprocal of its mean return time. States that the chain returns to quickly get lots of mass; states it takes ages to revisit get little.

Example 17.5.11 (Mean Return Times of the Two-State Chain)

For the two-state chain, the theorem predicts the mean return time

$$m_1 = \frac{1}{\pi_1} = \frac{\alpha + \beta}{\beta},$$

and this is easy enough to verify by hand – always a healthy exercise. Starting from state 1, either the chain stays put (probability $1 - \alpha$), in which case $T_1 = 1$; or it jumps to state 2 (probability α), where it must wait a $\text{Geometric}(\beta)$ number of steps to jump back, giving $T_1 = 1 + \text{Geometric}(\beta)$ in distribution. Hence

$$m_1 = (1 - \alpha) \cdot 1 + \alpha \left(1 + \frac{1}{\beta}\right) = 1 + \frac{\alpha}{\beta} = \frac{\alpha + \beta}{\beta},$$

exactly as predicted.

Before moving on, let's record one more consequence of the uniqueness theory, which gives the excursion counts ν_k a pleasingly simple form in the positive recurrent case.

Corollary 17.5.12 (Visits Per Excursion)

Let P be irreducible with invariant distribution π . Then for any states $x, y \in S$, the expected number of visits to y during an excursion from x is

$$\nu_x(y) = \frac{\pi_y}{\pi_x}.$$

Proof. By the theorem above, P is positive recurrent, and in particular recurrent, with $\pi_x > 0$ for all x . Now $\lambda_y = \pi_y/\pi_x$ defines an invariant measure with $\lambda_x = 1$, so by the uniqueness of invariant measures, $\lambda = \nu_x$. $\square$

So, for example, between successive rainy days, the expected number of sunny days is simply the ratio of the equilibrium probabilities of sun and rain. Equilibrium ratios really do describe the fine structure of the chain's excursions, not just long-run averages.

Let's see the trichotomy of positive recurrence, null recurrence and transience play out on our favourite examples.

Example 17.5.13 (The Simple Random Walk on $\mathbb{Z}$ is Null Recurrent)

We proved via Pólya's theorem that the simple random walk on $\mathbb{Z}$ is recurrent. Is it positive recurrent? If it were, it would have an invariant distribution π . Solving $\pi = \pi P$ means

$$\pi_i = \frac{1}{2}\pi_{i-1} + \frac{1}{2}\pi_{i+1} \quad \text{for all } i \in \mathbb{Z},$$

and we can spot the invariant measure $\lambda_i = 1$ for all i . By uniqueness of invariant measures (the walk is irreducible and recurrent), *every* invariant measure is a constant multiple of this one – and none of these have finite total mass, so no invariant distribution exists. Hence the walk is null recurrent: it always returns home, but the expected time to do so is infinite.

Remark. One might hope for a converse: does having a 'nice' invariant measure imply recurrence? No – the simple random walk on $\mathbb{Z}^3$ is transient, and yet $\lambda_i = 1$ is a perfectly good invariant measure for it (the transition matrix is doubly stochastic). Existence of invariant *measures* is cheap; it is invariant *distributions* that carry real information.

Example 17.5.14 (A Transient Chain with Many Invariant Measures)

Consider the asymmetric random walk on $\mathbb{Z}$ with $P_{i,i+1} = p$, $P_{i,i-1} = q$, where $p + q = 1$ and $p > q$. This chain is transient. Solving $\lambda = \lambda P$ gives the recurrence $\lambda_i = p\lambda_{i-1} + q\lambda_{i+1}$, with general solution

$$\lambda_i = a + b \left(\frac{p}{q}\right)^i, \quad a, b \geq 0.$$

These are genuinely different invariant measures (not multiples of each other), so the uniqueness theorem really does need recurrence. And again, none of them can be normalised, so there is no invariant distribution – as must be the case, since the chain is not positive recurrent.

Example 17.5.15 (A Positive Recurrent Walk)

Now consider a random walk on $S = \{0, 1, 2, \dots\}$ with a drift towards the origin: let $0 < p < 1/2$, and set

$$P_{i,i+1} = p, \quad P_{i,i-1} = q = 1 - p \quad \text{for } i \geq 1, \quad P_{0,1} = p, \quad P_{0,0} = q.$$

Rather than trying to compute mean return times directly, we look for an invariant distribution. The equations $\pi = \pi P$ read

$$\pi_0 = q\pi_0 + q\pi_1, \quad \pi_i = p\pi_{i-1} + q\pi_{i+1} \quad \text{for } i \geq 1.$$

The first equation gives $\pi_1 = (p/q)\pi_0$, and then induction on the second^a gives

$$\pi_i = \left(\frac{p}{q}\right)^i \pi_0.$$

Since $p < q$, this has finite total mass, and normalising gives the geometric distribution

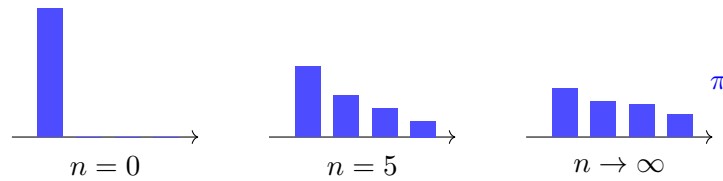
$$\pi_i = \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^i.$$

So by our theorem the chain is positive recurrent, with mean return times $m_i = 1/\pi_i$ – and we found this out without ever computing an expectation directly. This is the invariant distribution machinery at its best.

^aSubstituting $\pi_i = (p/q)^i \pi_0$ into the second equation: $p(p/q)^{i-1} + q(p/q)^{i+1} = (p/q)^i(q+p) = (p/q)^i$, as needed.

§17.6 Convergence to Equilibrium

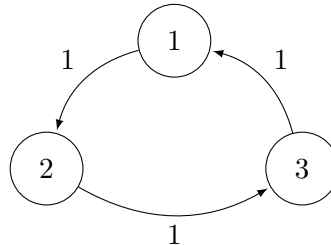
We now come to the central limit-theorem-flavoured result of this course: for a nice enough Markov chain, the distribution of X_n converges to the invariant distribution π , *regardless of the initial distribution*. Run the chain for long enough and it forgets where it started.



Of course, some hypotheses will be needed. We have already seen that we should assume irreducibility and positive recurrence, so that an invariant distribution exists and is unique. But there is one more, sneakier obstruction, which we address first.

§17.6.1 Aperiodicity

Consider the Markov chain that cycles deterministically around three states, as below.



This chain is irreducible and positive recurrent, and its invariant distribution is the uniform one, $\pi = (1/3, 1/3, 1/3)$. But started from state 1, we have $\mathbb{P}(X_n = 1) = 1$ when $3 \mid n$ and $\mathbb{P}(X_n = 1) = 0$ otherwise – the distribution of X_n just cycles, and certainly doesn't converge to π . The chain never forgets its starting point, because its position modulo 3 remembers it forever. To rule out this kind of parity problem, we introduce the notion of periodicity.

Definition 17.6.1 (Period)

For a state $i \in S$, the **period** of i is

$$d_i = \gcd\{n \geq 1 \mid p_{i,i}(n) > 0\}.$$

We say the state i is **aperiodic** if $d_i = 1$.

In the cycle above, each state has period 3. In the simple random walk on $\mathbb{Z}$, each state has period 2 – and indeed we took care to work with $p_{0,0}(2n)$ throughout our analysis of it. More generally, the simple random walk on any *bipartite* graph (one whose vertices can be 2-coloured so that every edge joins different colours) has period 2, since every closed walk has even length: the chain forever alternates between the two colour classes, and its colour at time n remembers the parity of n . On the other hand, note that a single self-loop anywhere in an irreducible chain is enough to give that state period 1 (as $p_{i,i}(1) > 0$), and we shall see shortly that aperiodicity then spreads to every state.

For our purposes, the importance of aperiodicity is the following characterisation.

Lemma 17.6.2 (Aperiodicity Criterion)

A state i is aperiodic if and only if $p_{i,i}(n) > 0$ for all sufficiently large n .

Proof. Let $D = \{n \geq 1 \mid p_{i,i}(n) > 0\}$. One direction is easy: if D contains all sufficiently large integers, then it certainly has greatest common divisor 1 (say, it contains two consecutive integers).

For the other direction, first note that D is closed under addition: if $a, b \in D$ then $p_{i,i}(a+b) \geq p_{i,i}(a)p_{i,i}(b) > 0$ by Chapman-Kolmogorov. Now suppose $\gcd D = 1$. Then there is some finite collection $n_1, \dots, n_r \in D$ with $\gcd\{n_1, \dots, n_r\} = 1$, and so by Bézout's lemma^a there are integers $a_1, \dots, a_r$ with $a_1 n_1 + \dots + a_r n_r = 1$. Collecting the positive and negative terms, we can write $M - N = 1$, where M and N are each non-negative integer combinations of the n_s , and hence (assuming $N \neq 0$; if $N = 0$ then $1 \in D$ and we're immediately done) both M and N lie in D by closure under addition.

So D contains the consecutive integers N and $N+1$. Now take any $n \geq N^2$, and write $n = qN + s$ with $0 \leq s < N$. Then $q \geq N > s$, so

$$n = (q-s)N + s(N+1)$$

expresses n as a non-negative integer combination of N and $N+1$. Hence $n \in D$, and D contains every integer at least N^2 . $\square$

^aStrictly, Bézout's lemma as usually stated handles two integers; the general case follows by induction on r .

Like recurrence and transience, aperiodicity turns out to be a class property, so for irreducible chains we can simply speak of *the chain* being aperiodic.

Lemma 17.6.3 (Aperiodicity is a Class Property)

Suppose P is irreducible and some state i is aperiodic. Then every state is aperiodic.

Proof. Let $j \in S$. By irreducibility there are n, m with $p_{j,i}(n) > 0$ and $p_{i,j}(m) > 0$. Then for any r ,

$$p_{j,j}(n+r+m) \geq p_{j,i}(n) p_{i,i}(r) p_{i,j}(m),$$

and by the previous lemma the middle factor is positive for all sufficiently large r . Hence $p_{j,j}(s) > 0$ for all sufficiently large s , and j is aperiodic. $\square$

§17.6.2 The Convergence Theorem

We can now state and prove the main theorem of this section.

Theorem 17.6.4 (Convergence to Equilibrium)

Let P be irreducible and aperiodic with invariant distribution π , and let X_n be Markov(λ, P) for any initial distribution λ . Then for all $j \in S$,

$$\mathbb{P}(X_n = j) \rightarrow \pi_j \quad \text{as } n \rightarrow \infty.$$

In particular, taking $\lambda = \delta_i$, we have $p_{i,j}(n) \rightarrow \pi_j$ for all $i, j \in S$.

The proof uses a technique known as *coupling*, which is worth dwelling on for a moment, as it's one of the loveliest ideas in probability. We want to compare the chain X_n with a chain in equilibrium. So we simply run a second, independent copy Y_n of the chain, started in the invariant distribution. The trick is then this: we watch the two chains until the (random) moment they happen to collide, and after that moment, we make X ‘copy’ Y . Copying Y doesn't change the distribution of X (by the strong Markov property, both chains restart identically from the collision point), but once they've collided, our chain literally *is* in equilibrium. So all we need is that the collision happens eventually, and this is a recurrence statement, which we have the tools to prove.

Proof. Let Y_n be Markov(π, P), independent of X_n .

Step 1: the product chain. Consider the process $W_n = (X_n, Y_n)$ on the state space $S \times S$. By independence, W_n is a Markov chain with initial distribution $\mu_{(x,y)} = \lambda_x \pi_y$ and transition matrix

$$\tilde{P}_{(x,y),(x',y')} = P_{x,x'} P_{y,y'}.$$

We claim $\tilde{P}$ is irreducible. Given states (x, y) and (x', y') , choose ℓ and m with $p_{x,x'}(\ell) > 0$ and $p_{y,y'}(m) > 0$. For any $r \geq 0$, we have $p_{x,x'}(\ell+r) \geq p_{x,x'}(\ell) p_{x',x'}(r)$, which by aperiodicity is positive for all sufficiently large r ; similarly $p_{y,y'}(m+r) > 0$ for all large r . So for sufficiently large n , both $p_{x,x'}(n)$ and $p_{y,y'}(n)$ are positive^a, and hence $\tilde{P}_{(x,y),(x',y')}(n) = p_{x,x'}(n) p_{y,y'}(n) > 0$.

Moreover, $\tilde{P}$ has an invariant distribution $\tilde{\pi}_{(x,y)} = \pi_x \pi_y$, as is easily checked using invariance of π in each coordinate. So by our existence and uniqueness theorem, the product chain is positive recurrent, and in particular recurrent.

Step 2: the coupling time. Fix any state $a \in S$, and let

$$T = \inf\{n \geq 1 \mid W_n = (a, a)\}$$

be the first time the two chains simultaneously sit at a . Since the product chain

is irreducible and recurrent, our ‘recurrent chains visit everything’ theorem gives $\mathbb{P}(T < \infty) = 1$.

Step 3: the swap. Now define

$$Z_n = \begin{cases} X_n & \text{if } n < T, \\ Y_n & \text{if } n \geq T. \end{cases}$$

We claim that Z_n has the same distribution as X_n , i.e. that Z is $\text{Markov}(\lambda, P)$. To see this, consider the swapped product process

$$W'_n = \begin{cases} (X_n, Y_n) & \text{if } n < T, \\ (Y_n, X_n) & \text{if } n \geq T. \end{cases}$$

The time T is a stopping time for W , and $W_T = (a, a)$. By the strong Markov property, conditional on $T < \infty$, the process $(W_{T+n})_{n \geq 0}$ is $\text{Markov}(\delta_{(a,a)}, \tilde{P})$, independent of $W_0, \dots, W_T$. But both the transition matrix $\tilde{P}$ and the state (a, a) are symmetric under swapping the two coordinates, so the coordinate-swapped process $((Y_{T+n}, X_{T+n}))_{n \geq 0}$ is *also* $\text{Markov}(\delta_{(a,a)}, \tilde{P})$, independent of the same past. Gluing this back onto the pre- T process, we conclude that W' is a Markov chain with the same initial distribution and transition matrix as W . In particular its first coordinate, which is exactly Z , is $\text{Markov}(\lambda, P)$ as claimed.

Step 4: the conclusion. Since $Y_0 \sim \pi$ and π is invariant, $\mathbb{P}(Y_n = j) = \pi_j$ for all n . Now for any $j \in S$,

$$\begin{aligned} |\mathbb{P}(X_n = j) - \pi_j| &= |\mathbb{P}(Z_n = j) - \mathbb{P}(Y_n = j)| \\ &= |\mathbb{P}(X_n = j, n < T) + \mathbb{P}(Y_n = j, n \geq T) \\ &\quad - \mathbb{P}(Y_n = j, n < T) - \mathbb{P}(Y_n = j, n \geq T)| \\ &= |\mathbb{P}(X_n = j, n < T) - \mathbb{P}(Y_n = j, n < T)| \\ &\leq \mathbb{P}(T > n). \end{aligned}$$

Since T is almost surely finite, $\mathbb{P}(T > n) \rightarrow 0$ as $n \rightarrow \infty$, and we are done. $\square$

^aThis is where aperiodicity earns its keep: without it, $p_{x,x'}(n)$ and $p_{y,y'}(n)$ might only be positive on disjoint sets of times (consider the 3-cycle), and the product chain fails to be irreducible.

It is worth appreciating how much of the theory we developed went into this proof: invariant distributions, positive recurrence, the classification of when invariant distributions exist, the strong Markov property, and aperiodicity all played essential roles.

Example 17.6.5 (The Two-State Chain, One Last Time)

The two-state chain with $\alpha, \beta \in (0, 1)$ is irreducible, and aperiodic since $p_{1,1}(1) = 1 - \alpha > 0$. So the convergence theorem applies, and predicts

$$p_{1,1}(n) \rightarrow \pi_1 = \frac{\beta}{\alpha + \beta},$$

which is precisely what our explicit formula $p_{1,1}(n) = \frac{\beta}{\alpha + \beta} + \frac{\alpha}{\alpha + \beta}(1 - \alpha - \beta)^n$ shows, since $|1 - \alpha - \beta| < 1$. It's pleasing to see the general theorem and the bare-hands

computation agree – and of course, the theorem’s real value is for chains where no explicit formula is available.

Example 17.6.6 (Convergence of the Reflected Biased Walk)

Recall the walk on $\{0, 1, 2, \dots\}$ with $P_{i,i+1} = p$ and $P_{i,i-1} = q$ for $i \geq 1$, and $P_{0,1} = p$, $P_{0,0} = q$, where $p < 1/2 < q$. We showed it is positive recurrent with geometric invariant distribution $\pi_i = (1 - p/q)(p/q)^i$. The chain is aperiodic thanks to the lazy loop at 0 ($P_{0,0} = q > 0$), so the convergence theorem applies:

$$\mathbb{P}(X_n = i) \rightarrow \left(1 - \frac{p}{q}\right) \left(\frac{p}{q}\right)^i \quad \text{for every } i,$$

no matter the initial distribution. Computing $p_{i,j}(n)$ explicitly for this chain would be thoroughly unpleasant – yet its limit comes almost for free. Note the contrast with the unreflected walk on $\mathbb{Z}$: one boundary and a drift towards it are enough to change the walk from null recurrent (or transient) to positive recurrent.

Remark (The Lazy Chain Trick). Periodicity, while a genuine obstruction to convergence, is a rather fragile one. Given any irreducible chain P , consider the *lazy chain* $P' = \frac{1}{2}(I + P)$, which at each step flips a fair coin and either stays put or takes a step of P . Then P' is aperiodic (each state has $P'_{i,i} \geq 1/2 > 0$), it is irreducible, and it has exactly the same invariant distributions as P , since $\pi P' = \frac{1}{2}\pi + \frac{1}{2}\pi P = \pi$ if and only if $\pi P = \pi$. So laziness destroys periodicity while preserving equilibrium – a trick worth remembering.

Aside: How Many Shuffles Suffice?

Convergence to equilibrium is not just an abstract limit theorem – it is the mathematics of *shuffling cards*. The arrangements of a deck of 52 cards form a set S of size $52!$, and a shuffling method (repeatedly cutting and riffing, say) defines a Markov chain on S : the next arrangement depends only on the current one and the randomness of the shuffle. Provided the shuffle is ‘random enough’ for the chain to be irreducible and aperiodic, our convergence theorem promises that the distribution of the deck’s arrangement converges to the unique invariant distribution – which, by the symmetry of the situation, is uniform on all $52!$ arrangements⁶. In other words, shuffle for long enough and the deck genuinely becomes perfectly mixed, regardless of how it started.

The really interesting question – *how fast?* – is beyond this course, but has a famously crisp answer: Bayer and Diaconis showed that for riffle shuffles of a 52-card deck, about *seven* shuffles are needed before the distribution is close to uniform, with a sharp ‘cut-off’ from poorly-mixed to well-mixed occurring around that point. The study of such mixing times is a thriving area of modern probability, and it all rests on the framework we have just built.

⁶A little more precisely: each shuffle composes the current arrangement with a random permutation, so the transition matrix satisfies $P_{\sigma,\tau} = P_{\rho\sigma,\rho\tau}$ for any permutation ρ , which forces each column of P to sum to 1 as well as each row. For such *doubly stochastic* matrices, the uniform distribution is invariant, since $\sum_{\sigma} \frac{1}{52!} P_{\sigma,\tau} = \frac{1}{52!}$.

§17.6.3 What Happens Without Positive Recurrence?

If the chain is irreducible but *not* positive recurrent, then there is no invariant distribution to converge to, and indeed the distribution of the chain flattens out entirely.

Theorem 17.6.7 (Null Recurrent and Transient Chains Forget Everything)

Let P be irreducible and aperiodic, and suppose P is not positive recurrent (that is, it is null recurrent or transient). Then for all $x, y \in S$,

$$p_{x,y}(n) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Proof. As in the convergence theorem, form the product chain on $S \times S$ with transition matrix $\tilde{P}_{(x,y),(x',y')} = P_{x,x'}P_{y,y'}$, which is irreducible by aperiodicity (this part of the earlier proof did not use positive recurrence). We consider two cases.

Case 1: $\tilde{P}$ is transient. For a transient chain, the expected total number of visits to any fixed state is finite from any starting point^a. Applying this to the product chain started at (x, x) , looking at visits to (y, y) ,

$$\sum_{n=0}^{\infty} \tilde{p}_{(x,x),(y,y)}(n) = \sum_{n=0}^{\infty} p_{x,y}(n)^2 < \infty,$$

so $p_{x,y}(n)^2 \rightarrow 0$, giving the result.

Case 2: $\tilde{P}$ is recurrent. Note that in this case P itself must be recurrent (if the product chain returns to (y, y) infinitely often, its first coordinate returns to y infinitely often), and hence P is null recurrent, and the invariant measure ν_y (for any fixed $y \in S$) has infinite total mass $\sum_z \nu_y(z) = m_y = \infty$.

Let $M > 0$. Since ν_y has infinite mass but is finite on each state, we can choose a *finite* set $A \subseteq S$ with $\nu_y(A) = \sum_{z \in A} \nu_y(z) > M$. Define a distribution μ by

$$\mu_z = \frac{\nu_y(z)}{\nu_y(A)} \mathbb{1}[z \in A].$$

The point of this construction is the following bound: for any $z \in S$ and $n \geq 0$, since $\mu \leq \nu_y/\nu_y(A)$ entrywise and ν_y is invariant,

$$(\mu P^n)_z \leq \frac{(\nu_y P^n)_z}{\nu_y(A)} = \frac{\nu_y(z)}{\nu_y(A)}.$$

In particular $(\mu P^n)_y \leq \nu_y(y)/\nu_y(A) = 1/\nu_y(A) < 1/M$.

Now we couple. Let (X_n, Y_n) be the product chain with initial distribution $\mu \times \delta_x$, that is, $X_0 \sim \mu$ and $Y_0 = x$ independently. Fix $s \in S$ and let $T = \inf\{n \geq 1 \mid (X_n, Y_n) = (s, s)\}$; since the product chain is irreducible and recurrent, $T < \infty$ almost surely. Defining Z_n to follow X before T and Y afterwards, exactly as in the convergence theorem, Z is Markov(μ, P), so $\mathbb{P}(Z_n = y) = (\mu P^n)_y < 1/M$. Then

$$\begin{aligned} p_{x,y}(n) &= \mathbb{P}(Y_n = y) \leq \mathbb{P}(Y_n = y, n \geq T) + \mathbb{P}(T > n) \\ &= \mathbb{P}(Z_n = y, n \geq T) + \mathbb{P}(T > n) \leq \frac{1}{M} + \mathbb{P}(T > n). \end{aligned}$$

Taking $n \rightarrow \infty$ gives $\limsup_n p_{x,y}(n) \leq 1/M$, and since M was arbitrary, $p_{x,y}(n) \rightarrow 0$. $\square$

^aIndeed, by the strong Markov property, the number of visits to state w' starting from w is bounded by a geometric random variable: $\mathbb{E}[V_{w'} \mid W_0 = w] = \mathbb{P}(T_{w'} < \infty \mid W_0 = w)/(1 - f_{w'}) < \infty$.

Example 17.6.8

For the simple random walk on $\mathbb{Z}$ (null recurrent, though periodic – consider instead its lazy version to meet the aperiodicity hypothesis) we can see this flattening concretely: our Stirling estimate gave

$$p_{0,0}(2n) \sim \frac{1}{\sqrt{\pi n}} \rightarrow 0.$$

The walk keeps coming back to 0 forever, but the *probability* of finding it at 0 (or any fixed state) at a given large time dwindles to nothing, as the walk spreads out over an ever-larger range of $\mathbb{Z}$.

§17.7 The Ergodic Theorem

The convergence theorem tells us about the distribution of the chain at a single large time n . In this section we will prove a different kind of limit theorem, about the behaviour of a single realisation of the chain over time: the *long-run proportion of time* spent in each state. This is the Markov chain analogue of the strong law of large numbers – and indeed the strong law is exactly what we will use to prove it. Let us recall the statement (it is proved in a course on probability theory, and we will treat it as a black box).

Fact 17.7.1 (Strong Law of Large Numbers)

Let $U_1, U_2, \dots$ be independent, identically distributed, non-negative random variables with $\mathbb{E}[U_1] = \mu \leq \infty$. Then

$$\mathbb{P}\left(\frac{U_1 + \dots + U_r}{r} \rightarrow \mu \text{ as } r \rightarrow \infty\right) = 1.$$

This holds also when $\mu = \infty$, in the sense that the averages tend to infinity almost surely.

We will apply this to the durations of the successive excursions of a Markov chain from a state. Since these excursions are independent (thanks to the strong Markov property), the total time taken for r excursions grows like $r \cdot m_i$, and inverting this relationship will give us the proportion of time spent at i . Let's set up notation: define the **number of visits to i before time n** as

$$V_i(n) = \sum_{m=0}^{n-1} \mathbb{1}[X_m = i],$$

so that $V_i(n)/n$ is the proportion of the first n time steps spent at i .

Theorem 17.7.2 (Ergodic Theorem)

Let P be irreducible, and let X_n be $\text{Markov}(\lambda, P)$ for any initial distribution λ . Then

$$\mathbb{P}\left(\frac{V_i(n)}{n} \rightarrow \frac{1}{m_i} \text{ as } n \rightarrow \infty\right) = 1,$$

where $m_i = \mathbb{E}[T_i \mid X_0 = i]$ is the mean return time, and we interpret $1/\infty = 0$. In particular, if P is positive recurrent with invariant distribution π , then the long-run proportion of time spent in each state i is almost surely π_i .

Note the hypotheses: there is no aperiodicity assumption here, and the result holds even in the null recurrent and transient cases. Time averages are more forgiving than distributional limits. To see the contrast at its starkest, consider our deterministic 3-cycle: the distribution of X_n cycles forever and never converges, but the proportion of time spent in each state converges beautifully to $1/3 = 1/m_i$, as each state is visited exactly once every three steps.

Proof. First suppose P is transient. Then the total number of visits $V_i = \lim_n V_i(n)$ is almost surely finite, so

$$\frac{V_i(n)}{n} \leq \frac{V_i}{n} \rightarrow 0,$$

and since transient states have $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$, certainly $m_i = \infty$ and $1/m_i = 0$, in agreement.

So suppose P is recurrent, and fix the state i . Let $T = T_i$ be the first passage time to i . By our ‘recurrent chains visit everything’ theorem, $T < \infty$ almost surely, and by the strong Markov property the process $(X_{T+n})_{n \geq 0}$ is $\text{Markov}(\delta_i, P)$. Moreover, discarding the initial segment of the chain before time T changes neither the limit of $V_i(n)/n$ (we discard only finitely many steps) nor $1/m_i$. So it suffices to prove the result when $\lambda = \delta_i$, i.e. assuming $X_0 = i$.

Now let $U_r = T_i^{(r)} - T_i^{(r-1)}$ be the length of the r -th excursion from i . By the strong Markov property applied at the stopping times $T_i^{(r-1)}$ (which are finite almost surely, by recurrence), the excursion lengths $U_1, U_2, \dots$ are independent and identically distributed with mean $\mathbb{E}[U_1] = m_i$. Writing $T_i^{(r)} = U_1 + \dots + U_r$, the strong law of large numbers gives

$$\frac{T_i^{(r)}}{r} \rightarrow m_i \quad \text{almost surely}$$

(including in the null recurrent case $m_i = \infty$).

The final step is to relate the passage times back to $V_i(n)$, which we do by a sandwiching argument. Since $X_0 = i$, at time $n - 1$ the chain has made exactly $V_i(n)$ visits to i (counting the one at time 0), so the $(V_i(n) - 1)$ -th return has happened by time $n - 1$, but the $V_i(n)$ -th has not:

$$T_i^{(V_i(n)-1)} \leq n - 1 < T_i^{(V_i(n))}.$$

Dividing through by $V_i(n)$,

$$\frac{T_i^{(V_i(n)-1)}}{V_i(n)-1} \cdot \frac{V_i(n)-1}{V_i(n)} \leq \frac{n}{V_i(n)} \leq \frac{T_i^{(V_i(n))}}{V_i(n)}.$$

By recurrence $V_i(n) \rightarrow \infty$ almost surely, so both the left and right hand sides converge almost surely to m_i (using our strong law statement along the random subsequence $r = V_i(n)$, which is valid since $V_i(n) \rightarrow \infty$). Hence $n/V_i(n) \rightarrow m_i$ almost surely, which is exactly the claim.

Finally, if P is positive recurrent, then $\pi_i = 1/m_i$ by our existence and uniqueness theorem. $\square$

Remark. The ergodic theorem is the theoretical justification for estimating π by *simulation*: run the chain for a long time and record the empirical proportions of time spent in each state. This idea underlies Markov chain Monte Carlo, one of the most widely used computational techniques in modern statistics – there, one cleverly *constructs* a Markov chain whose invariant distribution is a distribution one wishes to sample from. (The construction is usually done via detailed balance, which we discuss next.)

Example 17.7.3 (An Unreliable Printer)

A printer is either working or broken, and its state is inspected each morning. If it is working today, it breaks down overnight with probability α ; if it is broken, an engineer manages to fix it overnight with probability β . This is exactly our two-state chain, with invariant distribution

$$\pi = \left(\frac{\beta}{\alpha + \beta}, \frac{\alpha}{\alpha + \beta} \right)$$

on the states {working, broken}. By the ergodic theorem, the long-run *proportion of days* on which the printer works is, with probability one,

$$\frac{V_{\text{working}}(n)}{n} \rightarrow \frac{\beta}{\alpha + \beta}.$$

Note the subtle upgrade from the convergence theorem: not only is the printer working with probability close to $\beta/(\alpha + \beta)$ on any particular distant day, but the actual observed fraction of good days converges to this value on almost every realisation of history. If, say, each working day generates income c , then the long-run income rate is $c\beta/(\alpha + \beta)$ per day, almost surely – statements of this form are exactly what the ergodic theorem is for.

§17.8 Time Reversal

§17.8.1 Reversing a Markov Chain

Here is a curious question to end the course with: if we film a Markov chain in equilibrium and play the tape backwards, what do we see? The Markov property is symmetric in time to a greater extent than one might expect – conditional on the present, the past and future are independent, and this description doesn't distinguish a direction of time. Indeed, it turns out the reversed film shows another Markov chain, and we can say

exactly which one.

Theorem 17.8.1 (Time Reversal)

Let P be irreducible with invariant distribution π , and let X_n be $\text{Markov}(\pi, P)$. Fix $N \geq 1$, and define $Y_n = X_{N-n}$ for $0 \leq n \leq N$. Then $(Y_n)_{0 \leq n \leq N}$ is $\text{Markov}(\pi, \widehat{P})$, where

$$\widehat{P}_{i,j} = \frac{\pi_j}{\pi_i} P_{j,i}.$$

Moreover $\widehat{P}$ is also irreducible, with invariant distribution π .

The chain $\widehat{P}$ is called the **time reversal** of P . (Note $\pi_i > 0$ for all i by irreducibility and positive recurrence, so there are no division-by-zero worries here.)

Proof. First, $\widehat{P}$ is a genuine (stochastic) transition matrix: its entries are non-negative, and using $\pi = \pi P$,

$$\sum_{j \in S} \widehat{P}_{i,j} = \frac{1}{\pi_i} \sum_{j \in S} \pi_j P_{j,i} = \frac{\pi_i}{\pi_i} = 1.$$

Next we check that Y has the right finite-dimensional distributions. For any states $i_0, \dots, i_N$,

$$\begin{aligned} \mathbb{P}(Y_0 = i_0, \dots, Y_N = i_N) &= \mathbb{P}(X_0 = i_N, X_1 = i_{N-1}, \dots, X_N = i_0) \\ &= \pi_{i_N} P_{i_N, i_{N-1}} \cdots P_{i_1, i_0}. \end{aligned}$$

Now we repeatedly substitute using $\pi_j P_{j,i} = \pi_i \widehat{P}_{i,j}$, moving from left to right: first $\pi_{i_N} P_{i_N, i_{N-1}} = \pi_{i_{N-1}} \widehat{P}_{i_{N-1}, i_N}$, then combining $\pi_{i_{N-1}}$ with the next factor, and so on. After N substitutions we obtain

$$\mathbb{P}(Y_0 = i_0, \dots, Y_N = i_N) = \pi_{i_0} \widehat{P}_{i_0, i_1} \cdots \widehat{P}_{i_{N-1}, i_N},$$

which by our very first theorem (the probability of a sequence of states) says exactly that Y is $\text{Markov}(\pi, \widehat{P})$.

For invariance of π under $\widehat{P}$,

$$\sum_{i \in S} \pi_i \widehat{P}_{i,j} = \sum_{i \in S} \pi_j P_{j,i} = \pi_j.$$

Finally, for irreducibility, let $i, j \in S$. By irreducibility of P there is a path $i = i_0, i_1, \dots, i_k = j$ with $P_{i_0, i_1} P_{i_1, i_2} \cdots P_{i_{k-1}, i_k} > 0$. Then, reversing the path,

$$\widehat{P}_{i_k, i_{k-1}} \cdots \widehat{P}_{i_1, i_0} = \frac{\pi_{i_0}}{\pi_{i_k}} P_{i_0, i_1} \cdots P_{i_{k-1}, i_k} > 0,$$

(the intermediate π factors telescope), so $j \rightarrow i$ in the reversed chain, as required. $\square$

Remark. The hypothesis that the chain starts *in equilibrium* is doing real work here. If X_0 had some other distribution, the reversed process would generally fail to be a homogeneous Markov chain at all – for instance, if $X_0 = i$ is deterministic, then watching

the film backwards, the process is mysteriously compelled to arrive at state i at the final frame, which no homogeneous transition mechanism can arrange. Equilibrium is exactly the condition that makes the chain look statistically the same at every time, so that reversing time doesn't distinguish any particular moment.

Example 17.8.2 (Reversing a Biased Cycle)

Let $n \geq 3$ and consider the walk on $\mathbb{Z}_n$ (the integers modulo n) that steps clockwise with probability $2/3$ and anticlockwise with probability $1/3$. By symmetry, the invariant distribution is uniform, $\pi_i = 1/n$. The time reversal is then

$$\widehat{P}_{i,j} = \frac{\pi_j}{\pi_i} P_{j,i} = P_{j,i},$$

so $\widehat{P}_{i,i-1} = P_{i-1,i} = 2/3$ and $\widehat{P}_{i,i+1} = 1/3$: the reversed chain is the walk biased *anticlockwise* instead. This is exactly what intuition demands – a film of a clockwise-drifting walker, played backwards, shows an anticlockwise-drifting walker. In particular $\widehat{P} \neq P$ here, and the direction of drift is precisely the feature that betrays the direction of time.

§17.8.2 Detailed Balance and Reversibility

The most interesting case of time reversal is when running the film backwards is *undetectable* – when the reversed chain is statistically identical to the original.

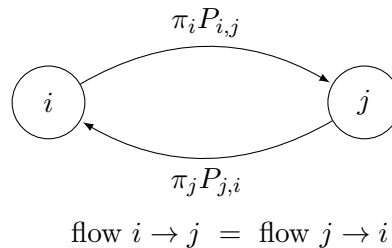
Definition 17.8.3 (Reversibility)

Let P be irreducible with invariant distribution π . The chain is **reversible** (in equilibrium) if $\widehat{P} = P$, that is, if

$$\pi_i P_{i,j} = \pi_j P_{j,i} \quad \text{for all } i, j \in S.$$

These equations are called the **detailed balance equations**.

The detailed balance equations have a very physical interpretation. In equilibrium, the quantity $\pi_i P_{i,j}$ is the probability of seeing the chain make the specific transition $i \rightarrow j$ at a given time step – a ‘flow of probability’ from i to j . Invariance ($\pi = \pi P$) says that the total flow into each state balances the total flow out. Detailed balance is a much stronger, local statement: the flow from i to j equals the flow from j to i , *along every edge individually*.



Why do we care? One excellent practical reason:

Lemma 17.8.4 (Detailed Balance Implies Invariance)

Let P be a transition matrix and μ a distribution satisfying the detailed balance equations $\mu_i P_{i,j} = \mu_j P_{j,i}$ for all i, j . Then μ is an invariant distribution for P .

Proof. We simply sum:

$$(\mu P)_j = \sum_{i \in S} \mu_i P_{i,j} = \sum_{i \in S} \mu_j P_{j,i} = \mu_j. \quad \square$$

The detailed balance equations are a system of ‘one equation per pair of states’, each involving only two unknowns, and are usually *far* easier to solve than $\pi = \pi P$. So a sensible strategy when hunting for an invariant distribution is: try detailed balance first. If it has a solution, you’ve found π (and shown the chain is reversible as a bonus). If not, the chain isn’t reversible, and you must go back to $\pi = \pi P$.

Example 17.8.5 (A Chain That is Not Reversible)

Let $n \geq 3$, and consider the random walk on $\mathbb{Z}_n$ (the integers modulo n) which steps clockwise with probability $2/3$ and anticlockwise with probability $1/3$:

$$P_{i,i+1} = \frac{2}{3}, \quad P_{i,i-1} = \frac{1}{3} \quad (\text{indices mod } n).$$

By symmetry we can check that $\pi_i = 1/n$ is invariant: each state receives flow $\frac{1}{n} \cdot \frac{2}{3}$ from its anticlockwise neighbour, and $\frac{1}{n} \cdot \frac{1}{3}$ from its clockwise neighbour, totalling $1/n$. But detailed balance fails:

$$\pi_i P_{i,i+1} = \frac{2}{3n} \neq \frac{1}{3n} = \pi_{i+1} P_{i+1,i}.$$

And this makes perfect sense – in equilibrium there is a net current of probability flowing clockwise around the cycle, and if we watched a film of this chain we could tell whether it was being played backwards, by watching which way the walker predominantly moves.

Example 17.8.6 (A Reversible Chain)

Now cut the cycle open: consider the same walk on $\{0, 1, \dots, n-1\}$, but where the walk cannot wrap around, say

$$P_{i,i+1} = \frac{2}{3} \quad (i < n-1), \quad P_{i,i-1} = \frac{1}{3} \quad (i > 0), \quad P_{0,0} = \frac{1}{3}, \quad P_{n-1,n-1} = \frac{2}{3}.$$

This time there is nowhere for a net current to circulate, and indeed detailed balance succeeds: the equations

$$\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i} \implies \pi_{i+1} = 2\pi_i,$$

give $\pi_i = c \cdot 2^i$, and normalising, $\pi_i = 2^i / (2^n - 1)$. So this chain is reversible, with an invariant distribution that piles up geometrically at the right-hand end.

The moral of the pair of examples: reversibility is characteristically obstructed by *cycles*

with a preferred direction. On tree-like or line-like state spaces, where every transition must eventually be retraced, chains tend to be reversible. In fact, for chains on the non-negative integers that only ever step to a neighbour, this is a theorem.

Proposition 17.8.7 (Birth-Death Chains are Reversible)

Let P be an irreducible birth-death chain on $S = \{0, 1, 2, \dots\}$ (or a finite interval of integers), i.e. $P_{i,j} = 0$ whenever $|i - j| > 1$, and suppose P has an invariant distribution π . Then the chain is reversible.

Proof. We must show $\pi_i P_{i,j} = \pi_j P_{j,i}$ for all i, j . This is trivial if $i = j$ or $|i - j| > 1$, so it suffices to show $\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i}$ for all i , and we induct on i . Consider the invariance equation at state 0. Since only states 0 and 1 can move to 0,

$$\pi_0 = \pi_0 P_{0,0} + \pi_1 P_{1,0},$$

and $P_{0,0} = 1 - P_{0,1}$, so rearranging gives $\pi_0 P_{0,1} = \pi_1 P_{1,0}$, the base case. For the inductive step, invariance at state $i + 1$ reads

$$\pi_{i+1} = \pi_i P_{i,i+1} + \pi_{i+1} P_{i+1,i+1} + \pi_{i+2} P_{i+2,i+1},$$

and substituting $P_{i+1,i+1} = 1 - P_{i+1,i} - P_{i+1,i+2}$ along with the inductive hypothesis $\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i}$, everything cancels down to

$$\pi_{i+1} P_{i+1,i+2} = \pi_{i+2} P_{i+2,i+1},$$

completing the induction. $\square$

So the Ehrenfest urn didn't just *happen* to satisfy detailed balance – any chain with that skeleton must. The probabilistic explanation is the one we gave above: to travel from i to j the chain must cross each intermediate edge, and every crossing of an edge in one direction must (in the long run) be matched by a crossing in the other direction, since the chain cannot go round any other way.

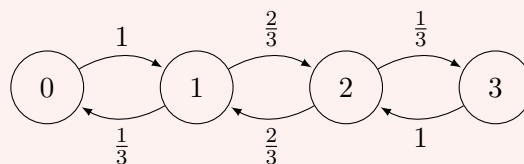
Let's do one more substantial example, which is a classical model from statistical physics.

Example 17.8.8 (The Ehrenfest Urn)

Two chambers are connected by a small hole, and contain N gas molecules between them. At each time step, a molecule is chosen uniformly at random and moves through the hole to the other chamber. Let X_n be the number of molecules in the left chamber, so X_n is a Markov chain on $\{0, 1, \dots, N\}$ with

$$P_{i,i-1} = \frac{i}{N}, \quad P_{i,i+1} = \frac{N-i}{N}.$$

For example, when $N = 3$ the chain looks as follows.



The state space is line-like, so we optimistically try detailed balance:

$$\pi_i P_{i,i+1} = \pi_{i+1} P_{i+1,i} \implies \pi_{i+1} = \pi_i \cdot \frac{N-i}{i+1}.$$

Iterating this down to π_0 gives

$$\pi_i = \pi_0 \cdot \frac{N(N-1) \cdots (N-i+1)}{i!} = \pi_0 \binom{N}{i},$$

and since $\sum_i \binom{N}{i} = 2^N$, normalising gives

$$\pi_i = \binom{N}{i} \frac{1}{2^N}.$$

So in equilibrium, the number of molecules in the left chamber is distributed as $\text{Bin}(N, 1/2)$ – exactly as if each molecule independently tossed a fair coin to choose its chamber. The chain is reversible: watching the molecules shuffle back and forth, we cannot tell which way time is flowing.

The mean return times contain a famous lesson. The mean return time to a half-full chamber is

$$m_{N/2} = \frac{1}{\pi_{N/2}} = \frac{2^N}{\binom{N}{N/2}} \approx \sqrt{\frac{\pi N}{2}},$$

(taking N even, and using Stirling), which is tiny. But the mean return time to the state where the left chamber is *empty* is

$$m_0 = \frac{1}{\pi_0} = 2^N,$$

which for N of the order of Avogadro's number is unimaginably vast. So the chain *is* recurrent – all the air in the room will eventually gather in one half of it! – but the expected time for this is so astronomically long that we will never observe it. This resolves (or at least, quantifies) the apparent conflict between microscopic reversibility and the macroscopic arrow of time.

§17.8.3 Random Walks on Graphs

We finish with a family of Markov chains where detailed balance works beautifully: random walks on graphs. Let $G = (V, E)$ be a finite connected graph. The **simple random walk on G** is the Markov chain with state space V in which, at each step, the walker moves to a uniformly random neighbour of its current vertex. Writing $d(v)$ for the degree of the vertex v , the transition matrix is

$$P_{u,v} = \begin{cases} \frac{1}{d(u)} & \text{if } uv \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Connectedness of G makes the chain irreducible. So, what is the invariant distribution? Following our own advice, we try detailed balance first. For an edge $uv \in E$, the equations read

$$\pi_u \cdot \frac{1}{d(u)} = \pi_v \cdot \frac{1}{d(v)},$$

which says exactly that $\pi_v \propto d(v)$. Since $\sum_v d(v) = 2|E|$ (each edge is counted from both ends), normalising gives the following.

Proposition 17.8.9 (Invariant Distribution for Random Walk on a Graph)

The simple random walk on a finite connected graph $G = (V, E)$ is reversible, with invariant distribution

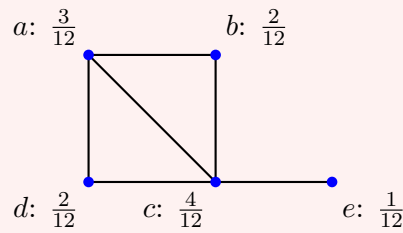
$$\pi_v = \frac{d(v)}{2|E|}, \quad v \in V.$$

In equilibrium, the walker is found at each vertex with probability proportional to its degree – well-connected vertices are visited more. A special case worth noting: if G is *regular* (all degrees equal), the invariant distribution is uniform. So a random walk on the vertices of a cube, or on a cycle, spends its time equally amongst all vertices in the long run – and, for instance, a walker on the corners of a cube takes on average $m_v = 1/\pi_v = 8$ steps to return to its starting corner.

Let's see the general case on a concrete graph.

Example 17.8.10 (A Walk on a Small Graph)

Consider the simple random walk on the graph below, where each vertex is labelled with its invariant probability $\pi_v = d(v)/2|E|$; here $|E| = 6$, so $2|E| = 12$.



The vertex c , with its four edges, carries the most mass, while the pendant vertex e carries the least. By our earlier theory, we can immediately read off facts that would be quite painful to compute directly: for example the mean return time to e is $m_e = 1/\pi_e = 12$, while the mean return time to c is $m_c = 12/4 = 3$.

Aside: The Random Surfer

A relative of the random walk on a graph powers one of the most consequential algorithms ever deployed: Google's *PageRank*. Imagine a 'random surfer' on the web, who at each step clicks a uniformly random link on their current page – a random walk on the (directed!) graph of the web. The importance of a page is then defined to be its invariant probability π_v : the long-run proportion of time the surfer spends there, which by the ergodic theorem is a genuinely meaningful quantity. A page is important if many important pages link to it, and this apparently circular definition is resolved exactly by the equation $\pi = \pi P$.

Because the web graph is directed, the lovely formula $\pi_v \propto d(v)$ no longer applies (detailed balance fails – links very much have a preferred direction), and the chain is far from irreducible. The fix is a version of the lazy chain trick: with small probability at each step, the surfer teleports to a uniformly random page. This makes the chain

irreducible and aperiodic, so a unique invariant distribution exists and can be found by simply iterating $\lambda \mapsto \lambda P$ – which, by convergence to equilibrium, approaches π from any starting guess.

Example 17.8.11 (The Knight on a Chessboard)

As a parting example: place a knight on the corner square of an otherwise empty chessboard, and let it make uniformly random legal moves. How long, on average, until it first returns to its starting corner?

This looks like a nightmare of a computation, but the theory makes it almost trivial. The knight performs a simple random walk on the graph whose vertices are the 64 squares, with edges given by legal knight moves. This graph is connected, so

$$m_{\text{corner}} = \frac{1}{\pi_{\text{corner}}} = \frac{2|E|}{d(\text{corner})} = \frac{\sum_v d(v)}{d(\text{corner})}.$$

Summing the degrees over the whole board^a gives 336, and a corner square has exactly 2 knight moves, so

$$m_{\text{corner}} = \frac{336}{2} = 168.$$

A lovely answer, obtained with essentially no computation – a fitting note on which to end the course.

^aThe degrees form the pattern: corner squares have 2 moves, squares adjacent to corners have 3 or 4, central squares have 8, and so on; carrying out the count gives $\sum_v d(v) = 336$.

Epilogue: The Big Picture

This is a good point to pause and assemble the whole story in one place. For an *irreducible* Markov chain, everything we have proved organises the possible behaviours into a clean trichotomy.

- *Transient*. The chain visits each state only finitely often, and escapes to infinity: $\mathbb{P}(T_i < \infty \mid X_0 = i) < 1$ and $\sum_n p_{i,i}(n) < \infty$. There is no invariant distribution, and $p_{i,j}(n) \rightarrow 0$. The long-run proportion of time in each state is 0. (Example: the simple random walk on $\mathbb{Z}^3$.)
- *Null recurrent*. The chain returns to each state infinitely often, but with infinite mean return time. Invariant *measures* exist and are unique up to scaling, but have infinite mass, so there is still no invariant *distribution*; again $p_{i,j}(n) \rightarrow 0$ (given aperiodicity) and the long-run proportions are 0. The chain is recurrent but only just. (Example: the simple random walk on $\mathbb{Z}$ or $\mathbb{Z}^2$.)
- *Positive recurrent*. Mean return times are finite, and there is a unique invariant distribution π , given by $\pi_i = 1/m_i$. The long-run proportion of time spent in state i is π_i almost surely (the ergodic theorem, no aperiodicity needed), and if the chain is additionally aperiodic then $\mathbb{P}(X_n = i) \rightarrow \pi_i$ from any initial distribution (convergence to equilibrium). (Examples: every finite irreducible chain, and drifting walks like the reflected biased walk.)

For finite state spaces, only the third behaviour is possible for irreducible chains, which

is why finite chains are so pleasant: an invariant distribution always exists, and irreducibility plus aperiodicity guarantees convergence to it.

Finally, the theory of time reversal gives a practical route to finding π : solve the detailed balance equations if you can (as for birth-death chains, the Ehrenfest urn, and random walks on graphs), and fall back on $\pi = \pi P$ when the chain carries a genuine probability current around a cycle. With these tools, remarkably many questions – mean return times, long-run proportions, limiting probabilities – reduce to a small amount of linear algebra.

Of course, this is only the beginning of the story of Markov processes. Natural next steps include chains that evolve in *continuous* time (the subject of the Part II course ‘Applied Probability’, where the exponential distribution takes centre stage), the quantitative study of *how fast* chains mix (as in the card shuffling aside), and the theory of martingales, which generalises the ‘restart and forget’ philosophy of the strong Markov property into one of the most powerful toolkits in modern probability. The reader is warmly encouraged to keep walking.

18 Methods

A large amount of physics (and indeed applied mathematics as a whole) is governed by a rather small collection of partial differential equations: the wave equation, the diffusion equation and Laplace's equation. The aim of this course is to build up a toolbox of techniques for solving these equations, and equations like them. The tools themselves – Fourier series, Sturm–Liouville theory, Green's functions, Fourier transforms and the method of characteristics – are united by a single underlying idea, that of *linearity*. Almost everything we do will amount to breaking a complicated problem into simple pieces (usually eigenfunctions or point sources), solving each piece, and adding up the results.

This article constitutes my notes for the 'Methods' course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§18.1 Fourier Series

§18.1.1 Periodic Functions

We will begin our study of methods and in particular Fourier series by considering some periodic functions.

Definition 18.1.1 (Periodic)

A function $f(x)$ is **periodic** if $f(x + T) = f(x)$ for all x , where T is the **period**.

Example 18.1.2 (Simple Harmonic Motion)

Many physical objects are described by *simple harmonic motion*, with the position given by

$$y = A \sin \omega t.$$

We call A the **amplitude**, and the period is $T = 2\pi/\omega$. The **frequency** is $1/T$.

Fourier series is all about trying to write periodic functions as particular sums of sines and cosines. Consider the set of functions

$$g_n(x) = \cos \frac{n\pi x}{L}, \quad \text{and} \quad h_n(x) = \sin \frac{n\pi x}{L},$$

where we take $n \in \mathbb{N}$. These functions are periodic with period $2L$.

You may recall the following set of identities:

$$\begin{aligned} \cos A \cos B &= \frac{1}{2} (\cos(A - B) + \cos(A + B)) \\ \sin A \sin B &= \frac{1}{2} (\cos(A - B) - \cos(A + B)) \\ \sin A \cos B &= \frac{1}{2} (\sin(A - B) + \sin(A + B)). \end{aligned}$$

We are going to try and define an inner product on this domain $[0, 2L]$, and using that we will be able to multiply these functions together and talk about their relative orthogonality.

Definition 18.1.3

We define the inner product $\langle f, g \rangle = \int_0^{2L} f(x)g(x) \, dx$.

We can then obtain some orthogonality conditions for h_n and g_n with respect to this inner product. We can compute for $n \neq m$

$$\begin{aligned} \langle h_n, h_m \rangle &= \int_0^{2L} \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} \, dx \\ &= \frac{1}{2} \int_0^{2L} \left(\cos \frac{(n-m)\pi}{L} x - \cos \frac{(n+m)\pi}{L} x \right) \, dx \\ &= \frac{1}{2} \frac{L}{\pi} \left[\frac{\sin(n-m)\pi x/L}{n-m} - \frac{\sin(n+m)\pi x/L}{n+m} \right]_0^{2L} \\ &= 0, \end{aligned}$$

and for $n = m$

$$\begin{aligned} \langle h_n, h_n \rangle &= \int_0^{2L} \sin^2 \frac{n\pi x}{L} \, dx \\ &= \int_0^{2L} \frac{1}{2} \left(1 - \cos \frac{2\pi n x}{L} \right) \, dx \\ &= L. \end{aligned}$$

Hence we obtain the orthogonality condition

$$\langle h_n, h_m \rangle = \begin{cases} L\delta_{mn} & \text{if } n, m \neq 0, \\ 0 & \text{if } m = 0. \end{cases}$$

Similarly, it's straightforward to check that

$$\langle g_n, g_m \rangle = \begin{cases} L\delta_{mn} & \text{if } n, m \neq 0, \\ 2L\delta_{0n} & \text{if } m = 0. \end{cases}$$

and

$$\langle h_n, g_m \rangle = 0.$$

These orthogonality conditions are important because we are going to use these functions as a complete orthogonal set which spans the space of ‘well-behaved periodic functions’.

§18.1.2 Definition of a Fourier Series

We can express any ‘well-behaved’ periodic function $f(x)$ with period $2L$ as

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L},$$

where a_n, b_n are constants such that the RHS is convergent for all x where f is continuous. At a discontinuity, the Fourier series approaches the midpoint of the upper and lower limits at that point.

Consider taking the inner product $\langle h_m, f \rangle$ and substitute the expression for f above, to get

$$\int_0^{2L} \sin \frac{m\pi x}{L} f(x) \, dx = \sum_{n=1}^{\infty} L b_n \delta_{nm} = L b_m.$$

Hence we find that (doing something similar with g_n)

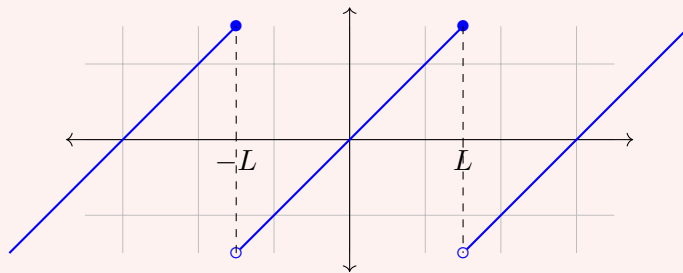
$$b_n = \frac{1}{L} \int_0^{2L} f(x) \sin \frac{n\pi x}{L} \, dx,$$

and $a_n = \frac{1}{L} \int_0^{2L} f(x) \cos \frac{n\pi x}{L} \, dx.$

Now, this expression for a_n includes the case $n = 0$, and says that $\frac{1}{2}a_0$ is the average value of the function. Also, the range of integration is one period, and we can equivalently integrate over $[-L, L]$ instead of $[0, 2L]$.

Example 18.1.4 (The Sawtooth Wave)

Consider the function $f(x) = x$ for $-L \leq x \leq L$, with the function being periodic elsewhere.



Here we have

$$a_n = \frac{1}{L} \int_{-L}^L x \cos \frac{n\pi x}{L} \, dx = 0, \quad (\text{integrating an odd function})$$

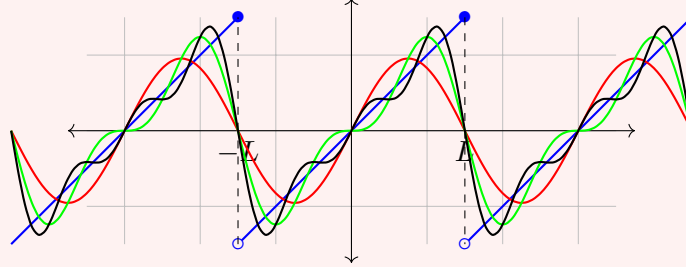
for all n , and

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L x \sin \frac{n\pi x}{L} \, dx \\ &= \frac{-2}{n\pi} \left[x \cos \frac{n\pi x}{L} \right]_0^L + \frac{2}{n\pi} \int_0^L \cos \frac{n\pi x}{L} \, dx \\ &= -\frac{2L}{n\pi} \cos n\pi + \frac{2L}{(n\pi)^2} \sin n\pi \\ &= \frac{2L}{n\pi} (-1)^{n+1}. \end{aligned}$$

So the sawtooth Fourier series is

$$2L \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n\pi} \sin\left(\frac{n\pi x}{L}\right) = \frac{2L}{\pi} \left[\sin\left(\frac{\pi x}{L}\right) - \frac{1}{2} \sin\left(\frac{2\pi x}{L}\right) + \frac{1}{3} \sin\left(\frac{3\pi x}{L}\right) + \cdots \right].$$

which is slowly convergent.



§18.1.3 The Dirichlet Conditions (Fourier's Theorem)

So we need 'well behaved' functions for what we have discussed about Fourier series to work, and for a Fourier series for f to be unique. But what exactly does it mean for a function to be 'well behaved' in this context? This is specified by the Dirichlet conditions, also known as Fourier's theorem.

Theorem 18.1.5 (The Dirichlet Conditions)

If f is a bounded periodic function with period $2L$ and finitely many minima, maxima and discontinuities on $0 \leq x \leq 2L$, then the Fourier series converges to $f(x)$ at all points where f is continuous, and at discontinuities converges to $\frac{1}{2}(f(x_+) + f(x_-))$.

One might note that these conditions are *much weaker* than those needed for Taylor series, but it does eliminate pathological functions such as $\sin 1/x$, $1/x$ and the indicator function on the rationals. The converse of this is *not* true, for example $\sin 1/x$ has a well defined Fourier series.

The proof of this result is *too* difficult, but is best done using complex methods so we will not dwell on it in this course.

The rate of convergence of the Fourier series depends on the 'smoothness' of the function.

Theorem 18.1.6 (Rate of Convergence of Fourier Series)

If $f(x)$ has continuous derivatives up to a p th derivative which is discontinuous, then the Fourier series converges as $\mathcal{O}(n^{-(p+1)})$ as $n \rightarrow \infty$.

Example 18.1.7 (Square Wave)

Consider the function

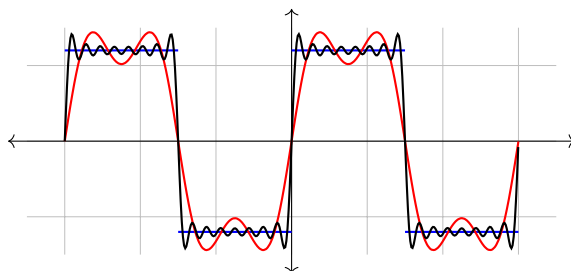
$$f(x) = \begin{cases} 1 & \text{if } 0 \leq x < 1, \\ -1 & \text{if } -1 \leq x < 0. \end{cases}$$

This function has discontinuous $p = 0$ th derivative (the function is discontinuous!),

and the Fourier series of this function is given by

$$f(x) = 4 \sum_{m=1}^{\infty} \frac{\sin(2m-1)\pi x}{(2m-1)\pi}.$$

It is worth pausing to look at how the partial sums of this series actually behave. Below is the square wave along with the partial sums taking $m \leq 2$ and $m \leq 8$.



Away from the discontinuities the partial sums settle down quite quickly, but near each jump they consistently *overshoot* by roughly 9% of the size of the jump, and this overshoot does not go away as we take more terms – it just gets squeezed into a narrower and narrower region around the discontinuity. This is the **Gibbs phenomenon**. It doesn't contradict the Dirichlet conditions (at each *fixed* x the series still converges), but it does tell us that the convergence is not uniform near a jump.

Integration of Fourier Series

As a general principle, it is *always* valid to integrate the Fourier series of $f(x)$ term-wise to obtain

$$F(x) = \int_{-L}^x f(u) \, du,$$

because $F(x)$ satisfies the Dirichlet conditions if $f(x)$ does.

Differentiation of Fourier Series

As a general principle, you must *take care* when differentiating a Fourier series term-wise. Let's look at an example.

Example 18.1.8 (Differentiating the Square Wave)

Consider the clearly non-differentiable square wave function, defined in [Example 18.1.7](#). We had

$$f(x) = 4 \sum_{m=1}^{\infty} \frac{\sin(2m-1)\pi x}{(2m-1)\pi}.$$

We can try and differentiate this term-wise to get

$$f'(x) \stackrel{?}{=} 4 \sum_{m=1}^{\infty} \cos(2m-1)\pi x,$$

but this series is unbounded!

So we need to be careful. Here's the result about what we can do with differentiation.

Theorem 18.1.9

If $f(x)$ is continuous and satisfies the Dirichlet conditions, and $f'(x)$ satisfies the Dirichlet conditions, then $f'(x)$ can be found by term-wise differentiation of the Fourier series of $f(x)$.

§18.1.4 Parseval's Theorem

Parseval's theorem is a relation between the integral of the square of a function and the square of the Fourier coefficients:

$$\begin{aligned} \int_0^{2L} [f(x)]^2 dx &= \int_0^{2L} dx \left[\frac{1}{2}a_0 + \sum_n a_n \cos \frac{n\pi x}{L} + \sum_n b_n \sin \frac{n\pi x}{L} \right]^2 \\ &= \int_0^{2L} dx \left[\frac{1}{4}a_0^2 + \sum_n a_n^2 \cos^2 \frac{n\pi x}{L} + \sum_n b_n^2 \sin^2 \frac{n\pi x}{L} \right] \\ &= L \left[\frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} (a_n^2 + b_n^2) \right], \end{aligned}$$

where the cross terms all vanish by orthogonality.

This result is also called the completeness relation, since the LHS $\geq$ RHS if any of the basis functions are missing.

Example 18.1.10 (The Basel Problem)

Consider the sawtooth wave $f(x) = x$ on $-L \leq x \leq L$. Then the LHS is

$$\int_{-L}^L x^2 dx = \frac{2}{3}L^3,$$

and the RHS is

$$L \sum_{n=1}^{\infty} \frac{4L^2}{n^2\pi^2} = \frac{4L^3}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2},$$

and thus $\sum_{n=1}^{\infty} 1/n^2 = \pi^2/6$.

§18.1.5 Alternative Fourier Series

There are some alternative forms of Fourier series.

Half-Range Series

Consider $f(x)$ defined only on $0 \leq x < L$. We can extend its range over $-L \leq x < L$ in two simple ways:

1. Require f to be odd, with $f(-x) = -f(x)$. Then considering its Fourier series we would have $a_n = 0$ for all n (as $\cos$ is even), and

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx.$$

This implies that $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$, which is a **Fourier sine series**.

2. Require f to be even, with $f(-x) = f(x)$, Then $b_n = 0$, and

$$a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx,$$

so $f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L}$, which is a **Fourier cosine series**.

Complex Representation

Recall that we can write

$$\cos \frac{n\pi x}{L} = \frac{1}{2} (e^{in\pi x/L} + e^{-in\pi x/L}) \quad \text{and} \quad \sin \frac{n\pi x}{L} = \frac{1}{2i} (e^{in\pi x/L} - e^{-in\pi x/L}).$$

So the Fourier series is

$$\begin{aligned} f(x) &= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} \\ &= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \frac{1}{2} (a_n - ib_n) e^{in\pi x/L} + \sum_{n=1}^{\infty} \frac{1}{2} (a_n + ib_n) e^{-in\pi x/L} \\ &= \sum_{m=-\infty}^{\infty} c_m e^{im\pi x/L}, \end{aligned}$$

where for $m > 0$, $c_m = \frac{1}{2}(a_m - ib_m)$, for $m < 0$, $c_m = \frac{1}{2}(a_{-m} + ib_{-m})$, and $c_0 = \frac{1}{2}a_0$. Equivalently,

$$c_m = \frac{1}{2L} \int_{-L}^L f(x) e^{-im\pi x/L} dx.$$

§18.1.6 Some Fourier Series Motivations – Self-Adjoint Matrices

We will now review a few results about matrices. Suppose that $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$ with inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\dagger \mathbf{v}$. The $n \times n$ matrix A is **self-adjoint** or **hermitian** if

$$\langle A\mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, A\mathbf{v} \rangle,$$

that is, if $A^\dagger = A$.

The eigenvalues λ_i and eigenvectors $\mathbf{v}_i$ satisfy

$$A\mathbf{v}_i = \lambda_i \mathbf{v}_i,$$

and have the following properties

- (i) The eigenvalues are real
- (ii) If $\lambda_i \neq \lambda_j$, then the corresponding eigenvectors are orthogonal with $\langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$
- (iii) We can rescale to create an orthonormal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$.

Given $\mathbf{b}$, we can solve for $\mathbf{x}$ in the equation

$$A\mathbf{x} = \mathbf{b}.$$

Express $\mathbf{b} = \sum_{i=1}^n b_i \mathbf{v}_i$, and we seek a solution $\mathbf{x} = \sum_{i=1}^n c_i \mathbf{v}_i$. Substituting this into the above equation, we get

$$A\mathbf{x} = \sum_i A c_i \mathbf{v}_i = \sum_i c_i \lambda_i \mathbf{v}_i = \sum_i b_i \mathbf{v}_i.$$

By orthogonality, we can equate coefficients to get $c_i \lambda_i = b_i$, so our solution is

$$\mathbf{x} = \sum_{i=1}^n \frac{b_i}{\lambda_i} \mathbf{v}_i.$$

So once we have a self-adjoint matrix A and have found all of the eigenvalues, we then have a general methodology for solving any matrix equation of the form $A\mathbf{x} = \mathbf{b}$. It turns out that you can do the same with Fourier series!

We can find a general solution to the differential equation for which cosines and sines are the eigenfunctions.

§18.1.7 Solving Inhomogeneous ODEs with Fourier Series

We wish to find $y(x)$ given $f(x)$ for the differential operator

$$\mathcal{L}[y] \equiv -\frac{d^2 y}{dx^2} = f(x).$$

To solve this, we need some boundary conditions, which we will take as $y(0) = y(L) = 0$.

The related eigenvalue problem is

$$\mathcal{L}[y_n] = \lambda_n y_n,$$

with $y_n(0) = y_n(L) = 0$, which has eigenfunctions and eigenvalues

$$y_n(x) = \sin \frac{n\pi x}{L}, \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2.$$

We seek solutions as half-range sine series. To do that, we will try substitute $y(x) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{L}$, and expand $f(x)$ in terms of its Fourier series, $f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$, with $b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$.

Substituting this into the differential equation, we get

$$\mathcal{L}[y] = -\frac{d^2}{dx^2} \left(\sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{L} \right) = \sum_{n=1}^{\infty} c_n \left(\frac{n\pi}{L} \right)^2 \sin \frac{n\pi x}{L}.$$

By orthogonality, we have $c_n \left(\frac{n\pi}{L} \right)^2 = b_n$, so the solution is

$$y(x) = \sum_{n=1}^{\infty} \frac{b_n}{\left(\frac{n\pi}{L} \right)^2} \sin \frac{n\pi x}{L} = \sum_{n=1}^{\infty} \frac{b_n}{\lambda_n} y_n.$$

Example 18.1.11

Consider the ‘square wave’ source given by $f(x) = 1$ for $0 \leq x < 1$, with the function being odd. This function has Fourier series

$$f(x) = 4 \sum_{m=1}^{\infty} \frac{\sin(2m-1)\pi x}{(2m-1)\pi}.$$

So the solution for y must then be

$$y(x) = \sum_n \frac{b_n}{\lambda_n} y_n = 4 \sum_m \frac{\sin(2m-1)\pi x}{((2m-1)\pi)^3},$$

but this is the Fourier series for $y(x) = \frac{1}{2}x(1-x)!$

§18.2 Sturm–Liouville Theory

In the last section we saw that the operator $\mathcal{L} = -d^2/dx^2$ behaves rather like a self-adjoint matrix: it has real eigenvalues, its eigenfunctions (sin and cos) are orthogonal, and expanding in those eigenfunctions lets us solve $\mathcal{L}[y] = f$ essentially by division. The natural question is: *which* differential operators behave like this? The answer is given by Sturm–Liouville theory, which identifies a large class of second-order operators for which the whole eigenfunction machinery goes through.

§18.2.1 Second Order Linear ODEs

We are going to be working with the general second order linear differential operator

$$\mathcal{L}y \equiv \alpha(x)y'' + \beta(x)y' + \gamma(x)y,$$

on an interval $a \leq x \leq b$, and we wish to solve equations of the form $\mathcal{L}y = f(x)$. Recall from Differential Equations that the general solution of such an equation is $y = y_c + y_p$, where the complementary function $y_c = Ay_1 + By_2$ is the general solution of the homogeneous equation $\mathcal{L}y = 0$, and y_p is any particular integral.

To pin down a unique solution we need two extra conditions. These may be **boundary conditions**, specifying something about y at both ends of the interval (say $y(a)$ and $y(b)$), or **initial conditions**, specifying $y(a)$ and $y'(a)$. Boundary conditions of the form $y(a) = y(b) = 0$ are called **homogeneous** boundary conditions. These are less restrictive than they look: if instead we were given $y(a) = c_1$, $y(b) = c_2$, we can subtract off any function u satisfying $\mathcal{L}u = 0$ (or indeed anything convenient) with $u(a) = c_1$, $u(b) = c_2$, and then $y - u$ satisfies homogeneous conditions. We will therefore usually assume our boundary conditions are homogeneous.

As in the previous section, our strategy for solving $\mathcal{L}y = f$ will be to first solve the related eigenvalue problem

$$\mathcal{L}y = \lambda w(x)y,$$

where $w(x)$ is some non-negative **weight function** which will appear naturally in what follows, and then expand everything in sight in terms of the eigenfunctions.

§18.2.2 Sturm–Liouville Form

For complex valued functions on $[a, b]$, we will use the inner product

$$\langle f, g \rangle = \int_a^b f^*(x)g(x) \, dx.$$

Definition 18.2.1 (Sturm–Liouville Form)

A second order differential operator is in **Sturm–Liouville form** if it is written as

$$\mathcal{L}y = -(p(x)y')' + q(x)y,$$

where p is differentiable and positive on (a, b) . The associated eigenvalue problem is $\mathcal{L}y = \lambda wy$, where the weight $w(x)$ is positive.

This might look like a rather restrictive shape for an operator, but it turns out essentially every second order operator we care about can be put in this form. Indeed, suppose we have the eigenvalue problem

$$\alpha(x)y'' + \beta(x)y' + \gamma(x)y = -\lambda\rho(x)y.$$

Multiply the whole equation by some function $F(x)$, which we will choose in a moment. The first two terms become

$$F\alpha y'' + F\beta y' = (F\alpha y')' - (F\alpha)'y' + F\beta y',$$

so if we choose F such that $(F\alpha)' = F\beta$, the y' terms cancel and we have something of Sturm–Liouville type. This condition rearranges to $F'/F = (\beta - \alpha')/\alpha$, which we can integrate to get the *integrating factor*

$$F(x) = \exp\left(\int^x \frac{\beta(t) - \alpha'(t)}{\alpha(t)} \, dt\right).$$

Note $F > 0$ automatically, being an exponential. With this choice,

$$-(F\alpha y')' - F\gamma y = \lambda F\rho y,$$

so we have Sturm–Liouville form with $p = -F\alpha$, $q = -F\gamma$ and $w = F\rho$ (with signs arranged as appropriate for $p, w > 0$).

Example 18.2.2 (The Hermite Equation)

Consider the equation

$$y'' - 2xy' + 2ny = 0.$$

Here $\alpha = 1$ and $\beta = -2x$, so

$$F = \exp\left(\int^x (-2t) \, dt\right) = e^{-x^2},$$

and multiplying through, the equation in Sturm–Liouville form is

$$-(e^{-x^2}y')' = 2ne^{-x^2}y,$$

an eigenvalue problem with eigenvalue $2n$ and weight function $w = e^{-x^2}$.

§18.2.3 Self-Adjoint Operators

We can now say what it means for a differential operator to be self-adjoint, in direct analogy with matrices.

Definition 18.2.3 (Self-Adjoint Operator)

The operator $\mathcal{L}$ is **self-adjoint** on $[a, b]$, for a given class of functions with given boundary conditions, if

$$\langle y_1, \mathcal{L}y_2 \rangle = \langle \mathcal{L}y_1, y_2 \rangle$$

for all y_1, y_2 in that class.

The point of Sturm–Liouville form is that it makes self-adjointness almost automatic. Take $\mathcal{L}y = -(py')' + qy$ with everything real, and compute

$$\begin{aligned} \langle y_1, \mathcal{L}y_2 \rangle - \langle \mathcal{L}y_1, y_2 \rangle &= \int_a^b [-y_1(py_2')' + y_1qy_2 + y_2(py_1')' - y_2qy_1] \, dx \\ &= \int_a^b [y_2(py_1')' - y_1(py_2')'] \, dx \\ &= \int_a^b [(py_1'y_2)' - (py_1y_2)'] \, dx \\ &= \left[p(y_1'y_2 - y_1y_2') \right]_a^b, \end{aligned}$$

where in the third line we added and subtracted $py_1'y_2'$ inside the integral. So $\mathcal{L}$ is self-adjoint precisely when the boundary term

$$\left[p(y_1'y_2 - y_1y_2') \right]_a^b = 0.$$

This vanishes for a number of natural types of boundary conditions:

- (i) $y(a) = y(b) = 0$, or $y'(a) = y'(b) = 0$, or more generally $y + ky' = 0$ at each endpoint (the **regular Sturm–Liouville problem**);
- (ii) periodic boundary conditions, where y, y' and p take the same values at a and b ;
- (iii) singular endpoints where $p(a) = p(b) = 0$, provided y stays bounded;
- (iv) any combination of the above, one condition at each end.

§18.2.4 Properties of the Eigenfunctions

Now suppose $\mathcal{L}$ is self-adjoint with the boundary conditions in play, and consider the eigenvalue problem

$$\mathcal{L}y_n = \lambda_n w y_n,$$

with $w > 0$. Just as for hermitian matrices, we get a package of results: real eigenvalues, orthogonal eigenfunctions, and (though we won't prove it) completeness.

Proposition 18.2.4 (Eigenvalues are Real)

The eigenvalues of a self-adjoint operator $\mathcal{L}$ (with weight w) are real, and the eigenfunctions can be taken to be real valued.

Proof. Suppose $\mathcal{L}y_n = \lambda_n w y_n$. Since $\mathcal{L}$ and w are real, taking complex conjugates gives $\mathcal{L}y_n^* = \lambda_n^* w y_n^*$. Then using self-adjointness,

$$0 = \langle y_n, \mathcal{L}y_n \rangle - \langle \mathcal{L}y_n, y_n \rangle = (\lambda_n - \lambda_n^*) \int_a^b w |y_n|^2 dx.$$

The integral is strictly positive since $w > 0$ and $y_n \not\equiv 0$, so $\lambda_n = \lambda_n^*$, that is, λ_n is real. Given this, the real and imaginary parts of y_n separately satisfy the eigenvalue equation, so we may take our eigenfunctions to be real. $\square$

Proposition 18.2.5 (Orthogonality of Eigenfunctions)

If $\mathcal{L}y_n = \lambda_n w y_n$ and $\mathcal{L}y_m = \lambda_m w y_m$ with $\lambda_n \neq \lambda_m$, then

$$\int_a^b w y_n y_m dx = 0.$$

Proof. By self-adjointness (taking everything real),

$$0 = \int_a^b (y_m \mathcal{L}y_n - y_n \mathcal{L}y_m) dx = (\lambda_n - \lambda_m) \int_a^b w y_n y_m dx,$$

and since $\lambda_n \neq \lambda_m$ the integral must vanish. $\square$

We say the eigenfunctions are **orthogonal with respect to the weight w** , and we write

$$\langle f, g \rangle_w = \int_a^b w f^* g dx$$

for the corresponding weighted inner product. It's frequently useful to normalise, defining

$$Y_n(x) = \frac{y_n(x)}{\left(\int_a^b w y_n^2 dx\right)^{1/2}}, \quad \text{so that} \quad \langle Y_n, Y_m \rangle_w = \delta_{nm}.$$

The final property, which we will assume without proof, is **completeness**: the eigenfunctions of a Sturm–Liouville problem span the space of ‘well-behaved’ functions on $[a, b]$. That is, any such f can be expanded as

$$f(x) = \sum_{n=1}^{\infty} a_n y_n(x).$$

Taking the weighted inner product with y_m and using orthogonality picks out the coefficients:

$$a_n = \frac{\int_a^b w y_n f dx}{\int_a^b w y_n^2 dx}.$$

Fourier series are exactly the special case $\mathcal{L} = -d^2/dx^2$, $w = 1$, with periodic boundary conditions.

Remark. For the regular Sturm–Liouville problem one can show a little more: the eigenvalues are simple (non-degenerate) and form an increasing sequence $\lambda_1 < \lambda_2 < \dots \rightarrow \infty$. The proof of simplicity is a nice exercise: if u, v are eigenfunctions with the same eigenvalue, then $(p(uv' - u'v))' = 0$, and evaluating the Wronskian-like quantity $p(uv' - u'v)$ at an endpoint shows it vanishes identically, forcing u and v to be proportional.

§18.2.5 Completeness and Parseval's Identity

Let's investigate what completeness really says. Define the partial sum

$$S_N(x) = \sum_{n=1}^N a_n y_n(x),$$

with a_n as above, and consider the mean-square error

$$\epsilon_N = \int_a^b w [f(x) - S_N(x)]^2 dx.$$

Expanding the square and using orthogonality,

$$\begin{aligned} \epsilon_N &= \int_a^b w f^2 dx - 2 \sum_{n=1}^N a_n \int_a^b w f y_n dx + \sum_{n=1}^N a_n^2 \int_a^b w y_n^2 dx \\ &= \int_a^b w f^2 dx - \sum_{n=1}^N a_n^2 \int_a^b w y_n^2 dx. \end{aligned}$$

Completeness is the statement that $\epsilon_N \rightarrow 0$ as $N \rightarrow \infty$; we say S_N **converges in the mean** to f . Note this is a ‘global’ notion of convergence, weaker than pointwise convergence. Sending $N \rightarrow \infty$ above gives the general form of Parseval's identity,

$$\int_a^b w f^2 dx = \sum_{n=1}^{\infty} a_n^2 \int_a^b w y_n^2 dx,$$

and if any eigenfunctions are missing from our set we only get the inequality $\geq$, which is **Bessel's inequality**.

There is one more thing worth extracting from this computation. Suppose we fix N and ask: what choice of coefficients c_n minimises the error $\int_a^b w (f - \sum_{n \leq N} c_n y_n)^2 dx$? Differentiating with respect to c_n ,

$$\frac{\partial \epsilon_N}{\partial c_n} = -2 \int_a^b w y_n \left(f - \sum_{m=1}^N c_m y_m \right) dx = -2 \left(\int_a^b w f y_n dx - c_n \int_a^b w y_n^2 dx \right),$$

which vanishes exactly when $c_n = a_n$ (and this is a minimum, since $\partial^2 \epsilon_N / \partial c_n^2 = 2 \int w y_n^2 dx > 0$). So truncated eigenfunction expansions are the *best possible* approximations in the mean-square sense – the first N coefficients don't need revising when you decide to take $N + 1$ terms.

§18.2.6 Legendre's Equation

Let's now meet the most important Sturm–Liouville problem in this course after Fourier series. **Legendre's equation** is

$$(1 - x^2)y'' - 2xy' + \lambda y = 0, \quad -1 \leq x \leq 1,$$

with the boundary conditions that y remains finite at $x = \pm 1$. This is already in Sturm–Liouville form:

$$-((1-x^2)y')' = \lambda y,$$

with $p = 1 - x^2$, $q = 0$ and $w = 1$. Note that $p(\pm 1) = 0$, so the endpoints are singular points of the equation, and the finiteness condition is exactly the sort of self-adjoint compatible boundary condition we saw earlier.

We solve by a power series about $x = 0$, writing $y = \sum_n c_n x^n$. Substituting in and comparing coefficients of x^n gives

$$(n+2)(n+1)c_{n+2} - n(n-1)c_n - 2nc_n + \lambda c_n = 0,$$

that is, the recursion

$$c_{n+2} = \frac{n(n+1) - \lambda}{(n+1)(n+2)} c_n.$$

Specifying c_0 and c_1 gives two independent solutions, one even and one odd:

$$y_{\text{even}} = c_0 \left[1 - \frac{\lambda}{2!}x^2 + \frac{(6-\lambda)(-\lambda)}{4!}x^4 + \cdots \right], \quad y_{\text{odd}} = c_1 \left[x + \frac{(2-\lambda)}{3!}x^3 + \cdots \right].$$

Now here's the crucial point. As $n \rightarrow \infty$ we have $c_{n+2}/c_n \rightarrow 1$, so these series behave like geometric series in x^2 and diverge at $x = \pm 1$ – one can check they blow up logarithmically, violating our boundary conditions. The only way to get an acceptable solution is for the series to *terminate*, which happens exactly when

$$\lambda = \ell(\ell+1) \quad \text{for some } \ell \in \mathbb{N},$$

killing the numerator at $n = \ell$. For each such λ , one of y_{even} , y_{odd} becomes a polynomial of degree ℓ (and the other still diverges, so we discard it).

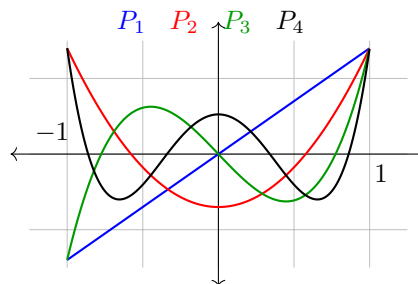
Definition 18.2.6 (Legendre Polynomials)

The **Legendre polynomial** $P_\ell(x)$ is the degree ℓ polynomial solution of Legendre's equation with $\lambda = \ell(\ell+1)$, normalised so that $P_\ell(1) = 1$.

The first few are

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{3x^2 - 1}{2}, \quad P_3 = \frac{5x^3 - 3x}{2}, \quad P_4 = \frac{35x^4 - 30x^2 + 3}{8}.$$

P_ℓ has the same parity as ℓ and has exactly ℓ zeros in $(-1, 1)$, so the polynomials oscillate more and more as ℓ grows – exactly like the higher Fourier modes.



Because they are eigenfunctions of a self-adjoint problem with distinct eigenvalues, the Legendre polynomials are automatically orthogonal:

$$\int_{-1}^1 P_n(x)P_m(x) \, dx = 0 \quad \text{for } n \neq m,$$

and one can compute the normalisation

$$\int_{-1}^1 P_n(x)^2 \, dx = \frac{2}{2n+1}.$$

There are a number of useful ways to generate the P_ℓ without solving the recursion each time. There is **Rodrigues' formula**

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} (x^2 - 1)^\ell,$$

and the **generating function**

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{\ell=0}^{\infty} P_\ell(x)t^\ell,$$

which we shall derive when we study Laplace's equation (it is secretly the potential of a point charge). There are also recursion relations such as

$$\begin{aligned} (\ell+1)P_{\ell+1}(x) &= (2\ell+1)xP_\ell(x) - \ell P_{\ell-1}(x), \\ \text{and } (2\ell+1)P_\ell(x) &= \frac{d}{dx} [P_{\ell+1}(x) - P_{\ell-1}(x)]. \end{aligned}$$

Since the Legendre polynomials form a complete orthogonal set on $[-1, 1]$, any well-behaved f on $[-1, 1]$ has a **Fourier–Legendre series**

$$f(x) = \sum_{\ell=0}^{\infty} a_\ell P_\ell(x), \quad a_\ell = \frac{2\ell+1}{2} \int_{-1}^1 f(x)P_\ell(x) \, dx.$$

Note that unlike Fourier series, no periodicity of f is required – the boundary conditions here were just finiteness.

§18.2.7 Solving Inhomogeneous ODEs by Eigenfunction Expansion

We now have everything needed to run the ‘self-adjoint matrix’ strategy in general. Suppose we want to solve

$$\mathcal{L}y = f(x) = w(x)F(x)$$

on $[a, b]$ with homogeneous boundary conditions, where $\mathcal{L}$ is in Sturm–Liouville form with eigenfunctions $\mathcal{L}y_n = \lambda_n w y_n$. Expand both the solution and the source:

$$y(x) = \sum_n c_n y_n(x), \quad F(x) = \sum_n a_n y_n(x),$$

where the a_n are known:

$$a_n = \frac{\int_a^b w F y_n \, dx}{\int_a^b w y_n^2 \, dx}.$$

Substituting,

$$\mathcal{L}y = \sum_n c_n \mathcal{L}y_n = w \sum_n c_n \lambda_n y_n = w \sum_n a_n y_n,$$

and by orthogonality we can equate coefficients, giving $c_n = a_n/\lambda_n$. So

$$y(x) = \sum_{n=1}^{\infty} \frac{a_n}{\lambda_n} y_n(x),$$

provided no $\lambda_n = 0$.

A common variant includes a driving term at fixed ‘frequency’: for the equation

$$\mathcal{L}y - \tilde{\lambda}wy = f(x),$$

with $\tilde{\lambda}$ a fixed constant, exactly the same computation gives

$$y(x) = \sum_{n=1}^{\infty} \frac{a_n}{\lambda_n - \tilde{\lambda}} y_n(x).$$

Notice what happens as $\tilde{\lambda}$ approaches an eigenvalue λ_n with $a_n \neq 0$: the response blows up. This is *resonance*, seen here in complete generality.

Towards Green’s Functions

Let’s rewrite our solution in a rather suggestive way. Substituting the integral formula for a_n into the solution, and writing $N_n = \int_a^b w y_n^2 dx$,

$$y(x) = \sum_{n=1}^{\infty} \frac{y_n(x)}{\lambda_n N_n} \int_a^b w(\xi) F(\xi) y_n(\xi) d\xi = \int_a^b \underbrace{\left(\sum_{n=1}^{\infty} \frac{y_n(x) y_n(\xi)}{\lambda_n N_n} \right)}_{G(x, \xi)} \underbrace{w(\xi) F(\xi)}_{f(\xi)} d\xi.$$

That is,

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi, \quad \text{where} \quad G(x, \xi) = \sum_{n=1}^{\infty} \frac{y_n(x) y_n(\xi)}{\lambda_n N_n}.$$

The function G , called the **Green’s function**, depends only on $\mathcal{L}$ and the boundary conditions – *not* on the source f . It acts as a kind of inverse operator

$$\mathcal{L}^{-1} = \int_a^b d\xi G(x, \xi),$$

in perfect analogy with $\mathbf{x} = A^{-1}\mathbf{b}$ for matrices. We will return to Green’s functions in earnest later in the course, where we will construct them directly (without infinite sums) and interpret them as responses to point sources.

§18.3 The Wave Equation

It’s time to put our machinery to work on an actual partial differential equation. The star of this section is the **wave equation**

$$\frac{1}{c^2} \frac{\partial^2 y}{\partial t^2} = \frac{\partial^2 y}{\partial x^2},$$

which governs (among many other things) the small vibrations of a stretched string.

§18.3.1 Waves on an Elastic String

Let's first see where the equation comes from. Consider a stretched elastic string with tension T and linear mass density μ , with ends fixed at $x = 0$ and $x = L$, and let $y(x, t)$ be the (small) transverse displacement. We make the key assumption that displacements are small, in the sense that $|\partial y / \partial x| \ll 1$.

Consider the segment of string between x and $x + \delta x$. Tension pulls the segment along the string at each end: at the right end at a small angle θ_2 to the horizontal, and at the left end at angle θ_1 . Resolving horizontally, since the string doesn't move sideways,

$$T_1 \cos \theta_1 = T_2 \cos \theta_2,$$

and as $\cos \theta = 1 + O(\theta^2)$, the tension is uniform along the string to leading order: $T_1 \approx T_2 = T$. Resolving vertically, using $\sin \theta \approx \tan \theta = \partial y / \partial x$ for small angles, the net transverse force is

$$T \left(\left. \frac{\partial y}{\partial x} \right|_{x+\delta x} - \left. \frac{\partial y}{\partial x} \right|_x \right) \approx T \frac{\partial^2 y}{\partial x^2} \delta x.$$

The mass of the segment is $\mu \delta x$, so Newton's second law (including gravity, with g downwards) gives

$$\mu \delta x \frac{\partial^2 y}{\partial t^2} = T \frac{\partial^2 y}{\partial x^2} \delta x - g \mu \delta x.$$

For a taut string the tension term utterly dominates gravity, so we neglect g , and dividing through by $\mu \delta x$ we obtain the wave equation

$$\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}, \quad \text{where } c = \sqrt{\frac{T}{\mu}}$$

is the **wave speed**. Alongside the boundary conditions $y(0, t) = y(L, t) = 0$, a full problem also specifies initial conditions

$$y(x, 0) = p(x), \quad \frac{\partial y}{\partial t}(x, 0) = q(x).$$

Note that we need *two* initial conditions, since the equation is second order in time.

§18.3.2 Separation of Variables

Our main technique for solving the wave equation (and later the diffusion and Laplace equations) is **separation of variables**: we look for solutions in the special product form

$$y(x, t) = X(x)T(t).$$

Substituting into the wave equation gives $\frac{1}{c^2} X \ddot{T} = X'' T$, and dividing by XT ,

$$\frac{1}{c^2} \frac{\ddot{T}}{T} = \frac{X''}{X}.$$

Now here's the trick: the left-hand side depends only on t , and the right-hand side only on x . The only way a function of t can equal a function of x for all x and t is if both are constant. Calling this **separation constant** $-\lambda$, we get two ODEs

$$X'' + \lambda X = 0, \quad \ddot{T} + \lambda c^2 T = 0.$$

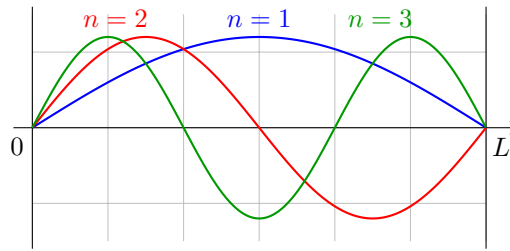
We solve the spatial problem first, since it carries the boundary conditions $X(0) = X(L) = 0$. This is exactly a Sturm–Liouville eigenvalue problem! Let's check the possible signs of λ .

- (i) If $\lambda < 0$, write $\chi^2 = -\lambda$; then $X = C \cosh \chi x + D \sinh \chi x$, and the boundary conditions force $C = D = 0$.
- (ii) If $\lambda = 0$, then $X = Ax + B$, and again only the trivial solution survives.
- (iii) If $\lambda > 0$, then $X = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x)$. Now $X(0) = 0$ gives $A = 0$, and $X(L) = 0$ requires $\sin(\sqrt{\lambda}L) = 0$, i.e. $\sqrt{\lambda}L = n\pi$.

So we find the eigenfunctions and eigenvalues

$$X_n(x) = \sin \frac{n\pi x}{L}, \quad \lambda_n = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, \dots$$

These spatial profiles are called the **normal modes** of the string: the shape in x is frozen, and only the amplitude oscillates in time. The $n = 1$ mode is the **fundamental**, and higher modes are *overtones*, with $n - 1$ stationary points ('nodes') in the interior.



With λ_n fixed, the temporal equation $\ddot{T} + (n\pi c/L)^2 T = 0$ has solutions

$$T_n(t) = C_n \cos \frac{n\pi ct}{L} + D_n \sin \frac{n\pi ct}{L},$$

so each normal mode oscillates at angular frequency $\omega_n = n\pi c/L$. Superposing everything (the equation is linear!), the general solution satisfying the boundary conditions is

$$y(x, t) = \sum_{n=1}^{\infty} \left(C_n \cos \frac{n\pi ct}{L} + D_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}.$$

It remains to impose the initial conditions. Setting $t = 0$,

$$y(x, 0) = p(x) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi x}{L},$$

which is precisely a half-range Fourier sine series, and differentiating in time first,

$$\frac{\partial y}{\partial t}(x, 0) = q(x) = \sum_{n=1}^{\infty} \frac{n\pi c}{L} D_n \sin \frac{n\pi x}{L}.$$

So by our Fourier coefficient formulas,

$$C_n = \frac{2}{L} \int_0^L p(x) \sin \frac{n\pi x}{L} dx, \quad D_n = \frac{2}{n\pi c} \int_0^L q(x) \sin \frac{n\pi x}{L} dx.$$

Example 18.3.1 (The Plucked String)

Take $L = 1$ and pluck the string at the point $x = \xi$: the initial condition is the

‘see-saw’ shape

$$y(x, 0) = p(x) = \begin{cases} x(1 - \xi) & \text{if } 0 \leq x < \xi, \\ \xi(1 - x) & \text{if } \xi \leq x < 1, \end{cases}$$

released from rest, $q(x) = 0$. We computed the Fourier series of this function before:

$$C_n = \frac{2 \sin n\pi\xi}{(n\pi)^2}, \quad D_n = 0,$$

so the subsequent motion is

$$y(x, t) = \sum_{n=1}^{\infty} \frac{2 \sin n\pi\xi}{(n\pi)^2} \sin(n\pi x) \cos(n\pi ct).$$

Notice that if you pluck at the midpoint $\xi = 1/2$, all the even harmonics vanish – plucking position controls the timbre.

Summarising, the general strategy of separation of variables is:

1. write down the linear PDE with its boundary and initial conditions;
2. separate variables to get decoupled ODEs;
3. impose the homogeneous boundary conditions to find the eigenvalues and eigenfunctions;
4. use the separation constants to solve for the remaining variables;
5. superpose the separable solutions into a general series solution;
6. fix the coefficients using the initial (or remaining boundary) conditions.

This exact recipe will reappear for the diffusion equation and Laplace’s equation.

§18.3.3 d’Alembert’s Solution and Characteristics

There’s a completely different way of looking at the wave equation, which doesn’t involve series at all. Using the sum formulas $\sin A \cos B = \frac{1}{2}(\sin(A - B) + \sin(A + B))$ and so on, each standing wave term in our series solution splits as

$$\begin{aligned} y(x, t) = & \frac{1}{2} \sum_{n=1}^{\infty} \left[C_n \sin \frac{n\pi}{L}(x - ct) + D_n \cos \frac{n\pi}{L}(x - ct) \right] \\ & + \frac{1}{2} \sum_{n=1}^{\infty} \left[C_n \sin \frac{n\pi}{L}(x + ct) - D_n \cos \frac{n\pi}{L}(x + ct) \right], \end{aligned}$$

that is, everything is a function of $x - ct$ plus a function of $x + ct$. This is no accident. Changing variables to $u = x - ct$, $v = x + ct$ transforms the wave equation into

$$\frac{\partial^2 y}{\partial u \partial v} = 0,$$

(we will verify this carefully when we study characteristics in general), which integrates immediately to give

$$y(x, t) = f(x - ct) + g(x + ct)$$

for arbitrary functions f and g : a right-moving wave and a left-moving wave, each travelling at speed c with unchanging shape.

Let's fit f and g to initial data on the whole line, $y(x, 0) = p(x)$ and $y_t(x, 0) = q(x)$. At $t = 0$,

$$f(x) + g(x) = p(x), \quad -cf'(x) + cg'(x) = q(x).$$

Differentiating the first equation and combining with the second,

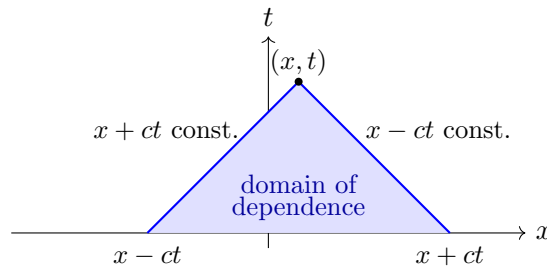
$$g'(x) = \frac{1}{2} \left(p'(x) + \frac{q(x)}{c} \right), \quad f'(x) = \frac{1}{2} \left(p'(x) - \frac{q(x)}{c} \right),$$

and integrating (the constants of integration cancel in $f + g$) we arrive at **d'Alembert's solution**

$$y(x, t) = \frac{1}{2} [p(x - ct) + p(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} q(s) \, ds.$$

In particular, a string released from rest ($q = 0$) evolves as $y = \frac{1}{2}[p(x - ct) + p(x + ct)]$ – the initial shape splits into two half-size copies moving apart.

d'Alembert's formula makes a physically important point manifest: the solution at position x and time t depends only on the initial data in the interval $[x - ct, x + ct]$. Information propagates no faster than speed c .



The lines $x \pm ct = \text{const}$ are called the **characteristics** of the wave equation. The shaded triangle is the **domain of dependence** of the point (x, t) ; dually, initial data at a point $(x_0, 0)$ can only ever influence the forward wedge $\{(x, t) : |x - x_0| \leq ct\}$, its **range of influence**.

§18.3.4 Energy of Oscillations

A vibrating string carries energy, and tracking it is both physically illuminating and a nice application of Parseval. The kinetic energy is

$$\text{KE} = \frac{1}{2} \mu \int_0^L \left(\frac{\partial y}{\partial t} \right)^2 dx,$$

and the potential energy is the work done stretching the string against tension. The stretched length element is $\sqrt{1 + (\partial y / \partial x)^2} dx$, so

$$\text{PE} = T \int_0^L \left(\sqrt{1 + \left(\frac{\partial y}{\partial x} \right)^2} - 1 \right) dx \approx \frac{T}{2} \int_0^L \left(\frac{\partial y}{\partial x} \right)^2 dx,$$

using the small-slope assumption. Hence, using $c^2 = T/\mu$, the total energy is

$$E = \frac{1}{2} \mu \int_0^L \left[\left(\frac{\partial y}{\partial t} \right)^2 + c^2 \left(\frac{\partial y}{\partial x} \right)^2 \right] dx.$$

Substituting our normal mode expansion and integrating using orthogonality (all the cross terms die, and $\int_0^L \sin^2 = \int_0^L \cos^2 = L/2$),

$$E = \frac{1}{4}\mu \sum_{n=1}^{\infty} \frac{n^2 \pi^2 c^2}{L} (C_n^2 + D_n^2),$$

where the time dependence has cancelled via $\cos^2 + \sin^2 = 1$. So the energy is *constant in time*, as it should be, and moreover it decomposes as a sum over normal modes – each mode holds its own energy and never trades it with the others. This is really Parseval’s theorem wearing a physics costume.

§18.3.5 Wave Reflection and Transmission

Let’s briefly consider what happens when a travelling wave meets a change in the medium. A **simple harmonic travelling wave** has the form

$$y = \text{Re} \left[A e^{i\omega(t-x/c)} \right] = |A| \cos(\omega(t-x/c) + \phi),$$

where the phase $\phi = \arg A$ and the wavelength is $2\pi c/\omega$.

Suppose the density of the string jumps at $x = 0$:

$$\mu = \begin{cases} \mu_- & \text{if } x < 0, \\ \mu_+ & \text{if } x > 0, \end{cases} \quad \text{so } c_{\pm} = \sqrt{T/\mu_{\pm}},$$

with the tension T constant throughout. Send in a wave $A e^{i\omega(t-x/c_-)}$ from the left. At the join, some of the wave is reflected, $B e^{i\omega(t+x/c_-)}$, and some is transmitted, $D e^{i\omega(t-x/c_+)}$. Note all three waves must share the frequency ω to have any hope of matching at $x = 0$ for all time.

The physical matching conditions at $x = 0$ are:

- (i) y is continuous (the string doesn’t break), giving $A + B = D$;
- (ii) the transverse force $T\partial y/\partial x$ is continuous (else a massless point would suffer infinite acceleration), giving

$$\frac{-i\omega A}{c_-} + \frac{i\omega B}{c_-} = \frac{-i\omega D}{c_+}.$$

Solving these two equations,

$$D = \frac{2c_+}{c_+ + c_-} A, \quad B = \frac{c_+ - c_-}{c_+ + c_-} A.$$

Various limits of this formula are illuminating:

- (i) If $c_- = c_+$ (no jump at all), then $D = A$ and $B = 0$: full transmission.
- (ii) If $\mu_+/\mu_- \rightarrow \infty$, the right half becomes infinitely heavy – effectively a fixed end (a *Dirichlet* boundary condition at $x = 0$). Then $c_+/c_- \rightarrow 0$, so $D = 0$ and $B = -A$: total reflection with phase flipped by π .
- (iii) If $\mu_+/\mu_- \rightarrow 0$, the right half is massless – a free end (a *Neumann* condition). Then $c_+/c_- \rightarrow \infty$, giving $B = A$: total reflection in phase (and the formula gives $D = 2A$, but this carries no energy as $\mu_+ \rightarrow 0$).

§18.3.6 The Wave Equation in Plane Polar Coordinates

So far our string has been one-dimensional. Let's now solve the two-dimensional wave equation on the unit disc – physically, a circular drum. We want $u(r, \theta, t)$ satisfying

$$\frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} = \nabla^2 u,$$

with boundary condition $u(1, \theta, t) = 0$ (the drum skin is clamped at the rim) and initial conditions

$$u(r, \theta, 0) = \phi(r, \theta), \quad \frac{\partial u}{\partial t}(r, \theta, 0) = \psi(r, \theta).$$

We separate off time first: substituting $u = T(t)V(r, \theta)$ gives

$$\ddot{T} + \lambda c^2 T = 0, \quad \nabla^2 V + \lambda V = 0.$$

In plane polars the spatial equation reads

$$\frac{\partial^2 V}{\partial r^2} + \frac{1}{r} \frac{\partial V}{\partial r} + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \lambda V = 0,$$

and separating again with $V = R(r)\Theta(\theta)$,

$$\Theta'' + \mu\Theta = 0, \quad r^2 R'' + rR' + (\lambda r^2 - \mu)R = 0.$$

On the disc, Θ must be 2π -periodic, which forces $\mu = m^2$ with $m \in \mathbb{N}$, and

$$\Theta_m(\theta) = A_m \cos m\theta + B_m \sin m\theta.$$

§18.3.7 Bessel's Equation

Now for the radial equation. Dividing by r puts it in Sturm–Liouville form,

$$-(rR')' + \frac{m^2}{r}R = \lambda rR,$$

with $p(r) = r$, $q(r) = m^2/r$ and weight $w(r) = r$. Note $p(0) = 0$, so $r = 0$ is a singular point, and the correct ‘boundary condition’ there is just that R stays bounded; at the rim we have $R(1) = 0$. Substituting $z = \sqrt{\lambda}r$ tidies away λ and yields **Bessel's equation**

$$z^2 \frac{d^2 R}{dz^2} + z \frac{dR}{dz} + (z^2 - m^2)R = 0.$$

Since $z = 0$ is a regular singular point, we solve by Frobenius' method, trying $R = z^p \sum_{n \geq 0} a_n z^n$. The lowest power of z gives the indicial equation

$$p^2 - m^2 = 0 \implies p = \pm m.$$

Taking the regular root $p = m$, comparing coefficients of z^{n+m} gives the recursion

$$((n+m)^2 - m^2)a_n + a_{n-2} = 0 \implies a_n = \frac{-a_{n-2}}{n(n+2m)},$$

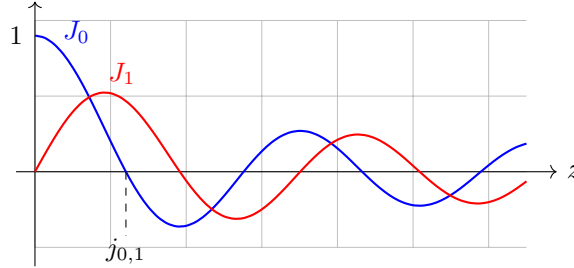
so only even n appear, and iterating,

$$a_{2k} = \frac{(-1)^k a_0}{2^{2k} k! (m+1)(m+2) \cdots (m+k)}.$$

With the conventional normalisation $a_0 = 1/(2^m m!)$ we obtain the **Bessel function of the first kind**

$$J_m(z) = \left(\frac{z}{2}\right)^m \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+m)!} \left(\frac{z}{2}\right)^{2k}.$$

The second solution (from $p = -m$, suitably interpreted) is the Bessel function of the second kind $Y_m(z)$, which diverges as $z \rightarrow 0$; boundedness at the origin will always eliminate it for us.



For small z , the leading behaviour is $J_0(z) \approx 1$ and $J_m(z) \approx \frac{1}{m!}(z/2)^m$. For large z , the Bessel functions become decaying oscillations:

$$J_m(z) \approx \sqrt{\frac{2}{\pi z}} \cos\left(z - \frac{m\pi}{2} - \frac{\pi}{4}\right).$$

In particular each J_m has infinitely many positive zeros. We write j_{mn} for the n th positive zero of J_m ; from the asymptotic form,

$$j_{mn} \approx n\pi + \frac{m\pi}{2} - \frac{\pi}{4} \quad \text{for large } n.$$

The zeros play exactly the role that $n\pi$ played for $\sin$: they are what the boundary condition will latch onto. (For reference, $j_{01} \approx 2.405$.)

§18.3.8 The Vibrating Drum

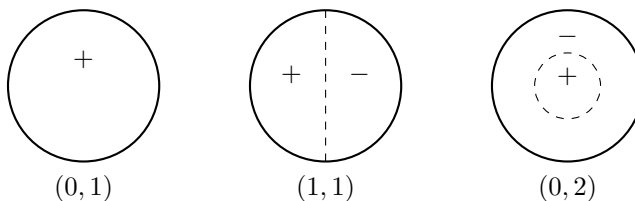
Let's assemble the solution of the drum problem. The radial solutions are $R(r) = AJ_m(\sqrt{\lambda}r) + BY_m(\sqrt{\lambda}r)$; boundedness at the centre gives $B = 0$, and the clamped rim $R(1) = 0$ requires

$$J_m(\sqrt{\lambda}) = 0 \implies \sqrt{\lambda} = j_{mn}, \quad \text{i.e. } \lambda_{mn} = j_{mn}^2.$$

So the spatial normal modes of the drum are

$$V_{mn}(r, \theta) = (A_{mn} \cos m\theta + B_{mn} \sin m\theta) J_m(j_{mn}r),$$

with frequencies $\omega_{mn} = j_{mn}c$. The mode (m, n) has m nodal diameters and $n - 1$ interior nodal circles.



The full solution is a superposition over all modes,

$$u(r, \theta, t) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} J_m(j_{mn}r) \left[(A_{mn} \cos m\theta + B_{mn} \sin m\theta) \cos(j_{mn}ct) \right. \\ \left. + (C_{mn} \cos m\theta + D_{mn} \sin m\theta) \sin(j_{mn}ct) \right],$$

and the coefficients are fixed by the initial conditions using the orthogonality of $\cos m\theta$, $\sin m\theta$ and of the Bessel functions themselves. Indeed, since the $J_m(j_{mn}r)$ for fixed m are eigenfunctions of a Sturm–Liouville problem with weight r , we have

$$\int_0^1 J_m(j_{mn}r) J_m(j_{mk}r) r \, dr = \frac{1}{2} [J_{m+1}(j_{mn})]^2 \delta_{nk}.$$

Example 18.3.2 (A Radially Symmetric Drum Strike)

Suppose the drum starts at rest from the radially symmetric profile $u(r, \theta, 0) = \phi(r) = 1 - r^2$. By symmetry only the $m = 0$ modes are excited, and starting from rest kills the $\sin(j_{0n}ct)$ terms, so

$$u(r, \theta, t) = \sum_{n=1}^{\infty} A_{0n} J_0(j_{0n}r) \cos(j_{0n}ct),$$

where

$$A_{0n} = \frac{\int_0^1 (1 - r^2) J_0(j_{0n}r) r \, dr}{\frac{1}{2} J_1(j_{0n})^2}.$$

Working out the integral using Bessel recursion relations, one finds

$$A_{0n} = \frac{8J_2(j_{0n})}{j_{0n}^2 J_1(j_{0n})^2}.$$

The fundamental frequency of a drum of diameter d is $\omega = j_{01}c \cdot 2/d \approx 4.8c/d$, compared with $\pi c/d \approx 3.1c/d$ for a string of length d . Moreover the overtone frequencies j_{0n} are *not* integer multiples of j_{01} – which is why a drum sounds like a thud rather than a note.

§18.4 The Diffusion Equation

The next equation in our collection is the **diffusion equation** (or *heat equation*)

$$\frac{\partial \theta}{\partial t} = D \nabla^2 \theta,$$

which describes how heat, or the concentration of diffusing particles, spreads out over time. It is first order in time – a fact with real consequences, as we’ll see: unlike the wave equation, diffusion has an arrow of time.

§18.4.1 Deriving the Diffusion Equation

Via Fourier's Law

Consider the temperature $\theta(\mathbf{x}, t)$ in a conducting body. The heat energy in a fixed volume V is

$$Q = \int_V c_V \rho \theta \, dV,$$

where c_V is the specific heat and ρ the density, so

$$\frac{dQ}{dt} = \int_V c_V \rho \frac{\partial \theta}{\partial t} \, dV.$$

Empirically, heat flows from hot to cold at a rate proportional to the temperature gradient. This is **Fourier's law**:

$$\mathbf{q} = -k \nabla \theta,$$

where $\mathbf{q}$ is the heat flux and k the conductivity. The heat leaving V through its boundary is then

$$-\frac{dQ}{dt} = \int_{\partial V} \mathbf{q} \cdot \hat{\mathbf{n}} \, dS = \int_{\partial V} (-k \nabla \theta) \cdot \hat{\mathbf{n}} \, dS = - \int_V k \nabla^2 \theta \, dV,$$

using the divergence theorem. Equating the two expressions,

$$\int_V \left(c_V \rho \frac{\partial \theta}{\partial t} - k \nabla^2 \theta \right) \, dV = 0,$$

and since V was arbitrary, the integrand must vanish identically. Writing $D = k/(c_V \rho)$, the **diffusion constant** (or diffusivity), we get the diffusion equation

$$\frac{\partial \theta}{\partial t} = D \nabla^2 \theta.$$

Via Random Walks

There is a rather different derivation which explains why the same equation governs diffusing gas particles. Suppose a particle takes a random step of displacement ξ every time interval Δt , where ξ is drawn from a probability density $p(\xi)$ with no directional bias, $\langle \xi \rangle = \int \xi p(\xi) \, d\xi = 0$. If $P(x, N\Delta t)$ is the probability density for the particle's position after N steps, then conditioning on the last step,

$$P(x, (N+1)\Delta t) = \int_{-\infty}^{\infty} p(\xi) P(x - \xi, N\Delta t) \, d\xi.$$

Taylor expanding P in its first argument about x ,

$$P(x, (N+1)\Delta t) \approx P(x, N\Delta t) - \langle \xi \rangle \frac{\partial P}{\partial x} + \frac{\langle \xi^2 \rangle}{2} \frac{\partial^2 P}{\partial x^2} = P(x, N\Delta t) + \frac{\langle \xi^2 \rangle}{2} \frac{\partial^2 P}{\partial x^2},$$

using $\int p = 1$ and $\langle \xi \rangle = 0$. So provided the variance scales with the time step as $\langle \xi^2 \rangle = 2D\Delta t$, taking Δt small gives

$$\frac{\partial P}{\partial t} = D \frac{\partial^2 P}{\partial x^2},$$

the diffusion equation again. The scaling $\langle \xi^2 \rangle \propto \Delta t$ is the fingerprint of diffusion: distance travelled grows like $\sqrt{t}$, not t .

§18.4.2 Similarity Solutions

That $\sqrt{t}$ scaling suggests that the combination $x/\sqrt{t}$ might be a natural variable, and indeed the diffusion equation admits **similarity solutions** depending only on the dimensionless variable

$$\eta = \frac{x}{2\sqrt{Dt}}.$$

Let's look for a solution $\theta(x, t) = \theta(\eta)$. Using the chain rule,

$$\frac{\partial \theta}{\partial t} = -\frac{1}{2} \frac{\eta}{t} \theta'(\eta), \quad D \frac{\partial^2 \theta}{\partial x^2} = D \left(\frac{1}{2\sqrt{Dt}} \right)^2 \theta''(\eta) = \frac{1}{4t} \theta''(\eta),$$

so the PDE collapses to the ODE

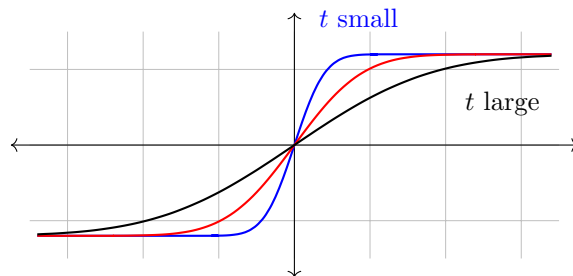
$$\theta'' = -2\eta \theta'.$$

Setting $\psi = \theta'$, we have $\psi'/\psi = -2\eta$, giving $\psi = Ce^{-\eta^2}$, and integrating once more (choosing the constant conveniently)

$$\theta(\eta) = \theta_0 \operatorname{erf}(\eta) = \theta_0 \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right), \quad \text{where} \quad \operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-u^2} du$$

is the **error function**.

This solution describes an initial step in temperature – say $\theta = \pm\theta_0$ for $x \gtrless 0$ at $t = 0^+$ – smoothing itself out, with the transition region spreading like $\sqrt{Dt}$:



Notice that at any $t > 0$, no matter how small, the solution is perfectly smooth and is nonzero at every x . Diffusion smooths instantly, and (formally) propagates information infinitely fast – in sharp contrast with the wave equation's strict speed limit c .

§18.4.3 Heat Conduction in a Finite Bar

Now let's do a full boundary value problem by separation of variables. Take a bar of length $2L$ occupying $-L \leq x \leq L$, with initial temperature a step,

$$\theta(x, 0) = H(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq L, \\ 0 & \text{if } -L \leq x < 0, \end{cases}$$

and ends held at fixed temperatures $\theta(L, t) = 1$ and $\theta(-L, t) = 0$.

There's an immediate obstacle: the boundary conditions aren't homogeneous, so our Sturm–Liouville machinery doesn't apply directly. The fix is to subtract off the **steady state** – the time-independent solution the bar will relax towards. Steady solutions of

the heat equation in one dimension satisfy $\theta_s'' = 0$, so $\theta_s = Ax + B$, and matching the boundary values,

$$\theta_s(x) = \frac{x + L}{2L}.$$

Now write $\hat{\theta}(x, t) = \theta(x, t) - \theta_s(x)$. Then $\hat{\theta}$ still satisfies the diffusion equation, but now with *homogeneous* boundary conditions $\hat{\theta}(\pm L, t) = 0$ and initial condition $\hat{\theta}(x, 0) = H(x) - \frac{x+L}{2L}$.

Separating variables, $\hat{\theta} = X(x)T(t)$ gives

$$X'' = -\lambda X, \quad \dot{T} = -D\lambda T.$$

The boundary conditions force $\lambda > 0$ as usual. Our initial condition $\hat{\theta}(x, 0)$ is an odd function of x (check this!), so we only need the odd eigenfunctions

$$X_n(x) = \sin \frac{n\pi x}{L}, \quad \lambda_n = \frac{n^2\pi^2}{L^2},$$

and the corresponding temporal solutions are decaying exponentials,

$$T_n(t) = \exp\left(-\frac{Dn^2\pi^2}{L^2}t\right).$$

So

$$\hat{\theta}(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L} \exp\left(-\frac{Dn^2\pi^2}{L^2}t\right).$$

At $t = 0$ this is a Fourier sine series, so

$$b_n = \frac{2}{L} \int_0^L \left(H(x) - \frac{x+L}{2L}\right) \sin \frac{n\pi x}{L} dx = \frac{2}{L} \int_0^L \frac{1}{2} \sin \frac{n\pi x}{L} dx - \frac{2}{L} \int_0^L \frac{x}{2L} \sin \frac{n\pi x}{L} dx.$$

The first integral is half a square wave, contributing $\frac{2}{n\pi}$ for odd n and 0 for even n ; the second is half a sawtooth, contributing $-\frac{(-1)^{n+1}}{n\pi}$. Rather pleasingly, in both the odd and even cases the total comes to the same thing:

$$b_n = \frac{1}{n\pi}.$$

Hence, restoring the steady state, the temperature of the bar is

$$\theta(x, t) = \frac{x+L}{2L} + \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin \frac{n\pi x}{L} \exp\left(-\frac{Dn^2\pi^2}{L^2}t\right).$$

There are two things worth noticing here. First, mode n decays at rate $Dn^2\pi^2/L^2$ – high harmonics die *extremely* fast, so the solution becomes smooth almost immediately and the late-time behaviour is dominated by the steady state plus the slowest mode. Secondly, for early times the series converges slowly, but there the similarity solution $\frac{1}{2} \left(1 + \operatorname{erf}\left(x/2\sqrt{Dt}\right)\right)$ is an excellent approximation, since the step hasn't yet 'felt' the boundaries. The two descriptions hand over to each other around $t \sim L^2/D$, the natural diffusion timescale.

§18.5 Laplace's Equation

If we look for *steady* states of the wave or diffusion equations – solutions with no time dependence at all – both reduce to **Laplace's equation**

$$\nabla^2 \phi = 0.$$

Solutions of Laplace's equation are called **harmonic functions**. The equation shows up everywhere: steady heat flow, electrostatic and gravitational potentials ($\mathbf{F} = -\nabla\phi$), and irrotational incompressible fluid flow ($\mathbf{v} = \nabla\phi$).

Laplace's equation is solved on a domain D , with data on the boundary ∂D rather than initial conditions. The two standard types are:

- (i) **Dirichlet conditions**: ϕ specified on ∂D ;
- (ii) **Neumann conditions**: the normal derivative $\partial\phi/\partial n = \hat{\mathbf{n}} \cdot \nabla\phi$ specified on ∂D .

§18.5.1 Uniqueness

Before solving anything, let's establish that boundary data pins the solution down.

Theorem 18.5.1 (Uniqueness for the Dirichlet Problem)

Suppose $\nabla^2 \phi = 0$ on a bounded domain D with $\phi = f$ given on ∂D . Then the solution, if it exists, is unique.

Proof. Suppose ϕ_1, ϕ_2 are both solutions and let $\psi = \phi_1 - \phi_2$, so that $\nabla^2 \psi = 0$ in D and $\psi = 0$ on ∂D . Consider

$$\int_D |\nabla\psi|^2 \, dV = \int_D [\nabla \cdot (\psi \nabla\psi) - \psi \nabla^2 \psi] \, dV = \int_{\partial D} \psi \frac{\partial\psi}{\partial n} \, dS = 0,$$

using the divergence theorem and then both $\nabla^2 \psi = 0$ and $\psi = 0$ on ∂D . Since $|\nabla\psi|^2 \geq 0$ and integrates to zero, $\nabla\psi \equiv 0$, so ψ is constant on D ; the boundary condition then forces $\psi \equiv 0$. $\square$

For Neumann conditions the same argument shows the solution is unique *up to an additive constant*, which is the best one could hope for.

The plan for the rest of this section is to run separation of variables in each of the standard coordinate systems – Cartesian, plane polar, cylindrical and spherical – and see which special functions each geometry summons.

§18.5.2 Cartesian Coordinates

In $\mathbb{R}^3$, Laplace's equation reads $\phi_{xx} + \phi_{yy} + \phi_{zz} = 0$. Seeking $\phi = X(x)Y(y)Z(z)$ and dividing by XYZ ,

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0.$$

Each ratio must be constant, and the three constants sum to zero – so they can't all have the same sign. If the boundary conditions make X and Y oscillatory, say $X''/X = -\lambda_\ell$

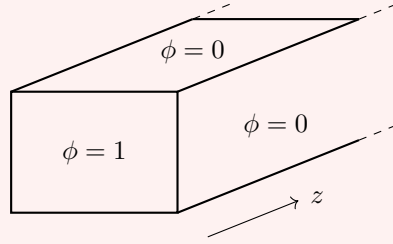
and $Y''/Y = -\lambda_m$ with $\lambda_\ell, \lambda_m > 0$, then necessarily

$$\frac{Z''}{Z} = \lambda_\ell + \lambda_m > 0,$$

and Z is a growing/decaying exponential. This is characteristic of Laplace's equation: it cannot oscillate in every direction at once.

Example 18.5.2 (Steady Heat Flow in a Semi-Infinite Bar)

Consider a bar with rectangular cross-section, occupying $0 \leq x \leq a$, $0 \leq y \leq b$, $z \geq 0$. The four long sides are held at $\phi = 0$, the end $z = 0$ is held at $\phi = 1$, and $\phi \rightarrow 0$ far down the bar as $z \rightarrow \infty$. What is the steady temperature inside?



The homogeneous conditions $X(0) = X(a) = 0$ and $Y(0) = Y(b) = 0$ give the familiar eigenfunctions

$$X_\ell(x) = \sin \frac{\ell\pi x}{a}, \quad \lambda_\ell = \frac{\ell^2\pi^2}{a^2}, \quad Y_m(y) = \sin \frac{m\pi y}{b}, \quad \lambda_m = \frac{m^2\pi^2}{b^2}.$$

Then $Z'' = (\lambda_\ell + \lambda_m)Z$, and discarding the growing exponential (since $\phi \rightarrow 0$ as $z \rightarrow \infty$),

$$Z_{\ell m}(z) = \exp \left[- \left(\frac{\ell^2}{a^2} + \frac{m^2}{b^2} \right)^{1/2} \pi z \right].$$

The general solution is

$$\phi = \sum_{\ell, m \geq 1} a_{\ell m} \sin \frac{\ell\pi x}{a} \sin \frac{m\pi y}{b} \exp \left[- \left(\frac{\ell^2}{a^2} + \frac{m^2}{b^2} \right)^{1/2} \pi z \right],$$

and imposing $\phi = 1$ at $z = 0$ is a double Fourier sine series:

$$a_{\ell m} = \left(\frac{2}{a} \int_0^a \sin \frac{\ell\pi x}{a} dx \right) \left(\frac{2}{b} \int_0^b \sin \frac{m\pi y}{b} dy \right) = \frac{4}{\ell\pi} \cdot \frac{4}{m\pi} = \frac{16}{\pi^2 \ell m}$$

for ℓ, m both odd, and 0 otherwise (each factor is the square wave computation from before). Note that far down the bar, the solution is utterly dominated by the $(\ell, m) = (1, 1)$ mode: boundary detail at $z = 0$ is exponentially forgotten, with the crudest features persisting longest.

§18.5.3 Plane Polar Coordinates

In two dimensions with polar coordinates (r, θ) , Laplace's equation is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} = 0.$$

Separating $\phi = R(r)\Theta(\theta)$ gives

$$\Theta'' + \mu\Theta = 0, \quad r(rR')' - \mu R = 0.$$

Periodicity of Θ forces $\mu = m^2$, $m \in \mathbb{N}$, with $\Theta_m = \cos m\theta, \sin m\theta$.

The radial equation this time is *not* Bessel's equation – there is no second separation constant (compare the wave equation on the disc, where λ appeared). It is an equidimensional equation, so we try $R = r^\beta$:

$$\beta^2 = m^2 \implies \beta = \pm m,$$

giving $R_m = r^m$ and r^{-m} for $m \geq 1$. For $m = 0$ we get $(rR')' = 0$, so $R' = c/r$ and

$$R_0 = \text{const}, \log r.$$

The general solution is therefore

$$\phi(r, \theta) = \frac{a_0}{2} + c_0 \log r + \sum_{m=1}^{\infty} (a_m \cos m\theta + b_m \sin m\theta) r^m + \sum_{m=1}^{\infty} (c_m \cos m\theta + d_m \sin m\theta) r^{-m}.$$

On a domain containing the origin, regularity kills $\log r$ and all the r^{-m} ; on an exterior domain, boundedness kills the r^m instead.

Example 18.5.3 (A Soap Film on a Distorted Wire)

Take a circular wire of radius 1, distorted slightly out of its plane so that its height is $f(\theta)$, and dip it in soap solution. The (small) height $\phi(r, \theta)$ of the resulting film satisfies Laplace's equation on the unit disc with $\phi(1, \theta) = f(\theta)$.

Regularity at the origin leaves

$$\phi(r, \theta) = \frac{a_0}{2} + \sum_{m=1}^{\infty} (a_m \cos m\theta + b_m \sin m\theta) r^m,$$

and at $r = 1$ this is exactly the Fourier series of f :

$$a_m = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \cos m\theta \, d\theta, \quad b_m = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin m\theta \, d\theta.$$

The factor r^m means the high harmonics of the wire's wobble decay rapidly towards the centre of the film: near $r = 0$ the film is almost flat at the average height $\frac{1}{2}a_0$. Once again, Laplace's equation smooths.

§18.5.4 Cylindrical Polar Coordinates

In cylindrical polars (r, θ, z) ,

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2} = 0.$$

Separating $\phi = R(r)\Theta(\theta)Z(z)$ gives

$$\Theta'' = -m^2\Theta, \quad Z'' = k^2Z, \quad r(rR')' + (k^2r^2 - m^2)R = 0.$$

Now the radial equation *is* Bessel's equation (in the variable kr), because the z separation supplies the second constant k^2 . The bounded-at-the-axis solutions are $R = J_m(kr)$.

If, say, we solve inside a cylinder of radius a with $\phi = 0$ on the curved surface, then $J_m(ka) = 0$ quantises $k = j_{mn}/a$, and the general solution decaying as $z \rightarrow \infty$ is

$$\phi(r, \theta, z) = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} (a_{mn} \cos m\theta + b_{mn} \sin m\theta) J_m\left(\frac{j_{mn}}{a}r\right) e^{-j_{mn}z/a},$$

with coefficients determined from the data on the end $z = 0$ using the orthogonality relation for Bessel functions from before. Structurally this is identical to the Cartesian semi-infinite bar: oscillation across the cross-section, exponential decay along the axis.

§18.5.5 Spherical Polar Coordinates

Finally, spherical polars (r, θ, φ) , where

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \phi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \phi}{\partial \varphi^2} = 0.$$

We will restrict to the **axisymmetric** case, where $\phi = \phi(r, \theta)$ has no φ dependence. Separating $\phi = R(r)\Theta(\theta)$,

$$(\sin \theta \Theta')' + \lambda \sin \theta \Theta = 0, \quad (r^2 R')' = \lambda R.$$

For the angular equation, substitute $x = \cos \theta$ (so $x \in [-1, 1]$, and $\frac{d}{d\theta} = -\sin \theta \frac{d}{dx}$):

$$\frac{d}{dx} \left[(1-x^2) \frac{d\Theta}{dx} \right] + \lambda \Theta = 0.$$

This is precisely Legendre's equation! Requiring finiteness at the poles $\theta = 0, \pi$ (i.e. $x = \pm 1$) gives

$$\lambda_\ell = \ell(\ell+1), \quad \Theta_\ell(\theta) = P_\ell(\cos \theta).$$

So this is where the Legendre polynomials live: they are the natural angular harmonics on the sphere.

The radial equation $(r^2 R')' = \ell(\ell+1)R$ is again equidimensional; trying $R = r^\beta$ gives

$$\beta(\beta+1) = \ell(\ell+1) \implies \beta = \ell \text{ or } -\ell-1,$$

so the general axisymmetric harmonic function is

$$\phi(r, \theta) = \sum_{\ell=0}^{\infty} \left(a_\ell r^\ell + \frac{b_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta).$$

Interior problems keep only r^ℓ ; exterior problems (with $\phi \rightarrow 0$ at infinity) keep only $r^{-\ell-1}$.

Example 18.5.4 (Boundary Data on the Unit Sphere)

Solve $\nabla^2 \phi = 0$ inside the unit sphere with $\phi(1, \theta) = f(\theta)$. Regularity gives $b_\ell = 0$, so writing $f(\theta) = F(\cos \theta)$,

$$F(x) = \sum_{\ell=0}^{\infty} a_\ell P_\ell(x) \implies a_\ell = \frac{2\ell+1}{2} \int_{-1}^1 F(x) P_\ell(x) dx,$$

a Fourier–Legendre series. For instance if $f(\theta) = \cos^2 \theta$, then $x^2 = \frac{1}{3}P_0(x) + \frac{2}{3}P_2(x)$, so

$$\phi(r, \theta) = \frac{1}{3} + \frac{2}{3}r^2 P_2(\cos \theta),$$

and no infinite series is needed at all.

The Generating Function for Legendre Polynomials

We can now pay off a debt from Sturm–Liouville theory. Place a unit point charge at the north pole of the unit sphere, $\mathbf{r}_0 = (0, 0, 1)$. Its potential at the point with spherical coordinates (r, θ) is

$$\Phi = \frac{1}{|\mathbf{r} - \mathbf{r}_0|} = \frac{1}{\sqrt{r^2 - 2r \cos \theta + 1}},$$

expanding $|\mathbf{r} - \mathbf{r}_0|^2 = r^2 \sin^2 \theta + (r \cos \theta - 1)^2$. Away from the charge this is an axisymmetric harmonic function, regular at the origin, so for $r < 1$ it *must* have an expansion

$$\frac{1}{\sqrt{r^2 - 2rx + 1}} = \sum_{\ell=0}^{\infty} a_{\ell} P_{\ell}(x) r^{\ell}, \quad x = \cos \theta.$$

To find the a_{ℓ} , look along the axis $x = 1$: the left side is $1/(1-r) = \sum_{\ell} r^{\ell}$, and $P_{\ell}(1) = 1$, so $a_{\ell} = 1$ for every ℓ . Hence

$$\frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{\ell=0}^{\infty} P_{\ell}(x) t^{\ell},$$

which is the generating function quoted earlier. Expanding the potential of a localised charge distribution in powers of $1/r$ using this identity gives the *multipole expansion* of electrostatics: monopole, dipole, quadrupole terms proportional to P_0, P_1, P_2 and so on.

§18.6 The Dirac Delta Function and Distributions

We have repeatedly hinted that the right way to think about Green’s functions is as responses to *point sources*: a unit impulse concentrated at a single point. To make sense of that we need a ‘function’ which is zero everywhere except one point, yet has total integral 1. No actual function does this, but the idea is far too useful to abandon – so in this section we develop the object that makes it precise.

§18.6.1 The Delta Function

Definition 18.6.1 (Dirac Delta)

The **Dirac delta function** $\delta(x - \xi)$ is the generalised function satisfying

- (i) $\delta(x - \xi) = 0$ for all $x \neq \xi$;
- (ii) $\int_{-\infty}^{\infty} \delta(x - \xi) dx = 1$.

Its defining role is the **sampling property**: for any f well-behaved near ξ ,

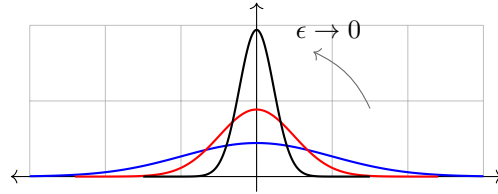
$$\int_{-\infty}^{\infty} f(x) \delta(x - \xi) dx = f(\xi).$$

Strictly, δ is not a function at all but a *distribution*, and it only really makes sense sitting inside an integral (though we will happily write equations involving δ outside integrals, with an integral always implied). Physically, $\delta(x - \xi)$ represents a unit point mass, charge, or impulse located at ξ .

A good mental model for δ is as the limit of increasingly tall, thin spikes of unit area. For instance, take the Gaussians

$$\delta_\epsilon(x) = \frac{1}{\epsilon\sqrt{\pi}} \exp\left(-\frac{x^2}{\epsilon^2}\right),$$

as $\epsilon \rightarrow 0$:



Let's verify the sampling property for this family. Substituting $y = x/\epsilon$,

$$\begin{aligned} \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \delta_\epsilon(x) f(x) \, dx &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-y^2} f(\epsilon y) \, dy \\ &= \lim_{\epsilon \rightarrow 0} \int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}} e^{-y^2} [f(0) + \epsilon y f'(0) + \cdots] \, dy = f(0), \end{aligned}$$

using $\int e^{-y^2} \, dy = \sqrt{\pi}$. The Gaussian family is not special: the *Dirichlet kernel* $\delta_n(x) = \frac{\sin nx}{\pi x}$ and spike families like $\frac{n}{2} \operatorname{sech}^2(nx)$ work equally well as $n \rightarrow \infty$. The limit object is the same δ in every case.

§18.6.2 Distributions and Test Functions

Let's spend a moment making this respectable. The trick is to stop asking for the value of δ at points, and instead define it by *how it acts on nice functions*.

Definition 18.6.2 (Test Function)

A **test function** $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is a smooth function which vanishes outside some bounded interval.

Definition 18.6.3 (Distribution)

A **distribution** (or **generalised function**) T is a continuous linear map from test functions to $\mathbb{R}$, written

$$T[\varphi] = \int_{-\infty}^{\infty} T(x) \varphi(x) \, dx,$$

where the integral notation is formal.

Every ordinary (integrable) function f defines a distribution via $\varphi \mapsto \int f \varphi \, dx$, but there are more distributions than functions: the delta function is the distribution

$$\delta[\varphi] = \varphi(0),$$

which is a perfectly well-defined linear map, even though no function represents it.

The real payoff is that distributions can always be differentiated. If T were an honest differentiable function, integration by parts would give $\int T'\varphi \, dx = -\int T\varphi' \, dx$ (the boundary terms vanish since φ has bounded support). So we simply *define* the **distributional derivative** by

$$T'[\varphi] = -T[\varphi'].$$

With this definition every distribution is infinitely differentiable, and we can differentiate functions with corners and jumps without guilt.

Example 18.6.4 (Derivative of the Heaviside Function)

The **Heaviside step function** is

$$H(x) = \begin{cases} 1 & \text{if } x \geq 0, \\ 0 & \text{if } x < 0. \end{cases}$$

As a distribution,

$$H'[\varphi] = -H[\varphi'] = -\int_0^\infty \varphi'(x) \, dx = \varphi(0) = \delta[\varphi],$$

so $H' = \delta$ in the sense of distributions – the jump differentiates to a spike. Consistently, $H(x) = \int_{-\infty}^x \delta(t) \, dt$.

Example 18.6.5 (Derivative of the Delta Function)

Similarly δ' is defined by

$$\int_{-\infty}^\infty \delta'(x - \xi) f(x) \, dx = -\int_{-\infty}^\infty \delta(x - \xi) f'(x) \, dx = -f'(\xi),$$

valid whenever f is smooth at ξ . One can visualise δ' as the limit of the derivatives of our Gaussian spikes: a violent up-down pair, which ‘measures’ minus the slope of f at the point.

§18.6.3 Properties of the Delta Function

The following properties get used constantly; each is proved by substitution inside an integral against a well-behaved f .

(i) (Sampling on an interval)

$$\int_a^b f(x) \delta(x - \xi) \, dx = \begin{cases} f(\xi) & \text{if } a < \xi < b, \\ 0 & \text{otherwise.} \end{cases}$$

(ii) (Evenness) $\delta(-(x - \xi)) = \delta(x - \xi)$, by the substitution $u = -(x - \xi)$.

(iii) (Scaling) For $a \neq 0$,

$$\delta(a(x - \xi)) = \frac{1}{|a|} \delta(x - \xi).$$

(iv) (Composition) If g has isolated simple roots $x_1, \dots, x_n$ (so $g'(x_i) \neq 0$), then

$$\delta(g(x)) = \sum_{i=1}^n \frac{\delta(x - x_i)}{|g'(x_i)|},$$

since near each root $g(x) \approx g'(x_i)(x - x_i)$ and we can apply scaling.

(v) (Isolation) If g is continuous at ξ then $g(x)\delta(x - \xi) = g(\xi)\delta(x - \xi)$.

§18.6.4 Eigenfunction Expansions of the Delta Function

Here is a question that will matter a lot for Green's functions: what is the Fourier series of $\delta(x)$? Working on $[-L, L]$ with the complex series $\delta(x) = \sum_n c_n e^{in\pi x/L}$, the coefficients are

$$c_n = \frac{1}{2L} \int_{-L}^L \delta(x) e^{-in\pi x/L} dx = \frac{1}{2L},$$

all equal! So

$$\delta(x) = \frac{1}{2L} \sum_{n=-\infty}^{\infty} e^{in\pi x/L}.$$

This series diverges wildly at $x = 0$, exactly as it should. It is 'equal' to δ in the distributional sense: integrated against any Fourier-expandable f , term-by-term integration gives $f(0)$. Extended periodically, this is a sum of delta spikes at $x = 2mL$ for $m \in \mathbb{Z}$, called the **Dirac comb**.

More generally, take any Sturm–Liouville system $\{y_n\}$ with weight w on $[a, b]$. Expanding $\delta(x - \xi) = \sum_n a_n y_n(x)$ using the standard coefficient formula,

$$a_n = \frac{\int_a^b w(x) y_n(x) \delta(x - \xi) dx}{\int_a^b w y_n^2 dx} = \frac{w(\xi) y_n(\xi)}{N_n}, \quad N_n = \int_a^b w y_n^2 dx,$$

so

$$\delta(x - \xi) = w(\xi) \sum_{n=1}^{\infty} \frac{y_n(\xi) y_n(x)}{N_n} = w(x) \sum_{n=1}^{\infty} \frac{y_n(\xi) y_n(x)}{N_n},$$

where swapping $w(\xi)$ for $w(x)$ is legitimate by the isolation property. This formula is the *completeness relation* in disguise – indeed, it's often taken as the definition of completeness of the family $\{y_n\}$.

Example 18.6.6

For the sine functions $y_n = \sin n\pi x$ on $[0, 1]$ (so $w = 1$, $N_n = 1/2$),

$$\delta(x - \xi) = 2 \sum_{n=1}^{\infty} \sin(n\pi\xi) \sin(n\pi x), \quad 0 < \xi < 1.$$

§18.7 Green's Functions for ODEs

We now come back to the idea we extracted from Sturm–Liouville theory: solving $\mathcal{L}y = f$ by finding an integral kernel G with

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi.$$

Armed with the delta function, we can now say precisely what G is, and construct it directly.

§18.7.1 Motivation: A Massive String

Consider a static string with tension T and mass density μ , suspended between fixed ends $y(0) = y(1) = 0$ and sagging under gravity. Balancing forces as in our wave equation derivation (but with no time dependence),

$$T \frac{d^2 y}{dx^2} - \mu g = 0, \quad \text{i.e.} \quad -y'' = f, \quad f = -\frac{\mu g}{T}.$$

We can of course just integrate twice: imposing the boundary conditions gives

$$y(x) = -\frac{\mu g}{T} \cdot \frac{1}{2} x(1-x).$$

But here is a more instructive route. Imagine replacing the continuous string by a very light string carrying a single point mass $\delta m = \mu \delta x$ at position ξ . The string is then straight on either side of the mass, and resolving forces at the mass point (small angles again) gives the tent-shaped displacement

$$y(x) = -\frac{\delta m g}{T} \times \begin{cases} x(1-\xi) & \text{if } x < \xi, \\ \xi(1-x) & \text{if } x > \xi. \end{cases}$$

Now, the equation is linear, so we can superpose: hanging point masses at positions $\xi_1, \xi_2, \dots$ adds the corresponding displacements. Taking the continuum limit, with $f(\xi) d\xi$ playing the role of the force from the mass element at ξ ,

$$y(x) = \int_0^1 f(\xi) G(x, \xi) d\xi, \quad \text{where } G(x, \xi) = \begin{cases} x(1-\xi) & \text{if } x < \xi, \\ \xi(1-x) & \text{if } x > \xi \end{cases}$$

is the response to a *unit* point force at ξ . You can check by direct integration that this reproduces $y = -\frac{\mu g}{2T} x(1-x)$ exactly. The function G – the shape of a string plucked by a unit point load – is the Green's function of the operator $-d^2/dx^2$ with these boundary conditions.

§18.7.2 Definition and the Jump Condition

Definition 18.7.1 (Green's Function)

Let $\mathcal{L}y = \alpha(x)y'' + \beta(x)y' + \gamma(x)y$ on $a \leq x \leq b$, with $\alpha \neq 0$ and α, β, γ continuous and bounded, with homogeneous boundary conditions $y(a) = y(b) = 0$. The **Green's function** of $\mathcal{L}$ is the solution $G(x, \xi)$, satisfying the boundary conditions, of

$$\mathcal{L}G(x, \xi) = \delta(x - \xi).$$

Given the Green's function, the solution of $\mathcal{L}y = f$ is immediate. Setting

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi,$$

we can check

$$\mathcal{L}y = \int_a^b \mathcal{L}G(x, \xi) f(\xi) d\xi = \int_a^b \delta(x - \xi) f(\xi) d\xi = f(x),$$

and y inherits the homogeneous boundary conditions from G . So G genuinely is the inverse operator: $\mathcal{L}^{-1} = \int_a^b d\xi G(x, \xi)$.

What does G look like? Away from $x = \xi$ the delta function vanishes, so G solves the *homogeneous* equation on each side:

$$G(x, \xi) = \begin{cases} G_1(x, \xi) & \text{if } a \leq x < \xi, \\ G_2(x, \xi) & \text{if } \xi < x \leq b, \end{cases} \quad \mathcal{L}G_1 = \mathcal{L}G_2 = 0,$$

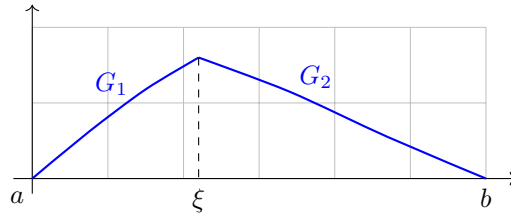
with $G_1(a, \xi) = 0$ and $G_2(b, \xi) = 0$. It remains to see how the two halves glue at $x = \xi$, and this is where the delta function makes itself felt:

- (i) *Continuity*: $G_1(\xi, \xi) = G_2(\xi, \xi)$;
- (ii) *Jump condition*: the derivative jumps by exactly

$$\left[\frac{\partial G}{\partial x} \right]_{\xi-}^{\xi+} = \frac{1}{\alpha(\xi)}.$$

To see why, integrate $\mathcal{L}G = \delta(x - \xi)$ over a shrinking interval $[\xi - \epsilon, \xi + \epsilon]$. The right side integrates to 1. On the left, $\int \alpha G'' dx \rightarrow \alpha(\xi)[G']_{\xi-}^{\xi+}$, while the $\beta G'$ and γG terms contribute nothing in the limit *provided* G is continuous (a jump in G itself would make G'' too singular – a δ' – which nothing on the right-hand side could match). Hence the two conditions above.

So the picture to hold in mind: $G(\cdot, \xi)$ is built from homogeneous solutions on either side, tied down at the ends, continuous at ξ , with a kink there of size $1/\alpha(\xi)$.



§18.7.3 Explicit Construction

Let's carry out the construction in general. Let y_1, y_2 be independent solutions of $\mathcal{L}y = 0$. Some linear combination $y_-(x)$ satisfies the left boundary condition $y_-(a) = 0$, and some combination $y_+(x)$ satisfies $y_+(b) = 0$. Then

$$G_1 = Cy_-(x), \quad G_2 = Dy_+(x),$$

for constants (in x !) C, D which may depend on ξ . The gluing conditions read

$$Cy_-(\xi) = Dy_+(\xi), \quad Dy'_+(\xi) - Cy'_-(\xi) = \frac{1}{\alpha(\xi)}.$$

Solving this pair of linear equations,

$$C(\xi) = \frac{y_+(\xi)}{\alpha(\xi)W(\xi)}, \quad D(\xi) = \frac{y_-(\xi)}{\alpha(\xi)W(\xi)},$$

where

$$W(\xi) = y_-(\xi)y'_+(\xi) - y_+(\xi)y'_-(\xi)$$

is the **Wronskian** of y_- and y_+ , which is nonzero as long as y_- and y_+ are independent.¹ Hence

$$G(x, \xi) = \begin{cases} \frac{y_-(x)y_+(\xi)}{\alpha(\xi)W(\xi)} & \text{if } a \leq x \leq \xi, \\ \frac{y_-(\xi)y_+(x)}{\alpha(\xi)W(\xi)} & \text{if } \xi \leq x \leq b, \end{cases}$$

and the solution to our boundary value problem is

$$y(x) = y_+(x) \int_a^x \frac{y_-(\xi)f(\xi)}{\alpha(\xi)W(\xi)} d\xi + y_-(x) \int_x^b \frac{y_+(\xi)f(\xi)}{\alpha(\xi)W(\xi)} d\xi.$$

(Note which piece of G applies in which integral: for $\xi < x$ we are to the right of the source, so $G = G_2$.)

A pleasant fact: if $\mathcal{L}$ is in Sturm–Liouville form, then $\beta = \alpha'$, and a quick computation shows $(\alpha W)' = \alpha W' + \alpha' W = 0$ using the ODE – so $\alpha(\xi)W(\xi)$ is a *constant*, and the Green's function is manifestly symmetric:

$$G(x, \xi) = G(\xi, x).$$

This is the analogue of a self-adjoint matrix having a symmetric inverse, and physically it is a reciprocity principle: the response at x to a source at ξ equals the response at ξ to a source at x .

Example 18.7.2

Let's solve $y'' - y = f(x)$ with $y(0) = y(1) = 0$. The homogeneous solutions are $e^{\pm x}$, and the combinations adapted to the boundary conditions are

$$y_-(x) = \sinh x, \quad y_+(x) = \sinh(1 - x).$$

Here $\alpha = 1$ and

$$\alpha W = \sinh x \cdot (-\cosh(1 - x)) - \sinh(1 - x) \cosh x = -\sinh 1,$$

using the addition formula (constant, as promised). So

$$G(x, \xi) = \begin{cases} -\frac{\sinh x \sinh(1 - \xi)}{\sinh 1} & \text{if } 0 \leq x \leq \xi, \\ -\frac{\sinh \xi \sinh(1 - x)}{\sinh 1} & \text{if } \xi \leq x \leq 1, \end{cases}$$

and

$$y(x) = -\frac{\sinh(1 - x)}{\sinh 1} \int_0^x \sinh \xi f(\xi) d\xi - \frac{\sinh x}{\sinh 1} \int_x^1 \sinh(1 - \xi) f(\xi) d\xi.$$

If the boundary conditions are *inhomogeneous*, say $y(0) = 0$, $y(1) = 1$, we handle them exactly as in the heat conduction problem: find any homogeneous solution y_p with $\mathcal{L}y_p = 0$ matching the boundary values (here $y_p = \sinh x / \sinh 1$), subtract it off, and solve the resulting homogeneous-boundary problem with the Green's function.

¹If y_- and y_+ are *not* independent, then there is a nontrivial homogeneous solution satisfying both boundary conditions – i.e. $\lambda = 0$ is an eigenvalue – and no Green's function exists. Compare trying to invert a singular matrix.

§18.7.4 Eigenfunction Expansion of the Green's Function

We derived at the end of Sturm–Liouville theory the formula

$$G(x, \xi) = \sum_{n=1}^{\infty} \frac{y_n(x)y_n(\xi)}{\lambda_n N_n}.$$

Let's re-derive it from our new definition, as a consistency check. Seek $G(x, \xi) = \sum_n A_n y_n(x)$ satisfying $\mathcal{L}G = \delta(x - \xi)$. Then

$$\mathcal{L}G = \sum_n A_n \lambda_n w(x) y_n(x),$$

while the delta function has the expansion $\delta(x - \xi) = w(x) \sum_n y_n(\xi) y_n(x) / N_n$ from the last section. Comparing coefficients,

$$A_n = \frac{y_n(\xi)}{\lambda_n N_n},$$

which is exactly the earlier formula. Note again that this only makes sense if no $\lambda_n = 0$.

§18.7.5 Initial Value Problems

Green's functions work just as well for initial value problems, with one striking new feature. Suppose we want to solve

$$\mathcal{L}y = f(t), \quad t \geq a, \quad y(a) = y'(a) = 0,$$

via $G(t, \tau)$ with $\mathcal{L}G = \delta(t - \tau)$ and $G(a, \tau) = G'(a, \tau) = 0$ (derivatives in t).

For $t < \tau$: G_1 is a homogeneous solution with $G_1(a) = G'_1(a) = 0$. But a homogeneous solution with both initial conditions zero is identically zero (writing $G_1 = Ay_1 + By_2$, the conditions form a linear system whose determinant is the Wronskian of y_1, y_2 at a , which is nonzero). So

$$G_1 \equiv 0 \quad \text{for } t < \tau:$$

nothing happens before the impulse.

For $t > \tau$: continuity forces $G_2(\tau, \tau) = 0$, so $G_2 = Dy_+(t)$ where y_+ is the homogeneous solution with $y_+(\tau) = 0$, and the jump condition $G'_2(\tau_+) = 1/\alpha(\tau)$ fixes $D = 1/(\alpha(\tau)y'_+(\tau))$. Hence

$$G(t, \tau) = \begin{cases} 0 & \text{if } t < \tau, \\ \frac{y_+(t)}{\alpha(\tau)y'_+(\tau)} & \text{if } t > \tau, \end{cases}$$

and

$$y(t) = \int_a^t G(t, \tau) f(\tau) \, d\tau.$$

Note the upper limit: only forcing *before* time t contributes. **Causality** is built into the structure of the Green's function – an idea that will return with force when we study signal processing and PDEs.

Example 18.7.3

Solve $y'' - y = f(t)$ with $y(0) = y'(0) = 0$. For $t > \tau$ we need the homogeneous solution vanishing at τ , namely $y_+ = \sinh(t - \tau)$, and the jump condition gives ($\alpha = 1$, $y'_+(\tau) = \cosh 0 = 1$) coefficient 1. So

$$y(t) = \int_0^t \sinh(t - \tau) f(\tau) \, d\tau.$$

Compare this with the boundary value problem for the same operator above – same equation, completely different Green's function. The Green's function belongs to the operator *together with* its boundary conditions.

Remark (Higher Order Equations). Everything generalises to an n th order operator $\mathcal{L}$ with leading coefficient $\alpha(x)$: the Green's function is built from homogeneous solutions satisfying the boundary conditions, with $G, G', \dots, G^{(n-2)}$ all continuous at $x = \xi$ and a jump

$$G^{(n-1)}(\xi_+) - G^{(n-1)}(\xi_-) = \frac{1}{\alpha(\xi)}$$

in the top derivative.

§18.8 Fourier Transforms

Fourier series decompose periodic functions into harmonics. But plenty of interesting functions – a solitary pulse, a decaying signal – are not periodic at all, and live on the whole real line. The **Fourier transform** is what Fourier series become when the period is sent to infinity: the discrete spectrum of harmonics merges into a continuum.

§18.8.1 Definition and the Inversion Theorem**Definition 18.8.1** (Fourier Transform)

The **Fourier transform** of $f(x)$ is

$$\tilde{f}(k) = \mathcal{F}[f](k) = \int_{-\infty}^{\infty} f(x) e^{-ikx} \, dx,$$

and the **inverse Fourier transform** is

$$f(x) = \mathcal{F}^{-1}[\tilde{f}](x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} \, dk.$$

(Beware: other books scatter the 2π differently – some split it symmetrically as $1/\sqrt{2\pi}$ on each – and any internally consistent convention is fine.)

Theorem 18.8.2 (Fourier Inversion Theorem)

If f and $\tilde{f}$ are absolutely integrable, then

$$\mathcal{F}^{-1}[\mathcal{F}[f]](x) = f(x)$$

at all points where f is continuous; at a jump the right-hand side gives $\frac{1}{2}(f(x_+) + f(x_-))$, just as for Fourier series.

Rather than prove this, let's see why it's plausible by deriving the transform as a limit of Fourier series. Take f on $[-L, L]$ with complex Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{ik_n x}, \quad k_n = \frac{n\pi}{L},$$

and write $\Delta k = \pi/L$ for the spacing of the wavenumbers. Then

$$c_n = \frac{1}{2L} \int_{-L}^L f(u) e^{-ik_n u} du = \frac{\Delta k}{2\pi} \int_{-L}^L f(u) e^{-ik_n u} du,$$

so

$$f(x) = \sum_{n=-\infty}^{\infty} \frac{\Delta k}{2\pi} e^{ik_n x} \int_{-L}^L f(u) e^{-ik_n u} du.$$

As $L \rightarrow \infty$, $\Delta k \rightarrow 0$, and the sum over n is a Riemann sum for an integral over k :

$$f(x) \rightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \left(\int_{-\infty}^{\infty} f(u) e^{-iku} du \right) dk = \mathcal{F}^{-1}[\mathcal{F}[f]](x).$$

Example 18.8.3 (The Gaussian)

Let $f(x) = \frac{1}{\sigma\sqrt{\pi}} e^{-x^2/\sigma^2}$. Since $\sin kx$ is odd, only the cosine part survives:

$$\tilde{f}(k) = \frac{1}{\sigma\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-x^2/\sigma^2} \cos(kx) dx.$$

Differentiating under the integral sign and integrating by parts,

$$\frac{d\tilde{f}}{dk} = \frac{-1}{\sigma\sqrt{\pi}} \int_{-\infty}^{\infty} x e^{-x^2/\sigma^2} \sin(kx) dx = -\frac{k\sigma^2}{2} \tilde{f}(k).$$

This little ODE gives $\tilde{f}(k) = C e^{-k^2\sigma^2/4}$, and setting $k = 0$ (where the integral is just the total area, 1) gives $C = 1$:

$$\tilde{f}(k) = e^{-k^2\sigma^2/4}.$$

So the transform of a Gaussian is another Gaussian, with *inverted* width: a narrow pulse has a broad spectrum and vice versa. (Readers who have met quantum mechanics will recognise the germ of the uncertainty principle.)

§18.8.2 Properties of the Fourier Transform

The Fourier transform converts awkward operations into easy ones. Throughout, $\tilde{f} = \mathcal{F}[f]$.

- (i) (Linearity) $\mathcal{F}[\lambda f + \mu g] = \lambda \tilde{f} + \mu \tilde{g}$.
- (ii) (Translation) $\mathcal{F}[f(x - \lambda)](k) = e^{-i\lambda k} \tilde{f}(k)$.

- (iii) (Frequency shift) $\mathcal{F}[e^{i\lambda x} f(x)](k) = \tilde{f}(k - \lambda)$.
- (iv) (Scaling) $\mathcal{F}[f(\lambda x)](k) = \frac{1}{|\lambda|} \tilde{f}(k/\lambda)$.
- (v) (Multiplication by x) $\mathcal{F}[xf(x)](k) = i\tilde{f}'(k)$.
- (vi) (Differentiation) $\mathcal{F}[f'(x)](k) = ik\tilde{f}(k)$.

Each is a one-line substitution or integration by parts; let's do the most important, (vi):

$$\int_{-\infty}^{\infty} f'(x)e^{-ikx} dx = \left[f(x)e^{-ikx} \right]_{-\infty}^{\infty} + ik \int_{-\infty}^{\infty} f(x)e^{-ikx} dx = ik\tilde{f}(k),$$

where the boundary term dies since $f \rightarrow 0$ at infinity (a consequence of absolute integrability). *Differentiation becomes multiplication by ik* – this single fact is why Fourier transforms solve differential equations.

There is also a pleasing **duality**: the inversion formula says $f(-x) = \frac{1}{2\pi} \int \tilde{f}(k)e^{-ikx} dk$, which upon relabelling $x \leftrightarrow k$ reads

$$\mathcal{F}[\tilde{f}](k) = 2\pi f(-k).$$

Transforming four times gets you home: $\mathcal{F}^4[f] = 4\pi^2 f$. In practice duality lets every transform we compute do double duty.

Example 18.8.4 (The Top Hat and sinc)

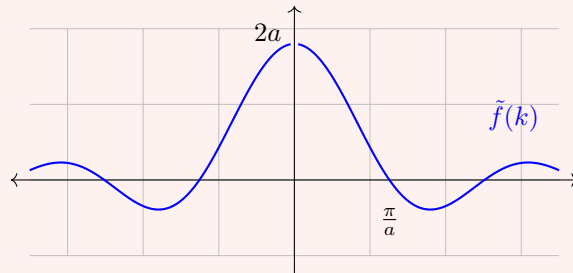
Let f be the ‘top hat’

$$f(x) = \begin{cases} 1 & \text{if } |x| \leq a, \\ 0 & \text{otherwise.} \end{cases}$$

Then

$$\tilde{f}(k) = \int_{-a}^a e^{-ikx} dx = \int_{-a}^a \cos(kx) dx = \frac{2 \sin ka}{k},$$

a scaled *sinc* function.



Notice that f has bounded support but $\tilde{f}$ spreads over all k , decaying only like $1/k$ – the slow decay is the price of the sharp edges. As a bonus, applying the inversion theorem at $x = 0$ (where f is continuous, $f(0) = 1$) gives

$$1 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2 \sin ka}{k} dk,$$

from which (relabelling and using evenness) we obtain the **Dirichlet discontinuous**

formula

$$\int_0^\infty \frac{\sin ax}{x} dx = \frac{\pi}{2} \operatorname{sgn} a = \begin{cases} \pi/2 & \text{if } a > 0, \\ 0 & \text{if } a = 0, \\ -\pi/2 & \text{if } a < 0. \end{cases}$$

§18.8.3 Convolution and Parseval's Theorem

Multiplication of transforms corresponds to a mixing operation on the original functions.

Definition 18.8.5 (Convolution)

The **convolution** of f and g is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y)g(x-y) dy.$$

Theorem 18.8.6 (Convolution Theorem)

$\mathcal{F}[f * g] = \tilde{f}(k) \tilde{g}(k)$. Dually, $\mathcal{F}[fg] = \frac{1}{2\pi} (\tilde{f} * \tilde{g})$.

Proof. Start from the product and invert:

$$\begin{aligned} \mathcal{F}^{-1}[\tilde{f}\tilde{g}](x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k)\tilde{g}(k)e^{ikx} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(y)e^{-iky} dy \right) \tilde{g}(k)e^{ikx} dk \\ &= \int_{-\infty}^{\infty} f(y) \left(\frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{g}(k)e^{ik(x-y)} dk \right) dy \\ &= \int_{-\infty}^{\infty} f(y) g(x-y) dy = (f * g)(x). \square \end{aligned}$$

Theorem 18.8.7 (Parseval's Theorem for Fourier Transforms)

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk.$$

Proof. First note that if $h(x) = g^*(-x)$ then $\tilde{h}(k) = \tilde{g}^*(k)$ (substitute and conjugate). Applying the convolution theorem to f and $g^*(-\cdot)$ and evaluating at $x = 0$,

$$\int_{-\infty}^{\infty} f(y) g^*(y) dy = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}(k) \tilde{g}^*(k) dk,$$

which says the Fourier transform preserves inner products up to the factor 2π : $\langle g, f \rangle = \frac{1}{2\pi} \langle \tilde{g}, \tilde{f} \rangle$. Taking $g = f$ gives the result. $\square$

§18.8.4 Fourier Transforms of Generalised Functions

The class of functions with honest Fourier transforms is a bit restrictive ($f \rightarrow 0$ at infinity, etc.), but distributions let us transform far more. The key identity comes from writing out the inversion theorem in full:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f(u) e^{-iku} \, du \right) e^{ikx} \, dk = \int_{-\infty}^{\infty} f(u) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-u)} \, dk \right] \, du.$$

For this to spit out $f(x)$ for every f , the bracket must be precisely a delta function:

$$\delta(x-u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-u)} \, dk.$$

This wildly divergent-looking integral is a perfectly good distributional statement, and it immediately gives a dictionary:

- (i) $\mathcal{F}[\delta(x)] = 1$, and more generally $\mathcal{F}[\delta(x-a)] = e^{-ika}$;
- (ii) $\mathcal{F}[1] = 2\pi\delta(k)$ (an infinitely spread function has an infinitely concentrated spectrum);
- (iii) $\mathcal{F}[\cos \omega x] = \pi(\delta(k-\omega) + \delta(k+\omega))$ and $\mathcal{F}[\sin \omega x] = i\pi(\delta(k+\omega) - \delta(k-\omega))$ – a pure tone has a spectrum concentrated at $\pm$ its frequency;
- (iv) from the Dirichlet discontinuous formula, $\mathcal{F}\left[\frac{1}{2} \operatorname{sgn} x\right] = \frac{1}{ik}$;
- (v) for the Heaviside function, writing $H(x) = \frac{1}{2} + \frac{1}{2} \operatorname{sgn} x$,

$$\mathcal{F}[H](k) = \pi\delta(k) + \frac{1}{ik}.$$

As a sanity check on (v): $H' = \delta$ requires $ik\tilde{H} = 1$, which holds since $k\delta(k) = 0$; and $H(x) + H(-x) = 1$ requires $\tilde{H}(k) + \tilde{H}(-k) = 2\pi\delta(k)$, which also checks out. Neither condition alone pins $\tilde{H}$ down – together they do.

§18.8.5 Solving Differential Equations with Fourier Transforms

Here's the general scheme: transform the equation (derivatives become multiplications), solve *algebraically*, and transform back (usually via the convolution theorem).

Example 18.8.8 (A Boundary Value Problem on the Line)

Solve $y'' - y = f(x)$ with $y \rightarrow 0$ as $x \rightarrow \pm\infty$. Transforming,

$$(ik)^2 \tilde{y} - \tilde{y} = \tilde{f} \implies \tilde{y}(k) = \frac{-\tilde{f}(k)}{1+k^2} = \tilde{f}(k) \tilde{g}(k), \quad \tilde{g}(k) = \frac{-1}{1+k^2}.$$

One can check (by computing the transform of $e^{-|x|}$) that $\tilde{g}$ is the transform of $g(x) = -\frac{1}{2}e^{-|x|}$. So by the convolution theorem,

$$y(x) = (f * g)(x) = -\frac{1}{2} \int_{-\infty}^{\infty} f(u) e^{-|x-u|} \, du.$$

This is exactly a Green's function solution, with $G(x, u) = -\frac{1}{2}e^{-|x-u|}$ – which is just

what we would have built with jump conditions, with the boundary conditions now imposed at $\pm\infty$. The Fourier transform found it for us automatically.

§18.8.6 Signal Processing and Transfer Functions

A common situation: a physical system (an amplifier, a suspension bridge, a circuit) receives an input signal $\mathcal{I}(t)$ and produces an output $\mathcal{O}(t)$, with the relationship linear and time-invariant. In the frequency domain such a system can only do one thing: multiply each frequency component by a fixed complex number. That is,

$$\tilde{\mathcal{O}}(\omega) = \tilde{\mathcal{R}}(\omega) \tilde{\mathcal{I}}(\omega),$$

where $\tilde{\mathcal{R}}(\omega)$ is the **transfer function** of the system. Its inverse transform $\mathcal{R}(t)$ is the **response function**, and by the convolution theorem

$$\mathcal{O}(t) = \int_{-\infty}^{\infty} \mathcal{I}(u) \mathcal{R}(t-u) du.$$

The response function is the output produced by a delta-function kick at $t=0$ – it is a Green's function by another name. For a physically realisable system, *causality* demands $\mathcal{R}(t) = 0$ for $t < 0$ (no output before the input), so if the input starts at $t=0$,

$$\mathcal{O}(t) = \int_0^t \mathcal{I}(u) \mathcal{R}(t-u) du,$$

which has precisely the shape of our initial value problem Green's function solutions.

Suppose the system is governed by an ODE,

$$\mathcal{L}\mathcal{O}(t) = \left(a_n \frac{d^n}{dt^n} + \cdots + a_1 \frac{d}{dt} + a_0 \right) \mathcal{O}(t) = \mathcal{I}(t).$$

Transforming, each derivative becomes $i\omega$, so

$$\tilde{\mathcal{R}}(\omega) = \frac{1}{a_0 + a_1(i\omega) + \cdots + a_n(i\omega)^n}.$$

To invert, factor the denominator over its roots and expand in partial fractions:

$$\tilde{\mathcal{R}}(\omega) = \sum_{j=1}^J \sum_{m=1}^{k_j} \frac{\Gamma_{jm}}{(i\omega - c_j)^m}.$$

The basic inverse transform (computable directly, for $\text{Re } c < 0$) is

$$\mathcal{F}^{-1} \left[\frac{1}{i\omega - c} \right] (t) = \begin{cases} e^{ct} & \text{if } t > 0, \\ 0 & \text{if } t < 0, \end{cases}$$

and repeatedly applying the 'multiplication by t ' rule gives

$$\mathcal{F}^{-1} \left[\frac{1}{(i\omega - c)^m} \right] (t) = \frac{t^{m-1}}{(m-1)!} e^{ct} \quad (t > 0).$$

So provided all roots satisfy $\text{Re } c_j < 0$ (a *stable* system – no exponentially growing responses), the response function is

$$\mathcal{R}(t) = \sum_{j=1}^J \sum_{m=1}^{k_j} \Gamma_{jm} \frac{t^{m-1}}{(m-1)!} e^{c_j t} \quad \text{for } t > 0,$$

and zero for $t < 0$. Causality has appeared automatically.

Example 18.8.9 (The Damped Oscillator)

Consider $y'' + 2py' + (p^2 + q^2)y = f(t)$ with $p > 0$ and $y(0) = y'(0) = 0$. The transfer function is

$$\tilde{\mathcal{R}}(\omega) = \frac{1}{(i\omega)^2 + 2p(i\omega) + p^2 + q^2} = \frac{1}{(i\omega + p)^2 + q^2},$$

with roots $i\omega = -p \pm iq$, both with negative real part. Partial fractions and the table above give

$$\mathcal{R}(t) = \frac{1}{q} e^{-pt} \sin(qt) \quad (t > 0),$$

a decaying oscillation, and the solution is

$$y(t) = \frac{1}{q} \int_0^t e^{-p(t-\tau)} \sin(q(t-\tau)) f(\tau) d\tau.$$

One can verify directly that $\mathcal{R}(t - \tau)$ satisfies $\mathcal{L}\mathcal{R} = \delta(t - \tau)$: it is the initial-value Green's function of the damped oscillator.

§18.8.7 Discrete Sampling and the Nyquist Frequency

In the real world we don't measure signals continuously; we *sample* them, at times $t_n = n\Delta$ for some fixed time step Δ . How much do we lose?

The natural frequency scale is the **Nyquist frequency**

$$f_c = \frac{1}{2\Delta},$$

the highest frequency the sampling can resolve – one sample per half-period. Consider sampling a pure tone $g_f(t) = A \cos(2\pi ft + \varphi)$. If $f = f_c$ exactly, then

$$g_{f_c}(t_n) = A \cos(\pi n + \varphi) = A \cos(\varphi) \cos(\pi n),$$

so the samples only see the combination $A \cos \varphi$: amplitude and phase have become degenerate, and half the information is gone. Worse, for a frequency *above* Nyquist, $f = f_c + \delta f$, the samples satisfy

$$g_{f_c+\delta f}(t_n) = g_{f_c-\delta f}(t_n) \quad (\text{up to phase}),$$

so the high frequency masquerades as the lower frequency $f_c - \delta f$. This is called **aliasing**, and it's why car wheels appear to spin backwards on film.

If, however, the signal contains *no* frequencies above some $\omega_{\max}$ (it is **bandwidth-limited**), sampling loses nothing at all:

Theorem 18.8.10 (Nyquist–Shannon Sampling Theorem)

Let g be bandwidth-limited with $\tilde{g}(\omega) = 0$ for $|\omega| > \omega_{\max}$, and sample at the Nyquist rate, $\Delta = \pi/\omega_{\max}$, giving samples $g_n = g(n\Delta)$. Then g is *exactly* reconstructible:

$$g(t) = \sum_{n=-\infty}^{\infty} g_n \frac{\sin(\omega_{\max} t - n\pi)}{\omega_{\max} t - n\pi}.$$

Proof. Since $\tilde{g}$ is supported on the bounded interval $[-\omega_{\max}, \omega_{\max}]$, we can expand it there in a *Fourier series* (in ω !): the coefficients of that series are, up to constants, exactly the samples g_n , since

$$g_n = \frac{1}{2\pi} \int_{-\omega_{\max}}^{\omega_{\max}} \tilde{g}(\omega) e^{i\pi n \omega / \omega_{\max}} d\omega.$$

So the samples determine the periodic extension $\tilde{g}_{\text{per}}(\omega) = \frac{\pi}{\omega_{\max}} \sum_n g_n e^{-i\pi n \omega / \omega_{\max}}$, and multiplying by the top hat on $[-\omega_{\max}, \omega_{\max}]$ recovers $\tilde{g}$ exactly. Inverting,

$$g(t) = \frac{1}{2\omega_{\max}} \sum_{n=-\infty}^{\infty} g_n \int_{-\omega_{\max}}^{\omega_{\max}} e^{i\omega(t-n\pi/\omega_{\max})} d\omega,$$

and each integral is a sinc, giving the stated formula. $\square$

The sinc functions are doing interpolation duty: the n th term equals g_n at $t = t_n$ and vanishes at every other sample point.

§18.8.8 The Discrete Fourier Transform

Finally, suppose we have only finitely many samples $h_m = h(t_m)$, $t_m = m\Delta$, $m = 0, \dots, N-1$. We estimate the Fourier transform at the N frequencies $f_n = \frac{n}{N\Delta}$ (which, thanks to aliasing, is the honest number of distinct frequencies available):

$$\tilde{h}(f_n) = \int_{-\infty}^{\infty} h(t) e^{-2\pi i f_n t} dt \approx \Delta \sum_{m=0}^{N-1} h_m e^{-2\pi i m n / N} = \Delta \tilde{h}_n.$$

Definition 18.8.11 (Discrete Fourier Transform)

The **discrete Fourier transform** of the vector $(h_0, \dots, h_{N-1})$ is

$$\tilde{h}_n = \sum_{m=0}^{N-1} h_m \omega^{mn}, \quad \omega = e^{-2\pi i / N},$$

i.e. $\tilde{h} = [\text{DFT}] h$ where $[\text{DFT}]_{nm} = \omega^{nm}$ is an $N \times N$ matrix of roots of unity.

The DFT matrix is (up to scaling) unitary: $[\text{DFT}]^{-1} = \frac{1}{N} [\text{DFT}]^\dagger$, so inversion is

$$h_m = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{h}_n e^{2\pi i m n / N}.$$

For example, with $N = 4$ we have $\omega = -i$ and

$$[\text{DFT}] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{pmatrix}.$$

The discrete analogues of our earlier theorems hold, e.g. Parseval becomes $\sum_m |h_m|^2 = \frac{1}{N} \sum_n |\tilde{h}_n|^2$, and convolution (defined cyclically) again maps to multiplication.

Remark (The Fast Fourier Transform). Computing the DFT naively costs $O(N^2)$ operations. The *fast Fourier transform* does it in $O(N \log N)$, by splitting the sum into even and odd samples: since $\omega_N^2 = \omega_{N/2}$,

$$\tilde{h}_k = \sum_{m=0}^{N/2-1} h_{2m} \omega_{N/2}^{mk} + \omega_N^k \sum_{m=0}^{N/2-1} h_{2m+1} \omega_{N/2}^{mk},$$

two DFTs of half the size, and recursing. This algorithm (this material is non-examinable, but too good to omit) is quietly running inside essentially every piece of modern signal-processing equipment.

§18.9 The Method of Characteristics

We now step back and develop a more geometric way of thinking about PDEs, which explains – among other things – where d’Alembert’s solution really came from, and why the wave, diffusion and Laplace equations behave so differently.

§18.9.1 Well-Posed Cauchy Problems

A **Cauchy problem** consists of a PDE for some function ϕ together with auxiliary data (values of ϕ and/or its derivatives) specified on some surface – in two dimensions, a curve. The data is called **Cauchy data**. Following Hadamard, we say the problem is **well-posed** if

- (i) a solution exists (we haven’t demanded too much),
- (ii) the solution is unique (we haven’t demanded too little), and
- (iii) the solution depends continuously on the data (small measurement errors don’t explode).

Which kinds of data make a problem well-posed depends dramatically on the equation: initial data suits the wave and diffusion equations, boundary data suits Laplace’s equation, and mixing these up goes badly wrong. The method of characteristics is the tool that explains why.

§18.9.2 Characteristics of First Order PDEs

Consider a curve C in the plane, parametrised as $(x(s), y(s))$, with tangent vector $\mathbf{u} = (dx/ds, dy/ds)$. The derivative of a function ϕ along the curve is

$$\left. \frac{d\phi}{ds} \right|_C = \frac{dx}{ds} \frac{\partial \phi}{\partial x} + \frac{dy}{ds} \frac{\partial \phi}{\partial y} = \mathbf{u} \cdot \nabla \phi.$$

Now consider the general homogeneous first order linear PDE

$$\alpha(x, y) \frac{\partial \phi}{\partial x} + \beta(x, y) \frac{\partial \phi}{\partial y} = 0.$$

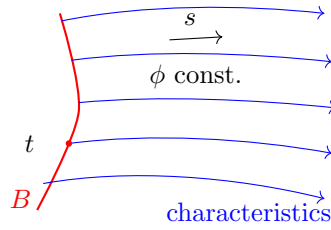
The left-hand side is exactly the directional derivative of ϕ along the vector field $\mathbf{u} = (\alpha, \beta)$! So the PDE says something beautifully simple: *ϕ is constant along the integral curves of $\mathbf{u}$* , i.e. the curves solving

$$\frac{dx}{ds} = \alpha(x, y), \quad \frac{dy}{ds} = \beta(x, y).$$

These curves are called the **characteristic curves** (or just characteristics) of the PDE. Solving a PDE has been reduced to solving a family of ODEs.

To pose a Cauchy problem, we specify $\phi = f(t)$ along some initial curve B , parametrised by t as $(x(t), y(t))$, chosen *transverse* to the characteristics (nowhere tangent to them). Each characteristic then picks up its constant value where it crosses B , and carries it across the plane: labelling characteristics by their crossing point t , with s the parameter along each (set $s = 0$ on B), we get coordinates (s, t) in which the solution is simply

$$\phi(s, t) = f(t).$$



Finally, provided the Jacobian $J = x_t y_s - x_s y_t$ of the map $(s, t) \mapsto (x, y)$ is nonzero (this is what transversality buys us), we can invert to find $s(x, y)$, $t(x, y)$ and hence $\phi(x, y) = f(t(x, y))$.

In summary, the recipe is:

1. write down the characteristic equations $dx/ds = \alpha$, $dy/ds = \beta$;
2. parametrise the initial curve B by $(x(t), y(t))$;
3. solve the characteristic equations with initial conditions on B at $s = 0$, obtaining $x(s, t)$, $y(s, t)$;
4. observe ϕ is constant along characteristics, so $\phi(s, t) = f(t)$;
5. invert to express (s, t) in terms of (x, y) .

Example 18.9.1

Solve $e^x \phi_x + \phi_y = 0$ with $\phi(x, 0) = \cosh x$. The characteristic equations are

$$\frac{dx}{ds} = e^x, \quad \frac{dy}{ds} = 1,$$

and the initial curve is $(x, y) = (t, 0)$ at $s = 0$. Solving: $-e^{-x} = s - e^{-t}$ and $y = s$. So along each characteristic, $e^{-x} + y = e^{-t}$, i.e. the characteristics are the curves $y = e^{-t} - e^{-x}$, and

$$t = -\log(y + e^{-x}).$$

Since ϕ is constant along characteristics, $\phi(s, t) = \cosh t$, so

$$\phi(x, y) = \cosh[\log(y + e^{-x})] = \frac{1}{2} \left(y + e^{-x} + \frac{1}{y + e^{-x}} \right).$$

One can check directly that this satisfies the PDE and the data.

§18.9.3 Inhomogeneous First Order PDEs

Now add a source:

$$\alpha(x, y)\phi_x + \beta(x, y)\phi_y = \gamma(x, y), \quad \phi = f(t) \text{ on } B.$$

The characteristics are unchanged, but ϕ is no longer constant along them – instead

$$\left. \frac{d\phi}{ds} \right|_C = \gamma(x(s), y(s)),$$

an ODE to be solved *along each characteristic*, with initial value $f(t)$ at $s = 0$. The geometry is the same; the data is now transported with modification rather than unchanged.

Example 18.9.2

Solve $\phi_x + 2\phi_y = ye^x$ with $\phi = \sin x$ on the line $y = x$. Characteristics: $dx/ds = 1$, $dy/ds = 2$, with $(x, y) = (t, t)$ at $s = 0$, giving

$$x = s + t, \quad y = 2s + t.$$

Along a characteristic,

$$\frac{d\phi}{ds} = ye^x = (2s + t)e^{s+t},$$

and integrating with respect to s (noting $\int 2se^s ds = 2(s-1)e^s$),

$$\phi(s, t) = (2s + t - 2)e^{s+t} + c(t).$$

The initial condition $\phi(0, t) = \sin t$ gives $c(t) = \sin t - (t-2)e^t$. Inverting $s = y - x$, $t = 2x - y$,

$$\phi(x, y) = (y - 2)e^x + (2 - 2x + y)e^{2x-y} + \sin(2x - y).$$

§18.9.4 Classification of Second Order PDEs

Let's now bring these ideas to second order equations. The general second order linear PDE in two variables is

$$a(x, y)\phi_{xx} + 2b(x, y)\phi_{xy} + c(x, y)\phi_{yy} + d\phi_x + e\phi_y + f\phi = 0,$$

and its behaviour is governed by the **principal part** – the top order terms – encoded in the symmetric matrix

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

Definition 18.9.3 (Classification)

The PDE is

- (i) **elliptic** if $b^2 - ac < 0$ (eigenvalues of A of the same sign) – e.g. Laplace's equation;
- (ii) **hyperbolic** if $b^2 - ac > 0$ (eigenvalues of opposite signs) – e.g. the wave equation;

(iii) **parabolic** if $b^2 - ac = 0$ (a zero eigenvalue) – e.g. the diffusion equation.

The terminology comes from the conic sections $ak_1^2 + 2bk_1k_2 + ck_2^2 = \text{const.}$ Note that the classification is pointwise: an equation with variable coefficients can be hyperbolic in one region and elliptic in another.

We say a curve along which $u(x, y)$ is constant is a **characteristic curve** of the equation if

$$\begin{pmatrix} u_x & u_y \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix} = 0.$$

(For first order equations this reduces to our earlier condition $\mathbf{u} \cdot \nabla u = 0$ along characteristics.) Writing the curve as $y = y(x)$ and using $u_x + u_y \frac{dy}{dx} = 0$ along it, the condition becomes a quadratic for the slope:

$$a \left(\frac{dy}{dx} \right)^2 - 2b \frac{dy}{dx} + c = 0 \implies \frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - ac}}{a}.$$

So we can read the classification off geometrically:

- (i) hyperbolic equations have *two* families of real characteristics;
- (ii) parabolic equations have exactly one;
- (iii) elliptic equations have none.

Characteristics are the curves along which information (and discontinuities in derivatives of the solution) propagate. The wave equation moves data along its two families of characteristics; Laplace's equation, with no real characteristics, moves nothing – its solutions are instantly smooth everywhere, and only boundary data makes sense.

§18.9.5 Characteristic Coordinates

For a hyperbolic equation, using the two families of characteristics as a coordinate system (u, v) kills both the ϕ_{uu} and ϕ_{vv} terms, reducing the equation to **canonical form**

$$\frac{\partial^2 \phi}{\partial u \partial v} + (\text{lower order terms}) = 0.$$

(For parabolic equations the canonical form is $\phi_{vv} + \dots = 0$, and for elliptic ones $\phi_{uu} + \phi_{vv} + \dots = 0$.)

Example 18.9.4

Consider $-y\phi_{xx} + \phi_{yy} = 0$. Here $a = -y$, $b = 0$, $c = 1$, so $b^2 - ac = y$: the equation is hyperbolic for $y > 0$, parabolic on $y = 0$ and elliptic for $y < 0$. In the hyperbolic region, the characteristic slopes are

$$\frac{dy}{dx} = \pm \frac{1}{\sqrt{y}} \implies \frac{2}{3}y^{3/2} \mp x = \text{const.},$$

so we take characteristic coordinates

$$u = \frac{2}{3}y^{3/2} + x, \quad v = \frac{2}{3}y^{3/2} - x.$$

Computing derivatives ($u_x = 1, u_y = \sqrt{y}, v_x = -1, v_y = \sqrt{y}$) and substituting, all

the ϕ_{uu} and ϕ_{vv} terms cancel, leaving

$$4\phi_{uv} + \frac{1}{6(u+v)}(\phi_u + \phi_v) = 0,$$

using $u + v = \frac{4}{3}y^{3/2}$. This is the canonical form.

§18.9.6 The General Solution of the Wave Equation

We can now settle a promise from earlier: the wave equation

$$\frac{1}{c^2}\phi_{tt} - \phi_{xx} = 0$$

has $a = 1/c^2$, $b = 0$, $c = -1$ (in variables (t, x)), so $b^2 - ac = 1/c^2 > 0$: hyperbolic, everywhere. The characteristic slopes are

$$\frac{dx}{dt} = \pm c,$$

giving the two straight-line families $x \mp ct = \text{const}$ and the characteristic coordinates

$$u = x - ct, \quad v = x + ct.$$

In these coordinates the chain rule gives $\phi_{tt} = c^2(\phi_{uu} - 2\phi_{uv} + \phi_{vv})$ and $\phi_{xx} = \phi_{uu} + 2\phi_{uv} + \phi_{vv}$, so the wave equation collapses to the canonical form

$$\frac{\partial^2 \phi}{\partial u \partial v} = 0.$$

Integrating with respect to v : $\phi_u = (\text{function of } u)$; integrating again:

$$\phi = G(u) + H(v) = G(x - ct) + H(x + ct),$$

confirming the travelling wave decomposition we used in §3, and fitting the initial data exactly as before recovers d'Alembert's formula

$$\phi(x, t) = \frac{1}{2} [f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(s) ds,$$

for data $\phi(x, 0) = f(x)$, $\phi_t(x, 0) = g(x)$. The domain of dependence picture from §3 is now revealed as a completely general fact about hyperbolic equations: data propagates along characteristics, and the solution at a point can only feel data between the two characteristics through that point.

§18.10 Green's Functions for PDEs

For our final act, we combine everything – Fourier transforms, delta functions, Green's functions and our knowledge of the classical equations – to construct Green's functions for PDEs: the response of the heat, wave and Laplace equations to point sources.

§18.10.1 The Fundamental Solution of the Diffusion Equation

Consider the heat equation on the whole line,

$$\frac{\partial \theta}{\partial t} - D \frac{\partial^2 \theta}{\partial x^2} = 0, \quad \theta(x, 0) = h(x), \quad \theta \rightarrow 0 \text{ as } x \rightarrow \pm\infty.$$

Take the Fourier transform in x . Then $\partial^2/\partial x^2$ becomes $-k^2$, and for each fixed k we have an ODE in time:

$$\frac{\partial \tilde{\theta}}{\partial t}(k, t) = -Dk^2 \tilde{\theta}(k, t) \implies \tilde{\theta}(k, t) = \tilde{h}(k) e^{-Dk^2 t}.$$

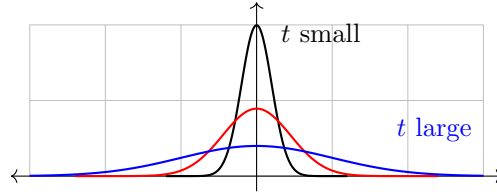
Each Fourier mode simply decays, at a rate proportional to k^2 – there is the smoothing of high frequencies again, now completely explicit. To return to real space, note $e^{-Dk^2 t}$ is a Gaussian in k , whose inverse transform (by our Gaussian example, suitably scaled) is another Gaussian. The convolution theorem then gives

$$\theta(x, t) = \int_{-\infty}^{\infty} h(u) S_d(x - u, t) du,$$

where

$$S_d(x, t) = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

is the **fundamental solution** (or *heat kernel*). If the initial condition is a point source of heat, $\theta(x, 0) = \theta_0 \delta(x)$, then $\theta = \theta_0 S_d(x, t)$ exactly: an ever-broadening Gaussian of constant area, with width $\sqrt{2Dt}$ – our similarity variable $\eta = x/2\sqrt{Dt}$ yet again.



Example 18.10.1 (An Initially Gaussian Temperature)

Suppose $\theta(x, 0) = \sqrt{\frac{a}{\pi}} \theta_0 e^{-ax^2}$, a Gaussian pulse of area θ_0 . The solution is the convolution of two Gaussians, which (completing the square in the exponent, or transforming and multiplying) is another Gaussian:

$$\theta(x, t) = \theta_0 \sqrt{\frac{a}{\pi(1 + 4aDt)}} \exp\left(\frac{-ax^2}{1 + 4aDt}\right).$$

The width grows like $\sqrt{t}$, the height falls correspondingly, and the area – the total heat – stays fixed at θ_0 .

§18.10.2 The Forced Diffusion Equation and Duhamel's Principle

Now add a source term:

$$\frac{\partial \theta}{\partial t} - D \frac{\partial^2 \theta}{\partial x^2} = f(x, t), \quad \theta(x, 0) = 0.$$

We build the Green's function $G(x, t; \xi, \tau)$, the response to a point source in both space and time:

$$\frac{\partial G}{\partial t} - D \frac{\partial^2 G}{\partial x^2} = \delta(x - \xi) \delta(t - \tau), \quad G = 0 \text{ at } t = 0.$$

Fourier transforming in x ,

$$\frac{\partial \tilde{G}}{\partial t} + Dk^2 \tilde{G} = e^{-ik\xi} \delta(t - \tau).$$

Multiplying by the integrating factor $e^{Dk^2 t}$ and integrating from 0 to t , the delta function fires only if $\tau < t$:

$$\tilde{G}(k, t; \xi, \tau) = e^{-ik\xi} e^{-Dk^2(t-\tau)} H(t - \tau).$$

But we recognise this: apart from the causality factor $H(t - \tau)$ and the shift $e^{-ik\xi}$, it's the transform of the heat kernel with t replaced by $t - \tau$. Inverting,

$$G(x, t; \xi, \tau) = H(t - \tau) S_d(x - \xi, t - \tau),$$

and the solution to the forced problem is

$$\theta(x, t) = \int_0^t \int_{-\infty}^{\infty} f(\xi, \tau) S_d(x - \xi, t - \tau) d\xi d\tau.$$

Look carefully at the structure of this answer. The inner integral is exactly the solution of the *unforced* heat equation started at time τ from initial data $f(\cdot, \tau)$. So the forced solution is a superposition, over all past times τ , of free evolutions seeded by the forcing at time τ : forcing acts as a continuous stream of fresh initial conditions. This correspondence between the forced problem with homogeneous initial data and the homogeneous problem with initial data is called **Duhamel's principle**.

§18.10.3 The Forced Wave Equation

The same programme for the wave equation:

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \frac{\partial^2 \phi}{\partial x^2} = f(x, t), \quad \phi(x, 0) = \phi_t(x, 0) = 0.$$

Transforming the Green's function equation in x ,

$$\frac{\partial^2 \tilde{G}}{\partial t^2} + c^2 k^2 \tilde{G} = e^{-ik\xi} \delta(t - \tau).$$

For each k this is an initial value problem for a simple harmonic oscillator kicked at $t = \tau$ – which we solved in §7! The causal solution is

$$\tilde{G} = e^{-ik\xi} \frac{\sin(kc(t - \tau))}{kc} H(t - \tau).$$

Inverting, with $A = x - \xi$ and $B = c(t - \tau)$, oddness kills the sine part of e^{ikA} and

$$\begin{aligned} G &= \frac{H(t - \tau)}{\pi c} \int_0^\infty \frac{\cos(kA) \sin(kB)}{k} dk \\ &= \frac{H(t - \tau)}{2\pi c} \int_0^\infty \frac{\sin k(A + B) - \sin k(A - B)}{k} dk, \end{aligned}$$

and by the Dirichlet discontinuous formula each integral is $\frac{\pi}{2} \operatorname{sgn}(\cdot)$, giving

$$G(x, t; \xi, \tau) = \frac{H(t - \tau)}{4c} [\operatorname{sgn}(A + B) - \operatorname{sgn}(A - B)] = \frac{1}{2c} H(c(t - \tau) - |x - \xi|).$$

The Green's function of the wave equation is a spreading 'top hat' of height $1/2c$: an impulse at (ξ, τ) affects exactly those points within distance $c(t - \tau)$ – the range of influence from the characteristics picture, reappearing from pure Fourier analysis. The solution is

$$\phi(x, t) = \frac{1}{2c} \int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} f(\xi, \tau) \, d\xi \, d\tau,$$

an integral of the forcing over the backwards light cone.

§18.10.4 Poisson's Equation and the Free-Space Green's Function

Finally, the elliptic case. **Poisson's equation** is the inhomogeneous Laplace equation

$$\nabla^2 \phi = -\rho(\mathbf{r}),$$

say in three dimensions (in electrostatics, ρ is the charge density up to constants). The delta function generalises to $\mathbb{R}^3$ in the obvious way: $\delta^{(3)}(\mathbf{r} - \mathbf{r}') = 0$ for $\mathbf{r} \neq \mathbf{r}'$ and $\int f(\mathbf{r}) \delta^{(3)}(\mathbf{r} - \mathbf{r}') \, dV = f(\mathbf{r}')$.

First we work on all of $\mathbb{R}^3$ with $\phi \rightarrow 0$ at infinity, seeking the **free-space Green's function**

$$\nabla^2 G_{\text{FS}}(\mathbf{r}; \mathbf{r}') = \delta^{(3)}(\mathbf{r} - \mathbf{r}').$$

Translating so $\mathbf{r}' = 0$, everything is spherically symmetric, so $G_{\text{FS}} = G(r)$. Integrate the defining equation over a ball B_r of radius r centred at the origin: the right side gives 1, while the divergence theorem turns the left side into a flux,

$$\int_{B_r} \nabla^2 G \, dV = \int_{\partial B_r} \nabla G \cdot \hat{\mathbf{n}} \, dS = 4\pi r^2 \frac{dG}{dr}.$$

Hence

$$\frac{dG}{dr} = \frac{1}{4\pi r^2} \implies G = -\frac{1}{4\pi r} + \text{const},$$

and decay at infinity kills the constant. Restoring the source point,

$$G_{\text{FS}}(\mathbf{r}; \mathbf{r}') = \frac{-1}{4\pi |\mathbf{r} - \mathbf{r}'|},$$

which is (minus) the Coulomb potential of a unit point charge. The solution of Poisson's equation in free space is therefore

$$\phi(\mathbf{r}) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, dV',$$

exactly the superposition-of-point-charges formula from electrostatics.

Remark. In two dimensions the same computation gives $G_{\text{FS}} = \frac{1}{2\pi} \log r$ – the $\log r$ solution we discarded by regularity when separating variables in plane polars was the point-source solution all along.

§18.10.5 Green's Identities and Boundary Value Problems

To handle Poisson's equation on a bounded domain D with boundary data, we need a few integral identities. For twice differentiable ϕ, ψ , the divergence theorem applied to $\phi \nabla \psi$ gives **Green's first identity**

$$\int_{\partial D} \phi \frac{\partial \psi}{\partial n} dS = \int_D (\phi \nabla^2 \psi + \nabla \phi \cdot \nabla \psi) dV,$$

and antisymmetrising in $\phi \leftrightarrow \psi$, **Green's second identity**

$$\int_{\partial D} \left(\phi \frac{\partial \psi}{\partial n} - \psi \frac{\partial \phi}{\partial n} \right) dS = \int_D (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV.$$

Now take $\psi = G_{\text{FS}}(\mathbf{r}; \mathbf{r}')$ and let ϕ solve $\nabla^2 \phi = -\rho$. There's a subtlety: G_{FS} is singular at $\mathbf{r}'$, so we excise a small ball $\mathcal{B}_\epsilon$ around $\mathbf{r}'$ and apply the second identity on $D \setminus \mathcal{B}_\epsilon$. On the small sphere $\partial \mathcal{B}_\epsilon$, we have $G_{\text{FS}} \sim -1/4\pi\epsilon$ and $\partial G_{\text{FS}}/\partial n \sim -1/4\pi\epsilon^2$ (with the normal pointing towards $\mathbf{r}'$), so as $\epsilon \rightarrow 0$,

$$\int_{\partial \mathcal{B}_\epsilon} \left(\phi \frac{\partial G_{\text{FS}}}{\partial n} - G_{\text{FS}} \frac{\partial \phi}{\partial n} \right) dS \rightarrow -\phi(\mathbf{r}'),$$

since the first term integrates ϕ against a shrinking sphere of total 'weight' -1 and the second vanishes like ϵ . The result is **Green's third identity**:

$$\phi(\mathbf{r}') = - \int_D G_{\text{FS}}(\mathbf{r}; \mathbf{r}') \rho(\mathbf{r}) dV + \int_{\partial D} \left(\phi \frac{\partial G_{\text{FS}}}{\partial n} - G_{\text{FS}} \frac{\partial \phi}{\partial n} \right) dS.$$

This expresses ϕ anywhere inside D in terms of the source and *both* ϕ and $\partial \phi / \partial n$ on the boundary. But for a Dirichlet problem we only *know* ϕ on the boundary – the normal derivative term is unwanted baggage.

The fix is elegant. Define the **Dirichlet Green's function** of the domain by

$$G(\mathbf{r}; \mathbf{r}') = G_{\text{FS}}(\mathbf{r}; \mathbf{r}') + H(\mathbf{r}; \mathbf{r}'),$$

where H is harmonic in D ($\nabla^2 H = 0$) and is chosen so that $G = 0$ for $\mathbf{r} \in \partial D$. Running Green's second identity with H (which has no singularity) and adding it to the third identity, all the H corrections combine and the $\partial \phi / \partial n$ term drops out (its coefficient G vanishes on ∂D). With Dirichlet data $\phi = h$ on ∂D :

$$\phi(\mathbf{r}') = - \int_D G(\mathbf{r}; \mathbf{r}') \rho(\mathbf{r}) dV + \int_{\partial D} h(\mathbf{r}) \frac{\partial G}{\partial n}(\mathbf{r}; \mathbf{r}') dS.$$

The first term is the familiar Green's function solution; the second implements the inhomogeneous boundary conditions.

§18.10.6 The Method of Images

Everything now rests on finding the harmonic correction H – and for symmetric domains there is a lovely trick: place fictitious **image sources** *outside* the domain, positioned so that their combined potential cancels on the boundary.

Example 18.10.2 (Laplace's Equation on a Half-Space)

Solve $\nabla^2 \phi = 0$ in the half-space $z > 0$, with $\phi(x, y, 0) = h(x, y)$ and $\phi \rightarrow 0$ at infinity. For a source at $\mathbf{r}' = (x', y', z')$ with $z' > 0$, place an equal and *opposite* image at the mirror point $\mathbf{r}'' = (x', y', -z')$:

$$G(\mathbf{r}; \mathbf{r}') = \frac{-1}{4\pi|\mathbf{r} - \mathbf{r}'|} + \frac{1}{4\pi|\mathbf{r} - \mathbf{r}''|}.$$

The image term is harmonic inside $z > 0$ (its singularity is at $z < 0$), and on the plane $z = 0$ every point is equidistant from $\mathbf{r}'$ and $\mathbf{r}''$, so $G = 0$ there: this *is* the Dirichlet Green's function. Computing the normal derivative on the boundary,

$$\left. \frac{\partial G}{\partial n} \right|_{z=0} = \frac{z'}{2\pi} [(x - x')^2 + (y - y')^2 + z'^2]^{-3/2},$$

so the solution of the boundary value problem is

$$\phi(x', y', z') = \frac{z'}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{h(x, y) \, dx \, dy}{[(x - x')^2 + (y - y')^2 + z'^2]^{3/2}}.$$

Example 18.10.3 (Images for the Wave Equation)

The method works for time-dependent problems too. To solve the wave equation on the half-line $x > 0$ with a fixed end $\phi(0, t) = 0$, take the free-space causal Green's function and subtract its mirror image:

$$G(x, t; \xi, \tau) = \frac{1}{2c} H(c(t - \tau) - |x - \xi|) - \frac{1}{2c} H(c(t - \tau) - |x + \xi|),$$

which vanishes at $x = 0$ for all time. (Adding instead of subtracting the image gives a free end – a Neumann condition.) Concretely, a leftward-travelling pulse started near $\xi > 0$ meets its opposite-signed image emerging from the boundary, and the result is the familiar picture of a wave reflecting off a fixed end with inverted phase – in agreement with the $B = -A$ limit of our reflection analysis in §3.

And with that, the toolbox is complete. It is worth stepping back to admire how the pieces interlock: separation of variables turns PDEs into Sturm–Liouville problems; completeness of eigenfunctions is the delta function's expansion; the delta function defines Green's functions; Fourier transforms construct them; and the characteristics of the equation dictate the causal structure that the Green's functions faithfully reproduce. Linearity, exploited from every possible direction.

19 Numerical Analysis

Almost none of the problems that we would like to solve can actually be solved. Most integrals have no closed form, most differential equations have no solution in terms of familiar functions, and a large linear system is not something one inverts by hand. Numerical analysis is the study of what to do about this: how to replace an infinite-dimensional problem by a finite-dimensional one whose answer we can compute, and – crucially – how to say something rigorous about the difference between the two.

So this is not a course about computers, and it is certainly not a course about programming. It is a course about *approximation*, and about the two questions that any approximation scheme must answer. How accurate is it? And how expensive is it? Almost every result in these notes is an answer to one of those two questions, and a surprising amount of quite beautiful classical mathematics – orthogonal polynomials, Rolle’s theorem, the maximum modulus principle – turns out to be exactly what is needed to answer them.

This article constitutes my notes for the ‘Numerical Analysis’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§19.1 Polynomial Interpolation

Polynomials are the simplest functions we know how to compute with: to evaluate one we need only additions and multiplications, and to integrate or differentiate one we need only the power rule. It is therefore natural that our first move, when confronted with an awkward function f , is to replace it by a polynomial that agrees with f at a handful of points. This is the *interpolation problem*, and it is the foundation on which almost everything else in this course is built – quadrature rules, multistep methods for differential equations and error estimates all come, in the end, from interpolating polynomials.

Throughout, we write $\mathcal{P}_n$ (or $\mathcal{P}_n[x]$ when we want to emphasise the variable) for the vector space of real polynomials of degree at most n . Note that $\dim \mathcal{P}_n = n + 1$, since a polynomial in $\mathcal{P}_n$ is determined by its $n + 1$ coefficients. That number, $n + 1$, is worth keeping in mind: it is exactly the number of data points that we shall be able to fit.

§19.1.1 Lagrange and Newton Polynomials

Let $f : [a, b] \rightarrow \mathbb{R}$ be a real-valued continuous function defined on some interval $[a, b]$ and let $(x_i)_{i=0}^n$ be $n + 1$ distinct points in $[a, b]$. We wish to construct a polynomial p of degree n which interpolates f at these points, i.e., satisfies

$$p(x_i) = f(x_i), \quad i = \overline{0, n}, \quad p \in \mathcal{P}_n$$

Theorem 19.1.1 (Existence and Uniqueness)

Given $f \in C[a, b]$ and $n + 1$ distinct points $(x_i)_{i=0}^n \in [a, b]$, there is exactly one

polynomial $p \in \mathcal{P}_n$ such that $p(x_i) = f(x_i)$ for all i .

Proof. Existence. There is at least one polynomial interpolant $p \in \mathcal{P}_n$, the one in the Lagrange form,

$$p(x) = \sum_{i=0}^n f(x_i) \ell_i(x) \quad \text{with} \quad \ell_i(x) := \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}, \quad i = 0, \dots, n, \quad (\dagger)$$

the ℓ_i s are called the **fundamental Lagrange polynomials**. Each ℓ_i is the product of n linear factors, hence $\ell_i \in \mathcal{P}_n$. It also equals 1 at x_i and vanishes at $x_j \neq x_i$, i.e., $\ell_i(x_j) = \delta_{ij}$. Therefore $p \in \mathcal{P}_n$ and

$$p(x_j) = \sum_{i=0}^n f(x_i) \ell_i(x_j) = f(x_j).$$

Uniqueness. There is at most one polynomial interpolant $p \in \mathcal{P}_n$ to f on $(x_i)_{i=0}^n$. For if there are two, $p, q \in \mathcal{P}_n$, then the polynomial $r := p - q$ is of degree n and vanishes at $n + 1$ points, whence $r \equiv 0$. $\square$

Let us introduce the so-called **nodal polynomial**

$$\omega(x) = \prod_{i=0}^n (x - x_i).$$

Then, in the expression $(\dagger)$ for ℓ_i , the numerator is simply $\omega(x)/(x - x_i)$ while the denominator is equal to $\omega'(x_i)$. With that we arrive to a compact Lagrange form

$$p(x) = \sum_{i=0}^n f(x_i) \ell_i(x) = \sum_{i=0}^n \frac{f(x_i)}{\omega'(x_i)} \frac{\omega(x)}{x - x_i}.$$

The lagrange forms above and in $(\dagger)$ for interpolating polynomials are easy to manipulate, but they are unsuitable for numerical evaluation. An alternative is the *Newton form* which has an *adaptive* nature.

Method 19.1.2 (Newton Form)

For $k = 0, 1, \dots, n$, let $p_k \in \mathcal{P}_k$ be the polynomial interpolant to f on $x_0, \dots, x_k$. Then two subsequent p_{k-1} and p_k interpolate the same values $f(x_i)$ for $i \leq k - 1$, hence their difference is a polynomial of degree k that vanishes at k points $x_0, \dots, x_{k-1}$. Thus

$$p_k(x) - p_{k-1}(x) = A_k \prod_{i=0}^{k-1} (x - x_i), \quad (19.1)$$

with some constant A_k which is seen to be equal to the leading coefficient of p_k .

It follows that $p = p_n$ can be built step by step as one constructs the sequence $(p_0, p_1, \dots)$, with p_k obtained from p_{k-1} by adding the term on the right-hand side

of (19.1), so that finally

$$p(x) = p_n(x) = p_0(x) + \sum_{k=1}^n [p_k(x) - p_{k-1}(x)] = \sum_{k=0}^n A_k \prod_{i=0}^{k-1} (x - x_i).$$

The coefficients A_k appearing here are important enough to be given a name of their own.

Definition 19.1.3 (Divided Difference)

Given $f \in C[a, b]$ and $k + 1$ distinct points $(x_i)_{i=0}^k \in [a, b]$, the **divided difference** $f[x_0, \dots, x_k]$ of order k is the leading coefficient of the polynomial $p_k \in \mathcal{P}_k$ which interpolates f at these points.

The divided difference is a symmetric function of the variables $[x_0, \dots, x_k]$, since the interpolating polynomial does not care about the order in which we list the interpolation points. Also, if $f(x) = x^m$ with $m \leq k$, then $p_k = f$ and so $f[x_0, \dots, x_k] = \delta_{km}$. The two lowest-order cases are worth recording: $f[x_0] = f(x_0)$, and $f[x_0, x_1]$ is the slope

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

of the straight line through $(x_0, f(x_0))$ and $(x_1, f(x_1))$, which is where the name ‘divided difference’ comes from.

With this definition we arrive at the **Newton formula** for the interpolating polynomial.

Theorem 19.1.4 (Newton Formula)

Given $n + 1$ distinct points $(x_i)_{i=0}^n$, let $p_n \in \mathcal{P}_n$ be the polynomial that interpolates f at these points. Then it may be written in the Newton form

$$p_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \dots \\ \dots + f[x_0, x_1, \dots, x_n](x - x_0)(x - x_1) \dots (x - x_{n-1}),$$

or, more compactly,

$$p_n(x) = \sum_{k=0}^n f[x_0, \dots, x_k] \prod_{i=0}^{k-1} (x - x_i)$$

To make this formula of any use, we need an expression for $f[x_0, \dots, x_k]$. One such can be derived from the Lagrange formula by identifying the leading coefficient of p . This turns out to be

$$f[x_0, \dots, x_n] = \sum_{i=0}^n \frac{f(x_i)}{\omega'(x_i)}, \quad \omega(x) := \prod_{i=0}^n (x - x_i).$$

However, this expression has computational disadvantages as the Lagrange form itself. A useful way to calculate divided difference is again an adaptive (or recurrence) approach.

Theorem 19.1.5 (Recurrence Relation)

For distinct $x_0, x_1, \dots, x_k$ with $k \geq 1$, we have

$$f[x_0, \dots, x_k] = \frac{f[x_1, \dots, x_k] - f[x_0, \dots, x_{k-1}]}{x_k - x_0}. \quad (19.2)$$

Proof. Let $q_0, q_1 \in \mathcal{P}_{k-1}$ be the polynomials such that q_0 interpolates f on $(x_0, x_1, \dots, x_{k-1})$ and q_1 interpolates on $(x_1, \dots, x_k)$. Consider the polynomial

$$p(x) = \frac{x - x_0}{x_k - x_0} q_1(x) + \frac{x_k - x}{x_k - x_0} q_0(x), \quad p \in \mathcal{P}_k.$$

One readily sees that $p(x_i) = f(x_i)$ for all i : at x_0 the first term vanishes and the second has coefficient 1, at x_k the reverse happens, and at an intermediate x_i both q_0 and q_1 return $f(x_i)$ while the two coefficients sum to 1. Hence p is the k -th degree interpolating polynomial to f . Moreover, the leading coefficient of p is equal to the difference of those of q_1 and q_0 divided by $x_k - x_0$, and that is exactly what the recurrence (19.2) says. $\square$

The recursive relation allows for the fast evaluation of the **divided difference table**

x_i	f_i	$f[*,*]$	$f[*,*,*]$	$\dots$	$f[*, \dots, *]$
x_0	$f[x_0]$				
		$f[x_0, x_1]$			
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$		
		$f[x_1, x_2]$		$\ddots$	
x_2	$f[x_2]$		$f[x_2, x_3, x_4]$	$\dots$	$f[x_0, x_1, \dots, x_n]$
		$f[x_2, x_3]$		$\ddots$	
x_3	$f[x_3]$		$\ddots$		
$\vdots$	$\vdots$	$\ddots$			
x_n	$f[x_n]$				

This can be done in $\mathcal{O}(n^2)$ operations and the outcome is the numbers $\{f[x_0, \dots, x_k]\}_{k=0}^n$ at the head of the columns which can be used in the Newton form.

Pictorially, each entry is obtained from its two neighbours to the left, in the pattern

$$\begin{array}{ccc}
 f[x_0, \dots, x_{k-1}] & \xrightarrow{-} & \\
 & \searrow & \\
 & & f[x_0, \dots, x_k] \\
 f[x_1, \dots, x_k] & \xrightarrow{+} & \text{then divide by } x_k - x_0
 \end{array}$$

so that the table is filled column by column, each column being one entry shorter than the last.

Finally evaluation of p at a given point x using Newton formula (provided that divided differences $A_k = f[x_0, \dots, x_k]$ are known) requires just n multiplications, as long as we

do it by the **Horner scheme**

$$p_n(x) = \{ \cdots \{ \{ A_n \times (x - x_{n-1}) + A_{n-1} \} \times (x - x_{n-2}) + A_{n-2} \} \\ \times (x - x_{n-3}) + \cdots + A_1 \} \times (x - x_0) + A_0.$$

It is worth pausing to compare the two forms. Both cost $\mathcal{O}(n^2)$ operations to set up. The Lagrange form is the one to reach for when we want to *manipulate* the interpolant symbolically – to integrate it, say, or to differentiate it – because f appears in it linearly, through the values $f(x_i)$, with coefficients ℓ_i that do not depend on f at all. This is exactly what we will exploit when we build quadrature rules in Section 3. The Newton form, on the other hand, is the one to reach for when we want to *compute*: adding a new interpolation point x_{n+1} costs only one extra term, whereas in the Lagrange form every single ℓ_i has to be recomputed from scratch.

§19.1.2 Examples

We will now look at some examples of these methods in use.

Example 19.1.6 (Interpolating Polynomial)

Given the data

x_i	0	1	2	3
$f(x_i)$	-3	-3	-1	9

we want to find the interpolating polynomial $p \in \mathcal{P}_3$ in both Lagrange and Newton forms.

Fundamental Lagrange polynomials. We have

$$\begin{aligned} \ell_0(x) &= \frac{(x-1)(x-2)(x-3)}{-6} = -\frac{1}{6}(x^3 - 6x^2 + 11x - 6), \\ \ell_1(x) &= \frac{x(x-2)(x-3)}{2} = \frac{1}{2}(x^3 - 5x^2 + 6x), \\ \ell_2(x) &= \frac{x(x-1)(x-3)}{-2} = -\frac{1}{2}(x^3 - 4x^2 + 3x), \\ \ell_3(x) &= \frac{x(x-1)(x-2)}{6} = \frac{1}{6}(x^3 - 3x^2 + 2x). \end{aligned}$$

Lagrange form. Using the polynomials above, we can write

$$\begin{aligned} p(x) &= (-3) \cdot \ell_0(x) + (-3) \cdot \ell_1(x) + (-1) \cdot \ell_2(x) + 9 \cdot \ell_3(x) \\ &= \left(\frac{1}{2} - \frac{3}{2} + \frac{1}{2} + \frac{3}{2} \right) x^3 + \left(-3 + \frac{15}{2} - 2 - \frac{9}{2} \right) x^2 + \left(\frac{11}{2} - 9 + \frac{3}{2} + 3 \right) x - 3 \\ &= x^3 - 2x^2 + x - 3. \end{aligned}$$

Newton form. Here we build the divided difference table, filling it in column by column using the recurrence (19.2). Since the nodes are spaced one apart, the denominators $x_k - x_0$ are 1, 2, 3 in the three successive columns.

x_i	$f[x_i]$	$f[*,*]$	$f[*,*,*]$	$f[*,*,*,*]$
0	-3			
		0		
1	-3		1	
		2		1
2	-1		4	
		10		
3	9			

Reading off the numbers at the head of each column gives

$$p(x) = -3 + 0 \cdot (x - 0) + 1 \cdot (x - 0)(x - 1) + 1 \cdot (x - 0)(x - 1)(x - 2),$$

and expanding this out we do indeed recover $p(x) = x^3 - 2x^2 + x - 3$, as we must.

Notice how much less work the second computation was. The advantage becomes more pronounced still if the data arrives one point at a time.

Example 19.1.7 (Adding a Point)

Suppose that to the data of the previous example we now append the point $x_4 = 4$, $f(x_4) = 33$. In the Lagrange form we would have to start again. In the Newton form we merely extend the table by one diagonal:

$$f[x_3, x_4] = \frac{33 - 9}{1} = 24, \quad f[x_2, x_3, x_4] = \frac{24 - 10}{2} = 7, \quad f[x_1, \dots, x_4] = \frac{7 - 4}{2} = \frac{3}{2},$$

and finally $f[x_0, \dots, x_4] = (\frac{3}{2} - 1)/4 = \frac{1}{8}$. Hence

$$p_4(x) = p_3(x) + \frac{1}{8}x(x - 1)(x - 2)(x - 3).$$

All the work done for p_3 has been retained.

Example 19.1.8 (A Divided Difference Table with Non-uniform Nodes)

Let $f(x) = \sqrt{x}$ and take the nodes $x_0 = 1$, $x_1 = 4$, $x_2 = 9$. Then

$$f[x_0, x_1] = \frac{2 - 1}{4 - 1} = \frac{1}{3}, \quad f[x_1, x_2] = \frac{3 - 2}{9 - 4} = \frac{1}{5},$$

and so

$$f[x_0, x_1, x_2] = \frac{\frac{1}{5} - \frac{1}{3}}{9 - 1} = -\frac{1}{60}.$$

The interpolant is therefore

$$p_2(x) = 1 + \frac{1}{3}(x - 1) - \frac{1}{60}(x - 1)(x - 4).$$

Evaluating at $x = 2$ gives $p_2(2) = 1 + \frac{1}{3} + \frac{1}{30} = 1.3\bar{6}$, against the true value $\sqrt{2} = 1.41421\dots$. The error is about 0.048, which – as we are about to see – is entirely predictable.

§19.1.3 The Error of Interpolation

An interpolant is only useful if we can say how far it is from f *between* the interpolation points. The following theorem, which is the workhorse of the whole subject, says that the error looks exactly like the next term of a Taylor series, with the monomial $(x-a)^{n+1}$ replaced by the nodal polynomial ω .

Theorem 19.1.9 (Interpolation Error)

Let $f \in C^{n+1}[a, b]$ and let $x_0, \dots, x_n \in [a, b]$ be distinct. Let $p \in \mathcal{P}_n$ interpolate f at these points. Then for every $x \in [a, b]$ there exists $\xi = \xi(x) \in (a, b)$ such that

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi) \omega(x), \quad \omega(x) = \prod_{i=0}^n (x - x_i).$$

Proof. If $x = x_i$ for some i then both sides vanish and there is nothing to prove, so fix $x \in [a, b]$ with $x \neq x_i$ for all i . In particular $\omega(x) \neq 0$, so we may define $\varphi : [a, b] \rightarrow \mathbb{R}$ by

$$\varphi(t) = f(t) - p(t) - (f(x) - p(x)) \frac{\omega(t)}{\omega(x)}.$$

This is a C^{n+1} function of t , and it has been rigged so that $\varphi(x) = 0$; it also vanishes at each of $x_0, \dots, x_n$, since there $f - p$ and ω both vanish. So φ has at least $n+2$ distinct zeros in $[a, b]$.

Now we apply Rolle's theorem repeatedly. Between consecutive zeros of φ there is a zero of φ' , so φ' has at least $n+1$ distinct zeros; between consecutive zeros of φ' there is a zero of φ'' , so φ'' has at least n distinct zeros; and continuing in this way, $\varphi^{(n+1)}$ has at least one zero, say $\xi \in (a, b)$.

But $p \in \mathcal{P}_n$ has $p^{(n+1)} \equiv 0$, while ω is monic of degree $n+1$ and so $\omega^{(n+1)} \equiv (n+1)!$. Hence

$$0 = \varphi^{(n+1)}(\xi) = f^{(n+1)}(\xi) - (f(x) - p(x)) \frac{(n+1)!}{\omega(x)},$$

and rearranging gives the result. $\square$

The same trick – count zeros, then apply Rolle – gives a very satisfying interpretation of divided differences: they are, up to a factorial, derivatives.

Theorem 19.1.10 (Divided Differences and Derivatives)

Let $x_0, \dots, x_n$ be distinct points and let $[a, b]$ be the smallest interval containing them. If $f \in C^n[a, b]$, then there is some $\xi \in (a, b)$ with

$$f[x_0, \dots, x_n] = \frac{1}{n!} f^{(n)}(\xi).$$

Proof. Let $p \in \mathcal{P}_n$ interpolate f at the $n+1$ points. Then $f - p$ vanishes at $n+1$ distinct points, so by Rolle applied n times, $(f - p)^{(n)}$ vanishes somewhere in (a, b) , say at ξ . But p has leading coefficient $f[x_0, \dots, x_n]$, so $p^{(n)} \equiv n! f[x_0, \dots, x_n]$, and thus

$$0 = f^{(n)}(\xi) - n! f[x_0, \dots, x_n]. \quad \square$$

Remark. This is a pleasing statement in its own right: it says that the divided difference is a ‘discrete derivative’, and it also lets us make sense of $f[x_0, \dots, x_n]$ when some of the points coalesce. Letting all the $x_i \rightarrow x$, the natural definition is $f[x, x, \dots, x] = f^{(n)}(x)/n!$, and with this convention the Newton formula becomes the Taylor expansion of f about x . Interpolation and Taylor expansion are, in this sense, the same thing.

Let us use Theorem 19.1.9 on the example above.

Example 19.1.11

With $f(x) = \sqrt{x}$ and nodes 1, 4, 9, we have $f'''(x) = \frac{3}{8}x^{-5/2}$ and $\omega(2) = (2-1)(2-4)(2-9) = 14$. So

$$\sqrt{2} - p_2(2) = \frac{1}{6} \cdot \frac{3}{8} \xi^{-5/2} \cdot 14 = \frac{7}{8} \xi^{-5/2}$$

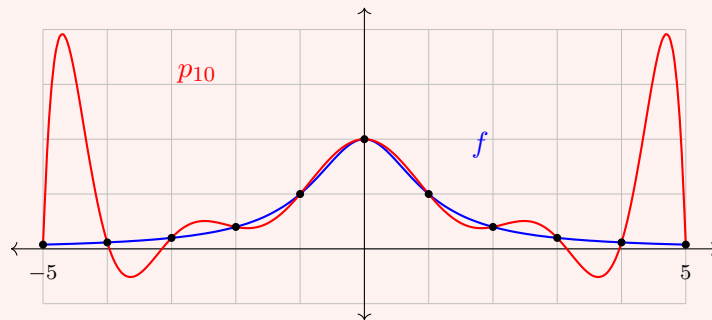
for some $\xi \in (1, 9)$. Since $\xi > 1$ we get $|\sqrt{2} - p_2(2)| \leq 7/8$, which is true but useless; a better bound comes from noticing that ξ is in practice near the interval where the action is. The exact error was 0.0476, corresponding to $\xi \approx 2.2$. The moral is that the theorem gives the *form* of the error very precisely, but a crude bound on $f^{(n+1)}$ can throw most of that precision away.

§19.1.4 The Runge Phenomenon

Theorem 19.1.9 contains two factors that we might hope to control. The factor $f^{(n+1)}(\xi)/(n+1)!$ is out of our hands – it is a property of f . But the nodal polynomial ω depends only on where we choose to put our interpolation points, and this we *can* control. The following famous example of Runge shows that the choice matters enormously, and that the obvious choice is a bad one.

Example 19.1.12 (Runge's Example)

Let $f(x) = (1+x^2)^{-1}$ on $[-5, 5]$, and interpolate at the $n+1$ equally spaced points $x_j = -5 + 10j/n$. Below is the case $n = 10$: the function f is in blue, the interpolant p_{10} in red, with the interpolation points marked.



The interpolant is faithful in the middle of the interval and catastrophically bad near the endpoints, where it oscillates with amplitude close to 2 even though $|f| \leq 1$ everywhere. Worse, this does not improve as n grows: one can show that

$$\max_{x \in [-5, 5]} |f(x) - p_n(x)| \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

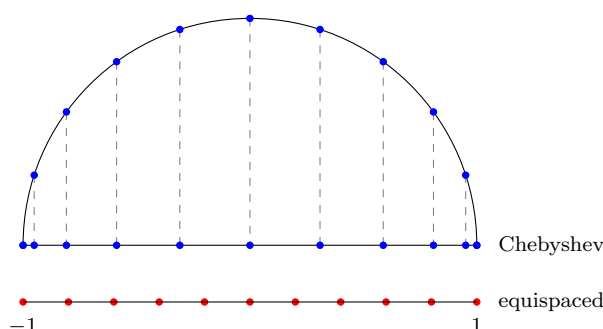
Interpolating at more equally spaced points makes matters *worse*, not better.

What has gone wrong? The culprit is the nodal polynomial. For equally spaced nodes on $[-1, 1]$, $|\omega|$ is tiny in the middle of the interval but grows by a factor exponential in n as we approach the ends, so the error bound is wildly non-uniform. If we want $|\omega|$ to be uniformly small, we should not space the nodes uniformly; we should cluster them near the endpoints.

The optimal choice is classical. On $[-1, 1]$ the **Chebyshev points**

$$x_j = \cos\left(\frac{(n-j)\pi}{n}\right), \quad j = 0, 1, \dots, n,$$

are obtained by taking $n+1$ equally spaced points on the upper half of the unit circle and projecting them down onto the real axis. The picture makes the clustering obvious.



With this choice one can show that the nodal polynomial satisfies

$$\max_{x \in [-1, 1]} |\omega(x)| = 2^{-n},$$

and that no other choice of $n+1$ nodes does better. (The nodal polynomial for the Chebyshev points is $2^{-n}T_{n+1}$, where T_{n+1} is the Chebyshev polynomial we shall meet in the next section; the optimality is a consequence of its equioscillation.) For a general interval $[a, b]$ we simply rescale, taking

$$x_j = \frac{a+b}{2} + \frac{b-a}{2} \cos\left(\frac{(n-j)\pi}{n}\right).$$

Repeating Runge's experiment with Chebyshev points, the interpolant converges uniformly to f , and rapidly at that.

Remark. This is a good point to pause and notice what the theory has bought us. Nothing about the function f has changed; all we have done is move the sampling points. But the same theorem that predicted disaster for equispaced nodes predicts success for Chebyshev nodes, because it isolates precisely the quantity (ω) that we are free to choose. This pattern – write down an exact error formula, look at which factor is under our control, and optimise it – recurs throughout the course, and is the whole idea behind Gaussian quadrature.

§19.2 Orthogonal Polynomials

We now change tack somewhat. Rather than asking for a polynomial that agrees with f at finitely many points, we ask for a polynomial that is close to f *on average*, in the sense

of least squares. This turns out to require a good basis for $\mathcal{P}_n$, and the right notion of ‘good’ is orthogonality with respect to a suitable inner product. Orthogonal polynomials are the central objects here, and they will reappear – somewhat unexpectedly – as the key to the most accurate quadrature rules there are.

§19.2.1 Inner Products on Function Spaces

Definition 19.2.1 (Inner Product)

Let V be a real vector space. An **inner product** (or scalar product) on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ which is

- (i) *symmetric*: $\langle f, g \rangle = \langle g, f \rangle$ for all $f, g \in V$;
- (ii) *bilinear*: linear in each argument;
- (iii) *positive definite*: $\langle f, f \rangle \geq 0$ for all f , with equality if and only if $f = 0$.

We say f and g are **orthogonal** if $\langle f, g \rangle = 0$, and we write $\|f\| = \langle f, f \rangle^{1/2}$ for the induced norm.

For us V will almost always be a space of functions on an interval, and the inner product will be of the following shape.

Definition 19.2.2 (Weight Function)

Let $[a, b] \subseteq \mathbb{R}$ (possibly infinite) and let $w : (a, b) \rightarrow \mathbb{R}$ be continuous, non-negative, and non-zero except at isolated points, with $\int_a^b w(x) dx < \infty$. We call w a **weight function**, and define

$$\langle f, g \rangle = \int_a^b w(x) f(x) g(x) dx.$$

The conditions on w are exactly what is needed for this to be an inner product on $C[a, b]$: symmetry and bilinearity are immediate, and if $\langle f, f \rangle = \int_a^b w f^2 = 0$ with $w f^2 \geq 0$ continuous, then $w f^2 \equiv 0$, and since w vanishes only at isolated points, $f \equiv 0$ by continuity.

A second, more down-to-earth example is available when we only know f at finitely many points.

Example 19.2.3 (Discrete Inner Product)

Let $\xi_1, \dots, \xi_m$ be distinct points and set

$$\langle f, g \rangle = \sum_{j=1}^m f(\xi_j) g(\xi_j).$$

This is symmetric and bilinear, and $\langle f, f \rangle = \sum_j f(\xi_j)^2 \geq 0$. It is *not* positive definite on all of $C[a, b]$ – a function vanishing at every ξ_j has zero norm – but it *is* positive definite on $\mathcal{P}_n$ provided $n < m$, since a nonzero polynomial of degree at most n cannot vanish at $m > n$ points. This is the inner product relevant to fitting a curve to experimental data.

Definition 19.2.4 (Orthogonal Polynomials)

A sequence $p_0, p_1, p_2, \dots$ of polynomials is a sequence of **orthogonal polynomials** with respect to $\langle \cdot, \cdot \rangle$ if $\deg p_n = n$ for every n , and $\langle p_n, p_m \rangle = 0$ whenever $n \neq m$.

Since $\deg p_n = n$ exactly, the polynomials $p_0, \dots, p_n$ are automatically linearly independent (they have distinct degrees), and so they form an orthogonal *basis* of $\mathcal{P}_n$. This is really the point of the definition: we are constructing a basis of $\mathcal{P}_n$ in which computations decouple.

Of course, orthogonal polynomials are only determined up to scaling, so to speak of *the* orthogonal polynomials we should normalise. The usual convention in this course is to take them **monic**, that is, with leading coefficient 1.

Theorem 19.2.5 (Existence and Uniqueness)

For any inner product on the space of polynomials there exists a unique sequence of monic orthogonal polynomials $p_0, p_1, \dots$, and these may be constructed by the Gram–Schmidt formula

$$p_{n+1}(x) = x^{n+1} - \sum_{k=0}^n \frac{\langle x^{n+1}, p_k \rangle}{\langle p_k, p_k \rangle} p_k(x).$$

Proof. We induct on n . For $n = 0$ the only monic polynomial of degree 0 is $p_0 \equiv 1$, so existence and uniqueness are clear. Suppose then that $p_0, \dots, p_n$ have been found. Define p_{n+1} by the displayed formula. The subtracted sum lies in $\mathcal{P}_n$, so p_{n+1} is monic of degree $n + 1$. For each $j \leq n$, using orthogonality of the p_k ,

$$\langle p_{n+1}, p_j \rangle = \langle x^{n+1}, p_j \rangle - \frac{\langle x^{n+1}, p_j \rangle}{\langle p_j, p_j \rangle} \langle p_j, p_j \rangle = 0,$$

so p_{n+1} is orthogonal to $p_0, \dots, p_n$ and hence to all of $\mathcal{P}_n$.

For uniqueness, suppose $\tilde{p}_{n+1}$ were another monic polynomial of degree $n + 1$ orthogonal to $\mathcal{P}_n$. Then $q = p_{n+1} - \tilde{p}_{n+1}$ has degree at most n , so $q \in \mathcal{P}_n$, and q is orthogonal to everything in $\mathcal{P}_n$ – in particular to itself. Hence $\langle q, q \rangle = 0$ and so $q = 0$. $\square$

Remark. Note the characterisation buried in that proof: p_n is the unique monic polynomial of degree n orthogonal to *every* polynomial of lower degree. We will use this form of the statement constantly.

Gram–Schmidt is a perfectly good proof, but it is a poor algorithm: computing p_{n+1} requires all of $p_0, \dots, p_n$, so the cost grows and rounding errors accumulate. For the inner products we care about there is a far better recurrence, which involves only the previous *two* polynomials.

§19.2.3 The Three-Term Recurrence

The key hypothesis is that multiplication by x is self-adjoint. For the weighted-integral inner product this is obvious:

$$\langle xf, g \rangle = \int_a^b w(x) xf(x)g(x) dx = \langle f, xg \rangle,$$

and it holds for the discrete inner product too.

Theorem 19.2.6 (Three-Term Recurrence)

Suppose the inner product satisfies $\langle xf, g \rangle = \langle f, xg \rangle$ for all polynomials f, g . Then the monic orthogonal polynomials satisfy

$$p_{n+1}(x) = (x - \alpha_n)p_n(x) - \beta_n p_{n-1}(x), \quad n \geq 0,$$

with the conventions $p_{-1} \equiv 0$, $p_0 \equiv 1$, β_0 arbitrary, and

$$\alpha_n = \frac{\langle p_n, xp_n \rangle}{\langle p_n, p_n \rangle}, \quad \beta_n = \frac{\langle p_n, p_n \rangle}{\langle p_{n-1}, p_{n-1} \rangle} > 0 \quad (n \geq 1).$$

Proof. Both p_{n+1} and xp_n are monic of degree $n+1$, so their difference lies in $\mathcal{P}_n$, and since $p_0, \dots, p_n$ is a basis of $\mathcal{P}_n$ we may write

$$p_{n+1}(x) - xp_n(x) = \sum_{k=0}^n \gamma_k p_k(x)$$

for some scalars γ_k . Taking the inner product with p_j for $j \leq n$ and using orthogonality of p_{n+1} against $\mathcal{P}_n$,

$$\gamma_j \langle p_j, p_j \rangle = -\langle xp_n, p_j \rangle = -\langle p_n, xp_j \rangle,$$

where the last step is the self-adjointness hypothesis. Now if $j \leq n-2$ then $xp_j \in \mathcal{P}_{n-1}$, which is orthogonal to p_n , so $\gamma_j = 0$. Only $j = n$ and $j = n-1$ survive, giving

$$p_{n+1}(x) = (x - \alpha_n)p_n(x) - \beta_n p_{n-1}(x), \quad \alpha_n = \frac{\langle p_n, xp_n \rangle}{\langle p_n, p_n \rangle}, \quad \beta_n = \frac{\langle p_n, xp_{n-1} \rangle}{\langle p_{n-1}, p_{n-1} \rangle}.$$

Finally, xp_{n-1} is monic of degree n , so $xp_{n-1} - p_n \in \mathcal{P}_{n-1}$ is orthogonal to p_n ; hence $\langle p_n, xp_{n-1} \rangle = \langle p_n, p_n \rangle$, which gives the stated β_n , manifestly positive. $\square$

This is a genuinely useful algorithm: each new polynomial costs $\mathcal{O}(1)$ inner products rather than $\mathcal{O}(n)$, and the recurrence is numerically stable. Let's look at the two examples that matter most.

Example 19.2.7 (Legendre Polynomials)

Take $[a, b] = [-1, 1]$ and $w \equiv 1$, so that $\langle f, g \rangle = \int_{-1}^1 fg dx$. The resulting monic orthogonal polynomials are the (monic) **Legendre polynomials**. By symmetry $\langle p_n, xp_n \rangle = 0$ for every n – the integrand is odd – so $\alpha_n = 0$ throughout, and the

recurrence collapses to $p_{n+1} = xp_n - \beta_n p_{n-1}$. Computing,

$$p_0 = 1, \quad p_1 = x, \quad p_2 = x^2 - \frac{1}{3}, \quad p_3 = x^3 - \frac{3}{5}x, \quad p_4 = x^4 - \frac{6}{7}x^2 + \frac{3}{35}.$$

For instance $\beta_1 = \langle p_1, p_1 \rangle / \langle p_0, p_0 \rangle = (2/3)/2 = 1/3$, giving $p_2 = x \cdot x - \frac{1}{3} = x^2 - \frac{1}{3}$ as claimed. In the literature one usually meets these normalised so that $P_n(1) = 1$; then $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$, and so on.

Example 19.2.8 (Chebyshev Polynomials)

Take $[a, b] = [-1, 1]$ with the weight $w(x) = (1 - x^2)^{-1/2}$, so

$$\langle f, g \rangle = \int_{-1}^1 \frac{f(x)g(x)}{\sqrt{1-x^2}} dx.$$

(The weight blows up at ± 1 , but integrably, so this is fine.) Define T_n by

$$T_n(\cos \theta) = \cos(n\theta).$$

That this really is a polynomial in $x = \cos \theta$ follows by induction from the identity $\cos((n+1)\theta) + \cos((n-1)\theta) = 2 \cos \theta \cos(n\theta)$, which gives at once the recurrence

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \quad T_0 = 1, \quad T_1 = x.$$

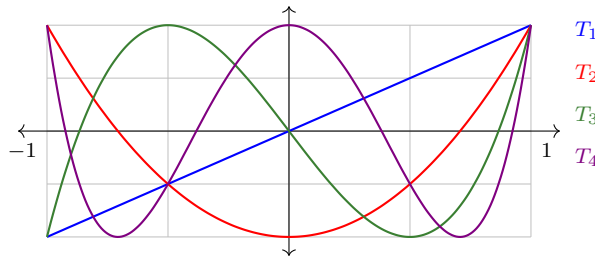
So $T_2 = 2x^2 - 1$, $T_3 = 4x^3 - 3x$, $T_4 = 8x^4 - 8x^2 + 1$, and in general T_n has leading coefficient 2^{n-1} for $n \geq 1$.

To see orthogonality, substitute $x = \cos \theta$, $dx = -\sin \theta d\theta$, $\sqrt{1-x^2} = \sin \theta$:

$$\langle T_n, T_m \rangle = \int_{-1}^1 \frac{T_n T_m}{\sqrt{1-x^2}} dx = \int_0^\pi \cos(n\theta) \cos(m\theta) d\theta = 0$$

for $n \neq m$, by the usual orthogonality of cosines. The monic Chebyshev polynomials are $2^{1-n}T_n$ for $n \geq 1$.

The Chebyshev polynomials are worth a picture, because their defining feature is visible at a glance: on $[-1, 1]$ they oscillate between -1 and $+1$, attaining those extreme values $n+1$ times.



This *equioscillation* is precisely why the Chebyshev points of the previous section are optimal interpolation nodes: among all monic polynomials of degree n , the monic Chebyshev polynomial $2^{1-n}T_n$ is the one with the smallest maximum modulus on $[-1, 1]$, namely 2^{1-n} . Any other monic q of degree n with $\|q\|_\infty < 2^{1-n}$ would have $2^{1-n}T_n - q$ alternating in sign at the $n+1$ extrema of T_n , hence having n zeros – impossible for a polynomial of degree $n-1$ that is not identically zero.

There are other classical families, each attached to its own weight, which we mention for completeness:

Family	Interval	Weight $w(x)$
Legendre	$[-1, 1]$	1
Chebyshev	$[-1, 1]$	$(1 - x^2)^{-1/2}$
Laguerre	$[0, \infty)$	e^{-x}
Hermite	$(-\infty, \infty)$	e^{-x^2}

§19.2.4 Least-Squares Approximation by Polynomials

We can now solve the approximation problem we set out with. Given f and an inner product, we seek the $p \in \mathcal{P}_n$ minimising $\|f - p\|$. Since $\mathcal{P}_n$ is a finite-dimensional subspace, the answer is of course the orthogonal projection of f onto it – and having an orthogonal basis to hand makes that projection completely explicit.

Theorem 19.2.9 (Least-Squares Approximation)

Let $p_0, p_1, \dots$ be the orthogonal polynomials for $\langle \cdot, \cdot \rangle$. Then the unique minimiser of $\|f - p\|$ over $p \in \mathcal{P}_n$ is

$$\hat{p}_n = \sum_{k=0}^n \frac{\langle f, p_k \rangle}{\langle p_k, p_k \rangle} p_k.$$

Proof. Any $p \in \mathcal{P}_n$ may be written $p = \sum_{k=0}^n c_k p_k$. Expanding and using orthogonality,

$$\begin{aligned} \|f - p\|^2 &= \langle f - \sum_k c_k p_k, f - \sum_k c_k p_k \rangle \\ &= \langle f, f \rangle - 2 \sum_{k=0}^n c_k \langle f, p_k \rangle + \sum_{k=0}^n c_k^2 \langle p_k, p_k \rangle. \end{aligned}$$

The cross terms $\langle p_j, p_k \rangle$ with $j \neq k$ have vanished, so this is a sum of $n + 1$ independent quadratics in the c_k . Completing the square in each,

$$\|f - p\|^2 = \langle f, f \rangle - \sum_{k=0}^n \frac{\langle f, p_k \rangle^2}{\langle p_k, p_k \rangle} + \sum_{k=0}^n \langle p_k, p_k \rangle \left(c_k - \frac{\langle f, p_k \rangle}{\langle p_k, p_k \rangle} \right)^2.$$

Since $\langle p_k, p_k \rangle > 0$, this is minimised precisely when every bracket vanishes, i.e. $c_k = \langle f, p_k \rangle / \langle p_k, p_k \rangle$, and the minimiser is unique. $\square$

Notice the structural virtue of this answer: the coefficient of p_k does not depend on n . If we decide we want a better approximation, we simply compute one more coefficient and append one more term; nothing already computed is wasted. This is exactly the adaptivity we admired in the Newton form.

Reading off the value at the minimum in the proof gives the error, and this is worth recording separately.

Corollary 19.2.10 (Bessel's Inequality and Pythagoras)

With $\hat{p}_n$ as above,

$$\|f - \hat{p}_n\|^2 = \langle f, f \rangle - \sum_{k=0}^n \frac{\langle f, p_k \rangle^2}{\langle p_k, p_k \rangle} = \|f\|^2 - \|\hat{p}_n\|^2.$$

In particular $\sum_{k=0}^n \langle f, p_k \rangle^2 / \langle p_k, p_k \rangle \leq \|f\|^2$, and $\langle f - \hat{p}_n, \hat{p}_n \rangle = 0$.

That last statement – the error is orthogonal to the approximation – is Pythagoras’ theorem, and it is the geometric content of the whole business. It also gives an alternative one-line proof of the theorem: for any $p \in \mathcal{P}_n$ we have $f - p = (f - \hat{p}_n) + (\hat{p}_n - p)$ with the two summands orthogonal, so $\|f - p\|^2 = \|f - \hat{p}_n\|^2 + \|\hat{p}_n - p\|^2 \geq \|f - \hat{p}_n\|^2$.

Since $\|f - \hat{p}_n\|$ is decreasing in n , it converges; the natural question is whether it converges to zero. It does, and the reason is a theorem from outside numerical analysis entirely.

Theorem 19.2.11 (Weierstrass Approximation Theorem)

Let $f \in C[a, b]$ with $[a, b]$ finite. For every $\varepsilon > 0$ there is a polynomial q with $|f(x) - q(x)| < \varepsilon$ for all $x \in [a, b]$.

We shall not prove this here – it is a piece of analysis rather than numerical analysis, and the reader will meet it elsewhere. What we want is the consequence.

Corollary 19.2.12 (Convergence of Least-Squares Approximation)

Let $\langle f, g \rangle = \int_a^b wfg \, dx$ with $[a, b]$ finite and w a weight function. Then for $f \in C[a, b]$ we have $\|f - \hat{p}_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Proof. Let $\varepsilon > 0$ and set $W = \int_a^b w(x) \, dx < \infty$. By Weierstrass there is a polynomial q , of degree m say, with $|f - q| < \varepsilon$ pointwise on $[a, b]$. Then

$$\|f - q\|^2 = \int_a^b w(x)(f(x) - q(x))^2 \, dx \leq \varepsilon^2 W.$$

For $n \geq m$ we have $q \in \mathcal{P}_n$, so by minimality $\|f - \hat{p}_n\| \leq \|f - q\| \leq \varepsilon \sqrt{W}$. As ε was arbitrary and $\|f - \hat{p}_n\|$ is non-increasing, the result follows. $\square$

Corollary 19.2.13 (Parseval’s Identity)

Under the hypotheses above,

$$\sum_{k=0}^{\infty} \frac{\langle f, p_k \rangle^2}{\langle p_k, p_k \rangle} = \langle f, f \rangle.$$

Proof. Immediate from $\|f\|^2 - \sum_{k=0}^n \langle f, p_k \rangle^2 / \langle p_k, p_k \rangle = \|f - \hat{p}_n\|^2 \rightarrow 0$. $\square$

Example 19.2.14 (Least Squares Fit of $|x|$)

Take $[a, b] = [-1, 1]$, $w \equiv 1$, and $f(x) = |x|$, and let us find the best quadratic approximation. Using the monic Legendre polynomials $p_0 = 1$, $p_1 = x$, $p_2 = x^2 - \frac{1}{3}$, we compute

$$\langle f, p_0 \rangle = \int_{-1}^1 |x| \, dx = 1, \quad \langle f, p_1 \rangle = \int_{-1}^1 |x| x \, dx = 0,$$

$$\langle f, p_2 \rangle = \int_{-1}^1 |x| \left(x^2 - \frac{1}{3}\right) dx = 2 \int_0^1 \left(x^3 - \frac{x}{3}\right) dx = 2 \left(\frac{1}{4} - \frac{1}{6}\right) = \frac{1}{6}.$$

Also $\langle p_0, p_0 \rangle = 2$ and $\langle p_2, p_2 \rangle = \int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx = \frac{8}{45}$. Hence

$$\hat{p}_2(x) = \frac{1}{2} + \frac{1/6}{8/45} \left(x^2 - \frac{1}{3}\right) = \frac{15}{16}x^2 + \frac{3}{16}.$$

A quick sanity check: $\hat{p}_2(\pm 1) = 18/16 > 1$ and $\hat{p}_2(0) = 3/16 > 0$, so the fit overshoots at the ends and at the kink, which is what one expects of a smooth curve chasing a corner.

§19.2.5 Least Squares with Discrete Data

In practice one rarely knows f everywhere; one knows m measurements $f(\xi_1), \dots, f(\xi_m)$ at distinct points and wants the $p \in \mathcal{P}_n$ minimising

$$\sum_{j=1}^m (f(\xi_j) - p(\xi_j))^2.$$

This is precisely the least-squares problem for the discrete inner product

$$\langle g, h \rangle = \sum_{j=1}^m g(\xi_j)h(\xi_j),$$

which as we noted is a genuine inner product on $\mathcal{P}_n$ as long as $n < m$. Everything above goes through verbatim: we build the orthogonal polynomials $p_0, \dots, p_n$ for this inner product using the three-term recurrence, which applies because multiplication by x is again self-adjoint, and then Theorem 19.2.9 hands us the answer.

Remark. The naive alternative is to write $p = \sum_k c_k x^k$ and solve the resulting $(n+1) \times (n+1)$ linear system for the c_k – the so-called *normal equations*. This works, but the matrix involved is close to the notorious Hilbert matrix and is horribly ill-conditioned even for modest n . Orthogonalising first turns the linear system into a diagonal one, and the improvement in accuracy is dramatic. We will meet the same phenomenon again, from a different direction, when we discuss the QR factorisation in Section 19.8.

§19.3 Approximation of Linear Functionals

Almost every quantity we might want to extract from a function – its value at a point, its derivative there, its integral over an interval – is a *linear* function of f . This section is about approximating such quantities using only the simplest data of all, namely point values of f .

Definition 19.3.1 (Linear Functional)

A **linear functional** on a real vector space V of functions is a linear map $L : V \rightarrow \mathbb{R}$.

We usually do not fuss over the space V ; we write down a formula for L and take V to be whatever space of functions makes the formula sensible.

Example 19.3.2

The following are all linear functionals.

- (i) $L(f) = f(\xi)$ for some fixed ξ , on $V = C[a, b]$.
- (ii) $L(f) = f'(\eta)$ for some fixed η , on $V = C^1[a, b]$.
- (iii) $L(f) = \int_a^b w(x)f(x) \, dx$, on $V = C[a, b]$.
- (iv) Any linear combination of the above, for instance

$$L(f) = f(\beta) - f(\alpha) - \frac{\beta - \alpha}{2}(f'(\beta) + f'(\alpha)).$$

Our aim is to approximate a complicated L (typically an integral) by a finite combination of simple ones,

$$L(f) \approx \sum_{i=0}^N a_i f(x_i),$$

where $x_0, \dots, x_N \in [a, b]$ are distinct. How should we choose the a_i , and the x_i ?

The natural idea is one we already have: replace f by its interpolating polynomial and apply L to that. Using the fundamental Lagrange polynomials for the nodes x_i ,

$$f(x) \approx \sum_{i=0}^N f(x_i) \ell_i(x) \quad \implies \quad L(f) \approx \sum_{i=0}^N L(\ell_i) f(x_i),$$

so we should take $a_i = L(\ell_i)$. With this choice the approximation is *exact* for $f \in \mathcal{P}_N$, since then the interpolant is f itself.

That is $N + 1$ free parameters buying exactness on an $(N + 1)$ -dimensional space, which sounds like a fair deal. But we have another $N + 1$ parameters that we have not used at all – the nodes x_i . If we choose those cleverly too, we might hope for exactness on $\mathcal{P}_{2N+1}$. Remarkably, for integrals we can achieve exactly that. This is Gaussian quadrature, and it is the subject of the rest of this section.

§19.3.1 Newton–Cotes Formulae

First, though, let us see what the naive choice gives. If we take the nodes to be equally spaced and $L(f) = \int_a^b f$, the resulting rules are the classical **Newton–Cotes formulae**.

Definition 19.3.3 (Newton–Cotes)

Fix $\nu \geq 1$ and let $x_i = a + i(b - a)/\nu$ for $i = 0, \dots, \nu$. The closed Newton–Cotes

rule of order ν is

$$\int_a^b f(x) \, dx \approx \sum_{i=0}^{\nu} a_i f(x_i), \quad a_i = \int_a^b \ell_i(x) \, dx.$$

Example 19.3.4 (The Trapezoidal Rule)

With $\nu = 1$ we have $\ell_0(x) = (b - x)/(b - a)$ and $\ell_1(x) = (x - a)/(b - a)$, each of which integrates to $(b - a)/2$. Hence

$$\int_a^b f(x) \, dx \approx \frac{b - a}{2} (f(a) + f(b)),$$

which is just the area of the trapezium under the chord joining the two endpoints.

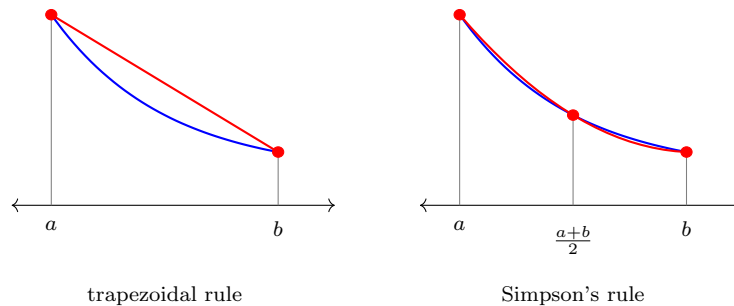
Example 19.3.5 (Simpson's Rule)

With $\nu = 2$ and nodes a , $m = (a + b)/2$, b , a short computation gives $a_0 = a_2 = (b - a)/6$ and $a_1 = 2(b - a)/3$, so

$$\int_a^b f(x) \, dx \approx \frac{b - a}{6} \left(f(a) + 4f\left(\frac{a + b}{2}\right) + f(b) \right).$$

By construction this is exact for $f \in \mathcal{P}_2$. In fact it is exact for $f \in \mathcal{P}_3$: taking $[a, b] = [-1, 1]$ for convenience, $\int_{-1}^1 x^3 \, dx = 0$ and the rule returns $\frac{1}{3}(-1 + 0 + 1) = 0$ as well. This extra order ‘for free’ is a general feature of Newton–Cotes rules with an even number of subintervals, and comes from the symmetry of the nodes.

Geometrically, the trapezoidal rule replaces f by a chord and Simpson's rule replaces it by a parabola through three points:



Remark. One might hope that taking ν larger and larger gives better and better rules. It does not: for $\nu \geq 8$ some of the weights a_i become negative, and by $\nu = 30$ they are enormous, so that rounding errors are amplified catastrophically. This is the Runge phenomenon again, wearing a different hat. In practice one instead uses a *composite* rule: chop $[a, b]$ into many small subintervals and apply a low-order rule (usually the trapezoidal or Simpson's rule) on each.

§19.3.2 Gaussian Quadrature

Now we use our remaining freedom. We work with a weighted integral

$$L(f) = \int_a^b w(x)f(x) \, dx,$$

and seek **weights** $b_1, \dots, b_\nu$ and **nodes** $c_1, \dots, c_\nu \in [a, b]$ such that

$$\int_a^b w(x)f(x) \, dx \approx \sum_{k=1}^{\nu} b_k f(c_k)$$

is exact for polynomials of as high a degree as possible. (The notation is traditional for this topic and clashes slightly with the rest of the course; we shall live with it.)

First, a limit on our ambitions.

Proposition 19.3.6

No choice of ν nodes and weights gives a formula exact for all $f \in \mathcal{P}_{2\nu}$.

Proof. Consider $q(x) = \prod_{k=1}^{\nu} (x - c_k)$, so that $q^2 \in \mathcal{P}_{2\nu}$. Then $\sum_k b_k q^2(c_k) = 0$, whereas

$$\int_a^b w(x)q(x)^2 \, dx > 0,$$

since $wq^2 \geq 0$ is continuous and not identically zero. $\square$

So $2\nu - 1$ is the best we could possibly hope for, and we shall achieve it. The clue is in the proof: we need the nodes to be the roots of a polynomial that is orthogonal to everything of low degree. Before that, we must check that such roots are usable at all – they had better be real, distinct, and inside (a, b) .

Theorem 19.3.7 (Zeros of Orthogonal Polynomials)

For $\nu \geq 1$, the monic orthogonal polynomial p_ν associated with $\langle f, g \rangle = \int_a^b wfg \, dx$ has ν distinct real zeros, all lying in (a, b) .

Proof. Since $p_0 \equiv 1$ and $\langle p_\nu, p_0 \rangle = 0$ for $\nu \geq 1$, we have

$$\int_a^b w(x)p_\nu(x) \, dx = 0,$$

and as $w \geq 0$ is not identically zero, p_ν must change sign somewhere in (a, b) .

Let $\varphi_1, \dots, \varphi_m$ be the points of (a, b) at which p_ν changes sign; we have just shown $m \geq 1$, and certainly $m \leq \nu$ since these are among the zeros of p_ν . Put

$$q(x) = \prod_{j=1}^m (x - \varphi_j) \in \mathcal{P}_m.$$

Then q changes sign at exactly the same points as p_ν , so qp_ν has constant sign on (a, b) and is not identically zero; hence $\langle q, p_\nu \rangle = \int_a^b wqp_\nu \, dx \neq 0$. But p_ν is

orthogonal to every polynomial of degree less than ν , so we must have $m \geq \nu$, and therefore $m = \nu$.

Thus p_ν has ν distinct sign changes in (a, b) , which accounts for all ν of its zeros; they are real, distinct, and lie in (a, b) . $\square$

Before choosing the nodes, let us record what the interpolatory choice of weights gives for *any* nodes.

Theorem 19.3.8 (Ordinary Quadrature)

Let $c_1, \dots, c_\nu \in [a, b]$ be distinct, with fundamental Lagrange polynomials $\ell_1, \dots, \ell_\nu$. If we choose

$$b_k = \int_a^b w(x) \ell_k(x) \, dx,$$

then the quadrature formula is exact for all $f \in \mathcal{P}_{\nu-1}$.

Proof. For $f \in \mathcal{P}_{\nu-1}$, interpolation at ν points reproduces f exactly, so $f = \sum_k f(c_k) \ell_k$. Integrating against w gives $\int_a^b w f = \sum_k f(c_k) b_k$. $\square$

And now the theorem that makes it all worthwhile.

Theorem 19.3.9 (Gaussian Quadrature)

Choose the nodes $c_1, \dots, c_\nu$ to be the zeros of the orthogonal polynomial p_ν , and the weights b_k as in Theorem 19.3.8. Then the quadrature formula

$$\int_a^b w(x) f(x) \, dx \approx \sum_{k=1}^{\nu} b_k f(c_k)$$

is exact for all $f \in \mathcal{P}_{2\nu-1}$. Moreover every weight b_k is strictly positive.

Proof. Let $f \in \mathcal{P}_{2\nu-1}$. Dividing by p_ν we may write

$$f = qp_\nu + r, \quad q, r \in \mathcal{P}_{\nu-1}.$$

Since p_ν is orthogonal to all of $\mathcal{P}_{\nu-1}$, we have $\int_a^b w qp_\nu \, dx = \langle q, p_\nu \rangle = 0$, so

$$\int_a^b w(x) f(x) \, dx = \int_a^b w(x) r(x) \, dx.$$

On the other side, each c_k is a zero of p_ν , so $f(c_k) = r(c_k)$ and hence

$$\sum_{k=1}^{\nu} b_k f(c_k) = \sum_{k=1}^{\nu} b_k r(c_k).$$

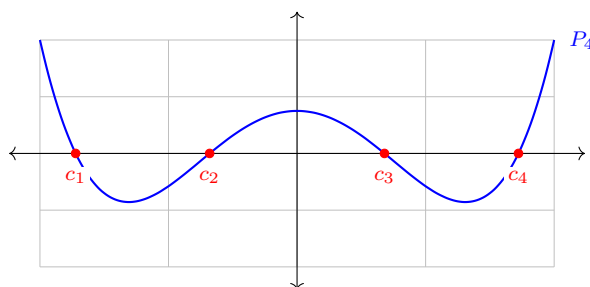
But $r \in \mathcal{P}_{\nu-1}$, so by Theorem 19.3.8 the two right-hand sides agree. Hence the formula is exact for f .

For positivity, apply exactness to $f = \ell_k^2 \in \mathcal{P}_{2\nu-2}$. Then

$$0 < \int_a^b w(x) \ell_k(x)^2 dx = \sum_{j=1}^{\nu} b_j \ell_k(c_j)^2 = \sum_{j=1}^{\nu} b_j \delta_{jk} = b_k. \quad \square$$

Together with Proposition 19.3.6, this says that Gaussian quadrature is optimal: $2\nu - 1$ is achieved and 2ν is impossible. The positivity of the weights is not a footnote either – it is what makes the rule numerically stable, since it means there is no cancellation between the terms and rounding errors cannot be amplified.

For the Legendre weight $w \equiv 1$ on $[-1, 1]$ the nodes are the zeros of the Legendre polynomials. Here is P_4 with its four zeros marked; these are the nodes of the four-point Gauss–Legendre rule, which is exact for all polynomials of degree up to 7.



Example 19.3.10 (Gauss–Legendre Rules)

For $w \equiv 1$ on $[-1, 1]$ the first few rules are as follows.

ν	nodes c_k	weights b_k	exact on
1	0	2	$\mathcal{P}_1$
2	$\pm 1/\sqrt{3}$	1, 1	$\mathcal{P}_3$
3	$0, \pm\sqrt{3/5}$	8/9, 5/9, 5/9	$\mathcal{P}_5$
4	$\pm 0.3399810, \pm 0.8611363$	0.6521452, 0.3478548	$\mathcal{P}_7$

Let us check the case $\nu = 2$ directly. The nodes are the zeros of $p_2 = x^2 - \frac{1}{3}$, namely $\pm 1/\sqrt{3}$. The rule $\int_{-1}^1 f \approx f(-1/\sqrt{3}) + f(1/\sqrt{3})$ certainly integrates 1 and x correctly. For x^2 it gives $2/3$, which is right; for x^3 it gives 0, which is right. And for x^4 it gives $2/9$ against the true value $2/5$ – so exactness stops at degree $3 = 2\nu - 1$, exactly as predicted.

Example 19.3.11 (Gauss Beats Simpson)

Let us approximate $\int_0^1 e^x dx = e - 1 = 1.718282$. Substituting $x = (t + 1)/2$ turns this into $\frac{1}{2} \int_{-1}^1 e^{(t+1)/2} dt$, and the two-point Gauss rule gives

$$\frac{1}{2} \left(e^{(1-1/\sqrt{3})/2} + e^{(1+1/\sqrt{3})/2} \right) = \frac{1}{2} (1.235345 + 2.200455) = 1.717900,$$

an error of 3.8×10^{-4} , using *two* evaluations of f . Simpson's rule, using *three* evaluations, gives

$$\frac{1}{6} \left(1 + 4e^{1/2} + e \right) = 1.718861,$$

an error of 5.8×10^{-4} . The trapezoidal rule gives 1.859141, an error of 0.14. Gaussian quadrature does more with less.

Remark. The price of Gaussian quadrature is that the nodes are irrational and must be looked up or computed (they are eigenvalues of the tridiagonal matrix formed from the three-term recurrence coefficients, which is how one actually computes them). Also, unlike a composite Newton–Cotes rule, refining from ν to $\nu + 1$ points reuses none of the old function evaluations. Both objections are real, and both have been engineered around; but the exactness result of Theorem 19.3.9 is why Gaussian quadrature remains the default for smooth integrands.

§19.4 Expressing Errors in Terms of Derivatives

We now have a supply of approximations to linear functionals, each of them exact on some space $\mathcal{P}_k$ of polynomials. What we do not yet have is a systematic way to say how large the error is when f is *not* a polynomial. The Peano kernel theorem provides exactly that: it converts the statement ‘the error annihilates $\mathcal{P}_k$ ’ into an exact integral formula for the error in terms of $f^{(k+1)}$.

The setup is as follows. We have a linear functional L , an approximation $\sum_{i=0}^n a_i L_i(f)$ built from simpler functionals L_i , and the error functional

$$e_L(f) = L(f) - \sum_{i=0}^n a_i L_i(f),$$

which is again linear. We suppose that $e_L(f) = 0$ for all $f \in \mathcal{P}_k$; we say that e_L **annihilates** $\mathcal{P}_k$.

Example 19.4.1

Suppose we approximate $L(f) = f(\beta)$ by the (rather silly) formula

$$L(f) \approx f(\alpha) + \frac{\beta - \alpha}{2} (f'(\alpha) + f'(\beta)), \quad \alpha \neq \beta.$$

Then

$$e_L(f) = f(\beta) - f(\alpha) - \frac{\beta - \alpha}{2} (f'(\alpha) + f'(\beta)).$$

One checks directly that this vanishes for $f = 1, x, x^2$: for $f = x^2$, the left-hand piece is $\beta^2 - \alpha^2$ and the right-hand piece is $(\beta - \alpha)(\alpha + \beta)$, which agree. It does not vanish for $f = x^3$. So e_L annihilates $\mathcal{P}_2$.

We want a bound of the form

$$|e_L(f)| \leq c_L \left\| f^{(k+1)} \right\|_{\infty},$$

with c_L as small as possible. For this we need $[a, b]$ to be finite, so that the suprema exist.

Definition 19.4.2 (Sharp Constant)

The constant c_L is **sharp** if for every $\varepsilon > 0$ there is some $f_\varepsilon \in C^{k+1}[a, b]$ with

$$|e_L(f_\varepsilon)| \geq (c_L - \varepsilon) \left\| f_\varepsilon^{(k+1)} \right\|_\infty.$$

§19.4.1 The Peano Kernel Theorem

The tool is Taylor's theorem with integral remainder,

$$f(x) = f(a) + (x-a)f'(a) + \cdots + \frac{(x-a)^k}{k!} f^{(k)}(a) + \frac{1}{k!} \int_a^x (x-\theta)^k f^{(k+1)}(\theta) d\theta.$$

The awkwardness is that x appears in the limit of the integral, which stops us from interchanging e_L with $\int$. We fix this with a piece of notation.

Definition 19.4.3 (Truncated Power Function)

For $k \geq 0$ we define

$$(x-\theta)_+^k = \begin{cases} (x-\theta)^k & x \geq \theta, \\ 0 & x < \theta. \end{cases}$$

Note that for $k \geq 1$ this is a C^{k-1} function of x : it is smooth except at $x = \theta$, where the first $k-1$ derivatives all vanish from both sides. With it, the Taylor remainder becomes

$$\frac{1}{k!} \int_a^b (x-\theta)_+^k f^{(k+1)}(\theta) d\theta,$$

where now the limits do not involve x .

Theorem 19.4.4 (Peano Kernel Theorem)

Let $[a, b]$ be finite and let e_L be a linear functional annihilating $\mathcal{P}_k$ which commutes with the integral in the sense below. Then for all $f \in C^{k+1}[a, b]$,

$$e_L(f) = \frac{1}{k!} \int_a^b K(\theta) f^{(k+1)}(\theta) d\theta, \quad K(\theta) = e_L((x-\theta)_+^k),$$

where e_L acts on $(x-\theta)_+^k$ as a function of x , with θ held fixed.

Proof. Write $f = T + R$ where $T \in \mathcal{P}_k$ is the Taylor polynomial of f about a of degree k and

$$R(x) = \frac{1}{k!} \int_a^b (x-\theta)_+^k f^{(k+1)}(\theta) d\theta$$

is the remainder. Since e_L annihilates $\mathcal{P}_k$ we have $e_L(T) = 0$, so by linearity $e_L(f) = e_L(R)$. Interchanging e_L with the integral,

$$e_L(f) = \frac{1}{k!} \int_a^b e_L((x-\theta)_+^k) f^{(k+1)}(\theta) d\theta = \frac{1}{k!} \int_a^b K(\theta) f^{(k+1)}(\theta) d\theta. \quad \square$$

Remark. The interchange of e_L with the integral is the only step needing care. For the functionals we use in this course – point values, point values of derivatives, and integrals, together with finite linear combinations of these – the interchange is valid and can be

verified directly in each case (for integrals it is Fubini, for point evaluations it is trivial, and for derivatives one differentiates under the integral sign). A sufficiently determined pure mathematician can construct linear functionals for which it fails, but we shall not meet any.

The point of the theorem is that K depends only on L and not at all on f . So having computed K once, we get a whole family of bounds by applying Hölder's inequality in different ways:

$$|e_L(f)| \leq \frac{1}{k!} \begin{cases} \left(\int_a^b |K(\theta)| \, d\theta \right) \|f^{(k+1)}\|_\infty, \\ \left(\int_a^b |K(\theta)|^2 \, d\theta \right)^{1/2} \|f^{(k+1)}\|_2, \\ \|K\|_\infty \|f^{(k+1)}\|_1. \end{cases}$$

These constants can be shown to be sharp. And do not forget the factor $1/k!$ when computing c_L – it is the most commonly dropped piece of the whole formula.

There is one very common situation in which we can say considerably more.

Proposition 19.4.5 (Sign-Definite Kernels)

If K does not change sign on $[a, b]$, then for every $f \in C^{k+1}[a, b]$ there is some $\xi \in (a, b)$, depending on f , with

$$e_L(f) = \frac{1}{k!} \left(\int_a^b K(\theta) \, d\theta \right) f^{(k+1)}(\xi).$$

Moreover the constant $c_L = \frac{1}{k!} \left| \int_a^b K \right|$ in the $\|\cdot\|_\infty$ bound is attained, by $f(x) = x^{k+1}$.

Proof. Since K has constant sign, the integral mean value theorem applies to $\int_a^b K f^{(k+1)}$ and yields $f^{(k+1)}(\xi) \int_a^b K$ for some ξ . For the second statement, $f = x^{k+1}$ has $f^{(k+1)} \equiv (k+1)!$ constant, so the inequality $|\int K f^{(k+1)}| \leq \int |K| \cdot \|f^{(k+1)}\|_\infty$ is an equality. $\square$

§19.4.2 Worked Kernels

Peano kernels are computed by brute force: substitute $(x - \theta)_+^k$ into the formula for e_L , and remember that the truncated power vanishes when its argument is negative, so the answer is piecewise.

Example 19.4.6 (The Trapezoidal Rule)

Take $[a, b] = [0, 1]$ and

$$e_L(f) = \int_0^1 f(x) \, dx - \frac{1}{2}(f(0) + f(1)),$$

which annihilates $\mathcal{P}_1$, so $k = 1$. Then for $\theta \in [0, 1]$,

$$K(\theta) = \int_0^1 (x - \theta)_+ \, dx - \frac{1}{2}((0 - \theta)_+ + (1 - \theta)_+) = \int_\theta^1 (x - \theta) \, dx - \frac{1}{2}(1 - \theta),$$

since $(0 - \theta)_+ = 0$ for $\theta \geq 0$. Hence

$$K(\theta) = \frac{(1 - \theta)^2}{2} - \frac{1 - \theta}{2} = -\frac{\theta(1 - \theta)}{2}.$$

This is a downward parabola vanishing at both endpoints, so $K \leq 0$ throughout and Proposition 19.4.5 applies. We have $\int_0^1 K = -\frac{1}{2}(\frac{1}{2} - \frac{1}{3}) = -\frac{1}{12}$, so

$$e_L(f) = -\frac{1}{12}f''(\xi), \quad |e_L(f)| \leq \frac{1}{12} \|f''\|_\infty,$$

which is the classical trapezoidal rule error, now derived rather than conjured.

Example 19.4.7 (Simpson's Rule)

Take $[a, b] = [-1, 1]$ and

$$e_L(f) = \int_{-1}^1 f(x) \, dx - \frac{1}{3}(f(-1) + 4f(0) + f(1)).$$

We saw that this annihilates $\mathcal{P}_3$, so we may take $k = 3$. For $\theta \in [0, 1]$ only the $f(1)$ term survives the truncation, and

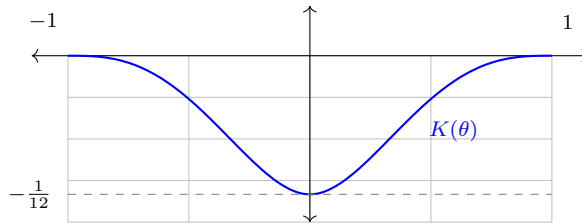
$$K(\theta) = \int_\theta^1 (x - \theta)^3 \, dx - \frac{1}{3}(1 - \theta)^3 = \frac{(1 - \theta)^4}{4} - \frac{(1 - \theta)^3}{3}.$$

For $\theta \in [-1, 0]$ the $f(0)$ term also contributes, giving

$$K(\theta) = \frac{(1 - \theta)^4}{4} - \frac{1}{3}(4(-\theta)^3 + (1 - \theta)^3).$$

One checks that the two branches agree at $\theta = 0$, both giving $\frac{1}{4} - \frac{1}{3} = -\frac{1}{12}$, and that $K(\pm 1) = 0$. In fact K is symmetric, $K(-\theta) = K(\theta)$, and negative on $(-1, 1)$.

Here is the kernel of Example 19.4.7, plotted on $[-1, 1]$.



Since $K \leq 0$ we may use Proposition 19.4.5. Computing the integral,

$$\int_{-1}^1 K(\theta) \, d\theta = 2 \int_0^1 \left(\frac{(1 - \theta)^4}{4} - \frac{(1 - \theta)^3}{3} \right) \, d\theta = 2 \left(\frac{1}{20} - \frac{1}{12} \right) = -\frac{1}{15},$$

and hence, with $k = 3$ so that $1/k! = 1/6$,

$$e_L(f) = -\frac{1}{90}f^{(4)}(\xi), \quad |e_L(f)| \leq \frac{1}{90} \|f^{(4)}\|_\infty$$

for some $\xi \in (-1, 1)$. Rescaling to a general interval gives the familiar $-\frac{(b-a)^5}{2880} f^{(4)}(\xi)$. The theorem applies just as well to functionals that have nothing to do with integration.

Example 19.4.8 (A Derivative Formula)

Suppose we approximate $f'(0)$ by the one-sided formula

$$f'(0) \approx -\frac{3}{2}f(0) + 2f(1) - \frac{1}{2}f(2),$$

so that $e_L(f) = f'(0) + \frac{3}{2}f(0) - 2f(1) + \frac{1}{2}f(2)$ on $[0, 2]$. Checking on $1, x, x^2$: for $f = 1$ we get $0 + \frac{3}{2} - 2 + \frac{1}{2} = 0$; for $f = x$ we get $1 + 0 - 2 + 1 = 0$; for $f = x^2$ we get $0 + 0 - 2 + 2 = 0$. For $f = x^3$ we get $0 + 0 - 2 + 4 = 2 \neq 0$. So e_L annihilates $\mathcal{P}_2$ and $k = 2$.

The kernel is $K(\theta) = e_L((x - \theta)_+^2)$. Since $\frac{d}{dx}(x - \theta)_+^2 = 2(x - \theta)_+$, the derivative term contributes $2(0 - \theta)_+ = 0$ for $\theta \geq 0$, as does the $f(0)$ term. So

$$K(\theta) = -2(1 - \theta)_+^2 + \frac{1}{2}(2 - \theta)_+^2 = \begin{cases} 2\theta - \frac{3}{2}\theta^2, & \theta \in [0, 1], \\ \frac{1}{2}(2 - \theta)^2, & \theta \in [1, 2]. \end{cases}$$

(The two branches agree at $\theta = 1$, both giving $\frac{1}{2}$.) Both branches are non-negative, so $K \geq 0$ and Proposition 19.4.5 applies. We have

$$\int_0^2 K(\theta) d\theta = \int_0^1 (2\theta - \frac{3}{2}\theta^2) d\theta + \int_1^2 \frac{1}{2}(2 - \theta)^2 d\theta = \frac{1}{2} + \frac{1}{6} = \frac{2}{3},$$

and hence, remembering the factor $1/k! = 1/2$,

$$e_L(f) = \frac{1}{3}f'''(\xi), \quad |e_L(f)| \leq \frac{1}{3} \|f'''\|_\infty$$

for some $\xi \in (0, 2)$.

Remark. A final subtlety. If e_L annihilates $\mathcal{P}_k$ then it also annihilates $\mathcal{P}_j$ for every $j < k$, so we get a *different* kernel $K_j(\theta) = e_L((x - \theta)_+^j)$ and a different error formula

$$e_L(f) = \frac{1}{j!} \int_a^b K_j(\theta) f^{(j+1)}(\theta) d\theta$$

valid for $f \in C^{j+1}[a, b]$. Usually taking $j = k$ is best, as it extracts the most information. But if f happens to be only C^{j+1} and not C^{k+1} – and integrands with limited smoothness are common in practice – then the lower-order kernel is the one that applies. Simpson's rule with a merely twice-differentiable integrand is a case in point.

§19.5 Ordinary Differential Equations

We now turn to what is probably the most important application of numerical analysis. We wish to approximate the solution $y : [0, \infty) \rightarrow \mathbb{R}^N$ of the initial value problem

$$y' = f(t, y), \quad y(0) = y_0,$$

where $f : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}^N$. Note that we allow y to be vector-valued, which costs us nothing notationally and buys us all higher-order scalar equations, since $y'' = g(t, y, y')$ becomes a first-order system in (y, y') .

We shall always assume that f is nice enough for the problem to make sense. In particular, we assume f is Lipschitz in its second argument, which by the Picard–Lindelöf theorem guarantees that a unique solution exists; and where we Taylor expand, we assume as much smoothness as the expansion requires. This is standard practice and is not usually where the interesting difficulties lie.

The idea is to replace the continuum of values $y(t)$ by a discrete sequence $y_0, y_1, y_2, \dots$ with $y_n \approx y(t_n)$, where $0 = t_0 < t_1 < \dots$. We shall almost always take a constant step size, $t_n = nh$. The two questions we must ask of any such scheme are: does y_n actually approach $y(t_n)$ as $h \rightarrow 0$, and how fast?

§19.5.1 One-Step Methods

Definition 19.5.1 (One-Step Method)

A **one-step method** is a recurrence of the form

$$y_{n+1} = \varphi_h(t_n, y_n),$$

in which y_{n+1} depends only on t_n, y_n, h and f .

The simplest of all is obtained by truncating the Taylor series of y after one term.

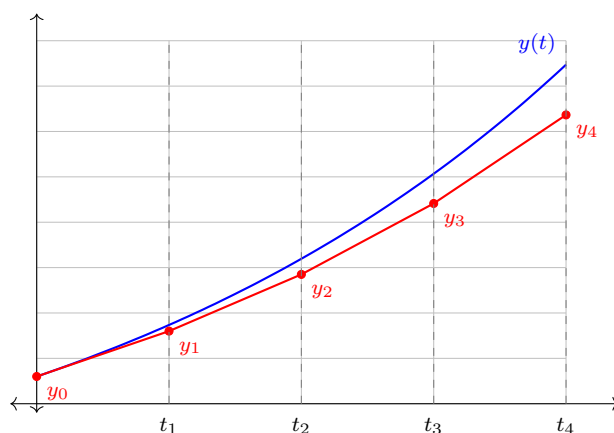
Definition 19.5.2 (Euler's Method)

Euler's method is the one-step method

$$y_{n+1} = y_n + hf(t_n, y_n).$$

The idea is transparent: from the point (t_n, y_n) , the differential equation tells us the slope of the solution curve through that point, so we follow that straight line for a time h and see where we end up. The resulting approximation is a polygon, and each segment of it is tangent to *some* solution curve of the equation – just not, in general, the one we want.

Here is Euler's method applied to $y' = y$, $y(0) = 1$, with $h = 1/4$. The true solution $y(t) = e^t$ is in blue and the Euler polygon in red.



Two things are visible in the picture. The Euler solution lags behind the true one, and the gap grows. What we need to know is whether the gap shrinks to nothing as we

shrink h .

Definition 19.5.3 (Convergence)

A method is **convergent** if, for every $t^* > 0$ and every initial value problem with sufficiently nice f ,

$$\lim_{h \rightarrow 0} \max_{n=0,1,\dots,\lfloor t^*/h \rfloor} \|y_n - y(t_n)\| = 0,$$

where y_n is the numerical solution computed with step size h .

Note what this is asking: not that a fixed number of steps become accurate, but that the error at a fixed *time* t^* , reached in $\lfloor t^*/h \rfloor$ steps, goes to zero. Since the number of steps grows like $1/h$, the errors made at each step must not be allowed to accumulate too badly. This is the whole content of the following theorem.

Theorem 19.5.4 (Convergence of Euler's Method)

Suppose f is Lipschitz in its second argument on $[0, t^*] \times \mathbb{R}^N$, that is, there is $\lambda \geq 0$ with

$$\|f(t, v) - f(t, w)\| \leq \lambda \|v - w\|$$

for all $t \in [0, t^*]$ and $v, w \in \mathbb{R}^N$. Then Euler's method converges, and indeed the error is $\mathcal{O}(h)$ uniformly on $[0, t^*]$.

Proof. The Lipschitz condition guarantees by Picard–Lindelöf that a unique solution y exists on $[0, t^*]$, so the statement makes sense. Write $e_n = y_n - y(t_n)$ for the (global) error at step n , and note $e_0 = 0$.

Taylor's theorem gives $y(t_{n+1}) = y(t_n) + hy'(t_n) + \mathcal{O}(h^2) = y(t_n) + hf(t_n, y(t_n)) + \mathcal{O}(h^2)$, where the $\mathcal{O}(h^2)$ is uniform in n because y'' is continuous on the compact interval $[0, t^*]$. Subtracting this from the Euler step,

$$\begin{aligned} \|e_{n+1}\| &= \|y_n + hf(t_n, y_n) - y(t_n) - hf(t_n, y(t_n)) - \mathcal{O}(h^2)\| \\ &\leq \|e_n\| + h \|f(t_n, y_n) - f(t_n, y(t_n))\| + ch^2 \\ &\leq (1 + h\lambda) \|e_n\| + ch^2 \end{aligned}$$

for some constant c independent of n and h . This is a linear recurrence, and unwinding it (using $e_0 = 0$) gives

$$\|e_{n+1}\| \leq ch^2 \sum_{j=0}^n (1 + h\lambda)^j = ch^2 \frac{(1 + h\lambda)^{n+1} - 1}{h\lambda} \leq \frac{ch}{\lambda} (1 + h\lambda)^{n+1}.$$

Now we use $1 + x \leq e^x$, valid for all real x , to get $(1 + h\lambda)^{n+1} \leq e^{(n+1)h\lambda} = e^{\lambda t_{n+1}} \leq e^{\lambda t^*}$. Hence

$$\|e_{n+1}\| \leq \frac{ce^{\lambda t^*}}{\lambda} h$$

for every n with $t_{n+1} \leq t^*$. The bound is independent of n and tends to 0 with h , which is convergence with $\mathcal{O}(h)$ error. $\square$

Remark. The factor $e^{\lambda t^*}$ is the price of accumulating $\mathcal{O}(1/h)$ errors of size $\mathcal{O}(h^2)$ each, with each error subsequently amplified by the flow. It is exponentially bad in t^* , which

is a genuine feature of the problem rather than an artefact of the proof: two nearby solutions of $y' = \lambda y$ really do separate exponentially. When $\lambda \rightarrow 0$ the bound looks singular, but $\frac{1}{\lambda}(e^{\lambda t^*} - 1) \rightarrow t^*$, so all is well.

The rate $\mathcal{O}(h)$ is what we mean by saying Euler's method has *order* 1. The general definition is local: we ask how much error a single step introduces, assuming that we started that step from the exact solution.

Definition 19.5.5 (Order)

A one-step method $y_{n+1} = \varphi_h(t_n, y_n)$ has **order** p if

$$y(t_{n+1}) - \varphi_h(t_n, y(t_n)) = \mathcal{O}(h^{p+1})$$

as $h \rightarrow 0$, for all sufficiently smooth f .

The exponent $p + 1$ rather than p is deliberate: as in the proof above, a *local* error of $\mathcal{O}(h^{p+1})$ made at each of $\mathcal{O}(1/h)$ steps accumulates to a *global* error of $\mathcal{O}(h^p)$. So a method of order p has global error $\mathcal{O}(h^p)$, and higher order means we can take larger steps for a given accuracy.

Example 19.5.6

For Euler's method, $y(t_{n+1}) - y(t_n) - hf(t_n, y(t_n)) = \frac{1}{2}h^2 y''(t_n) + \mathcal{O}(h^3) = \mathcal{O}(h^2)$, so $p = 1$.

§19.5.2 The Theta Method

We got Euler's method by evaluating f at the left end of the interval $[t_n, t_{n+1}]$. We could equally evaluate at the right end, or at any weighted average of the two.

Definition 19.5.7 (Theta Method)

For $\theta \in [0, 1]$, the **theta method** is

$$y_{n+1} = y_n + h \left(\theta f(t_n, y_n) + (1 - \theta) f(t_{n+1}, y_{n+1}) \right).$$

Taking $\theta = 1$ recovers Euler's method. Taking $\theta = 0$ gives the **backward Euler method**

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}),$$

and $\theta = \frac{1}{2}$ gives the **trapezoidal rule**

$$y_{n+1} = y_n + \frac{h}{2} (f(t_n, y_n) + f(t_{n+1}, y_{n+1})).$$

For $\theta \neq 1$ the unknown y_{n+1} appears on both sides, so the method is **implicit**: each step requires solving a (generally nonlinear) equation. That is a real cost, and one might wonder why we would pay it. There are two reasons. The first is order.

Proposition 19.5.8 (Order of the Theta Method)

The theta method has order 1, except when $\theta = \frac{1}{2}$, in which case it has order 2.

Proof. Assume $y_n = y(t_n)$ and expand everything about t_n , writing $y = y(t_n)$, $y' = y'(t_n)$ and so on. We have

$$y(t_{n+1}) = y + hy' + \frac{1}{2}h^2y'' + \frac{1}{6}h^3y''' + \mathcal{O}(h^4),$$

while $f(t_n, y(t_n)) = y'$ and, since $y'(t_{n+1}) = y' + hy'' + \frac{1}{2}h^2y''' + \mathcal{O}(h^3)$,

$$h\left(\theta y' + (1 - \theta)y'(t_{n+1})\right) = hy' + (1 - \theta)h^2y'' + \frac{1}{2}(1 - \theta)h^3y''' + \mathcal{O}(h^4).$$

Subtracting, the local error is

$$\left(\theta - \frac{1}{2}\right)h^2y'' + \left(\frac{\theta}{2} - \frac{1}{3}\right)h^3y''' + \mathcal{O}(h^4).$$

The h^2 term vanishes precisely when $\theta = \frac{1}{2}$; and when it does, the coefficient of h^3 is $\frac{1}{4} - \frac{1}{3} = -\frac{1}{12} \neq 0$. So the order is 1 in general and 2 for $\theta = \frac{1}{2}$. $\square$

The second reason for tolerating implicitness is *stability*, which will occupy us in Section 19.6. For the moment we simply record that implicit methods can be far better behaved on hard problems.

§19.5.3 Multistep Methods

Euler's method throws away everything it has computed except the most recent value. Since we have $y_0, \dots, y_n$ sitting in memory, it is natural to use several of them.

Definition 19.5.9 (Multistep Method)

An **s-step method** is a recurrence

$$\sum_{\ell=0}^s \rho_{\ell} y_{n+\ell} = h \sum_{\ell=0}^s \sigma_{\ell} f(t_{n+\ell}, y_{n+\ell}), \quad n = 0, 1, \dots,$$

normalised so that $\rho_s = 1$. The method is **explicit** if $\sigma_s = 0$ and **implicit** otherwise.

To start such a method we need s initial values $y_0, \dots, y_{s-1}$, of which the problem supplies only one; the rest are generated by some one-step method (typically a Runge–Kutta method of matching order). It is convenient to package the coefficients into two polynomials

$$\rho(w) = \sum_{\ell=0}^s \rho_{\ell} w^{\ell}, \quad \sigma(w) = \sum_{\ell=0}^s \sigma_{\ell} w^{\ell},$$

which will turn out to encode both the accuracy and the stability of the method.

Example 19.5.10

Euler's method is the one-step method with $\rho(w) = w - 1$, $\sigma(w) = 1$. Backward Euler has $\rho(w) = w - 1$, $\sigma(w) = w$. The trapezoidal rule has $\rho(w) = w - 1$, $\sigma(w) = \frac{1}{2}(w + 1)$. A genuine two-step example is the **Adams–Bashforth** method

$$y_{n+2} - y_{n+1} = h \left(\frac{3}{2}f(t_{n+1}, y_{n+1}) - \frac{1}{2}f(t_n, y_n) \right),$$

with $\rho(w) = w^2 - w$ and $\sigma(w) = \frac{3}{2}w - \frac{1}{2}$; and an implicit one is the two-step **Adams–Moulton** method

$$y_{n+2} - y_{n+1} = h \left(\frac{5}{12}f(t_{n+2}, y_{n+2}) + \frac{2}{3}f(t_{n+1}, y_{n+1}) - \frac{1}{12}f(t_n, y_n) \right).$$

Definition 19.5.11 (Order of a Multistep Method)

The multistep method above has **order** p if

$$\sum_{\ell=0}^s \rho_{\ell} y(t_{n+\ell}) - h \sum_{\ell=0}^s \sigma_{\ell} y'(t_{n+\ell}) = \mathcal{O}(h^{p+1})$$

as $h \rightarrow 0$, for all sufficiently smooth functions y .

Checking this by Taylor expansion each time would be tedious. Happily, the conditions can be packaged into a single elegant statement about ρ and σ .

Theorem 19.5.12 (Order Conditions)

The multistep method has order p if and only if

$$\rho(e^z) - z\sigma(e^z) = \mathcal{O}(z^{p+1}) \quad \text{as } z \rightarrow 0.$$

Proof. Expand everything about t_n . Writing $y^{(k)} = y^{(k)}(t_n)$ and $t_{n+\ell} = t_n + \ell h$,

$$y(t_{n+\ell}) = \sum_{k=0}^{\infty} \frac{(\ell h)^k}{k!} y^{(k)}, \quad y'(t_{n+\ell}) = \sum_{k=1}^{\infty} \frac{(\ell h)^{k-1}}{(k-1)!} y^{(k)}.$$

Substituting into the definition and collecting powers of h ,

$$\sum_{\ell} \rho_{\ell} y(t_{n+\ell}) - h \sum_{\ell} \sigma_{\ell} y'(t_{n+\ell}) = \left(\sum_{\ell} \rho_{\ell} \right) y + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\sum_{\ell} \ell^k \rho_{\ell} - k \sum_{\ell} \ell^{k-1} \sigma_{\ell} \right) h^k y^{(k)}.$$

Since y is arbitrary, this is $\mathcal{O}(h^{p+1})$ precisely when

$$\sum_{\ell} \rho_{\ell} = 0 \quad \text{and} \quad \sum_{\ell} \ell^k \rho_{\ell} = k \sum_{\ell} \ell^{k-1} \sigma_{\ell} \quad \text{for } k = 1, \dots, p. \quad (*)$$

On the other hand, substituting $e^{\ell z} = \sum_k (\ell z)^k / k!$,

$$\begin{aligned} \rho(e^z) - z\sigma(e^z) &= \sum_{\ell} \rho_{\ell} e^{\ell z} - z \sum_{\ell} \sigma_{\ell} e^{\ell z} \\ &= \left(\sum_{\ell} \rho_{\ell} \right) + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\sum_{\ell} \ell^k \rho_{\ell} - k \sum_{\ell} \ell^{k-1} \sigma_{\ell} \right) z^k, \end{aligned}$$

which is $\mathcal{O}(z^{p+1})$ under exactly the same conditions $(*)$. $\square$

Example 19.5.13

For the two-step Adams–Bashforth method, $\rho(w) = w^2 - w$ and $\sigma(w) = \frac{3}{2}w - \frac{1}{2}$. Then

$$\rho(e^z) = e^{2z} - e^z = z + \frac{3}{2}z^2 + \frac{7}{6}z^3 + \mathcal{O}(z^4),$$

$$z\sigma(e^z) = z\left(\frac{3}{2}e^z - \frac{1}{2}\right) = z + \frac{3}{2}z^2 + \frac{3}{4}z^3 + \mathcal{O}(z^4),$$

so $\rho(e^z) - z\sigma(e^z) = \frac{5}{12}z^3 + \mathcal{O}(z^4)$. Hence the method has order 2, and the coefficient $\frac{5}{12}$ is its *error constant*.

§19.5.4 The Root Condition and Dahlquist's Theorem

High order is not enough. Here is a cautionary example.

Example 19.5.14

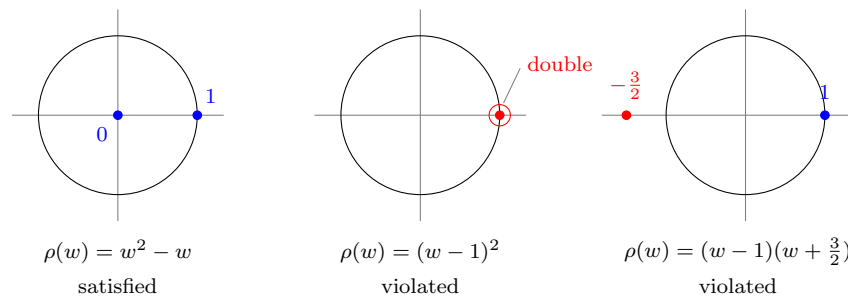
Consider the two-step method $y_{n+2} - 2y_{n+1} + y_n = 0$, that is $\rho(w) = (w - 1)^2$ and $\sigma \equiv 0$. Then $\rho(e^z) - z\sigma(e^z) = (e^z - 1)^2 = z^2 + \mathcal{O}(z^3)$, so the method has order 1. But it does not involve f at all – the numerical solution is the arithmetic progression $y_n = y_0 + n(y_1 - y_0)$ regardless of the differential equation. It certainly does not converge.

What is wrong is that the recurrence itself, ignoring the right-hand side, has solutions that grow. The homogeneous recurrence $\sum_{\ell} \rho_{\ell} y_{n+\ell} = 0$ has solutions $y_n = w^n$ where $\rho(w) = 0$, and repeated roots produce solutions nw^n , n^2w^n and so on. For these to stay bounded we need the following.

Definition 19.5.15 (Root Condition)

A polynomial ρ obeys the **root condition** if all of its zeros lie in the closed unit disc $|w| \leq 1$, and every zero on the unit circle $|w| = 1$ is simple.

Note that $\rho(1) = \sum_{\ell} \rho_{\ell} = 0$ is forced by order $p \geq 1$, so $w = 1$ is always a zero; the root condition says that it must be a simple one and that no other zero may escape the disc.



The remarkable fact is that order and the root condition are not merely necessary for convergence – together they are sufficient.

Theorem 19.5.16 (Dahlquist Equivalence Theorem)

A multistep method is convergent if and only if it has order $p \geq 1$ and its polynomial ρ obeys the root condition.

We shall not prove this; the proof is long and was not given in lectures. But it is worth

appreciating what it does for us. Convergence is a statement about all initial value problems and all step sizes – an infinite amount of checking. The theorem replaces it by two finite algebraic checks on two polynomials, both of which can be done by hand in a minute.

Example 19.5.17

The two-step Adams–Bashforth method has $\rho(w) = w^2 - w = w(w - 1)$, whose roots are 0 and 1; the root condition holds, and we computed the order to be 2. So it converges.

Example 19.5.18

The method $y_{n+2} - 3y_{n+1} + 2y_n = h(f(t_{n+1}, y_{n+1}) - 2f(t_n, y_n))$ has $\rho(w) = w^2 - 3w + 2 = (w - 1)(w - 2)$. The root $w = 2$ lies outside the unit disc, so the method fails the root condition and does not converge, however good its order may be. (In fact its order is 2.)

§19.5.5 Constructing Good Multistep Methods

We would like a recipe for producing convergent methods of high order. Theorem 19.5.12 tells us to make $\rho(e^z) - z\sigma(e^z)$ vanish to high order at $z = 0$; equivalently, substituting $w = e^z$ so that $z = \log w$, we want

$$\rho(w) - \sigma(w) \log w = \mathcal{O}(|w - 1|^{p+1}) \quad \text{as } w \rightarrow 1.$$

So: choose ρ first, subject to $\rho(1) = 0$ and the root condition, and then choose σ to be the truncation of the Taylor expansion of $\rho(w)/\log w$ about $w = 1$. If $s = \deg \rho$, we truncate to degree $s - 1$ for an explicit method and degree s for an implicit one.

Method 19.5.19 (Adams Methods)

Take $\rho(w) = w^{s-1}(w - 1)$, the simplest choice satisfying the root condition (roots 0 with multiplicity $s - 1$, and a simple root at 1). The corresponding methods have the form

$$y_{n+s} - y_{n+s-1} = h \sum_{\ell=0}^s \sigma_\ell f(t_{n+\ell}, y_{n+\ell}).$$

The explicit choice ($\sigma_s = 0$) gives the **Adams–Bashforth** methods, of order s ; the implicit choice gives the **Adams–Moulton** methods, of order $s + 1$.

To see the orders claimed, note that $\rho(w)/\log w$ is analytic near $w = 1$ (the zero of $\log w$ at $w = 1$ is cancelled by the factor $w - 1$). If σ is its Taylor polynomial of degree m about $w = 1$, then $\rho(w)/\log w - \sigma(w) = \mathcal{O}(|w - 1|^{m+1})$ and hence

$$\rho(w) - \sigma(w) \log w = \mathcal{O}(|w - 1|^{m+1} |\log w|) = \mathcal{O}(|w - 1|^{m+2}),$$

giving order $m + 1$. With $m = s - 1$ we get order s , and with $m = s$ order $s + 1$.

Example 19.5.20 (Small Adams Methods)

For $s = 1$: $\rho(w) = w - 1$, and Adams–Bashforth is Euler’s method (order 1), while Adams–Moulton is the trapezoidal rule (order 2).

For $s = 2$: $\rho(w) = w^2 - w$. The expansion of $\rho(w)/\log w$ about $w = 1$ begins $1 + \frac{3}{2}(w-1) + \frac{5}{12}(w-1)^2 + \dots$. Truncating to degree 1 gives $\sigma(w) = 1 + \frac{3}{2}(w-1) = \frac{3}{2}w - \frac{1}{2}$, which is the two-step Adams–Bashforth method of order 2; keeping the quadratic term gives the two-step Adams–Moulton method of order 3, which written out is

$$y_{n+2} - y_{n+1} = h \left(\frac{5}{12}f_{n+2} + \frac{2}{3}f_{n+1} - \frac{1}{12}f_n \right),$$

where we abbreviate $f_m = f(t_m, y_m)$.

There is a second family worth knowing, obtained by fixing σ instead of ρ .

Definition 19.5.21 (Backward Differentiation Formulae)

A **BDF method** is an s -step method of the form

$$\sum_{\ell=0}^s \rho_\ell y_{n+\ell} = h \sigma_s f(t_{n+s}, y_{n+s}),$$

so that f is evaluated at one point only, at the new time level.

Theorem 19.5.22 (BDF Coefficients)

The s -step BDF method of order s has

$$\rho(w) = \sigma_s \sum_{\ell=1}^s \frac{1}{\ell} w^{s-\ell} (w-1)^\ell, \quad \sigma_s = \left(\sum_{\ell=1}^s \frac{1}{\ell} \right)^{-1}.$$

Proof. We want $\rho(w) - \sigma_s w^s \log w = \mathcal{O}(|w-1|^{s+1})$. Now

$$\log w = -\log\left(\frac{1}{w}\right) = -\log\left(1 - \frac{w-1}{w}\right) = \sum_{\ell=1}^{\infty} \frac{1}{\ell} \left(\frac{w-1}{w}\right)^\ell,$$

so that

$$\sigma_s w^s \log w = \sigma_s \sum_{\ell=1}^{\infty} \frac{1}{\ell} w^{s-\ell} (w-1)^\ell.$$

Truncating this series after $\ell = s$ commits an error $\mathcal{O}(|w-1|^{s+1})$, and the truncation is exactly the stated ρ , which is a polynomial of degree s . Finally, $\rho_s = 1$ requires the coefficient of w^s to be 1; each term contributes σ_s/ℓ , giving $\sigma_s \sum_{\ell=1}^s 1/\ell = 1$. $\square$

Example 19.5.23

For $s = 1$ we get $\sigma_1 = 1$ and $\rho(w) = w - 1$: backward Euler. For $s = 2$, $\sigma_2 = (1 + \frac{1}{2})^{-1} = \frac{2}{3}$ and

$$\rho(w) = \frac{2}{3} \left(w(w-1) + \frac{1}{2}(w-1)^2 \right) = w^2 - \frac{4}{3}w + \frac{1}{3},$$

giving the method

$$y_{n+2} - \frac{4}{3}y_{n+1} + \frac{1}{3}y_n = \frac{2}{3}h f(t_{n+2}, y_{n+2}).$$

Its ρ has roots 1 and $\frac{1}{3}$, so the root condition holds and the method converges.

Remark. The BDF construction produces a ρ that we did not get to choose, so we must check the root condition afterwards – and it can fail. In fact the s -step BDF method satisfies the root condition precisely when $s \leq 6$. Nobody uses BDF7. The reason to care about these methods at all is their exceptional stability on stiff problems, which we come to shortly.

§19.5.6 Runge–Kutta Methods

Multistep methods get high order by remembering the past. Runge–Kutta methods get it a different way: they take several trial steps *within* the current interval, and combine them. The motivation comes from quadrature. If f happened not to depend on y at all, then

$$y(t_{n+1}) = y(t_n) + \int_{t_n}^{t_{n+1}} f(t) dt,$$

and we could apply any quadrature rule we liked to the integral – Gaussian quadrature, say – to get

$$y_{n+1} = y_n + h \sum_{\ell=1}^{\nu} b_{\ell} f(t_n + c_{\ell}h),$$

where the b_{ℓ} and c_{ℓ} come from the rule rescaled to $[0, 1]$. When f does depend on y we would need $y(t_n + c_{\ell}h)$, which we do not have. The Runge–Kutta idea is to manufacture approximations to those intermediate values out of the data we do have.

Definition 19.5.24 (Runge–Kutta Method)

A ν -stage Runge–Kutta method is

$$k_{\ell} = f \left(t_n + c_{\ell}h, y_n + h \sum_{j=1}^{\nu} a_{\ell j} k_j \right), \quad \ell = 1, \dots, \nu, \quad y_{n+1} = y_n + h \sum_{\ell=1}^{\nu} b_{\ell} k_{\ell},$$

where $c_{\ell} = \sum_{j=1}^{\nu} a_{\ell j}$. The method is **explicit** if $a_{\ell j} = 0$ whenever $j \geq \ell$, so that each k_{ℓ} is computed from those before it; otherwise it is implicit.

The coefficients are conventionally displayed in a **Butcher tableau**

$$\begin{array}{c|c} c & A \\ \hline & b^{\top} \end{array} = \begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1\nu} \\ c_2 & a_{21} & a_{22} & \cdots & a_{2\nu} \\ \vdots & \vdots & \vdots & & \vdots \\ c_{\nu} & a_{\nu 1} & a_{\nu 2} & \cdots & a_{\nu \nu} \\ \hline & b_1 & b_2 & \cdots & b_{\nu} \end{array}$$

in which the row sums of A are the entries of c . An explicit method has A strictly lower triangular, so its tableau has a triangle of zeros (usually left blank) on and above the diagonal.

Example 19.5.25 (Euler and the Trapezoidal Rule)

Euler's method is the one-stage explicit method with tableau

$$\begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}$$

and the trapezoidal rule is the two-stage implicit method

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}.$$

Let us see how the order conditions arise, in the smallest interesting case.

Example 19.5.26 (Two-Stage Explicit Methods)

Take $\nu = 2$ with $a_{21} = c_2$, so that

$$k_1 = f(t_n, y_n), \quad k_2 = f(t_n + c_2 h, y_n + c_2 h k_1),$$

and $y_{n+1} = y_n + h(b_1 k_1 + b_2 k_2)$. Assume $y_n = y(t_n)$ and expand. Taylor's theorem in two variables gives

$$k_2 = f + c_2 h (f_t + f_y f) + \mathcal{O}(h^2),$$

where f and its partials are evaluated at (t_n, y_n) . On the other hand, differentiating $y' = f(t, y)$ gives $y'' = f_t + f_y f$, so

$$y_n + h(b_1 k_1 + b_2 k_2) = y + h(b_1 + b_2)y' + h^2 b_2 c_2 y'' + \mathcal{O}(h^3),$$

whereas $y(t_{n+1}) = y + h y' + \frac{1}{2} h^2 y'' + \mathcal{O}(h^3)$. Matching coefficients, the local error is

$$h(1 - b_1 - b_2)y' + \frac{h^2}{2}(1 - 2b_2 c_2)y'' + \mathcal{O}(h^3),$$

so we obtain order 2 exactly when

$$b_1 + b_2 = 1, \quad b_2 c_2 = \frac{1}{2}.$$

Two popular solutions are $b = (\frac{1}{2}, \frac{1}{2})$, $c_2 = 1$ (the *improved Euler* or Heun method) and $b = (0, 1)$, $c_2 = \frac{1}{2}$ (the *midpoint* method).

Proposition 19.5.27

No two-stage explicit Runge–Kutta method has order 3.

Proof. It suffices to exhibit one problem for which order 3 fails. Take $y' = y$, $y(0) = 1$, with exact solution $y(t) = e^t$. Then $k_1 = e^{t_n}$ and $k_2 = e^{t_n}(1 + c_2 h)$, so the local error is

$$e^{t_{n+1}} - e^{t_n} (1 + h(b_1 + b_2) + h^2 b_2 c_2) = e^{t_n} (e^h - 1 - h(b_1 + b_2) - h^2 b_2 c_2).$$

Expanding e^h , the h^3 coefficient is $\frac{1}{6}e^{t_n}$, which no choice of b_ℓ, c_ℓ can remove since they do not appear in it. So the local error is $\Theta(h^3)$ at best. $\square$

The most celebrated Runge–Kutta method of all is the four-stage, fourth-order one, which people simply call ‘RK4’.

Example 19.5.28 (The Classical RK4)

The method

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f(t_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_1), \\ k_3 &= f(t_n + \tfrac{h}{2}, y_n + \tfrac{h}{2}k_2), \\ k_4 &= f(t_n + h, y_n + hk_3), \\ y_{n+1} &= y_n + \tfrac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \end{aligned}$$

has order 4. Its Butcher tableau is

$$\begin{array}{c|ccc} 0 & & & \\ \frac{1}{2} & \frac{1}{2} & & \\ \frac{1}{2} & 0 & \frac{1}{2} & \\ 1 & 0 & 0 & 1 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}.$$

Notice that the weights $\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6}$ and the nodes $0, \frac{1}{2}, \frac{1}{2}, 1$ are exactly Simpson’s rule on $[0, 1]$ (with the midpoint counted twice). This is no accident: if f is independent of y then $k_2 = k_3$ and the method *is* Simpson’s rule.

Remark. Verifying that RK4 has order 4 by direct Taylor expansion is a genuinely unpleasant computation – one must match multivariate derivatives of f up to third order – and we shall not do it here. The systematic treatment is due to Butcher and is organised by rooted trees. What is worth knowing is the general shape of the answer: an explicit ν -stage method has order at most ν , with equality possible only for $\nu \leq 4$. To get order 5 one needs 6 stages, which is one reason RK4 remains so popular.

Remark. Implicit Runge–Kutta methods can do much better. The ν -stage *Gauss–Legendre* method, whose nodes c_ℓ are the Gaussian quadrature nodes on $[0, 1]$, has order 2ν – the maximum possible – which is precisely the exactness result of Theorem 19.3.9 reappearing in a new guise. The two-stage version, with

$$c = \left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{6}, \tfrac{1}{2} + \tfrac{\sqrt{3}}{6}\right), \quad b = \left(\tfrac{1}{2}, \tfrac{1}{2}\right),$$

has order 4 using only two evaluations of f per step, at the cost of solving an implicit system.

§19.6 Stiff Equations and Stability

Everything so far has been about behaviour as $h \rightarrow 0$. In practice we do not take $h \rightarrow 0$; we take h as large as we dare, and the question is how large that is. For some problems the answer is dictated not by accuracy but by *stability*, and the discrepancy can be enormous.

Example 19.6.1

Consider the scalar problem $y' = -100y$, $y(0) = 1$, whose solution e^{-100t} has decayed to 10^{-44} by $t = 1$. Euler's method gives $y_n = (1-100h)^n$. If $h = 0.1$ then $y_n = (-9)^n$, which oscillates with wildly growing amplitude: at $t = 1$ we get $y_{10} = 3.5 \times 10^9$ instead of essentially zero. We need $|1 - 100h| < 1$, that is $h < 0.02$, purely to avoid nonsense – even though the solution itself is utterly featureless after $t = 0.05$ and any reasonable notion of accuracy would permit much larger steps.

Definition 19.6.2 (Stiffness)

An initial value problem is **stiff** if the step size of a numerical method must be kept much smaller than accuracy alone would demand, in order to keep the numerical solution bounded.

This is a deliberately informal definition, and stiffness is a property of the problem-and-method pair rather than of the problem alone. Stiffness typically arises in systems $y' = Ay$ whose eigenvalues have very different magnitudes: the fast-decaying components are numerically irrelevant after a moment, but they continue to dictate the step size forever. Such systems are the norm in chemical kinetics and in the method of lines for parabolic PDEs.

§19.6.1 Linear Stability Domains

To analyse this we use a test problem simple enough to compute with and rich enough to be informative.

Definition 19.6.3 (Linear Stability Domain)

Apply a numerical method with constant step h to the scalar test equation

$$y' = \lambda y, \quad y(0) = 1, \quad \lambda \in \mathbb{C}.$$

The **linear stability domain** of the method is

$$\mathcal{D} = \{h\lambda \in \mathbb{C} : y_n \rightarrow 0 \text{ as } n \rightarrow \infty\}.$$

The exact solution $e^{\lambda t}$ tends to 0 precisely when $\operatorname{Re} \lambda < 0$, so we would like $\mathcal{D}$ to contain the whole left half-plane.

Definition 19.6.4 (A-stability)

A method is **A-stable** if $\mathbb{C}^- = \{z \in \mathbb{C} : \operatorname{Re} z < 0\} \subseteq \mathcal{D}$.

Let us compute $\mathcal{D}$ for the theta method. Applying it to $y' = \lambda y$ gives

$$y_{n+1} = y_n + h\lambda(\theta y_n + (1-\theta)y_{n+1}) \implies y_{n+1} = \frac{1+\theta z}{1-(1-\theta)z} y_n, \quad z = h\lambda.$$

So $y_n = r(z)^n$ with $r(z) = \frac{1+\theta z}{1-(1-\theta)z}$, and $y_n \rightarrow 0$ if and only if $|r(z)| < 1$.

Example 19.6.5 (Euler's Method)

Here $\theta = 1$ and $r(z) = 1 + z$, so

$$\mathcal{D} = \{z : |1 + z| < 1\},$$

the open disc of radius 1 centred at -1 . This is a small set: it fails to contain, for instance, $z = -3$, which is exactly the instability we saw above. Euler's method is not A-stable.

Example 19.6.6 (Backward Euler)

Here $\theta = 0$ and $r(z) = (1 - z)^{-1}$, so

$$\mathcal{D} = \{z : |1 - z| > 1\},$$

the *exterior* of the open disc of radius 1 centred at $+1$. This contains the whole left half-plane, so backward Euler is A-stable. In fact $\mathcal{D}$ is much bigger than $\mathbb{C}^-$, which means backward Euler damps some components that the true solution does not damp – a form of inaccuracy, but a benign one.

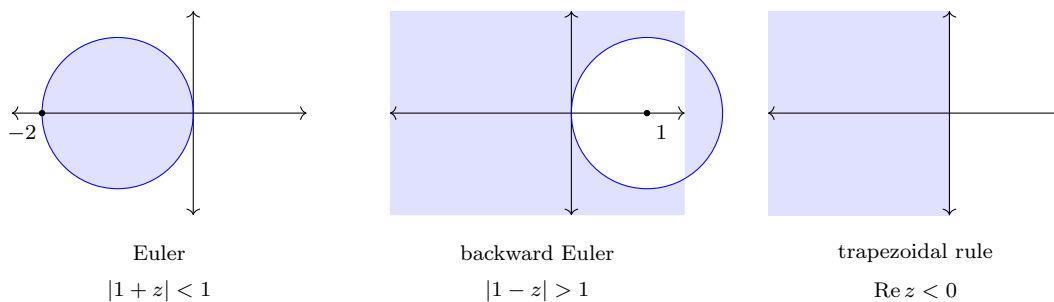
Example 19.6.7 (The Trapezoidal Rule)

Here $\theta = \frac{1}{2}$ and

$$r(z) = \frac{1 + z/2}{1 - z/2} = \frac{2 + z}{2 - z},$$

so $|r(z)| < 1$ if and only if $|2 + z| < |2 - z|$, that is, if and only if z is closer to -2 than to 2 . Hence $\mathcal{D} = \mathbb{C}^-$ exactly: the trapezoidal rule is A-stable, and its stability domain is precisely right – it damps exactly what should be damped.

Here are the three domains, with the stable region shaded.



The contrast is striking. The explicit method has a bounded stability domain, so for any fixed λ with $\operatorname{Re} \lambda \ll 0$ there is a hard ceiling on h ; the two implicit methods have no such ceiling at all. This is the pay-off that justifies the extra work of solving an implicit equation at each step, and it is why stiff problems are solved with implicit methods.

Remark. The trapezoidal rule is A-stable but its $r(z)$ satisfies $|r(z)| \rightarrow 1$ as $|z| \rightarrow \infty$, so very stiff components are barely damped – they merely oscillate in sign. Backward Euler has $r(z) \rightarrow 0$, and kills them outright. A method with $r(\infty) = 0$ is called *L-stable*, and for genuinely stiff problems this is what one wants.

§19.6.2 Stability of Multistep and Runge–Kutta Methods

For a multistep method the analysis is a little different, since the recurrence is of order s . Applying the method to $y' = \lambda y$ with $z = h\lambda$ gives

$$\sum_{\ell=0}^s \rho_{\ell} y_{n+\ell} = z \sum_{\ell=0}^s \sigma_{\ell} y_{n+\ell}, \quad \text{i.e.} \quad \sum_{\ell=0}^s (\rho_{\ell} - z\sigma_{\ell}) y_{n+\ell} = 0.$$

This is a linear recurrence with characteristic polynomial $\rho(w) - z\sigma(w)$, and its solutions tend to zero exactly when all the roots of that polynomial lie strictly inside the unit disc. So

$$\mathcal{D} = \{z \in \mathbb{C} : \text{all roots } w \text{ of } \rho(w) - z\sigma(w) \text{ satisfy } |w| < 1\}.$$

Notice that $z = 0$ gives back the root condition (in its non-strict form), which is reassuring: linear stability at $z = 0$ is exactly convergence.

This is harder to work with, and the news is bad.

Theorem 19.6.8 (Second Dahlquist Barrier)

No A-stable multistep method has order greater than 2. Among the A-stable multistep methods of order 2, the trapezoidal rule has the smallest error constant.

We do not prove this. It is a genuine obstruction rather than a failure of ingenuity, and it forces a choice: either accept low order, or weaken the demand for A-stability, or leave the multistep world altogether.

The second option is what is done in practice. Many stiff systems have eigenvalues that are not merely in $\mathbb{C}^-$ but far from the imaginary axis, so it suffices to have a stability domain containing a wedge around the negative real axis. A method whose stability domain contains such a wedge $\{z : |\arg(-z)| < \alpha\}$ is called $A(\alpha)$ -stable, and the BDF methods of order up to 6 all have this property with a decent α . This is why they exist.

The third option is Runge–Kutta. Applying a Runge–Kutta method to $y' = \lambda y$ always gives $y_{n+1} = r(z)y_n$ for a rational function r , so $\mathcal{D} = \{z : |r(z)| < 1\}$; and there is no barrier theorem here.

Example 19.6.9 (An A-stable Implicit Runge–Kutta Method)

Consider the two-stage implicit method with tableau

$$\begin{array}{c|cc} 0 & \frac{1}{4} & -\frac{1}{4} \\ \frac{2}{3} & \frac{1}{4} & \frac{5}{12} \\ \hline & \frac{1}{4} & \frac{3}{4} \end{array},$$

that is,

$$k_1 = f\left(t_n, y_n + \frac{h}{4}(k_1 - k_2)\right), \quad k_2 = f\left(t_n + \frac{2h}{3}, y_n + \frac{h}{12}(3k_1 + 5k_2)\right),$$

with $y_{n+1} = y_n + \frac{h}{4}(k_1 + 3k_2)$. Applying this to $y' = \lambda y$ and eliminating k_1, k_2 gives $y_{n+1} = r(z)y_n$ with

$$r(z) = \frac{6 + 2z}{6 - 4z + z^2}.$$

We claim $\mathcal{D} \supseteq \mathbb{C}^-$. The denominator $z^2 - 4z + 6$ has roots $2 \pm i\sqrt{2}$, both in the right half-plane, so r is analytic on $\overline{\mathbb{C}^-}$; and $r(z) \rightarrow 0$ as $|z| \rightarrow \infty$. By the maximum modulus principle, $|r|$ attains its maximum over the closed left half-plane on the boundary, so it suffices to check $|r(ix)| \leq 1$ for $x \in \mathbb{R}$. Now

$$|6 + 2ix|^2 = 36 + 4x^2, \quad |6 - x^2 - 4ix|^2 = (6 - x^2)^2 + 16x^2 = 36 + 4x^2 + x^4,$$

so $|r(ix)|^2 = \frac{36+4x^2}{36+4x^2+x^4} \leq 1$, with equality only at $x = 0$. Hence $|r(z)| < 1$ throughout $\mathbb{C}^-$, and the method is A-stable. It can be shown that it also has order 3 – beating the Dahlquist barrier comfortably.

Remark. The same argument shows the ν -stage Gauss–Legendre methods are all A-stable; the two-stage one has

$$r(z) = \frac{12 + 6z + z^2}{12 - 6z + z^2},$$

for which $|r(ix)| = 1$ for all real x , so its stability domain is exactly $\mathbb{C}^-$. Order 4 and exact stability, in two stages – at the price of solving a coupled implicit system at every step.

§19.6.3 Implementation

We close the discussion of differential equations with a few words on what one actually does, since the theory above leaves two practical questions unanswered: how big should h be, and how do we solve the implicit equations?

Choosing the step size

A user of an ODE solver does not want to specify h ; they want to specify an error tolerance ε and have the solver work out h for itself, varying it as the solution demands. To do that we need a cheap estimate of the local error at each step.

Method 19.6.10 (The Milne Device)

Run two multistep methods of the same order side by side, one explicit (the *predictor*) and one implicit (the *corrector*). If the two methods have error constants c_P and c_C , then

$$y_{n+1}^P \approx y(t_{n+1}) - c_P h^{p+1} y^{(p+1)}(t_n), \quad y_{n+1}^C \approx y(t_{n+1}) - c_C h^{p+1} y^{(p+1)}(t_n).$$

Subtracting eliminates the unknown derivative,

$$h^{p+1} y^{(p+1)}(t_n) \approx \frac{y_{n+1}^P - y_{n+1}^C}{c_C - c_P},$$

and substituting back gives the estimate

$$y_{n+1}^C - y(t_{n+1}) \approx \frac{c_C}{c_C - c_P} (y_{n+1}^P - y_{n+1}^C)$$

for the local error of the corrector, computable from quantities we have already.

Example 19.6.11

With the two-step Adams–Bashforth method as predictor ($c_P = \frac{5}{12}$) and the trapezoidal rule as corrector ($c_C = -\frac{1}{12}$), the estimate is

$$y_{n+1}^C - y(t_{n+1}) \approx -\frac{1}{6} (y_{n+1}^P - y_{n+1}^C).$$

The predictor costs one extra evaluation of f , which is cheap, and it doubles as an excellent starting guess for the implicit corrector equation.

Given the estimate, the logic is straightforward: if it exceeds the tolerance ε , reject the step and retry with smaller h ; if it is comfortably below (say below $\varepsilon/10$), accept the step and increase h ; otherwise accept and carry on unchanged.

For Runge–Kutta methods there is no single error constant to exploit, and the analogous device is *embedded* Runge–Kutta: one uses two methods, of orders p and $p+1$, sharing as many stages as possible so that the extra accuracy is nearly free.

Example 19.6.12 (An Embedded Pair)

With

$$k_1 = f(t_n, y_n), \quad k_2 = f\left(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1\right), \quad k_3 = f\left(t_n + h, y_n - hk_1 + 2hk_2\right),$$

the combination $y_{n+1}^{[1]} = y_n + hk_2$ has order 2, while $y_{n+1}^{[2]} = y_n + \frac{h}{6}(k_1 + 4k_2 + k_3)$ has order 3. The difference $y_{n+1}^{[1]} - y_{n+1}^{[2]}$ estimates the error in the lower-order formula, at the cost of three function evaluations in total.

Solving the implicit equations

An implicit method requires us to solve, at each step, an equation of the form

$$y_{n+s} = h\sigma_s f(t_{n+s}, y_{n+s}) + v,$$

where v collects everything already known. The simplest approach is **functional iteration**,

$$y_{n+s}^{[j+1]} = h\sigma_s f(t_{n+s}, y_{n+s}^{[j]}) + v,$$

started from the predictor. This is a contraction provided $h|\sigma_s|\lambda < 1$, where λ is the Lipschitz constant of f – but that is precisely the sort of restriction on h we adopted an implicit method to escape. So for stiff problems functional iteration is useless, and one uses the **Newton–Raphson** method instead: setting $\psi(y) = y - h\sigma_s f(t_{n+s}, y) - v$, iterate

$$y_{n+s}^{[j+1]} = y_{n+s}^{[j]} - \left(D\psi(y_{n+s}^{[j]})\right)^{-1} \psi(y_{n+s}^{[j]}),$$

which converges quadratically from a good starting guess. Evaluating and inverting the Jacobian $D\psi = I - h\sigma_s D_y f$ at every iteration is expensive, so in practice one freezes it,

$$y_{n+s}^{[j+1]} = y_{n+s}^{[j]} - \left(D\psi(y_{n+s}^{[0]})\right)^{-1} \psi(y_{n+s}^{[j]}),$$

recomputing only when convergence deteriorates. Either way we are left with a linear system to solve at every step, usually with the same matrix many times over – which is exactly the situation the next section is designed for.

§19.7 Numerical Linear Algebra

Solving $A\mathbf{x} = \mathbf{b}$ is the single most common task in scientific computing, and one that we have already run into twice in this course – in the normal equations for least-squares fitting, and in the Newton iteration for implicit ODE methods. In pure linear algebra the problem is considered solved: $\mathbf{x} = A^{-1}\mathbf{b}$, and Cramer’s rule even gives a formula. But Cramer’s rule for a 20×20 system requires more arithmetic than has been performed in the history of the world, so we shall have to do better.

The strategy throughout is *factorisation*: write A as a product of matrices for which the system is easy, solve a sequence of easy systems, and – crucially – reuse the factorisation if the same A appears again with a different $\mathbf{b}$.

§19.7.1 Triangular Systems

The easy case is a triangular matrix.

Definition 19.7.1

A square matrix L is **lower triangular** if $L_{ij} = 0$ for $j > i$, and **unit lower triangular** if in addition $L_{ii} = 1$ for all i . Similarly U is **upper triangular** if $U_{ij} = 0$ for $j < i$.

If L is lower triangular and nonsingular, the system $L\mathbf{y} = \mathbf{b}$ is solved by *forward substitution*: the first equation gives $y_1 = b_1/L_{11}$, and having found $y_1, \dots, y_{i-1}$ the i -th equation gives

$$y_i = \frac{1}{L_{ii}} \left(b_i - \sum_{j < i} L_{ij} y_j \right).$$

This costs $\mathcal{O}(n^2)$ operations, one multiplication for each nonzero entry of L . Upper triangular systems are solved the same way from the bottom up, by *back substitution*.

§19.7.2 LU Factorisation

Definition 19.7.2 (LU Factorisation)

An **LU factorisation** of a square matrix A is a factorisation $A = LU$ in which L is unit lower triangular and U is upper triangular.

Given such a factorisation, everything becomes cheap. To solve $A\mathbf{x} = \mathbf{b}$ we solve $L\mathbf{y} = \mathbf{b}$ by forward substitution and then $U\mathbf{x} = \mathbf{y}$ by back substitution, at a cost of $\mathcal{O}(n^2)$ for each new right-hand side. The determinant is

$$\det A = \det L \det U = \prod_{i=1}^n U_{ii},$$

so A is singular precisely when some $U_{ii} = 0$.

How do we compute it? Write $\mathbf{l}_1, \dots, \mathbf{l}_n$ for the columns of L and $\mathbf{u}_1^\top, \dots, \mathbf{u}_n^\top$ for the rows of U . Then

$$A = LU = \sum_{k=1}^n \mathbf{l}_k \mathbf{u}_k^\top,$$

an expression of A as a sum of n rank-one matrices. The point is that $\mathbf{l}_k$ has zeros in its first $k-1$ entries and $\mathbf{u}_k$ has zeros in its first $k-1$ entries, so $\mathbf{l}_k \mathbf{u}_k^\top$ has zeros in its first $k-1$ rows *and* its first $k-1$ columns. Consequently the first row and column of A come entirely from the $k=1$ term, and we can read them off.

Method 19.7.3 (LU by Outer Products)

Set $A_0 = A$. For $k = 1, 2, \dots, n$:

- (i) let $\mathbf{u}_k^\top$ be the k -th row of A_{k-1} ;
- (ii) let $\mathbf{l}_k$ be the k -th column of A_{k-1} , divided by $(A_{k-1})_{kk}$ so that $(\mathbf{l}_k)_k = 1$;
- (iii) set $A_k = A_{k-1} - \mathbf{l}_k \mathbf{u}_k^\top$.

Then $A = \sum_k \mathbf{l}_k \mathbf{u}_k^\top = LU$.

At step k the dominant cost is forming the outer product, which touches an $(n-k+1) \times (n-k+1)$ block, so the total cost is $\sum_k (n-k+1)^2 \sim \frac{1}{3}n^3$ operations. This is the same $\mathcal{O}(n^3)$ as Gaussian elimination – indeed, the algorithm *is* Gaussian elimination, with the multipliers recorded in L instead of discarded. The advantage of keeping them is that a second right-hand side then costs only $\mathcal{O}(n^2)$ rather than another $\mathcal{O}(n^3)$.

Example 19.7.4

Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{pmatrix}.$$

Step 1: $\mathbf{u}_1^\top = (2, 1, 1)$ and $\mathbf{l}_1 = (1, 2, 4)^\top$, so

$$A_1 = A - \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} (2, 1, 1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 3 & 5 \end{pmatrix}.$$

Step 2: $\mathbf{u}_2^\top = (0, 1, 1)$ and $\mathbf{l}_2 = (0, 1, 3)^\top$, giving

$$A_2 = A_1 - \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} (0, 1, 1) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Step 3: $\mathbf{u}_3^\top = (0, 0, 2)$, $\mathbf{l}_3 = (0, 0, 1)^\top$. Hence

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 3 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{pmatrix},$$

and $\det A = 2 \cdot 1 \cdot 2 = 4$.

The algorithm breaks down if $(A_{k-1})_{kk} = 0$, since we cannot divide by it. Sometimes this is unavoidable: the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has no LU factorisation at all, as one sees immediately by multiplying out $\begin{pmatrix} 1 & 0 \\ \ell & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix}$ and comparing top-left entries. The precise condition is the following.

Theorem 19.7.5 (Existence and Uniqueness of LU)

Let A be $n \times n$ and let $A^{(k)}$ denote its leading $k \times k$ principal submatrix. If $A^{(k)}$ is nonsingular for every $k = 1, \dots, n$, then A has an LU factorisation, and it is unique.

Proof. Uniqueness first, and note that it does not need the hypothesis beyond A being nonsingular. Suppose $A = LU = L'U'$ with L, L' unit lower triangular and U, U' upper triangular and nonsingular. Then $(L')^{-1}L = U'U^{-1}$. The left side is unit lower triangular (inverses and products of unit lower triangular matrices are unit lower triangular) and the right side is upper triangular. A matrix that is both is diagonal, and being unit lower triangular it must be I . Hence $L = L'$ and $U = U'$.

Existence, by induction on k , constructing the factorisation of $A^{(k)}$. For $k = 1$, take $L = (1)$ and $U = (A_{11})$, which is legitimate since $A^{(1)} \neq 0$. Suppose $A^{(k)} = L_k U_k$ with L_k unit lower triangular and U_k upper triangular; since $\det A^{(k)} = \det U_k \neq 0$, both factors are nonsingular. Write

$$A^{(k+1)} = \begin{pmatrix} A^{(k)} & \mathbf{b} \\ \mathbf{c}^\top & d \end{pmatrix},$$

and look for a factorisation

$$A^{(k+1)} = \begin{pmatrix} L_k & \mathbf{0} \\ \mathbf{x}^\top & 1 \end{pmatrix} \begin{pmatrix} U_k & \mathbf{y} \\ \mathbf{0}^\top & u \end{pmatrix} = \begin{pmatrix} L_k U_k & L_k \mathbf{y} \\ \mathbf{x}^\top U_k & \mathbf{x}^\top \mathbf{y} + u \end{pmatrix}.$$

The top-left block is already correct. The remaining equations are $L_k \mathbf{y} = \mathbf{b}$, $U_k^\top \mathbf{x} = \mathbf{c}$ and $u = d - \mathbf{x}^\top \mathbf{y}$, and each has a unique solution because L_k and U_k are nonsingular. This completes the induction, and taking $k = n$ gives the result. $\square$

Remark. The hypothesis is sufficient but not necessary: the zero matrix has an LU factorisation (many of them) while all its principal minors vanish. What the theorem really says is that if the leading minors are nonsingular, the naive algorithm never has to divide by zero – and the pivots it uses are exactly $U_{kk} = \det A^{(k)} / \det A^{(k-1)}$.

§19.7.3 Pivoting

A zero pivot is fatal, and – worse – a *small* pivot is nearly fatal, since dividing by it amplifies whatever rounding error is present. The cure for both is to reorder the rows.

Method 19.7.6 (LU with Partial Pivoting)

At step k , before doing anything else, choose $p \geq k$ maximising $|(A_{k-1})_{pk}|$ and let P_k be the permutation matrix exchanging rows k and p . Then take $\mathbf{u}_k^\top$ to be the k -th row of $P_k A_{k-1}$, take

$$\mathbf{l}_k = \frac{1}{(P_k A_{k-1})_{kk}} (k\text{-th column of } P_k A_{k-1}),$$

and set $A_k = P_k A_{k-1} - \mathbf{l}_k \mathbf{u}_k^\top$.

Choosing the largest available entry guarantees that every entry of $\mathbf{l}_k$ has modulus at

most 1, so the multipliers never magnify errors. Unwinding the algorithm, one finds

$$(P_{n-1} \cdots P_1)A = \tilde{\mathbf{l}}_1 \mathbf{u}_1^\top + \cdots + \tilde{\mathbf{l}}_n \mathbf{u}_n^\top, \quad \tilde{\mathbf{l}}_k = P_{n-1} \cdots P_{k+1} \mathbf{l}_k,$$

and since the permutations applied to $\mathbf{l}_k$ only touch entries from position $k+1$ onwards, the vectors $\tilde{\mathbf{l}}_k$ still have zeros in their first $k-1$ places. So they are the columns of a unit lower triangular matrix, and we have proved the following.

Theorem 19.7.7 (LU with Pivoting)

For any square matrix A there is a permutation matrix P such that $PA = LU$ with L unit lower triangular and U upper triangular.

Proof. Run the algorithm above, taking $P = P_{n-1} \cdots P_1$. The only remaining worry is a step at which the entire k -th column of A_{k-1} (from row k down) is zero; in that case there is nothing to eliminate, and we may simply take $P_k = I$, $\mathbf{l}_k = \mathbf{e}_k$ and $\mathbf{u}_k^\top$ the k -th row of A_{k-1} , which preserves everything we need. $\square$

To solve $A\mathbf{x} = \mathbf{b}$ we then solve $L\mathbf{y} = P\mathbf{b}$ and $U\mathbf{x} = \mathbf{y}$; permuting a vector is free.

§19.7.4 Symmetric Matrices and Cholesky

When A is symmetric it is wasteful to compute both L and U , since we ought to be able to get one from the other. The right factorisation is the following.

Definition 19.7.8

An **$LDL^\top$ factorisation** of a symmetric matrix A is a factorisation $A = LDL^\top$ with L unit lower triangular and D diagonal.

Setting $U = DL^\top$ recovers an LU factorisation, so by Theorem 19.7.5 the $LDL^\top$ factorisation is unique when it exists. It is computed by exactly the same outer-product scheme: writing $A = \sum_k D_{kk} \mathbf{l}_k \mathbf{l}_k^\top$, we set $A_0 = A$, take $D_{kk} = (A_{k-1})_{kk}$, take $\mathbf{l}_k$ to be the k -th column of A_{k-1} scaled so that $(\mathbf{l}_k)_k = 1$, and put $A_k = A_{k-1} - D_{kk} \mathbf{l}_k \mathbf{l}_k^\top$. Since we only need the lower half of each A_k , the work is halved.

Example 19.7.9

Take $A = \begin{pmatrix} 2 & 4 \\ 4 & 11 \end{pmatrix}$. Then $D_{11} = 2$ and $\mathbf{l}_1 = (1, 2)^\top$, so

$$A_1 = \begin{pmatrix} 2 & 4 \\ 4 & 11 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} (1, 2) = \begin{pmatrix} 0 & 0 \\ 0 & 3 \end{pmatrix},$$

whence $D_{22} = 3$, $\mathbf{l}_2 = (0, 1)^\top$ and

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

Definition 19.7.10

A symmetric matrix A is **positive definite** if $\mathbf{x}^\top A \mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$.

Theorem 19.7.11

A real symmetric matrix A is positive definite if and only if it has an $LDL^\top$ factorisation in which every diagonal entry of D is positive.

Proof. ($\Leftarrow$) If $A = LDL^\top$ with $D_{kk} > 0$, then for $\mathbf{x} \neq \mathbf{0}$ put $\mathbf{z} = L^\top \mathbf{x}$, which is nonzero since $L^\top$ is nonsingular. Then $\mathbf{x}^\top A \mathbf{x} = \mathbf{z}^\top D \mathbf{z} = \sum_k D_{kk} z_k^2 > 0$.

($\Rightarrow$) Suppose A is positive definite. Every leading principal submatrix $A^{(k)}$ is then positive definite (test with vectors supported on the first k coordinates), hence nonsingular, so by Theorem 19.7.5 the factorisation exists. Its diagonal entries satisfy $D_{kk} = \det A^{(k)} / \det A^{(k-1)}$, and all these determinants are positive because a positive definite matrix has positive determinant (its eigenvalues are all positive). Hence $D_{kk} > 0$. $\square$

Definition 19.7.12 (Cholesky Factorisation)

If A is symmetric positive definite, write $D^{1/2}$ for the diagonal matrix with entries $\sqrt{D_{kk}}$ and set $\tilde{L} = LD^{1/2}$. Then

$$A = \tilde{L} \tilde{L}^\top$$

with $\tilde{L}$ lower triangular with positive diagonal. This is the **Cholesky factorisation**.

Cholesky is the workhorse for symmetric positive definite systems. It needs no pivoting – the theorem guarantees the pivots are positive, and one can show they cannot grow – it uses half the storage and half the arithmetic of LU, and it fails, in a numerically detectable way, precisely when A is not positive definite. That last property makes it the standard test for positive definiteness.

§19.7.5 Banded and Sparse Matrices

Large matrices arising in practice are rarely dense. Discretising a differential equation, for instance, couples each unknown only to its neighbours, and the resulting matrix has almost all entries zero. It would be a shame to spend $\mathcal{O}(n^3)$ operations on such a matrix, and we need not.

Theorem 19.7.13 (Inheritance of Zeros)

Let $A = LU$ be an LU factorisation. If $A_{ij} = 0$ for all $j \leq m$ (that is, row i of A begins with m zeros, with $m < i$), then $L_{ij} = 0$ for $j \leq m$; and symmetrically, leading zeros in the columns of A above the diagonal are inherited by U .

Proof. We induct on j . Suppose $A_{i1} = 0$. Since $A_{i1} = \sum_k L_{ik} U_{k1} = L_{i1} U_{11}$ and $U_{11} \neq 0$, we get $L_{i1} = 0$. Now suppose $L_{i1} = \dots = L_{i,j-1} = 0$ and $A_{ij} = 0$ with

$j < i$. Then

$$0 = A_{ij} = \sum_{k \leq j} L_{ik} U_{kj} = L_{ij} U_{jj},$$

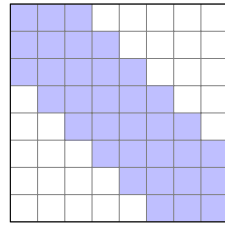
and $U_{jj} \neq 0$ gives $L_{ij} = 0$. The statement for U follows by applying this to $A^\top$. $\square$

Definition 19.7.14 (Banded Matrix)

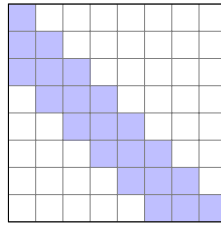
A matrix A is **banded** with bandwidth r if $A_{ij} = 0$ whenever $|i - j| > r$.

Combining the two: if A is banded with bandwidth r , then row i of A begins with $i - r - 1$ zeros and so does row i of L ; likewise for U . So L and U are banded with the same bandwidth r , and the factorisation can be computed by touching only the band. The cost drops from $\mathcal{O}(n^3)$ to $\mathcal{O}(r^2 n)$, which for a tridiagonal matrix ($r = 1$) is $\mathcal{O}(n)$.

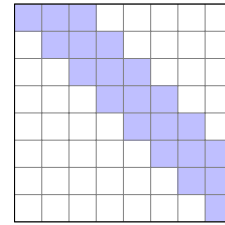
The picture is worth a glance: shaded cells are the ones that may be nonzero.



A (bandwidth $r = 2$)



L



U

Remark. For a general sparse matrix – one with many zeros but no particular pattern – the situation is more delicate, because elimination can create nonzeros where there were none. This is called *fill-in*, and minimising it by reordering rows and columns is a genuinely hard combinatorial problem, attacked in practice with graph-theoretic heuristics. Iterative methods, which never form a factorisation at all, are the other standard response. Both are beyond this course, and both are covered in Part II.

§19.8 QR Factorisation and Least Squares

The LU factorisation is fast, but it is not always well behaved: the factors can be much larger than A itself, and errors can grow accordingly. Orthogonal matrices, by contrast, preserve lengths exactly, and so cannot amplify anything. This makes them the tool of choice when accuracy matters more than speed – and, as we shall see, they are exactly what is needed for the least-squares problem.

Definition 19.8.1

A set of vectors $\{\mathbf{q}_i\}$ in $\mathbb{R}^m$ is **orthonormal** if $\mathbf{q}_i^\top \mathbf{q}_j = \delta_{ij}$. A square matrix Q is **orthogonal** if its columns are orthonormal, equivalently $Q^\top Q = I$, equivalently $Q^{-1} = Q^\top$.

The key property is that orthogonal matrices are isometries: $\|Q\mathbf{x}\|^2 = \mathbf{x}^\top Q^\top Q \mathbf{x} = \|\mathbf{x}\|^2$. In particular $|\det Q| = 1$, and multiplying by Q can neither magnify nor shrink an error.

Definition 19.8.2 (QR Factorisation)

Let A be $m \times n$. A **QR factorisation** of A is a factorisation $A = QR$ with Q an $m \times m$ orthogonal matrix and R an $m \times n$ upper triangular matrix. If $m \geq n$, a **reduced** (or *skinny*) QR factorisation is $A = QR$ with Q an $m \times n$ matrix with orthonormal columns and R an $n \times n$ upper triangular matrix.

If A is square and nonsingular and $A = QR$, then $A\mathbf{x} = \mathbf{b}$ becomes $R\mathbf{x} = Q^\top \mathbf{b}$, a triangular system. So QR does everything LU does; it just costs about twice as much.

To see what a reduced QR factorisation *means*, write $\mathbf{a}_1, \dots, \mathbf{a}_n$ for the columns of A and $\mathbf{q}_1, \dots, \mathbf{q}_n$ for those of Q . Comparing columns in $A = QR$ and using upper-triangularity,

$$\mathbf{a}_k = \sum_{j=1}^k R_{jk} \mathbf{q}_j.$$

So $\mathbf{q}_1, \dots, \mathbf{q}_k$ is an orthonormal basis for the span of $\mathbf{a}_1, \dots, \mathbf{a}_k$, for every k . A reduced QR factorisation is nothing other than an orthonormalisation of the columns of A , carried out one column at a time. That immediately suggests an algorithm.

§19.8.1 Gram–Schmidt**Method 19.8.3 (Gram–Schmidt)**

Assume $m \geq n$ and that the columns of A are linearly independent. Set

$$R_{11} = \|\mathbf{a}_1\|, \quad \mathbf{q}_1 = \mathbf{a}_1 / R_{11},$$

and having found $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$, set

$$R_{ij} = \mathbf{q}_i^\top \mathbf{a}_j \quad (i < j), \quad \mathbf{b}_j = \mathbf{a}_j - \sum_{i=1}^{j-1} R_{ij} \mathbf{q}_i, \quad R_{jj} = \|\mathbf{b}_j\|, \quad \mathbf{q}_j = \mathbf{b}_j / R_{jj}.$$

The vector $\mathbf{b}_j$ is $\mathbf{a}_j$ with its components along $\mathbf{q}_1, \dots, \mathbf{q}_{j-1}$ removed, so it is orthogonal to all of them; and it is nonzero because the columns of A are independent. The cost is $\mathcal{O}(mn^2)$.

This is the algorithm one meets in a first linear algebra course, and it is perfectly correct. Numerically, however, it is disappointing: the computed $\mathbf{q}_j$ drift out of orthogonality, because each is built from all its predecessors and their small errors compound. (Rearranging the arithmetic gives the ‘modified’ Gram–Schmidt process, which is noticeably better, but still not ideal.) The trouble is philosophical as much as practical – we are building an orthogonal matrix by a procedure that is not itself orthogonal.

The alternative is *orthogonal triangularisation*: rather than orthogonalising the columns of A , multiply A on the left by a sequence of orthogonal matrices, chosen to introduce zeros, until what is left is upper triangular. If

$$\Omega_k \cdots \Omega_1 A = R$$

with each Ω_i orthogonal, then $A = QR$ with $Q = (\Omega_k \cdots \Omega_1)^\top$, which is orthogonal by construction. We need a supply of orthogonal matrices that create zeros, and there are two standard choices.

§19.8.2 Givens Rotations

Definition 19.8.4 (Givens Rotation)

For $1 \leq p < q \leq m$ and $\theta \in (-\pi, \pi]$, the **Givens rotation** $\Omega^{[p,q]}(\theta)$ is the $m \times m$ matrix equal to the identity except that

$$\Omega_{pp} = \Omega_{qq} = \cos \theta, \quad \Omega_{pq} = \sin \theta, \quad \Omega_{qp} = -\sin \theta.$$

Geometrically this is a rotation of the plane spanned by $\mathbf{e}_p$ and $\mathbf{e}_q$, leaving everything else fixed; it is clearly orthogonal. For $m = 4$, say,

$$\Omega^{[1,2]} = \begin{pmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \Omega^{[2,4]} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & 0 & \sin \theta \\ 0 & 0 & 1 & 0 \\ 0 & -\sin \theta & 0 & \cos \theta \end{pmatrix}.$$

Proposition 19.8.5

Let A be $m \times n$, let $1 \leq p < q \leq m$ and $1 \leq j \leq n$. Then there is θ such that $(\Omega^{[p,q]}A)_{qj} = 0$. Moreover $\Omega^{[p,q]}A$ agrees with A in every row except the p -th and q -th, and those two rows are linear combinations of the p -th and q -th rows of A .

Proof. The second and third statements hold for every θ , directly from the definition. For the first, if $A_{pj} = A_{qj} = 0$ then any θ works. Otherwise take

$$\cos \theta = \frac{A_{pj}}{\sqrt{A_{pj}^2 + A_{qj}^2}}, \quad \sin \theta = \frac{A_{qj}}{\sqrt{A_{pj}^2 + A_{qj}^2}},$$

which is a legitimate choice since the two squares sum to 1. Then

$$(\Omega^{[p,q]}A)_{qj} = -\sin \theta A_{pj} + \cos \theta A_{qj} = \frac{-A_{qj}A_{pj} + A_{pj}A_{qj}}{\sqrt{A_{pj}^2 + A_{qj}^2}} = 0. \quad \square$$

To triangularise A we apply rotations that zero the subdiagonal entries one at a time, in an order that does not destroy the zeros already made. Because $\Omega^{[p,q]}$ only alters rows p and q , taking the pairs $[p, q]$ in lexicographic order works: we clear column 1 using $\Omega^{[1,2]}, \Omega^{[1,3]}, \dots, \Omega^{[1,m]}$, then column 2 using $\Omega^{[2,3]}, \dots$, and so on.

Example 19.8.6

For a 3×3 matrix, $\Omega^{[2,3]}\Omega^{[1,3]}\Omega^{[1,2]}A$ is upper triangular for suitable choices of the three angles.

The cost is again $\mathcal{O}(mn^2)$, so on a general matrix Givens has no advantage. Its virtue is *selectivity*: each rotation touches only two rows, so if A already has many zeros we need only a few rotations. For an $n \times n$ upper Hessenberg matrix (zero below the first subdiagonal) just $n - 1$ rotations suffice, giving a QR factorisation in $\mathcal{O}(n^2)$ operations. This is exactly the situation inside the QR algorithm for eigenvalues, which is why Givens rotations are not merely of historical interest.

§19.8.3 Householder Reflections

For a dense matrix, we would rather clear a whole column at once. Reflections do this.

Definition 19.8.7 (Householder Reflection)

For $\mathbf{u} \in \mathbb{R}^m \setminus \{\mathbf{0}\}$, the **Householder reflection** associated with $\mathbf{u}$ is

$$H = I - \frac{2\mathbf{u}\mathbf{u}^\top}{\mathbf{u}^\top\mathbf{u}}.$$

Proposition 19.8.8

H is symmetric and orthogonal, with $H\mathbf{u} = -\mathbf{u}$ and $H\mathbf{v} = \mathbf{v}$ for every $\mathbf{v}$ orthogonal to $\mathbf{u}$. Moreover, if $\mathbf{a}, \mathbf{b} \in \mathbb{R}^m$ satisfy $\|\mathbf{a}\| = \|\mathbf{b}\|$ and $\mathbf{u} = \mathbf{a} - \mathbf{b}$ is nonzero, then $H\mathbf{a} = \mathbf{b}$.

Proof. Symmetry is clear. For orthogonality, writing $\alpha = \mathbf{u}^\top\mathbf{u}$,

$$H^\top H = H^2 = I - \frac{4\mathbf{u}\mathbf{u}^\top}{\alpha} + \frac{4\mathbf{u}(\mathbf{u}^\top\mathbf{u})\mathbf{u}^\top}{\alpha^2} = I.$$

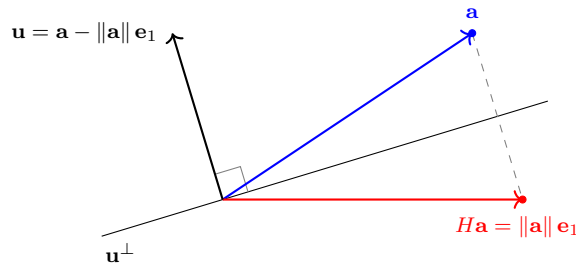
The formulas $H\mathbf{u} = -\mathbf{u}$ and $H\mathbf{v} = \mathbf{v}$ are immediate. For the last claim, put $\mathbf{u} = \mathbf{a} - \mathbf{b}$ and compute

$$\mathbf{u}^\top\mathbf{a} = \|\mathbf{a}\|^2 - \mathbf{b}^\top\mathbf{a}, \quad \mathbf{u}^\top\mathbf{u} = \|\mathbf{a}\|^2 - 2\mathbf{b}^\top\mathbf{a} + \|\mathbf{b}\|^2 = 2(\|\mathbf{a}\|^2 - \mathbf{b}^\top\mathbf{a}),$$

using $\|\mathbf{a}\| = \|\mathbf{b}\|$. Hence $2\mathbf{u}^\top\mathbf{a}/\mathbf{u}^\top\mathbf{u} = 1$ and

$$H\mathbf{a} = \mathbf{a} - \mathbf{u} = \mathbf{b}. \quad \square$$

So H is the reflection of $\mathbb{R}^m$ in the hyperplane $\mathbf{u}^\perp$, and the last part says that any two vectors of equal length are reflections of each other in the hyperplane bisecting them. That is precisely what we need: taking $\mathbf{b} = \pm \|\mathbf{a}\| \mathbf{e}_1$ turns any vector into a multiple of $\mathbf{e}_1$.



Method 19.8.9 (Householder QR)

At the first step, let $\mathbf{a}_1$ be the first column of A and take

$$\mathbf{u} = \mathbf{a}_1 + \text{sign}(A_{11}) \|\mathbf{a}_1\| \mathbf{e}_1,$$

so that $H\mathbf{a}_1 = -\text{sign}(A_{11}) \|\mathbf{a}_1\| \mathbf{e}_1$ and HA has zeros below the diagonal in its first column.

In general, suppose columns $1, \dots, k-1$ have been dealt with. Let $\tilde{\mathbf{a}}_k \in \mathbb{R}^{m-k+1}$ consist of the entries $A_{kk}, \dots, A_{mk}$ of the current k -th column, put

$$\tilde{\mathbf{u}} = \tilde{\mathbf{a}}_k + \text{sign}(A_{kk}) \|\tilde{\mathbf{a}}_k\| \mathbf{e}_1 \in \mathbb{R}^{m-k+1},$$

and apply the block-diagonal orthogonal matrix

$$\begin{pmatrix} I_{k-1} & 0 \\ 0 & \tilde{H} \end{pmatrix}, \quad \tilde{H} = I - \frac{2\tilde{\mathbf{u}}\tilde{\mathbf{u}}^\top}{\tilde{\mathbf{u}}^\top\tilde{\mathbf{u}}}.$$

The leading I_{k-1} leaves the first $k-1$ rows – and hence the zeros already created – untouched.

The choice of sign deserves comment. Either sign gives a valid reflection, but if A_{kk} and $\|\tilde{\mathbf{a}}_k\|$ nearly cancel then $\tilde{\mathbf{u}}$ is tiny and computed inaccurately. Choosing the sign of A_{kk} makes the two terms reinforce rather than cancel, which is the numerically safe option.

In practice one never forms $\tilde{H}$ explicitly. Applying it to a vector $\mathbf{v}$ is just

$$\tilde{H}\mathbf{v} = \mathbf{v} - \frac{2(\tilde{\mathbf{u}}^\top\mathbf{v})}{\tilde{\mathbf{u}}^\top\tilde{\mathbf{u}}}\tilde{\mathbf{u}},$$

an inner product and an axpy, costing $\mathcal{O}(m)$ rather than $\mathcal{O}(m^2)$. Similarly Q itself is normally stored as the list of vectors $\tilde{\mathbf{u}}$, and one computes $Q^\top\mathbf{b}$ by applying the reflections in turn. The total cost is $\mathcal{O}(mn^2)$.

Example 19.8.10

Suppose we have reached

$$A = \begin{pmatrix} 2 & 4 & 7 \\ 0 & 3 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -2 \end{pmatrix},$$

with the first two columns already processed, and we wish to clear the third. Here

$$\tilde{\mathbf{a}}_3 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \quad \|\tilde{\mathbf{a}}_3\| = 3, \quad \tilde{\mathbf{u}} = \tilde{\mathbf{a}}_3 + 3\mathbf{e}_1 = \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix},$$

and $\tilde{\mathbf{u}}^\top\tilde{\mathbf{u}} = 30$, $\tilde{\mathbf{u}}^\top\tilde{\mathbf{a}}_3 = 10 + 1 + 4 = 15$. Hence

$$\tilde{H}\tilde{\mathbf{a}}_3 = \tilde{\mathbf{a}}_3 - \frac{2 \cdot 15}{30}\tilde{\mathbf{u}} = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} - \begin{pmatrix} 5 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 0 \end{pmatrix},$$

and the result is

$$R = \begin{pmatrix} 2 & 4 & 7 \\ 0 & 3 & -1 \\ 0 & 0 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Remark. So which should one use? For a dense matrix, Householder: it clears an entire column with one reflection and is beautifully stable. For a matrix that is already nearly triangular, Givens: each rotation is cheap and touches only the rows that need touching. Gram–Schmidt survives mainly because it produces the columns of Q one at a time, which is occasionally what one wants – and because it generalises to infinite dimensions, where reflections do not obviously help.

§19.8.4 Linear Least Squares

We finish with the problem that motivated all of this. Let A be $m \times n$ with $m > n$ and let $\mathbf{b} \in \mathbb{R}^m$. The system $A\mathbf{x} = \mathbf{b}$ has more equations than unknowns and generically has no solution at all; this is the situation of fitting n parameters to m measurements. The sensible thing to ask for instead is the $\mathbf{x}$ minimising the residual.

Definition 19.8.11 (Least Squares Problem)

Given $A \in \mathbb{R}^{m \times n}$ and $\mathbf{b} \in \mathbb{R}^m$, the **linear least squares problem** is to find $\mathbf{x} \in \mathbb{R}^n$ minimising $\|A\mathbf{x} - \mathbf{b}\|_2$.

Theorem 19.8.12 (Normal Equations)

A vector $\mathbf{x} \in \mathbb{R}^n$ solves the least squares problem if and only if

$$A^\top(A\mathbf{x} - \mathbf{b}) = \mathbf{0}, \quad \text{equivalently} \quad A^\top A\mathbf{x} = A^\top \mathbf{b}.$$

Proof. ($\Leftarrow$) Suppose $A^\top(A\mathbf{x} - \mathbf{b}) = \mathbf{0}$ and let $\mathbf{v} \in \mathbb{R}^n$ be arbitrary. Writing $\mathbf{v} = \mathbf{x} + (\mathbf{v} - \mathbf{x})$ and expanding,

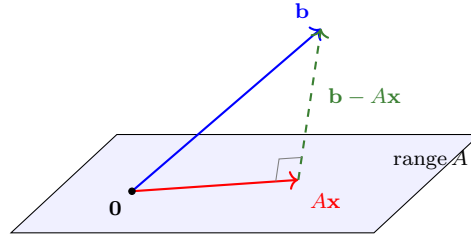
$$\begin{aligned} \|A\mathbf{v} - \mathbf{b}\|^2 &= \|(A\mathbf{x} - \mathbf{b}) + A(\mathbf{v} - \mathbf{x})\|^2 \\ &= \|A\mathbf{x} - \mathbf{b}\|^2 + 2(\mathbf{v} - \mathbf{x})^\top A^\top(A\mathbf{x} - \mathbf{b}) + \|A(\mathbf{v} - \mathbf{x})\|^2 \\ &= \|A\mathbf{x} - \mathbf{b}\|^2 + \|A(\mathbf{v} - \mathbf{x})\|^2 \geq \|A\mathbf{x} - \mathbf{b}\|^2. \end{aligned}$$

($\Rightarrow$) Conversely, if $\mathbf{x}$ is a minimiser then for every $\mathbf{v}$ the scalar function $t \mapsto \|A(\mathbf{x} + t\mathbf{v}) - \mathbf{b}\|^2$ has a minimum at $t = 0$. Differentiating,

$$0 = \left. \frac{d}{dt} \right|_{t=0} (\|A\mathbf{x} - \mathbf{b}\|^2 + 2t\mathbf{v}^\top A^\top(A\mathbf{x} - \mathbf{b}) + t^2 \|A\mathbf{v}\|^2) = 2\mathbf{v}^\top A^\top(A\mathbf{x} - \mathbf{b}),$$

and as $\mathbf{v}$ was arbitrary, $A^\top(A\mathbf{x} - \mathbf{b}) = \mathbf{0}$. □

The geometric content is that the residual must be orthogonal to the column space of A : the best approximation $A\mathbf{x}$ is the orthogonal projection of $\mathbf{b}$ onto $\text{range } A$. This is Pythagoras again, exactly as in Section 2.



So one could simply solve the $n \times n$ system $A^\top A \mathbf{x} = A^\top \mathbf{b}$, say by Cholesky, since $A^\top A$ is symmetric and (if A has full column rank) positive definite. This is what statisticians did for a century, and it is not wrong. But it has three defects that a numerical analyst notices immediately.

- (i) Forming $A^\top A$ costs $\mathcal{O}(mn^2)$ operations and, more importantly, destroys any sparsity that A possessed – a sparse A can easily have a dense $A^\top A$.
- (ii) The entries of $A^\top A$ are sums of products, so if the entries of A vary greatly in magnitude, forming them loses accuracy through cancellation.
- (iii) Most seriously, the conditioning of $A^\top A$ is the *square* of that of A . Squaring an already-large condition number is an excellent way to lose every digit one has.

The remedy is to use the QR factorisation, which lets us minimise the residual without ever forming $A^\top A$.

Theorem 19.8.13 (Least Squares via QR)

Let A be $m \times n$ with $m \geq n$ and linearly independent columns, and let $A = QR$ be a reduced QR factorisation, with Q of size $m \times n$ having orthonormal columns and R upper triangular and nonsingular. Then the least squares problem is solved by the unique solution of

$$R\mathbf{x} = Q^\top \mathbf{b}.$$

Proof. By Theorem 19.8.12, $\mathbf{x}$ is a solution if and only if $A^\top(A\mathbf{x} - \mathbf{b}) = \mathbf{0}$. Substituting $A = QR$ and using $Q^\top Q = I$,

$$A^\top(A\mathbf{x} - \mathbf{b}) = R^\top Q^\top(QR\mathbf{x} - \mathbf{b}) = R^\top(R\mathbf{x} - Q^\top \mathbf{b}).$$

As R is nonsingular, so is $R^\top$, and the above vanishes precisely when $R\mathbf{x} = Q^\top \mathbf{b}$. That triangular system has a unique solution, found by back substitution. $\square$

Remark. Using the full QR factorisation gives the same conclusion in a way that also exhibits the residual. If $A = QR$ with Q orthogonal $m \times m$ and $R = \begin{pmatrix} R_1 \\ 0 \end{pmatrix}$ with R_1 upper triangular $n \times n$, then since $Q^\top$ preserves norms,

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \|Q^\top A\mathbf{x} - Q^\top \mathbf{b}\|^2 = \|R_1\mathbf{x} - \mathbf{c}_1\|^2 + \|\mathbf{c}_2\|^2,$$

where $Q^\top \mathbf{b} = \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{pmatrix}$ is split after n entries. The second term does not involve $\mathbf{x}$ at all, so the minimum is attained by solving $R_1\mathbf{x} = \mathbf{c}_1$, and the minimum value of the residual is exactly $\|\mathbf{c}_2\|$. This is perhaps the cleanest way to see the whole result: an orthogonal change of coordinates turns an unsolvable system into a solvable one plus an irreducible remainder.

Example 19.8.14 (Fitting a Straight Line)

Suppose we wish to fit $y = c_0 + c_1 t$ to the four data points $(0, 1)$, $(1, 3)$, $(2, 4)$, $(3, 4)$. Then

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 4 \end{pmatrix},$$

and the normal equations read

$$\begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \end{pmatrix} = \begin{pmatrix} 12 \\ 23 \end{pmatrix},$$

with solution $c_1 = 1$, $c_0 = \frac{3}{2}$. So the best-fitting line is $y = \frac{3}{2} + t$, and the residual vector is $(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})^\top$, whose components sum to zero and which is orthogonal to $(0, 1, 2, 3)^\top$ – exactly the two normal equations, read backwards.

Remark. It is worth stepping back at the end. The least squares problem we have just solved is the same problem we solved in Section 2 with orthogonal polynomials, in a different costume: there the vector space was $C[a, b]$ or $\mathcal{P}_n$ and the orthogonal basis was $p_0, \dots, p_n$; here it is $\mathbb{R}^m$ and the orthogonal basis consists of the columns of Q . In both cases the answer is an orthogonal projection, in both cases the good algorithm is the one that builds an orthogonal basis first, and in both cases the reason is the same – orthogonality decouples the problem and refuses to amplify error. If there is one idea to take away from this course, it is probably that one.

20 Optimization

Optimization is the art of doing the best you can with what you have got. Almost every quantitative question you might ask about the world – how should a factory allocate its raw materials, how should goods be routed through a road network, how should two adversaries play a game – can be phrased as the problem of making some function as small (or as large) as possible, subject to a list of constraints on what you are allowed to do. The subject is about turning that vague desire into a precise mathematical problem, and then actually *solving* it.

Two themes run through everything we do. The first is *convexity*: for a convex problem, local information is global information, and this is what makes optimization tractable at all. The second is *duality*: every optimization problem casts a shadow, a second problem whose solution certifies the optimality of the first. Duality is what lets us know when we are done, and it appears in different disguises as complementary slackness, as shadow prices, as the max-flow min-cut theorem and as the minimax theorem of game theory. Along the way we will develop honest algorithms – gradient descent, the simplex method, the transportation algorithm, Ford–Fulkerson – that can be run by hand or by machine.

This article constitutes my notes for the ‘Optimization’ course, held in Easter 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§20.1 Convex Optimization

§20.1.1 Optimization Problems

In this course, we care about problems that look like ‘minimize $f(x)$ subject to $x \in X$ ’, or problems with constraints that look like ‘minimize¹ $f(x)$ subject to $x \in X$ where $h(x) = b$ ’.

In such problems, we call f the **objective function**, and if $x \in \mathbb{R}^n$, we call the components **decision variables**. Constraints such as the one above can be given as a **functional constraint** $h(x) = b$ and a **regional constraint** $x \in X$. If both of the constraints hold for a given x , we say it is in the **feasible set** $X(b)$. A problem is feasible if the feasible set is non-empty, and bounded if the minimum over this set is bounded.

A point $x^* \in X(b)$ is **optimal** if it minimizes f over $X(b)$. The value $f(x^*)$ is the **optimal cost**.

It is worth setting up some notation for all of this, since we are going to be writing down an awful lot of optimization problems. We will write a problem in the form

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & h(x) = b, \\ & x \in X,\end{array}$$

¹We will almost always talk about minimizing, since maximizing is basically the same thing: to maximize f we can minimize $-f$.

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $b \in \mathbb{R}^m$ and $X \subseteq \mathbb{R}^n$. So the feasible set is

$$X(b) = \{x \in X : h(x) = b\}.$$

Remark. The split between ‘functional’ and ‘regional’ constraints is not intrinsic to the problem – it is a choice that we make. The same set of feasible points can very often be described in several ways, and a large part of the skill in this course lies in choosing the description that makes the problem easiest to attack. Typically the regional constraint $x \in X$ will be something simple like $x \geq 0$ or $x \in \mathbb{Z}^n$, and the functional constraints will be the interesting ones that we handle with Lagrange multipliers.

At first glance it looks as though we have restricted ourselves to *equality* constraints, which would be a serious limitation. In fact we have not: an inequality constraint can always be converted into an equality constraint at the cost of introducing a new variable. If we want $h(x) \leq b$, we introduce a **slack variable** s and write

$$h(x) \leq b \quad \longleftrightarrow \quad h(x) + s = b, \quad s \geq 0.$$

The new variable s measures exactly how much ‘slack’ there is in the constraint, and the sign condition $s \geq 0$ is pushed into the regional constraint. This trick will be used constantly.

Example 20.1.1

Consider the problem of minimizing $x_1^2 + x_2^2$ subject to $x_1 + x_2 \geq 1$. Here $f(x) = x_1^2 + x_2^2$, and we can take $X = \mathbb{R}^2$ with the single functional constraint $x_1 + x_2 \geq 1$. Introducing a slack variable, this becomes: minimize $x_1^2 + x_2^2$ subject to $x_1 + x_2 - s = 1$ and $(x_1, x_2, s) \in \mathbb{R}^2 \times [0, \infty)$. The feasible set is a closed half-plane, and the optimum is at $x = (1/2, 1/2)$ with optimal cost $1/2$ – the point of the half plane closest to the origin.

§20.1.2 Convex Functions

We will first study optimization problems involving convex functions without constraints. In particular, we are going to study the problem:

Minimize $f(x)$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function.

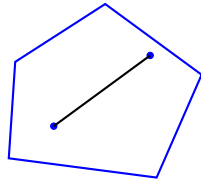
The reason we will care specifically about convex functions is, as you will see, that local information about the function gives us global information about the function.

It is likely that you are already familiar with the notions of convex sets and convex functions.

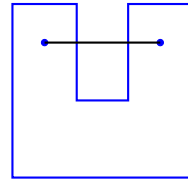
Definition 20.1.2 (Convex Set)

A set $S \subseteq \mathbb{R}^n$ is **convex** if for all $x, y \in S$ and all $\lambda \in [0, 1]$ we have $\lambda x + (1 - \lambda)y \in S$.

Informally, a set is convex if the line segment between any two points is totally contained in the set.



convex



not convex

We can also define convexity for functions.

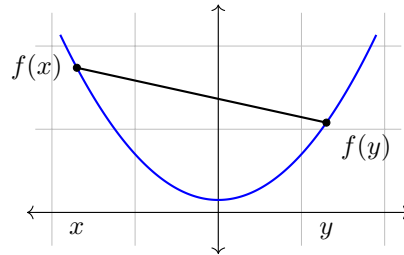
Definition 20.1.3 (Convex Function)

A function $f : S \rightarrow \mathbb{R}$ is **convex** if S is convex and for all $x, y \in S$ and $\lambda \in [0, 1]$ we have

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

If this inequality is strict (for $x \neq y$ and $\lambda \in (0, 1)$) then the function is **strictly convex**. A function is called **concave** if $-f$ is convex.

Again informally, this says that if we consider the graph of the function then the chord between two points will always lie above the function.



Note that a linear function $f(x) = c \cdot x$ is both convex and concave, since the defining inequality holds with equality. So linear programming, which we will spend a lot of time on later, is a special case of convex optimization.

In general, checking the definition by hand is hard, and we typically use some other conditions to check for convexity.

Proposition 20.1.4 (First-Order Conditions for Convexity)

A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if for all $x, y \in \mathbb{R}^n$ we have

$$f(y) \geq f(x) + (y - x) \cdot \nabla f(x).$$

Proof. We will first check that convexity implies that this condition holds by showing it holds for $n = 1$ and using that for the general case.

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex, and $x, y \in \mathbb{R}$ then for any $t \in (0, 1]$ we have

$$\begin{aligned} f((1 - t)x + ty) &= f(x + t(y - x)) \\ &\leq (1 - t)f(x) + tf(y) \\ \implies f(y) &\geq f(x) + \frac{f(x + t(y - x)) - f(x)}{t}. \end{aligned}$$

Taking the limit as $t \rightarrow 0$, the quotient tends to $f'(x)(y - x)$, and so $f(y) \geq f(x) + f'(x)(y - x)$.

Now for $f : \mathbb{R}^n \rightarrow \mathbb{R}$, taking any $x, y \in \mathbb{R}^n$ we can define the restricted function $g(t) = f((1 - t)x + ty)$. Since f is convex, g is also convex. We can then compute

$$g'(t) = (y - x) \cdot \nabla f((1 - t)x + ty).$$

So using the $n = 1$ case, we have $g(1) \geq g(0) + g'(0)(1 - 0)$, that is, $f(y) \geq f(x) + (y - x) \cdot \nabla f(x)$, so the condition holds as required.

Now we will show that the first-order conditions imply convexity. For some $t \in [0, 1]$, we define $x_t = (1 - t)x + ty$. Then we know that

$$\begin{aligned} f(x) &\geq f(x_t) + (x - x_t) \cdot \nabla f(x_t), \\ f(y) &\geq f(x_t) + (y - x_t) \cdot \nabla f(x_t). \end{aligned}$$

Multiplying the first by $(1 - t)$ and the second by t and summing, the gradient terms combine to give

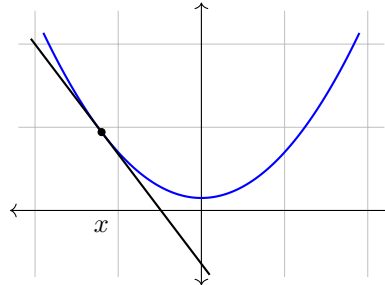
$$(1 - t)(x - x_t) + t(y - x_t) = (1 - t)x + ty - x_t = 0,$$

so we are left with

$$(1 - t)f(x) + tf(y) \geq f(x_t),$$

as required. $\square$

This condition has a very clean geometric meaning: the tangent plane to the graph of f at any point lies entirely *below* the graph.



This picture makes the key fact about convex minimization obvious. If $\nabla f(x) = 0$ at some point x , then the first-order condition reads $f(y) \geq f(x)$ for *all* y , so x is a global minimum. In other words, for a convex function there is no such thing as a bad local minimum: every stationary point is a global minimum. This is exactly the promised phenomenon of local information giving global information, and it is the reason we can hope to minimize convex functions by walking downhill.

A possibly more familiar condition for convexity is the second-order condition. This involves the *Hessian*, the generalization of the second derivative to $\mathbb{R}^n$.

Definition 20.1.5 (Hessian)

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice differentiable function, then the **Hessian matrix** $\nabla^2 f$ is

the symmetric matrix of second derivatives given by

$$(\nabla^2 f)_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

In particular, we care about a specific property of the Hessian matrix.

Definition 20.1.6 (Positive Semidefinite)

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is **positive semidefinite** if for all $x \in \mathbb{R}^n$ we have $x \cdot Ax \geq 0$; equivalently, if all of the eigenvalues of A are non-negative. We write $A \succeq 0$. We write $A \succeq B$ to mean $A - B \succeq 0$.

Proposition 20.1.7 (Second-Order Conditions for Convexity)

A twice differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $\nabla^2 f(x)$ is positive semidefinite at all $x \in \mathbb{R}^n$.

Proof. Using the Taylor expansion of f , we have

$$f(y) = f(x) + (y - x) \cdot \nabla f(x) + \frac{1}{2}(y - x) \cdot (\nabla^2 f(z)(y - x)),$$

where $z = (1 - \lambda)x + \lambda y$ for some $\lambda \in [0, 1]$. Since $\nabla^2 f(z) \succeq 0$, the final term is non-negative, and so

$$f(y) \geq f(x) + (y - x) \cdot \nabla f(x).$$

Then by the first-order conditions, f is convex. □

The converse also holds, but we will not need it.

Example 20.1.8

Let us see the conditions in action.

- (i) $f(x) = x \cdot Ax$ for a symmetric matrix A has $\nabla^2 f = 2A$, so f is convex exactly when $A \succeq 0$.
- (ii) $f(x_1, x_2) = x_1^2 + 100x_2^2$ has Hessian $\text{diag}(2, 200) \succeq 0$, so is convex.
- (iii) $f(x) = e^{c \cdot x}$ has $\nabla^2 f = (cc^T)e^{c \cdot x}$, and $y \cdot (cc^T)y = (c \cdot y)^2 \geq 0$, so f is convex for any c .
- (iv) $f(x_1, x_2) = x_1x_2$ has Hessian $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, whose eigenvalues are ± 1 . So this function is neither convex nor concave.

§20.1.3 Gradient Descent

To solve our problem of minimizing $f(x)$ where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function, we can begin by noting that convexity implies that a local minimum is a global minimum. So if we find a point where the gradient of the function is 0, then we would be finished. This suggests an extremely simple ‘greedy’ strategy: start somewhere, look around for a nearby point with a smaller value of f , move there, and repeat.

To make this concrete we need to know which direction to move in. Taking a small step of size ε in the direction v and Taylor expanding,

$$f(x + \varepsilon v) \approx f(x) + \varepsilon v \cdot \nabla f(x),$$

so any v with $v \cdot \nabla f(x) < 0$ will decrease f , at least for small enough ε . We call such a v a **descent direction**. The most natural choice is $v = -\nabla f(x)$, which gives

$$f(x - \varepsilon \nabla f(x)) \approx f(x) - \varepsilon \|\nabla f(x)\|^2 \leq f(x).$$

Definition 20.1.9 (Gradient Descent)

The **gradient descent** algorithm for minimizing f is as follows.

- (1) Choose a starting point x_0 and set $t = 0$.
- (2) If $\nabla f(x_t) = 0$ (or we have run out of patience), stop and return x_t .
- (3) Choose a step size $\eta_t > 0$ and set $x_{t+1} = x_t - \eta_t \nabla f(x_t)$.
- (4) Increment t and go to (2).

The whole art is in the choice of η_t . If it is too small we crawl; if it is too large we overshoot and oscillate. To say anything precise we need to control how fast the gradient can change.

Definition 20.1.10 (Smoothness)

A continuously differentiable $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **β -smooth** if ∇f is β -Lipschitz, that is

$$\|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|$$

for all x, y . If f is twice differentiable this is equivalent to $\nabla^2 f(x) \preceq \beta I$ for all x .

Smoothness says that the linear approximation to f cannot be too badly wrong.

Lemma 20.1.11 (Quadratic Upper Bound)

If f is β -smooth then for all x, y ,

$$f(y) \leq f(x) + (y - x) \cdot \nabla f(x) + \frac{\beta}{2} \|y - x\|^2.$$

Proof. By Taylor's theorem there is a point z on the segment from x to y with

$$f(y) = f(x) + (y - x) \cdot \nabla f(x) + \frac{1}{2} (y - x) \cdot (\nabla^2 f(z)(y - x)),$$

and since $\nabla^2 f(z) \preceq \beta I$ we have $(y - x) \cdot (\nabla^2 f(z)(y - x)) \leq \beta \|y - x\|^2$. □

Together with convexity this sandwiches f between two functions that agree to first order at x : a linear lower bound and a quadratic upper bound. The quadratic upper bound is the useful one, because we can minimize it exactly.

Corollary 20.1.12

If f is β -smooth and we take a step of size $1/\beta$, then

$$f\left(x - \frac{1}{\beta} \nabla f(x)\right) \leq f(x) - \frac{1}{2\beta} \|\nabla f(x)\|^2.$$

Proof. Minimizing the right-hand side of the quadratic upper bound over y , we differentiate to get $\nabla f(x) + \beta(y - x) = 0$, so the minimizer is $y = x - \frac{1}{\beta} \nabla f(x)$. Substituting this in,

$$\begin{aligned} f(y) &\leq f(x) - \frac{1}{\beta} \|\nabla f(x)\|^2 + \frac{\beta}{2} \cdot \frac{1}{\beta^2} \|\nabla f(x)\|^2 \\ &= f(x) - \frac{1}{2\beta} \|\nabla f(x)\|^2. \square \end{aligned}$$

So each step makes definite progress, provided the gradient is not small. But a small gradient does not by itself mean we are near the optimum – think of a function with a long, almost flat valley. To rule this out we need a lower bound on the curvature as well.

Definition 20.1.13 (Strong Convexity)

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is **α -strongly convex** if for all x, y ,

$$f(y) \geq f(x) + (y - x) \cdot \nabla f(x) + \frac{\alpha}{2} \|y - x\|^2.$$

For twice differentiable f this is equivalent to $\nabla^2 f(x) \succeq \alpha I$ for all x .

Lemma 20.1.14

Let f be α -strongly convex with minimum value p^* . Then for any x ,

$$p^* \geq f(x) - \frac{1}{2\alpha} \|\nabla f(x)\|^2.$$

Proof. Take the minimum over y of both sides of the strong convexity inequality. The left-hand side becomes p^* . The right-hand side is a quadratic in y , minimized where $\nabla f(x) + \alpha(y - x) = 0$, that is at $y = x - \frac{1}{\alpha} \nabla f(x)$, giving

$$p^* \geq f(x) - \frac{1}{\alpha} \|\nabla f(x)\|^2 + \frac{1}{2\alpha} \|\nabla f(x)\|^2 = f(x) - \frac{1}{2\alpha} \|\nabla f(x)\|^2. \quad \square$$

In other words, if $\|\nabla f(x)\| \leq \sqrt{2\alpha\varepsilon}$ then $f(x) \leq p^* + \varepsilon$: for a strongly convex function, a small gradient really does mean we are nearly done. Putting the two bounds together gives a genuine convergence guarantee.

Theorem 20.1.15 (Convergence of Gradient Descent)

Let f be β -smooth and α -strongly convex with $0 < \alpha \leq \beta$, and let x^* be its

minimizer. Then gradient descent with step size $1/\beta$ satisfies

$$f(x_T) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right)^T (f(x_0) - f(x^*)) \leq e^{-\alpha T/\beta} \cdot \frac{\beta}{2} \|x_0 - x^*\|^2.$$

Proof. Write $\delta_t = f(x_t) - f(x^*)$. By the corollary above and then the lemma above (applied at x_t , with $p^* = f(x^*)$),

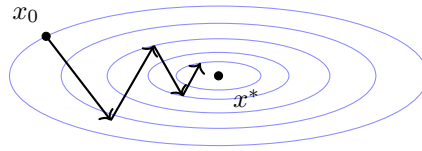
$$\delta_{t+1} \leq \delta_t - \frac{1}{2\beta} \|\nabla f(x_t)\|^2 \leq \delta_t - \frac{\alpha}{\beta} \delta_t = \left(1 - \frac{\alpha}{\beta}\right) \delta_t,$$

where the middle step used $\|\nabla f(x_t)\|^2 \geq 2\alpha\delta_t$. Induction gives the first inequality, and $1 - u \leq e^{-u}$ gives the second. Finally, β -smoothness applied at x^* (where $\nabla f(x^*) = 0$) gives $\delta_0 \leq \frac{\beta}{2} \|x_0 - x^*\|^2$. $\square$

The quantity $\kappa = \beta/\alpha \geq 1$ is called the **condition number** of f . To get $f(x_T) - f(x^*) \leq \varepsilon$ we need about $\kappa \log(1/\varepsilon)$ steps; each extra decimal place of accuracy costs a constant number of extra steps, which is called *linear convergence* and is very fast indeed. But the constant is κ , and if κ is huge then convergence is slow.

Example 20.1.16

Consider $f(x_1, x_2) = \frac{1}{2}(x_1^2 + 100x_2^2)$, whose Hessian is $\text{diag}(1, 100)$, so $\alpha = 1$, $\beta = 100$ and $\kappa = 100$. The level sets are extremely elongated ellipses, and the gradient at a typical point points almost straight at the long axis rather than at the origin. Gradient descent therefore zig-zags across the valley making very slow progress along it.



§20.1.4 Newton's Method and Barrier Methods

The problem in the example above is that gradient descent treats all coordinates the same way, whereas the function does not. The fix is to rescale by the Hessian. Recall that the second-order approximation to f near x_t is

$$f(x) \approx f(x_t) + (x - x_t) \cdot \nabla f(x_t) + \frac{1}{2}(x - x_t) \cdot (\nabla^2 f(x_t)(x - x_t)),$$

and minimizing the right-hand side over x gives $x = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t)$.

Definition 20.1.17 (Newton's Method)

Newton's method is the iteration

$$x_{t+1} = x_t - (\nabla^2 f(x_t))^{-1} \nabla f(x_t).$$

So instead of approximating f by a linear function and stepping a fixed distance, we approximate it by a paraboloid and jump straight to the bottom of that paraboloid.

Near the optimum this converges quadratically: one can show that

$$\|x_{t+1} - x^*\| \leq c \|x_t - x^*\|^2$$

once x_t is close enough to x^* , so the number of correct digits roughly doubles each step. The price is that we have to compute and invert an $n \times n$ matrix at every step.

Remark. In one dimension, Newton's method reads $x_{t+1} = x_t - f'(x_t)/f''(x_t)$, which is exactly the familiar Newton–Raphson iteration for finding a root of $g = f'$. Minimizing f and finding a zero of its derivative are of course the same problem.

Finally, a word about constraints. Suppose we want to minimize $f(x)$ subject to $f_i(x) \leq 0$ for $i = 1, \dots, m$. We would love to reduce this to an unconstrained problem, and the obvious way is to minimize

$$f(x) + \sum_{i=1}^m \phi(f_i(x)),$$

where $\phi(y) = 0$ for $y \leq 0$ and $\phi(y) = +\infty$ for $y > 0$. This is exactly the original problem, but ϕ is not remotely differentiable and so all of our machinery fails. Instead we use a smooth approximation.

Definition 20.1.18 (Logarithmic Barrier)

The **logarithmic barrier method** solves the unconstrained problem

$$\text{minimize } tf(x) - \sum_{i=1}^m \log(-f_i(x))$$

for a sequence of increasing values of $t > 0$, using the previous solution as the starting point for the next.

The function $-\log(-y)$ is $+\infty$ for $y \geq 0$ and finite for $y < 0$, rising steeply as $y \rightarrow 0^-$; it fences the iterates inside the feasible region. As $t \rightarrow \infty$ the objective f dominates the barrier, and the minimizer converges to the solution of the constrained problem. Each subproblem is solved by Newton's method. This is the basic idea behind *interior point methods*, which are what modern solvers actually use.

§20.2 Lagrangian Methods

Everything in the previous section was about *unconstrained* problems. Real problems come with constraints, and the central idea for handling them – one of the great ideas in the subject – is to not handle them at all. Instead we build the constraints into the objective function with a price attached, and then solve an unconstrained problem. If we get the prices right, the solution of the unconstrained problem automatically satisfies the constraints.

§20.2.1 The Lagrangian

Throughout this section we consider the problem

$$\begin{aligned} &\text{minimize} && f(x) \\ &\text{subject to} && h(x) = b, \end{aligned}$$

$$x \in X,$$

where $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$. We will refer to this as the **primal problem**, denoted (P) .

Definition 20.2.1 (Lagrangian)

The **Lagrangian** of the problem (P) is the function

$$L(x, \lambda) = f(x) - \lambda \cdot (h(x) - b),$$

where $\lambda \in \mathbb{R}^m$ is the vector of **Lagrange multipliers**.

The point is this. If x satisfies the functional constraint $h(x) = b$, then the second term vanishes and $L(x, \lambda) = f(x)$, whatever λ is. So on the feasible set the Lagrangian *is* the objective function. Off the feasible set, the term $-\lambda \cdot (h(x) - b)$ acts as a penalty (or a reward, depending on signs) for violating the constraints, and if we choose λ well then the penalty is enough to keep the unconstrained minimizer of L inside the feasible set. Making this precise gives the following theorem, which will be our main tool.

Theorem 20.2.2 (Lagrangian Sufficiency)

Suppose we can find $\lambda^* \in \mathbb{R}^m$ and $x^* \in X$ such that

- (i) x^* minimizes $L(\cdot, \lambda^*)$ over all of X , that is, $L(x^*, \lambda^*) = \min_{x \in X} L(x, \lambda^*)$; and
- (ii) x^* is feasible, that is, $x^* \in X(b)$.

Then x^* is optimal for (P) .

Proof. Since x^* is feasible we certainly have $f(x^*) \geq \min_{x \in X(b)} f(x)$, so we only need the reverse inequality. For this, note that if $x \in X(b)$ then $h(x) - b = 0$, so subtracting $\lambda^* \cdot (h(x) - b)$ changes nothing:

$$\begin{aligned} \min_{x \in X(b)} f(x) &= \min_{x \in X(b)} (f(x) - \lambda^* \cdot (h(x) - b)) \\ &\geq \min_{x \in X} (f(x) - \lambda^* \cdot (h(x) - b)) \\ &= \min_{x \in X} L(x, \lambda^*) \\ &= L(x^*, \lambda^*) \\ &= f(x^*) - \lambda^* \cdot (h(x^*) - b) = f(x^*). \end{aligned}$$

Here the inequality is because we are minimizing over a larger set $X \supseteq X(b)$, the next-to-last line is hypothesis (i), and the last line is hypothesis (ii). So $f(x^*) = \min_{x \in X(b)} f(x)$ as claimed. $\square$

This gives us a recipe. Rather than minimize over the awkward set $X(b)$, we minimize over the nice set X for each fixed λ , obtaining a candidate $x^*(\lambda)$, and then hunt for the value of λ making $x^*(\lambda)$ feasible. Notice that we have to be careful: for some λ the Lagrangian will not have a finite minimum at all. So we define

$$\Lambda = \left\{ \lambda \in \mathbb{R}^m : \inf_{x \in X} L(x, \lambda) > -\infty \right\},$$

and only consider $\lambda \in \Lambda$.

Example 20.2.3

Consider the problem

$$\begin{aligned} & \text{minimize} && -x_1 - x_2 + x_3 \\ & \text{subject to} && x_1^2 + x_2^2 = 4, \\ & && x_1 + x_2 + x_3 = 1, \end{aligned}$$

over $x \in \mathbb{R}^3$. So $X = \mathbb{R}^3$ and $b = (4, 1)$. The Lagrangian is

$$\begin{aligned} L(x, \lambda) &= -x_1 - x_2 + x_3 - \lambda_1(x_1^2 + x_2^2 - 4) - \lambda_2(x_1 + x_2 + x_3 - 1) \\ &= (-(1 + \lambda_2)x_1 - \lambda_1 x_1^2) + (-(1 + \lambda_2)x_2 - \lambda_1 x_2^2) + (1 - \lambda_2)x_3 + 4\lambda_1 + \lambda_2. \end{aligned}$$

We have arranged the terms so that each variable appears in exactly one bracket, and so we can minimize each bracket separately.

First, which λ lie in Λ ? The x_3 term is linear, so unless $\lambda_2 = 1$ we can send it to $-\infty$. The x_1 term is a quadratic with leading coefficient $-\lambda_1$, so we need $\lambda_1 \leq 0$; and if $\lambda_1 = 0$ then the remaining linear term $-(1 + \lambda_2)x_1 = -2x_1$ is unbounded below. So $\Lambda = \{\lambda : \lambda_1 < 0, \lambda_2 = 1\}$.

For such λ , minimizing each bracket by differentiating,

$$-(1 + \lambda_2) - 2\lambda_1 x_1 = 0 \implies x_1 = \frac{-2}{2\lambda_1} = \frac{-1}{\lambda_1},$$

and identically $x_2 = -1/\lambda_1$. Note x_3 is completely free (its coefficient is zero), which is exactly what we want – it leaves us room to satisfy the second constraint.

Now we impose feasibility. From $x_1^2 + x_2^2 = 4$ we get $2/\lambda_1^2 = 2$, so $\lambda_1 = -1/\sqrt{2}$ (taking the negative root as required) and $x_1 = x_2 = \sqrt{2}$. Then the second constraint gives $x_3 = 1 - 2\sqrt{2}$.

By Lagrangian sufficiency, $x^* = (\sqrt{2}, \sqrt{2}, 1 - 2\sqrt{2})$ is optimal, with optimal cost $-1 - 2\sqrt{2}$.

§20.2.2 Inequality Constraints and Complementary Slackness

Now suppose the constraints are inequalities:

$$\begin{aligned} & \text{minimize} && f(x) \\ & \text{subject to} && h(x) \leq b, \\ & && x \in X. \end{aligned}$$

We already know how to deal with this – add a slack variable $s \geq 0$ and write $h(x) + s = b$. The Lagrangian becomes

$$L(x, s, \lambda) = f(x) - \lambda \cdot (h(x) + s - b),$$

and we now minimize over $x \in X$ and $s \geq 0$. Writing the Lagrangian as

$$L(x, s, \lambda) = \underbrace{f(x) - \lambda \cdot (h(x) - b)}_{\text{does not involve } s} - \lambda \cdot s,$$

we can read off two things immediately.

Firstly, since s ranges over all of $[0, \infty)^m$, the term $-\lambda \cdot s$ is bounded below only if $\lambda \leq 0$. So $\lambda \in \Lambda$ forces $\lambda \leq 0$: with the sign convention above, *Lagrange multipliers for $\leq$ constraints are non-positive*.

Secondly, given $\lambda \leq 0$, the minimum over $s \geq 0$ of $-\lambda \cdot s$ is attained by taking $s_i = 0$ whenever $\lambda_i < 0$ (and s_i arbitrary when $\lambda_i = 0$). In either case the minimizing s satisfies $\lambda_i s_i = 0$ for every i . This is important enough to have a name.

Theorem 20.2.4 (Complementary Slackness)

At an optimum obtained by the Lagrangian method, for each i we have

$$\lambda_i s_i = 0, \quad \text{equivalently} \quad \lambda_i (h_i(x) - b_i) = 0.$$

So for each i , either the constraint is **tight** ($h_i(x) = b_i$) or the multiplier vanishes ($\lambda_i = 0$) – or both.

In words: a constraint that is not binding has zero multiplier. If we are not up against a constraint, then it is not costing us anything, so its price should be zero. We will see this interpretation made precise in a moment, and we will meet the same phenomenon again and again – in linear programming, in the transportation algorithm, and in game theory.

Complementary slackness is enormously useful in practice, because it lets us split into cases: for each constraint, either $s_i = 0$ or $\lambda_i = 0$, and often most of the cases can be eliminated quickly.

Example 20.2.5

Consider

$$\begin{aligned} &\text{minimize} && x_1 - 3x_2 \\ &\text{subject to} && x_1^2 + x_2^2 \leq 4, \\ &&& x_1 + x_2 \leq 2, \end{aligned}$$

over $x \in \mathbb{R}^2$. Adding slack variables $s_1, s_2 \geq 0$, the Lagrangian is

$$\begin{aligned} L &= (x_1 - 3x_2) - \lambda_1(x_1^2 + x_2^2 + s_1 - 4) - \lambda_2(x_1 + x_2 + s_2 - 2) \\ &= ((1 - \lambda_2)x_1 - \lambda_1 x_1^2) + ((-3 - \lambda_2)x_2 - \lambda_1 x_2^2) - \lambda_1 s_1 - \lambda_2 s_2 + 4\lambda_1 + 2\lambda_2. \end{aligned}$$

As discussed, $\lambda_1, \lambda_2 \leq 0$. Minimizing the two brackets,

$$(1 - \lambda_2) - 2\lambda_1 x_1 = 0, \quad (-3 - \lambda_2) - 2\lambda_1 x_2 = 0.$$

If $\lambda_1 = 0$ these read $\lambda_2 = 1$ and $\lambda_2 = -3$, which is a contradiction. So $\lambda_1 < 0$, and complementary slackness gives $s_1 = 0$, i.e. the first constraint is tight: $x_1^2 + x_2^2 = 4$.

Now suppose $\lambda_2 < 0$. Then $s_2 = 0$ too, so $x_1 + x_2 = 2$ as well. Combined with $x_1^2 + x_2^2 = 4$ this gives $(x_1, x_2) = (0, 2)$ or $(2, 0)$. Substituting into the two stationarity equations, the first case gives $(\lambda_1, \lambda_2) = (1, -3)$ and the second gives $(\lambda_1, \lambda_2) = (-1, 1)$; both have a positive multiplier, so both are impossible.

Hence $\lambda_2 = 0$, and the stationarity equations become $x_1 = 1/(2\lambda_1)$ and $x_2 = -3/(2\lambda_1)$. Substituting into $x_1^2 + x_2^2 = 4$,

$$\frac{1+9}{4\lambda_1^2} = 4 \implies \lambda_1^2 = \frac{5}{8} \implies \lambda_1 = -\sqrt{5/8},$$

giving

$$(x_1, x_2) = \left(-\sqrt{2/5}, 3\sqrt{2/5} \right).$$

Finally we check that this is feasible for the second constraint: $x_1 + x_2 = 2\sqrt{2/5} \approx 1.26 \leq 2$, so $s_2 = 2 - 2\sqrt{2/5} \geq 0$. By Lagrangian sufficiency this is optimal, with cost $-10\sqrt{2/5} = -2\sqrt{10}$.

Let us record the general procedure, since it is what we will do every time.

Remark (The Lagrangian Recipe). To solve ‘minimize $f(x)$ subject to $h(x) \leq b, x \in X$ ’:

- (1) Introduce slack variables to get equality constraints $h(x) + s = b, s \geq 0$.
- (2) Form the Lagrangian $L(x, s, \lambda) = f(x) - \lambda \cdot (h(x) + s - b)$.
- (3) Find Λ , the set of λ for which $\inf_{x \in X, s \geq 0} L$ is finite.
- (4) For each $\lambda \in \Lambda$, find the minimizers $x^*(\lambda), s^*(\lambda)$.
- (5) Find $\lambda^* \in \Lambda$ for which $x^*(\lambda^*), s^*(\lambda^*)$ is feasible. Then $x^*(\lambda^*)$ is optimal.

§20.2.3 Lagrangian Duality

The Lagrangian recipe raises an obvious worry: what if there is no λ^* that makes the minimizer feasible? Then the method fails, and we would like to understand when and why. The answer is the theory of duality, which is much more than a diagnostic tool – it is arguably the most important structure in the whole subject.

Definition 20.2.6 (Dual Function and Dual Problem)

The **dual function** associated with (P) is

$$g(\lambda) = \inf_{x \in X} L(x, \lambda),$$

defined for $\lambda \in \Lambda$. The **dual problem** (D) is

$$\begin{array}{ll} \text{maximize} & g(\lambda) \\ \text{subject to} & \lambda \in \Lambda. \end{array}$$

Note that g is an infimum of *affine* functions of λ (one for each x), and an infimum of affine functions is always concave. So the dual problem is always the maximization of a concave function over a convex set, i.e. always a convex problem, no matter how horrible the primal problem was. This is remarkable and is one reason duality is so useful.

Theorem 20.2.7 (Weak Duality)

If $x \in X(b)$ and $\lambda \in \Lambda$, then $f(x) \geq g(\lambda)$. In particular

$$\inf_{x \in X(b)} f(x) \geq \sup_{\lambda \in \Lambda} g(\lambda).$$

Proof. This is the first two lines of the proof of Lagrangian sufficiency. For $x \in X(b)$ we have $h(x) = b$, so

$$f(x) = f(x) - \lambda \cdot (h(x) - b) = L(x, \lambda) \geq \inf_{x' \in X} L(x', \lambda) = g(\lambda). \quad \square$$

So every feasible λ gives a *lower bound* on the optimal cost, and every feasible x gives an *upper bound*. This is already valuable: if we find x and λ with $f(x) = g(\lambda)$, we have proved that both are optimal without any further work. In general, though, there may be a gap.

Definition 20.2.8 (Duality Gap)

The **duality gap** is

$$\inf_{x \in X(b)} f(x) - \sup_{\lambda \in \Lambda} g(\lambda) \geq 0.$$

If the duality gap is zero we say that **strong duality** holds.

The relationship with the Lagrangian method is exactly what one would hope.

Proposition 20.2.9

The Lagrangian method succeeds – that is, there is a λ^* whose associated minimizer is feasible – if and only if strong duality holds.

Proof. If the method succeeds with multiplier λ^* and minimizer x^* , then by the proof of Lagrangian sufficiency, $g(\lambda^*) = L(x^*, \lambda^*) = f(x^*) = \min_{x \in X(b)} f(x)$, so the gap is zero.

Conversely if the gap is zero then there is λ^* (or a sequence approaching it) with $g(\lambda^*) = \inf_{x \in X(b)} f(x)$, and the chain of inequalities in the sufficiency proof collapses to equalities, so the minimizer of $L(\cdot, \lambda^*)$ is feasible and optimal. $\square$

So we should ask: when does strong duality hold? The answer is beautifully geometric.

§20.2.4 Supporting Hyperplanes

The key object is the following.

Definition 20.2.10 (Value Function)

The **value function** of the problem (P) is $\phi : \mathbb{R}^m \rightarrow \mathbb{R} \cup \{\pm\infty\}$ given by

$$\phi(c) = \inf_{x \in X(c)} f(x),$$

the optimal cost when the right-hand side of the functional constraint is changed from b to c .

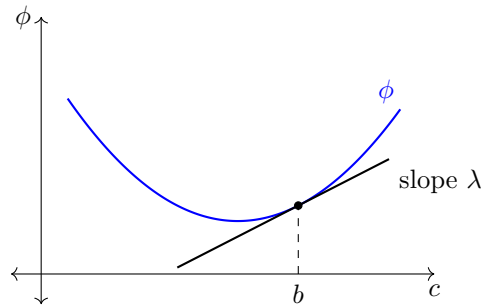
So $\phi(b)$ is the number we are trying to compute, and ϕ tells us how that number responds to perturbing the constraints. Now here is the definition that makes everything work.

Definition 20.2.11 (Supporting Hyperplane)

A function $\phi : \mathbb{R}^m \rightarrow \mathbb{R}$ has a **supporting hyperplane** at b if there is some $\lambda \in \mathbb{R}^m$ such that

$$\phi(c) \geq \phi(b) + \lambda \cdot (c - b) \quad \text{for all } c \in \mathbb{R}^m.$$

That is, there is a hyperplane through the point $(b, \phi(b))$ which lies entirely below the graph of ϕ . Compare this with the first-order condition for convexity, where the tangent plane at every point lies below the graph. Indeed one can show (we will not prove it) that ϕ is convex if and only if ϕ has a supporting hyperplane at every point.



Theorem 20.2.12 (Strong Duality)

Strong duality holds for (P) if and only if the value function ϕ has a supporting hyperplane at b .

Proof. Suppose first that ϕ has a supporting hyperplane at b , with normal given by λ . The trick is to compute $g(\lambda)$ by splitting the minimization over $x \in X$ into a minimization over the possible values $c = h(x)$, and then over the x achieving each value:

$$\begin{aligned} g(\lambda) &= \inf_{x \in X} (f(x) - \lambda \cdot (h(x) - b)) \\ &= \inf_{c \in \mathbb{R}^m} \inf_{x \in X(c)} (f(x) - \lambda \cdot (h(x) - c) - \lambda \cdot (c - b)) \\ &= \inf_{c \in \mathbb{R}^m} \left(\inf_{x \in X(c)} f(x) - \lambda \cdot (c - b) \right) \\ &= \inf_{c \in \mathbb{R}^m} (\phi(c) - \lambda \cdot (c - b)) \\ &\geq \phi(b), \end{aligned}$$

where in the third line we used that $h(x) = c$ for $x \in X(c)$, and the final inequality is precisely the supporting hyperplane property. But weak duality says $g(\lambda) \leq \phi(b)$, so $g(\lambda) = \phi(b)$ and the duality gap is zero.

Conversely, suppose strong duality holds, so there is λ with $g(\lambda) = \phi(b)$. Then for any c ,

$$\phi(b) = g(\lambda) = \inf_{x \in X} (f(x) - \lambda \cdot (h(x) - c) - \lambda \cdot (c - b))$$

$$\begin{aligned} &\leq \inf_{x \in X(c)} (f(x) - \lambda \cdot (h(x) - c)) - \lambda \cdot (c - b) \\ &= \phi(c) - \lambda \cdot (c - b), \end{aligned}$$

which rearranges to the supporting hyperplane condition. $\square$

This is a clean answer, but it is stated in terms of ϕ , which is the thing we do not know. Fortunately there is a checkable sufficient condition.

Theorem 20.2.13

Consider the problem ‘minimize $f(x)$ subject to $h(x) \leq b$, $x \in X$ ’, and suppose that X is convex, f is convex and h is convex (componentwise). Then the value function ϕ is convex, and hence strong duality holds.

Proof. Let $b_1, b_2 \in \mathbb{R}^m$ and $\delta \in [0, 1]$, and set $b_\delta = \delta b_1 + (1 - \delta)b_2$. Take any feasible $x_1 \in X(b_1)$ and $x_2 \in X(b_2)$, and let $x_\delta = \delta x_1 + (1 - \delta)x_2$, which lies in X by convexity of X . By convexity of h ,

$$h(x_\delta) \leq \delta h(x_1) + (1 - \delta)h(x_2) \leq \delta b_1 + (1 - \delta)b_2 = b_\delta,$$

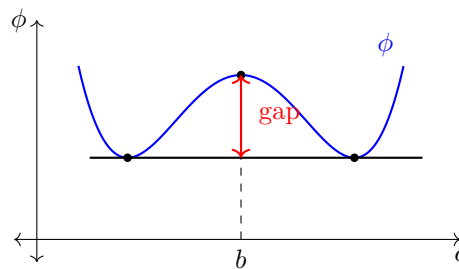
so $x_\delta \in X(b_\delta)$. Hence by convexity of f ,

$$\phi(b_\delta) \leq f(x_\delta) \leq \delta f(x_1) + (1 - \delta)f(x_2).$$

Taking the infimum over $x_1 \in X(b_1)$ and $x_2 \in X(b_2)$ gives $\phi(b_\delta) \leq \delta\phi(b_1) + (1 - \delta)\phi(b_2)$, so ϕ is convex. $\square$

Remark. It is worth stressing where convexity was used. We needed $h(x) \leq b$ rather than $h(x) = b$: with an equality constraint, $h(x_\delta) \leq b_\delta$ does not tell us $h(x_\delta) = b_\delta$, and the argument breaks. This is why in practice one always tries to write constraints as inequalities when hoping for strong duality.

Here, incidentally, is what failure looks like: if ϕ is not convex, there may be a ‘dent’ at b and no hyperplane through $(b, \phi(b))$ can stay below the graph. The best any hyperplane can do is to support the convex hull of the graph from below, and the vertical distance between $\phi(b)$ and that best hyperplane is exactly the duality gap.



§20.2.5 Shadow Prices

We finish this section with the interpretation of Lagrange multipliers that gives them their name in economics, and which explains the strange sign conventions.

Imagine a factory that makes n products out of m raw materials. If it makes quantities

$x = (x_1, \dots, x_n)$ then it makes profit $f(x)$ (we will minimize $-f$, but let us think in terms of maximizing profit), and producing x consumes $h_j(x)$ units of raw material j . The factory has b_j units of material j available. So the factory owner solves ‘maximize $f(x)$ subject to $h(x) \leq b$ ’.

Now a supplier turns up and offers a small extra quantity $\varepsilon = (\varepsilon_1, \dots, \varepsilon_m)$ of raw materials. How much should the owner be willing to pay? The new optimal profit is $\phi(b + \varepsilon)$, so the increase is

$$\phi(b + \varepsilon) - \phi(b) \approx \sum_{j=1}^m \frac{\partial \phi}{\partial b_j}(b) \varepsilon_j.$$

The quantity $\partial \phi / \partial b_j$ is the marginal value to this factory of one more unit of material j : it is called the **shadow price** of material j . It is a ‘shadow’ price because it is invisible from outside – it depends entirely on the internal state of the factory, not on the market.

Theorem 20.2.14

If ϕ is differentiable at b and has a supporting hyperplane at b with normal λ , then $\lambda = \nabla \phi(b)$.

Proof. Let $a \in \mathbb{R}^m$ be arbitrary and $\delta > 0$. The supporting hyperplane condition at $c = b + \delta a$ gives

$$\phi(b + \delta a) - \phi(b) \geq \delta \lambda \cdot a, \quad \text{so} \quad \frac{\phi(b + \delta a) - \phi(b)}{\delta} \geq \lambda \cdot a.$$

Letting $\delta \rightarrow 0$ and using differentiability, $\nabla \phi(b) \cdot a \geq \lambda \cdot a$. Since a was arbitrary we may also apply this with $-a$, giving $\nabla \phi(b) \cdot a = \lambda \cdot a$ for all a , and hence $\lambda = \nabla \phi(b)$. $\square$

Example 20.2.15

Let us verify this in a case where we can compute everything. Consider

$$\text{minimize } \frac{1}{2}(x_1^2 + x_2^2) \text{ subject to } x_1 + x_2 = b,$$

over $x \in \mathbb{R}^2$. The Lagrangian is

$$L(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) - \lambda(x_1 + x_2 - b) = \left(\frac{1}{2}x_1^2 - \lambda x_1\right) + \left(\frac{1}{2}x_2^2 - \lambda x_2\right) + \lambda b,$$

which is bounded below for every λ , so $\Lambda = \mathbb{R}$. Minimizing each bracket gives $x_1 = x_2 = \lambda$, and feasibility then forces $2\lambda = b$, so

$$\lambda^* = \frac{b}{2}, \quad x^* = \left(\frac{b}{2}, \frac{b}{2}\right), \quad \phi(b) = \frac{b^2}{4}.$$

And indeed $\phi'(b) = b/2 = \lambda^*$, exactly as the theorem promises. Concretely: relaxing the constraint from b to $b + \varepsilon$ increases the optimal cost by about $\lambda^* \varepsilon$.

So the Lagrange multipliers *are* the shadow prices. This makes complementary slackness completely intuitive: if a raw material is not fully used up ($s_j > 0$, the constraint is slack) then an extra unit of it is worth nothing, so its shadow price λ_j is zero. Conversely, if it

has a nonzero price then we must be using every last unit of it.

There is a matching interpretation of the dual problem. Think of it from the supplier's point of view. The supplier posts prices λ for the raw materials and offers to buy the finished products. Given prices λ , the factory owner will choose x to maximize

$$\underbrace{f(x)}_{\text{revenue}} - \underbrace{\lambda \cdot (h(x) - b)}_{\text{cost of materials}},$$

which is exactly the Lagrangian. The supplier, knowing this, chooses λ to make the factory owner's best achievable profit as small as possible – that is, the supplier solves the dual problem. Weak duality says the supplier can never squeeze the owner below the true optimum, and strong duality says that with the right prices the two coincide: at equilibrium there is no advantage to either party.

§20.3 Linear Programming

We now specialize to the case where everything in sight is linear. This may sound like a drastic restriction, and in a sense it is, but linear programs are astonishingly expressive – an enormous fraction of the optimization problems that actually arise in industry are linear programs, or are usefully approximated by them. More importantly for us, in this setting the general theory of the previous section becomes completely explicit and computable, and we will be able to write down an algorithm that solves any linear program by hand.

§20.3.1 Linear Programs

Definition 20.3.1 (Linear Program)

A **linear program** is an optimization problem in which the objective function and all of the constraints are linear (more precisely, affine) functions of the decision variables.

A general linear program might look something like

$$\begin{aligned} \text{minimize} \quad & 2x_1 - x_2 + 4x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_4 \leq 2, \\ & 3x_2 - x_3 = 5, \\ & x_3 + x_4 \geq 3, \\ & x_1 \geq 0, \ x_3 \leq 0, \end{aligned}$$

with x_2 and x_4 unconstrained in sign. This is a mess of different constraint types, and before doing anything we would like to tidy it up. Happily, all of the constraint types are interchangeable.

Definition 20.3.2 (Equivalent Problems)

Two optimization problems are **equivalent** if from a feasible solution to either one we can construct a feasible solution to the other with the same cost. Rewriting a problem as an equivalent one is called **reduction**.

The reductions we need are all elementary:

- $a \cdot x \leq b$ becomes $-a \cdot x \geq -b$;
- $a \cdot x \geq b$ becomes $a \cdot x - s = b$ with $s \geq 0$ (a *surplus* variable), and $a \cdot x \leq b$ becomes $a \cdot x + s = b$ with $s \geq 0$ (a *slack* variable);
- $a \cdot x = b$ becomes the pair $a \cdot x \geq b$ and $-a \cdot x \geq -b$;
- a variable $x_j \leq 0$ is replaced by $-x_j \geq 0$;
- a variable x_j of unrestricted sign is replaced by $x_j^+ - x_j^-$ with $x_j^+, x_j^- \geq 0$.

Applying these as needed, we can put any linear program into either of two standard shapes.

Definition 20.3.3 (General and Standard Form)

A linear program is in **general form** if it is written

$$\begin{array}{ll} \text{minimize} & c \cdot x \\ \text{subject to} & Ax \geq b, \end{array}$$

and in **standard form** if it is written

$$\begin{array}{ll} \text{minimize} & c \cdot x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array}$$

where in both cases $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$.

Standard form is what we will use for computation, since the constraint $Ax = b$ is just a linear system; general form is convenient for stating duality. Both are fully general: to get from general to standard form, first introduce surplus variables to turn $Ax \geq b$ into $Ax - s = b$ with $s \geq 0$, then split each unrestricted x_j as $x_j^+ - x_j^-$, giving

$$\begin{array}{ll} \text{minimize} & c \cdot x^+ - c \cdot x^- \\ \text{subject to} & Ax^+ - Ax^- - s = b, \\ & x^+, x^-, s \geq 0, \end{array}$$

which is in standard form in the variables $(x^+, x^-, s) \in \mathbb{R}_{\geq 0}^{2n+m}$.

Notice that reduction to standard form typically increases the number of variables. That is fine: what matters is that the constraint matrix stays sparse and the structure stays linear.

§20.3.2 A Two-Dimensional Example

Before doing any theory, it is worth solving a small linear program by drawing a picture, because the picture explains everything that follows.

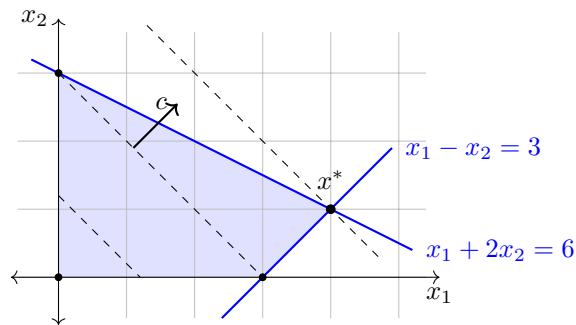
Example 20.3.4

Consider the problem

$$\text{maximize} \quad x_1 + x_2$$

$$\begin{aligned} \text{subject to } & x_1 + 2x_2 \leq 6, \\ & x_1 - x_2 \leq 3, \\ & x_1, x_2 \geq 0. \end{aligned}$$

The feasible set is the shaded polygon below: the intersection of four half-planes. The dashed lines are the level sets $x_1 + x_2 = \text{const}$ of the objective, which all have gradient $(1, 1)$. To maximize the objective, we slide the dashed line in the direction $(1, 1)$ as far as we can while still touching the feasible set.



The optimum is at $x^* = (4, 1)$, where the two constraint lines meet, with optimal value 5. Two features of this picture are completely general, and are the theoretical backbone of linear programming:

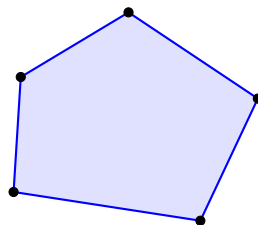
- (i) the feasible set is a *polyhedron* – an intersection of finitely many half-spaces; and
- (ii) the optimum is attained at a *corner* of that polyhedron.

If the objective happens to be parallel to one of the edges, then a whole edge is optimal – but even then, a corner is among the optimal points. So to solve a linear program, it suffices to search the (finitely many!) corners. Our next job is to say precisely what a corner is, algebraically.

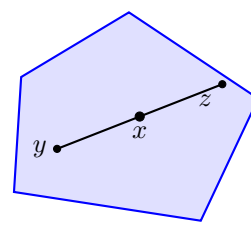
§20.3.3 Extreme Points

Definition 20.3.5 (Extreme Point)

A point x in a convex set C is an **extreme point** if it cannot be written as $x = \delta y + (1 - \delta)z$ for $y, z \in C$ with $y \neq z$ and $\delta \in (0, 1)$. That is, x does not lie strictly inside any line segment contained in C .



extreme points



x is not extreme

The relevance is the following general fact, which explains observation (ii) above.

Proposition 20.3.6

Let f be a convex function on a convex set C , and let $y, z \in C$. Then f attains its maximum over the segment $[y, z]$ at an endpoint. Consequently, if C is a bounded polyhedron then $\max_{x \in C} f(x)$ is attained at an extreme point of C .

Proof. For the first part, if $x = \delta y + (1 - \delta)z$ then by convexity

$$f(x) \leq \delta f(y) + (1 - \delta)f(z) \leq \max\{f(y), f(z)\}.$$

For the second, every point of a bounded polyhedron C can be written as a convex combination $x = \sum_{i=1}^k \delta_i x^{(i)}$ of extreme points $x^{(i)}$, with $\delta_i \geq 0$ and $\sum \delta_i = 1$. By convexity again,

$$f(x) \leq \sum_{i=1}^k \delta_i f(x^{(i)}) \leq \max_i f(x^{(i)}),$$

so no interior point beats the best extreme point. $\square$

Since a linear function is convex (and concave), the maximum *and* the minimum of a linear function on a bounded polyhedron are both attained at extreme points. So the corners really are all we need to look at. What we lack is a way of finding them without drawing a picture, which is hopeless in $\mathbb{R}^{100}$.

§20.3.4 Basic Solutions

Consider now a linear program in standard form,

$$\begin{aligned} &\text{minimize} && c \cdot x \\ &\text{subject to} && Ax = b, \\ &&& x \geq 0, \end{aligned}$$

with $A \in \mathbb{R}^{m \times n}$ and typically $n > m$, so that $Ax = b$ is an underdetermined system with a whole affine space of solutions. Intuitively, a corner of the feasible set is a point where as many constraints as possible are tight; since the only inequality constraints are $x_j \geq 0$, this means a point with as many zero coordinates as possible.

Definition 20.3.7 (Basic Solution)

A vector $x \in \mathbb{R}^n$ is a **basic solution** of $Ax = b$ if $Ax = b$ and x has at most m nonzero entries. If in addition $x \geq 0$, it is a **basic feasible solution** (bfs).

To construct basic solutions, we choose which m coordinates are allowed to be nonzero. We make two standing assumptions on A , both harmless.

- (A) The m rows of A are linearly independent. (Otherwise some constraint is redundant or inconsistent, and we can delete it.)
- (B) Every set of m columns of A is linearly independent. (This is a genericity assumption; we can always perturb b slightly to arrange it.)

Given a choice of indices $B = \{B(1), \dots, B(m)\} \subseteq \{1, \dots, n\}$, write A_j for the j th column of A and let

$$A_B = (A_{B(1)} \quad \cdots \quad A_{B(m)}) \in \mathbb{R}^{m \times m}$$

be the corresponding **basis matrix**. Setting $x_j = 0$ for $j \notin B$, the equation $Ax = b$ reduces to $A_B x_B = b$, where $x_B = (x_{B(1)}, \dots, x_{B(m)})$. By assumption (B), A_B is invertible, and so

$$x_B = A_B^{-1}b.$$

The indices in B are the **basic indices**, the corresponding variables are the **basic variables**, and the rest are the **non-basic variables** (all set to zero). If moreover $A_B^{-1}b \geq 0$, we have found a basic feasible solution.

(C) (**Non-degeneracy**) Every basic solution has *exactly* m nonzero entries.

This last assumption is not without loss of generality, and we will discuss what goes wrong without it in Section 20.4.3. Until then we assume it.

Now we can connect the algebra to the geometry.

Theorem 20.3.8 (Basic Feasible Solutions are Extreme Points)

Let $P = \{x : Ax = b, x \geq 0\}$. Then x is an extreme point of P if and only if x is a basic feasible solution.

Proof. Suppose first that x is a bfs, with basis B , and suppose $x = \delta y + (1 - \delta)z$ with $y, z \in P$ and $\delta \in (0, 1)$. For $j \notin B$ we have $x_j = 0$, so $\delta y_j + (1 - \delta)z_j = 0$; but $y_j, z_j \geq 0$ and $\delta \in (0, 1)$, so $y_j = z_j = 0$. Hence y and z are also supported on B , and $A_B y_B = b = A_B z_B$. As A_B is invertible, $y_B = z_B = A_B^{-1}b = x_B$, and so $y = z = x$. Thus x is extreme.

Conversely suppose $x \in P$ is not a bfs, so x has $r > m$ nonzero entries, say in positions $i_1, \dots, i_r$. The columns $A_{i_1}, \dots, A_{i_r}$ are $r > m$ vectors in $\mathbb{R}^m$, hence linearly dependent: there are weights $w_{i_1}, \dots, w_{i_r}$, not all zero, with

$$w_{i_1}A_{i_1} + \dots + w_{i_r}A_{i_r} = 0.$$

Extend w to a vector in $\mathbb{R}^n$ by setting $w_i = 0$ for $i \notin \{i_1, \dots, i_r\}$. Then $w \neq 0$ and $Aw = 0$, so $A(x \pm \varepsilon w) = b$ for every ε . Moreover w is supported where x is strictly positive, so for small enough $\varepsilon > 0$ we have $x \pm \varepsilon w \geq 0$, i.e. $x \pm \varepsilon w \in P$. But then x is the midpoint of these two distinct points of P , so x is not extreme. $\square$

Corollary 20.3.9

There are at most $\binom{n}{m}$ basic solutions, and hence at most $\binom{n}{m}$ extreme points of P .

Example 20.3.10

Let us see this explicitly on the example of Section 3.2. Writing it as a minimization and adding slack variables x_3, x_4 , the standard form is: minimize $-x_1 - x_2$ subject to

$$x_1 + 2x_2 + x_3 = 6, \quad x_1 - x_2 + x_4 = 3, \quad x \geq 0,$$

so $m = 2$, $n = 4$ and there are at most $\binom{4}{2} = 6$ basic solutions. Let us list them all.

basis	$x = (x_1, x_2, x_3, x_4)$	feasible?	cost
$\{3, 4\}$	$(0, 0, 6, 3)$	yes	0
$\{1, 3\}$	$(3, 0, 3, 0)$	yes	-3
$\{1, 4\}$	$(6, 0, 0, -3)$	no	—
$\{2, 3\}$	$(0, -3, 12, 0)$	no	—
$\{2, 4\}$	$(0, 3, 0, 6)$	yes	-3
$\{1, 2\}$	$(4, 1, 0, 0)$	yes	-5

For instance, for the basis $\{1, 2\}$ we set $x_3 = x_4 = 0$ and solve $x_1 + 2x_2 = 6$, $x_1 - x_2 = 3$, giving $(x_1, x_2) = (4, 1)$. Comparing with the picture in Section 3.2, the four basic feasible solutions are exactly the four corners $(0, 0)$, $(3, 0)$, $(0, 3)$ and $(4, 1)$ of the polygon, and the two infeasible basic solutions are the points $(6, 0)$ and $(0, -3)$ where two constraint lines meet *outside* the feasible region. The best cost among the feasible ones is -5 at $(4, 1)$, in agreement with what we found geometrically.

Theorem 20.3.11 (Fundamental Theorem of Linear Programming)

If a linear program in standard form is feasible and bounded, then there is an optimal solution which is a basic feasible solution.

Proof. Let x be an optimal solution with as few nonzero entries as possible, say r of them. If $r \leq m$ then x is a bfs and we are done, so suppose $r > m$. By the argument in the previous proof, there is $w \neq 0$ with $Aw = 0$, supported on the positions where x is nonzero.

Consider $x_\theta = x + \theta w$, which satisfies $Ax_\theta = b$ for all $\theta \in \mathbb{R}$, and lies in P for all θ in some interval around 0. Its cost is $c \cdot x + \theta(c \cdot w)$. If $c \cdot w \neq 0$ then choosing the sign of θ appropriately strictly decreases the cost, contradicting optimality of x . So $c \cdot w = 0$ and x_θ is optimal for every feasible θ .

Now increase $|\theta|$ (in either direction) until some coordinate of x_θ first hits zero – this happens for a finite θ since w is nonzero and supported where x is. The resulting point is optimal, feasible, and has strictly fewer than r nonzero entries, contradicting minimality of r . $\square$

This is genuinely good news: it reduces an optimization over a continuum to a search over a finite set. The bad news is that $\binom{n}{m}$ is astronomically large – for $n = 100$ and $m = 50$ it exceeds 10^{29} – so brute force is out of the question. The simplex method, which we come to shortly, is a clever way of walking from one basic feasible solution to a better one.

§20.3.5 Duality for Linear Programs

Before the algorithm, let us work out what the duality theory of Section 2 says here. It says something extremely concrete.

Consider a linear program in general form, with surplus variables added:

$$\begin{aligned} &\text{minimize} && c \cdot x \\ &\text{subject to} && Ax - s = b, \\ &&& s \geq 0, \end{aligned}$$

where x is unrestricted. The Lagrangian is

$$L(x, s, \lambda) = c \cdot x - \lambda \cdot (Ax - s - b) = (c - A^T \lambda) \cdot x + \lambda \cdot s + \lambda \cdot b.$$

Since x ranges over all of $\mathbb{R}^n$, this is bounded below only if $c - A^T \lambda = 0$; and since s ranges over $[0, \infty)^m$, we need $\lambda \geq 0$. Given both, the infimum is attained at $s = 0$ and equals $\lambda \cdot b$. Hence

$$\Lambda = \{\lambda : A^T \lambda = c, \lambda \geq 0\}, \quad g(\lambda) = \lambda \cdot b,$$

and we have proved the following.

Proposition 20.3.12 (Dual of a General-Form LP)

The dual of

$$\text{minimize } c \cdot x \quad \text{subject to } Ax \geq b$$

is

$$\text{maximize } \lambda \cdot b \quad \text{subject to } A^T \lambda = c, \lambda \geq 0.$$

Notice that the dual of a general-form problem is in standard form. The same computation for a standard-form primal gives the reverse.

Proposition 20.3.13 (Dual of a Standard-Form LP)

The dual of

$$\text{minimize } c \cdot x \quad \text{subject to } Ax = b, x \geq 0$$

is

$$\text{maximize } \lambda \cdot b \quad \text{subject to } A^T \lambda \leq c.$$

Proof. Here $X = \{x : x \geq 0\}$ and $L(x, \lambda) = c \cdot x - \lambda \cdot (Ax - b) = (c - A^T \lambda) \cdot x + \lambda \cdot b$. Since x ranges over the non-negative orthant, this is bounded below exactly when $c - A^T \lambda \geq 0$, and then the infimum is attained at $x = 0$ and equals $\lambda \cdot b$. $\square$

These two facts fit together neatly.

Proposition 20.3.14

The dual of the dual is the primal.

Proof. Start with the standard-form primal and its dual ‘maximize $\lambda \cdot b$ subject to $A^T \lambda \leq c$ ’. Rewrite this as a minimization in general form: putting $\mu = -\lambda$,

$$\text{minimize } \mu \cdot b \quad \text{subject to } A^T \mu \geq -c,$$

whose optimal value is the negative of the original. Applying the general-form dual

with data $(b, A^T, -c)$ in place of (c, A, b) , the dual is

$$\text{maximize } -\theta \cdot c \text{ subject to } A\theta = b, \theta \geq 0,$$

that is, minimize $c \cdot \theta$ subject to $A\theta = b, \theta \geq 0$, which is the original primal. $\square$

For a completely general linear program, the pattern of the dual is worth recording; each of these can be derived by the reductions above.

Proposition 20.3.15 (Dual of a General Linear Program)

The dual of

$$\begin{aligned} &\text{minimize} && c \cdot x \\ &\text{subject to} && a_i \cdot x \geq b_i \quad i \in M_1, \\ & && a_i \cdot x \leq b_i \quad i \in M_2, \\ & && a_i \cdot x = b_i \quad i \in M_3, \\ & && x_j \geq 0 \quad j \in N_1, \\ & && x_j \leq 0 \quad j \in N_2, \\ & && x_j \text{ free} \quad j \in N_3, \end{aligned}$$

is

$$\begin{aligned} &\text{maximize} && \lambda \cdot b \\ &\text{subject to} && \lambda_i \geq 0 \quad i \in M_1, \\ & && \lambda_i \leq 0 \quad i \in M_2, \\ & && \lambda_i \text{ free} \quad i \in M_3, \\ & && \lambda \cdot A_j \leq c_j \quad j \in N_1, \\ & && \lambda \cdot A_j \geq c_j \quad j \in N_2, \\ & && \lambda \cdot A_j = c_j \quad j \in N_3. \end{aligned}$$

The mnemonic is that constraints in the primal correspond to variables in the dual and vice versa. It is worth having the correspondence in a table:

primal (minimize)	dual (maximize)
i th constraint $\geq$	i th variable ≥ 0
i th constraint $\leq$	i th variable ≤ 0
i th constraint $=$	i th variable free
j th variable ≥ 0	j th constraint $\leq$
j th variable ≤ 0	j th constraint $\geq$
j th variable free	j th constraint $=$
objective vector c	right-hand side c
right-hand side b	objective vector b
constraint matrix A	constraint matrix A^T

There is an easy way to remember the direction of the inequalities: whichever choice makes weak duality come out right is the correct one. For instance in the standard-form case, a primal variable $x_j \geq 0$ contributes $c_j x_j$ to the objective, and we need $(c_j - \lambda \cdot A_j)x_j \geq 0$ for the bound $c \cdot x \geq \lambda \cdot b$ to work; with $x_j \geq 0$ this forces $\lambda \cdot A_j \leq c_j$, which is a $\leq$ constraint in the dual.

Since a linear program has a convex objective, convex constraints and convex feasible region, the theorem of Section 2.4 applies immediately.

Theorem 20.3.16 (Strong Duality for Linear Programs)

If a linear program is feasible and bounded, then so is its dual, and their optimal values are equal.

Proof. The value function $\phi(d) = \inf\{c \cdot x : Ax \geq d\}$ is convex by the theorem in Section 2.4 (taking $X = \mathbb{R}^n$, and noting that $f(x) = c \cdot x$ and $h(x) = -Ax$ are linear, hence convex). Hence ϕ has a supporting hyperplane at b , and strong duality holds. $\square$

§20.3.6 Complementary Slackness for Linear Programs

Strong duality gives us a certificate of optimality, and in the linear case that certificate takes an extremely usable form.

Theorem 20.3.17 (Optimality Conditions)

Let x be feasible for the primal and λ feasible for the dual, in the general setting of the previous proposition. Then x and λ are both optimal if and only if

- (i) $\lambda_i(a_i \cdot x - b_i) = 0$ for all i , and
- (ii) $(c_j - \lambda \cdot A_j)x_j = 0$ for all j .

Proof. Set $u_i = \lambda_i(a_i \cdot x - b_i)$ and $v_j = (c_j - \lambda \cdot A_j)x_j$. Feasibility of x and λ makes each $u_i \geq 0$ and each $v_j \geq 0$: for instance if $i \in M_1$ then $a_i \cdot x - b_i \geq 0$ and $\lambda_i \geq 0$, while if $i \in M_2$ both factors flip sign, and if $i \in M_3$ the second factor is zero. The same case analysis works for v_j . Now

$$\sum_i u_i = \lambda \cdot (Ax) - \lambda \cdot b, \quad \sum_j v_j = c \cdot x - \lambda \cdot (Ax),$$

so adding,

$$0 \leq \sum_i u_i + \sum_j v_j = c \cdot x - \lambda \cdot b.$$

If (i) and (ii) hold then all u_i, v_j vanish, so $c \cdot x = \lambda \cdot b$, and by weak duality both must be optimal. Conversely if both are optimal then strong duality gives $c \cdot x = \lambda \cdot b$, so $\sum u_i + \sum v_j = 0$; as all the terms are non-negative, they must all be zero. $\square$

Conditions (i) and (ii) are the **complementary slackness** conditions for a linear program. Condition (i) says: if the dual variable λ_i is nonzero then the i th primal constraint is tight. Condition (ii) says: if the primal variable x_j is nonzero then the j th dual constraint is tight. So the whole state of the optimum is captured by:

$$\text{primal feasibility} \quad + \quad \text{dual feasibility} \quad + \quad \text{complementary slackness}.$$

These three conditions are what the simplex method will maintain and enforce, and the algorithm can be understood entirely in terms of them: it keeps the first and third true at all times, and works to make the second true.

Example 20.3.18

Return to our two-dimensional example, written as a minimization in general form. We minimize $-x_1 - x_2$ subject to

$$\begin{pmatrix} -1 & -2 \\ -1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \geq \begin{pmatrix} -6 \\ -3 \\ 0 \\ 0 \end{pmatrix}.$$

The dual is: maximize $-6\lambda_1 - 3\lambda_2$ subject to $\lambda \geq 0$ and

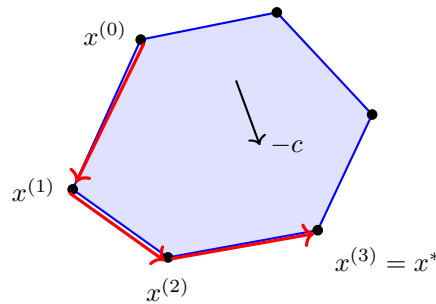
$$-\lambda_1 - \lambda_2 + \lambda_3 = -1, \quad -2\lambda_1 + \lambda_2 + \lambda_4 = -1.$$

At the primal optimum $x^* = (4, 1)$ the first two constraints are tight and the last two are slack, so complementary slackness forces $\lambda_3 = \lambda_4 = 0$. The remaining equations give $\lambda_1 + \lambda_2 = 1$ and $2\lambda_1 - \lambda_2 = 1$, so $\lambda = (2/3, 1/3, 0, 0)$, which is indeed non-negative. The dual objective is $-6 \cdot \frac{2}{3} - 3 \cdot \frac{1}{3} = -5$, matching the primal optimal value of -5 . So each certifies the other.

The shadow price interpretation is now concrete: $\lambda_1 = 2/3$ says that relaxing the first constraint from $x_1 + 2x_2 \leq 6$ to ≤ 7 would increase the maximum of $x_1 + x_2$ by $2/3$.

§20.4 The Simplex Method

We know that an optimal solution can be found among the basic feasible solutions, and we know that there are far too many of these to enumerate. The **simplex method**, due to Dantzig, resolves this by moving from one basic feasible solution to an adjacent one, always decreasing the cost, until no improving move exists. Geometrically, it walks along the edges of the feasible polyhedron from corner to corner, downhill all the way.

**§20.4.1 Reduced Costs**

To turn the picture into an algorithm we need to answer two questions: when is a basic feasible solution optimal, and if it is not, which neighbouring solution should we move to? Both answers come from duality.

Suppose we are at a basic feasible solution x with basis matrix $B = A_B$ (we will write B for both the index set and the matrix, as is traditional). We have primal feasibility for free. Let us try to construct a λ making complementary slackness hold, and then check whether it is dual feasible.

For a standard-form problem, the three optimality conditions read

- (primal feasibility) $Ax = b, x \geq 0$;
- (dual feasibility) $A^T \lambda \leq c$;
- (complementary slackness) $x \cdot (c - A^T \lambda) = 0$.

Since $x_j = 0$ for non-basic j , complementary slackness only constrains the basic coordinates, where $x_B > 0$; so it forces $c_B - B^T \lambda = 0$, that is

$$\lambda = (B^T)^{-1} c_B.$$

With this choice of λ , primal feasibility and complementary slackness both hold. Only dual feasibility $A^T \lambda \leq c$ remains, and this motivates the following definition.

Definition 20.4.1 (Reduced Costs)

Given a basis B , the vector of **reduced costs** is

$$\bar{c} = c - A^T (B^T)^{-1} c_B, \quad \text{componentwise} \quad \bar{c}_j = c_j - c_B \cdot (B^{-1} A_j).$$

By construction, dual feasibility is precisely the statement $\bar{c} \geq 0$, so we have proved most of the following.

Theorem 20.4.2 (Optimality Criterion)

Let x be a basic feasible solution with basis B and reduced cost vector $\bar{c}$. If $\bar{c} \geq 0$ then x is optimal.

Note that $\bar{c}_j = 0$ for every basic j , since $B^{-1} A_{B(i)} = e_i$ and $c_B \cdot e_i = c_{B(i)}$. So we only need to inspect the non-basic reduced costs.

The reduced costs have a direct meaning too, which tells us what to do when some of them are negative. Suppose j is non-basic and we try to increase x_j from 0 while keeping $Ax = b$. Formally, define the j th **basic direction** d by $d_j = 1$, $d_i = 0$ for all other non-basic i , and d_B determined by the requirement $Ad = 0$:

$$Bd_B + A_j = 0 \implies d_B = -B^{-1} A_j.$$

Then $A(x + \theta d) = b$ for all θ , and since $x_B > 0$ (non-degeneracy!) we have $x + \theta d \geq 0$ for all small enough $\theta > 0$. So d is a feasible direction. Its effect on the cost is

$$c \cdot (x + \theta d) = c \cdot x + \theta (c_j - c_B \cdot (B^{-1} A_j)) = c \cdot x + \theta \bar{c}_j.$$

So $\bar{c}_j$ is exactly the rate of change of the cost as we bring x_j into the solution. If $\bar{c}_j < 0$, moving in the j th basic direction reduces the cost, and we should do so as far as possible.

How far can we go? We need $x + \theta d \geq 0$. The non-basic coordinates are fine (x_j increases from 0, the others stay at 0). For the basic coordinates, writing $u = -d_B = B^{-1} A_j$, we need $x_{B(i)} - \theta u_i \geq 0$ for each i . If $u \leq 0$ this holds for all $\theta > 0$, and the cost is unbounded below. Otherwise the largest permissible step is

$$\theta^* = \min_{\{i: u_i > 0\}} \frac{x_{B(i)}}{u_i},$$

the so-called **ratio test**. Let ℓ attain the minimum. At $\theta = \theta^*$ the basic variable $x_{B(\ell)}$ becomes zero, so we have arrived at a new point with (again) m nonzero entries: x_j has entered the basis and $x_{B(\ell)}$ has left it.

Theorem 20.4.3

With the notation above, $y = x + \theta^* d$ is a basic feasible solution with basis obtained from B by replacing $A_{B(\ell)}$ with A_j , and $c \cdot y = c \cdot x + \theta^* \bar{c}_j < c \cdot x$.

Proof. We have $Ay = b$ and $y \geq 0$ by the choice of θ^* , and $y_{B(\ell)} = 0$ while $y_j = \theta^* > 0$, so exactly m entries are nonzero and they sit in the claimed positions. The cost formula was computed above, and $\theta^* > 0$, $\bar{c}_j < 0$ give the strict decrease. $\square$

Putting this together gives the algorithm.

Definition 20.4.4 (The Simplex Method)

To minimize $c \cdot x$ subject to $Ax = b$, $x \geq 0$:

- (1) Start at a basic feasible solution x with basis B .
- (2) Compute the reduced costs $\bar{c}$. If $\bar{c} \geq 0$, stop: x is optimal.
- (3) Otherwise choose j with $\bar{c}_j < 0$, and set $u = B^{-1}A_j$.
- (4) If $u \leq 0$, stop: the problem is unbounded.
- (5) Otherwise compute $\theta^* = \min\{x_{B(i)}/u_i : u_i > 0\}$, attained at $i = \ell$.
- (6) Replace $B(\ell)$ by j in the basis, update $x \mapsto x + \theta^* d$, and go to (2).

Since the cost strictly decreases at each step (under the non-degeneracy assumption) we never revisit a basis, and there are finitely many bases, so the algorithm terminates.

§20.4.2 The Simplex Tableau

Executing the algorithm as stated would require inverting B at every step, which is intolerable by hand. The **tableau** is a bookkeeping device that maintains all of the required quantities automatically using row operations.

Definition 20.4.5 (Simplex Tableau)

The **simplex tableau** for the basis B is the array

$-c_B \cdot x_B$	$\bar{c}^T$
$B^{-1}b$	$B^{-1}A$

or in more detail

$-c_B \cdot x_B$	$\bar{c}_1$	$\bar{c}_2$	$\cdots$	$\bar{c}_n$
$x_{B(1)}$	$\vdots$	$\vdots$		$\vdots$
$\vdots$	$B^{-1}A_1$	$B^{-1}A_2$	$\cdots$	$B^{-1}A_n$
$x_{B(m)}$	$\vdots$	$\vdots$		$\vdots$

So the top-left entry is minus the current cost, the top row holds the reduced costs, the

left column holds the current basic feasible solution, and the body holds $B^{-1}A$ – which is what the ratio test and basic directions need. Note that the basic columns of $B^{-1}A$ form an identity matrix, and the corresponding entries of the top row are zero; this is how we recognize the basis by looking at the tableau.

The remarkable fact – and it really is worth pausing over – is that the update in step (6) of the algorithm is achieved by pure Gaussian elimination on the tableau. The recipe is:

- (i) Choose a **pivot column** j with $\bar{c}_j < 0$.
- (ii) Choose the **pivot row** ℓ minimizing $x_{B(i)}/u_i$ among rows with $u_i > 0$, where u_i is the entry in row i of column j . The entry at (ℓ, j) is the **pivot**.
- (iii) Divide row ℓ by the pivot, then add multiples of row ℓ to every other row (including row 0) so that column j becomes the standard basis vector e_ℓ .

Why this works is not hard to see: multiplying on the left by $\bar{B}^{-1}B$ is exactly a sequence of row operations, and the row operations that turn column j into e_ℓ are precisely those which implement the change of basis.

Example 20.4.6

We solve

$$\begin{aligned} \text{minimize} \quad & -x_1 - x_2 - x_3 \\ \text{subject to} \quad & x_1 + 2x_2 + 2x_3 \leq 10, \\ & 2x_1 + x_2 + 2x_3 \leq 10, \\ & 2x_1 + 2x_2 + x_3 \leq 20, \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

Adding slack variables $x_4, x_5, x_6 \geq 0$ puts this in standard form with

$$A = \begin{pmatrix} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 10 \\ 10 \\ 20 \end{pmatrix}, \quad c = (-1, -1, -1, 0, 0, 0).$$

The slack variables give us a basic feasible solution for free: take $B = \{4, 5, 6\}$, so B is the identity matrix and $x = (0, 0, 0, 10, 10, 20)$. Since $c_B = 0$ we have $\bar{c} = c$ and cost 0, and the initial tableau is

0	-1	-1	-1	0	0	0
10	1	2	2	1	0	0
10	2	1	2	0	1	0
20	2	2	1	0	0	1

Since $\bar{c}_1 = -1 < 0$ we pivot in column 1. The ratios are $10/1 = 10$, $10/2 = 5$ and $20/2 = 10$, so the smallest is in row 2 and the pivot is the boxed 2. We perform

$$R_2 \mapsto \frac{1}{2}R_2, \quad R_0 \mapsto R_0 + \frac{1}{2}R_2^{\text{old}}, \quad R_1 \mapsto R_1 - \frac{1}{2}R_2^{\text{old}}, \quad R_3 \mapsto R_3 - R_2^{\text{old}},$$

obtaining

5	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$	0
5	0	$\frac{3}{2}$	1	1	$-\frac{1}{2}$	0
5	1	$\frac{1}{2}$	1	0	$\frac{1}{2}$	0
10	0	1	-1	0	-1	1

The basis is now $\{4, 1, 6\}$ and the current solution is $x = (5, 0, 0, 5, 0, 10)$ with cost -5 (remember the top-left entry is *minus* the cost).

Now $\bar{c}_2 = -\frac{1}{2} < 0$, so we pivot in column 2. The ratios are $5/\frac{3}{2} = \frac{10}{3}$, $5/\frac{1}{2} = 10$ and $10/1 = 10$, so we pivot on the boxed $\frac{3}{2}$ in row 1. We perform

$$R_1 \mapsto \frac{2}{3}R_1, \quad R_0 \mapsto R_0 + \frac{1}{3}R_1^{\text{old}}, \quad R_2 \mapsto R_2 - \frac{1}{3}R_1^{\text{old}}, \quad R_3 \mapsto R_3 - \frac{2}{3}R_1^{\text{old}},$$

giving

$\frac{20}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
$\frac{10}{3}$	0	1	$\frac{2}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	0
$\frac{10}{3}$	1	0	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{2}{3}$	0
$\frac{20}{3}$	0	0	$-\frac{5}{3}$	$-\frac{2}{3}$	$-\frac{2}{3}$	1

Every reduced cost is now non-negative, so we are optimal. Reading off the basis $\{2, 1, 6\}$ and the left-hand column, the optimal solution is

$$x^* = \left(\frac{10}{3}, \frac{10}{3}, 0, 0, 0, \frac{20}{3}\right)$$

with optimal cost $-\frac{20}{3}$.

We get the dual solution for free as well: the reduced costs of the slack variables are $\bar{c}_4 = \bar{c}_5 = \frac{1}{3}$, $\bar{c}_6 = 0$, and since $\bar{c}_{n+i} = -\lambda_i$ for slack variable i when the slack columns start as the identity, we read off $\lambda = (-\frac{1}{3}, -\frac{1}{3}, 0)$. The third constraint has zero multiplier, which is consistent with the fact that $x_6 = \frac{20}{3} > 0$ – that constraint is slack at the optimum.

Example 20.4.7 (An Unbounded Problem)

Consider the problem: minimize $-x_1 - x_2$ subject to $x_1 - x_2 \leq 1$ and $x_1, x_2 \geq 0$. Adding a slack variable, the standard form has

$$A = \begin{pmatrix} 1 & -1 & 1 \end{pmatrix}, \quad b = (1), \quad c = (-1, -1, 0),$$

and the basis $B = \{3\}$ gives the initial tableau

0	-1	-1	0
1	1	-1	1

Both $\bar{c}_1$ and $\bar{c}_2$ are negative. If we choose the pivot column $j = 2$, then $u = B^{-1}A_2 = (-1) \leq 0$, so no ratio test is possible and the problem is unbounded. Sure enough, $x = (t, t, 1)$ is feasible for every $t \geq 0$ and has cost $-2t \rightarrow -\infty$: the feasible region contains the ray in the direction $(1, 1)$, along which the objective decreases forever.

Note that had we chosen the pivot column $j = 1$ instead, we would have pivoted (with $\theta^* = 1$) to the basis $\{1\}$ and only discovered the unboundedness at the next step. Unboundedness is a property of the problem, not of the pivoting rule; the algorithm will always find it eventually.

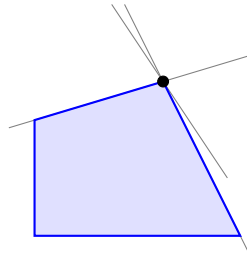
§20.4.3 Degeneracy and Bland's Rule

We have been assuming non-degeneracy throughout, and it is time to see what it was doing.

Definition 20.4.8 (Degeneracy)

A basic feasible solution is **degenerate** if it has fewer than m nonzero entries, so that some basic variable takes the value zero.

Geometrically, degeneracy happens when more than $n-m$ of the constraints pass through the same vertex, so that the vertex is described by more than one basis. In the tableau, degeneracy is visible as a zero in the left-hand column, or as a tie in the ratio test.



three constraints meeting at one vertex

If the ratio test is a tie, then θ^* may be zero: the pivot changes the basis but does *not* move the point. The cost then does not decrease, and it becomes possible in principle for the algorithm to return to a basis it has already visited, and loop forever. This is called **cycling**. It is rare in practice but it genuinely happens, and there are small explicit examples.

The fix is to break ties in a way that provably cannot cycle.

Definition 20.4.9 (Bland's Rule)

Bland's rule is to resolve both choices in the simplex method by smallest index:

- (i) among all j with $\bar{c}_j < 0$, choose the pivot column with the smallest index j ;
- (ii) among all rows attaining the minimum in the ratio test, choose the one whose *basic variable* $B(\ell)$ has the smallest index.

Theorem 20.4.10 (Bland)

The simplex method with Bland's rule terminates.

Proof sketch. Suppose the algorithm cycles. During a cycle the cost never changes, so every pivot has $\theta^* = 0$ and the point x is fixed throughout; let D be the set of

variables that enter and leave the basis during the cycle, and let q be the largest index in D . Consider the iteration at which q leaves the basis, say with entering variable $p \in D$ (so $p < q$), and the (later) iteration at which q enters. One shows by comparing the reduced costs at these two iterations that the choice of p at the first iteration and the choice of q at the second are inconsistent with the smallest-index rule: at the second iteration the variable p would still have had negative reduced cost, and $p < q$, so p would have been chosen instead of q . This contradiction shows no cycle can occur, and since there are finitely many bases the algorithm terminates. $\square$

An alternative and more common fix in practice is *lexicographic* pivoting, which amounts to perturbing b to $b + (\varepsilon, \varepsilon^2, \dots, \varepsilon^m)$ for a formal infinitesimal ε , thereby making every basic solution non-degenerate.

Remark (How fast is the simplex method?). We have shown that the simplex method terminates, but the bound we get is terrible: there are $\binom{n}{m}$ bases and in principle we might visit a large fraction of them. In fact Klee and Minty constructed, for each n , a linear program on a slightly squashed n -dimensional cube for which Dantzig's original pivoting rule visits all 2^n vertices. So the simplex method is not a polynomial-time algorithm, at least for that rule, and no pivoting rule is known to be polynomial.

Nevertheless the simplex method is superb in practice: on real problems it typically takes a number of pivots linear in m , and this behaviour can be made precise in various average-case and smoothed-analysis models. Meanwhile there *are* provably polynomial-time algorithms for linear programming – the ellipsoid method of Khachiyan and the interior point methods descended from Karmarkar's algorithm, which are essentially the barrier methods of Section 1.4 applied to linear constraints. Modern solvers implement both families and choose between them.

§20.4.4 The Two-Phase Simplex Method

There is one loose end: the simplex method has to start at a basic feasible solution, and in the example above we were handed one by the slack variables. That is a piece of luck peculiar to problems of the form $Ax \leq b$ with $b \geq 0$. In general – for instance with $\geq$ constraints, where the surplus variables would need to be negative – finding an initial bfs is itself a non-trivial problem.

The trick is that finding a feasible point is itself a linear program that we can solve by the simplex method, and *that* problem always has an obvious starting point. We introduce **artificial variables**.

Given the standard-form problem ‘minimize $c \cdot x$ subject to $Ax = b$, $x \geq 0$ ’, where without loss of generality $b \geq 0$ (multiply rows by -1 if needed), consider the auxiliary problem

$$\begin{aligned} \text{minimize} \quad & y_1 + \dots + y_m \\ \text{subject to} \quad & Ax + y = b, \\ & x, y \geq 0. \end{aligned}$$

This has the obvious basic feasible solution $x = 0$, $y = b$. Moreover its optimal value is 0 if and only if the original problem is feasible, and any solution with $y = 0$ gives a feasible x . So:

- **Phase I:** solve the auxiliary problem by the simplex method. If the optimal value is positive, the original problem is infeasible and we stop. Otherwise we finish with a basic feasible solution in which all the artificial variables are zero.
- **Phase II:** drop the artificial variables and the Phase I objective row, and run the simplex method on the original objective starting from the basis just found.

In practice we carry both objective rows through Phase I, so that the original objective row is correctly updated and ready for Phase II.

Example 20.4.11

Consider

$$\begin{aligned} &\text{minimize} && 6x_1 + 3x_2 \\ &\text{subject to} && x_1 + x_2 \geq 1, \\ &&& 2x_1 - x_2 \geq 1, \\ &&& 3x_2 \leq 2, \\ &&& x_1, x_2 \geq 0. \end{aligned}$$

Adding surplus variables z_1, z_2 and a slack z_3 ,

$$x_1 + x_2 - z_1 = 1, \quad 2x_1 - x_2 - z_2 = 1, \quad 3x_2 + z_3 = 2.$$

Setting $x = 0$ would force $z_1 = z_2 = -1$, which is infeasible, so we add artificial variables y_1, y_2 to the first two rows and minimize $y_1 + y_2$. The starting basis is $\{y_1, y_2, z_3\}$, giving the tableau (with the original objective in row 0 and the Phase I objective in row 0'):

	x_1	x_2	z_1	z_2	z_3	y_1	y_2
0	6	3	0	0	0	0	0
0	0	0	0	0	0	1	1
1	1	1	-1	0	0	1	0
1	2	-1	0	-1	0	0	1
2	0	3	0	0	1	0	0

This is not yet a valid tableau, because the reduced costs in the basic columns y_1, y_2 must be zero. Subtracting rows 1 and 2 from row 0' fixes this:

	x_1	x_2	z_1	z_2	z_3	y_1	y_2
0	6	3	0	0	0	0	0
-2	-3	0	1	1	0	0	0
1	1	1	-1	0	0	1	0
1	2	-1	0	-1	0	0	1
2	0	3	0	0	1	0	0

The Phase I cost is 2 (top-left of row 0' is -2). Pivoting on the boxed entry in

column x_1 (ratios $1/1 = 1$ and $1/2 = \frac{1}{2}$, the latter smaller):

	x_1	x_2	z_1	z_2	z_3	y_1	y_2
-3	0	6	0	3	0	0	-3
$-\frac{1}{2}$	0	$-\frac{3}{2}$	1	$-\frac{1}{2}$	0	0	$\frac{3}{2}$
$\frac{1}{2}$	0	$\frac{3}{2}$	-1	$\frac{1}{2}$	0	1	$-\frac{1}{2}$
$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$-\frac{1}{2}$	0	0	$\frac{1}{2}$
2	0	3	0	0	1	0	0

Now $\bar{c}_{x_2} = -\frac{3}{2} < 0$ in the Phase I row. The ratios are $\frac{1/2}{3/2} = \frac{1}{3}$ and $2/3$, so we pivot on the boxed $\frac{3}{2}$:

	x_1	x_2	z_1	z_2	z_3	y_1	y_2
-5	0	0	4	1	0	-4	-1
0	0	0	0	0	0	1	1
$\frac{1}{3}$	0	1	$-\frac{2}{3}$	$\frac{1}{3}$	0	$\frac{2}{3}$	$-\frac{1}{3}$
$\frac{2}{3}$	1	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	0	2	-1	1	-2	1

The Phase I objective is now 0 and the artificial variables have left the basis, so Phase I is complete. Deleting the artificial columns and the Phase I row leaves

	x_1	x_2	z_1	z_2	z_3
-5	0	0	4	1	0
$\frac{1}{3}$	0	1	$-\frac{2}{3}$	$\frac{1}{3}$	0
$\frac{2}{3}$	1	0	$-\frac{1}{3}$	$-\frac{1}{3}$	0
1	0	0	2	-1	1

All reduced costs in row 0 are non-negative, so we are already optimal – Phase II has nothing to do. The optimal solution is $x_1 = \frac{2}{3}$, $x_2 = \frac{1}{3}$, $z_3 = 1$, with optimal cost $6 \cdot \frac{2}{3} + 3 \cdot \frac{1}{3} = 5$, which is consistent with the top-left entry -5 .

Remark. It can happen that Phase I terminates with optimal value 0 but with an artificial variable still in the basis (at value zero). This is a degenerate situation; one can always pivot it out on any nonzero entry in a non-artificial column, or if the whole row is zero, delete the row as redundant.

§20.4.5 The Dual Simplex Method

The simplex method as described maintains primal feasibility and complementary slackness at every step, and works towards dual feasibility. There is a mirror-image algorithm, the **dual simplex method**, which maintains *dual* feasibility and complementary slackness and works towards primal feasibility. In tableau terms: we start with a tableau whose row 0 is non-negative but whose left-hand column has some negative entries, and we pivot to remove the negative entries while keeping row 0 non-negative.

Definition 20.4.12 (Dual Simplex Method)

Given a tableau with $\bar{c} \geq 0$:

- (1) If $B^{-1}b \geq 0$, stop: the solution is primal feasible, and hence optimal.
- (2) Otherwise choose a **pivot row** ℓ with $(B^{-1}b)_\ell < 0$.
- (3) If every entry of row ℓ (in the body) is non-negative, stop: the primal is infeasible.
- (4) Otherwise choose the **pivot column** j minimizing $|\bar{c}_j/a_{\ell j}|$ over all j with $a_{\ell j} < 0$.
- (5) Pivot on the entry $a_{\ell j}$ as usual, and go to (1).

The ratio test in step (4) is exactly what is needed to keep row 0 non-negative: after pivoting, $\bar{c}_k$ becomes $\bar{c}_k - (\bar{c}_j/a_{\ell j})a_{\ell k}$, and one checks that the minimizing choice keeps all of these non-negative. Notice that the roles of rows and columns, and of the two feasibility conditions, have swapped relative to the primal simplex method – which is precisely what one would expect from the fact that the dual of the dual is the primal.

The dual simplex method is extremely useful when we have already solved a problem and then change b (for instance by adding a new constraint). The old solution stays dual feasible but may become primal infeasible, so we can restart from it with the dual simplex method rather than solving from scratch.

Example 20.4.13

Consider

$$\begin{aligned} \text{minimize} \quad & 2x_1 + 3x_2 \\ \text{subject to} \quad & x_1 + x_2 \geq 2, \\ & x_1 + 3x_2 \geq 3, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Subtracting surplus variables and then multiplying both constraint rows by -1 to make the slack variables basic, we get $-x_1 - x_2 + z_1 = -2$ and $-x_1 - 3x_2 + z_2 = -3$ with $z_1, z_2 \geq 0$, and the basis $\{z_1, z_2\}$ gives the tableau

	x_1	x_2	z_1	z_2
0	2	3	0	0
-2	-1	-1	1	0
-3	-1	-3	0	1

Row 0 is non-negative (dual feasible) but the left column is not (primal infeasible), so we use the dual simplex method. The most negative entry is -3 in row 2, so we take $\ell = 2$. The ratios $|\bar{c}_j/a_{2j}|$ for the negative entries of that row are $|2/(-1)| = 2$

and $|3/(-3)| = 1$, so we pivot on the boxed -3 in column x_2 :

	x_1	x_2	z_1	z_2
-3	1	0	0	1
-1	$-\frac{2}{3}$	0	1	$-\frac{1}{3}$
1	$\frac{1}{3}$	1	0	$-\frac{1}{3}$

Still one negative entry in the left column, so $\ell = 1$. The negative entries in that row are $-\frac{2}{3}$ in column x_1 and $-\frac{1}{3}$ in column z_2 , with ratios $|1/(-\frac{2}{3})| = \frac{3}{2}$ and $|1/(-\frac{1}{3})| = 3$; so we pivot on the boxed $-\frac{2}{3}$:

	x_1	x_2	z_1	z_2
$-\frac{9}{2}$	0	0	$\frac{3}{2}$	$\frac{1}{2}$
$\frac{3}{2}$	1	0	$-\frac{3}{2}$	$\frac{1}{2}$
$\frac{1}{2}$	0	1	$\frac{1}{2}$	$-\frac{1}{2}$

Now both the left column and row 0 are non-negative, so we have an optimal solution $x^* = (\frac{3}{2}, \frac{1}{2})$ with optimal cost $\frac{9}{2}$. Note that we never needed a Phase I.

§20.5 Network Problems

A great many of the linear programs that arise in practice have a very particular shape: the variables are indexed by the edges of a graph, and the constraints say that whatever flows into a vertex must flow out of it again. These are the **network flow** problems. They are important both because they model an enormous range of real situations – shipping goods, routing traffic, scheduling, matching – and because their special structure lets us solve them far more efficiently than by general-purpose simplex.

§20.5.1 Flows in Networks

Definition 20.5.1 (Directed Graph)

A **directed graph** (or **digraph**) is a pair $G = (V, E)$ where V is a finite set of **vertices** and $E \subseteq V \times V$ is a set of **edges**. We think of $(i, j) \in E$ as an arrow from i to j .

Definition 20.5.2 (Flow)

Let $G = (V, E)$ be a digraph on n vertices. A **flow** on G is an assignment of a real number x_{ij} to each edge $(i, j) \in E$, thought of as the amount of some commodity travelling along that edge. The flow is subject to:

- a vector $b \in \mathbb{R}^n$, where b_i is the amount injected into the network at vertex i from outside. If $b_i > 0$ we call i a **source** and if $b_i < 0$ a **sink**;
- a **cost** c_{ij} per unit of flow along (i, j) ;
- capacities $\underline{m}_{ij} \leq x_{ij} \leq \overline{m}_{ij}$ on each edge.

Definition 20.5.3 (Minimum Cost Flow)

The **minimum cost flow problem** is the linear program

$$\begin{aligned} & \text{minimize} && \sum_{(i,j) \in E} c_{ij} x_{ij} \\ & \text{subject to} && b_i + \sum_{j: (j,i) \in E} x_{ji} = \sum_{j: (i,j) \in E} x_{ij} \quad \text{for all } i \in V, \\ & && \underline{m}_{ij} \leq x_{ij} \leq \overline{m}_{ij} \quad \text{for all } (i,j) \in E. \end{aligned}$$

The first family of constraints is **conservation of flow**: what goes in must come out.

Summing the conservation equations over all i , every x_{ij} appears once on each side, so we get $\sum_i b_i = 0$. This is a necessary condition for feasibility, and it is obvious: the network has no storage, so the total injected must equal the total removed.

The conservation constraints can be written compactly.

Definition 20.5.4 (Incidence Matrix)

The **incidence matrix** of G is the matrix $A \in \mathbb{R}^{|V| \times |E|}$ whose column indexed by the edge (i,j) has $+1$ in row i , -1 in row j , and zeros elsewhere.

With this notation the conservation constraints are exactly $Ax = b$, and the minimum cost flow problem is a linear program in the (almost) standard form we have been studying. What makes network problems special is that A is a very sparse matrix with entries in $\{0, \pm 1\}$, and this has strong consequences – notably, if b and the capacities are integers then there is an integral optimal solution.

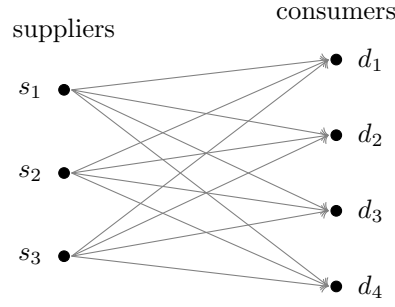
§20.5.2 The Transportation Problem

The most important special case is when the graph is bipartite.

Definition 20.5.5 (Transportation Problem)

Suppose we have n suppliers, where supplier i has s_i units available, and m consumers, where consumer j demands d_j units, with $\sum_i s_i = \sum_j d_j$. Shipping one unit from i to j costs c_{ij} . The **transportation problem** is

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^n \sum_{j=1}^m c_{ij} x_{ij} \\ & \text{subject to} && \sum_{j=1}^m x_{ij} = s_i \quad \text{for } i = 1, \dots, n, \\ & && \sum_{i=1}^n x_{ij} = d_j \quad \text{for } j = 1, \dots, m, \\ & && x \geq 0. \end{aligned}$$



This looks like a special case, but it is in fact completely general.

Theorem 20.5.6

Every minimum cost flow problem with finite capacities, or with non-negative costs, is equivalent to a transportation problem.

Proof. First we normalize. We may assume $\underline{m}_{ij} = 0$ for all edges: substituting $x_{ij} = \underline{m}_{ij} + \tilde{x}_{ij}$ turns the constraint into $0 \leq \tilde{x}_{ij} \leq \overline{m}_{ij} - \underline{m}_{ij}$ and the conservation equations into the same equations with

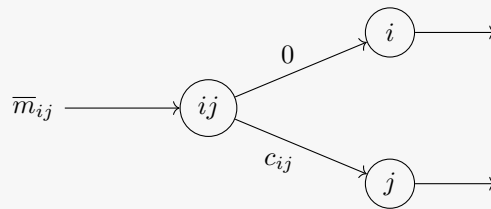
$$\tilde{b}_i = b_i + \sum_{j: (j,i) \in E} \underline{m}_{ji} - \sum_{j: (i,j) \in E} \underline{m}_{ij}.$$

(We are just secretly shipping the minimum amount along each edge before we start.) Next, if some capacity is infinite but all costs are non-negative, we may replace it by $\sum_i |b_i|$, since no optimal solution ever ships more than this along any edge (it would have to run round a cycle, and cycles cost nothing to remove when costs are non-negative). So assume all capacities are finite.

Now build a transportation problem. For each *vertex* i of the original graph, create a consumer with demand

$$\sum_{k: (i,k) \in E} \overline{m}_{ik} - b_i.$$

For each *edge* (i, j) of the original graph, create a supplier with supply $\overline{m}_{ij}$, joined to consumer i at cost 0 and to consumer j at cost c_{ij} .



Total supply is $\sum_{(i,j)} \overline{m}_{ij}$, and total demand is $\sum_i (\sum_k \overline{m}_{ik} - b_i) = \sum_{(i,j)} \overline{m}_{ij}$ since $\sum_i b_i = 0$; so the transportation problem is balanced.

Given a flow x in the original network, ship x_{ij} from supplier (i, j) to consumer j and the remaining $\overline{m}_{ij} - x_{ij}$ to consumer i . Then supplier (i, j) 's supply is exactly used up, and $0 \leq x_{ij} \leq \overline{m}_{ij}$ is exactly the requirement that both shipments are

non-negative. The total received by consumer i is

$$\sum_{k: (i,k) \in E} (\bar{m}_{ik} - x_{ik}) + \sum_{k: (k,i) \in E} x_{ki},$$

and setting this equal to the demand $\sum_k \bar{m}_{ik} - b_i$ rearranges precisely to the conservation equation at i . Finally, the cost of the transportation solution is $\sum_{(i,j)} c_{ij} x_{ij}$, since the edges into consumer i are free. So the two problems are equivalent. $\square$

§20.5.3 Optimality for the Transportation Problem

The transportation problem is a linear program, so we know exactly what optimality means: primal feasibility, dual feasibility and complementary slackness. Let us compute what these say. Since there are two families of constraints, we take two families of multipliers: $\lambda \in \mathbb{R}^n$ for the suppliers and $\mu \in \mathbb{R}^m$ for the consumers. The Lagrangian is

$$\begin{aligned} L(x, \lambda, \mu) &= \sum_{i,j} c_{ij} x_{ij} - \sum_i \lambda_i \left(\sum_j x_{ij} - s_i \right) - \sum_j \mu_j \left(\sum_i x_{ij} - d_j \right) \\ &= \sum_{i,j} (c_{ij} - \lambda_i - \mu_j) x_{ij} + \sum_i \lambda_i s_i + \sum_j \mu_j d_j. \end{aligned}$$

Since $x \geq 0$, this has a finite infimum exactly when $c_{ij} - \lambda_i - \mu_j \geq 0$ for all i, j .

Theorem 20.5.7 (Optimality Conditions)

A feasible x is optimal for the transportation problem if there exist $\lambda \in \mathbb{R}^n$, $\mu \in \mathbb{R}^m$ with

- (i) $\lambda_i + \mu_j \leq c_{ij}$ for all (i, j) (dual feasibility)
- (ii) $(c_{ij} - \lambda_i - \mu_j) x_{ij} = 0$ for all (i, j) (complementary slackness)

Proof. Immediate from the general optimality conditions for a linear program, applied to the Lagrangian above. $\square$

Condition (ii) says: wherever we actually ship something, the dual constraint is tight. Note that the multipliers are not unique – adding a constant k to every λ_i and subtracting it from every μ_j changes nothing. We therefore normalize by setting $\lambda_1 = 0$, which leaves $m + n - 1$ unknowns.

To get started we need an initial basic feasible solution, and here there is a very simple recipe: the **north-west corner rule**. Start at the top-left cell of the table, ship as much as possible (the smaller of the remaining supply and remaining demand), cross off whichever of the row or column is exhausted, and repeat.

Theorem 20.5.8

The set of cells with $x_{ij} > 0$ in a basic feasible solution of the transportation problem forms a spanning tree of the bipartite graph, and hence contains exactly $m + n - 1$ cells.

We will not prove this, but it is easy to believe: there are $m + n$ constraints of which

one is redundant (because $\sum s_i = \sum d_j$), so a basis has $m + n - 1$ elements; and a set of edges with no cycle and $m + n - 1$ elements on $m + n$ vertices is a spanning tree. The practical consequence is that the equations $\lambda_i + \mu_j = c_{ij}$ over the basic cells, together with $\lambda_1 = 0$, always have a unique solution – we can propagate along the tree.

§20.5.4 The Transportation Algorithm

We record the data in a **transportation tableau**, one cell per (i, j) , showing the current flow x_{ij} , the cost c_{ij} and the quantity $\lambda_i + \mu_j$, with the supplies down the right, the demands along the bottom, and λ, μ in the margins. The algorithm is then:

- (1) Find an initial basic feasible solution (e.g. by the north-west corner rule).
- (2) Solve $\lambda_i + \mu_j = c_{ij}$ over the basic cells, with $\lambda_1 = 0$.
- (3) If $\lambda_i + \mu_j \leq c_{ij}$ for every cell, stop – the solution is optimal.
- (4) Otherwise pick a cell (i, j) with $\lambda_i + \mu_j > c_{ij}$. Adding (i, j) to the spanning tree creates a unique cycle. Send θ units around this cycle (alternately $+\theta$ and $-\theta$), increasing θ until some basic cell is driven to zero. That cell leaves the basis and (i, j) enters.
- (5) Go to (2).

This is exactly the simplex method in disguise: step (3) is the check $\bar{c} \geq 0$, and step (4) is the ratio test.

Example 20.5.9

Take three suppliers with $s = (14, 10, 9)$ and four consumers with $d = (12, 5, 8, 8)$, and costs

$$C = \begin{pmatrix} 5 & 3 & 4 & 6 \\ 2 & 7 & 4 & 1 \\ 5 & 6 & 2 & 4 \end{pmatrix}.$$

The north-west corner rule gives

$$x = \begin{pmatrix} 12 & 2 & 0 & 0 \\ 0 & 3 & 7 & 0 \\ 0 & 0 & 1 & 8 \end{pmatrix},$$

with basic cells $(1, 1), (1, 2), (2, 2), (2, 3), (3, 3), (3, 4)$ – six of them, as promised. Complementary slackness gives

$$\lambda_1 + \mu_1 = 5, \quad \lambda_1 + \mu_2 = 3, \quad \lambda_2 + \mu_2 = 7, \quad \lambda_2 + \mu_3 = 4, \quad \lambda_3 + \mu_3 = 2, \quad \lambda_3 + \mu_4 = 4,$$

which with $\lambda_1 = 0$ solves to

$$\lambda = (0, 4, 2), \quad \mu = (5, 3, 0, 2).$$

It is convenient to record all of this in a single tableau, with the multipliers in the margins and each cell showing the current flow and the cost, written $x_{ij} | c_{ij}$:

	$\mu_1 = 5$	$\mu_2 = 3$	$\mu_3 = 0$	$\mu_4 = 2$	s_i
$\lambda_1 = 0$	12 5	2 3	0 4	0 6	14
$\lambda_2 = 4$	0 2	3 7	7 4	0 1	10
$\lambda_3 = 2$	0 5	0 6	1 2	8 4	9
d_j	12	5	8	8	

One checks at a glance that the rows sum to the supplies and the columns to the demands. Now we tabulate $\lambda_i + \mu_j$ against c_{ij} :

$\lambda_i + \mu_j$	$\mu_1 = 5$	$\mu_2 = 3$	$\mu_3 = 0$	$\mu_4 = 2$
$\lambda_1 = 0$	5 (5)	3 (3)	0 (4)	2 (6)
$\lambda_2 = 4$	9 (2)	7 (7)	4 (4)	6 (1)
$\lambda_3 = 2$	7 (5)	5 (6)	2 (2)	4 (4)

(costs c_{ij} in brackets). Cell (2, 1) violates the condition badly: $9 > 2$. Cell (2, 4) also violates it, as does (3, 1); we pick (2, 1).

Adding (2, 1) to the tree creates the cycle $(2, 1) \rightarrow (1, 1) \rightarrow (1, 2) \rightarrow (2, 2) \rightarrow (2, 1)$. Putting θ into (2, 1) forces $x_{11} \mapsto 12 - \theta$, $x_{12} \mapsto 2 + \theta$ and $x_{22} \mapsto 3 - \theta$. The binding constraint is $x_{22} \geq 0$, so $\theta = 3$ and cell (2, 2) leaves. The new solution is

$$x = \begin{pmatrix} 9 & 5 & 0 & 0 \\ 3 & 0 & 7 & 0 \\ 0 & 0 & 1 & 8 \end{pmatrix},$$

with basic cells (1, 1), (1, 2), (2, 1), (2, 3), (3, 3), (3, 4). Re-solving for the multipliers,

$$\lambda_1 + \mu_1 = 5, \lambda_1 + \mu_2 = 3, \lambda_2 + \mu_1 = 2, \lambda_2 + \mu_3 = 4, \lambda_3 + \mu_3 = 2, \lambda_3 + \mu_4 = 4,$$

giving $\lambda = (0, -3, -5)$ and $\mu = (5, 3, 7, 9)$. Checking again:

$\lambda_i + \mu_j$	$\mu_1 = 5$	$\mu_2 = 3$	$\mu_3 = 7$	$\mu_4 = 9$
$\lambda_1 = 0$	5 (5)	3 (3)	7 (4)	9 (6)
$\lambda_2 = -3$	2 (2)	0 (7)	4 (4)	6 (1)
$\lambda_3 = -5$	0 (5)	-2 (6)	2 (2)	4 (4)

Now (2, 4) violates: $6 > 1$. (So does (1, 3) and (1, 4).) Taking (2, 4), the cycle is $(2, 4) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (3, 4) \rightarrow (2, 4)$, so $x_{23} \mapsto 7 - \theta$, $x_{33} \mapsto 1 + \theta$, $x_{34} \mapsto 8 - \theta$. The binding constraint is $x_{23} \geq 0$ at $\theta = 7$, giving

$$x = \begin{pmatrix} 9 & 5 & 0 & 0 \\ 3 & 0 & 0 & 7 \\ 0 & 0 & 8 & 1 \end{pmatrix}.$$

The new multipliers solve to $\lambda = (0, -3, 0)$ and $\mu = (5, 3, 2, 4)$, and now

$\lambda_i + \mu_j$	$\mu_1 = 5$	$\mu_2 = 3$	$\mu_3 = 2$	$\mu_4 = 4$
$\lambda_1 = 0$	5 (5)	3 (3)	2 (4)	4 (6)
$\lambda_2 = -3$	2 (2)	0 (7)	-1 (4)	1 (1)
$\lambda_3 = 0$	5 (5)	3 (6)	2 (2)	4 (4)

Every entry satisfies $\lambda_i + \mu_j \leq c_{ij}$, so we are optimal. The cost is

$$9 \cdot 5 + 5 \cdot 3 + 3 \cdot 2 + 7 \cdot 1 + 8 \cdot 2 + 1 \cdot 4 = 45 + 15 + 6 + 7 + 16 + 4 = 93.$$

Remark (Transshipment). A variant that often occurs is the **transshipment problem**, in which goods may pass through intermediate depots rather than going directly from supplier to consumer. Formally this is just a minimum cost flow problem on a graph with some vertices having $b_i = 0$, and by the reduction theorem above it can be converted into a transportation problem. In practice a slicker route is available: replace each cost c_{ij} by the cost $\hat{c}_{ij}$ of the *cheapest path* from i to j in the network (computable by any shortest-path algorithm), and then solve the resulting transportation problem with costs $\hat{c}$. Any optimal solution of that problem is realized by routing each shipment along its cheapest path.

§20.5.5 The Assignment Problem

A further special case, and one of the classics, is the following.

Definition 20.5.10 (Assignment Problem)

Given n workers and n jobs, and a cost c_{ij} of assigning worker i to job j , the **assignment problem** is to find a bijection σ from workers to jobs minimizing $\sum_i c_{i\sigma(i)}$.

Written as an optimization over $\{0, 1\}$ -matrices this is a combinatorial problem with $n!$ candidates, which sounds hopeless. But it is exactly a transportation problem with $s_i = d_j = 1$ for all i, j :

$$\begin{aligned} &\text{minimize} && \sum_{i,j} c_{ij} x_{ij} \\ &\text{subject to} && \sum_j x_{ij} = 1 \text{ for each } i, \quad \sum_i x_{ij} = 1 \text{ for each } j, \quad x \geq 0. \end{aligned}$$

The feasible set of this linear program consists of the *doubly stochastic* matrices, and the corresponding combinatorial problem is recovered by the following classical fact, which we quote.

Theorem 20.5.11 (Birkhoff–von Neumann)

The extreme points of the set of doubly stochastic $n \times n$ matrices are exactly the permutation matrices.

Combining this with the fundamental theorem of linear programming, some optimal solution of the linear program is an extreme point, hence a permutation matrix, hence

an honest assignment. So we may simply solve the transportation problem and read off the answer – an $n!$ -sized search has been replaced by a polynomial-time algorithm. This phenomenon, where the linear relaxation of an integer problem automatically has integral optima, is exactly what makes network problems so tractable, and it recurs in the maximum flow problem below.

§20.5.6 The Maximum Flow Problem

We now turn to a different, and in some ways more fundamental, network problem. There are no costs; instead every edge has a capacity, there is a single source (vertex 1) and a single sink (vertex n), and we want to push as much as possible from source to sink.

Definition 20.5.12 (Maximum Flow Problem)

Given a digraph $G = (V, E)$ on vertices $\{1, \dots, n\}$ with capacities $c_{ij} \geq 0$, the **maximum flow problem** is

$$\begin{aligned} & \text{maximize} && \delta \\ & \text{subject to} && \sum_{j: (i,j) \in E} x_{ij} - \sum_{j: (j,i) \in E} x_{ji} = \begin{cases} \delta & i = 1 \\ -\delta & i = n \\ 0 & \text{otherwise,} \end{cases} \\ & && 0 \leq x_{ij} \leq c_{ij} \quad \text{for all } (i,j) \in E. \end{aligned}$$

We call δ the **value** of the flow.

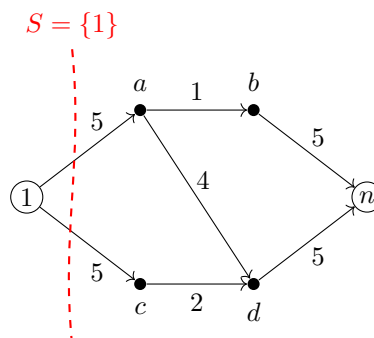
This is a minimum cost flow problem in disguise: add an edge from n back to 1 with cost -1 and infinite capacity, and set all $b_i = 0$; then minimizing the cost is the same as maximizing the flow round the added edge, which is δ . But there is a much better way to think about it.

Definition 20.5.13 (Cut)

A **cut** of G is a partition of V into two sets $(S, V \setminus S)$. Its **capacity** is

$$C(S) = \sum_{\substack{(i,j) \in E \\ i \in S, j \in V \setminus S}} c_{ij},$$

the total capacity of the edges crossing from S to $V \setminus S$.



The following inequality is the easy half of the story: no flow can beat any cut.

Proposition 20.5.14

Let x be a feasible flow with value δ , and let $(S, V \setminus S)$ be any cut with $1 \in S$ and $n \in V \setminus S$. Then $\delta \leq C(S)$.

Proof. For $X, Y \subseteq V$ write $f_x(X, Y) = \sum_{(i,j) \in E, i \in X, j \in Y} x_{ij}$ for the total flow from X to Y . Summing the conservation equations over $i \in S$, all terms vanish except the one at $i = 1$, so

$$\delta = \sum_{i \in S} \left(\sum_{j: (i,j) \in E} x_{ij} - \sum_{j: (j,i) \in E} x_{ji} \right) = f_x(S, V) - f_x(V, S).$$

Splitting V as $S \cup (V \setminus S)$ in both terms, the $f_x(S, S)$ contributions cancel, leaving

$$\delta = f_x(S, V \setminus S) - f_x(V \setminus S, S) \leq f_x(S, V \setminus S) \leq C(S),$$

using $f_x(V \setminus S, S) \geq 0$ and $x_{ij} \leq c_{ij}$. $\square$

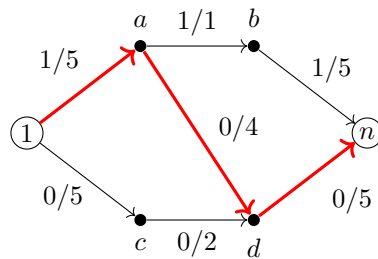
The remarkable fact is that this bound is always attained: some cut is exactly as good as the best flow. This is a theorem of Ford and Fulkerson, and it is a beautiful example of strong duality in a combinatorial setting – the maximum flow problem and the minimum cut problem are dual to each other.

Definition 20.5.15 (Augmenting Path)

Let x be a feasible flow. A path $v_0, v_1, \dots, v_k$ in the *undirected* sense – so consecutive vertices are joined by an edge in one direction or the other – is an **augmenting path** if

- $x_{v_i v_{i+1}} < c_{v_i v_{i+1}}$ for every edge traversed forwards, and
- $x_{v_{i+1} v_i} > 0$ for every edge traversed backwards.

The point is that along such a path we can increase the flow by some $\varepsilon > 0$ on the forward edges and decrease it by ε on the backward edges, and conservation still holds at every intermediate vertex. So if there is an augmenting path from 1 to n , the flow value can be increased and x was not optimal.



augmenting path 1, a, d, n

Theorem 20.5.16 (Max-Flow Min-Cut)

The maximum value of a feasible flow equals the minimum capacity of a cut separating 1 from n :

$$\delta^* = \min\{C(S) : S \subseteq V, 1 \in S, n \in V \setminus S\}.$$

Proof. By the proposition, $\delta^* \leq C(S)$ for every such cut, so it is enough to exhibit a cut with $C(S) = \delta^*$.

Let x be an optimal flow, and define

$$S = \{1\} \cup \{i \in V : \text{there is an augmenting path from 1 to } i\}.$$

Since x is optimal there is no augmenting path from 1 to n , so $n \notin S$ and $(S, V \setminus S)$ is a cut separating 1 from n . From the proof of the proposition,

$$\delta^* = f_x(S, V \setminus S) - f_x(V \setminus S, S),$$

and we claim both terms take their extreme values.

First, $f_x(V \setminus S, S) = 0$. Indeed, suppose $(v, u) \in E$ with $v \in V \setminus S$, $u \in S$ and $x_{vu} > 0$. Then we may extend an augmenting path from 1 to u by traversing (v, u) backwards, obtaining an augmenting path from 1 to v ; so $v \in S$, a contradiction.

Second, $f_x(S, V \setminus S) = C(S)$. Indeed, suppose $(u, v) \in E$ with $u \in S$, $v \in V \setminus S$ and $x_{uv} < c_{uv}$. Then we may extend an augmenting path from 1 to u by traversing (u, v) forwards, so $v \in S$, again a contradiction. Hence every edge from S to $V \setminus S$ is saturated.

Therefore $\delta^* = C(S) - 0 = C(S)$, as required. $\square$

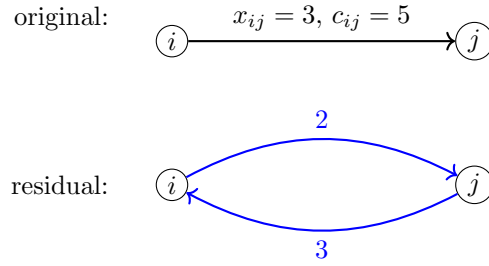
Notice that the proof is constructive: it tells us that if we cannot find an augmenting path, we are done, and it even hands us the certifying cut. This is the algorithm.

Definition 20.5.17 (Ford–Fulkerson Algorithm)

To find a maximum flow:

- (1) Start from any feasible flow, e.g. $x = 0$.
- (2) Search for an augmenting path from 1 to n . If there is none, stop – x is optimal.
- (3) Increase the flow along the path by as much as possible (the smallest of $c_{ij} - x_{ij}$ over forward edges and x_{ji} over backward edges), and go to (2).

The bookkeeping is easiest with the **residual network**: for each edge (i, j) we record a forward residual capacity $c_{ij} - x_{ij}$ and a backward residual capacity x_{ij} . An augmenting path is then just an ordinary directed path in the residual network, and the amount we can push is the smallest residual capacity along it.

**Example 20.5.18**

Consider the network drawn above, with capacities

$$c_{1a} = 5, \quad c_{ab} = 1, \quad c_{bn} = 5, \quad c_{1c} = 5, \quad c_{cd} = 2, \quad c_{dn} = 5, \quad c_{ad} = 4,$$

and start with $x = 0$.

- (i) The path $1, a, b, n$ is augmenting; the bottleneck is $c_{ab} = 1$, so we send 1 unit. Now $x_{1a} = x_{ab} = x_{bn} = 1$ and $\delta = 1$.
- (ii) The path $1, a, d, n$ is augmenting; the bottleneck is $c_{1a} - x_{1a} = 4$, so we send 4 units. Now $x_{1a} = 5$, $x_{ad} = 4$, $x_{dn} = 4$ and $\delta = 5$.
- (iii) The path $1, c, d, n$ is augmenting; the bottleneck is $c_{dn} - x_{dn} = 1$, so we send 1 unit. Now $x_{1c} = x_{cd} = 1$, $x_{dn} = 5$ and $\delta = 6$.

The final flow is $x_{1a} = 5$, $x_{ab} = 1$, $x_{bn} = 1$, $x_{ad} = 4$, $x_{1c} = 1$, $x_{cd} = 1$, $x_{dn} = 5$.

To see that this is optimal we compute the set S of vertices reachable from 1 by an augmenting path. The edge $1a$ is saturated, so a is not reachable that way; but $x_{1c} = 1 < 5$, so $c \in S$. From c , $x_{cd} = 1 < 2$, so $d \in S$. From d , the edge dn is saturated, but the edge ad carries flow $4 > 0$, so traversing it *backwards* puts $a \in S$. From a , the edge ab is saturated, so we get nothing new. Hence

$$S = \{1, a, c, d\},$$

and $n \notin S$, so this is a genuine cut. Its capacity is the total capacity of edges leaving S , namely ab and dn :

$$C(S) = c_{ab} + c_{dn} = 1 + 5 = 6 = \delta^*,$$

confirming optimality.

Remark. Finding the cut requires a little care: S consists of everything reachable by an augmenting path, and augmenting paths may traverse edges backwards, so one must chase the reachability to the end. When Ford–Fulkerson stops, the set of reachable vertices is automatically a minimum cut – that is exactly what the proof of the max-flow min-cut theorem showed.

Example 20.5.19

Here is a larger example in which a backward edge is genuinely needed. Take vertices $1, a, b, c, d, n$ with capacities

$$c_{1a} = 10, \quad c_{1c} = 10, \quad c_{ab} = 4, \quad c_{ac} = 2, \quad c_{ad} = 8, \quad c_{bn} = 10, \quad c_{cd} = 9, \quad c_{db} = 6, \quad c_{dn} = 10,$$

starting from $x = 0$.

- (i) Augment along $1, a, d, n$ by $\min(10, 8, 10) = 8$. Now $\delta = 8$, with $x_{1a} = 8$, $x_{ad} = 8$ (saturated) and $x_{dn} = 8$.
- (ii) Augment along $1, c, d, n$ by $\min(10, 9, 2) = 2$. Now $\delta = 10$, with $x_{1c} = 2$, $x_{cd} = 2$ and $x_{dn} = 10$ (saturated).
- (iii) Augment along $1, c, d, b, n$ by $\min(8, 7, 6, 10) = 6$. Now $\delta = 16$, with $x_{1c} = 8$, $x_{cd} = 8$, $x_{db} = 6$ (saturated) and $x_{bn} = 6$.
- (iv) Augment along $1, a, b, n$ by $\min(2, 4, 4) = 2$, the bottleneck being $c_{1a} - x_{1a} = 2$. Now $\delta = 18$, with $x_{1a} = 10$ (saturated), $x_{ab} = 2$ and $x_{bn} = 8$.

At this stage the only unsaturated edge out of 1 is $1c$, so $c \in S$; from c we have $x_{cd} = 8 < 9$, so $d \in S$; from d both dn and db are saturated, but $x_{ad} = 8 > 0$ so traversing ad backwards gives $a \in S$; from a we have $x_{ab} = 2 < 4$, so $b \in S$; and from b we have $x_{bn} = 8 < 10$, so $n \in S$. So we have found an augmenting path,

$$1 \rightarrow c \rightarrow d \xrightarrow{\text{backwards}} a \rightarrow b \rightarrow n,$$

with bottleneck $\min(2, 1, 8, 2, 2) = 1$, coming from the edge cd . Augmenting by 1 (and remembering to *decrease* x_{ad}) gives $\delta = 19$ and

$$x_{1a} = 10, x_{1c} = 9, x_{ab} = 3, x_{ac} = 0, x_{ad} = 7, x_{cd} = 9, x_{db} = 6, x_{bn} = 9, x_{dn} = 10.$$

Now $1a$ is saturated and $x_{1c} = 9 < 10$, so $c \in S$; but from c the edge cd is saturated and the only other edge at c is ac , which carries no flow. So $S = \{1, c\}$, and

$$C(S) = c_{1a} + c_{cd} = 10 + 9 = 19 = \delta^*.$$

So the flow of 19 is optimal, certified by the cut $\{1, c\}$. Notice that without the backward step in the fifth augmentation we would have stopped at $\delta = 18$: the flow of 8 units sent from a to d in step (i) was, with hindsight, a mistake, and the backward edge is precisely the mechanism by which the algorithm undoes it.

Theorem 20.5.20 (Termination)

If all capacities are integers, then the Ford–Fulkerson algorithm terminates in finitely many steps, and the maximum flow is integral.

Proof. If x is integral, then the bottleneck along any augmenting path is a positive integer, so the augmented flow is integral and the value increases by at least 1. Since δ is bounded above by the capacity of any cut, only finitely many augmentations can occur. Starting from $x = 0$, which is integral, the result follows. $\square$

The same argument works for rational capacities (clear denominators). For irrational capacities, the algorithm can in principle fail to terminate if the augmenting paths are chosen badly – choosing shortest augmenting paths always works, and gives the Edmonds–Karp algorithm.

§20.5.7 An Application: Bipartite Matching

The integrality in the theorem above is what makes max-flow such a powerful combinatorial tool: it turns questions about $\{0, 1\}$ -valued combinatorial objects into questions about flows.

Theorem 20.5.21

Every k -regular bipartite graph has a perfect matching.

Proof. Let G be a bipartite graph with parts L and R each of size N , with every vertex having degree k . (The parts have equal size because counting edges gives $k|L| = |E| = k|R|$.) Build a flow network: add a source 1 joined to every vertex of L with capacity 1 , add a sink n joined from every vertex of R with capacity 1 , and direct every edge of G from L to R with capacity 1 .

The cut $S = \{1\}$ has capacity N , so $\delta^* \leq N$. On the other hand, sending $1/k$ along every edge of G and 1 along every edge from the source and to the sink gives a feasible flow of value N (each vertex of L receives 1 and sends out $k \cdot \frac{1}{k} = 1$). So $\delta^* = N$.

By the previous theorem, since all capacities are integers, there is an *integral* flow of value N . Every edge then carries flow 0 or 1 , and every vertex of L has exactly one outgoing edge with flow 1 (it receives exactly 1 from the source), and similarly every vertex of R has exactly one incoming edge with flow 1 . The edges carrying flow 1 therefore form a perfect matching. $\square$

Remark. The same technique proves Hall's marriage theorem: a bipartite graph with parts L, R has a matching saturating L if and only if $|N(A)| \geq |A|$ for every $A \subseteq L$. One shows that a cut of capacity less than $|L|$ in the network above corresponds exactly to a violating set A , and then applies max-flow min-cut.

§20.6 Game Theory

We close with a topic that at first sight has nothing to do with optimization: what should you do when the thing you are optimizing against is another person, who is optimizing against you? The answer, remarkably, is that a two-player zero-sum game is exactly a pair of dual linear programs, and the fundamental theorem of such games – the minimax theorem – is exactly strong duality. This is one of the most satisfying appearances of the theory we have built.

§20.6.1 Two-Player Zero-Sum Games

Definition 20.6.1 (Zero-Sum Game)

A **two-person zero-sum game** consists of two players, P_1 and P_2 , with m and n available actions respectively, and a **payoff matrix** $A \in \mathbb{R}^{m \times n}$. If P_1 plays i and P_2 plays j , then P_1 wins the amount a_{ij} and P_2 loses the same amount.

So P_1 chooses a row and wants the entry to be large; P_2 chooses a column and wants it to be small. 'Zero-sum' means that one player's gain is exactly the other's loss – there is nothing to cooperate over.

Suppose first that the players move in sequence, and P_1 must move first. If P_1 plays row i then P_2 , seeing this, will respond with the column minimizing a_{ij} . So P_1 should solve

$$\max_i \min_j a_{ij}.$$

Symmetrically, if P_2 moves first then P_2 should solve

$$\min_j \max_i a_{ij}.$$

It is always the case that

$$\max_i \min_j a_{ij} \leq \min_j \max_i a_{ij},$$

since $\min_{j'} a_{ij'} \leq a_{ij} \leq \max_{i'} a_{i'j}$ for every i, j ; taking the max over i on the left and the min over j on the right preserves this. Moving second is (weakly) an advantage: you get to see what your opponent did.

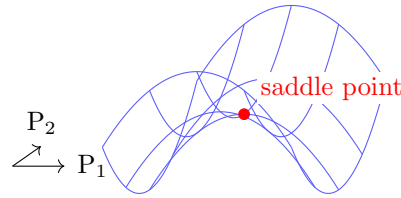
Definition 20.6.2 (Saddle Point)

An entry $a_{i^*j^*}$ is a **saddle point** if it is simultaneously the minimum of its row and the maximum of its column, that is

$$a_{i^*j} \geq a_{i^*j^*} \geq a_{ij^*} \quad \text{for all } i, j.$$

Equivalently, $\max_i \min_j a_{ij} = a_{i^*j^*} = \min_j \max_i a_{ij}$. In this case $a_{i^*j^*}$ is called the **value** of the game.

The name comes from the geometry: at a saddle point the payoff surface curves up in one direction and down in the other, like a mountain pass.



Example 20.6.3

Take

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Row minima are 1 and 3, so $\max_i \min_j a_{ij} = 3$. Column maxima are 3 and 4, so $\min_j \max_i a_{ij} = 3$. These agree, so $(2, 1)$ is a saddle point and the value of the game is 3. Both players should play their part of the saddle point regardless of who moves first, and neither can do better by deviating.

Example 20.6.4

Take

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}.$$

Row minima are 2 and 1, so $\max_i \min_j a_{ij} = 2$. Column maxima are 4 and 3, so $\min_j \max_i a_{ij} = 3$. These do not agree, and there is no saddle point.

In this game there is no stable pair of choices. If P_1 plays row 1 then P_2 prefers column 2; but against column 2, P_1 prefers row 2; against row 2, P_2 prefers column 1; against column 1, P_1 prefers row 1 – and we are back where we started. Whatever either player does, if the other knows it they can exploit it.

§20.6.2 Mixed Strategies

The resolution of the second example is one of the great ideas of game theory: the players should randomize. If P_1 picks a row at random according to a probability distribution, then P_2 cannot know which row will be played, only the distribution, and so cannot exploit the choice exactly.

Definition 20.6.5 (Mixed Strategy)

A **mixed strategy** for P_1 is a vector $p \in \mathbb{R}^m$ with $p \geq 0$ and $\sum_i p_i = 1$, where p_i is the probability of playing action i . Similarly a mixed strategy for P_2 is $q \in \mathbb{R}^n$ with $q \geq 0$, $\sum_j q_j = 1$. A strategy which plays one action with probability 1 is called **pure**.

If P_1 plays p and P_2 plays q , the expected payoff to P_1 is

$$\sum_{i,j} p_i a_{ij} q_j = p \cdot (Aq) = p^T Aq.$$

If P_1 plays p and P_2 responds with a pure strategy j , the expected payoff is $\sum_i a_{ij} p_i = (A^T p)_j$; and since a mixed strategy is an average of pure ones, P_2 can do no better than the best pure response. So P_1 wants to solve

$$\begin{aligned} & \text{maximize} && \min_j (A^T p)_j \\ & \text{subject to} && \sum_i p_i = 1, \quad p \geq 0. \end{aligned}$$

This is not yet a linear program, but the standard trick of introducing a variable for the minimum makes it one. Writing $e = (1, 1, \dots, 1)^T$ (of whatever length is needed),

$$\begin{aligned} (P_1) \quad & \text{maximize} && v \\ & \text{subject to} && A^T p \geq ve, \\ & && e \cdot p = 1, \quad p \geq 0. \end{aligned}$$

Similarly, P_2 wants to minimize the maximum expected loss, giving

$$\begin{aligned} (P_2) \quad & \text{minimize} && w \\ & \text{subject to} && Aq \leq we, \\ & && e \cdot q = 1, \quad q \geq 0. \end{aligned}$$

Both are honest linear programs, in the variables (p, v) and (q, w) respectively. And now the punchline.

Theorem 20.6.6

The problems (P_1) and (P_2) are dual to each other.

Proof. Introduce a slack variable $s \geq 0$ to write P_2 's constraints as $Aq + s = we$, and take multipliers $\lambda_1 \in \mathbb{R}^m$ for these and $\lambda_2 \in \mathbb{R}$ for the normalization $e \cdot q = 1$. The Lagrangian is

$$\begin{aligned} L(w, q, s, \lambda) &= w + \lambda_1 \cdot (Aq + s - we) - \lambda_2(e \cdot q - 1) \\ &= w(1 - \lambda_1 \cdot e) + (A^T \lambda_1 - \lambda_2 e) \cdot q + \lambda_1 \cdot s + \lambda_2. \end{aligned}$$

Now: w is unrestricted, so we need $\lambda_1 \cdot e = 1$; $q \geq 0$, so we need $A^T \lambda_1 - \lambda_2 e \geq 0$; and $s \geq 0$, so we need $\lambda_1 \geq 0$. Given all of these the infimum is λ_2 . Hence the dual problem is

$$\begin{aligned} &\text{maximize} && \lambda_2 \\ &\text{subject to} && A^T \lambda_1 \geq \lambda_2 e, \\ &&& e \cdot \lambda_1 = 1, \lambda_1 \geq 0, \end{aligned}$$

which is exactly (P_1) under the identification $\lambda_1 = p$, $\lambda_2 = v$. $\square$

Theorem 20.6.7 (Minimax Theorem)

For any payoff matrix A ,

$$\max_p \min_q p^T A q = \min_q \max_p p^T A q,$$

where the maximum and minimum are over mixed strategies. The common value v is called the **value** of the game.

Proof. The left-hand side is the optimal value of (P_1) and the right-hand side is the optimal value of (P_2) , and we have just shown these are dual linear programs. Both are feasible – take p and q uniform, and v, w suitably extreme – and both are bounded, since $\min_{i,j} a_{ij} \leq p^T A q \leq \max_{i,j} a_{ij}$ always. So strong duality applies and the optimal values agree. $\square$

This is a genuinely surprising theorem. It says that in a zero-sum game, once we allow randomization, there is no advantage to moving second: P_1 can guarantee an expected payoff of at least v no matter what P_2 does, and P_2 can guarantee losing at most v no matter what P_1 does. The game has a well-defined value, and there is no secret to keep.

The complementary slackness conditions for this dual pair are worth recording, since they are what we use to compute.

Theorem 20.6.8 (Optimality Conditions for Games)

Mixed strategies p, q are optimal for P_1 and P_2 respectively, with value v , if and only if

- (i) $A^T p \geq ve$, $e \cdot p = 1$, $p \geq 0$; (primal feasibility)
- (ii) $Aq \leq ve$, $e \cdot q = 1$, $q \geq 0$; (dual feasibility)
- (iii) $v = p^T A q$. (complementary slackness)

Proof. Complementary slackness for the dual pair above reads $p \cdot (Aq - we) = 0$ and $q \cdot (A^T p - ve) = 0$. Since $e \cdot p = e \cdot q = 1$, these say $p^T Aq = w$ and $p^T Aq = v$ respectively, so at an optimum $v = w = p^T Aq$. $\square$

The practical content of (iii) is the following pair of statements, which we will use in a moment:

- if $p_i > 0$ then the i th constraint of P_2 's problem is tight, i.e. $(Aq)_i = v$;
- if $q_j > 0$ then the j th constraint of P_1 's problem is tight, i.e. $(A^T p)_j = v$.

In words: every action that a player actually uses with positive probability must give exactly the value of the game against the opponent's optimal strategy. If some action gave strictly less, the player would drop it.

§20.6.3 Finding Optimal Strategies

In principle we can always solve the linear program by the simplex method, but for small games there are shortcuts.

- (1) **Look for a saddle point.** If the payoff matrix has one, the corresponding pure strategies are optimal for both players and we are done.
- (2) **Eliminate dominated actions.** If $a_{ij} \geq a_{i'j}$ for every j , then row i **dominates** row i' : P_1 can never do worse by playing i instead of i' , so there is an optimal strategy with $p_{i'} = 0$, and we may delete row i' . Similarly, if $a_{ij} \leq a_{ij'}$ for every i then column j dominates column j' (from P_2 's point of view, smaller is better), and we may delete column j' .
- (3) **Reduce to a small linear program.** A $2 \times n$ or $m \times 2$ game reduces to a one-dimensional problem that can be solved graphically.
- (4) **Use complementary slackness** to recover the other player's strategy once you have one of them.

Example 20.6.9

Let us return to the game with no saddle point,

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix},$$

and solve it properly. Write P_1 's mixed strategy as $p = (p, 1 - p)$. Against P_2 's two pure strategies the expected payoffs are

$$(A^T p)_1 = 4p + (1 - p) = 1 + 3p, \quad (A^T p)_2 = 2p + 3(1 - p) = 3 - p,$$

and P_1 wants to maximize the smaller of these. The first is increasing in p and the second decreasing, so the maximum of the minimum is where they cross:

$$1 + 3p = 3 - p \implies p = \frac{1}{2}, \quad v = \frac{5}{2}.$$

Similarly, writing $q = (q, 1 - q)$ for P_2 ,

$$(Aq)_1 = 4q + 2(1 - q) = 2 + 2q, \quad (Aq)_2 = q + 3(1 - q) = 3 - 2q,$$

and P_2 minimizes the larger of these, which happens where $2 + 2q = 3 - 2q$, that is $q = \frac{1}{4}$, again with value $\frac{5}{2}$.

So the value of the game is $\frac{5}{2}$, with $p^* = (\frac{1}{2}, \frac{1}{2})$ and $q^* = (\frac{1}{4}, \frac{3}{4})$. Notice that $\frac{5}{2}$ lies strictly between $\max_i \min_j a_{ij} = 2$ and $\min_j \max_i a_{ij} = 3$: randomizing has genuinely bought P_1 something over any pure strategy, and it has cost P_2 less than any pure strategy would have.

Notice also the structure of the answer: at the optimum, *both* of P_2 's pure strategies give P_1 the same expected payoff. This is complementary slackness at work – since $q_1, q_2 > 0$, both of P_1 's constraints must be tight. Equivalently, P_1 chooses p precisely so as to make P_2 indifferent between their options; if P_2 were not indifferent, they would have a strictly better reply and could exploit P_1 .

Example 20.6.10

Consider the game with payoff matrix

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & \frac{1}{2} \\ 1 & 3 & 2 \end{pmatrix}.$$

First, is there a saddle point? The row minima are $2, \frac{1}{2}, 1$, so $\max_i \min_j = 2$; the column maxima are $3, 3, 4$, so $\min_j \max_i = 3$. These differ, so there is no saddle point and we will need mixed strategies.

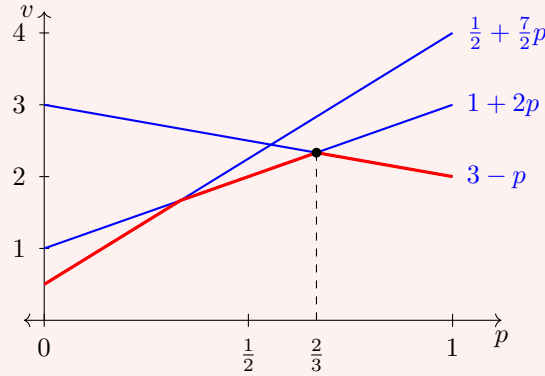
Next, look for domination. Row 1 is $(2, 3, 4)$ and row 3 is $(1, 3, 2)$; since $2 \geq 1$, $3 \geq 3$ and $4 \geq 2$, row 1 dominates row 3, so we may take $p_3 = 0$ and work with

$$\tilde{A} = \begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & \frac{1}{2} \end{pmatrix}.$$

Write P_1 's strategy as $p = (p, 1 - p, 0)$. Then P_1 's problem becomes

$$\begin{aligned} &\text{maximize} && v \\ &\text{subject to} && 2p + 3(1 - p) \geq v, \\ & && 3p + 1(1 - p) \geq v, \\ & && 4p + \frac{1}{2}(1 - p) \geq v, \\ & && 0 \leq p \leq 1, \end{aligned}$$

that is, $v \leq \min \{3 - p, 1 + 2p, \frac{1}{2} + \frac{7}{2}p\}$, and we want to choose p to make this minimum as large as possible. Since this is a one-dimensional problem we can just draw the three lines and look at their lower envelope.



The lower envelope, drawn in red, is maximized where the lines $3 - p$ and $1 + 2p$ cross, at $p = \frac{2}{3}$, giving $v = \frac{7}{3}$. (The third line, $\frac{1}{2} + \frac{7}{2}p$, takes the value $\frac{1}{2} + \frac{7}{3} = \frac{17}{6} > \frac{7}{3}$ there, so it is not binding.) So

$$p^* = \left(\frac{2}{3}, \frac{1}{3}, 0\right), \quad v = \frac{7}{3}.$$

Now we use complementary slackness to find P_2 's strategy q . Since $p_1, p_2 > 0$, the first two constraints of P_2 's problem must be tight:

$$(Aq)_1 = 2q_1 + 3q_2 + 4q_3 = \frac{7}{3}, \quad (Aq)_2 = 3q_1 + q_2 + \frac{1}{2}q_3 = \frac{7}{3}.$$

Also, the third constraint of P_1 's problem is slack (we computed $\frac{17}{6} > \frac{7}{3}$), so complementary slackness forces $q_3 = 0$. With $q_1 + q_2 = 1$, the first equation gives

$$2q_1 + 3(1 - q_1) = \frac{7}{3} \implies 3 - q_1 = \frac{7}{3} \implies q_1 = \frac{2}{3},$$

and one checks that the second equation is satisfied: $3 \cdot \frac{2}{3} + \frac{1}{3} = \frac{7}{3}$. So

$$q^* = \left(\frac{2}{3}, \frac{1}{3}, 0\right).$$

Finally we should sanity check the answer against the optimality conditions. We have $A^T p^* = \left(\frac{7}{3}, \frac{7}{3}, \frac{17}{6}\right) \geq \frac{7}{3}e$ and $Aq^* = \left(\frac{7}{3}, \frac{7}{3}, \frac{5}{3}\right) \leq \frac{7}{3}e$, and $p^{*T} Aq^* = \frac{7}{3}$. All three conditions hold, so p^* and q^* really are optimal and the value of the game is $\frac{7}{3}$.

Remark. It is worth noticing what this answer means. P_1 's value from playing the pure strategy 'always row 1' is only 2 (their opponent would always answer with column 1), and from 'always row 2' is only $\frac{1}{2}$. By mixing the two in the ratio 2 : 1, P_1 guarantees $\frac{7}{3} > 2$. Randomization is strictly better than any deterministic play. That this is possible at all, and that the resulting value is exactly matched by the opponent's best defence, is the content of the minimax theorem – which is to say, of linear programming duality.

21 Quantum Mechanics

Quantum mechanics is the mathematical framework used to describe nature at the scale of subatomic particles. These notes will primarily focus on building up this mathematical framework, and not so much on the actual applications or physical background to it.

This article constitutes my notes for the ‘Quantum Mechanics’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly¹ in quite a few areas (more so than some of my other notes). Still, all lectured material should be covered.

§21.1 A Quantum Prelude

Before we do any mathematics, it is worth spending a little time on the physics that forced quantum mechanics into existence. None of the material in this section is logically necessary for what follows – we could just write down the axioms and get on with it – but the axioms will look much less arbitrary if we first see the experiments they were invented to explain.

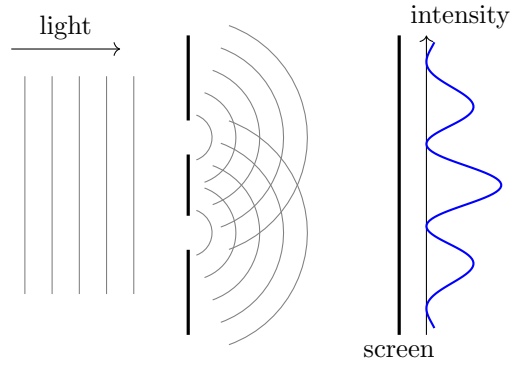
The story, compressed into a single sentence, is this: at the turn of the twentieth century it became clear that light (which everyone knew was a wave) sometimes behaves like a stream of particles, and that electrons (which everyone knew were particles) sometimes behave like waves.

§21.1.1 Light: Waves or Particles?

First, a quick reminder of the vocabulary of waves, which we will use constantly. A wave with period T in time has frequency $\nu = 1/T$ and **angular frequency** $\omega = 2\pi/T$; a wave with period λ in space (the **wavelength**) has **wavenumber** $k = 2\pi/\lambda$. The basic travelling wave in one dimension is $f(x, t) = Ae^{i(kx - \omega t)}$, with amplitude A , moving at speed ω/k ; in three dimensions this becomes the **plane wave** $Ae^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$, with **wavevector** $\mathbf{k}$ and $\lambda = 2\pi/|\mathbf{k}|$. A governing wave equation imposes a relation $\omega = \omega(\mathbf{k})$ between the two – the **dispersion relation** – and for any *linear* wave equation, solutions superpose: the sum of two solutions is a solution. All of the interference phenomena below, and indeed quantum mechanics itself, flow from this superposition property.

By 1900 the wave theory of light looked untouchable. Young’s double slit experiment (1801–03) showed that light passing through two narrow slits produces an interference pattern – alternating bright and dark fringes, from the two paths adding in or out of phase – exactly as water waves do, and Maxwell’s theory of electromagnetism (1862–64) identified light as an electromagnetic wave, predicting its speed from the constants of electricity and magnetism.

¹In particular, these notes are influenced by the ‘Principles of Quantum Mechanics’ course, along with numerous books.



The first cracks appeared with *black-body radiation*. A body heated to temperature T emits radiation, and the simplest system to study is a black body – a perfectly absorbing surface. Classical physics predicted that the intensity of emitted radiation should grow without bound as the frequency increases, which is plainly wrong (this is the ‘ultraviolet catastrophe’). The observed spectrum instead has a clear maximum, and Planck (1900) found that this curve could be derived exactly if one assumed that radiation of angular frequency ω is exchanged only in discrete lumps of energy

$$E = \hbar\omega,$$

where $\hbar \approx 1.05 \times 10^{-34}$ J s is the **reduced Planck constant**².

Einstein (1905) took this quantisation seriously in his analysis of the **photoelectric effect**. Take a metal surface in a vacuum and shine light of angular frequency ω on it; for suitable frequencies, electrons are emitted. If light were purely a wave, we would expect:

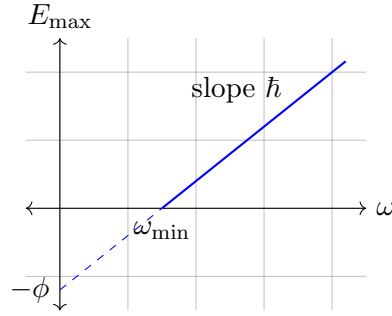
- (i) the energy of the emitted electrons to grow with the *intensity* of the light, since a more intense wave carries more energy;
- (ii) light of any frequency to eventually eject electrons, provided the intensity is high enough;
- (iii) a delay at low intensities, while enough energy is absorbed.

Every one of these expectations fails. What is actually observed is that the maximum kinetic energy of the emitted electrons depends only on the frequency,

$$E_{\max} = \hbar\omega - \phi,$$

where ϕ is a constant depending on the metal, that no electrons at all are emitted below the threshold frequency $\omega_{\min} = \phi/\hbar$ no matter how intense the light, and that emission is immediate but the *rate* of emission grows with intensity.

²One often also sees the ordinary Planck constant $h = 2\pi\hbar$. Quantum mechanics is much more pleasant when written in terms of $\hbar$, so that is what we will use throughout. Note that $\hbar$ has dimensions of energy $\times$ time, or equivalently position $\times$ momentum – the dimensions of *action*, and also of angular momentum.



Einstein's explanation is that light of angular frequency ω arrives in quanta – **photons** – each carrying energy $E = \hbar\omega$, and each photon can liberate at most one electron. An electron bound to the metal with binding energy ϕ then escapes with energy at most $\hbar\omega - \phi$, and increasing the intensity increases the number of photons (and hence the emission rate) but not the energy of each one.

Further confirmation came from *Compton scattering* (1923): X-rays scattered off electrons change frequency by an amount depending on the scattering angle, exactly as if each X-ray photon were a relativistic particle with energy $E = \hbar\omega$ and momentum

$$\mathbf{p} = \hbar\mathbf{k},$$

where $\mathbf{k}$ is the wavevector of the light. Indeed, treating the collision as relativistic two-particle scattering (a nice exercise with the kinematics of IA Dynamics and Relativity) and inserting these relations, one predicts the frequency shift

$$\frac{1}{\omega'} = \frac{1}{\omega} + \frac{\hbar}{m_e c^2} (1 - \cos \theta)$$

for X-rays scattered through an angle θ – which is exactly what Compton measured. So light carries energy and momentum in particle-like packets, while still interfering like a wave. Remarkably, G. I. Taylor (1909) repeated the double slit experiment with light so faint that only one photon was in the apparatus at a time – and the interference pattern still built up, photon by photon. Whatever a photon is, it is not simply a classical particle following one path or the other.

§21.1.2 Atoms and Their Spectra

A second, independent crisis was brewing inside the atom. Rutherford's scattering experiment (1911) – firing α particles at gold foil, and finding that most passed straight through while a few bounced almost straight back – showed that an atom is mostly empty space, with its positive charge concentrated in a tiny nucleus, orbited by electrons.

This 'solar system' picture has two fatal problems.

- (i) An orbiting electron is accelerating, and by classical electromagnetism an accelerating charge radiates. The electron should lose energy continuously and spiral into the nucleus in a fraction of a second. Atoms should not exist.
- (ii) Atoms absorb and emit light only at certain discrete frequencies – the *spectral lines*. For hydrogen these frequencies fit the remarkably simple empirical formula

$$\omega_{mn} = 2\pi c R_0 \left(\frac{1}{n^2} - \frac{1}{m^2} \right), \quad m > n,$$

where m, n are positive integers and $R_0 \approx 1.1 \times 10^7 \text{ m}^{-1}$ is the *Rydberg constant*. Nothing in classical mechanics singles out these frequencies.

(A word on how spectra are seen. Pass a current through a tube of hydrogen and the emitted light, split by a prism, is not a rainbow but a handful of sharp bright lines – the *emission spectrum*. Shine white light *through* cold hydrogen and the same frequencies appear as dark absorption lines cut out of the continuum. The frequencies are fingerprints of the element – this is how helium was discovered in the Sun before it was found on Earth – and any theory of the atom lives or dies by predicting them.)

Bohr (1913) proposed a fix that was as effective as it was unjustified: he postulated that the electron's orbital angular momentum is *quantised*,

$$L = m_e v r = n \hbar, \quad n = 1, 2, 3, \dots,$$

so that only certain orbits are allowed, and that radiation is emitted only when the electron jumps between them. Let's see what this gives us. For a circular orbit of radius r , the Coulomb force provides the centripetal acceleration:

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{m_e v^2}{r}.$$

Combining this with $v = n\hbar/m_e r$, we can solve for the allowed radii:

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} n^2 = a_0 n^2,$$

where $a_0 \approx 5.3 \times 10^{-11} \text{ m}$ is the **Bohr radius**. The energy of the n th orbit is then

$$E_n = \frac{1}{2} m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 a_0} \cdot \frac{1}{n^2} = -\frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \cdot \frac{1}{n^2}.$$

The lowest energy state, $n = 1$, has $E_1 \approx -13.6 \text{ eV}$, and the energy released in a jump from level m down to level n is carried away by a photon with $\hbar\omega = E_m - E_n$, which reproduces the Rydberg formula exactly, with the right value of R_0 in terms of fundamental constants.

This is a genuinely impressive result, and also clearly not the whole story – nothing explains *why* angular momentum should come in units of $\hbar$, and the model fails completely for atoms with more than one electron. We will return to hydrogen at the very end of these notes, and derive its energy levels honestly.

§21.1.3 Matter Waves

The missing idea came from de Broglie (1923–24), who proposed a striking symmetry: if light waves behave like particles, then particles should behave like waves. Specifically, associated to any particle with energy E and momentum $\mathbf{p}$ there is a wave with

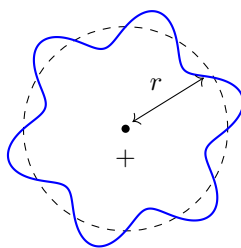
$$\omega = \frac{E}{\hbar}, \quad \mathbf{k} = \frac{\mathbf{p}}{\hbar},$$

that is, with wavelength $\lambda = 2\pi/|\mathbf{k}| = 2\pi\hbar/|\mathbf{p}|$.

This immediately demystifies Bohr's quantisation rule. If the electron is a wave running around a circular orbit of radius r , then for the wave to join up with itself we need a whole number of wavelengths to fit around the circumference: $2\pi r = n\lambda$. But then

$$L = pr = \hbar k \cdot \frac{n\lambda}{2\pi} = \hbar \cdot \frac{2\pi}{\lambda} \cdot \frac{n\lambda}{2\pi} = n\hbar,$$

which is exactly Bohr's condition – quantisation appears as a standing wave condition.



Is any of this true? Davisson and Germer, and independently G. P. Thomson (1927–28), fired electrons at crystals and observed diffraction patterns – interference fringes – with exactly the wavelength $\lambda = 2\pi\hbar/|\mathbf{p}|$ predicted by de Broglie³. Matter genuinely does interfere with itself.

So by the mid 1920s the experimental picture was clear: both light and matter propagate like waves but are detected like particles, energies and angular momenta of bound systems are quantised, and the classical description fails at atomic scales. Between 1925 and 1930, Heisenberg, Schrödinger, Born, Dirac, Pauli and others assembled these fragments into a coherent mathematical theory. Building up (the wave-mechanical formulation of) that theory is the business of the rest of these notes.

§21.2 Wavefunctions and States

§21.2.1 States in Classical Mechanics

Before setting out the framework of quantum mechanics, it is worth being clear about the abstract structure of the theory it replaces.

In classical mechanics, particle motion is described by Newton's laws. These are a set of second order differential equations, which when coupled with two initial conditions then specify the future motion of the particle for all time. These initial conditions are typically taken to be position and momentum $(x(t_0), p(t_0))$. So Newton's laws really say that the entire state of a particle is encoded by a vector $(x(t), p(t))$ in the *phase space* M of position and momentum. As time passes, the particle's state traces out a path in phase space, but the whole path is determined by the initial vector (along with the differential equation given by Newton's laws).

There is a way of packaging Newton's laws that makes the parallel with quantum mechanics especially close: Hamilton's formulation, in which the dynamics is generated by the total energy $H(x, p) = \frac{p^2}{2m} + U(x)$ through

$$\frac{dx}{dt} = \frac{\partial H}{\partial p} \quad \text{and} \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x}.$$

Keep this in mind – in quantum mechanics too, it will be the energy that generates time evolution (and Ehrenfest's theorem in Section 6 will make the correspondence exact, at the level of averages).

But of course a particle really corresponds to something physical – it's not really an element of an abstract vector space M . We need some way to go from this framework of

³There is a pleasing irony here: J. J. Thomson won a Nobel prize for showing the electron is a particle, and his son G. P. Thomson won one for showing it is a wave.

phase space to physical quantities. This is done by introducing *observables*, which are functions $O : M \rightarrow \mathbb{R}$.

Observables can be all sorts of things – position, kinetic energy, angular momentum, and everything else. These observables are defined for vectors in M , but we typically want to evaluate them at the vector corresponding to our particle’s state at some time t . Still, we don’t factor time into our definition of observables, it’s just baked into the point at which we evaluate them.

Classical mechanics is a game of juggling initial conditions, inputs to Newton’s laws (such as forces and potentials), and observables. We will see that in the abstract, quantum and classical mechanics are similar in this way: there will be a space of states, a notion of observables, and an equation of motion. What changes – completely – is what these things *are*.

One warning before we begin. Classical mechanics and quantum mechanics are fundamentally different – you can’t derive quantum mechanics from classical mechanics, and attempts to introduce it like that will fail. In reality, you should look at the mathematical framework of quantum mechanics and use that to build intuition. To help with this, looking at the *mathematical* framework of classical mechanics may also help, but it shouldn’t be where you derive your intuition.

§21.2.2 The Wavefunction and the Born Rule

So how do we describe a point particle? In classical mechanics we had the pair of vectors $(\mathbf{x}, \mathbf{p})$. The corresponding description in quantum mechanics is the *state* of the particle, which is described by a complex-valued *wavefunction*.

Definition 21.2.1 (Wavefunction)

A **wavefunction** is a complex-valued function $\psi(\mathbf{x}, t) : \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{C}$, satisfying certain mathematical properties dictated by its physical interpretation (described below). At each fixed time t , we think of $\psi(\cdot, t)$ as a function of position alone, with t an external parameter.

At first sight this is a strange object – we have replaced six real numbers with an entire complex function. The connection between the wavefunction and the underlying physical particle is given by the *Born rule*, and it is probabilistic.

Axiom 21.2.2 (The Born Rule)

The probability density for a particle in the state ψ to be found at position $\mathbf{x}$ at time t is

$$\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2.$$

That is, the probability of finding the particle in a small volume δV about $\mathbf{x}$ is $\rho(\mathbf{x}, t) \delta V$.

The Born rule immediately imposes a restriction on which functions can be wavefunctions. Because the particle has to be *somewhere* with probability 1, we need

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x}, t)|^2 dV = 1,$$

in which case we say ψ is **normalised**. It is frequently useful to work with unnormalised wavefunctions, for which we only demand that the above integral is finite and non-zero – such ψ are called **normalisable**, or *square-integrable*. Given a normalisable wavefunction with

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x}, t)|^2 dV = N,$$

the rescaled wavefunction $\psi/\sqrt{N}$ is then normalised, and it is this rescaled function that the Born rule speaks about.

Example 21.2.3 (Normalising a Gaussian)

Let's normalise the (one-dimensional, time-independent) wavefunction

$$\psi(x) = Ce^{-x^2/2\sigma^2}, \quad \sigma > 0.$$

We need $\int_{-\infty}^{\infty} |\psi|^2 dx = |C|^2 \int_{-\infty}^{\infty} e^{-x^2/\sigma^2} dx = |C|^2 \sigma \sqrt{\pi} = 1$, using the Gaussian integral $\int_{-\infty}^{\infty} e^{-u^2} du = \sqrt{\pi}$. So we may take

$$C = \frac{1}{(\pi\sigma^2)^{1/4}},$$

(or this multiplied by any phase). The particle is then found within $[-\sigma, \sigma]$ with probability $\int_{-\sigma}^{\sigma} |\psi|^2 dx \approx 0.84$, and the tails, while never exactly zero, are negligible beyond a few multiples of σ . Wavefunctions of exactly this shape will keep appearing throughout these notes – as wavepackets, as oscillator ground states, and as the states of minimal uncertainty.

Remark (For Pedants). If we have two wavefunctions which differ by a complex phase, $\tilde{\psi}(\mathbf{x}, t) = e^{i\theta} \psi(\mathbf{x}, t)$ for some constant $\theta \in \mathbb{R}$, then this does not change the probability density $\rho(\mathbf{x}, t)$ – or indeed anything else physically measurable. Because of this, we treat the two wavefunctions ψ and $\tilde{\psi}$ as corresponding to the same state. So states really correspond to rays in the vector space of wavefunctions. We will not fuss about this distinction, but it is worth keeping in mind that overall phases are unphysical, while *relative* phases in a superposition very much are physical, as we are about to see.

§21.2.3 Hilbert Space and Inner Products

The natural language of quantum mechanics is linear algebra. In a finite dimensional vector space, we have abstract vectors v , which can be represented concretely by considering some basis and writing say $v = (v_1, v_2, \dots, v_n)$. In quantum mechanics, we deal with the abstract object of a *state*, and it too can be represented concretely – the wavefunction $\psi(\mathbf{x}, t)$ should be thought of as the (continuous) collection of coefficients of the state in the ‘basis’ of positions⁴.

The set of states we work with is the following.

Definition 21.2.4 (Hilbert Space)

⁴This point of view is developed properly in the ‘Principles of Quantum Mechanics’ course, where states are vectors $|\psi\rangle$ in an abstract Hilbert space, and the wavefunction is the collection of components $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$. In this course we always work directly with the wavefunction.

The **Hilbert space** $\mathcal{H}$ is the vector space of normalisable (square-integrable) functions on $\mathbb{R}^3$,

$$\mathcal{H} = \left\{ \psi : \mathbb{R}^3 \rightarrow \mathbb{C} \mid 0 < \int_{\mathbb{R}^3} |\psi|^2 dV < \infty \right\} \cup \{0\}.$$

Calling this a vector space is a claim, albeit a small one: we need linear combinations of normalisable functions to be normalisable.

Proposition 21.2.5 (Superposition Principle)

If $\psi_1, \psi_2 \in \mathcal{H}$ and $\lambda_1, \lambda_2 \in \mathbb{C}$, then $\lambda_1\psi_1 + \lambda_2\psi_2 \in \mathcal{H}$.

Proof. For any complex numbers z_1, z_2 we have $2|z_1||z_2| \leq |z_1|^2 + |z_2|^2$, and so

$$|z_1 + z_2|^2 \leq |z_1|^2 + 2|z_1||z_2| + |z_2|^2 \leq 2|z_1|^2 + 2|z_2|^2.$$

Applying this pointwise with $z_i = \lambda_i\psi_i(\mathbf{x})$ and integrating,

$$\int_{\mathbb{R}^3} |\lambda_1\psi_1 + \lambda_2\psi_2|^2 dV \leq 2|\lambda_1|^2 \int_{\mathbb{R}^3} |\psi_1|^2 dV + 2|\lambda_2|^2 \int_{\mathbb{R}^3} |\psi_2|^2 dV < \infty,$$

as required. $\square$

This innocent-looking closure property is where quantum mechanics departs from everything classical: *the superposition of two states is a state*. If ψ_1 is ‘particle here’ and ψ_2 is ‘particle there’, then $(\psi_1 + \psi_2)/\sqrt{2}$ is a perfectly legitimate physical state, and it is not ‘here’, nor ‘there’, nor ‘here or there with 50/50 ignorance’ – as interference experiments show, it is something genuinely new. Nothing analogous holds for solutions of Newton’s equations.

The Hilbert space also carries an inner product, which will be the central computational tool of the whole course.

Definition 21.2.6 (Inner Product)

For wavefunctions $\psi, \phi \in \mathcal{H}$, we define

$$\langle \psi, \phi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) \phi(\mathbf{x}) dV.$$

Note carefully the complex conjugate on the first argument. This inner product has the properties you would hope for.

Proposition 21.2.7 (Properties of the Inner Product)

For $\psi, \phi \in \mathcal{H}$ and $\lambda_1, \lambda_2 \in \mathbb{C}$:

- (i) $\langle \psi, \phi \rangle$ is well defined (the integral converges);
- (ii) $\langle \psi, \phi \rangle^* = \langle \phi, \psi \rangle$;
- (iii) the inner product is linear in the second argument, $\langle \psi, \lambda_1\phi_1 + \lambda_2\phi_2 \rangle = \lambda_1\langle \psi, \phi_1 \rangle + \lambda_2\langle \psi, \phi_2 \rangle$, and antilinear in the first, $\langle \lambda_1\psi_1 + \lambda_2\psi_2, \phi \rangle = \lambda_1^*\langle \psi_1, \phi \rangle + \lambda_2^*\langle \psi_2, \phi \rangle$;

(iv) $\langle \psi, \psi \rangle \geq 0$, with equality (for continuous ψ) if and only if $\psi = 0$.

Proof. Properties (ii)–(iv) follow directly from the definition. For (i), we use $2|\psi^* \phi| \leq |\psi|^2 + |\phi|^2$ pointwise, so

$$\int_{\mathbb{R}^3} |\psi^* \phi| dV \leq \frac{1}{2} \int_{\mathbb{R}^3} |\psi|^2 dV + \frac{1}{2} \int_{\mathbb{R}^3} |\phi|^2 dV < \infty,$$

and the integral converges absolutely. $\square$

We write $\|\psi\|^2 = \langle \psi, \psi \rangle$, so that ψ is normalised exactly when $\|\psi\| = 1$. Two useful pieces of terminology, imported straight from linear algebra:

Definition 21.2.8 (Orthonormality and Completeness)

A set of wavefunctions $\{\psi_n\}$ is **orthonormal** if $\langle \psi_m, \psi_n \rangle = \delta_{mn}$. It is **complete** if every $\psi \in \mathcal{H}$ can be expanded as

$$\psi = \sum_n \lambda_n \psi_n$$

for some constants $\lambda_n \in \mathbb{C}$.

For a complete orthonormal set, the expansion coefficients can be extracted with the inner product, just as Fourier coefficients are.

Proposition 21.2.9 (Extracting Coefficients)

If $\{\psi_n\}$ is a complete orthonormal set and $\psi = \sum_n \lambda_n \psi_n$, then $\lambda_n = \langle \psi_n, \psi \rangle$.

Proof. Using linearity and orthonormality,

$$\langle \psi_n, \psi \rangle = \left\langle \psi_n, \sum_m \lambda_m \psi_m \right\rangle = \sum_m \lambda_m \langle \psi_n, \psi_m \rangle = \sum_m \lambda_m \delta_{nm} = \lambda_n. \quad \square$$

This should feel exactly like Fourier series – and indeed Fourier series will literally appear when we solve the particle in a box.

Example 21.2.10 (A Complete Orthonormal Set)

On the interval $[-a, a]$ (which will be the home of the ‘infinite square well’ later), the functions

$$\psi_n(x) = \frac{1}{\sqrt{a}} \begin{cases} \cos \frac{n\pi x}{2a} & n \text{ odd,} \\ \sin \frac{n\pi x}{2a} & n \text{ even,} \end{cases} \quad n = 1, 2, 3, \dots$$

form an orthonormal set: for instance, for $m \neq n$ both odd,

$$\frac{1}{a} \int_{-a}^a \cos \frac{m\pi x}{2a} \cos \frac{n\pi x}{2a} dx = \frac{1}{2a} \int_{-a}^a \left(\cos \frac{(m-n)\pi x}{2a} + \cos \frac{(m+n)\pi x}{2a} \right) dx = 0,$$

with the mixed sine-cosine integrals vanishing by parity. Completeness (for functions

vanishing at $\pm a$) is exactly the convergence of Fourier series, which we take on trust from Analysis. This set will turn out to be the set of energy eigenfunctions of a particle in a box, and the expansion coefficients will carry direct physical meaning.

§21.2.4 The Postulates of Quantum Mechanics

We can now clearly set out the mathematical framework that quantum mechanics uses. It's natural to think about answering the following questions: How is a quantum state represented mathematically? How do we take this representation and calculate physical quantities? Given the state at some time t_1 , how do we find the state at some time t_2 ? With answers to these we are well on our way.

The truth is that you have to start somewhere, and these postulates are where it's going to be. They cannot be derived – they are justified by the (superb) agreement of their consequences with experiment.

Axiom 21.2.11 (States)

The state of a quantum mechanical system at time t is given by a non-zero wavefunction $\psi(\cdot, t)$ in the Hilbert space $\mathcal{H}$, with ψ and $\tilde{\psi} = \lambda\psi$ (for $\lambda \in \mathbb{C} \setminus \{0\}$) describing the same state.

Axiom 21.2.12 (Observables)

Every observable quantity O of the system corresponds to a Hermitian linear operator $\hat{O} : \mathcal{H} \rightarrow \mathcal{H}$.

Here an **operator** is a linear map $\hat{O} : \mathcal{H} \rightarrow \mathcal{H}$, that is

$$\hat{O}(\lambda_1\psi_1 + \lambda_2\psi_2) = \lambda_1\hat{O}\psi_1 + \lambda_2\hat{O}\psi_2,$$

and *Hermitian* is a symmetry condition with respect to the inner product whose statement and consequences we will develop properly in Section 6 – for now, think ‘real eigenvalues guaranteed’. The observables we will meet most often are built from position and momentum:

$$\begin{aligned} \text{position} \quad \hat{x}_i\psi &= x_i\psi, \\ \text{momentum} \quad \hat{p}_i\psi &= -i\hbar\frac{\partial\psi}{\partial x_i}, \\ \text{energy} \quad \hat{H}\psi &= -\frac{\hbar^2}{2m}\nabla^2\psi + U(\mathbf{x})\psi, \end{aligned}$$

where the **Hamiltonian** $\hat{H}$ is the operator version of the classical energy $\frac{\mathbf{p}^2}{2m} + U(\mathbf{x})$ of a particle of mass m moving in a potential U – notice it is exactly what you get by substituting $\hat{p}_i$ for p_i . The momentum operator looks like a logical leap right now; we will motivate it when we discuss the Schrödinger equation, and again (better) when we compute expectation values.

Next, measurement. You'll notice that in classical mechanics, an observable is a function that takes a state and produces some numerical value – something that can be thought of as reading out a value from some measurement device. For quantum mechanics we work with states in a vector space, so if we had two different states with two different

values of an observable, then since the sum of states is also a state – what would the observable read on the sum? There is no consistent classical answer. The way quantum mechanics deals with this is as follows.

Axiom 21.2.13 (Measurement)

The possible outcomes of a measurement of the observable O are the eigenvalues of the operator $\hat{O}$. The states in which O has a definite, well-defined value are exactly the eigenfunctions of $\hat{O}$: if $\hat{O}\psi = \lambda\psi$, then measuring O in the state ψ returns λ with certainty.

Axiom 21.2.14 (The Born Rule, Amplitude Form)

Suppose ψ_1 and ψ_2 are normalised eigenfunctions of $\hat{O}$ with distinct eigenvalues $\lambda_1 \neq \lambda_2$, and consider the superposition

$$\psi = \alpha_1\psi_1 + \alpha_2\psi_2, \quad \alpha_1, \alpha_2 \in \mathbb{C}.$$

Then ψ does not have a well-defined value of O . If O is measured in this state, the result is λ_1 or λ_2 , with probabilities

$$\mathbb{P}(\lambda_1) = \frac{|\alpha_1|^2}{|\alpha_1|^2 + |\alpha_2|^2}, \quad \mathbb{P}(\lambda_2) = \frac{|\alpha_2|^2}{|\alpha_1|^2 + |\alpha_2|^2},$$

and immediately after a measurement returning λ_i , the state of the system is ψ_i .

The general form of this rule (for arbitrarily many eigenfunctions, and with degenerate eigenvalues) will be given in Section 6, once we have established that eigenfunctions of Hermitian operators really do form complete orthonormal sets. Note that this axiom implies that a global phase on a state makes no difference to any probability – but relative phases between the α_i *can* make a difference to the probabilities of other observables, which is precisely the interference we saw in the double slit experiment. Note also the final sentence – the notorious ‘collapse of the wavefunction’. Measurement changes the state, discontinuously.

Finally, dynamics.

Axiom 21.2.15 (Time Evolution)

Between measurements, the wavefunction evolves according to the **time-dependent Schrödinger equation**

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi,$$

where $\hat{H}$ is the Hamiltonian operator of the system.

The Hamiltonian is the distinguished observable of the theory: the energy determines the dynamics. The whole of the next section is devoted to this equation.

A final word on the shape of the framework. Where classical mechanics had (phase space, observable functions, Newton/Hamilton’s equations), quantum mechanics has (Hilbert space, Hermitian operators, the Schrödinger equation) – the parallel we promised. But the measurement axioms have no classical counterpart at all: they are the genuinely new

– and genuinely strange – ingredient, and interpreting what they ‘really mean’ (what counts as a measurement? what happens during collapse?) remains contested a century on. Mercifully, using the rules requires no philosophy, and their predictions have never once failed an experimental test. We shall calculate.

§21.3 The Schrödinger Equation

§21.3.1 The Time-Dependent Schrödinger Equation

For a particle of mass m moving in a potential $U(\mathbf{x})$, the time evolution axiom reads explicitly

$$i\hbar \frac{\partial \psi}{\partial t}(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + U(\mathbf{x})\psi(\mathbf{x}, t).$$

This equation is a postulate – it is not derived from anything – but we can at least see that it is a *sensible* postulate, by checking it against de Broglie’s matter waves.

Consider a free particle, $U = 0$, and try the plane wave

$$\psi(\mathbf{x}, t) = \exp(i(\mathbf{k} \cdot \mathbf{x} - \omega t)).$$

Substituting into the Schrödinger equation, the left hand side gives $\hbar\omega\psi$ and the right hand side gives $\frac{\hbar^2|\mathbf{k}|^2}{2m}\psi$, so the plane wave is a solution precisely when

$$\hbar\omega = \frac{\hbar^2|\mathbf{k}|^2}{2m}.$$

Using the de Broglie relations $E = \hbar\omega$ and $\mathbf{p} = \hbar\mathbf{k}$, this dispersion relation is exactly

$$E = \frac{|\mathbf{p}|^2}{2m},$$

the correct energy-momentum relation for a free non-relativistic particle. So the Schrödinger equation is precisely the wave equation whose dispersion relation encodes Newtonian kinematics via de Broglie’s dictionary. This also motivates the momentum operator: acting on the plane wave,

$$-i\hbar\nabla\psi = \hbar\mathbf{k}\psi = \mathbf{p}\psi,$$

so plane waves are eigenfunctions of $\hat{\mathbf{p}} = -i\hbar\nabla$ with eigenvalue exactly their de Broglie momentum.

Remark. Two structural features are worth noting immediately.

- (i) The equation is *first order in time*. Contrast Newton’s second law, which is second order: there, we needed both position and momentum as initial data. Here a single initial condition – the wavefunction $\psi(\mathbf{x}, t_0)$ – determines the entire future evolution. The wavefunction really is the whole state.
- (ii) Time and space appear asymmetrically (first derivative against second derivative). This is how it must be for a non-relativistic theory, but it means the Schrödinger equation cannot be compatible with special relativity, where space and time must be treated on an equal footing. The relativistic story (the Dirac equation, quantum field theory) is well beyond this course.

Remark. The equation is also *linear*: any complex linear combination of solutions is a solution. This is forced on us by the superposition principle, and it is what makes the linear algebra of the previous section the right toolkit.

Remark (What the Wavefunction is Not). It is tempting – and was tempting historically, to Schrödinger himself – to picture ψ as a physical wave rippling through space, like an electromagnetic field. The Born rule already strains this picture (the wave would have to ‘collapse’ instantaneously everywhere upon measurement), but the decisive objection is that for *two* particles the wavefunction is a function $\psi(\mathbf{x}_1, \mathbf{x}_2, t)$ on six-dimensional configuration space, not two waves in ordinary space – and it need not factorise as $\psi_1(\mathbf{x}_1)\psi_2(\mathbf{x}_2)$. (When it doesn’t, the particles are said to be *entangled*, which is the doorway to the strangest and most fertile parts of the subject – but multi-particle systems are beyond this course.) The wavefunction is best regarded as what it is: the mathematical repository of everything knowable about the system.

§21.3.2 Conservation of Probability

There is a consistency check we had better perform. The Born rule requires the total probability $\int |\psi|^2 dV$ to equal 1 at all times, but the Schrödinger equation tells the wavefunction how to evolve – these had better be compatible, i.e. the normalisation of the wavefunction had better be preserved by time evolution. It is, and the proof reveals a conserved current, in close analogy with the conservation of charge in electromagnetism or mass in fluid dynamics.

Theorem 21.3.1 (Conservation of Probability)

Let ψ solve the time-dependent Schrödinger equation with real potential U , and suppose $\psi \rightarrow 0$ suitably fast as $|\mathbf{x}| \rightarrow \infty$. Define the **probability current**

$$\mathbf{J}(\mathbf{x}, t) = -\frac{i\hbar}{2m} (\psi^* \nabla \psi - \psi \nabla \psi^*).$$

Then the probability density $\rho = |\psi|^2$ satisfies the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = 0,$$

and consequently $\int_{\mathbb{R}^3} |\psi(\mathbf{x}, t)|^2 dV$ is constant in time.

Proof. The Schrödinger equation and its complex conjugate give

$$\frac{\partial \psi}{\partial t} = \frac{i\hbar}{2m} \nabla^2 \psi - \frac{i}{\hbar} U \psi, \quad \frac{\partial \psi^*}{\partial t} = -\frac{i\hbar}{2m} \nabla^2 \psi^* + \frac{i}{\hbar} U \psi^*,$$

where we used that U is real. Then

$$\begin{aligned} \frac{\partial \rho}{\partial t} &= \frac{\partial}{\partial t} (\psi^* \psi) = \psi^* \frac{\partial \psi}{\partial t} + \psi \frac{\partial \psi^*}{\partial t} \\ &= \frac{i\hbar}{2m} (\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^*) + \frac{i}{\hbar} (-U \psi^* \psi + U \psi \psi^*) \\ &= \frac{i\hbar}{2m} \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) = -\nabla \cdot \mathbf{J}, \end{aligned}$$

where in the second-to-last step we used $\nabla \cdot (\psi^* \nabla \psi) = \psi^* \nabla^2 \psi + \nabla \psi^* \cdot \nabla \psi$, the cross terms cancelling. This is the continuity equation. Integrating over a large ball B_R

and applying the divergence theorem,

$$\frac{d}{dt} \int_{B_R} |\psi|^2 dV = - \int_{\partial B_R} \mathbf{J} \cdot d\mathbf{S} \rightarrow 0 \text{ as } R \rightarrow \infty,$$

since ψ (and hence $\mathbf{J}$) vanishes at infinity. So the total integral is conserved. $\square$

The continuity equation is a stronger statement than mere conservation of the total: it says probability is conserved *locally*. Probability cannot teleport from one region to another; it must flow through the intervening space, with flux $\mathbf{J}$. We will use the current $\mathbf{J}$ quantitatively when we study scattering.

Example 21.3.2 (Currents of Basic States)

Two instructive special cases (in one dimension, where $J = -\frac{i\hbar}{2m}(\psi^*\psi' - \psi\psi'^*)$):

- (i) *Real wavefunctions carry no current.* If $\psi = \psi^*$ then the two terms cancel and $J = 0$ – and more generally the same holds if ψ is real up to a constant phase. All our bound states below can be taken real, so they carry no flux anywhere: they really are ‘stationary’ in the strongest sense.

- (ii) *Plane waves carry uniform current.* For $\psi = Ae^{i(kx - \omega t)}$,

$$J = -\frac{i\hbar}{2m} (|A|^2(ik) - |A|^2(-ik)) = |A|^2 \frac{\hbar k}{m},$$

which is (density) $\times$ (velocity), with the velocity $p/m = \hbar k/m$. Probability flows steadily to the right (for $k > 0$), consistent with ρ being constant: inflow balances outflow everywhere.

Complexity of the wavefunction, in both senses of the word, is where motion lives.

§21.3.3 Stationary States and the Time-Independent Schrödinger Equation

Faced with a linear PDE, we separate variables. Try a solution of the form

$$\psi(\mathbf{x}, t) = \chi(\mathbf{x}) T(t).$$

Substituting into the time-dependent Schrödinger equation and dividing through by χT ,

$$i\hbar \frac{T'(t)}{T(t)} = \frac{\hat{H}\chi(\mathbf{x})}{\chi(\mathbf{x})}.$$

The left hand side depends only on t and the right hand side only on $\mathbf{x}$, so both are equal to a constant, E say. The time equation $i\hbar T' = ET$ immediately gives

$$T(t) = e^{-iEt/\hbar},$$

while the spatial equation is the following.

Definition 21.3.3 (Time-Independent Schrödinger Equation)

The **time-independent Schrödinger equation** is the eigenvalue equation

$$\hat{H}\chi = E\chi, \quad \text{i.e.} \quad -\frac{\hbar^2}{2m}\nabla^2\chi(\mathbf{x}) + U(\mathbf{x})\chi(\mathbf{x}) = E\chi(\mathbf{x}).$$

That is, χ is an eigenfunction of the Hamiltonian with eigenvalue E – by the measurement axiom, a state of definite energy E . (Since $\hat{H}$ is Hermitian, E is guaranteed to be real; a complex E would also make $|T(t)|$ blow up either forwards or backwards in time, destroying normalisation.)

Definition 21.3.4 (Stationary State)

A **stationary state** of energy E is a solution of the time-dependent Schrödinger equation of the form

$$\psi(\mathbf{x}, t) = \chi(\mathbf{x}) e^{-iEt/\hbar},$$

where χ satisfies the time-independent Schrödinger equation with eigenvalue E .

The name is explained by computing the probability density:

$$\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2 = |\chi(\mathbf{x})|^2 \left| e^{-iEt/\hbar} \right|^2 = |\chi(\mathbf{x})|^2,$$

which is independent of time. Nothing observable about a stationary state ever changes – the time dependence is a pure phase. This, incidentally, resolves the stability problem of the atom: an electron in a stationary state has literally nothing about it that evolves, so there is nothing to radiate.

Why do we care so much about this special class of solutions? Because of linearity, and a completeness fact we will establish in Section 6: *every* solution of the time-dependent Schrödinger equation is a superposition of stationary states. If the Hamiltonian has (say, discrete) eigenfunctions χ_n with eigenvalues E_n , the general solution is

$$\psi(\mathbf{x}, t) = \sum_n a_n \chi_n(\mathbf{x}) e^{-iE_n t/\hbar}, \quad a_n \in \mathbb{C},$$

and the coefficients are fixed by the initial data: $a_n = \langle \chi_n, \psi(\cdot, 0) \rangle$. Solving the time-independent equation is therefore not a warm-up problem – it is the whole problem. Once you know the spectrum of $\hat{H}$, time evolution is just attaching phases.

In general the Hamiltonian can also have a continuous family of eigenfunctions χ_α with energies E_α (we will meet this for the free particle), in which case sums are replaced by integrals:

$$\psi(\mathbf{x}, t) = \int A(\alpha) \chi_\alpha(\mathbf{x}) e^{-iE_\alpha t/\hbar} d\alpha.$$

It is worth seeing explicitly how superposition brings back genuine time dependence, given that each individual stationary state has none.

Example 21.3.5 (A Two-State Superposition)

Suppose χ_1, χ_2 are orthonormal (real) stationary states with distinct energies $E_1 \neq E_2$, and consider

$$\psi(\mathbf{x}, t) = \frac{1}{\sqrt{2}} \left(\chi_1(\mathbf{x}) e^{-iE_1 t/\hbar} + \chi_2(\mathbf{x}) e^{-iE_2 t/\hbar} \right).$$

The probability density is

$$|\psi|^2 = \frac{1}{2} \left(\chi_1^2 + \chi_2^2 + 2\chi_1\chi_2 \cos \left(\frac{(E_2 - E_1)t}{\hbar} \right) \right),$$

which *oscillates* in time, at the single angular frequency

$$\omega_{21} = \frac{E_2 - E_1}{\hbar}.$$

The probability sloshes back and forth between the regions where $\chi_1\chi_2 > 0$ and those where $\chi_1\chi_2 < 0$. Notice that this is precisely the Bohr transition frequency between the two levels – it is superpositions like this, driven by a coupling to the electromagnetic field, that radiate at the observed spectral frequencies. Note also that the *relative phase* between the two terms is doing real physical work here, as promised in Section 2: replacing $\chi_2 \mapsto e^{i\theta}\chi_2$ shifts the phase of the oscillation.

A large portion of these notes – all of the next two sections – consists of solving the time-independent Schrödinger equation for various potentials U , and interpreting the answers. It is remarkable how much physics is contained in a single ordinary differential equation.

§21.4 One-Dimensional Systems

We now restrict to one spatial dimension, where the time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\chi''(x) + U(x)\chi(x) = E\chi(x)$$

is an ordinary differential equation, and everything can be computed explicitly. This is not just a pedagogical simplification – the one-dimensional problems in this section are the hydrogen atoms of quantum mechanics, and (as we will see in Section 7) genuinely three-dimensional problems often reduce to one-dimensional ones.

§21.4.1 The Free Particle and Wavepackets

Let's start with the simplest possible potential: $U = 0$. The time-independent Schrödinger equation is $-\frac{\hbar^2}{2m}\chi'' = E\chi$, with solutions

$$\chi_k(x) = e^{ikx}, \quad E = \frac{\hbar^2 k^2}{2m},$$

for any $k \in \mathbb{R}$. Attaching the time dependence, we get the **de Broglie plane waves**

$$\psi_k(x, t) = \exp \left(ikx - \frac{i\hbar k^2}{2m}t \right),$$

which are momentum eigenstates: $\hat{p}\psi_k = \hbar k\psi_k$. Notice that the energy spectrum is *continuous* – any $E \geq 0$ occurs – in sharp contrast with the bound systems we study later.

There is, however, an embarrassment:

$$\int_{-\infty}^{\infty} |\psi_k(x, t)|^2 dx = \int_{-\infty}^{\infty} 1 dx = \infty,$$

so plane waves are not normalisable, and strictly speaking are not states in $\mathcal{H}$ at all⁵. There are two standard ways of dealing with this, and both are physically illuminating:

- (i) build normalisable solutions as superpositions of plane waves – *wavepackets*;
- (ii) keep the plane waves but reinterpret them as describing steady *beams* of particles rather than single particles (we will do this when we study scattering).

Let's carry out the first honestly. By the superposition principle, for any (reasonable) amplitude function $A(k)$,

$$\psi(x, t) = \int_{-\infty}^{\infty} A(k) \exp\left(ikx - \frac{i\hbar k^2}{2m}t\right) dk$$

solves the time-dependent Schrödinger equation. Choose a Gaussian profile peaked around a wavenumber k_0 :

$$A(k) = \exp\left(-\frac{\sigma}{2}(k - k_0)^2\right), \quad \sigma > 0.$$

The resulting solution is called the **Gaussian wavepacket**. The integral is doable by hand: the exponent is

$$F(k) = -\frac{\sigma}{2}(k - k_0)^2 + ikx - \frac{i\hbar k^2}{2m}t = -\frac{\alpha}{2}k^2 + \beta k + \delta,$$

where

$$\alpha = \sigma + \frac{i\hbar t}{m}, \quad \beta = \sigma k_0 + ix, \quad \delta = -\frac{\sigma}{2}k_0^2.$$

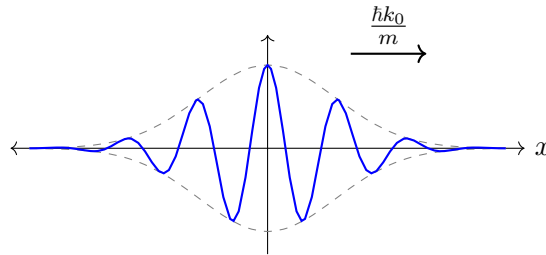
Completing the square, $F(k) = -\frac{\alpha}{2}\left(k - \frac{\beta}{\alpha}\right)^2 + \frac{\beta^2}{2\alpha} + \delta$, and shifting the contour of integration (justified since the integrand is entire and decays), we are left with a standard Gaussian integral $\int_{-\infty}^{\infty} e^{-\alpha u^2/2} du = \sqrt{2\pi/\alpha}$, giving

$$\psi(x, t) = \sqrt{\frac{2\pi}{\alpha}} \exp\left(\frac{\beta^2}{2\alpha} + \delta\right).$$

This looks opaque, but taking the modulus squared and simplifying (a worthwhile exercise in complex bookkeeping),

$$\rho(x, t) = |\bar{\psi}(x, t)|^2 = \sqrt{\frac{\sigma}{\pi\left(\sigma^2 + \frac{\hbar^2 t^2}{m^2}\right)}} \exp\left(-\frac{\sigma\left(x - \frac{\hbar k_0}{m}t\right)^2}{\sigma^2 + \frac{\hbar^2 t^2}{m^2}}\right),$$

where $\bar{\psi}$ is the normalised wavepacket. This is a Gaussian bump, and we can read off the physics directly.



⁵This is a general feature: normalisability requires $\chi \rightarrow 0$ at infinity, so any eigenfunction that remains oscillatory out to infinity – any ‘unbound’ state – fails to be normalisable. The plane wave is the extreme case.

The peak of the bump is at

$$\langle x \rangle = \int_{-\infty}^{\infty} x \rho(x, t) dx = \frac{\hbar k_0}{m} t,$$

so the packet moves with constant velocity $\hbar k_0/m = p_0/m$, exactly the classical velocity of a particle of momentum $p_0 = \hbar k_0$. The width of the bump,

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{1}{2} \left(\sigma + \frac{\hbar^2 t^2}{m^2 \sigma} \right)},$$

grows with time: wavepackets spread. This is dispersion – the different plane-wave components travel at different speeds, so the packet cannot hold itself together. One can similarly compute (using the momentum operator $\hat{p} = -i\hbar \frac{\partial}{\partial x}$)

$$\langle p \rangle = \int_{-\infty}^{\infty} \bar{\psi}^* \left(-i\hbar \frac{\partial}{\partial x} \right) \bar{\psi} dx = \hbar k_0, \quad \Delta p = \frac{\hbar}{\sqrt{2 \left(\sigma + \frac{\hbar^2 t^2}{m^2 \sigma} \right)}},$$

so momentum is sharp to precision Δp , which *shrinks* as the packet spreads. At $t = 0$ we find the tidy relation

$$\Delta x \Delta p = \frac{\hbar}{2}.$$

A tightly localised packet (small σ) has a wildly uncertain momentum, and vice versa – with the product bounded below. Compare the two extremes: a plane wave has $\Delta p = 0$ but $\Delta x = \infty$. This trade-off is not an artefact of the Gaussian: it is the *uncertainty principle*, which we will state and prove in general in Section 6 – and we will see there that the Gaussian is precisely the minimising shape.

§21.4.2 The Infinite Square Well

Our first genuinely quantum system is a particle trapped in a box: the **infinite square well**

$$U(x) = \begin{cases} 0 & |x| \leq a, \\ \infty & |x| > a. \end{cases}$$

The infinite potential should be understood as a limit (of the finite well of the next-but-one subsection, say): its effect is that the wavefunction must vanish wherever $U = \infty$, since otherwise the term $U\chi$ in the Schrödinger equation is infinite. So $\chi(x) = 0$ for $|x| > a$, and by continuity we get boundary conditions

$$\chi(\pm a) = 0.$$

(At an infinite jump in U we insist only that χ is continuous; χ' is allowed to jump. At any finite discontinuity of U , both χ and χ' must be continuous – we will use this repeatedly.)

Inside the well the equation is free:

$$\chi'' + k^2 \chi = 0, \quad k = \sqrt{\frac{2mE}{\hbar^2}},$$

with general solution $\chi(x) = A \sin kx + B \cos kx$. (One can check directly that $E \leq 0$ gives only the zero solution.) The boundary conditions read

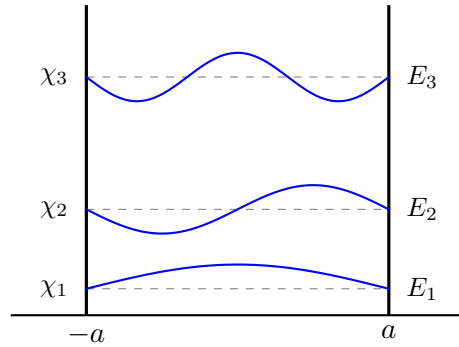
$$A \sin ka + B \cos ka = 0, \quad -A \sin ka + B \cos ka = 0,$$

adding and subtracting: $B \cos ka = 0$ and $A \sin ka = 0$. We cannot have both $A = B = 0$, and $\sin ka, \cos ka$ never vanish together, so exactly one of the following holds:

- (i) $A = 0$ and $\cos ka = 0$, giving $k = \frac{n\pi}{2a}$ with n odd, and even eigenfunctions $\chi \propto \cos kx$;
- (ii) $B = 0$ and $\sin ka = 0$, giving $k = \frac{n\pi}{2a}$ with n even, and odd eigenfunctions $\chi \propto \sin kx$.

Fixing the constants by normalisation, $\int_{-a}^a |\chi_n|^2 dx = 1$, we obtain a discrete family of stationary states, one for each positive integer n :

$$\chi_n(x) = \frac{1}{\sqrt{a}} \begin{cases} \cos \frac{n\pi x}{2a} & n \text{ odd,} \\ \sin \frac{n\pi x}{2a} & n \text{ even,} \end{cases} \quad E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 \pi^2 n^2}{8ma^2}.$$



This small computation displays most of the characteristic features of bound quantum systems, so let's pause and stare at it.

- (i) *The energy is quantised.* Only the discrete values $E_n \propto n^2$ occur – the boundary conditions act exactly like the ‘standing wave’ condition in de Broglie’s picture: an integer number of half-wavelengths must fit in the box.
- (ii) *The ground state energy is not zero.* The lowest possible energy is $E_1 = \hbar^2 \pi^2 / 8ma^2 > 0$: a confined quantum particle cannot sit still. This is forced by the uncertainty principle – confinement to $\Delta x \sim a$ requires $\Delta p \gtrsim \hbar/a$, and hence kinetic energy of order $\hbar^2/ma^2$.
- (iii) *Nodes.* The n th eigenfunction has $n - 1$ interior zeros, at which the particle is never found. It is genuinely strange that a particle can be found on both sides of a point it can never be found at; resist the temptation to think of the particle as ‘moving’ – the state is stationary.
- (iv) *The classical limit.* For large n , the density $|\chi_n|^2$ oscillates so rapidly that any realistic position measurement averages it out to the constant $1/2a$ – the uniform distribution a classical bouncing particle would give.

Remark. The eigenfunctions $\{\chi_n\}$ are precisely a Fourier basis for $[-a, a]$, so completeness of eigenfunctions of $\hat{H}$ is, in this example, exactly the classical statement that Fourier series converge. The general solution of the time-dependent problem in the box,

$$\psi(x, t) = \sum_{n=1}^{\infty} a_n \chi_n(x) e^{-iE_n t/\hbar},$$

is a Fourier series with rotating coefficients.

Example 21.4.1 (Position Statistics in the Ground State)

Let's compute the position statistics of the ground state $\chi_1 = \frac{1}{\sqrt{a}} \cos \frac{\pi x}{2a}$. By symmetry $\langle x \rangle = 0$ (the density is even), and

$$\langle x^2 \rangle = \frac{1}{a} \int_{-a}^a x^2 \cos^2 \frac{\pi x}{2a} dx = \frac{1}{2a} \int_{-a}^a x^2 \left(1 + \cos \frac{\pi x}{a}\right) dx = \frac{a^2}{3} - \frac{2a^2}{\pi^2},$$

where the second integral is done by integrating by parts twice. So

$$\Delta x = a \sqrt{\frac{1}{3} - \frac{2}{\pi^2}} \approx 0.36 a,$$

somewhat less than the value $a/\sqrt{3} \approx 0.58 a$ for a uniform distribution – the ground state huddles towards the middle of the box. For the momentum, $\langle p \rangle = 0$ (again by symmetry – or note χ_1 is real, so its current J vanishes), while $\langle p^2 \rangle = 2mE_1 = \hbar^2 \pi^2 / 4a^2$, since χ_1 is an eigenfunction of $\hat{p}^2 = 2m\hat{H}$ inside the well. Multiplying,

$$\Delta x \Delta p = \frac{\pi \hbar}{2} \sqrt{\frac{1}{3} - \frac{2}{\pi^2}} \approx 0.57 \hbar > \frac{\hbar}{2},$$

comfortably (but not extravagantly) consistent with the uncertainty principle we will prove in Section 6.

Exercise 21.4.2. Show that in the n th eigenstate, $\Delta x \Delta p = \frac{n\pi\hbar}{2} \sqrt{\frac{1}{3} - \frac{2}{n^2\pi^2}}$, which grows linearly in n – highly excited states are far from minimal uncertainty.

§21.4.3 Parity

You will have noticed that every eigenfunction of the infinite well was either even or odd. This is not a coincidence, and it is worth isolating the mechanism, since we will use it as a labour-saving device repeatedly.

Proposition 21.4.3 (Parity of Eigenfunctions)

Suppose the potential is even, $U(-x) = U(x)$, and that χ is an eigenfunction of $\hat{H}$ whose eigenvalue E is *non-degenerate* (no other linearly independent eigenfunction has the same eigenvalue). Then χ is either even or odd.

Proof. If U is even then the map $x \mapsto -x$ leaves the time-independent Schrödinger equation unchanged, so if $\chi(x)$ is a solution with eigenvalue E , so is $\tilde{\chi}(x) = \chi(-x)$. By non-degeneracy, $\tilde{\chi}$ must be proportional to χ , say $\chi(-x) = \alpha\chi(x)$. Applying this twice,

$$\chi(x) = \chi(-(-x)) = \alpha\chi(-x) = \alpha^2\chi(x),$$

so $\alpha^2 = 1$, i.e. $\alpha = \pm 1$. Then $\alpha = 1$ makes χ even and $\alpha = -1$ makes it odd. $\square$

Even when eigenvalues are degenerate, we can always *choose* a basis of even and odd eigenfunctions (take $\chi(x) \pm \chi(-x)$). The upshot for computations: for a symmetric potential we may search for even solutions and odd solutions separately, which roughly halves the work. In the language of operators, the **parity operator** $\hat{P}\chi(x) = \chi(-x)$ commutes with $\hat{H}$, and we are choosing simultaneous eigenfunctions of both – a point

we will return to in Section 6.

In fact, in one dimension the non-degeneracy hypothesis comes for free, which is why every bound state we compute in this section is automatically even or odd.

Proposition 21.4.4 (No Degeneracy in One Dimension)

Bound state energy levels of the one-dimensional Schrödinger equation are non-degenerate.

Proof. Suppose χ_1 and χ_2 are both normalisable eigenfunctions with the same eigenvalue E . Consider the Wronskian

$$W(x) = \chi_1 \chi_2' - \chi_2 \chi_1'.$$

Differentiating and using the Schrödinger equation for each ($\chi_i'' = \frac{2m}{\hbar^2}(U - E)\chi_i$),

$$W' = \chi_1 \chi_2'' - \chi_2 \chi_1'' = \frac{2m}{\hbar^2}(U - E)(\chi_1 \chi_2 - \chi_2 \chi_1) = 0,$$

so W is constant. Since bound states (and their derivatives) decay at infinity, $W \rightarrow 0$, so $W \equiv 0$. Then wherever the functions are non-zero,

$$\frac{\chi_1'}{\chi_1} = \frac{\chi_2'}{\chi_2},$$

and integrating gives $\ln \chi_1 = \ln \chi_2 + \text{const}$, i.e. $\chi_1 \propto \chi_2$. So the two eigenfunctions represent the same state. $\square$

(In three dimensions this argument collapses – there is no Wronskian – and degeneracy returns with a vengeance, as we will see: it is one of the main new phenomena of Section 7.)

Exercise 21.4.5. Show that the parity operator $\hat{P}$ is Hermitian, that $\hat{P}^2 = \hat{I}$, and deduce that its eigenvalues are ± 1 with the eigenfunctions being exactly the even and odd functions. Show also that $[\hat{H}, \hat{P}] = 0$ if and only if the potential is even.

§21.4.4 The Finite Square Well

Now let's make the walls finite: consider

$$U(x) = \begin{cases} 0 & |x| \leq a, \\ U_0 & |x| > a, \end{cases}$$

with $U_0 > 0$. We look for **bound states**: normalisable eigenfunctions, which requires $0 < E < U_0$ (for $E > U_0$ solutions oscillate at infinity, giving the scattering states of the next section). Classically a particle with $E < U_0$ is imprisoned in $|x| \leq a$ forever. Quantum mechanically, something more interesting happens.

By parity, we may look for even and odd solutions separately; we will do the even case in detail. Write

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}},$$

both real and positive. Inside the well, $\chi'' = -k^2\chi$, and the even solution is $\chi = B \cos kx$. Outside, $\chi'' = \kappa^2\chi$, with solutions $e^{\pm\kappa x}$; normalisability kills the growing exponential on each side, and evenness ties the two coefficients together, leaving

$$\chi(x) = \begin{cases} Ce^{\kappa x} & x < -a, \\ B \cos kx & |x| \leq a, \\ Ce^{-\kappa x} & x > a. \end{cases}$$

Since U has only a finite jump at $x = \pm a$, both χ and χ' must be continuous there. At $x = a$:

$$B \cos ka = Ce^{-\kappa a}, \quad -kB \sin ka = -\kappa Ce^{-\kappa a}.$$

Dividing the second equation by the first (both sides are non-zero for a non-trivial solution),

$$k \tan ka = \kappa.$$

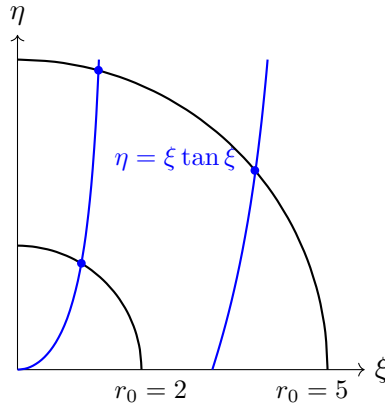
This, together with the defining relation

$$k^2 + \kappa^2 = \frac{2mU_0}{\hbar^2},$$

determines the allowed energies. These are transcendental equations, so we solve them graphically. Introduce the dimensionless variables $\xi = ka$ and $\eta = \kappa a$; then we need the intersections of

$$\eta = \xi \tan \xi \quad \text{and} \quad \xi^2 + \eta^2 = r_0^2, \quad r_0 = a\sqrt{\frac{2mU_0}{\hbar^2}},$$

in the quadrant $\xi, \eta > 0$: the branches of $\xi \tan \xi$ against a circle whose radius r_0 measures how ‘big’ the well is (in depth and width together).



Each intersection (ξ_n, η_n) gives a bound state with energy

$$E_n = \frac{\hbar^2 \xi_n^2}{2ma^2}.$$

Reading off the picture:

- (i) There is *always* at least one even bound state, no matter how small r_0 (i.e. how shallow or narrow the well): the first branch of $\xi \tan \xi$ starts at the origin, so the circle always crosses it.

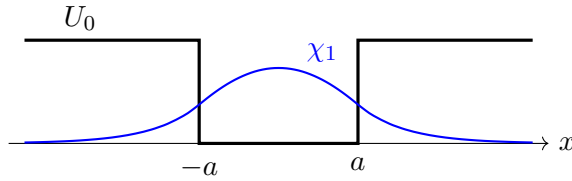
- (ii) There are only ever *finitely many* bound states: the circle has finite radius, and the branches of $\xi \tan \xi$ are spaced π apart. Deeper and wider wells (larger r_0) support more bound states – in the figure, $r_0 = 2$ gives one even bound state, while $r_0 = 5$ gives two.
- (iii) As $U_0 \rightarrow \infty$, $r_0 \rightarrow \infty$ and the intersections approach the asymptotes $\xi_n \rightarrow \frac{n\pi}{2}$ (n odd) – exactly the even levels of the infinite well, as they had better.

The odd solutions run identically: inside the well $\chi = A \sin kx$, outside the same decaying exponentials but now with opposite signs on the two sides, and the matching at $x = a$ gives

$$-k \cot ka = \kappa, \quad \text{i.e.} \quad -\xi \cot \xi = \eta.$$

The branches of $-\xi \cot \xi$ interleave with those of $\xi \tan \xi$ (the first becomes positive only for $\xi > \pi/2$), so the even and odd levels alternate up the spectrum, just as in the infinite well – and an odd bound state exists only once $r_0 > \pi/2$, i.e. only if the well is deep enough. This condition will return with new significance in the three-dimensional theory (§7.2).

Here is what the ground state actually looks like:



The striking feature is the exponential tails: $\chi \neq 0$ in the region $|x| > a$, which a classical particle of this energy can never enter. The particle has positive probability of being found in the *classically forbidden region*. The tails decay over the length scale $1/\kappa$; note that the closer E is to the top of the well, the smaller κ and the further the state leaks out. This leakage is the seed of *quantum tunnelling*, which we will meet properly in the next section.

§21.4.5 The Harmonic Oscillator

The final – and most important – one-dimensional system is the **harmonic oscillator**:

$$U(x) = \frac{1}{2}m\omega^2 x^2.$$

Its importance comes from a Taylor expansion: near a generic stable equilibrium x_0 of *any* potential, $U(x) \approx U(x_0) + \frac{1}{2}U''(x_0)(x - x_0)^2$, so every small oscillation in nature – molecular vibrations, phonons in crystals, modes of the electromagnetic field – is approximately a harmonic oscillator. Classically the particle oscillates as $x = A \cos(\omega t + \phi)$ with any amplitude, so any energy $E \geq 0$ is possible.

The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\chi''(x) + \frac{1}{2}m\omega^2 x^2 \chi(x) = E\chi(x).$$

First we tidy up: the natural length scale is $\sqrt{\hbar/m\omega}$, so substitute

$$\xi = \sqrt{\frac{m\omega}{\hbar}} x, \quad \varepsilon = \frac{2E}{\hbar\omega},$$

which reduces the equation to the dimensionless form

$$-\frac{d^2\chi}{d\xi^2} + \xi^2\chi = \varepsilon\chi.$$

Let's think about how solutions must behave for large $|\xi|$: there $\chi'' \approx \xi^2\chi$, which has solutions behaving like $e^{\pm\xi^2/2}$, and normalisability forces the decaying one. Indeed, we can spot one exact solution immediately: $\chi = e^{-\xi^2/2}$ satisfies $\chi'' = (\xi^2 - 1)\chi$, so it is an eigenfunction with $\varepsilon = 1$, that is

$$\chi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right), \quad E_0 = \frac{\hbar\omega}{2}.$$

A Gaussian! (And in fact this is the ground state, as the count of nodes will confirm.)

Guided by the asymptotics, for the general solution we make the ansatz

$$\chi(\xi) = f(\xi) e^{-\xi^2/2}.$$

Substituting in, f must satisfy **Hermite's equation**:

$$f'' - 2\xi f' + (\varepsilon - 1)f = 0.$$

The point $\xi = 0$ is regular, so we look for a power series solution $f(\xi) = \sum_{n \geq 0} a_n \xi^n$. Substituting and comparing coefficients of ξ^n :

$$(n+2)(n+1)a_{n+2} - 2na_n + (\varepsilon - 1)a_n = 0,$$

which gives the recurrence

$$a_{n+2} = \frac{2n+1-\varepsilon}{(n+1)(n+2)} a_n.$$

The even coefficients are determined by a_0 and the odd ones by a_1 ; by parity (the potential is even!) we may take exactly one of a_0, a_1 to be non-zero.

Now, the crucial question: does the series terminate? Here is why it must.

Proposition 21.4.6

If the series for f does not terminate, then $\chi = f e^{-\xi^2/2}$ is not normalisable.

Proof. Suppose the series is infinite. For large n the recurrence gives

$$\frac{a_{n+2}}{a_n} = \frac{2n+1-\varepsilon}{(n+1)(n+2)} \sim \frac{2}{n}.$$

Compare with the function $g(\xi) = e^{\xi^2} = \sum_m \frac{\xi^{2m}}{m!} = \sum_n b_n \xi^n$, whose coefficients satisfy

$$\frac{b_{n+2}}{b_n} = \frac{(n/2)!}{(n/2+1)!} = \frac{2}{n+2} \sim \frac{2}{n}.$$

So the tails of the two series agree asymptotically, and $f(\xi)$ grows like e^{ξ^2} as $|\xi| \rightarrow \infty$ (this comparison can be made rigorous). But then

$$\chi(\xi) = f(\xi) e^{-\xi^2/2} \sim e^{\xi^2/2},$$

which blows up, and is certainly not normalisable. $\square$

So f must be a polynomial: the series must terminate. Looking at the recurrence, the coefficient a_{N+2} vanishes (with $a_N \neq 0$) exactly when

$$\varepsilon = 2N + 1, \quad N = 0, 1, 2, \dots,$$

and undoing the substitution $\varepsilon = 2E/\hbar\omega$, we have proven the following famous result.

Theorem 21.4.7 (Spectrum of the Harmonic Oscillator)

The energy levels of the quantum harmonic oscillator are

$$E_N = \left(N + \frac{1}{2}\right) \hbar\omega, \quad N = 0, 1, 2, \dots$$

The energy levels form an exactly evenly-spaced ladder, with spacing $\hbar\omega$ – precisely the quantum of energy $\hbar\omega$ that Planck and Einstein needed for radiation of frequency ω . The ground state energy $\frac{1}{2}\hbar\omega$ is again non-zero (the ‘zero-point energy’).

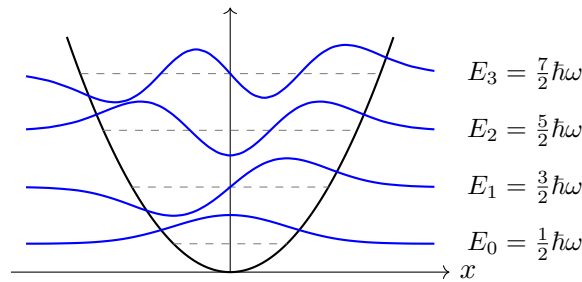
The corresponding polynomials $f_N =: H_N$, normalised conventionally, are the **Hermite polynomials**, the first few being

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad H_3(\xi) = 8\xi^3 - 12\xi,$$

and the eigenfunctions – the *Hermite functions* – are

$$\chi_N(\xi) = C_N H_N(\xi) e^{-\xi^2/2}, \quad C_N = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^N N!}},$$

each a polynomial oscillation inside a Gaussian envelope, with χ_N having exactly N nodes and parity $(-1)^N$. (Orthogonality of the χ_N , i.e. $\int H_M H_N e^{-\xi^2} d\xi = 2^N N! \sqrt{\pi} \delta_{MN}$, is automatic from the general theory of the next section: they are eigenfunctions of a Hermitian operator with distinct eigenvalues.)



Note that each χ_N leaks past the classical turning points (where the dashed level meets the parabola) into the classically forbidden region, just as in the finite well. And the classical limit works as it did in the box: a classical oscillator of amplitude A spends most time near the turning points (where it moves slowest), with position distribution $1/\pi\sqrt{A^2 - x^2}$, and for large N the rapidly-oscillating $|\chi_N|^2$ locally averages to exactly this – largest near the turning points, smallest in the middle – the opposite of the ground state, which piles up in the centre. The Hermite functions form a complete orthonormal set for $\mathcal{H}$ – another instance of the completeness of Hamiltonian eigenfunctions that we keep running into, and will address head-on in Section 6.

Example 21.4.8 (Half-Oscillator)

As a quick test of understanding, consider the potential $U(x) = \frac{1}{2}m\omega^2 x^2$ for $x > 0$ and $U(x) = \infty$ for $x \leq 0$. An eigenfunction must vanish at $x = 0$ and solve the oscillator equation for $x > 0$ – so the eigenfunctions are exactly the *odd* oscillator eigenfunctions restricted to $x > 0$, and the spectrum is

$$E = \left(2j + 1 + \frac{1}{2}\right) \hbar\omega, \quad j = 0, 1, 2, \dots,$$

i.e. every other level of the full oscillator, starting at $\frac{3}{2}\hbar\omega$.

Before moving on, it is worth collecting the qualitative features that all our bound state problems have shared, since they hold for essentially any one-dimensional potential well.

- (i) In the *classically allowed* region ($E > U(x)$), the Schrödinger equation reads $\chi'' = -k(x)^2\chi$: the eigenfunction oscillates, faster where $E - U$ is larger.
- (ii) In the *classically forbidden* region ($E < U(x)$), $\chi'' = \kappa(x)^2\chi$: the eigenfunction grows or decays roughly exponentially, and normalisability selects decay – giving the tunnelling tails we keep seeing.
- (iii) Matching an oscillation smoothly onto decaying tails at both ends is only possible for special values of E – this is, at bottom, *why* bound state energies are quantised.
- (iv) The n th bound state (ordering by energy) has $n - 1$ nodes: higher energy means faster oscillation means more sign changes. Together with non-degeneracy and parity, this gives a reliable sanity check on any claimed spectrum.

§21.5 Scattering in One Dimension

So far we have studied bound states – normalisable eigenfunctions with discrete energies. We now turn to the complementary situation: a particle fired *at* a potential from far away, which may bounce back or pass through. This is a scattering problem, and it is where the non-normalisable solutions we discarded earlier earn their keep.

§21.5.1 Beams and Probability Flux

The physically honest way to describe a localised incoming particle is a wavepacket, say a Gaussian wavepacket centred far to the left of the potential and moving right. At late times, the packet splits into a reflected packet moving left and a transmitted packet moving right, and we would define the **reflection and transmission coefficients**

$$R = \lim_{t \rightarrow \infty} \int_{-\infty}^0 |\psi(x, t)|^2 dx, \quad T = \lim_{t \rightarrow \infty} \int_0^{\infty} |\psi(x, t)|^2 dx,$$

the probabilities of each outcome, with $R + T = 1$ automatic from conservation of probability. This is doable but computationally painful. Happily, there is a much slicker approach: the **beam interpretation**.

Take the plane wave stationary state

$$\psi_k(x, t) = A e^{ikx} e^{-iE_k t/\hbar}, \quad E_k = \frac{\hbar^2 k^2}{2m},$$

and instead of apologising for $\rho_k = |A|^2$ being constant (non-normalisable), *reinterpret* it: ψ_k describes not one particle but a steady beam of independent particles, each with momentum $\hbar k$, with constant average density $|A|^2$ per unit length. The probability current of Section 3 becomes the particle flux of the beam:

$$J_k = -\frac{i\hbar}{2m} \left(\psi_k^* \frac{\partial \psi_k}{\partial x} - \psi_k \frac{\partial \psi_k^*}{\partial x} \right) = |A|^2 \frac{\hbar k}{m},$$

which is just (density) $\times$ (velocity p/m), exactly as for a classical beam. Reflection and transmission coefficients are then *ratios of fluxes*:

$$R = \frac{|J_{\text{ref}}|}{|J_{\text{inc}}|}, \quad T = \frac{|J_{\text{trans}}|}{|J_{\text{inc}}|}.$$

One can show (by building wavepackets out of these stationary states) that the two approaches agree, so we will happily use the easier one.

One structural fact makes flux bookkeeping in one dimension especially clean.

Proposition 21.5.1 (Flux is Constant in Stationary States)

For a stationary state in one dimension, the probability current $J(x)$ is independent of x .

Proof. For a stationary state, $\rho = |\chi|^2$ is independent of t , so the continuity equation $\partial_t \rho + \partial_x J = 0$ reduces to $\partial_x J = 0$. $\square$

So in any scattering problem, whatever flux flows in must flow out: evaluating J far to the left and far to the right of the potential and equating them will give us $R + T = 1$ automatically, whatever the shape of the obstacle in between.

§21.5.2 The Potential Step

Our first scattering problem is the **potential step**

$$U(x) = \begin{cases} 0 & x \leq 0, \\ U_0 & x > 0, \end{cases}$$

with a beam of particles of energy E incident from the left. There are two regimes, depending on whether the beam has enough energy to climb the step.

Case 1: $E > U_0$. Write

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \bar{k} = \sqrt{\frac{2m(E - U_0)}{\hbar^2}},$$

both real. The stationary Schrödinger equation solves separately on each side:

$$\chi(x) = \begin{cases} e^{ikx} + Be^{-ikx} & x \leq 0, \\ Ce^{i\bar{k}x} & x > 0. \end{cases}$$

We have already inserted the physics here, so let's spell it out. On the left, e^{ikx} is the incident beam (normalised to unit amplitude, which we may do) and Be^{-ikx} is

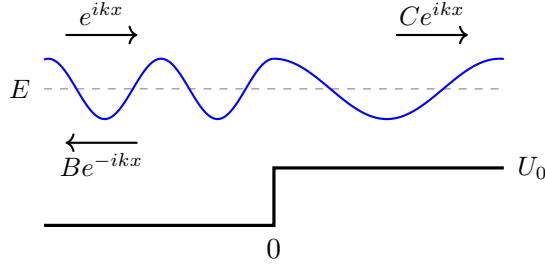
the reflected beam. On the right, $Ce^{i\bar{k}x}$ is the transmitted beam; a term $De^{-i\bar{k}x}$ would describe a beam coming in from $x = +\infty$, which our experimental setup does not contain, so $D = 0$.

U has a finite jump, so χ and χ' are continuous at $x = 0$:

$$1 + B = C, \quad ik(1 - B) = i\bar{k}C,$$

giving

$$B = \frac{k - \bar{k}}{k + \bar{k}}, \quad C = \frac{2k}{k + \bar{k}}.$$



Now compute fluxes. On the left the current splits into incident and reflected parts, $J_{\text{inc}} = \frac{\hbar k}{m}$ and $J_{\text{ref}} = -\frac{\hbar k}{m}|B|^2$; on the right $J_{\text{trans}} = \frac{\hbar \bar{k}}{m}|C|^2$. So

$$R = |B|^2 = \left(\frac{k - \bar{k}}{k + \bar{k}} \right)^2, \quad T = \frac{\bar{k}}{k}|C|^2 = \frac{4k\bar{k}}{(k + \bar{k})^2}.$$

Note the extra factor of $\bar{k}/k$ in T – transmission compares *fluxes*, not amplitudes, and the transmitted beam moves more slowly. You can check directly that

$$R + T = \frac{(k - \bar{k})^2 + 4k\bar{k}}{(k + \bar{k})^2} = 1,$$

as conservation of probability demands. Sanity checks: as $E \rightarrow U_0$ (so $\bar{k} \rightarrow 0$) we get $R \rightarrow 1$ – a barely-clearing beam is entirely reflected – and as $E \rightarrow \infty$ we get $T \rightarrow 1$. But notice that for any finite energy $R > 0$: *a quantum particle can be reflected by a step it has enough energy to climb*. This has no classical analogue.

Case 2: $E < U_0$. Now $\bar{k} = i\kappa$ is purely imaginary, with $\kappa = \sqrt{2m(U_0 - E)/\hbar^2} > 0$, and the solution for $x > 0$ becomes

$$\chi(x) = Ce^{-\kappa x},$$

where we discarded the growing exponential. The matching gives the same formulae with $\bar{k} = i\kappa$:

$$B = \frac{k - i\kappa}{k + i\kappa},$$

which has $|B| = 1$, so $R = 1$: total reflection, as a classical particle would do. But the wavefunction is *not* zero beyond the step – it decays exponentially, penetrating the forbidden region to depth $\sim 1/\kappa$, just like the tails of the finite-well bound states. There is no transmitted flux (χ real up to phase for $x > 0$ gives $J = 0$), so nothing is ‘getting through’ in steady state – but the non-zero amplitude in the wall is about to become very important.

Example 21.5.2 (Reflection by a Cliff)

Here is a variant that classical intuition gets badly wrong. Take a *downward* step, $U = 0$ for $x \leq 0$ and $U = -U_0 < 0$ for $x > 0$, with a beam incident from the left with energy $E > 0$. A classical particle simply speeds up over the edge, with certainty. Quantum mechanically the algebra of Case 1 applies verbatim with $\bar{k} = \sqrt{2m(E + U_0)}/\hbar > k$, so

$$R = \left(\frac{k - \bar{k}}{k + \bar{k}} \right)^2 > 0,$$

and in the low energy limit $E \ll U_0$ we have $\bar{k} \gg k$ and $R \rightarrow 1$: a slow particle is almost certainly *reflected by the attractive drop*. Reflection is caused by any sharp change in the potential – in either direction – because it is a wave phenomenon (an impedance mismatch), not a matter of insufficient energy. This is why cold neutrons bounce off material surfaces, which is quantitatively the basis of neutron mirrors and the storage of ultracold neutrons in bottles.

§21.5.3 The Potential Barrier and Tunnelling

Now make the wall thin: consider the **potential barrier**

$$U(x) = \begin{cases} U_0 & 0 < x < a, \\ 0 & \text{otherwise,} \end{cases}$$

with a beam incident from the left with energy $0 < E < U_0$ – classically, guaranteed to bounce off. Write as usual

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \quad \kappa = \sqrt{\frac{2m(U_0 - E)}{\hbar^2}}.$$

The solution has three pieces:

$$\chi(x) = \begin{cases} e^{ikx} + Ae^{-ikx} & x \leq 0, \\ Be^{-\kappa x} + Ce^{\kappa x} & 0 < x < a, \\ De^{ikx} & x \geq a. \end{cases}$$

Note that inside the barrier we must keep *both* exponentials – the barrier is finite, so neither blows up – and on the right we keep only the outgoing beam. Continuity of χ and χ' at $x = 0$ and $x = a$ gives four equations:

$$\begin{aligned} 1 + A &= B + C, & ik(1 - A) &= \kappa(C - B), \\ Be^{-\kappa a} + Ce^{\kappa a} &= De^{ika}, & \kappa(Be^{\kappa a} - Ce^{-\kappa a}) &= ikDe^{ika}, \end{aligned}$$

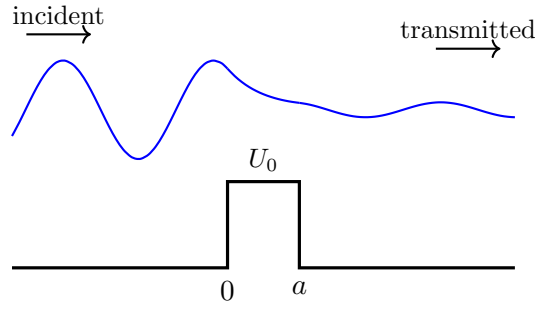
four linear equations in the four unknowns A, B, C, D . Solving (eliminate B and C first), one finds

$$D = \frac{-4i\kappa k e^{-ika}}{(\kappa - ik)^2 e^{\kappa a} - (\kappa + ik)^2 e^{-\kappa a}},$$

and the transmission coefficient $T = |D|^2$ (the incident and transmitted beams travel at the same speed) simplifies, after some algebra with $\sinh$, to

$$T = \frac{4k^2 \kappa^2}{(k^2 + \kappa^2)^2 \sinh^2(\kappa a) + 4k^2 \kappa^2}.$$

The key point: $T > 0$. The beam has energy strictly less than the barrier, and some of it gets through anyway. This is **quantum tunnelling**.



The figure shows $\text{Re } \chi$: an incident-plus-reflected standing wave pattern on the left, exponential decay through the barrier, and a small transmitted wave – of the same wavelength as the incident one, but much smaller amplitude – emerging on the right.

For a tall or wide barrier, $\kappa a \gg 1$, we have $\sinh(\kappa a) \approx \frac{1}{2}e^{\kappa a}$ and so

$$T \approx \frac{16k^2\kappa^2}{(k^2 + \kappa^2)^2} e^{-2\kappa a} \propto \exp\left(-\frac{2a}{\hbar} \sqrt{2m(U_0 - E)}\right).$$

The transmission probability decays *exponentially* with the barrier width and with the square root of its height. This exponential sensitivity is why tunnelling is invisible for everyday objects but ubiquitous at atomic scales, and it is quantitatively responsible for, among other things, α -decay of nuclei (the decay rate spans some twenty orders of magnitude across nuclei, tracking the exponential above), nuclear fusion in the Sun (protons tunnel through their Coulomb repulsion), and the scanning tunnelling microscope (where the exponential dependence of the tunnelling current on the tip–surface distance gives sub-atomic resolution).

Exercise 21.5.3. Put numbers in: for an electron tunnelling through a vacuum gap with $U_0 - E \approx 4$ eV (a typical metal work function), show that $2\kappa \approx 2 \text{ \AA}^{-1}$, so that increasing a scanning tunnelling microscope’s tip–surface gap by a single ångström suppresses the current by a factor of $e^2 \approx 7$. This is why the STM can resolve individual atoms.

Remark (Scattering Above the Barrier). The case $E > U_0$ is obtained from our formulae by the replacement $\kappa \mapsto i\bar{k}$ with $\bar{k} = \sqrt{2m(E - U_0)}/\hbar$, which turns $\sinh$ into $\sin$:

$$T = \frac{4k^2\bar{k}^2}{(k^2 - \bar{k}^2)^2 \sin^2(\bar{k}a) + 4k^2\bar{k}^2}.$$

Classically T would be 1; quantum mechanically $T < 1$ in general (reflection by a barrier you can clear, just as at the step) – *except* at the special energies where $\sin \bar{k}a = 0$, i.e. $\bar{k}a = n\pi$, where $T = 1$ exactly. These perfect transmissions are **resonances**: a whole number of half-wavelengths fits across the barrier, and the reflected waves from its two edges interfere destructively. The same effect for a potential *well* explains the Ramsauer–Townsend effect, in which noble gases are bizarrely transparent to electrons of one particular energy.

§21.6 Operators, Measurement and Uncertainty

We now pay our debts. Several times we have leaned on properties of Hermitian operators – real eigenvalues, orthogonal eigenfunctions, completeness – and on a still-unproven uncertainty principle. This section develops the operator theory of quantum mechanics properly, and it is here that the structure of the subject really lives.

§21.6.1 Expectation Values

The Born rule makes the outcome of a measurement a random variable, so the natural first question is: what is its mean? For position this needs no new ideas – the probability density is $|\psi|^2$, so (working in one dimension, with ψ normalised)

$$\langle \hat{x} \rangle_\psi = \int_{-\infty}^{\infty} x |\psi(x, t)|^2 dx = \int_{-\infty}^{\infty} \psi^* x \psi dx = \langle \psi, \hat{x} \psi \rangle.$$

The final expression is the one that generalises. For any observable with operator $\hat{A}$, we define the **expectation value** of $\hat{A}$ in the (normalised) state ψ as

$$\langle \hat{A} \rangle_\psi = \langle \psi, \hat{A} \psi \rangle = \int \psi^* (\hat{A} \psi) dx.$$

We will justify, in §6.3 below, that this really is the mean of the measurement distribution given by the Born rule. But first, let's use it to close a gap from Section 2: why is momentum represented by $-i\hbar\partial_x$? Here is a satisfying consistency check: *the expectation of momentum is the mass times the velocity of the expectation of position*. Assume ψ solves the Schrödinger equation. Then using the continuity equation $\partial_t |\psi|^2 = -\partial_x J$,

$$\begin{aligned} m \frac{d}{dt} \langle \hat{x} \rangle_\psi &= m \int_{-\infty}^{\infty} x \frac{\partial |\psi|^2}{\partial t} dx = -m \int_{-\infty}^{\infty} x \frac{\partial J}{\partial x} dx = m \int_{-\infty}^{\infty} J dx \\ &= -\frac{i\hbar}{2} \int_{-\infty}^{\infty} \left(\psi^* \frac{\partial \psi}{\partial x} - \psi \frac{\partial \psi^*}{\partial x} \right) dx = -i\hbar \int_{-\infty}^{\infty} \psi^* \frac{\partial \psi}{\partial x} dx = \langle \hat{p} \rangle_\psi, \end{aligned}$$

integrating by parts twice (boundary terms vanish for normalisable states). So with $\hat{p} = -i\hbar\partial_x$, the quantum expectations obey the classical kinematic relation $\langle p \rangle = m \frac{d}{dt} \langle x \rangle$ – this is the first instalment of Ehrenfest's theorem, coming shortly.

§21.6.2 Hermitian Operators

Definition 21.6.1 (Hermitian Conjugate)

The **Hermitian conjugate** (or adjoint) of an operator $\hat{A}$ is the operator $\hat{A}^\dagger$ defined by

$$\langle \hat{A}^\dagger \psi, \phi \rangle = \langle \psi, \hat{A} \phi \rangle \quad \text{for all } \psi, \phi \in \mathcal{H}.$$

Directly from the definition one checks the algebraic rules

$$(\lambda_1 \hat{A}_1 + \lambda_2 \hat{A}_2)^\dagger = \lambda_1^* \hat{A}_1^\dagger + \lambda_2^* \hat{A}_2^\dagger, \quad (\hat{A} \hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger.$$

Definition 21.6.2 (Hermitian Operator)

An operator $\hat{A}$ is **Hermitian** (self-adjoint) if $\hat{A}^\dagger = \hat{A}$, that is if

$$\langle \hat{A} \psi, \phi \rangle = \langle \psi, \hat{A} \phi \rangle \quad \text{for all } \psi, \phi \in \mathcal{H}.$$

Example 21.6.3 (The Basic Observables are Hermitian)

The position operator is Hermitian, essentially because x is real:

$$\langle \hat{x} \psi, \phi \rangle = \int (x\psi)^* \phi dx = \int \psi^* (x\phi) dx = \langle \psi, \hat{x} \phi \rangle.$$

The momentum operator is Hermitian, by integration by parts – and note how the factor of i is essential, absorbing the minus sign that integration by parts produces:

$$\langle \hat{p}\psi, \phi \rangle = \int \left(-i\hbar \frac{\partial \psi}{\partial x} \right)^* \phi dx = i\hbar \int \frac{\partial \psi^*}{\partial x} \phi dx = -i\hbar \int \psi^* \frac{\partial \phi}{\partial x} dx = \langle \psi, \hat{p}\phi \rangle,$$

the boundary terms vanishing since states decay at infinity. (So ∂_x alone is *anti*-Hermitian; the $-i\hbar$ fixes it.) Similarly, integrating by parts twice shows $\hat{p}^2$ (and hence the kinetic energy) is Hermitian, and multiplication by a real potential $U(x)$ is Hermitian just like $\hat{x}$ – so the Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m} + U(\hat{x})$ is Hermitian.

While we have the machinery out, here is a small but structurally important observation.

Proposition 21.6.4 (Energies are Bounded Below)

For any normalised state ψ , the kinetic energy has $\langle \hat{T} \rangle_\psi \geq 0$, and consequently every energy eigenvalue satisfies $E \geq \min U$.

Proof. Integrating by parts (boundary terms vanishing as usual),

$$\langle \hat{T} \rangle_\psi = -\frac{\hbar^2}{2m} \int \psi^* \frac{\partial^2 \psi}{\partial x^2} dx = \frac{\hbar^2}{2m} \int \left| \frac{\partial \psi}{\partial x} \right|^2 dx \geq 0.$$

If $\hat{H}\chi = E\chi$ with χ normalised, then $E = \langle \hat{T} \rangle_\chi + \langle U \rangle_\chi \geq 0 + \min U$. $\square$

This is the promise, made when we stated the axioms, that the Hamiltonian's spectrum is bounded below – there is a stable state of least energy, and matter cannot cascade down through an endless ladder of ever-lower energies. (It also confirms little facts we used along the way, e.g. that the infinite well has no eigenvalues with $E \leq 0$.)

Now the two fundamental structural facts, both with proofs that are one-line applications of Hermiticity.

Proposition 21.6.5 (Real Eigenvalues)

The eigenvalues of a Hermitian operator are real.

Proof. Let $\hat{A}\psi = a\psi$ with ψ normalised. Then

$$a = \langle \psi, \hat{A}\psi \rangle = \langle \hat{A}\psi, \psi \rangle = a^* \langle \psi, \psi \rangle = a^*,$$

using Hermiticity in the middle step. So $a \in \mathbb{R}$. $\square$

Proposition 21.6.6 (Orthogonal Eigenfunctions)

Eigenfunctions of a Hermitian operator with distinct eigenvalues are orthogonal.

Proof. Let $\hat{A}\psi_1 = a_1\psi_1$ and $\hat{A}\psi_2 = a_2\psi_2$ with $a_1 \neq a_2$. Then

$$a_1 \langle \psi_1, \psi_2 \rangle = \langle \hat{A}\psi_1, \psi_2 \rangle = \langle \psi_1, \hat{A}\psi_2 \rangle = a_2 \langle \psi_1, \psi_2 \rangle,$$

(using $a_1 \in \mathbb{R}$ for the first equality), so $(a_1 - a_2)\langle\psi_1, \psi_2\rangle = 0$, and since $a_1 \neq a_2$ we get $\langle\psi_1, \psi_2\rangle = 0$. $\square$

These are exactly the statements you know for real symmetric (or complex Hermitian) matrices, transplanted to infinite dimensions – along with their proofs. The remaining matrix fact, diagonalisability, is also what we want, but its infinite-dimensional proof is genuinely beyond this course, so we record it as an article of faith.

Fact 21.6.7 (Completeness of Eigenfunctions)

The eigenfunctions (discrete and continuous together) of a Hermitian operator form a complete basis of $\mathcal{H}$: every state can be expanded in them.

For every operator in this course, completeness can be checked concretely (e.g. Fourier series for the infinite well, Hermite functions for the oscillator), so nothing we do rests on an unproved generality.

Remark (Continuous Spectra and Improper Eigenfunctions). The caveat ‘discrete and continuous together’ is doing real work. The momentum operator has *no* normalisable eigenfunctions at all – its would-be eigenfunctions are the plane waves e^{ikx} , with a continuum of eigenvalues $\hbar k$ – and the position operator is even worse: $\hat{x}\psi = x_0\psi$ forces ψ to vanish everywhere except at the point x_0 , so its ‘eigenfunctions’ are the Dirac delta functions $\delta(x - x_0)$. Neither family lives in $\mathcal{H}$, but both are complete in the integral sense: the expansion of a state in position eigenfunctions is the tautology $\psi(x) = \int \psi(x_0)\delta(x - x_0)dx_0$, and the expansion in momentum eigenfunctions is precisely the *Fourier transform*,

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\psi}(p) e^{ipx/\hbar} dp.$$

The measurement rules extend with sums replaced by integrals: $|\psi(x_0)|^2$ is the probability density for position measurements – so the Born rule of Section 2 is exactly the continuous-spectrum case of the general measurement axioms – and $|\tilde{\psi}(p)|^2$ is the probability density for momentum measurements. Making all this rigorous takes functional analysis beyond this course, but using it takes only care.

Applied to the Hamiltonian, this fact is what justifies the claim of Section 3 that stationary states suffice:

Corollary 21.6.8 (General Solution of the Schrödinger Equation)

Every solution of the time-dependent Schrödinger equation may be written

$$\psi(x, t) = \sum_n a_n \chi_n(x) e^{-iE_n t/\hbar}, \quad a_n = \langle \chi_n, \psi(\cdot, 0) \rangle,$$

where $\{\chi_n\}$ is a complete orthonormal set of eigenfunctions of $\hat{H}$ (with integrals over any continuous part of the spectrum understood).

§21.6.3 Measurement

We can now state the measurement rules in their final form, generalising the two-state Born rule of Section 2. Let $\hat{O}$ be an observable with a complete orthonormal set of

eigenfunctions $\{\psi_n\}$ and eigenvalues $\{\lambda_n\}$, and let $\psi = \sum_n a_n \psi_n$ be a normalised state.

- (i) A measurement of O returns some eigenvalue of $\hat{O}$.
- (ii) If the eigenvalues are non-degenerate, the probability of the outcome λ_n is

$$\mathbb{P}(O = \lambda_n) = |a_n|^2 = |\langle \psi_n, \psi \rangle|^2.$$

- (iii) If an eigenvalue λ is degenerate, with orthonormal eigenfunctions $\{\psi_i\}_{i \in I}$ spanning its eigenspace, then $\mathbb{P}(O = \lambda) = \sum_{i \in I} |a_i|^2$.
- (iv) (*The projection postulate.*) If the measurement returns λ , the state immediately afterwards is the normalised projection of ψ onto the λ -eigenspace:

$$\psi \mapsto \frac{1}{\sqrt{\sum_{i \in I} |a_i|^2}} \sum_{i \in I} a_i \psi_i,$$

which for a non-degenerate eigenvalue is simply ψ_n . An immediately repeated measurement therefore returns λ again with certainty.

Let us sanity-check this. The probabilities sum to one:

$$\sum_n |a_n|^2 = \sum_{n,m} a_n^* a_m \langle \psi_n, \psi_m \rangle = \left\langle \sum_n a_n \psi_n, \sum_m a_m \psi_m \right\rangle = \langle \psi, \psi \rangle = 1,$$

using orthonormality – this is just Parseval's theorem. And the mean of the distribution is

$$\sum_n \lambda_n |a_n|^2 = \sum_{n,m} a_n^* \lambda_m a_m \delta_{nm} = \left\langle \sum_n a_n \psi_n, \sum_m a_m \lambda_m \psi_m \right\rangle = \langle \psi, \hat{O} \psi \rangle = \langle \hat{O} \rangle_\psi,$$

confirming that the expectation value defined in §6.1 is indeed the statistical mean of measurement outcomes. Notice how tightly the mathematics interlocks with the physics here: Hermiticity guarantees the outcomes (eigenvalues) are real numbers, orthogonality makes the probabilities well defined and additive, and completeness guarantees every state can actually be measured.

Let's do a full worked example, in the system where we can compute everything.

Example 21.6.9 (Measuring the Energy of a Particle in a Box)

Take the infinite square well of §4.2 (width $2a$, eigenfunctions χ_n , energies $E_n = \hbar^2 \pi^2 n^2 / 8ma^2$), and suppose the particle is prepared in the state

$$\psi = \frac{1}{\sqrt{6}} \chi_1 + \sqrt{\frac{2}{3}} \chi_2 + \frac{1}{\sqrt{6}} \chi_3.$$

(This is normalised: $\frac{1}{6} + \frac{2}{3} + \frac{1}{6} = 1$.) Measuring the energy:

- (i) The possible outcomes are E_1 , E_2 , E_3 only – no other value can ever occur, however many times we repeat the experiment on identically prepared systems.
- (ii) Their probabilities are $|a_n|^2$: we get E_1 with probability $\frac{1}{6}$, E_2 with probability $\frac{2}{3}$, and E_3 with probability $\frac{1}{6}$.

(iii) The mean outcome is

$$\langle \hat{H} \rangle_\psi = \frac{1}{6}E_1 + \frac{2}{3}E_2 + \frac{1}{6}E_3 = \left(\frac{1}{6} + \frac{8}{3} + \frac{3}{2} \right) E_1 = \frac{13}{3}E_1,$$

which – note well – is not itself an allowed outcome of any single measurement.

(iv) If the measurement returns E_2 , the state collapses to χ_2 , and all subsequent energy measurements return E_2 with certainty. The information about the original coefficients is irretrievably gone.

Suppose instead we had measured *position* first, getting some value near x_0 , and then energy. The position measurement collapses ψ to a state sharply peaked at x_0 , which is *not* an energy eigenstate but a fresh superposition of all the χ_n – so the subsequent energy measurement can now return values other than E_1, E_2, E_3 with non-zero probability. Measurements of incompatible observables interfere with each other's statistics; this is the operational meaning of non-commutation, made quantitative by the uncertainty principle below.

Example 21.6.10 (A Flat Start)

For a slightly less rigged example, prepare the particle in the box in the *uniform* state $\psi(x, 0) = 1/\sqrt{2a}$ (constant across the well). The expansion coefficients are genuine Fourier integrals: for n even, $a_n = 0$ by parity, while for n odd,

$$a_n = \langle \chi_n, \psi \rangle = \frac{1}{\sqrt{2a}} \cdot \frac{1}{\sqrt{a}} \int_{-a}^a \cos \frac{n\pi x}{2a} dx = \frac{\sqrt{2}}{a} \cdot \frac{2a}{n\pi} \sin \frac{n\pi}{2} = \pm \frac{2\sqrt{2}}{n\pi}.$$

So an energy measurement returns E_1 with probability $|a_1|^2 = 8/\pi^2 \approx 0.81$, E_3 with probability $8/9\pi^2 \approx 0.09$, and so on – one can check $\sum_{n \text{ odd}} 8/n^2\pi^2 = 1$ using $\sum_{n \text{ odd}} 1/n^2 = \pi^2/8$. Every possible outcome is an odd level, with rapidly decaying probabilities; and since the state is not an eigenstate, $\psi(x, t)$ genuinely evolves, sloshing around the box forever (with, pleasingly, exact revivals: all the phases $e^{-iE_n t/\hbar}$ realign at $t = 16ma^2/\pi\hbar$, since each $E_n/E_1 = n^2$ is an integer).

Remark (The Measurement Problem). It would be dishonest not to flag the elephant in the room. We have given the theory *two* rules of evolution: the Schrödinger equation (deterministic, linear, continuous) between measurements, and collapse (probabilistic, nonlinear, discontinuous) during them – without ever saying what physically counts as a ‘measurement’. A measuring device is itself made of atoms; applying the Schrödinger equation to the device plus particle produces a superposition of ‘pointer states’, not a definite outcome – this is the content of Schrödinger’s famous cat. How (or whether) collapse really happens is called the *measurement problem*, and proposed resolutions (decoherence, many-worlds, hidden variables, objective collapse, ...) remain genuinely contested. None of this ambiguity infects the predictions: for every experiment anyone has devised, the rules above say what the statistics will be, and they have always been right. We take the pragmatic road, as does the course.

§21.6.4 Commutators

Operators, unlike numbers, need not commute – and essentially all distinctively quantum phenomena trace back to this. The failure to commute is measured by the following bracket.

Definition 21.6.11 (Commutator)

The **commutator** of operators $\hat{A}$ and $\hat{B}$ is

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}.$$

The commutator is antisymmetric, $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$, linear in each slot, and satisfies the Leibniz-type rules

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}], \quad [\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B},$$

which are checked by expanding everything out (and are worth internalising – they let you avoid ever expanding everything out again).

Example 21.6.12 (The Canonical Commutation Relation)

The most important commutator in quantum mechanics is between position and momentum. Apply the operators to a test wavefunction (never manipulate bare operators when unsure!):

$$\begin{aligned} \hat{x}\hat{p}\psi &= x \left(-i\hbar \frac{\partial \psi}{\partial x} \right) = -i\hbar x \frac{\partial \psi}{\partial x}, \\ \hat{p}\hat{x}\psi &= -i\hbar \frac{\partial}{\partial x}(x\psi) = -i\hbar\psi - i\hbar x \frac{\partial \psi}{\partial x}. \end{aligned}$$

Subtracting, $[\hat{x}, \hat{p}]\psi = i\hbar\psi$ for every ψ , so

$$[\hat{x}, \hat{p}] = i\hbar,$$

(times the identity operator). This is the **canonical commutation relation**, and one can regard it as the mathematical core of quantum mechanics – everything in this section flows from it.

Commuting observables are exactly the simultaneously measurable ones, in the following sense.

Definition 21.6.13 (Simultaneously Diagonalisable)

Hermitian operators $\hat{A}$, $\hat{B}$ are **simultaneously diagonalisable** if there is a complete basis $\{\psi_i\}$ of joint eigenfunctions: $\hat{A}\psi_i = \lambda_i\psi_i$ and $\hat{B}\psi_i = \mu_i\psi_i$.

Theorem 21.6.14 (Compatibility Criterion)

Hermitian operators $\hat{A}$ and $\hat{B}$ are simultaneously diagonalisable if and only if $[\hat{A}, \hat{B}] = 0$.

Proof. ($\Rightarrow$) Given a complete joint eigenbasis, for each basis element

$$[\hat{A}, \hat{B}]\psi_i = \hat{A}(\mu_i\psi_i) - \hat{B}(\lambda_i\psi_i) = (\lambda_i\mu_i - \mu_i\lambda_i)\psi_i = 0,$$

and by linearity and completeness $[\hat{A}, \hat{B}]\psi = 0$ for every ψ .

($\Leftarrow$) Suppose $[\hat{A}, \hat{B}] = 0$, and let V_λ be the eigenspace of $\hat{A}$ for eigenvalue λ . For $\psi \in V_\lambda$,

$$\hat{A}(\hat{B}\psi) = \hat{B}(\hat{A}\psi) = \lambda(\hat{B}\psi),$$

so $\hat{B}$ maps V_λ into itself. The restriction of $\hat{B}$ to each V_λ is still Hermitian, so (by completeness applied within V_λ) we may pick a basis of each V_λ consisting of eigenfunctions of $\hat{B}$. These are joint eigenfunctions, and running over all λ they form a complete basis of $\mathcal{H}$. $\square$

If $[\hat{A}, \hat{B}] = 0$ we call the observables **compatible**: in a joint eigenstate both have definite values, so both can be known simultaneously with perfect precision, and measuring one does not disturb the (collapsed) eigenstate of the other. By contrast $[\hat{x}, \hat{p}] = i\hbar \neq 0$: position and momentum are incompatible, and no state whatsoever has definite values of both. The uncertainty principle below is the quantitative version of this statement. We have also already used this circle of ideas twice without saying so: parity commutes with a symmetric Hamiltonian (§4.3), and in Section 7 the set $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$ of mutually commuting operators will organise the whole three-dimensional theory.

Remark (Labelling States). Compatibility also solves a bookkeeping problem that degeneracy creates. If an eigenvalue of $\hat{A}$ is degenerate, the outcome of measuring A does not pin down the state – we need more information. The fix is to find further observables commuting with $\hat{A}$ (and each other), and keep adding them until the joint eigenspaces are one-dimensional; such a maximal collection is called a *complete set of commuting observables*, and the corresponding eigenvalues (‘quantum numbers’) then label a basis of states unambiguously. For the hydrogen atom the set will be $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$ with quantum numbers (N, ℓ, m) – this is precisely why atomic states are labelled by three quantum numbers, and the scheme survives (with spin added) into the periodic table.

§21.6.5 Ehrenfest’s Theorem

If quantum mechanics is right, it is right about tennis balls too, so classical mechanics must be hiding inside it somewhere. The precise statement is that *expectation values obey classical-looking equations of motion*, and the commutator controls the dynamics.

Theorem 21.6.15 (Ehrenfest’s Theorem)

Let ψ evolve by the Schrödinger equation, and let $\hat{A}$ be an observable with no explicit time dependence. Then

$$\frac{d}{dt}\langle \hat{A} \rangle_\psi = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle_\psi.$$

Proof. Differentiating under the integral,

$$\frac{d}{dt}\langle \psi, \hat{A}\psi \rangle = \left\langle \frac{\partial \psi}{\partial t}, \hat{A}\psi \right\rangle + \left\langle \psi, \hat{A} \frac{\partial \psi}{\partial t} \right\rangle.$$

The Schrödinger equation gives $\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar} \hat{H} \psi$, so, using antilinearity in the first argument and Hermiticity of $\hat{H}$,

$$\frac{d}{dt} \langle \hat{A} \rangle_\psi = -\frac{1}{i\hbar} \langle \hat{H} \psi, \hat{A} \psi \rangle + \frac{1}{i\hbar} \langle \psi, \hat{A} \hat{H} \psi \rangle = \frac{i}{\hbar} \left(\langle \psi, \hat{H} \hat{A} \psi \rangle - \langle \psi, \hat{A} \hat{H} \psi \rangle \right) = \frac{i}{\hbar} \langle [\hat{H}, \hat{A}] \rangle_\psi. \quad \square$$

(If $\hat{A}$ does depend explicitly on time, one simply adds $\langle \partial \hat{A} / \partial t \rangle_\psi$ to the right hand side; we will never need this.) Let's harvest the consequences, for a particle with $\hat{H} = \frac{\hat{p}^2}{2m} + U(\hat{x})$.

(i) *Energy conservation*: taking $\hat{A} = \hat{H}$, since $[\hat{H}, \hat{H}] = 0$,

$$\frac{d}{dt} \langle \hat{H} \rangle_\psi = 0.$$

(ii) *Velocity*: taking $\hat{A} = \hat{x}$, we need $[\hat{H}, \hat{x}] = \frac{1}{2m} [\hat{p}^2, \hat{x}]$. By the Leibniz rule and the canonical commutation relation,

$$[\hat{p}^2, \hat{x}] = \hat{p}[\hat{p}, \hat{x}] + [\hat{p}, \hat{x}]\hat{p} = -2i\hbar\hat{p},$$

so

$$\frac{d}{dt} \langle \hat{x} \rangle_\psi = \frac{i}{\hbar} \cdot \left(\frac{-2i\hbar}{2m} \right) \langle \hat{p} \rangle_\psi = \frac{\langle \hat{p} \rangle_\psi}{m},$$

recovering (now effortlessly) the relation we derived by hand in §6.1.

(iii) *Newton's second law*: taking $\hat{A} = \hat{p}$, only the potential contributes, and on a test function

$$[U(\hat{x}), \hat{p}] \psi = U(-i\hbar \psi') + i\hbar (U\psi)' = i\hbar U'(x) \psi,$$

so

$$\frac{d}{dt} \langle \hat{p} \rangle_\psi = -\langle U'(\hat{x}) \rangle_\psi.$$

Example 21.6.16 (The Oscillator Swings Classically)

For the harmonic oscillator, $U'(x) = m\omega^2 x$ is *linear*, so $\langle U'(\hat{x}) \rangle = m\omega^2 \langle \hat{x} \rangle$ exactly, and Ehrenfest's equations close up:

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{\langle \hat{p} \rangle}{m}, \quad \frac{d}{dt} \langle \hat{p} \rangle = -m\omega^2 \langle \hat{x} \rangle \implies \frac{d^2}{dt^2} \langle \hat{x} \rangle = -\omega^2 \langle \hat{x} \rangle.$$

So for *any* state whatsoever – however spread out, however wildly non-classical – the mean position of an oscillator swings sinusoidally at exactly the classical frequency ω . (In a stationary state, of course, $\langle \hat{x} \rangle = 0$ for all time: the oscillation only shows up in superpositions of neighbouring levels, which oscillate at frequency $(E_{N+1} - E_N)/\hbar = \omega$ – everything is consistent.)

Together, (ii) and (iii) say that $\langle \hat{x} \rangle$ and $\langle \hat{p} \rangle$ obey exactly Newton's equations, with the force averaged over the state⁶. Classical mechanics is the story of the means; quantum mechanics is the story of the whole distribution.

⁶Note the subtlety: the classical trajectory would require $\frac{d}{dt} \langle p \rangle = -U'(\langle x \rangle)$, but what we actually get is $-\langle U'(x) \rangle$. These agree when U' is linear (free particle, uniform force, harmonic oscillator) or when the state is sharply peaked – which is exactly the classical limit.

§21.6.6 The Uncertainty Principle

Having discussed the means of measurement distributions, we turn to their spreads.

Definition 21.6.17 (Uncertainty)

The **uncertainty** of the observable A in the (normalised) state ψ is $\Delta_\psi A \geq 0$, where

$$(\Delta_\psi A)^2 = \langle (\hat{A} - \langle \hat{A} \rangle_\psi)^2 \rangle_\psi = \langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2.$$

This is precisely the variance of the measurement distribution (the second equality is the usual expand-the-square manipulation, using linearity of expectation values). The uncertainty vanishes exactly on states where the observable has a definite value:

Lemma 21.6.18

$(\Delta_\psi A)^2 \geq 0$, with equality if and only if ψ is an eigenfunction of $\hat{A}$.

Proof. Write $\hat{A}' = \hat{A} - \langle \hat{A} \rangle_\psi$ (a Hermitian operator, since $\langle \hat{A} \rangle_\psi$ is real). By Hermiticity,

$$(\Delta_\psi A)^2 = \langle \psi, \hat{A}'^2 \psi \rangle = \langle \hat{A}' \psi, \hat{A}' \psi \rangle = \|\hat{A}' \psi\|^2 \geq 0,$$

with equality iff $\hat{A}' \psi = 0$, i.e. iff $\hat{A} \psi = \langle \hat{A} \rangle_\psi \psi$. $\square$

The engine of the uncertainty principle is the Cauchy–Schwarz inequality, which holds in any inner product space.

Theorem 21.6.19 (Cauchy–Schwarz Inequality)

For $\psi, \phi \in \mathcal{H}$,

$$|\langle \psi, \phi \rangle|^2 \leq \langle \psi, \psi \rangle \langle \phi, \phi \rangle,$$

with equality if and only if $\phi = \lambda \psi$ for some $\lambda \in \mathbb{C}$ (or $\psi = 0$).

Proof. We may assume $\psi \neq 0$. For any $\lambda \in \mathbb{C}$,

$$0 \leq \langle \phi - \lambda \psi, \phi - \lambda \psi \rangle = \langle \phi, \phi \rangle - \lambda \langle \phi, \psi \rangle - \lambda^* \langle \psi, \phi \rangle + |\lambda|^2 \langle \psi, \psi \rangle.$$

Choose the minimising value $\lambda = \langle \psi, \phi \rangle / \langle \psi, \psi \rangle$; the right hand side becomes

$$\langle \phi, \phi \rangle - \frac{|\langle \psi, \phi \rangle|^2}{\langle \psi, \psi \rangle} \geq 0,$$

which rearranges to the claim. Equality forces $\langle \phi - \lambda \psi, \phi - \lambda \psi \rangle = 0$, i.e. $\phi = \lambda \psi$. $\square$

Theorem 21.6.20 (Generalised Uncertainty Principle)

For observables A, B and any normalised state ψ ,

$$\Delta_\psi A \Delta_\psi B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle_\psi \right|.$$

Proof. Let $\hat{A}' = \hat{A} - \langle \hat{A} \rangle_\psi$ and $\hat{B}' = \hat{B} - \langle \hat{B} \rangle_\psi$, both Hermitian, and note $[\hat{A}', \hat{B}'] = [\hat{A}, \hat{B}]$ (the subtracted constants commute with everything). By the lemma above and Cauchy–Schwarz,

$$(\Delta_\psi A)^2 (\Delta_\psi B)^2 = \|\hat{A}'\psi\|^2 \|\hat{B}'\psi\|^2 \geq |\langle \hat{A}'\psi, \hat{B}'\psi \rangle|^2 = |\langle \psi, \hat{A}'\hat{B}'\psi \rangle|^2,$$

using Hermiticity of $\hat{A}'$ at the last step. Now split the product into commutator and anticommutator parts:

$$\hat{A}'\hat{B}' = \frac{1}{2}[\hat{A}', \hat{B}'] + \frac{1}{2}\{\hat{A}', \hat{B}'\}, \quad \{\hat{A}', \hat{B}'\} = \hat{A}'\hat{B}' + \hat{B}'\hat{A}'.$$

The anticommutator of Hermitian operators is Hermitian, so $\langle \{\hat{A}', \hat{B}'\} \rangle_\psi$ is *real*; the commutator is anti-Hermitian ($[\hat{A}', \hat{B}']^\dagger = -[\hat{A}', \hat{B}']$), so $\langle [\hat{A}', \hat{B}'] \rangle_\psi$ is purely *imaginary* (both by the argument in the proof that Hermitian operators have real expectations). Hence the two contributions are the real and imaginary parts of $\langle \psi, \hat{A}'\hat{B}'\psi \rangle$, and

$$|\langle \psi, \hat{A}'\hat{B}'\psi \rangle|^2 = \frac{1}{4}|\langle [\hat{A}, \hat{B}] \rangle_\psi|^2 + \frac{1}{4}|\langle \{\hat{A}', \hat{B}'\} \rangle_\psi|^2 \geq \frac{1}{4}|\langle [\hat{A}, \hat{B}] \rangle_\psi|^2.$$

Taking square roots gives the theorem. $\square$

Corollary 21.6.21 (Heisenberg's Uncertainty Principle)

For any normalised state ψ ,

$$\Delta_\psi x \Delta_\psi p \geq \frac{\hbar}{2}.$$

Proof. Take $\hat{A} = \hat{x}$, $\hat{B} = \hat{p}$ in the theorem: $[\hat{x}, \hat{p}] = i\hbar$, so the right hand side is $\frac{1}{2}|i\hbar| = \frac{\hbar}{2}$, independent of the state. $\square$

Let us be clear about what this does and does not say. It is a statement about *states*, not about clumsy measuring devices: no quantum state, however cleverly prepared, has both a sharp position and a sharp momentum. It underlies phenomena we have already met – zero-point energies (confinement forces momentum spread, hence kinetic energy), the stability of atoms (an electron squashed into the nucleus would have enormous momentum uncertainty), and the spreading of wavepackets.

Remark (Energy-Time Uncertainty). You will often see a companion relation $\Delta E \Delta t \gtrsim \hbar$ quoted. Beware: time is *not* an observable in quantum mechanics (there is no operator $\hat{t}$), so this cannot be an instance of the theorem above. The honest statement comes from Ehrenfest: for any observable A ,

$$\Delta_\psi H \cdot \Delta_\psi A \geq \frac{1}{2} |\langle [\hat{H}, \hat{A}] \rangle_\psi| = \frac{\hbar}{2} \left| \frac{d}{dt} \langle \hat{A} \rangle_\psi \right|,$$

so defining $\tau_A = \Delta_\psi A / \left| \frac{d}{dt} \langle \hat{A} \rangle_\psi \right|$ – the time it takes the statistics of A to change noticeably – we get $\Delta_\psi H \cdot \tau_A \geq \hbar/2$. States with sharp energy change slowly (a stationary state, with $\Delta H = 0$, never changes at all); fast dynamics requires energy spread. This is the sense in which short-lived excited states have broad spectral lines.

The uncertainty principle is not only a prohibition – used with a little courage, it is a computational tool.

Example 21.6.22 (Estimating the Ground State Energy of the Oscillator)

For the harmonic oscillator, the ground state must balance two competing costs: potential energy grows if the state spreads out, kinetic energy grows (via the uncertainty principle) if it is squeezed. Let's estimate the minimum. For a state centred at the origin with $\langle \hat{x} \rangle = \langle \hat{p} \rangle = 0$ (as holds for the ground state, by parity), we have $\langle \hat{x}^2 \rangle = (\Delta x)^2$ and $\langle \hat{p}^2 \rangle = (\Delta p)^2$, so

$$\langle \hat{H} \rangle = \frac{(\Delta p)^2}{2m} + \frac{1}{2}m\omega^2(\Delta x)^2 \geq \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2}m\omega^2(\Delta x)^2,$$

using $\Delta p \geq \hbar/2\Delta x$. Minimising over Δx (by calculus, or AM–GM), the minimum of the right hand side is at $(\Delta x)^2 = \hbar/2m\omega$, giving

$$\langle \hat{H} \rangle \geq \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2}.$$

This lower bound is exactly the true ground state energy $E_0 = \frac{1}{2}\hbar\omega$ – the bound is achieved because the true ground state is a Gaussian, i.e. a minimal uncertainty state (see below). The same crude argument applied to hydrogen (balancing $\hbar^2/2m_e r^2$ against $-e^2/4\pi\epsilon_0 r$) predicts both the Bohr radius and the -13.6 eV ground state energy up to factors of order one: the sizes and energies of atoms are set by the uncertainty principle.

Which states are the most classical, saturating the bound? Tracing through the proof, equality requires equality in Cauchy–Schwarz, $\hat{B}'\psi = \lambda\hat{A}'\psi$, and vanishing of the anticommutator term, which together force λ to be purely imaginary: $(\hat{p} - \langle p \rangle)\psi = ic(\hat{x} - \langle x \rangle)\psi$ for some real c . Centering things at the origin for simplicity, this is the ODE

$$-i\hbar\psi' = icx\psi \implies \psi(x) = C \exp\left(-\frac{cx^2}{2\hbar}\right),$$

a *Gaussian* (with $c > 0$ for normalisability). So the minimal-uncertainty states are exactly the Gaussian wavepackets of §4.1 – which is why the Gaussian packet computation gave $\Delta x \Delta p = \hbar/2$ exactly at $t = 0$ – and, pleasingly, why the oscillator ground state (a Gaussian!) is the physical state that comes closest to sitting classically still at the bottom of the well.

§21.7 Quantum Mechanics in Three Dimensions

Real particles live in three dimensions, so it is time to go back to the full Schrödinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + U(\mathbf{x})\psi,$$

with wavefunctions normalised over $\mathbb{R}^3$. Everything from Sections 2, 3 and 6 – inner products, Hermitian operators, measurement, stationary states – was set up in $\mathbb{R}^3$ from the start, so carries over verbatim. What is genuinely new in three dimensions is *angular momentum*, and the associated richness of degeneracy.

The observables now come in triples: $\hat{x}_i\psi = x_i\psi$ and $\hat{p}_i = -i\hbar\frac{\partial}{\partial x_i}$ for $i = 1, 2, 3$, with canonical commutation relations

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0.$$

Different coordinate directions are compatible: it is x_i and p_i *in the same direction* that refuse to commute.

§21.7.1 Separable Solutions

When the potential splits as a sum $U(\mathbf{x}) = U_1(x_1) + U_2(x_2) + U_3(x_3)$, the time-independent Schrödinger equation separates in Cartesian coordinates: substituting $\chi(\mathbf{x}) = \chi_1(x_1)\chi_2(x_2)\chi_3(x_3)$, each factor satisfies a one-dimensional Schrödinger equation with potential U_i , and the energies add. All our one-dimensional work is thus immediately recycled.

Example 21.7.1 (The Three-Dimensional Harmonic Oscillator)

For $U = \frac{1}{2}m\omega^2|\mathbf{x}|^2 = \sum_i \frac{1}{2}m\omega^2x_i^2$, the separable eigenfunctions are products of Hermite functions, with

$$E_{n_1n_2n_3} = (n_1 + n_2 + n_3 + \frac{3}{2})\hbar\omega, \quad n_i \in \{0, 1, 2, \dots\}.$$

The energy depends only on $n = n_1 + n_2 + n_3$, so the level $E = (n + \frac{3}{2})\hbar\omega$ has degeneracy equal to the number of ways of writing n as an ordered sum of three non-negative integers, namely $\binom{n+2}{2} = \frac{1}{2}(n+1)(n+2)$. This is our first meeting with large degeneracies – a symptom of the symmetry of the potential, as we are about to understand properly.

Example 21.7.2 (Particle in a Cubical Box)

Similarly, for a particle confined to a cube of side L (infinite walls, one corner at the origin), the eigenfunctions are products of one-dimensional box modes $\sin(n_i\pi x_i/L)$, with

$$E = \frac{\hbar^2\pi^2}{2mL^2} (n_1^2 + n_2^2 + n_3^2), \quad n_i \in \{1, 2, 3, \dots\}.$$

The ground state $(1, 1, 1)$ is unique, but the first excited level, $(2, 1, 1), (1, 2, 1), (1, 1, 2)$, is three-fold degenerate – the three states are related by rotating the box, so the degeneracy is again symmetry in action. (Higher up, ‘accidental’ coincidences like $1^2 + 1^2 + 5^2 = 3^2 + 3^2 + 3^2 = 27$ appear as well.) Counting the number of lattice points with $n_1^2 + n_2^2 + n_3^2 \leq 2mEL^2/\hbar^2\pi^2$ gives the *density of states* for a quantum gas – the starting point of statistical mechanics.

§21.7.2 Spherically Symmetric Potentials and the Radial Equation

The potentials nature actually likes – above all the Coulomb potential – are not sums but are **spherically symmetric**: $U = U(r)$ where $r = |\mathbf{x}|$. The right coordinates are spherical polars (r, θ, ϕ) ,

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta,$$

in which the Laplacian takes the form

$$\nabla^2 = \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{1}{r^2} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right].$$

As a first pass, let's look for eigenfunctions that share the symmetry of the potential: $\chi = \chi(r)$ only. Then only the radial part of the Laplacian acts,

$$\nabla^2 \chi(r) = \frac{1}{r} \frac{d^2}{dr^2} (r\chi(r)) = \chi'' + \frac{2}{r} \chi',$$

and the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m} \left(\chi'' + \frac{2}{r} \chi' \right) + U(r)\chi = E\chi.$$

There is a substitution that makes this look one-dimensional: let

$$\sigma(r) = r\chi(r).$$

Then the equation collapses to

$$-\frac{\hbar^2}{2m} \sigma''(r) + U(r)\sigma(r) = E\sigma(r),$$

which is *exactly* the one-dimensional Schrödinger equation – but on the half-line $r \geq 0$. The boundary conditions carry the memory of three dimensions: normalisability now reads

$$\int_{\mathbb{R}^3} |\chi|^2 dV = 4\pi \int_0^\infty |\chi(r)|^2 r^2 dr = 4\pi \int_0^\infty |\sigma(r)|^2 dr < \infty,$$

and for $\chi = \sigma/r$ to be well behaved at the origin we need

$$\sigma(0) = 0.$$

So: radial problems are one-dimensional problems on $r > 0$ with a hard wall at $r = 0$ – precisely the setting of the half-oscillator example of §4.5. A useful slogan: *the s-wave states of a 3D well are the odd states of the corresponding 1D well.*

Example 21.7.3 (The Spherical Potential Well)

Take $U(r) = 0$ for $r \leq a$ and $U(r) = U_0$ for $r > a$, and look for bound states $0 < E < U_0$. With the usual $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(U_0 - E)}/\hbar$, the equation for σ is the finite-well equation, and the condition $\sigma(0) = 0$ selects the *odd* solutions:

$$\sigma(r) = \begin{cases} A \sin kr & r \leq a, \\ B e^{-\kappa r} & r > a. \end{cases}$$

Matching σ and σ' at $r = a$ gives

$$-k \cot ka = \kappa,$$

to be solved against $k^2 + \kappa^2 = 2mU_0/\hbar^2$, i.e. in the dimensionless variables of §4.4, the intersections of $\eta = -\xi \cot \xi$ with the circle of radius r_0 . The first branch of $-\xi \cot \xi$ only becomes positive for $\xi > \pi/2$, so there is *no bound state at all* unless $r_0 > \pi/2$, i.e. unless

$$U_0 a^2 > \frac{\pi^2 \hbar^2}{8m}.$$

Contrast the one-dimensional finite well, which always had a bound state: in three dimensions, a well must be deep or wide enough to trap a particle. (This is physically relevant: the deuteron – the proton-neutron bound state – is only just bound, and

this criterion is essentially the estimate nuclear physicists use for the strength of the nuclear force.)

To go beyond spherically symmetric *states*, we need to understand the angular part of the Laplacian – and that operator turns out to be precisely the angular momentum.

§21.7.3 Angular Momentum

Classically, angular momentum $\mathbf{L} = \mathbf{x} \times \mathbf{p}$ is the conserved quantity associated with rotational symmetry, and conservation of $\mathbf{L}$ is what reduces classical central-force motion to a one-dimensional radial problem. We now build the quantum analogue, by direct substitution.

Definition 21.7.4 (Angular Momentum Operators)

The **angular momentum operators** are

$$\hat{\mathbf{L}} = \hat{\mathbf{x}} \times \hat{\mathbf{p}} = -i\hbar \mathbf{x} \times \nabla, \quad \text{i.e.} \quad \hat{L}_i = -i\hbar \epsilon_{ijk} x_j \frac{\partial}{\partial x_k},$$

together with the **total angular momentum**

$$\hat{L}^2 = \hat{L}_1^2 + \hat{L}_2^2 + \hat{L}_3^2.$$

(There is no ordering ambiguity in the definition: $\hat{L}_i$ involves only products like $\hat{x}_2 \hat{p}_3$ with different indices, which commute.) Each $\hat{L}_i$ is Hermitian, being built from commuting Hermitian factors, and so is $\hat{L}^2$.

The central computation is the commutator of two components. Let's do one honestly, acting on a test function and using $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$:

$$\begin{aligned} [\hat{L}_1, \hat{L}_2] &= [\hat{x}_2 \hat{p}_3 - \hat{x}_3 \hat{p}_2, \hat{x}_3 \hat{p}_1 - \hat{x}_1 \hat{p}_3] \\ &= [\hat{x}_2 \hat{p}_3, \hat{x}_3 \hat{p}_1] + [\hat{x}_3 \hat{p}_2, \hat{x}_1 \hat{p}_3] \\ &= \hat{x}_2 [\hat{p}_3, \hat{x}_3] \hat{p}_1 + \hat{x}_1 [\hat{x}_3, \hat{p}_3] \hat{p}_2 = i\hbar (\hat{x}_1 \hat{p}_2 - \hat{x}_2 \hat{p}_1) = i\hbar \hat{L}_3, \end{aligned}$$

where the two dropped commutators vanish because all their factors commute. The same computation with indices cycled gives the full set of **angular momentum commutation relations**:

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k.$$

So the three components of angular momentum are mutually *incompatible*: no state has definite values of two different components (except the states with $\hat{\mathbf{L}}\psi = 0$). We cannot know the angular momentum vector; the best we can hope for is one component, plus the total length. And indeed:

Proposition 21.7.5

$[\hat{L}^2, \hat{L}_i] = 0$ for each i .

Proof. Using the Leibniz rule and the commutation relations (take $i = 3$, the others

being identical):

$$\begin{aligned} [\hat{L}^2, \hat{L}_3] &= \sum_{j=1}^3 (\hat{L}_j [\hat{L}_j, \hat{L}_3] + [\hat{L}_j, \hat{L}_3] \hat{L}_j) \\ &= i\hbar (-\hat{L}_1 \hat{L}_2 - \hat{L}_2 \hat{L}_1 + \hat{L}_2 \hat{L}_1 + \hat{L}_1 \hat{L}_2) = 0, \end{aligned}$$

the $j = 3$ terms vanishing outright. $\square$

By convention we choose the compatible pair to be $\hat{L}^2$ and $\hat{L}_3 = \hat{L}_z$. Moreover, angular momentum is compatible with any spherically symmetric Hamiltonian.

Proposition 21.7.6

For $\hat{H} = \frac{\hat{p}^2}{2m} + U(r)$ with U spherically symmetric,

$$[\hat{H}, \hat{L}_i] = 0 \quad \text{and} \quad [\hat{H}, \hat{L}^2] = 0.$$

Proof (Sketch). First, using the Leibniz rules and the canonical commutation relations, one computes

$$[\hat{L}_i, \hat{x}_j] = i\hbar \epsilon_{ijk} \hat{x}_k, \quad [\hat{L}_i, \hat{p}_j] = i\hbar \epsilon_{ijk} \hat{p}_k$$

– both position and momentum transform as vectors. From these,

$$[\hat{L}_i, \hat{p}^2] = \sum_j (\hat{p}_j [\hat{L}_i, \hat{p}_j] + [\hat{L}_i, \hat{p}_j] \hat{p}_j) = i\hbar \epsilon_{ijk} (\hat{p}_j \hat{p}_k + \hat{p}_k \hat{p}_j) = 0,$$

since a symmetric expression is contracted against the antisymmetric ϵ_{ijk} . Identically $[\hat{L}_i, \hat{x}^2] = 0$, and more generally $\hat{L}_i$ commutes with any function of r alone (e.g. by noting $\hat{L}_i$ involves only angular derivatives, as the spherical coordinate expressions below make manifest). Hence $[\hat{H}, \hat{L}_i] = 0$, and then $[\hat{H}, \hat{L}^2] = 0$ follows by the Leibniz rule. $\square$

So rotations are symmetries, and angular momentum is conserved (by Ehrenfest's theorem, $\frac{d}{dt} \langle \hat{L}_i \rangle = 0$), exactly as classically. So for a central potential we have a triple of mutually commuting observables

$$\{\hat{H}, \hat{L}^2, \hat{L}_z\},$$

and by the compatibility theorem of §6.4 we can look for a complete set of *joint* eigenfunctions of all three at once. (This set is in fact maximal – there is no further independent operator commuting with all of them – so these three quantum numbers are a complete label for the states.) This is the quantum version of ‘use conservation of $\mathbf{L}$ to reduce to a radial problem’, and it is exactly how we will crack the hydrogen atom.

§21.7.4 Spherical Harmonics

To find the joint eigenfunctions, we express the angular momentum operators in spherical polar coordinates. For $\hat{L}_z$ the computation is short enough to do in full: holding r and θ fixed and varying ϕ , we have $\partial x / \partial \phi = -r \sin \theta \sin \phi = -y$ and $\partial y / \partial \phi = r \sin \theta \cos \phi = x$

(and $\partial z/\partial\phi = 0$), so by the chain rule

$$\frac{\partial}{\partial\phi} = x \frac{\partial}{\partial y} - y \frac{\partial}{\partial x},$$

which is exactly $\hat{L}_z/(-i\hbar)$. That is, we get the beautifully simple

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial\phi},$$

(rotation about the z -axis is translation in ϕ – compare $\hat{p} = -i\hbar\partial_x$!) and, by a longer but entirely similar computation,

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right].$$

Comparing with the Laplacian in spherical polars, we discover the identity

$$-\hbar^2 \nabla^2 = -\frac{\hbar^2}{r} \frac{\partial^2}{\partial r^2} r + \frac{\hat{L}^2}{r^2},$$

i.e. the angular part of the kinetic energy is $\hat{L}^2/2mr^2$ – exactly the centrifugal term of classical mechanics.

The operators $\hat{L}_z, \hat{L}^2$ act only on angles, so we look for their joint eigenfunctions as functions $Y(\theta, \phi)$ on the unit sphere:

$$\hat{L}_z Y = \hbar m Y, \quad \hat{L}^2 Y = \lambda Y.$$

Separating $Y(\theta, \phi) = P(\theta) \Phi(\phi)$, the $\hat{L}_z$ equation gives $-i\hbar\Phi' = \hbar m\Phi$, so

$$\Phi(\phi) = e^{im\phi}.$$

Now, the point with coordinates ϕ and $\phi + 2\pi$ is the *same point of space*, so single-valuedness of the wavefunction demands $e^{2\pi im} = 1$:

$$m \in \mathbb{Z}.$$

The eigenvalues of any angular momentum component are integer multiples of $\hbar$ – quantisation of angular momentum has just fallen out of periodicity, with no extra input. (m is called the **magnetic** (or azimuthal) **quantum number**.)

Substituting into the $\hat{L}^2$ equation gives an ODE in θ alone:

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{dP}{d\theta} \right) - \frac{m^2}{\sin^2\theta} P = -\frac{\lambda}{\hbar^2} P.$$

This is the *associated Legendre equation* (in the variable $u = \cos\theta$). The analysis is of the same flavour as the Hermite story – generic solutions blow up at the poles $\theta = 0, \pi$, and demanding regularity quantises the eigenvalue – so we just quote the result: regular solutions exist precisely when

$$\lambda = \hbar^2 \ell(\ell + 1), \quad \ell \in \{0, 1, 2, \dots\}, \quad |m| \leq \ell,$$

and are the *associated Legendre functions*

$$P_{\ell,m}(\cos\theta) = (\sin\theta)^{|m|} \frac{d^{|m|}}{d(\cos\theta)^{|m|}} P_{\ell}(\cos\theta),$$

where P_{ℓ} is the ℓ th Legendre polynomial. The constraint $|m| \leq \ell$ appears because P_{ℓ} has degree ℓ , so differentiating more than ℓ times gives zero – and it is the quantum counterpart of the classical triviality $|L_z| \leq |\mathbf{L}|$. We call ℓ the **total angular momentum quantum number**.

Definition 21.7.7 (Spherical Harmonics)

The joint eigenfunctions

$$Y_{\ell,m}(\theta, \phi) = P_{\ell,m}(\cos \theta) e^{im\phi}, \quad \hat{L}^2 Y_{\ell,m} = \hbar^2 \ell(\ell+1) Y_{\ell,m}, \quad \hat{L}_z Y_{\ell,m} = \hbar m Y_{\ell,m},$$

are called the **spherical harmonics**.

For each ℓ there are exactly $2\ell+1$ values of m , and the $Y_{\ell,m}$ form a complete orthogonal family on the sphere (as they must, being eigenfunctions of Hermitian operators with distinct eigenvalue pairs). The first few, normalised so that $\int_{S^2} |Y|^2 = 1$:

$$Y_{0,0} = \frac{1}{\sqrt{4\pi}}, \quad Y_{1,0} = \sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_{1,\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}.$$

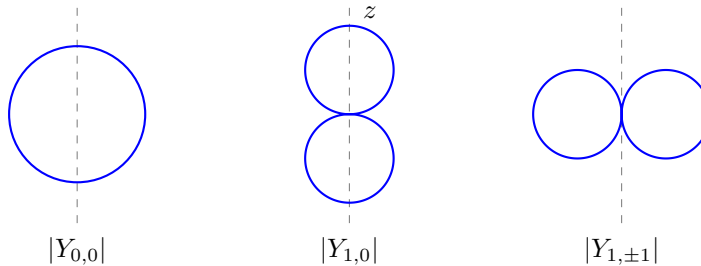
Example 21.7.8 (Verifying $Y_{1,0}$)

Let's check directly that $Y = \cos \theta$ does what we claim. It has no ϕ dependence, so $\hat{L}_z Y = -i\hbar \partial Y / \partial \phi = 0$: indeed $m = 0$. For $\hat{L}^2$:

$$\hat{L}^2 Y = -\frac{\hbar^2}{\sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cdot (-\sin \theta)) = \frac{\hbar^2}{\sin \theta} \cdot 2 \sin \theta \cos \theta = 2\hbar^2 \cos \theta,$$

so $\hat{L}^2 Y = \ell(\ell+1)\hbar^2 Y$ with $\ell = 1$, as claimed. A measurement of $\mathbf{L}^2$ in this state returns $2\hbar^2$ – note, *not* $\hbar^2$: the eigenvalues of $\hat{L}^2$ are $\ell(\ell+1)\hbar^2$, never a perfect square times $\hbar^2$, one of several places where the true quantum answer differs from the Bohr model's $L^2 = n^2\hbar^2$.

It is worth having a picture of these. Plotting $|Y_{\ell,m}|$ radially against direction (a cross-section through the z -axis is shown; the full shapes are these curves revolved about the vertical axis):



The state $Y_{0,0}$ is isotropic (no angular momentum, no preferred direction); $Y_{1,0}$ concentrates along the z -axis; $Y_{1,\pm 1}$ concentrate in the equatorial plane. Readers who have met atomic orbitals will recognise the s and p orbitals – chemistry's p_x, p_y, p_z are real linear combinations of $Y_{1,0}$ and $Y_{1,\pm 1}$.

Example 21.7.9 (Measuring L_z)

Suppose a particle's angular dependence is proportional to $\sin \theta \cos \phi$ (this is the ' p_x orbital' – and one can check it is orthogonal to $Y_{0,0}$ and $Y_{1,0}$ directly, e.g. $\int_{S^2} \cos \theta \sin \theta \cos \phi d\Omega = 0$ since the ϕ -integral of $\cos \phi$ vanishes). What does a measurement of L_z give? First expand in spherical harmonics: since $\cos \phi =$

$$\frac{1}{2}(e^{i\phi} + e^{-i\phi}),$$

$$\sin \theta \cos \phi = \frac{1}{2} \sin \theta e^{i\phi} + \frac{1}{2} \sin \theta e^{-i\phi} \propto Y_{1,1} - Y_{1,-1}$$

(up to the normalisation constants), an equal-weight superposition of $m = 1$ and $m = -1$. So by the measurement postulates, L_z returns $+\hbar$ or $-\hbar$, each with probability $\frac{1}{2}$ – and never 0, even though $\langle \hat{L}_z \rangle = 0$. A measurement of $\mathbf{L}^2$ on the same state returns $2\hbar^2$ with certainty, since only $\ell = 1$ appears. This is the whole apparatus of Section 6 – expansion, Born rule, compatible observables – running in its natural habitat.

Finally, let's put the pieces together for a general central potential. Since $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$ commute, we may take energy eigenfunctions of the separated form

$$\chi(r, \theta, \phi) = R(r) Y_{\ell, m}(\theta, \phi),$$

and substituting into $\hat{H}\chi = E\chi$ using $-\hbar^2\nabla^2 = -\frac{\hbar^2}{r}\partial_r^2 r + \frac{\hat{L}^2}{r^2}$, the spherical harmonic passes through everything and cancels, leaving the **radial Schrödinger equation**

$$-\frac{\hbar^2}{2m} \left(R'' + \frac{2}{r} R' \right) + \underbrace{\left(U(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right)}_{U_{\text{eff}}(r)} R = ER.$$

This is a one-dimensional problem in the *effective potential* U_{eff} : the true potential plus the centrifugal barrier $\hbar^2 \ell(\ell+1)/2mr^2$, which pushes states with angular momentum away from the origin. Note two facts worth savouring:

- (i) taking $\ell = 0$ recovers exactly the spherically symmetric analysis of §7.2 – the ‘s-wave’ states;
- (ii) m does not appear in the radial equation at all, so every energy level automatically comes with degeneracy at least $2\ell+1$: rotating a state costs no energy. Degeneracy is symmetry made visible.

§21.8 The Hydrogen Atom

We have assembled everything needed for the crown jewel of the course: the quantum mechanics of the hydrogen atom, the problem whose experimental fingerprints (the spectral lines of Section 1) forced the whole theory into being.

Hydrogen is one electron bound to one proton. The proton is nearly two thousand times heavier, so we model it as fixed at the origin⁷, creating the Coulomb potential in which the electron moves:

$$U(r) = -\frac{e^2}{4\pi\epsilon_0} \cdot \frac{1}{r}.$$

Since $U \rightarrow 0$ at infinity, bound states – normalisable eigenfunctions – must have $E < 0$, and these are what we look for.

⁷More honestly, one works in the centre-of-mass frame and replaces m_e by the reduced mass $\mu = m_e m_p / (m_e + m_p) \approx 0.9995 m_e$ – a correction well below the accuracy we care about here (though spectroscopists can see it: it is how deuterium was discovered).

§21.8.1 The s-wave States

Let's warm up with the spherically symmetric states, $\ell = 0$, following §7.2 – this case already contains the entire energy spectrum. The radial equation reads

$$-\frac{\hbar^2}{2m_e} \left(\chi'' + \frac{2}{r} \chi' \right) - \frac{e^2}{4\pi\epsilon_0 r} \chi = E\chi.$$

Tidy up by defining the (positive) constants

$$\nu^2 = -\frac{2m_e E}{\hbar^2}, \quad \beta = \frac{e^2 m_e}{2\pi\epsilon_0 \hbar^2},$$

so the equation becomes

$$\chi'' + \frac{2}{r} \chi' + \left(\frac{\beta}{r} - \nu^2 \right) \chi = 0.$$

As $r \rightarrow \infty$ this is $\chi'' \approx \nu^2 \chi$, so solutions behave like $e^{\pm \nu r}$ and normalisability selects the decaying exponential. As with the harmonic oscillator, we peel off the asymptotics: substitute

$$\chi(r) = f(r) e^{-\nu r},$$

giving

$$f'' + \frac{2}{r}(1 - \nu r)f' + \frac{1}{r}(\beta - 2\nu)f = 0.$$

Try a power series $f(r) = \sum_{n \geq 0} a_n r^n$ (one checks first, by trying $f \sim r^c$, that the indicial equation $c(c+1) = 0$ forces $c = 0$; the root $c = -1$ would make $\chi \sim 1/r$, which is not acceptable at the origin). Substituting and comparing coefficients of r^{n-1} :

$$n(n+1)a_n = (2\nu n - \beta)a_{n-1},$$

that is,

$$a_n = \frac{2\nu n - \beta}{n(n+1)} a_{n-1}.$$

Once again everything hinges on whether the series terminates. If it does not, then asymptotically $a_n/a_{n-1} \sim 2\nu/n$, which is precisely the growth rate of the coefficients of $e^{2\nu r}$ – so f would grow like $e^{2\nu r}$, overwhelming the decaying factor and making $\chi \sim e^{\nu r}$ non-normalisable (the same comparison argument as for the oscillator). So the series must terminate: there is an integer $N \geq 1$ with $a_N = 0$ and $a_{N-1} \neq 0$, which by the recurrence means

$$2\nu N = \beta, \quad \text{i.e.} \quad \nu = \frac{\beta}{2N}.$$

Unwinding the definitions of ν and β , we have found the energy levels:

$$E_N = -\frac{\hbar^2 \nu^2}{2m_e} = -\frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \cdot \frac{1}{N^2},$$

in exact agreement with the Bohr model, and hence with the observed spectrum – but this time derived from an honest wave equation, with no orbits postulated.

The eigenfunctions are polynomials (of degree $N-1$, the *Laguerre polynomials*) times a decaying exponential. In particular the ground state, $N=1$, is a pure exponential:

$$\chi_1(r) = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}, \quad a_0 = \frac{1}{\nu} \Big|_{N=1} = \frac{2}{\beta} = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2},$$

where the decay length is exactly the Bohr radius $a_0 \approx 5.3 \times 10^{-11}$ m of Section 1.

Example 21.8.1 (The Size of the Atom)

Where is the electron? Let's compute the expected radius in the ground state:

$$\langle r \rangle = \int_{\mathbb{R}^3} \chi_1^* r \chi_1 dV = \frac{4\pi}{\pi a_0^3} \int_0^\infty r^3 e^{-2r/a_0} dr = \frac{4}{a_0^3} \cdot \frac{3! a_0^4}{2^4} = \frac{3}{2} a_0,$$

using $\int_0^\infty r^3 e^{-\alpha r} dr = 3!/\alpha^4$. So the quantum atom is one and a half times larger, on average, than Bohr's orbit – a genuinely different (and experimentally correct) prediction. The electron is not on a circle of radius a_0 ; it is in a fuzzy exponential cloud.

Example 21.8.2 (Energy Bookkeeping in the Ground State)

In the same way, $\langle 1/r \rangle = 4\pi \cdot \frac{1}{\pi a_0^3} \int_0^\infty r e^{-2r/a_0} dr = 1/a_0$, so the expected potential energy in the ground state is

$$\langle U \rangle = -\frac{e^2}{4\pi\epsilon_0} \left\langle \frac{1}{r} \right\rangle = -\frac{e^2}{4\pi\epsilon_0 a_0} = 2E_1,$$

and hence the expected kinetic energy is $\langle T \rangle = E_1 - \langle U \rangle = -E_1 > 0$. So

$$\langle T \rangle = -\frac{1}{2} \langle U \rangle,$$

which is precisely the *virial theorem* for an inverse-square force law, in expectation – another classical relation surviving intact as a statement about means. Numerically: the electron has kinetic energy +13.6 eV and potential energy –27.2 eV, and it costs 13.6 eV to pull it away.

§21.8.2 The General Solution

Now the full problem, with angular momentum. By §7.4 we take

$$\chi(r, \theta, \phi) = R(r) Y_{\ell, m}(\theta, \phi),$$

where the radial function satisfies

$$R'' + \frac{2}{r} R' + \left(\frac{\beta}{r} - \nu^2 - \frac{\ell(\ell+1)}{r^2} \right) R = 0.$$

The large- r asymptotics are unchanged (the centrifugal term is subleading), so again $R = g(r)e^{-\nu r}$. Near the origin the centrifugal term dominates: trying $g \sim r^c$ gives the indicial equation

$$c(c+1) = \ell(\ell+1) \implies c = \ell \text{ or } c = -\ell - 1,$$

and regularity at the origin forces $c = \ell$ – states with angular momentum are suppressed like r^ℓ near the centre, pushed out by the centrifugal barrier. So we write

$$g(r) = r^\ell \sum_{n \geq 0} a_n r^n,$$

and the same machinery as before (substitute, compare coefficients) produces the recurrence

$$a_n = \frac{2\nu(n+\ell) - \beta}{n(n+2\ell+1)} a_{n-1}.$$

Again the series must terminate or R grows like $e^{\nu r}$; terminating at the $n_{\max}$ -th term requires

$$2\nu(n_{\max} + \ell) = \beta, \quad n_{\max} \geq 1.$$

Defining the **principal quantum number** $N = n_{\max} + \ell$, we get $\nu = \beta/2N$ exactly as before, and therefore *exactly the same energy levels*

$$E_N = -\frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \cdot \frac{1}{N^2} = -\frac{\hbar^2}{2m_e a_0^2} \cdot \frac{1}{N^2},$$

now with the constraint $\ell \leq N - 1$ (since $n_{\max} \geq 1$). The full set of stationary states is indexed by three quantum numbers:

$$\chi_{N,\ell,m}(r, \theta, \phi) = R_{N,\ell}(r) Y_{\ell,m}(\theta, \phi), \quad \begin{cases} N = 1, 2, 3, \dots \\ \ell = 0, 1, \dots, N - 1, \\ m = -\ell, \dots, \ell, \end{cases}$$

with $R_{N,\ell}(r) = r^\ell g_{N,\ell}(r) e^{-r/Na_0}$ and $g_{N,\ell}$ a polynomial of degree $N - \ell - 1$ (a *generalised Laguerre polynomial*). In spectroscopic notation, states with $\ell = 0, 1, 2, \dots$ are called $s, p, d, \dots$ states; the state $\chi_{2,1,m}$ is ‘ $2p$ ’, and so on.

Example 21.8.3 (The First Few Radial Functions)

The recurrence makes these easy to generate. For $N = 1$, $\ell = 0$ the polynomial is a constant, giving the ground state we already know. For $N = 2$, $\ell = 0$ we have $\nu = 1/2a_0$ and one step of the recurrence gives $a_1/a_0 = (2\nu - \beta)/2 = -\nu$, while for $N = 2$, $\ell = 1$ the polynomial is again constant. Normalising:

$$\begin{aligned} R_{1,0}(r) &= \frac{2}{a_0^{3/2}} e^{-r/a_0}, \\ R_{2,0}(r) &= \frac{1}{\sqrt{2} a_0^{3/2}} \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0}, \\ R_{2,1}(r) &= \frac{1}{2\sqrt{6} a_0^{3/2}} \cdot \frac{r}{a_0} e^{-r/2a_0}. \end{aligned}$$

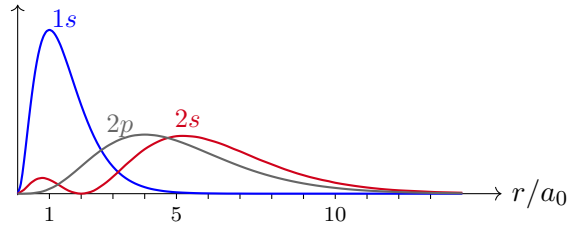
Note the general features on display: $N - \ell - 1$ radial nodes ($R_{2,0}$ vanishes at $r = 2a_0$; $R_{2,1}$ has none), the r^ℓ suppression at the origin, and the decay length Na_0 growing with the principal quantum number – excited atoms are physically bigger.

The energy depends on N *alone* – not on ℓ or m . So the level E_N is highly degenerate:

$$D(N) = \sum_{\ell=0}^{N-1} (2\ell + 1) = N^2.$$

The m -degeneracy is the rotational symmetry we understood in §7.4; but the ℓ -degeneracy – states with visibly different shapes and different angular momenta sharing an energy exactly – is special to the $1/r$ potential, and is called an *accidental* degeneracy. (It is not really an accident: the Coulomb problem has a hidden extra conserved quantity, the Runge–Lenz vector, just as in the classical Kepler problem where it is responsible for orbits being closed ellipses. But that story belongs to another course.)

It is illuminating to look at the *radial probability density* $P(r) = r^2 |R_{N,\ell}(r)|^2$ (the r^2 from the volume element: $P(r)\delta r$ is the probability of finding the electron at radius between r and $r + \delta r$):



The $1s$ density peaks exactly at $r = a_0$ (differentiate $r^2 e^{-2r/a_0}$ and see) – so Bohr’s orbit survives as the *most probable* radius of the ground state. The $2s$ state shows a radial node (a spherical shell where the electron is never found), while $2p$ has none but is pushed outwards by the centrifugal barrier; both are concentrated roughly four times (N^2) further out than $1s$.

Exercise 21.8.4. Show that the radial density of the $2p$ state, $P(r) \propto r^4 e^{-r/a_0}$, peaks at exactly $r = 4a_0 = 2^2 a_0$ – the Bohr orbit for $N = 2$ – and more generally that the ‘circular’ states ($\ell = N - 1$, whose radial functions $r^{N-1} e^{-r/Na_0}$ have no nodes) peak at exactly $r = N^2 a_0$. The Bohr model lives on inside quantum mechanics as the family of maximal-angular-momentum states.

§21.8.3 The Hydrogen Spectrum

Finally, let’s close the loop with Section 1. An atom in an excited state can drop to a lower level, emitting a photon carrying the energy difference⁸: a transition from level M down to level $N < M$ produces a photon of angular frequency

$$\hbar\omega = E_M - E_N = \frac{e^4 m_e}{32\pi^2 \epsilon_0^2 \hbar^2} \left(\frac{1}{N^2} - \frac{1}{M^2} \right),$$

which is precisely the Rydberg formula, with the Rydberg constant now expressed in fundamental constants (and matching its measured value). Transitions down to $N = 1$ form the *Lyman series* (ultraviolet), those down to $N = 2$ the *Balmer series* (the visible lines that gave the game away historically), down to $N = 3$ the *Paschen series* (infrared), and so on.

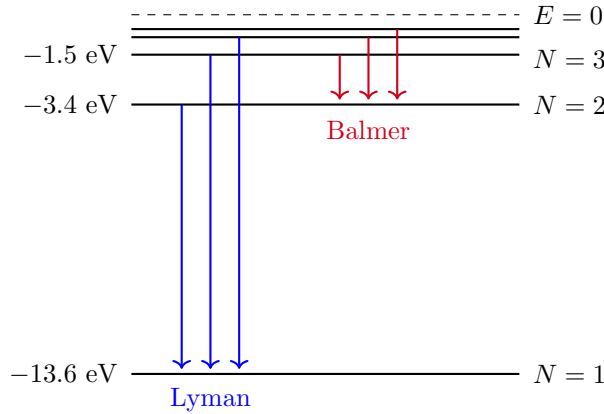
Example 21.8.5 (The Red Line of Hydrogen)

The lowest Balmer transition, $3 \rightarrow 2$, releases

$$E_3 - E_2 = 13.6 \left(\frac{1}{4} - \frac{1}{9} \right) \text{ eV} = 13.6 \cdot \frac{5}{36} \text{ eV} \approx 1.89 \text{ eV},$$

corresponding to a wavelength $\lambda = 2\pi\hbar c / (E_3 - E_2) \approx 656 \text{ nm}$ – the famous red line $H\alpha$, visible in a discharge tube in any teaching lab, and responsible for the red glow of emission nebulae across the night sky. It is hard to think of another place in physics where a short power series computation connects so directly to the colour of things.

⁸The mechanism of the transition – the coupling of the atom to the electromagnetic field – is outside our theory: strictly, our stationary states never change. Getting this right requires quantum electrodynamics; the energy bookkeeping, which is all we need for the spectrum, does not.



For $E > 0$ the spectrum of $\hat{H}$ is continuous – these are the scattering states of an ionised electron flying past a proton – which is why the discrete lines accumulate at the *ionisation energy* 13.6 eV and give way to a continuum beyond.

Let's take stock of what quantum mechanics has achieved here, compared with the Bohr model it replaces.

- (i) The energy levels and spectrum come out identically – and correctly – but now as eigenvalues of a differential operator, with the quantisation forced by normalisability, not decreed.
- (ii) The angular momentum is completely different, and Bohr's was wrong: the true eigenvalues of $\hat{L}^2$ are $\hbar^2\ell(\ell+1)$ with $\ell \leq N-1$, not $\hbar^2N^2$ – so the ground state has *zero* angular momentum (it agrees with Bohr's $L = N\hbar$ only in the limit of large ℓ). The electron does not orbit.
- (iii) The model explains what Bohr could not: the degeneracies N^2 (visible in the splitting of spectral lines in electric and magnetic fields), and the relative intensities of the lines, via the shapes of the wavefunctions.

§21.8.4 Beyond Hydrogen

It is natural to ask how far this machinery reaches. For an atom of atomic number Z – a nucleus of charge $+Ze$ with Z electrons – the honest Hamiltonian acts on a wavefunction $\psi(\mathbf{x}_1, \dots, \mathbf{x}_Z)$ of all the electron positions at once:

$$\hat{H} = \sum_{j=1}^Z \left(-\frac{\hbar^2}{2m_e} \nabla_j^2 - \frac{Ze^2}{4\pi\epsilon_0|\mathbf{x}_j|} \right) + \sum_{j<k} \frac{e^2}{4\pi\epsilon_0|\mathbf{x}_j - \mathbf{x}_k|}.$$

The last sum – the electron-electron repulsion – is what ruins everything: without it, the equation would separate, and the eigenfunctions would just be products of hydrogen-like states

$$\psi = \chi_{(1)}(\mathbf{x}_1)\chi_{(2)}(\mathbf{x}_2)\cdots\chi_{(Z)}(\mathbf{x}_Z), \quad E = \sum_j E_{(j)},$$

each factor a state $\chi_{N,\ell,m}$ in the (rescaled, $a_0 \mapsto a_0/Z$) Coulomb potential. With the repulsion, no exact solution is known for any $Z \geq 2$, and one falls back on treating the product form as an approximation in which each electron sees an averaged, screened nuclear charge. This is crude but structurally right, and two extra ingredients – both beyond this course, both essential – turn it into the periodic table:

- (i) electrons carry *spin*, an internal two-valued degree of freedom with no classical analogue, doubling the count of states at each level from N^2 to $2N^2$;
- (ii) identical electrons obey the *Pauli exclusion principle*: no two may occupy the same state.

Filling the one-electron levels from the bottom, two electrons at a time per orbital, gives shells of sizes $2, 8, 18, \dots = 2N^2$, and the chemistry of an element is governed by its outermost, partially-filled shell – the periodicity of the periodic table is the level structure of the hydrogen atom, doubled and stacked. (The screening breaks the accidental ℓ -degeneracy, which is why the shells actually fill in the slightly scrambled order familiar from chemistry.)

Getting all of this right – spin, identical particles, perturbative approximations, and the deeper structure (Dirac notation, abstract Hilbert spaces, symmetries) hiding beneath the wave mechanics we have used – is the business of the sequel courses, ‘Principles of Quantum Mechanics’ above all.

And that is a fitting place to stop. From five axioms about wavefunctions we have recovered – and corrected – the atom.

22 Statistics

Statistics is the science of making informed decisions. It is an area of mathematics that is widely applicable, and covers the design of experiments, the analysis of data, statistical inference, and the communication of uncertainty and risk. In this course we will focus on formal statistical inference.

Probability, as we met it in IA, starts from a fully specified model and asks what the data will look like. Statistics runs that arrow backwards: we are handed the data, and asked what the model must have been. That inversion is what gives the subject its character – there is never a single right answer, only estimates with quantified uncertainty, and procedures whose long-run behaviour we can guarantee. Almost everything in these notes is an attempt to make one of the words ‘estimate’, ‘uncertainty’ or ‘guarantee’ precise.

This article constitutes my notes for the ‘Statistics’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered. Note that familiarity with classical probability theory is assumed, and a full recap is not included – the reader is referred to my ‘Probability’ notes for anything in Section 1 that is unfamiliar.

§22.1 Introduction

§22.1.1 Parametric Inference

Let $X_1, \dots, X_n$ be i.i.d random variables. We will assume that the distribution of X_i belongs to some family with parameter $\theta \in \Theta$. For example, we could have $X_1 \sim \text{Poisson}(\mu)$ with $\theta = \mu \in \Theta = (0, \infty)$. We can also have a multi-dimensional parameter such as $X_1 \sim N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2) \in \Theta = \mathbb{R} \times (0, \infty)$. We will use the observed $X = (X_1, \dots, X_n)$ to make inferences about the parameter θ . Such inferences include:

1. Point estimate of $\hat{\theta}(X)$ of θ ;
2. An interval estimate $(\hat{\theta}_1(X), \hat{\theta}_2(X))$ of θ ;
3. Testing hypothesis about θ , that is, checking whether there is evidence in the data X against a given hypothesis.

In this course we will usually assume that the distribution family of $X_1, \dots, X_n$ is known, and that the parameter is unknown. From time to time we will be able to make slightly more general statements.

This setup is called **parametric** inference, because the unknown is a finite-dimensional parameter. It is a genuine modelling assumption, and a strong one: if the family is badly chosen then no amount of clever estimation will save us. In practice one checks the family afterwards, and Section 5 gives us the tools (goodness-of-fit tests) to do exactly that.

§22.1.2 Random Samples and Statistics

We should fix some language before we start.

Definition 22.1.1 (Random Sample)

A **random sample** of size n from a distribution f is a collection $X = (X_1, \dots, X_n)$ of independent random variables, each with distribution f . We write $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.

Definition 22.1.2 (Statistic)

A **statistic** is any function $T(X)$ of the data alone. In particular a statistic may *not* depend on the unknown parameter θ : it must be something we can actually compute once the data are in front of us.

The distribution of a statistic $T(X)$ – which of course will in general depend on θ – is called its **sampling distribution**. Most of statistics consists of choosing a statistic whose sampling distribution we understand well enough to say something useful.

The two statistics we will use over and over are the sample mean and the sum of squared deviations from it.

Definition 22.1.3 (Sample Mean and S_{XX})

Given a sample $X_1, \dots, X_n$ we write

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_{XX} = \sum_{i=1}^n (X_i - \bar{X})^2.$$

If $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X_1) = \sigma^2$ then linearity of expectation and bilinearity of variance immediately give

$$\mathbb{E}[\bar{X}] = \mu, \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n},$$

so the sample mean is ‘centred at the right place’ and gets more concentrated as n grows. It is worth recording the standardised version

$$Z_n = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma},$$

which has mean 0 and variance 1 whatever μ and σ are. The central limit theorem says Z_n is approximately $N(0, 1)$ for large n , and this single fact is responsible for a remarkable fraction of the approximations in this course.

There is one small wrinkle worth noticing now, since it will bite us later. A short computation gives

$$\begin{aligned} \mathbb{E}[S_{XX}] &= \mathbb{E} \left[\sum_i (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right] \\ &= n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2, \end{aligned}$$

so S_{XX}/n *underestimates* σ^2 , and it is $S_{XX}/(n-1)$ that is unbiased. The intuition is that we have used up one ‘degree of freedom’ estimating the mean, and the data look slightly less spread out about $\bar{X}$ than about the true μ . This phrase ‘degrees of freedom’ will be given a precise geometric meaning in Section 7.

§22.1.3 Sampling Distributions from the Normal

Almost all the exact distribution theory in this course comes from one place: sums of squares of independent standard normals. We collect the three families here, since we will want to refer to them from the very beginning. All three are already implicitly present in IA Probability as special cases or ratios of gamma distributions.

Recall that $X \sim \text{Gamma}(\alpha, \lambda)$ has density proportional to $x^{\alpha-1}e^{-\lambda x}$ on $(0, \infty)$, and that gammas with a common rate add: $\text{Gamma}(\alpha_1, \lambda) + \text{Gamma}(\alpha_2, \lambda) = \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$ for independent summands.

Definition 22.1.4 (Chi-Squared Distribution)

Let $Z_1, \dots, Z_k$ be i.i.d $N(0, 1)$. Then

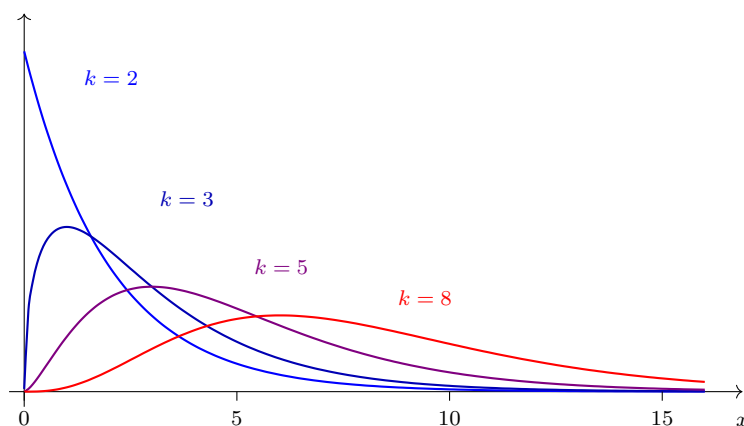
$$\chi_k^2 := Z_1^2 + \dots + Z_k^2$$

is said to have the **chi-squared distribution on k degrees of freedom**.

If $Z \sim N(0, 1)$ then $Z^2 \sim \text{Gamma}(1/2, 1/2)$ (a one-line change of variables), so by the addition rule $\chi_k^2 \sim \text{Gamma}(k/2, 1/2)$, with density

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}, \quad x > 0.$$

In particular $\mathbb{E}[\chi_k^2] = k$ and $\text{Var}(\chi_k^2) = 2k$. The shape changes character as k increases: for $k \leq 2$ the density is decreasing, and for $k \geq 3$ it has an interior mode at $k - 2$, drifting rightwards and flattening out.



We will write $\chi_k^2(\alpha)$ for the **upper α point** of the distribution, that is, the number with $\mathbb{P}(\chi_k^2 > \chi_k^2(\alpha)) = \alpha$. The same convention will be used for every distribution in these notes.

Definition 22.1.5 (Student's t -Distribution)

Let $Z \sim N(0, 1)$ and $V \sim \chi_k^2$ be independent. Then

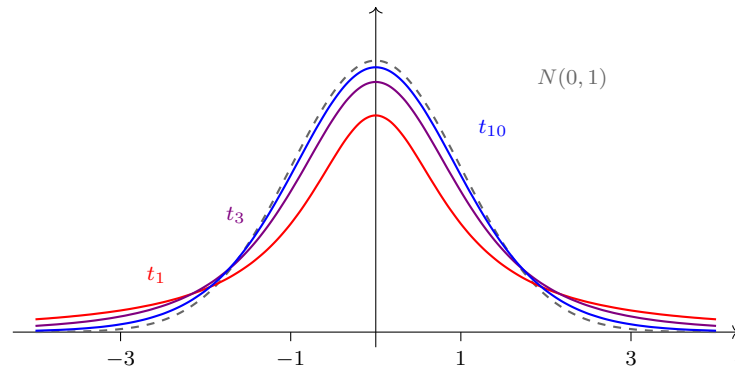
$$T = \frac{Z}{\sqrt{V/k}}$$

is said to have the **t -distribution on k degrees of freedom**, written $T \sim t_k$.

The point of this definition – which looks arbitrary until you see it used – is that it describes what happens when you standardise a normal mean but have to *estimate* the standard deviation rather than knowing it. We will prove exactly this in Section 6. One can compute the density to be

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\Gamma\left(\frac{k}{2}\right) \sqrt{\pi k}} \left(1 + \frac{t^2}{k}\right)^{-(k+1)/2},$$

which is symmetric and bell-shaped, but with polynomially decaying (hence much fatter) tails than the normal. Indeed $\mathbb{P}(T > t) > \mathbb{P}(Z > t)$ for every $t > 0$: having to estimate σ costs us, and the cost shows up precisely as extra probability far from the centre.



As $k \rightarrow \infty$ we have $V/k \rightarrow 1$ by the law of large numbers, so $t_k \rightarrow N(0, 1)$; the picture above shows this happening already by $k = 10$. For small k the tails really are heavy: t_1 is the Cauchy distribution, which has no mean at all. In general $\mathbb{E}[T] = 0$ for $k > 1$ and $\text{Var}(T) = k/(k-2)$ for $k > 2$, with the variance infinite at $k = 2$ and undefined at $k = 1$.

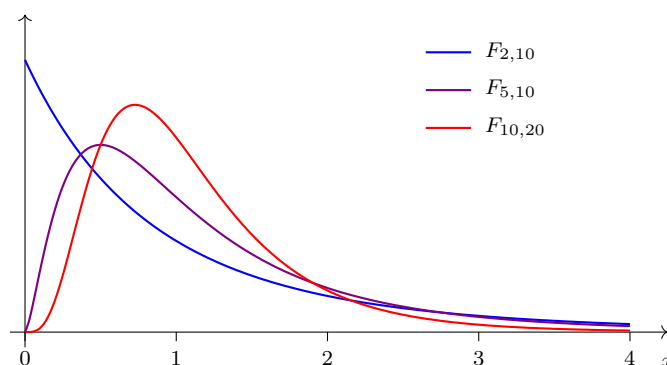
Definition 22.1.6 (*F*-Distribution)

Let $V \sim \chi_m^2$ and $W \sim \chi_n^2$ be independent. Then

$$F = \frac{V/m}{W/n}$$

is said to have the ***F*-distribution on m and n degrees of freedom**, written $F \sim F_{m,n}$.

This is what appears when we want to compare two variances, or – much more importantly – when we want to compare the fit of two nested models by comparing two sums of squares. It is supported on $(0, \infty)$ and concentrates around 1 as both $m, n \rightarrow \infty$, which is what one would expect of a ratio of two estimates of the same quantity.



Two relations are worth remembering, both immediate from the definitions. First, if $T \sim t_n$ then $T^2 \sim F_{1,n}$, since $T^2 = Z^2/(V/n)$ and $Z^2 \sim \chi_1^2$. Second, if $F \sim F_{m,n}$ then $1/F \sim F_{n,m}$. The first of these will let us identify a two-sided t -test with an F -test in Section 7.

§22.2 Estimation

Suppose $X_1, \dots, X_n$ are i.i.d observations with a pdf (or pmf) $f_X(x \mid \theta)$ where θ is an unknown parameter in a parameter space Θ . The first and most basic inferential question is: what single value of θ do the data suggest?

§22.2.1 Estimators, Bias and Mean Squared Error

Definition 22.2.1 (Estimator)

An **estimator** is a statistic or function of the data $T(X) = \hat{\theta}$ which does *not* depend on θ , and is used to approximate the true parameter θ .

The distribution of $T(X)$ is its **sampling distribution**.

We can then go on to define the *bias* of an estimator.

Definition 22.2.2 (Bias)

The **bias** of $\hat{\theta} = T(X)$ is

$$\text{bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta,$$

where $\mathbb{E}_\theta$ gives expectation in the model $X_i \sim f_X(\cdot \mid \theta)$.

Remark. In general the bias is a function of the true parameter θ , even though it is not explicit in the notation $\text{bias}(\hat{\theta})$.

Definition 22.2.3 (Unbiased Estimator)

We say that $\hat{\theta}$ is **unbiased** if $\text{bias}(\hat{\theta}) = 0$ for *all* values of the true parameter θ .

Example 22.2.4 (Defining an Estimator)

Consider the i.i.d random variables $X_1, \dots, X_n \sim N(\mu, 1)$. Then we can define the

estimator

$$\hat{\mu} = T(X) = \frac{1}{n} \sum_i X_i = \bar{X}.$$

The sampling distribution of $\hat{\mu}$ is $T(X) \sim N(\mu, 1/n)$.

We can see that $\hat{\mu}$ is an unbiased estimator since

$$\mathbb{E}_\theta(\hat{\mu}) = \mathbb{E}_\mu(\bar{X}) = \mu,$$

for all $\mu \in \mathbb{R}$.

Unbiasedness on its own is a very weak requirement – it says nothing about how far $\hat{\theta}$ typically is from θ , only that the errors average out. One way of measuring ‘how far’ $\hat{\theta}$ is from θ ‘on average’ is given by the *mean squared error* (mse).

Definition 22.2.5 (Mean Squared Error)

The **mean squared error** of $\hat{\theta}$ is defined to be

$$\text{mse}(\hat{\theta}) = \mathbb{E}_\theta \left[(\hat{\theta} - \theta)^2 \right].$$

Again, note that $\text{mse}(\hat{\theta})$ may depend on θ , even though this dependence is omitted from the notation.

Proposition 22.2.6 (Bias-Variance Decomposition)

For any estimator $\hat{\theta}$ with finite second moment,

$$\text{mse}(\hat{\theta}) = \text{Var}_\theta(\hat{\theta}) + \text{bias}^2(\hat{\theta}).$$

Proof. We add and subtract $\mathbb{E}_\theta \hat{\theta}$ inside the square:

$$\begin{aligned} \text{mse}(\hat{\theta}) &= \mathbb{E}_\theta \left[(\hat{\theta} - \mathbb{E}_\theta \hat{\theta} + \mathbb{E}_\theta \hat{\theta} - \theta)^2 \right] \\ &= \mathbb{E}_\theta \left[(\hat{\theta} - \mathbb{E}_\theta \hat{\theta})^2 \right] + 2(\mathbb{E}_\theta \hat{\theta} - \theta) \mathbb{E}_\theta [\hat{\theta} - \mathbb{E}_\theta \hat{\theta}] + (\mathbb{E}_\theta \hat{\theta} - \theta)^2 \\ &= \underbrace{\text{Var}_\theta(\hat{\theta})}_{\geq 0} + \underbrace{\text{bias}_\theta^2(\hat{\theta})}_{\geq 0}, \end{aligned}$$

since the cross term vanishes: $\mathbb{E}_\theta[\hat{\theta} - \mathbb{E}_\theta \hat{\theta}] = 0$. □

This decomposition is the single most useful identity in estimation theory, because it shows that there is a genuine *tradeoff*: we may be able to buy a large reduction in variance at the cost of a small bias, and come out ahead. Insisting on unbiasedness forbids that trade entirely. Let’s see this happen.

Example 22.2.7 (A Biased Estimator that Wins)

Consider the random variable $X \sim \text{Bin}(n, \theta)$, and suppose n is known, with $\theta \in [0, 1]$ being an unknown parameter.

The standard estimator for this example would be

$$T_U = \frac{X}{n},$$

which can be thought of as the ‘proportion of successes observed’. This estimator is clearly unbiased since

$$\mathbb{E}_\theta(T_U) = \frac{\mathbb{E}_\theta X}{n} = \frac{n\theta}{n} = \theta.$$

We can also compute that

$$\text{mse}(T_U) = \text{Var}_\theta(T_U) = \frac{\theta(1-\theta)}{n}.$$

Now consider another estimator

$$T_B = \frac{X+1}{n+2} = w \cdot \underbrace{\frac{X}{n}}_{T_U} + (1-w) \cdot \frac{1}{2}, \quad w = \frac{n}{n+2},$$

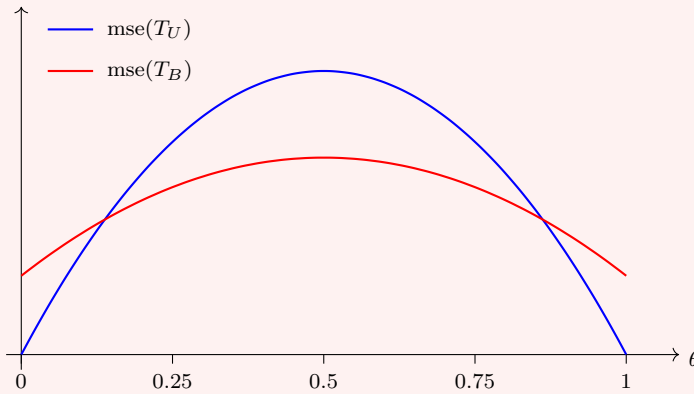
which shrinks the observed proportion towards $1/2$. Its bias is

$$\text{bias}(T_B) = \frac{n\theta + 1}{n+2} - \theta = \frac{1-2\theta}{n+2},$$

which vanishes only at $\theta = 1/2$, and its variance is $w^2\theta(1-\theta)/n$. So

$$\text{mse}(T_B) = w^2 \cdot \frac{\theta(1-\theta)}{n} + (1-w)^2 \left(\frac{1}{2} - \theta\right)^2.$$

Plotting these against θ for $n = 10$, we see that neither estimator dominates.



The biased estimator is better over a broad range of θ near $1/2$, and worse near the endpoints. Which one we prefer therefore depends on what we believe about θ before seeing the data – an observation which will lead us, in Section 4, straight to Bayesian inference.

It is also worth seeing that insisting on unbiasedness can produce estimators that are frankly absurd.

Example 22.2.8 (Unbiasedness Gone Wrong)

Suppose $X \sim \text{Poisson}(\lambda)$ and we wish to estimate $\theta = \mathbb{P}(X = 0)^2 = e^{-2\lambda}$. For $T(X)$ to be unbiased we need, for all $\lambda > 0$,

$$\mathbb{E}_\lambda[T(X)] = \sum_{x=0}^{\infty} T(x) \frac{\lambda^x e^{-\lambda}}{x!} = e^{-2\lambda},$$

that is, $\sum_{x \geq 0} T(x) \lambda^x / x! = e^{-\lambda}$. Comparing coefficients of λ^x with the power series for $e^{-\lambda}$ gives $T(x) = (-1)^x$.

So the *unique* unbiased estimator of θ takes only the values ± 1 , and estimates a quantity known to lie in $(0, 1)$ by -1 whenever we observe an odd number of events. Unbiasedness is clearly not, by itself, a guarantee of anything sensible.

§22.2.2 Sufficiency

The data $X = (X_1, \dots, X_n)$ typically contains far more detail than we need. If $X_1, \dots, X_n$ are i.i.d Bernoulli, it seems clear that the *order* of the successes tells us nothing about θ ; only the total number matters. Sufficiency is the notion that makes this precise.

Definition 22.2.9 (Sufficient Statistic)

A statistic $T(X)$ is **sufficient** for θ if the conditional distribution of X given $T(X)$ does not depend on θ .

In other words – once we know $T(X)$, the remaining randomness in X is pure noise, generated by a mechanism we could simulate ourselves without knowing θ . Nothing about θ is lost by throwing away X and keeping only $T(X)$. Note that θ and $T(X)$ may both be vector-valued, and need not have the same dimension.

Example 22.2.10 (Bernoulli Trials)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(θ) with $\theta \in [0, 1]$. The joint mass function is

$$f_X(x | \theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i},$$

which depends on x only through $t = \sum_i x_i$. Let $T(X) = \sum_i X_i$; since $T \sim \text{Bin}(n, \theta)$, for any x with $\sum_i x_i = t$ we get

$$\mathbb{P}_\theta(X = x | T(X) = t) = \frac{\mathbb{P}_\theta(X = x)}{\mathbb{P}_\theta(T = t)} = \frac{\theta^t (1 - \theta)^{n-t}}{\binom{n}{t} \theta^t (1 - \theta)^{n-t}} = \frac{1}{\binom{n}{t}},$$

and the probability is 0 if $\sum_i x_i \neq t$. This is free of θ , so T is sufficient: given the total, the particular arrangement of successes is uniform over all $\binom{n}{t}$ arrangements, regardless of θ .

Checking sufficiency from the definition means computing a conditional distribution, which is tedious. Happily there is a criterion that lets us read sufficiency straight off the likelihood.

Theorem 22.2.11 (Factorisation Criterion)

A statistic T is sufficient for θ if and only if we can write

$$f_X(x | \theta) = g(T(x), \theta) h(x)$$

for some functions g and h , where h does not depend on θ .

Proof. We give the proof in the discrete case; the continuous case is identical with sums replaced by integrals (and a little care about null sets).

Suppose first that the factorisation holds, and let $t = T(x)$. Then

$$\begin{aligned} \mathbb{P}_\theta(X = x | T(X) = t) &= \frac{\mathbb{P}_\theta(X = x, T(X) = t)}{\mathbb{P}_\theta(T(X) = t)} \\ &= \frac{g(T(x), \theta) h(x)}{\sum_{x': T(x') = t} g(T(x'), \theta) h(x')} \\ &= \frac{g(t, \theta) h(x)}{g(t, \theta) \sum_{x': T(x') = t} h(x')} = \frac{h(x)}{\sum_{x': T(x') = t} h(x')}, \end{aligned}$$

which does not depend on θ . (If $T(x) \neq t$ the conditional probability is 0, which certainly does not depend on θ .) So T is sufficient.

Conversely, suppose T is sufficient. Then

$$\begin{aligned} f_X(x | \theta) &= \mathbb{P}_\theta(X = x) = \mathbb{P}_\theta(X = x, T(X) = T(x)) \\ &= \underbrace{\mathbb{P}_\theta(X = x | T(X) = T(x))}_{=h(x)} \underbrace{\mathbb{P}_\theta(T(X) = T(x))}_{=g(T(x), \theta)}, \end{aligned}$$

where the first factor is free of θ precisely by sufficiency. $\square$

Let's put this to work.

Example 22.2.12 (Bernoulli, Again)

With $f_X(x | \theta) = \theta^{\sum x_i} (1-\theta)^{n-\sum x_i}$ we may take $T(x) = \sum_i x_i$, $g(t, \theta) = \theta^t (1-\theta)^{n-t}$ and $h(x) \equiv 1$. So T is sufficient, and we never had to compute a conditional distribution.

Example 22.2.13 (Uniform on $[0, \theta]$)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$ with $\theta > 0$. The joint density is

$$f_X(x | \theta) = \prod_{i=1}^n \frac{1}{\theta} \mathbb{1}\{0 \leq x_i \leq \theta\} = \underbrace{\frac{1}{\theta^n} \mathbb{1}\left\{\max_i x_i \leq \theta\right\}}_{g(T(x), \theta)} \cdot \underbrace{\mathbb{1}\left\{\min_i x_i \geq 0\right\}}_{h(x)},$$

so $T(X) = \max_i X_i$ is sufficient for θ . This is a good example of a sufficient statistic that is not a sum: only the largest observation constrains θ at all.

Example 22.2.14 (Normal with Both Parameters Unknown)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$ unknown. Expanding the exponent,

$$f_X(x | \theta) = (2\pi\sigma^2)^{-n/2} \exp \left(-\frac{1}{2\sigma^2} \sum_i x_i^2 + \frac{\mu}{\sigma^2} \sum_i x_i - \frac{n\mu^2}{2\sigma^2} \right),$$

which depends on x only through $(\sum_i x_i, \sum_i x_i^2)$. Taking $h \equiv 1$, this pair is sufficient for (μ, σ^2) .

§22.2.3 Minimal Sufficiency

Sufficient statistics are far from unique. The whole data $T(X) = X$ is always sufficient, and any injective function of a sufficient statistic is again sufficient. What we actually want is a sufficient statistic that compresses as much as possible.

Definition 22.2.15 (Minimal Sufficiency)

A sufficient statistic T is **minimal sufficient** if it is a function of every other sufficient statistic; equivalently, if S is sufficient then $S(x) = S(y)$ implies $T(x) = T(y)$.

Remark. If S and T are minimal sufficient then they are in bijection, that is, $T(x) = T(y)$ if and only if $S(x) = S(y)$. So we can think of minimal sufficient statistics as being unique up to bijection.

Theorem 22.2.16 (Criterion for Minimal Sufficiency)

Suppose that $f_X(x | \theta)/f_X(y | \theta)$ is constant in θ if and only if $T(x) = T(y)$. Then T is minimal sufficient.

Before proving this, it is worth understanding what the hypothesis says. Write $x \sim_1 y$ if the likelihood ratio $f_X(x | \theta)/f_X(y | \theta)$ is free of θ , and $x \sim_2 y$ if $T(x) = T(y)$. Both are equivalence relations on the sample space, and the hypothesis is exactly that they have the same equivalence classes. Note that this also gives a *recipe*: compute the classes of $\sim_1$, and define T to be any labelling of them. So minimal sufficient statistics always exist.

Proof. We first show T is sufficient. For each value t in the range of T pick a representative z_t with $T(z_t) = t$. Then for any x ,

$$f_X(x | \theta) = \underbrace{f_X(z_{T(x)} | \theta)}_{g(T(x), \theta)} \cdot \underbrace{\frac{f_X(x | \theta)}{f_X(z_{T(x)} | \theta)}}_{h(x)}.$$

The first factor depends on x only through $T(x)$, and the second is free of θ by hypothesis, since $T(x) = T(z_{T(x)})$. By the factorisation criterion, T is sufficient.

Now let S be any other sufficient statistic, so $f_X(x | \theta) = g_S(S(x), \theta)h_S(x)$. Suppose

$S(x) = S(y)$. Then

$$\frac{f_X(x | \theta)}{f_X(y | \theta)} = \frac{g_S(S(x), \theta) h_S(x)}{g_S(S(y), \theta) h_S(y)} = \frac{h_S(x)}{h_S(y)},$$

which is constant in θ . By hypothesis this forces $T(x) = T(y)$. So T is a function of S , and as S was arbitrary, T is minimal sufficient. $\square$

Example 22.2.17 (Normal Samples)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with both parameters unknown. Then

$$\frac{f_X(x | \mu, \sigma^2)}{f_X(y | \mu, \sigma^2)} = \exp \left(-\frac{1}{2\sigma^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{\mu}{\sigma^2} \left(\sum_i x_i - \sum_i y_i \right) \right).$$

This is constant in (μ, σ^2) if and only if both bracketed differences vanish, that is, if and only if $\sum_i x_i = \sum_i y_i$ and $\sum_i x_i^2 = \sum_i y_i^2$. Hence

$$T(X) = \left(\sum_i X_i, \sum_i X_i^2 \right)$$

is minimal sufficient, as is the more familiar bijective repackaging $(\bar{X}, S_{XX})$.

Example 22.2.18 (Dimension Mismatch)

The parameter and a minimal sufficient statistic need not have the same dimension. Take $X_1, \dots, X_n \sim N(\mu, \mu^2)$, so $\Theta = \mathbb{R} \setminus \{0\}$ is one-dimensional. Running the same computation with $\sigma^2 = \mu^2$, the ratio is

$$\exp \left(-\frac{1}{2\mu^2} \left(\sum_i x_i^2 - \sum_i y_i^2 \right) + \frac{1}{\mu} \left(\sum_i x_i - \sum_i y_i \right) \right),$$

and since $1/\mu$ and $1/\mu^2$ are not proportional as functions of μ , this is constant only when both differences vanish. So $(\bar{X}, S_{XX})$ is still minimal sufficient: a two-dimensional statistic for a one-dimensional parameter.

§22.2.4 The Rao-Blackwell Theorem

We now have a notion of ‘the data, compressed without loss’. The Rao-Blackwell theorem says that this compression is not merely lossless, but actively *useful*: averaging any estimator over the discarded information can only improve it.

From now on we will usually drop the subscript θ from $\mathbb{E}_\theta$ and $\mathbb{P}_\theta$, with the understanding that all expectations are taken in the model with true parameter θ .

Theorem 22.2.19 (Rao-Blackwell)

Let T be sufficient for θ , and let $\tilde{\theta}$ be an estimator with $\mathbb{E}[\tilde{\theta}^2] < \infty$ for all θ . Define

$$\hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T].$$

Then $\hat{\theta}$ is an estimator, and for all θ ,

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \leq \mathbb{E}[(\tilde{\theta} - \theta)^2],$$

with equality if and only if $\tilde{\theta}$ is (almost surely) a function of T .

Remark. It is essential that T be *sufficient*, and not merely any statistic. Otherwise $\hat{\theta}$ would generally depend on θ and so fail to be an estimator at all:

$$\hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T] = \int \tilde{\theta}(x) \underbrace{f_{X|T}(x \mid T)}_{\text{free of } \theta \text{ as } T \text{ is sufficient}} dx.$$

Sufficiency is exactly the condition that makes this conditional average computable from the data.

Proof. By the tower property,

$$\mathbb{E}[\hat{\theta}] = \mathbb{E}[\mathbb{E}[\tilde{\theta} \mid T]] = \mathbb{E}[\tilde{\theta}],$$

so $\hat{\theta}$ and $\tilde{\theta}$ have exactly the same bias. By the conditional variance formula,

$$\text{Var}(\tilde{\theta}) = \mathbb{E} \left[\underbrace{\text{Var}(\tilde{\theta} \mid T)}_{\geq 0} \right] + \underbrace{\text{Var}(\mathbb{E}[\tilde{\theta} \mid T])}_{=\text{Var}(\hat{\theta})} \geq \text{Var}(\hat{\theta}).$$

Now apply the bias-variance decomposition to both estimators: they have equal squared bias, and $\hat{\theta}$ has no larger variance, so $\text{mse}(\hat{\theta}) \leq \text{mse}(\tilde{\theta})$.

For the equality case, the inequality above is strict unless $\mathbb{E}[\text{Var}(\tilde{\theta} \mid T)] = 0$, that is, unless $\text{Var}(\tilde{\theta} \mid T) = 0$ almost surely, which says exactly that $\tilde{\theta}$ is a function of T . $\square$

So the recipe is: find *any* unbiased estimator, however crude, then condition it on a sufficient statistic. The result is unbiased and at least as good. Let's do this twice.

Example 22.2.20 (Rao-Blackwellising a Poisson Indicator)

Let $X_1, \dots, X_n$ be i.i.d $\text{Poisson}(\lambda)$, and suppose we wish to estimate $\theta = \mathbb{P}(X_1 = 0) = e^{-\lambda}$. The joint mass function is

$$f_X(x \mid \lambda) = \frac{e^{-n\lambda} \lambda^{\sum x_i}}{\prod_i x_i!},$$

so $T = \sum_i X_i$ is sufficient by the factorisation criterion (and note that re-parametrising by $\theta = e^{-\lambda}$ does not change this). Recall $T \sim \text{Poisson}(n\lambda)$.

A very crude unbiased estimator is

$$\tilde{\theta} = \mathbb{1}\{X_1 = 0\},$$

which throws away $X_2, \dots, X_n$ entirely, but does satisfy $\mathbb{E}[\tilde{\theta}] = \mathbb{P}(X_1 = 0) = \theta$. Conditioning on T ,

$$\begin{aligned} \hat{\theta} = \mathbb{E}[\tilde{\theta} \mid T = t] &= \mathbb{P}\left(X_1 = 0 \mid \sum_i X_i = t\right) \\ &= \frac{\mathbb{P}(X_1 = 0) \mathbb{P}\left(\sum_{i \geq 2} X_i = t\right)}{\mathbb{P}\left(\sum_{i \geq 1} X_i = t\right)} \\ &= \frac{e^{-\lambda} \cdot e^{-(n-1)\lambda} ((n-1)\lambda)^t / t!}{e^{-n\lambda} (n\lambda)^t / t!} = \left(\frac{n-1}{n}\right)^t. \end{aligned}$$

So the Rao-Blackwellised estimator is

$$\hat{\theta} = \left(1 - \frac{1}{n}\right)^{\sum_i X_i},$$

which is unbiased and has mean squared error no larger than $\tilde{\theta}$ for every θ . It is also a sensible-looking quantity: writing $\sum_i X_i = n\bar{X}$, we have $\hat{\theta} = (1 - 1/n)^{n\bar{X}} \rightarrow e^{-\bar{X}}$, and $\bar{X} \rightarrow \lambda$ almost surely by the strong law, so $\hat{\theta} \rightarrow e^{-\lambda} = \theta$.

Example 22.2.21 (Rao-Blackwellising for the Uniform)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$, and recall $T = \max_i X_i$ is sufficient. Since $\mathbb{E}[X_1] = \theta/2$, the estimator $\tilde{\theta} = 2X_1$ is unbiased. Conditioning on $T = t$ and splitting on whether X_1 achieves the maximum,

$$\begin{aligned} \hat{\theta} &= 2 \mathbb{E}\left[X_1 \mid \max_i X_i = t\right] \\ &= 2t \cdot \mathbb{P}\left(X_1 = \max_i X_i \mid \max_i X_i = t\right) \\ &\quad + 2 \mathbb{E}[X_1 \mid X_1 < t] \mathbb{P}\left(X_1 \neq \max_i X_i \mid \max_i X_i = t\right). \end{aligned}$$

By symmetry X_1 is the maximum with probability $1/n$, and given that it is not, X_1 is uniform on $[0, t]$, so has mean $t/2$. Hence

$$\hat{\theta} = \frac{2t}{n} + 2 \cdot \frac{t}{2} \cdot \frac{n-1}{n} = \frac{2t + t(n-1)}{n} = \frac{n+1}{n} \max_i X_i.$$

This is a dramatic improvement. One can compute directly that $\text{Var}(\tilde{\theta}) = \theta^2/3$, while

$$\text{Var}(\hat{\theta}) = \frac{\theta^2}{n(n+2)},$$

so the mean squared error drops from order 1 to order n^{-2} . That is much better than the usual n^{-1} rate, and reflects the fact that the maximum of a uniform sample converges to θ unusually quickly.

§22.2.5 Maximum Likelihood Estimation

We now come to the workhorse of the whole subject. The idea is disarmingly simple: having observed x , prefer the value of θ under which what we saw was most probable.

Definition 22.2.22 (Likelihood)

For fixed observed data x , the **likelihood function** $L : \Theta \rightarrow \mathbb{R}$ is

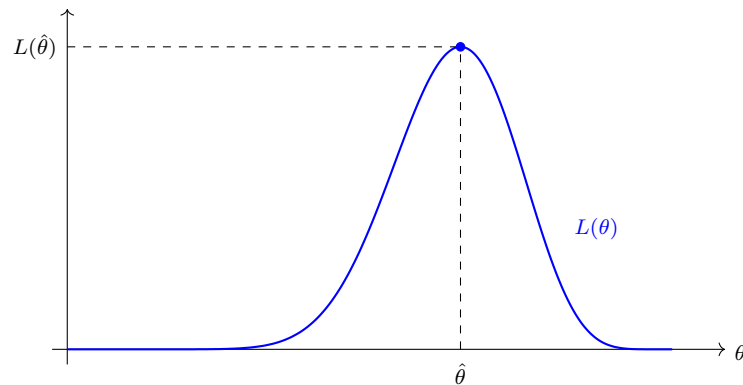
$$L(\theta) = f_X(x \mid \theta) = \prod_{i=1}^n f_{X_i}(x_i \mid \theta),$$

and the **log-likelihood** is $\ell(\theta) = \log L(\theta) = \sum_i \log f_{X_i}(x_i \mid \theta)$.

Note carefully that L is a function of θ , not of x : it is the joint density read ‘sideways’. It is not a probability distribution on Θ and does not integrate to 1.

Definition 22.2.23 (Maximum Likelihood Estimator)

A **maximum likelihood estimator** (MLE) $\hat{\theta}$ is any value of $\theta \in \Theta$ maximising L , or equivalently (as log is increasing) maximising ℓ .



In well-behaved cases Θ is an open interval and ℓ is differentiable and concave, so we may find $\hat{\theta}$ by solving $\ell'(\theta) = 0$. But this is not a definition, only a technique – and it fails exactly when the likelihood has its maximum on a boundary, which happens more often than one might expect.

Example 22.2.24 (Bernoulli)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(p). Then

$$\ell(p) = \sum_i [X_i \log p + (1 - X_i) \log(1 - p)] = \left(\sum_i X_i \right) \log p + \left(n - \sum_i X_i \right) \log(1 - p),$$

so

$$\ell'(p) = \frac{\sum_i X_i}{p} - \frac{n - \sum_i X_i}{1 - p},$$

which vanishes precisely at $p = \bar{X}$. As $\ell'' < 0$ throughout $(0, 1)$ this is the unique maximum, so $\hat{p} = \bar{X}$. This estimator is unbiased.

Example 22.2.25 (Normal with Both Parameters Unknown)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$. Then

$$\ell(\mu, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_i (X_i - \mu)^2.$$

Differentiating,

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_i (X_i - \mu),$$

which vanishes at $\mu = \bar{X}$ for any σ^2 . Substituting this and differentiating in σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_i (X_i - \bar{X})^2,$$

which vanishes at $\sigma^2 = S_{XX}/n$. So

$$(\hat{\mu}, \hat{\sigma}^2) = \left(\bar{X}, \frac{S_{XX}}{n} \right).$$

We computed in Section 1 that $\mathbb{E}[S_{XX}] = (n-1)\sigma^2$, so $\hat{\sigma}^2$ is biased downwards, with

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \sigma^2.$$

The bias vanishes as $n \rightarrow \infty$; we say $\hat{\sigma}^2$ is **asymptotically unbiased**. This behaviour is entirely typical: maximum likelihood estimators of variance parameters are usually a little too small, because they fit the mean to the data first.

Example 22.2.26 (Uniform on $[0, \theta]$: A Boundary Maximum)

Let $X_1, \dots, X_n$ be i.i.d $U[0, \theta]$. The likelihood is

$$L(\theta) = \frac{1}{\theta^n} \mathbb{1} \left\{ \theta \geq \max_i X_i \right\},$$

which is zero for $\theta < \max_i X_i$ and then strictly decreasing. So it is maximised at

$$\hat{\theta}_{\text{MLE}} = \max_i X_i,$$

and calculus was of no use whatsoever. Comparing with the Rao-Blackwellised estimator $\frac{n+1}{n} \max_i X_i$ obtained above, we see that the MLE is biased downwards, with $\mathbb{E}[\hat{\theta}_{\text{MLE}}] = \frac{n}{n+1} \theta$ – which is obvious in hindsight, since $\max_i X_i < \theta$ always.

Example 22.2.27 (Poisson)

Let $X_1, \dots, X_n$ be i.i.d $\text{Poisson}(\lambda)$. Then $\ell(\lambda) = -n\lambda + (\sum_i X_i) \log \lambda - \sum_i \log(X_i!)$, and $\ell'(\lambda) = -n + \sum_i X_i/\lambda$ vanishes at $\hat{\lambda} = \bar{X}$. Since $\ell''(\lambda) = -\sum_i X_i/\lambda^2 < 0$ this is the maximum, and it is unbiased.

§22.2.6 Properties of the Maximum Likelihood Estimator

Maximum likelihood is not just convenient; it has several structural properties that make it the default choice.

Proposition 22.2.28 (MLEs Depend Only on Sufficient Statistics)

If T is sufficient for θ , then any maximum likelihood estimator is a function of T .

Proof. By the factorisation criterion $L(\theta) = g(T(x), \theta)h(x)$, and $h(x)$ is a positive constant as far as maximisation over θ is concerned. So the maximiser depends on x only through $T(x)$. $\square$

Proposition 22.2.29 (Invariance)

Let H be a bijection of Θ onto some set Φ , and let $\varphi = H(\theta)$. If $\hat{\theta}$ is a maximum likelihood estimator for θ , then $H(\hat{\theta})$ is a maximum likelihood estimator for φ .

Proof. The likelihood for φ is $L^*(\varphi) = L(H^{-1}(\varphi))$, and as H is a bijection, L^* is maximised at $\varphi = H(\hat{\theta})$ exactly when L is maximised at $\hat{\theta}$. $\square$

This invariance is a genuine convenience: to estimate $e^{-\lambda}$ in the Poisson model we may simply write down $e^{-\bar{X}}$. (Note the contrast with unbiasedness, which has no such invariance – $\mathbb{E}[g(\hat{\theta})] \neq g(\mathbb{E}[\hat{\theta}])$ in general.)

Finally, and most importantly, maximum likelihood estimators are asymptotically normal.

Fact 22.2.30 (Asymptotic Normality of the MLE)

Under regularity conditions on the model (which we will not state precisely here), the maximum likelihood estimator $\hat{\theta} = \hat{\theta}_n$ based on n i.i.d observations satisfies

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, \Sigma)$$

in distribution as $n \rightarrow \infty$, for a limiting covariance matrix Σ determined by the model. Precisely, for all ‘nice’ sets A ,

$$\mathbb{P}_\theta(\sqrt{n}(\hat{\theta} - \theta) \in A) \rightarrow \mathbb{P}(Z \in A), \quad Z \sim N(0, \Sigma).$$

The matrix Σ is the inverse of the *Fisher information* per observation, and one can show that no reasonable estimator can asymptotically beat it – so maximum likelihood is, in a precise sense, asymptotically optimal. We will not prove any of this in this course, but the statement matters: it tells us that for large n the MLE is approximately unbiased, approximately normal, and has a variance of order $1/n$. That is exactly the input we need to construct approximate confidence intervals, which is where we go next.

Remark. For many models ℓ cannot be maximised in closed form, and $\hat{\theta}$ is found numerically (typically by Newton-Raphson on ℓ' , which converges quickly because ℓ is asymptotically quadratic near its maximum). The asymptotic theory above applies just the same.

§22.3 Confidence Intervals

A point estimate on its own is not much of an answer. If a poll reports that 52% of voters support a proposal, we want to know whether that number is accurate to within one percentage point or ten. An interval estimate answers this, and confidence intervals are the frequentist way of producing one.

§22.3.1 Definition and Construction

Definition 22.3.1 (Confidence Interval)

A $100\gamma\%$ **confidence interval** for a parameter θ is a pair of statistics $A(X) \leq B(X)$ such that

$$\mathbb{P}_\theta(A(X) \leq \theta \leq B(X)) = \gamma \quad \text{for all } \theta \in \Theta.$$

The reader should stare at that probability statement for a moment, because everything hinges on what is random in it. The parameter θ is a fixed unknown constant; it is the *endpoints* $A(X)$ and $B(X)$ that are random. We are not saying ‘ θ falls in a fixed interval 95% of the time’; we are saying ‘a random interval catches the fixed θ 95% of the time’. Note also that the requirement is uniform in θ : the coverage must be γ whatever the truth happens to be.

Example 22.3.2 (Normal Mean, Known Variance)

Let $X_1, \dots, X_n$ be i.i.d $N(\theta, 1)$, and let us find a 95% confidence interval for θ . We know that

$$\bar{X} \sim N\left(\theta, \frac{1}{n}\right), \quad \text{so} \quad Z = \sqrt{n}(\bar{X} - \theta) \sim N(0, 1).$$

The crucial point is that the distribution of Z is *the same for every* θ . So we may pick numbers $a < b$ with $\Phi(b) - \Phi(a) = 0.95$ once and for all, and then

$$0.95 = \mathbb{P}_\theta\left(a \leq \sqrt{n}(\bar{X} - \theta) \leq b\right) = \mathbb{P}_\theta\left(\bar{X} - \frac{b}{\sqrt{n}} \leq \theta \leq \bar{X} - \frac{a}{\sqrt{n}}\right).$$

Hence $(\bar{X} - b/\sqrt{n}, \bar{X} - a/\sqrt{n})$ is a 95% confidence interval, for *any* such a, b . To make the interval as short as possible we take it symmetric, $-a = b = z_{0.025} \approx 1.96$, where in general $z_\alpha = \Phi^{-1}(1 - \alpha)$ denotes the upper α point of $N(0, 1)$. This gives the familiar

$$\bar{X} \pm \frac{1.96}{\sqrt{n}}.$$

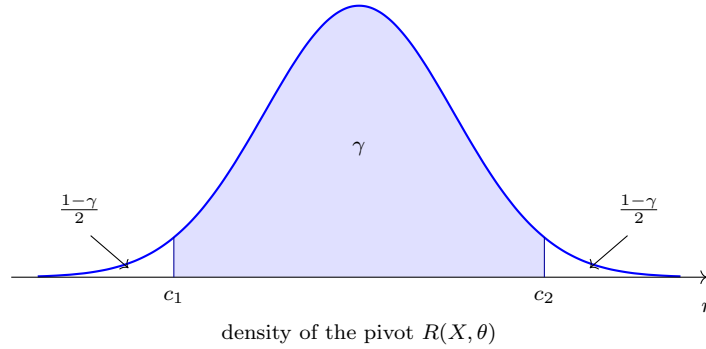
The technique used there deserves a name.

Definition 22.3.3 (Pivot)

A **pivot** is a function $R(X, \theta)$ of both the data and the parameter whose distribution under $\mathbb{P}_\theta$ is the same for every $\theta \in \Theta$.

A pivot is not a statistic (it involves θ), and its distribution is not a sampling distribution of anything we can compute – but because it does not depend on θ , we can look up its quantiles in advance. The general recipe is then:

- (i) Find a pivot $R(X, \theta)$.
- (ii) Choose constants $c_1 < c_2$ with $\mathbb{P}(c_1 \leq R \leq c_2) = \gamma$, using the (known) distribution of R .
- (iii) Rearrange the inequalities $c_1 \leq R(X, \theta) \leq c_2$ to isolate θ , obtaining $A(X) \leq \theta \leq B(X)$.



Step (iii) is where things can get awkward – the rearrangement need not be possible in closed form, and if R is not monotone in θ the resulting set may not even be an interval. When θ is vector-valued we simply drop the pretence and speak of a **confidence set**: any $A(X) \subseteq \Theta$ with $\mathbb{P}_\theta(\theta \in A(X)) = \gamma$ for all θ .

A useful closure property, immediate from the definition, is the following.

Proposition 22.3.4

If $(A(X), B(X))$ is a $100\gamma\%$ confidence interval for θ and T is a strictly increasing function, then $(T(A(X)), T(B(X)))$ is a $100\gamma\%$ confidence interval for $T(\theta)$.

Proof. Since T is strictly increasing, $A(X) \leq \theta \leq B(X)$ if and only if $T(A(X)) \leq T(\theta) \leq T(B(X))$, so the two events are identical and have the same probability. $\square$

§22.3.2 Standard Constructions

Example 22.3.5 (Normal Variance, Known Mean)

Let $X_1, \dots, X_n$ be i.i.d $N(0, \sigma^2)$ with σ^2 unknown. Since $X_i/\sigma \sim N(0, 1)$, a natural pivot is

$$R(X, \sigma^2) = \sum_{i=1}^n \frac{X_i^2}{\sigma^2} \sim \chi_n^2,$$

whose distribution is free of σ^2 as required. (We could have used the single observation X_1/σ , but that would waste $n - 1$ of our data points and give a hopelessly wide interval.) Taking $c_1 = \chi_n^2(0.975)$ and $c_2 = \chi_n^2(0.025)$ so that $\mathbb{P}(c_1 \leq R \leq c_2) = 0.95$, and rearranging,

$$0.95 = \mathbb{P}\left(c_1 \leq \frac{\sum_i X_i^2}{\sigma^2} \leq c_2\right) = \mathbb{P}\left(\frac{\sum_i X_i^2}{c_2} \leq \sigma^2 \leq \frac{\sum_i X_i^2}{c_1}\right).$$

Note that the interval is *not* symmetric about any natural estimator, because the χ^2 distribution is skewed.

Example 22.3.6 (A Bernoulli Proportion)

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(p) with n large. The MLE is $\hat{p} = \bar{X}$, and by the central limit theorem

$$\sqrt{n} \frac{\hat{p} - p}{\sqrt{p(1-p)}} \approx N(0, 1).$$

This is an approximate pivot, but rearranging the resulting inequalities for p requires solving a quadratic (since p appears in the denominator too). The standard dodge is to replace $p(1-p)$ by $\hat{p}(1-\hat{p})$ in the denominator, which is legitimate for large n since $\hat{p} \rightarrow p$. Then

$$\mathbb{P} \left(-z_{0.025} \leq \sqrt{n} \frac{\hat{p} - p}{\sqrt{\hat{p}(1-\hat{p})}} \leq z_{0.025} \right) \approx 0.95,$$

giving the approximate 95% confidence interval

$$\hat{p} \pm z_{0.025} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

Its width is largest when $\hat{p} = 1/2$, where it is about $2 \times 1.96 \times \frac{1}{2\sqrt{n}} \approx 1/\sqrt{n}$. Replacing $\hat{p}(1-\hat{p})$ by its maximum $1/4$ gives the **conservative** interval $\hat{p} \pm 1/\sqrt{n}$, which has coverage *at least* 95% for every p – this is where the familiar ‘margin of error of 3% on a poll of 1000 people’ comes from.

Example 22.3.7 (Poisson Rate)

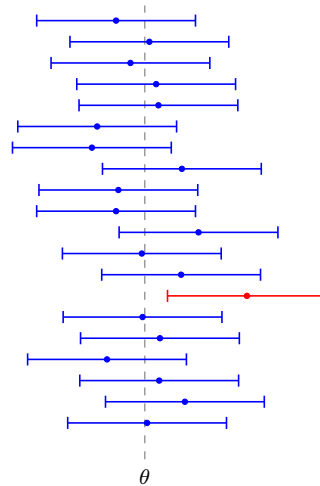
Let $X_1, \dots, X_n$ be i.i.d Poisson(μ) with n large. Since $\text{Var}(X_1) = \mu$, the central limit theorem gives

$$\sqrt{n} \frac{\bar{X} - \mu}{\sqrt{\mu}} \approx N(0, 1),$$

and substituting $\hat{\mu} = \bar{X}$ in the denominator as before yields the approximate 95% interval $\bar{X} \pm 1.96\sqrt{\bar{X}/n}$.

§22.3.3 Interpreting Confidence Intervals

The frequentist meaning of a confidence interval is a statement about the *procedure*, not about the particular interval we happen to have computed. If we were to repeat the experiment many times, each time constructing the interval by the same recipe, then about $100\gamma\%$ of those intervals would contain the true θ .



Twenty intervals from twenty repetitions of the same experiment are drawn above; one of them (in red) misses the true value. Before the data are seen, the probability that the interval will catch θ is 0.95. After the data are seen, the interval is a specific pair of numbers, and θ either is or is not inside it – there is no residual randomness left to have a probability about.

This is not mere pedantry. Here is an example where the naive reading is plainly absurd.

Example 22.3.8 (A 50% Interval We Sometimes Know Is Right)

Let X_1, X_2 be i.i.d uniform on $(\theta - \frac{1}{2}, \theta + \frac{1}{2})$. Then

$$\mathbb{P}\left(\min_i X_i \leq \theta \leq \max_i X_i\right) = \mathbb{P}(X_1 \leq \theta \leq X_2) + \mathbb{P}(X_2 \leq \theta \leq X_1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

since each X_i falls above or below θ with probability $1/2$, independently of the other. So the interval $(\min_i X_i, \max_i X_i)$ is a perfectly valid 50% confidence interval.

But now suppose we observe $x_1 = 0.1$ and $x_2 = 0.9$, so $|x_1 - x_2| = 0.8 > 1/2$. Since both observations lie within $1/2$ of θ , we have $\theta > 0.9 - 0.5 = 0.4$ and $\theta < 0.1 + 0.5 = 0.6$, so certainly $\theta \in (0.1, 0.9)$. We are 100% certain, not 50%! Conversely if $x_1 = 0.4$ and $x_2 = 0.45$ we have learned rather little, and 50% feels generous.

The coverage statement is still true – averaged over repetitions, half the intervals contain θ . It is simply not a statement about this particular data set.

If one genuinely wants a probability statement about θ given the data actually observed, one has to be willing to treat θ as a random variable in the first place. That is the Bayesian position, and it is where we turn next.

§22.4 Bayesian Inference

Everything so far has been **frequentist**: θ is a fixed unknown constant, and all probability statements are about the data, in the context of hypothetical repetitions of the experiment. Bayesian inference takes a different starting point. It treats θ itself as a random variable, with a distribution expressing our uncertainty about it, and then simply conditions on the data.

§22.4.1 Priors and Posteriors

Definition 22.4.1 (Prior and Posterior)

The **prior** distribution $\pi(\theta)$ on Θ represents our beliefs about θ before observing any data. Given data X with density (or mass function) $f_X(x | \theta)$, the **posterior** distribution is the conditional distribution of θ given $X = x$, written $\pi(\theta | x)$.

By Bayes' rule,

$$\pi(\theta | x) = \frac{\pi(\theta)f_X(x | \theta)}{f_X(x)}, \quad f_X(x) = \int_{\Theta} f_X(x | \theta)\pi(\theta) d\theta$$

(with the integral replaced by a sum if Θ is discrete). Since the denominator does not involve θ at all, it is nothing but a normalising constant, and in practice we almost always work with the proportionality

$$\pi(\theta | x) \propto \pi(\theta) f_X(x | \theta),$$

in words: *posterior* $\propto$ *prior* $\times$ *likelihood*. This is the entire machinery. The art lies in recognising the right-hand side as a standard distribution up to normalisation, which saves us from ever computing the integral.

Remark. If T is sufficient for θ , the posterior depends on the data only through T . Indeed by the factorisation criterion

$$\pi(\theta | x) \propto \pi(\theta)g(T(x), \theta)h(x) \propto \pi(\theta)g(T(x), \theta),$$

since $h(x)$ is constant in θ . So the Bayesian and frequentist notions of ‘the relevant information in the data’ agree exactly.

Example 22.4.2 (A Medical Test)

A patient is tested for a disease. Let $\theta = 1$ if they are infected and $\theta = 0$ otherwise, and let $X = 1$ if the test is positive. The likelihood is specified by the **sensitivity** $f_X(1 | 1)$ and the **specificity** $f_X(0 | 0)$ of the test. For the prior we might take $\pi(1)$ to be the prevalence of the disease in the population. Then

$$\pi(1 | X = 1) = \frac{\pi(1)f_X(1 | 1)}{\pi(1)f_X(1 | 1) + \pi(0)f_X(1 | 0)}.$$

If the disease is rare, so $\pi(1) \ll \pi(0)$, this can be small even for an excellent test: with sensitivity and specificity both 99% and prevalence 0.1%, a positive test gives

$$\pi(1 | X = 1) = \frac{0.001 \times 0.99}{0.001 \times 0.99 + 0.999 \times 0.01} \approx 0.09.$$

The false positives from the huge healthy population swamp the true positives. This is the classical illustration of why the prior cannot be ignored.

§22.4.2 Conjugate Priors

The computation is especially pleasant when prior and posterior belong to the same family.

Definition 22.4.3 (Conjugacy)

A family of priors is **conjugate** to a likelihood if the posterior always lies in the same family.

Example 22.4.4 (Beta-Binomial)

Let θ be the mortality rate for a particular operation, and model the number of deaths in n operations as $X \sim \text{Bin}(n, \theta)$. Suppose data from other hospitals suggest the rate lies between 3% and 20% with an average around 10%. A natural family of priors on $[0, 1]$ is the beta family,

$$\pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}, \quad \theta \sim \text{Beta}(a, b),$$

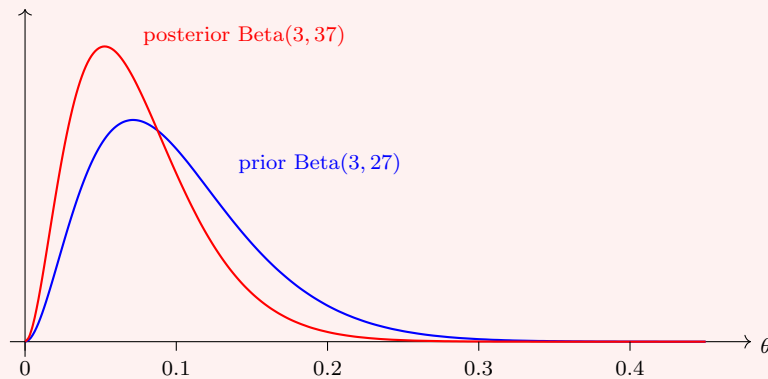
which has mean $a/(a+b)$. Taking $a = 3$, $b = 27$ gives $\mathbb{E}[\theta] = 0.1$ and $\mathbb{P}(0.03 < \theta < 0.2) \approx 0.9$, matching our beliefs well.

Now suppose we observe x deaths in n operations. Then

$$\pi(\theta | x) \propto \theta^{a-1}(1-\theta)^{b-1} \cdot \theta^x(1-\theta)^{n-x} = \theta^{x+a-1}(1-\theta)^{n-x+b-1},$$

which we recognise immediately as $\text{Beta}(a+x, b+n-x)$. So the beta family is conjugate to the binomial likelihood, and the update rule is beautifully simple: add the successes to a , add the failures to b .

With $n = 10$ operations and $x = 0$ deaths, the posterior is $\text{Beta}(3, 37)$.



The posterior has both a smaller mean ($3/40 = 0.075$ against $3/30 = 0.1$) and a smaller variance than the prior: observing no deaths has shifted our beliefs downwards and sharpened them. Notice also how modest the shift is – ten operations is not much data against a prior worth 30 pseudo-observations, and the Bayesian machinery weights them accordingly.

Example 22.4.5 (Gamma-Poisson)

Let $X_1, \dots, X_n$ be i.i.d $\text{Poisson}(\theta)$ with prior $\theta \sim \text{Gamma}(a, b)$, so $\pi(\theta) \propto \theta^{a-1}e^{-b\theta}$. Then

$$\pi(\theta | x) \propto \theta^{a-1}e^{-b\theta} \cdot \theta^{\sum x_i} e^{-n\theta} = \theta^{a+\sum x_i-1} e^{-(b+n)\theta},$$

so the posterior is $\text{Gamma}(a + \sum_i x_i, b + n)$: the gamma family is conjugate to the Poisson.

Example 22.4.6 (Normal-Normal)

Let $X_1, \dots, X_n$ be i.i.d $N(\theta, \sigma^2)$ with σ^2 known, and give θ the prior $N(\mu_0, \tau_0^2)$. Then

$$\pi(\theta | x) \propto \exp\left(-\frac{(\theta - \mu_0)^2}{2\tau_0^2}\right) \exp\left(-\frac{\sum_i (x_i - \theta)^2}{2\sigma^2}\right).$$

The exponent is a quadratic in θ , so the posterior is normal, and completing the square gives $N(\mu_1, \tau_1^2)$ with

$$\frac{1}{\tau_1^2} = \frac{1}{\tau_0^2} + \frac{n}{\sigma^2}, \quad \mu_1 = \tau_1^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right).$$

The interpretation is very clean if we think in terms of **precisions** (reciprocal variances): precisions add, and the posterior mean is the precision-weighted average of the prior mean and the sample mean. As $n \rightarrow \infty$ the data term dominates and $\mu_1 \rightarrow \bar{x}$, so the prior is eventually washed out – which is reassuring.

§22.4.3 Point Estimates as Optimal Decisions

The posterior is the complete Bayesian answer. But often we must actually *do* something – commit to a single number, or a single course of action. Decision theory tells us how.

We choose a decision δ from a set Δ of available actions, and specify a **loss function** $L(\theta, \delta)$ giving the penalty incurred for taking action δ when the truth is θ . Having observed x , we then choose δ to minimise the **posterior expected loss**

$$h(\delta) = \int_{\Theta} L(\theta, \delta) \pi(\theta | x) d\theta.$$

Definition 22.4.7 (Bayes Estimator)

When $\Delta = \Theta$ (so that our ‘action’ is to report an estimate of θ), the minimiser $\hat{\theta}^{(B)}$ of $h(\delta)$ is the **Bayes estimator** of θ for the loss L .

Different losses give different estimators, and pleasingly the three most natural losses give the three most natural summaries of the posterior.

Example 22.4.8 (Quadratic Loss Gives the Posterior Mean)

Take $L(\theta, \delta) = (\theta - \delta)^2$. Then

$$h(\delta) = \int_{\Theta} (\theta - \delta)^2 \pi(\theta | x) d\theta,$$

and differentiating under the integral,

$$h'(\delta) = -2 \int_{\Theta} (\theta - \delta) \pi(\theta | x) d\theta,$$

which vanishes when $\delta = \int_{\Theta} \theta \pi(\theta | x) d\theta$. As $h''(\delta) = 2 > 0$ this is the minimum. So the Bayes estimator under quadratic loss is the *posterior mean*.

Example 22.4.9 (Absolute Loss Gives the Posterior Median)

Take $L(\theta, \delta) = |\theta - \delta|$. Splitting the integral at δ ,

$$h(\delta) = \int_{-\infty}^{\delta} (\delta - \theta) \pi(\theta | x) d\theta + \int_{\delta}^{\infty} (\theta - \delta) \pi(\theta | x) d\theta,$$

and differentiating (the boundary terms cancel since the integrands vanish at $\theta = \delta$),

$$h'(\delta) = \int_{-\infty}^{\delta} \pi(\theta | x) d\theta - \int_{\delta}^{\infty} \pi(\theta | x) d\theta.$$

This vanishes when the two tails have equal mass, i.e. at the *posterior median*. Since h' is increasing, this is the minimum.

Example 22.4.10 (Zero-One Loss Gives the Posterior Mode)

For $\varepsilon > 0$ take $L(\theta, \delta) = \mathbb{1}\{|\theta - \delta| > \varepsilon\}$, which penalises us only if we miss by more than ε . Then

$$h(\delta) = 1 - \int_{\delta-\varepsilon}^{\delta+\varepsilon} \pi(\theta | x) d\theta,$$

which is minimised by centring the window on the region of highest posterior density. As $\varepsilon \rightarrow 0$ this converges to the *posterior mode*, also called the maximum a posteriori (MAP) estimate. Note that with a flat prior the posterior is proportional to the likelihood, so the MAP estimate is then exactly the maximum likelihood estimate – a satisfying point of contact between the two philosophies.

Example 22.4.11 (Beta-Binomial, Continued)

In the surgery example the posterior was $\text{Beta}(3, 37)$. Under quadratic loss the Bayes estimator is the posterior mean $3/40 = 0.075$, while the posterior mode is $(3 - 1)/(3 + 37 - 2) = 2/38 \approx 0.053$. Compare the maximum likelihood estimate $\hat{\theta} = x/n = 0$, which asserts that the operation is perfectly safe on the basis of ten cases. The prior is doing real work here, and doing it sensibly.

§22.4.4 Credible Intervals**Definition 22.4.12** (Credible Interval)

A $100\gamma\%$ **credible interval** for θ is an interval $(A(x), B(x))$ with

$$\mathbb{P}(A(x) \leq \theta \leq B(x) | X = x) = \int_{A(x)}^{B(x)} \pi(\theta | x) d\theta = \gamma.$$

Superficially this looks identical to a confidence interval, but the probability means something quite different – and much closer to what people usually want. Here θ is the random quantity and the data are fixed, so we really may say: ‘given the data I have, there is probability γ that θ lies in this interval’. That is precisely the statement that was illegitimate for confidence intervals.

The price, of course, is that the answer depends on the prior. Two statisticians with

different priors will report different credible intervals from the same data, and neither is wrong. Note also that credible intervals are not unique: any A, B capturing γ of the posterior mass will do. Common choices are the **equal-tailed** interval, running between the $\frac{1-\gamma}{2}$ and $\frac{1+\gamma}{2}$ posterior quantiles, and the **highest posterior density** interval, which is the shortest such interval and consists of the points with the largest posterior density.

Example 22.4.13

Continuing the surgery example, the equal-tailed 95% credible interval for the mortality rate under the $\text{Beta}(3, 37)$ posterior runs between the 2.5% and 97.5% points of that distribution, which are approximately 0.016 and 0.173. So having seen ten operations with no deaths, we would report a mortality rate of about 7.5%, with a credible interval of roughly (1.6%, 17.3%).

§22.5 Hypothesis Testing

Estimation asks ‘what is θ ?’. Hypothesis testing asks a cruder but often more useful question: ‘is there evidence in the data against a particular claim about θ ?’. This is the form most scientific questions actually take – does the drug work, is the coin fair, does the data fit the model – and the framework we build here is the one used, for better or worse, throughout the empirical sciences.

§22.5.1 Hypotheses, Tests and Errors

Definition 22.5.1 (Hypothesis)

A **hypothesis** is an assertion about the distribution of the data X . The **null hypothesis** H_0 is the default or baseline assertion; the **alternative hypothesis** H_1 is what we entertain if H_0 is found wanting.

A hypothesis is **simple** if it completely specifies the distribution of X , and **composite** otherwise.

Example 22.5.2

Let $X_1, \dots, X_n$ be i.i.d Bernoulli(θ). We might test $H_0 : \theta = \frac{1}{2}$ against $H_1 : \theta = \frac{3}{4}$, in which case both hypotheses are simple; or $H_0 : \theta = \frac{1}{2}$ against $H_1 : \theta \neq \frac{1}{2}$, in which case H_1 is composite.

Example 22.5.3

Suppose the X_i take values in $\{0, 1, 2, \dots\}$ and we test $H_0 : X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$ for *some* λ , against H_1 : the X_i have some other distribution. Here H_0 is composite (because λ is unspecified), and the test is a **goodness-of-fit** test. We will return to these later in this section.

Definition 22.5.4 (Test)

A **test** of H_0 is specified by a **critical region** (or rejection region) $C \subseteq \mathcal{X}$, where

$\mathcal{X}$ is the sample space. If $X \in C$ we **reject** H_0 ; otherwise we **fail to reject** H_0 . The complement $A = \mathcal{X} \setminus C$ is the **acceptance region**.

The asymmetric language is deliberate. Failing to reject H_0 is not evidence *for* H_0 – it may simply mean we had too little data to detect a departure. There are two ways to go wrong:

Definition 22.5.5 (Type I and Type II Errors)

A **type I error** (a false positive) is rejecting H_0 when it is true; a **type II error** (a false negative) is failing to reject H_0 when it is false. When H_0 and H_1 are simple we write

$$\alpha = \mathbb{P}_{H_0}(X \in C), \quad \beta = \mathbb{P}_{H_1}(X \notin C).$$

We call α the **size** of the test and $1 - \beta = \mathbb{P}_{H_1}(X \in C)$ its **power**.

The two error probabilities trade off against each other: shrinking C reduces α and increases β . The classical resolution of this tension is to fix an acceptable type I error rate α (conventionally 0.05 or 0.01) and then, subject to that constraint, make the power as large as possible. Note the asymmetry this builds in: H_0 is protected, and we reject it only on strong evidence. Note also that computing α requires only H_0 , whereas computing β requires us to specify H_1 precisely – which is one reason simple alternatives are so much easier to handle.

§22.5.2 The Neyman-Pearson Lemma

Suppose both hypotheses are simple, with X having density (or mass function) f_i under H_i . Which critical region should we choose?

Definition 22.5.6 (Likelihood Ratio Test)

The **likelihood ratio statistic** is

$$\Lambda_x(H_0; H_1) = \frac{f_1(x)}{f_0(x)},$$

and the **likelihood ratio test** has critical region $C = \{x : \Lambda_x(H_0; H_1) > k\}$ for some constant k .

In other words – reject H_0 exactly when the data are sufficiently more probable under H_1 than under H_0 . This is an obviously reasonable thing to do, and the remarkable content of the next result is that it is *optimal*.

Lemma 22.5.7 (Neyman-Pearson)

Let H_0 and H_1 be simple hypotheses with densities f_0, f_1 that are positive on the same set. Suppose that for some $k > 0$ the likelihood ratio test with critical region $C = \{x : f_1(x)/f_0(x) > k\}$ has size α . Then among all tests of size at most α , this test has the largest power.

Proof. Write $\overline{C}$ for the complement of C , and let

$$\alpha = \int_C f_0(x) \, dx, \quad \beta = \int_{\overline{C}} f_1(x) \, dx$$

be the error probabilities of the likelihood ratio test. Let C^* be the critical region of any other test, with error probabilities α^*, β^* , and suppose $\alpha^* \leq \alpha$. We must show $\beta \leq \beta^*$.

Cancelling the contribution from $\overline{C} \cap \overline{C}^*$, which is common to both,

$$\beta - \beta^* = \int_{\overline{C}} f_1 - \int_{\overline{C}^*} f_1 = \int_{\overline{C} \cap C^*} f_1 - \int_{C \cap \overline{C}^*} f_1.$$

Now on $\overline{C}$ we have $f_1 \leq k f_0$, and on C we have $f_1 \geq k f_0$. Applying these to the two integrals respectively (the first is over a subset of $\overline{C}$, the second over a subset of C),

$$\begin{aligned} \beta - \beta^* &\leq k \int_{\overline{C} \cap C^*} f_0 - k \int_{C \cap \overline{C}^*} f_0 \\ &= k \left[\int_{\overline{C} \cap C^*} f_0 + \int_{C \cap C^*} f_0 - \int_{C \cap C^*} f_0 - \int_{C \cap \overline{C}^*} f_0 \right] \\ &= k \left[\int_{C^*} f_0 - \int_C f_0 \right] = k [\alpha^* - \alpha] \leq 0, \end{aligned}$$

where in the middle line we added and subtracted the same quantity in order to reassemble the integrals over C^* and over C . Hence $\beta \leq \beta^*$, and the likelihood ratio test has power $1 - \beta \geq 1 - \beta^*$. $\square$

Remark. The hypothesis that a likelihood ratio test of *exactly* size α exists is a real restriction, and typically fails for discrete data, where the achievable sizes form a finite set. The fix is to allow **randomised tests**, which for certain boundary values of x reject H_0 with some probability strictly between 0 and 1; every size α is then achievable, and the lemma holds in that generality.

Example 22.5.8 (The Z-Test)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu = \mu_1$ where $\mu_1 > \mu_0$. The likelihood ratio is

$$\begin{aligned} \Lambda_x &= \frac{(2\pi\sigma_0^2)^{-n/2} \exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_1)^2\right)}{(2\pi\sigma_0^2)^{-n/2} \exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_0)^2\right)} \\ &= \exp\left(\underbrace{\frac{\mu_1 - \mu_0}{\sigma_0^2}}_{>0} n\bar{x} + \frac{n(\mu_0^2 - \mu_1^2)}{2\sigma_0^2}\right), \end{aligned}$$

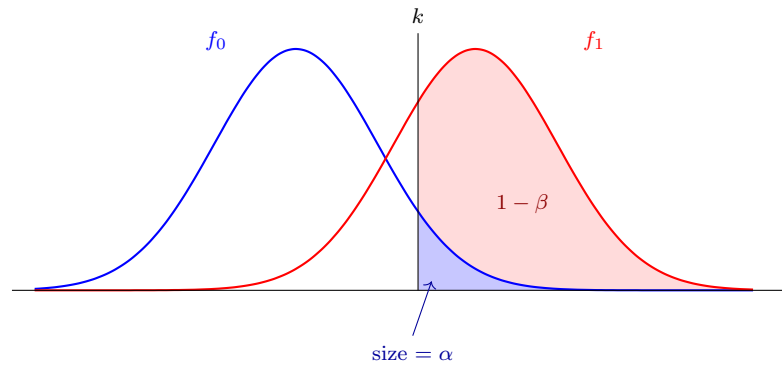
which depends on the data only through $\bar{x}$, and is strictly increasing in it. Hence

$\Lambda_x > k$ if and only if $\bar{x} > k'$, if and only if

$$Z = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > k''$$

for suitable constants. Under H_0 we have $Z \sim N(0, 1)$, so to achieve size α we take $k'' = z_\alpha = \Phi^{-1}(1 - \alpha)$. This is the **Z-test**, and by Neyman-Pearson it is the most powerful test of its size.

The geometry of what is going on is worth a picture. Below are the two densities of the test statistic Z under H_0 and H_1 , with the critical region $\{Z > k\}$ shaded under each.



Moving k to the right shrinks the blue shaded area (the size) but also shrinks the red one (the power). The Neyman-Pearson lemma says that among all ways of carving a region of blue area α out of the line, taking a right-hand tail of the likelihood ratio captures the most red.

§22.5.3 p -Values

Reporting only ‘reject at the 5% level’ throws away information: it does not distinguish a marginal result from an overwhelming one. The p -value records how extreme the observed data actually were.

Definition 22.5.9 (p -Value)

Suppose the critical region has the form $C = \{x : T(x) > k\}$ for a test statistic T , and let x^* be the observed data. The **p -value** is

$$p = \mathbb{P}_{H_0}(T(X) \geq T(x^*)).$$

Equivalently, the p -value is the smallest size α at which we would have rejected H_0 with these data. Small p means the observed data were surprising under H_0 .

Example 22.5.10

Continuing the Z -test with $\mu_0 = 5$, $\sigma_0 = 1$, $\alpha = 0.05$, suppose we observe $x^* = (5.1, 5.5, 4.9, 5.3)$. Then $\bar{x}^* = 5.2$ and $z^* = \sqrt{4}(5.2 - 5)/1 = 0.4$. The critical region is $\{Z > 1.645\}$, so we do not reject H_0 ; the p -value is $1 - \Phi(0.4) \approx 0.34$, so a value at least this extreme would arise about a third of the time even if H_0 were true.

Proposition 22.5.11

If the test statistic T has a continuous distribution, then under H_0 the p -value is uniformly distributed on $[0, 1]$.

Proof. Let F be the distribution function of T under H_0 , so $p = 1 - F(T)$. Then for $u \in (0, 1)$,

$$\begin{aligned}\mathbb{P}_{H_0}(p \leq u) &= \mathbb{P}_{H_0}(1 - F(T) \leq u) = \mathbb{P}_{H_0}(F(T) \geq 1 - u) \\ &= \mathbb{P}_{H_0}(T \geq F^{-1}(1 - u)) = 1 - F(F^{-1}(1 - u)) = u. \square\end{aligned}$$

This is a useful sanity check, and also a warning: under the null, p -values below 0.05 occur exactly 5% of the time by construction. If one tests twenty independent null hypotheses, one expects a ‘significant’ result by chance alone.

§22.5.4 Composite Hypotheses and Power Functions

Now let $X \sim f_X(\cdot | \theta)$ with $\theta \in \Theta$, and consider $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$, where Θ_0, Θ_1 are disjoint. The error probabilities now depend on which particular θ obtains, so we need a function rather than a number.

Definition 22.5.12 (Power Function, Size, UMP)

The **power function** of a test with critical region C is

$$W(\theta) = \mathbb{P}_\theta(X \in C), \quad \theta \in \Theta.$$

Its **size** is $\alpha = \sup_{\theta \in \Theta_0} W(\theta)$, the worst-case type I error rate. A test with power function W and size at most α is **uniformly most powerful** (UMP) of size α if for every other test with power function W^* and size at most α ,

$$W(\theta) \geq W^*(\theta) \quad \text{for all } \theta \in \Theta_1.$$

The word ‘uniformly’ is doing the work: we are asking for a single test that is simultaneously best against every alternative. Such tests need not exist – and typically do not, for two-sided alternatives, since the best test against $\theta > \theta_0$ points in the wrong direction for $\theta < \theta_0$. For one-sided alternatives in simple models, however, they often do.

Example 22.5.13 (One-Sided Test for a Normal Mean)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and consider

$$H_0 : \mu \leq \mu_0, \quad H_1 : \mu > \mu_0.$$

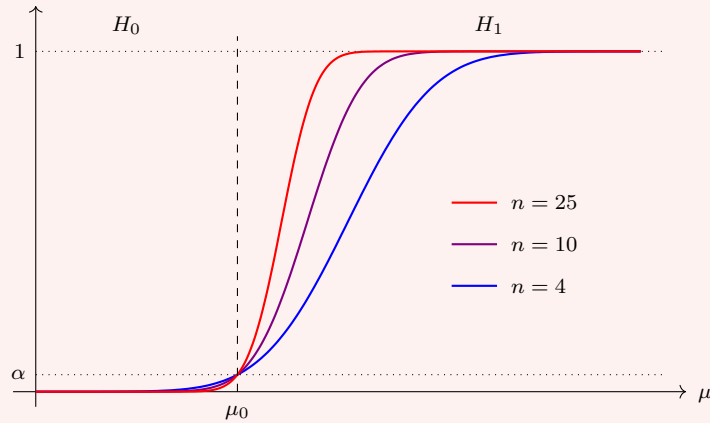
We claim the Z -test with $C = \{Z > z_\alpha\}$, where $Z = \sqrt{n}(\bar{X} - \mu_0)/\sigma_0$, is UMP of size α .

First we compute the power function. Under $\mathbb{P}_\mu$ we have $\sqrt{n}(\bar{X} - \mu)/\sigma_0 \sim N(0, 1)$,

so

$$\begin{aligned}
 W(\mu) &= \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > z_\alpha \right) \\
 &= \mathbb{P}_\mu \left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma_0} > z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right) \\
 &= 1 - \Phi \left(z_\alpha + \frac{\sqrt{n}(\mu_0 - \mu)}{\sigma_0} \right).
 \end{aligned}$$

This is strictly increasing in μ , so $\sup_{\mu \leq \mu_0} W(\mu) = W(\mu_0) = 1 - \Phi(z_\alpha) = \alpha$, and the test has size α as claimed.



The picture shows what a good power function looks like: it passes through α at the boundary μ_0 , stays below α on Θ_0 , and climbs towards 1 on Θ_1 , the more steeply the more data we have.

Now for optimality. The key observation is that the critical region C does not depend on μ_1 at all – it was determined by μ_0 and α alone. So fix any $\mu_1 > \mu_0$ and consider the simple hypotheses $H'_0 : \mu = \mu_0$ and $H'_1 : \mu = \mu_1$. By the previous example, C is precisely the likelihood ratio critical region for H'_0 against H'_1 , of size α .

Let C^* be any test of H_0 against H_1 with size at most α . Regarded as a test of H'_0 against H'_1 , its size is

$$W^*(\mu_0) \leq \sup_{\mu \leq \mu_0} W^*(\mu) \leq \alpha.$$

So by the Neyman-Pearson lemma, $W(\mu_1) \geq W^*(\mu_1)$. Since $\mu_1 > \mu_0$ was arbitrary, the test is UMP.

§22.5.5 Generalised Likelihood Ratio Tests

For genuinely composite hypotheses there is no likelihood ratio to speak of, since neither hypothesis specifies a single density. The natural fix is to compare the best each hypothesis can do.

Definition 22.5.14 (Generalised Likelihood Ratio)

Suppose the hypotheses are **nested**, $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$ with $\Theta_0 \subseteq \Theta_1$.

The **generalised likelihood ratio** is

$$\Lambda_x(H_0; H_1) = \frac{\sup_{\theta \in \Theta_1} f_X(x | \theta)}{\sup_{\theta \in \Theta_0} f_X(x | \theta)},$$

and the **generalised likelihood ratio test** rejects H_0 when Λ_x is large.

Since $\Theta_0 \subseteq \Theta_1$ we always have $\Lambda_x \geq 1$: the larger model can always fit at least as well. The question is whether it fits *enough* better to justify its extra parameters, and the test statistic $2 \log \Lambda_x$ measures exactly this.

Example 22.5.15 (Two-Sided Test for a Normal Mean)

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known, and test $H_0 : \mu = \mu_0$ against $H_1 : \mu \in \mathbb{R}$. The numerator is maximised at the MLE $\hat{\mu} = \bar{X}$, so

$$\Lambda_x = \frac{\exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \bar{x})^2\right)}{\exp\left(-\frac{1}{2\sigma_0^2} \sum_i (x_i - \mu_0)^2\right)}.$$

Expanding $\sum_i (x_i - \mu_0)^2 = \sum_i (x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$, everything cancels except

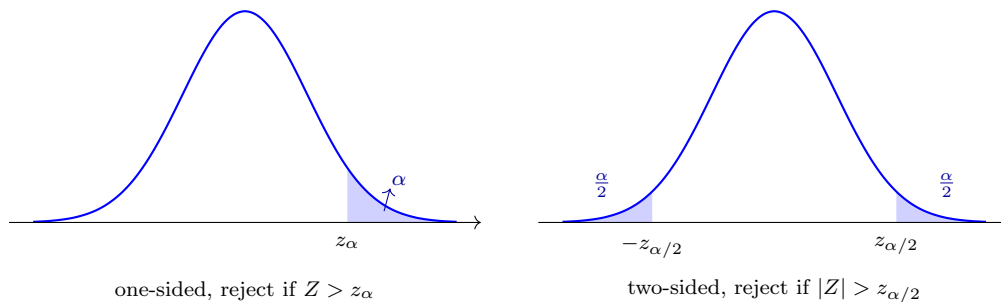
$$2 \log \Lambda_x = \frac{n(\bar{x} - \mu_0)^2}{\sigma_0^2} = \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} \right)^2.$$

Under H_0 the bracket is $N(0, 1)$, so $2 \log \Lambda_x \sim \chi_1^2$ *exactly*, and the size- α test rejects when

$$\frac{n(\bar{X} - \mu_0)^2}{\sigma_0^2} > \chi_1^2(\alpha), \quad \text{equivalently} \quad \left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \right| > z_{\alpha/2}.$$

This is a **two-sided** test: the critical region consists of both tails in terms of the original statistic.

It is worth seeing the one- and two-sided critical regions side by side, since choosing between them is a modelling decision that must be made *before* looking at the data.



In the example above the exact distribution of $2 \log \Lambda$ was available. Usually it is not, and we appeal to the following remarkable theorem, whose content is that the χ_1^2 we just found was no accident.

Definition 22.5.16 (Dimension of a Hypothesis)

The **dimension** of $H_0 : \theta \in \Theta_0$ is the number of free parameters in Θ_0 .

Example 22.5.17

If $\Theta_0 = \{\theta \in \mathbb{R}^k : \theta_1 = \dots = \theta_p = 0\}$ then $\dim \Theta_0 = k - p$. More generally if $A \in \mathbb{R}^{p \times k}$ has linearly independent rows and $\Theta_0 = \{\theta \in \mathbb{R}^k : A\theta = b\}$, then $\dim \Theta_0 = k - p$: each independent linear constraint costs one dimension.

Theorem 22.5.18 (Wilks)

Suppose $\Theta_0 \subseteq \Theta_1$ with $\dim \Theta_1 - \dim \Theta_0 = p$, and let $X = (X_1, \dots, X_n)$ be i.i.d from $f_X(\cdot | \theta)$ with the true θ in the interior of Θ_0 . Then under regularity conditions, as $n \rightarrow \infty$,

$$2 \log \Lambda_X \rightarrow \chi_p^2$$

in distribution; that is, $\mathbb{P}_\theta(2 \log \Lambda_X \leq \ell) \rightarrow \mathbb{P}(\Xi \leq \ell)$ for every $\ell > 0$, where $\Xi \sim \chi_p^2$.

We will not prove this (the proof is a Taylor expansion of ℓ about the MLE, combined with asymptotic normality of the MLE). But it is enormously useful: it says that whatever the model, the price of p extra parameters is measured on the same universal χ_p^2 scale. In particular the size- α generalised likelihood ratio test rejects when $2 \log \Lambda_X > \chi_p^2(\alpha)$, and this has size approximately α for large n .

Example 22.5.19

In the two-sided normal example, $\dim \Theta_1 = 1$ and $\dim \Theta_0 = 0$, so $p = 1$ and Wilks' theorem predicts $2 \log \Lambda \approx \chi_1^2$ – which we verified exactly, for every n .

§22.5.6 Goodness of Fit

We now apply this machinery to the question of whether a model fits at all. Let $X_1, \dots, X_n$ be i.i.d taking values in $\{1, \dots, k\}$, and write $p_i = \mathbb{P}(X_1 = i)$ and

$$N_i = \#\{j : X_j = i\},$$

so $\sum_i p_i = 1$ and $\sum_i N_i = n$. Then $(N_1, \dots, N_k) \sim \text{Multinomial}(n; p_1, \dots, p_k)$, and the parameter space $\Theta_1 = \{p : p_i \geq 0, \sum_i p_i = 1\}$ has dimension $k - 1$.

The **goodness-of-fit** problem is to test $H_0 : p = \tilde{p}$ for a specified vector $\tilde{p}$, against the unrestricted H_1 . The likelihood is

$$L(p) \propto p_1^{N_1} \dots p_k^{N_k}, \quad \ell(p) = \text{const} + \sum_i N_i \log p_i.$$

Lemma 22.5.20

The maximum likelihood estimator of p over Θ_1 is $\hat{p}_i = N_i/n$.

Proof. Maximise ℓ subject to $\sum_i p_i = 1$ by Lagrange multipliers: setting $\mathcal{L} =$

$\sum_i N_i \log p_i - \lambda(\sum_i p_i - 1)$ and differentiating gives $N_i/p_i = \lambda$, so $p_i = N_i/\lambda$. Summing over i and using the constraint gives $\lambda = n$. $\square$

So the generalised likelihood ratio statistic is

$$2 \log \Lambda = 2(\ell(\hat{p}) - \ell(\tilde{p})) = 2 \sum_{i=1}^k N_i \log \left(\frac{N_i}{n\tilde{p}_i} \right) = 2 \sum_{i=1}^k o_i \log \frac{o_i}{e_i},$$

writing $o_i = N_i$ for the **observed** counts and $e_i = n\tilde{p}_i$ for the counts **expected** under H_0 . Since $\dim \Theta_1 - \dim \Theta_0 = (k-1) - 0 = k-1$, Wilks' theorem tells us to refer this to χ_{k-1}^2 .

§22.5.7 Pearson's Chi-Squared Statistic

There is a classical approximation to $2 \log \Lambda$ which is easier to compute by hand and is what is usually reported.

Proposition 22.5.21 (Pearson's Statistic)

With $\delta_i = o_i - e_i$, and assuming each δ_i/e_i is small,

$$2 \log \Lambda \approx \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}.$$

Proof. Write $o_i = e_i + \delta_i$, so

$$2 \log \Lambda = 2 \sum_i (e_i + \delta_i) \log \left(1 + \frac{\delta_i}{e_i} \right).$$

Expanding $\log(1+u) = u - u^2/2 + O(u^3)$ with $u = \delta_i/e_i$,

$$2 \log \Lambda \approx 2 \sum_i (e_i + \delta_i) \left(\frac{\delta_i}{e_i} - \frac{\delta_i^2}{2e_i^2} \right) = 2 \sum_i \left(\delta_i + \frac{\delta_i^2}{e_i} - \frac{\delta_i^2}{2e_i} \right) + O\left(\sum_i \frac{\delta_i^3}{e_i^2} \right).$$

Now $\sum_i \delta_i = \sum_i o_i - \sum_i e_i = n - n = 0$, so the first term vanishes and we are left with $\sum_i \delta_i^2/e_i$. $\square$

This is **Pearson's χ^2 statistic**, and it is referred to the same χ_{k-1}^2 distribution. The rule of thumb is that the approximation (both to $2 \log \Lambda$ and of both to χ^2) is adequate provided every $e_i \geq 5$.

Example 22.5.22 (Mendel's Peas)

Mendel crossed pea plants and classified 556 offspring by shape and colour, obtaining

$$N = (\underbrace{315}_{\text{smooth green}}, \underbrace{108}_{\text{smooth yellow}}, \underbrace{102}_{\text{wrinkled green}}, \underbrace{31}_{\text{wrinkled yellow}}).$$

His theory of independent assortment of two dominant/recessive traits predicts

$$H_0 : \tilde{p} = \left(\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16} \right),$$

giving expected counts $e = (312.75, 104.25, 104.25, 34.75)$. Computing,

$$2 \log \Lambda = 0.618, \quad \sum_i \frac{(o_i - e_i)^2}{e_i} = 0.604,$$

which we refer to χ_3^2 . Since $\chi_3^2(0.05) = 7.815$, we do not reject H_0 at the 5% level. Indeed the p -value is $\mathbb{P}(\chi_3^2 > 0.60) \approx 0.90$: the fit is not merely acceptable, it is suspiciously good.

The null hypothesis need not fully specify $\tilde{p}$; often it says only that p lies in some smaller family.

Example 22.5.23 (Composite Null: Hardy-Weinberg)

Individuals have one of three genotypes with probabilities

$$(p_1, p_2, p_3) = (\theta^2, 2\theta(1 - \theta), (1 - \theta)^2)$$

for some allele frequency $\theta \in (0, 1)$, so $\dim \Theta_0 = 1$. The log-likelihood under H_0 is

$$\ell(\theta) = \text{const} + (2N_1 + N_2) \log \theta + (2N_3 + N_2) \log(1 - \theta),$$

which is maximised at

$$\hat{\theta} = \frac{2N_1 + N_2}{2n}$$

– the observed frequency of the allele, as one would hope. Then

$$2 \log \Lambda = 2 \left(\ell(\hat{p}) - \ell(\hat{\theta}) \right) = 2 \sum_i o_i \log \frac{o_i}{e_i} \approx \sum_i \frac{(o_i - e_i)^2}{e_i}, \quad e_i = n p_i(\hat{\theta}),$$

which is now referred to χ_p^2 with $p = \dim \Theta_1 - \dim \Theta_0 = 2 - 1 = 1$. The general rule: *each parameter estimated under the null costs one degree of freedom*.

§22.5.8 Contingency Tables

Two of the most common applications of the χ^2 test involve two-way tables of counts, and it is a pleasant surprise that they turn out to be the same test.

Testing Independence

Suppose we observe i.i.d pairs $(X_1, Y_1), \dots, (X_n, Y_n)$ where X_ℓ takes values in $\{1, \dots, r\}$ and Y_ℓ in $\{1, \dots, c\}$, and we wish to test whether X and Y are independent. The data are summarised by the [contingency table](#)

$$N_{ij} = \# \{ \ell : (X_\ell, Y_\ell) = (i, j) \},$$

and (N_{ij}) , flattened into a vector, is multinomial with probabilities (p_{ij}) . Writing $p_{i+} = \sum_j p_{ij}$ and $p_{+j} = \sum_i p_{ij}$ for the marginals, the hypotheses are

$$H_0 : p_{ij} = p_{i+} p_{+j} \text{ for all } i, j, \quad H_1 : p \text{ unrestricted.}$$

The unrestricted MLE is $\hat{p}_{ij} = N_{ij}/n$ as before. Under H_0 the free parameters are the two marginal vectors, each constrained to sum to 1, and a Lagrange multiplier computation

identical to the one above gives

$$\hat{p}_{i+} = \frac{N_{i+}}{n}, \quad \hat{p}_{+j} = \frac{N_{+j}}{n}.$$

Hence, with $o_{ij} = N_{ij}$ and $e_{ij} = n\hat{p}_{i+}\hat{p}_{+j} = N_{i+}N_{+j}/n$,

$$2 \log \Lambda = 2 \sum_{i,j} N_{ij} \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} = 2 \sum_{i,j} o_{ij} \log \frac{o_{ij}}{e_{ij}} \approx \sum_{i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}}.$$

The degrees of freedom are

$$p = \dim \Theta_1 - \dim \Theta_0 = (rc - 1) - [(r - 1) + (c - 1)] = (r - 1)(c - 1),$$

so we refer the statistic to $\chi^2_{(r-1)(c-1)}$.

Testing Homogeneity

Now suppose the row totals are fixed by design rather than random – for instance, we allocate a fixed number of patients to each of several treatments. Then the rows are independent multinomials,

$$(N_{i1}, \dots, N_{ic}) \sim \text{Multinomial}(n_{i+}; p_{i1}, \dots, p_{ic}),$$

and the natural question is whether the treatments differ:

$$H_0 : p_{1j} = p_{2j} = \dots = p_{rj} \text{ for all } j, \quad H_1 : \text{each row an arbitrary probability vector.}$$

The log-likelihood is $\ell(p) = \text{const} + \sum_{i,j} N_{ij} \log p_{ij}$, exactly as in the independence problem. Maximising row by row gives $\hat{p}_{ij} = N_{ij}/n_{i+}$ under H_1 ; under H_0 , writing p_j for the common value, $\ell = \text{const} + \sum_j N_{+j} \log p_j$ is maximised at $\hat{p}_j = N_{+j}/n_{++}$. So

$$2 \log \Lambda = 2 \sum_{i,j} N_{ij} \log \frac{N_{ij}/n_{i+}}{N_{+j}/n_{++}} = 2 \sum_{i,j} o_{ij} \log \frac{o_{ij}}{e_{ij}}, \quad e_{ij} = \frac{n_{i+}N_{+j}}{n_{++}},$$

which is *identical* to the statistic for independence, with identical expected counts. The degrees of freedom also agree:

$$p = r(c - 1) - (c - 1) = (r - 1)(c - 1).$$

So although the sampling schemes and the hypotheses are conceptually quite different, the two tests are operationally the same: we reject one exactly when we reject the other.

Example 22.5.24 (A Clinical Trial)

150 patients are randomly assigned in equal numbers to a placebo, a half dose and a full dose of a drug, with results

	improved	no difference	worse	
placebo	18	17	15	50
half dose	20	10	20	50
full dose	25	13	12	50
	63	40	47	150

Here the row totals are fixed at 50, so this is a test of homogeneity. The expected counts are $e_{ij} = 50 \times N_{+j}/150 = N_{+j}/3$, giving the first row of expectations

(21, 13.33, 15.67) and similarly for the others. Computing,

$$2 \log \Lambda = 5.129, \quad \sum_{i,j} \frac{(o_{ij} - e_{ij})^2}{e_{ij}} = 5.173.$$

With $(r-1)(c-1) = 4$ degrees of freedom and $\chi_4^2(0.05) = 9.488$, we do not reject H_0 at the 5% level: this trial provides no significant evidence that the drug does anything.

Remark. Two weaknesses of these tests are worth flagging.

- (i) The χ^2 approximation needs n large; the usual heuristic is that every *expected* count should be at least 5. When it is not, one should use an exact test (such as Fisher's exact test), which is non-examinable.
- (ii) The tests often have low power, because Θ_1 is enormous. A test that must guard against every possible departure from H_0 cannot be very sensitive to any particular one.

§22.5.9 Tests and Confidence Sets

We close this section by making precise a duality that has been lurking throughout: confidence sets and hypothesis tests are two views of the same object.

Theorem 22.5.25

Let $X \sim f_X(\cdot | \theta)$, $\theta \in \Theta$.

- (i) Suppose that for each $\theta_0 \in \Theta$ there is a size- α test of $H_0 : \theta = \theta_0$ with acceptance region $A(\theta_0)$. Then

$$I(X) = \{\theta : X \in A(\theta)\}$$

is a $100(1 - \alpha)\%$ confidence set for θ .

- (ii) Conversely, if $I(X)$ is a $100(1 - \alpha)\%$ confidence set for θ , then $A(\theta_0) = \{x : \theta_0 \in I(x)\}$ is the acceptance region of a size- α test of $H_0 : \theta = \theta_0$.

Proof. Both parts follow from the single observation that, by construction,

$$\theta_0 \in I(X) \iff X \in A(\theta_0),$$

so the two events have the same probability under $\mathbb{P}_{\theta_0}$. For (i), the test having size α means $\mathbb{P}_{\theta_0}(X \in A(\theta_0)) = 1 - \alpha$, whence $\mathbb{P}_{\theta_0}(\theta_0 \in I(X)) = 1 - \alpha$; as θ_0 was arbitrary this is exactly the confidence set property. Part (ii) is the same computation read backwards. $\square$

In words – a confidence set is the collection of parameter values that the data would not have rejected. This is often the quickest route to a test, or to an interval.

Example 22.5.26

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma_0^2)$ with σ_0^2 known. We found the $100(1 - \alpha)\%$ confi-

dence interval

$$I(X) = \left(\bar{X} \pm \frac{z_{\alpha/2} \sigma_0}{\sqrt{n}} \right).$$

By part (ii) of the theorem, a size- α test of $H_0 : \mu = \mu_0$ is obtained by rejecting when $\mu_0 \notin I(X)$, that is, when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} \right| > z_{\alpha/2},$$

which is precisely the two-sided test we derived from the generalised likelihood ratio. The two derivations agree, as they must.

§22.6 Normal Theory Tests

Every test we have constructed so far has assumed the variance known, which is unrealistic. In this section we develop the distribution theory that lets us drop that assumption, and then deploy it in the standard tests for normal data. Everything rests on one theorem – the joint distribution of $\bar{X}$ and S_{XX} – which we prove in two ways.

In this section vectors are columns, and we write $u^\top$ for the transpose.

§22.6.1 The Multivariate Normal Distribution

We recall the essentials from IA Probability. For a random vector $X = (X_1, \dots, X_n)^\top$ we write $\mathbb{E}[X]$ for the vector of means and

$$\text{Var}(X) = \mathbb{E} \left[(X - \mathbb{E}X)(X - \mathbb{E}X)^\top \right]$$

for the covariance matrix, so $\text{Var}(X)_{ij} = \text{Cov}(X_i, X_j)$. For constant $A \in \mathbb{R}^{k \times n}$ and $b \in \mathbb{R}^k$ we have the familiar

$$\mathbb{E}[AX + b] = A \mathbb{E}[X] + b, \quad \text{Var}(AX + b) = A \text{Var}(X) A^\top.$$

Definition 22.6.1 (Multivariate Normal)

A random vector X has a **multivariate normal** distribution if $t^\top X$ is univariate normal for every fixed $t \in \mathbb{R}^n$ (allowing degenerate normals of zero variance).

Proposition 22.6.2

If X is multivariate normal then so is $AX + b$, for any constant $A \in \mathbb{R}^{k \times n}$ and $b \in \mathbb{R}^k$.

Proof. For $t \in \mathbb{R}^k$, $t^\top (AX + b) = (A^\top t)^\top X + t^\top b$ is a univariate normal plus a constant. $\square$

Proposition 22.6.3

A multivariate normal distribution is determined by its mean vector μ and covariance

matrix Σ , and we write $X \sim N(\mu, \Sigma)$.

Proof. The moment generating function is $M_X(t) = \mathbb{E}[e^{t^\top X}]$. Since $t^\top X$ is univariate normal with mean $t^\top \mu$ and variance $t^\top \Sigma t$, evaluating the one-dimensional normal mgf at 1 gives

$$M_X(t) = \exp\left(t^\top \mu + \frac{1}{2} t^\top \Sigma t\right),$$

which depends only on μ and Σ . By uniqueness of moment generating functions, the distribution is determined. $\square$

An immediate corollary, which we will use constantly, is that for multivariate normal vectors *zero covariance implies independence*: if Σ is block diagonal, the mgf factorises accordingly.

§22.6.2 Orthogonal Projections

The geometry underlying normal theory is that of orthogonal projection, so we set up the linear algebra we need.

Definition 22.6.4 (Orthogonal Projection)

A matrix $P \in \mathbb{R}^{n \times n}$ is an **orthogonal projection** onto its column space $\text{col}(P)$ if $Pv = v$ for all $v \in \text{col}(P)$ and $Pw = 0$ for all $w \in \text{col}(P)^\perp$.

Proposition 22.6.5

P is an orthogonal projection if and only if P is symmetric and idempotent ($P^2 = P$).

Proof. Suppose P is symmetric and idempotent. If $v \in \text{col}(P)$ then $v = Pa$ for some a , so $Pv = P^2a = Pa = v$. If $w \in \text{col}(P)^\perp$ then $P^\top w = 0$ by definition of the orthogonal complement, and $P^\top = P$, so $Pw = 0$.

Conversely, suppose P is an orthogonal projection. Any $a \in \mathbb{R}^n$ decomposes uniquely as $a = v + w$ with $v \in \text{col}(P)$, $w \in \text{col}(P)^\perp$, and then $P^2a = P^2v + P^2w = Pv = P(v + w) = Pa$; as a was arbitrary, $P^2 = P$. For symmetry, note that for any u_1, u_2 we have $Pu_1 \in \text{col}(P)$ and $(I - P)u_2 \in \text{col}(P)^\perp$, so

$$u_1^\top P^\top (I - P)u_2 = (Pu_1) \cdot ((I - P)u_2) = 0.$$

Hence $P^\top (I - P) = 0$, i.e. $P^\top = P^\top P$. But $P^\top P$ is symmetric, so $P^\top$ is symmetric, so P is. $\square$

Corollary 22.6.6

If P is an orthogonal projection then so is $I - P$, projecting onto $\text{col}(P)^\perp$.

Proof. $I - P$ is symmetric, and $(I - P)^2 = I - 2P + P^2 = I - P$. $\square$

Proposition 22.6.7

If P is an orthogonal projection then $P = UU^\top$, where the columns of U form an orthonormal basis of $\text{col}(P)$. Consequently $\text{rank } P = \text{tr } P$.

Proof. Let U have as its columns an orthonormal basis of $\text{col}(P)$, so $U^\top U = I_r$ where $r = \dim \text{col}(P)$. Then $UU^\top$ is symmetric, and $(UU^\top)(UU^\top) = U(U^\top U)U^\top = UU^\top$, so it is an orthogonal projection; its column space is $\text{col}(U) = \text{col}(P)$, and two orthogonal projections with the same column space are equal (both act as the identity there and as zero on the complement).

For the rank, $\text{rank } P = r = \text{tr}(I_r) = \text{tr}(U^\top U) = \text{tr}(UU^\top) = \text{tr } P$, using $\text{tr}(AB) = \text{tr}(BA)$. $\square$

Now the key distributional result.

Theorem 22.6.8 (Projections of Spherical Normals)

Let $X \sim N(0, \sigma^2 I_n)$ and let P be an orthogonal projection of rank r . Then

- (i) $PX \sim N(0, \sigma^2 P)$ and $(I - P)X \sim N(0, \sigma^2(I - P))$, and these are independent;
- (ii) $\|PX\|^2 / \sigma^2 \sim \chi_r^2$.

Proof. The stacked vector $(PX, (I - P)X)^\top \in \mathbb{R}^{2n}$ is a linear function of X , hence multivariate normal, and is therefore determined by its mean and covariance. Its mean is 0, and its covariance is

$$\begin{pmatrix} P \\ I - P \end{pmatrix} \sigma^2 I \begin{pmatrix} P \\ I - P \end{pmatrix}^\top = \sigma^2 \begin{pmatrix} P^2 & P(I - P) \\ (I - P)P & (I - P)^2 \end{pmatrix} = \sigma^2 \begin{pmatrix} P & 0 \\ 0 & I - P \end{pmatrix},$$

using idempotence and $P(I - P) = P - P^2 = 0$. This identifies the marginals, and since the covariance is block diagonal and the vector is multivariate normal, the two blocks are independent.

For (ii), write $P = UU^\top$ as above. Then

$$\|PX\|^2 = X^\top P^\top PX = X^\top PX = X^\top UU^\top X = \|U^\top X\|^2,$$

and $U^\top X \sim N(0, \sigma^2 U^\top U) = N(0, \sigma^2 I_r)$. So the r components of $U^\top X / \sigma$ are i.i.d $N(0, 1)$, and

$$\frac{\|PX\|^2}{\sigma^2} = \sum_{i=1}^r \left(\frac{(U^\top X)_i}{\sigma} \right)^2 \sim \chi_r^2. \quad \square$$

§22.6.3 Normal Random Samples

We can now prove the central result of the section.

Theorem 22.6.9 (Joint Distribution of $\bar{X}$ and S_{XX})

Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$. Then

- (i) $\bar{X} \sim N(\mu, \sigma^2/n)$;
- (ii) $S_{XX}/\sigma^2 \sim \chi_{n-1}^2$;
- (iii) $\bar{X}$ and S_{XX} are independent.

Proof. Let P be the $n \times n$ matrix with every entry $1/n$. It is symmetric, and $P^2 = P$ since each entry of P^2 is $n \cdot \frac{1}{n^2} = \frac{1}{n}$; so P is an orthogonal projection, onto the one-dimensional space of constant vectors. Note $PX = (\bar{X}, \dots, \bar{X})^\top$ and $\text{tr } P = 1$, so $\text{rank } P = 1$ and $\text{rank}(I - P) = n - 1$.

Write $X = M + \varepsilon$ where $M = (\mu, \dots, \mu)^\top$ and $\varepsilon \sim N(0, \sigma^2 I)$. Since M is constant, $PM = M$ and $(I - P)M = 0$, so

$$\bar{X} = (PX)_1 = \mu + (P\varepsilon)_1$$

is a function of $P\varepsilon$, while

$$S_{XX} = \sum_i (X_i - \bar{X})^2 = \|X - PX\|^2 = \|(I - P)X\|^2 = \|(I - P)\varepsilon\|^2$$

is a function of $(I - P)\varepsilon$. By Theorem 22.6.8 these two are independent, giving (iii), and

$$\frac{S_{XX}}{\sigma^2} = \frac{\|(I - P)\varepsilon\|^2}{\sigma^2} \sim \chi_{n-1}^2,$$

giving (ii). Finally $\bar{X}$ is normal (a linear function of a normal vector) with mean μ and variance σ^2/n , giving (i). $\square$

Remark. This is where the mysterious $n - 1$ of Section 1 comes from. The data live in $\mathbb{R}^n$; fitting the mean uses up the one-dimensional space of constant vectors, and the residual $X - \bar{X}\mathbf{1}$ is confined to the orthogonal complement, of dimension $n - 1$. The ‘degrees of freedom’ really are dimensions of a subspace. It also explains why $\bar{X}$ and S_{XX} are independent, which is otherwise a rather surprising fact: they are functions of the components of ε in orthogonal subspaces.

There is a classical alternative proof, which is worth seeing because it makes the change of coordinates explicit.

Alternative proof, by an orthogonal change of basis. Let A be any $n \times n$ orthogonal matrix whose first row is $(1/\sqrt{n}, \dots, 1/\sqrt{n})$; such a matrix exists by extending that unit vector to an orthonormal basis. Put $Y = AX$, so

$$Y \sim N(AM, \sigma^2 AA^\top) = N(AM, \sigma^2 I),$$

and $AM = (\sqrt{n}\mu, 0, \dots, 0)^\top$. So $Y_1 \sim N(\sqrt{n}\mu, \sigma^2)$ and $Y_2, \dots, Y_n$ are i.i.d $N(0, \sigma^2)$, all independent. But $Y_1 = \frac{1}{\sqrt{n}} \sum_i X_i = \sqrt{n}\bar{X}$, giving (i), and

$$Y_2^2 + \dots + Y_n^2 = \|Y\|^2 - Y_1^2 = X^\top A^\top A X - n\bar{X}^2 = \sum_i X_i^2 - n\bar{X}^2 = S_{XX},$$

a sum of $n - 1$ independent $N(0, \sigma^2)$ squares, giving (ii). Independence of Y_1 from $Y_2, \dots, Y_n$ gives (iii). $\square$

§22.6.4 The One-Sample t -Test

Now we can finally handle unknown variance. Let $X_1, \dots, X_n$ be i.i.d $N(\mu, \sigma^2)$ with both parameters unknown, and write

$$\tilde{\sigma}^2 = \frac{S_{XX}}{n-1}$$

for the unbiased estimator of σ^2 . By Theorem 22.6.9,

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim N(0, 1), \quad V = \frac{S_{XX}}{\sigma^2} \sim \chi_{n-1}^2,$$

and these are independent. So by the definition of the t -distribution,

$$\frac{Z}{\sqrt{V/(n-1)}} = \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{S_{XX}/((n-1)\sigma^2)}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\tilde{\sigma}} \sim t_{n-1}.$$

The unknown σ has cancelled: this is a genuine pivot, and it is the reason the t -distribution was invented.

Proposition 22.6.10 (Confidence Interval for a Normal Mean)

With $X_1, \dots, X_n$ i.i.d $N(\mu, \sigma^2)$, both unknown, a $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm \frac{\tilde{\sigma}}{\sqrt{n}} t_{n-1}\left(\frac{\alpha}{2}\right).$$

Proof. Immediate from the pivot above and the symmetry of t_{n-1} about 0:

$$1 - \alpha = \mathbb{P}\left(-t_{n-1}\left(\frac{\alpha}{2}\right) \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\tilde{\sigma}} \leq t_{n-1}\left(\frac{\alpha}{2}\right)\right),$$

and rearranging for μ . □

Definition 22.6.11 (One-Sample t -Test)

To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ with σ^2 unknown, compute

$$t = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\tilde{\sigma}}$$

and reject H_0 at size α if $|t| > t_{n-1}(\alpha/2)$. For the one-sided alternative $H_1 : \mu > \mu_0$ we reject if $t > t_{n-1}(\alpha)$.

By the duality of Section 5.9 this is exactly the test that rejects when μ_0 falls outside the confidence interval above. Note that $t_{n-1}(\alpha/2) > z_{\alpha/2}$, so the interval is wider and the test more conservative than the corresponding Z -procedures: this is the price of not knowing σ .

Example 22.6.12

Suppose $n = 10$ observations give $\bar{x} = 4.2$ and $\tilde{\sigma} = 1.1$, and we test $H_0 : \mu = 3.5$

against $H_1 : \mu \neq 3.5$. Then

$$t = \frac{\sqrt{10}(4.2 - 3.5)}{1.1} = \frac{3.162 \times 0.7}{1.1} \approx 2.01.$$

Since $t_9(0.025) = 2.262$, we do not reject at the 5% level – though we would at the 10% level, as $t_9(0.05) = 1.833$.

§22.6.5 The Two-Sample t -Test

More often we wish to compare two groups. Let

$$X_1, \dots, X_m \stackrel{\text{iid}}{\sim} N(\mu_X, \sigma^2), \quad Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu_Y, \sigma^2)$$

be independent samples with a *common* unknown variance, and test $H_0 : \mu_X = \mu_Y$.

The natural estimator of the difference is $\bar{X} - \bar{Y}$, with

$$\bar{X} - \bar{Y} \sim N\left(\mu_X - \mu_Y, \sigma^2 \left(\frac{1}{m} + \frac{1}{n}\right)\right).$$

For the variance we pool the two sums of squares. Since $S_{XX}/\sigma^2 \sim \chi_{m-1}^2$ and $S_{YY}/\sigma^2 \sim \chi_{n-1}^2$ independently, and independent chi-squareds add,

$$\frac{S_{XX} + S_{YY}}{\sigma^2} \sim \chi_{m+n-2}^2,$$

and this is independent of $\bar{X} - \bar{Y}$ (each sum of squares is independent of its own sample mean, and the samples are independent of each other). Writing

$$\tilde{\sigma}^2 = \frac{S_{XX} + S_{YY}}{m + n - 2}$$

for the **pooled** variance estimator, we obtain the pivot

$$\frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}.$$

Definition 22.6.13 (Two-Sample t -Test)

Under the above assumptions, to test $H_0 : \mu_X = \mu_Y$ compute

$$t = \frac{\bar{X} - \bar{Y}}{\tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}}}$$

and reject at size α if $|t| > t_{m+n-2}(\alpha/2)$. Correspondingly,

$$(\bar{X} - \bar{Y}) \pm \tilde{\sigma} \sqrt{\frac{1}{m} + \frac{1}{n}} t_{m+n-2}\left(\frac{\alpha}{2}\right)$$

is a $100(1 - \alpha)\%$ confidence interval for $\mu_X - \mu_Y$.

The assumption of equal variances is essential to the pooling step. If it is doubtful, one should first test it (see below), or use an approximate procedure such as Welch's test, which is not part of this course.

§22.6.6 Paired Data

Suppose instead that the observations come in pairs (X_i, Y_i) , $i = 1, \dots, n$ – the same patient measured before and after treatment, say, or two measuring instruments applied to the same specimen. The two-sample test is then *wrong*, because it assumes the samples independent, and here they are emphatically not: a patient with a high ‘before’ reading will tend to have a high ‘after’ reading too.

The remedy is disarmingly simple. Set $D_i = X_i - Y_i$ and treat $D_1, \dots, D_n$ as a single sample. If the pairs are i.i.d then so are the D_i , and testing $H_0 : \mathbb{E}[D] = 0$ by the one-sample t -test on the differences is exactly right.

Definition 22.6.14 (Paired t -Test)

Given paired data, let $D_i = X_i - Y_i$ and suppose $D_1, \dots, D_n$ are i.i.d $N(\mu_D, \sigma_D^2)$. To test $H_0 : \mu_D = 0$, compute

$$t = \frac{\sqrt{n} \bar{D}}{\tilde{\sigma}_D}, \quad \tilde{\sigma}_D^2 = \frac{1}{n-1} \sum_i (D_i - \bar{D})^2,$$

and reject at size α if $|t| > t_{n-1}(\alpha/2)$.

Pairing is not merely a technical correction; it is usually a large *gain*. The subject-to-subject variability, which in the two-sample test is noise inflating $\tilde{\sigma}$, cancels entirely in the differences. When a paired design is available it will almost always detect a given effect with far fewer observations than an unpaired one.

Example 22.6.15

Eight subjects have their reaction times measured before and after a dose of caffeine, with differences (before minus after, in milliseconds)

$$D = (12, -3, 18, 7, 22, 5, 14, 9).$$

Here $\bar{D} = 10.5$ and $\sum_i (D_i - \bar{D})^2 = 430$, so $\tilde{\sigma}_D^2 = 430/7 \approx 61.4$ and $\tilde{\sigma}_D \approx 7.84$. Then

$$t = \frac{\sqrt{8} \times 10.5}{7.84} \approx 3.79,$$

against $t_7(0.025) = 2.365$. So we reject H_0 at the 5% level and conclude that caffeine has an effect.

§22.6.7 The F -Test for Equality of Variances

Finally, suppose we wish to compare two variances – either because it is the question of interest, or to check the pooling assumption of the two-sample t -test. Let $X_1, \dots, X_m$ be i.i.d $N(\mu_X, \sigma_X^2)$ and $Y_1, \dots, Y_n$ be i.i.d $N(\mu_Y, \sigma_Y^2)$, independent, and test $H_0 : \sigma_X^2 = \sigma_Y^2$.

We know $S_{XX}/\sigma_X^2 \sim \chi_{m-1}^2$ and $S_{YY}/\sigma_Y^2 \sim \chi_{n-1}^2$, independently. So by definition of the F -distribution,

$$\frac{S_{XX}/((m-1)\sigma_X^2)}{S_{YY}/((n-1)\sigma_Y^2)} = \frac{\tilde{\sigma}_X^2/\sigma_X^2}{\tilde{\sigma}_Y^2/\sigma_Y^2} \sim F_{m-1, n-1}.$$

This is a pivot for the ratio σ_X^2/σ_Y^2 , so it gives both a test and a confidence interval.

Definition 22.6.16 (*F-Test for Equality of Variances*)

Under $H_0 : \sigma_X^2 = \sigma_Y^2$ the statistic

$$F = \frac{\tilde{\sigma}_X^2}{\tilde{\sigma}_Y^2}$$

has the $F_{m-1, n-1}$ distribution. We reject H_0 at size α against the two-sided alternative if $F > F_{m-1, n-1}(\alpha/2)$ or $F < F_{m-1, n-1}(1 - \alpha/2)$.

Remark. Recall that $1/F \sim F_{n-1, m-1}$, so one may always arrange to have the larger sample variance on top and use only the upper tail (doubling the resulting p -value for a two-sided test). Note also that this test is notoriously sensitive to departures from normality – far more so than the t -test, which is quite robust. A significant F here may be telling us about the tails of the data rather than about the variances.

Example 22.6.17

Suppose $m = 11$ observations give $\tilde{\sigma}_X^2 = 24.5$, and $n = 16$ give $\tilde{\sigma}_Y^2 = 9.8$. Then $F = 24.5/9.8 = 2.5$ on $(10, 15)$ degrees of freedom. Since $F_{10, 15}(0.025) = 3.06$, we do not reject equality of variances at the 5% level, and pooling for a subsequent t -test would be defensible.

§22.7 The Linear Model

The linear model is the single most used statistical model in existence, and it is a fitting place to end, because it brings together everything we have done: least squares as maximum likelihood, projections and degrees of freedom, t -tests for individual parameters, F -tests from the generalised likelihood ratio, and confidence intervals from pivots.

§22.7.1 The Model

Suppose we observe pairs $(x_1, Y_1), \dots, (x_n, Y_n)$ where $Y_i \in \mathbb{R}$ and $x_i = (x_{i1}, \dots, x_{ip})^\top \in \mathbb{R}^p$. The Y_i are the **responses** (or dependent variables); the x_{ij} are the **predictors** (or covariates, or independent variables) and are treated as fixed and known. We model

$$Y_i = \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i,$$

where $\beta \in \mathbb{R}^p$ are unknown **coefficients** and ε is a random **noise** vector.

Example 22.7.1

Let Y_i be the number of insurance claims made by driver i in a year, and let the predictors record the driver's age, years of experience, and points on their licence. The model asserts that the expected number of claims is a linear combination of these.

An **intercept** is included by setting $x_{i1} = 1$ for all i , so that β_1 is a constant term; we will always assume this has been done unless stated otherwise. It is worth stressing that ‘linear’ refers to linearity in β , not in the predictors: the model

$$Y_i = a + bz_i + cz_i^2 + \varepsilon_i$$

is a perfectly good linear model with $x_i = (1, z_i, z_i^2)^\top$. This flexibility is a large part of why the model is so widely used.

The coefficient β_j is interpreted as the change in $\mathbb{E}[Y_i]$ produced by increasing x_{ij} by one while holding all other predictors fixed. This is emphatically *not* a causal statement unless the predictors were assigned by the experimenter – a point on which a great deal of applied statistics comes to grief.

§22.7.2 Matrix Formulation and Assumptions

Stacking everything up, let

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad X = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

where X is called the **design matrix**. The model is then simply

$$Y = X\beta + \varepsilon.$$

We make two **moment assumptions** on the noise:

- (i) $\mathbb{E}[\varepsilon] = 0$, so that $\mathbb{E}[Y_i] = x_i^\top \beta$ – the model is correctly specified;
- (ii) $\text{Var}(\varepsilon) = \sigma^2 I$, that is, $\text{Var}(\varepsilon_i) = \sigma^2$ for all i and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. Constant variance is called **homoscedasticity**.

Note that no distributional assumption has yet been made; we will add normality in Section 7.5.

Throughout we assume the design matrix X has full rank p , equivalently that its columns are linearly independent. Since $X \in \mathbb{R}^{n \times p}$ this forces $n \geq p$: we need at least as many observations as parameters, which is only sensible.

§22.7.3 Least Squares Estimation

Definition 22.7.2 (Least Squares Estimator)

The **least squares estimator** $\hat{\beta}$ minimises the **residual sum of squares**

$$S(\beta) = \|Y - X\beta\|^2 = \sum_{i=1}^n (Y_i - x_i^\top \beta)^2.$$

Proposition 22.7.3

If X has full column rank then S has the unique minimiser

$$\hat{\beta} = (X^\top X)^{-1} X^\top Y.$$

Moreover $\mathbb{E}[\hat{\beta}] = \beta$ and $\text{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$.

Proof. S is a positive semi-definite quadratic in β , so is minimised at a stationary point. Differentiating,

$$\frac{\partial S}{\partial \beta_k} = -2 \sum_{i=1}^n x_{ik} \left(Y_i - \sum_j x_{ij} \beta_j \right),$$

and setting all p of these to zero gives the **normal equations**

$$X^\top X \hat{\beta} = X^\top Y.$$

Since X has full column rank, $X^\top X$ is invertible: if $X^\top X v = 0$ then $\|Xv\|^2 = v^\top X^\top X v = 0$, so $Xv = 0$, so $v = 0$. Hence $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, and this is the unique stationary point.

As $\hat{\beta}$ is a linear function of Y ,

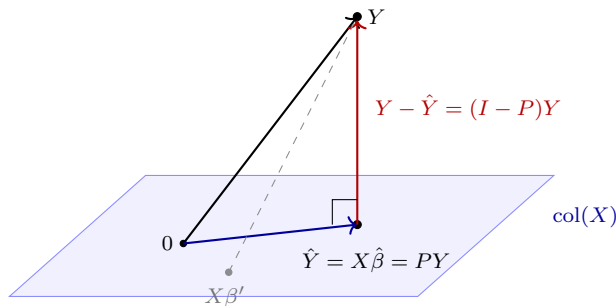
$$\mathbb{E}[\hat{\beta}] = (X^\top X)^{-1} X^\top \mathbb{E}[Y] = (X^\top X)^{-1} X^\top X \beta = \beta,$$

and writing $B = (X^\top X)^{-1} X^\top$,

$$\text{Var}(\hat{\beta}) = B \text{Var}(Y) B^\top = \sigma^2 B B^\top = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1}.$$

□

The geometry is much more illuminating than the algebra. The vector $X\beta$ ranges over $\text{col}(X)$, a p -dimensional subspace of $\mathbb{R}^n$, as β ranges over $\mathbb{R}^p$. Minimising $\|Y - X\beta\|$ therefore means finding the point of $\text{col}(X)$ closest to Y – which is exactly the orthogonal projection of Y onto $\text{col}(X)$.



Any other point $X\beta'$ of the subspace is further from Y , by Pythagoras. Let us record this projection formally.

Definition 22.7.4 (Fitted Values, Hat Matrix, Residuals)

The **fitted values** are $\hat{Y} = X\hat{\beta} = PY$ where

$$P = X (X^\top X)^{-1} X^\top$$

is the **hat matrix**. The **residuals** are $Y - \hat{Y} = (I - P)Y$.

Proposition 22.7.5

P is the orthogonal projection onto $\text{col}(X)$.

Proof. If $v \in \text{col}(X)$ then $v = Xb$ for some b , and $Pv = X(X^\top X)^{-1}X^\top Xb = Xb = v$. If $w \in \text{col}(X)^\perp$ then $X^\top w = 0$, so $Pw = 0$. (Equivalently: P is visibly symmetric, and $P^2 = X(X^\top X)^{-1}X^\top X(X^\top X)^{-1}X^\top = P$.) $\square$

In particular $\text{tr } P = \text{rank } P = p$ and $\text{rank}(I - P) = n - p$, and the residuals are orthogonal to every column of X . That last fact is worth remembering: least squares residuals are, by construction, uncorrelated with every predictor in the model.

§22.7.4 The Gauss-Markov Theorem

Least squares is not just geometrically natural; among a large class of estimators it is optimal, and remarkably this requires no normality assumption at all.

Theorem 22.7.6 (Gauss-Markov)

Let $\beta^* = CY$ be any estimator of β that is linear in Y and unbiased. Then for every $t \in \mathbb{R}^p$,

$$\text{Var}(t^\top \hat{\beta}) \leq \text{Var}(t^\top \beta^*),$$

where $\hat{\beta}$ is the least squares estimator. We say $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE).

Remark. Thinking of t as the predictor vector for a new observation, $t^\top \hat{\beta}$ is our prediction of its expected response. The theorem says least squares gives the most precise such prediction among all linear unbiased methods, simultaneously for every t . Taking $t = e_j$ recovers the statement that $\text{Var}(\hat{\beta}_j) \leq \text{Var}(\beta_j^*)$ for each coordinate.

Proof. Since $\text{Var}(t^\top \beta^*) - \text{Var}(t^\top \hat{\beta}) = t^\top [\text{Var}(\beta^*) - \text{Var}(\hat{\beta})] t$, it suffices to show that $\text{Var}(\beta^*) - \text{Var}(\hat{\beta})$ is positive semi-definite.

Put $A = C - (X^\top X)^{-1}X^\top$, so that $\beta^* = \hat{\beta} + AY$. Unbiasedness of both estimators gives $\mathbb{E}[AY] = 0$; but $\mathbb{E}[AY] = AX\beta$, and as this vanishes for every β we must have $AX = 0$. Now

$$\begin{aligned} \text{Var}(\beta^*) &= \text{Var}\left(\left[A + (X^\top X)^{-1}X^\top\right]Y\right) \\ &= \sigma^2 \left[A + (X^\top X)^{-1}X^\top\right] \left[A + (X^\top X)^{-1}X^\top\right]^\top \\ &= \sigma^2 \left(AA^\top + \underbrace{AX(X^\top X)^{-1}}_{=0} + (X^\top X)^{-1} \underbrace{X^\top A^\top}_{=0} + (X^\top X)^{-1}\right) \\ &= \sigma^2 AA^\top + \text{Var}(\hat{\beta}). \end{aligned}$$

So $\text{Var}(\beta^*) - \text{Var}(\hat{\beta}) = \sigma^2 AA^\top$, and $u^\top AA^\top u = \|A^\top u\|^2 \geq 0$ for every u . $\square$

§22.7.5 The Normal Linear Model

We now strengthen the moment assumptions to a full distributional one.

Definition 22.7.7 (Normal Linear Model)

The **normal linear model** is $Y = X\beta + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2 I)$, where β and $\sigma^2 > 0$ are both unknown.

The likelihood is

$$L(\beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2}\|Y - X\beta\|^2\right),$$

so

$$\ell(\beta, \sigma^2) = \text{const} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|Y - X\beta\|^2.$$

For any fixed σ^2 , maximising over β means minimising $\|Y - X\beta\|^2$. So:

Proposition 22.7.8

In the normal linear model the maximum likelihood estimator of β is the least squares estimator $\hat{\beta} = (X^\top X)^{-1} X^\top Y$, and

$$\hat{\sigma}^2 = \frac{1}{n} \|Y - X\hat{\beta}\|^2 = \frac{1}{n} \|(I - P)Y\|^2.$$

Proof. The first claim is the observation above. Substituting $\hat{\beta}$ and differentiating in σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|(I - P)Y\|^2,$$

which vanishes at the stated value, and this is a maximum. $\square$

That least squares and maximum likelihood coincide under normal errors is a happy accident of the exponential of a quadratic, but it is the reason least squares is so ubiquitous.

Theorem 22.7.9 (Distribution Theory for the Normal Linear Model)

In the normal linear model,

- (i) $\hat{\beta} \sim N(\beta, \sigma^2(X^\top X)^{-1})$;
- (ii) $n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$;
- (iii) $\hat{\beta}$ and $\hat{\sigma}^2$ are independent.

Proof. (i) We have already computed the mean and variance of $\hat{\beta}$, and it is a linear function of the multivariate normal Y , hence multivariate normal.

(ii) Since P projects onto $\text{col}(X)$ we have $(I - P)X = 0$, so

$$\frac{n\hat{\sigma}^2}{\sigma^2} = \frac{\|(I - P)Y\|^2}{\sigma^2} = \frac{\|(I - P)(X\beta + \varepsilon)\|^2}{\sigma^2} = \frac{\|(I - P)\varepsilon\|^2}{\sigma^2} \sim \chi_{\text{rank}(I-P)}^2$$

by Theorem 22.6.8, and $\text{rank}(I - P) = \text{tr } I - \text{tr } P = n - p$.

(iii) From (ii), $\hat{\sigma}^2$ is a function of $(I - P)\varepsilon$. On the other hand

$$\hat{\beta} = (X^\top X)^{-1} X^\top (X\beta + \varepsilon) = \beta + (X^\top X)^{-1} X^\top \varepsilon = \beta + (X^\top X)^{-1} X^\top P\varepsilon,$$

using $X^\top P = X^\top$, so $\hat{\beta}$ is a function of $P\epsilon$. These are independent by Theorem 22.6.8. $\square$

Since $\mathbb{E}[\chi_{n-p}^2] = n - p$ we get

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-p}{n}\sigma^2 < \sigma^2,$$

so as usual the maximum likelihood estimator of the variance is biased downwards. We therefore define the unbiased estimator

$$\tilde{\sigma}^2 = \frac{n\hat{\sigma}^2}{n-p} = \frac{\|(I-P)Y\|^2}{n-p},$$

and note that $\tilde{\sigma}^2$ is what one should always use in practice. Once again the divisor is the dimension of the space in which the residuals live.

§22.7.6 Inference for the Coefficients

Theorem 22.7.9 gives us everything we need to construct pivots.

Proposition 22.7.10 (Confidence Interval for a Single Coefficient)

In the normal linear model,

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\tilde{\sigma}^2 (X^\top X)^{-1}_{jj}}} \sim t_{n-p},$$

and hence

$$\hat{\beta}_j \pm t_{n-p}\left(\frac{\alpha}{2}\right) \sqrt{\tilde{\sigma}^2 (X^\top X)^{-1}_{jj}}$$

is a $100(1 - \alpha)\%$ confidence interval for β_j .

Proof. By part (i) of Theorem 22.7.9, the j th coordinate is distributed as

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 (X^\top X)^{-1}_{jj}\right),$$

so

$$U = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^\top X)^{-1}_{jj}}} \sim N(0, 1).$$

By parts (ii) and (iii), $V = n\hat{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$ independently of U . Hence

$$\frac{U}{\sqrt{V/(n-p)}} = \frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{(X^\top X)^{-1}_{jj}}} \cdot \frac{\sigma}{\tilde{\sigma}} = \frac{\hat{\beta}_j - \beta_j}{\tilde{\sigma} \sqrt{(X^\top X)^{-1}_{jj}}} \sim t_{n-p},$$

the unknown σ having cancelled. Rearranging the two-sided t inequality gives the interval. $\square$

By the test-interval duality, a size- α test of $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ rejects exactly when 0 falls outside this interval, i.e. when

$$\left| \frac{\hat{\beta}_j}{\tilde{\sigma} \sqrt{(X^\top X)^{-1}_{jj}}} \right| > t_{n-p}\left(\frac{\alpha}{2}\right).$$

This is the t -statistic that statistical software prints next to every coefficient.

Example 22.7.11 (The One-Sample t -Test as a Linear Model)

Take $p = 1$ with $X = (1, \dots, 1)^\top$, so the model is $Y_i = \mu + \varepsilon_i$, i.e. $Y_1, \dots, Y_n$ i.i.d $N(\mu, \sigma^2)$. Then $X^\top X = n$, $\hat{\beta} = \bar{Y}$, $(X^\top X)^{-1}_{11} = 1/n$ and $\tilde{\sigma}^2 = S_{YY}/(n-1)$, so the interval above is exactly

$$\bar{Y} \pm \frac{\tilde{\sigma}}{\sqrt{n}} t_{n-1} \left(\frac{\alpha}{2} \right),$$

recovering the one-sample t -interval of Section 6.4.

We can also give a confidence set for the whole vector β at once.

Proposition 22.7.12 (Confidence Set for β)

In the normal linear model,

$$\frac{\|X(\hat{\beta} - \beta)\|^2 / p}{\tilde{\sigma}^2} \sim F_{p, n-p},$$

so

$$\left\{ \beta' \in \mathbb{R}^p : \frac{\|X(\hat{\beta} - \beta')\|^2}{p \tilde{\sigma}^2} \leq F_{p, n-p}(\alpha) \right\}$$

is a $100(1 - \alpha)\%$ confidence set for β .

Proof. Since $\hat{\beta} - \beta \sim N(0, \sigma^2(X^\top X)^{-1})$, writing $(X^\top X)^{1/2}$ for the symmetric positive definite square root,

$$(X^\top X)^{1/2}(\hat{\beta} - \beta) \sim N(0, \sigma^2(X^\top X)^{1/2}(X^\top X)^{-1}(X^\top X)^{1/2}) = N(0, \sigma^2 I_p),$$

so its squared norm over σ^2 is χ_p^2 . But that squared norm is

$$(\hat{\beta} - \beta)^\top X^\top X (\hat{\beta} - \beta) = \|X(\hat{\beta} - \beta)\|^2.$$

This is a function of $\hat{\beta}$, hence independent of $\tilde{\sigma}^2$, and $(n-p)\tilde{\sigma}^2/\sigma^2 \sim \chi_{n-p}^2$. Taking the ratio of the two scaled chi-squareds gives the F distribution, the σ^2 cancelling. $\square$

Geometrically the confidence set is an ellipsoid centred at $\hat{\beta}$, with principal axes along the eigenvectors of $X^\top X$: directions in which the predictors carry little information give long axes. Note that both results in this subsection are *exact* – no asymptotics have been used anywhere.

§22.7.7 F -Tests and Model Comparison

Now suppose we wish to test whether a whole block of coefficients vanishes. Relabelling if necessary, take

$$H_0 : \beta_1 = \dots = \beta_{p_0} = 0, \quad H_1 : \beta \in \mathbb{R}^p.$$

Split $X = (X_0, X_1)$ with $X_0 \in \mathbb{R}^{n \times p_0}$ and $X_1 \in \mathbb{R}^{n \times (p-p_0)}$, and correspondingly $\beta = (\beta^0, \beta^1)$. Under H_0 the model reduces to $Y = X_1\beta^1 + \varepsilon$. Write

$$P = X(X^\top X)^{-1}X^\top, \quad P_1 = X_1(X_1^\top X_1)^{-1}X_1^\top$$

for the projections onto $\text{col}(X)$ and the smaller $\text{col}(X_1)$.

Lemma 22.7.13

$P - P_1$ is an orthogonal projection of rank p_0 , and $(I - P)(P - P_1) = 0$.

Proof. Since $\text{col}(X_1) \subseteq \text{col}(X)$, applying P to anything in $\text{col}(X_1)$ leaves it fixed, so $PP_1 = P_1$; taking transposes, $P_1P = P_1$. Hence $P - P_1$ is symmetric, and

$$(P - P_1)^2 = P^2 - PP_1 - P_1P + P_1^2 = P - P_1 - P_1 + P_1 = P - P_1,$$

so it is an orthogonal projection. Its rank is $\text{tr } P - \text{tr } P_1 = p - (p - p_0) = p_0$. Finally

$$(I - P)(P - P_1) = P - P_1 - P^2 + PP_1 = P - P_1 - P + P_1 = 0. \quad \square$$

Now we compute the generalised likelihood ratio. From the proposition above, the maximised log-likelihood in a normal linear model with projection P is

$$\ell(\hat{\beta}, \hat{\sigma}^2) = \text{const} - \frac{n}{2} \log \frac{\|(I - P)Y\|^2}{n},$$

so

$$2 \log \Lambda = n \left[\log \frac{\|(I - P_1)Y\|^2}{n} - \log \frac{\|(I - P)Y\|^2}{n} \right] = n \log \frac{\|(I - P_1)Y\|^2}{\|(I - P)Y\|^2}.$$

This is increasing in the ratio of sums of squares, which by Lemma 22.7.13 and Pythagoras satisfies

$$\begin{aligned} \frac{\|(I - P_1)Y\|^2}{\|(I - P)Y\|^2} &= \frac{\|(I - P)Y + (P - P_1)Y\|^2}{\|(I - P)Y\|^2} \\ &= \frac{\|(I - P)Y\|^2 + \|(P - P_1)Y\|^2}{\|(I - P)Y\|^2} = 1 + \frac{\|(P - P_1)Y\|^2}{\|(I - P)Y\|^2}, \end{aligned}$$

the cross term vanishing because $(I - P)(P - P_1) = 0$. So the generalised likelihood ratio test rejects for large values of

$$F = \frac{\|(P - P_1)Y\|^2/p_0}{\|(I - P)Y\|^2/(n - p)},$$

and, delightfully, we can find its exact distribution rather than appealing to Wilks.

Theorem 22.7.14

Under $H_0 : \beta_1 = \cdots = \beta_{p_0} = 0$ in the normal linear model,

$$F = \frac{\|(P - P_1)Y\|^2/p_0}{\|(I - P)Y\|^2/(n - p)} \sim F_{p_0, n-p}.$$

Proof. We know $\|(I - P)Y\|^2 = \|(I - P)\varepsilon\|^2 \sim \sigma^2 \chi_{n-p}^2$. Under H_0 we have $Y = X_1\beta^1 + \varepsilon$, and since both P and P_1 fix $\text{col}(X_1)$,

$$(P - P_1)Y = (P - P_1)X_1\beta^1 + (P - P_1)\varepsilon = (P - P_1)\varepsilon,$$

so by Theorem 22.6.8 and Lemma 22.7.13, $\|(P - P_1)Y\|^2 \sim \sigma^2 \chi_{p_0}^2$.

For independence, note that the stacked vector $((I - P)\varepsilon, (P - P_1)\varepsilon)$ is multivariate normal with mean zero and covariance

$$\sigma^2 \begin{pmatrix} (I - P)^2 & (I - P)(P - P_1) \\ (P - P_1)(I - P) & (P - P_1)^2 \end{pmatrix} = \sigma^2 \begin{pmatrix} I - P & 0 \\ 0 & P - P_1 \end{pmatrix},$$

which is block diagonal, so the two blocks are independent. Taking the ratio of the two scaled chi-squareds and cancelling σ^2 gives $F_{p_0, n-p}$. $\square$

So the size- α test rejects H_0 when $F > F_{p_0, n-p}(\alpha)$, and this is exact for every n, p, p_0 .

Remark. The numerator $\|(P - P_1)Y\|^2$ is the reduction in residual sum of squares achieved by adding the p_0 extra predictors, and the denominator estimates σ^2 . So F compares ‘variance explained per extra parameter’ against ‘unexplained variance per residual degree of freedom’. Adding predictors always reduces the residual sum of squares; the F -test asks whether it reduces it by more than chance would predict.

Proposition 22.7.15

When $p_0 = 1$, the F -test of $H_0 : \beta_1 = 0$ is identical to the two-sided t -test of the same hypothesis.

Proof sketch. Recall from Section 1 that $T \sim t_{n-p}$ implies $T^2 \sim F_{1, n-p}$, and correspondingly $t_{n-p}(\alpha/2)^2 = F_{1, n-p}(\alpha)$. So the t -test rejects exactly when

$$\frac{\hat{\beta}_1^2}{\tilde{\sigma}^2(X^\top X)_{11}^{-1}} > F_{1, n-p}(\alpha),$$

and since the denominators of the two statistics already agree, it suffices to check that

$$\frac{\hat{\beta}_1^2}{(X^\top X)_{11}^{-1}} = \|(P - P_1)Y\|^2.$$

When $p_0 = 1$, $P - P_1$ is the rank-one projection onto the span of $v = (I - P_1)X_0$, where X_0 is the first column of X . A computation using $X_0^\top(I - P_1) = v^\top = v^\top(P - P_1)$ gives

$$\|(P - P_1)Y\|^2 = \frac{(v^\top Y)^2}{\|v\|^2} = \frac{(X_0^\top(I - P_1)X\hat{\beta})^2}{\|(I - P_1)X_0\|^2} = \|(I - P_1)X_0\|^2 \hat{\beta}_1^2,$$

using $(I - P_1)X = [(I - P_1)X_0, 0, \dots, 0]$. Finally, blockwise matrix inversion gives $(X^\top X)_{11}^{-1} = \|(I - P_1)X_0\|^{-2}$, and the two expressions agree. $\square$

§22.7.8 Simple Linear Regression

The simplest interesting case is $p = 2$ with a single predictor. It is customary to centre the predictor, writing

$$Y_i = a + b(x_i - \bar{x}) + \varepsilon_i, \quad \bar{x} = \frac{1}{n} \sum_i x_i,$$

which makes the algebra much cleaner because it renders the two columns of the design matrix orthogonal.

Proposition 22.7.16

In the centred simple linear regression model,

$$\hat{a} = \bar{Y}, \quad \hat{b} = \frac{S_{xY}}{S_{xx}},$$

where $S_{xY} = \sum_i (x_i - \bar{x})(Y_i - \bar{Y})$ and $S_{xx} = \sum_i (x_i - \bar{x})^2$. Moreover $\hat{a}$ and $\hat{b}$ are independent, with

$$\hat{a} \sim N\left(a, \frac{\sigma^2}{n}\right), \quad \hat{b} \sim N\left(b, \frac{\sigma^2}{S_{xx}}\right).$$

Proof. Minimise $S(a, b) = \sum_i (Y_i - a - b(x_i - \bar{x}))^2$. Since $\sum_i (x_i - \bar{x}) = 0$,

$$\frac{\partial S}{\partial a} = -2 \sum_i (Y_i - a - b(x_i - \bar{x})) = -2 \sum_i (Y_i - a),$$

which vanishes at $a = \bar{Y}$, independently of b . Substituting,

$$\frac{\partial S}{\partial b} = -2 \sum_i (Y_i - \bar{Y} - b(x_i - \bar{x})) (x_i - \bar{x}),$$

which vanishes at $b = S_{xY}/S_{xx}$.

For the distributions, the design matrix is $X = [\mathbf{1}, x - \bar{x}\mathbf{1}]$ whose columns are orthogonal, so

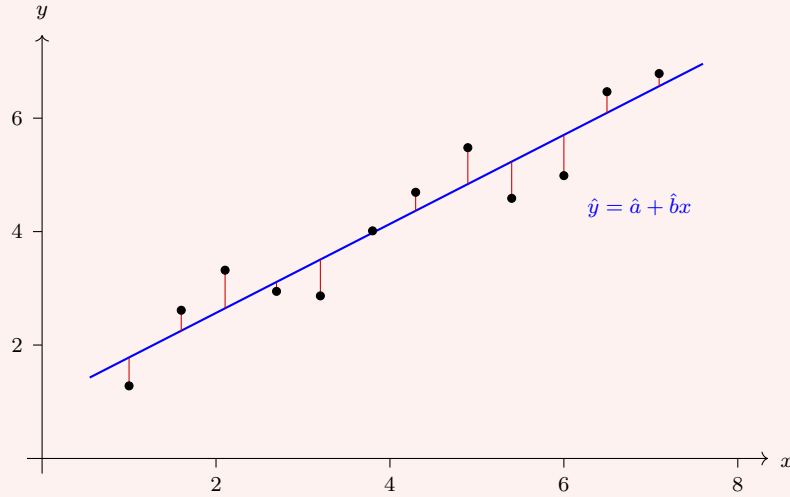
$$X^\top X = \begin{pmatrix} n & 0 \\ 0 & S_{xx} \end{pmatrix}, \quad (X^\top X)^{-1} = \begin{pmatrix} 1/n & 0 \\ 0 & 1/S_{xx} \end{pmatrix}.$$

By Theorem 22.7.9, $(\hat{a}, \hat{b})$ is bivariate normal with this covariance scaled by σ^2 ; the covariance is diagonal, so they are independent. $\square$

Note that S_{xY}/n is the sample covariance of x and Y , and S_{xx}/n the sample variance of x , so $\hat{b}$ is the sample covariance divided by the sample variance – exactly the formula for a regression coefficient one meets in elementary statistics.

Example 22.7.17 (A Worked Regression)

Twelve observations were taken, and the fitted line $\hat{y} = 0.99 + 0.79x$ obtained. The picture below shows the data, the fitted line, and the residuals as vertical segments.



Least squares chooses the line making the sum of the squared *vertical* red segments as small as possible. Note the asymmetry – vertical, not perpendicular, distance – which reflects the fact that the x values are treated as fixed and only Y is random.

§22.7.9 Prediction

Having fitted the model, we would like to use it. Let $x^* \in \mathbb{R}^p$ be a new vector of predictor values.

The **expected response** at x^* is $x^{*\top} \beta$, which we estimate by $x^{*\top} \hat{\beta}$. Since

$$x^{*\top} (\hat{\beta} - \beta) \sim N(0, \sigma^2 x^{*\top} (X^\top X)^{-1} x^*),$$

writing $\tau^2 = x^{*\top} (X^\top X)^{-1} x^*$ and dividing by the independent $\tilde{\sigma}$ as before,

$$\frac{x^{*\top} (\hat{\beta} - \beta)}{\tilde{\sigma} \tau} \sim t_{n-p}.$$

So a $100(1 - \alpha)\%$ confidence interval for the expected response is

$$x^{*\top} \hat{\beta} \pm \tilde{\sigma} \tau t_{n-p} \left(\frac{\alpha}{2} \right).$$

That, however, is not usually what we want. A new observation at x^* will be

$$Y^* = x^{*\top} \beta + \varepsilon^*, \quad \varepsilon^* \sim N(0, \sigma^2)$$

independent of the data, and there are now *two* sources of uncertainty: we do not know the line, and the new point will scatter about it anyway.

Definition 22.7.18 (Prediction Interval)

A $100(1 - \alpha)\%$ **prediction interval** for Y^* is a random interval $I(Y)$ with

$$\mathbb{P}(Y^* \in I(Y)) = 1 - \alpha,$$

where the probability is taken over the joint distribution of Y^* and $Y_1, \dots, Y_n$.

Proposition 22.7.19

$$x^{\star\top} \hat{\beta} \pm \tilde{\sigma} \sqrt{\tau^2 + 1} t_{n-p} \left(\frac{\alpha}{2} \right)$$

is a $100(1 - \alpha)\%$ prediction interval for Y^* .

Proof. Let $\hat{Y}^* = x^{\star\top} \hat{\beta}$. Then $\hat{Y}^* - Y^* = x^{\star\top} (\hat{\beta} - \beta) - \varepsilon^*$ has mean zero, and as the two terms are independent,

$$\text{Var}(\hat{Y}^* - Y^*) = \sigma^2 x^{\star\top} (X^\top X)^{-1} x^* + \sigma^2 = \sigma^2 (\tau^2 + 1).$$

Both terms are normal, so $\hat{Y}^* - Y^* \sim N(0, \sigma^2(\tau^2 + 1))$, and this is independent of $\tilde{\sigma}^2$. Dividing,

$$\frac{\hat{Y}^* - Y^*}{\tilde{\sigma} \sqrt{\tau^2 + 1}} \sim t_{n-p},$$

and rearranging gives the interval. $\square$

The extra $+1$ under the square root is the irreducible noise of the new observation. Because τ^2 is typically small (of order $1/n$) while the 1 is not, prediction intervals are much wider than confidence intervals for the expected response, and they do *not* shrink to zero as $n \rightarrow \infty$. Confusing the two is a common and serious error.

Example 22.7.20

In the regression example above we have $n = 12$, $\bar{x} = 4.05$, $S_{xx} = 43.4$ and $\tilde{\sigma} = 0.54$, and we wish to predict at $x^* = 6$. With the centred parametrisation $x^{\star\top} = (1, x^* - \bar{x})$ and $(X^\top X)^{-1} = \text{diag}(1/n, 1/S_{xx})$, so

$$\tau^2 = \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{xx}} = \frac{1}{12} + \frac{(1.95)^2}{43.4} = 0.083 + 0.088 = 0.171,$$

giving $\tau = 0.41$. The fitted value is $0.99 + 0.79 \times 6 = 5.73$, and with $t_{10}(0.025) = 2.228$ we obtain a 95% confidence interval for the expected response

$$5.73 \pm 0.54 \times 0.41 \times 2.228 = (5.24, 6.22),$$

but a 95% prediction interval

$$5.73 \pm 0.54 \times \sqrt{1.171} \times 2.228 = (4.43, 7.03),$$

which is more than twice as wide. Notice also that τ^2 grows quadratically as x^* moves away from $\bar{x}$: extrapolation is punished, as it should be.

§22.7.10 One-Way Analysis of Variance

Our final application shows that the linear model covers situations that do not look like regression at all. Suppose we compare J treatments, with N subjects receiving each, and model

$$Y_{ij} = \alpha + \mu_j + \varepsilon_{ij}, \quad i = 1, \dots, N, \quad j = 1, \dots, J,$$

with ε_{ij} i.i.d $N(0, \sigma^2)$. Here the predictor is **categorical**: it records only which group a subject is in.

As written the parameters are not identifiable, since adding a constant to α and subtracting it from every μ_j changes nothing. We impose the **corner point constraint** $\mu_1 = 0$, so that group 1 (typically the control) is the baseline and μ_j measures the effect of treatment j relative to it. The design matrix is then

$$X = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & & \vdots \end{pmatrix} = (X_1 \ X_0)$$

where X_1 is the column of ones and X_0 the $J - 1$ indicator columns. Here $n = JN$ and $p = J$.

Definition 22.7.21 (One-Way ANOVA)

The **one-way analysis of variance** test is the test of

$$H_0 : \mu_2 = \cdots = \mu_J = 0, \quad H_1 : \mu_2, \dots, \mu_J \in \mathbb{R}$$

in the above model; under H_0 all groups have the same mean $\mathbb{E}[Y_{ij}] = \alpha$.

This is exactly the F -test of Section 7.7 with $p_0 = J - 1$. To write it out concretely we need only identify the two projections. Since $\text{col}(X)$ is the space of vectors constant within each group, P replaces every observation by its group mean:

$$\bar{Y}_{\cdot j} = \frac{1}{N} \sum_{i=1}^N Y_{ij}, \quad PY = (\bar{Y}_{\cdot 1}, \dots, \bar{Y}_{\cdot 1}, \dots, \bar{Y}_{\cdot J}, \dots, \bar{Y}_{\cdot J})^\top,$$

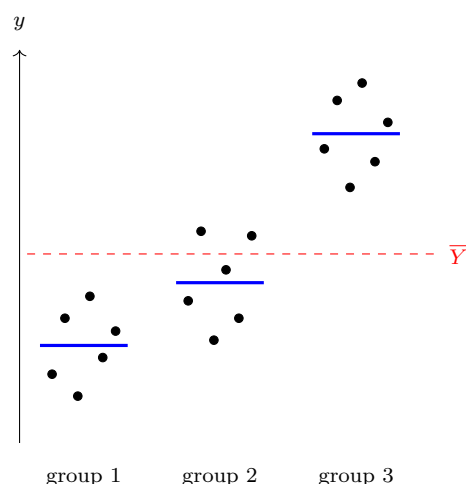
while $\text{col}(X_1)$ is the space of constant vectors, so P_1 replaces everything by the grand mean $\bar{Y}$. Therefore

$$\|(P - P_1)Y\|^2 = \sum_{j=1}^J N (\bar{Y}_{\cdot j} - \bar{Y})^2, \quad \|(I - P)Y\|^2 = \sum_{i,j} (Y_{ij} - \bar{Y}_{\cdot j})^2,$$

and the F -statistic is

$$F = \frac{\sum_j N (\bar{Y}_{\cdot j} - \bar{Y})^2 / (J - 1)}{\sum_{i,j} (Y_{ij} - \bar{Y}_{\cdot j})^2 / (JN - J)} = \frac{\text{variability between groups}}{\text{variability within groups}} \sim F_{J-1, JN-J}$$

under H_0 . This is where the name ‘analysis of variance’ comes from: the total sum of squares splits, by Pythagoras, into a between-groups part and a within-groups part, and we ask which dominates.



The blue bars are the group means $\bar{Y}_{\cdot j}$ and the red dashed line is the grand mean. The F -statistic is large when the blue bars are spread out relative to the scatter of the dots about their own bar. In the picture above they arguably are, and indeed a formal test would reject.

Remark. Everything here generalises. **Two-way** analysis of variance handles groups classified by two factors,

$$Y_{ijk} = \alpha + \mu_j + \lambda_k + \varepsilon_{ijk},$$

again with corner point constraints $\mu_1 = \lambda_1 = 0$, and one may test $H_0 : \mu_j \equiv 0$, or $H_0 : \lambda_k \equiv 0$, or both, by exactly the same F -test machinery. One can also add *interaction* terms γ_{jk} to allow the effect of one factor to depend on the level of the other. Unequal group sizes cause no difficulty in principle, only messier formulae. In every case the recipe is the same: write down the design matrix, project, and compare sums of squares.

This is a good point at which to stop. The linear model, and its relatives, occupy a remarkable position: general enough to describe an enormous range of experiments, yet special enough that every distributional question about it can be answered exactly, by projecting onto a subspace and counting dimensions.

23 Variational Principles

From finding geodesics to expressing the fundamental laws of nature, variational principles and the calculus of variations arise naturally in pure & applied mathematics, and theoretical physics. Here, we will study such ideas, with the underlying goal of considering problems involving minimizing (or maximizing) quantities that depend on an entire function.

This article constitutes my notes for the ‘Variational Principles’ course, held in Easter 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§23.1 Introduction

We will begin our study of variational principles by getting a feel for the type of problems that we care about. As a field, the study of variational principles can be traced back to a single problem, posed by Johann Bernoulli in 1696. It is there that we will begin.

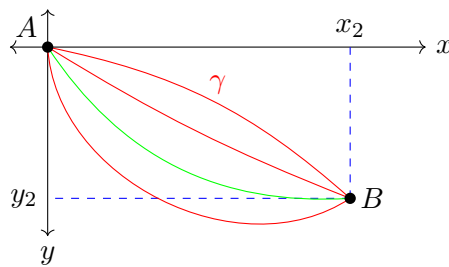
§23.1.1 A Motivating Problem – The Brachistochrone

One of the earliest problems in the calculus of variations is the **Brachistochrone** problem.

Example 23.1.1 (Brachistochrone)

Consider a particle sliding on a wire between two fixed points A and B , under the influence of gravity. Which shape of the wire will give the shortest travel time, when the particle starts from rest?

It is not immediately obvious what the answer to this problem is. We won’t solve it right now, but we can still take some steps in the right direction. We first setup a coordinate system, setting A to be the origin, and B to be some point (x_2, y_2) .



In this coordinate system, for a given curve γ , we can write down the travel time

$$T = \int_{\gamma} dt = \int_{\gamma} \frac{1}{v} ds,$$

where v is the velocity of the particle. Since the particle starts at $y = 0$, conservation of energy implies that

$$\frac{1}{2}mv^2 + mgy = 0$$

$$\implies v = \sqrt{-2gy}.$$

Thus to solve our problem, we really want to minimize

$$T[y] = \frac{1}{\sqrt{2g}} \int_0^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{-y}} dx,$$

subject to y being a smooth function with $y(0) = 0$, and $y(x_2) = y_2$.

§23.1.2 Problems of Interest

Let's jump to another problem from pure mathematics, finding geodesics.

Example 23.1.2 (Geodesic Problem)

Find the shortest path γ between two points on a surface Σ (if one exists).

If Σ is $\mathbb{R}^2$, we can again adopt a coordinate system, and try to find the shortest curve γ between the two points $A = (x_1, y_1)$ and $B = (x_2, y_2)$. To solve this problem, we really want to minimize

$$D[y] = \int_{x_1}^{x_2} \sqrt{1 + (y')^2} dx,$$

subject to y being a smooth function with $y(x_1) = y_1$ and $y(x_2) = y_2$.

Both of these problems have a similar flavour: we are trying to minimize or maximize something like

$$F[y] = \int_{x_1}^{x_2} f(x, y(x), y'(x)) dx, \quad (*)$$

among all functions such that $y(x_1) = y_1$ and $y(x_2) = y_2$. This quantity $(*)$ is called a **functional**, that is, a function on the space of functions¹.

In this article, we are going to study the *calculus of variations*, in which we will be finding extrema (minima, maxima, and saddle points) of functionals. We will start by looking back at and extending our knowledge of extrema in finite dimensions, before using this knowledge to approach problems involving functionals.

Notation. Throughout this article, we will frequently refer to different spaces of functions. We will let $C(\mathbb{R})$ be the space of continuous functions on $\mathbb{R}$, and $C^k(\mathbb{R})$ be the space of functions with continuous k th derivative. We will also write $C_{(\alpha, \beta)}^k(\mathbb{R})$ for the subset of functions f with $f(\alpha) = f(\beta)$.

Being careful about the space of functions we work in matters more than it might first appear. Just as in real analysis, where the domain of a function has to be specified before we can sensibly ask whether it attains a maximum, here we have to say which functions are ‘admissible’ before asking for the best one. Most of the time this will be some class of smooth functions with prescribed values at the endpoints, and we will not belabour the point – but the reader should be aware that some of the more delicate questions in this subject (does a minimiser exist at all?) are questions about the choice of function space, and are well beyond the scope of this article.

¹In standard calculus, we usually consider functions which map numbers $\rightarrow$ numbers. In the calculus of variations, we consider *functional* which map functions $\rightarrow$ numbers. An intuitive example is that of arc-length, which takes some curve and gives back a number corresponding to its length.

§23.1.3 Variational Principles in Nature

The phrase ‘variational principle’ refers to something slightly more specific than the calculus of variations: it is the assertion that some *law of nature* can be recast as the statement that a certain functional is stationary. This is a remarkable state of affairs, and it will be a recurring theme in what follows.

The oldest example is Fermat’s principle, which we will meet properly in [subsection 23.4.3](#).

Law 23.1.3 (Fermat’s Principle)

Light travelling between two points takes the path which requires the least time.

From this single sentence one can derive the law of reflection and Snell’s law of refraction, both of which were discovered empirically long before. Far more striking is that *all* of classical mechanics can be phrased this way.

Law 23.1.4 (Principle of Least Action)

A particle moving in a potential V with kinetic energy T travels between two fixed points in such a way that the [action](#)

$$S[\mathbf{x}] = \int_{t_1}^{t_2} (T - V) \, dt$$

is stationary.

We will see in [section 23.7](#) that this is exactly equivalent to Newton’s second law. It is worth pausing on how strange this is. Newton’s law is ‘local’ in time: it tells the particle how to accelerate right now, given the force right now. The principle of least action is ‘global’: it compares entire histories of the particle, from t_1 to t_2 , and picks out the one which is stationary. That these two descriptions agree is not obvious, and the machinery we are about to develop is precisely what makes the equivalence work.

There is a further payoff. Once a physical law is written as a variational principle, the *symmetries* of the functional turn directly into conserved quantities – this is Noether’s theorem, which we will prove in [subsection 23.7.3](#). Conservation of momentum, angular momentum and energy all fall out of the invariance of the action under translations, rotations, and time-shifts respectively. This is by some distance the most efficient route to the conservation laws.

§23.2 Calculus for Functions on $\mathbb{R}^n$

For this section, we are going to consider a function $f \in C^2(\mathbb{R}^n)$, that is, $f : \mathbb{R}^n \rightarrow \mathbb{R}$, with continuous second derivative.

§23.2.1 Stationary Points

Recall the definition of a stationary point for a function $f \in C^2(\mathbb{R}^n)$.

Definition 23.2.1 (Stationary Point)

We say that $a \in \mathbb{R}^n$ is a **stationary point** of f if $\nabla f(a) = 0$.

If a is a stationary point of f , then we can look at the Taylor expansion of f around a to determine the nature of the stationary point. For x near a , using the summation convention we have

$$f(x) = f(a) + \cancel{(x-a) \cdot \nabla f(a)} + \frac{1}{2}(x_i - a_i)(x_j - a_j) \partial_{ij}^2 f|_a + O(|x - a|^3),$$

where $\partial_i f = \partial f / \partial x_i$, and the linear term dies precisely because a is stationary. So near a stationary point, the entire behaviour of f is controlled by a quadratic form.

Now let $H_{ij} = \partial_{ij}^2 f$ be the **Hessian** matrix, which we note is symmetric since $f \in C^2$. If we shift the origin such that $a = 0$ and write $H = H(0)$, then being real symmetric it can be diagonalized by an orthogonal matrix R , giving

$$H' = R^T H(0) R = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}.$$

Writing $x' = R^T x$ for the coordinates in this new (rotated) frame, we have

$$f(x') - f(0) = \frac{1}{2} \sum_{i=1}^n \lambda_i (x'_i)^2 + O(|x'|^3).$$

The behaviour then depends on whether the eigenvalues of $H(0)$ are positive or negative:

- (i) If all $\lambda_i > 0$, then $f(x') > f(0)$ in all directions, and we have a local minimum.
- (ii) If all $\lambda_i < 0$, then $f(x') < f(0)$ in all directions, and we have a local maximum.
- (iii) If some of the eigenvalues are positive and some are negative, then $f(x')$ is increasing in some directions and is decreasing in others, and we have a saddle point.
- (iv) If some (or all) of the eigenvalues are 0, then we need to consider higher order derivatives in the Taylor expansion.

In the special case where $n = 2$, then $\det H = \lambda_1 \lambda_2$ and $\text{tr } H = \lambda_1 + \lambda_2$. From this we obtain the conditions:

- If $\det > 0$, $\text{tr} > 0$ we have a local minimum.
- If $\det > 0$, $\text{tr} < 0$ we have a local maximum.
- If $\det < 0$ we have a saddle point.
- If $\det = 0$, then we need to look at higher derivatives.

Remark. If a function $f : D \rightarrow \mathbb{R}$ is defined on some domain D , then the minimum or maximum could be obtained on the boundary of D , where we may not have $\nabla f = 0$. This is an easy thing to forget, and it will come back in a rather more dramatic form later on when we look at soap films.

Here is a pretty application of the $n = 2$ criteria. Suppose $f : D \rightarrow \mathbb{R}$ is **harmonic** on $D \subseteq \mathbb{R}^2$, meaning $\nabla^2 f = f_{xx} + f_{yy} = 0$. Then $\text{tr } H = 0$ everywhere, so at any

stationary point either both eigenvalues vanish or they have opposite signs. Hence a non-constant harmonic function has no interior maxima or minima at all: its extreme values are attained on ∂D . This is the *maximum principle*, which the reader may have seen in complex analysis.

Example 23.2.2 (Finding and Classifying Stationary Points)

Consider the function $f(x, y) = x^3 + y^3 - 3xy$. Then at a stationary point we have

$$\nabla f = \begin{pmatrix} 3x^2 - 3y \\ 3y^2 - 3x \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

These imply that at a stationary point, $y^4 = y$. Thus we have the points $(0, 0)$ and $(1, 1)$. We can write down the Hessian as

$$H = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}.$$

At $(0, 0)$, we have $\det H = -9 < 0$, so f has a saddle point there. At $(1, 1)$, we have $\det H = 27 > 0$, and $\text{tr } H = 12 > 0$, so f has a local minimum there.

The degenerate case really does need care, as the following pair shows.

Example 23.2.3 (Degenerate Hessians)

Consider $f(x, y) = x^4 + y^4$ and $g(x, y) = x^4 - y^4$. Both have their only stationary point at the origin, and for both the Hessian there is the zero matrix, so $\det H = \text{tr } H = 0$ and our criteria say nothing at all. But f clearly has a global minimum at the origin, while g has a saddle: it increases along the x -axis and decreases along the y -axis. There is no way to tell these apart without looking at the fourth-order terms.

This is exactly analogous to the one-dimensional situation, where $f''(0) = 0$ leaves us unable to distinguish x^4 from x^3 from $-x^4$.

§23.2.2 Constraints and Lagrange Multipliers

Consider the following problem, and two natural ways to solve it.

Example 23.2.4

Find the circle centered at $(0, 0)$ with smallest radius such that the circle intersects the parabola $y = x^2 - 1$.

Direct Solution. One way to solve this problem is to substitute the constraint $y = x^2 - 1$ into the expression we are trying to minimize.

$$r^2 = x^2 + y^2 = x^2 + (x^2 - 1)^2 = x^4 - x^2 + 1.$$

To have a stationary point, the derivative of the RHS must be zero, so $4x^3 - 2x = 0$. This gives us two solutions: $x = \pm 1/\sqrt{2}, y = -1/2$ and $x = 0, y = -1$. These give the two possible radii of $\sqrt{3}/2$ and 1, of which the first is the smallest.

While this solution works, it's not hard to imagine that we may not be able to solve the constraint for y – and even when we can, doing so may destroy a symmetry of the problem which we would rather keep. An alternate solution is given below.

Lagrange Multipliers Solution. Define the function $h(x, y, \lambda) = f(x, y) - \lambda g(x, y)$, where $f(x, y) = x^2 + y^2 = r^2$, $g(x, y) = 0$ is the constraint, and λ is the Lagrange multiplier. So $h = x^2 + y^2 - \lambda(y - x^2 + 1)$. We are going to extremize this over 3 variables with no constraints.

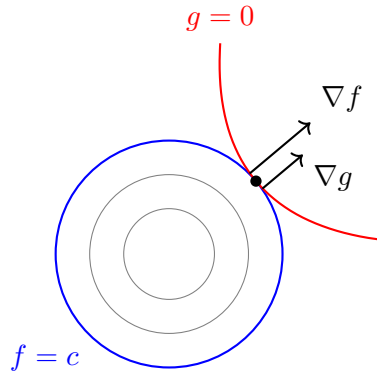
$$\begin{aligned}\frac{\partial h}{\partial x} &= 2x + 2\lambda x = 0, \\ \frac{\partial h}{\partial y} &= 2y - \lambda = 0, \\ \frac{\partial h}{\partial \lambda} &= -(y - x^2 + 1) = 0,\end{aligned}$$

where the third condition is exactly our constraint – this is the small miracle that makes the method work. From the first equation, either $x = 0$ or $\lambda = -1$. From these we get either $x = 0, y = -1$ with $f = 1, \lambda = -2$, or $y = -1/2, x = \pm 1/\sqrt{2}$ with $f = 3/4, \lambda = -1$, giving the radius $\sqrt{3}/2$ as before.

So why does the second solution work? Well suppose that we wanted to minimize a function f subject to $g = 0$. Then ∇g is perpendicular to the level set $g = 0$. Also, ∇f is perpendicular to the level set $f(x) = c$, for any constant c . Now imagine walking along the constraint curve. As long as ∇f has a nonzero component *along* the curve, we can decrease f by taking a step in the appropriate direction, and so we are not at an extremum. So at the extremum ∇f must be entirely perpendicular to the constraint curve; that is, parallel to ∇g , so

$$\nabla f - \lambda \nabla g = 0,$$

for some λ . Equivalently, the level curve of f through the extremum is *tangent* to the constraint curve. Thus it suffices to consider the extrema of $h = f - \lambda g$.



This picture also tells us when the method can fail: if $\nabla g = 0$ at the extremum, then $\nabla f = \lambda \nabla g$ carries no information. Such points are called *singular* and have to be checked separately.

This method generalizes to multiple variables and multiple constraints. For example, if we wanted to extremize $f : \mathbb{R}^n \rightarrow \mathbb{R}$, subject to $g_\alpha(x) = 0$, where $g_\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$ and $\alpha = 1, \dots, k$, we would define

$$h(x_1, \dots, x_n, \lambda_1, \dots, \lambda_k) = f - \sum_{\alpha=1}^k \lambda_\alpha g_\alpha,$$

and would extremize this function with respect to the $n + k$ variables. Geometrically, the constraints $g_\alpha = 0$ cut out an $(n - k)$ -dimensional surface, and the condition is that ∇f lies in the span of the normals $\nabla g_1, \dots, \nabla g_k$, that is, ∇f has no component tangent to the surface.

Example 23.2.5 (Two Constraints)

Let us maximize $f(x, y, z) = x + y + z$ on the circle cut out of the unit sphere by the plane $z = \frac{1}{2}$. So we take

$$g_1 = x^2 + y^2 + z^2 - 1, \quad g_2 = z - \frac{1}{2},$$

and set $h = f - \lambda g_1 - \mu g_2$. Then

$$\frac{\partial h}{\partial x} = 1 - 2\lambda x = 0, \quad \frac{\partial h}{\partial y} = 1 - 2\lambda y = 0, \quad \frac{\partial h}{\partial z} = 1 - 2\lambda z - \mu = 0,$$

together with the two constraints. The first two give $x = y = 1/(2\lambda)$, and the constraints then give $2x^2 = 1 - z^2 = 3/4$, so $x = y = \pm\sqrt{3}/(2\sqrt{2})$. The maximum is at the positive root, where

$$f = \frac{\sqrt{3}}{\sqrt{2}} + \frac{1}{2} = \sqrt{\frac{3}{2}} + \frac{1}{2}.$$

Note that μ was never needed: it only served to absorb the fact that the z -equation is not required to balance. This is typical – the multipliers are auxiliary unknowns which we are free to discard once they have done their job.

Although we have been treating λ as a bookkeeping device, it does have a meaning, and a rather useful one.

Proposition 23.2.6 (The Meaning of the Multiplier)

Suppose we extremize f subject to $g(x) = c$, and let $x^*(c)$ be the (assumed smooth) location of the extremum and $f^*(c) = f(x^*(c))$ its value. Then

$$\frac{df^*}{dc} = \lambda.$$

Proof. Differentiating $f^*(c) = f(x^*(c))$ with the chain rule,

$$\frac{df^*}{dc} = \nabla f(x^*) \cdot \frac{dx^*}{dc} = \lambda \nabla g(x^*) \cdot \frac{dx^*}{dc},$$

using the stationarity condition $\nabla f = \lambda \nabla g$. But differentiating the constraint $g(x^*(c)) = c$ gives $\nabla g(x^*) \cdot dx^*/dc = 1$, and the result follows. $\square$

In other words – λ measures how much the optimal value improves when we relax the constraint by one unit. In economics this is called a *shadow price*, and in physics the multiplier is very often a physically meaningful quantity in its own right (a force, a pressure, a chemical potential). We will see a striking example of this when we solve the isoperimetric problem in [Example 23.5.9](#), where the multiplier turns out to be the radius of the optimal circle.

§23.3 Convexity and the Legendre Transform

Before we leave finite dimensions, we should discuss a class of functions for which the story above becomes dramatically simpler: the **convex** functions. For these, every stationary point is a global minimum, so all the fiddling with Hessians disappears. Convexity is also exactly the hypothesis needed to make the *Legendre transform* – our second major tool – behave itself.

§23.3.1 Convex Sets and Convex Functions

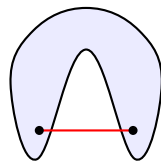
We begin with the underlying notion for sets.

Definition 23.3.1 (Convex Set)

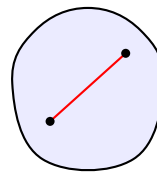
A set $S \subseteq \mathbb{R}^n$ is **convex** if for all $x, y \in S$ and all $t \in [0, 1]$ we have

$$(1 - t)x + ty \in S.$$

That is, the line segment joining any two points of S lies entirely within S .



not convex



convex

Definition 23.3.2 (Convex Function)

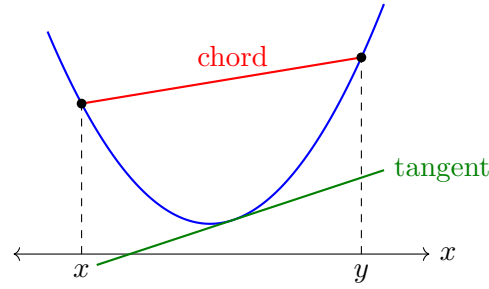
A function $f : S \rightarrow \mathbb{R}$ with $S \subseteq \mathbb{R}^n$ is **convex** if

- (i) its domain S is a convex set, and
- (ii) for all $x, y \in S$ and $t \in (0, 1)$,

$$f((1 - t)x + ty) \leq (1 - t)f(x) + tf(y).$$

If the inequality is strict whenever $x \neq y$, we say f is **strictly convex**. We say f is **concave** (respectively strictly concave) if $-f$ is convex (respectively strictly convex).

The second condition is best read geometrically. The right hand side, as t runs over $[0, 1]$, traces out the **chord** joining $(x, f(x))$ to $(y, f(y))$; the left hand side traces out the graph of f between the same two points. So f is convex exactly when its graph lies on or below all of its chords.



We have drawn a tangent line in as well, because it will feature shortly: a convex function also lies *above* all of its tangents.

Example 23.3.3

(i) $f(x) = x^2$ is strictly convex on $\mathbb{R}$. Indeed,

$$[(1-t)x + ty]^2 - (1-t)x^2 - ty^2 = -t(1-t)(x-y)^2 < 0$$

for $t \in (0, 1)$ and $x \neq y$, after expanding and collecting terms.

(ii) $f(x) = |x|$ is convex, by the triangle inequality, but not strictly so.

(iii) $f(x) = 1/x$ on the domain $\{x > 0\}$ is strictly convex.

(iv) $f(x) = 1/x$ on $\mathbb{R} \setminus \{0\}$ is *not* convex, for the trivial reason that its domain is not a convex set. This is not mere pedantry: if we tried to patch the definition by setting $f(0) = 0$, the chord from $(-1, -1)$ to $(1, 1)$ would lie below the graph. Condition (i) is doing real work.

§23.3.2 Conditions for Convexity

Checking convexity straight from the definition is unpleasant. If f is differentiable there is a much better test.

Theorem 23.3.4 (First-Order Convexity Condition)

Let f be differentiable on a convex domain S . Then f is convex if and only if

$$f(y) \geq f(x) + (y - x) \cdot \nabla f(x)$$

for all $x, y \in S$.

Proof. Suppose first that f is convex. Fix $x, y \in S$ and define, for $t \in [0, 1]$,

$$h(t) = (1-t)f(x) + tf(y) - f((1-t)x + ty).$$

Convexity says exactly that $h(t) \geq 0$, and clearly $h(0) = 0$. Hence

$$\frac{h(t) - h(0)}{t} \geq 0 \quad \text{for } t \in (0, 1),$$

and letting $t \rightarrow 0^+$ gives $h'(0) \geq 0$. On the other hand we may differentiate h

directly by the chain rule, giving

$$h'(t) = f(y) - f(x) - (y - x) \cdot \nabla f((1 - t)x + ty),$$

so that $h'(0) = f(y) - f(x) - (y - x) \cdot \nabla f(x)$. Combining these gives the inequality.

Conversely, suppose the inequality holds. Let $z = (1 - t)x + ty$. Applying the inequality twice, at the point z ,

$$\begin{aligned} f(x) &\geq f(z) + (x - z) \cdot \nabla f(z) = f(z) + t(x - y) \cdot \nabla f(z), \\ f(y) &\geq f(z) + (y - z) \cdot \nabla f(z) = f(z) - (1 - t)(x - y) \cdot \nabla f(z). \end{aligned}$$

Multiplying the first by $(1 - t)$, the second by t , and adding, the gradient terms cancel and we are left with

$$(1 - t)f(x) + tf(y) \geq f(z) = f((1 - t)x + ty),$$

which is convexity. $\square$

The geometric content is the one we advertised above: $f(x) + (y - x) \cdot \nabla f(x)$ is precisely the tangent plane to the graph of f at x , so the condition says that a convex function lies above all of its tangent planes. From this the promised simplification is immediate.

Corollary 23.3.5

Any stationary point of a convex function is a global minimum. If f is strictly convex, there is at most one such point.

Proof. Let $\nabla f(x_0) = 0$. Then [Theorem 23.3.4](#) gives $f(y) \geq f(x_0) + (y - x_0) \cdot 0 = f(x_0)$ for every y in the domain. For the uniqueness statement, if $x_0 \neq x_1$ were two global minima with common value m , then strict convexity would give $f(\frac{1}{2}(x_0 + x_1)) < m$, a contradiction. $\square$

By applying [Theorem 23.3.4](#) twice, once at x and once at y , and adding the two inequalities, we obtain a symmetric form of the condition:

$$(y - x) \cdot [\nabla f(y) - \nabla f(x)] \geq 0.$$

In one dimension this reads $(y - x)(f'(y) - f'(x)) \geq 0$, which just says f' is non-decreasing. Differentiating one more time gives the criterion that is usually the most convenient of all.

Theorem 23.3.6 (Second-Order Convexity Condition)

Let f be twice differentiable on a convex domain. Then f is convex if and only if the Hessian H is positive semi-definite at every point (all eigenvalues ≥ 0). If H is positive definite everywhere, then f is strictly convex.

Proof (sketch). Put $y = x + h$ in the symmetric first-order condition, so that

$$h \cdot [\nabla f(x + h) - \nabla f(x)] \geq 0.$$

Taylor expanding the bracket in h using the summation convention, this is

$$h_i [h_j \partial_j \partial_i f + O(|h|^2)] = h_i H_{ij} h_j + O(|h|^3) \geq 0.$$

Since h is arbitrary and may be taken small, this holds if and only if $h^T H h \geq 0$ for all h , i.e. H is positive semi-definite. $\square$

Remark. The converse of the last sentence is false: $f(x) = x^4$ is strictly convex, but $f''(0) = 0$. So positive definiteness is sufficient but not necessary for strictness.

Example 23.3.7

Let $f(x, y) = 1/(xy)$ on the domain $x, y > 0$. Then

$$H = \frac{1}{xy} \begin{pmatrix} 2/x^2 & 1/(xy) \\ 1/(xy) & 2/y^2 \end{pmatrix},$$

so that

$$\det H = \frac{1}{x^2 y^2} \left(\frac{4}{x^2 y^2} - \frac{1}{x^2 y^2} \right) = \frac{3}{x^4 y^4} > 0, \quad \text{tr } H = \frac{2}{xy} \left(\frac{1}{x^2} + \frac{1}{y^2} \right) > 0.$$

Both eigenvalues are therefore positive and f is strictly convex.

It is tempting to notice that we only ever used $xy > 0$, and conclude that f is convex on the larger domain $\{xy > 0\}$. This is wrong – that domain is not connected, let alone convex, and the definition fails at the first hurdle.

§23.3.3 Inequalities from Convexity

Convexity is a machine for producing inequalities, and it is worth seeing it run. The starting point is the extension of the defining inequality from two points to many.

Theorem 23.3.8 (Jensen's Inequality)

Let f be convex, let $x_1, \dots, x_n$ lie in its domain, and let $t_1, \dots, t_n \geq 0$ with $\sum_i t_i = 1$. Then

$$f \left(\sum_{i=1}^n t_i x_i \right) \leq \sum_{i=1}^n t_i f(x_i).$$

Proof. We induct on n . The case $n = 2$ is the definition (the cases $t_i \in \{0, 1\}$ being trivial), and $n = 1$ is vacuous. Suppose the result holds for $n - 1$ and let $t_n < 1$ (if $t_n = 1$ there is nothing to prove). Write $s = 1 - t_n = \sum_{i < n} t_i > 0$ and

$$z = \sum_{i=1}^{n-1} \frac{t_i}{s} x_i,$$

which lies in the domain since the domain is convex and the weights t_i/s sum to 1.

Then $\sum_i t_i x_i = sz + t_n x_n$, so by the $n = 2$ case and then the inductive hypothesis,

$$f\left(\sum_i t_i x_i\right) \leq sf(z) + t_n f(x_n) \leq s \sum_{i < n} \frac{t_i}{s} f(x_i) + t_n f(x_n) = \sum_{i=1}^n t_i f(x_i). \quad \square$$

There is a continuous version too, which is the one that connects to the rest of this article: if $\rho \geq 0$ has $\int \rho = 1$, then $f\left(\int \rho(x)u(x) \, dx\right) \leq \int \rho(x)f(u(x)) \, dx$, proved in the same way by approximating the integral by sums. In probabilistic language this is $f(\mathbb{E}[U]) \leq \mathbb{E}[f(U)]$, which is how the reader will have met it in Probability.

Corollary 23.3.9 (AM–GM Inequality)

For positive reals $a_1, \dots, a_n$,

$$\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}.$$

Proof. The function $f(x) = -\log x$ is convex on $x > 0$, since $f''(x) = 1/x^2 > 0$. Applying Jensen with $t_i = 1/n$ and $x_i = a_i$,

$$-\log\left(\frac{1}{n} \sum_i a_i\right) \leq -\frac{1}{n} \sum_i \log a_i = -\log\left(\prod_i a_i\right)^{1/n}.$$

Multiplying by -1 and exponentiating (both $\log$ and $\exp$ being increasing) gives the result. $\square$

The next inequality is really a statement about the Legendre transform in disguise, as we will see in a moment.

Proposition 23.3.10 (Young's Inequality)

Let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. The case $ab = 0$ is clear, so take $a, b > 0$. Applying Jensen to the convex function $-\log$ with weights $1/p, 1/q$ and points a^p, b^q ,

$$-\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \leq -\frac{1}{p} \log a^p - \frac{1}{q} \log b^q = -\log(ab),$$

and exponentiating gives the claim. $\square$

Corollary 23.3.11 (Hölder's Inequality)

Let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for reals u_i, v_i ,

$$\sum_{i=1}^n |u_i v_i| \leq \left(\sum_{i=1}^n |u_i|^p\right)^{1/p} \left(\sum_{i=1}^n |v_i|^q\right)^{1/q}.$$

Proof. Write $U = (\sum_i |u_i|^p)^{1/p}$ and $V = (\sum_i |v_i|^q)^{1/q}$, and assume both are nonzero (otherwise all the u_i or all the v_i vanish and the result is trivial). Apply Young's inequality with $a = |u_i|/U$ and $b = |v_i|/V$ and sum over i :

$$\frac{1}{UV} \sum_i |u_i v_i| \leq \frac{1}{p} \frac{\sum_i |u_i|^p}{U^p} + \frac{1}{q} \frac{\sum_i |v_i|^q}{V^q} = \frac{1}{p} + \frac{1}{q} = 1,$$

and multiplying by UV finishes the proof. $\square$

Taking $p = q = 2$ recovers the Cauchy–Schwarz inequality. We will meet a genuinely variational inequality later, when the Rayleigh–Ritz method in [section 23.8](#) lets us bound eigenvalues by making guesses.

§23.3.4 The Legendre Transform

We now come to a construction which, on first meeting, looks arbitrary, and which turns out to be one of the most useful things in mathematical physics. It appears in thermodynamics (relating energy, free energy and enthalpy), in mechanics (relating the Lagrangian and the Hamiltonian), and in optimisation (as duality).

Here is the motivation. Suppose we have a differentiable function $f(x)$, and for whatever reason we would rather work with the derivative

$$p = \frac{df}{dx}$$

as our independent variable. In physics p is usually the interesting quantity: for a Lagrangian it is a momentum, for the energy of a gas it is a temperature. The obvious thing to try, $g(p) = f(x(p))$, is a bad idea – it is not symmetric, it loses information when f has an inflection, and it has no nice properties whatsoever.

What we would *like* is a function f^* with the pleasing reciprocal property

$$\frac{df^*}{dp} = x,$$

so that if p is conjugate to x , then x is conjugate to p . In terms of differentials, we have $df = p dx$ and we want $df^* = x dp$. Since the product rule gives $d(xp) = x dp + p dx$, the choice

$$f^*(p) = xp - f(x)$$

does exactly what we want. We take this as our definition, but phrased so as not to require differentiability.

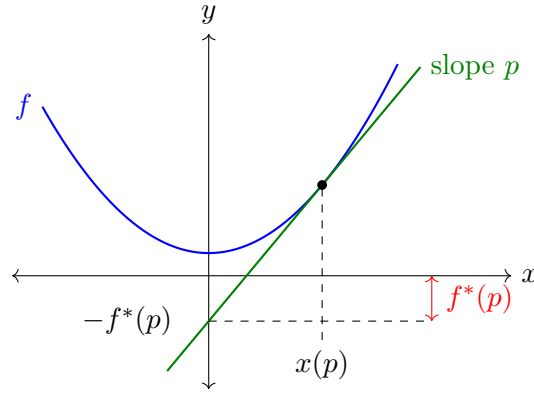
Definition 23.3.12 (Legendre Transform)

The **Legendre transform** of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the function f^* given by

$$f^*(p) = \sup_x (p \cdot x - f(x)).$$

The domain of f^* is taken to be the set of p for which the supremum is finite. We call p the **conjugate variable** to x .

In one dimension there is a very clean picture. For a fixed p , the quantity $px - f(x)$ is the vertical gap between the line $y = px$ and the graph $y = f(x)$, and $f^*(p)$ is the largest such gap. Equivalently – and this is the version worth remembering – draw the tangent line to f of slope p ; then $-f^*(p)$ is its intercept with the y -axis.



Notice that the supremum is finite exactly when the line of slope p can be pushed up against the graph from below, which is why convexity is about to become important.

Proposition 23.3.13

If the domain of f^* is non-empty then it is a convex set, and f^* is a convex function – regardless of whether f is convex.

Proof. Let p, q lie in the domain of f^* and let $t \in (0, 1)$. Then

$$\begin{aligned} f^*((1-t)p + tq) &= \sup_x [(1-t)p \cdot x + tq \cdot x - f(x)] \\ &= \sup_x [(1-t)(p \cdot x - f(x)) + t(q \cdot x - f(x))] \\ &\leq (1-t) \sup_x [p \cdot x - f(x)] + t \sup_x [q \cdot x - f(x)] \\ &= (1-t)f^*(p) + tf^*(q), \end{aligned}$$

using only that the supremum of a sum is at most the sum of the suprema. In particular the right hand side is finite, so $(1-t)p + tq$ is in the domain, which is therefore convex; and the inequality is exactly convexity of f^* . $\square$

In practice, if f is convex and differentiable, we do not compute a supremum at all: we differentiate. The supremum of $p \cdot x - f(x)$ occurs where

$$\nabla(p \cdot x - f(x)) = 0 \iff p = \nabla f(x),$$

and if f is strictly convex then ∇f is injective, so this can be solved uniquely for $x = x(p)$, giving

$$f^*(p) = p \cdot x(p) - f(x(p)).$$

Example 23.3.14

- (i) Let $f(x) = \frac{1}{2}ax^2$ with $a > 0$. Then $p = f'(x) = ax$, so $x = p/a$, and

$$f^*(p) = p \cdot \frac{p}{a} - \frac{1}{2}a \left(\frac{p}{a}\right)^2 = \frac{p^2}{2a}.$$

A parabola transforms to a parabola, with the coefficient inverted. Note that transforming again returns $\frac{1}{2}ax^2$, as we would hope.

- (ii) Let $f(v) = -\sqrt{1-v^2}$ on $|v| < 1$ (the lower half of a circle). Then

$$p = f'(v) = \frac{v}{\sqrt{1-v^2}} \implies v = \frac{p}{\sqrt{1+p^2}},$$

which exists for all $p \in \mathbb{R}$, and

$$f^*(p) = pv + \sqrt{1-v^2} = \frac{p^2}{\sqrt{1+p^2}} + \frac{1}{\sqrt{1+p^2}} = \sqrt{1+p^2}.$$

A circle transforms to a hyperbola. (The reader who has met special relativity may enjoy recognising this.)

- (iii) Let $f(x) = cx$ for a constant c . This is convex but not strictly. Then $px - f(x) = (p-c)x$, which is unbounded above unless $p = c$. So the domain of f^* is the single point $\{c\}$, and $f^*(c) = 0$. A line transforms to a point.

The last example makes the general phenomenon clear: the Legendre transform trades the ‘flatness’ of a function for the ‘sharpness’ of its transform, and vice versa. It also explains why convexity is exactly the right hypothesis for the following.

Theorem 23.3.15

If f is convex and differentiable, then $f^{**} = f$.

Proof. We have $f^*(p) = p \cdot x(p) - f(x(p))$, where $x(p)$ satisfies $p = \nabla f(x(p))$. Differentiating with respect to p_i and using the product and chain rules,

$$\begin{aligned} \partial_i f^*(p) &= x_i + p_j \partial_i x_j(p) - \partial_i x_j(p) \partial_j f(x) \\ &= x_i + p_j \partial_i x_j(p) - \partial_i x_j(p) p_j \\ &= x_i, \end{aligned}$$

where in the second line we used $p_j = \partial_j f(x)$. So $\nabla f^*(p) = x$: the conjugate variable to p is the original x . Therefore

$$f^{**}(x) = (x \cdot p - f^*(p))|_{p=p(x)} = x \cdot p - (p \cdot x - f(x)) = f(x). \quad \square$$

Convexity really is needed here. Since f^{**} is a Legendre transform it is automatically convex, so $f^{**} = f$ can only possibly hold for convex f . For a non-convex f , the double transform f^{**} turns out to be the largest convex function lying below f – its ‘convex hull’ – so the transform simply cannot see the non-convex parts.

Remark. Young’s inequality is now transparent. Take $f(x) = x^p/p$ for $x \geq 0$ and $p > 1$. Then $f^*(b) = \sup_x (bx - x^p/p) = b^q/q$ where $\frac{1}{p} + \frac{1}{q} = 1$, by the usual differentiation. But directly from the definition of a supremum, $f^*(b) \geq ab - f(a)$ for any a , which

rearranges to $ab \leq a^p/p + b^q/q$. Every Legendre transform gives an inequality in this way, the general statement $p \cdot x \leq f(x) + f^*(p)$ being called **Fenchel's inequality**.

§23.3.5 An Application to Thermodynamics

The reason a thermodynamics course is full of Legendre transforms is worth a paragraph. Consider a gas of a fixed number of particles in a container. In principle we could write down the equations of motion of all 10^{23} particles and solve them, which is not a plan. Instead we describe the gas by a handful of macroscopic quantities: the entropy S , volume V , pressure P and temperature T . The energy is naturally a function

$$E = E(S, V).$$

There are two ways to change it: we can do work on the gas by compressing it, contributing $-P \, dV$, or we can heat it, contributing $T \, dS$. So

$$dE = T \, dS - P \, dV,$$

and comparing with the chain rule,

$$T = \frac{\partial E}{\partial S}, \quad P = -\frac{\partial E}{\partial V}.$$

The trouble is that entropy is not something one can easily hold fixed in a laboratory, whereas temperature is. We would therefore like a function of T and V instead of S and V – and T is precisely the variable conjugate to S . Taking the (negative of the) Legendre transform in S gives the **Helmholtz free energy**

$$F(T, V) = \inf_S (E(S, V) - TS) = E - TS,$$

where the infimum is at $T = \partial E / \partial S$ as required. Its differential is

$$dF = dE - T \, dS - S \, dT = -S \, dT - P \, dV,$$

so that $S = -\partial F / \partial T$ and $P = -\partial F / \partial V$. This is the whole point: no information has been lost, and we can recover everything by differentiating F , which we could not do had we naively substituted $S = S(T, V)$ into E .

Transforming instead in V gives the **enthalpy** $H(S, P) = E + PV$, and transforming in both gives the **Gibbs free energy** $G(T, P) = E - TS + PV$. The convexity of E in its arguments – which is a physical statement about the stability of the gas – is what guarantees these transforms are invertible.

§23.4 The Euler-Lagrange Equations

We are now going to move on to deriving some necessary conditions for the extrema of a functional.

§23.4.1 Deriving the Euler-Lagrange Equations

We want to extremize the functional $(*)$,

$$F[y] = \int_{\alpha}^{\beta} f(x, y(x), y'(x)) \, dx,$$

where we assume that f is given, and that our competing functions have fixed endpoints $y(\alpha) = y_1$ and $y(\beta) = y_2$.

Recall how we found stationary points in $\mathbb{R}^n$: we perturbed our candidate point a by a small displacement, expanded to first order, and demanded that the first-order term vanish for *every* direction of displacement. We shall do precisely the same thing here. The only difference is that a ‘displacement’ is now a whole function, and there are infinitely many directions to worry about.

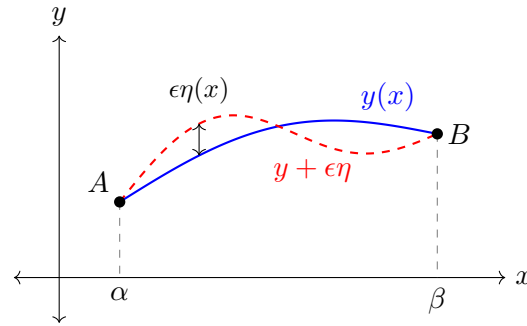
So suppose y is our candidate extremum, and perturb it to

$$y \mapsto y + \epsilon\eta(x),$$

where ϵ is a small real number and η is any (sufficiently smooth) function with

$$\eta(\alpha) = \eta(\beta) = 0.$$

The condition on η is forced on us: the endpoints of y are fixed, so any admissible competitor must agree with y there.



We will compute $F[y + \epsilon\eta]$ and expand in ϵ . To turn the resulting condition into a differential equation, we will need an important lemma.

Lemma 23.4.1 (Fundamental Lemma of Calculus of Variations)

If $g : [\alpha, \beta] \rightarrow \mathbb{R}$ is continuous on $[\alpha, \beta]$ and is such that

$$\int_{\alpha}^{\beta} g(x)\eta(x) \, dx = 0,$$

for all η continuous on $[\alpha, \beta]$ with $\eta(\alpha) = \eta(\beta) = 0$, then $g(x) = 0$ for all $x \in [\alpha, \beta]$.

Proof. Say (for a contradiction) that there exists some $\bar{x} \in (\alpha, \beta)$ such that $g(\bar{x}) \neq 0$. Without loss of generality say $g(\bar{x}) > 0$ (else replace g by $-g$). Then by continuity there exists an interval $[x_1, x_2] \subset (\alpha, \beta)$ containing $\bar{x}$ such that $g(x) > c$ on $[x_1, x_2]$ for some $c > 0$. Then set

$$\eta(x) = \begin{cases} (x - x_1)(x_2 - x) & \text{if } x \in [x_1, x_2], \\ 0 & \text{if } x \notin [x_1, x_2]. \end{cases}$$

This η is continuous, vanishes at α and β , is non-negative, and is strictly positive

on (x_1, x_2) . Since the integrand vanishes off $[x_1, x_2]$, we have

$$\int_{\alpha}^{\beta} g(x)\eta(x) \, dx = \int_{x_1}^{x_2} g(x)\eta(x) \, dx > c \int_{x_1}^{x_2} (x - x_1)(x_2 - x) \, dx = \frac{c(x_2 - x_1)^3}{6} > 0,$$

which is a contradiction. So g vanishes on (α, β) , and hence by continuity on all of $[\alpha, \beta]$. $\square$

Remark. In the proof above, we construct a *bump function* η . It is possible to make this smoother by taking various powers of the constructed function: replacing η by $[(x - x_1)(x_2 - x)]^{k+1}$ on $[x_1, x_2]$ gives a C^k bump. So the lemma remains true if we insist that our test functions be as smooth as we like, which is what we shall actually need.

The lemma is worth internalising in the following form: the only continuous function which is ‘invisible’ to every test function is the zero function. This is exactly the tool we need to convert an integral condition into a pointwise one.

Now let’s return to our computation. We have

$$\begin{aligned} F[y + \epsilon\eta] &= \int_{\alpha}^{\beta} f(x, y + \epsilon\eta, y' + \epsilon\eta') \, dx \\ &= F[y] + \epsilon \int_{\alpha}^{\beta} \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) \, dx + O(\epsilon^2), \end{aligned}$$

where the partial derivatives are all evaluated along the unperturbed path. We will return to the $O(\epsilon^2)$ term in [section 23.9](#), where it tells us about the *nature* of the stationary point. For now we note that for an extremum we want the first-order term to vanish for *every* admissible η , that is

$$\left. \frac{d}{d\epsilon} F[y + \epsilon\eta] \right|_{\epsilon=0} = \int_{\alpha}^{\beta} \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' \right) \, dx = 0.$$

This quantity is called the **first variation** of F at y .

We cannot apply the fundamental lemma to this directly, because the integrand involves η' as well as η . We now make a move which we will return to time and time again in the calculus of variations: we integrate the offending term by parts, so that only η appears. This gives

$$\begin{aligned} 0 &= \int_{\alpha}^{\beta} \left(\frac{\partial f}{\partial y} \eta - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \eta \right) \, dx + \left[\frac{\partial f}{\partial y'} \eta \right]_{\alpha}^{\beta} \\ &= \int_{\alpha}^{\beta} \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right) \eta \, dx, \end{aligned}$$

the boundary term vanishing precisely because $\eta(\alpha) = \eta(\beta) = 0$. Applying the fundamental lemma of the calculus of variations, we obtain the following necessary condition for an extremum.

Theorem 23.4.2 (Euler-Lagrange Equation)

Suppose $y \in C^2_{(\alpha, \beta)}(\mathbb{R})$ is a function with fixed endpoints that extremizes the functional

$$F[y] = \int_{\alpha}^{\beta} f(x, y(x), y'(x)) \, dx.$$

Then y must satisfy the **Euler-Lagrange** equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0. \quad (\dagger)$$

Notation. The LHS of $(\dagger)$ is denoted by $\frac{\delta F[y]}{\delta y(x)}$ and is the **functional derivative** of F . Writing $\delta y = \epsilon \eta$, the first variation then reads

$$\delta F[y] = \int_{\alpha}^{\beta} \frac{\delta F[y]}{\delta y(x)} \delta y(x) dx,$$

a pleasing analogue of $df = \nabla f \cdot dx$ from multivariable calculus. In this language, the Euler–Lagrange equation is simply the statement ‘ $\nabla F = 0$ ’.

A few comments are in order before we start solving things.

Remark.

- (i) Expanding $\frac{d}{dx} (\partial f / \partial y')$ by the chain rule shows that $(\dagger)$ is a *second-order* ODE for y ,

$$\frac{\partial^2 f}{\partial (y')^2} y'' + \frac{\partial^2 f}{\partial y \partial y'} y' + \frac{\partial^2 f}{\partial x \partial y'} - \frac{\partial f}{\partial y} = 0,$$

to be solved subject to the two conditions $y(\alpha) = y_1$ and $y(\beta) = y_2$. This is a *boundary* value problem rather than an initial value problem, and so, unlike for initial value problems, we get no general guarantee of existence or uniqueness. We will meet examples of both failures.

- (ii) The Euler–Lagrange equation is a *necessary* condition, not a sufficient one. Exactly like $f'(x) = 0$ in ordinary calculus, it is satisfied by minima, maxima and saddles alike. Determining which we have requires the second variation.
- (iii) When we write $\partial f / \partial y$, we are treating x , y and y' as three *independent* arguments of f . This is a genuine source of confusion, and should be carefully distinguished from the total derivative along a path,

$$\frac{d}{dx} = \frac{\partial}{\partial x} + y' \frac{\partial}{\partial y} + y'' \frac{\partial}{\partial y'}.$$

For instance, if $f(x, y, y') = x [(y')^2 - y^2]$ then $\partial f / \partial x = (y')^2 - y^2$, whereas

$$\frac{df}{dx} = (y')^2 - y^2 - 2xyy' + 2xy'y''.$$

These are very different objects.

- (iv) The boundary term $[\eta \partial f / \partial y']_{\alpha}^{\beta}$ vanished because we fixed the endpoints. It will not always do so, and when it does not we obtain extra ‘natural’ boundary conditions instead; see [subsection 23.5.4](#).

Before we get to the classical problems, let us do one entirely routine example, just to see the machinery turn over.

Example 23.4.3

Let us find the stationary points of

$$F[y] = \int_0^1 ((y')^2 + y^2) \, dx, \quad y(0) = 0, \, y(1) = 1.$$

Here $f = (y')^2 + y^2$, so

$$\frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial y'} = 2y',$$

and the Euler–Lagrange equation (†) is

$$2y - \frac{d}{dx}(2y') = 0 \implies y'' = y.$$

The general solution is $y = A \sinh x + B \cosh x$. The condition $y(0) = 0$ gives $B = 0$, and $y(1) = 1$ gives $A = 1/\sinh 1$, so

$$y = \frac{\sinh x}{\sinh 1}.$$

Substituting back (and integrating by parts once) gives the stationary value $F[y] = \coth 1 \approx 1.313$.

It is worth noticing that we could have got the same answer from the Beltrami identity below, since f has no explicit x -dependence: that would give $(y')^2 - y^2 = \text{constant}$, a first-order equation. Either route works, and in general one takes whichever is less painful.

§23.4.2 First Integrals of the Euler–Lagrange Equations

Of course, now that we have a necessary condition for extremality, our next concern is solving the Euler–Lagrange equation. The Euler–Lagrange equations give us a second-order ODE for y . It is sometimes possible to reduce the order of the ODE to get a first-order ODE, a **first integral**, which is what we will look at in this section. There are two cases where this happens, and they correspond to the two ‘missing variables’ the integrand might have.

The first case is easy. If f does not explicitly depend on y , then $\partial f/\partial y = 0$, and the Euler–Lagrange equation reads

$$\frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0 \implies \frac{\partial f}{\partial y'} = c,$$

for some constant c . We have integrated the equation once for free, and what remains is a first-order ODE. A variable which does not appear in the integrand (only its derivative does) is called **cyclic** or **ignorable**, and every cyclic variable gives us such a conserved quantity. In mechanics these will turn out to be the conserved momenta.

Example 23.4.4 (Geodesics on the Euclidean Plane)

Consider again the problem of finding geodesics on the Euclidean plane.

We want to minimize the functional

$$D[y] = \int_{x_1}^{x_2} \sqrt{1 + (y')^2} \, dx.$$

Using the Euler-Lagrange equations, we have $f(y') = \sqrt{1 + (y')^2}$, so $\partial f / \partial y = 0$, and thus by the result above

$$\frac{y'}{\sqrt{1 + (y')^2}} = c,$$

for some constant c . For this to be the case, we must have y' constant, so y is a straight line.

Example 23.4.5 (Geodesics on a Cylinder)

Consider the cylinder $x^2 + y^2 = a^2$ in $\mathbb{R}^3$, parameterized by an angle ϕ and a height z . The line element is

$$ds^2 = a^2 d\phi^2 + dz^2,$$

so thinking of z as a function of ϕ , we wish to extremize

$$F[z] = \int_{\phi_1}^{\phi_2} \sqrt{a^2 + (z')^2} \, d\phi.$$

Now z is cyclic, so

$$\frac{\partial f}{\partial z'} = \frac{z'}{\sqrt{a^2 + (z')^2}} = c,$$

which forces z' to be constant, say $z' = m$, and hence

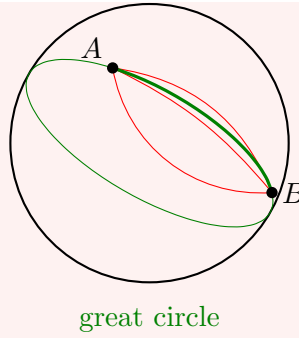
$$z = m\phi + z_0.$$

This is a *helix*. The special cases are worth noting: $m = 0$ gives a horizontal circle, and letting $m \rightarrow \infty$ (that is, treating ϕ as a function of z and finding it constant) gives a vertical straight line. All three are what one would draw by wrapping a piece of string around a tube.

Notice also that between two given points there are *infinitely many* helical geodesics – one for each number of times we wind around the cylinder – of which only one is shortest. Once again, the Euler–Lagrange equation cannot tell them apart.

Example 23.4.6 (Geodesics on the Sphere)

Consider the sphere embedded in three-dimensional Euclidean space, $S^2 \subset \mathbb{R}^3$, and two points A and B on the sphere. We want to find the shortest path between A and B .



We first parameterize the sphere, with

$$\begin{aligned}x &= \sin \theta \sin \phi, \\y &= \sin \theta \cos \phi, \\z &= \cos \theta,\end{aligned}$$

where $0 < \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. Then to find lengths of paths on the sphere, we use the line element

$$ds^2 = dx^2 + dy^2 + dz^2 = d\theta^2 + \sin^2 \theta d\phi^2.$$

Now we can parameterize a curve by thinking of ϕ as a function of θ . Then we have

$$ds = \sqrt{1 + \sin^2 \theta \cdot (\phi')^2} d\theta.$$

Thus the functional we want to extremize is

$$F[\phi] = \int_{\alpha}^{\beta} \sqrt{1 + \sin^2 \theta \cdot (\phi')^2} d\theta.$$

We note here that $\partial f / \partial \theta = 0$, so using the result we obtained above, $\partial f / \partial \phi' = \kappa$, for some constant κ . That is,

$$\frac{\sin^2 \theta \cdot \phi'}{\sqrt{1 + \sin^2 \theta \cdot (\phi')^2}} = \kappa.$$

Squaring this and solving for $(\phi')^2$, we get^a

$$(\phi')^2 = \frac{\kappa^2}{\sin^2 \theta (\sin^2 \theta - \kappa^2)} \implies \phi = \pm \int \frac{\kappa d\theta}{\sin \theta \sqrt{\sin^2 \theta - \kappa^2}}.$$

This looks unpleasant, but yields to the substitution $u = \cot \theta$, for which $du = -\csc^2 \theta d\theta$. Writing $\sin^2 \theta - \kappa^2 = \sin^2 \theta [(1 - \kappa^2) - \kappa^2 u^2]$ after dividing through by $\sin^2 \theta$ and using $\csc^2 \theta = 1 + u^2$, the integral becomes

$$\phi - \phi_0 = \mp \int \frac{\kappa du}{\sqrt{(1 - \kappa^2) - \kappa^2 u^2}} = \mp \arcsin \left(\frac{\kappa u}{\sqrt{1 - \kappa^2}} \right),$$

for some constant ϕ_0 . Rearranging, and absorbing the sign into ϕ_0 ,

$$\cot \theta = \frac{\sqrt{1 - \kappa^2}}{\kappa} \sin(\phi - \phi_0).$$

To see what this is, multiply through by $\sin \theta$ and expand the sine:

$$\cos \theta = \frac{\sqrt{1 - \kappa^2}}{\kappa} (\sin \theta \cos \phi \cos \phi_0 - \sin \theta \sin \phi \sin \phi_0).$$

In terms of our Cartesian coordinates this says $z = c_1 y + c_2 x$ for constants c_1, c_2 , which is the equation of a plane through the origin. So the geodesic lies on the intersection of the sphere with a plane through its centre: it is an arc of a **great circle**, as we expected.

Note that the Euler–Lagrange equation cannot distinguish the *minor* arc from the *major* arc – both are stationary. We will resolve this in [section 23.9](#) using the second variation, where we find that the major arc fails to be a minimum. Note also that if A and B are antipodal, there are infinitely many geodesics between them, which is our promised failure of uniqueness.

^aThere is a sign ambiguity, corresponding to going either forwards or backwards around the sphere.

The second case is subtler, and considerably more useful. Consider, for general $f(x, y, y')$, the quantity $f - y' \partial f / \partial y'$. Differentiating along a path with the total derivative,

$$\begin{aligned} \frac{d}{dx} \left(f - y' \frac{\partial f}{\partial y'} \right) &= \frac{\partial f}{\partial x} + y' \frac{\partial f}{\partial y} + y'' \frac{\partial f}{\partial y'} - y' \frac{\partial}{\partial y'} \left(y' \frac{\partial f}{\partial y'} \right) - y' \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \\ &= y' \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) + \frac{\partial f}{\partial x} \\ &= \frac{\partial f}{\partial x}, \end{aligned}$$

where in the last line we used the Euler–Lagrange equations to kill the bracket. So if f does not depend explicitly on x , we obtain another first integral.

Proposition 23.4.7 (Beltrami Identity)

If $f = f(y, y')$ has no explicit x -dependence, then along any solution of the Euler–Lagrange equation

$$f - y' \frac{\partial f}{\partial y'} = c$$

for some constant c .

This is a genuinely powerful trick: a great many of the classical problems have integrands with no explicit x , and the Beltrami identity reduces the second-order equation to a first-order separable one in one move. The reader should note that it also has a physical meaning which we will uncover in [section 23.7](#): if x is time, then $y' \partial f / \partial y' - f$ is exactly the *energy*, and its constancy is conservation of energy.

Example 23.4.8 (The Brachistochrone)

Consider again the problem of finding the shape of a wire which gives the shortest travel time for a particle sliding along it, beginning at rest.

We want to minimize the functional

$$T[y] = \frac{1}{\sqrt{2g}} \int_0^{x_2} \frac{\sqrt{1 + (y')^2}}{\sqrt{-y}} dx,$$

subject to $y(x_1) = y_1$ and $y(x_2) = y_2$. Here, f does not depend on x and thus we can apply the result above to get

$$\frac{\sqrt{1 + (y')^2}}{\sqrt{-y}} - y' \frac{y'}{\sqrt{1 + (y')^2} \sqrt{-y}} = \kappa,$$

for some constant κ . We can solve this differential equation by simplifying to get

$$\frac{1}{\sqrt{1 + (y')^2}} = \kappa \sqrt{-y},$$

and then we can separate variables to get

$$x = \pm \kappa \int \frac{\sqrt{-y}}{\sqrt{1 + \kappa^2 y}} dy.$$

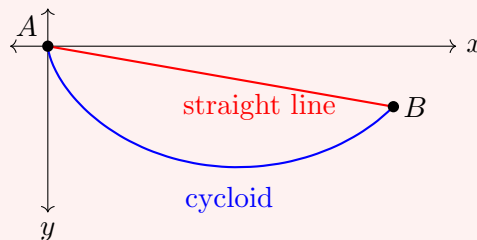
The substitution $y = -\frac{1}{\kappa^2} \sin^2 \frac{\theta}{2}$ then gives (after some algebra)

$$x = \pm \frac{1}{2\kappa^2} (\theta - \sin \theta) + C,$$

for some constant C . Our particle starts at the origin at $\theta = 0$, so $C = 0$; taking the positive root and writing $a = 1/(2\kappa^2)$, the solution is

$$\begin{cases} x = a(\theta - \sin \theta), \\ y = -a(1 - \cos \theta). \end{cases}$$

This is the parameterized equation of a **cycloid**: the curve traced by a point on the rim of a wheel of radius a rolling (here, on the underside of the x -axis) with constant speed. The constant a , and the final value of θ , are fixed by requiring the curve to pass through B .



The shape is worth a moment's thought. The optimal wire dives steeply at first, even though this takes the particle *below* B and forces it to climb back up at the end. The reason is that the early plunge buys speed, and the extra speed is worth more than the extra distance. This is the sort of answer one would be unlikely to guess.

§23.4.3 Fermat's Principle

We can now look properly at the first variational principle in this course, which describes the motion of light.

Law 23.4.9 (Fermat's Principle)

Light travels along the path between two points which requires the least time.

Suppose we had a ray with path $y = y(x)$, and suppose the speed of light at (x, y) is $c(x, y)$. Then we want to extremize the functional

$$F[y] = \int \frac{1}{c} ds = \int_{\alpha}^{\beta} \frac{\sqrt{1 + (y')^2}}{c(x, y)} dx.$$

If we assume that c depends only on x – so the medium is stratified into vertical layers – then we have $\partial f / \partial y = 0$, and we can use the results of the previous section to obtain

$$\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2} c(x)} = k$$

for some constant k .

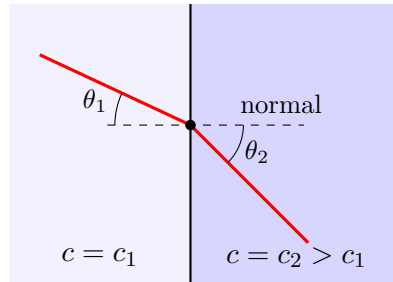
Now let θ be the angle the ray makes with the x -axis, so that $\tan \theta = y'$ and hence

$$\frac{y'}{\sqrt{1 + (y')^2}} = \frac{\tan \theta}{\sec \theta} = \sin \theta.$$

Our first integral therefore says that $\sin \theta / c(x)$ is constant along the ray:

$$\frac{\sin \theta_1}{c(x_1)} = \frac{\sin \theta}{c(x)},$$

which is exactly **Snell's law** in optics.



In the picture we have taken c to jump from c_1 to a larger value c_2 at an interface. Snell's law then reads $\sin \theta_1 / c_1 = \sin \theta_2 / c_2$, so $\theta_2 > \theta_1$: the ray bends away from the normal on entering the faster medium. This is the familiar refraction of light on passing from glass into air. More generally, in a medium where c increases with x , the angle θ increases too, and the ray curves – which is exactly the mechanism behind mirages, where hot air near the ground has a higher c .

§23.5 Extensions of the Euler-Lagrange Equations

We are now going to see how the Euler-Lagrange equations can be used in settings that are not just unconstrained functional extremization of a single function of a single variable. In each case the derivation is the same three-step dance – perturb, integrate by parts, apply the fundamental lemma – and only the bookkeeping changes.

§23.5.1 Several Dependent Variables

Suppose that instead of a single function y , we have a vector of them,

$$\mathbf{y}(x) = (y_1(x), \dots, y_n(x)),$$

and we wish to extremize

$$F[\mathbf{y}] = \int_{\alpha}^{\beta} f(x, y_1, \dots, y_n, y'_1, \dots, y'_n) \, dx,$$

over functions with all the y_i fixed at the endpoints. This is the situation for, say, a particle moving in three dimensions, where $\mathbf{y}$ is its position.

The derivation goes through unchanged. Perturb by $\mathbf{y} \mapsto \mathbf{y} + \epsilon \boldsymbol{\eta}$ where $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)$ vanishes at both endpoints, expand, and integrate each term by parts:

$$F[\mathbf{y} + \epsilon \boldsymbol{\eta}] - F[\mathbf{y}] = \epsilon \int_{\alpha}^{\beta} \sum_{i=1}^n \eta_i \left(\frac{\partial f}{\partial y_i} - \frac{d}{dx} \frac{\partial f}{\partial y'_i} \right) dx + O(\epsilon^2).$$

Now here is the one new idea. The η_i are independent of one another, so we are free to take $\eta_j \equiv 0$ for all $j \neq i$, leaving a single term to which the fundamental lemma applies. Doing this for each i in turn gives:

Theorem 23.5.1 (Euler–Lagrange Equations, Several Dependent Variables)

A stationary point of $F[\mathbf{y}]$ satisfies

$$\frac{\partial f}{\partial y_i} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'_i} \right) = 0, \quad i = 1, \dots, n.$$

This is a coupled system of n second-order ODEs. Both first integrals survive: if f has no explicit dependence on some particular y_j , then $\partial f / \partial y'_j$ is constant; and if f has no explicit dependence on x , then

$$f - \sum_{i=1}^n y'_i \frac{\partial f}{\partial y'_i} = \text{constant},$$

by exactly the computation we did for [Proposition 23.4.7](#).

Example 23.5.2 (Geodesics, Parameterized Properly)

Our earlier treatment of geodesics in the plane had a defect: by writing y as a function of x , we quietly assumed that the shortest curve never doubles back. We can avoid this by parameterizing the curve as $\mathbf{r}(t) = (x(t), y(t))$ for $t \in [0, 1]$, with $\mathbf{r}(0) = A$ and $\mathbf{r}(1) = B$, and minimising

$$L[x, y] = \int_0^1 \sqrt{\dot{x}^2 + \dot{y}^2} \, dt.$$

Neither x nor y appears explicitly, so both are cyclic and we get two first integrals:

$$\frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = c, \quad \frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = s.$$

These constants are not independent: squaring and adding gives $c^2 + s^2 = 1$, so we

may write $c = \cos \vartheta$ and $s = \sin \vartheta$. Then both equations reduce to $\dot{x} \sin \vartheta = \dot{y} \cos \vartheta$, which integrates to

$$y \cos \vartheta = x \sin \vartheta + A,$$

a straight line of gradient $\tan \vartheta$. Note how the redundancy in the parameterization showed up as a redundancy in the equations – this is characteristic of problems with a reparameterization symmetry.

§23.5.2 Several Independent Variables

Now we go the other way, and let our unknown be a function of several variables. Suppose $\phi = \phi(x, y, z)$ on a domain $\mathcal{D} \subseteq \mathbb{R}^3$, and consider

$$F[\phi] = \iiint_{\mathcal{D}} f(x, y, z, \phi, \phi_x, \phi_y, \phi_z) \, dx \, dy \, dz,$$

where $\phi_{x_i} = \partial\phi/\partial x_i$, and where ϕ is prescribed on the boundary $\partial\mathcal{D}$.

Perturb by $\phi \mapsto \phi + \epsilon\eta$ with $\eta = 0$ on $\partial\mathcal{D}$. The first-order term is

$$\epsilon \iiint_{\mathcal{D}} \left(\eta \frac{\partial f}{\partial \phi} + \eta_x \frac{\partial f}{\partial \phi_x} + \eta_y \frac{\partial f}{\partial \phi_y} + \eta_z \frac{\partial f}{\partial \phi_z} \right) dV.$$

The role of integration by parts is now played by the divergence theorem. Writing

$$\mathbf{w} = \left(\frac{\partial f}{\partial \phi_x}, \frac{\partial f}{\partial \phi_y}, \frac{\partial f}{\partial \phi_z} \right),$$

the last three terms are $\nabla \eta \cdot \mathbf{w} = \nabla \cdot (\eta \mathbf{w}) - \eta \nabla \cdot \mathbf{w}$, and the divergence term integrates to $\oint_{\partial\mathcal{D}} \eta \mathbf{w} \cdot d\mathbf{S} = 0$ since η vanishes on the boundary. So we are left with

$$\epsilon \iiint_{\mathcal{D}} \eta \left(\frac{\partial f}{\partial \phi} - \nabla \cdot \mathbf{w} \right) dV = 0,$$

and a version of the fundamental lemma in several variables (proved by exactly the same bump-function argument, in a small ball rather than an interval) gives the following.

Theorem 23.5.3 (Euler–Lagrange Equation, Several Independent Variables)

A stationary point of $F[\phi]$ satisfies

$$\frac{\partial f}{\partial \phi} - \nabla \cdot \left(\frac{\partial f}{\partial \phi_x}, \frac{\partial f}{\partial \phi_y}, \frac{\partial f}{\partial \phi_z} \right) = 0,$$

or in suffix notation with the summation convention,

$$\frac{\partial f}{\partial \phi} - \partial_i \left(\frac{\partial f}{\partial (\partial_i \phi)} \right) = 0.$$

Note that this is now a *partial* differential equation for ϕ . Nothing about the derivation used $n = 3$, and the same statement holds in any number of independent variables.

Example 23.5.4 (Laplace's Equation)

Consider the ‘energy’ functional

$$F[\phi] = \iint_{\mathcal{D} \subseteq \mathbb{R}^2} \frac{1}{2} (\phi_x^2 + \phi_y^2) \, dx \, dy = \iint_{\mathcal{D}} \frac{1}{2} |\nabla \phi|^2 \, dA,$$

which is (up to constants) the electrostatic energy of a potential ϕ , or the elastic energy of a stretched membrane with height ϕ . Here $\partial f / \partial \phi = 0$ and $\partial f / \partial \phi_x = \phi_x$, $\partial f / \partial \phi_y = \phi_y$, so the Euler–Lagrange equation is

$$\phi_{xx} + \phi_{yy} = 0,$$

which is Laplace’s equation. So harmonic functions are exactly the stationary points of the Dirichlet energy – a fact known as *Dirichlet’s principle*, and one of the classical bridges between the calculus of variations and PDE theory. The reader will recognise the equation, and the tools for solving it, from the Laplace’s equation chapter of my Methods notes.

§23.5.3 Higher Derivatives

Occasionally our integrand involves higher derivatives of y :

$$F[y] = \int_{\alpha}^{\beta} f(x, y, y', \dots, y^{(n)}) \, dx.$$

Now we need to fix more data at the endpoints – specifically $y, y', \dots, y^{(n-1)}$ – and correspondingly our variations η must satisfy $\eta^{(k)}(\alpha) = \eta^{(k)}(\beta) = 0$ for $k = 0, \dots, n-1$. The first variation is

$$\epsilon \int_{\alpha}^{\beta} \left(\frac{\partial f}{\partial y} \eta + \frac{\partial f}{\partial y'} \eta' + \dots + \frac{\partial f}{\partial y^{(n)}} \eta^{(n)} \right) dx,$$

and we now integrate the $\eta^{(k)}$ term by parts k times. Every boundary term produced involves some $\eta^{(j)}$ with $j \leq n-1$ evaluated at an endpoint, so every one of them vanishes. Each integration by parts contributes a sign, and applying the fundamental lemma gives

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} + \frac{d^2}{dx^2} \frac{\partial f}{\partial y''} - \dots + (-1)^n \frac{d^n}{dx^n} \frac{\partial f}{\partial y^{(n)}} = 0.$$

Example 23.5.5

Let us extremize

$$F[y] = \int_0^1 (y'')^2 \, dx, \quad y(0) = y'(0) = 0, \quad y(1) = 0, \quad y'(1) = 1.$$

Here $\partial f / \partial y = \partial f / \partial y' = 0$ and $\partial f / \partial y'' = 2y''$, so the equation is

$$-\frac{d^2}{dx^2} (2y'') = 0 \implies y''' = 0,$$

so y is a cubic. Imposing $y(0) = y'(0) = 0$ kills the constant and linear terms, so $y = ax^3 + bx^2$; then $y(1) = 0$ gives $a + b = 0$ and $y'(1) = 1$ gives $3a + 2b = 1$, whence

$a = 1, b = -1$:

$$y_0 = x^3 - x^2.$$

In fact we can show that this is an *absolute* minimum, and it is instructive to see how, since the argument does not involve second variations at all. Let η be *any* function (not necessarily small) with $\eta = \eta' = 0$ at both endpoints, and not identically zero. Then

$$F[y_0 + \eta] - F[y_0] = \int_0^1 (\eta'')^2 dx + 2 \int_0^1 y_0'' \eta'' dx.$$

The first integral is strictly positive. For the second, $y_0'' = 6x - 2$, so integrating by parts twice,

$$2 \int_0^1 (6x - 2) \eta'' dx = 2 [(6x - 2) \eta']_0^1 - 12 \int_0^1 \eta' dx = 2 [(6x - 2) \eta']_0^1 - 12 [\eta]_0^1 = 0,$$

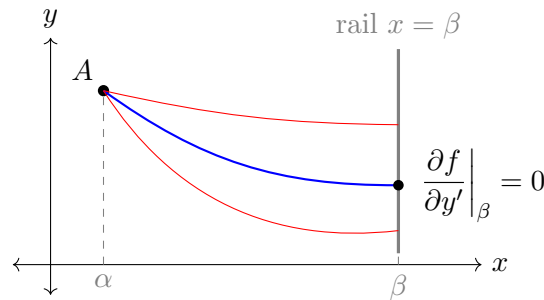
since η and η' vanish at both endpoints. Hence $F[y_0 + \eta] > F[y_0]$ for every admissible $\eta \neq 0$, and y_0 is the unique global minimiser. When such an argument is available it is much less work than computing second variations.

§23.5.4 Variable Endpoints and Natural Boundary Conditions

Throughout the derivation of the Euler–Lagrange equation we quietly used the fixed endpoints twice: once to argue that η must vanish there, and once to discard the boundary term

$$\left[\eta \frac{\partial f}{\partial y'} \right]_{\alpha}^{\beta}.$$

What if one of the endpoints is not fixed? Suppose, for instance, that $y(\alpha) = y_1$ is prescribed, but the value $y(\beta)$ is free – the curve must end somewhere on the vertical line $x = \beta$, but we do not care where. This is the situation of a bead threaded on a rail.



Repeating the derivation, the first variation is

$$\epsilon \int_{\alpha}^{\beta} \eta \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) dx + \epsilon \left[\eta \frac{\partial f}{\partial y'} \right]_{\alpha}^{\beta},$$

and we still have $\eta(\alpha) = 0$, so the boundary term is just $\epsilon \eta(\beta) \partial f / \partial y'|_{\beta}$.

Now we argue in two stages. First restrict to those η which *do* vanish at β ; for these the boundary term drops out and we conclude as before that y satisfies the Euler–Lagrange equation. But then, for a general η , the integral term vanishes automatically, and we are left with

$$\eta(\beta) \frac{\partial f}{\partial y'} \Big|_{x=\beta} = 0.$$

Since $\eta(\beta)$ is now arbitrary, we must have

Theorem 23.5.6 (Natural Boundary Condition)

If y extremizes $F[y] = \int_{\alpha}^{\beta} f(x, y, y') \, dx$ with $y(\alpha)$ fixed and $y(\beta)$ free, then y satisfies the Euler–Lagrange equation together with the **natural boundary condition**

$$\left. \frac{\partial f}{\partial y'} \right|_{x=\beta} = 0.$$

The name is apt: we did not impose this condition, the variational problem produced it for us. Notice that the count works out – we lost one boundary condition by freeing the endpoint, and gained exactly one back.

Example 23.5.7

Consider the shortest path from a fixed point $A = (0, y_1)$ to the line $x = \beta$. Here $f = \sqrt{1 + (y')^2}$, and we already know the Euler–Lagrange equation forces a straight line. The natural boundary condition is

$$\left. \frac{y'}{\sqrt{1 + (y')^2}} \right|_{\beta} = 0 \implies y'(\beta) = 0.$$

Since y is a straight line, $y' \equiv 0$: the shortest path is horizontal, meeting the rail at right angles. This is obvious geometrically, which is a good sanity check on the machinery.

Example 23.5.8

Here is a less obvious one. Consider

$$F[y] = \int_0^1 ((y')^2 + y^2) \, dx, \quad y(0) = 1, \quad y(1) \text{ free}.$$

As we computed earlier, the Euler–Lagrange equation is $y'' = y$, so $y = A \sinh x + B \cosh x$, and $y(0) = 1$ gives $B = 1$. The natural boundary condition is

$$2y'|_{x=1} = 0 \implies y'(1) = 0,$$

so $A \cosh 1 + \sinh 1 = 0$, giving $A = -\tanh 1$ and

$$y = \cosh x - \tanh 1 \sinh x = \frac{\cosh(1 - x)}{\cosh 1}.$$

Notice that the free endpoint has arranged itself so that the curve arrives with zero gradient. Had we instead fixed $y(1) = 0$, we would have obtained a quite different function.

If *both* endpoints are free, the argument runs the same way and we obtain two natural conditions, $\partial f / \partial y'|_{\alpha} = \partial f / \partial y'|_{\beta} = 0$. More generally, a mixture is possible: one endpoint fixed and one free, or a ‘spring’ term added to the functional at the endpoint, in which case the natural condition becomes a Robin-type condition rather than a Neumann one. The general principle is simply that the boundary term in the first variation must

vanish, however we arrange it.

More generally, if the right-hand endpoint is required only to lie on some curve $y = h(x)$, one finds after a slightly longer calculation the **transversality condition**

$$\left[f + (h' - y') \frac{\partial f}{\partial y'} \right] \Big|_{x=\beta} = 0.$$

For the arclength integrand this reduces to $1 + h'y' = 0$, that is, the extremal must meet the curve $y = h(x)$ orthogonally – again what one would guess.

Remark. There is a third possibility, which we have already used without comment: the curve might have *no* boundary at all, as for a closed curve. Then there is no boundary term to worry about and no boundary conditions to impose. This is what happens in the isoperimetric problem below.

§23.5.5 Euler-Lagrange Equations with Constraints

Suppose we wanted to extremize a functional $F[y]$ with

$$F[y] = \int_{\alpha}^{\beta} f(x, y, y') \, dx,$$

subject to the constraint functional

$$G[y] = \int_{\alpha}^{\beta} g(x, y, y') \, dx = k,$$

for a constant k . Such a problem is called **isoperimetric**, for historical reasons which will become clear in a moment. We are going to solve it by generalizing the method of Lagrange multipliers that we introduced in [subsection 23.2.2](#): we extremize the functional

$$\Phi[y, \lambda] = F[y] - \lambda (G[y] - k)$$

over both y and the number λ . Varying with respect to λ recovers the constraint $G[y] = k$, and varying with respect to y amounts to replacing f by $f - \lambda g$ in the Euler-Lagrange equations, giving

$$\frac{\partial}{\partial y} (f - \lambda g) - \frac{d}{dx} \left(\frac{\partial}{\partial y'} (f - \lambda g) \right) = 0.$$

As in finite dimensions, we now have one extra unknown (λ) and one extra equation (the constraint), so the count is right. The justification is the same as before: the constraint restricts us to a ‘surface’ in the space of functions, and stationarity along that surface says the functional derivative of F is proportional to that of G .

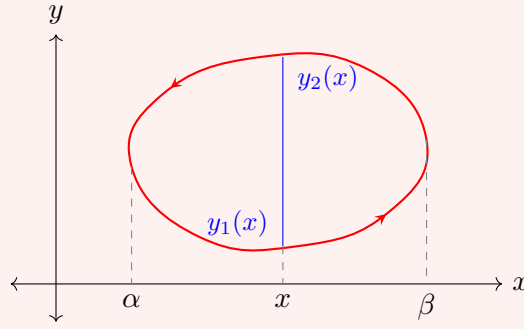
Example 23.5.9 (Isoperimetric / Dido’s Problem)

What simple closed plane curve of fixed length L maximizes the area enclosed by this curve?

The name comes from the legend of Dido, who on founding Carthage was promised as much land as she could enclose with the hide of an ox – and, cutting the hide into a long thin strip, was faced with exactly this problem.

We can immediately see that such a curve has to be convex: if it were not, reflecting a non-convex arc in the chord joining its endpoints would keep the perimeter the same while increasing the area. We set the problem up in Cartesian coordinates as

follows.



On the x -axis, we have minimal and maximal values α and β , and a simple closed curve C with the given orientation. The x coordinate monotonically increases from $\alpha \rightarrow \beta$ along the lower arc, and then decreases from $\beta \rightarrow \alpha$ along the upper arc.

Given $x \in (\alpha, \beta)$, there are two points of the curve above x , at heights $y_1(x) < y_2(x)$, and from this we can write down the area as

$$A[y] = \int_{\alpha}^{\beta} (y_2(x) - y_1(x)) \, dx = - \oint_C y \, dx,$$

the sign being fixed by the orientation, and our perimeter constraint as

$$L[y] = \oint_C d\ell = \oint_C \sqrt{1 + (y')^2} \, dx = L$$

being constant.

We now want to extremize $A - \lambda L$, and so we define

$$h = y - \lambda \sqrt{1 + (y')^2},$$

so that we can then try to solve the Euler-Lagrange equations (not worrying about the boundary term in the derivation of the Euler-Lagrange equations, as C has no boundary).

Since this has no dependence on x , we can use one of our first integrals. We know that $\partial h / \partial x = 0$, and also that

$$\begin{aligned} k &= h - y' \frac{\partial h}{\partial y'} \\ &= y - \lambda \sqrt{1 + (y')^2} + y' \lambda \frac{y'}{\sqrt{1 + (y')^2}} \\ &= y - \frac{\lambda}{\sqrt{1 + (y')^2}}, \end{aligned}$$

for a constant k . Thus

$$(y')^2 = \frac{\lambda^2}{(y - k)^2} - 1,$$

which is separable. Writing $u = y - k$ we get $dx = \pm u \, du / \sqrt{\lambda^2 - u^2}$, and integrating gives $x - x_0 = \mp \sqrt{\lambda^2 - u^2}$. Squaring and rearranging, the solution is

$$(x - x_0)^2 + (y - y_0)^2 = \lambda^2,$$

which is a circle of radius λ . Then $2\pi\lambda = L$ gives $\lambda = L/2\pi$. This is a necessary (but not sufficient) condition for our solution, so the circle is the only candidate.

Notice that the Lagrange multiplier is here exactly the radius of the optimal circle, and hence that $dA/dL = \lambda$ – which is easily checked directly, since $A = L^2/4\pi$. This is our general interpretation of the multiplier as a rate of change of the optimal value with the constraint, in action.

§23.5.6 Pointwise Constraints

The constraint in an isoperimetric problem is a single number. Sometimes instead we want a constraint to hold at *every* point of the curve – for instance, that the curve lie on a given surface. In that case a single multiplier is not enough; we need a multiplier *function* $\lambda(t)$, one for each point.

Example 23.5.10 (Geodesics on a General Surface)

Let $\Sigma = \{\mathbf{x} \in \mathbb{R}^3 : g(\mathbf{x}) = 0\}$ be a surface, and let $A, B \in \Sigma$. To find the geodesics between them, parameterize a path by $t \in [0, 1]$ with $\mathbf{x}(0) = A$, $\mathbf{x}(1) = B$, and extremize

$$\Phi[\mathbf{x}, \lambda] = \int_0^1 \left\{ \sqrt{\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2} - \lambda(t)g(\mathbf{x}) \right\} dt.$$

Treating λ as a fourth dependent variable, the Euler–Lagrange equation for λ is

$$\frac{d}{dt} \frac{\partial h}{\partial \dot{\lambda}} - \frac{\partial h}{\partial \lambda} = 0,$$

and since h does not involve $\dot{\lambda}$ at all, this just says $g(\mathbf{x}) = 0$ for all t – the constraint, as required. Varying the x_i gives

$$\frac{d}{dt} \left(\frac{\dot{x}_i}{\sqrt{\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2}} \right) + \lambda \frac{\partial g}{\partial x_i} = 0.$$

If we parameterize by arclength the square root is 1, and the equation says $\ddot{\mathbf{x}} = -\lambda \nabla g$: the acceleration of a geodesic is always *normal* to the surface. In other words, a geodesic is a curve which does not turn within the surface – it only bends as much as the surface forces it to. This is a very clean characterisation, and one which the reader will meet again in Differential Geometry.

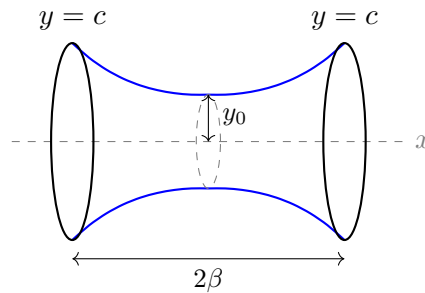
Of course, we could instead have solved the constraint $g = 0$ by parameterizing Σ explicitly, as we did for the sphere. Which approach is easier depends entirely on the surface.

§23.6 Soap Films and Hanging Chains

We now pause the general theory to work through two classical problems in detail. Both are genuinely physical, both are solved by the hyperbolic cosine, and both have subtleties that the Euler–Lagrange equation alone does not reveal.

§23.6.1 Minimal Surfaces of Revolution

Dip two coaxial circular wire rings into soapy water and pull them apart. The soap film that forms is a surface of revolution, and since surface tension makes the film minimise its area, it is a solution of a variational problem.



Take the axis of revolution to be the x -axis, and let the film be the surface obtained by rotating a curve $y = y(x) > 0$ about it, with the rings at $x = \pm\beta$ both of radius c . The area of the surface of revolution is

$$A[y] = \int_{-\beta}^{\beta} 2\pi y \sqrt{1 + (y')^2} \, dx,$$

subject to $y(\pm\beta) = c$. Discarding the 2π , the integrand $f = y\sqrt{1 + (y')^2}$ has no explicit x -dependence, so the Beltrami identity applies:

$$y\sqrt{1 + (y')^2} - y' \frac{yy'}{\sqrt{1 + (y')^2}} = \frac{y}{\sqrt{1 + (y')^2}} = y_0$$

for some constant $y_0 > 0$. Rearranging,

$$y' = \pm \frac{\sqrt{y^2 - y_0^2}}{y_0},$$

which separates. Substituting $y = y_0 \cosh u$ gives $dx = y_0 du$, so $u = (x - x_1)/y_0$ and

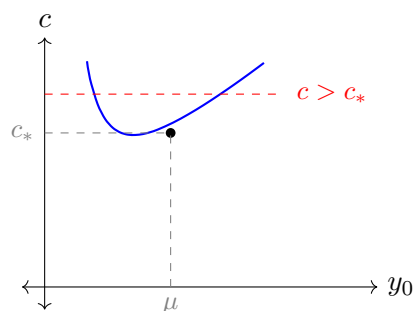
$$y = y_0 \cosh \left(\frac{x - x_1}{y_0} \right).$$

This curve is a **catenary**, and the surface it sweeps out is a **catenoid**. By the symmetry of our boundary conditions, $x_1 = 0$, and y_0 is the radius of the narrowest part of the film – its ‘waist’.

All that remains is to determine y_0 , from

$$c = y_0 \cosh \left(\frac{\beta}{y_0} \right).$$

This is where it gets interesting. Let us scale so that $\beta = 1$, and ask how many solutions y_0 this equation has for given c . Consider the function $\Psi(y_0) = y_0 \cosh(1/y_0)$: as $y_0 \rightarrow 0^+$ it blows up (since $\cosh$ beats the linear factor), and as $y_0 \rightarrow \infty$ it also blows up (since $\cosh \rightarrow 1$ and the linear factor wins). So Ψ has a minimum somewhere in between.



Differentiating,

$$\Psi'(y_0) = \cosh \frac{1}{y_0} - \frac{1}{y_0} \sinh \frac{1}{y_0},$$

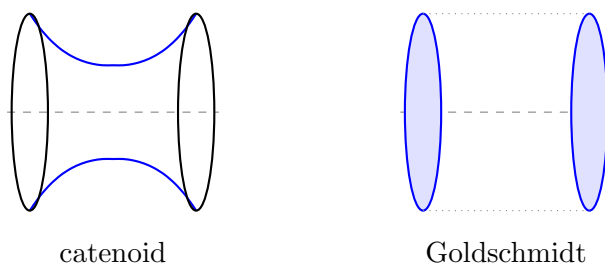
which vanishes when $\coth(1/y_0) = 1/y_0$. Numerically this gives $y_0 = \mu \approx 0.833$, and $c_* = \Psi(\mu) \approx 1.509$. Hence:

- If $c > c_*$, there are *two* catenoids meeting the boundary conditions: a fat one with $y_0 > \mu$, and a thin one with $y_0 < \mu$.
- If $c = c_*$, there is exactly one.
- If $c < c_*$ – that is, if the rings are pulled too far apart relative to their size – there is *no* solution at all.

This is our promised failure of existence and uniqueness for the boundary value problem, and it is physically real: pull the rings far enough apart and the soap film snaps.

§23.6.2 The Goldschmidt Solution

But what happens then? Physically, the film collapses into two flat discs, one spanning each ring. This is called the **Goldschmidt solution**, and it is a perfectly good competitor of area $2\pi c^2$ – but it is not a smooth surface of revolution, so the Euler–Lagrange equation never saw it.



Comparing the areas is a slightly delicate calculation, but the upshot is that as the rings are separated, the fat catenoid's area increases until, at a separation *strictly less* than the critical one, it exceeds $2\pi c^2$. Past that point the Goldschmidt solution is the true global minimum even though a catenoid still exists.

There are two morals here, both worth remembering.

- (i) The Euler–Lagrange equation only finds *smooth* stationary points. Minima can occur at the boundary of the space of admissible functions – just as, for functions on $\mathbb{R}^n$, minima can occur at the boundary of the domain.

- (ii) Of the two catenoids, the thin one ($y_0 < \mu$) turns out to be a saddle rather than a minimum: it is unstable, and a real soap film will never adopt it. We are not yet in a position to prove this, but we will develop the necessary tools in [section 23.9](#).

§23.6.3 The Hanging Chain

A closely related problem: what shape does a heavy flexible chain of fixed length L take when suspended from two points? The chain will hang so as to minimise its gravitational potential energy, subject to its length being fixed – so this is an isoperimetric problem, and we need a Lagrange multiplier.

If the chain has uniform mass per unit length ρ and hangs as $y = y(x)$ for $x \in [-\beta, \beta]$, its potential energy is

$$V[y] = \int_{-\beta}^{\beta} \rho g y \sqrt{1 + (y')^2} dx,$$

and the length constraint is

$$L[y] = \int_{-\beta}^{\beta} \sqrt{1 + (y')^2} dx = L.$$

So we extremize with the integrand

$$h = (\rho g y - \lambda) \sqrt{1 + (y')^2}.$$

But this is exactly the soap film integrand with y replaced by $y - \lambda/\rho g$! So the calculation is identical, and the answer is a catenary,

$$y = \frac{\lambda}{\rho g} + a \cosh\left(\frac{x - x_1}{a}\right),$$

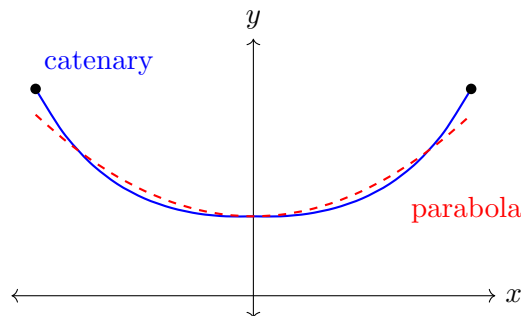
with a and λ fixed by the endpoint conditions and the length constraint. Here the multiplier λ has a direct physical meaning: it is (up to sign) the horizontal tension in the chain.

That the soap film and the hanging chain give the same curve is not a coincidence – both minimise $\int y d\ell$ – but it is a pleasing one, and it is the reason the curve is called a catenary (from *catena*, a chain).

It is a common misconception that a hanging chain is a parabola. It is not, though the two are close near the bottom: expanding,

$$a \cosh \frac{x}{a} = a + \frac{x^2}{2a} + \frac{x^4}{24a^3} + \cdots,$$

so they agree to second order and diverge thereafter. Galileo believed the curve was a parabola; the correct answer was found by Huygens, Leibniz and Johann Bernoulli in 1691.



§23.6.4 Plateau's Problem and the Minimal Surface Equation

Our soap film was a surface of revolution, which reduced the problem to one dependent variable. What if we dip an arbitrary closed wire loop into soapy water? The film that forms spans the loop and minimises area, and finding it is called **Plateau's problem**. This is a genuine multi-variable calculus of variations problem, and it is a good test of the machinery from the previous section.

Suppose the surface can be written as a graph $z = \phi(x, y)$ over a domain $\mathcal{D}$ in the plane, with ϕ prescribed on $\partial\mathcal{D}$ by the wire. Since $dz = \phi_x dx + \phi_y dy$, the line element on the surface is

$$ds^2 = (1 + \phi_x^2) dx^2 + 2\phi_x\phi_y dx dy + (1 + \phi_y^2) dy^2 = g_{ij} dx^i dx^j,$$

where

$$g = \begin{pmatrix} 1 + \phi_x^2 & \phi_x\phi_y \\ \phi_x\phi_y & 1 + \phi_y^2 \end{pmatrix}$$

is the **first fundamental form** of the surface. The area element is $dA = \sqrt{\det g} dx dy$, and $\det g = 1 + \phi_x^2 + \phi_y^2$ (the cross terms cancel), so we wish to extremize

$$A[\phi] = \iint_{\mathcal{D}} \sqrt{1 + \phi_x^2 + \phi_y^2} dx dy.$$

Writing h for the integrand, we have $\partial h / \partial \phi = 0$ and

$$\frac{\partial h}{\partial \phi_x} = \frac{\phi_x}{\sqrt{1 + \phi_x^2 + \phi_y^2}}, \quad \frac{\partial h}{\partial \phi_y} = \frac{\phi_y}{\sqrt{1 + \phi_x^2 + \phi_y^2}},$$

so the Euler–Lagrange equation is

$$\partial_x \left(\frac{\phi_x}{\sqrt{1 + \phi_x^2 + \phi_y^2}} \right) + \partial_y \left(\frac{\phi_y}{\sqrt{1 + \phi_x^2 + \phi_y^2}} \right) = 0.$$

Expanding the derivatives and clearing the denominator gives the **minimal surface equation**

$$(1 + \phi_y^2) \phi_{xx} - 2\phi_x\phi_y\phi_{xy} + (1 + \phi_x^2) \phi_{yy} = 0.$$

This is a nonlinear second-order PDE, and it is not one we can solve in general. Two observations, though.

First, if the surface is nearly flat, so that $|\nabla\phi| \ll 1$, we may drop the quadratic terms and are left with

$$\phi_{xx} + \phi_{yy} = 0,$$

Laplace's equation again. This is the same linearisation that turns the exact equations of a stretched membrane into the wave equation studied in the Methods course.

Second, we can solve it exactly in the circularly symmetric case, and recover the catenoid. Put $z = \phi(r)$ with $r = \sqrt{x^2 + y^2}$. Then $\phi_x = z'x/r$ and $\phi_y = z'y/r$, and computing the second derivatives and substituting reduces the minimal surface equation to

$$rz'' + z' + (z')^3 = 0.$$

Setting $w = z'$ and multiplying through by w , this becomes

$$\frac{1}{2}r \frac{d(w^2)}{dr} + w^2 + w^4 = 0,$$

and with $u = w^2$ this is separable:

$$\frac{du}{u(1+u)} = -\frac{2 dr}{r}.$$

Integrating with partial fractions gives $u/(1+u) = r_0^2/r^2$ for a constant r_0 , so $w^2 = r_0^2/(r^2 - r_0^2)$ and

$$\frac{dz}{dr} = \frac{r_0}{\sqrt{r^2 - r_0^2}} \implies z - z_0 = r_0 \operatorname{arcosh}\left(\frac{r}{r_0}\right).$$

Inverting,

$$r = r_0 \cosh\left(\frac{z - z_0}{r_0}\right),$$

which is exactly the catenoid we found in [subsection 23.6.1](#) – reassuringly, since the two calculations describe the same soap film. It is always worth doing a problem twice when the two routes are this different.

§23.7 Hamilton’s Principle and Lagrangian Mechanics

We now turn to the most celebrated application of everything we have built: the reformulation of classical mechanics as a variational principle. The payoff is enormous. Newton’s laws are tied to Cartesian coordinates and inertial frames, and become painful the moment we have constraints (a bead on a wire, a pendulum on a pendulum). The variational formulation is coordinate-free, handles constraints gracefully, and – through Noether’s theorem – hands us the conservation laws for free.

§23.7.1 The Principle of Least Action

Definition 23.7.1 (Lagrangian and Action)

For a particle of mass m moving in $\mathbb{R}^3$ under a potential $V(\mathbf{x})$, with kinetic energy $T = \frac{1}{2}m|\dot{\mathbf{x}}|^2$, the **Lagrangian** is

$$L(\mathbf{x}, \dot{\mathbf{x}}, t) = T - V,$$

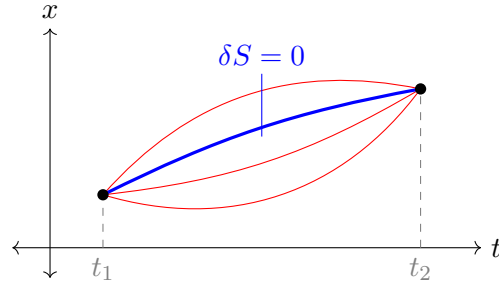
and the **action** of a path $\mathbf{x}(t)$ between times t_1 and t_2 is

$$S[\mathbf{x}] = \int_{t_1}^{t_2} L(\mathbf{x}, \dot{\mathbf{x}}, t) dt.$$

Law 23.7.2 (Hamilton’s Principle)

The path actually taken by the particle between two fixed configurations at times t_1 and t_2 is a stationary point of the action.

The difference $T - V$ looks strange – one would more naturally write down $T + V$, the energy. But it is $T - V$ that works, and we can check this immediately.



Theorem 23.7.3

Hamilton's principle is equivalent to Newton's second law $m\ddot{\mathbf{x}} = -\nabla V$.

Proof. This is the Euler–Lagrange equations for several dependent variables from subsection 23.5.1, with x replaced by t and y_i by x_i . We have

$$L = \frac{1}{2}m(\dot{x}_1^2 + \dot{x}_2^2 + \dot{x}_3^2) - V(\mathbf{x}),$$

so $\partial L/\partial \dot{x}_i = m\dot{x}_i$ and $\partial L/\partial x_i = -\partial V/\partial x_i$. The Euler–Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_i} \right) - \frac{\partial L}{\partial x_i} = 0$$

therefore read $m\ddot{x}_i = -\partial V/\partial x_i$, which is Newton's second law. $\square$

So the mysterious minus sign is exactly what is needed to produce ‘mass times acceleration equals force’. Two immediate remarks. First, ‘least action’ is a misnomer – the action is stationary, and is often a saddle rather than a minimum. Second, the Euler–Lagrange equations retain their form under *any* change of coordinates, since our derivation never used what the coordinates meant. This is what makes the formulation so powerful.

§23.7.2 Generalised Coordinates

Suppose we have N particles in $\mathbb{R}^3$. Rather than tracking N vectors, we regard the state of the whole system as a single point in $\mathbb{R}^{3N}$, called the **configuration space**. If the system is subject to constraints – rigid rods, beads on wires, and so on – the accessible configurations form a lower-dimensional surface inside $\mathbb{R}^{3N}$, and we may choose any convenient coordinates $q_1, \dots, q_n$ on it. These are the **generalised coordinates**, and n is the number of degrees of freedom.

The great advantage is that the constraint forces disappear entirely: we simply write the kinetic and potential energies in terms of the q_i , form

$$L = L(q_i, \dot{q}_i, t),$$

and write down the Euler–Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0.$$

The quantity $p_i = \partial L/\partial \dot{q}_i$ is called the **generalised** (or **conjugate**) **momentum** associated with q_i ; for a Cartesian coordinate it is the ordinary momentum $m\dot{x}$.

Example 23.7.4 (The Simple Pendulum)

A mass m swings on a light rod of length ℓ . There is one degree of freedom, the angle θ from the downward vertical. The speed of the mass is $\ell\dot{\theta}$ and its height above the lowest point is $\ell(1 - \cos \theta)$, so

$$L = \frac{1}{2}m\ell^2\dot{\theta}^2 - mg\ell(1 - \cos \theta).$$

Then $\partial L/\partial \dot{\theta} = m\ell^2\dot{\theta}$ and $\partial L/\partial \theta = -mg\ell \sin \theta$, so the equation of motion is

$$m\ell^2\ddot{\theta} = -mg\ell \sin \theta \implies \ddot{\theta} = -\frac{g}{\ell} \sin \theta,$$

in two lines and with no mention of the tension in the rod. Doing this with Newton's laws requires resolving forces along and perpendicular to the rod, and is distinctly more work.

Example 23.7.5 (Motion in a Central Force)

Consider a particle moving in a plane under a central potential $V(r)$. Using plane polar coordinates (r, θ) , the velocity has components $\dot{r}$ and $r\dot{\theta}$, so

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r).$$

Now θ is cyclic – it does not appear in L , only $\dot{\theta}$ does – so we immediately get a first integral,

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} = \text{constant},$$

which is conservation of angular momentum. Also L has no explicit t -dependence, so the Beltrami identity applies (in its several-variable form):

$$\begin{aligned} \text{constant} &= \dot{r} \frac{\partial L}{\partial \dot{r}} + \dot{\theta} \frac{\partial L}{\partial \dot{\theta}} - L \\ &= m\dot{r}^2 + mr^2\dot{\theta}^2 - \frac{1}{2}m\dot{r}^2 - \frac{1}{2}mr^2\dot{\theta}^2 + V(r) \\ &= \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + V(r), \end{aligned}$$

which is $T + V$: conservation of energy. So both conservation laws for the Kepler problem drop out of the structure of L without solving anything.

Example 23.7.6 (The Double Pendulum)

To see how little the method cares about complexity, consider a pendulum of length ℓ_1 carrying a mass m_1 , from which hangs a second pendulum of length ℓ_2 carrying m_2 . Let θ_1, θ_2 be the angles of the two rods from the vertical; these are our generalised coordinates, and there are two degrees of freedom.

The position of the first mass is $\ell_1(\sin \theta_1, -\cos \theta_1)$ and of the second is that plus

$\ell_2(\sin \theta_2, -\cos \theta_2)$. Differentiating and squaring,

$$T = \frac{1}{2}(m_1 + m_2)\ell_1^2\dot{\theta}_1^2 + \frac{1}{2}m_2\ell_2^2\dot{\theta}_2^2 + m_2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2),$$

while the potential energy is

$$V = -(m_1 + m_2)g\ell_1\cos\theta_1 - m_2g\ell_2\cos\theta_2.$$

Writing $L = T - V$ and grinding out $\frac{d}{dt}(\partial L/\partial\dot{\theta}_i) = \partial L/\partial\theta_i$ gives the (famously chaotic) equations of motion. The point is not the equations, which are unpleasant, but that we obtained them by writing down two energies and differentiating. There is no need to think about the forces in the rods at all, which is what makes a Newtonian treatment of this system so disagreeable.

Note also that L has no explicit t -dependence, so energy is conserved; but no coordinate is cyclic, so there is no second conserved quantity. Two degrees of freedom and only one conservation law is, roughly speaking, why the system is chaotic.

That last example is suggestive. A missing coordinate gave conservation of angular momentum; a missing time gave conservation of energy. There is a general principle lurking here, and it is due to Emmy Noether.

§23.7.3 Noether's Theorem

The key idea is to replace ‘the Lagrangian does not depend on θ ’ – which is a statement about a particular coordinate system – with the coordinate-free statement ‘the Lagrangian is invariant under rotations’. Symmetries, rather than missing variables.

Definition 23.7.7 (Continuous Symmetry)

Consider a functional $F[\mathbf{y}] = \int_{\alpha}^{\beta} f(y_i, y'_i, x) \, dx$. A **one-parameter family of transformations** is a smooth map

$$y_i(x) \mapsto Y_i(x, s), \quad Y_i(x, 0) = y_i(x),$$

for s in some interval around 0. It is a **continuous symmetry** of f if

$$\frac{\partial}{\partial s} f(Y_i(x, s), Y'_i(x, s), x) = 0$$

for all s , where $' = \partial/\partial x$ as usual.

Theorem 23.7.8 (Noether's Theorem)

Let $Y_i(x, s)$ be a continuous symmetry of f . Then, along any solution of the Euler–Lagrange equations,

$$\sum_{i=1}^n \frac{\partial f}{\partial y'_i} \frac{\partial Y_i}{\partial s} \bigg|_{s=0}$$

is constant, i.e. it is a first integral.

Proof. The proof is three lines of chain rule, and is worth following carefully. Using the summation convention, and evaluating everything at $s = 0$,

$$\begin{aligned}
 0 &= \left. \frac{\partial f}{\partial s} \right|_{s=0} \\
 &= \left. \frac{\partial f}{\partial y_i} \frac{\partial Y_i}{\partial s} \right|_{s=0} + \left. \frac{\partial f}{\partial y'_i} \frac{\partial Y'_i}{\partial s} \right|_{s=0} \\
 &= \left[\frac{d}{dx} \left(\frac{\partial f}{\partial y'_i} \right) \frac{\partial Y_i}{\partial s} + \frac{\partial f}{\partial y'_i} \frac{d}{dx} \left(\frac{\partial Y_i}{\partial s} \right) \right] \Big|_{s=0} \\
 &= \frac{d}{dx} \left[\frac{\partial f}{\partial y'_i} \frac{\partial Y_i}{\partial s} \right] \Big|_{s=0}.
 \end{aligned}$$

In the third line we used the Euler–Lagrange equation to replace $\partial f/\partial y_i$, and swapped the order of the mixed partial derivatives $\partial_s \partial_x = \partial_x \partial_s$; the last line is the product rule read backwards. Integrating, the bracket is constant. $\square$

The statement is short but the content is remarkable: *every continuous symmetry of a physical system yields a conserved quantity*. Let us see it work.

Example 23.7.9 (Conservation of Momentum)

Let $\mathbf{y} = (y, z)$ and

$$f = \frac{1}{2}(y')^2 + \frac{1}{2}(z')^2 - V(y - z),$$

which describes two particles interacting through a potential depending only on their separation. Consider translating both by the same amount:

$$Y = y + s, \quad Z = z + s.$$

Then $Y' = y'$, $Z' = z'$, and $Y - Z = y - z$, so f is unchanged and this is a symmetry. Since $\partial Y/\partial s = \partial Z/\partial s = 1$, Noether's theorem gives the first integral

$$\frac{\partial f}{\partial y'} \cdot 1 + \frac{\partial f}{\partial z'} \cdot 1 = y' + z' = \text{constant},$$

which is conservation of total momentum. The physical statement is that space is homogeneous – it does not matter *where* the experiment is performed.

Example 23.7.10 (Conservation of Angular Momentum)

Take the central force Lagrangian $L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r)$ and the rotation

$$R = r, \quad \Theta = \theta + s.$$

This leaves L unchanged since L does not involve θ , so it is a symmetry, and Noether's theorem gives

$$\frac{\partial L}{\partial \dot{r}} \cdot 0 + \frac{\partial L}{\partial \dot{\theta}} \cdot 1 = mr^2\dot{\theta} = \text{constant},$$

as we found before. The physical statement is that space is isotropic.

Remark. Conservation of energy also comes from a symmetry – invariance under time translation – but this is a symmetry of the *independent* variable rather than the dependent ones, and so is not literally covered by the version of Noether’s theorem we proved. In our framework it appears instead as the Beltrami identity, which is the same statement in disguise. A more general version of Noether’s theorem handles both at once.

§23.7.4 The Hamiltonian

We saw in Proposition 23.4.7 that when L has no explicit time dependence, the quantity

$$H = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L = \sum_i p_i \dot{q}_i - L$$

is conserved. But look at the right hand side: this is exactly the Legendre transform of L with respect to $\dot{\mathbf{q}}$.

Definition 23.7.11 (Hamiltonian)

The **Hamiltonian** is the Legendre transform of the Lagrangian in the velocities,

$$H(\mathbf{q}, \mathbf{p}, t) = \sup_{\mathbf{v}} (\mathbf{p} \cdot \mathbf{v} - L(\mathbf{q}, \mathbf{v}, t)) = \mathbf{p} \cdot \dot{\mathbf{q}} - L,$$

where $\dot{\mathbf{q}}$ is expressed in terms of $\mathbf{p}$ by inverting $p_i = \partial L / \partial \dot{q}_i$.

Convexity of L in $\dot{\mathbf{q}}$ – which for $T = \frac{1}{2}m|\dot{\mathbf{q}}|^2$ is automatic – is exactly what guarantees this inversion is possible, and this is why we spent time on the Legendre transform earlier.

Example 23.7.12

Take $L = \frac{1}{2}m|\dot{\mathbf{q}}|^2 - V(\mathbf{q})$. Then $\mathbf{p} = \partial L / \partial \dot{\mathbf{q}} = m\dot{\mathbf{q}}$, so $\dot{\mathbf{q}} = \mathbf{p}/m$, and

$$\begin{aligned} H &= \mathbf{p} \cdot \frac{\mathbf{p}}{m} - \left(\frac{1}{2}m \frac{|\mathbf{p}|^2}{m^2} - V(\mathbf{q}) \right) \\ &= \frac{|\mathbf{p}|^2}{2m} + V(\mathbf{q}) = T + V. \end{aligned}$$

So the Hamiltonian is the total energy – but expressed in terms of momentum rather than velocity, which is the whole point.

The change of variables from $(\mathbf{q}, \dot{\mathbf{q}})$ to $(\mathbf{q}, \mathbf{p})$ does more than tidy the energy; it transforms the equations of motion into a strikingly symmetric first-order system.

Theorem 23.7.13 (Hamilton’s Equations)

If L satisfies the Euler–Lagrange equations, then

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.$$

Proof. The slick proof is to compute the differential of $H = p_i \dot{q}_i - L$ in two ways. Regarding H as a function of $(\mathbf{q}, \mathbf{p}, t)$,

$$dH = \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt.$$

On the other hand, differentiating $p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t)$ directly,

$$dH = p_i d\dot{q}_i + \dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i - \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i - \frac{\partial L}{\partial t} dt.$$

The first and fourth terms cancel, since $p_i = \partial L / \partial \dot{q}_i$ – this cancellation is the magic of the Legendre transform, and it is why H genuinely is a function of $\mathbf{p}$ and not of $\dot{\mathbf{q}}$. For the third term, the Euler–Lagrange equation gives

$$\frac{\partial L}{\partial q_i} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = \dot{p}_i.$$

So

$$dH = \dot{q}_i dp_i - \dot{p}_i dq_i - \frac{\partial L}{\partial t} dt,$$

and comparing coefficients with the first expression gives the result. $\square$

Example 23.7.14 (The Harmonic Oscillator in Phase Space)

Take $L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}m\omega^2 q^2$. Then $p = m\dot{q}$, and

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2.$$

Hamilton's equations are

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial q} = -m\omega^2 q,$$

which combine to give $\ddot{q} = -\omega^2 q$ as expected. But the first-order form has an advantage: since H is conserved, every trajectory lies on a level set of H , that is, on an ellipse

$$\frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E$$

in the (q, p) plane. We can therefore read off the qualitative behaviour of every solution – they all go round and round – without solving anything at all. This is the typical use of the Hamiltonian formulation: it turns dynamics into geometry in phase space.

We have traded n second-order equations for $2n$ first-order ones, which carry the same amount of information (the same number of constants of integration). The space of pairs $(\mathbf{q}, \mathbf{p})$ is called **phase space**, and solutions of Hamilton's equations are *trajectories* in it. The symmetry between q and p in these equations – broken only by a minus sign – is the starting point of a very rich theory, and it is the form of mechanics which passes most directly into quantum mechanics.

Remark. Hamilton's equations can also be obtained variationally, which is rather sat-

isfying. Consider the functional on phase space

$$S[\mathbf{q}, \mathbf{p}] = \int_{t_1}^{t_2} (p_i \dot{q}_i - H(\mathbf{q}, \mathbf{p}, t)) \, dt,$$

and treat $\mathbf{q}$ and $\mathbf{p}$ as independent. Since the integrand contains no $\dot{p}_i$, the Euler–Lagrange equation for p_i is simply $\partial f / \partial p_i = 0$, giving $\dot{q}_i = \partial H / \partial p_i$. The equation for q_i gives $-\partial H / \partial q_i - dp_i / dt = 0$, which is the second of Hamilton’s equations.

§23.8 Eigenvalue Problems and the Rayleigh–Ritz Method

We now return to constrained variational problems, and to an application which is of great practical importance: estimating eigenvalues. The key observation is that eigenvalues of certain differential operators are the minima of an associated functional – and because minima are easy to bound from above (just guess something!), this gives a genuinely useful computational method.

§23.8.1 The Sturm–Liouville Problem

Example 23.8.1 (The Sturm–Liouville Problem)

Let $\rho(x) > 0$ be some known function for $x \in [\alpha, \beta]$, and let $\sigma(x)$ be another function. We consider the functional

$$F[y] = \int_{\alpha}^{\beta} [\rho \cdot (y')^2 + \sigma \cdot y^2] \, dx,$$

and we want to minimize F subject to the normalisation

$$G[y] = \int_{\alpha}^{\beta} y^2 \, dx = 1,$$

and the boundary conditions $y(\alpha) = y(\beta) = 0$.

We define the functional

$$\Phi[y; \lambda] = F[y] - \lambda (G[y] - 1),$$

whose integrand is $h = \rho(y')^2 + \sigma y^2 - \lambda y^2$ (plus a constant, which does not affect the equations). Then

$$\frac{\partial h}{\partial y'} = 2\rho y', \quad \frac{\partial h}{\partial y} = 2\sigma y - 2\lambda y,$$

and the Euler–Lagrange equation $\frac{d}{dx}(\partial h / \partial y') = \partial h / \partial y$ becomes, after dividing by 2,

$$-\frac{d}{dx} \left(\rho \frac{dy}{dx} \right) + \sigma y = \lambda y.$$

Writing $\mathcal{L}$ for the operator on the left, this is

$$\mathcal{L}y = \lambda y,$$

an eigenvalue problem, and $\mathcal{L}$ is the **Sturm–Liouville operator** familiar from the Methods course^a. So the Lagrange multiplier of our constrained problem is precisely the eigenvalue.

If $\rho = 1$ this is the time-independent Schrödinger equation with potential σ , which we will return to shortly.

^aThe reader who wants the full theory – self-adjointness, orthogonality and completeness of the eigenfunctions, and the eigenfunction expansion – is referred to the Sturm–Liouville chapter of my Methods notes. We will use those facts here without proof.

This connection is worth stating as a theorem, because it is the engine of everything that follows.

Theorem 23.8.2 (Eigenvalues as Minima)

Let $\mathcal{L}y = -(\rho y')' + \sigma y$ with $\rho > 0$, acting on functions with $y(\alpha) = y(\beta) = 0$, and let its eigenvalues be $\lambda_1 \leq \lambda_2 \leq \dots$ with eigenfunctions $y_1, y_2, \dots$. Define the **Rayleigh quotient**

$$\mathcal{R}[y] = \frac{F[y]}{G[y]} = \frac{\int_{\alpha}^{\beta} (\rho(y')^2 + \sigma y^2) \, dx}{\int_{\alpha}^{\beta} y^2 \, dx}.$$

Then $\lambda_1 = \min \mathcal{R}[y]$, the minimum being over all admissible $y \neq 0$, and it is attained at $y = y_1$.

Proof. First, note that $\mathcal{R}$ is unaffected by scaling y , so we may as well impose $G[y] = 1$; then $\mathcal{R}[y] = F[y]$.

Suppose y is a stationary point of F subject to $G = 1$, with multiplier λ . Multiplying the Euler–Lagrange equation $-(\rho y')' + \sigma y = \lambda y$ by y and integrating,

$$\int_{\alpha}^{\beta} (-y(\rho y')' + \sigma y^2) \, dx = \lambda \int_{\alpha}^{\beta} y^2 \, dx = \lambda.$$

Integrating the first term by parts,

$$\int_{\alpha}^{\beta} (\rho(y')^2 + \sigma y^2) \, dx - [y\rho y']_{\alpha}^{\beta} = \lambda,$$

the boundary term vanishing since $y(\alpha) = y(\beta) = 0$. So $F[y] = \lambda$: *the value of the functional at a stationary point is the corresponding eigenvalue*. Hence the smallest value of $\mathcal{R}$ over stationary points is the smallest eigenvalue.

It remains to see that the minimum over *all* y is attained at a stationary point, which follows by expanding an arbitrary y in the eigenfunctions. Writing $y = \sum_n c_n y_n$ with the y_n orthonormal (which we may do by the completeness of the eigenfunctions, proved in the Methods course), the same integration by parts gives

$$F[y] = \sum_n \lambda_n c_n^2, \quad G[y] = \sum_n c_n^2,$$

so that

$$\mathcal{R}[y] = \frac{\sum_n \lambda_n c_n^2}{\sum_n c_n^2} \geq \lambda_1,$$

since every $\lambda_n \geq \lambda_1$. Equality holds exactly when $c_n = 0$ for all n with $\lambda_n > \lambda_1$, i.e. when y is a ground state. $\square$

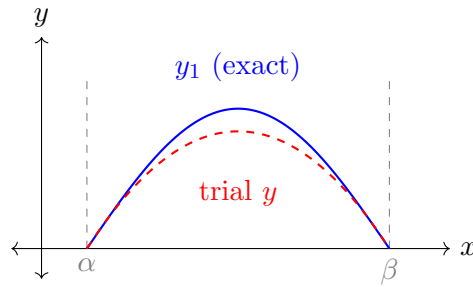
The last display deserves emphasis. It says that $\mathcal{R}[y]$ is a *weighted average* of the eigenvalues, with weights c_n^2 given by how much of each eigenfunction y contains. An average of numbers is at least the smallest of them, and that is the entire content of the variational principle.

§23.8.2 The Rayleigh–Ritz Method

Here is the practical consequence. Suppose we cannot solve $\mathcal{L}y = \lambda y$ exactly, but we would like to know λ_1 . By [Theorem 23.8.2](#), *any* admissible function y we care to write down gives an upper bound

$$\lambda_1 \leq \mathcal{R}[y].$$

So we take a **trial function** y depending on a few free parameters, compute $\mathcal{R}[y]$, and minimise over the parameters. This is the **Rayleigh–Ritz method**. The more parameters we allow, the better the bound – and the bound is always from above, so we always know which way we are wrong.



Example 23.8.3

Let us estimate the lowest eigenvalue of $-y'' = \lambda y$ on $[0, 1]$ with $y(0) = y(1) = 0$. Of course we know the answer is $\lambda_1 = \pi^2 \approx 9.8696$, which makes this a good test.

Take the simplest trial function vanishing at both ends, $y = x(1 - x)$. Then $y' = 1 - 2x$, and

$$\int_0^1 (y')^2 dx = \int_0^1 (1 - 2x)^2 dx = \frac{1}{3}, \quad \int_0^1 y^2 dx = \int_0^1 x^2(1 - x)^2 dx = \frac{1}{30}.$$

So

$$\lambda_1 \leq \frac{1/3}{1/30} = 10.$$

This is within 1.5% of the true value, from a one-line calculation with a function that is not even the right shape. The reason the method works so well is that $\mathcal{R}$ is *stationary* at y_1 , so an error of size ϵ in the trial function produces an error of only $O(\epsilon^2)$ in the estimate.

If we want to do better we can add a parameter. Taking $y = x(1-x)(1+ax)$ and minimising over a gives $\lambda_1 \leq 9.8697$, correct to four decimal places.

Remark. The method extends to higher eigenvalues. Since $\mathcal{R}[y] = \sum \lambda_n c_n^2 / \sum c_n^2$, if we restrict to trial functions orthogonal to y_1 (so $c_1 = 0$) then the same argument gives $\lambda_2 = \min \mathcal{R}[y]$ over that restricted class, and so on. In practice one rarely knows y_1 exactly, which limits the usefulness – but symmetry often helps. For instance, if the problem is symmetric about the midpoint then y_1 is even, so any odd trial function is automatically orthogonal to it and bounds λ_2 .

§23.8.3 Inequalities from Variational Principles

The Rayleigh quotient can be read in the other direction. Instead of using known eigenvalues to bound $\mathcal{R}$, we can use a known eigenvalue to prove an inequality about *all* functions – and some rather famous inequalities arise this way.

Theorem 23.8.4 (Poincaré–Wirtinger Inequality)

Let $y \in C^1[0, L]$ with $y(0) = y(L) = 0$. Then

$$\int_0^L (y')^2 dx \geq \frac{\pi^2}{L^2} \int_0^L y^2 dx,$$

with equality if and only if $y = A \sin(\pi x/L)$.

Proof. Take $\rho = 1$ and $\sigma = 0$ in Theorem 23.8.2, so that $\mathcal{L}y = -y''$. The eigenvalue problem $-y'' = \lambda y$ with $y(0) = y(L) = 0$ has solutions $y_n = \sin(n\pi x/L)$ with $\lambda_n = n^2\pi^2/L^2$, so the smallest eigenvalue is $\lambda_1 = \pi^2/L^2$. The theorem then says

$$\frac{\int_0^L (y')^2 dx}{\int_0^L y^2 dx} = \mathcal{R}[y] \geq \lambda_1 = \frac{\pi^2}{L^2}$$

for every admissible y , which is the claim. Equality requires y to be a ground state, i.e. a multiple of $\sin(\pi x/L)$. $\square$

This is a genuinely useful inequality – it says that a function pinned to zero at both ends cannot be large without also having a large derivative somewhere – and it would be distinctly awkward to prove by hand. The constant π^2/L^2 is sharp precisely because it is an eigenvalue.

Remark. The same trick with $\rho = 1$ and a nonzero σ gives a whole family of inequalities, one for each Sturm–Liouville problem whose spectrum we happen to know. It is also how one proves the isoperimetric inequality analytically: writing the perimeter and area of a closed curve as Fourier series and applying Wirtinger’s inequality on $[0, 2\pi]$ gives $L^2 \geq 4\pi A$ directly, with equality only for a circle – which is the rigorous version of the calculation we did in Example 23.5.9.

§23.8.4 The Variational Principle in Quantum Mechanics

The most celebrated application of all this is to quantum mechanics, where the eigenvalue problem is the time-independent Schrödinger equation

$$\hat{H}\psi = E\psi, \quad \hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}),$$

and the eigenvalues E_n are the allowed energies of the system. Taking $\rho = \hbar^2/2m$ and $\sigma = V$ in our Sturm–Liouville problem, the Rayleigh quotient is

$$\mathcal{R}[\psi] = \frac{\int \left(\frac{\hbar^2}{2m} |\nabla\psi|^2 + V|\psi|^2 \right) dV}{\int |\psi|^2 dV} = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle},$$

which the reader will recognise from the Quantum Mechanics course as the expectation value of the energy in the state ψ , written $\langle \hat{H} \rangle_\psi$ in the notation of my Quantum Mechanics notes.

Theorem 23.8.5 (Variational Principle for the Ground State)

For any normalisable ψ ,

$$E_0 \leq \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle},$$

where E_0 is the ground state energy.

The proof is [Theorem 23.8.2](#) verbatim. Physically the statement is almost obvious – the average energy of any state is at least the lowest available energy – but as a computational tool it is invaluable, since the Schrödinger equation can be solved exactly for only a handful of potentials.

Example 23.8.6 (A Bound for the Hydrogen Atom)

The ground state of hydrogen satisfies (in the s -wave sector, with $u = r\psi$)

$$-\frac{\hbar^2}{2m} \frac{d^2u}{dr^2} - \frac{e^2}{4\pi\epsilon_0 r} u = Eu, \quad u(0) = 0.$$

Take the trial function $u = re^{-r/a}$ with a a free parameter. The required integrals are all of the form $\int_0^\infty r^n e^{-2r/a} dr = n! (a/2)^{n+1}$, and after some computation one finds

$$\mathcal{R}[u] = \frac{\hbar^2}{2ma^2} - \frac{e^2}{4\pi\epsilon_0 a}.$$

Minimising over a gives $a = 4\pi\epsilon_0\hbar^2/me^2$, the Bohr radius, and

$$E_0 \leq -\frac{me^4}{2(4\pi\epsilon_0)^2\hbar^2},$$

which is in fact the exact ground state energy – because our trial function happened to be the exact eigenfunction. The lesson is that a good guess, informed by the physics, can be very good indeed; and even a poor guess gives a rigorous bound.

§23.9 The Second Variation

We have been calling solutions of the Euler–Lagrange equation ‘extrema’, but of course they are only stationary points. It is time to make good on the promise to determine their nature, by looking at the $O(\epsilon^2)$ term we discarded.

§23.9.1 Computing the Second Variation

Let y solve the Euler–Lagrange equation for $F[y] = \int_{\alpha}^{\beta} f(x, y, y') \, dx$, and perturb by $\epsilon\eta$ with η vanishing at the endpoints. Expanding f to second order in its last two arguments,

$$\begin{aligned} F[y + \epsilon\eta] - F[y] &= \epsilon \underbrace{\int_{\alpha}^{\beta} \eta \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \frac{\partial f}{\partial y'} \right) dx}_{=0 \text{ by Euler–Lagrange}} \\ &\quad + \frac{\epsilon^2}{2} \int_{\alpha}^{\beta} \left(\eta^2 \frac{\partial^2 f}{\partial y^2} + 2\eta\eta' \frac{\partial^2 f}{\partial y \partial y'} + (\eta')^2 \frac{\partial^2 f}{\partial (y')^2} \right) dx + O(\epsilon^3). \end{aligned}$$

The coefficient of ϵ^2 is called the **second variation**, written $\delta^2 F[y]$. We can simplify it: noting that $2\eta\eta' = \frac{d}{dx}(\eta^2)$ and integrating that term by parts (the boundary term vanishing as η does),

$$\int_{\alpha}^{\beta} \frac{d(\eta^2)}{dx} \frac{\partial^2 f}{\partial y \partial y'} \, dx = - \int_{\alpha}^{\beta} \eta^2 \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y \partial y'} \right) \, dx,$$

so that

$$\boxed{\delta^2 F[y] = \frac{1}{2} \int_{\alpha}^{\beta} (P(\eta')^2 + Q\eta^2) \, dx}$$

where

$$P = \frac{\partial^2 f}{\partial (y')^2}, \quad Q = \frac{\partial^2 f}{\partial y^2} - \frac{d}{dx} \left(\frac{\partial^2 f}{\partial y \partial y'} \right),$$

both evaluated along the solution y . This is the analogue of the Hessian from [subsection 23.2.2](#), and the same logic applies: if $\delta^2 F[y] > 0$ for all admissible $\eta \neq 0$, then y is a local minimum.

Proposition 23.9.1

If y solves the Euler–Lagrange equation and $P > 0$, $Q \geq 0$ along y , then y is a local minimum of F .

Proof. Under these hypotheses the integrand $P(\eta')^2 + Q\eta^2$ is non-negative, and vanishes identically only if $\eta' \equiv 0$; combined with $\eta(\alpha) = 0$ this forces $\eta \equiv 0$. So $\delta^2 F[y] > 0$ for every $\eta \neq 0$. $\square$

Example 23.9.2 (Geodesics in the Plane)

For $f = \sqrt{1 + (y')^2}$ we have $\partial f / \partial y = 0$, so $Q = 0$, and

$$P = \frac{\partial}{\partial y'} \left(\frac{y'}{\sqrt{1 + (y')^2}} \right) = \frac{1}{(1 + (y')^2)^{3/2}} > 0.$$

So every solution of the Euler–Lagrange equation – that is, every straight line – is a local minimum of length. This confirms what we always believed.

Example 23.9.3 (The Brachistochrone)

For $f = \sqrt{(1 + (y')^2)/(-y)}$ a slightly longer computation gives

$$P = \frac{1}{(1 + (y')^2)^{3/2} \sqrt{-y}} > 0, \quad Q = \frac{1}{4\sqrt{1 + (y')^2} (-y)^{5/2}} > 0$$

on the region $y < 0$ where the problem makes sense. So the cycloid really is a local minimum of the travel time, as Bernoulli claimed.

Example 23.9.4 (A Maximum)

It is worth seeing the other sign at least once. Consider

$$F[y] = \int_0^1 (y^2 - (y')^2) \, dx, \quad y(0) = y(1) = 0.$$

The Euler–Lagrange equation is $y'' + y = 0$, and with these boundary conditions on an interval of length $1 < \pi$ the only solution is $y \equiv 0$. Here

$$P = -2 < 0, \quad Q = 2 > 0,$$

so the Legendre condition for a minimum fails everywhere; applying the proposition above to $-F$ (for which $P = 2 > 0$ and $Q = -2$) is not immediately conclusive either. But we can settle it directly: for any η vanishing at the endpoints, the Poincaré–Wirtinger inequality of [section 23.8](#) gives $\int_0^1 (\eta')^2 \geq \pi^2 \int_0^1 \eta^2 > \int_0^1 \eta^2$, so

$$\delta^2 F[0] = \frac{1}{2} \int_0^1 (2\eta^2 - 2(\eta')^2) \, dx < 0$$

for every $\eta \neq 0$. So $y \equiv 0$ is a local *maximum*. Notice how the sharp constant π^2 did the work: had the interval been longer than π , the inequality would have gone the other way for some η and we would have had a saddle instead.

§23.9.2 The Legendre Condition

The examples above were lucky, in that P and Q were both of a good sign. What if $Q < 0$ and $P > 0$? It turns out that P is much more important than Q , for the following reason.

Proposition 23.9.5 (Legendre Condition)

If y_0 is a local minimum of F , then $P \geq 0$ along y_0 .

Proof (sketch). Suppose $P(x_0) < 0$ for some x_0 . We construct an η which is very small everywhere – so its $Q\eta^2$ contribution is negligible – but which oscillates extremely rapidly near x_0 , so that η' is large. Concretely, take η supported on a small

interval of width δ around x_0 and of amplitude δ ; then $\eta^2 \sim \delta^2$ while $(\eta')^2 \sim 1$, so on integrating over the interval, the Q term contributes $O(\delta^3)$ and the P term contributes $O(\delta) < 0$. For small enough δ the P term wins and $\delta^2 F[y_0] < 0$, contradicting minimality. $\square$

So $P \geq 0$ is *necessary* for a minimum. It is not sufficient, as the following example shows – and understanding why not is the last piece of the theory.

Example 23.9.6

Consider

$$F[y] = \int_0^\beta ((y')^2 - y^2) \, dx, \quad y(0) = y(\beta) = 0,$$

where we suppose $\beta \neq k\pi$ for integers k . The Euler–Lagrange equation is $y'' + y = 0$, whose general solution is $A \sin x + B \cos x$; imposing the boundary conditions with $\beta \neq k\pi$ forces $A = B = 0$, so the only stationary point is $y \equiv 0$, at which $F = 0$.

Here $P = 2 > 0$ and $Q = -2 < 0$, so the Legendre condition is satisfied but our sufficient condition is not. Is $y \equiv 0$ a minimum? Consider the trial variation $\eta = A \sin(\pi x/\beta)$, which vanishes at both ends. Then

$$\delta^2 F[0] = \int_0^\beta ((\eta')^2 - \eta^2) \, dx = A^2 \left(\frac{\pi^2}{\beta^2} - 1 \right) \frac{\beta}{2},$$

using $\int_0^\beta \sin^2(\pi x/\beta) \, dx = \int_0^\beta \cos^2(\pi x/\beta) \, dx = \beta/2$. This is negative precisely when $\beta > \pi$.

So $y \equiv 0$ fails to be a minimum as soon as the interval is *too long*. This is a genuinely new phenomenon: the local data (P and Q) is the same for all β , and yet the answer changes. What matters is a global property of the interval.

§23.9.3 The Jacobi Accessory Condition

Legendre attempted to prove that $P > 0$ implies local minimality; the example above shows this cannot work, but his method can be repaired into a genuine sufficient condition.

The idea is to complete the square. Let ϕ be any differentiable function on $[\alpha, \beta]$. Since η vanishes at both endpoints,

$$\int_\alpha^\beta \frac{d}{dx} (\phi \eta^2) \, dx = 0 \implies \int_\alpha^\beta (\phi' \eta^2 + 2\phi \eta \eta') \, dx = 0.$$

We may add this zero to the second variation without changing anything:

$$\delta^2 F[y] = \frac{1}{2} \int_\alpha^\beta (P(\eta')^2 + 2\phi \eta \eta' + (Q + \phi') \eta^2) \, dx.$$

Now suppose $P > 0$. Then we can complete the square in η' :

$$\delta^2 F[y] = \frac{1}{2} \int_\alpha^\beta \left(P \left(\eta' + \frac{\phi}{P} \eta \right)^2 + \left(Q + \phi' - \frac{\phi^2}{P} \right) \eta^2 \right) \, dx.$$

If we could choose ϕ to make the second bracket vanish identically, the integrand would be a perfect square, and we would be done – for then $\delta^2 F[y] \geq 0$, with equality only if

$$\eta' + \frac{\phi}{P}\eta \equiv 0.$$

This is a first-order linear ODE for η with $\eta(\alpha) = 0$, so by uniqueness of solutions $\eta \equiv 0$. Hence $\delta^2 F[y] > 0$ for all $\eta \neq 0$, and y is a local minimum.

So everything hinges on solving

$$\phi^2 = P(Q + \phi'),$$

a nonlinear (Riccati) equation for ϕ . The standard trick linearises it: substitute

$$\phi = -P \frac{u'}{u}$$

for some function u with $u \neq 0$ on $[\alpha, \beta]$. Then $\phi^2/P = P(u'/u)^2$ and

$$\phi' = -\left(\frac{Pu'}{u}\right)' = -\frac{(Pu')'}{u} + P\left(\frac{u'}{u}\right)^2,$$

so the equation $\phi^2/P = Q + \phi'$ becomes

$$P\left(\frac{u'}{u}\right)^2 = Q - \frac{(Pu')'}{u} + P\left(\frac{u'}{u}\right)^2,$$

and after cancelling and multiplying by u we are left with the following.

Theorem 23.9.7 (Jacobi Accessory Condition)

Let y solve the Euler–Lagrange equation with $P > 0$ along y . If the **Jacobi accessory equation**

$$-(Pu')' + Qu = 0$$

has a solution u with $u \neq 0$ everywhere on $[\alpha, \beta]$, then $\delta^2 F[y] > 0$ and y is a local minimum of F .

The reader will notice that the left hand side is $\mathcal{L}u$, where $\mathcal{L}$ is exactly the Sturm–Liouville operator from [section 23.8](#). This is not a coincidence: integrating the second variation by parts once more gives

$$\delta^2 F[y] = \frac{1}{2} \int_{\alpha}^{\beta} \eta [-(P\eta')' + Q\eta] \, dx = \frac{1}{2} \int_{\alpha}^{\beta} \eta \mathcal{L}\eta \, dx,$$

so the question of whether $\delta^2 F$ is positive is precisely the question of whether $\mathcal{L}$ has any negative eigenvalues. In particular, if there is an η with $\mathcal{L}\eta = -\omega^2\eta$ and $\eta(\alpha) = \eta(\beta) = 0$, then $\delta^2 F[y] = -\frac{\omega^2}{2} \int \eta^2 < 0$ and y is not a minimum.

The condition ‘ $u \neq 0$ on $[\alpha, \beta]$ ’ has a name: a point $x^* > \alpha$ at which the solution u with $u(\alpha) = 0$ vanishes again is called a **conjugate point** to α . The Jacobi condition says that a stationary path is a local minimum provided no conjugate point occurs before we reach β – which is exactly the ‘interval too long’ phenomenon of [Example 23.9.6](#).

Example 23.9.8

Return to $F[y] = \frac{1}{2} \int_{\alpha}^{\beta} ((y')^2 - y^2) \, dx$, so $P = 1$ and $Q = -1$. The Jacobi accessory equation is

$$u'' + u = 0 \implies u = A \sin x + B \cos x.$$

We need this to be nonvanishing on $[\alpha, \beta]$, that is, $\tan x \neq -B/A$ throughout. Since $\tan$ takes every value once per interval of length π , such a choice of A, B is possible if and only if

$$|\beta - \alpha| < \pi.$$

This is exactly the condition we found by hand in [Example 23.9.6](#), now derived systematically.

Example 23.9.9 (Geodesics on the Sphere, Revisited)

We can finally settle which arcs of a great circle are minimising. Parameterizing the sphere as before, the length functional is

$$F[\theta] = \int \sqrt{(\theta')^2 + \sin^2 \theta} \, d\phi,$$

where now θ is a function of ϕ . Every great circle can be rotated to the equator, so consider the stationary path $\theta_0 \equiv \pi/2$, and let ϕ run from ϕ_1 to ϕ_2 .

Writing $f = \sqrt{(\theta')^2 + \sin^2 \theta}$ and evaluating along $\theta_0 = \pi/2$ (where $\theta' = 0$ and $\sin \theta_0 = 1$), we get $f = 1$ and

$$P = \frac{\partial^2 f}{\partial (\theta')^2} = 1, \quad Q = \frac{\partial^2 f}{\partial \theta^2} = -1,$$

so that

$$\delta^2 F \left[\frac{\pi}{2} \right] = \frac{1}{2} \int_{\phi_1}^{\phi_2} ((\eta')^2 - \eta^2) \, d\phi,$$

which is exactly the functional of the previous example. Hence the arc is a local minimum if and only if

$$|\phi_2 - \phi_1| < \pi,$$

that is, if and only if it is the *minor* arc. The major arc is a stationary point which is not a minimum, exactly as we suspected. If $|\phi_2 - \phi_1| = \pi$ the endpoints are antipodal, the conjugate point is reached precisely at the end, and correspondingly there is a whole circle's worth of geodesics of the same length – the degeneracy showing up as the vanishing of the second variation along the family.

Example 23.9.10 (Soap Films Revisited)

We can now say something about the promise made in [subsection 23.6.1](#). There we found that for rings of radius $c > c_*$ there are two catenoids,

$$y = y_0 \cosh \left(\frac{x}{y_0} \right),$$

one with $y_0 > \mu$ and one with $y_0 < \mu$, where $\mu \approx 0.833$ is the location of the minimum of $\Psi(y_0) = y_0 \cosh(1/y_0)$.

For the area integrand $f = y\sqrt{1 + (y')^2}$ we compute

$$P = \frac{\partial^2 f}{\partial (y')^2} = \frac{y}{(1 + (y')^2)^{3/2}} > 0,$$

so the Legendre condition is satisfied by both catenoids, and we cannot distinguish them that way. The distinction lies in the Jacobi accessory equation, and it is exactly the conjugate point phenomenon: for the fat catenoid, the solution u of $-(Pu')' + Qu = 0$ vanishing at $x = -\beta$ stays nonzero all the way to $x = \beta$, whereas for the thin one a conjugate point occurs strictly inside the interval. Hence the fat catenoid is a local minimum and the thin one is not – it is a saddle, and any real soap film would collapse away from it.

One can make this concrete without solving the Jacobi equation, by noticing that the family of catenoids through the left-hand ring is a one-parameter family of solutions, and that a conjugate point is precisely a place where two nearby members of the family meet. Geometrically, the catenaries through a fixed point have an *envelope*, and the conjugate point is where the extremal touches it. Inside the envelope there are two catenaries to each point (our fat and thin solutions); outside there are none – which is exactly the trichotomy we found by looking at Ψ .

Remark. The reader who found the Jacobi condition to be a lot of machinery for a small return should compare it with the situation in finite dimensions. There, ‘ H positive definite at a stationary point’ is both necessary (up to semi-definiteness) and sufficient for a local minimum, and there is nothing more to say. The reason infinite dimensions are harder is precisely the phenomenon of [Example 23.9.6](#): the quadratic form $\delta^2 F$ can be positive on every ‘direction’ we happen to test and still fail to be positive definite, because the relevant operator has a continuum of directions to be negative in. Conjugate points are the bookkeeping which detects this.

This is a good place to stop. We began with a question about a bead sliding down a wire, and we have ended with a criterion, in terms of conjugate points on a sphere, for a path to be genuinely shortest. Along the way the same three-step argument – perturb, integrate by parts, apply the fundamental lemma – produced the equations of geometry, of optics, and of mechanics. That a single piece of calculus should reach so far is, to this author at least, the most attractive thing about the subject.

III

Part II

About Part II

Part II is where the degree stops being a degree and becomes a menu. There are around forty courses on offer and no obligation to take any particular one, so no two students in the year sit the same set. The twenty-one chapters here are the ones I took, and they are not a syllabus so much as a set of choices; another student's Part II could overlap with this one in three courses and be no less complete.

The change in character is real, and it is not only about choice. Part IB abstracted the objects of Part IA; Part II mostly stops abstracting and starts going deep. A typical course here takes one idea and follows it for twenty-four lectures until it arrives somewhere that would have been unimaginable two years earlier – Galois Theory begins with field extensions and ends by proving that the general quintic is not soluble by radicals; Algebraic Geometry begins with the zero set of a polynomial and ends at the Riemann–Roch theorem; Algebraic Topology begins with loops in a space and ends by proving that $\mathbb{R}^m \cong \mathbb{R}^n$ forces $m = n$. The courses also stop being polite to one another. In Part IB every course could reasonably assume every other; in Part II a lecturer has to assume a specific list, and the list is short.

The algebraic strand is the one that most obviously pays off. Groups, Rings and Modules splits into four: Galois Theory, which turns questions about polynomials into questions about finite groups; Representation Theory, which turns questions about finite groups into questions about matrices, and gets character tables and Burnside's theorem out of it; Number Fields, which repairs unique factorisation in rings of algebraic integers by replacing elements with ideals; and Algebraic Geometry, which builds a dictionary between ideals and varieties and then uses it relentlessly. Number Theory sits slightly below these, elementary in its methods and enormously useful in its results, and feeds both Number Fields and Coding and Cryptography.

The analytic strand pulls in the opposite direction, towards infinite dimensions. Linear Analysis is what happens when linear algebra is done on a Banach space, and its three great theorems – Hahn–Banach, Baire category and the consequences of completeness – are the tools of everything after it. Probability and Measure rebuilds the whole of Part IA Probability on a measure-theoretic foundation, which is what finally makes the strong law of large numbers and the central limit theorem provable rather than plausible. Analysis of Functions then combines the two, developing L^p spaces, weak topologies, distributions and Sobolev spaces, and ends by solving elliptic partial differential equations with them. Topics in Analysis is the outlier of the group and a delight: a collection of essentially unrelated beautiful results – fixed point theorems, Runge's theorem, Gaussian quadrature, the irrationality of e and π – held together by nothing but taste.

Geometry in Part II is where the various strands meet. Differential Geometry retells IB Geometry with more machinery and more nerve, arriving at Gauss–Bonnet for the second time in this book. Riemann Surfaces is the most cross-bred course of the year, needing complex analysis for its local theory, covering spaces from Algebraic Topology for its global theory, and pointing directly at Algebraic Geometry. Algebraic Topology itself develops the fundamental group and then simplicial homology, and is the course that best explains why anybody ever wanted a functor.

The mathematical physics courses continue the IB story. Principles of Quantum Mechanics recasts IB Quantum Mechanics in Dirac's language, where the harmonic oscillator, angular momentum and perturbation theory become algebra rather than differential equations, and it in turn supports Statistical Physics, which is the one course in the

book that genuinely derives thermodynamics rather than assuming it. Integrable Systems occupies strange and wonderful territory between the applied and the pure, using the inverse scattering transform to solve nonlinear PDEs exactly. Quantum Information and Computation needs almost nothing but linear algebra and a little quantum mechanics, and gets to Shor's and Grover's algorithms remarkably fast.

The remaining courses are the genuinely independent ones. Logic and Set Theory builds ordinals, cardinals and the axiom of choice and then proves Gödel's incompleteness theorem; Automata and Formal Languages does the computability half of the same story; Graph Theory is combinatorics with almost no prerequisites and some of the best proofs in the book; Coding and Cryptography turns Shannon's theorems into practical codes; Stochastic Financial Models takes the probability strand into continuous time and derives Black–Scholes. Any of these five can be read cold, given Part IA and a little patience.

If Part IA rewards being read in order and Part IB rewards being read in strands, Part II rewards being read out of curiosity. Pick the chapter whose first page interests you, and use the dependency graphs in the front matter to work out what you owe.

24 Algebraic Geometry

Algebraic geometry is the study of the shapes cut out by polynomial equations. Write down a system of polynomials, look at the set of points where they all vanish, and you have a geometric object; write down the ring of polynomial functions on that object, and you have an algebraic one. The whole subject is an extended meditation on how much each of these two things knows about the other. It turns out that they know almost everything about each other, and the translations back and forth are what give algebraic geometry its distinctive flavour – a topological question about a solution set becomes a question about prime ideals, and a question about ideals becomes a question about geometry.

We will spend the first half of the course setting up this dictionary carefully, first for affine varieties and then for projective ones, where the geometry is much better behaved. In the second half we specialise to curves, where the theory becomes remarkably complete: we will be able to attach an integer – the genus – to every smooth projective curve, and the Riemann–Roch theorem will let us compute with it.

This article constitutes my notes for the ‘Algebraic Geometry’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

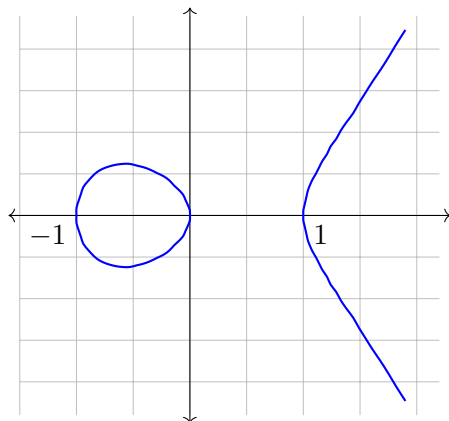
§24.1 Introduction

Let us begin with an example, since the whole subject grows out of examples of this kind. Consider the polynomial

$$f(X, Y) = Y^2 - X^3 + X \in \mathbb{Z}[X, Y],$$

and ask for its solution set. The answer depends rather dramatically on where we look for solutions.

If we look for real solutions, we get a subset of $\mathbb{R}^2$ that we can draw. Since $Y^2 = X(X - 1)(X + 1)$, a real point requires $X(X^2 - 1) \geq 0$, which happens exactly when $-1 \leq X \leq 0$ or $X \geq 1$. So the real locus has two pieces: a compact oval sitting over $[-1, 0]$, and an unbounded branch over $[1, \infty)$.



This picture is already suggestive. A ‘generic’ line meets the curve in three points, which is no accident: substituting $Y = mX + c$ into f produces a cubic in X , so there should be three roots. That there are sometimes fewer real roots than three is precisely the sort of irritation that pushes us towards algebraically closed fields.

If instead we look for solutions in $\mathbb{C}^2$, we can no longer draw the picture – the solution set is a real surface sitting inside a real four-dimensional space – but the algebra becomes uniform. Every line really does meet the curve three times. In fact, once we compactify appropriately, the solution set turns out to be a torus, and this is the first hint of the deep relationship between algebraic geometry and topology that the second half of these notes will explore.

Finally, if we look for solutions over $\mathbb{Q}$ or over $\mathbb{Z}$, we are doing number theory, and the questions become extremely hard. We will not do that here, but it is worth knowing that this is the same picture viewed through a different lens.

§24.1.1 Conventions

We will work throughout over the complex numbers. There are two properties of $\mathbb{C}$ that we care about:

- (i) $\mathbb{C}$ is algebraically closed – every non-constant polynomial in one variable has a root. This is what makes the dictionary between algebra and geometry work at all; without it, $\mathbb{V}(X^2 + 1) \subseteq \mathbb{R}^1$ is empty even though $X^2 + 1$ is a perfectly good non-unit.
- (ii) $\mathbb{C}$ has characteristic zero, which means we can differentiate polynomials without anything going wrong, and $\mathbb{C}$ is uncountable, which we will exploit once (in the proof of the Nullstellensatz).

Almost everything we do works verbatim over an arbitrary algebraically closed field, and the reader who prefers to write k everywhere is encouraged to do so. We will write $\mathbb{C}[\mathbf{X}] = \mathbb{C}[X_1, \dots, X_n]$ when the number of variables is clear from context.

Two facts from IB Groups, Rings and Modules will be used constantly, and it is worth recalling them now.

Theorem 24.1.1 (Hilbert Basis Theorem)

If R is a Noetherian ring then so is $R[X]$. In particular $\mathbb{C}[X_1, \dots, X_n]$ is Noetherian: every ideal of $\mathbb{C}[\mathbf{X}]$ is finitely generated.

Theorem 24.1.2

$\mathbb{C}[X_1, \dots, X_n]$ is a unique factorisation domain.

The first of these is what makes all of our geometric objects ‘finite’ in nature, and the second is what lets us talk about the irreducible polynomial cutting out a hypersurface.

§24.2 Affine Varieties

§24.2.1 Affine Space and Polynomial Functions

We begin with the ambient space.

Definition 24.2.1 (Affine Space)

Affine n -space over $\mathbb{C}$ is the set $\mathbb{A}^n = \mathbb{C}^n$. Its elements are called **points**, and we typically write a point as $P = (a_1, \dots, a_n)$.

Why the new notation for something we already have a name for? The point is one of emphasis. When we write $\mathbb{C}^n$ we are usually thinking of a vector space, with a distinguished origin and a linear structure. When we write $\mathbb{A}^n$ we are forgetting all of that and remembering only that $\mathbb{A}^n$ is the natural home of the polynomial ring: every $f \in \mathbb{C}[X_1, \dots, X_n]$ determines a function

$$f: \mathbb{A}^n \rightarrow \mathbb{A}^1, \quad P \mapsto f(P),$$

and it is this ring of functions, rather than any linear structure, that we regard as the essential data.

Definition 24.2.2 (Affine Subspace)

An **affine subspace** of $\mathbb{A}^n$ is a subset of the form $v + U$, where $U \leq \mathbb{C}^n$ is a linear subspace and $v \in \mathbb{C}^n$.

Note that a polynomial is determined by the function it induces on $\mathbb{A}^n$: if f and g agree at every point of $\mathbb{C}^n$ then $f = g$ as polynomials. (This uses that $\mathbb{C}$ is infinite; over $\mathbb{F}_2$ the polynomials X and X^2 induce the same function on $\mathbb{A}^1$.) So we may freely conflate polynomials with the functions they define.

§24.2.2 Vanishing Loci

Definition 24.2.3 (Vanishing Locus)

Let $S \subseteq \mathbb{C}[X_1, \dots, X_n]$ be any subset. The **vanishing locus** of S is

$$\mathbb{V}(S) = \{P \in \mathbb{A}^n : f(P) = 0 \text{ for all } f \in S\} \subseteq \mathbb{A}^n.$$

A subset of $\mathbb{A}^n$ of the form $\mathbb{V}(S)$ is called an **affine (algebraic) variety**.

A warning about terminology: some authors (and some textbooks you may consult) reserve the word ‘variety’ for what we will call an *irreducible* variety, and use ‘algebraic set’ for the general notion. We will not do that, but you should be alert to it.

Let us see some examples straight away.

Example 24.2.4 (i) Take $n = 1$. A single polynomial $f \in \mathbb{C}[X]$ has finitely many roots, so $\mathbb{V}(f)$ is a finite set; and conversely, any finite set $\{a_1, \dots, a_m\} \subseteq \mathbb{A}^1$ is $\mathbb{V}(f)$ for $f = \prod_i (X - a_i)$. Also $\mathbb{V}(0) = \mathbb{A}^1$. So the affine varieties in $\mathbb{A}^1$ are exactly the finite sets together with $\mathbb{A}^1$ itself.

(ii) A **hypersurface** in $\mathbb{A}^n$ is a variety of the form $\mathbb{V}(f)$ for a single non-constant

f. For $n = 2$ these are the **plane curves**.

(iii) Not all varieties are naturally presented by equations. The **twisted cubic** is

$$C = \{(t, t^2, t^3) : t \in \mathbb{C}\} \subseteq \mathbb{A}^3,$$

presented parametrically. It is nonetheless a variety: $C = \mathbb{V}(X_2 - X_1^2, X_3 - X_1^3)$. Indeed if $x_2 = x_1^2$ and $x_3 = x_1^3$ then the point is (t, t^2, t^3) for $t = x_1$.

(iv) Every affine subspace is a variety, being the vanishing locus of a collection of degree one polynomials. In particular a point $(a_1, \dots, a_n)$ is the variety $\mathbb{V}(X_1 - a_1, \dots, X_n - a_n)$.

Passing from a set S of polynomials to $\mathbb{V}(S)$ loses a lot of information, and the first thing to do is to understand exactly what it remembers. The following is the basic finiteness statement of the subject.

Theorem 24.2.5

Let $S \subseteq \mathbb{C}[\mathbf{X}]$ and let $I = (S)$ be the ideal it generates. Then

- (i) $\mathbb{V}(S) = \mathbb{V}(I)$;
- (ii) there is a *finite* subset $\{f_1, \dots, f_r\} \subseteq S$ with $\mathbb{V}(S) = \mathbb{V}(f_1, \dots, f_r)$.

Proof. For (i), certainly $\mathbb{V}(I) \subseteq \mathbb{V}(S)$ since $S \subseteq I$. Conversely if $P \in \mathbb{V}(S)$ and $g = \sum_j h_j f_j \in I$ with $f_j \in S$, then $g(P) = \sum_j h_j(P) f_j(P) = 0$, so $P \in \mathbb{V}(I)$.

For (ii), by the Hilbert Basis Theorem I is finitely generated, say $I = (g_1, \dots, g_m)$. Each g_i lies in the ideal generated by S , so is a finite combination $g_i = \sum_j h_{ij} f_{ij}$ with $f_{ij} \in S$. Let T be the (finite) set of all the f_{ij} occurring. Then $I = (g_1, \dots, g_m) \subseteq (T) \subseteq I$, so $(T) = I$ and by (i) we get $\mathbb{V}(S) = \mathbb{V}(I) = \mathbb{V}(T)$. $\square$

In other words, *every* system of polynomial equations, however large, is equivalent to a finite subsystem. This is the geometric content of the Hilbert Basis Theorem, and it is not at all obvious from the geometric side.

The operation $\mathbb{V}$ reverses inclusions and interacts with the lattice operations on ideals in a way we now record.

Proposition 24.2.6

Let $S, T \subseteq \mathbb{C}[\mathbf{X}]$ be subsets and let $(I_j)_{j \in J}$ be a family of ideals of $\mathbb{C}[\mathbf{X}]$. Then

- (i) if $S \subseteq T$ then $\mathbb{V}(T) \subseteq \mathbb{V}(S)$;
- (ii) $\mathbb{V}(0) = \mathbb{A}^n$ and $\mathbb{V}(\mathbb{C}[\mathbf{X}]) = \mathbb{V}(1) = \emptyset$;
- (iii) $\bigcap_{j \in J} \mathbb{V}(I_j) = \mathbb{V}(\sum_j I_j)$;
- (iv) $\mathbb{V}(I) \cup \mathbb{V}(J) = \mathbb{V}(I \cap J) = \mathbb{V}(IJ)$.

Proof. Parts (i) and (ii) are immediate from the definitions.

For (iii), a point lies in every $\mathbb{V}(I_j)$ if and only if it is killed by every element of

every I_j , if and only if it is killed by $\bigcup_j I_j$; and $\mathbb{V}$ of a set equals $\mathbb{V}$ of the ideal it generates, which is $\sum_j I_j$.

For (iv), we have $IJ \subseteq I \cap J \subseteq I$ and likewise for J , so by (i),

$$\mathbb{V}(I) \cup \mathbb{V}(J) \subseteq \mathbb{V}(I \cap J) \subseteq \mathbb{V}(IJ).$$

It remains to show $\mathbb{V}(IJ) \subseteq \mathbb{V}(I) \cup \mathbb{V}(J)$. Suppose $P \in \mathbb{V}(IJ)$ but $P \notin \mathbb{V}(I)$, so there is $g \in I$ with $g(P) \neq 0$. For any $f \in J$ we have $fg \in IJ$, so $f(P)g(P) = 0$, whence $f(P) = 0$. Thus $P \in \mathbb{V}(J)$. $\square$

§24.2.3 The Zariski Topology

Proposition 24.2.6 says exactly that the affine varieties are the closed sets of a topology: they contain $\emptyset$ and the whole space, and are closed under arbitrary intersections and finite unions.

Definition 24.2.7 (Zariski Topology)

The **Zariski topology** on $\mathbb{A}^n$ is the topology whose closed sets are precisely the affine varieties. If $V \subseteq \mathbb{A}^n$ is a variety, the Zariski topology on V is the subspace topology.

Definition 24.2.8 (Euclidean Topology)

The **Euclidean** or **analytic** topology on $\mathbb{A}^n = \mathbb{C}^n$ is the usual metric topology. Again, a variety inherits a subspace topology.

Every Zariski closed set is Euclidean closed, since polynomials are continuous; so the Zariski topology is *coarser* than the Euclidean one. It is in fact very much coarser, and the reader should be warned that it does not behave like the topologies met in IB Analysis and Topology.

Proposition 24.2.9

The Zariski topology on $\mathbb{A}^1$ is the cofinite topology. It is compact and irreducible, but not Hausdorff. The Euclidean topology on $\mathbb{A}^1$ is Hausdorff but not compact.

Proof. We saw that the varieties in $\mathbb{A}^1$ are the finite sets and $\mathbb{A}^1$ itself, which is the definition of the cofinite topology. Any two nonempty open sets are cofinite, hence meet (as $\mathbb{C}$ is infinite), so the space is not Hausdorff. Compactness is immediate: given an open cover, one member U is nonempty, and its complement is finite, so finitely many further members finish the job. $\square$

Remark. The Zariski topology on $\mathbb{A}^2$ is *not* the product topology of the Zariski topologies on $\mathbb{A}^1 \times \mathbb{A}^1$. In the product topology the closed sets are finite unions of sets of the form $A \times B$ with A, B closed; the diagonal $\mathbb{V}(X_1 - X_2)$ is Zariski closed in $\mathbb{A}^2$ but is not of this form.

Because the Zariski topology is so coarse, nonempty open sets are enormous. We will use this constantly: a Zariski open set is ‘almost all’ of the space, and a statement holding

on a nonempty open set is a statement holding ‘generically’.

§24.2.4 Irreducibility

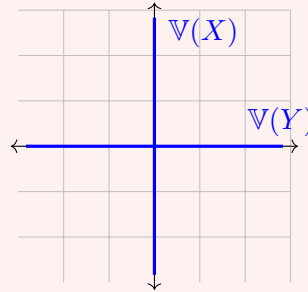
A cofinite topology has no interesting disconnections, but varieties can still fall apart in an algebraic sense, as the following notion makes precise.

Definition 24.2.10 (Irreducible)

A nonempty variety V is **irreducible** if whenever $V = V_1 \cup V_2$ with V_1, V_2 closed subsets of V , we have $V = V_1$ or $V = V_2$. Otherwise V is **reducible**.

Example 24.2.11

The variety $V = \mathbb{V}(XY) \subseteq \mathbb{A}^2$ is reducible: it is the union of the two coordinate axes $\mathbb{V}(X)$ and $\mathbb{V}(Y)$, neither of which is all of V .



By contrast, we shall see shortly that each axis is irreducible.

Note that irreducibility is a genuinely topological condition – it says that V is not the union of two proper closed subsets – but it is a condition no Hausdorff space with more than one point satisfies. It is a condition suited to the Zariski topology.

Proposition 24.2.12

Every affine variety is a finite union of irreducible varieties.

Proof. Suppose not, and let V be a variety that is not a finite union of irreducibles. In particular V is not itself irreducible, so $V = V_1 \cup W_1$ with V_1, W_1 proper closed subsets. If both V_1 and W_1 were finite unions of irreducibles then so would V be; so (relabelling) V_1 is not a finite union of irreducibles. Repeating, we obtain a strictly decreasing chain

$$V = V_0 \supsetneq V_1 \supsetneq V_2 \supsetneq \cdots$$

of varieties. Let $I_j = I(V_j)$ (defined below in Section 24.2.5); the containments reverse, giving a strictly increasing chain of ideals $I_0 \subsetneq I_1 \subsetneq \cdots$. But $\mathbb{C}[\mathbf{X}]$ is Noetherian, so no such chain exists, and we have our contradiction. $\square$

The reader will notice that we have used the ideal $I(V)$ of a variety before defining it; the reader who objects may instead run the argument with $\bigcup_j I_j$, which is an ideal and hence finitely generated, so equals $\sum_{j \leq N} I_j$ for some N , forcing the chain to stabilise. Either way, the essential point is the Noetherian property.

Definition 24.2.13 (Minimal Decomposition)

A **minimal decomposition** of a variety V is an expression $V = V_1 \cup \cdots \cup V_r$ with each V_i irreducible and $V_i \not\subseteq V_j$ for $i \neq j$.

Given any decomposition into irreducibles, we can throw away any piece contained in another and obtain a minimal one. What is not obvious is that the result does not depend on the choices made.

Proposition 24.2.14

Minimal decompositions are unique up to reordering.

Proof. Suppose $V = \bigcup_{i=1}^r V_i = \bigcup_{j=1}^s W_j$ are two minimal decompositions. Fix i . Then

$$V_i = V_i \cap V = \bigcup_{j=1}^s (V_i \cap W_j),$$

a finite union of closed subsets of V_i . By irreducibility, $V_i = V_i \cap W_j$ for some j , i.e. $V_i \subseteq W_j$. Symmetrically $W_j \subseteq V_k$ for some k , so $V_i \subseteq V_k$, and minimality forces $i = k$ and hence $V_i = W_j$. So each V_i appears among the W_j and vice versa. $\square$

Definition 24.2.15 (Irreducible Components)

The irreducible varieties appearing in the (unique) minimal decomposition of V are the **irreducible components** of V .

§24.2.5 The Ideal of a Variety

We now go the other way, from geometry back to algebra.

Definition 24.2.16

Let $X \subseteq \mathbb{A}^n$ be any subset. The **ideal of X** is

$$I(X) = \{f \in \mathbb{C}[\mathbf{X}] : f(P) = 0 \text{ for all } P \in X\}.$$

This really is an ideal: it is closed under addition, and if f vanishes on X then so does gf for any g . It is the largest set of polynomials with vanishing locus containing X .

Proposition 24.2.17

Let $V, W \subseteq \mathbb{A}^n$ be affine varieties and $S \subseteq \mathbb{C}[\mathbf{X}]$.

- (i) If $V = \mathbb{V}(S)$ then $S \subseteq I(V)$.
- (ii) $V = \mathbb{V}(I(V))$.
- (iii) $V \subseteq W$ if and only if $I(W) \subseteq I(V)$. In particular $V = W$ if and only if $I(V) = I(W)$.
- (iv) $I(V \cup W) = I(V) \cap I(W)$.

Proof. (i) is immediate. For (ii), $S \subseteq I(V)$ gives $\mathbb{V}(I(V)) \subseteq \mathbb{V}(S) = V$, and $V \subseteq \mathbb{V}(I(V))$ holds by definition of $I(V)$.

For (iii), if $V \subseteq W$ then any polynomial vanishing on W vanishes on V . Conversely suppose $V \not\subseteq W$ and pick $P \in V \setminus W$. As $W = \mathbb{V}(I(W))$ and $P \notin W$, there is $f \in I(W)$ with $f(P) \neq 0$; then $f \notin I(V)$.

(iv) is immediate from the definitions. $\square$

So I is an injection from varieties in $\mathbb{A}^n$ to ideals of $\mathbb{C}[\mathbf{X}]$, with $\mathbb{V}$ as a left inverse. It is certainly not surjective: the ideals (X) and (X^2) in $\mathbb{C}[X]$ have the same vanishing locus $\{0\}$, and $I(\{0\}) = (X)$. So the composite $I \circ \mathbb{V}$ is not the identity on ideals, and understanding exactly what it does is the content of the Nullstellensatz.

First, however, let us record the algebraic characterisation of irreducibility.

Proposition 24.2.18

A variety V is irreducible if and only if $I(V)$ is a prime ideal.

Recall that $I(V)$ prime means: whenever $f_1 f_2 \in I(V)$, either $f_1 \in I(V)$ or $f_2 \in I(V)$. Geometrically, the failure of primality is the existence of two functions, neither vanishing identically on V , whose product does; and such a pair splits V in two.

Proof. Suppose V is reducible, $V = V_1 \cup V_2$ with $V_i \subsetneq V$ closed. Then $I(V_i) \supsetneq I(V)$ strictly, and $V_1 \not\subseteq V_2$, $V_2 \not\subseteq V_1$ (else the decomposition would be trivial), so $I(V_1) \not\subseteq I(V_2)$ and vice versa. Choose $f_1 \in I(V_1) \setminus I(V_2)$ and $f_2 \in I(V_2) \setminus I(V_1)$. Neither lies in $I(V) = I(V_1) \cap I(V_2)$, but $f_1 f_2$ vanishes on $V_1 \cup V_2 = V$. So $I(V)$ is not prime.

Conversely suppose $I(V)$ is not prime, with $f_1 f_2 \in I(V)$ but $f_i \notin I(V)$. Set $V_i = V \cap \mathbb{V}(f_i)$, a closed subset of V . Since $f_i \notin I(V)$, f_i does not vanish identically on V , so $V_i \subsetneq V$. But for any $P \in V$ we have $f_1(P)f_2(P) = 0$, so $P \in V_1 \cup V_2$. Hence $V = V_1 \cup V_2$ is reducible. $\square$

Example 24.2.19

$\mathbb{V}(X) \subseteq \mathbb{A}^2$ is irreducible, since $I(\mathbb{V}(X)) = (X)$ and $\mathbb{C}[X, Y]/(X) \cong \mathbb{C}[Y]$ is an integral domain, so (X) is prime. Hence $\mathbb{V}(XY) = \mathbb{V}(X) \cup \mathbb{V}(Y)$ is the minimal decomposition of $\mathbb{V}(XY)$ into irreducible components.

Example 24.2.20

More generally, if $f \in \mathbb{C}[\mathbf{X}]$ is irreducible then (f) is prime (as $\mathbb{C}[\mathbf{X}]$ is a UFD), so the hypersurface $\mathbb{V}(f)$ is irreducible. If $f = f_1^{a_1} \cdots f_r^{a_r}$ is the factorisation of a general f into irreducibles, then $\mathbb{V}(f) = \mathbb{V}(f_1) \cup \cdots \cup \mathbb{V}(f_r)$ is the decomposition into irreducible components.

§24.2.6 Hilbert's Nullstellensatz

We now come to the theorem which makes the whole dictionary work. The name means ‘theorem of the zeros’, and it comes in two forms.

Theorem 24.2.21 (Weak Nullstellensatz) (i) Every maximal ideal of $\mathbb{C}[X_1, \dots, X_n]$ has the form $\mathfrak{m}_P = (X_1 - a_1, \dots, X_n - a_n)$ for some point $P = (a_1, \dots, a_n) \in \mathbb{A}^n$.
(ii) If $I \subseteq \mathbb{C}[\mathbf{X}]$ is a proper ideal then $\mathbb{V}(I) \neq \emptyset$.

Part (ii) is the statement that a system of polynomial equations with no ‘algebraic obstruction’ – that is, such that 1 is not a combination of the equations – really does have a solution. It is the many-variable generalisation of the fundamental theorem of algebra.

The proof we give uses that $\mathbb{C}$ is uncountable, and so is special to $\mathbb{C}$; the theorem is true over any algebraically closed field, but the general proof (via the Noether normalisation lemma, or via Zariski’s lemma) is more work.

Proof. Each ideal $\mathfrak{m}_P$ is maximal: the evaluation homomorphism $\mathbb{C}[\mathbf{X}] \rightarrow \mathbb{C}$, $f \mapsto f(P)$, is surjective with kernel exactly $\mathfrak{m}_P$, so $\mathbb{C}[\mathbf{X}]/\mathfrak{m}_P \cong \mathbb{C}$ is a field.

Conversely let $\mathfrak{m}$ be a maximal ideal and set $K = \mathbb{C}[\mathbf{X}]/\mathfrak{m}$, a field containing $\mathbb{C}$. Write a_i for the image of X_i in K , so that $K = \mathbb{C}[a_1, \dots, a_n]$. If every a_i lies in $\mathbb{C} \subseteq K$ then $X_i - a_i \in \mathfrak{m}$ for each i , so $\mathfrak{m} \supseteq \mathfrak{m}_P$ for $P = (a_1, \dots, a_n)$, and maximality of $\mathfrak{m}_P$ gives $\mathfrak{m} = \mathfrak{m}_P$ as required.

So suppose for contradiction that $K \supsetneq \mathbb{C}$, and pick $t \in K \setminus \mathbb{C}$. Since $\mathbb{C}$ is algebraically closed, t cannot be algebraic over $\mathbb{C}$, so t is transcendental and the elements $\{(t - c)^{-1} : c \in \mathbb{C}\}$ make sense in K . We claim they are linearly independent over $\mathbb{C}$. Indeed, a relation $\sum_{i=1}^m \lambda_i (t - c_i)^{-1} = 0$ with distinct c_i , cleared of denominators, gives $\sum_i \lambda_i \prod_{j \neq i} (t - c_j) = 0$; substituting the transcendental t into this polynomial identity and evaluating the polynomial at c_i shows $\lambda_i \prod_{j \neq i} (c_i - c_j) = 0$, so $\lambda_i = 0$.

Thus K contains an uncountable linearly independent set over $\mathbb{C}$, so $\dim_{\mathbb{C}} K$ is uncountable. On the other hand, let $U_m \subseteq K$ be the $\mathbb{C}$ -span of the monomials $a_1^{r_1} \cdots a_n^{r_n}$ with $\sum r_i \leq m$. Each U_m is finite-dimensional and $K = \bigcup_{m \geq 0} U_m$, so K has a countable spanning set and hence countable dimension. This is a contradiction.

For (ii), a proper ideal I is contained in some maximal ideal $\mathfrak{m}_P$ (using that $\mathbb{C}[\mathbf{X}]$ is Noetherian, or Zorn’s lemma), and then $P \in \mathbb{V}(\mathfrak{m}_P) \subseteq \mathbb{V}(I)$. $\square$

To state the strong form we need one more piece of algebra.

Definition 24.2.22 (Radical)

The **radical** of an ideal $I \subseteq R$ is

$$\sqrt{I} = \{f \in R : f^m \in I \text{ for some } m \geq 1\}.$$

An ideal is **radical** if $I = \sqrt{I}$.

That $\sqrt{I}$ is an ideal follows from the binomial theorem: if $f^a, g^b \in I$ then every term of $(f + g)^{a+b}$ contains f^a or g^b . Note $\sqrt{\sqrt{I}} = \sqrt{I}$, and prime ideals are radical. Since f and f^m have the same zeros, $\mathbb{V}(I) = \mathbb{V}(\sqrt{I})$ always.

Theorem 24.2.23 (Strong Nullstellensatz)

Let $I \trianglelefteq \mathbb{C}[X_1, \dots, X_n]$ be an ideal. Then

$$I(\mathbb{V}(I)) = \sqrt{I}.$$

Proof. The inclusion $\sqrt{I} \subseteq I(\mathbb{V}(I))$ is clear: if $f^m \in I$ then f^m , hence f , vanishes on $\mathbb{V}(I)$.

For the converse we use the *Rabinowitsch trick*. Let $f \in I(\mathbb{V}(I))$, and let $g_1, \dots, g_r$ generate I . Introduce a new variable T and consider the ideal

$$J = (g_1, \dots, g_r, 1 - Tf) \trianglelefteq \mathbb{C}[X_1, \dots, X_n, T].$$

I claim $\mathbb{V}(J) = \emptyset$ in $\mathbb{A}^{n+1}$. Indeed, if $(P, \tau) \in \mathbb{V}(J)$ then $P \in \mathbb{V}(I)$, so $f(P) = 0$, and then $1 - \tau f(P) = 1 \neq 0$, a contradiction.

By the weak Nullstellensatz, $J = \mathbb{C}[\mathbf{X}, T]$, so we may write

$$1 = \sum_{i=1}^r h_i(\mathbf{X}, T)g_i(\mathbf{X}) + h_0(\mathbf{X}, T)(1 - Tf(\mathbf{X})).$$

This is an identity in the polynomial ring, so we may substitute $T = 1/f$ and work in the field $\mathbb{C}(X_1, \dots, X_n)$ (note $f \neq 0$, else $f \in \sqrt{I}$ trivially). The last term vanishes, leaving

$$1 = \sum_{i=1}^r h_i(\mathbf{X}, 1/f) g_i(\mathbf{X}).$$

Multiplying by a sufficiently large power f^N to clear all denominators gives $f^N = \sum_i \tilde{h}_i g_i$ with $\tilde{h}_i \in \mathbb{C}[\mathbf{X}]$, so $f^N \in I$ and $f \in \sqrt{I}$. $\square$

§24.2.7 The Dictionary

We can now say precisely what I and $\mathbb{V}$ do.

Corollary 24.2.24

The maps $V \mapsto I(V)$ and $I \mapsto \mathbb{V}(I)$ are mutually inverse, inclusion-reversing bijections

$$\{\text{affine varieties in } \mathbb{A}^n\} \longleftrightarrow \{\text{radical ideals of } \mathbb{C}[X_1, \dots, X_n]\}.$$

Under this bijection, irreducible varieties correspond to prime ideals, and points correspond to maximal ideals.

Proof. We know $\mathbb{V}(I(V)) = V$ for varieties V , and $I(\mathbb{V}(I)) = \sqrt{I} = I$ for radical I . Also $I(V)$ is always radical, since if f^m vanishes on V then so does f . The correspondence for prime ideals is Proposition 24.2.18, and for maximal ideals it is Theorem 24.2.21(i). $\square$

This is the fundamental dictionary of the subject, and it is worth pausing over. Every geometric question about varieties in $\mathbb{A}^n$ can, in principle, be converted into an algebraic question about radical ideals of a polynomial ring. The rest of the course consists of doing exactly this, in increasingly refined ways.

Example 24.2.25

Let us see how the correspondence handles the classic example. In $\mathbb{C}[X]$, the ideal (X^2) is not radical, and $\sqrt{(X^2)} = (X)$. Correspondingly $\mathbb{V}(X^2) = \{0\} = \mathbb{V}(X)$: the variety cannot see the doubling. Later, when we come to intersection multiplicities and to schemes, we will want to remember the difference; for now we simply record that varieties forget it.

Example 24.2.26

Consider $I = (X^2 - Y^2, X^2 + Y^2 - 2) \subseteq \mathbb{C}[X, Y]$. Adding and subtracting, I contains $2X^2 - 2$ and $2Y^2 - 2$, so $I = (X^2 - 1, Y^2 - 1)$ and $\mathbb{V}(I) = \{(\pm 1, \pm 1)\}$, four points. The corresponding radical ideal is $I(\mathbb{V}(I)) = (X^2 - 1, Y^2 - 1)$, which is indeed radical – the quotient is $\mathbb{C}^4$ by the Chinese remainder theorem, and a finite product of fields has no nilpotents.

§24.3 Regular Functions and Morphisms

Having built our objects, we need the maps between them. As always in mathematics, the maps are more important than the objects, and here they will let us make precise the sense in which a variety ‘is’ its ring of functions.

§24.3.1 Coordinate Rings

Let $V \subseteq \mathbb{A}^n$ be a variety. Any polynomial $f \in \mathbb{C}[\mathbf{X}]$ restricts to a function $V \rightarrow \mathbb{C}$, but different polynomials may restrict to the same function. Precisely, f and g restrict to the same function on V if and only if $f - g$ vanishes on V , i.e. $f - g \in I(V)$. So the ring of functions $V \rightarrow \mathbb{C}$ obtained by restricting polynomials is the quotient $\mathbb{C}[\mathbf{X}]/I(V)$.

Definition 24.3.1 (Coordinate Ring)

Let $V \subseteq \mathbb{A}^n$ be an affine variety. Its **coordinate ring**, or **ring of regular functions**, is

$$\mathbb{C}[V] = \frac{\mathbb{C}[X_1, \dots, X_n]}{I(V)}.$$

Elements of $\mathbb{C}[V]$ are called **regular functions** on V . We sometimes write $\mathcal{O}(V)$ for $\mathbb{C}[V]$.

The coordinate ring is a finitely generated $\mathbb{C}$ -algebra (generated by the images x_i of the coordinate functions X_i), and it is *reduced*: it has no nonzero nilpotents, because $I(V)$ is radical. Conversely, every reduced finitely generated $\mathbb{C}$ -algebra arises this way, by writing it as $\mathbb{C}[\mathbf{X}]/I$ with I radical.

Corollary 24.3.2

A variety V is irreducible if and only if $\mathbb{C}[V]$ is an integral domain.

Proof. $\mathbb{C}[V]$ is a domain if and only if $I(V)$ is prime, which by Proposition 24.2.18

happens if and only if V is irreducible. $\square$

Example 24.3.3

$\mathbb{C}[\mathbb{A}^n] = \mathbb{C}[X_1, \dots, X_n]$, since $I(\mathbb{A}^n) = 0$.

Example 24.3.4

Let $V = \{0\} \subseteq \mathbb{A}^1$. We can present V as $\mathbb{V}(X^2)$, but $I(V) = \sqrt{(X^2)} = (X)$, so

$$\mathbb{C}[V] = \mathbb{C}[X]/(X) \cong \mathbb{C}.$$

The presentation as a doubled point is invisible to the coordinate ring. This is exactly the information that the theory of schemes is designed to retain: the ring $\mathbb{C}[X]/(X^2)$ is a perfectly good ring, with a nonzero element squaring to zero, and it remembers the doubling. We will not pursue this.

Example 24.3.5

Let $C = \mathbb{V}(Y - X^2) \subseteq \mathbb{A}^2$ be a parabola. Then $I(C) = (Y - X^2)$ (one checks that $Y - X^2$ is irreducible, so $(Y - X^2)$ is prime hence radical, and it certainly has the right vanishing locus). Hence

$$\mathbb{C}[C] = \mathbb{C}[X, Y]/(Y - X^2) \cong \mathbb{C}[X],$$

the isomorphism sending $Y \mapsto X^2$. So the parabola has the same coordinate ring as the affine line – and we will see in a moment that this means they are isomorphic as varieties.

Remark. One should be careful: $\mathbb{C}[V]$ on its own, as an abstract ring, does not remember the embedding $V \subseteq \mathbb{A}^n$. A surjection $\mathbb{C}[\mathbf{X}] \rightarrow \mathbb{C}[V]$ is the same thing as a presentation of V as a subvariety of $\mathbb{A}^n$, and there are many such. The parabola example above is a case in point: $\mathbb{C}[X] \cong \mathbb{C}[X, Y]/(Y - X^2)$ presents the same ring both as $\mathbb{A}^1 \subseteq \mathbb{A}^1$ and as a curve in $\mathbb{A}^2$.

§24.3.2 Morphisms

Definition 24.3.6 (Morphism)

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be affine varieties. A **morphism** (or **regular map**) $\varphi: V \rightarrow W$ is a function such that there exist $f_1, \dots, f_m \in \mathbb{C}[V]$ with

$$\varphi(P) = (f_1(P), \dots, f_m(P))$$

for all $P \in V$. We write $\text{Mor}(V, W)$ for the set of morphisms $V \rightarrow W$.

In other words, a morphism is a map given in coordinates by polynomials. Note the requirement that the image actually lands in W , which is a genuine condition on the f_i .

Example 24.3.7 (i) A morphism $V \rightarrow \mathbb{A}^1$ is precisely an element of $\mathbb{C}[V]$.

- (ii) Linear maps, translations, and more generally affine-linear maps $\mathbb{A}^n \rightarrow \mathbb{A}^m$ are morphisms.
- (iii) If $V \subseteq W$ are varieties in $\mathbb{A}^n$, the inclusion $V \hookrightarrow W$ is a morphism.
- (iv) The map $\mathbb{A}^1 \rightarrow \mathbb{A}^3$, $t \mapsto (t, t^2, t^3)$, is a morphism onto the twisted cubic.

Proposition 24.3.8

Compositions of morphisms are morphisms, so affine varieties and morphisms form a category.

Proof. A polynomial in polynomials is a polynomial. $\square$

Morphisms are automatically continuous for the Zariski topology: if $Z = \mathbb{V}(g_1, \dots, g_r) \subseteq W$ then $\varphi^{-1}(Z) = \mathbb{V}(g_1 \circ \varphi, \dots, g_r \circ \varphi)$ is closed in V . The converse is very far from true – for instance every bijection of $\mathbb{A}^1$ is a Zariski homeomorphism – so being a morphism is much stronger than being continuous.

§24.3.3 Pullbacks

The key construction is that a morphism induces a map on functions, going the other way.

Definition 24.3.9 (Pullback)

Let $\varphi: V \rightarrow W$ be a morphism and $g \in \mathbb{C}[W]$. The **pullback** of g is

$$\varphi^*(g) = g \circ \varphi: V \rightarrow \mathbb{C}.$$

If g is (the class of) a polynomial in the coordinates on $\mathbb{A}^m$, then $g \circ \varphi$ is a polynomial in the coordinates on $\mathbb{A}^n$, so $\varphi^*(g) \in \mathbb{C}[V]$. Thus $\varphi^*: \mathbb{C}[W] \rightarrow \mathbb{C}[V]$, and it is clearly a ring homomorphism which is the identity on the constants $\mathbb{C}$.

Definition 24.3.10

A ring homomorphism between $\mathbb{C}$ -algebras which restricts to the identity on $\mathbb{C}$ is a **$\mathbb{C}$ -algebra homomorphism**.

Theorem 24.3.11

Let $V \subseteq \mathbb{A}^n$ and $W \subseteq \mathbb{A}^m$ be affine varieties. The map $\varphi \mapsto \varphi^*$ is a bijection

$$\text{Mor}(V, W) \longleftrightarrow \{\mathbb{C}\text{-algebra homomorphisms } \mathbb{C}[W] \rightarrow \mathbb{C}[V]\}.$$

Proof. Let $y_1, \dots, y_m \in \mathbb{C}[W]$ be the coordinate functions, i.e. the restrictions to W of the standard coordinates on $\mathbb{A}^m$.

Injectivity. If $\varphi: V \rightarrow W$ is a morphism and $P \in V$, then

$$\varphi(P) = (y_1(\varphi(P)), \dots, y_m(\varphi(P))) = (\varphi^*(y_1)(P), \dots, \varphi^*(y_m)(P)),$$

so φ is determined by $\varphi^*(y_1), \dots, \varphi^*(y_m)$, hence by φ^* .

Surjectivity. Let $\lambda: \mathbb{C}[W] \rightarrow \mathbb{C}[V]$ be a $\mathbb{C}$ -algebra homomorphism, and put $f_j = \lambda(y_j) \in \mathbb{C}[V]$. Define $\varphi = (f_1, \dots, f_m): V \rightarrow \mathbb{A}^m$; this is given by polynomials, so we need only check its image lies in W . Let $P \in V$ and $g \in I(W)$. Since λ is a $\mathbb{C}$ -algebra homomorphism, it commutes with polynomial expressions, so

$$g(f_1(P), \dots, f_m(P)) = (g(\lambda(y_1), \dots, \lambda(y_m)))(P) = (\lambda(g(y_1, \dots, y_m)))(P).$$

But $g(y_1, \dots, y_m) = 0$ in $\mathbb{C}[W]$, because $g \in I(W)$. Hence $g(\varphi(P)) = 0$ for all $g \in I(W)$, so $\varphi(P) \in \mathbb{V}(I(W)) = W$. Finally $\varphi^*(y_j) = f_j = \lambda(y_j)$, and since the y_j generate $\mathbb{C}[W]$ as a $\mathbb{C}$ -algebra, $\varphi^* = \lambda$. $\square$

Definition 24.3.12 (Isomorphism)

Varieties V and W are **isomorphic** if there are morphisms $\varphi: V \rightarrow W$ and $\psi: W \rightarrow V$ with $\psi\varphi = \text{id}_V$ and $\varphi\psi = \text{id}_W$.

Corollary 24.3.13

$V \cong W$ as varieties if and only if $\mathbb{C}[V] \cong \mathbb{C}[W]$ as $\mathbb{C}$ -algebras.

Proof. Immediate from Theorem 24.3.11, since the bijection is compatible with composition (it is contravariantly functorial): $(\psi\varphi)^* = \varphi^*\psi^*$ and $\text{id}^* = \text{id}$. $\square$

This is the promised statement that a variety ‘is’ its coordinate ring. All geometric information about an affine variety is encoded in a finitely generated reduced $\mathbb{C}$ -algebra, and every such algebra arises. In the language of category theory, the functor $V \mapsto \mathbb{C}[V]$ is an anti-equivalence between the category of affine varieties and the category of finitely generated reduced $\mathbb{C}$ -algebras.

Example 24.3.14

The twisted cubic $C = \{(t, t^2, t^3)\}$ is isomorphic to $\mathbb{A}^1$. Indeed $I(C) = (X_2 - X_1^2, X_3 - X_1^3)$, and in $\mathbb{C}[C]$ we have $x_2 = x_1^2$, $x_3 = x_1^3$, so $\mathbb{C}[C] = \mathbb{C}[x_1] \cong \mathbb{C}[X]$.

Example 24.3.15

Not every bijective morphism is an isomorphism. Consider $\varphi: \mathbb{A}^1 \rightarrow C = \mathbb{V}(Y^2 - X^3) \subseteq \mathbb{A}^2$ given by $t \mapsto (t^2, t^3)$. This is a bijection (the inverse sends $(x, y) \neq (0, 0)$ to y/x , and $(0, 0)$ to 0). But $\varphi^*: \mathbb{C}[C] \rightarrow \mathbb{C}[t]$ has image $\mathbb{C}[t^2, t^3] \subsetneq \mathbb{C}[t]$, since $t \notin \mathbb{C}[t^2, t^3]$. So φ^* is not surjective and φ is not an isomorphism. Geometrically, the cuspidal cubic C has a singular point at the origin which the smooth line $\mathbb{A}^1$ does not; we will make this precise in Section 24.5.

Example 24.3.16

Let $V = \mathbb{V}(X_1X_2(X_1 - X_2)) \subseteq \mathbb{A}^2$, a union of three concurrent lines in the plane, and let $W = \mathbb{V}(X_1X_2, X_2X_3, X_3X_1) \subseteq \mathbb{A}^3$, the union of the three coordinate axes.

Both are unions of three lines meeting at a single point, and they are homeomorphic in the Euclidean topology. Nevertheless they are not isomorphic as varieties: the tangent space to V at the origin is two-dimensional (it sits inside a plane) while that of W is three-dimensional. We will be able to see this once we have defined tangent spaces.

§24.3.4 Rational Functions and Local Rings

Regular functions are rather rigid. On $\mathbb{A}^1$ the only regular functions are the polynomials, and there are many natural functions $-1/X$, say – that we would like to allow, provided we keep track of where they are undefined. The right framework is that of rational functions.

Throughout this subsection, let V be an *irreducible* affine variety, so that $\mathbb{C}[V]$ is an integral domain.

Definition 24.3.17 (Function Field)

The **function field** of an irreducible affine variety V is the field of fractions

$$\mathbb{C}(V) = \text{Frac}(\mathbb{C}[V]).$$

Its elements are called **rational functions** on V .

Definition 24.3.18

A rational function $\varphi \in \mathbb{C}(V)$ is **regular** (or **defined**) at $P \in V$ if it can be written as $\varphi = f/g$ with $f, g \in \mathbb{C}[V]$ and $g(P) \neq 0$. The set of points at which φ is regular is the **domain** of φ .

If $\varphi = f/g$ then φ genuinely is a function on the Zariski open set $V \setminus \mathbb{V}(g)$, and this set is nonempty (as $g \neq 0$ in $\mathbb{C}[V]$) hence dense, since V is irreducible. The domain of φ is the union of all such open sets, so it is a dense open subset of V . A rational function is thus a function defined ‘almost everywhere’.

Example 24.3.19

Consider $\varphi = X_1^2/X_2 \in \mathbb{C}(\mathbb{A}^2)$. As written it is defined off the locus $X_2 = 0$. Note that $X_1^3/(X_1X_2)$ is the *same* element of $\mathbb{C}(\mathbb{A}^2)$, even though the naive domain of the second expression is smaller. So an element of $\mathbb{C}(V)$ is not literally a function with a well-defined domain until we take the union over all representations. Here the domain really is $\mathbb{A}^2 \setminus \mathbb{V}(X_2)$.

Remark. A slicker way to say all this: a rational function on V is an equivalence class of pairs (U, f) where $U \subseteq V$ is nonempty open and $f: U \rightarrow \mathbb{C}$ is a regular function on U , and $(U, f) \sim (U', f')$ if $f = f'$ on some nonempty open subset of $U \cap U'$. Irreducibility guarantees that $U \cap U'$ is nonempty, so this really is an equivalence relation.

Just as regular functions assemble into morphisms, rational functions assemble into rational maps.

Definition 24.3.20 (Rational Map of Affine Varieties)

Let $V \subseteq \mathbb{A}^n$ be irreducible and $W \subseteq \mathbb{A}^m$ a variety. A **rational map** $\varphi: V \dashrightarrow W$ is a tuple $\varphi = (f_1, \dots, f_m)$ of rational functions $f_j \in \mathbb{C}(V)$ such that $\varphi(P) \in W$ for every P at which all the f_j are regular. The **domain** $\text{dom } \varphi$ is the set of such P , a dense open subset of V .

A rational map is thus a morphism defined on a dense open subset, and two rational maps agreeing on a dense open subset are equal. If $\varphi(\text{dom } \varphi)$ is dense in W we call φ **dominant**; in that case φ pulls back rational functions and gives an inclusion of fields $\varphi^*: \mathbb{C}(W) \hookrightarrow \mathbb{C}(V)$. We will develop all of this properly in the projective setting, where it belongs; for now, here is the example that everyone should have seen.

Example 24.3.21 (Rational parametrisation of the circle)

Let $C = \mathbb{V}(X^2 + Y^2 - 1) \subseteq \mathbb{A}^2$. Define $\varphi: C \dashrightarrow \mathbb{A}^1$ by

$$\varphi(x, y) = \frac{y}{1+x},$$

a rational map, regular away from the point $(-1, 0)$. Geometrically $\varphi(P)$ is the slope of the line joining $(-1, 0)$ to P . In the other direction, define $\psi: \mathbb{A}^1 \dashrightarrow C$ by

$$\psi(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right),$$

which is regular away from $t = \pm i$, and whose image does lie in C since $(1-t^2)^2 + (2t)^2 = (1+t^2)^2$. A direct substitution gives $\varphi\psi = \text{id}$ and $\psi\varphi = \text{id}$ wherever both sides are defined. So C and $\mathbb{A}^1$ are *birational*, even though φ is not defined everywhere: the conic and the line become isomorphic once we delete a point from each.

Reading the formula for ψ with $t = q/p$ rational and clearing denominators gives $(p^2 - q^2)^2 + (2pq)^2 = (p^2 + q^2)^2$, which is the classical parametrisation of Pythagorean triples. This is a good advertisement for the subject: a geometric statement about one conic contains an arithmetic statement about infinitely many integers.

The functions regular at a fixed point form a ring, and its structure is the local algebraic invariant we will use over and over again.

Definition 24.3.22 (Local Ring)

A ring R is a **local ring** if it has a unique maximal ideal.

Definition 24.3.23

Let V be an irreducible variety and $P \in V$. The **local ring of V at P** is

$$\mathcal{O}_{V,P} = \{\varphi \in \mathbb{C}(V) : \varphi \text{ is regular at } P\}.$$

Lemma 24.3.24

A ring R is a local ring if and only if $R \setminus R^\times$ is an ideal, in which case it is the unique maximal ideal.

Proof. An ideal $A \trianglelefteq R$ is proper if and only if A contains no unit, i.e. $A \subseteq R \setminus R^\times$. So if $R \setminus R^\times$ is an ideal, it is proper and contains every proper ideal, hence is the unique maximal ideal.

Conversely, suppose R is local with maximal ideal $\mathfrak{m}$. Then $\mathfrak{m} \subseteq R \setminus R^\times$. If $x \in R \setminus R^\times$ then $(x) \neq R$, so (x) is contained in a maximal ideal, which must be $\mathfrak{m}$. Hence $R \setminus R^\times = \mathfrak{m}$ is an ideal. $\square$

Proposition 24.3.25

$\mathcal{O}_{V,P}$ is a local ring, with maximal ideal

$$\mathfrak{m}_{V,P} = \{\varphi \in \mathcal{O}_{V,P} : \varphi(P) = 0\}.$$

Its units are exactly the φ with $\varphi(P) \neq 0$, and $\mathcal{O}_{V,P}/\mathfrak{m}_{V,P} \cong \mathbb{C}$.

Proof. If $\varphi = f/g \in \mathcal{O}_{V,P}$ with $g(P) \neq 0$, then φ is invertible in $\mathcal{O}_{V,P}$ if and only if $g/f \in \mathcal{O}_{V,P}$, which happens exactly when $f(P) \neq 0$, i.e. $\varphi(P) \neq 0$. So $\mathcal{O}_{V,P} \setminus \mathcal{O}_{V,P}^\times = \mathfrak{m}_{V,P}$, which is an ideal (being the kernel of the evaluation homomorphism $\varphi \mapsto \varphi(P)$). Lemma 24.3.24 applies. Evaluation is surjective onto $\mathbb{C}$, giving the last statement. $\square$

Example 24.3.26

Take $V = \mathbb{A}^1$ and $P = 0$. Then

$$\mathcal{O}_{\mathbb{A}^1,0} = \left\{ \frac{f(t)}{g(t)} : g(0) \neq 0 \right\} \subseteq \mathbb{C}(t),$$

with maximal ideal (t) . This ring sits inside the ring $\mathbb{C}[[t]]$ of formal power series: expanding $1/g(t)$ as a power series (possible since $g(0) \neq 0$) embeds $\mathcal{O}_{\mathbb{A}^1,0} \hookrightarrow \mathbb{C}[[t]]$. The ring $\mathbb{C}[[t]]$ is also local, with maximal ideal (t) , and one should think of $\mathcal{O}_{\mathbb{A}^1,0}$ as the ring of ‘algebraic’ functions with a power series expansion at 0.

The local ring $\mathcal{O}_{V,P}$ is precisely the localisation of $\mathbb{C}[V]$ at the prime ideal $I(P)/I(V)$, in the sense of commutative algebra. It records everything about V near P , and much of the second half of these notes consists of extracting geometry from these rings.

§24.4 Projective Varieties

Affine varieties are not quite the right objects. Two problems recur: distinct parallel lines in $\mathbb{A}^2$ fail to meet, and affine varieties are never compact (in the Euclidean topology) unless they are finite. Both are cured by working in projective space, which adds ‘points at infinity’ in a canonical way. The payoff is enormous: theorems become clean, intersection numbers become constant, and the geometry of curves becomes tractable.

§24.4.1 Projective Space

Definition 24.4.1 (Projectivisation)

Let U be a finite-dimensional complex vector space. The **projectivisation** $\mathbb{P}(U)$ is the set of one-dimensional linear subspaces ('lines through the origin') of U . We write $\mathbb{P}^n = \mathbb{P}(\mathbb{C}^{n+1})$, **projective n -space**.

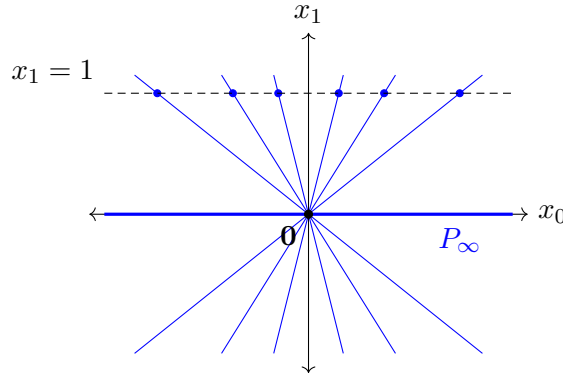
We index coordinates on $\mathbb{C}^{n+1}$ by $0, 1, \dots, n$. A line through the origin is spanned by some nonzero $(a_0, \dots, a_n)$, and two vectors span the same line exactly when they differ by a nonzero scalar. We write

$$(a_0 : a_1 : \dots : a_n) \in \mathbb{P}^n$$

for the point corresponding to the line spanned by $(a_0, \dots, a_n)$, and call these **homogeneous coordinates**. Thus

$$\mathbb{P}^n = \frac{\mathbb{C}^{n+1} \setminus \{0\}}{\mathbb{C}^\times},$$

where $\mathbb{C}^\times$ acts by scaling. For instance $(2 : 1 : -2) = (4 : 2 : -4) = (-1 : -\frac{1}{2} : 1)$ in $\mathbb{P}^2$.



The picture above (drawn over $\mathbb{R}$ for legibility) shows $\mathbb{P}^1$ as the set of lines through the origin in the plane. Almost every such line meets the horizontal line $x_1 = 1$ in exactly one point, which gives an identification of most of $\mathbb{P}^1$ with a copy of $\mathbb{A}^1$; the exception is the horizontal axis itself, which is 'the point at infinity'.

Let us record this decomposition in general. Splitting according to whether $a_0 = 0$,

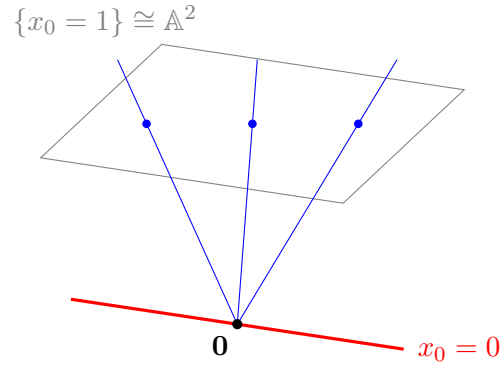
$$\begin{aligned} \mathbb{P}^n &= \{(a_0 : \dots : a_n) : a_0 \neq 0\} \sqcup \{(0 : a_1 : \dots : a_n)\} \\ &= \{(1 : b_1 : \dots : b_n) : b_i \in \mathbb{C}\} \sqcup \mathbb{P}^{n-1} \\ &\cong \mathbb{A}^n \sqcup \mathbb{P}^{n-1}, \end{aligned}$$

where the first identification uses that we may scale a_0 to 1, uniquely. Iterating,

$$\mathbb{P}^n = \mathbb{A}^n \sqcup \mathbb{A}^{n-1} \sqcup \dots \sqcup \mathbb{A}^1 \sqcup \{\text{pt}\}.$$

In particular $\mathbb{P}^1 = \mathbb{A}^1 \sqcup \{\infty\}$ is the Riemann sphere, and $\mathbb{P}^2$ is an affine plane together with a 'line at infinity'.

The same picture in one dimension higher is the one to carry around for the rest of the course. A point of $\mathbb{P}^2$ is a line through the origin in $\mathbb{C}^3$; such a line meets the affine plane $\{x_0 = 1\}$ in exactly one point unless it happens to lie in the plane $\{x_0 = 0\}$, in which case it contributes a point of the line at infinity.



The three blue lines are three points of $\mathbb{A}^2 \subseteq \mathbb{P}^2$; the red line, lying in the plane $x_0 = 0$ and so never meeting $\{x_0 = 1\}$, is one of the points of ℓ_∞ .

Definition 24.4.2 (Standard Affine Patches)

For $0 \leq j \leq n$ let $H_j = \{(a_0 : \cdots : a_n) : a_j = 0\}$, the *j th coordinate hyperplane*, and let $U_j = \mathbb{P}^n \setminus H_j$ be the *j th standard affine patch*.

There is a bijection $U_j \rightarrow \mathbb{A}^n$ given by

$$(a_0 : \cdots : a_n) \mapsto \left(\frac{a_0}{a_j}, \dots, \widehat{\frac{a_j}{a_j}}, \dots, \frac{a_n}{a_j} \right),$$

where the hat means ‘omit’, with inverse $(b_1, \dots, b_n) \mapsto (b_1 : \cdots : b_{j-1} : 1 : b_j : \cdots : b_n)$. Each H_j is a copy of $\mathbb{P}^{n-1}$. Since a point of $\mathbb{P}^n$ has some nonzero coordinate, $\{U_0, \dots, U_n\}$ covers $\mathbb{P}^n$: projective space is glued together out of $n + 1$ copies of affine space.

Definition 24.4.3

The **Zariski** (resp. **Euclidean**) topology on $\mathbb{P}^n$ is the quotient topology induced from the Zariski (resp. Euclidean) topology on $\mathbb{C}^{n+1} \setminus \{\mathbf{0}\}$ by the quotient map $\pi : \mathbb{C}^{n+1} \setminus \{\mathbf{0}\} \rightarrow \mathbb{P}^n$.

Proposition 24.4.4

$\mathbb{P}^n$ is compact in the Euclidean topology.

Proof. The unit sphere $S^{2n+1} \subseteq \mathbb{C}^{n+1} \setminus \{\mathbf{0}\}$ is compact, and every line through the origin meets it, so $\pi|_{S^{2n+1}}$ is surjective and continuous. A continuous image of a compact space is compact. $\square$

This is the second reason to prefer $\mathbb{P}^n$: it is compact, whereas $\mathbb{A}^n$ is not. Compactness will be responsible for a great deal of good behaviour, most notably the fact that the image of a projective variety under a morphism is closed.

§24.4.2 Homogeneous Ideals and Projective Varieties

We cannot simply take vanishing loci of arbitrary polynomials in $\mathbb{P}^n$.

Example 24.4.5

The polynomial $X_0 + 1 \in \mathbb{C}[X_0, X_1]$ does not define a function on $\mathbb{P}^1$, nor even a well-defined zero set: at the point $(-1 : 0) = (1 : 0)$ the representative $(-1, 0)$ gives 0 while $(1, 0)$ gives 2.

The fix is to restrict to polynomials whose vanishing is scale-invariant.

Definition 24.4.6 (Homogeneous Polynomial)

A **monomial** in $\mathbb{C}[X_0, \dots, X_n]$ is a product $X_0^{d_0} \cdots X_n^{d_n}$ with $d_i \geq 0$; its **degree** is $\sum_i d_i$. A polynomial is **homogeneous of degree d** if it is a $\mathbb{C}$ -linear combination of monomials of degree d .

Every $f \in \mathbb{C}[\mathbf{X}]$ decomposes uniquely as a finite sum $f = \sum_{r \geq 0} f_{[r]}$ with $f_{[r]}$ homogeneous of degree r ; we call $f_{[r]}$ the **homogeneous parts** of f .

Lemma 24.4.7

If F is homogeneous of degree d then $F(\lambda \mathbf{a}) = \lambda^d F(\mathbf{a})$. In particular, whether $F(\mathbf{a}) = 0$ depends only on the point $(\mathbf{a}) \in \mathbb{P}^n$, not on the representative.

Proof. Each monomial of degree d scales by λ^d . □

Definition 24.4.8 (Homogeneous Ideal)

An ideal $I \subseteq \mathbb{C}[X_0, \dots, X_n]$ is **homogeneous** if it can be generated by homogeneous polynomials (possibly of different degrees).

Lemma 24.4.9

An ideal I is homogeneous if and only if for every $f \in I$, all homogeneous parts $f_{[r]}$ lie in I .

Proof. ($\Leftarrow$) Take any generating set and replace each generator by its homogeneous parts.

($\Rightarrow$) Let I be generated by homogeneous $g_1, \dots, g_s$ with $\deg g_j = d_j$, and let $f = \sum_j h_j g_j \in I$. Splitting each h_j into homogeneous parts $h_{j[r]}$, the degree r part of f is $f_{[r]} = \sum_j h_{j[r-d_j]} g_j \in I$. □

Definition 24.4.10 (Projective Variety)

Let I be a homogeneous ideal of $\mathbb{C}[X_0, \dots, X_n]$. Its **vanishing locus** is

$$\mathbb{V}(I) = \{P \in \mathbb{P}^n : F(P) = 0 \text{ for all homogeneous } F \in I\}.$$

A **projective variety** is a subset of $\mathbb{P}^n$ of this form.

By Lemma 24.4.9 this is the same as requiring $f(P) = 0$ for all $f \in I$, interpreting

‘ $f(P) = 0$ ’ as ‘ $f(\mathbf{a}) = 0$ for one, equivalently every, representative $\mathbf{a}$ ’. As in the affine case, the projective varieties are the closed sets of a topology; we omit the (identical) verification.

Example 24.4.11

Let $U \leq \mathbb{C}^{n+1}$ be a linear subspace, say the kernel of a collection of linear forms L_j . Then $\mathbb{P}(U) = \mathbb{V}(\{L_j\}) \subseteq \mathbb{P}^n$ is a projective variety, called a **projective linear subspace**. Projective linear subspaces of $\mathbb{P}^n$ are in bijection with linear subspaces of $\mathbb{C}^{n+1}$; those of dimension 1 are called **lines**, and those of dimension $n - 1$ are **hyperplanes**.

Example 24.4.12

Two distinct lines in $\mathbb{P}^2$ always meet, in exactly one point. Indeed, they correspond to distinct 2-dimensional subspaces $U_1, U_2 \leq \mathbb{C}^3$, and $\dim(U_1 \cap U_2) \geq 2 + 2 - 3 = 1$, with equality since $U_1 \neq U_2$. This is the first fruit of working projectively: there are no parallel lines.

The group $\mathrm{GL}_{n+1}(\mathbb{C})$ acts on $\mathbb{P}^n$ by acting on homogeneous coordinates. Scalar matrices act trivially, so the action factors through

$$\mathrm{PGL}_{n+1}(\mathbb{C}) = \mathrm{GL}_{n+1}(\mathbb{C})/\mathbb{C}^\times,$$

and this group acts transitively on $\mathbb{P}^n$ (indeed on ordered $(n + 2)$ -tuples of points in general position). We refer to elements of $\mathrm{PGL}_{n+1}(\mathbb{C})$ as **projective changes of coordinates**, and we will use them freely to put varieties into convenient position.

§24.4.3 Affine Charts and Projective Closure

We have $\mathbb{P}^n = U_0 \cup \cdots \cup U_n$ with each $U_j \cong \mathbb{A}^n$. The relationship between projective varieties and their affine pieces goes both ways.

Suppose $V = \mathbb{V}(I) \subseteq \mathbb{P}^n$ is a projective variety. Identify $U_0 \cong \mathbb{A}^n$ with coordinates $Y_i = X_i/X_0$. Then

$$V \cap U_0 = \mathbb{V}(I_0), \quad I_0 = \{F(1, Y_1, \dots, Y_n) : F \in I \text{ homogeneous}\},$$

an affine variety. So *the intersection of a projective variety with an affine chart is an affine variety*: locally, projective varieties look affine.

Conversely, given an affine variety $W \subseteq \mathbb{A}^n \cong U_0$, we may take its closure in $\mathbb{P}^n$. The following makes this explicit.

Definition 24.4.13 (Homogenisation)

Let $f \in \mathbb{C}[Y_1, \dots, Y_n]$ have total degree d . Its **homogenisation** is

$$f^h(X_0, \dots, X_n) = X_0^d f\left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\right) \in \mathbb{C}[X_0, \dots, X_n],$$

which is homogeneous of degree d and not divisible by X_0 . For an ideal $I \subseteq \mathbb{C}[\mathbf{Y}]$

we write I^h for the ideal generated by $\{f^h : f \in I\}$; this is homogeneous. The **projective closure** of an affine variety $V \subseteq \mathbb{A}^n$ is $\mathbb{V}(I(V)^h) \subseteq \mathbb{P}^n$.

Example 24.4.14

If $f(Y_1, Y_2) = 1 + Y_1^2 + Y_1Y_2^2$ then $d = 3$ and $f^h = X_0^3 + X_0X_1^2 + X_1X_2^2$.

Remark. It is essential to homogenise *every* element of I , not merely a generating set. If $I = (f_1, \dots, f_r)$ then $(f_1^h, \dots, f_r^h) \subseteq I^h$ but the containment is often strict. (It is an equality when I is principal.) For example, in $\mathbb{C}[Y_1, Y_2, Y_3]$ take the ideal of the affine twisted cubic $I = (Y_2 - Y_1^2, Y_3 - Y_1^3)$; homogenising the two generators gives $X_0X_2 - X_1^2$ and $X_0^2X_3 - X_1^3$, whose common zero locus contains the whole line $\{X_0 = X_1 = 0\}$, which is not in the projective closure of the twisted cubic. One needs also the homogenisation of $Y_1Y_3 - Y_2^2 \in I$, namely $X_1X_3 - X_2^2$.

Proposition 24.4.15

Let $V \subseteq \mathbb{A}^n = U_0$ be an affine variety. Then the Zariski closure $\overline{V}$ of V in $\mathbb{P}^n$ equals the projective closure $\mathbb{V}(I(V)^h)$.

Proof. Write $I = I(V)$, which we may assume radical. Certainly $\mathbb{V}(I^h)$ is closed and contains V (as f^h restricted to U_0 is f), so $\overline{V} \subseteq \mathbb{V}(I^h)$.

Conversely, let $Y = \mathbb{V}(J) \supseteq V$ be any closed set, J homogeneous. Take a homogeneous $G \in J$; we may write $G = X_0^e f^h$ for some $f \in \mathbb{C}[\mathbf{Y}]$ and $e \geq 0$, by dehomogenising and re-homogenising. Now G vanishes on V , so $f = G(1, \mathbf{Y})$ vanishes on $V \subseteq \mathbb{A}^n$, so $f \in I(V) = I$. Hence $f^h \in I^h$ and so $G = X_0^e f^h \in I^h$. Therefore $J \subseteq I^h$ and $\mathbb{V}(I^h) \subseteq \mathbb{V}(J) = Y$. Taking $Y = \overline{V}$ gives the reverse inclusion. $\square$

Remark. The operations do not quite invert one another. If $V \subseteq \mathbb{P}^n$ is projective and $W = V \cap U_0$, then $\overline{W}$ need not be V : take $V = \mathbb{V}(X_0)$, so $W = \emptyset$ and $\overline{W} = \emptyset \neq V$. The problem is that V may have components entirely at infinity.

Example 24.4.16

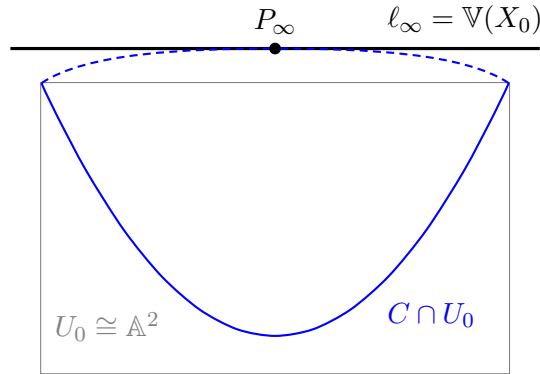
Let $V = \mathbb{V}(X_0X_1 - X_2^2) \subseteq \mathbb{P}^2$, a conic. Look at it in the three standard patches.

- In U_0 , setting $X_0 = 1$ and writing $(y_1, y_2) = (X_1/X_0, X_2/X_0)$, we get $y_1 = y_2^2$: a parabola.
- In U_1 we similarly get a parabola.
- In U_2 , setting $X_2 = 1$ and writing $(z_0, z_1) = (X_0/X_2, X_1/X_2)$, we get $z_0z_1 = 1$: a rectangular hyperbola.

So the parabola and the hyperbola are two views of the same projective curve, seen from different affine charts. The ellipse is a third. That the conics of school geometry become a single object is one of the most satisfying features of projective geometry.

It is worth drawing the picture that goes with this. Fix the chart $U_0 \cong \mathbb{A}^2$ and think of $\mathbb{P}^2$ as this affine plane together with the line at infinity $\ell_\infty = \mathbb{V}(X_0)$, whose points record the directions in which one can head off to infinity. A parabola in the affine plane

has both of its arms heading off in the same (vertical) direction, so in $\mathbb{P}^2$ they meet at a single point of ℓ_∞ , and the projective conic is a closed curve.



Here is the general classification, which is just the classification of quadratic forms from IB Linear Algebra.

Theorem 24.4.17

Let $Q = \mathbb{V}(F) \subseteq \mathbb{P}^n$ with F homogeneous of degree 2. After a projective change of coordinates, $F = X_0^2 + X_1^2 + \cdots + X_{r-1}^2$, where r is the rank of the quadratic form F .

Proof. Over $\mathbb{C}$ every quadratic form is equivalent to $\sum_{i < r} X_i^2$ where r is its rank. $\square$

§24.4.4 The Projective Nullstellensatz

Two homogeneous ideals may cut out the same projective variety for a trivial reason: the **irrelevant ideal** $\mathfrak{m}_0 = (X_0, \dots, X_n)$ has $\mathbb{V}(\mathfrak{m}_0) = \emptyset$ in $\mathbb{P}^n$ (its affine cone is just the origin, which we deleted). So the correct statement has to allow for this.

Definition 24.4.18

For $V \subseteq \mathbb{P}^n$, let $I^h(V) \trianglelefteq \mathbb{C}[X_0, \dots, X_n]$ be the ideal generated by all homogeneous polynomials vanishing on V .

Theorem 24.4.19 (Projective Nullstellensatz)

Let I be a homogeneous ideal of $\mathbb{C}[X_0, \dots, X_n]$.

- (i) $\mathbb{V}(I) = \emptyset$ in $\mathbb{P}^n$ if and only if $\sqrt{I} \supseteq (X_0, \dots, X_n)$.
- (ii) If $V = \mathbb{V}(I) \neq \emptyset$ then $I^h(V) = \sqrt{I}$.

Proof. Let $C(V) = \mathbb{V}(I) \subseteq \mathbb{A}^{n+1}$ denote the affine vanishing locus, the **affine cone**. Since I is homogeneous, $C(V)$ is stable under scaling and contains $\mathbf{0}$, and the quotient map restricts to a surjection $C(V) \setminus \{\mathbf{0}\} \rightarrow \mathbb{V}(I) \subseteq \mathbb{P}^n$. In particular $\mathbb{V}(I) = \emptyset$ if and only if $C(V) \subseteq \{\mathbf{0}\}$.

For (i), $C(V) \subseteq \{\mathbf{0}\}$ means $I(C(V)) \supseteq (X_0, \dots, X_n)$, and by the affine Nullstellen-

satz $I(C(V)) = \sqrt{I}$.

For (ii), the homogeneous polynomials vanishing on $\mathbb{V}(I) \subseteq \mathbb{P}^n$ are exactly the homogeneous polynomials vanishing on $C(V) \subseteq \mathbb{A}^{n+1}$, so $I^h(V) = I(C(V)) = \sqrt{I}$ by the affine Nullstellensatz again (using Lemma 24.4.9 to see that $I(C(V))$ is homogeneous). $\square$

Exactly as before, we get:

Proposition 24.4.20 (i) Every projective variety is a finite union of irreducible projective varieties, uniquely if the decomposition is minimal.

(ii) A projective variety V is irreducible if and only if $I^h(V)$ is prime.

Proof. (i) is verbatim the affine argument. For (ii), the only new point is that if a homogeneous ideal I is not prime then there exist *homogeneous* $F, G \notin I$ with $FG \in I$: take $f, g \notin I$ with $fg \in I$, and let F (resp. G) be a homogeneous part of f (resp. g) of largest degree not lying in I ; then the top-degree part of fg that could fail to lie in I is FG . With that in hand, the affine proof applies. $\square$

Proposition 24.4.21

Let $V \subseteq \mathbb{P}^n$ be irreducible and $W \subsetneq V$ a proper closed subvariety. Then $V \setminus W$ is dense in V .

Proof. Let F be homogeneous and vanish on $V \setminus W$. Since $W \neq V$, by the projective Nullstellensatz there is a homogeneous $G \in I^h(W) \setminus I^h(V)$. Then FG vanishes on W (because G does) and on $V \setminus W$ (because F does), hence on all of V . As $I^h(V)$ is prime and $G \notin I^h(V)$, we get $F \in I^h(V)$. So every homogeneous polynomial vanishing on $V \setminus W$ vanishes on V , which says exactly that $V \setminus W$ is dense. $\square$

This is the projective form of the principle that nonempty open subsets are large. Morally, W is lower-dimensional than V , and removing something lower-dimensional cannot disconnect or shrink an irreducible variety.

§24.4.5 Rational Functions on Projective Varieties

There is no useful analogue of the coordinate ring in the projective setting: a homogeneous polynomial has a well-defined zero set on $\mathbb{P}^n$ but not a well-defined value. However, a *ratio* of two homogeneous polynomials of the same degree *does* have a well-defined value wherever the denominator does not vanish, since the scaling factors λ^d cancel.

Definition 24.4.22 (Function Field)

Let $V \subseteq \mathbb{P}^n$ be an irreducible projective variety. Its **function field** is

$$\mathbb{C}(V) = \left\{ \frac{F}{G} : F, G \in \mathbb{C}[\mathbf{X}] \text{ homogeneous of the same degree, } G \notin I^h(V) \right\} / \sim,$$

where $F_1/G_1 \sim F_2/G_2$ if and only if $F_1G_2 - F_2G_1 \in I^h(V)$.

Lemma 24.4.23

The relation $\sim$ is an equivalence relation, and $\mathbb{C}(V)$ is a field.

Proof. Reflexivity and symmetry are clear. For transitivity, suppose $F_1G_2 \equiv F_2G_1$ and $F_2G_3 \equiv F_3G_2$ modulo $I^h(V)$. Working in the integral domain $R = \mathbb{C}[\mathbf{X}]/I^h(V)$ (a domain since $I^h(V)$ is prime), we get $\bar{F}_1\bar{G}_2\bar{G}_3 = \bar{F}_2\bar{G}_1\bar{G}_3 = \bar{F}_3\bar{G}_2\bar{G}_1$, so $\bar{G}_2(\bar{F}_1\bar{G}_3 - \bar{F}_3\bar{G}_1) = 0$. As $\bar{G}_2 \neq 0$ and R is a domain, $F_1G_3 - F_3G_1 \in I^h(V)$.

The field operations are the evident ones (note that $F_1/G_1 + F_2/G_2 = (F_1G_2 + F_2G_1)/G_1G_2$ is again a ratio of forms of equal degree), and F/G is invertible whenever $F \notin I^h(V)$. $\square$

Proposition 24.4.24

$\mathbb{C}(V)$ is a finitely generated field extension of $\mathbb{C}$.

Proof. Some coordinate, say X_0 , does not vanish identically on V . We claim $\mathbb{C}(V) = \mathbb{C}(x_1, \dots, x_n)$ where x_i is the class of X_i/X_0 . Given a ratio F/G of forms of degree d , divide numerator and denominator by X_0^d ; then F/X_0^d is a polynomial in the X_i/X_0 , and likewise for G . So F/G lies in the field generated by the x_i . $\square$

Corollary 24.4.25

Let $V \subseteq \mathbb{P}^n$ be irreducible with $V \not\subseteq \{X_0 = 0\}$, and let $V_0 = V \cap U_0$, an irreducible affine variety. Then $\mathbb{C}(V) \cong \mathbb{C}(V_0)$.

Proof. $\mathbb{C}[V_0] = \mathbb{C}[x_1, \dots, x_n]$ where x_i is the restriction of X_i/X_0 , so $\mathbb{C}(V_0) = \text{Frac } \mathbb{C}[V_0]$ is generated by the x_i , and the proof above shows $\mathbb{C}(V)$ is too. One checks the relations match, giving an isomorphism. $\square$

So the function field is a birational invariant that can be computed in any affine chart. Regularity and local rings are defined exactly as before: $\varphi \in \mathbb{C}(V)$ is **regular at P** if it can be written F/G with $G(P) \neq 0$, and $\mathcal{O}_{V,P} \subseteq \mathbb{C}(V)$ is the local ring of germs at P , with maximal ideal $\mathfrak{m}_P$. Under the isomorphism of Corollary 24.4.25, $\mathcal{O}_{V,P} \cong \mathcal{O}_{V_0,P}$ for $P \in V_0$.

§24.4.6 Rational Maps and Morphisms**Proposition 24.4.26**

Let $F_0, \dots, F_m \in \mathbb{C}[X_0, \dots, X_n]$ be homogeneous of the same degree d , not all zero. Then $\mathbf{F} = (F_0, \dots, F_m)$ descends to a well-defined map

$$\varphi: \mathbb{P}^n \setminus \mathbb{V}(F_0, \dots, F_m) \rightarrow \mathbb{P}^m, \quad P \mapsto (F_0(\mathbf{a}) : \dots : F_m(\mathbf{a})),$$

where $\mathbf{a}$ is any representative of P .

Proof. Replacing $\mathbf{a}$ by $\lambda\mathbf{a}$ multiplies every $F_j(\mathbf{a})$ by λ^d , which does not change the point of $\mathbb{P}^m$. The condition $P \notin \mathbb{V}(F_0, \dots, F_m)$ ensures not all coordinates are zero. $\square$

Two tuples can define the same map on the overlap of their domains: multiplying all F_j by a fixed homogeneous G gives a tuple defining the same map wherever both are defined, but on a smaller set. The following definition is designed to handle this.

Definition 24.4.27 (Rational Map)

Let $V \subseteq \mathbb{P}^n$ be irreducible. A **rational map** $\varphi: V \dashrightarrow \mathbb{P}^m$ is an equivalence class of tuples $(F_0, \dots, F_m)$ of homogeneous polynomials of a common degree, not all in $I^h(V)$, where

$$(F_0, \dots, F_m) \sim (G_0, \dots, G_m) \iff F_i G_j - F_j G_i \in I^h(V) \text{ for all } i, j.$$

A point $P \in V$ is a **regular point** of φ if some representative has $F_j(P) \neq 0$ for some j . The **domain** $\text{dom } \varphi$ is the set of regular points. If $\text{dom } \varphi = V$, we call φ a **morphism** and write $\varphi: V \rightarrow \mathbb{P}^m$.

If $W \subseteq \mathbb{P}^m$ is a projective variety, a rational map (or morphism) $V \dashrightarrow W$ is one $V \dashrightarrow \mathbb{P}^m$ whose image lies in W . An **isomorphism** is a morphism with a two-sided inverse morphism.

The subtlety here – that a rational map may be regular at a point without any *obvious* representative witnessing it – is important, and the next example is the one to remember.

Example 24.4.28 (Projection from a point)

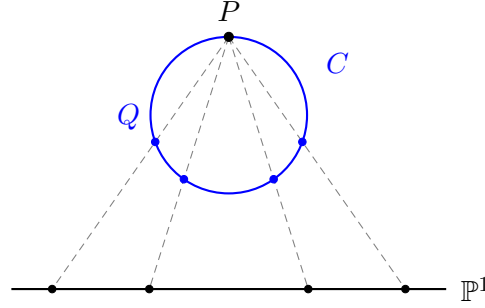
Let $P = (0 : 0 : 1) \in \mathbb{P}^2$ and define $\pi: \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$ by $\pi(a_0 : a_1 : a_2) = (a_0 : a_1)$. This is undefined exactly at P , and it is regular everywhere else. Geometrically, π sends a point $Q \neq P$ to the line PQ , thought of as a point of the $\mathbb{P}^1$ of lines through P .

Example 24.4.29

Let $C = \mathbb{V}(X_0 X_2 - X_1^2) \subseteq \mathbb{P}^2$ and restrict the projection π of the previous example to C . Note $P = (0 : 0 : 1) \in C$, so *a priori* $\pi|_C$ is undefined at P . But $\pi|_C$ is represented by (X_0, X_1) , and we may look for another representative (F_0, F_1) with

$$F_0 X_1 - F_1 X_0 \in I^h(C) = (X_0 X_2 - X_1^2).$$

The pair (X_1, X_2) works, since $X_1 \cdot X_1 - X_2 \cdot X_0 = -(X_0 X_2 - X_1^2)$. At $P = (0 : 0 : 1)$ we have $X_2(P) = 1 \neq 0$, so $\pi|_C$ is regular at P , with value $\pi(P) = (0 : 1)$. Hence $\pi|_C: C \rightarrow \mathbb{P}^1$ is a morphism.



The picture shows the construction: a point $Q \in C$ is sent to the point where the line PQ meets a fixed line. Since a line meets a conic in two points, and one of them is P , the other is determined, and this is why the map is a bijection.

Proposition 24.4.30

Let $C = \mathbb{V}(F) \subseteq \mathbb{P}^2$ with F an irreducible homogeneous quadratic. Then $C \cong \mathbb{P}^1$.

Proof. An irreducible quadratic form in three variables has rank 3; after a change of coordinates we may take $F = X_0X_2 - X_1^2$. By Example 24.4.29 we have a morphism $\pi: C \rightarrow \mathbb{P}^1$. Define $\mu: \mathbb{P}^1 \rightarrow \mathbb{P}^2$ by

$$\mu(Y_0 : Y_1) = (Y_0^2 : Y_0Y_1 : Y_1^2).$$

This is a morphism (the three forms have no common zero on $\mathbb{P}^1$), and its image lies in C since $Y_0^2 \cdot Y_1^2 - (Y_0Y_1)^2 = 0$. Now $\pi(\mu(Y_0 : Y_1)) = (Y_0^2 : Y_0Y_1) = (Y_0 : Y_1)$ whenever $Y_0 \neq 0$, and using the other representative (X_1, X_2) of π we get $(Y_0Y_1 : Y_1^2) = (Y_0 : Y_1)$ whenever $Y_1 \neq 0$; so $\pi\mu = \text{id}$. Conversely, for $(a_0 : a_1 : a_2) \in C$ we have $a_0a_2 = a_1^2$, so $\mu(a_0 : a_1) = (a_0^2 : a_0a_1 : a_1^2) = (a_0^2 : a_0a_1 : a_0a_2) = (a_0 : a_1 : a_2)$ when $a_0 \neq 0$, and similarly on the other patch. So $\mu\pi = \text{id}$. $\square$

So up to isomorphism there is only one smooth conic in $\mathbb{P}^2$, and it is a line. The distinction between lines and conics is a matter of how they are embedded, not of their intrinsic geometry. We will see in due course that this ceases to be true in degree 3 and beyond.

Example 24.4.31 (Veronese embedding)

Fix $n, d \geq 1$ and let $F_0, \dots, F_m$ be all the monomials of degree d in $X_0, \dots, X_n$, so $m = \binom{n+d}{d} - 1$. The **Veronese map** is

$$\nu_d: \mathbb{P}^n \rightarrow \mathbb{P}^m, \quad \mathbf{a} \mapsto (F_0(\mathbf{a}) : \dots : F_m(\mathbf{a})).$$

It is a morphism, since the F_j include X_i^d for each i and these have no common zero. One can show ν_d is injective with image a projective variety isomorphic to $\mathbb{P}^n$. The map μ used in Proposition 24.4.30 is ν_2 for $n = 1$. The point of the construction is that ν_d turns degree d hypersurfaces into hyperplane sections.

Example 24.4.32 (Segre embedding)

Products of projective spaces are not projective spaces: $\mathbb{P}^n \times \mathbb{P}^m \not\cong \mathbb{P}^{n+m}$. But they

are still projective varieties. The **Segre map** is

$$\sigma_{n,m}: \mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{nm+n+m}, \quad ((x_i), (y_j)) \mapsto (x_i y_j)_{i,j},$$

where we index the target coordinates by Z_{ij} , $0 \leq i \leq n$, $0 \leq j \leq m$. Note $(n+1)(m+1) - 1 = nm + n + m$.

Theorem 24.4.33

The Segre map $\sigma_{n,m}$ is a bijection onto the projective variety $\mathbb{V}(I)$, where I is generated by the quadrics $Z_{ij}Z_{pq} - Z_{iq}Z_{pj}$.

Proof. Clearly the image lies in $\mathbb{V}(I)$. Work on the patch $Z_{00} \neq 0$ and set $y_{ij} = Z_{ij}/Z_{00}$. The relations $Z_{ij}Z_{00} = Z_{i0}Z_{0j}$ become $y_{ij} = y_{i0}y_{0j}$, and one checks the remaining relations follow. So on this patch $\mathbb{V}(I)$ is the graph of the map $\mathbb{A}^n \times \mathbb{A}^m \rightarrow \mathbb{A}^{nm}$, $((y_{i0}), (y_{0j})) \mapsto (y_{i0}y_{0j})$, and hence

$$(y_{ij}) \longmapsto ((y_{10}, \dots, y_{n0}), (y_{01}, \dots, y_{0m}))$$

is a two-sided inverse to $\sigma_{n,m}$ there. The same works on each patch $Z_{ij} \neq 0$. $\square$

Concretely, $\sigma_{1,1}: \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$ has image the quadric surface $\mathbb{V}(Z_{00}Z_{11} - Z_{01}Z_{10}) \subseteq \mathbb{P}^3$: identifying $\mathbb{C}^4$ with 2×2 matrices, $\sigma_{1,1}$ is the outer product $(v, w) \mapsto vw^T$, whose image is the set of matrices of rank at most one, cut out by the vanishing of the determinant. As a corollary, products of projective varieties are projective varieties.

§24.4.7 Birational Maps

Composing rational maps is delicate: the image of $\text{dom } \varphi$ might land entirely inside the indeterminacy locus of ψ . The following condition rules this out.

Definition 24.4.34 (Dominant)

A rational map $\varphi: V \dashrightarrow W$ is **dominant** if $\varphi(\text{dom } \varphi)$ is Zariski dense in W .

Proposition 24.4.35

If $\varphi: V \dashrightarrow W$ is dominant then $\psi \circ \varphi$ is a well-defined rational map for any rational $\psi: W \dashrightarrow Z$.

Proof. $\text{dom } \psi$ is a nonempty open subset of W , and $\varphi(\text{dom } \varphi)$ is dense, so $\varphi^{-1}(\text{dom } \psi)$ is a nonempty open subset of $\text{dom } \varphi$. On it, $\psi \circ \varphi$ is given by composing tuples of homogeneous polynomials, hence is a rational map. $\square$

A dominant rational map $\varphi: V \dashrightarrow W$ induces a $\mathbb{C}$ -algebra map on function fields $\varphi^*: \mathbb{C}(W) \rightarrow \mathbb{C}(V)$ by $f \mapsto f \circ \varphi$; this is a map of fields, hence injective.

Definition 24.4.36 (Birational)

Irreducible varieties V, W are **birational** if there are dominant rational maps $\varphi: V \dashrightarrow W$ and $\psi: W \dashrightarrow V$ with $\psi\varphi$ and $\varphi\psi$ equal to the identity as rational maps. Such φ is called a **birational map**.

Equivalently, V and W are birational if they contain isomorphic nonempty open subsets. Every isomorphism is birational, but not conversely.

Example 24.4.37 (Cremona transformation)

Consider $\kappa: \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ given by

$$\kappa(X_0 : X_1 : X_2) = (X_1X_2 : X_0X_2 : X_0X_1),$$

which is $(x_0 : x_1 : x_2) \mapsto (1/x_0 : 1/x_1 : 1/x_2)$ cleared of denominators. It is undefined exactly at the three coordinate points $(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1)$. One computes $\kappa \circ \kappa = \text{id}$ as a rational map, so κ is birational; it is not an isomorphism, since it is not everywhere defined. The Cremona transformation sends a general line to a conic, so it certainly does not preserve degrees.

Remark. The group $\text{Bir}(\mathbb{P}^2)$ of birational automorphisms of $\mathbb{P}^2$ is called the *Cremona group*. A theorem of Noether and Castelnuovo says it is generated by $\text{PGL}_3(\mathbb{C})$ together with κ . The analogous statement in higher dimensions is false and the structure of $\text{Bir}(\mathbb{P}^n)$ for $n \geq 3$ remains mysterious.

Theorem 24.4.38

Let V, W be irreducible projective varieties. Then V and W are birational if and only if $\mathbb{C}(V) \cong \mathbb{C}(W)$ as $\mathbb{C}$ -algebras.

Proof. $(\Rightarrow)$ A birational map φ gives $\varphi^*: \mathbb{C}(W) \rightarrow \mathbb{C}(V)$ and φ^{-1} gives an inverse, so these are isomorphisms.

$(\Leftarrow)$ Suppose $\Theta: \mathbb{C}(W) \rightarrow \mathbb{C}(V)$ is a $\mathbb{C}$ -algebra isomorphism. We may assume $V \subseteq \mathbb{P}^n$ is not contained in $\{X_0 = 0\}$ and $W \subseteq \mathbb{P}^m$ is not contained in $\{Y_0 = 0\}$. As we showed, $\mathbb{C}(V) = \mathbb{C}(x_1, \dots, x_n)$ with $x_i = X_i/X_0$ and $\mathbb{C}(W) = \mathbb{C}(y_1, \dots, y_m)$ with $y_j = Y_j/Y_0$. Write $\Theta(y_j) = f_j(x_1, \dots, x_n)$ for rational functions f_j of n variables. Clearing denominators and homogenising with respect to X_0 , there are homogeneous $F_0, \dots, F_m$ of a common degree with

$$f_j\left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\right) = \frac{F_j(X_0, \dots, X_n)}{F_0(X_0, \dots, X_n)}.$$

Then $(F_0 : \dots : F_m)$ defines a rational map $\varphi: V \dashrightarrow \mathbb{P}^m$; its image lies in W , since any homogeneous $G \in I^h(W)$ pulls back to $G(f_1, \dots, f_m)$ up to a factor, which is Θ applied to the relation $G(y_1, \dots, y_m) = 0$, hence zero. Running the same argument with Θ^{-1} gives $\psi: W \dashrightarrow V$, and the composites induce the identity on function fields, hence are the identity as rational maps. $\square$

This should be compared with Corollary 24.3.13: coordinate rings classify affine varieties up to isomorphism, and function fields classify irreducible varieties up to birational equivalence. Classifying varieties up to birational equivalence is thus the same as classifying finitely generated field extensions of $\mathbb{C}$, and for curves we will be able to do this

completely.

§24.5 Tangent Spaces, Smoothness and Dimension

So far our varieties have been rather formless: we have no notion of how big one is, and no way to distinguish the smooth curve $y = x^2$ from the cuspidal one $y^2 = x^3$. Both defects are repaired at once by the tangent space, which is a purely algebraic construction imitating the derivative.

§24.5.1 The Tangent Space to a Hypersurface

Let $V = \mathbb{V}(f) \subseteq \mathbb{A}^n$ be a hypersurface and $P = (a_1, \dots, a_n) \in V$. A line through P has the form

$$L = \{(a_1 + b_1 t, \dots, a_n + b_n t) : t \in \mathbb{C}\}$$

for some $\mathbf{b} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$. Substituting into f gives a polynomial in one variable,

$$g(t) = f(a_1 + b_1 t, \dots, a_n + b_n t) = \sum_{r \geq 0} c_r t^r,$$

whose roots are the parameters at which L meets V . Since $P \in V$ we have $c_0 = 0$, so $t = 0$ is a root; and by the chain rule

$$c_1 = g'(0) = \sum_{i=1}^n b_i \frac{\partial f}{\partial X_i}(P).$$

The line meets V at P ‘to order at least two’ – geometrically, is tangent – exactly when $c_1 = 0$ as well.

Definition 24.5.1 (Tangent Space to a Hypersurface)

Let $V = \mathbb{V}(f) \subseteq \mathbb{A}^n$ and $P \in V$. The **(affine embedded) tangent space** to V at P is the affine subspace

$$T_{V,P}^{\text{aff}} = \mathbb{V} \left(\sum_{i=1}^n \frac{\partial f}{\partial X_i}(P) (X_i - a_i) \right) \subseteq \mathbb{A}^n.$$

A line L through P is **tangent** to V at P if $L \subseteq T_{V,P}^{\text{aff}}$.

The polynomial cutting out $T_{V,P}^{\text{aff}}$ is linear, so $T_{V,P}^{\text{aff}}$ is either a hyperplane (dimension $n - 1$), when some partial derivative is nonzero at P , or all of $\mathbb{A}^n$, when they all vanish.

Definition 24.5.2 (Smooth and Singular Points)

With the notation above, P is a **singular point** of V if $\frac{\partial f}{\partial X_i}(P) = 0$ for all i , equivalently $\dim T_{V,P}^{\text{aff}} = n$. Otherwise P is a **smooth** (or **nonsingular**) point. V is **smooth** if all its points are.

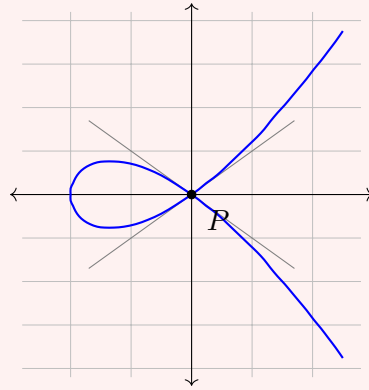
This is the **Jacobian criterion**: a point is singular precisely when the gradient of the defining equation vanishes there. Let us test it on the standard examples.

Example 24.5.3 (Nodal cubic)

Let $C = \mathbb{V}(f)$ with $f = Y^2 - X^2(X + 1) = Y^2 - X^3 - X^2$. Then

$$\frac{\partial f}{\partial X} = -3X^2 - 2X, \quad \frac{\partial f}{\partial Y} = 2Y.$$

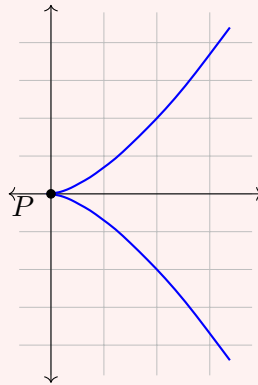
Both vanish when $Y = 0$ and $X(3X + 2) = 0$, i.e. at $(0, 0)$ and $(-2/3, 0)$; but $(-2/3, 0) \notin C$. So the unique singular point is the origin. Near the origin, $Y^2 \approx X^2$, and indeed the real picture shows two branches crossing transversally, with tangent lines $Y = \pm X$: this is a **node**.



The two grey lines are the tangent directions to the two branches; note that neither is ‘the’ tangent line, and that the tangent space at P is the whole plane.

Example 24.5.4 (Cuspidal cubic)

Let $C = \mathbb{V}(Y^2 - X^3)$. Then $\partial f/\partial X = -3X^2$ and $\partial f/\partial Y = 2Y$, both vanishing only at the origin, which lies on C . So again the origin is the unique singular point, but the singularity is of a different kind: the curve has a single branch with a **cusp**, parametrised by $t \mapsto (t^2, t^3)$.

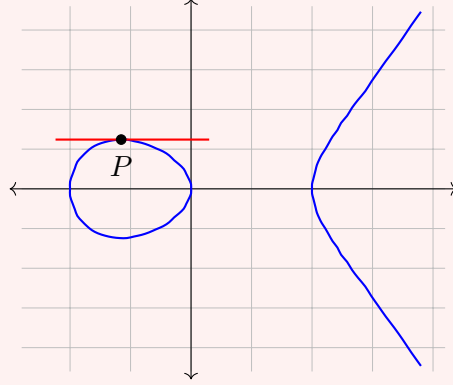
**Example 24.5.5 (A smooth point)**

Contrast the previous two examples with $C = \mathbb{V}(f)$, $f = Y^2 - X^3 + X$, at the point $P = (-1/\sqrt{3}, y_0) \approx (-0.577, 0.620)$, where $y_0 = \sqrt{2}3^{-3/4}$. Here $\partial f/\partial Y = 2y_0 \neq 0$, so P is a smooth point, the tangent space is a line, and there is a genuine tangent

line: differentiating implicitly, its slope is

$$\left. \frac{3X^2 - 1}{2Y} \right|_P = \frac{3 \cdot \frac{1}{3} - 1}{2y_0} = 0,$$

so the tangent at P is the horizontal line $Y = y_0$. In fact this curve is smooth everywhere: $\partial f / \partial Y = 2Y = 0$ forces $Y = 0$, and then $X^3 = X$ and $3X^2 = 1$ cannot both hold.



Here the tangent line touches the curve and stays on one side of it near P – compare the node, where no single line is tangent, and the cusp, where the tangent ‘line’ degenerates entirely.

§24.5.2 Tangent Spaces in General

For a general variety, cut out by several equations, we take the intersection of the tangent hyperplanes and record it as a vector subspace rather than an affine one.

Definition 24.5.6 (Tangent Space)

Let $V \subseteq \mathbb{A}^n$ be a variety and $P \in V$. The **tangent space** is

$$T_{V,P} = \left\{ \mathbf{v} \in \mathbb{C}^n : \sum_{i=1}^n v_i \frac{\partial f}{\partial X_i}(P) = 0 \text{ for all } f \in I(V) \right\} \leq \mathbb{C}^n.$$

This is a linear subspace of $\mathbb{C}^n$, and it suffices to impose the conditions for f running over a generating set of $I(V)$, by the product and sum rules for derivatives. Concretely, if $I(V) = (f_1, \dots, f_r)$ then $T_{V,P}$ is the kernel of the **Jacobian matrix**

$$J(P) = \left(\frac{\partial f_j}{\partial X_i}(P) \right)_{j,i},$$

so that

$$\dim T_{V,P} = n - \text{rank } J(P).$$

Example 24.5.7

Revisiting Example 24.3.16: for $V = \mathbb{V}(X_1 X_2 (X_1 - X_2)) \subseteq \mathbb{A}^2$ we have $I(V) = (X_1 X_2 (X_1 - X_2))$, whose gradient is a vector of quadratics, vanishing at the origin. So $T_{V,0} = \mathbb{C}^2$. For $W = \mathbb{V}(X_1 X_2, X_2 X_3, X_3 X_1) \subseteq \mathbb{A}^3$ every generator has vanishing

gradient at the origin, so $T_{W,0} = \mathbb{C}^3$. If V and W were isomorphic, some isomorphism would carry the origin to the origin (it is the unique point of each where the local structure is singular), and would induce a linear isomorphism of tangent spaces. Since $2 \neq 3$, $V \not\cong W$.

To transport tangent spaces along maps we need the derivative. Let $\varphi: V \dashrightarrow W$ be a rational map of irreducible varieties, $P \in \text{dom } \varphi$ and $Q = \varphi(P)$. Working in affine charts, φ is given by rational functions $f_1, \dots, f_m$ regular at P , and we set

$$d\varphi_P(\mathbf{v}) = \left(\sum_{i=1}^n v_i \frac{\partial f_j}{\partial X_i}(P) \right)_{j=1}^m,$$

a linear map $\mathbb{C}^n \rightarrow \mathbb{C}^m$.

Proposition 24.5.8

With the above notation:

- (i) $d\varphi_P(T_{V,P}) \subseteq T_{W,Q}$;
- (ii) $d\varphi_P$ depends only on φ , not on the chosen representatives;
- (iii) if $\psi: W \dashrightarrow Z$ is rational with $Q \in \text{dom } \psi$ then $d(\psi\varphi)_P = d\psi_Q \circ d\varphi_P$;
- (iv) if φ is birational and φ^{-1} is regular at Q , then $d\varphi_P: T_{V,P} \rightarrow T_{W,Q}$ is an isomorphism.

Proof. (i) Let $g \in I(W)$, with coordinates Y_j on W . Then $\varphi^*g = g(f_1, \dots, f_m)$ vanishes identically on V near P , so by the chain rule, for $\mathbf{v} \in T_{V,P}$,

$$0 = \sum_i v_i \frac{\partial(\varphi^*g)}{\partial X_i}(P) = \sum_j \frac{\partial g}{\partial Y_j}(Q) \left(\sum_i v_i \frac{\partial f_j}{\partial X_i}(P) \right),$$

which says exactly that $d\varphi_P(\mathbf{v}) \in T_{W,Q}$.

(ii) If f'_j is another representative, then $f_j - f'_j$ vanishes on V where defined, so $f_j - f'_j = p_j/q_j$ with $p_j \in I(V)$ and $q_j(P) \neq 0$. Differentiating by the quotient rule and using $p_j(P) = 0$,

$$\frac{\partial(f_j - f'_j)}{\partial X_i}(P) = \frac{1}{q_j(P)} \frac{\partial p_j}{\partial X_i}(P),$$

and since $p_j \in I(V)$, contracting with $\mathbf{v} \in T_{V,P}$ gives zero.

(iii) is the chain rule, and (iv) follows from (iii) applied to $\varphi^{-1}\varphi = \text{id}$. $\square$

For projective varieties we define $T_{V,P}$ by choosing an affine patch $U_j \ni P$ and setting $T_{V,P} = T_{V \cap U_j, P}$. Part (iv) of Proposition 24.5.8, applied to the transition maps between patches (which are birational and regular on the overlaps), shows this is independent of j up to canonical isomorphism.

There is also a convenient projective form of the tangent space to a hypersurface.

Definition 24.5.9

Let $V = \mathbb{V}(F) \subseteq \mathbb{P}^n$ with F homogeneous. The **projective tangent space** at $P \in V$ is

$$T_{V,P}^{\text{proj}} = \mathbb{V} \left(\sum_{i=0}^n \frac{\partial F}{\partial X_i}(P) X_i \right) \subseteq \mathbb{P}^n.$$

That P itself lies in this hyperplane follows from **Euler's relation**: for F homogeneous of degree d ,

$$\sum_{i=0}^n X_i \frac{\partial F}{\partial X_i} = d F,$$

which is checked on monomials. Dehomogenising shows $T_{V,P}^{\text{proj}} \cap U_0 = T_{V \cap U_0, P}^{\text{aff}}$, so the two notions agree.

Proposition 24.5.10

Let $V = \mathbb{V}(F) \subseteq \mathbb{P}^n$ be an irreducible hypersurface. Then the singular locus of V is a proper Zariski closed subset; in particular the smooth points are dense.

Proof. The singular locus is $V \cap \bigcap_i \mathbb{V}(\partial F / \partial X_i)$, which is closed. Suppose it were all of V . Then each $\partial F / \partial X_i$ vanishes on V , so lies in $I^h(V) = (F)$ (as F is irreducible, (F) is prime hence radical). But $\partial F / \partial X_i$ is homogeneous of degree $\deg F - 1 < \deg F$, so divisibility by F forces $\partial F / \partial X_i = 0$ for all i . In characteristic zero this means F is constant, so $V = \mathbb{P}^n$ or $\emptyset$, contradicting that V is a hypersurface. $\square$

§24.5.3 Dimension

We can now define dimension. The idea is that the tangent space has the ‘expected’ dimension at most points, and jumps up at bad points.

Definition 24.5.11 (Dimension)

Let V be an irreducible affine or projective variety. Its **dimension** is

$$\dim V = \min_{P \in V} \dim T_{V,P}.$$

A point $P \in V$ is **smooth** if $\dim T_{V,P} = \dim V$, and **singular** otherwise. For reducible V , we set $\dim V$ to be the maximum of the dimensions of the irreducible components.

A variety of dimension 1 is a **curve**, of dimension 2 a **surface**.

Theorem 24.5.12

Let V be a nonempty irreducible variety. The set of smooth points of V is a nonempty Zariski open subset, hence dense.

Proof. Nonemptiness is by definition, since the minimum is attained. We may work in an affine chart, so assume $V \subseteq \mathbb{A}^n$ with $I(V) = (f_1, \dots, f_r)$. Then $\dim T_{V,P} =$

$n - \text{rank } J(P)$ where J is the Jacobian. For any s ,

$$\{P \in V : \dim T_{V,P} \geq s\} = \{P \in V : \text{rank } J(P) \leq n - s\},$$

and the locus where a matrix has rank at most k is cut out by the vanishing of all its $(k+1) \times (k+1)$ minors, which are polynomials in the entries, hence in P . So this set is closed. Taking $s = \dim V + 1$, the singular locus is closed and proper, so the smooth locus is open and nonempty; density follows from Proposition 24.4.21. $\square$

Corollary 24.5.13

If V and W are irreducible and birational, then $\dim V = \dim W$.

Proof. A birational map restricts to an isomorphism between nonempty open subsets $U \subseteq V$ and $U' \subseteq W$. Shrinking, we may assume both consist of smooth points. Then by Proposition 24.5.8(iv) the tangent spaces are isomorphic at corresponding points, so the minima agree. $\square$

§24.5.4 Dimension via Transcendence Degree

The definition above is geometric; there is an equivalent algebraic description in terms of the function field, which is often much easier to compute with. Recall from Galois theory that if $K \subseteq L$ are fields, an element $\alpha \in L$ is **transcendental** over K if it satisfies no nonzero polynomial over K , and a subset $S \subseteq L$ is **algebraically independent** over K if there is no nonzero polynomial relation among its elements with coefficients in K .

Definition 24.5.14

An extension $K/\mathbb{C}$ is **purely transcendental** if $K = \mathbb{C}(x_1, \dots, x_m)$ for some algebraically independent $x_1, \dots, x_m$.

Proposition 24.5.15

Let $K/\mathbb{C}$ be a finitely generated field extension. Then there is a purely transcendental subextension $K_0 = \mathbb{C}(x_1, \dots, x_m) \subseteq K$ with K/K_0 finite. Moreover $K = K_0(y)$ for a single $y \in K$.

Proof. Write $K = \mathbb{C}(x_1, \dots, x_n)$. Choose a maximal algebraically independent subset of $\{x_1, \dots, x_n\}$, which after relabelling is $\{x_1, \dots, x_m\}$; put $K_0 = \mathbb{C}(x_1, \dots, x_m)$. By maximality, each of $x_{m+1}, \dots, x_n$ is algebraic over K_0 , so K/K_0 is finitely generated and algebraic, hence finite. Since $\mathbb{C}$ has characteristic zero, K/K_0 is separable, and the Primitive Element Theorem from II Galois Theory gives $K = K_0(y)$. $\square$

The integer m is independent of choices; it is the **transcendence degree** $\text{tr. deg}_{\mathbb{C}} K$, and the argument is an exchange lemma identical to the one for bases of vector spaces.

Proposition 24.5.16

Let $x_1, \dots, x_n$ be algebraically independent over $\mathbb{C}$, put $K = \mathbb{C}(x_1, \dots, x_n)$, and let

x_{n+1} be algebraic over K . Then

$$I = \{g \in \mathbb{C}[X_1, \dots, X_{n+1}] : g(x_1, \dots, x_{n+1}) = 0\}$$

is a principal ideal generated by an irreducible polynomial f .

Proof. Since the x_i are algebraically independent, $R = \mathbb{C}[x_1, \dots, x_n] \subseteq K$ is isomorphic to a polynomial ring, hence a UFD, with $\text{Frac } R = K$. As x_{n+1} is algebraic over K , it has a minimal polynomial $h \in K[T]$, monic and irreducible. Clearing denominators, let $b \in R$ be a common denominator of the coefficients of h and set $f = bh \in R[T]$. By Gauss's Lemma, f is irreducible in $R[T] \cong \mathbb{C}[X_1, \dots, X_{n+1}]$.

If $g \in I$ then $g(x_1, \dots, x_n, T) \in K[T]$ has x_{n+1} as a root, so is divisible by h in $K[T]$; by Gauss's Lemma again, $f \mid g$ in $R[T]$. Hence $I = (f)$. $\square$

Corollary 24.5.17

Every irreducible variety is birational to a hypersurface.

Proof. Let $K = \mathbb{C}(V)$. By Proposition 24.5.15 we may write $K = \mathbb{C}(x_1, \dots, x_n, x_{n+1})$ with $x_1, \dots, x_n$ algebraically independent and x_{n+1} algebraic over $\mathbb{C}(x_1, \dots, x_n)$. By Proposition 24.5.16,

$$\mathbb{C}[x_1, \dots, x_{n+1}] \cong \frac{\mathbb{C}[X_1, \dots, X_{n+1}]}{(f)}$$

for an irreducible f , and K is its field of fractions. So $\mathbb{C}(V) \cong \mathbb{C}(\mathbb{V}(f))$ for the hypersurface $\mathbb{V}(f) \subseteq \mathbb{A}^{n+1}$, and Theorem 24.4.38 gives the result. $\square$

Corollary 24.5.18

For an irreducible variety V we have $\dim V = \text{tr. deg}_{\mathbb{C}} \mathbb{C}(V)$. In particular an irreducible hypersurface in $\mathbb{A}^n$ or $\mathbb{P}^n$ has dimension $n - 1$, and $\dim \mathbb{A}^n = \dim \mathbb{P}^n = n$.

Proof. Both sides are birational invariants – the left by the corollary above, the right because birational varieties have isomorphic function fields. By Corollary 24.5.17 we may assume $V = \mathbb{V}(f) \subseteq \mathbb{A}^n$ is an irreducible hypersurface with f non-constant, say involving X_n . Then $\mathbb{C}(V) = \mathbb{C}(x_1, \dots, x_n)$ where $x_1, \dots, x_{n-1}$ are algebraically independent and x_n is algebraic over them, so $\text{tr. deg} = n - 1$. On the other hand some partial derivative of f is not identically zero on V (Proposition 24.5.10), so at a general point the Jacobian has rank 1 and $\dim T_{V,P} = n - 1$. Hence both sides equal $n - 1$. $\square$

This is the definition of dimension that we will actually use: *the dimension of an irreducible variety is the transcendence degree of its function field*. In particular a **curve** is an irreducible variety whose function field is a finite extension of $\mathbb{C}(t)$ for some transcendental t , and this is the description that will drive everything in the second half of these notes.

§24.6 Plane Curves and Bézout's Theorem

We now specialise to curves in $\mathbb{P}^2$, where we can be extremely concrete. The organising principle is Bézout's theorem: two plane curves of degrees m and n meet in exactly mn points, provided we work projectively, over $\mathbb{C}$, and count with multiplicity. Each of these three provisos is essential, and each corresponds to something we have set up.

§24.6.1 Plane Curves and their Degree

Definition 24.6.1 (Plane Curve)

A **projective plane curve** of **degree** d is $C = \mathbb{V}(F) \subseteq \mathbb{P}^2$ where $F \in \mathbb{C}[X_0, X_1, X_2]$ is homogeneous of degree d with no repeated irreducible factors. Curves of degree 1, 2, 3, 4 are called **lines**, **conics**, **cubics** and **quartics**.

Requiring no repeated factors makes F recoverable from C : by the Nullstellensatz $I^h(C) = \sqrt{(F)} = (F_{\text{red}})$, so the degree is well defined. C is irreducible if and only if F is.

Example 24.6.2

If $C = \mathbb{V}(f) \subseteq \mathbb{A}^2$ is an affine plane curve of degree d , its projective closure is $\mathbb{V}(f^h)$, a projective plane curve of degree d . The points at infinity are found by setting $X_0 = 0$: they are the zeros of the top-degree part $f_{[d]}$ of f , homogenised. For instance the affine cubic $Y^2 = X^3 - X$ has $f^h = X_0X_2^2 - X_1^3 + X_1X_0^2$, and setting $X_0 = 0$ gives $X_1^3 = 0$, so there is exactly one point at infinity, $(0 : 0 : 1)$, and it is a triple point of the intersection with the line at infinity. This point is where the two ends of the real picture in Section 1 join up.

§24.6.2 Intersection Multiplicity

The naive count of intersection points is not constant: a line meets a conic in two points usually, but in only one if it is tangent. To get a constant answer we must count tangential intersections twice.

Definition 24.6.3 (Intersection Multiplicity with a Line)

Let $C = \mathbb{V}(F)$ be a plane curve of degree d and L a line, $L \not\subseteq C$. Parametrise L by $s \mapsto \mathbf{a} + s\mathbf{b}$ (in homogeneous coordinates: $(s : t) \mapsto s\mathbf{a} + t\mathbf{b}$). Then $G(s, t) = F(s\mathbf{a} + t\mathbf{b})$ is a nonzero binary form of degree d , so factors as

$$G(s, t) = c \prod_k (\beta_k s - \alpha_k t)^{m_k}$$

with $(\alpha_k : \beta_k)$ distinct. The **intersection multiplicity** of C and L at the point P_k corresponding to $(\alpha_k : \beta_k)$ is $I_{P_k}(C, L) = m_k$, and $I_P(C, L) = 0$ for $P \notin C \cap L$.

Since a binary form of degree d over $\mathbb{C}$ factors into d linear factors, we get for free:

Proposition 24.6.4 (Bézout for a line)

Let C be a plane curve of degree d and L a line with $L \not\subseteq C$. Then

$$\sum_{P \in C \cap L} I_P(C, L) = d.$$

In particular $|C \cap L| \leq d$, with equality for all but finitely many lines.

Proof. The displayed identity is the factorisation of G above, as $\deg G = d$ (the coefficient of s^d is $F(\mathbf{a})$, which is nonzero for a suitable choice of $\mathbf{a} \in L$ unless $L \subseteq C$). $\square$

One checks directly that $I_P(C, L) \geq 2$ if and only if L is the tangent line to C at P , or P is a singular point of C ; this is exactly the computation with which we opened Section 24.5.

Definition 24.6.5 (Intersection Multiplicity, general)

For plane curves $C = \mathbb{V}(F)$ and $D = \mathbb{V}(G)$ with no common component and $P \in \mathbb{P}^2$, the **intersection multiplicity** is

$$I_P(C, D) = \dim_{\mathbb{C}} \frac{\mathcal{O}_{\mathbb{P}^2, P}}{(f, g)},$$

where f, g are the dehomogenisations of F, G in an affine chart containing P .

This is finite and independent of the chart, and it agrees with the multiplicity defined above when D is a line. We will not develop its properties in detail – the local algebra required is beyond this course – but it is worth recording the characterisation that makes it usable: $I_P(C, D)$ is the unique assignment of non-negative integers such that

- (i) $I_P(C, D) = I_P(D, C)$, and $I_P(C, D) = 0$ if and only if $P \notin C \cap D$;
- (ii) $I_P(C, D) = 1$ if and only if P is a smooth point of both curves and their tangent lines at P differ (a **transverse** intersection);
- (iii) I_P is additive: $I_P(C, D \cup D') = I_P(C, D) + I_P(C, D')$;
- (iv) $I_P(C, D) = I_P(C, D')$ whenever $G' = G + HF$ for some form H of the appropriate degree.

In practice one computes with the definition directly, and in small cases this is easy: the local ring $\mathcal{O}_{\mathbb{A}^2, P}$ is obtained from $\mathbb{C}[X, Y]$ by inverting the polynomials not vanishing at P , so any factor which is a unit at P may simply be deleted.

Example 24.6.6

Take $P = (0, 0)$ throughout, and write $\mathcal{O} = \mathcal{O}_{\mathbb{A}^2, P}$.

- (i) Let $C = \mathbb{V}(y - x^2)$ be a parabola and $L = \mathbb{V}(y)$ its tangent at P . Then

$$\frac{\mathcal{O}}{(y - x^2, y)} = \frac{\mathcal{O}}{(y, x^2)} \cong \frac{\mathbb{C}[x]_{(x)}}{(x^2)},$$

which has $\mathbb{C}$ -basis $\{1, x\}$, so $I_P(C, L) = 2$. The tangency is visible in the

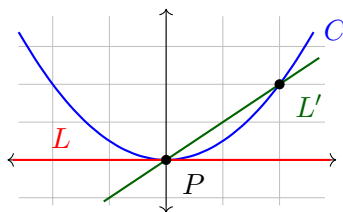
algebra as the nilpotent x .

- (ii) Let $L' = \mathbb{V}(y - x)$ instead. Substituting, $(y - x^2, y - x) = (y - x, x - x^2) = (y - x, x)$, since $1 - x$ is a unit in $\mathcal{O}$. So the quotient is $\mathbb{C}$ and $I_P(C, L') = 1$: the intersection is transverse. Of course L' also meets C at $(1, 1)$, and $1 + 1 = 2$ agrees with Bézout for a line and a conic.
- (iii) Let C be the nodal cubic $\mathbb{V}(y^2 - x^2(x + 1))$ and $L = \mathbb{V}(y)$, one of the two lines through the node that is *not* a branch tangent. Then

$$\frac{\mathcal{O}}{(y^2 - x^3 - x^2, y)} \cong \frac{\mathbb{C}[x]_{(x)}}{(x^2(x + 1))} \cong \frac{\mathbb{C}[x]_{(x)}}{(x^2)},$$

as $x + 1$ is a unit at P ; so $I_P(C, L) = 2$. The line meets the singular point twice, once on each branch.

- (iv) With the same C and the branch tangent $L'' = \mathbb{V}(y - x)$, substituting $y = x$ gives $x^2 - x^3 - x^2 = -x^3$, so $I_P(C, L'') = 3$. Indeed a line meets a cubic in three points, so L'' meets C nowhere else.



The tangent line L meets the parabola C only at P , but with multiplicity 2; the transverse line L' meets it at two distinct points, each with multiplicity 1. In both cases the total is $2 = 1 \cdot 2$, as Bézout demands.

§24.6.3 Bézout's Theorem

Theorem 24.6.7 (Bézout)

Let $C, D \subseteq \mathbb{P}^2$ be plane curves of degrees m and n with no common component. Then $C \cap D$ is finite and

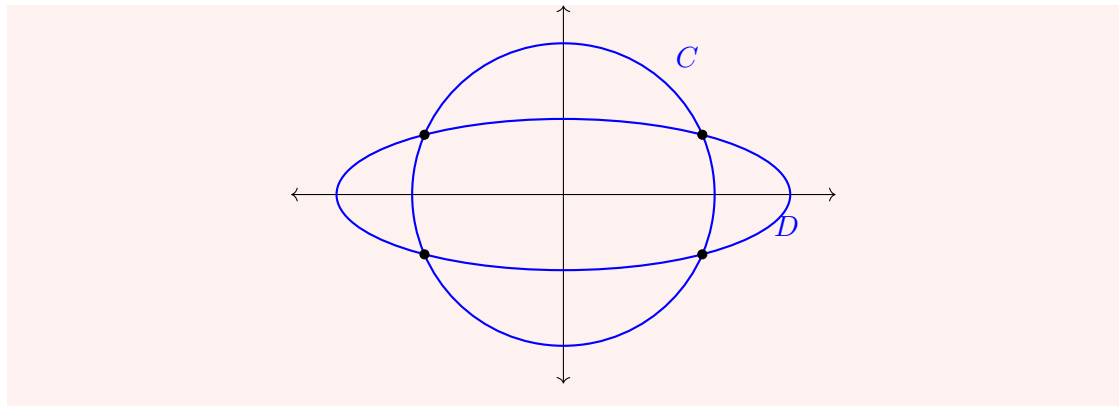
$$\sum_{P \in C \cap D} I_P(C, D) = mn.$$

In particular $|C \cap D| \leq mn$, with equality if and only if all intersections are transverse.

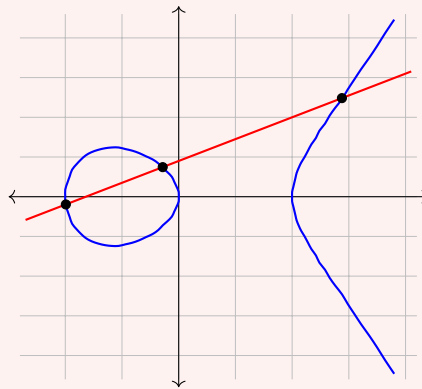
We will prove the weak form – the inequality $|C \cap D| \leq mn$ – in Section 24.8 using divisors, once we have the machinery of orders of vanishing. The strong form requires the full theory of intersection multiplicity. For now let us see what it says.

Example 24.6.8

A line and a conic meet in 2 points, a line and a cubic in 3, and two conics in 4. Here is the last of these: the circle $X^2 + Y^2 = 4$ and the ellipse $X^2/9 + Y^2 = 1$ meet in the four points $(\pm\sqrt{27}/\sqrt{8}, \pm\sqrt{5/8}) \approx (\pm 1.837, \pm 0.791)$.

**Example 24.6.9**

The line $Y = 0.55X + 0.45$ meets the cubic $Y^2 = X^3 - X$ in the three points with $X \approx 1.439, -0.995, -0.141$. All three are real here, and they are visible in the picture.

**Corollary 24.6.10** (Five points determine a conic)

Let $P_1, \dots, P_5 \in \mathbb{P}^2$ be five points, no three collinear. Then there is a unique conic through all five, and it is smooth.

Proof. A conic is $\mathbb{V}(F)$ for a homogeneous quadratic F in three variables; the space of such F is a 6-dimensional vector space, and $F(P_i) = 0$ is a linear condition. Five linear conditions on a 6-dimensional space have a nonzero solution, so a conic through the five points exists.

If two distinct conics C, D passed through all five points, then by Bézout they share a component (else $|C \cap D| \leq 4 < 5$). A shared component is a line or the whole conic; the latter is excluded as $C \neq D$, so $C = L \cup L'$ and $D = L \cup L''$ with L a common line. Then at least three of the five points lie on one line, contradiction. So the conic is unique.

Finally, a singular conic is a union of two lines (or a double line), which again would force three collinear points. So the conic is smooth. $\square$

§24.6.4 Inflection Points

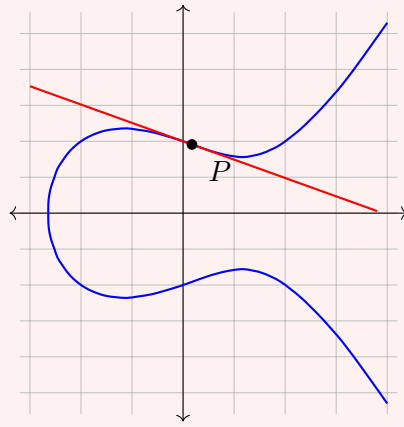
Definition 24.6.11 (Flex)

Let $C \subseteq \mathbb{P}^2$ be a plane curve and P a smooth point with tangent line L . Then P is a **point of inflection**, or **flex**, if $I_P(C, L) \geq 3$.

So a flex is a point where the tangent line meets the curve to order at least three – it ‘crosses over’ rather than touching. For a curve of degree 2 there are no flexes, since $I_P(C, L) \leq 2$ by Bézout with a line. For cubics, flexes are exactly what we need to normalise the equation.

Example 24.6.12

On $C: Y^2 = X^3 - X + 1$ there is a real flex at $P \approx (0.0871, 0.9558)$, with tangent line of slope ≈ -0.5112 . Substituting the tangent into the cubic gives a polynomial with a triple root at $x = 0.0871$.



Note that near P the tangent line crosses from one side of the curve to the other.

There is a beautiful way of detecting flexes algebraically.

Definition 24.6.13 (Hessian)

Let $F \in \mathbb{C}[X_0, X_1, X_2]$ be homogeneous of degree d . Its **Hessian** is

$$H_F = \det \left(\frac{\partial^2 F}{\partial X_i \partial X_j} \right)_{0 \leq i, j \leq 2},$$

a homogeneous polynomial of degree $3(d - 2)$.

Theorem 24.6.14

Let $C = \mathbb{V}(F) \subseteq \mathbb{P}^2$ be a smooth curve of degree $d \geq 2$ and $P \in C$. Then P is a flex of C if and only if $H_F(P) = 0$. Consequently, a smooth plane curve of degree $d \geq 3$ has at least one flex, and at most $3d(d - 2)$ of them; a smooth cubic has exactly 9.

Proof sketch. Move P to $(1 : 0 : 0)$ and arrange that the tangent line there is $\mathbb{V}(X_2)$. Dehomogenising in the chart $X_0 = 1$ with coordinates (x, y) , the curve is $y = ax^2 + (\text{higher order})$ near the origin, and P is a flex precisely when $a = 0$. A

direct computation using Euler's relation shows that $H_F(P)$ is a nonzero multiple of a , which gives the equivalence.

For the count: H_F has degree $3(d-2)$, so if $C \not\subseteq \mathbb{V}(H_F)$ then Bézout gives $|C \cap \mathbb{V}(H_F)| \leq 3d(d-2)$, with equality when all intersections are transverse. That $C \not\subseteq \mathbb{V}(H_F)$ for smooth C (in characteristic zero) is a separate check. For $d=3$ one shows all the intersections are transverse, giving exactly 9 flexes. $\square$

§24.6.5 Weierstrass Form

We can now put any smooth plane cubic into a standard shape. This is the calculation that connects the abstract theory to the concrete equations you have seen in number theory.

Theorem 24.6.15 (Weierstrass Normal Form)

Let $C \subseteq \mathbb{P}^2$ be a smooth plane cubic. After a projective change of coordinates,

$$C = \mathbb{V}(X_0X_2^2 - X_1^3 - aX_0^2X_1 - bX_0^3)$$

for some $a, b \in \mathbb{C}$ with $4a^3 + 27b^2 \neq 0$. Equivalently, in the affine chart $X_0 = 1$,

$$C: y^2 = x^3 + ax + b = (x - \lambda_1)(x - \lambda_2)(x - \lambda_3)$$

with $\lambda_1, \lambda_2, \lambda_3$ distinct, together with the single point $(0 : 0 : 1)$ at infinity.

Proof. By Theorem 24.6.14, C has a flex P_0 . Change coordinates so that $P_0 = (0 : 0 : 1)$ and the tangent line to C at P_0 is the line at infinity $\mathbb{V}(X_0)$. Write F for the defining cubic. Since P_0 is a flex, $I_{P_0}(C, \mathbb{V}(X_0)) = 3$, so by Proposition 24.6.4 the line at infinity meets C only at P_0 ; restricting F to $X_0 = 0$ therefore gives (up to scalar) X_1^3 . So F has no $X_1^2X_2$, $X_1X_2^2$ or X_2^3 terms except via X_0 , and after scaling

$$F = cX_0X_2^2 + X_0X_2\ell(X_0, X_1) - X_1^3 + X_0q(X_0, X_1)$$

for a linear form ℓ and quadratic q . Smoothness at $P_0 = (0 : 0 : 1)$ forces $c \neq 0$ (else $\partial F/\partial X_0, \partial F/\partial X_1, \partial F/\partial X_2$ all vanish there), so we may scale $c = 1$. Completing the square in X_2 – that is, replacing X_2 by $X_2 - \frac{1}{2}\ell(X_0, X_1)$, a linear change of coordinates – removes the middle term, leaving

$$F = X_0X_2^2 - X_1^3 + X_0\tilde{q}(X_0, X_1)$$

for a new quadratic $\tilde{q}$. Finally, replacing X_1 by $X_1 - \frac{1}{3}(\text{coefficient of } X_1^2 \text{ in } \tilde{q})X_0$ kills the $X_0X_1^2$ term, giving the stated form.

It remains to see that smoothness is equivalent to $x^3 + ax + b$ having distinct roots. In the affine chart, $f = y^2 - x^3 - ax - b$ has $\partial f/\partial y = 2y$ and $\partial f/\partial x = -(3x^2 + a)$. A singular point has $y = 0$ and x a common root of $x^3 + ax + b$ and its derivative, i.e. a repeated root; and $4a^3 + 27b^2$ is (up to sign) the discriminant of $x^3 + ax + b$, vanishing exactly when there is a repeated root. $\square$

We call the pair (C, P_0) , where $P_0 = (0 : 0 : 1)$ is the flex at infinity, an **elliptic curve**. (This is not quite the general definition – a genuinely intrinsic one will be given in

Section 24.11 – but for plane cubics it is equivalent.)

§24.6.6 The Group Law on a Cubic

Here is the most striking fact about plane cubics: the points of a smooth cubic form an abelian group, and the group operation is given by drawing lines.

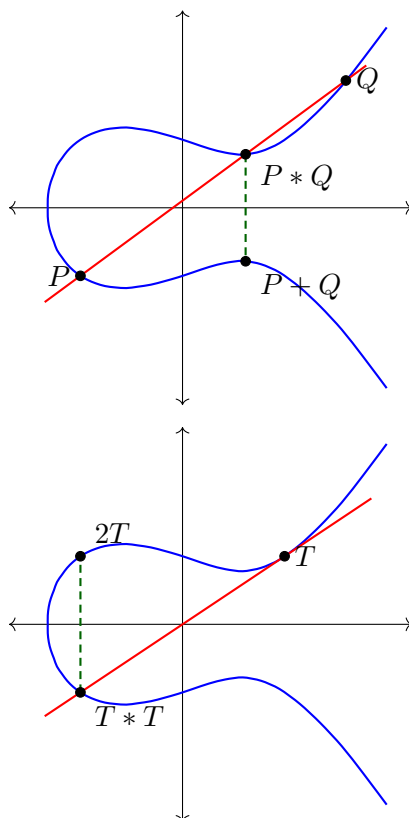
By Bézout, a line L meets a smooth cubic C in exactly three points counted with multiplicity. Given $P, Q \in C$, let L be the line through them (the tangent line at P if $P = Q$), and let $P * Q$ denote the third point of $C \cap L$.

Definition 24.6.16 (Chord-and-Tangent Law)

Let (C, P_0) be a smooth plane cubic with a chosen flex P_0 . For $P, Q \in C$ define

$$P + Q = P_0 * (P * Q).$$

In Weierstrass form, with $P_0 = (0 : 0 : 1)$ the point at infinity, the line through P_0 and any point $R = (x, y)$ is the vertical line through R , so $P_0 * R = (x, -y)$. Thus the recipe is: *draw the line through P and Q , find the third intersection with C , and reflect in the x -axis.*



The second picture shows the doubling: on $y^2 = x^3 - x + 1$ the tangent at $T = (1, 1)$ has slope 1 and meets the curve again at $(-1, -1)$, so $2T = (-1, 1)$.

Theorem 24.6.17

Let (C, P_0) be a smooth plane cubic with P_0 a flex. Then $(C, +)$ is an abelian group

with identity P_0 , and $-P = P_0 * P$. Moreover $P + Q + R = P_0$ if and only if P, Q, R are collinear.

Proof sketch. Commutativity is clear, since $P * Q = Q * P$. The identity: $P + P_0 = P_0 * (P * P_0)$, and $P_0 * (P * P_0) = P$ because the line through P_0 and $P * P_0$ is the line through P_0 and P , whose third point is P . Inverses: set $-P = P_0 * P$, so $P + (-P) = P_0 * (P * (P_0 * P)) = P_0 * P_0 = P_0$, the last equality because P_0 is a flex, so the tangent at P_0 meets C only at P_0 .

The collinearity statement follows: if P, Q, R are collinear then $R = P * Q$, so $P + Q = P_0 * R = -R$ and $P + Q + R = P_0$.

Associativity is the only difficulty. It can be proved by a direct (and unpleasant) computation with the Weierstrass equation, or by an application of Bézout's theorem for cubics – if two cubics meet in nine points and a third cubic passes through eight of them, it passes through the ninth – or, most cleanly, by identifying $(C, +)$ with the divisor class group, which we do in Theorem 24.11.12. $\square$

The last of these is the argument we will actually give, and it explains *why* there is a group law: the points of C are in bijection with the degree zero divisor classes, and those obviously form a group. Everything geometric in the chord-and-tangent construction is the statement that three points of C sum to zero in the class group exactly when they are cut out by a line.

§24.7 Curves and their Local Rings

From now on we specialise to curves, where the theory becomes both deeper and more complete than in higher dimensions. Recall that a curve is an irreducible variety of dimension 1; equivalently, by Corollary 24.5.18, one whose function field has transcendence degree 1 over $\mathbb{C}$.

Notation. Unless stated otherwise, from now on *curve* means a smooth irreducible projective curve. We will always have in mind some embedding $C \subseteq \mathbb{P}^n$, but the ambient space will play no essential role: everything we do is intrinsic.

§24.7.1 Closed Subsets of Curves

The first thing to establish is that curves really are one-dimensional in the naive sense.

Proposition 24.7.1

Let C be an irreducible curve and $D \subsetneq C$ a proper Zariski closed subset. Then D is a finite set of points.

Proof. Decomposing into irreducible components, it suffices to show that a proper irreducible closed $W \subsetneq V$ is a point, where we may pass to an affine chart and take $V \subseteq \mathbb{A}^n$ affine irreducible with $\text{tr. deg}_{\mathbb{C}} \mathbb{C}(V) = 1$.

The inclusion $\iota: W \hookrightarrow V$ gives a surjection $\iota^*: \mathbb{C}[V] \twoheadrightarrow \mathbb{C}[W]$ with kernel $I(W)/I(V)$, which is nonzero since $I(V) \subsetneq I(W)$ by the Nullstellensatz. Choose $x \in \mathbb{C}[V]$

nonzero with $\iota^*(x) = 0$.

Suppose for contradiction that $\mathbb{C}[W] \neq \mathbb{C}$, and pick $t \in \mathbb{C}[W] \setminus \mathbb{C}$; choose $y \in \mathbb{C}[V]$ with $\iota^*(y) = t$. I claim x and y are algebraically independent over $\mathbb{C}$. If not, there is a nonzero $g \in \mathbb{C}[S, T]$ with $g(x, y) = 0$; choose such a g of minimal degree in S . Applying ι^* , $g(0, t) = 0$ in $\mathbb{C}[W]$. Now t is transcendental over $\mathbb{C}$, since $\mathbb{C}$ is algebraically closed and any element of the domain $\mathbb{C}[W]$ that is algebraic over $\mathbb{C}$ must lie in $\mathbb{C}$. Hence $g(0, T)$ is the zero polynomial, i.e. $S \mid g$. Then $g = Sg_1$ and $xg_1(x, y) = 0$; as $\mathbb{C}[V]$ is a domain and $x \neq 0$, $g_1(x, y) = 0$, contradicting minimality.

So $\mathbb{C}(V)$ would contain two algebraically independent elements, which contradicts the fact that $\mathbb{C}(V)$ has transcendence degree 1. Hence $\mathbb{C}[W] = \mathbb{C}$, so W is a single point. $\square$

So the Zariski topology on a curve is the cofinite topology, exactly as on $\mathbb{A}^1$. The interesting structure is all in the local rings.

§24.7.2 Local Parameters

Theorem 24.7.2

Let P be a smooth point of an irreducible curve C . Then the maximal ideal $\mathfrak{m}_P \subseteq \mathcal{O}_{C,P}$ is principal.

Any generator π_P of $\mathfrak{m}_P$ is called a **local parameter**, **local coordinate** or **uniformiser** at P . It is unique up to multiplication by a unit of $\mathcal{O}_{C,P}$.

The proof rests on the following piece of commutative algebra, which one should think of as an algebraic substitute for the implicit function theorem.

Lemma 24.7.3 (Nakayama)

Let R be a local ring with maximal ideal $\mathfrak{m}$, and let M be a finitely generated R -module.

- (i) If $\mathfrak{m}M = M$ then $M = 0$.
- (ii) If $N \leq M$ is a submodule with $\mathfrak{m}M + N = M$, then $M = N$.

Proof. (i) Suppose $M \neq 0$ and let $b_1, \dots, b_k$ be a generating set of minimal size. Since $M = \mathfrak{m}M$, we may write $b_k = \sum_j a_j b_j$ with $a_j \in \mathfrak{m}$. Then $(1 - a_k)b_k = \sum_{j < k} a_j b_j$. As $a_k \in \mathfrak{m}$ and R is local, $1 - a_k$ is a unit, so b_k lies in the span of $b_1, \dots, b_{k-1}$, contradicting minimality.

(ii) Apply (i) to M/N , noting $\mathfrak{m}(M/N) = (\mathfrak{m}M + N)/N = M/N$. $\square$

Proof of Theorem 24.7.2. Pass to an affine chart and change coordinates so that $P = \mathbf{0} \in \mathbb{A}^n$. Write $\mathbb{C}[C] = \mathbb{C}[x_1, \dots, x_n]$ for the coordinate ring, where x_i is the image of X_i . Then

$$\mathcal{O}_P = \left\{ \frac{f}{g} : f, g \in \mathbb{C}[C], g(P) \neq 0 \right\}, \quad \mathfrak{m}_P = x_1 \mathcal{O}_P + \dots + x_n \mathcal{O}_P.$$

Note that every ideal $J \subseteq \mathcal{O}_P$ is finitely generated: J is determined by $J \cap \mathbb{C}[C]$,

which is finitely generated as $\mathbb{C}[C]$ is Noetherian.

Since P is smooth and $\dim C = 1$, the tangent space $T_{C,P}$ is a line; after a further linear change of coordinates we may assume $T_{C,P} = \{X_2 = \cdots = X_n = 0\}$. The tangent space is cut out by the linear parts at P of the elements of $I(C)$, so for each $j \geq 2$ there is $f_j \in I(C)$ whose linear part is X_j , say

$$f_j = X_j - h_j(X_1, \dots, X_n)$$

with h_j having no terms of degree < 2 . Passing to $\mathbb{C}[C]$, where $f_j = 0$,

$$x_j = h_j(x_1, \dots, x_n) \in \mathfrak{m}_P^2 \quad (j \geq 2).$$

Hence

$$\mathfrak{m}_P = \sum_{i=1}^n x_i \mathcal{O}_P \subseteq x_1 \mathcal{O}_P + \mathfrak{m}_P^2 \subseteq \mathfrak{m}_P.$$

Now apply Nakayama's lemma with $R = \mathcal{O}_P$, $M = \mathfrak{m}_P$ (finitely generated), $N = x_1 \mathcal{O}_P$, noting $\mathfrak{m}_P M = \mathfrak{m}_P^2$: we get $\mathfrak{m}_P = x_1 \mathcal{O}_P$, so $\pi_P = x_1$ is a local parameter. $\square$

Corollary 24.7.4

Let $C = \mathbb{V}(f) \subseteq \mathbb{A}^2$ be an irreducible plane curve and $P \in C$ a smooth point. Then the function $Q \mapsto X(Q) - X(P)$ is a local parameter at P if and only if $\frac{\partial f}{\partial Y}(P) \neq 0$; and $Q \mapsto Y(Q) - Y(P)$ is one if and only if $\frac{\partial f}{\partial X}(P) \neq 0$.

Proof. This is the proof above: if $\partial f / \partial Y(P) \neq 0$ then the tangent line is not vertical, so after translating P to the origin the roles of x_1 and x_2 in the argument are as stated. $\square$

Note that by smoothness at least one of the two partial derivatives is nonzero, so at least one of $x - x(P)$, $y - y(P)$ is always a local parameter. This will be used constantly in computations with plane curves.

§24.7.3 Orders of Vanishing

Once $\mathfrak{m}_P$ is principal, every element of the function field has a well-defined order of vanishing at P , and this is the single most useful tool we have.

Corollary 24.7.5 (Valuation)

Let P be a smooth point of a curve C . There is a surjective group homomorphism

$$\nu_P: \mathbb{C}(C)^\times \rightarrow \mathbb{Z}$$

such that

- (i) $\mathcal{O}_{C,P} = \{0\} \cup \{f \in \mathbb{C}(C)^\times : \nu_P(f) \geq 0\}$;
- (ii) $\mathfrak{m}_P = \{0\} \cup \{f \in \mathbb{C}(C)^\times : \nu_P(f) > 0\}$;
- (iii) for any local parameter π_P and any $f \in \mathbb{C}(C)^\times$ we may write $f = \pi_P^{\nu_P(f)} u$

with $u \in \mathcal{O}_{C,P}^\times$.

Moreover $\nu_P(f + g) \geq \min\{\nu_P(f), \nu_P(g)\}$.

Proof. Write $\mathfrak{m}_P = (\pi_P)$, so $\mathfrak{m}_P^n = (\pi_P^n)$. Let $J = \bigcap_{n \geq 0} \mathfrak{m}_P^n$. This is a finitely generated ideal with $\mathfrak{m}_P J = J$ (since $\pi_P \mathfrak{m}_P^n = \mathfrak{m}_P^{n+1}$), so $J = 0$ by Nakayama. Thus no nonzero element vanishes to infinite order, and for $f \in \mathcal{O}_{C,P} \setminus \{0\}$ there is a unique $n \geq 0$ with $f \in \mathfrak{m}_P^n \setminus \mathfrak{m}_P^{n+1}$; set $\nu_P(f) = n$. Writing $f = \pi_P^n g$, we have $g \notin \mathfrak{m}_P$, so g is a unit, giving (iii) for $f \in \mathcal{O}_{C,P}$.

For $f \in \mathbb{C}(C)^\times \setminus \mathcal{O}_{C,P}$, write $f = g/h$ with $g, h \in \mathcal{O}_{C,P}$; then $g = \pi_P^k u$, $h = \pi_P^\ell u'$ with u, u' units, so $f = \pi_P^{k-\ell}(u/u')$ with $k - \ell < 0$. Set $\nu_P(f) = k - \ell$. Multiplicativity, and hence the homomorphism property, is now clear from (iii), and surjectivity from $\nu_P(\pi_P^n) = n$. Properties (i) and (ii) restate the construction. The last inequality is immediate: if $\nu_P(f), \nu_P(g) \geq m$ then $f, g \in \mathfrak{m}_P^m$ and so $f + g \in \mathfrak{m}_P^m$. $\square$

Definition 24.7.6

If $\nu_P(f) = n > 0$ we say f has a **zero of order n** at P ; if $\nu_P(f) = -n < 0$, a **pole of order n** .

Example 24.7.7

Let $C = \mathbb{A}^1$ and $P = 0$, so $\mathcal{O}_{\mathbb{A}^1,0} = \{f/g : g(0) \neq 0\}$ and t is a local parameter. Then $\nu_0(f/g)$ is the multiplicity of $t = 0$ as a root of f . Concretely,

$$\nu_0\left(\frac{t^2 + 8t^3}{1 + t^4}\right) = 2.$$

Thinking of f/g as a power series in t (possible since $g(0) \neq 0$), ν_0 is the index of the first nonzero coefficient. This is exactly the order of vanishing familiar from IB Complex Analysis, and the analogy is not a coincidence.

Example 24.7.8

On $\mathbb{P}^1$ with coordinate $t = X_1/X_0$, a local parameter at $a \in \mathbb{C}$ is $t - a$, and a local parameter at $\infty = (0 : 1)$ is $1/t$. So for $f = \prod_i (t - a_i)^{n_i}$ we have $\nu_{a_i}(f) = n_i$ and $\nu_\infty(f) = -\sum_i n_i$.

Corollary 24.7.9

If C is a curve and $P \in C$ is smooth, then for any $f \in \mathbb{C}(C)^\times$, at least one of f, f^{-1} is regular at P .

Proof. $\nu_P(f)$ and $\nu_P(f^{-1}) = -\nu_P(f)$ cannot both be negative. $\square$

This fails badly in higher dimensions: on $\mathbb{A}^2$ neither X/Y nor Y/X is regular at the origin. It is a genuinely one-dimensional phenomenon, and it has the following striking consequence.

Corollary 24.7.10

Let C be a smooth curve. Then every rational map $\varphi: C \dashrightarrow \mathbb{P}^m$ is a morphism.

Proof. Fix $P \in C$. Reordering coordinates, assume the image of φ is not contained in $\{X_0 = 0\}$, so we may write $\varphi = (1 : g_1 : \cdots : g_m)$ with $g_j \in \mathbb{C}(C)$. If every $g_j \in \mathcal{O}_{C,P}$ then φ is visibly regular at P . Otherwise let $t = \min_j \nu_P(g_j) < 0$ and choose a local parameter π_P . Then

$$\varphi = (\pi_P^{-t} : \pi_P^{-t} g_1 : \cdots : \pi_P^{-t} g_m),$$

an equivalent representative, and now every entry lies in $\mathcal{O}_{C,P}$ while at least one has $\nu_P = 0$, hence is nonzero at P . So φ is regular at P . $\square$

Since every projective variety sits inside some $\mathbb{P}^m$, we conclude: *a rational map from a smooth curve to a projective variety is automatically a morphism.* There are no indeterminacy loci in dimension one. This is the technical fact that makes the theory of curves so much cleaner than that of surfaces.

§24.7.4 Morphisms of Curves. Degree and Ramification**Proposition 24.7.11**

Let $\varphi: C \rightarrow D$ be a non-constant morphism of irreducible projective curves. Then every fibre $\varphi^{-1}(Q)$ is finite, φ is dominant, and $\varphi^*: \mathbb{C}(D) \hookrightarrow \mathbb{C}(C)$ makes $\mathbb{C}(C)$ a finite extension of $\mathbb{C}(D)$.

Proof. $\varphi^{-1}(Q)$ is closed and, since φ is non-constant, proper, hence finite by Proposition 24.7.1. As C is infinite, $\varphi(C)$ is infinite, so its closure in D is all of D by Proposition 24.7.1 again; so φ is dominant and φ^* is defined and injective.

Pick $t \in \mathbb{C}(D) \setminus \mathbb{C}$ and put $x = \varphi^*t$. Both $\mathbb{C}(D)/\mathbb{C}(t)$ and $\mathbb{C}(C)/\mathbb{C}(x)$ are finite, since both fields have transcendence degree 1 and t, x are transcendental. Hence $\mathbb{C}(C)$ is finite over $\mathbb{C}(x) = \varphi^*\mathbb{C}(t) \subseteq \varphi^*\mathbb{C}(D)$, so a fortiori finite over $\varphi^*\mathbb{C}(D)$. $\square$

Definition 24.7.12 (Degree)

The **degree** of a non-constant morphism $\varphi: C \rightarrow D$ of curves is

$$\deg \varphi = [\mathbb{C}(C) : \varphi^*\mathbb{C}(D)].$$

Definition 24.7.13 (Ramification)

Let $\varphi: C \rightarrow D$ be a non-constant morphism of curves, $P \in C$ smooth, $Q = \varphi(P)$ smooth. The **ramification index** of φ at P is

$$e_P = e(\varphi, P) = \nu_P(\varphi^*\pi_Q),$$

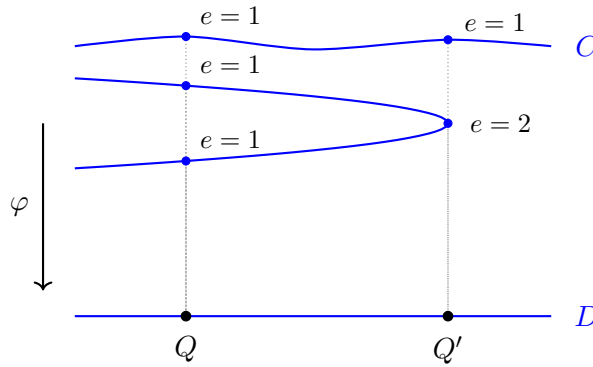
where π_Q is a local parameter at Q . We say φ is **ramified** at P if $e_P > 1$, and **unramified** otherwise.

This is well defined: another local parameter differs from π_Q by a unit of $\mathcal{O}_{D,Q}$, whose pullback is a unit of $\mathcal{O}_{C,P}$ and hence has $\nu_P = 0$.

Example 24.7.14

Consider $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^1$ given in the affine coordinate by $z \mapsto z^d$. On function fields, $\varphi^*: \mathbb{C}(Y) \rightarrow \mathbb{C}(X)$ sends $Y \mapsto X^d$, so $\varphi^*\mathbb{C}(Y) = \mathbb{C}(X^d)$ and $\deg \varphi = [\mathbb{C}(X) : \mathbb{C}(X^d)] = d$.

At $P = 0$ we have $Q = 0$ with local parameter Y , and $\varphi^*Y = X^d$, so $e_0 = \nu_0(X^d) = d$: the map is totally ramified at 0. The same holds at ∞ . At $P = 1$ we have $Q = 1$ with local parameter $Y - 1$, and $\varphi^*(Y - 1) = X^d - 1$, which has a simple zero at $X = 1$; so $e_1 = 1$. Indeed the fibre over 1 consists of the d distinct d th roots of unity, each with $e = 1$. Note $\sum_{P \in \varphi^{-1}(1)} e_P = d = \deg \varphi$, and also $\sum_{P \in \varphi^{-1}(0)} e_P = d$. This is a general phenomenon.



The picture is the one to keep in mind: a degree 3 map has three preimages over a general point, but the preimages can collide, and when two sheets come together the ramification index absorbs the deficit. The next theorem says the bookkeeping always balances.

Theorem 24.7.15

Let $\varphi: C \rightarrow D$ be a non-constant morphism of irreducible projective curves.

- (i) φ is surjective.
- (ii) If C and D are smooth then for every $Q \in D$,

$$\deg \varphi = \sum_{P \in \varphi^{-1}(Q)} e_P.$$

- (iii) $e_P = 1$ for all but finitely many $P \in C$.

We prove (i) now, and (iii) will follow from the theory of differentials in Section 24.9. Part (ii) requires a piece of commutative algebra (that $\mathcal{O}_{D,Q}$ is a principal ideal domain and the semilocal ring $\bigcap_{P \in \varphi^{-1}(Q)} \mathcal{O}_{C,P}$ is free of rank $\deg \varphi$ over it) which we take on trust. The proof of (i) rests on the following, which is the projective analogue of compactness.

Proposition 24.7.16 (Fundamental theorem of elimination theory)

The projection $\mathbb{P}^n \times \mathbb{A}^m \rightarrow \mathbb{A}^m$ is a closed map for the Zariski topology.

We do not prove this; it is a matter of showing that the condition on $\mathbf{b} \in \mathbb{A}^m$ that a system of homogeneous equations with coefficients depending on $\mathbf{b}$ has a nontrivial solution is itself given by polynomial equations in $\mathbf{b}$ (resultants). Note that it genuinely uses projectivity: $\mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ is not closed, since $\mathbb{V}(XY - 1)$ has image $\mathbb{A}^1 \setminus \{0\}$.

Proposition 24.7.17

Let $\varphi: V \rightarrow W$ be a morphism of quasi-projective varieties with V projective. Then φ is a closed map.

Here a **quasi-projective variety** is a Zariski open subset of a projective variety; all affine and all projective varieties are quasi-projective, and all our definitions (rational functions, morphisms, local rings) extend verbatim.

Proof. Factor φ as $V \rightarrow \Gamma_\varphi \subseteq V \times W \rightarrow W$, where $\Gamma_\varphi = \{(x, \varphi(x))\}$ is the graph. The graph is closed, being the preimage of the diagonal $\Delta_W \subseteq W \times W$ (which is closed) under $\varphi \times \text{id}$. So it suffices to show the projection $V \times W \rightarrow W$ is closed. As $V \subseteq \mathbb{P}^n$ is closed, it suffices to show $\mathbb{P}^n \times W \rightarrow W$ is closed; covering W by affine opens, it suffices to show $\mathbb{P}^n \times U \rightarrow U$ is closed for U affine; and as U is closed in some $\mathbb{A}^m$, it suffices to show $\mathbb{P}^n \times \mathbb{A}^m \rightarrow \mathbb{A}^m$ is closed, which is Proposition 24.7.16. $\square$

Proof of Theorem 24.7.15(i). $\varphi(C)$ is closed in D by Proposition 24.7.17, so by Proposition 24.7.1 it is either finite or all of D . As φ is non-constant and C is irreducible, $\varphi(C)$ is infinite, so $\varphi(C) = D$. $\square$

Here is the payoff, and it is the algebraic geometer's version of Liouville's theorem.

Corollary 24.7.18

Let C be a smooth projective irreducible curve and $f \in \mathbb{C}(C)^\times$.

- (i) If f is regular at every point of C , then f is constant.
- (ii) $\{P \in C : \nu_P(f) \neq 0\}$ is finite, and $\sum_{P \in C} \nu_P(f) = 0$.

Proof. Given f , consider the map $\varphi = (1 : f): C \rightarrow \mathbb{P}^1$, which is a morphism by Corollary 24.7.10. Write $0 = (1 : 0)$ and $\infty = (0 : 1)$ in $\mathbb{P}^1$. Then $\varphi(P) = 0$ iff $f(P) = 0$, and $\varphi(P) = \infty$ iff f has a pole at P .

(i) If f is everywhere regular then φ misses ∞ , so φ is not surjective, so by Theorem 24.7.15(i) it is constant, i.e. f is constant.

(ii) Assume f is non-constant, so φ is a non-constant morphism, and let $n = \deg \varphi$. The function $t = X_1/X_0$ is a local parameter at $0 \in \mathbb{P}^1$ and $\varphi^*t = f$; so for P with $f(P) = 0$ we get $e_P = \nu_P(f)$. Similarly $1/t$ is a local parameter at ∞ and $\varphi^*(1/t) = 1/f$, so for P a pole we get $e_P = -\nu_P(f)$. At all other P , $\nu_P(f) = 0$. The zeros and poles are the fibres over 0 and ∞ , hence finite, and Theorem 24.7.15(ii)

applied to $Q = 0$ and $Q = \infty$ gives

$$\sum_{f(P)=0} \nu_P(f) = n = \sum_{f(P)=\infty} (-\nu_P(f)),$$

so the total sum is zero. $\square$

So on a projective curve there are no non-constant ‘holomorphic’ functions, and every rational function has as many zeros as poles, counted with multiplicity. Both statements will be familiar to anyone who has met compact Riemann surfaces, and the analogy is exact.

§24.8 Divisors on Curves

The field $\mathbb{C}(C)$ is enormous and hard to study directly. Divisor theory is a device for cutting it into finite-dimensional pieces: instead of studying all rational functions at once, we study those whose poles are bounded by a prescribed amount at prescribed points.

Throughout, C is a smooth irreducible projective curve.

§24.8.1 Divisors, Degree and Linear Equivalence

Definition 24.8.1 (Divisor)

A **divisor** on C is an element of the free abelian group on the points of C ,

$$\text{Div}(C) = \bigoplus_{P \in C} \mathbb{Z}[P].$$

So a divisor is a finite formal sum $D = \sum_P n_P [P]$ with $n_P \in \mathbb{Z}$, almost all zero. Its **degree** is $\deg D = \sum_P n_P$. We write $\nu_P(D) = n_P$.

The degree map $\deg: \text{Div}(C) \rightarrow \mathbb{Z}$ is a surjective group homomorphism; its kernel is written $\text{Div}^0(C)$.

Definition 24.8.2

A divisor D is **effective**, written $D \geq 0$, if $\nu_P(D) \geq 0$ for all P . We write $D \geq D'$ for $D - D' \geq 0$.

Definition 24.8.3 (Divisor of a Function)

For $f \in \mathbb{C}(C)^\times$, the **divisor of f** is

$$\text{div}(f) = \sum_{P \in C} \nu_P(f) [P] \in \text{Div}(C).$$

Divisors of this form are called **principal**, and they form a subgroup $\text{Prin}(C) \leq \text{Div}(C)$.

That the sum is finite, and that $\deg \text{div}(f) = 0$, is Corollary 24.7.18. That $\text{Prin}(C)$ is a

subgroup follows from $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$, which is the multiplicativity of ν_P . So $\operatorname{Prin}(C) \leq \operatorname{Div}^0(C)$.

Definition 24.8.4 (Picard Group)

The **Picard group** or **divisor class group** of C is

$$\operatorname{Pic}(C) = \operatorname{Cl}(C) = \operatorname{Div}(C) / \operatorname{Prin}(C),$$

and $\operatorname{Pic}^0(C) = \operatorname{Cl}^0(C) = \operatorname{Div}^0(C) / \operatorname{Prin}(C)$. Divisors D, D' are **linearly equivalent**, written $D \sim D'$, if $D - D'$ is principal.

The degree descends to $\operatorname{Pic}(C) \rightarrow \mathbb{Z}$, so linearly equivalent divisors have the same degree. The group $\operatorname{Pic}^0(C)$ measures the failure of the converse, and it turns out to be a subtle and extremely interesting invariant.

Proposition 24.8.5

On $\mathbb{P}^1$, every degree zero divisor is principal; that is, $\operatorname{Pic}^0(\mathbb{P}^1) = 0$ and $\operatorname{Pic}(\mathbb{P}^1) \cong \mathbb{Z}$ via the degree.

Proof. Identify $\mathbb{P}^1 = \mathbb{C} \cup \{\infty\}$ with coordinate t , and let $D = \sum_{a \in \mathbb{C}} n_a [a] + n_\infty [\infty]$ have degree 0, so $n_\infty = -\sum_a n_a$. Put

$$f = \prod_{a \in \mathbb{C}} (t - a)^{n_a} \in \mathbb{C}(t)^\times,$$

a finite product. Then $\nu_a(f) = n_a$ for each $a \in \mathbb{C}$, since $t - a$ is a local parameter at a and the other factors are units there. At ∞ , using the local parameter $1/t$, each factor $(t - a)$ has $\nu_\infty = -1$, so $\nu_\infty(f) = -\sum_a n_a = n_\infty$. Hence $\operatorname{div}(f) = D$. $\square$

That this fails for other curves is the source of the whole theory. We will see, for instance, that on a smooth cubic $\operatorname{Pic}^0(C) \cong C$ as a set, which is a very large group indeed.

§24.8.2 Hyperplane Sections and the Degree of a Curve

There is a second, geometric, source of divisors.

Definition 24.8.6 (Hyperplane Section)

Let $C \subseteq \mathbb{P}^n$ be a curve and L a nonzero linear form with $C \not\subseteq \mathbb{V}(L)$. The **hyperplane section** of C by $\mathbb{V}(L)$ is

$$\operatorname{div}(L) = \sum_{P \in C} n_P [P], \quad n_P = \nu_P \left(\frac{L}{X_i} \right) \text{ for any } i \text{ with } X_i(P) \neq 0.$$

This is well defined: if $X_i(P), X_j(P) \neq 0$ then $X_i/X_j \in \mathcal{O}_{C,P}^\times$, so $\nu_P(X_i/X_j) = 0$ and $\nu_P(L/X_i) = \nu_P(L/X_j)$. Note $\operatorname{div}(L)$ is effective, since L/X_i is regular at every such P , and it is supported on $C \cap \mathbb{V}(L)$, a finite set.

Proposition 24.8.7

If L, L' are linear forms with $C \not\subseteq \mathbb{V}(L) \cup \mathbb{V}(L')$, then

$$\operatorname{div}(L) - \operatorname{div}(L') = \operatorname{div}\left(\frac{L}{L'}\right),$$

which is principal. In particular $\deg \operatorname{div}(L) = \deg \operatorname{div}(L')$.

Proof. At a point P with $X_i(P) \neq 0$ we have $\nu_P(L/L') = \nu_P(L/X_i) - \nu_P(L'/X_i)$, since ν_P is a homomorphism. The degrees agree because principal divisors have degree 0. $\square$

Definition 24.8.8 (Degree of a Curve)

The **degree** of a curve $C \subseteq \mathbb{P}^n$ is $\deg C = \deg \operatorname{div}(L)$ for any linear form L not vanishing on C .

So the degree of a curve counts, with multiplicity, the points in which a general hyperplane meets it. A line in $\mathbb{P}^2$ has degree 1, a smooth conic degree 2, and in general a plane curve $\mathbb{V}(F)$ with $\deg F = d$ has degree d by Proposition 24.6.4.

The construction generalises in two useful directions. First, if $\varphi: C \rightarrow \mathbb{P}^n$ is any non-constant morphism with $\varphi(C) \not\subseteq \mathbb{V}(L)$, we may set $\operatorname{div}(L) = \sum_P \nu_P(\varphi^*L/X_i)[P]$; taking φ to be an inclusion recovers the above. Secondly, if G is homogeneous of degree $m \geq 1$ and does not vanish on C , we set

$$\operatorname{div}(G) = \sum_{P \in C} \nu_P\left(\frac{\varphi^*G}{X_i^m}\right)[P],$$

again independent of the choice of i with $X_i(P) \neq 0$. The same argument as above shows $\operatorname{div}(G) \sim m \operatorname{div}(L)$ for any linear form L , because G/L^m is a rational function on C . This is exactly what we need to prove Bézout.

Theorem 24.8.9 (Weak Bézout)

Let $C, C' \subseteq \mathbb{P}^2$ be distinct smooth irreducible projective plane curves of degrees m and n . Then $|C \cap C'| \leq mn$.

Proof. Write $C = \mathbb{V}(F)$, $C' = \mathbb{V}(G)$ with $\deg F = m$, $\deg G = n$. Restricting G to C gives a divisor $\operatorname{div}(G)$ on C , which is effective and supported exactly on $C \cap C'$ – if $\nu_P(G/X_i^n) > 0$ then G vanishes at P . Since $C \neq C'$ and both are irreducible, G does not vanish identically on C , so $\operatorname{div}(G)$ is defined.

By the remark above, $\operatorname{div}(G) \sim n \operatorname{div}(L)$ for a linear form L , so

$$\deg \operatorname{div}(G) = n \deg \operatorname{div}(L) = n \deg C = nm.$$

As $\operatorname{div}(G)$ is effective with support $C \cap C'$, each point contributes at least 1 to the degree, so $|C \cap C'| \leq mn$. $\square$

The full Bézout theorem is the assertion that $\nu_P(G/X_i^n)$ is exactly the intersection multiplicity $I_P(C, C')$, so that the count is an equality when weighted correctly. With

the definition of I_P we gave, this is essentially a tautology; the work is in showing that definition has the expected properties.

§24.8.3 Riemann–Roch Spaces

Now we come to the central construction.

Definition 24.8.10 (Riemann–Roch Space)

Let $D \in \text{Div}(C)$. The **Riemann–Roch space** of D is

$$L(D) = \{f \in \mathbb{C}(C)^\times : \text{div}(f) + D \geq 0\} \cup \{0\},$$

that is, the set of f with $\nu_P(f) \geq -\nu_P(D)$ for all P . We write $\ell(D) = \dim_{\mathbb{C}} L(D)$.

Unwinding: if $\nu_P(D) = n > 0$, then f is allowed a pole of order at most n at P ; if $\nu_P(D) = -n < 0$, then f is required to vanish to order at least n at P ; and if $\nu_P(D) = 0$, then f must be regular at P . So $L(D)$ is the space of rational functions with poles bounded by D .

Proposition 24.8.11

$L(D)$ is a $\mathbb{C}$ -vector subspace of $\mathbb{C}(C)$.

Proof. Closure under scalars is clear, and $\nu_P(f + g) \geq \min\{\nu_P(f), \nu_P(g)\}$ from Corollary 24.7.5 gives closure under addition. $\square$

Example 24.8.12

Let $C = \mathbb{P}^1$, $\infty = (0 : 1)$, and $D = m[\infty]$ with $m \geq 0$. An element of $L(D)$ is a rational function with no poles in $\mathbb{C}$ – hence a polynomial in t , by Proposition 24.8.5 type reasoning – with a pole of order at most m at infinity, hence of degree at most m . So $L(D)$ is spanned by $1, t, t^2, \dots, t^m$ and

$$\ell(m[\infty]) = m + 1 = \deg D + 1.$$

Proposition 24.8.13

Let D be a divisor on C .

- (i) $L(0) = \mathbb{C}$, so $\ell(0) = 1$.
- (ii) If $\deg D < 0$ then $L(D) = 0$.
- (iii) For any $P \in C$, $\ell(D) \leq \ell(D - [P]) + 1$.
- (iv) If $\deg D \geq 0$ then $\ell(D) \leq \deg D + 1$. In particular $L(D)$ is always finite-dimensional.

Proof. (i) $L(0)$ is the set of everywhere-regular functions, which is $\mathbb{C}$ by Corollary 24.7.18(i).

(ii) If $f \in L(D)$ is nonzero then $\operatorname{div}(f) + D \geq 0$, so $0 \leq \deg(\operatorname{div} f + D) = \deg D$.

(iii) Let $n = \nu_P(D)$ and π_P a local parameter. Define

$$\operatorname{ev}_P: L(D) \rightarrow \mathbb{C}, \quad f \mapsto (\pi_P^n f)(P).$$

This makes sense: $f \in L(D)$ has $\nu_P(f) \geq -n$, so $\nu_P(\pi_P^n f) \geq 0$ and $\pi_P^n f$ is regular at P . The map is $\mathbb{C}$ -linear, and its kernel is $\{f : \nu_P(\pi_P^n f) > 0\} = \{f : \nu_P(f) \geq -n + 1\} \cap L(D) = L(D - [P])$. So $\ell(D) \leq \ell(D - [P]) + 1$.

(iv) Let $d = \deg D \geq 0$ and pick any P . Applying (iii) $d + 1$ times,

$$\ell(D) \leq \ell(D - (d + 1)[P]) + d + 1 = d + 1,$$

since $\deg(D - (d + 1)[P]) = -1 < 0$ and (ii) applies. $\square$

Intuitively, (iii) says that relaxing the pole bound at one point by one can only gain one dimension: you gain the possibility of one extra term in the local expansion.

Proposition 24.8.14

If $D \sim E$ then $L(D) \cong L(E)$ as $\mathbb{C}$ -vector spaces; in particular $\ell(D) = \ell(E)$.

Proof. Write $D - E = \operatorname{div}(g)$. Multiplication by g is a $\mathbb{C}$ -linear map $L(D) \rightarrow L(E)$: if $\operatorname{div}(f) + D \geq 0$ then $\operatorname{div}(fg) + E = \operatorname{div}(f) + \operatorname{div}(g) + E = \operatorname{div}(f) + D \geq 0$. Multiplication by g^{-1} is its inverse. $\square$

So ℓ is a function on $\operatorname{Pic}(C)$, and computing it is the central problem that Riemann–Roch solves.

§24.9 Differentials

We have produced divisors in two ways so far: from rational functions, and from hyperplane sections. There is a third source, which will turn out to be the most important of all: differentials. Their divisor class is intrinsic to the curve – it does not depend on any embedding – and the dimension of its Riemann–Roch space is the genus.

§24.9.1 Differentials of a Field Extension

The construction is formal, imitating the rules of calculus.

Definition 24.9.1 (Space of Differentials)

Let $K/\mathbb{C}$ be a field extension. The **space of differentials** $\Omega_{K/\mathbb{C}}$ is the quotient M/N , where M is the K -vector space with basis the symbols δx for $x \in K$, and $N \leq M$ is the subspace generated by

$$\delta(x + y) - \delta x - \delta y, \quad \delta(xy) - x \delta y - y \delta x, \quad \delta a$$

for $x, y \in K$ and $a \in \mathbb{C}$. We write dx for the image of δx , and $d: K \rightarrow \Omega_{K/\mathbb{C}}$ for the resulting map, the **exterior derivative**. We abbreviate $\Omega_{K/\mathbb{C}}$ to Ω_K .

So Ω_K is the universal recipient of a ‘derivative’.

Definition 24.9.2 (Derivation)

Let U be a K -vector space. A $\mathbb{C}$ -linear map $D: K \rightarrow U$ is a **derivation** if $D(xy) = xD(y) + yD(x)$ for all x, y .

Lemma 24.9.3 (Universal property)

A $\mathbb{C}$ -linear map $D: K \rightarrow U$ is a derivation if and only if there is a (necessarily unique) K -linear map $\lambda: \Omega_K \rightarrow U$ with $D = \lambda \circ d$.

Proof. If such a λ exists then D inherits the Leibniz rule from d , which satisfies it by construction. Conversely, given a derivation D , define $\tilde{\lambda}: M \rightarrow U$ on the basis by $\delta x \mapsto D(x)$ and extend K -linearly. The relations generating N are exactly the statements that D is additive, Leibniz and kills $\mathbb{C}$ (note $D(1) = D(1 \cdot 1) = 2D(1)$, so $D(1) = 0$, and $\mathbb{C}$ -linearity gives $D(a) = 0$). So $\tilde{\lambda}(N) = 0$ and $\tilde{\lambda}$ descends. $\square$

Remark. Every derivation obeys the quotient rule: if $y \neq 0$ then $D(x) = D(y \cdot \frac{x}{y}) = yD(\frac{x}{y}) + \frac{x}{y}D(y)$, so

$$D\left(\frac{x}{y}\right) = \frac{yD(x) - xD(y)}{y^2}.$$

Lemma 24.9.4 (i) Let $f = h/g \in \mathbb{C}(X_1, \dots, X_n)$ and let $x_1, \dots, x_n \in K$ be such that $y = f(x_1, \dots, x_n)$ is defined. Then

$$dy = \sum_{i=1}^n \frac{\partial f}{\partial X_i}(x_1, \dots, x_n) dx_i.$$

(ii) If $K = \mathbb{C}(x_1, \dots, x_n)$ then Ω_K is spanned over K by $dx_1, \dots, dx_n$.

Proof. (i) By $\mathbb{C}$ -linearity it suffices to treat monomials, where it is the Leibniz rule and induction, and then quotients, where it is the quotient rule. (ii) is immediate from (i), since every element of K is such an $f(x_1, \dots, x_n)$. $\square$

Theorem 24.9.5

Let $K/\mathbb{C}(t)$ be a finite extension with t transcendental over $\mathbb{C}$. Then Ω_K is a one-dimensional K -vector space, spanned by dt .

Proof. First suppose $K = \mathbb{C}(t)$. By Lemma 24.9.4(ii), Ω_K is spanned by dt , so it has dimension 0 or 1; we must rule out 0. By Lemma 24.9.3 it suffices to exhibit one nonzero derivation on K , and $\frac{d}{dt}: \mathbb{C}(t) \rightarrow \mathbb{C}(t)$ is one.

Now suppose $K \supsetneq K_0 = \mathbb{C}(t)$. Since $\mathbb{C}$ has characteristic zero, K/K_0 is separable, so $K = K_0(\alpha)$ for some α ; let $h \in K_0[T]$ be its minimal polynomial. As h is irreducible and K_0 has characteristic zero, h has no repeated roots, so $h'(\alpha) \neq 0$.

By Lemma 24.9.4, Ω_K is spanned by dt and $d\alpha$. Writing $D_t h$ for the polynomial obtained by differentiating the coefficients of h with respect to t , applying d to the relation $h(\alpha) = 0$ gives

$$0 = (D_t h)(\alpha) dt + h'(\alpha) d\alpha,$$

so $d\alpha = -\frac{(D_t h)(\alpha)}{h'(\alpha)} dt$ and Ω_K is spanned by dt alone.

It remains to show $\Omega_K \neq 0$, for which we exhibit a nonzero derivation. Define $D: K_0[T] \rightarrow K$ by

$$D\left(\sum_i c_i T^i\right) = \sum_i (D_t c_i) \alpha^i + \left(\sum_i i c_i \alpha^{i-1}\right) \cdot \left(\frac{-(D_t h)(\alpha)}{h'(\alpha)}\right).$$

One checks that D is a derivation and that $D(hK_0[T]) = 0$, using the displayed formula for $D(h)$; hence D descends to a nonzero derivation $K = K_0[T]/(h) \rightarrow K$. $\square$

Since the function field of a curve is exactly a finite extension of some $\mathbb{C}(t)$ with t transcendental (Proposition 24.5.15), the theorem applies to all our curves.

§24.9.2 Rational Differentials on a Curve

Definition 24.9.6

Let C be a curve. Write $\Omega_C = \Omega_{\mathbb{C}(C)/\mathbb{C}}$; its elements are **rational differentials** on C . A differential $\omega \in \Omega_C$ is **regular** at P if it can be written $\omega = \sum_i f_i dg_i$ with all $f_i, g_i \in \mathcal{O}_{C,P}$. We write

$$\Omega_{C,P} = \{\omega \in \Omega_C : \omega \text{ regular at } P\}.$$

Note $\Omega_{C,P}$ is an $\mathcal{O}_{C,P}$ -module but not a $\mathbb{C}(C)$ -subspace: multiplying a regular differential by a function with a pole destroys regularity.

Theorem 24.9.7

Let $P \in C$ be a point with local parameter π_P . Then $\Omega_{C,P}$ is a free $\mathcal{O}_{C,P}$ -module of rank one generated by $d\pi_P$; that is,

$$\Omega_{C,P} = \{f d\pi_P : f \in \mathcal{O}_{C,P}\}.$$

Proof. First, $\Omega_{C,P}$ is finitely generated over $\mathcal{O}_{C,P}$: choose an affine patch $C_0 \ni P$ with $\mathbb{C}[C_0] = \mathbb{C}[x_1, \dots, x_n]$. Given $f \in \mathcal{O}_{C,P}$, write $f = g/h$ with g, h polynomials in the x_i and $h(P) \neq 0$; by the quotient rule and Lemma 24.9.4,

$$df = \sum_{i=1}^n \frac{h \frac{\partial g}{\partial X_i} - g \frac{\partial h}{\partial X_i}}{h^2}(x_1, \dots, x_n) dx_i,$$

and every coefficient is regular at P since $h(P) \neq 0$. So $\Omega_{C,P}$ is generated by $dx_1, \dots, dx_n$.

Now clearly $\mathcal{O}_{C,P} d\pi_P \subseteq \Omega_{C,P}$. Conversely, for $f \in \mathcal{O}_{C,P}$ we have $f - f(P) \in \mathfrak{m}_P =$

(π_P) , so $f = f(P) + \pi_P g$ with $g \in \mathcal{O}_{C,P}$; the Leibniz rule gives

$$df = g d\pi_P + \pi_P dg \in \mathcal{O}_{C,P} d\pi_P + \mathfrak{m}_P \Omega_{C,P}.$$

Hence $\Omega_{C,P} = \mathcal{O}_{C,P} d\pi_P + \mathfrak{m}_P \Omega_{C,P}$, and Nakayama's lemma (Lemma 24.7.3(ii)) gives $\Omega_{C,P} = \mathcal{O}_{C,P} d\pi_P$. Freeness holds because $d\pi_P \neq 0$ in the one-dimensional space Ω_C and $\mathcal{O}_{C,P}$ is a domain. $\square$

Since Ω_C is one-dimensional over $\mathbb{C}(C)$, every $\omega \in \Omega_C$ can be written $\omega = f d\pi_P$ for a unique $f \in \mathbb{C}(C)$. This lets us define orders of vanishing for differentials.

Definition 24.9.8

For $\omega \in \Omega_C \setminus \{0\}$ and $P \in C$, write $\omega = f d\pi_P$ and set $\nu_P(\omega) = \nu_P(f)$.

This is well defined: if π' is another local parameter then $\pi' = u\pi_P$ with u a unit, so $d\pi' = u d\pi_P + \pi_P du$, and $du = h d\pi_P$ with h regular, so $d\pi' = (u + \pi_P h) d\pi_P$ with $u + \pi_P h$ a unit. Hence ν_P is unchanged.

Lemma 24.9.9

Let $\omega \in \Omega_C$ be nonzero. Then $\nu_P(\omega) \neq 0$ for only finitely many P .

Proof. Since $\nu_P(fdg) = \nu_P(f) + \nu_P(dg)$ and $\nu_P(f) = 0$ for all but finitely many P , it suffices to treat $\omega = dg$. Here g is non-constant (else $dg = 0$), so g is transcendental over $\mathbb{C}$ and $\mathbb{C}(C)/\mathbb{C}(g)$ is finite. Consider the morphism $\varphi = (1 : g) : C \rightarrow \mathbb{P}^1$. By Corollary 24.7.18 there are finitely many P with $g(P) = \infty$, and by Theorem 24.7.15(iii) finitely many with $e_P > 1$.

At any other P , put $Q = \varphi(P) = g(P) \in \mathbb{C}$. Then $t - g(P)$ is a local parameter at Q , and since $e_P = 1$, $\varphi^*(t - g(P)) = g - g(P)$ is a local parameter at P . Hence $dg = d(g - g(P))$ is d of a local parameter, so $\nu_P(dg) = 0$. $\square$

(Strictly, this argument uses Theorem 24.7.15(iii), which we have not yet proved. The logic is not circular: (iii) follows from the case $\omega = dt$ on $\mathbb{P}^1$, computed explicitly in Example 24.9.13 below, together with the ramification formula of Lemma 24.10.7. We record the dependency and move on.)

Definition 24.9.10 (Canonical Divisor)

For $\omega \in \Omega_C \setminus \{0\}$, set $\text{div}(\omega) = \sum_{P \in C} \nu_P(\omega)[P] \in \text{Div}(C)$.

Proposition 24.9.11

If ω, ω' are nonzero rational differentials on C , then $\text{div}(\omega) - \text{div}(\omega')$ is principal.

Proof. Ω_C is one-dimensional over $\mathbb{C}(C)$, so $\omega = f\omega'$ for some $f \in \mathbb{C}(C)^\times$, and then $\nu_P(\omega) = \nu_P(f) + \nu_P(\omega')$ at every P . $\square$

Definition 24.9.12 (Canonical Class)

The class $K_C = [\text{div}(\omega)] \in \text{Pic}(C)$, for any nonzero $\omega \in \Omega_C$, is the **canonical class** of C . Any divisor in this class is a **canonical divisor**.

Example 24.9.13

Let $C = \mathbb{P}^1$ with affine coordinate t . For $a \in \mathbb{C}$, $t - a$ is a local parameter at a and $dt = d(t - a)$, so $\nu_a(dt) = 0$. At ∞ , the local parameter is $s = 1/t$, and $t = 1/s$ gives $dt = -s^{-2}ds$, so

$$\nu_\infty(dt) = \nu_\infty(-s^{-2}) = -2.$$

Hence

$$\text{div}(dt) = -2[\infty], \quad \deg K_{\mathbb{P}^1} = -2.$$

In particular $K_{\mathbb{P}^1}$ has nonzero degree, so it is not principal: there is no rational differential on $\mathbb{P}^1$ with neither zeros nor poles.

§24.9.3 The Genus**Definition 24.9.14 (Genus)**

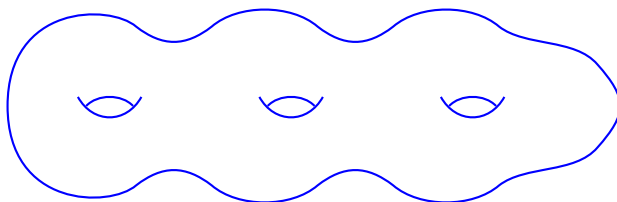
The **genus** of a curve C is $g(C) = \ell(K_C)$.

By Proposition 24.8.14 this is well defined, even though $L(K_C)$ itself depends on the choice of representative. Concretely, $L(K_C)$ is (isomorphic to) the space of **regular differentials** on C : for ω_0 a fixed nonzero differential, $f \mapsto f\omega_0$ identifies $L(\text{div } \omega_0)$ with $\{\omega \in \Omega_C : \text{div } \omega \geq 0\}$, the differentials with no poles. So the genus counts the everywhere-regular differentials.

Example 24.9.15

$g(\mathbb{P}^1) = 0$. Indeed $K_{\mathbb{P}^1} = -2[\infty]$ has degree $-2 < 0$, so $\ell(K_{\mathbb{P}^1}) = 0$ by Proposition 24.8.13(ii).

The genus is the invariant that will organise everything that follows. Topologically, a smooth projective curve over $\mathbb{C}$ is a compact orientable real surface, and g is the number of handles: $\mathbb{P}^1$ is a sphere ($g = 0$), a smooth cubic is a torus ($g = 1$), and in general the genus g surface looks like this.



a curve of genus 3

This is not a definition we can use – we have defined g algebraically – but it is the right picture, and the numerology will confirm it: we will show $\deg K_C = 2g - 2$, and $2 - 2g$ is precisely the Euler characteristic of a genus g surface.

§24.9.4 The Canonical Class of a Plane Curve

We can compute the canonical class of a smooth plane curve completely explicitly. The following calculation is the model for everything: choose a convenient differential, and compute its order of vanishing at every point using Corollary 24.7.4.

Example 24.9.16 (Smooth plane cubics)

Let $C = \mathbb{V}(F) \subseteq \mathbb{P}^2$ with $F = X_0X_2^2 - \prod_{i=1}^3(X_1 - \lambda_i X_0)$ and $\lambda_1, \lambda_2, \lambda_3$ distinct; this is the Weierstrass form of Theorem 24.6.15, and it is smooth. In the affine chart $X_0 = 1$ with coordinates $x = X_1/X_0$, $y = X_2/X_0$, the equation is

$$f(x, y) = y^2 - p(x) = 0, \quad p(x) = \prod_{i=1}^3 (x - \lambda_i).$$

Applying d gives the relation $2y dy = p'(x) dx$ in Ω_C . Take

$$\omega = \frac{dx}{y} = \frac{2 dy}{p'(x)}.$$

We compute $\text{div}(\omega)$.

If $P = (a, b)$ with $b \neq 0$, then $\partial f / \partial y = 2b \neq 0$, so $x - a$ is a local parameter at P by Corollary 24.7.4, and $\nu_P(dx) = 0$; also $\nu_P(1/y) = 0$. So $\nu_P(\omega) = 0$.

If $P = (\lambda_i, 0)$, then $\partial f / \partial x = -p'(\lambda_i) \neq 0$ (the λ_i are distinct), so y is a local parameter at P and $\nu_P(dy) = 0$; also $\nu_P(p'(x)) = 0$ since $p'(\lambda_i) \neq 0$. Using the second expression, $\nu_P(\omega) = 0$.

Finally, at the point at infinity $P_0 = (0 : 0 : 1)$: in the chart $X_2 = 1$, with $z = X_0/X_2 = 1/y$ and $w = X_1/X_2 = x/y$, the equation becomes $z = \prod_i (w - \lambda_i z)$, and one computes that w is a local parameter at P_0 with $\nu_{P_0}(z) = 3$, $\nu_{P_0}(x) = \nu_{P_0}(w/z) = -2$ and $\nu_{P_0}(y) = \nu_{P_0}(1/z) = -3$. Then $\nu_{P_0}(dx) = \nu_{P_0}(x) - 1 = -3$, so

$$\nu_{P_0}(\omega) = \nu_{P_0}(dx) - \nu_{P_0}(y) = -3 - (-3) = 0.$$

Hence $\text{div}(\omega) = 0$ and $K_C = 0$. Therefore

$$g(C) = \ell(K_C) = \ell(0) = 1.$$

Theorem 24.9.17

Every smooth plane cubic has genus 1. In particular no smooth plane cubic is isomorphic to $\mathbb{P}^1$.

Proof. By Theorem 24.6.15 we may change coordinates into the shape of Example 24.9.16, and the genus is an isomorphism invariant. Since $g(\mathbb{P}^1) = 0 \neq 1$, the last statement follows. $\square$

The general computation runs along exactly the same lines.

Theorem 24.9.18

Let $C = \mathbb{V}(F) \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d . Then

$$K_C = (d - 3)H,$$

where H is the class of a hyperplane section of C . In particular $\deg K_C = d(d - 3)$.

Proof. Change coordinates so that $(0 : 1 : 0) \notin C$. Let $x = X_1/X_0$ and $y = X_2/X_0$ in $\mathbb{C}(C)$, and set $f(X, Y) = F(1, X, Y)$, so $f(x, y) = 0$. Applying d ,

$$\frac{\partial f}{\partial X}(x, y) dx + \frac{\partial f}{\partial Y}(x, y) dy = 0 \quad \text{in } \Omega_C.$$

Choose

$$\omega = \frac{dx}{\frac{\partial f}{\partial Y}(x, y)} = \frac{-dy}{\frac{\partial f}{\partial X}(x, y)}.$$

If $P \in C \cap U_0$ has $\frac{\partial f}{\partial Y}(P) \neq 0$, then $x - x(P)$ is a local parameter and $\nu_P(\omega) = -\nu_P(\frac{\partial f}{\partial Y}) = 0$. Otherwise smoothness gives $\frac{\partial f}{\partial X}(P) \neq 0$, and the second expression gives $\nu_P(\omega) = 0$ likewise. So ω has neither zeros nor poles on the affine patch U_0 .

Since $(0 : 1 : 0) \notin C$, the points of C at infinity (i.e. with $X_0 = 0$) all have $X_2 \neq 0$, so lie in U_2 . Work there with $z = X_0/X_2$ and $w = X_1/X_2$, and $g(Z, W) = F(Z, W, 1)$, and set

$$\eta = \frac{dz}{\frac{\partial g}{\partial W}(z, w)} = \frac{-dw}{\frac{\partial g}{\partial Z}(z, w)}.$$

The same argument gives $\nu_P(\eta) = 0$ for all $P \in C \cap U_2$.

It remains to compare ω and η . We have $z = 1/y$, $w = x/y$, and F homogeneous of degree d gives $f(X, Y) = Y^d g(1/Y, X/Y)$. Differentiating this identity and substituting, a short computation yields

$$\omega = z^{d-3} \eta.$$

Hence for P with $X_2(P) \neq 0$,

$$\nu_P(\omega) = (d - 3)\nu_P(z) + \nu_P(\eta) = (d - 3)\nu_P(z),$$

and since $z = X_0/X_2$, the divisor $\sum_P \nu_P(z)[P]$ restricted to U_2 is exactly $\text{div}(X_0)$, the hyperplane section by $\mathbb{V}(X_0)$ (which is supported at infinity, where ω was not already computed to be zero). Combining the two patches,

$$\text{div}(\omega) = (d - 3)\text{div}(X_0),$$

which is $(d - 3)H$. □

Corollary 24.9.19

Smooth plane curves of distinct degrees $d, d' \geq 2$ are never isomorphic.

Proof. $\deg K_C = d(d - 3)$ depends only on the isomorphism class of C , and $d \mapsto d(d - 3)$ is injective on $\{d \geq 2\}$ (it is strictly increasing for $d \geq 2$ apart from $d(d - 3)$ taking the values $-2, 0, 4, 10, \dots$ at $d = 2, 3, 4, 5$, which are distinct). □

In particular there are infinitely many pairwise non-isomorphic smooth projective curves, which is already far from obvious.

Proposition 24.9.20

Let C be a smooth plane curve of degree $d \geq 3$ with affine equation $f(x, y) = 0$ as above. Then

$$\left\{ \frac{x^r y^s dx}{\frac{\partial f}{\partial Y}(x, y)} : r, s \geq 0, r + s \leq d - 3 \right\}$$

is a basis for $L(K_C)$, taking the representative $(d - 3)H$ with H the hyperplane at infinity. Consequently

$$g(C) = \#\{(r, s) : r, s \geq 0, r + s \leq d - 3\} = \binom{d-1}{2} = \frac{(d-1)(d-2)}{2}.$$

Proof sketch. The proof runs exactly as in Theorem 24.9.18: multiplying ω by the monomial $x^r y^s$ shifts the order of vanishing at infinity by a controlled amount, and the condition $r + s \leq d - 3$ is exactly what keeps $\text{div}(x^r y^s \omega) + (d - 3)H$ effective. Linear independence holds because the monomials $x^r y^s$ with $r + s \leq d - 3$ are linearly independent in $\mathbb{C}(C)$, as $\deg f = d$. We omit the details. $\square$

The formula $g = \frac{(d-1)(d-2)}{2}$ is the **genus–degree formula**. For $d = 1, 2$ it gives $g = 0$, consistent with lines and conics being isomorphic to $\mathbb{P}^1$; for $d = 3$ it gives $g = 1$, matching Theorem 24.9.17; for $d = 4$ it gives $g = 3$. We will re-derive it from Riemann–Roch in a moment, which is the more illuminating route.

§24.10 The Riemann–Roch Theorem

We have two invariants of a divisor D on a curve C : the degree $\deg D$, which is easy to compute, and the dimension $\ell(D)$, which is what we actually want. Proposition 24.8.13 gives the crude bound $\ell(D) \leq \deg D + 1$. Riemann–Roch says that this is an equality up to a correction term measured by the genus and a second Riemann–Roch space.

Theorem 24.10.1 (Riemann–Roch)

Let C be a smooth irreducible projective curve of genus g , and let $D \in \text{Div}(C)$. Let $K = K_C$ be a canonical divisor. Then

$$\ell(D) - \ell(K - D) = \deg D - g + 1.$$

The proof is beyond the scope of this course. It is worth knowing that it is a genuinely deep statement, and that it has several very different proofs: a sheaf-cohomological one, in which $\ell(D) = \dim H^0(C, \mathcal{O}(D))$ and $\ell(K - D) = \dim H^1(C, \mathcal{O}(D))$ and the theorem is a computation of Euler characteristics; an analytic one via Hodge theory; and Weil’s original adelic proof. It is closely related to Stokes’ theorem and the Gauss–Bonnet theorem, both of which relate a local differential-geometric quantity to a global topological one.

Let us extract its consequences, which is where all the fun is.

§24.10.1 First Consequences**Corollary 24.10.2**

$$\deg K_C = 2g - 2.$$

Proof. Apply Riemann–Roch with $D = K$:

$$\ell(K) - \ell(0) = \deg K - g + 1.$$

Now $\ell(K) = g$ by definition of the genus and $\ell(0) = 1$ by Proposition 24.8.13(i), so $g - 1 = \deg K - g + 1$, giving $\deg K = 2g - 2$. $\square$

Note that $2g - 2 = -\chi(C)$ where χ is the Euler characteristic of the associated real surface. This is the promised confirmation that our algebraic genus is the topological one.

Corollary 24.10.3

If $\deg D > 2g - 2$ then $\ell(D) = \deg D - g + 1$.

Proof. $\deg(K - D) = 2g - 2 - \deg D < 0$, so $\ell(K - D) = 0$ by Proposition 24.8.13(ii). $\square$

So for divisors of large degree, ℓ is completely determined by the degree and the genus. The ‘interesting’ range $0 \leq \deg D \leq 2g - 2$ is exactly where the geometry of the particular curve enters.

Corollary 24.10.4 (Genus–degree formula)

Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree d . Then

$$g(C) = \frac{(d-1)(d-2)}{2}.$$

Proof. For $d = 1, 2$ we know $C \cong \mathbb{P}^1$ (Proposition 24.4.30), so $g = 0$, matching the formula. For $d \geq 3$, Theorem 24.9.18 gives $K_C = (d-3)H$ with $\deg H = \deg C = d$, so $\deg K_C = d(d-3)$. By Corollary 24.10.2,

$$2g - 2 = d(d-3) \implies g = \frac{d^2 - 3d + 2}{2} = \frac{(d-1)(d-2)}{2}. \quad \square$$

Example 24.10.5

A smooth plane quartic has genus 3, a quintic genus 6, a sextic genus 10. Note that not every integer occurs: there is no smooth plane curve of genus 2. (There are plenty of genus 2 curves; they just do not embed smoothly in $\mathbb{P}^2$.)

Corollary 24.10.6

If $g(C) = 0$ then $C \cong \mathbb{P}^1$.

Proof. Pick any point $P \in C$ and let $D = [P]$, of degree $1 > -2 = 2g - 2$. By Corollary 24.10.3, $\ell(D) = 1 - 0 + 1 = 2$. So $L(D)$ contains a non-constant function f (as the constants give only a one-dimensional subspace), whose only possible pole is a simple pole at P . Consider $\varphi = (1 : f) : C \rightarrow \mathbb{P}^1$, a morphism by Corollary 24.7.10. Its degree equals the number of poles of f counted with multiplicity, which by the proof of Corollary 24.7.18 is $-\nu_P(f) = 1$. A degree 1 morphism induces an isomorphism of function fields, so $C \cong \mathbb{P}^1$ by Theorem 24.4.38; and a birational morphism of smooth projective curves is an isomorphism (its inverse is a rational map, hence a morphism by Corollary 24.7.10). $\square$

So genus 0 curves are exactly the rational ones, and there is only one of them.

§24.10.2 Riemann–Hurwitz

We now compare the genera of two curves related by a morphism. The key is that differentials pull back.

Let $\varphi : C \rightarrow D$ be a non-constant morphism of curves. If $\omega = f dt \in \Omega_D$, define

$$\varphi^*\omega = (\varphi^*f) d(\varphi^*t) \in \Omega_C.$$

This is well defined by Theorem 24.9.5 applied on both sides.

Lemma 24.10.7

Let $P \in C$, $Q = \varphi(P)$, with local parameters π_P, π_Q and ramification index e_P . Then for any nonzero $\omega \in \Omega_D$,

$$\nu_P(\varphi^*\omega) = e_P \nu_Q(\omega) + e_P - 1.$$

Proof. By definition of e_P we may write $\varphi^*\pi_Q = u\pi_P^{e_P}$ with $u \in \mathcal{O}_{C,P}^\times$. Differentiating,

$$\varphi^*(d\pi_Q) = d(u\pi_P^{e_P}) = \left(e_P u \pi_P^{e_P-1} + \pi_P^{e_P} \frac{du}{d\pi_P} \right) d\pi_P.$$

The bracket has $\nu_P = e_P - 1$: the first term has order exactly $e_P - 1$ (here we use characteristic zero, so $e_P \neq 0$ in $\mathbb{C}$), and the second has order at least e_P . So $\nu_P(\varphi^*d\pi_Q) = e_P - 1$, which is the case $\nu_Q(\omega) = 0$.

In general write $\omega = v\pi_Q^m d\pi_Q$ with $v \in \mathcal{O}_{D,Q}^\times$ and $m = \nu_Q(\omega)$, using Theorem 24.9.7. Then

$$\varphi^*\omega = (\varphi^*v)(\varphi^*\pi_Q)^m \varphi^*(d\pi_Q),$$

and taking ν_P gives $0 + me_P + (e_P - 1)$ as claimed. $\square$

The formula generalises the elementary fact that $\frac{d}{dx}x^n = nx^{n-1}$ drops the order by one; ramification of order e costs $e - 1$.

Theorem 24.10.8 (Riemann–Hurwitz)

Let $\varphi : C \rightarrow D$ be a non-constant morphism of smooth projective irreducible curves

of degree n . Then

$$2g(C) - 2 = n(2g(D) - 2) + \sum_{P \in C} (e_P - 1).$$

Proof. Choose a nonzero $\omega \in \Omega_D$. Then $\varphi^*\omega$ is a nonzero element of Ω_C , so $\text{div}(\varphi^*\omega)$ is a canonical divisor on C and by Corollary 24.10.2 its degree is $2g(C) - 2$. Now compute, grouping the points of C into fibres and using Lemma 24.10.7:

$$\begin{aligned} 2g(C) - 2 &= \sum_{P \in C} \nu_P(\varphi^*\omega) \\ &= \sum_{Q \in D} \sum_{P \in \varphi^{-1}(Q)} (e_P \nu_Q(\omega) + e_P - 1) \\ &= \sum_{Q \in D} \left(\nu_Q(\omega) \sum_{P \in \varphi^{-1}(Q)} e_P + \sum_{P \in \varphi^{-1}(Q)} (e_P - 1) \right) \\ &= \sum_{Q \in D} (n \nu_Q(\omega)) + \sum_{P \in C} (e_P - 1) \\ &= n \deg \text{div}(\omega) + \sum_{P \in C} (e_P - 1) \\ &= n(2g(D) - 2) + \sum_{P \in C} (e_P - 1), \end{aligned}$$

where the fourth line uses Theorem 24.7.15(ii). □

Note that the ramification term is a sum of non-negative integers, and (by Theorem 24.7.15(iii)) all but finitely many terms vanish, so the sum is finite. Conversely, running the argument in the case $D = \mathbb{P}^1$ shows that the sum *must* be finite, which is how one proves Theorem 24.7.15(iii).

Corollary 24.10.9

With notation as above, $g(C) \geq g(D)$. If $g(C) < g(D)$ then every rational map $C \dashrightarrow D$ is constant.

Proof. If $g(D) \geq 1$ then $2g(D) - 2 \geq 0$ and Riemann–Hurwitz gives $2g(C) - 2 \geq n(2g(D) - 2) \geq 2g(D) - 2$ since $n \geq 1$, so $g(C) \geq g(D)$. If $g(D) = 0$ there is nothing to prove. For the second statement, recall that a rational map from a smooth curve is a morphism (Corollary 24.7.10), so a non-constant one would contradict the first part. □

Example 24.10.10

There is no non-constant morphism $\mathbb{P}^1 \rightarrow C$ for any curve C of genus ≥ 1 . In particular a smooth plane cubic cannot be parametrised by rational functions – which is a purely algebraic statement with a genuinely geometric proof.

Example 24.10.11 (Hyperelliptic curves)

Consider the smooth projective model C of the affine curve $y^2 = p(x)$ where p has degree $2m$ with distinct roots, and let $\varphi: C \rightarrow \mathbb{P}^1$ be $(x, y) \mapsto x$. This has degree 2, since $\mathbb{C}(C) = \mathbb{C}(x)[y]/(y^2 - p(x))$ is a degree 2 extension of $\mathbb{C}(x)$.

Ramification occurs exactly where the two preimages coincide, i.e. at the $2m$ roots of p , and there $e_P = 2$; one checks the two points at infinity are unramified when $\deg p$ is even. So Riemann–Hurwitz gives

$$2g(C) - 2 = 2(-2) + 2m \cdot 1 = 2m - 4,$$

so $g(C) = m - 1$. Taking $m = 2$ (a quartic p) gives $g = 1$; $m = 3$ gives $g = 2$; and in general every genus is achieved. So genus 2 curves exist even though no plane curve has genus 2.

Example 24.10.12

The map $\varphi: \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $z \mapsto z^d$, has degree d and is ramified exactly at 0 and ∞ , each with $e = d$. Riemann–Hurwitz reads

$$-2 = d(-2) + 2(d - 1),$$

which is indeed an identity. This is a useful sanity check on the signs.

The next example is the classical one, and it is worth doing slowly because it shows Riemann–Hurwitz being used to count something one would not otherwise know how to count.

Example 24.10.13 (The class of a plane curve)

Let $C \subseteq \mathbb{P}^2$ be a smooth plane curve of degree $d \geq 2$, so $g = \frac{(d-1)(d-2)}{2}$ by Corollary 24.10.4. Fix a point $R \in \mathbb{P}^2 \setminus C$ and let $\pi: C \rightarrow \mathbb{P}^1$ be the projection from R of Example 24.4.28, restricted to C ; this is a morphism by Corollary 24.7.10. The fibre over a point of $\mathbb{P}^1$ is $C \cap L$ for the corresponding line L through R , so $\deg \pi = d$.

What is the ramification? Choose R general – in particular not on any tangent line of C that is tangent at two points, and not on any flex tangent. Then $e_P = I_P(C, L) \leq 2$ for the line $L = \overline{RP}$, with $e_P = 2$ exactly when L is tangent to C at P . So $\sum_P (e_P - 1)$ counts the points of C at which the tangent line passes through R . Riemann–Hurwitz gives

$$(d - 1)(d - 2) - 2 = d \cdot (-2) + \sum_P (e_P - 1),$$

so

$$\sum_P (e_P - 1) = d^2 - 3d + 2 - 2 + 2d = d(d - 1).$$

Thus from a general point of $\mathbb{P}^2$ there are exactly $d(d - 1)$ tangent lines to a smooth plane curve of degree d . This number is called the **class** of C . For $d = 2$ it says there are two tangents from an external point to a conic, which is familiar; for $d = 3$ it gives six tangents to a smooth cubic.

Example 24.10.14

Projecting from a point R on C instead, the line $\overline{RP}$ already meets C at R , so the fibres have size $d - 1$ and $\deg \pi = d - 1$. Riemann–Hurwitz now reads

$$(d - 1)(d - 2) - 2 = (d - 1)(-2) + \sum_P (e_P - 1),$$

giving $\sum_P (e_P - 1) = d^2 - d - 2$. For $d = 3$ this is 4: a smooth cubic, projected from a point on it, is a double cover of $\mathbb{P}^1$ branched over four points. That is exactly the degree 2 map $(x, y) \mapsto x$ on $y^2 = p(x)$ with $\deg p = 3$, branched at the three roots of p and at infinity.

Definition 24.10.15

A curve C with $g(C) \geq 2$ is **hyperelliptic** if it admits a morphism $C \rightarrow \mathbb{P}^1$ of degree 2.

The curves of Example 24.10.11 with $m \geq 3$ are hyperelliptic. One shows (it is on the last example sheet) that C is hyperelliptic if and only if there is a divisor D with $\deg D = 2$ and $\ell(D) = 2$; the degree 2 map is then φ_D , constructed below.

§24.11 Curves of Low Genus

We finish by using Riemann–Roch to answer concrete questions: which curves are there, and how do we embed them in projective space?

§24.11.1 Maps to Projective Space from Divisors

Riemann–Roch produces functions; functions produce maps to projective space. Here is the construction.

Definition 24.11.1

Let D be a divisor on C with $\ell(D) = n + 1 \geq 2$, and let $f_0, \dots, f_n$ be a basis of $L(D)$. The **morphism associated to D** is

$$\varphi_D: C \rightarrow \mathbb{P}^n, \quad P \mapsto (f_0(P) : \dots : f_n(P)).$$

This is a rational map, hence (Corollary 24.7.10) a morphism, and it is independent of the choice of basis up to an element of $\mathrm{PGL}_{n+1}(\mathbb{C})$. We say φ_D is an **embedding** if it is an isomorphism onto its image.

Definition 24.11.2

A divisor D satisfies condition $(\star)$ if for all $P, Q \in C$ (not necessarily distinct),

$$\ell(D - [P] - [Q]) = \ell(D) - 2.$$

By Proposition 24.8.13(iii), $\ell(D) - 2 \leq \ell(D - [P] - [Q]) \leq \ell(D)$ always, so $(\star)$ says the drop is as large as possible. Geometrically: taking $P \neq Q$, condition $(\star)$ says there is a

function in $L(D)$ vanishing at P but not at Q , i.e. φ_D separates P and Q ; taking $P = Q$, it says φ_D separates tangent directions at P , i.e. $d(\varphi_D)_P$ is injective.

Theorem 24.11.3

φ_D is an embedding if and only if D satisfies $(\star)$.

We omit the proof, which is a careful unwinding of the two conditions in the previous paragraph.

Corollary 24.11.4

If $\deg D > 2g$ then φ_D is an embedding.

Proof. Both $\deg D$ and $\deg(D - [P] - [Q]) = \deg D - 2$ exceed $2g - 2$, so Corollary 24.10.3 applies to both, giving

$$\ell(D) - \ell(D - [P] - [Q]) = \deg D - (\deg D - 2) = 2. \quad \square$$

Corollary 24.11.5

Every curve of genus g embeds in $\mathbb{P}^N$ for some N depending only on g .

Proof. If $g \geq 2$, take D in the class $2K_C$, of degree $4g - 4 > 2g$. If $g = 1$, take $D = 3[P_0]$ for any P_0 , of degree $3 > 2$. If $g = 0$, $C \cong \mathbb{P}^1$ already. In each case $\deg D > 2g$, so Corollary 24.11.4 applies, and $N = \ell(D) - 1$ is determined by g via Corollary 24.10.3. $\square$

Theorem 24.11.6 (Canonical embedding)

Let C be a non-hyperelliptic curve of genus $g \geq 3$. Then $\varphi_{K_C}: C \rightarrow \mathbb{P}^{g-1}$ is an embedding.

Proof. Note $\ell(K) = g \geq 3 \geq 2$, so φ_K is defined and maps to $\mathbb{P}^{g-1}$. Suppose K fails $(\star)$, so there are P, Q with $\ell(K - [P] - [Q]) \geq g - 1$. Apply Riemann–Roch to $D = [P] + [Q]$:

$$\ell(D) - \ell(K - D) = \deg D - g + 1 = 3 - g,$$

so $\ell(D) \geq (g - 1) + 3 - g = 2$. But $\ell(D) = 2$ with $\deg D = 2$ says exactly that C is hyperelliptic, contradiction. (And $\ell(D) \leq \deg D + 1 = 3$; if $\ell(D) = 3$ then by Corollary 24.10.6-type reasoning $C \cong \mathbb{P}^1$, excluded since $g \geq 3$.) $\square$

So every non-hyperelliptic curve has a canonical model in $\mathbb{P}^{g-1}$, determined intrinsically with no choices. For $g = 3$ this is a plane quartic, matching the genus–degree formula.

§24.11.2 Genus Zero: Rational Curves

We have already done this case, but let us collect the statements.

Theorem 24.11.7

For a smooth projective irreducible curve C , the following are equivalent:

- (i) $g(C) = 0$;
- (ii) $C \cong \mathbb{P}^1$;
- (iii) $\mathbb{C}(C) \cong \mathbb{C}(t)$;
- (iv) C has a divisor of degree 1 that is linearly equivalent to an effective divisor with $\ell = 2$;
- (v) $\text{Pic}^0(C) = 0$.

Proof. (i) $\Rightarrow$ (ii) is Corollary 24.10.6. (ii) $\Rightarrow$ (iii) is clear, and (iii) $\Rightarrow$ (ii) is Theorem 24.4.38 together with Corollary 24.7.10. (ii) $\Rightarrow$ (v) is Proposition 24.8.5. For (v) $\Rightarrow$ (i), if $g \geq 1$ then for distinct P, Q we have $\ell([P]) = 1$ by Corollary 24.10.3 if $g = 1$ (or Proposition 24.8.13 in general), so $[P] \not\sim [Q]$ and $[P] - [Q]$ is a nonzero element of $\text{Pic}^0(C)$. The equivalence with (iv) is the computation in Corollary 24.10.6. $\square$

Example 24.11.8

The conic $C = \mathbb{V}(X_0X_2 - X_1^2)$ has genus 0, so is isomorphic to $\mathbb{P}^1$; we checked this directly in Proposition 24.4.30, and the map $\mathbb{P}^1 \rightarrow C$ is φ_D for $D = 2[\infty]$, since $L(2[\infty])$ has basis $1, t, t^2$.

Example 24.11.9

The nodal cubic $\mathbb{V}(Y^2 - X^2(X + 1))$ is *not* smooth, so our theory does not directly apply. But it is birational to $\mathbb{P}^1$: parametrise by the slope $t = Y/X$ of the line through the node, giving $t^2 = X + 1$, so $X = t^2 - 1$ and $Y = t(t^2 - 1)$. This is a rational parametrisation, and the two values $t = \pm 1$ both map to the node. So the smooth projective curve birational to it is $\mathbb{P}^1$, and the node is what you get by gluing two points of $\mathbb{P}^1$ together. The same works for the cusp, with $t = Y/X$ giving (t^2, t^3) .

§24.11.3 Genus One: Elliptic Curves**Definition 24.11.10 (Elliptic Curve)**

An **elliptic curve** is a pair $E = (C, P_0)$ where C is a smooth projective irreducible curve of genus 1 and $P_0 \in C$.

The choice of base point is essential: in genus 1 there is no canonical point, and different choices give isomorphic curves but different groups structures (well, isomorphic ones, but the isomorphism is a translation).

Genus 1 is the first case where the interesting range $0 \leq \deg D \leq 2g - 2$ is nonempty – but only just, since $2g - 2 = 0$. This makes the arithmetic particularly clean.

Corollary 24.11.11

Let C have genus 1 and let D be a divisor with $\deg D > 0$. Then $\ell(D) = \deg D$.

Proof. K_C has degree $2g - 2 = 0$, so $\deg(K - D) < 0$ and $\ell(K - D) = 0$. Riemann–Roch gives $\ell(D) = \deg D - 1 + 1 = \deg D$. $\square$

Now here is the construction of the group law, done intrinsically.

Theorem 24.11.12

Let $E = (C, P_0)$ be an elliptic curve. Then the map

$$\beta: C \rightarrow \text{Pic}^0(C), \quad P \mapsto [P] - [P_0]$$

is a bijection. Transporting the group structure of $\text{Pic}^0(C)$ back along β makes C into an abelian group with identity P_0 .

Proof. Injectivity. If $\beta(P) = \beta(Q)$ then $[P] \sim [Q]$, so by Proposition 24.8.14, $\ell([P]) = \ell([Q])$ and there is $f \in \mathbb{C}(C)^\times$ with $\text{div}(f) = [P] - [Q]$. If $P \neq Q$ then f has a single simple pole, so $\varphi = (1 : f): C \rightarrow \mathbb{P}^1$ has degree 1 and is therefore an isomorphism, forcing $g(C) = 0$, a contradiction. Hence $P = Q$.

Surjectivity. Let $D \in \text{Div}^0(C)$. Then $\deg(D + [P_0]) = 1 > 0$, so $\ell(D + [P_0]) = 1$ by Corollary 24.11.11. Pick $0 \neq f \in L(D + [P_0])$; then $\text{div}(f) + D + [P_0]$ is effective of degree 1, hence equals $[P]$ for a unique point P . So $D + [P_0] \sim [P]$, i.e. $D \sim [P] - [P_0]$ and $D = \beta(P)$ in Pic^0 . $\square$

Definition 24.11.13

For $P, Q \in E$ define $P +_E Q$ to be the unique point R with $\beta(R) = \beta(P) + \beta(Q)$, i.e. with

$$[P] + [Q] - [P_0] \sim [R].$$

Theorem 24.11.14

Let $E = (C, P_0)$ be an elliptic curve realised as a smooth plane cubic with P_0 a flex. Then for $P, Q, R \in C$,

$$P +_E Q +_E R = P_0 \iff P, Q, R \text{ are collinear.}$$

Consequently $+_E$ agrees with the chord-and-tangent law of Definition 24.6.16, and in particular that law is associative.

Proof. Let L be a line, with $\text{div}(L) = [P] + [Q] + [R]$ its divisor on C (a degree 3 effective divisor by Bézout). Let L_0 be the tangent line to C at the flex P_0 , so $\text{div}(L_0) = 3[P_0]$. By Proposition 24.8.7,

$$\text{div}(L) - \text{div}(L_0) = \text{div}(L/L_0)$$

is principal, so $[P] + [Q] + [R] \sim 3[P_0]$, that is,

$$\beta(P) + \beta(Q) + \beta(R) = [[P] + [Q] + [R] - 3[P_0]] = 0.$$

Hence $P +_E Q +_E R = P_0$.

Conversely, suppose $P +_E Q +_E R = P_0$, and let L be the line through P and Q , meeting C again at R' . By the above, $P +_E Q +_E R' = P_0$, so $R = R'$ and P, Q, R are collinear.

Finally, the chord-and-tangent law sets $P + Q = P_0 * (P * Q)$; writing $S = P * Q$, the points P, Q, S are collinear so $P +_E Q +_E S = P_0$, and $P_0, S, P_0 * S$ are collinear so $S +_E (P_0 * S) = P_0$ (using $P_0 +_E X = X$). Hence $P_0 * S = -_E S = P +_E Q$. $\square$

This completes the proof of Theorem 24.6.17: associativity of the chord-and-tangent law is inherited from associativity of addition in $\text{Pic}^0(C)$, which is free.

Example 24.11.15

On $E: y^2 = x^3 - x + 1$ with P_0 the point at infinity, take $P = (-1, -1)$ and $Q = (1.6, 1.8698\dots)$. The line through them has slope $m = 1.1038\dots$, and substituting into the cubic gives $x^3 - m^2x^2 - (\dots)x - (\dots) = 0$ whose roots sum to m^2 . Hence

$$x_{P*Q} = m^2 - x_P - x_Q = 0.6183\dots,$$

and $P +_E Q = (0.6183\dots, -0.7862\dots)$. This is the first figure of Section 6.6. Similarly the tangent at $T = (1, 1)$ has slope $(3 \cdot 1 - 1)/2 = 1$, so $x_{2T} = 1 - 2 = -1$ and $2T = (-1, 1)$.

Remark. Over $\mathbb{C}$, a genus 1 curve is topologically a torus $S^1 \times S^1$, and in fact $C \cong \mathbb{C}/\Lambda$ for a lattice $\Lambda \leq \mathbb{C}$, as a complex manifold and as a group. The group law $+_E$ corresponds to addition in $\mathbb{C}/\Lambda$, and the Weierstrass form comes from the Weierstrass $\wp$ -function, which satisfies

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Over $\mathbb{Q}$, the group $E(\mathbb{Q})$ of rational points is finitely generated (Mordell's theorem), and this is where the subject meets number theory.

§24.11.4 Recovering the Weierstrass Equation

We defined an elliptic curve twice: once concretely, as a smooth plane cubic with a chosen flex (Theorem 24.6.15), and once abstractly, as a genus 1 curve with a chosen point. It is time to show that the two definitions agree, and the argument is a lovely illustration of how Riemann–Roch is used in practice: one writes down the spaces $L(n[P_0])$ for increasing n , watches their dimensions grow by one each time, and waits for the moment when there are more obvious elements than there is room for.

Theorem 24.11.16

Let $E = (C, P_0)$ be an elliptic curve. Then there exist $x, y \in \mathbb{C}(C)$ and $a, b \in \mathbb{C}$ with $4a^3 + 27b^2 \neq 0$ such that

$$\varphi = (1 : x : y) : C \longrightarrow \mathbb{P}^2$$

is an isomorphism onto the smooth plane cubic $\mathbb{V}(X_0X_2^2 - X_1^3 - aX_0^2X_1 - bX_0^3)$, carrying P_0 to the flex $(0 : 0 : 1)$.

Proof. Write $D_n = n[P_0]$. By Corollary 24.11.11, $\ell(D_n) = n$ for all $n \geq 1$. We build up bases.

First, $L(D_1) = \mathbb{C}$: a function with at most a simple pole at P_0 and no other poles would, if non-constant, give a degree 1 morphism $C \rightarrow \mathbb{P}^1$ and hence force $g = 0$. Since $\ell(D_2) = 2$, there is $x \in L(D_2) \setminus \mathbb{C}$, and x must have a pole of order exactly 2 at P_0 (order 1 is excluded as above). Since $\ell(D_3) = 3$, there is $y \in L(D_3) \setminus L(D_2)$, necessarily with a pole of order exactly 3. So

$$\nu_{P_0}(x) = -2, \quad \nu_{P_0}(y) = -3,$$

and $\{1, x\}$, $\{1, x, y\}$ are bases of $L(D_2), L(D_3)$.

Now count pole orders. The functions

$$1, x, y, x^2, xy, x^3, y^2$$

have poles at P_0 of orders 0, 2, 3, 4, 5, 6, 6 respectively, and no poles elsewhere; so all seven lie in $L(D_6)$. But $\ell(D_6) = 6$, so they satisfy a nontrivial linear relation

$$\alpha y^2 + \beta xy + \gamma y = \delta x^3 + \varepsilon x^2 + \zeta x + \eta.$$

Here $\alpha \neq 0$: otherwise the left side lies in $L(D_5)$ while the right has a pole of order exactly 6 unless $\delta = 0$ too, and then $1, x, y, x^2, xy$ would be dependent, which they are not (their pole orders 0, 2, 3, 4, 5 are distinct, and functions with distinct pole orders at a single point are linearly independent). The same argument gives $\delta \neq 0$. Rescaling $x \mapsto \lambda x$, $y \mapsto \mu y$ suitably, we may take $\alpha = \delta = 1$. Completing the square in y (replacing y by $y - \frac{1}{2}(\beta x + \gamma)$) and then the cube in x removes the remaining terms, leaving

$$y^2 = x^3 + ax + b.$$

It remains to identify the map. We have $\deg D_3 = 3 > 2 = 2g$, so by Corollary 24.11.4 the morphism $\varphi_{D_3} = (1 : x : y)$ is an embedding. Its image lies in the cubic $\mathbb{V}(F)$, $F = X_0X_2^2 - X_1^3 - aX_0^2X_1 - bX_0^3$, by the relation just derived, and the image is closed (Proposition 24.7.17) and irreducible of dimension 1. If F were reducible then the image, being irreducible, would lie in a component of degree 1 or 2, hence in a line or a conic; but those are isomorphic to $\mathbb{P}^1$ by Proposition 24.4.30, and $C \cong \varphi(C)$ has genus 1. So F is irreducible and the image is all of $\mathbb{V}(F)$, giving $C \cong \mathbb{V}(F)$; since C is smooth, so is $\mathbb{V}(F)$, and Theorem 24.6.15 says this forces $x^3 + ax + b$ to have distinct roots, i.e. $4a^3 + 27b^2 \neq 0$.

Finally, x and y have poles only at P_0 , so $\varphi(P_0)$ is the unique point of the image with $X_0 = 0$; comparing pole orders, $\varphi(P_0) = (0 : 0 : 1)$, which is the flex at infinity of the Weierstrass cubic. $\square$

So the two definitions of ‘elliptic curve’ coincide: a genus 1 curve with a marked point is the same thing as a plane cubic in Weierstrass form with its flex at infinity, and by Theorem 24.11.12 the group law we defined intrinsically is the chord-and-tangent construction.

Example 24.11.17

The recipe is completely explicit. On the curve $y^2 = x^3 - x + 1$, the space $L(2[P_0])$ is spanned by 1 and x , the space $L(3[P_0])$ by 1, x , y , and the space $L(4[P_0])$ by 1, x , y , x^2 . To see the last of these directly: a function with a pole of order at most 4 at infinity and no other poles is a polynomial in x and y of the shape $c_0 + c_1x + c_2y + c_3x^2$, since $\nu_{P_0}(x^iy^j) = -(2i + 3j)$ and the only (i, j) with $2i + 3j \leq 4$ and $j \leq 1$ (we may always reduce j using $y^2 = x^3 - x + 1$) are $(0, 0), (1, 0), (0, 1), (2, 0)$.

§24.11.5 Genus Two and Beyond**Proposition 24.11.18**

Every curve of genus 2 is hyperelliptic.

Proof. Take $D = K_C$, of degree $2g - 2 = 2$, with $\ell(D) = g = 2$. This is precisely the criterion for hyperellipticity noted after Example 24.10.11: φ_K is a morphism $C \rightarrow \mathbb{P}^1$, and by Theorem 24.11.3 it is not an embedding (it maps a genus 2 curve to a genus 0 one), so it has degree 2. $\square$

Example 24.11.19

Genus 2 curves are exactly the smooth projective models of $y^2 = p(x)$ with $\deg p = 6$ (or 5) and p squarefree, by Example 24.10.11 with $m = 3$. Up to isomorphism they are parametrised by the six branch points modulo $\mathrm{PGL}_2(\mathbb{C})$, giving a 3-dimensional family. In general, curves of genus $g \geq 2$ form a $(3g - 3)$ -dimensional family – the **moduli space** $\mathcal{M}_g$ – which one can already guess from this kind of count.

Example 24.11.20

Let us determine all the curves of genus 3. If C is non-hyperelliptic, the canonical map embeds C in $\mathbb{P}^{g-1} = \mathbb{P}^2$ as a curve of degree $\deg K = 2g - 2 = 4$, i.e. a smooth plane quartic; and conversely every smooth plane quartic has genus 3 by the genus–degree formula. If C is hyperelliptic, it is the smooth model of $y^2 = p(x)$ with $\deg p = 8$. So genus 3 curves are either plane quartics or hyperelliptic.

We close by recording the classification of maps in the simplest cases, which is a good test of everything we have set up.

Proposition 24.11.21

Let C be a curve of genus g .

- (i) If $g = 0$, the morphisms $C \rightarrow \mathbb{P}^1$ of degree d are exactly φ_D for D of degree d , and there is a $(2d + 1)$ -dimensional family of them.
- (ii) If $g \geq 1$, there is no morphism $C \rightarrow \mathbb{P}^1$ of degree 1.
- (iii) If $g \geq 2$, every morphism $C \rightarrow C$ of degree 1 is an automorphism, and the group of these is finite (of order at most $84(g - 1)$, by Hurwitz’s automorphism theorem).

Proof of (ii). A degree 1 morphism induces an isomorphism of function fields, hence is an isomorphism of curves, so $C \cong \mathbb{P}^1$ and $g = 0$. $\square$

Remark. Hurwitz's bound in (iii) is a beautiful application of Riemann–Hurwitz. If $G = \text{Aut}(C)$ is finite of order N , the quotient C/G is a curve D of some genus h , and $C \rightarrow D$ has degree N ; analysing the possible ramification data with Riemann–Hurwitz shows that

$$2g - 2 = N \left(2h - 2 + \sum_i \left(1 - \frac{1}{r_i} \right) \right)$$

for the ramification orders r_i over the branch points, and the smallest positive value the bracket can take is $1/42$, attained when $h = 0$ and $(r_1, r_2, r_3) = (2, 3, 7)$. Hence $N \leq 84(g - 1)$.

§24.11.6 A Summary

It is worth stepping back. We began with the dictionary

$$\{\text{affine varieties}\} \longleftrightarrow \{\text{reduced finitely generated } \mathbb{C}\text{-algebras}\},$$

refined it to

$$\left\{ \begin{array}{c} \text{irreducible varieties up to} \\ \text{birational equivalence} \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{finitely generated} \\ \text{field extensions of } \mathbb{C} \end{array} \right\},$$

and then specialised to transcendence degree 1, where the classification becomes tractable. For curves we found:

- every function field of transcendence degree 1 is the function field of a *unique* smooth projective curve, since birational smooth projective curves are isomorphic (Corollary 24.7.10);
- each such curve carries an intrinsic invariant, the genus $g = \ell(K_C)$, computable from any plane model by the genus–degree formula, and from any map to $\mathbb{P}^1$ by Riemann–Hurwitz;
- the finite-dimensional spaces $L(D)$ are computed by Riemann–Roch, and they in turn produce all the maps from C to projective space;
- genus 0 gives exactly $\mathbb{P}^1$; genus 1 gives the elliptic curves, whose points form a group; and genus ≥ 2 gives rigid objects with finite automorphism groups.

The natural next steps – sheaves and their cohomology, which make Riemann–Roch a formal consequence of a general index theorem, and schemes, which restore the nilpotents we threw away in Section 2 – are the subject of Part III.

25 Algebraic Topology

Topology is the study of spaces up to continuous deformation, and that is a slippery thing to get a handle on. Given two spaces, how could we possibly hope to show that there is *no* homeomorphism between them? There are infinitely many candidate maps, and no amount of failing to find one counts as a proof.

Algebraic topology answers this by translating the problem into algebra. We will build machines that eat a topological space and spit out a group, in such a way that homeomorphic (indeed, homotopy equivalent) spaces give isomorphic groups. Then two spaces with non-isomorphic groups are certainly not homeomorphic, and a question with infinitely many cases has become a finite computation. We will build two such machines: the *fundamental group*, which measures the failure of loops in a space to be contracted to a point, and *homology*, which detects holes of every dimension at once. Along the way we will get some genuinely striking theorems almost for free – that $\mathbb{R}^n \cong \mathbb{R}^m$ forces $n = m$, that every continuous map of a disc to itself has a fixed point, and that every non-constant complex polynomial has a root.

This article constitutes my notes for the ‘Algebraic Topology’ course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§25.1 The Fundamental Group

Throughout these notes, a **space** means a topological space, and a **map** means a continuous function – we will say ‘function’ when we do not want to assume continuity. We write $X \cong Y$ to mean that X and Y are homeomorphic, and $G \cong H$ to mean that groups G and H are isomorphic. A few spaces will appear often enough to deserve names:

- the one-point space $*$;
- the unit interval $I = [0, 1] \subseteq \mathbb{R}$, and the cube I^n ;
- the disc $D^n = \{v \in \mathbb{R}^n \mid \|v\| \leq 1\}$, which is homeomorphic to I^n ;
- the sphere $S^{n-1} = \{v \in \mathbb{R}^n \mid \|v\| = 1\}$.

We write id_X for the identity map on X , and c_p for a constant map with value p (the domain being clear from context).

The two results we are aiming at in the first instance are that $\mathbb{R}^n \cong \mathbb{R}^m$ implies $n = m$, and that every map $D^n \rightarrow D^n$ has a fixed point. For $n = 1$ both are easy: removing a point from $\mathbb{R}$ disconnects it, which distinguishes $\mathbb{R}$ from $\mathbb{R}^m$ for $m \geq 2$, and the fixed point statement is the intermediate value theorem. What we need is a way of running these arguments in higher dimensions, and that requires something finer than connectedness.

§25.1.1 Homotopy

The right notion of ‘sameness’ in this subject turns out not to be homeomorphism but something coarser. We want to regard two maps as the same if one can be continuously

deformed into the other.

Definition 25.1.1 (Homotopy)

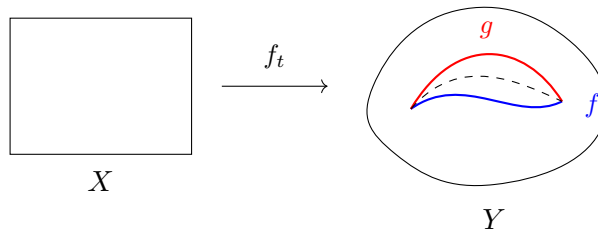
Let $f, g : X \rightarrow Y$ be maps. A **homotopy** from f to g is a map

$$H : X \times I \rightarrow Y$$

such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$.

If such an H exists, we say f is **homotopic** to g , and write $f \simeq g$ or $f \simeq_H g$.

We should think of the second coordinate as time: at each time $t \in I$ we have a map $f_t = H(-, t) : X \rightarrow Y$, and as t runs from 0 to 1 these interpolate between f and g . The content of continuity of H is that the family (f_t) varies continuously in t , so there is no jumping.



Often we want to deform f into g without moving certain points at all.

Definition 25.1.2 (Homotopy Relative to a Subspace)

Let $A \subseteq X$. We say f is homotopic to g **relative to A** , written $f \simeq g \text{ rel } A$, if there is a homotopy H from f to g with $H(a, t) = f(a) = g(a)$ for all $a \in A$ and all $t \in I$.

Taking $A = \emptyset$ recovers ordinary homotopy, so it costs nothing to prove things in this generality.

Proposition 25.1.3

For spaces X, Y and $A \subseteq X$, ‘homotopic rel A ’ is an equivalence relation on maps $X \rightarrow Y$. In particular, when $A = \emptyset$, homotopy is an equivalence relation.

Proof. We check the three properties.

1. *Reflexivity:* $f \simeq f$ since $H(x, t) = f(x)$ is a homotopy.
2. *Symmetry:* if $H(x, t)$ is a homotopy from f to g , then $H(x, 1-t)$ is a homotopy from g to f .
3. *Transitivity:* Suppose $f, g, h : X \rightarrow Y$ and $f \simeq_H g \text{ rel } A$, $g \simeq_{H'} h \text{ rel } A$. We want to show that $f \simeq h \text{ rel } A$. The idea is to ‘glue’ the two homotopies together: we know how to continuously deform f to g , and g to h , so we just do these one after another, at double speed. We define $H'' : X \times I \rightarrow Y$ by

$$H''(x, t) = \begin{cases} H(x, 2t) & 0 \leq t \leq \frac{1}{2} \\ H'(x, 2t - 1) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

This is well-defined since $H(x, 1) = g(x) = H'(x, 0)$, and continuous by the gluing lemma. It is easy to check that H'' is a homotopy rel A . $\square$

We used the gluing lemma there, and we will use it constantly, so let us record it.

Lemma 25.1.4 (Gluing Lemma)

Let $X = C_1 \cup C_2$ with C_1, C_2 closed in X , and let $f : X \rightarrow Y$ be a function such that $f|_{C_1}$ and $f|_{C_2}$ are both continuous. Then f is continuous.

Proof. It is enough to show that preimages of closed sets are closed. Let $K \subseteq Y$ be closed. Then $f^{-1}(K) \cap C_i = (f|_{C_i})^{-1}(K)$ is closed in C_i , and hence closed in X as C_i is closed in X . So $f^{-1}(K) = (f^{-1}(K) \cap C_1) \cup (f^{-1}(K) \cap C_2)$ is a union of two closed sets, and is closed. $\square$

Homotopy behaves well under composition, which is what makes the whole theory work.

Lemma 25.1.5

Consider the spaces and maps

$$\begin{array}{ccccc} X & \xrightarrow{f_0} & Y & \xrightarrow{g_0} & Z \\ & \searrow f_1 & & \searrow g_1 & \\ & & Y & & \end{array}$$

If $f_0 \simeq_H f_1$ and $g_0 \simeq_{H'} g_1$, then $g_0 \circ f_0 \simeq g_1 \circ f_1$.

Proof. We will show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$, and will then be done since homotopy is an equivalence relation. We write down two homotopies.

1. The composition

$$X \times I \xrightarrow{H} Y \xrightarrow{g_0} Z$$

is the first homotopy we need, showing $g_0 \circ f_0 \simeq g_0 \circ f_1$.

2. The following composition is a homotopy from $g_0 \circ f_1$ to $g_1 \circ f_1$:

$$X \times I \xrightarrow{f_1 \times \text{id}_I} Y \times I \xrightarrow{H'} Z$$

Indeed at $t = 0$ this is $H'(f_1(x), 0) = g_0(f_1(x))$ and at $t = 1$ it is $g_1(f_1(x))$. $\square$

Now that we have a notion of two maps being ‘the same’, we get a notion of two spaces being the same: we ask for maps in both directions whose composites are homotopic to, rather than equal to, the identity.

Definition 25.1.6 (Homotopy Equivalence)

A map $f : X \rightarrow Y$ is a **homotopy equivalence** if there exists a map $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. We call g a **homotopy inverse** for f .

If a homotopy equivalence $f : X \rightarrow Y$ exists, we say X and Y are **homotopy equivalent**, and write $X \simeq Y$.

Note the deliberate abuse: $\simeq$ means homotopy for maps and homotopy equivalence for spaces. Any homeomorphism is in particular a homotopy equivalence, so $X \cong Y$ implies $X \simeq Y$, but the converse fails badly – as we will see, $\mathbb{R}^2$ is homotopy equivalent to a point.

Proposition 25.1.7

Homotopy equivalence of spaces is an equivalence relation.

Proof. Symmetry and reflexivity are immediate from the definition. To show transitivity, let $f : X \rightarrow Y$ and $h : Y \rightarrow Z$ be homotopy equivalences, and $g : Y \rightarrow X$ and $k : Z \rightarrow Y$ be their homotopy inverses. We will show that $h \circ f : X \rightarrow Z$ is a homotopy equivalence with homotopy inverse $g \circ k$. Using Lemma 25.1.5 freely, we have

$$(h \circ f) \circ (g \circ k) = h \circ (f \circ g) \circ k \simeq h \circ \text{id}_Y \circ k = h \circ k \simeq \text{id}_Z.$$

Similarly,

$$(g \circ k) \circ (h \circ f) = g \circ (k \circ h) \circ f \simeq g \circ \text{id}_Y \circ f = g \circ f \simeq \text{id}_X,$$

as required. $\square$

Definition 25.1.8 (Contractible)

If $X \simeq *$, the one point space, then X is **contractible**.

Unwinding this, X is contractible exactly when $\text{id}_X \simeq c_p$ for some $p \in X$; the homotopy is a way of continuously shrinking the whole space down to a point inside itself.

Example 25.1.9

Any non-empty convex $Y \subseteq \mathbb{R}^n$ is contractible. Indeed, fix $p \in Y$ and define $H(y, t) = (1 - t)y + tp$, which lies in Y by convexity and is a homotopy from id_Y to c_p . In particular $\mathbb{R}^n$ and D^n and I^n are all contractible.

Proposition 25.1.10

Let Y be contractible. Then any two maps $f_0, f_1 : X \rightarrow Y$ are homotopic.

Proof. Say $\text{id}_Y \simeq c_p$. Then $f_0 = \text{id}_Y \circ f_0 \simeq c_p \circ f_0 = c_p$, and likewise $f_1 \simeq c_p$, so $f_0 \simeq f_1$ by transitivity. $\square$

Corollary 25.1.11

A contractible space is path connected.

Proof. Let $p, q \in Y$. Applying Proposition 25.1.10 with $X = *$, the constant maps $* \rightarrow Y$ at p and at q are homotopic via some $H : * \times I \rightarrow Y$. Then $t \mapsto H(*, t)$ is a path from p to q . $\square$

A very common way of exhibiting a homotopy equivalence is to squash a space down onto a subspace sitting inside it.

Definition 25.1.12 (Retraction)

Let $A \subseteq X$ be a subspace. A **retraction** $r : X \rightarrow A$ is a map such that $r \circ \iota = \text{id}_A$, where $\iota : A \hookrightarrow X$ is the inclusion. If such an r exists, we call A a **retract** of X .

The retraction r is a **deformation retraction** if $\iota \circ r \simeq \text{id}_X$. A deformation retraction is **strong** if we require this homotopy to be a homotopy rel A .

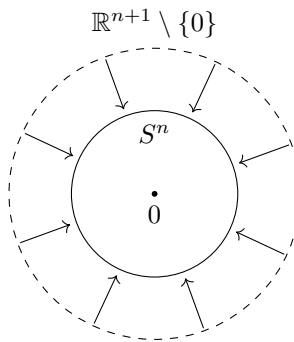
If $r : X \rightarrow A$ is a deformation retraction then r is by definition a homotopy equivalence, with homotopy inverse ι . So $A \simeq X$.

Example 25.1.13

The map $r : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow S^n$ given by $r(v) = v/\|v\|$ is a strong deformation retraction. Indeed $r \circ \iota = \text{id}_{S^n}$, and

$$H(v, t) = \frac{v}{(1-t) + t\|v\|}$$

is a homotopy from id to $\iota \circ r$ fixing S^n pointwise. So $\mathbb{R}^{n+1} \setminus \{0\} \simeq S^n$: the punctured space is homotopy equivalent to the sphere, even though the two are very far from homeomorphic.



§25.1.2 Paths

We now specialise to the maps that will give us our invariant. The idea is very simple: we probe a space X by mapping intervals into it, and see which of the resulting loops can be contracted.

Definition 25.1.14 (Path)

A **path** in a space X is a map $\gamma : I \rightarrow X$. If $\gamma(0) = x_0$ and $\gamma(1) = x_1$, we say γ is a path from x_0 to x_1 , and write $\gamma : x_0 \rightsquigarrow x_1$.

If $\gamma(0) = \gamma(1) = x_0$, we say γ is a **loop** (based at x_0).

We write $\Omega(X, x_0, x_1)$ for the set of paths $x_0 \rightsquigarrow x_1$ in X , and $\Omega(X, x_0) = \Omega(X, x_0, x_0)$ for the set of loops based at x_0 .

Definition 25.1.15 (Concatenation and Inverses)

If we have two paths γ_1 from x_0 to x_1 and γ_2 from x_1 to x_2 , we define the **concatenation** to be

$$(\gamma_1 \cdot \gamma_2)(t) = \begin{cases} \gamma_1(2t) & 0 \leq t \leq \frac{1}{2} \\ \gamma_2(2t - 1) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

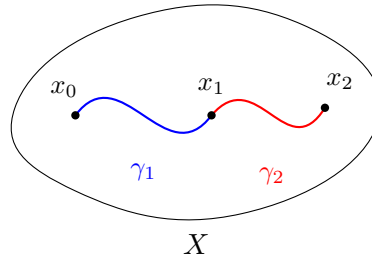
This is continuous by the gluing lemma.

The **inverse** of a path $\gamma : I \rightarrow X$ is defined by

$$\gamma^{-1}(t) = \gamma(1 - t).$$

The **constant path** at a point $x \in X$ is given by $c_x(t) = x$.

So $\gamma_1 \cdot \gamma_2$ traverses γ_1 at double speed, then γ_2 at double speed, and γ^{-1} runs γ backwards.

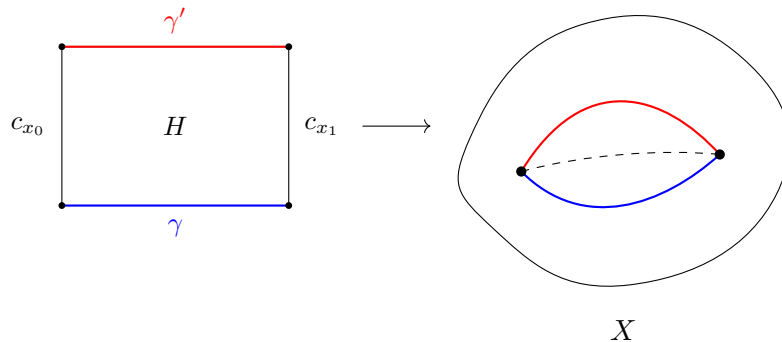


We would like to make $\Omega(X, x_0)$ into a group under concatenation, and the reader will immediately object that this cannot work: concatenation is not associative on the nose, since $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2$ and $\gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$ traverse the three paths at different speeds. The fix is to pass to homotopy classes, and here we must be careful to only allow homotopies which fix the endpoints – otherwise every loop could be slid off to a constant and nothing would be left.

Definition 25.1.16 (Homotopy of Paths)

Paths $\gamma, \gamma' : I \rightarrow X$ are **homotopic as paths** if they are homotopic $\text{rel}\{0, 1\} \subseteq I$, that is, if the endpoints stay fixed throughout. We write $\gamma \simeq \gamma'$.

Concretely, $\gamma \simeq \gamma'$ means there is a map $H : I \times I \rightarrow X$ with $H(s, 0) = \gamma(s)$, $H(s, 1) = \gamma'(s)$, and $H(0, t) = x_0$, $H(1, t) = x_1$ for all t . It is worth drawing the square: the bottom edge is γ , the top edge is γ' , and the two vertical edges are constant.



Lemma 25.1.17 (The Group Operation is Well-Defined)

Let $\gamma_1, \gamma_2 : I \rightarrow X$ be paths with $\gamma_1(1) = \gamma_2(0)$. Then if $\gamma_1 \simeq \gamma'_1$ and $\gamma_2 \simeq \gamma'_2$, we have $\gamma_1 \cdot \gamma_2 \simeq \gamma'_1 \cdot \gamma'_2$.

Proof. Suppose that $\gamma_1 \simeq_{H_1} \gamma'_1$ and $\gamma_2 \simeq_{H_2} \gamma'_2$. Then we can construct the homotopy

$$H(s, t) = \begin{cases} H_1(2s, t) & 0 \leq s \leq \frac{1}{2} \\ H_2(2s - 1, t) & \frac{1}{2} \leq s \leq 1, \end{cases}$$

which is well-defined since $H_1(1, t) = \gamma_1(1) = \gamma_2(0) = H_2(0, t)$, and continuous by the gluing lemma. It visibly fixes the endpoints. $\square$

Lemma 25.1.18 (Indeed a Group Operation)

Let $\gamma_0 : x_0 \rightsquigarrow x_1$, $\gamma_1 : x_1 \rightsquigarrow x_2$, $\gamma_2 : x_2 \rightsquigarrow x_3$ be paths. Then

1. $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2 \simeq \gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$.
2. $\gamma_0 \cdot c_{x_1} \simeq \gamma_0 \simeq c_{x_0} \cdot \gamma_0$.
3. $\gamma_0 \cdot \gamma_0^{-1} \simeq c_{x_0}$ and $\gamma_0^{-1} \cdot \gamma_0 \simeq c_{x_1}$.

Proof. We first note that the composition of any path γ with an order-preserving surjection $\phi : I \rightarrow I$ is homotopic as paths to the original path, via $H(s, t) = \gamma(t\phi(s) + (1-t)s)$. Such a ϕ is called a **reparameterisation**, and the homotopy fixes endpoints because $\phi(0) = 0$ and $\phi(1) = 1$.

1. Take ϕ to be the function

$$\phi(t) = \begin{cases} \frac{t}{2} & 0 \leq t \leq \frac{1}{2} \\ t - \frac{1}{4} & \frac{1}{2} \leq t \leq \frac{3}{4} \\ 2t - 1 & \frac{3}{4} \leq t \leq 1. \end{cases}$$

Then we have $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2 \simeq ((\gamma_0 \cdot \gamma_1) \cdot \gamma_2) \circ \phi = \gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$.

2. To see $\gamma_0 \cdot c_{x_1} \simeq \gamma_0$, reparameterise γ_0 with

$$t \mapsto \begin{cases} 2t & 0 \leq t \leq \frac{1}{2} \\ 1 & \frac{1}{2} \leq t \leq 1, \end{cases}$$

and the other side is analogous.

3. We have that $c_{x_0} \simeq \gamma_0 \cdot \gamma_0^{-1}$ via

$$H(s, t) = \begin{cases} \gamma_0(2s) & 0 \leq s \leq \frac{t}{2} \\ \gamma_0(t) & \frac{t}{2} \leq s \leq 1 - \frac{t}{2} \\ \gamma_0^{-1}(2s - 1) & 1 - \frac{t}{2} \leq s \leq 1. \end{cases}$$

At time t this path runs along γ_0 as far as $\gamma_0(t)$, waits there, and comes back; at $t = 0$ it is constant and at $t = 1$ it is $\gamma_0 \cdot \gamma_0^{-1}$. Applying this to γ_0^{-1} gives the other identity. $\square$

§25.1.3 The Fundamental Group

We can now define one of our central objects of study.

Definition 25.1.19 (Fundamental Group)

Let X be a space and $x_0 \in X$. The **fundamental group** of X (based at x_0), denoted $\pi_1(X, x_0)$, is the set of homotopy classes of loops in X based at x_0 , that is, $\pi_1(X, x_0) = \Omega(X, x_0) / \simeq$.

The group operation is defined by $[\gamma_0][\gamma_1] = [\gamma_0 \cdot \gamma_1]$, with $[\gamma]^{-1} = [\gamma^{-1}]$ and identity the class $e = [c_{x_0}]$ of the constant path.

Theorem 25.1.20

The fundamental group is a group.

Proof. The operation is well-defined by Lemma 25.1.17, and the group axioms hold by Lemma 25.1.18. $\square$

The definition depends on a choice of basepoint, and we will deal with that dependence shortly. First, let us see that π_1 is not just an invariant of spaces, but of maps as well – this is what makes it useful.

Definition 25.1.21 (Based Space)

A **based space** is a pair (X, x_0) of a space X and a point $x_0 \in X$, the **basepoint**. A **map of based spaces** $f : (X, x_0) \rightarrow (Y, y_0)$ is a map $f : X \rightarrow Y$ such that $f(x_0) = y_0$. A **based homotopy** is a homotopy $\text{rel}\{x_0\}$.

Proposition 25.1.22 (π_1 is a Functor)

To a based map $f : (X, x_0) \rightarrow (Y, y_0)$, there is an associated function $f_* = \pi_1(f) : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$, defined by $[\gamma] \mapsto [f \circ \gamma]$. Moreover it satisfies

1. $\pi_1(f)$ is a homomorphism of groups.
2. If $f \simeq f' \text{ rel}\{x_0\}$, then $\pi_1(f) = \pi_1(f')$.
3. For any based maps $(A, a) \xrightarrow{h} (B, b) \xrightarrow{k} (C, c)$, we have $\pi_1(k \circ h) = \pi_1(k) \circ \pi_1(h)$.
4. $\pi_1(\text{id}_X) = \text{id}_{\pi_1(X, x_0)}$.

Proof. First, f_* is well-defined: if $\gamma \simeq_H \gamma'$ as paths, then $f \circ H$ is a homotopy of paths from $f \circ \gamma$ to $f \circ \gamma'$.

For (i), observe that composing with f commutes with concatenation,

$$(f \circ (\gamma \cdot \gamma'))(t) = \begin{cases} f(\gamma(2t)) & t \leq \frac{1}{2} \\ f(\gamma'(2t-1)) & t \geq \frac{1}{2} \end{cases} = ((f \circ \gamma) \cdot (f \circ \gamma'))(t),$$

so $f_*([\gamma][\gamma']) = [f \circ (\gamma \cdot \gamma')] = [f \circ \gamma][f \circ \gamma'] = f_*[\gamma]f_*[\gamma']$.

For (ii), if H is a based homotopy from f to f' and γ is a loop at x_0 , then $(s, t) \mapsto H(\gamma(s), t)$ is a homotopy of paths from $f \circ \gamma$ to $f' \circ \gamma$: it fixes the endpoints since $\gamma(0) = \gamma(1) = x_0$ and $H(x_0, t) = y_0$ for all t .

Parts (iii) and (iv) are immediate: $(k \circ h)_*[\gamma] = [k \circ h \circ \gamma] = k_*[h \circ \gamma] = k_*h_*[\gamma]$, and $(\text{id}_X)_*[\gamma] = [\gamma]$. $\square$

In the language of categories, π_1 is a functor from based spaces to groups. The following diagram, expressing part (iii), commutes.

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(g \circ f)_*} & \pi_1(Z, z_0) \\ & \searrow f_* \quad \nearrow g_* & \\ & \pi_1(Y, y_0) & \end{array}$$

This gives us our first real payoff, essentially for free.

Corollary 25.1.23

If $f : X \rightarrow Y$ is a homeomorphism with $f(x_0) = y_0$, then $f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$ is an isomorphism.

Proof. Apply the functor to $f \circ f^{-1} = \text{id}_Y$ and $f^{-1} \circ f = \text{id}_X$ to get $f_* \circ (f^{-1})_* = \text{id}$ and $(f^{-1})_* \circ f_* = \text{id}$. $\square$

§25.1.4 Change of Basepoint

We want to remove the basepoint dependence of the fundamental group. Of course we cannot hope to compare $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ when x_0 and x_1 lie in different path components, since a loop at x_0 knows nothing about the rest of X . But if there is a path between them we can drag loops along it.

Proposition 25.1.24

A path $u : x_0 \rightsquigarrow x_1$ induces a group isomorphism $u_\# : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ by $[\gamma] \mapsto [u^{-1} \cdot \gamma \cdot u]$. This satisfies

1. If $u \simeq u'$, then $u_\# = u'_\#$.
2. $(c_{x_0})_\# = \text{id}_{\pi_1(X, x_0)}$.
3. If $v : x_1 \rightsquigarrow x_2$, then $(u \cdot v)_\# = v_\# \circ u_\#$.
4. If $f : X \rightarrow Y$ with $f(x_0) = y_0$ and $f(x_1) = y_1$, then

$$(f \circ u)_\# \circ f_* = f_* \circ u_\# : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_1).$$

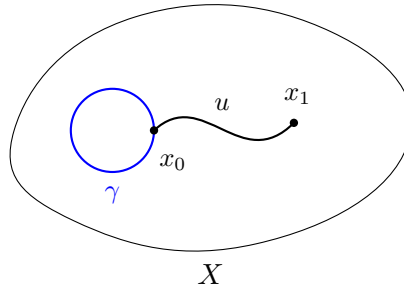
5. If $x_1 = x_0$, then $u_\#$ is the automorphism of $\pi_1(X, x_0)$ given by conjugation by $[u]$.

Proof. That $u_\#$ is well-defined follows from Lemma 25.1.17, and it is a homomorphism because

$$u_\#[\gamma] u_\#[\gamma'] = [u^{-1} \cdot \gamma \cdot u \cdot u^{-1} \cdot \gamma' \cdot u] = [u^{-1} \cdot \gamma \cdot \gamma' \cdot u] = u_\#([\gamma][\gamma']),$$

using $u \cdot u^{-1} \simeq c_{x_1}$. Properties (i), (ii) and (v) are immediate from the definitions, and (iii) follows from associativity together with $(u \cdot v)^{-1} \simeq v^{-1} \cdot u^{-1}$. For (iv) we compute both sides on $[\gamma]$: the left hand side gives $[(f \circ u)^{-1} \cdot (f \circ \gamma) \cdot (f \circ u)]$, and the right hand side gives $[f \circ (u^{-1} \cdot \gamma \cdot u)]$, and these agree since f commutes with concatenation and inversion of paths.

Finally, (ii) and (iii) give $(u^{-1})_\# \circ u_\# = (u \cdot u^{-1})_\# = (c_{x_0})_\# = \text{id}$ and similarly the other way, so $u_\#$ is an isomorphism with inverse $(u^{-1})_\#$. $\square$



So if x_0 and x_1 are in the same path connected component, then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$. For a path connected space we will often write $\pi_1(X)$, keeping in mind that the isomorphism depends on the choice of path, so this is only well-defined up to (not necessarily inner) automorphism.

The next lemma is the technical heart of the section: it says that a homotopy between maps which does *not* respect basepoints still induces the same map on π_1 , provided we correct by the change of basepoint isomorphism along the track of the basepoint.

Lemma 25.1.25

Let $f, g : X \rightarrow Y$ be maps with $f \simeq_H g$, let $x_0 \in X$, and let $u(t) = H(x_0, t)$, a path $f(x_0) \sim g(x_0)$. Then the following diagram commutes.

$$\begin{array}{ccc} & & \pi_1(Y, f(x_0)) \\ & \nearrow f_* & \downarrow u_\# \\ \pi_1(X, x_0) & & \pi_1(Y, g(x_0)) \\ & \searrow g_* & \end{array}$$

That is, $g_* = u_\# \circ f_*$.

Proof. Suppose we have a loop $\gamma : I \rightarrow X$ based at x_0 . We need to check that

$$g_*([\gamma]) = u_{\#} \circ f_*([\gamma]),$$

in other words that

$$g \circ \gamma \simeq u^{-1} \cdot (f \circ \gamma) \cdot u.$$

To prove this we want to build a homotopy. Consider the composition

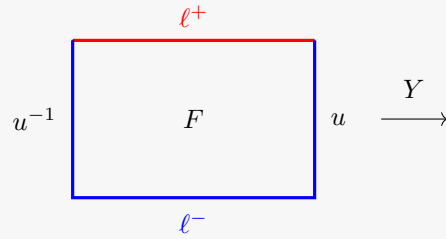
$$F : I \times I \xrightarrow{\gamma \times \text{id}_I} X \times I \xrightarrow{H} Y.$$

Our plan is to exhibit two homotopic paths ℓ^+ and ℓ^- in $I \times I$ such that

$$F \circ \ell^+ = g \circ \gamma, \quad F \circ \ell^- = u^{-1} \cdot (f \circ \gamma) \cdot u.$$

This is in general a good strategy: X is a complicated and horrible space we do not understand, so to construct a homotopy we make ourselves work in the much nicer space $I \times I$.

Our ℓ^+ and ℓ^- are defined in a rather simple way. Precisely, ℓ^+ is the path $s \mapsto (s, 1)$ along the top of the square, and ℓ^- is the concatenation of the paths $s \mapsto (0, 1-s)$, $s \mapsto (s, 0)$ and $s \mapsto (1, s)$, going down the left edge, along the bottom, and up the right edge.



Note that ℓ^+ and ℓ^- are homotopic as paths. If this is not obvious, we can manually check the homotopy

$$L(s, t) = t\ell^+(s) + (1-t)\ell^-(s),$$

which works because $I \times I$ is convex, and fixes the endpoints $(0, 1)$ and $(1, 1)$. Hence $F \circ \ell^+ \simeq F \circ \ell^-$ as paths, via $F \circ L$.

Now we check that the compositions $F \circ \ell^\pm$ are indeed what we want. We have

$$F \circ \ell^+(s) = H(\gamma(s), 1) = g \circ \gamma(s),$$

and along the three pieces of ℓ^- we get $H(x_0, 1-s) = u^{-1}(s)$, then $H(\gamma(s), 0) = f(\gamma(s))$, then $H(x_0, s) = u(s)$, so that

$$F \circ \ell^- = u^{-1} \cdot (f \circ \gamma) \cdot u$$

as required. □

Theorem 25.1.26

If $f : X \rightarrow Y$ is a homotopy equivalence, and $x_0 \in X$, then the induced map

$f_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, f(x_0))$ is an isomorphism.

Proof. Let $g : Y \rightarrow X$ be a homotopy inverse, so $f \circ g \simeq_H \text{id}_Y$ and $g \circ f \simeq_{H'} \text{id}_X$.

We have no guarantee that $g \circ f(x_0) = x_0$, but our homotopy H' gives us a path $u' = H'(x_0, \cdot) : x_0 \leadsto g \circ f(x_0)$.

Applying Lemma 25.1.25 with id_X playing the role of ' f ' and $g \circ f$ playing the role of ' g ', we get

$$u'_\# \circ (\text{id}_X)_* = (g \circ f)_*.$$

Using functoriality, this says

$$g_* \circ f_* = u'_\#.$$

However, we know that $u'_\#$ is an isomorphism. So f_* is injective and g_* is surjective.

Doing the same thing the other way round with $f \circ g$ instead of $g \circ f$, we learn that g_* is injective and f_* is surjective. So both are isomorphisms. $\square$

This is a substantial improvement on Corollary 25.1.23: the fundamental group cannot distinguish homotopy equivalent spaces, but that is exactly what makes it computable, since we may replace a space by any convenient space homotopy equivalent to it.

Corollary 25.1.27

If X is contractible, then $\pi_1(X, x_0)$ is trivial for every x_0 .

Proof. $\pi_1(*, *)$ is trivial, since there is only one loop, and $X \simeq *$. $\square$

§25.1.5 Simply Connected Spaces

Definition 25.1.28 (Simply Connected Space)

A space X is **simply connected** if it is path connected and $\pi_1(X, x_0)$ is trivial for some (equivalently, by Proposition 25.1.24, any) choice of $x_0 \in X$.

Lemma 25.1.29

A path connected space X is simply connected if and only if for any $x_0, x_1 \in X$, there is a unique homotopy class of paths $x_0 \leadsto x_1$.

Proof. Suppose X is simply connected, and let $u, v : x_0 \leadsto x_1$ be paths. Now $u \cdot v^{-1}$ is a loop based at x_0 , so it is homotopic to the constant path. So we have

$$u \simeq u \cdot v^{-1} \cdot v \simeq c_{x_0} \cdot v \simeq v.$$

On the other hand, suppose there is a unique homotopy class of paths $x_0 \leadsto x_1$ for all $x_0, x_1 \in X$. Then in particular there is a unique homotopy class of loops based at x_0 , namely that of c_{x_0} . So $\pi_1(X, x_0)$ is trivial. $\square$

There is a useful reformulation of triviality in π_1 in terms of extending maps over the disc. Recall that a loop $\gamma \in \Omega(X, x_0)$ has $\gamma(0) = \gamma(1)$, so it descends to a map $\bar{\gamma} : S^1 \rightarrow X$

with $\bar{\gamma}(e^{2\pi it}) = \gamma(t)$, using $I/\{0, 1\} \cong S^1$.

Definition 25.1.30

A map $f : X \rightarrow Y$ is **null-homotopic** if $f \simeq c_p$ for some $p \in Y$.

Lemma 25.1.31

A map $f : S^1 \rightarrow Y$ extends to a map $F : D^2 \rightarrow Y$ if and only if f is null-homotopic.

Proof. If F is such an extension, define $H(v, t) = F(tv)$ for $v \in S^1$. Then $H(v, 1) = F(v) = f(v)$ and $H(v, 0) = F(0)$, so $f \simeq c_{F(0)}$.

Conversely, suppose $f \simeq_H c_p$ with $H(v, 0) = p$ and $H(v, 1) = f(v)$. Define

$$F(v) = \begin{cases} H\left(\frac{v}{\|v\|}, \|v\|\right) & v \neq 0 \\ p & v = 0. \end{cases}$$

This is continuous away from 0, and continuous at 0 because $H(w, t) \rightarrow p$ uniformly in w as $t \rightarrow 0$ (using compactness of S^1). It restricts to f on S^1 . $\square$

Proposition 25.1.32

For a path connected space X , the following are equivalent.

1. X is simply connected.
2. Every map $f : S^1 \rightarrow X$ is null-homotopic.
3. Every map $f : S^1 \rightarrow X$ extends to a map $D^2 \rightarrow X$.

Proof. Parts (ii) and (iii) are equivalent by Lemma 25.1.31.

Suppose X is simply connected and let $f : S^1 \rightarrow X$, and set $x_0 = f(1)$ and $\gamma(t) = f(e^{2\pi it})$, a loop at x_0 with $\bar{\gamma} = f$. Since $[\gamma] = 1$ we have $\gamma \simeq c_{x_0}$ via some H fixing endpoints; then $\bar{H}(e^{2\pi is}, t) = H(s, t)$ is well-defined (as $H(0, t) = H(1, t) = x_0$) and continuous, and gives $f \simeq c_{x_0}$.

Conversely, suppose every map $S^1 \rightarrow X$ is null-homotopic and let γ be a loop at x_0 . Then $\bar{\gamma}$ extends to $F : D^2 \rightarrow X$, and $H(s, t) = F((1-t)e^{2\pi is} + t)$ is a homotopy of paths from γ to c_{x_0} : it takes the circle and shrinks it towards the basepoint $1 \in D^2$, staying inside D^2 by convexity, and it fixes $s = 0, 1$. $\square$

Example 25.1.33

Every convex subset of $\mathbb{R}^n$ is simply connected, being contractible. We will see in the next section that S^1 is *not* simply connected, which is the first genuinely non-trivial computation of the course, and that S^n for $n \geq 2$ is simply connected.

Example 25.1.34

If X and Y are spaces with basepoints x_0, y_0 , then

$$\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0).$$

Indeed a map $I \rightarrow X \times Y$ is exactly a pair of maps $I \rightarrow X$ and $I \rightarrow Y$, by the universal property of the product topology, and likewise for homotopies $I \times I \rightarrow X \times Y$. So the two projections induce a bijection, which is clearly a homomorphism.

§25.2 Covering Spaces

We now start to develop some machinery which will allow us to compute fundamental groups. So far we have no example of a space with a non-trivial fundamental group, for the good reason that showing a loop is *not* null-homotopic means ruling out all possible homotopies at once. The trick is to unroll the space: if we can find a simply connected space sitting above X in a suitably uniform way, then loops in X become paths upstairs, and paths are much easier to tell apart than loops.

§25.2.1 Definitions and Examples

Definition 25.2.1 (Evenly Covered Set)

Suppose $p : \hat{X} \rightarrow X$ is a map. We say that an open set $U \subseteq X$ is **evenly covered** by p if

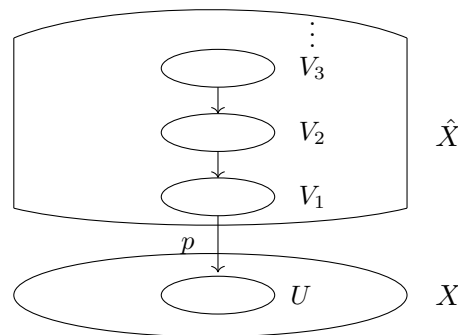
$$p^{-1}(U) = \bigsqcup_{\alpha \in A} V_{\alpha},$$

a disjoint union of open sets, where $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$ is a homeomorphism for each $\alpha \in A$.

The V_{α} are called the **sheets** over U . Note that the inverse $p_{\alpha}^{-1} = (p|_{V_{\alpha}})^{-1} : U \rightarrow \hat{X}$ is continuous, being the inverse of a homeomorphism followed by an inclusion; we will use this constantly.

Definition 25.2.2 (Covering Map)

A map $p : \hat{X} \rightarrow X$ is a **covering map** if for all $x \in X$ there exists an open neighbourhood U_x of x which is evenly covered by p . In this case, we call $\hat{X}$ a **covering space** for X .



Example 25.2.3

Let A be a set with the discrete topology. Then the projection $p : A \times X \rightarrow X$ is a covering map, since X itself is evenly covered: $p^{-1}(X) = \bigsqcup_{\alpha \in A} \{\alpha\} \times X$. This is the **trivial cover**, and the point of the definition is that a general covering map looks like this *locally* but need not do so globally.

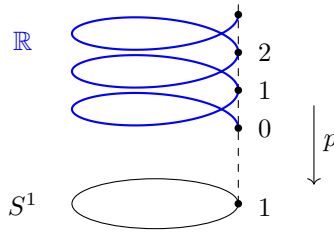
Example 25.2.4

The map $p : \mathbb{R} \rightarrow S^1$ given by $p(t) = e^{2\pi it}$ is a covering map. Indeed, if $V \subseteq \mathbb{R}$ is an open interval of length less than 1 and $U = p(V)$, then

$$p^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} (V + n),$$

and p restricted to each translate $V + n$ is a homeomorphism onto U . Every point of S^1 has such a neighbourhood.

The picture to have in mind is the helix: $\mathbb{R}$ winds around and around above the circle, and p is vertical projection.

**Example 25.2.5**

For $n \geq 1$, the map $p_n : S^1 \rightarrow S^1$, $z \mapsto z^n$, is a covering map. If $U \subseteq S^1$ is an open arc of length less than $2\pi/n$ then

$$p_n^{-1}(U) = \bigsqcup_{i \in \mathbb{Z}/n\mathbb{Z}} \omega^i V, \quad \omega = e^{2\pi i/n},$$

where V is one of the n arcs mapping homeomorphically onto U . So this is an n -sheeted cover.

Definition 25.2.6

The **n -dimensional real projective space** is $\mathbb{RP}^n = S^n / \sim$, where $\sim$ is generated by $x \sim -x$, with the quotient topology.

Example 25.2.7

The quotient map $p : S^n \rightarrow \mathbb{RP}^n$ is a covering map. Given $x \in S^n$, let V_x be the open hemisphere centred at x and $U_x = p(V_x)$. Since V_x contains no pair of antipodal points, $p^{-1}(U_x) = V_x \sqcup (-V_x)$ and p restricted to each is a homeomorphism onto U_x . This is a two-sheeted cover.

Example 25.2.8

The map $p \times p : \mathbb{R}^2 \rightarrow S^1 \times S^1 = T^2$ is a covering map of the torus, being a product of covering maps.

§25.2.2 Lifting

The whole theory rests on the following idea: given a map into X , we try to push it up into $\hat{X}$.

Definition 25.2.9 (Lift)

Suppose $p : \hat{X} \rightarrow X$ is a covering map and $f : Z \rightarrow X$ is a map. Then we say that $\hat{f} : Z \rightarrow \hat{X}$ is a **lift** of f if $p \circ \hat{f} = f$, that is, if

$$\begin{array}{ccc} & & \hat{X} \\ & \nearrow \hat{f} & \downarrow p \\ Z & \xrightarrow{f} & X \end{array}$$

commutes.

Notation. We say a map $f : Z \rightarrow X$ has the **lifting property** at $z_0 \in Z$ if for every $\hat{x}_0 \in p^{-1}(f(z_0))$ there is a lift $\hat{f}$ of f with $\hat{f}(z_0) = \hat{x}_0$, and the **unique lifting property** (ULP) at z_0 if this lift is moreover unique. For a path $\gamma : [a, b] \rightarrow X$ we say ‘ γ has the ULP’ to mean that it has the ULP at a .

The strategy for lifting a path is to chop it into small pieces, each of which lands in a single evenly covered set, and lift those one at a time. To chop it up we need the following, which the reader may have seen in a metric spaces course.

Lemma 25.2.10 (Lebesgue Covering Lemma)

Suppose X is a compact metric space and $\{U_\alpha\}_\alpha$ is an open cover of X . Then there exists $\delta > 0$ such that for all $x \in X$ we have $B_\delta(x) \subseteq U_\alpha$ for some α .

Proof. Given $x \in X$, let $\alpha(x)$ and $\delta(x) > 0$ be such that $B_{2\delta(x)}(x) \subseteq U_{\alpha(x)}$. Then $\{B_{\delta(x)}(x)\}_{x \in X}$ is an open cover of X , so by compactness there is a finite subcover $\{B_{\delta(x_i)}(x_i)\}_{i=1}^n$. Let $\delta = \min\{\delta(x_1), \dots, \delta(x_n)\} > 0$. Then given $y \in X$ we have $y \in B_{\delta(x_i)}(x_i)$ for some i , and so

$$B_\delta(y) \subseteq B_{\delta(x_i)+\delta}(x_i) \subseteq B_{2\delta(x_i)}(x_i) \subseteq U_{\alpha(x_i)}$$

as required. □

Lemma 25.2.11

Let $f : Z \rightarrow X$ with Z connected, and suppose $\text{im}(f) \subseteq U$ where U is evenly covered. Then f has the unique lifting property at every $z_0 \in Z$.

Proof. Since U is evenly covered, $p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha}$. Given $\hat{x}_0 \in p^{-1}(f(z_0))$ we have $\hat{x}_0 \in V_{\alpha_0}$ for some α_0 . Then $p_{\alpha_0}^{-1} = (p|_{V_{\alpha_0}})^{-1} : U \rightarrow \hat{X}$ is continuous with $p_{\alpha_0}^{-1}(f(z_0)) = \hat{x}_0$, so $\hat{f} = p_{\alpha_0}^{-1} \circ f$ is a lift of f with the right value at z_0 .

For uniqueness, notice that $p^{-1}(U) = V_{\alpha_0} \sqcup (\bigsqcup_{\alpha \neq \alpha_0} V_{\alpha})$ is a disconnection of $p^{-1}(U)$. Any lift $\hat{f}$ has connected image (as Z is connected) meeting V_{α_0} , so $\text{im}(\hat{f}) \subseteq V_{\alpha_0}$. But $p|_{V_{\alpha_0}}$ is a homeomorphism, so $\hat{f} = p_{\alpha_0}^{-1} \circ f$ is forced. $\square$

Lemma 25.2.12

Suppose $\gamma : [a, b] \rightarrow X$ and $a' \in [a, b]$. If $\gamma|_{[a, a']}$ has the ULP at a and $\gamma|_{[a', b]}$ has the ULP at a' , then γ has the ULP at a .

Proof. Given $\hat{x} \in p^{-1}(\gamma(a))$, we have a lift $\hat{\gamma}_1 : [a, a'] \rightarrow \hat{X}$ of $\gamma|_{[a, a']}$ with $\hat{\gamma}_1(a) = \hat{x}$, and then a lift $\hat{\gamma}_2 : [a', b] \rightarrow \hat{X}$ of $\gamma|_{[a', b]}$ with $\hat{\gamma}_2(a') = \hat{\gamma}_1(a')$. So the concatenation $\hat{\gamma}_1 \hat{\gamma}_2$, which is continuous by the gluing lemma, is a lift of γ at a .

For uniqueness, suppose $\hat{\eta}$ is any lift with $\hat{\eta}(a) = \hat{x}$. Then $\hat{\eta}|_{[a, a']}$ is a lift of $\gamma|_{[a, a']}$ starting at $\hat{x}$, so $\hat{\eta}|_{[a, a']} = \hat{\gamma}_1$ by uniqueness. In particular $\hat{\eta}(a') = \hat{\gamma}_1(a')$, so $\hat{\eta}|_{[a', b]} = \hat{\gamma}_2$ by uniqueness again. Hence $\hat{\eta} = \hat{\gamma}_1 \hat{\gamma}_2$. $\square$

Theorem 25.2.13 (Path Lifting)

Let $p : \hat{X} \rightarrow X$ be a covering map. Then any path $\gamma : I \rightarrow X$ has the unique lifting property.

Proof. Since p is a covering map, every $x \in X$ has an evenly covered neighbourhood U_x . Then $\{U_x \mid x \in X\}$ is an open cover of X , so $\{\gamma^{-1}(U_x) \mid x \in X\}$ is an open cover of I . Thus by the Lebesgue covering lemma there exists $\delta > 0$ such that for each t we have $B_{\delta}(t) \subseteq \gamma^{-1}(U_{x(t)})$ for some $x(t)$.

Choose n with $1/n < \delta$ and set $a_i = i/n$. Then $[a_i, a_{i+1}] \subseteq B_{\delta}(a_i)$, so $\gamma([a_i, a_{i+1}]) \subseteq U_{x(a_i)}$, which is evenly covered. As $[a_i, a_{i+1}]$ is connected, Lemma 25.2.11 says $\gamma|_{[a_i, a_{i+1}]}$ has the ULP at a_i . By induction on i using Lemma 25.2.12, $\gamma|_{[0, a_i]}$ has the ULP at 0 for every i , and in particular so does γ . $\square$

The same argument, run over a grid of small squares instead of a partition of the interval, lifts homotopies.

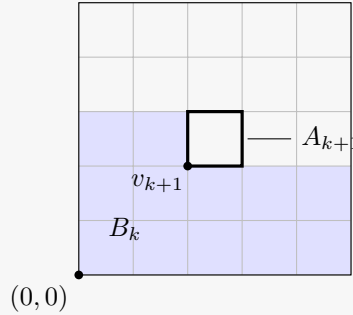
Theorem 25.2.14 (Homotopy Lifting)

Suppose $p : \hat{X} \rightarrow X$ is a covering map and $H : I \times I \rightarrow X$ is a homotopy. Then H has the lifting property at $(0, 0)$.

Proof. Let $\{U_x \mid x \in X\}$ be an open cover of X by evenly covered neighbourhoods. Since I^2 is compact, by the Lebesgue covering lemma there exists $\delta > 0$ such that $B_{\delta}(v) \subseteq H^{-1}(U_{x(v)})$ for each $v \in I^2$.

Choose n with $\sqrt{2}/n < \delta$ and divide I^2 into squares of side length $1/n$. Enumerate

them $A_1, A_2, \dots, A_{n^2}$, starting from the bottom left and going right along each row and then up. Label the bottom left corner of A_i as v_i , so that $v_1 = (0, 0)$. Since $A_i \subseteq B_\delta(v_i)$ we have $H(A_i) \subseteq U_{x(v_i)}$, which is evenly covered, and A_i is connected, so $H|_{A_i}$ has the ULP at v_i by Lemma 25.2.11. Let $B_k = \bigcup_{i=1}^k A_i$.



We prove by induction that $H|_{B_k}$ has the lifting property at $(0, 0)$. For $k = 1$ we have $B_1 = A_1$ and $v_1 = (0, 0)$, so this is what we already know. Now suppose $H|_{B_k}$ has a lift $\hat{H}_k : B_k \rightarrow \hat{X}$ with $\hat{H}_k(0, 0) = \hat{x}$. As $H|_{A_{k+1}}$ has the lifting property at v_{k+1} , we may choose a lift $\hat{h}_k : A_{k+1} \rightarrow \hat{X}$ with $\hat{h}_k(v_{k+1}) = \hat{H}_k(v_{k+1})$.

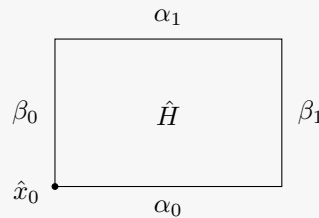
Now $B_k \cap A_{k+1}$ is either one or two edges of A_{k+1} , in either case a set homeomorphic to I containing v_{k+1} . By uniqueness of path lifting, $\hat{H}_k$ and $\hat{h}_k$ agree on $A_{k+1} \cap B_k$, since both are lifts of $H|_{A_{k+1} \cap B_k}$ agreeing at v_{k+1} . So by the gluing lemma we obtain a well-defined lift $\hat{H}_{k+1}$ of $H|_{B_{k+1}}$. Taking $k = n^2$ finishes the proof. $\square$

The point of all this is the following, which says that homotopic loops downstairs lift to paths with the same endpoint upstairs. This is the mechanism that will let us distinguish loops.

Proposition 25.2.15

Let $p : \hat{X} \rightarrow X$ be a covering map and suppose $\gamma_0, \gamma_1 \in \Omega(X, x_0, x_1)$ with $\gamma_0 \simeq \gamma_1$. Let $\hat{\gamma}_i$ be the lift of γ_i with $\hat{\gamma}_i(0) = \hat{x}_0$. Then $\hat{\gamma}_0 \simeq \hat{\gamma}_1$. In particular, $\hat{\gamma}_0(1) = \hat{\gamma}_1(1)$.

Proof. Suppose $H : I^2 \rightarrow X$ is a homotopy of paths between γ_0 and γ_1 . By homotopy lifting we have a lift $\hat{H} : I^2 \rightarrow \hat{X}$ with $\hat{H}(0, 0) = \hat{x}_0$. Let $\alpha_i(t) = \hat{H}(t, i)$ and $\beta_i(t) = \hat{H}(i, t)$ for $i = 0, 1$, the images of the four edges of the square.



Applying uniqueness of path lifting to each edge in turn:

1. β_0 is a lift of the constant path c_{x_0} starting at $\hat{x}_0$, so $\beta_0 = c_{\hat{x}_0}$;
2. α_0 is a lift of γ_0 with $\alpha_0(0) = \hat{x}_0$, so $\alpha_0 = \hat{\gamma}_0$;

3. α_1 is a lift of γ_1 with $\alpha_1(0) = \beta_0(1) = \hat{x}_0$, so $\alpha_1 = \hat{\gamma}_1$;
4. writing $\hat{x}_1 = \hat{\gamma}_0(1)$, β_1 is a lift of c_{x_1} starting at $\hat{x}_1$, so $\beta_1 = c_{\hat{x}_1}$.

So $\hat{H}$ is a homotopy of paths from $\hat{\gamma}_0$ to $\hat{\gamma}_1$, and in particular the two have the same endpoint $\hat{x}_1$. $\square$

Corollary 25.2.16

If $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is a covering map, then $p_* : \pi_1(\hat{X}, \hat{x}_0) \rightarrow \pi_1(X, x_0)$ is injective.

Proof. Suppose $p_*[\gamma_0] = p_*[\gamma_1]$ for loops γ_0, γ_1 at x_0 . Then $p \circ \gamma_0 \simeq p \circ \gamma_1$ as paths, so by Proposition 25.2.15 their lifts starting at $\hat{x}_0$ are homotopic as paths. But these lifts are γ_0 and γ_1 , by uniqueness of lifting. So $[\gamma_0] = [\gamma_1]$. $\square$

§25.2.3 The Fundamental Group of the Circle

Fix a covering map $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$. Given a loop γ at x_0 , let $\hat{\gamma}$ be its unique lift starting at $\hat{x}_0$. Then $p(\hat{\gamma}(1)) = \gamma(1) = x_0$, so $\hat{\gamma}(1)$ lies in the fibre $p^{-1}(x_0)$, and by Proposition 25.2.15 it depends only on the homotopy class of γ . We therefore get a well-defined function

$$\varepsilon_p : \pi_1(X, x_0) \longrightarrow p^{-1}(x_0), \quad [\gamma] \mapsto \hat{\gamma}(1).$$

This is the key construction: it turns a homotopy-theoretic object into a set of points.

Definition 25.2.17 (Universal Cover)

A covering map $p : \hat{X} \rightarrow X$ is a **universal cover** if $\hat{X}$ is simply connected.

Example 25.2.18

$p : \mathbb{R} \rightarrow S^1, t \mapsto e^{2\pi it}$, is a universal cover, since $\mathbb{R}$ is contractible. Likewise $\mathbb{R}^2 \rightarrow T^2$ is a universal cover of the torus.

Proposition 25.2.19

If $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ is a universal cover, then $\varepsilon_p : \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$ is a bijection.

Proof. Injectivity. Suppose $\varepsilon_p[\gamma_0] = \varepsilon_p[\gamma_1] = \hat{x}_1$. Then $\hat{\gamma}_0$ and $\hat{\gamma}_1$ are both paths from $\hat{x}_0$ to $\hat{x}_1$ in $\hat{X}$, which is simply connected, so $\hat{\gamma}_0 \simeq \hat{\gamma}_1$ as paths by Lemma 25.1.29. Composing the homotopy with p gives $\gamma_0 \simeq \gamma_1$, so $[\gamma_0] = [\gamma_1]$.

Surjectivity. Given $\hat{x} \in p^{-1}(x_0)$, use path connectedness of $\hat{X}$ to find a path $\eta : \hat{x}_0 \leadsto \hat{x}$. Then $\gamma = p \circ \eta$ is a loop at x_0 , and η is its lift starting at $\hat{x}_0$, so $\varepsilon_p[\gamma] = \eta(1) = \hat{x}$. $\square$

This is already enough to compute the fundamental group of the circle as a *set*: we have $p^{-1}(1) = \mathbb{Z}$ for the helix covering, so $\pi_1(S^1, 1)$ is countably infinite. To get the group

structure we have to work slightly harder, and the mechanism is that we can slide the covering space $\mathbb{R}$ along itself.

Theorem 25.2.20

$\varepsilon_p : \pi_1(S^1, 1) \rightarrow \mathbb{Z}$ is a group isomorphism. In particular $\pi_1(S^1) \cong \mathbb{Z}$.

Proof. It is a bijection by Proposition 25.2.19, using $p^{-1}(1) = \mathbb{Z} \subseteq \mathbb{R}$, so it suffices to check that it is a homomorphism.

For $n \in \mathbb{Z}$ define $\varphi_n : \mathbb{R} \rightarrow \mathbb{R}$ by $\varphi_n(x) = x + n$. Then $p \circ \varphi_n = p$, since $e^{2\pi i(x+n)} = e^{2\pi i x}$. So if $\hat{\gamma}$ is any lift of a path γ , then $\varphi_n \circ \hat{\gamma}$ is another lift, now starting at $\hat{\gamma}(0) + n$.

Suppose $\varepsilon_p[\gamma] = n$ and $\varepsilon_p[\gamma'] = n'$, so the lifts based at 0 satisfy $\hat{\gamma}(1) = n$ and $\hat{\gamma}'(1) = n'$. Then $\varphi_n \circ \hat{\gamma}'$ is a lift of γ' starting at $n = \hat{\gamma}(1)$, so the concatenation

$$\widehat{\gamma \cdot \gamma'} = \hat{\gamma} \cdot (\varphi_n \circ \hat{\gamma}')$$

is the lift of $\gamma \cdot \gamma'$ starting at 0. Evaluating at 1,

$$\varepsilon_p[\gamma \cdot \gamma'] = \varphi_n(\hat{\gamma}'(1)) = n + n' = \varepsilon_p[\gamma] + \varepsilon_p[\gamma'].$$

So ε_p is a homomorphism, and hence an isomorphism. $\square$

Concretely, the class corresponding to $n \in \mathbb{Z}$ is that of the loop $\gamma_n(t) = e^{2\pi i n t}$, which winds n times around the circle: its lift starting at 0 is $\hat{\gamma}_n(t) = nt$, ending at n . So the fundamental group of the circle counts winding number, exactly as intuition demands.

Corollary 25.2.21

S^1 is not contractible, and S^1 is not a retract of D^2 .

Proof. If S^1 were contractible then $\pi_1(S^1)$ would be trivial, but it is $\mathbb{Z}$.

Suppose $r : D^2 \rightarrow S^1$ were a retraction, with $\iota : S^1 \hookrightarrow D^2$ the inclusion. Then $r \circ \iota = \text{id}_{S^1}$, so $r_* \circ \iota_* = \text{id}$ on $\pi_1(S^1, 1) = \mathbb{Z}$. But ι_* factors through $\pi_1(D^2, 1)$, which is trivial as D^2 is convex. So the identity on $\mathbb{Z}$ factors through the trivial group, which is absurd. $\square$

Corollary 25.2.22 (Brouwer Fixed Point Theorem, $n = 2$)

Every map $f : D^2 \rightarrow D^2$ has a fixed point.

Proof. Suppose $f(x) \neq x$ for all x . Define $r(x) \in S^1$ to be the point where the ray from $f(x)$ through x meets S^1 . This is continuous (we will verify a formula for it in Theorem 25.6.13), and if $x \in S^1$ then $r(x) = x$. So r is a retraction $D^2 \rightarrow S^1$, contradicting the previous corollary. $\square$

Example 25.2.23

Since $\pi_1(X \times Y) \cong \pi_1(X) \times \pi_1(Y)$ (Example 25.1.34), the torus has

$$\pi_1(T^2) = \pi_1(S^1 \times S^1) \cong \mathbb{Z}^2.$$

This already tells us that $T^2 \not\cong S^2$ once we know that S^2 is simply connected, and that $T^2 \not\cong S^1 \vee S^1$ once we know the latter has non-abelian fundamental group.

§25.2.4 Degree and the Fundamental Theorem of Algebra

The isomorphism $\pi_1(S^1) \cong \mathbb{Z}$ lets us attach an integer to any self-map of the circle. There is one small nuisance: a map $f : S^1 \rightarrow S^1$ need not preserve the basepoint, so f_* lands in $\pi_1(S^1, f(1))$ rather than $\pi_1(S^1, 1)$. We fix this by transporting back along a path, and the following lemma says that the choice of path does not matter.

Lemma 25.2.24

Let $z \in S^1$ and let $u, v : z \rightsquigarrow 1$ be paths. Then $u_{\#} = v_{\#} : \pi_1(S^1, z) \rightarrow \pi_1(S^1, 1)$.

Proof. By Proposition 25.1.24, $(v^{-1})_{\#} \circ u_{\#}$ is the map $[\gamma] \mapsto [\eta][\gamma][\eta]^{-1}$ where $\eta = v \cdot u^{-1}$ is a loop at 1. But $\pi_1(S^1, 1) \cong \mathbb{Z}$ is abelian, so conjugation is trivial and $(v^{-1})_{\#} \circ u_{\#} = \text{id}$. Hence $u_{\#} = v_{\#}$. $\square$

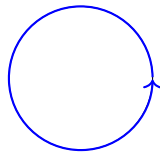
Definition 25.2.25 (Degree)

Let $f : S^1 \rightarrow S^1$ be a map, and set $z = f(1)$. Choose any path $u : z \rightsquigarrow 1$. Then

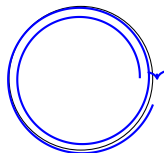
$$u_{\#} \circ f_* : \pi_1(S^1, 1) \rightarrow \pi_1(S^1, 1)$$

is a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$, so is multiplication by some integer. We define the **degree** $\deg f$ to be that integer, namely $(u_{\#} \circ f_*)(1)$.

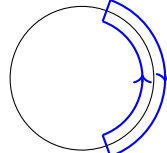
By Lemma 25.2.24 this is independent of the choice of u . Intuitively, $\deg f$ counts how many times f wraps the circle around itself, with sign.



$\deg = 1$



$\deg = 2$



$\deg = 0$

Proposition 25.2.26

Let $f_n : S^1 \rightarrow S^1$ be given by $z \mapsto z^n$. Then

1. $\deg f_n = n$;
2. maps $g_0, g_1 : S^1 \rightarrow S^1$ are homotopic if and only if $\deg g_0 = \deg g_1$;
3. $g : S^1 \rightarrow S^1$ extends to a map $D^2 \rightarrow S^1$ if and only if $\deg g = 0$.

Proof. (i) Since $f_n(1) = 1$ we may take u constant. Now $f_n \circ \gamma_1 = \gamma_n$, where $\gamma_n(t) = e^{2\pi i n t}$, so $f_{n*}[\gamma_1] = [\gamma_n]$, which corresponds to $n \in \mathbb{Z}$.

(ii) Suppose $g_0 \simeq_H g_1$, and let $u(t) = H(1, t)$, a path $g_0(1) \leadsto g_1(1)$. By Lemma 25.1.25, $g_{1*} = u_{\#} \circ g_{0*}$. Choose $v : g_1(1) \leadsto 1$; then $u \cdot v : g_0(1) \leadsto 1$ and

$$\deg g_1 = (v_{\#} \circ g_{1*})(1) = (v_{\#} \circ u_{\#} \circ g_{0*})(1) = ((u \cdot v)_{\#} \circ g_{0*})(1) = \deg g_0,$$

using $(u \cdot v)_{\#} = v_{\#} \circ u_{\#}$.

Conversely, it is enough to show $g \simeq f_{\deg g}$. First suppose $g(1) = 1$, and let $\gamma = g \circ \gamma_1$, so that $g = \bar{\gamma}$ and $\deg g = g_*(1) = [\gamma]$. If $\deg g = n$ then $\gamma \simeq \gamma_n$ as paths, whence $\bar{\gamma} \simeq \bar{\gamma}_n = f_n$ as maps $S^1 \rightarrow S^1$. In general, if $g(1) = e^{2\pi i x}$, then $g \simeq g_0$ where $g_0(z) = e^{-2\pi i x} g(z)$, via the homotopy $g_t(z) = e^{-2\pi i t x} g(z)$; and $g_0(1) = 1$, so we may apply the previous case.

(iii) By Lemma 25.1.31, g extends to D^2 if and only if g is null-homotopic, that is, if and only if $g \simeq c_1 = f_0$, which by (ii) happens if and only if $\deg g = 0$. $\square$

As an application, we can now prove the fundamental theorem of algebra — a statement with no topology in it whatsoever — by a winding number argument.

Let $P(w) = w^n + a_{n-1}w^{n-1} + \cdots + a_0 = w^n + q(w)$ be a monic complex polynomial of degree $n \geq 1$.

Lemma 25.2.27

Let $R_0 = \max\{1, \sum_{i=0}^{n-1} |a_i|\}$. Then $|w| > R_0$ implies $|w^n| > |q(w)|$.

Proof. If $|w| > R_0 \geq 1$ then $|w|^{i-n+1} \leq 1$ for $0 \leq i \leq n-1$, so

$$\frac{|q(w)|}{|w|^{n-1}} \leq \sum_{i=0}^{n-1} |a_i| |w|^{i-n+1} \leq \sum_{i=0}^{n-1} |a_i| \leq R_0.$$

Dividing by $|w|$ gives $|q(w)|/|w|^n \leq R_0/|w| < 1$. $\square$

Theorem 25.2.28 (Fundamental Theorem of Algebra)

Every complex polynomial of degree $n \geq 1$ has a root in $\mathbb{C}$.

Proof. We may assume P is monic, as above. Fix $R > R_0$ and consider the two maps $g_0, g_1 : S^1 \rightarrow \mathbb{C} \setminus \{0\}$ given by

$$g_0(z) = (Rz)^n, \quad g_1(z) = P(Rz).$$

These are homotopic via $g_t(z) = P_t(Rz)$ where $P_t(w) = w^n + tq(w)$: this really does avoid 0, since $|w| = R > R_0$ gives $|w^n| > |q(w)| \geq |tq(w)|$ by Lemma 25.2.27. Let $\pi : \mathbb{C} \setminus \{0\} \rightarrow S^1$ be the radial projection $w \mapsto w/|w|$. Then $\pi \circ g_0 \simeq \pi \circ g_1$ as maps $S^1 \rightarrow S^1$, and $\pi \circ g_0 = f_n$ up to the constant $R^n/|R^n| = 1$, so

$$\deg(\pi \circ g_1) = \deg(\pi \circ g_0) = n.$$

Now suppose for contradiction that P has no root. Then $P : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0\}$, so g_1 extends to $G_1 : D^2 \rightarrow \mathbb{C} \setminus \{0\}$ by $G_1(z) = P(Rz)$, and hence $\pi \circ g_1$ extends to $\pi \circ G_1 : D^2 \rightarrow S^1$. By Proposition 25.2.26(iii) this forces $\deg(\pi \circ g_1) = 0$, so $n = 0$, contradicting $n \geq 1$. $\square$

§25.2.5 The Lifting Criterion

We can lift paths and homotopies. When can we lift a map from a general space Z ? There are two obstructions, one of which we have already met.

Definition 25.2.29

A space X is **locally path connected** if for every open $U \subseteq X$ and every $x \in U$ there is a path connected open V with $x \in V \subseteq U$.

Example 25.2.30

Every open subset of $\mathbb{R}^n$ and every polyhedron is locally path connected. A standard example of a space which is not is the ‘comb’

$$X = (\mathbb{R} \times \{0\}) \cup \left(\bigcup_{n \geq 1} \{1/n\} \times \mathbb{R} \right) \cup (\{0\} \times \mathbb{R}) \subseteq \mathbb{R}^2 :$$

a small open neighbourhood of a point $(0, y)$ with $y \neq 0$ contains bits of infinitely many teeth, and cannot be shrunk to a path connected one.

Proposition 25.2.31 (Simply Connected Lifting)

Let $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ be a covering map, and let Z be simply connected and locally path connected. Then every map $f : (Z, z_0) \rightarrow (X, x_0)$ has a unique lift $\hat{f} : (Z, z_0) \rightarrow (\hat{X}, \hat{x}_0)$.

Proof. Uniqueness, and the formula. Suppose $\hat{f}$ is such a lift, and let $z \in Z$. Choose a path $\gamma : z_0 \leadsto z$, which exists as Z is path connected. Then $\hat{f} \circ \gamma$ is a lift of $f \circ \gamma$ starting at $\hat{x}_0$, so by uniqueness of path lifting it must be $\widehat{f \circ \gamma}$. Hence

$$\hat{f}(z) = (\hat{f} \circ \gamma)(1) = \widehat{f \circ \gamma}(1),$$

which determines $\hat{f}$ completely.

Existence. We take the displayed formula as a *definition* of $\hat{f}$. This is independent of the choice of γ : if $\gamma_0, \gamma_1 : z_0 \leadsto z$ then $\gamma_0 \simeq \gamma_1$ as Z is simply connected (Lemma 25.1.29), so $f \circ \gamma_0 \simeq f \circ \gamma_1$, so their lifts have the same endpoint by Proposition 25.2.15. Also $p(\hat{f}(z)) = p(\widehat{f \circ \gamma}(1)) = f(\gamma(1)) = f(z)$, so $\hat{f}$ is a lift as a function, and taking $\gamma = c_{z_0}$ gives $\hat{f}(z_0) = \hat{x}_0$.

Continuity. This is where local path connectedness enters. Let $z \in Z$ and let $U \subseteq \hat{X}$ be an open neighbourhood of $\hat{f}(z)$; we must find a neighbourhood V of z with $\hat{f}(V) \subseteq U$. Choose an evenly covered open $W \ni f(z)$, so $p^{-1}(W) = \bigsqcup_{\alpha} W_{\alpha}$,

and say $\hat{f}(z) \in W_{\alpha_0}$. Replacing U by $U \cap W_{\alpha_0}$, we may assume $U \subseteq W_{\alpha_0}$, and then $p(U)$ is open (as $p|_{W_{\alpha_0}}$ is a homeomorphism onto W) and contains $f(z)$.

By continuity of f there is an open $V' \ni z$ with $f(V') \subseteq p(U)$, and by local path connectedness there is a path connected open V with $z \in V \subseteq V'$. We claim $\hat{f}(V) \subseteq U$. Given $z' \in V$, choose a path $\gamma' : z \rightsquigarrow z'$ inside V . Then $f \circ \gamma'$ has image in $p(U) \subseteq W$, so $\tilde{\gamma}' = p_{\alpha_0}^{-1} \circ f \circ \gamma'$ is a lift of $f \circ \gamma'$ starting at $p_{\alpha_0}^{-1}(f(z)) = \hat{f}(z)$. Concatenating, $\widehat{f \circ (\gamma \cdot \gamma')}(1) = \widehat{f \circ \gamma} \cdot \tilde{\gamma}'$, and so

$$\hat{f}(z') = \widehat{f \circ (\gamma \cdot \gamma')}(1) = \tilde{\gamma}'(1) = p_{\alpha_0}^{-1}(f(z')) \in U$$

as required. $\square$

Simple connectedness is genuinely needed here, not just a convenience.

Example 25.2.32

The identity map $\text{id} : S^1 \rightarrow S^1$ does not lift along $p : \mathbb{R} \rightarrow S^1$. If it did, we would have a map $s : S^1 \rightarrow \mathbb{R}$ with $p \circ s = \text{id}$, and applying π_1 would factor $\text{id}_{\mathbb{Z}}$ through $\pi_1(\mathbb{R}) = 1$.

That computation suggests the general criterion: the obstruction is exactly the image of π_1 .

Theorem 25.2.33 (Lifting Criterion)

Let $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ be a covering map and let Z be path connected and locally path connected. A map $f : (Z, z_0) \rightarrow (X, x_0)$ lifts to $\hat{f} : (Z, z_0) \rightarrow (\hat{X}, \hat{x}_0)$ if and only if

$$f_*(\pi_1(Z, z_0)) \subseteq p_*(\pi_1(\hat{X}, \hat{x}_0)),$$

and in that case the lift is unique.

Proof. If $\hat{f}$ exists then $f_* = p_* \circ \hat{f}_*$, so the containment holds.

Conversely, suppose the containment holds. Uniqueness, and the formula $\hat{f}(z) = \widehat{f \circ \gamma}(1)$, are exactly as in Proposition 25.2.31; and continuity is proved verbatim. The only point needing a new argument is well-definedness. Suppose $\gamma_0, \gamma_1 : z_0 \rightsquigarrow z$. Then $\gamma_0 \cdot \gamma_1^{-1}$ is a loop at z_0 , so

$$f_*[\gamma_0 \cdot \gamma_1^{-1}] = p_*[\delta]$$

for some loop δ in $\hat{X}$ at $\hat{x}_0$. Thus $(f \circ \gamma_0) \cdot (f \circ \gamma_1)^{-1} \simeq p \circ \delta$, and lifting this homotopy at $\hat{x}_0$, the lift of $p \circ \delta$ is the loop δ , which ends at $\hat{x}_0$. Hence the lift of $(f \circ \gamma_0) \cdot (f \circ \gamma_1)^{-1}$ at $\hat{x}_0$ is a loop, which says precisely that $\widehat{f \circ \gamma_0}(1) = \widehat{f \circ \gamma_1}(1)$. $\square$

§25.2.6 Deck Transformations

Covering spaces of a fixed space X form a category, and it will pay to think of them that way.

Definition 25.2.34 (Covering Transformation)

Let $p_i : \hat{X}_i \rightarrow X$ be covering maps for $i = 1, 2$. A **covering transformation** (or morphism of covers) $\phi : (p_1, \hat{X}_1) \rightarrow (p_2, \hat{X}_2)$ is a map $\phi : \hat{X}_1 \rightarrow \hat{X}_2$ with $p_2 \circ \phi = p_1$, so that

$$\begin{array}{ccc} \hat{X}_1 & \xrightarrow{\quad \phi \quad} & \hat{X}_2 \\ & \searrow p_1 \quad \swarrow p_2 & \\ & X & \end{array}$$

commutes. It is a **covering isomorphism** if it is a bijection.

Equivalently, ϕ is a lift of p_1 along p_2 . So all the lifting theory applies at once.

Example 25.2.35

Let $p_1(z) = z^6$ and $p_2(z) = z^2$ be covering maps $S^1 \rightarrow S^1$. Then $\phi(z) = z^3$ is a covering transformation from (p_1, S^1) to (p_2, S^1) , since $(z^3)^2 = z^6$.

Lemma 25.2.36

A bijective covering map is a homeomorphism.

Proof. Let $p : \hat{Y} \rightarrow Y$ be a bijective covering map. Then p^{-1} exists as a function, and we must check it is continuous. Cover Y by evenly covered open sets U . As p is a bijection, $p^{-1}(U)$ consists of a single sheet, and $p^{-1}|_U$ is the inverse of a homeomorphism, hence continuous. Continuity is local, so p^{-1} is continuous. $\square$

Lemma 25.2.37

Let X be locally path connected, and let $\phi : (p_1, \hat{X}_1) \rightarrow (p_2, \hat{X}_2)$ be a surjective covering transformation. Then $\phi : \hat{X}_1 \rightarrow \hat{X}_2$ is itself a covering map.

Proof. Let $\hat{x}_2 \in \hat{X}_2$ and put $x = p_2(\hat{x}_2)$. Choose neighbourhoods U_1, U_2 of x evenly covered by p_1, p_2 respectively; then $U = U_1 \cap U_2$ is evenly covered by both, and by local path connectedness we may choose a path connected open V with $x \in V \subseteq U$. Write

$$p_1^{-1}(V) = \bigsqcup_{\alpha} V_{\alpha}, \quad p_2^{-1}(V) = \bigsqcup_{\beta} V'_{\beta},$$

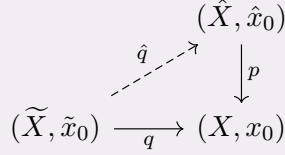
where each V_{α}, V'_{β} is homeomorphic to V and hence path connected. Since $p_2 \circ \phi = p_1$ we have $\phi(V_{\alpha}) \subseteq p_2^{-1}(V)$, and V_{α} is path connected, so $\phi(V_{\alpha}) \subseteq V'_{\beta(\alpha)}$ for a unique $\beta(\alpha)$. On V_{α} the relation $p_2 \circ \phi = p_1$ reads $\phi|_{V_{\alpha}} = (p_2|_{V'_{\beta(\alpha)}})^{-1} \circ p_1|_{V_{\alpha}}$, a composite of homeomorphisms. Hence

$$\phi^{-1}(V'_{\beta}) = \bigsqcup_{\alpha: \beta(\alpha)=\beta} V_{\alpha}$$

with each piece mapped homeomorphically onto V'_{β} , so V'_{β} is evenly covered by ϕ . As ϕ is surjective, these V'_{β} cover $\hat{X}_2$. $\square$

Proposition 25.2.38 (Universal Property of the Universal Cover)

Let X be locally path connected and let $q : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a universal cover. Then for any covering map $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ there is a unique covering transformation $\hat{q} : (\tilde{X}, \tilde{x}_0) \rightarrow (\hat{X}, \hat{x}_0)$ with $\hat{q}(\tilde{x}_0) = \hat{x}_0$.



Proof. $\tilde{X}$ is simply connected, and it is locally path connected because X is and q is a local homeomorphism. So this is exactly Proposition 25.2.31 applied to the map q . $\square$

Corollary 25.2.39 (Uniqueness of Universal Covers)

Let X be locally path connected. If $q : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ and $q' : (\tilde{X}', \tilde{x}'_0) \rightarrow (X, x_0)$ are universal covers, then there is a unique covering isomorphism between them taking $\tilde{x}_0$ to $\tilde{x}'_0$. In particular $\tilde{X} \cong \tilde{X}'$.

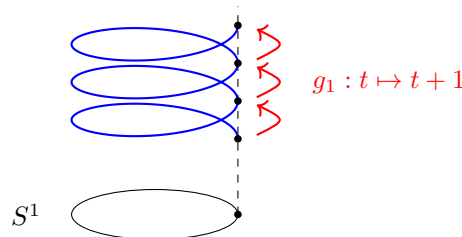
Proof. Proposition 25.2.38 gives covering transformations $\hat{q} : \tilde{X} \rightarrow \tilde{X}'$ and $\hat{q}' : \tilde{X}' \rightarrow \tilde{X}$ preserving basepoints. Their composites are covering transformations $\tilde{X} \rightarrow \tilde{X}$ fixing $\tilde{x}_0$, and the identity is another such, so by the uniqueness in Proposition 25.2.38 they are the identity. Hence $\hat{q}$ is a bijection, and a homeomorphism by Lemmas 25.2.37 and 25.2.36. $\square$

So we may speak of *the* universal cover of a (nice) space. We will not prove existence; suffice it to say that every path connected, locally path connected and *semi-locally simply connected* space has one, and all the spaces we care about (manifolds, polyhedra) qualify.

Definition 25.2.40 (Deck Group)

Let $p : \hat{X} \rightarrow X$ be a covering map. The **deck group** $G_D(p)$ is the group of covering isomorphisms $(p, \hat{X}) \rightarrow (p, \hat{X})$, under composition. Its elements are called **deck transformations**, and they act on $\hat{X}$ on the left.

A deck transformation is a symmetry of the covering: a homeomorphism of $\hat{X}$ which permutes each fibre $p^{-1}(x)$ among itself.



Example 25.2.41

For $p : \mathbb{R} \rightarrow S^1$, the deck group is $\{g_n : t \mapsto t + n \mid n \in \mathbb{Z}\} \cong \mathbb{Z}$. Indeed each g_n satisfies $p \circ g_n = p$; conversely a deck transformation g is a lift of p , determined by $g(0) \in p^{-1}(1) = \mathbb{Z}$, so $g = g_{g(0)}$. Note $G_D(p) \cong \mathbb{Z} \cong \pi_1(S^1)$.

Example 25.2.42

For $p : S^n \rightarrow \mathbb{RP}^n$, the deck group is $\{\text{id}, x \mapsto -x\} \cong \mathbb{Z}/2\mathbb{Z}$.

That the deck group of the helix covering agrees with $\pi_1(S^1)$ is no accident.

Theorem 25.2.43

Let X be locally path connected and let $q : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$ be a universal cover. Then $G_D(q) \cong \pi_1(X, x_0)$.

Proof. First, the evaluation map $G_D(q) \rightarrow q^{-1}(x_0)$, $g \mapsto g(\tilde{x}_0)$, is a bijection: given $\tilde{x} \in q^{-1}(x_0)$, Proposition 25.2.38 (applied with $\hat{X} = \tilde{X}$ and basepoint $\tilde{x}$) provides a unique covering transformation taking $\tilde{x}_0 \mapsto \tilde{x}$, and it is an isomorphism by the uniqueness corollary.

Composing with the bijection $\varepsilon_q^{-1} : q^{-1}(x_0) \rightarrow \pi_1(X, x_0)$ of Proposition 25.2.19, we get a bijection $\Theta : \pi_1(X, x_0) \rightarrow G_D(q)$, characterised by $\Theta([\gamma])(\tilde{x}_0) = \tilde{\gamma}(1)$, where $\tilde{\gamma}$ is the lift of γ at $\tilde{x}_0$.

It remains to check that Θ is a homomorphism. Write $g = \Theta[\gamma]$ and $g' = \Theta[\gamma']$. Since g is a deck transformation, $g \circ \tilde{\gamma}'$ is a lift of γ' starting at $g(\tilde{x}_0) = \tilde{\gamma}(1)$, so

$$\widetilde{\gamma \cdot \gamma'} = \tilde{\gamma} \cdot (g \circ \tilde{\gamma}'),$$

whence

$$\Theta[\gamma \cdot \gamma'](\tilde{x}_0) = \widetilde{\gamma \cdot \gamma'}(1) = g(\tilde{\gamma}'(1)) = g(g'(\tilde{x}_0)) = (g \circ g')(\tilde{x}_0).$$

Both $\Theta[\gamma \cdot \gamma']$ and $g \circ g'$ are deck transformations sending $\tilde{x}_0$ to the same point, so they are equal. $\square$

§25.2.7 The Galois Correspondence

We can now describe *all* the connected covers of a space in terms of the subgroup structure of its fundamental group. Throughout, let X be path connected and locally path connected with universal cover $q : (\tilde{X}, \tilde{x}_0) \rightarrow (X, x_0)$, and identify $G = \pi_1(X, x_0)$ with the deck group $G_D(q)$ via Theorem 25.2.43.

Given a subgroup $H \leq G$, the group H acts on $\tilde{X}$ by deck transformations, and we may form the quotient

$$X_H = H \backslash \tilde{X},$$

whose points are the orbits $H \cdot \tilde{x}$. Since deck transformations satisfy $q \circ h = q$, the map

q descends to $p_H : X_H \rightarrow X$, $H \cdot \tilde{x} \mapsto q(\tilde{x})$, and we have a tower

$$\begin{array}{ccc} \widetilde{X} & & 1 \\ \pi_H \downarrow & & \downarrow \\ X_H & & H \\ p_H \downarrow & & \downarrow \\ X & & G \end{array}$$

Proposition 25.2.44

π_H and p_H are covering maps. If $H = G$ then p_G is a covering isomorphism, so $X \cong G \backslash \widetilde{X}$.

Proof. Let $x \in X$ and choose a path connected evenly covered neighbourhood U of x , so $q^{-1}(U) = \bigsqcup_{\alpha} U_{\alpha}$. The deck group acts simply transitively on $q^{-1}(x)$, so the sheets are permuted simply transitively too, and we may index them as $q^{-1}(U) = \bigsqcup_{g \in G} g \cdot U_0$ for a fixed sheet U_0 . Then the H -orbits of sheets are indexed by the cosets gH , giving

$$p_H^{-1}(U) = \bigsqcup_{gH \in G/H} \pi_H(\bigcup_{h \in H} gh \cdot U_0),$$

and each piece maps homeomorphically onto U ; similarly π_H^{-1} of each piece is $\bigsqcup_{h \in H} gh \cdot U_0$. So U is evenly covered by p_H and each piece is evenly covered by π_H .

If $H = G$ then p_G is injective on fibres, hence bijective, hence a homeomorphism by Lemma 25.2.36. $\square$

In particular a universal covering map is exactly the quotient by the action of $\pi_1(X, x_0)$.

Definition 25.2.45

A covering map $p : \hat{X} \rightarrow X$ is a **normal** (or regular) cover if $G_D(p)$ acts transitively on the fibre $p^{-1}(x_0)$.

Now consider the two sets

$$S(X, x_0) = \{H \mid H \leq \pi_1(X, x_0)\},$$

$$C(X, x_0) = \{(p, \hat{X}, \hat{x}_0) \mid p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0) \text{ a covering map, } \hat{X} \text{ path connected}\} / \sim,$$

where $(p, \hat{X}, \hat{x}_0) \sim (p', \hat{X}', \hat{x}'_0)$ if there is a covering isomorphism between them matching the basepoints. Define

$$\alpha : S(X, x_0) \rightarrow C(X, x_0), \quad \alpha(H) = (p_H, X_H, \pi_H(\tilde{x}_0)),$$

$$\beta : C(X, x_0) \rightarrow S(X, x_0), \quad \beta(p, \hat{X}, \hat{x}_0) = p_*(\pi_1(\hat{X}, \hat{x}_0)),$$

noting that β lands in subgroups by Corollary 25.2.16.

Theorem 25.2.46 (Galois Correspondence)

α and β are mutually inverse bijections.

Proof. $\beta \circ \alpha = \text{id}$. Write $x_{0,H} = \pi_H(\tilde{x}_0)$. Given $[\gamma] \in H$ with lift $\tilde{\gamma}$ at $\tilde{x}_0$, the path $\pi_H \circ \tilde{\gamma}$ is a loop in X_H at $x_{0,H}$, since $\tilde{\gamma}(1) = [\gamma] \cdot \tilde{x}_0$ lies in the same H -orbit as $\tilde{x}_0$. Moreover

$$p_{H*}[\pi_H \circ \tilde{\gamma}] = [p_H \circ \pi_H \circ \tilde{\gamma}] = [q \circ \tilde{\gamma}] = [\gamma],$$

so $H \subseteq \beta(\alpha(H))$. Conversely, if $[\gamma] = p_{H*}[\delta]$ for a loop δ in X_H , then the lift of γ at $\tilde{x}_0$ ends at a point of $\pi_H^{-1}(x_{0,H}) = H \cdot \tilde{x}_0$, which says exactly that $[\gamma] \in H$.

$\alpha \circ \beta = \text{id}$. Let $p : (\hat{X}, \hat{x}_0) \rightarrow (X, x_0)$ be a path connected cover and $H = p_*(\pi_1(\hat{X}, \hat{x}_0))$. By Proposition 25.2.38 there is a covering transformation $\hat{q} : \hat{X} \rightarrow \hat{X}$ with $\hat{q}(\hat{x}_0) = \hat{x}_0$, and it is surjective since $\hat{X}$ is path connected. We claim $\hat{q}$ is constant on H -orbits. Indeed let $h = [p \circ \delta] \in H$ for a loop δ in $\hat{X}$ at $\hat{x}_0$, and let $\tilde{x} \in \hat{X}$ with a path $\eta : \tilde{x}_0 \leadsto \tilde{x}$. Recall from the proof of Proposition 25.2.31 that $\hat{q}(\tilde{x}) = \widehat{q \circ \eta}(1)$, the lift being taken at $\hat{x}_0$. Now $h \cdot \eta$ is a path from $h \cdot \tilde{x}_0$ to $h \cdot \tilde{x}$, so a path from $\tilde{x}_0$ to $h \cdot \tilde{x}$ is $\eta_h \cdot (h \circ \eta)$ where $\eta_h : \tilde{x}_0 \leadsto h \cdot \tilde{x}_0$ satisfies $q \circ \eta_h = p \circ \delta$. Hence

$$q \circ (\eta_h \cdot (h \circ \eta)) = \delta \cdot \widehat{q \circ \eta},$$

and evaluating at 1 gives $\hat{q}(h \cdot \tilde{x}) = \widehat{q \circ \eta}(1) = \hat{q}(\tilde{x})$, as claimed.

So $\hat{q}$ factors as $\hat{q} = p' \circ \pi_H$ for a map $p' : X_H \rightarrow \hat{X}$, which is a covering transformation over X taking $x_{0,H}$ to $\hat{x}_0$. It is surjective as $\hat{q}$ is, and injective because both X_H and $\hat{X}$ have fibres over x_0 in bijection with $H \backslash G$ — for X_H by construction, for $\hat{X}$ by the first half of the proof. So p' is a covering isomorphism, giving $\alpha(\beta(p, \hat{X}, \hat{x}_0)) = (p, \hat{X}, \hat{x}_0)$ in $C(X, x_0)$. $\square$

The dictionary this sets up is worth spelling out.

Remark. Under the correspondence of Theorem 25.2.46:

1. the whole group G corresponds to the trivial cover (id, X, x_0) , and the trivial subgroup 1 corresponds to the universal cover;
2. the index $[G : H]$ equals the number of sheets $|p_H^{-1}(x_0)|$;
3. changing the basepoint of a cover replaces H by a conjugate: if $g = [\gamma]$ and $\hat{\gamma}$ is the lift of γ to X_H at $x_{0,H}$, then $(p_H, X_H, \hat{\gamma}(1))$ corresponds to $g^{-1}Hg$. So unbased covers correspond to conjugacy classes of subgroups;
4. p_H is a normal cover if and only if $H \trianglelefteq G$, and in that case $G_D(p_H) \cong G/H$.

Example 25.2.47

Take $X = S^1$, so $G = \mathbb{Z}$. The subgroups are $n\mathbb{Z}$ for $n \geq 0$. The subgroup 0 gives the universal cover $\mathbb{R} \rightarrow S^1$, and $n\mathbb{Z}$ for $n \geq 1$ gives the n -sheeted cover $S^1 \rightarrow S^1$, $z \mapsto z^n$: indeed $p_{n*}(\pi_1(S^1)) = n\mathbb{Z}$, since p_n takes a generating loop to a loop of winding number n . As $\mathbb{Z}$ is abelian, every cover of S^1 is normal, with deck group $\mathbb{Z}/n\mathbb{Z}$. This is the complete list of connected covers of the circle.

Example 25.2.48

The two-sheeted cover $S^n \rightarrow \mathbb{RP}^n$ is universal for $n \geq 2$, since we will show in the next section that S^n is simply connected for $n \geq 2$. Hence

$$\pi_1(\mathbb{RP}^n) \cong G_D(p) \cong \mathbb{Z}/2\mathbb{Z} \quad (n \geq 2).$$

§25.3 The Seifert–van Kampen Theorem

Covering space theory computed $\pi_1(S^1)$, and with it π_1 of anything built out of circles by taking products or quotients by nice group actions. That is a rather short list. The Seifert–van Kampen theorem is the general computational tool: it tells us how to assemble $\pi_1(X)$ from π_1 of the pieces of an open cover of X by two sets. Before we can state it we need the group theory it takes values in.

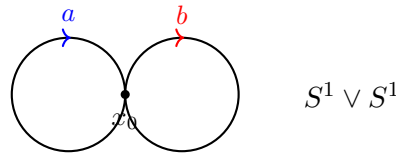
§25.3.1 Wedges and Free Groups**Definition 25.3.1 (Wedge)**

Let (X_i, x_i) for $1 \leq i \leq n$ be based spaces. Their **wedge** is

$$\bigvee_{i=1}^n (X_i, x_i) = \left(\prod_{i=1}^n X_i \right) / \sim,$$

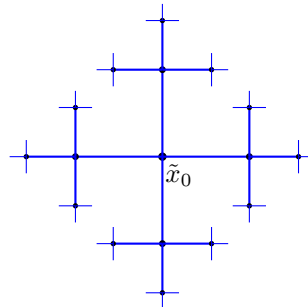
where $\sim$ is generated by $x_i \sim x_j$ for all i, j . That is, we glue the spaces together at their basepoints.

The example we care about is the **figure-eight** $S^1 \vee S^1$, and more generally the wedge of n circles.



Write $\iota_1, \iota_2 : (S^1, 1) \rightarrow (S^1 \vee S^1, x_0)$ for the two inclusions, and set $a = \iota_{1*}(1)$, $b = \iota_{2*}(1)$ in $\pi_1(S^1 \vee S^1, x_0)$. What group do a and b generate? We can answer this with covering spaces.

Let $T_\infty(4)$ be the infinite 4-valent tree: the connected graph in which every vertex has exactly four neighbours and there are no cycles, topologised as the union of its finite subtrees $T_n(4)$ (so U is open exactly when each $U \cap T_n(4)$ is).



Proposition 25.3.2

There is a covering map $q : T_\infty(4) \rightarrow S^1 \vee S^1$, and $T_\infty(4)$ is simply connected. So $T_\infty(4)$ is the universal cover of the figure-eight.

Proof. Colour the four edges at each vertex with the labels a, a^{-1}, b, b^{-1} , one each, in such a way that the a -edge out of v is the a^{-1} -edge out of its other endpoint, and similarly for b . Now map each a -labelled edge homeomorphically onto the first circle, respecting orientation, and each b -labelled edge onto the second, sending all vertices to x_0 . A point in the interior of an edge has a neighbourhood inside that edge which is evenly covered, and a small star-shaped neighbourhood of x_0 (a little arc of each circle either side) is evenly covered by the corresponding small stars around each vertex. So q is a covering map.

For simple connectedness: I is compact, so the image of any path lies in some finite subtree $T_n(4)$. A finite tree is contractible — collapse the leaves one at a time — and hence simply connected, so any loop in $T_\infty(4)$ is null-homotopic within $T_n(4)$, and so within $T_\infty(4)$. $\square$

Corollary 25.3.3

$\pi_1(S^1 \vee S^1, x_0)$ is not abelian.

Proof. By Proposition 25.2.19, $\varepsilon_q : \pi_1(S^1 \vee S^1, x_0) \rightarrow q^{-1}(x_0)$ is a bijection onto the vertex set of the tree. Lifting the loop $a \cdot b$ at $\tilde{x}_0$, we travel along the a -edge and then a b -edge; lifting $b \cdot a$, we travel along the b -edge and then an a -edge. These end at different vertices, so $ab \neq ba$. $\square$

The bijection in that proof does more: it identifies $\pi_1(S^1 \vee S^1)$ completely. A vertex $\tilde{x}$ of the tree is joined to $\tilde{x}_0$ by a unique shortest path, and recording the labels of the edges traversed gives an ‘address’ for $\tilde{x}$, namely a word $w = \ell_1 \ell_2 \cdots \ell_r$ in the letters $a^{\pm 1}, b^{\pm 1}$ containing no substring $aa^{-1}, a^{-1}a, bb^{-1}, b^{-1}b$ — a **reduced word**. Conversely every reduced word is the address of exactly one vertex. Under ε_q , multiplication in π_1 corresponds to concatenating words and then cancelling. This is precisely the free group.

Definition 25.3.4 (Free Group)

A **free group** on a set S is a group F_S together with a function $S \rightarrow F_S$ such that for any group G and any function $\varphi : S \rightarrow G$ there is a unique homomorphism $\Phi : F_S \rightarrow G$ restricting to φ on S .

$$\begin{array}{ccc} & & F_S \\ & \nearrow & \vdots \Phi \\ S & \xrightarrow{\varphi} & G \end{array}$$

Lemma 25.3.5

If F_S and F'_S are both free on S then $F_S \cong F'_S$; more generally a bijection $S \rightarrow T$ induces an isomorphism $F_S \rightarrow F_T$. So the isomorphism type of F_S depends only on $|S|$, and we write F_n for the free group on n generators.

Proof. Let $\varphi : S \rightarrow T$ be a bijection with inverse ψ , and let $\Phi : F_S \rightarrow F_T$ and $\Psi : F_T \rightarrow F_S$ be the induced homomorphisms. Then $\Psi \circ \Phi : F_S \rightarrow F_S$ restricts to the identity on S ; but so does id_{F_S} , and the universal property says there is only one such homomorphism. So $\Psi \circ \Phi = \text{id}$, and by symmetry $\Phi \circ \Psi = \text{id}$. $\square$

Theorem 25.3.6

$\pi_1(\bigvee_{i=1}^n S^1) \cong F_n$, free on the classes $a_1, \dots, a_n$ of the n circles.

Proof. Exactly as above, the universal cover of $\bigvee_{i=1}^n S^1$ is the infinite $2n$ -valent tree $T_\infty(2n)$, and ε_q identifies π_1 with the set of reduced words in $a_1^{\pm 1}, \dots, a_n^{\pm 1}$, with multiplication given by concatenate-and-cancel. It remains to check the universal property. Given $\varphi : \{a_1, \dots, a_n\} \rightarrow G$, extend it to inverses by $\varphi(a_i^{-1}) = \varphi(a_i)^{-1}$ and define

$$\Phi(\ell_1 \cdots \ell_r) = \varphi(\ell_1) \cdots \varphi(\ell_r).$$

This is a homomorphism, since concatenating two reduced words and cancelling a substring $\ell\ell^{-1}$ has no effect on the right hand side. It is the unique homomorphism restricting to φ , since the a_i generate. $\square$

We will express our answers as presentations, so let us fix notation. For $S \subseteq G$ we write $\langle S \rangle$ for the smallest subgroup containing S and $\langle\langle S \rangle\rangle$ for the smallest *normal* subgroup containing S (the intersection of all such, which is a normal subgroup).

Definition 25.3.7 (Presentation)

Given a set S and $R \subseteq F_S$, we write $\langle S \mid R \rangle = F_S / \langle\langle R \rangle\rangle$. If $G \cong \langle S \mid R \rangle$ we call $\langle S \mid R \rangle$ a **presentation** of G , with **generators** S and **relations** R .

Every group has a presentation: take $S = G$ and $R = \ker(\Phi_G : F_G \rightarrow G)$. Usually we want much smaller ones.

Example 25.3.8

$\langle a \mid \emptyset \rangle \cong \mathbb{Z}$, $\langle a \mid a^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$, $\langle a, b \mid \emptyset \rangle \cong F_2$, and $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$. Note that $\langle a, b \mid ab^{-3} \rangle \cong \mathbb{Z}$, since a can be eliminated.

The last example is an instance of a general and very useful move.

Proposition 25.3.9

Let $\langle S \mid R \rangle$ be a presentation, let $a \notin S$ and let $w \in F_S$. Then

$$\langle S \mid R \rangle \cong \langle S \cup \{a\} \mid R \cup \{aw^{-1}\} \rangle.$$

Proof. Define φ from left to right by $s \mapsto s$, and ψ from right to left by $s \mapsto s$ and $a \mapsto w$; both are well-defined as they respect the relations. They are mutually inverse. $\square$

We will also freely use that a relation may be replaced by a conjugate of itself, or multiplied by another relation, without changing the group.

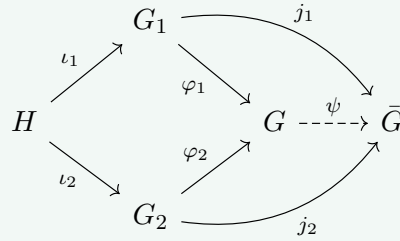
Remark. One should not get too optimistic about presentations. It is a theorem of Novikov and Boone that there is no algorithm which, given a finite presentation, decides whether the group it presents is trivial. So ‘compute π_1 ’ can only ever mean ‘write down a presentation’.

§25.3.2 Amalgamated Free Products

Definition 25.3.10 (Amalgamated Free Product)

Let $\iota_1 : H \rightarrow G_1$ and $\iota_2 : H \rightarrow G_2$ be group homomorphisms. A group G is an **amalgamated free product** of G_1 and G_2 along H if

1. there are homomorphisms $\varphi_i : G_i \rightarrow G$ with $\varphi_1 \circ \iota_1 = \varphi_2 \circ \iota_2$; and
2. this is universal: for any group $\bar{G}$ with homomorphisms $j_i : G_i \rightarrow \bar{G}$ satisfying $j_1 \circ \iota_1 = j_2 \circ \iota_2$, there is a unique $\psi : G \rightarrow \bar{G}$ with $\psi \circ \varphi_i = j_i$.



We write $G = G_1 *_H G_2$.

Categorically this is a pushout; concretely it is ‘ G_1 and G_2 glued along H , with no relations imposed beyond those forced’. As with all universal properties, uniqueness is automatic.

Proposition 25.3.11

If G and G' are both amalgamated free products of G_1, G_2 along H , then $G \cong G'$.

Proof. Universality gives $\alpha : G \rightarrow G'$ and $\beta : G' \rightarrow G$ compatible with the φ_i . Then $\beta \circ \alpha : G \rightarrow G$ is compatible with the φ_i , as is id_G , so uniqueness forces $\beta \circ \alpha = \text{id}_G$; likewise $\alpha \circ \beta = \text{id}_{G'}$. $\square$

Proposition 25.3.12

Amalgamated free products exist. Explicitly, if $G_i = \langle S_i \mid R_i \rangle$ and $H = \langle T \mid W \rangle$, then

$$G_1 *_H G_2 = \langle S_1 \sqcup S_2 \mid R_1 \cup R_2 \cup \{\iota_1(t)\iota_2(t)^{-1} \mid t \in T\} \rangle,$$

where $\iota_i(t)$ is read as a word in S_i .

Proof. Call the right hand side G , and let $\varphi_i : G_i \rightarrow G$ send $s \in S_i$ to s ; this is well-defined since the relations R_i hold in G . The added relations say exactly that $\varphi_1 \circ \iota_1$ and $\varphi_2 \circ \iota_2$ agree on the generators T of H , hence agree.

Given j_1, j_2 into $\bar{G}$ as in the definition, define ψ on generators by $\psi(s) = j_i(s)$ for $s \in S_i$. The relations in R_i are satisfied because j_i is a homomorphism, and the relation $\iota_1(t)\iota_2(t)^{-1}$ is satisfied because $j_1\iota_1(t) = j_2\iota_2(t)$. So ψ is a well-defined homomorphism, and it is the unique one with $\psi \circ \varphi_i = j_i$ since the S_i generate G . $\square$

Example 25.3.13

If $H = 1$ we write $G_1 * G_2$, the **free product**. For instance $F_m * F_n \cong F_{m+n}$, and $\mathbb{Z} * \mathbb{Z} \cong F_2$.

§25.3.3 The Theorem

We first prove the ‘easy half’, which is already useful on its own.

Theorem 25.3.14

Let $X = U_1 \cup U_2$ with U_1, U_2 open, and suppose $U_1 \cap U_2$ is path connected with $x_0 \in U_1 \cap U_2$. Let $j_i : U_i \hookrightarrow X$ be the inclusions. Then $\pi_1(X, x_0)$ is generated by

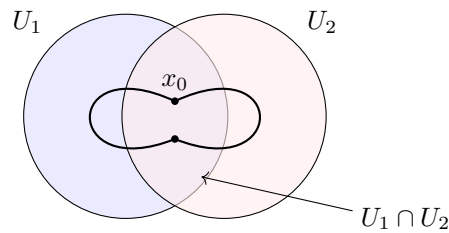
$$j_{1*}(\pi_1(U_1, x_0)) \cup j_{2*}(\pi_1(U_2, x_0)).$$

Proof. Let γ be a loop at x_0 . Then $\{\gamma^{-1}(U_1), \gamma^{-1}(U_2)\}$ is an open cover of I , so by the Lebesgue covering lemma there is n such that each $[\frac{j}{n}, \frac{j+1}{n}]$ maps into U_1 or into U_2 ; label each subinterval accordingly, choosing arbitrarily if both work. Amalgamating adjacent subintervals with the same label, we get $0 = t_0 < t_1 < \dots < t_k = 1$ and paths $\gamma_i = \gamma|_{[t_{i-1}, t_i]}$, each contained in a single $U_{c(i)}$, with consecutive labels different. In particular each $\gamma(t_i)$ for $0 < i < k$ lies in $U_1 \cap U_2$.

Since $U_1 \cap U_2$ is path connected, choose paths $\eta_i : \gamma(t_i) \leadsto x_0$ inside $U_1 \cap U_2$ for $1 \leq i \leq k-1$. Inserting $\eta_i \cdot \eta_i^{-1}$ at each break point,

$$\gamma \simeq (\gamma_1 \cdot \eta_1) \cdot (\eta_1^{-1} \cdot \gamma_2 \cdot \eta_2) \cdots (\eta_{k-1}^{-1} \cdot \gamma_k) = \delta_1 \cdot \delta_2 \cdots \delta_k,$$

where each δ_i is a loop at x_0 lying entirely inside $U_{c(i)}$. Hence $[\gamma] = [\delta_1] \cdots [\delta_k]$ is a product of elements of $j_{1*}(\pi_1(U_1, x_0)) \cup j_{2*}(\pi_1(U_2, x_0))$. $\square$



Corollary 25.3.15

If $X = U_1 \cup U_2$ with U_1, U_2 open and simply connected and $U_1 \cap U_2$ path connected and non-empty, then X is simply connected.

Proof. $\pi_1(X, x_0)$ is generated by the trivial group. $\square$

Example 25.3.16

S^n is simply connected for $n \geq 2$. Take $U^\pm = S^n \setminus \{\mp e_1\}$, the sphere with one point removed. Both are homeomorphic to $\mathbb{R}^n$ by stereographic projection, hence simply connected, and

$$U^+ \cap U^- \cong \mathbb{R}^n \setminus \{0\} \simeq S^{n-1},$$

which is path connected precisely when $n \geq 2$.

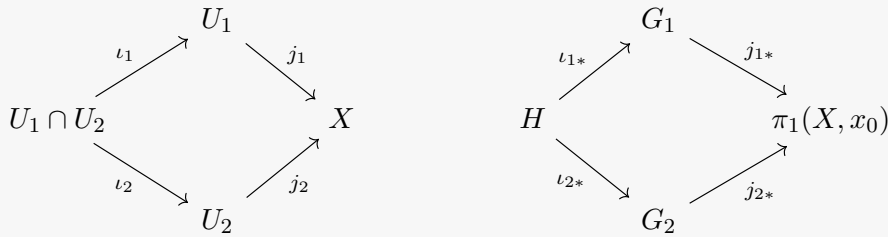
Note how badly this fails for $n = 1$: there $U^+ \cap U^-$ has two components, and indeed $\pi_1(S^1) = \mathbb{Z}$. The full theorem tells us exactly how the failure of $\pi_1(X)$ to be generated freely is controlled by $\pi_1(U_1 \cap U_2)$.

Theorem 25.3.17 (Seifert–van Kampen)

Let $X = U_1 \cup U_2$ with U_1, U_2 open and $U_1 \cap U_2$ path connected containing x_0 ; suppose also that U_1, U_2 are path connected. Write $G_i = \pi_1(U_i, x_0)$ and $H = \pi_1(U_1 \cap U_2, x_0)$, with $\iota_{i*} : H \rightarrow G_i$ induced by the inclusions. Then

$$\pi_1(X, x_0) \cong G_1 *_H G_2.$$

Proof. The square of inclusions



commutes, so applying π_1 gives a commuting square of groups. By the universal property of the amalgamated free product there is a unique homomorphism

$$\psi : G_1 *_H G_2 \rightarrow \pi_1(X, x_0)$$

with $\psi \circ \varphi_i = j_{i*}$. It is surjective by Theorem 25.3.14, since its image contains both $j_{i*}(G_i)$. The work is in showing that it is injective.

So suppose $g \in G_1 *_H G_2$ has $\psi(g) = 1$. Write $g = \varphi_{c(1)}(g_1) \cdots \varphi_{c(k)}(g_k)$ as a product of images of elements $g_i \in G_{c(i)}$, and represent each g_i by a loop δ_i in $U_{c(i)}$ based at x_0 . Then $\gamma = \delta_1 \cdots \delta_k$ is a loop in X with $[\gamma] = 1$, so there is a homotopy $F : I \times I \rightarrow X$ with $F(s, 0) = \gamma(s)$, $F(s, 1) = x_0$ and $F(0, t) = F(1, t) = x_0$.

Apply the Lebesgue covering lemma to the open cover $\{F^{-1}(U_1), F^{-1}(U_2)\}$ of I^2 to get n such that each small square $A = [\frac{i}{n}, \frac{i+1}{n}] \times [\frac{j}{n}, \frac{j+1}{n}]$ has $F(A)$ inside some

$U_{c(A)}$; refining n further we may assume the break points of γ occur at multiples of $1/n$. For each vertex v of the grid choose a path η_v from x_0 to $F(v)$, lying in $U_1 \cap U_2$ if $F(v) \in U_1 \cap U_2$, and in U_c if $F(v)$ only lies in U_c ; take η_v constant when $F(v) = x_0$. For each edge e of the grid with endpoints v, w , we get a loop

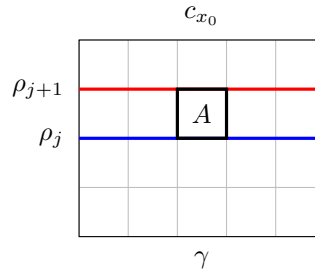
$$\lambda_e = \eta_v \cdot (F \circ e) \cdot \eta_w^{-1}$$

based at x_0 , lying in whichever U_c contains the squares adjacent to e ; if both indices are available, λ_e lies in $U_1 \cap U_2$ and so determines an element of H , whose images in G_1 and G_2 are identified in $G_1 *_H G_2$. In every case, then, λ_e determines a well-defined element $[\lambda_e] \in G_1 *_H G_2$.

Now let $\rho_j \in G_1 *_H G_2$ be the product of the $[\lambda_e]$ along the horizontal grid line at height j/n , read left to right. Two facts finish the proof.

1. $\rho_0 = g$ and $\rho_n = 1$. The bottom row is γ , cut at the break points, and since the η_v for v on the bottom row are constant we get exactly $\varphi_{c(1)}(g_1) \cdots \varphi_{c(k)}(g_k) = g$. The top row is constant at x_0 , so all its edge loops are trivial.
2. $\rho_j = \rho_{j+1}$ for each j . Consider a single square A between the two rows, with $F(A) \subseteq U_{c(A)}$. Its four edge loops λ_e all lie in $U_{c(A)}$, and $F|_A$ shows that the composite loop around ∂A is null-homotopic in $U_{c(A)}$, because A is convex. Hence in $G_{c(A)}$, and therefore in $G_1 *_H G_2$, the product of the bottom and right edge classes equals the product of the left and top edge classes. Multiplying these relations across the row, and cancelling the interior vertical edges in pairs, converts ρ_j into ρ_{j+1} .

Therefore $g = \rho_0 = \rho_n = 1$, and ψ is injective. $\square$



§25.3.4 Applications

The most common way to apply the theorem is to a space built by gluing a disc onto another space along its boundary circle.

Definition 25.3.18

Let $f : S^1 \rightarrow X$ be a map. The space obtained by **attaching a 2-cell** along f is

$$X \cup_f D^2 = (X \sqcup D^2) / \sim, \quad z \sim f(z) \text{ for } z \in S^1.$$

Proposition 25.3.19

Let X be path connected with $x_0 = f(1)$, and suppose $\pi_1(X, x_0) = \langle S \mid R \rangle$. Then

$$\pi_1(X \cup_f D^2, x_0) \cong \pi_1(X, x_0) / \langle\langle f_*(1) \rangle\rangle = \langle S \mid R \cup \{w_f\} \rangle,$$

where w_f is any word representing the class $f_*(1) \in \pi_1(X, x_0)$ of the attaching loop.

Proof. Let $\pi : X \sqcup D^2 \rightarrow X \cup_f D^2$ be the quotient map and set

$$U_1 = \pi(X \sqcup (D^2 \setminus \{0\})), \quad U_2 = \pi((D^2)^\circ).$$

These are open, $U_1 \cup U_2 = X \cup_f D^2$, and $U_1 \cap U_2 = \pi((D^2)^\circ \setminus \{0\})$ is path connected. Now $D^2 \setminus \{0\}$ strong deformation retracts onto S^1 , so U_1 strong deformation retracts onto X , giving $\pi_1(U_1) \cong \pi_1(X, x_0)$. Also $\pi_1(U_2) = 1$ as $(D^2)^\circ$ is convex, and $U_1 \cap U_2 \simeq S^1$, so $H = \pi_1(U_1 \cap U_2) \cong \mathbb{Z} = \langle t \rangle$, generated by a small loop around the puncture. Under ι_{1*} this maps to the class of the attaching loop $f_*(1)$, and under ι_{2*} to 1.

(Strictly, the basepoint must be moved into $U_1 \cap U_2$; we conjugate by a path to x_0 , which is harmless.) By Theorem 25.3.17 and Proposition 25.3.12,

$$\pi_1(X \cup_f D^2) \cong \langle S \mid R \rangle *_{\langle t \rangle} 1 = \langle S, t \mid R \cup \{t^{-1}w_f, t\} \rangle \cong \langle S \mid R \cup \{w_f\} \rangle. \quad \square$$

In words: attaching a disc kills the class of the loop you attach it along, and nothing else. We can now compute the fundamental groups of all the closed surfaces, because every closed surface is a polygon with its edges identified in pairs — which is to say, a wedge of circles with a single disc attached.

Example 25.3.20 (The Torus)

Take the square I^2 and identify its edges according to the word $aba^{-1}b^{-1}$, meaning that we glue the left edge to the right edge and the bottom to the top, both without a flip. The result is T^2 . Its 1-skeleton — the image of ∂I^2 — is $S^1 \vee S^1$, since all four corners are identified to a single point, and the interior of the square is a 2-cell attached along the loop $a \cdot b \cdot a^{-1} \cdot b^{-1}$. So by Theorem 25.3.6 and Proposition 25.3.19,

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2,$$

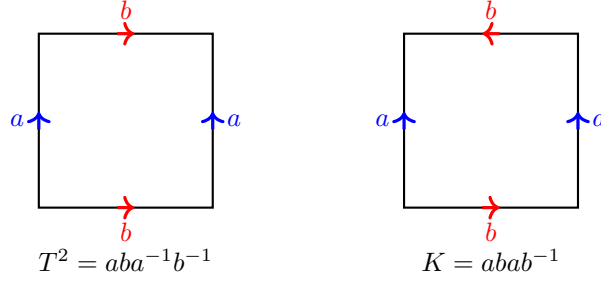
agreeing with the computation in Example 25.2.23.

Example 25.3.21 (The Klein Bottle)

Identify the edges of I^2 according to $abab^{-1}$: the left and right edges are glued as before, but the bottom and top are glued with a flip. The result is the Klein bottle K , and

$$\pi_1(K) \cong \langle a, b \mid abab^{-1} \rangle.$$

The relation rearranges to $bab^{-1} = a^{-1}$, so this group is the semidirect product $\mathbb{Z} \rtimes \mathbb{Z}$ in which the second factor acts on the first by inversion. In particular it is not abelian. Its abelianisation is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$, whereas $\pi_1(T^2)^{\text{ab}} = \mathbb{Z}^2$ is torsion-free, so $K \not\cong T^2$ and in particular the Klein bottle is not homeomorphic to the torus.

**Example 25.3.22 (The Projective Plane)**

Identifying the edges of I^2 by $abab$, equivalently identifying antipodal points on the boundary of a disc, gives $\mathbb{RP}^2$. The 1-skeleton is a single circle $\langle a \rangle$ and the disc is attached along a^2 , so

$$\pi_1(\mathbb{RP}^2) \cong \langle a \mid a^2 \rangle \cong \mathbb{Z}/2\mathbb{Z},$$

as we already knew from the covering $S^2 \rightarrow \mathbb{RP}^2$.

Example 25.3.23 (Surfaces of Genus g)

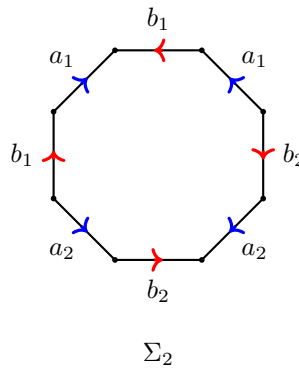
Let Σ_g be the closed orientable surface of genus g , obtained from a $4g$ -gon by identifying its edges according to the word

$$a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

All $4g$ vertices are identified to a point, so the 1-skeleton is $\bigvee_{i=1}^{2g} S^1$, and hence

$$\pi_1(\Sigma_g) \cong \left\langle a_1, b_1, \dots, a_g, b_g \mid \prod_{i=1}^g a_i b_i a_i^{-1} b_i^{-1} \right\rangle.$$

Abelianising gives $\mathbb{Z}^{2g}$, so the Σ_g are pairwise non-homotopy-equivalent.



Alternatively, we can cut a genus 2 surface along a circle into two one-holed tori, which gives $\pi_1(\Sigma_2) \cong \langle a_1, b_1 \rangle *_{\mathbb{Z}} \langle a_2, b_2 \rangle$, with the amalgamation identifying the two commutators.

Finally, here is a taste of where these techniques lead. A **knot** is an embedding $S^1 \hookrightarrow S^3$, and the basic problem is to tell knots apart. Since all knots are homeomorphic to S^1 as spaces, we must look instead at the complement.

Example 25.3.24 (Knot Complements)

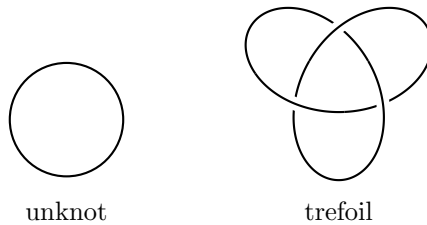
Let $U \subseteq S^3$ be the **unknot**, a round circle. Then $S^3 \setminus U$ deformation retracts onto a circle (spin around the axis of the unknot), so

$$\pi_1(S^3 \setminus U) \cong \mathbb{Z}.$$

Now let T be the **trefoil**. Cutting S^3 into the two solid tori on either side of the standard torus, on which T can be drawn as a curve of slope $3/2$, and applying Seifert–van Kampen gives

$$\pi_1(S^3 \setminus T) \cong \langle x, y \mid x^2 = y^3 \rangle.$$

This group surjects onto S_3 : send $x \mapsto (1\ 2)$ and $y \mapsto (1\ 2\ 3)$, and check $x^2 = e = y^3$. So $\pi_1(S^3 \setminus T)$ is non-abelian, and in particular not $\mathbb{Z}$. Hence the trefoil is genuinely knotted.

**§25.4 Simplicial Complexes**

We have squeezed a great deal out of the fundamental group, but it has a fundamental limitation: it only sees one-dimensional phenomena. We showed that S^n is simply connected for all $n \geq 2$, so π_1 cannot tell S^2 from S^3 . To detect higher-dimensional holes we need a new invariant, and to build it we will first cut our spaces into triangles.

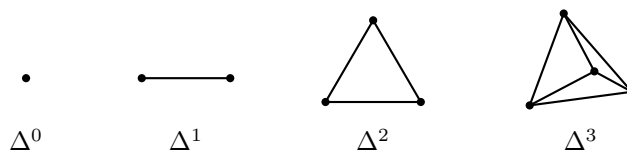
§25.4.1 Simplices**Definition 25.4.1 (Simplex)**

The **standard n -simplex** is

$$\Delta^n = \left\{ (x_0, \dots, x_n) \in \mathbb{R}^{n+1} \mid x_i \geq 0, \sum_{i=0}^n x_i = 1 \right\}$$

with the subspace topology.

So Δ^0 is a point, Δ^1 is homeomorphic to I , Δ^2 is a triangle and Δ^3 is a tetrahedron. Each Δ^n is closed and bounded in $\mathbb{R}^{n+1}$, hence compact, and the standard basis vectors $e_0, \dots, e_n$ are its **vertices**. By radial projection from the barycentre, $\Delta^n \cong D^n$.



Definition 25.4.2 (Face)

For $I \subseteq \{0, \dots, n\}$, the **I th face** of Δ^n is

$$e_I = \{x \in \Delta^n \mid x_i = 0 \text{ for } i \notin I\},$$

and $F(\Delta^n) = \{e_I \mid I \subseteq \{0, \dots, n\}\}$ is the set of faces.

We abbreviate $I = \{i_0 < \dots < i_k\}$ by the string $i_0 i_1 \dots i_k$, so for instance e_{02} is the edge of Δ^2 joining e_0 and e_2 . Note that $e_{\{i\}} = e_i$, that $\Delta^n = e_{0\dots n}$, that $e_\emptyset = \emptyset$, that $e_I \cong \Delta^{|I|-1}$, and crucially that

$$e_I \subseteq e_J \iff I \subseteq J, \quad e_I \cap e_J = e_{I \cap J}.$$

Definition 25.4.3

A map $|f| : \Delta^n \rightarrow \mathbb{R}^N$ is **affine linear** if it is the restriction of a linear map $\mathbb{R}^{n+1} \rightarrow \mathbb{R}^N$, equivalently if $|f|(\sum x_i e_i) = \sum x_i |f|(e_i)$. An affine linear map $|f| : \Delta^n \rightarrow \Delta^m$ is **simplicial** if it takes vertices to vertices, so that it is determined by a function $\hat{f} : \{0, \dots, n\} \rightarrow \{0, \dots, m\}$ via $|f|(e_i) = e_{\hat{f}(i)}$.

Affine linear maps are continuous and determined by their values on vertices, and a simplicial map satisfies $|f|(e_I) = e_{\hat{f}(I)}$.

Definition 25.4.4 (Affine Independence)

Vectors $v_0, \dots, v_n \in \mathbb{R}^N$ are **affine linearly independent** if whenever $\sum t_i v_i = 0$ with $\sum t_i = 0$, all $t_i = 0$. Equivalently, $v_1 - v_0, \dots, v_n - v_0$ are linearly independent, or the affine linear map $|f| : \Delta^n \rightarrow \mathbb{R}^N$ with $|f|(e_i) = v_i$ is injective.

In that case we write

$$[v_0, \dots, v_n] = \text{im } |f| = \left\{ \sum x_i v_i \mid x_i \geq 0, \sum x_i = 1 \right\}$$

and call it a **Euclidean simplex** of dimension n .

Since Δ^n is compact and $\mathbb{R}^N$ Hausdorff, $|f| : \Delta^n \rightarrow [v_0, \dots, v_n]$ is a homeomorphism. The following lemma says that a Euclidean simplex remembers its vertices, so ‘the vertices of σ ’ is well-defined; we write $V(\sigma)$ for this set.

Lemma 25.4.5

For $X \subseteq \mathbb{R}^N$ let $Z(X)$ be the set of $x \in X$ such that whenever $x = \sum t_i x_i$ with $x_i \in X$, $t_i > 0$ and $\sum t_i = 1$, we have $x_i = x$ for some i . Then $Z([v_0, \dots, v_n]) = \{v_0, \dots, v_n\}$.

Proof. If x is not a vertex then $x = \sum x_i v_i$ with at least two x_i positive, and writing x as a convex combination of the v_i involved exhibits $x \notin Z$. Conversely suppose $v_k = \sum_i t_i x_i$ with $t_i > 0$, $\sum t_i = 1$, and write $x_i = \sum_j s_{ij} v_j$ with $s_{ij} \geq 0$ and

$\sum_j s_{ij} = 1$. Then

$$v_k = \sum_j \left(\sum_i t_i s_{ij} \right) v_j,$$

and $\sum_j \sum_i t_i s_{ij} = 1$, so affine independence forces $\sum_i t_i s_{ij} = 0$ for $j \neq k$. As $t_i > 0$ and $s_{ij} \geq 0$, we get $s_{ij} = 0$ for all i and $j \neq k$, so every $x_i = v_k$. $\square$

§25.4.2 Simplicial Complexes

There are two ways to package a collection of simplices glued along faces — one combinatorial, one geometric — and we shall need both.

Definition 25.4.6 (Abstract Simplicial Complex)

An **abstract simplicial complex** in Δ^n is a subset $K \subseteq F(\Delta^n)$ which is downward closed: if $e_J \in K$ and $I \subseteq J$ then $e_I \in K$.

Its **polyhedron** is $|K| = \bigcup_{e_I \in K} e_I \subseteq \Delta^n$, with the subspace topology, and its **r -skeleton** is $K_r = \{e_I \in K \mid |I| \leq r+1\}$.

Abstract simplicial complexes carry no topology themselves; the topology lives on the polyhedron, which is compact and Hausdorff. We have a filtration $\{e_\emptyset\} = K_{-1} \subseteq K_0 \subseteq \cdots \subseteq K_n = K$, and $\dim K$ is the largest r with $K_r \neq K_{r-1}$.

Example 25.4.7

$\Delta^n = F(\Delta^n)$ is an abstract simplicial complex with $|\Delta^n| = \Delta^n \cong D^n$.

Example 25.4.8

$S^{n-1} = \Delta_{n-1}^n = \{e_I \mid I \subsetneq \{0, \dots, n\}\}$, obtained by throwing away the top face, is an abstract simplicial complex with polyhedron $\partial\Delta^n \cong S^{n-1}$: the boundary of the tetrahedron is a sphere.

Definition 25.4.9

A **simplicial map** $f : K \rightarrow L$ between abstract simplicial complexes in Δ^n and Δ^m is a map induced by some $\hat{f} : \{0, \dots, n\} \rightarrow \{0, \dots, m\}$ with the property that $e_I \in K$ implies $e_{\hat{f}(I)} \in L$; then $f(e_I) = e_{\hat{f}(I)}$. It is a **simplicial isomorphism** if it is a bijection, in which case $|f| : |K| \rightarrow |L|$ is a homeomorphism.

For the geometric version we work inside $\mathbb{R}^N$. Write $\mathcal{S}(\mathbb{R}^N)$ for the set of Euclidean simplices in $\mathbb{R}^N$.

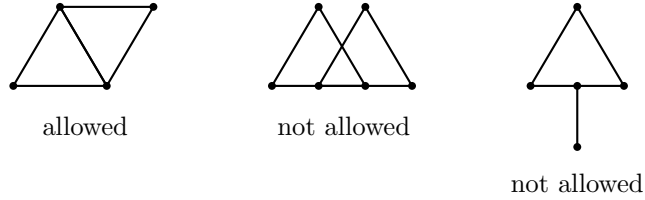
Definition 25.4.10 (Euclidean Simplicial Complex)

A finite set $K \subseteq \mathcal{S}(\mathbb{R}^N)$ is a **Euclidean simplicial complex** if

1. $\sigma \in K$ and τ a face of σ implies $\tau \in K$;
2. $\sigma_1, \sigma_2 \in K$ implies $\sigma_1 \cap \sigma_2$ is a face of each (and so lies in K).

Its polyhedron is $|K| = \bigcup_{\sigma \in K} \sigma \subseteq \mathbb{R}^N$ and its r -skeleton is $K_r = \{\sigma \in K \mid \dim \sigma \leq r\}$.

Condition (ii) is the one that does the work: it forbids the two configurations on the right below, in which simplices meet in something which is not a common face.



The two notions are equivalent, in the following sense.

Definition 25.4.11

Let K' be an abstract simplicial complex in Δ^n and $|\varphi| : \Delta^n \rightarrow \mathbb{R}^N$ affine linear with $|\varphi|$ injective on $|K'|$. Then $\varphi(K') = \{|\varphi|(e_I) \mid e_I \in K'\}$ is called a **realisation** of K' .

Proposition 25.4.12

A realisation $\varphi(K')$ is a Euclidean simplicial complex, and $|\varphi| : |K'| \rightarrow |\varphi(K')|$ is a homeomorphism. Conversely, every Euclidean simplicial complex is a realisation of some abstract simplicial complex, uniquely up to simplicial isomorphism.

Proof. For the first part, downward closure is inherited from K' , and for intersections, injectivity gives

$$|\varphi|(e_{I_1}) \cap |\varphi|(e_{I_2}) = |\varphi|(e_{I_1 \cap I_2}) = |\varphi|(e_{I_1 \cap I_2}),$$

which is a common face. That $|\varphi|$ is a homeomorphism follows from compactness of $|K'|$.

Conversely, let K be a Euclidean simplicial complex, and write its vertex set as $V(K) = \{v_0, \dots, v_n\}$. Define

$$K' = \{e_{i_0 \dots i_k} \in F(\Delta^n) \mid [v_{i_0}, \dots, v_{i_k}] \in K\}$$

and let $|\varphi|(e_i) = v_i$. We must check $|\varphi|$ is injective on $|K'|$. It is injective on each $e_I \in K'$, since the corresponding v_i are affine independent. Suppose $|\varphi|(p) = |\varphi|(q) = x$ with $p \in e_I$, $q \in e_J$, both in K' . Then x lies in $|\varphi|(e_I) \cap |\varphi|(e_J)$, which is a common face $|\varphi|(e_{I'})$ with $I' \subseteq I \cap J$; by injectivity on e_I and on e_J we get $p, q \in e_{I'}$, and then $p = q$ by injectivity on $e_{I'}$. $\square$

Definition 25.4.13

For a Euclidean simplex σ of dimension n , its **boundary complex** is $\partial\sigma = F(\sigma) \setminus \{\sigma\}$, with polyhedron $|\partial\sigma| \cong S^{n-1}$. The **open simplex** is $\sigma^\circ = \sigma \setminus |\partial\sigma|$.

Lemma 25.4.14

Let K be a simplicial complex and $x \in |K|$. Then there is a unique $\sigma \in K$ with $x \in \sigma^\circ$.

Proof. Work in the abstract picture, where $x \in \Delta^n$. Set $I = \{i \mid x_i \neq 0\}$. Then $x \in e_I^\circ$, and $x \in e_J$ if and only if $I \subseteq J$, which gives both existence and uniqueness; and $e_I \in K$ by downward closure, since $x \in e_J$ for some $e_J \in K$. $\square$

So the open simplices of K partition $|K|$.

§25.4.3 Barycentric Subdivision

Simplicial maps are very rigid: there are only finitely many of them between two given complexes, so most continuous maps $|K| \rightarrow |L|$ are not simplicial. The remedy is to chop K into finer and finer pieces, and the standard way to do this is barycentric subdivision. We first need cones.

Definition 25.4.15

Let $X \subseteq \mathbb{R}^N$ and $p \in \mathbb{R}^N$. We say p is **independent** of X if for every $x \in X$ the ray from p through x meets X only at x . In that case the **cone** is

$$C_p(X) = \{tp + (1-t)x \mid t \in [0, 1], x \in X\},$$

and for a Euclidean simplicial complex K with p independent of $|K|$ we set

$$C_p(K) = K \cup \{[v_0, \dots, v_j, p] \mid [v_0, \dots, v_j] \in K\}.$$

Lemma 25.4.16

If p is independent of $|K|$ then $C_p(K)$ is a Euclidean simplicial complex with $|C_p(K)| = C_p(|K|)$, and every $\sigma \in C_p(K)$ satisfies $\sigma \cap |K| \in K$.

Definition 25.4.17 (Barycentre)

The **barycentre** of $\sigma = [v_0, \dots, v_n]$ is $b_\sigma = \frac{1}{n+1} \sum_{i=0}^n v_i$.

One checks that b_σ is independent of $|\partial\sigma|$ and that $C_{b_\sigma}(|\partial\sigma|) = \sigma$. We may now define barycentric subdivision by a simultaneous induction on dimension: for a simplex σ we set

$$\beta(\sigma) = C_{b_\sigma}(\mathcal{B}(\partial\sigma)),$$

and for a Euclidean simplicial complex K we set

$$\mathcal{B}(K) = \bigcup_{\sigma \in K} \beta(\sigma),$$

starting from $\beta(\emptyset) = \{\emptyset\}$ and $\mathcal{B}(\{\emptyset\}) = \{\emptyset\}$. In words: subdivide the boundary, then cone off from the barycentre.

Theorem 25.4.18

For every n and every n -dimensional Euclidean simplex σ and n -dimensional Euclidean simplicial complex K :

1. $\beta(\sigma)$ is an n -dimensional Euclidean simplicial complex with $|\beta(\sigma)| = \sigma$, and if τ is a face of σ and $\sigma_1 \in \beta(\sigma)$ then $\sigma_1 \cap \tau \in \beta(\tau)$;
2. $\mathcal{B}(K)$ is an n -dimensional Euclidean simplicial complex with $|\mathcal{B}(K)| = |K|$.

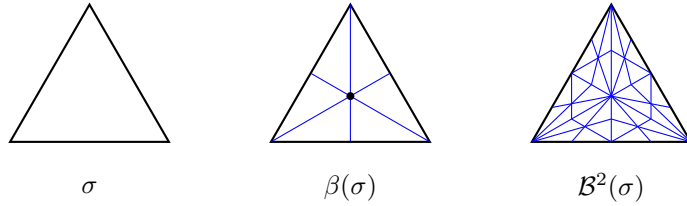
Proof. Both hold trivially in dimension -1 . Assume both in dimensions $< n$.

(i) Now $\partial\sigma$ has dimension $n-1$, so $\mathcal{B}(\partial\sigma)$ is a Euclidean simplicial complex with polyhedron $|\partial\sigma|$ by the inductive hypothesis (ii). As b_σ is independent of $|\partial\sigma|$, Lemma 25.4.16 says $\beta(\sigma) = C_{b_\sigma}(\mathcal{B}(\partial\sigma))$ is a Euclidean simplicial complex with polyhedron $C_{b_\sigma}(|\partial\sigma|) = \sigma$. The last claim also follows from Lemma 25.4.16.

(ii) $\mathcal{B}(K)$ is finite, and downward closed since each $\beta(\sigma)$ is. For intersections, let $\sigma_i \in \beta(\sigma'_i)$ with $\sigma'_i \in K$. Then $\sigma_1 \cap \sigma_2 \subseteq \sigma'_1 \cap \sigma'_2 = \tau \in K$. By (i), $\sigma_i \cap \tau \in \beta(\tau)$, and $\beta(\tau)$ is a simplicial complex, so

$$\sigma_1 \cap \sigma_2 = (\sigma_1 \cap \tau) \cap (\sigma_2 \cap \tau) \in \beta(\tau) \subseteq \mathcal{B}(K).$$

Finally $|\mathcal{B}(K)| = \bigcup_{\sigma \in K} |\beta(\sigma)| = \bigcup_{\sigma \in K} \sigma = |K|$. □



The reason for subdividing is that the pieces get smaller, and we can quantify this.

Definition 25.4.19 (Mesh)

The **mesh** of a Euclidean simplex σ is $\mu(\sigma) = \max_{v_i, v_j \in V(\sigma)} \|v_i - v_j\|$, and $\mu(K) = \max_{\sigma \in K} \mu(\sigma)$.

Lemma 25.4.20

If σ is a Euclidean simplex and $x, v \in \sigma$ then $\|v - x\| \leq \max_{v_i \in V(\sigma)} \|v - v_i\|$. Consequently $\mu(\sigma) = \max_{x, v \in \sigma} \|v - x\|$.

Proof. Write $x = \sum x_i v_i$ with $x_i \geq 0$ and $\sum x_i = 1$; then $v = \sum x_i v_i$, so

$$\|v - x\| = \left\| \sum x_i (v - v_i) \right\| \leq \sum x_i \|v - v_i\| \leq \max_i \|v - v_i\|. \quad \square$$

Corollary 25.4.21

If $\dim \sigma = n$ then $\mu(\beta(\sigma)) \leq \frac{n}{n+1} \mu(\sigma)$, and if $\dim K = n$ then $\mu(\mathcal{B}(K)) \leq \frac{n}{n+1} \mu(K)$.

Hence $\mu(\mathcal{B}^r(K)) \rightarrow 0$ as $r \rightarrow \infty$.

Proof. First, for $\sigma = [v_0, \dots, v_n]$ and any $v \in \sigma$, applying Lemma 25.4.20,

$$\|b_\sigma - v\| \leq \max_i \|b_\sigma - v_i\| = \max_i \frac{1}{n+1} \left\| \sum_{j \neq i} (v_j - v_i) \right\| \leq \frac{n}{n+1} \mu(\sigma).$$

Now let $\tau \in \beta(\sigma)$. If $\tau \in \mathcal{B}(\partial\sigma)$ then by induction $\mu(\tau) \leq \frac{n-1}{n} \mu(\partial\sigma) \leq \frac{n}{n+1} \mu(\sigma)$. Otherwise $\tau = [w_0, \dots, w_k, b_\sigma]$ with $[w_0, \dots, w_k] \in \mathcal{B}(\partial\sigma)$, and the distance between any two of its vertices is at most $\max\{\frac{n-1}{n} \mu(\sigma), \frac{n}{n+1} \mu(\sigma)\} = \frac{n}{n+1} \mu(\sigma)$. The statement for complexes follows by taking maxima, and iterating gives $\mu(\mathcal{B}^r(K)) \leq (\frac{n}{n+1})^r \mu(K) \rightarrow 0$. $\square$

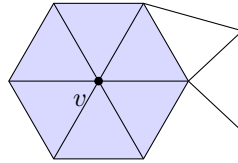
§25.4.4 Simplicial Approximation

We can now say what it means for a simplicial map to approximate a continuous one.

Definition 25.4.22 (Star)

Let K be a simplicial complex and $v \in V(K)$. The **star** of v is

$$\text{St}_K(v) = \bigcup_{\{\sigma \in K \mid v \in V(\sigma)\}} \sigma^\circ.$$



$\text{St}_K(v)$ is the shaded region, without its outer boundary

Lemma 25.4.23

Let K be a simplicial complex.

1. If $x \in \sigma^\circ$ for $\sigma \in K$, then $x \in \text{St}_K(v)$ if and only if $v \in V(\sigma)$.
2. $\text{St}_K(v) = |K| \setminus \bigcup \{\sigma \in K \mid v \notin V(\sigma)\}$, so it is open in $|K|$.
3. $\{\text{St}_K(v) \mid v \in V(K)\}$ is an open cover of $|K|$.

Proof. (i) is immediate from Lemma 25.4.14. For (ii), by (i) the complement of $\text{St}_K(v)$ consists of the open simplices σ° with $v \notin V(\sigma)$; and the union of these equals the union of the corresponding closed simplices, since faces of a simplex not containing v also do not contain v . That union is a finite union of compact sets, hence closed. (iii) follows since every x lies in some σ° , and σ has at least one vertex. $\square$

Definition 25.4.24 (Simplicial Approximation)

Let K, L be simplicial complexes and $f : |K| \rightarrow |L|$ continuous. A function $\hat{g} : V(K) \rightarrow V(L)$ is a **simplicial approximation** to f if

$$f(\text{St}_K(v)) \subseteq \text{St}_L(\hat{g}(v)) \quad \text{for all } v \in V(K).$$

Theorem 25.4.25

If $\hat{g}$ is a simplicial approximation to $f : |K| \rightarrow |L|$, then $\hat{g}$ determines a simplicial map $g : K \rightarrow L$, and $|g| \simeq f$.

Proof. Let $\sigma \in K$ and pick $x \in \sigma^\circ$; say $f(x) \in \tau^\circ$ with $\tau \in L$. Since $x \in \text{St}_K(v)$ for every $v \in V(\sigma)$, we get

$$f(x) \in \bigcap_{v \in V(\sigma)} f(\text{St}_K(v)) \subseteq \bigcap_{v \in V(\sigma)} \text{St}_L(\hat{g}(v)),$$

and by Lemma 25.4.23(i) this says $\hat{g}(v) \in V(\tau)$ for every $v \in V(\sigma)$. So the affine linear extension $|g|$ maps σ into τ , and $|g|(\sigma)$ is a face of τ , hence lies in L . So g is simplicial.

For the homotopy, define $H : |K| \times I \rightarrow \mathbb{R}^M$ by $H(x, t) = t|g|(x) + (1 - t)f(x)$. This is continuous, and we claim it lands in $|L|$. With x, σ, τ as above, $|g|(x) = \sum_i x_i |g|(v_i) \in \tau$ since each $|g|(v_i) \in V(\tau)$, and $f(x) \in \tau$; as τ is convex, $H(x, t) \in \tau \subseteq |L|$. So H is a homotopy $f \simeq |g|$ in $|L|$. $\square$

Theorem 25.4.26 (Simplicial Approximation Theorem)

Let K, L be simplicial complexes and $f : |K| \rightarrow |L|$ continuous. Then there exist $r \geq 0$ and a simplicial map $g : \mathcal{B}^r(K) \rightarrow L$ with $|g| \simeq f$.

Proof. By Lemma 25.4.23, $\{\text{St}_L(w) \mid w \in V(L)\}$ is an open cover of $|L|$, so

$$\{f^{-1}(\text{St}_L(w)) \mid w \in V(L)\}$$

is an open cover of $|K|$. Now $|K|$ is a compact metric space, so the Lebesgue covering lemma gives $\delta > 0$ such that each $B_\delta(x)$ lies in some $f^{-1}(\text{St}_L(w_x))$.

By Corollary 25.4.21, choose r with $\mu(\mathcal{B}^r(K)) < \delta$ and put $K' = \mathcal{B}^r(K)$, noting $|K'| = |K|$. If $x \in V(K')$ and $\sigma \in K'$ has $x \in V(\sigma)$ then $\sigma \subseteq B_\delta(x)$, since every point of σ is within $\mu(K') < \delta$ of x . Hence

$$\text{St}_{K'}(x) \subseteq \bigcup_{\{\sigma \mid x \in V(\sigma)\}} \sigma \subseteq B_\delta(x),$$

so $f(\text{St}_{K'}(x)) \subseteq f(B_\delta(x)) \subseteq \text{St}_L(w_x)$. Therefore $\hat{g}(x) = w_x$ defines a simplicial approximation to f , and Theorem 25.4.25 does the rest. $\square$

This is the technical engine of everything to come. Here is a first payoff, which already does something the fundamental group could not.

Corollary 25.4.27

Let K, L be simplicial complexes with $\dim K < \dim L$, and let $f : |K| \rightarrow |L|$ be continuous. Then $f \simeq |g|$ for some $|g|$ which is not surjective.

Proof. Take a simplicial $g : \mathcal{B}^r(K) \rightarrow L$ with $|g| \simeq f$. Simplicial maps do not raise dimension, so $|g|$ has image in $|L_k|$ where $k = \dim \mathcal{B}^r(K) = \dim K < \dim L$. As $|L_k| \subsetneq |L|$, this is not surjective. $\square$

Theorem 25.4.28

If $k < n$, every map $S^k \rightarrow S^n$ is null-homotopic.

Proof. We have $S^k \cong |\mathbb{S}^k|$ and $S^n \cong |\mathbb{S}^n|$ with $\dim \mathbb{S}^k = k < n = \dim \mathbb{S}^n$, so by Corollary 25.4.27, $f \simeq |g|$ where $|g|$ misses some point $p \in S^n$. But then $|g|$ factors through $S^n \setminus \{p\} \cong \mathbb{R}^n$, which is contractible, so $|g|$ is null-homotopic; hence so is f . $\square$

That gives one half of the statement $S^n \simeq S^m \implies n = m$: any map $S^k \rightarrow S^n$ with $k < n$ is null-homotopic, whereas we would need id_{S^n} not to be. To prove the latter we need homology.

§25.5 Simplicial Homology

Here is the idea. Having chopped a space into simplices, we form for each k the free abelian group on the k -simplices, and a boundary homomorphism d taking a simplex to the (signed) sum of its faces. A k -dimensional hole should be a k -cycle — something with no boundary — which is not itself the boundary of anything. Measuring how badly ‘no boundary’ fails to imply ‘is a boundary’ is exactly taking a quotient group, and that quotient is homology.

§25.5.1 Chain Complexes

Definition 25.5.1 (Chain Complex)

A **chain complex** $(C_\bullet, d)$ consists of

1. free abelian groups C_i for $i \in \mathbb{Z}$, all but finitely many of which are zero;
2. homomorphisms $d_i : C_i \rightarrow C_{i-1}$;
3. such that $d_{i-1} \circ d_i = 0$ for all i .

$$\cdots \rightarrow C_{i+1} \xrightarrow{d_{i+1}} C_i \xrightarrow{d_i} C_{i-1} \rightarrow \cdots$$

We write $C_\bullet = \bigoplus_i C_i$ and $d = \bigoplus_i d_i$, so that condition (iii) says exactly $d^2 = 0$.

Definition 25.5.2

Let $x \in C_k$. We say x is a **cycle** if $dx = 0$, and a **boundary** if $x = dy$ for some

$$y \in C_{k+1}.$$

Since $d^2 = 0$, every boundary is a cycle: $\text{im } d_{k+1} \subseteq \ker d_k$. The whole subject consists of measuring the failure of the converse.

Now we build the chain complex of a simplicial complex. Let K be an abstract simplicial complex in Δ^n .

Definition 25.5.3 (Simplicial Chain Complex)

The **reduced chain group** is the free abelian group

$$\widetilde{C}_k(K) = \langle e_I \mid e_I \in K, |I| = k + 1 \rangle,$$

with boundary map defined on basis elements by

$$d(e_I) = \sum_{j=0}^k (-1)^j e_{I_j}, \quad I_j = I \setminus \{i_j\},$$

where $I = i_0 i_1 \cdots i_k$ with $i_0 < \cdots < i_k$.

Note that $\widetilde{C}_{-1}(K) = \langle e_\emptyset \rangle \cong \mathbb{Z}$, generated by the empty face, and $d(e_i) = e_\emptyset$ for each vertex. The **unreduced** chain complex $C_\bullet(K)$ is obtained by throwing this away: $C_k(K) = \widetilde{C}_k(K)$ for $k \geq 0$ and $C_{-1}(K) = 0$, with the same d .

The alternating signs are what makes everything work.

Proposition 25.5.4

$$d^2 = 0.$$

Proof. It suffices to check this on basis elements. Expanding,

$$d^2(e_I) = \sum_{j < j'} c_{jj'} e_{I_{jj'}},$$

where $I_{jj'} = I \setminus \{i_j, i_{j'}\}$. There are two ways to arrive at this term: deleting i_j first and then $i_{j'}$, which now sits in position $j' - 1$, contributing $(-1)^j (-1)^{j'-1}$; or deleting $i_{j'}$ first and then i_j , still in position j , contributing $(-1)^{j'} (-1)^j$. So

$$c_{jj'} = (-1)^{j+j'-1} + (-1)^{j+j'} = 0. \quad \square$$

Example 25.5.5

Take $K = \Delta^2$, the full triangle. Then

$$\widetilde{C}_2 = \langle e_{012} \rangle, \quad \widetilde{C}_1 = \langle e_{01}, e_{02}, e_{12} \rangle, \quad \widetilde{C}_0 = \langle e_0, e_1, e_2 \rangle, \quad \widetilde{C}_{-1} = \langle e_\emptyset \rangle,$$

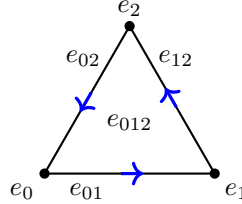
and all other $\widetilde{C}_i = 0$. The boundary maps are

$$d(e_{012}) = e_{12} - e_{02} + e_{01}, \quad d(e_{01}) = e_1 - e_0, \quad d(e_{02}) = e_2 - e_0, \quad d(e_{12}) = e_2 - e_1,$$

and $d(e_i) = e_\emptyset$. As a check,

$$d^2(e_{012}) = (e_2 - e_1) - (e_2 - e_0) + (e_1 - e_0) = 0.$$

The signs encode orientations: e_{01} is the edge from e_0 to e_1 , and $d(e_{012}) = e_{01} - e_{02} + e_{12}$ traverses the boundary of the triangle consistently anticlockwise.



§25.5.2 Homology Groups

Definition 25.5.6 (Homology)

Let $(C_\bullet, d)$ be a chain complex. Its **k th homology group** is

$$H_k(C_\bullet) = \frac{\ker d_k}{\operatorname{im} d_{k+1}}.$$

For a simplicial complex K we set $H_k(K) = H_k(C_\bullet(K))$ and $\widetilde{H}_k(K) = H_k(\widetilde{C}_\bullet(K))$, the **reduced homology**.

Homology groups are abelian, being quotients of subgroups of free abelian groups. For $x \in \ker d$ we write $[x]$ for its class in $H_\bullet$.

Example 25.5.7

Continuing Example 25.5.5, let us compute $\widetilde{H}_\bullet(\Delta^2)$.

- d_2 is injective, so $\ker d_2 = 0$ and $\widetilde{H}_2 = 0$.
- For $x = ae_{01} + be_{12} + ce_{02}$ we have $dx = -(a+c)e_0 + (a-b)e_1 + (b+c)e_2$, so $dx = 0$ iff $a = b = -c$. Hence $\ker d_1 = \langle e_{01} + e_{12} - e_{02} \rangle = \operatorname{im} d_2$, giving $\widetilde{H}_1 = 0$.
- $\ker d_0 = \{ae_0 + be_1 + ce_2 \mid a + b + c = 0\}$, which is spanned by $e_1 - e_0, e_2 - e_0$, and these lie in $\operatorname{im} d_1$. So $\widetilde{H}_0 = 0$.
- d_0 is surjective onto $\widetilde{C}_{-1}$, so $\widetilde{H}_{-1} = 0$.

So all the reduced homology of the full triangle vanishes. Passing to unreduced homology only changes degree 0:

$$H_k(\Delta^2) = \begin{cases} \mathbb{Z} & k = 0 \\ 0 & \text{otherwise.} \end{cases}$$

Example 25.5.8

Let $K = \{e_\emptyset, e_0, \dots, e_r\}$, so $|K|$ is $r + 1$ points. Then $\widetilde{C}_i(K) = 0$ for $i > 0$, and $\ker d_0 = \langle e_1 - e_0, \dots, e_r - e_0 \rangle \cong \mathbb{Z}^r$ with $\operatorname{im} d_1 = 0$. So $\widetilde{H}_0(K) \cong \mathbb{Z}^r$ and

$$H_0(K) \cong \mathbb{Z}^{r+1}.$$

That is a general phenomenon: H_0 counts path components, while $\widetilde{H}_0$ counts them minus one.

Example 25.5.9 (The Circle)

Let T_n be the boundary of a convex n -gon, realising the abstract complex

$$K = \{e_\emptyset, e_0, \dots, e_{n-1}, e_{01}, e_{12}, \dots, e_{(n-2)(n-1)}, e_{0(n-1)}\},$$

so that $|K| \cong S^1$. Then $C_1(K)$ is free of rank n and $C_0(K)$ is free of rank n , with $d(e_{i(i+1)}) = e_{i+1} - e_i$ and $d(e_{0(n-1)}) = e_{n-1} - e_0$. Setting

$$x = e_{01} + e_{12} + \dots + e_{(n-2)(n-1)} - e_{0(n-1)},$$

we get $dx = 0$, and a short computation shows $\ker d_1 = \langle x \rangle$. As $C_2 = 0$,

$$H_1(K) = \langle x \rangle \cong \mathbb{Z}.$$

Also $\ker d_0 = C_0$ and $\operatorname{im} d_1$ is spanned by the $e_{i+1} - e_i$, so $H_0(K) \cong \mathbb{Z}$. Note that the answer does not depend on n , even though the complexes do — as it must not, since all the $|T_n|$ are homeomorphic. That homology depends only on $|K|$ is a theorem we will prove in the next section.

§25.5.3 Chain Maps

We want simplicial maps to induce maps on homology, and for that we need the notion of a map of chain complexes.

Definition 25.5.10 (Chain Map)

Let $(C_\bullet, d)$ and $(C'_\bullet, d')$ be chain complexes. A **chain map** $f : C_\bullet \rightarrow C'_\bullet$ is a collection of homomorphisms $f_i : C_i \rightarrow C'_i$ with $f_{i-1} \circ d_i = d'_i \circ f_i$, that is, with $fd = d'f$.

Proposition 25.5.11

A chain map f induces a homomorphism $f_* : H_\bullet(C) \rightarrow H_\bullet(C')$ by $f_*[x] = [f(x)]$, and $(f \circ g)_* = f_* \circ g_*$.

Proof. If $dx = 0$ then $d'f(x) = f(dx) = 0$, so $f(\ker d) \subseteq \ker d'$; and if $x = dy$ then $f(x) = d'f(y)$, so $f(\operatorname{im} d) \subseteq \operatorname{im} d'$. Hence f descends to the quotients. Functoriality is immediate. $\square$

Now let $f : K \rightarrow L$ be a simplicial map, determined by $\hat{f} : \{0, \dots, n\} \rightarrow \{0, \dots, m\}$. The obvious guess is to define $f_\#(e_I) = e_{\hat{f}(I)}$, and this fails for two reasons.

First, if $\hat{f}$ is not injective on I then $e_{\hat{f}(I)}$ has smaller dimension than e_I : for $f : \Delta^1 \rightarrow \Delta^1$ with $e_0, e_1 \mapsto e_0$ we would need $f_\#(e_{01}) = e_0$, which does not preserve grading.

Second, the signs go wrong. For $f : \Delta^1 \rightarrow \Delta^1$ swapping e_0 and e_1 , we would have $f_\#(e_{01}) = e_{01}$, so $df_\#(e_{01}) = e_1 - e_0$, while $f_\#d(e_{01}) = f_\#(e_1 - e_0) = e_0 - e_1$.

The fix for both is to allow *ordered* index tuples and to build the signs in.

Definition 25.5.12

For $I = (i_0, \dots, i_k) \in \{0, \dots, n\}^{k+1}$, define $e_I \in \widetilde{C}_k(\Delta^n)$ by

$$e_I = \begin{cases} 0 & \text{if } I \text{ has a repeated entry,} \\ (-1)^{S(I)} e_{I'} & \text{otherwise,} \end{cases}$$

where I' is the increasing rearrangement of I and $(-1)^{S(I)}$ is the sign of the permutation taking I to I' .

So $e_{102} = -e_{012} = e_{210}$ and $e_{001} = 0$. With this convention the boundary formula holds in general.

Proposition 25.5.13

For any $I \in \{0, \dots, n\}^{k+1}$ we have $d(e_I) = \sum_{j=0}^k (-1)^j e_{I_j}$, where I_j omits the j th entry of the tuple.

Proof. If I has a repeated entry the claim is a direct check: if the repeat is at positions $j_1 \neq j_2$, all terms e_{I_j} with $j \notin \{j_1, j_2\}$ vanish, and the remaining two terms cancel by the sign rule. Otherwise let σ take I to I' ; if i_j sits in position j' in I' then removing it changes the number of inversions by $j - j'$, so $S(I_j) \equiv S(I) + j - j' \pmod{2}$. Hence

$$\begin{aligned} \sum_j (-1)^j e_{I_j} &= \sum_j (-1)^j (-1)^{S(I) + j - j'} e_{(I')_{j'}} \\ &= (-1)^{S(I)} \sum_{j'} (-1)^{j'} e_{(I')_{j'}} = (-1)^{S(I)} d(e_{I'}) = d(e_I). \quad \square \end{aligned}$$

Definition 25.5.14

For a simplicial map $f : K \rightarrow L$ define $f_\# : C_k(K) \rightarrow C_k(L)$ on basis elements by

$$f_\#(e_{i_0 \dots i_k}) = e_{\hat{f}(i_0) \dots \hat{f}(i_k)}.$$

This preserves grading, and collapses degenerate images to zero rather than dropping dimension.

Proposition 25.5.15

$f_\#$ is a chain map.

Proof. Using Proposition 25.5.13 twice,

$$d(f_{\#}(e_I)) = d(e_{f(I)}) = \sum_{j=0}^k (-1)^j e_{(f(I))_j} = f_{\#} \left(\sum_{j=0}^k (-1)^j e_{I_j} \right) = f_{\#}(d(e_I)). \quad \square$$

Example 25.5.16

For $f : \Delta^1 \rightarrow \Delta^1$ with $e_0, e_1 \mapsto e_0$ we get $f_{\#}(e_{01}) = e_{00} = 0$, and for f the swap we get $f_{\#}(e_{01}) = e_{10} = -e_{01}$, so $df_{\#}(e_{01}) = -(e_1 - e_0) = e_0 - e_1 = f_{\#}(de_{01})$. Both problems are fixed.

We write $f_* : H_{\bullet}(K) \rightarrow H_{\bullet}(L)$ for the induced map on homology.

§25.5.4 Chain Homotopies

Homotopic maps of spaces should induce the same map on homology. The algebraic shadow of a homotopy is the following.

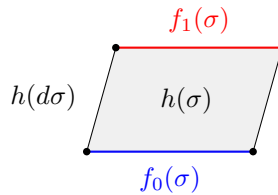
Definition 25.5.17 (Chain Homotopy)

Chain maps $f_0, f_1 : (C, d) \rightarrow (C', d')$ are **chain homotopic**, written $f_0 \simeq f_1$, if there are homomorphisms $h_i : C_i \rightarrow C'_{i+1}$ with

$$d'h + hd = f_0 - f_1.$$

We call h a **chain homotopy**.

The formula looks unmotivated until one draws the picture. If $f_0, f_1 : X \rightarrow Y$ are homotopic via H , then H sweeps each simplex σ out into a prism $\sigma \times I$, and the boundary of that prism consists of the two ends $f_0(\sigma)$ and $f_1(\sigma)$ together with the prism on $\partial\sigma$. Writing h for ‘sweep out into a prism’, that reads $dh + hd = f_1 - f_0$, up to signs.



Lemma 25.5.18

If $f_0 \simeq f_1$ then $f_{0*} = f_{1*} : H_{\bullet}(C) \rightarrow H_{\bullet}(C')$.

Proof. Let $[x] \in H_{\bullet}(C)$, so $dx = 0$. Then

$$f_{0*}[x] - f_{1*}[x] = [(f_0 - f_1)x] = [(d'h + hd)x] = [d'h(x)] = 0,$$

since $d'h(x) \in \text{im } d'$. □

Definition 25.5.19

A chain complex (C, d) is **contractible** if $\text{id}_C \simeq 0$.

Lemma 25.5.20

If (C, d) is contractible then $H_\bullet(C) = 0$.

Proof. $[x] = (\text{id})_*[x] = 0_*[x] = 0$ for every class. □

The basic example of a contractible chain complex comes from coning off.

Definition 25.5.21

Let K be an abstract simplicial complex in Δ^n not containing the vertex e_0 . Its **cone** is $C_{e_0}(K) = K \cup \{e_{0I} \mid e_I \in K\}$, an abstract simplicial complex.

Proposition 25.5.22

$\widetilde{C}_\bullet(C_{e_0}(K))$ is contractible, and hence $\widetilde{H}_\bullet(C_{e_0}(K)) = 0$.

Proof. Define $h : \widetilde{C}_k \rightarrow \widetilde{C}_{k+1}$ on basis elements by $h(e_I) = e_{0I}$, noting that this is zero when $0 \in I$. We check $dh + hd = \text{id}$.

If $0 \notin I$ then by Proposition 25.5.13,

$$dh(e_I) = d(e_{0I}) = e_I + \sum_{j=0}^k (-1)^{j+1} e_{0I_j} = e_I - h(de_I),$$

so $dh(e_I) + hd(e_I) = e_I$. If $0 \in I$, say $I = 0J$ with $0 \notin J$, then $h(e_I) = 0$ so $dh(e_I) = 0$, while

$$hd(e_{0J}) = h\left(e_J - \sum_j (-1)^j e_{0J_j}\right) = e_{0J} - 0 = e_I,$$

using $h(e_{0J_j}) = e_{00J_j} = 0$. So $dh + hd = \text{id}$ in both cases. □

Corollary 25.5.23

$H_i(C_{e_0}(K)) = \mathbb{Z}$ for $i = 0$ and 0 otherwise. In particular this holds for $\Delta^n = C_{e_0}(\Delta^n \text{ on } \{1, \dots, n\})$, so a simplex has the homology of a point.

Proof. Write $(\widetilde{C}, \widetilde{d})$ and (C, d) for the reduced and unreduced complexes. For $i > 0$ we have $\ker \widetilde{d}_i = \ker d_i$ and $\text{im } \widetilde{d}_{i+1} = \text{im } d_{i+1}$, so $H_i(C) = \widetilde{H}_i = 0$. In degree 0, $\widetilde{H}_{-1} = 0$ says $\widetilde{d}_0 : \widetilde{C}_0 \rightarrow \langle e_\emptyset \rangle \cong \mathbb{Z}$ is surjective, and $\widetilde{H}_0 = 0$ says $\ker \widetilde{d}_0 = \text{im } \widetilde{d}_1$. Hence

$$\mathbb{Z} \cong \widetilde{C}_0 / \ker \widetilde{d}_0 = \widetilde{C}_0 / \text{im } \widetilde{d}_1 = C_0 / \text{im } d_1 = \ker d_0 / \text{im } d_1 = H_0(C). \quad \square$$

§25.5.5 Exact Sequences

To compute anything beyond cones we need a way of relating the homology of a space to that of its pieces. The algebra behind this is the theory of exact sequences.

Definition 25.5.24 (Exact Sequence)

A sequence of abelian groups and homomorphisms

$$\cdots \rightarrow A_{k+1} \xrightarrow{f_{k+1}} A_k \xrightarrow{f_k} A_{k-1} \rightarrow \cdots$$

is **exact at A_k** if $\ker f_k = \operatorname{im} f_{k+1}$, and **exact** if it is exact everywhere. A **short exact sequence** is an exact sequence of the form

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0.$$

Unwinding: $0 \rightarrow A \rightarrow B$ is exact at A iff f is injective; $B \rightarrow C \rightarrow 0$ is exact at C iff g is surjective; and a short exact sequence says exactly that f is injective, g is surjective, and $C \cong B/f(A)$.

Definition 25.5.25

A **short exact sequence of chain complexes** is a pair of chain maps

$$0 \rightarrow A_\bullet \xrightarrow{f} B_\bullet \xrightarrow{g} C_\bullet \rightarrow 0$$

such that $0 \rightarrow A_k \rightarrow B_k \rightarrow C_k \rightarrow 0$ is exact for every k .

The following is the engine of every homology computation we will do.

Lemma 25.5.26 (Snake Lemma)

Given a short exact sequence of chain complexes as above, there are homomorphisms $\partial_k : H_k(C) \rightarrow H_{k-1}(A)$, the **connecting homomorphisms**, such that the sequence

$$\cdots \rightarrow H_k(A) \xrightarrow{f_*} H_k(B) \xrightarrow{g_*} H_k(C) \xrightarrow{\partial_k} H_{k-1}(A) \xrightarrow{f_*} \cdots$$

is exact. This is the **long exact sequence** of the short exact sequence.

Proof. Construction of ∂_k . Let $[c] \in H_k(C)$, so $dc = 0$.

1. g is surjective, so choose $b \in B_k$ with $g(b) = c$.
2. $g(db) = dg(b) = dc = 0$, so $db \in \ker g = \operatorname{im} f$; choose $a \in A_{k-1}$ with $f(a) = db$.
3. $f(da) = df(a) = d^2b = 0$, and f is injective, so $da = 0$.

We set $\partial_k[c] = [a]$.

$$\begin{array}{ccccccc}
0 & \longrightarrow & A_k & \xrightarrow{f} & B_k & \xrightarrow{g} & C_k \longrightarrow 0 \\
& & \downarrow d & & \downarrow d & \nearrow \partial_k & \downarrow d \\
0 & \longrightarrow & A_{k-1} & \xrightarrow{f} & B_{k-1} & \xrightarrow{g} & C_{k-1} \longrightarrow 0
\end{array}$$

Well-definedness. Suppose $g(b') = c$ too. Then $b - b' \in \ker g = \operatorname{im} f$, say $b - b' = f(\alpha)$. Then $f(a) - f(a') = db - db' = f(d\alpha)$, so $a - a' = d\alpha$ by injectivity of f , and $[a] = [a']$. Now suppose $[c] = [c']$, so $c - c' = d\gamma$ with $\gamma = g(\beta)$ by surjectivity. Replacing b by $b - d\beta$ we may assume b maps to c' , and $d(b - d\beta) = db$, so we get the same a .

Exactness at $H_k(C)$. If $\partial_k[c] = 0$ then $a = d\alpha$ for some $\alpha \in A_k$, and then $d(b - f(\alpha)) = db - f(d\alpha) = db - f(a) = 0$, so $[b - f(\alpha)] \in H_k(B)$ and $g_*[b - f(\alpha)] = [g(b)] = [c]$. Conversely if $[c] = g_*[b]$ with $db = 0$ then in the construction we may take $a = 0$.

Exactness at $H_k(B)$. Certainly $g_*f_* = (gf)_* = 0$. Conversely if $g_*[b] = 0$ then $g(b) = d\gamma$ for some $\gamma \in C_{k+1}$, and choosing β with $g(\beta) = \gamma$ we get $g(b - d\beta) = 0$, so $b - d\beta = f(\alpha)$ for some $\alpha \in A_k$. Then $f(d\alpha) = d(b - d\beta) = 0$, so $d\alpha = 0$ and $[b] = [b - d\beta] = f_*[\alpha]$.

Exactness at $H_{k-1}(A)$. If $[a] = \partial_k[c]$ then $f_*[a] = [db] = 0$. Conversely if $f_*[a] = 0$ then $f(a) = db$ for some $b \in B_k$, and $dg(b) = g(db) = g(f(a)) = 0$, so $[g(b)] \in H_k(C)$ and by construction $\partial_k[g(b)] = [a]$. $\square$

§25.5.6 The Mayer–Vietoris Sequence

Let K_1, K_2 be abstract simplicial complexes in Δ^n . Then $K_1 \cap K_2$ and $K_1 \cup K_2$ are also abstract simplicial complexes, and the inclusions give a commuting square, hence a commuting square of chain maps.

Proposition 25.5.27

The sequence

$$0 \longrightarrow C_\bullet(K_1 \cap K_2) \xrightarrow{i} C_\bullet(K_1) \oplus C_\bullet(K_2) \xrightarrow{j} C_\bullet(K_1 \cup K_2) \longrightarrow 0,$$

with $i(x) = (i_{1\#}x, i_{2\#}x)$ and $j(x, y) = j_{1\#}x - j_{2\#}y$, is a short exact sequence of chain complexes.

Proof. Each $C_\bullet(K)$ has as basis the $e_I \in K$, and the maps $i_\#, j_\#$ send basis elements to basis elements, so we may argue basiswise.

i is injective since $i_{1\#}$ is. Next, $ji = j_{1\#}i_{1\#} - j_{2\#}i_{2\#} = 0$ as the square commutes, so $\operatorname{im} i \subseteq \ker j$. Conversely if $j(x, y) = 0$ then $x = y$ as elements of $C_\bullet(K_1 \cup K_2)$, so all the simplices occurring lie in both K_1 and K_2 , giving $x = y \in C_\bullet(K_1 \cap K_2)$ and $(x, y) = i(x)$.

Finally, if $e_I \in K_1 \cup K_2$ then $e_I \in K_1$ or $e_I \in K_2$, and correspondingly $e_I = j(e_I, 0)$ or $e_I = j(0, -e_I)$. So j is surjective. $\square$

Theorem 25.5.28 (Mayer–Vietoris)

Let K_1, K_2 be abstract simplicial complexes in Δ^n . Then there is a long exact sequence

$$\begin{aligned} \cdots \rightarrow H_k(K_1 \cap K_2) &\xrightarrow{i_*} H_k(K_1) \oplus H_k(K_2) \xrightarrow{j_*} H_k(K_1 \cup K_2) \\ &\xrightarrow{\partial_k} H_{k-1}(K_1 \cap K_2) \xrightarrow{i_*} H_{k-1}(K_1) \oplus H_{k-1}(K_2) \rightarrow \cdots \end{aligned}$$

Proof. Apply the snake lemma to Proposition 25.5.27. $\square$

Example 25.5.29 (Disjoint Unions)

If K_1 and K_2 are disjoint complexes inside a common Δ^n , so that $K_1 \cap K_2 = \{e_\emptyset\}$ and hence $C_\bullet(K_1 \cap K_2) = 0$, the Mayer–Vietoris sequence degenerates to

$$0 \rightarrow H_k(K_1) \oplus H_k(K_2) \rightarrow H_k(K_1 \sqcup K_2) \rightarrow 0,$$

so $H_k(K_1 \sqcup K_2) \cong H_k(K_1) \oplus H_k(K_2)$: homology of a disjoint union is the direct sum.

We can already compute the homology of spheres, using nothing but the long exact sequence.

Theorem 25.5.30

For $n \geq 1$,

$$H_k(\mathbb{S}^n) = \begin{cases} \mathbb{Z} & k = 0, n \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Recall $\mathbb{S}^n = \Delta^{n+1} \setminus \{e_{0\dots(n+1)}\}$ has polyhedron S^n . We induct on n . For $n = 1$ this is Example 25.5.9, since $|\mathbb{S}^1|$ is the boundary of a triangle.

For the inductive step, write $\mathbb{S}^n = K_1 \cup K_2$ where

$$K_1 = \{e_I \mid I \subseteq \{0, \dots, n\}\}, \quad K_2 = \{e_I \mid I \subsetneq \{0, \dots, n+1\}, I \neq \{0, \dots, n\}\}.$$

Both are downward closed, and their union is all of $\mathbb{S}^n$. Now $K_1 = \Delta^n$ is a simplex, while $K_2 = C_{e_{n+1}}(\mathbb{S}^{n-1})$ is a cone on the vertex e_{n+1} , so both have the homology of a point by Corollary 25.5.23; geometrically these are the two hemispheres. Their intersection is $\{e_I \mid I \subsetneq \{0, \dots, n\}\} = \mathbb{S}^{n-1}$, the equator. For $k \geq 2$ the Mayer–Vietoris sequence gives

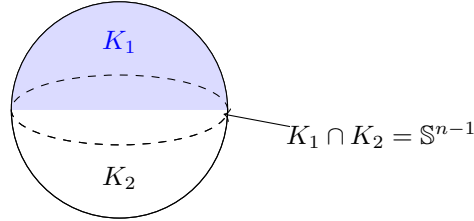
$$0 \rightarrow H_k(\mathbb{S}^n) \xrightarrow{\partial_k} H_{k-1}(\mathbb{S}^{n-1}) \rightarrow 0,$$

using $H_k(K_1) = H_k(K_2) = H_{k-1}(K_1) = H_{k-1}(K_2) = 0$ for $k \geq 2$. So $H_k(\mathbb{S}^n) \cong H_{k-1}(\mathbb{S}^{n-1})$, which by induction is $\mathbb{Z}$ if $k = n$ and 0 otherwise. For $k = 1$ and $n \geq 2$

the sequence reads

$$0 \rightarrow H_1(\mathbb{S}^n) \rightarrow H_0(\mathbb{S}^{n-1}) \xrightarrow{i_*} H_0(K_1) \oplus H_0(K_2),$$

and $i_* : \mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}$ is injective (it is $x \mapsto (x, x)$), so $H_1(\mathbb{S}^n) = 0$. Finally $H_0(\mathbb{S}^n) = \mathbb{Z}$ since $|\mathbb{S}^n|$ is connected. $\square$



§25.6 The Topological Invariance of Homology

So far homology is an invariant of a *simplicial complex*, not of a space: we have computed $H_\bullet(K)$, and we noticed in Example 25.5.9 that the answer for the n -gon did not depend on n , but we have not shown that this is a general phenomenon. In this section we show that $H_\bullet(K)$ depends only on the homotopy type of $|K|$, and that continuous maps induce homomorphisms. The tool is simplicial approximation.

§25.6.1 Approximations are Chain Homotopic

Lemma 25.6.1

Let $f_0, f_1 : K \rightarrow L$ be simplicial approximations to the same continuous map $F : |K| \rightarrow |L|$. Then for every $\sigma \in K$ there is a simplex $\tau \in L$ containing $f_0(v)$ and $f_1(v)$ for every vertex v of σ .

Proof. Pick $x \in \sigma^\circ$ and let $\tau \in L$ be the unique simplex with $F(x) \in \tau^\circ$. For each vertex v of σ we have $x \in \text{St}_K(v)$, so $F(x) \in \text{St}_L(f_i(v))$ for $i = 0, 1$; and by Lemma 25.4.23(i) this says $f_i(v) \in V(\tau)$. $\square$

Theorem 25.6.2

Let $f_0, f_1 : K \rightarrow L$ be simplicial approximations to the same $F : |K| \rightarrow |L|$. Then $f_{0\#} \simeq f_{1\#}$, and hence $f_{0*} = f_{1*} : H_\bullet(K) \rightarrow H_\bullet(L)$.

Proof. Define the **prism operator** $h : C_k(K) \rightarrow C_{k+1}(L)$ on basis elements by

$$h(e_{i_0 \dots i_k}) = \sum_{j=0}^k (-1)^j e_{\hat{f}_0(i_0) \dots \hat{f}_0(i_j) \hat{f}_1(i_j) \dots \hat{f}_1(i_k)}.$$

Each term is a legitimate element of $C_{k+1}(L)$: by Lemma 25.6.1 all the indices appearing lie in the vertex set of a single simplex $\tau \in L$, so the corresponding face of τ lies in L (or the term is zero, if an index repeats).

It is now a direct computation with the formula of Proposition 25.5.13 that

$$dh + hd = f_{1\#} - f_{0\#}.$$

Indeed, expanding $dh(e_I)$, the terms in which one deletes an index other than the two ‘middle’ ones $\hat{f}_0(i_j), \hat{f}_1(i_j)$ are precisely the terms of $-h(de_I)$, with matching signs. The terms where one of the middle indices is deleted telescope: deleting $\hat{f}_1(i_j)$ from the j th summand cancels against deleting $\hat{f}_0(i_{j+1})$ from the $(j+1)$ st, leaving only the two extreme terms, namely $e_{\hat{f}_1(i_0)\dots\hat{f}_1(i_k)} = f_{1\#}(e_I)$ from $j=0$ and $-e_{\hat{f}_0(i_0)\dots\hat{f}_0(i_k)} = -f_{0\#}(e_I)$ from $j=k$.

So h is a chain homotopy, and Lemma 25.5.18 gives $f_{0*} = f_{1*}$. $\square$

The picture is the same prism as before: $h(\sigma)$ is the simplicial subdivision of $\sigma \times I$, with $f_0(\sigma)$ at the bottom and $f_1(\sigma)$ at the top.

§25.6.2 Induced Maps

To define F_* for a continuous F , we approximate; but the approximation lives on a subdivision of K , so we need to know that subdividing does not change homology.

Theorem 25.6.3

Let K be a simplicial complex. Any simplicial approximation $g : \mathcal{B}(K) \rightarrow K$ to the identity map on $|K| = |\mathcal{B}(K)|$ induces an isomorphism

$$\nu_K = g_* : H_\bullet(\mathcal{B}(K)) \longrightarrow H_\bullet(K),$$

and ν_K does not depend on the choice of g .

Proof sketch. Independence of the choice is Theorem 25.6.2, since any two such g are simplicial approximations to the same map id . (Note that such a g exists: for each vertex w of $\mathcal{B}(K)$ we have $w \in \sigma^\circ$ for a unique $\sigma \in K$, and we may take $\hat{g}(w)$ to be any vertex of σ .)

For the isomorphism, one induces on the skeleta of K . For K a single simplex both $H_\bullet(\mathcal{B}(K))$ and $H_\bullet(K)$ are the homology of a point, by Corollary 25.5.23, and g_* is easily checked to be an isomorphism there. For the inductive step one writes K as the union of its $(n-1)$ -skeleton and its n -simplices, compares the two Mayer–Vietoris sequences using the fact that g is compatible with the decomposition, and applies the five lemma. $\square$

Writing $\nu_{K,r} : H_\bullet(\mathcal{B}^r(K)) \rightarrow H_\bullet(K)$ for the composite $\nu_K \circ \nu_{\mathcal{B}(K)} \circ \dots$, we can now make the key definition.

Definition 25.6.4

Let $F : |K| \rightarrow |L|$ be continuous. By the simplicial approximation theorem there is a simplicial $f : \mathcal{B}^r(K) \rightarrow L$ with $|f| \simeq F$. Define

$$F_* = f_* \circ \nu_{K,r}^{-1} : H_\bullet(K) \longrightarrow H_\bullet(L).$$

Theorem 25.6.5

F_* is well-defined, independent of the choice of r and f . Moreover $(\text{id})_* = \text{id}$ and $(F \circ G)_* = F_* \circ G_*$.

Proof sketch. If $f : \mathcal{B}^r(K) \rightarrow L$ and $f' : \mathcal{B}^{r'}(K) \rightarrow L$ are two simplicial approximations to F , with $r \leq r'$ say, choose a simplicial approximation $g : \mathcal{B}^{r'}(K) \rightarrow \mathcal{B}^r(K)$ to the identity. Then $f \circ g$ and f' are both simplicial approximations to F , so $f_*g_* = f'_*$ by Theorem 25.6.2; and $g_* = \nu_{\mathcal{B}^r(K), r'-r}$, so $f'_*\nu_{K, r'}^{-1} = f_*\nu_{K, r}^{-1}$.

Functoriality follows similarly: if f approximates F and g approximates G (after enough subdivision that the composite makes sense), then $f \circ g$ approximates $F \circ G$. $\square$

Theorem 25.6.6 (Homotopy Invariance)

If $F_0, F_1 : |K| \rightarrow |L|$ are continuous with $F_0 \simeq F_1$, then $F_{0*} = F_{1*}$.

Proof sketch. Let $H : |K| \times I \rightarrow |L|$ be a homotopy. Triangulate $|K| \times I$, which is the polyhedron of a simplicial complex M containing two copies K^0, K^1 of K at the ends. Applying simplicial approximation to H gives a simplicial map $\mathcal{B}^r(M) \rightarrow L$ restricting on the two ends to approximations of F_0 and F_1 ; the inclusions of the ends are homotopy equivalences onto $|M|$ inducing the same map on homology, by the argument of Theorem 25.6.2 applied to the prism. Hence $F_{0*} = F_{1*}$. $\square$

Corollary 25.6.7

If $|K| \simeq |L|$ then $H_\bullet(K) \cong H_\bullet(L)$.

Proof. Let $F : |K| \rightarrow |L|$ and $G : |L| \rightarrow |K|$ with $F \circ G \simeq \text{id}$ and $G \circ F \simeq \text{id}$. Then by Theorems 25.6.5 and 25.6.6, $F_*G_* = \text{id}$ and $G_*F_* = \text{id}$, so F_* is an isomorphism. $\square$

Definition 25.6.8

A space X is **triangulable** if $X \cong |K|$ for some simplicial complex K . In that case we define $H_\bullet(X) = H_\bullet(K)$, which by Corollary 25.6.7 is well-defined.

Not every space is triangulable, or even homotopy equivalent to a triangulable space — the infinite wedge $\bigvee_{i=1}^\infty S^1$ is an example — but every compact manifold and every CW complex we shall meet is.

§25.6.3 First Consequences**Proposition 25.6.9**

If $|K|$ is path connected then $H_0(K) \cong \mathbb{Z}$. In general $H_0(K) \cong \mathbb{Z}^k$ where k is the number of path components of $|K|$.

Proof. $C_0(K)$ is free on the vertices e_i , and every e_i is a cycle. For vertices e_i, e_j , let $F_i : \Delta^0 \rightarrow |K|$ be the map hitting e_i ; then $F_{i*}[e_0] = [e_i]$. If $|K|$ is path connected then $F_i \simeq F_j$, so $[e_i] = [e_j]$ by homotopy invariance. Hence $H_0(K)$ is generated by the single class $[e_0]$. It is non-trivial: the homomorphism $C_0(K) \rightarrow \mathbb{Z}$ sending each $e_i \mapsto 1$ kills $\text{im } d_1$ (as $d(e_{ij}) = e_j - e_i$), so it descends to a surjection $H_0(K) \rightarrow \mathbb{Z}$. Hence $H_0(K) \cong \mathbb{Z}$.

In general $|K|$ is the disjoint union of its path components, each of which is the polyhedron of a subcomplex, and Example 25.5.29 gives the direct sum decomposition. $\square$

Theorem 25.6.10

For $n \geq 1$,

$$H_k(S^n) = \begin{cases} \mathbb{Z} & k = 0, n \\ 0 & \text{otherwise,} \end{cases}$$

and $H_k(D^n) = \mathbb{Z}$ for $k = 0$, zero otherwise. Consequently $S^n \simeq S^m$ implies $n = m$.

Proof. We have $S^n \cong |\mathbb{S}^n|$ and $D^n \cong |\Delta^n|$, so this is Theorem 25.5.30 and Corollary 25.5.23. If $S^n \simeq S^m$ with $n \neq m$, comparing H_n gives $\mathbb{Z} \cong 0$ (taking $n \geq 1$; the case $n = 0$ is clear as S^0 is disconnected). $\square$

Corollary 25.6.11 (Invariance of Dimension)

If $\mathbb{R}^n \cong \mathbb{R}^m$ then $n = m$.

Proof. Take $n, m \geq 1$. A homeomorphism $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ restricts to a homeomorphism $\mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^m \setminus \{f(0)\}$. By Example 25.1.13 these are homotopy equivalent to S^{n-1} and S^{m-1} respectively, so $S^{n-1} \simeq S^{m-1}$ and hence $n = m$ by Theorem 25.6.10. $\square$

Corollary 25.6.12

For $n \geq 1$ there is no retraction $r : D^n \rightarrow S^{n-1}$.

Proof. Let $j : S^{n-1} \hookrightarrow D^n$ be the inclusion. If r were a retraction then $r \circ j = \text{id}$, so $r_* j_* = \text{id}$ on $H_{n-1}(S^{n-1})$. For $n \geq 2$ this is $\mathbb{Z}$, while $H_{n-1}(D^n) = 0$, so the identity on $\mathbb{Z}$ factors through 0, which is absurd. For $n = 1$ use H_0 : $H_0(S^0) = \mathbb{Z}^2$ and $H_0(D^1) = \mathbb{Z}$, and $\mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2$ cannot compose to the identity. $\square$

Theorem 25.6.13 (Brouwer Fixed Point Theorem)

Every continuous $F : D^n \rightarrow D^n$ has a fixed point.

Proof. Suppose not, so $F(x) \neq x$ for all x . Define $G : D^n \rightarrow S^{n-1}$ by letting $G(x)$ be the point where the ray from $F(x)$ through x meets S^{n-1} ; we must check this is continuous.

For $p \in D^n$ and $v \in S^{n-1}$, the ray $\{p + tv \mid t \geq 0\}$ meets S^{n-1} where $\|p + tv\|^2 = 1$, that is

$$t^2 + 2t\langle p, v \rangle + \|p\|^2 - 1 = 0, \quad t = -\langle p, v \rangle \pm \sqrt{\langle p, v \rangle^2 + 1 - \|p\|^2},$$

and the discriminant is non-negative since $\|p\| \leq 1$. Taking the larger root,

$$\tau(p, v) = -\langle p, v \rangle + \sqrt{\langle p, v \rangle^2 + 1 - \|p\|^2}$$

is continuous in (p, v) , and so is $P(p, v) = p + \tau(p, v)v$. Hence

$$G(x) = P\left(F(x), \frac{x - F(x)}{\|x - F(x)\|}\right)$$

is continuous, using $x \neq F(x)$. If $x \in S^{n-1}$ then x itself lies on the ray and on the sphere with $t > 0$, so $G(x) = x$. Thus G is a retraction $D^n \rightarrow S^{n-1}$, contradicting Corollary 25.6.12. $\square$

§25.7 Homology Calculations and Applications

We now have a working invariant. In this final section we put it to work: we define the degree of a map of spheres, compute the homology of all the closed surfaces, and establish the Euler characteristic and the Lefschetz fixed point theorem.

§25.7.1 Degree

Since $H_n(S^n) \cong \mathbb{Z}$ for $n \geq 1$, any map $f : S^n \rightarrow S^n$ induces a homomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$, which is multiplication by an integer.

Definition 25.7.1 (Degree)

For $n \geq 1$ and $f : S^n \rightarrow S^n$, the **degree** $\deg f \in \mathbb{Z}$ is defined by

$$f_*(x) = (\deg f) x \quad \text{for all } x \in H_n(S^n) \cong \mathbb{Z}.$$

Note that this is independent of the choice of generator, since -1 times a generator gives the same multiplier.

Proposition 25.7.2

For maps $f, g : S^n \rightarrow S^n$:

1. $\deg \text{id} = 1$ and $\deg(g \circ f) = \deg g \cdot \deg f$;
2. if $f \simeq g$ then $\deg f = \deg g$;
3. if f is constant then $\deg f = 0$;
4. if f is a homotopy equivalence then $\deg f = \pm 1$.

Proof. (i) is functoriality (Theorem 25.6.5) and (ii) is homotopy invariance (Theorem 25.6.6). For (iii), a constant map factors through a point, and $H_n(*) = 0$ for

$n \geq 1$. For (iv), if g is a homotopy inverse then $\deg g \deg f = \deg \text{id} = 1$ in $\mathbb{Z}$. $\square$

Corollary 25.7.3

id_{S^n} is not null-homotopic, so S^n is not contractible.

Proof. $\deg \text{id} = 1 \neq 0 = \deg(\text{constant})$. $\square$

Together with Theorem 25.4.28 this completes the story we began in Section 25.4: for $k < n$ every map $S^k \rightarrow S^n$ is null-homotopic, whereas id_{S^n} is not, so $S^k \not\cong S^n$.

For $n = 1$ this notion of degree agrees with the one we defined via π_1 in Section 2, since both are computed by the winding number of the image loop — for instance, $z \mapsto z^m$ has degree m in both senses.

Example 25.7.4

Let $r : S^n \rightarrow S^n$ be the reflection $(x_0, \dots, x_n) \mapsto (-x_0, x_1, \dots, x_n)$. Then $\deg r = -1$. To see this, triangulate S^n as $|\mathbb{S}^n|$ so that r is simplicial and swaps two vertices $e_0 \leftrightarrow e_1$, fixing the rest; then $r_\#$ acts on the top cycle by transposing two indices, hence by -1 . Consequently the antipodal map $a = (-1)^{\text{all coordinates}}$, being a composite of $n + 1$ reflections, has $\deg a = (-1)^{n+1}$.

§25.7.2 The Homology of Surfaces

We first record a splitting lemma, which lets us read off answers from short exact sequences.

Proposition 25.7.5

If $0 \rightarrow A \rightarrow B \rightarrow \mathbb{Z}^r \rightarrow 0$ is exact, then $B \cong A \oplus \mathbb{Z}^r$.

Proof. Choose $b_i \in B$ mapping to the standard basis vectors of $\mathbb{Z}^r$; this defines a homomorphism $s : \mathbb{Z}^r \rightarrow B$ splitting the surjection, since $\mathbb{Z}^r$ is free. Then $b \mapsto (b - s(g(b)), g(b))$ is an isomorphism $B \rightarrow A \oplus \mathbb{Z}^r$. $\square$

Example 25.7.6 (The Torus)

Triangulate T^2 as $|K|$ and write $K = K_1 \cup K_2$, where $|K_1|$ and $|K_2|$ are the two cylinders obtained by cutting the torus along two parallel circles. Then $|K_i| \simeq S^1$ and $|K_1 \cap K_2| \cong S^1 \sqcup S^1$.

Both inclusions $S^1 \hookrightarrow |K_i|$ of the two boundary circles are homotopy equivalences, and they are homotopic to each other. So on H_1 , writing c_1, c_2 for the classes of the two circles of the intersection, the map $\alpha_1 = i_*$ has matrix

$$\alpha_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2,$$

and the same is true of α_0 on H_0 . Hence $\ker \alpha_k \cong \mathbb{Z}$ and $\text{coker } \alpha_k \cong \mathbb{Z}$ for $k = 0, 1$.

Since $H_2(K_i) = H_2(S^1) = 0$, the Mayer–Vietoris sequence

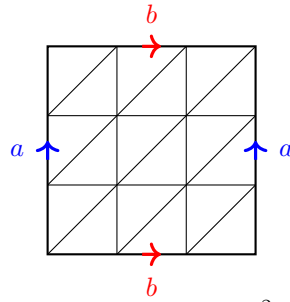
$$\begin{aligned} 0 \rightarrow H_2(K) \xrightarrow{\partial} H_1(K_1 \cap K_2) \xrightarrow{\alpha_1} H_1(K_1) \oplus H_1(K_2) \rightarrow H_1(K) \\ \xrightarrow{\partial} H_0(K_1 \cap K_2) \xrightarrow{\alpha_0} H_0(K_1) \oplus H_0(K_2) \rightarrow H_0(K) \rightarrow 0 \end{aligned}$$

breaks into short exact sequences

$$\begin{aligned} 0 \rightarrow H_2(K) \rightarrow \ker \alpha_1 \rightarrow 0, \quad 0 \rightarrow \operatorname{coker} \alpha_0 \rightarrow H_0(K) \rightarrow 0, \\ 0 \rightarrow \operatorname{coker} \alpha_1 \rightarrow H_1(K) \rightarrow \ker \alpha_0 \rightarrow 0. \end{aligned}$$

Therefore $H_2(T^2) \cong \mathbb{Z}$, and $H_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ by Proposition 25.7.5, and $H_0(T^2) \cong \mathbb{Z}$. In summary

$$H_k(T^2) = \begin{cases} \mathbb{Z} & k = 0, 2 \\ \mathbb{Z}^2 & k = 1 \\ 0 & \text{otherwise.} \end{cases}$$



a triangulation of T^2

For higher genus it is cleanest to build the surface out of a wedge of circles and a single disc, exactly as in Example 25.3.23. Write $\Sigma_{g,1}$ for the surface Σ_g with an open disc removed.

Lemma 25.7.7

$\Sigma_{g,1} \simeq \bigvee_{i=1}^{2g} S^1$, and hence

$$H_k(\Sigma_{g,1}) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}^{2g} & k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Realise Σ_g as a $4g$ -gon with its boundary identified, and remove an open disc from the interior. What is left deformation retracts radially onto the boundary of the polygon, which after the identifications is $\bigvee_{i=1}^{2g} S^1$. The homology of a wedge of m circles is $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^m$ and 0 above — by induction on m using Mayer–Vietoris with K_1 the first circle and K_2 the rest (thickened slightly so as to intersect in a contractible set). $\square$

Theorem 25.7.8

For the closed orientable surface of genus $g \geq 0$,

$$H_k(\Sigma_g) = \begin{cases} \mathbb{Z} & k = 0, 2 \\ \mathbb{Z}^{2g} & k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Proof. Write $\Sigma_g = K_1 \cup K_2$ with $|K_1| = \Sigma_{g,1}$ and $|K_2| = D^2$ the disc we removed, so $|K_1 \cap K_2| \cong S^1$. Mayer–Vietoris gives

$$\begin{aligned} 0 \rightarrow H_2(\Sigma_g) &\xrightarrow{\partial} H_1(S^1) \xrightarrow{\iota_*} H_1(\Sigma_{g,1}) \rightarrow H_1(\Sigma_g) \\ &\xrightarrow{\partial} H_0(S^1) \xrightarrow{\alpha_0} H_0(K_1) \oplus H_0(K_2), \end{aligned}$$

using $H_2(K_i) = 0$ from Lemma 25.7.7 and Corollary 25.5.23.

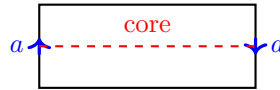
The key point is that $\iota_* : \mathbb{Z} \rightarrow \mathbb{Z}^{2g}$ is the zero map. Under the identification of Lemma 25.7.7, the boundary circle is the loop

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1},$$

whose class in $H_1(\bigvee S^1) = \mathbb{Z}^{2g}$ is obtained by adding up the signed number of times it traverses each circle. Each generator occurs once forwards and once backwards, so the total is 0.

Hence $\partial : H_2(\Sigma_g) \rightarrow \mathbb{Z}$ is an isomorphism onto $\ker \iota_* = \mathbb{Z}$, giving $H_2(\Sigma_g) \cong \mathbb{Z}$. Also $\mathbb{Z}^{2g} = \text{coker } \iota_* \hookrightarrow H_1(\Sigma_g)$, and $\alpha_0 : \mathbb{Z} \rightarrow \mathbb{Z}^2$ is injective (both components send a point to a point), so $\partial = 0$ on H_1 and the map $\mathbb{Z}^{2g} \rightarrow H_1(\Sigma_g)$ is onto. So $H_1(\Sigma_g) \cong \mathbb{Z}^{2g}$. Finally $H_0(\Sigma_g) \cong \mathbb{Z}$ by connectedness. For $g = 0$ this reduces to $H_\bullet(S^2)$, as it should. $\square$

The same argument handles the non-orientable surfaces. Let N_r be the closed surface obtained from a $2r$ -gon by identifying the edges according to the word $a_1^2 a_2^2 \cdots a_r^2$; equivalently, N_r is the connected sum of r copies of $\mathbb{RP}^2$. So $N_1 = \mathbb{RP}^2$ and $N_2 = K$, the Klein bottle, and $N_{r,1}$, the surface with an open disc removed, is homotopy equivalent to $\bigvee_{i=1}^r S^1$. For $r = 1$ the punctured surface $N_{1,1}$ is the Möbius band, and its boundary circle wraps *twice* around the core.



the Möbius band $N_{1,1}$

Theorem 25.7.9

For $r \geq 1$,

$$H_k(N_r) = \begin{cases} \mathbb{Z} & k = 0 \\ \mathbb{Z}^{r-1} \oplus \mathbb{Z}/2\mathbb{Z} & k = 1 \\ 0 & \text{otherwise.} \end{cases}$$

In particular $H_2(N_r) = 0$ and $H_1(\mathbb{RP}^2) = \mathbb{Z}/2\mathbb{Z}$.

Proof. As before, $N_r = N_{r,1} \cup D^2$ with $N_{r,1} \simeq \bigvee_{i=1}^r S^1$ and intersection S^1 . This time the attaching loop is $a_1^2 \cdots a_r^2$, whose class in $H_1 = \mathbb{Z}^r$ is $(2, 2, \dots, 2)$. So $\iota_* : \mathbb{Z} \rightarrow \mathbb{Z}^r$ is injective, whence $H_2(N_r) = \ker \iota_* = 0$, and

$$H_1(N_r) \cong \text{coker } \iota_* = \mathbb{Z}^r / \langle (2, \dots, 2) \rangle.$$

Changing basis to $u_1 = e_1 + \cdots + e_r$, $u_2 = e_2, \dots, u_r = e_r$, this is $\mathbb{Z}^r / \langle 2u_1 \rangle \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}^{r-1}$, as claimed. (As in the orientable case, α_0 is injective so the sequence terminates correctly.) $\square$

Notice that homology detects non-orientability through torsion, and through the vanishing of H_2 : a closed orientable surface has a fundamental class in H_2 , and a non-orientable one does not. In particular the Σ_g and N_r are pairwise non-homotopy-equivalent, which together with the classification of surfaces says that homology is a complete invariant for closed surfaces.

§25.7.3 The Euler Characteristic

Definition 25.7.10 (Euler Characteristic)

Let K be a simplicial complex. Its **Euler characteristic** is

$$\chi(K) = \sum_k (-1)^k \text{rank } C_k(K),$$

the alternating sum of the numbers of simplices in each dimension.

For a two-dimensional complex this is ‘vertices – edges + faces’, the classical Euler formula. It is not at all obvious from the definition that this depends only on $|K|$, but homology makes it so.

Theorem 25.7.11

$\chi(K) = \sum_k (-1)^k \text{rank } H_k(K)$. In particular $\chi(K)$ depends only on the homotopy type of $|K|$.

Proof. Write $Z_k = \ker d_k$ and $B_k = \text{im } d_{k+1}$, both subgroups of the free abelian group C_k , hence free. Rank-nullity applied to $d_k : C_k \rightarrow C_{k-1}$, whose image is B_{k-1} , gives

$$\text{rank } C_k = \text{rank } Z_k + \text{rank } B_{k-1},$$

and $H_k = Z_k/B_k$ gives $\text{rank } Z_k = \text{rank } H_k + \text{rank } B_k$. Hence

$$\sum_k (-1)^k \text{rank } C_k = \sum_k (-1)^k (\text{rank } H_k + \text{rank } B_k + \text{rank } B_{k-1}),$$

and the B terms cancel in pairs, leaving $\sum_k (-1)^k \text{rank } H_k$. $\square$

Example 25.7.12

From Theorems 25.7.8 and 25.7.9,

$$\chi(\Sigma_g) = 1 - 2g + 1 = 2 - 2g, \quad \chi(N_r) = 1 - (r - 1) + 0 = 2 - r,$$

using that $\mathbb{Z}/2\mathbb{Z}$ has rank 0. So $\chi(S^2) = 2$, $\chi(T^2) = 0$, $\chi(\mathbb{RP}^2) = 1$ and $\chi(K) = 0$ for the Klein bottle. Also $\chi(S^n) = 1 + (-1)^n$.

§25.7.4 The Lefschetz Fixed Point Theorem

Brouwer's theorem said that a self-map of a disc has a fixed point. There is a beautiful generalisation, which computes an obstruction to being fixed-point-free out of the induced maps on homology. For this it is convenient to work with rational coefficients: replacing each $C_k(K) \cong \mathbb{Z}^{n_k}$ by $\mathbb{Q}^{n_k}$ with the same boundary matrices gives a chain complex of $\mathbb{Q}$ -vector spaces, and all the theory goes through verbatim, with $H_k(K; \mathbb{Q})$ a vector space of dimension rank $H_k(K)$.

Definition 25.7.13 (Lefschetz Number)

Let $f : C \rightarrow C$ be a chain map of a finitely generated chain complex of $\mathbb{Q}$ -vector spaces. Its **Lefschetz number** is

$$L(f) = \sum_k (-1)^k \operatorname{tr}(f_k : C_k \rightarrow C_k),$$

and likewise $L(f_*) = \sum_k (-1)^k \operatorname{tr}(f_{k*} : H_k(C) \rightarrow H_k(C))$.

Proposition 25.7.14

$L(f) = L(f_*)$.

Proof. Write $U_k = \operatorname{im} d_{k+1} \subseteq \ker d_k \subseteq C_k$, and choose complements so that $\ker d_k = U_k \oplus V_k$ and $C_k = U_k \oplus V_k \oplus U'_k$. Then d maps U'_k isomorphically onto U_{k-1} and kills $U_k \oplus V_k$, so in block form

$$d = \begin{pmatrix} 0 & 0 & I \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Since f is a chain map it preserves $\ker d$ and $\operatorname{im} d$, so

$$f_k = \begin{pmatrix} A_k & X_k & * \\ 0 & B_k & * \\ 0 & 0 & A'_k \end{pmatrix},$$

and comparing $df = fd$ in the top-right block gives $A_{k-1} = A'_k$. Now $H_k(C) = (U_k \oplus V_k)/U_k \cong V_k$, on which f_{k*} acts by B_k . Hence

$$L(f) = \sum_k (-1)^k (\operatorname{tr} A_k + \operatorname{tr} B_k + \operatorname{tr} A_{k-1}) = \sum_k (-1)^k \operatorname{tr} B_k = L(f_*),$$

because the A terms cancel in consecutive pairs. □

Corollary 25.7.15

$\chi(K) = L(\text{id}) = L(\text{id}_{H_\bullet})$, recovering Theorem 25.7.11.

Definition 25.7.16

For a continuous $F : |K| \rightarrow |K|$, its **Lefschetz number** is $L(F) = L(F_*)$, where $F_* : H_\bullet(K; \mathbb{Q}) \rightarrow H_\bullet(K; \mathbb{Q})$.

Theorem 25.7.17 (Lefschetz Fixed Point Theorem)

Let K be a simplicial complex and $F : |K| \rightarrow |K|$ continuous. If $L(F) \neq 0$ then F has a fixed point.

Proof sketch. Suppose F has no fixed point. As $|K|$ is compact and $x \mapsto \|F(x) - x\|$ is continuous and positive, there is $\varepsilon > 0$ with $\|F(x) - x\| \geq \varepsilon$ for all x .

Subdivide until $\mu(\mathcal{B}^r K) < \varepsilon/3$, and take a simplicial approximation $f : \mathcal{B}^{r+s}(K) \rightarrow \mathcal{B}^r(K)$ to F for suitable s . Then for each simplex σ of $\mathcal{B}^{r+s}(K)$, the image $f(\sigma)$ is at distance at least $\varepsilon/3$ from σ , so $f(\sigma) \neq \sigma$; that is, no basis element of the chain complex is sent to a multiple of itself. Hence every diagonal entry of every matrix f_k vanishes, so $L(f) = 0$, and by Proposition 25.7.14 and the identification of f_* with F_* we get $L(F) = 0$. $\square$

Example 25.7.18

If $|K|$ is contractible then $H_0 = \mathbb{Q}$ and $H_k = 0$ for $k > 0$, and $F_{0*} = \text{id}$ on H_0 since $|K|$ is connected. So $L(F) = 1 \neq 0$ for every F , and every self-map of a contractible polyhedron has a fixed point. Taking $|K| = D^n$ recovers the Brouwer fixed point theorem.

Example 25.7.19

Let $F : S^n \rightarrow S^n$. Then $L(F) = 1 + (-1)^n \deg F$. So if $\deg F \neq (-1)^{n+1}$ then F has a fixed point. In particular any map $S^n \rightarrow S^n$ without a fixed point has the degree of the antipodal map, and any map of S^{2n} of degree 1 — for instance any orientation-preserving homeomorphism — has a fixed point.

Example 25.7.20

Let $F : \Sigma_g \rightarrow \Sigma_g$ with $g \geq 2$, and suppose F acts as the identity on H_1 . Then

$$L(F) = 1 - 2g + \text{tr}(F_{2*}),$$

where F_{2*} is multiplication by the degree d of F . So $L(F) = 1 - 2g + d$, and F has a fixed point unless d happens to equal $2g - 1$. For example, any such F of degree 1 has a fixed point once $g \geq 2$.

We finish with a remark on where all this goes. Simplicial homology is concrete, and that is its virtue for a first course, but the price is the enormous labour of the invariance

theorems: we had to subdivide, approximate, and compare, simply to know that $H_\bullet(K)$ depends only on $|K|$. *Singular* homology, in which one takes the free abelian group on *all* continuous maps $\Delta^n \rightarrow X$, is manifestly a topological invariant and is defined for every space; the price there is that the chain groups are enormous and nothing can be computed directly. The two theories agree on triangulable spaces, and the modern approach is to use singular homology for the theory and simplicial (or cellular) homology for the computations — which is, in effect, what we have been doing all along.

26 Analysis of Functions

Analysis of functions is the study of *spaces* of functions, and of the operations that act on them. The change of viewpoint is a small one to state and an enormous one to work with: instead of asking what a single function does at a single point, we regard a function as one point in an infinite-dimensional space, and we ask about the geometry and topology of that space.

Once we do this, a great deal becomes possible. Questions about partial differential equations turn into questions about linear operators between Banach spaces. Questions about convergence of Fourier series turn into questions about orthonormal bases in a Hilbert space. And the classical obstruction that non-differentiable functions cannot be differentiated simply dissolves: we enlarge the notion of a function until every distribution has derivatives of every order, and then we work out afterwards which of our generalised objects were honest functions all along.

The unifying technique of the course is *compactness by weakening*. In infinite dimensions the closed unit ball is never compact in the norm topology, so bounded sequences need not have convergent subsequences and the classical method of extracting limits fails. The remedy is to weaken the topology until compactness returns, at the price that the limits we obtain are much worse behaved. Recovering the good behaviour afterwards – through convexity, lower semicontinuity, elliptic regularity – is the real work, and the pay-off is the existence of solutions to nonlinear partial differential equations that no explicit formula will ever reach.

This article constitutes my notes for the ‘Analysis of Functions’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§26.1 Introduction

Let me try to say in slightly more detail what the plan is.

We begin, in Section 26.2, with a review of measure and integration. Almost none of this is new – it is the Probability and Measure course, with the probabilistic language stripped out – but we do need it in a hurry and in a slightly different key, so it is worth setting down. The one genuinely new item is Littlewood’s three principles, which are the informal slogans that make measure theory feel like analysis rather than bookkeeping.

Section 26.3 builds the spaces L^p . These are our first examples of Banach spaces made of functions, and essentially everything later in the course is a statement about them or about their relatives. We prove Hölder’s and Minkowski’s inequalities, completeness, and – most importantly for what follows – a collection of density theorems, culminating in the fact that any L^p function ($p < \infty$) can be approximated in norm by smooth compactly supported functions. The tool that does this is *mollification*: convolution with a shrinking bump. It is worth understanding it thoroughly, since it is used on almost every subsequent page.

Section 26.5 is about duality. We compute the dual of L^p , prove the Radon–Nikodým theorem (a measure-theoretic statement, via a Hilbert space argument – an early sign

of how the subject works), and state the Riesz representation theorem which identifies measures as functionals on continuous functions. Section 26.6 then introduces the weak and weak-* topologies, proves Hahn–Banach and Banach–Alaoglu, and so gets us the compactness we were after.

With compactness in hand, Section 26.7 sets out the direct method of the calculus of variations in the abstract: a coercive, lower semicontinuous functional on a suitable space attains its infimum. The delicate point is that coercivity wants a weak topology and lower semicontinuity wants a strong one, and *convexity* is what reconciles them.

Sections 26.8 and 26.9 develop the Fourier transform and distribution theory in parallel, since each is much improved by the other. The Fourier transform is an isometry (up to constants) of L^2 , an isomorphism of the Schwartz space, and – once we have tempered distributions – a bijection of a class of objects large enough to include every polynomial, every bounded function, and the Dirac delta.

Finally Sections 26.10 and 26.11 put everything to work. Sobolev spaces are the function spaces adapted to differential operators; the Sobolev embedding theorem says that enough weak differentiability forces classical differentiability, and the Rellich–Kondrachov theorem provides the compactness that lets the direct method run. We finish by solving the Dirichlet problem, describing the eigenvalues of the Laplacian on a bounded domain, and proving existence of minimisers for a class of variational problems whose Euler–Lagrange equations we could not hope to solve by hand.

§26.2 Measure and Integration

We assume the reader has met the Lebesgue integral. This section is a review: it fixes notation, records the results we shall use constantly, and adds the three theorems of Littlewood that describe how far measurable objects are from classical ones. Everything up to and including the convergence theorems is proved in my Probability and Measure notes, and I shall not repeat those proofs here; the same applies to the relations between the various modes of convergence (almost everywhere, in measure, in L^p), which are set out there in detail and which we use without comment.

§26.2.1 Measure spaces

We recall the basic framework.

Definition 26.2.1 (σ -algebra)

Let E be a set. A collection $\mathcal{E}$ of subsets of E is a **σ -algebra** on E if

- (i) $\emptyset \in \mathcal{E}$;
- (ii) $A \in \mathcal{E} \implies A^c = E \setminus A \in \mathcal{E}$;
- (iii) if $(A_n)_{n \in \mathbb{N}}$ is a sequence in $\mathcal{E}$ then $\bigcup_{n \in \mathbb{N}} A_n \in \mathcal{E}$.

The pair $(E, \mathcal{E})$ is a **measurable space**, and elements of $\mathcal{E}$ are **measurable sets**.

Definition 26.2.2 (Measure)

A **measure** on a measurable space $(E, \mathcal{E})$ is a function $\mu : \mathcal{E} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ which is **countably additive**: if $(A_n)_{n \in \mathbb{N}}$ are disjoint elements of $\mathcal{E}$ then

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

We call $(E, \mathcal{E}, \mu)$ a **measure space**.

Countable additivity is the whole content of the definition; everything else about measures is a consequence of it, together with the elementary manipulations of set theory. In particular a measure is monotone, countably subadditive, and continuous along increasing and (given finiteness) decreasing sequences of sets.

Example 26.2.3

On any set E we may take $\mathcal{E} = 2^E$ and $\mu(A) = \text{card}(A) \in [0, \infty]$, the counting measure. This is not a frivolous example: with $E = \mathbb{N}$ the L^p spaces of the counting measure are the sequence spaces ℓ^p , and every theorem we prove about L^p spaces has a sequence space avatar which is usually easier to think about.

Definition 26.2.4 (Generated σ -algebra)

Given a collection $\mathcal{A}$ of subsets of E , the σ -algebra **generated** by $\mathcal{A}$ is

$$\sigma(\mathcal{A}) = \bigcap_{\substack{\mathcal{E} \supseteq \mathcal{A} \\ \mathcal{E} \text{ a } \sigma\text{-algebra}}} \mathcal{E},$$

the smallest σ -algebra containing $\mathcal{A}$. If $(E, \mathcal{T})$ is a topological space then $\mathcal{B}(E) = \sigma(\mathcal{T})$ is the **Borel σ -algebra**.

The example we care about above all others is Lebesgue measure. Its existence was the main construction of the measure theory course, and we simply quote it.

Theorem 26.2.5 (Lebesgue measure)

There is a σ -algebra $\mathcal{M}$ of subsets of $\mathbb{R}^n$ and a measure λ on $(\mathbb{R}^n, \mathcal{M})$ such that

- (i) $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}$;
- (ii) if $A = (a_1, b_1] \times \cdots \times (a_n, b_n]$ is a box, with $-\infty < a_i \leq b_i < \infty$, then

$$\lambda(A) = (b_1 - a_1) \cdots (b_n - a_n);$$

- (iii) (**Borel regularity**) $A \in \mathcal{M}$ if and only if for every $\varepsilon > 0$ there are a closed set C and an open set O with $C \subseteq A \subseteq O$ and $\lambda(O \setminus C) < \varepsilon$.

These conditions determine $\mathcal{M}$ and λ uniquely.

We write dx for $d\lambda$ and $|A|$ for $\lambda(A)$ when no confusion can arise. Property (iii) deserves emphasis, because it is the one we shall reach for repeatedly and it is easy to forget that it is available. It says that a Lebesgue measurable set is, up to arbitrarily small error, both an open set and a closed set. This is the rigorous form of Littlewood's first principle, and

it is the engine behind almost every approximation argument in the course: if we want to prove something for all measurable sets, it is usually enough to prove it for boxes and then turn the handle.

Definition 26.2.6 (Measurable function)

Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measurable spaces. A function $f : E \rightarrow G$ is **measurable** if $f^{-1}(A) \in \mathcal{E}$ for all $A \in \mathcal{G}$.

When $(G, \mathcal{G}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ or $(\mathbb{C}, \mathcal{B}(\mathbb{C}))$ we just say ‘ f is measurable on E ’; when $(G, \mathcal{G}) = ([0, \infty], \mathcal{B}([0, \infty]))$ we say f is a non-negative measurable function. Recall that measurability is preserved by the vector space operations, by products, by countable suprema and infima, and by pointwise (indeed almost everywhere) limits. This last stability property is exactly what the Riemann theory lacks and is the reason we work with the Lebesgue integral at all.

§26.2.2 The integral

Definition 26.2.7 (Simple function)

A **simple function** on $(E, \mathcal{E})$ is a function of the form $f = \sum_{k=1}^N a_k \mathbb{1}_{A_k}$ with a_k scalars and $A_k \in \mathcal{E}$.

For a non-negative simple function (which we may always write with $a_k \in [0, \infty]$ and A_k disjoint) we define

$$\mu(f) = \int_E f \, d\mu = \sum_{k=1}^N a_k \mu(A_k),$$

with the convention $0 \cdot \infty = \infty \cdot 0 = 0$. For general non-negative measurable f we set

$$\mu(f) = \int_E f \, d\mu = \sup\{\mu(g) : 0 \leq g \leq f, \, g \text{ simple}\},$$

and finally a measurable f is **integrable** if $\mu(|f|) < \infty$, in which case writing $f = f^+ - f^-$ with $f^\pm \geq 0$ we set $\mu(f) = \mu(f^+) - \mu(f^-)$. Complex-valued f are handled by taking real and imaginary parts.

The integral so defined is linear, monotone, and additive over disjoint sets. The two theorems that make it usable are the following.

Theorem 26.2.8 (Monotone convergence)

If f_n are non-negative measurable with $f_n \uparrow f$ almost everywhere, then $\int_E f_n \, d\mu \uparrow \int_E f \, d\mu$.

Theorem 26.2.9 (Dominated convergence)

Suppose $f_n \rightarrow f$ almost everywhere and $|f_n| \leq g$ almost everywhere for some integrable g . Then f is integrable and $\int_E f_n \, d\mu \rightarrow \int_E f \, d\mu$.

We shall also use Fatou’s lemma ($\int \liminf f_n \leq \liminf \int f_n$ for non-negative f_n) and Fubini–Tonelli, which lets us exchange the order of a double integral provided either the

integrand is non-negative or the double integral converges absolutely. These will be used without comment.

Remark. A remark on the meaning of ' $f \in L^p$ ' that is worth making early. Two functions agreeing almost everywhere have the same integral against everything, and there is no measure-theoretic way to tell them apart. We therefore work throughout with equivalence classes of functions modulo equality almost everywhere, while continuing to write $f(x)$ as though f were a function. This costs us nothing until we want to restrict f to a set of measure zero – for instance to a hypersurface, when solving a boundary value problem – and then it costs us a whole theorem (the trace theorem, Section 26.10).

§26.2.3 Littlewood's three principles

Littlewood wrote in 1944 that there are three principles, roughly expressible as follows: *every measurable set is nearly a finite union of intervals; every measurable function is nearly continuous; every convergent sequence of measurable functions is nearly uniformly convergent.* The word 'nearly' means 'up to a set of arbitrarily small measure'. Each principle is a theorem, and it is instructive to see them proved, both because we use them and because their proofs are a compendium of the standard tricks.

The first principle is Borel regularity, essentially by definition of Lebesgue measure.

Proposition 26.2.10

Let $A \subseteq \mathbb{R}^n$ be measurable with $|A| < \infty$. Then for every $\varepsilon > 0$ there is a finite union B of boxes with $|A \Delta B| < \varepsilon$.

Proof. By Theorem 26.2.5(iii) choose an open $O \supseteq A$ with $|O \setminus A| < \varepsilon/2$. Every open subset of $\mathbb{R}^n$ is a countable disjoint union of half-open dyadic boxes $\bigcup_k Q_k$; since $|O| \leq |A| + \varepsilon/2 < \infty$, the tail $\sum_{k>N} |Q_k|$ can be made smaller than $\varepsilon/2$. Taking $B = \bigcup_{k \leq N} Q_k$ gives

$$|A \Delta B| \leq |O \setminus A| + |O \setminus B| < \varepsilon. \quad \square$$

The third principle is Egorov's theorem. Note that some hypothesis of finite measure is necessary: on $\mathbb{R}$ the functions $f_n = \mathbb{1}_{[n, n+1]}$ converge pointwise to 0 but not uniformly off any set of finite measure.

Theorem 26.2.11 (Egorov)

Let $E \subseteq \mathbb{R}^d$ be measurable with $|E| < \infty$, and let $f_n, f : E \rightarrow \mathbb{C}$ be measurable with $f_n \rightarrow f$ almost everywhere on E . Then for every $\varepsilon > 0$ there is a closed set $A_\varepsilon \subseteq E$ with $|E \setminus A_\varepsilon| < \varepsilon$ and $f_n \rightarrow f$ uniformly on A_ε .

Proof. Discarding a null set, we may assume $f_n \rightarrow f$ everywhere on E . For $n, k \in \mathbb{N}$ put

$$E_k^n = \{x \in E : |f_j(x) - f(x)| < 1/n \text{ for all } j > k\}.$$

For fixed n these increase in k , and their union over k is all of E by pointwise convergence. Since $|E| < \infty$ we have $|E_k^n| \uparrow |E|$, so we may choose k_n with $|E \setminus E_{k_n}^n| < 2^{-n}\varepsilon$.

Set $A' = \bigcap_{n \geq 1} E_{k_n}^n$. Then

$$|E \setminus A'| \leq \sum_{n \geq 1} |E \setminus E_{k_n}^n| < \varepsilon,$$

after replacing ε by $\varepsilon/2$ if we like. And the convergence is uniform on A' : given $\delta > 0$ pick $n > 1/\delta$; then for every $x \in A' \subseteq E_{k_n}^n$ and every $j > k_n$ we have $|f_j(x) - f(x)| < 1/n < \delta$, and the choice of k_n did not depend on x . Finally, Borel regularity lets us shrink A' to a closed A_ε with $|A' \setminus A_\varepsilon|$ as small as we please. $\square$

The second principle is Lusin's theorem, and its proof uses the third.

Theorem 26.2.12 (Lusin)

Let $E \subseteq \mathbb{R}^d$ be measurable with $|E| < \infty$ and let f be measurable and almost everywhere finite on E . Then for every $\varepsilon > 0$ there is a closed $F_\varepsilon \subseteq E$ with $|E \setminus F_\varepsilon| < \varepsilon$ such that $f|_{F_\varepsilon}$ is continuous.

Proof. Suppose first that $f = \sum_{n=1}^m a_n \mathbb{1}_{A_n}$ is simple, with the A_n a partition of E . By Borel regularity choose compact $K_n \subseteq A_n$ with $|A_n \setminus K_n| < \varepsilon/m$, and set $B = \bigcup_{n=1}^m K_n$, so that $|E \setminus B| < \varepsilon$. The sets K_n are compact and disjoint, hence pairwise at positive distance; as f is constant on each K_n it is continuous on B .

For general f , choose simple $f_n \rightarrow f$ almost everywhere, and by the previous paragraph choose measurable C_n with $|C_n| < 2^{-n}$ and f_n continuous on $E \setminus C_n$. By Egorov there is $A \subseteq E$ with $|E \setminus A| < \varepsilon/3$ and $f_n \rightarrow f$ uniformly on A . Pick N with $\sum_{n \geq N} 2^{-n} < \varepsilon/3$ and set

$$F' = A \setminus \bigcup_{n \geq N} C_n,$$

so $|E \setminus F'| < 2\varepsilon/3$. On F' each f_n ($n \geq N$) is continuous and $f_n \rightarrow f$ uniformly, so $f|_{F'}$ is continuous. Borel regularity again produces a closed $F_\varepsilon \subseteq F'$ of nearly the same measure. $\square$

Remark. A word of warning, because this trips up everyone at least once. To say ' $f|_F$ is continuous' is a statement about f restricted to F with its subspace topology; it is emphatically *not* the statement that f , as a function on E , is continuous at each point of F . For a spectacular example take $f = \mathbb{1}_{\mathbb{Q}}$ on $\mathbb{R}$. Then f is continuous at no point whatever, but $f|_{\mathbb{R} \setminus \mathbb{Q}} \equiv 0$ is as continuous as one could wish.

§26.3 L^p Spaces

Every measure space carries with it a family of Banach spaces, one for each $p \in [1, \infty]$, and these are the spaces in which we shall spend the rest of the course. Their theory is a rare thing in analysis: it is genuinely uniform in p , so that a single argument handles a continuum of spaces, and yet the endpoints $p = 1$ and $p = \infty$ are always exceptional, and the exceptions are always interesting.

§26.3.1 Definitions

Definition 26.3.1 (L^p norm)

Let $(E, \mathcal{E}, \mu)$ be a measure space and $f : E \rightarrow \mathbb{C}$ measurable. For $1 \leq p < \infty$ we define

$$\|f\|_{L^p} = \left(\int_E |f|^p d\mu \right)^{1/p},$$

and we define the **essential supremum**

$$\|f\|_{L^\infty} = \operatorname{ess\,sup}_E |f| = \inf \{K \geq 0 : |f| \leq K \text{ almost everywhere}\}.$$

For $1 \leq p \leq \infty$, the L^p **space** over E is

$$L^p(E, \mu) = \{f : E \rightarrow \mathbb{C} \text{ measurable} : \|f\|_{L^p} < \infty\} / \sim,$$

where $f \sim g$ if $f = g$ almost everywhere.

When $(E, \mathcal{E}, \mu) = (\mathbb{R}^n, \mathcal{M}, \lambda)$ we write simply $L^p(\mathbb{R}^n)$, and when $\Omega \subseteq \mathbb{R}^n$ is measurable, $L^p(\Omega)$ means $L^p(\Omega, \lambda|_\Omega)$. The quotient by almost-everywhere equality is not a technicality we can dispense with: without it $\|\cdot\|_{L^p}$ is only a seminorm, since a function supported on a null set has norm zero.

Note that the infimum in the definition of $\operatorname{ess\,sup}$ is attained – if $|f| \leq K_n$ a.e. and $K_n \downarrow K$ then $|f| \leq K$ off the union of countably many null sets – so $|f| \leq \|f\|_{L^\infty}$ almost everywhere, as one would hope.

Example 26.3.2

Let $E = \mathbb{N}$ with counting measure. Then $L^p(\mathbb{N})$ is the sequence space ℓ^p of sequences (a_n) with $\sum |a_n|^p < \infty$ (with the obvious modification at $p = \infty$). Since no non-empty set has measure zero, there is no quotient to take, and ‘almost everywhere’ means ‘everywhere’.

Example 26.3.3

Let $E = [0, 1]$ with Lebesgue measure and $f(x) = x^{-\alpha}$ for $\alpha > 0$. Then $\int_0^1 x^{-\alpha p} dx < \infty$ if and only if $\alpha p < 1$, so $f \in L^p([0, 1])$ exactly when $p < 1/\alpha$. Thus by varying α we can produce a function lying in L^p for all p below any prescribed threshold and in no L^p above it. The spaces really are different.

§26.3.2 Hölder and Minkowski

The two fundamental inequalities both come from convexity. We say $p, q \in [1, \infty]$ are **conjugate exponents** if

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with the convention $1/\infty = 0$; so 1 and ∞ are conjugate, and 2 is self-conjugate.

Lemma 26.3.4 (Young’s inequality)

Let $p, q \in (1, \infty)$ be conjugate and $a, b \geq 0$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality if and only if $a^p = b^q$.

Proof. If a or b is zero this is clear, so assume both positive. The logarithm is strictly concave, so

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \log(a^p) + \frac{1}{q} \log(b^q) = \log a + \log b = \log(ab),$$

using $1/p + 1/q = 1$, and strict concavity gives equality precisely when $a^p = b^q$. Exponentiating gives the result. $\square$

Theorem 26.3.5 (Hölder's inequality)

Let $p, q \in [1, \infty]$ be conjugate and let $f \in L^p(E, \mu)$, $g \in L^q(E, \mu)$. Then $fg \in L^1(E, \mu)$ and

$$\|fg\|_{L^1} \leq \|f\|_{L^p} \|g\|_{L^q}.$$

Proof. The cases $p = 1, q = \infty$ (and its mirror) are immediate from $|fg| \leq |f| \|g\|_{L^\infty}$ almost everywhere. So suppose $1 < p < \infty$. If either norm is zero the corresponding function vanishes a.e. and there is nothing to prove; and if either is infinite there is nothing to prove either. So normalise: let $F = |f|/\|f\|_{L^p}$ and $G = |g|/\|g\|_{L^q}$, so that $\|F\|_{L^p} = \|G\|_{L^q} = 1$. By Lemma 26.3.4 applied pointwise,

$$FG \leq \frac{F^p}{p} + \frac{G^q}{q},$$

and integrating,

$$\int_E FG \, d\mu \leq \frac{1}{p} \int_E F^p \, d\mu + \frac{1}{q} \int_E G^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1.$$

Multiplying through by $\|f\|_{L^p} \|g\|_{L^q}$ gives the claim. $\square$

Hölder's inequality is the reason conjugate exponents are worth naming. It says the pairing $(f, g) \mapsto \int fg$ is bounded on $L^p \times L^q$, and we shall see in Section 26.5 that it accounts for *all* bounded linear functionals on L^p when $p < \infty$.

Theorem 26.3.6 (Minkowski's inequality)

Let $1 \leq p \leq \infty$ and $f, g \in L^p(E, \mu)$. Then $f + g \in L^p(E, \mu)$ and

$$\|f + g\|_{L^p} \leq \|f\|_{L^p} + \|g\|_{L^p}.$$

Proof. The cases $p = 1$ and $p = \infty$ are immediate from the pointwise triangle inequality, so take $1 < p < \infty$. Since $|f + g|^p \leq (2 \max(|f|, |g|))^p \leq 2^p(|f|^p + |g|^p)$,

the left-hand side is finite. Now write

$$\int_E |f + g|^p \, d\mu \leq \int_E |f| |f + g|^{p-1} \, d\mu + \int_E |g| |f + g|^{p-1} \, d\mu,$$

and apply Hölder to each term, noting that $|f + g|^{p-1} \in L^q$ with

$$\| |f + g|^{p-1} \|_{L^q} = \left(\int_E |f + g|^{(p-1)q} \, d\mu \right)^{1/q} = \|f + g\|_{L^p}^{p/q}$$

since $(p-1)q = p$. This gives

$$\|f + g\|_{L^p}^p \leq (\|f\|_{L^p} + \|g\|_{L^p}) \|f + g\|_{L^p}^{p/q}.$$

If $\|f + g\|_{L^p} = 0$ we are done; otherwise divide by $\|f + g\|_{L^p}^{p/q}$ and use $p - p/q = 1$. $\square$

So $L^p(E, \mu)$ is a normed vector space. We will also want the following integral form of Minkowski's inequality, which says that 'the norm of an integral is at most the integral of the norms' – a continuous version of the triangle inequality, with the sum replaced by an integral.

Lemma 26.3.7 (Minkowski's integral inequality)

Let $F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{C}$ be measurable with $F(\cdot, y) \in L^p$ for a.e. y , and let $1 \leq p < \infty$. Then

$$\left(\int_{\mathbb{R}^n} \left| \int_{\mathbb{R}^n} F(x, y) \, dy \right|^p \, dx \right)^{1/p} \leq \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |F(x, y)|^p \, dx \right)^{1/p} \, dy.$$

Proof. For $p = 1$ this is exactly Tonelli's theorem, so suppose $1 < p < \infty$, and write $G(x) = \int_{\mathbb{R}^n} |F(x, y)| \, dy$, which dominates the modulus of the inner integral on the left. Suppose first that $\|G\|_{L^p} < \infty$. Then, using Tonelli to exchange the order of integration and then Hölder in the x variable,

$$\begin{aligned} \int_{\mathbb{R}^n} G(x)^p \, dx &= \int_{\mathbb{R}^n} G(x)^{p-1} \int_{\mathbb{R}^n} |F(x, y)| \, dy \, dx \\ &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} G(x)^{p-1} |F(x, y)| \, dx \right) \, dy \\ &\leq \int_{\mathbb{R}^n} \|G^{p-1}\|_{L^q} \|F(\cdot, y)\|_{L^p} \, dy = \|G\|_{L^p}^{p/q} \int_{\mathbb{R}^n} \|F(\cdot, y)\|_{L^p} \, dy, \end{aligned}$$

since $(p-1)q = p$. If $\|G\|_{L^p} = 0$ there is nothing to prove; otherwise divide by $\|G\|_{L^p}^{p/q}$ and use $p - p/q = 1$.

It remains to remove the assumption $\|G\|_{L^p} < \infty$. Replace F by $F \mathbb{1}_{K \times K'}$ for compact sets K, K' and by $\min(|F|, m)$: for the truncated function G is bounded with bounded support, so the above applies, and the right-hand side only increases if we untruncate. Letting $m \rightarrow \infty$ and $K, K' \uparrow \mathbb{R}^n$, monotone convergence gives the general case. $\square$

§26.3.3 Completeness

The next theorem is the reason L^p spaces are useful rather than merely definable.

Theorem 26.3.8 (Riesz–Fischer)

For every $1 \leq p \leq \infty$, the space $(L^p(E, \mu), \|\cdot\|_{L^p})$ is a Banach space.

Proof. Recall that a normed space is complete if and only if every absolutely convergent series converges. So let (f_k) be a sequence in L^p with $\sum_k \|f_k\|_{L^p} = M < \infty$.

Take first $p < \infty$. Set $g_N = \sum_{k=1}^N |f_k|$ and $g = \sum_{k=1}^\infty |f_k| \in [0, \infty]$, so $g_N \uparrow g$ pointwise. By Minkowski $\|g_N\|_{L^p} \leq M$ for every N , and by monotone convergence applied to $g_N^p \uparrow g^p$,

$$\int_E g^p \, d\mu = \lim_N \int_E g_N^p \, d\mu \leq M^p.$$

In particular $g < \infty$ almost everywhere, so the series $\sum_k f_k(x)$ converges absolutely for almost every x ; call its sum $f(x)$ (and set $f = 0$ on the null set where it does not converge). Then $|f| \leq g$, so $f \in L^p$. Moreover

$$\left| f - \sum_{k=1}^N f_k \right|^p \leq \left(\sum_{k>N} |f_k| \right)^p \leq g^p \in L^1,$$

and the left-hand side tends to 0 pointwise a.e., so dominated convergence gives $\left\| f - \sum_{k \leq N} f_k \right\|_{L^p} \rightarrow 0$ as required.

For $p = \infty$, let $A_k = \{|f_k| > \|f_k\|_{L^\infty}\}$, a null set, and let $A = \bigcup_k A_k$, also null. Off A the series $\sum_k f_k$ converges uniformly by the Weierstrass M -test, to a bounded limit f , and uniform convergence off a null set is exactly convergence in L^∞ . $\square$

The $p = 2$ case is special, because the norm comes from an inner product

$$\langle f, g \rangle_{L^2} = \int_E \bar{f} g \, d\mu,$$

which is finite by Hölder (this is Cauchy–Schwarz). So $L^2(E, \mu)$ is a Hilbert space, and we may use all the geometry – orthogonal projection, orthonormal bases, the Riesz representation theorem – that Hilbert space affords. Much of what we do later works only for $p = 2$ for exactly this reason.

§26.3.4 Inclusions between L^p spaces

It is natural to ask how the spaces $L^p(E, \mu)$ sit relative to one another as p varies. The answer depends entirely on the measure space, and the two extreme cases are worth having clearly in mind.

Proposition 26.3.9

Suppose $\mu(E) < \infty$ and $1 \leq p \leq q \leq \infty$. Then $L^q(E, \mu) \subseteq L^p(E, \mu)$, and

$$\|f\|_{L^p} \leq \mu(E)^{\frac{1}{p} - \frac{1}{q}} \|f\|_{L^q}.$$

Proof. If $q = \infty$ then $\int |f|^p \leq \|f\|_{L^\infty}^p \mu(E)$ directly. Otherwise apply Hölder to the pair $|f|^p \in L^{q/p}$ and $1 \in L^r$ where r is conjugate to q/p :

$$\int_E |f|^p d\mu \leq \left(\int_E |f|^q d\mu \right)^{p/q} \mu(E)^{1-p/q},$$

and take p -th roots. □

On a space of infinite measure the situation is completely different: there are *no* inclusions at all between distinct $L^p(\mathbb{R}^n)$. A function can fail to be in L^p for two quite different reasons – it can be too singular locally, or it can decay too slowly at infinity – and raising p helps with one while hurting the other.

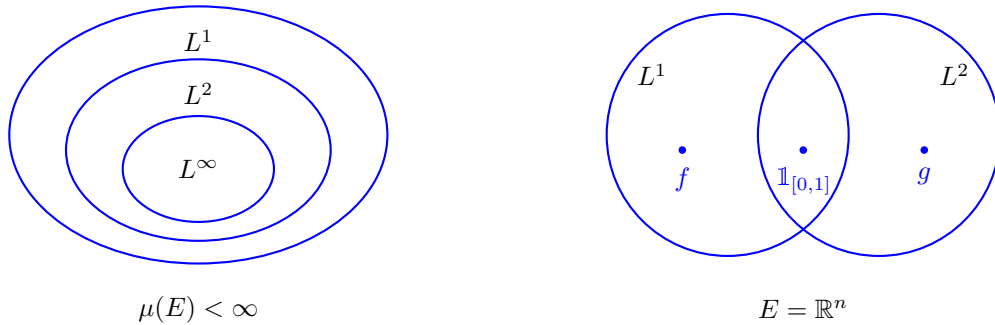
Example 26.3.10

Work on $\mathbb{R}$, and let $1 \leq p < q < \infty$. Then

$$f(x) = |x|^{-1/q} \mathbb{1}_{\{|x|<1\}} \in L^p \setminus L^q, \quad g(x) = |x|^{-1/p} \mathbb{1}_{\{|x|>1\}} \in L^q \setminus L^p.$$

Indeed $\int_0^1 x^{-p/q} dx < \infty$ since $p/q < 1$, while $\int_0^1 x^{-1} dx = \infty$; and $\int_1^\infty x^{-q/p} dx < \infty$ since $q/p > 1$, while $\int_1^\infty x^{-1} dx = \infty$.

The following picture summarises the two situations. On the left, a finite measure space, where the L^p spaces are nested and higher exponents mean smaller spaces; on the right, $\mathbb{R}^n$, where the spaces genuinely overlap and neither contains the other.



There is one general statement that survives on any measure space, and it is a useful one: the L^p norms are logarithmically convex in $1/p$.

Proposition 26.3.11 (Interpolation)

Let $1 \leq p < r < q \leq \infty$ and write $\frac{1}{r} = \frac{\theta}{p} + \frac{1-\theta}{q}$ for some $\theta \in (0, 1)$. Then for measurable f ,

$$\|f\|_{L^r} \leq \|f\|_{L^p}^\theta \|f\|_{L^q}^{1-\theta}.$$

In particular $L^p \cap L^q \subseteq L^r$.

Proof. Suppose that $q < \infty$; the case $q = \infty$ is a similar but easier variant. We factorise $|f|^r = |f|^{\theta r} |f|^{(1-\theta)r}$ and apply Hölder with the pair $p/(\theta r)$ and $q/((1-\theta)r)$,

which are indeed conjugate by the definition of θ :

$$\int |f|^r \leq \left(\int |f|^p \right)^{\theta r/p} \left(\int |f|^q \right)^{(1-\theta)r/q}.$$

Taking r -th roots gives the result. $\square$

§26.3.5 Density of simple functions

Every approximation theorem in this course begins with simple functions, so let us record the basic statement.

Proposition 26.3.12

Let $1 \leq p < \infty$. The set of simple functions s with $\mu(\{s \neq 0\}) < \infty$ is dense in $L^p(E, \mu)$. For $p = \infty$, the simple functions are dense, but the finite-support condition cannot be imposed.

Proof. Let $f \in L^p$ and, splitting into real and imaginary and then positive and negative parts, assume $f \geq 0$. Choose simple $0 \leq s_n \uparrow f$ pointwise (the standard dyadic approximation does this). Then $|f - s_n|^p \leq f^p \in L^1$ and $|f - s_n|^p \rightarrow 0$ pointwise, so $\|f - s_n\|_{L^p} \rightarrow 0$ by dominated convergence. Each s_n has $\mu(\{s_n \neq 0\}) < \infty$: indeed $s_n \leq f$, so on $\{s_n \geq \varepsilon\}$ we have $\mu(\{s_n \geq \varepsilon\}) \leq \varepsilon^{-p} \|f\|_{L^p}^p < \infty$ by Chebyshev, and s_n takes finitely many values.

For $p = \infty$ the dyadic approximation converges uniformly on the set where f is bounded, which is almost everything; but a simple function of finite support cannot approximate the constant function 1 on $\mathbb{R}^n$ in L^∞ . $\square$

Simple functions are, as their name suggests, simple, but they are not *pleasant*: they are discontinuous, and one cannot differentiate them. In Section 26.4 we shall upgrade this density statement dramatically, to smooth compactly supported functions. That is a much more useful statement, and the mechanism which produces it – mollification – is one of the central tools of the course.

§26.3.6 Separability

Definition 26.3.13

A metric space is **separable** if it has a countable dense subset.

Separability is a smallness condition, and we shall need it in two places: to run a diagonal argument in the proof of Banach–Alaoglu, and to know that L^2 has a countable orthonormal basis.

Proposition 26.3.14

For $1 \leq p < \infty$, the space $L^p(\mathbb{R}^n)$ is separable. The space $L^\infty(\mathbb{R}^n)$ is not.

Proof. For separability, consider the countable family $\mathcal{C}$ of finite rational linear combinations of indicators of boxes with rational vertices. Given $f \in L^p$ and $\varepsilon > 0$,

Proposition 26.3.12 produces a simple $s = \sum_{k \leq N} a_k \mathbb{1}_{A_k}$ with finite-measure A_k and $\|f - s\|_{L^p} < \varepsilon/2$. Each $\mathbb{1}_{A_k}$ is approximated in L^p by an indicator of a finite union of rational boxes, by Proposition 26.2.10 together with $\|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} = |A \triangle B|^{1/p}$; and each a_k may be replaced by a nearby rational. So $\mathcal{C}$ is dense.

For L^∞ , note that $\|\mathbb{1}_{B_r(0)} - \mathbb{1}_{B_s(0)}\|_{L^\infty} = 1$ whenever $r \neq s$. This is an uncountable family of points pairwise at distance 1, and no countable set can be dense in such a space (each point of the family needs its own approximant). $\square$

§26.4 Convolution and Mollification

We now come to the single most useful technique in the course. The idea is very simple. Averaging a function against a small smooth bump produces a smooth function, and if the bump is small enough the average is close to the original. So every L^p function ($p < \infty$) is a limit of smooth functions, and any statement which is stable under L^p limits need only be checked for smooth functions.

§26.4.1 Convolution

Definition 26.4.1 (Convolution)

For measurable $f, g : \mathbb{R}^n \rightarrow \mathbb{C}$ we define the **convolution**

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x - y) \, dy,$$

whenever the integral exists.

The integral certainly exists for every x when, say, $f \in L^1(\mathbb{R}^n)$ and $g \in L^\infty(\mathbb{R}^n)$, or when $f, g \in C_c(\mathbb{R}^n)$; we shall see in a moment exactly how much integrability is needed. The following bookkeeping lemma records the algebraic properties.

Lemma 26.4.2

For $f, g, h \in C_c^\infty(\mathbb{R}^n)$ we have $f * g = g * f$ and $(f * g) * h = f * (g * h)$. Moreover

$$\int_{\mathbb{R}^n} f * g = \left(\int_{\mathbb{R}^n} f \right) \left(\int_{\mathbb{R}^n} g \right).$$

Proof. Commutativity is the substitution $z = x - y$; associativity and the last identity are Fubini together with a change of variables, all of which is justified since everything in sight is continuous and compactly supported. (The hypotheses can be weakened considerably – for instance the last identity holds for $f, g \in L^1$ – but this version suffices for us.) $\square$

Before turning to regularity we should record when the defining integral converges, and in which space the answer lives. The following is the basic estimate, and the case $q = 1$ of it is the one we shall use on almost every subsequent page.

Theorem 26.4.3 (Young's convolution inequality)

Let $1 \leq p \leq \infty$.

- (i) If $f \in L^p(\mathbb{R}^n)$ and $g \in L^1(\mathbb{R}^n)$ then $(f * g)(x)$ converges absolutely for almost every x , and $f * g \in L^p(\mathbb{R}^n)$ with

$$\|f * g\|_{L^p} \leq \|f\|_{L^p} \|g\|_{L^1}.$$

- (ii) If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$ with q conjugate to p , then $(f * g)(x)$ converges absolutely for every x , and $f * g \in L^\infty(\mathbb{R}^n)$ with $\|f * g\|_{L^\infty} \leq \|f\|_{L^p} \|g\|_{L^q}$.

Proof. (i) For $p = \infty$ this is immediate, so take $p < \infty$ and set $F(x, y) = f(x - y)g(y)$. Then for each fixed y ,

$$\|F(\cdot, y)\|_{L^p} = |g(y)| \left(\int_{\mathbb{R}^n} |f(x - y)|^p dx \right)^{1/p} = |g(y)| \|f\|_{L^p}$$

by translation invariance of Lebesgue measure. Minkowski's integral inequality (Lemma 26.3.7) applied to $|F|$ gives

$$\left(\int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} |f(x - y)g(y)| dy \right)^p dx \right)^{1/p} \leq \int_{\mathbb{R}^n} |g(y)| \|f\|_{L^p} dy = \|f\|_{L^p} \|g\|_{L^1}.$$

In particular the inner integral is finite for almost every x , which is the absolute convergence, and the resulting function dominates $|f * g|$.

- (ii) This is Hölder applied at each fixed x , using that $y \mapsto f(x - y)$ has the same L^p norm as f . $\square$

Remark. The two cases are the endpoints of the general statement: if $1 \leq p, q, r \leq \infty$ satisfy $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$ then $\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}$. One can deduce this from the two cases above by interpolation, or prove it directly by splitting $|f(x - y)g(y)|$ into three factors and applying Hölder with three exponents. We shall not need it.

Convolution is very much a smoothing operation, in a sense the next result makes precise: $f * g$ inherits the *best* regularity of its two factors, not the worst. To state it we need the standard notation for higher derivatives.

Definition 26.4.4 (Multi-index)

An (n) -**multi-index** is an element $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_{\geq 0}^n$. We write $|\alpha| = \alpha_1 + \dots + \alpha_n$ and $\alpha! = \alpha_1! \dots \alpha_n!$, and for $x \in \mathbb{R}^n$ we write $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$. For $f \in C^k(\mathbb{R}^n)$ with $k \geq |\alpha|$ we write

$$D^\alpha f = \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_n^{\alpha_n}}.$$

Definition 26.4.5

We say $f \in L_{\text{loc}}^p(\mathbb{R}^n)$ if $f \mathbb{1}_K \in L^p(\mathbb{R}^n)$ for every compact $K \subseteq \mathbb{R}^n$.

Theorem 26.4.6

Suppose $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $g \in C^k_c(\mathbb{R}^n)$ for some $k \geq 0$. Then $f * g \in C^k(\mathbb{R}^n)$, and $D^\alpha(f * g) = f * D^\alpha g$ for every multi-index α with $|\alpha| \leq k$.

Proof. Write $\tau_\xi f(x) = f(x - \xi)$ for translation. Note that $\tau_\xi(f * g) = f * (\tau_\xi g)$ straight from the definition.

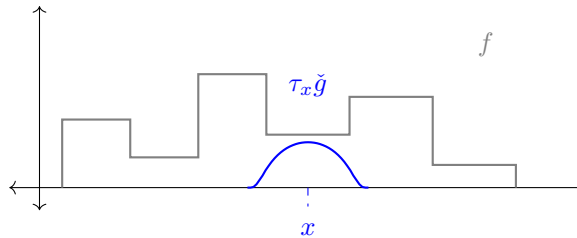
Take $k = 0$ first. Choose R so large that $\text{supp } g \subseteq B_{R-1}(0)$. As $|\xi| \rightarrow 0$ we have $\tau_\xi g \rightarrow g$ pointwise, and $|\tau_\xi g| \leq (\sup |g|)\mathbb{1}_{B_R(0)}$ for $|\xi| < 1$; since f is locally integrable, $|f|(\sup |g|)\mathbb{1}_{B_R(0)}$ is an integrable dominating function on the region that matters. Dominated convergence therefore gives $\tau_\xi(f * g) \rightarrow f * g$ pointwise, i.e. $f * g$ is continuous.

Take $k = 1$. Let $\Delta_i^h g(x) = h^{-1}(g(x + he_i) - g(x))$, so $\Delta_i^h g \rightarrow D_i g$ pointwise as $h \rightarrow 0$. By the mean value theorem $\Delta_i^h g(x) = D_i g(x + te_i)$ for some $|t| \leq |h|$, so $|\Delta_i^h g| \leq (\sup |D_i g|)\mathbb{1}_{B_R(0)}$ for $|h| < 1$, again a legitimate dominating function. Hence, by dominated convergence,

$$\frac{(f * g)(x + he_i) - (f * g)(x)}{h} = (f * \Delta_i^h g)(x) \rightarrow (f * D_i g)(x).$$

So $f * g$ has partial derivatives $f * D_i g$, and these are continuous by the case $k = 0$; hence $f * g \in C^1$. Induction on k finishes the proof. $\square$

The picture to have in mind is that of a fixed bump g being slid across f : the value $(f * g)(x)$ is a weighted average of f over a window centred at x , and moving the window moves the average smoothly, no matter how badly behaved f is inside it.

**§26.4.2 Approximate identities****Definition 26.4.7 (Mollifier)**

Fix $\phi \in C^\infty_c(\mathbb{R}^n)$ with $\phi \geq 0$ and $\int_{\mathbb{R}^n} \phi = 1$, supported in the unit ball. For $\varepsilon > 0$ set

$$\phi_\varepsilon(y) = \varepsilon^{-n} \phi(y/\varepsilon).$$

The family $(\phi_\varepsilon)_{\varepsilon > 0}$ is called a **mollifier**, or an **approximate identity**.

Each ϕ_ε is smooth, non-negative, supported in $B_\varepsilon(0)$, and has integral 1 (the factor ε^{-n} is exactly what is needed for this). So as $\varepsilon \downarrow 0$ the graphs become taller and narrower, converging in a sense we will make precise later to the Dirac delta.

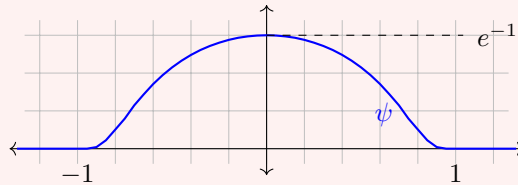
Of course we should check such a ϕ exists, and the standard construction is worth seeing because the same function will reappear when we build test functions in Section 26.9.

Example 26.4.8 (The standard bump)

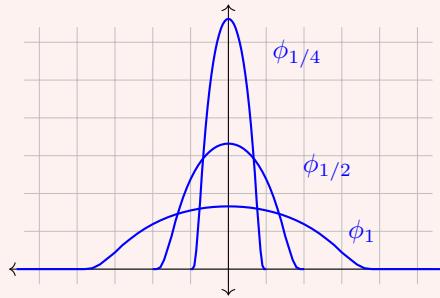
Define $\psi : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\psi(x) = \begin{cases} \exp\left(\frac{-1}{1-x^2}\right) & |x| < 1, \\ 0 & |x| \geq 1. \end{cases}$$

This is smooth. Away from $x = \pm 1$ this is clear. At $x = \pm 1$, every derivative of ψ on $(-1, 1)$ is of the form $R(x)(1-x^2)^{-m}e^{-1/(1-x^2)}$ for a polynomial R and integer m , and $t^{-m}e^{-1/t} \rightarrow 0$ as $t \downarrow 0$ since the exponential beats every power. So every derivative extends continuously by 0, and $\psi \in C_c^\infty(\mathbb{R})$ with $\text{supp } \psi = [-1, 1]$.



Normalising, $\phi = \psi / \int_{\mathbb{R}} \psi$ is a legitimate mollifier on $\mathbb{R}$; in $\mathbb{R}^n$ one takes $\psi(|x|^2)$ suitably scaled, or simply a product of one-dimensional bumps. The rescaled family ϕ_ε concentrates as $\varepsilon \downarrow 0$:



Each of these has the same area, namely 1.

Before proving that mollification converges, we need one lemma, which says that translation acts continuously on L^p . It is intuitively obvious and genuinely needs proof.

Lemma 26.4.9 (Continuity of translation)

Let $1 \leq p < \infty$ and $g \in L^p(\mathbb{R}^n)$. Then $\|\tau_z g - g\|_{L^p} \rightarrow 0$ as $z \rightarrow 0$.

Proof. This is a density argument of exactly the shape we shall use again and again. First suppose $g = \mathbb{1}_R$ for a box R . Then $\|\tau_z \mathbb{1}_R - \mathbb{1}_R\|_{L^p}^p = |R \Delta (R+z)| \rightarrow 0$ as $z \rightarrow 0$, by direct computation with the side lengths. By linearity the result follows for finite disjoint unions of boxes.

Next let B be measurable with $|B| < \infty$. Given $\varepsilon > 0$, Proposition 26.2.10 gives a finite union of boxes A with $\|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} = |A \Delta B|^{1/p} < \varepsilon$. Since translation is an isometry of L^p ,

$$\|\tau_z \mathbb{1}_B - \mathbb{1}_B\|_{L^p} \leq \|\tau_z \mathbb{1}_B - \tau_z \mathbb{1}_A\|_{L^p} + \|\tau_z \mathbb{1}_A - \mathbb{1}_A\|_{L^p} + \|\mathbb{1}_A - \mathbb{1}_B\|_{L^p} < 3\varepsilon$$

for $|z|$ small enough. Hence the result holds for indicators of finite-measure sets, so for simple functions of finite support, so – again by the isometry of translation and a three-term estimate – for all of L^p by Proposition 26.3.12. $\square$

Remark. This fails for $p = \infty$: take $g = \mathbb{1}_{[0,\infty)}$, for which $\|\tau_z g - g\|_{L^\infty} = 1$ for every $z \neq 0$. The failure is the same one that will prevent us from approximating L^∞ functions by continuous ones.

Theorem 26.4.10 (Mollification)

Let (ϕ_ε) be a mollifier and let $g \in L^p(\mathbb{R}^n)$ with $1 \leq p < \infty$. Then $\phi_\varepsilon * g \in C^\infty(\mathbb{R}^n)$, $\|\phi_\varepsilon * g\|_{L^p} \leq \|g\|_{L^p}$, and

$$\phi_\varepsilon * g \rightarrow g \quad \text{in } L^p(\mathbb{R}^n) \text{ as } \varepsilon \rightarrow 0.$$

Proof. Smoothness is Theorem 26.4.6: the convolution is defined at every point because $\int_{B_\varepsilon(x)} |g| \leq \|g\|_{L^p} \|\mathbb{1}_{B_\varepsilon(x)}\|_{L^q} < \infty$ by Hölder, so $g \in L^1_{\text{loc}}$, and $\phi_\varepsilon \in C_c^\infty$.

For the convergence, we estimate pointwise. Substituting $y = \varepsilon z$ and using $\int \phi = 1$,

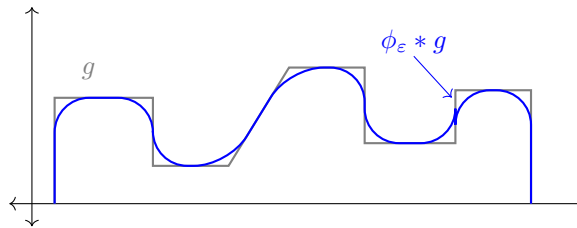
$$\begin{aligned} |(\phi_\varepsilon * g)(x) - g(x)| &= \left| \int_{\mathbb{R}^n} \phi_\varepsilon(y) g(x - y) \, dy - g(x) \right| \\ &= \left| \int_{\mathbb{R}^n} \phi(z) (g(x - \varepsilon z) - g(x)) \, dz \right| \\ &\leq \int_{\mathbb{R}^n} \phi(z) |\tau_{\varepsilon z} g(x) - g(x)| \, dz. \end{aligned}$$

Now take L^p norms in x and apply Minkowski's integral inequality (Lemma 26.3.7) to move the norm inside the z -integral:

$$\|\phi_\varepsilon * g - g\|_{L^p} \leq \int_{\mathbb{R}^n} \phi(z) \|\tau_{\varepsilon z} g - g\|_{L^p} \, dz.$$

The integrand is bounded by $2\phi(z) \|g\|_{L^p} \in L^1$ and tends to 0 pointwise in z as $\varepsilon \rightarrow 0$, by Lemma 26.4.9. So the integral tends to 0 by dominated convergence. The norm bound $\|\phi_\varepsilon * g\|_{L^p} \leq \|g\|_{L^p}$ follows from the same application of Minkowski's integral inequality without the subtraction. $\square$

Here is the picture. Convolving a rough function with ϕ_ε rounds off its corners and jumps on the scale ε , and leaves it alone where it was already smooth on that scale.



Corollary 26.4.11

For $1 \leq p < \infty$, the space $C_c^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$.

Proof. Given $g \in L^p$ and $\varepsilon > 0$, first truncate: $g \mathbb{1}_{B_R(0)} \rightarrow g$ in L^p as $R \rightarrow \infty$ by dominated convergence, so we may assume g has compact support. Now $\phi_\delta * g \rightarrow g$ in L^p by Theorem 26.4.10, and $\phi_\delta * g$ is smooth with support contained in the δ -neighbourhood of $\text{supp } g$, hence compact. $\square$

This is not true for $p = \infty$, and the reason is worth stating: a uniform limit of continuous functions is continuous, so no sequence in $C(\mathbb{R}^n)$ can converge in L^∞ to a function with a genuine jump. The space L^∞ is simply too big – it is not separable either – and this is the first of several places where it will have to be treated separately.

Corollary 26.4.12

Suppose $f \in L^1_{\text{loc}}(\Omega)$ for $\Omega \subseteq \mathbb{R}^n$ open and

$$\int_{\Omega} f \varphi \, dx = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

Then $f = 0$ almost everywhere.

Proof. Fix a ball B with $\overline{B} \subseteq \Omega$; it suffices to show $f = 0$ a.e. on B . For $x \in B$ and ε smaller than the distance from B to $\partial\Omega$, the function $y \mapsto \phi_\varepsilon(x - y)$ lies in $C_c^\infty(\Omega)$, so the hypothesis gives $(\phi_\varepsilon * f)(x) = 0$. But $f \mathbb{1}_B \in L^1$, so $\phi_\varepsilon * (f \mathbb{1}_B) \rightarrow f \mathbb{1}_B$ in L^1 by Theorem 26.4.10, and for x well inside B these agree. Hence $f = 0$ a.e. on B . $\square$

This innocuous corollary – often called the fundamental lemma of the calculus of variations – is what will allow us to pass from ‘the equation holds when tested against every smooth function’ to ‘the equation holds’. It is used every time we solve a PDE weakly.

§26.4.3 The Lebesgue differentiation theorem

Mollification says that averages of f over small balls converge to f in L^p . It is natural to ask whether they converge *pointwise*, and the answer is yes, almost everywhere. This is a genuinely non-trivial theorem, and the standard proof introduces a tool – the maximal function – which is the beginning of a whole subject.

Definition 26.4.13 (Hardy–Littlewood maximal function)

For $f \in L^1(\mathbb{R}^n)$, define $Mf : \mathbb{R}^n \rightarrow [0, \infty]$ by

$$Mf(x) = \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)| \, dy.$$

The maximal function is never in L^1 unless $f = 0$ (it decays like $|x|^{-n}$ at infinity, which is not integrable), so the best we can hope for is the following *weak type* bound.

Lemma 26.4.14

Let $f \in L^1(\mathbb{R}^n)$. Then Mf is measurable and finite almost everywhere, and there

is a constant C_n depending only on n with

$$|\{Mf > \lambda\}| \leq \frac{C_n \|f\|_{L^1}}{\lambda} \quad \text{for all } \lambda > 0.$$

Proof. Write $A_\lambda = \{Mf > \lambda\}$. For each $x \in A_\lambda$ there is $r_x > 0$ with

$$\frac{1}{|B_{r_x}(x)|} \int_{B_{r_x}(x)} |f| > \lambda.$$

We first check A_λ is open (which gives measurability of Mf , since A_λ is a superlevel set). If $x_m \rightarrow x$ with $x \in A_\lambda$, then $\mathbb{1}_{B_{r_x}(x_m)} \rightarrow \mathbb{1}_{B_{r_x}(x)}$ almost everywhere and dominated convergence gives

$$\frac{1}{|B_{r_x}(x_m)|} \int_{B_{r_x}(x_m)} |f| \rightarrow \frac{1}{|B_{r_x}(x)|} \int_{B_{r_x}(x)} |f| > \lambda,$$

so $x_m \in A_\lambda$ for large m . Thus the complement of A_λ is closed under limits, i.e. A_λ is open.

Now let $K \subseteq A_\lambda$ be compact. The balls $\{B_{r_x}(x) : x \in A_\lambda\}$ cover K , so finitely many $B_1, \dots, B_N$ do. By the Wiener covering lemma there is a disjoint subcollection $B_{i_1}, \dots, B_{i_k}$ with $|\bigcup_{i \leq N} B_i| \leq 3^n \sum_{j \leq k} |B_{i_j}|$. Hence

$$|K| \leq 3^n \sum_{j=1}^k |B_{i_j}| \leq \frac{3^n}{\lambda} \sum_{j=1}^k \int_{B_{i_j}} |f| \leq \frac{3^n}{\lambda} \int_{\mathbb{R}^n} |f| = \frac{3^n \|f\|_{L^1}}{\lambda},$$

using disjointness in the last inequality. Taking the supremum over compact $K \subseteq A_\lambda$ and using inner regularity gives $|A_\lambda| \leq 3^n \|f\|_{L^1} / \lambda$. Finally $\{Mf = \infty\} \subseteq A_\lambda$ for every λ , so it is null. $\square$

Theorem 26.4.15 (Lebesgue differentiation theorem)

Let $f \in L^1(\mathbb{R}^n)$. Then for almost every $x \in \mathbb{R}^n$,

$$\lim_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| \, dy = 0.$$

Such x are called **Lebesgue points** of f .

The idea of the proof is a beautiful one and is used throughout analysis: the statement is trivially true for continuous functions, continuous functions are dense, and the maximal inequality controls the error made in passing to the limit. This last ingredient is essential – density alone is never enough for a pointwise statement.

Proof. For $\lambda > 0$ let

$$A_\lambda = \left\{ x \in \mathbb{R}^n : \limsup_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| \, dy > 2\lambda \right\};$$

it is enough to show $|A_\lambda| = 0$ for every $\lambda > 0$.

Fix $\varepsilon > 0$ and, by Corollary 26.4.11, choose $g \in C_c^\infty(\mathbb{R}^n)$ with $\|f - g\|_{L^1} < \varepsilon$. For any x and r , the triangle inequality gives

$$\begin{aligned} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| \, dy &\leq \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - g(y)| \, dy \\ &\quad + \frac{1}{|B_r(x)|} \int_{B_r(x)} |g(y) - g(x)| \, dy + |g(x) - f(x)|. \end{aligned}$$

The first term is at most $M_{f-g}(x)$ by definition, and the second tends to 0 as $r \rightarrow 0$ by continuity of g . So

$$\limsup_{r \rightarrow 0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y) - f(x)| \, dy \leq M_{f-g}(x) + |f(x) - g(x)|.$$

Consequently, if $x \in A_\lambda$ then either $M_{f-g}(x) > \lambda$ or $|f(x) - g(x)| > \lambda$. By Lemma 26.4.14 the first set has measure at most $C_n \varepsilon / \lambda$, and by Chebyshev's inequality the second has measure at most ε / λ . Hence

$$|A_\lambda| \leq \frac{(1 + C_n)\varepsilon}{\lambda},$$

and since $\varepsilon > 0$ was arbitrary, $|A_\lambda| = 0$. $\square$

Corollary 26.4.16

For $f \in L^1(\mathbb{R}^n)$ and any mollifier (ϕ_ε) , we have $\phi_\varepsilon * f \rightarrow f$ almost everywhere as $\varepsilon \rightarrow 0$.

Proof. At a Lebesgue point x , bounding $\phi_\varepsilon \leq \|\phi\|_{L^\infty} \varepsilon^{-n} \mathbb{1}_{B_\varepsilon(0)}$ gives

$$|(\phi_\varepsilon * f)(x) - f(x)| \leq \int \phi_\varepsilon(y) |f(x - y) - f(x)| \, dy \leq \frac{C}{|B_\varepsilon(x)|} \int_{B_\varepsilon(x)} |f(y) - f(x)| \, dy,$$

which tends to 0. $\square$

Corollary 26.4.17

Let $g \in L^1(\mathbb{R})$ and set $G(x) = \int_{-\infty}^x g(t) \, dt$. Then G is differentiable at almost every x , with $G'(x) = g(x)$.

Proof. At a Lebesgue point x of g we have, for $h > 0$,

$$\begin{aligned} \left| \frac{G(x+h) - G(x)}{h} - g(x) \right| &= \left| \frac{1}{h} \int_x^{x+h} (g(t) - g(x)) \, dt \right| \\ &\leq \frac{2}{|B_h(x)|} \int_{B_h(x)} |g - g(x)| \rightarrow 0, \end{aligned}$$

and similarly for $h < 0$. $\square$

Example 26.4.18

Take $g = \mathbb{1}_{(0,\infty)}$ on $\mathbb{R}$ (which is in L^1_{loc} , and the theorem applies locally). Every $x \neq 0$ is a Lebesgue point, and indeed

$$G(x) = \int_{-\infty}^x g = \begin{cases} 0 & x \leq 0 \\ x & x > 0 \end{cases}$$

is differentiable exactly off 0, with $G' = g$ there. The point $x = 0$ is not a Lebesgue point, and this single bad point is the seed from which the whole theory of distributions grows: we would very much like to say that $G' = g$ and $g' = \delta_0$, and presently we shall.

§26.5 Hilbert Space and Duality

We turn now to the linear structure. The spaces L^p are Banach spaces, and L^2 is a Hilbert space; the point of this section is to see what that buys us. The headline results are a complete description of the dual space of L^p for $p < \infty$, and – as a by-product of the Hilbert space geometry – a proof of the Radon–Nikodým theorem, which is a theorem of measure theory obtained by functional-analytic means. That last is characteristic of the subject: once the spaces are set up correctly, hard-looking measure theory becomes soft linear algebra.

§26.5.1 Orthogonal systems of functions

Throughout, $\mathcal{H}$ denotes a Hilbert space over $\mathbb{C}$ with inner product $\langle \cdot, \cdot \rangle$, conjugate-linear in the first variable. The model to keep in mind is $\mathcal{H} = L^2(E, \mu)$ with $\langle f, g \rangle = \int_E \bar{f}g \, d\mu$.

Definition 26.5.1

A subset $S = \{u_j\}_{j \in J} \subseteq \mathcal{H}$ is **orthogonal** if $\langle u_i, u_j \rangle = 0$ whenever $i \neq j$, and **orthonormal** if in addition $\|u_j\| = 1$ for all j . We say S is **complete** if $\overline{\text{span } S} = \mathcal{H}$, and an orthonormal complete set is an **orthonormal basis**. A Hilbert space is **separable** exactly when it admits a countable orthonormal basis.

If $\{e_n\}_{n \in \mathbb{N}}$ is a countable orthonormal basis then every $x \in \mathcal{H}$ has the expansion

$$x = \sum_{n \in \mathbb{N}} \langle e_n, x \rangle e_n, \quad \|x\|^2 = \sum_{n \in \mathbb{N}} |\langle e_n, x \rangle|^2,$$

the series converging in norm; this is Parseval's identity, and it says that $\mathcal{H} \cong \ell^2$ as soon as $\mathcal{H}$ is separable and infinite-dimensional. Abstractly, then, there is only one separable infinite-dimensional Hilbert space. What makes the subject non-trivial is that the *same* space carries many different natural bases, and that choosing a good one is often the whole problem.

Example 26.5.2

Some orthonormal bases worth knowing.

- (i) The trigonometric system $\{x \mapsto e^{-2\pi i n x} : n \in \mathbb{Z}\}$ is an orthonormal basis of

$L^2((0, 1))$. Orthonormality is a computation; completeness follows either from Stone–Weierstrass (the trigonometric polynomials are dense in the continuous periodic functions in the uniform norm) or from Corollary 26.4.11. This is the classical theory of Fourier series, in one line.

(ii) Let

$$\psi(x) = \begin{cases} 1 & 0 \leq x < 1/2, \\ -1 & 1/2 \leq x < 1, \\ 0 & \text{otherwise,} \end{cases}$$

and set $\psi_{n,k}(x) = 2^{n/2}\psi(2^n x - k)$. Then $\{\psi_{n,k} : n, k \in \mathbb{Z}\}$ is an orthonormal basis of $L^2(\mathbb{R})$, the **Haar system**. It is the simplest example of a wavelet basis, and unlike the trigonometric system its elements are localised in space as well as in frequency.

(iii) Let $\mathcal{H} = L^2(\mathbb{R}, e^{-x^2} dx)$, with inner product $\langle f, g \rangle = \int_{\mathbb{R}} \bar{f} g e^{-x^2} dx$. Applying Gram–Schmidt to the linearly independent family $1, x, x^2, \dots$ produces orthogonal polynomials H_k with $\deg H_k = k$; normalised so that the coefficient of x^k in H_k is 2^k , these are the **Hermite polynomials**, and they form a complete orthogonal set.

The result from Linear Analysis we shall use over and over is the following.

Theorem 26.5.3 (Riesz representation, Hilbert space form)

Let $\mathcal{H}$ be a Hilbert space and $\Lambda : \mathcal{H} \rightarrow \mathbb{C}$ a bounded linear functional. Then there is a unique $w \in \mathcal{H}$ with

$$\Lambda u = \langle w, u \rangle \quad \text{for all } u \in \mathcal{H},$$

and $\|\Lambda\|_{\mathcal{H}'} = \|w\|$.

Proof. We recall the argument. If $\Lambda = 0$ take $w = 0$. Otherwise $K = \ker \Lambda$ is a proper closed subspace, so there is $v \in K^\perp$ with $\|v\| = 1$ (take any $z \notin K$ and subtract its orthogonal projection onto K , which exists because K is closed and convex). For any u , the vector $u - \frac{\Lambda u}{\Lambda v} v$ lies in K , hence is orthogonal to v , so $\langle v, u \rangle = \frac{\Lambda u}{\Lambda v}$. Thus $w = \overline{\Lambda v} v$ works. Uniqueness and the norm identity are immediate. $\square$

§26.5.2 The Radon–Nikodým theorem

Here is a first illustration of the power of Theorem 26.5.3.

Definition 26.5.4

Let μ, ν be measures on a measurable space $(E, \mathcal{E})$. We say ν is **absolutely continuous** with respect to μ , written $\nu \ll \mu$, if $\mu(A) = 0 \implies \nu(A) = 0$ for all $A \in \mathcal{E}$. We say μ and ν are **mutually singular**, written $\mu \perp \nu$, if there is a measurable Z with $\mu(Z) = 0 = \nu(Z^c)$.

Theorem 26.5.5 (Radon–Nikodým)

Let μ, ν be finite measures on $(E, \mathcal{E})$ with $\nu \ll \mu$. Then there is a non-negative $w \in L^1(E, \mu)$ such that

$$\nu(A) = \int_A w \, d\mu \quad \text{for all } A \in \mathcal{E};$$

equivalently, $\int_E F \, d\nu = \int_E Fw \, d\mu$ for every measurable $F : E \rightarrow [0, \infty]$.

Proof. The trick is to introduce two auxiliary measures which are comparable to both μ and ν . Set $\alpha = \mu + 2\nu$ and $\beta = \nu + 2\mu$, both finite. Since $\mu \leq \alpha$ and $\nu \leq \alpha$, we have

$$\mathcal{H} := L^2(E, \alpha) \subseteq L^2(E, \mu) \cap L^2(E, \nu),$$

and every $f \in \mathcal{H}$ is β -integrable. Consider the linear functional

$$\Lambda(f) = \int_E f \, d\beta, \quad f \in \mathcal{H}.$$

It is bounded, since

$$|\Lambda(f)| \leq \int_E |f| \, d\beta = 2 \int_E |f| \, d\mu + \int_E |f| \, d\nu \leq 2 \int_E |f| \, d\alpha \leq 2\sqrt{\alpha(E)} \|f\|_{L^2(\alpha)}$$

by Cauchy–Schwarz. By Theorem 26.5.3 there is $g \in \mathcal{H}$ with

$$\int_E f \, d\beta = \int_E gf \, d\alpha \quad \text{for all } f \in \mathcal{H}.$$

Unwinding the definitions of α and β , this says

$$\int_E (2g - 1)f \, d\nu = \int_E (2 - g)f \, d\mu \quad \text{for all } f \in \mathcal{H}. \quad (*)$$

We claim $\frac{1}{2} \leq g \leq 2$ almost everywhere with respect to both μ and ν . Indeed, let $A_j = \{g < \frac{1}{2} - \frac{1}{j}\}$ and put $f = \mathbb{1}_{A_j}$ in (*):

$$\frac{3}{2}\mu(A_j) \leq \int_E (2 - g)\mathbb{1}_{A_j} \, d\mu = \int_E (2g - 1)\mathbb{1}_{A_j} \, d\nu \leq -\frac{2}{j}\nu(A_j) \leq 0,$$

forcing $\mu(A_j) = 0$ and then $\nu(A_j) = 0$. Letting $j \rightarrow \infty$ gives $g \geq 1/2$ a.e.; the argument at the other end, with $A_j = \{g > 2 + 1/j\}$, gives $g \leq 2$ a.e.

Now let $Z = \{g = 1/2\}$. Putting $f = \mathbb{1}_Z$ in (*) gives $0 = \int_Z (2 - g) \, d\mu \geq \frac{3}{2}\mu(Z)$, so $\mu(Z) = 0$, and hence $\nu(Z) = 0$ by absolute continuity. Given measurable $F : E \rightarrow [0, \infty]$, define on Z^c

$$f = \frac{F}{2g - 1}, \quad w = \frac{2 - g}{2g - 1},$$

and $f = w = 0$ on Z . Both are well defined and bounded multiples of F and 1 respectively, since $2g - 1 \in [0, 3]$ is bounded away from 0 off Z after a further

exhaustion, and applying $(*)$ (first for bounded F , then in general by monotone convergence) gives

$$\int_E F \, d\nu = \int_{Z^c} (2g - 1)f \, d\nu = \int_{Z^c} (2 - g)f \, d\mu = \int_E Fw \, d\mu$$

as required. Taking $F = 1$ shows $w \in L^1(E, \mu)$, and taking $F = \mathbb{1}_A$ gives the stated form. $\square$

§26.5.3 Dual spaces

Definition 26.5.6 (Dual space)

Let X be a topological vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. The **(topological) dual** X' is the vector space of continuous linear functionals $\Lambda : X \rightarrow \mathbb{F}$. If X is normed, we norm X' by

$$\|\Lambda\|_{X'} = \sup_{\|x\|_X \leq 1} |\Lambda(x)|.$$

The space X' is always a Banach space, complete regardless of whether X is, since completeness is inherited from the target field. What is not at all obvious is that X' contains anything: the definition alone does not produce a single non-zero functional. That is the business of the Hahn–Banach theorem, which we prove in Section 26.6; for now we record what we shall need.

Lemma 26.5.7

Let X be a normed space and $x \neq y$ in X . Then there is $\Lambda \in X'$ with $\Lambda(x) \neq \Lambda(y)$.

We defer the proof to Corollary 26.6.9. Granting it, for each $x \in X$ the evaluation map $f_x : X' \rightarrow \mathbb{F}$, $\Lambda \mapsto \Lambda(x)$, is linear and bounded (with $|f_x(\Lambda)| \leq \|x\| \|\Lambda\|$), hence $f_x \in X'' = (X')'$; and Lemma 26.5.7 says $x \mapsto f_x$ is injective. In fact it is isometric, again by Hahn–Banach.

Definition 26.5.8 (Reflexive)

A Banach space X is **reflexive** if the canonical map $X \rightarrow X''$, $x \mapsto f_x$, is surjective (hence an isometric isomorphism), in which case we write $X = X''$.

For a Hilbert space, Theorem 26.5.3 gives a conjugate-linear isometry $\mathcal{H} \rightarrow \mathcal{H}'$; applying it twice shows that every Hilbert space is reflexive. This will matter enormously later: reflexivity is what makes bounded sequences have weakly convergent subsequences.

§26.5.4 The dual of L^p

Fix a measure space $(E, \mathcal{E}, \mu)$ and conjugate exponents p, q . Hölder's inequality says that for $g \in L^q$ the map

$$\Lambda_g(f) = \int_E gf \, d\mu$$

is a bounded linear functional on L^p with $\|\Lambda_g\|_{(L^p)'} \leq \|g\|_{L^q}$. In fact equality holds.

Lemma 26.5.9

For $1 \leq p \leq \infty$ and $g \in L^q$ we have $\|\Lambda_g\|_{(L^p)'} = \|g\|_{L^q}$. Hence $\kappa : L^q \rightarrow (L^p)'$, $g \mapsto \Lambda_g$, is a linear isometry.

Proof. Only ' $\geq$ ' needs proof. Suppose $1 < q < \infty$ and $g \neq 0$; take

$$f = \frac{|g|^{q-1} \overline{\text{sgn } g}}{\|g\|_{L^q}^{q-1}}, \quad \text{so that} \quad \|f\|_{L^p}^p = \frac{\int |g|^{(q-1)p}}{\|g\|_{L^q}^{(q-1)p}} = 1$$

since $(q-1)p = q$. Then $\Lambda_g(f) = \|g\|_{L^q}$. For $q = 1$ take $f = \overline{\text{sgn } g}$, of L^∞ norm 1. For $q = \infty$, given $\varepsilon > 0$ let $A = \{|g| > \|g\|_{L^\infty} - \varepsilon\}$, which has positive measure; assuming μ σ -finite we may shrink it to have finite positive measure, and $f = \mu(A)^{-1} \mathbb{1}_A \overline{\text{sgn } g}$ has $\|f\|_{L^1} = 1$ and $\Lambda_g(f) \geq \|g\|_{L^\infty} - \varepsilon$. $\square$

So L^q sits isometrically inside $(L^p)'$. The theorem is that for $p < \infty$ it is everything.

Theorem 26.5.10

Let $(E, \mathcal{E}, \mu)$ be σ -finite, let $1 \leq p < \infty$ and let q be conjugate to p . Then $\kappa : L^q(E, \mu) \rightarrow L^p(E, \mu)'$ is an isometric isomorphism. We write $L^p(E, \mu)' = L^q(E, \mu)$.

Proof. By Lemma 26.5.9 it remains to prove surjectivity, and by Lemma 26.5.9 again it is enough to produce, for each $\Lambda \in (L^p)'$, some $g \in L^q$ with $\Lambda = \Lambda_g$.

Reduction to real scalars and positive functionals. Writing $f = f_r + if_i$ and $\Lambda_r(f) = \text{Re } \Lambda(f)$, $\Lambda_i(f) = \text{Im } \Lambda(f)$ for real f , a complex functional is determined by a pair of real ones, so we may take the scalars real. A real-linear bounded $\Lambda : L^p(E; \mathbb{R}) \rightarrow \mathbb{R}$ is called **positive** if $\Lambda(f) \geq 0$ whenever $f \geq 0$; one checks (this is on the example sheet, and is the Jordan decomposition in disguise) that every bounded real-linear functional is a difference $\Lambda = \Lambda^+ - \Lambda^-$ of positive bounded ones. So assume Λ positive.

The finite measure case. Suppose first $\mu(E) < \infty$. Define $\nu(A) = \Lambda(\mathbb{1}_A)$ for $A \in \mathcal{E}$. Then $\nu \geq 0$ by positivity, $\nu(\emptyset) = 0$, and ν is finitely additive by linearity. It is countably additive: if $B = \bigcup_{n \geq 1} A_n$ with the A_n disjoint, and $B_k = \bigcup_{n \leq k} A_n$, then

$$\|\mathbb{1}_B - \mathbb{1}_{B_k}\|_{L^p} = \mu(B \setminus B_k)^{1/p} \rightarrow 0,$$

so $\nu(B) - \nu(B_k) = \Lambda(\mathbb{1}_B - \mathbb{1}_{B_k}) \rightarrow 0$ by continuity of Λ . The same computation shows $\mu(A) = 0 \implies \nu(A) = 0$, i.e. $\nu \ll \mu$. By Theorem 26.5.5 there is a non-negative $g \in L^1(E, \mu)$ with $\Lambda(\mathbb{1}_A) = \int_A g \, d\mu$ for all A ; by linearity $\Lambda(f) = \int_E fg \, d\mu$ for all simple f , and hence for all bounded measurable f by dominated convergence.

g lies in L^q . Suppose $1 < p < \infty$. For $N > 0$ let $g_N = \min(g, N)$ and $f_N = g_N^{q-1}$, a bounded function. Then

$$\int_E g_N^q \, d\mu \leq \int_E f_N g \, d\mu = \Lambda(f_N) \leq \|\Lambda\| \|f_N\|_{L^p} = \|\Lambda\| \left(\int_E g_N^q \, d\mu \right)^{1/p},$$

using $(q-1)p = q$. Dividing (the integral is finite as g_N is bounded and $\mu(E) < \infty$) gives $\|g_N\|_{L^q} \leq \|\Lambda\|$, and monotone convergence as $N \rightarrow \infty$ gives $\|g\|_{L^q} \leq \|\Lambda\| < \infty$. For $p = 1$ one argues instead that $\mu(\{g > \|\Lambda\| + \varepsilon\}) = 0$, directly from $\Lambda(\mathbb{1}_A) \leq \|\Lambda\| \mu(A)$, so $g \in L^\infty$. Finally, since $g \in L^q$ the functional Λ_g is bounded on L^p and agrees with Λ on the dense set of simple functions (Proposition 26.3.12), so $\Lambda = \Lambda_g$.

The σ -finite case. Write $E = \bigcup_k E_k$ with $E_k \uparrow$ and $\mu(E_k) < \infty$. Applying the above on each E_k produces $g_k \in L^q(E_k)$ representing Λ on functions supported in E_k ; by uniqueness these are consistent, so they patch to a measurable g on E with $\|g\mathbb{1}_{E_k}\|_{L^q} \leq \|\Lambda\|$ for every k . Monotone convergence gives $g \in L^q$ with $\|g\|_{L^q} \leq \|\Lambda\|$, and $\Lambda = \Lambda_g$ on the dense set of functions supported in some E_k , hence everywhere. $\square$

Corollary 26.5.11

For $1 < p < \infty$, $L^p(\mathbb{R}^n)$ is reflexive.

Proof. $(L^p)'' = (L^q)' = L^p$, and one checks that the composite isomorphism is the canonical embedding. $\square$

The theorem genuinely fails at $p = \infty$: the dual of L^∞ is much larger than L^1 . To see this, note that $C([0, 1]) \subseteq L^\infty([0, 1])$ and $f \mapsto f(0)$ is a bounded functional on $C([0, 1])$; Hahn–Banach extends it to L^∞ , and no $g \in L^1$ can represent it (it vanishes on every f vanishing near 0, which forces $g = 0$ a.e., but the functional is not zero). Correspondingly neither L^1 nor L^∞ is reflexive. This is not a technicality to be regretted: it is why the natural home for many variational problems is L^p with $1 < p < \infty$, and why problems formulated in L^1 (or in the space of measures) are genuinely harder.

§26.5.5 The Riesz representation theorem for measures

There is one more duality worth recording, and it goes the other way: instead of describing functionals in terms of functions, it describes them in terms of *measures*. This is the sense in which a measure ‘is’ a way of integrating continuous functions.

Definition 26.5.12

Let E be a topological space and $\mathcal{E} \supseteq \mathcal{B}(E)$ a σ -algebra. A measure μ on $(E, \mathcal{E})$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are a closed C and an open O with $C \subseteq A \subseteq O$ and $\mu(O \setminus C) < \varepsilon$.

Lebesgue measure on $(\mathbb{R}^n, \mathcal{M})$ is regular, by Theorem 26.2.5(iii). Now if μ is a finite regular measure on $\mathbb{R}^n$, then every $f \in C_c^0(\mathbb{R}^n; \mathbb{R})$ is measurable and bounded, so

$$\Lambda(f) = \int_{\mathbb{R}^n} f \, d\mu$$

defines a positive bounded linear functional on $C_c^0(\mathbb{R}^n; \mathbb{R})$. The question is whether μ can be recovered from Λ , and whether every such Λ arises this way.

Lemma 26.5.13

A finite regular measure μ on $\mathbb{R}^n$ is determined by the functional $\Lambda(f) = \int f \, d\mu$ on $C_c^0(\mathbb{R}^n; \mathbb{R})$.

Proof. We would like to take $f = \mathbb{1}_O$, but indicators are not continuous, so we approximate. Let O be open, and for $k \in \mathbb{N}$ set $O_k = O \cap \{|x| < k\}$ and

$$\chi_k(x) = \begin{cases} 1 & x \in O_k, \text{ dist}(x, O_k^c) \geq 1/k, \\ k \text{ dist}(x, O_k^c) & x \in O_k, \text{ dist}(x, O_k^c) < 1/k, \\ 0 & x \notin O_k. \end{cases}$$

Each χ_k is continuous with compact support (distance functions are 1-Lipschitz) and $\chi_k \uparrow \mathbb{1}_O$ pointwise. By monotone convergence $\Lambda(\chi_k) \rightarrow \mu(O)$. So Λ determines μ on open sets, and hence on all of $\mathcal{E}$ by regularity. $\square$

Theorem 26.5.14 (Riesz–Markov)

Let $\Lambda : C_c^0(\mathbb{R}^n; \mathbb{R}) \rightarrow \mathbb{R}$ be a positive bounded linear functional. Then there is a σ -algebra $\mathcal{M} \supseteq \mathcal{B}(\mathbb{R}^n)$ and a unique finite regular measure μ on it with

$$\Lambda(f) = \int_{\mathbb{R}^n} f \, d\mu \quad \text{for all } f \in C_c^0(\mathbb{R}^n; \mathbb{R}).$$

The proof is a construction of μ from Λ by an outer measure argument – one defines $\mu(O) = \sup\{\Lambda(f) : 0 \leq f \leq 1, \text{supp } f \subseteq O\}$ for open O and extends by regularity – and it is genuinely technical. We shall take it on trust.

Example 26.5.15

Take $\Lambda(f) = f(0)$. This is positive and bounded, and the measure it represents is $\mu = \delta_0$, the Dirac point mass at the origin. The reader who feels that δ_0 ‘should’ be a function will be pleased to know that in Section 26.9 it becomes a perfectly respectable distribution, and that this example is the prototype.

§26.6 Weak Topologies and Compactness

In finite dimensions, the Bolzano–Weierstrass theorem says that bounded sequences have convergent subsequences, and this single fact underlies most existence proofs in analysis: to produce an object with a desired property, construct a bounded sequence of approximations and pass to a limit. In infinite dimensions this fails completely. In ℓ^2 the orthonormal sequence (e_n) is bounded but $\|e_n - e_m\| = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy; indeed the closed unit ball of a normed space is compact if and only if the space is finite-dimensional.

The remedy is to keep the sequence and change the topology. If we declare fewer sets to be open, more sets become compact, and with luck we can weaken the topology enough to restore Bolzano–Weierstrass while keeping enough separation to make the limits useful. This section carries that out. We need the Hahn–Banach theorem first, because a weak topology is built from linear functionals and is useless unless there are plenty of them.

§26.6.1 The Hahn–Banach theorem

Hahn–Banach is an extension theorem: a bounded functional defined on a subspace extends to the whole space with the same bound. We work over $\mathbb{R}$; the complex case follows by taking real parts, since a complex-linear Λ is determined by $\ell = \operatorname{Re} \Lambda$ via $\Lambda(x) = \ell(x) - i\ell(ix)$.

For the sake of generality we replace the norm bound by something weaker.

Definition 26.6.1 (Sublinear functional)

A **sublinear functional** on a real vector space X is a map $p : X \rightarrow \mathbb{R}$ with

$$p(x + y) \leq p(x) + p(y), \quad p(tx) = tp(x)$$

for all $x, y \in X$ and $t \geq 0$.

Example 26.6.2

Every seminorm is sublinear, and so is $|\ell|$ for any linear ℓ . But sublinear functionals need not be non-negative or symmetric: on $\mathbb{R}$, $p(x) = x$ is sublinear.

Note that a one-sided bound $\ell(x) \leq p(x)$ automatically gives a two-sided one, since $-\ell(x) = \ell(-x) \leq p(-x)$, whence $-p(-x) \leq \ell(x) \leq p(x)$.

Lemma 26.6.3 (Banach extension lemma)

Let X be a real vector space, p a sublinear functional on X , $M \leq X$ a subspace, and $\ell : M \rightarrow \mathbb{R}$ linear with $\ell \leq p$ on M . Then for any $x \in X \setminus M$ there is a linear $\tilde{\ell}$ on $\widetilde{M} = \operatorname{span}(M \cup \{x\})$ extending ℓ with $\tilde{\ell} \leq p$ on $\widetilde{M}$.

Proof. Every $w \in \widetilde{M}$ is uniquely $w = \lambda x + y$ with $y \in M$, $\lambda \in \mathbb{R}$, so $\tilde{\ell}$ is determined by the single number $a = \tilde{\ell}(x)$, and the problem is to choose a compatibly.

For $y, z \in M$ we have

$$\ell(y) + \ell(z) = \ell(y + z) \leq p(y + z) \leq p(y - x) + p(z + x),$$

so that $\ell(y) - p(y - x) \leq p(z + x) - \ell(z)$ for all $y, z \in M$. Hence

$$a := \sup_{y \in M} (\ell(y) - p(y - x)) \leq \inf_{z \in M} (p(z + x) - \ell(z)) < \infty,$$

and both bounds are finite. By construction $\ell(y) - a \leq p(y - x)$ and $\ell(z) + a \leq p(z + x)$ for all $y, z \in M$. Now take $w = \lambda x + y$. For $\lambda > 0$, apply the second inequality with $z = \lambda^{-1}y$ and multiply by λ ; for $\lambda < 0$, apply the first with $y' = -\lambda^{-1}y$ and multiply by $-\lambda$; for $\lambda = 0$ there is nothing to prove. In each case we get $\tilde{\ell}(w) \leq p(w)$. $\square$

If X/M is finite-dimensional we can now finish by induction, and if X is separable we can finish by a countable induction plus a limiting argument. In general we need a form of the axiom of choice.

Proposition 26.6.4 (Zorn's lemma)

Let $(S, \leq)$ be a non-empty partially ordered set in which every totally ordered subset has an upper bound. Then S has a maximal element.

Theorem 26.6.5 (Hahn–Banach)

Let X be a real vector space, p a sublinear functional on X , $M \leq X$ and $\ell : M \rightarrow \mathbb{R}$ linear with $\ell \leq p$ on M . Then there is a linear $\tilde{\ell} : X \rightarrow \mathbb{R}$ extending ℓ with $\tilde{\ell} \leq p$ on X .

Proof. Let S be the set of pairs (N, ℓ^*) where $M \leq N \leq X$ and $\ell^* : N \rightarrow \mathbb{R}$ is linear, extends ℓ , and satisfies $\ell^* \leq p$ on N . Order S by $(N_1, \ell_1^*) \leq (N_2, \ell_2^*)$ if $N_1 \leq N_2$ and ℓ_2^* extends ℓ_1^* . Then $S \neq \emptyset$ (it contains (M, ℓ)), and any chain $T \subseteq S$ has the upper bound (N_T, ℓ_T) with $N_T = \bigcup_{(N, \ell^*) \in T} N$ – a subspace because T is totally ordered – and ℓ_T the common value. By Proposition 26.6.4 there is a maximal $(\tilde{N}, \tilde{\ell})$; and if $\tilde{N} \neq X$ then Lemma 26.6.3 would extend it further, contradicting maximality. So $\tilde{N} = X$. $\square$

Corollary 26.6.6

Let X be a normed space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $M \leq X$, and $\Lambda : M \rightarrow \mathbb{F}$ a bounded linear functional. Then there is $\tilde{\Lambda} \in X'$ extending Λ with $\|\tilde{\Lambda}\|_{X'} = \|\Lambda\|_{M'}$.

Proof. For $\mathbb{F} = \mathbb{R}$ apply Theorem 26.6.5 with $p(x) = \|\Lambda\|_{M'} \|x\|$. For $\mathbb{F} = \mathbb{C}$, write $\Lambda(x) = \ell(x) - i\ell(ix)$ with $\ell = \operatorname{Re} \Lambda$ real-linear; since for each x we may pick θ with $|\Lambda(x)| = \ell(e^{i\theta}x)$, the real and complex norms agree, and extending ℓ then setting $\tilde{\Lambda}(x) = \tilde{\ell}(x) - i\tilde{\ell}(ix)$ does the job. $\square$

Corollary 26.6.7

For any $x \neq 0$ in a normed space X there is $\Lambda \in X'$ with $\|\Lambda\|_{X'} = 1$ and $\Lambda(x) = \|x\|$. Consequently $\|x\| = \sup_{\|\Lambda\|_{X'} \leq 1} |\Lambda(x)|$, and the canonical map $X \rightarrow X''$ is isometric.

Proof. Define $\Lambda(tx) = t\|x\|$ on $M = \mathbb{F}x$, of norm 1, and extend by Corollary 26.6.6. $\square$

§26.6.2 Geometric forms

Hahn–Banach has a geometric face which is often more useful. Recall that $A \subseteq X$ is **convex** if $tx + (1-t)y \in A$ for all $x, y \in A$ and $t \in [0, 1]$. In $\mathbb{R}^n$ one knows that disjoint compact convex sets can be separated by a hyperplane; Hahn–Banach extends this to infinite dimensions, with the linear functional playing the role of the hyperplane.

Theorem 26.6.8 (Geometric Hahn–Banach)

Let A, B be disjoint, non-empty, convex subsets of a real or complex Banach space

X .

(a) If A is open, there are $\Lambda \in X'$ and $\gamma \in \mathbb{R}$ with

$$\operatorname{Re} \Lambda(x) < \gamma \leq \operatorname{Re} \Lambda(y) \quad \text{for all } x \in A, y \in B.$$

If B is also open, the second inequality is strict too.

(b) If A is compact and B is closed, there are $\Lambda \in X'$ and $\gamma_1 < \gamma_2$ with

$$\operatorname{Re} \Lambda(x) < \gamma_1 < \gamma_2 < \operatorname{Re} \Lambda(y) \quad \text{for all } x \in A, y \in B.$$

Proof. We may take X real.

(a) Pick $a_0 \in A$, $b_0 \in B$ and set $x_0 = b_0 - a_0$, and let $C = A - B + x_0$. Then C is convex (a sum of convex sets), open (a union of translates of the open A), and contains 0; and $x_0 \notin C$ since $A \cap B = \emptyset$. Define the **Minkowski gauge**

$$p(x) = \inf\{t > 0 : t^{-1}x \in C\}.$$

One checks (example sheet) that p is sublinear, that $p(x) \leq k\|x\|$ for some $k > 0$ (because C contains a ball about 0), and that $p(y) < 1$ for $y \in C$; since $x_0 \notin C$ we get $p(x_0) \geq 1$.

On $M = \mathbb{R}x_0$ define $f(tx_0) = t$. Then $f \leq p$ on M : for $t > 0$, $f(tx_0) = t \leq tp(x_0) = p(tx_0)$, and for $t \leq 0$, $f(tx_0) = t \leq 0 \leq p(tx_0)$. By Theorem 26.6.5 extend f to a linear Λ on X with $\Lambda \leq p$; then $-k\|x\| \leq -p(-x) \leq \Lambda(x) \leq p(x) \leq k\|x\|$, so $\Lambda \in X'$.

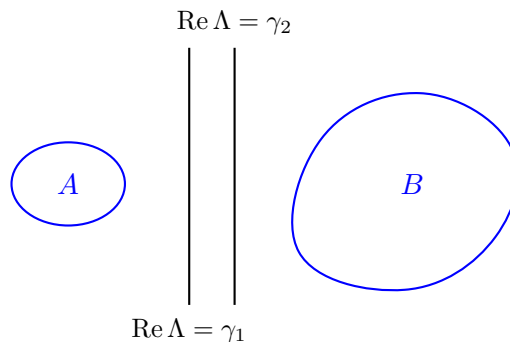
For $a \in A$, $b \in B$ we have $a - b + x_0 \in C$, so

$$\Lambda(a) - \Lambda(b) + 1 = \Lambda(a - b + x_0) \leq p(a - b + x_0) < 1,$$

giving $\Lambda(a) < \Lambda(b)$. Thus $\Lambda(A)$ and $\Lambda(B)$ are convex subsets of $\mathbb{R}$ (that is, intervals) with $\sup \Lambda(A) \leq \inf \Lambda(B)$. As Λ is a non-zero continuous linear map it is open, so $\Lambda(A)$ is an open interval and $\gamma = \sup \Lambda(A)$ is not attained; this is the claim.

(b) Since A is compact, B closed and they are disjoint, $d = \inf\{\|a - b\| : a \in A, b \in B\} > 0$. Let $V = B_{d/2}(0)$; then $A + V$ is open, convex, and disjoint from B , so (a) provides Λ with $\sup \Lambda(A + V) \leq \inf \Lambda(B)$. Since $\Lambda(A)$ is a compact subset of the open set $\Lambda(A + V)$ we may insert the two constants $\gamma_1 < \gamma_2$. $\square$

Pictorially: a point outside a closed convex set can be cut off from it by a hyperplane, with room to spare.



Corollary 26.6.9

For distinct x, y in a normed space X there is $\Lambda \in X'$ with $\Lambda(x) \neq \Lambda(y)$.

Proof. Apply Theorem 26.6.8(b) with $A = \{x\}$ and $B = \{y\}$ (or Corollary 26.6.7 to $x - y$). $\square$

Corollary 26.6.10

Let $M \leq X$ be a subspace with $\overline{M} \neq X$, and let $x_0 \in X \setminus \overline{M}$. Then there is $\Lambda \in X'$ with $\Lambda(x_0) = 1$ and $\Lambda|_M = 0$.

Proof. Take $A = \{x_0\}$, $B = \overline{M}$ in Theorem 26.6.8(b) to get Λ_0 with $\operatorname{Re} \Lambda_0(x_0) < \gamma_1 < \gamma_2 < \operatorname{Re} \Lambda_0(y)$ for $y \in \overline{M}$. Since $\overline{M}$ is a subspace, $\Lambda_0(\overline{M})$ is a subspace of $\mathbb{F}$ bounded away from $\operatorname{Re} \Lambda_0(x_0)$, which forces $\Lambda_0|_M = 0$; then rescale. $\square$

Remark. Corollary 26.6.10 is the standard tool for proving *density*: to show a subspace M is dense in X , it suffices to show that the only $\Lambda \in X'$ vanishing on M is $\Lambda = 0$. This is usually much easier, because we have a concrete description of X' to work with. We shall use exactly this manoeuvre when studying Sobolev spaces.

§26.6.3 Consequences of Baire category

We record, without proof, three theorems from Linear Analysis which all follow from the Baire category theorem (a complete metric space is not a countable union of nowhere dense sets). Their common feature is that they turn a pointwise or algebraic hypothesis into a uniform or topological conclusion. Proofs of all three, and of Baire's theorem itself, are in my Linear Analysis notes; the same is true of the Hilbert space material of Section 26.5 and of the spectral theorem for compact self-adjoint operators, which we shall borrow in Section 26.11.

Theorem 26.6.11 (Principle of uniform boundedness)

Let X be a Banach space, Y a normed space, and $\mathcal{T} \subseteq \mathcal{B}(X, Y)$ a family of bounded operators with $\sup_{T \in \mathcal{T}} \|Tx\| < \infty$ for every $x \in X$. Then $\sup_{T \in \mathcal{T}} \|T\| < \infty$.

Theorem 26.6.12 (Open mapping theorem)

A surjective bounded linear map between Banach spaces is open. In particular a bijective bounded linear map has bounded inverse.

Theorem 26.6.13 (Closed graph theorem)

A linear map between Banach spaces whose graph is closed is bounded.

The consequence we need immediately is the following, which says that weak convergence – defined in a moment – cannot produce unbounded sequences.

Proposition 26.6.14

Let X be a Banach space and (x_n) a sequence with $\Lambda(x_n)$ convergent for every $\Lambda \in X'$. Then $(\|x_n\|)$ is bounded.

Proof. View each x_n as an element f_{x_n} of $X'' = \mathcal{B}(X', \mathbb{F})$, noting X' is a Banach space. For each Λ the sequence $f_{x_n}(\Lambda) = \Lambda(x_n)$ is convergent, hence bounded. By Theorem 26.6.11, $\sup_n \|f_{x_n}\|_{X''} < \infty$; and $\|f_{x_n}\|_{X''} = \|x_n\|$ by Corollary 26.6.7. $\square$

§26.6.4 Seminorms and locally convex spaces

To define the weak topologies properly we need topologies generated not by a norm but by a family of seminorms.

Definition 26.6.15 (Seminorm)

A **seminorm** on a vector space X over $\mathbb{F}$ is a map $p : X \rightarrow \mathbb{R}$ with $p(x + y) \leq p(x) + p(y)$ and $p(\lambda x) = |\lambda|p(x)$.

Taking $\lambda = 0$ gives $p(0) = 0$, and then $0 = p(0) \leq 2p(x)$ shows $p \geq 0$. So a seminorm is a norm except that $p(x) = 0$ need not force $x = 0$. A single seminorm is therefore too coarse to separate points; the idea is to use many of them.

Definition 26.6.16

A family $\mathcal{P}$ of seminorms on X is **separating** if for every $x \neq 0$ there is $p \in \mathcal{P}$ with $p(x) \neq 0$.

Given a separating family $\mathcal{P}$, set $V(p, n) = \{x : p(x) < 1/n\}$ for $p \in \mathcal{P}$ and $n \in \mathbb{N}$, let $\dot{\beta}$ be the collection of finite intersections of such sets, and let $\beta = \{x + B : x \in X, B \in \dot{\beta}\}$.

Theorem 26.6.17

β is a base for a Hausdorff topology $\tau_{\mathcal{P}}$ on X , with respect to which addition, scalar multiplication and every $p \in \mathcal{P}$ are continuous.

Definition 26.6.18

A space of the form $(X, \tau_{\mathcal{P}})$ is a **locally convex topological vector space**, or LCTVS. If $\mathcal{P} = \{p_n\}_{n \in \mathbb{N}}$ is countable, $\tau_{\mathcal{P}}$ is metrisable by

$$d(x, y) = \sum_{n=1}^{\infty} 2^{-n} \frac{p_n(x - y)}{1 + p_n(x - y)},$$

and if this metric is complete we call X a **Fréchet space**.

‘Locally convex’ refers to the fact that the basic neighbourhoods $V(p, n)$ are convex, which is exactly the hypothesis needed for Hahn–Banach separation arguments to work. All the spaces in this course are locally convex; we shall meet Fréchet spaces again when we build test functions.

§26.6.5 Weak and weak-* topologies

Let X be a Banach space. Taking $\mathcal{P} = \{\|\cdot\|\}$ recovers the norm topology, which for contrast we now call the **strong topology** τ_s ; convergence in it, written $x_n \rightarrow x$, means $\|x_n - x\| \rightarrow 0$.

Definition 26.6.19 (Weak topology)

For $\Lambda \in X'$ the map $p_\Lambda(x) = |\Lambda(x)|$ is a seminorm, and the family $\{p_\Lambda : \Lambda \in X'\}$ is separating by Corollary 26.6.9. The topology it generates is the **weak topology** τ_w on X . Thus $x_n \rightharpoonup x$ (x_n converges weakly to x) if and only if

$$\Lambda(x_n) \rightarrow \Lambda(x) \quad \text{for all } \Lambda \in X'.$$

Definition 26.6.20 (Weak-* topology)

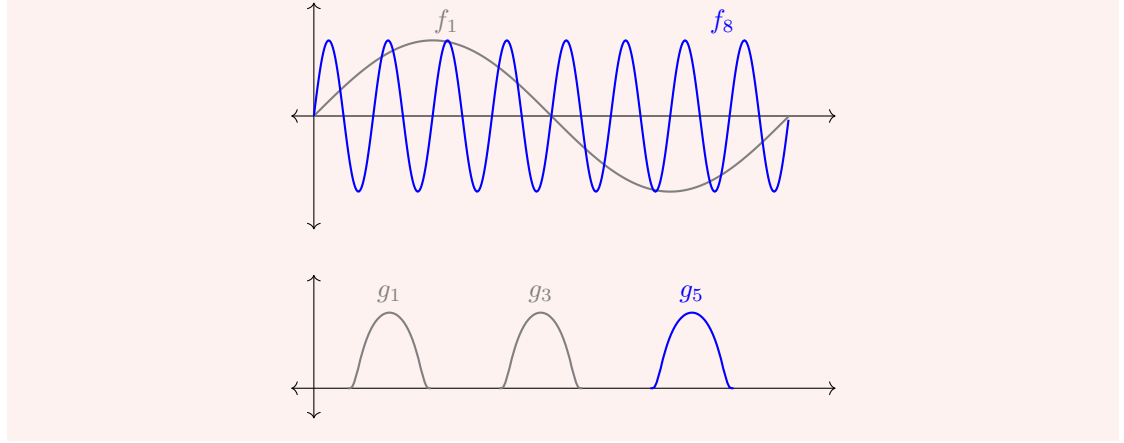
The dual X' carries its own strong and weak topologies, but also a third: the family of seminorms $\{p_x : \Lambda \mapsto |\Lambda(x)| : x \in X\}$ generates the **weak-* topology** τ_{w*} on X' . Thus $\Lambda_n \xrightarrow{*} \Lambda$ if and only if $\Lambda_n(x) \rightarrow \Lambda(x)$ for every $x \in X$: weak-* convergence is pointwise convergence of functionals.

Since $X \subseteq X''$, the weak-* topology on X' is coarser than its weak topology, and the two coincide exactly when X is reflexive. Strong convergence implies weak convergence implies weak-* convergence, and none of the implications reverses.

Example 26.6.21

Two examples of weak convergence to 0 without strong convergence, which between them illustrate the two mechanisms available.

- (i) (*Oscillation.*) In $L^2((0, 2\pi))$ let $f_n(x) = \sin(nx)$. Then $\|f_n\|_{L^2} = \sqrt{\pi}$ for all n , so $f_n \not\rightarrow 0$ strongly. But $\int_0^{2\pi} g(x) \sin(nx) \, dx \rightarrow 0$ for every $g \in L^2$: this is true for g in the dense set of trigonometric polynomials by orthogonality, hence for all g by density and uniform boundedness. So $f_n \rightharpoonup 0$. The mass does not go away; it is redistributed into ever higher frequencies, and every fixed test function averages it out.
- (ii) (*Escape to infinity.*) In $L^2(\mathbb{R})$ let $g_n(x) = \phi(x-n)$ for a fixed non-zero bump ϕ . Again $\|g_n\|_{L^2} = \|\phi\|_{L^2}$ is constant, but for $h \in C_c(\mathbb{R})$ the supports eventually separate, so $\int h g_n = 0$ for large n , and density gives $g_n \rightharpoonup 0$. The mass is neither destroyed nor spread out – it simply escapes to infinity.

**Example 26.6.22**

Specialising to our spaces: for $1 \leq p < \infty$ and $f_i, f \in L^p(\mathbb{R}^n)$, we have $f_i \rightharpoonup f$ if and only if

$$\int_{\mathbb{R}^n} g f_i \longrightarrow \int_{\mathbb{R}^n} g f \quad \text{for all } g \in L^q(\mathbb{R}^n),$$

by Theorem 26.5.10. When $1 < p < \infty$ the space is reflexive and this is the same as weak-* convergence. For $f_i \in L^\infty(\mathbb{R}^n) = L^1(\mathbb{R}^n)'$, weak-* convergence means testing against all $g \in L^1$, which is strictly weaker than weak convergence, since $(L^\infty)'$ is strictly bigger than L^1 .

Oscillation is by far the more common of the two mechanisms, and it is worth having a general statement. The following says that a rapidly oscillating periodic function converges weakly to its mean – the oscillations are not damped, they are simply averaged out by any fixed test function.

Proposition 26.6.23 (Oscillating sequences)

Let $g \in L^\infty(\mathbb{R})$ be 1-periodic with mean $\bar{g} = \int_0^1 g(t) \, dt$, and set $g_m(x) = g(mx)$. Then $g_m \xrightarrow{*} \bar{g}$ in $L^\infty(\mathbb{R}) = L^1(\mathbb{R})'$; that is,

$$\int_{\mathbb{R}} h(x) g(mx) \, dx \longrightarrow \bar{g} \int_{\mathbb{R}} h(x) \, dx \quad \text{for every } h \in L^1(\mathbb{R}).$$

Proof. Each g_m has $\|g_m\|_{L^\infty} = \|g\|_{L^\infty}$, so the functionals $\Lambda_m(h) = \int h g_m$ have uniformly bounded norms on $L^1(\mathbb{R})$. A uniformly bounded family of functionals which converges on a dense subset converges everywhere: if $\|\Lambda_m\| \leq C$ and $\Lambda_m(h_0)$ converges for h_0 in a dense set, then for general h and $\varepsilon > 0$ we may pick h_0 with $\|h - h_0\|_{L^1} < \varepsilon$ and estimate $|\Lambda_m(h) - \Lambda_{m'}(h)| \leq 2C\varepsilon + |\Lambda_m(h_0) - \Lambda_{m'}(h_0)|$, so $(\Lambda_m(h))$ is Cauchy.

So it is enough to treat $h = \mathbb{1}_{[a,b]}$, since finite linear combinations of such indicators are dense in $L^1(\mathbb{R})$. Substituting $t = mx$,

$$\int_a^b g(mx) \, dx = \frac{1}{m} \int_{ma}^{mb} g(t) \, dt.$$

The interval $[ma, mb]$ has length $m(b-a)$ and so contains $N_m = \lfloor m(b-a) \rfloor$ complete periods of g , each contributing $\bar{g}$, together with a leftover interval of length at most 1 on which the integral is at most $\|g\|_{L^\infty}$ in modulus. Hence

$$\left| \frac{1}{m} \int_{ma}^{mb} g - \frac{N_m}{m} \bar{g} \right| \leq \frac{\|g\|_{L^\infty}}{m} \rightarrow 0,$$

and $N_m/m \rightarrow b-a$, giving the claim. $\square$

Example 26.6.24

Take $g(x) = \sin(2\pi x)$, so $\bar{g} = 0$ and $u_m(x) = \sin(2\pi mx) \xrightarrow{*} 0$. But now consider $u_m^2 = \sin^2(2\pi mx)$: this is again of the form $G(mx)$ for the 1-periodic $G = \sin^2(2\pi \cdot)$, whose mean is $1/2$, so

$$u_m \xrightarrow{*} 0 \quad \text{while} \quad u_m^2 \xrightarrow{*} \frac{1}{2} \neq 0 = (\text{weak limit of } u_m)^2.$$

This little computation deserves to be remembered, because it is the whole difficulty of the last third of the course in miniature. Weak limits do not commute with nonlinear operations. Any functional built by integrating a nonlinear function of u will therefore fail to be weakly *continuous*, and the best we can hope for is an inequality in one direction. Note which direction we got here: $\frac{1}{2} > 0$, so the limit of $\int u_m^2$ is *larger* than $\int u^2$. That is lower semicontinuity, and Section 26.7 explains why convexity of the integrand always makes it fall this way.

Remark. Two warnings. First, the weak topology is *not* metrisable on an infinite-dimensional space (though it is metrisable on bounded sets when X' is separable), so one must be careful with sequential arguments; we shall always be in the situation where sequences suffice. Second, weak convergence is strictly weaker than strong convergence, but by Proposition 26.6.14 a weakly convergent sequence is still bounded, and one always has

$$\|x\| \leq \liminf_{n \rightarrow \infty} \|x_n\| \quad \text{when } x_n \rightharpoonup x,$$

since $\|x\| = \sup_{\|\Lambda\| \leq 1} |\Lambda(x)| = \sup_{\Lambda} \lim_n |\Lambda(x_n)| \leq \liminf_n \|x_n\|$. The norm can drop in the limit – it did in both parts of Example 26.6.21, from a positive constant to 0 – but it cannot jump up. This inequality, **weak lower semicontinuity of the norm**, is the single most useful property of weak convergence, and everything in Section 26.7 is built on it.

§26.6.6 Banach–Alaoglu

Now the pay-off. It is worth recalling first the one compactness theorem for function spaces that we already have, from the topology of $C(K)$: it buys strong (uniform) convergence, but at the price of a hypothesis – equicontinuity – that is much too strong to verify in practice.

Theorem 26.6.25 (Arzelà–Ascoli, a weak form)

Let (f_i) be a sequence of functions $[0, 1] \rightarrow \mathbb{C}$ which is bounded and equicontinuous. Then (f_i) has a uniformly convergent subsequence.

What follows is the trade we advertised: we drop equicontinuity entirely, keep only boundedness, and accept a much weaker form of convergence in exchange.

Theorem 26.6.26 (Banach–Alaoglu)

Let X be a normed space and $\overline{B}' = \{\Lambda \in X' : \|\Lambda\|_{X'} \leq 1\}$ the closed unit ball of the dual. Then $\overline{B}'$ is compact in the weak-* topology.

The general statement is proved by embedding $\overline{B}'$ into the product $\prod_{x \in X} \{z : |z| \leq \|x\|\}$ and applying Tychonoff's theorem, and therefore uses the axiom of choice in an essential way. For our purposes a sequential version on separable spaces is enough, and it can be proved by hand with a diagonal argument.

Theorem 26.6.27 (Banach–Alaoglu, sequential form)

Let X be a separable Banach space and let $(\Lambda_j)_{j=1}^\infty$ be a sequence in $\overline{B}' \subseteq X'$. Then there is a subsequence (Λ_{j_k}) and $\Lambda \in \overline{B}'$ with $\Lambda_{j_k} \xrightarrow{*} \Lambda$.

Proof. Let $D = \{x_1, x_2, \dots\}$ be a countable dense subset of X .

Diagonal extraction. The sequence $(\Lambda_j(x_1))_j$ is bounded in $\mathbb{F}$, since $|\Lambda_j(x_1)| \leq \|x_1\|$. By Bolzano–Weierstrass some subsequence converges, to a limit we call $\Lambda(x_1)$; write $\Lambda_{1,k}$ for this subsequence. Extract from it a further subsequence $\Lambda_{2,k}$ along which $\Lambda_{2,k}(x_2)$ converges, to $\Lambda(x_2)$; and so on. The diagonal sequence $\Lambda_{j,j}$ is eventually a subsequence of each $\Lambda_{l,k}$, so $\Lambda_{j,j}(x) \rightarrow \Lambda(x)$ for every $x \in D$, with $|\Lambda(x)| \leq \|x\|$.

Extension to X . The map $\Lambda : D \rightarrow \mathbb{F}$ is Lipschitz: given $x, y \in D$, choose k so large that $|\Lambda_{k,k}(x) - \Lambda(x)|$ and $|\Lambda_{k,k}(y) - \Lambda(y)|$ are both less than ε ; then

$$|\Lambda(x) - \Lambda(y)| \leq |\Lambda(x) - \Lambda_{k,k}(x)| + |\Lambda_{k,k}(x - y)| + |\Lambda_{k,k}(y) - \Lambda(y)| \leq 2\varepsilon + \|x - y\|.$$

As ε was arbitrary, Λ is 1-Lipschitz on D and so extends uniquely to a continuous map on $X = \overline{D}$.

Linearity. Let $x, y \in X$ and $a \in \mathbb{F}$, and set $z = x + ay$. Choose $x', y', z' \in D$ close to x, y, z respectively with $z' = x' + ay'$ approximately; then

$$\begin{aligned} |\Lambda(z) - \Lambda(x) - a\Lambda(y)| &\leq |\Lambda(z) - \Lambda(z')| + |\Lambda(x) - \Lambda(x')| + |a||\Lambda(y) - \Lambda(y')| \\ &\quad + |\Lambda(z') - \Lambda_{k,k}(z')| + |\Lambda(x') - \Lambda_{k,k}(x')| \\ &\quad + |a||\Lambda(y') - \Lambda_{k,k}(y')| \\ &\quad + |\Lambda_{k,k}(z' - x' - ay')|, \end{aligned}$$

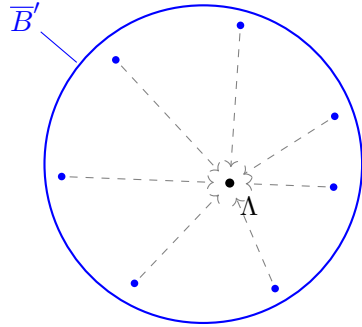
and each line is small: the first by continuity of Λ and the choice of x', y', z' , the second by taking k large, the third because $\Lambda_{k,k}$ is linear of norm at most 1 and $z' - x' - ay'$ is small. So Λ is linear, and being 1-Lipschitz, $\Lambda \in \overline{B}'$.

Convergence. Finally, for arbitrary $x \in X$ pick $x' \in D$ with $\|x - x'\|$ small; then

$$|\Lambda_{j,j}(x) - \Lambda(x)| \leq |\Lambda_{j,j}(x - x')| + |\Lambda_{j,j}(x') - \Lambda(x')| + |\Lambda(x' - x)|,$$

where the outer terms are at most $\|x - x'\|$ each and the middle tends to 0. Hence $\Lambda_{j,j} \xrightarrow{*} \Lambda$. $\square$

The theorem is worth pausing over. The closed unit ball of X' is never norm-compact in infinite dimensions, and yet it is always weak-* compact. Nothing has been gained for free: what we get in exchange for compactness is that the limit is only a weak-* limit, so that the convergence carries very little information. The whole art of the subject is to use structure – convexity, an equation, an estimate – to upgrade the weak limit afterwards.



$$\Lambda_j \xrightarrow{*} \Lambda, \text{ but } \|\Lambda_j - \Lambda\|_{X'} \not\rightarrow 0$$

Corollary 26.6.28

Let $1 < p \leq \infty$ and let (f_j) be a sequence in $L^p(\mathbb{R}^n)$ with $\|f_j\|_{L^p} \leq K$ for all j . Then there is a subsequence (f_{j_k}) and $f \in L^p(\mathbb{R}^n)$ with $\|f\|_{L^p} \leq K$ and

$$\int_{\mathbb{R}^n} f_{j_k} g \, dx \longrightarrow \int_{\mathbb{R}^n} f g \, dx \quad \text{for all } g \in L^q(\mathbb{R}^n).$$

Proof. $L^p(\mathbb{R}^n) = L^q(\mathbb{R}^n)'$ by Theorem 26.5.10, and $L^q(\mathbb{R}^n)$ is separable since $q < \infty$ (Proposition 26.3.14). Apply Theorem 26.6.27 to $K^{-1}f_j$. $\square$

Corollary 26.6.29

In a reflexive Banach space X (in particular in any Hilbert space, or in L^p for $1 < p < \infty$), every bounded sequence has a weakly convergent subsequence.

Proof. Apply Theorem 26.6.27 to the sequence viewed in $X'' = (X')'$, using that weak-* and weak convergence coincide on a reflexive space. (For the separability hypothesis one restricts to the closed span of the sequence, which is a separable reflexive space.) $\square$

Notice that $p = 1$ is excluded, and this is not an oversight. In $L^1(\mathbb{R})$ the sequence $f_n = n\mathbb{1}_{[0,1/n]}$ is bounded, but it has no weakly convergent subsequence: the only candidate limit is the Dirac mass at 0, which is not a function. Failures of weak compactness in L^1 are always of this type – concentration – and controlling them is the subject of the Dunford–Pettis theorem, which we shall not need.

§26.7 Convexity and the Direct Method

We now set out, in the abstract, the method by which we shall later prove that partial differential equations have solutions. The idea goes back to Hilbert and is disarmingly simple: to find a minimiser of a functional S , take a sequence u_n along which $S[u_n]$ tends to the infimum, extract a convergent subsequence, and hope that the limit is a minimiser. Every step of this can fail, and the interest is entirely in the conditions which make it work.

§26.7.1 Lower semicontinuity

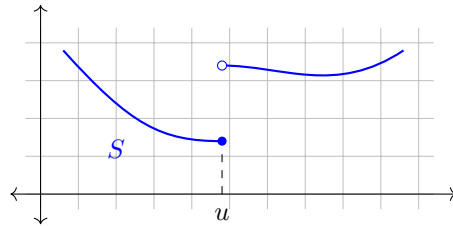
Definition 26.7.1

Let X be a topological space. A functional $S : X \rightarrow \mathbb{R} \cup \{+\infty\}$ is **(sequentially) lower semicontinuous** if

$$u_n \rightarrow u \implies S[u] \leq \liminf_{n \rightarrow \infty} S[u_n].$$

It is **coercive** if every sublevel set $\{u : S[u] \leq K\}$ is sequentially relatively compact – that is, if $S[u_n] \leq K$ for all n then (u_n) has a convergent subsequence.

Continuity implies lower semicontinuity, but not conversely. The point of the weaker hypothesis is that the value of S is allowed to *drop* in a limit but not to jump up; and dropping is harmless when we are minimising.



Theorem 26.7.2 (The direct method)

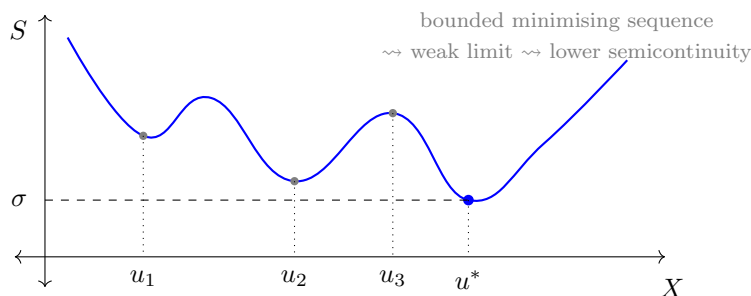
Let X be a topological space and $S : X \rightarrow \mathbb{R} \cup \{+\infty\}$ coercive and lower semicontinuous. Then S attains its infimum on X .

Proof. Let $\sigma = \inf\{S[u] : u \in X\} \in [-\infty, \infty)$ and choose a **minimising sequence** (u_n) with $S[u_n] \rightarrow \sigma$. In particular $(S[u_n])$ is bounded above, say by K , so coercivity provides a subsequence $u_{n_j} \rightarrow u^*$ for some $u^* \in X$. By lower semicontinuity,

$$S[u^*] \leq \liminf_{j \rightarrow \infty} S[u_{n_j}] = \sigma.$$

As $S[u^*] \geq \sigma$ by definition of the infimum, we get $S[u^*] = \sigma$; in particular $\sigma > -\infty$. $\square$

That is the entire method, and as stated it is nearly vacuous. All the work lies in verifying the two hypotheses, and here we run into a fundamental tension. Coercivity asks for compactness, which wants a *weak* topology; lower semicontinuity asks for the functional to behave well under limits, which wants a *strong* topology. Any given functional will typically satisfy one and fail the other. Resolving this is the content of the rest of the section, and the resolution is convexity.



§26.7.2 Coercivity in the weak topology

From now on let X be a reflexive separable Banach space. On such a space, coercivity in the weak topology is nothing more than an a priori bound.

Lemma 26.7.3

Let X be reflexive and suppose $S : X \rightarrow \mathbb{R}$ has the property that $S[u] \leq K$ implies $\|u\| \leq \tilde{K}$ for some $\tilde{K}$ depending only on K . Then S is coercive for the weak topology of X .

Proof. Immediate from Corollary 26.6.29: a bounded sequence has a weakly convergent subsequence. $\square$

So on a reflexive space, coercivity is free – it is just the statement that sublevel sets of S are bounded, which for a concrete functional is a routine estimate. The difficulty has been entirely displaced onto lower semicontinuity, which we must now verify for the weak topology. Since weak convergence is a much weaker hypothesis on (u_n) , weak lower semicontinuity is a much stronger conclusion, and it is simply false in general. (Consider $S[u] = -\|u\|^2$ on a Hilbert space and $u_n = e_n$: then $u_n \rightharpoonup 0$ but $S[u_n] = -1 < 0 = S[0]$.)

§26.7.3 Convexity saves the day

Lemma 26.7.4 (Mazur)

Let X be a Banach space and $U \subseteq X$ convex and strongly closed. Then U is weakly closed.

Proof. Let $x \in U^c$. Applying Theorem 26.6.8(b) to the compact convex set $\{x\}$ and the closed convex set U gives $\Lambda \in X'$ and $\gamma_1 < \gamma_2$ with $\operatorname{Re} \Lambda(x) < \gamma_1 < \gamma_2 < \operatorname{Re} \Lambda(y)$ for all $y \in U$. The set $\{z : \operatorname{Re} \Lambda(z) < \gamma_1\}$ is weakly open (it is a preimage of an open set under a weakly continuous map), contains x , and is disjoint from U . So U^c is weakly open. $\square$

The reader should appreciate how much this says. A strongly closed set need not be weakly closed – the unit *sphere* of a Hilbert space is strongly closed but its weak closure is the whole closed unit ball, since $e_n \rightharpoonup 0$. Convexity rules out exactly this kind of behaviour.

Theorem 26.7.5

Let X be a Banach space and $S : X \rightarrow \mathbb{R}$ convex and lower semicontinuous for the strong topology. Then S is lower semicontinuous for the weak topology.

Proof. For each $K \in \mathbb{R}$ the sublevel set $E_K = \{u : S[u] \leq K\}$ is convex (by convexity of S) and strongly closed (by strong lower semicontinuity), hence weakly closed by Lemma 26.7.4.

Now suppose $u_n \rightharpoonup u^*$, and set $\ell = \liminf_n S[u_n]$; we may assume $\ell < \infty$. Passing to a subsequence we may assume $S[u_{n_j}] \rightarrow \ell$. Given $\varepsilon > 0$, there is J with $S[u_{n_j}] \leq \ell + \varepsilon$ for $j \geq J$, i.e. $u_{n_j} \in E_{\ell+\varepsilon}$ for $j \geq J$. Since $E_{\ell+\varepsilon}$ is weakly closed and $u_{n_j} \rightharpoonup u^*$, we get $u^* \in E_{\ell+\varepsilon}$, that is $S[u^*] \leq \ell + \varepsilon$. As $\varepsilon > 0$ was arbitrary, $S[u^*] \leq \ell$. $\square$

Putting the pieces together we obtain the theorem we shall use.

Corollary 26.7.6

Let X be a reflexive Banach space and $S : X \rightarrow \mathbb{R}$ convex, strongly lower semicontinuous, and with bounded sublevel sets. Then S attains its infimum on X .

Proof. Combine Lemma 26.7.3, Theorem 26.7.5 and Theorem 26.7.2, all applied to the weak topology. (Strictly, Theorem 26.7.2 was stated for sequential compactness and sequential lower semicontinuity, which is exactly what the three results provide.) $\square$

Example 26.7.7

The norm itself is convex and continuous, so $u \mapsto \|u\|$ is weakly lower semicontinuous: this recovers the inequality $\|u^*\| \leq \liminf \|u_n\|$ for $u_n \rightharpoonup u^*$ that we saw directly earlier. More usefully, so is $u \mapsto \|u\|^2 - 2 \operatorname{Re} \Lambda(u)$ for a fixed $\Lambda \in X'$, and this functional has bounded sublevel sets since

$$\|u\|^2 - 2 \operatorname{Re} \Lambda(u) \geq \|u\|^2 - 2 \|\Lambda\| \|u\| \rightarrow \infty$$

as $\|u\| \rightarrow \infty$. So it attains its infimum. In a Hilbert space this is a long-winded proof of the Riesz representation theorem; on a Sobolev space it will be a proof that the Dirichlet problem has a solution.

Convexity is not a technical convenience that a cleverer argument might remove. Here is the standard example showing that without it the conclusion is simply false.

Example 26.7.8 (Bolza)

Work on $X = H_0^1((0, 1))$ – a space we shall construct properly in Section 26.10, but for now the reader may think of piecewise smooth functions on $[0, 1]$ vanishing at the endpoints. Define

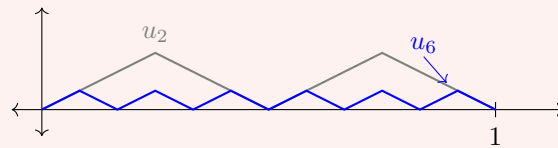
$$S[u] = \int_0^1 \left((u'(x)^2 - 1)^2 + u(x)^2 \right) dx.$$

Clearly $S \geq 0$. I claim $\inf S = 0$, but that no u attains it.

That the infimum is not attained is easy: $S[u] = 0$ would force $u = 0$ almost everywhere, hence $u' = 0$ almost everywhere, and then the first term of the integrand equals 1 everywhere, so $S[u] = 1$. Contradiction.

That the infimum is nevertheless 0 is the interesting half. Let u_m be the sawtooth with m teeth: $u_m(0) = 0$ and $u'_m = \pm 1$, alternating in sign on the $2m$ intervals $[j/(2m), (j+1)/(2m)]$. Then $u'_m(x)^2 = 1$ everywhere except at the finitely many corners, so the first term of the integrand vanishes identically, while $0 \leq u_m \leq 1/(2m)$ pointwise, so

$$S[u_m] = \int_0^1 u_m^2 \, dx \leq \frac{1}{4m^2} \rightarrow 0.$$



The minimising sequence oscillates faster and faster; by Proposition 26.6.23 it converges weakly (indeed uniformly) to 0, and $S[0] = 1 > 0 = \lim S[u_m]$. So S is *not* weakly lower semicontinuous, and the direct method breaks at exactly the step Theorem 26.7.5 was designed to supply. The culprit is visible in the integrand: $z \mapsto (z^2 - 1)^2$ has two wells and is not convex, so the gradient prefers to spend its time at the two values ± 1 rather than at their average 0.

Remark. It is worth isolating the shape of the argument, because it recurs in every application. To minimise S over X :

- (i) show S is bounded below and take a minimising sequence (u_n) ;
- (ii) derive an a priori bound $\|u_n\| \leq K$ from $S[u_n] \leq \sigma + 1$ – this is a *coercivity* estimate and is where the structure of S enters;
- (iii) extract $u_{n_j} \rightharpoonup u^*$ by reflexivity;
- (iv) show $S[u^*] \leq \liminf S[u_{n_j}]$ using convexity;
- (v) conclude that u^* is a minimiser, and derive its Euler–Lagrange equation by differentiating $t \mapsto S[u^* + tv]$ at $t = 0$.

Step (v) is what connects minimisation to differential equations, and we carry it out in Section 26.11.

§26.8 The Fourier Transform

The Fourier transform converts differentiation into multiplication, and so turns a differential equation with constant coefficients into algebra. That single fact is worth a whole course. What we want here is a clean account in the language we have built: on which spaces is the transform defined, what are its mapping properties, and can it be inverted.

§26.8.1 The transform on L^1

Definition 26.8.1 (Fourier transform)

For $f \in L^1(\mathbb{R}^n)$ we define $\mathcal{F}f = \hat{f} : \mathbb{R}^n \rightarrow \mathbb{C}$ by

$$\hat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-i(x \cdot \xi)} dx.$$

Since $|f(x)e^{-i(x \cdot \xi)}| = |f(x)| \in L^1$, the integral converges absolutely for every ξ , and $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}$. (There are several conventions in circulation, differing by factors of 2π in the exponent or in front of the integral. We use this one; it makes the differentiation rule clean at the cost of a constant $(2\pi)^n$ in the inversion formula.)

Lemma 26.8.2 (Riemann–Lebesgue)

Let $f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^0(\mathbb{R}^n)$ with $\|\hat{f}\|_{L^\infty} \leq \|f\|_{L^1}$, and $\hat{f}(\xi) \rightarrow 0$ as $|\xi| \rightarrow \infty$.

Proof. If $\xi_m \rightarrow \xi$ then $f(x)e^{-i(x \cdot \xi_m)} \rightarrow f(x)e^{-i(x \cdot \xi)}$ pointwise, dominated by $|f| \in L^1$, so $\hat{f}(\xi_m) \rightarrow \hat{f}(\xi)$ by dominated convergence; and the sup bound is immediate. For the decay, let $\varepsilon > 0$ and use Corollary 26.4.11 to find $f_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ with $\|f - f_\varepsilon\|_{L^1} < \varepsilon$. Integrating by parts (there are no boundary terms, as f_ε has compact support),

$$\hat{f}_\varepsilon(\xi) = \int_{\mathbb{R}^n} f_\varepsilon(x) \operatorname{div}_x \left(\frac{\xi}{-i|\xi|^2} e^{-i(x \cdot \xi)} \right) dx = \int_{\mathbb{R}^n} \frac{\xi \cdot Df_\varepsilon(x)}{i|\xi|^2} e^{-i(x \cdot \xi)} dx,$$

so that $|\hat{f}_\varepsilon(\xi)| \leq |\xi|^{-1} \|Df_\varepsilon\|_{L^1} < \varepsilon$ once $|\xi|$ is large enough. Then $|\hat{f}(\xi)| \leq |\hat{f} - \hat{f}_\varepsilon| + |\hat{f}_\varepsilon| < 2\varepsilon$ for such ξ . $\square$

So $\mathcal{F}$ maps $L^1(\mathbb{R}^n)$ into the space $C_0(\mathbb{R}^n)$ of continuous functions vanishing at infinity. It is not onto – the image is a dense proper subspace – and this is one reason L^1 is not the natural home for the transform.

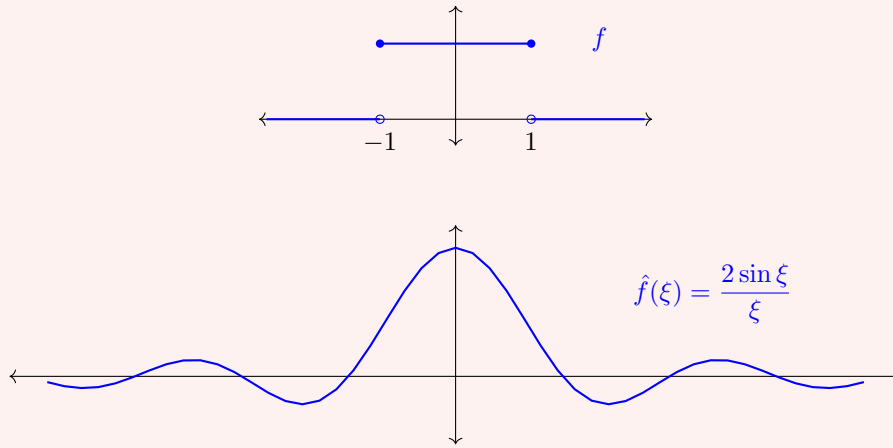
Example 26.8.3

Take $n = 1$.

- (i) $f = \mathbb{1}_{(-1,1)}$ has $\hat{f}(\xi) = \int_{-1}^1 e^{-ix\xi} dx = \frac{2 \sin \xi}{\xi}$ for $\xi \neq 0$, and $\hat{f}(0) = 2$.
- (ii) $f(x) = e^{-|x|}$ has $\hat{f}(\xi) = \frac{2}{1 + \xi^2}$.
- (iii) $f(x) = \frac{1}{1 + x^2}$ has $\hat{f}(\xi) = \pi e^{-|\xi|}$ – the transform of (ii), up to constants, which is the inversion theorem at work.
- (iv) $f(x) = e^{-x^2/2}$ has $\hat{f}(\xi) = \sqrt{2\pi} e^{-\xi^2/2}$. To see this, note $f' = -xf$, so by Theorem 26.8.5 below $i\xi \hat{f} = -(\hat{f})' \cdot (-1)^{-1}$, giving $(\hat{f})' = -\xi \hat{f}$; solving this ODE with $\hat{f}(0) = \int e^{-x^2/2} = \sqrt{2\pi}$ gives the claim. The Gaussian is (up to scaling) the unique fixed point of $\mathcal{F}$, which is why it appears everywhere in this subject.

The first example is worth drawing, since it shows both the smoothing and the

delocalisation.



Note that $\hat{f}$ is smooth (indeed real-analytic) but decays only like $|\xi|^{-1}$, and in particular is not integrable. Compact support has been traded for smoothness, and vice versa; this trade-off is a theme.

§26.8.2 Basic properties

Write $e_y(x) = e^{i(x \cdot y)}$ and recall $\tau_y f(x) = f(x - y)$ and $\check{f}(x) = f(-x)$.

Lemma 26.8.4

Let $f, g \in L^1(\mathbb{R}^n)$, $y \in \mathbb{R}^n$ and $\mu > 0$.

- (i) Writing $f_\mu(x) = \mu^{-n} f(\mu^{-1}x)$, we have $\widehat{f_\mu}(\xi) = \hat{f}(\mu\xi)$.
- (ii) $\widehat{e_y f} = \tau_y \hat{f}$ and $\widehat{\tau_y f} = e_{-y} \hat{f}$.
- (iii) $f * g \in L^1(\mathbb{R}^n)$ and $\widehat{f * g} = \hat{f} \hat{g}$.

Proof. (i) and (ii) are changes of variable. For (iii), Tonelli gives $\|f * g\|_{L^1} \leq \|f\|_{L^1} \|g\|_{L^1}$, so $f * g \in L^1$, and then by Fubini

$$\begin{aligned} \widehat{f * g}(\xi) &= \iint f(y)g(x - y)e^{-i(x \cdot \xi)} \, dy \, dx \\ &= \int f(y)e^{-i(y \cdot \xi)} \left(\int g(z)e^{-i(z \cdot \xi)} \, dz \right) \, dy = \hat{f}(\xi)\hat{g}(\xi). \square \end{aligned}$$

Part (iii) is the reason the Fourier transform is useful for differential equations: it turns the awkward, non-local operation of convolution into pointwise multiplication.

Theorem 26.8.5 (i) Suppose $f \in C^1(\mathbb{R}^n)$ with $f, D_j f \in L^1(\mathbb{R}^n)$ for each j . Then $\widehat{D_j f}(\xi) = i\xi_j \hat{f}(\xi)$.

(ii) Suppose $(1 + |x|)f \in L^1(\mathbb{R}^n)$. Then $\hat{f} \in C^1(\mathbb{R}^n)$ and $D_j \hat{f}(\xi) = \widehat{-ix_j f}(\xi)$.

Proof. (i) Given $\varepsilon > 0$ choose $f_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ with $\|f - f_\varepsilon\|_{L^1} + \sum_j \|D_j f - D_j f_\varepsilon\|_{L^1} < \varepsilon$ (mollify and cut off). For compactly supported smooth functions, integration by parts gives

$$\widehat{D_j f_\varepsilon}(\xi) = \int e^{-i(x \cdot \xi)} D_j f_\varepsilon(x) \, dx = i\xi_j \int e^{-i(x \cdot \xi)} f_\varepsilon(x) \, dx = i\xi_j \hat{f}_\varepsilon(\xi).$$

Hence

$$|\widehat{D_j f}(\xi) - i\xi_j \hat{f}(\xi)| \leq \|D_j f - D_j f_\varepsilon\|_{L^1} + |\xi| \|f - f_\varepsilon\|_{L^1} < (1 + |\xi|)\varepsilon,$$

and letting $\varepsilon \rightarrow 0$ with ξ fixed gives the identity.

(ii) By hypothesis $x_j f \in L^1$, so $\widehat{-ix_j f}$ is defined and continuous. For $h \neq 0$,

$$\frac{\hat{f}(\xi + he_j) - \hat{f}(\xi)}{h} = \int f(x) e^{-i(x \cdot \xi)} \frac{e^{-ix_j h} - 1}{h} \, dx,$$

and the integrand converges pointwise to $-ix_j f(x) e^{-i(x \cdot \xi)}$, dominated by $|x_j f(x)| \in L^1$ since $|e^{i\theta} - 1| = 2|\sin(\theta/2)| \leq |\theta|$. Dominated convergence finishes the proof. $\square$

In words: *decay of f controls smoothness of $\hat{f}$, and smoothness of f controls decay of $\hat{f}$.* A space on which both hold to all orders would be mapped to itself by $\mathcal{F}$, and this is exactly the definition of the next space.

§26.8.3 The Schwartz space

Definition 26.8.6 (Schwartz space)

A function $\varphi \in C^\infty(\mathbb{R}^n)$ is **rapidly decreasing** if

$$p_N(\varphi) = \sup_{x \in \mathbb{R}^n, |\alpha| \leq N} |(1 + |x|)^N D^\alpha \varphi(x)| < \infty$$

for every $N \in \mathbb{N}$. The set of such functions is the **Schwartz space** $\mathcal{S}(\mathbb{R}^n)$, and we equip it with the topology generated by the separating family of seminorms $\{p_N : N \in \mathbb{N}\}$.

Since the family is countable, $\mathcal{S}(\mathbb{R}^n)$ is metrisable, and in fact it is complete, so it is a Fréchet space. Concretely, $\varphi_j \rightarrow 0$ in $\mathcal{S}(\mathbb{R}^n)$ if and only if $\sup_x |(1 + |x|)^N D^\alpha \varphi_j(x)| \rightarrow 0$ for every N and α . One may equally use the seminorms $\sup_{|\alpha|, |\beta| \leq N} |x^\beta D^\alpha \varphi(x)|$; they generate the same topology.

Example 26.8.7

$\varphi(x) = e^{-|x|^2}$ is in $\mathcal{S}(\mathbb{R}^n)$, and so is every element of $C_c^\infty(\mathbb{R}^n)$. The function $(1 + |x|^2)^{-1001}$ is not, since it fails to decay faster than every polynomial. Thus

$$C_c^\infty(\mathbb{R}^n) \subsetneq \mathcal{S}(\mathbb{R}^n) \subsetneq C^\infty(\mathbb{R}^n),$$

with each inclusion continuous.

Corollary 26.8.8

$\mathcal{F}$ maps $\mathcal{S}(\mathbb{R}^n)$ continuously into itself.

Proof. First note the elementary bound, for any f with $(1 + |x|)^{n+1}f$ bounded,

$$\|f\|_{L^1} = \int_{\mathbb{R}^n} \frac{|f|(1 + |x|)^{n+1}}{(1 + |x|)^{n+1}} dx \leq C_n \sup_x (|f(x)|(1 + |x|)^{n+1}),$$

where $C_n = \int_{\mathbb{R}^n} (1 + |x|)^{-(n+1)} dx < \infty$. If $\varphi \in \mathcal{S}(\mathbb{R}^n)$ then $D^\alpha(x^\beta \varphi) \in L^1(\mathbb{R}^n)$ for all multi-indices, so applying Theorem 26.8.5 repeatedly gives $\left| \widehat{D^\alpha(x^\beta \varphi)}(\xi) \right| = |\xi^\alpha D^\beta \hat{\varphi}(\xi)|$ up to signs, whence

$$\sup_\xi |\xi^\alpha D^\beta \hat{\varphi}(\xi)| \leq \|D^\alpha(x^\beta \varphi)\|_{L^1} \leq C_{\alpha,\beta} \sup_{x, |\delta| \leq |\alpha|} (1 + |x|)^{|\beta|+n+1} |D^\delta \varphi(x)|.$$

The right-hand side is finite, so $\hat{\varphi} \in \mathcal{S}(\mathbb{R}^n)$; and if $\varphi_j \rightarrow 0$ in $\mathcal{S}$ then the same estimate gives $\hat{\varphi}_j \rightarrow 0$ in $\mathcal{S}$, which is continuity. $\square$

§26.8.4 Inversion**Theorem 26.8.9 (Fourier inversion)**

Suppose $f, \hat{f} \in L^1(\mathbb{R}^n)$. Then for almost every x ,

$$f(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi) e^{i(x \cdot \xi)} d\xi.$$

If f is in addition continuous, the identity holds for every x .

Proof. The naive computation – substitute the definition of $\hat{f}$ and swap the integrals – fails, because $e^{i((x-y) \cdot \xi)}$ is not integrable in ξ . The standard repair is to insert a Gaussian convergence factor.

Let $G(\xi) = e^{-|\xi|^2/2}$ and recall from Example 26.8.3(iv) (applied in each variable) that $\hat{G} = (2\pi)^{n/2}G$. For $\varepsilon > 0$ consider

$$I_\varepsilon(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi) G(\varepsilon \xi) e^{i(x \cdot \xi)} d\xi.$$

Since $|\hat{f}(\xi)G(\varepsilon \xi)| \leq |\hat{f}(\xi)| \in L^1$ and $G(\varepsilon \xi) \rightarrow 1$ pointwise, dominated convergence gives $I_\varepsilon(x) \rightarrow (2\pi)^{-n} \int \hat{f}(\xi) e^{i(x \cdot \xi)} d\xi$ as $\varepsilon \rightarrow 0$, for every x .

On the other hand, everything in sight is now absolutely integrable in (ξ, y) jointly, so Fubini applies:

$$I_\varepsilon(x) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} f(y) \left(\int_{\mathbb{R}^n} G(\varepsilon \xi) e^{i((x-y) \cdot \xi)} d\xi \right) dy = \int_{\mathbb{R}^n} f(y) k_\varepsilon(x - y) dy,$$

where, by Lemma 26.8.4(i) and $\hat{G} = (2\pi)^{n/2}G$,

$$k_\varepsilon(z) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} G(\varepsilon\xi) e^{i(z \cdot \xi)} d\xi = \frac{1}{(2\pi)^{n/2} \varepsilon^n} e^{-|z|^2/(2\varepsilon^2)}.$$

This is precisely an approximate identity: $k_\varepsilon \geq 0$, $\int k_\varepsilon = 1$, and k_ε concentrates at the origin as $\varepsilon \rightarrow 0$. (It is not compactly supported, but the proof of Theorem 26.4.10 used only the normalisation and the concentration, and goes through verbatim.) Hence $I_\varepsilon = k_\varepsilon * f \rightarrow f$ in L^1 , and so along a subsequence almost everywhere. Comparing the two limits gives the result, and if f is continuous then both sides are continuous and agree a.e., hence everywhere. $\square$

Corollary 26.8.10

$\mathcal{F} : \mathcal{S}(\mathbb{R}^n) \rightarrow \mathcal{S}(\mathbb{R}^n)$ is a topological isomorphism, with $\mathcal{F}^2 f = (2\pi)^n \check{f}$ and $\mathcal{F}^4 = (2\pi)^{2n} \text{id}$.

Proof. If $\varphi \in \mathcal{S}$ then $\varphi, \hat{\varphi} \in L^1$ and φ is continuous, so Theorem 26.8.9 applies at every point; rewriting the inversion formula as $\varphi(x) = (2\pi)^{-n} \hat{\hat{\varphi}}(-x)$ gives $\mathcal{F}^2 \varphi = (2\pi)^n \hat{\varphi}$, and squaring gives $\mathcal{F}^4 = (2\pi)^{2n} \text{id}$. So $\mathcal{F}$ is a bijection of $\mathcal{S}$ with continuous inverse $(2\pi)^{-n} \mathcal{F}^3$, using Corollary 26.8.8. $\square$

§26.8.5 Plancherel and the transform on L^2

The Fourier transform is not defined on L^2 by the integral formula – an L^2 function need not be integrable – but it extends there by density, and on L^2 it is at its best: it becomes (a multiple of) a unitary map.

Theorem 26.8.11 (Plancherel/Parseval)

Let $f, g \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. Then $\hat{f}, \hat{g} \in L^2(\mathbb{R}^n)$ and

$$\langle f, g \rangle_{L^2} = (2\pi)^{-n} \langle \hat{f}, \hat{g} \rangle_{L^2}, \quad \text{in particular} \quad \|\hat{f}\|_{L^2} = (2\pi)^{n/2} \|f\|_{L^2}.$$

Proof. The Schwartz case. Let $f, g \in \mathcal{S}(\mathbb{R}^n)$ and set $h(y) = \overline{g(-y)}$, so that $h \in \mathcal{S}(\mathbb{R}^n)$ too, and

$$\hat{h}(\xi) = \int_{\mathbb{R}^n} \overline{g(-y)} e^{-i(y \cdot \xi)} dy = \overline{\int_{\mathbb{R}^n} g(z) e^{-i(z \cdot \xi)} dz} = \overline{\hat{g}(\xi)},$$

substituting $z = -y$. Now put $k = f * h$, which lies in $\mathcal{S}(\mathbb{R}^n)$ (it is smooth by Theorem 26.4.6, and rapidly decreasing because each derivative is a convolution of two rapidly decreasing functions). On the one hand, $\hat{k} = \hat{f} \hat{h} = \hat{f} \overline{\hat{g}}$ by Lemma 26.8.4(iii), so evaluating the inversion formula of Theorem 26.8.9 at $x = 0$ gives

$$k(0) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{k}(\xi) d\xi = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} \hat{f}(\xi) \overline{\hat{g}(\xi)} d\xi.$$

On the other hand, directly from the definition of the convolution,

$$k(0) = \int_{\mathbb{R}^n} f(y)h(-y) \, dy = \int_{\mathbb{R}^n} f(y)\overline{g(y)} \, dy.$$

Comparing the two, and conjugating to match our convention $\langle f, g \rangle_{L^2} = \int \overline{f}g$, gives $\langle f, g \rangle_{L^2} = (2\pi)^{-n} \langle \hat{f}, \hat{g} \rangle_{L^2}$ for Schwartz f, g .

The general case. Let $f \in L^1 \cap L^2$. Truncating and mollifying as in Corollary 26.4.11 produces $f_m \in C_c^\infty(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$ with $f_m \rightarrow f$ in L^1 and in L^2 simultaneously: indeed $f \mathbb{1}_{B_R(0)} \rightarrow f$ in both norms by dominated convergence, and $\phi_\varepsilon * (f \mathbb{1}_{B_R(0)}) \rightarrow f \mathbb{1}_{B_R(0)}$ in both norms by Theorem 26.4.10. By the Schwartz case applied to differences, $\|\hat{f}_m - \hat{f}_l\|_{L^2} = (2\pi)^{n/2} \|f_m - f_l\|_{L^2}$, so $(\hat{f}_m)$ is Cauchy in L^2 and converges to some $F \in L^2$. But $\|\hat{f}_m - \hat{f}\|_{L^\infty} \leq \|f_m - f\|_{L^1} \rightarrow 0$, so $\hat{f}_m \rightarrow \hat{f}$ uniformly, and hence (along a subsequence, using L^2 convergence) $F = \hat{f}$ almost everywhere. Therefore $\hat{f} \in L^2$ with $\|\hat{f}\|_{L^2} = \lim_m \|\hat{f}_m\|_{L^2} = (2\pi)^{n/2} \lim_m \|f_m\|_{L^2} = (2\pi)^{n/2} \|f\|_{L^2}$, and the polarised identity for $\langle f, g \rangle$ follows in the same way. $\square$

So $(2\pi)^{-n/2} \mathcal{F}$ is an isometry from $L^1 \cap L^2$, which is dense in L^2 , into L^2 . A densely defined isometry into a complete space extends uniquely to an isometry of the closure, and we obtain:

Definition 26.8.12

The unique continuous extension $\overline{\mathcal{F}} : L^2(\mathbb{R}^n) \rightarrow L^2(\mathbb{R}^n)$ of $\mathcal{F}|_{L^1 \cap L^2}$ is the **Fourier–Plancherel transform**. We write $\mathcal{F}$ for it as well.

Proposition 26.8.13

$\overline{\mathcal{F}}$ is a bijection of $L^2(\mathbb{R}^n)$ with $\|\overline{\mathcal{F}}f\|_{L^2} = (2\pi)^{n/2} \|f\|_{L^2}$, and for $f \in L^2(\mathbb{R}^n)$,

$$\overline{\mathcal{F}}f = \lim_{R \rightarrow \infty} \int_{B_R(0)} f(x) e^{-i(x \cdot \xi)} \, dx$$

with the limit taken in $L^2(\mathbb{R}^n)$.

Proof. Set $f_R = f \mathbb{1}_{B_R(0)}$. Then $f_R \in L^1 \cap L^2$ by Hölder, and $f_R \rightarrow f$ in L^2 ; since $\overline{\mathcal{F}}$ is continuous, $\hat{f}_R \rightarrow \overline{\mathcal{F}}f$ in L^2 . Surjectivity follows from $\mathcal{F}^4 = (2\pi)^{2n} \text{id}$ on $\mathcal{S}$, which passes to the closure. $\square$

Remark. The Fourier transform therefore acts on L^1 (mapping into C_0), on L^2 (bijectively, isometrically up to constants), and by interpolation on L^p for $1 \leq p \leq 2$ (the Hausdorff–Young inequality, $\|\hat{f}\|_{L^q} \leq C \|f\|_{L^p}$ for conjugate p, q with $p \leq 2$). For $p > 2$ it does not map L^p into any L^q , and the transform of such a function only makes sense as a distribution. This is one of several reasons we now build distribution theory.

§26.9 Distributions

In the study of differential equations one constantly meets objects which behave like functions but are not.

Example 26.9.1

Consider

$$G : \mathbb{R}^3 \setminus \{0\} \rightarrow \mathbb{R}, \quad x \mapsto \frac{-1}{4\pi|x|}.$$

Every physicist knows that $\nabla^2 G = \delta$, the point mass at the origin, and this fact is the basis of the theory of the Laplace and Poisson equations. But G is not twice differentiable at 0, δ is not a function, and the equation as written is meaningless.

The theory of distributions makes such statements rigorous, and it does so by a strategy we have already used twice: instead of studying an object directly, study how it pairs against nice test objects. A locally integrable function f is determined by the numbers $\int f\varphi$ for φ ranging over a suitable class; so let us simply *define* a generalised function to be a rule assigning a number to each φ . The class of φ must be small enough that many rules qualify, and large enough that the rule is determined; the balance is struck by taking φ smooth and compactly supported.

§26.9.1 Spaces of test functions

Let $\Omega \subseteq \mathbb{R}^n$ be open. We have three test function spaces, in increasing order of size.

The space $C_c^\infty(\Omega)$ of smooth functions with compact support in Ω carries a natural vector space structure, and it can be given a topology making it a locally convex topological vector space. The construction (an inductive limit over compact exhaustions) is not especially illuminating and we omit it; what matters is the following list of properties, which is all we shall ever use.

Theorem 26.9.2

The space $C_c^\infty(\Omega)$ carries a topology τ such that:

- (i) $(C_c^\infty(\Omega), \tau)$ is a locally convex topological vector space, which we denote $\mathcal{D}(\Omega)$;
- (ii) a sequence (φ_j) converges to 0 in $\mathcal{D}(\Omega)$ if and only if there is a single compact $K \subseteq \Omega$ with $\text{supp } \varphi_j \subseteq K$ for all j , and $\sup_{x \in K} |D^\alpha \varphi_j(x)| \rightarrow 0$ for every multi-index α ;
- (iii) if Y is a locally convex topological vector space and $\Lambda : \mathcal{D}(\Omega) \rightarrow Y$ is linear, then Λ is continuous if and only if it is sequentially continuous.

Property (ii) is a strong requirement – the supports must not run away, and *all* derivatives must converge uniformly. Property (iii) is what makes the space usable in practice, since it lets us verify continuity one sequence at a time.

Example 26.9.3

Let $\varphi \in C_c^\infty(\mathbb{R})$ be non-zero.

- (a) $\varphi_j(x) = e^{-j}\varphi(jx)$ converges to 0 in $\mathcal{D}(\mathbb{R})$: the supports shrink to $\{0\}$, and although $D^\alpha\varphi_j$ picks up a factor $j^{|\alpha|}$, the factor e^{-j} swamps it.
- (b) $\varphi_j(x) = j^{-1000}\varphi(jx)$ does *not* converge to 0 in $\mathcal{D}(\mathbb{R})$: for $|\alpha| > 1000$ the derivatives blow up.
- (c) $\varphi_j(x) = e^{-j}\varphi(x - j)$ does not converge to 0 in $\mathcal{D}(\mathbb{R})$, even though every derivative tends to 0 uniformly, because the supports escape to infinity and no single compact set contains them all.

The second space is built on $C^\infty(\Omega)$, with no support restriction. Choose an exhaustion of Ω by compact sets K_i with $K_i \subseteq K_{i+1}^\circ$ and $\bigcup_i K_i = \Omega$, and for $\varphi \in C^\infty(\Omega)$ set

$$p_N(\varphi) = \sup_{x \in K_N, |\alpha| \leq N} |D^\alpha \varphi(x)|.$$

These form a countable separating family of seminorms, whose topology is complete; the resulting Fréchet space is denoted $\mathcal{E}(\Omega)$.

Theorem 26.9.4

$\varphi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ if and only if $\sup_{x \in K} |D^\alpha \varphi_j(x)| \rightarrow 0$ for every compact $K \subseteq \Omega$ and every multi-index α ; that is, if and only if all derivatives converge to 0 uniformly on compact sets.

Example 26.9.5

If $\varphi \in C_c^\infty(\mathbb{R})$ is non-zero then $\varphi_j(x) = \varphi(x - j) \rightarrow 0$ in $\mathcal{E}(\mathbb{R})$, since on any fixed compact set φ_j eventually vanishes identically. As we saw, this sequence does not converge in $\mathcal{D}(\mathbb{R})$: the topology of $\mathcal{E}$ is strictly weaker.

The third space is the Schwartz space $\mathcal{S}(\mathbb{R}^n)$ of Section 26.8, which sits between the other two:

$$\mathcal{D}(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n) \subseteq \mathcal{E}(\mathbb{R}^n),$$

with both inclusions continuous (and, one checks, with dense image).

§26.9.2 Distributions

Definition 26.9.6 (Distribution)

Let $\Omega \subseteq \mathbb{R}^n$ be open. The space of **distributions** $\mathcal{D}'(\Omega)$ is the continuous dual of $\mathcal{D}(\Omega)$: the space of linear maps $u : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ such that $u[\varphi_j] \rightarrow 0$ whenever $\varphi_j \rightarrow 0$ in $\mathcal{D}(\Omega)$. We equip $\mathcal{D}'(\Omega)$ with the weak-* topology, so that

$$u_j \rightarrow u \text{ in } \mathcal{D}'(\Omega) \iff u_j[\varphi] \rightarrow u[\varphi] \text{ for every } \varphi \in \mathcal{D}(\Omega).$$

We write $u[\varphi]$ rather than $u(\varphi)$ to emphasise that u is not a function of x .

Example 26.9.7 (a) For $x \in \Omega$, the evaluation $\delta_x : \varphi \mapsto \varphi(x)$ is a distribution, the **Dirac delta** at x . Continuity is clear from Theorem 26.9.2(ii).

(b) For $f \in L^1_{\text{loc}}(\Omega)$, the map

$$T_f : \varphi \mapsto \int_{\Omega} f \varphi \, dx$$

is a distribution. Indeed if $\varphi_j \rightarrow 0$ in $\mathcal{D}(\Omega)$ with supports in a compact K , then $\varphi_j \rightarrow 0$ uniformly, so $|f \mathbb{1}_K \varphi_j| \leq C|f| \mathbb{1}_K \in L^1$ and dominated convergence gives $T_f[\varphi_j] \rightarrow 0$.

The following characterisation of continuity is the practical test.

Lemma 26.9.8

A linear map $u : \mathcal{D}(\Omega) \rightarrow \mathbb{C}$ is continuous if and only if for every compact $K \subseteq \Omega$ there are $N_K \in \mathbb{N}$ and $C_K > 0$ with

$$|u[\varphi]| \leq C_K \sup_{x \in K} \sum_{|\alpha| \leq N_K} |D^\alpha \varphi(x)| \quad \text{for all } \varphi \in C_c^\infty(K).$$

Proof. ($\Leftarrow$) If $\varphi_j \rightarrow 0$ in $\mathcal{D}(\Omega)$ then the supports lie in a common compact K and every $D^\alpha \varphi_j \rightarrow 0$ uniformly on K ; the estimate then forces $u[\varphi_j] \rightarrow 0$.

($\Rightarrow$) Suppose the estimate fails for some compact K . Then for each j there is $\varphi_j \in C_c^\infty(K)$ with

$$|u[\varphi_j]| > j \sup_{x \in K} \sum_{|\alpha| \leq j} |D^\alpha \varphi_j(x)|.$$

Put $\psi_j = \varphi_j / |u[\varphi_j]|$. Then for any fixed β and all $j \geq |\beta|$,

$$\sup_K |D^\beta \psi_j| = \frac{\sup_K |D^\beta \varphi_j|}{|u[\varphi_j]|} \leq \frac{1}{j} \rightarrow 0,$$

so $\psi_j \rightarrow 0$ in $\mathcal{D}(\Omega)$ – but $|u[\psi_j]| = 1$ for all j , contradicting continuity. $\square$

Definition 26.9.9 (Order)

If in Lemma 26.9.8 the integer N_K can be chosen independently of K , we say u has **finite order**, and the least such N is the **order** of u .

Example 26.9.10

δ_x and T_f have order 0; $\varphi \mapsto \varphi'(x)$ has order 1; and

$$u[\varphi] = \sum_{i=1}^{\infty} \varphi^{(i)}(i)$$

defines a distribution on $\mathbb{R}$ (the sum is finite for each φ) which is continuous but has no finite order.

The map $f \mapsto T_f$ is injective on $L^1_{\text{loc}}(\Omega)$: if $T_f = T_g$ then $\int (f - g)\varphi = 0$ for all $\varphi \in C_c^\infty(\Omega)$, so $f = g$ almost everywhere by Corollary 26.4.12. We may therefore regard $L^1_{\text{loc}}(\Omega)$ as a subspace of $\mathcal{D}'(\Omega)$, and from now on we abuse notation by writing f for T_f .

Distributions are ‘generalised functions’ in exactly this sense.

§26.9.3 Operations on distributions

The rule for extending an operation from functions to distributions is always the same: find an identity, valid for smooth functions, which expresses the operation in terms of the pairing, and promote it to a definition.

Multiplication by a smooth function. If $a \in C^\infty(\Omega)$ and $f \in L^1_{\text{loc}}$, then $T_{af}[\varphi] = T_f[a\varphi]$. Since $a\varphi \in \mathcal{D}(\Omega)$ whenever φ is, we define

$$(au)[\varphi] = u[a\varphi], \quad u \in \mathcal{D}'(\Omega).$$

Note that there is no way to multiply two distributions: δ_0^2 is not defined, and the failure is not a defect of the theory but a genuine obstruction.

Differentiation. If $f \in C^1$ then integration by parts gives $T_{D_i f}[\varphi] = -T_f[D_i \varphi]$, with no boundary term since φ has compact support. We therefore define, for any multi-index α ,

$$D^\alpha u[\varphi] = (-1)^{|\alpha|} u[D^\alpha \varphi].$$

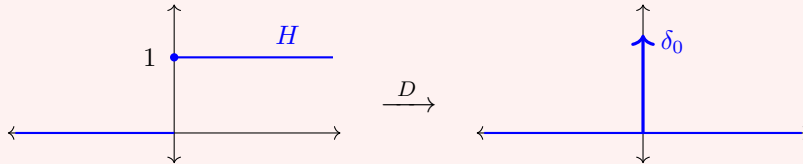
This is again a distribution, by Lemma 26.9.8. So *every distribution is infinitely differentiable*, and in particular every locally integrable function has derivatives of all orders in $\mathcal{D}'$. The price is that $D^\alpha u$ need not lie in L^1_{loc} even if u does.

Example 26.9.11

Let $H = \mathbb{1}_{[0, \infty)}$ be the Heaviside function on $\mathbb{R}$, which is certainly in L^1_{loc} . For $\varphi \in \mathcal{D}(\mathbb{R})$,

$$DH[\varphi] = -H[\varphi'] = -\int_0^\infty \varphi'(x) \, dx = \varphi(0) = \delta_0[\varphi],$$

so $DH = \delta_0$ in $\mathcal{D}'(\mathbb{R})$. Differentiating again, $D^2H = D\delta_0$ is the distribution $\varphi \mapsto -\varphi'(0)$, which is not even a measure.



Here is a second example, slightly more delicate, which shows that distributions can also repair *divergent* integrals and not merely non-existent derivatives.

Example 26.9.12 (Principal value)

The function $1/x$ is not locally integrable on $\mathbb{R}$, so it does not define a distribution by integration. Nevertheless the symmetric divergences at 0^+ and 0^- cancel, and

$$\text{p.v.}\left(\frac{1}{x}\right)[\varphi] = \lim_{\varepsilon \downarrow 0} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} \, dx$$

does define a distribution on $\mathbb{R}$. To see that the limit exists, split the integral at $|x| = 1$. The outer part $\int_{|x| > 1} \varphi(x)/x \, dx$ converges absolutely, since $|1/x| \leq 1$ there and φ has compact support. For the inner part, the oddness of $1/x$ gives

$\int_{\varepsilon < |x| < 1} dx/x = 0$, so we may subtract $\varphi(0)$ for free:

$$\int_{\varepsilon < |x| < 1} \frac{\varphi(x)}{x} dx = \int_{\varepsilon < |x| < 1} \frac{\varphi(x) - \varphi(0)}{x} dx,$$

and now the integrand is bounded in modulus by $\sup |\varphi'|$ by the mean value theorem, so the limit exists by dominated convergence. The same estimate bounds

$$\left| \text{p.v.} \left(\frac{1}{x} \right) [\varphi] \right| \leq 2 \sup_{\mathbb{R}} |\varphi'| + \left(\int_{|x| > 1, x \in K} \frac{dx}{|x|} \right) \sup_{\mathbb{R}} |\varphi|$$

for φ supported in a compact K , so Lemma 26.9.8 applies and the principal value is a distribution of order (at most) 1. It is genuinely of order 1, and in particular is not a measure.

It is also, as one would hope, a derivative. The function $\log |x|$ is locally integrable, and I claim $D \log |x| = \text{p.v.}(1/x)$. Indeed, for $\varphi \in \mathcal{D}(\mathbb{R})$, integrating by parts on $\{x > \varepsilon\}$ and on $\{x < -\varepsilon\}$ separately,

$$- \int_{|x| > \varepsilon} \log |x| \varphi'(x) dx = \log \varepsilon (\varphi(\varepsilon) - \varphi(-\varepsilon)) + \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx.$$

Since $|\varphi(\varepsilon) - \varphi(-\varepsilon)| \leq 2\varepsilon \sup |\varphi'|$ and $\varepsilon \log \varepsilon \rightarrow 0$, the boundary term vanishes in the limit, and dominated convergence handles the left-hand side. So $D \log |x|[\varphi] = - \int \log |x| \varphi' = \text{p.v.}(1/x)[\varphi]$, as claimed. The reader who has met contour integration will recognise the principal value as the real part of the Sokhotski–Plemelj formula.

With these operations we can make sense of linear partial differential equations in $\mathcal{D}'(\Omega)$: given $a_\alpha \in C^\infty(\Omega)$ and $w \in \mathcal{D}'(\Omega)$, we may ask for $u \in \mathcal{D}'(\Omega)$ with

$$\sum_{|\alpha| \leq k} a_\alpha D^\alpha u = w.$$

Solutions in this sense are called **weak** or **distributional** solutions, and the whole point of the last part of the course is that they are much easier to find than classical ones – after which one works to prove that they are classical after all.

Example 26.9.13

Mollifiers converge to the delta: for $\varphi \in \mathcal{D}(\mathbb{R}^n)$,

$$\phi_\varepsilon[\varphi] = \int \phi_\varepsilon(x) \varphi(x) dx \rightarrow \varphi(0) = \delta_0[\varphi]$$

by the same computation as in Theorem 26.4.10. So $\phi_\varepsilon \rightarrow \delta_0$ in $\mathcal{D}'(\mathbb{R}^n)$, which is the precise sense in which the pictures in Section 26.4 ‘converge to a spike’.

§26.9.4 Compactly supported and tempered distributions

The duals of the other two test spaces give distinguished subspaces of $\mathcal{D}'$.

Lemma 26.9.14

A linear $u : \mathcal{E}(\Omega) \rightarrow \mathbb{C}$ is continuous if and only if there are a compact $K \subseteq \Omega$, $N \in \mathbb{N}$ and $C > 0$ with

$$|u[\varphi]| \leq C \sup_{x \in K, |\alpha| \leq N} |D^\alpha \varphi(x)| \quad \text{for all } \varphi \in \mathcal{E}(\Omega).$$

Proof. ($\Leftarrow$) If $\varphi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ then in particular the right-hand side tends to 0.

($\Rightarrow$) Suppose no such (K, N, C) exists. Take a compact exhaustion (K_j) of Ω ; then for each j there is $\varphi_j \in \mathcal{E}(\Omega)$ with $|u[\varphi_j]| \geq j \sup_{x \in K_j, |\alpha| \leq j} |D^\alpha \varphi_j(x)|$. Setting $\psi_j = \varphi_j / |u[\varphi_j]|$, for any compact $\tilde{K}$ and $\tilde{N}$ we may choose $J > \tilde{N}$ with $\tilde{K} \subseteq K_j$ for $j \geq J$, and then

$$\sup_{x \in \tilde{K}, |\alpha| \leq \tilde{N}} |D^\alpha \psi_j(x)| \leq \frac{1}{j} \rightarrow 0.$$

So $\psi_j \rightarrow 0$ in $\mathcal{E}(\Omega)$ while $|u[\psi_j]| = 1$: contradiction. $\square$

Definition 26.9.15

A distribution $u \in \mathcal{D}'(\Omega)$ has **compact support** if there is a compact $K \subseteq \Omega$ with $u[\varphi] = 0$ whenever $\text{supp } \varphi \subseteq \Omega \setminus K$.

The continuous inclusion $\mathcal{D}(\Omega) \subseteq \mathcal{E}(\Omega)$ dualises to a map $\mathcal{E}'(\Omega) \rightarrow \mathcal{D}'(\Omega)$, which is injective because $\mathcal{D}(\Omega)$ is dense in $\mathcal{E}(\Omega)$. Lemma 26.9.14 shows that its image consists of compactly supported distributions; conversely, if $u \in \mathcal{D}'(\Omega)$ is supported in a compact K , choose $\chi \in C_c^\infty(\Omega)$ with $\chi \equiv 1$ near K and set $\tilde{u}[\varphi] = u[\chi\varphi]$, which is independent of the choice of χ . So

$$\mathcal{E}'(\Omega) = \{\text{distributions of compact support}\}.$$

Example 26.9.16

$\delta_x \in \mathcal{E}'(\Omega)$; and if $f \in L^1(\Omega)$ vanishes a.e. outside a compact set, $T_f \in \mathcal{E}'(\Omega)$. But $u[\varphi] = \sum_{n \in \mathbb{Z}} \varphi(n)$ on $\mathbb{R}$ does not have compact support.

Definition 26.9.17 (Tempered distribution)

The continuous dual $\mathcal{S}'(\mathbb{R}^n)$ of the Schwartz space is the space of **tempered distributions**.

The chain $\mathcal{D}(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n) \subseteq \mathcal{E}(\mathbb{R}^n)$ of continuous dense inclusions dualises to injections

$$\mathcal{E}'(\mathbb{R}^n) \hookrightarrow \mathcal{S}'(\mathbb{R}^n) \hookrightarrow \mathcal{D}'(\mathbb{R}^n).$$

So the tempered distributions form a class intermediate between compactly supported distributions and all distributions. The name is a mnemonic: a distribution is tempered when it does not grow faster than polynomially.

Example 26.9.18 (i) If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ satisfies $\int (1+|x|)^{-N} |f(x)| \, dx < \infty$ for some N , then $T_f \in \mathcal{S}'(\mathbb{R}^n)$. Indeed for $\varphi \in \mathcal{S}$,

$$\begin{aligned} |T_f[\varphi]| &= \left| \int (1+|x|)^{-N} f(x) \cdot (1+|x|)^N \varphi(x) \, dx \right| \\ &\leq \left(\int \frac{|f|}{(1+|x|)^N} \right) \sup_x (1+|x|)^N |\varphi(x)|, \end{aligned}$$

which is a bound of the required form. In particular every L^p function and every polynomial is a tempered distribution.

(ii) $f(x) = e^{|x|^2}$ gives $T_f \in \mathcal{D}'(\mathbb{R}^n) \setminus \mathcal{S}'(\mathbb{R}^n)$: it grows too fast to be paired with a Gaussian.

(iii) $u[\varphi] = \sum_{m \in \mathbb{Z}} m^{50} \varphi(m)$ lies in $\mathcal{S}'(\mathbb{R})$ but not $\mathcal{E}'(\mathbb{R})$.

§26.9.5 Convolution and fundamental solutions

Recall $\check{\varphi}(y) = \varphi(-y)$, so that $\tau_x \check{\varphi}(y) = \varphi(x-y)$. For $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ and $\varphi \in \mathcal{D}(\mathbb{R}^n)$,

$$(f * \varphi)(x) = \int f(y) \varphi(x-y) \, dy = T_f[\tau_x \check{\varphi}],$$

which suggests the definition.

Definition 26.9.19

For $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{D}(\mathbb{R}^n)$ set $(u * \varphi)(x) = u[\tau_x \check{\varphi}]$.

This is a genuine function of x , bilinear in (u, φ) . Note that $(u * \check{\varphi})(0) = u[\varphi]$, so the family of functions $u * \varphi$ determines u completely.

Example 26.9.20

$\delta_0 * \varphi = \varphi$, since $(\delta_0 * \varphi)(x) = \tau_x \check{\varphi}(0) = \varphi(x)$. So δ_0 is an identity for convolution, which is the reason for the name ‘approximate identity’.

Lemma 26.9.21

Let $u \in \mathcal{D}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{D}(\mathbb{R}^n)$. Then

- (i) $u * \varphi \in C^\infty(\mathbb{R}^n)$, and $D^\alpha(u * \varphi) = (D^\alpha u) * \varphi = u * (D^\alpha \varphi)$;
- (ii) if $u \in \mathcal{E}'(\mathbb{R}^n)$ then $u * \varphi \in \mathcal{D}(\mathbb{R}^n)$.

Proof. (i) We have

$$\frac{(u * \varphi)(x + h e_i) - (u * \varphi)(x)}{h} = u \left[\frac{\tau_{x+h e_i} \check{\varphi} - \tau_x \check{\varphi}}{h} \right],$$

and one checks (example sheet) that the argument converges in $\mathcal{D}(\mathbb{R}^n)$ to $\tau_x((D_i \varphi)^\vee)$ as $h \rightarrow 0$; continuity of u then gives $D_i(u * \varphi) = u * D_i \varphi$, and iterating gives

smoothness. For the other identity, note

$$D_y^\alpha (\tau_x \check{\varphi})(y) = \frac{\partial^{|\alpha|}}{\partial y^\alpha} \varphi(x-y) = (-1)^{|\alpha|} (D^\alpha \varphi)(x-y) = (-1)^{|\alpha|} \tau_x((D^\alpha \varphi)^\vee)(y),$$

so that

$$((D^\alpha u) * \varphi)(x) = (-1)^{|\alpha|} u[D^\alpha \tau_x \check{\varphi}] = u[\tau_x((D^\alpha \varphi)^\vee)] = (u * D^\alpha \varphi)(x).$$

(ii) If u vanishes on functions supported outside a compact K , then for $|x|$ large the support of $\tau_x \check{\varphi}$ misses K , so $(u * \varphi)(x) = 0$. $\square$

Using the fact that u is determined by the functions $u * \varphi$, one may define the convolution of $u_1 \in \mathcal{D}'(\mathbb{R}^n)$ with $u_2 \in \mathcal{E}'(\mathbb{R}^n)$ by requiring

$$(u_1 * u_2) * \varphi = u_1 * (u_2 * \varphi) \quad \text{for all } \varphi \in \mathcal{D}(\mathbb{R}^n),$$

which makes sense because $u_2 * \varphi \in \mathcal{D}(\mathbb{R}^n)$ by Lemma 26.9.21(ii). One checks $u * \delta_0 = u$, and:

Lemma 26.9.22

For $u_1 \in \mathcal{D}'(\mathbb{R}^n)$ and $u_2 \in \mathcal{E}'(\mathbb{R}^n)$, $D^\alpha(u_1 * u_2) = (D^\alpha u_1) * u_2 = u_1 * (D^\alpha u_2)$.

Proof. Test against $\varphi \in \mathcal{D}(\mathbb{R}^n)$ and use Lemma 26.9.21(i) repeatedly:

$$\begin{aligned} (D^\alpha(u_1 * u_2)) * \varphi &= (u_1 * u_2) * D^\alpha \varphi = u_1 * (u_2 * D^\alpha \varphi) \\ &= u_1 * ((D^\alpha u_2) * \varphi) = (u_1 * D^\alpha u_2) * \varphi, \end{aligned}$$

and the other equality is the same computation with the roles reversed. $\square$

This is exactly what is needed to invert constant-coefficient differential operators.

Definition 26.9.23 (Fundamental solution)

Let $L = \sum_{|\alpha| \leq N} a_\alpha D^\alpha$ with $a_\alpha \in \mathbb{C}$. A **fundamental solution** of L is a distribution $G \in \mathcal{D}'(\mathbb{R}^n)$ with $LG = \delta_0$.

Theorem 26.9.24

If G is a fundamental solution of L and $u_0 \in \mathcal{E}'(\mathbb{R}^n)$, then $u = G * u_0$ satisfies $Lu = u_0$.

Proof. By Lemma 26.9.22,

$$L(G * u_0) = \sum_{\alpha} a_\alpha D^\alpha (G * u_0) = \left(\sum_{\alpha} a_\alpha D^\alpha G \right) * u_0 = (LG) * u_0 = \delta_0 * u_0 = u_0. \quad \square$$

Example 26.9.25

Return to Example 26.9.1. Take $L = -\nabla^2 = -\sum_{i=1}^3 \partial^2/\partial x_i^2$ on $\mathbb{R}^3$ and let

$$g(x) = \frac{1}{4\pi|x|} \quad (x \neq 0), \quad g(0) = 0.$$

Then $g \in L^1_{\text{loc}}(\mathbb{R}^3)$ (the singularity $|x|^{-1}$ is integrable in three dimensions) and $G = T_g$ is a fundamental solution of L : one computes $\nabla^2 g = 0$ away from the origin and, cutting out a small ball and integrating by parts twice, finds that the boundary terms converge to $\varphi(0)$. So for $f \in C_c^\infty(\mathbb{R}^3)$,

$$u(x) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{f(y)}{|x-y|} dy$$

is a smooth solution of $-\nabla^2 u = f$. The physicists were right all along; it just took a duality to say so.

§26.9.6 Fourier transforms of tempered distributions

For $f \in L^1(\mathbb{R}^n)$ and $\varphi \in \mathcal{S}(\mathbb{R}^n)$, Fubini gives

$$T_{\hat{f}}[\varphi] = \int \hat{f}(x)\varphi(x) dx = \iint \varphi(x)f(y)e^{-i(y \cdot x)} dy dx = \int f(y)\hat{\varphi}(y) dy = T_f[\hat{\varphi}].$$

As $\mathcal{F}$ preserves $\mathcal{S}(\mathbb{R}^n)$ (Corollary 26.8.8), the right-hand side makes sense for any tempered distribution.

Definition 26.9.26

For $u \in \mathcal{S}'(\mathbb{R}^n)$, the **Fourier transform** $\hat{u} \in \mathcal{S}'(\mathbb{R}^n)$ is defined by $\hat{u}[\varphi] = u[\hat{\varphi}]$.

Note that this really does require tempered distributions: $\mathcal{F}$ does not preserve $\mathcal{D}(\mathbb{R}^n)$ (a compactly supported function has an analytic transform, which cannot have compact support unless it vanishes), so there is no Fourier transform on $\mathcal{D}'$.

Example 26.9.27

Fix $\xi \in \mathbb{R}^n$ and let $u = \delta_\xi$. Then

$$\hat{u}[\varphi] = \hat{\varphi}(\xi) = \int \varphi(x)e^{-i(x \cdot \xi)} dx = T_{e_{-\xi}}[\varphi],$$

so $\hat{\delta}_\xi = T_{e_{-\xi}}$; in particular $\hat{\delta}_0 = 1$. Dually, using Theorem 26.8.9,

$$\widehat{T_{e_x}[\varphi]} = \int e^{i(x \cdot y)} \hat{\varphi}(y) dy = (2\pi)^n \varphi(x) = (2\pi)^n \delta_x[\varphi],$$

so $\widehat{T_{e_x}} = (2\pi)^n \delta_x$. The delta and the constant function are Fourier transforms of one another, which is the rigorous version of the formal identity $\int e^{ix\xi} d\xi = 2\pi\delta(x)$ familiar from methods courses.

Lemma 26.9.28

Let $u \in \mathcal{S}'(\mathbb{R}^n)$, $x \in \mathbb{R}^n$ and α a multi-index. Writing X^α for the function $x \mapsto x^\alpha$,

and $\tau_x u[\varphi] = u[\tau_{-x}\varphi]$, $\hat{u}[\varphi] = u[\check{\varphi}]$, we have

$$\widehat{e_x u} = \tau_x \hat{u}, \quad \widehat{\tau_x u} = e_{-x} \hat{u}, \quad \widehat{D^\alpha u} = i^{|\alpha|} X^\alpha \hat{u}, \quad D^\alpha \hat{u} = (-i)^{|\alpha|} \widehat{X^\alpha u},$$

and $\hat{\hat{u}} = (2\pi)^n \check{u}$.

Proof. Each is the corresponding identity for Schwartz functions, transposed. For instance

$$\widehat{e_x u}[\varphi] = e_x u[\hat{\varphi}] = u[e_x \hat{\varphi}] = u[\widehat{\tau_{-x}\varphi}] = \hat{u}[\tau_{-x}\varphi] = \tau_x \hat{u}[\varphi],$$

and for the derivative,

$$\widehat{D^\alpha u}[\varphi] = (-1)^{|\alpha|} u[D^\alpha \hat{\varphi}] = (-1)^{|\alpha|} u[\widehat{(-i)^{|\alpha|} X^\alpha \varphi}] = i^{|\alpha|} \hat{u}[X^\alpha \varphi] = i^{|\alpha|} X^\alpha \hat{u}[\varphi].$$

The last identity follows from Corollary 26.8.10. $\square$

Lemma 26.9.29

$\mathcal{F} : \mathcal{S}'(\mathbb{R}^n) \rightarrow \mathcal{S}'(\mathbb{R}^n)$ is a homeomorphism for the weak-* topology.

Proof. Linearity is clear. If $u_j \xrightarrow{*} u$ then $\hat{u}_j[\varphi] = u_j[\hat{\varphi}] \rightarrow u[\hat{\varphi}] = \hat{u}[\varphi]$ for every φ , so $\hat{u}_j \xrightarrow{*} \hat{u}$: the map is sequentially continuous. Since $\mathcal{F}^4 = (2\pi)^{2n} \text{id}$, the inverse is $(2\pi)^{-2n} \mathcal{F}^3$, which is continuous by the same argument. $\square$

§26.9.7 Periodic distributions

As an application, and as preparation for Fourier series in the distributional setting, we look at periodic distributions.

Definition 26.9.30

$u \in \mathcal{D}'(\mathbb{R}^n)$ is **periodic** if $\tau_y u = u$ for all $y \in \mathbb{Z}^n$.

Example 26.9.31

If $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ is $\mathbb{Z}^n$ -periodic then T_f is periodic. More interestingly, if $v \in \mathcal{E}'(\mathbb{R}^n)$ then $u = \sum_{y \in \mathbb{Z}^n} \tau_y v$ is a periodic distribution (for each $\varphi \in \mathcal{D}$ the sum is finite).

For a periodic smooth function we often want the average over a fundamental cell $q = \{x : -1/2 \leq x_i < 1/2\}$, namely $M[f] = \int_q f$. We would like to write $M[f] = T_f[\mathbb{1}_q]$, but $\mathbb{1}_q$ is not a test function. The fix is a smooth partition of unity.

Lemma 26.9.32

Let $Q = \{x : -1 \leq x_i < 1\}$. There is $\psi \in C^\infty(\mathbb{R}^n)$ with $\psi \geq 0$, $\text{supp } \psi \subseteq Q^\circ$, and $\sum_{y \in \mathbb{Z}^n} \tau_y \psi = 1$. Moreover if ψ, ψ' are two such functions and $u \in \mathcal{D}'(\mathbb{R}^n)$ is periodic, then $u[\psi] = u[\psi']$.

Proof. Pick $\psi_0 \in C^\infty(\mathbb{R}^n)$ supported in Q° with $\psi_0 \geq 0$ and $\psi_0 \equiv 1$ on q (a smoothed indicator, built from the bump of Example 26.4.8). Set $s(x) = \sum_{y \in \mathbb{Z}^n} \psi_0(x - y)$; this sum is locally finite, so $s \in C^\infty$, it is $\mathbb{Z}^n$ -periodic, and $s \geq 1$ since every x has some translate in q . Then $\psi = \psi_0/s$ works.

For the second part, using periodicity of u and of the partitions,

$$u[\psi] = u\left[\psi \sum_y \tau_y \psi'\right] = \sum_y u[\psi \tau_y \psi'] = \sum_y \tau_{-y} u[\psi' \tau_{-y} \psi] = \sum_y u[\psi' \tau_y \psi] = u[\psi'],$$

all sums being finite. $\square$

So we may define the **mean** $M[u] = u[\psi]$ of a periodic distribution, and if $u = T_f$ for periodic $f \in L^1_{\text{loc}}$ this agrees with $\int_q f$.

Lemma 26.9.33

If $v \in \mathcal{E}'(\mathbb{R}^n)$ then $u = \sum_{y \in \mathbb{Z}^n} \tau_y v$ converges weakly-* in $\mathcal{S}'(\mathbb{R}^n)$. Conversely every periodic distribution has this form; in particular every periodic distribution is tempered.

Proof. For the first part, Lemma 26.9.14 gives compact $K \subseteq B_R(0)$, $k \in \mathbb{N}$ and $C > 0$ with $|v[\varphi]| \leq C \sup_{x \in K, |\alpha| \leq k} |D^\alpha \varphi(x)|$. Then $|\tau_y v[\varphi]| \leq C \sup_{x \in K, |\alpha| \leq k} |D^\alpha \varphi(x + y)|$. For $x \in K$ we have $1 + |y| \leq (1 + R)(1 + |x + y|)$, so for any m ,

$$|\tau_y v[\varphi]| \leq C \frac{(1 + R)^m}{(1 + |y|)^m} \sup_{x' \in \mathbb{R}^n, |\alpha| \leq k} |(1 + |x'|)^m D^\alpha \varphi(x')|.$$

Choosing $m \geq n+1$ makes $\sum_{y \in \mathbb{Z}^n} (1 + |y|)^{-m}$ finite, so the series converges absolutely, uniformly over φ in a bounded set of $\mathcal{S}$.

Conversely let u be periodic and ψ a periodic partition of unity. For $\varphi \in \mathcal{D}(\mathbb{R}^n)$,

$$u[\varphi] = u\left[\varphi \sum_y \tau_y \psi\right] = \sum_y u[\varphi \tau_y \psi],$$

and $u[\varphi \tau_y \psi] = \tau_y u[\varphi \tau_y \psi] = (\psi u)[\tau_{-y} \varphi] = \tau_y(\psi u)[\varphi]$. So $v = \psi u$ works, and v has compact support since $v[\varphi] = u[\psi \varphi] = 0$ whenever $\text{supp } \varphi \cap \text{supp } \psi = \emptyset$. $\square$

Lemma 26.9.34

Suppose $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies $(e_{-y} - 1)u = 0$ for all $y \in \mathbb{Z}^n$. Then

$$u = \sum_{y \in \mathbb{Z}^n} c_y \delta_{2\pi y}$$

for constants c_y with $|c_y| \leq C(1 + |y|)^N$ for some $C > 0$ and $N \in \mathbb{N}$, the sum converging in $\mathcal{S}'(\mathbb{R}^n)$.

Proof. Write $\Lambda^* = 2\pi\mathbb{Z}^n$. If $\varphi \in \mathcal{D}(\mathbb{R}^n)$ has $\text{supp } \varphi \cap \Lambda^* = \emptyset$ then $(e_{-y} - 1)$ is non-vanishing on $\text{supp } \varphi$ for suitable y , so $(e_{-y} - 1)^{-1}\varphi$ is a legitimate test function and $u[\varphi] = (e_{-y} - 1)u[(e_{-y} - 1)^{-1}\varphi] = 0$. Hence u is supported on Λ^* .

Take a periodic partition of unity ψ and set $\tilde{\psi}(x) = \psi(x/2\pi)$, so that $\sum_y \tau_{2\pi y} \tilde{\psi} = 1$. Put $v_y = (\tau_{2\pi y} \tilde{\psi})u$, supported at the single point $2\pi y$, with $\sum_y v_y = u$. From $(e_{-y'} - 1)v_y = 0$ and $e^{-i(x_j - 2\pi y_j)} - 1 = (x_j - 2\pi y_j)\kappa(x_j)$ with κ smooth and non-vanishing near $2\pi y_j$, we get $(x_j - 2\pi y_j)v_y = 0$ for each j . Taylor expanding $\varphi(x) = \varphi(2\pi y) + \sum_j (x_j - 2\pi y_j)\varphi_j(x)$ with φ_j smooth, and using that v_y has compact support so extends to $\mathcal{E}$, gives $v_y[\varphi] = \varphi(2\pi y)v_y[1]$; that is, $v_y = c_y \delta_{2\pi y}$ with $c_y = u[\tau_{2\pi y} \tilde{\psi}]$.

For the bound, Lemma 26.9.14 applied in $\mathcal{S}'$ supplies N, k, C such that

$$|u[\varphi]| \leq C \sup_{x, |\alpha| \leq k} \left| (1 + |x|)^N D^\alpha \varphi(x) \right|,$$

whence

$$|c_y| \leq C \sup_{x, |\alpha| \leq k} \left| (1 + |x + 2\pi y|)^N D^\alpha \tilde{\psi}(x) \right| \leq C'(1 + |y|)^N. \quad \square$$

Theorem 26.9.35 (Fourier series of a periodic distribution)

Let $u \in \mathcal{D}'(\mathbb{R}^n)$ be periodic. Then there are $c_y \in \mathbb{C}$ with

$$u = \sum_{y \in \mathbb{Z}^n} c_y T_{e_{2\pi y}}$$

converging in $\mathcal{S}'(\mathbb{R}^n)$, where $c_y = M[e_{-2\pi y}u]$ satisfies $|c_y| \leq C(1 + |y|)^N$ for some C, N .

Proof. Periodicity says $\tau_{y'}u = u$ for $y' \in \mathbb{Z}^n$, and u is tempered by Lemma 26.9.33, so taking Fourier transforms and using Lemma 26.9.28 gives $(e_{-y'} - 1)\hat{u} = 0$. By Lemma 26.9.34,

$$\hat{u} = (2\pi)^n \sum_{y \in \mathbb{Z}^n} c_y \delta_{2\pi y},$$

and applying the inversion formula of Lemma 26.9.28 together with Example 26.9.27 gives $u = \sum_y c_y T_{e_{2\pi y}}$. Finally $M[e_{-2\pi y} T_{e_{2\pi y'}}] = \delta_{yy'}$ by direct computation, and $u \mapsto M[e_{-2\pi y}u]$ is continuous on $\mathcal{S}'$, so $M[e_{-2\pi y}u] = c_y$. $\square$

We usually abuse notation and write $u(x) = \sum_{y \in \mathbb{Z}^n} c_y e^{2\pi i(y \cdot x)}$. The theorem says that *every* periodic distribution has a Fourier series, with coefficients of at most polynomial growth, and that the series converges in $\mathcal{S}'$.

Example 26.9.36 (Poisson summation)

Let $u = \sum_{y \in \mathbb{Z}^n} \delta_y$, the ‘Dirac comb’. Its Fourier coefficients are

$$c_y = M[e_{-2\pi y}u] = u[\psi e_{2\pi y}] = \sum_{y' \in \mathbb{Z}^n} \psi(y') e^{-2\pi i(y \cdot y')} = \sum_{y' \in \mathbb{Z}^n} \psi(y') = 1,$$

since $e^{-2\pi i(y \cdot y')} = 1$ for $y, y' \in \mathbb{Z}^n$ and $\sum_{y'} \psi(y') = 1$ by the partition of unity property. Theorem 26.9.35 therefore gives **Poisson's summation formula**

$$\sum_{y \in \mathbb{Z}^n} \delta_y = \sum_{y \in \mathbb{Z}^n} T_{e^{2\pi i y}}, \quad \text{or informally} \quad \sum_{y \in \mathbb{Z}^n} \delta(x - y) = \sum_{y \in \mathbb{Z}^n} e^{2\pi i(y \cdot x)}.$$

Pairing both sides with a Schwartz function φ recovers the classical statement $\sum_y \varphi(y) = \sum_y \hat{\varphi}(2\pi y)$, a formula with a great many applications in number theory and in the study of lattices.

§26.10 Sobolev Spaces

Distribution theory tells us that every locally integrable function can be differentiated, but the derivative may be a wild object. Sobolev spaces impose the reasonable extra requirement that the derivatives be functions again, and indeed functions in L^p . The resulting spaces are the natural setting for the theory of partial differential equations: they are complete, they have good compactness properties, and – the content of the embedding theorem – an element with enough weak derivatives is automatically a classical function.

§26.10.1 Weak derivatives

Definition 26.10.1 (Sobolev space)

Let $\Omega \subseteq \mathbb{R}^n$ be open, $k \in \mathbb{Z}_{\geq 0}$ and $1 \leq p \leq \infty$. We say $f \in L^p(\Omega)$ belongs to the **Sobolev space** $W^{k,p}(\Omega)$ if for every multi-index α with $|\alpha| \leq k$ there is $f^\alpha \in L^p(\Omega)$ with

$$D^\alpha T_f = T_{f^\alpha} \quad \text{in } \mathcal{D}'(\Omega),$$

that is,

$$\int_{\Omega} f^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_{\Omega} f D^\alpha \varphi \, dx \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

We call f^α the **α -th weak derivative** of f , and write $D^\alpha f$ for it.

The weak derivative is unique when it exists, by Corollary 26.4.12, and agrees with the classical derivative when that exists and is continuous. From now on we drop the distinction between f and T_f entirely.

Definition 26.10.2

$W^{k,p}(\Omega)$ is normed by

$$\|f\|_{W^{k,p}} = \left(\sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^p}^p \right)^{1/p} \quad (p < \infty),$$

and by $\|f\|_{W^{k,\infty}} = \max_{|\alpha| \leq k} \|D^\alpha f\|_{L^\infty}$ when $p = \infty$. We write $H^k(\Omega) = W^{k,2}(\Omega)$, which is a Hilbert space with $\langle f, g \rangle_{H^k} = \sum_{|\alpha| \leq k} \langle D^\alpha f, D^\alpha g \rangle_{L^2}$.

Proposition 26.10.3

$W^{k,p}(\Omega)$ is a Banach space.

Proof. Let (f_j) be Cauchy. Then each $(D^\alpha f_j)$ is Cauchy in L^p , so converges to some $g_\alpha \in L^p$ by Theorem 26.3.8. Write $f = g_0$. For $\varphi \in C_c^\infty(\Omega)$, Hölder gives $\int D^\alpha f_j \varphi \rightarrow \int g_\alpha \varphi$ and $\int f_j D^\alpha \varphi \rightarrow \int f D^\alpha \varphi$, so passing to the limit in the defining identity shows $g_\alpha = D^\alpha f$ weakly. Hence $f \in W^{k,p}$ and $f_j \rightarrow f$ in the norm. $\square$

Weak derivatives obey the rules one expects, provided one does not ask for too much. They are linear, and they commute with each other (both facts are inherited from $\mathcal{D}'(\Omega)$, where they are immediate from the definition). The product rule holds against a smooth factor, which is exactly the case we shall need when localising with a cut-off function.

Lemma 26.10.4 (Leibniz rule)

Let $u \in W^{1,p}(\Omega)$ and $\chi \in C^\infty(\Omega)$ with χ and $D\chi$ bounded. Then $\chi u \in W^{1,p}(\Omega)$ and

$$D_i(\chi u) = \chi D_i u + (D_i \chi)u.$$

Proof. Both terms on the right lie in $L^p(\Omega)$ by boundedness of χ and $D\chi$. Let $\varphi \in C_c^\infty(\Omega)$; then $\chi\varphi \in C_c^\infty(\Omega)$ is a legitimate test function, and $D_i(\chi\varphi) = \chi D_i \varphi + (D_i \chi)\varphi$ classically. So

$$\begin{aligned} \int_\Omega \chi u D_i \varphi \, dx &= \int_\Omega u (D_i(\chi\varphi) - (D_i \chi)\varphi) \, dx \\ &= - \int_\Omega (D_i u) \chi \varphi \, dx - \int_\Omega u (D_i \chi) \varphi \, dx, \end{aligned}$$

using the definition of the weak derivative of u against the test function $\chi\varphi$. This is precisely the assertion. $\square$

Iterating gives the corresponding statement for higher derivatives: if $u \in W^{k,p}(\Omega)$ and $\chi \in C_c^\infty(\Omega)$ then $\chi u \in W^{k,p}(\Omega)$, with $D^\alpha(\chi u)$ given by the usual Leibniz expansion. What one *cannot* do is multiply two Sobolev functions and expect the product rule, or apply the chain rule with a merely continuous outer function; these require separate arguments and extra hypotheses.

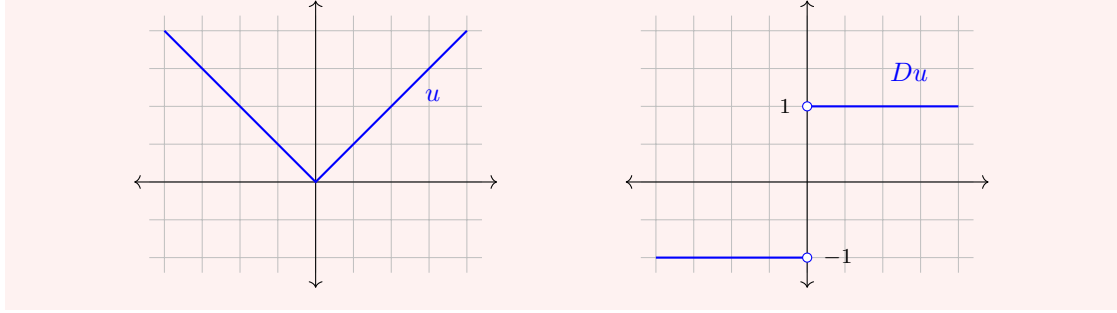
Example 26.10.5 (i) Let $u(x) = |x|$ on $(-2, 2)$. Then u is not differentiable at 0, but for $\varphi \in C_c^\infty((-2, 2))$,

$$- \int_{-2}^2 |x| \varphi'(x) \, dx = \int_{-2}^0 \varphi'(x) \, dx - \int_0^2 x \varphi'(x) \, dx = \int_{-2}^2 \operatorname{sgn}(x) \varphi(x) \, dx$$

after integrating by parts on each side (the boundary terms at 0 cancel). So $Du = \operatorname{sgn}$ weakly, and $u \in W^{1,p}((-2, 2))$ for every p .

(ii) By contrast, the Heaviside function $H = \mathbb{1}_{[0,\infty)}$ is in no $W^{1,p}$: its distributional derivative is δ_0 , which is not a function.

So a kink is permitted – it costs one derivative – but a jump is not.



§26.10.2 Sobolev spaces via the Fourier transform

On $\Omega = \mathbb{R}^n$ with $p = 2$ we can say something much sharper, because the Fourier transform turns differentiation into multiplication. Indeed for $f \in L^2(\mathbb{R}^n)$ we have $D^\alpha f \in L^2(\mathbb{R}^n)$ if and only if $\xi^\alpha \hat{f}(\xi) \in L^2(\mathbb{R}^n)$, by Lemma 26.9.28 and Plancherel. Since $\sum_{|\alpha| \leq k} |\xi^\alpha|^2$ is comparable to $(1 + |\xi|^2)^k$, this suggests a definition which makes sense for non-integer orders.

Definition 26.10.6

For $s \in \mathbb{R}$, we say $u \in \mathcal{S}'(\mathbb{R}^n)$ belongs to $H^s(\mathbb{R}^n)$ if $\hat{u} \in L^2_{\text{loc}}(\mathbb{R}^n)$ and

$$\|u\|_{H^s}^2 = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{u}(\xi)|^2 d\xi < \infty.$$

This makes $H^s(\mathbb{R}^n)$ a Hilbert space with $\langle u, v \rangle_{H^s} = (2\pi)^{-n} \int \hat{u} \bar{\hat{v}} (1 + |\xi|^2)^s d\xi$.

For $s \in \mathbb{Z}_{\geq 0}$ we have $H^s(\mathbb{R}^n) = W^{s,2}(\mathbb{R}^n)$ with equivalent norms, by the remarks above. But now s may be any real number: fractional orders measure fractional differentiability, and negative orders describe distributions rougher than functions. For instance $\delta_0 \in H^s(\mathbb{R}^n)$ exactly when $s < -n/2$, since $\hat{\delta}_0 = 1$ and $\int (1 + |\xi|^2)^s d\xi < \infty$ iff $2s < -n$.

The Fourier description makes the following density statement almost trivial, and we shall use it repeatedly to reduce statements about H^s to computations with Schwartz functions.

Lemma 26.10.7

For every $s \in \mathbb{R}$, the space $\mathcal{S}(\mathbb{R}^n)$ is dense in $H^s(\mathbb{R}^n)$.

Proof. Let $u \in H^s(\mathbb{R}^n)$, so that $w = (1 + |\xi|^2)^{s/2} \hat{u} \in L^2(\mathbb{R}^n)$. By Corollary 26.4.11 choose $\psi \in C_c^\infty(\mathbb{R}^n)$ with $\|w - \psi\|_{L^2} < (2\pi)^{n/2} \varepsilon$, and set

$$v = \mathcal{F}^{-1}[(1 + |\xi|^2)^{-s/2} \psi].$$

The function $\xi \mapsto (1 + |\xi|^2)^{-s/2}$ is smooth on all of $\mathbb{R}^n$, so $(1 + |\xi|^2)^{-s/2} \psi \in C_c^\infty(\mathbb{R}^n) \subseteq \mathcal{S}(\mathbb{R}^n)$, whence $v \in \mathcal{S}(\mathbb{R}^n)$ by Corollary 26.8.10. And

$$\|u - v\|_{H^s}^2 = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{u} - \hat{v}|^2 d\xi = \frac{1}{(2\pi)^n} \|w - \psi\|_{L^2}^2 < \varepsilon^2. \quad \square$$

§26.10.3 The Sobolev embedding theorem

Here is the theorem that makes Sobolev spaces worth the trouble.

Theorem 26.10.8 (Sobolev embedding)

Let $k \in \mathbb{Z}_{\geq 0}$ and $s > k + n/2$. If $f \in H^s(\mathbb{R}^n)$ then there is $f^* \in C^k(\mathbb{R}^n)$ with $f = f^*$ almost everywhere, and

$$\sup_{x \in \mathbb{R}^n, |\alpha| \leq k} |D^\alpha f^*(x)| \leq c_{n,k,s} \|f\|_{H^s}.$$

We write $H^s(\mathbb{R}^n) \subseteq C^k(\mathbb{R}^n)$.

Proof. Suppose first $f \in \mathcal{S}(\mathbb{R}^n)$. By the inversion theorem and Lemma 26.9.28,

$$D^\alpha f(x) = \frac{i^{|\alpha|}}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i(x \cdot \xi)} \xi^\alpha \hat{f}(\xi) \, d\xi,$$

so by Cauchy–Schwarz, inserting and removing a factor $(1 + |\xi|^2)^{s/2}$,

$$|D^\alpha f(x)| \leq \frac{1}{(2\pi)^n} \left(\int_{\mathbb{R}^n} (1 + |\xi|^2)^s |\hat{f}(\xi)|^2 \, d\xi \right)^{1/2} \left(\int_{\mathbb{R}^n} \frac{|\xi^\alpha|^2}{(1 + |\xi|^2)^s} \, d\xi \right)^{1/2}.$$

Since $|\alpha| \leq k$ we have $|\xi^\alpha|^2 \leq c_k(1 + |\xi|^2)^k$, so the second integral is at most

$$c_k \int_{\mathbb{R}^n} \frac{d\xi}{(1 + |\xi|^2)^{s-k}} = c_{k,n,s}^2 < \infty$$

precisely because $2(s - k) > n$. Hence $\sup_{x, |\alpha| \leq k} |D^\alpha f(x)| \leq c_{n,k,s} \|f\|_{H^s}$ for $f \in \mathcal{S}(\mathbb{R}^n)$.

For general $f \in H^s(\mathbb{R}^n)$, approximate: by Lemma 26.10.7 there are $f_m \in \mathcal{S}(\mathbb{R}^n)$ with $f_m \rightarrow f$ in H^s , hence in L^2 , and so (after passing to a subsequence) almost everywhere. The sequence (f_m) is Cauchy in H^s , hence by the estimate Cauchy in $C^k(\mathbb{R}^n)$ with the uniform norm on all derivatives up to order k ; that space is complete, so $f_m \rightarrow f^*$ in C^k . Since also $f_m \rightarrow f$ a.e., $f = f^*$ a.e. $\square$

The moral is that *differentiability can be bought with integrability, at an exchange rate of $n/2$ derivatives*. It is a genuinely surprising statement: membership of H^s is a statement about integrals, and yet for s large enough it forces pointwise regularity. Note also that the threshold is sharp: in $\mathbb{R}^n$ with $n \geq 2$, the function $\log \log(1 + 1/|x|)$ near the origin is in $H^{n/2}$ but unbounded, so $s = k + n/2$ will not do.

The general L^p statement, which we quote, is a chart of the same phenomenon at other exponents.

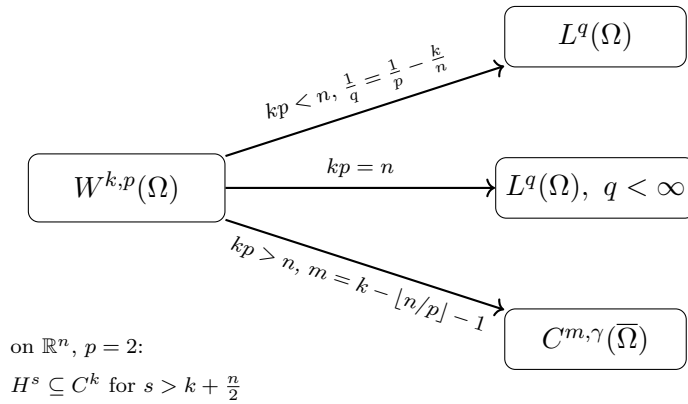
Theorem 26.10.9 (General Sobolev embeddings)

Let $\Omega \subseteq \mathbb{R}^n$ be a bounded domain with C^1 boundary, $k \geq 1$ and $1 \leq p < \infty$.

- (i) If $kp < n$ then $W^{k,p}(\Omega) \subseteq L^q(\Omega)$ for $\frac{1}{q} = \frac{1}{p} - \frac{k}{n}$, continuously.
- (ii) If $kp > n$ then $W^{k,p}(\Omega) \subseteq C^{m,\gamma}(\overline{\Omega})$, where $m = k - \lfloor n/p \rfloor - 1$ and $\gamma \in (0, 1)$

is determined by n, k, p .

- (iii) If $kp = n$ then $W^{k,p}(\Omega) \subseteq L^q(\Omega)$ for every $q < \infty$, but not in general for $q = \infty$.



§26.10.4 The spaces $H_0^1(\Omega)$ and traces

For boundary value problems we need to make sense of the restriction of a Sobolev function to $\partial\Omega$, which is a set of measure zero. Since elements of H^s are only defined up to null sets, this requires an argument; the answer is that it can be done as soon as $s > 1/2$.

Theorem 26.10.10 (Trace theorem)

Let $s > 1/2$. There is a bounded surjective linear map

$$T : H^s(\mathbb{R}^n) \rightarrow H^{s-1/2}(\mathbb{R}^{n-1})$$

with $Tf = f|_{\{x_n=0\}}$ for every $f \in \mathcal{S}(\mathbb{R}^n)$.

The proof (example sheet) is a computation with the Fourier transform: one shows that the partial transform of the restriction is obtained by integrating $\hat{f}$ in the ξ_n variable, and estimates the result with Cauchy–Schwarz exactly as in Theorem 26.10.8. Note the loss of half a derivative, and note that $s > 1/2$ is needed: an $H^{1/2}$ function has no well-defined trace. By combining with local coordinate changes one obtains a trace map onto sufficiently regular hypersurfaces, in particular onto the boundary of a Lipschitz domain.

Definition 26.10.11

Let $\Omega \subseteq \mathbb{R}^n$ be open. Extending by zero, we regard $C_c^\infty(\Omega) \subseteq H^1(\mathbb{R}^n)$, and define $H_0^1(\Omega)$ to be the closure of $C_c^\infty(\Omega)$ in $H^1(\mathbb{R}^n)$.

For $f \in C_c^\infty(\Omega)$, Plancherel gives the convenient formula

$$\|f\|_{H^1}^2 = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} (1 + |\xi|^2) |\hat{f}(\xi)|^2 \, d\xi = \int_{\Omega} (|Df(x)|^2 + |f(x)|^2) \, dx,$$

so $H_0^1(\Omega)$ is a Hilbert space with inner product

$$\langle u, v \rangle_{H^1} = \int_{\Omega} (Du \cdot \overline{Dv} + \overline{u}v) \, dx.$$

Elements of $H_0^1(\Omega)$ vanish outside Ω and have zero trace on $\partial\Omega$. Indeed, if $\varphi \in C_c^\infty((\Omega^c)^\circ)$ then $\Lambda_\varphi(u) = \int_{\mathbb{R}^n} \varphi u$ vanishes for $u \in C_c^\infty(\Omega)$ and is continuous on $H^1(\mathbb{R}^n)$, so it vanishes on $H_0^1(\Omega)$ – that is, every $u \in H_0^1(\Omega)$ vanishes a.e. on Ω^c . And if $\partial\Omega$ is regular enough for the trace map $T : H^1 \rightarrow H^{1/2}(\partial\Omega)$ to be defined and bounded, then $Tu = 0$ for $u \in C_c^\infty(\Omega)$ and hence for all $u \in H_0^1(\Omega)$. So, informally:

$$H_0^1(\Omega) = \{u \in H^1 : u = 0 \text{ on } \partial\Omega \text{ and outside } \Omega\},$$

and this is the space in which we shall pose the Dirichlet problem.

§26.10.5 Poincaré's inequality

On $H_0^1(\Omega)$ with Ω bounded, the full H^1 norm is controlled by the gradient alone. This is a genuinely useful fact: it says that a function which vanishes on the boundary cannot be large without having a large derivative somewhere.

Theorem 26.10.12 (Poincaré)

Let $\Omega \subseteq \mathbb{R}^n$ be contained in a slab $\{0 < x_1 < d\}$. Then

$$\|u\|_{L^2(\Omega)} \leq d \|Du\|_{L^2(\Omega)} \quad \text{for all } u \in H_0^1(\Omega).$$

Proof. By density it is enough to treat $u \in C_c^\infty(\Omega)$, which we extend by zero to the slab. For such u , the fundamental theorem of calculus in the x_1 direction gives

$$u(x) = \int_0^{x_1} \partial_1 u(t, x_2, \dots, x_n) dt,$$

so by Cauchy–Schwarz

$$|u(x)|^2 \leq x_1 \int_0^d |\partial_1 u(t, x')|^2 dt \leq d \int_0^d |\partial_1 u(t, x')|^2 dt,$$

where $x' = (x_2, \dots, x_n)$. Integrating in x_1 over $(0, d)$ and then in x' over $\mathbb{R}^{n-1}$,

$$\int_\Omega |u|^2 dx \leq d^2 \int_\Omega |\partial_1 u|^2 dx \leq d^2 \int_\Omega |Du|^2 dx. \quad \square$$

Corollary 26.10.13

For Ω bounded, $\|Du\|_{L^2(\Omega)}$ is a norm on $H_0^1(\Omega)$ equivalent to $\|u\|_{H^1}$.

Proof. Clearly $\|Du\|_{L^2} \leq \|u\|_{H^1}$, and conversely $\|u\|_{H^1}^2 = \|Du\|_{L^2}^2 + \|u\|_{L^2}^2 \leq (1 + d^2) \|Du\|_{L^2}^2$. $\square$

Poincaré's inequality fails on all of $H^1(\Omega)$ – constants are a counterexample – and it fails on $H_0^1(\mathbb{R}^n) = H^1(\mathbb{R}^n)$, where there is no bounded direction. Boundedness of the domain, or at least of one direction, is essential.

§26.10.6 Rellich–Kondrachov

Finally we come to the compactness theorem. Let Ω be open and bounded and let (u_m) be a bounded sequence in $H_0^1(\Omega)$, say $\|u_m\|_{H^1} \leq K$. Since $H_0^1(\Omega)$ is a Hilbert space, hence reflexive, Corollary 26.6.29 gives a subsequence with $u_m \rightharpoonup u$ in $H_0^1(\Omega)$, and $\|u\|_{H^1} \leq K$. Moreover for $w \in L^2(\Omega)$ we have $|\langle w, v \rangle_{L^2}| \leq \|w\|_{L^2} \|v\|_{H^1}$, so $v \mapsto \langle w, v \rangle_{L^2}$ is a bounded functional on $H_0^1(\Omega)$ and therefore $u_m \rightharpoonup u$ in $L^2(\Omega)$ as well. The theorem is that this convergence is in fact *strong*.

Theorem 26.10.14 (Rellich–Kondrachov)

Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. Suppose $(u_m) \subseteq H_0^1(\Omega)$ with $\|u_m\|_{H^1} \leq K$ and $u_m \rightharpoonup u$ in $L^2(\Omega)$ for some $u \in H_0^1(\Omega)$. Then $u_m \rightarrow u$ strongly in $L^2(\Omega)$.

Proof. Regard everything as defined on $\mathbb{R}^n$ by extension by zero. By Plancherel,

$$\|u_m - u\|_{L^2}^2 = \frac{1}{(2\pi)^n} \int_{|\xi| < R} |\hat{u}_m - \hat{u}|^2 d\xi + \frac{1}{(2\pi)^n} \int_{|\xi| \geq R} |\hat{u}_m - \hat{u}|^2 d\xi,$$

and we treat the two ranges differently. This is the standard division into low and high frequencies, and it is worth remembering: the high frequencies are killed by the derivative bound, the low frequencies by the weak convergence.

High frequencies. For $|\xi| \geq R$ we have $1 \leq (1 + |\xi|^2)/R^2$, so

$$\begin{aligned} \int_{|\xi| \geq R} |\hat{u}_m - \hat{u}|^2 d\xi &\leq \frac{2}{R^2} \int_{\mathbb{R}^n} (1 + |\xi|^2) (|\hat{u}_m|^2 + |\hat{u}|^2) d\xi \\ &\leq \frac{C}{R^2} (\|u_m\|_{H^1}^2 + \|u\|_{H^1}^2) \leq \frac{CK^2}{R^2}, \end{aligned}$$

which we may make as small as we please by choosing R large – *uniformly in m* .

Low frequencies. Fix such an R . Since $|\Omega| < \infty$, the function $x \mapsto e^{-i(x \cdot \xi)}$ lies in $L^2(\Omega)$ for each ξ , and $\hat{u}_m(\xi) = \langle e_\xi, u_m \rangle_{L^2(\Omega)}$. So weak convergence in $L^2(\Omega)$ gives $\hat{u}_m(\xi) \rightarrow \hat{u}(\xi)$ pointwise. Moreover, by the Riemann–Lebesgue bound and Cauchy–Schwarz on the bounded set Ω ,

$$|\hat{u}_m(\xi) - \hat{u}(\xi)|^2 \leq 2 (\|u_m\|_{L^1}^2 + \|u\|_{L^1}^2) \leq 2|\Omega| (\|u_m\|_{L^2}^2 + \|u\|_{L^2}^2) \leq C',$$

a constant, which is integrable on the ball $\{|\xi| < R\}$. Dominated convergence therefore gives

$$\int_{|\xi| < R} |\hat{u}_m(\xi) - \hat{u}(\xi)|^2 d\xi \rightarrow 0 \quad \text{as } m \rightarrow \infty.$$

Combining, $\limsup_m \|u_m - u\|_{L^2}^2 \leq CK^2/R^2$ for every R , so the limit is 0. $\square$

Corollary 26.10.15

Let Ω be open and bounded and let $(u_m) \subseteq H_0^1(\Omega)$ with $\|u_m\|_{H^1} \leq K$. Then there is a subsequence and $u \in H_0^1(\Omega)$ with $u_m \rightharpoonup u$ in H^1 and $u_m \rightarrow u$ strongly in L^2 .

Corollary 26.10.16

Let Ω be open and bounded and let $A : L^2(\Omega) \rightarrow H_0^1(\Omega)$ be bounded and linear. Then A , viewed as a map $L^2(\Omega) \rightarrow L^2(\Omega)$, is compact.

Proof. If $\|u_m\|_{L^2} \leq K$ then $\|Au_m\|_{H^1} \leq \|A\| K$, so (Au_m) has an L^2 -convergent subsequence by Corollary 26.10.15. $\square$

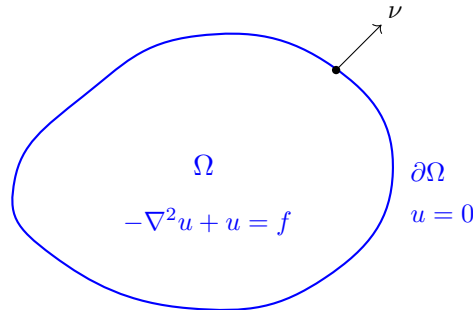
The inclusion $H_0^1(\Omega) \hookrightarrow L^2(\Omega)$ is thus a compact operator: the unit ball of H_0^1 is precompact in L^2 . This is the compactness that we shall use to solve elliptic problems and to diagonalise the Laplacian. Note where boundedness of Ω entered – twice, in fact – and note that the result is false on $\mathbb{R}^n$, where the translates $\phi(\cdot - m)$ of a fixed bump are bounded in H^1 with no L^2 -convergent subsequence.

§26.11 Applications to Elliptic Equations

We can now do what the whole course has been building towards. In this section $\Omega \subseteq \mathbb{R}^n$ is open and bounded, and we study the boundary value problem

$$\begin{cases} -\nabla^2 u + u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Classically one would ask for $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ satisfying this pointwise, and proving that such a u exists is hard. The modern strategy is to relax the problem until existence is easy, and then to work to show that the object produced is better than it had to be.

**§26.11.1 Weak solutions**

Suppose for a moment that u is a classical solution and $v \in C_c^\infty(\Omega)$. Multiplying by $\bar{v}$ and integrating by parts (no boundary term, as v has compact support),

$$\int_{\Omega} (Du \cdot \overline{Dv} + u\bar{v}) \, dx = \int_{\Omega} f\bar{v} \, dx,$$

that is, $\langle v, u \rangle_{H^1} = \langle v, f \rangle_{L^2}$ in our notation. The left side makes sense for $u \in H^1$, and by continuity the identity extends from $v \in C_c^\infty(\Omega)$ to $v \in H_0^1(\Omega)$. This motivates the following.

Definition 26.11.1 (Weak solution)

Let $f \in L^2(\Omega)$. We say $u \in H_0^1(\Omega)$ is a **weak solution** of the boundary value problem if

$$\langle u, v \rangle_{H^1} = \langle f, v \rangle_{L^2} \quad \text{for all } v \in H_0^1(\Omega).$$

Note that the boundary condition is encoded in the requirement $u \in H_0^1(\Omega)$, not in a separate equation. Conversely, if u is a weak solution then testing against $v \in C_c^\infty(\Omega)$ and integrating by parts shows that $-\nabla^2 u + u = f$ holds in $\mathcal{D}'(\Omega)$. So the notions match up as they should.

Lemma 26.11.2

For every $f \in L^2(\Omega)$ there is a unique weak solution $u \in H_0^1(\Omega)$, and $\|u\|_{H^1} \leq \|f\|_{L^2}$.

Proof. The map $\Lambda : v \mapsto \langle f, v \rangle_{L^2}$ is linear, and bounded on $H_0^1(\Omega)$ since

$$|\langle f, v \rangle_{L^2}| \leq \|f\|_{L^2} \|v\|_{L^2} \leq \|f\|_{L^2} \|v\|_{H^1}.$$

As $H_0^1(\Omega)$ is a Hilbert space, Theorem 26.5.3 provides a unique $u \in H_0^1(\Omega)$ with $\langle u, v \rangle_{H^1} = \Lambda(v)$ for all v , and $\|u\|_{H^1} = \|\Lambda\| \leq \|f\|_{L^2}$. $\square$

That is the entire existence proof. It is worth pausing to see how much was smuggled in: the completeness of H_0^1 (Riesz–Fischer), the identification of the H^1 inner product with the Dirichlet form (Plancherel), and the interpretation of the boundary condition (density of C_c^∞). Each was a theorem, and together they reduce a partial differential equation to the observation that a Hilbert space is its own dual.

Definition 26.11.3

Write $A : L^2(\Omega) \rightarrow H_0^1(\Omega)$ for the map $f \mapsto u$ given by Lemma 26.11.2, the **solution operator**.

Proposition 26.11.4

A is linear and bounded with $\|Af\|_{H^1} \leq \|f\|_{L^2}$, and, regarded as a map $L^2(\Omega) \rightarrow L^2(\Omega)$, it is compact and Hermitian.

Proof. Linearity: if $u = Af$, $w = Ag$ then for all v ,

$$\langle u + aw, v \rangle_{H^1} = \langle f, v \rangle_{L^2} + \bar{a} \langle g, v \rangle_{L^2} = \langle f + ag, v \rangle_{L^2},$$

so $A(f + ag) = Af + aAg$ by uniqueness. Boundedness is Lemma 26.11.2; compactness follows from Corollary 26.10.16. For the Hermitian property, with $u = Af$, $w = Ag$,

$$\langle f, Ag \rangle_{L^2} = \langle f, w \rangle_{L^2} = \overline{\langle w, f \rangle_{L^2}} = \overline{\langle w, u \rangle_{H^1}} = \langle u, w \rangle_{H^1} = \langle Af, g \rangle_{L^2}. \quad \square$$

§26.11.2 The Lax–Milgram theorem

Lemma 26.11.2 used the fact that the Dirichlet form is exactly the H^1 inner product. For a general elliptic operator this fails: the associated bilinear form is neither symmetric nor equal to an inner product. The following generalisation of the Riesz representation theorem is the standard remedy.

Theorem 26.11.5 (Lax–Milgram)

Let $\mathcal{H}$ be a Hilbert space and $B : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ a sesquilinear form which is

- (i) **bounded**: $|B(u, v)| \leq C \|u\| \|v\|$ for some $C > 0$, and
- (ii) **coercive**: $|B(u, u)| \geq c \|u\|^2$ for some $c > 0$.

Then for every $\Lambda \in \mathcal{H}'$ there is a unique $u \in \mathcal{H}$ with $B(u, v) = \Lambda(v)$ for all $v \in \mathcal{H}$, and $\|u\| \leq c^{-1} \|\Lambda\|$.

Proof. For fixed u , the map $v \mapsto \overline{B(u, v)}$ is a bounded linear functional, so by Theorem 26.5.3 there is a unique $Su \in \mathcal{H}$ with $B(u, v) = \langle v, Su \rangle^-$; more conveniently, define $S : \mathcal{H} \rightarrow \mathcal{H}$ by $\langle Su, v \rangle = B(u, v)$. Then S is linear and bounded with $\|S\| \leq C$, by (i).

Coercivity gives $c \|u\|^2 \leq |B(u, u)| = |\langle Su, u \rangle| \leq \|Su\| \|u\|$, so $\|Su\| \geq c \|u\|$. Hence S is injective with closed range: if $Su_m \rightarrow w$ then (u_m) is Cauchy, so $u_m \rightarrow u$ and $Su = w$. If the range were not everything, there would be $0 \neq z \perp \text{ran } S$, and then $c \|z\|^2 \leq |B(z, z)| = |\langle Sz, z \rangle| = 0$, a contradiction. So S is a bijection.

Given Λ , Theorem 26.5.3 gives w with $\Lambda(v) = \langle w, v \rangle$, and $u = S^{-1}w$ satisfies $B(u, v) = \langle Su, v \rangle = \langle w, v \rangle = \Lambda(v)$. Uniqueness is injectivity of S , and the bound follows from $\|u\| \leq c^{-1} \|Su\| = c^{-1} \|w\| = c^{-1} \|\Lambda\|$. $\square$

Example 26.11.6

Let a_{ij}, b, c be bounded measurable on Ω with $\sum_{ij} a_{ij}(x) \xi_i \xi_j \geq \theta |\xi|^2$ for some $\theta > 0$ (**uniform ellipticity**), and consider

$$Lu = - \sum_{i,j} D_i(a_{ij} D_j u) + cu.$$

The associated form is $B(u, v) = \int_{\Omega} \left(\sum_{ij} a_{ij} D_j u \overline{D_i v} + cu \overline{v} \right)$, which is bounded on $H_0^1(\Omega)$, and coercive as soon as $c \geq 0$ (or, using Poincaré, whenever $c > -\theta/d^2$). So Theorem 26.11.5 gives a unique weak solution of $Lu = f$ in $H_0^1(\Omega)$ for every $f \in L^2(\Omega)$. Note that B need not be symmetric – $a_{ij} \neq a_{ji}$ is allowed – so the Riesz theorem alone would not have sufficed.

§26.11.3 Elliptic regularity

Our weak solution has, so far, one weak derivative. The equation says more.

Proposition 26.11.7

Let $s \in \mathbb{R}$ and $f \in H^s(\mathbb{R}^n)$. Then there is a unique $u \in H^{s+2}(\mathbb{R}^n)$ with $-\nabla^2 u + u = f$ in $\mathcal{S}'(\mathbb{R}^n)$, and $\|u\|_{H^{s+2}} = \|f\|_{H^s}$.

Proof. Taking Fourier transforms, the equation reads $(|\xi|^2 + 1)\hat{u}(\xi) = \hat{f}(\xi)$, so necessarily $\hat{u} = \hat{f}/(1 + |\xi|^2)$, which defines a tempered distribution and hence a unique u . Then

$$\|u\|_{H^{s+2}}^2 = \frac{1}{(2\pi)^n} \int (1 + |\xi|^2)^{s+2} \frac{|\hat{f}(\xi)|^2}{(1 + |\xi|^2)^2} d\xi = \|f\|_{H^s}^2. \quad \square$$

So on $\mathbb{R}^n$ the operator $-\nabla^2 + 1$ gains exactly two derivatives, which is the best possible and is the defining feature of a second-order elliptic operator. On a domain we obtain the same gain locally, by cutting off.

Definition 26.11.8

For $s > 0$ let $H_{\text{loc}}^s(\Omega) = \{u \in L_{\text{loc}}^2(\Omega) : \chi u \in H^s(\mathbb{R}^n) \text{ for all } \chi \in C_c^\infty(\Omega)\}$.

Note that if $s > n/2 + k$ and $u \in H_{\text{loc}}^s(\Omega)$ then $u \in C^k(\Omega)$: given U with $\bar{U} \subseteq \Omega$ compact, choose $\chi \in C_c^\infty(\Omega)$ with $\chi \equiv 1$ on U ; then $\chi u \in H^s(\mathbb{R}^n) \subseteq C^k(\mathbb{R}^n)$ by Theorem 26.10.8, and $\chi u = u$ on U .

Theorem 26.11.9 (Interior elliptic regularity)

Let $u \in H_0^1(\Omega)$ be a weak solution of $-\nabla^2 u + u = f$ with $f \in L^2(\Omega)$. Then $u \in H_{\text{loc}}^2(\Omega)$. If moreover $f \in H_{\text{loc}}^k(\Omega) \cap L^2(\Omega)$ then $u \in H_{\text{loc}}^{k+2}(\Omega)$; and if $f \in C^\infty(\Omega) \cap L^2(\Omega)$ then $u \in C^\infty(\Omega)$ and the equation holds classically.

Proof. Fix a compact $K \subseteq \Omega$ and $\chi_K \in C_c^\infty(\Omega)$ with $\chi_K \equiv 1$ on K , and set $w = \chi_K u$. For any $\varphi \in \mathcal{S}(\mathbb{R}^n)$ the function $v = \chi_K \varphi$ lies in $C_c^\infty(\Omega) \subseteq H_0^1(\Omega)$, so it is admissible as a test function. Substituting and moving the cut-off across the derivatives, which is legitimate by the Leibniz rule of Lemma 26.10.4, gives

$$\int_{\mathbb{R}^n} \bar{w}(-\nabla^2 \varphi + \varphi) dx = \int_{\mathbb{R}^n} \bar{g} \varphi dx, \quad g = f \chi_K - 2Du \cdot D\chi_K - u \nabla^2 \chi_K.$$

So $w \in H^1(\mathbb{R}^n)$ solves $-\nabla^2 w + w = g$ in $\mathcal{S}'(\mathbb{R}^n)$, and $g \in L^2 = H^0$ since $u \in H^1$ and $f \in L^2$. By Proposition 26.11.7, $w \in H^2(\mathbb{R}^n)$. As K was arbitrary, $u \in H_{\text{loc}}^2(\Omega)$.

Now bootstrap. If additionally $f \in H_{\text{loc}}^1$, then the formula for g shows $g \in H^1(\mathbb{R}^n)$ (each term now has one more derivative available), so $w \in H^3(\mathbb{R}^n)$ and $u \in H_{\text{loc}}^3(\Omega)$. Iterating gives $u \in H_{\text{loc}}^{k+2}(\Omega)$ whenever $f \in H_{\text{loc}}^k(\Omega) \cap L^2(\Omega)$. If f is smooth this holds for every k , so $u \in \bigcap_k C^k(\Omega) = C^\infty(\Omega)$ by the remark above, and a smooth distributional solution is a classical solution. $\square$

This gain of regularity – for free, from the equation alone – is called **elliptic regularity**, and it is the reason the weak formulation is not a cheat. We solved a weakened problem and got a classical solution anyway.

Remark. It is emphatically a property of elliptic operators, not of differential operators in general. The wave equation $u_{tt} - u_{xx} = 0$ has the general solution $u(x, t) = u_+(x + t) + u_-(x - t)$, and if $u_\pm$ are merely continuous this is a perfectly good weak solution

which is nowhere differentiable. Hyperbolic equations propagate singularities rather than smoothing them.

§26.11.4 The Fredholm alternative and eigenvalues of the Laplacian

Consider now the problem with a potential:

$$\begin{cases} -\nabla^2 u + Vu = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

with $V : \Omega \rightarrow \mathbb{R}$ smooth and bounded. The weak formulation is

$$\int_{\Omega} (Du \cdot \overline{Dv} + Vu\overline{v}) \, dx = \int_{\Omega} f\overline{v} \, dx \quad \text{for all } v \in H_0^1(\Omega),$$

but now the left-hand side need not be an inner product, since V may be negative. Instead of appealing to Lax–Milgram (which would need V bounded below appropriately) we rewrite:

$$\int_{\Omega} (Du \cdot \overline{Dv} + u\overline{v}) \, dx = \int_{\Omega} ((1 - V)u + f)\overline{v} \, dx,$$

which says precisely $u = A(f + (1 - V)u)$, or

$$(I - \mathcal{K})u = Af, \quad \mathcal{K}u = A((1 - V)u).$$

Since $u \mapsto (1 - V)u$ is bounded on $L^2(\Omega)$ and $A : L^2 \rightarrow H_0^1$ is bounded, $\mathcal{K} : L^2(\Omega) \rightarrow L^2(\Omega)$ is compact by Corollary 26.10.16. The Fredholm theory of compact operators now applies verbatim:

Theorem 26.11.10

Let Ω be open and bounded and V smooth and bounded. Then exactly one of the following holds.

- (i) There is a non-zero $w \in H_0^1(\Omega) \cap C^\infty(\Omega)$ with $-\nabla^2 w + Vw = 0$.
- (ii) For every $f \in L^2(\Omega)$ there is a unique weak solution $u \in H_0^1(\Omega)$ of the boundary value problem.

Proof. By the Fredholm alternative for the compact operator $\mathcal{K}$, either $(I - \mathcal{K})w = 0$ for some $w \neq 0$, or $I - \mathcal{K}$ is a bijection of $L^2(\Omega)$. In the first case $w = A((1 - V)w)$, so $w \in H_0^1(\Omega)$, and w is a weak solution of the homogeneous problem; Theorem 26.11.9 (applied with the potential term absorbed into the right-hand side, and bootstrapped) gives $w \in C^\infty(\Omega)$. In the second case, given f set $u = (I - \mathcal{K})^{-1}Af$. $\square$

The same circle of ideas produces the spectral decomposition of the Laplacian, which is one of the most-used theorems in mathematical physics.

Theorem 26.11.11

Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. There is an orthonormal basis $(w_k)_{k \geq 1}$ of $L^2(\Omega)$ with $w_k \in H_0^1(\Omega) \cap C^\infty(\Omega)$ and

$$-\nabla^2 w_k = \lambda_k w_k \quad \text{in } \Omega,$$

where $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$ and $\lambda_k \rightarrow \infty$.

Proof. By Proposition 26.11.4, $A : L^2(\Omega) \rightarrow L^2(\Omega)$ is compact and Hermitian. By the spectral theorem for such operators, $\sigma(A) = \{0, \mu_1, \mu_2, \dots\}$ with μ_k real, accumulating only at 0, and $L^2(\Omega)$ has an orthonormal basis of eigenvectors of A .

Suppose $Aw = \mu w$ with $w \neq 0$. Then $w = \mu^{-1}Aw \in H_0^1(\Omega)$ provided $\mu \neq 0$, and indeed $\mu \neq 0$: for any $v \in H_0^1(\Omega)$ we have $\langle w, v \rangle_{L^2} = \langle Aw, v \rangle_{H^1} = \mu \langle w, v \rangle_{H^1}$, and taking $v = w$ gives $\|w\|_{L^2}^2 = \mu \|w\|_{H^1}^2$, which forces $\mu > 0$. So $w \in H_0^1(\Omega)$ is a weak solution of $-\nabla^2 w + w = \mu^{-1}w$, and elliptic regularity (Theorem 26.11.9, bootstrapping through $H_{\text{loc}}^1, H_{\text{loc}}^3, H_{\text{loc}}^5, \dots$) gives $w \in C^\infty(\Omega)$. Rewriting with $\lambda = \mu^{-1} - 1$ gives $-\nabla^2 w = \lambda w$, and $\lambda \geq 0$ since $\mu \|w\|_{H^1}^2 = \|w\|_{L^2}^2 \leq \|w\|_{H^1}^2$. Finally $\mu_k \rightarrow 0$ translates into $\lambda_k \rightarrow \infty$. $\square$

Corollary 26.11.12

$(\sqrt{2} \sin(k\pi x))_{k=1}^\infty$ is an orthonormal basis of $L^2((0, 1))$.

Proof. Take $\Omega = (0, 1)$, where the eigenvalue problem $-w'' = \lambda w$, $w(0) = w(1) = 0$ has exactly the solutions $w_k = \sqrt{2} \sin(k\pi x)$ with $\lambda_k = k^2\pi^2$. $\square$

That the classical Fourier sine series is an orthonormal basis is thus a corollary of Rellich–Kondrachov and the spectral theorem. This is a pleasingly indirect route to a classical fact, and unlike the classical proof it generalises: on any bounded domain whatever, however irregular, the Dirichlet eigenfunctions form a basis.

§26.11.5 The direct method in action

Finally we return to the variational point of view of Section 26.7, and show that the solution of the Dirichlet problem is a minimiser of an energy.

Proposition 26.11.13

Let $f \in L^2(\Omega)$ and define $S : H_0^1(\Omega) \rightarrow \mathbb{R}$ by

$$S[u] = \int_{\Omega} (|Du|^2 + |u|^2 - \bar{f}u - f\bar{u}) \, dx = \|u\|_{H^1}^2 - 2 \operatorname{Re} \langle f, u \rangle_{L^2}.$$

Then S attains its infimum, at a unique w , and w is precisely the weak solution of the Dirichlet problem.

Proof. Bounded below and coercive. By Cauchy–Schwarz and $2ab \leq a^2 + b^2$,

$$S[u] \geq \|u\|_{H^1}^2 - 2 \|u\|_{L^2} \|f\|_{L^2} \geq \|u\|_{H^1}^2 - \frac{1}{2} \|u\|_{L^2}^2 - 2 \|f\|_{L^2}^2 \geq \frac{1}{2} \|u\|_{H^1}^2 - 2 \|f\|_{L^2}^2.$$

So $\sigma = \inf S > -\infty$, and any u with $S[u] \leq K$ has $\|u\|_{H^1}^2 \leq 2K + 4 \|f\|_{L^2}^2$.

Minimising sequence and weak limit. Take u_m with $S[u_m] \rightarrow \sigma$. By the above the sequence is bounded in $H_0^1(\Omega)$, so by Corollary 26.6.29 we may pass to a subsequence with $u_m \rightharpoonup w$ in $H_0^1(\Omega)$.

Lower semicontinuity. The norm is weakly lower semicontinuous, so $\|w\|_{H^1} \leq \liminf \|u_m\|_{H^1}$; and $\langle f, \cdot \rangle_{L^2}$ is a bounded functional on $H_0^1(\Omega)$, so $\operatorname{Re}\langle f, u_m \rangle \rightarrow \operatorname{Re}\langle f, w \rangle$. Hence

$$S[w] = \|w\|_{H^1}^2 - 2 \operatorname{Re}\langle f, w \rangle \leq \liminf_m \left(\|u_m\|_{H^1}^2 - 2 \operatorname{Re}\langle f, u_m \rangle \right) = \liminf_m S[u_m] = \sigma,$$

and as $S[w] \geq \sigma$ we get $S[w] = \sigma$.

Euler–Lagrange equation. For $v \in H_0^1(\Omega)$ and $t \in \mathbb{R}$, expanding,

$$S[w + tv] = S[w] + 2t (\operatorname{Re}\langle w, v \rangle_{H^1} - \operatorname{Re}\langle f, v \rangle_{L^2}) + t^2 \|v\|_{H^1}^2.$$

Since $t = 0$ is a minimum of this quadratic in t , the linear coefficient vanishes: $\operatorname{Re}\langle w, v \rangle_{H^1} = \operatorname{Re}\langle f, v \rangle_{L^2}$ for all v , and replacing v by iv gives the imaginary parts too. So w is a weak solution, and conversely reading the expansion backwards shows any weak solution is a minimiser. Uniqueness now follows from Lemma 26.11.2, or directly from strict convexity of S . $\square$

The last computation is the general phenomenon: *critical points of an energy are weak solutions of its Euler–Lagrange equation*. What makes this valuable is that the direct method applies to functionals far more general than quadratic ones, where no Riesz representation theorem is available.

Theorem 26.11.14

Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded, and let $L : \mathbb{C}^n \rightarrow \mathbb{R}$ be convex with

$$L(z) \geq \gamma|z|^2 - C \quad \text{for all } z,$$

for some $\gamma, C > 0$. Then $S[u] = \int_{\Omega} L(Du) \, dx$ attains its infimum on $H_0^1(\Omega)$.

Proof. The growth condition gives $S[u] \geq \gamma \|Du\|_{L^2}^2 - C|\Omega|$, so sublevel sets of S are bounded in $H_0^1(\Omega)$ by Poincaré’s inequality (Corollary 26.10.13), and S is bounded below. Convexity of L makes $u \mapsto S[u]$ convex, and it is strongly lower semicontinuous by Fatou’s lemma applied along an L^2 -convergent subsequence of gradients. Corollary 26.7.6 applies. $\square$

Example 26.11.15

The functional

$$u \mapsto \int_{\Omega} (1 + |Du|^4)^{1/4} \, dx$$

is not covered by any of our linear theory: its Euler–Lagrange equation is a quasilinear second-order PDE with no explicit solution and no useful integral representation. But the integrand is convex with the required growth (after adjusting for the quartic power on a bounded domain), so a minimiser exists. This is the direct method’s *raison d’être*: it produces solutions to problems that cannot be solved.

Remark. Let us close by recalling the pattern of the argument, since it is the moral of the course. We wanted a solution to an equation. We could not find one directly, so we

- (i) reformulated the problem in a Banach space of functions, chosen so that the rele-

- vant estimate is an inequality between norms (Sobolev spaces);
- (ii) enlarged the notion of solution until existence became a compactness argument (weak solutions, weak topologies, Banach–Alaoglu);
 - (iii) used convexity to ensure the limit object was still a solution (lower semicontinuity, the direct method);
 - (iv) and then used the equation itself to prove that the object we had constructed was much better behaved than the construction guaranteed (elliptic regularity).

Every step required a theorem, and every theorem required the machinery of measure, integration and functional analysis assembled in the earlier sections. That is what analysis of functions is for.

27 Automata and Formal Languages

We all know an algorithm when we see one, and for most of mathematics that is enough. But some questions cannot be answered without saying exactly what an algorithm *is*: if we want to prove that no algorithm solves a certain problem, we have to quantify over all possible algorithms, and to do that we need a definition. This course provides one – in fact several, which turn out to agree – and then uses it to show that a great many natural problems have no algorithmic solution at all.

Running alongside this is the theory of *formal languages*. A language here is just a set of finite strings over a fixed alphabet, and this is a flexible enough notion to encode almost anything: sets of numbers, sets of graphs, sets of programs. Chomsky’s insight was that languages come in a hierarchy, graded by how complicated a machine is needed to recognise them, and much of what follows is an exploration of the bottom two levels of that hierarchy – the regular languages, recognised by finite automata, and the context-free languages, recognised by automata with a stack – before we climb to the top and meet the languages recognised by a genuinely general-purpose computer.

This article constitutes my notes for the ‘Automata and Formal Languages’ course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§27.1 Introduction

§27.1.1 Algorithms and the Entscheidungsproblem

There is a difference between knowing that something exists and being able to find it. Compare the two statements

‘every polynomial of degree $n \geq 1$ over $\mathbb{C}$ has a root’

and

‘there is a procedure which, given such a polynomial, produces a root’.

The first is the fundamental theorem of algebra. The second is a genuinely different assertion, and there are plenty of situations in mathematics where the existence proof is available but no procedure is.

Historically, the demand for procedures came from Hilbert. In his 1900 address in Paris he set out a list of problems for the coming century, the tenth of which asks for a procedure which, given a Diophantine equation (a polynomial with integer coefficients), decides whether it has an integer solution. In 1928, in the book *Grundzüge der theoretischen Logik* of Hilbert and Ackermann, the [Entscheidungsproblem](#) (‘decision problem’) was posed: give a procedure which, given a formula of predicate logic, decides whether it is a tautology.

In both cases Hilbert expected a positive answer, and here is the crucial asymmetry. To answer such a question positively one does not need a definition of ‘algorithm’ at all: one simply writes an algorithm down, and everyone agrees on seeing it that it is one. To answer it negatively, one must rule out every conceivable algorithm, and for

that a precise definition is unavoidable. Both of Hilbert's questions turned out to have negative answers – Church and Turing settled the Entscheidungsproblem in 1936, and Matiyasevich settled the tenth problem in 1970 – and neither answer could have been given before the work we do in this course.

§27.1.2 Alphabets, Words and Languages

What are the objects on which computation is performed? One's first guess is 'numbers', but neither of the two problems above has numbers as its input: one takes polynomials and the other takes formulae. What is common to them is that the inputs can be written down as finite strings of symbols drawn from a fixed finite stock. This is how all modern computation works, and it is the setting we adopt.

Definition 27.1.1 (Strings)

Let Ω be a set, whose elements we call **symbols**. For $n \in \mathbb{N}$ we write Ω^n for the set of functions $n \rightarrow \Omega$, thinking of the natural number n as the set $\{0, 1, \dots, n-1\}$ in the usual von Neumann way, so that an element of Ω^n is a sequence $(x_0, \dots, x_{n-1})$ of symbols. We write

$$\Omega^* = \bigcup_{n \in \mathbb{N}} \Omega^n$$

for the set of all finite sequences of symbols, which we call **Ω -strings**.

We will suppress the sequence notation entirely and write strings by juxtaposition, so that (x_0, x_1, x_2) is written $x_0x_1x_2$. The following conventions will be used without comment.

- The **length** of a string $\alpha \in \Omega^n$ is $|\alpha| = n$.
- Ω^0 has exactly one element, the **empty string** ε .
- Given $\alpha \in \Omega^n$ and $k \leq n$, the **truncation** $\alpha \upharpoonright_k$ is the unique element of Ω^k which is an initial segment of α .
- Given $\alpha \in \Omega^m$ and $\beta \in \Omega^n$, their **concatenation** $\alpha\beta \in \Omega^{m+n}$ is defined in the obvious way. Concatenation is associative with identity ε , so Ω^* is a monoid.
- We define $\alpha^0 = \varepsilon$ and $\alpha^{n+1} = \alpha\alpha^n$.
- We identify a symbol $x \in \Omega$ with the string of length one that it determines.
- If $Y, Z \subseteq \Omega^*$ then $YZ = \{\alpha\beta \mid \alpha \in Y, \beta \in Z\}$, and if $Y = \{\alpha\}$ we simply write αZ .
- A function $f: \Omega \rightarrow \Omega'$ lifts to a function $\Omega^* \rightarrow \Omega'^*$ acting symbolwise; we write this lift as f as well.

Now, not every string is a sensible object. If Ω is a stock of letters then Ω^* contains a great many strings which are not words of English; if Ω is a stock of words then Ω^* contains a great many strings which are not sentences. So we will want to designate a subset of Ω^* as the *well-formed* strings, and it is these subsets which are the objects of study.

Definition 27.1.2 (Alphabet, word, language)

An **alphabet** is a finite nonempty set Σ , whose elements we call **letters** and denote $a, b, c, \dots$. We write

$$\mathbb{W} = \Sigma^*$$

for the set of **words**, denoted $u, v, w, \dots$, and $\mathbb{W}^+ = \mathbb{W} \setminus \{\varepsilon\}$ for the set of nonempty words. A **language** over Σ is a subset $L \subseteq \mathbb{W}$.

Because a language is simply a set of words, and words can encode anything, this is a very general notion indeed. The set of prime numbers written in binary is a language over $\{0, 1\}$; so is the set of (codes of) computer programs that never terminate. The whole of this course can be read as the study of how complicated a language can be.

§27.1.3 Counting Languages

We record here some counting facts from Numbers and Sets which will be used repeatedly, usually to demonstrate that some class of languages cannot possibly contain all of them. Recall that X is **countable** if there is a surjection $\mathbb{N} \rightarrow X$ (or $X = \emptyset$), and that X is **infinite** if there is an injection $\mathbb{N} \rightarrow X$.

Proposition 27.1.3

If X is nonempty and countable then X^* is countably infinite.

Proof. Pick $x \in X$. The map $n \mapsto x^n$ is an injection $\mathbb{N} \rightarrow X^*$, so X^* is infinite.

For countability, let $\pi: \mathbb{N} \rightarrow X$ be a surjection. Every $k \in \mathbb{N}$ with $k > 0$ has a unique prime factorisation $k = \prod_i p_i^{k_i}$, where $p_0 = 2, p_1 = 3, p_2 = 5, \dots$ enumerates the primes. Read k_0 as a length and send k to the string

$$\pi(k_1) \pi(k_2) \cdots \pi(k_{k_0}) \in X^*,$$

sending 0 to ε as well. Given any string $\alpha = y_1 \cdots y_n \in X^*$, choose m_j with $\pi(m_j) = y_j$ and take $k = 2^n 3^{m_1} 5^{m_2} \cdots$; this maps onto α . So the map is a surjection $\mathbb{N} \rightarrow X^*$. $\square$

Proposition 27.1.4

If X is countable then the set $\text{Fin}(X)$ of finite subsets of X is countable.

Proof. The ‘forgetful’ map $X^* \rightarrow \text{Fin}(X)$ sending $y_1 \cdots y_n$ to $\{y_1, \dots, y_n\}$ is a surjection: given a finite $F \subseteq X$, list its elements in any order. Composing with the surjection of Proposition 27.1.3 gives a surjection $\mathbb{N} \rightarrow \text{Fin}(X)$. $\square$

On the other hand, by Cantor’s theorem $\mathcal{P}(X)$ is uncountable whenever X is infinite. Since $\mathbb{W}$ is countably infinite, we obtain the following, which is the first thing worth knowing about languages.

Corollary 27.1.5

There are uncountably many languages over any alphabet.

Every method we shall meet for *specifying* a language – a grammar, an automaton, a regular expression, a register machine – is a finite object over some finite alphabet, and so there are only countably many of them. Consequently *every* such method fails to describe almost every language. This is not a defect of any particular formalism; it is a fact about cardinality, and it is worth keeping in mind throughout.

§27.2 Rewrite Systems and Grammars

§27.2.1 Rewrite Systems

The most primitive way of generating strings is to start with one and repeatedly replace pieces of it. This is the notion of a rewrite system, and everything in the first half of the course is a special case of it.

Definition 27.2.1 (Rewrite system)

Let Ω be a finite set of symbols. An element (α, β) of $\Omega^* \times \Omega^*$ is called a **rewrite rule** or **production rule**, and is written $\alpha \rightarrow \beta$. A pair $R = (\Omega, P)$ is a **rewrite system** if P is a *finite* set of rewrite rules.

The intended reading of $\alpha \rightarrow \beta$ is ‘wherever you see α , you may replace it by β ’. Let us make that precise.

Definition 27.2.2 (Derivation)

Let $R = (\Omega, P)$ be a rewrite system and let $\sigma, \tau \in \Omega^*$. We write

$$\sigma \Rightarrow_R \tau$$

and say that **R produces τ from σ in one step**, if there are $\alpha, \beta, \gamma, \delta \in \Omega^*$ with $\sigma = \alpha\gamma\beta$, $\tau = \alpha\delta\beta$ and $\gamma \rightarrow \delta \in P$. We write $\Rightarrow_R^*$ for the reflexive transitive closure of $\Rightarrow_R$. A sequence

$$\sigma_0 \Rightarrow_R \sigma_1 \Rightarrow_R \cdots \Rightarrow_R \sigma_n$$

is an **R -derivation of σ_n from σ_0 of length n** , and we write

$$\mathcal{D}(R, \sigma) = \{\tau \in \Omega^* \mid \sigma \Rightarrow_R^* \tau\}$$

for the set of strings derivable from σ .

Note that α and β here are the *context* in which the rewriting takes place; the rule fires regardless of what surrounds the occurrence of γ . We will meet later a restricted notion in which the context is allowed to constrain the rule, and this is the origin of the word ‘context-free’.

Since a rewrite system is a finite object over a finite alphabet, Propositions 27.1.3 and 27.1.4 give the following immediately.

Proposition 27.2.3

For fixed finite Ω , there are only countably many rewrite systems on Ω .

Proof. Ω^* is countable, hence so is $\Omega^* \times \Omega^*$, hence so is $\text{Fin}(\Omega^* \times \Omega^*)$, and every P lies in the latter. $\square$

§27.2.2 Grammars

The application we care about is to languages. Chomsky's idea was that a natural language should be modelled not by a finite list of well-formed sentences, but by a finite list of *rules* which generate infinitely many sentences. The reason for insisting on infinitude is **linguistic recursion**: if X is a grammatical English sentence then so is 'Alice believes that X ', and if we impose a maximum sentence length we are obliged to cut this off at some arbitrary point.

Here is a toy example of the sort of thing Chomsky had in mind. Take the rules

$$\begin{array}{lll} S \rightarrow \text{NP VP}, & \text{NP} \rightarrow \text{Adj NP}, & \text{NP} \rightarrow \text{Noun}, \\ \text{VP} \rightarrow \text{Verb}, & \text{VP} \rightarrow \text{Verb Adv}, & \end{array}$$

together with rules turning Adj into an adjective, Noun into a noun and so on. Starting from S one can derive 'colourless green ideas sleep furiously', Chomsky's famous example of a sentence that is grammatical but meaningless; one cannot derive 'furiously sleep ideas green colourless', which is neither. The moral is that grammaticality is a formal, checkable property, whereas meaningfulness is not something a rewrite system can see.

The symbols in this example fall into two classes: those like S and NP which are scaffolding, and those like 'ideas' which are part of the final product. This distinction is built into the definition.

Definition 27.2.4 (Grammar)

Let Σ be an alphabet and let V be a nonempty finite set disjoint from Σ , whose elements we call **variables** or **nonterminal symbols**, denoted $A, B, C, \dots$. Write $\Omega = \Sigma \cup V$. A **grammar** is a tuple $G = (\Sigma, V, P, S)$ where (Ω, P) is a rewrite system and $S \in V$ is the **start symbol**.

We use the notation for rewrite systems for the associated rewrite system, writing $\Rightarrow_G$, $\Rightarrow_G^*$ and $\mathcal{D}(G, \sigma)$. The **language generated by G** is

$$\mathcal{L}(G) = \mathcal{D}(G, S) \cap \mathbb{W}.$$

So a word lies in $\mathcal{L}(G)$ exactly when it can be derived from the start symbol and contains no variables. Two easy ways for this to fail are worth noting straight away.

Example 27.2.5

If P contains no rule of the form $S \rightarrow \alpha$, then $\mathcal{D}(G, S) = \{S\}$, which contains no words, so $\mathcal{L}(G) = \emptyset$. Similarly, if no rule has a right-hand side lying in $\mathbb{W}$, then nothing in $\mathcal{D}(G, S)$ can be a word, and again $\mathcal{L}(G) = \emptyset$.

Example 27.2.6

Let $\Sigma = \{a\}$, $V = \{S\}$ and $P_0 = \{S \rightarrow aaS, S \rightarrow a\}$. We claim that

$$\mathcal{L}(G_0) = \{a^{2n+1} \mid n \in \mathbb{N}\}.$$

Each rule replaces a string by one of length two greater, or of the same length, so by induction on the length of a derivation every element of $\mathcal{D}(G_0, S)$ has odd length; in particular every word produced is a^{2n+1} for some n . Conversely, applying $S \rightarrow aaS$ exactly n times and then $S \rightarrow a$ produces a^{2n+1} .

Observe that the only feature of P_0 used in the argument was that every rule preserves the parity of the length. So the rule sets

$$\begin{aligned} P_1 &= \{S \rightarrow aSa, S \rightarrow a\}, & P_2 &= \{S \rightarrow Saa, S \rightarrow a\}, \\ P_3 &= \{S \rightarrow aaS, S \rightarrow aaSaa, S \rightarrow a\} \end{aligned}$$

all generate the same language. Different grammars can generate the same language, and this is the phenomenon we look at next.

§27.2.3 Equivalence of Grammars

Definition 27.2.7 (Equivalent grammars)

Grammars G and G' over the same alphabet are **equivalent** if $\mathcal{L}(G) = \mathcal{L}(G')$.

Our aim in this subsection is to show that, for a fixed alphabet Σ , there are only countably many languages of the form $\mathcal{L}(G)$. The point requiring care is that V is allowed to be an arbitrary finite set, so there is a proper class of grammars to begin with; we must cut this down to a set. The tool is the observation that the names of the variables do not matter.

Definition 27.2.8 (Isomorphism of grammars)

Let $G = (\Sigma, V, P, S)$ and $G' = (\Sigma, V', P', S')$ be grammars over the same alphabet. A function $f: \Omega \rightarrow \Omega'$ is an **isomorphism** if

- (i) $f \upharpoonright_{\Sigma} = \text{id}$;
- (ii) $f(S) = S'$;
- (iii) $f \upharpoonright_V$ is a bijection $V \rightarrow V'$;
- (iv) $\alpha \rightarrow \beta \in P$ if and only if $f(\alpha) \rightarrow f(\beta) \in P'$,

where in (iv) we use the lift of f to Ω^* .

Proposition 27.2.9

Isomorphic grammars are equivalent.

Proof. If f is an isomorphism $G \rightarrow G'$ then f^{-1} is an isomorphism $G' \rightarrow G$, so it suffices to prove $\mathcal{L}(G) \subseteq \mathcal{L}(G')$. Let $w \in \mathcal{L}(G)$, with derivation

$$S = \sigma_0 \Rightarrow_G \sigma_1 \Rightarrow_G \cdots \Rightarrow_G \sigma_n = w.$$

Apply f throughout. By (ii) the first term is S' ; by (i) the last term is still w , since w contains only letters; and by (iv) each single step remains a legal step, since the decomposition $\sigma_i = \alpha\gamma\beta$ maps to $f(\sigma_i) = f(\alpha)f(\gamma)f(\beta)$. So $S' \Rightarrow_{G'}^* w$ and $w \in \mathcal{L}(G')$. $\square$

Proposition 27.2.10

Let $G = (\Sigma, V, P, S)$ be a grammar and let V' be any set with $|V| = |V'|$ and $V' \cap \Sigma = \emptyset$. Then there are P' and S' with $G' = (\Sigma, V', P', S')$ equivalent to G .

Proof. Choose a bijection $f: V \rightarrow V'$ and extend it to Ω by the identity on Σ , so that (i) and (iii) hold. Set $S' = f(S)$ and $P' = \{f(\alpha) \rightarrow f(\beta) \mid \alpha \rightarrow \beta \in P\}$, so that (ii) and (iv) hold. Then $G \cong G'$, so the two are equivalent. $\square$

Proposition 27.2.11

For a fixed alphabet Σ , there are only countably many languages of the form $\mathcal{L}(G)$ for a grammar G .

Proof. Fix a finite set V disjoint from Σ . There are only countably many rewrite systems on $\Omega = \Sigma \cup V$, and only finitely many choices of start symbol, so the collection $\mathcal{G}_V$ of grammars with this variable set is countable, and hence so is $\mathcal{L}_V = \{\mathcal{L}(G) \mid G \in \mathcal{G}_V\}$.

By Proposition 27.2.10, $\mathcal{L}_V$ depends only on $|V|$, so we may write $\mathcal{L}_n$ for the value at any n -element V , and every language generated by a grammar lies in some $\mathcal{L}_n$. Then $\bigcup_{n>0} \mathcal{L}_n$ is a countable union of countable sets and is therefore countable. $\square$

Combining this with Corollary 27.1.5, almost every language is generated by no grammar at all. Grammars are nevertheless the right thing to study, because the languages they generate are exactly the ones we can hope to compute with; we shall see in Section 27.8 that they are precisely the *computably enumerable* languages.

§27.2.4 The Chomsky Hierarchy

Restricting the *shape* of the allowed production rules restricts the languages that can be generated, and Chomsky's classification is obtained by imposing four increasingly severe such restrictions.

Definition 27.2.12 (Types of production rule)

Let $\alpha \rightarrow \beta$ be a production rule. We call it

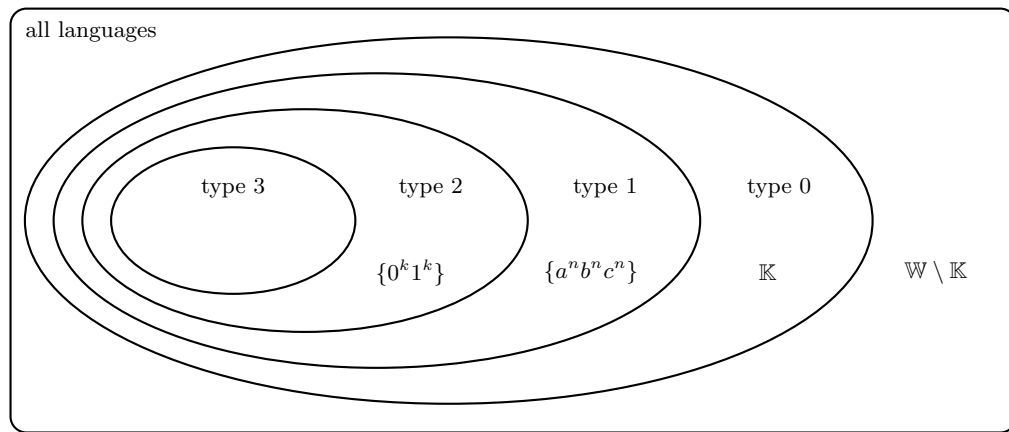
- (i) **noncontracting** if $|\alpha| \leq |\beta|$;
- (ii) **context-sensitive** if there are $A \in V$, $\gamma, \delta \in \Omega^*$ and $\eta \in \Omega^*$ with $|\eta| > 0$ such that $\alpha = \gamma A \delta$ and $\beta = \gamma \eta \delta$;
- (iii) **context-free** if $\alpha = A \in V$ and $|\beta| > 0$;
- (iv) **regular** if $\alpha = A \in V$ and β is either a or aB for some $a \in \Sigma$ and $B \in V$.

A grammar has one of these properties if all of its rules do, and a language has one of these properties if it is generated by some grammar which has it.

A context-sensitive rule may replace A by η , but only when A occurs in the context γ_δ ; the surrounding material is not allowed to change. A context-free rule may replace A by β wherever it occurs. A regular rule may only replace a variable by a letter, or by a letter followed by a single variable, so that a derivation grows strictly from left to right.

Immediately from the definitions, $\text{regular} \Rightarrow \text{context-free} \Rightarrow \text{context-sensitive} \Rightarrow \text{non-contracting}$. (For the middle implication, take $\gamma = \delta = \varepsilon$.) In Chomsky's numbering, a language L is

- **type 0** if $L = \mathcal{L}(G)$ for some grammar G ;
- **type 1** if $L = \mathcal{L}(G)$ for some context-sensitive G ;
- **type 2** if $L = \mathcal{L}(G)$ for some context-free G ;
- **type 3** if $L = \mathcal{L}(G)$ for some regular G .



The picture is the one to keep in mind, and proving that the inclusions are all strict – that the sample languages really do sit where they are drawn – occupies a large part of the course. We will show that $\{0^k 1^k \mid k > 0\}$ is context-free but not regular, and that $\{a^n b^n c^n \mid n > 0\}$ is context-sensitive but not context-free; the halting set $\mathbb{K}$ and its complement will be dealt with in Section 27.8.

The following theorem of Chomsky says that two of our four conditions define the same class of languages, even though they do not define the same class of grammars. We shall not prove it.

Theorem 27.2.13 (Chomsky)

A language is noncontracting if and only if it is context-sensitive.

§27.2.5 Decision Problems

Alongside the question ‘which languages arise?’ we ask ‘what can we work out about them?’. Three questions will accompany us throughout.

Definition 27.2.14 (The three decision problems) (i) The **word problem** takes as

input a grammar G and a word w , and asks whether $w \in \mathcal{L}(G)$.

- (ii) The **emptiness problem** takes as input a grammar G and asks whether $\mathcal{L}(G) = \emptyset$.
- (iii) The **equivalence problem** takes as input grammars G and G' and asks whether $\mathcal{L}(G) = \mathcal{L}(G')$.

A problem is **solvable** if there is an algorithm giving the correct answer for every input, and **unsolvable** otherwise.

For the moment ‘algorithm’ is informal; in Section 27.8.7 we will replace it by a precise definition and revisit all three problems. It will turn out that all three are unsolvable for grammars in general. Lower down the hierarchy things are better, and we can already deal with the word problem.

Lemma 27.2.15

Let G be a noncontracting grammar and let $w \in \mathbb{W}$. There is a bound N , depending only on $|w|$ and $|\Omega|$, such that $w \in \mathcal{L}(G)$ if and only if w has a G -derivation of length at most N .

Proof. Let $S = \sigma_0 \Rightarrow_G \cdots \Rightarrow_G \sigma_n = w$ be a derivation of minimal length. As G is noncontracting, the sequence of lengths $1 = |\sigma_0| \leq |\sigma_1| \leq \cdots \leq |\sigma_n| = |w|$ is nondecreasing. If $\sigma_r = \sigma_s$ for some $r < s$ then deleting $\sigma_{r+1}, \dots, \sigma_s$ leaves a shorter derivation, so by minimality all the σ_i are distinct. But every σ_i is a string over Ω of length at most $|w|$, and there are only

$$N = \sum_{\ell=1}^{|w|} |\Omega|^\ell$$

of these, so $n \leq N$ by pigeonhole. $\square$

Corollary 27.2.16

The word problem is solvable for noncontracting, context-sensitive, context-free and regular grammars.

Proof. Given G and w , compute N as above. There are only finitely many derivations of length at most N , since at each step only finitely many rules can be applied at finitely many positions, and they can be enumerated; check each one. $\square$

This is a good illustration of the difference between solvability and practicality. The bound N is astronomically large, and the algorithm above is of no practical use whatsoever; but solvability is a statement about existence, and existence is what we get.

§27.2.6 Closure Properties

The other standard question about a class of languages is which operations it is closed under. Given languages L and M over Σ , the operations of interest are

$$LM, \quad L \cup M, \quad L \cap M, \quad L \setminus M, \quad \mathbb{W}^+ \setminus L.$$

We use $\mathbb{W}^+ \setminus L$ rather than $\mathbb{W} \setminus L$ for the complement, because (as we check below) a noncontracting grammar can never produce ε , so no class below type 0 could possibly be closed under an operation that introduces the empty word.

Some closure properties come for free from others: for instance, closure under complement and union implies closure under intersection, by De Morgan.

Definition 27.2.17 (Union and concatenation grammars)

Let $G = (\Sigma, V, P, S)$ and $G' = (\Sigma, V', P', S')$ be grammars, and let T be a variable in neither V nor V' . The **concatenation grammar** is

$$H = (\Sigma, V \cup V' \cup \{T\}, P \cup P' \cup \{T \rightarrow SS'\}, T),$$

and the **union grammar** is

$$H' = (\Sigma, V \cup V' \cup \{T\}, P \cup P' \cup \{T \rightarrow S, T \rightarrow S'\}, T).$$

It is clear that $\mathcal{L}(G)\mathcal{L}(G') \subseteq \mathcal{L}(H)$ and $\mathcal{L}(G) \cup \mathcal{L}(G') \subseteq \mathcal{L}(H')$. The reverse inclusions, however, are *not* automatic. Even after relabelling so that $V \cap V' = \emptyset$, the two rule sets can interact: a rule of P whose left-hand side contains letters can fire on a string produced by P' . The way to rule this out is to arrange that no rule mentions a letter on its left.

Definition 27.2.18 (Variable-based)

A rule $\alpha \rightarrow \beta$ is **variable-based** if every symbol occurring in α is a variable, and a grammar is variable-based if all of its rules are.

Regular and context-free grammars are automatically variable-based, since their rules have $\alpha = A \in V$. Context-sensitive grammars need not be.

Lemma 27.2.19

Every grammar is equivalent to a variable-based grammar. Moreover, if the original grammar is context-free, context-sensitive or noncontracting, then so is the new one.

Proof. Let $G = (\Sigma, V, P, S)$. For each letter $a \in \Sigma$ introduce a fresh variable X_a , and let $X: \Omega \rightarrow \Omega$ send $a \mapsto X_a$ for $a \in \Sigma$ and fix V pointwise; lift X to Ω^* . Put

$$P' = \{X(\alpha) \rightarrow X(\beta) \mid \alpha \rightarrow \beta \in P\}, \quad P'' = \{X_a \rightarrow a \mid a \in \Sigma\},$$

and let $G' = (\Sigma, V \cup \{X_a \mid a \in \Sigma\}, P' \cup P'', S)$. Every rule of P' has only variables on the left by construction, and every rule of P'' has a single variable on the left, so G' is variable-based. The added rules $X_a \rightarrow a$ are regular, so they preserve context-freeness, context-sensitivity and noncontraction.

It remains to check $\mathcal{L}(G) = \mathcal{L}(G')$. Applying X to a G -derivation of w gives a G' -derivation of $X(w)$, and then applying the rules $X_a \rightarrow a$ once for each letter of w gives w ; so $\mathcal{L}(G) \subseteq \mathcal{L}(G')$.

Conversely let $S = \sigma_0 \Rightarrow_{G'} \cdots \Rightarrow_{G'} \sigma_m = w$ be a G' -derivation. Apply X to every term to get a sequence $\tau_0, \dots, \tau_m$ with $\tau_0 = S$ and $\tau_m = X(w)$. If the step

$\sigma_i \Rightarrow_{G'} \sigma_{i+1}$ used a rule $X(\alpha) \rightarrow X(\beta)$, then since X fixes such strings we still have $\tau_i \Rightarrow_{G'} \tau_{i+1}$ by the same rule, which is a rule of G after erasing X . If instead it used a rule $X_a \rightarrow a$, then $\tau_i = \tau_{i+1}$, and we delete the repetition. Deleting all such repetitions leaves a genuine G -derivation of w (after erasing X throughout, which is legitimate since no string in the resulting sequence contains any X_a). Hence $w \in \mathcal{L}(G)$. $\square$

Theorem 27.2.20

Let G and G' be variable-based grammars over Σ with $V \cap V' = \emptyset$. Then

$$\mathcal{L}(H) = \mathcal{L}(G)\mathcal{L}(G'), \quad \mathcal{L}(H') = \mathcal{L}(G) \cup \mathcal{L}(G').$$

Consequently the classes of context-free, context-sensitive and noncontracting languages are closed under union and concatenation, and the class of regular languages is closed under union.

Proof. One inclusion in each case has already been observed. For the other, consider a derivation in H . The first step must be $T \rightarrow SS'$, since T occurs in no other rule's right-hand side and appears nowhere else. Thereafter every rule applied has a variable-only left-hand side, so it lies entirely within the V -part or entirely within the V' -part of the string, these being disjoint. Thus the derivation splits into a G -derivation acting on the first component and a G' -derivation acting on the second, and any word produced is of the form vw with $v \in \mathcal{L}(G)$ and $w \in \mathcal{L}(G')$. The argument for H' is the same, the first step being $T \rightarrow S$ or $T \rightarrow S'$.

For the closure statements, given two grammars of the relevant type we first pass to equivalent variable-based grammars with disjoint variable sets, using Lemma 27.2.19 and Proposition 27.2.10, and then apply the above. The union grammar of two regular grammars is not regular, because of the rules $T \rightarrow S$; we shall prove closure of the regular languages under union and concatenation by other means in Section 27.3. $\square$

§27.2.7 The Empty Word

Remark. No noncontracting grammar produces ε : every string in a derivation from S has length at least $|S| = 1$. In particular no type 1, 2 or 3 language contains the empty word, which is why $\mathbb{W}^+$ rather than $\mathbb{W}$ is the correct ambient set for those classes.

One might hope simply to add the rule $S \rightarrow \varepsilon$ to put ε into a language, but this can have side effects: if S occurs on the right-hand side of some rule then the new rule can fire in the middle of a derivation and delete a symbol that was doing work.

Definition 27.2.21

A rule $\alpha \rightarrow \beta$ is **S -safe** if S does not occur in β . A grammar is **ε -adequate** if all of its rules are S -safe.

If G is ε -adequate then S can only ever appear as the very first string of a derivation, so adding $S \rightarrow \varepsilon$ produces a grammar for $\mathcal{L}(G) \cup \{\varepsilon\}$ and nothing else. Moreover every

grammar is equivalent to an ε -adequate one: replace $G = (\Sigma, V, P, S)$ by

$$G' = (\Sigma, V \cup \{T\}, P \cup \{T \rightarrow S\}, T)$$

for a fresh variable T . This will be used without comment whenever it is convenient.

§27.3 Regular Languages

The lowest level of the hierarchy is the most thoroughly understood, and it is where we start. We will see that regular languages admit four completely different descriptions – by grammars, by deterministic automata, by nondeterministic automata, and by regular expressions – and that all four are effectively interchangeable. We will also obtain a tool (the pumping lemma) for proving that a language is *not* regular, and a canonical minimal automaton for each regular language.

§27.3.1 Regular Derivations

Recall that a regular rule has the form $A \rightarrow a$ or $A \rightarrow aB$. We call the first kind **terminal** and the second kind **nonterminal**. The shape of regular derivations is extremely rigid, which is what makes the theory work.

Lemma 27.3.1

Let G be a regular grammar. If $S \Rightarrow_G^* \alpha$ then $\alpha \in \mathbb{W} \cup \mathbb{W}V$; that is, α is a word, possibly followed by a single variable.

Proof. Induction on the length of the derivation. The empty derivation gives $\alpha = S = \varepsilon S \in \mathbb{W}V$. Suppose $S \Rightarrow_G^* \beta \Rightarrow_G \alpha$ with $\beta \in \mathbb{W} \cup \mathbb{W}V$. If $\beta \in \mathbb{W}$ then β contains no variable, but every rule rewrites a variable, so no step is possible; so $\beta = wA$ with $w \in \mathbb{W}$, $A \in V$. The applicable rules are $A \rightarrow a$ and $A \rightarrow aB$, giving $\alpha = wa \in \mathbb{W}$ or $\alpha = waB \in \mathbb{W}V$ respectively. $\square$

Lemma 27.3.2

Let G be regular and $w \in \mathbb{W}$ with $S \Rightarrow_G^* w$. Then every derivation of w from S has length exactly $|w|$, and consists of $|w| - 1$ nonterminal rules followed by a single terminal rule.

Proof. A nonterminal rule increases the length of the string by one and leaves the number of variables at one; a terminal rule leaves the length unchanged and reduces the number of variables from one to zero. Starting from S , of length 1 with one variable, we therefore need exactly $|w| - 1$ nonterminal steps to reach length $|w|$. Once the variable count reaches zero no further rule applies, so the terminal rule must come last, and there is exactly one of them. $\square$

Note that the derivation is nonetheless not *unique*, since there may be several rules $A \rightarrow aB$ with the same A and a . This is precisely the nondeterminism that we will have to deal with below.

As a first application, we can now handle concatenation, which Theorem 27.2.20 did not give us for regular languages.

Lemma 27.3.3

Regular languages are closed under concatenation.

Proof. Let $G = (\Sigma, V, P, S)$ and $G' = (\Sigma, V', P', S')$ be regular, with $V \cap V' = \emptyset$ after relabelling. Let P^* be obtained from P by replacing every terminal rule $A \rightarrow a$ by the nonterminal rule $A \rightarrow aS'$, and set $H = (\Sigma, V \cup V', P^* \cup P', S)$. This is regular. We claim $\mathcal{L}(H) = \mathcal{L}(G)\mathcal{L}(G')$.

If $S \Rightarrow_G^* v$ and $S' \Rightarrow_{G'}^* w$, then running the G -derivation in H and using the modified last rule yields $S \Rightarrow_H^* vS'$, and then the G' -derivation gives $S \Rightarrow_H^* vw$.

Conversely suppose $S \Rightarrow_H^* u$ with $u \in \mathbb{W}$. By Lemma 27.3.1 every intermediate string is $w_i X_i$ with $X_i \in V \cup V'$. Initially $X_0 = S \in V$. The only rules taking a variable of V to a variable of V' are the modified rules $A \rightarrow aS'$, and no rule of P' mentions a variable of V ; so once the derivation enters V' it stays there. Splitting the derivation at the unique step where this happens gives $u = vw$ with $S \Rightarrow_G^* v$ (using the original terminal rule in place of the modified one) and $S' \Rightarrow_{G'}^* w$. $\square$

§27.3.2 Deterministic Automata

We now come to the machine model. A deterministic automaton reads a word one letter at a time, from left to right, keeping track of nothing except which of finitely many states it is in.

Definition 27.3.4 (Deterministic automaton)

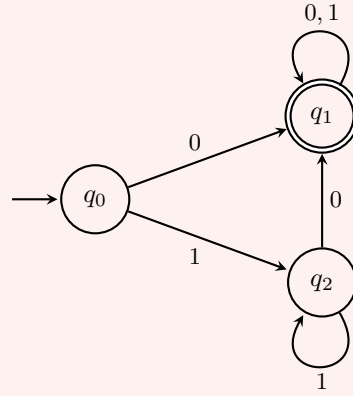
Let Σ be an alphabet. A **deterministic automaton** is a tuple $D = (\Sigma, Q, \delta, q_0, F)$ where Q is a finite set of **states**, $q_0 \in Q$ is the **start state**, $F \subseteq Q \setminus \{q_0\}$ is the set of **accept states**, and $\delta: Q \times \Sigma \rightarrow Q$ is the **transition function**.

The requirement $q_0 \notin F$ is a convention forced on us by the fact that regular grammars cannot generate ε ; it will mean that ε never lies in the language of an automaton, matching the grammar side exactly. It is a harmless restriction, and we will return to it when we discuss the Kleene star.

We draw an automaton as a labelled directed graph: one node per state, drawn as a circle, or as a double circle if the state is accepting; an unlabelled incoming arrow marks the start state; and for each q and a there is an arrow from q to $\delta(q, a)$ labelled a . Every node therefore has exactly $|\Sigma|$ outgoing arrows, though several may be drawn as one arrow with several labels.

Example 27.3.5

Let $\Sigma = \{0, 1\}$ and consider the following automaton.



Reading 110 we pass $q_0 \rightarrow q_2 \rightarrow q_2 \rightarrow q_1$ and finish in an accept state; reading 111 we pass $q_0 \rightarrow q_2 \rightarrow q_2 \rightarrow q_2$ and do not. We shall verify below that this machine accepts exactly the words containing at least one 0.

To say precisely what a machine accepts we extend δ from letters to words.

Definition 27.3.6 (Language of an automaton)

Define $\hat{\delta}: Q \times \mathbb{W} \rightarrow Q$ by recursion on the length of the word:

$$\hat{\delta}(q, \varepsilon) = q, \quad \hat{\delta}(q, aw) = \hat{\delta}(\delta(q, a), w).$$

The **language accepted by D** is

$$\mathcal{L}(D) = \{w \in \mathbb{W} \mid \hat{\delta}(q_0, w) \in F\}.$$

The **state sequence** of D on input $w = a_0a_1 \cdots a_{n-1}$ is the sequence $q_0, q_1, \dots, q_n$ with $q_{i+1} = \delta(q_i, a_i)$; it is uniquely determined and has length $|w| + 1$.

Note that $\varepsilon \notin \mathcal{L}(D)$ for every D , since $\hat{\delta}(q_0, \varepsilon) = q_0 \notin F$.

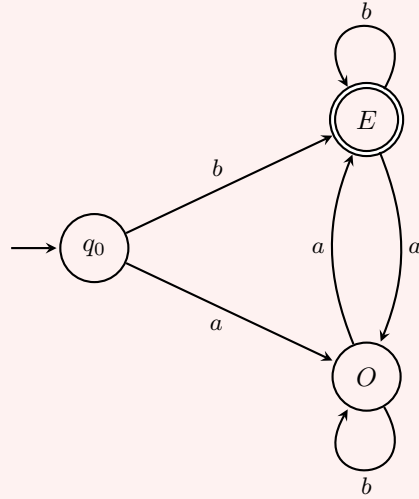
Example 27.3.7

We verify the claim of Example 27.3.5: $\mathcal{L}(D)$ is the set of words containing at least one 0.

Every 0-labelled arrow in the diagram leads to q_1 , and both arrows out of q_1 return to q_1 . So if w contains a 0, then after reading the first 0 the machine is in q_1 and stays there, and w is accepted. If w contains no 0, then $w = 1^k$; by induction $\hat{\delta}(q_0, 1^k) = q_2$ for $k \geq 1$ and $= q_0$ for $k = 0$, and neither is accepting. So w is rejected.

Example 27.3.8

Let $\Sigma = \{a, b\}$. The following automaton accepts the nonempty words containing an even number of a 's.



The states E and O record the parity of the number of a 's read so far, and q_0 is a copy of E which exists only to keep ε out of the language. Thus b is accepted ($q_0 \rightarrow E$), aa is accepted ($q_0 \rightarrow O \rightarrow E$), and a is rejected ($q_0 \rightarrow O$), as is ε .

The last example is slightly unsatisfying: the states q_0 and E do the same job. We will make this observation precise in Section 27.3.8, where we show that every automaton has a canonical smallest equivalent form. To do that we need a notion of map between automata.

Definition 27.3.9 (Homomorphism)

Let $D = (\Sigma, Q, \delta, q_0, F)$ and $D' = (\Sigma, Q', \delta', q'_0, F')$. A function $f: Q \rightarrow Q'$ is a **homomorphism** if

- (i) $\delta'(f(q), a) = f(\delta(q, a))$ for all $q \in Q, a \in \Sigma$;
- (ii) $f(q_0) = q'_0$;
- (iii) $q \in F$ if and only if $f(q) \in F'$.

A bijective homomorphism is an **isomorphism**; its inverse is then a homomorphism too.

Condition (i) says that f commutes with a single step, and a trivial induction upgrades this to $\hat{\delta}'(f(q), w) = f(\hat{\delta}(q, w))$ for all words w .

Proposition 27.3.10

If there is a homomorphism $f: D \rightarrow D'$ then $\mathcal{L}(D) = \mathcal{L}(D')$.

Proof. For any word w we have

$$\hat{\delta}(q_0, w) \in F \iff f(\hat{\delta}(q_0, w)) \in F' \iff \hat{\delta}'(q'_0, w) \in F'$$

using (iii), then (i) and (ii). □

So a homomorphism, even a very non-injective and non-surjective one, preserves the language. This tells us what failures of bijectivity mean. If f is not surjective, the

missing states of D' are not reachable from q'_0 . If f is not injective, say $f(p) = f(q)$ with $p \neq q$, then for every word w

$$f(\hat{\delta}(p, w)) = \hat{\delta}'(f(p), w) = \hat{\delta}'(f(q), w) = f(\hat{\delta}(q, w)),$$

so $\hat{\delta}(p, w) \in F$ if and only if $\hat{\delta}(q, w) \in F$: the states p and q cannot be told apart by any word. Both phenomena are forms of redundancy, and eliminating them is exactly what minimisation does.

§27.3.3 From Automata to Grammars

Theorem 27.3.11

If D is a deterministic automaton then $\mathcal{L}(D)$ is regular.

Proof. Let $D = (\Sigma, Q, \delta, q_0, F)$ and define a grammar $G = (\Sigma, Q, P, q_0)$ with variable set Q and

$$P = \{p \rightarrow aq \mid \delta(p, a) = q\} \cup \{p \rightarrow a \mid \delta(p, a) \in F\}.$$

Every rule is regular. Now let $w = a_0a_1 \cdots a_n$ be a nonempty word, and let $q_0, q_1, \dots, q_{n+1}$ be the state sequence of D on w . Then

$$\begin{aligned} w \in \mathcal{L}(D) &\iff q_{n+1} \in F \\ &\iff q_i \rightarrow a_i q_{i+1} \in P \text{ for } i < n, \text{ and } q_n \rightarrow a_n \in P \\ &\iff q_0 \Rightarrow_G a_0 q_1 \Rightarrow_G \cdots \Rightarrow_G a_0 \cdots a_{n-1} q_n \Rightarrow_G w, \end{aligned}$$

where the middle equivalence holds because the state sequence is uniquely determined by w . By Lemma 27.3.2 every derivation of w in G has this shape, so the last line holds if and only if $w \in \mathcal{L}(G)$. Finally ε lies in neither language. $\square$

We would like the converse. The obvious attempt – take $Q = V$ and read off δ from the rules – fails immediately, because a regular grammar may contain both $A \rightarrow aB$ and $A \rightarrow aB'$, and then there is no single value for $\delta(A, a)$. The standard remedy is to allow the machine to be in a *set* of states at once.

§27.3.4 Nondeterministic Automata and the Subset Construction

Definition 27.3.12 (Nondeterministic automaton)

A **nondeterministic automaton** is a tuple $N = (\Sigma, Q, \delta, q_0, F)$ exactly as before, except that

$$\delta: Q \times \Sigma \rightarrow \mathcal{P}(Q).$$

We read $\delta(q, a)$ as the set of states the machine *may* move to on reading a in state q ; if $\delta(q, a) = \emptyset$ then the computation gets stuck. The graphical convention is unchanged, except that a node may now have several, or no, outgoing arrows for a given letter.

Definition 27.3.13

For a nondeterministic automaton define $\hat{\delta}: Q \times \mathbb{W} \rightarrow \mathcal{P}(Q)$ by

$$\hat{\delta}(q, \varepsilon) = \{q\}, \quad \hat{\delta}(q, wa) = \bigcup_{p \in \hat{\delta}(q, w)} \delta(p, a),$$

and set

$$\mathcal{L}(N) = \{w \in \mathbb{W} \mid \hat{\delta}(q_0, w) \cap F \neq \emptyset\}.$$

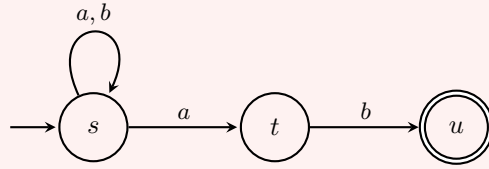
So N accepts w if *some* run on w ends in an accept state. A deterministic automaton is the special case in which every $\delta(q, a)$ is a singleton, and the two definitions of $\hat{\delta}$ then agree.

Example 27.3.14

Let $\Sigma = \{a, b\}$ and consider the nondeterministic automaton N with states s (start), t and u (accept), and

$$\delta(s, a) = \{s, t\}, \quad \delta(s, b) = \{s\}, \quad \delta(t, b) = \{u\},$$

all other values being $\emptyset$.



The machine guesses when it has reached the penultimate letter. It accepts exactly the words ending in ab : on input aab we get $\hat{\delta}(s, a) = \{s, t\}$, then $\hat{\delta}(s, aa) = \{s, t\}$, then $\hat{\delta}(s, aab) = \{s\} \cup \{u\} = \{s, u\}$, which meets F ; whereas on input aba we get $\{s, t\}$, then $\{s, u\}$, then $\{s, t\}$, which does not.

The nondeterminism is only apparent, and this is the fundamental theorem about finite automata.

Theorem 27.3.15 (Subset construction)

Let N be a nondeterministic automaton. Then there is a deterministic automaton D with $\mathcal{L}(N) = \mathcal{L}(D)$.

Proof. Let $N = (\Sigma, Q, \delta, q_0, F)$ and define

$$D = (\Sigma, \mathcal{P}(Q), \Delta, \{q_0\}, \mathcal{F}), \quad \Delta(X, a) = \bigcup_{q \in X} \delta(q, a),$$

$$\mathcal{F} = \{X \subseteq Q \mid X \cap F \neq \emptyset\}.$$

(If $\{q_0\} \in \mathcal{F}$, that is if $q_0 \in F$, then F was not a legal accept set for N in the first place; so the convention $\mathcal{F} \subseteq \mathcal{P}(Q) \setminus \{\{q_0\}\}$ holds automatically.)

Fix a word $w = a_0 \cdots a_{n-1}$. The state sequence of D on w is $X_0 = \{q_0\}$ and $X_{i+1} = \bigcup_{q \in X_i} \delta(q, a_i)$; the sequence of state-sets of N is $Y_0 = \{q_0\}$ and $Y_{i+1} = \bigcup_{q \in Y_i} \delta(q, a_i)$.

These recursions are identical, so $X_i = Y_i$ for all i , whence

$$w \in \mathcal{L}(D) \iff X_n \cap F \neq \emptyset \iff \hat{\delta}(q_0, w) \cap F \neq \emptyset \iff w \in \mathcal{L}(N). \quad \square$$

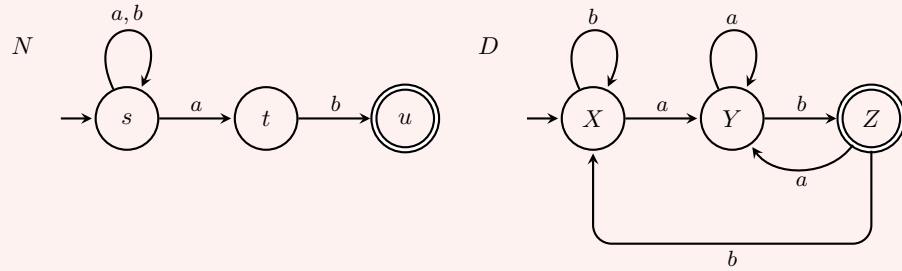
Remark. The construction turns an n -state machine into one with 2^n states, and this exponential blow-up cannot in general be avoided. In practice one only ever builds the subsets *reachable* from $\{q_0\}$, which is often far fewer.

Example 27.3.16

Let us carry out the subset construction on the automaton N of Example 27.3.14. Starting from $\{s\}$ and closing off under Δ :

$$\begin{aligned} \Delta(\{s\}, a) &= \{s, t\}, & \Delta(\{s\}, b) &= \{s\}, & \Delta(\{s, t\}, a) &= \{s, t\}, & \Delta(\{s, t\}, b) &= \{s, u\}, \\ \Delta(\{s, u\}, a) &= \{s, t\}, & \Delta(\{s, u\}, b) &= \{s\}. \end{aligned}$$

Only three of the eight subsets are reachable, and only $\{s, u\}$ meets $F = \{u\}$. Writing $X = \{s\}$, $Y = \{s, t\}$, $Z = \{s, u\}$ we obtain the deterministic automaton D drawn on the right below, beside the original N .



Tracing $abab$ through D : $X \rightarrow Y \rightarrow Z \rightarrow Y \rightarrow Z$, ending in the accept state, and indeed $abab$ ends in ab . Tracing abb : $X \rightarrow Y \rightarrow Z \rightarrow X$, rejected, and indeed abb does not end in ab .

Theorem 27.3.17

If G is a regular grammar then there is a nondeterministic automaton N with $\mathcal{L}(G) = \mathcal{L}(N)$.

Proof. Let $G = (\Sigma, V, P, S)$ and let H be a fresh symbol, the **halt state**. Put $Q = V \cup \{H\}$ and define $N = (\Sigma, Q, \delta, S, \{H\})$ where, for $A \in V$,

$$\delta(A, a) = \begin{cases} \{B \mid A \rightarrow aB \in P\} & \text{if } A \rightarrow a \notin P, \\ \{B \mid A \rightarrow aB \in P\} \cup \{H\} & \text{if } A \rightarrow a \in P, \end{cases}$$

and $\delta(H, a) = \emptyset$. Note $H \neq S$, so the convention $F \subseteq Q \setminus \{q_0\}$ holds.

Let $w = a_0 \cdots a_n$ be a nonempty word. By Lemma 27.3.2, $w \in \mathcal{L}(G)$ if and only if there are variables $S = A_0, A_1, \dots, A_n$ with $A_i \rightarrow a_i A_{i+1} \in P$ for $i < n$ and $A_n \rightarrow a_n \in P$. By the definition of δ , this holds if and only if there are states with $A_{i+1} \in \delta(A_i, a_i)$ for $i < n$ and $H \in \delta(A_n, a_n)$, which holds if and only if $H \in \hat{\delta}(S, w)$, i.e. $w \in \mathcal{L}(N)$. Neither language contains ε . $\square$

Assembling Theorems 27.3.11, 27.3.15 and 27.3.17, we have the following, which is the first genuinely satisfying theorem of the course.

Corollary 27.3.18

For a language L , the following are equivalent:

- (i) L is regular, i.e. $L = \mathcal{L}(G)$ for a regular grammar G ;
- (ii) $L = \mathcal{L}(N)$ for a nondeterministic automaton N ;
- (iii) $L = \mathcal{L}(D)$ for a deterministic automaton D .

Moreover, each translation is effective: given one of the three objects we can compute the other two.

§27.3.5 The Pumping Lemma for Regular Languages

We now have three ways to show a language *is* regular. To show that one is not, the standard tool is the following. It expresses in combinatorial terms the fact that a machine with finitely many states must, on a long enough word, revisit a state, and can then be made to go round the resulting loop as often as we please.

Definition 27.3.19 (Regular pumping lemma)

A language L **satisfies the regular pumping lemma with pumping number n** if every $w \in L$ with $|w| \geq n$ can be written as $w = xyz$ with

$$|y| > 0, \quad |xy| \leq n, \quad xy^kz \in L \text{ for all } k \in \mathbb{N}.$$

We call y a **pump** for w .

Theorem 27.3.20 (Pumping lemma)

Every regular language satisfies the regular pumping lemma, with pumping number the number of states of any deterministic automaton accepting it.

Proof. Let $L = \mathcal{L}(D)$ with $D = (\Sigma, Q, \delta, q_0, F)$, and put $n = |Q|$. Let $w \in L$ with $|w| \geq n$, say $w = a_0a_1 \cdots a_{n-1}v$.

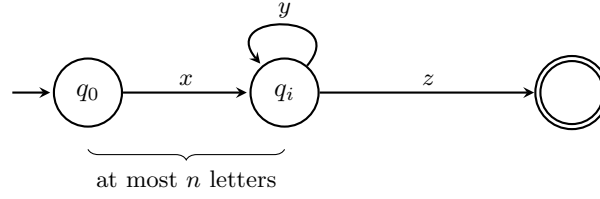
Let $q_0, q_1, \dots, q_n$ be the states visited while reading $a_0 \cdots a_{n-1}$. This is a list of $n+1$ states drawn from a set of size n , so by pigeonhole there are $i < j \leq n$ with $q_i = q_j$. Put

$$x = a_0 \cdots a_{i-1}, \quad y = a_i \cdots a_{j-1}, \quad z = a_j \cdots a_{n-1}v,$$

so that $xyz = w$, $|y| = j - i > 0$ and $|xy| = j \leq n$.

Now $\hat{\delta}(q_0, x) = q_i$ and $\hat{\delta}(q_i, y) = q_j = q_i$, so by induction on k we get $\hat{\delta}(q_0, xy^k) = q_i$ for every $k \in \mathbb{N}$. Finally $\hat{\delta}(q_i, z) = \hat{\delta}(q_j, z) = \hat{\delta}(q_0, w) \in F$, so $\hat{\delta}(q_0, xy^kz) \in F$ and $xy^kz \in L$. $\square$

The picture behind the proof is worth drawing: the accepting path through the machine, of length at least n , must contain a loop within its first n steps, and the loop can be traversed any number of times.

**Example 27.3.21**

The language $L = \{0^k 1^k \mid k > 0\}$ is not regular.

Suppose it were, with pumping number N . The word $w = 0^N 1^N$ lies in L and has length $2N \geq N$, so it can be written $w = xyz$ with $|xy| \leq N$ and $|y| = \ell > 0$. Since $|xy| \leq N$, the pump y lies within the first N letters of w , all of which are 0s; so $y = 0^\ell$. Pumping down with $k = 0$ gives $xz = 0^{N-\ell} 1^N \in L$, which is false since $\ell > 0$. So L is not regular.

We will see in Section 27.4 that $L = \{0^k 1^k\}$ is context-free, so this example already shows that the inclusion of type 3 in type 2 is strict.

Example 27.3.22

Fix $n \geq 2$ and let $L = \{0^n w \mid w \in \mathbb{W}\}$, the words beginning with n zeroes. This is regular: take $V = \{S, X_0, \dots, X_{n-1}\}$ and

$$P = \{S \rightarrow 0X_0\} \cup \{X_i \rightarrow 0X_{i+1} \mid i \leq n-3\} \cup \{X_{n-2} \rightarrow 0X_{n-1}, X_{n-2} \rightarrow 0\} \\ \cup \{X_{n-1} \rightarrow 0, X_{n-1} \rightarrow 1, X_{n-1} \rightarrow 0X_{n-1}, X_{n-1} \rightarrow 1X_{n-1}\}.$$

But no deterministic automaton with at most n states accepts L . Indeed, such an automaton would give L pumping number n by Theorem 27.3.20, and then the word $0^n \in L$ of length n could be pumped down to a word with fewer than n zeroes, which is not in L .

It is essential to remember that the pumping lemma is a necessary but *not* a sufficient condition for regularity. The next example makes this vivid.

Example 27.3.23

Let $\Sigma = \{0, 1\}$. For a word w containing at least one 0, define the **tail** of w to be the number of 1s following the last 0. Given $X \subseteq \mathbb{N}$, let

$$L_X = \{w \in \mathbb{W}^+ \mid w \text{ contains no } 0\} \cup \{w \mid w \text{ has a tail lying in } X\}.$$

If $X \neq Y$ then $L_X \neq L_Y$: taking m in the symmetric difference, the word 01^m distinguishes them. So $X \mapsto L_X$ is injective, and since $\mathcal{P}(\mathbb{N})$ is uncountable while there are only countably many regular languages, some L_X is not regular.

Nevertheless *every* L_X satisfies the pumping lemma with pumping number 2. Let $w \in L_X$ with $|w| \geq 2$.

- If $w = 0z$, take $x = \varepsilon$, $y = 0$. Pumping up prepends more 0s, which changes neither the presence of a 0 nor the tail. Pumping down gives z : either z still contains a 0, in which case the tail is unchanged, or it does not, in which case

$z \in 1^*$ and again lies in L_X (note $z \neq \varepsilon$ as $|w| \geq 2$).

- If $w = 1z$, take $x = \varepsilon$, $y = 1$. If z contains no 0 then neither does $1^k z$, so all pumps lie in L_X . If z contains a 0 then the tail of $1^k z$ equals the tail of z for every k , so again all pumps lie in L_X .

Hence there are uncountably many languages satisfying the regular pumping lemma, and in particular the pumping lemma does not characterise the regular languages.

§27.3.6 Closure Properties

We have already proved closure under union (Theorem 27.2.20 together with the automaton translation, or directly) and concatenation (Lemma 27.3.3). Automata make the Boolean operations easy.

Proposition 27.3.24

The regular languages are closed under complement (relative to $\mathbb{W}^+$), intersection and difference.

Proof. Let $L = \mathcal{L}(D)$ with $D = (\Sigma, Q, \delta, q_0, F)$. We may assume that no transition enters the start state, that is, that $\delta(q, a) \neq q_0$ for all q and a . Indeed, if this fails, adjoin a fresh state q'_0 , put $\delta(q'_0, a) = \delta(q_0, a)$ for every letter a , and take q'_0 as the new start state, demoting q_0 to an ordinary state (which is not accepting, since $q_0 \notin F$). No transition enters q'_0 , and $\hat{\delta}(q'_0, w) = \hat{\delta}(q_0, w)$ for every nonempty w , so the language is unchanged. We rename q'_0 back to q_0 .

Now set $D' = (\Sigma, Q, \delta, q_0, Q \setminus (F \cup \{q_0\}))$. For a nonempty word w we have $\hat{\delta}(q_0, w) \neq q_0$, so $\hat{\delta}(q_0, w) \in Q \setminus (F \cup \{q_0\})$ if and only if $\hat{\delta}(q_0, w) \notin F$. Hence $\mathcal{L}(D') = \mathbb{W}^+ \setminus L$.

Intersection and difference now follow from De Morgan, since $L \cap M = \mathbb{W}^+ \setminus ((\mathbb{W}^+ \setminus L) \cup (\mathbb{W}^+ \setminus M))$ and $L \setminus M = L \cap (\mathbb{W}^+ \setminus M)$. $\square$

There is a more direct construction for intersection and union which is worth knowing, because it is the one that fails for the higher classes.

Definition 27.3.25 (Product automaton)

Given $D = (\Sigma, Q, \delta, q_0, F)$ and $D' = (\Sigma, Q', \delta', q'_0, F')$, the **product automaton** is

$$D \times D' = (\Sigma, Q \times Q', \delta \times \delta', (q_0, q'_0), F''), \quad (\delta \times \delta')((q, q'), a) = (\delta(q, a), \delta'(q', a)),$$

where F'' is either $\{(q, q') \mid q \in F \text{ and } q' \in F'\}$ or $\{(q, q') \mid q \in F \text{ or } q' \in F'\}$.

An immediate induction gives $(\widehat{\delta \times \delta'})((q_0, q'_0), w) = (\hat{\delta}(q_0, w), \hat{\delta}'(q'_0, w))$, so the first choice of F'' yields $\mathcal{L}(D) \cap \mathcal{L}(D')$ and the second yields $\mathcal{L}(D) \cup \mathcal{L}(D')$. In words: the product machine runs both automata side by side on the same input, which it can afford to do because each has only finitely many states. We will see in Section 27.4 that no such construction can exist for context-free languages.

§27.3.7 Regular Expressions and Kleene's Theorem

There is a fourth description of the regular languages, and it is the one familiar from programming.

Definition 27.3.26 (Kleene star)

For a language L , the **Kleene star** of L is

$$L^* = \{w_1 w_2 \cdots w_k \mid k \in \mathbb{N}, w_i \in L\},$$

the set of all concatenations of finitely many words of L ; in particular $\varepsilon \in L^*$, taking $k = 0$. The **Kleene plus** is $L^+ = \{w_1 \cdots w_k \mid k \geq 1, w_i \in L\}$, so that $L^* = L^+ \cup \{\varepsilon\}$.

Definition 27.3.27 (Regular expression)

The **regular expressions** over Σ are defined inductively:

- (i) $\emptyset$ is a regular expression;
- (ii) ε is a regular expression;
- (iii) a is a regular expression for each $a \in \Sigma$;
- (iv) if R, S are regular expressions then so is $(R + S)$;
- (v) if R, S are regular expressions then so is (RS) ;
- (vi) if R is a regular expression then so are R^* and R^+ ,

and nothing else is. To each regular expression E we assign a language $\mathcal{L}(E)$ by the matching recursion:

$$\begin{aligned} \mathcal{L}(\emptyset) &= \emptyset, & \mathcal{L}(\varepsilon) &= \{\varepsilon\}, & \mathcal{L}(a) &= \{a\}, \\ \mathcal{L}(R + S) &= \mathcal{L}(R) \cup \mathcal{L}(S), & \mathcal{L}(RS) &= \mathcal{L}(R)\mathcal{L}(S), \\ \mathcal{L}(R^*) &= \mathcal{L}(R)^*, & \mathcal{L}(R^+) &= \mathcal{L}(R)^+. \end{aligned}$$

We drop parentheses when no ambiguity results, giving concatenation a higher binding power than $+$, so that $RS + T$ means $((RS) + T)$.

The only irritation is the empty word, which regular expressions can produce and regular grammars cannot. We deal with it by a definition.

Definition 27.3.28

A language L is **essentially regular** if L or $L \setminus \{\varepsilon\}$ is regular; equivalently, if there is a regular L' with $L = L'$ or $L = L' \cup \{\varepsilon\}$.

Theorem 27.3.29

If E is a regular expression then $\mathcal{L}(E)$ is essentially regular.

Proof. We induct on the structure of E . The base cases are clear: $\emptyset$ and $\{a\}$ are regular (take $P = \emptyset$ and $P = \{S \rightarrow a\}$ respectively) and $\{\varepsilon\} = \emptyset \cup \{\varepsilon\}$ is essentially regular.

For the induction step it is enough to show that essentially regular languages are closed under union, concatenation and Kleene plus, since $\mathcal{L}(E^*) = \mathcal{L}(E + E^+)$.

Union and concatenation of regular languages are regular by Theorem 27.2.20 and Lemma 27.3.3, and the extension to essentially regular languages is a short case analysis: for instance $(L \cup \{\varepsilon\})(M \cup \{\varepsilon\}) = LM \cup L \cup M \cup \{\varepsilon\}$, and $LM \cup L \cup M$ is regular.

So let $G = (\Sigma, V, P, S)$ be regular; we must show $\mathcal{L}(G)^+$ is regular. We may assume G is ε -adequate, so S never occurs on the right of a rule. Let

$$P^+ = P \cup \{A \rightarrow aS \mid A \rightarrow a \in P\}, \quad G^+ = (\Sigma, V, P^+, S).$$

This is regular; we claim $\mathcal{L}(G^+) = \mathcal{L}(G)^+$.

($\supseteq$) Let $w = w_0w_1 \cdots w_n$ with each $w_i \in \mathcal{L}(G)$. Induct on n . For $n = 0$ any G -derivation is a G^+ -derivation. For $n > 0$, by induction $S \Rightarrow_{G^+}^* w_0 \cdots w_{n-1}$; this derivation ends with a terminal rule $A \rightarrow a$, and replacing that final step by $A \rightarrow aS$ gives $S \Rightarrow_{G^+}^* w_0 \cdots w_{n-1}S$. Appending a G -derivation of w_n gives w .

($\subseteq$) Let $S \Rightarrow_{G^+}^* w$ and let $n \geq 1$ be the number of strings in the derivation in which S occurs (the first one always does). Induct on n . If $n = 1$ then S occurs only at the start, so no added rule was used and $S \Rightarrow_G^* w$, giving $w \in \mathcal{L}(G) \subseteq \mathcal{L}(G)^+$. If $n > 1$, write the derivation as $S \Rightarrow_{G^+}^* vS \Rightarrow_{G^+}^* w$ where vS is the *last* string containing S . The step producing vS used an added rule $A \rightarrow aS$; replacing it by $A \rightarrow a$ gives a derivation $S \Rightarrow_{G^+}^* v$ with $n - 1$ occurrences of S , so $v \in \mathcal{L}(G)^+$ by induction. From vS onward S never reappears, so no added rule fires, and $w = vw'$ with $S \Rightarrow_G^* w'$, i.e. $w' \in \mathcal{L}(G)$. Hence $w \in \mathcal{L}(G)^+$. $\square$

The converse is Kleene's theorem proper, and it is usually proved by an algorithm which eliminates the states of an automaton one at a time. We give the closely related dynamic-programming version.

Theorem 27.3.30 (Kleene)

Let D be a deterministic automaton. Then there is a regular expression E with $\mathcal{L}(E) = \mathcal{L}(D)$. Consequently a language is essentially regular if and only if it is $\mathcal{L}(E)$ for some regular expression E .

Proof. Write $Q = \{q_0, q_1, \dots, q_{m-1}\}$. For $i, j < m$ and $0 \leq k \leq m$, let

$$R_{i,j}^k = \{w \in \mathbb{W} \mid \hat{\delta}(q_i, w) = q_j, \text{ all states strictly inside}$$

$$\text{the run lying in } \{q_0, \dots, q_{k-1}\}\},$$

where 'strictly inside' means excluding the first and last states of the run. We show by induction on k that each $R_{i,j}^k$ is $\mathcal{L}(E)$ for a regular expression $E_{i,j}^k$.

For $k = 0$ no intermediate states are allowed, so $R_{i,j}^0$ consists of ε (if $i = j$) together

with the letters a with $\delta(q_i, a) = q_j$. This is a finite set of words of length at most 1, hence is $\mathcal{L}(E)$ for a finite sum of the base expressions.

For the induction step, a run from q_i to q_j through $\{q_0, \dots, q_k\}$ either avoids q_k altogether, or visits q_k ; in the latter case, splitting at the first and last visits to q_k decomposes the word as a run $q_i \rightarrow q_k$ avoiding q_k internally, then finitely many runs $q_k \rightarrow q_k$ avoiding q_k internally, then a run $q_k \rightarrow q_j$ avoiding q_k internally. Hence

$$R_{i,j}^{k+1} = R_{i,j}^k \cup R_{i,k}^k (R_{k,k}^k)^* R_{k,j}^k,$$

and we may take $E_{i,j}^{k+1} = E_{i,j}^k + E_{i,k}^k (E_{k,k}^k)^* E_{k,j}^k$.

Finally $\mathcal{L}(D) = \bigcup_{q_j \in F} R_{0,j}^m$, so we take $E = \sum_{q_j \in F} E_{0,j}^m$. Since $q_0 \notin F$, the word ε does not lie in this language, so the resulting expression describes $\mathcal{L}(D)$ exactly.

For the last sentence, combine this with Theorem 27.3.29: if L is essentially regular then L is $\mathcal{L}(E)$ or $\mathcal{L}(E + \varepsilon)$ for a suitable E . $\square$

Example 27.3.31

The automaton of Example 27.3.5 accepts the words containing at least one 0, and indeed

$$\mathcal{L}(D) = \mathcal{L}(1^*0(0+1)^*).$$

The automaton of Example 27.3.16 accepts $\mathcal{L}((a+b)^*ab)$.

§27.3.8 Minimisation

We saw in Example 27.3.8 an automaton with two states doing the same job. We now make this precise and show that every regular language has a canonical smallest automaton.

Definition 27.3.32

A state q is **accessible** if $q = \hat{\delta}(q_0, w)$ for some $w \in \mathbb{W}$, and **inaccessible** otherwise.

States q, q' are **distinguished by w** if exactly one of $\hat{\delta}(q, w), \hat{\delta}(q', w)$ lies in F . They are **distinguishable** if some word distinguishes them, and **indistinguishable** otherwise; we write $q \sim q'$ in the latter case.

D is **irreducible** if every state is accessible and every pair of distinct states is distinguishable.

Suppose $f: D \rightarrow D'$ is a homomorphism. If p, q are distinguishable by w then, since f commutes with $\hat{\delta}$ and preserves acceptance, $f(p)$ and $f(q)$ are distinguished by w , so $f(p) \neq f(q)$. And if $q' \in Q'$ is accessible, say $q' = \hat{\delta}'(q'_0, w)$, then $q' = f(\hat{\delta}(q_0, w))$ lies in the image of f . Hence:

Proposition 27.3.33

Any homomorphism between irreducible automata is an isomorphism.

Clearly $\sim$ is an equivalence relation on Q ; write $[q]$ for the class of q .

Definition 27.3.34 (Quotient automaton)

Define

$$D/\sim = (\Sigma, Q/\sim, [\delta], [q_0], [F]), \quad [\delta]([q], a) = [\delta(q, a)], \quad [F] = \{[q] \mid q \in F\}.$$

Two checks are needed. First, being accepting is a class property: if $q \sim q'$ then ε does not distinguish them, so $q \in F \iff q' \in F$; in particular $[F]$ really is a set of classes, and $[q_0] \notin [F]$ since $q_0 \notin F$. Second, $[\delta]$ is well defined: if $q \sim q'$ but $\delta(q, a) \not\sim \delta(q', a)$, witnessed by a word w , then aw would distinguish q from q' .

Lemma 27.3.35

The quotient map $q \mapsto [q]$ is a surjective homomorphism, so $\mathcal{L}(D) = \mathcal{L}(D/\sim)$. Moreover distinct states of $D/\sim$ are distinguishable, and if every state of D is accessible then so is every state of $D/\sim$.

Proof. That the quotient map is a homomorphism is exactly the two checks above; it is surjective by construction, so accessibility is inherited and Proposition 27.3.10 gives the languages agree.

An easy induction gives $[\hat{\delta}][q], w = [\hat{\delta}(q, w)]$. Now suppose $[q] \neq [q']$, so $q \not\sim q'$; take w distinguishing them, say $\hat{\delta}(q, w) \in F$ and $\hat{\delta}(q', w) \notin F$. Then $[\hat{\delta}][q], w = [\hat{\delta}(q, w)] \in [F]$ while $[\hat{\delta}][q'], w \notin [F]$, so w distinguishes $[q]$ from $[q']$. $\square$

Theorem 27.3.36

For every deterministic automaton D there is an irreducible deterministic automaton I with $\mathcal{L}(D) = \mathcal{L}(I)$ and at most as many states.

Proof. Let $A \subseteq Q$ be the set of accessible states; note $q_0 \in A$ and δ restricts to A . Put $D^* = (\Sigma, A, \delta \upharpoonright_{A \times \Sigma}, q_0, F \cap A)$. The inclusion $A \hookrightarrow Q$ is a homomorphism, so $\mathcal{L}(D^*) = \mathcal{L}(D)$, and every state of D^* is accessible. Now take $I = D^*/\sim$ and apply Lemma 27.3.35. $\square$

Theorem 27.3.37

If I and I' are irreducible with $\mathcal{L}(I) = \mathcal{L}(I')$, then $I \cong I'$.

Proof. By Proposition 27.3.33 it suffices to construct a homomorphism $I \rightarrow I'$. Write $I = (\Sigma, Q, \delta, q_0, F)$ and $I' = (\Sigma, Q', \delta', q'_0, F')$ with $Q \cap Q' = \emptyset$. Extend $\sim$ to $Q \cup Q'$ by declaring $q \sim q'$ (for $q \in Q, q' \in Q'$) when for all w , $\hat{\delta}(q, w) \in F \iff \hat{\delta}'(q', w) \in F'$; this is again an equivalence relation. Since the languages agree, $q_0 \sim q'_0$.

Every $q \in Q$ is $\sim$ -equivalent to some $q' \in Q'$. Let $\text{sp}(q)$ be the length of a shortest word w with $\hat{\delta}(q_0, w) = q$, which exists by accessibility. Induct on $\text{sp}(q)$. If $\text{sp}(q) = 0$ then $q = q_0 \sim q'_0$. If $\text{sp}(q) = k + 1$, choose p and a with $\delta(p, a) = q$ and $\text{sp}(p) = k$; by induction pick $p' \sim p$ and set $q' = \delta'(p', a)$. If w is any word then $\hat{\delta}(q, w) = \hat{\delta}(p, aw)$

and $\hat{\delta}'(q', w) = \hat{\delta}'(p', aw)$, so $p \sim p'$ gives $q \sim q'$.

The witness is unique. If $q' \sim q \sim p'$ with $q', p' \in Q'$ then $q' \sim p'$ by transitivity, and irreducibility of I' forces $q' = p'$.

So there is a well-defined $f: Q \rightarrow Q'$ with $q \sim f(q)$. Then $f(q_0) = q'_0$; taking $w = \varepsilon$ in the definition of $\sim$ gives $q \in F \iff f(q) \in F'$; and for any a , the computation above shows $\delta(q, a) \sim \delta'(f(q), a)$, so $f(\delta(q, a)) = \delta'(f(q), a)$ by uniqueness. Thus f is a homomorphism. $\square$

Corollary 27.3.38

Every regular language is accepted by an irreducible automaton, unique up to isomorphism, and this automaton has fewer states than any other automaton accepting the language.

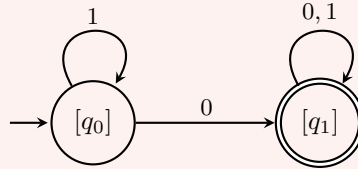
Proof. Existence and uniqueness are Theorems 27.3.36 and 27.3.37. If D is any automaton for L , Theorem 27.3.36 produces an irreducible I with at most $|Q|$ states, and I is isomorphic to the canonical one. $\square$

Example 27.3.39

Return to Example 27.3.5. Is that automaton irreducible? All three states are accessible. The state q_1 is distinguished from the others by ε . But for any word w ,

$$\hat{\delta}(q_0, w) \in F \iff w \text{ contains a } 0 \iff \hat{\delta}(q_2, w) \in F,$$

so $q_0 \sim q_2$ and the automaton is not irreducible. Quotienting merges them, and we obtain the two-state machine



which is irreducible. Notice that this still satisfies our convention that the start state is not accepting, as it must, since ε contains no 0. Applying the same analysis to Example 27.3.8 merges q_0 with E and produces the expected two-state parity machine.

§27.3.9 The Myhill–Nerode Theorem

Where do the states of the minimal automaton come from? The answer is that they are determined by the language alone, with no reference to any machine.

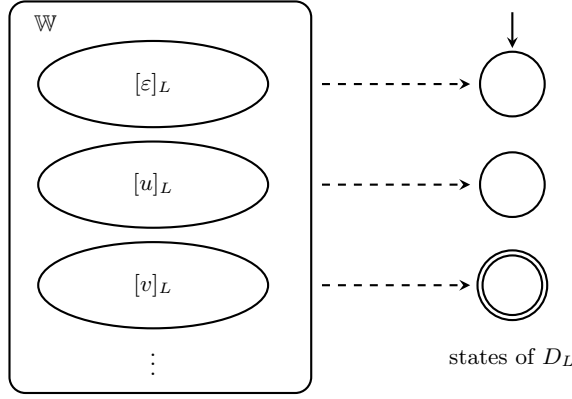
Definition 27.3.40 (Nerode equivalence)

Let $L \subseteq \mathbb{W}$ be any language. For $u, v \in \mathbb{W}$ write

$$u \sim_L v \quad \text{if and only if} \quad \text{for all } w \in \mathbb{W}, \quad uw \in L \iff vw \in L.$$

This is an equivalence relation on $\mathbb{W}$; write $[u]_L$ for the class of u and $\mathbb{W}/\sim_L$ for the set of classes. It is a **right congruence**: if $u \sim_L v$ then $ua \sim_L va$ for every letter a .

The relation $\sim_L$ says that u and v have the same *future*: no continuation can tell them apart. If a machine is reading L and has seen u , the only thing it can usefully remember is $[u]_L$.



Theorem 27.3.41 (Myhill–Nerode)

Let $L \subseteq \mathbb{W}$ with $\varepsilon \notin L$. Then L is regular if and only if $\sim_L$ has finitely many equivalence classes. In that case the number of classes is the number of states of the minimal automaton for L .

Proof. ($\Rightarrow$) Let $L = \mathcal{L}(D)$ for a deterministic D with state set Q . If $\hat{\delta}(q_0, u) = \hat{\delta}(q_0, v)$ then for every w

$$\begin{aligned} uw \in L &\iff \hat{\delta}(q_0, uw) \in F \iff \hat{\delta}(\hat{\delta}(q_0, u), w) \in F \\ &\iff \hat{\delta}(\hat{\delta}(q_0, v), w) \in F \iff vw \in L, \end{aligned}$$

so $u \sim_L v$. Hence $u \mapsto \hat{\delta}(q_0, u)$ induces a well-defined surjection from a subset of Q onto $\mathbb{W}/\sim_L$, and there are at most $|Q|$ classes.

($\Leftarrow$) Suppose there are finitely many classes. Define

$$D_L = (\Sigma, \mathbb{W}/\sim_L, \delta_L, [\varepsilon]_L, F_L), \quad \delta_L([u]_L, a) = [ua]_L, \quad F_L = \{[u]_L \mid u \in L\}.$$

The transition function is well defined because $\sim_L$ is a right congruence, and F_L is a union of classes because $u \sim_L v$ with $w = \varepsilon$ gives $u \in L \iff v \in L$. Since $\varepsilon \notin L$ we have $[\varepsilon]_L \notin F_L$, so this is a legal automaton. An induction on $|w|$ gives $\hat{\delta}_L([\varepsilon]_L, w) = [w]_L$, whence

$$w \in \mathcal{L}(D_L) \iff [w]_L \in F_L \iff w \in L.$$

For the last claim: D_L is irreducible. Every state is accessible, since $[w]_L = \hat{\delta}_L([\varepsilon]_L, w)$. And if $[u]_L \neq [v]_L$ then some w has (say) $uw \in L$ and $vw \notin L$, and this w distinguishes the two states. So D_L is the minimal automaton, by Theorem 27.3.37. $\square$

Remark. The hypothesis $\varepsilon \notin L$ is forced on us only by the convention $F \subseteq Q \setminus \{q_0\}$;

without it the theorem holds verbatim for the usual definition of automaton in which the start state may accept.

Myhill–Nerode gives a second, often more convenient, method of proving non-regularity: exhibit infinitely many pairwise inequivalent words.

Example 27.3.42

Once more let $L = \{0^k 1^k \mid k > 0\}$. For $i \neq j$ the word 1^i distinguishes 0^i from 0^j , since $0^i 1^i \in L$ but $0^j 1^i \notin L$. So the words $0^0, 0^1, 0^2, \dots$ lie in pairwise distinct classes, $\sim_L$ has infinitely many classes, and L is not regular.

Example 27.3.43

Let L be the set of nonempty words over $\{0, 1\}$ containing at least one 0. If u contains a 0 then $uw \in L$ for every w ; if u does not, then $uw \in L$ if and only if w contains a 0. So there are exactly two classes, and the minimal automaton has two states – exactly as we found by minimising.

Notice how much less work the second example was than the direct minimisation. This is typical: Myhill–Nerode lets us compute the minimal automaton by thinking about the language rather than by manipulating a machine.

§27.3.10 Decision Problems for Regular Languages

We can now settle all three of our decision problems for regular grammars. The word problem was dealt with in Corollary 27.2.16 (or, more sensibly, by running the automaton). The other two require a little work, and both are consequences of the pumping lemma and the finiteness of the machinery.

Lemma 27.3.44

Let L be a nonempty regular language with pumping number n . Then L contains a word of length less than n .

Proof. Take $w \in L$ of minimal length. If $|w| \geq n$ then $w = xyz$ with $|y| > 0$ and $xz = xy^0z \in L$, which is shorter, contradicting minimality. $\square$

Corollary 27.3.45

The emptiness problem for regular grammars is solvable.

Proof. Given a regular grammar G , construct a deterministic automaton for $\mathcal{L}(G)$ and let n be its number of states, a pumping number by Theorem 27.3.20. By Lemma 27.3.44, $\mathcal{L}(G) = \emptyset$ if and only if $\mathcal{L}(G)$ contains no word of length less than n . There are finitely many such words, and the word problem is solvable, so we can check them all. $\square$

Corollary 27.3.46

The finiteness problem for regular grammars – given G , is $\mathcal{L}(G)$ finite? – is solvable.

Proof. With n as above, we claim $\mathcal{L}(G)$ is infinite if and only if it contains a word w with $n \leq |w| < 2n$. If such a w exists then it can be pumped, and $\{xy^kz \mid k \in \mathbb{N}\}$ is an infinite subset of $\mathcal{L}(G)$. Conversely, if $\mathcal{L}(G)$ is infinite, take $w \in \mathcal{L}(G)$ of minimal length at least n ; writing $w = xyz$ as in the pumping lemma we have $|y| \leq |xy| \leq n$, so $xz \in \mathcal{L}(G)$ has length $|w| - |y| \geq |w| - n$, and by minimality $|w| - n < n$. Again there are finitely many words to check. $\square$

For the equivalence problem the plan is to build the minimal automaton for each language and check whether they are isomorphic, which by Theorem 27.3.37 settles the question. We must check that both steps can actually be carried out in finitely many steps.

Proposition 27.3.47

Given a deterministic automaton D and a state q , it is solvable whether q is accessible.

Proof. If $\hat{\delta}(q_0, w) = q$ for some w , then taking w of minimal length, the states visited must all be distinct (else we could excise a loop), so $|w| < |Q|$. There are finitely many such words, and we can compute $\hat{\delta}$ on each. $\square$

Theorem 27.3.48 (The table-filling algorithm)

Given a deterministic automaton D and states q, q' , it is solvable whether $q \sim q'$.

Proof. Form a table A indexed by unordered pairs of distinct states. Each entry is either blank or holds a word. The intended invariant is that $A_{q,q'}$ holds a word distinguishing q and q' .

Initialise. If exactly one of q, q' lies in F , set $A_{q,q'} = \varepsilon$.

Iterate. Repeatedly do the following sweep: for each blank entry $A_{q,q'}$ and each $a \in \Sigma$, if $\delta(q, a) \neq \delta(q', a)$ and the entry $A_{\delta(q,a), \delta(q',a)}$ holds a word w , then set $A_{q,q'} = aw$. Each sweep takes at most $|Q|^2|\Sigma|$ operations, and fills at least one blank entry or else nothing changes; since there are at most $|Q|^2$ entries, after at most $|Q|^2$ sweeps the table is stable.

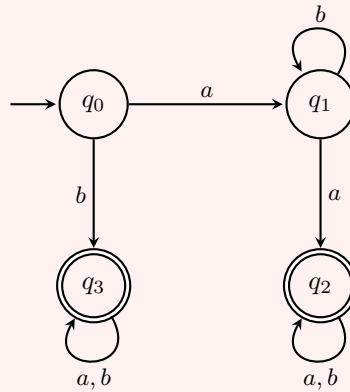
Correctness. If $A_{q,q'} = w$ then w genuinely distinguishes q and q' : this is true at initialisation, and is preserved by the iteration, since $\hat{\delta}(q, aw) = \hat{\delta}(\delta(q, a), w)$.

Conversely suppose some distinguishable pair is left blank; among all such pairs choose q, q' with a distinguishing word w of minimal length. If $w = \varepsilon$ then the pair was filled at initialisation, a contradiction. So $w = av$. Then v distinguishes $\delta(q, a)$ from $\delta(q', a)$, and v is shorter than w , so by minimality that pair is not blank in the final table. But then the sweep would have filled $A_{q,q'}$, contradicting stability.

So on termination, $A_{q,q'}$ is blank precisely when $q \sim q'$. $\square$

Example 27.3.49

Consider the automaton



After initialisation, the pairs involving exactly one of the accept states q_2, q_3 are marked with ε :

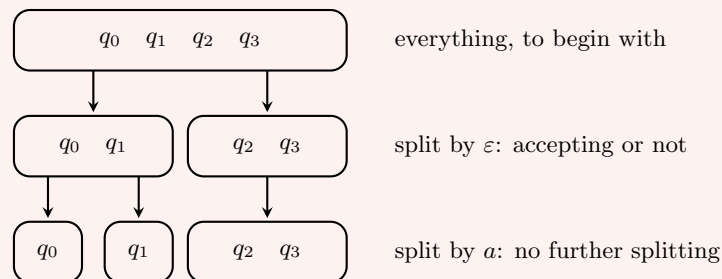
	q_0	q_1	q_2	q_3
q_0			ε	ε
q_1			ε	ε
q_2				
q_3				

In the first sweep, $\delta(q_0, a) = q_1$ and $\delta(q_1, a) = q_2$, and the entry A_{q_1, q_2} is marked, so we mark A_{q_0, q_1} with a :

	q_0	q_1	q_2	q_3
q_0		a	ε	ε
q_1			ε	ε
q_2				
q_3				

The only remaining blank is A_{q_2, q_3} , and it stays blank: both q_2 and q_3 are absorbing accept states, so $\delta(q_2, a) = q_2$, $\delta(q_3, a) = q_3$ and the entry A_{q_2, q_3} can only ever be filled from itself. So $q_2 \sim q_3$, and the minimal automaton has three states $[q_0], [q_1], [q_2]$.

It is illuminating to read the same computation backwards, as a **refinement of partitions**. We begin with the partition of Q into one block, and each time an entry is filled we split the block containing that pair:



The process must stop, since each round strictly refines the partition and Q is finite; when it stops, the blocks are exactly the $\sim$ -classes. (Directly: $\mathcal{L}(D) = \mathcal{L}(b(a+b)^* + ab^*a(a+b)^*)$, and one can check by hand that the three classes are ‘nothing decided yet’, ‘an a has been seen’, and ‘accepted forever’.)

Corollary 27.3.50

The equivalence problem for regular grammars is solvable.

Proof. Given regular grammars G, G' , construct deterministic automata for them; delete inaccessible states, which we can identify; and quotient by $\sim$, which we can compute by Theorem 27.3.48. This yields irreducible automata I, I' . Now $\mathcal{L}(G) = \mathcal{L}(G')$ if and only if $I \cong I'$ by Theorems 27.3.36 and 27.3.37, and since I and I' are finite there are only finitely many candidate bijections to test. $\square$

For regular languages, then, everything is as good as it could be: all the closure properties hold, all three decision problems are solvable, and there is a canonical smallest machine. As we ascend the hierarchy, each of these will fail in turn.

§27.4 Context-Free Languages

We move one step up. A context-free rule replaces a single variable A by an arbitrary nonempty string, with no restriction on the shape of the right-hand side. We already know from Example 27.3.21 that this buys us something, since $\{0^k 1^k \mid k > 0\}$ is generated by the context-free grammar

$$S \rightarrow 0S1, \quad S \rightarrow 01$$

but is not regular. The essential new feature is that a context-free derivation is no longer linear. In a regular grammar every intermediate string had the form wA , so a derivation was a list; here several variables can be present at once, and the natural object recording a derivation is a tree.

§27.4.1 Trees

We set up trees in the way that is most convenient for the arguments below, namely as sets of finite sequences of naturals: a node is identified with the route taken to reach it from the root.

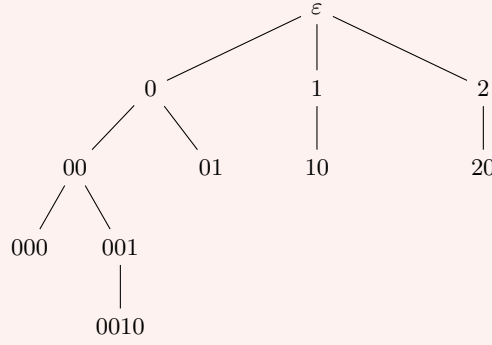
Definition 27.4.1 (Tree)

A set $T \subseteq \mathbb{N}^*$ is a **(finitely branching) tree** if it is closed under initial segments, and for every $t \in T$ there is a **branching number** $n \in \mathbb{N}$ such that $tk \in T$ if and only if $k < n$.

The empty sequence is the **root**, and lies in every nonempty tree. A node with branching number 0 is a **leaf**. The **level** of t is $|t|$, and if T is finite the maximum level is the **height** of T . The **branch** to t is the sequence $t \restriction_0, t \restriction_1, \dots, t \restriction_{|t|} = t$ of its initial segments.

Example 27.4.2

The set $T = \{\varepsilon, 0, 1, 2, 00, 01, 10, 20, 000, 001, 0010\}$ is a tree, drawn below with the root at the top.



Its height is 4; its leaves are 01, 10, 20, 000 and 0010; and the branch to 0010 is ε , 0, 00, 001, 0010.

Definition 27.4.3 (Subtree)

For $t \in T$, the **subtree at t** is $T_t = \{s \mid ts \in T\}$, which is again a tree.

Definition 27.4.4 (Left-to-right order)

Define $t < s$ for $t \neq s$ in T if, letting k_0 be least with $t(k_0) \neq s(k_0)$ (assuming such k_0 exists), we have $t(k_0) < s(k_0)$.

This is a partial order: it does not compare two nodes lying on the same branch, such as 0 and 00. It does, however, totally order the nodes at any fixed level, and in particular it totally orders the leaves. Reading the leaves in this order is what turns a tree into a string.

§27.4.2 Parse Trees

Definition 27.4.5 (Parse tree)

Let $G = (\Sigma, V, P, S)$ be a context-free grammar. A pair $\mathbb{T} = (T, \ell)$ is a **G -parse tree** if T is a finite tree and $\ell: T \rightarrow \Omega$ is a **labelling** such that

- (i) $\ell(\varepsilon) \in V$, and we say that $\mathbb{T}$ **starts with** $\ell(\varepsilon)$;
- (ii) if $\ell(t) \in \Sigma$ then t is a leaf;
- (iii) if $\ell(t) = A \in V$ and t has branching number $n+1$, then $A \rightarrow \ell(t_0)\ell(t_1) \cdots \ell(t_n)$ is a rule of P .

If $t_0 < t_1 < \cdots < t_m$ are the leaves of T in left-to-right order, the **string parsed by $\mathbb{T}$** is

$$\sigma_{\mathbb{T}} = \ell(t_0)\ell(t_1) \cdots \ell(t_m).$$

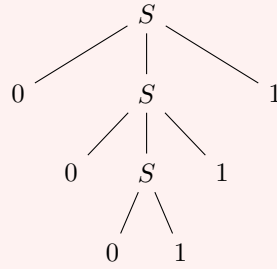
Note that a variable labelling a node of branching number 0 is permitted; such a leaf contributes a variable to $\sigma_{\mathbb{T}}$, and corresponds to a derivation that has not been finished. If $t \in T$, splitting the leaves into those to the left of t , those below t and those to the right of t gives

$$\sigma_{\mathbb{T}} = \alpha \sigma_{\mathbb{T}_t} \beta$$

for some strings α, β , where $\mathbb{T}_t = (T_t, \ell_t)$ with $\ell_t(s) = \ell(ts)$.

Example 27.4.6

Let G be the grammar $S \rightarrow 0S1 \mid 01$ over $\Sigma = \{0, 1\}$. The parse tree for 000111 is the following.



Its leaves in left-to-right order are the six letters 0, 0, 0, 1, 1, 1, so $\sigma_{\mathbb{T}} = 000111$ as promised. The corresponding derivation is $S \Rightarrow 0S1 \Rightarrow 00S11 \Rightarrow 000111$.

Proposition 27.4.7

Let G be context-free and $w \in \mathbb{W}$. Then $w \in \mathcal{L}(G)$ if and only if there is a G -parse tree $\mathbb{T}$ starting with S and with $\sigma_{\mathbb{T}} = w$.

Proof. Call a sequence $\mathbb{T}_0, \dots, \mathbb{T}_n$ of G -parse trees **derivative** if $\mathbb{T}_0 = (\{\varepsilon\}, \varepsilon \mapsto S)$, and each $\mathbb{T}_{i+1}$ is obtained from $\mathbb{T}_i$ by choosing a leaf t with $\ell_i(t) = A \in V$ and a rule $A \rightarrow x_0 \cdots x_n \in P$, and adjoining successors $t0, \dots, tn$ with $\ell_{i+1}(tk) = x_k$.

Derivative sequences correspond exactly to derivations from S : adjoining the successors of t replaces the occurrence of A contributed by t in $\sigma_{\mathbb{T}_i}$ by $x_0 \cdots x_n$, and leaves the rest of the string alone, which is precisely one application of $A \rightarrow x_0 \cdots x_n$. Hence $\sigma_{\mathbb{T}_0} \Rightarrow_G \sigma_{\mathbb{T}_1} \Rightarrow_G \cdots \Rightarrow_G \sigma_{\mathbb{T}_n}$, and conversely each derivation determines such a sequence by recording where each rule was applied.

So if there is a derivative sequence ending at $\mathbb{T}$, then $S \Rightarrow_G^* \sigma_{\mathbb{T}}$. It remains to show every parse tree $\mathbb{T}$ starting with S arises this way. Set $\mathbb{T}_0$ to be the one-node tree. Given $\mathbb{T}_i$ with $T_i \subsetneq T$, pick $t \in T \setminus T_i$ of minimal length; then its predecessor $t \upharpoonright_{|t|-1}$ is a leaf of T_i but not of T , and is labelled by a variable, so we may adjoin all of its T -successors to form T_{i+1} . Since T is finite this terminates with $T_n = T$, giving a derivative sequence. $\square$

The following surgery on parse trees will be needed for the pumping lemma.

Definition 27.4.8 (Grafting)

Let $\mathbb{T} = (T, \ell)$ be a parse tree, $t \in T$ with $\ell(t) = A$, and $\mathbb{T}' = (T', \ell')$ a parse tree starting with A . Then $\text{graft}(\mathbb{T}, t, \mathbb{T}') = (S, \ell^*)$ where

$$S = \{s \in T \mid t \not\subseteq s\} \cup \{tu \mid u \in T'\}, \quad \ell^*(s) = \begin{cases} \ell(s) & t \not\subseteq s, \\ \ell'(u) & s = tu, u \in T'. \end{cases}$$

That is, we cut off the subtree at t and glue $\mathbb{T}'$ in its place; the result is again a parse tree, because the only condition involving t is that the rule applied at t matches its

successors, and this is inherited from $\mathbb{T}'$. If $\sigma_{\mathbb{T}} = \alpha \sigma_{\mathbb{T}_t} \beta$ then

$$\sigma_{\text{graft}(\mathbb{T}, t, \mathbb{T}')} = \alpha \sigma_{\mathbb{T}'} \beta.$$

§27.4.3 Ambiguity

A word may have several parse trees, and this matters a great deal in practice, since the parse tree of a program is what determines its meaning.

Definition 27.4.9

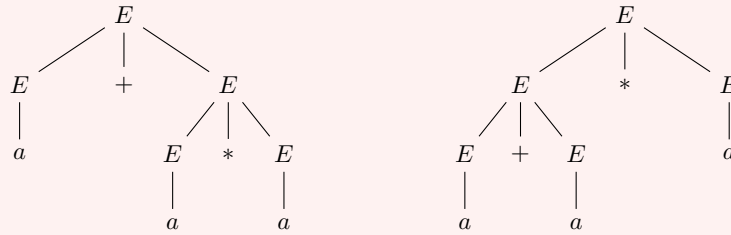
A context-free grammar G is **ambiguous** if some $w \in \mathcal{L}(G)$ has two distinct parse trees starting with S .

Example 27.4.10

Let $\Sigma = \{a, +, *, (,)\}$, $V = \{E\}$ and

$$E \rightarrow E + E \mid E * E \mid (E) \mid a.$$

The word $a + a * a$ has the two parse trees



both parsing $a + a * a$. The left tree groups the expression as $a + (a * a)$ and the right as $(a + a) * a$; if the letters were numbers these would evaluate differently. So G is ambiguous. One can find an unambiguous grammar for the same language by introducing variables for ‘term’ and ‘factor’, which is what forces the usual precedence rules; but there are context-free languages for which *every* grammar is ambiguous, and these are called **inherently ambiguous**.

§27.4.4 Chomsky Normal Form

Parse trees for general context-free grammars can have arbitrarily large branching, which makes counting arguments awkward. It is therefore very convenient that we may always assume the branching is at most 2.

Definition 27.4.11 (Chomsky normal form)

A grammar is in **Chomsky normal form** if every rule is of the form $A \rightarrow BC$ with $B, C \in V$ (a **binary** rule) or $A \rightarrow a$ with $a \in \Sigma$ (a **unary** rule).

Every grammar in Chomsky normal form is context-free. In such a grammar, parse trees are binary and derivations have a fixed length.

Lemma 27.4.12

Let G be in Chomsky normal form and $w \in \mathcal{L}(G)$ with $|w| = n$. Then every G -derivation of w from S has length exactly $2n - 1$.

Proof. A binary rule increases the length by one and the variable count by one; a unary rule leaves the length alone and decreases the variable count by one. We begin with the string S , of length 1 with one variable, and end with w , of length n with no variables. If p binary and q unary rules are used, then $1 + p = n$ and $1 + p - q = 0$, so $p = n - 1$ and $q = n$, giving $2n - 1$ steps. $\square$

Theorem 27.4.13 (Chomsky)

Every context-free grammar is equivalent to one in Chomsky normal form, and such a grammar can be computed from the original.

Proof. There are three kinds of rule obstructing Chomsky normal form:

- (a) rules $A \rightarrow \alpha$ with $|\alpha| \geq 2$ where α contains a letter;
- (b) rules $A \rightarrow B$ with $B \in V$, called **unit rules**;
- (c) rules $A \rightarrow \alpha$ with $|\alpha| > 2$ where α contains only variables.

We remove them in turn.

(a) For each $a \in \Sigma$ introduce a fresh variable X_a and the rule $X_a \rightarrow a$, and replace every offending rule $A \rightarrow \alpha$ by $A \rightarrow X(\alpha)$, where X replaces each letter a by X_a . This is exactly the construction of Lemma 27.2.19 applied to the right-hand sides, and the same argument shows the language is unchanged. Afterwards, the only rules whose right-hand side contains a letter are the unary rules.

(b) Call G **unit closed** if whenever $A \rightarrow B$ and $B \rightarrow \alpha$ lie in P , so does $A \rightarrow \alpha$. Every grammar can be made unit closed by repeatedly adding such rules; at most $|V| \cdot |P|$ new rules can be added, so this terminates, and it clearly does not change the language.

Now suppose G is unit closed, and let G' be G with all unit rules deleted. Certainly $\mathcal{L}(G') \subseteq \mathcal{L}(G)$. Conversely let $w \in \mathcal{L}(G)$ and take a G -derivation of w of minimal length. Suppose it uses a unit rule, and consider the *last* such use:

$$S \Rightarrow_G^* \alpha A \beta \Rightarrow_G \alpha B \beta \Rightarrow_G^* w.$$

Since w contains no variables, the occurrence of B just created must eventually be rewritten; let

$$S \Rightarrow_G^* \alpha A \beta \Rightarrow_G \alpha B \beta \Rightarrow_G^* \gamma B \delta \Rightarrow_G \gamma \zeta \delta \Rightarrow_G^* w$$

where $B \rightarrow \zeta$ is the first rule applied to that occurrence. The segment $\alpha B \beta \Rightarrow_G^* \gamma B \delta$ does not touch that occurrence of B , so the same rules apply verbatim to $\alpha A \beta$, giving $\alpha A \beta \Rightarrow_G^* \gamma A \delta$. By unit closure $A \rightarrow \zeta \in P$, so we may continue $\gamma A \delta \Rightarrow_G \gamma \zeta \delta$, and we have produced a strictly shorter derivation of w – contradicting minimality. Hence the minimal derivation uses no unit rule, and is a G' -derivation.

(c) Finally consider $A \rightarrow A_1 A_2 \cdots A_n$ with $n > 2$ and all $A_i \in V$. Introduce fresh variables $X_1, \dots, X_{n-2}$ and replace this rule by

$$A \rightarrow A_1 X_1, \quad X_1 \rightarrow A_2 X_2, \quad \dots, \quad X_{n-3} \rightarrow A_{n-2} X_{n-2}, \quad X_{n-2} \rightarrow A_{n-1} A_n.$$

The new variables occur in no other rule, so the only strings derivable using them are the intermediate stages of the replacement, and the language is unchanged.

Applying (a), then (b), then (c) leaves only binary and unary rules. $\square$

§27.4.5 The Pumping Lemma for Context-Free Languages

The pumping lemma for regular languages came from a repeated state on a path. Its analogue here comes from a repeated *variable* on a branch of a parse tree, and the resulting pump has two pieces – one on either side of the repeated subtree.

Definition 27.4.14

A language L satisfies the **context-free pumping lemma with pumping number n** if every $w \in L$ with $|w| \geq n$ can be written $w = xuyvz$ with

$$|uyv| \leq n, \quad |uv| > 0, \quad xu^k y v^k z \in L \text{ for all } k \in \mathbb{N}.$$

We call the pair (u, v) the **pump**.

Remark. Note the differences from the regular version. The pump has two parts, of which one may be empty (only their total length is required to be positive). There is no constraint on where the pump sits in the word: x and z may be arbitrarily long. But the pump together with the material between its two halves is bounded.

If L satisfies the regular pumping lemma with number n then it satisfies the context-free one with the same n , taking $v = \varepsilon$; so by Example 27.3.23 there are uncountably many languages satisfying the context-free pumping lemma, and it cannot possibly characterise the countably many context-free languages.

Theorem 27.4.15

Every context-free language satisfies the context-free pumping lemma for some n .

Proof. Let L be context-free. By Theorem 27.4.13 take a Chomsky normal form grammar G with $\mathcal{L}(G) = L$, put $m = |V|$ and $n = 2^m$.

Claim: if $\mathbb{T}$ is a G -parse tree of height $h + 1$ whose string is a word, then $|\sigma_{\mathbb{T}}| \leq 2^h$. Since every node has branching number 1 or 2, and since $\sigma_{\mathbb{T}}$ is a word, the leaves are exactly the nodes labelled by letters and each arises from a unary rule applied to its parent. Deleting the leaves therefore gives a binary tree of height h , whose leaves are the parents; there are at most 2^h of these, and each has exactly one child. So $|\sigma_{\mathbb{T}}| \leq 2^h$.

Now let $w \in L$ with $|w| \geq 2^m$ and let $\mathbb{T}$ be a parse tree for w . By the claim, the height of $\mathbb{T}$ is at least $m + 1$. Let t be a node at maximal level, so the branch to t has $h + 1 \geq m + 2$ nodes; all but the last are labelled by variables, the last by a

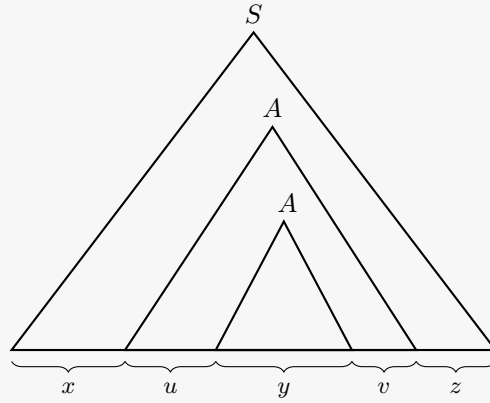
letter.

Let s be the node on that branch for which the subtree $\mathbb{T}_s$ has height exactly $m + 1$. By the claim, $|\sigma_{\mathbb{T}_s}| \leq 2^m = n$. The portion of the branch from s to t has $m + 2$ nodes, of which $m + 1$ are labelled by variables; as there are only m variables, by pigeonhole two of them carry the same label. Say $t_0 \subsetneq t_1$ lie on this branch with $\ell(t_0) = \ell(t_1) = A$.

Decompose the string according to these nodes:

$$\sigma_{\mathbb{T}} = x' \sigma_{\mathbb{T}_s} z', \quad \sigma_{\mathbb{T}_s} = x'' \sigma_{\mathbb{T}_{t_0}} z'', \quad \sigma_{\mathbb{T}_{t_0}} = u \sigma_{\mathbb{T}_{t_1}} v, \quad \sigma_{\mathbb{T}_{t_1}} = y,$$

and set $x = x'x''$ and $z = z''z'$, so that $w = xyvz$. Since $\sigma_{\mathbb{T}_{t_0}} = u y v$ is a substring of $\sigma_{\mathbb{T}_s}$, we get $|uyv| \leq n$. And $t_0 \neq t_1$ with G in Chomsky normal form means the subtree at t_0 strictly contains that at t_1 together with at least one further leaf, so $|uv| > 0$.



Finally we pump. Both $\mathbb{T}_{t_0}$ and $\mathbb{T}_{t_1}$ are parse trees starting with A , so grafting is available. Set

$$\mathbb{T}_{(0)} = \mathbb{T}_{t_1}, \quad \mathbb{T}_{(i+1)} = \text{graft}(\mathbb{T}_{t_0}, t_1, \mathbb{T}_{(i)}),$$

so that by induction $\sigma_{\mathbb{T}_{(i)}} = u^i y v^i$. Then $\text{graft}(\mathbb{T}, t_0, \mathbb{T}_{(k)})$ is a parse tree starting with S whose string is $x u^k y v^k z$, so $x u^k y v^k z \in L$ for every k . (For $k = 0$ this is grafting $\mathbb{T}_{t_1}$ in place of $\mathbb{T}_{t_0}$, which is the ‘pump down’.) $\square$

Example 27.4.16

The language $L = \{a^n b^n c^n \mid n > 0\}$ is not context-free.

Suppose it were, with pumping number N . Consider $a^N b^N c^N \in L$ and write it as $x y v z$ with $|uyv| \leq N$ and $|uv| > 0$. Since uyv is a substring of length at most N , and the blocks of a ’s, b ’s and c ’s each have length N , the substring uyv cannot contain letters of all three kinds. Pumping up to $k = 2$ increases the number of occurrences of at least one letter (as $|uv| > 0$) but leaves the count of at least one other letter unchanged. The result is therefore not of the form $a^m b^m c^m$, contradiction.

Remark. The language $\{a^n b^n c^n\}$ is context-sensitive; a grammar for it may be built from rules such as $S \rightarrow abc \mid aAbc$, $Ab \rightarrow bA$, $Ac \rightarrow Bbcc$, $bB \rightarrow Bb$, $aB \rightarrow aa \mid aaA$, and one checks that this generates exactly L . So the inclusion of type 2 in type 1 is

strict, as drawn in our picture of the hierarchy.

§27.4.6 Closure Properties

The context-free languages inherit closure under union and concatenation from Theorem 27.2.20, since context-free grammars are variable-based and the added rules $T \rightarrow SS'$, $T \rightarrow S$, $T \rightarrow S'$ are context-free. They are also closed under Kleene plus: given G , take a fresh T and put $T \rightarrow S \mid TS$.

The good news stops there.

Proposition 27.4.17

The context-free languages are not closed under intersection, complement or difference.

Proof. Let

$$L_0 = \{a^n b^n c^k \mid n, k > 0\}, \quad L_1 = \{a^k b^n c^n \mid n, k > 0\}.$$

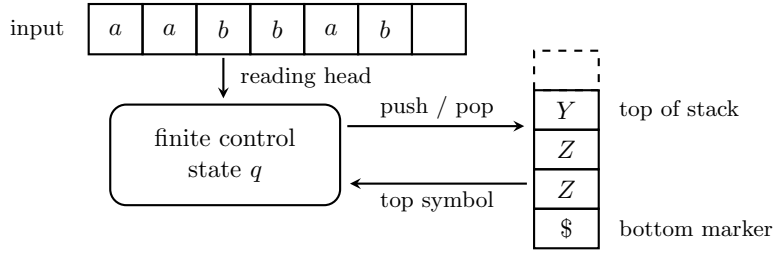
Both are context-free: L_0 is the concatenation of $\{a^n b^n\}$ and $\{c^k\}$, and L_1 is the concatenation of $\{a^k\}$ and $\{b^n c^n\}$, and each factor is context-free: the two-letter blocks by the grammar of Example 27.4.6 with the letters renamed, and $\{c^k\}$ and $\{a^k\}$ by regular grammars. But $L_0 \cap L_1 = \{a^n b^n c^n \mid n > 0\}$, which is not context-free by Example 27.4.16.

Since the class is closed under union, closure under complement would give closure under intersection by De Morgan; so complement fails too, as does difference, since $L \cap M = L \setminus (L \setminus M)$. $\square$

This has an important consequence for the machine model we are about to meet: *what-ever* machines characterise the context-free languages, there can be no product construction for them. Two such machines cannot in general be run side by side on the same input. In hindsight the reason is easy to see, and it is that the machine will have an unbounded memory, so the pair of memories cannot be packaged into a single memory of the same kind.

§27.4.7 Pushdown Automata

We describe, without proofs, the machine model corresponding to context-free languages. A **pushdown automaton** is a finite automaton equipped with a **stack**: an unbounded sequence of symbols of which only the topmost is visible. Two operations are available, **push** (place a new symbol on top) and **pop** (remove the topmost symbol). There is no random access; a symbol placed on the stack can only be read after everything placed on top of it has been removed, so symbols come off in the reverse of the order in which they went on.



Formally, the transition function of such a machine takes as an extra input the symbol currently on top of the stack, and produces as an extra output an instruction to push a symbol, to pop, or to do nothing:

$$\delta: Q \times \Sigma \times \Gamma \longrightarrow \mathcal{P}(Q \times \Gamma^*),$$

where Γ is a finite **stack alphabet**. The machine accepts if some run consumes the whole input and finishes in an accept state.

Theorem 27.4.18

A language is context-free if and only if it is accepted by a pushdown automaton.

It is worth seeing informally why the stack is exactly the right amount of extra power for $\{0^k 1^k\}$. Reading the 0s, the machine pushes one stack symbol per 0; reading the 1s, it pops one per 1; and it accepts if the stack empties exactly as the input runs out. The stack cannot do this twice over – once emptied, the information is gone – which is the informal reason that $\{a^n b^n c^n\}$ is beyond its reach.

Note also that, unlike the case of finite automata, *nondeterministic* pushdown automata are strictly more powerful than deterministic ones: the deterministic pushdown automata accept a proper subclass of the context-free languages (for example the language of even-length palindromes is context-free but not deterministic context-free). So the pleasant state of affairs of Theorem 27.3.15 does not survive the move up the hierarchy either.

§27.4.8 Decision Problems for Context-Free Languages

Proposition 27.4.19

The word problem for context-free grammars is solvable.

Proof. Context-free grammars are noncontracting, so this is Corollary 27.2.16. Alternatively, and much more efficiently: convert to Chomsky normal form, whereupon by Lemma 27.4.12 every derivation of w has length exactly $2|w| - 1$, and check all of them. (The standard polynomial-time algorithm, due to Cocke, Younger and Kasami, computes for each substring of w the set of variables deriving it, by dynamic programming.) $\square$

Proposition 27.4.20

The emptiness problem for context-free grammars is solvable.

Proof. Exactly as for regular languages. Let n be a context-free pumping number for $\mathcal{L}(G)$, computed as $2^{|V|}$ from a Chomsky normal form grammar. If $\mathcal{L}(G) \neq \emptyset$, take $w \in \mathcal{L}(G)$ of minimal length; if $|w| \geq n$ then $w = xuyvz$ can be pumped down to xyz , which is shorter and still in $\mathcal{L}(G)$, a contradiction. So $\mathcal{L}(G)$ is nonempty if and only if it contains a word of length less than n , and there are finitely many of those to test. $\square$

Remark. The equivalence problem for context-free grammars, by contrast, is *unsolvable*; so is the question of whether a given context-free grammar is ambiguous, and whether the intersection of two context-free languages is empty. We shall not prove these, but the first taste of such a result is the unsolvability of the corresponding problems for type 0 grammars in Section 27.8.7.

We can summarise the situation so far in a table.

	union	concatenation	intersection	complement
regular	yes	yes	yes	yes
context-free	yes	yes	no	no
	word	emptiness	equivalence	
regular	yes	yes	yes	
context-free	yes	yes	no	

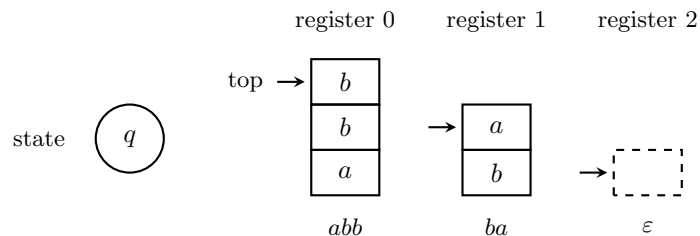
Everything above the line is settled; to go further we need to say what an algorithm is, and that is what we do next.

§27.5 Register Machines

We now give a precise definition of ‘algorithm’. The model we use is the *register machine*: a finite automaton equipped not with one stack, but with finitely many *registers*, each holding a word $w \in \mathbb{W}$. The machine can inspect the last letter of a register, delete it, or append a new one – so a register behaves as a stack, with the right-hand end of the word as the top.

§27.5.1 The Definition

A register machine has an alphabet Σ , a finite set Q of states, and finitely many *registers*, each holding a word $w \in \mathbb{W}$. The machine can inspect the last letter of a register, delete it, or append a new one – so a register behaves as a stack, with the right-hand end of the word as the top.



A snapshot of the whole machine is called a configuration.

Definition 27.5.1 (Configuration)

A **configuration** (or **snapshot**) of length $n+1$ is a tuple $(q, w_0, \dots, w_n) \in Q \times \mathbb{W}^{n+1}$.

One might be tempted to define the dynamics by a transition function $\delta: Q \times \mathbb{W}^{n+1} \rightarrow Q \times \mathbb{W}^{n+1}$, but this is far too permissive: there are uncountably many such functions, and they can do anything at all to the registers in one step. Instead we allow only four kinds of elementary instruction.

Definition 27.5.2 (Instruction)

Let Σ be an alphabet and Q a nonempty finite set of states. A (Σ, Q) -**instruction** is a tuple of one of the forms

$$\begin{aligned} (0, k, a, q) &\in \mathbb{N} \times \mathbb{N} \times \Sigma \times Q, & (1, k, a, q, q') &\in \mathbb{N} \times \mathbb{N} \times \Sigma \times Q \times Q, \\ (2, k, q, q') &\in \mathbb{N} \times \mathbb{N} \times Q \times Q, & (3, k, q, q') &\in \mathbb{N} \times \mathbb{N} \times Q \times Q. \end{aligned}$$

We write these more legibly as

$$+(k, a, q), \quad ?(k, a, q, q'), \quad ?(k, \varepsilon, q, q'), \quad -(k, q, q')$$

respectively, and we write $\text{Instr}(\Sigma, Q)$ for the set of all of them.

Their intended meanings are as follows.

- $+(k, a, q)$: push the letter a onto register k , then move to state q .
- $?(k, a, q, q')$: if the top letter of register k is a , move to state q ; otherwise move to q' .
- $?(k, \varepsilon, q, q')$: if register k is empty, move to state q ; otherwise move to q' .
- $-(k, q, q')$: pop the top letter of register k . If the register was already empty, move to state q ; otherwise move to q' .

Note that only $+$ and $-$ change the registers, and each changes exactly one register by exactly one letter.

Definition 27.5.3 (Register machine)

A tuple $M = (\Sigma, Q, P)$ is a Σ -**register machine** if Σ is an alphabet, Q is a finite set of states containing two distinguished states $q_S \neq q_H$ (the **start state** and the **halt state**), and

$$P: Q \rightarrow \text{Instr}(\Sigma, Q)$$

is the **program**. Writing $Q = \{q_0, \dots, q_n\}$, we present P as a finite list of **program lines** $q_i \mapsto P(q_i)$. Only finitely many register indices k occur in P ; the largest is the **upper register index** of M .

Definition 27.5.4 (Computation)

Let M have upper register index m and let $C = (q, w_0, \dots, w_m)$ be a configuration. We say M **transforms** C into C' if one of the following holds:

- $P(q) = +(k, a, q')$ and $C' = (q', w_0, \dots, w_{k-1}, w_k a, w_{k+1}, \dots, w_m)$;
- $P(q) = ?(k, a, q', q'')$ and either $w_k = wa$ for some w and $C' = (q', w_0, \dots, w_m)$, or w_k does not end in a and $C' = (q'', w_0, \dots, w_m)$;

- $P(q) = ?(k, \varepsilon, q', q'')$ and either $w_k = \varepsilon$ and $C' = (q', w_0, \dots, w_m)$, or $w_k \neq \varepsilon$ and $C' = (q'', w_0, \dots, w_m)$;
- $P(q) = -(k, q', q'')$ and either $w_k = \varepsilon$ and $C' = (q', w_0, \dots, w_m)$, or $w_k = wa$ and $C' = (q'', w_0, \dots, w_{k-1}, w, w_{k+1}, \dots, w_m)$.

Given $\mathbf{w} \in \mathbb{W}^{m+1}$, the **computation sequence** of M on input $\mathbf{w}$ is defined by recursion:

$$C(0, M, \mathbf{w}) = (q_S, \mathbf{w}), \quad C(k+1, M, \mathbf{w}) = \text{the transform of } C(k, M, \mathbf{w}).$$

If $\mathbf{w}$ is shorter than $m+1$ we pad it on the right with copies of ε ; if it is longer we simply carry the extra registers along untouched.

Remark. Every configuration has a transform, so computation sequences as defined are always infinite. Halting is not the absence of a next step, but the arrival at a designated state.

Definition 27.5.5 (Halting)

The computation of M on input $\mathbf{w}$ **halts at time k** if k is least with $C(k, M, \mathbf{w}) = (q_H, \mathbf{v})$ for some $\mathbf{v}$; in this case $\mathbf{v}$ is the **output**. If no such k exists, the computation **does not halt**.

§27.5.2 Strong Equivalence

Definition 27.5.6

Register machines M and M' are **strongly equivalent** if for every input $\mathbf{w}$ and every k , the configurations $C(k, M, \mathbf{w})$ and $C(k, M', \mathbf{w})$ have the same register contents, and M halts on $\mathbf{w}$ at time k if and only if M' does.

So strongly equivalent machines do literally the same thing, step by step; only their state names may differ. Relabelling states shows that if $|Q| = |Q'|$ then every (Σ, Q, P) has a strongly equivalent (Σ, Q', P') .

Proposition 27.5.7 (Padding lemma)

Every register machine is strongly equivalent to infinitely many distinct register machines.

Proof. Let $M = (\Sigma, Q, P)$ and add a fresh state $\hat{q}$ to Q , with $P(\hat{q})$ any instruction whatsoever. Since $\hat{q}$ occurs in no other program line, it never appears in any computation sequence, so the resulting machine is strongly equivalent to M . Repeating gives infinitely many. $\square$

Proposition 27.5.8

Up to strong equivalence there are only countably many register machines over a fixed Σ .

Proof. Up to strong equivalence only $|Q|$ matters, so fix a set Q with $|Q| = n$ and bound the register index by k . Counting the four kinds of instruction,

$$|\text{Instr}(\Sigma, Q)| = (k+1)n|\Sigma| + (k+1)n^2|\Sigma| + (k+1)n^2 + (k+1)n^2 =: N_{n,k},$$

which is finite, so there are $N_{n,k}^n$ programs. The class $M_{n,k}$ of such machines is therefore finite, and every register machine lies in some $M_{n,k}$; so $\bigcup_{n,k} M_{n,k}$ is a countable union of finite sets. $\square$

This is the source of all the negative results in the course. There are only countably many algorithms, and uncountably many things one might want to compute.

§27.5.3 Operations and Questions

Rather than reason about programs line by line, we build up a library of things register machines can do, and thereafter argue at a higher level. There are two kinds of task.

Definition 27.5.9 (Operation)

An **operation** is a partial function $f: \mathbb{W}^{m+1} \rightarrow \mathbb{W}^{m+1}$. We write $f(\mathbf{w}) \downarrow$ if $\mathbf{w} \in \text{dom } f$ (f **converges** at $\mathbf{w}$) and $f(\mathbf{w}) \uparrow$ otherwise (f **diverges**).

A register machine M **performs** f if for every $\mathbf{w}$ we have $f(\mathbf{w}) \downarrow$ if and only if M halts on $\mathbf{w}$, and in that case the output of M is $f(\mathbf{w})$.

Example 27.5.10

The operation ‘never halt’ is the empty function. It is performed by any program which never mentions q_H on the right-hand side of a line, for instance

$$q_S \mapsto +(0, a, q_S), \quad q_H \mapsto +(0, a, q_S).$$

The operation ‘halt immediately, changing nothing’ is the identity function on $\mathbb{W}^{m+1}$, performed by $q_S \mapsto ?(0, a, q_H, q_H)$: the test is performed, both branches lead to q_H , and nothing is written.

Definition 27.5.11 (Question)

A **question with $k+1$ answers** is a partition of $\mathbb{W}^{m+1}$ into sets $A_0, \dots, A_k$. A register machine **answers** the question if it has distinguished **answer states** $\bar{q}_0, \dots, \bar{q}_k$ such that on input $\mathbf{w}$ the computation reaches, after finitely many steps, the configuration $(\bar{q}_i, \mathbf{w})$, where i is the index with $\mathbf{w} \in A_i$.

So a question is answered without disturbing the registers, and the answer is recorded in the state.

Example 27.5.12

‘Is register i empty?’ is answered by the single line $q_S \mapsto ?(i, \varepsilon, \bar{q}_Y, \bar{q}_N)$, and ‘does register i end in the letter a ?’ by $q_S \mapsto ?(i, a, \bar{q}_Y, \bar{q}_N)$.

The next two lemmas are what allow us to program at a high level: they say that

machines can be run one after another, and that they can be selected between.

Lemma 27.5.13 (Concatenation)

If M performs F and M' performs F' , then there is a register machine $\hat{M}$ performing $F' \circ F$.

Proof. Assume the state sets are disjoint. Let P^* be P with the line for q_H deleted and with every occurrence of q_H replaced by q'_S , and set $\hat{M} = (\Sigma, (Q \setminus \{q_H\}) \cup Q', P^* \cup P')$, with start state q_S and halt state q'_H . Running $\hat{M}$ on $\mathbf{w}$ reproduces the computation of M exactly until the moment M would halt, at which point $\hat{M}$ is in state q'_S with registers $F(\mathbf{w})$, and thereafter reproduces the computation of M' on that input. So $\hat{M}$ halts precisely when both $F(\mathbf{w}) \downarrow$ and $F'(F(\mathbf{w})) \downarrow$, with output $F'(F(\mathbf{w}))$. $\square$

Lemma 27.5.14 (Case distinction)

Let Q be a question with answers $A_0, \dots, A_k$, answered by a machine M ; and let F_i be operations performed by machines M_i . Then some register machine performs

$$G(\mathbf{w}) = F_i(\mathbf{w}) \quad \text{when } \mathbf{w} \in A_i.$$

Proof. Assume Q is disjoint from each Q_i and that the Q_i meet only in a common halt state q_H . Let P_i^* be P_i with every occurrence of $q_{S,i}$ replaced by the answer state $\bar{q}_i$ of M , and delete the line for $q_{S,i}$. Put

$$Q^* = Q \cup \bigcup_{i \leq k} (Q_i \setminus \{q_{S,i}\}), \quad P^* = P \cup \bigcup_{i \leq k} P_i^*.$$

On input $\mathbf{w}$, the machine first runs M , which reaches $\bar{q}_i$ with the registers untouched, and from there runs M_i . $\square$

§27.5.4 A Register Machine API

We may now assemble a library of operations, each built out of the previous ones by concatenation and case distinction. A register is **unused** if no program line mentions it, and **empty** if it contains ε ; registers used only for intermediate work are called **scratch registers**. Since any given machine mentions only finitely many registers, an unused empty register is always available.

- ‘Delete the last letter of register i ’: $q_S \mapsto -(i, q_H, q_H)$.
- ‘Push a onto register i ’: $q_S \mapsto +(i, a, q_H)$.
- ‘Empty register i ’: $q_S \mapsto -(i, q_H, q_S)$. This pops repeatedly, halting exactly when the register was found to be empty.
- ‘Halt if register i is nonempty, otherwise diverge’: answer ‘is register i empty?’, then apply case distinction with the operations ‘never halt’ and ‘halt immediately’.
- ‘Append the fixed word $w = a_0 a_1 \dots a_\ell$ to register i ’: concatenate the $\ell + 1$ pushes.
- ‘Replace register i by the fixed word w ’: empty it, then append w .

- ‘What is the last letter of register i ?’: this question has $|\Sigma| + 1$ answers, one per letter and one for ‘empty’. Ask ‘does register i end in a_0 ?’; if yes go to $\bar{q}_0$, if no ask about a_1 , and so on; if every answer is no, the register is empty and we go to $\bar{q}_\varepsilon$.
- ‘Copy the last letter of register i onto register j ’: ask the previous question and push the corresponding letter in each case.
- ‘Move the last letter of register i onto register j ’: copy it, then delete it from i .
- ‘Move register i into register j in reverse order’: repeat the previous operation until register i is empty.
- ‘Move register i into register j in the correct order’: move i into an unused empty register k in reverse order, then move k into j in reverse order. Reversing twice restores the order.
- ‘Reverse register i ’: move it in reverse into an unused empty register, then move that back in the correct order – or equivalently, move it out and back once each.
- ‘Move register i into both j and k in reverse order’: copy the last letter into j , then into k , then delete it; repeat.
- ‘Copy register i into register j ’: move i into j and a scratch register k simultaneously (in reverse), then move k back into i (in reverse). Composing with a reversal fixes the order if required.
- ‘Is register i equal to the fixed word $w = a_0 \cdots a_\ell$?’: run subroutines $S_\ell, S_{\ell-1}, \dots, S_0$, where S_r asks whether the last letter of register i is a_r ; if not, go to a state q_N ; if so, move that letter to a scratch register k and run S_{r-1} , or go to q_Y if $r = 0$. At q_N and at q_Y , move the contents of k back into i , restoring it, and then answer $\bar{q}_N$ or $\bar{q}_Y$ as appropriate. The scratch register must be restored, since a question is not allowed to alter the registers.

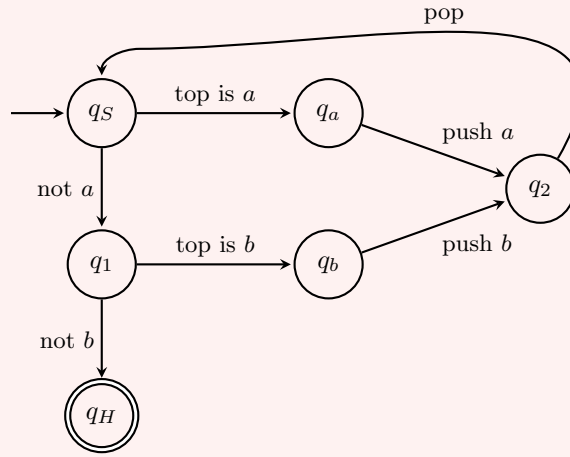
It is worth writing out one of these in full, to see that the formalism really does what we claim.

Example 27.5.15

Take $\Sigma = \{a, b\}$. The following five-line program moves the contents of register 1 into register 2 in reverse order, leaving register 1 empty.

$$\begin{array}{lll} q_S \mapsto ?(1, a, q_a, q_1), & q_1 \mapsto ?(1, b, q_b, q_H), & q_a \mapsto +(2, a, q_2), \\ q_b \mapsto +(2, b, q_2), & q_2 \mapsto -(1, q_S, q_S). & \end{array}$$

As a flow diagram:



Let us trace it with register 1 containing ab and register 2 empty. In q_S , the top letter is b , not a , so we move to q_1 ; there the top letter is b , so we move to q_b and push b onto register 2; in q_2 we pop, leaving register 1 holding a , and return to q_S . Now the top letter is a , so we go to q_a , push a onto register 2 (which now holds ba), and pop, leaving register 1 empty. Back at q_S : the top letter is not a (the register is empty), so we go to q_1 ; it is not b either, so we halt. Register 2 holds ba , the reverse of ab , as claimed.

From now on we shall specify register machines by describing what they do in English, in terms of the library above, exactly as one writes a program in terms of library functions rather than machine code. This is legitimate precisely because Lemmas 27.5.13 and 27.5.14 let us assemble the pieces.

§27.6 Computable Functions and Sets

§27.6.1 Computable Functions

Most computations need scratch space, and we do not want the final answer to depend on the litter left behind. So we declare that register 0 holds the output and that everything else is scratch.

Definition 27.6.1

Let M be a register machine and $k \in \mathbb{N}$. Define $f_{M,k}: \mathbb{W}^k \rightarrow \mathbb{W}$ by

$$f_{M,k}(\mathbf{w}) = \begin{cases} v_0 & \text{if } M \text{ halts on } \mathbf{w} \text{ with output } \mathbf{v}, \\ \uparrow & \text{if } M \text{ does not halt on } \mathbf{w}. \end{cases}$$

A partial function $f: \mathbb{W}^k \rightarrow \mathbb{W}$ is **computable** if $f = f_{M,k}$ for some register machine M . For $k = 1$ we write $W_M = \text{dom } f_{M,1}$.

If M and M' are strongly equivalent then $f_{M,k} = f_{M',k}$, though the converse fails badly: two machines may compute the same function in completely different ways.

Remark. By Proposition 27.5.8 there are only countably many computable functions, while there are uncountably many partial functions $\mathbb{W} \rightarrow \mathbb{W}$. So almost every function is

not computable. On the other hand, by the padding lemma each computable function is computed by infinitely many machines. And by Lemmas 27.5.13 and 27.5.14, computable functions are closed under composition and case distinction.

Example 27.6.2

The identity $\mathbb{W} \rightarrow \mathbb{W}$ is computable (halt immediately). Each constant function $c(\mathbf{w}) = v$ is computable (replace register 0 by v). Each projection $\pi_i(\mathbf{w}) = w_i$ is computable (empty register 0, then move register i into register 0).

§27.6.2 Computable and Computably Enumerable Sets

To speak of computable *sets* we use characteristic functions. Since our values are words rather than truth values, we fix once and for all a letter $a \in \Sigma$ to stand for ‘true’, with ε standing for ‘false’.

Definition 27.6.3

Let $X \subseteq \mathbb{W}^k$. A total $f: \mathbb{W}^k \rightarrow \mathbb{W}$ is **a characteristic function of X** if $f(\mathbf{w}) \neq \varepsilon$ exactly when $\mathbf{w} \in X$; it is **the characteristic function χ_X** if moreover $f(\mathbf{w}) = a$ for $\mathbf{w} \in X$. We call X **computable** if χ_X is computable.

A partial f is **a pseudocharacteristic function of X** if $\text{dom } f = X$; it is **the pseudocharacteristic function ψ_X** if additionally $f(\mathbf{w}) = a$ for $\mathbf{w} \in X$. We call X **computably enumerable** if ψ_X is computable.

A computable set is one we can *decide*: given $\mathbf{w}$, the machine always halts and tells us the answer. A computably enumerable set is one we can only *semi-decide*: the machine halts when the answer is yes, and runs forever when the answer is no. Since a language is a set of words, both notions apply to languages.

Proposition 27.6.4

Let $X \subseteq \mathbb{W}^k$.

- (i) X is computable if and only if its complement is computable.
- (ii) X is computably enumerable if and only if $X = \text{dom } f_{M,k}$ for some register machine M .
- (iii) If X is computable then X is computably enumerable.

Proof. Write the proof for $k = 1$; the general case is identical. By case distinction, if g, h are computable then so is the function equal to $g(w)$ when $w \neq \varepsilon$ and $h(w)$ when $w = \varepsilon$, since ‘is register 0 empty?’ is answerable.

- (i) Let f_1 be the computable function given in this way by $g(w) = \varepsilon$ and $h(w) = a$; that is, f_1 swaps a and ε . Then $\chi_{X^c} = f_1 \circ \chi_X$ and conversely.
- (ii) If $X = \text{dom } f$ with f computable, let f_2 be given by $g(w) = a$, $h(w) = a$, so f_2 is the total constant function a ; then $\psi_X = f_2 \circ f$, which is computable with domain X . Conversely ψ_X is itself computable with domain X .

(iii) Let f_3 be given by $g(w) = a$ and $h(w) \uparrow$ (halt if register 0 is nonempty, else diverge). Then $\psi_X = f_3 \circ \chi_X$. $\square$

The converse of (iii) is false, and proving so is the content of the halting problem below. First, a sanity check that our new notion is at least as strong as the old ones.

Theorem 27.6.5

Every regular language is computable.

Proof. Let $L = \mathcal{L}(D)$ with $D = (\Sigma, Q, \delta, q_0, F)$. We describe a register machine with input in register 0.

First, move register 0 into register 1 in reverse order. (This is necessary because a register machine sees the *last* letter of a word, whereas an automaton reads from the left; reversing makes the automaton's first letter the topmost.)

For each $q \in Q$ our machine has a block of states Q_q , meaning ‘we are simulating D in state q ’. Move into Q_{q_0} .

On entering Q_q : ask what the last letter of register 1 is. If register 1 is empty, move to a state $\bar{q}_{acc}$ if $q \in F$, and to $\bar{q}_{rej}$ otherwise. Otherwise, if the last letter is b , delete it and move into $Q_{\delta(q,b)}$, repeating.

Since each pass deletes a letter, the loop terminates after $|w|$ passes. At $\bar{q}_{acc}$, empty register 0, push a , and halt; at $\bar{q}_{rej}$, empty register 0 and halt. The resulting machine computes χ_L . $\square$

§27.6.3 The Shortlex Ordering

Our functions take words as inputs, but recursion – both the definition of recursive functions in Section 27.7 and simple counting arguments – needs a notion of ‘the next one’. We therefore transport the order type of $\mathbb{N}$ onto $\mathbb{W}$.

Definition 27.6.6 (Shortlex)

Fix a total order on Σ . The **shortlex order** on $\mathbb{W}$ is defined by $w < v$ if

- (i) $|w| < |v|$; or
- (ii) $|w| = |v|$, $w \neq v$, and at the least position where they differ, the letter of w precedes that of v .

So we sort by length first, and break ties lexicographically. This is a total order with least element ε .

Example 27.6.7

For $\Sigma = \{0, 1\}$ with $0 < 1$, the shortlex order begins

$\varepsilon, 0, 1, 00, 01, 10, 11, 000, 001, 010, 011, 100, 101, 110, 111, 0000, \dots$

Identifying each word with its index (counting from 0), we have $\#\varepsilon = 0$, $\#10 = 5$, $\#01 = 4$, and since $\#010 = 9 = 5 + 4$ we may write $10 + 01 = 010$. Transporting

the semiring structure of $\mathbb{N}$ in this way makes $\mathbb{W}$ into a commutative semiring; there are $|\Sigma|^k$ words of length k , so for $|\Sigma| = 2$ the index of 0^k is $2^k - 1$.

Theorem 27.6.8

$(\mathbb{W}, <)$ is order-isomorphic to $(\mathbb{N}, <)$.

Proof. For a fixed w the set $\{v \mid v < w\}$ is finite, since it consists of words of length at most $|w|$. So $\#(w) = |\{v \mid v < w\}|$ is a well-defined function $\mathbb{W} \rightarrow \mathbb{N}$, and it is clearly order-preserving. It is surjective, since the words listed in increasing order take every index in turn, and order-preserving maps between total orders are injective. $\square$

Theorem 27.6.9

The set $\{(v, w) \mid v < w\}$ is computable, and the **successor function** $s: \mathbb{W} \rightarrow \mathbb{W}$ determined by $\#s(w) = \#w + 1$ is computable.

Proof. To compare v and w : copy them into scratch registers and repeatedly delete the last letter of each until one is empty; whichever empties first is shorter. If they empty simultaneously, copy them again in reverse order (so that we may read them from the left) and delete letters in step until they differ, then compare those letters in the fixed order on Σ ; if no difference is found the words are equal. All of this uses only operations from Section 27.5.4.

For the successor: locate the last letter of w which is not the largest letter of Σ , replace it by its successor in Σ , and replace everything after it by copies of the least letter. If every letter of w is the largest letter (including the case $w = \varepsilon$), output the least letter repeated $|w| + 1$ times. $\square$

§27.6.4 Merging and Splitting Words

We shall repeatedly need to package a pair of words into a single word. Recall the **Cantor zigzag function**

$$z(i, j) = \frac{(i + j)(i + j + 1)}{2} + j,$$

a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$; it enumerates the lattice points of the first quadrant along successive anti-diagonals.

Definition 27.6.10

For $v, w \in \mathbb{W}$, the **merge** $v * w$ is the unique word with $\#(v * w) = z(\#v, \#w)$. Conversely each w **splits** uniquely as $\#w = z(\#w_{(0)}, \#w_{(1)})$, and we call $w_{(0)}, w_{(1)}$ the components of w .

Since addition and multiplication of indices are computable (they can be built from the successor function by recursion, as we will see in Section 27.7), merging is computable; strictly speaking splitting is an *operation* rather than a computable function, since it

has two outputs, but it is performed by a register machine which writes the components into two registers.

The point of merging is that a single existential quantifier over $\mathbb{W}$ can simulate two. We shall use this so often that it acquires a name: the **zigzag argument**.

§27.6.5 Coding of Programs

We now take the decisive step and treat programs as data. Let Σ be an alphabet and let Σ' be Σ together with ten new symbols

$$\mathbf{0} \ \mathbf{1} \ + \ - \ ? \ (\) \ , \ \mapsto \ \square.$$

For a mathematical object o we write $\text{code}(o) \in \Sigma'^*$ for its code, defined as follows.

- Naturals are written in binary using $\mathbf{0}$ and $\mathbf{1}$, so $\text{code}(19) = \mathbf{10011}$.
- For $Q = \{q_0, \dots, q_n\}$ we set $\text{code}(q_k) = \text{code}(k)$.
- Instructions are coded using the punctuation, for instance

$$\text{code}(+(k, a, \ell)) = +(\text{code}(k), a, \text{code}(\ell)),$$

and similarly for $?$ and $-$.

- A program line is coded by $\text{code}(q \mapsto I) = \text{code}(q) \mapsto \text{code}(I)$, and a machine by listing its lines separated by $,$.
- A tuple of words is coded by $\text{code}(\mathbf{w}) = \square w_0 \square w_1 \square \dots \square w_k \square$, and a configuration by $\text{code}(q, \mathbf{w}) = \text{code}(q) \text{code}(\mathbf{w})$.

Not every Σ' -word is a code, but it is a purely syntactic matter to check whether a given word is one, and this check can plainly be carried out by a register machine using the operations of Section 27.5.4.

Remark. We have enlarged the alphabet, which looks like cheating. It is not: if $\Sigma \subseteq \Sigma'$ then every Σ -machine is a Σ' -machine, and conversely one can encode each letter of Σ' as a fixed-length block of letters of Σ provided $|\Sigma| \geq 2$, translating Σ' -machines into Σ -machines. So computability over any alphabet with at least two letters is the same notion, and we may enlarge the alphabet whenever convenient.

§27.6.6 The Universal Register Machine

Lemma 27.6.11

The function

$$h(w, u, v) = \begin{cases} \text{code}(C(\#v, M, \mathbf{w})) & \text{if } w = \text{code}(M), u = \text{code}(\mathbf{w}) \text{ for some } M, \mathbf{w}, \\ \uparrow & \text{otherwise} \end{cases}$$

is computable.

Proof. First check syntactically that w codes a machine and u codes a tuple; if not, diverge. Then define h by recursion on v :

$$h(w, u, \varepsilon) = \text{code}(q_S) u, \quad h(w, u, s(v)) = \text{code}(C'),$$

where C' is the transform, under the machine coded by w , of the configuration coded by $h(w, u, v)$. Computing C' from $\text{code}(C)$ and $\text{code}(M)$ is a finite syntactic manipulation: read off the current state from $\text{code}(C)$, look up the corresponding line of $\text{code}(M)$, and edit the appropriate register block accordingly. All of these are operations from Section 27.5.4, and the recursion itself is available because computable functions are closed under recursion (Theorem 27.7.8 below, whose proof does not depend on this lemma). $\square$

Corollary 27.6.12

For each M and k the **truncated computation function**

$$t_{M,k}(\mathbf{w}, v) = \begin{cases} a & \text{if } M \text{ has halted on } \mathbf{w} \text{ by time } \#v, \\ \varepsilon & \text{otherwise} \end{cases}$$

is computable, and it is total.

Proof. Using h and recursion on v , examine the configurations at all times $\#u < \#v$; output a if any of them has state q_H , and ε otherwise. Every step of this terminates, so $t_{M,k}$ is total. $\square$

The point of $t_{M,k}$ is that it converts a statement about halting – which we cannot decide – into a statement about halting *within a given time*, which we can.

Theorem 27.6.13 (The software principle)

The function

$$g(v, u) = \begin{cases} f_{M,k}(\mathbf{w}) & \text{if } v = \text{code}(M), u = \text{code}(\mathbf{w}) \text{ with } |\mathbf{w}| = k, \\ \uparrow & \text{otherwise} \end{cases}$$

is computable.

Proof. Check syntactically that v and u are codes of the right kind; if not, diverge. Let $f'(v, u, x)$ be the computable function returning a if the configuration $h(v, u, x)$ has state q_H , and ε otherwise; this is computable by Lemma 27.6.11.

Apply minimisation to f' (Theorem 27.7.8) to obtain a computable $\mu(v, u)$ returning the least x with $f'(v, u, x) = a$, and diverging if there is none. If M never halts on $\mathbf{w}$ then $\mu(v, u) \uparrow$, which is exactly what we want. If M does halt, then $\mu(v, u)$ converges to the halting time; compute $h(v, u, \mu(v, u))$, read off the block coding register 0, and write it into register 0. $\square$

Definition 27.6.14

A register machine U with $f_{U,2} = g$ is called a **universal register machine**.

A universal register machine is a machine with a fixed, finite number of states and registers which can nonetheless simulate any register machine whatsoever, however many states and registers that machine has. This is the mathematical content of the statement

that a computer can run any program: hardware is fixed, software is data.

It also lets us drop machines from the notation entirely. For $v \in \mathbb{W}$ write

$$f_{v,k}(\mathbf{w}) = f_{U,2}(v, \text{code}(\mathbf{w})),$$

so that $f_{v,k} = f_{M,k}$ whenever $v = \text{code}(M)$, and $f_{v,k}$ is the empty function when v is not a code. Write $W_v = \text{dom } f_{v,1}$; then $\{W_v \mid v \in \mathbb{W}\}$ is exactly the collection of computably enumerable subsets of $\mathbb{W}$.

§27.6.7 The s - m - n Theorem

The next result says that we may freeze one argument of a computable function of two arguments, and that a code for the resulting function can be computed from the frozen value. This operation is **currying**, after Haskell Curry.

Theorem 27.6.15 (s - m - n theorem, parameter theorem)

Let $g: \mathbb{W}^{k+1} \rightarrow \mathbb{W}$ be computable. Then there is a *total* computable $h: \mathbb{W} \rightarrow \mathbb{W}$ such that

$$f_{h(v),k}(\mathbf{w}) = g(\mathbf{w}, v) \quad \text{for all } \mathbf{w}, v.$$

Proof. Fix v . The operation $\mathbf{w} \mapsto (\mathbf{w}, v)$ is performed by the register machine M_v which simply writes v into register k ; and $\text{code}(M_v)$ is computable from v , since M_v has $|v| + 1$ program lines each of which is written down mechanically from a letter of v . Let M be a machine with $f_{M,k+1} = g$.

By Lemma 27.5.13 the machine $M \circ M_v$ (first M_v , then M) performs the composite, and inspecting the proof of that lemma shows that $\text{code}(M \circ M_v)$ is computable from $\text{code}(M_v)$ and $\text{code}(M)$: one relabels states and concatenates lists of lines, both syntactic operations. So

$$h(v) = \text{code}(M \circ M_v)$$

is total and computable, and

$$f_{h(v),k}(\mathbf{w}) = f_{M \circ M_v,k}(\mathbf{w}) = g(\mathbf{w}, v). \quad \square$$

Remark. The content is the computability of h . That *some* function h with $f_{h(v),k} = g(\cdot, v)$ exists is trivial, since $g(\cdot, v)$ is computable for each v and so has *some* code; the theorem says that a code can be found uniformly and effectively.

§27.6.8 The Halting Problem

Definition 27.6.16

Define

$$\mathbb{K}_0 = \{(w, v) \mid f_{w,1}(v) \downarrow\}, \quad \mathbb{K} = \{w \mid f_{w,1}(w) \downarrow\}.$$

So $\mathbb{K}_0$ is the set of pairs (program, input) for which the program halts, and $\mathbb{K}$ is its diagonal.

Theorem 27.6.17

$\mathbb{K}_0$ and $\mathbb{K}$ are computably enumerable.

Proof. By Proposition 27.6.4(ii) it suffices to exhibit them as domains of computable functions. By the software principle, $f_{U,2}(w, v) = f_{w,1}(v)$, so $\text{dom } f_{U,2} = \mathbb{K}_0$. The diagonal map $\Delta(w) = (w, w)$ is computable (copy register 0 into register 1), so $f_{U,2} \circ \Delta$ is computable with domain $\mathbb{K}$. $\square$

Theorem 27.6.18 (Unsolvability of the halting problem)

Neither $\mathbb{K}_0$ nor $\mathbb{K}$ is computable.

Proof. Suppose $\mathbb{K}_0$ were computable, so that $\chi_{\mathbb{K}_0}$ is a computable total function. Define

$$f(w) = \begin{cases} \uparrow & \text{if } \chi_{\mathbb{K}_0}(w, w) = a, \\ \varepsilon & \text{if } \chi_{\mathbb{K}_0}(w, w) = \varepsilon. \end{cases}$$

This is computable, by case distinction on the value of $\chi_{\mathbb{K}_0}(w, w)$ composed with the machines ‘never halt’ and ‘halt immediately’. So $f = f_{d,1}$ for some $d \in \mathbb{W}$. But then

$$f(d) \downarrow \iff f_{d,1}(d) \downarrow \iff (d, d) \in \mathbb{K}_0 \iff \chi_{\mathbb{K}_0}(d, d) = a \iff f(d) \uparrow,$$

a contradiction. The argument for $\mathbb{K}$ is identical with $\chi_{\mathbb{K}}(w)$ in place of $\chi_{\mathbb{K}_0}(w, w)$. $\square$

Corollary 27.6.19

There is a computably enumerable set which is not computable, so the converse of Proposition 27.6.4(iii) fails.

This is the archetype of all unsolvability results, and its shape should be recognised: we assume a decision procedure exists, build from it a machine which is designed to disagree with itself, and then feed that machine its own code. The reader will recognise the diagonal argument of Cantor’s theorem, and the self-reference of Gödel’s incompleteness theorem, in the same shape.

§27.7 Recursive Functions

Register machines are one definition of computability. Church’s, which came first, is completely different in spirit: instead of describing a machine, one specifies a small stock of obviously computable functions and a small stock of obviously computability-preserving constructions, and takes the closure. The remarkable fact – and the main theorem of this section – is that the two definitions agree exactly.

§27.7.1 Primitive Recursion and Minimisation

We work with partial functions $\mathbb{W}^k \rightharpoonup \mathbb{W}$ throughout, using the shortlex successor s of Theorem 27.6.9 in place of the successor on $\mathbb{N}$.

Definition 27.7.1 (Basic functions)

The **basic functions** are

$$\begin{aligned}\pi_{k,i}: \mathbb{W}^k &\rightarrow \mathbb{W}, & \pi_{k,i}(\mathbf{w}) &= w_i && \text{(the **projections**)}, \\ c_{k,\varepsilon}: \mathbb{W}^k &\rightarrow \mathbb{W}, & c_{k,\varepsilon}(\mathbf{w}) &= \varepsilon && \text{(the **constant** functions),} \\ s: \mathbb{W} &\rightarrow \mathbb{W}, & \#s(w) &= \#w + 1 && \text{(the **successor**)}.\end{aligned}$$

Definition 27.7.2 (The three constructions)

Let $f, g, g_1, \dots, g_m$ be partial functions.

- (i) (**Composition**) If $f: \mathbb{W}^m \rightharpoonup \mathbb{W}$ and $g_1, \dots, g_m: \mathbb{W}^k \rightharpoonup \mathbb{W}$, their composition is

$$h(\mathbf{w}) = f(g_1(\mathbf{w}), \dots, g_m(\mathbf{w})),$$

which is defined exactly when all the $g_i(\mathbf{w})$ are and f is defined at the resulting tuple.

- (ii) (**Recursion**) If $f: \mathbb{W}^k \rightharpoonup \mathbb{W}$ and $g: \mathbb{W}^{k+2} \rightharpoonup \mathbb{W}$, the function $h: \mathbb{W}^{k+1} \rightharpoonup \mathbb{W}$ defined by

$$h(\mathbf{w}, \varepsilon) = f(\mathbf{w}), \quad h(\mathbf{w}, s(v)) = g(\mathbf{w}, v, h(\mathbf{w}, v))$$

is said to be defined by recursion from f and g .

- (iii) (**Minimisation**) If $f: \mathbb{W}^{k+1} \rightharpoonup \mathbb{W}$, the function $h: \mathbb{W}^k \rightharpoonup \mathbb{W}$ defined by

$$h(\mathbf{w}) = \begin{cases} v & \text{if } f(\mathbf{w}, u) \downarrow \text{ for all } u \leq v, \text{ and } v \text{ is least with } f(\mathbf{w}, v) = \varepsilon, \\ \uparrow & \text{if no such } v \text{ exists} \end{cases}$$

is said to be defined by minimisation from f .

A class $\mathcal{C}$ of partial functions is **closed** under one of these if the result of applying it to members of $\mathcal{C}$ again lies in $\mathcal{C}$.

Minimisation is the operation which introduces genuine partiality: it says ‘search upwards through $\mathbb{W}$ for the first v making f vanish’, and the search may never terminate. Note the careful clause requiring $f(\mathbf{w}, u)$ to converge for all $u \leq v$: the search must actually be carried out, and it cannot skip over a value at which f diverges.

Definition 27.7.3

A partial function is **recursive** (or **partial recursive**) if it lies in the smallest class containing the basic functions and closed under composition, recursion and minimisation. It is **primitive recursive** if it lies in the smallest class containing the basic functions and closed under composition and recursion.

The class of all partial functions is closed under all three constructions, so the definition makes sense.

Remark. Any class containing the basic functions and closed under composition contains all constant functions $c_{k,v}(\mathbf{w}) = v$: if $\#v = n$ then $c_{k,v} = s^n \circ c_{k,\varepsilon}$.

Every primitive recursive function is total, by induction: the basic functions are total, and composition and recursion of total functions are total. So minimisation is essential if we want to capture partial functions – and, as it happens, if we want to capture all total computable functions either, though we shall not prove that.

Example 27.7.4

Addition is primitive recursive. Interpreting the semiring structure on $\mathbb{W}$ transported from $\mathbb{N}$, we want h with $\#h(w, v) = \#w + \#v$. Take

$$h(w, \varepsilon) = \pi_{1,0}(w) = w, \quad h(w, s(v)) = (s \circ \pi_{3,2})(w, v, h(w, v)) = s(h(w, v)).$$

Here $f = \pi_{1,0}$ is basic and $g = s \circ \pi_{3,2}$ is a composition of basic functions, so h is primitive recursive.

Multiplication is then obtained by recursion from addition, and exponentiation by recursion from multiplication, in the same way. With a little more work, so are the predecessor, truncated subtraction, comparison, the Cantor zigzag function and its inverses, and bounded sums, products and searches.

§27.7.2 Recursion Trees

The definition of ‘recursive’ is a closure, so proofs about recursive functions go by induction on the way a function is built. It is convenient to make the construction into a concrete object.

Attach to each construction a label, an **arity** and a **branching number**:

	label	arity	branching
projection	$B_{k,i}^\pi$	k	0
constant	B_k^c	k	0
successor	B^s	1	0
composition	$C_{n,k}$	k	$n + 1$
recursion	R_k	$k + 1$	2
minimisation	M_k	k	1

Definition 27.7.5 (Recursion tree)

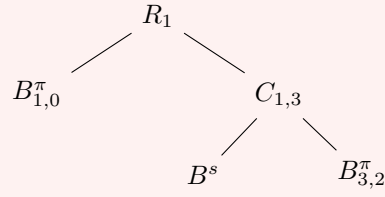
A labelled tree (T, ℓ) is a **recursion tree** if the branching number of each node equals the branching number of its label, and

- (i) if $\ell(t) = C_{n,k}$, the first successor of t has a label of arity n and the remaining n successors have labels of arity k ;
- (ii) if $\ell(t) = R_k$, the first successor has arity k and the second has arity $k + 2$;
- (iii) if $\ell(t) = M_k$, the unique successor has arity $k + 1$.

A recursion tree is **primitive** if no node is labelled M_k .

Example 27.7.6

The recursion tree for the addition function of Example 27.7.4 is



It has no minimisation nodes, so it is primitive, confirming that addition is primitive recursive.

Theorem 27.7.7

A partial function f is recursive if and only if $f = f_{T,\ell}$ for some recursion tree (T, ℓ) , where $f_{T,\ell}$ is computed by reading the tree from the leaves upwards. It is primitive recursive if and only if it admits a primitive such tree.

Proof. Given a recursion tree, define $f_{T,\ell}$ by induction on the height of T : a leaf is labelled by a basic function, and an internal node applies the corresponding construction to the functions attached to its successors, the arity conditions guaranteeing that this makes sense.

Conversely, the class of functions of the form $f_{T,\ell}$ contains the basic functions (one-node trees) and is closed under the three constructions (join trees under a new root with the appropriate label), so it contains all recursive functions. The primitive case is identical. $\square$

§27.7.3 Recursive Functions Are Computable**Theorem 27.7.8**

The computable functions contain the basic functions and are closed under composition, recursion and minimisation. In particular every partial recursive function is computable.

Proof. The basic functions are computable: the projections and constants were dealt with in Section 27.6, and s in Theorem 27.6.9. Closure under composition is Lemma 27.5.13 (with a little bookkeeping to evaluate the g_i into separate scratch registers before applying f).

Recursion. Let f, g be computable and let h be defined by recursion from them; we describe a machine computing $h(\mathbf{w}, v)$. Let k and ℓ be unused empty registers; k will hold a counter and ℓ the current value.

- (i) Compute $f(\mathbf{w})$ and store it in register ℓ . (If $f(\mathbf{w}) \uparrow$ the machine diverges here, which is correct, since then $h(\mathbf{w}, v) \uparrow$ for every v .)
- (ii) If $v = \varepsilon$, output the content of ℓ and halt.
- (iii) Otherwise repeat the following. Let u be the content of ℓ and x the content of

k . Compute $g(\mathbf{w}, x, u)$ and overwrite ℓ with the result. Then compare x with v : if $\#x = \#v - 1$, that is if $s(x) = v$, output ℓ and halt; otherwise apply s to register k and repeat.

After the j th pass through the loop, register ℓ holds $h(\mathbf{w}, s^j(\varepsilon))$, by induction; so the machine halts with $h(\mathbf{w}, v)$ if all the intermediate values converge, and diverges otherwise, as required.

Minimisation. Let f be computable and h obtained from it by minimisation. Let k be an unused empty register, holding ε initially. Repeat: compute $f(\mathbf{w}, u)$ where u is the content of k . If this diverges then the machine diverges, which is correct. If the result is ε , output u and halt. Otherwise apply s to register k and repeat. $\square$

Remark. The proof gives more than the statement: the computable functions are *themselves* closed under recursion and minimisation, so we may use those constructions freely when building register machines. This is what we did in Lemma 27.6.11, Corollary 27.6.12 and Theorem 27.6.13, and there is no circularity, since the proof above uses none of those results.

§27.7.4 Computable Functions Are Recursive

The converse takes more work, and we sketch it; the details are routine but lengthy.

Theorem 27.7.9

Every computable function is partial recursive.

Proof sketch. Fix a register machine M with upper register index m , and code configurations as words as in Section 27.6; write $\gamma(k) = \text{code}(C(k, M, \mathbf{w}))$ for the code of the configuration at time k on input $\mathbf{w}$.

Step 1: the transition is primitive recursive. The function taking $\text{code}(C)$ to $\text{code}(C')$, where M transforms C into C' , is built from operations on words – extracting the state block, comparing it against each of the finitely many program lines of M , and appending or deleting a letter in one register block. Each of these is a bounded search or a case distinction over a fixed finite set, applied to substrings of a word whose length is bounded by that of the input, so each is primitive recursive. (Concretely, one codes a word by its shortlex index and uses the primitive recursive functions of Example 27.7.4 together with bounded minimisation, which preserves primitive recursiveness.)

Step 2: the whole computation is primitive recursive. The function $(\mathbf{w}, v) \mapsto \gamma(\#v)$ is obtained by recursion on v from Step 1, with the base case $\gamma(0) = \text{code}(q_S, \mathbf{w})$, which is primitive recursive. Consequently

$$\theta(\mathbf{w}, v) = \begin{cases} \varepsilon & \text{if the state of } \gamma(\#v) \text{ is } q_H, \\ a & \text{otherwise} \end{cases}$$

is primitive recursive, since checking the state block is a comparison.

Step 3: one minimisation. Applying minimisation to θ yields $\mu(\mathbf{w})$, the least v such that M has reached q_H at time $\#v$, and $\mu(\mathbf{w}) \uparrow$ exactly when M does not halt on

w. Finally

$$f_{M,k}(\mathbf{w}) = \rho(\gamma(\#\mu(\mathbf{w}))),$$

where ρ extracts the block coding register 0, and ρ is primitive recursive. So $f_{M,k}$ is partial recursive. $\square$

Remark. The proof exhibits every computable function in the form

$$f(\mathbf{w}) = \rho(U(\mathbf{w}, \mu v. \theta(\mathbf{w}, v)))$$

with ρ, U, θ primitive recursive and a *single* application of minimisation. This is Kleene's normal form theorem, and it says that all the unboundedness of computation is concentrated in one unbounded search.

§27.7.5 The Church–Turing Thesis

There are other definitions of computability. **Turing machines** have a single two-way infinite tape and a read-write head which may move one cell at a time; **while programs** have variables and the control structures of an ordinary programming language, and halt simply when the instructions run out (so their computation sequences may be finite, unlike ours). Both give rise to notions of computable partial function.

Theorem 27.7.10

For a partial function $f: \mathbb{W}^k \rightarrow \mathbb{W}$, the following are equivalent:

- (i) f is computable by a register machine;
- (ii) f is partial recursive;
- (iii) f is Turing computable;
- (iv) f is while computable.

We have proved (i) $\iff$ (ii); the remaining equivalences are proved by simulation arguments of exactly the same character.

These four models look nothing like each other. One is a machine with stacks, one is a purely syntactic closure of a handful of functions, one is a machine with a tape, and one is a programming language. That they define the same class of functions is striking, and it is not an isolated coincidence: every model of computation which anyone has proposed and which is ‘reasonable’ – meaning, roughly, that each step is finitely describable and effects a bounded change – has turned out to define the same class.

Definition 27.7.11

The **Church–Turing thesis** is the assertion that any function computable by any effective procedure whatsoever is computable in the above sense.

This is not a mathematical statement and cannot be proved, since ‘effective procedure’ is not a mathematical notion; it is a claim that our formal definition has correctly captured an informal one. The evidence for it is the theorem above, together with ninety years of failure to find a counterexample. Its practical use is that, having accepted it, we may describe algorithms in English and conclude that the corresponding functions are

computable, without ever writing down a register machine again. We shall do this freely from now on.

Remark. The thesis also lets us make our decision problems precise, and this is what we do in Section 27.8.7: a problem is *solvable* exactly when the associated set of codes is computable.

§27.8 Computably Enumerable Sets

§27.8.1 Sets with Quantifiers

There is a purely logical description of the computably enumerable sets, and it turns out to be the most useful way of recognising them in practice.

Definition 27.8.1

A set $X \subseteq \mathbb{W}^k$ is

- Σ_1 if there is a computable $Y \subseteq \mathbb{W}^{k+1}$ with

$$\mathbf{w} \in X \iff \exists y ((\mathbf{w}, y) \in Y);$$

we call $X = p(Y) = \{\mathbf{w} \mid \exists y (\mathbf{w}, y) \in Y\}$ the **projection** of Y ;

- Π_1 if its complement is Σ_1 ;
- Δ_1 if it is both Σ_1 and Π_1 .

Remark. The Σ is for the existential quantifier and the Π for the universal, following the analogy between sums and existentials, products and universals. The Δ is for *Durchschnitt*, ‘intersection’. The clash with our use of Σ for the alphabet is unfortunate but standard; the subscript should keep them apart.

Proposition 27.8.2

Every computable set is Δ_1 .

Proof. Computable sets are closed under complement, so it suffices to show every computable X is Σ_1 . Take $Y = \{(\mathbf{w}, y) \mid \mathbf{w} \in X\}$, which is computable (ignore y), and note $p(Y) = X$: we have added a vacuous existential quantifier. $\square$

Theorem 27.8.3

The computably enumerable sets are exactly the Σ_1 sets.

Proof. ($\Rightarrow$) Let X be computably enumerable, say $\psi_X = f_{M,k}$. Let

$$Y = \{(\mathbf{w}, y) \mid t_{M,k}(\mathbf{w}, y) = a\},$$

where $t_{M,k}$ is the truncated computation function of Corollary 27.6.12. Then $t_{M,k}$

is total computable and is a characteristic function of Y , so Y is computable. And

$$\mathbf{w} \in X \iff \psi_X(\mathbf{w}) \downarrow \iff M \text{ halts on } \mathbf{w} \text{ at some time} \iff \exists y ((\mathbf{w}, y) \in Y),$$

so $X = p(Y)$ is Σ_1 .

($\Leftarrow$) Let $X = p(Y)$ with Y computable. Then Y^c is computable, so χ_{Y^c} is computable and total. Apply minimisation to χ_{Y^c} : the resulting $h(\mathbf{w})$ is the least y with $\chi_{Y^c}(\mathbf{w}, y) = \varepsilon$, i.e. the least y with $(\mathbf{w}, y) \in Y$, and diverges if there is none. So h is computable with $\text{dom } h = p(Y) = X$, and X is computably enumerable by Proposition 27.6.4(ii). $\square$

The utility of this is that quantifiers can be manipulated where machines cannot. Here is the basic move.

Example 27.8.4 (The zigzag argument)

Let $f: \mathbb{W}^2 \rightarrow \mathbb{W}$ be computable. Then $X = \{w \mid \exists v f(w, v) \downarrow\}$ is computably enumerable.

Note first that ' $f(w, v) \downarrow$ ' is *not* a computable predicate, so we cannot simply quote the definition. Let $f = f_{M,2}$ and set

$$Z = \{(w, v_0, v_1) \mid t_{M,2}(w, v_0, v_1) = a\},$$

which is computable. Now merge the two existential quantifiers into one by merging the words:

$$Y = \{(w, u) \mid (w, u_{(0)}, u_{(1)}) \in Z\},$$

which is computable since splitting is performable. Then

$$\begin{aligned} w \in X &\iff \exists v_0 \exists v_1 (w, v_0, v_1) \in Z \\ &\iff \exists u (w, u_{(0)}, u_{(1)}) \in Z \\ &\iff \exists u (w, u) \in Y \iff w \in p(Y), \end{aligned}$$

so X is Σ_1 .

The name comes from the picture: to search over pairs (v_0, v_1) we enumerate them along the anti-diagonals of $\mathbb{N}^2$, which lets us make progress on infinitely many computations at once by running each of them for a while. Any two consecutive existential quantifiers can be merged this way, and we use the technique without further comment.

Corollary 27.8.5

The computable sets are exactly the Δ_1 sets.

Proof. One direction is the proposition above. Conversely let X be Δ_1 , so both X and X^c are computably enumerable, say $\psi_X = f_{M,k}$ and $\psi_{X^c} = f_{M',k}$. By Theorem 27.8.3,

$$\mathbf{w} \in X \iff \exists v t_{M,k}(\mathbf{w}, v) = a, \quad \mathbf{w} \notin X \iff \exists v t_{M',k}(\mathbf{w}, v) = a.$$

Define

$$f(\mathbf{w}, v) = \begin{cases} t_{M,k}(\mathbf{w}, v_{(1)}) & \text{if } \#v_{(0)} \text{ is even,} \\ t_{M',k}(\mathbf{w}, v_{(1)}) & \text{if } \#v_{(0)} \text{ is odd,} \end{cases}$$

which is total and computable. Every $\mathbf{w}$ lies in exactly one of X , X^c , so at least one of the two searches succeeds; hence minimising f (with the roles of ε and a interchanged, so that we search for the first v with $f(\mathbf{w}, v) \neq \varepsilon$) always terminates, giving a total computable h . Output a if $\#h(\mathbf{w})_{(0)}$ is even and ε otherwise; this is χ_X . $\square$

The idea is worth restating: if we can semi-decide both a set and its complement, we can decide it, by running both semi-decision procedures in parallel and waiting for one of them to answer. This is sometimes called *Post's theorem*.

Corollary 27.8.6

Σ_1 is not closed under complement.

Proof. $\mathbb{K}$ is Σ_1 but not computable, hence not Δ_1 by Corollary 27.8.5, hence not Π_1 . So $\mathbb{W} \setminus \mathbb{K}$ is Π_1 but not Σ_1 . $\square$

§27.8.2 Closure Properties

Proposition 27.8.7

The computable sets are closed under union, intersection, complement, difference and concatenation.

Proof. For A, B computable, the function

$$\chi_{A \cap B}(\mathbf{w}) = \begin{cases} a & \chi_A(\mathbf{w}) = a \text{ and } \chi_B(\mathbf{w}) = a, \\ \varepsilon & \text{otherwise} \end{cases}$$

is computable: evaluate χ_A into a scratch register, then χ_B , then combine by case distinction. Both terminate, since characteristic functions are total. Complement was Proposition 27.6.4(i), and union and difference follow.

For concatenation, let $A, B \subseteq \mathbb{W}$ and $w \in \mathbb{W}$. There are exactly $|w| + 1$ ways to write $w = vu$, and we can enumerate them; for each, evaluate $\chi_A(v)$ and $\chi_B(u)$, both of which terminate. Output a if some decomposition works. This is a bounded search and always terminates. $\square$

Proposition 27.8.8

The computably enumerable sets are closed under union, intersection and concatenation, but not under complement or difference.

Proof. Failure of complement is Corollary 27.8.6; and since $\mathbb{W}$ is computably enumerable and $\mathbb{W} \setminus A$ is a difference, difference fails too.

Intersection: the function equal to a when $\psi_A(\mathbf{w}) \downarrow$ and $\psi_B(\mathbf{w}) \downarrow$, and undefined

otherwise, is computed by running the two machines one after the other; if either diverges the composite diverges, which is exactly right.

Union: here we may *not* run them one after the other, since if ψ_A diverges we never reach ψ_B . Instead observe that

$$\mathbf{w} \in A \cup B \iff \exists v (t_{M,k}(\mathbf{w}, v) = a \text{ or } t_{M',k}(\mathbf{w}, v) = a),$$

with M, M' machines for ψ_A, ψ_B ; the bracketed set is computable, so $A \cup B$ is Σ_1 . Informally: run both machines in parallel, alternating steps, and halt as soon as either does.

Concatenation: let Z be the set of triples (w, v, u) such that v is an initial segment of w , say $w = vv'$, and both $\psi_A(v)$ and $\psi_B(v')$ have halted by time $\#u$. This is computable, using truncated computation functions. Setting $Y = \{(w, u) \mid (w, u_{(0)}, u_{(1)}) \in Z\}$ and zigzagging, $w \in AB$ if and only if $w \in p(Y)$. $\square$

Proposition 27.8.9

$X \subseteq \mathbb{W}$ is computably enumerable if and only if $X = \text{Im } f$ for some partial computable f .

Proof. If X is computably enumerable then

$$f(w) = \begin{cases} w & \psi_X(w) \downarrow, \\ \uparrow & \text{otherwise} \end{cases}$$

is computable with image X .

Conversely let $X = \text{Im } f$ with $f = f_{c,1}$. Let Z be the set of triples (w, v, u) such that $t_{c,1}(v, u) = a$ and the output of the machine coded by c on input v , at the halting time, is w ; this is computable, since given a bound on the running time we can simply run the machine. Then $w \in X$ if and only if $\exists v \exists u (w, v, u) \in Z$, and zigzagging gives a Σ_1 definition. $\square$

Remark. A stronger statement is true: a nonempty X is computably enumerable if and only if $X = \text{Im } f$ for a *total* computable f , so that $f(\varepsilon), f(s(\varepsilon)), f(s^2(\varepsilon)), \dots$ literally enumerates X , possibly with repetitions. This is what the name ‘computably enumerable’ refers to. The idea is to enumerate pairs (v, u) by zigzag and output the result of f on v if it has halted by time $\#u$, and some fixed element of X otherwise.

§27.8.3 Type 0 Languages

We can now identify the very top of the Chomsky hierarchy.

Theorem 27.8.10

Every type 0 language is computably enumerable.

Proof. Let $G = (\Sigma, V, P, S)$ and enlarge the alphabet to $\Sigma' = \Sigma \cup \{\rightarrow\}$. Code a

derivation $\sigma_0, \sigma_1, \dots, \sigma_n$ as the Σ' -word

$$\sigma_0 \rightarrow \sigma_1 \rightarrow \dots \rightarrow \sigma_n,$$

and call a word of this shape a **derivation code** if consecutive σ_i really are related by a single application of a rule of P . Whether a given word is a derivation code is decidable: split it at the $\rightarrow$ symbols and, for each consecutive pair, search over the finitely many rules and the finitely many positions at which each could have fired. Let

$$Y = \{(w, v) \mid v \text{ is a derivation code with initial string } S \text{ and final string } w\},$$

which is therefore computable. Then $w \in \mathcal{L}(G)$ if and only if $\exists v (w, v) \in Y$, so $\mathcal{L}(G)$ is Σ_1 . $\square$

Theorem 27.8.11

Every computably enumerable $X \subseteq \mathbb{W}$ is a type 0 language.

Proof sketch. Use Theorem 27.7.10 and work with a Turing machine M computing ψ_X ; arrange (by adding a clean-up phase) that the accepting computation on input w takes the tape from $q_S \square w \square$ to $q_H \square a \square$, with the head returned to the front.

A Turing machine *is* a rewrite system: each instruction ‘in state q reading b , write c , move right, go to q' ’ is a rewrite rule on strings which contain a single state symbol marking the head position. So there is a rewrite system R with

$$q_S \square w \square \Rightarrow_R^* q_H \square a \square \iff w \in X.$$

Now build a grammar which runs this backwards: start from S , produce $q_H \square a \square$, apply all the reversed instructions of M (which are again rewrite rules), and when the symbol q_S appears, delete everything except the word w sitting between the two $\square$ s. Then $\mathcal{L}(G) = X$. Working backwards is essential, because a grammar derives *from* a start symbol; the details, which we omit, may be found in Salomaa’s *Formal Languages*. $\square$

Corollary 27.8.12

The type 0 languages are exactly the computably enumerable languages, that is, exactly the Σ_1 languages.

§27.8.4 Many-One Reducibility

Most of our unsolvability proofs so far have had the same shape: ‘if this problem were solvable, so would be the halting problem’. We now name that shape.

Definition 27.8.13 (Reduction)

Let $A, B \subseteq \mathbb{W}$. A total computable $f: \mathbb{W} \rightarrow \mathbb{W}$ is a **reduction** from A to B if

$$w \in A \iff f(w) \in B \quad \text{for all } w \in \mathbb{W},$$

that is, if $A = f^{-1}(B)$. We write $A \leq_m B$ if such an f exists, and say A is **many-one reducible** to B .

The subscript m records that f need not be injective. Intuitively $A \leq_m B$ says that A is no more complicated than B : any method for deciding membership in B yields one for A , by first applying f .

Proposition 27.8.14

The relation $\leq_m$ is reflexive and transitive, and respects complements: $A \leq_m B$ implies $\mathbb{W} \setminus A \leq_m \mathbb{W} \setminus B$. Moreover, if $A \leq_m B$ then

- (i) if B is computable then so is A ;
- (ii) if B is computably enumerable then so is A .

Proof. Reflexivity uses the identity, transitivity composition. The same f witnesses the statement about complements, since $w \notin A \iff f(w) \notin B$. For (i) and (ii), if f is the reduction then $\chi_A = \chi_B \circ f$ and $\psi_A = \psi_B \circ f$, and composites of computable functions are computable. $\square$

Contrapositively – and this is how the proposition is used – if $A \leq_m B$ and A is not computable then B is not computable. In particular:

$$\begin{aligned} \mathbb{K} \leq_m A &\implies A \text{ is not computable,} \\ \mathbb{W} \setminus \mathbb{K} \leq_m A &\implies A \text{ is not computably enumerable.} \end{aligned}$$

Note that $\leq_m$ is *not* antisymmetric, so it is a preorder rather than a partial order. As usual we may quotient: writing $A \equiv_m B$ when $A \leq_m B$ and $B \leq_m A$, the relation $\leq_m$ induces a genuine partial order on the equivalence classes, which are called the **many-one degrees**.

Proposition 27.8.15

Let A be computable and let $B \neq \emptyset, \mathbb{W}$. Then $A \leq_m B$.

Proof. Choose $v \in B$ and $u \notin B$. Since A is computable,

$$f(w) = \begin{cases} v & w \in A, \\ u & w \notin A \end{cases}$$

is total computable, and $w \in A \iff f(w) \in B$. $\square$

So the computable sets (other than $\emptyset$ and $\mathbb{W}$, which are degenerate: nothing reduces to them but themselves) form the bottom degree.

§27.8.5 Degrees of Unsolvability

Definition 27.8.16

Let $\mathcal{C}$ be a class of subsets of $\mathbb{W}$. A set A is **$\mathcal{C}$ -hard** if $B \leq_m A$ for every $B \in \mathcal{C}$, and **$\mathcal{C}$ -complete** if in addition $A \in \mathcal{C}$.

A $\mathcal{C}$ -complete set is a hardest member of $\mathcal{C}$: solving it solves everything in $\mathcal{C}$.

Corollary 27.8.17

Every computable A with $A \neq \emptyset, \mathbb{W}$ is Δ_1 -complete.

Proof. The Δ_1 sets are the computable sets by Corollary 27.8.5, so this is Proposition 27.8.15. $\square$

Theorem 27.8.18

$\mathbb{K}$ is Σ_1 -complete.

Proof. We know $\mathbb{K} \in \Sigma_1$. Let X be any Σ_1 set, so X is computably enumerable, say $X = \text{dom } f$ with f computable. Consider $g(w, u) = f(w)$, which is computable (ignore the second argument). By the s - m - n theorem there is a total computable h with

$$f_{h(w),1}(u) = g(w, u) = f(w).$$

We claim h reduces X to $\mathbb{K}$.

If $w \in X$ then $f(w) \downarrow$, so $f_{h(w),1}$ is the total constant function with value $f(w)$; in particular $f_{h(w),1}(h(w)) \downarrow$, so $h(w) \in \mathbb{K}$.

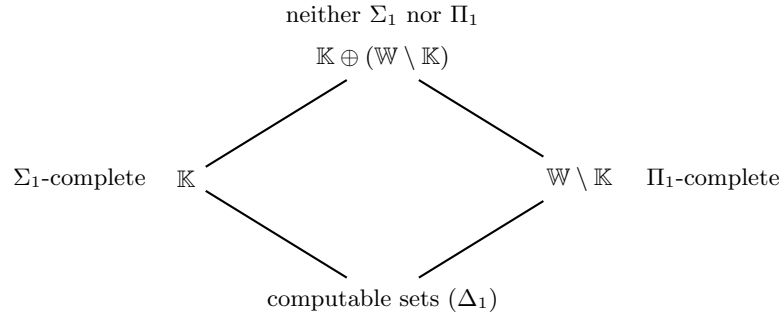
If $w \notin X$ then $f(w) \uparrow$, so $f_{h(w),1}(u) \uparrow$ for every u ; in particular $f_{h(w),1}(h(w)) \uparrow$, so $h(w) \notin \mathbb{K}$. $\square$

Dually, $\mathbb{W} \setminus \mathbb{K}$ is Π_1 -complete. And the two are genuinely different: $\mathbb{W} \setminus \mathbb{K} \not\leq_m \mathbb{K}$, since otherwise $\mathbb{W} \setminus \mathbb{K}$ would be computably enumerable by Proposition 27.8.14; and $\mathbb{K} \not\leq_m \mathbb{W} \setminus \mathbb{K}$ by taking complements. So there are at least three distinct degrees so far. There are more.

Definition 27.8.19

Suppose $\{0, 1\} \subseteq \Sigma$. The **join** of $A, B \subseteq \mathbb{W}$ is $A \oplus B = 0A \cup 1B$.

Then $A \leq_m A \oplus B$ via $w \mapsto 0w$, and likewise $B \leq_m A \oplus B$; and if $A \leq_m C$ and $B \leq_m C$ then $A \oplus B \leq_m C$, by case distinction on the first letter. So $\oplus$ computes least upper bounds in the partial order of degrees, and in particular $\mathbb{K} \oplus (\mathbb{W} \setminus \mathbb{K})$ is a set which is neither Σ_1 nor Π_1 but is above both.



Iterating the construction produces infinitely many degrees, and in fact the structure of the many-one degrees is extremely rich; this is the subject of computability theory proper, which we do not pursue.

§27.8.6 Rice's Theorem

The halting problem is unsolvable. Rice's theorem says, spectacularly, that *every* non-trivial question about the behaviour of a program is unsolvable.

Definition 27.8.20

Words v, u are **weakly equivalent**, written $v \sim u$, if $W_v = W_u$; that is, if the machines they code halt on exactly the same inputs. A set $I \subseteq \mathbb{W}$ is an **index set** if it is closed under weak equivalence.

An index set is thus a set of programs specified purely by the *function* they compute – more precisely, by its domain – rather than by any feature of the code. Index sets correspond exactly to properties of computably enumerable sets.

Example 27.8.21

$\emptyset$ and $\mathbb{W}$ are the trivial index sets. Others are

$$\begin{aligned} \mathbf{Emp} &= \{v \mid W_v = \emptyset\}, & \mathbf{Fin} &= \{v \mid W_v \text{ is finite}\}, \\ \mathbf{Inf} &= \{v \mid W_v \text{ is infinite}\}, & \mathbf{Tot} &= \{v \mid W_v = \mathbb{W}\}. \end{aligned}$$

The emptiness problem for programs is precisely **Emp**.

The key construction is the following. Fix $w \in \mathbb{W}$ and consider

$$g(u, v) = \begin{cases} f_{w,1}(v) & \text{if } u \in \mathbb{K}, \\ \uparrow & \text{if } u \notin \mathbb{K}. \end{cases}$$

This looks like an illegal case distinction, since membership of $\mathbb{K}$ is not computable. But it *is* computable, because we do not need to decide the case: on input (u, v) , run the machine coded by u on input u ; if and when it halts, run the machine coded by w on input v and return the answer. If $u \notin \mathbb{K}$ the first phase never terminates, which produces exactly the divergence we want. This is a trick worth remembering: an uncomputable case distinction is harmless when the ‘false’ branch is divergence.

By the s - m - n theorem there is a total computable h with $f_{h(u),1}(v) = g(u, v)$, so

$$W_{h(u)} = \begin{cases} W_w & u \in \mathbb{K}, \\ \emptyset & u \notin \mathbb{K}. \end{cases}$$

Theorem 27.8.22 (Rice)

No non-trivial index set is computable.

Proof. Let I be an index set with $I \neq \emptyset, \mathbb{W}$. Fix $e \in \mathbb{W}$ with $W_e = \emptyset$ (for instance the code of a machine which never halts). Either $e \in I$ or not.

Case $e \in I$. Since $I \neq \mathbb{W}$, choose $w \notin I$; then $W_w \neq \emptyset$, since $W_w = \emptyset$ would give $w \sim e$ and hence $w \in I$. Build h as above from this w . If $u \in \mathbb{K}$ then $W_{h(u)} = W_w$, so $h(u) \sim w$ and $h(u) \notin I$; if $u \notin \mathbb{K}$ then $W_{h(u)} = \emptyset = W_e$, so $h(u) \sim e$ and $h(u) \in I$. Hence

$$u \in \mathbb{W} \setminus \mathbb{K} \iff h(u) \in I,$$

so $\mathbb{W} \setminus \mathbb{K} \leq_m I$ and I is not computable.

Case $e \notin I$. Since $I \neq \emptyset$, choose $w \in I$; again $W_w \neq \emptyset$. With the same h : if $u \in \mathbb{K}$ then $h(u) \sim w$ so $h(u) \in I$, and if $u \notin \mathbb{K}$ then $h(u) \sim e$ so $h(u) \notin I$. Hence $\mathbb{K} \leq_m I$, and again I is not computable. $\square$

Remark. The proof gives more than the statement. If $e \in I$ then $\mathbb{W} \setminus \mathbb{K} \leq_m I$, so I is not even computably enumerable. If $e \notin I$ then $\mathbb{K} \leq_m I$, so I is not Π_1 ; but this leaves open whether I is computably enumerable.

Applying this: $e \in \mathbf{Emp}$ and $e \in \mathbf{Fin}$, so neither **Emp** nor **Fin** is computably enumerable. On the other hand $e \notin \mathbf{Inf}$ and $e \notin \mathbf{Tot}$, so we obtain only $\mathbb{K} \leq_m \mathbf{Inf}$ and $\mathbb{K} \leq_m \mathbf{Tot}$.

Corollary 27.8.23

None of **Emp**, **Fin**, **Inf**, **Tot** is computable.

Theorem 27.8.24

Fin is neither Σ_1 nor Π_1 .

Proof. By Remark , $\mathbb{W} \setminus \mathbb{K} \leq_m \mathbf{Fin}$, so **Fin** is not Σ_1 . For the other half we show $\mathbb{K} \leq_m \mathbf{Fin}$. Consider

$$g(w, v) = \begin{cases} \uparrow & \text{if } t_{w,1}(w, v) = a, \\ \varepsilon & \text{otherwise,} \end{cases}$$

which is computable: compute the (total) truncated computation function and then either diverge or halt. By the s - m - n theorem take total computable h with $f_{h(w),1}(v) = g(w, v)$.

If $w \in \mathbb{K}$, let v_0 be the halting time of $f_{w,1}(w)$. Then $t_{w,1}(w, v) = a$ for all v with $\#v > \#v_0$, so $f_{h(w),1}(v) \uparrow$ for all such v ; hence $W_{h(w)}$ is contained in the finite set

of v with $\#v \leq \#v_0$, and is finite. So $h(w) \in \mathbf{Fin}$.

If $w \notin \mathbb{K}$ then $t_{w,1}(w, v) = \varepsilon$ for every v , so $f_{h(w),1}$ is the total constant function ε and $W_{h(w)} = \mathbb{W}$, which is infinite. So $h(w) \notin \mathbf{Fin}$.

Hence h is a reduction from $\mathbb{K}$ to $\mathbf{Fin}$, and $\mathbf{Fin}$ is not Π_1 . $\square$

Remark. By contrast, one can show that $\mathbf{Emp} \equiv_m \mathbb{W} \setminus \mathbb{K}$, so $\mathbf{Emp}$ is Π_1 -complete, while $\mathbf{Tot}$ and $\mathbf{Inf}$ are, like $\mathbf{Fin}$, in neither class. The hierarchy of Σ_n and Π_n sets obtained by allowing n alternating quantifiers – the *arithmetical hierarchy* – is where these sets properly live.

§27.8.7 Solvability of Decision Problems

We can now return to the decision problems of Section 27.4 with a precise definition in hand. Fix a coding of grammars such that every word w codes a grammar G_w and every grammar is G_w for some w ; this is arranged by decreeing that words which are not syntactically valid code some fixed trivial grammar. Then

- (i) the word problem is the set $\{(w, v) \mid w \in \mathcal{L}(G_v)\}$;
- (ii) the emptiness problem is the set $\{w \mid \mathcal{L}(G_w) = \emptyset\}$;
- (iii) the equivalence problem is the set $\{(w, v) \mid \mathcal{L}(G_w) = \mathcal{L}(G_v)\}$;

and each is **solvable** exactly when the corresponding set is computable.

Theorem 27.8.25

The word problem for type 0 grammars is unsolvable.

Proof. Let $W = \{(w, v) \mid w \in \mathcal{L}(G_v)\}$ and suppose it computable. Define

$$f(w) = \begin{cases} \uparrow & w \in \mathcal{L}(G_w), \\ a & w \notin \mathcal{L}(G_w). \end{cases}$$

Since W is computable this is a legitimate case distinction, so f is computable, and $\text{dom } f$ is computably enumerable. By Theorem 27.8.11 there is a grammar G with $\mathcal{L}(G) = \text{dom } f$, say $G = G_d$. But then

$$d \in \mathcal{L}(G_d) \iff d \in \text{dom } f \iff f(d) \downarrow \iff d \notin \mathcal{L}(G_d),$$

a contradiction. $\square$

Corollary 27.8.26

The emptiness problem for type 0 grammars is unsolvable.

Proof. By Theorems 27.8.10 and 27.8.11, $\{\mathcal{L}(G_w) \mid w \in \mathbb{W}\}$ is exactly the collection of computably enumerable sets, and translating between a grammar and a machine for the same language is effective in both directions. So the emptiness problem for grammars is many-one equivalent to $\mathbf{Emp}$, which is not computable by Rice's theorem. $\square$

Corollary 27.8.27

The equivalence problem for type 0 grammars is unsolvable.

Proof. Consider $\mathbf{Eq} = \{(w, v) \mid W_w = W_v\}$, and fix e with $W_e = \emptyset$. The function $g(w) = (w, e)$ is total computable, and

$$w \in \mathbf{Emp} \iff W_w = \emptyset = W_e \iff (w, e) \in \mathbf{Eq},$$

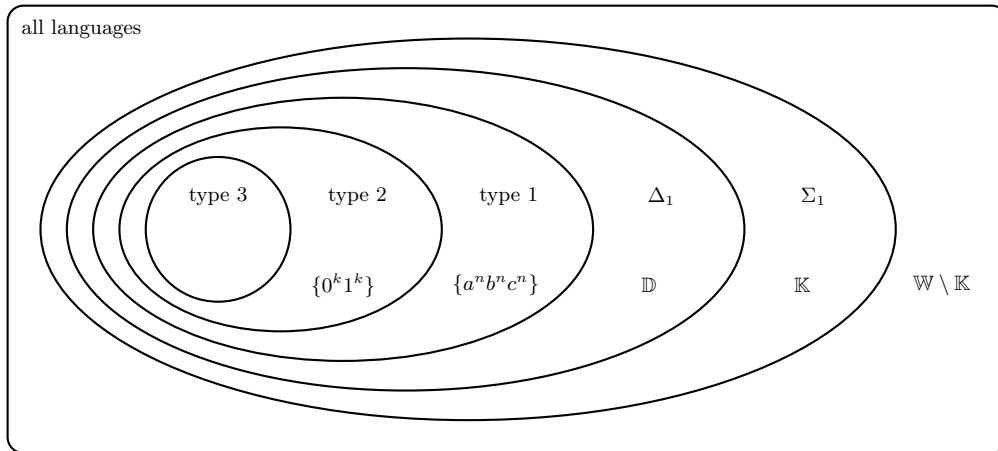
so $\mathbf{Emp} \leq_m \mathbf{Eq}$ and $\mathbf{Eq}$ is not computable. Translating to grammars as above gives the result. $\square$

So all three of our problems, solvable for regular grammars and two of them for context-free grammars, become unsolvable at the top of the hierarchy.

§27.9 The Relationship Between the Language Classes

We close by drawing the whole picture together. We have met the following classes of languages over a fixed alphabet Σ , in increasing order:

$$\begin{aligned} \text{regular} &\subseteq \text{context-free} \subseteq \text{context-sensitive} \\ &\subseteq \text{computable} \subseteq \text{computably enumerable} \subseteq \mathcal{P}(\mathbb{W}). \end{aligned}$$



Here type 3, 2, 1 are the regular, context-free and context-sensitive languages, Δ_1 the computable languages, and Σ_1 the computably enumerable languages, which are exactly the type 0 languages.

Every inclusion is strict, and each strictness has been (or can be) witnessed:

- $\{0^k 1^k \mid k > 0\}$ is context-free but not regular, by the pumping lemma for regular languages (Example 27.3.21) or by Myhill–Nerode.
- $\{a^n b^n c^n \mid n > 0\}$ is context-sensitive but not context-free, by the pumping lemma for context-free languages (Example 27.4.16).
- There is a computable language $\mathbb{D}$ which is not context-sensitive. Indeed, the context-sensitive grammars can be effectively enumerated as $G_0, G_1, \dots$, and the word problem for each is uniformly solvable by Corollary 27.2.16; so, listing the

words of $\mathbb{W}$ in shortlex order as $w_0, w_1, \dots$, the diagonal language

$$\mathbb{D} = \{w_n \mid w_n \notin \mathcal{L}(G_n)\}$$

is computable but differs from every $\mathcal{L}(G_n)$.

- $\mathbb{K}$ is computably enumerable but not computable (Theorem 27.6.18).
- $\mathbb{W} \setminus \mathbb{K}$ is a language which is not computably enumerable, hence not of the form $\mathcal{L}(G)$ for any grammar at all (Corollary 27.8.6).
- Finally, there are uncountably many languages and only countably many grammars, so almost every language lies outside the outermost ellipse.

Closure properties and decision problems degrade as we ascend:

	union	concat.	Kleene +	intersection	complement
regular	yes	yes	yes	yes	yes
context-free	yes	yes	yes	<i>no</i>	<i>no</i>
context-sensitive	yes	yes	yes	yes	yes
computable	yes	yes	yes	yes	yes
computably enumerable	yes	yes	yes	yes	<i>no</i>

	word problem	emptiness	equivalence
regular	solvable	solvable	solvable
context-free	solvable	solvable	<i>unsolvable</i>
context-sensitive	solvable	<i>unsolvable</i>	<i>unsolvable</i>
computably enumerable	<i>unsolvable</i>	<i>unsolvable</i>	<i>unsolvable</i>

The context-free row is anomalous in the first table: it is the only class in the list failing to be closed under intersection, and we saw in Section 27.4 that this is forced by the pushdown model, whose single stack cannot be duplicated. The last row of the second table is the content of Section 27.8.7.

There is a pleasing overall moral. Each class is characterised in two ways, by a grammar and by a machine:

class	grammars	machines
regular	regular grammars	finite automata
context-free	context-free grammars	pushdown automata
context-sensitive	noncontracting grammars	linear bounded automata
computably enumerable	arbitrary grammars	register (or Turing) machines

and the amount of memory available to the machine – none, a stack, a tape of length proportional to the input, an unbounded tape – is exactly what determines the class. That is the shape of the subject: a hierarchy of languages, a matching hierarchy of machines, and at the top, an unavoidable wall of undecidability.

28 Coding and Cryptography

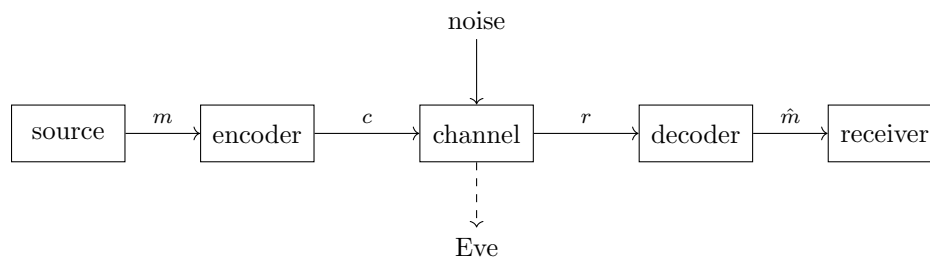
Communication is a mathematical problem. Somebody wants to say something to somebody else; between them lies a channel that is expensive, or unreliable, or overheard, or all three at once. Coding theory is the study of how to package a message so that it survives the journey, and cryptography is the study of how to package it so that only the intended recipient can read it. Both subjects turn out to be, at bottom, about the same thing: the arithmetic of finite sets, and the surprising leverage that a little algebra and a little probability give us over them.

The most striking result in the course is due to Shannon, and it says something that ought to be false. If a channel corrupts one bit in ten, entirely at random, one might reasonably guess that transmitting a long message without error requires sending it over and over again, so that the rate at which information gets through must fall to zero. It does not. There is a positive number – the *capacity* of the channel – below which arbitrarily reliable communication is possible, and Shannon’s theorem tells us exactly what that number is. What makes the proof so memorable is that it establishes the existence of extremely good codes by choosing one at random, without ever exhibiting a single one.

This article constitutes my notes for the ‘Coding and Cryptography’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§28.1 Modelling Communication

Before we can prove anything we need a model, and the one we shall use is the one Shannon drew in 1948. There is a **source** which produces a message; an **encoder** which turns that message into something suitable for transmission; a **channel** down which the transmission travels and in which things go wrong; a **decoder** which tries to undo the damage; and finally a **receiver**, for whom the whole business was arranged.



By long-standing convention the source is called Alice, the receiver is called Bob, and any third party listening in on the channel is called Eve. The two halves of the course correspond to the two things that can go wrong in the middle of that diagram. In coding theory the enemy is the noise, and we want the message to arrive *economically* and *reliably*. In cryptography the enemy is Eve, and we want the message to arrive *privately*.

It is worth separating two quite different problems which are often confused. **Noiseless coding**, or source coding, is about compression: we want to spend as few symbols as

possible on a message, and it is adapted to the *source*. Morse code is the canonical example – the letter E, the commonest letter in English, gets the single dot, and the letter Q gets four symbols. **Noisy coding**, or channel coding, is about redundancy: we want to add enough extra symbols that corruption can be spotted and undone, and it is adapted to the *channel*.

Example 28.1.1 (ISBNs)

Every book (before 2007) carried a ten-digit International Standard Book Number $a_1a_2 \dots a_{10}$, where $a_1, \dots, a_9 \in \{0, 1, \dots, 9\}$ and $a_{10} \in \{0, 1, \dots, 9, X\}$, subject to the single condition

$$\sum_{j=1}^{10} ja_j \equiv 0 \pmod{11}.$$

Here X stands for 10, and the modulus 11 is prime, which is what makes everything work. Suppose one digit is mistyped, say a_k becomes a'_k . Then the check sum changes by $k(a_k - a'_k)$, and since $1 \leq k \leq 10$ and $|a_k - a'_k| \leq 10$, neither factor is divisible by 11, so the sum is no longer 0 modulo 11 and the error is detected. Similarly, if two adjacent digits are transposed, say a_k and a_{k+1} swap, the sum changes by $a_{k+1} - a_k$, again nonzero modulo 11 unless the digits were equal in the first place.

The channel here is a human being reading a number off a page and typing it into a computer, and the code is beautifully adapted to the errors that channel makes.

We now make the notion of a channel precise.

Definition 28.1.2 (Communication Channel)

A **communication channel** accepts strings of symbols from a finite **input alphabet** $\mathcal{A} = \{a_1, \dots, a_r\}$ and outputs strings of symbols from a finite **output alphabet** $\mathcal{B} = \{b_1, \dots, b_s\}$. It is specified by the probabilities

$$\mathbb{P}(y_1 \dots y_n \text{ received} \mid x_1 \dots x_n \text{ sent}).$$

In this generality nothing can be said, because the channel is allowed to remember everything it has ever done. Almost always we assume much more.

Definition 28.1.3 (Discrete Memoryless Channel)

A **discrete memoryless channel** (DMC) is a channel for which the probabilities

$$p_{ij} = \mathbb{P}(b_j \text{ received} \mid a_i \text{ sent})$$

are the same on every use of the channel, and are independent of all past and future uses. The $r \times s$ matrix $P = (p_{ij})$ is the **channel matrix**; each of its rows sums to 1.

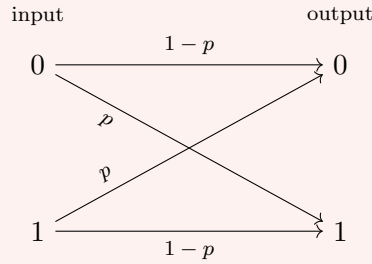
Two examples will accompany us for the rest of the course.

Example 28.1.4 (Binary Symmetric Channel)

The **binary symmetric channel** (BSC) with error probability $p \in [0, 1]$ has $\mathcal{A} = \mathcal{B} = \{0, 1\}$ and channel matrix

$$P = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}.$$

Each bit is flipped independently with probability p , and transmitted faithfully with probability $1 - p$.



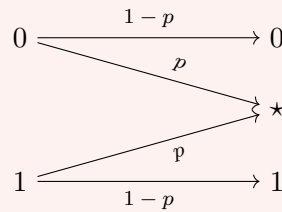
We shall almost always assume $p < \frac{1}{2}$. This costs nothing: a channel with $p > \frac{1}{2}$ becomes one with error probability $1 - p$ if Bob simply inverts everything he receives, and we shall see shortly that a channel with $p = \frac{1}{2}$ is useless, in that it carries no information whatsoever.

Example 28.1.5 (Binary Erasure Channel)

The **binary erasure channel** (BEC) has $\mathcal{A} = \{0, 1\}$ and $\mathcal{B} = \{0, 1, \star\}$, with channel matrix

$$P = \begin{pmatrix} 1-p & 0 & p \\ 0 & 1-p & p \end{pmatrix}.$$

Here p is the probability that the symbol arrives illegibly; receiving $\star$ is called a **splurge error**. This channel never lies to us, it merely occasionally declines to speak.



To use a channel n times we pass to the **n th extension**, whose input alphabet is $\mathcal{A}^n$, whose output alphabet is $\mathcal{B}^n$, and whose channel matrix (for a DMC) is

$$\mathbb{P}(y_1 \dots y_n \mid x_1 \dots x_n) = \prod_{i=1}^n \mathbb{P}(y_i \mid x_i).$$

Definition 28.1.6 (Code, Rate, Error Rate)

A **code** C of length n is a function $\mathcal{M} \rightarrow \mathcal{A}^n$, where $\mathcal{M}$ is a set of messages; implicitly we also fix a decoding rule $\mathcal{B}^n \rightarrow \mathcal{M}$. We usually identify C with its image, a subset of $\mathcal{A}^n$.

- The **size** of C is $m = |\mathcal{M}|$.
- The **information rate** of C is $\rho(C) = \frac{1}{n} \log_2 m$.
- The **error rate** of C is $\hat{e}(C) = \max_{x \in \mathcal{M}} \mathbb{P}(\text{error} \mid x \text{ sent})$.

Throughout these notes $\log$ means $\log_2$ unless we say otherwise; this is the convention that makes the answers come out in bits.

Definition 28.1.7 (Capacity)

A channel can **transmit reliably at rate R** if there is a sequence of codes $(C_n)_{n \geq 1}$, with C_n of length n , such that

$$\lim_{n \rightarrow \infty} \rho(C_n) = R \quad \text{and} \quad \lim_{n \rightarrow \infty} \hat{e}(C_n) = 0.$$

The **capacity** of the channel is the supremum of all rates at which it can transmit reliably.

Note carefully what is being asked. We are not asking for one good code; we are asking for a whole family of codes whose rate stays bounded away from zero while the error probability is driven to zero. It is far from obvious that this can be done at all for a channel that makes errors, and the fact that it can – that the binary symmetric channel with $p < \frac{1}{2}$ has strictly positive capacity – is Shannon’s noisy coding theorem, which we prove in Section 28.4.7.

§28.2 Noiseless Coding

We begin with the easier half of the problem, in which the channel is perfect and the only question is economy. A source emits symbols from an alphabet $\mathcal{A}$, some more often than others, and we want to represent them as short strings over a transmission alphabet $\mathcal{B}$.

§28.2.1 Codes and Decipherability

Let $\mathcal{B}$ be a finite alphabet. Write $\mathcal{B}^* = \bigcup_{n \geq 0} \mathcal{B}^n$ for the set of all finite strings over $\mathcal{B}$, including the empty string. The **concatenation** of $x = x_1 \dots x_r$ and $y = y_1 \dots y_s$ is $xy = x_1 \dots x_r y_1 \dots y_s$.

Definition 28.2.1 (Code)

Let $\mathcal{A}$ and $\mathcal{B}$ be alphabets. A **code** is a function $c: \mathcal{A} \rightarrow \mathcal{B}^*$. The elements of the image of c are the **codewords**.

We write $m = |\mathcal{A}|$ and $a = |\mathcal{B}|$, and call c an **a -ary code of size m** . A 2-ary code is *binary*, a 3-ary code is *ternary*.

Example 28.2.2 (The Greek Fire Code)

Take $\mathcal{A} = \{\alpha, \beta, \dots, \omega\}$, the 24 letters of the Greek alphabet, and $\mathcal{B} = \{1, 2, 3, 4, 5\}$. Arrange the letters in a 5×5 square and encode each by its coordinates, so $c(\alpha) = 11$,

$c(\beta) = 12$, and so on. To transmit the pair xy one holds up x torches on one hilltop and y torches on another. This scheme is described by Polybius, and is very probably the oldest code in these notes by some two thousand years.

A code is only useful if we can send more than one symbol. Given a message $x_1x_2 \dots x_n \in \mathcal{A}^*$ we transmit $c(x_1)c(x_2) \dots c(x_n)$, so that c extends to a map

$$c^*: \mathcal{A}^* \rightarrow \mathcal{B}^*.$$

Definition 28.2.3 (Decipherable)

A code c is **decipherable**, or **uniquely decodable**, if c^* is injective.

It is not enough for c itself to be injective, as the following shows.

Example 28.2.4

Let $\mathcal{A} = \{1, 2, 3, 4\}$ and $\mathcal{B} = \{0, 1\}$, and set

$$c(1) = 0, \quad c(2) = 1, \quad c(3) = 00, \quad c(4) = 01.$$

This is injective, but $c^*(114) = 0001 = 0001$ and $c^*(312) = 0001 = 0001$, so c^* is not. On receiving 0001 Bob has no way to tell which message was meant.

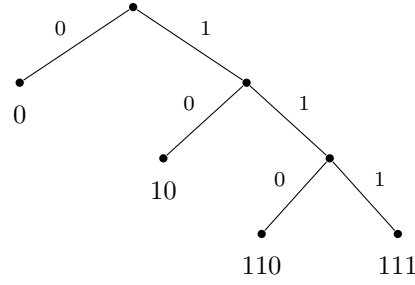
There are three easy ways of guaranteeing decipherability, assuming c injective.

- (i) A **block code**, in which all codewords have the same length. The Greek fire code is of this kind.
- (ii) A **comma code**, which reserves one letter of $\mathcal{B}$ to mark the end of a codeword and never uses it otherwise.
- (iii) A **prefix-free code**, in which no codeword is an initial segment of another.

Block codes and comma codes are both special cases of prefix-free codes. Prefix-free codes have the pleasant property that Bob can decode as he reads: the moment the symbols he has seen form a codeword, he knows that codeword has ended, because no longer codeword begins that way. For this reason they are also called **instantaneous** codes.

Not every decipherable code is prefix-free. For instance $c(1) = 0$, $c(2) = 01$, $c(3) = 011$, $c(4) = 0111$ is decipherable – each 0 marks the start of a new codeword – but $c(1)$ is a prefix of $c(2)$. Here Bob must look ahead to the next 0 before committing. Remarkably, we shall see in a moment that allowing such lookahead buys us precisely nothing.

The right way to picture a prefix-free binary code is as a rooted binary tree. Reading a codeword left to right tells us which way to turn at each vertex, and the prefix-free condition says exactly that all the codewords sit at *leaves*.



§28.2.2 Kraft's Inequality

How short can the codewords of a prefix-free code be? Not arbitrarily short: the tree picture makes it clear that a short codeword uses up a lot of the tree. The following inequality measures exactly how much.

Definition 28.2.5 (Kraft's Inequality)

Let $c: \mathcal{A} \rightarrow \mathcal{B}^*$ be a code with $|\mathcal{A}| = m$ and $|\mathcal{B}| = a$, whose codewords have lengths $\ell_1, \dots, \ell_m$. We say that c satisfies **Kraft's inequality** if

$$\sum_{i=1}^m a^{-\ell_i} \leq 1.$$

Theorem 28.2.6 (Kraft)

Given lengths $\ell_1, \dots, \ell_m$, there is a prefix-free a -ary code with these codeword lengths if and only if Kraft's inequality holds.

Proof. Let $s = \max_i \ell_i$ and let n_ℓ be the number of i with $\ell_i = \ell$, so that Kraft's inequality reads $\sum_{\ell=1}^s n_\ell a^{-\ell} \leq 1$, or after multiplying by a^s ,

$$n_1 a^{s-1} + n_2 a^{s-2} + \dots + n_{s-1} a + n_s \leq a^s. \quad (*)$$

Suppose first that c is prefix-free. A codeword of length ℓ is a prefix of exactly $a^{s-\ell}$ strings of length s , and by the prefix-free condition no string of length s has two different codewords as prefixes. So the left hand side of $(*)$ counts, without repetition, some of the strings of length s , of which there are a^s in all. Hence $(*)$ holds.

Conversely suppose $(*)$ holds; we build a prefix-free code with n_ℓ codewords of length ℓ for each ℓ , by induction on s . If $s = 1$ the inequality says $n_1 \leq a$ and we simply choose n_1 of the a letters. For $s > 1$, the inductive hypothesis applied to the lengths at most $s - 1$ gives a prefix-free code $\hat{c}$ with n_ℓ codewords of length ℓ for every $\ell < s$. Now the strings of length s having some codeword of $\hat{c}$ as a prefix number exactly

$$n_1 a^{s-1} + \dots + n_{s-1} a,$$

so by $(*)$ at least n_s strings of length s remain untouched. Choosing n_s of them and adjoining them to $\hat{c}$ gives a code which is still prefix-free, since none of the new

codewords extends an old one (they were chosen to avoid this) and none of the old codewords extends a new one (they are shorter). $\square$

Remark. The proof is constructive, and gives a perfectly practical algorithm: list the required lengths in increasing order and greedily assign codewords, at each stage taking any string of the right length that does not extend a codeword already chosen. Kraft's inequality guarantees that we never run out.

§28.2.3 McMillan's Inequality

Now we come to the result promised above, which says that lookahead is worthless.

Theorem 28.2.7 (McMillan)

Every decipherable code satisfies Kraft's inequality.

Proof. Let $c: \mathcal{A} \rightarrow \mathcal{B}^*$ be decipherable with word lengths $\ell_1, \dots, \ell_m$, and set $s = \max_i \ell_i$. The trick is to consider the R th power of the quantity in question. For any $R \in \mathbb{N}$,

$$\left(\sum_{i=1}^m a^{-\ell_i} \right)^R = \sum_{\ell=R}^{Rs} b_\ell a^{-\ell},$$

where b_ℓ is the number of ways of choosing an ordered R -tuple of codewords whose total length is ℓ ; this is just what happens when we expand the product. Now a tuple of R codewords of total length ℓ concatenates to a string in $\mathcal{B}^\ell$, and because c is decipherable, distinct tuples give distinct strings. Hence

$$b_\ell \leq |\mathcal{B}^\ell| = a^\ell,$$

and so each term $b_\ell a^{-\ell}$ is at most 1. There are at most Rs terms, so

$$\left(\sum_{i=1}^m a^{-\ell_i} \right)^R \leq Rs, \quad \text{that is} \quad \sum_{i=1}^m a^{-\ell_i} \leq (Rs)^{1/R}.$$

Letting $R \rightarrow \infty$, the right hand side tends to 1, and Kraft's inequality follows. $\square$

The proof is a small piece of magic: an inequality that fails to hold is amplified by exponentiation into an absurdity, while the polynomial factor Rs is powerless to keep up.

Corollary 28.2.8

A decipherable code with word lengths $\ell_1, \dots, \ell_m$ exists if and only if a prefix-free code with the same word lengths exists.

Proof. A prefix-free code is decipherable. Conversely, a decipherable code satisfies Kraft's inequality by McMillan, and then Kraft's theorem produces a prefix-free code with the same lengths. $\square$

So from the point of view of word lengths – which is the only point of view that matters for economy – we may as well restrict attention to prefix-free codes, and we shall.

§28.2.4 Entropy

We now need a way of saying how much information a source actually produces, so that we know what we are aiming at. The right notion is Shannon's **entropy**. Informally, the entropy of a random variable X is the expected number of fair coin tosses needed to determine its value.

Example 28.2.9

Suppose X takes four values with probability $\frac{1}{4}$ each. Identifying the values with 00, 01, 10, 11, two coin tosses always suffice and never fewer, so the entropy should be 2.

Suppose instead X takes four values with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$. Identify them with the strings 0, 10, 110, 111: toss a coin, and if it comes up 0 we are done, otherwise toss again, and so on. The expected number of tosses is

$$\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3 = \frac{7}{4}.$$

The second variable is in this sense less random than the first.

Both examples are instances of the following.

Definition 28.2.10 (Entropy)

Let X be a random variable taking finitely many values $x_1, \dots, x_n$ with probabilities $p_1, \dots, p_n$. The **entropy** of X is

$$H(X) = H(p_1, \dots, p_n) = - \sum_{i=1}^n p_i \log p_i = \mathbb{E}[-\log p(X)],$$

where $\log = \log_2$ and we adopt the convention $0 \log 0 = 0$.

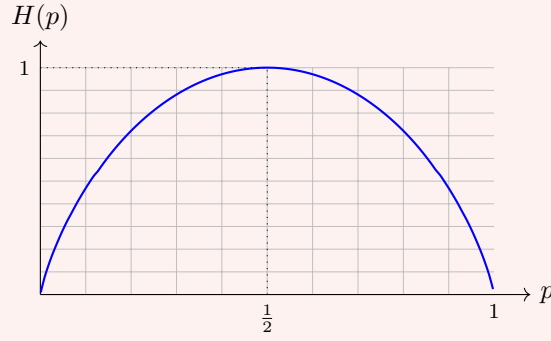
The convention is forced on us by continuity, since $x \log x \rightarrow 0$ as $x \rightarrow 0^+$. Entropy is measured in **bits**. Clearly $H(X) \geq 0$, with equality if and only if some $p_i = 1$, that is, if and only if X is constant.

Example 28.2.11 (The Binary Entropy Function)

For a biased coin with probability p of heads we write

$$H(p) = H(p, 1-p) = -p \log p - (1-p) \log(1-p).$$

Differentiating, $H'(p) = \log \frac{1-p}{p}$, which is positive for $p < \frac{1}{2}$ and negative for $p > \frac{1}{2}$; and $H''(p) < 0$ throughout, so H is strictly concave with a maximum of 1 at $p = \frac{1}{2}$, vanishing at $p = 0$ and $p = 1$.



This little function is going to appear everywhere, so it is worth committing its shape to memory: symmetric about $\frac{1}{2}$, with infinite slope at the two ends.

The following inequality is the workhorse of the whole subject.

Proposition 28.2.12 (Gibbs' Inequality)

Let $(p_1, \dots, p_n)$ and $(q_1, \dots, q_n)$ be probability distributions. Then

$$-\sum_{i=1}^n p_i \log p_i \leq -\sum_{i=1}^n p_i \log q_i,$$

with equality if and only if $p_i = q_i$ for all i .

The right hand side is the **cross entropy** of q relative to p ; the difference between the two sides is called the relative entropy, or Kullback–Leibler divergence, and Gibbs' inequality says it is non-negative.

Proof. Since $\log x = \ln x / \ln 2$ we may work with natural logarithms throughout. Let $I = \{i : p_i \neq 0\}$. The elementary inequality $\ln x \leq x - 1$ for $x > 0$, with equality only at $x = 1$, gives for $i \in I$

$$\ln \frac{q_i}{p_i} \leq \frac{q_i}{p_i} - 1,$$

and multiplying by $p_i > 0$ and summing over I ,

$$\sum_{i \in I} p_i \ln \frac{q_i}{p_i} \leq \sum_{i \in I} q_i - \sum_{i \in I} p_i \leq 1 - 1 = 0,$$

using $\sum_{i \in I} p_i = 1$ and $\sum_{i \in I} q_i \leq 1$. Therefore

$$-\sum_{i=1}^n p_i \ln p_i = -\sum_{i \in I} p_i \ln p_i \leq -\sum_{i \in I} p_i \ln q_i \leq -\sum_{i=1}^n p_i \ln q_i,$$

the last step because the omitted terms have $p_i = 0$. If equality holds throughout we need $\sum_{i \in I} q_i = 1$ and $q_i/p_i = 1$ for every $i \in I$; the first forces $q_i = 0$ for $i \notin I$, and the second gives $p_i = q_i$ on I . $\square$

Corollary 28.2.13

$H(p_1, \dots, p_n) \leq \log n$, with equality if and only if $p_1 = \dots = p_n = \frac{1}{n}$.

Proof. Take $q_i = \frac{1}{n}$ in Gibbs' inequality: the right hand side is $-\sum p_i \log \frac{1}{n} = \log n$. $\square$

In other words, the uniform distribution is the most random one, which is what one would hope.

§28.2.5 Optimal Codes and the Noiseless Coding Theorem

Fix an alphabet $\mathcal{A} = \{\mu_1, \dots, \mu_m\}$ of $m \geq 2$ messages and a transmission alphabet $\mathcal{B}$ with $a = |\mathcal{B}| \geq 2$. Let X be a random variable on $\mathcal{A}$ with $\mathbb{P}(X = \mu_i) = p_i$. A code c with word lengths $\ell_1, \dots, \ell_m$ has expected word length

$$\mathbb{E}[S] = \sum_{i=1}^m p_i \ell_i.$$

Definition 28.2.14 (Optimal Code)

A code $c: \mathcal{A} \rightarrow \mathcal{B}^*$ is **optimal** if it has the smallest expected word length among all decipherable codes for X .

We can now say precisely how well one can do.

Theorem 28.2.15 (Shannon's Noiseless Coding Theorem)

The expected word length $\mathbb{E}[S]$ of an optimal code satisfies

$$\frac{H(X)}{\log a} \leq \mathbb{E}[S] < \frac{H(X)}{\log a} + 1.$$

Moreover the lower bound holds for *every* decipherable code, and is attained exactly when $p_i = a^{-\ell_i}$ for integers ℓ_i (so that in particular $\sum_i a^{-\ell_i} = 1$).

Proof. The lower bound. Let c be an arbitrary decipherable code, with word lengths $\ell_1, \dots, \ell_m$. Put $D = \sum_i a^{-\ell_i}$ and $q_i = a^{-\ell_i}/D$, so that (q_i) is a probability distribution. By Gibbs' inequality,

$$\begin{aligned} H(X) &\leq -\sum_i p_i \log q_i = -\sum_i p_i (-\ell_i \log a - \log D) \\ &= \log D + \log a \sum_i p_i \ell_i. \end{aligned}$$

By McMillan's inequality $D \leq 1$, so $\log D \leq 0$, and therefore

$$H(X) \leq \log a \cdot \mathbb{E}[S],$$

which is the lower bound. Equality forces $D = 1$ and $p_i = q_i = a^{-\ell_i}$.

The upper bound. It suffices to exhibit *some* decipherable code beating the bound, since an optimal code can only be better. Set

$$\ell_i = \lceil -\log_a p_i \rceil,$$

so that $-\log_a p_i \leq \ell_i < -\log_a p_i + 1$. From the first inequality $p_i \geq a^{-\ell_i}$, and summing, $\sum_i a^{-\ell_i} \leq 1$. So Kraft's inequality holds and there is a prefix-free code c with these word lengths. From the second inequality,

$$\mathbb{E}[S] = \sum_i p_i \ell_i < \sum_i p_i (-\log_a p_i + 1) = \frac{H(X)}{\log a} + 1. \quad \square$$

The code constructed in the second half is worth a name.

Definition 28.2.16 (Shannon–Fano Coding)

The **Shannon–Fano code** for probabilities $p_1, \dots, p_m$ takes word lengths $\ell_i = \lceil -\log_a p_i \rceil$ and then assigns codewords greedily in order of increasing length, as in the proof of Kraft's theorem.

Example 28.2.17

Take $a = 2$ and the source

i	1	2	3	4	5	6
p_i	0.30	0.22	0.18	0.14	0.10	0.06

The entropy is

$$H(X) = -\sum_i p_i \log p_i \approx 2.4198.$$

The Shannon–Fano lengths are $\ell_i = \lceil -\log_2 p_i \rceil$, namely

$$\ell = (2, 3, 3, 3, 4, 5),$$

and $\sum_i 2^{-\ell_i} = \frac{1}{4} + 3 \cdot \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = 0.71875 \leq 1$ as it must be. Assigning codewords greedily gives

i	1	2	3	4	5	6
$c(i)$	00	010	011	100	1010	10110

with expected word length

$$\mathbb{E}[S] = 0.30 \cdot 2 + 0.22 \cdot 3 + 0.18 \cdot 3 + 0.14 \cdot 3 + 0.10 \cdot 4 + 0.06 \cdot 5 = 2.92.$$

This does obey $H(X) \leq \mathbb{E}[S] < H(X) + 1$, but only just, and it is visibly wasteful: nothing at all uses the string 11, and we could shorten $c(6)$ to 1011 without breaking anything. Shannon–Fano coding is not optimal.

§28.2.6 Huffman Coding

Huffman's algorithm produces a genuinely optimal code, and it does so by a beautifully simple greedy rule. We take $a = 2$ and $\mathcal{B} = \{0, 1\}$ for simplicity; everything generalises.

Order the messages so that $p_1 \geq p_2 \geq \dots \geq p_m$. We define the code recursively.

Definition 28.2.18 (Huffman Code)

If $m = 2$, take the codewords 0 and 1. If $m > 2$, form the source on $m - 1$ symbols

$\mu_1, \dots, \mu_{m-2}, \nu$ with probabilities $p_1, \dots, p_{m-2}, p_{m-1} + p_m$, take a **Huffman code** for it, and then set

$$c(\mu_{m-1}) = c(\nu)0, \quad c(\mu_m) = c(\nu)1,$$

keeping all other codewords unchanged.

In other words: repeatedly merge the two least likely symbols into one, until only two remain; then read the resulting binary tree from the root outwards. Huffman codes are prefix-free by construction, since every codeword sits at a leaf. They are not unique – ties in the probabilities force arbitrary choices – but we shall see that all the choices are equally good.

Example 28.2.19

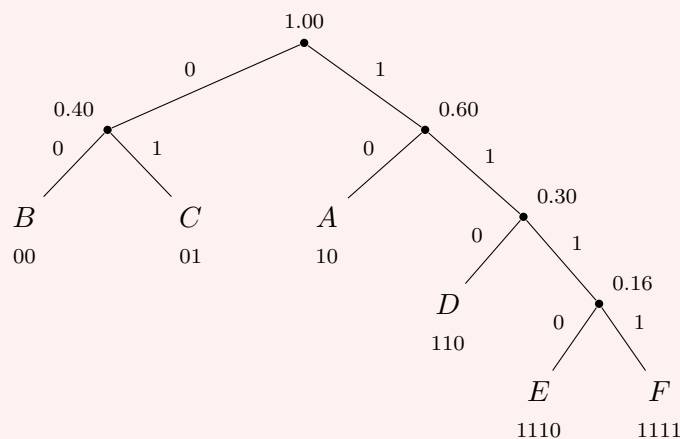
Take the same source as Example 28.2.17, writing the messages as $A, \dots, F$:

	A	B	C	D	E	F
p	0.30	0.22	0.18	0.14	0.10	0.06

We merge repeatedly.

- The two smallest are E and F ; merge them into X with probability 0.16. Remaining: A 0.30, B 0.22, C 0.18, X 0.16, D 0.14.
- The two smallest are D and X ; merge them into Y with probability 0.30. Remaining: A 0.30, Y 0.30, B 0.22, C 0.18.
- The two smallest are C and B ; merge them into Z with probability 0.40. Remaining: A 0.30, Y 0.30, Z 0.40.
- The two smallest are A and Y ; merge them into W with probability 0.60.
- Finally merge Z and W .

The resulting tree is as follows, with 0 on the left branch and 1 on the right.



Reading off, the word lengths are 2, 2, 2, 3, 4, 4 and

$$\mathbb{E}[S] = 0.30 \cdot 2 + 0.22 \cdot 2 + 0.18 \cdot 2 + 0.14 \cdot 3 + 0.10 \cdot 4 + 0.06 \cdot 4 = 2.46,$$

comfortably better than the 2.92 of Shannon–Fano, and within 0.05 of the entropy bound 2.4198.

To prove optimality we need two structural facts about optimal codes, both of which are obvious once stated.

Lemma 28.2.20

Let c be an optimal prefix-free code for messages $\mu_1, \dots, \mu_m$ with probabilities $p_1 \geq \dots \geq p_m$, and word lengths $\ell_1, \dots, \ell_m$. Then

- (i) if $p_i > p_j$ then $\ell_i \leq \ell_j$;
- (ii) among the codewords of maximal length, there are two which differ only in their last digit.

Proof. (i). Suppose $p_i > p_j$ but $\ell_i > \ell_j$. Swapping the i th and j th codewords leaves the code prefix-free and changes the expected word length by

$$(p_i \ell_j + p_j \ell_i) - (p_i \ell_i + p_j \ell_j) = (p_i - p_j)(\ell_j - \ell_i) < 0,$$

contradicting optimality.

(ii). Let $\ell = \max_i \ell_i$, and suppose no two codewords of length ℓ agree in their first $\ell - 1$ digits. Deleting the final digit of every codeword of length ℓ then leaves all these codewords distinct; and the code is still prefix-free, because a shortened codeword y could only be a prefix of some other codeword z if $y0$ or $y1$ were, and z has length at most ℓ , so $z \in \{y0, y1\}$, which our assumption excludes. But this strictly decreases the expected word length, contradicting optimality. $\square$

Theorem 28.2.21

Huffman codes are optimal.

Proof. We induct on m . For $m = 2$ the Huffman code has both word lengths equal to 1, which is clearly best possible.

Let $m > 2$ and let c_m be the Huffman code for X_m taking values $\mu_1, \dots, \mu_m$ with probabilities $p_1 \geq \dots \geq p_m$. By construction c_m is built from a Huffman code c_{m-1} for the merged source X_{m-1} on $\mu_1, \dots, \mu_{m-2}, \nu$ with probabilities $p_1, \dots, p_{m-2}, p_{m-1} + p_m$, by extending the codeword of ν by one digit in each of two ways. Comparing expected lengths term by term,

$$\mathbb{E}[S_m] = \mathbb{E}[S_{m-1}] + p_{m-1} + p_m. \quad (1)$$

Now let c'_m be an optimal code for X_m ; by the corollary to McMillan's theorem we may take it prefix-free. By Lemma 28.2.20(i) the two longest codewords belong to two of the least likely messages, and by (ii) we may assume, after relabelling messages of equal probability and swapping codewords of equal length, that

$$c'_m(\mu_{m-1}) = y0, \quad c'_m(\mu_m) = y1$$

for some string y . Define a code c'_{m-1} for X_{m-1} by

$$c'_{m-1}(\mu_i) = c'_m(\mu_i) \ (i \leq m-2), \quad c'_{m-1}(\nu) = y.$$

This is prefix-free: no codeword of c'_m other than $y0, y1$ can have y as a prefix, else it would have $y0$ or $y1$ as a prefix. Exactly as in (1),

$$\mathbb{E}[S'_m] = \mathbb{E}[S'_{m-1}] + p_{m-1} + p_m. \quad (2)$$

By the inductive hypothesis c_{m-1} is optimal for X_{m-1} , so $\mathbb{E}[S_{m-1}] \leq \mathbb{E}[S'_{m-1}]$. Combining with (1) and (2),

$$\mathbb{E}[S_m] \leq \mathbb{E}[S'_m],$$

and since c'_m was optimal, so is c_m . $\square$

Remark. Not every optimal code is a Huffman code – one can always relabel $0 \leftrightarrow 1$ in a subtree, for instance. What the theorem really gives us is that the Huffman word lengths are optimal word lengths.

Huffman coding is optimal but not perfect: it can never do better than the entropy bound, and if one symbol has probability close to 1 it must still spend a whole bit on it. The remedy is to Huffman-code blocks of N symbols at a time, which as we shall see in Section 28.4 drives the cost per symbol down to the entropy.

§28.2.7 Joint Entropy

Finally in this section we record how entropy behaves for pairs of random variables, which we shall need repeatedly later on.

If X, Y take values in $\mathcal{A}, \mathcal{B}$ then (X, Y) takes values in $\mathcal{A} \times \mathcal{B}$, and its entropy

$$H(X, Y) = - \sum_{x \in \mathcal{A}} \sum_{y \in \mathcal{B}} \mathbb{P}(X = x, Y = y) \log \mathbb{P}(X = x, Y = y)$$

is called the **joint entropy**. The same definition works for any finite tuple.

Lemma 28.2.22 (Subadditivity)

$H(X, Y) \leq H(X) + H(Y)$, with equality if and only if X and Y are independent.

Proof. Write $\mathcal{A} = \{x_1, \dots, x_m\}$, $\mathcal{B} = \{y_1, \dots, y_n\}$, and set $p_{ij} = \mathbb{P}(X = x_i, Y = y_j)$, $p_i = \mathbb{P}(X = x_i)$, $q_j = \mathbb{P}(Y = y_j)$. Both (p_{ij}) and $(p_i q_j)$ are probability distributions on $\mathcal{A} \times \mathcal{B}$, so Gibbs' inequality applies and gives

$$\begin{aligned} H(X, Y) &= - \sum_{i,j} p_{ij} \log p_{ij} \leq - \sum_{i,j} p_{ij} \log(p_i q_j) \\ &= - \sum_i \left(\sum_j p_{ij} \right) \log p_i - \sum_j \left(\sum_i p_{ij} \right) \log q_j \\ &= - \sum_i p_i \log p_i - \sum_j q_j \log q_j = H(X) + H(Y). \end{aligned}$$

Equality in Gibbs holds precisely when $p_{ij} = p_i q_j$ for all i, j , which is exactly independence. $\square$

Intuitively: knowing two things at once is never harder than knowing each separately, and it is exactly as hard only when the two are unrelated.

§28.3 Noisy Channels and Error-Correcting Codes

We now switch the channel back on. Throughout this section, unless we say otherwise, the channel is the binary symmetric channel with error probability $p < \frac{1}{2}$, and a code is a subset of $\mathbb{F}_2^n$.

Definition 28.3.1 (Binary Code)

A **binary** $[n, m]$ -**code** is a subset $C \subseteq \{0, 1\}^n$ with $|C| = m$. We call n the **length** of the code and the elements of C its **codewords**.

Such a code lets us send one of m messages using n bits, that is, using the channel n times. Since $1 \leq m \leq 2^n$, the information rate $\rho(C) = \frac{1}{n} \log m$ satisfies $0 \leq \rho(C) \leq 1$, with $\rho(C) = 0$ for the useless code $|C| = 1$ and $\rho(C) = 1$ for the code $C = \{0, 1\}^n$ which does no error correction at all. All the interest lies in between.

§28.3.1 Decoding Rules

Suppose Bob receives $x \in \{0, 1\}^n$. What should he guess was sent? There are three natural answers.

Definition 28.3.2 (Decoding Rules)

Let C be a binary $[n, m]$ -code.

- The **ideal observer** rule decodes x as the codeword $c \in C$ maximising $\mathbb{P}(c \text{ sent} \mid x \text{ received})$.
- The **maximum likelihood** rule decodes x as the $c \in C$ maximising the probability $\mathbb{P}(x \text{ received} \mid c \text{ sent})$.
- The **minimum distance** rule decodes x as the $c \in C$ minimising the Hamming distance $d(x, c)$,

$$d(x, y) = |\{i : x_i \neq y_i\}|.$$

The first is what we actually want, but it requires knowing the prior distribution on messages. The third is what we can actually compute. Happily, under mild hypotheses they all coincide.

Lemma 28.3.3

Let C be a binary $[n, m]$ -code used over a BSC with error probability p .

- (i) If all messages are equally likely, the ideal observer and maximum likelihood rules agree.
- (ii) If $p < \frac{1}{2}$, the maximum likelihood and minimum distance rules agree.

Proof. (i). By Bayes' theorem,

$$\mathbb{P}(c \text{ sent} \mid x \text{ received}) = \frac{\mathbb{P}(c \text{ sent})}{\mathbb{P}(x \text{ received})} \mathbb{P}(x \text{ received} \mid c \text{ sent}).$$

For fixed x , the denominator does not depend on c , and by hypothesis neither does $\mathbb{P}(c \text{ sent})$. So the two quantities are maximised together.

(ii). If $d(x, c) = r$ then exactly r of the n transmitted bits were flipped, so

$$\mathbb{P}(x \text{ received} \mid c \text{ sent}) = p^r (1-p)^{n-r} = (1-p)^n \left(\frac{p}{1-p} \right)^r.$$

As $p < \frac{1}{2}$ we have $\frac{p}{1-p} < 1$, so this is a strictly decreasing function of r . Maximising it is therefore the same as minimising $r = d(x, c)$. $\square$

The hypothesis in (i) is reasonable in practice, because if we first compress the message using a good noiseless code the resulting bits look close to uniform. The hypothesis in (ii) is harmless, as discussed earlier. Channels with $p = 0$ are called **lossless**, and channels with $p = \frac{1}{2}$ are called **useless**.

It is worth seeing that the rules really can differ.

Example 28.3.4

Let $C = \{000, 111\}$, sent with probabilities $\alpha = \frac{9}{10}$ and $1 - \alpha = \frac{1}{10}$ respectively, over a BSC with $p = \frac{1}{4}$. Suppose 110 is received. Then

$$\mathbb{P}(000 \text{ sent} \mid 110) = \frac{\alpha p^2 (1-p)}{\alpha p^2 (1-p) + (1-\alpha) p (1-p)^2} = \frac{\frac{9}{10} \cdot \frac{1}{16} \cdot \frac{3}{4}}{\frac{9}{10} \cdot \frac{1}{16} \cdot \frac{3}{4} + \frac{1}{10} \cdot \frac{1}{4} \cdot \frac{9}{16}} = \frac{3}{4},$$

so $\mathbb{P}(111 \text{ sent} \mid 110) = \frac{1}{4}$. The ideal observer decodes 110 as 000, since 000 was so much more likely *a priori*. But $d(110, 111) = 1 < 2 = d(110, 000)$, so minimum distance decoding – and hence maximum likelihood decoding – gives 111.

Remark. Minimum distance decoding is conceptually simple but potentially expensive: naively one compares the received word against every codeword, which is hopeless once $|C|$ is large. Much of Section 28.5 is devoted to making it cheap for structured codes. One should also fix a convention for ties: either make an arbitrary choice, or declare a failure and ask for retransmission.

From now on we use minimum distance decoding.

§28.3.2 Error Detection and Correction

Definition 28.3.5

A binary $[n, m]$ -code C is

- **d -error detecting** if changing at most d digits of a codeword can never produce another codeword;
- **e -error correcting** if, knowing that $x \in \mathbb{F}_2^n$ differs from some codeword in at most e places, we can determine that codeword.

Example 28.3.6 (Repetition Code)

The **repetition code** of length n has codewords $00 \dots 0$ and $11 \dots 1$. It is an $[n, 2]$ -code, it is $(n - 1)$ -error detecting and $\lfloor \frac{n-1}{2} \rfloor$ -error correcting, and its information rate is $\frac{1}{n}$, which tends to 0. Reliability is bought at the price of everything else.

Example 28.3.7 (Simple Parity Check Code)

The **simple parity check code** (or **paper tape code**) of length n is

$$C = \left\{ (x_1, \dots, x_n) \in \mathbb{F}_2^n : \sum_{i=1}^n x_i = 0 \right\}.$$

This is an $[n, 2^{n-1}]$ -code, it is 1-error detecting and 0-error correcting, and its information rate is $\frac{n-1}{n} \rightarrow 1$. Economy is bought at the price of everything else.

Between these two extremes sits the first genuinely clever code.

Example 28.3.8 (Hamming's Original Code)

Define $C \subseteq \mathbb{F}_2^7$ by the three conditions

$$c_1 + c_3 + c_5 + c_7 = 0, \quad c_2 + c_3 + c_6 + c_7 = 0, \quad c_4 + c_5 + c_6 + c_7 = 0.$$

The digits c_3, c_5, c_6, c_7 may be chosen freely and then c_1, c_2, c_4 are determined, so $|C| = 2^4 = 16$ and C is a $[7, 16]$ -code with rate $\frac{4}{7}$.

Suppose $x \in \mathbb{F}_2^7$ is received. Form the **syndrome** $z = (z_1, z_2, z_4)$ where

$$z_1 = x_1 + x_3 + x_5 + x_7, \quad z_2 = x_2 + x_3 + x_6 + x_7, \quad z_4 = x_4 + x_5 + x_6 + x_7.$$

If $x \in C$ then $z = (0, 0, 0)$. If instead $x = c + e_i$, a codeword with the i th digit flipped, then by linearity $z_x = z_{e_i}$, and one checks case by case that

$$z_1 + 2z_2 + 4z_4 = i$$

as an ordinary integer. For instance e_5 appears in the first and third conditions but not the second, giving $z = (1, 0, 1)$ and $1 + 0 + 4 = 5$. So the syndrome does not merely tell us that an error has occurred, it tells us *where*: the code is 1-error correcting. This is a remarkably efficient piece of bookkeeping, and we will see in a moment that in a precise sense it cannot be improved on.

To analyse such codes properly we need the geometry of $\mathbb{F}_2^n$.

Lemma 28.3.9

The Hamming distance d is a metric on $\mathbb{F}_2^n$.

Proof. Certainly $d(x, y) \geq 0$ with equality if and only if $x = y$, and $d(x, y) = d(y, x)$.

For the triangle inequality, note that if $x_i \neq z_i$ then $x_i \neq y_i$ or $y_i \neq z_i$, so

$$\{i : x_i \neq z_i\} \subseteq \{i : x_i \neq y_i\} \cup \{i : y_i \neq z_i\},$$

and taking sizes gives $d(x, z) \leq d(x, y) + d(y, z)$. $\square$

Equivalently, $d(x, y) = \sum_i d_1(x_i, y_i)$ where d_1 is the discrete metric on $\mathbb{F}_2$; the triangle inequality is inherited coordinatewise.

Definition 28.3.10 (Minimum Distance)

The **minimum distance** of a code C with $|C| \geq 2$ is

$$d(C) = \min\{d(c_1, c_2) : c_1, c_2 \in C, c_1 \neq c_2\}.$$

An $[n, m]$ -code with minimum distance d is called an $[n, m, d]$ -code.

Lemma 28.3.11

Let C be a code with minimum distance d . Then

- (i) C is $(d - 1)$ -error detecting, but cannot detect all sets of d errors;
- (ii) C is $\lfloor \frac{d-1}{2} \rfloor$ -error correcting, but cannot correct all sets of $\lfloor \frac{d-1}{2} \rfloor + 1$ errors.

Proof. (i). If $c \in C$ and $1 \leq d(x, c) \leq d - 1$ then $x \notin C$, since every other codeword is at distance at least d from c ; so the corruption is detected. On the other hand if c_1, c_2 are codewords at distance exactly d , then d well-chosen errors carry c_1 to c_2 and nothing at all is noticed.

(ii). Put $e = \lfloor \frac{d-1}{2} \rfloor$, so that $2e < d$. Suppose $x \in \mathbb{F}_2^n$ and $c_1 \in C$ with $d(x, c_1) \leq e$. For any other codeword c_2 , the triangle inequality gives

$$d(x, c_2) \geq d(c_1, c_2) - d(x, c_1) \geq d - e > e,$$

so c_1 is the unique codeword within distance e of x and we recover it. For the second half, choose c_1, c_2 with $d(c_1, c_2) = d$ and let x agree with c_1 except in $e + 1$ of the places where c_1 and c_2 differ. Then $d(x, c_1) = e + 1$ while $d(x, c_2) = d - (e + 1) \leq e + 1$, using $d \leq 2(e + 1)$; so x is at least as close to c_2 as to c_1 and we cannot reliably decode. $\square$

Example 28.3.12

The repetition code of length n is an $[n, 2, n]$ -code. The simple parity check code of length n is an $[n, 2^{n-1}, 2]$ -code, since flipping any two digits preserves the parity. The trivial code $\mathbb{F}_2$ is an $[n, 2^n, 1]$ -code. Hamming's original code is 1-error correcting so has $d \geq 3$ by Lemma 28.3.11; and 0000000 and 1110000 are both codewords, so $d = 3$ exactly, making it a $[7, 16, 3]$ -code.

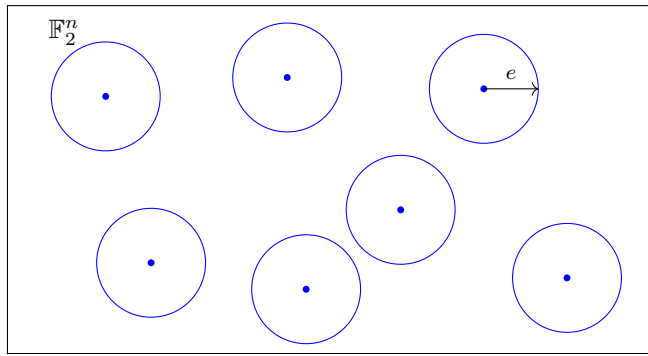
§28.3.3 Covering Estimates and Perfect Codes

The picture behind Lemma 28.3.11 is that an e -error correcting code is a packing of disjoint balls of radius e into $\mathbb{F}_2^n$. Counting gives an immediate upper bound on how many codewords there can be.

Definition 28.3.13

For $x \in \mathbb{F}_2^n$ and $r \geq 0$, write $B(x, r) = \{y \in \mathbb{F}_2^n : d(x, y) \leq r\}$ for the closed **Hamming ball** of radius r . Its size does not depend on x , and we write

$$V(n, r) = |B(x, r)| = \sum_{i=0}^r \binom{n}{i}.$$



The dots are the codewords; the discs are the balls of radius e inside which minimum distance decoding succeeds. They must be disjoint, and they all live inside a space of size 2^n .

Lemma 28.3.14 (Hamming's Bound; the Sphere-Packing Bound)

An e -error correcting code C of length n satisfies

$$|C| \leq \frac{2^n}{V(n, e)}.$$

Proof. If $c_1 \neq c_2$ are codewords then $B(c_1, e) \cap B(c_2, e) = \emptyset$, since a point in the intersection could not be decoded. Hence

$$\sum_{c \in C} |B(c, e)| \leq |\mathbb{F}_2^n|, \quad \text{that is} \quad |C| V(n, e) \leq 2^n. \quad \square$$

Definition 28.3.15 (Perfect Code)

An e -error correcting code C of length n with $|C| = \frac{2^n}{V(n, e)}$ is called **perfect**.

Remark. Equivalently, C is perfect if the balls $B(c, e)$ for $c \in C$ partition $\mathbb{F}_2^n$: every word in the space is within distance e of exactly one codeword. Perfect codes are wonderfully efficient but also brittle, since any $e+1$ errors are guaranteed to be decoded to the wrong codeword, with no warning.

Example 28.3.16

Hamming's $[7, 16, 3]$ -code is 1-error correcting, and

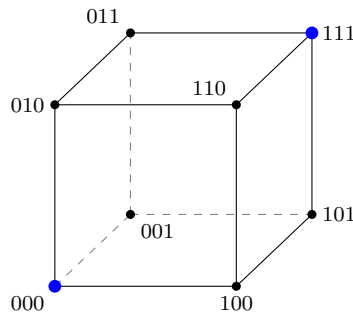
$$\frac{2^7}{V(7, 1)} = \frac{128}{1 + 7} = 16 = |C|,$$

so it is perfect. Indeed, the syndrome map sends $\mathbb{F}_2^7$ onto $\mathbb{F}_2^3$, and the eight syndromes correspond precisely to 'no error' and the seven possible single-bit errors.

Example 28.3.17

The repetition code of length n is perfect if and only if n is odd. If $n = 2k + 1$ it is k -error correcting and $V(n, k) = \sum_{i \leq k} \binom{n}{i} = 2^{n-1}$ by symmetry, so $2^n / V(n, k) = 2 = |C|$. If n is even the count fails.

The smallest interesting case is worth drawing. Here is $\mathbb{F}_2^3$ as the vertices of a cube, with the repetition code $\{000, 111\}$ marked.



Every vertex is adjacent to exactly one of the two blue vertices, which is the statement that the $[3, 2, 3]$ repetition code is perfect.

Remark. If $\frac{2^n}{V(n, e)}$ is not an integer then certainly no perfect e -error correcting code of length n exists. The converse is false: with $n = 90$ and $e = 2$ the quotient is an integer, but no such code exists.

Definition 28.3.18

$A(n, d)$ denotes the largest m for which an $[n, m, d]$ -code exists.

Determining $A(n, d)$ in general is a hard open problem, and only small cases are known.

Example 28.3.19

$A(n, 1) = 2^n$, achieved by the trivial code. $A(n, 2) = 2^{n-1}$, achieved by the simple parity check code (the upper bound follows from the next theorem). $A(n, n) = 2$, achieved by the repetition code.

Lemma 28.3.20

$A(n, d + 1) \leq A(n, d)$.

Proof. Let $m = A(n, d + 1)$ and let C be an $[n, m, d + 1]$ -code. Choose distinct $c_1, c_2 \in C$ with $d(c_1, c_2) = d + 1$, and let c'_1 differ from c_1 in exactly one of the places where c_1 and c_2 differ, so $d(c'_1, c_2) = d$. For any other $c \in C$,

$$d + 1 \leq d(c, c_1) \leq d(c, c'_1) + d(c'_1, c_1) = d(c, c'_1) + 1,$$

so $d(c, c'_1) \geq d$. Hence $C' = (C \setminus \{c_1\}) \cup \{c'_1\}$ has m elements, length n , and minimum distance exactly d . $\square$

Corollary 28.3.21

$A(n, d) = \max\{m : \text{there is an } [n, m, d']\text{-code for some } d' \geq d\}.$

Now we can sandwich $A(n, d)$ between two natural bounds.

Theorem 28.3.22 (Hamming and GSV Bounds)

$$\frac{2^n}{V(n, d-1)} \leq A(n, d) \leq \frac{2^n}{V(n, \lfloor \frac{d-1}{2} \rfloor)}.$$

The upper bound is Hamming's bound, a *packing* estimate. The lower bound is the **Gilbert–Shannon–Varshamov** (GSV) bound, and is a *covering* estimate: it says that a maximal code cannot leave any large empty region.

Proof. The upper bound is Lemma 28.3.14 together with Lemma 28.3.11(ii).

For the lower bound, let $m = A(n, d)$ and let C be an $[n, m, d]$ -code. There can be no $x \in \mathbb{F}_2^n$ with $d(x, c) \geq d$ for all $c \in C$, for then $C \cup \{x\}$ would be an $[n, m + 1, d]$ -code, contradicting maximality. Hence every x lies within distance $d - 1$ of some codeword, that is,

$$\mathbb{F}_2^n = \bigcup_{c \in C} B(c, d - 1),$$

and counting gives $2^n \leq m V(n, d - 1)$. $\square$

There is a third bound, of a completely different flavour, which is often the sharpest of all for large d .

Theorem 28.3.23 (Singleton Bound)

For any $[n, m, d]$ -code, $m \leq 2^{n-d+1}$.

Proof. Let C be an $[n, m, d]$ -code and consider the map $\pi: C \rightarrow \mathbb{F}_2^{n-d+1}$ which deletes the last $d - 1$ coordinates. If $c_1 \neq c_2$ had the same image then they would agree in the first $n - d + 1$ places and so differ in at most $d - 1$ places, contradicting $d(C) = d$. So π is injective and $m = |C| \leq 2^{n-d+1}$. $\square$

A code attaining the Singleton bound is called **maximum distance separable**, or MDS. Over $\mathbb{F}_2$ these are rare, but we shall meet a large family of them over larger alphabets in Section 28.6.

Example 28.3.24

Take $n = 10$, $d = 3$. Then $V(10, 1) = 11$ and $V(10, 2) = 1 + 10 + 45 = 56$, so Theorem 28.3.22 gives

$$\frac{1024}{56} \leq A(10, 3) \leq \frac{1024}{11}, \quad \text{that is} \quad 19 \leq A(10, 3) \leq 93.$$

Singleton gives only $A(10, 3) \leq 2^8 = 256$, which is worse here. The true value, established by computer search, is $A(10, 3) = 72$; so the GSV bound in particular is far from sharp.

§28.3.4 Asymptotics

For applications we care about long codes, and about how the rate behaves when we insist that the minimum distance grows proportionally to the length. So fix $\delta \in (0, \frac{1}{2})$ and study

$$\frac{1}{n} \log A(n, \lfloor n\delta \rfloor)$$

as $n \rightarrow \infty$. The answer is governed, once again, by the binary entropy function.

Proposition 28.3.25

Let $0 < \delta < \frac{1}{2}$. Then

- (i) $\log V(n, \lfloor n\delta \rfloor) \leq nH(\delta)$;
- (ii) $\frac{1}{n} \log A(n, \lfloor n\delta \rfloor) \geq 1 - H(\delta)$.

Proof. (i) implies (ii). By the GSV bound and monotonicity of V in its second argument,

$$A(n, \lfloor n\delta \rfloor) \geq \frac{2^n}{V(n, \lfloor n\delta \rfloor - 1)} \geq \frac{2^n}{V(n, \lfloor n\delta \rfloor)},$$

so taking logarithms and using (i),

$$\frac{1}{n} \log A(n, \lfloor n\delta \rfloor) \geq 1 - \frac{\log V(n, \lfloor n\delta \rfloor)}{n} \geq 1 - H(\delta).$$

Proof of (i). Since H is increasing on $(0, \frac{1}{2})$ and $V(n, \lfloor n\delta \rfloor) = V(n, \lfloor n\delta' \rfloor)$ for $\delta' = \lfloor n\delta \rfloor / n \leq \delta$, we may assume $n\delta$ is an integer. Now expand $1 = (\delta + (1 - \delta))^n$ and throw away all but the first $n\delta + 1$ terms:

$$\begin{aligned} 1 &= \sum_{i=0}^n \binom{n}{i} \delta^i (1 - \delta)^{n-i} \geq \sum_{i=0}^{n\delta} \binom{n}{i} \delta^i (1 - \delta)^{n-i} \\ &= (1 - \delta)^n \sum_{i=0}^{n\delta} \binom{n}{i} \left(\frac{\delta}{1 - \delta} \right)^i \geq (1 - \delta)^n \left(\frac{\delta}{1 - \delta} \right)^{n\delta} \sum_{i=0}^{n\delta} \binom{n}{i} \\ &= \delta^{n\delta} (1 - \delta)^{n(1-\delta)} V(n, n\delta), \end{aligned}$$

where the second inequality uses $\frac{\delta}{1-\delta} < 1$ (as $\delta < \frac{1}{2}$), so that each of the terms

$\left(\frac{\delta}{1-\delta}\right)^i$ with $i \leq n\delta$ is at least $\left(\frac{\delta}{1-\delta}\right)^{n\delta}$. Taking logarithms,

$$0 \geq n\delta \log \delta + n(1-\delta) \log(1-\delta) + \log V(n, n\delta) = -nH(\delta) + \log V(n, n\delta),$$

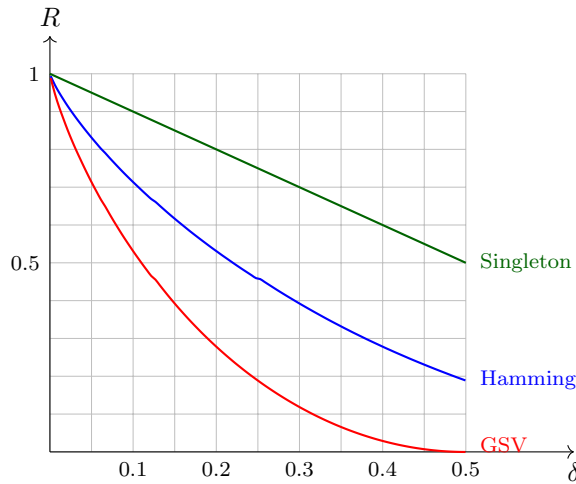
which is (i). $\square$

Lemma 28.3.26

For $0 < \delta < \frac{1}{2}$, $\lim_{n \rightarrow \infty} \frac{\log V(n, \lfloor n\delta \rfloor)}{n} = H(\delta)$.

Proof. The upper bound is Proposition 28.3.25(i). For the lower bound it suffices to keep the single term $\binom{n}{\lfloor n\delta \rfloor}$ and apply Stirling's formula, which gives $\frac{1}{n} \log \binom{n}{\lfloor n\delta \rfloor} \rightarrow H(\delta)$. $\square$

So the constant $H(\delta)$ in the proposition cannot be improved. It is instructive to plot the three bounds we now have, expressed as constraints on the rate $R = \frac{1}{n} \log |C|$ in terms of the relative distance $\delta = d/n$. The Hamming bound becomes $R \leq 1 - H(\delta/2)$, the Singleton bound becomes $R \leq 1 - \delta$, and the GSV bound becomes $R \geq 1 - H(\delta)$.



Anything achievable lies below the blue and green curves and, by GSV, at least the region below the red curve *is* achievable. The gap between red and blue is enormous, and closing it – determining the true asymptotic trade-off between rate and distance – remains one of the central open problems of coding theory.

§28.3.5 Constructing New Codes from Old

We finish this section with three easy operations on codes; each is worth knowing because they let us move around the parameter space by hand. Let C be an $[n, m, d]$ -code.

Example 28.3.27 (Parity Check Extension)

The **parity check extension** of C is

$$C^+ = \left\{ \left(c_1, \dots, c_n, \sum_{i=1}^n c_i \right) : (c_1, \dots, c_n) \in C \right\},$$

an $[n+1, m, d']$ -code where $d' = d+1$ if d is odd and $d' = d$ if d is even. (All codewords of C^+ have even weight, so distances between them are even; hence an odd d must increase.)

Example 28.3.28 (Punctured Code)

Deleting the i th digit from every codeword gives the **punctured code** C^- . If $d \geq 2$ this is an $[n-1, m, d']$ -code with $d' \in \{d-1, d\}$; the codewords remain distinct because they differed in at least two places.

Example 28.3.29 (Shortened Code)

Fix i and $\alpha \in \mathbb{F}_2$, and let

$$C' = \{ (c_1, \dots, c_{i-1}, c_{i+1}, \dots, c_n) : (c_1, \dots, c_{i-1}, \alpha, c_{i+1}, \dots, c_n) \in C \}.$$

This is the **shortened code**. It has length $n-1$ and minimum distance $d' \geq d$, and for a suitable choice of α (namely, the more popular value of the i th digit) it has size $m' \geq \frac{m}{2}$.

§28.4 Information Theory

We now have two loose ends to tie up. In Section 2 we measured the cost of encoding a *single* symbol and found the entropy $H(X)$ standing in the way; but a real source emits a long stream of symbols, possibly with dependence between them, and we should ask what the cost per symbol is in the long run. In Section 3 we defined the capacity of a channel and did not compute it for a single example. Both questions are answered by the same circle of ideas, and the answers are Shannon's two coding theorems.

§28.4.1 Sources and Information Rate

Definition 28.4.1 (Source)

A **source** is a sequence of random variables $X_1, X_2, \dots$ taking values in an alphabet $\mathcal{A}$.

Example 28.4.2

A **Bernoulli** or **memoryless** source is one in which the X_i are independent and identically distributed. English text is emphatically not of this kind – the letter u is much more likely after a q – but it is the case in which everything can be computed.

Definition 28.4.3 (Reliable Encoding)

A source $X_1, X_2, \dots$ is **reliably encodable at rate r** if there are subsets $A_n \subseteq \mathcal{A}^n$ with

- (i) $\lim_{n \rightarrow \infty} \frac{\log |A_n|}{n} = r$, and
- (ii) $\lim_{n \rightarrow \infty} \mathbb{P}((X_1, \dots, X_n) \in A_n) = 1$.

The **information rate** H of the source is the infimum of all rates at which it is reliably encodable.

The idea is that we agree to encode faithfully only the messages in A_n , which takes $\log |A_n|$ bits, and accept a vanishing probability of complete failure otherwise. Since $A_n = \mathcal{A}^n$ always works, $H \leq \log |\mathcal{A}|$; and clearly $H \geq 0$. Both extremes occur.

Before going further, a piece of notation which is easy to misread. If X is a discrete random variable with probability mass function $p_X(x) = \mathbb{P}(X = x)$, then $p(X)$ denotes the composite $p_X \circ X$, that is, the random variable

$$\omega \mapsto \mathbb{P}(X = X(\omega)).$$

It is the ‘probability of what actually happened’. Similarly, given a tuple of random variables $X^{(n)} = (X_1, \dots, X_n)$, we write $p(X^{(n)})$ for

$$\omega \mapsto \mathbb{P}(X_1 = X_1(\omega), \dots, X_n = X_n(\omega)).$$

With this notation, $H(X) = \mathbb{E}[-\log p(X)]$.

Example 28.4.4

Let $\mathcal{A} = \{A, B, C\}$ and suppose that $X^{(2)} = (X_1, X_2)$ takes the six values

$$AB, \quad AC, \quad BC, \quad BA, \quad CA, \quad CB$$

with probabilities 0.3, 0.1, 0.1, 0.2, 0.25, 0.05 respectively. Then $p_{X^{(2)}}(AB) = 0.3$ and so on, so the random variable $p(X^{(2)})$ takes the value 0.3 with probability 0.3, the value 0.1 with probability 0.2 (there are two outcomes of probability 0.1), the value 0.2 with probability 0.2, and so on.

Recall from IA Probability that $X_n \rightarrow L$ **in probability**, written $X_n \xrightarrow{\mathbb{P}} L$, if $\mathbb{P}(|X_n - L| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$, and that the weak law of large numbers says that if $Y_1, Y_2, \dots$ are i.i.d. real random variables with finite mean then $\frac{1}{n} \sum_{i \leq n} Y_i \xrightarrow{\mathbb{P}} \mathbb{E}[Y_1]$.

Example 28.4.5

Let $X_1, X_2, \dots$ be a Bernoulli source. Then $p(X_1), p(X_2), \dots$ are i.i.d., and by independence

$$p(X_1, \dots, X_n) = p(X_1)p(X_2) \cdots p(X_n).$$

Applying the weak law to the i.i.d. variables $-\log p(X_i)$,

$$-\frac{1}{n} \log p(X_1, \dots, X_n) = -\frac{1}{n} \sum_{i=1}^n \log p(X_i) \xrightarrow{\mathbb{P}} \mathbb{E}[-\log p(X_1)] = H(X_1).$$

This last observation is the key to everything in this section, and it deserves a name; but first let us see that it gives the right answer for Bernoulli sources by an elementary route.

Lemma 28.4.6

The information rate of a Bernoulli source $X_1, X_2, \dots$ is at most the expected word length of an optimal binary code c for X_1 .

Proof. Let L_i be the length of the codeword $c(X_i)$, so the L_i are i.i.d. with mean $\mathbb{E}[L_1]$. Fix $\varepsilon > 0$ and let

$$A_n = \{x \in \mathcal{A}^n : c^*(x) \text{ has length} < n(\mathbb{E}[L_1] + \varepsilon)\}.$$

Then

$$\mathbb{P}((X_1, \dots, X_n) \in A_n) = \mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n L_i < \mathbb{E}[L_1] + \varepsilon\right) \rightarrow 1$$

by the weak law. Since c is decipherable, c^* is injective, and there are fewer than $2^{n(\mathbb{E}[L_1] + \varepsilon) + 1}$ binary strings of length less than $n(\mathbb{E}[L_1] + \varepsilon)$; so

$$|A_n| \leq 2^{n(\mathbb{E}[L_1] + \varepsilon) + 1}.$$

Enlarging A_n if necessary so that $|A_n| = \lfloor 2^{n(\mathbb{E}[L_1] + \varepsilon)} \rfloor$ (which only helps condition (ii)), we get $\frac{1}{n} \log |A_n| \rightarrow \mathbb{E}[L_1] + \varepsilon$. Hence the source is reliably encodable at rate $\mathbb{E}[L_1] + \varepsilon$ for every $\varepsilon > 0$, and the result follows. $\square$

Corollary 28.4.7

A Bernoulli source has information rate $H < H(X_1) + 1$.

Proof. Combine Lemma 28.4.6 with the upper bound in Shannon's noiseless coding theorem (Theorem 28.2.15). $\square$

The '+1' is the same waste we met before, and the same trick removes it. Suppose we encode in blocks of size N : set $Y_1 = (X_1, \dots, X_N)$, $Y_2 = (X_{N+1}, \dots, X_{2N})$ and so on, giving a source over the alphabet $\mathcal{A}^N$. One checks directly from the definition that if (X_i) has information rate H then (Y_j) has information rate NH .

Proposition 28.4.8

The information rate H of a Bernoulli source satisfies $H \leq H(X_1)$.

Proof. The blocked source (Y_j) is again Bernoulli, so Corollary 28.4.7 applies to it:

$$NH < H(Y_1) + 1 = H(X_1, \dots, X_N) + 1 = NH(X_1) + 1,$$

the last equality by independence and Lemma 28.2.22 (with equality since the X_i are independent). Dividing by N gives $H < H(X_1) + \frac{1}{N}$ for every N . $\square$

§28.4.2 The Asymptotic Equipartition Property

Example 28.4.5 isolates exactly the property we need, and it is worth abstracting.

Definition 28.4.9 (AEP)

A source $X_1, X_2, \dots$ satisfies the **asymptotic equipartition property** (AEP) with constant $H \geq 0$ if

$$-\frac{1}{n} \log p(X_1, \dots, X_n) \xrightarrow{\mathbb{P}} H.$$

Example 28.4.10

Toss a biased coin with $\mathbb{P}(\text{head}) = q$, and let $X_1, X_2, \dots$ record the results. After N tosses we expect about qN heads, and any *particular* sequence with exactly qN heads has probability

$$q^{qN} (1-q)^{(1-q)N} = 2^{N(q \log q + (1-q) \log(1-q))} = 2^{-NH(q)}.$$

Not every sequence looks like this; but the atypical ones carry vanishing total probability. So with high probability we see a ‘typical sequence’, and typical sequences are all roughly *equally likely*, each with probability about $2^{-NH(q)}$. This is what the name is describing: the probability mass is asymptotically spread evenly over about 2^{NH} sequences.

That reformulation is worth recording precisely.

Lemma 28.4.11

A source satisfies the AEP with constant H if and only if for every $\varepsilon > 0$ there is n_0 such that for all $n \geq n_0$ there is a **typical set** $T_n \subseteq \mathcal{A}^n$ with

- (i) $\mathbb{P}((X_1, \dots, X_n) \in T_n) > 1 - \varepsilon$;
- (ii) $2^{-n(H+\varepsilon)} \leq p(x_1, \dots, x_n) \leq 2^{-n(H-\varepsilon)}$ for all $(x_1, \dots, x_n) \in T_n$.

Proof. Suppose the AEP holds and let $\varepsilon > 0$. Define

$$T_n = \left\{ (x_1, \dots, x_n) : \left| -\frac{1}{n} \log p(x_1, \dots, x_n) - H \right| \leq \varepsilon \right\},$$

which is precisely the set on which (ii) holds. By convergence in probability, $\mathbb{P}((X_1, \dots, X_n) \in T_n) \rightarrow 1$, so (i) holds for n large.

Conversely, given such sets T_n , for any $\varepsilon > 0$ we have

$$\mathbb{P}\left(\left|-\frac{1}{n} \log p(X_1, \dots, X_n) - H\right| \leq \varepsilon\right) \geq \mathbb{P}((X_1, \dots, X_n) \in T_n) > 1 - \varepsilon$$

for $n \geq n_0(\varepsilon)$, which is convergence in probability. $\square$

§28.4.3 Shannon's First Coding Theorem

Theorem 28.4.12 (Shannon's First Coding Theorem)

A source satisfying the AEP with constant H has information rate exactly H .

Proof. Rate H is achievable. Let $\varepsilon > 0$ and take typical sets T_n as in Lemma 28.4.11. For $n \geq n_0(\varepsilon)$ and $(x_1, \dots, x_n) \in T_n$ we have $p(x_1, \dots, x_n) \geq 2^{-n(H+\varepsilon)}$, so

$$1 \geq \mathbb{P}(T_n) \geq |T_n| 2^{-n(H+\varepsilon)}, \quad \text{giving} \quad \frac{1}{n} \log |T_n| \leq H + \varepsilon.$$

Taking $A_n = T_n$ (padded if necessary so that the limit in (i) exists and equals $H + \varepsilon$) shows that the source is reliably encodable at rate $H + \varepsilon$. As $\varepsilon > 0$ was arbitrary, the information rate is at most H .

No smaller rate is achievable. If $H = 0$ there is nothing to prove, so assume $H > 0$ and let $0 < \varepsilon < \frac{H}{2}$. Suppose for contradiction that the source is reliably encodable at rate $H - 2\varepsilon$, witnessed by sets A_n . Let T_n be typical sets for this ε . For $(x_1, \dots, x_n) \in T_n$ we have $p(x_1, \dots, x_n) \leq 2^{-n(H-\varepsilon)}$, so

$$\mathbb{P}((X_1, \dots, X_n) \in A_n \cap T_n) \leq |A_n| 2^{-n(H-\varepsilon)},$$

and hence

$$\frac{1}{n} \log \mathbb{P}(A_n \cap T_n) \leq -(H - \varepsilon) + \frac{1}{n} \log |A_n| \rightarrow -(H - \varepsilon) + (H - 2\varepsilon) = -\varepsilon.$$

Therefore $\log \mathbb{P}(A_n \cap T_n) \rightarrow -\infty$, so $\mathbb{P}(A_n \cap T_n) \rightarrow 0$. But

$$\mathbb{P}(T_n) \leq \mathbb{P}(A_n \cap T_n) + \mathbb{P}(\mathcal{A}^n \setminus A_n) \rightarrow 0 + 0 = 0,$$

contradicting $\mathbb{P}(T_n) > 1 - \varepsilon$. So the information rate is at least H . $\square$

Corollary 28.4.13

A Bernoulli source has information rate exactly $H(X_1)$.

Proof. Example 28.4.5 verifies the AEP with constant $H(X_1)$. $\square$

Remark. The theorem is not merely an existence statement; it comes with a coding scheme. Encode the typical sequences with a block code of length $\lceil n(H + \varepsilon) \rceil$, and encode the atypical sequences however you like (say, by a reserved escape symbol followed by the raw string). The scheme fails with vanishing probability and costs $H + \varepsilon$ bits per symbol.

Many non-Bernoulli sources also satisfy the AEP. Under suitable hypotheses – for in-

stance for a stationary ergodic source – the sequence $\frac{1}{n}H(X_1, \dots, X_n)$ is decreasing, and the AEP holds with $H = \lim_n \frac{1}{n}H(X_1, \dots, X_n)$. This is what makes the theory applicable to real text.

§28.4.4 Capacity of the Binary Symmetric Channel

We now turn to the channel side. Recall the definitions from Section 1: a code of length n is $C \subseteq \mathcal{A}^n$, its rate is $\rho(C) = \frac{\log|C|}{n}$, its error rate is $\hat{e}(C) = \max_{c \in C} \mathbb{P}(\text{error} \mid c \text{ sent})$, and the capacity is the supremum of rates at which reliable transmission is possible.

As a warm-up – and because it shows exactly where the GSV bound earns its keep – let us prove directly that a reasonably good binary symmetric channel has positive capacity.

Lemma 28.4.14

Let $\varepsilon > 0$ and let a BSC with error probability p be used n times. Then

$$\lim_{n \rightarrow \infty} \mathbb{P}(\text{the channel makes at least } n(p + \varepsilon) \text{ errors}) = 0.$$

Proof. Let U_i be the indicator that the i th digit is mistransmitted. The U_i are i.i.d. with $\mathbb{E}[U_i] = p$, and

$$\mathbb{P}\left(\sum_{i=1}^n U_i \geq n(p + \varepsilon)\right) \leq \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n U_i - p\right| \geq \varepsilon\right) \rightarrow 0$$

by the weak law of large numbers. □

Proposition 28.4.15

A binary symmetric channel with error probability $p < \frac{1}{4}$ has nonzero capacity.

Proof. Choose δ with $2p < \delta < \frac{1}{2}$; this is possible precisely because $p < \frac{1}{4}$. We claim we can transmit reliably at rate $R = 1 - H(\delta) > 0$.

Let C_n be an $[n, A(n, \lfloor n\delta \rfloor), \lfloor n\delta \rfloor]$ -code of maximal size. By the GSV bound and Proposition 28.3.25,

$$|C_n| = A(n, \lfloor n\delta \rfloor) \geq 2^{n(1-H(\delta))} = 2^{nR},$$

and passing to a subcode we may assume $|C_n| = \lfloor 2^{nR} \rfloor$, still with minimum distance at least $\lfloor n\delta \rfloor$. Using minimum distance decoding, an error occurs only if the channel makes more than $\lfloor \frac{\lfloor n\delta \rfloor - 1}{2} \rfloor$ errors, so

$$\hat{e}(C_n) \leq \mathbb{P}\left(\text{at least } \frac{n\delta - 1}{2} \text{ errors in } n \text{ uses}\right).$$

Now pick $\varepsilon > 0$ with $p + \varepsilon < \frac{\delta}{2}$, which is possible as $2p < \delta$. For large n ,

$$\frac{n\delta - 1}{2} = n\left(\frac{\delta}{2} - \frac{1}{2n}\right) > n(p + \varepsilon),$$

so $\hat{e}(C_n) \leq \mathbb{P}(\text{at least } n(p + \varepsilon) \text{ errors}) \rightarrow 0$ by Lemma 28.4.14. Since $\rho(C_n) \rightarrow R$, we are done. $\square$

This is already remarkable, but the constant is wrong: the answer is not $1 - H(\delta)$ for some $\delta > 2p$, and the restriction $p < \frac{1}{4}$ is an artefact. To get the truth we need a more delicate measure of how much a channel destroys.

§28.4.5 Conditional Entropy and Fano's Inequality

Definition 28.4.16 (Conditional Entropy)

Let X, Y take values in $\mathcal{A}, \mathcal{B}$. The **conditional entropy** of X given $Y = y$ is

$$H(X | Y = y) = - \sum_{x \in \mathcal{A}} \mathbb{P}(X = x | Y = y) \log \mathbb{P}(X = x | Y = y),$$

and the **conditional entropy** of X given Y is

$$H(X | Y) = \sum_{y \in \mathcal{B}} \mathbb{P}(Y = y) H(X | Y = y).$$

Clearly $H(X | Y) \geq 0$. It measures how much uncertainty about X remains once Y is known.

Lemma 28.4.17 (Chain Rule)

$$H(X, Y) = H(X | Y) + H(Y).$$

Proof. Expanding the definition and writing $\mathbb{P}(X = x | Y = y)\mathbb{P}(Y = y) = \mathbb{P}(X = x, Y = y)$,

$$\begin{aligned} H(X | Y) &= - \sum_y \sum_x \mathbb{P}(X = x, Y = y) \log \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)} \\ &= - \sum_y \sum_x \mathbb{P}(X = x, Y = y) \log \mathbb{P}(X = x, Y = y) \\ &\quad + \sum_y \left(\sum_x \mathbb{P}(X = x, Y = y) \right) \log \mathbb{P}(Y = y) \\ &= H(X, Y) + \sum_y \mathbb{P}(Y = y) \log \mathbb{P}(Y = y) = H(X, Y) - H(Y). \square \end{aligned}$$

Example 28.4.18

Let X be uniform on $\{1, \dots, 6\}$, modelling a die, and let Y be 0 if X is even and 1 if X is odd. Then Y is a function of X , so $H(X, Y) = H(X) = \log 6$, and $H(Y) = \log 2 = 1$. The chain rule gives $H(X | Y) = \log 6 - \log 2 = \log 3$, which is right: knowing the parity leaves three equally likely values. Applying the chain rule the other way round, $H(Y | X) = H(X, Y) - H(X) = 0$, as it must be.

Corollary 28.4.19

$H(X | Y) \leq H(X)$, with equality if and only if X and Y are independent.

Proof. Combine the chain rule with subadditivity (Lemma 28.2.22): $H(X | Y) = H(X, Y) - H(Y) \leq H(X)$, with the same equality condition. $\square$

Everything extends to tuples: replacing X by $X^{(r)} = (X_1, \dots, X_r)$ and Y by $Y^{(s)}$, we may speak of $H(X_1, \dots, X_r | Y_1, \dots, Y_s)$. One must read such expressions carefully: $H(X, Y | Z)$ is the entropy of the *pair* (X, Y) given Z , not the entropy of X together with the entropy of Y given Z .

Lemma 28.4.20

$H(X | Y) \leq H(X | Y, Z) + H(Z)$.

Proof. Expand $H(X, Y, Z)$ in two ways using the chain rule:

$$H(Z | X, Y) + \underbrace{H(X | Y) + H(Y)}_{H(X, Y)} = H(X, Y, Z) = H(X | Y, Z) + \underbrace{H(Z | Y) + H(Y)}_{H(Y, Z)}.$$

Since $H(Z | X, Y) \geq 0$ we get $H(X | Y) \leq H(X | Y, Z) + H(Z | Y)$, and $H(Z | Y) \leq H(Z)$ by Corollary 28.4.19. $\square$

Now for the key estimate. Think of X as what was sent and Y as what was decoded; then $p = \mathbb{P}(X \neq Y)$ is the probability of error, and Fano's inequality bounds how much uncertainty can remain.

Proposition 28.4.21 (Fano's Inequality)

Let X, Y take values in an alphabet $\mathcal{A}$ with $|\mathcal{A}| = m \geq 2$, and let $p = \mathbb{P}(X \neq Y)$. Then

$$H(X | Y) \leq H(p) + p \log(m - 1).$$

Proof. Let Z be the indicator of the event $X \neq Y$, so $H(Z) = H(p)$. By Lemma 28.4.20 it suffices to show $H(X | Y, Z) \leq p \log(m - 1)$.

If $Z = 0$ then $X = Y$, so $H(X | Y = y, Z = 0) = 0$. If $Z = 1$ then X is one of the $m - 1$ values other than y , so $H(X | Y = y, Z = 1) \leq \log(m - 1)$. Hence

$$\begin{aligned} H(X | Y, Z) &= \sum_{y \in \mathcal{A}} \sum_{z \in \{0,1\}} \mathbb{P}(Y = y, Z = z) H(X | Y = y, Z = z) \\ &\leq \sum_{y \in \mathcal{A}} \mathbb{P}(Y = y, Z = 1) \log(m - 1) \\ &= \mathbb{P}(Z = 1) \log(m - 1) = p \log(m - 1). \square \end{aligned}$$

The two terms have a clean reading: $H(p)$ is the information needed to say *whether* an error occurred, and $p \log(m - 1)$ is the information needed to say *which* error it was, in the worst case.

§28.4.6 Mutual Information and Information Capacity

Definition 28.4.22 (Mutual Information)

The **mutual information** of X and Y is

$$I(X; Y) = H(X) - H(X | Y).$$

By the chain rule $I(X; Y) = H(X) + H(Y) - H(X, Y)$, from which it is visibly symmetric in X and Y , and non-negative by subadditivity, with $I(X; Y) = 0$ if and only if X and Y are independent. It measures how much knowing one variable tells us about the other – exactly the quantity a channel ought to be judged by.

Definition 28.4.23 (Information Capacity)

Let a discrete memoryless channel have input alphabet $\mathcal{A}$ and output alphabet $\mathcal{B}$. For a random variable X on $\mathcal{A}$, let Y be the resulting output. The **information capacity** of the channel is

$$C = \max_X I(X; Y),$$

the maximum being over all distributions of X on $\mathcal{A}$.

This maximum really is attained, since I is a continuous function of the input distribution $(p_1, \dots, p_m)$, which ranges over the compact simplex $\{p_i \geq 0, \sum p_i = 1\}$. Note that the information capacity depends only on the channel matrix, and is a finite computation; whereas the (operational) capacity of Section 1 is defined by a limiting process over all possible codes. The content of Shannon's second theorem is that they agree.

Theorem 28.4.24 (Shannon's Noisy Coding Theorem)

For a discrete memoryless channel, the operational capacity equals the information capacity.

We shall prove the inequality 'operational $\leq$ information' in general, and the reverse inequality for the binary symmetric channel. First, let us see what the theorem buys us.

Example 28.4.25 (i) *The binary symmetric channel with error probability p .* Let $\mathbb{P}(X = 0) = \alpha$. Given X , the output differs from the input with probability p regardless, so $H(Y | X = 0) = H(Y | X = 1) = H(p)$ and hence $H(Y | X) = H(p)$. Also $\mathbb{P}(Y = 0) = \alpha(1 - p) + (1 - \alpha)p$. Therefore

$$\begin{aligned} C &= \max_{\alpha} [H(Y) - H(Y | X)] \\ &= \max_{\alpha} [H(\alpha(1 - p) + (1 - \alpha)p) - H(p)] = 1 - H(p), \end{aligned}$$

the maximum being at $\alpha = \frac{1}{2}$, when the argument of the outer H is $\frac{1}{2}$. So

$$C = 1 + p \log p + (1 - p) \log(1 - p).$$

If $p = 0$ or $p = 1$ then $C = 1$; if $p = \frac{1}{2}$ then $C = 0$, confirming that such a channel is useless.

- (ii) *The binary erasure channel with erasure probability p .* Here it is easier to compute $I(Y; X) = H(X) - H(X | Y)$. With $\mathbb{P}(X = 0) = \alpha$, receiving 0 or 1 determines X , so $H(X | Y = 0) = H(X | Y = 1) = 0$, while $H(X | Y = \star) = H(\alpha)$. As $\mathbb{P}(Y = \star) = p$,

$$C = \max_{\alpha} [H(\alpha) - pH(\alpha)] = (1 - p) \max_{\alpha} H(\alpha) = 1 - p,$$

again at $\alpha = \frac{1}{2}$. This is exactly what one would guess: a fraction p of the transmissions is simply lost.

Note the strength of the first computation. A channel that flips one bit in ten has capacity $1 - H(0.1) = 0.531$, so we can push more than half a bit of genuine information through each use of it, with arbitrarily small error probability.

Lemma 28.4.26

If a discrete memoryless channel has information capacity C , its n th extension has information capacity nC .

Proof. Let $X_1, \dots, X_n$ be the input and $Y_1, \dots, Y_n$ the output. Because the channel is memoryless, Y_i depends only on X_i , so

$$H(Y_1, \dots, Y_n | X_1, \dots, X_n) = \sum_{i=1}^n H(Y_i | X_1, \dots, X_n) = \sum_{i=1}^n H(Y_i | X_i).$$

Hence, using subadditivity for the first term,

$$\begin{aligned} I(X_1, \dots, X_n; Y_1, \dots, Y_n) &= H(Y_1, \dots, Y_n) - H(Y_1, \dots, Y_n | X_1, \dots, X_n) \\ &\leq \sum_{i=1}^n H(Y_i) - \sum_{i=1}^n H(Y_i | X_i) = \sum_{i=1}^n I(X_i; Y_i) \leq nC. \end{aligned}$$

Conversely, take the X_i independent and each achieving $I(X_i; Y_i) = C$. Then the Y_i are independent too, so $H(Y_1, \dots, Y_n) = \sum_i H(Y_i)$ and the first inequality is an equality. Hence the maximum is exactly nC . $\square$

§28.4.7 Proof of the Noisy Coding Theorem

We begin with the converse, which holds for any discrete memoryless channel.

Proposition 28.4.27 (Converse to the Noisy Coding Theorem)

The operational capacity of a discrete memoryless channel is at most its information capacity C .

Proof. Suppose, for contradiction, that reliable transmission is possible at some rate $R > C$. Then there are codes C_n of length n and size $\lfloor 2^{nR} \rfloor$ with $\rho(C_n) \rightarrow R$ and $\hat{e}(C_n) \rightarrow 0$.

Introduce the **average error rate**

$$e(C_n) = \frac{1}{|C_n|} \sum_{c \in C_n} \mathbb{P}(\text{error} \mid c \text{ sent}) \leq \hat{e}(C_n),$$

so $e(C_n) \rightarrow 0$ as well. Let X be uniform on C_n , let Y be the channel output, and let $\hat{X}$ be the decoded codeword; write $p = e(C_n) = \mathbb{P}(X \neq \hat{X})$.

Since X is uniform, $H(X) = \log|C_n| = \log\lfloor 2^{nR} \rfloor \geq nR - 1$ for n large. By Fano's inequality applied to X and $\hat{X}$, and then the fact that $\hat{X}$ is a function of Y (so conditioning on Y can only be more informative),

$$H(X \mid Y) \leq H(X \mid \hat{X}) \leq H(p) + p \log(|C_n| - 1) \leq 1 + pnR.$$

Therefore, using Lemma 28.4.26 for the n th extension,

$$nC \geq I(X; Y) = H(X) - H(X \mid Y) \geq (nR - 1) - (1 + pnR),$$

so that

$$pnR \geq n(R - C) - 2, \quad \text{that is} \quad p \geq \frac{n(R - C) - 2}{nR} \rightarrow \frac{R - C}{R} > 0.$$

This contradicts $p = e(C_n) \rightarrow 0$. Hence no rate above C is achievable. $\square$

The direct half is the celebrated one, and Shannon's proof is a masterpiece of the probabilistic method: rather than constructing a good code, we pick one at random and show that the average is good enough.

Proposition 28.4.28

Let a BSC have error probability p , and let $R < 1 - H(p)$. Then there are codes C_n of length n and size $\lfloor 2^{nR} \rfloor$ with $\rho(C_n) \rightarrow R$ and $e(C_n) \rightarrow 0$.

Note that this concerns the *average* error rate e ; we shall upgrade to $\hat{e}$ afterwards.

Proof. Without loss of generality $p < \frac{1}{2}$. Choose $\varepsilon > 0$ small enough that $p + \varepsilon < \frac{1}{2}$ and $R < 1 - H(p + \varepsilon)$; this is possible by continuity of H . Put $m = \lfloor 2^{nR} \rfloor$ and $r = \lfloor n(p + \varepsilon) \rfloor$. We use minimum distance decoding, breaking ties arbitrarily.

Let $\mathcal{C}$ be the set of all $[n, m]$ -codes, so $|\mathcal{C}| = \binom{2^n}{m}$, and choose $C = \{c_1, \dots, c_m\}$ from $\mathcal{C}$ uniformly at random. Choose $i \in \{1, \dots, m\}$ uniformly at random too, send c_i , and let Y be the output. Then

$$\mathbb{P}(Y \text{ not decoded as } c_i) = \frac{1}{|\mathcal{C}|} \sum_{C \in \mathcal{C}} e(C),$$

so there exists some $C_n \in \mathcal{C}$ with $e(C_n)$ at most this average. It therefore suffices to show that the left hand side tends to 0.

If $B(Y, r) \cap C = \{c_i\}$ then Y is certainly decoded as c_i , so

$$\mathbb{P}(Y \text{ not decoded as } c_i) \leq \mathbb{P}(c_i \notin B(Y, r)) + \mathbb{P}(B(Y, r) \cap C \supsetneq \{c_i\}),$$

and we handle the two terms separately.

The first term is $\mathbb{P}(d(c_i, Y) > r)$, which is the probability that the channel makes more than $n(p + \varepsilon)$ errors; this tends to 0 by Lemma 28.4.14.

For the second term, condition on Y and on c_i . Given that $c_i \in B(Y, r)$, each of the other $m - 1$ codewords is uniform over the remaining $2^n - 1$ words, so

$$\mathbb{P}(c_j \in B(Y, r) \mid c_i \in B(Y, r)) = \frac{V(n, r) - 1}{2^n - 1} \leq \frac{V(n, r)}{2^n}$$

for $j \neq i$. Hence, by a union bound and Proposition 28.3.25(i),

$$\begin{aligned} \mathbb{P}(B(Y, r) \cap C \not\supseteq \{c_i\}) &\leq \sum_{j \neq i} \mathbb{P}(c_j \in B(Y, r) \mid c_i \in B(Y, r)) \\ &\leq (m - 1) \frac{V(n, r)}{2^n} \leq 2^{nR} 2^{nH(p+\varepsilon)} 2^{-n} \\ &= 2^{-n(1-H(p+\varepsilon)-R)} \longrightarrow 0, \end{aligned}$$

because $R < 1 - H(p + \varepsilon)$. This completes the proof. $\square$

Proposition 28.4.29

In Proposition 28.4.28 we may replace e by $\hat{e}$.

Proof. Choose R' with $R < R' < 1 - H(p)$ and apply the previous proposition to R' , obtaining codes C'_n of size $\lfloor 2^{nR'} \rfloor$ with $e(C'_n) \rightarrow 0$. Order the codewords of C'_n by their individual error probabilities and delete the worse half, giving a code C''_n . If more than half the codewords had error probability exceeding $2e(C'_n)$ the average would exceed $e(C'_n)$; so $\hat{e}(C''_n) \leq 2e(C'_n) \rightarrow 0$. Now C''_n has size $\frac{1}{2} \lfloor 2^{nR'} \rfloor \geq 2^{nR}$ for large n , since $2^{nR'-1} = 2^{n(R'-1/n)} \geq 2^{nR}$ eventually. Passing to a subcode of size exactly $\lfloor 2^{nR} \rfloor$ can only decrease $\hat{e}$, and gives $\rho \rightarrow R$. $\square$

Putting the two halves together: the binary symmetric channel with error probability p has operational capacity exactly $1 - H(p)$, which by Example 28.4.25 is also its information capacity. This is Theorem 28.4.24 for the BSC.

Remark. The proof is purely existential. It tells us that almost every code of the right size is good, but it gives no way of writing one down, and no way of decoding one efficiently even if we could – minimum distance decoding over 2^{nR} codewords is hopeless. Finding explicit families of codes approaching capacity, with practical decoding algorithms, took another fifty years. The rest of these notes on coding theory is about algebraic constructions which are far from optimal but eminently practical.

§28.4.8 An Aside: the Kelly Criterion

Entropy turns up in places that have nothing obviously to do with communication. Here is one, included because it is charming and because it was lectured.

Suppose a biased coin with $\mathbb{P}(\text{head}) = p$ is tossed repeatedly, and that before each toss we may stake money on heads at odds u : a stake of k returns ku if the coin comes up heads and nothing otherwise. Start with a bankroll $X_0 = 1$ and bet a fixed fraction

$w \in [0, 1)$ of the current bankroll each time, so

$$X_{n+1} = \begin{cases} X_n(wu + (1-w)) & \text{if toss } n+1 \text{ is a head,} \\ X_n(1-w) & \text{otherwise.} \end{cases}$$

Then $Y_{n+1} = X_{n+1}/X_n$ are i.i.d., and $\log X_n = \sum_{i=1}^n \log Y_i$ is a sum of i.i.d. terms.

Lemma 28.4.30

Let $\mu = \mathbb{E}[\log Y_1]$ and $\sigma^2 = \text{Var}(\log Y_1)$. Then for $a > 0$,

$$\mathbb{P}\left(\left|\frac{\log X_n}{n} - \mu\right| \geq a\right) = \mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n \log Y_i - \mu\right| \geq a\right) \leq \frac{\sigma^2}{na^2}.$$

In particular, for any $\varepsilon, \delta > 0$ there is N with $\mathbb{P}(|\frac{\log X_n}{n} - \mu| \geq \delta) \leq \varepsilon$ for all $n \geq N$.

Proof. Immediate from Chebyshev's inequality applied to the sample mean. $\square$

So the bankroll behaves like $2^{n\mu}$, and to do well in the long run we should maximise μ rather than the expected return. Now

$$\mu = \mathbb{E}[\log Y] = p \log(1 + (u-1)w) + (1-p) \log(1-w).$$

Differentiating with respect to w and setting the result to zero, one finds that μ is maximised at $w = 0$ if $up \leq 1$ (the bet is not worth taking) and at

$$w = \frac{up - 1}{u - 1}$$

if $up > 1$. This is **Kelly's criterion**. Writing $q = 1 - p$ and substituting the optimal w ,

$$\mathbb{E}[\log Y] = p \log p + q \log q + \log u - q \log(u-1) = -H(p) + \log u - q \log(u-1).$$

Entropy has appeared, and it is exactly the penalty paid for the randomness of the coin. Betting below the Kelly fraction still grows the bankroll, but more slowly; betting sufficiently far above it makes the bankroll tend to zero almost surely, however favourable the odds.

§28.5 Linear Codes

Shannon's theorem guarantees good codes but produces none, and even if it did we could not use them, because minimum distance decoding over an unstructured code of size 2^{nR} is computationally hopeless. The way out is to insist that our codes be *linear subspaces*, at which point all the machinery of IB Linear Algebra becomes available and both encoding and decoding become matrix arithmetic.

§28.5.1 Linear Codes, Generator and Parity Check Matrices

Definition 28.5.1 (Linear Code)

A binary code $C \subseteq \mathbb{F}_2^n$ is **linear** if $0 \in C$ and $x + y \in C$ whenever $x, y \in C$; equivalently, if C is a vector subspace of $\mathbb{F}_2^n$.

Definition 28.5.2 (Rank)

The **rank** of a linear code C is its dimension as an $\mathbb{F}_2$ -vector space. A linear code of length n and rank k is an **(n, k) -code**; if its minimum distance is d it is an **(n, k, d) -code**.

We use round brackets for linear codes and square brackets for general ones. If $v_1, \dots, v_k$ is a basis then every codeword is uniquely $\sum \lambda_i v_i$ with $\lambda_i \in \mathbb{F}_2$, so $|C| = 2^k$. Thus an (n, k) -code is an $[n, 2^k]$ -code, and its information rate is $\frac{k}{n}$ – a pleasantly simple formula.

Definition 28.5.3 (Weight)

The **weight** of $x \in \mathbb{F}_2^n$ is $w(x) = d(x, 0)$, the number of nonzero digits.

Lemma 28.5.4

The minimum distance of a linear code equals the minimum weight of a nonzero codeword.

Proof. For $x, y \in \mathbb{F}_2^n$ we have $d(x, y) = d(x + y, 0) = w(x + y)$, since over $\mathbb{F}_2$ subtraction is addition. If C is linear and $x, y \in C$ are distinct then $x + y$ is a nonzero codeword, and conversely every nonzero codeword z arises as $z = z + 0$. So the two minima are over the same set of numbers. $\square$

This is already a large saving: computing $d(C)$ naively requires $\binom{2^k}{2}$ comparisons, but for a linear code only $2^k - 1$ weights.

Definition 28.5.5 (Dot Product, Dual Code)

For $x, y \in \mathbb{F}_2^n$ set $x \cdot y = \sum_{i=1}^n x_i y_i \in \mathbb{F}_2$; this is symmetric and bilinear. Given $P \subseteq \mathbb{F}_2^n$, the **parity check code** defined by P is

$$C = \{x \in \mathbb{F}_2^n : p \cdot x = 0 \text{ for all } p \in P\}.$$

For a linear code C , the **dual code** is

$$C^\perp = \{x \in \mathbb{F}_2^n : x \cdot y = 0 \text{ for all } y \in C\}.$$

A warning: this bilinear form is not an inner product, since $x \cdot x = 0$ for any x of even weight. In particular $C \cap C^\perp$ need not be $\{0\}$, and a code can even be *self-dual*.

Example 28.5.6 (i) $P = \{11 \dots 1\}$ gives the simple parity check code.

(ii) $P = \{1010101, 0110011, 0001111\}$ gives Hamming's original $[7, 16, 3]$ -code of Example 28.3.8.

(iii) If C is linear then so are the parity check extension C^+ and any punctured code C^- .

Lemma 28.5.7

Every parity check code is linear.

Proof. $p \cdot 0 = 0$, and if $p \cdot x = p \cdot y = 0$ then $p \cdot (x + y) = 0$ by bilinearity. $\square$

In particular $C^\perp$ is always linear. We shall shortly see the converse: every linear code is a parity check code.

Definition 28.5.8 (Generator and Parity Check Matrices)

Let C be an (n, k) -code. A **generator matrix** for C is a $k \times n$ matrix G whose rows form a basis of C . A **parity check matrix** for C is a generator matrix H for $C^\perp$; we shall see it is an $(n - k) \times n$ matrix.

The two matrices give two descriptions of the same code: codewords are the linear combinations of rows of G , and also the linear dependence relations among the columns of H ,

$$C = \{x \in \mathbb{F}_2^n : Hx = 0\},$$

where we regard x as a column vector. Encoding uses G and decoding uses H .

Definition 28.5.9 (Syndrome)

Let C be a linear code with parity check matrix H . The **syndrome** of $x \in \mathbb{F}_2^n$ is $Hx \in \mathbb{F}_2^{n-k}$.

Here is the point. Suppose we receive $x = c + z$ where $c \in C$ and z is the error pattern. Since $Hc = 0$,

$$Hx = Hz,$$

so the syndrome depends only on the error, not on the codeword that was sent. If C is e -error correcting we may precompute Hx for every x with $w(x) \leq e$ – there are $V(n, e)$ of them, far fewer than 2^k codewords in general – and store the results in a table. On receiving x we compute Hx , look it up, and decode as $c = x - z$. The cost of decoding is one matrix multiplication and one table lookup.

More generally, two words have the same syndrome exactly when they differ by a codeword, so the syndromes are in bijection with the cosets of C in $\mathbb{F}_2^n$. Minimum distance decoding amounts to choosing, in each coset, a word of least weight – a **coset leader** – and subtracting it. Displaying the cosets as the rows of an array gives the classical picture.

z	coset $z + C$				$H z$
0	0	c_2	$\cdots$	c_m	0
z_2	z_2	$z_2 + c_2$	$\cdots$	$z_2 + c_m$	s_2
z_3	z_3	$z_3 + c_2$	$\cdots$	$z_3 + c_m$	s_3
$\vdots$	$\vdots$	$\vdots$		$\vdots$	$\vdots$
z_t	z_t	$z_t + c_2$	$\cdots$	$z_t + c_m$	s_t

The first row is the code itself; the leftmost column holds the coset leaders; the rightmost column holds the syndromes, and it is the only column we need to store. On receiving x we find the row with syndrome Hx and subtract its leader.

Definition 28.5.10 (Equivalence)

Codes $C_1, C_2 \subseteq \mathbb{F}_2^n$ are **equivalent** if some permutation of the n coordinates carries C_1 onto C_2 .

Equivalent codes have the same length, size and minimum distance, so we generally work up to equivalence.

Lemma 28.5.11

Every (n, k) -code is equivalent to one with generator matrix in the **standard form**

$$G = (I_k \mid B)$$

for some $k \times (n - k)$ matrix B .

Proof. Let G be any generator matrix. Row operations do not change the row space, so by Gaussian elimination we may put G in row echelon form, with leading 1s in columns $\ell(1) < \ell(2) < \cdots < \ell(k)$. Permuting columns replaces C by an equivalent code, so we may assume $\ell(i) = i$. Further row operations clear the entries above the leading 1s, leaving $G = (I_k \mid B)$. $\square$

With G in standard form, a message $y \in \mathbb{F}_2^k$ (a row vector) is encoded as

$$yG = (y \mid yB),$$

that is, the message itself followed by $n - k$ **check digits**. Such an encoding is called **systematic**, and is convenient because the message can be read straight off the code-word.

Lemma 28.5.12

For a linear code $C \leq \mathbb{F}_2^n$, $\text{rank } C + \text{rank } C^\perp = n$.

Proof. This is the annihilator statement from IB Linear Algebra, but here is a direct argument. Passing to an equivalent code we may assume C has generator matrix $G = (I_k \mid B)$. Consider the linear map $\gamma: \mathbb{F}_2^n \rightarrow \mathbb{F}_2^k$, $x \mapsto Gx$. Since G contains I_k , its rows are linearly independent, so γ is surjective. Its kernel is exactly the set of x orthogonal to every row of G , that is, $C^\perp$. By rank-nullity,

$$n = \dim \ker \gamma + \dim \operatorname{img} \gamma = \operatorname{rank} C^\perp + k. \quad \square$$

Corollary 28.5.13

$(C^\perp)^\perp = C$. In particular every linear code is a parity check code, namely the one defined by $C^\perp$.

Proof. If $x \in C$ then $x \cdot y = 0$ for all $y \in C^\perp$ by definition, so $C \subseteq (C^\perp)^\perp$. Applying Lemma 28.5.12 twice,

$$\operatorname{rank}(C^\perp)^\perp = n - \operatorname{rank} C^\perp = n - (n - \operatorname{rank} C) = \operatorname{rank} C,$$

so the inclusion is an equality. $\square$

Lemma 28.5.14

If C has generator matrix $G = (I_k \mid B)$ then $H = (B^\top \mid I_{n-k})$ is a parity check matrix for C .

Proof. We compute

$$GH^\top = (I_k \mid B) \begin{pmatrix} B \\ I_{n-k} \end{pmatrix} = B + B = 0$$

over $\mathbb{F}_2$. So every row of H is orthogonal to every row of G , giving $\operatorname{rank} H \leq \dim C^\perp$ as subspaces; but H contains I_{n-k} , so $\operatorname{rank} H = n - k = \operatorname{rank} C^\perp$ by Lemma 28.5.12. Hence the rows of H are a basis of $C^\perp$. $\square$

The minimum distance can also be read off from H , which is often the most convenient way to compute it.

Lemma 28.5.15

Let C be a linear code with parity check matrix H . Then $d(C) = d$ if and only if some d columns of H are linearly dependent, but no $d - 1$ columns are.

Proof. A word x lies in C if and only if $Hx = 0$, that is, if and only if the columns of H indexed by the support of x sum to zero. So there is a nonzero codeword of weight w exactly when some w columns of H are linearly dependent. Now apply Lemma 28.5.4. $\square$

Example 28.5.16

Let C be the $(5, 2)$ -code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = (I_2 \ B), \quad B = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its four codewords are 00000, 10110, 01011, 11101, of weights 0, 3, 3, 4, so $d(C) = 3$ by Lemma 28.5.4 and C is a $(5, 2, 3)$ -code, correcting one error. By Lemma 28.5.14,

$$H = (B^\top \ I_3) = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix},$$

and indeed no two columns of H are equal (so no two are dependent), while columns 1, 3, 4 sum to zero – these being the positions of the codeword 10110 – confirming $d = 3$ via Lemma 28.5.15.

The syndromes of the five single-bit errors are just the columns of H :

z	00000	10000	01000	00100	00010	00001	00101	10001
Hx	000	110	011	100	010	001	101	111

There are 8 syndromes and only 6 patterns of weight at most 1, so two cosets need leaders of weight 2; the last two columns give such a choice. Note the choice is not unique – the syndrome 101 also arises from 11000, and 111 from 01100 – which is Lemma 28.3.11 telling us that a code with $d = 3$ cannot correct two errors.

Let us decode. Suppose we receive $x = 10111$. Since Hx is the sum of the columns of H in the positions where x is nonzero, namely 1, 3, 4, 5, we compute

$$Hx = (1, 1, 0) + (1, 0, 0) + (0, 1, 0) + (0, 0, 1) = (0, 0, 1).$$

Looking up 001 in the table gives $z = 00001$, so we decode x as $10111 + 00001 = 10110$, which is a codeword as required.

§28.5.2 Hamming Codes

Lemma 28.5.15 makes the construction of 1-error correcting codes almost automatic. We need $d = 3$, so we need a parity check matrix whose columns are pairwise linearly independent; over $\mathbb{F}_2$ that just means the columns are nonzero and distinct. To get as many columns as possible for a given number of rows, we should use *all* of them.

Definition 28.5.17 (Hamming Code)

Let $d \geq 2$ and $n = 2^d - 1$. Let H be the $d \times n$ matrix whose columns are the n nonzero elements of $\mathbb{F}_2^d$, in some order. The **Hamming $(n, n - d)$ -code** is the linear code with parity check matrix H .

Lemma 28.5.18

The Hamming $(n, n - d)$ -code has minimum distance 3 and is a perfect 1-error correcting code.

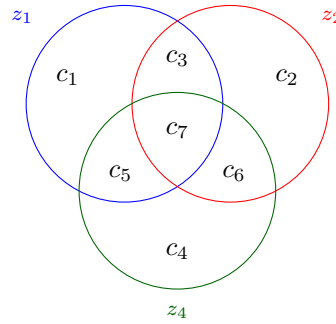
Proof. The columns of H are distinct and nonzero, so no two are dependent; and any three columns $u, v, u + v$ are dependent, and such a triple exists as soon as $d \geq 2$. By Lemma 28.5.15, $d(C) = 3$, so C corrects $\lfloor \frac{3-1}{2} \rfloor = 1$ error. Finally

$$\frac{2^n}{V(n, 1)} = \frac{2^n}{1 + n} = \frac{2^n}{2^d} = 2^{n-d} = |C|,$$

so C is perfect. $\square$

If the columns of H are listed in the order $1, 2, \dots, n$ read as binary numbers, then the syndrome of a single error in position i is the binary expansion of i , and decoding is instantaneous: compute the syndrome, read it as a number, and flip that bit. Taking $d = 3$ recovers Hamming's original $(7, 4, 3)$ -code of Example 28.3.8.

That code has a famous picture. Draw three overlapping circles, one for each parity check, and place the seven digits in the seven regions so that digit i lies in circle j exactly when the j th binary digit of i is 1.



Each circle must contain an even number of 1s: those are exactly the three parity conditions

$$c_1 + c_3 + c_5 + c_7 = 0, \quad c_2 + c_3 + c_6 + c_7 = 0, \quad c_4 + c_5 + c_6 + c_7 = 0.$$

Now the decoding rule becomes visual. A single error breaks precisely the parity of those circles containing it, and since every region is determined by the set of circles containing it, the broken circles identify the erroneous digit uniquely. If, for instance, only the blue and green circles fail, the culprit is c_5 , which lies in those two and no other.

§28.5.3 Weight Enumerators

The weights of the codewords of a linear code contain a great deal of information, and it is convenient to package them into a polynomial.

Definition 28.5.19 (Weight Enumerator)

The **weight enumerator** of a linear code $C \leq \mathbb{F}_2^n$ is the homogeneous polynomial

$$W_C(x, y) = \sum_{c \in C} x^{n-w(c)} y^{w(c)} = \sum_{i=0}^n A_i x^{n-i} y^i,$$

where A_i is the number of codewords of weight i .

Note $W_C(1, 1) = |C|$ and $W_C(1, 0) = 1$, and that $d(C)$ is the least $i > 0$ with $A_i \neq 0$. If C is used over a BSC with error probability p , then the probability that a transmitted codeword is received as another codeword – an undetected error – is $W_C(1-p, p) - (1-p)^n$, so the weight enumerator is exactly what one needs to analyse error detection.

Example 28.5.20

The repetition code of length n has $W(x, y) = x^n + y^n$. The simple parity check code of length n has $W(x, y) = \sum_{i \text{ even}} \binom{n}{i} x^{n-i} y^i = \frac{1}{2} [(x+y)^n + (x-y)^n]$. Hamming's $(7, 4, 3)$ -code has weight distribution $A_0 = 1$, $A_3 = A_4 = 7$, $A_7 = 1$, so

$$W(x, y) = x^7 + 7x^4y^3 + 7x^3y^4 + y^7.$$

(One checks $A_3 = 7$ directly: a weight-3 codeword corresponds to a triple of columns of H summing to zero, and each of the 7 nonzero syndromes u determines such a triple. The remaining counts follow since C contains 1111111, so $A_i = A_{7-i}$.)

There is a beautiful and useful relation between the weight enumerator of a code and that of its dual, which comes from Fourier analysis on $\mathbb{F}_2^n$.

Lemma 28.5.21 (Poisson Summation)

Let $C \leq \mathbb{F}_2^n$ be linear and let $f: \mathbb{F}_2^n \rightarrow \mathbb{R}$ have Fourier transform

$$\hat{f}(v) = \sum_{u \in \mathbb{F}_2^n} (-1)^{u \cdot v} f(u).$$

Then

$$\sum_{v \in C} \hat{f}(v) = |C| \sum_{u \in C^\perp} f(u).$$

Proof. Exchanging the order of summation,

$$\sum_{v \in C} \hat{f}(v) = \sum_{v \in C} \sum_u (-1)^{u \cdot v} f(u) = \sum_u f(u) \sum_{v \in C} (-1)^{u \cdot v}.$$

The inner sum is $|C|$ if $u \in C^\perp$. Otherwise the map $v \mapsto u \cdot v$ is a surjective linear map $C \rightarrow \mathbb{F}_2$, so its kernel has index 2 in C and the sum is $\frac{1}{2}|C| - \frac{1}{2}|C| = 0$. $\square$

Theorem 28.5.22 (MacWilliams Identity)

For a linear code $C \leq \mathbb{F}_2^n$,

$$W_{C^\perp}(x, y) = \frac{1}{|C|} W_C(x+y, x-y).$$

Proof. Apply Lemma 28.5.21 to $f(u) = x^{n-w(u)} y^{w(u)}$, which factorises as $f(u) =$

$\prod_{i=1}^n g(u_i)$ with $g(0) = x$, $g(1) = y$. Then the Fourier transform factorises too:

$$\hat{f}(v) = \sum_{u \in \mathbb{F}_2^n} \prod_{i=1}^n (-1)^{u_i v_i} g(u_i) = \prod_{i=1}^n \left(\sum_{u_i \in \mathbb{F}_2} (-1)^{u_i v_i} g(u_i) \right) = \prod_{i=1}^n (x + (-1)^{v_i} y),$$

which is $(x + y)^{n-w(v)}(x - y)^{w(v)}$. Summing over $v \in C$ gives $W_C(x + y, x - y)$ on the left of Lemma 28.5.21, and $|C| W_{C^\perp}(x, y)$ on the right. $\square$

Example 28.5.23

Take C the simple parity check code of length n , with $W_C(x, y) = \frac{1}{2}[(x+y)^n + (x-y)^n]$ and $|C| = 2^{n-1}$. Then

$$W_{C^\perp}(x, y) = \frac{1}{2^{n-1}} \cdot \frac{1}{2} [(2x)^n + (2y)^n] = x^n + y^n,$$

which is the weight enumerator of the repetition code. And indeed the dual of the parity check code is the repetition code, as one sees directly.

§28.5.4 Reed–Muller Codes

Our final family of linear codes is built out of the geometry of $\mathbb{F}_2^d$, and includes several codes we have already met. It is the family used by NASA to transmit the Mariner photographs of Mars.

Let $X = \{p_1, \dots, p_n\}$ be a set of size n . There is a chain of bijections

$$\mathcal{P}(X) \xrightarrow{A \mapsto \mathbb{1}_A} \{f: X \rightarrow \mathbb{F}_2\} \xrightarrow{f \mapsto (f(p_1), \dots, f(p_n))} \mathbb{F}_2^n,$$

under which the symmetric difference $A \triangle B$ corresponds to vector addition, and the intersection $A \cap B$ corresponds to the **wedge product**

$$x \wedge y = (x_1 y_1, \dots, x_n y_n).$$

So we may think of vectors in $\mathbb{F}_2^n$ as subsets of X , and we shall pass between the two pictures freely.

Now take $X = \mathbb{F}_2^d$, so that $n = |X| = 2^d$. Let $v_0 = (1, 1, \dots, 1) = \mathbb{1}_X$, and for $1 \leq i \leq d$ let

$$v_i = \mathbb{1}_{H_i}, \quad H_i = \{p \in X : p_i = 0\},$$

the indicator of the i th coordinate hyperplane.

Definition 28.5.24 (Reed–Muller Code)

For $0 \leq r \leq d$, the **Reed–Muller code** $RM(d, r)$ of **order** r and length 2^d is the linear code spanned by v_0 together with all wedge products of at most r of the v_i , $1 \leq i \leq d$. (By convention the empty wedge product is v_0 .)

Example 28.5.25

Let $d = 3$ and list $X = \mathbb{F}_2^3$ in binary order. Then the spanning vectors are

X	000	001	010	011	100	101	110	111
v_0	1	1	1	1	1	1	1	1
v_1	1	1	1	1	0	0	0	0
v_2	1	1	0	0	1	1	0	0
v_3	1	0	1	0	1	0	1	0
$v_1 \wedge v_2$	1	1	0	0	0	0	0	0
$v_1 \wedge v_3$	1	0	1	0	0	0	0	0
$v_2 \wedge v_3$	1	0	0	0	1	0	0	0
$v_1 \wedge v_2 \wedge v_3$	1	0	0	0	0	0	0	0

Notice that deleting the first column and keeping only the rows v_0, v_1, v_2, v_3 leaves a 4×7 matrix, and the code it generates has weight distribution $A_0 = 1$, $A_3 = A_4 = 7$, $A_7 = 1$. So it is a $(7, 4, 3)$ -code – indeed it is Hamming’s original code, punctured out of $RM(3, 1)$.

We can identify the small cases directly. $RM(3, 0)$ is spanned by v_0 alone, so it is the repetition code of length 8. $RM(3, 1)$ is spanned by v_0, v_1, v_2, v_3 , and is equivalent to the parity check extension of Hamming’s $(7, 4)$ -code. $RM(3, 2)$ is an $(8, 7)$ -code, equivalent to the simple parity check code of length 8. And $RM(3, 3)$ is the whole of $\mathbb{F}_2^8$.

Theorem 28.5.26 (i) The 2^d vectors $v_{i_1} \wedge \cdots \wedge v_{i_s}$ with $i_1 < \cdots < i_s$ and $0 \leq s \leq d$ form a basis of $\mathbb{F}_2^n$, where $n = 2^d$.

(ii) $\text{rank } RM(d, r) = \sum_{s=0}^r \binom{d}{s}$.

Proof. (i). There are $\sum_{s=0}^d \binom{d}{s} = 2^d = n$ vectors in the list, so it suffices to show they span, that is, that $RM(d, d) = \mathbb{F}_2^n$. Fix $p \in X$ and set

$$y_i = \begin{cases} v_i & \text{if } p_i = 0, \\ v_0 + v_i & \text{if } p_i = 1. \end{cases}$$

In the set picture y_i is the indicator of $\{q : q_i = p_i\}$, so

$$y_1 \wedge \cdots \wedge y_d = \mathbb{1}_{\{p\}}.$$

Expanding the left hand side by distributivity of $\wedge$ over $+$ writes $\mathbb{1}_{\{p\}}$ as a sum of wedge products of the v_i , hence $\mathbb{1}_{\{p\}} \in RM(d, d)$. As p was arbitrary and the $\mathbb{1}_{\{p\}}$ span $\mathbb{F}_2^n$, we are done.

(ii). $RM(d, r)$ is spanned by those basis vectors with $s \leq r$, of which there are $\sum_{s \leq r} \binom{d}{s}$, and these are linearly independent by (i). $\square$

To compute the minimum distance we use a construction which is of independent interest.

Definition 28.5.27 (Bar Product)

Let $C_2 \subseteq C_1$ be linear codes of length n . Their **bar product** is

$$C_1 \mid C_2 = \{(x \mid x + y) : x \in C_1, y \in C_2\},$$

a linear code of length $2n$.

Lemma 28.5.28 (i) $\text{rank}(C_1 \mid C_2) = \text{rank } C_1 + \text{rank } C_2$;

(ii) $d(C_1 \mid C_2) = \min\{2d(C_1), d(C_2)\}$.

Proof. (i). If $x_1, \dots, x_k$ is a basis of C_1 and $y_1, \dots, y_\ell$ a basis of C_2 , then

$$\{(x_i \mid x_i) : i \leq k\} \cup \{(0 \mid y_j) : j \leq \ell\}$$

spans $C_1 \mid C_2$, and is independent: a vanishing combination must have zero first half, forcing all x -coefficients to vanish, and then the y -coefficients too.

(ii). Let $(x \mid x + y)$ be a nonzero element. If $y \neq 0$ then, since weight is subadditive and $w(u) + w(u + v) \geq w(v)$,

$$w(x \mid x + y) = w(x) + w(x + y) \geq w(y) \geq d(C_2).$$

If $y = 0$ then $x \neq 0$ and $w(x \mid x) = 2w(x) \geq 2d(C_1)$. Hence $d(C_1 \mid C_2) \geq \min\{2d(C_1), d(C_2)\}$. For the reverse, choose $x \in C_1$ of weight $d(C_1)$ to get a codeword $(x \mid x)$ of weight $2d(C_1)$, and choose $y \in C_2$ of weight $d(C_2)$ to get $(0 \mid y)$ of weight $d(C_2)$. $\square$

Theorem 28.5.29 (i) $RM(d, r) = RM(d - 1, r) \mid RM(d - 1, r - 1)$ for $0 < r < d$;

(ii) $RM(d, r)$ has minimum distance 2^{d-r} .

Proof. (i). Order $X = \mathbb{F}_2^d$ so that the points with last coordinate 0 come first, identifying the two halves with $\mathbb{F}_2^{d-1}$. Under this splitting, the vectors $v_0, v_1, \dots, v_{d-1}$ for $\mathbb{F}_2^d$ become $(w \mid w)$ where w is the corresponding vector for $\mathbb{F}_2^{d-1}$, while $v_d = (\mathbf{1} \mid \mathbf{0})$. A wedge product of at most r of the v_i with $i < d$ therefore has the form $(w \mid w)$ with $w \in RM(d - 1, r)$; and a wedge product involving v_d has the form $(w \mid 0)$ with $w \in RM(d - 1, r - 1)$, since one of its r factors has been used up. As $(w \mid 0) = (w \mid w + w)$, every generator lies in $RM(d - 1, r) \mid RM(d - 1, r - 1)$; and comparing ranks using Theorem 28.5.26(ii), Lemma 28.5.28(i) and Pascal's rule

$$\sum_{s \leq r} \binom{d}{s} = \sum_{s \leq r} \binom{d-1}{s} + \sum_{s \leq r-1} \binom{d-1}{s}$$

shows the two codes have the same dimension, hence are equal.

(ii). If $r = 0$ then $RM(d, 0)$ is the repetition code of length 2^d , of minimum distance 2^d . If $r = d$ then $RM(d, d) = \mathbb{F}_2^{2^d}$, of minimum distance $1 = 2^{d-d}$. For $0 < r < d$ we induct on d , using (i) and Lemma 28.5.28(ii):

$$d(RM(d, r)) = \min\{2 \cdot 2^{(d-1)-r}, 2^{(d-1)-(r-1)}\} = \min\{2^{d-r}, 2^{d-r}\} = 2^{d-r}. \quad \square$$

Example 28.5.30

$RM(5, 1)$ is a $(32, 6)$ -code with minimum distance $2^4 = 16$, so it corrects 7 errors at rate $\frac{6}{32}$. This is the code used by Mariner 9 in 1971 to send back pictures of Mars: each pixel was one of 64 grey levels, encoded in 32 bits, and the channel was extremely noisy.

§28.6 Cyclic Codes

Linear codes made decoding into linear algebra. Imposing one further symmetry – invariance under cyclic shifts – turns them into ideals in a polynomial ring, and then all of the theory of finite fields from IB Groups, Rings and Modules comes into play. The reward is a family of codes with prescribed minimum distance and an efficient decoding algorithm.

§28.6.1 Cyclic Codes and Generator Polynomials**Definition 28.6.1 (Cyclic Code)**

A linear code $C \subseteq \mathbb{F}_2^n$ is **cyclic** if

$$(a_0, a_1, \dots, a_{n-1}) \in C \implies (a_{n-1}, a_0, \dots, a_{n-2}) \in C.$$

Identify $\mathbb{F}_2^n$ with the quotient ring

$$R = \mathbb{F}_2[X]/(X^n - 1)$$

via $\pi(a_0 + a_1X + \dots + a_{n-1}X^{n-1}) = (a_0, \dots, a_{n-1})$; this is a bijection because every class modulo $X^n - 1$ has a unique representative of degree less than n . Under this identification, cyclic shift becomes multiplication by X :

$$X \cdot (a_0 + a_1X + \dots + a_{n-1}X^{n-1}) = a_{n-1} + a_0X + \dots + a_{n-2}X^{n-1} \pmod{X^n - 1}.$$

Lemma 28.6.2

A code $C \subseteq \mathbb{F}_2^n$ is cyclic if and only if the corresponding subset $\mathcal{C} \subseteq R$ is an ideal of R .

Proof. Linearity of C says exactly that $\mathcal{C}$ is an additive subgroup, and the computation above says that closure under cyclic shift is closure under multiplication by X . Since $\mathcal{C}$ is closed under addition and multiplication by X , it is closed under multiplication by every polynomial, that is, it is an ideal; and conversely. $\square$

From now on we identify C with $\mathcal{C}$. Ideals of $R = \mathbb{F}_2[X]/(X^n - 1)$ correspond to ideals of $\mathbb{F}_2[X]$ containing $X^n - 1$; and $\mathbb{F}_2[X]$ is a principal ideal domain, so these correspond to the divisors of $X^n - 1$. This gives a complete classification.

Theorem 28.6.3 (Generator Polynomial)

Let C be a cyclic code of length n . Then there is a unique polynomial $g(X) \in \mathbb{F}_2[X]$ with

- (i) $C = (g)$, the ideal generated by g ;
- (ii) $g(X) \mid X^n - 1$.

In particular, $p(X)$ represents a codeword if and only if $g \mid p$. We call g the **generator polynomial** of C .

Proof. Assume $C \neq \{0\}$ and let g be a nonzero polynomial of least degree representing a codeword, so $\deg g < n$. Certainly $(g) \subseteq C$ since C is an ideal. Conversely let p represent a codeword; by the division algorithm in $\mathbb{F}_2[X]$, $p = qg + r$ with $\deg r < \deg g$. Then $r = p - qg \in C$, and minimality of $\deg g$ forces $r = 0$. So $C = (g)$. Taking $p = X^n - 1$, which represents $0 \in C$, gives $g \mid X^n - 1$, which is (ii).

For uniqueness, if $(g_1) = (g_2)$ then $g_1 \mid g_2$ and $g_2 \mid g_1$ in $\mathbb{F}_2[X]$ (using the divisibility of $X^n - 1$ to work in $\mathbb{F}_2[X]$ rather than the quotient), so $g_1 = cg_2$ with $c \in \mathbb{F}_2^\times = \{1\}$. $\square$

Lemma 28.6.4

Let C be cyclic of length n with generator polynomial $g(X) = a_0 + a_1X + \cdots + a_kX^k$, $a_k \neq 0$. Then

$$g, Xg, X^2g, \dots, X^{n-k-1}g$$

is a basis for C , so $\text{rank } C = n - k$.

Proof. These $n - k$ polynomials have degrees $k, k+1, \dots, n-1$, all less than n , so they are genuine representatives and are linearly independent (distinct leading terms). They span: any codeword is qg for some q , and reducing q modulo X^{n-k} we may assume $\deg q < n - k$, so qg is a combination of the listed elements. $\square$

Corollary 28.6.5

With g as above, a generator matrix for C is the $(n - k) \times n$ matrix

$$G = \begin{pmatrix} a_0 & a_1 & \cdots & a_k & 0 & \cdots & 0 \\ 0 & a_0 & \cdots & a_{k-1} & a_k & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \\ 0 & 0 & \cdots & a_0 & a_1 & \cdots & a_k \end{pmatrix}.$$

Definition 28.6.6 (Parity Check Polynomial)

Let g be the generator polynomial of a cyclic code C of length n . The **parity check polynomial** is the polynomial h with $g(X)h(X) = X^n - 1$.

Corollary 28.6.7

Writing $h(X) = b_0 + b_1X + \cdots + b_{n-k}X^{n-k}$, a parity check matrix for C is the $k \times n$

matrix

$$H = \begin{pmatrix} b_{n-k} & b_{n-k-1} & \cdots & b_0 & 0 & \cdots & 0 \\ 0 & b_{n-k} & \cdots & b_1 & b_0 & \cdots & 0 \\ \vdots & & \ddots & & & \ddots & \vdots \\ 0 & 0 & \cdots & b_{n-k} & b_{n-k-1} & \cdots & b_0 \end{pmatrix}.$$

Proof. The inner product of the i th row of G with the j th row of H is the coefficient of $X^{n-k-i+j}$ in $g(X)h(X) = X^n - 1$. Since $1 \leq i \leq n-k$ and $1 \leq j \leq k$, we have $0 < n-k-i+j < n$, and all such coefficients of $X^n - 1$ vanish. So the rows of H lie in $C^\perp$; and $b_{n-k} \neq 0$ gives $\text{rank } H = k = \text{rank } C^\perp$. $\square$

Remark. For $f(X) = \sum_{i=0}^m f_i X^i$ of degree m , the **reverse** polynomial is

$$\check{f}(X) = f_m + f_{m-1}X + \cdots + f_0 X^m = X^m f(1/X).$$

The matrix H above is exactly the generator matrix built from $\check{h}$, so the dual of a cyclic code is the cyclic code generated by $\check{h}$. In particular the dual of a cyclic code is cyclic.

Lemma 28.6.8

If n is odd then $X^n - 1 = f_1(X) \cdots f_t(X)$ with the f_i distinct irreducible polynomials in $\mathbb{F}_2[X]$. Consequently there are exactly 2^t cyclic codes of length n .

Proof sketch. $X^n - 1$ and its formal derivative $nX^{n-1} = X^{n-1}$ (as n is odd) are coprime, so $X^n - 1$ is separable and hence has no repeated irreducible factors. The cyclic codes correspond to the divisors of $X^n - 1$, of which there are 2^t . $\square$

The hypothesis is needed: over $\mathbb{F}_2$ we have $X^2 - 1 = (X - 1)^2$. From now on n is odd.

§28.6.2 BCH Codes

We now recall what we need about finite fields, all of which is from IB Groups, Rings and Modules. If $f \in \mathbb{F}_p[X]$ is irreducible then $\mathbb{F}_p[X]/(f)$ is a field with $p^{\deg f}$ elements, and every finite field arises this way; for each prime power $q = p^\alpha$ there is a unique field $\mathbb{F}_q$ of order q up to isomorphism (note $\mathbb{F}_q \not\cong \mathbb{Z}/q\mathbb{Z}$ when $\alpha > 1$). The multiplicative group $\mathbb{F}_q^\times$ is cyclic of order $q - 1$, and a generator is called a **primitive element**.

Fix an odd n , and choose $r \geq 1$ with $2^r \equiv 1 \pmod{n}$; such r exists since 2 is a unit modulo n . Let $K = \mathbb{F}_{2^r}$ and set

$$\mu_n(K) = \{x \in K : x^n = 1\} \leq K^\times.$$

As $n \mid 2^r - 1 = |K^\times|$ and $K^\times$ is cyclic, $\mu_n(K)$ is cyclic of order exactly n , say

$$\mu_n(K) = \{1, \alpha, \alpha^2, \dots, \alpha^{n-1}\}$$

for a primitive n th root of unity $\alpha \in K$. Since n is odd, $X^n - 1$ has n distinct roots in K , namely these, so K contains a splitting field for $X^n - 1$.

Definition 28.6.9 (Defining Set)

Let $A \subseteq \mu_n(K)$. The cyclic code of length n with **defining set** A is

$$C = \{f(X) \in \mathbb{F}_2[X]/(X^n - 1) : f(a) = 0 \text{ for all } a \in A\}.$$

This really is cyclic: the conditions $f(a) = 0$ are $\mathbb{F}_2$ -linear and are preserved under multiplication by X . Its generator polynomial is the nonzero polynomial in $\mathbb{F}_2[X]$ of least degree vanishing on A , that is, the least common multiple of the minimal polynomials over $\mathbb{F}_2$ of the elements of A .

Definition 28.6.10 (BCH Code)

The cyclic code of length n with defining set $\{\alpha, \alpha^2, \dots, \alpha^{\delta-1}\}$ is the **BCH code** with **design distance** δ . (The name is for Bose, Ray-Chaudhuri and Hocquenghem.)

The point of the definition is the following bound, which lets us specify the minimum distance in advance – something no construction so far has allowed.

Lemma 28.6.11 (Vandermonde)

$$\det \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{pmatrix} = \prod_{1 \leq j < i \leq n} (x_i - x_j).$$

In particular the determinant is nonzero whenever the x_i are distinct.

Theorem 28.6.12 (BCH Bound)

A BCH code C with design distance δ has $d(C) \geq \delta$.

Proof. Consider the $(\delta - 1) \times n$ matrix over K

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \alpha^4 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{\delta-1} & \alpha^{2(\delta-1)} & \cdots & \alpha^{(\delta-1)(n-1)} \end{pmatrix}.$$

A word $c \in \mathbb{F}_2^n$ lies in C precisely when $Hc = 0$, since the j th row of Hc is $c(\alpha^j)$. Take any $\delta - 1$ columns of H , say those indexed by $i_1 < \cdots < i_{\delta-1}$. Dividing the j th row by nothing and factoring α^{i_s} out of the s th column, the resulting $(\delta - 1) \times (\delta - 1)$ matrix is a Vandermonde matrix in the distinct elements $\alpha^{i_1}, \dots, \alpha^{i_{\delta-1}}$, hence invertible by Lemma 28.6.11. So no $\delta - 1$ columns of H are linearly dependent, and by the argument of Lemma 28.5.15 every nonzero codeword has weight at least δ . $\square$

Remark. The matrix H is not a parity check matrix in our earlier sense, since its entries lie in K rather than $\mathbb{F}_2$. Nevertheless the dependence argument goes through verbatim, because a codeword is a vector over $\mathbb{F}_2 \subseteq K$ and linear dependence over K is what we tested.

Example 28.6.13

Take $n = 7$. Over $\mathbb{F}_2$,

$$X^7 - 1 = (X + 1)(X^3 + X + 1)(X^3 + X^2 + 1),$$

so by Lemma 28.6.8 there are $2^3 = 8$ cyclic codes of length 7. Take $g(X) = X^3 + X + 1$, so

$$h(X) = (X + 1)(X^3 + X^2 + 1) = X^4 + X^2 + X + 1,$$

and $\text{rank } C = 7 - 3 = 4$. The parity check matrix built from $\check{h}(X) = X^4 + X^3 + X^2 + 1$ is

$$H = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix},$$

whose columns are precisely the seven nonzero elements of $\mathbb{F}_2^3$. So C is (equivalent to) the Hamming (7, 4)-code.

Let us see this again from the BCH point of view. Take $K = \mathbb{F}_8 = \mathbb{F}_2[\beta]/(\beta^3 + \beta + 1)$, so β is a root of g and a primitive 7th root of unity. The powers of β are

i	0	1	2	3	4	5	6
β^i	1	β	β^2	$\beta + 1$	$\beta^2 + \beta$	$\beta^2 + \beta + 1$	$\beta^2 + 1$

Squaring is a field automorphism of K fixing $\mathbb{F}_2$, so $g(\beta) = 0$ gives $g(\beta^2) = g(\beta)^2 = 0$ and likewise $g(\beta^4) = 0$. Hence the BCH code with defining set $\{\beta, \beta^2\}$ has generator polynomial exactly g , and its design distance is $\delta = 3$. So $d(C) \geq 3$; and we know Hamming's code has $d = 3$ exactly.

§28.6.3 Decoding BCH Codes

Let C be a BCH code of length n with defining set $\{\alpha, \dots, \alpha^{\delta-1}\}$, so $d(C) \geq \delta$ and C should correct

$$t = \lfloor \frac{\delta - 1}{2} \rfloor$$

errors. Suppose $c \in C$ is sent and $r = c + e$ is received, where the error pattern e has at most t nonzero digits. Writing $r(X), c(X), e(X)$ for the corresponding polynomials, $c(\alpha^j) = 0$ for $1 \leq j \leq \delta - 1$ and so

$$r(\alpha^j) = e(\alpha^j), \quad 1 \leq j \leq \delta - 1.$$

These $\delta - 1$ elements of K are the **syndromes**, and everything must be recovered from them.

Definition 28.6.14 (Error Locator Polynomial)

Let $\mathcal{E} = \{i : e_i = 1\}$ be the set of error positions. The **error locator polynomial** is

$$\sigma(X) = \prod_{i \in \mathcal{E}} (1 - \alpha^i X) \in K[X].$$

The roots of σ are exactly the α^{-i} for $i \in \mathcal{E}$, so knowing σ is the same as knowing where the errors are. Note $\sigma(0) = 1$ and $\deg \sigma = |\mathcal{E}| \leq t$.

Theorem 28.6.15

Suppose $|\mathcal{E}| \leq t$ where $2t + 1 \leq \delta$. Then $\sigma(X)$ is the unique polynomial of least degree in $K[X]$ such that

(i) $\sigma(0) = 1$;

(ii) $\sigma(X) \sum_{j=1}^{2t} r(\alpha^j) X^j \equiv \omega(X) \pmod{X^{2t+1}}$ for some $\omega \in K[X]$ with $\deg \omega \leq t$.

Proof. Existence. Define the **error co-locator** $\omega(X) = -X\sigma'(X)$, so by the product rule

$$\omega(X) = \sum_{i \in \mathcal{E}} \alpha^i X \prod_{j \in \mathcal{E}, j \neq i} (1 - \alpha^j X),$$

which has degree $\deg \sigma = |\mathcal{E}| \leq t$. Work in the ring $K[[X]]$ of formal power series, in which $1 - \alpha^i X$ is invertible. A partial fractions computation gives

$$\frac{\omega(X)}{\sigma(X)} = \sum_{i \in \mathcal{E}} \frac{\alpha^i X}{1 - \alpha^i X} = \sum_{i \in \mathcal{E}} \sum_{j \geq 1} (\alpha^i X)^j = \sum_{j \geq 1} X^j \sum_{i \in \mathcal{E}} (\alpha^j)^i = \sum_{j \geq 1} e(\alpha^j) X^j,$$

the last step because $e(X) = \sum_{i \in \mathcal{E}} X^i$. Hence

$$\sigma(X) \sum_{j \geq 1} e(\alpha^j) X^j = \omega(X)$$

exactly, in $K[[X]]$. Since $2t \leq \delta - 1$ we have $r(\alpha^j) = e(\alpha^j)$ for $1 \leq j \leq 2t$, and truncating modulo X^{2t+1} gives (ii).

Uniqueness. Suppose $\tilde{\sigma}, \tilde{\omega}$ also satisfy (i) and (ii) with $\deg \tilde{\sigma} \leq \deg \sigma$. The roots of σ are distinct and nonzero, and $\omega(X) = -X\sigma'(X)$ does not vanish at them (differentiate σ at a simple root); so σ and ω are coprime. From (ii) for both pairs,

$$\tilde{\sigma}(X)\omega(X) \equiv \sigma(X)\tilde{\omega}(X) \pmod{X^{2t+1}},$$

and both sides have degree at most $2t$, so this is an equality of polynomials. Since σ and ω are coprime, $\sigma \mid \tilde{\omega}$; combined with $\deg \tilde{\sigma} \leq \deg \sigma$ this gives $\tilde{\sigma} = \lambda\sigma$, and (i) forces $\lambda = 1$. $\square$

This yields an algorithm. On receiving $r(X)$:

- (1) Compute the syndromes $r(\alpha^j)$ for $1 \leq j \leq 2t$ and form $S(X) = \sum_{j=1}^{2t} r(\alpha^j) X^j$.
- (2) Write $\sigma(X) = 1 + \sigma_1 X + \cdots + \sigma_t X^t$ with unknown coefficients. Condition (ii) says the coefficients of X^i in $\sigma(X)S(X)$ vanish for $t+1 \leq i \leq 2t$, giving t linear equations

$$\sum_{j=0}^t \sigma_j r(\alpha^{i-j}) = 0, \quad \sigma_0 = 1.$$

- (3) Solve these over K , keeping a solution of least degree.
- (4) Compute $\mathcal{E} = \{i : \sigma(\alpha^{-i}) = 0\}$ and check $|\mathcal{E}| = \deg \sigma$.
- (5) Set $e(X) = \sum_{i \in \mathcal{E}} X^i$, put $c(X) = r(X) + e(X)$, and check that c is a codeword.

Example 28.6.16

Continue Example 28.6.13: $n = 7$, $g(X) = X^3 + X + 1$, defining set $\{\beta, \beta^2\}$, $\delta = 3$, $t = 1$. Take the codeword $c(X) = g(X) = 1 + X + X^3$, that is $c = 1101000$, and suppose an error occurs in position 5, so

$$r(X) = 1 + X + X^3 + X^5.$$

We compute the syndromes, using the table of powers of β above:

$$r(\beta) = 1 + \beta + (\beta + 1) + (\beta^2 + \beta + 1) = 1 + \beta + \beta^2 = \beta^5,$$

and $r(\beta^2) = r(\beta)^2 = \beta^{10} = \beta^3$. So $S(X) = \beta^5 X + \beta^3 X^2$.

With $t = 1$ we set $\sigma(X) = 1 + \sigma_1 X$; the coefficient of X^2 in $\sigma(X)S(X)$ is $\beta^3 + \sigma_1 \beta^5$, which must vanish, so

$$\sigma_1 = \beta^{3-5} = \beta^{-2} = \beta^5.$$

Thus $\sigma(X) = 1 + \beta^5 X$, whose root is $X = \beta^{-5} = \beta^2$. Solving $\beta^{-i} = \beta^2$ gives $i \equiv -2 \equiv 5 \pmod{7}$, so $\mathcal{E} = \{5\}$ and $|\mathcal{E}| = 1 = \deg \sigma$ as required. Correcting, $c(X) = r(X) + X^5 = 1 + X + X^3$, which is indeed a codeword. Everything is consistent.

Example 28.6.17

For a code that genuinely corrects two errors, take $n = 15$ and $K = \mathbb{F}_{16} = \mathbb{F}_2[\alpha]/(\alpha^4 + \alpha + 1)$, so α is a primitive 15th root of unity. The minimal polynomial of α is $X^4 + X + 1$ (with roots $\alpha, \alpha^2, \alpha^4, \alpha^8$) and that of α^3 is $X^4 + X^3 + X^2 + X + 1$ (with roots $\alpha^3, \alpha^6, \alpha^{12}, \alpha^9$). The BCH code with defining set $\{\alpha, \alpha^2, \alpha^3, \alpha^4\}$ therefore has

$$g(X) = (X^4 + X + 1)(X^4 + X^3 + X^2 + X + 1)$$

of degree 8, giving a $(15, 7)$ -code with design distance 5; in fact $d = 5$ exactly, so it is a $(15, 7, 5)$ -code correcting two errors.

§28.6.4 Reed–Solomon Codes

Everything above works over any finite field, and one particularly clean case deserves its own name. Let q be a prime power, let α be a primitive element of $\mathbb{F}_q$, and take $n = q - 1$, so that α is a primitive n th root of unity *in the field $\mathbb{F}_q$ itself*. Codes are now subsets of $\mathbb{F}_q^n$; Hamming distance, weight and linearity all make sense verbatim.

Definition 28.6.18 (Reed–Solomon Code)

Let $1 \leq k \leq n = q - 1$. The **Reed–Solomon code** $RS(n, k)$ is the cyclic code over $\mathbb{F}_q$ of length n with generator polynomial

$$g(X) = (X - \alpha)(X - \alpha^2) \cdots (X - \alpha^{n-k}).$$

Equivalently it is the BCH code with defining set $\{\alpha, \dots, \alpha^{n-k}\}$ and design distance $\delta = n - k + 1$; the difference from the binary case is that the defining set already consists of elements of the ground field, so no extension is needed and the degree of g is exactly

$n - k$.

Proposition 28.6.19

$RS(n, k)$ has rank k and minimum distance exactly $n - k + 1$. In particular it is MDS: it attains the Singleton bound.

Proof. Since $\deg g = n - k$, Lemma 28.6.4 gives $\text{rank} = n - (n - k) = k$. The BCH bound gives $d \geq \delta = n - k + 1$, and the Singleton bound (Theorem 28.3.23, whose proof works over any alphabet, giving $|C| \leq q^{n-d+1}$) gives $q^k \leq q^{n-d+1}$, that is, $d \leq n - k + 1$. $\square$

Example 28.6.20

Take $q = 5$ and $\alpha = 2$, a primitive root modulo 5 since 2, 4, 3, 1 are its successive powers. With $n = 4$ and $k = 2$,

$$g(X) = (X - 2)(X - 4) = X^2 - 6X + 8 = X^2 + 4X + 3 \in \mathbb{F}_5[X],$$

and indeed $g \mid X^4 - 1 = (X - 1)(X - 2)(X - 3)(X - 4)$. The code is

$$RS(4, 2) = \{(a + bX)g(X) : a, b \in \mathbb{F}_5\},$$

of size 25. For instance g itself is the codeword $(3, 4, 1, 0)$, of weight 3, and one checks that every nonzero codeword has weight at least $3 = n - k + 1$. So this is a $(4, 2, 3)$ MDS code over $\mathbb{F}_5$.

Remark. Reed–Solomon codes are ubiquitous in practice – they are what protects the data on a compact disc, in a QR code, and on the Voyager spacecraft. Their great virtue is that a symbol of $\mathbb{F}_{2^m}$ is a block of m bits, so a single symbol error absorbs a whole *burst* of up to m consecutive bit errors. Real channels tend to produce errors in bursts rather than independently, which is exactly the situation the binary symmetric channel fails to model.

§28.6.5 Linear Feedback Shift Registers

We now change tack, apparently completely, and consider a piece of hardware. The connection to what we have just done will emerge shortly, and it will also be the basis of the stream ciphers of Section 28.7.

Definition 28.6.21 (Feedback Shift Register)

A **(general) feedback shift register** of length d is a map $f: \mathbb{F}_2^d \rightarrow \mathbb{F}_2^d$ of the form

$$f(x_0, \dots, x_{d-1}) = (x_1, \dots, x_{d-1}, C(x_0, \dots, x_{d-1}))$$

for some function $C: \mathbb{F}_2^d \rightarrow \mathbb{F}_2$. Given an **initial fill** $(y_0, \dots, y_{d-1})$, the associated **stream** is the sequence $y_0, y_1, \dots$ defined by

$$y_n = C(y_{n-d}, \dots, y_{n-1}), \quad n \geq d.$$

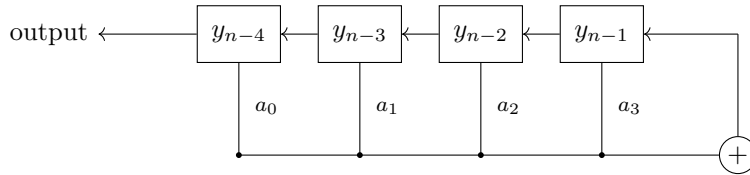
Definition 28.6.22 (LFSR)

A feedback shift register is a **linear feedback shift register** (LFSR) if C is linear, say

$$C(x_0, \dots, x_{d-1}) = \sum_{i=0}^{d-1} a_i x_i.$$

We always take $a_0 = 1$, since otherwise the register really has length less than d .

Physically it is a row of d memory cells; at each tick every cell passes its contents to its left neighbour, the leftmost cell is output, and the rightmost cell receives the sum of the tapped cells.



For an LFSR the stream satisfies the linear recurrence

$$y_n = \sum_{i=0}^{d-1} a_i y_{n-d+i}, \quad n \geq d,$$

and it is natural to encode the coefficients in a polynomial.

Definition 28.6.23 (Auxiliary and Feedback Polynomials)

The **auxiliary polynomial** of the LFSR is

$$P(X) = X^d + a_{d-1}X^{d-1} + \dots + a_1X + a_0 = \sum_{i=0}^d a_i X^i \quad (a_d := 1),$$

and the **feedback polynomial** is its reverse

$$\check{P}(X) = a_0X^d + a_1X^{d-1} + \dots + a_{d-1}X + 1 = \sum_{i=0}^d a_{d-i}X^i.$$

The **generating function** of a stream (y_n) is $G(X) = \sum_{j \geq 0} y_j X^j \in \mathbb{F}_2[[X]]$.

Theorem 28.6.24

A stream $(y_n)_{n \geq 0}$ arises from an LFSR with auxiliary polynomial P if and only if its generating function has the form

$$G(X) = \frac{A(X)}{\check{P}(X)}$$

for some $A \in \mathbb{F}_2[X]$ with $\deg A < \deg \check{P} = d$.

Proof. The condition is that

$$G(X)\check{P}(X) = \left(\sum_{j \geq 0} y_j X^j\right) \left(\sum_{i=0}^d a_{d-i} X^i\right)$$

is a polynomial of degree less than d , that is, that the coefficient of X^n vanishes for all $n \geq d$. That coefficient is

$$\sum_{i=0}^d a_{d-i} y_{n-i} = y_n + \sum_{i=1}^d a_{d-i} y_{n-i},$$

using $a_d = 1$. Setting this to zero and reindexing gives exactly $y_n = \sum_{i=0}^{d-1} a_i y_{n-d+i}$. $\square$

So streams from LFSRs of length d are precisely the sequences whose generating function is a rational function with denominator of degree d – which is the same problem we solved when decoding BCH codes, where we had to recognise a power series as ω/σ . The two subjects are the same subject.

Example 28.6.25

Take $d = 4$ with $a_0 = a_1 = 1$ and $a_2 = a_3 = 0$, so

$$y_n = y_{n-4} + y_{n-3}, \quad P(X) = X^4 + X + 1, \quad \check{P}(X) = X^4 + X^3 + 1.$$

With initial fill 1000 the stream begins

$$1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, \dots$$

and one checks that $y_{15+r} = y_r$, so the period is $15 = 2^4 - 1$, the largest possible. (This happens exactly when P is a primitive polynomial, meaning that its roots are primitive elements of $\mathbb{F}_{16}$.)

§28.6.6 The Berlekamp–Massey Method

Suppose now that we are handed a stream (x_n) which we are told comes from *some* LFSR, but we know neither its length d nor its coefficients. Recovering them is the problem of **linear complexity**, and it has a clean linear-algebraic solution.

We seek d and $a_0, \dots, a_{d-1}$ (with $a_d = 1$) such that

$$x_n + \sum_{i=1}^d a_{d-i} x_{n-i} = 0 \quad \text{for all } n \geq d.$$

Writing out these equations for $n = d, d+1, \dots, 2d$ gives the matrix equation

$$\underbrace{\begin{pmatrix} x_d & x_{d-1} & \cdots & x_1 & x_0 \\ x_{d+1} & x_d & \cdots & x_2 & x_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{2d} & x_{2d-1} & \cdots & x_{d+1} & x_d \end{pmatrix}}_{A_d} \begin{pmatrix} a_d \\ a_{d-1} \\ \vdots \\ a_1 \\ a_0 \end{pmatrix} = 0.$$

This is a system of $d+1$ homogeneous equations in $d+1$ unknowns, and it has a nonzero solution – our sought-after coefficients, with $a_d = 1$ – so $\det A_d = 0$.

The **Berlekamp–Massey method** exploits this. Compute $\det A_0, \det A_1, \det A_2, \dots$ in turn (starting at A_r if we already know $d \geq r$).

- If $\det A_i \neq 0$ then $d \neq i$, and we move on.
- If $\det A_i = 0$ we tentatively assume $d = i$, solve the system for $a_0, \dots, a_{i-1}$, and test the resulting recurrence against as many further terms of the stream as we like. If it fails, we move on.

The first i that survives the test is the answer.

Example 28.6.26

Apply this to the stream of Example 28.6.25, pretending we do not know where it came from:

$$x = 1, 0, 0, 0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, \dots$$

- $A_0 = (x_0) = (1)$ has $\det = 1 \neq 0$, so $d \neq 0$.
- $A_1 = \begin{pmatrix} x_1 & x_0 \\ x_2 & x_1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $\det = 0$. Solving with $a_1 = 1$: the first row gives $0 \cdot 1 + a_0 = 0$, so $a_0 = 0$, and the recurrence $x_n = 0$ fails already at $x_4 = 1$. So $d \neq 1$.
- $A_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$ has a zero row, so $\det = 0$. Solving with $a_2 = 1$: row 1 gives $a_0 = 0$, row 3 gives $x_4 + a_1x_3 + a_0x_2 = 1 \neq 0$. The system is inconsistent, so $d \neq 2$.
- $A_3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ is a permutation matrix, so $\det = 1 \neq 0$ and $d \neq 3$.
- $A_4 = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}$. Solving with $a_4 = 1$: the rows give in turn $a_0 = 1$, $a_3 = 0$, $a_2 = 0$, $a_1 = 1$, and the last row $1 + a_3 + a_0 = 1 + 0 + 1 = 0$ is consistent. (So $\det A_4 = 0$, as it must be.)

We conclude that $d = 4$ with $(a_0, a_1, a_2, a_3) = (1, 1, 0, 0)$, that is, $y_n = y_{n-4} + y_{n-3}$, which is exactly the register we started with. Checking against the later terms of the stream confirms it.

Remark. The moral for cryptography is discouraging. A stream produced by a linear feedback shift register of length d is completely determined by $2d$ of its terms, recoverable by solving a linear system. Any cipher whose key stream is generated linearly is therefore broken by a modest amount of known plaintext, as we shall see in Section 28.7.

§28.7 Cryptography

We now turn from noise to malice. Alice wants to send Bob a message, and Eve is listening. The techniques of the previous sections were designed against an opponent – the channel – who was random but not clever; now our opponent is clever, and we have to design against the worst case rather than the average one.

§28.7.1 Cryptosystems and Classical Ciphers

Alice and Bob share some secret information called the **key**, drawn from a set $\mathcal{K}$. The unencrypted message, the **plaintext**, lies in a set $\mathcal{M}$, and the encrypted message, the **ciphertext**, lies in a set $\mathcal{C}$.

Definition 28.7.1 (Cryptosystem)

A **cryptosystem** consists of sets $(\mathcal{M}, \mathcal{K}, \mathcal{C})$ together with an **encryption** function $e: \mathcal{M} \times \mathcal{K} \rightarrow \mathcal{C}$ and a **decryption** function $d: \mathcal{C} \times \mathcal{K} \rightarrow \mathcal{M}$ satisfying

$$d(e(m, k), k) = m \quad \text{for all } m \in \mathcal{M}, k \in \mathcal{K}.$$

Example 28.7.2 (Substitution and Vigenère Ciphers)

Let $\Sigma = \{A, B, \dots, Z\}$ and $\mathcal{M} = \mathcal{C} = \Sigma^*$.

- The **simple substitution cipher** takes $\mathcal{K}$ to be the set of permutations of Σ ; each plaintext letter is replaced by its image under the chosen $\pi \in \mathcal{K}$. There are $26! \approx 4 \times 10^{26}$ keys, which sounds like a lot.
- The **Vigenère cipher** takes $\mathcal{K} = \Sigma^d$ for some d . Identify Σ with $\mathbb{Z}/26\mathbb{Z}$, write the key repeatedly beneath the plaintext, and add. For instance, the plaintext ATTACKATDAWN with key LEMON gives

ATTACKATDAWN
LEMONLEMONLE
LXFOPVEFRNHR

The case $d = 1$ is the **Caesar cipher**.

Neither of these is secure, and it is worth being precise about what security would mean. We assume throughout **Kerckhoffs' principle**: Eve knows the functions e and d and the probability distributions on $\mathcal{K}$ and $\mathcal{M}$; the only thing she does not know is the key itself. There is then a hierarchy of attacks.

Level 1. *Ciphertext only.* Eve knows some ciphertext and nothing else.

Level 2. *Known plaintext.* Eve knows a substantial stretch of plaintext together with its ciphertext, but not the key.

Level 3. *Chosen plaintext.* Eve can obtain the ciphertext of any plaintext she likes.

Remark. Both examples above fail at Level 1 against English plaintext of any length, by frequency analysis: in the substitution cipher the letter frequencies are merely permuted, and in the Vigenère cipher the same is true within each residue class modulo d (and d

itself is found by looking for repeated ciphertext fragments). Even against random-looking plaintext, both fall immediately at Level 2, since a single plaintext–ciphertext pair reveals the key. Modern applications demand Level 3 security.

§28.7.2 Perfect Secrecy and the One-Time Pad

To do better we should say what perfect security means, and information theory gives us the language. Model the key and message as *independent* random variables K, M on $\mathcal{K}, \mathcal{M}$, and let $C = e(M, K)$ be the ciphertext.

Definition 28.7.3 (Perfect Secrecy)

A cryptosystem has **perfect secrecy** if $H(M | C) = H(M)$; equivalently $I(M; C) = 0$; equivalently M and C are independent.

In other words, seeing the ciphertext tells Eve nothing at all about the message that she did not already know. This is an extremely strong requirement, and it is expensive.

Proposition 28.7.4 (Shannon)

Suppose a cryptosystem has perfect secrecy, and that every message in $\mathcal{M}$ and every ciphertext in $\mathcal{C}$ occurs with positive probability. Then $|\mathcal{K}| \geq |\mathcal{C}| \geq |\mathcal{M}|$. In particular the key is at least as long as the message.

Proof. Fix a key k . The map $m \mapsto e(m, k)$ is injective, since $d(\cdot, k)$ inverts it; hence $|\mathcal{C}| \geq |\mathcal{M}|$.

Now fix a ciphertext $c \in \mathcal{C}$ and a message $m \in \mathcal{M}$. By perfect secrecy M and C are independent, so

$$\mathbb{P}(C = c | M = m) = \mathbb{P}(C = c) > 0,$$

and therefore there is at least one key k with $e(m, k) = c$. Fix c and let m range over $\mathcal{M}$: the keys so obtained are distinct, because a single key k can satisfy $e(m, k) = c$ for at most one m , again by injectivity. Hence $|\mathcal{K}| \geq |\mathcal{M}|$. Applying the same argument with m fixed and c ranging over $\mathcal{C}$ – for each c there is a key carrying m to c , and distinct c need distinct keys since $e(m, \cdot)$ is a function – gives $|\mathcal{K}| \geq |\mathcal{C}|$. $\square$

So perfect secrecy demands a key space at least as large as the message space: a fresh $\log|\mathcal{M}|$ bits of shared secret for every message sent. That is a heavy price, and we shall see at the end of this section that the one-time pad pays exactly it. Before that, it is worth quantifying how far short of perfection an imperfect system falls, and for that we need two more pieces of vocabulary.

Definition 28.7.5

The **message equivocation** is $H(M | C)$ and the **key equivocation** is $H(K | C)$.

Lemma 28.7.6

$$H(M | C) \leq H(K | C).$$

Proof. Since $M = d(C, K)$, knowing C and K determines M , so $H(M | C, K) = 0$ and hence $H(M, C, K) = H(C, K)$. Now expand by the chain rule in two ways:

$$\begin{aligned} H(K | C) &= H(K, C) - H(C) = H(M, C, K) - H(C) \\ &= H(K | M, C) + H(M, C) - H(C) = H(K | M, C) + H(M | C), \end{aligned}$$

and $H(K | M, C) \geq 0$. □

So determining the key is at least as hard as determining the message – which is reassuring, since it means we may safely study the key equivocation instead.

Suppose $\mathcal{M} = \mathcal{C} = \mathcal{A}$ and we send n messages $M^{(n)} = (M_1, \dots, M_n)$ using the *same* key, obtaining $C^{(n)}$.

Definition 28.7.7 (Unicity Distance)

The **unicity distance** is the least n for which $H(K | C^{(n)}) = 0$, that is, the least amount of ciphertext which determines the key uniquely.

Repeating the computation of the lemma with $M^{(n)}$ in place of M , and using independence of K and $M^{(n)}$,

$$H(K | C^{(n)}) = H(K) + H(M^{(n)}) - H(C^{(n)}).$$

Make three natural assumptions: all keys are equally likely, so $H(K) = \log|\mathcal{K}|$; the message source has information rate H , so $H(M^{(n)}) \approx nH$ for large n ; and all ciphertexts look equally likely, so $H(C^{(n)}) = n \log|\mathcal{A}|$. Then

$$H(K | C^{(n)}) \approx \log|\mathcal{K}| + nH - n \log|\mathcal{A}|,$$

which is non-negative precisely when

$$n \leq U = \frac{\log|\mathcal{K}|}{\log|\mathcal{A}| - H} = \frac{\log|\mathcal{K}|}{R \log|\mathcal{A}|}, \quad R = 1 - \frac{H}{\log|\mathcal{A}|},$$

where R is the **redundancy** of the source. So U is our estimate for the unicity distance. To make it large we may either enlarge the key space or, more cheaply, reduce the redundancy of the plaintext – which is to say, *compress before encrypting*. This is a genuinely practical moral, and it is pleasing that the two halves of the course meet here.

Definition 28.7.8 (One-Time Pad)

Let the plaintext, key and ciphertext be streams $p_0, p_1, \dots$, $k_0, k_1, \dots$ and $z_0, z_1, \dots$ in $\mathbb{F}_2$ with $z_n = p_n + k_n$. The **one-time pad** is the cryptosystem in which the k_i are independent and uniform on $\{0, 1\}$.

Then each $z_n = p_n + k_n$ is uniform on $\{0, 1\}$ and independent of everything else, so C is independent of M : the one-time pad has perfect secrecy, and its unicity distance is infinite. Eve, given the ciphertext, learns literally nothing.

The catch is in the name. The key must be as long as the message, must be truly random, and must never be reused (if $z = p + k$ and $z' = p' + k$ then $z + z' = p + p'$, and Eve is back to a classical cipher). Generating and distributing such a key is precisely the problem we were trying to solve. The one-time pad is therefore used only where the stakes justify the cost.

§28.7.3 Stream Ciphers

The practical compromise is to share a short key and use it to *generate* a long key stream deterministically, by means of a shift register. The security now rests on Eve being unable to predict the stream.

Lemma 28.7.9

Let $x_0, x_1, \dots$ be a stream produced by a feedback shift register of length d . Then there are $M, N \leq 2^d$ with $x_{N+r} = x_r$ for all $r \geq M$.

Proof. Let $f: \mathbb{F}_2^d \rightarrow \mathbb{F}_2^d$ be the register and $v_i = (x_i, \dots, x_{i+d-1})$, so $f(v_i) = v_{i+1}$. The $2^d + 1$ vectors $v_0, \dots, v_{2^d}$ cannot all be distinct, so $v_a = v_b$ for some $a < b \leq 2^d$. Put $M = a$ and $N = b - a$; then $v_M = v_{M+N}$ and applying f repeatedly gives $v_r = v_{r+N}$ for all $r \geq M$. $\square$

Remark. So the maximum period of a general feedback shift register of length d is 2^d . For a *linear* register the all-zero fill is a fixed point, so the maximum period is $2^d - 1$, attained exactly when the auxiliary polynomial is primitive – as in Example 28.6.25.

Stream ciphers built from LFSRs are cheap, fast, and error-tolerant: encryption and decryption happen bit by bit with no need to buffer the message, and a corrupted bit affects only itself. They are also, as we saw at the end of Section 28.6, completely broken at Level 2. Given $2d$ bits of known plaintext, Eve recovers $2d$ bits of key stream, runs Berlekamp–Massey, and obtains the register.

One might try to fix this by combining several registers. It does not help very much.

Lemma 28.7.10

Let (x_n) and (y_n) be streams from LFSRs of lengths M and N . Then

- (i) $(x_n + y_n)$ comes from an LFSR of length at most $M + N$;
- (ii) $(x_n y_n)$ comes from an LFSR of length at most MN .

Proof sketch, non-examinable. Assume for simplicity that the auxiliary polynomials P, Q have distinct roots $\alpha_1, \dots, \alpha_M$ and $\beta_1, \dots, \beta_N$ in some field $K \supseteq \mathbb{F}_2$. Solving the recurrences, $x_n = \sum_i \lambda_i \alpha_i^n$ and $y_n = \sum_j \mu_j \beta_j^n$ for constants in K . Then $x_n + y_n$ is a combination of the α_i^n and β_j^n , so satisfies the recurrence with auxiliary polynomial $P(X)Q(X)$, of degree $M + N$; and $x_n y_n = \sum_{i,j} \lambda_i \mu_j (\alpha_i \beta_j)^n$ satisfies the one with auxiliary polynomial $\prod_{i,j} (X - \alpha_i \beta_j)$, of degree MN . $\square$

Adding streams therefore buys nothing: the sum is no harder to analyse than a single register of length $M + N$ would have been. Multiplying does increase the effective length, but $x_n y_n = 0$ whenever either factor vanishes, so the output is heavily biased towards 0 and leaks information in other ways. Genuinely nonlinear feedback functions are used in practice, but they are hard to analyse – and the danger is that Eve may understand the register better than Alice and Bob do.

§28.7.4 Public Key Cryptography

Stream ciphers are **symmetric**: the decryption procedure is the encryption procedure, or is trivially deduced from it. Every symmetric system suffers the key exchange problem: before Alice and Bob can talk, they must somehow already have talked.

In an **asymmetric** or **public key** cryptosystem the key splits in two. There is a **public key**, used for encryption and broadcast freely, and a **private key**, used for decryption and known only to its owner. Knowing the algorithms and the public key, it should still be infeasible to find the private key or to decrypt messages. This automatically gives Level 3 security – Eve can encrypt whatever she likes, since so can anyone – and it removes the key exchange problem entirely.

Such systems are built on computational problems believed to be hard. Two dominate.

- (i) *Factoring*. Given $N = pq$ with p, q large primes, find p and q .
- (ii) *The discrete logarithm problem*. Given a large prime p , a primitive root g modulo p , and $x \in \mathbb{F}_p^\times$, find a with $x \equiv g^a \pmod{p}$.

Definition 28.7.11 (Polynomial Time)

An algorithm runs in **polynomial time** if the number of bit operations it performs on an input of size N bits is at most cN^d for constants c, d .

Note the input size for a problem about N is $\log_2 N$, the number of bits of N ; this is what makes trial division a bad algorithm.

Example 28.7.12

The following are polynomial time: addition, multiplication and division of integers; Euclid's algorithm and hence computation of greatest common divisors and modular inverses; modular exponentiation $x^a \bmod N$ by repeated squaring; and primality testing (see II Number Theory for the Miller–Rabin and Solovay–Strassen tests, which are fast and probabilistic, and note that a deterministic polynomial time test also exists).

No polynomial time algorithm is known for (i) or (ii). The naive methods take exponential time:

- For factoring, dividing N by successive primes up to $\sqrt{N}$ succeeds in $O(\sqrt{N}) = O(2^{B/2})$ steps where $B = \log_2 N$.
- For discrete logarithms there is the **baby-step giant-step** method. Set $m = \lceil \sqrt{p} \rceil$ and write $a = qm + r$ with $0 \leq q, r < m$. Then $x \equiv g^{qm+r}$ gives $g^{qm} \equiv g^{-r}x \pmod{p}$. Tabulate the m values g^{qm} and the m values $g^{-r}x$, sort, and look for a match; this takes $O(\sqrt{p} \log p)$ steps.

The best known general methods for both are factor base methods, culminating in the number field sieve, with running time

$$O\left(\exp\left(c(\log N)^{1/3}(\log \log N)^{2/3}\right)\right)$$

for a known constant c – subexponential, but still far from polynomial.

§28.7.5 The Rabin Cryptosystem

We recall from IA Numbers and Sets and II Number Theory that **Euler's totient function** $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ counts the integers below n coprime to it, that $\varphi(pq) = (p-1)(q-1)$ for distinct primes, and that by Lagrange's theorem

$$a^{\varphi(N)} \equiv 1 \pmod{N} \quad \text{whenever } (a, N) = 1,$$

the Fermat–Euler theorem; when $N = p$ is prime this is Fermat's little theorem.

Lemma 28.7.13

Let $p = 4k - 1$ be prime and $d \in \mathbb{Z}$. If $x^2 \equiv d \pmod{p}$ is soluble, then $x \equiv d^k$ is a solution.

Proof. Let x_0 be a solution; we may assume $p \nmid x_0$, else $d \equiv 0$ and the claim is trivial. Then

$$d^{2k-1} \equiv x_0^{2(2k-1)} = x_0^{4k-2} = x_0^{p-1} \equiv 1 \pmod{p},$$

$$\text{so } (d^k)^2 = d^{2k} \equiv d \cdot d^{2k-1} \equiv d.$$

□

So modulo a prime $p \equiv 3 \pmod{4}$ we can extract square roots in polynomial time. This is the basis of Rabin's scheme.

The private key is a pair of large distinct primes $p, q \equiv 3 \pmod{4}$; the public key is $N = pq$. We take $\mathcal{M} = \mathcal{C} = (\mathbb{Z}/N\mathbb{Z})^\times$ and encrypt m as

$$c \equiv m^2 \pmod{N}.$$

To decrypt, Bob uses Lemma 28.7.13 to find x_1 with $x_1^2 \equiv c \pmod{p}$ and x_2 with $x_2^2 \equiv c \pmod{q}$, then combines them with the Chinese Remainder Theorem: running Euclid's algorithm on p, q gives r, s with $rp + sq = 1$, and then

$$x = sqx_1 + rpx_2$$

satisfies $x \equiv x_1 \pmod{p}$ and $x \equiv x_2 \pmod{q}$, so $x^2 \equiv c \pmod{N}$.

Lemma 28.7.14 (i) Let p be an odd prime and $(d, p) = 1$. Then $x^2 \equiv d \pmod{p}$ has either no solutions or exactly two.

(ii) Let $N = pq$ with p, q distinct odd primes and $(d, N) = 1$. Then $x^2 \equiv d \pmod{N}$ has either no solutions or exactly four.

Proof. (i). $x^2 \equiv y^2$ if and only if $p \mid (x - y)(x + y)$, so $x \equiv \pm y$; and $y \not\equiv -y$ as p is odd.

(ii). If x_0 is a solution then by (i) and the Chinese Remainder Theorem the solutions are exactly those x with $x \equiv \pm x_0 \pmod{p}$ and $x \equiv \pm x_0 \pmod{q}$, and the four sign choices give four distinct residues modulo N . □

So Bob must sift four candidate plaintexts, and the message needs enough built-in redundancy (a checksum, say) to identify the right one. In exchange, we get a security guarantee of a kind that is unavailable for RSA.

Theorem 28.7.15

Breaking the Rabin cryptosystem is essentially as hard as factoring N .

Proof. If we can factor N then we can decrypt, as above. Conversely, suppose we have an algorithm which extracts square roots modulo N . Choose x modulo N at random and feed x^2 to the algorithm, obtaining some y with $y^2 \equiv x^2$. By the lemma there are four square roots, and the algorithm cannot know which one we started from, so with probability $\frac{1}{2}$ we get $y \not\equiv \pm x$. In that case $N \mid (x - y)(x + y)$ but N divides neither factor, so $\gcd(N, x - y)$ is a nontrivial factor of N . Repeating r times, we factor N with probability $1 - 2^{-r}$. $\square$

Example 28.7.16

Take $p = 7$, $q = 11$, so $N = 77$ (and indeed $7 \equiv 11 \equiv 3 \pmod{4}$). To encrypt $m = 13$: $c \equiv 13^2 = 169 \equiv 15 \pmod{77}$.

To decrypt, note $7 = 4 \cdot 2 - 1$ so $k = 2$ and $x_1 \equiv 15^2 = 225 \equiv 1 \pmod{7}$; and $11 = 4 \cdot 3 - 1$ so $k = 3$ and $x_2 \equiv 15^3 \equiv 4^3 = 64 \equiv 9 \pmod{11}$. Checking, $1^2 = 1 \equiv 15$ and $9^2 = 81 \equiv 4 \equiv 15$ modulo 7 and 11 respectively. Combining, $x \equiv 1 \pmod{7}$ and $x \equiv 9 \pmod{11}$ gives $x = 64$; and $64^2 = 4096 = 53 \cdot 77 + 15$, so indeed $64^2 \equiv 15$. The four square roots of 15 modulo 77 are $\{13, 20, 57, 64\}$, of which 13 was our message.

§28.7.6 The RSA Cryptosystem

RSA is the same idea with a general exponent in place of squaring. We begin with a factoring result which will do double duty.

Let $N = pq$ with p, q distinct odd primes, and suppose we know some multiple m of $\varphi(N)$. Write $m = 2^a b$ with $a \geq 1$ and b odd, write $o_p(x)$ for the multiplicative order of x modulo p , and set

$$X = \{x \in (\mathbb{Z}/N\mathbb{Z})^\times : o_p(x^b) \neq o_q(x^b)\}.$$

Theorem 28.7.17 (i) If $x \in X$ then there is $0 \leq t < a$ such that $\gcd(x^{2^t b} - 1, N)$ is a nontrivial factor of N .

(ii) $|X| \geq \frac{1}{2}\varphi(N) = \frac{1}{2}|(\mathbb{Z}/N\mathbb{Z})^\times|$.

Proof. (i). By Fermat–Euler, $x^{\varphi(N)} \equiv 1$, hence $x^m \equiv 1 \pmod{N}$. Put $y = x^b$, so $y^{2^a} \equiv 1$ and therefore $o_p(y)$ and $o_q(y)$ are both powers of 2. Since $x \in X$ they differ, so without loss of generality $o_p(y) < o_q(y)$; write $o_p(y) = 2^t$, so $0 \leq t < a$. Then $y^{2^t} \equiv 1 \pmod{p}$ but $y^{2^t} \not\equiv 1 \pmod{q}$, so

$$\gcd(y^{2^t} - 1, N) = p.$$

(ii). The Chinese Remainder Theorem gives a group isomorphism

$$(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times \times (\mathbb{Z}/q\mathbb{Z})^\times, \quad x \mapsto (x \bmod p, x \bmod q).$$

We claim that if we partition $(\mathbb{Z}/p\mathbb{Z})^\times$ according to the value of $o_p(x^b)$, then one part has size exactly $\frac{1}{2}(p-1)$. Let g be a primitive root modulo p . As $g^{p-1} \equiv 1$ and $(p-1) \mid m = 2^a b$, we get $(g^b)^{2^a} \equiv 1$, so $o_p(g^b) = 2^t$ for some t . Writing $x = g^\kappa$ we have $x^b = (g^b)^\kappa$, so

$$o_p(x^b) = \frac{2^t}{\gcd(2^t, \kappa)} = \begin{cases} 2^t & \kappa \text{ odd,} \\ < 2^t & \kappa \text{ even,} \end{cases}$$

and exactly half of the $\kappa \in \{0, \dots, p-2\}$ are odd. This proves the claim, and the same holds modulo q .

Now fix $y \in (\mathbb{Z}/q\mathbb{Z})^\times$ and consider the possible $x \in (\mathbb{Z}/p\mathbb{Z})^\times$. At most half of them have $o_p(x^b)$ equal to the particular value $o_q(y^b)$, by the claim applied to whichever part that is. Hence at least $\frac{1}{2}(p-1)$ choices of x give $o_p(x^b) \neq o_q(y^b)$. Summing over the $q-1$ choices of y and using the isomorphism,

$$|X| \geq \frac{1}{2}(p-1)(q-1) = \frac{1}{2}\varphi(N). \quad \square$$

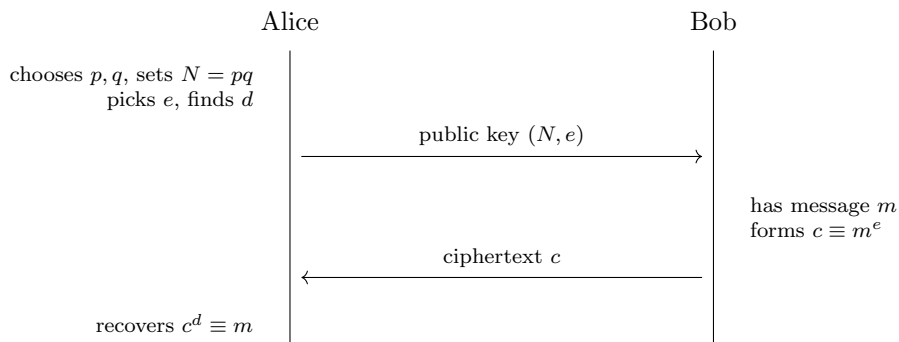
So knowing $\varphi(N)$, or any multiple of it, lets us factor N quickly: pick x at random, and with probability at least $\frac{1}{2}$ some $\gcd(x^{2^t b} - 1, N)$ splits N .

We can now describe the scheme. Alice chooses large distinct primes p, q at random and sets $N = pq$. She chooses an **encrypting exponent** e with $\gcd(e, \varphi(N)) = 1$ – for instance a prime larger than both p and q – and uses Euclid's algorithm to find d, k with

$$de - k\varphi(N) = 1;$$

d is the **decrypting exponent**. The public key is (N, e) and the private key is (N, d) . Encryption and decryption are

$$c \equiv m^e \pmod{N}, \quad m \equiv c^d \pmod{N}.$$



Proposition 28.7.18

Decryption inverts encryption: $c^d \equiv m \pmod{N}$.

Proof. We have $de = 1 + k\varphi(N)$, so if $\gcd(m, N) = 1$ then by Fermat–Euler

$$c^d \equiv m^{de} = m \cdot (m^{\varphi(N)})^k \equiv m \pmod{N}.$$

(The probability that a random m fails $\gcd(m, N) = 1$ is $\frac{p+q-1}{N}$, utterly negligible for large p, q ; in fact the congruence still holds in that case, by checking modulo p and q separately.) $\square$

Example 28.7.19

Take $p = 17$, $q = 23$, so $N = 391$ and $\varphi(N) = 16 \cdot 22 = 352$. Take $e = 3$, which is coprime to 352. Euclid gives $3 \cdot 235 = 705 = 2 \cdot 352 + 1$, so $d = 235$.

To encrypt $m = 100$: $100^2 = 10000 = 25 \cdot 391 + 225$, so $100^2 \equiv 225$, and $100^3 \equiv 225 \cdot 100 = 22500 = 57 \cdot 391 + 213$, so $c = 213$. Decrypting, one computes $213^{235} \equiv 100 \pmod{391}$ by repeated squaring, as promised.

Corollary 28.7.20

Finding the RSA private key (N, d) from the public key (N, e) is essentially as hard as factoring N .

Proof. If we can factor N we compute $\varphi(N)$ and then d . Conversely, suppose we have an algorithm producing d from (N, e) . Then $de \equiv 1 \pmod{\varphi(N)}$, so $m := de - 1$ is a multiple of $\varphi(N)$, and Theorem 28.7.17 factors N with probability at least $\frac{1}{2}$ per random trial, hence with probability $1 - 2^{-r}$ after r trials. $\square$

Remark. Note carefully what this does *not* say. It says that recovering the private key is as hard as factoring. Whether *decrypting a single message* is as hard as factoring is unknown, and for the Rabin scheme the corresponding statement is a theorem – which is one reason Rabin’s scheme is theoretically more satisfying, even though RSA is what the world uses.

RSA also has the drawback of being slow, since modular exponentiation with 2048-bit numbers is expensive. In practice one uses RSA once, to agree on a key for a fast symmetric cipher, and then switches.

§28.7.7 Attacks on RSA

Correct mathematics is not enough; the scheme must also be used correctly. Here are some ways it goes wrong.

Example 28.7.21 (Homomorphism Attack)

The map $m \mapsto m^e$ is multiplicative, and an attacker can exploit that. Suppose a bank sends instructions of the form (M_1, M_2) , where M_1 names a client and M_2 is an amount to be credited, encrypted as $(Z_1, Z_2) = (M_1^e, M_2^e)$. A client C deposits £100 and observes the resulting ciphertext (Z_1, Z_2) . Now C replays the message (Z_1, Z_2^3) , which decrypts to (M_1, M_2^3) – a credit of £10⁶ – without C having broken RSA at all. More crudely still, C could simply resend (Z_1, Z_2) repeatedly; that particular

attack is defeated by timestamping messages, but the first is not.

Example 28.7.22 (Authenticity via RSA)

The same algebra can be used constructively. If Alice encrypts m with her *private* key (N, d) , then anyone can decrypt it with her public key, since $(m^d)^e = (m^e)^d = m$; but nobody else could have produced it. So if Bob sends Alice a random message m and receives back something which, decrypted with her public key, ends in m , he can be confident that Alice is on the other end.

The lesson is that we should encrypt and authenticate deliberately rather than incidentally, which brings us to signatures.

§28.7.8 Signatures and Hash Functions

Consider a message m from Alice to Bob. There are four distinct goals one might have.

- (i) *Secrecy*: no third party can read m .
- (ii) *Integrity*: no third party can alter m undetectably.
- (iii) *Authenticity*: Bob is sure that Alice sent m .
- (iv) *Non-repudiation*: Bob can prove to a third party that Alice sent m .

Encryption addresses the first; signatures address the rest.

Definition 28.7.23 (Signature)

A message m is **signed** as a pair (m, s) , where the **signature** $s = s(m, k)$ depends on m and on the private key k . The signature function should be such that without the private key one cannot produce valid signatures, but anyone can check them using the public key.

Note that the signature depends on the message, not just on the sender; a signature that did not would simply be copied onto other messages.

Example 28.7.24 (RSA Signatures)

Alice has private key (N, d) and public key (N, e) . She signs m as (m, s) with $s \equiv m^d \pmod{N}$, and Bob verifies by checking $s^e \equiv m$.

For instance with $N = 391$, $e = 3$, $d = 235$ as above, Alice signs $m = 100$ with $s = 100^{235} \bmod 391 = 94$, and Bob checks $94^3 = 830584 = 2124 \cdot 391 + 100$, so $94^3 \equiv 100$ as required.

This naive scheme has two flaws. It inherits the homomorphism attack, since $s_1 s_2$ signs $m_1 m_2$. And it admits **existential forgery**: an attacker chooses s first and then computes $m = s^e \bmod N$, obtaining a validly signed pair (m, s) for a message she did not choose. One hopes that such an m is meaningless, but hope is not a security argument.

The standard fix is to sign a **digest** of the message rather than the message itself.

Definition 28.7.25 (Hash Function)

A **hash function** is a publicly known function $h: \mathcal{M} \rightarrow \{1, \dots, N-1\}$, from a large space of messages to a small space of digests, for which it is computationally infeasible to find a **collision**, that is, a pair $m \neq m'$ with $h(m) = h(m')$. One also asks that given y it should be infeasible to find m with $h(m) = y$.

Collisions certainly exist, by the pigeonhole principle; the requirement is that no one can find them. Signing $(m, h(m)^d)$ rather than (m, m^d) defeats both attacks above: the homomorphism attack fails because $h(m_1 m_2) \neq h(m_1)h(m_2)$, and existential forgery fails because the attacker would have to find a message with a prescribed digest. Widely used hash functions include the (now broken) MD5 and the SHA family.

§28.7.9 Diffie–Hellman Key Exchange

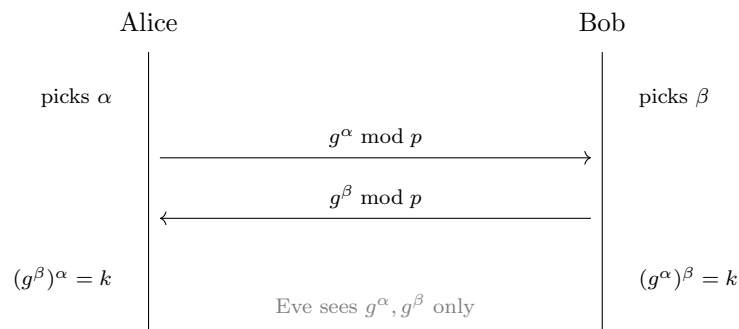
The discrete logarithm problem gives a completely different route to public key cryptography, beginning with an idea of striking simplicity: Alice and Bob can agree on a shared secret in public.

Example 28.7.26 (Diffie–Hellman Key Exchange)

Let p be a large prime and g a primitive root modulo p , both public. Alice chooses a secret exponent α and sends $g^\alpha \bmod p$; Bob chooses a secret β and sends $g^\beta \bmod p$. Each now raises what they received to their own secret exponent, and both obtain

$$k = g^{\alpha\beta} \bmod p,$$

which they use as a shared key. Eve has seen g , g^α and g^β and must produce $g^{\alpha\beta}$; Diffie and Hellman conjectured that this is as hard as the discrete logarithm problem.

**Example 28.7.27**

Take $p = 23$ and $g = 5$. Alice picks $\alpha = 6$ and sends $5^6 = 15625 \equiv 8$. Bob picks $\beta = 15$ and sends $5^{15} \equiv 19$. Alice computes $19^6 \equiv 2$ and Bob computes $8^{15} \equiv 2$, and the shared key is $k = 2$.

Remark. Diffie–Hellman is vulnerable to an active attacker who can intercept and replace messages – the *man in the middle* – who simply runs the protocol separately with each of Alice and Bob and relays traffic between the two sessions. Defeating this needs authentication, which is what signatures are for.

Example 28.7.28 (Shamir's Padlock Scheme)

Here is a related idea requiring no shared key at all. Work modulo a prime p . Alice picks a coprime to $p - 1$ and computes a' with $aa' \equiv 1 \pmod{p - 1}$; Bob does the same with b, b' . To send m , Alice sends m^a ; Bob returns $(m^a)^b$; Alice sends back $(m^{ab})^{a'}$; and Bob computes

$$(m^{aba'})^{b'} = m^{aa'bb'} \equiv m \pmod{p}$$

by Fermat's little theorem, since $aa'bb' \equiv 1 \pmod{p - 1}$. The picture is that Alice puts her padlock on a box and sends it; Bob adds his padlock and sends it back; Alice removes hers; Bob removes his.

$$m \xrightarrow{A} m^a \xrightarrow{B} m^{ab} \xrightarrow{A} m^{aba'} \xrightarrow{B} m$$

It costs three transmissions rather than one, which is why it is a curiosity rather than a standard.

§28.7.10 The ElGamal Scheme

Diffie–Hellman can be turned into a genuine public key cryptosystem.

Definition 28.7.29 (ElGamal Encryption)

Alice chooses a large prime p , a primitive root g modulo p , and a secret u with $1 < u < p - 1$. Her public key is (p, g, y) with $y \equiv g^u$, and her private key is u . To send Alice a message m , Bob chooses a random k and sends the pair

$$(c_1, c_2) = (g^k, my^k) \pmod{p}.$$

Alice recovers $m = c_2 c_1^{-u} \pmod{p}$, since $c_1^u = g^{ku} = y^k$.

Example 28.7.30

Take $p = 23$, $g = 5$, $u = 7$, so $y = 5^7 \equiv 17$. To send $m = 10$, Bob picks $k = 3$: then $c_1 = 5^3 = 125 \equiv 10$ and $c_2 = 10 \cdot 17^3 \equiv 10 \cdot 14 = 140 \equiv 2$. So the ciphertext is $(10, 2)$. Alice computes $c_1^u = 10^7 \equiv 14$, inverts it ($14 \cdot 5 = 70 \equiv 1$, so $14^{-1} = 5$), and finds $m = 2 \cdot 5 = 10$.

Note that a fresh k is used for every message, so the same plaintext encrypts differently each time – a genuine advantage over textbook RSA. The same construction gives a signature scheme.

Definition 28.7.31 (ElGamal Signature Scheme)

With p, g, u, y as above and a collision-resistant hash function $h: \mathcal{M} \rightarrow \{1, \dots, p - 1\}$, Alice signs m as follows. She chooses a random k with $1 \leq k \leq p - 2$ and $\gcd(k, p - 1) = 1$, and computes r, s with

$$r \equiv g^k \pmod{p}, \quad h(m) \equiv ur + ks \pmod{p - 1}.$$

The congruence for s is soluble because k is invertible modulo $p - 1$. The signature is (r, s) .

Verification is a one-line computation:

$$g^{h(m)} \equiv g^{ur+ks} \equiv (g^u)^r (g^k)^s \equiv y^r r^s \pmod{p},$$

so Bob accepts (r, s) if and only if $g^{h(m)} \equiv y^r r^s$. Forging a signature by the obvious routes requires solving a discrete logarithm.

Example 28.7.32

Take $p = 23$, $g = 5$, $u = 7$, $y = 17$, and suppose $h(m) = 9$. Alice picks $k = 5$, which is coprime to 22. Then $r = 5^5 = 3125 \equiv 20$. Solving $9 \equiv 7 \cdot 20 + 5s \pmod{22}$, that is $5s \equiv 9 - 140 \equiv 1 \pmod{22}$, gives $s = 9$ (as $5 \cdot 9 = 45 = 2 \cdot 22 + 1$). Checking: $g^{h(m)} = 5^9 \equiv 11$, and $y^r r^s = 17^{20} \cdot 20^9 \equiv 11$. The signature verifies.

It is vital that Alice choose a fresh k for each message. Suppose she signs m_1 and m_2 with the same k , giving signatures (r, s_1) and (r, s_2) – note r depends only on k , so it repeats, which is how Eve spots the mistake. Then

$$h(m_1) \equiv ur + ks_1, \quad h(m_2) \equiv ur + ks_2 \pmod{p-1},$$

so subtracting,

$$h(m_1) - h(m_2) \equiv k(s_1 - s_2) \pmod{p-1}.$$

We need one small fact to finish.

Lemma 28.7.33

The congruence $ax \equiv b \pmod{n}$ has either no solutions or exactly $\gcd(a, n)$ solutions modulo n .

Proof. Let $\delta = \gcd(a, n)$. If $\delta \nmid b$ there is no solution, since δ divides the left hand side for any x . If $\delta \mid b$, the congruence is equivalent to $\frac{a}{\delta}x \equiv \frac{b}{\delta} \pmod{\frac{n}{\delta}}$, and $\frac{a}{\delta}, \frac{n}{\delta}$ are coprime, so this has a unique solution modulo $\frac{n}{\delta}$ and hence δ solutions modulo n . $\square$

By Lemma 28.7.33 there are $\gcd(p-1, s_1 - s_2)$ candidate values of k ; Eve tests each against $r \equiv g^k$ and keeps the right one. Then from $ur \equiv h(m_1) - ks_1$ she gets $\gcd(p-1, r)$ candidates for u , and tests each against $y \equiv g^u$. So a single repeated k hands over Alice's private key.

§28.7.11 The Digital Signature Algorithm

The **Digital Signature Algorithm** (DSA), developed by the NSA, is a variant of the ElGamal signature scheme designed to keep signatures short by working in a small subgroup.

The public parameters are (p, q, g) constructed as follows.

- p is a prime of exactly N bits, where N is a multiple of 64 with $512 \leq N \leq 1024$.
- q is a prime of 160 bits with $q \mid p - 1$.

- $g \equiv h^{(p-1)/q} \pmod{p}$, where h is a primitive root modulo p ; thus g has order exactly q in $\mathbb{F}_p^\times$.
- Alice chooses a private key x with $1 < x < q$ and publishes $y \equiv g^x \pmod{p}$.

To sign a message m with $0 \leq m < q$, Alice chooses a random k with $1 < k < q$ and computes

$$s_1 \equiv (g^k \bmod p) \bmod q, \quad s_2 \equiv k^{-1}(m + xs_1) \bmod q.$$

The signature is (s_1, s_2) , a pair of 160-bit numbers however large p is. To verify, Bob computes

$$w \equiv s_2^{-1}, \quad u_1 \equiv mw, \quad u_2 \equiv s_1w \pmod{q}, \quad v = (g^{u_1}y^{u_2} \bmod p) \bmod q,$$

and accepts if $v = s_1$.

Proposition 28.7.34

A correctly signed and unaltered message is accepted.

Proof. From the definition of s_2 we get $(m + xs_1)w \equiv (m + xs_1)s_2^{-1} \equiv k \pmod{q}$. Since g has order q modulo p , exponents of g only matter modulo q , so

$$\begin{aligned} v &= (g^{mw}g^{xs_1w} \bmod p) \bmod q = (g^{(m+xs_1)w} \bmod p) \bmod q \\ &= (g^k \bmod p) \bmod q = s_1. \square \end{aligned}$$

As with ElGamal, reusing k is fatal: if Alice signs m_1 and m_2 with the same k then s_1 repeats and subtracting the two congruences for s_2 recovers k , hence x .

§28.7.12 Commitment Schemes

Suppose Alice wants to commit to a bit $m \in \{0, 1\}$ so that

- Bob cannot determine m without Alice's help, and
- Alice cannot change m after committing.

This is **bit commitment**, and it is what one needs to toss a coin over the telephone, or to run a poll whose results stay hidden until everyone has voted.

The first solution is immediate given any secure cipher. Alice commits by sending $c = e_A(m)$, using a key known only to her; Bob cannot read it. To reveal, she sends her key, and Bob decrypts. He can also check that e_A and d_A really are mutually inverse, so she cannot pass off a bogus key that decrypts c to the other bit.

The second is prettier and uses coding theory. Suppose Alice and Bob have two channels: a clear one and a binary symmetric channel with error probability $p \in (0, \frac{1}{2})$ which neither of them can influence. Bob publishes a binary linear code C of length N and minimum distance d , and Alice publishes a random nontrivial linear map $\theta: C \rightarrow \mathbb{F}_2$. To commit to m , Alice picks a random $c \in C$ with $\theta(c) = m$ and sends c down the *noisy* channel; Bob receives $r = c + e$.

Now $\mathbb{E}[d(r, c)] = Np$, and we choose N so that $Np \gg d$. Then the received word r is far further from c than codewords are from each other, so Bob has no idea which codeword was sent and hence no idea of $\theta(c) = m$. That is (i).

To reveal, Alice sends c down the clear channel, and Bob checks that $d(c, r) \approx Np$. She cannot cheat: she knows c but not e , so if she tries to reveal some $c' \neq c$ she must arrange $d(r, c') \approx Np$ without knowing r . The only way to be confident of that is to take c' very close to c ; but distinct codewords are at distance at least d , and the probability that a codeword at distance $\geq d$ from c happens to be at distance about Np from r is negligible for suitable N and d . That is (ii). (One should choose N and d so that the probability of an *honest* commitment being rejected, because the channel happened to make far too many or too few errors, is also negligible.)

§28.7.13 Secret Sharing

We finish with a scheme of a rather different flavour. Suppose entry to a bunker is controlled by a secret positive integer S , known only to the Leader. Each of the n members of the Faculty is to be given a **shadow** or **share**, in such a way that any k of them together can reconstruct S , but no $k - 1$ of them can learn anything about it at all.

Definition 28.7.35 (Threshold Scheme)

Let $k \leq n$. A **(k, n) -threshold scheme** is a method of distributing a secret S among n participants so that any k of them can reconstruct S , but no smaller subset can.

Shamir's construction is a one-line application of Lagrange interpolation: a polynomial of degree $k - 1$ is determined by its values at k points, and completely undetermined by its values at $k - 1$ points.

Let $0 \leq S \leq M$ be the secret. The Leader chooses and publishes a prime $p > n, M$, chooses independent uniform random coefficients $a_1, \dots, a_{k-1}$ in $\{0, \dots, p-1\}$, sets $a_0 = S$, and chooses distinct $x_1, \dots, x_n$ in $\{1, \dots, p-1\}$. Writing

$$P(x) = a_0 + a_1x + \dots + a_{k-1}x^{k-1} \pmod{p},$$

the r th participant receives the pair $(x_r, P(x_r))$, and the Leader destroys all working.

Now suppose k participants assemble, with pairs (y_j, z_j) , $1 \leq j \leq k$, where the y_j are distinct. They must solve

$$\begin{pmatrix} 1 & y_1 & \cdots & y_1^{k-1} \\ 1 & y_2 & \cdots & y_2^{k-1} \\ \vdots & & \ddots & \vdots \\ 1 & y_k & \cdots & y_k^{k-1} \end{pmatrix} \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{k-1} \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_k \end{pmatrix}$$

for $(c_0, \dots, c_{k-1})$ over $\mathbb{F}_p$. By Lemma 28.6.11 the matrix is Vandermonde with distinct nodes, so it is invertible and the solution is unique; and $(a_0, \dots, a_{k-1})$ is a solution by construction. Hence $c_0 = a_0 = S$.

If instead only $k - 1$ participants assemble, the corresponding system has $k - 1$ equations in k unknowns. Moving the c_0 column to the right hand side, the remaining matrix

$$\begin{pmatrix} y_1 & y_1^2 & \cdots & y_1^{k-1} \\ \vdots & & \ddots & \vdots \\ y_{k-1} & y_{k-1}^2 & \cdots & y_{k-1}^{k-1} \end{pmatrix}$$

has determinant $y_1 \cdots y_{k-1} \prod_{j < i} (y_i - y_j) \neq 0$, so for *every* choice of c_0 there is exactly one consistent completion. The $k - 1$ participants therefore learn nothing whatsoever about S : all p values remain exactly equally likely.

Example 28.7.36

Take $k = 3$, $p = 17$ and secret $S = 5$, with random coefficients $a_1 = 3$, $a_2 = 2$, so $P(x) = 5 + 3x + 2x^2$. The shares at $x = 1, \dots, 5$ are

$$(1, 10), \quad (2, 2), \quad (3, 15), \quad (4, 15), \quad (5, 2)$$

working modulo 17. Suppose the first three participants meet. Lagrange interpolation at 0 gives

$$S = 10 \cdot \frac{(0-2)(0-3)}{(1-2)(1-3)} + 2 \cdot \frac{(0-1)(0-3)}{(2-1)(2-3)} + 15 \cdot \frac{(0-1)(0-2)}{(3-1)(3-2)} = 10 \cdot 3 + 2 \cdot (-3) + 15 \cdot 1,$$

which is $30 - 6 + 15 = 39 \equiv 5 \pmod{17}$, recovering the secret.

Remark. The scheme has several pleasant features. The shares can be refreshed at any time without changing S , by choosing a new random polynomial with the same constant term; an eavesdropper who has slowly accumulated shares from the old polynomial learns nothing about the new one. And the shares are no larger than the secret, which is optimal.

Example 28.7.37 (Hierarchical Schemes)

In a $(3, n)$ -threshold scheme, give ordinary employees one share each, vice-presidents two, and the Leader three. Then the secret can be recovered by any three employees, or by a vice-president together with any one employee, or by the Leader alone. The number of shares held encodes seniority.

Example 28.7.38 (Reliable Storage)

Alice wishes to store a private key both securely and reliably. She uses a $(k, 2k - 1)$ -threshold scheme and puts the $2k - 1$ shares in different places. She can recover the key as long as she does not lose more than half of them, so the scheme is reliable; and a thief must steal more than half of them, so it is secure. This is exactly the same trade-off between redundancy and information rate that occupied us in Section 3, in a new costume.

29 Differential Geometry

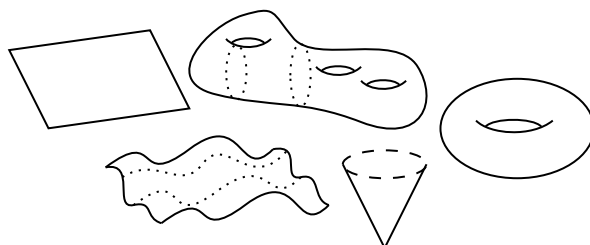
Differential geometry is what happens when you take the calculus you learned in first year and try to do it on something that isn't flat. The objects of study are *smooth manifolds* – spaces which, close up, look like a piece of Euclidean space, but which globally may be curved, knotted or twisted in interesting ways – and the tools of study are the derivative and the integral, suitably reinterpreted. The central discovery of the subject, and the one which gives it its character, is that curvature is not merely a feature of how a space sits inside a bigger space: a two-dimensional creature confined to a surface can measure the curvature of its world without ever leaving it. This is Gauss's *Theorema Egregium*, and everything in the second half of this course flows from it, culminating in the Gauss–Bonnet theorem, which ties the total curvature of a compact surface to a purely topological invariant.

This article constitutes my notes for the 'Differential Geometry' course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§29.1 Differential Topology

§29.1.1 Manifolds

The idea of a manifold is to generalize our typical boring and flat 'Euclidean space' with an object where everything isn't necessarily flat. For example, we can take some surfaces as drawn below.



We really care about surfaces where, in a sufficiently small region around each point, things look like we are in normal flat Euclidean space. This, along with a desire for things to be nice and smooth (as we shall define) will manifest into our basic notion of a *manifold*.

Definition 29.1.1 (Smooth)

For an open subset $U \subseteq \mathbb{R}^n$, we say a function $f : U \rightarrow \mathbb{R}^m$ is **smooth**^a if it has continuous partial derivatives of all orders.

^aFor a function $f : X \rightarrow \mathbb{R}^m$ with X not open, we say that f is smooth if the above definition holds for some open neighborhood V_x containing x for each $x \in X$.

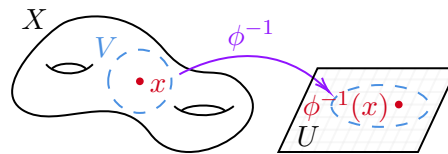
Definition 29.1.2 (Diffeomorphism)

A smooth function $f : X \rightarrow Y$ is a **diffeomorphism** if it's a bijection with smooth inverse. If a diffeomorphism between X and Y exists we say they are **diffeomorphic**.

Definition 29.1.3 (Manifold)

We say that $X \subseteq \mathbb{R}^n$ is a **k -dimensional manifold** if each $x \in X$ has a neighborhood $V \subseteq X$ diffeomorphic to an open set of $\mathbb{R}^k$.

With neighborhoods around each point looking like $\mathbb{R}^k$, we can sensibly imagine that one could locally put a coordinate system on such neighborhoods, one coming from the diffeomorphism to $\mathbb{R}^k$. This motivates the definition of a *chart* on a manifold.



Definition 29.1.4 (Chart)

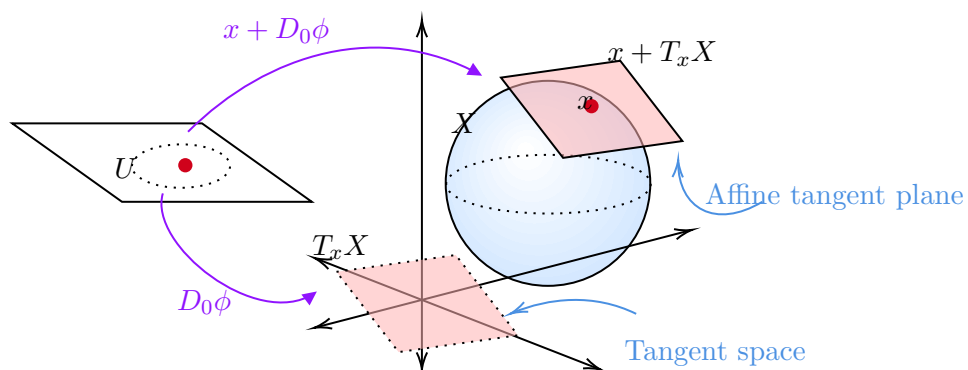
A diffeomorphism $\phi : U \rightarrow V$, where U is an open set of $\mathbb{R}^k$ is a **parameterisation** of the neighborhood V . The inverse diffeomorphism $\phi^{-1} : V \rightarrow U$ is called a **chart** on V .

Of course, if we have manifolds X and Z with $Z \subseteq X$ we say that Z is a **submanifold** of X . In this case, the **codimension** of Z in X is $\dim X - \dim Z$.

§29.1.2 Tangent Spaces and Derivatives

This is a course in differential geometry, and we've pinned down the 'geometry' part, now it's time to pin down the 'differential part'. We want to generalise the notion of a derivative for smooth functions on manifolds.

The idea you should have in your head of derivatives are that they are the best linear approximation to a given function. From this perspective, defining the derivative on manifolds requires us to answer two questions: what space should be our 'linear approximation' to a manifold that our derivative should act on, and then what is our linear approximation? Answering the first of these questions will lead us to the notion of **tangent spaces**.



Definition 29.1.5 (Tangent Space of a Manifold)

Let $X \subseteq \mathbb{R}^n$ be an arbitrary smooth manifold, and let $x \in X$. Choose a parameterisation $\phi : U \rightarrow X$ with $\phi(0) = x$ where U is an open set of $\mathbb{R}^k$. Then thinking of ϕ as a map from U to $\mathbb{R}^N$, we define the **tangent space** $T_x X$ to be the image of the derivative of ϕ at 0, $D_0\phi(\mathbb{R}^k)$.

In this definition we had to pick some parameterisation ϕ , but one would hope that our resulting space is independent of this choice. Indeed, this is the case.

Lemma 29.1.6 (Tangent Spaces are Parameterisation Independent)

$T_x X$ does not depend on the parameterisation ϕ chosen.

Proof. Suppose $\psi : V \rightarrow X$ is another choice of parameterisation, and suppose without loss of generality that $\phi(0) = \psi(0) = x$. By shrinking both U and V , we may assume that $\phi(U) = \psi(V)$. Then the map $h = \psi^{-1} \circ \phi : U \rightarrow V$ is a diffeomorphism. Now we write $\phi = \psi \circ h$ and we differentiate. Using the chain rule we have: $D_0\phi = D_0\psi \circ D_0h$. Since D_0h is an invertible linear map, it follows at once that $D_0\phi$ and $D_0\psi$ have the same image. $\square$

We are now ready to define the derivative of a smooth map $f : X \rightarrow Y$ between arbitrary manifolds, and we will see that it comes rather naturally. We would hope that for X and Y open subsets of euclidean space, our new definition should coincide with our previous definition of the derivative. Also, we would expect that the chain rule should still hold. This is going to force our definition.

Let $\phi : U \rightarrow X$ and $\psi : V \rightarrow Y$ be parameterisations, with $\phi(0) = x$, $\psi(0) = y$ and U, V being open subsets of $\mathbb{R}^k$ and $\mathbb{R}^l$ respectively. Then for small enough U , the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \phi \uparrow & & \uparrow \psi \\ U & \xrightarrow{h=\psi^{-1} \circ f \circ \phi} & V \end{array}$$

Then $D_0\psi$, $D_0\phi$, D_0h are all well defined, and as we said previously we would hope the chain rule still holds, which implies that the following square of linear maps commutes:

$$\begin{array}{ccc} T_x X & \xrightarrow{df_x} & T_y Y \\ d\phi_0 \uparrow & & \uparrow d\psi_0 \\ \mathbb{R}^k & \xrightarrow{dh_0} & \mathbb{R}^l \end{array}$$

Then since $D_0\phi$ is an isomorphism, only one definition of $D_x f$ will meet our requirements, namely

$$D_x f = D_0\psi \circ D_0h \circ D_0\phi^{-1}.$$

Again, one would hope that this definition of $D_x f$ is independent of ϕ and ψ , and this is the case (and the same proof as before works). Also, we have forced the chain rule still to hold.

Theorem 29.1.7 (Chain Rule)

If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are smooth maps between manifolds, then

$$D_x(g \circ f) = D_{f(x)}g \circ D_xf.$$

Proof. This is immediate from the ordinary chain rule. Choose smooth extensions F of f and G of g to open neighbourhoods of x and $f(x)$ in the ambient Euclidean spaces, arranged so that $G \circ F$ is defined; then $G \circ F$ is a smooth extension of $g \circ f$, and

$$D_x(g \circ f) = D(G \circ F)|_x|_{T_xX} = (DG|_{f(x)} \circ DF|_x)|_{T_xX} = D_{f(x)}g \circ D_xf,$$

where the middle equality is the chain rule in Euclidean space and the last uses that D_xf maps T_xX into $T_{f(x)}Y$. $\square$

The Inverse Function Theorem

Let $f : X \rightarrow Y$ be a smooth map between manifolds. We say that f is a **local diffeomorphism** at x if f maps a neighbourhood of x diffeomorphically onto a neighbourhood of $f(x)$.

Theorem 29.1.8 (Inverse Function Theorem)

Suppose that $f : X \rightarrow Y$ is a smooth map whose derivative df_x at the point x is an isomorphism. Then f is a local diffeomorphism at x .

Proof Sketch. This can be proven using the inverse function theorem for smooth functions between open sets of Euclidean space, as was shown in Analysis II. $\square$

§29.1.3 Regular Values and Sard's Theorem

Let $f : X \rightarrow Y$ be a smooth map between manifolds. Let C be the set of all points $x \in X$ such that $df_x : T_xX \rightarrow T_{f(x)}Y$ is not surjective.

Definition 29.1.9

A point in C will be called a **critical point**. A point in $f(C)$ will be called a **critical value**. A point in the complement of $f(C)$ will be called a **regular value**.

Remark. Note that if $\dim X < \dim Y$, then $C = X$ and the preimage of a regular value is the empty set.

Theorem 29.1.10 (Preimage Theorem)

Let y be a regular value of $f : X \rightarrow Y$ with $\dim X \geq \dim Y$. Then the set $f^{-1}(y)$ is a submanifold of X with $\dim f^{-1}(y) = \dim X - \dim Y$.

Proof. Let $x \in f^{-1}(y)$. Since y is a regular value, the derivative df_x maps T_xX onto T_yY . The kernel of df_x is a subspace K of T_xX of dimension $p := \dim X - \dim Y$.

Suppose $X \subset \mathbb{R}^N$ and let $T : \mathbb{R}^N \rightarrow \mathbb{R}^p$ be any linear map such that $\text{Ker}(T) \cap K = \{0\}$. Consider the map $F : X \rightarrow Y \times \mathbb{R}^p$ given by

$$F(z) = (f(z), T(z)).$$

The derivative of F is given by

$$dF_x(v) = (df_x(v), T(v))$$

which is clearly nonsingular by our choice of T . By the inverse function theorem, F is a local diffeomorphism at x , i.e. F maps some neighbourhood U of x diffeomorphically onto a neighbourhood V of $(y, T(x))$. Hence F maps $f^{-1}(y) \cap U$ diffeomorphically onto $(\{y\} \times \mathbb{R}^p) \cap V$ which proves that $f^{-1}(y)$ is a manifold with $\dim f^{-1}(y) = p$. $\square$

The case $\dim X = \dim Y$ is particularly important. The theorem says that if y is a regular value of f , then $f^{-1}(y)$ is a 0-dimensional manifold, i.e. collection of points. If X is compact, this collection must be finite (why?) so we obtain:

Corollary 29.1.11

Let $f : X \rightarrow Y$ be a smooth map between manifolds of the same dimension. If X is compact and y is a regular value, $f^{-1}(y)$ consists of a finite set of points.

We can actually say a bit more:

Theorem 29.1.12 (Stack of Records Theorem)

Let $f : X \rightarrow Y$ be a smooth map between manifolds of the same dimension with X compact. Let y be a regular value of f and write $f^{-1}(y) = \{x_1, \dots, x_k\}$. Then there exists a neighbourhood U of y in Y such that $f^{-1}(U)$ is a disjoint union $V_1 \cup \dots \cup V_k$, where V_i is an open neighbourhood of x_i and f maps each V_i diffeomorphically onto U .

Proof. By the inverse function theorem we can pick disjoint neighbourhoods W_i of x_i such that f maps W_i diffeomorphically onto a neighbourhood of y . Observe that $f(X - \cup_i W_i)$ is a compact set which does not contain y . Now take

$$U := \cap_i f(W_i) - f(X - \cup_i W_i).$$

$\square$

If we let $\#f^{-1}(y)$ be the cardinality of $f^{-1}(y)$, the theorem implies that the function $y \mapsto \#f^{-1}(y)$ is locally constant as y ranges over regular values of f .

The Preimage theorem gives a particularly nice way of producing manifolds. For example, if we consider the map $f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ given by $f(x) = |x|^2$ we can check easily that 1 is a regular value of f (do it!). Hence $S^n = f^{-1}(1)$ is a smooth manifold of dimension n . By switching our mental wavelength a little we can produce quite interesting examples as follows:

Example 29.1.13 (Orthogonal Group)

Let $O(n)$ be the group of orthogonal matrices of size $n \times n$, i.e., a matrix $A \in O(n)$ if and only if $AA^t = I$, where I is the identity matrix. The space of all $n \times n$ matrices $M(n)$ is just $\mathbb{R}^{n^2}$, so we can think of $O(n)$ as living inside $\mathbb{R}^{n^2}$. We will show that $O(n)$ is a manifold of dimension $\frac{n(n-1)}{2}$.

Let $S(n) \subset M(n)$ be the space of all symmetric matrices. Since it is a vector space, is clearly a submanifold of $M(n)$ and it has dimension $\frac{n(n+1)}{2}$.

Let $f : M(n) \rightarrow S(n)$ be the smooth map $f(A) = AA^t$. Since $O(n) = f^{-1}(I)$ it suffices to show that I is a regular value of f . We compute:

$$\begin{aligned} df_A(H) &= \lim_{s \rightarrow 0} \frac{f(A + sH) - f(A)}{s} \\ &= \lim_{s \rightarrow 0} \frac{(A + sH)(A + sH)^t - AA^t}{s} \\ &= HA^t + AH^t. \end{aligned}$$

Fix A with $AA^t = I$. We must prove that given any $C \in S(n) = T_I S(n)$ there is $H \in M(n) = T_A M(n)$ such that

$$df_A(H) = HA^t + AH^t = C.$$

Since C is symmetric, we can write $C = \frac{1}{2}C + \frac{1}{2}C^t$ so if we can solve $HA^t = \frac{1}{2}C$ we are done. Multiplying by A on the right and using $AA^t = I$ we obtain $H = \frac{1}{2}CA$ which solves $df_A(H) = C$. Thus I is a regular value of f .

Note that $O(n)$ is both a group and a manifold. In fact, the group operations are smooth, that is, the maps $(A, B) \mapsto AB$ and $A \mapsto A^{-1} = A^t$ are smooth. (Why?) A group that is manifold, and whose group operations are smooth, is called a Lie group.

Sard's Theorem

The Preimage theorem raises the question: is it easy to find regular values? How are abundant are they?

Recall that an arbitrary set A in $\mathbb{R}^n$ has measure zero if it can be covered by a countable number of rectangular solids with arbitrary small total volume. In other words given $\varepsilon > 0$, there exists a countable collection $\{R_1, R_2, \dots\}$ of rectangular solids in $\mathbb{R}^n$, such that A is contained in $\cup_i R_i$ and

$$\sum_i \text{Vol}(R_i) < \varepsilon.$$

Let X be manifold. An arbitrary subset $A \subset X$ has **measure zero** if, for every local parametrization ϕ of X , $\phi^{-1}(A)$ has measure zero in Euclidean space.

The following deep theorem, tells us that there are plenty of regular values.

Theorem 29.1.14 (Sard's Theorem, 1942)

The set of critical values of a smooth map $f : X \rightarrow Y$ has measure zero.

Since a set of measure zero cannot contain a non-empty open set we obtain:

Corollary 29.1.15

The regular values of any smooth map $f : X \rightarrow Y$ are dense in Y .

Morse Functions

A particularly useful class of maps to a one-dimensional target are those whose critical points are as simple as possible.

Definition 29.1.16

Let X be a smooth manifold. A smooth function $f : X \rightarrow \mathbb{R}$ is a **Morse function** if at every critical point x of f the Hessian of f , computed in any chart, is non-singular. Equivalently, the critical points of f are non-degenerate, and in particular isolated.

Sard's theorem is exactly what one needs to show that Morse functions are abundant: for $X \subseteq \mathbb{R}^N$ and almost every $a \in \mathbb{R}^N$, the function $f_a(x) = f(x) + \langle a, x \rangle$ is Morse. This, and the Morse lemma (near a non-degenerate critical point one can choose coordinates in which f is a quadratic form), are developed on the first example sheet; they are the starting point of Morse theory, which reconstructs the topology of X from the critical points of a single function on it.

§29.1.4 Transversality

We know that the solutions of the equation $f(x) = y$ form a smooth manifold, provided that y is a regular value of $f : X \rightarrow Y$. Suppose now that we replace y by a submanifold $Z \subset Y$ and we ask, when is $f^{-1}(Z)$ a submanifold of X ?

Definition 29.1.17 (Transversal)

A smooth map $f : X \rightarrow Y$ is said to be **transversal** to a submanifold $Z \subset Y$ if for every $x \in f^{-1}(Z)$ we have

$$\text{Image}(df_x) + T_{f(x)}Z = T_{f(x)}Y.$$

We write $f \pitchfork Z$.

Note that if $Z = \{y\}$, the notion of transversality reduces to the notion of regular value.

With this definition, we can now state a more general version of the Preimage theorem.

Theorem 29.1.18

If the smooth map $f : X \rightarrow Y$ is transversal to a submanifold $Z \subset Y$, then $f^{-1}(Z)$ is submanifold of X . Moreover, $f^{-1}(Z)$ and Z have the same codimension.

Proof (Non-examinable, sketch only). It is not hard to see that Z can be written in a neighbourhood of a point $y = f(x)$ as the zero set of a collection of functions $h_1, \dots, h_r$, where r is the codimension of Z in Y . Let $H := (h_1, \dots, h_r)$. Then near x , $f^{-1}(Z)$ is the zero set of the function $H \circ f$. Thus if $0 \in \mathbb{R}^r$ is a regular value of

$H \circ f$ we are done. But $dH_y \circ df_x$ is surjective if and only if

$$\text{Image}(df_x) + T_{f(x)}Z = T_{f(x)}Y$$

since $dH_y : T_yY \rightarrow \mathbb{R}^r$ is onto with kernel T_yZ . $\square$

Example 29.1.19

Consider $f : \mathbb{R} \rightarrow \mathbb{R}^2$ given by $f(t) = (0, t)$ and let Z be the x axis in $\mathbb{R}^2$. Then f is transversal to Z , but for example the map $h(t) = (t, t^2)$ is not.

An important special case occurs when f is the inclusion of a submanifold X of Y and Z is another submanifold of Y . In this case the condition of transversality reduces to

$$T_xX + T_xZ = T_xY$$

for every $x \in X \cap Z$. This condition is quite easy to visualize and when it holds we say that X and Z are transversal (we write $X \pitchfork Z$). We now have:

Theorem 29.1.20

The intersection of two transversal submanifolds of Y is a submanifold of codimension given by

$$\text{codim}(X \cap Z) = \text{codim } X + \text{codim } Z.$$

One of the main virtues of the notion of transversality is its stability i.e. it survives after small perturbations of the map f . You can convince yourself of this property by looking at pictures of transversal submanifolds of Euclidean space. Another very important virtue is genericity: smooth maps may be deformed by arbitrary small amounts into a map that is transversal to Z . (This is non-obvious and we refer to Chapter 2 in the book by Guillemin and Pollack for details.)

§29.1.5 Manifolds with Boundary

Consider the closed half-space

$$\mathbb{H}^k := \{(x_1, \dots, x_k) \in \mathbb{R}^k : x_k \geq 0\}.$$

The boundary $\partial\mathbb{H}^n$ is defined to be the hyperplane $x_k = 0$ in $\mathbb{R}^k$.

Definition 29.1.21

A subset $X \subset \mathbb{R}^N$ is called a smooth k -manifold with boundary if each $x \in X$ has a neighbourhood diffeomorphic to an open set in $\mathbb{H}^k$. As before, such a diffeomorphism is called a chart on X . The boundary of X , denoted ∂X , is given by the set of points that belong to the image of $\partial\mathbb{H}^k$ under some local parametrization. Its complement is called the interior of X , $\text{Int}(X) = X - \partial X$.

Remark (Warning). Do not confuse the boundary or interior of X as defined above with the topological notions of interior and boundary as a subset of $\mathbb{R}^N$.

The tangent space is defined as before, so that $T_x X$ is a k -dimensional vector space even for points $x \in \partial X$. The interior of X is a k -manifold without boundary and ∂X is a manifold without boundary of dimension $k - 1$ (this requires a proof!).

Here is an easy way of generating examples.

Lemma 29.1.22

Let X be a manifold without boundary and let $f : X \rightarrow \mathbb{R}$ be a smooth function with 0 as a regular value. Then the subset $\{x \in X : f(x) \geq 0\}$ is a smooth manifold with boundary equal to $f^{-1}(0)$.

Proof. The set where $f > 0$ is open in X and is therefore a submanifold of the same dimension as X . For a point $x \in X$ with $f(x) = 0$, the same proof as in Theorem 29.1.10 shows that x has a neighbourhood diffeomorphic to a neighbourhood of a point in $\mathbb{H}^k$. $\square$

As an easy application of the lemma, consider the unit ball B^k given by all $x \in \mathbb{R}^k$ such that $|x| \leq 1$. By considering the function $f(x) = 1 - |x|^2$ it follows that B^k is a smooth manifold with boundary S^{k-1} .

Theorem 29.1.23

Let $f : X \rightarrow Y$ be a smooth map from an m -manifold with boundary to an n -manifold, where $m > n$. If y is a regular value, both for f and for the restriction of f to ∂X , then $f^{-1}(y)$ is a smooth $(m - n)$ -manifold with boundary equal to $f^{-1}(y) \cap \partial X$.

Proof. Recall that being a submanifold is a local property so without loss of generality we can suppose that $f : \mathbb{H}^m \rightarrow \mathbb{R}^n$ with $y \in \mathbb{R}^n$ a regular value. Consider $z \in f^{-1}(y)$. If z belongs to the interior of $\mathbb{H}^m$, then as in Theorem 29.1.10, $f^{-1}(y)$ is a smooth manifold near z .

Let z be now in $\partial \mathbb{H}^m$. Since f is smooth, there is a neighbourhood U of z in $\mathbb{R}^m$ and a smooth map $F : U \rightarrow \mathbb{R}^n$ such that F restricted to $U \cap \mathbb{H}^m$ is f . By shrinking U if necessary we can assume that F has no critical points in U (why?). Hence $F^{-1}(y)$ is a smooth manifold of dimension $m - n$.

Let $\pi : F^{-1}(y) \rightarrow \mathbb{R}$ be the projection $(x_1, \dots, x_m) \mapsto x_m$. Observe that the tangent space of $F^{-1}(y)$ at a point $x \in \pi^{-1}(0)$ is equal to the kernel of

$$dF_x = df_x : \mathbb{R}^m \rightarrow \mathbb{R}^n.$$

Hence 0 must be a regular value of π since we are assuming that y is a regular value of f restricted to $\partial \mathbb{H}^m$.

But $F^{-1}(y) \cap \mathbb{H}^m = f^{-1}(y) \cap U$ is the set of all $x \in F^{-1}(y)$ with $\pi(x) \geq 0$ and by Lemma 29.1.22, is a smooth manifold with boundary equal to $\pi^{-1}(0)$. $\square$

It is not hard to guess what is the appropriate version of this theorem for the more general case of a map $f : X \rightarrow Y$ and a submanifold Z of Y . Suppose X has boundary, but Y and Z are boundaryless. The next theorem is stated without proof.

Theorem 29.1.24

Suppose that both $f : X \rightarrow Y$ and $f|_{\partial X} : \partial X \rightarrow Y$ are transversal to Z . Then $f^{-1}(Z)$ is a manifold with boundary given by $f^{-1}(Z) \cap \partial X$ and codimension equal to the codimension of Z .

§29.1.6 Degree Modulo 2

Let X be a smooth boundaryless manifold. Then $X \times [0, 1]$ is a manifold with boundary $\partial X = X \times \{0\} \cup X \times \{1\}$.

Definition 29.1.25

Two maps $f, g : X \rightarrow Y$ are called **smoothly homotopic** if there exists a smooth map $F : X \times [0, 1] \rightarrow Y$ with $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in X$. The map F is called a **smooth homotopy** between f and g . The relation of smooth homotopy is an equivalence relation. (Why?) The equivalence class to which a map belongs is its **homotopy** class. Let $f_t : X \rightarrow Y$ be the 1-parameter family of maps given by $f_t(x) = F(x, t)$.

Definition 29.1.26

The diffeomorphism f is **smoothly isotopic** to g if there exists a smooth homotopy $F : X \times [0, 1] \rightarrow Y$ from f to g such that for each $t \in [0, 1]$, f_t maps X diffeomorphically onto Y .

Lemma 29.1.27 (Homotopy Lemma)

Let $f, g : X \rightarrow Y$ be smooth maps which are smoothly homotopic. Suppose X is compact, has the same dimension as Y and $\partial X = \emptyset$. If y is a regular value for both f and g , then

$$\#f^{-1}(y) = \#g^{-1}(y) \pmod{2}.$$

Proof. The proof relies on the following important fact, the ‘Classification of 1-manifolds’: Every compact connected 1-manifold is diffeomorphic to $[0, 1]$ or S^1 .

As every compact manifold is the disjoint union of finitely many compact connected manifolds we have that the boundary of any compact 1-manifold consists of an even number of points.

Let $F : X \times [0, 1] \rightarrow Y$ be a smooth homotopy between f and g . Assume for the time being that y is also a regular value for F . Then $F^{-1}(y)$ is a compact 1-manifold with boundary equal to

$$F^{-1}(y) \cap (X \times \{0\} \cup X \times \{1\}) = f^{-1}(y) \times \{0\} \cup g^{-1}(y) \times \{1\}.$$

Thus the cardinality of the boundary of $F^{-1}(y)$ is just $\#f^{-1}(y) + \#g^{-1}(y)$. By the corollary above

$$\#f^{-1}(y) = \#g^{-1}(y) \pmod{2}.$$

If y is not a regular value of F we proceed as follows. From Theorem 29.1.12 we

know that $w \mapsto \#f^{-1}(w), w \mapsto \#g^{-1}(w)$ are locally constant as w ranges over regular values. Thus there are neighbourhoods V and W of y , consisting of regular values of f and g respectively for which

$$\#f^{-1}(w) = \#f^{-1}(y)$$

for all $w \in V$, and

$$\#g^{-1}(w) = \#g^{-1}(y)$$

for all $w \in W$. By Sard's theorem we can choose a regular value z of F in $V \cap W$. Then

$$\#f^{-1}(y) = \#f^{-1}(z) = \#g^{-1}(z) = \#g^{-1}(y) \pmod{2}$$

as desired. $\square$

Lemma 29.1.28 (Homogeneity Lemma)

Let X be a smooth connected manifold, possibly with boundary. Let y and z be points in $\text{Int}(X)$. Then there exists a diffeomorphism $h : X \rightarrow X$ smoothly isotopic to the identity such that $h(y) = z$.

Proof. Let us call two points y and z “isotopic” if there exists a diffeomorphism h isotopic to the identity that maps y to z . It is evident that this is an equivalence relation. Since X is connected, it suffices to show that each equivalence class is a open set.

To prove that equivalence classes are open we will construct an isotopy h_t of $\mathbb{R}^k$ such that h_0 is the identity, each h_t is the identity outside the open unit ball around the origin, and $h_1(0)$ is any specified point in the open unit ball. This will suffice, since we can now parametrize a small neighbourhood of y in $\text{Int}(X)$ and use h_t to construct an isotopy that will move y to any point nearby.

Let $\varphi : \mathbb{R}^k \rightarrow \mathbb{R}$ be a smooth function such that $\varphi(x) > 0$ for $|x| < 1$ and $\varphi(x) = 0$ for $|x| \geq 1$ (why do they exist?). Given a unit vector $u \in \mathbb{R}^k$ consider the ordinary differential equation in $\mathbb{R}^k$ given by

$$\frac{dx}{dt} = u\varphi(x).$$

Let $F_t : \mathbb{R}^k \rightarrow \mathbb{R}^k$ be the flow of this differential equation, i.e., for each $x \in \mathbb{R}^k$, the curve $t \mapsto F_t(x)$ is the unique solution passing through x . Standard theorems on differential equations tell us that:

1. F_t is defined for all $x \in \mathbb{R}^k$ and $t \in \mathbb{R}$ and smooth;
2. F_0 is the identity;
3. $F_{t+s} = F_t \circ F_s$.

Clearly for each t , F_t leaves all points outside the unit ball fixed. For appropriate choices of u and t , F_t will map the origin to any point in the open unit ball. (I may replace this proof by the one given by Guillemin and Pollack on page 142 of their book which does not use flows.) $\square$

In what follows suppose that X is compact and without boundary and Y is connected

and with the same dimension as X . Let $f : X \rightarrow Y$ be a smooth map.

Theorem 29.1.29 (Degree mod 2)

If y and z are regular values of f then

$$\#f^{-1}(y) = \#f^{-1}(z) \pmod{2}.$$

This common residue class is called the **degree mod 2** of f (denoted $\deg_2(f)$) and only depends on the homotopy class of f .

Proof. Given y and z , let h be the diffeomorphism smoothly isotopic to the identity such that $h(y) = z$ given by the Homogeneity Lemma. Observe that z is also a regular value of $h \circ f$. Since $h \circ f$ is homotopic to f , the Homotopy Lemma tells us that

$$\#(h \circ f)^{-1}(z) = \#f^{-1}(z) \pmod{2}.$$

But $(h \circ f)^{-1}(z) = f^{-1}(h^{-1}(z)) = f^{-1}(y)$ and therefore

$$\#f^{-1}(y) = \#f^{-1}(z) \pmod{2}$$

as desired. Let g be smoothly homotopic to f . By Sard's theorem, there is a point $y \in Y$ which is a regular value for both f and g (why?). By the Homotopy Lemma

$$\deg_2(f) = \#f^{-1}(y) \pmod{2} = \#g^{-1}(y) \pmod{2} = \deg_2(g)$$

which completes the proof. $\square$

Example 29.1.30

The identity map of a compact boundaryless manifold X has $\deg_2 = 1$, since any point has exactly one preimage, while a constant map has $\deg_2 = 0$, since a point other than the image value has none. Since $\deg_2$ is a homotopy invariant, the identity is never homotopic to a constant map.

This innocuous observation has real force.

Corollary 29.1.31

There is no smooth map $f : B^{k+1} \rightarrow S^k$ restricting to the identity on $S^k = \partial B^{k+1}$; that is, the sphere is not a smooth retract of the ball.

Proof. If such an f existed, then $F : S^k \times [0, 1] \rightarrow S^k$ given by $F(x, t) = f(tx)$ would be a smooth homotopy between the constant map $x \mapsto f(0)$ and the identity, contradicting the example above. $\square$

The previous example yields the smooth version of a famous fixed point theorem.

Theorem 29.1.32 (Smooth Brouwer Fixed Point Theorem)

Any smooth map $f : B^k \rightarrow B^k$ has a fixed point.

Proof. Suppose f has no fixed point. We will construct a map $g : B^k \rightarrow S^{k-1}$ which restricts to the identity on S^{k-1} which contradicts Corollary 29.1.31. For $x \in B^k$, let $g(x) \in S^{k-1}$ be the point where the line segment starting at $f(x)$ and passing through x hits the boundary. One can write a formula for g to show smoothness. $\square$

In fact, any continuous map $f : B^k \rightarrow B^k$ has a fixed point (Brouwer fixed point theorem). One can prove this using the smooth version approximating f by polynomials. Indeed, by the Weierstrass approximation theorem, given $\varepsilon > 0$, there is a polynomial P such that $|f(x) - P(x)| < \varepsilon$ for all $x \in B^k$. Set $Q(x) = P(x)/(1 + \varepsilon)$. Now Q maps B^k into B^k and $|f(x) - Q(x)| < 2\varepsilon$ for all $x \in B^k$. Suppose $f(x) \neq x$ for all x . Then the continuous function $|f(x) - x|$ must take a positive minimum τ on B^k . If now let $\varepsilon = \tau/2$ the Q from above does not have a fixed point, which contradicts the smooth Brouwer fixed point theorem.

Intersection Numbers Modulo 2

What we said above about degree modulo 2 can be generalized to the case of a smooth map $f : X \rightarrow Y$ and Z a submanifold of Y . Suppose that:

1. X is compact without boundary;
2. Z is closed and without boundary;
3. $f \pitchfork Z$
4. $\dim X + \dim Z = \dim Y$.

Under these conditions $f^{-1}(Z)$ is a closed 0-dimensional submanifold of X and hence it consists of finitely many points. Define the **mod 2 intersection number** of the map f with Z , $I_2(f, Z)$, to be the cardinality of $f^{-1}(Z)$ modulo 2. An analogue of the Homotopy Lemma (for intersection numbers) also holds here: if f_0 and f_1 are maps transversal to Z and homotopic, then $I_2(f_0, Z) = I_2(f_1, Z)$. In fact, we can define $I_2(f, Z)$ for a map f which is not necessarily transversal to Z . Using that transversality is generic, we can find a map g homotopic to f such that $g \pitchfork Z$ and we now set $I_2(f, Z) := I_2(g, Z)$. It does not matter which g we choose as long as g is homotopic to f , thanks to the Homotopy Lemma for intersection numbers.

Let us have a closer look at the important special case in which X itself is a compact submanifold of Y and Z a closed submanifold of complementary dimension. In this case f is the inclusion map $X \hookrightarrow Y$, and if $X \pitchfork Z$, $I_2(X, Z) := I_2(f, Z)$ is just $\#(X \cap Z) \bmod 2$. If $I_2(X, Z) \neq 0$, it means that no matter how X is deformed, we cannot move it away completely from Z .

Example 29.1.33

Let Y be the 2-torus $S^1 \times S^1$. Let $X = S^1 \times \{1\}$ and $Z = \{1\} \times S^1$. Then $I_2(X, Z) = 1$ and we cannot smoothly pull the circles apart. When $\dim X = \frac{1}{2} \dim Y$, we can consider $I_2(X, X)$ the self-intersection number modulo 2. If X is the central curve in a Möbius band, then $I_2(X, X) = 1$.

§29.1.7 Abstract Manifolds and Whitney's Embedding Theorem

One can actually define manifolds without making any reference to the ambient space $\mathbb{R}^N$. You will study manifolds in this more abstract setting in Part III and in the Riemann Surfaces course. In any case, here is the definition:

Definition 29.1.34

An n -dimensional smooth manifold is a second countable Hausdorff space X together with a collection of maps called **charts** such that:

1. a chart is a homeomorphism $\phi : U \rightarrow \phi(U)$, where U is open in X and $\phi(U)$ is open in $\mathbb{R}^n$
2. each point $x \in X$ belongs to the domain of some chart;
3. for charts $\phi : U \rightarrow \phi(U) \subset \mathbb{R}^n$ and $\psi : V \rightarrow \phi(V) \subset \mathbb{R}^n$, the map $\phi \circ \psi^{-1} : \psi(U \cap V) \rightarrow \phi(U \cap V)$ is smooth;
4. the collection of charts is maximal with respect to the properties above.

A set of charts satisfying the first three properties is called an **atlas**. An atlas can always be enlarged uniquely to give a maximal atlas as in the definition. If in the definition we require the maps $\phi \circ \psi^{-1}$ just to be of class C^k ($k \geq 1$) then we say that we have a manifold of class C^k . (Recall that a map is of class C^k if it has continuous partial derivatives up to order k .)

The definition is set up so that it is plain how to define a smooth map between manifolds. An **immersion** is a smooth map $f : X \rightarrow Y$ such that df_x is injective for all $x \in X$. A **submersion** is a smooth map $f : X \rightarrow Y$ such that df_x is surjective for all $x \in X$. An **embedding** is a smooth map $f : X \rightarrow Y$ which is an immersion and a homeomorphism onto its image.

The next theorem tells us that we did not lose much by restricting our attention to manifolds as subsets of Euclidean space.

Theorem 29.1.35 (Whitney's embedding theorem)

A smooth n -manifold X can be embedded into $\mathbb{R}^{2n+1}$.

The proof of this theorem can be found in most books about manifolds. In fact, Whitney proved the much harder result that X can be embedded in $\mathbb{R}^{2n}$.

§29.2 Curves in $\mathbb{R}^3$

Before we look at surfaces it is worth spending some time on the simplest interesting manifolds of all: one-dimensional ones, that is, curves. Curves are simple enough that we can say everything about them, and the story we will tell – attach a moving frame to the object, differentiate it, and read off the geometry from the coefficients that appear – is exactly the story we will tell for surfaces later, but with all the technicalities stripped away.

The punchline is that a curve in $\mathbb{R}^3$ has precisely two local invariants, its *curvature* and its *torsion*, and that these determine the curve completely up to a rigid motion of space.

Curvature says how fast the curve is turning, and torsion says how fast it is failing to be planar.

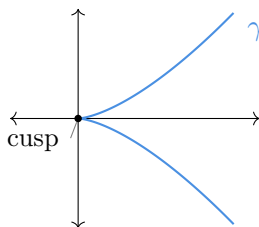
§29.2.1 Regular Curves and Arc Length

We should first be careful about what we mean by a curve. There is a difference between a curve as a *map* and a curve as a *subset* of $\mathbb{R}^3$, and it is the map we want.

Definition 29.2.1 (Smooth Curve)

A **smooth curve** in $\mathbb{R}^n$ is a smooth map $\gamma : (a, b) \rightarrow \mathbb{R}^n$, where $-\infty \leq a < b \leq \infty$. Its image $\gamma((a, b))$ is called the **trace** of the curve. We say γ is **regular** if $\gamma'(t) \neq 0$ for all $t \in (a, b)$.

Regularity is exactly the condition that γ is an immersion, and it is what stops the trace from having corners. It really is needed: the map $\gamma(t) = (t^2, t^3)$ is a perfectly smooth map $\mathbb{R} \rightarrow \mathbb{R}^2$, but $\gamma'(0) = 0$ and the trace has a cusp at the origin.



For a regular curve there is a preferred way to parametrise: by the distance travelled along it.

Definition 29.2.2 (Arc Length)

Let $\gamma : (a, b) \rightarrow \mathbb{R}^n$ be a smooth curve and fix $t_0 \in (a, b)$. The **arc length** of γ starting at t_0 is

$$s(t) = \int_{t_0}^t \|\gamma'(u)\| \, du.$$

We say γ is parametrised by arc length, or has **unit speed**, if $\|\gamma'(t)\| = 1$ for all t .

Lemma 29.2.3 (Unit Speed Reparametrisation)

Every regular curve admits a unit speed reparametrisation. That is, if $\gamma : (a, b) \rightarrow \mathbb{R}^n$ is regular then there is a diffeomorphism φ from an interval (c, d) onto (a, b) such that $\tilde{\gamma} = \gamma \circ \varphi$ has unit speed.

Proof. With s as above we have $s'(t) = \|\gamma'(t)\| > 0$ by regularity, so s is strictly increasing and smooth, hence a diffeomorphism from (a, b) onto an interval (c, d) ; its inverse $\varphi = s^{-1}$ is smooth by the inverse function theorem. Setting $\tilde{\gamma} = \gamma \circ \varphi$, the chain rule gives

$$\tilde{\gamma}'(s) = \gamma'(\varphi(s))\varphi'(s) = \frac{\gamma'(\varphi(s))}{\|\gamma'(\varphi(s))\|},$$

which has unit norm. $\square$

From now on, when a curve is unit speed we will write s for the parameter and use a dot for d/ds ; for a general parameter we write t and use a prime. Everything we define will be checked to be independent of the parametrisation, so we may always compute in whichever one is convenient.

§29.2.2 Curvature and the Frenet–Serret Formulae

Let γ be a unit speed curve. Its velocity $T = \dot{\gamma}$ is a unit vector, the **unit tangent**. Differentiating $\langle T, T \rangle = 1$ gives

$$\langle \dot{T}, T \rangle = 0,$$

so the acceleration of a unit speed curve is always perpendicular to its velocity. All the acceleration goes into turning, none into speeding up, and the size of that turning is our first invariant.

Definition 29.2.4 (Curvature)

The **curvature** of a unit speed curve γ is

$$\kappa(s) = \|\ddot{\gamma}(s)\| = \|\dot{T}(s)\| \geq 0.$$

For a curve given in an arbitrary regular parametrisation it would be painful to have to find the arc length reparametrisation before computing anything, so it is worth recording a direct formula.

Proposition 29.2.5

If γ is a regular curve in $\mathbb{R}^3$ with arbitrary parameter t , then

$$\kappa = \frac{\|\gamma' \times \gamma''\|}{\|\gamma'\|^3}.$$

Proof. Write $v = \|\gamma'\|$ and let s be arc length, so $ds/dt = v$ and $\gamma' = vT$. Differentiating,

$$\gamma'' = v'T + v \frac{dT}{dt} = v'T + v^2 \dot{T}.$$

Since $T \times T = 0$ and $\dot{T} \perp T$ with $\|\dot{T}\| = \kappa$, we get

$$\gamma' \times \gamma'' = vT \times (v'T + v^2 \dot{T}) = v^3(T \times \dot{T}),$$

and $\|T \times \dot{T}\| = \|T\| \|\dot{T}\| = \kappa$ as the two are orthogonal. Rearranging gives the result. $\square$

Example 29.2.6 (Lines and Circles)

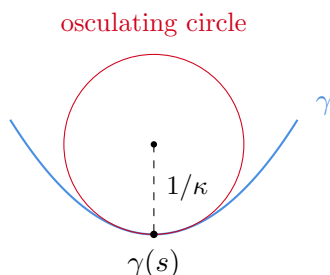
A straight line $\gamma(s) = p + sq$ with $\|q\| = 1$ has $\ddot{\gamma} = 0$, so $\kappa \equiv 0$. Conversely if $\kappa \equiv 0$ then $\ddot{\gamma} \equiv 0$, so $\gamma(s) = p + sq$ for constants p, q : a unit speed curve is a straight line if and only if its curvature vanishes identically.

The circle of radius a traversed at unit speed is $\gamma(s) = (a \cos(s/a), a \sin(s/a))$, and

$$\ddot{\gamma}(s) = -\frac{1}{a} (\cos(s/a), \sin(s/a)),$$

so $\kappa \equiv 1/a$. Small circles are very curved, large circles are nearly straight, which is as it should be.

The circle example gives the standard geometric interpretation of κ : at a point where $\kappa(s) > 0$ there is a unique circle which agrees with γ to second order at $\gamma(s)$, namely the circle of radius $1/a = 1/\kappa(s)$ lying in the plane spanned by $\dot{\gamma}$ and $\ddot{\gamma}$ and centred on the concave side. This is the **osculating circle**, and $1/\kappa(s)$ is the **radius of curvature**.



Now suppose $\kappa(s) > 0$ everywhere. Then $\dot{T}/\kappa$ is a unit vector orthogonal to T , and we can complete to an orthonormal basis of $\mathbb{R}^3$.

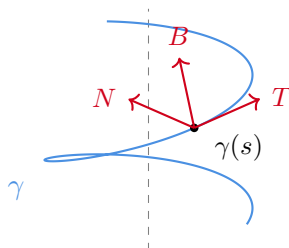
Definition 29.2.7 (Frenet Frame)

Let γ be a unit speed curve in $\mathbb{R}^3$ with $\kappa > 0$ everywhere. The **principal normal** and **binormal** are

$$N = \frac{\dot{T}}{\kappa}, \quad B = T \times N.$$

The triple (T, N, B) is a right-handed orthonormal basis of $\mathbb{R}^3$ at each point, called the **Frenet frame** of γ .

The plane spanned by T and N is the **osculating plane**; it is the plane the curve is doing its best to lie in. The whole of the local theory of curves comes from differentiating the frame and expressing the answer back in terms of the frame.



Theorem 29.2.8 (Frenet–Serret Formulae)

Let γ be a unit speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then there is a smooth function τ , the **torsion** of γ , such that

$$\begin{aligned} \dot{T} &= \kappa N, \\ \dot{N} &= -\kappa T + \tau B, \end{aligned}$$

$$\dot{B} = -\tau N.$$

Equivalently, writing the frame as the rows of a matrix,

$$\frac{d}{ds} \begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}.$$

Proof. Since (T, N, B) is an orthonormal basis at each point we may expand each derivative in it, so we must simply identify the nine coefficients $\langle \dot{T}, T \rangle$, $\langle \dot{T}, N \rangle$ and so on. Differentiating the six relations $\langle X, Y \rangle = \delta_{XY}$ for $X, Y \in \{T, N, B\}$ gives

$$\langle \dot{X}, Y \rangle + \langle X, \dot{Y} \rangle = 0,$$

so the matrix of coefficients is antisymmetric and only three entries need to be found. By definition $\dot{T} = \kappa N$, which gives the first row and, by antisymmetry, the $\langle \dot{N}, T \rangle = -\kappa$ entry and $\langle \dot{B}, T \rangle = 0$. Defining $\tau := \langle \dot{N}, B \rangle$ fills in the rest, and τ is smooth since N and B are. $\square$

Remark. Note the sign convention: some books define τ with the opposite sign. Ours is the one for which the right-handed helix (below) has positive torsion.

Torsion measures the rate at which the osculating plane rotates, or equivalently the rate at which the curve pulls out of that plane. So it should vanish exactly for planar curves, and it does.

Proposition 29.2.9

A unit speed curve γ in $\mathbb{R}^3$ with $\kappa > 0$ has $\tau \equiv 0$ if and only if its trace is contained in a plane.

Proof. If $\tau \equiv 0$ then $\dot{B} = -\tau N = 0$, so B is a constant vector b . Fix s_0 and consider $f(s) = \langle \gamma(s) - \gamma(s_0), b \rangle$. Then $f(s_0) = 0$ and $\dot{f} = \langle T, b \rangle = 0$, so $f \equiv 0$ and γ lies in the plane through $\gamma(s_0)$ with normal b .

Conversely, suppose γ lies in a plane with unit normal b . Differentiating $\langle \gamma - \gamma(s_0), b \rangle = 0$ twice gives $\langle T, b \rangle = 0$ and $\langle \dot{T}, b \rangle = 0$, so b is orthogonal to both T and N and hence $B = \pm b$ is constant. Therefore $\dot{B} = 0$ and, since $N \neq 0$, $\tau \equiv 0$. $\square$

As with curvature, it is convenient to have a formula valid in an arbitrary parametrisation.

Proposition 29.2.10

If γ is a regular curve in $\mathbb{R}^3$ with $\gamma' \times \gamma'' \neq 0$, then

$$\tau = \frac{\langle \gamma' \times \gamma'', \gamma''' \rangle}{\|\gamma' \times \gamma''\|^2}.$$

Proof. Again put $v = \|\gamma'\|$. We computed $\gamma' = vT$ and $\gamma'' = v'T + v^2\kappa N$, so

differentiating once more and collecting only the terms we need,

$$\gamma''' = (\text{multiple of } T) + (\text{multiple of } N) + v^3 \kappa \tau B,$$

using $\dot{N} = -\kappa T + \tau B$. From the proof of the curvature formula, $\gamma' \times \gamma'' = v^3 \kappa B$. Hence

$$\langle \gamma' \times \gamma'', \gamma''' \rangle = v^3 \kappa \cdot v^3 \kappa \tau = v^6 \kappa^2 \tau = \|\gamma' \times \gamma''\|^2 \tau,$$

as claimed. $\square$

Example 29.2.11 (The Helix)

Let $a > 0$, $b \neq 0$ and consider the helix $\gamma(t) = (a \cos t, a \sin t, bt)$. We have

$$\gamma' = (-a \sin t, a \cos t, b), \quad \gamma'' = (-a \cos t, -a \sin t, 0), \quad \gamma''' = (a \sin t, -a \cos t, 0),$$

so $\|\gamma'\| = \sqrt{a^2 + b^2}$ and

$$\gamma' \times \gamma'' = (ab \sin t, -ab \cos t, a^2), \quad \|\gamma' \times \gamma''\| = a\sqrt{a^2 + b^2}.$$

Therefore

$$\kappa = \frac{a\sqrt{a^2 + b^2}}{(a^2 + b^2)^{3/2}} = \frac{a}{a^2 + b^2}, \quad \tau = \frac{\langle \gamma' \times \gamma'', \gamma''' \rangle}{a^2(a^2 + b^2)} = \frac{a^2 b}{a^2(a^2 + b^2)} = \frac{b}{a^2 + b^2}.$$

Both are constant. Letting $b \rightarrow 0$ we recover the circle of radius a , with $\kappa = 1/a$ and $\tau = 0$.

The helix is in fact the *only* curve with constant curvature and torsion, which is a consequence of the following, the fundamental theorem of the local theory of curves. It says that (κ, τ) is a complete set of local invariants.

Theorem 29.2.12 (Fundamental Theorem of Curve Theory)

Let $\kappa : (a, b) \rightarrow (0, \infty)$ and $\tau : (a, b) \rightarrow \mathbb{R}$ be smooth. Then there is a unit speed curve $\gamma : (a, b) \rightarrow \mathbb{R}^3$ with curvature κ and torsion τ , and any two such curves differ by a rigid motion of $\mathbb{R}^3$ (that is, by a map $x \mapsto Ax + c$ with $A \in \text{SO}(3)$).

Proof. Existence. The Frenet–Serret formulae are a linear system of ordinary differential equations for the nine components of (T, N, B) , with smooth coefficients. By Picard’s theorem there is a unique solution on all of (a, b) with any prescribed initial value at s_0 ; take the initial value to be the standard basis (e_1, e_2, e_3) .

We must check the solution stays orthonormal. Consider the six functions $\langle X, Y \rangle$ for $X, Y \in \{T, N, B\}$. Differentiating and substituting the equations shows these six functions themselves satisfy a linear system of ODEs, one solution of which is the constant δ_{XY} ; by uniqueness, and since they agree at s_0 , they are equal to it for all s . So (T, N, B) remains orthonormal, and by continuity it stays right-handed.

Now set $\gamma(s) = \int_{s_0}^s T(u) du$. Then $\dot{\gamma} = T$ has unit length, $\ddot{\gamma} = \dot{T} = \kappa N$ has norm κ since $\kappa > 0$, so the curvature of γ is κ and its principal normal is N ; the binormal is then $T \times N = B$ (right-handedness) and $\langle \dot{N}, B \rangle = \tau$, as required.

Uniqueness. Suppose γ_1, γ_2 both have curvature κ and torsion τ , with frames (T_i, N_i, B_i) . The two frames at s_0 are right-handed orthonormal bases, so there is a unique $A \in \text{SO}(3)$ carrying the first to the second; replacing γ_1 by $A\gamma_1 + (\gamma_2(s_0) - A\gamma_1(s_0))$ – a rigid motion, which changes neither κ nor τ – we may assume the two curves and their frames agree at s_0 . Consider

$$f(s) = \|T_1 - T_2\|^2 + \|N_1 - N_2\|^2 + \|B_1 - B_2\|^2.$$

Differentiating and using Frenet–Serret for both curves, every term cancels: for instance

$$\frac{1}{2} \frac{d}{ds} \|T_1 - T_2\|^2 = \langle \kappa N_1 - \kappa N_2, T_1 - T_2 \rangle,$$

and adding the analogous expressions for N and B the κ and τ terms cancel in pairs, giving $\dot{f} = 0$. Since $f(s_0) = 0$ and $f \geq 0$ we get $f \equiv 0$, so $T_1 = T_2$ everywhere and hence $\gamma_1 - \gamma_2$ is constant; as they agree at s_0 they are equal. $\square$

Corollary 29.2.13

A unit speed curve in $\mathbb{R}^3$ with constant curvature $\kappa_0 > 0$ and constant torsion τ_0 is a circular helix (a circle if $\tau_0 = 0$).

Proof. Solve $a/(a^2 + b^2) = \kappa_0$, $b/(a^2 + b^2) = \tau_0$ for $a > 0$ and b : taking $a = \kappa_0/(\kappa_0^2 + \tau_0^2)$ and $b = \tau_0/(\kappa_0^2 + \tau_0^2)$ works. The corresponding helix, reparametrised at unit speed, has the given curvature and torsion, so by uniqueness our curve is a rigid motion of it. $\square$

§29.2.3 Plane Curves

For curves in the plane the theory simplifies, but in one respect it improves: a plane curve has a preferred normal direction available whether or not κ vanishes, and so we can attach a *sign* to the curvature.

Definition 29.2.14 (Signed Curvature)

Let $\gamma : (a, b) \rightarrow \mathbb{R}^2$ be a unit speed plane curve with tangent $T = \dot{\gamma}$. Let N_s be T rotated anticlockwise by $\pi/2$, so that (T, N_s) is a positively oriented orthonormal basis of $\mathbb{R}^2$. The **signed curvature** κ_s is defined by

$$\dot{T} = \kappa_s N_s.$$

This makes sense because $\dot{T} \perp T$, so $\dot{T}$ is automatically a multiple of N_s . Differentiating $\langle N_s, N_s \rangle = 1$ and $\langle T, N_s \rangle = 0$ gives $\dot{N}_s = -\kappa_s T$, so we have a two-dimensional Frenet system

$$\dot{T} = \kappa_s N_s, \quad \dot{N}_s = -\kappa_s T,$$

and clearly $\kappa = |\kappa_s|$. The sign records which way the curve is turning: anticlockwise where $\kappa_s > 0$, clockwise where $\kappa_s < 0$. Reversing the orientation of the curve reverses the sign.

Since T is a unit vector in the plane, we can write it as an angle, and it is worth knowing that the angle can be chosen smoothly.

Lemma 29.2.15 (Turning Angle)

Let $\gamma : (a, b) \rightarrow \mathbb{R}^2$ be a unit speed curve. Then there is a smooth function $\theta : (a, b) \rightarrow \mathbb{R}$ with

$$T(s) = (\cos \theta(s), \sin \theta(s)),$$

unique up to adding a constant in $2\pi\mathbb{Z}$, and $\dot{\theta} = \kappa_s$.

Proof. Write $T = (T_1, T_2)$. Pick θ_0 with $T(s_0) = (\cos \theta_0, \sin \theta_0)$ and define

$$\theta(s) = \theta_0 + \int_{s_0}^s (T_1 \dot{T}_2 - T_2 \dot{T}_1) \, du,$$

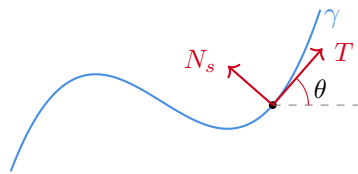
which is smooth. Using $T_1^2 + T_2^2 = 1$ one checks that both

$$\frac{d}{ds} (T_1 \cos \theta + T_2 \sin \theta) \quad \text{and} \quad \frac{d}{ds} (T_2 \cos \theta - T_1 \sin \theta)$$

vanish identically; evaluating at s_0 then gives $T_1 \cos \theta + T_2 \sin \theta \equiv 1$ and $T_2 \cos \theta - T_1 \sin \theta \equiv 0$, which forces $T = (\cos \theta, \sin \theta)$. Finally

$$\dot{T} = \dot{\theta}(-\sin \theta, \cos \theta) = \dot{\theta}N_s,$$

so $\dot{\theta} = \kappa_s$. Uniqueness is clear since two such angle functions differ by a continuous $2\pi\mathbb{Z}$ -valued function. $\square$



Now let $\gamma : [0, L] \rightarrow \mathbb{R}^2$ be a **closed** unit speed curve, meaning that γ and all its derivatives agree at 0 and L (so that γ extends to an L -periodic smooth curve on $\mathbb{R}$). Then $T(0) = T(L)$, so $\theta(L) - \theta(0) \in 2\pi\mathbb{Z}$.

Definition 29.2.16 (Turning Number)

The **turning number** of a closed plane curve γ of length L is the integer

$$n(\gamma) = \frac{1}{2\pi} (\theta(L) - \theta(0)) = \frac{1}{2\pi} \int_0^L \kappa_s \, ds.$$

The circle traversed anticlockwise has turning number 1, and traversed k times has turning number k ; a figure eight has turning number 0. What is intuitively obvious but genuinely delicate to prove is that a curve which does not cross itself cannot do anything exotic.

Theorem 29.2.17 (Hopf's Umlaufsatz)

A simple closed plane curve has turning number ± 1 , with the sign $+$ if it is traversed anticlockwise.

Proof Sketch. Translate so that γ lies in the upper half plane with $\gamma(0)$ on the x -axis, so the tangent at $s = 0$ is horizontal. On the triangle $\Delta = \{(s_1, s_2) : 0 \leq s_1 \leq s_2 \leq L\}$ define the *secant map*

$$\Psi(s_1, s_2) = \frac{\gamma(s_2) - \gamma(s_1)}{\|\gamma(s_2) - \gamma(s_1)\|}$$

off the diagonal and the two corners, extended by $\Psi(s, s) = T(s)$ and $\Psi(0, L) = -T(0)$. Simplicity of γ is exactly what makes Ψ well defined and continuous on Δ . The turning number is the degree of Ψ restricted to the diagonal edge, and since Δ is convex we may deform that edge to the other two edges without changing the degree. Along those two edges the secant vector sweeps monotonically through a half turn each (the curve lies on one side of the x -axis), giving a total of one full turn. $\square$

§29.2.4 The Isoperimetric Inequality

We now turn to two global results about plane curves: statements which involve the curve as a whole rather than a neighbourhood of a point. The first is the oldest theorem in the subject, going back to the legend of Dido: among all closed curves of a given length, the circle encloses the most area.

The proof is a beautiful piece of Fourier analysis, resting on the following inequality, which says that a mean-zero periodic function cannot oscillate slower than the fundamental frequency of its period.

Lemma 29.2.18 (Wirtinger's Inequality)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function of period L with $\int_0^L f \, dt = 0$. Then

$$\int_0^L |f'|^2 \, dt \geq \frac{4\pi^2}{L^2} \int_0^L |f|^2 \, dt,$$

with equality if and only if $f(t) = a \cos(2\pi t/L) + b \sin(2\pi t/L)$ for some constants a, b .

Proof. Write $f(t) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n t/L}$ and $f'(t) = \sum_{n \in \mathbb{Z}} b_n e^{2\pi i n t/L}$ for the Fourier series of f and f' ; these converge absolutely and uniformly since f is C^1 . The hypothesis $\int_0^L f = 0$ says $a_0 = 0$, and periodicity of f gives $b_0 = \frac{1}{L} \int_0^L f' = 0$. Integrating by parts,

$$b_n = \frac{1}{L} \int_0^L f'(t) e^{-2\pi i n t/L} \, dt = \frac{2\pi i n}{L} \cdot \frac{1}{L} \int_0^L f(t) e^{-2\pi i n t/L} \, dt = \frac{2\pi i n}{L} a_n.$$

Now Parseval's identity gives

$$\int_0^L |f'|^2 \, dt = L \sum_{n \neq 0} |b_n|^2 = \frac{4\pi^2}{L} \sum_{n \neq 0} n^2 |a_n|^2 \geq \frac{4\pi^2}{L} \sum_{n \neq 0} |a_n|^2 = \frac{4\pi^2}{L^2} \int_0^L |f|^2 \, dt,$$

using $n^2 \geq 1$. Equality holds if and only if $a_n = 0$ for every $|n| \geq 2$, which is the stated form. $\square$

Theorem 29.2.19 (Isoperimetric Inequality)

Let $\Omega \subseteq \mathbb{R}^2$ be a bounded domain whose boundary $\partial\Omega$ is a simple closed C^1 curve, of length L and enclosing area A . Then

$$L^2 \geq 4\pi A,$$

with equality if and only if Ω is a disc.

Proof. Parametrise $\partial\Omega$ by arc length $s \in [0, L]$, writing the position vector as $X(s) = (x(s), y(s))$, and let ν be the outward unit normal along $\partial\Omega$. Translating Ω changes neither L nor A , so we may assume $\int_0^L X \, ds = 0$.

Applied to the position vector field $X(x, y) = (x, y)$, whose divergence is 2, the divergence theorem gives

$$2A = \int_{\Omega} \nabla \cdot X \, dA = \int_{\partial\Omega} \langle X, \nu \rangle \, ds \leq \int_0^L \|X\| \, ds \leq \left(\int_0^L \|X\|^2 \, ds \right)^{1/2} L^{1/2},$$

the last step by Cauchy–Schwarz. Now x and y are C^1 and L -periodic with vanishing mean, so Wirtinger’s inequality applies to each, and since $(x')^2 + (y')^2 = 1$,

$$\int_0^L \|X\|^2 \, ds \leq \frac{L^2}{4\pi^2} \int_0^L ((x')^2 + (y')^2) \, ds = \frac{L^3}{4\pi^2}.$$

Substituting, $2A \leq (L^{3/2}/2\pi) \cdot L^{1/2} = L^2/2\pi$, which is the inequality.

If equality holds then in particular equality holds in Cauchy–Schwarz, so $\|X\|$ is constant along $\partial\Omega$: every boundary point is the same distance from the origin, and $\partial\Omega$ is a circle. $\square$

Remark. The same argument in fact shows a little more. If we had used Green’s theorem in the form $A = \int_0^L xy' \, ds$ instead, adding and subtracting x^2 gives the identity

$$\frac{L^2}{2\pi} - 2A = \int_0^L \left((x')^2 - \frac{4\pi^2}{L^2} x^2 \right) \, ds + \int_0^L \left(\frac{2\pi}{L} x - y' \right)^2 \, ds$$

after rescaling the parameter to have period 2π , which exhibits the deficit $L^2 - 4\pi A$ as a sum of two explicitly non-negative quantities.

§29.2.5 The Four Vertex Theorem

Our second global result concerns the curvature function of a closed plane curve. On a circle κ_s is constant; on any other simple closed curve it must vary, and the theorem says it must do so in a reasonably complicated way.

Definition 29.2.20 (Vertex)

A **vertex** of a plane curve γ is a point where $\kappa'_s = 0$, that is, a stationary point of the signed curvature.

An ellipse which is not a circle has exactly four vertices, the ends of the two axes, so the following is sharp.

Theorem 29.2.21 (Four Vertex Theorem)

Every simple closed plane curve has at least four vertices.

We will prove this for *convex* curves, meaning that the curve bounds a convex region; equivalently (after choosing the anticlockwise orientation) $\kappa_s \geq 0$ everywhere. The general case is true but substantially harder, and was only settled in full by Osserman in 1985.

Proof for convex curves. Parametrise γ by arc length on $[0, L]$. Since κ_s is a smooth function on a circle it attains a maximum and a minimum, so there are at least two vertices. Vertices where κ_s changes from increasing to decreasing alternate with those where it changes from decreasing to increasing, so the number of vertices, if finite, is even; hence it suffices to rule out the case of exactly two.

So suppose the only vertices are $p = \gamma(s_1)$ and $q = \gamma(s_2)$, at which κ_s attains its maximum and minimum. Then κ_s is strictly increasing on one of the two arcs of γ between p and q and strictly decreasing on the other. Let ℓ be the line through p and q . As γ is convex, ℓ meets the curve only at p and q , and the two arcs lie strictly on opposite sides of ℓ . Choose coordinates so that ℓ is the x -axis, and write $\gamma = (x, y)$; then $\kappa'_s \cdot y$ has a constant sign along the curve, and is not identically zero, so

$$\int_0^L \kappa'_s(s)y(s) \, ds \neq 0.$$

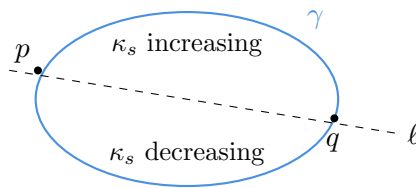
On the other hand, the Frenet equations in the plane read $\dot{T} = \kappa_s N_s$, that is, with $T = (\dot{x}, \dot{y})$ and $N_s = (-\dot{y}, \dot{x})$,

$$\ddot{x} = -\kappa_s \dot{y}, \quad \ddot{y} = \kappa_s \dot{x}.$$

Integrating by parts around the closed curve (all boundary terms vanish by periodicity),

$$\int_0^L \kappa'_s y \, ds = - \int_0^L \kappa_s \dot{y} \, ds = \int_0^L \ddot{x} \, ds = [\dot{x}]_0^L = 0,$$

a contradiction. □



Remark. A converse also holds (Gluck): any strictly positive smooth periodic function with at least two maxima and two minima is the curvature function of a simple closed convex curve.

§29.3 Surfaces, Length and Area

We now specialise the general machinery of Section 1 to the case that interests us most: two-dimensional manifolds sitting inside $\mathbb{R}^3$. This is where differential geometry began, with Gauss's work on surveying, and it remains the setting in which every important

idea in the subject can be seen in its simplest non-trivial form.

Everything in Section 1 applies verbatim. But we are now going to do *geometry* rather than topology: we will measure lengths, angles and areas on our surfaces, and then measure how they curve. The extra structure needed to do this comes from the ambient Euclidean inner product, and it will be packaged into two quadratic forms, the first and second fundamental forms.

§29.3.1 Smooth Surfaces

Definition 29.3.1 (Smooth Surface)

A **smooth surface** is a 2-dimensional smooth manifold $\Sigma \subseteq \mathbb{R}^3$. As in Section 1, a diffeomorphism $\sigma : V \rightarrow U \subseteq \Sigma$ from an open set $V \subseteq \mathbb{R}^2$ onto an open subset U of Σ is called an **allowable parametrisation** (or *allowable parameterisation*) of U .

We will consistently write $\sigma : V \rightarrow U \subseteq \Sigma$ for an allowable parametrisation, with coordinates (u, v) on V , and abbreviate the partial derivatives of σ by

$$\sigma_u = \frac{\partial \sigma}{\partial u}, \quad \sigma_v = \frac{\partial \sigma}{\partial v}, \quad \sigma_{uu} = \frac{\partial^2 \sigma}{\partial u^2}, \quad \text{and so on.}$$

Regularity of σ – which is part of it being a parametrisation – says exactly that σ_u and σ_v are linearly independent at every point, so $\{\sigma_u, \sigma_v\}$ is a basis for the tangent plane.

It is worth recording the standard ways in which surfaces turn up, all of which we already have the tools to handle.

Example 29.3.2 (Ways to Build Surfaces) (i) *Graphs.* If $f : V \rightarrow \mathbb{R}$ is smooth on an open $V \subseteq \mathbb{R}^2$ then $\sigma(u, v) = (u, v, f(u, v))$ parametrises the graph of f , which is therefore a smooth surface. Every surface looks like this locally.

(ii) *Level sets.* If $f : W \rightarrow \mathbb{R}$ is smooth on an open $W \subseteq \mathbb{R}^3$ and c is a regular value, then $f^{-1}(c)$ is a smooth surface by the preimage theorem. For instance $f(x, y, z) = x^2 + y^2 + z^2$ gives the spheres, and $f(x, y, z) = x^2 + y^2 - z^2$ gives the hyperboloids.

(iii) *Surfaces of revolution.* Let $\eta(u) = (f(u), 0, g(u))$ be a regular injective curve in the xz -plane with $f > 0$. Rotating about the z -axis gives

$$\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u)).$$

(iv) *Ruled surfaces.* Given a curve η and a nowhere-zero vector field w along it, $\sigma(u, v) = \eta(u) + v w(u)$ sweeps out a surface (where regular): a union of straight lines. Cylinders, cones and the helicoid all arise this way.

§29.3.2 Orientability and Tangent Planes

Let's now use the framework we have developed to discuss a geometric property of surfaces: *orientability*.

Definition 29.3.3 (Orientation-Preserving Maps)

Let V, V' be open sets in $\mathbb{R}^2$, and let $f : V \rightarrow V'$ be a diffeomorphism. We say that f is **orientation-preserving** if $Df|_x$ has positive determinant for all $x \in V$.

Definition 29.3.4 (Orientability)

An abstract smooth surface Σ is **orientable** if it admits an atlas $\{(U_i, \phi_i)\}$ where the transition maps are all orientation-preserving. Such an atlas is a **orientation** of Σ .

We will now show that orientability is a ‘topological’ property of surfaces, in that it is preserved by diffeomorphisms.

Theorem 29.3.5

If Σ_1 and Σ_2 are diffeomorphic abstract smooth surfaces, then Σ_1 is orientable if and only if Σ_2 is orientable.

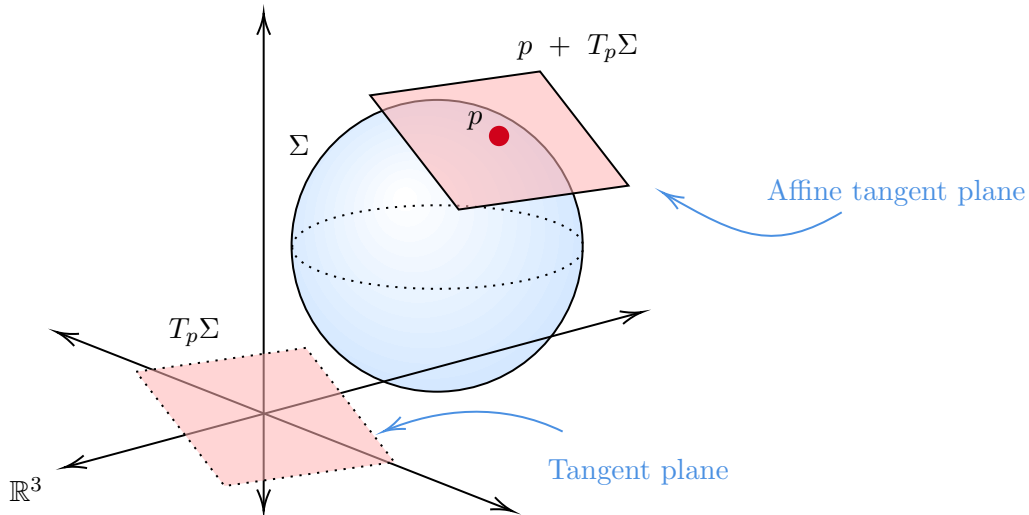
Proof. Let $f : \Sigma_1 \rightarrow \Sigma_2$ be diffeomorphism. Suppose Σ_2 is orientable and equipped with an oriented smooth atlas. Consider the atlas on Σ_1 of charts of the form $(f^{-1}(U), \phi \circ f|_{f^{-1}(U)})$, where (U, ψ) is a chart at $f(p)$ in the oriented atlas for Σ_2 . Then the transition map between two such charts is exactly a transition map between charts in the Σ_2 atlas. $\square$

So what does orientability *feel like*, in a geometric sense? To discuss this, we are going to need the idea of *tangent planes*.

Definition 29.3.6 (Tangent Plane)

Let Σ be a smooth surface in $\mathbb{R}^3$, and $p \in \Sigma$. Let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parameterisation of Σ near p , so V is an open subset of $\mathbb{R}^2$ and U is open in Σ , such that $\sigma(0) = p$. The **tangent plane** $T_p\Sigma$ to p at Σ is the image of $D\sigma|_0 \subseteq \mathbb{R}^3$, which is a two-dimensional vector subspace of $\mathbb{R}^3$. The **affine tangent plane** is $p + T_p\Sigma$, which is an affine^a subspace of $\mathbb{R}^3$.

^aAn *affine* subspace of a vector space is a translate of a linear subspace.



We should check that this notion makes geometric sense, in that it's independent of the choice of parametrisation used.

Lemma 29.3.7

$T_p\Sigma$ is independent of the choice of allowable parametrisation.

Proof. Suppose $\sigma : V \rightarrow U$ and $\tilde{\sigma} : \tilde{V} \rightarrow \tilde{U}$ are allowable parametrisations with $\sigma(0) = \tilde{\sigma}(0) = p$. Then there exists a transition map $\sigma^{-1} \circ \tilde{\sigma}$ which is a diffeomorphism of open sets in $\mathbb{R}^2$. So $D(\sigma^{-1} \circ \tilde{\sigma})|_0$ is an isomorphism, and since $\tilde{\sigma} = \sigma \circ (\sigma^{-1} \circ \tilde{\sigma})$ the images of $D\tilde{\sigma}|_0$ and $D\sigma|_0$ agree. $\square$

We can then sensibly define the *normal* to a smooth surface.

Definition 29.3.8

If Σ is a smooth surface in $\mathbb{R}^3$ and $p \in \Sigma$, the **normal direction** to Σ at p is $(T_p\Sigma)^\perp$, the Euclidean orthogonal complement to the tangent plane at p .

We can see that for every $p \in \Sigma$, there exist exactly two normalized normal vectors, and from this, we finally get our interpretation of orientability.

Definition 29.3.9

A smooth surface in $\mathbb{R}^3$ is **two-sided** if it admits a continuous global choice of unit normal vector.

Lemma 29.3.10 (Interpretation of Orientability)

A smooth surface in $\mathbb{R}^3$ is orientable if and only if it is two-sided.

Proof. Suppose Σ is orientable, with an atlas $\{(U_i, \sigma_i)\}$ whose transition maps all

have positive Jacobian determinant. On each U_i define

$$n_i = \frac{(\sigma_i)_u \times (\sigma_i)_v}{\|(\sigma_i)_u \times (\sigma_i)_v\|},$$

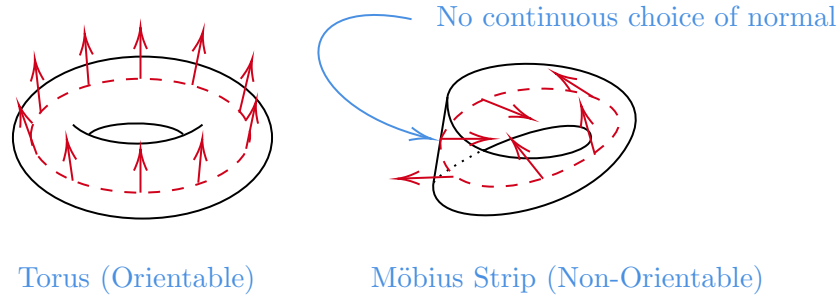
which is smooth and unit. If $p \in U_i \cap U_j$ and φ is the transition map, then as in the proof that area is well defined we have $(\sigma_j)_u \times (\sigma_j)_v = \det(D\varphi) (\sigma_i)_u \times (\sigma_i)_v$, and $\det(D\varphi) > 0$, so $n_i = n_j$ on the overlap. Hence the n_i glue to a global continuous (indeed smooth) unit normal, and Σ is two-sided.

Conversely suppose n is a global continuous unit normal. Take any atlas, and for each chart discard it in favour of the chart with u and v swapped if necessary, so that

$$\frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = n$$

on that patch – possible since the left hand side is a continuous unit normal, hence equal to $\pm n$ on each connected component, and swapping the coordinates changes its sign. For two such charts the same identity as above gives $\det(D\varphi) > 0$, so this atlas is an orientation. $\square$

Some illustrations of an orientable and non-orientable surface are shown below.



§29.3.3 The First Fundamental Form

Having pinned down what our surfaces are, we can start doing geometry on them: measuring lengths, angles and areas.

Recall that for a curve $\gamma : (a, b) \rightarrow \mathbb{R}^3$, the length of γ is given by

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt,$$

and that this is independent of the parametrisation.¹ Now suppose the curve lies on a surface. We would like to compute its length without ever leaving the surface, that is, in terms of the coordinates (u, v) of a parametrisation.

So let Σ be a smooth surface in $\mathbb{R}^3$, let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parametrisation, and let $\gamma : (a, b) \rightarrow \mathbb{R}^3$ be a smooth curve whose image lies in U . Writing $\gamma(t) = \sigma(u(t), v(t))$, the chain rule gives $\gamma'(t) = \sigma_u u'(t) + \sigma_v v'(t)$, and so

$$\|\gamma'(t)\|^2 = Eu'(t)^2 + 2Fu'(t)v'(t) + Gv'(t)^2,$$

where

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle = \langle \sigma_v, \sigma_u \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle,$$

¹Feel free to check this – it is just the substitution rule.

and $\langle \cdot, \cdot \rangle$ is the Euclidean inner product. So three functions on V are all we need.

Definition 29.3.11 (First Fundamental Form)

The **first fundamental form** of Σ in the parametrisation σ is the expression

$$I = E \, du^2 + 2F \, du \, dv + G \, dv^2,$$

equivalently the symmetric matrix $\begin{pmatrix} E & F \\ F & G \end{pmatrix} = (D\sigma)^\top (D\sigma)$ representing the restriction of the Euclidean inner product to $T_p\Sigma$ in the basis $\{\sigma_u, \sigma_v\}$.

Since the inner product is positive definite and $\{\sigma_u, \sigma_v\}$ is a basis, we always have $E > 0$, $G > 0$ and $EG - F^2 > 0$. In terms of the first fundamental form, if γ has image in the image of σ then

$$L(\gamma) = \int_a^b \sqrt{E(u')^2 + 2Fu'v' + G(v')^2} \, dt,$$

and the angle θ between two tangent vectors $a\sigma_u + b\sigma_v$ and $c\sigma_u + d\sigma_v$ is given by

$$\cos \theta = \frac{Eac + F(ad + bc) + Gbd}{\sqrt{Ea^2 + 2Fab + Gb^2} \sqrt{Ec^2 + 2Fcd + Gd^2}}.$$

So *all* of the metric geometry of the surface, as far as a curve-measuring inhabitant of it is concerned, is contained in E , F and G .

Example 29.3.12 (Some First Fundamental Forms) (i) The plane $\sigma(u, v) = (u, v, 0)$ has $\sigma_u = (1, 0, 0)$ and $\sigma_v = (0, 1, 0)$, so $I = du^2 + dv^2$, as it must.

(ii) The cylinder $\sigma(u, v) = (a \cos u, a \sin u, v)$ has $\sigma_u = a(-\sin u, \cos u, 0)$ and $\sigma_v = (0, 0, 1)$, so $I = a^2 du^2 + dv^2$.

(iii) A surface of revolution $\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u))$ has

$$\sigma_u = (f' \cos v, f' \sin v, g'), \quad \sigma_v = (-f \sin v, f \cos v, 0),$$

so $E = (f')^2 + (g')^2$, $F = 0$ and $G = f^2$. If the profile curve $\eta = (f, 0, g)$ is unit speed then this is simply

$$I = du^2 + f(u)^2 dv^2.$$

(iv) The unit sphere, parametrised by latitude and longitude as

$$\sigma(u, v) = (\cos u \cos v, \cos u \sin v, \sin u),$$

is the surface of revolution with $f = \cos u$ and $g = \sin u$, so

$$I = du^2 + \cos^2 u \, dv^2.$$

(v) The torus $\sigma(u, v) = ((a + r \cos u) \cos v, (a + r \cos u) \sin v, r \sin u)$ (with $a > r > 0$) is the surface of revolution with $f = a + r \cos u$, $g = r \sin u$, so $E = r^2$, $F = 0$, $G = (a + r \cos u)^2$.

§29.3.4 Isometries and Conformal Maps

With length encapsulated by the first fundamental form, we get a nice way to capture when two surfaces have the same geometry in a *length* sense.

Definition 29.3.13 (Isometric)

Let Σ, Σ' be smooth surfaces in $\mathbb{R}^3$. A diffeomorphism $f : \Sigma \rightarrow \Sigma'$ is an **isometry** if $Df|_p : T_p\Sigma \rightarrow T_{f(p)}\Sigma'$ is a linear isometry for every p ; equivalently, if f preserves the length of all curves. If such an f exists we say Σ and Σ' are **isometric**.

Definition 29.3.14 (Locally Isometric)

We say that Σ, Σ' are **locally isometric** near points $p \in \Sigma$ and $q \in \Sigma'$ if there exist open neighbourhoods U of p and U' of q which are isometric.

The whole point of the first fundamental form is the following, which turns a geometric question into a computation.

Lemma 29.3.15

Smooth surfaces Σ, Σ' in $\mathbb{R}^3$ are locally isometric near $p \in \Sigma$ and $q \in \Sigma'$ if and only if there exist allowable parametrisations $\sigma : V \rightarrow U \subseteq \Sigma$ and $\sigma' : V \rightarrow U' \subseteq \Sigma'$, with the same domain V , whose first fundamental forms are equal.

Proof. Suppose such parametrisations exist and set $f = \sigma' \circ \sigma^{-1} : U \rightarrow U'$, a diffeomorphism. In the bases $\{\sigma_u, \sigma_v\}$ and $\{\sigma'_u, \sigma'_v\}$, $Df|_p$ is the identity matrix, and the two first fundamental forms are equal, so $Df|_p$ carries an inner product to the same inner product, and is therefore a linear isometry.

Conversely, suppose $f : U \rightarrow U'$ is an isometry and let $\sigma : V \rightarrow U$ be any allowable parametrisation. Then $\sigma' = f \circ \sigma$ is an allowable parametrisation of U' on the same domain, and $\sigma'_u = Df|(\sigma_u)$, $\sigma'_v = Df|(\sigma_v)$. Since $Df|$ preserves inner products, $E' = E$, $F' = F$ and $G' = G$. $\square$

Example 29.3.16 (The Plane and the Cylinder)

Parametrise the plane by $\sigma(u, v) = (au, v, 0)$, so that $I = a^2 du^2 + dv^2$, and compare with the cylinder $\tilde{\sigma}(u, v) = (a \cos u, a \sin u, v)$ computed above. On $V = (0, 2\pi) \times \mathbb{R}$ the two first fundamental forms agree, so a cylinder is locally isometric to a plane: you really can roll a sheet of paper into a tube without stretching it. They are not globally isometric, since one is simply connected and the other is not.

Example 29.3.17 (The Sphere Cannot be Flattened)

No open subset of S^2 is isometric to an open subset of the plane. This is not obvious from the definition – it follows immediately from the Theorema Egregium of Section 5 – but it is worth flagging now as the reason that every world map distorts distances.

A weaker but very useful notion asks only that angles, not lengths, be preserved.

Definition 29.3.18 (Conformal)

A diffeomorphism $f : \Sigma \rightarrow \Sigma'$ is **conformal** if for every p there is $\lambda(p) > 0$ with

$$\langle Df|_p(w_1), Df|_p(w_2) \rangle = \lambda(p) \langle w_1, w_2 \rangle$$

for all $w_1, w_2 \in T_p\Sigma$. Equivalently, in corresponding parametrisations $I' = \lambda I$.

Since the angle formula above is unchanged when E, F, G are all multiplied by the same positive function, a conformal map is precisely one that preserves angles (though not orientation of angles unless we also ask $\det Df > 0$). A parametrisation is called **isothermal** if in it

$$I = \lambda(u, v)^2 (du^2 + dv^2),$$

that is, if σ is a conformal map from a piece of the plane to the surface. Isothermal parametrisations always exist locally, though this is a genuinely hard theorem; they will be crucial when we come to minimal surfaces.

Example 29.3.19 (Stereographic Projection)

Stereographic projection $\pi : S^2 \setminus \{N\} \rightarrow \mathbb{R}^2$ from the north pole is conformal, with the inverse parametrisation

$$\sigma(u, v) = \frac{1}{1 + u^2 + v^2} (2u, 2v, u^2 + v^2 - 1)$$

having first fundamental form

$$I = \frac{4}{(1 + u^2 + v^2)^2} (du^2 + dv^2),$$

so it is isothermal. This is why nautical charts made by stereographic projection get the angles right, and it is the reason for the appearance of $\pi \circ \mathcal{N}$ later in the Weierstrass representation.

§29.3.5 Area

With length pinned down by the first fundamental form, it is not too hard to pin down area.

Recall that the parallelogram spanned by two vectors w_1, w_2 has area $\|w_1 \times w_2\|$, and that

$$\|w_1 \times w_2\|^2 = \langle w_1, w_1 \rangle \langle w_2, w_2 \rangle - \langle w_1, w_2 \rangle^2.$$

So if $\sigma : V \rightarrow U \subseteq \Sigma$ is an allowable parametrisation, the infinitesimal parallelogram spanned by $\sigma_u du$ and $\sigma_v dv$ has area

$$\|\sigma_u \times \sigma_v\| du dv = \sqrt{EG - F^2} du dv.$$

This gives us a sensible way to define area, but we had better check it does not depend on the parametrisation.

Lemma 29.3.20

Let $\Omega \subseteq \Sigma$ be a bounded domain contained in the images of two allowable parametrisations.

sations $\sigma : V \rightarrow U$ and $\tilde{\sigma} : \tilde{V} \rightarrow \tilde{U}$. Then

$$\int_{\sigma^{-1}(\Omega)} \sqrt{EG - F^2} \, du \, dv = \int_{\tilde{\sigma}^{-1}(\Omega)} \sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} \, d\tilde{u} \, d\tilde{v}.$$

Proof. Let $\varphi = \sigma^{-1} \circ \tilde{\sigma}$ be the transition map, a diffeomorphism between open subsets of $\mathbb{R}^2$, and let $J = \det D\varphi$ be its Jacobian determinant. Since $\tilde{\sigma} = \sigma \circ \varphi$, the chain rule gives $D\tilde{\sigma} = D\sigma \circ D\varphi$, and hence

$$\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}} = J(\sigma_u \times \sigma_v),$$

since taking the cross product of the two columns of a 3×2 matrix and then post-composing with a 2×2 matrix scales by its determinant. Taking norms, $\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = |J|\sqrt{EG - F^2} \circ \varphi$, and the claim is exactly the change of variables formula. $\square$

Definition 29.3.21 (Area)

Let Σ be a smooth surface in $\mathbb{R}^3$, let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parametrisation and let $\Omega \subseteq U$ be a bounded domain. Then

$$\text{area}(\Omega) = \int_{\sigma^{-1}(\Omega)} \sqrt{EG - F^2} \, du \, dv.$$

More generally, for continuous $h : \Sigma \rightarrow \mathbb{R}$ we write

$$\int_{\Omega} h \, dA = \int_{\sigma^{-1}(\Omega)} h(\sigma(u, v)) \sqrt{EG - F^2} \, du \, dv,$$

and call $dA = \sqrt{EG - F^2} \, du \, dv$ the **area element** of Σ .

Remark. For regions not contained in a single parametrised patch, one patches these integrals together using a partition of unity. In practice we can nearly always find a single parametrisation covering everything except a finite union of curves, which contribute no area, and so we will not fuss about this.

Example 29.3.22 (Area of the Torus)

For the torus with $E = r^2$, $F = 0$, $G = (a + r \cos u)^2$ we have $\sqrt{EG - F^2} = r(a + r \cos u)$, so

$$\text{area} = \int_0^{2\pi} \int_0^{2\pi} r(a + r \cos u) \, du \, dv = 4\pi^2 ra.$$

This is Pappus's theorem: the area is the circumference $2\pi r$ of the generating circle times the distance $2\pi a$ travelled by its centre.

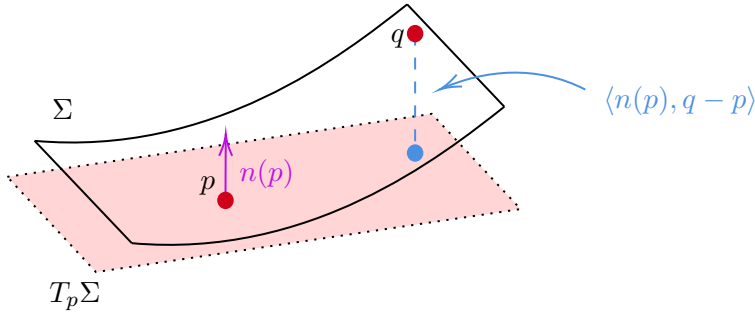
Remark. Isometries preserve area, since they preserve E, F, G . The converse is false: an area-preserving map need not be an isometry. Archimedes' theorem gives a beautiful example – the projection of the unit sphere onto the circumscribed cylinder, radially away from the axis, preserves area but certainly not lengths.

§29.4 The Gauss Map and Curvature

§29.4.1 The Second Fundamental Form

Intuitively, the first fundamental form pins down how length works along our surface. We now, through the second fundamental form, will try to pin down how *curved* our surface is (as an embedding of $\mathbb{R}^3$).

We are (informally) going to look at curvature at a point as measuring how much we deviate from the tangent plane by moving a small amount away from our point.



Let Σ be a smooth surface in $\mathbb{R}^3$, and let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parameterisation. Consider a point $p = \sigma(u, v)$, and some nearby point $q = \sigma(u + h, v + \ell)$. Then by Taylor's theorem we have

$$\begin{aligned} \sigma(u + h, v + \ell) &= \sigma(u, v) + h\sigma_u(u, v) + \ell\sigma_v(u, v) \\ &\quad + \frac{1}{2} (h^2\sigma_{uu}(u, v) + 2h\ell\sigma_{uv}(u, v) + \ell^2\sigma_{vv}(u, v)) \\ &\quad + O(h^3, \ell^3). \end{aligned}$$

Recalling that the tangent plane is given by $T_p\Sigma = \text{span}\{\sigma_u, \sigma_v\}$, we get the perpendicular distance from $q = \sigma(u + h, v + \ell)$ to the affine tangent plane $T_p\Sigma + p$ is given by

$$\begin{aligned} \langle n, \sigma(u + h, v + \ell) - \sigma(u, v) \rangle &= \frac{1}{2} (\langle n, \sigma_{uu} \rangle h^2 + 2\langle n, \sigma_{uv} \rangle h\ell + \langle n, \sigma_{vv} \rangle \ell^2) \\ &\quad + O(h^3, \ell^3). \end{aligned}$$

and so we can see that these inner products give us this difference. This inspires our definition of the second fundamental form.

Definition 29.4.1 (Second Fundamental Form)

The **second fundamental form** of Σ in the allowable parameterisation σ is

$$L \, du^2 + 2M \, du \, dv + N \, dv^2$$

where $L = \langle n, \sigma_{uu} \rangle$, $M = \langle n, \sigma_{uv} \rangle$, and $N = \langle n, \sigma_{vv} \rangle$, and n is the unit normal given by $n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}$.

We will say more about how this relates to curvature in the coming sections. It's good to know (but not proven in this course) that the two fundamental forms completely determine a smooth oriented connected surface in $\mathbb{R}^3$, up to rigid motion. This is the *fundamental theorem of surfaces in $\mathbb{R}^3$* .

§29.4.2 The Gauss Map and the Shape Operator

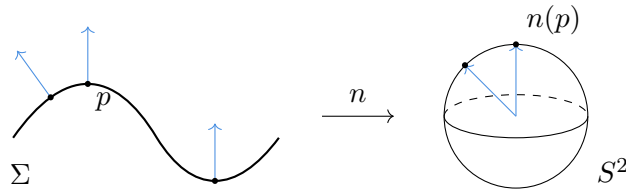
We are now going to define what our notion of curvature actually is. This is done using the *Gauss map*, which packages the unit normal into a map to the sphere.

Definition 29.4.2 (Gauss Map)

Let Σ be a smooth oriented surface in $\mathbb{R}^3$. The **Gauss map** is

$$n : \Sigma \rightarrow S^2, \quad p \mapsto n(p),$$

sending each point to its positively oriented unit normal, viewed as a point of the unit sphere.



Lemma 29.4.3

The Gauss map is smooth.

Proof. Smoothness is a local property, so we may check it in an allowable parametrisation σ compatible with the orientation, where

$$n \circ \sigma = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}.$$

The numerator is smooth and the denominator is smooth and non-vanishing (since σ_u, σ_v are linearly independent), so $n \circ \sigma$ is smooth. $\square$

The reason the Gauss map is so useful is a small miracle of dimension counting. Its derivative at p is a linear map

$$Dn|_p : T_p\Sigma \rightarrow T_{n(p)}S^2,$$

but $T_{n(p)}S^2$ is the plane orthogonal to $n(p)$ in $\mathbb{R}^3$, which is precisely $T_p\Sigma$. So $Dn|_p$ is an *endomorphism of the tangent plane*, and we may take its determinant, its trace, and its eigenvalues.

Definition 29.4.4 (Shape Operator)

The **shape operator** (or Weingarten map) of Σ at p is the linear map

$$S_p = -Dn|_p : T_p\Sigma \rightarrow T_p\Sigma.$$

The sign is a convention chosen so that the sphere with its outward normal has positive curvature. The key algebraic fact is the following.

Lemma 29.4.5

The shape operator is self-adjoint with respect to the first fundamental form, and

$$\Pi_p(w_1, w_2) = \langle S_p(w_1), w_2 \rangle.$$

In particular the second fundamental form is a symmetric bilinear form.

Proof. Work in an allowable parametrisation and abbreviate n for $n \circ \sigma$. Applying the chain rule to a curve $\sigma(u(t), v(t))$ shows that $Dn|_p(\sigma_u) = n_u$ and $Dn|_p(\sigma_v) = n_v$, so it is enough to check the claim on the basis $\{\sigma_u, \sigma_v\}$.

Differentiating the identities $\langle n, \sigma_u \rangle = 0$ and $\langle n, \sigma_v \rangle = 0$ gives

$$\begin{aligned} \langle n_u, \sigma_u \rangle &= -\langle n, \sigma_{uu} \rangle = -L, & \langle n_u, \sigma_v \rangle &= -\langle n, \sigma_{uv} \rangle = -M, \\ \langle n_v, \sigma_u \rangle &= -\langle n, \sigma_{vu} \rangle = -M, & \langle n_v, \sigma_v \rangle &= -\langle n, \sigma_{vv} \rangle = -N. \end{aligned}$$

The two middle expressions agree because mixed partials commute, which gives $\langle Dn|_p \sigma_u, \sigma_v \rangle = \langle \sigma_u, Dn|_p \sigma_v \rangle$: self-adjointness. Reading the same four identities the other way round says exactly that the matrix of Π in the basis $\{\sigma_u, \sigma_v\}$ is the matrix of $\langle S_p(\cdot), \cdot \rangle$. $\square$

The identities in that proof deserve to be displayed on their own, since they are the computational heart of the subject:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = - \begin{pmatrix} \langle n_u, \sigma_u \rangle & \langle n_u, \sigma_v \rangle \\ \langle n_v, \sigma_u \rangle & \langle n_v, \sigma_v \rangle \end{pmatrix} = -(Dn)^\top(D\sigma),$$

in perfect analogy with $I = (D\sigma)^\top(D\sigma)$.

§29.4.3 Normal and Principal Curvatures

The second fundamental form has a direct geometric meaning in terms of curves on the surface.

Definition 29.4.6 (Normal Curvature)

Let γ be a unit speed curve on Σ with $\gamma(0) = p$. The **normal curvature** of γ at p is

$$\kappa_n = \langle \ddot{\gamma}(0), n(p) \rangle,$$

the component of the acceleration in the normal direction.

Proposition 29.4.7

The normal curvature of a unit speed curve γ on Σ at p depends only on the tangent vector $\dot{\gamma}(0)$, and

$$\kappa_n = \Pi_p(\dot{\gamma}(0), \dot{\gamma}(0)).$$

Proof. Along γ we have $\langle n(\gamma(t)), \dot{\gamma}(t) \rangle = 0$. Differentiating at $t = 0$,

$$\langle Dn|_p(\dot{\gamma}(0)), \dot{\gamma}(0) \rangle + \langle n(p), \ddot{\gamma}(0) \rangle = 0,$$

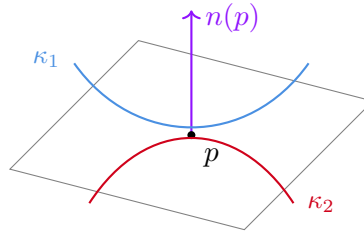
so $\kappa_n = \langle S_p(\dot{\gamma}(0)), \dot{\gamma}(0) \rangle = \Pi_p(\dot{\gamma}(0), \dot{\gamma}(0))$, which depends only on $\dot{\gamma}(0)$. $\square$

This is sometimes called Meusnier's theorem. It says that all curves on Σ through p with the same tangent direction bend away from the tangent plane at the same rate; in particular we may as well compute with the most symmetric such curve.

Definition 29.4.8 (Normal Section)

Given a unit vector $w \in T_p\Sigma$, let Π be the plane through p spanned by w and $n(p)$. Then $\Pi \cap \Sigma$ is, near p , a smooth curve (the two are transversal), called the **normal section** of Σ at p in the direction w .

A normal section is a plane curve lying in Π , and its principal normal at p is $\pm n(p)$; so its signed curvature there is exactly $\Pi_p(w, w)$. Normal curvature is genuinely the curvature of the slice you get by cutting the surface with a plane containing the normal.



Since S_p is self-adjoint, the spectral theorem applies.

Definition 29.4.9 (Principal Curvatures)

The eigenvalues $\kappa_1 \geq \kappa_2$ of the shape operator S_p are the **principal curvatures** of Σ at p , and the corresponding unit eigenvectors are the **principal directions**. A point where $\kappa_1 = \kappa_2$ is called **umbilic**.

Corollary 29.4.10

There is an orthonormal basis $\{e_1, e_2\}$ of $T_p\Sigma$ with $S_p(e_i) = \kappa_i e_i$, and κ_1, κ_2 are the maximum and minimum values of the normal curvature over all unit tangent directions.

Proof. The first statement is the spectral theorem for a self-adjoint operator on a two-dimensional inner product space. For the second, write a unit vector as $w = \cos \theta e_1 + \sin \theta e_2$; then

$$\Pi_p(w, w) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta,$$

which is a convex combination of κ_1 and κ_2 , attaining each at $\theta = 0, \pi/2$. $\square$

The formula in that proof is *Euler's theorem*, and it is worth remembering: the normal curvature in the direction making angle θ with the first principal direction is $\kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta$.

Definition 29.4.11 (Gaussian and Mean Curvature)

Let Σ be a smooth surface in $\mathbb{R}^3$. The **Gaussian curvature** and **mean curvature** of Σ at p are

$$K(p) = \det S_p = \det (Dn|_p) = \kappa_1 \kappa_2, \quad H(p) = \frac{1}{2} \operatorname{tr} S_p = \frac{1}{2}(\kappa_1 + \kappa_2).$$

Note that reversing the orientation replaces n by $-n$, hence S_p by $-S_p$; so κ_1, κ_2 and H change sign, but K does not. In particular K is well defined even on a non-orientable surface, since every surface is locally orientable. Both are smooth functions of p , being polynomial expressions in the derivatives of a smooth parametrisation.

Remark. The principal curvatures are the roots of $t^2 - 2Ht + K = 0$, so $\kappa_{1,2} = H \pm \sqrt{H^2 - K}$ and in particular $H^2 \geq K$ always, with equality exactly at umbilic points.

§29.4.4 Curvature in Local Coordinates

We can now compute K and H from the two fundamental forms. This is the formula that makes everything computable.

Proposition 29.4.12

In any allowable parametrisation,

$$K = \frac{LN - M^2}{EG - F^2}, \quad H = \frac{LG - 2MF + NE}{2(EG - F^2)}.$$

Proof. Write the matrix of S_p in the basis $\{\sigma_u, \sigma_v\}$ as $A = (a_{ij})$, so that

$$-n_u = a_{11}\sigma_u + a_{21}\sigma_v, \quad -n_v = a_{12}\sigma_u + a_{22}\sigma_v.$$

Taking the inner product of each with σ_u and with σ_v , and using the identities $-\langle n_u, \sigma_u \rangle = L$ and so on, we obtain the matrix identity

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} A,$$

that is, $A = I^{-1}II$. Taking determinants gives

$$K = \det A = \frac{LN - M^2}{EG - F^2},$$

and computing $A = \frac{1}{EG-F^2} \begin{pmatrix} G & -F \\ -F & E \end{pmatrix} \begin{pmatrix} L & M \\ M & N \end{pmatrix}$ explicitly and taking half the trace gives the stated H . $\square$

Example 29.4.13 (The Sphere)

For the unit sphere with outward normal, $n(p) = p$, so $Dn|_p = \operatorname{id}$ and $S_p = -\operatorname{id}$. Hence $\kappa_1 = \kappa_2 = -1$, $K = 1$ and $H = -1$. Every point is umbilic. With the inward normal we would get $\kappa_i = +1$; either way $K = 1$. For the sphere of radius a the same computation gives $K = 1/a^2$: big spheres are less curved.

Example 29.4.14 (The Cylinder)

For the cylinder $\sigma(u, v) = (a \cos u, a \sin u, v)$ we found $E = a^2$, $F = 0$, $G = 1$. Also $\sigma_{uv} = \sigma_{vu} = 0$ and $\sigma_{uu} = (-a \cos u, -a \sin u, 0)$, while $n = (\cos u, \sin u, 0)$ is the outward normal, so

$$L = -a, \quad M = N = 0.$$

Therefore $K = 0$ and $H = -1/(2a)$. The principal curvatures are $-1/a$ (around the cylinder) and 0 (along it). So the cylinder is *flat* – consistent with the fact that it is locally isometric to the plane, as we saw, and a first hint of the Theorema Egregium.

Example 29.4.15 (The Torus)

For the torus $\sigma(u, v) = ((a + r \cos u) \cos v, (a + r \cos u) \sin v, r \sin u)$ we have $E = r^2$, $F = 0$, $G = (a + r \cos u)^2$, and one computes

$$L = r, \quad M = 0, \quad N = (a + r \cos u) \cos u,$$

so

$$K = \frac{r(a + r \cos u) \cos u}{r^2(a + r \cos u)^2} = \frac{\cos u}{r(a + r \cos u)}.$$

Thus the outer half of the torus ($|u| < \pi/2$) has $K > 0$, the inner half has $K < 0$, and the top and bottom circles $u = \pm\pi/2$ have $K = 0$. This matches the intuition that the outside of a doughnut is dome-like and the inside is saddle-like.

Example 29.4.16 (Surfaces of Revolution)

Let $\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u))$ with unit speed profile, so $(f')^2 + (g')^2 = 1$ and $I = du^2 + f^2 dv^2$. A short computation gives

$$L = f'g'' - f''g', \quad M = 0, \quad N = fg',$$

and hence

$$K = \frac{(f'g'' - f''g')g'}{f}.$$

Differentiating $(f')^2 + (g')^2 = 1$ gives $f'f'' + g'g'' = 0$, from which $(f'g'' - f''g')g' = -f''$, and so the memorable formula

$$K = -\frac{f''}{f}.$$

§29.4.5 Elliptic, Hyperbolic and Parabolic Points

The sign of the Gaussian curvature controls the local shape of the surface, via the Taylor expansion we used to motivate the second fundamental form.

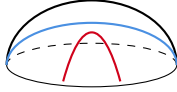
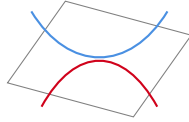
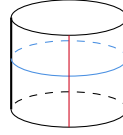
Definition 29.4.17

A smooth surface in $\mathbb{R}^3$ with $K \equiv 0$ is called **flat**.

Definition 29.4.18

Let Σ be a smooth surface in $\mathbb{R}^3$, and let $p \in \Sigma$. We say p is

- (i) **elliptic** if $K(p) > 0$;
- (ii) **hyperbolic** if $K(p) < 0$;
- (iii) **parabolic** if $K(p) = 0$ but $S_p \neq 0$;
- (iv) **planar** if $S_p = 0$.

elliptic: $K > 0$ hyperbolic: $K < 0$ parabolic: $K = 0$ **Proposition 29.4.19**

Let $p \in \Sigma$ be an elliptic point. Then there is a neighbourhood of p in Σ lying strictly on one side of the affine tangent plane $p + T_p\Sigma$ (apart from p itself). If p is hyperbolic, then every neighbourhood of p contains points on both sides.

Proof. Take an allowable parametrisation with $\sigma(0,0) = p$. From the Taylor expansion at the start of this section, the signed distance from $\sigma(h,\ell)$ to the affine tangent plane is

$$\langle \sigma(h,\ell) - p, n(p) \rangle = \frac{1}{2} (Lh^2 + 2Mh\ell + N\ell^2) + O((h^2 + \ell^2)^{3/2}),$$

that is, $\frac{1}{2}\Pi_p(w, w) + o(\|w\|^2)$ where $w = h\sigma_u + \ell\sigma_v$.

If $K(p) > 0$ then $LN - M^2 > 0$, so the quadratic form Π_p is definite, and hence $|\Pi_p(w, w)| \geq c\|w\|^2$ for some $c > 0$ by compactness of the unit circle. The error term is $o(\|w\|^2)$, so for $\|w\|$ small the sign of the whole expression is that of Π_p , which is constant.

If $K(p) < 0$ then Π_p is indefinite, so there are unit vectors $w_{\pm}$ with $\Pi_p(w_+, w_+) > 0 > \Pi_p(w_-, w_-)$, and the same estimate applied along the two directions produces points on both sides arbitrarily close to p . $\square$

Example 29.4.20 (The Monkey Saddle)

Let Σ be the graph of $z = x^3 - 3xy^2$. At the origin all first and second derivatives of the height function vanish, so $\Pi_0 = 0$ and the origin is a planar point. Writing $x = r \cos \theta$, $y = r \sin \theta$ gives $z = r^3 \cos 3\theta$, so the surface crosses its tangent plane six times around the origin – a saddle with room for two legs and a tail.

§29.4.6 Curvature as Area Distortion

The determinant of a linear map is the factor by which it scales areas, and the Gauss map's derivative is a map between two-dimensional spaces. That gives the following interpretation of K , which is essentially Gauss's original definition.

Theorem 29.4.21

Let Σ be a smooth surface in $\mathbb{R}^3$ and $p \in \Sigma$ with $K(p) \neq 0$. Let (A_i) be a decreasing sequence of neighbourhoods of p shrinking to p , in the sense that for every $\varepsilon > 0$ we have $A_i \subseteq B(p, \varepsilon)$ for all large i . Then

$$|K(p)| = \lim_{i \rightarrow \infty} \frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)}.$$

Proof. Fix an allowable parametrisation σ near p . The area element of Σ is $\|\sigma_u \times \sigma_v\|$, and the composite $n \circ \sigma$ parametrises the image on the sphere, with area element $\|n_u \times n_v\|$, both with respect to $du dv$. But $n_u = -S(\sigma_u)$ and $n_v = -S(\sigma_v)$, so

$$n_u \times n_v = \det(S) \sigma_u \times \sigma_v = K \sigma_u \times \sigma_v.$$

Hence

$$\frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)} = \frac{\int_{\sigma^{-1}(A_i)} |K| \|\sigma_u \times \sigma_v\|}{\int_{\sigma^{-1}(A_i)} \|\sigma_u \times \sigma_v\|},$$

a weighted average of $|K|$ over a region shrinking to p . Since K is continuous and $K(p) \neq 0$ (so n is a local diffeomorphism near p and the areas above are genuinely areas of the image), this tends to $|K(p)|$. $\square$

Informally, then, the Gaussian curvature is an infinitesimal measure of how much the Gauss map distorts area. On a sphere of radius a a patch of area A maps to a patch of area A/a^2 ; on a cylinder every patch maps into a single curve, of area zero, which is why $K = 0$.

§29.4.7 Compact Surfaces Have an Elliptic Point

Here is a first genuinely global statement, and a good illustration of a technique we will use again: put the surface inside the smallest sphere you can and look at where they touch.

Theorem 29.4.22

Every compact smooth surface $\Sigma \subseteq \mathbb{R}^3$ has an elliptic point. In particular, no compact surface in $\mathbb{R}^3$ has $K \leq 0$ everywhere.

Proof. The function $p \mapsto \|p\|^2$ is continuous on the compact set Σ , so it attains its maximum at some p_0 , at distance $R = \|p_0\|$ from the origin. Then Σ is contained in the closed ball of radius R and touches the sphere S_R of radius R at p_0 .

Let γ be any unit speed curve in Σ with $\gamma(0) = p_0$. The function $h(t) = \|\gamma(t)\|^2$ has a maximum at $t = 0$, so $\dot{h}(0) = 2\langle \gamma(0), \dot{\gamma}(0) \rangle = 0$ – which says p_0 is orthogonal to $T_{p_0}\Sigma$, so the unit normal there is $n(p_0) = \pm p_0/R$; choose the orientation so that $n(p_0) = -p_0/R$ (the inward normal). Also

$$0 \geq \frac{1}{2}\ddot{h}(0) = \langle \dot{\gamma}(0), \dot{\gamma}(0) \rangle + \langle \gamma(0), \ddot{\gamma}(0) \rangle = 1 + \langle p_0, \ddot{\gamma}(0) \rangle.$$

Now the normal curvature of γ at p_0 is $\kappa_n = \langle \ddot{\gamma}(0), n(p_0) \rangle = -\langle \ddot{\gamma}(0), p_0 \rangle/R$, so the

displayed inequality reads $R\kappa_n \geq 1$, i.e.

$$\kappa_n \geq \frac{1}{R}$$

for every unit tangent direction at p_0 . In particular both principal curvatures at p_0 are at least $1/R > 0$, so

$$K(p_0) = \kappa_1\kappa_2 \geq \frac{1}{R^2} > 0,$$

and p_0 is elliptic. □

Remark. This is the germ of several deeper global theorems. For instance, a compact surface with K constant must have $K > 0$ by the above, and one can then show it is a sphere; and Hilbert's theorem states that there is no complete surface in $\mathbb{R}^3$ with constant negative curvature, so the hyperbolic plane of Section 10 can never be realised as a surface in $\mathbb{R}^3$. We will not prove either here.

§29.5 The Theorema Egregium

The Gaussian curvature was defined using the Gauss map, which needs a normal vector, which needs an ambient $\mathbb{R}^3$. So on the face of it K measures how the surface sits inside space. Gauss's remarkable theorem – *theorema egregium* really does mean ‘remarkable theorem’, and he named it himself – says that this appearance is deceptive: K can be computed from the first fundamental form alone, and so is unchanged by any isometry.

The strategy is exactly the one that worked for curves. There we differentiated the Frenet frame and expressed the answer back in the frame; here we differentiate the frame $\{\sigma_u, \sigma_v, n\}$ and express the answer back in that frame. The coefficients which appear are the Christoffel symbols, and the compatibility conditions coming from equality of mixed partials are the Gauss and Codazzi–Mainardi equations.

§29.5.1 Christoffel Symbols

Let $\sigma : V \rightarrow U \subseteq \Sigma$ be an allowable parametrisation. At each point $\{\sigma_u, \sigma_v, n\}$ is a basis of $\mathbb{R}^3$, so we may expand the second derivatives of σ in it.

Definition 29.5.1 (Christoffel Symbols)

The **Christoffel symbols** of Σ in the parametrisation σ are the functions Γ_{ij}^k defined by

$$\begin{aligned}\sigma_{uu} &= \Gamma_{11}^1\sigma_u + \Gamma_{11}^2\sigma_v + Ln, \\ \sigma_{uv} &= \Gamma_{12}^1\sigma_u + \Gamma_{12}^2\sigma_v + Mn, \\ \sigma_{vv} &= \Gamma_{22}^1\sigma_u + \Gamma_{22}^2\sigma_v + Nn.\end{aligned}$$

The normal components are the coefficients of the second fundamental form, by definition. Note that $\sigma_{uv} = \sigma_{vu}$, so the symbols are symmetric in their lower indices, $\Gamma_{12}^k = \Gamma_{21}^k$: there are six of them in all.

The whole point is that the Christoffel symbols are *intrinsic*.

Lemma 29.5.2

The Christoffel symbols are determined by E, F, G and their first partial derivatives.

Proof. Take the inner product of each of the three defining equations with σ_u and with σ_v ; the normal terms drop out. Using

$$\begin{aligned}\langle \sigma_{uu}, \sigma_u \rangle &= \frac{1}{2}E_u, & \langle \sigma_{uu}, \sigma_v \rangle &= F_u - \frac{1}{2}E_v, & \langle \sigma_{uv}, \sigma_u \rangle &= \frac{1}{2}E_v, \\ \langle \sigma_{uv}, \sigma_v \rangle &= \frac{1}{2}G_u, & \langle \sigma_{vv}, \sigma_u \rangle &= F_v - \frac{1}{2}G_u, & \langle \sigma_{vv}, \sigma_v \rangle &= \frac{1}{2}G_v,\end{aligned}$$

each of which follows by differentiating the definitions of E, F, G , we obtain three linear systems

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma_{11}^1 \\ \Gamma_{11}^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}E_u \\ F_u - \frac{1}{2}E_v \end{pmatrix}, \quad \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma_{12}^1 \\ \Gamma_{12}^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}E_v \\ \frac{1}{2}G_u \end{pmatrix},$$

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma_{22}^1 \\ \Gamma_{22}^2 \end{pmatrix} = \begin{pmatrix} F_v - \frac{1}{2}G_u \\ \frac{1}{2}G_v \end{pmatrix}.$$

Since $EG - F^2 > 0$ the matrix is invertible, and Cramer's rule solves each system in terms of E, F, G and their first derivatives. $\square$

Example 29.5.3 (Orthogonal Parametrisations)

When $F = 0$ the systems above are diagonal and we can read off everything at once:

$$\Gamma_{11}^1 = \frac{E_u}{2E}, \quad \Gamma_{11}^2 = -\frac{E_v}{2G}, \quad \Gamma_{12}^1 = \frac{E_v}{2E}, \quad \Gamma_{12}^2 = \frac{G_u}{2G}, \quad \Gamma_{22}^1 = -\frac{G_u}{2E}, \quad \Gamma_{22}^2 = \frac{G_v}{2G}.$$

§29.5.2 The Gauss and Codazzi–Mainardi Equations

We have written down $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ and (from the shape operator) n_u, n_v in terms of the frame. The only constraint left to impose is that mixed partial derivatives commute, and this is where all the content is.

Theorem 29.5.4 (Gauss Equation)

In any allowable parametrisation,

$$EK = (\Gamma_{11}^2)_v - (\Gamma_{12}^2)_u + \Gamma_{11}^1 \Gamma_{12}^2 - \Gamma_{12}^1 \Gamma_{11}^2 + \Gamma_{11}^2 \Gamma_{22}^2 - (\Gamma_{12}^2)^2.$$

Proof. Write $n_u = a_{11}\sigma_u + a_{21}\sigma_v$ and $n_v = a_{12}\sigma_u + a_{22}\sigma_v$; from the previous section the matrix (a_{ij}) of $Dn|_p = -S_p$ is $-\mathbf{I}^{-1}\mathbf{II}$, so

$$a_{21} = \frac{FL - EM}{EG - F^2}, \quad a_{22} = \frac{FM - EN}{EG - F^2}.$$

Now differentiate the first Christoffel identity with respect to v and the second with respect to u :

$$\sigma_{uuv} = (\Gamma_{11}^1)_v \sigma_u + \Gamma_{11}^1 \sigma_{uv} + (\Gamma_{11}^2)_v \sigma_v + \Gamma_{11}^2 \sigma_{vv} + L_v n + L n_v,$$

$$\sigma_{uvu} = (\Gamma_{12}^1)_u \sigma_u + \Gamma_{12}^1 \sigma_{uu} + (\Gamma_{12}^2)_u \sigma_v + \Gamma_{12}^2 \sigma_{uv} + M_u n + M n_u.$$

These are equal. Expanding $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}, n_u, n_v$ in the frame and comparing the coefficients of σ_v (which is legitimate since the frame is a basis) gives

$$(\Gamma_{11}^2)_v + \Gamma_{11}^1 \Gamma_{12}^2 + \Gamma_{11}^2 \Gamma_{22}^2 + L a_{22} = (\Gamma_{12}^2)_u + \Gamma_{12}^1 \Gamma_{11}^2 + (\Gamma_{12}^2)^2 + M a_{21}.$$

Finally

$$M a_{21} - L a_{22} = \frac{M(FL - EM) - L(FM - EN)}{EG - F^2} = \frac{E(LN - M^2)}{EG - F^2} = EK,$$

and rearranging gives the claim. $\square$

Theorem 29.5.5 (Theorema Egregium)

The Gaussian curvature of a smooth surface in $\mathbb{R}^3$ depends only on the first fundamental form. Consequently, if $f : \Sigma \rightarrow \Sigma'$ is an isometry then $K' \circ f = K$; that is, Gaussian curvature is an isometry invariant.

Proof. The right hand side of the Gauss equation involves only Christoffel symbols and their derivatives, hence only E, F, G and their derivatives up to second order. Since $E = \langle \sigma_u, \sigma_u \rangle > 0$ we may divide, and K is expressed intrinsically.

For the second statement, if f is an isometry and σ is an allowable parametrisation of Σ , then $f \circ \sigma$ is an allowable parametrisation of Σ' with the same first fundamental form, so the two curvature computations produce the same function. $\square$

Comparing the coefficients of n instead of σ_v , and doing the same with $\sigma_{uvv} = \sigma_{vvu}$, gives the remaining compatibility conditions.

Theorem 29.5.6 (Codazzi–Mainardi Equations)

In any allowable parametrisation,

$$\begin{aligned} L_v - M_u &= L\Gamma_{12}^1 + M(\Gamma_{12}^2 - \Gamma_{11}^1) - N\Gamma_{11}^2, \\ M_v - N_u &= L\Gamma_{22}^1 + M(\Gamma_{22}^2 - \Gamma_{12}^1) - N\Gamma_{12}^2. \end{aligned}$$

Proof. Compare the coefficients of n in the two expansions of $\sigma_{uvv} = \sigma_{vvu}$ displayed above: this gives $\Gamma_{11}^1 M + \Gamma_{11}^2 N + L_v = \Gamma_{12}^1 L + \Gamma_{12}^2 M + M_u$, which rearranges to the first equation. Expanding $\sigma_{uvv} = \sigma_{vvu}$ in the same way and comparing n coefficients gives $\Gamma_{12}^1 M + \Gamma_{12}^2 N + M_v = \Gamma_{22}^1 L + \Gamma_{22}^2 M + N_u$, which is the second. $\square$

Remark. The Gauss and Codazzi–Mainardi equations are precisely the integrability conditions for the system defining σ , and one can prove a converse: given smooth functions E, F, G (with $E, G, EG - F^2 > 0$) and L, M, N on a simply connected open $V \subseteq \mathbb{R}^2$ satisfying these equations, there is a smooth surface, unique up to a rigid motion of $\mathbb{R}^3$, with these as its fundamental forms. This is the *fundamental theorem of surface theory*, the exact analogue of the fundamental theorem of curve theory from Section 2. We will not prove it.

§29.5.3 Useful Special Cases

In practice one almost always arranges $F = 0$, and then the Gauss equation collapses to something usable.

Corollary 29.5.7 (Orthogonal Gauss Formula)

If σ is an orthogonal parametrisation ($F \equiv 0$) then

$$K = -\frac{1}{2\sqrt{EG}} \left[\left(\frac{E_v}{\sqrt{EG}} \right)_v + \left(\frac{G_u}{\sqrt{EG}} \right)_u \right].$$

Proof. Substitute the orthogonal Christoffel symbols computed above into the Gauss equation and simplify; it is a slightly tedious but entirely mechanical calculation. $\square$

Corollary 29.5.8

If $I = du^2 + G(u, v) dv^2$, then with $h = \sqrt{G}$,

$$K = -\frac{h_{uu}}{h}.$$

Proof. Put $E = 1$ in the orthogonal Gauss formula: the first bracket vanishes and $G_u/\sqrt{G} = 2h_u$, so $K = -(2h_u)_u/(2h) = -h_{uu}/h$. $\square$

Corollary 29.5.9 (Isothermal Case)

If σ is isothermal, so $E = G = \lambda^2$ and $F = 0$, then

$$K = -\frac{1}{\lambda^2} \nabla^2 (\log \lambda),$$

where $\nabla^2 = \partial_u^2 + \partial_v^2$ is the flat Laplacian.

Proof. Here $\sqrt{EG} = \lambda^2$, so $E_v/\sqrt{EG} = 2\lambda_v/\lambda = 2(\log \lambda)_v$ and likewise $G_u/\sqrt{EG} = 2(\log \lambda)_u$. Substituting into the orthogonal Gauss formula gives the result. $\square$

Example 29.5.10 (The Sphere, Intrinsically)

The unit sphere has $I = du^2 + \cos^2 u dv^2$ in latitude-longitude coordinates. So $h = \cos u$ and

$$K = -\frac{h_{uu}}{h} = -\frac{-\cos u}{\cos u} = 1,$$

with not a single normal vector in sight.

Example 29.5.11 (Surfaces of Revolution, Intrinsically)

A surface of revolution with unit speed profile has $I = du^2 + f(u)^2 dv^2$, so immediately $K = -f''/f$, recovering the formula of the previous section by a purely intrinsic route.

Example 29.5.12 (Why the Sphere Cannot be Flattened)

A piece of the plane has $K \equiv 0$ and a piece of the unit sphere has $K \equiv 1$. By the Theorema Egregium no open subset of the sphere is isometric to an open subset of the plane. Every flat map of the world lies about distances – and now we know it is not the cartographers' fault.

Example 29.5.13 (The Cylinder Again)

The cylinder has $I = a^2 du^2 + dv^2$, with constant coefficients, so $K \equiv 0$: the cylinder is intrinsically flat, exactly as its local isometry with the plane demands. Note that the *mean* curvature does not have this property: the plane has $H \equiv 0$ and the cylinder has $H = -1/2a \neq 0$, so H is emphatically not intrinsic.

§29.6 Geodesics

§29.6.1 Length, Energy and Variations

We are now going to discuss a particular geometric phenomenon, *geodesics*, which are curves that are (in some sense) the shortest path between points.

Fix a smooth surface Σ and points $p, q \in \Sigma$, and let $\Omega(p, q)$ denote the set of smooth curves $\gamma : [a, b] \rightarrow \Sigma$ with $\gamma(a) = p$ and $\gamma(b) = q$. We would like to find the critical points of the length functional L on $\Omega(p, q)$. The trouble is that L has a square root in it, which makes differentiating it unpleasant, and it is also insensitive to reparametrisation, so its critical points are never isolated. Both problems are fixed by working with a different functional.

Definition 29.6.1 (Energy)

The **energy** of a curve $\gamma : [a, b] \rightarrow \mathbb{R}^3$ is given by

$$\mathcal{E}(\gamma) = \int_a^b \|\gamma'(t)\|^2 dt.$$

We write $\mathcal{E}$ rather than E to avoid a clash with the first fundamental form. Energy is sensitive to reparametrisation, but the two functionals are related by Cauchy–Schwarz.

Lemma 29.6.2

For any $\gamma \in \Omega(p, q)$ we have $L(\gamma)^2 \leq (b - a)\mathcal{E}(\gamma)$, with equality if and only if γ has constant speed.

Proof. Apply Cauchy–Schwarz to $\|\gamma'\|$ and the constant function 1 on $[a, b]$:

$$L(\gamma)^2 = \left(\int_a^b \|\gamma'\| \cdot 1 dt \right)^2 \leq (b - a) \int_a^b \|\gamma'\|^2 dt = (b - a)\mathcal{E}(\gamma),$$

with equality exactly when $\|\gamma'\|$ is a constant multiple of 1. $\square$

Since every curve can be reparametrised to have constant speed without changing its

length, minimising L is the same problem as minimising $\mathcal{E}$ over constant speed curves, and the latter is the more tractable one.

Definition 29.6.3 (One-Parameter Variation)

Let $\gamma : [a, b] \rightarrow \Sigma$, where Σ is a smooth surface in $\mathbb{R}^3$. A **one-parameter variation** (with fixed endpoints) of γ is a smooth map $\Gamma : (-\varepsilon, \varepsilon) \times [a, b] \rightarrow \Sigma$ such that, writing $\gamma_s = \Gamma(s, \cdot)$, we have $\gamma_0 = \gamma$ and $\gamma_s(a), \gamma_s(b)$ are independent of s .

Definition 29.6.4 (Geodesic)

A smooth curve $\gamma : [a, b] \rightarrow \Sigma$ is a **geodesic** if, for every variation (γ_s) of γ with fixed endpoints, we have $\left. \frac{d}{ds} \right|_{s=0} \mathcal{E}(\gamma_s) = 0$; that is, if γ is a critical point of the energy functional on $\Omega(\gamma(a), \gamma(b))$.

§29.6.2 The Geodesic Equations

So how do we actually find geodesics? It turns out to be *relatively* easy, using the geodesic equations.

Suppose we have some curve γ with image contained in the image of an allowable parametrisation σ . Then for sufficiently small s we can write $\gamma_s(t) = \sigma(u(s, t), v(s, t))$. Let the first fundamental form with respect to σ be

$$E du^2 + 2F du dv + G dv^2.$$

Then by definition,

$$\mathcal{E}(\gamma_s) = \int_a^b (E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2) dt,$$

where the dots are $\partial/\partial t$ and E, F, G are evaluated at $(u(s, t), v(s, t))$. Differentiating under the integral sign and integrating by parts – the boundary terms vanish because the endpoints are fixed – we get

$$\left. \frac{d}{ds} \mathcal{E}(\gamma_s) \right|_{s=0} = \int_a^b \left(A \frac{\partial u}{\partial s} + B \frac{\partial v}{\partial s} \right) dt \Big|_{s=0}$$

where

$$A = E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2 - 2 \frac{d}{dt} (E\dot{u} + F\dot{v}),$$

$$B = E_v \dot{u}^2 + 2F_v \dot{u}\dot{v} + G_v \dot{v}^2 - 2 \frac{d}{dt} (F\dot{u} + G\dot{v}).$$

Since $\partial u/\partial s$ and $\partial v/\partial s$ at $s = 0$ can be prescribed to be any pair of smooth functions vanishing at a and b , the integral vanishes for all variations exactly when $A = B = 0$. These give us the *geodesic equations*.

Theorem 29.6.5 (Geodesic Equations)

A smooth curve $\gamma(t) = \sigma(u(t), v(t))$ is a geodesic if and only if it satisfies the geodesic equations

$$\frac{d}{dt} (E\dot{u} + F\dot{v}) = \frac{E_u \dot{u}^2 + 2F_u \dot{u}\dot{v} + G_u \dot{v}^2}{2},$$

$$\frac{d}{dt}(F\dot{u} + G\dot{v}) = \frac{E_v\dot{u}^2 + 2F_v\dot{u}\dot{v} + G_v\dot{v}^2}{2}.$$

Unlike length, energy is sensitive to reparameterisation, but Cauchy–Schwarz gives us that $L(\gamma)^2 \leq \mathcal{E}(\gamma)(b-a)$, with equality if and only if γ is parametrised proportional to arc length.

Corollary 29.6.6

If γ has constant speed and locally minimises length, then it is a geodesic. Further, if γ minimises energy globally, then it minimises length globally and is parametrised with constant speed.

Expanding the derivatives on the left and using the formulae for the Christoffel symbols, the geodesic equations take an equivalent and often more convenient form.

Corollary 29.6.7

A curve $\gamma(t) = \sigma(u(t), v(t))$ is a geodesic if and only if

$$\begin{aligned}\ddot{u} + \Gamma_{11}^1\dot{u}^2 + 2\Gamma_{12}^1\dot{u}\dot{v} + \Gamma_{22}^1\dot{v}^2 &= 0, \\ \ddot{v} + \Gamma_{11}^2\dot{u}^2 + 2\Gamma_{12}^2\dot{u}\dot{v} + \Gamma_{22}^2\dot{v}^2 &= 0.\end{aligned}$$

In this form the geodesic equations are visibly a second order system of ODEs, with smooth coefficients depending only on the first fundamental form. Picard’s theorem therefore gives us existence, uniqueness and smooth dependence on initial data all at once.

Proposition 29.6.8 (Local Existence and Uniqueness of Geodesics)

Let Σ be a smooth surface, $p \in \Sigma$ and $w \in T_p\Sigma$. Then there is $\varepsilon > 0$ and a unique geodesic $\gamma : (-\varepsilon, \varepsilon) \rightarrow \Sigma$ with $\gamma(0) = p$ and $\dot{\gamma}(0) = w$. Moreover γ depends smoothly on (p, w) .

Proof. Choose an allowable parametrisation about p . In it the geodesic equations are a second order ODE system $\ddot{x} = h(x, \dot{x})$ with h smooth, which is equivalent to a first order system in $(x, \dot{x})$ with smooth right hand side. Picard’s theorem gives a unique local solution with any prescribed initial position and velocity, and the standard theory gives smooth dependence on those initial conditions. $\square$

Note also the important fact that geodesics automatically have constant speed. This will follow immediately from the characterisation in the next subsection.

§29.6.3 Geodesics Are Straightest

In $\mathbb{R}^2$, the geodesics are not just locally shortest, they are also locally *straightest*. We would intuitively expect this to hold on other surfaces too: the change in the tangent vector should be as small as possible subject to the constraint that we lie on the surface. That is, the acceleration should be entirely used up in staying on the surface.

Theorem 29.6.9

Let Σ be a smooth surface in $\mathbb{R}^3$. A smooth curve $\gamma : [a, b] \rightarrow \Sigma$ is a geodesic if and only if $\ddot{\gamma}(t)$ is everywhere normal to the surface Σ , that is, $\ddot{\gamma}(t) \perp T_{\gamma(t)}\Sigma$ for all t .

Proof. Let Γ be a variation of γ with fixed endpoints and put $W(t) = \frac{\partial \Gamma}{\partial s} \Big|_{s=0}$, a vector field along γ tangent to Σ and vanishing at the endpoints. Differentiating under the integral sign,

$$\frac{d}{ds} \Big|_{s=0} \mathcal{E}(\gamma_s) = \int_a^b 2 \left\langle \frac{\partial^2 \Gamma}{\partial s \partial t}, \frac{\partial \Gamma}{\partial t} \right\rangle_{s=0} dt = 2 \int_a^b \langle \dot{W}, \dot{\gamma} \rangle dt = -2 \int_a^b \langle W, \ddot{\gamma} \rangle dt,$$

integrating by parts and using $W(a) = W(b) = 0$.

If $\ddot{\gamma} \perp T_{\gamma}\Sigma$ everywhere then the integrand vanishes for every variation, so γ is a geodesic. Conversely, suppose γ is a geodesic but $\ddot{\gamma}(t_0)$ has a non-zero tangential part Z for some t_0 . Choose a smooth bump function $\chi \geq 0$ supported in a small neighbourhood of t_0 with $\chi(t_0) = 1$, and let W be the tangential vector field along γ obtained by extending χZ (in a parametrisation, take $W = \chi(t)(\alpha\sigma_u + \beta\sigma_v)$ with α, β the constant coordinates of Z). Then W arises from a genuine variation – take $\Gamma(s, t) = \sigma(u(t) + s\alpha\chi(t), v(t) + s\beta\chi(t))$ – and $\int \langle W, \ddot{\gamma} \rangle > 0$ if the support is small enough, contradicting criticality. $\square$

Corollary 29.6.10

Every geodesic has constant speed.

Proof. $\frac{d}{dt} \langle \dot{\gamma}, \dot{\gamma} \rangle = 2 \langle \ddot{\gamma}, \dot{\gamma} \rangle = 0$, since $\ddot{\gamma}$ is normal and $\dot{\gamma}$ is tangent. $\square$

This characterisation makes many examples immediate.

Example 29.6.11 (Geodesics in the Plane)

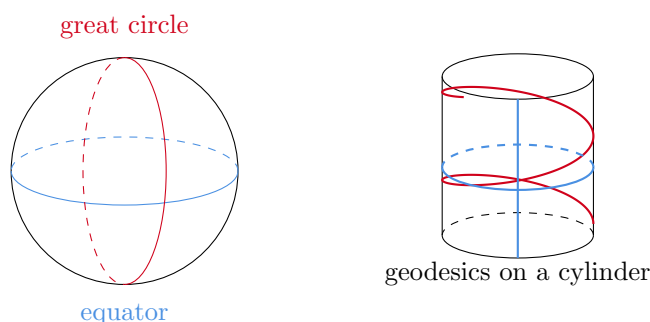
For a curve in a plane, $\ddot{\gamma}$ is normal to the plane if and only if $\ddot{\gamma} = 0$, so the geodesics are exactly the constant speed straight lines.

Example 29.6.12 (Geodesics on the Sphere)

Let γ be a unit speed geodesic on S^2 . The normal at p is p itself, so $\ddot{\gamma} = \lambda\gamma$ for some function λ ; taking the inner product with γ and using $\langle \gamma, \gamma \rangle = 1$, $\langle \dot{\gamma}, \gamma \rangle = 0$ and $\langle \ddot{\gamma}, \gamma \rangle = -1$ gives $\lambda = -1$. So $\ddot{\gamma} = -\gamma$, whose solutions are

$$\gamma(t) = \cos t \gamma(0) + \sin t \dot{\gamma}(0),$$

which traverses the great circle cut out by the plane through the origin spanned by $\gamma(0)$ and $\dot{\gamma}(0)$. So the geodesics on the sphere are exactly the constant speed parametrisations of great circles.



Example 29.6.13 (Geodesics on the Cylinder)

Since the cylinder is locally isometric to the plane and geodesics are determined by the first fundamental form, the geodesics of the cylinder are the images of straight lines under the rolling-up map. These are the vertical rulings, the horizontal circles, and the helices. Alternatively, unroll: on the cylinder $I = a^2 du^2 + dv^2$ has constant coefficients, so the geodesic equations read $\ddot{u} = \ddot{v} = 0$, giving u, v affine in t .

§29.6.4 Planes of Symmetry

There is a useful symmetry principle which lets us find geodesics with no computation at all.

Proposition 29.6.14

Let Σ be a smooth surface in $\mathbb{R}^3$ and let Π be a plane such that reflection in Π maps Σ to itself. Then every constant speed parametrisation of a connected component of $\Sigma \cap \Pi$ is a geodesic.

Proof. Let R denote reflection in Π and let γ be a unit speed parametrisation of a component of $\Sigma \cap \Pi$. Since R is an isometry of $\mathbb{R}^3$ fixing Σ setwise and fixing γ pointwise, R maps Σ to Σ and hence maps $T_{\gamma(t)}\Sigma$ to itself, and $R(\ddot{\gamma}) = (R\ddot{\gamma}) = \ddot{\gamma}$, so $\ddot{\gamma}$ lies in Π .

Now decompose $\ddot{\gamma}$ into its tangential and normal parts. The tangential part is orthogonal to $\dot{\gamma}$ (as γ has unit speed), and it lies in Π ; but $T_{\gamma(t)}\Sigma \cap \Pi$ is the line spanned by $\dot{\gamma}(t)$, because Σ and Π meet transversally along γ . So the tangential part is both parallel and orthogonal to $\dot{\gamma}$, hence zero, and γ is a geodesic. $\square$

§29.6.5 Surfaces of Revolution

One common family of surfaces is the surfaces of revolution, obtained by rotating $\eta(u) = (f(u), 0, g(u))$ – where η is smooth and injective and $f(u) > 0$ – about the z -axis. We will assume throughout that η has unit speed, so that

$$I = du^2 + f(u)^2 dv^2.$$

Definition 29.6.15

A circle obtained by rotating a point of η is called a **parallel**. A curve obtained by

rotating η itself by a fixed angle about the z -axis is a **meridian**.

Every plane in $\mathbb{R}^3$ containing the z -axis is a plane of symmetry of Σ , and meets Σ exactly in a pair of meridians. So the previous proposition gives:

Corollary 29.6.16

All meridians of a surface of revolution, parametrised at constant speed, are geodesics.

Not all parallels are geodesics however – think of a circle of latitude on the sphere near the north pole, which is visibly not straight.

Lemma 29.6.17

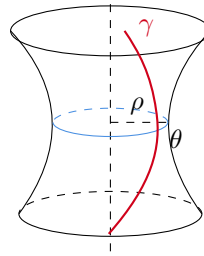
The parallel $u = u_0$, parametrised at constant speed, is a geodesic if and only if $f'(u_0) = 0$.

Proof. With $E = 1$, $F = 0$, $G = f^2$ the geodesic equations become

$$\ddot{u} = f f' \dot{v}^2, \quad \frac{d}{dt} (f^2 \dot{v}) = 0.$$

Along the parallel $u \equiv u_0$ with constant speed, $\dot{v}$ is a non-zero constant, so the second equation holds automatically and the first reduces to $0 = f(u_0) f'(u_0) \dot{v}^2$. As $f > 0$ and $\dot{v} \neq 0$, this holds precisely when $f'(u_0) = 0$: the parallel must sit at a critical radius. $\square$

Now consider a unit speed curve $\gamma(t)$ on Σ , making an angle θ with the parallel through each of its points, and let $\rho = f(u)$ be the radius of that parallel. There is a striking conservation law.



Theorem 29.6.18 (Clairaut's Relation)

If γ is a unit speed geodesic on a surface of revolution, then $\rho \cos \theta$ is constant along γ .

Proof. Write $\gamma(t) = \sigma(u(t), v(t))$, so $\dot{\gamma} = \sigma_u \dot{u} + \sigma_v \dot{v}$. The tangent to the parallel is σ_v , of length $\sqrt{G} = f = \rho$. Since $F = 0$ and $\|\dot{\gamma}\| = 1$,

$$\cos \theta = \frac{\langle \sigma_v, \dot{\gamma} \rangle}{\|\sigma_v\| \|\dot{\gamma}\|} = \frac{G \dot{v}}{\rho} = \rho \dot{v}.$$

So $\rho \cos \theta = \rho^2 \dot{v} = f^2 \dot{v}$, and the second geodesic equation says precisely that this is constant. $\square$

Example 29.6.19 (Trapped Geodesics)

Let γ be a geodesic on an ellipsoid of revolution which is not a meridian, so that $c = \rho \cos \theta > 0$. Since $|\cos \theta| \leq 1$ we always have $\rho \geq c$, so the geodesic can never enter the polar caps where the parallels are narrower than c : it is trapped in a band between two parallels, oscillating between them forever. In particular, any geodesic passing through a pole must have $c = 0$ and hence be a meridian.

§29.6.6 The Covariant Derivative and Parallel Transport

The characterisation ‘ $\ddot{\gamma}$ is normal’ can be phrased more elegantly. The obstruction to $\dot{\gamma}$ being constant is entirely in the tangential direction, so let us give that tangential part a name.

Definition 29.6.20 (Vector Field Along a Curve)

Let $\gamma : I \rightarrow \Sigma$ be a smooth curve. A **vector field along γ** is a smooth map $W : I \rightarrow \mathbb{R}^3$ with $W(t) \in T_{\gamma(t)}\Sigma$ for every t .

Definition 29.6.21 (Covariant Derivative)

The **covariant derivative** of a vector field W along γ is

$$\frac{DW}{dt} = \pi_{T_{\gamma(t)}\Sigma} \left(\frac{dW}{dt} \right),$$

the orthogonal projection of dW/dt onto the tangent plane. We say W is **parallel** along γ if $DW/dt \equiv 0$.

With this language, γ is a geodesic precisely when $D\dot{\gamma}/dt = 0$: a geodesic is a curve whose velocity field is parallel along it. In the plane a vector field is parallel exactly when it is constant, so this really is the right generalisation of ‘moving in a straight line at constant speed’.

Proposition 29.6.22

If V, W are parallel vector fields along γ then $\langle V(t), W(t) \rangle$ is constant. In particular parallel vector fields have constant length, and the angle between two of them is constant.

Proof. $\frac{d}{dt} \langle V, W \rangle = \langle \dot{V}, W \rangle + \langle V, \dot{W} \rangle$. Now $\dot{V}$ differs from $DV/dt = 0$ by a normal vector, and W is tangent, so $\langle \dot{V}, W \rangle = 0$; similarly for the other term. $\square$

The covariant derivative looks as though it depends on the embedding – we projected onto a tangent plane sitting in $\mathbb{R}^3$. It does not.

Lemma 29.6.23

The covariant derivative depends only on the first fundamental form. Explicitly, if

$\gamma(t) = \sigma(u(t), v(t))$ and $W = a\sigma_u + b\sigma_v$, then

$$\begin{aligned} \frac{DW}{dt} &= (\dot{a} + \Gamma_{11}^1 a\dot{u} + \Gamma_{12}^1 (a\dot{v} + b\dot{u}) + \Gamma_{22}^1 b\dot{v}) \sigma_u \\ &\quad + (\dot{b} + \Gamma_{11}^2 a\dot{u} + \Gamma_{12}^2 (a\dot{v} + b\dot{u}) + \Gamma_{22}^2 b\dot{v}) \sigma_v. \end{aligned}$$

Proof. Differentiating $W = a\sigma_u + b\sigma_v$ along γ gives

$$\frac{dW}{dt} = \dot{a}\sigma_u + \dot{b}\sigma_v + a(\sigma_{uu}\dot{u} + \sigma_{uv}\dot{v}) + b(\sigma_{vu}\dot{u} + \sigma_{vv}\dot{v}).$$

Now expand $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ using the Christoffel symbols and discard the n components, which is exactly what the projection does. Collecting terms gives the stated formula, and the Christoffel symbols depend only on E, F, G . $\square$

Since $DW/dt = 0$ is a linear first order ODE system in (a, b) , Picard's theorem gives at once:

Proposition 29.6.24

Let $\gamma : I \rightarrow \Sigma$ be a smooth curve, $t_0 \in I$ and $w_0 \in T_{\gamma(t_0)}\Sigma$. Then there is a unique parallel vector field W along γ with $W(t_0) = w_0$, defined on all of I .

Definition 29.6.25 (Parallel Transport)

With notation as above, the map

$$\mathbf{p} : T_{\gamma(t_0)}\Sigma \rightarrow T_{\gamma(t_1)}\Sigma, \quad w_0 \mapsto W(t_1)$$

is called **parallel transport** along γ from $\gamma(t_0)$ to $\gamma(t_1)$.

Parallel transport is linear, since the defining ODE is linear, and it is a linear isometry by the proposition on inner products above. It is emphatically *path dependent*: transporting a vector around a closed loop on the sphere returns it rotated by an angle, and Gauss–Bonnet will tell us that the angle is the enclosed curvature.

§29.6.7 The Exponential Map

Geodesics give us a canonical way to move away from a point in a given direction, and hence a canonical way to put coordinates near a point.

Given $p \in \Sigma$ and $w \in T_p\Sigma$, let γ_w be the unique geodesic with $\gamma_w(0) = p$ and $\dot{\gamma}_w(0) = w$.

Definition 29.6.26 (Exponential Map)

The **exponential map** at p is

$$\exp_p(w) = \gamma_w(1),$$

defined for those $w \in T_p\Sigma$ for which γ_w exists up to time 1, and with $\exp_p(0) = p$.

Proposition 29.6.27

$\exp_p$ is defined and smooth on a neighbourhood of 0 in $T_p\Sigma$.

Proof. By the local existence theorem there is $\varepsilon > 0$ such that γ_w is defined on $(-2, 2)$ for every w in the closed disc $\{\|w\| \leq \varepsilon\}$: indeed, existence for a fixed time on a compact set of initial conditions follows from the local statement plus a compactness argument, and geodesics rescale, $\gamma_{\lambda w}(t) = \gamma_w(\lambda t)$, so shrinking ε buys more time. Smoothness is the smooth dependence of the solution of an ODE on its initial conditions. $\square$

Proposition 29.6.28

$\exp_p$ is a diffeomorphism from a neighbourhood of $0 \in T_p\Sigma$ onto its image.

Proof. By the inverse function theorem it suffices to show $D\exp_p|_0$ is invertible. Fix $w \in T_p\Sigma$ and consider the curve $\alpha(t) = tw$ in $T_p\Sigma$. Then

$$\exp_p(\alpha(t)) = \exp_p(tw) = \gamma_{tw}(1) = \gamma_w(t),$$

so differentiating at $t = 0$ gives $D\exp_p|_0(w) = \dot{\gamma}_w(0) = w$. So $D\exp_p|_0$ is the identity. $\square$

Definition 29.6.29 (Normal Neighbourhood)

The image of such a restriction of $\exp_p$ is a **normal neighbourhood** of p , and $\exp_p^{-1}$ gives **geodesic normal coordinates** on it.

Remark. The name comes from Lie theory. If $X \subseteq M(n)$ is a matrix group, with the inner product $\langle A, B \rangle = \text{tr}(AB^\top)$, then for $A \in T_I X$ the curve $t \mapsto e^{tA}$ satisfies $\ddot{\alpha} = A^2\alpha$, which one checks is normal to X ; so it is a geodesic, and $\exp_I(A) = e^A$ really is the matrix exponential.

§29.6.8 Geodesic Polar Coordinates

Composing the exponential map with polar coordinates on $T_p\Sigma$ gives an especially convenient parametrisation.

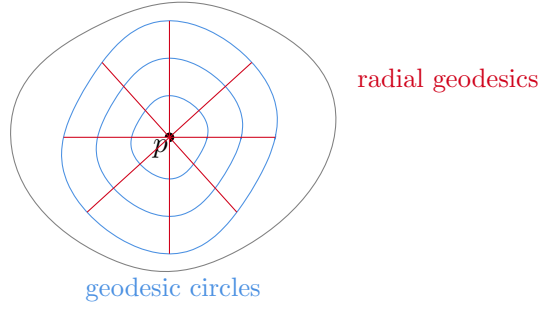
Definition 29.6.30 (Geodesic Polar Coordinates)

Choose an orthonormal basis of $T_p\Sigma$ and let $w(\theta)$ be the unit vector at angle θ . For ε small enough that $\exp_p$ is a diffeomorphism on the disc of radius ε , define

$$\sigma(r, \theta) = \exp_p(r w(\theta)) = \gamma_{w(\theta)}(r), \quad 0 < r < \varepsilon.$$

These are **geodesic polar coordinates** centred at p . The curves $\theta = \text{const}$ are the **radial geodesics** and the curves $r = \text{const}$ are the **geodesic circles** about p .

Note that σ is a genuine allowable parametrisation away from $r = 0$ (the origin, as always with polar coordinates, is a removable defect).

**Proposition 29.6.31 (The Gauss Lemma)**

In geodesic polar coordinates the first fundamental form is

$$I = dr^2 + G(r, \theta) d\theta^2,$$

that is, $E \equiv 1$ and $F \equiv 0$. Moreover

$$G(0, \theta) = 0, \quad (\sqrt{G})_r(0, \theta) = 1, \quad K = -\frac{(\sqrt{G})_{rr}}{\sqrt{G}}.$$

Proof. For fixed θ , $r \mapsto \sigma(r, \theta)$ is the unit speed geodesic $\gamma_{w(\theta)}$, so $E = \langle \sigma_r, \sigma_r \rangle = 1$. For F , note σ_{rr} is the acceleration of a geodesic, hence normal to Σ , so $\langle \sigma_{rr}, \sigma_\theta \rangle = 0$, and

$$F_r = \langle \sigma_{rr}, \sigma_\theta \rangle + \langle \sigma_r, \sigma_{\theta r} \rangle = 0 + \frac{1}{2} \frac{\partial}{\partial \theta} \langle \sigma_r, \sigma_r \rangle = \frac{1}{2} E_\theta = 0.$$

So F is independent of r . Writing $\sigma_\theta = r D \exp_p|_{rw(\theta)}(w'(\theta))$, we see $\sigma_\theta \rightarrow 0$ as $r \rightarrow 0$, hence $F \rightarrow 0$; since F is constant in r this gives $F \equiv 0$. The same expression gives $G = r^2 \|D \exp_p|_{rw}(w')\|^2$, so $G(0, \theta) = 0$ and

$$(\sqrt{G})_r(0, \theta) = \|D \exp_p|_0(w'(\theta))\| = \|w'(\theta)\| = 1.$$

Finally the formula for K is the corollary of the Gauss equation for a metric of the form $dr^2 + G d\theta^2$. $\square$

The geometric content of $F \equiv 0$ is worth stating: *geodesic circles meet radial geodesics at right angles*. This is what makes the next result work.

Proposition 29.6.32 (Geodesics Locally Minimise Length)

Let q lie in a normal neighbourhood of p , and let γ be the radial geodesic from p to q . Then every piecewise smooth curve in Σ from p to q has length at least $L(\gamma)$, with equality only if it is a reparametrisation of γ .

Proof. Say $q = \sigma(r_0, \theta_0)$, so $L(\gamma) = r_0$. Let α be any piecewise smooth curve from p to q . Consider first the case that α stays inside the normal neighbourhood; write

$\alpha(t) = \sigma(r(t), \theta(t))$ (away from p). Then by the Gauss lemma

$$L(\alpha) = \int \sqrt{\dot{r}^2 + G\dot{\theta}^2} dt \geq \int |\dot{r}| dt \geq \left| \int \dot{r} dt \right| = r_0,$$

with equality only if $\dot{\theta} \equiv 0$ and $\dot{r}$ never changes sign, i.e. only if α is a monotone reparametrisation of the radial geodesic.

If α leaves the normal neighbourhood, then it must cross the geodesic circle of radius r_1 for every $r_1 < \varepsilon$, and the same estimate applied to the initial segment shows $L(\alpha) \geq r_1$ for all such r_1 , hence $L(\alpha) \geq \varepsilon > r_0$. $\square$

Remark. ‘Locally’ is essential. On the sphere, the major arc of a great circle is a geodesic but is nowhere near the shortest path; and a geodesic on a cylinder can wind around many times. Geodesics are critical points of energy, and critical points need not be minima.

§29.6.9 Surfaces of Constant Curvature

Geodesic polar coordinates let us classify surfaces of constant curvature, locally.

Theorem 29.6.33 (Minding)

Let Σ be a smooth surface in $\mathbb{R}^3$ with constant Gaussian curvature K . Then Σ is locally isometric to

- (i) the plane with $I = du^2 + dv^2$, if $K = 0$;
- (ii) the sphere of radius $1/\sqrt{K}$, with $I = du^2 + \frac{1}{K} \cos^2(\sqrt{K}u) dv^2$, if $K > 0$;
- (iii) the hyperbolic plane of curvature K , with $I = du^2 + \frac{1}{|K|} \cosh^2(\sqrt{|K|}u) dv^2$, if $K < 0$.

In particular any two surfaces with the same constant curvature are locally isometric.

Proof. Work in geodesic polar coordinates about a point, so $I = dr^2 + G d\theta^2$ and, writing $h = \sqrt{G}$, the Gauss lemma gives

$$h_{rr} + Kh = 0, \quad h(0, \theta) = 0, \quad h_r(0, \theta) = 1.$$

This is a linear ODE in r for each fixed θ with prescribed initial data, so its solution is unique and independent of θ :

$$h(r) = \begin{cases} r & K = 0, \\ \frac{1}{\sqrt{K}} \sin(\sqrt{K}r) & K > 0, \\ \frac{1}{\sqrt{|K|}} \sinh(\sqrt{|K|}r) & K < 0. \end{cases}$$

So the first fundamental form in geodesic polar coordinates is completely determined by K , and any two surfaces of the same constant curvature have parametrisations with the same first fundamental form – hence are locally isometric. Identifying the resulting metrics with the standard models (after the substitution taking polar to the coordinates listed) gives the stated forms. $\square$

Remark. Globally the situation is quite different. There are plenty of flat surfaces which are not isometric to the plane – the cylinder and the flat torus, for example – and Gauss–Bonnet will explain why the sphere is the only compact one with $K \equiv 1$. Meanwhile Hilbert’s theorem says that no complete surface in $\mathbb{R}^3$ has constant negative curvature at all, so case (iii) can only ever be realised locally, on pieces such as the pseudosphere.

§29.7 Minimal Surfaces

Geodesics arose as the critical points of the length functional on curves. The obvious two-dimensional analogue is to look for critical points of the *area* functional on surfaces. The Euler–Lagrange equations are now a partial differential equation rather than an ordinary one, which makes the theory harder – but also much richer, because complex analysis turns out to solve it.

The physical model to keep in mind is a soap film spanning a wire loop: surface tension makes it minimise area subject to the boundary condition.

§29.7.1 The First Variation of Area

Let $\sigma : V \rightarrow \Sigma$ be an allowable parametrisation, let $D \subseteq V$ be a bounded domain with $\overline{D} \subseteq V$, and let $h : \overline{D} \rightarrow \mathbb{R}$ be smooth. We deform the surface in the normal direction, with the amount of deformation controlled by h .

Definition 29.7.1 (Normal Variation)

The **normal variation** of $\sigma(\overline{D})$ determined by h is the map

$$\sigma^t(u, v) = \sigma(u, v) + t h(u, v) n(\sigma(u, v)), \quad t \in (-\varepsilon, \varepsilon).$$

For small t the vectors

$$\sigma_u^t = \sigma_u + t h n_u + t h_u n, \quad \sigma_v^t = \sigma_v + t h n_v + t h_v n$$

remain linearly independent, so σ^t parametrises a genuine surface.

Proposition 29.7.2 (First Variation of Area)

With notation as above,

$$\left. \frac{d}{dt} \right|_{t=0} \text{area}(\sigma^t(D)) = -2 \int_D h H \sqrt{EG - F^2} \, du \, dv,$$

where H is the mean curvature of Σ .

Proof. Compute the first fundamental form of σ^t . Using $\langle n, \sigma_u \rangle = \langle n, \sigma_v \rangle = 0$ and the identities $\langle n_u, \sigma_u \rangle = -L$, $\langle n_u, \sigma_v \rangle = \langle n_v, \sigma_u \rangle = -M$, $\langle n_v, \sigma_v \rangle = -N$, we get

$$E^t = E - 2thL + O(t^2),$$

$$F^t = F - 2thM + O(t^2),$$

$$G^t = G - 2thN + O(t^2),$$

where the $O(t^2)$ terms are smooth in (u, v, t) . Hence

$$\begin{aligned} E^t G^t - (F^t)^2 &= EG - F^2 - 2th(EN - 2FM + GL) + O(t^2) \\ &= (EG - F^2)(1 - 4thH) + O(t^2), \end{aligned}$$

using $H = (LG - 2MF + NE)/2(EG - F^2)$. Taking square roots and differentiating under the integral sign at $t = 0$,

$$\left. \frac{d}{dt} \right|_{t=0} \int_D \sqrt{E^t G^t - (F^t)^2} \, du \, dv = \int_D \frac{-4hH}{2} \sqrt{EG - F^2} \, du \, dv,$$

which is the claim. $\square$

Definition 29.7.3 (Minimal Surface)

A smooth surface in $\mathbb{R}^3$ is **minimal** if its mean curvature vanishes identically.

Corollary 29.7.4

A surface is minimal if and only if it is a critical point of area under all normal variations, i.e. the first variation vanishes for every choice of D and h .

Proof. If $H \equiv 0$ the first variation formula gives zero. Conversely, if $H(q) \neq 0$ for some q , choose by continuity a small domain D around $\sigma^{-1}(q)$ on which H does not vanish and put $h = H$; then

$$\left. \frac{d}{dt} \right|_{t=0} \text{area} = -2 \int_D H^2 \sqrt{EG - F^2} \, du \, dv < 0,$$

so the surface is not critical. $\square$

Remark. As with geodesics, ‘minimal’ means *critical*, not minimising. A minimal surface need not minimise area even among nearby competitors, though small enough pieces of one always do.

The condition $H = 0$ says $\kappa_1 = -\kappa_2$, so at every point of a minimal surface the two principal curvatures are equal and opposite: minimal surfaces curve up as much as they curve down. In particular $K = \kappa_1 \kappa_2 = -\kappa_1^2 \leq 0$ everywhere, so every point of a minimal surface is hyperbolic or planar. Combined with the fact that every compact surface has an elliptic point, we get:

Corollary 29.7.5

There are no compact minimal surfaces in $\mathbb{R}^3$.

§29.7.2 Examples

Example 29.7.6 (The Plane)

A plane has $S_p = 0$ and so $H \equiv 0$. It is the trivial minimal surface, and the question that started the subject (asked by Lagrange in 1762) was whether there is any other.

Example 29.7.7 (The Catenoid)

Meusnier answered Lagrange in 1776. Consider the surface of revolution generated by the catenary $x = a \cosh(z/a)$, that is,

$$\sigma(u, v) = (a \cosh v \cos u, a \cosh v \sin u, av).$$

One computes $E = G = a^2 \cosh^2 v$ and $F = 0$, so this parametrisation is isothermal, and

$$L = -a, \quad M = 0, \quad N = a,$$

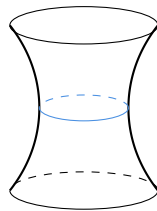
giving $H = (LG + NE)/2EG = 0$. So the catenoid is minimal. It is in fact the only minimal surface of revolution other than the plane – a fact one can prove by solving $H = 0$ for a profile curve, which reduces to $ff'' = 1 + (f')^2$ and hence to the catenary.

Example 29.7.8 (The Helicoid)

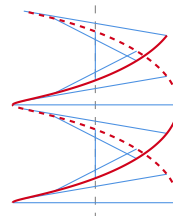
The helicoid is the ruled surface

$$\sigma(u, v) = (a \sinh v \cos u, a \sinh v \sin u, au),$$

swept out by a horizontal line rotating at constant speed as it rises. Again $E = G = a^2 \cosh^2 v$ and $F = 0$, so it too is isothermal with the same first fundamental form as the catenoid – and hence the two are locally isometric. It is minimal for the same reason. Physically, one can bend a catenoid soap film continuously into a helicoid without stretching it anywhere; the pictures of this deformation are worth looking up.



catenoid



helicoid

Example 29.7.9 (Ruled Surfaces)

A **ruled surface** $\sigma(u, v) = \eta(u) + vw(u)$ is a union of straight lines. Every straight line on a surface is a geodesic and has vanishing normal curvature, so at every point of a ruled surface some normal curvature is zero, hence $\kappa_1 \kappa_2 \leq 0$ and $K \leq 0$. The plane and the helicoid are the only ruled minimal surfaces in $\mathbb{R}^3$, a theorem of Catalan.

§29.7.3 Isothermal Coordinates and Harmonicity

The reason complex analysis enters is the following identity.

Proposition 29.7.10

Let σ be an isothermal parametrisation, so $E = G = \lambda^2$ and $F = 0$. Then

$$\nabla^2 \sigma := \sigma_{uu} + \sigma_{vv} = 2\lambda^2 H n.$$

Proof. Differentiating $\langle \sigma_u, \sigma_u \rangle = \langle \sigma_v, \sigma_v \rangle$ with respect to u gives $\langle \sigma_{uu}, \sigma_u \rangle = \langle \sigma_{uv}, \sigma_v \rangle$, and differentiating $\langle \sigma_u, \sigma_v \rangle = 0$ with respect to v gives $\langle \sigma_{uv}, \sigma_v \rangle = -\langle \sigma_u, \sigma_{vv} \rangle$. Adding,

$$\langle \sigma_{uu} + \sigma_{vv}, \sigma_u \rangle = 0,$$

and symmetrically $\langle \sigma_{uu} + \sigma_{vv}, \sigma_v \rangle = 0$. So $\nabla^2 \sigma$ is normal. Its normal component is $L + N$, and in the isothermal case

$$H = \frac{LG + NE}{2EG} = \frac{L + N}{2\lambda^2},$$

so $L + N = 2\lambda^2 H$ as claimed. $\square$

Corollary 29.7.11

An isothermally parametrised surface is minimal if and only if the three coordinate functions of σ are harmonic.

Since every minimal surface admits isothermal coordinates – for minimal surfaces this is much easier than the general theorem – minimal surfaces are locally exactly the harmonic isothermal immersions. Because a non-constant harmonic function on a connected open set has no interior maximum, this gives another proof that no minimal surface is compact.

§29.7.4 The Weierstrass Representation

We can now write down *all* minimal surfaces, locally, in terms of holomorphic data.

Proposition 29.7.12

Let $\sigma(u, v) = (x, y, z)$ be a parametrisation of a surface in $\mathbb{R}^3$ and set

$$\theta_1 = x_u - ix_v, \quad \theta_2 = y_u - iy_v, \quad \theta_3 = z_u - iz_v.$$

Then

- (i) σ is isothermal if and only if $\theta_1^2 + \theta_2^2 + \theta_3^2 = 0$;
- (ii) if σ is isothermal, then the surface is minimal if and only if $\theta_1, \theta_2, \theta_3$ are holomorphic in $\xi = u + iv$.

Proof. For (i), expand:

$$\sum_j \theta_j^2 = \sum_j (x_u^{(j)})^2 - \sum_j (x_v^{(j)})^2 - 2i \sum_j x_u^{(j)} x_v^{(j)} = E - G - 2iF,$$

which vanishes exactly when $E = G$ and $F = 0$.

For (ii), the Cauchy–Riemann equations for $\theta_1 = x_u - ix_v$ are $x_{uu} = -x_{vv}$ and $x_{uv} = x_{vu}$. The second always holds and the first says x is harmonic, so θ_1 is

holomorphic if and only if x is harmonic; likewise for θ_2, θ_3 . Now apply the previous corollary. $\square$

So the problem becomes: describe all triples of holomorphic functions with $\theta_1^2 + \theta_2^2 + \theta_3^2 = 0$. This is a conic, and conics can be rationally parametrised.

Lemma 29.7.13

Let $D \subseteq \mathbb{C}$ be a domain, g meromorphic on D and f holomorphic on D , such that f has a zero of order at least $2k$ wherever g has a pole of order k . Then

$$\theta_1 = \frac{1}{2}f(1 - g^2), \quad \theta_2 = \frac{i}{2}f(1 + g^2), \quad \theta_3 = fg$$

are holomorphic on D and satisfy $\theta_1^2 + \theta_2^2 + \theta_3^2 = 0$. Conversely, every triple of holomorphic functions satisfying this equation arises in this way, apart from the degenerate case $\theta_1 = i\theta_2, \theta_3 = 0$.

Proof. For the first part, the pole conditions make each θ_j holomorphic, and

$$\theta_1^2 + \theta_2^2 = \frac{f^2}{4} [(1 - g^2)^2 - (1 + g^2)^2] = -f^2 g^2 = -\theta_3^2.$$

Conversely, given holomorphic θ_j with $\sum \theta_j^2 = 0$, suppose $\theta_1 \neq i\theta_2$ and set

$$f = \theta_1 - i\theta_2, \quad g = \frac{\theta_3}{f}.$$

Then $fg = \theta_3$, and from $\theta_1^2 + \theta_2^2 = -\theta_3^2$ we get $(\theta_1 - i\theta_2)(\theta_1 + i\theta_2) = -\theta_3^2$, so $\theta_1 + i\theta_2 = -fg^2$; solving the two linear equations for θ_1, θ_2 gives the stated formulae. Holomorphy of $\theta_1 + i\theta_2 = -fg^2$ is exactly the condition on the orders of zeros and poles. $\square$

Proposition 29.7.14

Let $D \subseteq \mathbb{C}$ be simply connected, fix $\xi_0 \in D$, and let $\theta_1, \theta_2, \theta_3$ be holomorphic on D with $\sum \theta_j^2 = 0$. Define

$$x(u, v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_1 \, d\xi, \quad y(u, v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_2 \, d\xi, \quad z(u, v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_3 \, d\xi.$$

Then $x_u - ix_v = \theta_1$, and similarly for y, z . Consequently, wherever $\sigma = (x, y, z)$ is an immersion, its image is a minimal surface.

Proof. Simple connectedness makes the integrals well defined and holomorphic. Write $\chi(\omega) = \int_{\xi_0}^{\omega} \theta_1 \, d\xi = x + i\beta$; the Cauchy–Riemann equations for χ give $\beta_u = -x_v$, so

$$\theta_1(\omega) = \chi'(\omega) = x_u + i\beta_u = x_u - ix_v,$$

as claimed. The previous two results then say that σ is isothermal with harmonic components, hence minimal. $\square$

Definition 29.7.15 (Weierstrass Data)

The pair (f, g) above is the **Weierstrass representation** of the resulting minimal surface.

By the existence of isothermal coordinates, *every* minimal surface has a local Weierstrass representation. Checking that a given (f, g) yields an embedding can be delicate; checking that it yields an immersion is easy, and amounts to asking that the zeros of f occur exactly at the poles of g and with exactly twice the order.

Example 29.7.16 (Enneper's Surface)

Take $D = \mathbb{C}$, $f(\xi) = 1$ and $g(\xi) = \xi$, so $\theta_1 = \frac{1}{2}(1 - \xi^2)$, $\theta_2 = \frac{i}{2}(1 + \xi^2)$, $\theta_3 = \xi$. Integrating from 0 and taking real parts gives

$$\sigma(u, v) = \frac{1}{2} \left(u - \frac{u^3}{3} + uv^2, -v - \frac{v^3}{3} - u^2v, u^2 - v^2 \right),$$

which is an immersion everywhere and an embedding near the origin. Its Gaussian curvature works out to $K = -16/(1 + u^2 + v^2)^4$.

Example 29.7.17 (Scherk's Surface)

Take $D = \{|\xi| < 1\}$, $f(\xi) = 4/(1 - \xi^4)$ and $g(\xi) = \xi$. The resulting surface satisfies the beautiful equation

$$\cos y = e^z \cos x,$$

is defined over the open square $(-\pi/2, \pi/2)^2$, and repeats periodically in a checker-board pattern. It was the first new minimal surface found after the catenoid and helicoid, in 1835.

§29.7.5 Interpreting the Weierstrass Data

The function g has a lovely geometric meaning: it is the Gauss map, read through stereographic projection.

Proposition 29.7.18

Let σ be a minimal immersion with Weierstrass data (f, g) , and let $\pi : S^2 \rightarrow \mathbb{C} \cup \{\infty\}$ be stereographic projection from the north pole. Then

$$g = \pi \circ n \circ \sigma,$$

and the conformal factor is $\lambda = \frac{1}{2}|f|(1 + |g|^2)$.

Proof. Since $\sigma_u - i\sigma_v = (\theta_1, \theta_2, \theta_3)$ and σ is isothermal,

$$\lambda^2 = E = \frac{1}{2} \langle \sigma_u - i\sigma_v, \sigma_u + i\sigma_v \rangle = \frac{1}{2} (|\theta_1|^2 + |\theta_2|^2 + |\theta_3|^2) = \left(\frac{|f|(1+|g|^2)}{2} \right)^2$$

after substituting the formulae for θ_j . A direct computation of the cross product

gives

$$\sigma_u \times \sigma_v = \frac{|f|^2(1 + |g|^2)}{4} (2 \operatorname{Re} g, 2 \operatorname{Im} g, |g|^2 - 1),$$

and normalising, the Gauss map is

$$n \circ \sigma = \frac{1}{1 + |g|^2} (2 \operatorname{Re} g, 2 \operatorname{Im} g, |g|^2 - 1),$$

which is exactly $\pi^{-1}(g)$. □

Corollary 29.7.19

If $K \not\equiv 0$ on a connected minimal surface, then the zeros of K are isolated.

Proof. On the example sheet one computes $K = - (4|g'| / (|f|(1 + |g|^2)^2))^2$, so K vanishes exactly where g' does. Since g is meromorphic and non-constant, g' has isolated zeros by the principle of isolated zeros. □

Theorem 29.7.20

Let σ be a minimal immersion defined on all of $\mathbb{C}$. Then either its image lies in a plane, or its Gauss map omits at most two points of S^2 .

Proof. If the degenerate case $\theta_1 = i\theta_2, \theta_3 = 0$ occurs then z is constant and the surface lies in a plane; so assume the Weierstrass data (f, g) exists. If g is constant then n is constant and again the image is planar. Otherwise $g : \mathbb{C} \rightarrow \mathbb{C} \cup \{\infty\}$ is a non-constant meromorphic function, so by Picard's theorem it omits at most two values of $\mathbb{C} \cup \{\infty\}$. Since $g = \pi \circ n \circ \sigma$ and π is a bijection, the Gauss map omits at most two points of S^2 . □

Example 29.7.21

Enneper's surface has $g(\xi) = \xi$, which omits only ∞ ; its Gauss map misses exactly one point. The catenoid has $g(\xi) = \xi$ on $\mathbb{C} \setminus \{0\}$, whose Gauss map misses two points, the north and south poles. So the bound is achieved.

Remark. For complete non-planar minimal surfaces, Fujimoto proved in 1988 that the Gauss map can omit at most four points, sharpening earlier work of Xavier; and four is attained. This is a striking instance of the general principle that minimal surfaces inherit the rigidity of holomorphic functions.

§29.8 The Gauss–Bonnet Theorem

We come to the theorem the whole course has been building towards. Gauss–Bonnet relates an analytic quantity – the integral of the curvature – to a purely topological one, the Euler characteristic. Curvature is a delicate local invariant which changes if you dent the surface with a finger; the Euler characteristic is a crude count of holes which does not. The theorem says that if you push curvature down in one place, it has to pop

up somewhere else.

We begin with the local statement, for which we need to measure how much a curve on a surface turns.

§29.8.1 Geodesic Curvature

Let Σ be an oriented surface and γ a unit speed curve on it. We have already seen that $\ddot{\gamma}$ splits into a normal part – the normal curvature $\kappa_n = \langle \ddot{\gamma}, n \rangle$ – and a tangential part, and that γ is a geodesic exactly when the tangential part vanishes. Since $\ddot{\gamma} \perp \dot{\gamma}$, the tangential part is a multiple of $n \times \dot{\gamma}$.

Definition 29.8.1 (Geodesic Curvature)

The **geodesic curvature** of a unit speed curve γ on an oriented surface Σ is

$$\kappa_g = \langle \ddot{\gamma}, n \times \dot{\gamma} \rangle.$$

So $\ddot{\gamma} = \kappa_g (n \times \dot{\gamma}) + \kappa_n n$, and therefore

$$\kappa^2 = \|\ddot{\gamma}\|^2 = \kappa_g^2 + \kappa_n^2 :$$

the curvature of a curve on a surface splits into the part forced on it by the surface bending, and the part due to the curve turning within the surface. Immediately from the definitions:

Proposition 29.8.2

A unit speed curve on Σ is a geodesic if and only if $\kappa_g \equiv 0$.

Example 29.8.3

For a curve in a plane, n is constant and κ_g is the signed curvature κ_s of Section 2. On the sphere, a circle of latitude at angle u from the equator has $\kappa_g = \tan u$: zero at the equator (a great circle) and blowing up at the poles.

The sign of κ_g depends on the orientation. More generally, given a unit vector field W along γ , the derivative $\dot{W}$ is orthogonal to W , so its tangential part is a multiple of $n \times W$; we write

$$\left[\frac{DW}{dt} \right] = \langle \dot{W}, n \times W \rangle,$$

the **algebraic value** of the covariant derivative, so that $DW/dt = [DW/dt](n \times W)$. Then $\kappa_g = [D\dot{\gamma}/dt]$. The following lemma is the technical heart of the local Gauss–Bonnet theorem: it says that the algebraic value measures the rate of turning relative to a reference field.

Lemma 29.8.4

Let V, W be smooth unit vector fields along a curve $\gamma : I \rightarrow \Sigma$, and let ψ be a smooth determination of the angle from V to W , that is, a smooth function with

$$W = \cos \psi V + \sin \psi (n \times V).$$

(Such a ψ exists and is unique up to adding a constant in $2\pi\mathbb{Z}$.) Then

$$\left[\frac{DW}{dt} \right] - \left[\frac{DV}{dt} \right] = \dot{\psi}.$$

Proof. Write $W = aV + b(n \times V)$ with $a^2 + b^2 = 1$; existence of a smooth ψ follows by setting $\psi(t) = \psi_0 + \int_{t_0}^t (a\dot{b} - b\dot{a})$ and checking as in the plane curve case. Abbreviate $iV = n \times V$, so that $n \times W = -\sin \psi V + \cos \psi (iV)$. Then

$$\dot{W} = -\dot{\psi} \sin \psi V + \dot{\psi} \cos \psi (iV) + \cos \psi \dot{V} + \sin \psi \frac{d(iV)}{dt},$$

and taking the inner product with $n \times W$, the first two terms contribute $\dot{\psi}(\sin^2 \psi + \cos^2 \psi) = \dot{\psi}$ while the last two contribute

$$\cos^2 \psi \langle \dot{V}, iV \rangle - \sin^2 \psi \langle V, \frac{d}{dt}(iV) \rangle = (\cos^2 \psi + \sin^2 \psi) \left[\frac{DV}{dt} \right],$$

using $\langle V, iV \rangle = 0$. Adding gives the result. $\square$

Proposition 29.8.5

Let σ be an orthogonal parametrisation compatible with the orientation, and let $\gamma(t) = \sigma(u(t), v(t))$ be a unit speed curve in its image. Then

$$\kappa_g = \frac{G_u \dot{v} - E_v \dot{u}}{2\sqrt{EG}} + \dot{\psi},$$

where ψ is a smooth determination of the angle from σ_u to $\dot{\gamma}$.

Proof. Put $e_1 = \sigma_u / \sqrt{E}$ and $e_2 = \sigma_v / \sqrt{G}$, a positively oriented orthonormal frame. By the lemma it is enough to compute $[De_1 / dt] = \langle \dot{e}_1, e_2 \rangle$. Now

$$\langle (e_1)_u, e_2 \rangle = \frac{1}{\sqrt{EG}} \langle \sigma_{uu}, \sigma_v \rangle = \frac{1}{\sqrt{EG}} (F_u - \frac{1}{2}E_v) = -\frac{E_v}{2\sqrt{EG}},$$

using $F \equiv 0$, and similarly

$$\langle (e_1)_v, e_2 \rangle = \frac{1}{\sqrt{EG}} \langle \sigma_{uv}, \sigma_v \rangle = \frac{G_u}{2\sqrt{EG}}.$$

Hence $[De_1 / dt] = \dot{u} \langle (e_1)_u, e_2 \rangle + \dot{v} \langle (e_1)_v, e_2 \rangle$ is the first term, and the lemma supplies the $\dot{\psi}$. $\square$

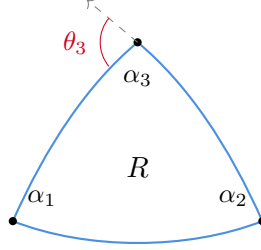
That first term is exactly the sort of thing Green's theorem eats, and the orthogonal Gauss formula is exactly what comes out. That is the whole proof of the local theorem.

§29.8.2 The Local Gauss–Bonnet Theorem

Definition 29.8.6 (Simple Region)

Let Σ be an oriented surface. A region $\bar{R} \subseteq \Sigma$ is **simple** if it is homeomorphic to

a closed disc and its boundary is the image of a simple closed curve $\alpha : [0, L] \rightarrow \Sigma$, parametrised by arc length, which is smooth except at finitely many points called **vertices**. The **exterior angle** θ_i at a vertex is the angle from the incoming tangent to the outgoing tangent, taken in $(-\pi, \pi)$. We say α is **positively oriented** if $n \times \dot{\alpha}$ points into R .



We will use, without proof, the following fact, which is the surface version of Hopf's Umlaufsatz: if $\bar{R}$ is a simple region contained in the image of a parametrisation whose domain is a disc, and ψ is a piecewise smooth determination of the angle from σ_u to $\dot{\alpha}$ which jumps by θ_i at the i th vertex, then the total change is

$$\psi(L) - \psi(0) = 2\pi.$$

Theorem 29.8.7 (Local Gauss–Bonnet)

Let $\bar{R}$ be a simple region contained in the image of an orthogonal parametrisation compatible with the orientation, whose domain is a disc, and let $\alpha : [0, L] \rightarrow \Sigma$ be a positively oriented arc length parametrisation of ∂R with exterior angles $\theta_1, \dots, \theta_k$. Then

$$\int_0^L \kappa_g \, ds + \int_R K \, dA = 2\pi - \sum_{i=1}^k \theta_i.$$

Proof. Away from the vertices we have, by the proposition above,

$$\kappa_g = \frac{G_u \dot{v} - E_v \dot{u}}{2\sqrt{EG}} + \frac{d\psi}{ds}.$$

Integrating over the smooth arcs and summing, the last term contributes $\psi(L) - \psi(0)$ together with the jumps, that is, $2\pi - \sum_i \theta_i$ by the fact quoted above. For the first term, Green's theorem in the (u, v) -plane gives

$$\int_0^L \frac{G_u \dot{v} - E_v \dot{u}}{2\sqrt{EG}} \, ds = \int_{\sigma^{-1}(R)} \left[\left(\frac{G_u}{2\sqrt{EG}} \right)_u + \left(\frac{E_v}{2\sqrt{EG}} \right)_v \right] \, du \, dv,$$

and by the orthogonal Gauss formula the integrand is $-K\sqrt{EG}$. Since $F = 0$ we have $\sqrt{EG} = \sqrt{EG - F^2}$, so this is $-\int_R K \, dA$. Rearranging gives the theorem. $\square$

Corollary 29.8.8 (Geodesic Triangles)

Let T be a geodesic triangle – a simple region whose three boundary arcs are

geodesics – with interior angles $\alpha_1, \alpha_2, \alpha_3$. Then

$$\int_T K \, dA = \alpha_1 + \alpha_2 + \alpha_3 - \pi.$$

Proof. The geodesic curvature vanishes on the boundary, and the exterior angles are $\theta_i = \pi - \alpha_i$. Substituting, $\int_T K \, dA = 2\pi - \sum(\pi - \alpha_i) = \sum \alpha_i - \pi$. $\square$

So the angle sum of a geodesic triangle exceeds π by exactly the total curvature it encloses. On the unit sphere this recovers the classical formula: the area of a spherical triangle is its angle excess. In the plane we recover Euclid.

§29.8.3 Triangulations and the Euler Characteristic

To globalise, we chop the surface into small pieces and add up.

Definition 29.8.9 (Triangulation)

A **triangulation** of a compact surface Σ (possibly with boundary) is a finite collection of topological triangles $T_1, \dots, T_F$ covering Σ , such that any two distinct triangles are either disjoint, meet in exactly one common vertex, or meet in exactly one common edge.

Definition 29.8.10 (Euler Characteristic)

The **Euler characteristic** of a triangulation is $\chi = V - E + F$, where V, E, F count vertices, edges and faces.

We will use the following facts from topology, which we do not prove.

Fact 29.8.11

Every compact smooth surface admits a triangulation whose edges are smooth curves, and every triangulation can be refined to a smooth one.

Fact 29.8.12 (Classification of Compact Surfaces)

Every compact connected orientable surface without boundary is diffeomorphic to a sphere with $g \geq 0$ handles attached, and this surface has Euler characteristic $2 - 2g$. In particular the Euler characteristic determines the surface up to diffeomorphism.

For the proof we will need triangulations by small triangles. Note first that subdividing a triangle into four by joining the midpoints of its edges adds 3 vertices, 6 edges and 3 faces, so it leaves χ unchanged. Repeating this finitely often makes all triangles as small as we like.

Proposition 29.8.13

Every point of a smooth surface has a neighbourhood admitting an orthogonal parametrisation.

Proof. Fix p and a unit speed geodesic γ through p with $\gamma(0) = p$. For each v , let γ^v be the unit speed geodesic with $\gamma^v(0) = \gamma(v)$ whose initial velocity is orthogonal to $\dot{\gamma}(v)$ and positively oriented. Define

$$\sigma(u, v) = \gamma^v(u).$$

By smooth dependence of solutions of the geodesic equations on initial conditions this is smooth near $(0, 0)$, and $\sigma_u(0, 0), \sigma_v(0, 0)$ are orthonormal, so by the inverse function theorem σ is a parametrisation near p . Finally $E = \|\sigma_u\|^2 = 1$ since the u -curves are unit speed geodesics, and

$$F_u = \langle \sigma_{uu}, \sigma_v \rangle + \langle \sigma_u, \sigma_{uv} \rangle = 0 + \frac{1}{2}E_v = 0,$$

so F is independent of u ; and $F(0, v) = 0$ by construction. Hence $F \equiv 0$. These are called **geodesic normal** (or Fermi) coordinates. $\square$

§29.8.4 The Global Gauss–Bonnet Theorem

Theorem 29.8.14 (Global Gauss–Bonnet)

Let Σ be a compact orientable smooth surface without boundary. Then for any triangulation of Σ with Euler characteristic χ ,

$$\int_{\Sigma} K \, dA = 2\pi\chi.$$

Proof. Cover Σ by finitely many open sets carrying orthogonal parametrisations, using the previous proposition and compactness, and let ε be a Lebesgue number for this cover. Take any smooth triangulation and subdivide repeatedly until every triangle has diameter less than ε ; as noted, this does not change χ . Now every triangle T_i lies inside a single orthogonal patch, so local Gauss–Bonnet applies to it:

$$\int_{T_i} K \, dA + \int_{\partial T_i} \kappa_g \, ds = \alpha_{i,1} + \alpha_{i,2} + \alpha_{i,3} - \pi,$$

with ∂T_i positively oriented. Sum over i . On the left, the curvature integrals add up to $\int_{\Sigma} K \, dA$. The geodesic curvature integrals cancel in pairs: every edge is shared by exactly two triangles and, because Σ is orientable, we may orient all the triangles compatibly, so each edge is traversed once in each direction and κ_g changes sign.

On the right, the angles at each vertex sum to 2π , so $\sum_{i,j} \alpha_{i,j} = 2\pi V$, and we subtract π once per face. Hence

$$\int_{\Sigma} K \, dA = 2\pi V - \pi F.$$

Finally, each face has three edges and each edge lies in two faces, so $3F = 2E$, giving $\pi F = 3\pi F - 2\pi F = 2\pi E - 2\pi F$ and therefore

$$\int_{\Sigma} K \, dA = 2\pi V - 2\pi E + 2\pi F = 2\pi\chi. \quad \square$$

Corollary 29.8.15

The Euler characteristic of a compact orientable smooth surface does not depend on the choice of triangulation.

Proof. The left hand side of Gauss–Bonnet does not mention the triangulation. $\square$

This is a genuinely startling deduction: a fact of pure combinatorial topology, proved by integrating curvature.

Theorem 29.8.16 (Gauss–Bonnet with Boundary)

Let R be a region with compact closure in an oriented surface Σ , whose boundary consists of n piecewise smooth simple closed curves $\alpha_1, \dots, \alpha_n$, positively oriented and parametrised by arc length, with exterior angles θ_j at the vertices. Then

$$\int_R K \, dA + \sum_{i=1}^n \int_{\alpha_i} \kappa_g \, ds = 2\pi\chi(R) - \sum_j \theta_j.$$

Proof. Identical to the previous proof, except that the geodesic curvature integrals along the boundary edges have no partner to cancel with, and the exterior angles at boundary vertices survive. $\square$

§29.8.5 Applications**Example 29.8.17 (Total Curvature of Standard Surfaces)**

The sphere has $\chi = 2$, so $\int_{S^2} K \, dA = 4\pi$ – which we can check directly, since $K \equiv 1$ and the area is 4π . The torus has $\chi = 0$, so $\int K \, dA = 0$: the positive curvature on the outside exactly cancels the negative curvature on the inside, however you deform it. A surface of genus $g \geq 2$ has $\chi = 2 - 2g < 0$, so it must have negative curvature somewhere.

Corollary 29.8.18

A compact orientable surface with $K > 0$ everywhere is diffeomorphic to S^2 .

Proof. By Gauss–Bonnet, $2\pi\chi = \int_{\Sigma} K \, dA > 0$, so $\chi > 0$. But $\chi = 2 - 2g$ for some $g \geq 0$, so $\chi = 2$ and $g = 0$. By the classification of compact surfaces, Σ is diffeomorphic to S^2 . $\square$

Corollary 29.8.19

A compact orientable surface with $K \leq 0$ everywhere has genus at least 1; and if $K < 0$ everywhere it has genus at least 2.

Proof. $2\pi(2 - 2g) = \int K \, dA \leq 0$ forces $g \geq 1$, and the strict inequality forces $2 - 2g < 0$, i.e. $g \geq 2$. $\square$

Proposition 29.8.20

On a compact orientable surface with $K > 0$ everywhere, any two simple closed geodesics must intersect.

Proof. Suppose γ_1, γ_2 are disjoint simple closed geodesics. By the previous corollary Σ is a sphere, and two disjoint simple closed curves on a sphere bound a region R homeomorphic to an annulus, which has $\chi(R) = 0$. Applying Gauss–Bonnet with boundary to R , and noting that $\kappa_g = 0$ along both boundary curves and there are no vertices,

$$\int_R K \, dA = 2\pi \cdot 0 = 0,$$

contradicting $K > 0$. □

Example 29.8.21 (The Catenoid)

The catenoid is homeomorphic to a cylinder and has $K < 0$ everywhere. A similar Gauss–Bonnet argument shows that such a surface carries at most one simple closed geodesic; and indeed the catenoid has exactly one, the circle at its waist.

Remark (Parallel Transport and Holonomy). Local Gauss–Bonnet also computes the angle by which parallel transport around a closed curve rotates a tangent vector. If γ bounds a simple region R and V is parallel along γ , then the lemma above with $W = \dot{\gamma}$ shows that the total turning of V relative to $\dot{\gamma}$ is $\int \kappa_g$, and the computation in the proof shows that transporting once around γ rotates a vector by exactly $\int_R K \, dA$. Carry a vector around a spherical triangle and it comes back rotated by the area of the triangle – this is what a Foucault pendulum is measuring.

§29.9 Total Curvature of Space Curves

We finish with two beautiful applications of the global theory to curves in $\mathbb{R}^3$. Both work by fattening the curve into a surface – a tube – and then applying what we know about the Gauss map.

Definition 29.9.1 (Total Curvature)

The **total curvature** of a closed unit speed curve $\alpha : [0, L] \rightarrow \mathbb{R}^3$ is $\int_0^L \kappa(s) \, ds$.

For a plane curve, the total curvature is $\int |\kappa_s| \geq |\int \kappa_s| = 2\pi$ if the curve is simple, by Hopf’s Umlaufsatz, with equality if and only if κ_s never changes sign, that is, if and only if the curve is convex. Fenchel’s theorem says the same bound holds in space, and that only plane curves achieve it.

§29.9.1 Tubes

Definition 29.9.2 (Tube)

Let $\alpha : [0, L] \rightarrow \mathbb{R}^3$ be a closed unit speed curve with $\kappa > 0$ everywhere, with Frenet frame (T, N, B) . For $r > 0$, the **tube** of radius r about α is the surface parametrised

by

$$\sigma(s, v) = \alpha(s) + r(N(s) \cos v + B(s) \sin v).$$

For r small enough the tube is an embedded surface. Its geometry is entirely explicit.

Lemma 29.9.3

For r small enough that $r\kappa < 1$ everywhere, the tube T_r is a smooth compact surface with

$$EG - F^2 = r^2(1 - r\kappa \cos v)^2, \quad n = -(N \cos v + B \sin v),$$

and Gaussian curvature

$$K = -\frac{\kappa \cos v}{r(1 - r\kappa \cos v)}.$$

Proof. Differentiating and using Frenet–Serret,

$$\sigma_s = (1 - r\kappa \cos v)T + r\tau(-N \sin v + B \cos v), \quad \sigma_v = r(-N \sin v + B \cos v).$$

These are linearly independent when $r\kappa < 1$, and a short computation gives $E = (1 - r\kappa \cos v)^2 + r^2\tau^2$, $F = r^2\tau$, $G = r^2$, whence $EG - F^2 = r^2(1 - r\kappa \cos v)^2$. Also $\sigma_s \times \sigma_v = -r(1 - r\kappa \cos v)(N \cos v + B \sin v)$, giving the stated unit normal. Differentiating n and using Frenet–Serret again gives

$$n_s \times n_v = \kappa \cos v (N \cos v + B \sin v) = -\frac{\kappa \cos v}{r(1 - r\kappa \cos v)} \sigma_s \times \sigma_v,$$

and since $n_s \times n_v = K \sigma_s \times \sigma_v$, the formula for K follows. $\square$

Note that $K \geq 0$ exactly when $\cos v \leq 0$, that is, on the ‘outer’ half of the tube – exactly as for the torus, which is the special case α a circle. We will also use, without proof, the following consequence of the stack of records theorem and Sard’s theorem: for a compact oriented surface Σ in $\mathbb{R}^3$,

$$\int_{\Sigma} |K| \, dA = \int_{S^2} |n^{-1}(y)| \, dy,$$

which is to say that $\int |K|$ counts, with multiplicity, how often the Gauss map covers the sphere.

§29.9.2 Fenchel’s Theorem

Theorem 29.9.4 (Fenchel)

Let α be a simple closed regular curve in $\mathbb{R}^3$ with nowhere vanishing curvature. Then its total curvature is at least 2π , with equality if and only if α is a convex plane curve.

Proof. Form the tube T_r for small r and let $R = \{p \in T_r : K(p) \geq 0\}$, which by the

lemma is the set where $\cos v \leq 0$, i.e. $v \in [\pi/2, 3\pi/2]$. Then

$$\begin{aligned} \int_R K \, dA &= \int_0^L \int_{\pi/2}^{3\pi/2} \frac{-\kappa \cos v}{r(1 - r\kappa \cos v)} \cdot r(1 - r\kappa \cos v) \, dv \, ds \\ &= - \int_0^L \kappa \, ds \int_{\pi/2}^{3\pi/2} \cos v \, dv, \end{aligned}$$

and since $\int_{\pi/2}^{3\pi/2} \cos v \, dv = -2$, this is $2 \int_0^L \kappa \, ds$.

Now the Gauss map restricted to R is surjective onto S^2 : given any unit vector y , the height function $s \mapsto \langle \alpha(s), y \rangle$ attains a maximum, and at that point the corresponding point of the tube in the direction y has outward normal y and non-negative curvature. Hence

$$2 \int_0^L \kappa \, ds = \int_R K \, dA = \int_{S^2} |(n|_R)^{-1}(y)| \, dy \geq \text{area}(S^2) = 4\pi,$$

giving $\int \kappa \geq 2\pi$.

Suppose equality holds. Then n is injective on the set where $K > 0$, so distinct points of that set have distinct normals. It follows that at every such point p the whole tube lies on one side of the tangent plane at p – otherwise, using that n is a local diffeomorphism near p and that the zero set of K is nowhere dense, one produces two distinct points with positive curvature and the same normal. Taking limits, the same holds at the points where $K = 0$, which are exactly the points of the tube in the directions $\pm B$.

Fix s_0 and look at the circle $s = s_0$ of the tube. The two points on it in the directions $\pm B(s_0)$ have parallel tangent planes, and the tube is squeezed between them; hence α lies in the plane Π midway between. (If not, take s_1 maximising the distance from Π ; then $T(s_1)$ is parallel to Π , and the circle $s = s_1$ then pokes out of the slab, a contradiction.) So α is a plane curve, and by the plane case discussed above, equality forces it to be convex. $\square$

§29.9.3 The Fáry–Milnor Theorem

If a closed curve is knotted, it has to bend more. This was conjectured by Borsuk and proved independently by Fáry (1949) and Milnor (1950); Milnor was an undergraduate at the time.

Definition 29.9.5 (Unknotted)

A regular simple closed curve in $\mathbb{R}^3$ is **unknotted** if its image is the boundary of an embedded disc, and **knotted** otherwise.

Theorem 29.9.6 (Fáry–Milnor)

A knotted regular simple closed curve in $\mathbb{R}^3$ with nowhere vanishing curvature has total curvature at least 4π .

Proof. We prove the contrapositive: if the total curvature is less than 4π then the curve is unknotted.

Form the tube T_r as before. Computing $\int_{T_r} |K| \, dA$ exactly as in Fenchel's proof, but now over all of $[0, 2\pi]$ in v and with $|\cos v|$ in place of $-\cos v$, gives

$$\int_{S^2} |n^{-1}(y)| \, dy = \int_{T_r} |K| \, dA = 4 \int_0^L \kappa \, ds.$$

If $\int_0^L \kappa \, ds < 4\pi$ then the right hand side is less than $16\pi = 4 \operatorname{area}(S^2)$, so the average number of preimages is less than 4. By Sard's theorem we may pick a regular value $y \in S^2$ of n with $|n^{-1}(y)| \leq 3$.

Now consider the height function $h(s) = \langle y, \alpha(s) \rangle$. We have $h'(s) = \langle y, T(s) \rangle$, and one checks from the formula for n that $h'(s) = 0$ precisely when the two points of the tube over s in the $\pm$ directions of the plane orthogonal to $T(s)$ hit y ; so h has at most three critical points. But h is a smooth function on a circle, so it has at least one maximum and one minimum, and its critical points alternate between maxima and minima, hence are even in number. So h has exactly two critical points, one maximum and one minimum.

Therefore α consists of exactly two arcs joining the maximum to the minimum, on each of which h is strictly monotone. For each value c strictly between the minimum and the maximum, the plane $h = c$ meets α in exactly two points; joining them by a straight segment and letting c vary sweeps out an embedded disc with boundary α . Hence α is unknotted. $\square$

Remark. Milnor proved the stronger statement that the total curvature of a knotted curve is *strictly* greater than 4π , and that 4π is the best possible bound: a suitably flattened trefoil has total curvature as close to 4π as we please.

§29.10 Abstract Surfaces

So far, the geometry of our surfaces has been induced by the space that surface has been embedded in (which has so far been $\mathbb{R}^3$). Of course, it's possible to do geometry on surfaces that do not arise in this way. One way of specifying the geometry of a surface is with a *Riemannian metric*.

The point of doing this is not merely generality. We saw that constant curvature $K < 0$ is realised locally by pieces of surfaces in $\mathbb{R}^3$, but Hilbert's theorem says that no *complete* surface of constant negative curvature exists in $\mathbb{R}^3$. The hyperbolic plane, one of the three classical two-dimensional geometries, therefore only exists as an abstract surface – and once we allow abstract surfaces, it is easy to write down.

§29.10.1 Riemannian Metrics

Definition 29.10.1 (Riemannian Metric)

Let $V \subseteq \mathbb{R}^2$ be an open set. An (abstract) **Riemannian metric** on V is a smooth

map from V to the set of positive definite symmetric bilinear forms, given by

$$p \mapsto \begin{pmatrix} E(p) & F(p) \\ F(p) & G(p) \end{pmatrix},$$

with $E, G, EG - F^2 > 0$.

If w is a tangent vector at $p \in V$, we can compute its infinitesimal length by

$$\|w\|^2 = w^\top \begin{pmatrix} E(p) & F(p) \\ F(p) & G(p) \end{pmatrix} w.$$

So, if $\gamma : [a, b] \rightarrow V$ is smooth,

$$L(\gamma) = \int_a^b \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2} dt,$$

where $\gamma(t) = (u(t), v(t))$. Areas are computed by $\int \sqrt{EG - F^2} du dv$, exactly as before.

Definition 29.10.2 (Abstract Riemannian Surface)

An **abstract Riemannian surface** is an abstract smooth surface Σ together with a Riemannian metric in each chart, such that the metrics agree on overlaps: if φ is a transition map then

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = (D\varphi)^\top \begin{pmatrix} E & F \\ F & G \end{pmatrix} (D\varphi).$$

Everything intrinsic that we have done goes through unchanged: the Christoffel symbols were defined purely in terms of E, F, G , hence so were the covariant derivative, parallel transport, the geodesic equations, and – by the Theorema Egregium – the Gaussian curvature, which we now simply *define* by the Gauss equation. Geodesic polar coordinates, the local and global Gauss–Bonnet theorems, and all their corollaries hold verbatim. What we lose is the second fundamental form, the mean curvature and the Gauss map, all of which needed an ambient space.

Example 29.10.3 (Surfaces in $\mathbb{R}^3$)

A smooth surface in $\mathbb{R}^3$ with its first fundamental form is an abstract Riemannian surface, and the transformation rule above is exactly how E, F, G transform.

Example 29.10.4 (The Flat Torus)

Take $\mathbb{R}^2$ with the Euclidean metric $du^2 + dv^2$ and quotient by the lattice $\mathbb{Z}^2$. The result is a torus with a flat metric, $K \equiv 0$, which is consistent with Gauss–Bonnet since $\chi = 0$. This surface is *not* isometric to any torus of revolution in $\mathbb{R}^3$, since we showed such a torus has points of positive and of negative curvature. Indeed, no compact flat surface embeds in $\mathbb{R}^3$ at all, since compact surfaces in $\mathbb{R}^3$ have an elliptic point.

§29.10.2 The Length Metric

An abstract Riemannian surface carries a genuine metric space structure, obtained in the obvious way.

Definition 29.10.5 (Length Metric)

Let Σ be a connected abstract Riemannian surface. For $p, q \in \Sigma$ define

$$d(p, q) = \inf \{L(\gamma) : \gamma \text{ a piecewise smooth curve in } \Sigma \text{ from } p \text{ to } q\}.$$

Proposition 29.10.6

d is a metric on Σ , and it induces the original topology.

Proof. Symmetry and the triangle inequality are clear from concatenating and reversing curves, and $d(p, p) = 0$. The only issue is that $d(p, q) > 0$ for $p \neq q$. Take geodesic polar coordinates about p on a normal neighbourhood of radius ε small enough not to contain q . Any curve from p to q must cross every geodesic circle about p of radius less than ε , and by the argument used to prove that geodesics locally minimise length, such a curve has length at least ε . So $d(p, q) \geq \varepsilon > 0$. The same argument shows the metric balls of small radius about p are exactly the images under $\exp_p$ of the Euclidean balls, so the topologies agree. $\square$

Remark. A curve realising the infimum, if one exists, is necessarily a geodesic; but such a curve need not exist – delete a point from the plane and there is no shortest path between two points on opposite sides of the hole. The Hopf–Rinow theorem says that this is the only obstruction: if (Σ, d) is complete as a metric space then any two points are joined by a minimising geodesic.

§29.10.3 The Hyperbolic Plane

The simplest abstract Riemannian surface which is not a surface in $\mathbb{R}^3$ is also the most important.

Definition 29.10.7 (The Upper Half-Plane Model)

The **hyperbolic plane** is the upper half plane

$$\mathbb{H} = \{(u, v) \in \mathbb{R}^2 : v > 0\}$$

equipped with the Riemannian metric

$$I = \frac{du^2 + dv^2}{v^2}.$$

This is an isothermal metric with conformal factor $\lambda = 1/v$, so the corollary to the Gauss equation computes its curvature instantly.

Proposition 29.10.8

The hyperbolic plane has constant curvature $K \equiv -1$.

Proof. With $\lambda = 1/v$ we have $\log \lambda = -\log v$, so $\nabla^2 \log \lambda = -\partial_v^2 \log v = 1/v^2$. Hence

$$K = -\frac{1}{\lambda^2} \nabla^2 (\log \lambda) = -v^2 \cdot \frac{1}{v^2} = -1. \quad \square$$

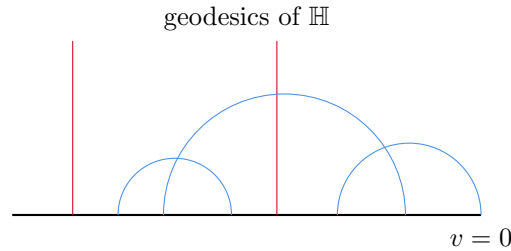
Proposition 29.10.9

The geodesics of $\mathbb{H}$ are the vertical half-lines and the semicircles orthogonal to the real axis.

Proof. Here $E = G = 1/v^2$ and $F = 0$, and the geodesic equations read

$$\frac{d}{dt} \left(\frac{\dot{u}}{v^2} \right) = 0, \quad \frac{d}{dt} \left(\frac{\dot{v}}{v^2} \right) = -\frac{\dot{u}^2 + \dot{v}^2}{v^3}.$$

The vertical line $u \equiv u_0$, $v = e^t$ satisfies both, so vertical lines are geodesics. For the general case, one can either integrate directly, or – more slickly – observe that the Möbius maps $z \mapsto (az + b)/(cz + d)$ with a, b, c, d real and $ad - bc > 0$ act on $\mathbb{H}$ by isometries (a computation with $\text{Im} \left(\frac{az+b}{cz+d} \right) = \frac{\text{Im } z}{|cz+d|^2}$), that this action is transitive on pairs (point, direction), and that such maps take lines and circles to lines and circles preserving angles. Since a vertical line is a geodesic and the images of vertical lines under this group are exactly the vertical lines and the semicircles meeting $\mathbb{R}$ at right angles, these are all the geodesics. $\square$



Theorem 29.10.10 (Area of a Hyperbolic Triangle)

A hyperbolic triangle with interior angles α, β, γ has area $\pi - \alpha - \beta - \gamma$.

Proof. Apply local Gauss–Bonnet to the triangle, whose sides are geodesics so $\kappa_g = 0$, with exterior angles $\pi - \alpha$ and so on:

$$\int_T K \, dA = \alpha + \beta + \gamma - \pi.$$

But $K \equiv -1$, so the left hand side is $-\text{area}(T)$, giving the result. $\square$

In particular the angle sum of a hyperbolic triangle is always less than π , and no hyperbolic triangle has area more than π . Compare the sphere, where $K \equiv 1$ and the area is the angle *excess*, and the plane, where $K \equiv 0$ and the angle sum is exactly π . All three statements are the same statement, with the sign of the curvature switched.

Remark. By Minding's theorem, any surface of constant curvature -1 is locally isometric to $\mathbb{H}$. The pseudosphere – the surface of revolution generated by the tractrix – is such a surface in $\mathbb{R}^3$, but it is incomplete: it has a singular circle where it cannot be continued. Hilbert's theorem says this is unavoidable, and that no complete surface of constant negative curvature exists in $\mathbb{R}^3$. So the hyperbolic plane is an object we can only reach by allowing abstract metrics – which is a fair summary of why one bothers with them.

Where Next

We have really only scratched the surface. The obvious next step is to allow more dimensions: the Christoffel symbols, the covariant derivative and parallel transport all generalise to n -dimensional Riemannian manifolds, with the Gaussian curvature replaced by the Riemann curvature tensor, and this is the subject of Part III Differential Geometry and of General Relativity. In another direction, the Gauss–Bonnet theorem is the first instance of an index theorem, expressing an analytic invariant in terms of a topological one; the Atiyah–Singer index theorem is the vast generalisation. And the complex-analytic side of minimal surface theory opens into the theory of Riemann surfaces. All of these begin exactly where we have stopped.

30 Galois Theory

Galois theory is the study of the symmetries of the solutions of polynomial equations. The central discovery of the subject – and it really is one of the most beautiful discoveries in all of mathematics – is that the messy analytic question ‘can I write down the roots of this polynomial using $+$, $-$, $\times$, $\div$ and $\sqrt[n]{}$?’ is exactly equivalent to a clean, finite, combinatorial question about a group. Galois’ insight was to stop looking at the roots one at a time, and instead to look at the group of symmetries that permutes them without disturbing any of the algebra they satisfy.

Modern treatments phrase all of this in terms of *field extensions*, and the heart of the course is a dictionary – the *Galois correspondence* – translating between the subfields of a field extension and the subgroups of its symmetry group. Once we have that dictionary in hand, an enormous amount falls out: the impossibility of trisecting an angle, the classification of finite fields, the constructibility of the regular 17-gon, the fundamental theorem of algebra, and of course the insolubility of the general quintic.

This article constitutes my notes for the ‘Galois Theory’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§30.1 Introduction

§30.1.1 A Bit of History

Historically the subject arose from looking at solutions to polynomial equations in one variable over $\mathbb{C}$. The question arose as to whether polynomials could be solved by a formula involving the coefficients and taking roots (‘soluble by radicals’). From school, we know that we can solve quadratics in this way, with

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

For a long time it has been known that cubics and quartics have similar, albeit rather more complicated, formulae for their roots. These were studied by Lagrange in the 1770s, who was the first to notice that what really mattered was how certain expressions in the roots behaved when the roots were permuted.

In 1799 Ruffini claimed that there was no general formula in the case of quintics, however there was a gap in his proof. It wasn’t until Abel in 1824 that there was a complete accepted proof, using permutations of roots of polynomials. This was the start of group theory.

Galois gave the first explanation of *when* a quintic was soluble by radicals or not in 1831, using the structure of a group of permutations of the roots, and in particular the importance of normal subgroups. Galois’ work was not known until Liouville published his papers in 1846. Liouville realised the connection with Cauchy’s work on permutations, but didn’t realise the importance of the group-theoretic structure, and in fact few of the contemporary mathematicians did so.

Galois entered his papers for various competitions and also for the entrance process for the École Polytechnique in Paris. He didn't get in however, and went to another university in Paris, where he got involved in politics and was eventually killed in a duel at the age of twenty. The night before the duel he left a $6\frac{1}{2}$ page manuscript setting out his ideas about the future development of the theory. His papers have been carefully studied by Peter Neumann, in *The Mathematical Writings of Évariste Galois* (History of European Mathematics, European Mathematical Society).

There is a historical introduction in I. Stewart's *Galois Theory*, which is very readable but doesn't quite cover the syllabus. Other books worth looking at are Artin's *Galois Theory*, Van der Waerden's *Modern Algebra*, Lang's *Algebra*, and Kaplansky's *Fields and Rings*.

§30.1.2 Solving Cubics and Quartics

Before we get going, it is worth reminding ourselves what these classical formulae actually look like, because the shape of the answer will turn out to be a very good guide to the shape of the theory.

Consider a cubic $f(x) = x^3 + ax^2 + bx + c$. Substituting $x \mapsto x - a/3$ (which is legitimate provided $\text{char } K \neq 3$) we may assume $a = 0$, so

$$f(x) = x^3 + px + q.$$

Cardano's trick is to write a root as $x = u + v$. Then

$$x^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvx,$$

so x is a root of f provided that $u^3 + v^3 = -q$ and $3uv = -p$. But then u^3 and v^3 are two numbers with known sum $-q$ and known product $-p^3/27$, so they are the roots of a *quadratic*,

$$t^2 + qt - \frac{p^3}{27} = 0.$$

Solving that quadratic and taking cube roots gives the roots of f . What we should notice here is the intrusion of an auxiliary quadratic – the **resolvent** – and the fact that solving a cubic requires taking a square root and then a cube root, in that order. The quartic behaves similarly: an auxiliary *cubic* appears, and the whole thing unravels into a chain of extractions of roots. The whole point of the course will be to explain *why* such a chain must exist for degrees 2, 3, 4, and why no such chain can exist for degree 5.

§30.1.3 Notation and Revision

Everything we do will take place inside rings and fields, and we will lean quite heavily on the Groups, Rings and Modules course. Let us fix our conventions and remind ourselves of what we need.

In this course a **ring** always means a commutative ring with a 1, and ring homomorphisms are required to send $1 \mapsto 1$. A **field** is a ring in which every non-zero element is a unit, for instance $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for p prime. For a ring R we write $R^\times$ for its group of units, so if K is a field then $K^\times = K \setminus \{0\}$.

We write $R[x]$ for the ring of polynomials in an indeterminate x with coefficients in R , that is, the set of formal sums $\sum_{i \geq 0} r_i x^i$ with all but finitely many r_i equal to zero.¹ If K is a field, then $K(x)$ denotes the **field of fractions** of $K[x]$, whose elements are equivalence classes of formal quotients f/g with $g \neq 0$, where $f/g = r/s$ exactly when $fs = gr$. Its elements are called **rational functions**.

The single most important fact we import from Groups, Rings and Modules is that polynomial rings over a field are as well-behaved as we could possibly hope.

Theorem 30.1.1 (Division Algorithm)

Let K be a field and let $a, b \in K[x]$ with $b \neq 0$. Then there exist $q, r \in K[x]$ with

$$a = qb + r, \quad \deg r < \deg b.$$

In other words, $K[x]$ is a Euclidean domain.

Everything else we need is a formal consequence.

Corollary 30.1.2

Let K be a field. Then:

- (i) $K[x]$ is a principal ideal domain;
- (ii) $K[x]$ is a unique factorisation domain;
- (iii) for $f \in K[x]$ non-constant, f is irreducible $\iff f$ is prime $\iff (f)$ is maximal $\iff K[x]/(f)$ is a field;
- (iv) for $a, b \in K[x]$, the ideal $(a) + (b)$ equals (g) for some g , and this g is a greatest common divisor of a and b , unique up to a unit. In particular $\gcd(a, b) = pa + qb$ for some $p, q \in K[x]$;
- (v) if $f \in K[x] \setminus \{0\}$ then f has at most $\deg f$ roots in K .^a

^aAs usual, α is a **root** of f if $f(\alpha) = 0$, and by the division algorithm this happens if and only if $(x - \alpha) \mid f$.

Proof. All of these were proved in Groups, Rings and Modules; (i) follows by taking a non-zero element of least degree in an ideal and dividing, (ii) follows since every PID is a UFD, (iii) is the standard characterisation of irreducibles in a PID, (iv) is immediate from (i), and (v) follows from (ii) together with the factor theorem. $\square$

The point of (iii) that we will use again and again is the following: *if $f \in K[x]$ is irreducible, then $K[x]/(f)$ is a field*. This is the engine that manufactures new fields out of old ones, and it is where the whole subject starts.

We will also want the standard irreducibility criteria from Groups, Rings and Modules, namely Gauss's lemma (a primitive polynomial in $\mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$) and Eisenstein's criterion.

¹It is a good exercise to check that if R is an integral domain then so is $R[x]$, and that in that case $\deg(fg) = \deg f + \deg g$.

§30.2 Field Extensions

§30.2.1 Extensions and Degree

The basic object of the course is not a field but a *pair* of fields, one sitting inside the other.

Definition 30.2.1 (Field Extension)

If $K \subseteq L$ where L is a field and K is a subring of L which is itself a field, then we say L is an **extension** of K , and write L/K .^a

^aThe notation L/K has nothing whatsoever to do with quotients. It is unfortunate, but universal.

For example $\mathbb{C}/\mathbb{R}$, $\mathbb{R}/\mathbb{Q}$ and $K(x)/K$ are all field extensions, as is L/K where $L = K[x]/(f)$ for $f \in K[x]$ irreducible (here we regard K as sitting inside L as the constant polynomials).

Now here is the observation that makes the whole theory work. If L/K is a field extension, then L is in particular a vector space over K : we can add elements of L , and we can multiply them by scalars from K , and all of the vector space axioms are instances of the field axioms in L . So we may bring the whole of linear algebra to bear.

Definition 30.2.2 (Degree)

The **degree** of a field extension L/K is

$$[L : K] = \dim_K L,$$

the dimension of L as a K -vector space. If $[L : K] < \infty$ we say L/K is a **finite** extension, and otherwise an **infinite** extension.

For instance $[\mathbb{C} : \mathbb{R}] = 2$, since $\{1, i\}$ is an $\mathbb{R}$ -basis of $\mathbb{C}$. On the other hand $[\mathbb{R} : \mathbb{Q}] = \infty$, and $[K(x) : K] = \infty$ since $1, x, x^2, \dots$ are linearly independent over K .

Before going further, let us record the smallest subfield that any field contains.

Definition 30.2.3 (Characteristic)

Let K be a field. There is a unique ring homomorphism $\mathbb{Z} \rightarrow K$, sending $1 \mapsto 1$. If it is injective then K contains a copy of $\mathbb{Q}$ and we say K has **characteristic** zero, written $\text{char } K = 0$. Otherwise its kernel is a non-zero prime ideal (p) , so that $\underbrace{1 + 1 + \dots + 1}_p = 0$ in K ; then K contains a copy of $\mathbb{F}_p$ and we say $\text{char } K = p$.

The kernel must be prime because its quotient embeds in the integral domain K . In either case the image of $\mathbb{Z} \rightarrow K$ generates a smallest subfield of K , called the **prime subfield**, which is $\mathbb{Q}$ or $\mathbb{F}_p$ accordingly. For instance $\text{char } \mathbb{F}_p(x) = p$, since $p \cdot 1 = 0$ already in $\mathbb{F}_p$.

We say K is a **finite field** if $\#K < \infty$, where $\#K$ denotes the number of elements of K . Already we can say something about what sizes are possible.

Lemma 30.2.4

If F is a finite field then $\text{char } F = p$ for some prime p , and $\#F = p^n$ where $n = [F : \mathbb{F}_p]$.

Proof. If $\#F < \infty$ then the map $\mathbb{Z} \rightarrow F$ cannot be injective, so $\text{char } F = p$ for some prime p and $\mathbb{F}_p \subseteq F$. Then F is a vector space over $\mathbb{F}_p$, necessarily finite-dimensional, say of dimension n . Choosing a basis gives an isomorphism of $\mathbb{F}_p$ -vector spaces $F \cong \mathbb{F}_p^n$, and hence $\#F = p^n$. $\square$

We will see later that, conversely, for every prime p and every $n \geq 1$ there is a field with p^n elements, and that it is unique up to isomorphism.

§30.2.2 The Tower Law

Degrees behave exactly as you would want them to in a tower of extensions – they multiply. This little fact is used constantly, often several times on the same page.

Theorem 30.2.5 (Tower Law)

Let $M/L/K$ be field extensions. Then M/K is finite if and only if both M/L and L/K are finite, and in that case

$$[M : K] = [M : L][L : K].$$

It is no harder, and rather more useful, to prove the following slightly more general statement about vector spaces.

Proposition 30.2.6

Let L/K be a finite field extension and let V be a vector space over L . Then V is finite dimensional over L if and only if it is finite dimensional over K , and in that case

$$\dim_K V = \dim_L V \cdot [L : K].$$

Proof. Suppose $\dim_L V = d < \infty$, and take an L -basis $\alpha_1, \dots, \alpha_d$ of V and a K -basis $\ell_1, \dots, \ell_n$ of L . We claim that the nd products

$$\{\ell_i \alpha_j : 1 \leq i \leq n, 1 \leq j \leq d\}$$

form a K -basis of V .

They span: any $v \in V$ can be written $v = \sum_j \mu_j \alpha_j$ with $\mu_j \in L$, and each $\mu_j = \sum_i \lambda_{ij} \ell_i$ with $\lambda_{ij} \in K$, so $v = \sum_{i,j} \lambda_{ij} \ell_i \alpha_j$.

They are independent: if $\sum_{i,j} \lambda_{ij} \ell_i \alpha_j = 0$ with $\lambda_{ij} \in K$, then collecting terms gives $\sum_j (\sum_i \lambda_{ij} \ell_i) \alpha_j = 0$, and the inner brackets lie in L , so by L -independence of the α_j each $\sum_i \lambda_{ij} \ell_i = 0$, and then by K -independence of the ℓ_i each $\lambda_{ij} = 0$.

Hence $\dim_K V = nd = \dim_L V \cdot [L : K] < \infty$.

Conversely if $\dim_K V < \infty$ then certainly $\dim_L V < \infty$, since a K -spanning set is

in particular an L -spanning set. $\square$

Proof of the Tower Law. If L/K is infinite then $L \subseteq M$ forces M/K to be infinite, and if M/L is infinite then so is M/K by the same argument applied to a set of elements of M independent over L . Otherwise apply Proposition 30.2.6 with $V = M$. $\square$

It is worth pausing to appreciate how much arithmetic information the tower law encodes. If we can find a subfield of L containing K , its degree over K must divide $[L : K]$; contrapositively, if $[L : K]$ is prime then there are no intermediate fields at all. This kind of numerical reasoning is remarkably powerful, and we will use it immediately.

§30.2.3 Algebraic and Transcendental Elements

Given a field extension L/K and an element $\alpha \in L$, we write $K[\alpha]$ for the smallest subring of L containing both K and α , and $K(\alpha)$ for the smallest subfield. Concretely,

$$K[\alpha] = \left\{ \sum_{i=0}^N r_i \alpha^i : r_i \in K, N \in \mathbb{N} \right\}, \quad K(\alpha) = \{f/g : f, g \in K[\alpha], g \neq 0\}.$$

More generally we write $K(\alpha_1, \dots, \alpha_n)$ for the smallest subfield of L containing K and all the α_i ; note that $K(\alpha, \beta) = (K(\alpha))(\beta)$, so we can adjoin elements one at a time.

Example 30.2.7

Take $L = \mathbb{C}$, $K = \mathbb{Q}$ and $\alpha = i$. Then

$$\mathbb{Q}[i] = \{a_0 + a_1 i + a_2 i^2 + \dots + a_n i^n : a_j \in \mathbb{Q}\} = \{a + bi : a, b \in \mathbb{Q}\},$$

since $i^2 = -1$ lets us collapse all the higher powers. This is already a field, because $(a + bi)^{-1} = (a - bi)/(a^2 + b^2)$, and so $\mathbb{Q}(i) = \mathbb{Q}[i]$.

That last coincidence – that $K[\alpha]$ happened to be a field already – is not an accident, and understanding exactly when it happens is the content of this subsection. The right way to see it is through the evaluation map.

Fix $\alpha \in L$ and consider the ring homomorphism

$$\phi : K[x] \rightarrow L, \quad x \mapsto \alpha,$$

which is to say $\phi(\sum r_i x^i) = \sum r_i \alpha^i$. Its image is exactly $K[\alpha]$. There are then precisely two possibilities, according to whether ϕ is injective.

Definition 30.2.8 (Algebraic and Transcendental)

With notation as above, we say α is **transcendental** over K if ϕ is injective, and **algebraic** over K if ϕ is not injective. Equivalently, α is algebraic over K if and only if $f(\alpha) = 0$ for some non-zero $f \in K[x]$.

Suppose α is algebraic. Then $\ker \phi$ is a non-zero ideal of the PID $K[x]$, so $\ker \phi = (f)$ for some non-zero f , unique up to a unit; normalising so that f is monic pins it down completely.

Definition 30.2.9 (Minimal Polynomial)

Let $\alpha \in L$ be algebraic over K . The **minimal polynomial** of α over K is the unique monic generator f of $\ker(\phi : K[x] \rightarrow L)$. Equivalently, it is the monic polynomial of least degree in $K[x]$ having α as a root, and it divides every polynomial in $K[x]$ with α as a root. We write $\deg_K \alpha = \deg f$ for the **degree** of α over K .

Lemma 30.2.10

The minimal polynomial of α over K is irreducible in $K[x]$.

Proof. Write f for the minimal polynomial, and suppose $f = gh$ with $\deg g, \deg h < \deg f$. Then $g(\alpha)h(\alpha) = f(\alpha) = 0$ in L , and L is a field and hence an integral domain, so $g(\alpha) = 0$ or $h(\alpha) = 0$. Either way we have found a non-zero polynomial of smaller degree than f vanishing at α , contradicting minimality. $\square$

Notice a small but useful consequence: since f is irreducible and $\deg f \geq 1$, its constant term a_0 is non-zero (else $x \mid f$). Writing $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ and rearranging $f(\alpha) = 0$ gives

$$\alpha \cdot \left(-\frac{1}{a_0} (\alpha^{n-1} + a_{n-1}\alpha^{n-2} + \cdots + a_1) \right) = 1,$$

so $\alpha^{-1} \in K[\alpha]$. This is the first hint that $K[\alpha]$ is a field whenever α is algebraic; here is the full statement.

Proposition 30.2.11

Let L/K be a field extension and $\alpha \in L$. The following are equivalent:

- (i) α is algebraic over K ;
- (ii) $[K(\alpha) : K] < \infty$;
- (iii) $\dim_K K[\alpha] < \infty$;
- (iv) $K[\alpha] = K(\alpha)$;
- (v) $K[\alpha]$ is a field.

When these hold, $[K(\alpha) : K] = \deg_K \alpha$, and $1, \alpha, \dots, \alpha^{d-1}$ is a K -basis of $K(\alpha)$, where $d = \deg_K \alpha$.

Proof. Suppose (i), and let f be the minimal polynomial of α , of degree d . Since ϕ has kernel (f) , the first isomorphism theorem gives $K[\alpha] \cong K[x]/(f)$, which is a field because f is irreducible; this is (v), and (iv) follows since $K(\alpha)$ is the smallest field containing K and α . Moreover $1, \alpha, \dots, \alpha^{d-1}$ span $K[\alpha]$ (divide by f and take the remainder) and are linearly independent (a linear relation would be a polynomial of degree $< d$ killing α), so $\dim_K K[\alpha] = d$, giving (iii) and (ii).

Conversely suppose (iii), so $\dim_K K[\alpha] = m < \infty$. Then $1, \alpha, \dots, \alpha^m$ are $m+1$ elements of an m -dimensional space, hence linearly dependent over K , and a non-trivial dependence is precisely a non-zero polynomial vanishing at α . So (iii) $\Rightarrow$ (i). Since (ii) $\Rightarrow$ (iii) (as $K[\alpha] \subseteq K(\alpha)$) and (v) $\Rightarrow$ (iv) $\Rightarrow$ (i) (if $K[\alpha]$ is a field then

$\alpha^{-1} \in K[\alpha]$, so $\alpha^{-1} = \sum r_i \alpha^i$ and hence α satisfies $\sum r_i x^{i+1} - 1 = 0$, all five are equivalent. $\square$

The transcendental case is much less interesting, precisely because there is only one of them.

Proposition 30.2.12

α is transcendental over K if and only if $\phi : K[x] \rightarrow K[\alpha]$ is an isomorphism, in which case it extends to an isomorphism $K(x) \rightarrow K(\alpha)$. In particular, any two simple transcendental extensions of K are K -isomorphic, both being isomorphic to the rational function field $K(x)$.

Proof. The map ϕ is surjective onto $K[\alpha]$ by definition, so it is an isomorphism precisely when it is injective, which is the definition of transcendence. An isomorphism of integral domains extends uniquely to their fields of fractions, giving $K(x) \cong K(\alpha)$. $\square$

Remark. Both ‘algebraic’ and ‘transcendental’, and the minimal polynomial itself, depend very much on the base field K , and it is worth being careful about this.

- (i) $2\pi i \in \mathbb{C}$ is algebraic over $\mathbb{R}$, with minimal polynomial $x^2 + 4\pi^2$, but it is transcendental over $\mathbb{Q}$ (this is Lindemann’s theorem, and is decidedly not easy).
- (ii) Let $\alpha = \sqrt{i} = \frac{\sqrt{2}}{2}(1 + i)$. Over $\mathbb{Q}$ the minimal polynomial of α is $x^4 + 1$, whereas over $\mathbb{Q}(i)$ it is $x^2 - i$. Note that

$$[\mathbb{Q}(i) : \mathbb{Q}] = 2, \quad [\mathbb{Q}(\alpha) : \mathbb{Q}(i)] = 2, \quad [\mathbb{Q}(\alpha) : \mathbb{Q}] = 4,$$

which is a first illustration of the tower law in action.

§30.2.4 Working with Degrees

Combining the tower law with Proposition 30.2.11 gives a very effective toolkit. Here is the basic divisibility statement.

Corollary 30.2.13

If L/K is a finite extension and $\alpha \in L$, then α is algebraic over K and $\deg_K \alpha \mid [L : K]$.

Proof. Since $K \subseteq K(\alpha) \subseteq L$ and $[L : K] < \infty$, we get $[K(\alpha) : K] < \infty$, so α is algebraic. The tower law applied to $L/K(\alpha)/K$ gives $[L : K] = [L : K(\alpha)] \cdot \deg_K \alpha$. $\square$

The following picture is the one to keep in mind: an element $\alpha \in L$ interposes an intermediate field, and the two degrees multiply to give the degree of the whole tower.

$$[L : K] \left\{ \begin{array}{l} L \\ | \\ [L : K(\alpha)] \\ K(\alpha) \\ | \\ \deg_K \alpha \\ K \end{array} \right.$$

Example 30.2.14

Let us see the toolkit at work.

- (i) Suppose $[L : K] = p$ is prime. Then for every $\alpha \in L \setminus K$ we have $L = K(\alpha)$. Indeed $[K(\alpha) : K]$ divides p , so is 1 or p ; it is not 1 since $\alpha \notin K$; so $[K(\alpha) : K] = p$ and hence $[L : K(\alpha)] = 1$.
- (ii) Every irreducible polynomial $f \in \mathbb{R}[x]$ has degree 1 or 2. Granting that $\mathbb{C}$ is algebraically closed, f has a root $\alpha \in \mathbb{C}$, and $\deg f = [\mathbb{R}(\alpha) : \mathbb{R}]$ divides $[\mathbb{C} : \mathbb{R}] = 2$.
- (iii) Let $L = \mathbb{Q}(\sqrt[3]{2}, \sqrt[4]{5})$. We claim $[L : \mathbb{Q}] = 12$. Certainly $L \supseteq \mathbb{Q}(\sqrt[3]{2})$, and $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$ and hence $3 \mid [L : \mathbb{Q}]$. Similarly $x^4 - 5$ is irreducible by Eisenstein at 5, so $4 \mid [L : \mathbb{Q}]$, and therefore $12 \mid [L : \mathbb{Q}]$. On the other hand $\sqrt[4]{5}$ satisfies $x^4 - 5$ over $\mathbb{Q}(\sqrt[3]{2})$, so $[L : \mathbb{Q}(\sqrt[3]{2})] \leq 4$ and hence $[L : \mathbb{Q}] \leq 12$. So $[L : \mathbb{Q}] = 12$.
- (iv) Let p be an odd prime, let $\omega = e^{2\pi i/p}$ and let $\alpha = \omega + \omega^{-1} = 2\cos(2\pi/p)$. What is $\deg_{\mathbb{Q}} \alpha$? Now ω is a root of

$$f(x) = 1 + x + \cdots + x^{p-1} = \frac{x^p - 1}{x - 1},$$

which is irreducible over $\mathbb{Q}$: applying Eisenstein at p to $f(x+1)$ does the job, since $f(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1}x^{p-2} + \cdots + \binom{p}{p-1}$. Hence $[\mathbb{Q}(\omega) : \mathbb{Q}] = p-1$. Clearly $\alpha \in \mathbb{Q}(\omega)$, so $\mathbb{Q}(\omega)/\mathbb{Q}(\alpha)/\mathbb{Q}$ is a tower and $\deg_{\mathbb{Q}} \alpha \mid p-1$. Now $\alpha\omega = \omega^2 + 1$, so ω is a root of $x^2 - \alpha x + 1 \in \mathbb{Q}(\alpha)[x]$, giving $[\mathbb{Q}(\omega) : \mathbb{Q}(\alpha)] \leq 2$. It is not 1, since $\alpha \in \mathbb{R}$ but $\omega \notin \mathbb{R}$, so $[\mathbb{Q}(\omega) : \mathbb{Q}(\alpha)] = 2$ and hence

$$\deg_{\mathbb{Q}} \alpha = \frac{p-1}{2}.$$

Next, a fact that ought to be surprising: the algebraic elements are closed under all the field operations, even though there is no obvious way to write down the minimal polynomial of a sum.

Corollary 30.2.15

Let L/K be a field extension.

- (i) Elements $\alpha_1, \dots, \alpha_n \in L$ are all algebraic over K if and only if $[K(\alpha_1, \dots, \alpha_n) : K] < \infty$.
- (ii) If $\alpha, \beta \in L$ are algebraic over K then so are $\alpha \pm \beta$, $\alpha\beta$ and α/β (the last provided $\beta \neq 0$).

Proof. For (i), suppose first that $[K(\alpha_1, \dots, \alpha_n) : K] < \infty$. Each α_i lies in this finite extension, so is algebraic over K by Corollary 30.2.13. Conversely, suppose each α_i is algebraic over K . Then α_n is in particular algebraic over the larger field $K(\alpha_1, \dots, \alpha_{n-1})$ (its minimal polynomial over K is still a polynomial there), so

$$[K(\alpha_1, \dots, \alpha_n) : K(\alpha_1, \dots, \alpha_{n-1})] < \infty,$$

and the result follows by induction and the tower law, as a product of finitely many finite numbers.

For (ii), all four elements lie in $K(\alpha, \beta)$, which is a finite extension of K by (i), so all four are algebraic by Corollary 30.2.13. $\square$

Corollary 30.2.16

The elements of L which are algebraic over K form a subfield of L .

It is instructive to see how one might find an explicit polynomial in a simple case – this is the sort of thing that Galois theory will eventually let us do systematically.

Example 30.2.17

Let $a, b \in K$ and let $\alpha = \sqrt{a}$, $\beta = \sqrt{b}$ in some extension. Put $\gamma = \alpha + \beta$. We compute, using $\alpha^2 = a$ and $\beta^2 = b$:

$$\begin{aligned}\gamma^2 &= a + b + 2\alpha\beta, \\ \gamma^4 &= (a + b)^2 + 4\alpha\beta(a + b) + 4ab,\end{aligned}$$

and $2(a + b)\gamma^2 = 2(a + b)^2 + 4\alpha\beta(a + b)$. Subtracting kills the awkward $\alpha\beta$ term:

$$\gamma^4 - 2(a + b)\gamma^2 = 4ab - (a + b)^2 = -(a - b)^2.$$

So γ is a root of $x^4 - 2(a + b)x^2 + (a - b)^2$.

More generally, if $\deg_K \alpha = m$ and $\deg_{K(\alpha)} \beta = n$, then $K(\alpha, \beta)$ is spanned over K by the mn monomials $\alpha^i \beta^j$ with $0 \leq i < m$, $0 \leq j < n$. So for any $\gamma \in K(\alpha, \beta)$, the $mn + 1$ elements $1, \gamma, \dots, \gamma^{mn}$ must be linearly dependent over K , which produces a polynomial of degree at most mn satisfied by γ . Be warned that the polynomial obtained this way is in general *not* the minimal polynomial.

Exercise 30.2.18. Show that if $m, n, mn \in \mathbb{Q}$ are all non-squares, then $[\mathbb{Q}(\sqrt{m} + \sqrt{n}) : \mathbb{Q}] = 4$.

§30.2.5 Algebraic Extensions

Definition 30.2.19 (Algebraic Extension)

An extension L/K is an **algebraic extension** if every $\alpha \in L$ is algebraic over K .

By Corollary 30.2.13, every finite extension is algebraic. The converse fails.

Example 30.2.20

Let $\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} : \alpha \text{ is algebraic over } \mathbb{Q}\}$, the field of **algebraic numbers** (a field, by Corollary 30.2.15). Then $\overline{\mathbb{Q}}/\mathbb{Q}$ is algebraic by construction, but it is not finite: $\sqrt[n]{2} \in \overline{\mathbb{Q}}$ has degree n over $\mathbb{Q}$ for every n , so $[\overline{\mathbb{Q}} : \mathbb{Q}] \geq n$ for all n .

Proposition 30.2.21 (i) Every finite extension is algebraic.

(ii) If $M/L/K$ are field extensions, then M/K is algebraic if and only if M/L and L/K are algebraic.

Proof. Part (i) is Corollary 30.2.13.

For (ii), suppose first M/K is algebraic. If $\alpha \in M$ then α is algebraic over K , hence over the larger field L ; and $L \subseteq M$, so every element of L is algebraic over K too.

Conversely suppose M/L and L/K are algebraic, and let $\alpha \in M$. Then α is algebraic over L , so

$$r_0 + r_1\alpha + \cdots + r_d\alpha^d = 0$$

for some $r_0, \dots, r_d \in L$ not all zero. Put $L_0 = K(r_0, \dots, r_d)$. Each r_j is algebraic over K since L/K is algebraic, so $[L_0 : K] < \infty$ by Corollary 30.2.15. Also α is algebraic over L_0 by the displayed equation, so $[L_0(\alpha) : L_0] < \infty$. By the tower law

$$[L_0(\alpha) : K] = [L_0(\alpha) : L_0][L_0 : K] < \infty,$$

and α lies in this finite extension of K , hence is algebraic over K . □

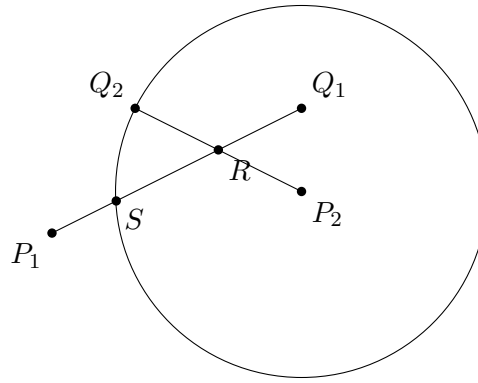
In other words, an extension L/K is algebraic exactly when L is a union of subfields each of which is finite over K .

§30.3 Ruler and Compass Constructions

This section is an interlude, but a very satisfying one. The Greeks asked a number of geometric questions which resisted attack for two thousand years: can we trisect a general angle with ruler and compass? Can we construct a square with the same area as a given circle? Can we construct a cube with double the volume of a given cube? With the tower law in hand, all three become easy.

§30.3.1 What Is Constructible?

We begin with two marked points in the plane, which we declare to be **constructed**. From a set of constructed points we may produce more, as follows: given constructed points we may draw the line through any two of them, and the circle centred at one of them passing through another, and then any point of intersection of two such lines, of two such circles, or of a line and a circle, is again constructed. A point is **constructible** if it can be obtained from the two initial points by finitely many such steps.



In the picture, R is the intersection of the lines P_1Q_1 and P_2Q_2 , and S is one of the two intersections of the line P_1Q_1 with the circle centred at P_2 passing through Q_2 ; both are therefore constructible from P_1, P_2, Q_1, Q_2 .

There are three standard constructions that we shall use freely, all of which are done in school.

- (i) Given a constructed point P and a constructed line ℓ , we may drop or erect the perpendicular to ℓ through P . (Draw a circle centred at P meeting ℓ in two points A, B ; then circles of equal radius centred at A and B meet in two points, and the line through those two points is the required perpendicular.)
- (ii) Given a constructed point P and a constructed line ℓ , we may draw the line through P parallel to ℓ : erect the perpendicular m to ℓ through P , then the perpendicular to m through P .
- (iii) Given constructed points a, b and a constructed point P on a constructed line ℓ , we may mark off the length $|ab|$ along ℓ starting from P : this is what a compass is for.

As a consequence of (i) and (ii) we may set up Cartesian coordinates, declaring our two initial points to be $(0, 0)$ and $(1, 0)$, and constructing the two coordinate axes.

Definition 30.3.1 (Constructible Number)

A real number λ is **constructible** if $|\lambda|$ is the distance between two constructible points.

Lemma 30.3.2

A point $P = (a, b)$ is constructible if and only if a and b are constructible real numbers.

Proof. If P is constructible, drop perpendiculars from P onto the two axes; the feet are constructible points, at distances $|a|$ and $|b|$ from the origin. Conversely, given constructible a and b , mark off a along the x -axis and b along the y -axis, erect perpendiculars there, and take their intersection. $\square$

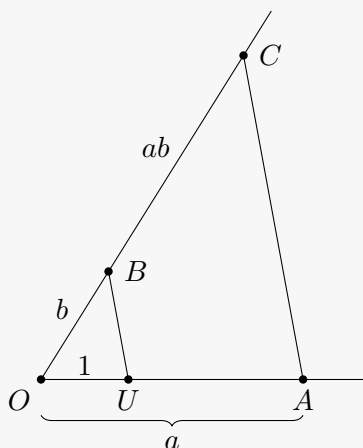
§30.3.2 Constructible Numbers Form a Field

Proposition 30.3.3

The set of constructible real numbers is a subfield of $\mathbb{R}$, closed moreover under taking square roots of positive elements.

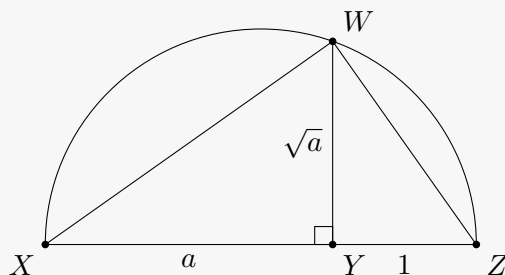
Proof. Let a, b be constructible. Then $a + b$ and $-a$ are constructible by marking off lengths along a line, so we have an additive subgroup containing $\mathbb{Q}$.

For products and quotients we use similar triangles. Draw two rays from a point O . Along the first mark points U and A at distances 1 and a ; along the second mark B at distance b . Draw the line UB , and then the line through A parallel to it, meeting the second ray at C . By similarity $|OC|/|OB| = |OA|/|OU|$, that is $|OC| = ab$.



Exactly the same picture read backwards gives a/b for $b \neq 0$: mark b and 1 on the first ray and a on the second.

Finally, suppose $a > 0$ is constructible. Mark off a segment of length a followed by a segment of length 1 along a line, giving a segment XZ of length $a + 1$ with the intermediate point Y . Draw the semicircle on XZ as diameter, and erect the perpendicular to XZ at Y , meeting the semicircle at W . The angle XWZ is a right angle (angle in a semicircle), so the triangles XYW and WYZ are similar, giving $|YW|^2 = |XY||YZ| = a$, so $|YW| = \sqrt{a}$.



□

That is the easy half. The point of the theory is the other half: nothing else is constructible.

Theorem 30.3.4

A real number λ is constructible if and only if there is a chain of subfields

$$\mathbb{Q} = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n = F \subseteq \mathbb{R}$$

with $\lambda \in F$, where for each j we have $F_{j+1} = F_j(\sqrt{r_j})$ for some $r_j \in F_j$ with $r_j > 0$.

Proof. If such a chain exists, then every element of F is constructible by the previous proposition (each stage only requires the field operations and one square root).

Conversely, suppose λ is constructible, via a sequence of constructions producing points $P_1, P_2, \dots, P_N$. We build the chain inductively, maintaining the invariant that the coordinates of $P_1, \dots, P_k$ all lie in $F_{j(k)}$ for some field in the chain. Initially the two starting points have coordinates in $\mathbb{Q} = F_0$. At each step, a new point arises in one of three ways.

- As the intersection of two lines through points with coordinates in F : solving two simultaneous linear equations with coefficients in F gives coordinates in F , so no extension is needed.
- As an intersection of a line and a circle, or of two circles, with all data in F . Subtracting the two circle equations reduces the second case to the first, and substituting the equation of the line into that of the circle gives a quadratic over F . Its roots lie in $F(\sqrt{r})$, where $r \in F$ is the discriminant, which is non-negative because the intersection is assumed to exist.

So each construction step either keeps us in the current field or extends it by one square root, and after finitely many steps we obtain the desired chain. $\square$

Corollary 30.3.5

If $\lambda \in \mathbb{R}$ is constructible then λ is algebraic over $\mathbb{Q}$, and $\deg_{\mathbb{Q}} \lambda = 2^m$ for some $m \geq 0$.

Proof. Each step of the chain has $[F_{j+1} : F_j] \in \{1, 2\}$, so by the tower law $[F : \mathbb{Q}] = 2^k$ for some k . Since $\lambda \in F$, Corollary 30.2.13 gives $\deg_{\mathbb{Q}} \lambda \mid 2^k$. $\square$

§30.3.3 The Classical Impossibilities**Example 30.3.6** (Duplicating the cube)

It is impossible to construct a cube of volume 2 from a cube of volume 1; that would require constructing $\sqrt[3]{2}$, and $\deg_{\mathbb{Q}} \sqrt[3]{2} = 3$, which is not a power of 2.

Example 30.3.7 (Trisecting the angle)

It is impossible to trisect a general angle. It suffices to exhibit one angle that cannot be trisected. We can certainly construct 60° , since $\cos 60^\circ = 1/2 \in \mathbb{Q}$. If we could trisect it we could construct 20° and hence $\alpha = \cos 20^\circ$. But the triple angle formula

$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ with $\theta = 20^\circ$ gives

$$8\alpha^3 - 6\alpha - 1 = 0.$$

The cubic $8x^3 - 6x - 1$ has no rational root (by the rational root test the only candidates are $\pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}$, and none work), so it is irreducible over $\mathbb{Q}$, and $\deg_{\mathbb{Q}} \alpha = 3$ is not a power of 2.

Example 30.3.8 (Squaring the circle)

It is impossible to construct a square of the same area as a given circle, since this requires constructing $\sqrt{\pi}$ and hence π , which is transcendental over $\mathbb{Q}$ by Lindemann's theorem – unless of course you are in Indiana in 1897.

Example 30.3.9 (Regular polygons)

A regular p -gon, for p an odd prime, is constructible only if $p - 1$ is a power of 2. Constructing the polygon is the same as constructing $\cos(2\pi/p)$, which we computed above to have degree $(p - 1)/2$ over $\mathbb{Q}$; for this to be a power of 2 we need $p - 1$ to be a power of 2.

Notice that Corollary 30.3.5 gives a necessary condition, not a sufficient one – being of degree 2^m over $\mathbb{Q}$ does not obviously produce a chain of quadratic extensions. Remarkably, once we have the Galois correspondence we will be able to prove the converse in the cases of interest, and settle exactly which regular polygons are constructible. That is Gauss's theorem, which we will reach in Section 30.10.

§30.4 Splitting Fields and Normal Extensions

§30.4.1 Adjoining a Root

We now start manufacturing extensions. Let $f \in K[x]$ be irreducible. Then (f) is maximal, so

$$K_f := K[x]/(f)$$

is a field, and K_f/K is a field extension (identifying K with the constant polynomials). Write $\alpha = x + (f)$ for the image of x . If $d = \deg f$, then $1, \alpha, \dots, \alpha^{d-1}$ is a K -basis of K_f , and

$$f(\alpha) = f(x) + (f) = 0 \text{ in } K_f,$$

so α is a root of f in K_f and $K_f = K(\alpha)$. We say that K_f is obtained from K by **adjoining a root of f** .

This is a genuinely magical construction: it produces a root of f out of nothing, by fiat. The price is that the root we get is not canonical – and understanding exactly how non-canonical it is turns out to be the whole of Galois theory.

To speak precisely about this we need a notion of a map that respects the base field.

Definition 30.4.1 (K -homomorphism)

Let L/K and M/K be extensions of the same field K . A **K -homomorphism** is a field homomorphism $\phi : L \rightarrow M$ with $\phi(a) = a$ for all $a \in K$. We write $\text{Aut}(L/K)$ for the group of K -isomorphisms $L \rightarrow L$.

Recall that any homomorphism of fields is automatically injective, since its kernel is an ideal not containing 1, and the only such ideal is $\{0\}$. For instance complex conjugation $\mathbb{C} \rightarrow \mathbb{C}$ is an $\mathbb{R}$ -homomorphism but not a $\mathbb{C}$ -homomorphism.

The following observation, though it takes only a few lines to prove, is the technical heart of the first half of the course. We shall refer to it as the Key Lemma.

Lemma 30.4.2 (Key Lemma)

Let L/K be a field extension and let $f \in K[x]$ be irreducible. Then there is a bijection

$$\{K\text{-homomorphisms } \phi : K_f \rightarrow L\} \longleftrightarrow \{\tilde{\alpha} \in L : f(\tilde{\alpha}) = 0\},$$

given by $\phi \mapsto \phi(\alpha)$ where $\alpha = x + (f)$.

Proof. Giving a K -homomorphism $K[x]/(f) \rightarrow L$ is the same as giving a ring homomorphism $\psi : K[x] \rightarrow L$ with $\psi|_K = \text{id}$ and $\ker \psi \supseteq (f)$. Such a ψ is determined by $\psi(x)$, since $\psi(\sum r_i x^i) = \sum r_i \psi(x)^i$; conversely any choice of $\psi(x) \in L$ determines such a ψ . Note that this formula says exactly that $\psi(g) = g(\psi(x))$ for every $g \in K[x]$.

So $\ker \psi \supseteq (f)$ if and only if $\psi(f) = 0$, that is $f(\psi(x)) = 0$, that is $\psi(x)$ is a root of f in L . (And in that case $\ker \psi = (f)$, since (f) is maximal and $\ker \psi \neq K[x]$.) The bijection sends ϕ to $\phi(\alpha) = \psi(x)$, and distinct roots evidently give distinct homomorphisms. $\square$

In particular the number of K -homomorphisms $K_f \rightarrow L$ is finite and equal to the number of *distinct* roots of f in L , which is at most $\deg f$. The possibility that f has repeated roots is exactly what will later be ruled out by *separability*, and the possibility that f has no roots at all in L is exactly what will be ruled out by *normality*. Those two conditions together will define a Galois extension.

Corollary 30.4.3

Let L/K be an extension and let $\alpha, \beta \in L$ be algebraic over K . Then α and β have the same minimal polynomial over K if and only if there is a K -isomorphism $K(\alpha) \rightarrow K(\beta)$ with $\alpha \mapsto \beta$.

Proof. Let f be the minimal polynomial of α , so that we have a K -isomorphism $K_f \xrightarrow{\sim} K(\alpha)$ sending $x + (f) \mapsto \alpha$. If β has the same minimal polynomial then likewise $K_f \xrightarrow{\sim} K(\beta)$, and composing one with the inverse of the other gives the required K -isomorphism.

Conversely, if $\theta : K(\alpha) \rightarrow K(\beta)$ is a K -isomorphism with $\theta(\alpha) = \beta$, then applying θ to $f(\alpha) = 0$ gives $f(\beta) = 0$, so the minimal polynomial g of β divides f . Applying the same argument to θ^{-1} gives $f \mid g$, and both are monic, so $f = g$. $\square$

Example 30.4.4

Let $f(x) = x^3 - 2 \in \mathbb{Q}[x]$, which is irreducible by Eisenstein. Let $\alpha = \sqrt[3]{2}$ be the real cube root and $\omega = e^{2\pi i/3}$, so that the roots of f in $\mathbb{C}$ are $\alpha, \omega\alpha, \omega^2\alpha$. By the corollary there is a $\mathbb{Q}$ -isomorphism

$$\mathbb{Q}(\alpha) \xrightarrow{\sim} \mathbb{Q}(\omega\alpha), \quad \alpha \mapsto \omega\alpha,$$

even though $\mathbb{Q}(\alpha) \subseteq \mathbb{R}$ while $\mathbb{Q}(\omega\alpha) \not\subseteq \mathbb{R}$. This is not a contradiction: an isomorphism of fields sees only the internal algebraic structure, and knows nothing about how the fields happen to sit inside $\mathbb{C}$. In particular this isomorphism is wildly discontinuous.

§30.4.2 Splitting Fields

Adjoining one root of f need not produce all of them. The right object is the following.

Definition 30.4.5 (Splitting Field)

Let $f \in K[x]$ be non-constant. A **splitting field** for f over K is an extension L/K such that

- (i) f splits into linear factors in $L[x]$, say $f = c(x - \alpha_1) \cdots (x - \alpha_d)$; and
- (ii) $L = K(\alpha_1, \dots, \alpha_d)$.

Equivalently, (ii) says that f does not split in any proper subfield of L containing K .

Example 30.4.6

Take $K = \mathbb{Q}$.

- (i) $f = x^2 + 1 = (x + i)(x - i)$, so $\mathbb{Q}(i)$ is a splitting field for f .
- (ii) $f = x^3 - 2$. A splitting field is $\mathbb{Q}(\alpha, \omega\alpha) = \mathbb{Q}(\alpha, \omega)$, where $\alpha = \sqrt[3]{2}$ and $\omega = e^{2\pi i/3}$. Its degree is divisible by $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$ and by $[\mathbb{Q}(\omega) : \mathbb{Q}] = 2$, hence by 6; and it is at most 6 since $\{1, \omega\}\{1, \alpha, \alpha^2\}$ spans. So $[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}] = 6$. Note in particular that adjoining a *single* root of f gives only a degree 3 extension, which is therefore not a splitting field.
- (iii) The splitting field of any quadratic is obtained by adjoining a single root, since if $x^2 + bx + c = (x - \alpha)(x - \alpha')$ then $\alpha' = -b - \alpha$ lies in $K(\alpha)$ automatically.
- (iv) $f = \frac{x^p - 1}{x - 1}$ with $\omega = e^{2\pi i/p}$. The roots are $\omega, \omega^2, \dots, \omega^{p-1}$, all of which lie in $\mathbb{Q}(\omega)$, so $\mathbb{Q}(\omega)$ is a splitting field.

Theorem 30.4.7 (Existence of splitting fields)

Let K be a field and $f \in K[x]$ non-constant. Then a splitting field for f over K exists, and $[L : K] \leq (\deg f)!$.

Proof. We induct on $\deg f$, the statement being for all fields simultaneously. If $\deg f = 1$ then f is already linear and $L = K$ works. Otherwise let g be an irreducible factor of f and put $K_1 = K_g = K(\alpha)$, so that $g(\alpha) = 0$ and hence $f(\alpha) = 0$ in K_1 , with $[K_1 : K] = \deg g \leq \deg f$. Then in $K_1[x]$ we may write $f(x) = (x - \alpha)f_1(x)$ with $\deg f_1 = \deg f - 1$. By induction f_1 has a splitting field $L = K_1(\alpha_2, \dots, \alpha_d)$ over K_1 with $[L : K_1] \leq (\deg f - 1)!$. Then f splits in L as $(x - \alpha)(x - \alpha_2) \cdots (x - \alpha_d)$ (up to the leading coefficient), and $L = K(\alpha, \alpha_2, \dots, \alpha_d)$, so L is a splitting field for f over K . The degree bound follows from the tower law. $\square$

Uniqueness is more delicate, and more interesting. Splitting fields are unique up to isomorphism, but – crucially – *not up to a unique isomorphism*. Here is a very concrete illustration.

Example 30.4.8

Take $K = \mathbb{R}$ and let us build $\mathbb{C}$. On the one hand $f(x) = x^2 + 1$ gives $\mathbb{R}_f = \mathbb{R}[x]/(x^2 + 1)$, with a distinguished element i , the image of x . On the other hand $g(y) = y^2 + 2y + 2 = (y + 1 - i)(y + 1 + i)$ gives $\mathbb{R}_g = \mathbb{R}[y]/(y^2 + 2y + 2)$, with its own distinguished element, the image of y . There are two $\mathbb{R}$ -isomorphisms $\mathbb{R}_g \rightarrow \mathbb{R}_f$, given by $y \mapsto -1 + i$ and $y \mapsto -1 - i$, and there is absolutely no reason to prefer one over the other. The fields are isomorphic; the isomorphism is not canonical.

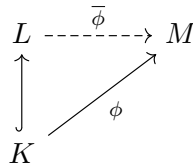
The following theorem does uniqueness and, at the same time, counts the isomorphisms – and it is that count which we will later turn into the order of the Galois group.

Theorem 30.4.9 (Uniqueness of splitting fields)

Let L be a splitting field for $f \in K[x]$ over K , and let $\phi : K \rightarrow M$ be a field homomorphism such that $\phi(f)$ splits into linear factors in $M[x]$. Then:

- (i) ϕ extends to a homomorphism $\bar{\phi} : L \rightarrow M$;
- (ii) the number of such extensions is at most $[L : K]$, with equality if f has no repeated roots in M ;
- (iii) if M is a splitting field for $\phi(f)$ over $\phi(K)$, then every such $\bar{\phi}$ is an isomorphism.

In particular any two splitting fields for f over K are K -isomorphic.



Proof. We induct on $[L : K]$. If $[L : K] = 1$ then $L = K$ and there is exactly one extension, namely ϕ itself, so (i) and (ii) hold.

Otherwise choose a root $\alpha \in L \setminus K$ of f , and let $g \in K[x]$ be the minimal polynomial of α over K , so that $g \mid f$ and $\deg g = [K(\alpha) : K] > 1$. Since $\phi(f)$ splits in M and $\phi(g) \mid \phi(f)$, the polynomial $\phi(g)$ also splits in M . By the Key Lemma applied to ϕ (that is, regarding M as an extension of $\phi(K) \cong K$), the extensions

of ϕ to $K(\alpha)$ correspond bijectively to the roots of $\phi(g)$ in M ; there are at most $\deg g = [K(\alpha) : K]$ of them, with equality if $\phi(g)$ has no repeated roots.

Now fix one such extension $\tilde{\phi} : K(\alpha) \rightarrow M$. Since L is a splitting field for f over $K(\alpha)$ as well, and $[L : K(\alpha)] < [L : K]$, the inductive hypothesis says that $\tilde{\phi}$ extends to $\bar{\phi} : L \rightarrow M$, in at most $[L : K(\alpha)]$ ways, with equality when there are no repeated roots. Multiplying up and using the tower law, the total number of extensions of ϕ to L is at most

$$[L : K(\alpha)] \cdot [K(\alpha) : K] = [L : K],$$

with equality when f has no repeated roots. This proves (i) and (ii).

For (iii), suppose M is a splitting field for $\phi(f)$, say $M = \phi(K)(\beta_1, \dots, \beta_d)$ where the β_j are the roots of $\phi(f)$. If $\alpha_1, \dots, \alpha_d$ are the roots of f in L , then $\bar{\phi}(\alpha_1), \dots, \bar{\phi}(\alpha_d)$ are roots of $\phi(f)$ in M , and they are distinct since $\bar{\phi}$ is injective; so they are exactly $\beta_1, \dots, \beta_d$. Since $L = K(\alpha_1, \dots, \alpha_d)$, the image $\bar{\phi}(L)$ contains $\phi(K)$ and all the β_j , hence is all of M . So $\bar{\phi}$ is surjective, and it is injective as it is a field homomorphism.

Finally, taking ϕ to be the inclusion $K \hookrightarrow M$ where M is another splitting field gives the last statement. $\square$

We record the count separately, since we will use it constantly.

Corollary 30.4.10

If L is a splitting field for f over K , then $\# \text{Aut}(L/K) \leq [L : K]$, with equality if f has no repeated roots in L .

Proof. Apply Theorem 30.4.9 with $M = L$ and ϕ the inclusion; every extension $\bar{\phi} : L \rightarrow L$ is an isomorphism by part (iii). $\square$

§30.4.3 Normal Extensions

Splitting fields are characterised by an intrinsic property, which does not mention any particular polynomial.

Definition 30.4.11 (Normal Extension)

An algebraic extension L/K is **normal** if for every $\alpha \in L$, the minimal polynomial of α over K splits into linear factors in $L[x]$.

In other words: if an irreducible polynomial over K has *one* root in L , it has *all* of them. For instance $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not normal, since $x^3 - 2$ has one root in $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$ but its other two roots are not real. On the other hand any extension of degree 2 is normal, by the quadratic case above.

Theorem 30.4.12

Let L/K be a finite extension. Then L/K is normal if and only if L is a splitting field over K for some $f \in K[x]$.

Proof. Suppose L/K is normal. Since L/K is finite we may write $L = K(\alpha_1, \dots, \alpha_n)$. Let f_i be the minimal polynomial of α_i over K ; by normality each f_i splits in L . Then $f = f_1 \cdots f_n$ splits in L , and L is generated over K by some of its roots, hence by all of them. So L is a splitting field for f .

Conversely suppose L is a splitting field for $f \in K[x]$. Let $\alpha \in L$ with minimal polynomial g , and let $M \supseteq L$ be a splitting field for g over L ; take $\beta \in M$ a root of g . We must show $\beta \in L$.

By Corollary 30.4.3 there is a K -isomorphism $\theta : K(\alpha) \rightarrow K(\beta)$ with $\alpha \mapsto \beta$. Now L is a splitting field for f over $K(\alpha)$, and $L(\beta)$ is a splitting field for f over $K(\beta)$. Applying Theorem 30.4.9(iii) to θ , we get an isomorphism $L \rightarrow L(\beta)$, so

$$[L : K(\alpha)] = [L(\beta) : K(\beta)].$$

But $[K(\alpha) : K] = [K(\beta) : K] = \deg g$, so multiplying up gives $[L : K] = [L(\beta) : K]$, whence $[L(\beta) : L] = 1$ and $\beta \in L$. $\square$

Corollary 30.4.13 (Normal closure)

Let L/K be a finite extension. Then there is a finite extension N/L such that N/K is normal, and such that no proper intermediate field $L \subseteq N' \subsetneq N$ is normal over K . Moreover any two such N are L -isomorphic.

Proof. Write $L = K(\alpha_1, \dots, \alpha_n)$, let f_i be the minimal polynomial of α_i over K and set $f = f_1 \cdots f_n$. Let N be a splitting field for f over L . Since the α_i are among the roots of f , in fact N is a splitting field for f over K , so N/K is normal by Theorem 30.4.12.

If $L \subseteq N' \subseteq N$ with N'/K normal, then each $\alpha_i \in N'$, so each f_i splits in N' , so f splits in N' and hence $N' \supseteq N$ by minimality of splitting fields; that is $N' = N$. Uniqueness up to L -isomorphism is Theorem 30.4.9(iii). $\square$

We call N the **normal closure** of L/K ; it is the smallest normal extension of K containing L .

§30.5 Separability

The Key Lemma told us that K -homomorphisms out of $K(\alpha)$ correspond to the *distinct* roots of the minimal polynomial of α . If that polynomial has repeated roots we lose homomorphisms, and everything we are going to build will collapse. So we must understand when an irreducible polynomial can have a repeated root. Over $\mathbb{Q}$ or $\mathbb{R}$ or $\mathbb{C}$ this never happens, and it takes a little imagination to see how it could – but in characteristic p it genuinely does.

§30.5.1 The Derivative Criterion

The tool is differentiation, but of course we cannot take limits in an abstract field, so we define the derivative formally.

Definition 30.5.1 (Formal Derivative)

Let K be a field. The **formal derivative** is the K -linear map $\frac{d}{dx} : K[x] \rightarrow K[x]$ determined by $x^n \mapsto nx^{n-1}$; that is,

$$\left(\sum_i a_i x^i \right)' = \sum_{i \geq 1} i a_i x^{i-1}.$$

Proposition 30.5.2

For $f, g \in K[x]$ we have $(fg)' = f'g + fg'$ and $(f \circ g)' = (f' \circ g) \cdot g'$.

Proof. Both sides of each identity are additive in f (and in g , for the first), so by linearity it suffices to check them on monomials, where they are immediate from the binomial theorem. $\square$

Lemma 30.5.3

Let L/K be a field extension, $f \in K[x]$ and $\alpha \in L$ with $f(\alpha) = 0$. Then α is a simple root of f (that is, $(x - \alpha)^2 \nmid f$ in $L[x]$) if and only if $f'(\alpha) \neq 0$.

Proof. Write $f(x) = (x - \alpha)g(x)$ in $L[x]$. Then $f'(x) = g(x) + (x - \alpha)g'(x)$, so $f'(\alpha) = g(\alpha)$. Now α is a repeated root exactly when $(x - \alpha) \mid g$, that is $g(\alpha) = 0$. $\square$

Corollary 30.5.4

A polynomial $f \in K[x]$ has a repeated root in some extension of K if and only if $\deg \gcd(f, f') > 0$.

Proof. Work in a splitting field L for f . If α is a repeated root then $(x - \alpha)$ divides both f and f' by the lemma, so $\gcd(f, f') \neq 1$. (Note that the gcd may be computed in $K[x]$ or in $L[x]$ with the same answer; see Lemma 30.5.5 below.) Conversely if $\gcd(f, f')$ has positive degree then it has a root α in L , and then $f(\alpha) = f'(\alpha) = 0$, so α is a repeated root of f . $\square$

We should check the claim about the gcd, since it is used repeatedly and is a nice illustration of the power of Bézout.

Lemma 30.5.5

Let L/K be a field extension and $f, g \in K[x]$, not both zero. Then $\gcd(f, g)$ is the same whether computed in $K[x]$ or in $L[x]$. Consequently the same is true of $\text{lcm}(f, g) = fg/\gcd(f, g)$.

Proof. Let $h = \gcd_K(f, g)$ and $h' = \gcd_L(f, g)$, both taken monic. Since $h \mid f$ and $h \mid g$ in $K[x]$, the same divisibilities hold in $L[x]$, so $h \mid h'$. Conversely, by Bézout we may write $h = pf + qg$ with $p, q \in K[x]$; since $h' \mid f$ and $h' \mid g$ in $L[x]$, we get $h' \mid h$. As both are monic, $h = h'$. $\square$

§30.5.2 Separable Polynomials and Perfect Fields

Definition 30.5.6 (Separable Polynomial)

A non-constant polynomial $f \in K[x]$ is **separable** if it has no repeated roots in a splitting field, that is, if it splits into *distinct* linear factors there. Otherwise it is **inseparable**.

Equivalently, by Corollary 30.5.4, f is separable if and only if $\gcd(f, f') = 1$. Note that this does not depend on the choice of splitting field, by uniqueness.

Proposition 30.5.7

Let $f \in K[x]$ be irreducible.

- (i) f is separable if and only if $f' \neq 0$.
- (ii) If $\text{char } K = 0$, then f is separable. Indeed every irreducible polynomial over a field of characteristic zero is separable.
- (iii) If $\text{char } K = p$, then f is inseparable if and only if $f(x) = g(x^p)$ for some $g \in K[x]$.

Proof. (i) We may take f monic. As f is irreducible, $\gcd(f, f')$ is 1 or f , since it divides f . If $f' = 0$ then $\gcd(f, f') = f$, which has positive degree, so f is inseparable. If $f' \neq 0$ then $\deg f' < \deg f$, so $\gcd(f, f') \neq f$ and hence equals 1, so f is separable.

(ii) Write $f = \sum a_i x^i$ with $\deg f = d \geq 1$. If $f' = 0$ then $ia_i = 0$ for all $i \geq 1$; in characteristic zero this forces $a_i = 0$ for all $i \geq 1$, so f is constant, contradicting irreducibility. So $f' \neq 0$ and we are done by (i).

(iii) As above $f' = 0$ if and only if $ia_i = 0$ for all $i \geq 1$, which in characteristic p says precisely that $a_i = 0$ whenever $p \nmid i$. That is exactly the statement that $f(x) = \sum_j a_{pj} x^{pj} = g(x^p)$ with $g(x) = \sum_j a_{pj} x^j$. $\square$

So inseparability is a characteristic p phenomenon, and even there it takes some work to produce. The reason is the following remarkable map.

Proposition 30.5.8 (Frobenius)

Let R be a ring with $\text{char } R = p$ prime. Then

$$\text{Frob} : R \rightarrow R, \quad x \mapsto x^p$$

is a ring homomorphism, called the **Frobenius map**.

Proof. Clearly $\text{Frob}(1) = 1$ and $\text{Frob}(xy) = \text{Frob}(x)\text{Frob}(y)$. For addition, the binomial theorem gives

$$(x + y)^p = x^p + \binom{p}{1} x^{p-1} y + \cdots + \binom{p}{p-1} x y^{p-1} + y^p,$$

and $p \mid \binom{p}{i}$ for $1 \leq i \leq p-1$ (the numerator $p!$ contains a factor p which is not cancelled by $i!(p-i)!$), so all the middle terms vanish. $\square$

In particular, if K is a field of characteristic p then $\text{Frob} : K \rightarrow K$ is injective, so it is an automorphism as soon as K is finite. Fields where it is surjective are exactly the ones with no inseparability.

Definition 30.5.9 (Perfect Field)

A field K is **perfect** if $\text{char } K = 0$, or if $\text{char } K = p$ and every element of K is a p th power.

Proposition 30.5.10

K is perfect if and only if every irreducible polynomial in $K[x]$ is separable. In particular every finite field is perfect, and so is every field of characteristic zero.

Proof. Characteristic zero is Proposition 30.5.7(ii), so suppose $\text{char } K = p$.

Suppose K is perfect and $f \in K[x]$ is irreducible but inseparable. By Proposition 30.5.7(iii), $f(x) = \sum_j a_j x^{pj}$. Write $a_j = b_j^p$, possible by perfection. Then

$$f(x) = \sum_j b_j^p x^{pj} = \left(\sum_j b_j x^j \right)^p$$

by the Frobenius homomorphism applied in $K[x]$, contradicting irreducibility.

Conversely, suppose $b \in K$ is not a p th power, and consider $f(x) = x^p - b$. Let α be a root of f in a splitting field, so $\alpha^p = b$ and

$$x^p - b = x^p - \alpha^p = (x - \alpha)^p,$$

using Frobenius again. So f is very far from separable. It remains to see that f is irreducible. If not, then some monic factor $(x - \alpha)^m$ with $0 < m < p$ lies in $K[x]$; expanding, its coefficient of x^{m-1} is $-m\alpha \in K$, and $m \neq 0$ in K , so $\alpha \in K$ and hence $b = \alpha^p$ is a p th power, a contradiction.

Finally, a finite field K of characteristic p is perfect because Frob is an injective map from the finite set K to itself, hence surjective. $\square$

Example 30.5.11

The smallest imperfect field is $K = \mathbb{F}_p(t)$, the field of rational functions over $\mathbb{F}_p$. Here t is not a p th power in K (a p th power in $\mathbb{F}_p(t)$ is a ratio of polynomials in t^p), so $x^p - t$ is irreducible and inseparable, and

$$\mathbb{F}_p(t^{1/p})/\mathbb{F}_p(t)$$

is a degree p extension in which the minimal polynomial of the generator has a single root of multiplicity p . This is a genuinely strange extension: by the Key Lemma there is exactly one K -homomorphism $K(t^{1/p}) \rightarrow K(t^{1/p})$, namely the identity, even

though the extension has degree p .

§30.5.3 Separable Extensions and the Separable Degree

Definition 30.5.12 (Separable Element and Extension)

Let L/K be an extension and $\alpha \in L$ algebraic over K . We say α is **separable** over K if its minimal polynomial over K is separable. An algebraic extension L/K is **separable** if every $\alpha \in L$ is separable over K .

So by Proposition 30.5.10, every algebraic extension of a perfect field – in particular of $\mathbb{Q}$, of $\mathbb{C}$, or of any finite field – is separable. The reader who is content to work in characteristic zero may safely read ‘separable’ as ‘no condition at all’.

The right way to organise the theory is to count homomorphisms.

Definition 30.5.13 (Separable Degree)

Let L/K be a finite extension and let N be a normal extension of K containing L (for instance a normal closure). The **separable degree** $[L : K]_s$ is the number of K -homomorphisms $L \rightarrow N$.

Proposition 30.5.14

With notation as above:

- (i) $[L : K]_s$ does not depend on the choice of N ;
- (ii) $[K(\alpha) : K]_s$ is the number of distinct roots of the minimal polynomial of α over K , and so $[K(\alpha) : K]_s = [K(\alpha) : K]$ if and only if α is separable over K ;
- (iii) $[L : K]_s \leq [L : K]$;
- (iv) if $M/L/K$ are finite extensions then $[M : K]_s = [M : L]_s [L : K]_s$.

Proof. (ii) is the Key Lemma 30.4.2, once we note that the minimal polynomial of α splits in N by normality.

For (iv), let N be normal over K containing M . Each K -homomorphism $\sigma : L \rightarrow N$ has image in N , and N is normal over $\sigma(L)$ too, so by Theorem 30.4.9(i) (applied to M , which is generated over L by finitely many algebraic elements, so is contained in a splitting field over L) each σ extends to M . The number of extensions of a given σ is $[M : L]_s$: transporting along the isomorphism $L \cong \sigma(L)$, this is the number of $\sigma(L)$ -homomorphisms $\sigma(M) \rightarrow N$. Since every K -homomorphism $M \rightarrow N$ restricts to some σ on L , the count is $[M : L]_s [L : K]_s$.

Now (iii) follows by writing $L = K(\alpha_1, \dots, \alpha_n)$, applying (iv) repeatedly, and using (ii) together with the fact that the number of distinct roots of a polynomial is at most its degree.

Finally (i): the count in (ii) does not depend on N , and the multiplicativity in (iv) then determines $[L : K]_s$ from a chain of simple extensions. $\square$

Theorem 30.5.15

A finite extension L/K is separable if and only if $[L : K]_s = [L : K]$.

Proof. Suppose $[L : K]_s = [L : K]$ and let $\alpha \in L$. By multiplicativity,

$$[L : K(\alpha)]_s [K(\alpha) : K]_s = [L : K(\alpha)] [K(\alpha) : K],$$

and each separable degree is at most the corresponding degree, so we must have equality in each; in particular $[K(\alpha) : K]_s = [K(\alpha) : K]$, so α is separable by Proposition 30.5.14(ii).

Conversely suppose L/K is separable and write $L = K(\alpha_1, \dots, \alpha_n)$. Set $K_i = K(\alpha_1, \dots, \alpha_i)$. Each α_i is separable over K , so its minimal polynomial over K is separable; the minimal polynomial over K_{i-1} divides it, so is separable too. Hence $[K_i : K_{i-1}]_s = [K_i : K_{i-1}]$ for each i , and multiplying up gives the result. $\square$

Corollary 30.5.16 (i) If α is separable over K then $K(\alpha)/K$ is a separable extension.

(ii) If M/L and L/K are finite separable extensions then so is M/K .

(iii) If L/K is finite and $L = K(\alpha_1, \dots, \alpha_n)$ with each α_i separable over K , then L/K is separable.

Proof. All three are immediate from Theorem 30.5.15 and the multiplicativity of both the degree and the separable degree; for (i) and (iii) use the argument in the second paragraph of that proof. $\square$

One further small lemma will be needed when we assemble separable polynomials.

Lemma 30.5.17

The lowest common multiple of finitely many separable polynomials is separable.

Proof. By Lemma 30.5.5 we may compute the lcm in any extension we like, so pass to a field in which all the polynomials split. There each is a product of distinct linear factors, and the lcm is the product of the distinct linear factors occurring in any of them, which is again separable. $\square$

§30.5.4 The Primitive Element Theorem

We now come to a genuinely useful structural theorem: a finite separable extension is always generated by a *single* element. This is not at all obvious, and the proof is a pretty piece of trickery.

Theorem 30.5.18 (Primitive Element Theorem)

Let $L = K(\alpha_1, \dots, \alpha_n, \beta)$ be a finite extension of K in which $\alpha_1, \dots, \alpha_n$ are separable over K (and β is arbitrary). Then $L = K(\gamma)$ for some $\gamma \in L$. Such a γ is called a

primitive element.

Proof. First suppose K is finite. Then L is finite too, so $L^\times$ is a cyclic group (Proposition 30.6.1 below, whose proof does not use this theorem), and taking γ to be a generator gives $L = K(\gamma)$ immediately.

So suppose K is infinite. By induction on n it suffices to treat the case $L = K(\alpha, \beta)$ with α separable over K ; indeed, once we know $K(\alpha_1, \beta) = K(\beta_1)$ for some β_1 , we may write $K(\alpha_1, \alpha_2, \beta) = K(\alpha_2, \beta_1)$ and continue.

We shall show that for all but finitely many $c \in K$, the element $\gamma = \beta + c\alpha$ generates L . It is enough to show $\alpha \in K(\gamma)$, since then $\beta = \gamma - c\alpha \in K(\gamma)$ as well and so $K(\gamma) = L$.

Let $f, g \in K[x]$ be the minimal polynomials of α, β respectively, and let M be a splitting field for fg over L , in which

$$f(x) = (x - \alpha_1) \cdots (x - \alpha_m), \quad g(x) = (x - \beta_1) \cdots (x - \beta_k),$$

with $\alpha = \alpha_1$ and $\beta = \beta_1$. Since α is separable, the α_i are distinct.

Now consider

$$h(x) = g(\gamma - cx) \in K(\gamma)[x].$$

We have $h(\alpha) = g(\gamma - c\alpha) = g(\beta) = 0$, and also $f(\alpha) = 0$, so $(x - \alpha)$ divides $\gcd(f, h)$ in $M[x]$. Suppose we can arrange that

$$\gcd(f, h) = x - \alpha.$$

By Lemma 30.5.5 this gcd may be computed in $K(\gamma)[x]$, so $x - \alpha \in K(\gamma)[x]$ and hence $\alpha \in K(\gamma)$, as required.

So we need $h(\alpha_i) \neq 0$ for $i = 2, \dots, m$. Now $h(\alpha_i) = 0$ if and only if $\gamma - c\alpha_i = \beta_j$ for some j , that is, since $\gamma = \beta + c\alpha$,

$$\beta_j - \beta = c(\alpha_i - \alpha).$$

As $\alpha_i \neq \alpha$ for $i \geq 2$, this says

$$c = \frac{\beta_j - \beta}{\alpha_i - \alpha}$$

for some $i \geq 2$ and some j . There are only finitely many such values, and K is infinite, so we may choose $c \in K$ avoiding them all. $\square$

Corollary 30.5.19

Every finite separable extension is simple. In particular, if $\text{char } K = 0$ and L/K is finite then $L = K(\gamma)$ for some γ .

Example 30.5.20

Take $K = \mathbb{Q}(i, \sqrt[3]{2})$, so $[K : \mathbb{Q}] = 6$. In the notation of the proof, $\beta_1 = i$, $\beta_2 = -i$, and $\alpha_1 = \sqrt[3]{2}$, $\alpha_2 = \omega\sqrt[3]{2}$, $\alpha_3 = \omega^2\sqrt[3]{2}$ where $\omega = e^{2\pi i/3}$. The bad values of c are the

numbers

$$\frac{\pm i - i}{\sqrt[3]{2}(\omega^{\pm 1} - 1)},$$

none of which is a non-zero rational. So $K = \mathbb{Q}(i + c\sqrt[3]{2})$ for *every* non-zero rational c . As one would expect, most elements of K generate it; for instance $i\sqrt[3]{2}$ works too.

Example 30.5.21

Separability really is needed. Let $K = \mathbb{F}_p(x, y)$ and $L = K(x^{1/p}, y^{1/p})$, so $[L : K] = p^2$. We claim that L is not simple. Indeed, if $\gamma \in L$ then we may write $\gamma = \sum_{i,j} a_{ij} x^{i/p} y^{j/p}$ with $a_{ij} \in K$ and $0 \leq i, j < p$, and applying Frobenius,

$$\gamma^p = \sum_{i,j} a_{ij}^p x^i y^j \in K.$$

So γ satisfies $t^p - \gamma^p \in K[t]$, whence $[K(\gamma) : K] \leq p < p^2 = [L : K]$, and $K(\gamma) \neq L$.

§30.6 Finite Fields

We already know that a finite field has p^n elements for some prime p and $n \geq 1$. In this section we prove the converse – for each such $q = p^n$ there is exactly one field with q elements – and work out its subfields. Everything hinges on one polynomial, $x^q - x$.

§30.6.1 The Multiplicative Group

Proposition 30.6.1

Let K be a field and let $G \leq K^\times$ be a finite subgroup. Then G is cyclic.

Proof. G is a finite abelian group, so by the structure theorem

$$G \cong \mathbb{Z}/m_1 \times \mathbb{Z}/m_2 \times \cdots \times \mathbb{Z}/m_r$$

with $m_1 \mid m_2 \mid \cdots \mid m_r$, and $\#G = m_1 m_2 \cdots m_r$. Every element of each factor is killed by m_r , so every $\alpha \in G$ satisfies $\alpha^{m_r} = 1$, that is, every element of G is a root of $x^{m_r} - 1$. But a polynomial of degree m_r over a field has at most m_r roots, so

$$m_1 m_2 \cdots m_r = \#G \leq m_r,$$

forcing $m_1 = \cdots = m_{r-1} = 1$. Hence $G \cong \mathbb{Z}/m_r$ is cyclic. $\square$

Example 30.6.2

$\mathbb{F}_7^\times = \langle 3 \rangle$, since the powers of 3 are 3, 2, 6, 4, 5, 1. Note that there is no canonical generator, and correspondingly no canonical isomorphism $\mathbb{F}_q^\times \cong \mathbb{Z}/(q-1)$; finding a generator (a ‘primitive root’) for a given p is, as you saw in Number Theory, a genuinely awkward computational problem.

Lemma 30.6.3 (Fermat's Little Theorem)

Let K be a finite field with $\#K = q$. Then every $\alpha \in K$ satisfies $\alpha^q = \alpha$. Consequently

$$x^q - x = \prod_{\alpha \in K} (x - \alpha)$$

in $K[x]$; in particular $x^q - x$ splits into distinct linear factors over K .

Proof. This is clear for $\alpha = 0$. Otherwise $\alpha \in K^\times$, a group of order $q - 1$, so $\alpha^{q-1} = 1$ by Lagrange, and hence $\alpha^q = \alpha$. So every one of the q elements of K is a root of the degree q polynomial $x^q - x$, which therefore factors as claimed. $\square$

So a finite field K with q elements is precisely a splitting field for $x^q - x$ over $\mathbb{F}_p$: it contains all q roots, and is generated by them since they are everything. That immediately gives uniqueness, by uniqueness of splitting fields. For existence we must construct the splitting field of $x^q - x$ and check it has the right size – which is where separability earns its keep.

§30.6.2 Existence, Uniqueness and Subfields**Theorem 30.6.4** (Classification of finite fields)

Let p be prime, $n \geq 1$ and $q = p^n$. Then:

- (i) there exists a field $\mathbb{F}_q$ with exactly q elements, namely the splitting field of $x^q - x$ over $\mathbb{F}_p$;
- (ii) any two fields with q elements are isomorphic;
- (iii) $\mathbb{F}_q$ contains a subfield with p^k elements if and only if $k \mid n$, and then that subfield is unique.

Proof. (i) Let K be a splitting field for $f(x) = x^q - x$ over $\mathbb{F}_p$, and let

$$S = \{\alpha \in K : \alpha^q = \alpha\}$$

be the set of roots. We claim S is a subfield. Indeed the map $\alpha \mapsto \alpha^q$ is the n -fold composite Frob^n of the Frobenius homomorphism, hence is itself a ring homomorphism, and S is exactly the set where it agrees with the identity; so S is closed under $+$, $-$, $\times$, contains 0 and 1, and is closed under inverses since $(\alpha^{-1})^q = (\alpha^q)^{-1} = \alpha^{-1}$. Since K is generated over $\mathbb{F}_p$ by S , and S is a field, $K = S$.

It remains to check that $\#S = q$. Now $f'(x) = qx^{q-1} - 1 = -1$, because $q = p^n = 0$ in characteristic p . So $\gcd(f, f') = 1$ and f is separable, hence has exactly q distinct roots in K . So $\#K = \#S = q$.

(ii) If $\#K = q$ then by Lemma 30.6.3 K is a splitting field for $x^q - x$ over its prime subfield $\mathbb{F}_p$, so any two such are isomorphic by Theorem 30.4.9.

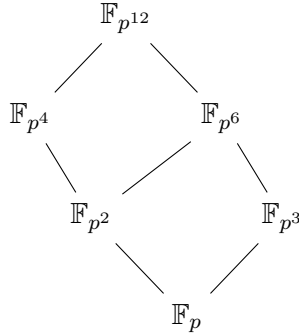
(iii) If $K \subseteq \mathbb{F}_q$ is a subfield with p^k elements, then $[\mathbb{F}_q : \mathbb{F}_p] = [\mathbb{F}_q : K][K : \mathbb{F}_p]$ gives $k \mid n$.

Conversely suppose $k \mid n$, say $n = k\ell$. We claim $x^{p^k} - x$ divides $x^{p^n} - x$. First, for

integers $a, b \geq 1$ with $a \mid b$ we have $y^a - 1 \mid y^b - 1$, since $y^a - 1$ divides $(y^a)^{b/a} - 1$ by the factorisation $z^m - 1 = (z - 1)(z^{m-1} + \cdots + 1)$. Applying this with $y = p$, $a = k$, $b = n$ gives $p^k - 1 \mid p^n - 1$; applying it again with $y = x$ gives $x^{p^k-1} - 1 \mid x^{p^n-1} - 1$, and multiplying by x gives the claim.

Hence every root of $x^{p^k} - x$ lies in $\mathbb{F}_q$, and $\{\alpha \in \mathbb{F}_q : \alpha^{p^k} = \alpha\}$ is a subfield with exactly p^k elements by the argument in (i). It is unique, since any subfield of order p^k consists of roots of $x^{p^k} - x$, of which there are only p^k in total. $\square$

The subfield structure is therefore a perfect copy of the divisibility lattice of n . Here it is for $n = 12$.



Note that $\mathbb{F}_{p^4}$ and $\mathbb{F}_{p^6}$ are incomparable, as $4 \nmid 6$ and $6 \nmid 4$; and their intersection is $\mathbb{F}_{p^2}$, matching $\gcd(4, 6) = 2$.

Example 30.6.5

Let us do the two smallest interesting cases explicitly, over $\mathbb{F}_2$.

- $q = 4$: $x^4 - x = x(x-1)(x^2 + x + 1)$, and $x^2 + x + 1$ is irreducible over $\mathbb{F}_2$ (it has no root in $\mathbb{F}_2$), so

$$\mathbb{F}_4 = \mathbb{F}_2[x]/(x^2 + x + 1).$$

- $q = 8$: over $\mathbb{F}_2$ we have

$$x^8 - x = x(x+1)(x^3 + x + 1)(x^3 + x^2 + 1),$$

and the two cubics are precisely the irreducible cubics over $\mathbb{F}_2$. So the six elements of $\mathbb{F}_8 \setminus \mathbb{F}_2$ split into two classes of three, according to which cubic they satisfy. Taking β with $\beta^3 = \beta + 1$, we get $\mathbb{F}_8 = \mathbb{F}_2(\beta)$ with basis $1, \beta, \beta^2$.

Note that $[\mathbb{F}_8 : \mathbb{F}_2] = 3$ and $[\mathbb{F}_4 : \mathbb{F}_2] = 2$, and $2 \nmid 3$, so $\mathbb{F}_4 \not\subseteq \mathbb{F}_8$ – despite $4 < 8$.

§30.6.3 Irreducible Polynomials over $\mathbb{F}_p$

The same divisibility yields a clean description of the irreducible polynomials over a finite field.

Proposition 30.6.6

Let $f \in \mathbb{F}_p[x]$ be monic irreducible of degree d , and let $q = p^n$. Then

$$f \mid x^q - x \iff d \mid n.$$

Consequently $x^{p^n} - x$ is the product of all monic irreducible polynomials in $\mathbb{F}_p[x]$ whose degree divides n , each occurring exactly once.

Proof. Let $K = \mathbb{F}_p[x]/(f) = \mathbb{F}_p(\alpha)$, a field with p^d elements. If $d \mid n$ then K embeds in $\mathbb{F}_q$ by Theorem 30.6.4(iii), so $\alpha \in \mathbb{F}_q$ satisfies $\alpha^q = \alpha$, and f , being the minimal polynomial of α , divides $x^q - x$.

Conversely if $f \mid x^q - x$ then α is a root of $x^q - x$ in some extension, so $\mathbb{F}_p(\alpha)$ is contained in a field with q elements, and $d = [\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ divides n .

For the last statement, $x^q - x$ is separable so has no repeated factors, and every monic irreducible factor has degree dividing n by the above; conversely every monic irreducible of degree dividing n divides it. $\square$

Counting degrees in the last statement gives the pretty identity

$$p^n = \sum_{d \mid n} d \cdot N_d,$$

where N_d is the number of monic irreducible polynomials of degree d over $\mathbb{F}_p$; Möbius inversion then gives $N_n = \frac{1}{n} \sum_{d \mid n} \mu(n/d) p^d$, which is positive for all n . In particular irreducible polynomials of every degree exist over $\mathbb{F}_p$ – though notice how much easier it was to construct $\mathbb{F}_q$ as a splitting field than to write down an irreducible polynomial of degree n by hand.

§30.7 Algebraic Closure

We have been rather carefully constructing splitting fields one polynomial at a time. It is natural to ask for a single field in which *every* polynomial splits.

Definition 30.7.1 (Algebraically Closed)

A field K is **algebraically closed** if every non-constant $f \in K[x]$ has a root in K (and hence, by induction, splits into linear factors in $K[x]$). An **algebraic closure** of K is an algebraic extension $\overline{K}/K$ with $\overline{K}$ algebraically closed.

Lemma 30.7.2

For a field K the following are equivalent:

- (i) K is algebraically closed;
- (ii) whenever L/K is an extension and $\alpha \in L$ is algebraic over K , we have $\alpha \in K$;
- (iii) the only algebraic extension of K is K itself.

Proof. (i) $\Rightarrow$ (ii): the minimal polynomial of α is irreducible over K , but every irreducible polynomial over an algebraically closed field is linear, so $\alpha \in K$. (ii) $\Rightarrow$ (iii) is immediate, and (iii) $\Rightarrow$ (i) follows since for non-constant f the extension K_g/K is algebraic, where g is an irreducible factor of f , so $K_g = K$ and g is linear. $\square$

Two examples we know: $\mathbb{C}$ is algebraically closed (we will prove this in Section 30.14 using Galois theory), and hence so is $\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} : \alpha \text{ algebraic over } \mathbb{Q}\}$, which is therefore an algebraic closure of $\mathbb{Q}$. Here is a more hands-on construction.

Example 30.7.3 (An algebraic closure of $\mathbb{F}_p$)

Choose integers $r_1 \mid r_2 \mid r_3 \mid \cdots$ such that every $n \in \mathbb{N}$ divides some r_i – for instance $r_i = i!$. By Theorem 30.6.4 we may choose embeddings

$$\mathbb{F}_{p^{r_1}} \hookrightarrow \mathbb{F}_{p^{r_2}} \hookrightarrow \mathbb{F}_{p^{r_3}} \hookrightarrow \cdots$$

and set $\overline{\mathbb{F}}_p = \bigcup_i \mathbb{F}_{p^{r_i}}$, which is a field since the union is increasing. It is algebraic over $\mathbb{F}_p$ since each $\mathbb{F}_{p^{r_i}}$ is finite over $\mathbb{F}_p$.

It is algebraically closed: given non-constant $f \in \overline{\mathbb{F}}_p[x]$, its finitely many coefficients all lie in some $\mathbb{F}_{p^{r_i}}$, and a splitting field for f over $\mathbb{F}_{p^{r_i}}$ is a finite field $\mathbb{F}_{p^N}$ with $r_i \mid N$. Choosing j with $N \mid r_j$, we may embed $\mathbb{F}_{p^N}$ into $\mathbb{F}_{p^{r_j}} \subseteq \overline{\mathbb{F}}_p$ over $\mathbb{F}_{p^{r_i}}$, so f splits in $\overline{\mathbb{F}}_p$.

The general case works the same way, but the induction has to be transfinite.

Theorem 30.7.4

Every field K has an algebraic closure $\overline{K}$. Moreover if L_1, L_2 are two algebraic closures of K then there is a K -isomorphism $L_1 \rightarrow L_2$, though it is very far from unique.

Proof sketch. One way to organise the construction is as follows. Consider the polynomial ring $R = K[x_f : f \in K[x] \text{ monic irreducible}]$ in one indeterminate for each monic irreducible polynomial, and let $I \leq R$ be the ideal generated by all $f(x_f)$. One checks that $I \neq R$ (a relation $\sum g_i f_i(x_{f_i}) = 1$ involves finitely many f_i , and evaluating in a splitting field for $\prod f_i$ gives $1 = 0$), so by Zorn's lemma I is contained in a maximal ideal $\mathfrak{m}$, and $K_1 = R/\mathfrak{m}$ is a field in which every monic irreducible polynomial over K acquires a root. Iterating gives a chain $K \subseteq K_1 \subseteq K_2 \subseteq \cdots$, and the union is algebraically closed; the subfield of elements algebraic over K is then an algebraic closure.

For uniqueness, consider the set of pairs (M, ϕ) where $K \subseteq M \subseteq L_1$ and $\phi : M \rightarrow L_2$ is a K -homomorphism, partially ordered by extension. Every chain has an upper bound (take the union), so by Zorn's lemma there is a maximal element (M, ϕ) . If $M \neq L_1$, pick $\alpha \in L_1 \setminus M$; its minimal polynomial over M has a root in L_2 since L_2 is algebraically closed, so by the Key Lemma ϕ extends to $M(\alpha)$, contradicting maximality. So $M = L_1$, and $\phi(L_1)$ is algebraically closed and L_2 is algebraic over it, so $\phi(L_1) = L_2$. $\square$

We will not use the general existence theorem in what follows; for us ‘a field in which f

splits' is always enough, and that we constructed by hand.

§30.8 The Galois Correspondence

Everything so far has concerned a single element α and the maps out of $K(\alpha)$ determined by the roots of its minimal polynomial. We now ask what happens when we look at all the roots at once, and how they interact. The answer is the Galois correspondence, and it is the heart of the course.

§30.8.1 Automorphism Groups

Recall that for an extension L/K we write

$$\text{Aut}(L/K) = \{\sigma : L \rightarrow L \mid \sigma \text{ is a field isomorphism with } \sigma|_K = \text{id}\},$$

which is a group under composition. It measures precisely the failure of uniqueness in our isomorphism theorems: if $\phi : M \rightarrow L$ is a K -isomorphism and $\sigma \in \text{Aut}(L/K)$, then $\sigma\phi$ is another one; and conversely if $\phi_1, \phi_2 : M \rightarrow L$ are K -isomorphisms then $\phi_1\phi_2^{-1} \in \text{Aut}(L/K)$, and it is non-trivial exactly when $\phi_1 \neq \phi_2$.

Lemma 30.8.1

Let L/K be an extension and $\sigma \in \text{Aut}(L/K)$.

- (i) If $f \in K[x]$ and $\alpha \in L$ with $f(\alpha) = 0$, then $f(\sigma\alpha) = 0$. So σ permutes the roots in L of any polynomial over K .
- (ii) If $L = K(\alpha_1, \dots, \alpha_n)$ and $\sigma\alpha_i = \alpha_i$ for all i , then $\sigma = \text{id}$.
- (iii) If L is a splitting field over K for a polynomial f with roots $\alpha_1, \dots, \alpha_n$, then restriction to $\{\alpha_1, \dots, \alpha_n\}$ gives an injective homomorphism $\text{Aut}(L/K) \hookrightarrow S_n$.

Proof. (i) Applying σ to $\sum a_i \alpha^i = 0$ and using $\sigma(a_i) = a_i$ gives $\sum a_i (\sigma\alpha)^i = 0$.

(ii) The set of elements fixed by σ is a subfield containing K and all the α_i , hence is L .

(iii) By (i) each σ permutes the roots, giving a homomorphism to S_n ; by (ii) it is injective since $L = K(\alpha_1, \dots, \alpha_n)$. $\square$

Part (iii) is worth dwelling on, because it is exactly Galois' original point of view: the Galois group of a polynomial is a group of *permutations of its roots*, and in particular is a rather small and concrete object. Let us compute a few.

Example 30.8.2 (i) Let $[L : K] = 2$ with $\text{char } K \neq 2$, so $L = K(\alpha)$ for any $\alpha \in L \setminus K$, with minimal polynomial $f(x) = x^2 + bx + c = (x - \alpha)(x - \alpha')$ in a splitting field. Since $\alpha + \alpha' = -b \in K$, in fact $\alpha' = -b - \alpha \in L$, so L is already a splitting field for f . By the Key Lemma there is a K -homomorphism $\sigma : L \rightarrow L$ with $\alpha \mapsto \alpha'$, which is an isomorphism, and $\sigma \neq \text{id}$ unless $\alpha = \alpha'$, that is $2\alpha = -b$, that is $\alpha = -b/2 \in K$, which is false. So

$$\text{Aut}(L/K) \cong \mathbb{Z}/2.$$

This covers $\text{Aut}(\mathbb{C}/\mathbb{R})$, $\text{Aut}(\mathbb{Q}(i)/\mathbb{Q})$ and $\text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$. It is worth a warning: $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{R}$, but the non-trivial automorphism $\sqrt{2} \mapsto -\sqrt{2}$ is wildly discontinuous for the topology induced from $\mathbb{R}$. We are looking at fields as algebraic objects only.

- (ii) Let $L = \mathbb{Q}(\sqrt[3]{2})$. The minimal polynomial of $\alpha = \sqrt[3]{2}$ is $x^3 - 2$, whose other roots $\omega\alpha, \omega^2\alpha$ are not real and hence not in L . So by the Key Lemma the only $\mathbb{Q}$ -homomorphism $L \rightarrow L$ is the identity, and

$$\text{Aut}(L/\mathbb{Q}) = 1,$$

even though $[L : \mathbb{Q}] = 3$. This is the failure of *normality*.

- (iii) If $\text{char } K = p$ and $L = K(t^{1/p})$ with $t \in K$ not a p th power, then again $\text{Aut}(L/K) = 1$ while $[L : K] = p$. This is the failure of *separability*.

- (iv) Let $L = K(x)$. Then $\text{Aut}(L/K) \cong \text{PGL}_2(K)$, acting by Möbius substitutions $x \mapsto \frac{ax+b}{cx+d}$. (A pleasant exercise.)

Example 30.8.3 (Biquadratic extensions)

Suppose $\text{char } K \neq 2$ and $[L : K] = 4$ with $L = K(\alpha, \beta)$, $\alpha^2 = a \in K$, $\beta^2 = b \in K$. Since $L \supseteq K(\alpha)$ and $\beta \notin K(\alpha)$, there is a unique $K(\alpha)$ -automorphism σ_β of L with $\beta \mapsto -\beta$, and similarly σ_α . These commute, both square to the identity, and neither is trivial, so

$$\mathbb{Z}/2 \times \mathbb{Z}/2 \leq \text{Aut}(L/K).$$

Conversely any K -automorphism must send $\alpha \mapsto \pm\alpha$ and $\beta \mapsto \pm\beta$, and is determined by those choices, so

$$\text{Aut}(L/K) \cong \mathbb{Z}/2 \times \mathbb{Z}/2.$$

The standard example is $L = \mathbb{Q}(i, \sqrt{2})$, with $\sigma_1 : i \mapsto -i$ and $\sigma_2 : \sqrt{2} \mapsto -\sqrt{2}$.

§30.8.2 Artin's Theorem

We now reverse the point of view. Instead of starting with an extension and forming its automorphism group, we start with a group of automorphisms and form its fixed field.

Definition 30.8.4 (Fixed Field)

Let L be a field and $G \leq \text{Aut}(L)$ a group of automorphisms. The **fixed field** is

$$L^G = \{\alpha \in L : \sigma\alpha = \alpha \text{ for all } \sigma \in G\},$$

which is easily checked to be a subfield of L .

Throughout this subsection, let $G \leq \text{Aut}(L)$ be a *finite* group of automorphisms and write $K = L^G$. Everything follows from two lemmas about orbits.

Lemma 30.8.5

Let $\gamma \in L$ and let $G\gamma = \{\gamma_1, \dots, \gamma_r\}$ be its orbit (with the γ_i distinct). Then γ is algebraic and separable over K , with minimal polynomial

$$g(x) = \prod_{i=1}^r (x - \gamma_i),$$

and in particular $\deg_K \gamma = \#G\gamma \leq \#G$.

Proof. Each $\sigma \in G$ permutes the orbit $G\gamma$, hence permutes the linear factors of g , hence fixes each coefficient of g . Therefore $g \in L^G[x] = K[x]$. By construction $g(\gamma) = 0$ and g has distinct roots, so γ is algebraic and its minimal polynomial f over K divides g and hence is separable too.

In fact $f = g$, because g is irreducible over K : if $g = g_1 g_2$ non-trivially in $K[x]$, then over L each g_i is a product of some of the $(x - \gamma_i)$, splitting $G\gamma$ into two non-empty sets A, B . But the coefficients of g_1 lie in K , so A is stable under G , which is impossible for a proper non-empty subset of a single orbit.

Finally $\#G\gamma \leq \#G$ by orbit-stabiliser. $\square$

In particular L/K is a separable extension, and every element of L has degree at most $\#G$ over K . That bound is enough to make the primitive element theorem bite.

Lemma 30.8.6

With $K = L^G$ as above, $[L : K] \leq \#G < \infty$, and $L = K(\gamma)$ for some $\gamma \in L$.

Proof. By Lemma 30.8.5 every element of L has degree at most $\#G$ over K , so we may choose $\gamma \in L$ with $[K(\gamma) : K]$ maximal. We claim $K(\gamma) = L$.

Let $\beta \in L$ be arbitrary. Then $[K(\gamma, \beta) : K] \leq [K(\gamma) : K][K(\beta) : K] < \infty$, and $K(\gamma, \beta)/K$ is separable since L/K is. By the Primitive Element Theorem 30.5.18, $K(\gamma, \beta) = K(\delta)$ for some δ . But then

$$[K(\delta) : K] \geq [K(\gamma) : K],$$

and by maximality equality holds, so $K(\delta) = K(\gamma)$ and hence $\beta \in K(\gamma)$. So $L = K(\gamma)$, and $[L : K] = \deg_K \gamma \leq \#G$. $\square$

Theorem 30.8.7 (Artin)

Let L be a field and $G \leq \text{Aut}(L)$ a finite group of automorphisms, and put $K = L^G$. Then

$$[L : K] = \#G \quad \text{and} \quad \text{Aut}(L/K) = G.$$

Proof. By Lemma 30.8.6 we have $L = K(\gamma)$ for some γ , and by Lemma 30.8.5,

$$[L : K] = \deg_K \gamma = \#G\gamma = \frac{\#G}{\#\text{Stab}_G(\gamma)}.$$

But if $\sigma \in \text{Stab}_G(\gamma)$ then σ fixes K and fixes γ , hence fixes $K(\gamma) = L$ pointwise, so

$\sigma = \text{id}$. Therefore $\text{Stab}_G(\gamma)$ is trivial and $[L : K] = \#G$.

For the second claim, certainly $G \leq \text{Aut}(L/K)$ by definition of $K = L^G$. Conversely any $\sigma \in \text{Aut}(L/K)$ is determined by $\sigma(\gamma)$, which must be a root of the minimal polynomial of γ , and there are only $[L : K]$ of those. So

$$\#G \leq \# \text{Aut}(L/K) \leq [L : K] = \#G,$$

forcing $G = \text{Aut}(L/K)$. □

Artin's theorem is a powerful *computational* tool, because it lets us identify fixed fields by a degree count alone.

Example 30.8.8

Let $L = \mathbb{C}(y)$ and let $\sigma, \tau \in \text{Aut}(L)$ be given by $\sigma(y) = i/y$ and $\tau(y) = -y$ (fixing $\mathbb{C}$). One checks $\sigma^2 = \tau^2 = \text{id}$ and $\sigma\tau = \tau\sigma$, so $G = \langle \sigma, \tau \rangle \cong \mathbb{Z}/2 \times \mathbb{Z}/2$. What is $K = L^G$?

The orbit of y is $Gy = \{y, -y, i/y, -i/y\}$, of size 4, so by Lemma 30.8.5 the minimal polynomial of y over K is

$$(x - y)(x + y)(x - i/y)(x + i/y) = (x^2 - y^2)(x^2 + y^{-2}) = x^4 + (y^{-2} - y^2)x^2 - 1.$$

So $\eta = y^2 - y^{-2} \in K$, and $[L : \mathbb{C}(\eta)] \leq 4$ since y satisfies a quartic over $\mathbb{C}(\eta)$. But $[L : K] = \#G = 4$ by Artin, and $\mathbb{C}(\eta) \subseteq K$, so

$$K = \mathbb{C}(y^2 - y^{-2}).$$

Example 30.8.9

Let $L = \mathbb{F}_{q^n}$ and $K = \mathbb{F}_q$, where $q = p^r$. Let $\varphi : L \rightarrow L$ be $\varphi(x) = x^q$, a field automorphism (an iterate of Frobenius). It fixes K pointwise by Fermat's little theorem, so $\varphi \in \text{Aut}(L/K)$. Its order is exactly n : we have $\varphi^n(x) = x^{q^n} = x$ for all x , so the order divides n ; and if $\varphi^m = \text{id}$ for $m < n$ then every element of L satisfies $x^{q^m} = x$, so $\#L \leq q^m < q^n$, a contradiction. Moreover $L^{\langle \varphi \rangle} = \{x : x^q = x\} = \mathbb{F}_q = K$. So by Artin's theorem

$$\text{Aut}(\mathbb{F}_{q^n}/\mathbb{F}_q) = \langle \varphi \rangle \cong \mathbb{Z}/n,$$

generated by the Frobenius. This is a complete description of the symmetries of finite fields, and it is very simple: *cyclic, generated by Frobenius*.

§30.8.3 Galois Extensions

We saw in Example 30.8.2 that $\# \text{Aut}(L/K)$ can be much smaller than $[L : K]$. It can never be larger.

Proposition 30.8.10

Let L/K be a finite extension. Then $\# \text{Aut}(L/K)$ divides $[L : K]$.

Proof. Put $G = \text{Aut}(L/K)$, which is finite (it injects into the roots of the minimal polynomial of a primitive element of any simple subextension; more simply, $\#G \leq [L : K]_s \leq [L : K]$). Let $M = L^G$, so $K \subseteq M \subseteq L$. By Artin's theorem $[L : M] = \#G$, and by the tower law

$$[L : K] = [L : M][M : K] = \# \text{Aut}(L/K) \cdot [M : K]. \quad \square$$

Definition 30.8.11 (Galois Extension)

A finite extension L/K is a **Galois extension** if

$$\# \text{Aut}(L/K) = [L : K].$$

In that case we write $\text{Gal}(L/K) = \text{Aut}(L/K)$ and call it the **Galois group**.

The following theorem gives four different faces of the same idea, and in practice one uses whichever is most convenient.

Theorem 30.8.12

Let L/K be a finite extension. The following are equivalent:

- (i) L/K is Galois;
- (ii) $K = L^G$ for some finite subgroup $G \leq \text{Aut}(L)$;
- (iii) L is a splitting field over K of some separable polynomial in $K[x]$;
- (iv) L/K is normal and separable.

Proof. (i) $\Rightarrow$ (ii). Take $G = \text{Aut}(L/K)$ and $M = L^G$. As in the proof of Proposition 30.8.10, $[L : M] = \#G = [L : K]$, so $[M : K] = 1$ and $K = L^G$.

(ii) $\Rightarrow$ (i). This is Artin's theorem: $[L : K] = \#G = \# \text{Aut}(L/K)$.

(ii) $\Rightarrow$ (iv). Lemma 30.8.5 says that for every $\gamma \in L$ the minimal polynomial over K is $\prod_{\delta \in G\gamma} (x - \delta)$, which is separable and splits in L . So L/K is separable and normal.

(iv) $\Rightarrow$ (iii). Write $L = K(\alpha_1, \dots, \alpha_n)$ and let f_i be the minimal polynomial of α_i over K . By separability each f_i is separable, and by normality each splits in L . Put $f = \text{lcm}(f_1, \dots, f_n)$, which is separable by Lemma 30.5.17 and splits in L ; and L is generated by its roots since the α_i are among them. So L is a splitting field of the separable polynomial f .

(iii) $\Rightarrow$ (i). This is Corollary 30.4.10: if L is a splitting field for f over K and f has no repeated roots, then $\# \text{Aut}(L/K) = [L : K]$. $\square$

Corollary 30.8.13

If L/K is Galois and $K \subseteq M \subseteq L$, then L/M is Galois.

Proof. Property (iv) of Theorem 30.8.12 passes to L/M : the minimal polynomial over M of $\alpha \in L$ divides the minimal polynomial over K , hence is separable and splits in L . $\square$

Corollary 30.8.14

Every finite separable extension L/K sits inside a Galois extension N/K , and there is a smallest such N , namely the normal closure of L/K .

Proof. Let N be the normal closure from Corollary 30.4.13, that is the splitting field of $f = f_1 \cdots f_n$ where f_i is the minimal polynomial of a generator α_i of L . Replacing f by $\text{lcm}(f_1, \dots, f_n)$ makes it separable without changing the splitting field, so N/K is Galois by Theorem 30.8.12(iii). $\square$

Note that M/K need *not* be Galois for intermediate M : for instance $\mathbb{Q} \subseteq \mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{Q}(\sqrt[3]{2}, \omega)$, where the big extension is Galois but the middle one is not. Exactly which intermediate fields are Galois over K is the second half of the fundamental theorem.

§30.8.4 The Fundamental Theorem

Theorem 30.8.15 (Fundamental Theorem of Galois Theory)

Let L/K be a finite Galois extension with $G = \text{Gal}(L/K)$. Then there are mutually inverse bijections

$$\{\text{subgroups } H \leq G\} \longleftrightarrow \{\text{fields } M \text{ with } K \subseteq M \subseteq L\},$$

given by $H \mapsto L^H$ and $M \mapsto \text{Gal}(L/M)$. In particular there are only finitely many intermediate fields. Moreover:

- (i) the correspondence is inclusion-reversing: $H \leq H'$ if and only if $L^{H'} \subseteq L^H$;
- (ii) $[L : L^H] = \#H$ and $[L^H : K] = (G : H)$;
- (iii) $H \trianglelefteq G$ if and only if L^H/K is a Galois extension, and in that case restriction gives an isomorphism

$$\text{Gal}(L^H/K) \cong G/H.$$

Proof. First, both maps are well defined: L^H is a field between K and L , and $\text{Gal}(L/M)$ makes sense because L/M is Galois by Corollary 30.8.13. We check both composites are the identity.

Starting from a subgroup: $H \mapsto L^H \mapsto \text{Aut}(L/L^H) = H$, by Artin's Theorem 30.8.7.

Starting from a field: $M \mapsto \text{Gal}(L/M) \mapsto L^{\text{Gal}(L/M)}$. Since L/M is Galois, Theorem 30.8.12(i) $\Rightarrow$ (ii) (applied with base field M) gives $L^{\text{Gal}(L/M)} = M$.

Statement (i) is immediate from the definitions, in both directions (for the 'if', apply the bijection). Statement (ii) is Artin's theorem together with the tower law.

For (iii), suppose first that $H \trianglelefteq G$. For $\sigma \in G$, $\eta \in H$ and $\ell \in L^H$ we have

$$\eta(\sigma\ell) = \sigma(\sigma^{-1}\eta\sigma)(\ell) = \sigma(\eta'\ell) = \sigma\ell,$$

where $\eta' = \sigma^{-1}\eta\sigma \in H$ by normality. So $\sigma(L^H) \subseteq L^H$, and by applying the same to σ^{-1} , in fact $\sigma(L^H) = L^H$. Hence restriction gives a homomorphism

$$\rho : G \rightarrow \text{Aut}(L^H/K), \quad \sigma \mapsto \sigma|_{L^H},$$

whose kernel is $\{\sigma \in G : \sigma|_{L^H} = \text{id}\} = \text{Gal}(L/L^H) = H$. So $G/H \hookrightarrow \text{Aut}(L^H/K)$, giving

$$\#\text{Aut}(L^H/K) \geq (G : H) = [L^H : K].$$

The reverse inequality is Proposition 30.8.10, so we have equality: L^H/K is Galois with $\text{Gal}(L^H/K) \cong G/H$.

Conversely suppose $M = L^H$ is Galois over K . Let $\sigma \in G$ and $\alpha \in M$. The minimal polynomial of α over K splits in M by normality of M/K , and $\sigma\alpha$ is one of its roots by Lemma 30.8.1(i), so $\sigma\alpha \in M$. Hence $\sigma(M) = M$ for all $\sigma \in G$. But then for $\eta \in H$ and $\alpha \in M$ we get $\sigma^{-1}\eta\sigma(\alpha) = \sigma^{-1}\sigma\alpha = \alpha$, so $\sigma^{-1}H\sigma$ fixes M pointwise, that is $\sigma^{-1}H\sigma \leq \text{Gal}(L/M) = H$. So $H \trianglelefteq G$. $\square$

Remark. It is worth recording the mnemonic: *big subgroups correspond to small fields*. The whole group G corresponds to K , and the trivial subgroup to L . A subgroup of index 2 gives a quadratic subextension of K , not of L .

§30.8.5 Worked Examples

Example 30.8.16

If $\text{Gal}(L/K) \cong \mathbb{Z}/p$ with p prime, then there are no proper intermediate fields, since $\mathbb{Z}/p$ has no proper non-trivial subgroups.

Example 30.8.17 (Finite fields)

Let $L = \mathbb{F}_{q^n}$ and $K = \mathbb{F}_q$. We saw in Example 30.8.9 that L/K is Galois with $\text{Gal}(L/K) = \langle \varphi \rangle \cong \mathbb{Z}/n$. The subgroups of $\mathbb{Z}/n$ are $\langle \varphi^m \rangle$ for $m \mid n$, of order n/m , so the intermediate fields are exactly the

$$L^{\langle \varphi^m \rangle} = \{x : x^{q^m} = x\} = \mathbb{F}_{q^m}, \quad m \mid n,$$

recovering Theorem 30.6.4(iii) at a stroke.

Example 30.8.18

Let $L = \mathbb{Q}(i, \sqrt{2})$ over $K = \mathbb{Q}$. By Example 30.8.3, $G = \mathbb{Z}/2 \times \mathbb{Z}/2 = \{1, \sigma_1, \sigma_2, \sigma_1\sigma_2\}$ where $\sigma_1 : i \mapsto -i$ and $\sigma_2 : \sqrt{2} \mapsto -\sqrt{2}$. The proper non-trivial subgroups are $\langle \sigma_1 \rangle, \langle \sigma_2 \rangle, \langle \sigma_1\sigma_2 \rangle$, so there are exactly three intermediate fields, namely

$$L^{\langle \sigma_1 \rangle} = \mathbb{Q}(\sqrt{2}), \quad L^{\langle \sigma_2 \rangle} = \mathbb{Q}(i), \quad L^{\langle \sigma_1\sigma_2 \rangle} = \mathbb{Q}(i\sqrt{2}).$$

Since G is abelian all subgroups are normal, so all three are Galois over $\mathbb{Q}$ – as they

must be, being quadratic.

The Galois correspondence also gives a very efficient way to compute minimal polynomials: by Lemma 30.8.5, the minimal polynomial of γ over K is $\prod_{\delta \in G\gamma} (x - \delta)$.

Example 30.8.19

In $L = \mathbb{Q}(i, \sqrt{2})$, take $\gamma = 1 + i + \sqrt{2}$. Its orbit under G is $\{1 \pm i \pm \sqrt{2}\}$, so its minimal polynomial is

$$(x - 1 - i - \sqrt{2})(x - 1 + i - \sqrt{2})(x - 1 - i + \sqrt{2})(x - 1 + i + \sqrt{2}) \in \mathbb{Q}[x],$$

which expands to $x^4 - 4x^3 + 4x^2 + 4x - 7$. Doing this by elimination would be painful; knowing the Galois group makes it a two-line computation.

Now let us do the example properly – the one that is worth committing to memory.

Example 30.8.20 (The subfields of $\mathbb{Q}(\sqrt[4]{2}, i)$)

Let $\alpha = \sqrt[4]{2} \in \mathbb{R}$ and $L = \mathbb{Q}(\alpha, i)$. Since $x^4 - 2$ has roots $\pm\alpha, \pm i\alpha$, the field L is a splitting field for $x^4 - 2$ over $\mathbb{Q}$, hence Galois (characteristic zero, so separability is automatic). Its degree is 8: indeed $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$ by Eisenstein, and $i \notin \mathbb{R} \supseteq \mathbb{Q}(\alpha)$ so $[L : \mathbb{Q}(\alpha)] = 2$.

So $\#G = 8$ where $G = \text{Gal}(L/\mathbb{Q})$, and G embeds into S_4 as permutations of the four roots. Let

$$r : \alpha \mapsto i\alpha, \quad i \mapsto i, \quad s : \alpha \mapsto \alpha, \quad i \mapsto -i.$$

Then r cycles the roots as $\alpha \rightarrow i\alpha \rightarrow -\alpha \rightarrow -i\alpha$, so has order 4, while s is complex conjugation, fixing $\pm\alpha$ and swapping $\pm i\alpha$, so has order 2. One checks $srs^{-1} = r^{-1}$, and so

$$G = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle \cong D_8,$$

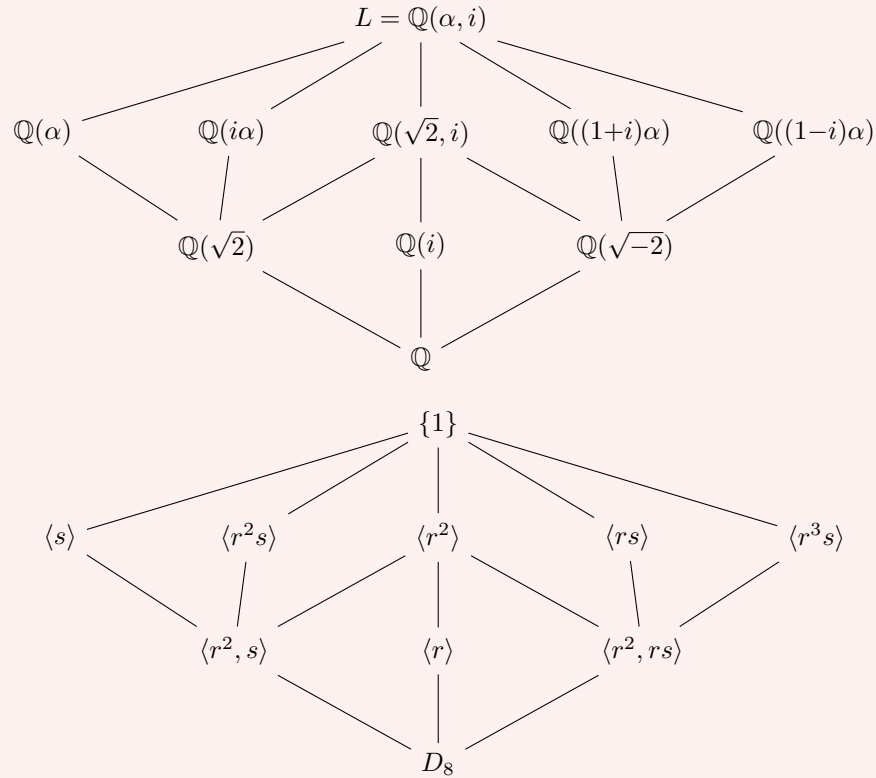
the dihedral group of order 8 (the symmetries of a square, the square in question having the four roots as its vertices in $\mathbb{C}$).

Now D_8 has exactly ten subgroups: itself, the trivial one, three of order 4 (namely $\langle r \rangle$, $\langle r^2, s \rangle$ and $\langle r^2, rs \rangle$) and five of order 2 (namely $\langle r^2 \rangle$, $\langle s \rangle$, $\langle r^2s \rangle$, $\langle rs \rangle$ and $\langle r^3s \rangle$). So L has exactly ten subfields, and we can name them all:

$$\begin{aligned} L^{\langle s \rangle} &= \mathbb{Q}(\alpha), & L^{\langle r^2s \rangle} &= \mathbb{Q}(i\alpha), & L^{\langle r^2 \rangle} &= \mathbb{Q}(\sqrt{2}, i), \\ L^{\langle rs \rangle} &= \mathbb{Q}((1+i)\alpha), & L^{\langle r^3s \rangle} &= \mathbb{Q}((1-i)\alpha), & L^{\langle r \rangle} &= \mathbb{Q}(i), \\ L^{\langle r^2, s \rangle} &= \mathbb{Q}(\sqrt{2}), & L^{\langle r^2, rs \rangle} &= \mathbb{Q}(\sqrt{-2}). \end{aligned}$$

Each is easily verified: for instance rs sends $\alpha \mapsto i\alpha$ and $i \mapsto -i$, so it fixes $(1+i)\alpha \mapsto (1-i)(i\alpha) = (1+i)\alpha$, and $\mathbb{Q}((1+i)\alpha)$ has degree 4 over $\mathbb{Q}$, matching $[L : L^{\langle rs \rangle}] = 2$.

Here are the two lattices. The subgroup lattice has been drawn *upside down*, with G at the bottom, so that corresponding objects sit in corresponding places; remember that the correspondence reverses inclusions.



Reading off part (iii) of the fundamental theorem: the normal subgroups of D_8 are $\{1\}$, $\langle r^2 \rangle$, the three subgroups of index 2, and D_8 itself. Correspondingly, the subfields Galois over $\mathbb{Q}$ are exactly

$$\mathbb{Q}, \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(i), \quad \mathbb{Q}(\sqrt{-2}), \quad \mathbb{Q}(\sqrt{2}, i), \quad L,$$

and the four degree-4 fields $\mathbb{Q}(\alpha), \mathbb{Q}(i\alpha), \mathbb{Q}((1 \pm i)\alpha)$ are *not* normal over $\mathbb{Q}$ – as we can see directly, since for instance $x^4 - 2$ has only two of its four roots inside $\mathbb{Q}(\alpha)$. Note also that $\langle s \rangle$ and $\langle r^2s \rangle$ are conjugate in D_8 (indeed $r\langle s \rangle r^{-1} = \langle r^2s \rangle$), which matches the fact that $\mathbb{Q}(\alpha)$ and $\mathbb{Q}(i\alpha)$ are abstractly isomorphic fields sitting differently inside L .

This example repays study. Almost every phenomenon in the subject is visible in it: non-normal intermediate fields, conjugate subgroups giving isomorphic-but-distinct subfields, and the way that a single degree-8 extension can hide ten different fields.

§30.9 Galois Groups of Polynomials

We now return to Galois' original setting: a polynomial and the permutations of its roots.

Definition 30.9.1 (Galois Group of a Polynomial)

Let $f \in K[x]$ be a separable polynomial and let L be a splitting field for f over K . Then L/K is Galois, and we write

$$\text{Gal}(f/K) = \text{Gal}(L/K).$$

If f has roots $\alpha_1, \dots, \alpha_n$ in L , then by Lemma 30.8.1 we may regard $\text{Gal}(f/K) \leq S_n$ as a group of permutations of the roots.

The identification with a subgroup of S_n depends on the ordering of the roots; changing the ordering conjugates the subgroup. So $\text{Gal}(f/K)$ is well defined as a subgroup of S_n only up to conjugacy, which is all we shall ever need.

Proposition 30.9.2

Let $f \in K[x]$ be separable of degree n . Then f is irreducible over K if and only if $\text{Gal}(f/K)$ acts transitively on the roots of f .

Proof. Suppose f is irreducible and let α, β be roots. By Corollary 30.4.3 there is a K -isomorphism $K(\alpha) \rightarrow K(\beta)$ with $\alpha \mapsto \beta$, and by Theorem 30.4.9(i) it extends to $\sigma : L \rightarrow L$, which is an automorphism. So the action is transitive.

Conversely if $f = gh$ non-trivially, then no automorphism can carry a root of g to a root of h , by Lemma 30.8.1(i), so the action is not transitive. $\square$

§30.9.1 Symmetric Polynomials

The coefficients of a polynomial are symmetric functions of its roots, so any expression in the roots that is invariant under all permutations ought to be expressible in the coefficients. That vague statement has a clean and useful theorem behind it.

Let R be a ring, and let S_n act on $R[z_1, \dots, z_n]$ by permuting the variables, $w \cdot z_i = z_{w(i)}$. The invariant ring $R[z_1, \dots, z_n]^{S_n}$ consists of the **symmetric polynomials**. Among them are the **elementary symmetric polynomials**

$$e_1 = \sum_i z_i, \quad e_2 = \sum_{i < j} z_i z_j, \quad \dots, \quad e_n = z_1 z_2 \cdots z_n.$$

Theorem 30.9.3 (Fundamental Theorem of Symmetric Polynomials)

The ring homomorphism

$$R[w_1, \dots, w_n] \rightarrow R[z_1, \dots, z_n]^{S_n}, \quad w_i \mapsto e_i$$

is an isomorphism. That is, every symmetric polynomial is a polynomial in $e_1, \dots, e_n$, uniquely so.

Proof. For $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{N}^n$ write $z^\lambda = z_1^{\lambda_1} \cdots z_n^{\lambda_n}$; these monomials form an R -basis of $R[z_1, \dots, z_n]$. Order $\mathbb{N}^n$ lexicographically. If λ is *decreasing*, meaning $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, then $\lambda \geq w\lambda$ for all $w \in S_n$, so every S_n -orbit in $\mathbb{N}^n$ contains a unique decreasing element, which is its maximum. It follows that the orbit sums $\sum_{\mu \in S_n \lambda} z^\mu$, indexed by decreasing λ , form an R -basis of the ring of symmetric polynomials.

Surjectivity. Let f be symmetric and non-zero, and let z^λ be its lexicographically

largest monomial, with coefficient c ; by symmetry λ is decreasing. Consider

$$E_\lambda = e_1^{\lambda_1 - \lambda_2} e_2^{\lambda_2 - \lambda_3} \cdots e_{n-1}^{\lambda_{n-1} - \lambda_n} e_n^{\lambda_n}.$$

Expanding, the lexicographically largest monomial of e_k is $z_1 z_2 \cdots z_k$, so the lexicographically largest monomial of E_λ is

$$z_1^{\lambda_1 - \lambda_2} (z_1 z_2)^{\lambda_2 - \lambda_3} \cdots (z_1 \cdots z_n)^{\lambda_n} = z^\lambda.$$

Hence $f - cE_\lambda$ is symmetric with strictly smaller leading monomial, and the process terminates by induction (there are finitely many monomials of any given degree).

Injectivity. We induct on n ; for $n = 1$ there is nothing to prove. Suppose $g \in R[w_1, \dots, w_n]$ is non-zero of minimal degree with $g(e_1, \dots, e_n) = 0$. Setting $z_n = 0$ turns e_i into the i th elementary symmetric polynomial in $z_1, \dots, z_{n-1}$ for $i < n$, and turns e_n into 0. So

$$g(e'_1, \dots, e'_{n-1}, 0) = 0 \text{ in } R[z_1, \dots, z_{n-1}],$$

whence by induction $g(w_1, \dots, w_{n-1}, 0) = 0$ identically, so $w_n \mid g$, say $g = w_n h$. Then $e_n h(e_1, \dots, e_n) = 0$, and $e_n = z_1 \cdots z_n \neq 0$ in the integral domain $R[z_1, \dots, z_n]$ (if R is a domain; in general compare coefficients), so $h(e_1, \dots, e_n) = 0$ with $\deg h < \deg g$, contradicting minimality. $\square$

Now let $f \in K[x]$ be monic of degree n with roots $\alpha_1, \dots, \alpha_n$ in a splitting field, so that

$$f(x) = x^n - a_1 x^{n-1} + a_2 x^{n-2} - \cdots + (-1)^n a_n$$

with $a_i = e_i(\alpha_1, \dots, \alpha_n)$. The theorem says that *any* symmetric expression in the roots is a polynomial in the coefficients $a_1, \dots, a_n$, with the same coefficient ring. This is the machine we now use.

§30.9.2 The Discriminant

Definition 30.9.4 (Discriminant)

Let $f \in K[x]$ be monic of degree n with roots $\alpha_1, \dots, \alpha_n$ in a splitting field. The **discriminant** of f is

$$\text{Disc}(f) = \prod_{i < j} (\alpha_i - \alpha_j)^2.$$

We also write $\delta = \prod_{i < j} (\alpha_i - \alpha_j)$, so that $\text{Disc}(f) = \delta^2$.

Proposition 30.9.5

With notation as above, and assuming $\text{char } K \neq 2$:

- (i) $\text{Disc}(f) \in K$, and indeed $\text{Disc}(f)$ is a universal polynomial with integer coefficients in the coefficients of f ;
- (ii) $\text{Disc}(f) \neq 0$ if and only if f is separable;
- (iii) if f is separable, then $\text{Gal}(f/K) \leq A_n$ if and only if $\text{Disc}(f)$ is a square in K .

Proof. (i) $\text{Disc}(f)$ is a symmetric polynomial in $\alpha_1, \dots, \alpha_n$ with integer coefficients, so by Theorem 30.9.3 it is a polynomial with integer coefficients in $e_1, \dots, e_n$, that is in the coefficients of f .

(ii) is immediate, as $\text{Disc}(f) = 0$ if and only if two roots coincide.

(iii) Let $G = \text{Gal}(f/K) \leq S_n$. Any $\sigma \in G$ permutes the α_i and hence permutes the factors $(\alpha_i - \alpha_j)$ up to sign, with the sign being $\text{sign}(\sigma)$; that is,

$$\sigma(\delta) = \text{sign}(\sigma)\delta.$$

So $G \leq A_n$ if and only if $\sigma\delta = \delta$ for all $\sigma \in G$, which by the Galois correspondence holds if and only if $\delta \in L^G = K$, that is, if and only if $\text{Disc}(f) = \delta^2$ is a square in K . (We used $\text{char } K \neq 2$ to know that $\delta \neq -\delta$.) $\square$

There is a slicker formula for the discriminant in terms of the derivative, which is often the quickest way to compute it.

Proposition 30.9.6

Let f be monic of degree n with roots $\alpha_1, \dots, \alpha_n$. Then

$$\prod_{i=1}^n f'(\alpha_i) = (-1)^{n(n-1)/2} \text{Disc}(f).$$

Proof. Differentiating $f(x) = \prod_j (x - \alpha_j)$ by the product rule gives

$$f'(x) = \sum_i \prod_{j \neq i} (x - \alpha_j),$$

and evaluating at α_i kills all the terms but one:

$$f'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j).$$

Multiplying over i gives $\prod_{i \neq j} (\alpha_i - \alpha_j)$. Pairing off (i, j) with (j, i) introduces a factor (-1) for each of the $\binom{n}{2}$ pairs, giving $(-1)^{n(n-1)/2} \prod_{i < j} (\alpha_i - \alpha_j)^2$. $\square$

Example 30.9.7

For a quadratic $f = x^2 + bx + c$ we have $\text{Disc}(f) = (\alpha_1 - \alpha_2)^2 = (\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2 = b^2 - 4c$, as it should be. Then $\text{Gal}(f/K) \leq A_2 = 1$ if and only if $b^2 - 4c$ is a square, which is exactly the condition for the roots to lie in K .

Example 30.9.8

For a cubic $f = x^3 + px + q$ we claim

$$\text{Disc}(f) = -4p^3 - 27q^2.$$

One can get this cheaply. Since $e_1 = 0$, the discriminant is a polynomial in $e_2 = p$

and $e_3 = -q$ only; and Disc is homogeneous of degree 6 in the roots, while p has degree 2 and q degree 3, so

$$\text{Disc}(f) = cp^3 + dq^2$$

for some integers c, d . Taking $f = (x - \alpha)^2(x + 2\alpha)$, which has $p = -3\alpha^2$, $q = 2\alpha^3$ and discriminant 0, gives $-27c\alpha^6 + 4d\alpha^6 = 0$, so $4d = 27c$. Taking $f = x^3 - x$, with roots $0, \pm 1$, gives $\text{Disc} = (0 - 1)^2(0 + 1)^2(1 + 1)^2 = 4$, so $-c = 4$; hence $c = -4$ and $d = -27$.

§30.9.3 Cubics

Let $f \in K[x]$ be a separable cubic with $\text{char } K \neq 2, 3$; after substituting $x \mapsto x - a_1/3$ we may take $f = x^3 + px + q$. Let L be a splitting field and $G = \text{Gal}(f/K) \leq S_3$.

If f is reducible then G is trivial (three roots in K) or $\cong S_2$ (one root in K and an irreducible quadratic factor). So suppose f is irreducible. Then G acts transitively on three points, so $3 \mid \#G$, forcing $G = A_3$ or $G = S_3$; and by Proposition 30.9.5, $G = A_3$ exactly when $\text{Disc}(f) = -4p^3 - 27q^2$ is a square in K .

The Galois correspondence for the intermediate field $K(\delta) = K(\sqrt{\text{Disc } f})$ then reads as follows.

$$\begin{array}{ccc}
 L = K(\alpha_1, \alpha_2, \alpha_3) & \longleftrightarrow & \{1\} \\
 \downarrow 3 & & \downarrow \text{index 3} \\
 K(\delta) = L^{G \cap A_3} & \longleftrightarrow & G \cap A_3 \\
 \downarrow 1 \text{ or } 2 & & \downarrow \text{index 1 or 2} \\
 K & \longleftrightarrow & G
 \end{array}$$

So the splitting field of an irreducible cubic is obtained from K by first adjoining a square root (of the discriminant) and then a cube root – which is exactly the shape of Cardano's formula. Let us see the cube root appear.

Example 30.9.9

Suppose $f = x^3 + px + q$ is irreducible over K , that $\text{char } K \neq 2, 3$, and that K contains $\delta = \sqrt{\text{Disc } f}$ and a primitive cube root of unity ω . Then $G = \text{Gal}(L/K) = A_3 = \langle \sigma \rangle \cong \mathbb{Z}/3$, where we may take σ to be the 3-cycle $\alpha_1 \rightarrow \alpha_2 \rightarrow \alpha_3 \rightarrow \alpha_1$. Define the **Lagrange resolvents**

$$\beta = \alpha_1 + \omega\alpha_2 + \omega^2\alpha_3, \quad \gamma = \alpha_1 + \omega^2\alpha_2 + \omega\alpha_3.$$

Then $\sigma\beta = \alpha_2 + \omega\alpha_3 + \omega^2\alpha_1 = \omega^{-1}\beta$ and similarly $\sigma\gamma = \omega\gamma$, so $\sigma(\beta^3) = \beta^3$ and $\sigma(\gamma^3) = \gamma^3$; hence $\beta^3, \gamma^3 \in L^G = K$. Also $\alpha_1 + \alpha_2 + \alpha_3 = 0$, so

$$\alpha_1 = \frac{1}{3}(\beta + \gamma), \quad \alpha_2 = \frac{1}{3}(\omega^2\beta + \omega\gamma), \quad \alpha_3 = \frac{1}{3}(\omega\beta + \omega^2\gamma).$$

It remains to compute β^3 and γ^3 . Using $\sum \alpha_i = 0$, $\sum_{i < j} \alpha_i \alpha_j = p$ and $1 + \omega + \omega^2 = 0$:

$$\beta\gamma = \sum_i \alpha_i^2 + (\omega + \omega^2) \sum_{i < j} \alpha_i \alpha_j = \left(\sum_i \alpha_i \right)^2 - 3 \sum_{i < j} \alpha_i \alpha_j = -3p,$$

so $\beta^3\gamma^3 = -27p^3$. A similar (slightly longer) expansion gives $\beta^3 + \gamma^3 = -27q$. Hence β^3, γ^3 are the roots of

$$t^2 + 27qt - 27p^3 = 0,$$

that is $\beta^3 = \frac{1}{2}(-27q + 3\sqrt{-3\text{Disc } f})$, and then $\gamma = -3p/\beta$. This is Cardano's formula, derived rather than guessed.

§30.9.4 Quartics and the Resolvent Cubic

Now let $f \in K[x]$ be an irreducible separable quartic, with $\text{char } K \neq 2, 3$, splitting field L and $G = \text{Gal}(f/K) \leq S_4$. Translating, we may assume $\sum \alpha_i = 0$, so

$$f(x) = x^4 + px^2 + qx + r.$$

Since G is transitive, $4 \mid \#G$, and the transitive subgroups of S_4 are (up to conjugacy)

$$S_4, \quad A_4, \quad D_8 = \langle (1234), (13) \rangle, \quad \mathbb{Z}/4 = \langle (1234) \rangle,$$

$$V = \{e, (12)(34), (13)(24), (14)(23)\}.$$

The key structural fact is that the Klein four-group V is normal in S_4 , with $S_4/V \cong S_3$; concretely, S_4 acts on the three partitions

$$(12 \mid 34), \quad (13 \mid 24), \quad (14 \mid 23)$$

of $\{1, 2, 3, 4\}$ into two pairs, and the kernel of the resulting homomorphism $S_4 \rightarrow S_3$ is exactly V . Since $\#V \cdot \#S_3 = 24$, the map is surjective.

Under the Galois correspondence, $G \cap V \trianglelefteq G$ corresponds to an intermediate field $M = L^{G \cap V}$ with $\text{Gal}(M/K) = G/(G \cap V) \hookrightarrow S_4/V \cong S_3$. So M should be the splitting field of a *cubic*, and we can find it.

Put

$$y_{12} = \alpha_1 + \alpha_2 = -(\alpha_3 + \alpha_4) = -y_{34},$$

and similarly y_{13}, y_{14} . Each element of V either fixes y_{12} or sends it to $y_{34} = -y_{12}$, so $y_{12}^2, y_{13}^2, y_{14}^2$ are fixed by V and hence lie in M . They are distinct: if $y_{12}^2 = y_{13}^2$ then either $y_{12} = y_{13}$, giving $\alpha_2 = \alpha_3$, or $y_{12} = -y_{13}$, giving $2\alpha_1 + \alpha_2 + \alpha_3 = 0$ and hence $\alpha_1 = \alpha_4$; both contradict separability. Finally S_4 permutes $\{y_{12}^2, y_{13}^2, y_{14}^2\}$, so their elementary symmetric functions lie in K .

Definition 30.9.10 (Resolvent Cubic)

The **resolvent cubic** of $f = x^4 + px^2 + qx + r$ is

$$g(x) = (x - y_{12}^2)(x - y_{13}^2)(x - y_{14}^2) \in K[x].$$

Proposition 30.9.11

With notation as above, $g(x) = x^3 + 2px^2 + (p^2 - 4r)x - q^2$, and $M = L^{G \cap V}$ is a splitting field for g over K . Moreover $L = M(y_{12}, y_{13})$, and $\alpha_1 = \frac{1}{2}(y_{12} + y_{13} + y_{14})$ with $y_{12}y_{13}y_{14} = -q$.

Proof. The formula for g is a direct expansion using the symmetric function identities $\sum \alpha_i = 0$, $\sum_{i < j} \alpha_i \alpha_j = p$, $\sum_{i < j < k} \alpha_i \alpha_j \alpha_k = -q$, $\alpha_1 \alpha_2 \alpha_3 \alpha_4 = r$; for instance

$$y_{12}^2 + y_{13}^2 + y_{14}^2 = -((\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4) + \dots) = -2p.$$

That $M \supseteq K(y_{12}^2, y_{13}^2, y_{14}^2)$ we have already seen; equality follows from Artin's theorem, since $\text{Aut}(L/K(y_{12}^2, y_{13}^2, y_{14}^2)) = G \cap V$ (an automorphism fixing all three squares must preserve each partition, hence lie in V).

For the last statements, $\alpha_1 = \frac{1}{2}((\alpha_1 + \alpha_2) + (\alpha_1 + \alpha_3) + (\alpha_1 + \alpha_4))$ since $\sum \alpha_i = 0$, and $y_{12}y_{13}y_{14} = -q$ by another expansion. Hence $L = K(\alpha_1, \dots, \alpha_4) = M(y_{12}, y_{13}, y_{14}) = M(y_{12}, y_{13})$. $\square$

So a quartic is solved by first solving its resolvent cubic and then extracting two square roots – and the whole tower is

$$K \subseteq K(\delta) \subseteq M \subseteq L$$

of degrees dividing 2, 3, 4 respectively. Comparing G with $G \cap A_4$ and $G \cap V$ lets us determine G completely.

Proposition 30.9.12

Let f be an irreducible separable quartic over K with resolvent cubic g . Then $\text{Gal}(f/K)$ is determined by the following table, where $\delta = \sqrt{\text{Disc } f}$.

g irreducible over K ?	$\delta \in K$?	G
yes	yes	A_4
yes	no	S_4
no, splits completely	yes	V
no, one root in K	no	D_8 or $\mathbb{Z}/4$

Proof. We have $[M : K] = \#(G/(G \cap V))$, which is the order of the image of G in S_3 , and this image is $\text{Gal}(g/K)$. Also $\delta \in K$ if and only if $G \leq A_4$. Running through the five transitive subgroups: $G = S_4$ gives image S_3 and $\delta \notin K$; $G = A_4$ gives image A_3 (so g is irreducible) and $\delta \in K$; $G = V$ gives trivial image, so g splits, and $V \leq A_4$ so $\delta \in K$; $G = D_8$ or $\mathbb{Z}/4$ gives image of order 2, so g has exactly one root in K , and neither is contained in A_4 . $\square$

Distinguishing D_8 from $\mathbb{Z}/4$ requires slightly more work (one asks whether f remains irreducible over $K(\delta)$), and we shall not pursue it.

§30.9.5 Reduction Modulo a Prime

For polynomials over $\mathbb{Z}$ there is a beautiful and very practical technique: reduce mod p , where finite field theory tells us everything, and transfer the information back.

Recall from Example 30.8.9 that $\text{Gal}(\bar{f}/\mathbb{F}_p)$ is cyclic, generated by Frobenius. Its cycle type is easy to read off.

Proposition 30.9.13

Let $\bar{f} \in \mathbb{F}_p[x]$ be separable of degree n with irreducible factors of degrees $n_1, \dots, n_r$. Then $\text{Gal}(\bar{f}/\mathbb{F}_p)$ is cyclic, generated by an element of cycle type $(n_1, \dots, n_r)$, and so has order $\text{lcm}(n_1, \dots, n_r)$.

Proof. Let L be a splitting field. Then $\text{Gal}(L/\mathbb{F}_p) = \langle \varphi \rangle$ with $\varphi(x) = x^p$. If α is a root of the irreducible factor g_i , then the roots of g_i are exactly the φ -orbit of α , which therefore has size $\deg g_i = n_i$. So φ acts on the n roots as a product of disjoint cycles of lengths $n_1, \dots, n_r$. $\square$

Theorem 30.9.14 (Dedekind)

Let $f \in \mathbb{Z}[x]$ be monic and separable of degree n , and let p be a prime such that the reduction $\bar{f} \in \mathbb{F}_p[x]$ is also separable. Then $\text{Gal}(\bar{f}/\mathbb{F}_p)$ is (conjugate in S_n to) a subgroup of $\text{Gal}(f/\mathbb{Q})$.

Proof sketch. Let L be a splitting field for f over $\mathbb{Q}$, with roots $\alpha_1, \dots, \alpha_n$, and put $R = \mathbb{Z}[\alpha_1, \dots, \alpha_n]$. Since f is monic, R is finitely generated as a $\mathbb{Z}$ -module. It is torsion-free, so it is free, say $R \cong \mathbb{Z}^M$. Choose a maximal ideal $P \supseteq pR$ of R ; then $F = R/P$ is a finite field of characteristic p , generated by the images $\bar{\alpha}_i$, and in $F[x]$ we have

$$\bar{f} = \prod_i (x - \bar{\alpha}_i).$$

As $\bar{f}$ is separable the $\bar{\alpha}_i$ are distinct, so F is a splitting field for $\bar{f}$.

The group $G = \text{Gal}(f/\mathbb{Q})$ permutes the α_i and hence acts on R ; let $H = \text{Stab}_G(P)$. Then H acts on $F = R/P$, permuting the $\bar{\alpha}_i$ exactly as it permutes the α_i , giving an injection $H \hookrightarrow \text{Gal}(F/\mathbb{F}_p)$. A counting argument using the Chinese remainder theorem on the G -orbit of P shows this injection is onto, which gives the result. $\square$

Corollary 30.9.15

With f and p as above, if $\bar{f}$ factors into distinct irreducibles of degrees $n_1, \dots, n_r$ over $\mathbb{F}_p$, then $\text{Gal}(f/\mathbb{Q})$ contains a permutation of cycle type $(n_1, \dots, n_r)$.

Example 30.9.16

Let $f = x^4 - 3x + 1$.

- Mod 2: $\bar{f} = x^4 + x + 1$, which has no root in $\mathbb{F}_2$ and is not divisible by $x^2 + x + 1$ (the only irreducible quadratic), so it is irreducible. Hence $\text{Gal}(f/\mathbb{Q})$ contains a 4-cycle.
- Mod 5: $\bar{f} = (x + 1)(x^3 - x^2 + x + 1)$, a product of an irreducible linear and an irreducible cubic. Hence $\text{Gal}(f/\mathbb{Q})$ contains a 3-cycle.

So $12 \mid \# \text{Gal}(f/\mathbb{Q})$, and $\text{Gal}(f/\mathbb{Q}) \leq S_4$, so it is A_4 or S_4 . But a 4-cycle is odd, so it is not in A_4 . Hence $\text{Gal}(f/\mathbb{Q}) = S_4$.

Note also that reduction mod p is a good irreducibility test: if $\bar{f}$ is irreducible for some p then f is irreducible over $\mathbb{Z}$ and hence over $\mathbb{Q}$. More subtly, a factorisation over $\mathbb{Z}$ must refine to a factorisation over $\mathbb{F}_p$ for every p , so if the factorisation shapes mod two different primes are incompatible, f is irreducible. For example $f = x^4 + 5x^2 - 2x - 3$ factors as $(x^2 + x + 1)^2 \pmod{2}$ and as $x(x^3 + 2x + 1) \pmod{3}$; a factorisation over $\mathbb{Z}$ would have to have a factor of degree 2 (from mod 2) and a factor of degree 1 (from mod 3) compatible with each other, which is impossible, so f is irreducible.

§30.9.6 The Generic Polynomial

Finally, symmetric functions let us produce a polynomial whose Galois group is as large as possible.

Theorem 30.9.17

Let k be a field and let $L = k(z_1, \dots, z_n)$ be the field of rational functions, with S_n acting by permuting the variables. Then

$$L^{S_n} = k(e_1, \dots, e_n),$$

and L/L^{S_n} is a Galois extension of degree $n!$ with Galois group S_n .

Proof. Certainly $k(e_1, \dots, e_n) \subseteq L^{S_n}$. Conversely let $\gamma = f/g \in L^{S_n}$ with $f, g \in R = k[z_1, \dots, z_n]$. Put $\mu = \prod_{\sigma \in S_n} \sigma g$, which is symmetric, so $\mu \in R^{S_n} = k[e_1, \dots, e_n]$ by Theorem 30.9.3. Then

$$\gamma = \frac{f}{g} = \frac{f \prod_{\sigma \neq 1} \sigma g}{\mu},$$

whose numerator is $\gamma\mu$, a symmetric element of R (it is symmetric since γ and μ are, and it lies in R). So $\gamma \in \text{Frac}(R^{S_n}) = k(e_1, \dots, e_n)$.

Since S_n acts faithfully on L , Artin's Theorem 30.8.7 gives $[L : L^{S_n}] = n!$ with $\text{Gal}(L/L^{S_n}) = S_n$. $\square$

Taking $K = k(e_1, \dots, e_n)$ and $f(x) = x^n - e_1 x^{n-1} + \dots + (-1)^n e_n \in K[x]$, we have $f(x) = \prod_i (x - z_i)$ in L , so L is a splitting field for f and

$$\text{Gal}(f/K) = S_n.$$

Informally: *the general polynomial of degree n has Galois group S_n* . This will be exactly what we need for the insolubility of the quintic.

Corollary 30.9.18

Every finite group G occurs as the Galois group of some Galois extension of fields.

Proof. By Cayley's theorem G embeds in S_n for some n (acting on itself by left

multiplication). Take $L = k(z_1, \dots, z_n)$ and apply Artin's theorem to $G \leq S_n \leq \text{Aut}(L)$: then L/L^G is Galois with group G . $\square$

Whether every finite group occurs as $\text{Gal}(L/\mathbb{Q})$ is the **inverse Galois problem**, and it is open. It is known for all soluble groups (Shafarevich) and for most finite simple groups, including – thanks to Thompson – the Monster.

§30.10 Cyclotomic Extensions

We now study the extension obtained by adjoining roots of unity. These are the simplest interesting Galois extensions of $\mathbb{Q}$, and they are also the raw material out of which radical extensions are built, so we shall need to understand them thoroughly.

§30.10.1 Roots of Unity

Let K be a field and $n \geq 1$ with $n \neq 0$ in K (so $\text{char } K = 0$, or $\text{char } K = p \nmid n$). Then $x^n - 1$ has derivative nx^{n-1} , and $\gcd(x^n - 1, nx^{n-1}) = 1$, so $x^n - 1$ is separable. Let L be a splitting field and

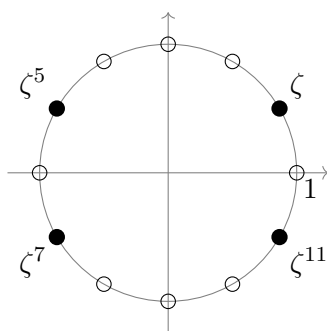
$$\mu_n = \{\zeta \in L : \zeta^n = 1\},$$

a group of order exactly n under multiplication. By Proposition 30.6.1 it is cyclic.

Definition 30.10.1 (Primitive Root of Unity)

A generator of μ_n is called a **primitive n th root of unity**, usually written ζ_n . There are $\phi(n)$ of them, namely ζ_n^a for $\gcd(a, n) = 1$.

Over $\mathbb{C}$ we can draw the picture: the n th roots of unity are the vertices of a regular n -gon inscribed in the unit circle, and the primitive ones are those ‘visible from the origin’ in the sense of being $e^{2\pi ia/n}$ with a coprime to n . Here is $n = 12$, with the four primitive twelfth roots of unity drawn as solid dots.



Proposition 30.10.2

Let $\zeta = \zeta_n$ be a primitive n th root of unity in a splitting field L of $x^n - 1$ over K . Then $L = K(\zeta)$, the extension L/K is Galois, and there is an injective group homomorphism

$$\chi : \text{Gal}(L/K) \hookrightarrow (\mathbb{Z}/n\mathbb{Z})^\times.$$

In particular $\text{Gal}(L/K)$ is abelian.

Proof. Every n th root of unity is a power of ζ , so $L = K(\zeta)$; and L is the splitting field of the separable polynomial $x^n - 1$, so L/K is Galois. Any $\sigma \in \text{Gal}(L/K)$ sends ζ to another root of $x^n - 1$, and must send a generator of μ_n to a generator, so $\sigma(\zeta) = \zeta^{a_\sigma}$ for a unique $a_\sigma \in (\mathbb{Z}/n\mathbb{Z})^\times$. If $\tau(\zeta) = \zeta^{a_\tau}$ then

$$\sigma\tau(\zeta) = \sigma(\zeta^{a_\tau}) = \zeta^{a_\sigma a_\tau},$$

so $\chi(\sigma) = a_\sigma$ is a homomorphism, and it is injective since σ is determined by $\sigma(\zeta)$. $\square$

For general K the map χ need not be surjective – for instance if K already contains ζ then $\text{Gal}(L/K)$ is trivial. The whole content of the next subsection is that over $\mathbb{Q}$ it is surjective.

§30.10.2 Cyclotomic Polynomials

If $d \mid n$ then $x^d - 1$ divides $x^n - 1$. Each $\zeta \in \mu_n$ is a primitive d th root of unity for exactly one $d \mid n$, namely $d = \text{ord}(\zeta)$. This suggests dividing out all the obvious factors.

Definition 30.10.3 (Cyclotomic Polynomial)

The n th **cyclotomic polynomial** is

$$\Phi_n(x) = \prod_{\substack{\zeta \in \mu_n \\ \text{ord}(\zeta)=n}} (x - \zeta) = \prod_{\substack{1 \leq a \leq n \\ \gcd(a,n)=1}} (x - \zeta_n^a),$$

a monic polynomial of degree $\phi(n)$.

Partitioning μ_n by the order of its elements gives

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x), \quad \text{equivalently} \quad \Phi_n(x) = \frac{x^n - 1}{\prod_{d \mid n, d \neq n} \Phi_d(x)},$$

which is an inductive recipe for computing Φ_n .

Proposition 30.10.4

$\Phi_n(x) \in \mathbb{Z}[x]$.

Proof. Induct on n . For $n = 1$, $\Phi_1 = x - 1$. In general $g = \prod_{d \mid n, d \neq n} \Phi_d$ lies in $\mathbb{Z}[x]$ by induction and is monic, so dividing $x^n - 1$ by g using the division algorithm in $\mathbb{Z}[x]$ (legitimate as g is monic) gives a quotient in $\mathbb{Z}[x]$; and that quotient must be Φ_n , since the division is the same computed in any larger ring. $\square$

Example 30.10.5

The first few are $\Phi_1 = x - 1$, $\Phi_2 = x + 1$, $\Phi_3 = x^2 + x + 1$, $\Phi_4 = x^2 + 1$. For p prime,

$\Phi_p = 1 + x + \cdots + x^{p-1}$. For $n = 8$ and $n = 6$:

$$\begin{aligned} x^8 - 1 &= \underbrace{(x-1)}_{\Phi_1} \underbrace{(x+1)}_{\Phi_2} \underbrace{(x^2+1)}_{\Phi_4} \underbrace{(x^4+1)}_{\Phi_8}, \\ x^6 - 1 &= \underbrace{(x-1)}_{\Phi_1} \underbrace{(x+1)}_{\Phi_2} \underbrace{(x^2+x+1)}_{\Phi_3} \underbrace{(x^2-x+1)}_{\Phi_6}. \end{aligned}$$

It is tempting to guess that the coefficients of Φ_n are always $0, \pm 1$. This is true for $n < 105$, and false for $n = 105$.

Theorem 30.10.6 (Gauss, Dedekind)

Φ_n is irreducible in $\mathbb{Q}[x]$.

Proof. By Gauss's lemma, since $\Phi_n \in \mathbb{Z}[x]$ is monic, it suffices to show Φ_n does not factor properly in $\mathbb{Z}[x]$. Suppose $\Phi_n = fg$ with $f, g \in \mathbb{Z}[x]$ monic and f irreducible of positive degree.

Let ξ be a root of f ; it is a primitive n th root of unity, and every primitive n th root of unity is ξ^a with $\gcd(a, n) = 1$. Since every such a is a product of primes not dividing n , it is enough to prove:

$$\text{if } f(\xi) = 0 \text{ and } p \text{ is a prime with } p \nmid n, \text{ then } f(\xi^p) = 0.$$

Granting this, every root of Φ_n is a root of f , so $\deg f = \phi(n)$ and g is constant, as required.

Suppose not, so $f(\xi^p) \neq 0$. Since ξ^p is a primitive n th root of unity, $\Phi_n(\xi^p) = 0$, so $g(\xi^p) = 0$. Thus ξ is a root of both $f(x)$ and $g(x^p)$; as f is irreducible and monic it is the minimal polynomial of ξ over $\mathbb{Q}$, so

$$f(x) \mid g(x^p) \text{ in } \mathbb{Z}[x]$$

(the quotient lies in $\mathbb{Z}[x]$ by Gauss's lemma, as f is monic).

Now reduce modulo p , writing $\bar{h}$ for the image of h in $\mathbb{F}_p[x]$. By the Frobenius homomorphism, $\bar{g}(x^p) = \bar{g}(x)^p$. So

$$\bar{f} \mid \bar{g}^p \text{ in } \mathbb{F}_p[x].$$

Let γ be an irreducible factor of $\bar{f}$; then $\gamma \mid \bar{g}$, so $\gamma^2 \mid \bar{f}\bar{g} = \overline{\Phi_n}$, and hence $\overline{\Phi_n}$ has a repeated root. But $\overline{\Phi_n} \mid x^n - 1$ in $\mathbb{F}_p[x]$, and $x^n - 1$ is separable over $\mathbb{F}_p$ because $p \nmid n$. This is a contradiction. $\square$

Corollary 30.10.7

$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n)$, and

$$\chi : \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^\times$$

is an isomorphism.

Proof. Φ_n is the minimal polynomial of ζ_n over $\mathbb{Q}$ by Theorem 30.10.6, so $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n) = \#(\mathbb{Z}/n\mathbb{Z})^\times$. The map χ of Proposition 30.10.2 is injective between groups of the same finite order, hence bijective. $\square$

§30.10.3 Subfields of Cyclotomic Fields

Since $(\mathbb{Z}/n\mathbb{Z})^\times$ is abelian, every subgroup is normal, so *every* subfield of $\mathbb{Q}(\zeta_n)$ is Galois over $\mathbb{Q}$ with abelian Galois group. (The converse – that every abelian extension of $\mathbb{Q}$ lies inside a cyclotomic field – is the Kronecker–Weber theorem, well beyond this course.) The Galois correspondence therefore turns questions about subfields of $\mathbb{Q}(\zeta_n)$ into questions about subgroups of $(\mathbb{Z}/n\mathbb{Z})^\times$.

Lemma 30.10.8

Let $n \geq 3$. Then the fixed field of complex conjugation in $\mathbb{Q}(\zeta_n)$ is the maximal real subfield

$$\mathbb{Q}(\zeta_n + \zeta_n^{-1}) = \mathbb{Q}(2 \cos(2\pi/n)),$$

and $[\mathbb{Q}(\zeta_n) : \mathbb{Q}(\zeta_n + \zeta_n^{-1})] = 2$. In particular $2 \mid \phi(n)$.

Proof. Complex conjugation restricts to an element of $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ corresponding to $-1 \in (\mathbb{Z}/n\mathbb{Z})^\times$, which is not the identity since $n \geq 3$; so it has order 2 and its fixed field has index 2. Now $\eta = \zeta + \zeta^{-1} = 2 \cos(2\pi/n)$ is real, so $\mathbb{Q}(\eta)$ lies in that fixed field; and ζ is a root of $x^2 - \eta x + 1 \in \mathbb{Q}(\eta)[x]$, so $[\mathbb{Q}(\zeta) : \mathbb{Q}(\eta)] \leq 2$, with equality since $\zeta \notin \mathbb{R}$. Comparing degrees, $\mathbb{Q}(\eta)$ is exactly the fixed field. $\square$

Remark. When $\text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ is not cyclic there may be several subfields of index 2; for instance $\mathbb{Q}(\zeta_8)$ has degree 4 over $\mathbb{Q}$ and contains all three of $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt{-2})$. The maximal real subfield is the one fixed by conjugation, here $\mathbb{Q}(\sqrt{2})$.

Example 30.10.9 ($n = 5$)

Here $(\mathbb{Z}/5)^\times \cong \mathbb{Z}/4$, so there is a unique intermediate field of degree 2 over $\mathbb{Q}$, namely $\mathbb{Q}(\eta)$ with $\eta = \zeta_5 + \zeta_5^{-1}$. Since $\zeta + \zeta^2 + \zeta^3 + \zeta^4 = -1$, we get $\eta + \eta' = -1$ where $\eta' = \zeta^2 + \zeta^3$, and $\eta\eta' = -1$, so η satisfies $x^2 + x - 1$ and

$$\eta = \frac{-1 + \sqrt{5}}{2}, \quad \mathbb{Q} \subsetneq \mathbb{Q}(\sqrt{5}) \subsetneq \mathbb{Q}(\zeta_5).$$

Example 30.10.10 ($n = 7$)

Now $(\mathbb{Z}/7)^\times \cong \mathbb{Z}/6$, generated by 3; write $\sigma = \chi^{-1}(3)$, so $\sigma(\zeta) = \zeta^3$. The subgroups of $\mathbb{Z}/6$ are $\{1\}$, $\langle \sigma^3 \rangle \cong \mathbb{Z}/2$, $\langle \sigma^2 \rangle \cong \mathbb{Z}/3$ and the whole group, so $\mathbb{Q}(\zeta_7)$ has exactly four subfields.

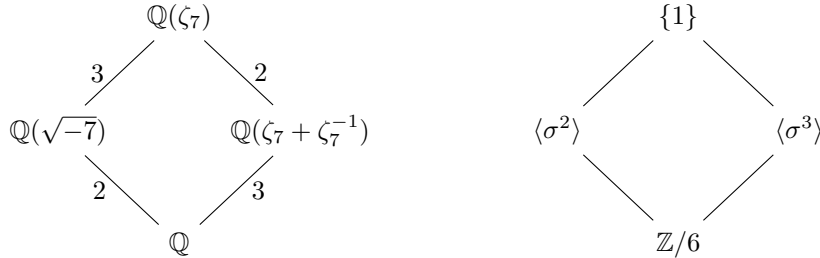
The subgroup $\langle \sigma^3 \rangle$ has order 2, so its fixed field has degree 3. Since $\sigma^3(\zeta) = \zeta^{27} = \zeta^{-1}$, this is $\mathbb{Q}(\eta)$ with $\eta = \zeta + \zeta^{-1}$, and the orbit of η under $\langle \sigma \rangle$ is $\{\eta_1, \eta_2, \eta_3\}$ where $\eta_j = \zeta^j + \zeta^{-j}$. So the minimal polynomial of η is

$$(x - \eta_1)(x - \eta_2)(x - \eta_3) = x^3 + x^2 - 2x - 1.$$

The subgroup $\langle \sigma^2 \rangle$ has order 3, so its fixed field is quadratic. Put $\varepsilon = \zeta + \zeta^2 + \zeta^4$ (the orbit of ζ under $\langle \sigma^2 \rangle$, since $\sigma^2(\zeta) = \zeta^2$). Then $\varepsilon' = \zeta^3 + \zeta^5 + \zeta^6$ is its conjugate, and

$$\varepsilon + \varepsilon' = -1, \quad \varepsilon\varepsilon' = 2,$$

so ε satisfies $x^2 + x + 2$, whose discriminant is -7 . Hence the quadratic subfield is $\mathbb{Q}(\sqrt{-7})$.



The pattern in these examples is general.

Proposition 30.10.11

Let p be an odd prime. Then the unique quadratic subfield of $\mathbb{Q}(\zeta_p)$ is $\mathbb{Q}(\sqrt{p^*})$, where

$$p^* = (-1)^{(p-1)/2}p.$$

Proof. Since $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong \mathbb{Z}/(p-1)$ is cyclic of even order, it has a unique subgroup of index 2, so there is a unique quadratic subfield F .

Now apply Proposition 30.9.6 to $f(x) = x^p - 1$, whose roots are the p th roots of unity and whose derivative is px^{p-1} :

$$\text{Disc}(f) = (-1)^{p(p-1)/2} \prod_{j=0}^{p-1} p\zeta^j = (-1)^{(p-1)/2} p^p \zeta^{(p-1) \cdot \frac{p(p-1)}{2}} = (-1)^{(p-1)/2} p^p,$$

using that p is odd and $\zeta^p = 1$. Since p is odd, p^p and p differ by the square p^{p-1} , so

$$\mathbb{Q}(\sqrt{\text{Disc } f}) = \mathbb{Q}\left(\sqrt{(-1)^{(p-1)/2}p}\right) = \mathbb{Q}(\sqrt{p^*}).$$

But $\delta = \sqrt{\text{Disc } f}$ lies in the splitting field $\mathbb{Q}(\zeta_p)$ and generates a quadratic extension of $\mathbb{Q}$ (it is irrational since p^* is not a square). By uniqueness, $F = \mathbb{Q}(\sqrt{p^*})$. $\square$

Corollary 30.10.12

$\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \not\cong A_p$, when regarded as permutations of the p th roots of unity.

This proposition is the first step of one of the standard Galois-theoretic proofs of quadratic reciprocity, which you may have seen in Number Theory.

§30.10.4 Constructible Polygons

We can now settle the question left open in Section 30.3.

Theorem 30.10.13 (Gauss)

The regular n -gon is constructible with ruler and compass if and only if

$$n = 2^k p_1 p_2 \cdots p_r$$

where the p_i are distinct primes of the form $2^{2^\ell} + 1$.

Primes of the form $2^m + 1$ are called **Fermat primes**; note that m must itself be a power of 2, since if m has an odd factor $t > 1$ then $2^{m/t} + 1$ divides $2^m + 1$. The only known Fermat primes are 3, 5, 17, 257 and 65537.

Proof. Constructing the regular n -gon is equivalent to constructing $\cos(2\pi/n)$, and since $[\mathbb{Q}(\zeta_n) : \mathbb{Q}(\cos(2\pi/n))] = 2$ by Lemma 30.10.8, this is equivalent to constructing ζ_n , that is, to every element of $\mathbb{Q}(\zeta_n)$ being constructible.

($\Rightarrow$) If ζ_n is constructible then by Corollary 30.3.5 its degree over $\mathbb{Q}$ is a power of 2, that is $\phi(n) = 2^s$. Writing $n = \prod_i p_i^{\alpha_i}$ we have $\phi(n) = \prod_i p_i^{\alpha_i-1}(p_i - 1)$, so no odd prime may occur with $\alpha_i \geq 2$, and each odd p_i must have $p_i - 1$ a power of 2. That is exactly the stated form.

($\Leftarrow$) Suppose $\phi(n) = 2^s$, so $G = \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ is an abelian group of order 2^s . Any such group has a chain of subgroups

$$\{1\} = A_0 < A_1 < \cdots < A_s = G$$

with $A_j/A_{j-1} \cong \mathbb{Z}/2$ (take a chain through the factors given by the structure theorem, or simply induct: an abelian 2-group has a subgroup of index 2). By the Galois correspondence this gives a tower

$$\mathbb{Q} = F_0 \subset F_1 \subset \cdots \subset F_s = \mathbb{Q}(\zeta_n)$$

with each $[F_j : F_{j-1}] = 2$, so each $F_j = F_{j-1}(\sqrt{r_j})$ for some $r_j \in F_{j-1}$. By Theorem 30.3.4, everything in $\mathbb{Q}(\zeta_n)$ is constructible. $\square$

Example 30.10.14 (The 17-gon)

Here $(\mathbb{Z}/17)^\times \cong \mathbb{Z}/16$, which has the chain $\{1\} < \mathbb{Z}/2 < \mathbb{Z}/4 < \mathbb{Z}/8 < \mathbb{Z}/16$, so

$$\mathbb{Q} \subset K_1 \subset K_2 \subset K_3 \subset \mathbb{Q}(\zeta_{17})$$

with each step quadratic, and $K_3 = \mathbb{Q}(\cos(2\pi/17))$. So $\cos(2\pi/17)$ is obtained from $\mathbb{Q}$ by extracting four successive square roots, and the regular 17-gon is constructible. Gauss discovered this at the age of nineteen, and it is said to have decided him on a career in mathematics.

§30.11 Kummer Theory and Cyclic Extensions

Cyclotomic extensions are the extensions generated by roots of $x^n - 1$. We now consider the general $x^n - a$, and discover that – once enough roots of unity are present – these are exactly the cyclic extensions. This is the last piece of machinery we need for the insolubility of the quintic.

§30.11.1 Linear Independence of Characters

Theorem 30.11.1 (Dedekind, Artin)

Let G be a group, L a field, and $\sigma_1, \dots, \sigma_n : G \rightarrow L^\times$ distinct group homomorphisms. Then $\sigma_1, \dots, \sigma_n$ are linearly independent over L as functions $G \rightarrow L$.

Proof. Induct on n ; the case $n = 1$ is clear since σ_1 is nowhere zero. Suppose $y_1, \dots, y_n \in L$ satisfy

$$y_1\sigma_1(g) + \dots + y_n\sigma_n(g) = 0 \quad \text{for all } g \in G. \quad (*)$$

As $\sigma_1 \neq \sigma_n$, choose $h \in G$ with $\sigma_1(h) \neq \sigma_n(h)$. Substituting hg for g in $(*)$ and using multiplicativity,

$$y_1\sigma_1(h)\sigma_1(g) + \dots + y_n\sigma_n(h)\sigma_n(g) = 0.$$

Subtracting $\sigma_n(h)$ times $(*)$ kills the last term:

$$\sum_{i=1}^{n-1} y_i(\sigma_i(h) - \sigma_n(h))\sigma_i(g) = 0 \quad \text{for all } g.$$

By induction all the coefficients vanish; in particular $y_1(\sigma_1(h) - \sigma_n(h)) = 0$, so $y_1 = 0$. Now $(*)$ has only $n - 1$ terms, and induction finishes the job. $\square$

Corollary 30.11.2 (Linear independence of embeddings)

Let K, L be fields and $\sigma_1, \dots, \sigma_n : K \rightarrow L$ distinct field homomorphisms. Then they are linearly independent over L .

Proof. Apply the theorem with $G = K^\times$; a field homomorphism restricts to a group homomorphism $K^\times \rightarrow L^\times$, and distinct field homomorphisms restrict to distinct group homomorphisms. $\square$

§30.11.2 Kummer Extensions

Throughout this subsection, let $n > 1$ with $n \neq 0$ in K , and suppose K contains a primitive n th root of unity $\zeta = \zeta_n$.

Theorem 30.11.3

Let $L = K(y)$ where $y^n = a \in K^\times$. Then L is a splitting field for $x^n - a$ over K , the extension L/K is Galois, and $\text{Gal}(L/K)$ is cyclic. Moreover $[L : K]$ is the least $m \geq 1$ with $y^m \in K$.

Proof. The roots of $x^n - a$ are $y, \zeta y, \dots, \zeta^{n-1}y$, which are n distinct elements of L since $\zeta \in K$. So $x^n - a$ is separable and L is its splitting field, hence L/K is Galois.

For $\sigma \in G = \text{Gal}(L/K)$ we have $\sigma(y)^n = a$, so $\sigma(y)/y \in \mu_n$. Define

$$\theta : G \rightarrow \mu_n, \quad \theta(\sigma) = \sigma(y)/y.$$

This is a homomorphism: since $\mu_n \subseteq K$ is fixed pointwise by G ,

$$\theta(\tau\sigma) = \frac{\tau\sigma(y)}{y} = \tau\left(\frac{\sigma(y)}{y}\right) \cdot \frac{\tau(y)}{y} = \theta(\sigma)\theta(\tau).$$

It is injective, since $\theta(\sigma) = 1$ means $\sigma(y) = y$, and σ is determined by $\sigma(y)$ as $L = K(y)$. So G embeds in the cyclic group $\mu_n \cong \mathbb{Z}/n$, hence is cyclic.

Finally, since L/K is Galois, $y^m \in K$ if and only if $\sigma(y^m) = y^m$ for all σ , if and only if $\theta(\sigma)^m = 1$ for all σ , if and only if $\#G = \#\theta(G)$ divides m . So the least such m is $\#G = [L : K]$. $\square$

Corollary 30.11.4

Let $a \in K^\times$. Then $x^n - a$ is irreducible over K if and only if a is not a d th power in K for any $d > 1$ dividing n .

Proof. Let y be a root and $L = K(y)$; then $x^n - a$ is irreducible if and only if $[L : K] = n$. Write $n = md$ with $d > 1$. Since $\zeta_n \in K$, one checks that a is a d th power in K if and only if $y^m \in K$: indeed if $y^m = c \in K$ then $a = y^n = c^d$, and conversely if $a = b^d$ then $(y^m/b)^d = 1$ so $y^m = \zeta_d^j b \in K$. By the theorem this holds if and only if $[L : K] \mid m$, that is $[L : K] < n$. $\square$

Remark. The hypothesis $\zeta_n \in K$ is essential here. Over $\mathbb{Q}$, the polynomial $x^4 + 4$ is reducible ($x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2)$) even though -4 is not a square or fourth power in $\mathbb{Q}$.

The converse of Theorem 30.11.3 is where the Lagrange resolvent comes into its own.

Theorem 30.11.5

Let K contain a primitive n th root of unity ζ , and let L/K be Galois with $\text{Gal}(L/K)$ cyclic of order n . Then $L = K(y)$ for some y with $y^n \in K$.

Proof. Write $\text{Gal}(L/K) = \langle \sigma \rangle$ with σ of order n . For $z \in L$ define the **Lagrange resolvent**

$$R(z) = \sum_{j=0}^{n-1} \zeta^{-j} \sigma^j(z) \in L.$$

Then, since $\zeta \in K$ is fixed by σ and $\sigma^n = 1$,

$$\sigma(R(z)) = \sum_{j=0}^{n-1} \zeta^{-j} \sigma^{j+1}(z) = \zeta \sum_{k=0}^{n-1} \zeta^{-k} \sigma^k(z) = \zeta R(z).$$

Hence $\sigma(R(z)^n) = \zeta^n R(z)^n = R(z)^n$, so $R(z)^n \in L^{\langle \sigma \rangle} = K$.

By Corollary 30.11.2 the distinct automorphisms $1, \sigma, \dots, \sigma^{n-1}$ are linearly independent over L , and the coefficients ζ^{-j} are not all zero, so R is not identically zero; choose z with $y = R(z) \neq 0$.

Then $\sigma^i(y) = \zeta^i y$ for each i , and these are n distinct elements, so the orbit of y has

size n and hence $[K(y) : K] = \deg_K y = n = [L : K]$ by Lemma 30.8.5. Therefore $L = K(y)$, and $y^n \in K$. $\square$

Definition 30.11.6 (Kummer Extension)

An extension of the form $L = K(y)$ with $y^n = a \in K$ and $\zeta_n \in K$ is called a **Kummer extension**.

So, in the presence of enough roots of unity, *cyclic* = *Kummer*. The two theorems above are exactly the bridge between the group theory (cyclic groups) and the algebra (extracting n th roots) that we will use to prove the theorem of Galois.

Example 30.11.7

With $n = 2$ and $\text{char } K \neq 2$, we have $\zeta_2 = -1 \in K$ automatically, and the theorems say: every quadratic extension of K is $K(\sqrt{a})$ for some non-square $a \in K^\times$. This we knew.

Example 30.11.8

Let $L/\mathbb{Q}$ be Galois of degree 3, for instance the splitting field of $x^3 - 3x + 1$ (whose discriminant is 81, a square, so its Galois group is A_3). Since $\zeta_3 \notin \mathbb{Q}$, this is *not* a Kummer extension, and indeed L is not of the form $\mathbb{Q}(\sqrt[3]{a})$ – such a field is not even normal over $\mathbb{Q}$. To apply Kummer theory one must first adjoin ζ_3 , which is exactly what we did in Example 30.9.9.

§30.11.3 Artin–Schreier Extensions

Kummer theory says nothing about cyclic extensions of degree p in characteristic p , since there are no primitive p th roots of unity there ($x^p - 1 = (x - 1)^p$). That case has its own, equally clean, description.

Theorem 30.11.9 (Artin–Schreier)

Let $\text{char } K = p$.

- (i) Let $a \in K$ and $f(x) = x^p - x - a$. Then either f splits completely in K , or f is irreducible; in the latter case $K(\alpha)/K$ is Galois of degree p with cyclic Galois group, where $f(\alpha) = 0$.
- (ii) Conversely, if L/K is Galois with $\text{Gal}(L/K) \cong \mathbb{Z}/p$, then $L = K(\alpha)$ where $\alpha^p - \alpha = a$ for some $a \in K$.

Proof. (i) If α is a root of f in a splitting field, then for $c \in \mathbb{F}_p \subseteq K$ we have

$$(\alpha + c)^p - (\alpha + c) - a = \alpha^p + c^p - \alpha - c - a = f(\alpha) = 0,$$

using $c^p = c$. So the roots of f are $\alpha, \alpha+1, \dots, \alpha+p-1$, all distinct, and f is separable with splitting field $K(\alpha)$. Hence $K(\alpha)/K$ is Galois. Any $\sigma \in \text{Gal}(K(\alpha)/K)$ sends $\alpha \mapsto \alpha + c_\sigma$ for some $c_\sigma \in \mathbb{F}_p$, and $\sigma \mapsto c_\sigma$ is an injective homomorphism into $(\mathbb{F}_p, +) \cong \mathbb{Z}/p$. So the Galois group is trivial or all of $\mathbb{Z}/p$; in the first case $\alpha \in K$

and f splits in K , and in the second $[K(\alpha) : K] = p$ so f is irreducible.

(ii) Write $\text{Gal}(L/K) = \langle \sigma \rangle$ of order p , and consider the trace map $\text{Tr} = \sum_{j=0}^{p-1} \sigma^j : L \rightarrow K$. By Corollary 30.11.2 the σ^j are linearly independent, so Tr is not identically zero; and it is K -linear, so there is $\theta \in L$ with $\text{Tr}(\theta) = 1$. Put

$$\alpha = \sum_{j=0}^{p-1} j \sigma^j(\theta).$$

Then, reindexing and using $\sigma^p = 1$,

$$\sigma(\alpha) = \sum_{j=0}^{p-1} j \sigma^{j+1}(\theta) = \sum_{k=1}^{p-1} (k-1) \sigma^k(\theta) + (p-1)\theta = \alpha - \sum_{k=0}^{p-1} \sigma^k(\theta) = \alpha - 1.$$

Hence $\sigma(\alpha^p - \alpha) = (\alpha - 1)^p - (\alpha - 1) = \alpha^p - 1 - \alpha + 1 = \alpha^p - \alpha$, so $a := \alpha^p - \alpha \in L^{\langle \sigma \rangle} = K$. Finally $\sigma(\alpha) \neq \alpha$, so $\alpha \notin K$, and since $[L : K] = p$ is prime, $L = K(\alpha)$. $\square$

§30.12 Solubility by Radicals

We have arrived at the theorem the subject was invented for. The plan is:

- define precisely what a ‘formula involving radicals’ is;
- define the corresponding class of groups;
- prove that the two match up;
- exhibit a quintic whose Galois group is not in the class.

Throughout this section we work in characteristic zero, which spares us a certain amount of fuss about separability and about $\text{char } K$ dividing the degrees involved. Everything goes through if one assumes $\text{char } K = 0$ or $\text{char } K > \deg f$.

§30.12.1 Soluble Groups

Definition 30.12.1 (Soluble Group)

A finite group G is **soluble** if there is a chain of subgroups

$$\{1\} = G_r \trianglelefteq G_{r-1} \trianglelefteq \dots \trianglelefteq G_1 \trianglelefteq G_0 = G$$

such that each G_{i+1} is normal in G_i and each quotient G_i/G_{i+1} is abelian. (We do *not* require $G_i \trianglelefteq G$.)

Lemma 30.12.2

A finite group G is soluble if and only if it admits such a chain with all quotients *cyclic*.

Proof. Cyclic groups are abelian, so one direction is trivial. Conversely, given a chain with abelian quotients, refine it: if $A = G_i/G_{i+1}$ is abelian then by the

structure theorem $A \cong \mathbb{Z}/a_1 \times \cdots \times \mathbb{Z}/a_m$, and

$$1 \leq \mathbb{Z}/a_1 \leq \mathbb{Z}/a_1 \times \mathbb{Z}/a_2 \leq \cdots \leq A$$

is a chain of subgroups (all normal, as A is abelian) with cyclic quotients. Pulling back along $G_i \rightarrow A$ inserts extra terms into the original chain, with cyclic quotients. $\square$

Example 30.12.3

Every abelian group is soluble. S_3 is soluble via $S_3 \triangleright A_3 \triangleright 1$, with quotients $\mathbb{Z}/2$ and $\mathbb{Z}/3$. S_4 is soluble via

$$S_4 \triangleright A_4 \triangleright V \triangleright 1,$$

with quotients $\mathbb{Z}/2$, $\mathbb{Z}/3$ and $\mathbb{Z}/2 \times \mathbb{Z}/2$. Every p -group is soluble, since a p -group has a non-trivial centre.

Here are the two derived pictures we care about, side by side. On the left is S_4 , which is soluble; on the right is S_5 , which as we shall see is not, because the chain grinds to a halt at the simple non-abelian group A_5 .

$$\begin{array}{ccc} S_4 & & S_5 \\ | \mathbb{Z}/2 & & | \mathbb{Z}/2 \\ A_4 & & A_5 \\ | \mathbb{Z}/3 & & \vdots A_5, \text{ simple and non-abelian} \\ V & & \{1\} \\ | \mathbb{Z}/2 \times \mathbb{Z}/2 & & \\ \{1\} & & \end{array}$$

Proposition 30.12.4

Let G be a finite group.

- (i) If G is soluble and $H \leq G$, then H is soluble.
- (ii) If G is soluble and $N \trianglelefteq G$, then G/N is soluble.
- (iii) If $N \trianglelefteq G$ with N and G/N soluble, then G is soluble.

Proof. (i) Let $G = G_0 \triangleright G_1 \triangleright \cdots \triangleright G_r = 1$ have abelian quotients and put $H_i = H \cap G_i$. If $h \in H_i$ then $hG_{i+1}h^{-1} \subseteq G_{i+1}$ and $hHh^{-1} = H$, so $H_{i+1} \trianglelefteq H_i$. Moreover the map

$$H_i/H_{i+1} \rightarrow G_i/G_{i+1}, \quad hH_{i+1} \mapsto hG_{i+1}$$

is a well-defined injective homomorphism (its kernel is $(H \cap G_i) \cap G_{i+1}/H_{i+1}$, which is trivial), so H_i/H_{i+1} is abelian.

(ii) Let $\pi : G \rightarrow G/N$ be the quotient map. Then $\pi(G_{i+1}) \trianglelefteq \pi(G_i)$, and G_i/G_{i+1} surjects onto $\pi(G_i)/\pi(G_{i+1})$, which is therefore abelian.

(iii) Concatenate a chain for N with the preimage under $G \rightarrow G/N$ of a chain for G/N ; the quotients are unchanged by the third isomorphism theorem. $\square$

Corollary 30.12.5

For $n \geq 5$, the groups A_n and S_n are insoluble.

Proof. Both contain a copy of A_5 , so by Proposition 30.12.4(i) it suffices to show A_5 is insoluble. But A_5 is simple (proved in Groups, Rings and Modules) and non-abelian, so the only chain available is $A_5 \triangleright 1$, whose quotient A_5 is not abelian. $\square$

§30.12.2 Radical Extensions**Definition 30.12.6** (Radical Extension)

An extension L/K is a **radical extension** if there is a chain

$$K = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_r = L$$

with $L_i = L_{i-1}(y_i)$ for some y_i with $y_i^{n_i} \in L_{i-1}$, for some $n_i \geq 1$.

Definition 30.12.7 (Soluble by Radicals)

A polynomial $f \in K[x]$ is **soluble by radicals** over K if its splitting field L is contained in some radical extension of K .

That is the correct formalisation of ‘there is a formula for the roots involving only the field operations and extraction of n th roots’. Note that we allow the radical extension to be strictly bigger than L : Cardano’s formula for a cubic with three real roots famously passes through complex numbers, so we must permit detours.

Lemma 30.12.8 (i) If L/K is radical and $\sigma : L \rightarrow M$ is a K -homomorphism, then $\sigma(L)/K$ is radical.

(ii) If L_1/K and L_2/K are radical subextensions of some M/K , then L_1L_2/K is radical, where L_1L_2 denotes the subfield of M generated by both.

(iii) If L/K is radical and N is a normal closure of L/K , then N/K is radical.

Proof. (i) Apply σ to the chain: $\sigma(L_i) = \sigma(L_{i-1})(\sigma y_i)$ and $(\sigma y_i)^{n_i} = \sigma(y_i^{n_i}) \in \sigma(L_{i-1})$.

(ii) Concatenate: take a chain for L_1/K , then adjoin the generators of a chain for L_2/K one at a time on top.

(iii) The normal closure N is generated over K by the images $\sigma(L)$ as σ ranges over the (finitely many) K -homomorphisms $L \rightarrow N$, since N is a splitting field for the product of the minimal polynomials of generators of L . Each $\sigma(L)/K$ is radical by (i), so N/K is radical by (ii). $\square$

§30.12.3 Galois’ Theorem**Theorem 30.12.9** (Galois)

Let $\text{char } K = 0$, let $f \in K[x]$ and let L be a splitting field for f over K . Then f is soluble by radicals over K if and only if $\text{Gal}(L/K)$ is a soluble group.

Proof. ($\Leftarrow$) Suppose $G = \text{Gal}(L/K)$ is soluble, and let $n = \#G$. Adjoin a primitive n th root of unity: $K(\zeta_n)/K$ is radical (it is $K(\zeta_n)$ with $\zeta_n^n = 1 \in K$).

Consider $L(\zeta_n)/K(\zeta_n)$. It is the splitting field of f over $K(\zeta_n)$, hence Galois, and restriction gives a homomorphism

$$\text{Gal}(L(\zeta_n)/K(\zeta_n)) \rightarrow \text{Gal}(L/K), \quad \sigma \mapsto \sigma|_L,$$

which makes sense because σ permutes the roots of f and so preserves L . It is injective: if $\sigma|_L = \text{id}$ then σ fixes L and $K(\zeta_n)$, hence fixes $L(\zeta_n)$. So $H := \text{Gal}(L(\zeta_n)/K(\zeta_n))$ embeds in G and is therefore soluble by Proposition 30.12.4(i).

By Lemma 30.12.2 choose a chain $H = H_0 \triangleright H_1 \triangleright \cdots \triangleright H_s = 1$ with cyclic quotients, and let $F_i = L(\zeta_n)^{H_i}$, so that

$$K(\zeta_n) = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_s = L(\zeta_n)$$

and, by the fundamental theorem applied to the Galois extension $L(\zeta_n)/F_i$, each F_{i+1}/F_i is Galois with cyclic Galois group of order $m_i = \#(H_i/H_{i+1})$. Now m_i divides $\#H$, which divides n , so $\zeta_{m_i} \in K(\zeta_n) \subseteq F_i$. By Kummer's Theorem 30.11.5, $F_{i+1} = F_i(y_i)$ with $y_i^{m_i} \in F_i$.

So $L(\zeta_n)/K(\zeta_n)$ is radical, and hence so is $L(\zeta_n)/K$. As $L \subseteq L(\zeta_n)$, the polynomial f is soluble by radicals.

($\Rightarrow$) Suppose $L \subseteq M$ with M/K radical, via a chain with exponents $n_1, \dots, n_r$; put $n = \text{lcm}(n_i)$. Replacing M by $M(\zeta_n)$ (still radical) and then by its normal closure over K (still radical by Lemma 30.12.8(iii)), we may assume that M/K is Galois, radical, and contains ζ_n . Rearranging the chain so that ζ_n is adjoined first, we have

$$K = M_0 \subseteq M_1 = K(\zeta_n) \subseteq M_2 \subseteq \cdots \subseteq M_t = M,$$

with $M_i = M_{i-1}(y_i)$ and $y_i^{n_i} \in M_{i-1}$ for $i \geq 2$, and $n_i \mid n$.

Now M_1/M_0 is Galois with abelian Galois group, by Proposition 30.10.2. And for $i \geq 2$, since $\zeta_{n_i} \in M_1 \subseteq M_{i-1}$, Kummer's Theorem 30.11.3 says M_i/M_{i-1} is Galois with cyclic Galois group.

Set $G_i = \text{Gal}(M/M_i)$, so

$$\text{Gal}(M/K) = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_t = \{1\}.$$

Applying the fundamental theorem to the Galois extension M/M_{i-1} : since M_i/M_{i-1} is Galois, $G_i \trianglelefteq G_{i-1}$ with

$$G_{i-1}/G_i \cong \text{Gal}(M_i/M_{i-1}),$$

which is abelian. So $\text{Gal}(M/K)$ is soluble.

Finally, L/K is normal (being a splitting field), so $\text{Gal}(M/L) \trianglelefteq \text{Gal}(M/K)$ with

$$\text{Gal}(L/K) \cong \text{Gal}(M/K) / \text{Gal}(M/L),$$

a quotient of a soluble group, hence soluble by Proposition 30.12.4(ii). $\square$

§30.12.4 The Insolubility of the Quintic

Theorem 30.12.10

Let k be a field of characteristic zero and $n \geq 5$. The general polynomial of degree n ,

$$f(x) = x^n - e_1x^{n-1} + e_2x^{n-2} - \cdots + (-1)^ne_n \in K[x], \quad K = k(e_1, \dots, e_n),$$

is not soluble by radicals over K .

Proof. By Theorem 30.9.17, $\text{Gal}(f/K) = S_n$, which is insoluble for $n \geq 5$ by Corollary 30.12.5. Now apply Theorem 30.12.9. $\square$

So there is no formula, in the coefficients, for the roots of the general quintic. But this leaves open the possibility that every *particular* quintic with rational coefficients might still be soluble – after all, the general quadratic, cubic and quartic all are. It is not so.

Theorem 30.12.11

The polynomial $f(x) = x^5 - 6x + 3 \in \mathbb{Q}[x]$ has Galois group S_5 , and hence is not soluble by radicals over $\mathbb{Q}$.

Proof. First, f is irreducible over $\mathbb{Q}$ by Eisenstein at 3. So $G = \text{Gal}(f/\mathbb{Q})$ acts transitively on the five roots, and hence $5 \mid \#G$ by orbit-stabiliser. By Cauchy's theorem G contains an element of order 5, which in S_5 must be a 5-cycle.

Next we count real roots. We have $f'(x) = 5x^4 - 6$, which vanishes at $x = \pm(6/5)^{1/4} \approx \pm 1.047$. Evaluating, $f(-1.047) \approx 8.02 > 0$ and $f(1.047) \approx -2.02 < 0$, while $f(x) \rightarrow \pm\infty$ as $x \rightarrow \pm\infty$. So f has exactly three real roots and one pair of complex conjugate roots.

Complex conjugation restricts to an element of G which fixes the three real roots and swaps the two complex ones – that is, a transposition.

Finally, a 5-cycle and a transposition generate S_5 . (Relabelling, the 5-cycle is $\sigma = (12345)$ and the transposition is $\tau = (ab)$; replacing σ by a suitable power we may assume $\tau = (12)$, and then $\sigma^j \tau \sigma^{-j}$ gives all the transpositions $(i, i+1)$, which generate S_5 .) So $G = S_5$, which is insoluble, and Theorem 30.12.9 applies. $\square$

Remark. Note how much the argument used: irreducibility (an algebraic input), the intermediate value theorem (an analytic input) and the classification of transitive subgroups of S_5 (a group-theoretic input). This mixture is entirely typical of Galois theory in practice.

§30.13 Trace and Norm

There are two very useful maps attached to a finite extension, obtained by remembering that L is a vector space over K and that multiplication by an element of L is a K -linear map.

Definition 30.13.1 (Trace and Norm)

Let L/K be a finite extension of degree n and let $\alpha \in L$. Multiplication by α ,

$$U_\alpha : L \rightarrow L, \quad U_\alpha(y) = \alpha y,$$

is K -linear. We define the **trace** and **norm** of α relative to L/K by

$$\mathrm{Tr}_{L/K}(\alpha) = \mathrm{tr}(U_\alpha) \in K, \quad \mathrm{N}_{L/K}(\alpha) = \det(U_\alpha) \in K,$$

and the **characteristic polynomial** of α by $f_{\alpha, L/K}(x) = \det(x \mathrm{id} - U_\alpha) \in K[x]$.

Example 30.13.2

Take $L = \mathbb{Q}(\sqrt{d})$ over $K = \mathbb{Q}$, with basis $1, \sqrt{d}$, and $\alpha = a + b\sqrt{d}$. Then $\alpha \cdot 1 = a + b\sqrt{d}$ and $\alpha \cdot \sqrt{d} = bd + a\sqrt{d}$, so

$$U_\alpha = \begin{pmatrix} a & bd \\ b & a \end{pmatrix},$$

giving $\mathrm{Tr}(\alpha) = 2a$ and $\mathrm{N}(\alpha) = a^2 - db^2$. For $d = -1$ this is the familiar $|a + bi|^2$.

Lemma 30.13.3

Let L/K be finite of degree n , $\alpha, \beta \in L$ and $c \in K$. Then

- (i) $\mathrm{Tr}_{L/K}$ is K -linear, and $\mathrm{Tr}_{L/K}(c) = nc$;
- (ii) $\mathrm{N}_{L/K}(\alpha\beta) = \mathrm{N}_{L/K}(\alpha)\mathrm{N}_{L/K}(\beta)$ and $\mathrm{N}_{L/K}(c\alpha) = c^n \mathrm{N}_{L/K}(\alpha)$;
- (iii) $\mathrm{N}_{L/K}(\alpha) = 0$ if and only if $\alpha = 0$, so $\mathrm{N}_{L/K} : L^\times \rightarrow K^\times$ is a group homomorphism.

Proof. The map $\alpha \mapsto U_\alpha$ is a K -algebra homomorphism from L to $\mathrm{End}_K(L)$, so (i) and (ii) follow from linearity and multiplicativity of trace and determinant. For (iii), $\det U_\alpha \neq 0$ if and only if U_α is invertible, which holds if and only if $\alpha \neq 0$ since L is a field. $\square$

Proposition 30.13.4

Let $L = K(\alpha)$ with minimal polynomial $m(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$. Then $f_{\alpha, L/K} = m$, and

$$\mathrm{Tr}_{L/K}(\alpha) = -c_{n-1}, \quad \mathrm{N}_{L/K}(\alpha) = (-1)^n c_0.$$

Proof. In the basis $1, \alpha, \dots, \alpha^{n-1}$, the matrix of U_α is the companion matrix of m

(multiplication sends $\alpha^i \mapsto \alpha^{i+1}$ for $i < n-1$, and $\alpha^{n-1} \mapsto -c_0 - c_1\alpha - \cdots - c_{n-1}\alpha^{n-1}$), whose characteristic polynomial is m . The formulae for the trace and determinant are the standard expressions for the coefficients of a characteristic polynomial. $\square$

For a Galois extension there is a much more useful description.

Proposition 30.13.5

Let L/K be a finite separable extension of degree n , and let $\sigma_1, \dots, \sigma_n : L \rightarrow N$ be the distinct K -homomorphisms into a normal closure N . Then for all $\alpha \in L$,

$$\mathrm{Tr}_{L/K}(\alpha) = \sum_{i=1}^n \sigma_i(\alpha), \quad \mathrm{N}_{L/K}(\alpha) = \prod_{i=1}^n \sigma_i(\alpha), \quad f_{\alpha, L/K}(x) = \prod_{i=1}^n (x - \sigma_i(\alpha)).$$

In particular, if L/K is Galois then $\mathrm{Tr}_{L/K}(\alpha) = \sum_{\sigma \in \mathrm{Gal}(L/K)} \sigma(\alpha)$ and $\mathrm{N}_{L/K}(\alpha) = \prod_{\sigma} \sigma(\alpha)$.

Proof. It suffices to prove the statement about the characteristic polynomial. Let $e_1, \dots, e_n$ be a K -basis of L and let $P = (\sigma_i(e_j)) \in \mathrm{Mat}_n(N)$. Then P is invertible: otherwise there is a non-zero row vector (y_i) over N with $\sum_i y_i \sigma_i(e_j) = 0$ for all j , and hence $\sum_i y_i \sigma_i = 0$ on all of L by K -linearity, contradicting Corollary 30.11.2.

Let $A = (a_{ij})$ be the matrix of U_α , so $\alpha e_j = \sum_r a_{rj} e_r$. Applying σ_i ,

$$\sigma_i(\alpha) \sigma_i(e_j) = \sum_r \sigma_i(e_r) a_{rj},$$

which says $SP = PA$ where $S = \mathrm{diag}(\sigma_1(\alpha), \dots, \sigma_n(\alpha))$. So A and S are conjugate, hence have the same characteristic polynomial, namely $\prod_i (x - \sigma_i(\alpha))$. $\square$

Proposition 30.13.6 (Transitivity)

Let $M/L/K$ be finite extensions. Then for all $\alpha \in M$,

$$\mathrm{Tr}_{L/K}(\mathrm{Tr}_{M/L}(\alpha)) = \mathrm{Tr}_{M/K}(\alpha), \quad \mathrm{N}_{L/K}(\mathrm{N}_{M/L}(\alpha)) = \mathrm{N}_{M/K}(\alpha).$$

Proof. We do the trace. Let $u_1, \dots, u_m$ be an L -basis of M and $v_1, \dots, v_n$ a K -basis of L , so the $u_i v_j$ form a K -basis of M . Let (a_{ik}) be the matrix of $U_{\alpha, M/L}$ over L , so $\alpha u_k = \sum_i a_{ik} u_i$, and let $A_{ik} \in \mathrm{Mat}_n(K)$ be the matrix of multiplication by a_{ik} on L . Then, with respect to the basis $\{u_i v_j\}$ ordered by i first, the matrix of $U_{\alpha, M/K}$ is the block matrix $(A_{ik})_{i,k}$. Taking traces,

$$\mathrm{Tr}_{M/K}(\alpha) = \sum_i \mathrm{tr}(A_{ii}) = \sum_i \mathrm{Tr}_{L/K}(a_{ii}) = \mathrm{Tr}_{L/K} \left(\sum_i a_{ii} \right) = \mathrm{Tr}_{L/K}(\mathrm{Tr}_{M/L}(\alpha)).$$

The norm is similar, using multiplicativity of the determinant of a block matrix. $\square$

The trace is a K -linear map $L \rightarrow K$, so it is either identically zero or surjective. Which one happens is exactly a question of separability.

Theorem 30.13.7

Let L/K be a finite extension. Then L/K is separable if and only if $\text{Tr}_{L/K}$ is surjective (equivalently, not identically zero).

Proof. Suppose L/K is separable, with K -homomorphisms $\sigma_1, \dots, \sigma_n$ into a normal closure. By Proposition 30.13.5, $\text{Tr}_{L/K} = \sum_i \sigma_i$, which is not identically zero by linear independence of embeddings (Corollary 30.11.2).

Conversely suppose L/K is inseparable, so $\text{char } K = p$ and there is $\alpha \in L$ with $\alpha \notin K(\alpha^p)$. Put $F = K(\alpha^p)$ and $F' = F(\alpha)$, so $[F' : F] = p$ (the minimal polynomial of α over F is $x^p - \alpha^p$). For any $\beta \in F'$ we have $\beta^p \in F$, so either $\beta \in F$, in which case $\text{Tr}_{F'/F}(\beta) = p\beta = 0$, or $F(\beta) = F'$ and the minimal polynomial of β over F is $x^p - \beta^p$, whose x^{p-1} coefficient is 0, so again $\text{Tr}_{F'/F}(\beta) = 0$ by Proposition 30.13.4. So $\text{Tr}_{F'/F} \equiv 0$, and by transitivity

$$\text{Tr}_{L/K} = \text{Tr}_{F/K} \circ \text{Tr}_{F'/F} \circ \text{Tr}_{L/F'} \equiv 0. \quad \square$$

Example 30.13.8

For the extension $\mathbb{F}_{q^n}/\mathbb{F}_q$, which is separable, there is therefore some α with $\text{Tr}(\alpha) = 1$. Concretely $\text{Tr}_{\mathbb{F}_{q^n}/\mathbb{F}_q}(\alpha) = \alpha + \alpha^q + \dots + \alpha^{q^{n-1}}$, and this map is surjective; the fibres all have size q^{n-1} .

Remark. This gives a second proof that a tower of separable extensions is separable: if M/L and L/K are separable then $\text{Tr}_{M/L}$ and $\text{Tr}_{L/K}$ are surjective, hence so is their composite $\text{Tr}_{M/K}$.

§30.14 Some Further Results**§30.14.1 The Fundamental Theorem of Algebra**

We have used several times that $\mathbb{C}$ is algebraically closed. Here is a proof which is essentially algebraic; the only analysis used is the intermediate value theorem, which is unavoidable since the real numbers are defined analytically. This subsection is non-examinable.

Theorem 30.14.1 (Fundamental Theorem of Algebra)

$\mathbb{C}$ is algebraically closed.

Proof. We use four facts:

- (a) every polynomial of odd degree over $\mathbb{R}$ has a real root (intermediate value theorem);
- (b) every element of $\mathbb{C}$ has a square root in $\mathbb{C}$, so every quadratic over $\mathbb{C}$ splits;
- (c) every finite group G has a subgroup H with $(G : H)$ odd and $\#H$ a power of 2 (a Sylow 2-subgroup);

- (d) every non-trivial finite p -group has a subgroup of index p (as it has non-trivial centre, one may induct).

It suffices to show that $\mathbb{C}$ has no non-trivial finite extension. So let $K/\mathbb{C}$ be finite; enlarging K , let L be a normal closure of $K/\mathbb{R}$, so that $L/\mathbb{R}$ is Galois (we are in characteristic zero, so separability is free) and $\mathbb{C} \subseteq L$. Put $G = \text{Gal}(L/\mathbb{R})$.

Let $H \leq G$ be a Sylow 2-subgroup, and consider L^H . By the fundamental theorem, $[L^H : \mathbb{R}] = (G : H)$, which is odd. But if $\alpha \in L^H$, then the minimal polynomial of α over $\mathbb{R}$ has odd degree (it divides $[L^H : \mathbb{R}]$), so by (a) it has a real root; being irreducible, it must be linear, so $\alpha \in \mathbb{R}$. Hence $L^H = \mathbb{R}$, so $H = G$ and G is a 2-group.

Now let $G_1 = \text{Gal}(L/\mathbb{C}) \leq G$, also a 2-group. If G_1 were non-trivial, by (d) it would have a subgroup G_2 of index 2, and then $[L^{G_2} : \mathbb{C}] = 2$, giving a quadratic extension of $\mathbb{C}$ – impossible by (b). So $G_1 = \{1\}$ and $L = \mathbb{C}$. Hence $K = \mathbb{C}$. $\square$

§30.14.2 Invariant Theory and the Inverse Galois Problem

Artin's theorem 30.8.7 is a machine for producing Galois extensions from the top down: fix a big field L and a finite group G of automorphisms, and L/L^G is automatically Galois with group G . Theorem 30.9.17 was the prototype, with $L = k(z_1, \dots, z_n)$ and $G = S_n$. Studying R^G for a ring R and a group $G \leq \text{Aut}(R)$ is the subject of **invariant theory**.

Example 30.14.2

For $G = A_n$ acting on $R = k[z_1, \dots, z_n]$ with $\text{char } k \neq 2$, one has

$$R^{A_n} = k[e_1, \dots, e_n, \delta], \quad \delta = \prod_{i < j} (z_i - z_j),$$

which is why the discriminant is the right tool for detecting whether a Galois group lies in A_n .

Example 30.14.3

Let $R = k[z_1, z_2]$ with $\text{char } k \neq 2$ and let $G = \{1, \sigma\}$ with $\sigma(z_i) = -z_i$. Then

$$R^G = k[z_1^2, z_2^2, z_1 z_2] \cong k[Y_1, Y_2, Y_3]/(Y_1 Y_2 - Y_3^2).$$

Over $\mathbb{R}$ the surface $Y_1 Y_2 = Y_3^2$ is a double cone, and the point where the two cones meet is a *singularity*. Invariant rings are a standard source of interesting singularities in algebraic geometry.

Given K and G , one cannot always find a Galois extension L/K with $\text{Gal}(L/K) \cong G$: if K is algebraically closed there are no non-trivial Galois extensions at all, and if $K = \mathbb{F}_p$ then every Galois group is cyclic. Over $\mathbb{Q}$, the **inverse Galois problem** asks whether every finite group arises. This is open. It is known for every finite soluble group (Shafarevich), for every abelian group (an exercise, using cyclotomic fields and Dirichlet's theorem), and for most finite simple groups.

A better question may be to understand the single enormous group $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the

absolute Galois group of $\mathbb{Q}$; the inverse Galois problem is the question of whether every finite group is a quotient of it. Understanding its representations is one of the central goals of the Langlands programme.

§30.14.3 A Geometric Postscript

Finally, here is a hint of what Galois theory looks like in geometry. Let $K = \mathbb{C}(x)$, the field of rational functions, whose elements we may regard as functions $\mathbb{C} \setminus P \rightarrow \mathbb{C}$ for a finite set of poles P . Consider

$$L = K(\sqrt{x^3 - x}) = \mathbb{C}(x)[y]/(y^2 - x^3 + x),$$

a quadratic extension, so L/K is Galois with group $\mathbb{Z}/2$ generated by $\sigma : y \mapsto -y, x \mapsto x$.

Now $L = \text{Frac}(R)$ where $R = \mathbb{C}[x, y]/(y^2 - x^3 + x)$, and R is the ring of polynomial functions on the curve

$$E = \{(x, y) \in \mathbb{C}^2 : y^2 = x^3 - x\}.$$

The inclusion $\mathbb{C}[x] \hookrightarrow R$ corresponds to the projection $p : E \rightarrow \mathbb{C}, (x, y) \mapsto x$. For each $x \notin \{0, \pm 1\}$ there are exactly two square roots of $x^3 - x$, so p is two-to-one away from three points, and the Galois group $\langle \sigma \rangle$ acts on E by deck transformations with $E/\langle \sigma \rangle = \mathbb{C}$.

In other words, the Galois correspondence for function fields is the theory of covering spaces in disguise, with the Galois group playing the role of the group of deck transformations, and the ramification points $\{0, \pm 1\}$ playing the role of the branch points. This analogy – fields and covering spaces, Galois groups and fundamental groups – runs very deep, and is the starting point of a great deal of modern algebraic geometry and number theory.

31 Graph Theory

Graph theory is an area of combinatorics which has lots of fun problems and plenty of interesting theorems.

This article constitutes my notes for the ‘Graph Theory’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§31.1 Introduction

For many people, ‘Graph Theory’ is a first course in combinatorics. It’s an area with a big focus on problem solving, and it can give a perspective on many other areas of mathematics.

§31.1.1 Definitions

We will begin this handout on graph theory naturally by defining what a graph is.

Definition 31.1.1 (Graph)

A **graph** is an ordered pair $G = (V, E)$ where V is the set of **vertices**, and $E \subseteq \{\{x, y\} \mid x, y \in V, x \neq y\}$ is a set of unordered pairs of vertices called **edges**.

We have a natural way of drawing a graph. For each vertex we have a point in the plane, and for each edge we draw a line between the corresponding pair of vertices.

Example 31.1.2 (Example of a Graph)

The ordered pair (V, E) where $V = \{1, 2, \dots, 6\}$ and $E = \{\{1, 2\}, \{2, 3\}, \dots, \{5, 6\}\}$ is a graph.



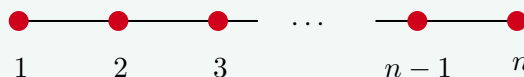
This graph is known as P_6 , a path on 6 vertices.

Common Graphs

There are some graphs that will appear repeatedly when working on problems involving graph theory, and we will define them now.

Definition 31.1.3 (Path)

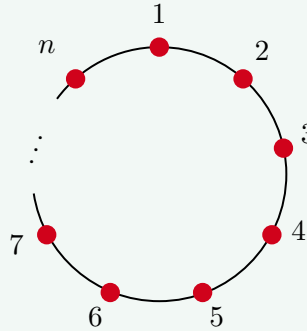
We define P_n to be the graph $V = \{1, \dots, n\}$, $E = \{\{1, 2\}, \{2, 3\}, \dots, \{n-1, n\}\}$ as shown.



We call this a **path** on n vertices, and say it has **length** $n - 1$.

Definition 31.1.4 (Cycle)

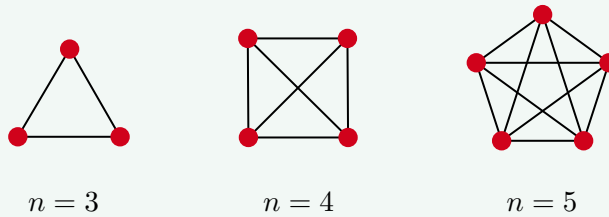
We define C_n (for $n \geq 3$) to be the graph $V = \{1, \dots, n\}$, $E = \{\{1, 2\}, \dots, \{n - 1, n\}, \{n, 1\}\}$ as shown.



We call this the **cycle** on n vertices.

Definition 31.1.5 (Complete Graph)

The **complete graph** on n vertices K_n is the graph $\{1, \dots, n\}$ and $E = \{\{i, j\} \mid i \neq j \in V\}$.

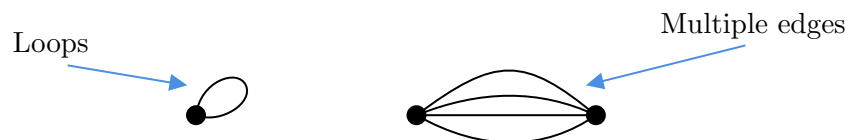


Note that there is an edge between every pair of vertices.

Definition 31.1.6 (Empty Graph)

We define the **empty graph** on n vertices $\overline{K}_n$ to have $V = \{1, \dots, n\}$ but $E = \emptyset$.

Remark. In our definition of a graph, we *don't allow*¹ loops, and there *cannot* be multiple edges between the same set of vertices.



You can define graphs where such things are allowed, but for now we will outlaw them. We also note that edges are *unordered pairs*, so for now edges have no direction.

To be slightly more succinct, we will use some shorthand notation.

Notation. If $G = (V, E)$ is a graph, and we have some edge $\{x, y\} \in E$, we will denote

¹These limitations are inherent in our definition, where we use sets rather than multisets.

it by xy . We will also define $|G| = |V|$, and $e(G) = |E|$.

Example 31.1.7 (Vertices and Edges of K_n)

Consider the graph K_n . We have $|K_n| = n$, and $e(K_n) = \binom{n}{2}$, as there is an edge between any pair of vertices.

Subgraphs

Now we will define the notion of a *subgraph*, in the natural way.

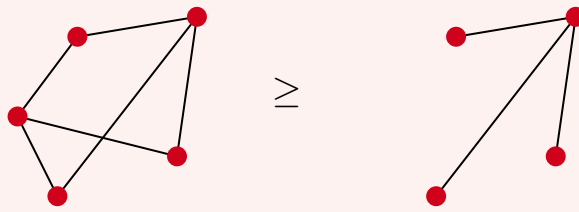
Definition 31.1.8 (Subgraph)

We say that $H = (V', E')$ is a **subgraph** of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$.

Informally, H is a subgraph of G if we can remove vertices and edges from G to get H . Let's look at some examples.

Example 31.1.9 (Example of a Subgraph)

The graph on the right is a subgraph of the graph on the left.



We are also going to use some notation for removing an edge or a vertex from a graph. Of course, when removing a vertex you also have to remove the edges connecting to it.

Notation (Adding/Removing Vertices & Edges). For an edge xy or a vertex x , we define $G - xy$ to be the graph G with the edge xy removed, and $G - x$ to be G with vertex x removed, along with all edges incident to x . We will also define $G + xy$ to be G with the edge xy , and $G + x$ to be G with the vertex x .

An easy way to get a subgraph is by taking a subset of the vertices and seeing what edges you get from the original graph.

Definition 31.1.10 (Induced Subgraph)

If $G = (V, E)$ is a graph and $X \subseteq V$, the **subgraph induced by X** is defined to be $G[X] = (X, \{xy \in E \mid x, y \in X\})$.

Graph Isomorphism

Now that we have defined graphs, it's natural to define some notion of isomorphism.

Definition 31.1.11 (Graph Isomorphism)

Let $G = (V, E)$ and $H = (V', E')$ be graphs. We say that $f : V \rightarrow V'$ is a **graph isomorphism** if $f(u)f(v) \in E' \iff uv \in E$.

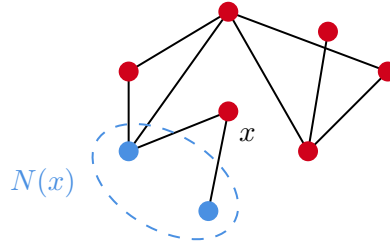
If there is a graph isomorphism between G and H then we say they are **isomorphic**.

Now for the following discussion, fix some graph $G = (V, E)$, and let $x \in V$.

Definition 31.1.12 (Neighbourhood)

If $xy \in E$, then we say that x and y are **adjacent**. We define the **neighborhood** of x to be the set $N(x) = \{y \in V \mid xy \in E\}$ of all vertices adjacent to x .

Note that as in the diagram below, x is not in its own neighborhood.



Definition 31.1.13 (Degree)

We define the **degree** of a vertex x to be $d(x) = |N(x)|$. This is equal to the number of edges that are incident to x .

Definition 31.1.14 (Regularity)

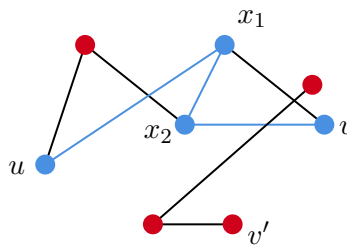
A graph G is said to be **regular** if all of the degrees are the same. We say G is k -regular if $d(x) = k$ for all $x \in V$.

Example 31.1.15 (Regular and Non-Regular Graphs)

The graphs K_n is $n - 1$ regular, and C_n is 2-regular. The graph P_n is not regular.

Connectivity

We now want to define some notion of *connectivity*, where a vertex u is connected to vertex v if you can follow some path in the graph to get from u to v .



For example, in the graph above we want to say somehow that u and v are connected, but u and v' are not. To do this, we will introduce some more definitions.

Definition 31.1.16 (uv Path)

A **uv path** is a sequence $x_1, x_2, \dots, x_l$ where $x_1, \dots, x_l$ are distinct, $x_1 = u$, $x_l = v$ and $x_i x_{i+1} \in E$.

In the example above, ux_1x_2v is a uv path.

The slight subtlety in this condition is the *distinctness* condition. For example, if $x_1 \dots x_l$ is a uv path and $y_1 \dots y_{l'}$ is a vw path, then $x_1 \dots x_l y_1 \dots y_{l'}$ may *not* be a uw path since we may have reused an edge. Of course, we can just not reuse edges by avoiding cycles.

Proposition 31.1.17 (Joining Paths)

If $x_1 \dots x_l$ is a uv path and $y_1 \dots y_{l'}$ is a vw path, then $x_1 \dots x_l y_1 \dots y_{l'}$ contains a uw path.

Proof. Choose a minimal subsequence $w_1 \dots w_r$ of $x_1 \dots x_l y_1 \dots y_{l'}$ such that

1. $w_i w_{i+1} \in E$.
2. $w_1 = u, w_r = w$.

We now claim that $w_1 \dots w_r$ is a uw path. If this was not the case, then it must fail on distinctness, so there would exist some z such that the sequence is

$$w_1 \dots w_a z w_{a+2} \dots w_b z w_{b+2} \dots w_r,$$

but now note that

$$w_1 \dots w_a z w_{b+2} \dots w_r$$

also satisfies the conditions for the subsequence, but is strictly shorter length. This contradicts the minimality condition. $\square$

Now given $G = (V, E)$, let's define an equivalence relation $\sim$ on V , where

$$x \sim y \iff \text{there exists an } xy \text{ path in } G.$$

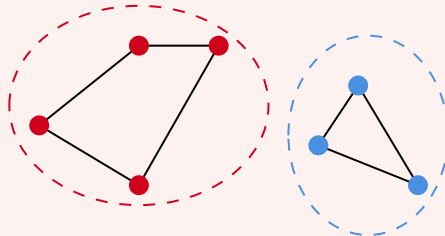
Proposition 31.1.18

$\sim$ is an equivalence relation.

Proof. Note that $\sim$ is reflexive and symmetric, and we get transitivity from our previous proposition. $\square$

Example 31.1.19

In the graph below, the vertices that are the same colour are in the same equivalence class under $\sim$.



Definition 31.1.20 (Connected Graph)

If there is a path between any two vertices in G then we say that G is **connected**.

Definition 31.1.21 (Connected Components)

We call the equivalence classes of $\sim$ on G the **components** or **connected components** of G .

Edges and Distance

We can also introduce some useful definitions relating to the edges of a graph.

Definition 31.1.22 (Minimum/Maximum Degree)

Let G be a graph. The **maximum degree** of G , $\Delta(G)$ is defined to be $\Delta(G) = \max_{x \in V} d(x)$. Similarly, we define the **minimum degree** of G , $\delta(G)$ to be $\delta(G) = \min_{x \in V} d(x)$.

In a k -regular graph as mentioned above, we have $\Delta(G) = \delta(G) = k$.

Definition 31.1.23 (Graph Distance)

Let $G = (V, E)$ be a graph. The associated **graph distance** $d : V \times V \rightarrow \mathbb{R}^{\geq 0} \cup \{\infty\}$ is defined so that $d(x, y)$ is the minimum path length from x to y if it exists, and ∞ otherwise.

Proposition 31.1.24 (Graph Distance is a Metric)

Let $G = (V, E)$ be a connected graph and let d be the associated graph distance. Then (V, d) defines a metric space.

Proof Sketch. We have $d(x, y) = 0$, and $d(x, y) = d(y, x)$ (taking the shortest path in the opposite direction), and $d(x, z) \leq d(x, y) + d(y, z)$ as we can find path from x to z by taking paths from x to y and y to z and adjoining them, and this puts an upper bound on $d(x, z)$. $\square$

§31.1.2 Trees

We will now discuss a special class of graph called *trees*. This class is quite restrictive (yet is quite useful), and they have some nice properties.

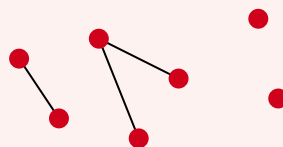
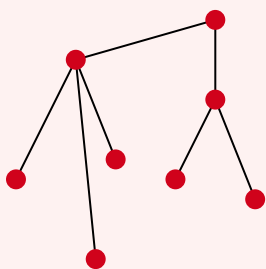
To define what a tree is, we first need a notion of when a graph is acyclic.

Definition 31.1.25 (Acyclic)

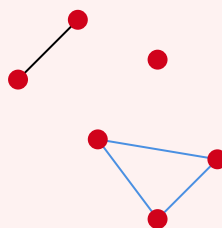
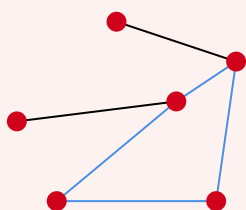
A graph G is said to be **acyclic** if it does not contain any subgraph isomorphic to a cycle, C_n .

Example 31.1.26 (Example of Acyclic/Non-Acyclic Graphs)

In the example below, the two graphs are both *acyclic*.



Two *non-acyclic* graphs are shown below. The subgraphs isomorphic to C_4 and C_3 are highlighted.

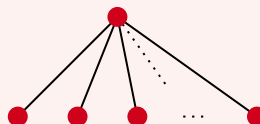
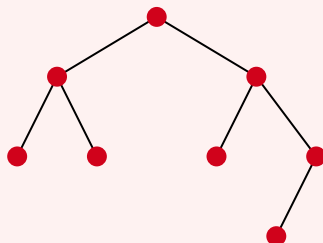
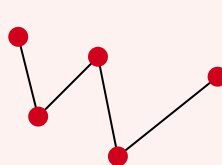


Definition 31.1.27 (Tree)

A **tree** is a connected, acyclic graph.

Example 31.1.28 (Examples of Trees)

The following three graphs are trees.



Proposition 31.1.29 (Characterising Trees)

The following are equivalent.

- (a) G is a tree.
- (b) G is a maximal acyclic graph (adding any edge creates a cycle).
- (c) G is a minimal connected graph (removing any edge disconnects the graph).

Proof. (a) $\implies$ (b). By definition G is acyclic. Let $x, y \in V$ such that $xy \notin E$. As G is connected, there is an xy path P . So xPy then defines a cycle.

(b) $\implies$ (a). By definition G is acyclic. So for a contradiction assume G is not connected and let x, y be vertices from different components. Now note $G + xy$ is

acyclic, but this contradicts the claim that G is maximally acyclic.

(a) $\implies$ (c). By definition G is connected. Suppose, for a contradiction, that there exists some vertices $x, y \in E$ with $x \neq y$ and $G - xy$ is connected. But then there is some xy path P that does not use the edge xy , so xPy is then a cycle, contradicting that G is acyclic.

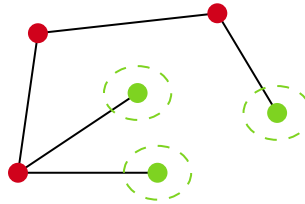
(c) $\implies$ (a). By definition G is connected. Again for a contradiction, assume that G contains a cycle C . Then let xy be an edge on C . We claim $G - xy$ is still connected. If $u, v \in V(G - xy)$ then let P be a path in G from u to v . If xy does not appear as consecutive vertices on this path, then u is connected to v . Otherwise, we can consider a new path where we replace x, y with the other vertices in $C - xy$ in order. Thus u and v are still connected. This contradicts the minimal connectedness of G .

□

Definition 31.1.30 (Leaf)

Let G be a graph. A vertex $v \in V(G)$ is a **leaf** if $d(v) = 1$.

For example, the tree below has three leaves.



In general, trees has a leaf.

Proposition 31.1.31 (Trees Have Leaves)

Every tree T with $|T| \geq 2$ has a leaf.

Proof. Let T be a tree with $|T| \geq 2$, and let P be a path of maximum length in T , with $P = x_1 \dots x_k$. We claim that $d(x_k) = 1$. Observe that $\deg(x_k) \geq 1$, since $x_k x_{k-1} \in E$. If x_k is adjacent to another vertex $y \neq x_{k-1}$, then either $y \in \{x_1, \dots, x_{k-2}\}$, which would imply that T contains a cycle, or $y \notin \{x_1, \dots, x_{k-2}\}$, then $x_1 \dots x_k y$ is a path longer than P , which violates its maximality. □

Remark. This proof gives us two leaves in T , which is the best we can hope for considering P_n is a tree with exactly two leaves.

Proposition 31.1.32 (Edges of a Tree)

Let T be a tree. Then $e(T) = |T| - 1$.

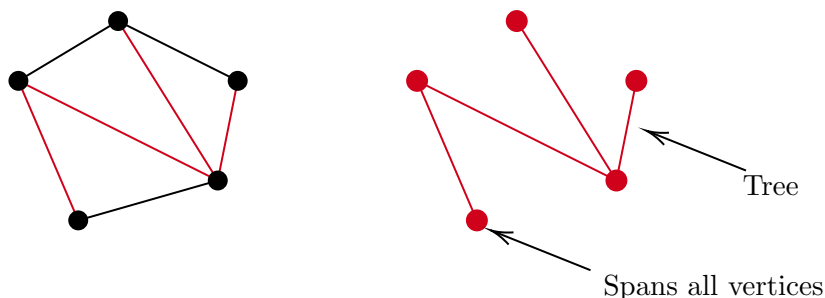
Proof. We will do induction on $n = |T|$. If $n = 1$, this is trivial as there is only one edge. Now given T with at least 2 vertices, let x be a leaf in T , and define $T' = T - x$.

T' must be acyclic, since we have only removed vertices. T' must also be connected since for all $u, v \in V(T')$ there exists a path from u to v in T that does not use x , so it is also a path from u to v in T' . Thus T' is a tree. Thus by induction, T' has $n - 2$ edges, and $e(T) = e(T') + 1 = |T| - 1$. $\square$

Now let's think about trees as subgraphs of other graphs.

Definition 31.1.33 (Spanning Tree)

Let G be a graph. We say T is a **spanning tree** of G if T is a tree on $V(G)$ and is a subgraph of G .



Spanning trees are useful in a number of contexts, one of which is giving a sensible ordering to the vertices of a graph. They are particularly useful because of the following result.

Proposition 31.1.34 (Connected Graphs have Spanning Trees)

Every connected graph contains a spanning tree.

Proof. A tree is a minimal connected graph. So take the connected graph and remove edges until it becomes a minimal connected graph. Then this will be a subgraph of the original graph, and will thus be a spanning tree. $\square$

§31.1.3 Bipartite Graphs

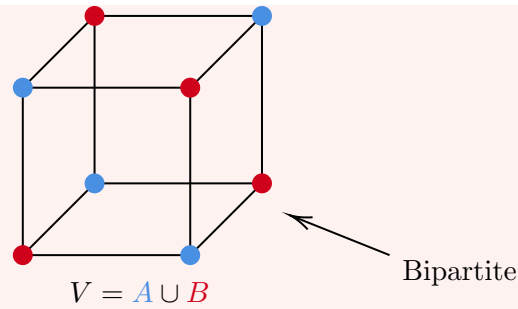
The next type of graph we will look at is *bipartite* graphs.

Definition 31.1.35 (Bipartite Graphs)

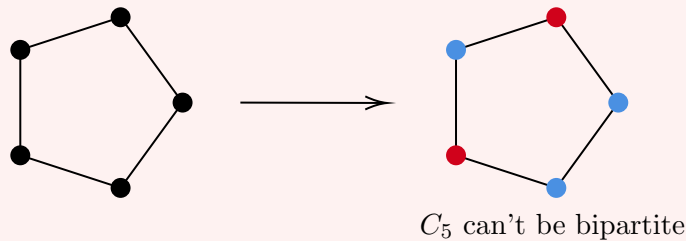
A graph $G = (V, E)$ is **bipartite** if $V = A \cup B$ where $A \cap B = \emptyset$ and all edges $xy \in E$ have either $x \in A, y \in B$ or $x \in B, y \in A$.

Example 31.1.36 (Example of Bipartite Graphs)

The graph below is bipartite, with vertices in the set A being coloured red and vertices in the set B being coloured blue.



An example of a non-bipartite graph is C_5 . To see this, we can start by choosing a vertex to be in A (without loss of generality), then the adjacent vertices must be in B , but then their adjacent vertices must be in A , but then there is an edge between two vertices in A . This is shown below.



The argument given for C_5 works in general.

Proposition 31.1.37 (Bipartite Cyclic Graphs)

The cycle C_{2k+1} is *not* bipartite, and the cycle C_{2k} is bipartite.

Proof. Assume that C_{2k+1} is bipartite. Then there must be disjoint sets A and B , and as $2k + 1$ is odd, we must have (without loss of generality), $|A| > |B|$. Now let's count the edges between A and B . This must be $2|A|$ and also $2|B|$, as every vertex has degree 2. But then $|A| = |B|$, which is a contradiction.

For C_{2n} , we can let $v_i \in A$ if i is even and $v_i \in B$ if i is odd. Then a vertex i only has edges to vertices $i - 1$ and $i + 1 \pmod{2}$, which have opposite parity. Thus all edges are between A and B , as required. $\square$

There is then a natural question: given some arbitrary graph G , how do we determine if a given graph is bipartite? It turns out that there is a nice check for 'bipartness'. We will state the result and then do some setup before we prove it.

Proposition 31.1.38 (Bipartite Criterion)

A graph G is bipartite if and only if G contains no odd cycles.

We need to first develop some theory regarding *circuits*. Informally, a circuit is like a cycle where we can revisit vertices.

Definition 31.1.39 (Circuit)

A circuit is a sequence $x_1 \dots x_l$ where $x_1 = x_l$ and $x_i x_{i+1} \in E$. The **length** of the circuit is $l - 1$, the number of edges traversed in the circuit.

Definition 31.1.40 (Odd Circuits)

If the length of a circuit is odd, then we say it is an **odd circuit**.

Proposition 31.1.41

An odd circuit contains an odd cycle.

Proof. We will prove this by induction on the length of the circuit. For a circuit of length 3, the circuit must be a cycle. In general, let $C = x_1 \dots x_l$ be our circuit. If $x_1, \dots, x_{l-1}$ are distinct, then C is a cycle and we are done.

Otherwise, there exists some $z \in C$ that is repeated. We write

$$C = x_1 \dots x_a z x_{a+2} \dots x_b z x_{b+2} \dots x_l.$$

We define $C' = x_1 \dots x_a z x_{b+2} \dots x_l$ and $C'' = z x_{a+2} \dots x_b z$. The length of C' and C'' is strictly less than the length of C . One of these circuits must have odd length, and by induction that odd circuit contains an odd cycle. $\square$

We can now prove our original bipartiteness criterion, that a graph is bipartite if and only if it contains no odd cycles.

Proof (Bipartite Criterion). If G was bipartite and contained an odd cycle, then there exists an odd cycle that is bipartite. But this is a contradiction.

Now if G is not bipartite, we can induct on the number of vertices. For $|G| = 1$, this holds. Now if G is not connected, let $C_1, \dots, C_k$ be the components of G . We may now apply our induction to each component of G to obtain a bipartition $V(C_i) = A_i \cup B_i$ for each $i \in 1, \dots, k$. Then $A = A_1 \cup \dots \cup A_k$ and $B = B_1 \cup \dots \cup B_k$ is a bipartition for the whole graph.

We may now assume without loss of generality that G is connected. Fix some vertex $v \in V$, and define

$$A = \{u \in V \mid d(u, v) \text{ is odd}\}$$

$$B = \{u \in V \mid d(u, v) \text{ is even}\}$$

We claim that $A \cup B$ is a bipartition.

Assume (for a contradiction) that u_1 is adjacent to u_2 and $d(u_1, v) \not\equiv d(u_2, v) \pmod{2}$. Then there exists paths P_1 from v to u_1 and P_2 from u_2 to v with $|P_1| \equiv |P_2| \pmod{2}$. But this implies that $vP_1u_1u_2P_2v$ defines an odd circuit in G . Therefore, by our previous proposition, G contains an odd cycle, which is a contradiction. $\square$

§31.2 Hall's Theorem

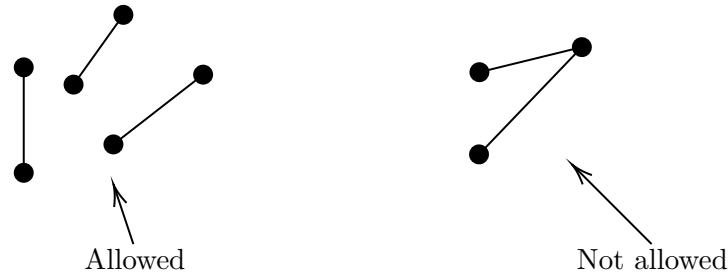
In this section, we will build up our knowledge of *matchings* so that we can prove the first theorem of the course – Hall's theorem.

§31.2.1 Matchings

An appropriate place to start is probably by defining what a matching is.

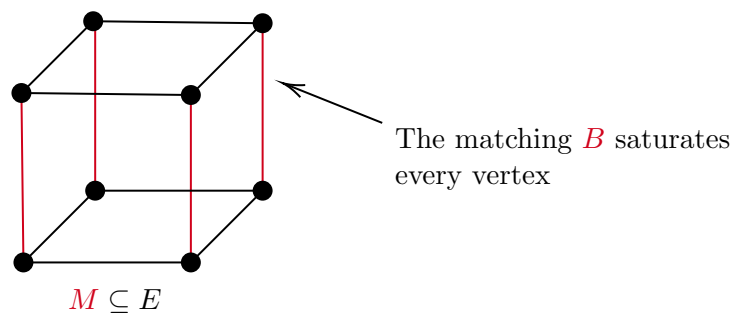
Definition 31.2.1 (Matching)

A **matching** of G is a collection of edges $M \subseteq E$ so that $\forall e_1, e_2 \in M$ with $e_1 \neq e_2$ has $e_1 \cap e_2 = \emptyset$.



Definition 31.2.2 (Saturated)

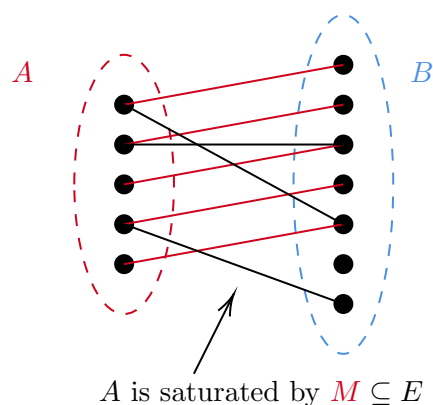
Given a graph $G = (V, E)$ and a matching M in G , we say that a vertex $v \in V$ is **saturated by M** if there exists an edge in M containing v .



A matching where every vertex is saturated (such as the above) is known as a **perfect matching**. However, there is no restriction in general on how many vertices are saturated by a matching.

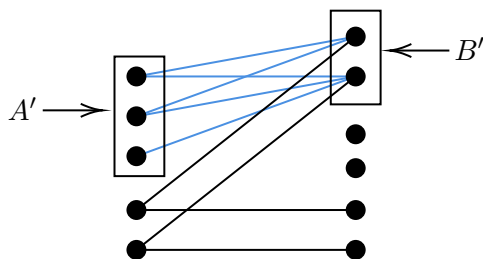
§31.2.2 Matching in Bipartite Graphs

Matchings are particularly interesting in bipartite graphs. We will be interested in the following question: given a bipartite graph $G = (A \cup B, E)$, when can I find a matching saturating A ?



This question is the same as asking when is there a function $f : A \rightarrow B$ where $xf(x) \in E$ that is an injection.

In trying to answer this question, we might try and think about why it may not be possible. The simplest reason is when B isn't large enough to have an injection, when $|B| \leq |A|$. In a similar way, we might have a graph is big enough, but that isn't true for a small part of the graph, like below.



What Hall's theorem says is that this issue is the only obstruction to creating such a matching.

Definition 31.2.3 (Neighbourhood of a Vertex Set)

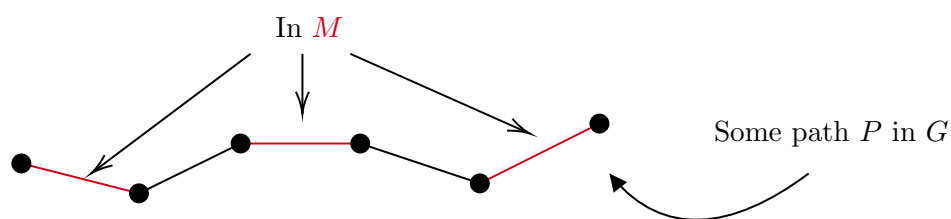
If $X \subseteq V$, then we define $N(X)$, the **neighbourhood of X** to be $N(X) = \bigcup_{x \in X} N(x)$.

With the notion of the neighborhood of a set of vertices, we can rephrase the issue mentioned about as $|N(A')| < |A'|$ for some subset $A' \subseteq A$. We will use this notation in our statement for Hall's theorem.

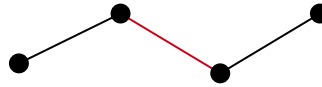
Before we prove the theorem, we will need to prepare a little bit.

Definition 31.2.4 (M -Alternating Path)

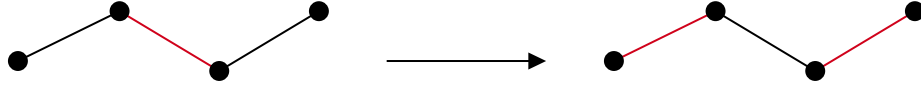
Let $G = (V, E)$ be a graph with a matching M in G . We say that a path $P = x_1 \dots x_l$ is **M -alternating** if $x_i x_{i+1}$ is alternately in M and not in M .



Another example of an M -alternating path is the one below.



If we saw the path above in the graph, and we knew that the end vertices were not saturated, then we could change the edges that are in M , so we would still have a matching. This move will be key in our proof of Hall's theorem.



We are going to call this configuration an *augmented path*.

Definition 31.2.5 (M -Augmenting Path)

Given a graph $G = (V, E)$ and a matching M in G , an M -alternating path $P = x_1 \dots x_l$ is said to be **M -augmenting** if x_1 and x_l are not saturated.

Proposition 31.2.6

If M is a matching in G of maximum size, then there are no M -augmenting paths.

Proof. If there is an M -augmenting path in G , then we can flip the edges of M along P to find a strictly larger matching. $\square$

An important observation is that an M -alternating path in a bipartite graph $G = (A \cup B, E)$ with $P = x_1 \dots x_l$, with $x_1 \in A$ and $x_1x_2 \in M$, then $x_{2k+1}x_{2k+2} \in M$, and vice versa.

§31.2.3 Hall's Theorem

We are now ready to state and prove Hall's matching theorem, which formalises the ideas mentioned in the previous section.

Theorem 31.2.7 (Hall's Theorem)

Let $G = (A \cup B, E)$ be a bipartite graph. Then there exists a matching saturating A if and only if every subset $A' \subseteq A$ satisfies $|N(A')| \geq |A'|$.

Proof. First we prove the forward (and easy) direction. Let $A' = \{x_1, \dots, x_t\} \subseteq A$. We have matching edges $x_1y_1, \dots, x_ty_t \in M \subseteq E$, and thus $\{y_1, \dots, y_t\} \subseteq N(A')$, and thus $|N(A')| \geq |A'|$.

Now for the other (harder) direction, which is that this condition implies the existence of such a matching. Choose a matching M in G with $|M|$ maximized. For a contradiction, assume there is some vertex $a_0 \in A$ that is not saturated by M .

We inductively define sets $A_i \subseteq A$, $B_t \subseteq B$ by setting $A_0 = \{a_0\}$, $B_0 = \emptyset$. We will maintain, for all t , that

1. $|A_t| = t + 1$, $|B_t| = t$.
2. Every vertex in $A_t \cup B_t$ is the endpoint of an alternating path that started at a_0 .
3. $A_t \setminus \{a_0\}$ is matched to B_t .

So given A_t, B_t , we need to define A_{t+1}, B_{t+1} .

First consider $N(A_t)$. We have $|N(A_t)| \geq |A_t| = t + 1 > |B_t|$. So $N(A_t) \setminus B_t$ is non-empty. So let $b_{t+1} \in N(A_t) \setminus B_t$. Observe that b_{t+1} can be reached along an alternating path which started at a_0 . Call $y \in A_t$ such that $y b_{t+1} \in E$. Then y can be reached along an alternating path starting at a_0 . Let $P = a_0 x_1 \dots x_l y$ be such an alternating path. Since $a_0 x_1 \notin M$, $x_l y \in M$. Hence $P b_{t+1}$ is a alternating path.

Now, b_{t+1} is saturated by M , as otherwise $P b_{t+1}$ would be an M -augmenting path, which is a contradiction. So let $a_{t+1} b_{t+1} \in M$. We claim $a_{t+1} \notin A_t$, since A_t is matched to B_t , and $b_{t+1} \notin B_t$. So we may define $A_{t+1} = A_t \cup \{a_{t+1}\}$ and $B_{t+1} = B_t \cup \{b_{t+1}\}$. We can check that what we claimed before holds.

Since $a_{t+1} \notin A_t$, and $b_{t+1} \notin B_t$, we have $|A_{t+1}| = |A_t| + 1$ and $|B_{t+1}| = |B_t| + 1$. Also every vertex is the endpoint of an alternating path starting at a_0 by construction. Also $A_{t+1} \setminus \{a_0\}$ is matched to B_{t+1} since $A_t \setminus \{a_0\}$ is matched to B_t , and by construction.

This completes the construction of A_t, B_t for all t . Then if $t > |A|$, then $|A_t| > |A|$, but $A_t \subseteq A$, which is a contradiction. $\square$

§31.2.4 Corollaries of Hall's Theorem

So Hall's theorem gives us a nice way to guarantee the existence of matchings that saturate a part of a bipartite graph. We are going to use Hall's theorem to prove some interesting results.

Corollary 31.2.8

A k -regular bipartite graph contains a perfect matching.

Proof. Let $G = (A \cup B, E)$, and let $A' \subseteq A$. We want to show that $|N(A')| \geq |A'|$ so that we may apply Hall's theorem.

We will count the number of edges between A' and $N(A')$ in two different ways. We know that each vertex in A' has degree k , so there is $k|A'|$ edges. But on the other hand, the number of edges in $N(A')$ is $|N(A')|k$. Thus $|N(A')|k \geq |A'|k$, so $|N(A')| \geq |A'|$, and by Hall's theorem there exists a matching saturating A' .

We claim that $|B| = |A|$. Indeed, the number of edges between A and B is $k|A|$ and is also $k|B|$, and thus $|A| = |B|$. So a matching saturating A also saturates B . So there exists a perfect matching. $\square$

We can also consider an extension of Hall's theorem. Hall's theorem tells us when, in a bipartite graph $G = (A \cup B, E)$, there is a matching saturating A , but what if we only wanted a matching of size k ?

Let's say that a matching in G has **deficiency** d if it saturates $|A| - d$ vertices.

Corollary 31.2.9

Let G be a bipartite graph. Then G contains a matching saturating $|A| - d$ vertices in A if and only if for all $A' \subseteq A$, we have $|N(A')| \geq |A'| - d$.

Proof. If we have a matching with deficiency d , then this clearly holds.

Now if this condition holds for some graph $G = (A \cup B, E)$, we define a new graph $\tilde{G}$ by setting $\tilde{B} = B \cup \{z_1, \dots, z_d\}$ where $z_1, \dots, z_d$ are distinct from $x \in A \cup B$, and we define $\tilde{E} = E \cup \{e_i a \mid i \in \{1, \dots, d\}, a \in A\}$. So $\tilde{G} = (A \cup \tilde{B}, \tilde{E})$.

This is a bipartite graph, and we observe that it satisfies Hall's condition. Thus $\tilde{G}$ has a matching M saturating A . So if we define M' by removing all of the edges with an endpoint in $\{z_1, \dots, z_d\}$, then M' saturates all but at most d vertices in G . $\square$

Another way of stating Hall's theorem is with set systems.

Definition 31.2.10 (System of Distinct Representatives)

Given sets $S_1, \dots, S_n \subset X$ where S_i is finite for all i . Then we say that $x_1, \dots, x_n \in X$ is a **system of distinct representatives** (or SDR) if they are distinct and $x_i \in S_i$.

The question is, for what set systems does there exist a system of distinct representatives? The answer is if they satisfy some Hall-like condition.

Corollary 31.2.11 (Existence of SDRs)

Given $S_1, \dots, S_n \subseteq X$, with S_i finite, then $S_1, \dots, S_n$ has a system of distinct representatives if and only if

$$\left| \bigcup_{i \in I} S_i \right| \geq |I|,$$

for all $I \in \{1, \dots, n\}$.

Proof. If such an SDR exists, then this condition clearly holds.

In the other direction, given that this condition holds, we define a graph G by setting $A = \{S_1, \dots, S_n\}$ and $B = \bigcup_{i=1}^n S_i$, with $E = \{\{S_i, x\} \mid x \in A, i \in \{1, \dots, n\}, x \in S_i\}$. Then $G = (A \cup B, E)$. Now observe that a SDR is exactly a matching in G that saturates A .

We check Hall's condition. Given $A' \subseteq A$, then $N(A') = \bigcup_{S_i \in A'} S_i$. Thus

$$|N(A')| = \left| \bigcup_{S_i \in A'} S_i \right| \geq |A'|,$$

by our condition. So such a matching exists. $\square$

Remark. The existence of SDRs is equivalent to Hall's theorem.

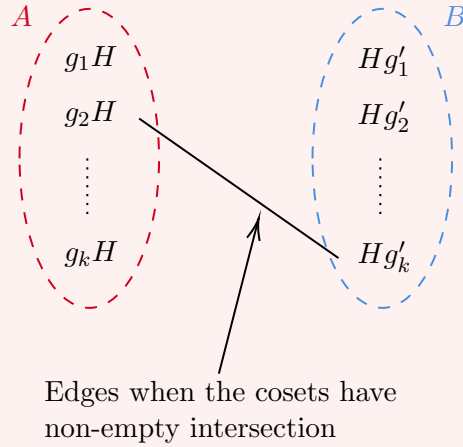
The last example of using Hall's theorem will be an application to group theory.

Example 31.2.12 (Left and Right Coset Representatives)

Let G be a group and let $H < G$, with G finite. We have the left cosets $g_1H, \dots, g_kH$ and right cosets $Hg'_1, \dots, Hg'_k$.

We want to know does there exists $h_1, \dots, h_k$ such that $h_1H, \dots, h_kH$ are all of the left cosets and $Hh_1, \dots, Hh_k$ are all of the right cosets. It turns out that this question is entirely combinatorial, and we can use Hall's theorem.

Let's form the bipartite graph $(A \cup B, E)$ with $A = \{g_1H, \dots, g_kH\}$ and $B = \{Hg'_1, \dots, Hg'_k\}$ so that there is an edge between g_iH and Hg'_j if the cosets have non empty intersection.



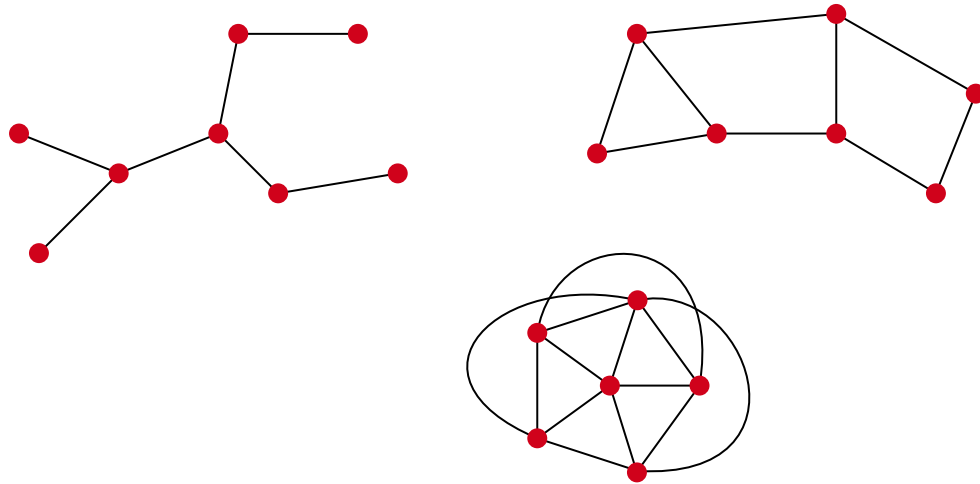
If we had a perfect matching in this graph, then we could pick an element h_i in the (non-empty) intersection of the matched left and right cosets, and then $h_1, \dots, h_k$ would be a set of coset representatives for all of the k left and right cosets.

So we want to show that Hall's theorem is satisfied. Given $I \subseteq \{1, \dots, k\}$, we want to show that the number of right cosets intersecting $\bigcup_{i \in I} g_iH$ is at least $|I|$.

Observe that $|\bigcup_{i \in I} g_iH| = |H||I|$. Since the right cosets partition G , and each coset has size $|H|$, so the number of right cosets intersecting $\bigcup g_iH$ is at least $|I|$. Thus $|N(\{g_iH \mid i \in I\})| \geq |I|$, and thus Hall's theorem is satisfied, and we have a perfect matching in the graph.

§31.3 Connectivity

We have already defined what it means for a graph to be connected, but consider the following connected graphs:



Clearly these are connected, but they are all ‘connected to different extents’. For example, in the first graph, removing any vertex disconnects the graph. In the second graph, any vertex could be removed and the graph would stay connected. We can also see that in the first graph, there’s only one path from one vertex to another, whereas in the third graph there is many. This also seems to correlate with ‘how connected’ a graph is.

§31.3.1 Measuring Connectivity

So looking at the graphs above, we two natural notions for ‘how connected a graph is’ can be informally described as

1. A ‘deletion notion’ of connectivity, where we consider how connected the graph is after some vertices are removed.
2. A ‘paths notion’ of connectivity, where we consider how many independent paths there is between vertices.

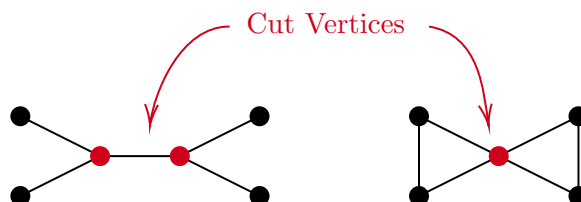
One of the main goals of this section will be turning this vague notion into a concrete concept, and proving an interesting result about how the two notions relate to each other.

Notation (Removing Vertex Sets). In this section (and for the rest of this article), if $G = (V, E)$ is a graph and $S \subseteq V$, we define $G - S = G - x_1 - x_2 - \cdots - x_l$ where $S = \{x_1, \dots, x_l\}$.

We will begin with a definition.

Definition 31.3.1 (Cut Vertex)

Let $G = (V, E)$ be a connected graph. We say that $v \in V$ is a **cut vertex** if $G - v$ is disconnected.



Definition 31.3.2 (Separator)

If $G = (G, V)$ is a connected graph, we say that a subset $S \subseteq V$ is a **separator** (or separating set) if $G - S$ is disconnected.

With these concepts defined, we can define our ‘deletion’ notion of connectivity.

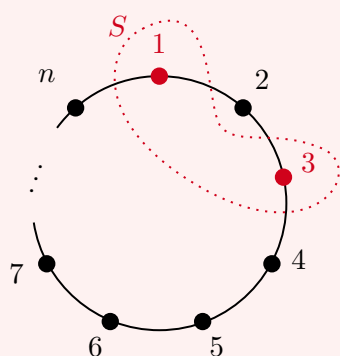
Definition 31.3.3 (Connectivity)

Let $G = (V, E)$ be a graph. The **connectivity** of G , denoted $\kappa(G)$, is the size of the smallest set $S \subseteq V$ such that $G - S$ is disconnected, or just a single vertex^a.

^aThis is needed for the case of a complete graph.

Example 31.3.4 (Connectivity of C_n)

Consider the graph C_n for $n \geq 3$.

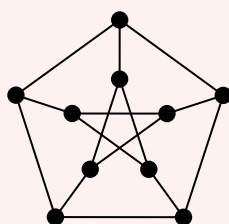


Removing one vertex will never disconnect C_n .
However, removing two vertices can disconnect it.

$$\implies \kappa(C_n) = 2$$

Example 31.3.5 (Connectivity of the Petersen Graph)

We define the **Petersen graph** with 10 vertices and 15 edges as shown below.



We can see that the connectivity is at most 3, since that is the degree of each vertex, and also removing two vertices won't disconnect the graph. Thus the connectivity of the Petersen graph is exactly 3.

Definition 31.3.6 (k -Connected)

We say that a graph G is **k -connected** if $\kappa(G) \geq k$. In particular, any set $S \subseteq V(G)$ with $|S| < k$ will have $G - S$ connected.

We note this immediately implies that G is 1-connected if and only if it's connected, and it is 2-connected if and only if it has no cut vertex.

We can note some basic properties of connectivity.

Lemma 31.3.7 (Increasing/Reducing Connectivity)

If $G = (V, E)$ is a k -connected graph and $v \in V$ then $G - v$ is $(k - 1)$ -connected^a. Also, if we have some $e \in E$, then $G - e$ is $(k - 1)$ -connected.

^aNote that it is also possible that $G - v$ is $(k + 1)$ -connected, so connectivity can also increase by deleting a vertex.

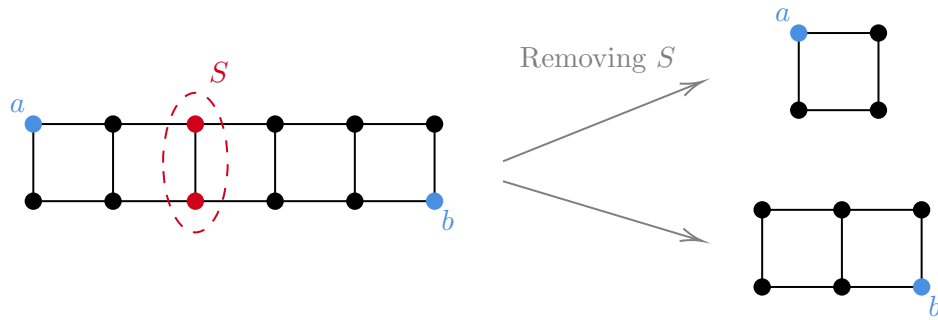
Proof Sketch. Check definitions. □

§31.3.2 Preparing for Menger's Theorem

We are going to prove Menger's theorem about connectivity measures, but we have to do some setting up beforehand.

Definition 31.3.8 (ab -Seperator)

If $G = (V, E)$ is a graph and $a, b \in V$ are distinct vertices, we say that S is an **ab seperator** if a and b lie in different components of $G - S$ (so $a, b \notin S$).



Definition 31.3.9 (Seperator)

If $G = (V, E)$ is a graph and $F \subseteq E$, we say that F is a **seperator** of G if $G - F$ is disconnected.

We then have a number of useful lemmas, that we will employ later on.

Notation. We let $\kappa_{a,b}(G)$ be the minimum size of an ab seperator.

Lemma 31.3.10

Let $G = (V, E)$ be a graph. Then $\kappa_{a,b}(G) \geq \kappa_{a,b}(G - v)$, where $v \in V$ and $v \neq a, b$.

Proof. If $G - v$ has an ab seperator S then $S \cup \{v\}$ is an ab seperator in G . □

Lemma 31.3.11

For a graph $G = (V, E)$, $\kappa_{a,b}(G - e) \geq \kappa_{a,b}(G) = 1$, for $e \in E$ and $a \sim b$.

Proof. If $G - e$ has an ab separator S then $S \cup \{x\}$ and $S \cup \{y\}$ are ab separators in G , where $e = xy$. $\square$

Lemma 31.3.12

Let $G = (V, E)$ be a graph with distinct non-adjacent vertices $a, b \in V$. Also let $\kappa_{a,b}(G) \geq k$. Let S be an ab separator in G , and say $G - S = A \cup C$, where A is the connected component containing a . Then define $\tilde{G}$ as the induced graph $G[A \cup S]$ with a vertex x joined to all of S . Then $\kappa_{a,x}(\tilde{G}) \geq k$.

Proof Sketch. Check definitions. $\square$

We can now formalize our ‘independent paths’ notion of connectivity.

Definition 31.3.13 (Independent Paths)

We say that $P_1, \dots, P_k$ are **independent ab paths** if each of $P_1, \dots, P_k$ are ab -paths and all of the vertices in these paths are distinct, apart from a and b .

§31.3.3 Actually Menger’s Theorem

Finally, we can state and prove Menger’s theorem.

Theorem 31.3.14 (Menger’s Theorem, First Form)

Let $G = (V, E)$ be a graph, with distinct and non-adjacent $a, b \in V$. If every ab separator in G has size at least k then we can find k independent ab paths.

Proof. Suppose for a contradiction, assume this is false and let G be the counterexample that

- (1) Minimizes $\kappa_{a,b}(G)$
- (2) Subject to (1), minimizes the number of edges in the graph.

Then we observe that $\kappa_{a,b}(G - e) = \kappa_{a,b}(G) - 1$ for any edge $e \in E$. We will let $k = \kappa_{a,b}(G)$.

Claim. There exists an ab separator S with $|S| = k$ so that $S \not\subseteq N(a)$ and $S \not\subseteq N(b)$.

We first observe that $N(a) \cap N(b)$ is empty. To see this, let $x \in N(a) \cap N(b)$ and $G' = G - x$. Then $\kappa_{a,b}(G') \geq \kappa_{a,b}(G) - 1$, thus there are $k - 1$ independent paths $P_1, \dots, P_{k-1}$ in G' , then $P_1, \dots, P_{k-1}, axb$ are k independent paths in G , and our graph would then not be a counterexample.

Now choose a shortest path $P = ax_1 \dots x_l b$. Let $G' = G - x_1 x_2$. We know that $\kappa_{a,b}(G') = k - 1$. Therefore there exists an ab separator S' of size $k - 1$. Let A be the component of a in $G' - S'$, and B be the component of b in $G' - S'$. Note that $x_1 x_2$ must have $x_1 \in A$ and $x_2 \in B$.

Note $x_1 \sim a$, and $x_2 \not\sim a$ since P is the shortest path. Also note that $x_2 \neq b$, since $N(a) \cap N(b) = \emptyset$.

If $S' \subseteq N(a)$, then $S' \cup \{x_2\}$ is an ab separator of size k , and $S \not\subseteq N(a), N(b)$. If $S' \subseteq N(b)$, then $S = S' \cup \{x_1\}$ is an ab separator of size k , and $S \not\subseteq N(b), N(a)$. Lastly, if $S' \not\subseteq N(a)$ and $S' \not\subseteq N(b)$, then $S = S' \cup \{x_1\}$ is an ab separator of size k that is not in $N(a)$ or $N(b)$. So our claim holds.

So let S be an ab separator with $|S| = k$ and $S \not\subseteq N(a), N(b)$. Define A, B to be the components containing a and b in $G - S$. We also define $\tilde{G}_a$ as $G[A \cup S]$ with a vertex x that joins to all of S . define $\tilde{G}_b$ likewise. We now have

$$e(\tilde{G}_a) < e(G), \quad e(\tilde{G}_b) < e(G).$$

We also have $\kappa_{a,x}(\tilde{G}_a) = k = \kappa_{b,x}(\tilde{G}_b)$. Thus $\tilde{G}_a$ and $\tilde{G}_b$ satisfy the theorem, by minimality.

So we can find independent ax paths $P_1, \dots, P_k \in \tilde{G}_a$ and yb paths $Q_1, \dots, Q_k \in \tilde{G}_b$. Thus we can find k independent ab paths by concatenation and reordering in G , as desired. $\square$

Remark. It should be noted that we need the non-adjacent condition, otherwise there is no ab separator. Before we write down the proof, we will isolate some notable facts about connectivity. Also, this result implies Hall's theorem.

Another form of Menger's theorem is more common.

Theorem 31.3.15 (Menger's Theorem, Second Form)

Let $G = (V, E)$ be a graph. Then G is k -connected if and only for all $u, v \in V$ with $u \neq v$, there exists k independent uv -paths.

Proof. If u is not adjacent to v , then apply Menger's theorem (first form) to find k -independent uv paths. If they are adjacent, then $G' = G - uv$ is $k - 1$ connected, thus Menger's theorem (first form) tells us that there are uv independent paths $P_1, \dots, P_{k-1}$ in $G - uv$. Thus $P_1, \dots, P_{k-1}, uv$ are k independent paths. The other direction is straightforward. $\square$

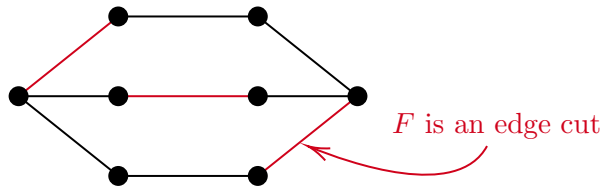
§31.3.4 Edge Connectivity

We have so far been looking at connectivity related to removing vertices and considering independent paths. We will now look at connectivity related to removing edges, and we will see that Menger's theorem is still useful.

Notation. If $G = (V, E)$ is a graph and $F \subseteq E$, define $G - F = (V, E \setminus F)$.

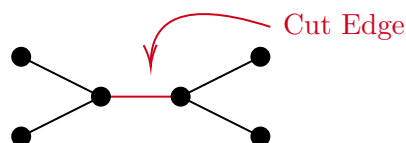
Definition 31.3.16 (Edge Cut)

An **edge cut** in a graph $G = (V, E)$ is a set $F \subseteq E$ so that $G - F$ is disconnected.



Definition 31.3.17 (Cut Edge)

A **cut edge** $e \in E$ is an edge so that $G - e$ is disconnected.

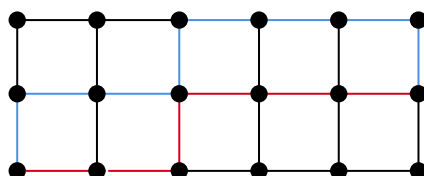


Now similarly to how we had independent paths before (that didn't share vertices), we can define a notion of edge independent paths.

Definition 31.3.18 (Edge Disjoint Paths)

We say that the uv paths $P_1, \dots, P_k$ are **edge disjoint** if $E(P_i) \cap E(P_j) = \emptyset$ for all $i \neq j$.

The **blue** and **red** paths are edge disjoint



We can then define edge connectivity (our deletion notion).

Definition 31.3.19 (Edge Connectivity)

Define the **edge connectivity** of G , $\lambda(G)$ to be the smallest $|S|$ with $S \subseteq E$ such that $G - S$ is disconnected.

Definition 31.3.20 (k -Edge-Connected)

We say a graph G is **k -edge-connected** if $\lambda(G) \geq k$. In other words, $G - F$ is connected for all $|F| \leq k - 1$, with $F \subseteq E$.

We note that G is 1-edge-connected if and only if it is connected, and it is 2-edge-connected if and only if there is no cut edge.

We have a Menger's theorem for this notion of connectivity also.

Theorem 31.3.21 (Menger's Theorem, Edge Version)

Let $G = (V, E)$ be a graph, and u, v be distinct vertices of G . If every set of edges $F \subseteq E$ that separates u from v has size greater than or equal to k , then there exists k edge disjoint paths from u to v .

We also have a similar second version.

Theorem 31.3.22 (Menger's Theorem, Edge Version Two)

Let $G = (V, E)$ be a graph. Then G is k -edge-connected if and only if for every

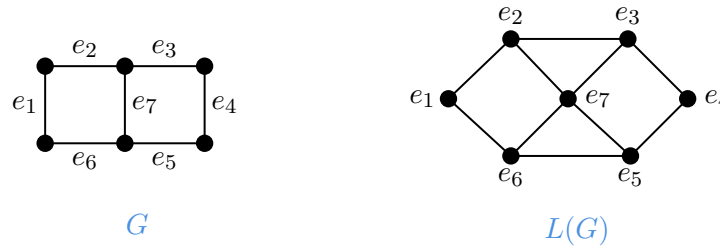
$u, v \in V$ with $u \neq v$ there exists k edge disjoint uv -paths $P_1, \dots, P_k$.

We are going to prove this by constructing a graph that we can then get the required result from by applying vertex Menger. The construction will be based on the idea of a *line graph*.

Definition 31.3.23 (Line Graph)

Given a graph $G = (V, E)$, we define $L(G)$ to be the **line graph** as follows. $V(L(G)) = E$, and for $e, f \in E$, we have $ef \in E(L(G))$ if $e \cap f \neq \emptyset$.

An example of a graph and its corresponding line graph is shown below.



We can now prove the edge version of Menger's theorem.

Proof (Menger's Theorem, Edge Version). We are given a graph $G = (V, E)$ and distinct vertices $u, v \in V$ such that every u, v separator has size $\geq k$.

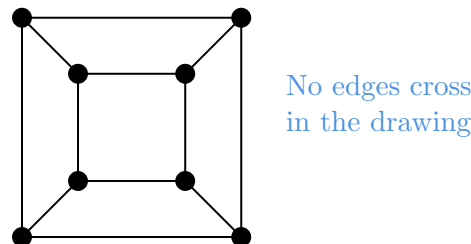
Construct the graph $\tilde{G}$ which is $L(G)$ along with vertices u and v with u joined to all $e \in V(L(G))$ such that $u \in e$, and likewise for v .

Then applying Menger's theorem (the vertex version, from one) to $\tilde{G}$ with u, v as the distinguished vertices. $\square$

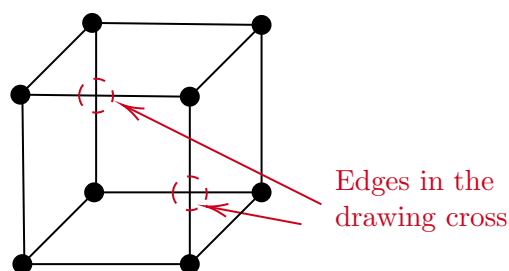
§31.4 Planar Graphs

Informally, a graph is **planar** if it can be drawn in the plane without any pair of edges crossing.

For example the cube graph we mentioned earlier is planar, as we can draw it as below.



Of course, a graph is planar only if *there is some drawing* where the edges don't intersect. For example, we could draw the cube graph as below (where edges intersect), and the graph would still be planar.



§31.4.1 Defining Planar Graphs

Let's formalize the notion of 'planar graphs' a bit. First we will define (a somewhat obvious notion) what we mean for a graph to be in a plane.

Definition 31.4.1 (Plane Graph)

A **plane graph** is a finite set of points $V \subseteq \mathbb{R}^2$ and a collection of disjoint polygonal curves (representing the edges) with start and endpoints in V .

Then we can define what it means for a graph to be *planar*.

Definition 31.4.2 (Planar Graph)

A graph G is **planar** if there exists a graph isomorphism from G to some plane graph.

And again, informally this says that a graph is planar if there is some way to draw it in the plane so that edges don't intersect.

We can also define the notion of 'faces' of a plane graph, by looking at the components.

Definition 31.4.3 (Faces)

The **faces** of a plane graph G are the connected components of $\mathbb{R}^2 - G$.

With this we get a nice relation between the vertices, edges and faces of a plane graph.

Theorem 31.4.4 (Euler's Formula)

Let G be a connected plane graph with V vertices, E edges and F faces. Then

$$V - E + F = 2.$$

Proof. We apply induction on the number of edges of G . The base case is $E = 0$, and then since G is connected, $V = 1$, and $F = 1$, and the formula holds. Now if G contains a cycle C , then let e be on C . Then $G' = G - e$ is still connected, and the number of faces increases by 1. The face enclosed is lost. So considering G' and applying Euler's formula, $V - (E - 1) + (F - 1) = 2$, and $V - E + F = 2$, as required.

If G does not contain a cycle, then G is acyclic and connected and is then a tree. Then $E = V - 1$, $F = 1$, and $V - E + F = 2$, as required. $\square$

Corollary 31.4.5 (Planar Graphs are Sparse)

Let $G = (V, E)$ be a planar graph, with $|V| \geq 3$. Then $|E| \leq 3|V| - 6$. This bound is also sharp^a.

^aBy considering plane graphs where every face is a triangle.

Proof. We may assume without loss of generality that G is connected (if not, then we can add edges until it is connected). Also draw G in the plane so that there is no edge crossings. Then by Euler's formula we have $V - E + F = 2$.

Now since every face has at least three edges on its boundary and every edge on the boundary is incident to at most two faces, we obtain

$$3F \leq |\{(e', f') \mid e' \in E, f' \text{ is a face, and } e' \text{ is on the boundary of } f'\}| \leq 2E.$$

Thus $3F \leq 2E$, so $3(2 - V + E) \leq 2E$, and $E \leq 3V - 6$, as required. $\square$

§31.4.2 Which Graphs are Planar?

In this section we really care about the following question:

Which graphs are planar?

The theorems proved at the end of the last section already give us a small amount of control over planar graphs. For example, we can show that K_5 is not planar.

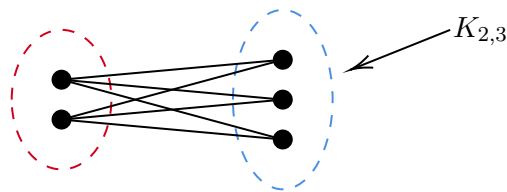
Example 31.4.6 (K_5 is Non-Planar)

We will show that K_5 is not planar. Since the number of edges is 10, and the number of vertices is 5, by the previous result we would need $10 \leq 3 \cdot 5 - 6$, which is not the case.

Let's consider another type of graph.

Definition 31.4.7 (Bipartite Complete Graph)

Define the **bipartite complete graph** $K_{n,m}$ to be the graph with vertex set $V = A \cup B$ where $A \cap B = \emptyset$ so that $|A| = n$, $|B| = m$, and $E(K_{n,m}) = \{ab \mid a \in A, b \in B\}$.



If you play around a bit, you can see that $K_{3,2}$ is planar, but we get a problem if we try use $K_{3,3}$.

Example 31.4.8 ($K_{3,3}$ is Non-Planar)

We will show that $K_{3,3}$ is not planar. Note that there is 9 edges, and 6 vertices. Then by the result in the previous section, we would need $9 \leq 3 \cdot 6 - 6$, which holds.

However, recall in the proof that of our result that we had each face having at least three edges on its boundary. But that for bipartite graphs, we can get something stronger, as each face has at least four edges on its boundary (since there is no cycles of length 3).

So if we repeat the proof of the previous theorem with the stronger bound, we obtain $E \leq 2V - 4$, which does not hold for this graph. Thus $K_{3,3}$ is not planar.

Now these two examples were interesting, but of course we care about whether *any* graph is planar. It turns out though that these are (in some sense) the *only* fundamentally non-planar graphs, in that any non planar graph will have one of these graphs ‘behind it’.

To look at this formally, we need to look at the idea of a subdivision.

Definition 31.4.9 (Subdivision)

A **subdivision** of a graph G is a graph $\tilde{G}$, obtained by replacing the edges of G with disjoint paths.

Lemma 31.4.10

If G is non-planar, then a subdivision of G is non-planar also.

Proof. Given a plane drawing of the subdivided graph, then by disregarding the vertices on the subdivided paths, we obtain a plane drawing of G (which is a contradiction). $\square$

Corollary 31.4.11

Subdivisions of $K_{3,3}$ and K_5 are non-planar.

Proof. Follows directly from the previous lemma. $\square$

What ties all of this together is Kuratowski’s theorem, which gives us a nice necessary and sufficient condition for a graph to be planar.

Theorem 31.4.12 (Kuratowski’s Theorem)

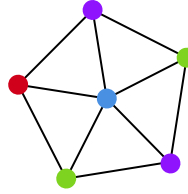
A graph G is planar if and only if G contains no subdivisions of $K_{3,3}$ or K_5 .

Proof. Omitted.

We will leave the topic of planar graphs here (for now), but we note that there are some other interesting notions that are related to planarity. For example, for what graphs is it possible to draw on a torus with no edge crossings?

§31.5 Graph Colouring

Informally, a graph colouring is just a way of colouring in different vertices of a graph, so that adjacent vertices are different colours. An example of a graph colouring is shown below.



§31.5.1 Basic Concepts

Of course, we need to define what this means in a slightly more mathematical sense, so we will define a colouring as follows.

Notation. We will write $[n] = \{1, 2, \dots, n\}$.

Definition 31.5.1 (r -Colouring)

Let $G = (V, E)$ be a graph. An r -colouring of G is a function $c : V \rightarrow [r]$ that satisfies $xy \in E \implies c(x) \neq c(y)$.

An r -colouring divides up a graph into r different colour classes, where there is only edges between the different colour classes.

Definition 31.5.2 (Chromatic Number)

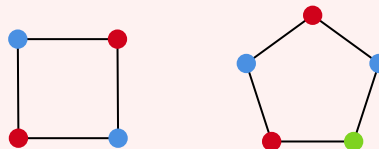
The **chromatic number** of a graph G , denoted $\chi(G)$ is the smallest r for which there exists an r -colouring.

Example 31.5.3 (Examples of Chromatic Numbers)

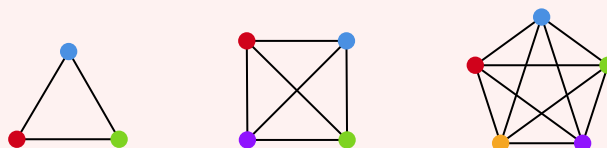
In the graph P_n , we have $\chi(P_n) = 2$ for $n \geq 2$, and $\chi(P_1) = 1$.



The graph C_n has $\chi(C_n) = 2$ if n is even, and $\chi(C_n) = 3$ if n is odd.



The complete graph K_n has $\chi(K_n) = n$.



The case for C_{2n} follows from a more general fact about bipartite graphs.

Proposition 31.5.4 (Chromatic Number of Bipartite Graphs)

If G is a bipartite graph, then $\chi(G) \leq 2$, and indeed $\chi(G) = 2$ unless $E = \emptyset$.

Proof Sketch. Colour the vertices in each part separately. □

Indeed, one way to think about chromatic number is as a generalization of bipartite graphs to multiple parts.

A straightforward observation to make is that the maximum degree of a graph (Δ) puts a bound on the chromatic number.

Proposition 31.5.5 (Degree Bound for Chromatic Number)

For a graph G , we have $\chi(G) \leq \Delta(G) + 1$. This bound is also sharp^a.

^aTake the complete graph or odd cycles. These are the only such examples though

To prove this, we are going to use a type of greedy algorithm.

Definition 31.5.6 (Greedy Colouring)

Given a graph $G = (V, E)$ with vertices $V = \{v_1, v_2, \dots, v_n\}$, the **greedy colouring** of G is a function $c_g : V \rightarrow \mathbb{N}$ defined inductively by

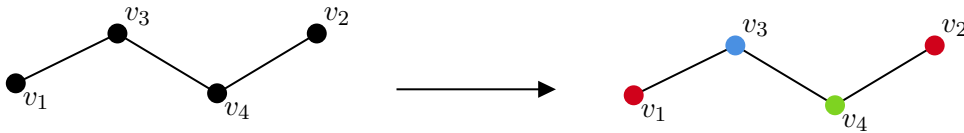
- $c_g(v_1) = 1$,
- Given coloured $v_1, \dots, v_t$, we have $c_g(v_{t+1}) = \min(\mathbb{N} \setminus \{c(v_i) \mid v_i \sim v_{t+1}, i \leq t\})$.

Proof of Proposition 31.5.5. Apply the greedy colouring to G with an arbitrary vertex ordering $v_1, \dots, v_n$. Then we note that

$$|\{c(v_i) \mid v_i \sim v_{t+1}, i \leq t\}| \leq \Delta(G),$$

and thus $c_g(v_{t+1}) \in [\Delta + 1]$. □

The ‘greedy’ approach need not give you any colouring that’s in any way optimal. For example, if we have the graph P_4 with vertices labelled as below, we get the following colouring from our greedy approach.



This is clearly not optimal.

§31.5.2 Brooks’ Theorem

One step up from the greedy approach to obtaining a colouring comes in the form of Brooks’ Theorem. Before we look at that, we will make an observation.

Proposition 31.5.7

Let G be a connected graph for which $\delta(G) < \Delta(G)$. Then $\chi(G) \leq \Delta(G)$.

Proof. We find a better ordering to apply the greedy colouring to. First define $v_n = v$, where $d(v) \leq \Delta(G) - 1$. Now choose an ordering of $v_1, \dots, v_{n-1}$ so that

$$d(v_1, v) \geq d(v_2, v) \geq \dots \geq d(v_{n-1}, v).$$

We claim that each vertex v_i with $i \in [n-1]$ has at most $\Delta(G) - 1$ neighbors in $v_1, \dots, v_{i-1}$.

This is true for v_n by definition. For v_i with $i \neq n$, we observe that a shortest path P from v_i to v_n contains a neighbor v_j of v_i with $d(v_j, v) < d(v_i, v)$. Thus v_j comes later in the ordering (that is, $j > i$).

So the greedy colouring gives each vertex one of $\Delta(G)$ colours. $\square$

We can now extend this idea to prove Brook's theorem.

Theorem 31.5.8 (Brooks' Theorem)

Let G be a connected graph that is not complete or an odd cycle. Then $\chi(G) \leq \Delta(G)$.

Proof. Let G be a counterexample with a minimal number of edges. We may assume G is a regular graph (Δ -regular), and also that $\Delta \geq 3$.

Since G is not complete, there exists a vertex v so that $G[N(v)]$ is not complete, so let $x, y \in N(v)$ where $x \not\sim y$. Colour these two vertices x, y the same colour.

If G is 3-connected then $G' = G - x - y$ is connected. In this case, we order the vertices of G' , labelled $v_3, \dots, v_n = v$ so that

$$d_{G'}(v_3, v) \geq d_{G'}(v_4, v) \geq \dots \geq d_{G'}(v_{n-1}, v).$$

The greedy colouring with the ordering $v_3, \dots, v_n$ will give us a Δ -colouring of G . Additionally, it's possible to ensure that each $w \in G'$ adjacent to either v_1 or v_2 is coloured differently from their common colour (check!). The key idea is that keeping G' connected allows the sequence of decreasing distances to be constructed.

If G has a cut vertex $w \in V$, then let $G - w = C_1 \cup \dots \cup C_k$ be the components, and $G_i = G[C_i \cup \{w\}]$. The maximum degree in G_i is bounded by Δ , and therefore G_i has a Δ -colouring for each i , by the minimality of the counter example. Note that G_i cannot be a complete graph on $\Delta + 1$ vertices since $d_{G_i}(w) \leq \Delta - 1$. By permuting the colours, we can assume each colouring gives the vertex w colour 1. Thus we have a colouring of G with Δ colours, which is a contradiction.

If $G - \{w_1, w_2\}$ is disconnected for $w_1 \neq w_2$ with $w_1, w_2 \in V$, then let $C_1, \dots, C_k$ be the components of $G - \{w_1, w_2\}$. We may assume that G has no cut vertex by the previous case. Now let $G_i = G[C_i \cup \{w_1, w_2\}] + w_1 w_2$. Note that $e(G_i) < e(G)$ since w_1, w_2 both send at least one edge to each component $C_1, \dots, C_k$. Also $d_{G_i}(w_1) \leq \Delta$ and likewise $d_{G_i}(w_2) \leq \Delta$, thus the maximum degree $G_i \leq \Delta$ and thus we can find a Δ -colouring of G_i for each i . That gives w_1, w_2 different colours. After permuting

these colours, I can assume w_1 is coloured 1 in each, and w_2 is coloured 2 in each. Thus we can put all of the colourings together to obtain a colouring of G . $\square$

§31.5.3 Colouring Planar Graphs

We will now consider colourings on graphs that are planar. It is in this section that the theorem ever-present in the popular maths psyche is kept (without proof).

Theorem 31.5.9 (Four-Colour Theorem)

Let G be a planar graph. Then $\chi(G) \leq 4$.

Or informally: any map can be coloured with only four colours, where neighboring regions have different colours. Note that this is the dual form of the theorem above. This theorem was proved in 1976 by Appel & Haken and centers around reducing the theorem to a large number of cases, that were checked by computer.

We are going to prove two slightly weaker versions, that are still quite interesting.

Theorem 31.5.10 (Six Colour Theorem – Warmup)

Let G be a planar graph. Then $\chi(G) \leq 6$.

Proof. We will use induction on $n = |G|$. For $n = 1$, this is trivial. Now inductively, we claim that there is a vertex v with $\deg(v) \leq 5$. We note

$$\frac{1}{n} \left[\sum_{x \in V} d(x) \right] = \frac{2E}{n} \leq 6 - \frac{12}{n} < 6.$$

Thus there is a vertex v with $\deg(v) \leq 5$. By induction, $G - v$ is 6-colourable, and since v has at most 5 neighbors, there is a colour in $[6]$ that does not appear in $N(v)$. Colouring v with this colour, we get that the graph is 6-colourable. $\square$

Now we are going to kick it up a notch shortly, by introducing one more ingredient.

Definition 31.5.11

Given a graph G and an r -colouring of G , let $v \in V(G)$, and define the **$\{i, j\}$ -component of v** to be all of the vertices that can be reached starting at v along a path using only colours i and j .

We make the following observation, which gives us an extra ‘move’ to use in the stronger proof

Proposition 31.5.12

Given a graph G with an r -colouring c , and for $i, j \in [r]$ with $i \neq j$, we can *swap* the colour on an $\{i, j\}$ -component to obtain a new colouring.

Proof Sketch. This works because of the ‘being reached’ condition in the $\{i, j\}$ -component definition. $\square$

Now we can prove the five colour theorem, using a lot of the ideas from the proof of the six colour theorem.

Theorem 31.5.13 (Five Colour Theorem)

Let G be a planar graph. Then $\chi(G) \leq 5$.

Proof. By induction on $n = |G|$, we note $n = 1$ is trivial. Now we proceed with the induction step. Let v be a vertex with $d(v) \leq 5$. Then apply induction to get a 5-colouring of $G - v$. Let $N(v) = \{x_1, \dots, x_5\}$, where $x_1, \dots, x_5$ are arranged in a clockwise manner. We may assume that $c(x_i) = i$ (otherwise we can colour v with the missing colour).

Now consider the $\{1, 3\}$ -component containing x_1 . If x_3 is not in this component, we can swap colours on this $\{1, 3\}$ component so that x_1 is colored with 3, and then colour v with 1.

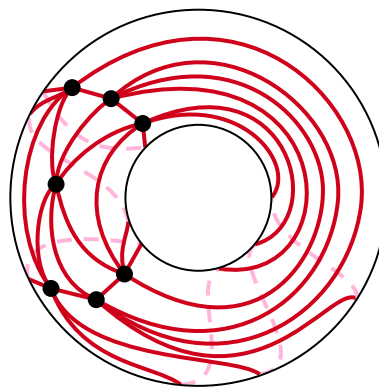
So we may assume there exists a path $x_1 \rightarrow x_3$ using colours 1 and 3 only. By the same argument there exists a path $x_2 \rightarrow x_4$ using colours 2 and 4 only. But then these paths must share a vertex, which is a contradiction. $\square$

§31.5.4 Colouring Graphs on (Other) Surfaces

Following on from the last section, we will consider a related guiding question:

If G is a graph, drawn on the torus, what can we say about $\chi(G)$?

More generally, if G is a graph drawn on a surface of genus g , what can we say about $\chi(G)$? For example, the graph K_7 can be drawn on a torus without edge crossings.



We may wonder if there is an Euler's formula for surfaces, and indeed there is, but instead of equality, we get a bound.

Theorem 31.5.14 (Euler's Formula for Surfaces)

If G is drawn in a surface of genus g , then $V - E + F \geq 2 - 2g$, where F is the number of connected components of $(\text{Surface} - G)$.

Proof Sketch. A similar inductive proof to the planar case. $\square$

Remark. By a surface of genus g , we mean a compact orientable surface of genus g . Informally this is the surface formed from taking a sphere and adding g ‘handles’ to it. Also note that $2 - 2g$ is the **Euler characteristic** of a surface of genus g .

We can then use this to get a bound on the edges of a graph drawn on a surface with no edge crossings.

Proposition 31.5.15

If $G = (V, E)$ is a graph drawn on a surface of genus g , then $|E| \leq 3(|G| - (2 - 2g))$.

Proof Sketch. We have $3F \leq 2E$, then apply Euler’s formula for surfaces. $\square$

We can now get a bound (similarly to the planar case) on the chromatic number of a graph drawn on a surface.

Theorem 31.5.16 (Heawood’s Theorem)

If G is a graph drawn on a surface of Euler characteristic E , with $E \leq 0^a$, then

$$\chi(G) \leq \left\lfloor \frac{7 + \sqrt{49 - 24E}}{2} \right\rfloor.$$

^aThis condition really is needed, as otherwise for the planar case we get $\chi(G) \leq 4$, which is correct but this is not a proof for it!

Proof. Let $G = (V, E)$ be a given graph with $\chi(G) = k$. We may assume that G has the minimum number of edges, subject to $\chi(G) = k$. Observe that $\delta(G) \geq k - 1$, as otherwise there would be a vertex v with $d(v) = k - 1$, and thus $\chi(G - v) = k - 1$, and thus $\chi(G) = k - 1$. Also $k \leq n$, where $n = |V|$. Now the average degree of each vertex is

$$\frac{1}{n} \left[\sum_{v \in V} d(v) \right] = \frac{2e}{n} \leq 6 \left(1 - \frac{E}{n} \right),$$

where e is the number of edges and E is the Euler characteristic.

Thus

$$k - 1 \leq \delta(G) \leq \text{avg degree} \leq 6 \left(1 - \frac{E}{n} \right) \leq 6 \left(1 - \frac{E}{k} \right),$$

and so $k^2 - k \leq 6(k - E)$, then solving gives the required result. $\square$

Remark. It turns out that this estimate is sharp. Calling $H(E) = \left\lfloor \frac{7 + \sqrt{49 - 24E}}{2} \right\rfloor$, we find that $K_{H(E)}$ can be drawn on a surface of Euler characteristic E .

§31.5.5 Edge Colouring

Now instead of assigning vertices colours, we will assign them to edges.

Definition 31.5.17 (Edge Colouring)

Let $G = (V, E)$ be a graph. A **k -edge colouring** is a function $c : E \rightarrow [k]$ so that $c(e) \neq c(f)$ for all $e, f \in E$ such that $e \cap f \neq \emptyset$.

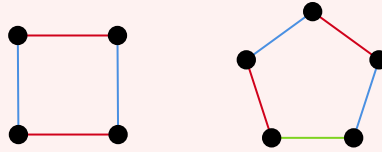
We have a similar notion to chromatic number for edge colourings.

Definition 31.5.18 (Edge Chromatic Number/Chromatic Index)

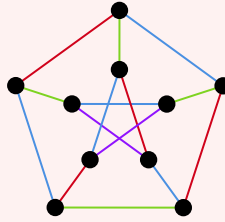
The **edge chromatic number** or **chromatic index** of a graph G is $\chi'(G)$, the minimum k for which a k -edge colouring exists.

Example 31.5.19 (Examples of Edge Chromatic Number)

The graph C_n has $\chi'(C_n) = 2$ if n is even, and $\chi'(C_n) = 3$ for n odd.



The Petersen graph G has $\chi'(G) = 4$.



In contrast to the chromatic number, it is much easier to get a handle on the edge chromatic number, as we will see in the following theorem.

Theorem 31.5.20 (Vizing's Theorem)

Let G be a graph. Then $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$.

Proof. We proceed by induction on $e(G)$. The basis step is trivial, then for the inductive step we are given a graph G , and let's assume for a contradiction that $\chi'(G) > \Delta + 1$.

By induction, $G - e$ has a $\Delta + 1$ colouring, so let's write $e = xy$ with $x \neq y \in G$. We define vertices $y_1, \dots, y_k$ inductively by setting $y_1 = y$. Now assume $y_1, \dots, y_t$ are defined, and the colours missing from $y_1, \dots, y_t$ are $c_1, \dots, c_t$ respectively. Then if $c_t \notin \{c_1, \dots, c_{t-1}\}$, then let y_{t+1} be so that xy_{t+1} receives colour c_t . Note that such a vertex exists as otherwise we could recolour xy_1 with c_1 , xy_2 with c_2 , and so on until xy_t is with c_t , to obtain a $\Delta + 1$ colouring, which is a contradiction.

Now if $c_t \in \{c_1, \dots, c_{t-1}\}$, then stop. Say we stop after k steps, so I have defined $y_1, \dots, y_k$ with missing colours $c_1, \dots, c_k$ and $c_k = c_i$ for some $i < k$. We may assume that $i = 1$ (otherwise uncolour the edge xy_i and recolour xy_1 with c_1 , and so on until xy_{i-1} with c_{i-1}).

Let's call the colour missing at x , c_0 . Consider the $\{c_0, c_1\}$ component C , containing y_1 . If $x \notin C$, then we can flip colours on C so that c_0 is missing at y_1 , then colour xy_1 with c_0 . Likewise, the $\{c_0, c_1\}$ -component containing y_k must contain x , as otherwise we flip colours on this component so that the colour c_0 is missing at y_k . Then recolour xy_k to c_0 and xy_i to c_i for $i < k$.

Thus $x, y_1, y_k \in C$, the $\{c_0, c_1\}$ -component. But this is impossible since x, y_1, y_k all have one of the colours $\{c_0, c_1\}$ missing, thus $d_C(x), d_C(y_1), d_C(y_k) \leq 1$. But this is impossible for a path or cycle. $\square$

Remark. This theorem is *not* true if we generalize to multigraphs, which are graphs that have multiple edges between vertices.

§31.6 Extremal Graph Theory

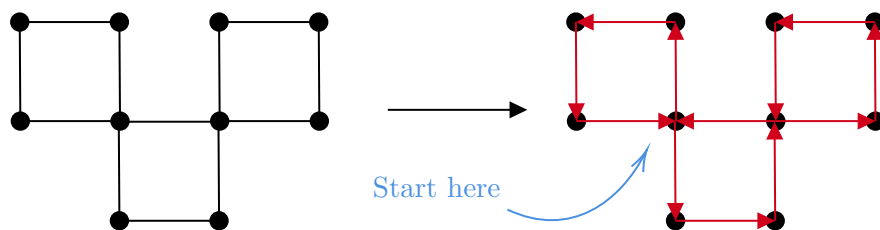
Welcome to the section on extremal graph theory, where we think about problems involving things with an 'extreme' flavour.

§31.6.1 Eulerian Circuits and Hamiltonian Cycles

We are going to begin our chapter on extremal graph theory with something that's not *really* extremal graph theory. Still, it will lead us nicely into the more extreme parts of this chapter and the course. Let's state a definition.

Definition 31.6.1 (Eulerian Circuit)

An **Eulerian circuit** is a circuit in a graph G that crosses each edge exactly once. If a G has an Eulerian circuit, we say that the graph is **Eulerian**.



What's nice is that there is a straightforward characterisation of Eulerian graphs.

Theorem 31.6.2 (Euler's Theorem)

A connected graph has an Eulerian circuit if and only if every vertex has even degree.

Proof. If a graph G has an Eulerian circuit, then the degree of each vertex must be even. This is because a fixed vertex x is entered and exited a fixed number of times.

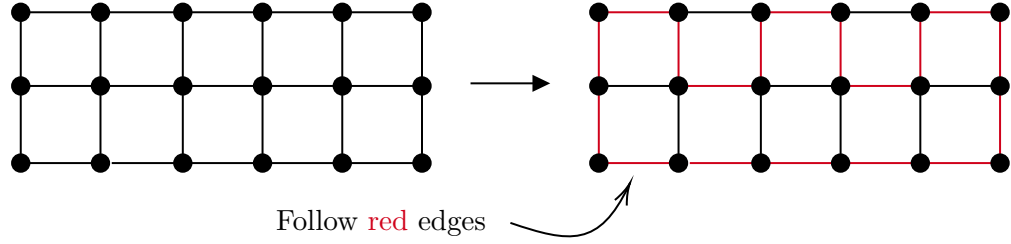
Now if each vertex of a graph G has even degree, we will apply induction on $e(G)$. If $e(G) = 0$, then we are done. Now if $d(x) \geq 1$ for all vertices x , then $d(x) \geq 2$, and G contains a cycle C . Define $G' = G - E(C)$. Let $G_1, \dots, G_k$ be the components of G' . The degree of each G_i has all degrees even. Thus by induction, there is an Eulerian circuit $W_1, \dots, W_k$ for each $G_1, \dots, G_k$ respectively. Thus we can combine

C with $W_1, \dots, W_k$ to obtain an Eulerian circuit for all of G . $\square$

Now we are going to define a similar looking notion, but it will not end up being so well behaved as Eulerian circuits.

Definition 31.6.3 (Hamiltonian Cycle)

Let G be a graph, then a **Hamiltonian cycle** in G is a cycle that visits each vertex exactly once. We say G is **Hamiltonian** if it contains a Hamiltonian cycle.



Our first question might be ‘is there an iff type condition for Hamiltonian cycles’, and so far there is no such property known. However there are some results that give us information about whether a graph is Hamiltonian.

Theorem 31.6.4 (Dirac’s Theorem)

Let G be a graph with $n \geq 3$ such that $\delta(G) \geq n/2$. Then G contains a Hamiltonian cycle.

Proof. For a contradiction, assume that G is a counterexample to the theorem. Note that G is connected, since $x \not\sim y$ then $N(x), N(y) \subset V \setminus \{x, y\}$ and $|N(x)|, |N(y)| \geq \frac{n}{2}$, and thus $N(x) \cap N(y) \neq \emptyset$, and thus $d(x, y) \leq 2$.

Let $x_1, \dots, x_\ell$ be the longest path in G . Note that $x_1, \dots, x_\ell$ does not form a cycle, as otherwise if $\ell = n$, then we have a contradiction, and if $\ell < n$ then there exists $y \notin \{x_1, \dots, x_\ell\}$ that is $y \sim x_i$. Thus we can find a longer path (which is also a contradiction).

Now if there exists $i \in \{1, \dots, \ell - 1\}$ so that $x_i \sim x_\ell$ and $x_{i+1} \sim x_1$, then the vertices $x_1, \dots, x_\ell$ form a cycle. This contradicts the above.

Define

$$N^+(x_\ell) = \{x_i \mid x_{i-1} \in N(x_\ell), 2 \leq i \leq \ell\}.$$

We have $N^+(x_\ell) \cap N(x_1) = \emptyset$, but this is impossible since $|N^+(x_\ell)| \geq n/2$, $|N(x_1)| \geq n/2$, and $N^+(x_\ell), N(x_1) \subseteq \{x_2, \dots, x_\ell\}$. Thus we have a contradiction. $\square$

Remark. We never *really* used this $\delta(G) \geq n/2$ condition fully. It suffices to have $d(x) + d(y) \geq n$ for $x \not\sim y$.

We can use this same argument to prove a more general result about paths.

Proposition 31.6.5

Let G be a connected graph. Let $k < n$ and assume $\delta(G) \geq k/2$. Then $G \geq P_{k+1}$.

Proof Sketch. Similar to Dirac's theorem. Choose a longest path $x_1, \dots, x_\ell$ in G . They don't form a cycle, and then we can force a configuration like $x_1 \sum x_{i+1}$, $x_\ell \sum x_i$ by using the minimum degree condition. $\square$

We may wonder in the above if it's possible to replace 'path' with 'cycle' in the above. The answer is, sadly, no.

A natural question is to wonder how many edges are needed to 'force' a (for example) triangle. For a Hamilton cycle, this sort of question would have been a bit strange, and a more natural question would be what is the minimum value of $\delta(G)$ to ensure that G contains a Hamilton cycle. In the next theorem, we will give a result that answers a simpler sort of question.

Theorem 31.6.6

Let G be a graph. If $e(G) > \frac{n}{2}(k-1)$, then G contains a path of length k .

Proof. We will prove the contrapositive: If G is P_{k+1} free, then $e(G) \leq \frac{n}{2}(k-1)$.

We will apply induction on n . This is true for $n = 2$. Then given a graph G on $|G| = n \geq 3$ vertices, if G is disconnected let $G_1, \dots, G_k$ be the components of G . Then each of these has $G_i \not\supseteq P_{k+1}$. Then by induction we have $e(G_i) \leq \frac{n(G_i)(k-1)}{2}$. Thus

$$e(G) = \sum_{i=1}^k e(G_i) \leq \sum \frac{n(G_i)(k-1)}{2} = \left(\frac{k-1}{2}\right)n,$$

and we are done.

Now if there is a vertex v with $d(v) \leq \frac{k-1}{2}$, then consider $G - v$. Then $e(G - v) = e(G) - d(v)$, and also $G - v$ does not contain P_{k+1} . Thus $e(G - v) \leq \frac{n-1}{2}(k-1)$. So

$$e(G) = e(G - v) + d(v) \leq \frac{n-1}{2}(k-1) + \frac{k-1}{2} \leq \frac{n}{2}(k-1).$$

Thus we can assume G is connected and $d(v) \geq \frac{k}{2}$. We can apply [Proposition 31.6.5](#) to find a path of length k , that is, $P_{k+1} \subseteq G$, which is a contradiction.

[We may also assume $k < n$ for if $k = n$ then $e(G) \leq \frac{k}{2}(n-1)$, which reduces to showing $e(G) \leq \binom{n}{2}$, which is trivial]. $\square$

§31.6.2 Complete Subgraphs and Turan's Theorem

Let's return to the question of how many edges are needed in a graph to guarantee the existence of a triangle. Later on, we will look more generally at the question of the number of edges needed to guarantee a K_n subgraph.

Theorem 31.6.7 (Mantel's Theorem)

If $e(G) > \frac{n^2}{4}$, then $G \supset K_3$, and this is sharp.

Proof. Given a K_3 -free graph G and $x, y \in V$ such that $x \sim y$ and $d(x) + d(y) \leq n$,

and let $m = e(G)$. Then summing we get

$$\sum_{xy \in E} d(x) + d(y) \leq mn.$$

We have that this is $\sum_x \sum_y d(x) \mathbb{1}(xy \in E) = \sum_{x \in V} (d(x))^2$. Now $\sum_{x \in V} d(x) = 2m$, and by Cauchy-Schwarz we have $(\sum d(x))^2 \leq n \sum d(x)^2$, and thus

$$mn^2 \geq (2m)^2 \implies \frac{n^2}{4} = e(G).$$

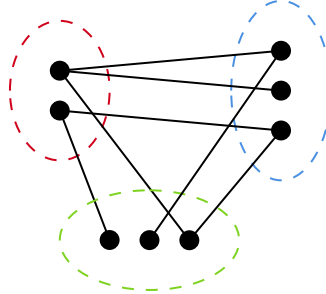
To see that this is sharp, consider the complete bipartite graph with n vertices. Then this has $n^2/4$ edges, but no triangle. Thus this is sharp. $\square$

The next question we are going to answer is how many edges are needed to guarantee the existence of a K_n subgraph. In the previous theorem, to see that our result was sharp, we considered the complete bipartite graph on n vertices. To discuss this more general question, a natural thing to look at is a generalisation of bipartite graphs.

Definition 31.6.8 (r -Partite Graph)

An **r -partite graph** is a graph of the form $G = (V, E)$ where the vertices are partitioned into r subsets $V = V_1 \cup \dots \cup V_r$ with $V_i \cap V_j = \emptyset$ if $i \neq j$, and $xy \notin E$ for all $x, y \in V_i$.

For example, the graph below is 3-partite.



It's worth noting that saying a graph is r -partite is the same as saying the graph has chromatic number of at most r . We care particularly about the case of an r -partite graph where all possible edges are included.

Definition 31.6.9 (Complete r -Partite Graph)

We say an r -partite graph is **complete** if no edge can be added with the graph remaining r -partite.

Now if we consider a complete r -partite graph G on n vertices, with all parts of size n/r , then we would have

$$e(G) = \binom{n}{r} \binom{r}{2} = \left(1 - \frac{1}{r}\right) \frac{n^2}{2}.$$

The graph G can also have no K_{r+1} subgraph, since that would imply there's an edge between two vertices in the same part. This gives us a lower bound on the number of edges needed to have a K_{r+1} subgraph, and it turns out that bound is sharp.

Theorem 31.6.10 (Turan's Theorem)

If $e(G) > \left(1 - \frac{1}{r}\right) \frac{n^2}{2}$, then $G \supset K_{r+1}$, and this is sharp.

Proof. Suppose we have some graph G on n vertices that had no K_{r+1} subgraph, and also that the result holds up to n, r . If $r \geq n$ then we are clearly done, so we may suppose that $n > r$. Let A be a K_r subgraph in G , which must exist by our assumption. Let $B = V - A$. Then we have $e(G) = e(A) + e(B) + e(A, B)$, where $e(A, B)$ denotes the number of edges between vertices in A and B . We have an upper bound of

$$e(G) \leq \binom{r}{2} + \left(1 - \frac{1}{r}\right) \frac{(n-r)^2}{2} + (n-r)(r-1),$$

where $e(A, B) \leq (n-r)(r-1)$ since each vertex in B has at most $r-1$ neighbours in A . We can rewrite this as

$$e(G) \leq \frac{1}{2} \left(1 - \frac{1}{r}\right) (r^2 + (n-r)^2 + (n-r)r) = \frac{1}{2} \left(1 - \frac{1}{r}\right) n^2,$$

as required. □

There's another non-examinable proof called Zykov Symmetrisation

§31.6.3 The Zarankiewicz Problem

Turan's theorem answers the question 'how many edges can a graph have without having a K_t subgraph'. A natural follow on question is 'how many edges can a bipartite graph have without having a $K_{t,t}$ subgraph'. This is the Zarankiewicz problem, and unfortunately does not yet have a definite answer like the previous question. We can however obtain some bounds.

Definition 31.6.11

We define $Z(n, t)$ to be the maximum number of edges in a bipartite graph with n vertices in each part and no $K_{t,t}$ subgraph.

We really care about $Z(n, t)$ where t is fixed and n is large. We will prove the following theorem.

Theorem 31.6.12

We have $Z(n, t) \leq t^{1/t} n^{2-1/t} + tn$ for all n .

Proof. Given a graph $G = (A \cup B, E)$ where $|A| = |B| = n$, and $G \not\supset K_{t,t}$, we want to show that $m = e(G) \leq t^{1/t} n^{2-1/t} + tn$.

For some set $S \subseteq A$, and $|S| = t$, we have

$$\left| \bigcap_{x \in S} N(x) \right| \leq t - 1,$$

as otherwise we would have a $K_{t,t}$ subgraph. Then averaging over such subsets S , we have

$$\binom{n}{t}^{-1} \sum_{\substack{S \subseteq A, \\ |S|=t}} \left| \bigcap_{x \in S} N(x) \right| \leq t - 1.$$

This sum can be written as

$$\begin{aligned} \sum_{\substack{S \subseteq A, \\ |S|=t}} \left| \bigcap_{x \in S} N(x) \right| &= \sum_{\substack{S=\{x_1, \dots, x_t\}, \\ S \subseteq A}} \sum_y \mathbb{1}(y \sim x_1, y \sim x_2, \dots, y \sim x_t) \\ &= \sum_y \sum_{\substack{S=\{x_1, \dots, x_t\}, \\ S \subseteq A}} \mathbb{1}(y \sim x_1, y \sim x_2, \dots, y \sim x_t) \\ &= \sum_y \binom{d(y)}{t}. \end{aligned}$$

We may assume that $d(y) \geq t - 1$ for all y , as otherwise we can add an edge incident with y and not create a $K_{t,t}$ subgraph. Then by convexity and since $d(y) \geq t - 1$, we have

$$\sum_y \binom{d(y)}{t} \geq \sum_{y \in B} \binom{d}{t} = n \binom{d}{t},$$

where $d = \frac{1}{n} \sum_{y \in B} d(y) = m/n$. Combining this inequality with what we obtained previously, we get the bound

$$t - 1 \geq \frac{n \binom{d}{t}}{\binom{n}{t}} = n \frac{d(d-1) \cdots (d-t+1)}{(n-1) \cdots (n-t+1)} \geq \frac{(d-t+1)^t}{n^{t-1}},$$

thus $(t-1)^{1/t} n^{t-1} \geq d-t+1$, and $t^{1/t} n^{2-1/t} + tn \geq m$, as required. $\square$

While we do have this upper bound, we don't know $Z(n, t)$ for most values of n . We do know, for example, that

$$\begin{aligned} cn^{3/2} &\leq Z(n, 2) \leq 2n^{3/2}, \\ c'n^{5/2} &\leq Z(n, 3) \leq 2n^{5/3}, \end{aligned}$$

for all large n , and some constants c, c' . Still we don't even know $t = 4$. The constructions are based on finite geometry. As an example, we will sketch the construction that shows the bound $cn^{3/2} \leq Z(n, 2)$.

Theorem 31.6.13

For infinitely many n we have $Z(n, 2) \geq cn^{3/2}$ for $c > 0$.

Proof Sketch. Let p be a prime, and consider^a $(\mathbb{Z}/p\mathbb{Z})^2$. Define a line $L = \{(x, ax + b) \mid x \in \mathbb{Z}/p\mathbb{Z}\}$, for some $a, b \in \mathbb{Z}/p\mathbb{Z}$ with $a \neq 0$.

We need the following straightforward facts:

1. Two distinct lines intersect in at most one point.
2. Each line contains p points.
3. There are p^2 points, $p(p-1) \approx p^2$ lines^b.

We are going to construct a graph bipartite graph $G = (A \cup B, E)$ where every vertex in A corresponds to a line and every vertex in B corresponds to a point. There's approximately p^2 vertices in each of these parts, so we will take n to be about p^2 . Then for $\ell \in A$ and $x \in B$, we define $\ell \sim x$ if $x \in \ell$. Then $e(G) \approx p^2 \cdot p = p^3 = n^{3/2}$. To see this graph works, if G contained a $K_{2,2}$ then there would exist two lines ℓ_1, ℓ_2 with $|\ell_1 \cap \ell_2| \geq 2$, which is a contradiction. $\square$

^aYou can consider this to be like a torus.

^bTo make this a proper proof, we would have to fix all of the approximations

§31.6.4 The Erdős-Stone Theorem

So far we have been interested in questions of the form ‘how many edges do I need in a graph before I force a subgraph H ’. We’ve been able to prove a variety of interesting but relatively scattered results of this form, but it turns out that if we consider the question asymptotically, we can obtain a more general unifying result.

We begin by defining the following function.

Definition 31.6.14 (Extremal Function)

Let H be a fixed graph. We define the **extremal function** $\text{ex}(n, H)$ to be

$$\text{ex}(n, H) = \max\{e(G) \mid |G| = n, G \not\supseteq H\}.$$

Using this function we can restate some of the results we previously obtained, such as Mantel’s theorem giving us $\text{ex}(n, K_3) \leq n^2/4$, and Turan’s theorem giving us $\text{ex}(n, K_{r+1}) \leq (1 - 1/r)n^2/2$.

Looking a bit more at Turan’s theorem, if we consider things asymptotically on the scale of n^2 , we get a weaker version of Turan that looks like

$$\lim_{n \rightarrow \infty} \frac{\text{ex}(n, K_{r+1})}{\binom{n}{2}} = \left(1 - \frac{1}{r}\right).$$

The Erdős-Stone Theorem allows us to get these types of results generally, using just the chromatic number of the graph.

Theorem 31.6.15 (Erdős-Stone Theorem)

Let H be a graph with $\chi(G) = r$, and $r \geq 2$. Then

$$\lim_{n \rightarrow \infty} \frac{\text{ex}(n, H)}{\binom{n}{2}} = \left(1 - \frac{1}{r-1}\right).$$

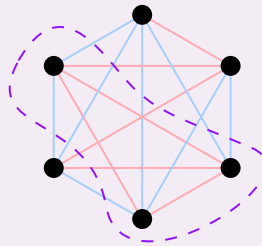
Proof. Omitted

§31.7 Ramsey Theory

We are going to jump into Ramsey theory, an area of graph theory that has a similar flavour to previous section. Ramsey theory is often described as the collection of results showing ‘complete disorder is impossible’, but its easiest to see what we study with an example.

Proposition 31.7.1 (Monochromatic Triangles)

Colour each of the edges of K_6 red or blue^a. Then there must be a monochromatic triangle.



^aNote that we don't have any restrictions on the colouring.

Proof. Consider some vertex v in the graph. We know that at least three incident edges must be the same colour, so suppose (without loss of generality) that they are red, with endpoints v_1, v_2, v_3 . Then either $v_1v_2v_3$ is a monochromatic blue triangle, or some v_iv_j is red, in which case vv_iv_j is a monochromatic red triangle. $\square$

Straight away we might wonder about K_4 . Does there exist some large enough n such that every 2-colouring of K_n contains a monochromatic K_4 ? This is the type of problem we will consider in Ramsey theory.

§31.7.1 Ramsey Numbers

We will begin our discussion of Ramsey theory by introducing the *Ramsey numbers*.

Definition 31.7.2 (Ramsey Numbers)

For $t \in \mathbb{N}$, we define $R(t)$ the **t th Ramsey number** to be the smallest n for which every 2-colouring of K_n contains a monochromatic K_t .

We already know that $R(3) = 6$ (since we showed there's always a monochromatic

triangle in K_6 , and it's not hard to find a colouring of K_5 that doesn't have one). We haven't yet proved though that such an $R(t)$ exists though, so we will clear this up first.

Theorem 31.7.3 (Ramsey's Theorem)

For all $t \in \mathbb{N}$, $R(t)$ is finite and $R(t) \leq 4^t$.

Proof. We will begin by proving a small lemma. For $s, t \geq 2$, we define the Ramsey number $R(s, t)$ to be the smallest n such that every red/blue colouring of K_n either contains a red K_s or a blue K_t .

We will prove that for all $s, t \geq 2$,

$$R(s, t) \leq \binom{s+t-2}{s-1}. \quad (\dagger)$$

Claim. Assume that $R(s-1, t)$ and $R(s, t-1)$ exist. Then $R(s, t)$ exists and $R(s, t) \leq R(s-1, t) + R(s, t-1)$.

Let $a = R(s-1, t)$ and $b = R(s, t-1)$ and let c be a red/blue colouring of K_{a+b} . Let x be a vertex in this graph, and we note $d(x) = a + b - 1$. Then x has either

- (i) a red neighbors
- (ii) b blue neighbors.

In case (i), let N_r be the red neighbors of x , and note $|N_r| \geq a$. Then the colouring induced on N_r contains either a K_{s-1} in red or a K_t in blue. In the latter case we are done, and in the former case we can add x to K_{s-1} to finish. Case (ii) is symmetric, and thus our claim is true.

Now we can return to showing $(\dagger)$. We are going to induct on $s+t$. Note that $R(s, 2) = s$ and $R(2, t) = t$. Inductively assume that $R(s-1, t), R(s, t-1)$ exist and satisfy this relation, then

$$R(s, t) \leq R(s-1, t) + R(s, t-1) \leq \binom{s+t-2}{s-2} + \binom{s+t-3}{s-1} \leq \binom{s+t-2}{s-1},$$

as desired. Taking $R(t) = R(t, t)$ gives $R(t) \leq \binom{2t-2}{t-1} \leq 4^t$. $\square$

Ramsey numbers are generally quite mysterious. We know for example that $R(3, 3) = 6$, $R(4, 4) = 18$, and $R(3, 7) = 23$, but even something like $R(5, 5)$ is unknown, and our best bounds are $43 \leq R(5, 5) \leq 48$. The numbers $R(3, t)$ are quite well understood, and for $\epsilon > 0$ we have (for sufficiently large t)

$$\left(\frac{1}{4} + \epsilon\right) \frac{t^2}{\log t} \leq R(3, t) \leq (1 + \epsilon) \frac{t^2}{\log t}.$$

A big question though is getting a lower bound on $R(t)$, which is currently nowhere as close as our upper bound.

§31.7.2 Infinite Ramsey

Now instead of finite complete graphs, we are going to consider an infinite graph.

Definition 31.7.4 (Complete Countable Graph)

We define the **complete countable graph** $G = (\mathbb{N}, \mathbb{N}^{(2)})$.

Notation. For a set X , we let $X^{(r)} = \{A \subseteq X \mid |A| = r\}$, the set of r element subsets of X .

Our motivating question will be the following: given a 2-colouring of the complete countable graph, what can we say about the monochromatic structures that appear in the colouring? Obviously by Ramsey's theorem we can find arbitrarily large monochromatic complete graphs, but is it possible to find a monochromatic complete countable subgraph?

Example 31.7.5 (Complete Countable Graph with Complete Countable Subgraph)

Consider the colouring of a complete countable graph where xy is red if $x + y$ is even, and blue if $x + y$ is odd. Then the subgraph induced on the vertices $\{2, 4, 8, \dots\}$ is a monochromatic complete countable subgraph.

Now consider instead the colouring where $x + y$ is red if it has an even number of prime factors, and $x + y$ is blue if it has an odd number of prime factors. Can you find a monochromatic complete countable subgraph?

It turns out that by another theorem of Ramsey, it is always possible to find such a subgraph.

Theorem 31.7.6 (Infinite Ramsey's Theorem)

Let $G = (\mathbb{N}, \mathbb{N}^{(2)})$, the complete countable graph. Then for every 2-colouring of G there exists an infinite set $X \subseteq \mathbb{N}$ so that $X^{(2)}$ is monochromatic.

Proof. We begin by inductively defining vertices $x_1, x_2, \dots, x_t$ and colours $c_1, c_2, \dots, c_t$ such that $N_{c_i}(x_i) \supset \{x_{i+1}, \dots, x_t\}$, and $\bigcap_{i=1}^t N_{c_i}(x_i)$ is infinite, where $N_{c_i}(x_i)$ is the neighborhood of x_i in the colour c_i .

We can do this by taking $x_1, \dots, x_t$ and then choosing any $x_{t+1} \in \bigcap_{i=1}^t N_{c_i}(x_i)$. Then we can define c_i accordingly, since at least one of $N_{\text{red}}(x_{t+1}) \cap \left(\bigcap_{i=1}^t N_{c_i}(x_i)\right)$ is infinite, or $N_{\text{blue}}(x_{t+1}) \cap \left(\bigcap_{i=1}^t N_{c_i}(x_i)\right)$ is infinite.

This then gives us an infinite collection $x_1, x_2, \dots$ so that $x_i x_j$ with $i < j$ are given colour c_i . Then since both

$$\{x_i \mid c_i = \text{red}\} \quad \text{and} \quad \{x_i \mid c_i = \text{blue}\}$$

are monochromatic cliques, at least one of them is infinite, giving a monochromatic complete countable subgraph. $\square$

Remark. By keeping track of the sizes of sets at each step, we can adapt this proof to give us a different proof of the finite case.

§31.7.3 Ramsey Numbers for Hypergraphs

Before, we were colouring $X^{(2)}$, the pairs of a set X (the vertices of a graph). Of course, we might wonder what happens when we colour r -sets?

Definition 31.7.7 (Hypergraph Ramsey Numbers)

We define $R^{(r)}(s, t)$ to be the smallest n such that every red/blue colouring of $[n]^{(r)}$ either contains a set $X \subseteq [n]$ with $|X| = s$ and $X^{(r)}$ red, or there exists $Y \subseteq [n]$ with $|Y| = t$ and $Y^{(r)}$ blue.

And of course, it's not guaranteed that these exist, but it is possible to show that they do.

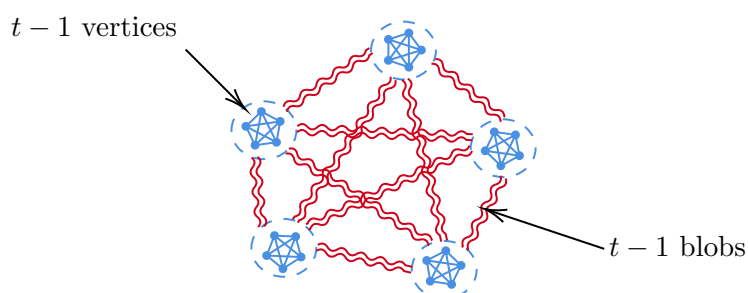
Theorem 31.7.8 (Hypergraph Ramsey Theorem)

For all $r, s, t \geq 2$, $R^{(r)}(s, t)$ exists.

Proof. Omitted.

§31.7.4 Lower Bounds for Ramsey Numbers

We said at the end of [subsection 31.7.1](#) that we were left to get a good lower bound on the Ramsey numbers $R(t)$. One way to do that might be by considering the following extremal-seeming construction:



This clearly doesn't have a monochromatic K_t subgraph, and we can compute the number of vertices is $(t-1)^2$, giving

$$R(t) \geq (t-1)^2.$$

However, it turns out that this quadratic bound is, quite frankly, awful. To show something better, we are going to need a new idea – considering random graphs and using probability.

Theorem 31.7.9 (Erdős)

$$R(t) \geq 2^{\frac{t}{2}}.$$

Proof. We will show that for a random colouring of the edges of K_n ,

$$\mathbb{P}(\text{there exists a monochromatic } K_t \subset K_n) \leq 2 \binom{n}{t} 2^{-\binom{t}{2}}.$$

Once we have shown this, we will then note for $n = 2^{t/2}$ that the RHS of this inequality is strictly less than 1, which implies that is *some* colouring of K_n with no monochromatic K_t .

To prove this inequality, we first fix some $A \subseteq K_n$ with $|A| = t$, and note that

$$\mathbb{P}(A^{(2)} \text{ is monochromatic}) = 2 \cdot 2^{-\binom{t}{2}}.$$

Applying the union bound then gives us our desired result

$$\mathbb{P}(\text{there exists a monochromatic } K_t \subset K_n) \leq \sum_{A \in K_n^{(t)}} 2 \cdot 2^{-\binom{t}{2}} = 2 \binom{n}{t} 2^{-\binom{t}{2}}.$$

Then to show that this is less than 1 for $n = 2^{t/2}$, we can take

$$2 \binom{n}{t} 2^{-\binom{t}{2}} \leq 2 \frac{n^t}{t!} 2^{-\frac{t(t-1)}{2}} = \left(\frac{2^{1/t}}{t^{1/t}} \cdot n 2^{-\frac{t-1}{2}} \right)^t,$$

and thus it is indeed less than 1 for $n = 2^{t/2}$. □

The proof idea we employed is known as the *probabilistic method*, and is an immensely powerful technique that can be used to show all kinds of interesting results.

Getting much better bounds on $R(t)$ is still a major open problem, and surprisingly the bounds we have proved so far aren't far from state of the art – the base of the exponential term is still the same in best known results. Also, there's currently no known construction that gives an exponential lower bound directly.

§31.8 The Probabilistic Method

At the end of our discussion on Ramsey numbers, we used an argument that combines probability and random graphs to prove a result. In this section we are going to look a bit more closely at the type of argument that we used.

§31.8.1 Random Graphs

We first are going to come up with a probability space involving a graph where each edge is included independently at random with probability p .

Definition 31.8.1 (Binomial Random Graph)

Let $n \in \mathbb{N}$ and $p \in [0, 1]$, let $\mathcal{G}(n, p)$ be the probability space on all graphs with vertex set $\{1, 2, \dots, n\}$, with $\mathbb{P}(ij \in E(\mathcal{G})) = p$, and every edge included/excluded independently.

Remark. For our ' $R(t) \geq 2^{t/2}$ ' result, we were working with $G(n, 1/2)$, which is the same as sampling uniformly on a graph on $\{1, \dots, n\}$.

Another use of random graphs is in the Zarankiewicz problem. Recall our definition of $Z(n, t)$. We proved already that $Z(n, t) \leq 2n^{2-1/t}$, and we hadn't yet proved a lower bound. We are able to come up with one using the probabilistic method.

Theorem 31.8.2

We have $Z(n, t) \geq \frac{1}{4}n^{2-\frac{2}{t+1}}$.

Proof. Let G be a random bipartite graph with parts A, B and $|A| = |B| = n$. Then, define $\widetilde{G}$ to be G with an edge removed from each $K_{t,t}$ in G . Then

$$e(\widetilde{G}) \geq e(G) - \#(K_{t,t}\text{'s in } G),$$

and by linearity of expectation

$$\mathbb{E}[e(\widetilde{G})] \geq \mathbb{E}[e(G)] - \mathbb{E}[\#(K_{t,t}\text{'s in } G)].$$

We have $\mathbb{E}[e(G)] = pn^2$, and also $\mathbb{E}[\#(K_{t,t}\text{'s in } G)]$ can be written using indicator functions as

$$\mathbb{E}\left[\sum_{A', B'} \mathbb{1}(A', B' \text{ induce a } K_{t,t})\right] = \binom{n}{t}^2 p^{t^2},$$

where we sum over $A' \leq A$, $B' \leq B$ with $|A'| = |B'| = t$. Thus

$$\mathbb{E}[e(\widetilde{G})] \geq pn^2 - \binom{n}{t}^2 p^{t^2}. \quad (*)$$

We want the RHS of this to be positive, so it's enough to have

$$pn^2 \geq n^{2t}p \iff n^{-\frac{2}{t+1}} \geq p,$$

so if $p = n^{-2/(t+1)}$ then $(*) \geq pn^2/2 \geq n^2/2 - 2/(t+1)$, as desired. $\square$

This proof is an example of the ‘modification method’.

§31.8.2 Random Graphs and Chromatic Number

One big picture question in graph theory is ‘what makes $\chi(G)$ large?’ A possible answer is that G contains a large clique, but that’s not the whole story. For example, there’s a triangle free graph with arbitrarily large χ . Another possible answer is that G has lots of short cycles, but it turns out that this is false too.

Definition 31.8.3 (Girth)

The **girth** of a graph G is the length of the shortest cycle in G , denoted $\text{girth}(G)$.

Definition 31.8.4 (Independent Set)

An **independent set** in a graph is a subset of vertices no two of which are adjacent. We will denote the size of the largest independent set $\alpha(G)$.

We can prove that graphs with large girth can also have arbitrarily large chromatic number. We will first need to note that if G is a graph, then

$$\chi(G) \geq \frac{n}{\alpha(G)}.$$

Theorem 31.8.5

For every $k, G \in \mathbb{N}$, there exists a graph G with $\text{girth}(G) \geq g$ and $\chi(G) \geq k$.

Proof. Let G be a random graph sampled from $\mathcal{G}(n, p)$ where $p = n^{\frac{1}{g}-1}$. We form the new graph $\widetilde{G}$ be the graph where we remove a vertex from each cycle of length $g - 1$.

We are going to do a few steps.

1. We want to show that $\mathbb{P}(|\widetilde{G}| \geq n/2) \rightarrow 1$ as $n \rightarrow \infty$.
2. We want to use $\chi(\widetilde{G}) \geq |\widetilde{G}|/\alpha(\widetilde{G})$, noting that $\alpha(\widetilde{G}) \leq \alpha(G)$. To do this, we will prove $\mathbb{P}(\alpha(G) \leq n/2k) \rightarrow 1$ as $n \rightarrow \infty$.

Step 1. Let X_i be the number of cycles of length i in G , and let $X = \sum_{i=1}^{g-1} X_i$. We can write

$$X_i = \sum_{x_1, \dots, x_i \in [n]} \mathbb{1}(x_1 \dots x_i \text{ are a cycle in } G).$$

Taking expectations,

$$\mathbb{E}[X_i] = \sum_{x_1, \dots, x_i \in [n]} \mathbb{P}(x_1 \dots x_i \text{ are a cycle in } G) \leq n^i p^i.$$

Using this we have

$$\mathbb{E}[\# \text{ of cycles of length } \leq g-1] = \mathbb{E}[X] = \sum_{i=1}^{g-1} (np)^i \leq gn^{\frac{g-1}{g}}.$$

Using Markov's inequality,

$$\mathbb{P}(X > n/2) \leq \frac{\mathbb{E}[X]}{n/2} \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

thus $\mathbb{P}(|\widetilde{G}| \geq n/2) \rightarrow 1$, as we needed.

Step 2. Let Y_t be the number of independent sets of size $t = n/2k$. Then

$$Y_t = \sum_{I \in [n]^{(t)}} \mathbb{1}(I \text{ is independent}),$$

and taking expectations

$$\mathbb{E}[Y_t] = \sum_{I \in [n]^{(t)}} \mathbb{P}(I \text{ is independent}) = \binom{n}{t} (1-p)^{\binom{t}{2}} \leq n^t e^{-p \binom{t}{2}} = \left(n e^{-p \frac{t-1}{2}} \right)^t,$$

and with $p = n^{1/g-1}$ and $t = n/2k$, so the RHS tends to 0 as n tends to infinity. Then

$$\mathbb{P}(Y_t \geq 1) \leq \mathbb{E}[Y_t] \rightarrow 0,$$

and $\mathbb{P}(Y_t \geq 1) = \mathbb{P}(\alpha(G) \geq t) = \mathbb{P}(\alpha(G) \geq n/2k)$. Putting that together, $\chi(\widetilde{G}) \geq \frac{|\widetilde{G}|}{\alpha(\widetilde{G})} \geq \frac{|\widetilde{G}|}{\alpha(G)} \geq \frac{n/2}{n/2k} \geq k$ holds with probability tending to 1 as n tends to infinity, finishing our proof. $\square$

§31.8.3 Structure of Random Graphs

So what does a typical graph sampled from $\mathcal{G}(n, p)$ look like? What does its structure look like as p varies from 0 to 1? As a first example, we will consider the following more precise question:

For what p does G sampled from $\mathcal{G}(n, p)$ contain a K_t ?

Theorem 31.8.6

Let G be sampled from $\mathcal{G}(n, p(n))$, and let $t \in \mathbb{N}$. Then $\lim_{n \rightarrow \infty} \mathbb{P}(G \text{ contains a } K_t) \rightarrow 1$ if $p(n)n^{\frac{2}{t-1}} \rightarrow \infty$. Moreover, $\lim_{n \rightarrow \infty} \mathbb{P}(G \text{ contains } K_t) \rightarrow 0$ if $p(n)n^{\frac{2}{t-1}} \rightarrow 0$.

Proof. Define X_t to be the number of K_t s in G . Then

$$X_t = \sum_{A \in [n]^{(t)}} \mathbb{1}(G[A] \text{ is complete}),$$

and

$$\mathbb{E}[X_t] = \sum_{A \in [n]^{(t)}} \mathbb{P}(G[A] \text{ is complete}) = \binom{n}{t} p^{\binom{t}{2}}.$$

We note that $\binom{n}{t} p^{\binom{t}{2}} \leq n^t p^{\binom{t}{2}} = \left(n^{\frac{2}{t-1}} p\right)^{\binom{t}{2}}$, and this tends to 0 or ∞ if $n^{\frac{2}{t-1}} p$ tends to 0 or ∞ respectively. So

$$\mathbb{P}(G \text{ contains a } K_t) = \mathbb{P}(X_t \geq 1) \leq \mathbb{E}[X_t] \rightarrow 0 \quad \text{if} \quad n^{\frac{2}{t-1}} p \rightarrow 0.$$

We now need to show that $\mathbb{P}(G \supseteq K_t) \rightarrow 1$ if $p n^{\frac{2}{t-1}} \rightarrow \infty$. We will use $\mathbb{P}(G \not\supseteq K_t) = \mathbb{P}(X_t = 0) \leq \mathbb{P}(|X_t - \mathbb{E}[X_t]| \geq \mathbb{E}[X_t])$. By Chebyshev, we have

$$\mathbb{P}(|X_t - \mathbb{E}[X_t]| \geq \mathbb{E}[X_t]) \leq \frac{\text{Var}(X_t)}{\mathbb{E}[X_t]^2},$$

so it suffices to show that this tends to 0.

We calculate the variance as

$$\begin{aligned} \text{Var}(X_t) &= \mathbb{E}[X_t^2] - (\mathbb{E}[X_t])^2 \\ &= \sum_{A, B \in [n]^{(t)}} [\mathbb{P}(G[A], G[B] \text{ complete}) - \mathbb{P}(G[A] \text{ complete})\mathbb{P}(G[B] \text{ complete})], \end{aligned}$$

which we calculate by hand as

$$\text{Var}(X_t) = \sum_{s=1}^t \binom{n}{t} \binom{t}{s} \binom{n-t}{t-s} \left[p^{2\binom{t}{2} - \binom{s}{2} - \binom{t-s}{2}} - p^{2\binom{t}{2}} \right],$$

and noting that $\binom{n-t}{t-s}/\binom{n}{t} \leq t!/n^s$ for n large, we have

$$\begin{aligned} \frac{\text{Var}(X_t)}{(\mathbb{E}[X])^2} &= \sum_{s=1}^t \frac{\binom{n}{t} \binom{t}{s} \binom{n-t}{t-s} [p^{2\binom{t}{2}-\binom{s}{2}} - p^{2\binom{t}{2}}]}{\binom{n}{t} p^{2\binom{t}{2}}}, \\ &\leq 2^{t+1} t! \sum_{s=1}^t \frac{1}{n^s} \cdot \frac{1}{p^{\binom{s}{2}}} \\ &\leq (t!)^2 \sum_{s=1}^t \left(\frac{1}{n^{\frac{s}{s-1}}} p \right)^{\binom{s}{2}}. \end{aligned}$$

Since $n^{\frac{2}{t-1}} p \rightarrow \infty$, then $n^{\frac{2}{s-1}} p \rightarrow \infty$ for each $s \leq t$, this goes to zero as required. $\square$

This is an example of a threshold result, where almost suddenly when p goes to a certain value, we start to see some consistent structure in the random graphs. Another somewhat similar question is:

What is the threshold for the graph to be connected?

We will find that this ‘transition’ will be much faster than with the previous example.

Theorem 31.8.7

Let G be sampled from $\mathcal{G}(n, p(n))$, then $\lim_{n \rightarrow \infty} \mathbb{P}(G \text{ is connected}) = 1$ if $p > (1 + \epsilon) \frac{\log n}{n}$ and $\lim_{n \rightarrow \infty} \mathbb{P}(G \text{ is disconnected}) = 0$ if $p < (1 - \epsilon) \frac{\log n}{n}$.

Proof. For the ‘first direction’ we will show that there is an isolated vertex with probability going to 1 if $p < (1 - \epsilon) \frac{\log n}{n}$. Let I be the number of isolated vertices. We are going to use a similar strategy to before, where we calculate the variance and apply Chebyshev.

We have

$$\mathbb{E}[I] = n(1-p)^{n-1} \quad \text{and} \quad \mathbb{E}[I^2] = \sum_{u,v \in [n]} \mathbb{P}(u, v \text{ are isolated}),$$

which gives us the variance

$$\begin{aligned} \text{Var}(I) &= \sum_{u,v \in [n]} [\mathbb{P}(u, v \text{ isolated}) - \mathbb{P}(u \text{ isolated})\mathbb{P}(v \text{ isolated})] \\ &= \left(\sum_{u \in [n]} \mathbb{P}(u \text{ iso}) - \mathbb{P}(u \text{ iso})^2 \right) + \left(\sum_{u \neq v \in [n]} \mathbb{P}(u, v \text{ iso}) - \mathbb{P}(u \text{ iso})\mathbb{P}(v \text{ iso}) \right) \\ &= (n(1-p)^{n-1}) + \left(\sum_{u \neq v \in [n]} (1-p)^{2(n-1)-1} - (1-p)^{2(n-1)} \right) \\ &= n(1-p)^{n-1} + n(n-1)(1-p)^{2(n-1)-1} p. \end{aligned}$$

Then we can apply Chebyshev to get

$$\begin{aligned}\mathbb{P}(I = 0) &\leq \mathbb{P}(|I - \mathbb{E}[I]| \geq \mathbb{E}[I]) \leq \frac{\text{Var}(I)}{\mathbb{E}[I]^2} \\ &= \frac{n(1-p)^{n-1} + n(n-1)(1-p)^{2(n-1)-1}}{n^2(1-p)^{2(n-1)}} \\ &= \frac{1}{(1-p)^{n-1}n} + p \rightarrow 0,\end{aligned}$$

as required.

Now for the ‘other direction’, we want to show that the $\mathbb{P}(G \text{ is disconnected}) \rightarrow 0$. We can write

$$\begin{aligned}\mathbb{P}(G \text{ is disconnected}) &= \mathbb{P}\left(\bigcup_{A \subseteq [n], 0 < |A| \leq n/2} \{\text{no edges between } A \text{ and } A'\}\right) \\ &\leq \sum_{A \subseteq [n], 0 < |A| \leq n/2} \mathbb{P}(\text{no edges between } A \text{ and } A') \\ &= \sum_{s=1}^{n/2} \binom{n}{s} (1-p)^{s(n-s)},\end{aligned}$$

so we want to show that this tends to zero. We can break up this sum as

$$\begin{aligned}\sum_{s=1}^{n/2} \binom{n}{s} (1-p)^{s(n-s)} &\leq \sum_{s=1}^{\epsilon n/2} \left(ne^{-p(n-s)}\right)^s + \sum_{\epsilon n/2 \leq s \leq n/2} \binom{n}{s} e^{-ps(n-s)} \\ &\leq \sum_{s=1}^{\epsilon n/2} \left(\frac{1}{n^{\epsilon/4}}\right)^s + \left(\sum_{\epsilon n/2 \leq s \leq n/2} \binom{n}{s}\right) e^{-\epsilon n \log n/8} \\ &\leq \sum_{s=1}^{\epsilon n/2} \left(\frac{1}{n^{\epsilon/4}}\right)^s + \left(2e^{-\epsilon \log n/8}\right)^n \rightarrow 0,\end{aligned}$$

thus $\mathbb{P}(G \text{ is disconnected}) \rightarrow 0$, as required. $\square$

§31.9 Algebraic Methods in Graph Theory

The final aspect of graph theory we are going to look at is the use of linear algebraic methods to solve problems. This is a naturally occurring technique, as will be shown by the problem below.

§31.9.1 A Motivating Problem

The **diameter** of a graph is the maximum distance between two vertices in a graph. We are going to try and attack the problem below:

How many vertices can a graph with fixed Δ have, given its diameter is at most 2?

By just thinking a little bit, it's not hard to come up with the bound

$$n \leq 1 + \Delta + (\Delta - 1)\Delta = 1 + \Delta^2,$$

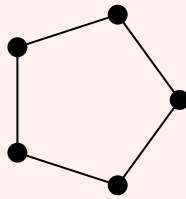
but is it possible to have a graph that attains this bound? We are going to call such graphs *Moore graphs*.

Definition 31.9.1 (Moore Graph)

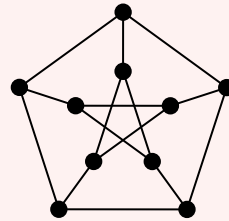
A **Moore graph** is a k -regular graph on $k^2 + 1$ vertices that has diameter 2.

Example 31.9.2 (Examples of Moore Graphs)

Below are Moore graphs for $k = 2$ and $k = 3$.



$k = 2$



$k = 3$

Note that the graph for $k = 3$ is the Petersen graph. Playing around a bit, you might find that such a graph for $k = 4$ *does not exist*, so it seems that Moore graphs are somewhat special.

Remark. A Moore graph is a k -regular graph for which any $x \neq y$ with $x, y \in V(G)$ has exactly one path of length ≤ 2 between x and y .

So Moore graphs don't always exist, leading to a natural question – does there exist infinitely many of them?

§31.9.2 The Adjacency Matrix

One way to encode a graph is using an adjacency matrix.

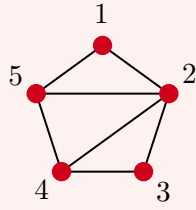
Definition 31.9.3 (Adjacency Matrix)

Given a graph $G = (V, E)$ on n vertices, we define the **adjacency matrix** of G to be the $n \times n$ matrix

$$A_{ij} = \begin{cases} 1 & \text{if } ij \in E, \\ 0 & \text{otherwise.} \end{cases}$$

Example 31.9.4 (Example of an Adjacency Matrix)

An graph and its associated adjacency matrix are given below.



Adjacency Matrix

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Now while this is just a straightforward representation of a graph, the fact that it's a matrix means that we can start asking questions about it, even if it's not immediately obvious that they have any meaning. For example, what do the eigenvalues of this matrix look like? What about the eigenvectors?

We can observe that the adjacency matrix is symmetric, which we know implies that it has real eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$, and an associated eigenbasis $v_1, \dots, v_n$ which is orthonormal.

We can also observe that for a graph G with adjacency matrix A , since there's no edge between a vertex and itself, that there is only zeros along the diagonal of A , so it has zero trace and

$$\sum_{i=1}^n \lambda_i = \text{tr}(A) = 0.$$

Also, if we call $\lambda_1 = \lambda_{\max}$ and $\lambda_n = \lambda_{\min}$, then

$$\lambda_{\max} = \max_{x:|x|^2=1} x^T A x, \quad \text{and} \quad \lambda_{\min} = \min_{x:|x|^2=1} x^T A x$$

Somewhat surprisingly, we are going to be able to connect all of these algebraic facts back to our original graph.

Proposition 31.9.5

Let $G = (V, E)$ be a graph with adjacency matrix A . Then

- (i) $\frac{1}{n} \sum_{x \in V} d(x) \leq \lambda_{\max}(G) \leq \Delta(G)$.
- (ii) $\lambda_{\max}(G) = \Delta(G)$ if and only if G is Δ -regular.
- (iii) $\lambda_{\min}(G) = \Delta(G)$ if and only if G is Δ -regular and bipartite.

Proof. Part (i). For the lower bound, let $w = \frac{1}{\sqrt{n}}(1, \dots, 1)$. Then we observe by interpreting the adjacency matrix in a graph theoretic sense that

$$\lambda_{\max} = \max_{x:|x|^2=1} x^T A x \geq w^T A w = \frac{1}{n} \sum_{x \in V} d(x).$$

For the upper bound, let $x = (x_1, \dots, x_n)$ be an eigenvector corresponding to $\lambda_{\max}$. Assume WLOG that $|x_1| \geq |x_i|$, for all i . Then

$$\lambda x_1 = (\lambda_{\max} x)_1 = (Ax)_1 = \sum_{i: v_i \sim v_1} x_i,$$

which gives us

$$|\lambda_{max}x_1| \leq \sum_{i:v_i \sim v_1} |x_i| \leq |x_1|\Delta,$$

so $|\lambda_{max}| \leq \Delta$, and $\lambda_{max} \leq \Delta$.

Part (ii). If G is Δ -regular, then consider $\mathbb{1} = (1, \dots, 1)$. Then

$$A\mathbb{1} = \Delta\mathbb{1},$$

thus $\lambda_{max} \geq \Delta$, so $\lambda_{max} = \Delta$ by our previous result.

If $\lambda_{max} = \Delta$, then let $x = (x_1, \dots, x_n)$ be an associated eigenvector, and as before assume $|x_1| \geq |x_i|$ for all i . Then

$$\Delta x_1 = (\Delta x)_1 = (Ax)_1 = \sum_{i:v_i \sim v_1} x_i,$$

therefore $x_i = x$ for all i where $v_i \sim v_1$. So now we repeat the argument for all $v_i \in N(v_1)$ to learn that all x_j with $d(v_j, v_1) \leq 2$ has $x_j = x_1$ and so on, until we get $x_1 = x_i$ for all v_j in the component of v_i . We then repeat for each component, to see that x is constant on each component, and thus G must be a regular graph.

Part (iii). Omitted. □

Another fact that will bring us closer to Moore graphs concerns the interpretation of the square of an adjacency matrix.

Proposition 31.9.6

Let G be a graph and let A be its adjacency matrix. Then $(A^2)_{ij}$ is the number of walks of length 2 between i, j .

Proof Sketch. Consider how A^2 is calculated. □

§31.9.3 Moore Graphs

Now we can apply our newfound algebraic tools to our question about Moore graphs and graph diameter. We will see that our result really does show that Moore graphs are pretty special.

Lemma 31.9.7 (Necessary Condition for the Existence of Moore Graphs)

Let G be a Moore graph of degree k , then

$$\frac{1}{2} \left(k^2 \pm \frac{k^2 - 2k}{\sqrt{4k - 3}} \right) \text{ are integers.}$$

Proof. Let G be a Moore graph of degree k on n vertices, and let A be the adjacency

matrix of G . Then we have

$$(A^2)_{ij} = \begin{cases} k & \text{if } i = j, \\ 1 & \text{if } i \neq j \text{ and } ij \notin E, \\ 0 & \text{if } ij \in E. \end{cases}$$

Thus $A^2 = (J - A) + (k - 1)I$, where J is the matrix of all 1s. So we have

$$A^2 + A - (k - 1)I = J.$$

Now let $\lambda_{\max}$ be a maximum eigenvalue of A , and let x be a corresponding eigenvector. Since G is k -regular, $\mathbb{1}$ is an eigenvector of A , and thus we know that $x \perp \mathbb{1}$, since $\lambda_{\max} \neq \Delta$. Then

$$(A^2 + A - (k - 1)I)x = Jx \implies (\lambda_{\max}^2 + \lambda_{\max} - (k - 1))x = 0,$$

so $\lambda_{\max}^2 + \lambda_{\max} - (k - 1) = 0$. Thus

$$\lambda_{\max} = \left(\frac{-1 \pm \sqrt{1 + 4(k - 1)}}{2} \right) = \left(\frac{-1 \pm \sqrt{4k - 3}}{2} \right).$$

So A has eigenvalues k with multiplicity 1, $\lambda_{\max}$ with multiplicity say r , and μ with multiplicity s , with $r + s = n - 1$. We also know that

$$\sum_{i=1}^n \lambda_i = 0 = k + r\lambda + s\mu,$$

and solving this for integers r and s and using $n = k^2 + 1$ gives the desired result. $\square$

With this lemma, we can heavily restrict the possible Moore graphs.

Theorem 31.9.8 (Hoffman-Singleton Theorem)

Let G be a Moore graph of degree k . Then $k \in \{2, 3, 7, 57\}$.

Proof. We need to find out when

$$\frac{1}{2} \left(k^2 \pm \frac{k^2 - 2k}{\sqrt{4k - 3}} \right)$$

are integers. One case is when $k^2 - 2k = 0$, implying that $k = 2$. Another case is when $4k - 3 = t^2$, where $t \mid k^2 - 2k$. Then we have

$$t \mid \left(\frac{t^2 + 3}{4} \right)^2 - 2 \left(\frac{t^3 + 3}{4} \right),$$

and multiplying by 16,

$$t \mid t^4 - 2t^2 - 15,$$

so $t \mid 15$, and thus $t \in \{3, 5, 15\}$, and since $4k - 3 = t^2$, this shows that $k \in \{3, 7, 57\}$. $\square$

Remark. For $k = 2, 3$ and 7 , the graphs are known to exist, but a graph for $k = 57$ is

has not yet been found.

32 Integrable Systems

Most differential equations cannot be solved. This is not a statement about our own cleverness – it is a statement about differential equations, the overwhelming majority of which have solutions that admit no description more economical than the equation itself. Every so often, though, we stumble across an equation that can be ‘integrated up’: one whose solutions can be written down in closed form, or reduced to quadratures, or generated from nothing by an algebraic recipe. Such equations are called *integrable*, and they are the subject of this course.

There is a curious tension here. On the one hand, integrable systems ought to be vanishingly rare, in the same way that a randomly chosen polynomial is unlikely to factorise nicely. On the other hand, integrable equations turn up all over physics – the harmonic oscillator, the Kepler problem, the spinning top, waves in shallow water, light in an optical fibre. Nature seems unreasonably fond of them. The explanation, when we find it, is always the same: integrability is a symptom of hidden structure. An integrable system has more conserved quantities than it has any right to have, and those conserved quantities conspire to fill up phase space with tori on which the motion is nothing more than uniform rotation.

The plan of the course is to take this idea and push it as far as it will go. We begin with ordinary differential equations, where the story is essentially complete: the Arnold–Liouville theorem tells us precisely what it means for a Hamiltonian system to be integrable, and how to integrate it. We then move to partial differential equations, which we can think of as infinite-dimensional dynamical systems. Here the theory is much wilder and much more beautiful. We will meet the Korteweg–de Vries equation and its *solitons* – solitary waves that collide and emerge unscathed, behaving far more like particles than like waves – and we will spend the heart of the course developing the *inverse scattering transform*, an extraordinary machine which solves a nonlinear PDE by turning it into a linear scattering problem, letting the scattering data evolve trivially, and then reconstructing the answer.

This article constitutes my notes for the ‘Integrable Systems’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered¹.

§32.1 Integrability of Ordinary Differential Equations

We begin with ordinary differential equations. The eventual goal of this section is the Arnold–Liouville theorem, which says that a Hamiltonian system possessing enough conserved quantities can be solved completely, and moreover tells us how. Before we get there we need to set up some machinery: flows, Poisson brackets, and canonical transformations.

¹We will lean fairly heavily on IB Variational Principles (Hamilton’s equations, the Legendre transform, Noether’s theorem) and on IB Methods (Fourier transforms, Green’s functions, Sturm–Liouville theory). A passing acquaintance with IB Quantum Mechanics – specifically reflection and transmission coefficients for a one-dimensional potential – will make Section 3 feel much more familiar.

§32.1.1 Vector Fields and Flow Maps

An m -dimensional first-order ODE is specified by a **vector field** $\mathbf{V} : \mathbb{R}^m \rightarrow \mathbb{R}^m$ together with an initial condition $\mathbf{x}_0 \in \mathbb{R}^m$. We look for a curve $\mathbf{x}(t) \in \mathbb{R}^m$, defined for t in some interval containing 0, satisfying

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \quad \mathbf{x}(0) = \mathbf{x}_0.$$

Restricting to first-order equations costs us nothing: an n th order ODE in one variable becomes a first-order system in n variables in the usual way, by treating the derivatives as new coordinates.

Throughout this course we will be cavalier about regularity, and assume that all our vector fields are as nice as we need them to be.

Fact 32.1.1

For a sufficiently nice vector field $\mathbf{V}$ and any initial condition $\mathbf{x}_0$, the problem $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$, $\mathbf{x}(0) = \mathbf{x}_0$ has a unique solution, and that solution depends smoothly on both t and $\mathbf{x}_0$.

Because the solution is unique, it is legitimate to package all of the solutions together into a single object.

Definition 32.1.2 (Flow map)

The **flow map** of a vector field $\mathbf{V}$ is the family of maps $g^t : \mathbb{R}^m \rightarrow \mathbb{R}^m$ defined by

$$\mathbf{x}(t) = g^t \mathbf{x}_0,$$

where $\mathbf{x}(t)$ solves $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$ with $\mathbf{x}(0) = \mathbf{x}_0$.

The flow map is smooth, and satisfies exactly the properties one would hope for.

Proposition 32.1.3

The flow map obeys

- (i) $g^0 = \text{id}$;
- (ii) $g^{t+s} = g^t g^s$;
- (iii) $(g^t)^{-1} = g^{-t}$.

Proof. The first is immediate from the definition, and the third follows from the first two on setting $s = -t$. For the second, fix $\mathbf{x}_0$ and s , and regard both $t \mapsto g^{t+s} \mathbf{x}_0$ and $t \mapsto g^t(g^s \mathbf{x}_0)$ as functions of t . Both solve

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \quad \mathbf{x}(0) = g^s \mathbf{x}_0,$$

and so by uniqueness of solutions they coincide. □

In the language of IA Groups, this says that $t \mapsto g^t$ is a homomorphism from $(\mathbb{R}, +)$ into the group of diffeomorphisms of $\mathbb{R}^m$. We say that $\mathbf{V}$ is the **infinitesimal generator** of

the flow, which is justified by Taylor expanding:

$$g^\varepsilon \mathbf{x}_0 = \mathbf{x}(0) + \varepsilon \dot{\mathbf{x}}(0) + o(\varepsilon) = \mathbf{x}_0 + \varepsilon \mathbf{V}(\mathbf{x}_0) + o(\varepsilon).$$

So to first order in ε , flowing for time ε is just moving along $\mathbf{V}$.

Suppose now that we have two vector fields $\mathbf{V}_1, \mathbf{V}_2$ generating flows g_1^t and g_2^s . A natural question is whether the flows *commute*, that is, whether

$$g_1^t g_2^s \mathbf{x}_0 = g_2^s g_1^t \mathbf{x}_0$$

for all $\mathbf{x}_0, s, t$. Asked directly this is a hopeless question, since it requires us to solve both ODEs. But there is an infinitesimal criterion, and infinitesimal criteria are always easier.

Definition 32.1.4 (Commutator)

Given vector fields $\mathbf{V}_1, \mathbf{V}_2 : \mathbb{R}^m \rightarrow \mathbb{R}^m$, their **commutator** is the vector field

$$[\mathbf{V}_1, \mathbf{V}_2] = \left(\mathbf{V}_1 \cdot \frac{\partial}{\partial \mathbf{x}} \right) \mathbf{V}_2 - \left(\mathbf{V}_2 \cdot \frac{\partial}{\partial \mathbf{x}} \right) \mathbf{V}_1,$$

where $\frac{\partial}{\partial \mathbf{x}} = (\partial_{x_1}, \dots, \partial_{x_m})^T$. Componentwise,

$$[\mathbf{V}_1, \mathbf{V}_2]_i = \sum_{j=1}^m \left((\mathbf{V}_1)_j \frac{\partial (\mathbf{V}_2)_i}{\partial x_j} - (\mathbf{V}_2)_j \frac{\partial (\mathbf{V}_1)_i}{\partial x_j} \right).$$

Proposition 32.1.5

Let $\mathbf{V}_1, \mathbf{V}_2$ be vector fields generating flows g_1^t, g_2^s . Then

$$[\mathbf{V}_1, \mathbf{V}_2] = 0 \iff g_1^t g_2^s = g_2^s g_1^t \text{ for all } s, t.$$

We will not prove this (it is a standard exercise, and appears on the first example sheet), but the idea is worth recording: expanding $g_1^{-t} g_2^{-s} g_1^t g_2^s \mathbf{x}_0$ to second order in the small parameters, everything cancels except a term $st [\mathbf{V}_1, \mathbf{V}_2](\mathbf{x}_0)$. The commutator measures precisely the failure of the two flows to commute, and it does so at the level of the vector fields, where we can compute.

§32.1.2 Hamiltonian Systems

We now specialise to a very particular class of ODEs. In IB Variational Principles we met Hamilton's equations, obtained from a Lagrangian by a Legendre transform; here we take them as our starting point.

The setting is a **phase space** $M = \mathbb{R}^{2n}$, with coordinates

$$(\mathbf{q}, \mathbf{p}) = (q_1, \dots, q_n, p_1, \dots, p_n),$$

which we think of as generalised positions and generalised momenta. We will usually write

$$\mathbf{x} = (\mathbf{q}, \mathbf{p})^T \in M.$$

The crucial point is that M is not just $\mathbb{R}^{2n}$: the coordinates come *paired up*, with q_i married to p_i . This pairing is the extra structure that distinguishes a phase space from a bare vector space, and we encode it in the $2n \times 2n$ antisymmetric matrix

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

called the **symplectic form**. Note that $J^T = -J$ and $J^2 = -I_{2n}$.

Almost everything in this section can be phrased in terms of J alone. The first example is the Poisson bracket.

Definition 32.1.6 (Poisson bracket)

For smooth $f, g : M \rightarrow \mathbb{R}$, the **Poisson bracket** is

$$\{f, g\} = \frac{\partial f}{\partial \mathbf{x}} \cdot J \frac{\partial g}{\partial \mathbf{x}} = \frac{\partial f}{\partial \mathbf{q}} \cdot \frac{\partial g}{\partial \mathbf{p}} - \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{\partial g}{\partial \mathbf{q}}.$$

The two expressions agree by direct computation with the block form of J . The Poisson bracket has a list of properties, most of which are obvious and one of which is not.

Proposition 32.1.7

The Poisson bracket satisfies, for all smooth $f, g, h : M \rightarrow \mathbb{R}$:

- (i) it is bilinear;
- (ii) it is antisymmetric, $\{f, g\} = -\{g, f\}$;
- (iii) it obeys the Leibniz rule $\{f, gh\} = \{f, g\}h + \{f, h\}g$;
- (iv) it obeys the Jacobi identity

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0;$$

- (v) on the coordinate functions,

$$\{q_i, q_j\} = \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}.$$

Proof. All of these are immediate from the definition except the Jacobi identity, which one proves by writing out all twenty-four terms and watching them cancel. The only structural remark worth making is that each of the three double brackets contains second derivatives of exactly one of f, g, h , and these second-derivative terms cancel in pairs because J is antisymmetric and second partial derivatives commute. $\square$

The last of these is called the **canonical commutation relation**, and it is the algebraic shadow of the pairing between q_i and p_i .

Definition 32.1.8 (Hamilton's equations)

Given a function $H : M \rightarrow \mathbb{R}$, called the **Hamiltonian**, **Hamilton's equations**

are

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}. \quad (*)$$

In terms of J these collapse into the single equation

$$\dot{\mathbf{x}} = J \frac{\partial H}{\partial \mathbf{x}}.$$

This is worth pausing over. To specify a general ODE on $\mathbb{R}^{2n}$ we must supply a vector field, which is $2n$ functions. To specify a Hamiltonian system we need only supply the *single* function H . The symplectic structure does the rest of the work. This is our first hint that Hamiltonian systems are special.

Definition 32.1.9 (Hamiltonian vector field)

Given $f : M \rightarrow \mathbb{R}$, the associated **Hamiltonian vector field** is

$$\mathbf{V}_f = J \frac{\partial f}{\partial \mathbf{x}}.$$

Thus Hamilton's equations say precisely that the motion is the flow of $\mathbf{V}_H$.

Now suppose we are watching a particle move according to $(*)$, and we are interested in some observable $f : M \rightarrow \mathbb{R}$ – its energy, its angular momentum, its distance from the origin. How does f change along the motion? The chain rule gives

$$\frac{df}{dt} = \frac{d}{dt}f(\mathbf{x}(t)) = \frac{\partial f}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} = \frac{\partial f}{\partial \mathbf{x}} \cdot J \frac{\partial H}{\partial \mathbf{x}} = \{f, H\}.$$

We record this, because it is the single most useful formula in the subject.

Proposition 32.1.10

If $\mathbf{x}(t)$ evolves according to Hamilton's equations with Hamiltonian H , then for any smooth $f : M \rightarrow \mathbb{R}$,

$$\frac{df}{dt} = \{f, H\}.$$

In other words – f is conserved if and only if it Poisson commutes with H . This is genuinely powerful. Without it, deciding whether some quantity is conserved means solving the equations of motion and staring at the answer, or else differentiating and hoping that things cancel. With it, we simply compute a bracket. And immediately we get one conserved quantity for free:

$$\frac{dH}{dt} = \{H, H\} = 0$$

by antisymmetry. Energy is conserved.

Example 32.1.11 (Particle in a potential)

Consider a particle of unit mass with Cartesian position $\mathbf{q} = (q_1, q_2, q_3)$ moving in a potential $U(\mathbf{q})$. Newton's second law reads

$$\ddot{\mathbf{q}} = -\frac{\partial U}{\partial \mathbf{q}}.$$

Introducing $\mathbf{p} = \dot{\mathbf{q}}$, we have

$$\dot{\mathbf{x}} = \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{pmatrix} = \begin{pmatrix} \mathbf{p} \\ -\partial U/\partial \mathbf{q} \end{pmatrix} = J \frac{\partial H}{\partial \mathbf{x}}, \quad H = \frac{1}{2}|\mathbf{p}|^2 + U(\mathbf{q}),$$

since $\partial H/\partial \mathbf{p} = \mathbf{p}$ and $\partial H/\partial \mathbf{q} = \partial U/\partial \mathbf{q}$. So Newtonian mechanics in a potential is a Hamiltonian system, and the Hamiltonian is the total energy.

Example 32.1.12 (Angular momentum in the two-body problem)

Fix the Sun at the origin and let a planet have Cartesian coordinates $\mathbf{q}$. The equation of motion is

$$\ddot{\mathbf{q}} = -\frac{\mathbf{q}}{|\mathbf{q}|^3},$$

which by the previous example is Hamiltonian with $\mathbf{p} = \dot{\mathbf{q}}$ and

$$H = \frac{1}{2}|\mathbf{p}|^2 - \frac{1}{|\mathbf{q}|}.$$

The angular momentum is $\mathbf{L} = \mathbf{q} \wedge \mathbf{p}$, or in components $L_i = \varepsilon_{ijk} q_j p_k$. Let us check that it is conserved, using summation convention throughout:

$$\begin{aligned} \{L_i, H\} &= \frac{\partial L_i}{\partial q_\ell} \frac{\partial H}{\partial p_\ell} - \frac{\partial L_i}{\partial p_\ell} \frac{\partial H}{\partial q_\ell} \\ &= \varepsilon_{ijk} \delta_{j\ell} p_k \cdot p_\ell - \varepsilon_{ijk} q_j \delta_{k\ell} \cdot \frac{q_\ell}{|\mathbf{q}|^3} \\ &= \varepsilon_{ijk} \left(p_j p_k - \frac{q_j q_k}{|\mathbf{q}|^3} \right) \\ &= 0, \end{aligned}$$

since in each term we are contracting the antisymmetric ε_{ijk} against something symmetric in $j \leftrightarrow k$. So all three components of $\mathbf{L}$ are conserved, as is H : this system has rather a lot of conserved quantities, which is exactly why the Kepler problem can be solved.

There is one more piece of structure worth recording. We now have two ways of combining functions to produce vector fields: we can take the Poisson bracket $\{f, g\}$ and then form its Hamiltonian vector field, or we can form $\mathbf{V}_f$ and $\mathbf{V}_g$ and take their commutator. These agree up to sign.

Proposition 32.1.13

For smooth $f, g : M \rightarrow \mathbb{R}$ we have $[\mathbf{V}_f, \mathbf{V}_g] = -\mathbf{V}_{\{f, g\}}$.

The proof is a computation using the Jacobi identity, and is left to the first example sheet. The content of the statement is that $f \mapsto \mathbf{V}_f$ is (up to a sign) a homomorphism of Lie algebras, from functions-under-Poisson-bracket to vector-fields-under-commutator. We will use it in the proof of the Arnold–Liouville theorem, where it will tell us that flows generated by commuting conserved quantities themselves commute.

§32.1.3 First Integrals and Liouville Integrability

Definition 32.1.14 (First integral)

Let (M, H) be a Hamiltonian system. A function $f : M \rightarrow \mathbb{R}$ is a **first integral** if

$$\{f, H\} = 0,$$

equivalently, if f is constant along every trajectory.

The terminology is historical. When we solve a differential equation by successive integrations, each integration produces a constant; the constant produced by the first integration was called the first integral. For our purposes, a first integral is simply a constant of the motion.

Our goal for the rest of this section is the following. If a Hamiltonian system has *enough* first integrals, then there is a change of coordinates in which the equations of motion become trivial, and we can integrate them up by inspection. To make ‘enough’ precise we need two conditions.

Definition 32.1.15 (Involution)

Two first integrals F, G are **in involution** if $\{F, G\} = 0$. We also say that they **Poisson commute**.

Definition 32.1.16 (Independence)

Functions $f_1, \dots, f_n : M \rightarrow \mathbb{R}$ are **independent** if at each point $\mathbf{x} \in M$ the gradients $\frac{\partial f_i}{\partial \mathbf{x}}$, $i = 1, \dots, n$, are linearly independent vectors in $\mathbb{R}^{2n}$.

Definition 32.1.17 (Liouville integrable)

A $2n$ -dimensional Hamiltonian system (M, H) is **integrable** (in the Liouville sense) if there exist n first integrals $f_1, \dots, f_n$ which are independent and in involution, that is $\{f_i, f_j\} = 0$ for all i, j .

By convention we always take $f_1 = H$, which is legitimate since $\{H, H\} = 0$.

The independence requirement is what stops the definition being vacuous. Without it we could always cheat, taking $f_1 = H$, $f_2 = H^2$, $f_3 = e^H$, and so on: these all Poisson commute with everything in sight, but they carry no information beyond H itself. Independence says that each new first integral genuinely cuts down the available motion.

The involutivity requirement is subtler and less obviously natural, but it is exactly what makes the proof of the Arnold–Liouville theorem work. It is worth noting that a system can have many first integrals without being integrable in this sense, if the integrals refuse to Poisson commute.

Example 32.1.18

Every two-dimensional Hamiltonian system ($n = 1$) is integrable: we need one first integral, and H itself will do. This is a restatement of the familiar fact that a single particle moving in one dimension in a potential can always be solved by energy

conservation and a quadrature.

Example 32.1.19

Consider the two-body problem above. We have H , and the three components of $\mathbf{L}$. However

$$\{L_1, L_2\} = L_3 \neq 0,$$

so the L_i are not in involution with each other, and we cannot simply take all four. But H , L_3 and $|\mathbf{L}|^2$ are pairwise in involution and independent, and the system is six-dimensional, so $n = 3$ and the Kepler problem is integrable.

§32.1.4 Canonical Transformations

We want to find coordinates in which an integrable system becomes trivial. But we cannot use arbitrary coordinates: the whole structure above depended on the pairing of q_i with p_i , and a general change of variables will destroy it. So we must decide which coordinate changes are allowed. There are several equivalent ways to do this; we take the most pragmatic, and simply demand that Hamilton's equations keep their form.

Definition 32.1.20 (Canonical transformation)

A coordinate change $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{Q}(\mathbf{q}, \mathbf{p}), \mathbf{P}(\mathbf{q}, \mathbf{p}))$ is **canonical** if it preserves the form of Hamilton's equations: that is, if

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}} \quad \Longleftrightarrow \quad \dot{\mathbf{Q}} = \frac{\partial \tilde{H}}{\partial \mathbf{P}}, \quad \dot{\mathbf{P}} = -\frac{\partial \tilde{H}}{\partial \mathbf{Q}},$$

where $\tilde{H}(\mathbf{Q}, \mathbf{P}) = H(\mathbf{q}, \mathbf{p})$.

Writing $\mathbf{x} = (\mathbf{q}, \mathbf{p})$ and $\mathbf{y} = (\mathbf{Q}, \mathbf{P})$, this says

$$\dot{\mathbf{x}} = J \frac{\partial H}{\partial \mathbf{x}} \quad \Longleftrightarrow \quad \dot{\mathbf{y}} = J \frac{\partial \tilde{H}}{\partial \mathbf{y}}.$$

Example 32.1.21

Swapping $\mathbf{q}$ and $\mathbf{p}$ is *not* canonical: it introduces a sign error, turning $\dot{\mathbf{Q}} = \partial \tilde{H} / \partial \mathbf{P}$ into $\dot{\mathbf{Q}} = -\partial \tilde{H} / \partial \mathbf{P}$. The map $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{p}, -\mathbf{q})$, on the other hand, is canonical.

Let us do the linear case honestly, since it tells us what the general answer must look like.

Example 32.1.22 (Linear canonical transformations)

Consider $\mathbf{x} \mapsto \mathbf{y} = A\mathbf{x}$ for a constant matrix A . We claim this is canonical if and only if $AJA^T = J$, in which case A is called **symplectic**.

Indeed, by linearity $\dot{\mathbf{y}} = A\dot{\mathbf{x}} = AJ \frac{\partial H}{\partial \mathbf{x}}$. Setting $\tilde{H}(\mathbf{y}) = H(\mathbf{x})$ and using the chain

rule,

$$\frac{\partial H}{\partial x_i} = \frac{\partial y_j}{\partial x_i} \frac{\partial \tilde{H}}{\partial y_j} = A_{ji} \frac{\partial \tilde{H}}{\partial y_j} = \left[A^T \frac{\partial \tilde{H}}{\partial \mathbf{y}} \right]_i.$$

Substituting back,

$$\dot{\mathbf{y}} = A J A^T \frac{\partial \tilde{H}}{\partial \mathbf{y}},$$

which is Hamilton's equation for $\tilde{H}$ precisely when $A J A^T = J$.

Since a differentiable map is locally linear, and Hamilton's equations are a local condition, the general statement is the obvious one.

Proposition 32.1.23

A smooth map $\mathbf{x} \mapsto \mathbf{y}(\mathbf{x})$ is canonical if and only if its Jacobian $D\mathbf{y}$ is symplectic at every point:

$$D\mathbf{y} J (D\mathbf{y})^T = J.$$

This is a clean criterion, but it is a criterion for *checking* that a transformation is canonical, not for *producing* one. For that we use generating functions.

Generating Functions

There are four standard flavours of generating function, differing in which pair of old and new variables one takes as independent. We will need only one of them.

Let $S : \mathbb{R}^{2n} \rightarrow \mathbb{R}$ be a function which we write as $S(\mathbf{q}, \mathbf{P})$ – old positions and new momenta – and define a transformation implicitly by

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}}, \quad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}}.$$

These equations are to be read as follows. The first is n equations relating $\mathbf{p}, \mathbf{q}, \mathbf{P}$; we solve them for $\mathbf{P}$ in terms of $(\mathbf{q}, \mathbf{p})$. The second then delivers $\mathbf{Q}$ in terms of $(\mathbf{q}, \mathbf{P})$, hence in terms of $(\mathbf{q}, \mathbf{p})$. Checking that the result is always canonical is another careful application of the chain rule, which we omit.

In practice we usually already have a candidate for $\mathbf{P}$ in mind – typically some collection of first integrals – and we hunt for an S making the first equation hold. The second equation then *tells* us what the conjugate variable $\mathbf{Q}$ has to be. This is exactly how the Arnold–Liouville theorem will proceed.

Example 32.1.24

Take $S(\mathbf{q}, \mathbf{P}) = \mathbf{q} \cdot \mathbf{P}$. Then

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}} = \mathbf{P}, \quad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}} = \mathbf{q},$$

so S generates the identity transformation.

Example 32.1.25

In two dimensions, take $S(q, P) = qP + q^2$. Then

$$p = \frac{\partial S}{\partial q} = P + 2q, \quad Q = \frac{\partial S}{\partial P} = q,$$

so the transformation is $(Q, P) = (q, p - 2q)$, or in matrix form

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix} = A \begin{pmatrix} q \\ p \end{pmatrix}.$$

We can verify directly that this is canonical:

$$AJA^T = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J,$$

as promised.

§32.1.5 The Arnold–Liouville Theorem

We can now state the central result about integrable ODEs. Informally it says: if a Hamiltonian system is integrable, then there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\phi, \mathbf{I})$ after which the new Hamiltonian depends only on $\mathbf{I}$. If we can achieve this then Hamilton's equations become

$$\dot{\mathbf{I}} = -\frac{\partial \tilde{H}}{\partial \phi} = 0, \quad \dot{\phi} = \frac{\partial \tilde{H}}{\partial \mathbf{I}},$$

so $\mathbf{I}(t) = \mathbf{I}_0$ is constant, and since the right-hand side of the second equation depends on $\mathbf{I}$ alone, $\dot{\phi}$ is constant too. Hence

$$\phi(t) = \phi_0 + \Omega t, \quad \Omega = \frac{\partial \tilde{H}}{\partial \mathbf{I}}(\mathbf{I}_0),$$

and we have solved the system completely. All the difficulty has been pushed into finding the transformation.

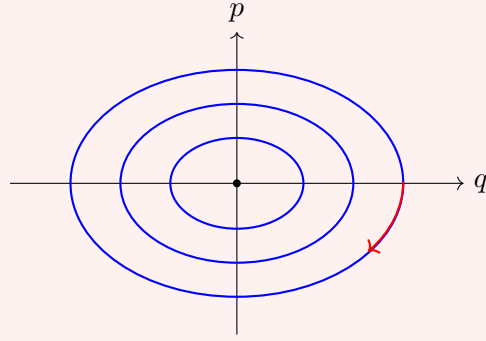
Before proving anything, let us see what the transformation looks like in the simplest possible example, where we can produce it by divine inspiration and then reverse-engineer where it came from.

Example 32.1.26 (Harmonic oscillator, first pass)

Take the harmonic oscillator on $M = \mathbb{R}^2$, with

$$H(q, p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2.$$

This is a two-dimensional system, so it is automatically integrable, with the single first integral $f_1 = H$. The level sets of H are ellipses:



Each of these is homeomorphic to a circle S^1 , and this is not an accident – it is the first half of the Arnold–Liouville theorem in miniature.

Now introduce new coordinates (ϕ, I) by

$$q = \sqrt{\frac{2I}{\omega}} \sin \phi, \quad p = \sqrt{2I\omega} \cos \phi.$$

One can check by computing the Jacobian that this is canonical. In these coordinates,

$$\tilde{H}(\phi, I) = \frac{1}{2} \cdot 2I\omega \cos^2 \phi + \frac{1}{2}\omega^2 \cdot \frac{2I}{\omega} \sin^2 \phi = \omega I.$$

There is no ϕ ! So Hamilton's equations read

$$\dot{\phi} = \frac{\partial \tilde{H}}{\partial I} = \omega, \quad \dot{I} = -\frac{\partial \tilde{H}}{\partial \phi} = 0,$$

giving $\phi(t) = \phi_0 + \omega t$ and $I(t) = I_0$. The oscillator goes round its ellipse at constant angular rate ω , which is of course the right answer.

Where could that transformation have come from? Here is a clue. Let us compute, for no obvious reason, the area enclosed by a level set of H , divided by 2π :

$$\begin{aligned} \frac{1}{2\pi} \oint p \, dq &= \frac{1}{2\pi} \int_0^{2\pi} p(\phi, I) \left(\frac{\partial q}{\partial \phi} d\phi + \frac{\partial q}{\partial I} dI \right) \\ &= \frac{1}{2\pi} \int_0^{2\pi} p(\phi, I) \frac{\partial q}{\partial \phi} d\phi \\ &= \frac{1}{2\pi} \int_0^{2\pi} \sqrt{2I\omega} \cos \phi \cdot \sqrt{\frac{2I}{\omega}} \cos \phi d\phi \\ &= \frac{2I}{2\pi} \int_0^{2\pi} \cos^2 \phi d\phi = I, \end{aligned}$$

using $dI = 0$ along a level set. So the new momentum I is nothing but the enclosed area over 2π – a quantity we could have computed knowing nothing whatsoever about the transformation.

There are two lessons here, and they are exactly the two halves of the theorem: the motion takes place on a circle, and the right momentum coordinate is $\frac{1}{2\pi} \oint p \, dq$.

Theorem 32.1.27 (Arnold–Liouville)

Let (M, H) be an integrable $2n$ -dimensional Hamiltonian system with independent, involutive first integrals $f_1, \dots, f_n$, where $f_1 = H$. For $\mathbf{c} \in \mathbb{R}^n$ set

$$M_{\mathbf{c}} = \{(\mathbf{q}, \mathbf{p}) \in M : f_i(\mathbf{q}, \mathbf{p}) = c_i, i = 1, \dots, n\}.$$

Then:

- (i) $M_{\mathbf{c}}$ is a smooth n -dimensional surface in M , invariant under the flow. If $M_{\mathbf{c}}$ is compact and connected, it is diffeomorphic to the n -torus

$$T^n = \underbrace{S^1 \times \dots \times S^1}_n.$$

- (ii) If $M_{\mathbf{c}}$ is compact and connected, then locally there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\boldsymbol{\phi}, \mathbf{I})$ to **action–angle coordinates**, in which the ϕ_k are angular coordinates on $M_{\mathbf{c}}$, the actions I_k are first integrals, and $\tilde{H}$ is independent of $\boldsymbol{\phi}$. Consequently

$$\dot{\mathbf{I}} = 0, \quad \dot{\boldsymbol{\phi}} = \frac{\partial \tilde{H}}{\partial \mathbf{I}} = \text{constant}.$$

The first part is pure differential geometry, and the second is where the work is. We give the argument in outline; readers who have not met the preimage theorem or the orbit–stabiliser theorem in this setting may take the geometric steps on faith.

Proof sketch. Part (i). That $M_{\mathbf{c}}$ is a smooth n -dimensional surface is the preimage theorem (a consequence of the inverse function theorem from IB Analysis and Topology) applied to $\mathbf{f} = (f_1, \dots, f_n) : M \rightarrow \mathbb{R}^n$. What makes it apply is precisely the independence of the f_i , which says that $\mathbf{c}$ is a regular value.

Next we show that $M_{\mathbf{c}}$ is a torus when compact and connected. Consider the Hamiltonian vector fields

$$\mathbf{V}_{f_i} = J \frac{\partial f_i}{\partial \mathbf{x}}.$$

We claim each is tangent to $M_{\mathbf{c}}$. It suffices to show that the derivative of each f_j along $\mathbf{V}_{f_i}$ vanishes, and indeed

$$\left(\mathbf{V}_{f_i} \cdot \frac{\partial}{\partial \mathbf{x}} \right) f_j = \frac{\partial f_j}{\partial \mathbf{x}} \cdot J \frac{\partial f_i}{\partial \mathbf{x}} = \{f_j, f_i\} = 0$$

by involutivity. So the flows g_i^t generated by the $\mathbf{V}_{f_i}$ map $M_{\mathbf{c}}$ into itself. Moreover these flows commute, since

$$[\mathbf{V}_{f_i}, \mathbf{V}_{f_j}] = -\mathbf{V}_{\{f_i, f_j\}} = -\mathbf{V}_0 = 0.$$

So writing $\mathbf{t} = (t_1, \dots, t_n) \in \mathbb{R}^n$ and

$$g^{\mathbf{t}} = g_1^{t_1} g_2^{t_2} \dots g_n^{t_n},$$

commutativity gives $g^{\mathbf{t}_1 + \mathbf{t}_2} = g^{\mathbf{t}_1} g^{\mathbf{t}_2}$, so we have an action of the group $(\mathbb{R}^n, +)$ on $M_{\mathbf{c}}$.

Fix $\mathbf{x} \in M_{\mathbf{c}}$ and let

$$\text{Stab}(\mathbf{x}) = \{\mathbf{t} \in \mathbb{R}^n : g^{\mathbf{t}}\mathbf{x} = \mathbf{x}\}.$$

By the orbit–stabiliser theorem, $\mathbf{t} \mapsto g^{\mathbf{t}}\mathbf{x}$ descends to a bijection $\mathbb{R}^n / \text{Stab}(\mathbf{x}) \rightarrow \text{Orb}(\mathbf{x})$, and one checks that the orbit of $\mathbf{x}$ is the connected component of $\mathbf{x}$ in $M_{\mathbf{c}}$ – so if $M_{\mathbf{c}}$ is connected, the orbit is everything. Now $\text{Stab}(\mathbf{x})$ is a subgroup of $\mathbb{R}^n$, and (provided the flows are non-degenerate) it is discrete, hence isomorphic to $\mathbb{Z}^k$ for some $0 \leq k \leq n$. Therefore

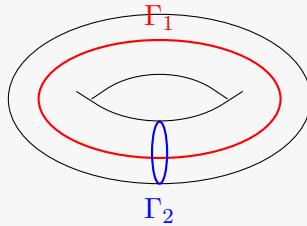
$$M_{\mathbf{c}} \cong \mathbb{R}^n / \mathbb{Z}^k \cong T^k \times \mathbb{R}^{n-k}.$$

If $M_{\mathbf{c}}$ is compact there can be no $\mathbb{R}$ factors, so $k = n$ and $M_{\mathbf{c}} \cong T^n$.

Part (ii). We construct the action–angle coordinates. For clarity of presentation we take $n = 2$; the general case is identical except that we need a higher-dimensional version of Green’s theorem, which we do not have to hand.

It is trivially easy to find *some* coordinates $(\mathbf{Q}, \mathbf{P})$ with $\mathbf{P}$ constant along the flow and $\mathbf{Q}$ angular: just set $\mathbf{P} = \mathbf{c}$ and use the diffeomorphism $M_{\mathbf{c}} \cong T^n$ to get angles. The problem is that such a transformation will almost never be canonical. So we must be cleverer about the choice of $\mathbf{P}$.

On the torus T^n there are n topologically distinct ‘basic’ loops $\Gamma_1, \dots, \Gamma_n$ – under the identification $M_{\mathbf{c}} \cong S^1 \times \dots \times S^1$, the loop Γ_j runs once around the j th factor and is constant on the others. For $n = 2$ these are the two obvious circles:



We now define the **actions**

$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q},$$

generalising the formula we found for the harmonic oscillator. The first thing to check is that this is well defined, since Γ_j is really a whole homotopy class of loops rather than a single one. So suppose Γ_2 and Γ'_2 are two such loops, bounding between them an annular region A on the torus.

On $M_{\mathbf{c}}$ the equations $f_i(\mathbf{q}, \mathbf{p}) = c_i$ hold identically, and we assume we may invert them locally to write $\mathbf{p} = \mathbf{p}(\mathbf{q}, \mathbf{c})$; the condition for this is $\det(\partial f_i / \partial p_j) \neq 0$. Differentiating the identity $f_i(\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c})) = c_i$ with respect to q_k gives

$$\frac{\partial f_i}{\partial q_k} + \frac{\partial f_i}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} = 0$$

on $M_{\mathbf{c}}$. Now use involutivity: on $M_{\mathbf{c}}$,

$$0 = \{f_i, f_j\}$$

$$\begin{aligned}
&= \frac{\partial f_i}{\partial q_k} \frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \frac{\partial f_j}{\partial q_k} \\
&= \left(-\frac{\partial f_i}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} \right) \frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k} \left(-\frac{\partial f_j}{\partial p_\ell} \frac{\partial p_\ell}{\partial q_k} \right) \\
&= \frac{\partial f_i}{\partial p_k} \left(\frac{\partial p_\ell}{\partial q_k} - \frac{\partial p_k}{\partial q_\ell} \right) \frac{\partial f_j}{\partial p_\ell},
\end{aligned}$$

where in the last line we relabelled dummy indices in the first term. Since the matrices $\partial f_i/\partial p_k$ are invertible, the matrix in the middle must vanish:

$$\frac{\partial p_\ell}{\partial q_k} = \frac{\partial p_k}{\partial q_\ell}.$$

For $n = 2$ the only content of this is $\partial p_1/\partial q_2 = \partial p_2/\partial q_1$. But then by Green's theorem

$$\left(\oint_{\Gamma_2} - \oint_{\Gamma'_2} \right) \mathbf{p} \cdot d\mathbf{q} = \oint_{\partial A} \mathbf{p} \cdot d\mathbf{q} = \iint_A \left(\frac{\partial p_2}{\partial q_1} - \frac{\partial p_1}{\partial q_2} \right) dq_1 dq_2 = 0.$$

So I_j depends only on the homotopy class of Γ_j , and hence $\mathbf{I} = \mathbf{I}(\mathbf{c})$ is a function of $\mathbf{c}$ alone. These are our new momenta. We assume $\mathbf{I}(\mathbf{c})$ is invertible, so that we may also write $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

It remains to produce the conjugate angles, and for this we use a generating function. Pick a base point $\mathbf{x}_0 \in M_{\mathbf{c}}$ and set

$$S(\mathbf{q}, \mathbf{I}) = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I})) \cdot d\mathbf{q}',$$

where $\mathbf{x} = (\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c}(\mathbf{I})))$ and the integral is along any path in $M_{\mathbf{c}}$. Is this well defined? If we deform the path slightly, sweeping out a region B , then S changes by $\oint_{\partial B} \mathbf{p} \cdot d\mathbf{q}$, which vanishes by exactly the Green's theorem computation above. But we are on a torus, and there are paths that cannot be deformed into one another: a path may wind around one of the cycles Γ_j on its way. Adding such a loop changes S by $\oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q} = 2\pi I_j$. So the total ambiguity is

$$S(\mathbf{q}, \mathbf{I}) \mapsto S(\mathbf{q}, \mathbf{I}) + 2\pi m_j I_j, \quad m_j \in \mathbb{Z},$$

and nothing worse. Differentiating with respect to $\mathbf{I}$, we conclude that

$$\phi = \frac{\partial S}{\partial \mathbf{I}}$$

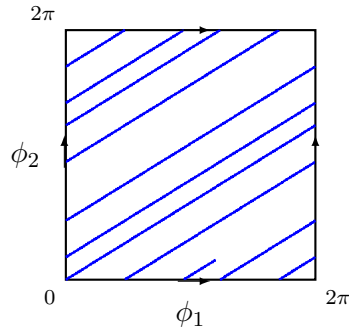
is well defined *modulo* 2π . These are the angles, and the ambiguity is exactly the ambiguity one expects of an angular coordinate.

Finally we should check that S really is a generating function in the sense of the previous subsection, i.e. that $\partial S/\partial \mathbf{q} = \mathbf{p}$. Writing $S = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{x}'$ with $\mathbf{F} = (\mathbf{p}, \mathbf{0})$, the fundamental theorem of calculus gives $\partial S/\partial \mathbf{x} = \mathbf{F}$, and in particular $\partial S/\partial \mathbf{q} = \mathbf{p}$ as required.

So S generates a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\boldsymbol{\phi}, \mathbf{I})$. Since $\mathbf{I} = \mathbf{I}(\mathbf{c}) = \mathbf{I}(\mathbf{f}(\mathbf{x}))$ is a function of the first integrals alone, we have $\dot{\mathbf{I}} = 0$; and since $\dot{\mathbf{I}} = -\partial\tilde{H}/\partial\boldsymbol{\phi}$, the new Hamiltonian is independent of $\boldsymbol{\phi}$. Integrating up,

$$\boldsymbol{\phi}(t) = \boldsymbol{\phi}_0 + \boldsymbol{\Omega}t, \quad \mathbf{I}(t) = \mathbf{I}_0, \quad \boldsymbol{\Omega} = \frac{\partial\tilde{H}}{\partial\mathbf{I}}(\mathbf{I}_0). \quad \square$$

The picture that emerges is a very appealing one. Phase space is foliated by invariant tori $M_{\mathbf{c}}$, one for each value of the conserved quantities, and on each torus the motion is uniform straight-line motion in the angle variables – a winding line, going round the j th cycle at frequency Ω_j . Drawing the torus in the usual way as a square with opposite edges identified, the trajectory is simply a straight line of slope Ω_2/Ω_1 :



If the frequencies $\Omega_1, \dots, \Omega_n$ are rationally related the trajectory closes up and the motion is periodic; if they are rationally independent the trajectory winds around forever, densely filling the torus, and the motion is called **quasi-periodic**.

§32.1.6 Action–Angle Variables

The proof above is constructive, and gives us a recipe. To integrate up an integrable Hamiltonian system:

- (1) Identify the n cycles $\Gamma_1, \dots, \Gamma_n$ on the invariant torus $M_{\mathbf{c}}$, and solve $f_i(\mathbf{q}, \mathbf{p}) = c_i$ for $\mathbf{p} = \mathbf{p}(\mathbf{q}, \mathbf{c})$.
- (2) Compute the actions

$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot d\mathbf{q},$$

and invert to obtain $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

- (3) Form the generating function

$$S(\mathbf{q}, \mathbf{I}) = \int^{\mathbf{q}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I})) \cdot d\mathbf{q}'$$

and read off the angles $\boldsymbol{\phi} = \partial S / \partial \mathbf{I}$.

- (4) Write $\tilde{H} = \tilde{H}(\mathbf{I})$ and read off the frequencies $\boldsymbol{\Omega} = \partial\tilde{H}/\partial\mathbf{I}$.

In practice step (4) is often all one wants: the frequencies of an integrable system can be extracted without ever computing the angles, since $\tilde{H}(\mathbf{I})$ is determined by step (2) alone.

Let us run the recipe on the harmonic oscillator, and see the transformation appear without any divine intervention.

Example 32.1.28 (Harmonic oscillator, second pass)

With $H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$ we have

$$M_c = \{(q, p) : \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 = c\},$$

an ellipse, which is indeed diffeomorphic to $T^1 = S^1$ as the theorem promises. There is only one cycle, the ellipse itself. Solving for p ,

$$p = p(q, c) = \pm \sqrt{2c - \omega^2 q^2}.$$

The action is the enclosed area over 2π . The ellipse $\frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 = c$ has semi-axes $\sqrt{2c}/\omega$ and $\sqrt{2c}$, so its area is $\pi \cdot \frac{\sqrt{2c}}{\omega} \cdot \sqrt{2c} = 2\pi c/\omega$ and

$$I = \frac{1}{2\pi} \oint p \, dq = \frac{c}{\omega}.$$

Inverting, $c = c(I) = \omega I$, so immediately

$$\tilde{H} = c = \omega I, \quad \Omega = \frac{d\tilde{H}}{dI} = \omega.$$

The frequency is ω , independent of amplitude – the defining feature of the harmonic oscillator.

For the angle, take the base point to be $q = 0$ with $p > 0$. Then

$$S(q, I) = \int_0^q \sqrt{2\omega I - \omega^2 q'^2} \, dq',$$

and differentiating under the integral sign,

$$\phi = \frac{\partial S}{\partial I} = \int_0^q \frac{\omega \, dq'}{\sqrt{2\omega I - \omega^2 q'^2}} = \int_0^q \frac{dq'}{\sqrt{2I/\omega - q'^2}} = \sin^{-1} \left(\sqrt{\frac{\omega}{2I}} q \right).$$

Sure enough, this is well defined only up to multiples of 2π . Inverting,

$$q = \sqrt{\frac{2I}{\omega}} \sin \phi, \quad p = \frac{\partial S}{\partial q} = \sqrt{2\omega I - \omega^2 q^2} = \sqrt{2I\omega} \cos \phi,$$

which is exactly the transformation we pulled out of thin air earlier.

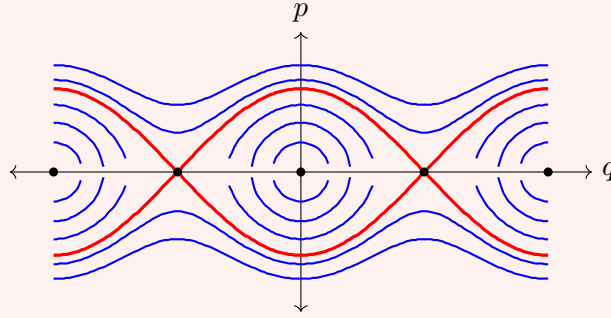
The next example is the one to remember, because it is the simplest system in which the frequency actually depends on the action.

Example 32.1.29 (The simple pendulum)

A pendulum of unit mass and unit length in unit gravity has Hamiltonian

$$H(q, p) = \frac{1}{2}p^2 - \cos q,$$

where q is the angle from the downward vertical. The phase portrait is the following, with the equilibria marked: centres at $q = 0, \pm 2\pi, \dots$ (the pendulum hanging down) and saddles at $q = \pm \pi$ (the pendulum balanced upright).



There are three qualitatively different kinds of motion, separated by the thick red curve.

- (i) For $-1 < E < 1$ the level set $\{H = E\}$ is a closed curve encircling the origin. This is **libration**: the pendulum swings back and forth about the bottom, with q bounded by the turning points $\pm q_0$ where $\cos q_0 = -E$.
- (ii) For $E > 1$ the level set consists of two curves, one with $p > 0$ throughout and one with $p < 0$. This is **rotation**: the pendulum has enough energy to go over the top, and q increases (or decreases) without bound.
- (iii) At $E = 1$ we have the **separatrix**, $p = \pm 2 \cos(q/2)$, on which the pendulum takes infinite time to creep up to the inverted position $q = \pi$.

Note that the phase space here is really a cylinder rather than a plane, since q and $q + 2\pi$ describe the same configuration; on the cylinder the libration curves are circles, and so are the rotation curves, but the two families of circles are not homotopic to one another.

Consider the librating region. Solving for p ,

$$p = \pm \sqrt{2(E + \cos q)},$$

and the action is

$$I(E) = \frac{1}{2\pi} \oint p \, dq = \frac{1}{\pi} \int_{-q_0}^{q_0} \sqrt{2(E + \cos q)} \, dq,$$

where $q_0 = \cos^{-1}(-E)$ and the factor of two from the upper and lower halves of the curve has been absorbed. Using $\cos q = 1 - 2 \sin^2(q/2)$ and writing $k^2 = (1 + E)/2 \in (0, 1)$, this becomes

$$I(E) = \frac{4}{\pi} \int_0^{q_0} \sqrt{k^2 - \sin^2(q/2)} \, dq = \frac{8}{\pi} [\mathcal{E}(k) - (1 - k^2)\mathcal{K}(k)],$$

where $\mathcal{K}$ and $\mathcal{E}$ are the complete elliptic integrals of the first and second kind. This is not an elementary function, and that is the point: the pendulum is integrable, but its action-angle variables involve elliptic functions. The frequency is

$$\Omega(I) = \frac{dE}{dI} = \left(\frac{dI}{dE} \right)^{-1} = \frac{\pi}{2\mathcal{K}(k)},$$

using $dI/dE = \frac{2}{\pi}\mathcal{K}(k)$, which follows from differentiating the integral for I under the integral sign.

It is worth sanity-checking the two limits. As $E \rightarrow -1^+$ we have $k \rightarrow 0$, and $\mathcal{K}(0) = \pi/2$, so $\Omega \rightarrow 1$: small oscillations have unit frequency, in agreement with linearising $\ddot{q} = -\sin q \approx -q$. As $E \rightarrow 1^-$ we have $k \rightarrow 1$ and $\mathcal{K}(k) \rightarrow \infty$, so $\Omega \rightarrow 0$: the period diverges as we approach the separatrix, which is the statement that the pendulum takes forever to reach the top. The pendulum is thus the standard example of an integrable system whose frequency depends on its amplitude.

Our final example has two degrees of freedom, so that we get an honest two-torus and two genuinely different cycles.

Example 32.1.30 (A particle in a central potential)

Consider a particle of unit mass moving in a plane under a central potential $U(r)$. Using polar coordinates (r, θ) , the Hamiltonian is

$$H = \frac{1}{2} \left(p_r^2 + \frac{p_\theta^2}{r^2} \right) + U(r),$$

with the canonical pairs (r, p_r) and (θ, p_θ) . This is a four-dimensional phase space, so we need two first integrals in involution, and we have them: $f_1 = H$ and $f_2 = p_\theta$, the angular momentum. They are in involution because H does not depend on θ :

$$\{p_\theta, H\} = -\frac{\partial H}{\partial \theta} = 0.$$

So the system is integrable. Fix $\mathbf{c} = (E, L)$. The invariant surface $M_{\mathbf{c}}$ is described by

$$p_\theta = L, \quad p_r = \pm \sqrt{2(E - U(r)) - \frac{L^2}{r^2}},$$

and provided the effective potential $U_{\text{eff}}(r) = U(r) + L^2/2r^2$ has a well, so that r oscillates between turning points r_- and r_+ , this really is a torus: θ runs around a circle, and (r, p_r) traces a closed curve in its own plane.

The two cycles are then evident. Taking Γ_1 to be the loop on which θ increases by 2π at fixed r , and Γ_2 the loop on which (r, p_r) traverses its closed curve at fixed θ , we get

$$I_1 = \frac{1}{2\pi} \oint_{\Gamma_1} p_\theta \, d\theta = L, \quad I_2 = \frac{1}{2\pi} \oint_{\Gamma_2} p_r \, dr = \frac{1}{\pi} \int_{r_-}^{r_+} \sqrt{2(E - U(r)) - \frac{L^2}{r^2}} \, dr.$$

The first action is just the angular momentum, as one would hope. The second is the **radial action**, and computing it gives E as a function of (I_1, I_2) and hence both frequencies at once.

For the Kepler potential $U(r) = -1/r$ this integral can be done by contour methods, with the answer

$$I_2 = -I_1 + \frac{1}{\sqrt{-2E}}, \quad \text{i.e.} \quad \tilde{H} = E = -\frac{1}{2(I_1 + I_2)^2}.$$

The remarkable thing here is that $\tilde{H}$ depends on I_1 and I_2 only through the combination $I_1 + I_2$, so that

$$\Omega_1 = \frac{\partial \tilde{H}}{\partial I_1} = \frac{\partial \tilde{H}}{\partial I_2} = \Omega_2 = \frac{1}{(I_1 + I_2)^3}.$$

The two frequencies are equal, so the winding line on the torus closes up after one circuit: Kepler orbits are closed curves, which is the statement that planetary orbits are ellipses that do not precess. This **degeneracy** is special to the $1/r$ potential (and to the harmonic oscillator), and its breaking by relativistic corrections is what produces the precession of the perihelion of Mercury.

This is a good point to pause and take stock. We have a complete and satisfying theory: a Hamiltonian system with enough commuting conserved quantities lives on invariant tori, and on those tori the motion is trivial. For the rest of the course we ask whether anything like this survives when we pass to partial differential equations. The answer, remarkably, is yes – but almost every ingredient will have to be reinvented.

§32.2 Integrable PDEs and Solitons

For the rest of the course we turn to partial differential equations. It is helpful to think of a PDE as an infinite-dimensional dynamical system: instead of finitely many coordinates $x_i(t)$ we have a whole function $u(x, t)$, with the spatial variable x playing the role of the index i .

What should we expect of an integrable PDE? If a $2n$ -dimensional ODE is integrable when it has n first integrals, and our phase space is now infinite-dimensional, then – since half of infinity is still infinity – we should expect an integrable PDE to possess *infinitely many* conserved quantities. We should also expect some magic transformation, analogous to the passage to action–angle variables, which turns the equation into something trivial. Both of these expectations turn out to be correct. What is much less obvious is how to find the transformation, and we will spend the whole of Section 3 constructing it for one particular equation.

We will not attempt to define integrability for PDEs. Nobody has managed to do so in a way that is both precise and useful, and we will be content to work with examples.

§32.2.1 Dispersion and Nonlinearity

The equation at the centre of this course is the Korteweg–de Vries equation

$$u_t + u_{xxx} - 6uu_x = 0.$$

Before meeting it properly, it pays to understand the two terms u_{xxx} and $-6uu_x$ separately, because each on its own does something violent to a wave, and the whole point of KdV is that the two violences cancel.

Example 32.2.1 (Dispersion)

Consider the *linear* equation

$$u_t + u_{xxx} = 0.$$

Being linear and constant-coefficient, it is solved by Fourier methods (IB Methods). Plane wave modes $u = e^{i(kx - \omega t)}$ solve it provided ω satisfies the **dispersion relation**

$$\omega = \omega(k) = -k^3.$$

The general solution is then a superposition,

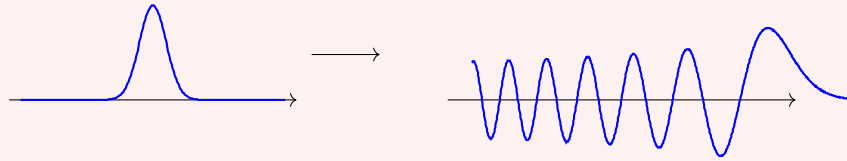
$$u(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega(k)t)} dk,$$

with $A(k) = \hat{u}_0(k)/2\pi$ determined by the initial data.

Writing a single mode as

$$u(x, t) = \exp\left(ik\left(x - \frac{\omega(k)}{k}t\right)\right),$$

we see that it travels with phase speed $\omega/k = -k^2$. The crucial point is that this depends on k : modes of different wavelength travel at different speeds. So an initial profile, which is a superposition of many modes, does not translate rigidly – it spreads out, with the short wavelengths racing off in one direction and the long wavelengths lagging behind. This is **dispersion**, and it is the work of the u_{xxx} term. A localised initial bump decays into an oscillatory wave train:



Had we written $u_t + u_x = 0$ instead, the dispersion relation would be $\omega = k$, the phase speed would be 1 for every mode, and the initial profile would translate rigidly. It is the *third* derivative that does the damage.

Example 32.2.2 (Wave breaking)

Now consider the *nonlinear* equation

$$u_t - 6uu_x = 0.$$

This looks worse, being nonlinear, but in fact it can be solved exactly for any initial data $u(x, 0) = f(x)$ by the method of characteristics. The characteristics are the curves along which u is constant, and since the equation says u is transported at speed $-6u$, they are straight lines. The solution is given implicitly by

$$u(x, t) = f(\xi), \quad \xi = x - 6tf(\xi).$$

This is fine as long as the implicit relation can be solved for ξ . Let us see when it fails. Differentiating,

$$u_x = f'(\xi) \frac{\partial \xi}{\partial x}, \quad \frac{\partial \xi}{\partial x} = 1 - 6tf'(\xi) \frac{\partial \xi}{\partial x},$$

so

$$\frac{\partial \xi}{\partial x} = \frac{1}{1 + 6tf'(\xi)}, \quad u_x = \frac{f'(\xi)}{1 + 6tf'(\xi)}.$$

Whenever f' takes a negative value somewhere, this blows up at the finite time $t = -1/(6f'(\xi))$: the slope becomes infinite, and after that the implicit formula would give a multi-valued ‘solution’.



This is **wave breaking**, and it is what the nonlinear term $-6uu_x$ does on its own: taller parts of the wave move faster, overtake the parts in front of them, and the wave topples over.

So dispersion spreads a wave out and destroys it, and nonlinearity steepens a wave and destroys it. What happens if we include both?

§32.2.2 The Korteweg–de Vries Equation

Definition 32.2.3 (KdV equation)

The **Korteweg–de Vries equation**, or **KdV equation**, is

$$u_t + u_{xxx} - 6uu_x = 0.$$

The equation has a distinguished history. In 1834 the Scottish engineer John Scott Russell was watching a boat being drawn along a canal by a pair of horses. When the boat stopped suddenly, the water it had been pushing rolled forward as a single smooth heap, and Russell – to his eternal credit – got on his horse and chased it for a mile or two along the canal, watching it travel without change of shape or speed. He called it the ‘wave of translation’. It took until 1895 for Korteweg and de Vries to write down the equation governing it.

Let us see, at least at the level of scaling, where the equation comes from.

Example 32.2.4 (Shallow water waves)

Consider small-amplitude surface waves on a layer of water of undisturbed depth h , and let $u(x, t)$ denote the surface elevation. Linear water wave theory gives the exact dispersion relation

$$\omega^2 = gk \tanh(kh).$$

For long waves, $kh \ll 1$, we expand $\tanh z = z - z^3/3 + O(z^5)$ to get

$$\omega^2 = gk \left(kh - \frac{(kh)^3}{3} + \dots \right) = c^2 k^2 \left(1 - \frac{(kh)^2}{3} + \dots \right), \quad c = \sqrt{gh},$$

so that for right-moving waves

$$\omega = ck \left(1 - \frac{k^2 h^2}{6} + \dots \right) = ck - \frac{ch^2}{6} k^3 + \dots.$$

A wave packet whose modes obey $\omega = ck - \beta k^3$ satisfies the linear PDE

$$u_t + cu_x + \beta u_{xxx} = 0, \quad \beta = \frac{ch^2}{6},$$

since $\partial_t \rightarrow -i\omega$ and $\partial_x \rightarrow ik$.

Now put back the leading nonlinearity. A finite-amplitude long wave rides on water of local depth $h + u$, so it travels a little faster where it is taller; carrying out the expansion properly gives a correction to the propagation speed of $\frac{3}{2}cu/h$, and hence a term γuu_x with $\gamma = 3c/2h$. Putting both effects together,

$$u_t + cu_x + \gamma uu_x + \beta u_{xxx} = 0.$$

Finally we normalise. Passing to a frame moving at speed c kills the cu_x term, and setting $u = Av(y, s)$ with $y = x - ct$ and $s = t/\beta$ gives

$$v_s + \frac{\gamma A}{\beta} vv_y + v_{yyy} = 0.$$

Choosing $A = -6\beta/\gamma$ produces

$$v_s + v_{yyy} - 6vv_y = 0,$$

which is KdV. So the equation is not some arbitrary combination of terms: it is what one gets from *any* weakly nonlinear, weakly dispersive, unidirectional wave system at leading order. This universality is why KdV turns up in plasma physics, in nonlinear optics, in blood flow and in lattice dynamics as well as in canals.

Some structural remarks are worth making before we solve anything. The KdV equation is invariant under

- (i) translations $x \mapsto x + a$, $t \mapsto t + b$;
- (ii) the scaling $u \mapsto \lambda^2 u$, $x \mapsto \lambda^{-1} x$, $t \mapsto \lambda^{-3} t$;
- (iii) the Galilean boost $u(x, t) \mapsto u(x - 6ct, t) + c$ for constant c .

The third is easy to miss and will matter later: adding a constant to u is the same as moving to a uniformly translating frame. The scaling symmetry tells us that there is really only a one-parameter family of solitary waves, all obtainable from one another by rescaling.

§32.2.3 Travelling Waves and the One-Soliton Solution

We now look for travelling wave solutions of KdV, and we shall find Russell's wave of translation. Let

$$u(x, t) = f(\xi), \quad \xi = x - ct,$$

for some speed c , and impose the boundary conditions $f, f', f'' \rightarrow 0$ as $|\xi| \rightarrow \infty$ (we want a localised wave on still water). Substituting,

$$-cf' + f''' - 6ff' = 0.$$

Every term is a derivative, so we can integrate once:

$$-cf + f'' - 3f^2 = A$$

for a constant A , and letting $\xi \rightarrow \infty$ forces $A = 0$. Now multiply by the integrating factor f' :

$$-cf f' + f'' f' - 3f^2 f' = 0,$$

and integrate again, noting that $f'' f' = \frac{1}{2} \frac{d}{d\xi} (f')^2$:

$$-\frac{c}{2} f^2 + \frac{1}{2} (f')^2 - f^3 = B,$$

and again $B = 0$ by the boundary conditions. Rearranging,

$$(f')^2 = 2f^3 + cf^2 = f^2(2f + c). \quad (\dagger)$$

This is a first-order ODE which we could grind through by separation of variables. It is quicker to guess. Since $(f')^2 \geq 0$ we need $2f + c \geq 0$ wherever $f \neq 0$; if we look for a solution with $f < 0$ (which is the natural sign for KdV in this normalisation) this forces $c > 0$ and $-c/2 \leq f \leq 0$. A function which decays at both ends and has a minimum of $-c/2$ suggests sech^2 , so we try

$$f(\xi) = -\frac{c}{2} \text{sech}^2(\alpha \xi)$$

and see what α must be. Using $\frac{d}{d\theta} \text{sech } \theta = -\text{sech } \theta \tanh \theta$,

$$f'(\xi) = c\alpha \text{sech}^2(\alpha \xi) \tanh(\alpha \xi), \quad (f')^2 = c^2 \alpha^2 \text{sech}^4 \tanh^2.$$

On the other side of $(\dagger)$,

$$f^2(2f + c) = \frac{c^2}{4} \text{sech}^4 \cdot c(1 - \text{sech}^2) = \frac{c^3}{4} \text{sech}^4 \tanh^2,$$

using $1 - \text{sech}^2 = \tanh^2$. So $(\dagger)$ holds precisely when $c^2 \alpha^2 = c^3/4$, that is $\alpha = \frac{1}{2} \sqrt{c}$. We have found our solution.

Proposition 32.2.5 (One-soliton solution)

For any $c > 0$, the function

$$u(x, t) = -\frac{c}{2} \text{sech}^2 \left(\frac{\sqrt{c}}{2} (x - ct - x_0) \right)$$

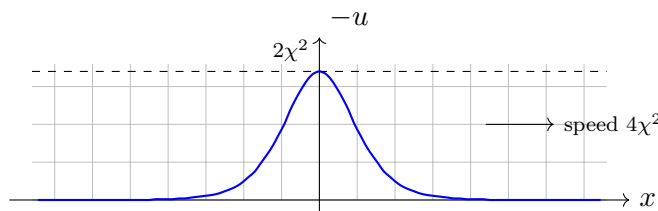
solves the KdV equation. Writing $c = 4\chi^2$ with $\chi > 0$, this takes the standard form

$$u(x, t) = -2\chi^2 \text{sech}^2 (\chi(x - 4\chi^2 t - x_0)).$$

We call this a **soliton**. Look at what it says: the amplitude is $2\chi^2$, the speed is $4\chi^2$, and the width is $1/\chi$. So

taller solitons are faster and narrower.

This single fact is responsible for almost everything interesting that follows.



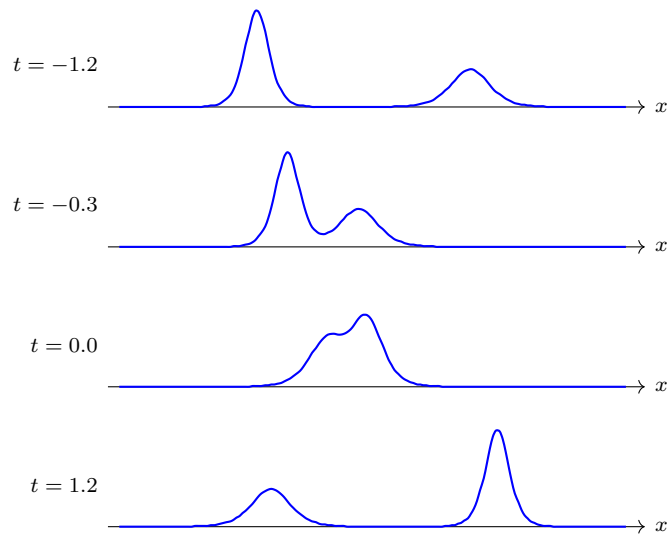
The balance is delicate. The dispersive term wants to spread the wave out; the nonlinear term wants to steepen its leading edge. For this particular profile, at this particular relation between height and speed, the two exactly cancel and the wave propagates unchanged forever.

Remark. The parameter χ is not a typographical accident. When we come to the inverse scattering transform we will find that $-\chi^2$ is precisely the eigenvalue of the Schrödinger operator $-\partial_x^2 + u$ associated with this potential. A one-soliton solution is the potential with exactly one bound state.

§32.2.4 Two Solitons, and Why Solitons Are Particles

Now for the interesting question. Suppose we start with two solitons, well separated, the taller one behind. Since the taller one is faster, it will catch up with the shorter one, and at some point they will overlap. The KdV equation is nonlinear, so we cannot simply add the two solutions; and we have every reason to expect that the collision will be messy, generating radiation and destroying both waves.

Here is what actually happens. (We will derive the exact formula in Section 3; for now we simply plot it, with $\chi_1 = 1$ and $\chi_2 = 1.6$. The window follows the interaction – both humps are in fact moving steadily to the right, and we plot $-u$ so that the solitons point upwards.)



The two solitons pass straight through one another and re-emerge with exactly their original shapes and speeds. Nothing is radiated away, nothing is lost. The only trace of the interaction is a **phase shift**: the taller soliton ends up slightly ahead of where it would have been, and the shorter one slightly behind. One can show that these shifts are

$$\Delta_2 = +\frac{1}{\chi_2} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1}, \quad \Delta_1 = -\frac{1}{\chi_1} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1},$$

for the taller and shorter solitons respectively, so the taller one gains and the shorter one loses.

This is extraordinary behaviour for a nonlinear PDE, and it is the reason for the name: a ‘soliton’ is a solitary wave which behaves like a *particle* in collisions. The phenomenon was discovered numerically by Zabusky and Kruskal in 1965, who were simulating KdV as a continuum model for the Fermi–Pasta–Ulam problem and were startled to find their

pulses surviving collisions intact. It was their work that started the modern theory of integrable systems.

Note also that in the middle panel, at the moment of collision, the combined profile is not the sum of the two solitons, nor even taller than the larger one – the interaction is genuinely nonlinear. Superposition is violated at every instant; it is only the asymptotics that are clean.

§32.2.5 Conservation Laws

If solitons survive collisions, something must be preventing energy from leaking into other modes. The something is an infinite family of conservation laws. Let us find the first few by hand.

We will always assume that u and its derivatives decay rapidly as $|x| \rightarrow \infty$, so that we may integrate by parts freely and discard boundary terms.

Definition 32.2.6 (Conservation law)

A **conservation law** for an evolution equation is a relation of the form

$$\frac{\partial T}{\partial t} + \frac{\partial X}{\partial x} = 0,$$

where the **density** T and the **flux** X are functions of x, t, u and the derivatives of u . If such a relation holds for all solutions, then

$$\frac{d}{dt} \int_{-\infty}^{\infty} T \, dx = -[X]_{-\infty}^{\infty} = 0,$$

so $\int T \, dx$ is a first integral.

Proposition 32.2.7

The KdV equation possesses the conservation laws

$$\begin{aligned} \partial_t(u) + \partial_x(u_{xx} - 3u^2) &= 0, \\ \partial_t(u^2) + \partial_x(2uu_{xx} - u_x^2 - 4u^3) &= 0, \\ \partial_t(u^3 + \tfrac{1}{2}u_x^2) + \partial_x(\cdots) &= 0. \end{aligned}$$

Hence

$$I_1 = \int u \, dx, \quad I_2 = \int u^2 \, dx, \quad I_3 = \int (u^3 + \tfrac{1}{2}u_x^2) \, dx$$

are first integrals of KdV.

Proof. The first is just KdV rewritten: $u_t = -u_{xxx} + 6uu_x = -\partial_x(u_{xx} - 3u^2)$.

For the second, multiply KdV by $2u$:

$$(u^2)_t = 2uu_t = 2u(-u_{xxx} + 6uu_x) = -2uu_{xxx} + 12u^2u_x.$$

We must write the right-hand side as $-\partial_x X$. Observe that

$$\partial_x(2uu_{xx} - u_x^2 - 4u^3) = 2u_x u_{xx} + 2u u_{xxx} - 2u_x u_{xx} - 12u^2 u_x = 2u u_{xxx} - 12u^2 u_x,$$

which is exactly $-(u^2)_t$, as required.

For the third we argue slightly differently, since guessing the flux is tedious. Set $I_3 = \int (u^3 + \frac{1}{2}u_x^2) dx$ and differentiate:

$$\begin{aligned} \frac{dI_3}{dt} &= \int (3u^2 u_t + u_x u_{xt}) dx \\ &= \int (3u^2 - u_{xx}) u_t dx, \end{aligned}$$

integrating the second term by parts. Now substitute $u_t = \partial_x(3u^2 - u_{xx})$ from the first conservation law, and write $w = 3u^2 - u_{xx}$. Then

$$\frac{dI_3}{dt} = \int w w_x dx = \frac{1}{2} \int \partial_x(w^2) dx = 0. \quad \square$$

Physically, I_1 is the mass, I_2 is the momentum, and I_3 will turn out to be the Hamiltonian for KdV, which we may reasonably call the energy. The pattern continues: there is a conserved density involving u^4 , another involving u^5 , and so on forever. Finding them one at a time by inspection quickly becomes hopeless, and in Section 3 we will produce the whole infinite family in one stroke using the scattering problem.

Remark. Not every conservation law is interesting. For instance $\partial_t(u_x) + \partial_x(u_{xx} - 3u^2)_x = 0$ gives the first integral $\int u_x dx = 0$, which is true but vacuous, since it holds for any decaying function whatsoever. We will meet a systematic supply of such trivial laws later, and we will have to learn to discard them.

§32.2.6 The Miura Transform

We finish this section with a piece of magic whose significance will only become clear in Section 3. Alongside KdV sits its close relative.

Definition 32.2.8 (mKdV)

The **modified KdV equation** is

$$v_t + v_{xxx} - 6v^2 v_x = 0.$$

The two equations look similar but are genuinely different (mKdV, for instance, is invariant under $v \mapsto -v$, which KdV is not). Nevertheless, they are related.

Proposition 32.2.9 (Miura transform)

Define the **Miura transform**

$$u = v^2 + v_x.$$

Then

$$u_t + u_{xxx} - 6uu_x = (2v + \partial_x)(v_t + v_{xxx} - 6v^2 v_x).$$

In particular, if v solves mKdV then $u = v^2 + v_x$ solves KdV.

Proof. This is a direct computation. From $u = v^2 + v_x$ we get

$$\begin{aligned} u_t &= 2vv_t + v_{xt}, \\ u_x &= 2vv_x + v_{xx}, \\ u_{xxx} &= 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx}, \\ 6uu_x &= 6(v^2 + v_x)(2vv_x + v_{xx}) = 12v^3v_x + 6v^2v_{xx} + 12vv_x^2 + 6v_xv_{xx}. \end{aligned}$$

Adding up,

$$\begin{aligned} u_t + u_{xxx} - 6uu_x &= 2vv_t + v_{xt} + 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx} \\ &\quad - 12v^3v_x - 6v^2v_{xx} - 12vv_x^2 - 6v_xv_{xx} \\ &= 2v(v_t + v_{xxx} - 6v^2v_x) + (v_{xt} + v_{xxxx} - 6v^2v_{xx} - 12vv_x^2). \end{aligned}$$

On the other hand, writing $M = v_t + v_{xxx} - 6v^2v_x$,

$$\partial_x M = v_{xt} + v_{xxxx} - 6(2vv_x \cdot v_x + v^2v_{xx}) = v_{xt} + v_{xxxx} - 12vv_x^2 - 6v^2v_{xx},$$

which is exactly the bracketed expression above. So the two sides agree. $\square$

This is already useful – it lets us manufacture solutions of KdV from solutions of mKdV. But the reason it is a landmark result is the following observation. The relation

$$u = v^2 + v_x$$

is a **Riccati equation** for v , and the standard trick for Riccati equations is to substitute

$$v = \frac{\psi_x}{\psi}.$$

Then

$$v_x = \frac{\psi_{xx}}{\psi} - \frac{\psi_x^2}{\psi^2}, \quad v^2 = \frac{\psi_x^2}{\psi^2},$$

so $u = \psi_{xx}/\psi$, that is

$$-\psi_{xx} + u\psi = 0.$$

Better still, KdV has the Galilean symmetry noted above, so if u solves KdV then so does $u + \lambda$ suitably translated; applying the Miura transform to the shifted equation gives

$$-\psi_{xx} + u\psi = \lambda\psi.$$

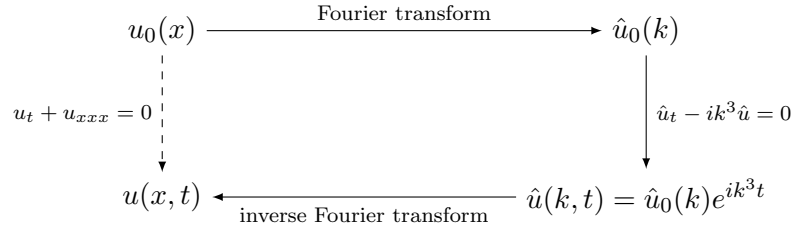
Out of nowhere, the time-independent Schrödinger equation has appeared, with the solution of KdV playing the role of the potential. Gardner, Greene, Kruskal and Miura took this hint seriously in 1967, and what came out of it is the subject of the next section.

§32.3 The Inverse Scattering Transform

We now come to the heart of the course. In IB Methods we learned to solve linear PDEs with the Fourier transform: to solve

$$u_t + u_{xxx} = 0,$$

we transform in x to get $\hat{u}_t - ik^3\hat{u} = 0$, which is a trivial ODE in t for each k ; we solve it, and transform back.



The whole scheme rests on the fact that the transform diagonalises ∂_x , converting a differential equation into an algebraic one.

The inverse scattering transform follows exactly the same three-step pattern, but with a much cleverer transform. Given $u(x, t)$, for each fixed t we regard u as the potential in a Schrödinger operator and compute its *scattering data*. The KdV equation then dictates how the scattering data evolves, and – astonishingly – the evolution is linear and explicit. Finally we invert the transform to recover u .

The remarkable thing is that every step of this process is linear, and yet the equation we are solving is not.

We build the machine in pieces: first the forward scattering problem (Section 3.1), then the inverse problem (Section 3.2), then the mechanism – Lax pairs – which links the evolution of u to the evolution of the scattering data (Sections 3.3–3.4), and finally the payoff (Sections 3.5–3.6).

§32.3.1 The Forward Scattering Problem

Throughout this section L denotes the Schrödinger operator

$$L = -\frac{\partial^2}{\partial x^2} + u(x),$$

where the potential u has compact support, so that $u = 0$ for $|x|$ sufficiently large. What we actually need is only that u decays fast enough at infinity, but assuming compact support saves us from having to be careful, and costs nothing.

For a given u , the **forward scattering problem** is to find the eigenvalues and eigenfunctions of L , that is, to solve

$$L\psi = \lambda\psi.$$

The *inverse* problem, which we address next, is to reconstruct u from that data.

There are two regimes to consider, corresponding to $\lambda > 0$ and $\lambda < 0$.

The Continuous Spectrum

Take $\lambda = k^2 > 0$ with k real. Since u vanishes for large $|x|$, any solution satisfies

$$\psi_{xx} + k^2\psi = 0$$

out there, so is asymptotically a combination of $e^{\pm ikx}$. We single out a particular solution $\varphi = \varphi(x, k)$ by the boundary condition

$$\varphi(x, k) = e^{-ikx} \quad \text{as } x \rightarrow -\infty,$$

and then there must exist coefficients $a(k)$ and $b(k)$ with

$$\varphi(x, k) = a(k)e^{-ikx} + b(k)e^{ikx} \quad \text{as } x \rightarrow +\infty.$$

Definition 32.3.1 (Reflection and transmission coefficients)

With φ as above, set

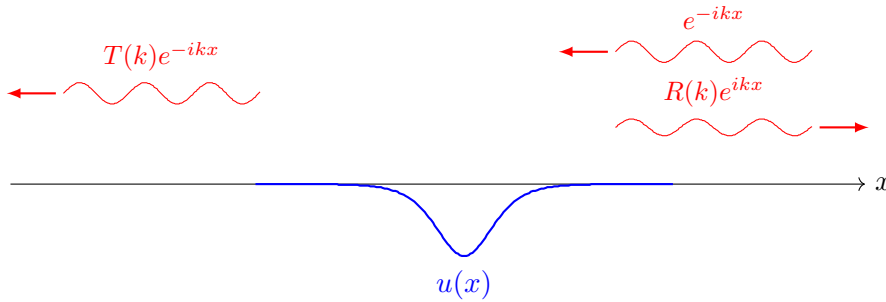
$$\Phi(x, k) = \frac{\varphi(x, k)}{a(k)}, \quad R(k) = \frac{b(k)}{a(k)}, \quad T(k) = \frac{1}{a(k)}.$$

We call R the **reflection coefficient** and T the **transmission coefficient**.

By construction,

$$\Phi(x, k) = \begin{cases} T(k)e^{-ikx} & x \rightarrow -\infty, \\ e^{-ikx} + R(k)e^{ikx} & x \rightarrow +\infty. \end{cases}$$

The physical reading is exactly the one from IB Quantum Mechanics. Thinking of e^{-ikx} as a leftward-moving wave and e^{ikx} as a rightward-moving one, we have a unit-amplitude wave incident from the right, of which a portion $R(k)e^{ikx}$ is reflected back to the right and a portion $T(k)e^{-ikx}$ is transmitted through to the left.



Conservation of probability flux (equivalently, constancy of the Wronskian) gives

$$|R(k)|^2 + |T(k)|^2 = 1,$$

which is on the first example sheet. One expects, and it is true, that $T(k) \rightarrow 1$ and $R(k) \rightarrow 0$ as $k \rightarrow \infty$: a sufficiently energetic wave barely notices the potential.

So far this is all hypothetical – we have described what a solution would look like if it existed. Let us check that it does.

The difficulty with existence questions for differential equations is that differentiation makes functions worse: it can destroy differentiability, or even continuity, so a differential operator may fling us out of whatever function space we started in. Integration, by contrast, makes functions better. So the standard move is to convert the differential equation into an integral equation, and work there.

Consider therefore the integral equation for $f = f(x, k)$,

$$f(x, k) = f_0(x, k) + \int_{-\infty}^{\infty} G(x - y, k)u(y)f(y, k) \, dy,$$

where f_0 is any solution of $(\partial_x^2 + k^2)f_0 = 0$ and G is a Green's function for $\partial_x^2 + k^2$, so that

$$(\partial_x^2 + k^2)G(x, k) = \delta(x).$$

Any solution of the integral equation solves the eigenvalue problem, since

$$(\partial_x^2 + k^2)f = (\partial_x^2 + k^2)f_0 + \int_{-\infty}^{\infty} (\partial_x^2 + k^2)G(x - y, k)u(y)f(y, k) \, dy$$

$$\begin{aligned}
&= 0 + \int_{-\infty}^{\infty} \delta(x-y)u(y)f(y,k) \, dy \\
&= u(x)f(x,k),
\end{aligned}$$

which rearranges to $Lf = k^2 f$.

To get the boundary condition we want, we choose $f_0 = e^{-ikx}$ and take the retarded Green's function

$$G(x,k) = \begin{cases} 0 & x < 0, \\ \frac{1}{k} \sin(kx) & x \geq 0. \end{cases}$$

Then the integrand vanishes unless $y \leq x$, so for x far to the left of the support of u the integral is zero and $f(x,k) = e^{-ikx}$ exactly, as required.

It remains to show that the integral equation has a solution. Write it abstractly as

$$(I - K)f = f_0, \quad (Kf)(x) = \int_{-\infty}^{\infty} G(x-y,k)u(y)f(y,k) \, dy,$$

so that formally

$$f = (I - K)^{-1} f_0 = (I + K + K^2 + \cdots) f_0.$$

This is a **Neumann series**, and one shows on the second example sheet that it converges (the point being that G is bounded and u has compact support, so K is a contraction after finitely many iterations). Given convergence, it visibly solves the equation:

$$(I - K)(f_0 + Kf_0 + K^2f_0 + \cdots) = f_0 + Kf_0 + \cdots - (Kf_0 + K^2f_0 + \cdots) = f_0.$$

We will use this series once more, for a completely different purpose, in Section 3.4; the key consequence to remember is that *the solution with $f_0 = 0$ is $f = 0$* .

Bound States

Now take $\lambda = -\kappa^2 < 0$, and look for solutions of

$$L\psi_\kappa = -\kappa^2 \psi_\kappa$$

which are normalisable,

$$\|\psi_\kappa\|^2 = \int_{-\infty}^{\infty} \psi_\kappa(x)^2 \, dx = 1.$$

These are the **bound states**, and we take them real without loss of generality.

Again, since u has compact support, any solution obeys $\varphi_{xx} - \kappa^2 \varphi = 0$ for large $|x|$, so is a combination of $e^{\pm \kappa x}$ out there. Fix φ_κ by

$$\varphi_\kappa(x) = e^{-\kappa x} \quad \text{as } x \rightarrow +\infty,$$

so that it decays to the right, and write

$$\varphi_\kappa(x) = \alpha(\kappa)e^{\kappa x} + \beta(\kappa)e^{-\kappa x} \quad \text{as } x \rightarrow -\infty.$$

For *any* κ we can solve the ODE and find such a φ_κ . What we cannot do for any κ is make it normalisable: the term $\beta(\kappa)e^{-\kappa x}$ blows up as $x \rightarrow -\infty$, so we need

$$\beta(\kappa) = 0.$$

This is a genuine constraint, and it is what quantises the spectrum. One can show that β has only finitely many zeros, which we label

$$\chi_1 > \chi_2 > \cdots > \chi_N > 0.$$

The corresponding normalised bound states are

$$\psi_n(x) = c_n \varphi_{\chi_n}(x),$$

with the constants c_n chosen so that $\|\psi_n\| = 1$. Equivalently, c_n is determined by the large- x behaviour

$$\psi_n(x) = c_n e^{-\chi_n x} \quad \text{as } x \rightarrow +\infty.$$

We call the c_n the **normalisation constants**.

The Scattering Data

Assembling everything, the outcome of the forward problem is the following package.

Definition 32.3.2 (Scattering data)

The **scattering data** of the operator $L = -\partial_x^2 + u$ is

$$S = \left\{ \{\chi_n, c_n\}_{n=1}^N, R(k), T(k) \right\}.$$

Let us compute it in an example where everything can be done by hand.

Example 32.3.3 (A delta-function potential)

Take $u(x) = -2\alpha\delta(x)$ with $\alpha > 0$. (This is not of compact support, but it decays even faster, so all is well.)

Continuous spectrum. Since u vanishes away from the origin, we have exactly

$$\Phi(x, k) = \begin{cases} T(k)e^{-ikx} & x < 0, \\ e^{-ikx} + R(k)e^{ikx} & x > 0. \end{cases}$$

Continuity at $x = 0$ gives

$$T = 1 + R.$$

For the second condition, integrate $L\Phi = k^2\Phi$ over $(-\varepsilon, \varepsilon)$ and let $\varepsilon \rightarrow 0$. The $k^2\Phi$ term contributes nothing in the limit, and we obtain the jump condition

$$\left[\Phi_x \right]_{0^-}^{0^+} = -2\alpha\Phi(0),$$

that is

$$(-ik + ikR) - (-ikT) = -2\alpha T.$$

Substituting $T = 1 + R$ and solving the two equations gives

$$R(k) = \frac{i\alpha}{k - i\alpha}, \quad T(k) = \frac{k}{k - i\alpha}.$$

As a check,

$$|R|^2 + |T|^2 = \frac{\alpha^2 + k^2}{k^2 + \alpha^2} = 1,$$

and as $k \rightarrow \infty$ we do indeed find $R \rightarrow 0$, $T \rightarrow 1$.

Bound states. Away from the origin $-\psi_{xx} + \chi^2\psi = 0$, and decay at both ends forces

$$\psi(x) = c e^{-\chi|x|}.$$

The same jump condition now reads

$$-c\chi - c\chi = -2\alpha c \implies \chi = \alpha.$$

So there is exactly one bound state, with $\chi_1 = \alpha$. Normalising,

$$1 = c_1^2 \int_{-\infty}^{\infty} e^{-2\alpha|x|} dx = \frac{c_1^2}{\alpha} \implies c_1 = \sqrt{\alpha}.$$

Hence the scattering data is

$$S = \left\{ \{\alpha, \sqrt{\alpha}\}, \frac{i\alpha}{k - i\alpha}, \frac{k}{k - i\alpha} \right\}.$$

The next example is the one that matters most, because it is a soliton.

Example 32.3.4 (The soliton potential)

Take $u(x) = -2\operatorname{sech}^2 x$, which is the one-soliton profile with $\chi = 1$ at time $t = 0$.

Continuous spectrum. We claim that

$$f(x, k) = e^{ikx} (\tanh x - ik)$$

satisfies $Lf = k^2 f$. Writing $T = \tanh x$, so that $T' = \operatorname{sech}^2 x = 1 - T^2$ and $T'' = -2TT'$,

$$\begin{aligned} f' &= e^{ikx} [ik(T - ik) + T'], \\ f'' &= e^{ikx} [-k^2(T - ik) + 2ikT' + T''], \end{aligned}$$

and hence

$$\begin{aligned} -f'' - 2T'f &= e^{ikx} [k^2(T - ik) - 2ikT' - T'' - 2T'(T - ik)] \\ &= e^{ikx} [k^2(T - ik) - 2ikT' - T'' - 2TT' + 2ikT'] \\ &= e^{ikx} k^2(T - ik), \end{aligned}$$

since $-T'' - 2TT' = 2TT' - 2TT' = 0$. So $Lf = k^2 f$ as claimed.

Replacing k by $-k$ gives another solution,

$$g(x, k) = e^{-ikx} (\tanh x + ik),$$

whose limits are

$$g \rightarrow e^{-ikx}(ik - 1) \text{ as } x \rightarrow -\infty, \quad g \rightarrow e^{-ikx}(ik + 1) \text{ as } x \rightarrow +\infty.$$

So the normalised Jost solution is $\varphi = g/(ik-1)$, and comparing with the definitions,

$$a(k) = \frac{ik+1}{ik-1}, \quad b(k) = 0.$$

Therefore

$$R(k) = 0, \quad T(k) = \frac{ik-1}{ik+1}.$$

The reflection coefficient vanishes identically! A soliton is completely transparent: waves of every frequency pass through it without any reflection at all. (Note $|T| = 1$, consistent with $|R|^2 + |T|^2 = 1$.)

Bound states. We look for a normalisable solution of $-\psi_{xx} - 2\operatorname{sech}^2 x \psi = -\chi^2 \psi$. Trying $\psi = \operatorname{sech} x$, and using $(\operatorname{sech} x)'' = \operatorname{sech} x - 2\operatorname{sech}^3 x$, we get

$$-\psi_{xx} - 2\operatorname{sech}^2 x \psi = -\operatorname{sech} x + 2\operatorname{sech}^3 x - 2\operatorname{sech}^3 x = -\operatorname{sech} x,$$

so $\psi = \operatorname{sech} x$ is a bound state with $\chi_1 = 1$. Normalising, $\int \operatorname{sech}^2 x \, dx = 2$, so the normalised state is $\psi_1 = \frac{1}{\sqrt{2}} \operatorname{sech} x$. Since $\operatorname{sech} x \sim 2e^{-x}$ as $x \rightarrow +\infty$,

$$\psi_1(x) \sim \sqrt{2} e^{-x}, \quad \text{so} \quad c_1 = \sqrt{2}.$$

One can check there are no others. So the scattering data is simply

$$S = \{ \{1, \sqrt{2}\}, R(k) = 0 \}.$$

Definition 32.3.5 (Reflectionless potential)

A potential u with $R(k) \equiv 0$ is called **reflectionless**.

We have just seen that the one-soliton potential is reflectionless. This is not a coincidence, and it is the key to everything: reflectionless potentials are exactly the potentials for which the inverse problem can be solved in closed form, and they will turn out to be exactly the N -soliton solutions.

§32.3.2 The Inverse Scattering Problem

Now we go backwards. Given scattering data S , can we reconstruct the potential u that produced it? The answer is yes, and remarkably the transmission coefficient is not even needed.

Theorem 32.3.6 (Gelfand–Levitan–Marchenko)

A potential $u = u(x)$ decaying rapidly as $|x| \rightarrow \infty$ is completely determined by the scattering data

$$S = \{ \{ \chi_n, c_n \}_{n=1}^N, R(k) \}.$$

Specifically, define

$$F(x) = \sum_{n=1}^N c_n^2 e^{-\chi_n x} + \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} R(k) \, dk,$$

and let $K(x, y)$ be the unique solution of the **GLM equation**

$$K(x, y) + F(x + y) + \int_x^\infty K(x, z)F(z + y) \, dz = 0, \quad y > x.$$

Then

$$u(x) = -2 \frac{d}{dx} K(x, x).$$

Proof. Beyond the scope of this course. □

The GLM equation looks forbidding but is not. For fixed x it is a linear integral equation in y , of the abstract form

$$\mathbf{K} + \mathbf{b} + \mathcal{A}\mathbf{K} = 0$$

for a known vector $\mathbf{b}$ and known linear operator $\mathcal{A}$. As with any linear equation, one can guess a solution and then verify it. In the reflectionless case, which is the one we shall care about, the integral equation degenerates into a finite-dimensional matrix equation and can be solved exactly.

Notice that the function F has a very transparent structure: the sum records the bound states, and the integral is the Fourier transform of the reflection coefficient. If there are no bound states and no reflection, $F = 0$, hence $K = 0$, hence $u = 0$ – reassuringly, the empty scattering data corresponds to the zero potential.

Example 32.3.7 (Reconstructing the soliton)

Let us check the theorem on the data we computed above,

$$S = \left\{ \{\chi_1, c_1\} = \{1, \sqrt{2}\}, R = 0 \right\},$$

and see whether we recover $u = -2 \operatorname{sech}^2 x$. Here

$$F(x) = c_1^2 e^{-\chi_1 x} = 2e^{-x}.$$

Since F is a single exponential in x , we look for a solution of the same shape,

$$K(x, y) = K_1(x)e^{-y}.$$

Substituting into the GLM equation,

$$K_1(x)e^{-y} + 2e^{-x}e^{-y} + \int_x^\infty K_1(x)e^{-z} \cdot 2e^{-z}e^{-y} \, dz = 0.$$

Every term carries a factor e^{-y} , which we cancel, and the remaining integral is

$$2K_1(x) \int_x^\infty e^{-2z} \, dz = K_1(x)e^{-2x}.$$

So

$$K_1(x) (1 + e^{-2x}) = -2e^{-x}, \quad K_1(x) = \frac{-2e^{-x}}{1 + e^{-2x}}.$$

Hence, writing $w = e^{-2x}$,

$$K(x, x) = K_1(x)e^{-x} = \frac{-2w}{1+w},$$

and since $dw/dx = -2w$,

$$\frac{d}{dx}K(x, x) = -2 \cdot \frac{w'(1+w) - ww'}{(1+w)^2} = \frac{-2w'}{(1+w)^2} = \frac{4w}{(1+w)^2}.$$

Therefore

$$u(x) = -2 \frac{d}{dx}K(x, x) = \frac{-8w}{(1+w)^2} = \frac{-8e^{-2x}}{(1+e^{-2x})^2} = \frac{-8}{(e^x + e^{-x})^2} = -2 \operatorname{sech}^2 x,$$

exactly as it should be.

We can now go in both directions between potentials and scattering data. That, on its own, is a piece of nineteenth-century-flavoured mathematical physics. What makes it explosive is the next ingredient.

§32.3.3 Lax Pairs

We need to connect the time evolution of u to the time evolution of the scattering data. The mechanism is due to Peter Lax.

Recall from Section 1 that under a Hamiltonian flow, functions evolve according to

$$\frac{df}{dt} = \{f, H\},$$

so H ‘generates’ the time evolution. In quantum mechanics the corresponding statement in the Heisenberg picture is

$$i\hbar \frac{dL}{dt} = [L, H],$$

for operators L . We want something of exactly this shape: an operator whose commutator with L generates the time evolution of L .

Definition 32.3.8 (Lax pair)

Let

$$L = a_m(x, t) \frac{\partial^m}{\partial x^m} + \cdots + a_1(x, t) \frac{\partial}{\partial x} + a_0(x, t)$$

be a time-dependent self-adjoint linear operator, and write

$$L_t = \dot{a}_m \frac{\partial^m}{\partial x^m} + \cdots + \dot{a}_0$$

for its derivative with respect to t (differentiating the coefficients only). If there is an operator A with

$$L_t = LA - AL = [L, A],$$

then (L, A) is called a **Lax pair**, and the equation $L_t = [L, A]$ is called **Lax’s equation**.

The reason to care is the following.

Theorem 32.3.9 (Isospectral flow)

Let (L, A) be a Lax pair, with A antisymmetric. Then the discrete eigenvalues of L are independent of time. Moreover if $L\psi = \lambda\psi$ then

$$L\tilde{\psi} = \lambda\tilde{\psi}, \quad \text{where } \tilde{\psi} = \psi_t + A\psi.$$

The word **isospectral** means precisely this: the operator changes with time, but its spectrum does not.

Proof. We assume that eigenvalues vary smoothly, so that an eigenvalue λ_0 with normalised eigenfunction ψ_0 at $t = 0$ extends to a family $\lambda(t)$, $\psi(x, t)$ with

$$L(t)\psi(x, t) = \lambda(t)\psi(x, t), \quad \|\psi\| = 1.$$

Differentiate with respect to t :

$$L_t\psi + L\psi_t = \lambda_t\psi + \lambda\psi_t,$$

so, substituting Lax's equation and using $L\psi = \lambda\psi$,

$$\begin{aligned} \lambda_t\psi &= L_t\psi + L\psi_t - \lambda\psi_t \\ &= LA\psi - AL\psi + L\psi_t - \lambda\psi_t \\ &= LA\psi - \lambda A\psi + L\psi_t - \lambda\psi_t \\ &= (L - \lambda)(\psi_t + A\psi). \end{aligned}$$

Now take the inner product with ψ and use self-adjointness of $L - \lambda$:

$$\lambda_t = \langle \psi, \lambda_t\psi \rangle = \langle \psi, (L - \lambda)(\psi_t + A\psi) \rangle = \langle (L - \lambda)\psi, \psi_t + A\psi \rangle = 0,$$

since $(L - \lambda)\psi = 0$. So λ is constant, and the displayed equation reduces to $(L - \lambda)\tilde{\psi} = 0$. $\square$

So far this is abstract. The whole thing turns on being able to find, for a given equation, an operator A making Lax's equation equivalent to it. For KdV such an A exists, and here it is.

Proposition 32.3.10 (Lax pair for KdV)

Let

$$L = -\partial_x^2 + u(x, t), \quad A = 4\partial_x^3 - 3(u\partial_x + \partial_x u).$$

Then A is antisymmetric, and

$$L_t = [L, A] \iff u_t + u_{xxx} - 6uu_x = 0.$$

Proof. Throughout, $\partial = \partial_x$, and we treat everything as operators acting on a test function; in particular ∂u means the operator $\psi \mapsto (u\psi)_x$, so that

$$\partial u = u\partial + u_x.$$

Hence

$$A = 4\partial^3 - 3u\partial - 3(u\partial + u_x) = 4\partial^3 - 6u\partial - 3u_x.$$

For antisymmetry, recall $\partial^\dagger = -\partial$ and $u^\dagger = u$, so

$$A^\dagger = -4\partial^3 - 3(u\partial)^\dagger - 3u_x = -4\partial^3 + 3\partial u - 3u_x = -4\partial^3 + 3u\partial + 3u_x - 3u_x,$$

giving $A^\dagger = -4\partial^3 + 6u\partial + 3u_x = -A$ once we recombine (the middle term arises twice, once from each half of $u\partial + \partial u$).

Now for the commutator. We will repeatedly use the operator identities

$$\partial^2 u \partial = u \partial^3 + 2u_x \partial^2 + u_{xx} \partial, \quad \partial^2 u_x = u_x \partial^2 + 2u_{xx} \partial + u_{xxx},$$

$$\partial^3 u = u \partial^3 + 3u_x \partial^2 + 3u_{xx} \partial + u_{xxx},$$

each of which is just the Leibniz rule. First,

$$\begin{aligned} LA &= (-\partial^2 + u)(4\partial^3 - 6u\partial - 3u_x) \\ &= -4\partial^5 + 6\partial^2 u \partial + 3\partial^2 u_x + 4u\partial^3 - 6u^2 \partial - 3uu_x \\ &= -4\partial^5 + 6(u\partial^3 + 2u_x \partial^2 + u_{xx} \partial) + 3(u_x \partial^2 + 2u_{xx} \partial + u_{xxx}) \\ &\quad + 4u\partial^3 - 6u^2 \partial - 3uu_x \\ &= -4\partial^5 + 10u\partial^3 + 15u_x \partial^2 + 12u_{xx} \partial - 6u^2 \partial + 3u_{xxx} - 3uu_x. \end{aligned}$$

Similarly,

$$\begin{aligned} AL &= (4\partial^3 - 6u\partial - 3u_x)(-\partial^2 + u) \\ &= -4\partial^5 + 4\partial^3 u + 6u\partial^3 - 6u\partial u + 3u_x \partial^2 - 3uu_x \\ &= -4\partial^5 + 4(u\partial^3 + 3u_x \partial^2 + 3u_{xx} \partial + u_{xxx}) + 6u\partial^3 \\ &\quad - 6(u^2 \partial + uu_x) + 3u_x \partial^2 - 3uu_x \\ &= -4\partial^5 + 10u\partial^3 + 15u_x \partial^2 + 12u_{xx} \partial - 6u^2 \partial + 4u_{xxx} - 9uu_x. \end{aligned}$$

Every differential term cancels in the difference, and we are left with the multiplication operator

$$[L, A] = LA - AL = (3u_{xxx} - 3uu_x) - (4u_{xxx} - 9uu_x) = -u_{xxx} + 6uu_x.$$

On the other hand $L_t = u_t$, since only the coefficient u depends on t . So Lax's equation $L_t = [L, A]$ says exactly

$$u_t = -u_{xxx} + 6uu_x,$$

which is KdV. □

This is the miracle on which everything rests. It deserves a moment's contemplation: a fifth-order operator equation collapses, after everything cancels, to a scalar equation – and that scalar equation happens to be KdV. Combined with the isospectral flow theorem, we already know something remarkable.

Corollary 32.3.11

If $u(x, t)$ evolves according to KdV, then the discrete eigenvalues $-\chi_n^2$ of the operator $L = -\partial_x^2 + u(x, t)$ are constant in time. In particular the number N of bound states never changes.

Since the χ_n are the soliton amplitudes and speeds, this already tells us that solitons cannot be created or destroyed by the evolution.

§32.3.4 Evolution of the Scattering Data

We now let $u = u(x, t)$ evolve by KdV and work out how the rest of the scattering data moves. Since u has compact support, note first that

$$A = 4\partial_x^3 \quad \text{as } |x| \rightarrow \infty.$$

Continuous Spectrum

For each fixed t construct, as in Section 3.1, the solution φ of $L\varphi = k^2\varphi$ with

$$\varphi(x, t) = \begin{cases} e^{-ikx} & x \rightarrow -\infty, \\ a(k, t)e^{-ikx} + b(k, t)e^{ikx} & x \rightarrow +\infty. \end{cases}$$

Here k is fixed: solutions exist for every real k regardless of u , so there is no question of k moving.

Exactly as in the proof of the isospectral flow theorem (but now with $\lambda = k^2$ constant by fiat rather than by conclusion), differentiating $L\varphi = k^2\varphi$ gives

$$0 = (L - k^2)(\varphi_t + A\varphi),$$

so $\tilde{\varphi} = \varphi_t + A\varphi$ is another solution of $L\tilde{\varphi} = k^2\tilde{\varphi}$. We now compute its asymptotics, using $A = 4\partial_x^3$ out there:

$$\tilde{\varphi}(x, t) = \begin{cases} 4ik^3e^{-ikx} & x \rightarrow -\infty, \\ (a_t + 4ik^3a)e^{-ikx} + (b_t - 4ik^3b)e^{ikx} & x \rightarrow +\infty, \end{cases}$$

since $\partial_x^3 e^{\mp ikx} = \pm ik^3 e^{\mp ikx}$.

Here is the clever step. Consider

$$\theta = 4ik^3\varphi - \tilde{\varphi}.$$

By linearity $L\theta = k^2\theta$, and by construction $\theta \rightarrow 0$ as $x \rightarrow -\infty$. But we showed in Section 3.1 that the solution of $Lf = k^2f$ with prescribed behaviour f_0 at $-\infty$ is $f = (I + K + K^2 + \cdots)f_0$; with $f_0 = 0$ this gives $f = 0$. Hence

$$\theta \equiv 0, \quad \text{that is} \quad \tilde{\varphi} = 4ik^3\varphi.$$

Comparing coefficients of e^{-ikx} and e^{ikx} as $x \rightarrow +\infty$,

$$\begin{aligned} a_t + 4ik^3a &= 4ik^3a, \\ b_t - 4ik^3b &= 4ik^3b, \end{aligned}$$

that is $a_t = 0$ and $b_t = 8ik^3b$. These we can solve:

$$a(k, t) = a(k, 0), \quad b(k, t) = b(k, 0)e^{8ik^3t},$$

and therefore

$$R(k, t) = R(k, 0)e^{8ik^3t}, \quad T(k, t) = T(k, 0).$$

Stop and appreciate this. We assumed that u evolves by a thoroughly nasty nonlinear PDE, and concluded that its reflection coefficient simply rotates in the complex plane at a constant rate, and that its transmission coefficient does not move at all.

Discrete Spectrum

By the isospectral flow theorem the χ_n are constant. For each t we have normalised bound states $\psi_n(x, t)$ with

$$L\psi_n = -\chi_n^2\psi_n, \quad \|\psi_n\| = 1, \quad \psi_n(x, t) = c_n(t)e^{-\chi_n x} \text{ as } x \rightarrow +\infty.$$

By the isospectral flow theorem again,

$$\tilde{\psi}_n = \partial_t \psi_n + A\psi_n$$

also satisfies $L\tilde{\psi}_n = -\chi_n^2\tilde{\psi}_n$. Since the eigenvalue $-\chi_n^2$ is simple (a standard Wronskian argument shows bound states of a one-dimensional Schrödinger operator are non-degenerate), $\tilde{\psi}_n$ must be proportional to ψ_n . But

$$\begin{aligned} \langle \psi_n, \tilde{\psi}_n \rangle &= \langle \psi_n, \partial_t \psi_n \rangle + \langle \psi_n, A\psi_n \rangle \\ &= \frac{1}{2} \frac{\partial}{\partial t} \langle \psi_n, \psi_n \rangle + \langle \psi_n, A\psi_n \rangle \\ &= 0, \end{aligned}$$

because $\|\psi_n\|$ is constant and A is antisymmetric (so $\langle \psi, A\psi \rangle = -\langle A\psi, \psi \rangle = -\langle \psi, A\psi \rangle$). A multiple of ψ_n orthogonal to ψ_n is zero, so

$$\tilde{\psi}_n \equiv 0.$$

Now look at the large- x behaviour. There $A = 4\partial_x^3$ and $\psi_n = c_n(t)e^{-\chi_n x}$, so

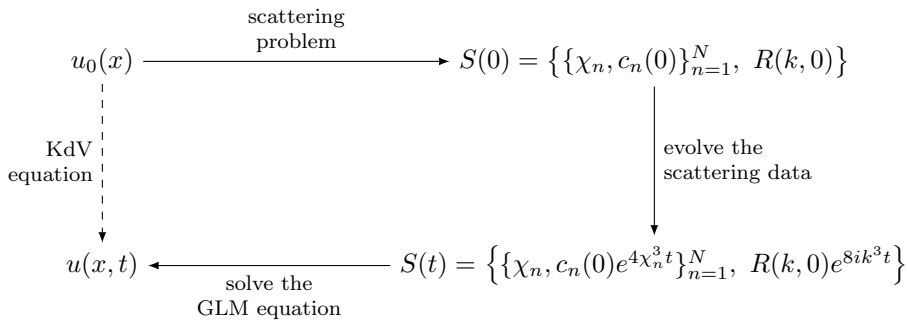
$$\tilde{\psi}_n(x, t) = (\dot{c}_n - 4\chi_n^3 c_n) e^{-\chi_n x} \text{ as } x \rightarrow +\infty.$$

Since $\tilde{\psi}_n \equiv 0$ we must have $\dot{c}_n = 4\chi_n^3 c_n$, and hence

$$c_n(t) = c_n(0)e^{4\chi_n^3 t}.$$

The Transform

We can now assemble the machine. Suppose u evolves by KdV with initial data $u_0(x) = u(x, 0)$.



That is: compute the scattering data of u_0 ; evolve it by the explicit formulae above; feed the result into the GLM equation; and read off $u(x, t)$. Compare this with the Fourier transform diagram at the start of this section – it is the same three steps, with a more sophisticated transform. All the nonlinearity of KdV has been absorbed into the (nonlinear, but classical) map $u \mapsto S$.

The one thing that makes it work is the equivalence

$$u_t + u_{xxx} - 6uu_x = 0 \iff L_t = [L, A],$$

so the whole method hinges on finding a Lax pair.

Remark. This is not special to KdV. Choosing different operators A in the Lax equation with the same L produces an infinite hierarchy of equations – the **KdV hierarchy** – all of which are solvable by the same transform and all of which share the same conserved quantities. Choosing a different L altogether produces different integrable equations; the choice

$$L = i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \partial_x + \begin{pmatrix} 0 & q \\ \bar{q} & 0 \end{pmatrix}$$

leads, by the same route, to the nonlinear Schrödinger equation, which we meet in Section 5.

§32.3.5 Reflectionless Potentials and N -Solitons

Now we cash in. The GLM equation is hard in general, but there is one case in which it can be solved completely, and it is exactly the case that produces solitons.

Recall that a potential is reflectionless if $R(k, 0) = 0$. Since

$$R(k, t) = R(k, 0)e^{8ik^3t},$$

a potential which starts reflectionless stays reflectionless forever. So this is a class of initial data preserved by the KdV flow, and within it the machinery becomes elementary.

Suppose then that $R \equiv 0$, so that

$$F(x) = \sum_{n=1}^N c_n^2 e^{-\chi_n x}.$$

(We suppress the time dependence for the moment, restoring it at the very end.) Since F is a finite sum of exponentials, we look for K of the same shape,

$$K(x, y) = \sum_{m=1}^N K_m(x) e^{-\chi_m y};$$

one shows on the second example sheet that K *must* have this form. Substituting into the GLM equation,

$$\sum_{m=1}^N K_m(x) e^{-\chi_m y} + \sum_{n=1}^N c_n^2 e^{-\chi_n x} e^{-\chi_n y} + \sum_{n,m} c_n^2 K_m(x) \int_x^\infty e^{-\chi_m z} e^{-\chi_n(z+y)} dz = 0.$$

The integral evaluates to

$$\int_x^\infty e^{-(\chi_n + \chi_m)z} dz = \frac{e^{-(\chi_n + \chi_m)x}}{\chi_n + \chi_m},$$

so, collecting everything over the common index n ,

$$\sum_{n=1}^N \left[c_n^2 e^{-\chi_n x} + K_n(x) + \sum_{m=1}^N \frac{c_n^2 K_m(x)}{\chi_n + \chi_m} e^{-(\chi_n + \chi_m)x} \right] e^{-\chi_n y} = 0. \quad (*)$$

The functions $e^{-\chi_n y}$, $n = 1, \dots, N$, are linearly independent, so each bracket must vanish separately. Thus for each n ,

$$K_n(x) + \sum_{m=1}^N \frac{c_n^2 e^{-(\chi_n + \chi_m)x}}{\chi_n + \chi_m} K_m(x) = -c_n^2 e^{-\chi_n x}.$$

For each fixed x this is a linear system! Setting

$$\begin{aligned} \mathbf{c} &= (c_1^2 e^{-\chi_1 x}, \dots, c_N^2 e^{-\chi_N x})^T, \\ \mathbf{K} &= (K_1(x), \dots, K_N(x))^T, \\ A_{nm} &= \delta_{nm} + \frac{c_n^2 e^{-(\chi_n + \chi_m)x}}{\chi_n + \chi_m}, \end{aligned}$$

the system reads

$$A\mathbf{K} = -\mathbf{c}.$$

So we have reduced an integral equation to inverting an $N \times N$ matrix. Even better, we do not need $\mathbf{K}$ itself, only the diagonal value

$$K(x, x) = \sum_{m=1}^N K_m(x) e^{-\chi_m x} = \sum_{m,n} (A^{-1})_{mn} (-\mathbf{c})_n e^{-\chi_m x}.$$

Now observe that

$$A'_{nm}(x) = \frac{d}{dx} A_{nm}(x) = -c_n^2 e^{-\chi_n x} e^{-\chi_m x} = (-\mathbf{c})_n e^{-\chi_m x},$$

which is exactly the combination appearing above. Hence

$$K(x, x) = \sum_{m,n} (A^{-1})_{mn} A'_{nm} = \text{tr} (A^{-1} A').$$

Finally we use the identity (an exercise on the second example sheet, proved by expanding the determinant)

$$\text{tr} (A^{-1} A') = \frac{1}{\det A} \frac{d}{dx} (\det A) = \frac{d}{dx} \log (\det A),$$

which gives

$$u(x) = -2 \frac{d}{dx} K(x, x) = -2 \frac{d^2}{dx^2} \log (\det A).$$

Restoring the time dependence via $c_n(t)^2 = c_n(0)^2 e^{8\chi_n^3 t}$, we have proved the following.

Theorem 32.3.12 (N -soliton solution)

Let $\chi_1 > \dots > \chi_N > 0$ and $c_1(0), \dots, c_N(0)$ be given. Then

$$u(x, t) = -2 \frac{\partial^2}{\partial x^2} \log (\det A(x, t)),$$

where A is the $N \times N$ matrix

$$A_{nm}(x, t) = \delta_{nm} + \frac{c_n(0)^2 e^{8\chi_n^3 t} e^{-(\chi_n + \chi_m)x}}{\chi_n + \chi_m},$$

is a solution of the KdV equation. It is the N -soliton solution, and describes N solitary waves of amplitudes $2\chi_n^2$ and speeds $4\chi_n^2$.

We have obtained a closed-form solution to a nonlinear PDE, involving nothing worse than a determinant. Let us unpack it in the two simplest cases.

Example 32.3.13 ($N = 1$)

With a single bound state, A is a 1×1 matrix,

$$\det A = 1 + \frac{c_1(0)^2 e^{8\chi_1^3 t} e^{-2\chi_1 x}}{2\chi_1} = 1 + e^{-2\chi_1(x - 4\chi_1^2 t - \delta)},$$

where the constant δ absorbs $c_1(0)$ via $e^{2\chi_1 \delta} = c_1(0)^2 / 2\chi_1$. Writing $\xi = x - 4\chi_1^2 t - \delta$ and $w = e^{-2\chi_1 \xi}$,

$$\log \det A = \log(1 + w),$$

and differentiating twice (using $w' = -2\chi_1 w$),

$$\frac{\partial}{\partial x} \log(1 + w) = \frac{-2\chi_1 w}{1 + w}, \quad \frac{\partial^2}{\partial x^2} \log(1 + w) = \frac{4\chi_1^2 w}{(1 + w)^2}.$$

Hence

$$\begin{aligned} u &= -\frac{8\chi_1^2 w}{(1 + w)^2} = -\frac{8\chi_1^2}{(w^{1/2} + w^{-1/2})^2} = -\frac{8\chi_1^2}{4 \cosh^2(\chi_1 \xi)} \\ &= -2\chi_1^2 \operatorname{sech}^2(\chi_1(x - 4\chi_1^2 t - \delta)), \end{aligned}$$

which is precisely the one-soliton solution of Section 2.3. So a single bound state means a single soliton, with χ_1 determining both its height and its speed, and $c_1(0)$ determining only where it is.

Example 32.3.14 ($N = 2$)

With two bound states, write

$$E_n = E_n(x, t) = \frac{c_n(0)^2 e^{8\chi_n^3 t} e^{-2\chi_n x}}{2\chi_n},$$

so that $A_{nn} = 1 + E_n$ and

$$A_{12}A_{21} = \frac{c_1^2 c_2^2 e^{-2(\chi_1 + \chi_2)x}}{(\chi_1 + \chi_2)^2} = \frac{4\chi_1 \chi_2}{(\chi_1 + \chi_2)^2} E_1 E_2.$$

Therefore

$$\det A = (1 + E_1)(1 + E_2) - \frac{4\chi_1 \chi_2}{(\chi_1 + \chi_2)^2} E_1 E_2 = 1 + E_1 + E_2 + \left(\frac{\chi_2 - \chi_1}{\chi_2 + \chi_1} \right)^2 E_1 E_2,$$

using $1 - \frac{4\chi_1\chi_2}{(\chi_1+\chi_2)^2} = \frac{(\chi_2-\chi_1)^2}{(\chi_2+\chi_1)^2}$. Substituting into the theorem gives the two-soliton solution plotted in Section 2.4.

The interesting thing is what happens at large $|t|$, and this we can extract without any further computation. Write

$$\delta = \frac{\chi_2 - \chi_1}{\chi_2 + \chi_1} \in (0, 1),$$

and follow the taller soliton by fixing $\xi = x - 4\chi_2^2 t$. In this frame

$$E_2 \propto e^{8\chi_2^3 t - 2\chi_2(4\chi_2^2 t + \xi)} = e^{-2\chi_2 \xi},$$

which is constant in t , while

$$E_1 \propto e^{8\chi_1^3 t - 2\chi_1(4\chi_2^2 t + \xi)} = e^{8\chi_1(\chi_1^2 - \chi_2^2)t} e^{-2\chi_1 \xi} \longrightarrow \begin{cases} 0 & t \rightarrow +\infty, \\ \infty & t \rightarrow -\infty, \end{cases}$$

since $\chi_1 < \chi_2$. So:

- As $t \rightarrow +\infty$ we have $\det A \rightarrow 1 + E_2$, and by the $N = 1$ computation u tends to a soliton of parameter χ_2 sitting at $\xi = \delta_2$ for some fixed δ_2 .
- As $t \rightarrow -\infty$ the term E_1 dominates, and

$$\det A \approx E_1 (1 + \delta^2 E_2).$$

Now $\log E_1$ is a linear function of x , so it is killed by ∂_x^2 and contributes nothing. We are left with $u \rightarrow -2\partial_x^2 \log(1 + \delta^2 E_2)$, which is again a soliton of parameter χ_2 – but with E_2 replaced by $\delta^2 E_2$.

Since $E_2 \propto e^{-2\chi_2 \xi}$, multiplying E_2 by δ^2 is the same as translating ξ by

$$\Delta \quad \text{where} \quad e^{2\chi_2 \Delta} = \delta^2, \quad \text{i.e.} \quad \Delta = \frac{1}{\chi_2} \log \delta = -\frac{1}{\chi_2} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1} < 0.$$

So long before the collision the tall soliton is at $\xi = \delta_2 + \Delta$, and long afterwards it is at $\xi = \delta_2$: it has been pushed *forward* by

$$|\Delta| = \frac{1}{\chi_2} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1}.$$

Repeating the argument in the frame of the shorter soliton (where now E_2 is the one that blows up) gives a shift backwards by $\frac{1}{\chi_1} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1}$. These are the phase shifts we quoted in Section 2.4.

Two remarks are in order. First, the number of solitons is the number of bound states, so we can read off the eventual fate of *any* initial profile from a linear eigenvalue problem: count the bound states of $-\partial_x^2 + u_0$, and that is how many solitons will emerge. Everything else – the reflection coefficient – disperses away, and one can show that it contributes a decaying oscillatory tail travelling in the opposite direction. So the general picture is:

any decaying initial profile resolves into N solitons plus radiation.

Second, the total absence of any interaction term in the final answer – the shifts Δ

depend only on χ_1 and χ_2 , not on when or where the collision happened – is precisely the sense in which solitons behave like particles.

§32.3.6 Infinitely Many Conservation Laws

We promised at the start of Section 2 that an integrable PDE should have infinitely many conserved quantities, and in Section 2.5 we found three of them by inspection. Now we can produce them all at once.

The idea is that we have already found something conserved: the quantity $a(k, t) = a(k, 0)$ is independent of time, for every k . If we can expand $a(k)$ in powers of $1/k$, then every coefficient in that expansion must be conserved as well, and we will have infinitely many first integrals in a single stroke.

Recall the setup: $\varphi = \varphi(x, k, t)$ solves $L\varphi = k^2\varphi$ with

$$\varphi = \begin{cases} e^{-ikx} & x \rightarrow -\infty, \\ a(k, t)e^{-ikx} + b(k, t)e^{ikx} & x \rightarrow +\infty. \end{cases}$$

Multiplying by e^{ikx} ,

$$e^{ikx}\varphi(x, k, t) = a(k) + b(k, t)e^{2ikx} \quad \text{as } x \rightarrow +\infty.$$

To isolate a we average out the oscillating term. Over the window $[R, 2R]$,

$$\frac{1}{R} \int_R^{2R} a(k) \, dx = a(k), \quad \frac{1}{R} \int_R^{2R} b(k, t)e^{2ikx} \, dx = O\left(\frac{1}{R}\right),$$

so

$$a(k) = \lim_{R \rightarrow \infty} \frac{1}{R} \int_R^{2R} e^{ikx}\varphi(x, k, t) \, dx = \lim_{R \rightarrow \infty} \int_1^2 e^{ikRx}\varphi(Rx, k, t) \, dx,$$

substituting $x \mapsto Rx$.

Now we need a good representation of φ . Since $\varphi = e^{-ikx}$ as $x \rightarrow -\infty$, it is natural to write

$$\varphi(x, k, t) = \exp\left(-ikx + \int_{-\infty}^x S(y, k, t) \, dy\right)$$

for some function S . Feeding this into the formula above and passing to the limit,

$$a(k) = \lim_{R \rightarrow \infty} \int_1^2 \exp\left(\int_{-\infty}^{Rx} S(y, k, t) \, dy\right) \, dx = \exp\left(\int_{-\infty}^{\infty} S(y, k, t) \, dy\right). \quad (\dagger)$$

The left-hand side has no t in it; the right-hand side appears to. That is where our conserved quantities will come from.

It remains to find S . Substituting the exponential ansatz into $L\varphi = k^2\varphi$, and using

$$\varphi_x = (S - ik)\varphi, \quad \varphi_{xx} = S_x\varphi + (S - ik)^2\varphi,$$

we find, after dividing by φ ,

$$S_x - 2ikS + S^2 = -u.$$

The function φ has disappeared entirely, and we are left with a **Riccati equation** for S . (This is the Riccati equation of the Miura transform in disguise, which is no coincidence.)

We solve it as an asymptotic series in $1/k$:

$$S(x, k, t) = \sum_{n=1}^{\infty} \frac{S_n(x, t)}{(2ik)^n}.$$

Substituting and comparing coefficients of k^{-n} gives the recursion

$$S_1 = -u, \quad S_{n+1} = \frac{dS_n}{dx} + \sum_{m=1}^{n-1} S_m S_{n-m}.$$

This is entirely mechanical, and the first few terms are

$$S_1 = -u, \quad S_2 = -u_x, \quad S_3 = -u_{xx} + u^2, \quad S_4 = -u_{xxx} + 4uu_x, \dots$$

Now put this back into (†):

$$\log a(k) = \int_{-\infty}^{\infty} S(x, k, t) dx = \sum_{n=1}^{\infty} \frac{1}{(2ik)^n} \int_{-\infty}^{\infty} S_n(x, t) dx.$$

The left-hand side is independent of t , and this holds for all k , so every coefficient must be independent of t separately. We conclude:

Theorem 32.3.15

If u evolves by KdV then, for every $n \geq 1$,

$$\int_{-\infty}^{\infty} S_n(x, t) dx$$

is a first integral, where the S_n are given by the recursion above.

Let us look at the first few.

(i) $n = 1$ gives

$$\int_{-\infty}^{\infty} u dx,$$

conservation of mass – our I_1 .

(ii) $n = 2$ gives $\int u_x dx$, which vanishes identically because u decays. This is a *trivial* conservation law: true, but empty. In fact one can show that S_n is a total derivative for every even n , so the even members of the family give us nothing.

(iii) $n = 3$ gives

$$\int_{-\infty}^{\infty} (-u_{xx} + u^2) dx = \int_{-\infty}^{\infty} u^2 dx,$$

conservation of momentum – our I_2 .

(iv) $n = 5$ gives $S_5 = -u_{xxxx} + 5u_x^2 + 6uu_{xx} - 2u^3$. Integrating, the first term vanishes and $\int 6uu_{xx} dx = -\int 6u_x^2 dx$ by parts, leaving

$$\int_{-\infty}^{\infty} S_5 dx = -\int_{-\infty}^{\infty} (u_x^2 + 2u^3) dx = -2I_3,$$

so we recover the energy.

So half of the family is trivial. But half of infinity is still infinity, and we have produced infinitely many genuine first integrals of KdV, exactly as the finite-dimensional theory led us to expect.

Remark. It is worth noticing where the conserved quantities came from: they are the coefficients in an expansion of the transmission data. The scattering data $\{\chi_n\}$ and $R(k)$ are, in a real sense, the action variables of KdV; the phases of c_n and R are the angles, moving at constant rates $4\chi_n^3$ and $8k^3$. In the next section we make this analogy considerably more precise.

§32.4 The Hamiltonian Structure of Integrable PDEs

The integrable ODEs of Section 1 were not arbitrary ODEs – they were Hamiltonian systems, and their integrability was expressed in terms of Poisson-commuting first integrals. If we really want to think of PDEs as infinite-dimensional dynamical systems, we should ask whether they too are Hamiltonian, and whether the conserved quantities we found are in involution. They are, and putting the theory in this language explains where the infinite family of conservation laws comes from.

§32.4.1 Infinite-Dimensional Hamiltonian Systems

Recall the finite-dimensional set-up: a phase space $\mathbb{R}^{2n}$, a non-degenerate antisymmetric matrix J , a Hamiltonian $H : \mathbb{R}^{2n} \rightarrow \mathbb{R}$, and the equation of motion

$$\frac{d\mathbf{x}}{dt} = J \frac{\partial H}{\partial \mathbf{x}}.$$

We now promote each ingredient in turn. The rule of thumb is that the discrete index i becomes the continuous variable x , sums become integrals, and partial derivatives become functional derivatives.

First, the inner product. In finite dimensions we had $\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$; here we replace the sum by an integral.

Notation. For functions u, v on $\mathbb{R}$ we write

$$\langle u, v \rangle = \int_{\mathbb{R}} u(x)v(x) \, dx.$$

If u and v also depend on t , then so does the inner product.

Secondly, functions on phase space. In finite dimensions the Hamiltonian was a function $H(\mathbf{x})$; here the analogue is not a function of u at each point, but a *functional*.

Definition 32.4.1 (Functional)

A **functional** is a real-valued map on a space of functions of the form

$$F[u] = \int_{\mathbb{R}} f(x, u, u_x, u_{xx}, \dots) \, dx.$$

The function f is called the **density**.

Thirdly, differentiation with respect to phase-space coordinates. We met this in IB Variational Principles.

Definition 32.4.2 (Functional derivative)

The **functional derivative** (or Euler–Lagrange derivative) of F at u is the unique function δF with

$$\langle \delta F, \eta \rangle = \lim_{\varepsilon \rightarrow 0} \frac{F[u + \varepsilon \eta] - F[u]}{\varepsilon}$$

for all smooth η of compact support; equivalently

$$F[u + \varepsilon \eta] = F[u] + \varepsilon \langle \delta F, \eta \rangle + o(\varepsilon).$$

Note that δF is itself a function of x , depending on u . We often write $\delta F / \delta u$ for it.

Example 32.4.3

Let $F[u] = \frac{1}{2} \int u_x^2 dx$. Then

$$\begin{aligned} F[u + \varepsilon \eta] &= \frac{1}{2} \int (u_x + \varepsilon \eta_x)^2 dx \\ &= \frac{1}{2} \int u_x^2 dx + \varepsilon \int u_x \eta_x dx + o(\varepsilon) \\ &= F[u] + \varepsilon \langle u_x, \eta_x \rangle + o(\varepsilon). \end{aligned}$$

This is not yet of the required form, since we want η rather than η_x in the second slot. When in doubt, integrate by parts – and since η has compact support there are no boundary terms:

$$F[u + \varepsilon \eta] = F[u] + \varepsilon \langle -u_{xx}, \eta \rangle + o(\varepsilon).$$

So $\delta F = -u_{xx}$.

The general formula, from IB Variational Principles, is

$$\delta F = \frac{\partial f}{\partial u} - D_x \left(\frac{\partial f}{\partial u_x} \right) + D_x^2 \left(\frac{\partial f}{\partial u_{xx}} \right) - \cdots,$$

where D_x is the *total* derivative, not the partial one.

Definition 32.4.4 (Total derivative)

For a function $f(x, u, u_x, \dots)$ and a given $u = u(x)$, the **total derivative** is

$$D_x f = \frac{\partial f}{\partial x} + u_x \frac{\partial f}{\partial u} + u_{xx} \frac{\partial f}{\partial u_x} + \cdots.$$

Example 32.4.5

For $f = xu$ we have $\partial f / \partial x = u$ but $D_x f = u + xu_x$. The partial derivative treats u as an independent coordinate; the total derivative remembers that u is a function of x .

Finally we need an analogue of J . It is helpful first to notice that in finite dimensions we can think of J in two ways: as an antisymmetric bilinear form, or – since we also have a dot product – as an antisymmetric linear *map*, applied via $\mathbf{v} \cdot J\mathbf{w}$. It is the

second picture that generalises: we look for an antisymmetric linear operator $\mathcal{J}$ acting on functions, and define

$$\{F, G\} = \langle \delta F, \mathcal{J} \delta G \rangle = \int \delta F(x) \mathcal{J} \delta G(x) \, dx.$$

Bilinearity and antisymmetry are then automatic. The Jacobi identity, however, is *not*: it constrains the choice of $\mathcal{J}$.

Definition 32.4.6 (Hamiltonian operator)

A **Hamiltonian operator** is a linear, antisymmetric operator $\mathcal{J}$ on functions such that the induced bracket

$$\{F, G\} = \langle \delta F, \mathcal{J} \delta G \rangle$$

satisfies the Jacobi identity.

The simplest antisymmetric operator one can write down is $\mathcal{J} = \partial_x$ (antisymmetric because $\langle u_x, v \rangle = -\langle u, v_x \rangle$ after integrating by parts). One can check – the proof is straightforward but tedious – that it is Hamiltonian.

Definition 32.4.7 (Hamiltonian form)

An evolution equation for $u(x, t)$ is in **Hamiltonian form** if it can be written

$$u_t = \mathcal{J} \frac{\delta H}{\delta u}$$

for some functional $H[u]$ and some Hamiltonian operator $\mathcal{J}$.

Everything from Section 1 now goes through verbatim.

Proposition 32.4.8

If $u_t = \mathcal{J} \delta H$ and $I = I[u]$ is a functional, then

$$\frac{dI}{dt} = \{I, H\}.$$

In particular I is a first integral if and only if $\{I, H\} = 0$.

Proof. Directly from the definition of the functional derivative,

$$\frac{dI}{dt} = \lim_{\varepsilon \rightarrow 0} \frac{I[u + \varepsilon u_t] - I[u]}{\varepsilon} = \langle \delta I, u_t \rangle = \langle \delta I, \mathcal{J} \delta H \rangle = \{I, H\}. \quad \square$$

In summary, the dictionary is:

<i>2n-dimensional phase space</i>	<i>infinite-dimensional phase space</i>
$x_i(t), i = 1, \dots, 2n$	$u(x, t), x \in \mathbb{R}$
$\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$	$\langle u, v \rangle = \int uv \, dx$
d/dt	$\partial/\partial t$
$\partial/\partial \mathbf{x}$	$\delta/\delta u$
antisymmetric matrix J	antisymmetric operator $\mathcal{J}$
functions $f(\mathbf{x})$	functionals $F[u]$

§32.4.2 Bi-Hamiltonian Systems

So far this is only a change of language. The interesting phenomenon is that the *same* equation may be Hamiltonian in more than one way.

Definition 32.4.9 (Bi-Hamiltonian)

An evolution equation is **bi-Hamiltonian** if it can be written in Hamiltonian form for two different Hamiltonian operators,

$$u_t = \mathcal{J}_0 \delta H_0 = \mathcal{J}_1 \delta H_1.$$

Example 32.4.10 (KdV is bi-Hamiltonian)

Take

$$\mathcal{J}_1 = \partial_x, \quad H_1[u] = \int \left(\frac{1}{2} u_x^2 + u^3 \right) dx.$$

Then the density is $f = \frac{1}{2} u_x^2 + u^3$, and

$$\delta H_1 = \frac{\partial f}{\partial u} - D_x \left(\frac{\partial f}{\partial u_x} \right) = 3u^2 - u_{xx},$$

so

$$\mathcal{J}_1 \delta H_1 = \partial_x (3u^2 - u_{xx}) = 6uu_x - u_{xxx} = u_t,$$

which is KdV. So H_1 – which is precisely the first integral I_3 from Section 2.5 – is the Hamiltonian for KdV.

But KdV is also Hamiltonian with respect to a completely different structure. Take

$$\mathcal{J}_0 = -\partial_x^3 + 4u\partial_x + 2u_x, \quad H_0[u] = \int \frac{1}{2} u^2 dx.$$

Now $\delta H_0 = u$, and

$$\mathcal{J}_0 \delta H_0 = -u_{xxx} + 4uu_x + 2u_x u = -u_{xxx} + 6uu_x = u_t,$$

which is KdV again. Here $H_0 = \frac{1}{2} I_2$ is the momentum. One checks that $\mathcal{J}_0$ is antisymmetric (the operators ∂_x^3 and $u\partial_x + \partial_x u$ both are) and that it satisfies the Jacobi identity, which is less obvious.

Why does this matter? Because two compatible Hamiltonian structures generate an infinite hierarchy of conserved quantities, all in involution, for free. Given the two structures, define a sequence of functionals $H_0, H_1, H_2, \dots$ recursively by

$$\mathcal{J}_1 \delta H_{n+1} = \mathcal{J}_0 \delta H_n.$$

(One has to show that this can always be solved for H_{n+1} ; we take this on faith.) Then:

Theorem 32.4.11 (Magri)

Suppose $u_t = \mathcal{J}_0 \delta H_0 = \mathcal{J}_1 \delta H_1$ is bi-Hamiltonian, and let $\{H_n\}_{n \geq 0}$ be defined by the recursion above. Then all the H_n are first integrals, and they are in involution:

$$\{H_n, H_m\} = 0 \quad \text{for all } n, m \geq 0,$$

where the bracket is taken with respect to $\mathcal{J}_1$.

Proof. The whole proof rests on one observation. For $m \geq 1$,

$$\begin{aligned}
 \{H_n, H_m\} &= \langle \delta H_n, \mathcal{J}_1 \delta H_m \rangle \\
 &= \langle \delta H_n, \mathcal{J}_0 \delta H_{m-1} \rangle && (\text{recursion}) \\
 &= -\langle \mathcal{J}_0 \delta H_n, \delta H_{m-1} \rangle && (\mathcal{J}_0 \text{ antisymmetric}) \\
 &= -\langle \mathcal{J}_1 \delta H_{n+1}, \delta H_{m-1} \rangle && (\text{recursion}) \\
 &= \langle \delta H_{n+1}, \mathcal{J}_1 \delta H_{m-1} \rangle && (\mathcal{J}_1 \text{ antisymmetric}) \\
 &= \{H_{n+1}, H_{m-1}\}.
 \end{aligned}$$

So we may shift an index from one slot to the other. Iterating m times gives

$$\{H_n, H_m\} = \{H_{n+m}, H_0\},$$

and iterating n times in the other direction gives $\{H_n, H_m\} = \{H_0, H_{n+m}\}$. Hence $\{H_n, H_m\} = \{H_m, H_n\}$; but the bracket is antisymmetric, so both sides vanish.

Finally, taking $m = 1$ shows $\{H_n, H_1\} = 0$ for all n , and H_1 is the Hamiltonian, so every H_n is a first integral. $\square$

This is a genuinely miraculous piece of bookkeeping: the mere existence of a second Hamiltonian structure forces infinitely many commuting conservation laws into existence. For KdV the resulting H_n are (up to normalisation) the integrals $\int S_{2n+1} dx$ we constructed from the scattering problem in Section 3.6. And the flows they generate,

$$u_{t_n} = \mathcal{J}_1 \delta H_n,$$

are the members of the KdV hierarchy: $n = 0$ gives $u_t = u_x$ (translation), $n = 1$ gives KdV, $n = 2$ gives a fifth-order equation, and so on. Since the Hamiltonians are in involution, all these flows commute, exactly as the Hamiltonian flows on a Liouville torus did in Section 1.

§32.4.3 Zero-Curvature Representation

There is a second, more geometric way to package integrability, which is often more flexible than the Lax formulation.

Given a function $u(x, t)$, suppose we can construct $N \times N$ matrices $U = U(\lambda)$ and $V = V(\lambda)$ depending on λ , on u and on its derivatives. Here λ is a **spectral parameter**, playing the same role as the λ in $L\varphi = \lambda\varphi$. Consider the linear system

$$\frac{\partial}{\partial x} \mathbf{v} = U(\lambda) \mathbf{v}, \quad \frac{\partial}{\partial t} \mathbf{v} = V(\lambda) \mathbf{v}, \quad (\dagger)$$

for an N -component vector $\mathbf{v} = \mathbf{v}(x, t; \lambda)$.

This is an overdetermined system – we have twice as many equations as unknowns – so there is a compatibility condition, obtained from $\mathbf{v}_{xt} = \mathbf{v}_{tx}$:

$$\begin{aligned}
 0 &= \frac{\partial}{\partial t} (U \mathbf{v}) - \frac{\partial}{\partial x} (V \mathbf{v}) \\
 &= U_t \mathbf{v} + U \mathbf{v}_t - V_x \mathbf{v} - V \mathbf{v}_x
 \end{aligned}$$

$$\begin{aligned}
&= U_t \mathbf{v} + UV \mathbf{v} - V_x \mathbf{v} - VU \mathbf{v} \\
&= (U_t - V_x + [U, V]) \mathbf{v}.
\end{aligned}$$

So if (†) is to have a solution for every initial $\mathbf{v}_0$, we need

Definition 32.4.12 (Zero-curvature equation)

The **zero-curvature equation** is

$$\frac{\partial U}{\partial t} - \frac{\partial V}{\partial x} + [U, V] = 0.$$

A theorem of Frobenius says the converse also holds: if the zero-curvature equation is satisfied, then solutions of (†) exist. So we have a correspondence between the solvability of a linear system and a nonlinear equation for U, V .

The name comes from differential geometry. The curvature of a connection A has components

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} + [A_i, A_j],$$

which is exactly the shape of our equation with $(A_1, A_2) = (U, V)$ and $(x_1, x_2) = (x, t)$. So the equation says that the connection built from U and V is flat.

Example 32.4.13 (Sine–Gordon)

Take

$$U(\lambda) = \frac{i}{2} \begin{pmatrix} 2\lambda & u_x \\ u_x & -2\lambda \end{pmatrix}, \quad V(\lambda) = \frac{1}{4i\lambda} \begin{pmatrix} \cos u & -i \sin u \\ i \sin u & -\cos u \end{pmatrix}.$$

A computation shows that the zero-curvature equation is equivalent to

$$u_{xt} = \sin u,$$

the sine–Gordon equation in light-cone coordinates (Section 5.1).

The geometric picture pays off immediately. Curvature is an intrinsic object: if it vanishes in one gauge it vanishes in all of them. So given one pair (U, V) solving the zero-curvature equation, a gauge transformation

$$U \mapsto \tilde{U} = gUg^{-1} + g_x g^{-1}, \quad V \mapsto \tilde{V} = gVg^{-1} + g_t g^{-1}$$

produces a new pair solving the same equation, and hence potentially a new integrable equation, or new solutions of the same one. This is explored on the last example sheet.

Zero curvature also supports its own version of inverse scattering. In the sine–Gordon example above, if $u_x \rightarrow 0$ as $|x| \rightarrow \infty$ then writing $\mathbf{v} = (\psi_1, \psi_2)^T$ the first equation of (†) becomes

$$\frac{\partial}{\partial x} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = i\lambda \begin{pmatrix} \psi_1 \\ -\psi_2 \end{pmatrix},$$

so that

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \rightarrow A \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{i\lambda x} + B \begin{pmatrix} 0 \\ 1 \end{pmatrix} e^{-i\lambda x}$$

as $|x| \rightarrow \infty$. The coefficients A and B are the scattering data, and the second equation of (†) tells us how they evolve in t . This is the AKNS scheme, and it is how one solves sine–Gordon and the nonlinear Schrödinger equation.

§32.4.4 From Lax Pairs to Zero Curvature

The two formulations are closely related, and in fact every Lax pair gives rise to a zero-curvature representation.

Recall that in the isospectral flow theorem we found two equations holding simultaneously:

$$L\psi = \lambda\psi, \quad \psi_t + A\psi = 0,$$

the second because $\tilde{\psi}$ vanished. Conversely, suppose we *impose* these two equations and demand $\lambda_t = 0$. Differentiating the first and substituting the second gives

$$L_t\psi + L\psi_t = \lambda\psi_t \implies L_t\psi = (L - \lambda)(-\psi_t) = (L - \lambda)A\psi = LA\psi - AL\psi,$$

that is $L_t = [L, A]$. So Lax's equation is exactly the compatibility condition for the pair of linear equations above – the same logical role played by the zero-curvature equation for (†).

To make the correspondence explicit, suppose

$$L = \partial_x^n + \sum_{j=0}^{n-1} u_j(x, t) \partial_x^j, \quad A = \partial_x^m + \sum_{j=0}^{m-1} v_j(x, t) \partial_x^j.$$

The equation $L\psi = \lambda\psi$ lets us express every derivative of order $\geq n$ in terms of the lower ones:

$$\partial_x^n \psi = \lambda\psi - \sum_{j=0}^{n-1} u_j \partial_x^j \psi,$$

and differentiating this repeatedly handles the higher derivatives. So introducing the vector

$$\Psi = (\psi, \partial_x \psi, \dots, \partial_x^{n-1} \psi)^T,$$

the equation $L\psi = \lambda\psi$ becomes a first-order system $\Psi_x = U(\lambda)\Psi$ with companion matrix

$$U(\lambda) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \lambda - u_0 & -u_1 & -u_2 & \cdots & -u_{n-1} \end{pmatrix}.$$

Similarly, differentiating $\psi_t + A\psi = 0$ a total of $i - 1$ times with respect to x and using the above to eliminate high derivatives, we get

$$(\partial_x^{i-1} \psi)_t + \sum_{j=1}^n V_{ij}(x, t) \partial_x^{j-1} \psi = 0$$

for suitable V_{ij} built from the u_j, v_j and their derivatives, that is $\Psi_t = -V\Psi$ (after an overall sign). So

$$L_t = [L, A] \iff \begin{cases} L\psi = \lambda\psi \\ \psi_t + A\psi = 0 \end{cases} \iff \begin{cases} \Psi_x = U(\lambda)\Psi \\ \Psi_t = V(\lambda)\Psi \end{cases} \iff U_t - V_x + [U, V] = 0.$$

The zero-curvature formulation is strictly more general – there are integrable equations with a zero-curvature representation but no scalar Lax pair – and it is the one that generalises most readily to other settings.

§32.5 Further Integrable Equations

KdV is not alone. In this section we meet two more equations which are integrable in the same sense – they have Lax pairs, infinitely many conservation laws and soliton solutions – and which are at least as important in applications.

§32.5.1 The Sine–Gordon Equation

Definition 32.5.1 (Sine–Gordon equation)

The **sine–Gordon equation** is

$$u_{tt} - u_{xx} + \sin u = 0.$$

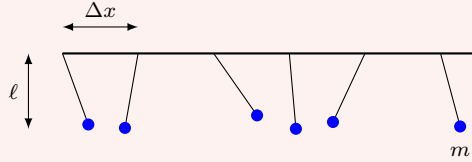
The name is a joke. The **Klein–Gordon equation** of relativistic quantum mechanics is

$$u_{tt} - u_{xx} + u = 0,$$

and replacing the u by $\sin u$ turns Klein into sine.

Example 32.5.2 (A chain of pendulums)

Here is a mechanical system governed by the equation. Take a horizontal rod, and hang from it a row of pendulums of length ℓ and mass m , spaced Δx apart, each free to rotate in the vertical plane perpendicular to the rod. Let $\theta_i(t)$ be the angle of the i th pendulum from the downward vertical, and connect neighbouring masses by torsional springs.



Gravity exerts a torque $-m\ell g \sin \theta_i$ on the i th pendulum, and the springs exert torques

$$\frac{K(\theta_{i+1} - \theta_i)}{\Delta x}, \quad \frac{K(\theta_{i-1} - \theta_i)}{\Delta x}$$

from its two neighbours. Newton's second law for rotation, with moment of inertia $m\ell^2$, gives

$$m\ell^2 \frac{d^2 \theta_i}{dt^2} = -m\ell g \sin \theta_i + \frac{K(\theta_{i+1} - 2\theta_i + \theta_{i-1}))}{\Delta x}.$$

Now divide by Δx and take the continuum limit $\Delta x \rightarrow 0$, holding $M = m/\Delta x$ fixed and writing $\theta_i(t) = \theta(i\Delta x, t)$. The second difference over Δx^2 tends to θ_{xx} , so

$$M\ell^2 \frac{\partial^2 \theta}{\partial t^2} = -M\ell g \sin \theta + K \frac{\partial^2 \theta}{\partial x^2}.$$

Rescaling x and t to remove the constants gives $u_{tt} - u_{xx} + \sin u = 0$.

There is also a differential-geometric origin: solutions of the sine–Gordon equation correspond to *pseudospherical* surfaces in $\mathbb{R}^3$, that is surfaces of constant negative Gaussian curvature. This is the historical source of the equation, and of the Bäcklund transformations of Section 6.

It is often convenient to pass to **light-cone coordinates**

$$\xi = \frac{1}{2}(x - t), \quad \tau = \frac{1}{2}(x + t),$$

in which $\partial_x = \frac{1}{2}(\partial_\xi + \partial_\tau)$ and $\partial_t = \frac{1}{2}(\partial_\tau - \partial_\xi)$, so that $\partial_t^2 - \partial_x^2 = -\partial_\xi \partial_\tau$. The equation becomes

$$\frac{\partial^2 u}{\partial \xi \partial \tau} = \sin u,$$

which is the form we will mostly use.

Kinks and Antikinks

Let us find the travelling waves. Put $u(x, t) = f(\gamma(x - vt))$ with $|v| < 1$ and $\gamma = (1 - v^2)^{-1/2}$, and write $\theta = \gamma(x - vt)$. Then $u_{tt} = v^2 \gamma^2 f''$ and $u_{xx} = \gamma^2 f''$, so

$$\gamma^2(v^2 - 1)f'' + \sin f = 0, \quad \text{i.e.} \quad f'' = \sin f,$$

since $\gamma^2(v^2 - 1) = -1$. This is the equation of a pendulum with the sign of gravity reversed – which is no coincidence, given where the equation came from. Multiplying by f' and integrating,

$$\frac{1}{2}(f')^2 = -\cos f + C,$$

and imposing $f \rightarrow 0, f' \rightarrow 0$ as $\theta \rightarrow -\infty$ gives $C = 1$. Hence

$$(f')^2 = 2(1 - \cos f) = 4 \sin^2(f/2), \quad f' = 2 \sin(f/2),$$

taking the positive root. Separating variables and substituting $g = f/2$,

$$\int \frac{dg}{\sin g} = \int d\theta \implies \log \tan \frac{g}{2} = \theta + \text{const},$$

so $\tan(f/4) = Ae^\theta$. Absorbing A into a translation we obtain the following.

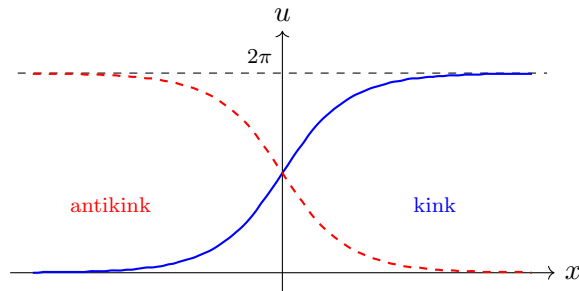
Proposition 32.5.3 (Kink solution)

For any $|v| < 1$,

$$u(x, t) = 4 \tan^{-1} \left(\exp \left(\frac{x - vt}{\sqrt{1 - v^2}} \right) \right)$$

solves the sine–Gordon equation. It is called a **kink**. Replacing $x - vt$ by $-(x - vt)$ gives the **antikink**.

The kink interpolates between $u = 0$ at $x = -\infty$ and $u = 2\pi$ at $x = +\infty$; the antikink runs the other way.



Note the Lorentz factor γ : a fast kink is narrower than a slow one, exactly as a moving rod is contracted in special relativity. This is not an accident, since the sine–Gordon equation is Lorentz invariant.

In the pendulum picture, a kink is a single full twist of the chain, propagating along it. Now remember that u is an *angle*: the values 0 and 2π describe the same physical configuration. So both the kink and the trivial solution $u = 0$ satisfy the same boundary conditions in the physical sense. Nevertheless the kink can never evolve into the trivial solution, and here is why.

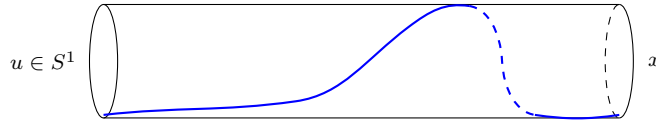
Definition 32.5.4 (Topological charge)

For a solution with $u \rightarrow 2\pi n_{\pm}$ as $x \rightarrow \pm\infty$, the **topological charge** is

$$Q = \frac{1}{2\pi} [u]_{x=-\infty}^{x=+\infty} = n_+ - n_- \in \mathbb{Z}.$$

Since u evolves continuously and Q takes values in $\mathbb{Z}$, the charge cannot change: it is conserved for purely topological reasons, with no differential equation required. The kink has $Q = 1$, the antikink $Q = -1$, and the vacuum $Q = 0$. So a kink is stable, not because it sits at a minimum of some energy, but because there is no continuous path from it to anything else.

If we think of u as taking values on the circle S^1 , the picture is that the solution is a path in the cylinder $\mathbb{R} \times S^1$ which winds Q times around:



No amount of continuous deformation, subject to the boundary conditions, can unwind it.

Remark. Sine–Gordon has multi-soliton solutions too, obtained (as we shall see in Section 6) by Bäcklund transformations. Besides kink–kink and kink–antikink scattering solutions there is a genuinely new object, the **breather**

$$u(x, t) = 4 \tan^{-1} \left(\frac{\sqrt{1 - \omega^2}}{\omega} \frac{\sin(\omega t)}{\cosh(\sqrt{1 - \omega^2} x)} \right),$$

a kink and antikink bound together, oscillating in place with frequency ω . It has $Q = 0$, and yet it is perfectly stable – a periodic, localised, nonlinear excitation with no linear analogue.

§32.5.2 The Nonlinear Schrödinger Equation

Our final example is the equation governing envelope waves: pulses in optical fibres, wave packets on deep water, and Bose–Einstein condensates.

Definition 32.5.5 (Nonlinear Schrödinger equation)

The **nonlinear Schrödinger equation** (NLS) for a complex-valued $q(x, t)$ is

$$iq_t + q_{xx} + 2|q|^2q = 0.$$

The sign of the nonlinear term matters. With $+2|q|^2q$ the equation is **focusing**, and supports *bright* solitons – localised pulses on a zero background. With $-2|q|^2q$ it is **defocusing**, and supports *dark* solitons – localised dips in a non-zero background. We treat the focusing case.

The equation looks like the linear Schrödinger equation of IB Quantum Mechanics with a potential $-2|q|^2$ generated by the wavefunction itself, which is exactly the physical situation in a nonlinear optical medium whose refractive index depends on the intensity of the light passing through it.

Let us find the soliton. Look first for a standing solution

$$q(x, t) = f(x)e^{i\theta t}$$

with f real. Then $iq_t = -\theta f e^{i\theta t}$, and the equation becomes

$$f'' = \theta f - 2f^3.$$

Multiplying by f' and integrating, with $f, f' \rightarrow 0$ at infinity,

$$(f')^2 = \theta f^2 - f^4 = f^2(\theta - f^2).$$

This has exactly the shape of the KdV travelling-wave equation, and the same guess works. Trying $f = a \operatorname{sech}(ax)$ we get

$$(f')^2 = a^4 \operatorname{sech}^2(ax) \tanh^2(ax) = a^4 \operatorname{sech}^2 - a^4 \operatorname{sech}^4,$$

while

$$f^2(\theta - f^2) = a^2 \theta \operatorname{sech}^2 - a^4 \operatorname{sech}^4,$$

so the two agree precisely when $\theta = a^2$.

Finally we set the soliton moving. NLS is Galilean invariant: if $q(x, t)$ is a solution then so is

$$\tilde{q}(x, t) = e^{i\left(\frac{v}{2}x - \frac{v^2}{4}t\right)} q(x - vt, t),$$

as one checks by direct substitution. Applying this to the standing solution gives the following.

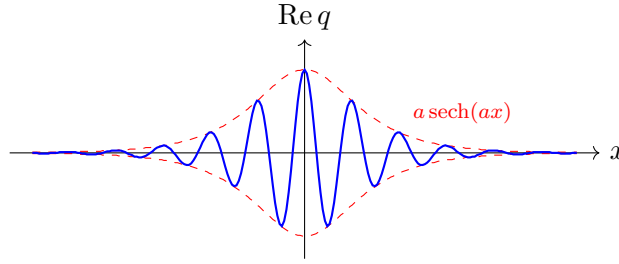
Proposition 32.5.6 (Bright soliton)

For any $a > 0$ and $v \in \mathbb{R}$,

$$q(x, t) = a \operatorname{sech}(a(x - vt)) \exp\left(i\left[\frac{v}{2}x + \left(a^2 - \frac{v^2}{4}\right)t\right]\right)$$

solves the focusing NLS equation.

So the modulus $|q| = a \operatorname{sech}(a(x - vt))$ is a solitary wave of amplitude a and width $1/a$, travelling at speed v , while the phase oscillates underneath it:



Note the crucial difference from KdV: here the amplitude and the speed are *independent* parameters. A tall NLS soliton is narrow, but it need not be fast.

Like KdV, the NLS equation has a Lax pair and is solvable by inverse scattering; the relevant linear problem is the 2×2 Zakharov–Shabat system,

$$\frac{\partial}{\partial x} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} -i\lambda & q \\ -\bar{q} & i\lambda \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix},$$

which is exactly of the zero-curvature type $\Psi_x = U(\lambda)\Psi$ discussed in Section 4.3. Each discrete eigenvalue of this system corresponds to a soliton, with the real and imaginary parts of the eigenvalue giving the soliton’s amplitude and velocity respectively. NLS solitons, like KdV solitons, pass through one another with nothing but a phase shift to show for it.

Finally, NLS is a Hamiltonian system in the sense of Section 4.1, with

$$iq_t = \frac{\delta H}{\delta \bar{q}}, \quad H[q] = \int (|q_x|^2 - |q|^4) \, dx,$$

and it has the conserved quantities

$$\int |q|^2 \, dx, \quad \frac{i}{2} \int (q\bar{q}_x - \bar{q}q_x) \, dx, \quad H[q]$$

– the number of photons, the momentum, and the energy – together with infinitely many more.

§32.6 Bäcklund Transformations

For a linear PDE we have the principle of superposition: add two solutions and get a third. For a nonlinear PDE this fails completely, and it is precisely this failure that makes nonlinear equations hard. Bäcklund transformations are a partial substitute – a way of manufacturing new solutions from old, not by adding them, but by solving an auxiliary system of first-order equations.

The idea originated in differential geometry, where Bäcklund studied transformations taking one pseudospherical surface to another. We will only look at what it does to PDEs. The general definition is a little slippery, so we start with an example that will be familiar.

Example 32.6.1 (Cauchy–Riemann)

Consider the Cauchy–Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

From IB Complex Analysis we know that a pair (u, v) satisfies these if and only if both u and v are harmonic, that is

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0.$$

(One direction is immediate: $u_{xx} + u_{yy} = (v_y)_x + (-v_x)_y = 0$.)

So suppose we have found some harmonic function v . We may then try to *solve* the Cauchy–Riemann equations for u , and whatever we get is guaranteed to be harmonic as well.

For instance take $v = 2xy$, which is harmonic. The Cauchy–Riemann equations become

$$u_x = 2x, \quad u_y = -2y,$$

which we integrate to get

$$u(x, y) = x^2 - y^2 + C$$

for an arbitrary constant C . And indeed $x^2 - y^2$ is harmonic.

So the Cauchy–Riemann equations act as a machine for turning solutions of Laplace’s equation into other solutions of Laplace’s equation, at the cost of solving a first-order system. This is exactly what a Bäcklund transformation does.

Definition 32.6.2 (Bäcklund transformation)

A **Bäcklund transformation** is a system of equations relating solutions of one PDE to

- (i) solutions of a *different* PDE; or
- (ii) other solutions of the *same* PDE.

In the second case we call it an **auto-Bäcklund transformation**.

Transformations of the first kind are useful when one of the two equations is easier than the other.

Example 32.6.3 (The Liouville equation)

The equation $u_{xt} = e^u$ is related to the trivial equation $v_{xt} = 0$ by the Bäcklund transformation

$$u_x + v_x = \sqrt{2} \exp\left(\frac{u - v}{2}\right), \quad u_t - v_t = \sqrt{2} \exp\left(\frac{u + v}{2}\right).$$

Verifying this is an exercise on the first example sheet. Since $v_{xt} = 0$ has the general solution $v = f(x) + g(t)$, this reduces the nonlinear Liouville equation to a pair of first-order ODEs.

§32.6.1 An Auto-Bäcklund Transformation for Sine–Gordon

The most important example, and the one we will use, is for the sine–Gordon equation in light-cone coordinates,

$$\varphi_{\xi\tau} = \sin \varphi.$$

Proposition 32.6.4

Fix a non-zero constant ε , and suppose φ and ψ are related by

$$\frac{\partial}{\partial \xi} (\varphi - \psi) = 2\varepsilon \sin \left(\frac{\varphi + \psi}{2} \right), \quad (\text{B1})$$

$$\frac{\partial}{\partial \tau} (\varphi + \psi) = \frac{2}{\varepsilon} \sin \left(\frac{\varphi - \psi}{2} \right). \quad (\text{B2})$$

Then φ solves the sine-Gordon equation if and only if ψ does.

Proof. Differentiate (B1) with respect to τ and use (B2):

$$\begin{aligned} (\varphi - \psi)_{\xi\tau} &= 2\varepsilon \cos \left(\frac{\varphi + \psi}{2} \right) \cdot \frac{1}{2} (\varphi + \psi)_\tau \\ &= \varepsilon \cos \left(\frac{\varphi + \psi}{2} \right) \cdot \frac{2}{\varepsilon} \sin \left(\frac{\varphi - \psi}{2} \right) \\ &= 2 \cos \left(\frac{\varphi + \psi}{2} \right) \sin \left(\frac{\varphi - \psi}{2} \right) \\ &= \sin \varphi - \sin \psi, \end{aligned}$$

using the identity $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$.

Similarly, differentiating (B2) with respect to ξ and using (B1),

$$\begin{aligned} (\varphi + \psi)_{\xi\tau} &= \frac{2}{\varepsilon} \cos \left(\frac{\varphi - \psi}{2} \right) \cdot \frac{1}{2} (\varphi - \psi)_\xi \\ &= \frac{1}{\varepsilon} \cos \left(\frac{\varphi - \psi}{2} \right) \cdot 2\varepsilon \sin \left(\frac{\varphi + \psi}{2} \right) \\ &= \sin \varphi + \sin \psi. \end{aligned}$$

Adding the two displayed results gives $2\varphi_{\xi\tau} = 2 \sin \varphi$, and subtracting them gives $2\psi_{\xi\tau} = 2 \sin \psi$. So in fact both equations hold, and in particular each implies the other. $\square$

Note that we have a whole *family* of auto-Bäcklund transformations, one for each $\varepsilon \neq 0$. This parameter will be essential.

§32.6.2 Generating Solitons from Nothing

Let us put the transformation to work. We start with the most boring solution imaginable, $\psi \equiv 0$, and see what comes out. The equations (B1), (B2) become

$$\varphi_\xi = 2\varepsilon \sin \frac{\varphi}{2}, \quad \varphi_\tau = \frac{2}{\varepsilon} \sin \frac{\varphi}{2}.$$

There is an evident symmetry between ξ and τ here, which suggests the ansatz

$$\varphi(\xi, \tau) = 2\chi(z), \quad z = \varepsilon\xi + \frac{\tau}{\varepsilon}.$$

Then $\varphi_\xi = 2\varepsilon\chi'$ and $\varphi_\tau = (2/\varepsilon)\chi'$, and *both* equations reduce to the same ODE,

$$\frac{d\chi}{dz} = \sin \chi.$$

This we can separate:

$$\int \frac{d\chi}{\sin \chi} = \int dz \implies \log \tan \frac{\chi}{2} = z + \text{const},$$

so $\tan(\chi/2) = Ae^z$ and $\chi = 2 \tan^{-1}(Ae^z)$. Therefore

$$\varphi(\xi, \tau) = 4 \tan^{-1} \left(A \exp \left(\varepsilon \xi + \frac{\tau}{\varepsilon} \right) \right).$$

Let us see what this is. Substituting $\xi = \frac{1}{2}(x - t)$ and $\tau = \frac{1}{2}(x + t)$,

$$\varepsilon \xi + \frac{\tau}{\varepsilon} = \frac{\varepsilon^2 + 1}{2\varepsilon} x + \frac{1 - \varepsilon^2}{2\varepsilon} t = \gamma(x - vt),$$

where

$$\gamma = \frac{\varepsilon^2 + 1}{2\varepsilon}, \quad v = \frac{\varepsilon^2 - 1}{\varepsilon^2 + 1}.$$

A quick check confirms these are consistent with $\gamma = (1 - v^2)^{-1/2}$:

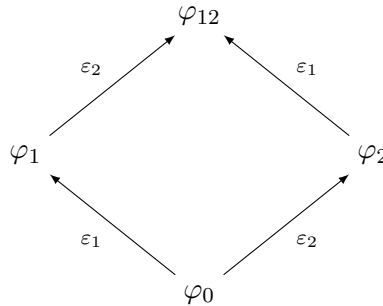
$$1 - v^2 = \frac{(\varepsilon^2 + 1)^2 - (\varepsilon^2 - 1)^2}{(\varepsilon^2 + 1)^2} = \frac{4\varepsilon^2}{(\varepsilon^2 + 1)^2}.$$

So φ is exactly the kink of Section 5.1, and the Bäcklund parameter ε is nothing but a relabelling of the kink's velocity. We have generated a soliton out of the vacuum, by solving two first-order ODEs.

The obvious next move is to apply the transformation again, to the kink instead of the vacuum, and get a two-soliton solution. This works, but it requires solving another pair of PDEs, and the algebra gets unpleasant. Remarkably, there is a way to avoid it entirely.

§32.6.3 Bianchi Permutability

Suppose we start from a solution φ_0 and apply the Bäcklund transformation twice, with two different parameters ε_1 and ε_2 , in the two possible orders. Write $\varphi_1 = B_{\varepsilon_1}(\varphi_0)$ and $\varphi_2 = B_{\varepsilon_2}(\varphi_0)$. The claim is that we may then choose $B_{\varepsilon_2}(\varphi_1)$ and $B_{\varepsilon_1}(\varphi_2)$ to be the *same* solution φ_{12} – the transformations commute:



This is the **Bianchi permutability theorem**, and its real content is that φ_{12} can be written down *algebraically*, with no integration at all.

Theorem 32.6.5 (Bianchi permutability)

With notation as above,

$$\tan\left(\frac{\varphi_{12} - \varphi_0}{4}\right) = \frac{\varepsilon_1 + \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \tan\left(\frac{\varphi_1 - \varphi_2}{4}\right).$$

Proof. We use only (B1), applied along each of the four edges of the diagram:

$$\begin{aligned} (\varphi_1 - \varphi_0)_\xi &= 2\varepsilon_1 \sin\left(\frac{\varphi_1 + \varphi_0}{2}\right), & (\varphi_2 - \varphi_0)_\xi &= 2\varepsilon_2 \sin\left(\frac{\varphi_2 + \varphi_0}{2}\right), \\ (\varphi_{12} - \varphi_1)_\xi &= 2\varepsilon_2 \sin\left(\frac{\varphi_{12} + \varphi_1}{2}\right), & (\varphi_{12} - \varphi_2)_\xi &= 2\varepsilon_1 \sin\left(\frac{\varphi_{12} + \varphi_2}{2}\right). \end{aligned}$$

Adding the first and third gives an expression for $(\varphi_{12} - \varphi_0)_\xi$; adding the second and fourth gives another. Equating them,

$$\varepsilon_1 \sin\left(\frac{\varphi_1 + \varphi_0}{2}\right) + \varepsilon_2 \sin\left(\frac{\varphi_{12} + \varphi_1}{2}\right) = \varepsilon_2 \sin\left(\frac{\varphi_2 + \varphi_0}{2}\right) + \varepsilon_1 \sin\left(\frac{\varphi_{12} + \varphi_2}{2}\right),$$

that is

$$\varepsilon_1 \left[\sin\left(\frac{\varphi_1 + \varphi_0}{2}\right) - \sin\left(\frac{\varphi_{12} + \varphi_2}{2}\right) \right] = \varepsilon_2 \left[\sin\left(\frac{\varphi_2 + \varphi_0}{2}\right) - \sin\left(\frac{\varphi_{12} + \varphi_1}{2}\right) \right].$$

Now apply $\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$ to each side. The crucial point is that the *same* quantity

$$\frac{A+B}{2} = \frac{\varphi_0 + \varphi_1 + \varphi_2 + \varphi_{12}}{4}$$

appears on both sides, so the cosine factor cancels (generically it is non-zero). We are left with

$$\varepsilon_1 \sin\left(\frac{\varphi_0 + \varphi_1 - \varphi_2 - \varphi_{12}}{4}\right) = \varepsilon_2 \sin\left(\frac{\varphi_0 - \varphi_1 + \varphi_2 - \varphi_{12}}{4}\right).$$

Write

$$P = \frac{\varphi_0 - \varphi_{12}}{4}, \quad Q = \frac{\varphi_1 - \varphi_2}{4},$$

so this reads $\varepsilon_1 \sin(P+Q) = \varepsilon_2 \sin(P-Q)$. Expanding both sides,

$$\varepsilon_1 (\sin P \cos Q + \cos P \sin Q) = \varepsilon_2 (\sin P \cos Q - \cos P \sin Q),$$

so

$$(\varepsilon_1 - \varepsilon_2) \sin P \cos Q = -(\varepsilon_1 + \varepsilon_2) \cos P \sin Q,$$

and dividing,

$$\tan P = -\frac{\varepsilon_1 + \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \tan Q.$$

Since $P = -(\varphi_{12} - \varphi_0)/4$, this is the stated formula. $\square$

This is a genuine **nonlinear superposition principle**. Given the vacuum $\varphi_0 = 0$ and two one-soliton solutions φ_1, φ_2 , it delivers the two-soliton solution

$$\varphi_{12} = 4 \tan^{-1} \left(\frac{\varepsilon_1 + \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \tan \left(\frac{\varphi_1 - \varphi_2}{4} \right) \right)$$

by pure algebra. Iterating produces the whole soliton hierarchy of sine–Gordon: from

φ_{12} and φ_{13} we get φ_{123} , and so on. Every multi-soliton solution of the sine–Gordon equation can be reached this way, starting from nothing.

Example 32.6.6

Take $\varphi_0 = 0$ and two kinks

$$\varphi_j = 4 \tan^{-1}(\exp(\gamma_j(x - v_j t))), \quad j = 1, 2,$$

with parameters $\varepsilon_1, \varepsilon_2$ as above. Writing $E_j = \exp(\gamma_j(x - v_j t))$ and using

$$\tan\left(\frac{\varphi_1 - \varphi_2}{4}\right) = \frac{\tan(\varphi_1/4) - \tan(\varphi_2/4)}{1 + \tan(\varphi_1/4)\tan(\varphi_2/4)} = \frac{E_1 - E_2}{1 + E_1 E_2},$$

the theorem gives the closed form

$$\varphi_{12}(x, t) = 4 \tan^{-1}\left(\frac{\varepsilon_1 + \varepsilon_2}{\varepsilon_1 - \varepsilon_2} \cdot \frac{E_1 - E_2}{1 + E_1 E_2}\right).$$

Depending on the signs and magnitudes of the parameters this describes kink–kink scattering, kink–antikink scattering, or – for complex conjugate ε ’s – the breather of Section 5.1. As with KdV, the only lasting effect of the collision is a phase shift.

Remark. It is no coincidence that we found a nonlinear superposition principle here and a determinant formula for KdV in Section 3.5. Both are shadows of the same underlying structure, and for KdV there is a Bäcklund transformation too, namely

$$w_{1,x} + w_{2,x} = \lambda - \frac{1}{2}(w_1 - w_2)^2$$

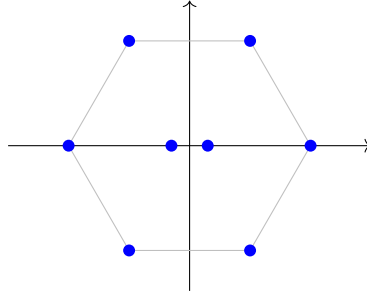
relating potentials $u_j = w_{j,x}$. Its Bianchi permutability formula reproduces exactly the N -soliton solutions we obtained from the GLM equation.

§32.7 Symmetry Methods

We finish with a topic which is both older and more general than everything else in this course: the systematic use of symmetry to solve differential equations. This is Lie’s programme, and it was the original reason Sophus Lie invented what we now call Lie groups.

Since differential equations are continuous objects, the symmetries we care about come in continuous families, and to handle these we need groups equipped with a smooth structure.

At this point a reader of applied inclination may be tempted to skip anything with the word ‘group’ in it. It is worth resisting the temptation, and here is a cautionary tale (whose historical accuracy I will not vouch for). In the 1960s Gell-Mann was trying to make sense of the bewildering zoo of apparently fundamental particles. He plotted them according to two quantum numbers, isospin and hypercharge, and obtained the following picture:



Clearly there is a pattern, but it is not at all clear where it comes from or how to extend it. The story goes that a pure mathematician wandered into the physics department, saw the diagram, and asked ‘so you’re interested in the eight-dimensional adjoint representation of $\mathfrak{su}(3)$ too?’. He was not. But the weight diagram of that representation looks exactly like the picture, and once the correspondence between representations of $\mathfrak{su}(3)$ and quantum numbers was understood, the whole subject fell into place.

§32.7.1 Lie Groups and Lie Algebras

We begin by recalling, from IA Groups, what a group is.

Definition 32.7.1 (Group)

A **group** is a set G with a binary operation $(g_1, g_2) \mapsto g_1 g_2$ which is associative, has an identity e , and in which every element has an inverse.

Definition 32.7.2 (Group action)

A group G **acts** on a set X if there is a map $G \times X \rightarrow X$, $(g, x) \mapsto g(x)$, with

$$g(h(x)) = (gh)(x), \quad e(x) = x$$

for all $g, h \in G$ and $x \in X$.

We now want groups whose elements depend smoothly on parameters. The honest definition involves manifolds; we adopt a working version which is really the definition of a *local* Lie group, and which is all we shall need.

Definition 32.7.3 (Lie group)

An **m -dimensional Lie group** is a group whose elements depend continuously on m real parameters, in such a way that the maps $(g_1, g_2) \mapsto g_1 g_2$ and $g \mapsto g^{-1}$ are smooth functions of those parameters.

In practice it is enough to check that $(g_1, g_2) \mapsto g_1 g_2^{-1}$ is smooth. We write elements as $g(\mathbf{t})$ with $\mathbf{t} \in \mathbb{R}^m$, normalised so that $g(\mathbf{0}) = e$.

Example 32.7.4

Every element of $G = \text{SO}(2)$ can be written as

$$g(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}, \quad t \in \mathbb{R}.$$

Since $g(t)^{-1} = g(-t)$, we have $g(t_1)g(t_2)^{-1} = g(t_1 - t_2)$, so in terms of parameters the map is $(t_1, t_2) \mapsto t_1 - t_2$, which is smooth. So $\text{SO}(2)$ is a one-dimensional Lie group.

Example 32.7.5 (The Heisenberg group)

Consider the matrices

$$g(\mathbf{t}) = \begin{pmatrix} 1 & t_1 & t_3 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{t} \in \mathbb{R}^3.$$

Multiplying,

$$g(\mathbf{a})g(\mathbf{b}) = \begin{pmatrix} 1 & a_1 + b_1 & a_3 + b_3 + a_1b_2 \\ 0 & 1 & a_2 + b_2 \\ 0 & 0 & 1 \end{pmatrix},$$

so this is closed under multiplication, and

$$g(\mathbf{b})^{-1} = \begin{pmatrix} 1 & -b_1 & b_1b_2 - b_3 \\ 0 & 1 & -b_2 \\ 0 & 0 & 1 \end{pmatrix}.$$

Combining,

$$(\mathbf{a}, \mathbf{b}) \mapsto (a_1 - b_1, a_2 - b_2, a_3 - b_3 + b_1b_2 - a_1b_2),$$

which is a polynomial, hence smooth. So the **Heisenberg group** is a three-dimensional Lie group.

Lie groups are still complicated. As with flows in Section 1.1, the trick is to look at the infinitesimal picture instead, which is linear and therefore easy. We do this informally, and only for matrix groups, so that we may freely add and differentiate group elements.

Let $A(\varepsilon)$ be a smooth curve in G with $A(0) = I$, and set $a = A'(0)$, so that

$$A(\varepsilon) = I + \varepsilon a + o(\varepsilon).$$

Define $\mathfrak{g}$ to be the set of all such ‘leading order terms’ a . We claim $\mathfrak{g}$ is a vector space, and more.

If $B(\varepsilon) = I + \varepsilon b + o(\varepsilon)$ is another such curve, then $\varepsilon \mapsto A(\varepsilon)B(\varepsilon)$ is a curve in G with

$$A(\varepsilon)B(\varepsilon) = (I + \varepsilon a + o(\varepsilon))(I + \varepsilon b + o(\varepsilon)) = I + \varepsilon(a + b) + o(\varepsilon),$$

so $a + b \in \mathfrak{g}$. And $\varepsilon \mapsto A(\lambda\varepsilon)$ gives $\lambda a \in \mathfrak{g}$ for any $\lambda \in \mathbb{R}$. So $\mathfrak{g}$ is a vector space, which is already interesting: out of a group, with its awkward non-commutative multiplication, we have extracted a vector space.

But there is more. Note first that $A(\varepsilon)^{-1} = I - \varepsilon a + o(\varepsilon)$, since if $A(\varepsilon)^{-1} = I + \varepsilon \tilde{a} + o(\varepsilon)$ then

$$I = A(\varepsilon)A(\varepsilon)^{-1} = I + \varepsilon(a + \tilde{a}) + o(\varepsilon),$$

forcing $\tilde{a} = -a$. Now consider the curve

$$C(\varepsilon) = A(\sqrt{\varepsilon})B(\sqrt{\varepsilon})A(\sqrt{\varepsilon})^{-1}B(\sqrt{\varepsilon})^{-1}.$$

Expanding and keeping track of the second-order terms (which is a slightly fiddly exercise), the first-order terms cancel and one finds

$$C(\varepsilon) = I + \varepsilon(ab - ba) + o(\varepsilon).$$

So $\mathfrak{g}$ is closed under the commutator. The moral is that although the product of two elements infinitesimally close to the identity need not be infinitesimally close in the same sense, the group-theoretic commutator always is.

Definition 32.7.6 (Lie algebra)

A **Lie algebra** is a vector space $\mathfrak{g}$ equipped with a bilinear, antisymmetric bracket $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying the **Jacobi identity**

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0.$$

Example 32.7.7

$\mathfrak{g} = \mathbb{R}^3$ with $[\mathbf{a}, \mathbf{b}] = \mathbf{a} \wedge \mathbf{b}$ is a Lie algebra.

Example 32.7.8

Let M be a phase space and $\mathfrak{g}$ the space of smooth functions $M \rightarrow \mathbb{R}$. Then

$$[f, g] = \{f, g\}$$

is a Lie algebra, by the properties of the Poisson bracket in Section 1.2. So Section 1 was secretly about Lie algebras all along.

Example 32.7.9

Let $G = \mathrm{SO}(n) = \{A \in \mathrm{Mat}_n(\mathbb{R}) : AA^T = I, \det A = 1\}$, and let $A(\varepsilon)$ be a curve with $A(0) = I$. Then

$$I = A(\varepsilon)A(\varepsilon)^T = (I + \varepsilon a + o(\varepsilon))(I + \varepsilon a^T + o(\varepsilon)) = I + \varepsilon(a + a^T) + o(\varepsilon),$$

so a must be antisymmetric. The determinant condition gives

$$1 = \det(I + \varepsilon a + o(\varepsilon)) = 1 + \varepsilon \operatorname{tr} a + o(\varepsilon),$$

so $\operatorname{tr} a = 0$, which antisymmetry already guarantees. Conversely every antisymmetric a arises, from the curve $A(t) = \exp(at)$. So

$$\mathfrak{so}(n) = \{a \in \mathrm{Mat}_n(\mathbb{R}) : a + a^T = 0\},$$

and one checks directly that it is closed under the commutator:

$$[a, b]^T = (ab - ba)^T = b^T a^T - a^T b^T = ba - ab = -[a, b].$$

The convention is that a group written in capitals has a Lie algebra written with the same letters in lower-case fraktur: $\mathrm{SO}(n)$ has $\mathfrak{so}(n)$, $\mathrm{SL}(n)$ has $\mathfrak{sl}(n)$, and so on. Choosing a basis $\{a_i\}$ of a finite-dimensional $\mathfrak{g}$, the bracket is determined by the **structure constants** c_{ij}^k in

$$[a_i, a_j] = \sum_k c_{ij}^k a_k.$$

§32.7.2 One-Parameter Groups of Transformations

Ultimately we want groups acting on the space in which our differential equation lives, giving new coordinates $\tilde{\mathbf{x}} = g(\mathbf{x})$. Happily, the action of a whole Lie group can be reduced to the study of one-parameter families, which we already understand from Section 1.1.

Definition 32.7.10 (One-parameter group of transformations)

A smooth family of maps $g^\varepsilon : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a **one-parameter group of transformations** (1.p.g.t.) if

$$g^0 = \mathrm{id}, \quad g^{\varepsilon_1} g^{\varepsilon_2} = g^{\varepsilon_1 + \varepsilon_2}.$$

It is **generated by** the vector field

$$\mathbf{V}(\mathbf{x}) = \left. \frac{d}{d\varepsilon} (g^\varepsilon \mathbf{x}) \right|_{\varepsilon=0},$$

so that $g^\varepsilon \mathbf{x} = \mathbf{x} + \varepsilon \mathbf{V}(\mathbf{x}) + o(\varepsilon)$. Conversely every vector field generates a 1.p.g.t. by solving

$$\frac{d\tilde{\mathbf{x}}}{d\varepsilon} = \mathbf{V}(\tilde{\mathbf{x}}), \quad \tilde{\mathbf{x}}(0) = \mathbf{x}.$$

This is exactly the flow-map picture of Section 1.1, with ε in place of t . Differential geometers write vector fields in a way that looks odd at first but is extremely convenient.

Notation. A vector field $\mathbf{V} = (V_1, \dots, V_n)^T$ is identified with the differential operator

$$V = V_1 \frac{\partial}{\partial x_1} + \dots + V_n \frac{\partial}{\partial x_n},$$

and we think of the $\partial/\partial x_i$ as a basis. We will freely confuse the two.

Example 32.7.11

The vector field $\mathbf{V} = (x^2 + y, yx)$ is written $V = (x^2 + y)\partial_x + yx\partial_y$.

The reason for the notation is that the commutator becomes literally the commutator of operators. Indeed, computing $(VW - WV)f$ for a test function f ,

$$\begin{aligned} (VW - WV)(f) &= V_j \frac{\partial}{\partial x_j} \left(W_i \frac{\partial f}{\partial x_i} \right) - W_j \frac{\partial}{\partial x_j} \left(V_i \frac{\partial f}{\partial x_i} \right) \\ &= \left(V_j \frac{\partial W_i}{\partial x_j} - W_j \frac{\partial V_i}{\partial x_j} \right) \frac{\partial f}{\partial x_i} + (V_j W_i - W_j V_i) \frac{\partial^2 f}{\partial x_i \partial x_j}, \end{aligned}$$

and the second-derivative terms cancel by symmetry, leaving exactly the commutator of vector fields from Section 1.1. So in this notation

$$[V, W] = VW - WV.$$

Let us see some examples of the correspondence between vector fields and transformations.

Example 32.7.12

Let $V = x\partial_x + \partial_y$. The 1.p.g.t. it generates solves

$$\frac{d\tilde{x}}{d\varepsilon} = \tilde{x}, \quad \frac{d\tilde{y}}{d\varepsilon} = 1, \quad (\tilde{x}(0), \tilde{y}(0)) = (x, y),$$

so

$$g^\varepsilon(x, y) = (xe^\varepsilon, y + \varepsilon).$$

Example 32.7.13

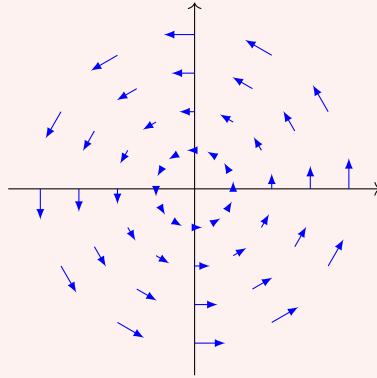
The natural action of $SO(2)$ on $\mathbb{R}^2$ is

$$g^\varepsilon(x, y) = (x \cos \varepsilon - y \sin \varepsilon, x \sin \varepsilon + y \cos \varepsilon),$$

which satisfies $g^0 = \text{id}$ and $g^{\varepsilon_1}g^{\varepsilon_2} = g^{\varepsilon_1+\varepsilon_2}$. Its generator is

$$V = \left. \frac{d\tilde{x}}{d\varepsilon} \right|_0 \partial_x + \left. \frac{d\tilde{y}}{d\varepsilon} \right|_0 \partial_y = -y\partial_x + x\partial_y,$$

which we can draw:



The arrows are tangent to circles about the origin, and grow with the radius, as they should for a rotation.

Example 32.7.14

$V = \alpha\partial_x$ generates $g^\varepsilon x = x + \alpha\varepsilon$, a translation at constant speed; while $V = \beta x\partial_x$ generates $g^\varepsilon x = e^{\beta\varepsilon}x$, an exponential dilation.

The relevance of all this to Lie groups is the following. If a Lie group G acts on $\mathbb{R}^n$, it typically contains many one-parameter subgroups, and any $g(\mathbf{t}) \in G$ sufficiently close to the identity can be written as a product

$$g(\mathbf{t}) = g_{i_1}^{\varepsilon_1} g_{i_2}^{\varepsilon_2} \cdots g_{i_N}^{\varepsilon_N}$$

of elements of such subgroups. So to understand the action of G it suffices to understand its one-parameter subgroups, and to understand those it suffices to understand the vector

fields generating them – that is, the Lie algebra. Once more, the infinitesimal picture is enough.

§32.7.3 Symmetries of Differential Equations

We write a general differential equation as

$$\Delta[\mathbf{x}, u, u_{\mathbf{x}}, u_{\mathbf{x}\mathbf{x}}, \dots] = 0,$$

where Δ may be vector-valued (so that we can handle systems), and a solution is a function $u = u(\mathbf{x})$ satisfying it.

Definition 32.7.15 (Lie point symmetry)

Let g^ε be a 1.p.g.t. generated by a vector field V , acting on both the independent and dependent variables:

$$(\tilde{\mathbf{x}}, \tilde{u}) = g^\varepsilon(\mathbf{x}, u).$$

We call g^ε a **Lie point symmetry** of Δ if

$$\Delta[\mathbf{x}, u, u_{\mathbf{x}}, \dots] = 0 \implies \Delta[\tilde{\mathbf{x}}, \tilde{u}, \tilde{u}_{\tilde{\mathbf{x}}}, \dots] = 0,$$

that is, if it takes solutions to solutions. We say the symmetry is *generated by* V .

Note that we are allowed to mix up the domain and the codomain – the new independent variable may depend on u , and vice versa. This freedom is what makes the theory powerful.

Example 32.7.16

Consider KdV, $\Delta = u_t + u_{xxx} - 6uu_x$. Translation in time,

$$g^\varepsilon(x, t, u) = (x, t + \varepsilon, u),$$

is a Lie point symmetry generated by $V = \partial_t$. Indeed, by the chain rule

$$\frac{\partial \tilde{u}}{\partial \tilde{t}} = \frac{\partial t}{\partial \tilde{t}} \frac{\partial u}{\partial t} + \frac{\partial x}{\partial \tilde{t}} \frac{\partial u}{\partial x} = \frac{\partial u}{\partial t},$$

and similarly $\tilde{u}_{\tilde{x}} = u_x$ and $\tilde{u}_{\tilde{x}\tilde{x}\tilde{x}} = u_{xxx}$, so $\Delta[\tilde{x}, \tilde{t}, \tilde{u}] = \Delta[x, t, u]$.

Symmetries obviously give us new solutions from old ones. Much more importantly, they let us *solve* equations.

Example 32.7.17 (Reduction of a first-order ODE)

Consider the homogeneous ODE

$$\frac{du}{dx} = F\left(\frac{u}{x}\right).$$

Both sides only see the combination u/x , so we should expect

$$g^\varepsilon(x, u) = (e^\varepsilon x, e^\varepsilon u)$$

to be a symmetry, generated by

$$V = x \frac{\partial}{\partial x} + u \frac{\partial}{\partial u}.$$

Now the trick: find coordinates (s, t) in which

$$V(s) = 0, \quad V(t) = 1,$$

so that in the new coordinates $V = \partial_t$. Since the equation is invariant under the flow of V , and the flow is now just translation in t , we expect the variable t not to appear explicitly.

Such coordinates are easy to find here:

$$s = \frac{u}{x}, \quad t = \log |x|,$$

for which $V(s) = x \cdot (-u/x^2) + u \cdot (1/x) = 0$ and $V(t) = x \cdot (1/x) = 1$. Inverting, $x = e^t$ and $u = se^t$. Then

$$du = e^t ds + se^t dt, \quad dx = e^t dt,$$

so

$$\frac{du}{dx} = \frac{ds}{dt} + s.$$

The equation therefore becomes

$$\frac{ds}{dt} = F(s) - s, \quad \text{i.e.} \quad \frac{dt}{ds} = \frac{1}{F(s) - s},$$

which – as promised – contains no t . So we can integrate it up:

$$t = C + \int^s \frac{ds'}{F(s') - s'},$$

and returning to the original variables,

$$\log |x| = C + \int^{u/x} \frac{ds}{F(s) - s}.$$

Knowing a single Lie point symmetry has reduced the ODE to a quadrature.

The same works in general. For an n th order ODE

$$\Delta [x, u, u', \dots, u^{(n)}] = 0$$

admitting a Lie point symmetry generated by

$$V = \xi(x, u) \frac{\partial}{\partial x} + \eta(x, u) \frac{\partial}{\partial u},$$

we introduce invariant coordinates $s = s(x, u)$, $t = t(x, u)$ with $V = \partial_t$. The equation becomes

$$\Delta [s, t', \dots, t^{(n)}] = 0$$

with no explicit t , and setting $r = t'$ reduces the order by one:

$$\Delta [s, r, r', \dots, r^{(n-1)}] = 0.$$

Now rinse and repeat, if we have more symmetries.

§32.7.4 Jets and Prolongations

This is all very well, but so far we have found our symmetries by inspection. We need a systematic method.

Start with the trivial case, a ‘zeroth order equation’ $\Delta[x, u] = 0$. Then g^ε is a symmetry if $\Delta = 0$ implies $\Delta \circ g^\varepsilon = 0$. Provided Δ is of **maximal rank** – meaning the matrix $\partial\Delta_j/\partial y_i$ has maximal rank, where y_i ranges over all the coordinates – one can show (example sheet) that this is equivalent to the infinitesimal condition

$$\Delta = 0 \implies V(\Delta) = 0,$$

where V acts as a differential operator. The maximal rank hypothesis rules out degenerate cases like $\Delta = x^2$, which vanishes only where its derivative does. The implication is only required to hold *on* the solution set, a restriction we call an **on-shell** condition.

This is a wonderful reduction, because the condition $V(\Delta) = 0$ is purely *algebraic* in the coefficients of V . The problem is that a genuine differential equation involves u_x , u_{xx} and so on, and V only knows what to do to x and u . The remedy is to extend its action to the derivatives, which is called *prolongation*.

Example 32.7.18

Take $g^\varepsilon : (x, u) \mapsto (e^\varepsilon x, e^{-\varepsilon} u)$, generated by $V = x\partial_x - u\partial_u$. Then by the chain rule

$$\frac{d\tilde{u}}{d\tilde{x}} = \frac{d\tilde{u}/dx}{d\tilde{x}/dx} = \frac{e^{-\varepsilon}u_x}{e^\varepsilon} = e^{-2\varepsilon}u_x,$$

so the induced transformation on the triple (x, u, u_x) is

$$(\tilde{x}, \tilde{u}, \tilde{u}_{\tilde{x}}) = (e^\varepsilon x, e^{-\varepsilon} u, e^{-2\varepsilon} u_x),$$

which is again a 1.p.g.t., now on a three-dimensional space, generated by

$$\text{pr}^{(1)} V = x \frac{\partial}{\partial x} - u \frac{\partial}{\partial u} - 2u_x \frac{\partial}{\partial u_x}.$$

Repeating for the second derivative gives $\tilde{u}_{\tilde{x}\tilde{x}} = e^{-3\varepsilon}u_{xx}$ and

$$\text{pr}^{(2)} V = x \frac{\partial}{\partial x} - u \frac{\partial}{\partial u} - 2u_x \frac{\partial}{\partial u_x} - 3u_{xx} \frac{\partial}{\partial u_{xx}}.$$

The space with coordinates (x, u, u_x) is called the **first jet space**, and $(x, u, u_x, \dots, u^{(n)})$ the **n th jet space**. Any function $u = u(x)$ determines a curve in it. The operator $\text{pr}^{(n)} V$ above is the **n th prolongation** of V .

If it feels strange to treat u_x as an independent coordinate, one may relabel it p_1 , and $u^{(n)}$ as p_n ; nothing changes. An n th order ODE is then an algebraic equation on the n th jet space, and we are back to the situation we could handle:

$$g^\varepsilon \text{ is a Lie point symmetry } \iff \text{pr}^{(n)} V(\Delta) = 0 \text{ whenever } \Delta = 0.$$

The catch, of course, is that we must restrict attention to points of the jet space which come from actual functions, and to transformations which are prolongations of transformations of the base space.

So we need a formula for the prolongation of an arbitrary vector field. Note how much easier this is at the level of the Lie *algebra*: prolonging the group action would be a horrible mess, whereas prolonging the vector field is a one-line recursion.

Write

$$g^\varepsilon(x, u) = (x + \varepsilon \xi(x, u), u + \varepsilon \eta(x, u)) + o(\varepsilon),$$

and seek the prolongation in the form

$$\text{pr}^{(n)} V = V + \sum_{k=1}^n \eta_k \frac{\partial}{\partial u^{(k)}},$$

so that $\tilde{u}^{(k)} = u^{(k)} + \varepsilon \eta_k + o(\varepsilon)$. To find η_1 we use the **contact condition**

$$d\tilde{u} = \tilde{u}_{\tilde{x}} d\tilde{x},$$

which is nothing but the statement that $\tilde{u}_{\tilde{x}}$ really is the derivative of $\tilde{u}$ with respect to $\tilde{x}$. Substituting the expansions,

$$du + \varepsilon d\eta = \tilde{u}_{\tilde{x}} (dx + \varepsilon d\xi) + o(\varepsilon).$$

We express everything in terms of dx . Since $du = u_x dx$ and

$$d\eta = \frac{\partial \eta}{\partial x} dx + \frac{\partial \eta}{\partial u} du = \left(\frac{\partial \eta}{\partial x} + u_x \frac{\partial \eta}{\partial u} \right) dx = (D_x \eta) dx,$$

with D_x the total derivative of Section 4.1, and similarly $d\xi = (D_x \xi) dx$, we get

$$(u_x + \varepsilon D_x \eta) dx = \tilde{u}_{\tilde{x}} (1 + \varepsilon D_x \xi) dx + o(\varepsilon).$$

Hence

$$\tilde{u}_{\tilde{x}} = \frac{u_x + \varepsilon D_x \eta}{1 + \varepsilon D_x \xi} + o(\varepsilon) = u_x + \varepsilon (D_x \eta - u_x D_x \xi) + o(\varepsilon),$$

so

$$\eta_1 = D_x \eta - u_x D_x \xi.$$

Exactly the same argument applied to the contact condition $d\tilde{u}^{(k)} = \tilde{u}^{(k+1)} d\tilde{x}$ gives the recursion in general.

Proposition 32.7.19 (Prolongation formula)

Let

$$V = \xi(x, u) \frac{\partial}{\partial x} + \eta(x, u) \frac{\partial}{\partial u}.$$

Then

$$\text{pr}^{(n)} V = V + \sum_{k=1}^n \eta_k \frac{\partial}{\partial u^{(k)}},$$

where

$$\eta_0 = \eta, \quad \eta_{k+1} = D_x \eta_k - u^{(k+1)} D_x \xi.$$

Example 32.7.20

For $g^\varepsilon : (x, u) \mapsto (e^\varepsilon x, e^{-\varepsilon} u)$ we have $\xi = x$ and $\eta = -u$, so

$$\eta_1 = D_x(-u) - u_x D_x(x) = -u_x - u_x = -2u_x,$$

in agreement with the direct computation above.

Example 32.7.21 (Checking a symmetry)

Let us verify by this method that $V = x\partial_x + u\partial_u$ is a symmetry of $\Delta = u_x - F(u/x)$. Here $\xi = x$, $\eta = u$, so

$$\eta_1 = D_x(u) - u_x D_x(x) = u_x - u_x = 0,$$

and $\text{pr}^{(1)} V = x\partial_x + u\partial_u$. Applying it,

$$\text{pr}^{(1)} V(\Delta) = x \cdot \left(\frac{u}{x^2} F' \left(\frac{u}{x} \right) \right) + u \cdot \left(-\frac{1}{x} F' \left(\frac{u}{x} \right) \right) + 0 = \frac{u}{x} F' - \frac{u}{x} F' = 0,$$

identically, so it is a symmetry – and not merely on shell.

In general, imposing $\text{pr}^{(n)} V(\Delta) = 0$ on shell produces an over-determined linear system of PDEs for the two unknown functions $\xi(x, u)$ and $\eta(x, u)$, the **determining equations**. These are usually easy to solve, and their solution space is exactly the Lie algebra of symmetries. An analogous formula for PDEs, with several independent variables, appears on the last example sheet.

§32.7.5 Group-Invariant Solutions and ODE Reductions

Symmetries do more than generate new solutions: they let us look for solutions which are themselves invariant.

Suppose a PDE admits a Lie point symmetry generated by V . A solution is **group-invariant** if it is unchanged by the symmetry. To find such solutions we introduce coordinates invariant under V , and look for solutions depending only on those. Since the number of invariant coordinates is one fewer than the number of variables, this reduces the PDE to an equation in one fewer variable – typically an ODE.

Example 32.7.22 (Travelling waves)

The most familiar case. Any translation-invariant equation admits $V = \partial_x + c\partial_t$, whose invariants are $z = x - ct$ and u . Looking for solutions $u = f(z)$ is exactly the travelling-wave ansatz we used for KdV, sine-Gordon and NLS. So all our soliton computations were symmetry reductions.

Example 32.7.23 (Sine-Gordon)

Take sine-Gordon in light-cone coordinates, $u_{xt} = \sin u$. This admits the scaling symmetry

$$g^\varepsilon(x, t, u) = (e^\varepsilon x, e^{-\varepsilon} t, u), \quad V = x\partial_x - t\partial_t.$$

An invariant coordinate is $z = xt$, since $V(z) = xt - tx = 0$; and u itself is invariant. So we look for

$$u(x, t) = F(z), \quad z = xt.$$

Then $u_x = tF'(z)$ and $u_{xt} = F'(z) + xtF''(z) = F' + zF''$, so the PDE becomes the ODE

$$zF'' + F' = \sin F.$$

This does not look like anything standard, but substituting $w = e^{iF}$ transforms it. Writing $F = -i \log w$, we have $F' = -iw'/w$ and $F'' = -i(w''/w - (w')^2/w^2)$, while $\sin F = (w - w^{-1})/2i$. Substituting and simplifying gives

$$\frac{d^2w}{dz^2} = \frac{1}{w} \left(\frac{dw}{dz} \right)^2 - \frac{1}{z} \frac{dw}{dz} + \frac{1}{z} \left(\frac{w^2}{2} - \frac{1}{2} \right),$$

which is the third Painlevé equation with $\alpha = \frac{1}{2}$, $\beta = -\frac{1}{2}$, $\gamma = \delta = 0$.

Example 32.7.24 (KdV)

The KdV equation admits a rather less obvious Lie point symmetry,

$$g^\varepsilon(x, t, u) = \left(x + \varepsilon t + \frac{1}{2}\varepsilon^2, \quad t + \varepsilon, \quad u - \frac{1}{6}\varepsilon \right),$$

generated by

$$V = t \frac{\partial}{\partial x} + \frac{\partial}{\partial t} - \frac{1}{6} \frac{\partial}{\partial u}.$$

(It is a combination of a Galilean boost and a time translation.) Invariant coordinates are

$$z = x - \frac{1}{2}t^2, \quad w = u + \frac{1}{6}t,$$

since $V(z) = t - t = 0$ and $V(w) = -\frac{1}{6} + \frac{1}{6} = 0$. So we seek a solution

$$u(x, t) = -\frac{1}{6}t + w(z).$$

Then

$$u_t = -\frac{1}{6} - tw'(z), \quad u_x = w'(z), \quad u_{xxx} = w'''(z),$$

and substituting into KdV,

$$0 = u_t + u_{xxx} - 6uu_x = -\frac{1}{6} - tw' + w''' - 6\left(-\frac{1}{6}t + w\right)w' = -\frac{1}{6} + w''' - 6ww'.$$

The t 's have cancelled, as they had to: our coordinates were built to be invariant, so nothing non-invariant can survive. Integrating once,

$$w'' - 3w^2 = \frac{1}{6}z + \text{const},$$

and after rescaling w and z and translating z to absorb the constant, this is the first Painlevé equation

$$\frac{d^2W}{dZ^2} = 6W^2 + Z.$$

Both reductions have landed us on a Painlevé equation, and this is not a coincidence.

§32.7.6 The Painlevé Test

Suppose someone hands us a PDE and asks whether it is integrable. We have no definition of integrability to appeal to, but we do have a *necessary condition* which can be checked.

Definition 32.7.25 (Singularity)

A **singularity** of a complex-valued function $w = w(z)$ is a point at which it fails to be analytic.

For a *linear* ODE

$$\frac{d^2w}{dz^2} + p(z)\frac{dw}{dz} + q(z)w = 0,$$

a theorem of complex analysis says that any singularity of a solution must already be a singularity of p or q . In particular the *locations* of the singularities do not depend on the initial conditions.

For nonlinear equations this fails spectacularly. Consider

$$\frac{dw}{dz} + w^2 = 0,$$

whose solution is

$$w(z) = \frac{1}{z - z_0},$$

with z_0 determined by the initial data. The singularity has moved: we call it a **movable** singularity.

Movable singularities are unavoidable in nonlinear problems, but their *type* turns out to be a very sensitive indicator.

Definition 32.7.26 (Painlevé property)

An ODE of the form

$$\frac{d^n w}{dz^n} = F\left(\frac{d^{n-1}w}{dz^{n-1}}, \dots, w, z\right)$$

has the **Painlevé property** if all of its movable singularities are at worst poles.

Example 32.7.27

The equation $w' + w^2 = 0$ has solution $w = (z - z_0)^{-1}$, whose movable singularity is a simple pole. So it has the Painlevé property.

Example 32.7.28

The equation $w' + w^3 = 0$ has solution

$$w(z) = \frac{1}{\sqrt{2(z - z_0)}},$$

whose movable singularity is a branch point, not a pole. So it does not have the Painlevé property.

Painlevé set out to classify all equations of the form

$$\frac{d^2w}{dz^2} = F\left(\frac{dw}{dz}, w, z\right),$$

with F rational, possessing this property. He found fifty of them, up to simple changes of coordinates. Of these, forty-four can be solved in terms of functions already known – elementary functions, elliptic functions, Bessel functions and the like. The remaining six define genuinely new functions, the **Painlevé transcendents**. The first two are

$$\begin{aligned} \text{(PI)} \quad & \frac{d^2w}{dz^2} = 6w^2 + z, \\ \text{(PII)} \quad & \frac{d^2w}{dz^2} = 2w^3 + zw + \alpha, \end{aligned}$$

and the remaining four (PIII)–(PVI) are increasingly baroque rational expressions; PIII is the one that appeared in the sine–Gordon reduction above,

$$\frac{d^2w}{dz^2} = \frac{1}{w} \left(\frac{dw}{dz} \right)^2 + \frac{1}{z} \left(-\frac{dw}{dz} + \alpha w^2 + \beta \right) + \gamma w^3 + \frac{\delta}{w}.$$

Remark. Painlevé was also twice prime minister of France, for nine weeks in 1917 and seven months in 1925. It is not recorded whether this affected his classification.

The connection with our subject is the following.

Conjecture 32.7.29 (Ablowitz–Ramani–Segur, 1980). Every ODE reduction of an integrable PDE has the Painlevé property.

This remains a conjecture, since (as we have said) integrability has no agreed definition; but it has been proved in every setting where a precise definition has been pinned down, and it is overwhelmingly supported by examples. Two of those examples are the reductions we performed in the previous subsection, both of which landed on Painlevé equations.

This gives us a practical test.

The Painlevé test.

- (1) Find all Lie point symmetries of the PDE, using the prolongation formula.
- (2) Compute the corresponding ODE reductions.
- (3) Test each reduction for the Painlevé property.

If any reduction fails, the equation is not integrable. If all of them pass, we have learned nothing definitive – but we have found strong evidence, and in practice a good reason to start hunting for a Lax pair.

That is a fitting place to stop. We began with ODEs, where integrability is completely understood: enough commuting conserved quantities, and the motion is uniform winding on a torus. We end with PDEs, where the corresponding picture is only visible in outline. The KdV equation has its infinitely many conserved quantities, its action–angle variables in the guise of scattering data, and its solitons; but what integrability *is*, in general, remains a question with no satisfactory answer. That there should be so much beautiful and useful mathematics hanging on an undefined term is perhaps the most remarkable feature of the whole subject.

33 Linear Analysis

Linear analysis is what happens when you take the linear algebra you know and let the dimension become infinite. Almost nothing survives unscathed. A linear map need not be continuous. A subspace need not be closed. A bounded sequence need not have a convergent subsequence, and the closed unit ball – that most innocuous of objects – is never compact. The whole edifice of ‘pick a basis and write down a matrix’ collapses, and we need genuinely new tools to replace it.

The tools we get are marvellous. The Hahn–Banach theorem manufactures linear functionals out of thin air, so that a normed space always has a rich supply of them. The Baire category theorem, an almost embarrassingly soft statement about complete metric spaces, yields three of the great theorems of the subject in a single stroke. And in Hilbert space – where we have not just a length but an angle – geometry comes flooding back, and we recover a spectral theorem that would make any linear algebraist feel at home.

This article constitutes my notes for the ‘Linear Analysis’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§33.1 Normed Spaces

Throughout this course, $\mathbb{F}$ will denote either $\mathbb{R}$ or $\mathbb{C}$, and all vector spaces will be over $\mathbb{F}$ unless we say otherwise. When a result works for both fields without change we will just write $\mathbb{F}$; when the distinction matters – and it does matter, for instance in the Hahn–Banach theorem – we will be careful to say so.

In IB Linear Algebra we studied vector spaces armed with nothing but their algebraic structure, and for finite-dimensional spaces this was enough: choose a basis, and everything becomes a matrix. In IB Analysis and Topology we studied metric spaces, where we could speak of convergence and continuity but had no algebra to play with. Linear analysis is what you get when you insist on having both at once, in a way that makes them compatible.

§33.1.1 Norms

The compatibility is achieved by the following definition, which is really just an abstraction of ‘length of a vector’ in $\mathbb{R}^n$.

Definition 33.1.1 (Norm)

Let V be a vector space over $\mathbb{F}$. A **norm** on V is a function $\|\cdot\| : V \rightarrow \mathbb{R}$ such that

- (i) (*Positivity*) $\|v\| \geq 0$ for all $v \in V$, with $\|v\| = 0$ if and only if $v = 0$;
- (ii) (*Homogeneity*) $\|\lambda v\| = |\lambda| \|v\|$ for all $\lambda \in \mathbb{F}$ and $v \in V$;
- (iii) (*Triangle inequality*) $\|v + w\| \leq \|v\| + \|w\|$ for all $v, w \in V$.

The pair $(V, \|\cdot\|)$ is called a **normed space**.

If we drop the ‘only if’ in (i), so that non-zero vectors are allowed to have zero length, we get what is called a **seminorm**. Seminorms turn up naturally – for instance $p(f) = |\int_0^1 f|$ on $C[0, 1]$ – and we will meet a very important one in the guise of the Minkowski functional later on.

The first thing to notice is that a norm gives us a metric for free.

Proposition 33.1.2

Let $(V, \|\cdot\|)$ be a normed space. Then $d(v, w) = \|v - w\|$ defines a metric on V .

Proof. Symmetry is homogeneity applied with $\lambda = -1$, and $d(v, w) = 0$ if and only if $v - w = 0$ by positivity. The triangle inequality for d reads $\|v - w\| \leq \|v - u\| + \|u - w\|$, which is the triangle inequality for the norm applied to $(v - u) + (u - w)$. $\square$

So all of the language of IB Analysis and Topology is available to us: open balls

$$B(v, r) = \{w \in V : \|v - w\| < r\}, \quad \bar{B}(v, r) = \{w \in V : \|v - w\| \leq r\},$$

open and closed sets, convergence, continuity, completeness, compactness. We will write B_V for the open unit ball $B(0, 1)$ and $\bar{B}_V$ for the closed unit ball when the space needs emphasising.

This metric is not just any old metric: it interacts well with the vector space structure. It is *translation invariant*, $d(v + u, w + u) = d(v, w)$, and it *scales*, $d(\lambda v, \lambda w) = |\lambda|d(v, w)$. In particular every ball is a translated dilate of the unit ball,

$$B(v, r) = v + rB_V,$$

so that the geometry of a normed space is completely determined by the shape of its unit ball. This is a genuinely useful principle, and worth keeping in mind.

Proposition 33.1.3

In a normed space V , the maps $+: V \times V \rightarrow V$ and $\cdot: \mathbb{F} \times V \rightarrow V$ are continuous, and so is $\|\cdot\|: V \rightarrow \mathbb{R}$.

Proof. If $\|v - v'\|, \|w - w'\| < \varepsilon/2$ then $\|(v + w) - (v' + w')\| < \varepsilon$, giving continuity of addition. For scalar multiplication, we compute

$$\|\lambda v - \mu w\| \leq |\lambda| \|v - w\| + |\lambda - \mu| \|w\|,$$

which is small when λ is close to μ and v close to w . Finally the reverse triangle inequality

$$|\|v\| - \|w\|| \leq \|v - w\|$$

shows that the norm is 1-Lipschitz, hence continuous. $\square$

The reverse triangle inequality above is worth recording separately – we will use it constantly. It follows from $\|v\| \leq \|v - w\| + \|w\|$ and the same with v, w swapped.

§33.1.2 Examples of Normed Spaces

Let’s look at some examples. The whole subject would be rather empty without a good stock of these, and most of the theorems we prove will be tested against this list.

Example 33.1.4 (Finite-dimensional ℓ^p norms)

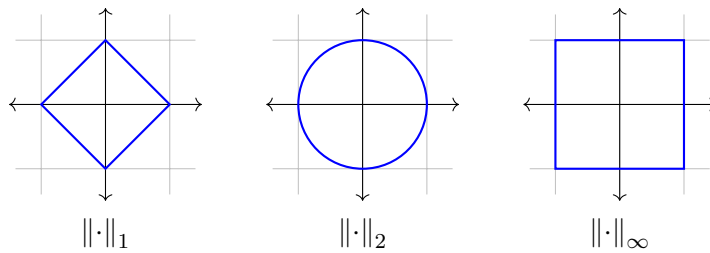
On $\mathbb{F}^n$, for $1 \leq p < \infty$, define

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

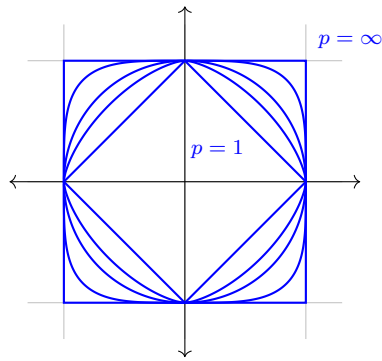
The case $p = 2$ is the usual Euclidean norm. We write ℓ_p^n for $\mathbb{F}^n$ equipped with $\|\cdot\|_p$.

Positivity and homogeneity are clear. The triangle inequality for $p = 1$ and $p = \infty$ is immediate, and for general p it is Minkowski's inequality, which we prove below.

Since the geometry of a normed space is the geometry of its unit ball, it is worth seeing what these unit balls actually look like. In $\mathbb{R}^2$ we get a diamond, a disc and a square.



As p increases from 1 to ∞ the unit ball swells continuously from the diamond out to the square, always staying convex and always containing the same four points $(\pm 1, 0)$, $(0, \pm 1)$. (Convexity of the ball is exactly the triangle inequality: if $\|v\|, \|w\| \leq 1$ and $t \in [0, 1]$ then $\|tv + (1-t)w\| \leq t + (1-t) = 1$. This is why $\|\cdot\|_p$ is *not* a norm for $p < 1$ – the ‘ball’ becomes a four-pointed star, and it is not convex.)



The curves drawn are $p = 1, \frac{3}{2}, 2, 4, \infty$, and they nest outwards as p increases.

Before we can prove Minkowski's inequality we need Hölder's, which is one of the workhorses of the subject. We say that $p, q \in [1, \infty]$ are **conjugate exponents** if

$$\frac{1}{p} + \frac{1}{q} = 1,$$

with the convention $1/\infty = 0$, so that 1 and ∞ are conjugate to each other and 2 is conjugate to itself.

Lemma 33.1.5 (Young's Inequality)

Let $p, q \in (1, \infty)$ be conjugate. Then for all $a, b \geq 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. If a or b is zero this is clear, so assume both are positive. The logarithm is concave, so

$$\log\left(\frac{a^p}{p} + \frac{b^q}{q}\right) \geq \frac{1}{p} \log(a^p) + \frac{1}{q} \log(b^q) = \log a + \log b = \log(ab),$$

and we exponentiate. $\square$

Proposition 33.1.6 (Hölder's Inequality)

Let $p, q \in [1, \infty]$ be conjugate exponents. Then for $x, y \in \mathbb{F}^n$,

$$\sum_{i=1}^n |x_i y_i| \leq \|x\|_p \|y\|_q.$$

Proof. The cases $p = 1, q = \infty$ (and vice versa) are immediate, since $\sum |x_i y_i| \leq (\max_i |y_i|) \sum_i |x_i|$. So suppose $1 < p < \infty$. If either x or y is zero there is nothing to prove; otherwise, by homogeneity we may rescale and assume $\|x\|_p = \|y\|_q = 1$. Then Young's inequality gives

$$\sum_{i=1}^n |x_i y_i| \leq \sum_{i=1}^n \left(\frac{|x_i|^p}{p} + \frac{|y_i|^q}{q} \right) = \frac{1}{p} + \frac{1}{q} = 1 = \|x\|_p \|y\|_q,$$

as required. $\square$

Proposition 33.1.7 (Minkowski's Inequality)

Let $1 \leq p \leq \infty$. Then for $x, y \in \mathbb{F}^n$,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Proof. Again $p = 1, \infty$ are clear, so take $1 < p < \infty$ and let q be conjugate to p . We may assume $\|x + y\|_p \neq 0$. Then

$$\begin{aligned} \|x + y\|_p^p &= \sum_{i=1}^n |x_i + y_i| \cdot |x_i + y_i|^{p-1} \\ &\leq \sum_{i=1}^n |x_i| \cdot |x_i + y_i|^{p-1} + \sum_{i=1}^n |y_i| \cdot |x_i + y_i|^{p-1} \\ &\leq (\|x\|_p + \|y\|_p) \left(\sum_{i=1}^n |x_i + y_i|^{(p-1)q} \right)^{1/q}, \end{aligned}$$

using Hölder on each sum. Since $(p-1)q = p$, the last bracket is $\|x + y\|_p^{p/q}$. Dividing through by $\|x + y\|_p^{p/q}$ and noting $p - p/q = 1$ gives the result. $\square$

Exactly the same arguments work for infinite sums and for integrals, and we will use them in those settings without further comment.

Example 33.1.8 (Sequence spaces)

For $1 \leq p < \infty$ let

$$\ell^p = \left\{ x = (x_1, x_2, \dots) : x_i \in \mathbb{F}, \sum_{i=1}^{\infty} |x_i|^p < \infty \right\}, \quad \|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p \right)^{1/p},$$

and let

$$\ell^\infty = \left\{ x : \sup_i |x_i| < \infty \right\}, \quad \|x\|_\infty = \sup_i |x_i|.$$

Minkowski's inequality (applied to partial sums and passing to the limit) shows both that ℓ^p is a vector space and that $\|\cdot\|_p$ is a norm on it.

Inside ℓ^∞ sit two further spaces which will be useful for producing counterexamples:

$$c_0 = \{x : x_i \rightarrow 0\}, \quad c_{00} = \{x : x_i = 0 \text{ for all but finitely many } i\}.$$

We give both the $\|\cdot\|_\infty$ norm. Note $c_{00} \subseteq c_0 \subseteq \ell^\infty$, and in fact c_{00} is dense in c_0 but not in ℓ^∞ .

Example 33.1.9 (Spaces of continuous functions)

Let K be a compact topological space (usually for us a closed bounded subset of $\mathbb{R}^n$, or a compact metric space). Write

$$C(K) = \{f : K \rightarrow \mathbb{F} \text{ continuous}\}, \quad \|f\|_\infty = \sup_{x \in K} |f(x)|.$$

The supremum is finite because a continuous function on a compact space is bounded, and it is attained. This is called the **uniform** or **supremum norm**, and convergence in it is exactly uniform convergence, as studied in IB Analysis and Topology.

More generally, for any set S we may take $B(S)$, the space of all bounded functions $S \rightarrow \mathbb{F}$, with the same norm.

Example 33.1.10 (L^p spaces)

For $1 \leq p < \infty$ define on $C[0, 1]$ the norm

$$\|f\|_{L^p} = \left(\int_0^1 |f(x)|^p dx \right)^{1/p}.$$

The integral version of Minkowski's inequality gives the triangle inequality, and $\|f\|_{L^p} = 0$ forces $f = 0$ because a non-zero continuous function is bounded away from zero on some interval. So this is a norm.

It is, however, a rather badly behaved one: as we will see, $(C[0, 1], \|\cdot\|_{L^p})$ is not complete. One writes $L^p[0, 1]$ for its completion. In IID Probability and Measure you will see that this completion can be described concretely as the space of p -integrable measurable functions modulo equality almost everywhere, but we will not need that here – for us L^p is simply ‘the completion of the continuous functions’.

Example 33.1.11 (Subspaces and products)

Any linear subspace W of a normed space V is a normed space with the restricted norm. If V, W are normed spaces then $V \oplus W$ becomes one via, say, $\|(v, w)\| = \|v\|_V + \|w\|_W$, and the topology this induces is the product topology.

There is one more construction we will want, and it is a little less obvious.

Proposition 33.1.12 (Quotient norm)

Let V be a normed space and $W \subseteq V$ a *closed* subspace. Then

$$\|v + W\|_{V/W} = \inf_{w \in W} \|v + w\| = \text{dist}(v, W)$$

defines a norm on the quotient vector space V/W .

Proof. Homogeneity and the triangle inequality are straightforward: for the latter, given $\varepsilon > 0$ pick $w_1, w_2 \in W$ with $\|v_i + w_i\| \leq \|v_i + W\| + \varepsilon$, and then $\|(v_1 + v_2) + (w_1 + w_2)\| \leq \|v_1 + W\| + \|v_2 + W\| + 2\varepsilon$.

For positivity, suppose $\|v + W\| = 0$. Then there are $w_n \in W$ with $\|v + w_n\| \rightarrow 0$, so $-w_n \rightarrow v$, and since W is closed we get $v \in W$, i.e. $v + W = 0$ in V/W . This is exactly where closedness is needed: without it we would only get a seminorm. $\square$

§33.1.3 Completeness and Banach Spaces

The single most important extra hypothesis in this course is completeness. Recall from IB Analysis and Topology that a metric space is complete if every Cauchy sequence converges.

Definition 33.1.13 (Banach space)

A normed space which is complete in the induced metric is called a **Banach space**.

Completeness is what lets us construct things: we build a candidate solution as a limit of approximations, and completeness guarantees that the limit exists. Almost every theorem in this course that produces something out of nothing uses completeness somewhere.

Let’s check our examples.

Proposition 33.1.14

For any set S , the space $B(S)$ of bounded functions $S \rightarrow \mathbb{F}$ with the supremum norm is a Banach space. Consequently so is $C(K)$ for K compact, and so is ℓ^∞ .

Proof. Let (f_n) be Cauchy in $B(S)$. For each fixed $x \in S$ we have $|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty$, so $(f_n(x))$ is Cauchy in $\mathbb{F}$ and therefore converges; call the limit $f(x)$.

Given $\varepsilon > 0$, choose N with $\|f_n - f_m\|_\infty < \varepsilon$ for $n, m \geq N$. Fixing $n \geq N$ and $x \in S$ and letting $m \rightarrow \infty$ in $|f_n(x) - f_m(x)| < \varepsilon$ gives $|f_n(x) - f(x)| \leq \varepsilon$. As x was arbitrary, $\|f_n - f\|_\infty \leq \varepsilon$ for all $n \geq N$. In particular f is bounded (being within ε of the bounded function f_N) and $f_n \rightarrow f$ in $B(S)$.

For $C(K)$: it is a subspace of $B(K)$, and a closed one, because a uniform limit of continuous functions is continuous. A closed subspace of a complete space is complete. Finally $\ell^\infty = B(\mathbb{N})$. $\square$

Proposition 33.1.15

For $1 \leq p < \infty$, ℓ^p is a Banach space. So are c_0 and c (the space of convergent sequences), with the supremum norm.

Proof. Let $(x^{(n)})$ be Cauchy in ℓ^p . For each coordinate i we have $|x_i^{(n)} - x_i^{(m)}| \leq \|x^{(n)} - x^{(m)}\|_p$, so each coordinate sequence is Cauchy in $\mathbb{F}$; let $x_i = \lim_n x_i^{(n)}$ and $x = (x_1, x_2, \dots)$.

Let $\varepsilon > 0$ and pick N with $\|x^{(n)} - x^{(m)}\|_p < \varepsilon$ for $n, m \geq N$. For any fixed $K \in \mathbb{N}$ and $n, m \geq N$,

$$\sum_{i=1}^K |x_i^{(n)} - x_i^{(m)}|^p < \varepsilon^p.$$

Letting $m \rightarrow \infty$ (a finite sum, so no issue) gives $\sum_{i=1}^K |x_i^{(n)} - x_i|^p \leq \varepsilon^p$, and now letting $K \rightarrow \infty$ gives $\|x^{(n)} - x\|_p \leq \varepsilon$. In particular $x^{(n)} - x \in \ell^p$, so $x = x^{(n)} - (x^{(n)} - x) \in \ell^p$, and $x^{(n)} \rightarrow x$.

For c_0 and c : both are subspaces of ℓ^∞ , and both are closed, since a uniform limit of sequences tending to 0 tends to 0, and a uniform limit of convergent sequences is convergent (this is the same interchange-of-limits argument as for continuity). $\square$

Example 33.1.16 (An incomplete normed space)

The space c_{00} of finitely supported sequences with $\|\cdot\|_\infty$ is *not* complete. Indeed take

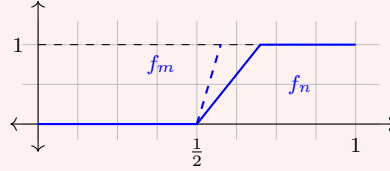
$$x^{(n)} = (1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}, 0, 0, \dots).$$

Then $\|x^{(n)} - x^{(m)}\|_\infty = 1/(\min(n, m) + 1) \rightarrow 0$, so the sequence is Cauchy, but the only possible limit is $(1, \frac{1}{2}, \frac{1}{3}, \dots)$, which is not in c_{00} . The point is that c_{00} is a dense but proper subspace of c_0 .

Example 33.1.17 ($C[0, 1]$ with the L^1 norm is incomplete)

Consider the continuous functions

$$f_n(x) = \begin{cases} 0 & 0 \leq x \leq \frac{1}{2}, \\ n(x - \frac{1}{2}) & \frac{1}{2} \leq x \leq \frac{1}{2} + \frac{1}{n}, \\ 1 & \frac{1}{2} + \frac{1}{n} \leq x \leq 1. \end{cases}$$



For $m > n$ we have $\|f_n - f_m\|_{L^1} \leq 1/n \rightarrow 0$, so (f_n) is Cauchy. If $f_n \rightarrow f$ in L^1 with f continuous, then on $[0, \frac{1}{2}]$ we would need $\int_0^{1/2} |f| = 0$, so $f \equiv 0$ there, and on $[\frac{1}{2} + \delta, 1]$ we would need $f \equiv 1$ for every $\delta > 0$. No continuous function does both. So $(C[0, 1], \|\cdot\|_{L^1})$ is not complete.

§33.1.4 Series in Normed Spaces

In a normed space we can add up infinitely many vectors, provided the partial sums converge. It turns out that the interaction between two natural notions of convergence for series characterises completeness exactly, which is a pleasant surprise and often the most convenient way to check that a space is Banach.

Definition 33.1.18 (Convergence of series)

Let V be a normed space and (v_n) a sequence in V . We say $\sum_{n=1}^{\infty} v_n$ **converges** to $v \in V$ if the partial sums $s_N = \sum_{n=1}^N v_n$ satisfy $s_N \rightarrow v$. We say the series is **absolutely convergent** if $\sum_{n=1}^{\infty} \|v_n\| < \infty$.

In $\mathbb{R}$, absolute convergence implies convergence, and the proof uses the completeness of $\mathbb{R}$ in an essential way. The same is true in any Banach space, and conversely.

Theorem 33.1.19 (Series criterion for completeness)

A normed space V is a Banach space if and only if every absolutely convergent series in V converges.

Proof. ($\Rightarrow$) Suppose V is complete and $\sum \|v_n\| < \infty$. For $M > N$,

$$\|s_M - s_N\| = \left\| \sum_{n=N+1}^M v_n \right\| \leq \sum_{n=N+1}^M \|v_n\| \leq \sum_{n>N} \|v_n\|,$$

which tends to 0 as $N \rightarrow \infty$ since the tail of a convergent series of reals tends to zero. So (s_N) is Cauchy, hence convergent.

($\Leftarrow$) Suppose every absolutely convergent series converges, and let (x_n) be a Cauchy sequence in V . Choose a subsequence as follows: pick $n_1 < n_2 < \dots$ such that

$$\|x_n - x_m\| < 2^{-k} \quad \text{for all } n, m \geq n_k.$$

Set $v_1 = x_{n_1}$ and $v_k = x_{n_k} - x_{n_{k-1}}$ for $k \geq 2$, so that $\sum_{k=1}^K v_k = x_{n_K}$. Then $\|v_k\| < 2^{-(k-1)}$ for $k \geq 2$, so $\sum \|v_k\| < \infty$ and by hypothesis $\sum v_k$ converges, i.e. $x_{n_K} \rightarrow x$ for some $x \in V$.

Finally, a Cauchy sequence with a convergent subsequence converges: given $\varepsilon > 0$, take N with $\|x_n - x_m\| < \varepsilon/2$ for $n, m \geq N$ and K with $n_K \geq N$ and $\|x_{n_K} - x\| < \varepsilon/2$; then $\|x_n - x\| < \varepsilon$ for $n \geq N$. $\square$

Remark. The subsequence trick in the second half is worth remembering in its own right: *every Cauchy sequence has a rapidly Cauchy subsequence*, one where the successive differences are summable. It reappears in measure theory and in the proof of the open mapping theorem.

Here is a typical application, which also gives us a supply of Banach spaces we did not have before.

Proposition 33.1.20

Let V be a Banach space and $W \subseteq V$ a closed subspace. Then W and V/W are both Banach spaces.

Proof. That W is complete is standard (a closed subset of a complete metric space is complete). For V/W we use the series criterion. Let $\sum_n \|v_n + W\| < \infty$. For each n pick $w_n \in W$ with

$$\|v_n + w_n\| \leq \|v_n + W\| + 2^{-n}.$$

Then $\sum_n \|v_n + w_n\| < \infty$, so by completeness of V the series $\sum_n (v_n + w_n)$ converges to some $v \in V$. Since the quotient map $q : V \rightarrow V/W$ satisfies $\|q(u)\| \leq \|u\|$, it is continuous, and hence

$$\sum_{n=1}^N (v_n + W) = q \left(\sum_{n=1}^N (v_n + w_n) \right) \rightarrow q(v)$$

as $N \rightarrow \infty$. So the series converges in V/W . $\square$

§33.1.5 Completions

We remarked above that $L^p[0, 1]$ is ‘the completion’ of $C[0, 1]$. Let us make sure that makes sense.

Theorem 33.1.21 (Completion)

Let V be a normed space. Then there is a Banach space $\bar{V}$ and a linear isometry $\iota : V \rightarrow \bar{V}$ with $\iota(V)$ dense in $\bar{V}$. Moreover $\bar{V}$ is unique up to isometric isomorphism.

Proof sketch. One route is the standard metric space completion from IB Analysis and Topology: take Cauchy sequences in V modulo the equivalence $(x_n) \sim (y_n)$ iff $\|x_n - y_n\| \rightarrow 0$. The vector space operations and the norm $\|[(x_n)]\| = \lim_n \|x_n\|$ descend to this set, and one checks the axioms.

A slicker route, available once we have the Hahn–Banach theorem, is to embed V isometrically into the Banach space $\ell^\infty(\bar{B}_{V^*})$ and take the closure of the image; we will see this idea again when we discuss the double dual.

Uniqueness: if $\iota_1 : V \rightarrow X_1$ and $\iota_2 : V \rightarrow X_2$ are two such, then $\iota_2 \circ \iota_1^{-1}$ is an isometry defined on the dense subspace $\iota_1(V)$ of X_1 , and a uniformly continuous map into a complete space extends uniquely to the closure. $\square$

We will use completions sparingly, but it is comforting to know they exist: it means that when we prove a theorem about Banach spaces, we have not narrowed our attention as much as it might appear.

§33.1.6 Finite-Dimensional Spaces

Finite-dimensional normed spaces are, from the point of view of this course, thoroughly boring – and it is worth understanding exactly *how* boring, because that tells us what to expect (or rather, what not to expect) in infinite dimensions. The headline is that a finite-dimensional space has essentially only one norm.

Definition 33.1.22 (Equivalent norms)

Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space V are **equivalent** if there is a constant $C \geq 1$ with

$$C^{-1} \|v\|_b \leq \|v\|_a \leq C \|v\|_b \quad \text{for all } v \in V.$$

This is an equivalence relation, and equivalent norms have the same open sets, the same convergent sequences, the same Cauchy sequences and the same compact sets – essentially, everything we care about is unaffected.

Theorem 33.1.23 (Equivalence of norms in finite dimensions)

Let V be a finite-dimensional vector space over $\mathbb{F}$. Then any two norms on V are equivalent.

Proof. Fix a basis $e_1, \dots, e_n$ of V and let $\|\cdot\|_1$ denote the associated ℓ^1 norm, $\|\sum_i \lambda_i e_i\|_1 = \sum_i |\lambda_i|$. Since equivalence is transitive, it suffices to show that an arbitrary norm $\|\cdot\|$ is equivalent to $\|\cdot\|_1$.

One direction is easy: writing $v = \sum_i \lambda_i e_i$,

$$\|v\| \leq \sum_{i=1}^n |\lambda_i| \|e_i\| \leq C \|v\|_1, \quad C := \max_i \|e_i\|,$$

which is finite as it is a maximum over a finite set.

For the other direction, let $S = \{v \in V : \|v\|_1 = 1\}$. The inequality just proved shows that $\|\cdot\| : (V, \|\cdot\|_1) \rightarrow \mathbb{R}$ is continuous, since

$$|\|v\| - \|w\|| \leq \|v - w\| \leq C \|v - w\|_1.$$

Also S is compact in $(V, \|\cdot\|_1)$: identifying V with $\mathbb{F}^n$ via the basis, S is the unit

sphere of ℓ_1^n , which is closed and bounded, hence compact by Bolzano–Weierstrass (applied coordinate by coordinate, extracting subsequences n times).

Therefore $\|\cdot\|$ attains its infimum on S , say at v_* . Since $v_* \in S$ we have $v_* \neq 0$, so $c := \|v_*\| > 0$, and $\|v\| \geq c$ for all $v \in S$. For general $v \neq 0$, applying this to $v/\|v\|_1 \in S$ gives $\|v\| \geq c\|v\|_1$. Taking $\max(C, c^{-1})$ finishes the proof. $\square$

Several pleasant consequences follow immediately.

Corollary 33.1.24

Let V be a finite-dimensional normed space. Then:

- (i) V is a Banach space;
- (ii) the closed unit ball $\bar{B}_V$ is compact;
- (iii) any linear map from V to a normed space W is continuous;
- (iv) any finite-dimensional subspace of a normed space is closed.

Proof. By the theorem we may compute in ℓ_1^n , which is complete and in which closed bounded sets are compact; this gives (i) and (ii). For (iii), if $T : V \rightarrow W$ is linear then $\|v\|' := \|v\|_V + \|Tv\|_W$ is a norm on V , hence equivalent to $\|\cdot\|_V$, so $\|Tv\|_W \leq \|v\|' \leq C\|v\|_V$; and a linear map satisfying such a bound is continuous, as we check in the next section. For (iv), a complete subspace of a metric space is closed. $\square$

Part (ii) is the interesting one, because it turns out to *characterise* finite dimensionality. The key is the following geometric lemma, which says that from any proper closed subspace we can always step almost perpendicularly away.

Lemma 33.1.25 (Riesz's Lemma)

Let X be a normed space and $Y \subsetneq X$ a closed proper subspace. Then for every $\theta \in (0, 1)$ there exists $x \in X$ with $\|x\| = 1$ and

$$\text{dist}(x, Y) = \inf_{y \in Y} \|x - y\| \geq \theta.$$

Proof. Pick any $z \in X \setminus Y$. Since Y is closed, $d := \text{dist}(z, Y) > 0$. As $\theta < 1$ we have $d < d/\theta$, so by definition of the infimum there is some $y_0 \in Y$ with

$$d \leq \|z - y_0\| \leq \frac{d}{\theta}.$$

Set

$$x = \frac{z - y_0}{\|z - y_0\|},$$

so $\|x\| = 1$. For any $y \in Y$,

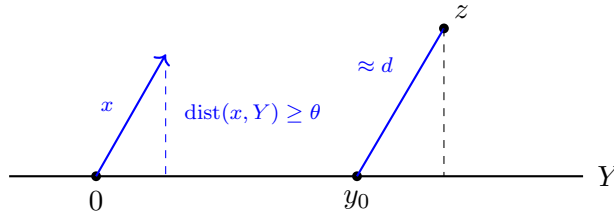
$$\|x - y\| = \frac{1}{\|z - y_0\|} \|z - y_0 - \|z - y_0\| y\| = \frac{\|z - y'\|}{\|z - y_0\|}$$

where $y' = y_0 + \|z - y_0\| y \in Y$. Hence $\|z - y'\| \geq d$, and so

$$\|x - y\| \geq \frac{d}{\|z - y_0\|} \geq \frac{d}{d/\theta} = \theta,$$

as required. $\square$

Geometrically, we take a point z off the subspace, find a point $y_0 \in Y$ that very nearly realises the distance from z to Y , and then rescale the difference to have length one. The resulting unit vector sticks out of Y by almost its full length.



Theorem 33.1.26 (Riesz)

Let X be a normed space. Then the closed unit ball $\bar{B}_X$ is compact if and only if X is finite-dimensional.

Proof. We have already seen $(\Leftarrow)$. For $(\Rightarrow)$, suppose X is infinite-dimensional. We build a sequence (x_n) of unit vectors with $\|x_n - x_m\| \geq \frac{1}{2}$ for $n \neq m$; such a sequence has no convergent subsequence, so $\bar{B}_X$ is not (sequentially, hence not) compact.

Take any unit vector x_1 . Having chosen $x_1, \dots, x_n$, let $Y_n = \text{span}\{x_1, \dots, x_n\}$. This is finite-dimensional, hence closed, and it is proper because X is infinite-dimensional. By Riesz's lemma with $\theta = \frac{1}{2}$ there is a unit vector x_{n+1} with $\text{dist}(x_{n+1}, Y_n) \geq \frac{1}{2}$, and in particular $\|x_{n+1} - x_j\| \geq \frac{1}{2}$ for all $j \leq n$. $\square$

Example 33.1.27

In ℓ^p (any $1 \leq p \leq \infty$) the standard unit vectors $e_n = (0, \dots, 0, 1, 0, \dots)$ satisfy $\|e_n\|_p = 1$ and $\|e_n - e_m\|_p = 2^{1/p}$ for $n \neq m$ (interpreted as 1 when $p = \infty$). So we can see the failure of compactness with our bare hands, without invoking Riesz's lemma.

This is the first genuinely infinite-dimensional phenomenon we have met, and it is a serious one. In $\mathbb{R}^n$, 'closed and bounded' is the same as 'compact', and compactness is what makes extremal problems solvable – continuous functions attain their bounds, sequences have convergent subsequences. In infinite dimensions all of that is gone. A great deal of this course consists of finding substitutes.

§33.2 Bounded Linear Operators

Having set up our objects, we now turn to the maps between them. In linear algebra the maps of interest were the linear ones; here we want them to be linear *and* continuous. The pleasant surprise is that for linear maps, continuity has a very concrete reformulation.

33.2.1 Boundedness and the Operator Norm

Definition 33.2.1 (Bounded linear map)

Let X, Y be normed spaces. A linear map $T : X \rightarrow Y$ is **bounded** if there is a constant $C \geq 0$ with

$$\|Tx\|_Y \leq C \|x\|_X \quad \text{for all } x \in X.$$

We write $\mathcal{B}(X, Y)$ for the set of bounded linear maps $X \rightarrow Y$, and $\mathcal{B}(X) = \mathcal{B}(X, X)$.

Note that ‘bounded’ here does *not* mean ‘has bounded image’ – no non-zero linear map has bounded image. It means ‘bounded on the unit ball’, which by linearity is the same as the displayed condition.

Proposition 33.2.2

Let $T : X \rightarrow Y$ be linear. The following are equivalent:

- (i) T is continuous;
- (ii) T is continuous at 0;
- (iii) T is bounded;
- (iv) T is Lipschitz.

Proof. (i) $\Rightarrow$ (ii) is trivial and (iv) $\Rightarrow$ (i) is standard.

(ii) $\Rightarrow$ (iii). By continuity at 0 there is $\delta > 0$ such that $\|x\| \leq \delta$ implies $\|Tx\| \leq 1$. For general $x \neq 0$, apply this to $\delta x / \|x\|$ to get $\|Tx\| \leq \delta^{-1} \|x\|$.

(iii) $\Rightarrow$ (iv). If $\|Tx\| \leq C \|x\|$ then $\|Tx - Tx'\| = \|T(x - x')\| \leq C \|x - x'\|$. $\square$

So continuity is a purely quantitative matter, and we may as well keep track of the best constant.

Definition 33.2.3 (Operator norm)

For $T \in \mathcal{B}(X, Y)$, the **operator norm** of T is

$$\|T\| = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X} = \sup_{\|x\| \leq 1} \|Tx\|_Y = \sup_{\|x\|=1} \|Tx\|_Y.$$

The three expressions agree by homogeneity (the last requires $X \neq \{0\}$). By construction $\|Tx\| \leq \|T\| \|x\|$, and $\|T\|$ is the least constant with this property.

Proposition 33.2.4

$\mathcal{B}(X, Y)$ is a vector space and $\|\cdot\|$ is a norm on it. Moreover if $T \in \mathcal{B}(X, Y)$ and $S \in \mathcal{B}(Y, Z)$ then $ST \in \mathcal{B}(X, Z)$ with

$$\|ST\| \leq \|S\| \|T\|.$$

Proof. If $\|T\| = 0$ then $Tx = 0$ for all x , so $T = 0$. Homogeneity is clear, and for the triangle inequality,

$$\|(S + T)x\| \leq \|Sx\| + \|Tx\| \leq (\|S\| + \|T\|) \|x\|,$$

so $\|S + T\| \leq \|S\| + \|T\|$. Finally $\|STx\| \leq \|S\| \|Tx\| \leq \|S\| \|T\| \|x\|$. $\square$

The last inequality says that $\mathcal{B}(X)$ is a **normed algebra**: it is an algebra under composition, and the norm is submultiplicative. If X is a Banach space it is a **Banach algebra**, by the next result.

Theorem 33.2.5

If Y is a Banach space then $\mathcal{B}(X, Y)$ is a Banach space, for any normed space X .

Proof. Let (T_n) be Cauchy in $\mathcal{B}(X, Y)$. For each $x \in X$,

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \|x\|,$$

so $(T_n x)$ is Cauchy in Y , hence converges; define $Tx = \lim_n T_n x$. The map T is linear, being a pointwise limit of linear maps.

Cauchy sequences are bounded, say $\|T_n\| \leq M$; then $\|Tx\| = \lim_n \|T_n x\| \leq M \|x\|$, so T is bounded.

Finally, given $\varepsilon > 0$ take N with $\|T_n - T_m\| < \varepsilon$ for $n, m \geq N$. For $\|x\| \leq 1$ and $n \geq N$,

$$\|T_n x - Tx\| = \lim_{m \rightarrow \infty} \|T_n x - T_m x\| \leq \varepsilon,$$

and taking the supremum over such x gives $\|T_n - T\| \leq \varepsilon$. So $T_n \rightarrow T$ in operator norm. $\square$

Note carefully that completeness of X played no role. This is a recurring theme: it is the *target* that needs to be complete.

Example 33.2.6 (i) Let $X = \ell^p$ and let $\lambda = (\lambda_n)$ be a bounded sequence. The **diagonal operator** $M_\lambda x = (\lambda_n x_n)$ is bounded with $\|M_\lambda\| = \|\lambda\|_\infty$. Indeed $\|M_\lambda x\|_p \leq \|\lambda\|_\infty \|x\|_p$, and testing on e_n gives $\|M_\lambda\| \geq |\lambda_n|$ for each n .

(ii) The **right shift** $R : \ell^p \rightarrow \ell^p$, $R(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$, is an isometry, so $\|R\| = 1$. It is injective but not surjective. The **left shift** $L(x_1, x_2, \dots) = (x_2, x_3, \dots)$ has $\|L\| = 1$, and is surjective but not injective. In finite dimensions no linear map can behave like either of these; this is the rank-nullity theorem failing to have an infinite-dimensional analogue.

(iii) Let $k \in C([0, 1]^2)$ and define $T : C[0, 1] \rightarrow C[0, 1]$ by

$$(Tf)(x) = \int_0^1 k(x, y) f(y) \, dy.$$

This is the **integral operator** with kernel k . It is bounded, with $\|T\| \leq \sup_x \int_0^1 |k(x, y)| \, dy \leq \|k\|_\infty$. These operators will be our main source of ex-

amples of compact operators.

- (iv) Differentiation $\frac{d}{dx} : C^1[0, 1] \rightarrow C[0, 1]$ is *unbounded* when $C^1[0, 1]$ is given the sup norm: take $f_n(x) = \sin(n^2x)/n$, so $\|f_n\|_\infty \leq 1/n \rightarrow 0$ but $\|f'_n\|_\infty = n \rightarrow \infty$. Unboundedness of differentiation is the reason a great deal of analysis is hard.

Proposition 33.2.7

Let X be a normed space, Y a Banach space and $Z \subseteq X$ a dense subspace. Then every $T \in \mathcal{B}(Z, Y)$ extends uniquely to $\tilde{T} \in \mathcal{B}(X, Y)$, and $\|\tilde{T}\| = \|T\|$.

Proof. Given $x \in X$ pick $z_n \in Z$ with $z_n \rightarrow x$. Then (z_n) is Cauchy, so $\|Tz_n - Tz_m\| \leq \|T\| \|z_n - z_m\|$ shows (Tz_n) is Cauchy in Y , and we set $\tilde{T}x = \lim_n Tz_n$. This is independent of the choice of sequence (interleave two such sequences), and linearity passes to the limit. Also $\|\tilde{T}x\| = \lim \|Tz_n\| \leq \|T\| \lim \|z_n\| = \|T\| \|x\|$, so $\|\tilde{T}\| \leq \|T\|$; the reverse is clear as $\tilde{T}$ extends T . Uniqueness holds because two continuous maps agreeing on a dense set agree everywhere. $\square$

§33.2.2 Isomorphisms

Definition 33.2.8 (Isomorphism)

A linear map $T : X \rightarrow Y$ between normed spaces is an **isomorphism** if it is a bijection and both T and T^{-1} are bounded. If in addition $\|Tx\| = \|x\|$ for all x , we call T an **isometric isomorphism** and say X and Y are **isometrically isomorphic**, written $X \cong Y$.

Being an isomorphism is equivalent to there existing $c, C > 0$ with $c\|x\| \leq \|Tx\| \leq C\|x\|$ and T surjective. The lower bound alone says T is **bounded below**, which is equivalent to T being injective with closed range.

Proposition 33.2.9

Let X be a Banach space and $T \in \mathcal{B}(X)$ with $\|T\| < 1$. Then $I - T$ is invertible in $\mathcal{B}(X)$, with

$$(I - T)^{-1} = \sum_{n=0}^{\infty} T^n,$$

the series converging in operator norm. Moreover $\|(I - T)^{-1}\| \leq (1 - \|T\|)^{-1}$.

Proof. Since $\|T^n\| \leq \|T\|^n$ and $\|T\| < 1$, the series $\sum_n T^n$ is absolutely convergent in $\mathcal{B}(X)$, which is a Banach space; so it converges, to S say, and $\|S\| \leq \sum_n \|T\|^n = (1 - \|T\|)^{-1}$. Writing $S_N = \sum_{n=0}^N T^n$ we have

$$(I - T)S_N = S_N(I - T) = I - T^{N+1} \longrightarrow I,$$

since $\|T^{N+1}\| \rightarrow 0$. As multiplication in $\mathcal{B}(X)$ is continuous, letting $N \rightarrow \infty$ gives $(I - T)S = S(I - T) = I$. $\square$

This is the **Neumann series**, and it is the workhorse of spectral theory. An immediate consequence is that the invertible elements of $\mathcal{B}(X)$ form an open set: if T is invertible and $\|S - T\| < \|T^{-1}\|^{-1}$ then $S = T(I - T^{-1}(T - S))$ is invertible.

§33.2.3 Dual Spaces

Among all bounded operators, those with target $\mathbb{F}$ deserve a special name.

Definition 33.2.10 (Dual space)

Let X be a normed space. Its **dual space** is

$$X^* = \mathcal{B}(X, \mathbb{F}),$$

whose elements are called **bounded linear functionals** on X .

Since $\mathbb{F}$ is complete, X^* is *always* a Banach space, whether or not X is. This is a very useful piece of leverage: even for a horrible normed space X , the dual is a nice object.

In IB Linear Algebra we studied the dual V' of a finite-dimensional space and found $\dim V' = \dim V$. In infinite dimensions the *algebraic* dual (all linear functionals) is enormous and useless; it is the topological dual X^* that carries information. It is not even obvious at this stage that $X^* \neq \{0\}$ for a general normed space X – that will be a consequence of the Hahn–Banach theorem.

Let's compute some duals.

Proposition 33.2.11

For $1 < p < \infty$ with conjugate exponent q , we have $(\ell^p)^* \cong \ell^q$. Also $(\ell^1)^* \cong \ell^\infty$ and $(c_0)^* \cong \ell^1$.

Proof. We do the case $1 \leq p < \infty$, the argument for c_0 being similar. Define $\Phi : \ell^q \rightarrow (\ell^p)^*$ by

$$(\Phi y)(x) = \sum_{n=1}^{\infty} x_n y_n.$$

By Hölder this converges with $|(\Phi y)(x)| \leq \|x\|_p \|y\|_q$, so $\Phi y \in (\ell^p)^*$ with $\|\Phi y\| \leq \|y\|_q$. It is clearly linear in y .

Φ is *isometric*. Suppose first $1 < p < \infty$. Given $y \in \ell^q$ non-zero, set $x_n = |y_n|^{q-1} \overline{\text{sgn}(y_n)}$ where $\text{sgn}(y) = y/|y|$ (and $x_n = 0$ if $y_n = 0$). Then $|x_n|^p = |y_n|^{(q-1)p} = |y_n|^q$, so $x \in \ell^p$ with $\|x\|_p = \|y\|_q^{q/p}$, while

$$(\Phi y)(x) = \sum_n |y_n|^q = \|y\|_q^q.$$

Hence $\|\Phi y\| \geq \|y\|_q^{q-q/p} = \|y\|_q$. For $p = 1, q = \infty$, test instead on $x = e_n$ to get $\|\Phi y\| \geq |y_n|$ for each n .

Φ is *surjective*. Let $f \in (\ell^p)^*$ and put $y_n = f(e_n)$. For any N , applying f to

$x = \sum_{n \leq N} |y_n|^{q-1} \overline{\text{sgn}(y_n)} e_n$ and using $|f(x)| \leq \|f\| \|x\|_p$ exactly as above gives

$$\left(\sum_{n \leq N} |y_n|^q \right)^{1/q} \leq \|f\|.$$

Letting $N \rightarrow \infty$ shows $y \in \ell^q$ with $\|y\|_q \leq \|f\|$. Finally, f and Φy are bounded functionals agreeing on each e_n , hence on c_{00} , which is dense in ℓ^p for $p < \infty$; so $f = \Phi y$. $\square$

Remark. The density of c_{00} is what fails for $p = \infty$, and indeed $(\ell^\infty)^* \neq \ell^1$: the dual of ℓ^∞ is strictly bigger, containing exotic objects called *Banach limits* which one constructs with Hahn–Banach. So duality is not symmetric: we have $(\ell^1)^* = \ell^\infty$ but $(\ell^\infty)^* \not\cong \ell^1$.

Example 33.2.12

If X is finite-dimensional with basis $e_1, \dots, e_n$, then every linear functional is bounded, and X^* is spanned by the dual basis $e_1^*, \dots, e_n^*$. In particular $(\ell_p^n)^* \cong \ell_q^n$ isometrically, by the same computation as above.

The dual of $C(K)$ is also computable, but the answer – the space of regular Borel measures on K , by the Riesz–Markov theorem – needs measure theory, so we will leave it aside. The one dual we will compute completely, and rather beautifully, is that of a Hilbert space.

§33.2.4 Adjoints

Every bounded operator induces one on the duals, going the other way.

Definition 33.2.13 (Adjoint)

Let $T \in \mathcal{B}(X, Y)$. The **adjoint** (or **dual map**) $T^* : Y^* \rightarrow X^*$ is defined by

$$T^*f = f \circ T, \quad \text{i.e. } (T^*f)(x) = f(Tx).$$

This is the same construction as the dual of a linear map in IB Linear Algebra, where it corresponded to transposing the matrix. Note that the Hilbert space adjoint we meet later is a slightly different (though closely related) object: it maps $H \rightarrow H$ rather than $H^* \rightarrow H^*$, and it is conjugate-linear in T .

Proposition 33.2.14

For $T \in \mathcal{B}(X, Y)$ we have $T^* \in \mathcal{B}(Y^*, X^*)$ with $\|T^*\| \leq \|T\|$. Moreover $(ST)^* = T^*S^*$ and $(S + T)^* = S^* + T^*$.

Proof. Linearity of T^* is immediate, and

$$\|T^*f\| = \sup_{\|x\| \leq 1} |f(Tx)| \leq \|f\| \sup_{\|x\| \leq 1} \|Tx\| = \|f\| \|T\|.$$

The algebraic identities are direct computations. $\square$

In fact $\|T^*\| = \|T\|$, but proving this requires us to know that Y^* has enough functionals to detect the norm of a vector in Y – and that is precisely what Hahn–Banach provides. So we turn to that now.

§33.2.5 The Double Dual

Since X^* is a normed space, it too has a dual, $X^{**} = (X^*)^*$, called the **double dual** or **bidual**. There is a completely canonical map from X into it.

Definition 33.2.15 (Canonical embedding)

Define $\iota : X \rightarrow X^{**}$ by

$$\iota(x)(f) = f(x) \quad (x \in X, f \in X^*).$$

That is, we let a vector act on functionals by evaluation. This is manifestly linear in x , and

$$|\iota(x)(f)| = |f(x)| \leq \|f\| \|x\|,$$

so $\iota(x) \in X^{**}$ with $\|\iota(x)\| \leq \|x\|$; hence $\iota \in \mathcal{B}(X, X^{**})$ with $\|\iota\| \leq 1$. Whether ι is *isometric* – equivalently, whether there are enough functionals to see the norm of every vector – is again a question for Hahn–Banach. We will see that it is, and this leads to the following notion.

Definition 33.2.16 (Reflexive)

A normed space X is **reflexive** if $\iota : X \rightarrow X^{**}$ is surjective (hence an isometric isomorphism).

Reflexivity is a strong condition. We will see that ℓ^p is reflexive for $1 < p < \infty$, that ℓ^1 and c_0 are not, and that every Hilbert space is. Note that a reflexive space must be a Banach space, since X^{**} always is.

Remark. It is essential that reflexivity is defined via the *canonical* map ι , and not merely by the existence of *some* isomorphism $X \cong X^{**}$. R. C. James constructed a Banach space which is isometrically isomorphic to its double dual but for which ι is not surjective; so the distinction is real, if rather subtle.

§33.3 The Hahn–Banach Theorem

We have repeatedly deferred questions to ‘Hahn–Banach’: does X^* separate points? Is ι isometric? Is $\|T^*\| = \|T\|$? All of these ask whether a normed space has *enough* bounded linear functionals, and all are answered by one theorem, which is really a theorem about extending functionals from a subspace to the whole space without increasing their size.

§33.3.1 Zorn’s Lemma

The extension will be carried out one dimension at a time, and to get from ‘one dimension at a time’ to ‘all at once’ we need a transfinite tool. Recall a **partial order** on a set P is a reflexive, antisymmetric, transitive relation $\leq$; a **chain** is a subset on which $\leq$ is a

total order; an **upper bound** for $S \subseteq P$ is $u \in P$ with $s \leq u$ for all $s \in S$; and $m \in P$ is **maximal** if $m \leq p$ implies $p = m$.

Theorem 33.3.1 (Zorn's Lemma)

Let $(P, \leq)$ be a non-empty partially ordered set in which every chain has an upper bound. Then P has a maximal element.

Zorn's lemma is equivalent to the axiom of choice (and to the well-ordering principle), as you will see in II Logic and Set Theory. We will simply use it.

Proposition 33.3.2

Every vector space has a basis, and every linearly independent set extends to a basis.

Proof. Let S be linearly independent in V and let P be the set of linearly independent subsets of V containing S , ordered by inclusion. It is non-empty ($S \in P$). Given a chain $\mathcal{C} \subseteq P$, the union $\bigcup \mathcal{C}$ is linearly independent – any linear dependence involves finitely many vectors, which all lie in a single member of the chain – so it is an upper bound. By Zorn there is a maximal linearly independent $B \supseteq S$. If $\text{span } B \neq V$, pick $v \notin \text{span } B$; then $B \cup \{v\}$ is linearly independent, contradicting maximality. So B is a basis. $\square$

Such a basis is called a **Hamel basis**, to distinguish it from the orthonormal bases we will meet in Hilbert space (where infinite sums are permitted). Hamel bases of infinite-dimensional Banach spaces are always uncountable and always useless – we will prove the first assertion with the Baire category theorem.

§33.3.2 The Analytic Form

The theorem is about domination by a sublinear functional.

Definition 33.3.3 (Sublinear functional)

Let V be a real vector space. A map $p : V \rightarrow \mathbb{R}$ is **sublinear** if

$$p(v + w) \leq p(v) + p(w) \quad \text{and} \quad p(\lambda v) = \lambda p(v)$$

for all $v, w \in V$ and $\lambda > 0$.

A norm is sublinear, but so is (say) $p(x) = \max_i x_i$ on $\mathbb{R}^n$, and so is the Minkowski functional of a convex set containing the origin. The extra generality is what makes the geometric forms of the theorem available.

Lemma 33.3.4 (One-step extension)

Let V be a real vector space, p sublinear on V , $W \subseteq V$ a subspace and $f : W \rightarrow \mathbb{R}$ linear with $f \leq p$ on W . Let $v_0 \in V \setminus W$. Then f extends to a linear $g : W \oplus \mathbb{R}v_0 \rightarrow \mathbb{R}$ with $g \leq p$.

Proof. Any linear extension is determined by the value $c = g(v_0)$, via $g(w + tv_0) = f(w) + tc$. We must choose c so that

$$f(w) + tc \leq p(w + tv_0) \quad \text{for all } w \in W, t \in \mathbb{R}.$$

For $t = 0$ this is the hypothesis. For $t > 0$, dividing by t and replacing w/t by w , the condition becomes

$$c \leq p(w + v_0) - f(w) \quad \text{for all } w \in W.$$

For $t < 0$, writing $t = -s$ with $s > 0$ and dividing by s , it becomes

$$c \geq f(w') - p(w' - v_0) \quad \text{for all } w' \in W.$$

So a suitable c exists precisely when

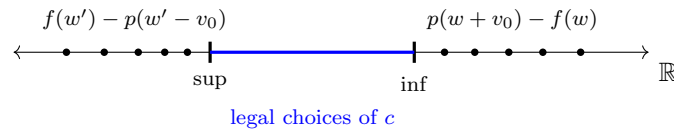
$$\sup_{w' \in W} (f(w') - p(w' - v_0)) \leq \inf_{w \in W} (p(w + v_0) - f(w)).$$

But for any $w, w' \in W$,

$$f(w') + f(w) = f(w' + w) \leq p(w' + w) \leq p(w' - v_0) + p(w + v_0),$$

and rearranging gives exactly $f(w') - p(w' - v_0) \leq p(w + v_0) - f(w)$. So the supremum is at most the infimum, and any c in between will do. $\square$

The heart of the matter is the last display. The positive values of t produce an upper bound for c and the negative values produce a lower bound, and sublinearity of p is exactly what guarantees that the lower bounds all sit below all of the upper bounds – so that there is a non-empty window of legal choices.



Theorem 33.3.5 (Hahn–Banach, analytic form)

Let V be a real vector space, $p : V \rightarrow \mathbb{R}$ sublinear, $W \subseteq V$ a subspace and $f : W \rightarrow \mathbb{R}$ linear with $f(w) \leq p(w)$ for all $w \in W$. Then there is a linear $\tilde{f} : V \rightarrow \mathbb{R}$ with $\tilde{f}|_W = f$ and $\tilde{f}(v) \leq p(v)$ for all $v \in V$.

Proof. Let P be the set of pairs (U, g) where $W \subseteq U \subseteq V$ is a subspace, $g : U \rightarrow \mathbb{R}$ is linear, $g|_W = f$ and $g \leq p$ on U . Order P by

$$(U_1, g_1) \leq (U_2, g_2) \iff U_1 \subseteq U_2 \text{ and } g_2|_{U_1} = g_1.$$

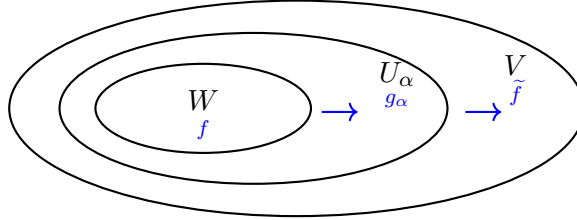
This is a partial order, and $P \neq \emptyset$ as $(W, f) \in P$.

Let $\mathcal{C} = \{(U_i, g_i)\}_{i \in I}$ be a chain in P . Put $U = \bigcup_i U_i$ and define g on U by $g(u) = g_i(u)$ whenever $u \in U_i$; this is well defined by the compatibility built into the order. Since $\mathcal{C}$ is a chain, any two elements of U lie in a common U_i , so U is a subspace and g is linear; and $g \leq p$ since each $g_i \leq p$. So $(U, g) \in P$ is an upper

bound for $\mathcal{C}$.

By Zorn's lemma, P has a maximal element $(U_{\max}, \tilde{f})$. If $U_{\max} \neq V$, pick $v_0 \in V \setminus U_{\max}$ and apply the one-step extension lemma to obtain a strictly larger element of P , contradicting maximality. Hence $U_{\max} = V$. $\square$

Notice that the picture of the extension is exactly a growing domain: we start on W , and repeatedly (transfinitely) enlarge the domain by one dimension, always keeping underneath the ceiling p .



For the complex case, and for the applications we actually want, the following symmetric version is more convenient. Recall that a **seminorm** is a map $p : V \rightarrow [0, \infty)$ with $p(v + w) \leq p(v) + p(w)$ and $p(\lambda v) = |\lambda|p(v)$ for all scalars λ .

Theorem 33.3.6 (Hahn–Banach, seminorm form)

Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, let p be a seminorm on V , let $W \subseteq V$ be a subspace and let $f : W \rightarrow \mathbb{F}$ be linear with $|f(w)| \leq p(w)$ on W . Then there is a linear $\tilde{f} : V \rightarrow \mathbb{F}$ with $\tilde{f}|_W = f$ and $|\tilde{f}(v)| \leq p(v)$ for all $v \in V$.

Proof. Real case. A seminorm is sublinear, so the analytic form gives an extension with $\tilde{f} \leq p$. Then also $-\tilde{f}(v) = \tilde{f}(-v) \leq p(-v) = p(v)$, so $|\tilde{f}| \leq p$.

Complex case. Let $g = \operatorname{Re} f : W \rightarrow \mathbb{R}$, which is $\mathbb{R}$ -linear, and note that

$$f(w) = g(w) - ig(iw),$$

since writing $f(w) = g(w) + ih(w)$ and using $f(iw) = if(w)$ gives $h(w) = -g(iw)$. Now $|g| \leq |f| \leq p$, and regarding V as a real vector space, the real case gives an $\mathbb{R}$ -linear extension $\tilde{g} : V \rightarrow \mathbb{R}$ with $|\tilde{g}| \leq p$. Define

$$\tilde{f}(v) = \tilde{g}(v) - i\tilde{g}(iv).$$

This is $\mathbb{R}$ -linear, and $\tilde{f}(iv) = \tilde{g}(iv) - i\tilde{g}(-v) = i(\tilde{g}(v) - i\tilde{g}(iv)) = i\tilde{f}(v)$, so it is $\mathbb{C}$ -linear. It restricts to f on W by the identity above.

Finally, fix v and choose $\theta \in \mathbb{R}$ with $e^{i\theta}\tilde{f}(v) = |\tilde{f}(v)|$. Then

$$|\tilde{f}(v)| = \tilde{f}(e^{i\theta}v) = \operatorname{Re} \tilde{f}(e^{i\theta}v) = \tilde{g}(e^{i\theta}v) \leq p(e^{i\theta}v) = p(v),$$

using that $|\tilde{f}(v)|$ is real so equals its own real part. $\square$

§33.3.3 Consequences

Now we harvest. Everything in this subsection follows by applying Hahn–Banach with $p = \|\cdot\|$ or a scaled version of it.

Corollary 33.3.7 (Norm-preserving extension)

Let X be a normed space, $W \subseteq X$ a subspace and $f \in W^*$. Then there is $\tilde{f} \in X^*$ with $\tilde{f}|_W = f$ and $\|\tilde{f}\|_{X^*} = \|f\|_{W^*}$.

Proof. Apply the seminorm form with $p(x) = \|f\|_{W^*} \|x\|$, which is a seminorm (indeed a norm if $f \neq 0$) and dominates $|f|$ on W . We obtain $\tilde{f}$ with $|\tilde{f}(x)| \leq \|f\| \|x\|$, so $\|\tilde{f}\| \leq \|f\|$; and $\|\tilde{f}\| \geq \|f\|$ since $\tilde{f}$ extends f . $\square$

Corollary 33.3.8 (Support functionals)

Let X be a normed space and $x_0 \in X$ with $x_0 \neq 0$. Then there is $f \in X^*$ with $\|f\| = 1$ and $f(x_0) = \|x_0\|$.

Proof. Let $W = \mathbb{F}x_0$ and define $f_0(\lambda x_0) = \lambda \|x_0\|$. Then $|f_0(\lambda x_0)| = \|\lambda x_0\|$, so $\|f_0\|_{W^*} = 1$. Extend by the previous corollary. $\square$

Such an f is called a **support functional** at x_0 . It is the abstract version of ‘the tangent hyperplane to the unit ball at $x_0/\|x_0\|$ ’.

Corollary 33.3.9

Let X be a normed space. Then:

- (i) if $X \neq \{0\}$ then $X^* \neq \{0\}$;
- (ii) X^* separates points: if $x \neq y$ there is $f \in X^*$ with $f(x) \neq f(y)$;
- (iii) for every $x \in X$,

$$\|x\| = \sup\{|f(x)| : f \in X^*, \|f\| \leq 1\},$$

and the supremum is attained;

- (iv) the canonical embedding $\iota : X \rightarrow X^{**}$ is an isometry.

Proof. (i) and (ii) follow from the existence of support functionals (for (ii) apply it to $x - y$). For (iii), the inequality $\leq$ is the support functional and $\geq$ is $|f(x)| \leq \|f\| \|x\|$. Finally (iv) is a restatement of (iii), since $\|\iota(x)\| = \sup_{\|f\| \leq 1} |\iota(x)(f)| = \sup_{\|f\| \leq 1} |f(x)|$. $\square$

Part (iv) is worth pausing on. It says that *every normed space embeds isometrically into a Banach space*, namely the closure of $\iota(X)$ in X^{**} . This gives the slick construction of the completion promised earlier.

Corollary 33.3.10

For $T \in \mathcal{B}(X, Y)$ we have $\|T^*\| = \|T\|$.

Proof. We know $\|T^*\| \leq \|T\|$. Conversely, given $\varepsilon > 0$ pick x with $\|x\| \leq 1$ and $\|Tx\| \geq \|T\| - \varepsilon$, and let $f \in Y^*$ be a support functional at Tx , so $\|f\| = 1$ and $f(Tx) = \|Tx\|$. Then

$$\|T^*\| \geq \|T^*f\| \geq |(T^*f)(x)| = |f(Tx)| = \|Tx\| \geq \|T\| - \varepsilon. \quad \square$$

Corollary 33.3.11 (Separating a point from a subspace)

Let $W \subseteq X$ be a closed subspace and $x_0 \in X \setminus W$. Then there is $f \in X^*$ with $\|f\| = 1$, $f|_W = 0$ and $f(x_0) = \text{dist}(x_0, W) > 0$.

Proof. Work in the quotient X/W , which is a normed space since W is closed, and let $q : X \rightarrow X/W$ be the quotient map, which satisfies $\|q\| \leq 1$. As $x_0 \notin W$ we have $q(x_0) \neq 0$, so there is a support functional $g \in (X/W)^*$ with $\|g\| = 1$ and $g(q(x_0)) = \|q(x_0)\| = \text{dist}(x_0, W)$. Put $f = g \circ q$. Then $\|f\| \leq 1$, f kills W , and $f(x_0) = \text{dist}(x_0, W)$. For $\|f\| = 1$, note $\|f\| \geq \text{dist}(x_0, W)/\|x_0 - w\|$ for every $w \in W$, and take the infimum over w . $\square$

Corollary 33.3.12

A subspace $W \subseteq X$ is dense if and only if the only $f \in X^*$ vanishing on W is $f = 0$.

Proof. If W is dense and $f|_W = 0$ then $f = 0$ by continuity. Conversely if $\bar{W} \neq X$ pick $x_0 \notin \bar{W}$ and apply the previous corollary to the closed subspace $\bar{W}$. $\square$

This last statement is one we will use again and again: to prove something lies in the closure of a subspace, it suffices to show every functional killing the subspace kills it too. It converts an approximation problem into a duality problem.

Example 33.3.13 (Reflexivity of ℓ^p)

Let $1 < p < \infty$ and q its conjugate. We computed $(\ell^p)^* \cong \ell^q$ and $(\ell^q)^* \cong \ell^p$, so $(\ell^p)^{**} \cong \ell^p$. Chasing the identifications shows the composite is exactly ι , so ℓ^p is reflexive.

On the other hand c_0 is not reflexive: $(c_0)^* \cong \ell^1$ and $(\ell^1)^* \cong \ell^\infty$, so $c_0^{**} \cong \ell^\infty \supsetneq c_0$. Nor is ℓ^1 reflexive, since $(\ell^1)^{**} = (\ell^\infty)^*$ is strictly larger than ℓ^1 .

Hahn–Banach is not merely a device for extending functionals we already have; it can conjure up functionals with prescribed and rather magical properties. Here is the standard example.

Example 33.3.14 (Banach limits)

We construct a linear functional Λ on the real space ℓ^∞ such that

- (i) $\liminf_n x_n \leq \Lambda(x) \leq \limsup_n x_n$ for every $x \in \ell^\infty$;
- (ii) $\Lambda(Lx) = \Lambda(x)$, where L is the left shift.

In particular Λ agrees with $\lim$ on the convergent sequences, but is defined on *all*

bounded sequences, and is shift-invariant. So it assigns a ‘limit’ to $(1, 0, 1, 0, \dots)$ – necessarily $\frac{1}{2}$, by (i) and (ii) applied to $x + Lx = (1, 1, 1, \dots)$.

For the construction, define

$$p(x) = \lim_{n \rightarrow \infty} \sup_{k \geq 1} \frac{x_k + x_{k+1} + \dots + x_{k+n-1}}{n}.$$

The limit exists because the quantity $a_n = \sup_k (x_k + \dots + x_{k+n-1})$ is subadditive in n , so a_n/n converges by Fekete’s lemma. One checks that p is sublinear and that $p(x) \leq \limsup_n x_n$, with $p(Lx - x) \leq 0$ and $p(x - Lx) \leq 0$.

Apply Hahn–Banach to the zero functional on the zero subspace, dominated by p : we get a linear $\Lambda \leq p$ on ℓ^∞ . Then $\Lambda(x) \leq p(x) \leq \limsup x_n$, and $-\Lambda(x) = \Lambda(-x) \leq p(-x) \leq -\liminf x_n$, giving (i). And $\Lambda(Lx - x) \leq p(Lx - x) \leq 0$ together with the same for $x - Lx$ gives (ii).

Such a Λ is called a **Banach limit**. It is an element of $(\ell^\infty)^*$ which does not come from ℓ^1 : it vanishes on c_{00} but not on ℓ^∞ . This is the promised proof that $(\ell^\infty)^* \supsetneq \ell^1$, and hence that ℓ^1 is not reflexive.

§33.3.4 The Geometric Forms

The analytic form is about functionals dominated by a sublinear p ; the geometric form is about hyperplanes separating convex sets. The bridge between them is the following construction, which turns a convex set into a sublinear functional.

Definition 33.3.15 (Minkowski functional)

Let V be a vector space and $C \subseteq V$ a convex set with $0 \in C$. The **Minkowski functional** (or **gauge**) of C is

$$\mu_C(v) = \inf\{t > 0 : v \in tC\} \in [0, \infty].$$

Lemma 33.3.16

Let X be a normed space and C an open convex set with $0 \in C$. Then μ_C is sublinear, $C = \{v : \mu_C(v) < 1\}$, and there is $M > 0$ with $\mu_C(v) \leq M \|v\|$.

Proof. Positive homogeneity is immediate from the definition. For subadditivity, suppose $v \in sC$ and $w \in tC$, say $v = sc_1$, $w = tc_2$ with $c_i \in C$. Then

$$\frac{v+w}{s+t} = \frac{s}{s+t}c_1 + \frac{t}{s+t}c_2 \in C$$

by convexity, so $\mu_C(v+w) \leq s+t$; taking infima gives $\mu_C(v+w) \leq \mu_C(v) + \mu_C(w)$.

Since C is open and $0 \in C$, there is $r > 0$ with $B(0, r) \subseteq C$, whence $\mu_C(v) \leq \|v\|/r$; take $M = 1/r$. If $v \in C$ then by openness $(1+\varepsilon)v \in C$ for small ε , so $\mu_C(v) \leq (1+\varepsilon)^{-1} < 1$; conversely $\mu_C(v) < 1$ gives $v \in tC \subseteq C$ for some $t < 1$ (using $0 \in C$ and convexity). $\square$

Theorem 33.3.17 (Hahn–Banach, separation form)

Let X be a real normed space and let $A, B \subseteq X$ be non-empty disjoint convex sets.

- (i) If A is open, there exist $f \in X^* \setminus \{0\}$ and $\alpha \in \mathbb{R}$ with

$$f(a) < \alpha \leq f(b) \quad \text{for all } a \in A, b \in B.$$

- (ii) If A is closed and B is compact, there exist $f \in X^* \setminus \{0\}$ and $\alpha_1 < \alpha_2$ with

$$f(a) \leq \alpha_1 < \alpha_2 \leq f(b) \quad \text{for all } a \in A, b \in B.$$

Proof. (i) Fix $a_0 \in A$, $b_0 \in B$ and set $v_0 = b_0 - a_0$ and

$$C = A - B + v_0 = \{a - b + v_0 : a \in A, b \in B\}.$$

Then C is convex (a sum of convex sets), open (a union of translates of the open set A), and contains 0 (take $a = a_0, b = b_0$). Also $v_0 \notin C$: otherwise $v_0 = a - b + v_0$ for some $a \in A, b \in B$, forcing $a = b$, contradicting disjointness.

Let $\mu = \mu_C$, which is sublinear and bounded by $M \|\cdot\|$. On the line $W = \mathbb{R}v_0$ define $g(tv_0) = t$. For $t > 0$ we have $g(tv_0) = t \leq t\mu(v_0) = \mu(tv_0)$, since $\mu(v_0) \geq 1$ as $v_0 \notin C$; for $t \leq 0$, $g(tv_0) \leq 0 \leq \mu(tv_0)$. So $g \leq \mu$ on W , and Hahn–Banach gives a linear $f : X \rightarrow \mathbb{R}$ with $f \leq \mu$ and $f(v_0) = 1$. Since $f(x) \leq \mu(x) \leq M \|x\|$ and $-f(x) = f(-x) \leq M \|x\|$, we get $f \in X^*$, and $f \neq 0$.

Now for $a \in A, b \in B$, the point $a - b + v_0 \in C$ so $\mu(a - b + v_0) < 1$, giving

$$f(a) - f(b) + 1 = f(a - b + v_0) \leq \mu(a - b + v_0) < 1,$$

i.e. $f(a) < f(b)$. Hence $\alpha := \inf_{b \in B} f(b)$ satisfies $f(a) \leq \alpha \leq f(b)$ for all a, b . Finally $f(a) < \alpha$ strictly: A is open and $f \neq 0$, so $f(A)$ is an open subset of $\mathbb{R}$ (pick u with $f(u) = 1$; then $f(a + su) = f(a) + s$ covers a neighbourhood of $f(a)$), and an open set cannot contain its own supremum.

(ii) Since A is closed, B compact and $A \cap B = \emptyset$, the distance $\delta = \text{dist}(A, B)$ is strictly positive. (If $\text{dist}(a_n, b_n) \rightarrow 0$, pass to a convergent subsequence $b_n \rightarrow b \in B$; then $a_n \rightarrow b$, so $b \in \bar{A} = A$, a contradiction.) Let

$$A_\delta = A + B(0, \delta/2),$$

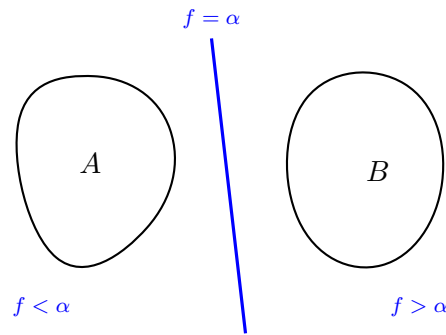
which is open and convex, and disjoint from B by the choice of δ . By (i) there is $f \neq 0$ with $f(a') < f(b)$ for $a' \in A_\delta, b \in B$. For $a \in A$ and $\|u\| < \delta/2$ we get $f(a) + f(u) < f(b)$, and taking the supremum over such u gives

$$f(a) + \frac{\delta}{2} \|f\| \leq f(b)$$

for all $a \in A, b \in B$. As $\|f\| > 0$ we may take $\alpha_1 = \sup_A f$ and $\alpha_2 = \alpha_1 + \frac{\delta}{2} \|f\| > \alpha_1$. $\square$

Geometrically, a functional f and a level α determine an affine **hyperplane** $H = f^{-1}(\alpha)$, which is closed exactly when f is continuous, and which divides X into two half-spaces $\{f < \alpha\}$ and $\{f > \alpha\}$. The theorem says that two disjoint convex sets can be put on

opposite sides of a hyperplane, strictly so if one is closed and the other compact.



Corollary 33.3.18

Let X be a real normed space, $C \subseteq X$ a closed convex set and $x_0 \notin C$. Then there is $f \in X^*$ with $\sup_{c \in C} f(c) < f(x_0)$. In particular, a closed convex set is the intersection of the closed half-spaces containing it.

Proof. Apply part (ii) with $A = C$ and $B = \{x_0\}$, which is compact. The final statement follows since any $x_0 \notin C$ is excluded by one of these half-spaces. $\square$

Remark. Convexity is essential. In $\mathbb{R}^2$ we cannot separate a point at the centre of an annulus from the annulus itself. The theorem is a genuinely convex-geometric statement, and it is the starting point of convex analysis and of the theory of weak topologies.

§33.4 The Baire Category Theorem

The Hahn–Banach theorem was a statement about the algebraic and convex structure of a normed space, and its proof used no completeness at all. We now turn to the other pillar of the subject, which is entirely about completeness, and which is in some ways even more remarkable: a very soft topological statement whose consequences include three of the fundamental theorems of functional analysis.

§33.4.1 Category

Recall that a subset A of a topological space X is **dense** if $\bar{A} = X$, equivalently if A meets every non-empty open set.

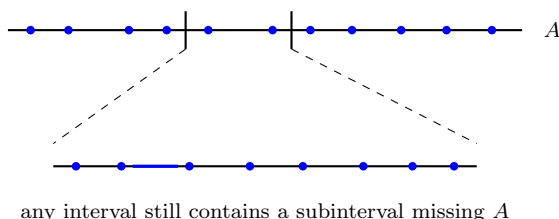
Definition 33.4.1 (Nowhere dense)

A subset A of a topological space X is **nowhere dense** (or **rare**) if $\bar{A}$ has empty interior, i.e. $\bar{A}$ contains no non-empty open set.

Equivalently, A is nowhere dense if and only if $X \setminus \bar{A}$ is dense: every open set contains an open subset missing A entirely. Think of it as being ‘full of holes at every scale’.

Example 33.4.2 (i) A single point in $\mathbb{R}$ is nowhere dense, as is any finite set, as is $\{1/n : n \in \mathbb{N}\} \cup \{0\}$.

- (ii) $\mathbb{Z}$ is nowhere dense in $\mathbb{R}$, but $\mathbb{Q}$ is *not* – its closure is all of $\mathbb{R}$.
- (iii) The Cantor set is nowhere dense in $[0, 1]$: it is closed, and contains no interval since it has empty interior by construction.
- (iv) In $C[0, 1]$, the set of polynomials of degree at most d is nowhere dense (it is a finite-dimensional, hence closed, proper subspace, and a proper subspace has empty interior – if it contained a ball it would contain a translate of a ball about 0, hence everything).



Definition 33.4.3 (Meagre)

A subset of a topological space is **meagre** (or **of first category**) if it is a countable union of nowhere dense sets, and **of second category** otherwise. A set whose complement is meagre is called **residual** or **comeagre**.

‘Meagre’ is a topological notion of smallness. It behaves as one would hope: a countable union of meagre sets is meagre, and a subset of a meagre set is meagre. The content of the Baire category theorem is that in a complete space, ‘meagre’ really does mean small – in particular, the whole space is not meagre in itself.

Theorem 33.4.4 (Baire Category Theorem)

Let (X, d) be a complete non-empty metric space. Then:

- (i) if $(U_n)_{n \geq 1}$ are open and dense in X , then $\bigcap_{n \geq 1} U_n$ is dense in X ;
- (ii) X is not meagre in itself. Equivalently, if $X = \bigcup_{n \geq 1} F_n$ with each F_n closed, then some F_n has non-empty interior.

Proof. (i) Let $W \subseteq X$ be non-empty and open; we must show $W \cap \bigcap_n U_n \neq \emptyset$. We construct a nested sequence of closed balls.

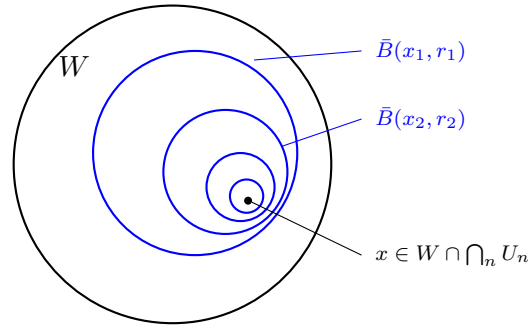
Since U_1 is dense and open, $W \cap U_1$ is non-empty and open, so it contains a closed ball $\bar{B}(x_1, r_1)$ with $0 < r_1 < 1$. Inductively, given $\bar{B}(x_n, r_n)$, the set $B(x_n, r_n) \cap U_{n+1}$ is non-empty (density of U_{n+1}) and open, so it contains a closed ball $\bar{B}(x_{n+1}, r_{n+1})$ with $0 < r_{n+1} < 2^{-n}$.

By construction $\bar{B}(x_1, r_1) \supseteq \bar{B}(x_2, r_2) \supseteq \dots$ and $r_n \rightarrow 0$. For $m > n$ we have $x_m \in \bar{B}(x_n, r_n)$, so $d(x_n, x_m) \leq r_n \rightarrow 0$ and (x_n) is Cauchy. By completeness $x_n \rightarrow x$ for some $x \in X$, and since $x_m \in \bar{B}(x_n, r_n)$ for all $m \geq n$ and this ball is closed, $x \in \bar{B}(x_n, r_n)$ for every n . Hence

$$x \in \bar{B}(x_1, r_1) \subseteq W \cap U_1 \quad \text{and} \quad x \in \bar{B}(x_{n+1}, r_{n+1}) \subseteq U_{n+1}$$

for every n , so $x \in W \cap \bigcap_n U_n$ as required.

(ii) Suppose $X = \bigcup_n A_n$ with each A_n nowhere dense. Then $U_n = X \setminus \bar{A}_n$ is open and dense, and $\bigcap_n U_n = X \setminus \bigcup_n \bar{A}_n = \emptyset$, contradicting (i) (as $X \neq \emptyset$). The reformulation follows by taking F_n closed with empty interior, i.e. nowhere dense. $\square$



Before the big applications, here are two quick ones that show the flavour.

Proposition 33.4.5

$\mathbb{Q}$ is not the intersection of countably many open subsets of $\mathbb{R}$.

Proof. Suppose $\mathbb{Q} = \bigcap_n U_n$ with U_n open. Each $U_n \supseteq \mathbb{Q}$ is dense. Enumerate $\mathbb{Q} = \{q_1, q_2, \dots\}$; then each $V_n = \mathbb{R} \setminus \{q_n\}$ is open and dense. So

$$\bigcap_n U_n \cap \bigcap_n V_n = \mathbb{Q} \setminus \mathbb{Q} = \emptyset,$$

contradicting Baire applied to the complete space $\mathbb{R}$. $\square$

Proposition 33.4.6

An infinite-dimensional Banach space cannot have a countable Hamel basis.

Proof. Suppose X has Hamel basis $\{e_1, e_2, \dots\}$, and let $X_n = \text{span}\{e_1, \dots, e_n\}$. Each X_n is finite-dimensional, hence closed, and proper (as X is infinite-dimensional), hence has empty interior. But $X = \bigcup_n X_n$, contradicting Baire. $\square$

Remark. Hence, for instance, c_{00} (which *does* have the countable Hamel basis $\{e_n\}$) admits no norm making it complete. That is a strong statement, and it would be painful to prove directly.

Theorem 33.4.7 (Osgood)

Let X be a complete metric space and $f_n : X \rightarrow \mathbb{R}$ continuous with $f_n \rightarrow f$ pointwise. Then the set of points at which f is continuous is residual, in particular dense.

Proof. For $\varepsilon > 0$ and $N \in \mathbb{N}$ put

$$F_{N,\varepsilon} = \{x \in X : |f_n(x) - f_m(x)| \leq \varepsilon \text{ for all } n, m \geq N\},$$

which is closed by continuity of the f_n . Pointwise convergence gives $X = \bigcup_N F_{N,\varepsilon}$,

so

$$U_\varepsilon = \bigcup_N \text{int } F_{N,\varepsilon}$$

is open, and it is dense: for any open ball B , applying Baire inside the complete space $\bar{B}$ to the sets $F_{N,\varepsilon} \cap \bar{B}$ produces some N for which $F_{N,\varepsilon} \cap \bar{B}$ has interior, so U_ε meets B .

Now let $D = \bigcap_{k \geq 1} U_{1/k}$, a residual set. We claim f is continuous at each $x \in D$. Given $\varepsilon > 0$ choose k with $3/k < \varepsilon$. As $x \in U_{1/k}$, there is N and an open $V \ni x$ with $V \subseteq F_{N,1/k}$. Letting $m \rightarrow \infty$ in the defining inequality gives $|f_N(y) - f(y)| \leq 1/k$ for all $y \in V$. Shrinking V so that $|f_N(y) - f_N(x)| < 1/k$ on it, we get

$$|f(y) - f(x)| \leq |f(y) - f_N(y)| + |f_N(y) - f_N(x)| + |f_N(x) - f(x)| \leq \frac{3}{k} < \varepsilon. \quad \square$$

So a pointwise limit of continuous functions, while not continuous in general, cannot be discontinuous everywhere – which incidentally shows that the indicator of $\mathbb{Q}$ is not a pointwise limit of continuous functions, though it is a pointwise limit of pointwise limits.

§33.4.2 A Continuous Nowhere Differentiable Function

Here is the classical showpiece application. Weierstrass constructed such a function explicitly in 1872, to general consternation; Banach observed in 1931 that Baire category makes the existence almost obvious – and in fact shows that such functions are the *typical* continuous function, the differentiable ones forming a meagre set.

Theorem 33.4.8

There is a continuous function $f : [0, 1] \rightarrow \mathbb{R}$ which is differentiable at no point of $[0, 1]$. Indeed, the set of such f is residual in $C[0, 1]$.

Proof. Work in the Banach space $X = C[0, 1]$ with the sup norm. For $n \in \mathbb{N}$ let

$$A_n = \{f \in X : \exists x \in [0, 1] \text{ with } |f(x) - f(y)| \leq n|x - y| \text{ for all } y \in [0, 1]\}.$$

If f is differentiable at some point x , then f is Lipschitz at x in this sense with some constant, so $f \in A_n$ for some n . Thus it suffices to show each A_n is closed with empty interior; then $\bigcup_n A_n$ is meagre and any f in its (non-empty, by Baire) complement works.

A_n is closed. Let $f_k \in A_n$ with $f_k \rightarrow f$ uniformly, and let x_k witness $f_k \in A_n$. By compactness of $[0, 1]$, pass to a subsequence with $x_k \rightarrow x$. For any $y \in [0, 1]$,

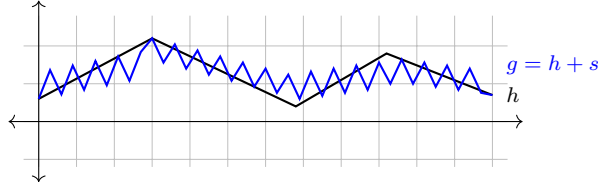
$$|f(x) - f(y)| \leq |f(x) - f(x_k)| + |f(x_k) - f_k(x_k)| + |f_k(x_k) - f_k(y)| + |f_k(y) - f(y)|.$$

The second and fourth terms are at most $\|f - f_k\|_\infty \rightarrow 0$; the first tends to 0 by continuity of f and $x_k \rightarrow x$; and the third is at most $n|x_k - y| \rightarrow n|x - y|$. Hence $|f(x) - f(y)| \leq n|x - y|$, so $f \in A_n$.

A_n has empty interior. Let $f \in X$ and $\varepsilon > 0$; we find g with $\|f - g\|_\infty < \varepsilon$ and $g \notin A_n$. By the Weierstrass approximation theorem (or just uniform continuity and piecewise-linear interpolation), choose a piecewise linear h with $\|f - h\|_\infty < \varepsilon/2$.

Let M be the maximum absolute slope of h . Let s be a ‘sawtooth’ function of amplitude $\varepsilon/2$ whose slopes all have absolute value Λ , where $\Lambda > M + n$; explicitly, s is the $\varepsilon/2$ -scaled zigzag of period ε/Λ . Put $g = h + s$.

Then $\|f - g\|_\infty \leq \|f - h\|_\infty + \|s\|_\infty < \varepsilon$. At every point x , and for y close enough to x on a suitable side, the difference quotient of g has absolute value at least $\Lambda - M > n$, since s contributes slope $\pm\Lambda$ and h at most M . Hence $g \notin A_n$. $\square$



The picture is the point of the proof: adding a tiny but very steep sawtooth changes the function hardly at all in the uniform norm, while destroying any chance of a bounded difference quotient. Uniform closeness is simply blind to derivatives.

§33.4.3 The Principle of Uniform Boundedness

The second application is the one that gets used most often in practice.

Theorem 33.4.9 (Banach–Steinhaus / Uniform Boundedness Principle)

Let X be a Banach space, Y a normed space, and $\{T_i\}_{i \in I} \subseteq \mathcal{B}(X, Y)$ a family of bounded operators which is *pointwise bounded*:

$$\sup_{i \in I} \|T_i x\| < \infty \quad \text{for every } x \in X.$$

Then it is *uniformly bounded*: $\sup_{i \in I} \|T_i\| < \infty$.

Proof. For $n \in \mathbb{N}$ let

$$F_n = \{x \in X : \|T_i x\| \leq n \text{ for all } i \in I\} = \bigcap_{i \in I} \{x : \|T_i x\| \leq n\}.$$

Each set in the intersection is closed (preimage of a closed set under the continuous map $x \mapsto \|T_i x\|$), so F_n is closed. By pointwise boundedness $X = \bigcup_n F_n$.

Since X is complete, Baire gives some N with F_N having non-empty interior, say $\bar{B}(x_0, r) \subseteq F_N$ with $r > 0$. Now for any x with $\|x\| \leq 1$ and any i ,

$$\|T_i x\| = \frac{1}{r} \|T_i(x_0 + rx) - T_i x_0\| \leq \frac{1}{r} (\|T_i(x_0 + rx)\| + \|T_i x_0\|) \leq \frac{2N}{r},$$

since both $x_0 + rx$ and x_0 lie in $\bar{B}(x_0, r) \subseteq F_N$. Hence $\|T_i\| \leq 2N/r$ for all i . $\square$

The theorem is genuinely surprising: pointwise information about a family of operators upgrades, for free, to uniform information. Completeness of X is essential – on c_{00} , the functionals $f_n(x) = nx_n$ are pointwise bounded but have $\|f_n\| = n$.

Corollary 33.4.10

Let X be a Banach space, Y a normed space and $T_n \in \mathcal{B}(X, Y)$ with $T_n x \rightarrow Tx$ for every $x \in X$. Then T is linear and bounded, with

$$\|T\| \leq \liminf_n \|T_n\| < \infty.$$

Proof. Linearity is clear. Each convergent sequence $(T_n x)$ is bounded, so $\{T_n\}$ is pointwise bounded, hence $\sup_n \|T_n\| = M < \infty$ by uniform boundedness. Then $\|Tx\| = \lim \|T_n x\| \leq M \|x\|$. The sharper bound follows by taking a subsequence realising the $\liminf$. $\square$

Corollary 33.4.11 (Weak boundedness implies boundedness)

Let X be a normed space and $S \subseteq X$. If $f(S)$ is bounded in $\mathbb{F}$ for every $f \in X^*$, then S is bounded in norm.

Proof. Apply uniform boundedness in the Banach space X^* to the family $\{\iota(x) : x \in S\} \subseteq X^{**}$. The hypothesis says exactly that this family is pointwise bounded on X^* , so $\sup_{x \in S} \|\iota(x)\| < \infty$; and $\|\iota(x)\| = \|x\|$ by Hahn–Banach. $\square$

Note the slight sleight of hand: X itself need not be complete, but X^* always is, so we may apply Baire there. This trick – move the problem into a dual space to gain completeness – is worth remembering.

Example 33.4.12 (Divergence of Fourier series)

Here is a striking application. For $f \in C(\mathbb{T})$, the continuous 2π -periodic functions with the sup norm, write

$$S_N f(x) = \sum_{|k| \leq N} \hat{f}(k) e^{ikx}, \quad \hat{f}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-ikt} dt.$$

We claim that there is a continuous function whose Fourier series diverges at 0; indeed the set of such functions is residual.

Consider the functionals $\Lambda_N : C(\mathbb{T}) \rightarrow \mathbb{C}$, $\Lambda_N f = S_N f(0)$. A computation with the geometric series gives

$$\Lambda_N f = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) D_N(t) dt, \quad D_N(t) = \sum_{|k| \leq N} e^{-ikt} = \frac{\sin((N + \frac{1}{2})t)}{\sin(t/2)},$$

where D_N is the **Dirichlet kernel**. One checks that

$$\|\Lambda_N\| = \frac{1}{2\pi} \int_{-\pi}^{\pi} |D_N(t)| dt =: L_N,$$

the **Lebesgue constant**: the bound $\leq$ is clear, and it is attained in the limit by

taking f to approximate $\operatorname{sgn} D_N$ continuously. Now

$$L_N \geq \frac{1}{\pi} \int_0^\pi \frac{|\sin((N + \frac{1}{2})t)|}{t} dt \geq \frac{1}{\pi} \sum_{k=1}^N \frac{1}{k} \int_0^\pi |\sin u| \frac{du}{\pi} \cdot c \longrightarrow \infty$$

for a suitable constant $c > 0$; the essential point is the divergence of the harmonic series, giving $L_N \sim \frac{4}{\pi^2} \log N$.

So $\|\Lambda_N\| \rightarrow \infty$. Since $C(\mathbb{T})$ is a Banach space, uniform boundedness says the family $\{\Lambda_N\}$ cannot be pointwise bounded: there is $f \in C(\mathbb{T})$ with $\sup_N |S_N f(0)| = \infty$, so the Fourier series of f diverges at 0.

Remark. This example is worth contrasting with what happens in L^2 , where we will prove that the Fourier series of *every* function converges (in the L^2 norm). Pointwise convergence is a much more delicate matter; Carleson proved in 1966 that the Fourier series of an L^2 function converges almost everywhere, which is a very deep theorem.

§33.4.4 The Open Mapping Theorem

Recall that a map $T : X \rightarrow Y$ is **open** if $T(U)$ is open in Y whenever U is open in X . This is a strong requirement, but for surjective bounded operators between Banach spaces it comes for free.

Theorem 33.4.13 (Open Mapping Theorem)

Let X, Y be Banach spaces and $T \in \mathcal{B}(X, Y)$ surjective. Then T is an open map. Equivalently, there is $c > 0$ with

$$T(B_X) \supseteq cB_Y,$$

where B_X, B_Y are the open unit balls.

Proof. First, the equivalence. If $T(B_X) \supseteq cB_Y$ then for $U \subseteq X$ open and $x \in U$, choose r with $x + rB_X \subseteq U$; then

$$T(U) \supseteq Tx + rT(B_X) \supseteq Tx + rcB_Y,$$

so $T(U)$ is open. Conversely if T is open then $T(B_X)$ is an open set containing 0.

Step 1: $\overline{T(B_X)}$ contains a ball about 0. Since T is surjective and $X = \bigcup_n nB_X$,

$$Y = \bigcup_{n=1}^{\infty} T(nB_X) = \bigcup_{n=1}^{\infty} nT(B_X) \subseteq \bigcup_{n=1}^{\infty} n\overline{T(B_X)}.$$

As Y is complete, Baire gives an n for which $n\overline{T(B_X)}$ has non-empty interior, hence so does $K := \overline{T(B_X)}$: say $B(y_0, r) \subseteq K$. Now K is convex and symmetric ($-K = K$), being the closure of the convex symmetric set $T(B_X)$. So $B(-y_0, r) = -B(y_0, r) \subseteq K$, and for $\|y\| < r$,

$$y = \frac{1}{2}((y_0 + y) + (-y_0 + y)) \in \frac{1}{2}(K + K) = K$$

by convexity. Hence $rB_Y \subseteq \overline{T(B_X)}$.

Step 2: removing the closure. We claim $\frac{r}{2}B_Y \subseteq T(B_X)$. By scaling, Step 1 gives

$$2^{-k}rB_Y \subseteq \overline{T(2^{-k}B_X)} \quad \text{for all } k \geq 0.$$

Let $y \in Y$ with $\|y\| < r/2$. Since $y \in \overline{T(\frac{1}{2}B_X)}$, there is x_1 with $\|x_1\| < \frac{1}{2}$ and $\|y - Tx_1\| < r/4$. Since $y - Tx_1 \in \overline{T(\frac{1}{4}B_X)}$, there is x_2 with $\|x_2\| < \frac{1}{4}$ and $\|y - Tx_1 - Tx_2\| < r/8$. Continuing, we obtain x_k with $\|x_k\| < 2^{-k}$ and

$$\left\| y - T \left(\sum_{k=1}^n x_k \right) \right\| < 2^{-(n+1)}r.$$

Now $\sum_k \|x_k\| < 1$, so by completeness of X the series $\sum_k x_k$ converges to some x with $\|x\| < 1$, and by continuity of T we get $Tx = y$. Hence $y \in T(B_X)$. $\square$

Both completeness hypotheses are used, and both are needed: completeness of Y for Baire in Step 1, completeness of X for the series in Step 2.

Theorem 33.4.14 (Inverse Mapping Theorem)

Let X, Y be Banach spaces and $T \in \mathcal{B}(X, Y)$ a bijection. Then $T^{-1} \in \mathcal{B}(Y, X)$; that is, T is an isomorphism.

Proof. T is open by the open mapping theorem, and an open bijection has continuous inverse: $(T^{-1})^{-1}(U) = T(U)$ is open for every open U . $\square$

This is a genuinely useful tool. It says that for operators between Banach spaces, algebraic invertibility automatically gives topological invertibility – so to check that an equation $Tx = y$ has a solution depending continuously on the data, we need only check existence and uniqueness.

Corollary 33.4.15

Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on a vector space X , each making X complete. If there is C with $\|x\|_2 \leq C\|x\|_1$ for all x , then the norms are equivalent.

Proof. The identity map $(X, \|\cdot\|_1) \rightarrow (X, \|\cdot\|_2)$ is a bounded bijection between Banach spaces, hence has bounded inverse. $\square$

Corollary 33.4.16

Let X be a Banach space and $X = Y \oplus Z$ an algebraic direct sum with Y, Z closed subspaces. Then the projection $P : X \rightarrow Y$ along Z is bounded.

Proof. $Y \oplus Z$ with the norm $\|(y, z)\| = \|y\| + \|z\|$ is a Banach space (as Y, Z are closed in the complete X), and the addition map $(y, z) \mapsto y + z$ is a bounded bijection onto X . Its inverse $x \mapsto (Px, x - Px)$ is bounded by the inverse mapping theorem, so P is bounded. $\square$

§33.4.5 The Closed Graph Theorem

To prove an operator is bounded we must show $x_n \rightarrow x$ implies $Tx_n \rightarrow Tx$. The closed graph theorem says that – between Banach spaces – we may *assume* that Tx_n converges, and only check that its limit is right. In practice this halves the work.

Definition 33.4.17 (Graph)

Let $T : X \rightarrow Y$ be linear. Its **graph** is

$$\Gamma(T) = \{(x, Tx) : x \in X\} \subseteq X \oplus Y.$$

We say T is **closed** if $\Gamma(T)$ is a closed subspace of $X \oplus Y$ (with the norm $\|(x, y)\| = \|x\| + \|y\|$).

Unwinding the definition, T is closed precisely when: whenever $x_n \rightarrow x$ in X and $Tx_n \rightarrow y$ in Y , we have $y = Tx$.

Theorem 33.4.18 (Closed Graph Theorem)

Let X, Y be Banach spaces and $T : X \rightarrow Y$ linear. Then T is bounded if and only if it is closed.

Proof. ($\Rightarrow$) If T is bounded and $(x_n, Tx_n) \rightarrow (x, y)$ then $Tx_n \rightarrow Tx$ by continuity, so $y = Tx$.

($\Leftarrow$) Suppose $\Gamma = \Gamma(T)$ is closed. Then Γ is a closed subspace of the Banach space $X \oplus Y$, hence itself a Banach space. Consider the two coordinate projections restricted to Γ :

$$\pi_X : \Gamma \rightarrow X, (x, Tx) \mapsto x, \quad \pi_Y : \Gamma \rightarrow Y, (x, Tx) \mapsto Tx.$$

Both are bounded, with norm at most 1. Moreover π_X is a bijection. By the inverse mapping theorem, $\pi_X^{-1} : X \rightarrow \Gamma$ is bounded. But then

$$T = \pi_Y \circ \pi_X^{-1}$$

is bounded, being a composite of bounded maps. □

Example 33.4.19

Let X be a Banach space and $T : X \rightarrow X$ linear with the property that $f \circ T \in X^*$ for every $f \in X^*$. Then T is bounded. Indeed, suppose $x_n \rightarrow x$ and $Tx_n \rightarrow y$. For any $f \in X^*$,

$$f(y) = \lim_n f(Tx_n) = \lim_n (f \circ T)(x_n) = (f \circ T)(x) = f(Tx),$$

using continuity of $f \circ T$. Since X^* separates points, $y = Tx$, so T is closed, hence bounded.

Example 33.4.20 (Where it fails)

Differentiation $D : C^1[0, 1] \rightarrow C[0, 1]$, with $C^1[0, 1]$ given the sup norm, *is* closed: if $f_n \rightarrow f$ uniformly and $f'_n \rightarrow g$ uniformly then f is differentiable with $f' = g$, by the theorem on differentiating uniform limits from IB Analysis and Topology. But we saw that D is unbounded. There is no contradiction: $(C^1[0, 1], \|\cdot\|_\infty)$ is not complete. This shows that completeness of the *domain* really is needed.

§33.5 The Topology of $C(K)$

Of all our examples, the space $C(K)$ of continuous functions on a compact space is the most important, and it deserves a section of its own. We ask three questions. Are there enough continuous functions on K to be interesting? Which subsets of $C(K)$ are compact? And which subalgebras are dense?

Throughout, K is a compact Hausdorff space, and $C(K)$ carries the sup norm.

§33.5.1 Normality and Urysohn's Lemma

Definition 33.5.1 (Normal space)

A topological space X is **normal** if for any two disjoint closed sets $C_1, C_2 \subseteq X$ there are disjoint open sets $U_1 \supseteq C_1$ and $U_2 \supseteq C_2$.

Proposition 33.5.2

Every compact Hausdorff space is normal. So is every metric space.

Proof. Let X be compact Hausdorff and C_1, C_2 disjoint closed sets, hence compact.

First suppose $C_2 = \{y\}$ is a point. For each $x \in C_1$ use the Hausdorff property to find disjoint open $U_x \ni x$ and $V_x \ni y$. The U_x cover C_1 , so finitely many suffice, say $U_{x_1}, \dots, U_{x_n}$; then $U = \bigcup_i U_{x_i}$ and $V = \bigcap_i V_{x_i}$ are disjoint open sets separating C_1 from y .

In general, for each $y \in C_2$ we obtain from the above disjoint open $U_y \supseteq C_1$ and $V_y \ni y$. The V_y cover C_2 , so finitely many suffice, and $\bigcap_i U_{y_i} \supseteq C_1$ and $\bigcup_i V_{y_i} \supseteq C_2$ are disjoint and open.

For a metric space, take $U_i = \{x : \text{dist}(x, C_i) < \text{dist}(x, C_{3-i})\}$; these are open by continuity of the distance functions, disjoint, and contain C_1, C_2 respectively. $\square$

Normality is precisely the hypothesis under which $C(X)$ is rich.

Lemma 33.5.3 (Urysohn's Lemma)

Let X be normal and $C_0, C_1 \subseteq X$ disjoint non-empty closed sets. Then there is a continuous $f : X \rightarrow [0, 1]$ with $f|_{C_0} = 0$ and $f|_{C_1} = 1$.

Proof. First note a reformulation of normality: if $C \subseteq U$ with C closed and U open, then there is an open V with

$$C \subseteq V \subseteq \bar{V} \subseteq U.$$

(Apply normality to C and $X \setminus U$ to get disjoint open $V \supseteq C$ and $W \supseteq X \setminus U$; then $\bar{V} \subseteq X \setminus W \subseteq U$.)

Let $D = \{k/2^n : n \geq 0, 0 \leq k \leq 2^n\}$ be the dyadic rationals in $[0, 1]$. We construct open sets U_r for $r \in D$ with

$$C_0 \subseteq U_0, \quad U_1 = X \setminus C_1, \quad \text{and} \quad \bar{U}_r \subseteq U_s \quad \text{whenever} \quad r < s.$$

Start by setting $U_1 = X \setminus C_1$ and using the reformulation to pick U_0 with $C_0 \subseteq U_0 \subseteq \bar{U}_0 \subseteq U_1$. Inductively, having defined U_r for all $r \in D$ with denominator 2^n , and given a new dyadic $r = (2k+1)/2^{n+1}$ lying between the old neighbours $a = k/2^n$ and $b = (k+1)/2^n$, we have $\bar{U}_a \subseteq U_b$ and use the reformulation to choose U_r with

$$\bar{U}_a \subseteq U_r \subseteq \bar{U}_r \subseteq U_b.$$

This maintains the nesting property.

Now define

$$f(x) = \inf\{r \in D : x \in U_r\},$$

with $f(x) = 1$ if x lies in no U_r . Then $f = 0$ on C_0 and $f = 1$ on C_1 , and $0 \leq f \leq 1$.

It remains to check continuity. The sets $[0, a)$ and $(a, 1]$ generate the topology of $[0, 1]$, so it is enough to see that $f^{-1}([0, a))$ and $f^{-1}((a, 1])$ are open. For the first,

$$f(x) < a \iff \exists r \in D, r < a, x \in U_r,$$

so $f^{-1}([0, a)) = \bigcup_{r < a} U_r$ is open. For the second, we claim

$$f(x) > a \iff \exists r \in D, r > a, x \notin \bar{U}_r,$$

giving $f^{-1}((a, 1]) = \bigcup_{r > a} (X \setminus \bar{U}_r)$, which is open. Indeed if $x \notin \bar{U}_r$ with $r > a$ then $x \notin U_s$ for all $s \leq r$ (as $U_s \subseteq \bar{U}_s \subseteq U_r$ for $s < r$), so $f(x) \geq r > a$. Conversely if $f(x) > a$, pick dyadics $a < r < s < f(x)$; then $x \notin U_s \supseteq \bar{U}_r$. $\square$

Corollary 33.5.4

If K is compact Hausdorff then $C(K)$ separates the points of K .

Proof. Points are closed in a Hausdorff space, so apply Urysohn to $\{x\}$ and $\{y\}$ for $x \neq y$. $\square$

Theorem 33.5.5 (Tietze Extension Theorem)

Let X be normal, $C \subseteq X$ closed and $f : C \rightarrow \mathbb{R}$ continuous and bounded. Then there is a continuous $\tilde{f} : X \rightarrow \mathbb{R}$ with $\tilde{f}|_C = f$ and $\|\tilde{f}\|_\infty = \|f\|_\infty$.

Proof. By rescaling assume $\|f\|_\infty \leq 1$. We claim: given continuous $g : C \rightarrow \mathbb{R}$ with $\|g\|_\infty \leq M$, there is a continuous $h : X \rightarrow \mathbb{R}$ with $\|h\|_\infty \leq M/3$ and $\|g - h|_C\|_\infty \leq 2M/3$.

To see this, let $A = g^{-1}([-M, -M/3])$ and $B = g^{-1}([M/3, M])$, which are disjoint closed subsets of C , hence closed in X . If both are non-empty, Urysohn gives a continuous $u : X \rightarrow [0, 1]$ with $u|_A = 0$, $u|_B = 1$, and we set $h = \frac{M}{3}(2u - 1)$, which takes values in $[-M/3, M/3]$. On A , $h = -M/3$ and $g \in [-M, -M/3]$, so $|g - h| \leq 2M/3$; similarly on B ; and on the rest of C both g and h lie in $[-M/3, M/3]$. (If A or B is empty, take h constant appropriately.)

Now iterate. Put $g_0 = f$ and $M_0 = 1$. Given g_n with $\|g_n\|_\infty \leq (2/3)^n$, obtain h_n on X with $\|h_n\|_\infty \leq \frac{1}{3}(2/3)^n$ and set $g_{n+1} = g_n - h_n|_C$, so $\|g_{n+1}\|_\infty \leq (2/3)^{n+1}$. Since

$$\sum_{n=0}^{\infty} \|h_n\|_\infty \leq \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = 1 < \infty,$$

the series $\tilde{f} = \sum_n h_n$ converges uniformly on X (as $C(X)$ with the sup norm, or rather $B(X)$, is complete), so $\tilde{f}$ is continuous with $\|\tilde{f}\|_\infty \leq 1$. On C we have $\sum_{n \leq N} h_n = f - g_{N+1} \rightarrow f$, so $\tilde{f}|_C = f$. Finally $\|\tilde{f}\|_\infty \geq \|f\|_\infty$ trivially, giving equality. $\square$

§33.5.2 The Arzelà–Ascoli Theorem

We know that closed bounded subsets of $C(K)$ are never compact (the space is infinite-dimensional). So what is the right extra condition? The answer is a uniformity in the modulus of continuity.

Definition 33.5.6 (Totally bounded)

A subset S of a metric space is **totally bounded** if for every $\varepsilon > 0$ there are finitely many points $x_1, \dots, x_n$ with $S \subseteq \bigcup_i B(x_i, \varepsilon)$. Such a set $\{x_i\}$ is called an **ε -net**.

Proposition 33.5.7

Let (X, d) be a complete metric space and $S \subseteq X$. Then $\bar{S}$ is compact if and only if S is totally bounded.

Proof. ($\Rightarrow$) If $\bar{S}$ is compact, cover it by the balls $B(x, \varepsilon)$, $x \in \bar{S}$, and extract a finite subcover.

($\Leftarrow$) In a metric space, compactness is equivalent to sequential compactness, so let $(y_n) \subseteq \bar{S}$; we find a Cauchy subsequence, which converges by completeness (and its limit is in $\bar{S}$, which is closed). By total boundedness of S , hence of $\bar{S}$, cover $\bar{S}$ by finitely many balls of radius 1; one of them, B_1 say, contains y_n for infinitely many n . Cover $\bar{S}$ by finitely many balls of radius $\frac{1}{2}$; one of them, B_2 , contains y_n for infinitely many of those n . Continuing and taking a diagonal subsequence gives (y_{n_k}) with $y_{n_k} \in B_1 \cap \dots \cap B_k$ for each k , so $d(y_{n_k}, y_{n_l}) \leq 2/k$ for $l \geq k$. This subsequence is Cauchy. $\square$

Definition 33.5.8 (Equicontinuity)

Let K be a compact Hausdorff space. A family $\mathcal{F} \subseteq C(K)$ is

- (i) **equibounded** if $\sup_{f \in \mathcal{F}} \|f\|_\infty < \infty$;
- (ii) **equicontinuous at $x \in K$** if for every $\varepsilon > 0$ there is an open $U \ni x$ with $|f(y) - f(x)| < \varepsilon$ for all $y \in U$ and all $f \in \mathcal{F}$;
- (iii) **equicontinuous** if it is equicontinuous at every point.

The point is the order of quantifiers: one neighbourhood works for the whole family at once. For $K \subseteq \mathbb{R}$ a typical sufficient condition is a uniform Lipschitz bound, $|f(x) - f(y)| \leq L|x - y|$ for all $f \in \mathcal{F}$.

Theorem 33.5.9 (Arzelà–Ascoli)

Let K be a compact Hausdorff space and $\mathcal{F} \subseteq C(K)$. Then $\mathcal{F}$ is relatively compact (i.e. $\bar{\mathcal{F}}$ is compact) if and only if $\mathcal{F}$ is equibounded and equicontinuous.

Proof. ($\Rightarrow$) Suppose $\bar{\mathcal{F}}$ is compact, so $\mathcal{F}$ is totally bounded. Equiboundedness is immediate. For equicontinuity at x , let $\varepsilon > 0$ and take an $\varepsilon/3$ -net $f_1, \dots, f_n$ for $\mathcal{F}$. Each f_i is continuous, so there is an open $U \ni x$ with $|f_i(y) - f_i(x)| < \varepsilon/3$ for $y \in U$ and all i (intersect the finitely many neighbourhoods). Given $f \in \mathcal{F}$, pick i with $\|f - f_i\|_\infty < \varepsilon/3$; then for $y \in U$,

$$|f(y) - f(x)| \leq |f(y) - f_i(y)| + |f_i(y) - f_i(x)| + |f_i(x) - f(x)| < \varepsilon.$$

($\Leftarrow$) Since $C(K)$ is complete, it suffices to show $\mathcal{F}$ is totally bounded. Let $\varepsilon > 0$. By equicontinuity, each $x \in K$ has an open $U_x \ni x$ with $|f(y) - f(x)| < \varepsilon/4$ for all $y \in U_x$, $f \in \mathcal{F}$. By compactness, $K = U_{x_1} \cup \dots \cup U_{x_m}$.

Consider the map $\Phi : \mathcal{F} \rightarrow \mathbb{R}^m$, $\Phi(f) = (f(x_1), \dots, f(x_m))$. By equiboundedness the image lies in a bounded subset of $\mathbb{R}^m$, which is totally bounded, so there are $f_1, \dots, f_n \in \mathcal{F}$ such that every $f \in \mathcal{F}$ has some j with $|f(x_i) - f_j(x_i)| < \varepsilon/4$ for all i .

We claim $\{f_j\}$ is an ε -net. Given f , choose j as above, and let $y \in K$, say $y \in U_{x_i}$. Then

$$|f(y) - f_j(y)| \leq |f(y) - f(x_i)| + |f(x_i) - f_j(x_i)| + |f_j(x_i) - f_j(y)| < \frac{\varepsilon}{4} + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} < \varepsilon.$$

So $\|f - f_j\|_\infty \leq \varepsilon$, as needed. $\square$

Example 33.5.10

Let $k \in C([0, 1]^2)$ and let $T : C[0, 1] \rightarrow C[0, 1]$ be the integral operator with kernel k . Then $T(\bar{B})$ is relatively compact. Indeed for $\|f\|_\infty \leq 1$ we have $\|Tf\|_\infty \leq \|k\|_\infty$, giving equiboundedness, and

$$|Tf(x) - Tf(x')| \leq \int_0^1 |k(x, y) - k(x', y)| dy \leq \sup_y |k(x, y) - k(x', y)|,$$

which is small when $|x - x'|$ is small, uniformly in f , since k is uniformly continuous

on the compact set $[0, 1]^2$. So T is a *compact operator* in the sense of the last section of these notes.

Theorem 33.5.11 (Cauchy–Peano)

Let $\varphi : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Then the initial value problem $y' = \varphi(t, y)$, $y(t_0) = y_0$ has a solution on some interval around t_0 .

Proof sketch. On a suitable rectangle, $|\varphi| \leq M$. Build approximate solutions y_n by Euler’s method with step $1/n$, or by delaying the argument. Each y_n is M -Lipschitz and uniformly bounded, so the family is equibounded and equicontinuous; by Arzelà–Ascoli a subsequence converges uniformly to some y , and passing to the limit in the integral equation $y_n(t) = y_0 + \int_{t_0}^t \varphi(s, y_n(s)) \, ds + o(1)$ shows y solves the problem. $\square$

Note the contrast with the Picard–Lindelöf theorem from IB Analysis and Topology, which assumed φ Lipschitz in y and gave existence *and uniqueness* via the contraction mapping theorem. Continuity alone buys existence only: $y' = \sqrt{|y|}$, $y(0) = 0$ has both $y \equiv 0$ and $y = t^2/4$ as solutions.

§33.5.3 The Stone–Weierstrass Theorem

Finally, density. Weierstrass proved that polynomials are dense in $C[0, 1]$; Stone found the right generality.

Definition 33.5.12 (Subalgebra)

A **subalgebra** $\mathcal{A} \subseteq C(K)$ is a linear subspace closed under pointwise multiplication. It **separates points** if for all $x \neq y$ in K there is $f \in \mathcal{A}$ with $f(x) \neq f(y)$.

Theorem 33.5.13 (Stone–Weierstrass, real case)

Let K be compact Hausdorff and $\mathcal{A} \subseteq C_{\mathbb{R}}(K)$ a subalgebra which separates points and contains the constant functions. Then $\mathcal{A}$ is dense in $C_{\mathbb{R}}(K)$.

Proof. Replacing $\mathcal{A}$ by $\bar{\mathcal{A}}$ (still a subalgebra, since multiplication is continuous on bounded sets), we may assume $\mathcal{A}$ is closed, and must show $\mathcal{A} = C_{\mathbb{R}}(K)$.

Step 1: $\mathcal{A}$ is a lattice. We claim $f \in \mathcal{A} \Rightarrow |f| \in \mathcal{A}$. By scaling assume $\|f\|_{\infty} \leq 1$. The function $t \mapsto |t|$ is a uniform limit on $[-1, 1]$ of polynomials p_n with $p_n(0) = 0$: for instance take the binomial series for $\sqrt{s}$ about $s = 1$, evaluated at $s = t^2$, which converges uniformly on $[-1, 1]$; alternatively use $\sqrt{t^2 + \varepsilon^2}$ and expand. Then $p_n(f) \in \mathcal{A}$ (as $\mathcal{A}$ is an algebra containing constants) and $p_n(f) \rightarrow |f|$ uniformly, so $|f| \in \mathcal{A}$ as $\mathcal{A}$ is closed. Consequently

$$\max(f, g) = \frac{1}{2}(f + g + |f - g|), \quad \min(f, g) = \frac{1}{2}(f + g - |f - g|)$$

lie in $\mathcal{A}$ whenever f, g do.

Step 2: two-point interpolation. Given $x \neq y$ in K and $\alpha, \beta \in \mathbb{R}$, there is $h \in \mathcal{A}$

with $h(x) = \alpha$, $h(y) = \beta$: take $g \in \mathcal{A}$ with $g(x) \neq g(y)$ and set

$$h = \alpha + (\beta - \alpha) \frac{g - g(x)}{g(y) - g(x)},$$

which is in $\mathcal{A}$ since $\mathcal{A}$ contains the constants.

Step 3: approximation. Let $f \in C_{\mathbb{R}}(K)$ and $\varepsilon > 0$. Fix $x \in K$. For each $y \in K$ choose $h_{x,y} \in \mathcal{A}$ with $h_{x,y}(x) = f(x)$ and $h_{x,y}(y) = f(y)$. The set $V_y = \{z : h_{x,y}(z) < f(z) + \varepsilon\}$ is open and contains y , so finitely many $V_{y_1}, \dots, V_{y_n}$ cover K . Put

$$h_x = \min(h_{x,y_1}, \dots, h_{x,y_n}) \in \mathcal{A}.$$

Then $h_x < f + \varepsilon$ everywhere and $h_x(x) = f(x)$.

Now the set $W_x = \{z : h_x(z) > f(z) - \varepsilon\}$ is open and contains x , so finitely many $W_{x_1}, \dots, W_{x_m}$ cover K . Put $h = \max(h_{x_1}, \dots, h_{x_m}) \in \mathcal{A}$. Then $f - \varepsilon < h < f + \varepsilon$ everywhere, i.e. $\|f - h\|_{\infty} \leq \varepsilon$. As $\mathcal{A}$ is closed, $f \in \mathcal{A}$. $\square$

Theorem 33.5.14 (Stone–Weierstrass, complex case)

Let K be compact Hausdorff and $\mathcal{A} \subseteq C_{\mathbb{C}}(K)$ a subalgebra which separates points, contains the constants, and is closed under complex conjugation. Then $\mathcal{A}$ is dense.

Proof. Let $\mathcal{A}_{\mathbb{R}} = \{f \in \mathcal{A} : f \text{ real-valued}\}$. Since $\mathcal{A}$ is conjugation-closed, $\operatorname{Re} f = \frac{1}{2}(f + \bar{f})$ and $\operatorname{Im} f = \frac{1}{2i}(f - \bar{f})$ lie in $\mathcal{A}_{\mathbb{R}}$ for every $f \in \mathcal{A}$. Then $\mathcal{A}_{\mathbb{R}}$ is a real subalgebra containing the real constants, and it separates points: if $f(x) \neq f(y)$ then $\operatorname{Re} f$ or $\operatorname{Im} f$ separates them. By the real case, $\mathcal{A}_{\mathbb{R}}$ is dense in $C_{\mathbb{R}}(K)$, and hence $\mathcal{A} \supseteq \mathcal{A}_{\mathbb{R}} + i\mathcal{A}_{\mathbb{R}}$ is dense in $C_{\mathbb{C}}(K)$. $\square$

Conjugation-closedness cannot be dropped: on the closed unit disc $\bar{D} \subseteq \mathbb{C}$, the polynomials in z form a separating subalgebra containing the constants, but their closure consists of functions holomorphic on the interior – nothing like all of $C(\bar{D})$.

Example 33.5.15 (i) Polynomials are dense in $C[a, b]$: they form a subalgebra containing the constants, and $x \mapsto x$ separates points. This is the Weierstrass approximation theorem.

(ii) Trigonometric polynomials $\operatorname{span}\{e^{ikx} : k \in \mathbb{Z}\}$ are dense in $C(\mathbb{T})$: this is a conjugation-closed subalgebra ($\overline{e^{ikx}} = e^{-ikx}$) containing the constants, and e^{ix} separates points of the circle. We will use this to prove completeness of the Fourier basis in L^2 .

(iii) Functions of the form $\sum_j f_j(x)g_j(y)$ are dense in $C([0, 1]^2)$, which gives Fubini's theorem for continuous functions almost for free.

Remark. If $\mathcal{A}$ separates points but does *not* contain the constants, the conclusion becomes: $\bar{\mathcal{A}}$ is either all of $C(K)$ or the set $\{f : f(x_0) = 0\}$ for some $x_0 \in K$. Applying the theorem to $\mathcal{A} + \mathbb{C}$ and chasing through gives this refinement.

§33.6 Hilbert Spaces

So far we have had a notion of length but not of angle. If we add one – that is, if the norm comes from an inner product – an enormous amount of Euclidean geometry survives into infinite dimensions, and the theory becomes vastly more powerful. These are the Hilbert spaces, and they are the setting in which quantum mechanics, Fourier analysis and the spectral theorem all live.

§33.6.1 Inner Products

Definition 33.6.1 (Inner product)

Let V be a vector space over $\mathbb{F}$. An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ such that for all $u, v, w \in V$ and $\lambda \in \mathbb{F}$:

- (i) (*Linearity in the first slot*) $\langle \lambda u + v, w \rangle = \lambda \langle u, w \rangle + \langle v, w \rangle$;
- (ii) (*Conjugate symmetry*) $\langle v, w \rangle = \overline{\langle w, v \rangle}$;
- (iii) (*Positive definiteness*) $\langle v, v \rangle \geq 0$, with equality if and only if $v = 0$.

The pair $(V, \langle \cdot, \cdot \rangle)$ is an **inner product space**.

Note that (i) and (ii) together force conjugate-linearity in the second slot: $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$. (This is the convention of the physicists' second slot; some books put the conjugate-linearity in the first slot instead. Nothing of substance depends on the choice, but one must be consistent.) Over $\mathbb{R}$ the inner product is just a symmetric positive definite bilinear form, as in IB Linear Algebra.

Example 33.6.2 (i) $\mathbb{F}^n$ with $\langle x, y \rangle = \sum_i x_i \bar{y}_i$, giving $\|\cdot\|_2$.

(ii) ℓ^2 with $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i \bar{y}_i$, which converges absolutely by Cauchy–Schwarz.

(iii) $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 f \bar{g} \, dx$, giving $\|\cdot\|_{L^2}$.

(iv) $C(\mathbb{T})$ with $\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f \bar{g} \, dx$.

Proposition 33.6.3 (Cauchy–Schwarz)

Let V be an inner product space. Then for all $v, w \in V$,

$$|\langle v, w \rangle| \leq \langle v, v \rangle^{1/2} \langle w, w \rangle^{1/2},$$

with equality if and only if v, w are linearly dependent.

Proof. If $w = 0$ both sides vanish. Otherwise, for $\lambda \in \mathbb{F}$,

$$0 \leq \langle v - \lambda w, v - \lambda w \rangle = \langle v, v \rangle - \bar{\lambda} \langle v, w \rangle - \lambda \overline{\langle v, w \rangle} + |\lambda|^2 \langle w, w \rangle.$$

Choosing $\lambda = \langle v, w \rangle / \langle w, w \rangle$ gives

$$0 \leq \langle v, v \rangle - \frac{|\langle v, w \rangle|^2}{\langle w, w \rangle},$$

which rearranges to the result. Equality forces $v = \lambda w$. $\square$

Corollary 33.6.4

$\|v\| = \langle v, v \rangle^{1/2}$ defines a norm on V .

Proof. Positivity and homogeneity are clear. For the triangle inequality,

$$\|v + w\|^2 = \|v\|^2 + 2 \operatorname{Re} \langle v, w \rangle + \|w\|^2 \leq \|v\|^2 + 2 \|v\| \|w\| + \|w\|^2 = (\|v\| + \|w\|)^2. \quad \square$$

Definition 33.6.5 (Hilbert space)

A **Hilbert space** is an inner product space which is complete in the induced norm.

So $\mathbb{F}^n$, ℓ^2 and $L^2[0, 1]$ are Hilbert spaces; $C[0, 1]$ with the L^2 inner product is an inner product space which is not complete.

Which norms come from an inner product? There is a clean answer.

Proposition 33.6.6 (Parallelogram law)

In an inner product space,

$$\|v + w\|^2 + \|v - w\|^2 = 2\|v\|^2 + 2\|w\|^2.$$

Conversely (Jordan–von Neumann), if a norm on V satisfies this identity for all v, w , then it comes from an inner product.

Proof. Expanding both left-hand terms, the cross terms $\pm 2 \operatorname{Re} \langle v, w \rangle$ cancel.

For the converse, the inner product must be given by the **polarisation identity**

$$\langle v, w \rangle = \frac{1}{4} (\|v + w\|^2 - \|v - w\|^2)$$

in the real case, and

$$\langle v, w \rangle = \frac{1}{4} \sum_{k=0}^3 i^k \|v + i^k w\|^2$$

in the complex case. One then verifies the inner product axioms directly, using the parallelogram law repeatedly; additivity in the first slot is a short computation, homogeneity over $\mathbb{Q}$ follows by induction, and over $\mathbb{R}$ by continuity. We omit the (unenlightening) details. $\square$

Example 33.6.7

The parallelogram law gives a quick way to see which of our spaces are not inner product spaces. In ℓ^p take $v = e_1$, $w = e_2$; then $\|v \pm w\|_p = 2^{1/p}$ and $\|v\|_p = \|w\|_p = 1$, so the law demands $2 \cdot 2^{2/p} = 4$, i.e. $p = 2$. Similarly $C[0, 1]$ with the sup norm is not an inner product space: take $f(x) = 1$ and $g(x) = x$, so $\|f\|_\infty = \|g\|_\infty = 1$, $\|f + g\|_\infty = 2$ and $\|f - g\|_\infty = 1$, and $4 + 1 \neq 2 + 2$.

Definition 33.6.8 (Orthogonality)

Vectors v, w in an inner product space are **orthogonal**, written $v \perp w$, if $\langle v, w \rangle = 0$. For $S \subseteq V$ the **orthogonal complement** is

$$S^\perp = \{v \in V : \langle v, s \rangle = 0 \text{ for all } s \in S\}.$$

A set is **orthonormal** if its elements are pairwise orthogonal and each has norm 1.

Proposition 33.6.9

$S^\perp$ is a closed subspace of V , and $S^\perp = (\overline{\text{span } S})^\perp$.

Proof. It is the intersection over $s \in S$ of the kernels of the continuous linear functionals $v \mapsto \langle v, s \rangle$, hence a closed subspace. The second claim holds since $\langle v, \cdot \rangle$ is continuous and conjugate-linear, so vanishing on S forces vanishing on $\overline{\text{span } S}$. $\square$

Proposition 33.6.10 (Pythagoras)

If $v \perp w$ then $\|v + w\|^2 = \|v\|^2 + \|w\|^2$. More generally, if $e_1, \dots, e_n$ are orthonormal and $v \in V$, then

$$\|v\|^2 = \left\| v - \sum_{i=1}^n \langle v, e_i \rangle e_i \right\|^2 + \sum_{i=1}^n |\langle v, e_i \rangle|^2.$$

Proof. The first claim is immediate on expanding. For the second, let $u = \sum_i \langle v, e_i \rangle e_i$. Then for each j ,

$$\langle v - u, e_j \rangle = \langle v, e_j \rangle - \langle v, e_j \rangle = 0,$$

so $v - u \perp u$, and Pythagoras with $\|u\|^2 = \sum_i |\langle v, e_i \rangle|^2$ gives the identity. $\square$

Corollary 33.6.11 (Bessel's Inequality)

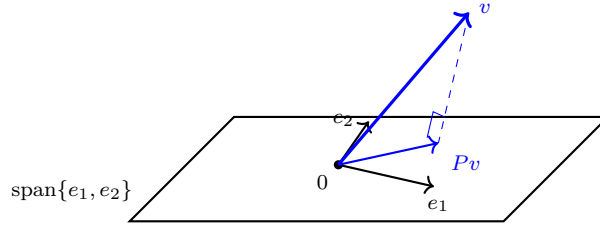
If $(e_i)_{i \in I}$ is an orthonormal family in V and $v \in V$, then

$$\sum_{i \in I} |\langle v, e_i \rangle|^2 \leq \|v\|^2,$$

where the sum is the supremum over finite subfamilies. In particular only countably many terms are non-zero.

Proof. Each finite partial sum is at most $\|v\|^2$ by the previous proposition. For the last claim, the set of i with $|\langle v, e_i \rangle| > 1/n$ has at most $n^2 \|v\|^2$ elements. $\square$

The geometry here is exactly what you would draw in $\mathbb{R}^3$: the vector $\sum_i \langle v, e_i \rangle e_i$ is the orthogonal projection of v onto the span of the e_i , and Bessel says a projection cannot be longer than the original.



Now the fundamental construction. In a general normed space, a closed convex set need not contain a point closest to a given point; in a Hilbert space it always does, and uniquely.

Theorem 33.6.12 (Projection onto a closed convex set)

Let H be a Hilbert space and $C \subseteq H$ a non-empty closed convex set. Then for every $v \in H$ there is a unique $P_C v \in C$ with

$$\|v - P_C v\| = \text{dist}(v, C) = \inf_{c \in C} \|v - c\|.$$

Moreover $P_C v$ is characterised among points of C by

$$\text{Re}\langle c - P_C v, v - P_C v \rangle \leq 0 \quad \text{for all } c \in C,$$

and P_C is 1-Lipschitz.

Proof. Existence. Let $d = \text{dist}(v, C)$ and pick $c_n \in C$ with $\|v - c_n\| \rightarrow d$. Apply the parallelogram law to $v - c_n$ and $v - c_m$:

$$\|c_m - c_n\|^2 = 2\|v - c_n\|^2 + 2\|v - c_m\|^2 - 4\left\|v - \frac{c_n + c_m}{2}\right\|^2.$$

By convexity $\frac{c_n + c_m}{2} \in C$, so the last term is at least $4d^2$, giving

$$\|c_m - c_n\|^2 \leq 2\|v - c_n\|^2 + 2\|v - c_m\|^2 - 4d^2 \rightarrow 0.$$

So (c_n) is Cauchy; by completeness it converges to some $c \in C$ (C is closed), and $\|v - c\| = d$.

Uniqueness. If c, c' both realise the distance, the same computation gives $\|c - c'\|^2 \leq 2d^2 + 2d^2 - 4d^2 = 0$.

Variational characterisation. Write $u = P_C v$. For $c \in C$ and $t \in (0, 1]$, convexity gives $u + t(c - u) \in C$, so

$$\|v - u\|^2 \leq \|v - u - t(c - u)\|^2 = \|v - u\|^2 - 2t \text{Re}\langle c - u, v - u \rangle + t^2 \|c - u\|^2.$$

Hence $2 \text{Re}\langle c - u, v - u \rangle \leq t \|c - u\|^2$ for all $t \in (0, 1]$; letting $t \rightarrow 0$ gives $\text{Re}\langle c - u, v - u \rangle \leq 0$. Conversely, if $u \in C$ satisfies this then for any $c \in C$,

$$\|v - c\|^2 = \|(v - u) - (c - u)\|^2 = \|v - u\|^2 - 2 \text{Re}\langle c - u, v - u \rangle + \|c - u\|^2 \geq \|v - u\|^2,$$

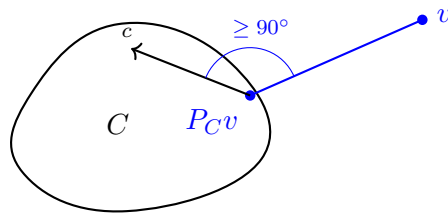
so u minimises.

Lipschitz. Let $u = P_C v$, $u' = P_C v'$. Adding the variational inequalities with $c = u'$ and $c = u$ respectively gives

$$\operatorname{Re}\langle u' - u, v - u \rangle \leq 0, \quad \operatorname{Re}\langle u - u', v' - u' \rangle \leq 0,$$

and adding, $\operatorname{Re}\langle u' - u, (v - v') - (u - u') \rangle \leq 0$, i.e. $\|u - u'\|^2 \leq \operatorname{Re}\langle u - u', v - v' \rangle \leq \|u - u'\| \|v - v'\|$. $\square$

The variational condition says that the vector from the closest point to v makes an obtuse (or right) angle with every direction into C . If you draw it, the reason is obvious: if the angle were acute, we could move a little way into C and get closer.



Applying this to a closed subspace gives the decomposition that makes Hilbert space theory work.

Theorem 33.6.13 (Orthogonal decomposition)

Let H be a Hilbert space and $M \subseteq H$ a closed subspace. Then

$$H = M \oplus M^\perp,$$

and the projection P_M onto M is linear, bounded with $\|P_M\| \leq 1$, satisfies $P_M^2 = P_M$, and has $\ker P_M = M^\perp$, $\operatorname{img} P_M = M$. Moreover $(M^\perp)^\perp = M$.

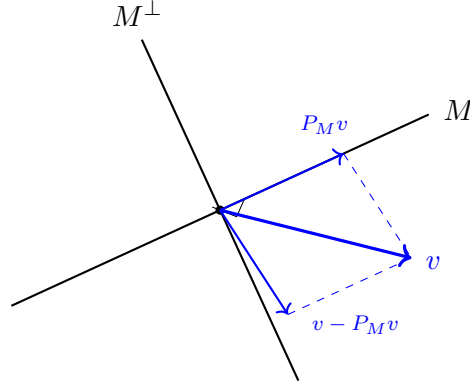
Proof. A subspace is convex and, being closed, the projection theorem applies. Let $v \in H$ and $u = P_M v$. The variational inequality reads $\operatorname{Re}\langle m - u, v - u \rangle \leq 0$ for all $m \in M$; as M is a subspace, $m - u$ ranges over all of M , and replacing $m - u$ by $\lambda(m - u)$ for arbitrary $\lambda \in \mathbb{F}$ we conclude

$$\langle m, v - u \rangle = 0 \quad \text{for all } m \in M,$$

i.e. $v - u \in M^\perp$. Hence $v = u + (v - u) \in M + M^\perp$. The sum is direct since $M \cap M^\perp = \{0\}$ (a vector orthogonal to itself is zero).

Uniqueness of the decomposition gives linearity of P_M , and Pythagoras gives $\|v\|^2 = \|P_M v\|^2 + \|v - P_M v\|^2 \geq \|P_M v\|^2$, so $\|P_M\| \leq 1$ (with equality unless $M = \{0\}$). Clearly $P_M^2 = P_M$, $\operatorname{img} P_M = M$ and $\ker P_M = M^\perp$.

Finally $M \subseteq (M^\perp)^\perp$ is clear. Conversely if $v \in (M^\perp)^\perp$, write $v = u + w$ with $u \in M$, $w \in M^\perp$; then $w = v - u \in (M^\perp)^\perp$ too, and $w \in M^\perp \cap (M^\perp)^\perp = \{0\}$, so $v = u \in M$. $\square$

**Corollary 33.6.14**

Let $M \subseteq H$ be a subspace of a Hilbert space. Then M is dense if and only if $M^\perp = \{0\}$.

Proof. Apply the theorem to $\bar{M}$: we have $\bar{M} = (\bar{M}^\perp)^\perp = (M^\perp)^\perp$, which is H exactly when $M^\perp = \{0\}$. $\square$

§33.6.3 The Riesz Representation Theorem

We can now compute the dual of a Hilbert space completely, and the answer is as beautiful as could be hoped: a Hilbert space is its own dual.

Theorem 33.6.15 (Riesz–Fréchet Representation Theorem)

Let H be a Hilbert space and $f \in H^*$. Then there is a unique $w \in H$ with

$$f(v) = \langle v, w \rangle \quad \text{for all } v \in H,$$

and $\|f\| = \|w\|$. The map $w \mapsto \langle \cdot, w \rangle$ is a conjugate-linear isometric bijection $H \rightarrow H^*$.

Proof. Uniqueness. If $\langle v, w \rangle = \langle v, w' \rangle$ for all v then $w - w' \perp H$, so $w = w'$.

Existence. If $f = 0$ take $w = 0$. Otherwise $M = \ker f$ is a proper closed subspace, so by orthogonal decomposition $M^\perp \neq \{0\}$; pick $u \in M^\perp$ with $\|u\| = 1$, and note $f(u) \neq 0$ (else $u \in M \cap M^\perp$).

For any $v \in H$, the vector

$$v - \frac{f(v)}{f(u)}u$$

lies in $\ker f = M$, hence is orthogonal to u . Taking the inner product with u gives

$$\langle v, u \rangle - \frac{f(v)}{f(u)} = 0, \quad \text{i.e.} \quad f(v) = f(u)\langle v, u \rangle = \langle v, \overline{f(u)}u \rangle.$$

So $w = \overline{f(u)}u$ works.

Isometry. $|f(v)| = |\langle v, w \rangle| \leq \|v\| \|w\|$ by Cauchy–Schwarz, so $\|f\| \leq \|w\|$; and $f(w) = \|w\|^2$ gives $\|f\| \geq \|w\|$. Conjugate-linearity of $w \mapsto \langle \cdot, w \rangle$ is immediate. $\square$

Corollary 33.6.16

Every Hilbert space is reflexive.

Proof. Write $\Phi : H \rightarrow H^*$ for the conjugate-linear bijection above. Then H^* is itself a Hilbert space with $\langle \Phi v, \Phi w \rangle_{H^*} = \langle w, v \rangle_H$ (one checks the axioms; conjugating swaps the slots). Applying Riesz again gives a conjugate-linear bijection $\Psi : H^* \rightarrow H^{**}$, and $\Psi \circ \Phi : H \rightarrow H^{**}$ is a linear isometric bijection. A short computation shows $\Psi \circ \Phi = \iota$, the canonical embedding: for $v \in H$ and $f = \Phi w \in H^*$,

$$(\Psi \Phi v)(f) = \langle f, \Phi v \rangle_{H^*} = \langle v, w \rangle_H = f(v) = \iota(v)(f).$$

So ι is surjective. $\square$

§33.6.4 Orthonormal Bases

We now build coordinates. In a Hilbert space we want to allow infinite linear combinations, which changes the notion of ‘basis’ completely from the Hamel bases of the previous sections.

Proposition 33.6.17 (Gram–Schmidt)

Let $(v_n)_{n \geq 1}$ be a linearly independent sequence in an inner product space. Then there is an orthonormal sequence (e_n) with $\text{span}\{e_1, \dots, e_n\} = \text{span}\{v_1, \dots, v_n\}$ for every n .

Proof. Inductively set $e_1 = v_1 / \|v_1\|$ and

$$u_n = v_n - \sum_{i=1}^{n-1} \langle v_n, e_i \rangle e_i, \quad e_n = \frac{u_n}{\|u_n\|}.$$

Then $u_n \neq 0$ by linear independence, $u_n \perp e_i$ for $i < n$ by the computation in Pythagoras’ theorem, and the spans agree by construction. $\square$

Example 33.6.18 (Legendre polynomials)

Take $H = L^2[-1, 1]$ with $\langle f, g \rangle = \int_{-1}^1 f \bar{g} \, dx$, and apply Gram–Schmidt to the monomials $1, x, x^2, \dots$, which are linearly independent. We get $e_0 = 1/\sqrt{2}$, and then

$$u_1 = x - \langle x, e_0 \rangle e_0 = x, \quad e_1 = \sqrt{\frac{3}{2}} x,$$

since $\int_{-1}^1 x \, dx = 0$ and $\int_{-1}^1 x^2 \, dx = \frac{2}{3}$. Next,

$$u_2 = x^2 - \langle x^2, e_0 \rangle e_0 - \langle x^2, e_1 \rangle e_1 = x^2 - \frac{1}{3},$$

using $\int_{-1}^1 x^2 \, dx = \frac{2}{3}$ and $\int_{-1}^1 x^3 \, dx = 0$; normalising, $e_2 = \sqrt{\frac{45}{8}}(x^2 - \frac{1}{3})$.

Up to normalisation these are the **Legendre polynomials** P_n of IB Methods, with the standard normalisation $P_n(1) = 1$ giving $P_0 = 1$, $P_1 = x$, $P_2 = \frac{1}{2}(3x^2 - 1)$ and $e_n = \sqrt{\frac{2n+1}{2}}P_n$. That they form an orthonormal *basis* of $L^2[-1, 1]$ follows from Stone–Weierstrass: the polynomials are dense in $C[-1, 1]$ for the sup norm, hence in L^2 , and their closed span is the closed span of the e_n .

So the Legendre expansion of a function – which in Methods appeared as a somewhat mysterious consequence of Sturm–Liouville theory – is just the expansion in an orthonormal basis, and Parseval gives

$$\int_{-1}^1 |f|^2 dx = \sum_{n \geq 0} \frac{2n+1}{2} \left| \int_{-1}^1 f P_n dx \right|^2.$$

Definition 33.6.19 (Orthonormal basis)

An orthonormal family $(e_i)_{i \in I}$ in a Hilbert space H is an **orthonormal basis** (or **complete orthonormal system**, or **Hilbert basis**) if $\overline{\text{span}\{e_i\}} = H$.

Proposition 33.6.20

For an orthonormal family $(e_i)_{i \in I}$ in a Hilbert space H , the following are equivalent:

- (i) (e_i) is an orthonormal basis;
- (ii) (e_i) is maximal among orthonormal families;
- (iii) $\{e_i\}^\perp = \{0\}$;
- (iv) $v = \sum_{i \in I} \langle v, e_i \rangle e_i$ for every $v \in H$;
- (v) (*Parseval*) $\|v\|^2 = \sum_{i \in I} |\langle v, e_i \rangle|^2$ for every $v \in H$.

Proof. (i) $\iff$ (iii) is the density criterion proved above. (ii) $\iff$ (iii): a unit vector orthogonal to all e_i would enlarge the family, and conversely.

(i) $\Rightarrow$ (iv). By Bessel, only countably many coefficients are non-zero, so list them as $i_1, i_2, \dots$ and set $s_n = \sum_{k \leq n} \langle v, e_{i_k} \rangle e_{i_k}$. For $m < n$, Pythagoras gives $\|s_n - s_m\|^2 = \sum_{m < k \leq n} |\langle v, e_{i_k} \rangle|^2$, which tends to 0 by Bessel; so (s_n) is Cauchy and converges to some $s \in H$. Now $v - s$ is orthogonal to every e_i (for $i = i_k$ this is a limit of the finite computation, and for other i the coefficient is zero), so $v = s$ by (iii). The value of the sum does not depend on the ordering, since any rearrangement converges to a vector with the same inner products against all e_i .

(iv) $\Rightarrow$ (v) is Pythagoras applied to the partial sums and passing to the limit, and (v) $\Rightarrow$ (iii) is immediate: if $v \perp$ all e_i then $\|v\| = 0$.

(iv) $\Rightarrow$ (i) is clear. □

Proposition 33.6.21

Every Hilbert space has an orthonormal basis, and H is separable if and only if it has a countable orthonormal basis.

Proof. Existence: Zorn's lemma applied to the set of orthonormal families ordered by inclusion; the union of a chain of orthonormal families is orthonormal, and a maximal element is an orthonormal basis by (ii) above.

If H is separable with dense countable set $\{v_n\}$, discard those v_n lying in the span of the earlier ones and apply Gram–Schmidt; the resulting orthonormal sequence has the same closed span, namely H . Conversely, given a countable orthonormal basis, the rational (or rational-complex) linear combinations of the e_n form a countable dense set. $\square$

Theorem 33.6.22 (Classification of separable Hilbert spaces)

Every infinite-dimensional separable Hilbert space is isometrically isomorphic to ℓ^2 .

Proof. Let $(e_n)_{n \geq 1}$ be a countable orthonormal basis of H and define

$$U : H \rightarrow \ell^2, \quad Uv = (\langle v, e_n \rangle)_{n \geq 1}.$$

By Bessel this lands in ℓ^2 , and by Parseval $\|Uv\|_2 = \|v\|$, so U is a linear isometry, in particular injective.

For surjectivity, let $a = (a_n) \in \ell^2$. The partial sums $s_N = \sum_{n \leq N} a_n e_n$ satisfy $\|s_N - s_M\|^2 = \sum_{M < n \leq N} |a_n|^2 \rightarrow 0$, so by completeness of H they converge to some v , and $\langle v, e_n \rangle = a_n$ by continuity of the inner product. So $Uv = a$. $\square$

This is a striking statement: up to isometric isomorphism there is only *one* infinite-dimensional separable Hilbert space. Compare the situation for Banach spaces, where the spaces ℓ^1 , ℓ^2 , c_0 , $C[0, 1]$ and $L^1[0, 1]$ are all separable and pairwise non-isomorphic. Hilbert space is a very rigid object.

Example 33.6.23 (Fourier series)

Take $H = L^2(\mathbb{T})$, the completion of $C(\mathbb{T})$ under $\langle f, g \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} f \bar{g}$. The family $e_k(x) = e^{ikx}$, $k \in \mathbb{Z}$, is orthonormal:

$$\langle e_k, e_l \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-l)x} dx = \delta_{kl}.$$

It is an orthonormal basis: by Stone–Weierstrass the trigonometric polynomials are dense in $C(\mathbb{T})$ for the sup norm, hence for the L^2 norm (as $\|\cdot\|_{L^2} \leq \|\cdot\|_{\infty}$ here), and $C(\mathbb{T})$ is dense in $L^2(\mathbb{T})$ by construction.

So for every $f \in L^2(\mathbb{T})$ the Fourier series converges in L^2 ,

$$\left\| f - \sum_{|k| \leq N} \hat{f}(k) e_k \right\|_{L^2} \rightarrow 0,$$

and Parseval reads

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} |f|^2 dx = \sum_{k \in \mathbb{Z}} |\hat{f}(k)|^2.$$

This is the theorem that was quoted without proof in IB Methods, and here it is at last, as a special case of a general principle: $L^2(\mathbb{T}) \cong \ell^2(\mathbb{Z})$ via the Fourier coefficients.

§33.6.5 Adjoints on Hilbert Space

Riesz representation lets us define an adjoint that lives on H itself rather than on H^* , and this is the version that generalises the conjugate transpose from IB Linear Algebra.

Proposition 33.6.24

Let H be a Hilbert space and $T \in \mathcal{B}(H)$. Then there is a unique $T^* \in \mathcal{B}(H)$ with

$$\langle Tv, w \rangle = \langle v, T^*w \rangle \quad \text{for all } v, w \in H,$$

and $\|T^*\| = \|T\|$.

Proof. Fix w . The map $v \mapsto \langle Tv, w \rangle$ is a bounded linear functional of norm at most $\|T\| \|w\|$, so by Riesz there is a unique vector, call it T^*w , with $\langle Tv, w \rangle = \langle v, T^*w \rangle$ for all v . Uniqueness of the Riesz vector makes T^* well defined, and gives linearity: $T^*(\lambda w_1 + w_2)$ and $\lambda T^*w_1 + T^*w_2$ represent the same functional. Also $\|T^*w\| \leq \|T\| \|w\|$, so $\|T^*\| \leq \|T\|$; applying the same to T^* and noting $T^{**} = T$ gives equality. $\square$

Proposition 33.6.25

For $S, T \in \mathcal{B}(H)$ and $\lambda \in \mathbb{F}$:

- (i) $(S + T)^* = S^* + T^*$, $(\lambda T)^* = \bar{\lambda}T^*$, $(ST)^* = T^*S^*$, $T^{**} = T$;
- (ii) $\|T^*T\| = \|T\|^2$;
- (iii) $\ker T = (\text{img } T^*)^\perp$ and $\overline{\text{img } T} = (\ker T^*)^\perp$.

Proof. (i) All are routine verifications from the defining identity.

(ii) $\|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2$, and conversely

$$\|Tv\|^2 = \langle Tv, Tv \rangle = \langle T^*Tv, v \rangle \leq \|T^*T\| \|v\|^2,$$

so $\|T\|^2 \leq \|T^*T\|$. (This identity is the C^* -identity, and it is the reason $\mathcal{B}(H)$ is such a rigid algebra.)

(iii) $v \in \ker T$ iff $Tv = 0$ iff $\langle Tv, w \rangle = 0$ for all w iff $\langle v, T^*w \rangle = 0$ for all w iff $v \perp \text{img } T^*$. Taking orthogonal complements and using $(M^\perp)^\perp = \bar{M}$ for a subspace gives the second identity applied to T^* . $\square$

Definition 33.6.26

Let $T \in \mathcal{B}(H)$. We say T is

- (i) **self-adjoint** (or Hermitian) if $T = T^*$;

- (ii) **unitary** if $T^*T = TT^* = I$;
- (iii) **normal** if $T^*T = TT^*$.

Self-adjoint and unitary operators are normal. Unitaries are exactly the surjective isometries of H : $\|Uv\|^2 = \langle U^*Uv, v \rangle = \|v\|^2$.

Example 33.6.27 (i) On $\mathbb{F}^n$, the adjoint of a matrix A is the conjugate transpose $\bar{A}^T$, and self-adjoint means Hermitian.

- (ii) On ℓ^2 , the left and right shifts satisfy $L = R^*$: indeed $\langle Rx, y \rangle = \sum_{n \geq 2} x_{n-1} \bar{y}_n = \sum_{n \geq 1} x_n \bar{y}_{n+1} = \langle x, Ly \rangle$. Neither is normal.
- (iii) The diagonal operator M_λ on ℓ^2 has $M_\lambda^* = M_{\bar{\lambda}}$, so it is normal, and self-adjoint exactly when all λ_n are real.
- (iv) The integral operator on $L^2[0, 1]$ with kernel k has adjoint the integral operator with kernel $k^*(x, y) = \overline{k(y, x)}$. It is self-adjoint when k is Hermitian, $k(y, x) = \overline{k(x, y)}$; this is the Hilbert space form of the symmetric Sturm–Liouville kernels from IB Methods.

Proposition 33.6.28

Let $T \in \mathcal{B}(H)$ with H complex.

- (i) T is self-adjoint if and only if $\langle Tv, v \rangle \in \mathbb{R}$ for all v .
- (ii) If T is self-adjoint then $\|T\| = \sup_{\|v\| \leq 1} |\langle Tv, v \rangle|$.
- (iii) If T is normal then $\ker T = \ker T^*$, and eigenvectors for distinct eigenvalues are orthogonal.

Proof. (i) If $T = T^*$ then $\langle Tv, v \rangle = \langle v, Tv \rangle = \overline{\langle Tv, v \rangle}$. Conversely, suppose $\langle Tv, v \rangle \in \mathbb{R}$ for all v . Then $\langle (T - T^*)v, v \rangle = 0$ for all v , and over $\mathbb{C}$ this forces $T = T^*$: expanding $\langle S(v + w), v + w \rangle = 0$ and $\langle S(v + iw), v + iw \rangle = 0$ for $S = T - T^*$ gives $\langle Sv, w \rangle = 0$ for all v, w .

(ii) Write M for the supremum. Clearly $M \leq \|T\|$. For the converse, note that for self-adjoint T ,

$$\langle T(v + w), v + w \rangle - \langle T(v - w), v - w \rangle = 4 \operatorname{Re} \langle Tv, w \rangle.$$

The left-hand side is at most $M(\|v + w\|^2 + \|v - w\|^2) = 2M(\|v\|^2 + \|w\|^2)$ in absolute value. Taking $\|v\| = \|w\| = 1$ gives $\operatorname{Re} \langle Tv, w \rangle \leq M$, and choosing a phase, $|\langle Tv, w \rangle| \leq M$; taking $w = Tv / \|Tv\|$ gives $\|Tv\| \leq M$.

(iii) If T is normal then $\|Tv\|^2 = \langle T^*Tv, v \rangle = \langle TT^*v, v \rangle = \|T^*v\|^2$, so the kernels agree. If $Tv = \lambda v$ and $Tw = \mu w$ with $\lambda \neq \mu$, note $T - \lambda I$ is normal so $T^*v = \bar{\lambda}v$, and then

$$\lambda \langle v, w \rangle = \langle Tv, w \rangle = \langle v, T^*w \rangle = \langle v, \bar{\mu}w \rangle = \bar{\mu} \langle v, w \rangle,$$

forcing $\langle v, w \rangle = 0$. □

§33.7 Spectral Theory

In IB Linear Algebra, the key to understanding a linear map on a finite-dimensional space was its eigenvalues. In infinite dimensions eigenvalues are not enough – an operator can fail to be invertible without having any eigenvalues at all – and the right generalisation is the *spectrum*.

Throughout this section X is a complex Banach space (complex is essential: over $\mathbb{R}$ even a rotation of the plane has empty spectrum) and $T \in \mathcal{B}(X)$. We abbreviate $T - \lambda I$ to $T - \lambda$.

§33.7.1 The Spectrum and the Resolvent

Definition 33.7.1 (Spectrum and resolvent)

Let $T \in \mathcal{B}(X)$. The **resolvent set** of T is

$$\rho(T) = \{\lambda \in \mathbb{C} : T - \lambda \text{ is a bijection } X \rightarrow X\},$$

and the **spectrum** is $\sigma(T) = \mathbb{C} \setminus \rho(T)$. For $\lambda \in \rho(T)$ the **resolvent** is

$$R_\lambda = R_T(\lambda) = (T - \lambda)^{-1} \in \mathcal{B}(X).$$

Two remarks on the definition. First, we only demanded that $T - \lambda$ be a bijection; by the inverse mapping theorem the inverse is then automatically bounded, so we get $R_\lambda \in \mathcal{B}(X)$ for free. This is a nice payoff from the Baire category theorem. Second, λ is an eigenvalue exactly when $T - \lambda$ fails to be *injective*; but it can also fail to be *surjective*, and that also puts λ in the spectrum. So eigenvalues form only part of the spectrum.

Definition 33.7.2 (Parts of the spectrum)

Let $\lambda \in \sigma(T)$. We say λ lies in

- (i) the **point spectrum** $\sigma_p(T)$ if $T - \lambda$ is not injective; the elements of $\ker(T - \lambda) \setminus \{0\}$ are **eigenvectors** and λ an **eigenvalue**;
- (ii) the **continuous spectrum** $\sigma_c(T)$ if $T - \lambda$ is injective with dense but proper range;
- (iii) the **residual spectrum** $\sigma_r(T)$ if $T - \lambda$ is injective with non-dense range.

We also define the **approximate point spectrum**

$$\sigma_{\text{ap}}(T) = \{\lambda : \exists v_n \text{ with } \|v_n\| = 1 \text{ and } \|(T - \lambda)v_n\| \rightarrow 0\}.$$

These three parts are disjoint and cover $\sigma(T)$. Note $\sigma_p \subseteq \sigma_{\text{ap}}$, and $\lambda \in \sigma_{\text{ap}}$ exactly when $T - \lambda$ is not bounded below.

Theorem 33.7.3

Let $T \in \mathcal{B}(X)$ with X a complex Banach space. Then:

- (i) $\rho(T)$ is open and $\lambda \mapsto R_\lambda$ is analytic on it;
- (ii) $\sigma(T) \subseteq \{\lambda : |\lambda| \leq \|T\|\}$, so $\sigma(T)$ is compact;

(iii) $\sigma(T) \neq \emptyset$.

Proof. (ii) If $|\lambda| > \|T\|$ then $\|T/\lambda\| < 1$, so by the Neumann series $I - T/\lambda$ is invertible, hence so is $T - \lambda = -\lambda(I - T/\lambda)$. Moreover

$$R_\lambda = -\frac{1}{\lambda} \sum_{n=0}^{\infty} \left(\frac{T}{\lambda}\right)^n, \quad \|R_\lambda\| \leq \frac{1}{|\lambda| - \|T\|}. \quad (*)$$

So $\sigma(T)$ is contained in the closed disc of radius $\|T\|$, and it is closed by (i).

(i) Let $\lambda_0 \in \rho(T)$. For λ near λ_0 write

$$T - \lambda = (T - \lambda_0) - (\lambda - \lambda_0) = (T - \lambda_0)(I - (\lambda - \lambda_0)R_{\lambda_0}).$$

If $|\lambda - \lambda_0| < \|R_{\lambda_0}\|^{-1}$ the bracket is invertible by the Neumann series, so $\lambda \in \rho(T)$ and

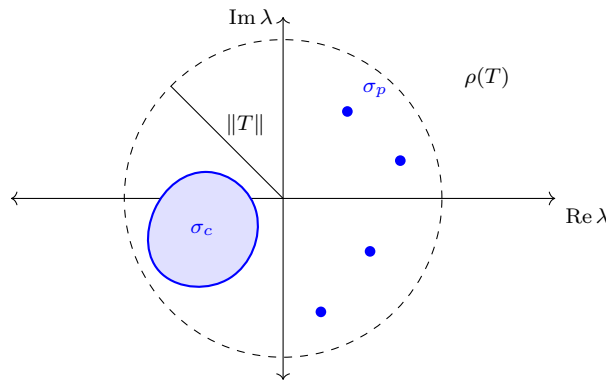
$$R_\lambda = \left(\sum_{n=0}^{\infty} (\lambda - \lambda_0)^n R_{\lambda_0}^n \right) R_{\lambda_0} = \sum_{n=0}^{\infty} R_{\lambda_0}^{n+1} (\lambda - \lambda_0)^n.$$

Thus $\rho(T)$ is open and R_λ is given locally by a convergent power series in λ with coefficients in $\mathcal{B}(X)$: it is analytic, with $\frac{d}{d\lambda} R_\lambda = R_\lambda^2$.

(iii) Suppose for contradiction $\sigma(T) = \emptyset$, so R_λ is defined and analytic on all of $\mathbb{C}$. Fix $\varphi \in \mathcal{B}(X)^*$ and let $g(\lambda) = \varphi(R_\lambda)$. Composing an analytic $\mathcal{B}(X)$ -valued function with a bounded functional gives an analytic function $\mathbb{C} \rightarrow \mathbb{C}$, so g is entire. By (*), $\|R_\lambda\| \rightarrow 0$ as $|\lambda| \rightarrow \infty$, hence $g(\lambda) \rightarrow 0$; in particular g is bounded. By Liouville's theorem from IB Complex Analysis, g is constant, and the constant is 0.

This holds for every $\varphi \in \mathcal{B}(X)^*$, so by Hahn–Banach (which says the dual separates points) $R_\lambda = 0$ for every λ . But R_λ is a bijection and $X \neq \{0\}$, so this is absurd. $\square$

Remark. The proof of (iii) is where complex scalars are indispensable. Over $\mathbb{R}$, the rotation by 90° on $\mathbb{R}^2$ has $T - \lambda$ invertible for every real λ .



Example 33.7.4 (Multiplication operators)

Let $\lambda = (\lambda_n) \in \ell^\infty$ and M_λ the diagonal operator on ℓ^2 . Then $M_\lambda - \mu$ is the diagonal operator with entries $\lambda_n - \mu$, which is invertible precisely when $\inf_n |\lambda_n - \mu| > 0$. Hence

$$\sigma(M_\lambda) = \overline{\{\lambda_n : n \in \mathbb{N}\}},$$

with $\sigma_p = \{\lambda_n\}$ and the remaining limit points forming the continuous spectrum. In particular every non-empty compact subset of $\mathbb{C}$ arises as a spectrum: take $\{\lambda_n\}$ to be a countable dense subset of it.

Example 33.7.5 (Shift operators)

Consider L, R on ℓ^2 , with $L = R^*$.

For R : it is an isometry, so $\|Rx - \mu x\| \geq (1 - |\mu|)\|x\|$ shows $R - \mu$ is bounded below for $|\mu| < 1$; but its range is not dense, since $\text{img}(R - \mu)^\perp = \ker(L - \bar{\mu})$ and $Ly = \bar{\mu}y$ is solved by $y = (1, \bar{\mu}, \bar{\mu}^2, \dots) \in \ell^2$. So $\{|\mu| < 1\} \subseteq \sigma_r(R)$. Also R has no eigenvalues: $Rx = \mu x$ forces $x = 0$ by comparing coordinates. Since $\sigma(R)$ is closed and contained in the closed unit disc, $\sigma(R) = \bar{D}$, with the open disc residual and the unit circle continuous spectrum.

For L : for $|\mu| < 1$ the vector $x = (1, \mu, \mu^2, \dots)$ lies in ℓ^2 and satisfies $Lx = \mu x$. So the open disc is point spectrum, and again $\sigma(L) = \bar{D}$, with the circle as continuous spectrum.

These are worth internalising. R is injective but not surjective, L is surjective but not injective, and their spectra are nonetheless the same closed disc.

Proposition 33.7.6

Let $T \in \mathcal{B}(X)$. Then $\partial\sigma(T) \subseteq \sigma_{\text{ap}}(T)$; in particular $\sigma_{\text{ap}}(T) \neq \emptyset$.

Proof. Let $\lambda \in \partial\sigma(T)$ and take $\lambda_n \in \rho(T)$ with $\lambda_n \rightarrow \lambda$. We claim $\|R_{\lambda_n}\| \rightarrow \infty$. If not, pass to a subsequence with $\|R_{\lambda_n}\| \leq M$; then for n large, $|\lambda - \lambda_n| < M^{-1}$, and the Neumann series argument used to prove openness of $\rho(T)$ shows $\lambda \in \rho(T)$, a contradiction.

So choose unit vectors w_n with $\|R_{\lambda_n} w_n\| \geq \frac{1}{2} \|R_{\lambda_n}\|$ and set $v_n = R_{\lambda_n} w_n / \|R_{\lambda_n} w_n\|$. Then $\|v_n\| = 1$ and

$$\|(T - \lambda_n)v_n\| = \frac{\|w_n\|}{\|R_{\lambda_n} w_n\|} \leq \frac{2}{\|R_{\lambda_n}\|} \rightarrow 0,$$

so $\|(T - \lambda)v_n\| \leq \|(T - \lambda_n)v_n\| + |\lambda_n - \lambda| \rightarrow 0$. Hence $\lambda \in \sigma_{\text{ap}}(T)$. The last claim holds since $\sigma(T)$ is a non-empty compact set, so has non-empty boundary. $\square$

Remark (Spectral radius). Define the **spectral radius** $r(T) = \sup_{\lambda \in \sigma(T)} |\lambda|$, which we have shown satisfies $r(T) \leq \|T\|$. In fact Gelfand's formula holds:

$$r(T) = \lim_{n \rightarrow \infty} \|T^n\|^{1/n},$$

the limit existing by subadditivity of $n \mapsto \log \|T^n\|$. The proof runs by considering the Laurent expansion $R_\lambda = -\sum_{n \geq 0} T^n \lambda^{-n-1}$, valid for $|\lambda| > \|T\|$ and, by analyticity of the resolvent, for $|\lambda| > r(T)$; comparing radii of convergence gives the formula. For a self-adjoint operator on Hilbert space, $\|T^2\| = \|T\|^2$ by the C^* -identity, so $\|T^{2^k}\| = \|T\|^{2^k}$ and hence $r(T) = \|T\|$ – recovering a fact we proved by hand above.

Proposition 33.7.7

Let $T \in \mathcal{B}(X)$. Then $\sigma(T^*) = \sigma(T)$, where T^* is the Banach space adjoint. If H is a complex Hilbert space and $T \in \mathcal{B}(H)$, then $\sigma(T^*) = \overline{\sigma(T)} = \{\bar{\lambda} : \lambda \in \sigma(T)\}$.

Proof. $T - \lambda$ is invertible if and only if $(T - \lambda)^* = T^* - \lambda$ is, since taking adjoints is multiplicative and $I^* = I$; and if S is a two-sided inverse of $T - \lambda$ then S^* is one for $T^* - \lambda$. Conversely if $T^* - \lambda$ is invertible then so is $T^{**} - \lambda$, and $T - \lambda$ is invertible by restricting to the image of the canonical embedding. On Hilbert space the adjoint is conjugate-linear in the scalar, whence the conjugation. $\square$

§33.7.2 The Spectrum of a Self-Adjoint Operator

For self-adjoint operators on a Hilbert space the spectrum is far better behaved, and the picture becomes much closer to the finite-dimensional one: the spectrum is real, there is no residual part, and the extreme points of the spectrum are exactly the extreme values of the quadratic form $v \mapsto \langle Tv, v \rangle$.

Proposition 33.7.8

Let H be a complex Hilbert space and $T \in \mathcal{B}(H)$ self-adjoint. Then

- (i) $\sigma(T) \subseteq \mathbb{R}$;
- (ii) $\sigma_r(T) = \emptyset$;
- (iii) writing $m = \inf_{\|v\|=1} \langle Tv, v \rangle$ and $M = \sup_{\|v\|=1} \langle Tv, v \rangle$, we have $\sigma(T) \subseteq [m, M]$ and $m, M \in \sigma(T)$.

Proof. (i) Let $\lambda = a + ib$ with $b \neq 0$. For any v ,

$$\|(T - \lambda)v\| \|v\| \geq |\langle (T - \lambda)v, v \rangle| \geq |\operatorname{Im} \langle (T - \lambda)v, v \rangle| = |b| \|v\|^2,$$

using that $\langle Tv, v \rangle$ is real. So $T - \lambda$ is bounded below, hence injective with closed range. The same estimate applies to $T - \bar{\lambda} = (T - \lambda)^*$, so $\ker(T - \lambda)^* = \{0\}$, and therefore $\operatorname{img}(\overline{T - \lambda}) = (\ker(T - \lambda)^*)^\perp = H$. A closed dense range is everything, so $T - \lambda$ is a bijection.

(ii) If $T - \lambda$ has non-dense range then $\ker(T - \bar{\lambda}) = \ker(T - \lambda)^* \neq \{0\}$; by (i) λ is real, so λ is an eigenvalue and lies in σ_p , not σ_r .

(iii) Let $\lambda > M$ be real and set $\delta = \lambda - M > 0$. Then

$$\langle (\lambda - T)v, v \rangle = \lambda \|v\|^2 - \langle Tv, v \rangle \geq \delta \|v\|^2,$$

so $\|(T - \lambda)v\| \geq \delta \|v\|$ by Cauchy–Schwarz, and as in (i) we conclude $T - \lambda$ is invertible. Similarly for $\lambda < m$. For the last claim, suppose $M \in \rho(T)$; then $S = M - T$ is a positive self-adjoint operator ($\langle Sv, v \rangle \geq 0$) which is invertible and bounded below by some $\delta > 0$. But $\inf_{\|v\|=1} \langle Sv, v \rangle = 0$ by definition of M , and for a positive operator the Cauchy–Schwarz inequality for the semi-inner product

$(v, w) \mapsto \langle Sv, w \rangle$ gives

$$\|Sv\|^2 = \langle Sv, Sv \rangle \leq \langle Sv, v \rangle^{1/2} \langle S(Sv), Sv \rangle^{1/2} \leq \langle Sv, v \rangle^{1/2} \|S\|^{3/2} \|v\|,$$

so $\langle Sv, v \rangle$ small forces $\|Sv\|$ small – contradicting boundedness below. Hence $M \in \sigma(T)$, and similarly m . $\square$

Corollary 33.7.9

If T is self-adjoint then $\|T\| = \max(|m|, |M|) = \sup_{\lambda \in \sigma(T)} |\lambda|$.

Proof. We proved $\|T\| = \sup_{\|v\| \leq 1} |\langle Tv, v \rangle| = \max(|m|, |M|)$ earlier, and $m, M \in \sigma(T) \subseteq [m, M]$. $\square$

§33.8 Compact Operators

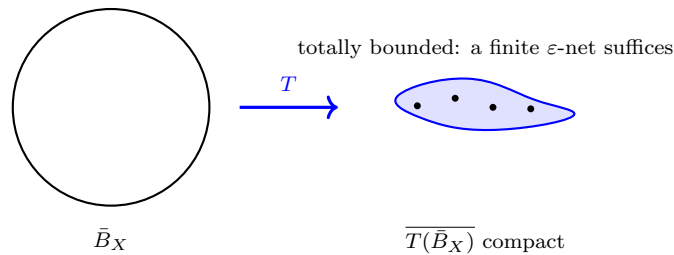
The failure of compactness of the unit ball was the first infinite-dimensional pathology we met. Compact operators are those that repair it: they are the operators which do send the unit ball into something with compact closure. They behave, in many respects, like matrices, and they have a genuinely satisfactory spectral theory.

§33.8.1 Definition and Examples

Definition 33.8.1 (Compact operator)

Let X, Y be normed spaces. A linear map $T : X \rightarrow Y$ is **compact** if $\overline{T(\bar{B}_X)}$ is compact in Y . We write $\mathcal{K}(X, Y)$ for the set of compact operators, and $\mathcal{K}(X) = \mathcal{K}(X, X)$.

Equivalently (when Y is complete, and using that compactness and sequential compactness agree in metric spaces), T is compact if and only if every bounded sequence (x_n) in X has a subsequence with (Tx_{n_k}) convergent. A compact operator is automatically bounded, since a compact set is bounded.



Example 33.8.2 (i) Any bounded operator with finite-dimensional range is compact: $T(\bar{B}_X)$ is a bounded subset of a finite-dimensional space, whose closure is compact by Riesz. We call such T a **finite rank** operator.

(ii) The identity on X is compact if and only if $\dim X < \infty$, by Riesz's theorem.

(iii) The diagonal operator M_λ on ℓ^2 is compact if and only if $\lambda_n \rightarrow 0$. If $\lambda_n \rightarrow 0$,

then M_λ is the operator-norm limit of the finite rank truncations $M_{\lambda^{(N)}}$, where $\lambda^{(N)}$ agrees with λ up to N and vanishes after; indeed $\|M_\lambda - M_{\lambda^{(N)}}\| = \sup_{n>N} |\lambda_n| \rightarrow 0$. Compactness then follows from the closedness theorem below. Conversely, if $|\lambda_{n_k}| \geq \delta > 0$ then $(M_\lambda e_{n_k})$ has no convergent subsequence, as $\|M_\lambda e_{n_k} - M_\lambda e_{n_l}\| \geq \delta\sqrt{2}$.

- (iv) The integral operator with continuous kernel k on $C[0, 1]$ is compact – this is exactly what we proved with Arzelà–Ascoli. The same operator on $L^2[0, 1]$ is compact too.

Theorem 33.8.3

Let X be a normed space and Y a Banach space. Then $\mathcal{K}(X, Y)$ is a closed linear subspace of $\mathcal{B}(X, Y)$. Moreover $\mathcal{K}(X)$ is a two-sided ideal in $\mathcal{B}(X)$: if T is compact and S bounded, then ST and TS are compact.

Proof. Subspace. If S, T are compact and (x_n) is bounded, extract a subsequence along which Sx_n converges, then a further subsequence along which Tx_n converges; then $(S + \lambda T)x_n$ converges along it.

Closed. Let $T_n \in \mathcal{K}(X, Y)$ with $T_n \rightarrow T$ in operator norm. We show $T(\bar{B}_X)$ is totally bounded; since Y is complete, this gives compact closure. Let $\varepsilon > 0$ and choose n with $\|T - T_n\| < \varepsilon/2$. As $T_n(\bar{B}_X)$ is totally bounded, there are $y_1, \dots, y_m$ forming an $\varepsilon/2$ -net for it. Then for $\|x\| \leq 1$ pick j with $\|T_n x - y_j\| < \varepsilon/2$, and

$$\|Tx - y_j\| \leq \|Tx - T_n x\| + \|T_n x - y_j\| < \varepsilon.$$

So $\{y_j\}$ is an ε -net for $T(\bar{B}_X)$.

Ideal. If T is compact and S bounded: $ST(\bar{B}_X) \subseteq \overline{S(T(\bar{B}_X))}$, the continuous image of a compact set, hence has compact closure. And $TS(\bar{B}_X) \subseteq \|S\| T(\bar{B}_X)$, which has compact closure. $\square$

Corollary 33.8.4

An operator-norm limit of finite rank operators is compact.

In a Hilbert space, the converse holds too, and this is a very useful way to think about compact operators.

Theorem 33.8.5 (Finite rank approximation)

Let H be a Hilbert space and $T \in \mathcal{B}(H)$. Then T is compact if and only if T is the operator-norm limit of finite rank operators.

Proof. $(\Leftarrow)$ is the corollary above.

$(\Rightarrow)$ Let T be compact and $\varepsilon > 0$. By total boundedness there are $v_1, \dots, v_n$ with $T(\bar{B}_H) \subseteq \bigcup_j B(Tv_j, \varepsilon)$ – we may take the net to consist of points of the image. Let $M = \text{span}\{Tv_1, \dots, Tv_n\}$, a finite-dimensional (hence closed) subspace, and let P be the orthogonal projection onto M . Set $T_\varepsilon = PT$, which has finite rank.

For $\|v\| \leq 1$, choose j with $\|Tv - Tv_j\| < \varepsilon$. Since $Tv_j \in M$ and P is the nearest-point projection onto M ,

$$\|Tv - PTv\| = \text{dist}(Tv, M) \leq \|Tv - Tv_j\| < \varepsilon.$$

Hence $\|T - T_\varepsilon\| \leq \varepsilon$. $\square$

Remark. This is false in general Banach spaces. Per Enflo constructed in 1973 a Banach space with a compact operator that is not a limit of finite rank operators – resolving the ‘approximation problem’ negatively, and famously earning him a live goose from Mazur.

Proposition 33.8.6

Let H be a Hilbert space and $T \in \mathcal{B}(H)$. Then T is compact if and only if T^* is.

Proof. If T is compact, write $T = \lim T_n$ with T_n finite rank. Then T_n^* has finite rank (its range is contained in $\text{img } T_n^*$, and $\text{rank } T_n^* = \text{rank } T_n$ since $\ker T_n^* = (\text{img } T_n)^\perp$), and $\|T^* - T_n^*\| = \|T - T_n\| \rightarrow 0$. So T^* is compact. The converse follows as $T^{**} = T$. $\square$

Definition 33.8.7 (Hilbert–Schmidt operator)

Let H be a separable Hilbert space with orthonormal basis (e_n) . An operator $T \in \mathcal{B}(H)$ is **Hilbert–Schmidt** if

$$\|T\|_{\text{HS}} = \left(\sum_n \|Te_n\|^2 \right)^{1/2} < \infty.$$

Proposition 33.8.8

The quantity $\|T\|_{\text{HS}}$ is independent of the choice of orthonormal basis, $\|T\| \leq \|T\|_{\text{HS}}$, and every Hilbert–Schmidt operator is compact.

Proof. If (f_m) is another orthonormal basis then by Parseval, applied twice,

$$\sum_n \|Te_n\|^2 = \sum_n \sum_m |\langle Te_n, f_m \rangle|^2 = \sum_m \sum_n |\langle e_n, T^* f_m \rangle|^2 = \sum_m \|T^* f_m\|^2,$$

and the right-hand side does not involve (e_n) ; applying the same argument to T^* shows it equals $\sum_m \|T f_m\|^2$. For the norm bound, write $v = \sum_n \langle v, e_n \rangle e_n$ with $\|v\| \leq 1$; then by Cauchy–Schwarz $\|Tv\| \leq \sum_n |\langle v, e_n \rangle| \|Te_n\| \leq \|v\| \|T\|_{\text{HS}}$.

For compactness, let P_N be the projection onto $\text{span}\{e_1, \dots, e_N\}$. Then TP_N has finite rank and $\|T - TP_N\| \leq \|T - TP_N\|_{\text{HS}} = (\sum_{n>N} \|Te_n\|^2)^{1/2} \rightarrow 0$, so T is a limit of finite rank operators. $\square$

Example 33.8.9

The diagonal operator M_λ on ℓ^2 is Hilbert–Schmidt exactly when $\lambda \in \ell^2$, compact exactly when $\lambda \in c_0$, and bounded exactly when $\lambda \in \ell^\infty$. So the Hilbert–Schmidt

operators form a proper subclass of the compact operators; taking $\lambda_n = 1/\log(n+1)$ gives a compact operator that is not Hilbert–Schmidt. The integral operator on $L^2[0, 1]$ with kernel $k \in L^2([0, 1]^2)$ is always Hilbert–Schmidt, with $\|T\|_{\text{HS}} = \|k\|_{L^2}$.

§33.8.2 Spectral Theory of Compact Operators

Compact operators have spectra that look very much like those of matrices.

Theorem 33.8.10

Let X be a complex Banach space, $T \in \mathcal{K}(X)$ and $\lambda \neq 0$. Then:

- (i) $\ker(T - \lambda)$ is finite-dimensional;
- (ii) $\text{img}(T - \lambda)$ is closed;
- (iii) $T - \lambda$ is injective if and only if it is surjective.

Consequently $\sigma(T) \setminus \{0\}$ consists of eigenvalues of finite multiplicity.

Proof. (i) Let $N = \ker(T - \lambda)$. On N we have $T = \lambda I$, so the restriction of T to N is λ times the identity, which is compact (a restriction of a compact operator to a closed invariant subspace is compact). As $\lambda \neq 0$, the identity on N is compact, so $\dim N < \infty$ by Riesz.

(ii) By (i), N is finite-dimensional, hence complemented: $X = N \oplus Z$ for some closed subspace Z (choose a basis $x_1, \dots, x_k$ of N , extend the coordinate functionals to X^* by Hahn–Banach, and let Z be the intersection of their kernels). Then $T - \lambda$ restricted to Z is injective with the same range.

We claim $S := (T - \lambda)|_Z$ is bounded below. If not, there are $z_n \in Z$ with $\|z_n\| = 1$ and $Sz_n \rightarrow 0$. By compactness pass to a subsequence with $Tz_n \rightarrow y$. Then $\lambda z_n = Tz_n - Sz_n \rightarrow y$, so $z_n \rightarrow y/\lambda =: z \in Z$ with $\|z\| = 1$, and by continuity $Sz = 0$ – contradicting injectivity on Z . So S is bounded below, and a bounded-below operator has closed range: if $Sz_n \rightarrow w$ then (z_n) is Cauchy, so $z_n \rightarrow z$ and $w = Sz$.

(iii) Suppose $T - \lambda$ is injective but not surjective. Set $X_n = \text{img}(T - \lambda)^n$. Each X_n is closed (by (ii) and induction, since $(T - \lambda)^n = (-\lambda)^n + (\text{compact})$, using that $\mathcal{K}$ is an ideal), and

$$X \supsetneq X_1 \supsetneq X_2 \supsetneq \cdots,$$

the inclusions being strict because $T - \lambda$ is injective (if $X_{n+1} = X_n$ then applying injectivity n times gives $X_1 = X$).

By Riesz’s lemma choose $x_n \in X_n$ with $\|x_n\| = 1$ and $\text{dist}(x_n, X_{n+1}) \geq \frac{1}{2}$. For $m > n$,

$$Tx_n - Tx_m = \lambda x_n - ((T - \lambda)x_n - (T - \lambda)x_m + \lambda x_m),$$

and the bracketed term lies in X_{n+1} (as $(T - \lambda)x_n \in X_{n+1}$, and $x_m, (T - \lambda)x_m \in X_m \subseteq X_{n+1}$). Hence $\|Tx_n - Tx_m\| \geq |\lambda| \text{dist}(x_n, X_{n+1}) \geq |\lambda|/2$, so (Tx_n) has no convergent subsequence, contradicting compactness.

Conversely suppose $T - \lambda$ is surjective but not injective. Now set $Y_n = \ker(T - \lambda)^n$, an increasing chain of finite-dimensional (hence closed) subspaces, strictly increasing

by surjectivity, and run the same Riesz's lemma argument with $\text{dist}(y_{n+1}, Y_n) \geq \frac{1}{2}$.
 The final statement: if $\lambda \in \sigma(T)$ and $\lambda \neq 0$, then $T - \lambda$ is not bijective, so by (iii) it is not injective, i.e. λ is an eigenvalue, with finite-dimensional eigenspace by (i). $\square$

Theorem 33.8.11 (Fredholm Alternative)

Let X be a complex Banach space, $T \in \mathcal{K}(X)$ and $\lambda \neq 0$. Then exactly one of the following holds:

- (a) the homogeneous equation $Tx = \lambda x$ has a non-zero solution;
- (b) for every $y \in X$ the equation $Tx - \lambda x = y$ has a unique solution, depending continuously on y .

Proof. This is a restatement of part (iii) above: either $T - \lambda$ fails to be injective, which is (a), or it is injective, hence surjective, hence a bijection, and its inverse is bounded by the inverse mapping theorem, which is (b). $\square$

This is exactly the dichotomy familiar from linear algebra – for a square matrix A , either $Ax = 0$ has a non-trivial solution or $Ax = y$ is always uniquely soluble – and it is remarkable that it survives into infinite dimensions for compact perturbations of λI . It fails badly without compactness: for the right shift R and λ with $|\lambda| < 1$, the equation $Rx - \lambda x = 0$ has only the zero solution, yet $R - \lambda$ is not surjective.

Theorem 33.8.12

Let $T \in \mathcal{K}(X)$ with X an infinite-dimensional complex Banach space. Then $0 \in \sigma(T)$, and $\sigma(T)$ is either finite or a sequence of eigenvalues tending to 0 (together with 0).

Proof. If $0 \notin \sigma(T)$ then T is invertible, so $I = T^{-1}T$ is compact, forcing $\dim X < \infty$.

For the second claim, it suffices to show that for each $\delta > 0$ there are only finitely many $\lambda \in \sigma(T)$ with $|\lambda| \geq \delta$. Suppose not, and let $\lambda_1, \lambda_2, \dots$ be distinct eigenvalues with $|\lambda_n| \geq \delta$, with eigenvectors v_n . Eigenvectors for distinct eigenvalues are linearly independent (the usual Vandermonde argument from IB Linear Algebra), so setting $V_n = \text{span}\{v_1, \dots, v_n\}$ we get a strictly increasing chain of finite-dimensional subspaces.

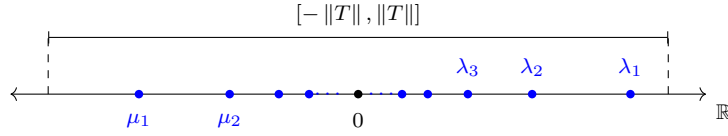
By Riesz's lemma pick $w_n \in V_n$ with $\|w_n\| = 1$ and $\text{dist}(w_n, V_{n-1}) \geq \frac{1}{2}$. For $m < n$,

$$Tw_n - Tw_m = \lambda_n w_n - ((\lambda_n - T)w_n + Tw_m).$$

Now $(\lambda_n - T)V_n \subseteq V_{n-1}$: writing $w_n = \sum_{k \leq n} a_k v_k$, the coefficient of v_n in $(\lambda_n - T)w_n$ is $a_n(\lambda_n - \lambda_n) = 0$. And $Tw_m \in V_m \subseteq V_{n-1}$. So the bracket lies in V_{n-1} , giving

$$\|Tw_n - Tw_m\| \geq |\lambda_n| \text{dist}(w_n, V_{n-1}) \geq \frac{\delta}{2}.$$

So (Tw_n) has no convergent subsequence, contradicting compactness. $\square$



§33.8.3 The Spectral Theorem

We can now prove the theorem that this whole course has been building towards: compact self-adjoint operators on a Hilbert space can be diagonalised, exactly as Hermitian matrices can.

Lemma 33.8.13

Let H be a complex Hilbert space and $T \in \mathcal{K}(H)$ self-adjoint and non-zero. Then $\|T\|$ or $-\|T\|$ is an eigenvalue of T .

Proof. Recall $\|T\| = \sup_{\|v\|=1} |\langle Tv, v \rangle|$, and that $\langle Tv, v \rangle$ is real. So there is a sequence (v_n) of unit vectors with $\langle Tv_n, v_n \rangle \rightarrow \lambda$, where $\lambda = \pm \|T\|$. Then

$$\begin{aligned} \|Tv_n - \lambda v_n\|^2 &= \|Tv_n\|^2 - 2\lambda \langle Tv_n, v_n \rangle + \lambda^2 \\ &\leq \|T\|^2 - 2\lambda \langle Tv_n, v_n \rangle + \lambda^2 \longrightarrow 2\lambda^2 - 2\lambda^2 = 0. \end{aligned}$$

By compactness, pass to a subsequence with $Tv_n \rightarrow w$. Then $\lambda v_n = Tv_n - (Tv_n - \lambda v_n) \rightarrow w$, so $v_n \rightarrow w/\lambda$ (using $\lambda \neq 0$, which holds as $T \neq 0$), and $\|w/\lambda\| = 1$. By continuity $T(w/\lambda) = \lambda(w/\lambda)$, so λ is an eigenvalue. $\square$

Theorem 33.8.14 (Spectral theorem for compact self-adjoint operators)

Let H be a complex Hilbert space and $T \in \mathcal{K}(H)$ self-adjoint. Then there is a finite or countable orthonormal system (e_n) in H and real numbers (λ_n) with $|\lambda_1| \geq |\lambda_2| \geq \dots$ and $\lambda_n \rightarrow 0$ (if the system is infinite), such that

$$Tv = \sum_n \lambda_n \langle v, e_n \rangle e_n \quad \text{for all } v \in H.$$

Equivalently, H has an orthonormal basis consisting of eigenvectors of T , and

$$H = \ker T \oplus \bigoplus_n \ker(T - \lambda_n)$$

as an orthogonal direct sum, each $\ker(T - \lambda_n)$ being finite-dimensional.

Proof. We construct the e_n inductively. Let $H_1 = H$ and $T_1 = T$. Given a closed T -invariant subspace H_n with $T_n = T|_{H_n}$, note that T_n is again compact and self-adjoint on H_n . If $T_n = 0$ we stop. Otherwise, by the lemma there is a unit vector $e_n \in H_n$ and $\lambda_n = \pm \|T_n\|$ with $Te_n = \lambda_n e_n$. Set

$$H_{n+1} = H_n \cap \{e_n\}^\perp,$$

which is closed, and T -invariant: if $v \perp e_n$ then $\langle Tv, e_n \rangle = \langle v, Te_n \rangle = \bar{\lambda}_n \langle v, e_n \rangle = 0$,

using self-adjointness and $\lambda_n \in \mathbb{R}$.

By construction the e_n are orthonormal, and $|\lambda_n| = \|T_n\|$ is non-increasing since $H_{n+1} \subseteq H_n$.

The λ_n tend to 0. If not, then $|\lambda_n| \geq \delta > 0$ for all n , and for $m \neq n$, orthogonality gives

$$\|Te_n - Te_m\|^2 = \|\lambda_n e_n - \lambda_m e_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\delta^2,$$

so (Te_n) has no convergent subsequence, contradicting compactness.

The formula. Let P_n be the orthogonal projection onto $\text{span}\{e_1, \dots, e_n\}$, so $P_n v = \sum_{k \leq n} \langle v, e_k \rangle e_k$ and $v - P_n v \in H_{n+1}$. Then

$$\left\| Tv - \sum_{k \leq n} \lambda_k \langle v, e_k \rangle e_k \right\| = \|T(v - P_n v)\| = \|T_{n+1}(v - P_n v)\| \leq |\lambda_{n+1}| \|v\|,$$

using $\|T_{n+1}\| = |\lambda_{n+1}|$ and $\|v - P_n v\| \leq \|v\|$. As $\lambda_{n+1} \rightarrow 0$ the left-hand side tends to 0, which is the required formula. (If the process terminates at stage n then $T_n = 0$ and the formula is exact.)

The decomposition. Grouping the e_n by their eigenvalue gives the eigenspaces $\ker(T - \lambda)$ for the distinct non-zero eigenvalues λ ; these are finite-dimensional as T is compact and $\lambda \neq 0$, and mutually orthogonal as T is self-adjoint. The formula shows that $Tv = 0$ whenever $v \perp$ all e_n , so the orthogonal complement of $\overline{\text{span}\{e_n\}}$ is exactly $\ker T$. Adjoining an orthonormal basis of $\ker T$ gives an orthonormal basis of H of eigenvectors. $\square$

Remark. The estimate in the proof gives more: writing $T_{(n)}v = \sum_{k \leq n} \lambda_k \langle v, e_k \rangle e_k$ for the truncation, we showed $\|T - T_{(n)}\| \leq |\lambda_{n+1}|$. So the eigenvalue expansion converges in operator norm, and exhibits T explicitly as a limit of finite rank operators.

Corollary 33.8.15 (Singular value decomposition)

Let $T \in \mathcal{K}(H_1, H_2)$ between Hilbert spaces. Then there are orthonormal systems (e_n) in H_1 , (f_n) in H_2 and $\mu_n \geq 0$ with $\mu_n \rightarrow 0$ such that $Tv = \sum_n \mu_n \langle v, e_n \rangle f_n$.

Proof. T^*T is compact, self-adjoint and positive, so by the spectral theorem $T^*Tv = \sum_n \mu_n^2 \langle v, e_n \rangle e_n$ with $\mu_n^2 \geq 0$ its eigenvalues and (e_n) orthonormal. Set $f_n = Te_n / \mu_n$ for $\mu_n \neq 0$; these are orthonormal since $\langle Te_n, Te_m \rangle = \langle T^*Te_n, e_m \rangle = \mu_n^2 \delta_{nm}$. $\square$

§33.8.4 Applications to Integral Equations

The theory was invented for a purpose, so let us see it applied. Consider the integral equation

$$f(x) - \mu \int_0^1 k(x, y) f(y) dy = g(x),$$

where k is a continuous Hermitian kernel, g is given, and f is to be found. Writing K for the integral operator, this is $(I - \mu K)f = g$, i.e. $(K - \mu^{-1})f = -\mu^{-1}g$.

Since K is compact and self-adjoint, the Fredholm alternative applies immediately.

Proposition 33.8.16

Let k be a continuous Hermitian kernel and $\mu \neq 0$. Exactly one of the following holds:

- (a) μ^{-1} is an eigenvalue of K , in which case the homogeneous equation $f = \mu K f$ has a non-trivial (finite-dimensional) space of solutions, and the inhomogeneous equation is soluble precisely when $g \perp \ker(K - \mu^{-1})$;
- (b) μ^{-1} is not an eigenvalue, in which case the equation has a unique solution for every g , and $\|f\| \leq C \|g\|$ for a constant depending only on k and μ .

Proof. The dichotomy is the Fredholm alternative. In case (a), the solubility criterion follows from $\overline{\text{img}(K - \mu^{-1})} = \ker(K - \mu^{-1})^\perp$ together with the fact (proved above) that the range is closed. $\square$

Moreover, the spectral theorem gives the solution explicitly. Let (e_n) be an orthonormal basis of eigenvectors with $Ke_n = \lambda_n e_n$. Expanding $f = \sum_n \hat{f}_n e_n$ and $g = \sum_n \hat{g}_n e_n$, the equation $(I - \mu K)f = g$ becomes

$$(1 - \mu\lambda_n)\hat{f}_n = \hat{g}_n \quad \text{for every } n,$$

so that, provided $\mu\lambda_n \neq 1$ for all n ,

$$f = \sum_n \frac{\hat{g}_n}{1 - \mu\lambda_n} e_n.$$

This converges because $\lambda_n \rightarrow 0$, so the denominators are eventually close to 1 and in particular bounded away from 0.

Example 33.8.17 (Sturm–Liouville theory)

In IB Methods we studied the eigenvalue problem

$$\mathcal{L}y = -(py')' + qy = \lambda wy \quad \text{on } [a, b], \quad y(a) = y(b) = 0,$$

and asserted – without proof – that the eigenvalues are real and discrete, the eigenfunctions are orthogonal with respect to the weight w , and the eigenfunctions form a complete system, so that any reasonable function can be expanded in them. We can now see where these facts come from.

Assume $p > 0$ and $q \geq 0$ on $[a, b]$, and $w > 0$. The operator $\mathcal{L}$ is unbounded, so we cannot apply our theory to it directly – this is the usual difficulty with differential operators. Instead we invert it. Using the Green's function $G(x, y)$ for $\mathcal{L}$ with the given boundary conditions (constructed in IB Methods from two solutions of the homogeneous equation and a jump condition), the map

$$(Kf)(x) = \int_a^b G(x, y)f(y) \, dy$$

is a two-sided inverse to $\mathcal{L}$ on the relevant space. The Green's function is continuous and symmetric, $G(x, y) = G(y, x)$, so K is a compact self-adjoint operator on $L^2([a, b], w \, dx)$.

Now apply the spectral theorem to K . It gives an orthonormal basis (y_n) of eigenfunctions, $Ky_n = \kappa_n y_n$ with κ_n real and $\kappa_n \rightarrow 0$. Since K is injective (as $\mathcal{L}Kf = f$), we have $\kappa_n \neq 0$, and applying $\mathcal{L}$ gives

$$\mathcal{L}y_n = \lambda_n w y_n, \quad \lambda_n = \kappa_n^{-1}.$$

Thus: the eigenvalues λ_n are real, they form a discrete sequence with $|\lambda_n| \rightarrow \infty$ (since $\kappa_n \rightarrow 0$), the eigenfunctions are orthogonal in the w -weighted inner product, and – this is the part that was simply asserted in Methods – they are *complete*, so that every $f \in L^2$ has a convergent eigenfunction expansion

$$f = \sum_n \langle f, y_n \rangle_w y_n.$$

The eigenfunction expansions of IB Methods, and the Fourier series that are their prototype, are all instances of the spectral theorem for compact self-adjoint operators.

Remark. This is a good place to stop and look back. We began by observing that the closed unit ball of an infinite-dimensional space is never compact, and that consequently the finite-dimensional theory breaks down. We then spent the course building substitutes: Hahn–Banach to supply functionals, Baire category to supply boundedness and openness for free, orthogonal projection to supply geometry, and finally compactness – reintroduced not as a property of the space but of the *operator* – to recover the spectral theorem. That last step is the pattern of much of modern analysis: when the space will not be compact, ask the operator to be.

34 Logic and Set Theory

Mathematics is done by writing down statements and proving them. In this course we take that activity itself as our object of study: we write down what a ‘statement’ is as a finite string of symbols, we write down what a ‘proof’ is as a finite list of such strings obeying some rules, and then we prove theorems about the resulting system. The most striking of these – the completeness theorem – says that the two ways we have of saying that something follows from something else, the semantic one (‘it is true in every situation where the hypotheses hold’) and the syntactic one (‘there is a proof of it’), give exactly the same answer.

Alongside this we develop set theory, which serves as the arena in which all of the above takes place. We will meet the ordinals, which extend counting past infinity and which give us induction and recursion of unlimited length; Zorn’s lemma, which is the form in which the axiom of choice is usually applied; and finally the axioms of Zermelo–Fraenkel set theory, together with a picture of the universe of sets that they describe. The two halves of the course meet at the very end, when we use a set-theoretic construction to write down a sentence of arithmetic that is true but unprovable.

This article constitutes my notes for the ‘Logic and Set Theory’ course, held in Lent 2023 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§34.1 Propositional Logic

We begin our study of logic by looking at *propositional logic*, which is the logic of statements built up out of atomic statements using connectives like ‘and’, ‘or’ and ‘implies’. It is a small enough system that we can prove everything we would want to about it, and it will act as a template for the more elaborate predicate logic that we meet in Section 34.4.

§34.1.1 Languages

The first thing to do is to say precisely what a statement *is*. It will be a finite string of symbols, built up according to some grammar from a supply of atoms.

Definition 34.1.1 (Language)

Let P be a set of **primitive propositions**. Unless otherwise stated we take $P = \{p_1, p_2, \dots\}$. The **language** L , or $L(P)$, is defined inductively by

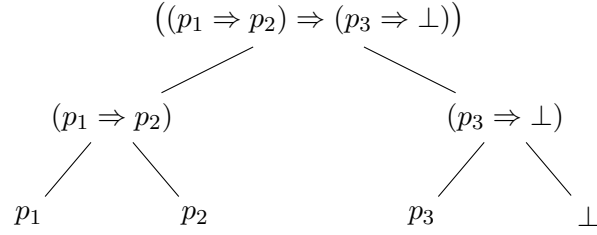
- (i) If $p \in P$, then $p \in L$;
- (ii) $\perp \in L$ (where we read $\perp$ as ‘false’);
- (iii) If $p, q \in L$ then $(p \Rightarrow q) \in L$.

Every **proposition** (that is, member of L) is a finite string of symbols from the alphabet $\{(\,, \Rightarrow, \perp, p_1, p_2, \dots\}$, satisfying some grammar. It is worth being clear about what the

words ‘defined inductively’ mean above, since we will use the same device repeatedly.

We let $L_1 = P \cup \{\perp\}$, and define $L_{n+1} = L_n \cup \{(p \Rightarrow q) \mid p, q \in L_n\}$. Then $L = \bigcup_{n=1}^{\infty} L_n$. The rules building up the language are injective and have disjoint ranges, and so there is exactly *one* way in which any element of the language can be constructed from the rules above – a fact we will lean on constantly, since it is what lets us define things on L by recursion.

This unique readability is best seen by drawing a proposition as a tree, with the primitives and $\perp$ at the leaves and each internal node an application of $\Rightarrow$.



The proposition drawn above is $\neg p_3$ implied by $p_1 \Rightarrow p_2$, once we introduce the standard abbreviations $\neg$ (not), $\wedge$ (and) and $\vee$ (or), defined by

$$\neg p = (p \Rightarrow \perp); \quad p \wedge q = \neg(p \Rightarrow \neg q); \quad p \vee q = (\neg p) \Rightarrow q.$$

These are abbreviations only – our language really does have just the two constructors $\perp$ and $\Rightarrow$. This is purely for convenience: every proof below would go through with more connectives, at the cost of more cases.

§34.1.2 Semantic Implication

We now assign ‘true’ or ‘false’ values to propositions. Since a proposition is just a string of symbols, this has to be done by hand.

Definition 34.1.2 (Valuation)

A **valuation** is a function $v : L \rightarrow \{0, 1\}$ (where we think of 0 as false and 1 as true) such that

- (i) $v(\perp) = 0$;
- (ii) $v(p \Rightarrow q) = 0$ if $v(p) = 1$ and $v(q) = 0$, and 1 otherwise.

One would imagine that it is enough to know how a valuation acts on the primitives, since they are what our language is built out of – in the same way that a linear map is pinned down by its action on a basis. This is indeed the case.

Proposition 34.1.3 (Valuations are Determined on the Primitives)

- (i) Let $v, v' : L \rightarrow \{0, 1\}$ be valuations that agree on the primitives p_i . Then $v = v'$.
- (ii) Any function $w : P \rightarrow \{0, 1\}$ extends to a valuation.

Proof. (i) Clearly v, v' agree on L_1 . Then if they agree on p and q , they agree on $(p \Rightarrow q)$. So by induction they agree on L_n for all n , and hence on L .

(ii) Let $v(p) = w(p)$ for all $p \in P$ and $v(\perp) = 0$, to obtain v on the set L_1 . Assuming v is defined on p, q we can define it at $(p \Rightarrow q)$ in the obvious way. Since each element of L is built up in exactly one way, this consistently defines v on all of L . $\square$

In other words, valuations on L correspond exactly to functions $P \rightarrow \{0, 1\}$, which is what makes truth tables work.

Definition 34.1.4 (Tautology)

A **tautology** is an element $t \in L$ such that $v(t) = 1$ for every valuation v . We write $\models t$.

Example 34.1.5 (Some Tautologies)

Each of the following is a tautology, as one checks by running through the (at most eight) possibilities for the values of the primitives involved.

- (i) $p \Rightarrow (q \Rightarrow p)$. If $v(p) = 0$ then the whole thing is 1 immediately, and if $v(p) = 1$ then $v(q \Rightarrow p) = 1$.
- (ii) $\neg\neg p \Rightarrow p$, that is, $((p \Rightarrow \perp) \Rightarrow \perp) \Rightarrow p$. If $v(p) = 1$ this is 1, and if $v(p) = 0$ then $v(p \Rightarrow \perp) = 1$ so $v(\neg\neg p) = 0$.
- (iii) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$. The only way for this to fail is $v(p) = v(q) = 1$ and $v(r) = 0$, and then the antecedent is already 0.

On the other hand $p \Rightarrow q$ is not a tautology, since we may take $v(p) = 1$ and $v(q) = 0$.

In practice one checks a tautology by drawing a truth table, listing the values of the primitives down the side and computing the value of each subformula.

Example 34.1.6 (A Truth Table)

We verify that $(\neg p \Rightarrow \neg q) \Rightarrow (q \Rightarrow p)$ is a tautology. Writing $v(p)$ and $v(q)$ in the first two columns:

$v(p)$	$v(q)$	$v(\neg p)$	$v(\neg q)$	$v(\neg p \Rightarrow \neg q)$	$v(q \Rightarrow p)$
0	0	1	1	1	1
0	1	1	0	0	0
1	0	0	1	1	1
1	1	0	0	1	1

In every row the last two columns agree, so the implication between them takes the value 1 throughout.

Remark. Since a valuation is determined by its values on the primitives, a formula in n primitives determines a function $\{0, 1\}^n \rightarrow \{0, 1\}$. Conversely every such function arises from a formula: one may write down a disjunction of conjunctions, one for each row of the truth table taking the value 1. So our two connectives $\perp$ and $\Rightarrow$ are already *adequate*, in the sense that every truth function is expressible.

Usually we are not interested in statements that hold outright, but in statements that

hold given some hypotheses.

Definition 34.1.7 (Semantic Implication)

Let $S \subseteq L$ and $t \in L$. We say S **entails** or **semantically implies** t , written $S \models t$, if $v(t) = 1$ whenever $v(s) = 1$ for all $s \in S$.

Definition 34.1.8 (Model)

We say that v is a **model** of S in L if $v(s) = 1$ for all $s \in S$.

So the statement $S \models t$ is precisely the statement that every model of S is also a model of t , and the notation $\models t$ is consistent with what we had before, being the same as $\emptyset \models t$.

Example 34.1.9

We have $\{p \vee q, \neg p\} \models q$: if $v(\neg p) = 1$ then $v(p) = 0$, and $p \vee q$ abbreviates $\neg p \Rightarrow q$, so $v(q) = 1$. On the other hand $\{p \Rightarrow q\} \not\models p$, as witnessed by the valuation sending every primitive to 0.

§34.1.3 Syntactic Implication

Semantic implication asks us to check every valuation, of which there are infinitely many. What we do in practice instead is give a *proof*. To make that notion precise we need a system of axioms and deduction rules.

Axiom 34.1.10 (Axiom Scheme for Propositional Logic)

- (1) $p \Rightarrow (q \Rightarrow p)$ for all $p, q \in L$;
- (2) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$ for all $p, q, r \in L$;
- (3) $(\neg\neg p) \Rightarrow p$ for all $p \in L$.

Each of these is really an axiom *scheme*, since each line above is a whole set of axioms, one for each choice of the propositions involved. Notice also that each of them is a tautology – as one would hope, and as we will need for soundness.

For our deduction rules we take only **modus ponens**: from p and $p \Rightarrow q$ we may deduce q .

Definition 34.1.11 (Syntactic Implication)

For $S \subseteq L$ and $t \in L$, we say S **proves** or **syntactically implies** t , written $S \vdash t$, if there is a finite sequence $t_1, \dots, t_n = t$ of elements of L such that each t_i is either an axiom, or a member of S , or is obtained as q where $t_j = p$ and $t_k = (p \Rightarrow q)$ for some $j, k < i$.

We call S the set of **premises** or **hypotheses**, and t the **conclusion**.

Example 34.1.12

We will prove $\{p \Rightarrow q, q \Rightarrow r\} \vdash p \Rightarrow r$.

1. $q \Rightarrow r$ (hypothesis)
2. $(q \Rightarrow r) \Rightarrow (p \Rightarrow (q \Rightarrow r))$ (axiom 1)
3. $p \Rightarrow (q \Rightarrow r)$ (modus ponens on 1, 2)
4. $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$ (axiom 2)
5. $(p \Rightarrow q) \Rightarrow (p \Rightarrow r)$ (modus ponens on 3, 4)
6. $p \Rightarrow q$ (hypothesis)
7. $p \Rightarrow r$ (modus ponens on 5, 6)

Definition 34.1.13 (Theorem)

If $\emptyset \vdash t$, we say that t is a **theorem**, and write $\vdash t$.

Example 34.1.14

We will prove the theorem $\vdash (p \Rightarrow p)$. This looks like it should be trivial, and the fact that it is not is a good illustration of how weak our axiom system looks from the inside.

1. $(p \Rightarrow ((p \Rightarrow p) \Rightarrow p)) \Rightarrow ((p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p))$ (axiom 2)
2. $p \Rightarrow ((p \Rightarrow p) \Rightarrow p)$ (axiom 1)
3. $(p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p)$ (modus ponens on 1, 2)
4. $p \Rightarrow (p \Rightarrow p)$ (axiom 1)
5. $p \Rightarrow p$ (modus ponens on 3, 4)

Two easy remarks that we will use without comment. First, proofs are *finite*, so if $S \vdash t$ then $S' \vdash t$ for some finite $S' \subseteq S$. Second, if $S \subseteq S'$ and $S \vdash t$ then $S' \vdash t$, since a proof from S is already a proof from S' .

Example 34.1.15 (A Theorem Needing Axiom 3)

Not everything provable is provable from axioms 1 and 2 alone; axiom 3 is what makes our system *classical* rather than intuitionistic. For instance $\vdash (\perp \Rightarrow p)$ for every p : by the deduction theorem below it suffices to show $\{\perp\} \vdash p$, and from $\perp$ we get $(\neg p) \Rightarrow \perp$ by axiom 1, which is $\neg\neg p$, and then p by axiom 3. Without axiom 3 there is no proof of this.

§34.1.4 The Deduction Theorem

Writing out proofs like the one above by hand is unpleasant, and we would rather not have to. The following theorem is the tool that makes syntactic implication usable: it says that the connective ' $\Rightarrow$ ' inside the language really does correspond to implication in the metalanguage.

Theorem 34.1.16 (Deduction Theorem)

Let $S \subseteq L$ and $p, q \in L$. Then $S \vdash (p \Rightarrow q)$ if and only if $S \cup \{p\} \vdash q$.

Proof. Given a proof of $p \Rightarrow q$ from S , we add the line p (as a hypothesis) and deduce q by modus ponens, obtaining a proof of q from $S \cup \{p\}$.

Conversely, suppose we have a proof of q from $S \cup \{p\}$, with lines $t_1, \dots, t_n$. We show that $S \vdash (p \Rightarrow t_i)$ for all i , by induction on i . There are four cases.

- If t_i is an axiom, we write t_i (axiom); $t_i \Rightarrow (p \Rightarrow t_i)$ (axiom 1); $p \Rightarrow t_i$ (modus ponens).
- If $t_i \in S$, we write t_i (hypothesis); $t_i \Rightarrow (p \Rightarrow t_i)$ (axiom 1); $p \Rightarrow t_i$ (modus ponens).
- If $t_i = p$, we write out the proof of $\vdash p \Rightarrow p$ given above.
- If t_i is obtained by modus ponens from t_j and $t_k = (t_j \Rightarrow t_i)$, then by induction we have $S \vdash p \Rightarrow t_j$ and $S \vdash p \Rightarrow (t_j \Rightarrow t_i)$. We then write $(p \Rightarrow (t_j \Rightarrow t_i)) \Rightarrow ((p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i))$ (axiom 2); $(p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i)$ (modus ponens); $p \Rightarrow t_i$ (modus ponens).

Taking $i = n$ gives $S \vdash (p \Rightarrow q)$. □

Notice that axioms 1 and 2 were exactly what was needed to run this argument – and, conversely, that this is really the only thing we ever want to do with them.

Example 34.1.17 (Using the Deduction Theorem)

To show $\{p \Rightarrow q, q \Rightarrow r\} \vdash p \Rightarrow r$, the deduction theorem says it is enough to show that $\{p, p \Rightarrow q, q \Rightarrow r\} \vdash r$, which is just modus ponens twice. Compare this with the seven-line proof above.

§34.1.5 Soundness and Adequacy

We now have two relations between a set of hypotheses and a conclusion, $\models$ and $\vdash$, defined in completely different ways. The **completeness theorem** says that they agree: $S \models t$ if and only if $S \vdash t$.

This splits into two halves. The **soundness** statement, $S \vdash t \Rightarrow S \models t$, informally says ‘our axioms and deduction rule are not silly’. The **adequacy** statement, $S \models t \Rightarrow S \vdash t$, says ‘our axioms are strong enough to deduce from S every semantic consequence of S ’. Soundness is easy.

Proposition 34.1.18 (Soundness)

Let $S \subseteq L$ and $t \in L$. Then $S \vdash t$ implies $S \models t$.

Proof. Let v be a model of S , and let $t_1, \dots, t_n = t$ be a proof of t from S . We show $v(t_i) = 1$ for all i by induction. If t_i is an axiom then $v(t_i) = 1$ as each axiom is a tautology; if $t_i \in S$ then $v(t_i) = 1$ as v is a model of S ; and if t_i follows by modus ponens from earlier lines p and $p \Rightarrow q$, then $v(p) = v(p \Rightarrow q) = 1$ forces $v(q) = 1$ by

the definition of a valuation. Hence $v(t) = 1$. $\square$

Adequacy is the substantial half. The key is to notice that a special case of it implies the whole thing. Let us say that S is **consistent** if $S \not\vdash \perp$.

Definition 34.1.19 (Consistent)

A set $S \subseteq L$ is **consistent** if $S \not\vdash \perp$.

The case $t = \perp$ of adequacy says: $S \models \perp$ implies $S \vdash \perp$. Taking the contrapositive, and noting that $S \models \perp$ says exactly that S has no model, this reads:

$$S \text{ consistent} \implies S \text{ has a model.}$$

This special case implies adequacy in general. Indeed, suppose $S \models t$. Then $S \cup \{\neg t\}$ has no model, so by the above it is inconsistent, i.e. $S \cup \{\neg t\} \vdash \perp$. By the deduction theorem $S \vdash ((\neg t) \Rightarrow \perp)$, that is, $S \vdash \neg\neg t$. Axiom 3 gives $S \vdash ((\neg\neg t) \Rightarrow t)$, so $S \vdash t$ by modus ponens.

So we must build a model of S knowing only that S is consistent. The naive attempt is to set $v(t) = 1$ exactly when $t \in S$. This fails for $S = \{p_1, p_1 \Rightarrow p_2\}$, which would make p_2 false while p_1 and $p_1 \Rightarrow p_2$ are true. The problem is that S need not contain everything it proves.

Definition 34.1.20 (Deductive Closure)

A set $S \subseteq L$ is **deductively closed** if $p \in S$ whenever $S \vdash p$. The **deductive closure** of S is the set $\{t \in L \mid S \vdash t\}$, which is deductively closed, and consistent if S is.

Passing to the deductive closure fixes the example above, but not everything: if neither p nor $\neg p$ is provable from S , our recipe makes both false, which no valuation does. Remarkably, that is the only remaining obstruction, and we can remove it by ‘swallowing up’ one of $p, \neg p$ for each p .

Theorem 34.1.21 (Model Existence Lemma)

Every consistent $S \subseteq L$ has a model.

Proof. First we note that for any consistent $S \subseteq L$ and any $p \in L$, at least one of $S \cup \{p\}$ and $S \cup \{\neg p\}$ is consistent^a.

Since L is countable (each proposition is a finite string from a countable alphabet), we may list it as $t_1, t_2, \dots$. Let $S_0 = S$, and having chosen S_n consistent, let S_{n+1} be whichever of $S_n \cup \{t_{n+1}\}$ and $S_n \cup \{\neg t_{n+1}\}$ is consistent. Set $\bar{S} = \bigcup_{n \geq 0} S_n$. Then for all $t \in L$ we have $t \in \bar{S}$ or $(\neg t) \in \bar{S}$.

The set $\bar{S}$ is consistent: a proof of $\perp$ from $\bar{S}$ is finite, so it is a proof of $\perp$ from some S_n , contradicting consistency of S_n . It is also deductively closed: if $\bar{S} \vdash t$ but $t \notin \bar{S}$, then $\neg t \in \bar{S}$ and so $\bar{S} \vdash \perp$.

So we define $v : L \rightarrow \{0, 1\}$ by

$$v(t) = \begin{cases} 1 & t \in \bar{S}, \\ 0 & \text{otherwise,} \end{cases}$$

and check that v is a valuation. Certainly $v(\perp) = 0$, since $\bar{S}$ is consistent and so $\perp \notin \bar{S}$. Now let $p, q \in L$.

Suppose $v(p) = 1$ and $v(q) = 0$, so $p \in \bar{S}$ and $q \notin \bar{S}$; we want $(p \Rightarrow q) \notin \bar{S}$. If not, then $(p \Rightarrow q) \in \bar{S}$ and $p \in \bar{S}$ give $\bar{S} \vdash q$, so $q \in \bar{S}$ as $\bar{S}$ is deductively closed, a contradiction.

Now suppose $v(q) = 1$, so $q \in \bar{S}$; we want $(p \Rightarrow q) \in \bar{S}$. Indeed $\bar{S} \vdash q$, and by axiom 1 $\bar{S} \vdash q \Rightarrow (p \Rightarrow q)$, so $\bar{S} \vdash (p \Rightarrow q)$ and hence $(p \Rightarrow q) \in \bar{S}$ by deductive closure.

Finally suppose $v(p) = 0$, so $p \notin \bar{S}$ and hence $\neg p \in \bar{S}$; we want $(p \Rightarrow q) \in \bar{S}$. It is enough to show that $\{p \Rightarrow \perp\} \vdash (p \Rightarrow q)$, which by the deduction theorem is equivalent to $\{p, p \Rightarrow \perp\} \vdash q$, and hence (after one modus ponens) to $\{\perp\} \vdash q$. But from $\perp$, axiom 1 gives $(\neg q) \Rightarrow \perp$, which is $\neg \neg q$, and then axiom 3 gives q . $\square$

^aIf not, then $S \cup \{p\} \vdash \perp$ and $S \cup \{\neg p\} \vdash \perp$. The first gives $S \vdash (p \Rightarrow \perp)$ by the deduction theorem, that is, $S \vdash (\neg p)$; feeding this into the second gives $S \vdash \perp$.

Corollary 34.1.22 (Adequacy)

Let $S \subseteq L$ and $t \in L$. Then $S \models t$ implies $S \vdash t$.

Proof. This is the argument given above, using Theorem 34.1.21. $\square$

Putting the two halves together we have our first main theorem.

Theorem 34.1.23 (Completeness Theorem for Propositional Logic)

Let $S \subseteq L$ and $t \in L$. Then $S \models t$ if and only if $S \vdash t$.

Proof. Soundness and adequacy. $\square$

The proof above assumed P , and hence L , to be countable, since we enumerated L . We will remove that restriction in Section 34.3.3, once Zorn's lemma is available.

§34.1.6 Compactness and Decidability

Directly from completeness we get some genuinely useful consequences, which are not at all obvious from the semantic definitions.

Theorem 34.1.24 (Compactness Theorem)

Let $S \subseteq L$ and $t \in L$ with $S \models t$. Then there is a finite subset $S' \subseteq S$ with $S' \models t$.

Proof. By completeness $S \vdash t$, and a proof is a finite object, so it uses only finitely many members S' of S . Then $S' \vdash t$, so $S' \models t$. $\square$

Corollary 34.1.25 (Compactness Theorem, Equivalent Form)

Let $S \subseteq L$ be such that every finite subset of S has a model. Then S has a model.

Proof. If S had no model then $S \models \perp$, so $S' \models \perp$ for some finite $S' \subseteq S$, meaning S' has no model. $\square$

It is worth pausing on how strange this is: the hypothesis only ever talks about finitely many propositions at a time, and yet it produces a single valuation making all of S true at once. Here is a typical application.

Example 34.1.26 (Compactness and Graph Colouring)

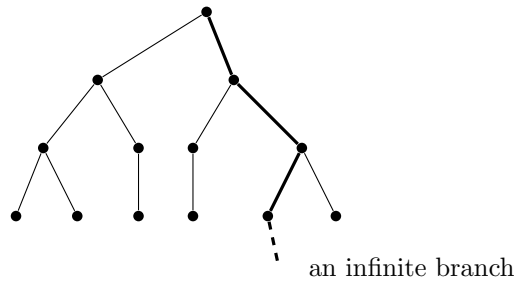
Let G be a graph, possibly infinite, such that every finite subgraph of G is k -colourable. Then G is k -colourable.

Take primitive propositions $p_{v,i}$ for v a vertex of G and $1 \leq i \leq k$, to be read as ‘ v has colour i ’. Let S consist of the propositions

- (i) $p_{v,1} \vee \cdots \vee p_{v,k}$ for each vertex v (‘every vertex has a colour’);
- (ii) $\neg(p_{v,i} \wedge p_{v,j})$ for each vertex v and $i \neq j$ (‘no vertex has two colours’);
- (iii) $\neg(p_{u,i} \wedge p_{v,i})$ for each edge uv and each i (‘adjacent vertices differ’).

A model of S is precisely a k -colouring of G . Any finite subset S' of S mentions only finitely many vertices, and a k -colouring of the (finite) subgraph they span gives a model of S' once we colour every other vertex arbitrarily. So by compactness S has a model.

The same argument, run on a tree instead of a graph, is exactly König’s lemma: an infinite, finitely branching tree has an infinite branch. The picture to keep in mind is that each finite level can be satisfied, and compactness threads these finite solutions together into an infinite one.



Finally, in propositional logic we can settle the decision problem completely.

Theorem 34.1.27 (Decidability)

Let $S \subseteq L$ be finite and $t \in L$. Then there is an algorithm to decide, in finite time, whether $S \vdash t$.

Proof. By completeness it is enough to decide whether $S \models t$, and only the finitely many primitives appearing in $S \cup \{t\}$ matter. So we simply draw the truth table, which has finitely many rows. $\square$

Searching for a proof directly would give no such algorithm, since there are infinitely many candidate proofs; it is completeness that converts the problem into a finite one. We will see in Section 34.4 that the corresponding question for predicate logic has no such easy answer, and in Section 34.7 that some closely related questions have no algorithm at all.

§34.2 Well-Orderings and Ordinals

We now jump away from logic for a while and look at some set theory. Everything in this section is really about a single idea: we want to be able to do induction and recursion along things much longer than $\mathbb{N}$, and the orderings that let us do that are the well-orderings.

§34.2.1 Well-Orderings

Definition 34.2.1 (Total Order)

A **total order** (or **linear order**) is a pair $(X, <)$ where X is a set and $<$ is a relation on X that is

- (i) *irreflexive*: for all $x \in X$, $x \not< x$;
- (ii) *transitive*: for all $x, y, z \in X$, $x < y$ and $y < z$ implies $x < z$;
- (iii) *trichotomous*: for all $x, y \in X$, exactly one of $x < y$, $y < x$, $x = y$ holds.

We use the standard notation $y > x$ for $x < y$, and $x \leq y$ for ‘ $x < y$ or $x = y$ ’. We could instead have defined a total order in terms of $\leq$ in the obvious way, with reflexivity, transitivity, antisymmetry and trichotomy.

Example 34.2.2 (Total Orders)

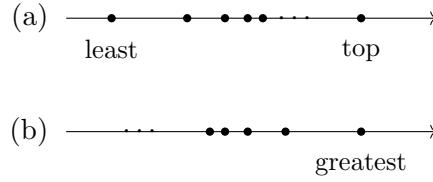
Each of $(\mathbb{N}, <)$, $(\mathbb{Z}, <)$, $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ is a total order. On the other hand $(\mathbb{N}^+, \text{‘divides’})$ is not one, as neither of 2, 3 divides the other, and nor is $(\mathcal{P}(S), \subseteq)$ once $|S| \geq 2$.

Definition 34.2.3 (Well-Ordering)

A total order $(X, <)$ is a **well-ordering** if every non-empty subset of X has a least element.

Example 34.2.4 (Well-Orderings and Non-Examples)

$(\mathbb{N}, <)$ is a well-ordering. $(\mathbb{Z}, <)$, $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are not, since $\mathbb{Z}$ itself, or $(0, 1) \subseteq \mathbb{R}$, has no least element. The set $\{1/2, 2/3, 3/4, \dots\} \subseteq \mathbb{R}$ is well-ordered (it has order type ω , in the language below), and so is $\{1/2, 2/3, \dots\} \cup \{1\}$, where the extra point sits above everything else. But $\{\dots, -1/3, -1/2, -1\} \subseteq \mathbb{R}$ is not, having no least element at all.



In (a) above every non-empty subset has a least element, and the order type is $\omega + 1$; in (b) the whole set has no least element, so this is not a well-ordering, even though it is a perfectly good total order. The difference is exactly that in (b) we can descend forever.

Proposition 34.2.5 (Descending Sequence Criterion)

A total order $(X, <)$ is a well-ordering if and only if there is no infinite strictly decreasing sequence $x_1 > x_2 > \dots$ in X .

Proof. If $x_1 > x_2 > \dots$ is such a sequence then $\{x_1, x_2, \dots\}$ has no least element. Conversely, if $S \subseteq X$ is non-empty with no least element, pick $x_1 \in S$; as x_1 is not least there is $x_2 \in S$ with $x_2 < x_1$, and continuing gives an infinite decreasing sequence. $\square$

The reverse direction of this used the axiom of choice, in that we chose each x_{n+1} from an infinite supply of candidates. This is one of only a handful of places in this section where choice appears, and we will discuss the axiom itself properly in Section 34.3.

The point of well-orderings is that they support induction.

Proposition 34.2.6 (Principle of Induction)

Let X be a well-ordering and let $S \subseteq X$ be such that whenever $y \in S$ for all $y < x$, we have $x \in S$. Then $S = X$.

Equivalently, if p is a property such that $p(y)$ holding for all $y < x$ implies $p(x)$, then $p(x)$ holds for all $x \in X$.

Proof. Suppose $S \neq X$. Then $X \setminus S$ is non-empty, so it has a least element x . Then every $y < x$ lies in S , so $x \in S$ by hypothesis, a contradiction. $\square$

Note that there is no separate base case: the case $x = \min X$ is the case where the hypothesis ' $y \in S$ for all $y < x$ ' is vacuous.

We will want to regard well-orderings that look the same as being the same.

Definition 34.2.7 (Order Isomorphism)

Two total orders X and Y are **order isomorphic** if there is a bijection $f : X \rightarrow Y$ with $x < y \iff f(x) < f(y)$. We call such an f an **isomorphism**.

For general total orders there can be many isomorphisms between two orders – consider $\mathbb{Z} \rightarrow \mathbb{Z}, x \mapsto x + n$. For well-orderings this cannot happen, which is the first sign of how rigid they are.

Proposition 34.2.8 (Uniqueness of Isomorphisms)

Let X, Y be isomorphic well-orderings. Then there is a unique isomorphism between them.

Proof. Let $f, g : X \rightarrow Y$ be isomorphisms. We show $f(x) = g(x)$ for all x by induction on x , using Proposition 34.2.6.

So suppose $f(y) = g(y)$ for all $y < x$, and let $S = \{f(y) \mid y < x\}$, which by assumption equals $\{g(y) \mid y < x\}$. As f is an order isomorphism, $f(x)$ is not in S , so $Y \setminus S$ is non-empty and has a least element a . Now every element of Y below $f(x)$ is of the form $f(y)$ with $y < x$, so lies in S ; hence $f(x) = a$. The identical argument gives $g(x) = a$, so $f(x) = g(x)$. $\square$

§34.2.2 Initial Segments and Recursion

Given an ordered set we can chop off the end and keep the beginning. What we are left with is an initial segment.

Definition 34.2.9 (Initial Segment)

A subset I of a totally ordered set X is an **initial segment** if $x \in I$ implies $y \in I$ for all $y < x$.

For any $x \in X$, the set $I_x = \{y \in X \mid y < x\}$ is an initial segment. Not every initial segment is of this form in a general total order: in $\mathbb{R}$ the set $\{x \mid x \leq 3\}$ is an initial segment which is not any I_x , and in $\mathbb{Q}$ so is $\{x \mid x \leq 0 \text{ or } x^2 < 2\}$. In a well-ordering, however, this is the only shape available.

Proposition 34.2.10 (Initial Segments of Well-Orderings)

Every proper initial segment of a well-ordered set X is of the form I_x for some $x \in X$.

Proof. Let $I \subsetneq X$ be an initial segment and let $x = \min(X \setminus I)$. If $y < x$ then $y \in I$ by minimality of x , and if $y \in I$ then $y < x$, since $y \geq x$ would put $x \in I$. So $I = I_x$. $\square$

We would now like to show that every subset of a well-ordering X is isomorphic to an initial segment of X . To construct the isomorphism we need to be able to define functions by recursion along X , which is the following theorem. Here $\mathcal{P}(X \times Y)$ is being used as ‘the set of all partial functions and more’, and G should be thought of as a rule telling us what to do next given everything we have done so far.

Theorem 34.2.11 (Definition by Recursion)

Let X be a well-ordering, let Y be a set, and let $G : \mathcal{P}(X \times Y) \rightarrow Y$ be any function. Then there is a unique function $f : X \rightarrow Y$ such that

$$f(x) = G(f|_{I_x}) \quad \text{for all } x \in X.$$

Proof. Say that h is an *attempt* if $h : I \rightarrow Y$ for some initial segment I of X , and $h(x) = G(h|_{I_x})$ for all $x \in I$.

If h, h' are attempts both defined at x , then $h(x) = h'(x)$, by induction on x : if they agree below x then $h|_{I_x} = h'|_{I_x}$, so $h(x) = G(h|_{I_x}) = G(h'|_{I_x}) = h'(x)$.

Next, for every x there *is* an attempt defined at x , again by induction on x . Indeed, suppose for each $y < x$ there is an attempt h_y defined at y . Let $h = \bigcup_{y < x} h_y$; by the previous paragraph the h_y agree wherever two of them are defined, so h is a function, and it is an attempt with domain I_x . Then $h' = h \cup \{(x, G(h))\}$ is an attempt defined at x .

So we may define $f : X \rightarrow Y$ by letting $f(x)$ be the common value $h(x)$ over all attempts h defined at x . This is well-defined by the first paragraph, and satisfies the recursion by construction.

For uniqueness, if f, f' both satisfy the recursion and agree below x , then $f(x) = G(f|_{I_x}) = G(f'|_{I_x}) = f'(x)$, so $f = f'$ by induction. $\square$

Before proving the result we were after, here is a small lemma that we will use several times.

Lemma 34.2.12

Let X be a well-ordering and $f : X \rightarrow X$ order-preserving. Then $f(x) \geq x$ for all $x \in X$.

Proof. If not, take the least x with $f(x) < x$. Applying f gives $f(f(x)) < f(x)$, so $f(x)$ is a smaller counterexample, contradicting minimality. $\square$

Corollary 34.2.13

A well-ordering X is never isomorphic to a proper initial segment of itself.

Proof. Suppose $f : X \rightarrow I_x$ is an isomorphism, where I_x is a proper initial segment. Regarding f as an order-preserving map $X \rightarrow X$, Lemma 34.2.12 gives $f(x) \geq x$. But $f(x) \in I_x$, so $f(x) < x$, a contradiction. $\square$

Now we can prove our result, by recursively sending the least unused element to the least unused element.

Proposition 34.2.14 (Subset Collapse)

Any subset Y of a well-ordering X is isomorphic to a unique initial segment of X .

Proof. If $f : Y \rightarrow X$ is to be an order-preserving bijection onto an initial segment of X , then it must send each x to the least element not yet used, that is,

$$f(x) = \min(X \setminus \{f(y) \mid y \in Y, y < x\}).$$

So define f by this recursion; it is well-defined, since one shows by induction that

$f(y) \leq y$ for all y , and hence x itself lies in the set we are minimising over, so that set is non-empty.

This f is order-preserving: if $y < x$ then $f(y)$ is the minimum of a set containing everything $f(x)$ is minimised over, and $f(y)$ itself is excluded from the latter, so $f(y) < f(x)$. Its image is an initial segment: if $z < f(x)$ then z is not in the set we minimised over to get $f(x)$, so $z = f(y)$ for some $y < x$. Finally, uniqueness of the initial segment follows from Corollary 34.2.13: two distinct initial segments of X are never isomorphic, since one is a proper initial segment of the other. $\square$

§34.2.3 Comparing Well-Orderings

We now know enough to compare any two well-orderings.

Definition 34.2.15 (Comparing Well-Orderings)

For well-orderings X, Y we write $X \leq Y$ if X is isomorphic to an initial segment of Y , and $X < Y$ if moreover X is not isomorphic to Y (equivalently, if X is isomorphic to a *proper* initial segment of Y).

By subset collapse, $X \leq Y$ if and only if X is isomorphic to *some subset* of Y , which is often the easier condition to check.

Proposition 34.2.16 (Trichotomy for Well-Orderings)

Let X and Y be well-orderings. Then $X \leq Y$ or $Y \leq X$; and if both, then X and Y are isomorphic.

Proof. Consider $f : X \rightarrow Y$ defined by recursion by $f(x) = \min(Y \setminus \{f(y) \mid y < x\})$. There are two cases. If f is everywhere defined, then exactly as in Proposition 34.2.14 it is an isomorphism from X onto an initial segment of Y , so $X \leq Y$. Otherwise there is a least x at which the recipe fails, meaning $\{f(y) \mid y < x\} = Y$; but then f restricted to I_x is an isomorphism $I_x \rightarrow Y$, so $Y \leq X$.

Now suppose both $X \leq Y$ and $Y \leq X$, via isomorphisms f onto an initial segment of Y and g onto an initial segment of X . Then $g \circ f$ is an isomorphism of X onto an initial segment of X , which by Corollary 34.2.13 must be all of X . So $g \circ f$ is onto X , forcing g to be onto X , and hence g (and likewise f) is an isomorphism. $\square$

§34.2.4 Constructing Larger Well-Orderings

Given some well-orderings, there are two natural ways to build a longer one: add an element at the top, or stack a nested family of them.

Definition 34.2.17 (Successor)

Given a well-ordering X , choose some $x \notin X$ and well-order $X \cup \{x\}$ by setting $y < x$ for all $y \in X$. This is the **successor** of X , written X^+ .

Clearly X^+ is a well-ordering (any non-empty subset either meets X , and takes its least element there, or is $\{x\}$) and $X < X^+$, since $X = I_x$ inside X^+ .

Definition 34.2.18 (Extensions and Nested Families)

Given well-orderings $(X, <_X)$ and $(Y, <_Y)$, we say Y **extends** X if X is an initial segment of Y and $<_X, <_Y$ agree where both are defined.

A family of well-orderings $\{X_i \mid i \in I\}$ is **nested** if for all i, j , one of X_i, X_j extends the other.

Proposition 34.2.19 (Unions of Nested Well-Orderings)

Let $\{X_i \mid i \in I\}$ be a nested family of well-orderings. Then there is a well-ordering X with $X_i \leq X$ for all i .

Proof. Let $X = \bigcup_i X_i$ with the ordering $<_X = \bigcup_i <_i$; that is, $x < y$ in X if $x, y \in X_i$ and $x <_i y$ for some i . Nestedness makes this a well-defined total order.

Given non-empty $S \subseteq X$, we have $S \cap X_i \neq \emptyset$ for some i ; let x be its least element under $<_i$. Then x is least in all of S : any $y \in S$ with $y < x$ would lie in X_i , since X_i is an initial segment of X by nestedness, and then $y \in S \cap X_i$ contradicts minimality. So X is a well-ordering, and each X_i is an initial segment of it. $\square$

§34.2.5 Ordinals

It is inconvenient to keep saying ‘a well-ordering, up to isomorphism’. So we give these isomorphism classes a name.

Definition 34.2.20 (Ordinal)

An **ordinal** is a well-ordered set, where we regard two ordinals as equal if they are isomorphic.

This is a definition to be taken in a slightly informal spirit – the isomorphism class of a well-ordering is far too big to be a set. In Section 34.5 we will give a genuine definition inside ZF (an ordinal is a transitive set totally ordered by $\in$) and check it does the same job. For now, nothing we say will depend on the difference.

Definition 34.2.21 (Order Type)

If a well-ordering X corresponds to the ordinal α , we say X has **order type** α , and write $\text{otp}(X) = \alpha$.

The order type of the unique well-ordering on a set of $k \in \mathbb{N}$ points is called k , and the order type of $(\mathbb{N}, <)$ is called ω .

Example 34.2.22 (Order Types)

In $\mathbb{R}$, the subset $\{-2, 3, \pi, 5\}$ has order type 4, the subset $\{1/2, 2/3, 3/4, \dots\}$ has order type ω , and $\{1/2, 2/3, \dots\} \cup \{1\}$ has order type $\omega + 1$ (a copy of ω with one extra point on top).

Example 34.2.23 (Countable Ordinals Live Inside $\mathbb{Q}$)

Every countable ordinal is the order type of a subset of $\mathbb{Q}$. Indeed, if α is countable then I_α is a countable well-ordering, and one may embed it into $\mathbb{Q}$ by recursion: having placed I_β inside $\mathbb{Q}$, place β above everything used so far if there is room, and otherwise squeeze it into a gap, which density allows. So the enormous ordinals of the previous list all fit inside the rationals – what changes as we ascend is not the size of the set but the shape of the ordering.

For ordinals α, β we write $\alpha \leq \beta$ if $X \leq Y$ for some (equivalently, any) X of order type α and Y of order type β ; this is well-defined since any two representatives are isomorphic. Proposition 34.2.16 says that any two ordinals are comparable, and that $\alpha \leq \beta \leq \alpha$ implies $\alpha = \beta$. We also write α^+ for the order type of X^+ .

Proposition 34.2.24 (I_α Has Order Type α)

For any ordinal α , the ordinals $\beta < \alpha$ form a set $I_\alpha = \{\beta \mid \beta < \alpha\}$, well-ordered by $<$, of order type α .

Proof. Let X be a well-ordering with $\text{otp}(X) = \alpha$. The well-orderings which are $< X$ are precisely, up to isomorphism, the proper initial segments of X , which by Proposition 34.2.10 are exactly the I_x for $x \in X$. Distinct x give non-isomorphic I_x by Corollary 34.2.13, so $x \mapsto \text{otp}(I_x)$ is an order-preserving bijection from X to I_α . $\square$

Proposition 34.2.25 (The Ordinals are Well-Ordered)

Every non-empty set S of ordinals has a least element.

Proof. Choose $\alpha \in S$. If α is least in S we are done. Otherwise $S \cap I_\alpha$ is non-empty, and I_α is well-ordered by Proposition 34.2.24, so it has a least element β . Any $\gamma \in S$ has $\gamma \geq \alpha > \beta$ or $\gamma \in I_\alpha$, in which case $\gamma \geq \beta$. So β is least in S . $\square$

So the ordinals behave in every respect like a well-ordering. They cannot literally be one, however.

Theorem 34.2.26 (Burali-Forti Paradox)

The ordinals do not form a set.

Proof. Suppose X were the set of all ordinals. By Proposition 34.2.25 it is well-ordered, so it has an order type, say α . But then X is isomorphic to I_α by Proposition 34.2.24, which is a proper initial segment of X (it omits α), contradicting Corollary 34.2.13. $\square$

This is not a paradox in the sense of an inconsistency – it is a theorem, and a useful one. We will write On for the class of all ordinals, and treat statements about it as abbreviations, exactly as we will do for classes in general in Section 34.5.

Definition 34.2.27 (Supremum)

Let $S = \{\alpha_i \mid i \in I\}$ be a set of ordinals. Then $\sup S$ is the least upper bound of S .

This exists: applying Proposition 34.2.19 to the nested family $\{I_{\alpha_i}\}$ gives an upper bound α , and then the set of upper bounds which are $\leq \alpha$ is a non-empty subset of I_α^+ , so has a least element by Proposition 34.2.25. For example $\sup\{2, 4, 6, \dots\} = \omega$, and $\sup\{\alpha\} = \alpha$.

§34.2.6 Some Ordinals

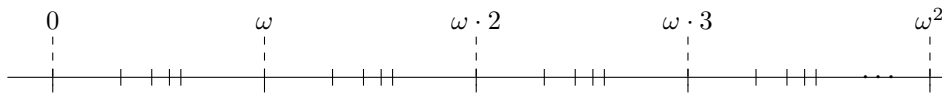
We know $0, 1, 2, \dots$ and ω . Writing $\alpha+1$ for α^+ , the successor and supremum operations already let us generate a great many ordinals:

$$\omega + 1, \omega + 2, \dots, \omega + \omega = \omega \cdot 2, \omega \cdot 2 + 1, \dots, \omega \cdot 3, \dots, \omega \cdot \omega = \omega^2, \dots$$

where each ‘ $\dots$ ’ is resolved by taking a supremum, for instance $\omega \cdot 2 = \sup\{\omega, \omega + 1, \omega + 2, \dots\}$ and $\omega^2 = \sup\{\omega, \omega \cdot 2, \omega \cdot 3, \dots\}$. The notation is justified in Section 34.2.9. Carrying on:

0	$\omega \cdot 2 + 1$	$\omega^2 + 1$	$\omega^2 \cdot 3$	$\omega^{\omega+2}$	$\varepsilon_0 + 1$
1	$\omega \cdot 2 + 2$	$\omega^2 + 2$	$\omega^2 \cdot 4$	$\vdots$	$\vdots$
2	$\omega \cdot 2 + 3$	$\omega^2 + 3$	$\omega^2 \cdot 5$	$\omega^{\omega \cdot 2}$	$\varepsilon_0 \cdot 2$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
ω	$\omega \cdot 3$	$\omega^2 + \omega$	ω^3	ω^{ω^2}	ε_0^2
$\omega + 1$	$\omega \cdot 4$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\omega + 2$	$\omega \cdot 5$	$\omega^2 + \omega \cdot 2$	ω^ω	ω^{ω^ω}	$\varepsilon_0^{\varepsilon_0}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\omega + \omega$ $= \omega \cdot 2$	$\omega \cdot \omega$ $= \omega^2$	$\omega^2 \cdot 2$	$\omega^{\omega+1}$	$\omega^{\omega^{\dots}}$ $= \varepsilon_0$	$\varepsilon_0^{\varepsilon_0^{\dots}}$ $= \varepsilon_1$

It helps to have a picture. Drawing each ordinal as a point on a line, with the tick marks bunching up as we approach a limit, the beginning of the list looks like this.



Every ordinal in the table above is *countable*: at each step we either add one element or take a countable supremum of countable ordinals, and a countable union of countable sets is countable. It is natural to ask whether we ever escape. Since a well-ordering of a set X of order type α is the same thing as an injection of X into I_α hitting an initial segment, this is essentially the question of whether we can well-order $\mathbb{R}$. The answer is yes, and remarkably we can prove it without ever writing such an ordering down.

§34.2.7 Uncountable Ordinals and Hartogs' Lemma**Theorem 34.2.28 (Existence of an Uncountable Ordinal)**

There is an uncountable ordinal.

Proof. Let

$$A = \{R \in \mathcal{P}(\omega \times \omega) \mid R \text{ is a well-ordering of a subset of } \omega\},$$

which is a set by separation from $\mathcal{P}(\omega \times \omega)$. Then $B = \{\text{otp}(R) \mid R \in A\}$ is a set (it is the image of A under a function-class, using replacement, which we discuss in Section 34.5), and B is precisely the set of all countable ordinals – every countable ordinal is the order type of a well-ordering of a subset of ω , and conversely.

Let $\omega_1 = \sup B$. If ω_1 were countable then $\omega_1 \in B$, and so $\omega_1^+ \in B$ as well (a successor of a countable ordinal is countable), giving $\omega_1^+ \leq \sup B = \omega_1$, which is absurd. So ω_1 is uncountable. $\square$

Alternatively, and more slickly: B cannot be all of the ordinals, since the ordinals do not form a set, so some ordinal is uncountable.

By construction ω_1 is the *least* uncountable ordinal, and it has some remarkable properties. Every proper initial segment I_α of ω_1 is countable, while ω_1 itself is not; so ω_1 is uncountable but is ‘built out of’ countable pieces. Also, every sequence $\alpha_1, \alpha_2, \dots$ of ordinals below ω_1 is bounded below ω_1 , namely by $\sup\{\alpha_1, \alpha_2, \dots\}$, which is countable as a countable union of countable sets. So ω_1 cannot be reached from below by any sequence – it is very far from being ω .

The same argument, run with an arbitrary set in place of ω , gives the following, which is the workhorse of the next section.

Theorem 34.2.29 (Hartogs’ Lemma)

For every set X there is an ordinal α that does not inject into X . The least such ordinal is written $\gamma(X)$, read ‘**Hartogs’ ordinal** of X ’.

Proof. Let A be the set of pairs (Y, R) with $Y \subseteq X$ and R a well-ordering of Y ; this is a set, being a subset of $\mathcal{P}(X) \times \mathcal{P}(X \times X)$. Let $B = \{\text{otp}(R) \mid (Y, R) \in A\}$, a set by replacement, and let $\gamma = \sup\{\beta^+ \mid \beta \in B\}$.

If γ injected into X , say with image Y , then transporting the ordering of I_γ across gives a well-ordering of $Y \subseteq X$ of order type γ , so $\gamma \in B$ and hence $\gamma^+ \leq \gamma$, a contradiction. So γ does not inject into X , and by Proposition 34.2.25 there is a least such ordinal. $\square$

So $\gamma(\omega) = \omega_1$. Note that $\gamma(X)$ is a genuine ‘size barrier’ for X : no ordinal $\geq \gamma(X)$ injects into X either, since it has $\gamma(X)$ as an initial segment.

§34.2.8 Successors and Limits

Ordinals come in two flavours.

Definition 34.2.30 (Limits and Successors)

An ordinal α is a **successor** if $\alpha = \beta^+$ for some ordinal β . Otherwise α is a **limit**.

An ordinal (viewed as a well-ordered set I_α) has a greatest element if and only if it is a successor. So α is a limit precisely when it has no greatest element, and then $\alpha = \sup I_\alpha$.

We typically write λ for a limit ordinal.

Example 34.2.31

$5 = 4^+$ is a successor, and $\omega + 2 = (\omega^+)^+$ is a successor. Both ω and $\omega \cdot 2$ are limits, as is 0 (vacuously – it is not a successor of anything). This is why the clause for limits in the recursive definitions below always says ‘non-zero limit’.

Consequently, when defining something by recursion on the ordinals we give three clauses: the value at 0, the value at α^+ in terms of the value at α , and the value at a non-zero limit λ in terms of the values below λ . This is **transfinite recursion**, and it is justified by Theorem 34.2.11 applied to I_β for each β (we cannot recurse on ‘all the ordinals’ at once, since they do not form a set, so officially we define the function on I_β for each β and use uniqueness to glue).

§34.2.9 Ordinal Arithmetic

We now make formal sense of the notation $\omega + \omega$, $\omega \cdot 2$ and so on used above.

Definition 34.2.32 (Ordinal Addition, Inductive)

We define $\alpha + \beta$ by recursion on β , keeping α fixed:

$$\begin{aligned}\alpha + 0 &= \alpha, \\ \alpha + \beta^+ &= (\alpha + \beta)^+, \\ \alpha + \lambda &= \sup\{\alpha + \gamma \mid \gamma < \lambda\} \quad \text{for } \lambda \text{ a non-zero limit.}\end{aligned}$$

Proposition 34.2.33 (Addition is Not Commutative)

We have $1 + \omega \neq \omega + 1$.

Proof. On one hand $\omega + 1 = \omega + 0^+ = (\omega + 0)^+ = \omega^+$. On the other hand, ω is a non-zero limit, so $1 + \omega = \sup\{1 + n \mid n < \omega\} = \sup\{1, 2, 3, \dots\} = \omega$. And $\omega < \omega^+$. $\square$

The asymmetry comes from our decision to recurse on the right-hand argument. Addition is, however, associative.

Proposition 34.2.34 (Addition is Associative)

For all ordinals α, β, γ we have $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$.

Proof. We induct on γ , keeping α, β fixed. If $\gamma = 0$ then $\alpha + (\beta + 0) = \alpha + \beta = (\alpha + \beta) + 0$.

If $\gamma = \delta^+$ is a successor, then

$$\alpha + (\beta + \delta^+) = \alpha + (\beta + \delta)^+ = [\alpha + (\beta + \delta)]^+ = [(\alpha + \beta) + \delta]^+ = (\alpha + \beta) + \delta^+,$$

using the induction hypothesis in the third equality.

If $\gamma = \lambda$ is a non-zero limit, then

$$(\alpha + \beta) + \lambda = \sup\{(\alpha + \beta) + \gamma \mid \gamma < \lambda\} = \sup\{\alpha + (\beta + \gamma) \mid \gamma < \lambda\}.$$

We claim that $\beta + \lambda$ is a limit. Indeed $\beta + \lambda = \sup\{\beta + \gamma \mid \gamma < \lambda\}$, and since λ is a limit, for every $\gamma < \lambda$ there is γ' with $\gamma < \gamma' < \lambda$, whence $\beta + \gamma < \beta + \gamma'$. So this set has no greatest element.

Hence $\alpha + (\beta + \lambda) = \sup\{\alpha + \delta \mid \delta < \beta + \lambda\}$, and we must show

$$\sup\{\alpha + \delta \mid \delta < \beta + \lambda\} = \sup\{\alpha + (\beta + \gamma) \mid \gamma < \lambda\}.$$

Each element of the right-hand set is an element of the left-hand set (as $\beta + \gamma < \beta + \lambda$), so the left supremum is $\geq$ the right. Conversely, given $\delta < \beta + \lambda = \sup\{\beta + \gamma \mid \gamma < \lambda\}$ we have $\delta < \beta + \gamma$ for some $\gamma < \lambda$, so $\alpha + \delta \leq \alpha + (\beta + \gamma)$. So the left supremum is $\leq$ the right, and they are equal. $\square$

There is an alternative definition of addition in terms of actual well-orderings, obtained by writing out all the elements of α and then all the elements of β after them:

$$\alpha + \beta = \underbrace{\hspace{1.5cm}}_{\alpha} \underbrace{\hspace{1.5cm}}_{\beta}$$

Definition 34.2.35 (Ordinal Addition, Synthetic)

$\alpha + \beta$ is the order type of $\alpha \sqcup \beta$ (say $\alpha \times \{0\} \cup \beta \times \{1\}$) with all of α before all of β .

From this picture, associativity is immediate, and it is obvious why $1 + \omega = \omega$ (a dot followed by ω dots is just ω dots) while $\omega + 1 \neq \omega$ (ω dots followed by a dot has a greatest element). Reassuringly, the two definitions agree.

Proposition 34.2.36 (The Two Definitions of Addition Agree)

The inductive and synthetic definitions of addition coincide.

Proof. We write $+$ for the inductive definition and $+'$ for the synthetic one, and induct on β with α fixed.

(i) *Zero:* $\alpha + 0 = \alpha = \alpha +' 0$.

(ii) *Successor:* $\alpha + \beta^+ = (\alpha + \beta)^+ = (\alpha +' \beta)^+$, which is the order type of

$$\underbrace{\hspace{1.5cm}}_{\alpha} \underbrace{\hspace{1.5cm}}_{\beta} \underbrace{\hspace{0.5cm}}_{\bullet}$$

and that is precisely $\alpha +' \beta^+$.

(iii) *Non-zero limit:* $\alpha + \lambda = \sup\{\alpha + \gamma \mid \gamma < \lambda\} = \sup\{\alpha +' \gamma \mid \gamma < \lambda\} = \alpha +' \lambda$, where the last step is because the well-orderings $\alpha \sqcup \gamma$ for $\gamma < \lambda$ are nested, so their supremum is their union, which is $\alpha \sqcup \lambda$. $\square$

Multiplication works in exactly the same way.

Definition 34.2.37 (Ordinal Multiplication, Inductive)

We define $\alpha \cdot \beta$ by recursion on β , keeping α fixed:

$$\begin{aligned}\alpha \cdot 0 &= 0, \\ \alpha \cdot \beta^+ &= \alpha \cdot \beta + \alpha, \\ \alpha \cdot \lambda &= \sup\{\alpha \cdot \gamma \mid \gamma < \lambda\} \quad \text{for } \lambda \text{ a non-zero limit.}\end{aligned}$$

Definition 34.2.38 (Ordinal Multiplication, Synthetic)

$\alpha \cdot \beta$ is the order type of $\alpha \times \beta$, ordered by $(x, y) < (x', y')$ if $y < y'$, or if $y = y'$ and $x < x'$.

That is, β copies of α laid end to end:

$$\alpha \cdot \beta = \underbrace{\alpha \quad \alpha \quad \cdots \quad \alpha}_{\beta \text{ copies}}.$$

As with addition, the two definitions agree, by an induction identical in shape to the one above. Multiplication is associative and again not commutative, and the synthetic picture shows why very clearly.

$$\begin{array}{l} \omega \cdot 2: \quad \underbrace{\quad \quad \quad}_{\omega} \quad \underbrace{\quad \quad \quad}_{\omega} \\ 2 \cdot \omega: \quad \underbrace{\quad \quad}_{2} \quad \underbrace{\quad \quad}_{2} \quad \underbrace{\quad \quad}_{2} \quad \cdots \end{array}$$

Reading the top line, $\omega \cdot 2 = \omega + \omega$ is genuinely longer than ω ; reading the bottom line, $2 \cdot \omega$ is ω copies of a two-element set, which is just ω . So $2 \cdot \omega = \omega \neq \omega \cdot 2$.

Definition 34.2.39 (Ordinal Exponentiation)

We define α^β by recursion on β :

$$\begin{aligned}\alpha^0 &= 1, \\ \alpha^{\beta^+} &= \alpha^\beta \cdot \alpha, \\ \alpha^\lambda &= \sup\{\alpha^\gamma \mid \gamma < \lambda\} \quad \text{for } \lambda \text{ a non-zero limit.}\end{aligned}$$

Example 34.2.40

$\omega^1 = \omega^0 \cdot \omega = \omega$ and $\omega^2 = \omega \cdot \omega$, as the notation in our table suggested. But $2^\omega = \sup\{2^n \mid n < \omega\} = \sup\{1, 2, 4, 8, \dots\} = \omega$, which is countable. So ordinal exponentiation is emphatically *not* cardinal exponentiation – we will meet the latter in Section 34.6, where $2^{\aleph_0}$ is uncountable.

§34.2.10 Normal Functions and ε -Numbers

The definitions above all have the same shape: increasing, and continuous at limits. This is worth isolating.

Definition 34.2.41 (Normal Function)

A function $f : \text{On} \rightarrow \text{On}$ is **normal** if $f(\alpha) < f(\alpha^+)$ for every α , and $f(\lambda) = \sup\{f(\gamma) \mid \gamma < \lambda\}$ for every non-zero limit λ .

For example $\alpha \mapsto \beta + \alpha$ is normal for any β , and $\alpha \mapsto \beta^\alpha$ is normal for any $\beta > 1$; both are immediate from the recursive definitions.

Lemma 34.2.42

A normal function f is strictly increasing, and satisfies $f(\alpha) \geq \alpha$ for all α .

Proof. For the first part, fix α and induct on $\beta > \alpha$. If $\beta = \alpha^+$ this is the definition. If $\beta = \gamma^+$ with $\gamma > \alpha$ then $f(\alpha) < f(\gamma) < f(\gamma^+)$ by induction. If β is a limit then $f(\beta) = \sup\{f(\gamma) \mid \gamma < \beta\} \geq f(\alpha^+) > f(\alpha)$, as $\alpha^+ < \beta$.

For the second, induct on α . It is clear for 0. For successors, $f(\alpha^+) > f(\alpha) \geq \alpha$ gives $f(\alpha^+) \geq \alpha^+$. For a non-zero limit λ , $f(\lambda) = \sup\{f(\gamma) \mid \gamma < \lambda\} \geq \sup\{\gamma \mid \gamma < \lambda\} = \lambda$. $\square$

Lemma 34.2.43 (Fixed Point Lemma)

Let f be a normal function. Then for every ordinal α there is $\beta \geq \alpha$ with $f(\beta) = \beta$.

Proof. Let $\beta = \sup\{\alpha, f(\alpha), f(f(\alpha)), \dots\}$. If the sequence is eventually constant we have found a fixed point. Otherwise the sequence is strictly increasing, so β is a limit and

$$f(\beta) = \sup\{f(\gamma) \mid \gamma < \beta\} = \sup\{f(\alpha), f(f(\alpha)), \dots\} = \beta,$$

using Lemma 34.2.42 for the middle equality. In either case $\beta \geq \alpha$. $\square$

Lemma 34.2.44 (Division for Normal Functions)

Let f be a normal function and α an ordinal with $f(0) \leq \alpha$. Then there is a greatest γ with $f(\gamma) \leq \alpha$.

Proof. Let $\gamma = \sup\{\beta \mid f(\beta) \leq \alpha\}$, which is a supremum of a set of ordinals since all such β satisfy $\beta \leq f(\beta) \leq \alpha$ by Lemma 34.2.42. If the set has a greatest element we are done. Otherwise γ is a limit and $f(\gamma) = \sup\{f(\beta) \mid \beta < \gamma\} \leq \alpha$, so γ itself is the greatest such ordinal. $\square$

Applying this to the normal function $\alpha \mapsto \omega^\alpha$ gives the **ε -numbers**: the ordinals ε with $\omega^\varepsilon = \varepsilon$. The least of them is

$$\varepsilon_0 = \sup\{\omega, \omega^\omega, \omega^{\omega^\omega}, \dots\},$$

which is where the fifth column of our table was heading. There is nothing special about ε_0 – since the supremum of a set of fixed points is again a fixed point, the fixed points themselves are enumerated by a normal function $\alpha \mapsto \varepsilon_\alpha$, which then has its own fixed points, and so on. All of these ordinals are still countable, and so are dwarfed by ω_1 .

§34.2.11 Further Ordinal Arithmetic

We record a few more facts about ordinal arithmetic, all proved by inductions of the kind we have already seen, and finish with a normal form for ordinals.

Proposition 34.2.45 (Strict Monotonicity on the Right)

Let α, β, γ be ordinals with $\beta < \gamma$. Then $\alpha + \beta < \alpha + \gamma$. Consequently $\alpha + \beta = \alpha + \gamma$ implies $\beta = \gamma$.

Proof. We induct on $\gamma > \beta$. If $\gamma = \delta^+$ then $\beta \leq \delta$, so $\alpha + \beta \leq \alpha + \delta < (\alpha + \delta)^+ = \alpha + \gamma$, using the induction hypothesis if $\beta < \delta$. If γ is a limit then $\beta^+ < \gamma$, so $\alpha + \beta < \alpha + \beta^+ \leq \sup\{\alpha + \delta \mid \delta < \gamma\} = \alpha + \gamma$.

The cancellation statement follows since $\beta \neq \gamma$ would give $\alpha + \beta \neq \alpha + \gamma$ by trichotomy. $\square$

Monotonicity on the *left* is false: $1 < 2$, but $1 + \omega = 2 + \omega = \omega$. So we may cancel on the left but not on the right.

Proposition 34.2.46 (Ordinal Subtraction)

Let $\alpha \leq \beta$. Then there is a unique ordinal γ with $\alpha + \gamma = \beta$.

Proof. Take a well-ordering Y of order type β . Since $\alpha \leq \beta$, there is an initial segment I of Y of order type α , and we let $\gamma = \text{otp}(Y \setminus I)$. By the synthetic definition of addition, $\alpha + \gamma = \text{otp}(Y) = \beta$. Uniqueness is Proposition 34.2.45. $\square$

Proposition 34.2.47 (Distributivity)

For all ordinals, $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$.

Proof. Use the synthetic definitions: $\alpha \cdot (\beta + \gamma)$ is $\beta + \gamma$ copies of α laid end to end, which is β copies followed by γ copies. $\square$

Distributivity on the other side fails, and the smallest counterexample is a good test of the definitions: $(1 + 1) \cdot \omega = 2 \cdot \omega = \omega$, whereas $1 \cdot \omega + 1 \cdot \omega = \omega + \omega = \omega \cdot 2$.

Proposition 34.2.48 (Division Algorithm)

Let α be an ordinal and $\beta > 0$. Then there are unique ordinals γ and δ with

$$\alpha = \beta \cdot \gamma + \delta, \quad \delta < \beta.$$

Proof. The map $\gamma \mapsto \beta \cdot \gamma$ is normal, so by Lemma 34.2.44 there is a greatest γ with $\beta \cdot \gamma \leq \alpha$. By subtraction there is a unique δ with $\beta \cdot \gamma + \delta = \alpha$, and $\delta < \beta$, since $\delta \geq \beta$ would give $\beta \cdot \gamma^+ = \beta \cdot \gamma + \beta \leq \alpha$, contradicting maximality of γ . Uniqueness follows from that of γ together with Proposition 34.2.45. $\square$

Applying this repeatedly with β a power of ω gives a normal form, exactly analogous to writing a natural number in base 10.

Theorem 34.2.49 (Cantor Normal Form)

Every ordinal $\alpha > 0$ can be written uniquely as

$$\alpha = \omega^{\beta_1} \cdot n_1 + \omega^{\beta_2} \cdot n_2 + \cdots + \omega^{\beta_k} \cdot n_k$$

with $k \geq 1$, ordinals $\beta_1 > \beta_2 > \cdots > \beta_k$ and positive integers $n_1, \dots, n_k$.

Proof sketch. Since $\alpha \mapsto \omega^\alpha$ is normal, Lemma 34.2.44 gives a greatest β_1 with $\omega^{\beta_1} \leq \alpha$. Dividing by ω^{β_1} using Proposition 34.2.48 gives $\alpha = \omega^{\beta_1} \cdot \gamma + \delta$ with $\delta < \omega^{\beta_1}$, and maximality of β_1 forces γ to be a positive integer n_1 . Now $\delta < \alpha$, so we may repeat; the process terminates because the ordinals δ produced are strictly decreasing, and there is no infinite strictly decreasing sequence of ordinals. Uniqueness follows by induction, using the fact that $\omega^\beta \cdot n < \omega^{\beta'}$ whenever $\beta < \beta'$ and $n < \omega$. $\square$

Example 34.2.50 (Arithmetic in Normal Form)

Let us compute $(\omega^2 + \omega \cdot 3) + (\omega \cdot 2 + 5)$. Addition is associative, and $\omega \cdot 3 + \omega \cdot 2 = \omega \cdot 5$ by distributivity, so this is $\omega^2 + \omega \cdot 5 + 5$. On the other hand $(\omega + 1) \cdot \omega = \sup\{(\omega + 1) \cdot n \mid n < \omega\} = \sup\{\omega, \omega \cdot 2 + 1, \omega \cdot 3 + 2, \dots\} = \omega^2$, so the 1 is entirely swallowed. In general, when adding ordinals in normal form, every term of the left-hand ordinal of degree less than the leading degree of the right-hand one simply disappears.

Example 34.2.51

$\omega^2 \cdot 3 + \omega \cdot 5 + 7$ is in Cantor normal form. On the other hand ε_0 is not expressible in this way in terms of *smaller* ordinals: its normal form is ω^{ε_0} , which is the statement that ε_0 is a fixed point of $\alpha \mapsto \omega^\alpha$. So the ε -numbers are precisely the ordinals which the normal form fails to simplify.

§34.3 Posets and Zorn's Lemma

Well-orderings are very rigid objects. In this section we look at the opposite extreme – orderings where elements are allowed to be incomparable – and arrive at Zorn's lemma, which is the form in which the axiom of choice is used in practice throughout mathematics.

§34.3.1 Partial Orders and Hasse Diagrams

Definition 34.3.1 (Poset)

A **partially ordered set**, or **poset**, is a pair $(X, \leq)$ where X is a set and $\leq$ is a relation on X that is reflexive, transitive and antisymmetric (if $x \leq y$ and $y \leq x$ then $x = y$). We write $x < y$ if $x \leq y$ and $x \neq y$; in terms of $<$, a poset is exactly an irreflexive transitive relation.

Example 34.3.2 (Posets)

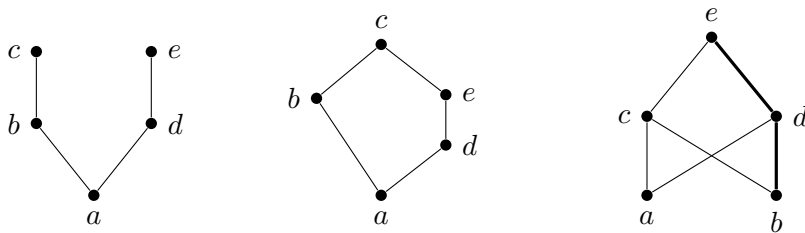
- (i) Any total order, for instance $(\mathbb{R}, \leq)$ or any well-ordering.
- (ii) $(\mathbb{N}^+, \text{'divides'})$, in which 2 and 3 are incomparable.
- (iii) For any set S , the power set $\mathcal{P}(S)$ ordered by $A \leq B$ if $A \subseteq B$.
- (iv) Any subset $X \subseteq \mathcal{P}(S)$ with the same ordering. Most of our applications will be of this form: X will be the set of subsets of S with some property, ordered by inclusion.
- (v) The five-element poset with $a \leq b \leq c$ and $a \leq d \leq e$ and nothing else (beyond what transitivity and reflexivity force). So $a \leq c$, but b and d are incomparable.

Small posets are best understood by drawing them.

Definition 34.3.3 (Hasse Diagram)

A **Hasse diagram** of a poset X consists of a drawing of the points of X in the plane, with an upward line from x to y whenever y **covers** x , meaning $y > x$ and there is no z with $y > z > x$.

Here are the poset (v) above on the left, and two more on the right.



The middle diagram is a warning: there is no notion of ‘height’ or ‘rank’ in a poset, since b is covered by c directly while d, e take two steps to get from a to c . In the right-hand diagram the thickened path $b < d < e$ is a chain, in the following sense.

Definition 34.3.4 (Chain and Antichain)

A subset S of a poset X is a **chain** if any two of its elements are comparable, that is, if S is totally ordered by $\leq$. It is an **antichain** if no two distinct members of S are comparable.

Example 34.3.5

In a total order, every subset is a chain and every antichain has at most one element. In $(\mathbb{N}^+, \text{'divides'})$ the set $\{1, 2, 4, 8, \dots\}$ is a chain and the set of primes is an antichain. In $\mathcal{P}(\mathbb{N})$, the sets $A_x = \{q \in \mathbb{Q} \mid q < x\}$ for $x \in \mathbb{R}$ form a chain of size $2^{\aleph_0}$.

Definition 34.3.6 (Upper Bound and Supremum)

For $S \subseteq X$, an element $x \in X$ is an **upper bound** for S if $y \leq x$ for all $y \in S$. It is a **least upper bound** or **supremum** for S if in addition $x \leq y$ for every upper bound y of S , and we then write $x = \sup S$.

Suprema are unique when they exist, by antisymmetry.

Definition 34.3.7 (Complete Poset)

A poset X is **complete** if every subset $S \subseteq X$ has a supremum.

In a complete poset there is automatically a greatest element $\sup X$ and a least element $\sup \emptyset$ (every element is vacuously an upper bound for $\emptyset$). The example to keep in mind is $\mathcal{P}(S)$, where $\sup$ is union. Note that $(\mathbb{R}, \leq)$ is *not* complete in this sense – unbounded sets have no supremum – but $[0, 1]$ is.

Definition 34.3.8 (Order-Preserving Map)

For a poset X , a function $f : X \rightarrow X$ is **order-preserving** if $x \leq y$ implies $f(x) \leq f(y)$.

Theorem 34.3.9 (Knaster–Tarski Fixed Point Theorem)

Let X be a complete poset and $f : X \rightarrow X$ order-preserving. Then f has a fixed point.

Proof. Let $E = \{x \in X \mid x \leq f(x)\}$ and $s = \sup E$. We claim $f(s) = s$.

To show $s \leq f(s)$, it is enough to show that $f(s)$ is an upper bound for E , since s is the *least* upper bound. And indeed if $x \in E$ then $x \leq s$, so $f(x) \leq f(s)$, so $x \leq f(x) \leq f(s)$.

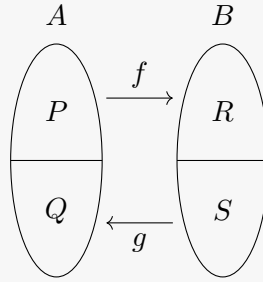
To show $f(s) \leq s$, it is enough to show $f(s) \in E$, since s is an upper bound for E . But we have just shown $s \leq f(s)$, so $f(s) \leq f(f(s))$ as f is order-preserving, which says exactly that $f(s) \in E$. $\square$

Note that no choice, and no ordinals, were used here. As an application we get a proof of the Schröder–Bernstein theorem, which we will want in Section 34.6.

Corollary 34.3.10 (Schröder–Bernstein Theorem)

Let A, B be sets and let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. Then there is a bijection from A to B .

Proof. We aim to partition A into P and Q , and B into R and S , in such a way that $f(P) = R$ and $g(S) = Q$. Given such a partition we may define a bijection $h : A \rightarrow B$ by taking $h = f$ on P and $h = g^{-1}$ on Q .



Since we need $S = B \setminus R = B \setminus f(P)$ and $Q = A \setminus P = g(S)$, what we want is a set $P \subseteq A$ with

$$P = A \setminus g(B \setminus f(P)).$$

Now $\mathcal{P}(A)$ is a complete poset under inclusion, and the map $P \mapsto A \setminus g(B \setminus f(P))$ is order-preserving from $\mathcal{P}(A)$ to itself, since it applies the order-reversing operation ‘complement’ twice. So it has a fixed point by Knaster–Tarski, and we are done. $\square$

§34.3.2 Zorn’s Lemma

Now to the heart of the matter. In a poset there may be no greatest element, but we can still hope for maximal ones.

Definition 34.3.11 (Maximal Element)

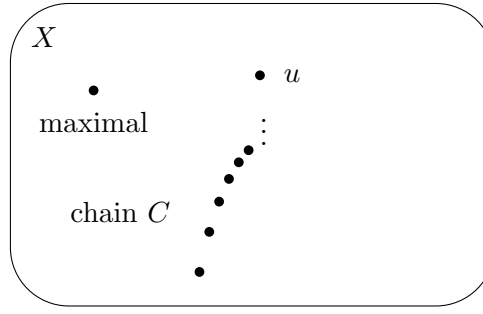
In a poset X , an element $x \in X$ is **maximal** if no $y \in X$ has $y > x$.

Take care to distinguish this from a *maximum* (greatest) element: in the third Hasse diagram above, e is a maximum, whereas in the first diagram c and e are both maximal and there is no maximum. Some posets have no maximal element at all, for instance $(\mathbb{R}, \leq)$ or $(\mathbb{N}^+, \text{‘divides’})$. In both of those examples there is also a chain with no upper bound ($\mathbb{N} \subseteq \mathbb{R}$, and the powers of 2 respectively), and Zorn’s lemma says that this is the only obstruction.

Theorem 34.3.12 (Zorn’s Lemma)

Assume the axiom of choice. Let X be a poset in which every chain has an upper bound. Then X has a maximal element.

Note that the empty chain must have an upper bound, so the hypothesis forces X to be non-empty; equivalently one can state the lemma for a non-empty poset in which every non-empty chain has an upper bound. Pictorially, the hypothesis says that whenever we have a chain marching upwards through X , there is something sitting above all of it, so that no chain can ‘escape’ – and then some element must be at the top.



Proof. Suppose X has no maximal element. Then for each $x \in X$ we may choose $x' \in X$ with $x' > x$, and for each chain C we may choose an upper bound $u(C)$; both of these use the axiom of choice.

Let $\gamma = \gamma(X)$ be the Hartogs' ordinal of X from Theorem 34.2.29. Pick any $x \in X$ and define x_α for $\alpha < \gamma$ by transfinite recursion:

$$\begin{aligned} x_0 &= x, \\ x_{\alpha+} &= (x_\alpha)', \\ x_\lambda &= u(\{x_\alpha \mid \alpha < \lambda\}) \quad \text{for } \lambda \text{ a non-zero limit.} \end{aligned}$$

For this to make sense we need $\{x_\alpha \mid \alpha < \lambda\}$ to be a chain, which follows by induction: at each stage we only ever add an element above everything so far.

But then $\alpha \mapsto x_\alpha$ is strictly increasing, hence injective, so we have injected γ into X , contradicting the definition of $\gamma(X)$. $\square$

The proof is short, but it should not be mistaken for an easy one: it rests on the whole apparatus of well-orderings, transfinite recursion, ordinals and Hartogs' lemma. The axiom of choice is used only in the very first line.

Example 34.3.13 (The Hypothesis of Zorn's Lemma)

- (i) In $\mathcal{P}(S)$ under inclusion, every chain has an upper bound (its union), and indeed there is a maximal element, namely S .
- (ii) In the poset of *finite* subsets of an infinite S , the chain $\{1\} \subseteq \{1, 2\} \subseteq \dots$ has no upper bound, and there is no maximal element.
- (iii) In $[0, 1)$ with the usual order, every chain has an upper bound in $[0, 1]$ but not always in $[0, 1)$, and there is no maximal element. This shows how carefully the upper bound must be checked to lie in the poset itself – it is the step that fails most often in practice.

§34.3.3 Applications of Zorn's Lemma

Proofs using Zorn's lemma nearly always look the same. We choose a poset whose elements are 'partial solutions' to our problem, usually a family of subsets ordered by inclusion; we check that the union of a chain of partial solutions is a partial solution; and then we check that a maximal partial solution is a solution.

Theorem 34.3.14 (Every Vector Space has a Basis)

Every vector space V has a basis.

Proof. Let X be the set of all linearly independent subsets of V , ordered by inclusion. Note $X \neq \emptyset$ as $\emptyset \in X$.

We claim a maximal element A of X is a basis. It is linearly independent by definition, and if it did not span then there would be some $v \in V$ outside its span, and $A \cup \{v\}$ would be linearly independent, contradicting maximality.

So we apply Zorn's lemma. Let $\{A_i \mid i \in I\}$ be a chain in X and let $A = \bigcup_{i \in I} A_i$. Then $A \supseteq A_i$ for all i , so we need only check $A \in X$, i.e. that A is linearly independent. Suppose we had a linear dependence $\lambda_1 x_1 + \cdots + \lambda_n x_n = 0$ with $x_1, \dots, x_n \in A$ and the λ_i not all zero. Each x_j lies in some A_{i_j} , and since the A_i form a chain, one of $A_{i_1}, \dots, A_{i_n}$ contains all the others, and hence contains all of $x_1, \dots, x_n$. That contradicts its linear independence. $\square$

Observe that linear algebra was used only at the very last step, to see that a maximal element does the job. This is typical. Note also that the theorem is highly non-constructive: it tells us that $\mathbb{R}$ has a basis as a vector space over $\mathbb{Q}$ (a *Hamel basis*), but no such basis can be written down.

Theorem 34.3.15 (Every Ring has a Maximal Ideal)

Let R be a non-trivial ring with 1. Then R has a maximal ideal.

Proof. Let X be the set of proper ideals of R , ordered by inclusion; it is non-empty as $\{0\} \in X$. Given a non-empty chain $\{I_j \mid j \in J\}$ in X , the union $I = \bigcup_j I_j$ is an ideal (any two elements of I lie in a common I_j , so I is closed under addition, and it is clearly closed under multiplication by R), and it is proper, since $1 \notin I_j$ for any j and so $1 \notin I$. So I is an upper bound in X . By Zorn's lemma X has a maximal element, which is by definition a maximal ideal. $\square$

A third application, this time with no algebra in sight at all.

Theorem 34.3.16 (Order-Extension Principle)

Every partial order $(X, \leq)$ can be extended to a total order on X ; that is, there is a total order $\preceq$ on X with $x \leq y$ implying $x \preceq y$.

Proof. Regard a partial order on X as a subset of $X \times X$, and let P be the set of all partial orders on X containing $\leq$, ordered by inclusion. It is non-empty as $\leq \in P$.

Given a non-empty chain in P , its union is again a partial order: reflexivity is clear, and given two or three pairs in the union we may find a single member of the chain containing all of them, in which transitivity and antisymmetry hold. So it is an upper bound, and Zorn's lemma gives a maximal $\preceq \in P$.

Finally, $\preceq$ is total. If x and y were incomparable, consider

$$R = \preceq \cup \{(a, b) \mid a \preceq x \text{ and } y \preceq b\}.$$

One checks that R is a partial order (transitivity is a short case-check, and antisymmetry uses the incomparability of x and y) which strictly contains $\preceq$, contradicting maximality. $\square$

We can also use Zorn to dispense with the countability assumption in our discussion of propositional logic, where we needed to enumerate L .

Theorem 34.3.17 (Model Existence Lemma, Uncountable Case)

Let P be any set of primitive propositions and let $S \subseteq L(P)$ be consistent. Then S has a model.

Proof. As in the proof of Theorem 34.1.21, it is enough to find a consistent $\bar{S} \supseteq S$ such that for every $t \in L$, either $t \in \bar{S}$ or $\neg t \in \bar{S}$; the valuation $v(t) = \mathbb{1}[t \in \bar{S}]$ then works verbatim.

We claim that a *maximal* consistent $\bar{S} \supseteq S$ has this property. Indeed if $t \notin \bar{S}$, then $\bar{S} \cup \{t\}$ is inconsistent by maximality, so $\bar{S} \cup \{t\} \vdash \perp$, so $\bar{S} \vdash \neg t$ by the deduction theorem. Then $\bar{S} \cup \{\neg t\}$ is consistent (anything it proves, $\bar{S}$ already proves), so $\neg t \in \bar{S}$ by maximality again.

So we must show a maximal consistent extension exists. Let $X = \{T \subseteq L \mid T \supseteq S, T \text{ consistent}\}$, ordered by inclusion, which is non-empty as $S \in X$. Let $\{T_i \mid i \in I\}$ be a non-empty chain in X and let $T = \bigcup_i T_i$. Certainly $T \supseteq S$. To see T is consistent, suppose $T \vdash \perp$. As proofs are finite, there are $t_1, \dots, t_n \in T$ with $\{t_1, \dots, t_n\} \vdash \perp$, and as the T_i are nested, some T_k contains all of the t_j – making T_k inconsistent, a contradiction. So $T \in X$ is an upper bound for the chain, and Zorn's lemma provides a maximal element of X . $\square$

§34.3.4 The Well-Ordering Principle

Zorn's lemma also delivers the result promised in Section 34.2.9, that every set can be well-ordered – including $\mathbb{R}$, for which no explicit well-ordering can be described.

Theorem 34.3.18 (Well-Ordering Principle)

Assume Zorn's lemma. Then every set S can be well-ordered.

Proof. Let X be the set of pairs (A, R) where $A \subseteq S$ and R is a well-ordering of A , ordered by $(A, R) \leq (A', R')$ if (A', R') extends (A, R) – that is, A is an initial segment of A' and R' agrees with R on A . Note $X \neq \emptyset$ as $(\emptyset, \emptyset) \in X$.

Given a non-empty chain $\{(A_i, R_i) \mid i \in I\}$ in X , the family $\{A_i\}$ is nested, so by Proposition 34.2.19 the pair $(\bigcup_i A_i, \bigcup_i R_i)$ is a well-ordering, and it clearly extends each (A_i, R_i) . So it is an upper bound.

By Zorn's lemma there is a maximal $(A, R) \in X$. If $A \neq S$ we may take $x \in S \setminus A$ and well-order $A \cup \{x\}$ by making x greater than everything in A ; this extends (A, R) strictly, contradicting maximality. So $A = S$. $\square$

§34.3.5 Zorn's Lemma and the Axiom of Choice

We assumed the axiom of choice to prove Zorn's lemma, and Zorn's lemma to prove the well-ordering principle. The circle closes with the following.

Theorem 34.3.19 (The Axiom of Choice)

Assume the well-ordering principle. Then for any family $\{A_i \mid i \in I\}$ of non-empty sets there is a **choice function** $f : I \rightarrow \bigcup_{i \in I} A_i$ with $f(i) \in A_i$ for all i .

Proof. Well-order $\bigcup_{i \in I} A_i$, and define $f(i)$ to be the least element of A_i . □

So the axiom of choice, Zorn's lemma and the well-ordering principle are all equivalent, over the other axioms of set theory. It is worth remarking on where choice was really needed: making *one* arbitrary choice is not an axiom at all – a set being non-empty means precisely that it has an element – and by induction, finitely many choices are also free. It is only the passage to infinitely many simultaneous arbitrary choices that requires an axiom.

The axiom of choice is unlike the other axioms in that the object it produces is not determined by its inputs: the union of A and B is a specific set, whereas a choice function for the family $\{1, 2\}$ could be either of two things. This is why proofs using it are non-constructive. That said, plenty of proofs that do not use choice are also non-constructive, so this should not be overstated.

§34.3.6 The Bourbaki–Witt Theorem

We close the section with a result in the same family as Zorn's lemma, but which needs no choice at all. Call a poset X **chain-complete** if every non-empty chain in X has a supremum.

Theorem 34.3.20 (Bourbaki–Witt)

Let X be a non-empty chain-complete poset and let $f : X \rightarrow X$ be *inflationary*, meaning $f(x) \geq x$ for all x . Then f has a fixed point.

Given Zorn's lemma this is immediate: a chain-complete poset satisfies the hypotheses of Zorn (a non-empty chain has its supremum as an upper bound, and the empty chain has any element of the non-empty X), so X has a maximal element x , and then $f(x) \geq x$ forces $f(x) = x$. What is interesting is that no choice is required.

Proof without choice. Fix $x_0 \in X$ and let $Y \subseteq X$ be the least subset with $x_0 \in Y$ that is closed under f and under suprema of non-empty chains (namely, the intersection of all such subsets). One checks by induction on this closure that Y is itself a chain: the point is that f and suprema are the only ways to produce new elements, and neither can create incomparability, though verifying this carefully takes some work.

Granting that Y is a chain, it has a supremum $s = \sup Y \in Y$. Then $f(s) \in Y$, so $f(s) \leq s$, while $f(s) \geq s$ as f is inflationary. So $f(s) = s$. □

The contrast with Zorn's lemma is instructive: there we had to *choose* some $x' > x$ at

each stage, and it is exactly that choice which the axiom of choice supplies. Here the function f tells us what to do next, so there is nothing to choose.

§34.4 Predicate Logic

Propositional logic cannot express any of the statements we actually care about, because it has no way of talking about *objects*. We now upgrade to predicate logic, also called *first-order logic*, in which we may quantify over the elements of a structure. Everything from Section 1 will reappear – a language, valuations (now structures), axioms, proofs, soundness, completeness, compactness – but each with more moving parts.

§34.4.1 Languages

Definition 34.4.1 (Language)

Let Ω (the **function symbols**) and Π (the **relation symbols**) be disjoint sets, and let $\alpha : \Omega \cup \Pi \rightarrow \mathbb{N}$ be a function, called the **arity**.

The **language** $L = L(\Omega, \Pi, \alpha)$ is the set of formulae, defined as follows.

- *Variables.* We have variables $x_1, x_2, \dots$ (also written $x, y, z, \dots$).
- *Terms.* These are defined inductively by
 - (i) every variable is a term;
 - (ii) if $f \in \Omega$ with $\alpha(f) = n$, and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term.
- *Atomic formulae.* There are three sorts:
 - (i) $\perp$;
 - (ii) $(s = t)$ for any terms s, t ;
 - (iii) $\phi(t_1, \dots, t_n)$ for any $\phi \in \Pi$ with $\alpha(\phi) = n$ and terms $t_1, \dots, t_n$.
- *Formulae.* These are defined inductively by
 - (i) atomic formulae are formulae;
 - (ii) $(p \Rightarrow q)$ is a formula for any formulae p, q ;
 - (iii) $(\forall x)p$ is a formula for any formula p and variable x .

A function symbol of arity 0 is a **constant**. As before, we use the abbreviations $\neg, \wedge, \vee$, and we write $(\exists x)p$ for $\neg(\forall x)(\neg p)$.

Example 34.4.2 (The Language of Groups)

Take $\Omega = \{m, i, e\}$ and $\Pi = \emptyset$, with $\alpha(m) = 2$, $\alpha(i) = 1$ and $\alpha(e) = 0$. Then $e, x_1, m(x_1, x_2)$ and $i(m(x_1, x_1))$ are terms, and $(m(e, e) = e)$ and $(\forall x)(m(x, i(x)) = e)$ are formulae. The theory of groups is the set consisting of the associativity axiom, the identity axioms and the inverse axioms, written out in this language.

Example 34.4.3 (Other Languages)

For posets, take $\Omega = \emptyset$ and $\Pi = \{\leq\}$ with $\alpha(\leq) = 2$. For fields, take $\Omega = \{+, \cdot, -, 0, 1\}$ with the evident arities and $\Pi = \emptyset$. For graphs, take $\Omega = \emptyset$ and $\Pi = \{\sim\}$ with $\alpha(\sim) = 2$.

It cannot be stressed enough that a formula is at this stage a string of meaningless symbols; it makes no sense to ask whether it is true. In particular the symbols in Ω and Π carry no meaning beyond their arity.

Definition 34.4.4 (Closed Term)

A term is **closed** if it contains no variables.

Definition 34.4.5 (Free and Bound Variables)

An occurrence of a variable x in a formula p is **bound** if it lies inside the brackets of a quantifier $(\forall x)$, and **free** otherwise.

Definition 34.4.6 (Sentence)

A **sentence** is a formula with no free variables.

For instance, in the language of groups $(\forall x)(m(x, y) = e)$ has x bound and y free, so it is not a sentence; it says something only once we know what y is.

Definition 34.4.7 (Substitution)

For a formula p , a variable x and a term t , the **substitution** $p[t/x]$ is the formula obtained by replacing each free occurrence of x in p by t .

§34.4.2 Semantic Entailment

In propositional logic we interpreted the language in $\{0, 1\}$ using valuations. Now we must interpret it in a set carrying operations and relations of the right arities.

Definition 34.4.8 (Structure)

An **L -structure** is a non-empty set A together with a function $f_A : A^n \rightarrow A$ for each $f \in \Omega$ with $\alpha(f) = n$, and a relation $\phi_A \subseteq A^n$ for each $\phi \in \Pi$ with $\alpha(\phi) = n$.

Remark. We explicitly forbid A from being empty, which makes life much easier later on (in an empty structure $(\forall x)p$ would hold vacuously for every p , including $p = \perp$).

Example 34.4.9

In the language of groups, an L -structure is a set with a binary operation, a unary operation and a distinguished element – no axioms required. Every group is an L -structure, but so is $\mathbb{N}$ with $m(x, y) = x + y$, $i(x) = x$ and $e = 0$.

We now define what it means for a sentence to hold in a structure. The definition is a

mouthful, but after it is made we can largely forget it.

Definition 34.4.10 (Interpretation)

For each sentence p and L -structure A , the **interpretation** $p_A \in \{0, 1\}$ is defined inductively as follows.

- *Closed terms.* Define $t_A \in A$ for each closed term t by

$$(f(t_1, \dots, t_n))_A = f_A((t_1)_A, \dots, (t_n)_A)$$

for $f \in \Omega$ with $\alpha(f) = n$ and $t_1, \dots, t_n$ closed terms.

- *Atomic formulae.* We take

$$\begin{aligned} \perp_A &= 0, \\ (s = t)_A &= \mathbb{1}[s_A = t_A], \\ (\phi(t_1, \dots, t_n))_A &= \mathbb{1}[(t_1)_A, \dots, (t_n)_A \in \phi_A]. \end{aligned}$$

- *Sentences.* We take

$$\begin{aligned} (p \Rightarrow q)_A &= \mathbb{1}[\text{not } (p_A = 1 \text{ and } q_A = 0)], \\ ((\forall x)p)_A &= \mathbb{1}[p[\bar{a}/x]_{\bar{A}} = 1 \text{ for all } a \in A], \end{aligned}$$

where for $a \in A$ we form a new language L' by adding a constant $\bar{a}$, and make A into an L' -structure $\bar{A}$ by setting $\bar{a}_{\bar{A}} = a$.

The last clause is the only awkward one, and the awkwardness is unavoidable: to say that $(\forall x)p$ holds we must talk about p with x replaced by an *element* of A , and elements of A are not symbols of the language until we make them so.

Definition 34.4.11 (Theory)

A **theory** is a set of sentences.

Definition 34.4.12 (Model)

If $p_A = 1$ we say p **holds** in A , or is **true** in A , or that A is a **model** of p . For a theory S , a model of S is a structure which is a model of every $s \in S$.

Definition 34.4.13 (Semantic Entailment)

For a theory S and a sentence t , we say S **entails** t , written $S \models t$, if every model of S is a model of t . We call t a **tautology**, written $\models t$, if $\emptyset \models t$.

The word *first-order* refers to the fact that our variables range over *elements* of the structure, not over subsets of it. This restriction is what makes the theory work, and also what makes it weak, as we will see.

§34.4.3 Syntactic Implication

As before we need axioms and deduction rules. We keep the three propositional axioms and add two describing equality and two describing $\forall$.

Axiom 34.4.14 (Axioms of Predicate Logic)

- (1) $p \Rightarrow (q \Rightarrow p)$ for any formulae p, q ;
- (2) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$ for any formulae p, q, r ;
- (3) $(\neg\neg p) \Rightarrow p$ for any formula p ;
- (4) $(\forall x)(x = x)$ for any variable x ;
- (5) $(\forall x)(\forall y)[(x = y) \Rightarrow (p \Rightarrow p[y/x])]$ for any variables x, y and formula p with y not occurring bound in p ;
- (6) $[(\forall x)p] \Rightarrow p[t/x]$ for any formula p , variable x and term t , with no free variable of t occurring bound in p ;
- (7) $[(\forall x)(p \Rightarrow q)] \Rightarrow [p \Rightarrow (\forall x)q]$ for any formulae p, q with x not occurring free in p .

The side conditions in (5), (6) and (7) all guard against accidental capture of variables. For example, without the condition in (6), taking p to be $(\exists y)(x \neq y)$ and t to be y would let us deduce the false statement $(\exists y)(y \neq y)$ from the true statement $(\forall x)(\exists y)(x \neq y)$.

Definition 34.4.15 (Deduction Rules)

The **deduction rules** are

- (i) *modus ponens*: from p and $p \Rightarrow q$ we may deduce q ;
- (ii) *generalisation*: from p we may deduce $(\forall x)p$, provided that no premise used in the proof of p so far has x as a free variable.

The proviso on generalisation is essential: from the premise ‘ x is even’ we must not be allowed to deduce that everything is even.

Definition 34.4.16 (Proof and Syntactic Implication)

A **proof** of p from S is a finite sequence of formulae, each of which is an axiom, or a member of S , or obtained from earlier lines by modus ponens or generalisation, and whose last line is p . If such a proof exists we write $S \vdash p$, and say S **proves** p . If $\emptyset \vdash p$ we call p a **theorem**.

As in the propositional case we say S is **consistent** if $S \not\vdash \perp$.

Example 34.4.17 (A Proof in Predicate Logic)

We show $\vdash [(\forall x)p] \Rightarrow (\exists x)p$, that is, $\vdash [(\forall x)p] \Rightarrow \neg(\forall x)(\neg p)$. By the deduction

theorem it is enough to show

$$\{(\forall x)p, (\forall x)(\neg p)\} \vdash \perp.$$

From $(\forall x)p$ and axiom (6) with $t = x$ we get p ; from $(\forall x)(\neg p)$ and axiom (6) we get $\neg p$, that is, $p \Rightarrow \perp$; and modus ponens gives $\perp$. Note where non-emptiness of structures is hiding: semantically, $[(\forall x)p] \Rightarrow (\exists x)p$ would fail in an empty structure.

Example 34.4.18 (Why Generalisation Needs its Proviso)

Without the proviso we could argue: from the premise $p(x)$ (' x is even'), deduce $(\forall x)p(x)$. But $\{p(x)\} \not\models (\forall x)p(x)$, as we see by taking the structure $\mathbb{N}$ with p interpreted as 'is even': the premise holds for some elements and the conclusion for none. Soundness would fail immediately.

§34.4.4 The Deduction Theorem

Proposition 34.4.19 (Deduction Theorem)

Let S be a set of formulae and p, q formulae. Then $S \vdash (p \Rightarrow q)$ if and only if $S \cup \{p\} \vdash q$.

Proof. The proof is exactly that of Theorem 34.1.16, except that in the backwards direction we must also handle lines obtained by generalisation.

So suppose the line $(\forall x)r$ was obtained from an earlier line r by generalisation, and by induction we have a proof of $p \Rightarrow r$ from S ; we want $S \vdash p \Rightarrow (\forall x)r$.

By the condition on generalisation, no premise used in the proof of r from $S \cup \{p\}$ has x free. Hence no premise used in our proof of $p \Rightarrow r$ from S has x free, so we may apply generalisation to get $S \vdash (\forall x)(p \Rightarrow r)$.

If x is not free in p , axiom (7) and modus ponens give $S \vdash p \Rightarrow (\forall x)r$ as required. If x is free in p , then p was not among the premises used in the proof of r from $S \cup \{p\}$, so in fact $S \vdash r$; then generalisation gives $S \vdash (\forall x)r$, and axiom (1) gives $S \vdash p \Rightarrow (\forall x)r$. $\square$

§34.4.5 Soundness

Proposition 34.4.20 (Soundness)

Let S be a theory and p a sentence. Then $S \vdash p$ implies $S \models p$.

Proof. Let A be a model of S and let $t_1, \dots, t_n = p$ be a proof of p from S . As in the propositional case we check by induction that each t_i holds in A – interpreting a formula with free variables as its universal closure. Each axiom holds in every structure (this is a routine but tedious check, one case per axiom), members of S hold as $A \models S$, and each of the two deduction rules preserves holding in A : modus ponens by the definition of $(p \Rightarrow q)_A$, and generalisation by the definition of $((\forall x)p)_A$ together with the proviso. $\square$

§34.4.6 The Model Existence Lemma

The interesting direction, as before, is adequacy, and again it reduces to the statement that a consistent theory has a model. But now we have to build an entire structure out of nothing, and the only raw material available is the language itself.

Before the proof, here are the four ideas that go into it, since the proof is otherwise unmotivated.

- (i) *Build the model out of the language.* Take the underlying set to be the set of closed terms of L , with each function symbol acting in the obvious way: $f([t_1], \dots, [t_n]) = [f(t_1, \dots, t_n)]$.
- (ii) *Quotient.* If S is the theory of fields, then $S \vdash (1 + 0 = 1)$, but ‘ $1 + 0$ ’ and ‘ 1 ’ are different strings. So we quotient the closed terms by $s \sim t$ if $S \vdash (s = t)$.
- (iii) *Make S complete.* Let S be the field axioms together with the sentence $(1 + 1 = 0) \vee (1 + 1 + 1 = 0)$. Then S proves neither disjunct, so in our quotient $[1 + 1] \neq [0]$ and $[1 + 1 + 1] \neq [0]$, and the structure we build is a field of neither characteristic 2 nor 3 – so not a model. We fix this by extending S to a *maximal* consistent theory, using Zorn’s lemma exactly as in Theorem 34.3.17.
- (iv) *Add witnesses.* Let S be the field axioms together with $(\exists x)(x \cdot x = 1 + 1)$. There need be no closed term t with $[t \cdot t] = [1 + 1]$, so again our structure fails. We fix this by adding, for each sentence $(\exists x)p$ in our theory, a brand new constant c to the language and the sentence $p[c/x]$ to the theory.

Unfortunately (iv) enlarges the language, which may destroy the maximality won in (iii), so we must alternate between the two infinitely often and take the union.

Theorem 34.4.21 (Model Existence Lemma)

Every consistent theory has a model.

Proof. Let S be consistent in the language $L = L(\Omega, \Pi)$.

Step 1: the theory. Extend S to a maximal consistent theory S_1 in L , using Zorn’s lemma as in Theorem 34.3.17. Such a theory is **complete**: for every sentence p , either $p \in S_1$ or $\neg p \in S_1$. Now add witnesses: for each sentence $(\exists x)p \in S_1$ introduce a new constant $c_{(\exists x)p}$, let C_1 be the set of these, and let

$$T_1 = S_1 \cup \{p[c_{(\exists x)p}/x] \mid (\exists x)p \in S_1\}$$

in the language $L_1 = L(\Omega \cup C_1, \Pi)$. Then T_1 is consistent: a proof of $\perp$ from T_1 uses finitely many of the new sentences, and replacing each new constant throughout the proof by a fresh variable and using $(\exists x)p \in S_1$ finitely many times converts it into a proof of $\perp$ from S_1 .

Repeat: extend T_1 to a maximal consistent S_2 in L_1 , add witnesses to get T_2 in L_2 , and so on. Let $\bar{S} = \bigcup_{n \geq 1} S_n$ in the language $\bar{L} = L(\Omega \cup \bigcup_n C_n, \Pi)$.

We claim $\bar{S}$ is consistent, complete and has witnesses. It is consistent since a proof of $\perp$ from it is finite, so comes from some S_n . It is complete: a sentence p of $\bar{L}$ is a finite string, so lies in some L_n , and S_{n+1} is complete in L_n , so proves p or $\neg p$.

And it has witnesses: if $(\exists x)p \in \bar{S}$ then $(\exists x)p \in S_n$ for some n , and T_n supplied a witness.

Step 2: the structure. On the set of closed terms of $\bar{L}$ define $s \sim t$ if $\bar{S} \vdash (s = t)$. This is an equivalence relation, by axioms (4) and (5). Let A be the set of equivalence classes, which is non-empty as $\bar{L}$ has at least one constant (if Ω had none, the witnesses supplied one – and if $\bar{S}$ has no existential sentences at all we may add a constant by hand). Make A into an $\bar{L}$ -structure by

$$f_A([t_1], \dots, [t_n]) = [f(t_1, \dots, t_n)],$$

$$\phi_A = \{([t_1], \dots, [t_n]) \mid \bar{S} \vdash \phi(t_1, \dots, t_n)\},$$

both of which are well-defined by axiom (5).

Step 3: A is a model. We claim that for every sentence p of $\bar{L}$,

$$p_A = 1 \iff \bar{S} \vdash p,$$

which finishes the proof, since $S \subseteq \bar{S}$ so every $p \in S$ holds in A (and A is an L -structure by forgetting the new constants). We induct on the structure of p .

If p is $\perp$ then $\perp_A = 0$ and $\bar{S} \nvdash \perp$. If p is $(s = t)$ for closed terms s, t , then $\bar{S} \vdash (s = t)$ if and only if $[s] = [t]$ by the definition of $\sim$, if and only if $s_A = t_A$. Relations are the same by the definition of ϕ_A .

If p is $(q \Rightarrow r)$, then since $\bar{S}$ is complete and consistent, $\bar{S} \vdash (q \Rightarrow r)$ if and only if $\bar{S} \vdash \neg q$ or $\bar{S} \vdash r$. (If $\bar{S} \nvdash \neg q$ and $\bar{S} \nvdash r$ then $\bar{S} \vdash q$ by completeness, so $\bar{S} \vdash r$ by modus ponens, a contradiction.) By induction this holds if and only if $q_A = 0$ or $r_A = 1$, which is the definition of $(q \Rightarrow r)_A = 1$.

Finally consider $(\exists x)q$, which suffices since $(\forall x)q$ is $\neg(\exists x)\neg q$. If $\bar{S} \vdash (\exists x)q$ then $(\exists x)q \in \bar{S}$ by completeness, so there is a witness c with $q[c/x] \in \bar{S}$, and by induction $q[c/x]_A = 1$, so $(\exists x)q$ holds in A . Conversely if $(\exists x)q$ holds in A , then since every element of A is $[t]$ for some closed term t , we have $q[t/x]_A = 1$ for some closed term t , so $\bar{S} \vdash q[t/x]$ by induction, and hence $\bar{S} \vdash (\exists x)q$. $\square$

Corollary 34.4.22 (Adequacy)

Let S be a theory and p a sentence. Then $S \models p$ implies $S \vdash p$.

Proof. If $S \models p$ then $S \cup \{\neg p\}$ has no model, so is inconsistent by Theorem 34.4.21. Then $S \cup \{\neg p\} \vdash \perp$, so $S \vdash \neg\neg p$ by the deduction theorem, so $S \vdash p$ by axiom (3). $\square$

Theorem 34.4.23 (Gödel's Completeness Theorem)

Let S be a theory and p a sentence. Then $S \models p$ if and only if $S \vdash p$.

Proof. Soundness and adequacy. $\square$

Remark. If L is countable, Zorn's lemma is not needed above: we can enumerate the sentences and extend S one at a time, as we did in the propositional case.

Notice that there is no analogue of the decidability theorem here: to check $S \models p$ directly we would have to inspect every L -structure, of which there are a proper class. We will see in Section 34.7 that this is not merely a failure of imagination.

§34.4.7 Compactness and the Löwenheim–Skolem Theorems

Theorem 34.4.24 (Compactness Theorem)

Let S be a theory such that every finite subset of S has a model. Then S has a model.

Proof. Replacing ‘has a model’ by ‘is consistent’ using the completeness theorem, this says that if every finite subset of S is consistent then S is consistent, which is immediate since proofs are finite. $\square$

This has some startling consequences, all obtained by the same trick: write down a theory whose finite fragments are obviously satisfiable, and conclude that the whole thing is.

Corollary 34.4.25 (Finiteness is Not First-Order)

There is no theory S in the language of groups whose models are precisely the finite groups.

Proof. Suppose S were such a theory. Let σ_k be the sentence saying ‘there are at least k elements’, namely

$$(\exists x_1) \cdots (\exists x_k) \left(\bigwedge_{i < j} x_i \neq x_j \right),$$

and let $T = S \cup \{\sigma_k \mid k \in \mathbb{N}\}$. Any finite subset of T mentions only finitely many σ_k , so has a model, namely a cyclic group of large enough order. By compactness T has a model, which is a finite group with at least k elements for every k – a contradiction. $\square$

Corollary 34.4.26

Let S be a theory with arbitrarily large finite models. Then S has an infinite model.

Proof. Add the sentences σ_k as above and apply compactness. $\square$

Corollary 34.4.27 (Connectedness is Not First-Order)

There is no theory in the language of graphs whose models are precisely the connected graphs.

Proof. Suppose S were such a theory. Add two constants a, b to the language and let

$$T = S \cup \{\delta_k \mid k \in \mathbb{N}\},$$

where δ_k is the sentence saying ‘there is no path from a to b of length k ’, which is a first-order sentence since it quantifies over the $k - 1$ intermediate vertices. Any finite subset of T mentions finitely many δ_k , and has a model: take a long path, with a and b its endpoints. So by compactness T has a model, which is a connected graph containing two vertices joined by no path at all – a contradiction. $\square$

Theorem 34.4.28 (Upward Löwenheim–Skolem)

Let S be a theory with an infinite model. Then S has an uncountable model.

Proof. Add constants $\{c_i \mid i \in I\}$ to L for some uncountable set I , and let

$$T = S \cup \{(c_i \neq c_j) \mid i, j \in I, i \neq j\}.$$

Any finite $T' \subseteq T$ mentions only finitely many of the c_i , so an infinite model of S is a model of T' once we interpret those finitely many constants as distinct elements. By compactness T has a model, which has at least $|I|$ elements. $\square$

Theorem 34.4.29 (Downward Löwenheim–Skolem)

Let L be a countable language, that is, with Ω and Π countable. Then if a theory S in L has a model, it has a countable model.

Proof. If S has a model then S is consistent by soundness, and the model constructed in the proof of Theorem 34.4.21 is countable: at each stage we add countably many constants to a countable language, and a countable union of countable sets is countable. So the set of closed terms, and hence its quotient, is countable. $\square$

Putting these together gives *Skolem’s paradox*: the theory ZF of Section 34.5 is written in a countable language, so if it has a model at all then it has a *countable* model – even though ZF proves the existence of uncountable sets. The resolution is that ‘uncountable’ is interpreted inside the model: a set x in the model is uncountable if the model contains no bijection between x and ω , and there is no reason for the model to contain every bijection that exists outside it.

§34.4.8 Categoricity

Since a first-order theory can never pin down the size of its models, we may ask for the next best thing: that it pin down the model once the size is fixed.

Definition 34.4.30 (Categoricity)

A theory S is κ -**categorical** if it has a model of cardinality κ and any two such models are isomorphic.

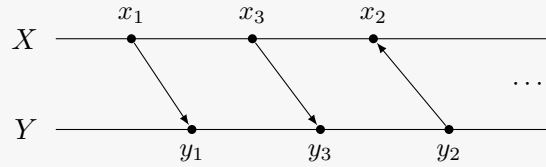
By the Löwenheim–Skolem theorems, a theory with an infinite model has models of many different sizes, so no such theory can be categorical outright. But $\aleph_0$ -categoricity does occur.

Theorem 34.4.31 (Cantor)

The theory of dense linear orders without endpoints is $\aleph_0$ -categorical. In particular, every countable dense linear order without endpoints is isomorphic to $(\mathbb{Q}, <)$.

Proof sketch. Let $X = \{x_1, x_2, \dots\}$ and $Y = \{y_1, y_2, \dots\}$ be two such orders, enumerated (not in order!). We build an isomorphism by a *back-and-forth* construction, extending a finite order-preserving partial map one point at a time, and alternating between the two sides so that every point gets used.

At an odd stage we take the least-indexed x not yet in the domain and find a partner for it in Y : the finitely many points already matched cut Y into finitely many intervals, and we need a point of Y in the interval corresponding to the position of x , which exists by density (or by the absence of endpoints, if x lies beyond everything matched so far). At an even stage we do the same in reverse, taking the least-indexed y not yet in the image.



Going forth at odd stages guarantees the map is defined everywhere; going back at even stages guarantees it is onto. The union of all these finite partial maps is therefore an isomorphism. $\square$

Categoricity is useful because of the following test, which we state without proof: if a theory S has no finite models and is κ -categorical for some infinite κ , then S is **complete**, meaning that for every sentence p either $S \vdash p$ or $S \vdash \neg p$. So the theory of dense linear orders without endpoints is complete, and in particular any sentence true in $\mathbb{Q}$ is provable from those axioms. The theory of the next section is very far from having this property.

§34.4.9 Peano Arithmetic and Non-Standard Models

Let us apply all this to arithmetic. Take $\Omega = \{0, s, +, \cdot\}$ with arities 0, 1, 2, 2, and $\Pi = \emptyset$. Here s is the successor function.

Definition 34.4.32 (Peano Arithmetic)

Peano arithmetic, written PA, is the theory with axioms

- (i) $(\forall x)(s(x) \neq 0)$;
- (ii) $(\forall x)(\forall y)(s(x) = s(y) \Rightarrow x = y)$;
- (iii) $(\forall y_1) \cdots (\forall y_n)[(p[0/x] \wedge (\forall x)(p \Rightarrow p[s(x)/x])) \Rightarrow (\forall x)p]$ for each formula p with free variables $x, y_1, \dots, y_n$;
- (iv) $(\forall x)(x + 0 = x)$;
- (v) $(\forall x)(\forall y)(x + s(y) = s(x + y))$;

- (vi) $(\forall x)(x \cdot 0 = 0)$;
 (vii) $(\forall x)(\forall y)(x \cdot s(y) = x \cdot y + x)$.

Axiom (iii) is the induction scheme, and the variables y_i are called *parameters*; without them we could not induct on a property like ' $x \geq y$ ' with y held fixed. Note that $\mathbb{N}$ is a model of PA, so PA is consistent.

The crucial point is that (iii) is a scheme with one instance per *formula*, and there are only countably many formulae. So PA asserts induction only for the countably many definable subsets of $\mathbb{N}$, and not for all of them. This is not a defect we can repair: a first-order language simply has no way to quantify over subsets.

The consequences are immediate. As PA has an infinite model, the upward Löwenheim–Skolem theorem gives an uncountable model of PA, which certainly is not isomorphic to $\mathbb{N}$. Such a model is called a **non-standard model of arithmetic**, and we can see what one looks like using compactness.

Example 34.4.33 (A Non-Standard Model of Arithmetic)

Add a constant c to the language and let

$$T = \text{PA} \cup \{c \neq 0, c \neq s(0), c \neq s(s(0)), \dots\}.$$

Any finite subset of T has a model, namely $\mathbb{N}$ with c interpreted as a large enough natural number. So by compactness T has a model A . Inside A the element c is greater than every $s^n(0)$, so A contains a copy of $\mathbb{N}$ followed by extra elements – and since $A \models \text{PA}$, every non-zero element has a predecessor, so those extra elements come in $\mathbb{Z}$ -shaped blocks, ordered densely. Every theorem of PA, and indeed every first-order sentence true in $\mathbb{N}$ that we care to add to T , holds in this strange structure too.

The same trick applied to the theory of ordered fields containing $\mathbb{R}$, with a constant ε and the sentences $0 < \varepsilon < 1/n$ for all n , produces an ordered field containing $\mathbb{R}$ with a non-zero infinitesimal element – a non-standard model of analysis, and the starting point for non-standard analysis.

Since the standard model $\mathbb{N}$ is not pinned down by PA, one may ask which sentences PA does prove. The completeness theorem tells us that $\text{PA} \vdash p$ if and only if p holds in *every* model, including the non-standard ones. So there is room for a sentence to be true in $\mathbb{N}$ but unprovable, and Section 34.7 shows that there is one.

§34.5 Set Theory

We have been using sets throughout, informally. We now write down axioms for them, in the first-order language of Section 34.4, and then see what the resulting universe looks like.

§34.5.1 The Axioms of ZF

Definition 34.5.1 (Zermelo–Fraenkel Set Theory)

Zermelo–Fraenkel set theory, ZF, is a first-order theory in the language with $\Omega = \emptyset$ and $\Pi = \{\in\}$, where $\alpha(\in) = 2$.

A *universe of sets* will mean a model of these axioms: a pair $(V, \in)$ where V is a set and $\in$ is a binary relation on V making the axioms true. Everything below is officially a statement of the form ‘let $(V, \in)$ be a model of ZF; then ...’, or equivalently, by the completeness theorem, ‘ZF $\vdash$...’.

Axiom 34.5.2 (Extension)

‘Sets with the same members are equal’:

$$(\forall x)(\forall y)[(\forall z)(z \in x \Leftrightarrow z \in y) \Rightarrow x = y].$$

Axiom 34.5.3 (Separation)

‘For a set x and a property p , we can form $\{z \in x \mid p(z)\}$ ’:

$$(\forall t_1) \cdots (\forall t_n)(\forall x)(\exists y)(\forall z)(z \in y \Leftrightarrow (z \in x \wedge p))$$

for each formula p with free variables $t_1, \dots, t_n, z$.

This is an axiom *scheme*, with one instance per formula p . The parameters t_i are needed to form sets such as $\{z \in x \mid z \in t\}$ for a variable t . Note carefully that we may only carve subsets out of a set we already have; the unrestricted version, allowing $\{z \mid p(z)\}$ for any p , is inconsistent by Russell’s paradox, taking $p(z)$ to be $z \notin z$.

Axiom 34.5.4 (Empty Set)

‘There is an empty set’:

$$(\exists x)(\forall y)(y \notin x).$$

We write $\emptyset$ for this set, which is unique by extension.

Axiom 34.5.5 (Pair Set)

‘We can form $\{x, y\}$ ’:

$$(\forall x)(\forall y)(\exists z)(\forall t)(t \in z \Leftrightarrow (t = x \vee t = y)).$$

We write $\{x, y\}$ for this z , and $\{x\}$ for $\{x, x\}$.

With pairs we can already encode ordered pairs and hence functions, entirely inside the language.

Definition 34.5.6 (Ordered Pair)

The **ordered pair** is $(x, y) = \{\{x\}, \{x, y\}\}$. One checks that $(x, y) = (z, t)$ if and only if $x = z$ and $y = t$, which is all we ever want from ordered pairs.

Definition 34.5.7 (Function)

We say f is a **function** if

$$(\forall x)(x \in f \Rightarrow x \text{ is an ordered pair}) \\ \wedge (\forall x)(\forall y)(\forall z)[((x, y) \in f \wedge (x, z) \in f) \Rightarrow y = z],$$

where ‘ x is an ordered pair’ abbreviates $(\exists y)(\exists z)(x = (y, z))$. We say x is the **domain** of f , written $x = \text{dom}(f)$, if f is a function and $(\forall y)(y \in x \Leftrightarrow (\exists z)((y, z) \in f))$. Then $f : x \rightarrow y$ abbreviates $(f \text{ is a function}) \wedge (x = \text{dom}(f)) \wedge (\forall z)(\forall t)((z, t) \in f \Rightarrow t \in y)$.

Back to the axioms.

Axiom 34.5.8 (Union)

‘We can form unions’:

$$(\forall x)(\exists y)(\forall z)(z \in y \Leftrightarrow (\exists t)(t \in x \wedge z \in t)).$$

We write $y = \bigcup x$; note this is the union of the members of x , so that $A \cup B$ is $\bigcup\{A, B\}$.

Intersections need no axiom: $\bigcap x$ is a subset of any member of x , so separation provides it.

Axiom 34.5.9 (Power Set)

‘We can form power sets’:

$$(\forall x)(\exists y)(\forall z)(z \in y \Leftrightarrow z \subseteq x),$$

where $z \subseteq x$ abbreviates $(\forall t)(t \in z \Rightarrow t \in x)$. We write $\mathcal{P}(x)$ for this y .

Axiom 34.5.10 (Infinity)

‘There is an infinite set’:

$$(\exists x)(\emptyset \in x \wedge (\forall y)(y \in x \Rightarrow y^+ \in x)),$$

where $y^+ = y \cup \{y\}$. A set satisfying this is called a **successor set**.

The intersection of all successor sets contained in a given one is again a successor set, so there is a least successor set; call it ω . Its members are

$$\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}, \dots$$

which we name $0, 1, 2, \dots$, so that $n = \{0, 1, \dots, n-1\}$. Minimality of ω gives induction: if $x \subseteq \omega$ is a successor set then $x = \omega$, that is,

$$(\forall x)[(x \subseteq \omega \wedge \emptyset \in x \wedge (\forall y)(y \in x \Rightarrow y^+ \in x)) \Rightarrow x = \omega].$$

We can now define ‘ x is finite’ to mean $(\exists y)(y \in \omega \wedge x \text{ bijects with } y)$, and ‘ x is countable’ to mean ‘ x is finite or x bijects with ω ’.

Axiom 34.5.11 (Foundation)

‘Every non-empty set has an $\in$ -minimal element’:

$$(\forall x)(x \neq \emptyset \Rightarrow (\exists y)(y \in x \wedge (\forall z)(z \in x \Rightarrow z \notin y))).$$

Foundation is the axiom that says sets are built up out of simpler sets: it forbids $x \in x$ (consider $\{x\}$) and infinite descending $\in$ -chains. It plays for $\in$ exactly the role that the well-ordering condition played for $<$.

Proposition 34.5.12

There is no set of all sets.

Proof. Suppose V were a set. Then by separation $R = \{x \in V \mid x \notin x\}$ is a set, and $R \in R$ if and only if $R \notin R$, a contradiction. $\square$

So separation really is the correct restriction of the naive comprehension principle: it is strong enough for everything we do, and weak enough to survive Russell’s argument. (In the presence of foundation we could argue differently, since then $x \notin x$ always, so $R = V$; but the proof above needs no such assumption.)

§34.5.2 A Digression on Classes

One axiom remains, and to state it we need to be able to say things like ‘for each $i \in I$ we have a set A_i ; now form $\{A_i \mid i \in I\}$ ’. The trouble is that the assignment $i \mapsto A_i$ may not be a function in V – it may be given by a formula rather than a set of ordered pairs.

Definition 34.5.13 (Class)

Let $(V, \in)$ be a model of ZF. A **class** is a collection C of points of V such that, for some formula p with free variable x (and possibly parameters), we have $x \in C$ if and only if $p(x)$ holds.

So classes are exactly the collections of the form $\{x \in V \mid p(x)\}$. Every set is a class, by separation. A class is a **proper class** if it is not a set, that is, if

$$\neg(\exists y)(\forall x)(x \in y \Leftrightarrow p).$$

For example V itself is a proper class, as is On by the Burali-Forti paradox. Statements about classes are always to be read as abbreviations for statements about the formulae defining them; classes are not objects of our theory.

Definition 34.5.14 (Function-Class)

A **function-class** F is a collection of ordered pairs for which there is a formula p with free variables x, y (and possibly parameters) such that $(x, y) \in F$ if and only if p holds, and such that $(x, y) \in F \wedge (x, z) \in F \Rightarrow y = z$.

Axiom 34.5.15 (Replacement)

‘The image of a set under a function-class is a set’. This is an axiom scheme, with an instance for each formula p :

$$\underbrace{(\forall t_1) \dots (\forall t_n)}_{\text{parameters}} \left[\underbrace{(\forall x)(\forall y)(\forall z)(p \wedge p[z/y] \Rightarrow y = z)}_{p \text{ is a function class}} \Rightarrow (\forall x) \underbrace{(\exists y)(\forall z)(z \in y \Leftrightarrow (\exists t)(t \in x \wedge p[t/x, z/y]))}_{y \text{ is the image of } x} \right].$$

That completes the list: **ZF** consists of extension, separation, empty set, pair set, union, power set, infinity, foundation and replacement. Notice that we have not included the axiom of choice.

Definition 34.5.16 (ZFC)

ZFC is **ZF** together with **AC**, the axiom of choice, ‘every family of non-empty sets has a choice function’:

$$(\forall f) \left[(\forall x)(x \in \text{dom } f \Rightarrow f(x) \neq \emptyset) \Rightarrow (\exists g) (\text{dom } g = \text{dom } f \wedge (\forall x)(x \in \text{dom } g \Rightarrow g(x) \in f(x))) \right],$$

where a family of sets $\{A_i \mid i \in I\}$ is officially a function $f : I \rightarrow V$ with $f(i) = A_i$.

§34.5.3 Transitive Sets

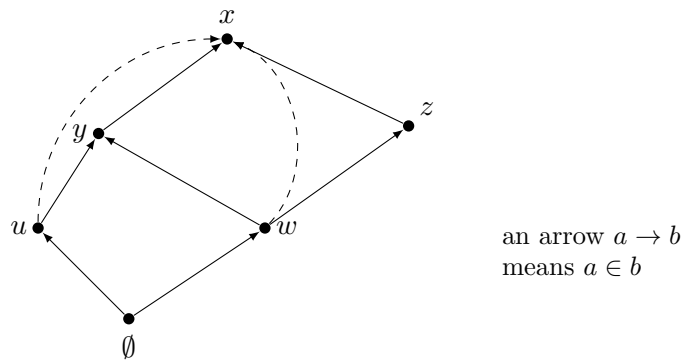
We now start to investigate what V must look like. The key notion is the following.

Definition 34.5.17 (Transitive Set)

A set x is **transitive** if every member of a member of x is a member of x :

$$(\forall y)(\forall z)((y \in z \wedge z \in x) \Rightarrow y \in x).$$

Equivalently, $y \in x$ implies $y \subseteq x$. For example $\emptyset$, $\{\emptyset\}$, $\{\emptyset, \{\emptyset\}\}$ and ω are transitive, whereas $\{\{\emptyset\}\}$ is not.



In the picture, the solid arrows are the memberships we started with, and x is transitive precisely when all the dashed arrows are present as well. The other thing the picture

shows is that following arrows downwards must terminate, which is exactly foundation, and this is what lets us do induction.

Lemma 34.5.18 (Transitive Closure)

Every set x is contained in a transitive set.

Proof. We would like to form

$$x \cup \left(\bigcup x\right) \cup \left(\bigcup\bigcup x\right) \cup \left(\bigcup\bigcup\bigcup x\right) \cup \cdots,$$

which is clearly transitive and contains x . By the union axiom, it is enough to obtain the set $\{x, \bigcup x, \bigcup\bigcup x, \dots\}$, and for that it is enough by replacement to check that $n \mapsto \bigcup^n x$ is a function-class on ω .

Define ‘ f is an attempt’ to mean

$$(f \text{ is a function}) \wedge (\text{dom } f \in \omega) \wedge (\text{dom } f \neq \emptyset) \wedge (f(0) = x) \\ \wedge (\forall n)(n \in \text{dom } f \wedge n \neq 0 \Rightarrow f(n) = \bigcup f(n-1)).$$

By the usual induction on ω , any two attempts agree wherever both are defined, and for each $n \in \omega$ there is an attempt defined at n (extend an attempt defined at $n-1$ by one step, using the union and pair set axioms). So the formula

$$p(y, z) = (\exists f)(f \text{ is an attempt} \wedge y \in \text{dom } f \wedge f(y) = z)$$

defines a function-class on ω , and we are done. $\square$

Since any intersection of transitive sets is transitive, it follows that x is contained in a *least* transitive set, called the **transitive closure** of x and written $\text{TC}(x)$.

§34.5.4 $\in$ -Induction and Foundation

We want foundation to give us the principle: if $p(x)$ holds whenever $p(y)$ holds for all $y \in x$, then p holds everywhere. This is $\in$ -induction.

Theorem 34.5.19 (Principle of $\in$ -Induction)

For each formula p with free variables $t_1, \dots, t_n, x$,

$$(\forall t_1) \cdots (\forall t_n) [(\forall x)((\forall y)(y \in x \Rightarrow p(y)) \Rightarrow p(x)) \Rightarrow (\forall x)p(x)].$$

Proof. Fix $t_1, \dots, t_n$ and suppose $\neg(\forall x)p(x)$, so $\neg p(x)$ for some x .

We would like to choose an $\in$ -minimal x with $\neg p(x)$ and apply the hypothesis, but $\{x \mid \neg p(x)\}$ need not be a set. Instead let $t = \text{TC}(\{x\})$ and $u = \{y \in t \mid \neg p(y)\}$, which is a set by separation. Then $u \neq \emptyset$ as $x \in u$, so by foundation u has an $\in$ -minimal member y . Now $\neg p(y)$; but if $z \in y$ then $z \in t$ by transitivity of t , and $z \notin u$ by minimality, so $p(z)$. So $p(z)$ holds for all $z \in y$, whence $p(y)$ by hypothesis, a contradiction. $\square$

Theorem 34.5.20

The axiom of foundation and the principle of $\in$ -induction are equivalent, in the presence of the other axioms of ZF.

Proof. We have just derived $\in$ -induction from foundation. Conversely, say that x is *regular* if

$$(\forall y)(x \in y \Rightarrow y \text{ has an } \in\text{-minimal member}).$$

Foundation says exactly that every x is regular (given non-empty y , take $x \in y$). To prove this by $\in$ -induction, it is enough to show that if every $y \in x$ is regular then x is regular.

So let z be a set with $x \in z$. If x is $\in$ -minimal in z we are done. Otherwise there is some $y \in x$ with $y \in z$; but y is regular and $y \in z$, so z has an $\in$ -minimal member. $\square$

§34.5.5 $\in$ -Recursion and Well-Founded Relations

Having induction, we want recursion: to define $F(x)$ in terms of the values $F(y)$ for $y \in x$.

Theorem 34.5.21 ($\in$ -Recursion Theorem)

Let G be an everywhere-defined function-class. Then there is a unique everywhere-defined function-class F such that

$$(\forall x)(F(x) = G(F|_x)),$$

where $F|_x = \{(y, F(y)) \mid y \in x\}$.

Proof. Note first that $F|_x$ is a set by replacement, so the statement makes sense.

Say ‘ f is an attempt’ if

$$(f \text{ is a function}) \wedge (\text{dom } f \text{ is transitive}) \wedge (\forall x)(x \in \text{dom } f \Rightarrow f(x) = G(f|_x)),$$

which makes sense as $\text{dom } f$ is transitive, so $x \in \text{dom } f$ gives $x \subseteq \text{dom } f$.

Any two attempts agree where both are defined, by $\in$ -induction: if f, f' agree at all $y \in x$ then $f|_x = f'|_x$, so $f(x) = G(f|_x) = G(f'|_x) = f'(x)$.

For each x there is an attempt defined at x , again by $\in$ -induction. Indeed, suppose each $y \in x$ has an attempt defined at y ; then for each such y there is a unique attempt f_y with domain $\text{TC}(\{y\})$, and $\{f_y \mid y \in x\}$ is a set by replacement. Put $f = \bigcup \{f_y \mid y \in x\}$, a function by the previous paragraph, and then $f' = f \cup \{(x, G(f|_x))\}$ is an attempt defined at x .

So we may define F by the formula

$$q(x, y) = (\exists f)(f \text{ is an attempt} \wedge x \in \text{dom } f \wedge f(x) = y).$$

Uniqueness of F is another $\in$ -induction. $\square$

It is worth asking what properties of the relation $\in$ we actually used. There are two.

- (i) It is **well-founded**: every non-empty set has an $\in$ -minimal member.
- (ii) It is **local**: for each y , the collection $\{x \mid x \in y\}$ is a set, which is what let us build transitive closures.

So for any well-founded, local relation given by a formula $r(x, y)$ we get r -induction and r -recursion by identical arguments. If r is a relation on a set a then locality is automatic, so we only need r to be well-founded. In particular, the induction and recursion theorems of Section 2 for well-orderings are special cases.

§34.5.6 Mostowski Collapse

One result about well-orderings remains to be generalised, namely subset collapse. Its analogue answers the question: when can a relation on a set be modelled by $\in$?

Definition 34.5.22 (Extensional Relation)

A relation r on a set a is **extensional** if

$$(\forall x, y \in a) [(\forall z \in a)(z r x \Leftrightarrow z r y) \Rightarrow x = y],$$

that is, if it obeys the axiom of extension.

Theorem 34.5.23 (Mostowski Collapse Theorem)

Let r be a well-founded extensional relation on a set a . Then there is a transitive set b and a bijection $f : a \rightarrow b$ with

$$(\forall x, y \in a)(x r y \Leftrightarrow f(x) \in f(y)).$$

Moreover b and f are unique.

Proof. Define f by r -recursion on a by

$$f(x) = \{f(y) \mid y r x\},$$

which is a set by replacement, and let $b = \{f(x) \mid x \in a\}$, again a set by replacement.

Then b is transitive by the definition of f (any member of $f(x)$ is some $f(y)$, which is in b), and f is surjective onto b by the definition of b . So it remains to show f is injective, and then $x r y \Leftrightarrow f(x) \in f(y)$ follows.

We show $(\forall y \in a)(f(x) = f(y) \Rightarrow x = y)$ for each $x \in a$, by r -induction on x . So suppose the claim for all $t r x$, and let $y \in a$ with $f(x) = f(y)$. By the definition of f we have $\{f(t) \mid t r x\} = \{f(u) \mid u r y\}$, so by the induction hypothesis $\{t \mid t r x\} = \{u \mid u r y\}$, and hence $x = y$ by extensionality.

For uniqueness, if f, f' both work then $f(x) = f'(x)$ for all $x \in a$ by r -induction. $\square$

This lets us give an honest definition of the ordinals, replacing the ‘isomorphism class’ fudge of Section 2.

Definition 34.5.24 (Ordinal, in ZF)

An **ordinal** is a transitive set that is totally ordered by $\in$.

Such a set is automatically well-ordered by $\in$, by foundation. For example $\emptyset$, $\{\emptyset\}$ and $\{\emptyset, \{\emptyset\}\}$ are ordinals, as is every $n \in \omega$ (recall $n = \{0, 1, \dots, n-1\}$), as is ω itself. Note the pleasant consequence that each ordinal *is* the set of ordinals below it, so $I_\alpha = \alpha$.

Now $\in$ on an ordinal is well-founded and extensional, so by Mostowski every well-ordering is order-isomorphic to a unique ordinal, which we call its **order type**. Hence two well-orderings are isomorphic if and only if they have the same order type, which is exactly what we wanted from ordinals in the first place.

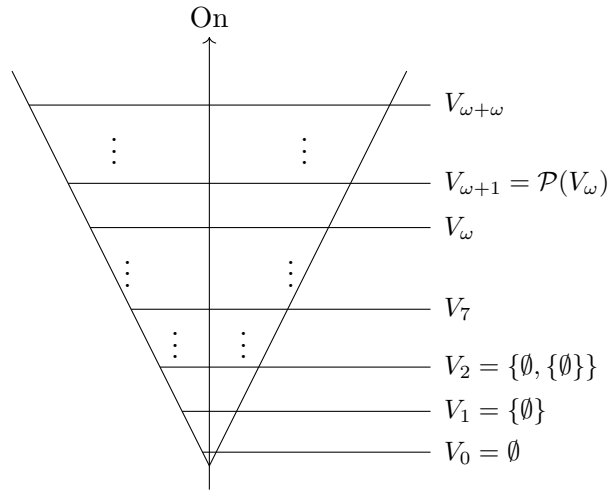
§34.5.7 The Cumulative Hierarchy

Finally we can draw the universe. The idea is to build V up in layers, starting from nothing and taking power sets, running along all the ordinals.

Definition 34.5.25 (Von Neumann Hierarchy)

Define sets V_α for $\alpha \in \text{On}$ by transfinite recursion:

$$\begin{aligned} V_0 &= \emptyset, \\ V_{\alpha+} &= \mathcal{P}(V_\alpha), \\ V_\lambda &= \bigcup \{V_\gamma \mid \gamma < \lambda\} \quad \text{for } \lambda \text{ a non-zero limit.} \end{aligned}$$



Every set will turn out to appear somewhere in this cone. Note first that $x \subseteq V_\alpha$ if and only if $x \in V_{\alpha+}$, straight from the definition.

Lemma 34.5.26

Each V_α is transitive.

Proof. Induct on α . Certainly $V_0 = \emptyset$ is transitive. If V_α is transitive and $x \in y \in \mathcal{P}(V_\alpha)$, then $y \subseteq V_\alpha$, so $x \in V_\alpha$, so $x \subseteq V_\alpha$ by transitivity, so $x \in \mathcal{P}(V_\alpha)$. Finally a union of transitive sets is transitive, which handles limits. $\square$

Lemma 34.5.27

If $\alpha \leq \beta$ then $V_\alpha \subseteq V_\beta$.

Proof. Fix α and induct on $\beta \geq \alpha$. The case $\beta = \alpha$ is trivial. Given $V_\alpha \subseteq V_\beta$, we have $V_\beta \subseteq \mathcal{P}(V_\beta)$ – since $x \in V_\beta$ implies $x \subseteq V_\beta$ by Lemma 34.5.26 – so $V_\alpha \subseteq \mathcal{P}(V_\beta) = V_{\beta+}$. Limits are immediate from the definition. $\square$

Theorem 34.5.28 (Every Set has a Rank)

Every set x belongs to some V_α .

Proof. We show $(\forall x)(\exists \alpha)(x \in V_\alpha)$ by $\in$ -induction on x . So we may assume that each $y \in x$ lies in some V_α , and hence that each $y \in x$ has a least α with $y \subseteq V_\alpha$; call it $\text{rank}(y)$, so that $y \in V_{\text{rank}(y)+}$.

Let $\alpha = \sup\{\text{rank}(y)^+ \mid y \in x\}$, which is a set by replacement. Then $y \in V_\alpha$ for every $y \in x$ by Lemma 34.5.27, so $x \subseteq V_\alpha$ and therefore $x \in V_{\alpha+}$. $\square$

So $V = \bigcup_{\alpha \in \text{On}} V_\alpha$: the picture above really is the whole universe. This is the sense in which the axiom of foundation succeeds in saying that every set is built out of simpler sets.

Definition 34.5.29 (Rank)

The **rank** of a set x is defined by $\in$ -recursion by

$$\text{rank}(x) = \sup\{\text{rank}(y)^+ \mid y \in x\}.$$

Proposition 34.5.30

$\text{rank}(x)$ is the least α with $x \subseteq V_\alpha$, equivalently the least α with $x \in V_{\alpha+}$.

Proof. This is precisely what the proof of Theorem 34.5.28 showed, by $\in$ -induction. $\square$

For example $\text{rank}(\emptyset) = 0$, $\text{rank}(n) = n$ for $n \in \omega$, $\text{rank}(\omega) = \omega$, and $\text{rank}(\alpha) = \alpha$ for every ordinal α . Note also that V_ω is precisely the collection of *hereditarily finite* sets – those which are finite, whose members are finite, and so on – and it is a model of every axiom of ZF except infinity.

Example 34.5.31 (Some Ranks)

We have $\text{rank}(\emptyset) = 0$, and $\text{rank}(\{x\}) = \text{rank}(x)^+$. So $\text{rank}(n) = n$ for $n \in \omega$, and more generally $\text{rank}(\alpha) = \alpha$ for every ordinal α : by $\in$ -induction, $\text{rank}(\alpha) = \sup\{\text{rank}(\beta)^+ \mid \beta \in \alpha\} = \sup\{\beta^+ \mid \beta < \alpha\} = \alpha$. Also $\text{rank}(\{\omega, \{\omega\}\}) = \omega + 2$.

Example 34.5.32 (V_ω and $V_{\omega+\omega}$)

The set V_ω consists of the **hereditarily finite** sets: those which are finite, all of whose members are finite, and so on. It is a model of every axiom of ZF except infinity, so infinity is not a consequence of the others. Similarly $V_{\omega+\omega}$ satisfies every axiom except replacement – the function-class $n \mapsto V_{\omega+n}$ has image unbounded in $V_{\omega+\omega}$ – so replacement does not follow from the rest either. Note that $V_{\omega+\omega}$ is a perfectly comfortable place to do most of mathematics: it contains $\mathbb{N}$, $\mathbb{R}$, all the usual function spaces, and so on.

Remark. The axiom of foundation is not needed for ordinary mathematics – one never wants a set with $x \in x$ – and indeed ZF without foundation is consistent if ZF is. What foundation buys us is precisely the picture above: it says that V has no ‘extra’ sets floating outside the cumulative hierarchy. Without it, $\bigcup_\alpha V_\alpha$ would still be a perfectly good model of the other axioms, in which foundation then holds.

§34.5.8 The Reflection Principle

We close this section with a striking consequence of replacement, which says that any finite piece of the theory of V is already visible in some V_α .

Theorem 34.5.33 (Reflection Principle)

Let $p_1, \dots, p_n$ be formulae. Then for every ordinal β there is $\alpha > \beta$ such that each p_i holds in V_α (with all parameters taken from V_α) if and only if it holds in V .

Proof sketch. We may assume the p_i are closed under subformulae. For each subformula of the form $(\exists x)q(x, \vec{y})$ define a function-class sending $\vec{y}$ to the least γ such that, if such an x exists at all, one exists in V_γ ; this is legitimate by replacement, since the ordinals are well-ordered. Given $\alpha_0 = \beta$, let α_{k+1} be an ordinal above α_k large enough that $V_{\alpha_{k+1}}$ contains a witness for every existential subformula with parameters in V_{α_k} , and set $\alpha = \sup_k \alpha_k$. Then V_α has a witness for each existential subformula whose parameters lie in V_α , and an induction on the structure of the formulae shows that each p_i holds in V_α if and only if it holds in V . $\square$

One consequence is worth noting: assuming ZF is consistent, it is not finitely axiomatizable. For suppose ZF were equivalent to a finite list of sentences $p_1, \dots, p_n$. Applying reflection to these gives an ordinal α with each p_i holding in V_α , so ZF would prove that some *set* is a model of ZF; by the soundness theorem this gives $\text{ZF} \vdash \text{Con}(\text{ZF})$, contradicting Gödel’s second incompleteness theorem in Section 34.7.

§34.6 Cardinals

Ordinals measure the *length* of a well-ordering. Cardinals measure the *size* of a set. We work in ZFC throughout, and note where choice is used.

§34.6.1 Cardinality

We write $x \leftrightarrow y$ to mean $(\exists f)(f \text{ is a bijection from } x \text{ to } y)$. We would like to attach to each set x an object $\text{card}(x)$ such that $\text{card}(x) = \text{card}(y)$ if and only if $x \leftrightarrow y$. The

obvious candidate, the collection of all sets bijecting with x , is a proper class, so we must be cleverer. In ZFC there is an easy fix.

Definition 34.6.1 (Cardinality)

The **cardinality** of a set x , written $\text{card}(x)$, is the least ordinal α with $x \leftrightarrow \alpha$.

This exists by the well-ordering principle: some well-ordering of x has an order type α with $x \leftrightarrow \alpha$, and then Proposition 34.2.25 gives a least one.

Remark (Scott's Trick). In ZF alone there may be no ordinal bijecting with x . Instead define the **essential rank** of x to be the least α such that some y with $y \leftrightarrow x$ has $\text{rank}(y) = \alpha$, and set

$$\text{card}(x) = \{y \in V_{\alpha+} \mid y \leftrightarrow x\},$$

which is a non-empty set with the required property. This device – replacing an equivalence class by its members of least rank – works for any equivalence relation on V .

Definition 34.6.2 (Cardinal)

A set m is a **cardinal** if $m = \text{card}(x)$ for some set x .

We use lower case letters m, n for cardinals and the corresponding upper case letters M, N for sets having those cardinalities.

Definition 34.6.3 (Cardinal Inequality)

For cardinals m, n we write $m \leq n$ if M injects into N , and $m < n$ if $m \leq n$ and $m \neq n$.

This is well-defined (independent of representatives), and it is a partial order: antisymmetry is precisely the Schröder–Bernstein theorem, Corollary 34.3.10. In ZFC it is a total order, since any two sets can be well-ordered and then compared by Proposition 34.2.16.

§34.6.2 The Alephs

Which ordinals are cardinals? Since $\omega \leftrightarrow \omega + 1 \leftrightarrow \omega \cdot 2 \leftrightarrow \omega^2$, most of them are not.

Definition 34.6.4 (Initial Ordinal)

An ordinal α is **initial** if no $\beta < \alpha$ has $\beta \leftrightarrow \alpha$; that is, if it is the least ordinal of its cardinality.

So the cardinals which are ordinals are exactly the initial ordinals. Every finite ordinal is initial, as are ω and $\gamma(x)$ for any set x (by minimality in Hartogs' lemma).

Definition 34.6.5 (The Omegas)

Define ω_α for $\alpha \in \text{On}$ by transfinite recursion:

$$\begin{aligned}\omega_0 &= \omega, \\ \omega_{\alpha+} &= \gamma(\omega_\alpha),\end{aligned}$$

$$\omega_\lambda = \sup\{\omega_\alpha \mid \alpha < \lambda\} \quad \text{for } \lambda \text{ a non-zero limit.}$$

Proposition 34.6.6

Each ω_α is initial, and every infinite initial ordinal is some ω_α .

Proof. That each ω_α is initial follows by induction: γ of an ordinal is initial, and a supremum of initial ordinals is initial (if $\beta < \omega_\lambda$ bijected with ω_λ then $\beta < \omega_\alpha$ for some $\alpha < \lambda$, and ω_λ would inject into ω_α).

Conversely let $\delta \geq \omega$ be initial. By induction $\omega_\alpha \geq \alpha$ for every α , so the ω_α are unbounded and there is a least ϵ with $\delta < \omega_\epsilon$. This ϵ is not a limit (else $\delta < \omega_\alpha$ for some $\alpha < \epsilon$) and not zero (as $\delta \geq \omega = \omega_0$), so $\epsilon = \alpha^+$ and $\omega_\alpha \leq \delta < \omega_{\alpha^+} = \gamma(\omega_\alpha)$. Then δ injects into ω_α , so being initial forces $\delta = \omega_\alpha$. $\square$

Definition 34.6.7 (Aleph)

We write $\aleph_\alpha = \text{card}(\omega_\alpha)$.

We use a separate symbol because card may be defined by Scott's trick rather than as an ordinal. From the proposition, the $\aleph_\alpha$ are precisely the cardinalities of the infinite sets – or, in ZF without choice, of the infinite *well-orderable* sets.

§34.6.3 Cardinal Arithmetic**Definition 34.6.8 (Cardinal Arithmetic)**

For cardinals m, n we define

$$m + n = \text{card}(M \sqcup N), \quad mn = \text{card}(M \times N), \quad m^n = \text{card}(M^N),$$

where $M^N = \{f \mid f \text{ is a function } N \rightarrow M\}$.

These do not depend on the choice of representatives. Unlike ordinal arithmetic, cardinal addition and multiplication are commutative and associative, and satisfy the usual laws such as $(m^n)^p = m^{np}$ – the last because a function $P \rightarrow (N \rightarrow M)$ is the same thing as a function $N \times P \rightarrow M$. Note that the notation clashes with ordinal arithmetic: $2^\omega = \omega$ as ordinals, but $2^{\aleph_0} > \aleph_0$ as cardinals. Which is meant is always determined by whether the symbols are ordinals or cardinals.

Example 34.6.9

$\mathbb{R} \leftrightarrow \mathcal{P}(\omega) \leftrightarrow \{0, 1\}^\omega$, so $\text{card}(\mathbb{R}) = \text{card}(\mathcal{P}(\omega)) = 2^{\aleph_0}$.

Example 34.6.10 (How Many Sequences of Reals?)

A real sequence is a function $\omega \rightarrow \mathbb{R}$, so the answer is

$$\text{card}(\mathbb{R}^\omega) = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 \cdot \aleph_0} = 2^{\aleph_0} = \text{card}(\mathbb{R}),$$

using $\aleph_0 \aleph_0 = \aleph_0$, which holds since $\omega \times \omega \leftrightarrow \omega$.

Theorem 34.6.11 (Cantor)

For every cardinal m we have $m < 2^m$.

Proof. Certainly $m \leq 2^m$, as $x \mapsto \{x\}$ injects M into $\mathcal{P}(M)$. Suppose $f : M \rightarrow \mathcal{P}(M)$ were a bijection, and let $A = \{x \in M \mid x \notin f(x)\}$. Then $A = f(a)$ for some $a \in M$, and now $a \in A$ if and only if $a \notin f(a) = A$, which is absurd. $\square$

Example 34.6.12 (More Cardinal Computations)

- (i) There are $2^{\aleph_0}$ continuous functions $\mathbb{R} \rightarrow \mathbb{R}$. Each is determined by its restriction to $\mathbb{Q}$, so there are at most $\text{card}(\mathbb{R}^{\mathbb{Q}}) = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ of them, and at least that many since the constants are continuous.
- (ii) There are $2^{2^{\aleph_0}}$ functions $\mathbb{R} \rightarrow \mathbb{R}$ altogether, namely $\text{card}(\mathbb{R}^{\mathbb{R}}) = (2^{\aleph_0})^{2^{\aleph_0}} = 2^{\aleph_0 \cdot 2^{\aleph_0}} = 2^{2^{\aleph_0}}$. So ‘almost no’ function is continuous.
- (iii) The set of finite sequences from an infinite set X has the same cardinality as X , since $\sum_{n < \omega} m^n = \aleph_0 \cdot m = m$ using Theorem 34.6.13. In particular there are only countably many formulae in a countable language, which is the fact we used repeatedly in Sections 34.4 and 34.7.

That last fact generalises completely, and it makes cardinal addition and multiplication of infinite cardinals utterly uninteresting.

Theorem 34.6.13

For every ordinal α we have $\aleph_\alpha \aleph_\alpha = \aleph_\alpha$.

Proof. We induct on α . Define a well-ordering of $\omega_\alpha \times \omega_\alpha$ by ‘going up in squares’: set $(x, y) < (z, w)$ if

- $\max(x, y) < \max(z, w)$; or
- $\max(x, y) = \max(z, w) = \beta$, and one of: $y < \beta$ and $z < \beta$; or $x = z = \beta$ and $y < w$; or $y = w = \beta$ and $x < z$.

In words, we enumerate the pairs by sweeping out successively larger square blocks, and within the β -th block we go up the left edge and then along the top.

Given $\delta \in \omega_\alpha \times \omega_\alpha$ we have $\delta \in \beta \times \beta$ for some $\beta < \omega_\alpha$, as ω_α is a limit. By the induction hypothesis $\beta \times \beta \leftrightarrow \beta$ (or β is finite), so the initial segment I_δ is contained in $\beta \times \beta$ and hence

$$\text{card}(I_\delta) \leq \text{card}(\beta \times \beta) = \text{card}(\beta) < \text{card}(\omega_\alpha).$$

So every proper initial segment of our well-ordering is smaller than ω_α , meaning its order type is at most ω_α . Hence $\omega_\alpha \times \omega_\alpha$ injects into ω_α , and the reverse injection

is obvious, so they biject by Schröder–Bernstein. $\square$

Corollary 34.6.14

For ordinals $\alpha \leq \beta$ we have $\aleph_\alpha + \aleph_\beta = \aleph_\alpha \aleph_\beta = \aleph_\beta$.

Proof. $\aleph_\beta \leq \aleph_\alpha + \aleph_\beta \leq 2\aleph_\beta \leq \aleph_\alpha \aleph_\beta \leq \aleph_\beta^2 = \aleph_\beta$. $\square$

So, for instance, $X \sqcup X \leftrightarrow X$ and $X \times X \leftrightarrow X$ for every infinite set X – results which in ZF alone are not provable.

Cardinal *exponentiation*, by contrast, is genuinely hard. We know $m < 2^m$ for all m by Cantor’s diagonal argument, so $\aleph_0 < 2^{\aleph_0}$, and $\aleph_1 \leq 2^{\aleph_0}$ since $2^{\aleph_0}$ is the cardinality of an uncountable well-orderable set. Whether equality holds – the **Continuum Hypothesis** – is independent of ZFC: Gödel showed it is consistent with ZFC, and Cohen showed its negation is too. ZFC does not even decide whether $2^{\aleph_0} < 2^{\aleph_1}$.

It is convenient to name the cardinals obtained by iterating the power set.

Definition 34.6.15 (Beth Numbers)

Define $\beth_0 = \aleph_0$, $\beth_{\alpha+} = 2^{\beth_\alpha}$, and $\beth_\lambda = \sup\{\beth_\alpha \mid \alpha < \lambda\}$ for a non-zero limit λ .

So $\beth_1 = 2^{\aleph_0} = \text{card}(\mathbb{R})$, and by Cantor’s theorem $\beth_\alpha \geq \aleph_\alpha$ for all α . The **generalised continuum hypothesis** is the assertion that $\beth_\alpha = \aleph_\alpha$ for every α , that is, that $2^{\aleph_\alpha} = \aleph_{\alpha+}$; like the continuum hypothesis it is independent of ZFC. Note that $\beth_\omega$ is a cardinal with $\beth_\omega = \sup_n \beth_n$, so it is a limit of cardinals reached from below by a sequence of length ω – which brings us to our last topic.

§34.6.4 Cofinality

There is, however, one genuine constraint on cardinal exponentiation, and it comes from the following notion.

Definition 34.6.16 (Cofinality)

The **cofinality** $\text{cf}(\alpha)$ of a limit ordinal α is the least order type of a subset of α which is unbounded in α . An infinite cardinal ω_α is **regular** if $\text{cf}(\omega_\alpha) = \omega_\alpha$, and **singular** otherwise.

Example 34.6.17

$\text{cf}(\omega) = \omega$, so ω is regular. Also $\text{cf}(\omega_1) = \omega_1$: we saw in Section 34.2.9 that every countable subset of ω_1 is bounded, so no subset of order type less than ω_1 is unbounded. On the other hand $\text{cf}(\omega \cdot 2) = \omega$, and $\text{cf}(\aleph_\omega) = \omega$ since $\aleph_\omega = \sup\{\aleph_n \mid n < \omega\}$, so $\aleph_\omega$ is singular.

The argument for ω_1 generalises: for any α , the cardinal $\aleph_{\alpha+}$ is regular, since a union of $\aleph_\alpha$ sets each of size $\aleph_\alpha$ has size $\aleph_\alpha$ by Theorem 34.6.13. So successor cardinals are regular (in ZFC), and only limit cardinals such as $\aleph_\omega$ can be singular.

Proposition 34.6.18

For any limit ordinal α , the ordinal $\text{cf}(\alpha)$ is a regular cardinal.

Proof. Let $\gamma = \text{cf}(\alpha)$ and let $S \subseteq \alpha$ be unbounded of order type γ . If $T \subseteq \gamma$ were unbounded in γ of order type $\delta < \gamma$, then the corresponding subset of S would be unbounded in α and have order type δ , contradicting minimality of γ . So $\text{cf}(\gamma) = \gamma$. In particular γ is initial, since a bijection with a smaller ordinal would give an unbounded subset of order type less than γ . $\square$

Cofinality bounds exponentiation from below.

Theorem 34.6.19 (König)

For every infinite cardinal m we have $\text{cf}(2^m) > m$.

Proof sketch. Suppose $2^m = \sup\{n_i \mid i \in I\}$ with $|I| \leq m$ and each $n_i < 2^m$. Writing 2^m as the set of functions $M \rightarrow \{0, 1\}$ and splitting it into $|I|$ pieces of size n_i , a diagonal argument produces a function differing from everything in every piece, which is a contradiction. $\square$

In particular $2^{\aleph_0} \neq \aleph_\omega$, since $\text{cf}(\aleph_\omega) = \omega$. So although ZFC is nearly silent about $2^{\aleph_0}$, it is not completely silent.

§34.7 Incompleteness

We finish by returning to Peano arithmetic. Our aim is to show that PA is *incomplete*: there is a sentence p with $\text{PA} \not\vdash p$ and $\text{PA} \not\vdash \neg p$. Since one of $p, \neg p$ must hold in $\mathbb{N}$, this produces a sentence which is true (in $\mathbb{N}$) but unprovable.

Note that this is not in tension with the completeness theorem, which says that $\text{PA} \vdash p$ whenever p holds in every model of PA. What it will show is that the non-standard models of Section 34.4 cannot be legislated away.

The strategy is beautiful and simple to state: we find a sentence p which asserts ‘ p is not provable in PA’. Then p is true if and only if p is unprovable. If p were false, it would be provable and hence true by soundness, a contradiction; so p is true, and therefore unprovable. All the work is in making a sentence of arithmetic talk about sentences of arithmetic.

Throughout, ‘true’ means true in $\mathbb{N}$ and ‘provable’ means provable in PA.

§34.7.1 Definability

Definition 34.7.1 (Definable)

A subset $S \subseteq \mathbb{N}$ is **definable** if there is a formula p with one free variable x such that for all $m \in \mathbb{N}$, we have $m \in S$ if and only if $p[m/x]$ is true. A function $f : \mathbb{N} \rightarrow \mathbb{N}$ is **definable** if there is a formula p with free variables x, y such that $f(m) = n$ if and only if $p[m/x, n/y]$ is true.

Example 34.7.2

The set of squares is definable by $(\exists y)(y \cdot y = x)$. The set of primes is definable by

$$x \neq 1 \wedge (\forall y)(\forall z)(y \cdot z = x \Rightarrow (y = 1 \vee z = 1)).$$

The function $x \mapsto \lfloor x/2 \rfloor$ is definable by $(x = 2y) \vee (x = 2y + 1)$. The set of powers of 2 is definable by ‘every prime dividing x equals 2’.

Since there are only countably many formulae, only countably many subsets of $\mathbb{N}$ are definable. The fact we need, which we shall not prove, is a converse of sorts: *every function or set given by an algorithm is definable in PA*.

The precise statement runs through primitive recursion.

Definition 34.7.3 (Primitive Recursive)

The **primitive recursive** functions $\mathbb{N}^k \rightarrow \mathbb{N}$ are those obtained from

- (i) the constant 0, the successor function s , and the projections $(x_1, \dots, x_k) \mapsto x_i$, by finitely many applications of
- (ii) *composition*: from g and $h_1, \dots, h_m$ form $\vec{x} \mapsto g(h_1(\vec{x}), \dots, h_m(\vec{x}))$;
- (iii) *primitive recursion*: from g and h form the function f satisfying $f(\vec{x}, 0) = g(\vec{x})$ and $f(\vec{x}, n^+) = h(\vec{x}, n, f(\vec{x}, n))$.

Addition, multiplication, exponentiation, factorials, ‘the n th prime’, bounded sums and products, and the characteristic functions of ‘ x divides y ’ and ‘ x is prime’ are all primitive recursive, by writing down the recursions. One then shows that every primitive recursive function is definable in PA, essentially by coding the finite sequence of intermediate values with the prime-power trick of the next section. Every function we need below is primitive recursive, so we may take definability for granted.

§34.7.2 Coding

The language of PA has the symbols

$$0, s, +, \cdot, =, \perp, \Rightarrow, (,), \forall, x, ',$$

where variables are $x, x', x'', \dots$. Assign to each symbol a number, say $v(0) = 1$, $v(s) = 2$, $\dots$, $v(') = 12$. Then encode a formula p with symbols $\sigma_1 \sigma_2 \dots \sigma_n$ by

$$c(p) = 2^{v(\sigma_1)} 3^{v(\sigma_2)} 5^{v(\sigma_3)} \dots (n\text{th prime})^{v(\sigma_n)}.$$

symbols	(	∀	x	)	(	x	=	0	)
	↓	↓	↓	↓	↓	↓	↓	↓	↓
codes	8	10	11	9	8	11	5	1	9
					↓				

$$c(p) = 2^8 \cdot 3^{10} \cdot 5^{11} \cdot 7^9 \cdot 11^8 \cdot 13^{11} \cdot 17^5 \cdot 19^1 \cdot 23^9$$

By unique factorisation, we can recover p from $c(p)$. Not every number codes a formula, and we write S_n for the formula coded by n , taking $S_n = \perp$ if n codes nothing.

The point of all this is that the syntactic notions we care about become arithmetical statements about codes, each of which is decided by an algorithm and hence is definable:

- (i) ‘ n codes a formula’;
- (ii) ‘ n codes a formula with exactly one free variable’;
- (iii) ‘ l, m, n code formulae and S_n follows from S_l, S_m by modus ponens’, and similarly for generalisation;
- (iv) ‘ n codes a logical axiom or an axiom of PA’.

Sequences of formulae are coded the same way, by

$$s(p_1, \dots, p_n) = 2^{c(p_1)} 3^{c(p_2)} \dots (n\text{th prime})^{c(p_n)},$$

and then ‘ n codes a proof’ and ‘ n codes a proof of S_m ’ are definable too. Let $\theta(m, n)$ be a formula defining ‘ n codes a proof of S_m ’. Then

$$\phi(m) = (\exists n) \theta(m, n)$$

defines ‘ S_m is provable’.

§34.7.3 Gödel’s Incompleteness Theorems

Now for the clever bit. Consider the statement

$$\chi(m) = \text{‘}m \text{ codes a formula } S_m \text{ with one free variable, and } S_m[m/x] \text{ is unprovable’}.$$

This is again decided by an algorithm applied to the definable ingredients above, so it is definable: there is a formula p with one free variable such that $\chi(m)$ holds if and only if $p[m/x]$ is true.

Let $N = c(p)$, so $S_N = p$. Then $p[N/x]$ says precisely that N codes a formula with one free variable whose substitution instance at N is unprovable. But that substitution instance is $S_N[N/x] = p[N/x]$ itself. So

$$p[N/x] \text{ asserts that } p[N/x] \text{ is unprovable,}$$

which is exactly the self-referential sentence we wanted.

Theorem 34.7.4 (Gödel’s First Incompleteness Theorem)

PA is incomplete: there is a sentence which is true in $\mathbb{N}$ but not provable in PA.

Proof. Let q be the sentence $p[N/x]$ constructed above, so that q is true if and only if q is unprovable. If q is false then q is provable, hence true by soundness – a contradiction. So q is true, and therefore unprovable. As q is true, $\neg q$ is false and so also unprovable, again by soundness. $\square$

Note that nothing in this argument is special to PA: it applies to any consistent theory strong enough to code its own syntax, and to which we have added any true sentences we like. So we cannot repair PA by adding axioms, provided that we can still recognise the axioms algorithmically.

Theorem 34.7.5 (Gödel's Second Incompleteness Theorem)

$\text{PA} \not\vdash \text{Con}(\text{PA})$, where $\text{Con}(\text{PA})$ is the sentence $(\forall n)(n \text{ does not code a proof of } \perp)$.

Proof sketch. Examining the proof above, the only external ingredient was soundness, which we used via the existence of a model of PA – and any model would do. So the argument formalises inside PA itself, to give

$$\text{PA} \cup \{\text{Con}(\text{PA})\} \vdash q.$$

If $\text{PA} \vdash \text{Con}(\text{PA})$ we would then have $\text{PA} \vdash q$, contradicting the first incompleteness theorem. $\square$

Finally, let T be the set of all sentences true in $\mathbb{N}$. This is a complete theory extending PA , so the argument above must break down for it – and the only step which can fail is the definability of the syntactic notions. So we obtain the following.

Theorem 34.7.6 (Undecidability of Truth)

There is no algorithm to decide whether a sentence of arithmetic is true in $\mathbb{N}$. Equivalently, the set $\{c(p) \mid p \in T\}$ is not definable.

Remark (The Entscheidungsproblem). In Section 34.4 we noted that there is no analogue of the decidability theorem for predicate logic. The methods of this section are what settle the matter: the set of theorems of predicate logic in a language with a single binary relation symbol is not decidable, since one may code arithmetic into it. What *is* true is that the theorems are recursively enumerable – by the completeness theorem we may simply enumerate all proofs – so validity is semi-decidable. Truth in $\mathbb{N}$, by the theorem above, is not even that.

The same phenomena occur higher up. ZFC proves that PA has a model, namely ω , so $\text{ZFC} \vdash \text{Con}(\text{PA})$ – which by the second incompleteness theorem shows that ZFC is strictly stronger than PA . But ZFC can code its own syntax just as well as PA can, so if ZFC is consistent then ZFC is incomplete, and $\text{ZFC} \not\vdash \text{Con}(\text{ZFC})$. There is no escape by moving to a stronger theory: any consistent, recognisable set of axioms strong enough to do arithmetic will leave some true statement unproved.

35 Number Fields

Number theory begins with the integers, and the single most important fact about the integers is that every one of them factors into primes in exactly one way. An enormous amount of elementary number theory is really just this statement being pushed around. So it is natural, when we want to solve an equation like $y^2 + 2 = x^3$ or $x^2 + y^2 = p$, to enlarge $\mathbb{Z}$ a little – to work in $\mathbb{Z}[\sqrt{-2}]$ or $\mathbb{Z}[i]$ – and hope that unique factorisation survives the move. Sometimes it does, and then the equation falls apart in a few lines. The trouble is that usually it does not: in $\mathbb{Z}[\sqrt{-5}]$ we have the notorious

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

and all four factors are irreducible. Unique factorisation of *elements* is simply false in general.

The great insight, due to Kummer and then Dedekind, is that nothing is really lost – we were just looking at the wrong objects. If instead of elements we factor *ideals*, unique factorisation comes back, exactly and in complete generality, in the ring of integers of any number field. The extent to which ideals fail to be principal – that is, the extent to which the ideal-theoretic picture differs from the element-theoretic one – is measured by a finite abelian group, the **class group**, and computing this group for a given field is one of the central occupations of the subject. The other central object is the group of units, whose structure is pinned down by Dirichlet’s unit theorem and which is intimately bound up with Pell’s equation.

Both of the two big finiteness theorems – finiteness of the class group, and the rank of the unit group – are proved by the same startling device: we embed our number field into a Euclidean space so that its ring of integers becomes a *lattice*, and then apply a piece of convex geometry due to Minkowski. Arithmetic questions become questions about counting lattice points in convex bodies. That this works at all is one of the small miracles of mathematics.

This article constitutes my notes for the ‘Number Fields’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§35.1 Introduction

§35.1.1 A Motivating Problem

Which integers can be written as $x^2 + y^2$? The answer is classical and completely satisfying: an odd prime p is a sum of two squares exactly when $p \equiv 1 \pmod{4}$. The slick proof, which we will give properly in Section 35.10, goes as follows. Factor

$$p = x^2 + y^2 = (x + iy)(x - iy)$$

in the ring $\mathbb{Z}[i]$ of Gaussian integers. This ring turns out to be Euclidean, hence a principal ideal domain, hence a unique factorisation domain; and in a UFD one can chase the factorisation of p around and read off the answer from whether -1 is a square mod p .

Emboldened, we might ask: which integers are of the form $x^2 + 5y^2$? Now we would like to factor

$$n = x^2 + 5y^2 = (x + y\sqrt{-5})(x - y\sqrt{-5})$$

in $\mathbb{Z}[\sqrt{-5}]$, and the argument collapses – because $\mathbb{Z}[\sqrt{-5}]$ is *not* a unique factorisation domain:

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

with all four factors irreducible and no two associate. This is a genuine obstruction, not an artefact of a bad proof, and understanding it is the whole point of this course.

§35.1.2 Kummer's Idea

The resolution, due to Kummer in the 1840s and refined by Dedekind, is that unique factorisation has not really failed: we have simply been factoring the wrong objects. Kummer imagined ‘ideal numbers’ – hypothetical divisors, not present in the ring, which would refine 2, 3 and $1 \pm \sqrt{-5}$ into common building blocks. Dedekind made this precise by replacing an ideal number with the set of things it divides, that is, with an *ideal* of the ring. In $\mathbb{Z}[\sqrt{-5}]$ the relevant ideals are

$$\mathfrak{p} = (2, 1 + \sqrt{-5}), \quad \mathfrak{q}_1 = (3, 1 + \sqrt{-5}), \quad \mathfrak{q}_2 = (3, 1 - \sqrt{-5}),$$

and one computes that

$$(2) = \mathfrak{p}^2, \quad (3) = \mathfrak{q}_1 \mathfrak{q}_2, \quad (1 + \sqrt{-5}) = \mathfrak{p} \mathfrak{q}_1, \quad (1 - \sqrt{-5}) = \mathfrak{p} \mathfrak{q}_2.$$

So the two apparently different factorisations of 6 are both the same factorisation

$$(6) = \mathfrak{p} \cdot \mathfrak{p} \cdot \mathfrak{q}_1 \cdot \mathfrak{q}_2,$$

merely bracketed in two ways. Unique factorisation was there all along; it just could not be seen at the level of elements. The main theorem of Section 35.4 is that this always happens: in the ring of integers of *any* number field, every nonzero ideal factors uniquely into prime ideals.

Of course, we then want to know how badly the ideals fail to be principal – because that is what stops us from converting statements about ideals back into statements about elements. The answer is a group, the class group, and one of the two big theorems of the course is that it is finite.

§35.1.3 What We Will Do

The plan of the course is as follows.

- Sections 35.2–35.3 set up the basic objects: number fields K , their rings of integers $\mathcal{O}_K$, and the numerical invariants attached to them – norms, traces, integral bases and the discriminant d_K .
- Sections 35.4–35.5 develop the ideal theory: $\mathcal{O}_K$ is a Dedekind domain, ideals factor uniquely into primes, and the failure of principality is measured by Cl_K .
- Section 35.6 answers the question ‘what are the prime ideals?’. They all sit above rational primes, and Dedekind’s criterion reduces finding them to factoring a polynomial mod p . The primes that behave badly – the *ramified* ones – are exactly the divisors of d_K .

- Section 35.7 is the geometric heart. We embed $\mathcal{O}_K$ into $\mathbb{R}^n$ as a lattice and apply Minkowski's convex body theorem to deduce that Cl_K is finite, with an explicit bound. Section 35.8 puts this to work computing class groups.
- Section 35.9 determines the unit group $\mathcal{O}_K^\times$ by the same geometric method: Dirichlet's unit theorem. In the real quadratic case this is the theory of Pell's equation.
- Section 35.10 solves some Diophantine equations, and Section 35.11 (non-examinable) shows how the class number and the regulator appear together as the residue of the Dedekind zeta function.

Throughout, the reader should keep two examples in mind: the quadratic fields $\mathbb{Q}(\sqrt{d})$, where everything can be computed by hand, and the cyclotomic fields $\mathbb{Q}(\zeta_m)$, which are where the subject came from and where it is still headed.

§35.2 Algebraic Integers

§35.2.1 Number Fields

We will work throughout with finite extensions of $\mathbb{Q}$. Recall from Galois Theory that if $K \subseteq L$ are fields with $\dim_K L < \infty$, we write $[L : K]$ for this dimension and call L/K a finite extension; every element of L is then algebraic over K .

Definition 35.2.1 (Number Field)

A **number field** is a finite extension of $\mathbb{Q}$. Its **degree** is $[K : \mathbb{Q}]$.

So $\mathbb{Q}$, $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt[3]{2})$ and $\mathbb{Q}(\zeta_5)$ are all number fields, of degrees 1, 2, 2, 3 and 4 respectively. Since $\mathbb{Q}$ has characteristic zero, every number field is a separable extension of $\mathbb{Q}$, and so by the primitive element theorem every number field has the form $K = \mathbb{Q}(\alpha)$ for a single α . We will use this constantly.

Inside $\mathbb{Q}$ sits $\mathbb{Z}$, and it is the arithmetic of $\mathbb{Z}$ that we care about. What we want, then, is the analogue of $\mathbb{Z}$ inside a general number field K . There is an obvious wrong answer – take everything that is a root of *some* polynomial over $\mathbb{Q}$, which gives all of K – and the correct answer comes from insisting on *monic integer* polynomials.

Definition 35.2.2 (Algebraic Integer)

Let K be a number field. An element $\alpha \in K$ is an **algebraic integer** if there is a monic polynomial $f \in \mathbb{Z}[x]$ with $f(\alpha) = 0$. We write

$$\mathcal{O}_K = \{\alpha \in K : \alpha \text{ is an algebraic integer}\}$$

for the set of algebraic integers of K , called the **ring of integers** of K .

Calling $\mathcal{O}_K$ a ring is jumping the gun – that it is closed under addition and multiplication is the first thing we have to prove, and it is not obvious. But let us first make sure that the definition does the right thing over $\mathbb{Q}$.

Lemma 35.2.3

$$\mathcal{O}_{\mathbb{Q}} = \mathbb{Z}.$$

Proof. If $a \in \mathbb{Z}$ then $x - a$ is monic with integer coefficients and kills a , so $\mathbb{Z} \subseteq \mathcal{O}_{\mathbb{Q}}$. Conversely suppose $\alpha = r/s \in \mathbb{Q}$ with r, s coprime and $s \geq 1$, and suppose

$$\alpha^n + a_{n-1}\alpha^{n-1} + \cdots + a_0 = 0$$

with all $a_i \in \mathbb{Z}$. Multiplying through by s^n gives

$$r^n = -s(a_{n-1}r^{n-1} + a_{n-2}r^{n-2}s + \cdots + a_0s^{n-1}),$$

so $s \mid r^n$. If $s > 1$, pick a prime $p \mid s$; then $p \mid r^n$, so $p \mid r$, contradicting coprimality. Hence $s = 1$ and $\alpha \in \mathbb{Z}$. $\square$

This is a reassuring start, and it also explains a piece of terminology: elements of $\mathbb{Z}$ are sometimes called **rational integers** when we need to distinguish them from algebraic integers. The picture to keep in mind is a square, with $\mathcal{O}_K$ playing the role for K that $\mathbb{Z}$ plays for $\mathbb{Q}$.

$$\begin{array}{ccc} & \text{field of fractions} & \\ \mathcal{O}_K & \hookrightarrow & K \\ \uparrow & & \uparrow \\ \mathbb{Z} & \hookrightarrow & \mathbb{Q} \\ & \text{field of fractions} & \end{array}$$

We will justify both ‘field of fractions’ labels shortly. Note straight away that $\alpha \in \mathcal{O}_K$ does *not* imply $1/\alpha \in \mathcal{O}_K$ – for instance $2 \in \mathcal{O}_{\mathbb{Q}} = \mathbb{Z}$ but $1/2 \notin \mathbb{Z}$. Understanding exactly which α have $1/\alpha \in \mathcal{O}_K$ is the theory of units, which we take up in Section 35.9.

§35.2.2 Integrality

The obstacle to showing $\mathcal{O}_K$ is a ring is that the Galois-theoretic argument – that $\alpha \pm \beta$ and $\alpha\beta$ are algebraic whenever α, β are – produces polynomials that need not be monic over $\mathbb{Z}$. We need a genuinely different idea, and the right one is a piece of commutative algebra which we may as well set up in general.

Definition 35.2.4 (Integral)

Let $R \subseteq S$ be commutative rings with 1.

- (i) An element $\alpha \in S$ is **integral over R** if there is a monic $f \in R[x]$ with $f(\alpha) = 0$.
- (ii) S is **integral over R** if every $\alpha \in S$ is integral over R .
- (iii) S is **finitely generated as an R -module** if there are $\alpha_1, \dots, \alpha_n \in S$ such that every element of S is an R -linear combination of the α_i ; equivalently, the map $R^n \rightarrow S$, $(r_1, \dots, r_n) \mapsto \sum r_i \alpha_i$, is surjective.

So $\alpha \in K$ is an algebraic integer precisely when it is integral over $\mathbb{Z}$. If $\alpha_1, \dots, \alpha_r \in S$ we write $R[\alpha_1, \dots, \alpha_r]$ for the subring of S generated by R and the α_i ; this is the image of the evaluation map $R[x_1, \dots, x_r] \rightarrow S$ sending $x_i \mapsto \alpha_i$. Beware that being finitely generated as a *module* is much stronger than being finitely generated as a *ring*: $\mathbb{Z}[x]$ is

generated as a ring by one element over $\mathbb{Z}$, but is very far from being a finitely generated $\mathbb{Z}$ -module.

The two notions are linked by the following pair of results, and the link is the whole point.

Proposition 35.2.5

Let $R \subseteq S$ be rings and let $s \in S$ be integral over R . Then $R[s]$ is a finitely generated R -module. More generally, if $s_1, \dots, s_n \in S$ are each integral over R , then $R[s_1, \dots, s_n]$ is a finitely generated R -module.

Proof. The ring $R[s]$ is spanned as an R -module by $1, s, s^2, \dots$. By hypothesis there are $a_0, \dots, a_{n-1} \in R$ with $s^n = a_{n-1}s^{n-1} + \dots + a_0$. Let $M = R \cdot 1 + Rs + \dots + Rs^{n-1}$. Then $sM \subseteq M$: indeed $s \cdot s^i = s^{i+1} \in M$ for $i < n-1$, and $s \cdot s^{n-1} = s^n \in M$ by the displayed relation. Hence M contains $s^n, s^{n+1}, \dots$ by induction, so $M = R[s]$, which is therefore generated by the n elements $1, s, \dots, s^{n-1}$.

For the general statement, first note the transitivity of finite generation: if $A \subseteq B \subseteq C$ with B a finitely generated A -module, say with generators b_i , and C a finitely generated B -module, say with generators c_j , then the products $b_i c_j$ generate C as an A -module. Now put $S_i = R[s_1, \dots, s_i]$, so $S_0 = R$ and $S_{i+1} = S_i[s_{i+1}]$. Each s_{i+1} is integral over R and hence over the larger ring S_i , so S_{i+1} is a finitely generated S_i -module by the first part. Transitivity gives the result. $\square$

Now for the converse direction, which is the clever one. The proof is essentially the proof of the Cayley–Hamilton theorem, and it is worth appreciating that it is entirely constructive: it hands us the monic polynomial explicitly, as a characteristic polynomial.

Theorem 35.2.6

Let $R \subseteq S$ be rings with S a finitely generated R -module. Then S is integral over R .

Proof. Let $\alpha_1, \dots, \alpha_n$ generate S as an R -module; since $1 \in S$ we may assume that $\alpha_1 = 1$ after enlarging the generating set. Fix $s \in S$. Each $s\alpha_i$ lies in S , so we may write

$$s\alpha_i = \sum_{j=1}^n b_{ij}\alpha_j$$

for some $b_{ij} \in R$. Setting $B = (b_{ij}) \in \text{Mat}_n(R)$ and $v = (\alpha_1, \dots, \alpha_n)^T$, this says

$$(sI - B)v = 0.$$

Recall that for any square matrix X over a commutative ring we have $\text{adj}(X)X = (\det X)I$. Multiplying on the left by $\text{adj}(sI - B)$ therefore gives $\det(sI - B)v = 0$; looking at the first coordinate and using $\alpha_1 = 1$ yields

$$\det(sI - B) = 0.$$

But $f(t) = \det(tI - B)$ is a monic polynomial of degree n with coefficients in R , and $f(s) = 0$. So s is integral over R . $\square$

Combining the two results gives exactly what we wanted.

Corollary 35.2.7

Let K be a number field. Then $\mathcal{O}_K$ is a subring of K .

Proof. Let $\alpha, \beta \in \mathcal{O}_K$. By Proposition 35.2.5, $\mathbb{Z}[\alpha, \beta]$ is a finitely generated $\mathbb{Z}$ -module, so by Theorem 35.2.6 every element of $\mathbb{Z}[\alpha, \beta]$ is integral over $\mathbb{Z}$. In particular $\alpha \pm \beta$ and $\alpha\beta$ are algebraic integers, and $0, 1 \in \mathcal{O}_K$ trivially. $\square$

The same argument gives transitivity of integrality, which we will want later.

Corollary 35.2.8 (Transitivity of Integrality)

Let $A \subseteq B \subseteq C$ be rings with B integral over A and C integral over B . Then C is integral over A .

Proof. Let $c \in C$ and let $c^n + b_{n-1}c^{n-1} + \cdots + b_0 = 0$ with $b_i \in B$. Put $B_0 = A[b_0, \dots, b_{n-1}]$ and $C_0 = B_0[c]$. Then B_0 is a finitely generated A -module, since each b_i is integral over A ; and C_0 is a finitely generated B_0 -module, since c is integral over B_0 (the displayed relation has coefficients in B_0). By transitivity of finite generation, C_0 is a finitely generated A -module, so Theorem 35.2.6 shows c is integral over A . $\square$

Remark. The passage to the finitely generated subring C_0 is essential: C itself might be enormous. For example the set of *all* algebraic integers in $\mathbb{C}$ is a ring which is integral over $\mathbb{Z}$, but it is nowhere near being a finitely generated $\mathbb{Z}$ -module. This kind of manoeuvre – reduce to a finitely generated situation, then apply the determinant trick – is one of the standard moves of commutative algebra, and it is worth internalising.

§35.2.3 Minimal Polynomials

To decide whether a given α is an algebraic integer, we would rather not have to hunt for a monic integer polynomial. Happily we do not have to: it is enough to look at the minimal polynomial.

Recall that for α algebraic over a field K , the **minimal polynomial** $p_\alpha \in K[x]$ is the monic polynomial of least degree with $p_\alpha(\alpha) = 0$, and that it divides every polynomial in $K[x]$ killing α (divide with remainder and use minimality); in particular it is unique and irreducible.

Proposition 35.2.9

Let K be a number field and $\alpha \in K$, with minimal polynomial $p_\alpha \in \mathbb{Q}[x]$. Then $\alpha \in \mathcal{O}_K$ if and only if $p_\alpha \in \mathbb{Z}[x]$.

Proof. If $p_\alpha \in \mathbb{Z}[x]$ then α is integral over $\mathbb{Z}$ by definition, since p_α is monic.

Conversely suppose $\alpha \in \mathcal{O}_K$, say $h(\alpha) = 0$ with $h \in \mathbb{Z}[x]$ monic. Then $p_\alpha \mid h$ in $\mathbb{Q}[x]$. Let M be a splitting field for p_α containing K , and let $\alpha_1, \dots, \alpha_m \in M$ be the roots of p_α . Each α_i is a root of h , hence is an algebraic integer. But the coefficients

of $p_\alpha = \prod_i (x - \alpha_i)$ are (up to sign) the elementary symmetric functions of the α_i , hence are algebraic integers by Corollary 35.2.7 applied in M . They also lie in $\mathbb{Q}$, so by Lemma 35.2.3 they lie in $\mathbb{Z}$. $\square$

Remark. There is a second proof, from IB Groups, Rings and Modules: p_α is monic and divides the monic integer polynomial h in $\mathbb{Q}[x]$, so writing $h = p_\alpha g$ with $g \in \mathbb{Q}[x]$ monic, Gauss's lemma forces both p_α and g to lie in $\mathbb{Z}[x]$. Either argument is fine; the first has the advantage of not needing h to be around.

This makes it easy to check membership of $\mathcal{O}_K$ in practice. For example $\frac{1}{2}(1 + \sqrt{5})$ has minimal polynomial $x^2 - x - 1 \in \mathbb{Z}[x]$, so it is an algebraic integer, whereas $\frac{1}{2}(1 + \sqrt{3})$ has minimal polynomial $x^2 - x - \frac{1}{2} \notin \mathbb{Z}[x]$ and so is not.

We can also now justify the ‘field of fractions’ claim from the picture above. The point is that any element of K becomes an algebraic integer after multiplication by a suitable rational integer – we can always clear denominators.

Lemma 35.2.10

Let K be a number field and $\alpha \in K$. Then there is a nonzero $n \in \mathbb{Z}$ with $n\alpha \in \mathcal{O}_K$. Consequently $\text{Frac}(\mathcal{O}_K) = K$.

Proof. Let $g(x) = x^m + c_{m-1}x^{m-1} + \cdots + c_0 \in \mathbb{Q}[x]$ be the minimal polynomial of α , and choose $n \in \mathbb{Z}_{>0}$ with $nc_i \in \mathbb{Z}$ for all i . Then

$$h(x) = n^m g\left(\frac{x}{n}\right) = x^m + nc_{m-1}x^{m-1} + n^2c_{m-2}x^{m-2} + \cdots + n^mc_0$$

is monic with integer coefficients (as $n^{m-i}c_i \in \mathbb{Z}$ for $i \leq m-1$), and $h(n\alpha) = n^m g(\alpha) = 0$. So $n\alpha \in \mathcal{O}_K$.

For the consequence, $\text{Frac}(\mathcal{O}_K) \subseteq K$ is clear, and any $\alpha \in K$ can be written as $(n\alpha)/n$ with $n\alpha \in \mathcal{O}_K$ and $n \in \mathbb{Z} \subseteq \mathcal{O}_K$. $\square$

§35.2.4 Quadratic Fields

Before developing more theory it is worth computing $\mathcal{O}_K$ in the simplest interesting case, both because we will use the answer constantly and because it already contains a surprise.

Definition 35.2.11 (Quadratic Field)

A **quadratic field** is a number field of degree 2.

Every quadratic field is of the form $\mathbb{Q}(\sqrt{d})$ for a unique squarefree $d \in \mathbb{Z} \setminus \{0, 1\}$. Indeed if $[K : \mathbb{Q}] = 2$ and $\alpha \in K \setminus \mathbb{Q}$ then $K = \mathbb{Q}(\alpha)$ and α satisfies a quadratic over $\mathbb{Q}$; completing the square writes $K = \mathbb{Q}(\sqrt{c})$ for some $c \in \mathbb{Q}$, and scaling c by squares of rationals reduces to squarefree $d \in \mathbb{Z}$. We call K **real quadratic** if $d > 0$ and **imaginary quadratic** if $d < 0$.

Lemma 35.2.12

Let $d \in \mathbb{Z}$ be squarefree, $d \neq 0, 1$, and let $K = \mathbb{Q}(\sqrt{d})$. Then

$$\mathcal{O}_K = \begin{cases} \mathbb{Z}[\sqrt{d}] & \text{if } d \equiv 2, 3 \pmod{4}, \\ \mathbb{Z}\left[\frac{1}{2}(1 + \sqrt{d})\right] & \text{if } d \equiv 1 \pmod{4}. \end{cases}$$

Proof. Write $\alpha = x + y\sqrt{d}$ with $x, y \in \mathbb{Q}$. If $y = 0$ then $\alpha \in \mathcal{O}_K$ iff $\alpha \in \mathbb{Z}$ by Lemma 35.2.3. If $y \neq 0$ then the minimal polynomial of α is

$$p_\alpha(t) = (t - x)^2 - dy^2 = t^2 - 2xt + (x^2 - dy^2),$$

so by Proposition 35.2.9, $\alpha \in \mathcal{O}_K$ if and only if $2x \in \mathbb{Z}$ and $x^2 - dy^2 \in \mathbb{Z}$.

Suppose these hold. Then $4x^2 - 4dy^2 \in \mathbb{Z}$ and $4x^2 \in \mathbb{Z}$, so $4dy^2 \in \mathbb{Z}$. Write $y = r/s$ in lowest terms; then $s^2 \mid 4dr^2$, so $s^2 \mid 4d$, and as d is squarefree this forces $s \mid 2$. So we may write $x = u/2$ and $y = v/2$ with $u, v \in \mathbb{Z}$, and the conditions become

$$u^2 - dv^2 \equiv 0 \pmod{4}.$$

Squares are 0 or 1 mod 4. If u, v are both even we get $x, y \in \mathbb{Z}$. If u is odd then $u^2 \equiv 1$, so we need $dv^2 \equiv 1 \pmod{4}$, forcing v odd and $d \equiv 1 \pmod{4}$. If u is even and v odd then $d \equiv 0 \pmod{4}$, impossible as d is squarefree.

So if $d \equiv 2, 3 \pmod{4}$ we must have u, v both even, giving $\mathcal{O}_K = \mathbb{Z} \oplus \mathbb{Z}\sqrt{d} = \mathbb{Z}[\sqrt{d}]$. If $d \equiv 1 \pmod{4}$ then $u \equiv v \pmod{2}$ is exactly the condition, and $\{u/2 + (v/2)\sqrt{d} : u \equiv v \pmod{2}\} = \mathbb{Z} \oplus \mathbb{Z} \cdot \frac{1}{2}(1 + \sqrt{d})$, which is the ring $\mathbb{Z}[\frac{1}{2}(1 + \sqrt{d})]$: note $\frac{1}{2}(1 + \sqrt{d})$ has minimal polynomial $t^2 - t + \frac{1}{4}(1 - d) \in \mathbb{Z}[x]$ precisely because $d \equiv 1 \pmod{4}$. $\square$

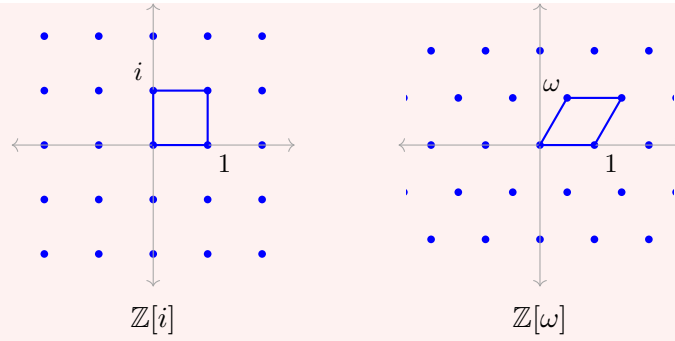
The case $d \equiv 1 \pmod{4}$ is the surprise: the naive guess $\mathbb{Z}[\sqrt{d}]$ is too small, and misses half of the integers. This will matter enormously.

Example 35.2.13

Let us record the two most important cases.

- (i) $d = -1$: here $-1 \equiv 3 \pmod{4}$, so $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$, the **Gaussian integers**. Embedded in $\mathbb{C}$ this is the square lattice.
- (ii) $d = -3$: here $-3 \equiv 1 \pmod{4}$, so $\mathcal{O}_{\mathbb{Q}(\sqrt{-3})} = \mathbb{Z}[\omega]$ where $\omega = \frac{1}{2}(1 + \sqrt{-3}) = e^{i\pi/3}$. These are the **Eisenstein integers**, and embedded in $\mathbb{C}$ they form the triangular lattice – strictly bigger than $\mathbb{Z}[\sqrt{-3}]$, which is a rectangular sublattice of index 2.

Here are the two pictures, drawn in $\mathbb{C}$.



In both cases $\mathcal{O}_K$ sits inside $\mathbb{C}$ as a lattice; that this always happens, in a suitable sense, is the content of Section 35.7.

Example 35.2.14

$\mathcal{O}_{\mathbb{Q}(\sqrt{5})} = \mathbb{Z}[\frac{1}{2}(1 + \sqrt{5})] = \mathbb{Z}[\varphi]$, where φ is the golden ratio, with $\varphi^2 = \varphi + 1$. On the other hand $\mathcal{O}_{\mathbb{Q}(\sqrt{2})} = \mathbb{Z}[\sqrt{2}]$ and $\mathcal{O}_{\mathbb{Q}(\sqrt{-5})} = \mathbb{Z}[\sqrt{-5}]$, since $2, 3 \equiv 2, 3 \pmod{4}$.

§35.3 Norms, Traces and Discriminants

§35.3.1 The Norm and the Trace

Two extremely useful invariants attach to an element of a field extension, and we met both in Galois Theory. The idea is to remember that L is a vector space over K , and that multiplication by an element of L is a K -linear map.

Definition 35.3.1 (Norm and Trace)

Let L/K be a finite extension of degree n and let $\alpha \in L$. Multiplication by α ,

$$m_\alpha : L \rightarrow L, \quad m_\alpha(y) = \alpha y,$$

is a K -linear endomorphism. We define the **norm** and **trace** of α relative to L/K by

$$N_{L/K}(\alpha) = \det(m_\alpha) \in K, \quad \text{Tr}_{L/K}(\alpha) = \text{tr}(m_\alpha) \in K.$$

When $K = \mathbb{Q}$ and L is understood we simply write $N(\alpha)$ and $\text{Tr}(\alpha)$.

Since $\alpha \mapsto m_\alpha$ is a K -algebra homomorphism $L \rightarrow \text{End}_K(L)$, the norm is multiplicative and the trace is K -linear:

$$N_{L/K}(\alpha\beta) = N_{L/K}(\alpha) N_{L/K}(\beta), \quad \text{Tr}_{L/K}(\alpha + \beta) = \text{Tr}_{L/K}(\alpha) + \text{Tr}_{L/K}(\beta).$$

Also $N_{L/K}(\alpha) = 0$ if and only if $\alpha = 0$, so $N_{L/K} : L^\times \rightarrow K^\times$ is a group homomorphism.

Example 35.3.2

Let $L = K(\sqrt{d})$ with $d \in K$ not a square, and let $\alpha = x + y\sqrt{d}$. In the basis $1, \sqrt{d}$

we have $\alpha \cdot 1 = x + y\sqrt{d}$ and $\alpha \cdot \sqrt{d} = dy + x\sqrt{d}$, so

$$m_\alpha = \begin{pmatrix} x & dy \\ y & x \end{pmatrix},$$

giving

$$\text{Tr}_{L/K}(x + y\sqrt{d}) = 2x, \quad \text{N}_{L/K}(x + y\sqrt{d}) = x^2 - dy^2.$$

Notice that these are exactly $\alpha + \bar{\alpha}$ and $\alpha\bar{\alpha}$, where $\bar{\alpha} = x - y\sqrt{d}$ is the conjugate. For $d = -1$ the norm is the familiar $|x + iy|^2$.

Recall from Galois Theory the relationship between m_α and the minimal polynomial: if p_α is the minimal polynomial of α over K , then the characteristic polynomial of m_α on L is

$$\det(xI - m_\alpha) = p_\alpha(x)^{[L:K(\alpha)]}.$$

(The reason is that L is a free $K(\alpha)$ -module of rank $[L : K(\alpha)]$, and on $K(\alpha)$ with basis $1, \alpha, \dots, \alpha^{\deg p_\alpha - 1}$ the matrix of m_α is the companion matrix of p_α .) Comparing coefficients gives, when $p_\alpha(t) = \prod_{i=1}^r (t - \alpha_i)$ splits in some extension,

$$\text{N}_{L/K}(\alpha) = \left(\prod_{i=1}^r \alpha_i \right)^{[L:K(\alpha)]}, \quad \text{Tr}_{L/K}(\alpha) = [L : K(\alpha)] \sum_{i=1}^r \alpha_i.$$

This has an immediate consequence for algebraic integers, which will be our main computational tool for recognising them.

Lemma 35.3.3

Let K be a number field and $\alpha \in \mathcal{O}_K$. Then $\text{N}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$ and $\text{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$.

Proof. By Proposition 35.2.9 the minimal polynomial p_α has integer coefficients, and the characteristic polynomial of m_α is a power of p_α , hence also lies in $\mathbb{Z}[x]$. The norm and trace are, up to sign, coefficients of this characteristic polynomial. $\square$

Remark. The converse is false in general, but true in the quadratic case: if $[K : \mathbb{Q}] = 2$ and $\alpha \in K \setminus \mathbb{Q}$, then the characteristic polynomial of m_α is the minimal polynomial, and it equals $x^2 - \text{Tr}(\alpha)x + \text{N}(\alpha)$. So $\alpha \in \mathcal{O}_K$ if and only if $\text{Tr}(\alpha), \text{N}(\alpha) \in \mathbb{Z}$. This is exactly what we used in the proof of Lemma 35.2.12.

§35.3.2 Embeddings

For number fields there is a much better description of the norm and trace, in terms of the embeddings into $\mathbb{C}$. Let $K = \mathbb{Q}(\theta)$ be a number field of degree n , with θ having minimal polynomial p_θ of degree n , so that $K \cong \mathbb{Q}[x]/(p_\theta)$ and $1, \theta, \dots, \theta^{n-1}$ is a $\mathbb{Q}$ -basis of K .

Lemma 35.3.4

There are exactly n field embeddings $K \hookrightarrow \mathbb{C}$.

Proof. As p_θ is irreducible over $\mathbb{Q}$ and $\mathbb{Q}$ has characteristic zero, $\gcd(p_\theta, p'_\theta) = 1$, so p_θ has n distinct roots $\theta_1, \dots, \theta_n \in \mathbb{C}$. A field homomorphism $\mathbb{Q}[x]/(p_\theta) \rightarrow \mathbb{C}$ is automatically $\mathbb{Q}$ -linear, hence determined by the image of x , which must be a root of p_θ ; and each root gives such a homomorphism. These n maps are distinct since they differ on θ . $\square$

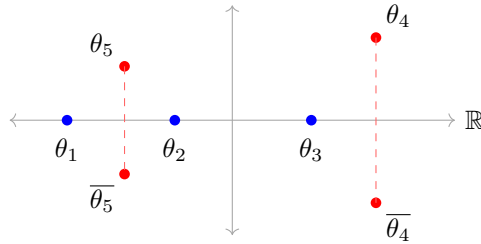
Some of these embeddings land in $\mathbb{R}$ and some do not, and the ones that do not come in conjugate pairs since complex conjugation permutes the roots of a real polynomial.

Definition 35.3.5 (Signature)

Let K be a number field of degree n . Let r be the number of embeddings $K \hookrightarrow \mathbb{R}$ and let s be the number of conjugate pairs of non-real embeddings $K \hookrightarrow \mathbb{C}$, so that $r + 2s = n$. We call (r, s) the **signature** of K , and label the embeddings

$$\sigma_1, \dots, \sigma_r : K \rightarrow \mathbb{R}, \quad \sigma_{r+1}, \overline{\sigma_{r+1}}, \dots, \sigma_{r+s}, \overline{\sigma_{r+s}} : K \rightarrow \mathbb{C}.$$

Since r counts the real roots of p_θ , and r can also be described as the number of embeddings of K into $\mathbb{R}$ (a description not mentioning θ), the pair (r, s) is an invariant of K and not of the chosen generator. Here is the picture: the conjugates of θ sit in $\mathbb{C}$, some on the real axis and the rest in mirror-image pairs.



Here $n = 7$, $r = 3$ and $s = 2$. Each root gives one embedding $\sigma_i : K \rightarrow \mathbb{C}$, determined by $\theta \mapsto \theta_i$.

Lemma 35.3.6

Let K be a number field of degree n with embeddings $\sigma_1, \dots, \sigma_n : K \rightarrow \mathbb{C}$. Then for all $\beta \in K$,

$$\mathrm{Tr}_{K/\mathbb{Q}}(\beta) = \sum_{i=1}^n \sigma_i(\beta), \quad \mathrm{N}_{K/\mathbb{Q}}(\beta) = \prod_{i=1}^n \sigma_i(\beta).$$

We call the numbers $\sigma_i(\beta)$ the **conjugates** of β .

Proof. Let p_β have degree r and roots $\beta_1, \dots, \beta_r$, and let $m = [K : \mathbb{Q}(\beta)]$, so $n = rm$. Each embedding $\mathbb{Q}(\beta) \rightarrow \mathbb{C}$ sends β to some β_j , and each of these r embeddings extends in exactly m ways to K (by Lemma 35.3.4 applied to $K/\mathbb{Q}(\beta)$, or by counting: there are n embeddings of K in total, distributed among the r embeddings of $\mathbb{Q}(\beta)$). So the multiset $\{\sigma_i(\beta)\}$ is $\{\beta_1, \dots, \beta_r\}$ with each element repeated m times, and the result follows from the formulae recorded above. $\square$

Example 35.3.7

If $K = \mathbb{Q}(\sqrt{d})$ with d squarefree, the two embeddings send $\sqrt{d} \mapsto \pm\sqrt{d}$, so the conjugates of $a + b\sqrt{d}$ are $a \pm b\sqrt{d}$, recovering Example 35.3.2. If $d > 0$ the signature is $(2, 0)$; if $d < 0$ it is $(0, 1)$. For $K = \mathbb{Q}(\sqrt[3]{2})$ the conjugates of $\sqrt[3]{2}$ are $\sqrt[3]{2}, \omega\sqrt[3]{2}, \omega^2\sqrt[3]{2}$ with $\omega = e^{2\pi i/3}$, so the signature is $(1, 1)$. For $K = \mathbb{Q}(\zeta_m)$ with $m > 2$ there are no real embeddings at all, so the signature is $(0, \varphi(m)/2)$.

§35.3.3 The Discriminant of a Basis

We now build a numerical invariant out of the trace. The construction is a quadratic form, and the crucial fact is that it is nondegenerate.

Definition 35.3.8 (Discriminant of a Tuple)

Let L/K be a finite extension of degree n and let $\alpha_1, \dots, \alpha_n \in L$. Define

$$\Delta(\alpha_1, \dots, \alpha_n) = \det (\mathrm{Tr}_{L/K}(\alpha_i \alpha_j))_{i,j} \in K,$$

the determinant of the Gram matrix of the **trace form** $(x, y) \mapsto \mathrm{Tr}_{L/K}(xy)$.

Proposition 35.3.9

Let L/K be a finite separable extension of degree n with distinct K -embeddings $\sigma_1, \dots, \sigma_n$ into an algebraic closure $\overline{K}$, and let $\alpha_1, \dots, \alpha_n \in L$. Then

$$\Delta(\alpha_1, \dots, \alpha_n) = \det (\sigma_i(\alpha_j))^2.$$

Moreover $\Delta(\alpha_1, \dots, \alpha_n) \neq 0$ if and only if $\alpha_1, \dots, \alpha_n$ is a K -basis of L ; equivalently, the trace form is nondegenerate.

Proof. Let $S = (\sigma_i(\alpha_j))_{i,j}$. Then the (i, j) entry of $S^T S$ is

$$\sum_{k=1}^n \sigma_k(\alpha_i) \sigma_k(\alpha_j) = \sum_{k=1}^n \sigma_k(\alpha_i \alpha_j) = \mathrm{Tr}_{L/K}(\alpha_i \alpha_j),$$

using Lemma 35.3.6 (whose proof works verbatim for any finite separable extension). Taking determinants gives the first claim.

For the second, first take the basis $1, \theta, \dots, \theta^{n-1}$ where $L = K(\theta)$, which exists by the primitive element theorem. Then S is the Vandermonde matrix $(\sigma_i(\theta)^{j-1})$, so

$$\Delta(1, \theta, \dots, \theta^{n-1}) = \prod_{i < j} (\sigma_i(\theta) - \sigma_j(\theta))^2,$$

which is nonzero: if $\sigma_i(\theta) = \sigma_j(\theta)$ then $\sigma_i = \sigma_j$ on $K(\theta) = L$, so $i = j$. Now if $\alpha_1, \dots, \alpha_n$ is any other basis, say $\alpha_i = \sum_j a_{ij} \theta^{j-1}$ with $A = (a_{ij}) \in \mathrm{GL}_n(K)$, then a direct computation with the Gram matrices gives

$$\Delta(\alpha_1, \dots, \alpha_n) = (\det A)^2 \Delta(1, \theta, \dots, \theta^{n-1}) \neq 0.$$

Finally, if the α_i are linearly dependent then so are the columns of the Gram matrix, and $\Delta = 0$. $\square$

Remark. Taking $\alpha_i = \theta^{i-1}$ we recover the discriminant of a polynomial from Galois Theory: if $p_\theta(t) = \prod_i (t - \sigma_i(\theta))$ then

$$\text{disc}(p_\theta) = \prod_{i < j} (\sigma_i(\theta) - \sigma_j(\theta))^2 = \Delta(1, \theta, \dots, \theta^{n-1}).$$

So the two uses of the word ‘discriminant’ agree, and we write $\text{disc}(\theta)$ for this common value. In practice it is computed from the resultant formula

$$\text{disc}(f) = (-1)^{n(n-1)/2} N(f'(\theta)).$$

The relevance to us is the following, which is the engine of everything in this section.

Lemma 35.3.10

Let K be a number field of degree n and let $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$. Then $\Delta(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}$. If moreover the α_i form a $\mathbb{Q}$ -basis of K , then $\Delta(\alpha_1, \dots, \alpha_n)$ is a nonzero integer.

Proof. Each $\alpha_i \alpha_j$ lies in $\mathcal{O}_K$, so $\text{Tr}(\alpha_i \alpha_j) \in \mathbb{Z}$ by Lemma 35.3.3; hence the Gram matrix has integer entries and its determinant is an integer. Nonvanishing is Proposition 35.3.9. $\square$

We record for reference two further standard properties, both proved in Galois Theory.

Proposition 35.3.11 (Transitivity)

Let $M/L/K$ be finite extensions. Then for all $\alpha \in M$,

$$\text{Tr}_{L/K}(\text{Tr}_{M/L}(\alpha)) = \text{Tr}_{M/K}(\alpha), \quad N_{L/K}(N_{M/L}(\alpha)) = N_{M/K}(\alpha).$$

In particular, if $\alpha \in L$ then $N_{M/K}(\alpha) = N_{L/K}(\alpha)^{[M:L]}$ and $\text{Tr}_{M/K}(\alpha) = [M : L] \text{Tr}_{L/K}(\alpha)$.

Example 35.3.12 (Discriminant of a Cubic)

For a cubic $f(x) = x^3 + ax + b$ (any cubic can be brought to this form by a substitution $x \mapsto x - c$, which does not change the discriminant), one has

$$\text{disc}(f) = -4a^3 - 27b^2.$$

So $\text{disc}(x^3 - 2) = -27 \cdot 4 = -108$, matching Example 35.3.17, and $\text{disc}(x^3 - x - 1) = 4 - 27 = -23$. Note also that the sign of $\text{disc}(f)$ detects the signature: if f has three real roots then $\prod_{i < j} (\theta_i - \theta_j)^2 > 0$, while if it has one real root and a conjugate pair the product is negative. So a cubic field has signature $(3, 0)$ or $(1, 1)$ according as $d_K > 0$ or $d_K < 0$. More generally $\text{sign}(d_K) = (-1)^s$ for any number field.

§35.3.4 Integral Bases

Definition 35.3.13 (Integral Basis)

Let K be a number field of degree n . An **integral basis** for K is a tuple $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ such that

$$\mathcal{O}_K = \mathbb{Z}\alpha_1 \oplus \dots \oplus \mathbb{Z}\alpha_n.$$

Note that an integral basis is automatically a $\mathbb{Q}$ -basis of K : it spans K over $\mathbb{Q}$ by Lemma 35.2.10, and n spanning elements in an n -dimensional space are independent. So an integral basis exhibits $\mathcal{O}_K$ as a free abelian group of rank n sitting inside K in the tidiest possible way. The following existence theorem is the reason the theory works at all.

Theorem 35.3.14 (Existence of Integral Bases)

Every number field K of degree n has an integral basis. In particular $\mathcal{O}_K \cong \mathbb{Z}^n$ as an abelian group.

Proof. Start with any $\mathbb{Q}$ -basis $\beta_1, \dots, \beta_n$ of K ; by Lemma 35.2.10 we may scale each β_i by a nonzero integer so that all $\beta_i \in \mathcal{O}_K$, without destroying the basis property. So the set

$$\mathcal{B} = \{(\alpha_1, \dots, \alpha_n) : \alpha_i \in \mathcal{O}_K \text{ forming a } \mathbb{Q}\text{-basis of } K\}$$

is nonempty, and by Lemma 35.3.10 the quantity $|\Delta(\alpha_1, \dots, \alpha_n)|$ is a positive integer for each element of $\mathcal{B}$. Choose $(\alpha_1, \dots, \alpha_n) \in \mathcal{B}$ minimising $|\Delta(\alpha_1, \dots, \alpha_n)|$.

We claim this is an integral basis. Certainly $\bigoplus \mathbb{Z}\alpha_i \subseteq \mathcal{O}_K$. Suppose not conversely, so there is $x \in \mathcal{O}_K$ with $x = \sum_i \lambda_i \alpha_i$, $\lambda_i \in \mathbb{Q}$, and some $\lambda_i \notin \mathbb{Z}$; relabelling, say $\lambda_1 \notin \mathbb{Z}$. Write $\lambda_1 = m + \varepsilon$ with $m \in \mathbb{Z}$ and $0 < \varepsilon < 1$, and put

$$\alpha'_1 = x - m\alpha_1 = \varepsilon\alpha_1 + \lambda_2\alpha_2 + \dots + \lambda_n\alpha_n \in \mathcal{O}_K.$$

Then $\alpha'_1, \alpha_2, \dots, \alpha_n$ is again a $\mathbb{Q}$ -basis of K consisting of algebraic integers (the change of basis matrix from $\alpha_1, \dots, \alpha_n$ is lower triangular with diagonal $\varepsilon, 1, \dots, 1$, so has determinant $\varepsilon \neq 0$), so lies in $\mathcal{B}$, and by the transformation rule in the proof of Proposition 35.3.9,

$$\Delta(\alpha'_1, \alpha_2, \dots, \alpha_n) = \varepsilon^2 \Delta(\alpha_1, \dots, \alpha_n).$$

Since $0 < \varepsilon^2 < 1$ this has strictly smaller absolute value, contradicting minimality. $\square$

Integral bases are far from unique: if $\alpha_1, \dots, \alpha_n$ is one, and $g \in \text{GL}_n(\mathbb{Z})$, then applying g gives another, and all integral bases arise this way. But $\det g = \pm 1$ for $g \in \text{GL}_n(\mathbb{Z})$, so $(\det g)^2 = 1$ and the discriminant does not notice. This gives us a genuine invariant of the field.

Definition 35.3.15 (Discriminant of a Number Field)

Let K be a number field with integral basis $\alpha_1, \dots, \alpha_n$. The **discriminant** of K is

$$d_K = \Delta(\alpha_1, \dots, \alpha_n) \in \mathbb{Z} \setminus \{0\},$$

which by the above does not depend on the choice of integral basis.

The discriminant is one of the two or three most important invariants attached to a number field. It knows about ramification (Section 35.6), it controls the size of the class group (Section 35.7), and – via the theorem of Hermite and Minkowski – there are only finitely many number fields with any given discriminant.

Example 35.3.16 (Quadratic Fields)

Let $K = \mathbb{Q}(\sqrt{d})$ with d squarefree. If $d \equiv 2, 3 \pmod{4}$ then $1, \sqrt{d}$ is an integral basis and

$$d_K = \det \begin{pmatrix} 1 & \sqrt{d} \\ 1 & -\sqrt{d} \end{pmatrix}^2 = (-2\sqrt{d})^2 = 4d.$$

If $d \equiv 1 \pmod{4}$ then $1, \frac{1}{2}(1 + \sqrt{d})$ is an integral basis and

$$d_K = \det \begin{pmatrix} 1 & \frac{1}{2}(1 + \sqrt{d}) \\ 1 & \frac{1}{2}(1 - \sqrt{d}) \end{pmatrix}^2 = (-\sqrt{d})^2 = d.$$

So $d_K \in \{d, 4d\}$. It is often cleaner to phrase results about quadratic fields in terms of d_K : note that $K = \mathbb{Q}(\sqrt{d_K})$ in both cases, that $d_K \equiv 0, 1 \pmod{4}$, and that

$$1, \frac{1}{2}(d_K + \sqrt{d_K})$$

is an integral basis in both cases. Integers arising as d_K for a quadratic field are called **fundamental discriminants**.

Remark (Stickelberger). Not every integer is a discriminant. Writing $\det(\sigma_i(\alpha_j)) = P - N$, where P is the sum of the terms of the determinant coming from even permutations and N those from odd ones, we have $d_K = (P - N)^2 = (P + N)^2 - 4PN$. Both $P + N$ and PN are fixed by every permutation of the embeddings, hence are rational; and they are algebraic integers, hence lie in $\mathbb{Z}$. So

$$d_K \equiv (P + N)^2 \equiv 0 \text{ or } 1 \pmod{4}.$$

This is **Stickelberger's criterion**, and it is consistent with what we found for quadratic fields, where d_K is d (with $d \equiv 1 \pmod{4}$) or $4d$.

Example 35.3.17 (Cubic Fields)

Let $K = \mathbb{Q}(\sqrt[3]{2})$. We have $\text{disc}(x^3 - 2) = -108$. If $1, \sqrt[3]{2}, \sqrt[3]{4}$ were not an integral basis, then $\mathbb{Z}[\sqrt[3]{2}] \subsetneq \mathcal{O}_K$ would have some index $m > 1$, and (as we shall see in Lemma 35.3.19) $\text{disc}(\sqrt[3]{2}) = m^2 d_K$, so $m^2 \mid 108$, giving $m \in \{2, 3, 6\}$. One can rule these out directly; the upshot is $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$ and $d_K = -108$.

Contrast this with $K = \mathbb{Q}(\alpha)$ where $\alpha^3 - \alpha^2 - 2\alpha - 8 = 0$. Here $\text{disc}(\alpha) = -2012 = -4 \cdot 503$, and 503 is prime, so the index of $\mathbb{Z}[\alpha]$ in $\mathcal{O}_K$ is 1 or 2. In fact it is 2: the element

$$\beta = \frac{1}{2}(\alpha^2 + \alpha)$$

has minimal polynomial $x^3 - 3x^2 - 10x - 8 \in \mathbb{Z}[x]$, so $\beta \in \mathcal{O}_K \setminus \mathbb{Z}[\alpha]$. Thus $d_K = -503$ and $1, \alpha, \beta$ is an integral basis. We will meet this field again in Section 35.6: it is the standard example of a field which is *not monogenic*, meaning $\mathcal{O}_K \neq \mathbb{Z}[\gamma]$ for every γ .

Example 35.3.18 (Cyclotomic Fields)

Let p be an odd prime and $K = \mathbb{Q}(\zeta_p)$ where $\zeta_p = e^{2\pi i/p}$. From Galois Theory, ζ_p has minimal polynomial the cyclotomic polynomial

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = 1 + x + \cdots + x^{p-1},$$

irreducible by Eisenstein applied to $\Phi_p(x+1)$, so $[K : \mathbb{Q}] = p-1$ and the signature is $(0, (p-1)/2)$. One shows (see Section 35.6.5) that $\mathcal{O}_K = \mathbb{Z}[\zeta_p]$ and

$$d_K = (-1)^{(p-1)/2} p^{p-2}.$$

For instance $d_{\mathbb{Q}(\zeta_5)} = 5^3 = 125$ and $d_{\mathbb{Q}(\zeta_7)} = -7^5 = -16807$, which one can check directly by computing $\text{disc}(\Phi_5)$ and $\text{disc}(\Phi_7)$. For $p = 3$ this gives $d_{\mathbb{Q}(\zeta_3)} = -3$, consistent with $\mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$ and Example 35.3.16.

The following criterion, which we will prove in more generality in Section 35.5, is the standard practical tool: it says that a squarefree discriminant certifies an integral basis for free.

Lemma 35.3.19 (Index and Discriminant)

Let K be a number field of degree n and let $\alpha_1, \dots, \alpha_n \in \mathcal{O}_K$ be a $\mathbb{Q}$ -basis of K . Let $M = \bigoplus_i \mathbb{Z}\alpha_i \subseteq \mathcal{O}_K$. Then $[\mathcal{O}_K : M]$ is finite and

$$\Delta(\alpha_1, \dots, \alpha_n) = [\mathcal{O}_K : M]^2 d_K.$$

In particular, if $\Delta(\alpha_1, \dots, \alpha_n)$ is squarefree then $M = \mathcal{O}_K$ and $d_K = \Delta(\alpha_1, \dots, \alpha_n)$.

Proof. Let $\gamma_1, \dots, \gamma_n$ be an integral basis, and write $\alpha_i = \sum_j a_{ij} \gamma_j$ with $A = (a_{ij}) \in \text{Mat}_n(\mathbb{Z})$. Since both are $\mathbb{Q}$ -bases of K , $\det A \neq 0$, so M has finite index in $\mathcal{O}_K$. By Smith normal form there are $P, Q \in \text{GL}_n(\mathbb{Z})$ with PAQ diagonal, say with entries $d_1, \dots, d_n$; changing bases on both sides accordingly, $\mathcal{O}_K/M \cong \bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$, so $[\mathcal{O}_K : M] = |d_1 \cdots d_n| = |\det A|$. Now the transformation rule gives

$$\Delta(\alpha_1, \dots, \alpha_n) = (\det A)^2 \Delta(\gamma_1, \dots, \gamma_n) = [\mathcal{O}_K : M]^2 d_K. \quad \square$$

Example 35.3.20

The polynomial $x^3 - x - 1$ has discriminant -23 , which is squarefree. Hence if α is a root, $\mathcal{O}_{\mathbb{Q}(\alpha)} = \mathbb{Z}[\alpha]$ and $d_{\mathbb{Q}(\alpha)} = -23$. Similarly $x^3 + x + 1$ has discriminant -31 , giving a cubic field of discriminant -31 .

§35.4 Ideals and Unique Factorisation**§35.4.1 Units and Irreducibles**

Let us begin with the multiplicative structure of $\mathcal{O}_K$. Recall that $u \in \mathcal{O}_K$ is a **unit** if $u^{-1} \in \mathcal{O}_K$, and we write $\mathcal{O}_K^\times$ for the group of units. There is a beautifully simple criterion.

Lemma 35.4.1

Let K be a number field and $x \in \mathcal{O}_K$. Then $x \in \mathcal{O}_K^\times$ if and only if $N_{K/\mathbb{Q}}(x) = \pm 1$.

Proof. If x is a unit then $1 = N(1) = N(x)N(x^{-1})$ with both factors in $\mathbb{Z}$ by Lemma 35.3.3, so $N(x) = \pm 1$.

Conversely suppose $N(x) = \pm 1$. Fix the embedding σ_1 and identify K with $\sigma_1(K) \subseteq \mathbb{C}$. By Lemma 35.3.6,

$$\pm 1 = N(x) = x \sigma_2(x) \cdots \sigma_n(x),$$

so $x^{-1} = \pm \prod_{i=2}^n \sigma_i(x)$. Each $\sigma_i(x)$ is a root of the minimal polynomial of x , which lies in $\mathbb{Z}[x]$, hence is an algebraic integer; so x^{-1} is an algebraic integer. It also lies in K , so $x^{-1} \in \mathcal{O}_K$. $\square$

Example 35.4.2

Let $K = \mathbb{Q}(\sqrt{d})$ with $d < 0$ squarefree. If $d \equiv 2, 3 \pmod{4}$ then $N(a + b\sqrt{d}) = a^2 + |d|b^2 = 1$ forces $b = 0$, $a = \pm 1$ unless $d = -1$, where we also get $a = 0$, $b = \pm 1$. So

$$\mathcal{O}_{\mathbb{Q}(i)}^\times = \{\pm 1, \pm i\}, \quad \mathcal{O}_{\mathbb{Q}(\sqrt{d})}^\times = \{\pm 1\} \text{ for } d \leq -2, d \equiv 2, 3 \pmod{4}.$$

If $d \equiv 1 \pmod{4}$ the norm of $\frac{1}{2}(u + v\sqrt{d})$ is $\frac{1}{4}(u^2 + |d|v^2)$, and $u^2 + |d|v^2 = 4$ has extra solutions only for $d = -3$, giving the six sixth-roots of unity. So imaginary quadratic fields have finite unit group – and, as we shall see, they are essentially the only fields with this property.

Example 35.4.3

By contrast, in $\mathbb{Z}[\sqrt{2}]$ we have $N(1 + \sqrt{2}) = 1 - 2 = -1$, so $1 + \sqrt{2}$ is a unit – and so are all its powers $(1 + \sqrt{2})^n$, which are pairwise distinct real numbers. So $\mathcal{O}_{\mathbb{Q}(\sqrt{2})}^\times$ is infinite. Finding solutions of $N(x) = \pm 1$ in a real quadratic field is exactly solving **Pell's equation** $a^2 - db^2 = \pm 1$, and we will return to this in Section 35.9.

Recall that $x \in \mathcal{O}_K$ is **irreducible** if x is not zero or a unit, and $x = ab$ implies a or b is a unit; and that x, y are **associates** if $x = uy$ for a unit u . Since $|N(x)|$ is a positive integer for $x \neq 0$, and $|N(x)| = 1$ exactly for units, an easy induction on $|N(x)|$ shows that every nonzero non-unit of $\mathcal{O}_K$ is a product of irreducibles. The trouble is uniqueness.

§35.4.2 The Failure of Unique Factorisation**Example 35.4.4**

Let $K = \mathbb{Q}(\sqrt{-5})$, so $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$ and $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Then

$$6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}).$$

The norms of the four factors are 4, 9, 6, 6. All four are irreducible: for instance if $2 = \alpha\beta$ with neither a unit then $4 = N(\alpha)N(\beta)$ with both factors > 1 , so $N(\alpha) = 2$, i.e. $a^2 + 5b^2 = 2$, which has no integer solutions. The same argument rules out norms

3 and 6 being $N(\alpha)N(\beta)$ nontrivially, since $a^2 + 5b^2 \in \{2, 3\}$ is insoluble. Moreover 2 and $1 \pm \sqrt{-5}$ are not associates, as the units are just ± 1 and the norms differ. So $\mathcal{O}_K$ is not a UFD.

Example 35.4.5

Similarly in $\mathbb{Z}[\sqrt{-5}]$ we have $21 = 3 \cdot 7 = (1 + 2\sqrt{-5})(1 - 2\sqrt{-5})$, with norms 9, 49, 21, 21; again all four factors are irreducible since $a^2 + 5b^2 \in \{3, 7\}$ is insoluble.

Recall from IB Groups, Rings and Modules that $\text{ED} \Rightarrow \text{PID} \Rightarrow \text{UFD}$, and that in a UFD irreducibles are prime. So $\mathbb{Z}[\sqrt{-5}]$ is not a PID either – indeed $(2, 1 + \sqrt{-5})$ is not principal.

Now, here is Kummer’s idea. In Example 35.4.4 the element 2 ought morally to factor further, into ‘ideal numbers’ which are not elements of the ring. Dedekind’s formulation replaces these ideal numbers with honest *ideals*. Watch what happens if we set

$$\mathfrak{p} = (2, 1 + \sqrt{-5}), \quad \mathfrak{q}_1 = (3, 1 + \sqrt{-5}), \quad \mathfrak{q}_2 = (3, 1 - \sqrt{-5}).$$

We will verify shortly that

$$\mathfrak{p}^2 = (2), \quad \mathfrak{q}_1 \mathfrak{q}_2 = (3), \quad \mathfrak{p} \mathfrak{q}_1 = (1 + \sqrt{-5}), \quad \mathfrak{p} \mathfrak{q}_2 = (1 - \sqrt{-5}),$$

so that the two factorisations of 6 become

$$(6) = \mathfrak{p}^2 \mathfrak{q}_1 \mathfrak{q}_2 = (\mathfrak{p} \mathfrak{q}_1)(\mathfrak{p} \mathfrak{q}_2),$$

which are the *same* factorisation into the ideals $\mathfrak{p}, \mathfrak{q}_1, \mathfrak{q}_2$, just bracketed differently. Unique factorisation was there all along; we simply could not see the factors.

§35.4.3 Arithmetic of Ideals

Let us set up the algebra. All our ideals will be ideals of $\mathcal{O}_K$, and we use the standard fraktur letters $\mathfrak{a}, \mathfrak{b}, \mathfrak{p}$ for them. Given $x_1, \dots, x_n \in \mathcal{O}_K$ we write $(x_1, \dots, x_n) = \sum_i x_i \mathcal{O}_K$ for the ideal they generate.

Definition 35.4.6 (Sum and Product of Ideals)

For ideals $\mathfrak{a}, \mathfrak{b} \subseteq \mathcal{O}_K$, define

$$\mathfrak{a} + \mathfrak{b} = \{x + y : x \in \mathfrak{a}, y \in \mathfrak{b}\}, \quad \mathfrak{a} \mathfrak{b} = \left\{ \sum_{i=1}^m x_i y_i : m \geq 0, x_i \in \mathfrak{a}, y_i \in \mathfrak{b} \right\}.$$

Both are ideals, the product is associative and commutative, and $\mathcal{O}_K = (1)$ is an identity for it. Concretely, $(x_1, \dots, x_n)(y_1, \dots, y_m) = (\{x_i y_j\}_{i,j})$; in particular $(x)(y) = (xy)$, so the principal ideals form a submonoid isomorphic to $\mathcal{O}_K \setminus \{0\}$ modulo units. Note also $\mathfrak{a} \mathfrak{b} \subseteq \mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}$.

Example 35.4.7

Let us do the promised calculation in $\mathbb{Z}[\sqrt{-5}]$. With $\mathfrak{p} = (2, 1 + \sqrt{-5})$,

$$\mathfrak{p}^2 = (4, 2(1 + \sqrt{-5}), (1 + \sqrt{-5})^2) = (4, 2 + 2\sqrt{-5}, -4 + 2\sqrt{-5}).$$

Every generator is divisible by 2, so $\mathfrak{p}^2 \subseteq (2)$; and $(2 + 2\sqrt{-5}) - (-4 + 2\sqrt{-5}) = 6$, so

$\mathfrak{p}^2 \ni 6 - 4 = 2$, giving $\mathfrak{p}^2 = (2)$. Similarly with $\mathfrak{q}_1 = (3, 1 + \sqrt{-5})$, $\mathfrak{q}_2 = (3, 1 - \sqrt{-5})$,

$$\mathfrak{q}_1 \mathfrak{q}_2 = (9, 3(1 - \sqrt{-5}), 3(1 + \sqrt{-5}), 6) = (3),$$

since $9 - 6 = 3$ lies in the left-hand ideal and every generator is a multiple of 3. Finally

$$\mathfrak{p} \mathfrak{q}_1 = (6, 2(1 + \sqrt{-5}), 3(1 + \sqrt{-5}), (1 + \sqrt{-5})^2),$$

which contains $3(1 + \sqrt{-5}) - 2(1 + \sqrt{-5}) = 1 + \sqrt{-5}$; and each generator is a multiple of $1 + \sqrt{-5}$ (as $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$), so $\mathfrak{p} \mathfrak{q}_1 = (1 + \sqrt{-5})$.

Recall that an ideal $\mathfrak{p} \trianglelefteq R$ is **prime** if $R/\mathfrak{p}$ is an integral domain, equivalently $\mathfrak{p} \neq R$ and $xy \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. *Throughout this course we additionally require prime ideals to be nonzero*, since the zero ideal is a nuisance and plays no role in factorisation.

§35.4.4 $\mathcal{O}_K$ is a Dedekind Domain

Everything we prove about factorisation rests on four properties of $\mathcal{O}_K$. We isolate them first. The key preliminary observation is that quotients of $\mathcal{O}_K$ are finite.

Lemma 35.4.8

Let K be a number field of degree n and let $\mathfrak{a} \trianglelefteq \mathcal{O}_K$ be a nonzero ideal. Then $\mathfrak{a} \cap \mathbb{Z} \neq \{0\}$, and $\mathcal{O}_K/\mathfrak{a}$ is finite. Moreover $\mathfrak{a} \cong \mathbb{Z}^n$ as an abelian group.

Proof. Pick $0 \neq \alpha \in \mathfrak{a}$ with minimal polynomial $x^m + a_{m-1}x^{m-1} + \cdots + a_0 \in \mathbb{Z}[x]$; note $a_0 \neq 0$ since the minimal polynomial is irreducible and $\alpha \neq 0$. Rearranging,

$$a_0 = -\alpha (\alpha^{m-1} + a_{m-1}\alpha^{m-2} + \cdots + a_1),$$

where the bracket lies in $\mathcal{O}_K$ and $\alpha \in \mathfrak{a}$, so $a_0 \in \mathfrak{a} \cap \mathbb{Z}$ and this is nonzero.

Now $a_0 \mathcal{O}_K \subseteq \mathfrak{a} \subseteq \mathcal{O}_K$, so $\mathcal{O}_K/a_0 \mathcal{O}_K$ surjects onto $\mathcal{O}_K/\mathfrak{a}$. But by Theorem 35.3.14, $\mathcal{O}_K \cong \mathbb{Z}^n$ as an abelian group, so

$$\mathcal{O}_K/a_0 \mathcal{O}_K \cong \mathbb{Z}^n/a_0 \mathbb{Z}^n \cong (\mathbb{Z}/a_0 \mathbb{Z})^n$$

is finite of order $|a_0|^n$. Hence $\mathcal{O}_K/\mathfrak{a}$ is finite. Finally $\mathfrak{a}$ is a subgroup of $\mathbb{Z}^n$, hence free of rank at most n ; and since $\mathcal{O}_K/\mathfrak{a}$ is finite the rank is exactly n , so $\mathfrak{a} \cong \mathbb{Z}^n$. $\square$

Proposition 35.4.9

Let K be a number field. Then

- (i) $\mathcal{O}_K$ is an integral domain;
- (ii) $\mathcal{O}_K$ is Noetherian;
- (iii) $\mathcal{O}_K$ is **integrally closed** in $K = \text{Frac}(\mathcal{O}_K)$: if $x \in K$ is integral over $\mathcal{O}_K$, then $x \in \mathcal{O}_K$;
- (iv) every nonzero prime ideal of $\mathcal{O}_K$ is maximal.

Proof. (i) $\mathcal{O}_K$ is a subring of the field K .

(ii) By Theorem 35.3.14, $\mathcal{O}_K \cong \mathbb{Z}^n$ is a Noetherian $\mathbb{Z}$ -module (every subgroup of $\mathbb{Z}^n$ is generated by at most n elements), so every ideal of $\mathcal{O}_K$ is finitely generated even as an abelian group, a fortiori as an ideal.

(iii) $\mathcal{O}_K$ is integral over $\mathbb{Z}$ by definition. If $x \in K$ is integral over $\mathcal{O}_K$ then x is integral over $\mathbb{Z}$ by transitivity (Corollary 35.2.8), so x is an algebraic integer in K , i.e. $x \in \mathcal{O}_K$.

(iv) Let $\mathfrak{p}$ be a nonzero prime. By Lemma 35.4.8, $\mathcal{O}_K/\mathfrak{p}$ is a finite integral domain, hence a field (for $0 \neq a \in \mathcal{O}_K/\mathfrak{p}$, the map $x \mapsto ax$ is injective hence surjective, so a is invertible). So $\mathfrak{p}$ is maximal. $\square$

Definition 35.4.10 (Dedekind Domain)

A **Dedekind domain** is a Noetherian, integrally closed integral domain in which every nonzero prime ideal is maximal.

So the content of Proposition 35.4.9 is that $\mathcal{O}_K$ is a Dedekind domain, and almost everything we do in this section works verbatim for a general Dedekind domain. The class of examples to have in mind besides $\mathcal{O}_K$ is the coordinate rings of smooth affine curves.

Remark. Condition (iv) is a dimension condition: it says that $\mathcal{O}_K$ has **Krull dimension** 1, i.e. the longest chain of prime ideals is $(0) \subsetneq \mathfrak{p}$. It genuinely fails in higher dimension: in $\mathbb{C}[x, y]$ the ideal (x) is prime but is properly contained in the prime (x, y) .

§35.4.5 Fractional Ideals

The goal is now unique factorisation of ideals into primes. The heart of the matter is showing that ideals have *inverses*, which forces us out of $\mathcal{O}_K$ and into K . First, two lemmas.

Lemma 35.4.11

Let $\mathfrak{p}$ be a prime ideal of a ring R and let $\mathfrak{a}, \mathfrak{b} \subseteq R$. If $\mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$ then $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$.

Proof. If not, pick $a \in \mathfrak{a} \setminus \mathfrak{p}$ and $b \in \mathfrak{b} \setminus \mathfrak{p}$. Then $ab \in \mathfrak{a}\mathfrak{b} \subseteq \mathfrak{p}$, contradicting primality. $\square$

Lemma 35.4.12

Let $\mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal. Then $\mathfrak{a}$ contains a product of nonzero prime ideals.

Proof. Suppose not. Since $\mathcal{O}_K$ is Noetherian, the nonempty set of nonzero ideals failing the conclusion has a maximal element $\mathfrak{a}$. Now $\mathfrak{a}$ is not prime (a prime contains itself) and $\mathfrak{a} \neq \mathcal{O}_K$ (which contains any prime). So there are $x, y \in \mathcal{O}_K$ with $xy \in \mathfrak{a}$ but $x, y \notin \mathfrak{a}$. Then $\mathfrak{a} \subsetneq \mathfrak{a} + (x)$ and $\mathfrak{a} \subsetneq \mathfrak{a} + (y)$, so by maximality each of these

contains a product of primes, say $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq \mathfrak{a} + (x)$ and $\mathfrak{q}_1 \cdots \mathfrak{q}_s \subseteq \mathfrak{a} + (y)$. But

$$\mathfrak{p}_1 \cdots \mathfrak{p}_r \mathfrak{q}_1 \cdots \mathfrak{q}_s \subseteq (\mathfrak{a} + (x))(\mathfrak{a} + (y)) \subseteq \mathfrak{a} + (xy) = \mathfrak{a},$$

since $xy \in \mathfrak{a}$. This contradicts the choice of $\mathfrak{a}$. $\square$

The next lemma is the technical crux. Part (ii) says that the would-be inverse of a proper ideal genuinely contains something new.

Lemma 35.4.13

Let K be a number field.

(i) Let $\mathfrak{a} \leq \mathcal{O}_K$ be a nonzero ideal and $x \in K$ with $x\mathfrak{a} \subseteq \mathfrak{a}$. Then $x \in \mathcal{O}_K$.

(ii) Let $\mathfrak{a} \leq \mathcal{O}_K$ be a nonzero proper ideal. Then

$$\mathfrak{a}^* := \{y \in K : y\mathfrak{a} \subseteq \mathcal{O}_K\}$$

strictly contains $\mathcal{O}_K$.

Proof. (i) By Lemma 35.4.8, $\mathfrak{a}$ is a free abelian group with basis $\alpha_1, \dots, \alpha_n$. Since $x\mathfrak{a} \subseteq \mathfrak{a}$ we may write $x\alpha_i = \sum_j a_{ij}\alpha_j$ with $a_{ij} \in \mathbb{Z}$, so with $A = (a_{ij})$ we get $(xI - A)(\alpha_1, \dots, \alpha_n)^T = 0$. As the α_i are not all zero and we are working inside the field K , we get $\det(xI - A) = 0$. This is a monic integer polynomial relation for x , so $x \in \mathcal{O}_K$.

(ii) If the conclusion holds for $\mathfrak{a}$ it holds for any nonzero ideal contained in $\mathfrak{a}$ (as $\mathfrak{a}' \subseteq \mathfrak{a}$ gives $\mathfrak{a}^* \subseteq (\mathfrak{a}')^*$). So we may assume $\mathfrak{a} = \mathfrak{p}$ is maximal, hence prime. Pick $0 \neq \alpha \in \mathfrak{p}$. By Lemma 35.4.12 there are primes with $\mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq (\alpha) \subseteq \mathfrak{p}$; choose such a product with r minimal. By Lemma 35.4.11 some $\mathfrak{p}_i \subseteq \mathfrak{p}$, say $\mathfrak{p}_1 \subseteq \mathfrak{p}$; but $\mathfrak{p}_1$ is maximal, so $\mathfrak{p}_1 = \mathfrak{p}$.

By minimality of r we have $\mathfrak{p}_2 \cdots \mathfrak{p}_r \not\subseteq (\alpha)$ (interpreting the empty product as $\mathcal{O}_K$), so pick $\beta \in \mathfrak{p}_2 \cdots \mathfrak{p}_r \setminus (\alpha)$. Then

$$\beta\mathfrak{p} \subseteq \mathfrak{p} \mathfrak{p}_2 \cdots \mathfrak{p}_r = \mathfrak{p}_1 \cdots \mathfrak{p}_r \subseteq (\alpha),$$

so $(\beta/\alpha)\mathfrak{p} \subseteq \mathcal{O}_K$, i.e. $\beta/\alpha \in \mathfrak{p}^*$. But $\beta \notin (\alpha)$ means $\beta/\alpha \notin \mathcal{O}_K$. $\square$

Definition 35.4.14 (Fractional Ideal)

A **fractional ideal** of K is a nonzero finitely generated $\mathcal{O}_K$ -submodule of K .

To distinguish them, honest ideals $\mathfrak{a} \leq \mathcal{O}_K$ are called **integral ideals**. Products of fractional ideals are defined exactly as before.

Lemma 35.4.15

A nonzero $\mathcal{O}_K$ -submodule $\mathfrak{q} \subseteq K$ is a fractional ideal if and only if there is $0 \neq c \in \mathcal{O}_K$ with $c\mathfrak{q} \subseteq \mathcal{O}_K$ an integral ideal. In that case $\mathfrak{q} \cong \mathbb{Z}^n$ as an abelian group.

Proof. If $\mathfrak{q}$ is finitely generated, say by $x_1, \dots, x_r \in K$, write $x_i = y_i/n_i$ with $y_i \in \mathcal{O}_K$ and $n_i \in \mathbb{Z}_{\neq 0}$ (Lemma 35.2.10), and take $c = n_1 \cdots n_r$; then $c\mathfrak{q} \subseteq \mathcal{O}_K$ is an $\mathcal{O}_K$ -submodule of $\mathcal{O}_K$, i.e. an ideal. Conversely if $c\mathfrak{q}$ is a nonzero ideal then it is finitely generated (indeed $\cong \mathbb{Z}^n$ by Lemma 35.4.8), and multiplication by c is an $\mathcal{O}_K$ -module isomorphism $\mathfrak{q} \rightarrow c\mathfrak{q}$, so $\mathfrak{q}$ is finitely generated and $\mathfrak{q} \cong \mathbb{Z}^n$. $\square$

In particular, for a nonzero integral ideal $\mathfrak{a}$ the set $\mathfrak{a}^* = \{y \in K : y\mathfrak{a} \subseteq \mathcal{O}_K\}$ is a fractional ideal: it is an $\mathcal{O}_K$ -module, and $\alpha\mathfrak{a}^* \subseteq \mathcal{O}_K$ for any $0 \neq \alpha \in \mathfrak{a}$.

Theorem 35.4.16 (Invertibility of Fractional Ideals)

Every fractional ideal $\mathfrak{q}$ of K is invertible: there is a fractional ideal $\mathfrak{r}$ with $\mathfrak{q}\mathfrak{r} = \mathcal{O}_K$. Moreover

$$\mathfrak{q}^{-1} = \{x \in K : x\mathfrak{q} \subseteq \mathcal{O}_K\}.$$

Proof. By Lemma 35.4.15 we may write $\mathfrak{q} = \frac{1}{c}\mathfrak{a}$ with $\mathfrak{a}$ integral, and $\mathfrak{q}$ is invertible if and only if $\mathfrak{a}$ is (multiply an inverse by c). So it suffices to treat integral ideals.

Suppose for contradiction that some nonzero integral ideal is not invertible. As $\mathcal{O}_K$ is Noetherian, choose $\mathfrak{a}$ maximal among non-invertible nonzero integral ideals; note $\mathfrak{a} \neq \mathcal{O}_K$, which is its own inverse. Let $\mathfrak{b} = \mathfrak{a}^*$, a fractional ideal. Since $1 \in \mathcal{O}_K \subseteq \mathfrak{b}$ we have $\mathfrak{a} \subseteq \mathfrak{a}\mathfrak{b} \subseteq \mathcal{O}_K$, the last inclusion by definition of $\mathfrak{a}^*$.

If $\mathfrak{a} = \mathfrak{a}\mathfrak{b}$ then every $x \in \mathfrak{b}$ satisfies $x\mathfrak{a} \subseteq \mathfrak{a}$, so $\mathfrak{b} \subseteq \mathcal{O}_K$ by Lemma 35.4.13(i) – contradicting Lemma 35.4.13(ii). Hence $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{b} \subseteq \mathcal{O}_K$. If $\mathfrak{a}\mathfrak{b} = \mathcal{O}_K$ then $\mathfrak{a}$ is invertible, a contradiction; so $\mathfrak{a} \subsetneq \mathfrak{a}\mathfrak{b} \subsetneq \mathcal{O}_K$, and by maximality $\mathfrak{a}\mathfrak{b}$ is invertible, say $(\mathfrak{a}\mathfrak{b})\mathfrak{c} = \mathcal{O}_K$. But then $\mathfrak{a}(\mathfrak{b}\mathfrak{c}) = \mathcal{O}_K$, so $\mathfrak{a}$ is invertible after all. Contradiction.

Finally, writing $X = \{x \in K : x\mathfrak{q} \subseteq \mathcal{O}_K\}$, certainly $\mathfrak{q}^{-1} \subseteq X$, so

$$\mathcal{O}_K = \mathfrak{q}\mathfrak{q}^{-1} \subseteq \mathfrak{q}X \subseteq \mathcal{O}_K,$$

forcing $\mathfrak{q}X = \mathcal{O}_K$ and hence $X = \mathfrak{q}^{-1}\mathfrak{q}X = \mathfrak{q}^{-1}$. $\square$

Remark. Unwinding, the theorem says exactly: *for every nonzero ideal $\mathfrak{a} \leq \mathcal{O}_K$ there is an ideal $\mathfrak{b}$ with $\mathfrak{a}\mathfrak{b}$ principal*. That is a very concrete statement, and it is the one to remember. In Example 35.4.7 we saw it in action: $\mathfrak{p}\mathfrak{p} = (2)$, $\mathfrak{q}_1\mathfrak{q}_2 = (3)$.

§35.4.6 To Contain is to Divide

Definition 35.4.17 (Divisibility of Ideals)

For ideals $\mathfrak{a}, \mathfrak{b} \leq \mathcal{O}_K$ we say $\mathfrak{a}$ **divides** $\mathfrak{b}$, written $\mathfrak{a} \mid \mathfrak{b}$, if there is an ideal $\mathfrak{c}$ with $\mathfrak{b} = \mathfrak{a}\mathfrak{c}$.

Since $\mathfrak{a}\mathfrak{c} \subseteq \mathfrak{a}$, divisibility implies containment. The converse is the slogan of the subject, and it is now a short argument.

Corollary 35.4.18 (To Contain is to Divide)

Let $\mathfrak{a}, \mathfrak{b}, \mathfrak{c} \leq \mathcal{O}_K$ be nonzero integral ideals. Then

- (i) $\mathfrak{b} \subseteq \mathfrak{a} \iff \mathfrak{b}\mathfrak{c} \subseteq \mathfrak{a}\mathfrak{c}$;

$$(ii) \mathfrak{a} \mid \mathfrak{b} \iff \mathfrak{ac} \mid \mathfrak{bc};$$

$$(iii) \mathfrak{a} \mid \mathfrak{b} \iff \mathfrak{b} \subseteq \mathfrak{a}.$$

Proof. In (i) and (ii) the forward implications are immediate, and the backward ones follow by multiplying by $\mathfrak{c}^{-1}$ and using Theorem 35.4.16. For (iii), we have already seen $\Rightarrow$. Conversely suppose $\mathfrak{b} \subseteq \mathfrak{a}$. Choose $\mathfrak{c}$ with $\mathfrak{ac} = (\alpha)$ principal (Theorem 35.4.16 and the remark following it). By (i), $\mathfrak{bc} \subseteq (\alpha)$. Writing $\mathfrak{bc} = (\beta_1, \dots, \beta_r)$, each $\beta_i \in (\alpha)$, so $\beta_i = \alpha\beta'_i$ with $\beta'_i \in \mathcal{O}_K$, giving

$$\mathfrak{bc} = (\beta'_1, \dots, \beta'_r)(\alpha) = (\beta'_1, \dots, \beta'_r)\mathfrak{ac}.$$

Cancelling $\mathfrak{c}$ by (ii) gives $\mathfrak{b} = (\beta'_1, \dots, \beta'_r)\mathfrak{a}$, so $\mathfrak{a} \mid \mathfrak{b}$. $\square$

We can now harvest the main theorem.

Theorem 35.4.19 (Unique Factorisation of Ideals)

Every nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ can be written as a product of prime ideals,

$$\mathfrak{a} = \mathfrak{p}_1^{a_1} \cdots \mathfrak{p}_r^{a_r}$$

with $\mathfrak{p}_i$ distinct primes and $a_i \geq 1$, and this expression is unique up to reordering. (The empty product is $\mathcal{O}_K$.)

Proof. Existence. Suppose some nonzero ideal is not a product of primes; by Noetherianity choose $\mathfrak{a}$ maximal such. Then $\mathfrak{a} \neq \mathcal{O}_K$ (the empty product) and $\mathfrak{a}$ is not prime, so $\mathfrak{a}$ is not maximal; choose an ideal $\mathfrak{b}$ with $\mathfrak{a} \subsetneq \mathfrak{b} \subsetneq \mathcal{O}_K$. By Corollary 35.4.18(iii), $\mathfrak{a} = \mathfrak{bc}$ for some ideal $\mathfrak{c}$, and $\mathfrak{c} \supsetneq \mathfrak{a}$ with $\mathfrak{c} \neq \mathcal{O}_K$ (else $\mathfrak{ba} = \mathfrak{a}$ would give $\mathfrak{b} = \mathcal{O}_K$ on cancelling). So $\mathfrak{a} \subsetneq \mathfrak{b}$ and $\mathfrak{a} \subsetneq \mathfrak{c}$, and by maximality both $\mathfrak{b}$ and $\mathfrak{c}$ are products of primes – hence so is $\mathfrak{a}$, a contradiction.

Uniqueness. Note first that if $\mathfrak{p}$ is prime and $\mathfrak{p} \mid \mathfrak{ab}$, then $\mathfrak{ab} \subseteq \mathfrak{p}$, so $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$ by Lemma 35.4.11, i.e. $\mathfrak{p} \mid \mathfrak{a}$ or $\mathfrak{p} \mid \mathfrak{b}$. So primes behave like primes. Now suppose $\mathfrak{p}_1 \cdots \mathfrak{p}_r = \mathfrak{q}_1 \cdots \mathfrak{q}_s$ with all factors prime (repetitions allowed). Then $\mathfrak{p}_1$ divides the right-hand side, so $\mathfrak{p}_1 \mid \mathfrak{q}_i$ for some i , say $i = 1$; then $\mathfrak{q}_1 \subseteq \mathfrak{p}_1$, and as $\mathfrak{q}_1$ is maximal, $\mathfrak{q}_1 = \mathfrak{p}_1$. Multiplying by $\mathfrak{p}_1^{-1}$ and inducting on r gives the result. $\square$

Corollary 35.4.20

The nonzero fractional ideals of K form an abelian group I_K under multiplication, and I_K is free abelian on the set of nonzero prime ideals of $\mathcal{O}_K$. A fractional ideal $\prod_{\mathfrak{p}} \mathfrak{p}^{a_{\mathfrak{p}}}$ is integral if and only if all $a_{\mathfrak{p}} \geq 0$.

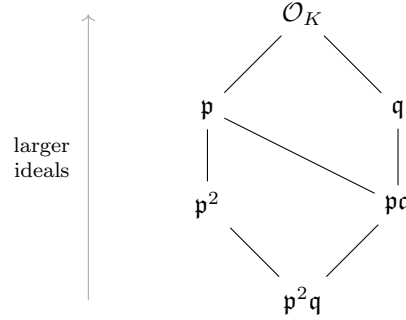
Proof. It is a group by Theorem 35.4.16. Any fractional ideal is $\mathfrak{ab}^{-1}$ with $\mathfrak{a}, \mathfrak{b}$ integral (Lemma 35.4.15), so the primes generate; and freeness is the uniqueness in Theorem 35.4.19. The last statement follows since $\prod \mathfrak{p}^{a_{\mathfrak{p}}} \subseteq \mathcal{O}_K$ if and only if $\mathcal{O}_K$ divides it. $\square$

So the arithmetic of ideals in $\mathcal{O}_K$ looks exactly like the arithmetic of positive integers, with the primes of $\mathbb{Z}$ replaced by the prime ideals of $\mathcal{O}_K$. For example, given $\mathfrak{a} = \prod \mathfrak{p}^{a_{\mathfrak{p}}}$

and $\mathfrak{b} = \prod \mathfrak{p}^{b_p}$, we have $\mathfrak{a} \mid \mathfrak{b}$ iff $a_p \leq b_p$ for all $\mathfrak{p}$, and

$$\mathfrak{a} + \mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{\min(a_p, b_p)} = \gcd(\mathfrak{a}, \mathfrak{b}), \quad \mathfrak{a} \cap \mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{\max(a_p, b_p)} = \text{lcm}(\mathfrak{a}, \mathfrak{b}),$$

the first because $\mathfrak{a} + \mathfrak{b}$ is the smallest ideal containing both, i.e. the largest dividing both. The divisors of an ideal form the same sort of lattice as the divisors of an integer. Here, for instance, is the divisor lattice of $\mathfrak{a} = \mathfrak{p}^2 \mathfrak{q}$, drawn with the larger ideals – that is, the divisors – higher up.



Example 35.4.21

In $\mathbb{Z}[\sqrt{-5}]$, Example 35.4.7 gives the prime factorisations

$$(6) = \mathfrak{p}^2 \mathfrak{q}_1 \mathfrak{q}_2, \quad (2) = \mathfrak{p}^2, \quad (3) = \mathfrak{q}_1 \mathfrak{q}_2, \quad (1 + \sqrt{-5}) = \mathfrak{p} \mathfrak{q}_1, \quad (1 - \sqrt{-5}) = \mathfrak{p} \mathfrak{q}_2,$$

so that the two ‘different’ factorisations $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ correspond to the two ways of splitting $\{\mathfrak{p}, \mathfrak{p}, \mathfrak{q}_1, \mathfrak{q}_2\}$ into two pairs. (That $\mathfrak{p}, \mathfrak{q}_1, \mathfrak{q}_2$ really are prime we will see in Section 35.6; for now, note that $\mathfrak{p}^2 = (2)$ has $\mathcal{O}_K/\mathfrak{p}$ of size 2, hence a field.) Unique factorisation of ideals holds; it is unique factorisation of *elements* that fails, because not every ideal is principal.

§35.5 The Class Group and the Ideal Norm

§35.5.1 The Class Group

We have a map

$$K^\times \longrightarrow I_K, \quad x \longmapsto (x),$$

sending an element to the principal fractional ideal it generates. It is a group homomorphism since $(x)(y) = (xy)$. Its kernel is $\{x \in K^\times : (x) = \mathcal{O}_K\} = \mathcal{O}_K^\times$, and its image is the subgroup $P_K \leq I_K$ of principal fractional ideals. The cokernel is the object we want.

Definition 35.5.1 (Class Group)

The **ideal class group** of a number field K is

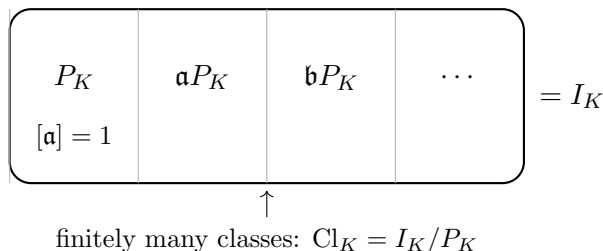
$$\text{Cl}_K = I_K / P_K.$$

We write $[\mathfrak{a}]$ for the class of $\mathfrak{a}$, and $h_K = |\text{Cl}_K|$ for the **class number**.

Thus $[\mathfrak{a}] = [\mathfrak{b}]$ if and only if $\mathfrak{b} = \gamma \mathfrak{a}$ for some $\gamma \in K^\times$, and $[\mathfrak{a}] = 1$ if and only if $\mathfrak{a}$ is principal. Everything fits into a single exact sequence, which is worth committing to memory:

$$1 \longrightarrow \mathcal{O}_K^\times \longrightarrow K^\times \longrightarrow I_K \longrightarrow \mathrm{Cl}_K \longrightarrow 1.$$

The two ends of this sequence are the two main theorems of the course: Cl_K is *finite*, and $\mathcal{O}_K^\times$ is finitely generated of a rank we can write down. Pictorially, Cl_K chops the group of fractional ideals into finitely many cosets of the principal ones.



The class group measures precisely the failure of unique factorisation.

Theorem 35.5.2

For a number field K the following are equivalent:

- (i) $\mathcal{O}_K$ is a principal ideal domain;
- (ii) $\mathcal{O}_K$ is a unique factorisation domain;
- (iii) Cl_K is trivial.

Proof. (i) $\iff$ (iii) is the definition, and (i) $\Rightarrow$ (ii) is a general fact from IB Groups, Rings and Modules.

For (ii) $\Rightarrow$ (i), let $\mathfrak{p}$ be a nonzero prime ideal and pick $0 \neq x \in \mathfrak{p}$. Factor $x = \alpha_1 \cdots \alpha_r$ into irreducibles. Since $\mathfrak{p}$ is prime, some $\alpha_i \in \mathfrak{p}$, so $(\alpha_i) \subseteq \mathfrak{p}$. In a UFD irreducibles are prime, so (α_i) is a nonzero prime ideal, hence maximal by Proposition 35.4.9(iv); as $\mathfrak{p} \neq \mathcal{O}_K$ we get $(\alpha_i) = \mathfrak{p}$. So every prime ideal is principal, and by Theorem 35.4.19 every ideal is a product of primes, hence principal. $\square$

Remark. This is a genuinely useful equivalence, and not one that holds in general rings: an arbitrary UFD need not be a PID ($\mathbb{Z}[x]$, say). What makes it work is Dedekind-ness, specifically that nonzero primes are maximal.

Example 35.5.3

In $\mathbb{Z}[\sqrt{-5}]$, the ideal $\mathfrak{p} = (2, 1 + \sqrt{-5})$ is not principal: if $\mathfrak{p} = (\beta)$ then $\mathfrak{p}^2 = (2)$ would give $\beta^2 = 2u$ for a unit $u = \pm 1$, so $N(\beta)^2 = 4$ and $N(\beta) = 2$; but $a^2 + 5b^2 = 2$ is insoluble. So $[\mathfrak{p}] \neq 1$ while $[\mathfrak{p}]^2 = [(2)] = 1$, and $\mathrm{Cl}_{\mathbb{Q}(\sqrt{-5})}$ contains an element of order 2. We will see in Section 35.8 that in fact $\mathrm{Cl}_{\mathbb{Q}(\sqrt{-5})} \cong \mathbb{Z}/2\mathbb{Z}$.

Example 35.5.4

Here is a useful trick in quadratic fields. Let $K = \mathbb{Q}(\sqrt{d})$ and write $\bar{\cdot}$ for the nontrivial automorphism. For an ideal $\mathfrak{a}$, the conjugate $\bar{\mathfrak{a}} = \{\bar{\alpha} : \alpha \in \mathfrak{a}\}$ is again an ideal, and $[\bar{\mathfrak{a}}] = [\mathfrak{a}]^{-1}$, because $\mathfrak{a}\bar{\mathfrak{a}}$ is always principal. Indeed one can always write $\mathfrak{a} = (b, \alpha)$

with $b \in \mathbb{Z}$, and then

$$\mathfrak{a}\bar{\mathfrak{a}} = (b^2, b\alpha, b\bar{\alpha}, \alpha\bar{\alpha}) = (b^2, b\alpha, b\operatorname{Tr}(\alpha), N(\alpha)) = (b\alpha, c),$$

where $c = \gcd(b^2, b\operatorname{Tr}(\alpha), N(\alpha))$, using $b\bar{\alpha} = b\operatorname{Tr}(\alpha) - b\alpha$. Now $x = b\alpha/c$ has $\operatorname{Tr}(x) = b\operatorname{Tr}(\alpha)/c \in \mathbb{Z}$ and $N(x) = (b^2/c)(N(\alpha)/c) \in \mathbb{Z}$, so $x \in \mathcal{O}_K$ and $c \mid b\alpha$. Hence $\mathfrak{a}\bar{\mathfrak{a}} = (c)$ is principal. In particular, in Example 35.4.7, $\mathfrak{q}_2 = \bar{\mathfrak{q}}_1$ and indeed $[\mathfrak{q}_2] = [\mathfrak{q}_1]^{-1}$.

§35.5.2 The Norm of an Ideal

We now attach a positive integer to each ideal, generalising $|N(\alpha)|$.

Definition 35.5.5 (Ideal Norm)

Let $\mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal. Its **norm** is

$$N(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}| = [\mathcal{O}_K : \mathfrak{a}],$$

which is finite by Lemma 35.4.8.

By Lagrange's theorem $N(\mathfrak{a}) \cdot 1 = 0$ in $\mathcal{O}_K/\mathfrak{a}$, so $N(\mathfrak{a}) \in \mathfrak{a} \cap \mathbb{Z}$; equivalently $\mathfrak{a} \mid (N(\mathfrak{a}))$.

Example 35.5.6

For p a rational prime and $n = [K : \mathbb{Q}]$, we have $\mathcal{O}_K/(p) \cong (\mathbb{Z}/p\mathbb{Z})^n$, so $N((p)) = p^n$. More generally $N((m)) = |m|^n$ for $0 \neq m \in \mathbb{Z}$.

The essential property is multiplicativity, which is what makes the norm a useful book-keeping device.

Proposition 35.5.7 (Multiplicativity of the Norm)

Let $\mathfrak{a}, \mathfrak{b} \subseteq \mathcal{O}_K$ be nonzero ideals. Then $N(\mathfrak{a}\mathfrak{b}) = N(\mathfrak{a})N(\mathfrak{b})$.

Proof. By unique factorisation and induction it suffices to treat $\mathfrak{b} = \mathfrak{p}$ prime. From the tower $\mathfrak{a}\mathfrak{p} \subseteq \mathfrak{a} \subseteq \mathcal{O}_K$ and the third isomorphism theorem,

$$N(\mathfrak{a}\mathfrak{p}) = [\mathcal{O}_K : \mathfrak{a}\mathfrak{p}] = [\mathcal{O}_K : \mathfrak{a}] [\mathfrak{a} : \mathfrak{a}\mathfrak{p}] = N(\mathfrak{a}) |\mathfrak{a}/\mathfrak{a}\mathfrak{p}|,$$

so it suffices to show $|\mathfrak{a}/\mathfrak{a}\mathfrak{p}| = N(\mathfrak{p})$.

By unique factorisation $\mathfrak{a} \neq \mathfrak{a}\mathfrak{p}$, so choose $\alpha \in \mathfrak{a} \setminus \mathfrak{a}\mathfrak{p}$. We claim the map of abelian groups

$$\phi : \mathcal{O}_K/\mathfrak{p} \rightarrow \mathfrak{a}/\mathfrak{a}\mathfrak{p}, \quad x + \mathfrak{p} \mapsto \alpha x + \mathfrak{a}\mathfrak{p}$$

is a well-defined isomorphism; this gives the result. It is well defined since $\alpha\mathfrak{p} \subseteq \mathfrak{a}\mathfrak{p}$.

Injectivity. Since $(\alpha) \subseteq \mathfrak{a}$, Corollary 35.4.18 gives $(\alpha) = \mathfrak{a}\mathfrak{c}$ for some ideal $\mathfrak{c}$. Suppose $\alpha x \in \mathfrak{a}\mathfrak{p}$. Then $x\mathfrak{a}\mathfrak{c} = x(\alpha) \subseteq \mathfrak{a}\mathfrak{p}$, so multiplying by $\mathfrak{a}^{-1}$, $x\mathfrak{c} \subseteq \mathfrak{p}$. As $\mathfrak{p}$ is prime, either $x \in \mathfrak{p}$ or $\mathfrak{c} \subseteq \mathfrak{p}$. But $\mathfrak{c} \subseteq \mathfrak{p}$ would give $(\alpha) = \mathfrak{a}\mathfrak{c} \subseteq \mathfrak{a}\mathfrak{p}$, contradicting the choice of α . So $x \in \mathfrak{p}$, and ϕ is injective.

Surjectivity. We must show $(\alpha) + \mathfrak{a}\mathfrak{p} = \mathfrak{a}$. We know $\mathfrak{a}\mathfrak{p} \subsetneq (\alpha) + \mathfrak{a}\mathfrak{p} \subseteq \mathfrak{a}$, and multiplying by $\mathfrak{a}^{-1}$ gives

$$\mathfrak{p} \subsetneq ((\alpha) + \mathfrak{a}\mathfrak{p}) \mathfrak{a}^{-1} \subseteq \mathcal{O}_K.$$

Since $\mathfrak{p}$ is maximal, the middle term is $\mathcal{O}_K$, so $(\alpha) + \mathfrak{a}\mathfrak{p} = \mathfrak{a}$ as required. $\square$

Remark. The conceptual proof runs as follows. By the Chinese remainder theorem $\mathcal{O}_K / \prod \mathfrak{p}_i^{a_i} \cong \prod \mathcal{O}_K / \mathfrak{p}_i^{a_i}$, so we reduce to prime powers; and then

$$|\mathcal{O}_K / \mathfrak{p}^e| = |\mathcal{O}_K / \mathfrak{p}| \cdot |\mathfrak{p} / \mathfrak{p}^2| \cdots |\mathfrak{p}^{e-1} / \mathfrak{p}^e| = N(\mathfrak{p})^e,$$

because each $\mathfrak{p}^i / \mathfrak{p}^{i+1}$ is a 1-dimensional vector space over the field $\mathcal{O}_K / \mathfrak{p}$ (it is nonzero by unique factorisation, and any $x \in \mathfrak{p}^i \setminus \mathfrak{p}^{i+1}$ generates, since $(x) + \mathfrak{p}^{i+1} = \mathfrak{p}^i$ by the maximality argument above). This last point is special to Dedekind domains: in $\mathbb{F}_p[x, y]$ with $\mathfrak{p} = (x, y)$ the quotient $\mathfrak{p} / \mathfrak{p}^2$ is 2-dimensional. The proof we gave packages both steps at once.

The norm of an ideal and the norm of an element agree where they should.

Lemma 35.5.8

Let $0 \neq \alpha \in \mathcal{O}_K$. Then $N((\alpha)) = |N_{K/\mathbb{Q}}(\alpha)|$.

Proof. Let $\gamma_1, \dots, \gamma_n$ be an integral basis of $\mathcal{O}_K$; then $\alpha\gamma_1, \dots, \alpha\gamma_n$ is a $\mathbb{Z}$ -basis of (α) . Using Proposition 35.3.9,

$$\begin{aligned} \Delta(\alpha\gamma_1, \dots, \alpha\gamma_n) &= \det(\sigma_i(\alpha\gamma_j))^2 = \det(\sigma_i(\alpha)\sigma_i(\gamma_j))^2 \\ &= \left(\prod_i \sigma_i(\alpha) \right)^2 \det(\sigma_i(\gamma_j))^2 = N_{K/\mathbb{Q}}(\alpha)^2 d_K. \end{aligned}$$

On the other hand, by Lemma 35.3.19 applied to the $\mathbb{Z}$ -basis $\alpha\gamma_i$ of (α) ,

$$\Delta(\alpha\gamma_1, \dots, \alpha\gamma_n) = [\mathcal{O}_K : (\alpha)]^2 d_K = N((\alpha))^2 d_K.$$

Comparing and taking positive square roots gives the result. $\square$

More generally the same computation identifies the discriminant of any ideal.

Lemma 35.5.9

Let $\mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal and let $\alpha_1, \dots, \alpha_n$ be a $\mathbb{Z}$ -basis of $\mathfrak{a}$ (which exists by Lemma 35.4.8). Then the α_i form a $\mathbb{Q}$ -basis of K and

$$\Delta(\alpha_1, \dots, \alpha_n) = N(\mathfrak{a})^2 d_K.$$

Proof. This is exactly Lemma 35.3.19 with $M = \mathfrak{a}$, once we note that $[\mathcal{O}_K : \mathfrak{a}] = N(\mathfrak{a})$ and that a $\mathbb{Z}$ -basis of $\mathfrak{a}$ is a $\mathbb{Q}$ -basis of K (since $\mathfrak{a} \supseteq m\mathcal{O}_K$ for some $m \neq 0$, and $\mathfrak{a}$ has rank n). $\square$

We record two consequences that we shall use constantly.

Corollary 35.5.10

Let K be a number field.

- (i) For each $m \geq 1$ there are only finitely many ideals $\mathfrak{a} \leq \mathcal{O}_K$ with $N(\mathfrak{a}) = m$.
- (ii) If $N(\mathfrak{a})$ is prime then $\mathfrak{a}$ is a prime ideal.
- (iii) If $\mathfrak{a} \subseteq \mathfrak{b}$ and $N(\mathfrak{a}) = N(\mathfrak{b})$ then $\mathfrak{a} = \mathfrak{b}$.

Proof. (i) If $N(\mathfrak{a}) = m$ then $m \in \mathfrak{a}$, so $\mathfrak{a} \mid (m)$; and (m) has only finitely many divisors by unique factorisation. (Alternatively: such $\mathfrak{a}$ correspond to ideals of the finite ring $\mathcal{O}_K/m\mathcal{O}_K$.)

(ii) $\mathcal{O}_K/\mathfrak{a}$ has prime order, so is $\mathbb{Z}/p\mathbb{Z}$, a field; hence $\mathfrak{a}$ is maximal, so prime.

(iii) By Corollary 35.4.18, $\mathfrak{a} = \mathfrak{b}\mathfrak{c}$ for some ideal $\mathfrak{c}$, and taking norms $N(\mathfrak{c}) = 1$, so $\mathfrak{c} = \mathcal{O}_K$. $\square$

§35.6 Factorisation of Primes

Unique factorisation of ideals reduces all questions about ideals to questions about prime ideals, so we had better understand what the prime ideals of $\mathcal{O}_K$ look like. It turns out that they all sit above rational primes, and that finding them amounts to factoring polynomials mod p .

§35.6.1 Primes Above p **Lemma 35.6.1**

Let $\mathfrak{p} \leq \mathcal{O}_K$ be a nonzero prime ideal. Then there is a unique rational prime p with $\mathfrak{p} \mid (p)$, namely the one with $\mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$. Moreover $N(\mathfrak{p}) = p^f$ for some $1 \leq f \leq n = [K : \mathbb{Q}]$.

Proof. $\mathfrak{p} \cap \mathbb{Z}$ is an ideal of $\mathbb{Z}$, hence $= m\mathbb{Z}$ for some $m \geq 0$, and $m \neq 0$ by Lemma 35.4.8. If $m = ab$ with $a, b \in \mathbb{Z}$, then $ab \in \mathfrak{p}$ so $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$, giving $m \mid a$ or $m \mid b$; hence $m = p$ is prime. Then $p \in \mathfrak{p}$, so $(p) \subseteq \mathfrak{p}$ and $\mathfrak{p} \mid (p)$ by Corollary 35.4.18. Uniqueness is clear since $\mathfrak{p} \cap \mathbb{Z}$ determines p , and if $\mathfrak{p} \mid (q)$ for a prime $q \neq p$ then $q \in \mathfrak{p} \cap \mathbb{Z} = p\mathbb{Z}$, absurd.

Writing $(p) = \mathfrak{p}\mathfrak{a}$ and taking norms, $p^n = N(\mathfrak{p})N(\mathfrak{a})$, so $N(\mathfrak{p}) = p^f$ with $1 \leq f \leq n$ (note $N(\mathfrak{p}) > 1$ as $\mathfrak{p} \neq \mathcal{O}_K$). $\square$

So to find all the prime ideals of $\mathcal{O}_K$, it is enough to factorise (p) for each rational prime p . This motivates the following language.

Definition 35.6.2 (Ramification Index and Residue Degree)

Let p be a rational prime and let

$$p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g}$$

be its factorisation into distinct primes. We say $\mathfrak{p}_i$ **lies above** p , written $\mathfrak{p}_i \mid p$. The integer e_i is the **ramification index** of $\mathfrak{p}_i$ over p , and f_i defined by $N(\mathfrak{p}_i) = p^{f_i}$ is the **residue degree**. Equivalently $f_i = [\mathcal{O}_K/\mathfrak{p}_i : \mathbb{F}_p]$, since $\mathcal{O}_K/\mathfrak{p}_i$ is a finite field containing $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

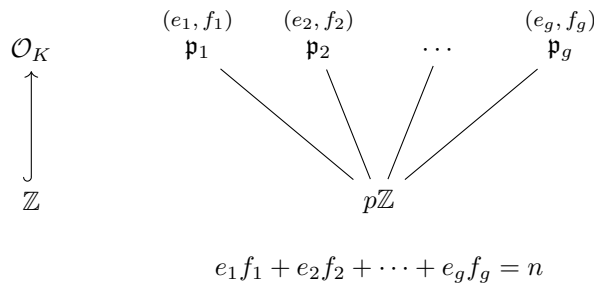
Theorem 35.6.3 (The Fundamental Identity)

With the notation above, $\sum_{i=1}^g e_i f_i = n = [K : \mathbb{Q}]$.

Proof. Take norms in $p\mathcal{O}_K = \prod_i \mathfrak{p}_i^{e_i}$ and use Proposition 35.5.7:

$$p^n = N((p)) = \prod_i N(\mathfrak{p}_i)^{e_i} = p^{\sum_i e_i f_i}. \quad \square$$

The picture to have in mind is a tower, with p at the bottom in $\mathbb{Z}$ and the primes above it in $\mathcal{O}_K$, each labelled with its e and f .



Definition 35.6.4 (Split, Inert, Ramified)

Let p be a rational prime and $p\mathcal{O}_K = \prod_i \mathfrak{p}_i^{e_i}$ as above. We say p is

- **ramified** in K if $e_i > 1$ for some i , and **unramified** otherwise;
- **inert** in K if $g = 1$ and $e_1 = 1$, so that (p) stays prime;
- **totally split** (or **split completely**) if $g = n$, so that $e_i = f_i = 1$ for all i ;
- **totally ramified** if $g = 1$ and $e_1 = n$.

We will see that only finitely many primes ramify, namely the divisors of d_K .

In the Galois case the picture is much more uniform, because the Galois group shuffles the primes above p among themselves.

Proposition 35.6.5 (Splitting in Galois Extensions)

Let $K/\mathbb{Q}$ be Galois with group G , and let p be a rational prime with $p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g}$. Then G acts transitively on $\{\mathfrak{p}_1, \dots, \mathfrak{p}_g\}$. Consequently $e_1 = \cdots = e_g =: e$ and $f_1 = \cdots = f_g =: f$, and

$$efg = n = [K : \mathbb{Q}].$$

Proof. First note that G preserves $\mathcal{O}_K$: if α satisfies a monic $h \in \mathbb{Z}[x]$ and $\tau \in G$, then $h(\tau\alpha) = \tau(h(\alpha)) = 0$, so $\tau\alpha \in \mathcal{O}_K$. Hence τ permutes the ideals of $\mathcal{O}_K$, preserving primality and fixing (p) , so it permutes $\{\mathfrak{p}_1, \dots, \mathfrak{p}_g\}$.

Suppose the action is not transitive, so some $\mathfrak{q} = \mathfrak{p}_j$ is not of the form $\tau(\mathfrak{p}_1)$ for any τ . By the Chinese remainder theorem (the $\tau(\mathfrak{p}_1)$ and $\mathfrak{q}$ being distinct maximal ideals) pick $x \in \mathcal{O}_K$ with

$$x \equiv 0 \pmod{\mathfrak{q}}, \quad x \equiv 1 \pmod{\tau(\mathfrak{p}_1)} \text{ for all } \tau \in G.$$

Then $N_{K/\mathbb{Q}}(x) = \prod_{\tau \in G} \tau(x) \in \mathfrak{q} \cap \mathbb{Z} = p\mathbb{Z} \subseteq \mathfrak{p}_1$, so some $\tau(x) \in \mathfrak{p}_1$, i.e. $x \in \tau^{-1}(\mathfrak{p}_1)$ – contradicting $x \equiv 1$ there.

Finally, applying τ to the factorisation of (p) permutes the $\mathfrak{p}_i$ while fixing (p) , so by uniqueness the exponents match up: $e_i = e_j$ whenever $\tau(\mathfrak{p}_i) = \mathfrak{p}_j$. And τ induces an isomorphism $\mathcal{O}_K/\mathfrak{p}_i \rightarrow \mathcal{O}_K/\tau(\mathfrak{p}_i)$, so the f_i agree too. Transitivity then makes all e_i and all f_i equal, and Theorem 35.6.3 becomes $efg = n$. $\square$

Example 35.6.6

In $\mathbb{Q}(i)$, $n = 2$, so (e, f, g) is $(1, 1, 2)$ for split primes, $(1, 2, 1)$ for inert primes and $(2, 1, 1)$ for the ramified prime 2 – exactly the trichotomy of Proposition 35.6.11. In $\mathbb{Q}(\zeta_7)$, $n = 6$, and the possible unramified behaviours are $(1, f, 6/f)$ for $f \mid 6$. Contrast this with $\mathbb{Q}(\sqrt[3]{2})$, which is *not* Galois: there we found $(5) = \mathfrak{p}_1\mathfrak{p}_2$ with $f_1 = 1$ and $f_2 = 2$, which is impossible in a Galois extension.

§35.6.2 Dedekind's Criterion

Suppose we can write $\mathcal{O}_K = \mathbb{Z}[\alpha]$ for some α with minimal polynomial $g \in \mathbb{Z}[x]$. Then

$$\mathcal{O}_K/p\mathcal{O}_K \cong \mathbb{Z}[x]/(g(x), p) \cong \mathbb{F}_p[x]/(\bar{g}(x)),$$

and the ideals of $\mathcal{O}_K$ containing p correspond to the ideals of this ring, which correspond in turn to the monic factors of $\bar{g}$ in $\mathbb{F}_p[x]$. So factoring (p) becomes factoring a polynomial mod p – a completely mechanical task. Remarkably, this works even when $\mathbb{Z}[\alpha] \neq \mathcal{O}_K$, provided p does not divide the index.

Theorem 35.6.7 (Dedekind's Criterion)

Let K be a number field, let $\alpha \in \mathcal{O}_K$ with $K = \mathbb{Q}(\alpha)$ and minimal polynomial $g \in \mathbb{Z}[x]$, and let p be a prime with $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$. Let

$$\bar{g} = \bar{\varphi}_1^{e_1} \cdots \bar{\varphi}_g^{e_g}$$

be the factorisation of $\bar{g} = g \bmod p$ into distinct monic irreducibles in $\mathbb{F}_p[x]$, and lift each $\bar{\varphi}_i$ to a monic $\varphi_i \in \mathbb{Z}[x]$ of the same degree. Then

$$p\mathcal{O}_K = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g}, \quad \mathfrak{p}_i = (p, \varphi_i(\alpha)),$$

the $\mathfrak{p}_i$ are distinct primes, and $N(\mathfrak{p}_i) = p^{f_i}$ with $f_i = \deg \varphi_i$.

Proof. Step 1: the ideals $\mathfrak{q}_i = (p, \varphi_i(\alpha)) \trianglelefteq \mathbb{Z}[\alpha]$ are prime with $\mathbb{Z}[\alpha]/\mathfrak{q}_i \cong \mathbb{F}_p[x]/(\bar{\varphi}_i)$. Consider the surjective ring homomorphism $\mathbb{Z}[x] \rightarrow \mathbb{F}_p[x]/(\bar{\varphi}_i)$ given by reduction mod p followed by reduction mod $\bar{\varphi}_i$. Its kernel contains p and φ_i ; conversely, if f maps to zero then $\bar{\varphi}_i \mid \bar{f}$, so $\bar{f} = \bar{\varphi}_i \bar{h}$ for some $\bar{h}$, and lifting $\bar{h}$ to $h \in \mathbb{Z}[x]$ gives $f - \varphi_i h \in p\mathbb{Z}[x]$. So the kernel is exactly (p, φ_i) . Since $g(\alpha) = 0$ and $\bar{\varphi}_i \mid \bar{g}$, the map factors through $\mathbb{Z}[x]/(g) \cong \mathbb{Z}[\alpha]$, and the same argument shows that the induced map $\mathbb{Z}[\alpha] \rightarrow \mathbb{F}_p[x]/(\bar{\varphi}_i)$ has kernel $\mathfrak{q}_i = (p, \varphi_i(\alpha))$. As $\bar{\varphi}_i$ is irreducible, $\mathbb{F}_p[x]/(\bar{\varphi}_i) \cong \mathbb{F}_{p^{f_i}}$ is a field, so $\mathfrak{q}_i$ is prime with $|\mathbb{Z}[\alpha]/\mathfrak{q}_i| = p^{f_i}$.

Step 2: $\iota : \mathbb{Z}[\alpha]/p\mathbb{Z}[\alpha] \rightarrow \mathcal{O}_K/p\mathcal{O}_K$ is an isomorphism. Write $M = \mathcal{O}_K/\mathbb{Z}[\alpha]$, a finite abelian group of order coprime to p by hypothesis. Multiplication by p on M is injective (its kernel is a p -group inside a group of order prime to p), hence bijective as M is finite. Now the kernel of ι is $(\mathbb{Z}[\alpha] \cap p\mathcal{O}_K)/p\mathbb{Z}[\alpha]$, and an element of this is $x \in \mathbb{Z}[\alpha]$ with $x = py$, $y \in \mathcal{O}_K$; then $py \equiv 0$ in M , so $y \in \mathbb{Z}[\alpha]$ and $x \in p\mathbb{Z}[\alpha]$. So ι is injective. For surjectivity we need $\mathcal{O}_K = \mathbb{Z}[\alpha] + p\mathcal{O}_K$, i.e. $pM = M$, which is the surjectivity of multiplication by p on M . Comparing orders, ι is an isomorphism.

Step 3. By Step 2, ideals of $\mathcal{O}_K$ containing p correspond to ideals of $\mathbb{Z}[\alpha]$ containing p , with $\mathfrak{q} \mapsto \mathfrak{q}\mathcal{O}_K$ and $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathbb{Z}[\alpha]$, and the quotients agree. So each $\mathfrak{p}_i := \mathfrak{q}_i\mathcal{O}_K = (p, \varphi_i(\alpha))$ is a prime ideal of $\mathcal{O}_K$ with $N(\mathfrak{p}_i) = p^{f_i}$, and $\mathfrak{p}_i \neq \mathfrak{p}_j$ for $i \neq j$ since $\bar{\varphi}_i, \bar{\varphi}_j$ are coprime, whence $\mathfrak{q}_i + \mathfrak{q}_j = \mathbb{Z}[\alpha]$.

Step 4: the factorisation. Since $(p, \varphi_i(\alpha))^{e_i} \subseteq (p, \varphi_i(\alpha)^{e_i})$, we get

$$\mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_g^{e_g} \subseteq (p, \varphi_1(\alpha)^{e_1} \cdots \varphi_g(\alpha)^{e_g}).$$

Now $\overline{\varphi_1^{e_1} \cdots \varphi_g^{e_g}} = \bar{g}$, so $\varphi_1^{e_1} \cdots \varphi_g^{e_g} = g + ps$ for some $s \in \mathbb{Z}[x]$, and evaluating at α (where $g(\alpha) = 0$) shows the right-hand ideal is $(p, ps(\alpha)) = (p)$. So $\prod_i \mathfrak{p}_i^{e_i} \subseteq (p)$. But $\deg g = n$ gives $\sum_i e_i f_i = n$, so

$$N\left(\prod_i \mathfrak{p}_i^{e_i}\right) = p^{\sum_i e_i f_i} = p^n = N((p)),$$

and Corollary 35.5.10(iii) forces equality. $\square$

Remark. The hypothesis $p \nmid [\mathcal{O}_K : \mathbb{Z}[\alpha]]$ is essential but mild in practice: for any fixed α generating K with $\alpha \in \mathcal{O}_K$, only finitely many primes divide the index, so Dedekind's criterion factors (p) for all but finitely many p . Different lifts φ_i give the same ideal, since they differ by a multiple of $p \in \mathfrak{p}_i$.

Example 35.6.8

Let $K = \mathbb{Q}(\sqrt[3]{2})$, so $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{2}]$ and $g = x^3 - 2$; the index is 1, so Dedekind applies to every p .

- (i) $p = 2$: $\bar{g} = x^3$, so $(2) = \mathfrak{p}^3$ with $\mathfrak{p} = (2, \sqrt[3]{2}) = (\sqrt[3]{2})$. So 2 is totally ramified, with $e = 3, f = 1$.
- (ii) $p = 3$: $\bar{g} = x^3 - 2 = x^3 + 1 = (x + 1)^3$ in $\mathbb{F}_3[x]$, so $(3) = (3, 1 + \sqrt[3]{2})^3$; again totally ramified.
- (iii) $p = 5$: 2 is a cube mod 5 (namely $3^3 = 27 \equiv 2$), and $x^3 - 2 = (x - 3)(x^2 + 3x + 4)$ in $\mathbb{F}_5[x]$, the quadratic factor being irreducible as its discriminant $9 - 16 =$

$-7 \equiv 3$ is not a square mod 5. So

$$(5) = (5, \sqrt[3]{2} - 3) (5, \sqrt[3]{4} + 3\sqrt[3]{2} + 4),$$

with $(e, f) = (1, 1)$ and $(1, 2)$; indeed $1 \cdot 1 + 1 \cdot 2 = 3$.

(iv) $p = 7$: the cubes mod 7 are $\{0, 1, 6\}$, so $x^3 - 2$ has no root mod 7; being cubic it is irreducible, and 7 is inert with $f = 3$.

(v) $p = 31$: here $x^3 - 2 \equiv (x - 4)(x - 7)(x - 20)$, so 31 is totally split.

Notice that 2 and 3 ramify, and $d_K = -108 = -2^2 \cdot 3^3$. This is not a coincidence.

Example 35.6.9 (A Non-Monogenic Field)

Return to $K = \mathbb{Q}(\alpha)$ with $\alpha^3 - \alpha^2 - 2\alpha - 8 = 0$, where $d_K = -503$ and $[\mathcal{O}_K : \mathbb{Z}[\alpha]] = 2$ (Example 35.3.17). Dedekind's criterion does *not* apply at $p = 2$; and indeed $\bar{g} = x^2(x + 1)$ in $\mathbb{F}_2[x]$ would predict $(2) = \mathfrak{p}^2\mathfrak{q}$, which is wrong. In fact 2 splits completely, $(2) = \mathfrak{p}_1\mathfrak{p}_2\mathfrak{p}_3$ with all $e_i = f_i = 1$ (one checks this using the integral basis $1, \alpha, \beta$).

This has a striking consequence: for *every* $\gamma \in \mathcal{O}_K \setminus \mathbb{Z}$, the index $[\mathcal{O}_K : \mathbb{Z}[\gamma]]$ is even, so $\mathcal{O}_K \neq \mathbb{Z}[\gamma]$. Indeed, if $2 \nmid [\mathcal{O}_K : \mathbb{Z}[\gamma]]$ then Dedekind's criterion would say that the minimal polynomial of γ has three distinct roots in $\mathbb{F}_2$ – but $\mathbb{F}_2$ only has two elements. So K has no integral basis of the form $1, \gamma, \gamma^2$: it is not **monogenic**. The same argument gives a general obstruction.

Corollary 35.6.10

Let $p < n = [K : \mathbb{Q}]$ be a prime and suppose $\mathcal{O}_K = \mathbb{Z}[\alpha]$ for some α . Then p does not split completely in K .

Proof. If p split completely then, by Dedekind's criterion, the minimal polynomial of α would have n distinct roots in $\mathbb{F}_p$; but $|\mathbb{F}_p| = p < n$. $\square$

§35.6.3 Primes in Quadratic Fields

The quadratic case is worth doing completely, because it is where we shall do most of our computations, and because the answer is a beautiful reformulation of quadratic reciprocity.

Proposition 35.6.11

Let $K = \mathbb{Q}(\sqrt{d})$ with d squarefree, and let p be an odd prime. Then

$$p\mathcal{O}_K = \begin{cases} \mathfrak{p}_1\mathfrak{p}_2 \text{ with } \mathfrak{p}_1 \neq \mathfrak{p}_2 & \text{if } \left(\frac{d}{p}\right) = 1 \quad (p \text{ splits}), \\ \mathfrak{p} \text{ prime} & \text{if } \left(\frac{d}{p}\right) = -1 \quad (p \text{ inert}), \\ \mathfrak{p}^2 & \text{if } p \mid d \quad (p \text{ ramifies}), \end{cases}$$

where $\left(\frac{\cdot}{p}\right)$ is the Legendre symbol. Explicitly, if $r^2 \equiv d \pmod{p}$ then $\mathfrak{p}_{1,2} = (p, \sqrt{d} \mp r)$.

$r)$, and in the ramified case $\mathfrak{p} = (p, \sqrt{d})$.

Proof. Whether $d \equiv 1 \pmod{4}$ or not, $\mathbb{Z}[\sqrt{d}] \subseteq \mathcal{O}_K$ has index 1 or 2, which is coprime to the odd prime p . So Dedekind's criterion applies with $\alpha = \sqrt{d}$, $g = x^2 - d$, and we simply factor $x^2 - d$ in $\mathbb{F}_p[x]$: it has two distinct roots, no roots, or a repeated root according as d is a nonzero square, a nonsquare, or zero mod p . $\square$

The prime 2 needs separate treatment, and here the $d \pmod{4}$ dichotomy resurfaces.

Lemma 35.6.12

Let $K = \mathbb{Q}(\sqrt{d})$ with d squarefree. Then 2 splits if $d \equiv 1 \pmod{8}$, is inert if $d \equiv 5 \pmod{8}$, and ramifies if $d \equiv 2, 3 \pmod{4}$.

Proof. If $d \equiv 1 \pmod{4}$ then $\mathcal{O}_K = \mathbb{Z}[\alpha]$ with $\alpha = \frac{1}{2}(1 + \sqrt{d})$, whose minimal polynomial is $x^2 - x + \frac{1}{4}(1 - d)$; the index is 1 so Dedekind applies. Modulo 2 this is $x^2 + x + c$ where $c = \frac{1}{4}(1 - d) \pmod{2}$. If $d \equiv 1 \pmod{8}$ then $c = 0$ and we get $x(x + 1)$, so 2 splits. If $d \equiv 5 \pmod{8}$ then $c = 1$ and $x^2 + x + 1$ is irreducible over $\mathbb{F}_2$, so 2 is inert.

If $d \equiv 2, 3 \pmod{4}$ then $\mathcal{O}_K = \mathbb{Z}[\sqrt{d}]$ and $x^2 - d$ reduces mod 2 to x^2 or $x^2 + 1 = (x + 1)^2$, so 2 ramifies. $\square$

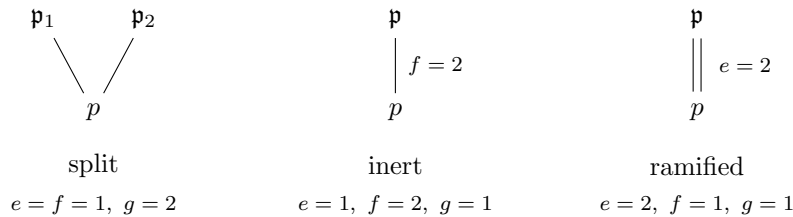
Comparing with Example 35.3.16 – where $d_K = d$ if $d \equiv 1 \pmod{4}$ and $4d$ otherwise – gives a clean statement.

Corollary 35.6.13

Let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field. A prime p ramifies in K if and only if $p \mid d_K$.

Proof. For odd p : p ramifies iff $p \mid d$, and since $d_K \in \{d, 4d\}$ this holds iff $p \mid d_K$. For $p = 2$: by Lemma 35.6.12, 2 ramifies iff $d \equiv 2, 3 \pmod{4}$, which is exactly when $d_K = 4d$ is even; and if $d \equiv 1 \pmod{4}$ then $d_K = d$ is odd. $\square$

Here are the three behaviours side by side, drawn as towers.



Example 35.6.14

Let $K = \mathbb{Q}(\sqrt{-5})$, so $d_K = -20$.

- $p = 2$: $-5 \equiv 3 \pmod{4}$, so 2 ramifies: $(2) = (2, 1 + \sqrt{-5})^2$, as we computed directly in Example 35.4.7.
- $p = 3$: $x^2 + 5 \equiv x^2 - 1 = (x - 1)(x + 1) \pmod{3}$, so $(3) = (3, \sqrt{-5} - 1)(3, \sqrt{-5} + 1)$

1).

- $p = 5$: $5 \mid d_K$, so 5 ramifies: $(5) = (\sqrt{-5})^2$.
- $p = 7$: the squares mod 7 are 1, 2, 4, and $-5 \equiv 2$, so 7 splits: indeed $x^2 + 5 \equiv (x-3)(x+3) \pmod{7}$.
- $p = 11$: $-5 \equiv 6 \pmod{11}$, and 6 is not a square mod 11, so 11 is inert.

This confirms in particular that the ideals $\mathfrak{p}, \mathfrak{q}_1, \mathfrak{q}_2$ of Example 35.4.7 really are prime.

Example 35.6.15

Let $K = \mathbb{Q}(\sqrt{-11})$, so $-11 \equiv 1 \pmod{4}$ and $\mathcal{O}_K = \mathbb{Z}[\frac{1}{2}(1 + \sqrt{-11})]$, in which $\mathbb{Z}[\sqrt{-11}]$ has index 2. To factor (5) we may use $\alpha = \sqrt{-11}$, since $5 \nmid 2$: modulo 5 we have $x^2 + 11 \equiv x^2 + 1 = (x-2)(x+2)$, so

$$(5) = (5, \sqrt{-11} - 2)(5, \sqrt{-11} + 2).$$

§35.6.4 Ramification and the Discriminant

Corollary 35.6.13 is a special case of a general and important theorem.

Theorem 35.6.16 (Ramification and the Discriminant)

Let K be a number field and p a rational prime. Then p ramifies in K if and only if $p \mid d_K$. In particular only finitely many primes ramify.

We shall not give the full proof, but the idea in the direction we use most is clear from Dedekind's criterion: if $\mathcal{O}_K = \mathbb{Z}[\alpha]$ with minimal polynomial g , then p ramifies exactly when $\bar{g}$ has a repeated factor in $\mathbb{F}_p[x]$, which happens exactly when $\text{disc}(\bar{g}) = 0$ in $\mathbb{F}_p$, i.e. when $p \mid \text{disc}(g) = d_K$. The general case is handled by reducing to this one after localising.

Example 35.6.17

In Example 35.6.8 we found that exactly 2 and 3 ramify in $\mathbb{Q}(\sqrt[3]{2})$, and $d_K = -108 = -2^2 3^3$. In Example 35.6.9, $d_K = -503$ is prime, so 503 is the only ramified prime; in particular 2 is unramified, consistent with $(2) = \mathfrak{p}_1 \mathfrak{p}_2 \mathfrak{p}_3$.

The theorem has a famous consequence, once we know (Section 35.7) that $|d_K| > 1$ whenever $K \neq \mathbb{Q}$: *every number field other than $\mathbb{Q}$ has at least one ramified prime*. Equivalently, $\mathbb{Q}$ admits no unramified extensions. This is the arithmetic analogue of the fact that the sphere is simply connected.

§35.6.5 Cyclotomic Fields

We finish the section with the other family of number fields that everyone should know. Let $\zeta = \zeta_m = e^{2\pi i/m}$ and $K = \mathbb{Q}(\zeta_m)$, the m th **cyclotomic field**.

Proposition 35.6.18

Let $K = \mathbb{Q}(\zeta_m)$. Then

- (i) $[K : \mathbb{Q}] = \varphi(m)$, and $K/\mathbb{Q}$ is Galois with $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$, the class of r acting by $\zeta_m \mapsto \zeta_m^r$;
- (ii) $\mathcal{O}_K = \mathbb{Z}[\zeta_m]$;
- (iii) a prime p ramifies in K if and only if $p \mid m$ (for $m \not\equiv 2 \pmod{4}$);
- (iv) if $p \nmid m$, then $p\mathcal{O}_K$ is a product of $\varphi(m)/f$ distinct primes each of residue degree f , where f is the multiplicative order of p modulo m . In particular p splits completely if and only if $p \equiv 1 \pmod{m}$.

Proof. Parts (i) is from Galois Theory: the minimal polynomial of ζ_m is the cyclotomic polynomial Φ_m , of degree $\varphi(m)$. Part (ii) we take on trust in general (for $m = p$ prime it is a standard exercise, sketched below), and (iii) follows from (ii) and Theorem 35.6.16.

For (iv), assume $p \nmid m$; by (ii) and Theorem 35.6.7 we must factor Φ_m in $\mathbb{F}_p[x]$. Since $p \nmid m$, the polynomial $x^m - 1$ is separable mod p , so $\bar{\Phi}_m = \bar{\gamma}_1 \cdots \bar{\gamma}_g$ with distinct irreducible factors $\bar{\gamma}_i$ (this re-proves that p is unramified). The Galois group $G \cong (\mathbb{Z}/m\mathbb{Z})^\times$ preserves $\mathcal{O}_K$, hence acts on the set of primes above p , and acts transitively on the roots of Φ_m ; so it permutes the $\bar{\gamma}_i$ transitively and they all have the same degree f' , say, with $gf' = \varphi(m)$.

It remains to show $f' = f = \text{ord}_m(p)$. In $\mathbb{F}_p[x]/(\bar{\gamma}_1) \cong \mathbb{F}_{p^{f'}}$, the image of x is the image of ζ_m , which is a root of $x^m - 1$ of exact order m (as $x^m - 1$ is separable mod p and $\Phi_m \mid x^m - 1$). So m divides $|\mathbb{F}_{p^{f'}}^\times| = p^{f'} - 1$, giving $p^{f'} \equiv 1 \pmod{m}$ and hence $f \mid f'$. Conversely the Frobenius $\alpha \mapsto \alpha^p$ generates $\text{Gal}(\mathbb{F}_{p^{f'}}/\mathbb{F}_p)$, and it acts on ζ_m as the element $p \in (\mathbb{Z}/m\mathbb{Z})^\times$, which has order f ; so $f' \mid f$. Hence $f = f'$. $\square$

Example 35.6.19 (The Prime Case)

Let p be an odd prime and $K = \mathbb{Q}(\zeta_p)$, so $[K : \mathbb{Q}] = p - 1$. Put $\lambda = 1 - \zeta_p$. Since

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = \prod_{k=1}^{p-1} (x - \zeta_p^k),$$

setting $x = 1$ gives

$$p = \Phi_p(1) = \prod_{k=1}^{p-1} (1 - \zeta_p^k) = N_{K/\mathbb{Q}}(\lambda).$$

Moreover $(1 - \zeta_p^k)/(1 - \zeta_p) = 1 + \zeta_p + \cdots + \zeta_p^{k-1} \in \mathbb{Z}[\zeta_p]$ is a unit for $1 \leq k \leq p - 1$ (its inverse is $(1 - \zeta_p)/(1 - \zeta_p^k)$, of the same shape after writing $\zeta_p = (\zeta_p^k)^{k'}$ with $kk' \equiv 1$). Hence all the $1 - \zeta_p^k$ are associates and

$$(p) = (\lambda)^{p-1}.$$

So p is totally ramified in $\mathbb{Q}(\zeta_p)$, with $e = p - 1$, $f = g = 1$, and $N((\lambda)) = p$, so (λ) is prime.

One deduces $\mathcal{O}_K = \mathbb{Z}[\zeta_p]$ as follows. From $\text{disc}(\Phi_p) = (-1)^{(p-1)/2} p^{p-2}$, the index $m = [\mathcal{O}_K : \mathbb{Z}[\zeta_p]]$ satisfies $m^2 \mid p^{p-2}$, so m is a power of p ; and one checks using the totally ramified prime λ that $p \nmid m$. Hence $m = 1$ and $d_K = (-1)^{(p-1)/2} p^{p-2}$.

Example 35.6.20

Take $m = 8$, so $\Phi_8(x) = x^4 + 1$ and $K = \mathbb{Q}(\zeta_8)$ has degree 4 with $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z})^2$. Then 2 is totally ramified. For odd p , the order of $p \bmod 8$ is 1 if $p \equiv 1 \pmod{8}$ and 2 otherwise. So $p \equiv 1 \pmod{8}$ splits completely into four primes, and every other odd p factors into two primes of residue degree 2. For instance $x^4 + 1$ factors into four linear factors mod 17, and into two irreducible quadratics mod 3.

Example 35.6.21

Take $m = 7$, so $K = \mathbb{Q}(\zeta_7)$ has degree 6 and $\text{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times$ is cyclic of order 6, generated by the class of 3.

- $p = 7$ is totally ramified: $(7) = (1 - \zeta_7)^6$, by Example 35.6.19.
- $p = 2$: the powers of 2 mod 7 are 2, 4, 1, so $f = 3$ and $(2) = \mathfrak{p}_1 \mathfrak{p}_2$ with $N(\mathfrak{p}_i) = 8$. Correspondingly Φ_7 factors mod 2 into two irreducible cubics, namely $(x^3 + x + 1)(x^3 + x^2 + 1)$.
- $p = 3$: the class of 3 generates the whole group, so $f = 6$ and 3 is inert; equivalently Φ_7 is irreducible mod 3.
- $p = 29 \equiv 1 \pmod{7}$: here $f = 1$, so 29 splits completely into six primes of norm 29.

In each case $\sum e_i f_i = 6$, as it must.

Remark (Cyclotomic Fields and Reciprocity). Since $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ is cyclic of order $p - 1$, it has a unique subgroup of index 2, so $\mathbb{Q}(\zeta_p)$ contains a unique quadratic subfield. Comparing discriminants (only p ramifies, and the discriminant of $\mathbb{Q}(\sqrt{d})$ must therefore be $\pm p$) shows that this subfield is

$$\mathbb{Q}(\sqrt{p^*}), \quad p^* = (-1)^{(p-1)/2} p.$$

Now Proposition 35.6.18(iv) says that the splitting of q in $\mathbb{Q}(\zeta_p)$ depends only on $q \bmod p$; hence so does its splitting in the subfield $\mathbb{Q}(\sqrt{p^*})$, which by Proposition 35.6.11 is governed by $\left(\frac{p^*}{q}\right)$. So $\left(\frac{p^*}{q}\right)$ depends only on $q \bmod p$ – and unwinding this is precisely the law of quadratic reciprocity. This is the first hint of *class field theory*: abelian extensions of $\mathbb{Q}$ are exactly the subfields of cyclotomic fields, and splitting in them is governed by congruences.

§35.7 The Geometry of Numbers

We now come to the geometric heart of the course. The idea is due to Minkowski, and it is disarmingly simple: realise $\mathcal{O}_K$ as a lattice in a Euclidean space, and use the fact

that a large enough convex body must contain a lattice point. From this one deduces that every ideal class contains an ideal of small norm, and hence that the class group is finite.

§35.7.1 Lattices

Definition 35.7.1 (Discrete)

A subset $X \subseteq \mathbb{R}^n$ is **discrete** if $X \cap C$ is finite for every compact $C \subseteq \mathbb{R}^n$; equivalently, if every $x \in X$ has $\varepsilon > 0$ with $B_\varepsilon(x) \cap X = \{x\}$.

Proposition 35.7.2

Let $\Lambda \subseteq \mathbb{R}^n$ be a subgroup of $(\mathbb{R}^n, +)$. Then Λ is discrete if and only if

$$\Lambda = \mathbb{Z}x_1 \oplus \cdots \oplus \mathbb{Z}x_m$$

for some $x_1, \dots, x_m \in \mathbb{R}^n$ linearly independent over $\mathbb{R}$.

Proof. ($\Leftarrow$) Extend $x_1, \dots, x_m$ to a basis of $\mathbb{R}^n$ and let $g \in \mathrm{GL}_n(\mathbb{R})$ take x_i to the standard basis vector e_i . Discreteness is preserved by g , and $\bigoplus_{i \leq m} \mathbb{Z}e_i$ is visibly discrete.

($\Rightarrow$) Choose $y_1, \dots, y_m \in \Lambda$ that are $\mathbb{R}$ -linearly independent with m maximal, so $m \leq n$, and let $V = \mathrm{span}_{\mathbb{R}}(y_1, \dots, y_m)$; by maximality $\Lambda \subseteq V$. Let

$$X = \{\sum_i \lambda_i y_i : \lambda_i \in [0, 1]\},$$

a compact set, so $X \cap \Lambda$ is finite. Put $\Lambda_0 = \bigoplus_i \mathbb{Z}y_i \subseteq \Lambda$. Any $\lambda \in \Lambda$ can be written as $\lambda = \lambda_0 + \lambda_1$ with $\lambda_0 \in \Lambda_0$ (the integer parts of the coordinates) and $\lambda_1 \in X \cap \Lambda$ (the fractional parts). Hence $[\Lambda : \Lambda_0] \leq |X \cap \Lambda| < \infty$.

Let $D = [\Lambda : \Lambda_0]$. By Lagrange, $D\Lambda \subseteq \Lambda_0$, so

$$\Lambda_0 \subseteq \Lambda \subseteq \frac{1}{D}\Lambda_0 \cong \mathbb{Z}^m.$$

So Λ is a subgroup of a free abelian group of rank m , hence free of rank at most m ; and it contains Λ_0 of rank m , so has rank exactly m . A $\mathbb{Z}$ -basis $x_1, \dots, x_m$ of Λ then spans V over $\mathbb{R}$, so is $\mathbb{R}$ -linearly independent. $\square$

Definition 35.7.3 (Lattice, Covolume)

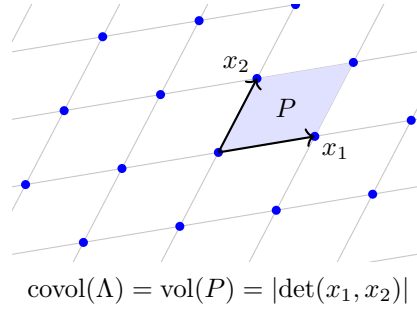
A **lattice** in $\mathbb{R}^n$ is a discrete subgroup $\Lambda \subseteq \mathbb{R}^n$ of rank n , i.e. one with $\Lambda = \bigoplus_{i=1}^n \mathbb{Z}x_i$ for an $\mathbb{R}$ -basis $x_1, \dots, x_n$ of $\mathbb{R}^n$. The **fundamental parallelotope** is

$$P = \{\sum_i \lambda_i x_i : \lambda_i \in [0, 1)\},$$

and the **covolume** of Λ is $\mathrm{covol}(\Lambda) = \mathrm{vol}(P) = |\det A|$, where A is the matrix whose rows are the x_i .

If $x'_1, \dots, x'_n$ is another $\mathbb{Z}$ -basis then the change of basis matrix lies in $\mathrm{GL}_n(\mathbb{Z})$ and has determinant ± 1 , so the covolume is well defined. The point of P is that it is a

fundamental domain: the translates $\gamma + P$ for $\gamma \in \Lambda$ tile $\mathbb{R}^n$ exactly.



Example 35.7.4

$\mathbb{Z}^n \subseteq \mathbb{R}^n$ is a lattice of covolume 1. On the other hand $\mathbb{Z}\sqrt{2} + \mathbb{Z}\sqrt{3} \subseteq \mathbb{R}$ is a subgroup that is *not* discrete – it is dense, by an argument with continued fractions – and correspondingly it is not free of rank 1.

§35.7.2 Minkowski's Theorem

Theorem 35.7.5 (Minkowski)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $S \subseteq \mathbb{R}^n$ measurable.

- (i) If $\text{vol}(S) > \text{covol}(\Lambda)$ then there exist $x \neq y$ in S with $x - y \in \Lambda$.
- (ii) Suppose in addition that S is convex and symmetric about 0 (that is, $s \in S \Rightarrow -s \in S$). If either
 - (a) $\text{vol}(S) > 2^n \text{covol}(\Lambda)$, or
 - (b) $\text{vol}(S) \geq 2^n \text{covol}(\Lambda)$ and S is closed and bounded,
 then S contains a nonzero point of Λ .

Proof. (i) Let P be a fundamental parallelotope. Since the translates $\gamma + P$ ($\gamma \in \Lambda$) partition $\mathbb{R}^n$,

$$\text{vol}(S) = \sum_{\gamma \in \Lambda} \text{vol}(S \cap (P + \gamma)) = \sum_{\gamma \in \Lambda} \text{vol}((S - \gamma) \cap P),$$

by translation invariance of volume. If the sets $(S - \gamma) \cap P$ were pairwise disjoint, they would all sit inside P and we would get $\text{vol}(P) \geq \text{vol}(S)$, contrary to hypothesis. So there are $\gamma \neq \mu$ in Λ and a point in $(S - \gamma) \cap (S - \mu)$, i.e. there are $x, y \in S$ with $x - \gamma = y - \mu$. Then $x - y = \gamma - \mu \in \Lambda \setminus \{0\}$, so in particular $x \neq y$.

(ii)(a) Apply (i) to $S' = \frac{1}{2}S$, which has $\text{vol}(S') = 2^{-n} \text{vol}(S) > \text{covol}(\Lambda)$. So there are $y \neq z \in S'$ with $y - z \in \Lambda \setminus \{0\}$. But $2y, 2z \in S$, so $-2z \in S$ by symmetry, and

$$y - z = \frac{1}{2}(2y) + \frac{1}{2}(-2z) \in S$$

by convexity. So $y - z$ is a nonzero lattice point in S .

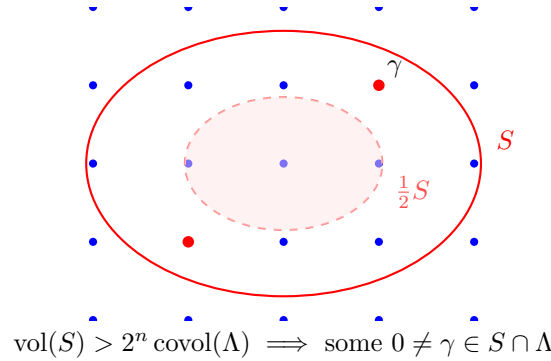
(ii)(b) For each integer $m \geq 1$ apply (a) to $S_m = (1 + \frac{1}{m})S$, which is convex, symmetric and of volume $(1 + \frac{1}{m})^n \text{vol}(S) > 2^n \text{covol}(\Lambda)$. This gives $\gamma_m \in (S_m \cap \Lambda) \setminus$

$\{0\}$. Now $S_m \subseteq S_1 = 2S$ for all m (by convexity and $0 \in S$), and $S_1 \cap \Lambda$ is finite since S_1 is compact and Λ discrete. So some value γ occurs infinitely often among the γ_m , and then

$$\gamma \in \bigcap_{m \geq 1} S_m = \overline{S} = S$$

as S is closed. Hence $\gamma \in (S \cap \Lambda) \setminus \{0\}$. $\square$

Here is part (ii)(a) in a picture. The shaded region is $S' = \frac{1}{2}S$; if its area exceeds the covolume then two of its points differ by a lattice vector, and doubling turns that into a lattice point inside S itself.



Remark. The bounds are sharp. Take $\Lambda = \mathbb{Z}^n$, so $\text{covol}(\Lambda) = 1$, and $S = (-1, 1)^n$, which is open, convex, symmetric, of volume exactly 2^n , and contains no nonzero lattice point. Closing it up to $[-1, 1]^n$ makes case (b) apply, and indeed $\pm e_i \in S \cap \Lambda$.

§35.7.3 The Minkowski Embedding

We now put K into a Euclidean space. Let K have degree n and signature (r, s) , with embeddings $\sigma_1, \dots, \sigma_r : K \rightarrow \mathbb{R}$ and $\sigma_{r+1}, \dots, \sigma_{r+s} : K \rightarrow \mathbb{C}$ (one from each conjugate pair). Define

$$\sigma : K \longrightarrow \mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^{r+2s} = \mathbb{R}^n,$$

$$\sigma(x) = (\sigma_1(x), \dots, \sigma_r(x), \text{Re } \sigma_{r+1}(x), \text{Im } \sigma_{r+1}(x), \dots, \text{Re } \sigma_{r+s}(x), \text{Im } \sigma_{r+s}(x)).$$

This is an injective $\mathbb{Q}$ -linear map, called the **Minkowski embedding**. Its whole purpose is the following.

Lemma 35.7.6

Let $\mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal. Then $\sigma(\mathfrak{a})$ is a lattice in $\mathbb{R}^n$, with

$$\text{covol}(\sigma(\mathfrak{a})) = 2^{-s} |d_K|^{1/2} N(\mathfrak{a}).$$

In particular $\text{covol}(\sigma(\mathcal{O}_K)) = 2^{-s} |d_K|^{1/2}$.

Proof. Let $\gamma_1, \dots, \gamma_n$ be a $\mathbb{Z}$ -basis of $\mathfrak{a}$. Then $\sigma(\mathfrak{a}) = \bigoplus_i \mathbb{Z} \sigma(\gamma_i)$, so it suffices to compute $|\det(\sigma(\gamma_1), \dots, \sigma(\gamma_n))|$ and check it is nonzero.

Consider first the matrix $S = (\sigma_i(\gamma_j))$ where now $\sigma_1, \dots, \sigma_n$ runs over *all* n embed-

dings. By Lemma 35.5.9,

$$(\det S)^2 = \Delta(\gamma_1, \dots, \gamma_n) = N(\mathfrak{a})^2 d_K, \quad \text{so } |\det S| = N(\mathfrak{a}) |d_K|^{1/2} \neq 0.$$

The matrix of σ differs from S by replacing each conjugate pair of rows $(z, \bar{z})$ by $(\operatorname{Re} z, \operatorname{Im} z)$, and

$$\begin{pmatrix} \operatorname{Re} z \\ \operatorname{Im} z \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} z \\ \bar{z} \end{pmatrix},$$

a matrix of determinant $\frac{1}{2} \cdot \frac{1}{2}(i + i) = \frac{i}{2}$, of absolute value $\frac{1}{2}$. Doing this for each of the s pairs multiplies the absolute determinant by 2^{-s} , giving

$$\operatorname{covol}(\sigma(\mathfrak{a})) = 2^{-s} N(\mathfrak{a}) |d_K|^{1/2} \neq 0,$$

which in particular shows $\sigma(\gamma_1), \dots, \sigma(\gamma_n)$ are $\mathbb{R}$ -independent, so $\sigma(\mathfrak{a})$ is a lattice. $\square$

Example 35.7.7

For K imaginary quadratic, $(r, s) = (0, 1)$ and σ is just the inclusion $K \subseteq \mathbb{C} = \mathbb{R}^2$ (after choosing a square root of d), which is what we drew for $\mathbb{Z}[i]$ and $\mathbb{Z}[\omega]$ in Section 1. The lemma says $\operatorname{covol}(\mathcal{O}_K) = \frac{1}{2} |d_K|^{1/2}$: for $\mathbb{Z}[i]$, $d_K = -4$ gives covolume 1, correct for the unit square lattice; for $\mathbb{Z}[\omega]$, $d_K = -3$ gives $\frac{\sqrt{3}}{2}$, correct for the triangular lattice with unit side.

For $K = \mathbb{Q}(\sqrt{d})$ real quadratic, $(r, s) = (2, 0)$ and $\sigma(x) = (x, \bar{x})$ using the two real embeddings; here $\operatorname{covol}(\mathcal{O}_K) = |d_K|^{1/2}$. For instance $\sigma(\mathbb{Z}[\sqrt{2}])$ is the lattice spanned by $(1, 1)$ and $(\sqrt{2}, -\sqrt{2})$, of covolume $2\sqrt{2} = \sqrt{8} = |d_K|^{1/2}$.

The reason this embedding is the right one is that the norm becomes a product of coordinates:

$$N_{K/\mathbb{Q}}(x) = \prod_{i=1}^r \sigma_i(x) \prod_{j=1}^s |\sigma_{r+j}(x)|^2,$$

so bounding $|N(x)|$ is bounding a product of coordinates – exactly the sort of thing the AM–GM inequality handles.

§35.7.4 The Minkowski Bound

Proposition 35.7.8 (Minkowski Bound)

Let K be a number field of degree n and signature (r, s) , and let $\mathfrak{a} \subseteq \mathcal{O}_K$ be a nonzero ideal. Then there is $\alpha \in \mathfrak{a}$, $\alpha \neq 0$, with

$$|N_{K/\mathbb{Q}}(\alpha)| \leq C_K N(\mathfrak{a}), \quad \text{where } C_K = \left(\frac{4}{\pi}\right)^s \frac{n!}{n^n} |d_K|^{1/2}$$

is the **Minkowski constant** of K .

Proof. For $t > 0$ let

$$B(t) = \left\{ (y_1, \dots, y_r, z_1, \dots, z_s) \in \mathbb{R}^r \times \mathbb{C}^s : \sum_{i=1}^r |y_i| + 2 \sum_{j=1}^s |z_j| \leq t \right\}.$$

This set is closed, bounded, convex and symmetric about 0, and (see below) has volume

$$\text{vol}(B(t)) = 2^r \left(\frac{\pi}{2}\right)^s \frac{t^n}{n!}.$$

Choose t so that $\text{vol}(B(t)) = 2^n \text{covol}(\sigma(\mathfrak{a})) = 2^n 2^{-s} |d_K|^{1/2} N(\mathfrak{a})$; solving,

$$t^n = \left(\frac{4}{\pi}\right)^s n! |d_K|^{1/2} N(\mathfrak{a}),$$

using $n - r - s = s$. By Minkowski's theorem 35.7.5(ii)(b) there is a nonzero $\alpha \in \mathfrak{a}$ with $\sigma(\alpha) \in B(t)$.

Write $\sigma(\alpha) = (y_1, \dots, y_r, z_1, \dots, z_s)$. Then $|N(\alpha)| = \prod_i |y_i| \prod_j |z_j|^2$, a product of n nonnegative reals whose sum is at most t . By AM–GM,

$$|N(\alpha)|^{1/n} \leq \frac{1}{n} \left(\sum_i |y_i| + 2 \sum_j |z_j| \right) \leq \frac{t}{n},$$

so $|N(\alpha)| \leq t^n / n^n = C_K N(\mathfrak{a})$. □

It remains to justify the volume formula, which we do by induction.

Lemma 35.7.9

Write $B_{r,s}(t)$ for the set above, sitting in $\mathbb{R}^r \times \mathbb{C}^s \cong \mathbb{R}^n$ with $n = r + 2s$. Then

$$\text{vol}(B_{r,s}(t)) = 2^r \left(\frac{\pi}{2}\right)^s \frac{t^n}{n!}.$$

Proof. We induct on $r + s$. If $r + s = 1$ there are two cases. For $(r, s) = (1, 0)$, $B_{1,0}(t) = [-t, t]$ has length $2t$, matching $2^1 t^1 / 1!$. For $(r, s) = (0, 1)$, $B_{0,1}(t) = \{z \in \mathbb{C} : 2|z| \leq t\}$ is a disc of radius $t/2$ and area $\pi t^2 / 4$, matching $(\pi/2) t^2 / 2!$.

For the inductive step in the real direction, slice along the new real coordinate y : the slice at height y is $B_{r,s}(t - |y|)$, so

$$\begin{aligned} \text{vol}(B_{r+1,s}(t)) &= \int_{-t}^t \text{vol}(B_{r,s}(t - |y|)) \, dy = 2 \int_0^t 2^r \left(\frac{\pi}{2}\right)^s \frac{(t - y)^n}{n!} \, dy \\ &= 2^{r+1} \left(\frac{\pi}{2}\right)^s \frac{t^{n+1}}{(n+1)!}, \end{aligned}$$

which is the formula in dimension $n + 1$.

In the complex direction, slice along the new complex coordinate z and write $z = \frac{1}{2} \rho e^{i\theta}$, so that the constraint reads $\rho \leq t$ and the slice is $B_{r,s}(t - \rho)$. The area

element is $|z| \, d|z| \, d\theta = \frac{1}{4} \rho \, d\rho \, d\theta$, so

$$\text{vol}(B_{r,s+1}(t)) = \int_0^{2\pi} \int_0^t \frac{\rho}{4} \, \text{vol}(B_{r,s}(t-\rho)) \, d\rho \, d\theta = \frac{2\pi}{4} \cdot 2^r \left(\frac{\pi}{2}\right)^s \int_0^t \frac{\rho(t-\rho)^n}{n!} \, d\rho.$$

Now $\int_0^t \rho(t-\rho)^n \, d\rho = t^{n+2}/((n+1)(n+2))$ by the substitution $\rho \mapsto t-\rho$ and the beta integral, so the right-hand side is $2^r(\pi/2)^{s+1}t^{n+2}/(n+2)!$, as required. $\square$

§35.7.5 Finiteness of the Class Group

Theorem 35.7.10 (Finiteness of the Class Group)

Let K be a number field. Then every ideal class of K contains an integral ideal $\mathfrak{b}$ with $N(\mathfrak{b}) \leq C_K$. Consequently Cl_K is finite, and it is generated by the classes of the prime ideals $\mathfrak{p}$ with $N(\mathfrak{p}) \leq C_K$.

Proof. Let $c \in \text{Cl}_K$ and pick an integral ideal $\mathfrak{a}$ with $[\mathfrak{a}] = c^{-1}$ (possible: any fractional ideal in c^{-1} can be scaled into $\mathcal{O}_K$). By Proposition 35.7.8 there is $0 \neq \alpha \in \mathfrak{a}$ with $|N(\alpha)| \leq C_K N(\mathfrak{a})$. Since $\alpha \in \mathfrak{a}$ we have $(\alpha) \subseteq \mathfrak{a}$, so $(\alpha) = \mathfrak{a}\mathfrak{b}$ for some integral ideal $\mathfrak{b}$ (Corollary 35.4.18). Taking norms and using Lemma 35.5.8,

$$N(\mathfrak{a}) N(\mathfrak{b}) = N((\alpha)) = |N(\alpha)| \leq C_K N(\mathfrak{a}),$$

so $N(\mathfrak{b}) \leq C_K$. And $[\mathfrak{b}] = [\mathfrak{a}]^{-1} = c$ since $\mathfrak{a}\mathfrak{b}$ is principal.

For finiteness: by Corollary 35.5.10(i) there are only finitely many ideals of norm at most C_K , so there are only finitely many classes. For the generation statement, if $N(\mathfrak{b}) \leq C_K$ and $\mathfrak{b} = \prod \mathfrak{p}_i^{a_i}$, then each $N(\mathfrak{p}_i) \leq N(\mathfrak{b}) \leq C_K$ by multiplicativity, so $[\mathfrak{b}]$ is a product of classes of small-norm primes. $\square$

This is a genuinely remarkable theorem, and it gives a completely effective algorithm: to compute Cl_K , list the primes $p \leq C_K$, factor each (p) using Dedekind's criterion, and then find the relations among the resulting classes. We will do this in Section 35.8.

Example 35.7.11

For K imaginary quadratic, $n = 2$, $s = 1$ and $C_K = \frac{2}{\pi}|d_K|^{1/2}$. For K real quadratic, $s = 0$ and $C_K = \frac{1}{2}|d_K|^{1/2}$. For $K = \mathbb{Q}(\sqrt{-7})$, $d_K = -7$ and $C_K = 2\sqrt{7}/\pi \approx 1.68 < 2$, so there are no primes of norm $\leq C_K$ at all and Cl_K is trivial: $\mathbb{Z}[\frac{1}{2}(1 + \sqrt{-7})]$ is a PID. The same works for $d = -1, -2, -3$.

The Minkowski bound also gives a lower bound for the discriminant, since applying it to the trivial class produces an ideal of norm at least 1.

Theorem 35.7.12 (Hermite–Minkowski)

Let K be a number field of degree $n \geq 2$. Then

$$|d_K| \geq \frac{\pi}{3} \left(\frac{3\pi}{4}\right)^{n-1} > 1.$$

In particular $|d_K| > 1$, so by Theorem 35.6.16 at least one prime ramifies in K .

Proof. Applying Theorem 35.7.10 to the trivial class gives an ideal $\mathfrak{b}$ with $1 \leq N(\mathfrak{b}) \leq C_K$, so $C_K \geq 1$, i.e.

$$|d_K|^{1/2} \geq \left(\frac{\pi}{4}\right)^s \frac{n^n}{n!} \geq \left(\frac{\pi}{4}\right)^{n/2} \frac{n^n}{n!} =: a_n^{1/2},$$

using $\pi/4 < 1$ and $s \leq n/2$. Now $a_n = \left(\frac{\pi}{4}\right)^n \left(\frac{n^n}{n!}\right)^2$ satisfies $a_2 = \frac{\pi^2}{4}$ and

$$\frac{a_{n+1}}{a_n} = \frac{\pi}{4} \left(1 + \frac{1}{n}\right)^{2n} \geq \frac{\pi}{4} \cdot 4 = \pi > \frac{3\pi}{4},$$

since $(1 + 1/n)^n \geq 2$ for $n \geq 1$. Hence

$$a_n \geq \frac{\pi^2}{4} \left(\frac{3\pi}{4}\right)^{n-2} = \frac{\pi}{3} \left(\frac{3\pi}{4}\right)^{n-1},$$

and $\frac{\pi}{3} \cdot \frac{3\pi}{4} = \frac{\pi^2}{4} > 2$, so the bound exceeds 1 for all $n \geq 2$. $\square$

Remark. Minkowski's theorem also implies (with more work) **Hermite's theorem**: for each D there are only finitely many number fields with $|d_K| \leq D$. So the discriminant is a genuinely rigid invariant. The smallest discriminants in absolute value in low degree are -3 (for $\mathbb{Q}(\sqrt{-3})$), -23 (for the cubic field $\mathbb{Q}(\alpha)$ with $\alpha^3 = \alpha + 1$, whose discriminant we computed in Section 2) and 117 in degree 4; compare these with the bound above, which gives $|d_K| \geq 2.4, 5.8, 13.7$ for $n = 2, 3, 4$.

§35.8 Computing Class Groups

Theorem 35.7.10 turns the computation of Cl_K into a finite, mechanical procedure. Let us set out the recipe and then run it several times.

- (1) Compute d_K and the Minkowski constant $C_K = \left(\frac{4}{\pi}\right)^s \frac{n!}{n^n} |d_K|^{1/2}$.
- (2) For each rational prime $p \leq C_K$, factor $p\mathcal{O}_K$ using Dedekind's criterion. The primes $\mathfrak{p}$ with $N(\mathfrak{p}) \leq C_K$ obtained this way generate Cl_K .
- (3) Decide which of these $\mathfrak{p}$ are principal, usually by asking whether there is $\alpha \in \mathcal{O}_K$ with $|N(\alpha)| = N(\mathfrak{p})$ – a norm-form equation.
- (4) Find relations. The most efficient source of relations is to factor (α) for elements α of small norm: each such factorisation is a relation in Cl_K .

Throughout, we use freely that in a quadratic field $[\bar{\mathfrak{a}}] = [\mathfrak{a}]^{-1}$ (Example 35.5.4), and that $N(a + b\sqrt{d}) = a^2 - db^2$.

Example 35.8.1 ($K = \mathbb{Q}(\sqrt{-5})$)

Here $d_K = -20$ and $C_K = \frac{2}{\pi} \sqrt{20} \approx 2.847$. So Cl_K is generated by primes of norm ≤ 2 , i.e. primes above 2. By Example 35.6.14, $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, 1 + \sqrt{-5})$ of norm 2.

Is $\mathfrak{p}$ principal? If $\mathfrak{p} = (\beta)$ then $N(\beta) = \pm 2$, i.e. $a^2 + 5b^2 = 2$, which is insoluble. So

$[\mathfrak{p}] \neq 1$, while $[\mathfrak{p}]^2 = [(2)] = 1$. Hence

$$\text{Cl}_{\mathbb{Q}(\sqrt{-5})} \cong \mathbb{Z}/2\mathbb{Z}, \quad h_K = 2.$$

In particular $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, which we knew, and the failure is as mild as it could possibly be.

Example 35.8.2 ($K = \mathbb{Q}(\sqrt{-23})$)

Here $-23 \equiv 1 \pmod{4}$, so $d_K = -23$, $\mathcal{O}_K = \mathbb{Z}[\alpha]$ with $\alpha = \frac{1}{2}(1 + \sqrt{-23})$ of minimal polynomial $x^2 - x + 6$, and $C_K = \frac{2}{\pi}\sqrt{23} \approx 3.05$. So we need the primes above 2 and 3.

Modulo 2: $x^2 - x + 6 \equiv x^2 + x = x(x+1)$, so $(2) = \mathfrak{p}\bar{\mathfrak{p}}$ with $\mathfrak{p} = (2, \alpha)$, $\bar{\mathfrak{p}} = (2, \alpha - 1)$, each of norm 2. Modulo 3: $x^2 - x + 6 \equiv x^2 - x = x(x-1)$, so $(3) = \mathfrak{q}\bar{\mathfrak{q}}$ with $\mathfrak{q} = (3, \alpha)$, $\bar{\mathfrak{q}} = (3, \alpha - 1)$, each of norm 3.

Norms of elements: $N(\frac{1}{2}(u + v\sqrt{-23})) = \frac{1}{4}(u^2 + 23v^2)$ with $u \equiv v \pmod{2}$. So $N(\gamma) = m$ requires $u^2 + 23v^2 = 4m$. For $m = 2$: $u^2 + 23v^2 = 8$ is insoluble, so $\mathfrak{p}$ is not principal. For $m = 3$: $u^2 + 23v^2 = 12$ is insoluble, so $\mathfrak{q}$ is not principal. For $m = 4$: $u^2 + 23v^2 = 16$ forces $v = 0, u = \pm 4$, giving $\gamma = \pm 2$ and $(\gamma) = (2) = \mathfrak{p}\bar{\mathfrak{p}}$; so $\mathfrak{p}^2$ is not principal either (if it were, $\mathfrak{p}^2 = (\gamma)$ with $N(\gamma) = 4$, forcing $\mathfrak{p}^2 = (2) = \mathfrak{p}\bar{\mathfrak{p}}$ and so $\mathfrak{p} = \bar{\mathfrak{p}}$, false).

Now for a relation. Take $m = 6$: $u^2 + 23v^2 = 24$ has the solution $u = v = 1$, giving

$$\gamma = \frac{1}{2}(1 + \sqrt{-23}) = \alpha, \quad N(\alpha) = 6.$$

Since $\alpha \in \mathfrak{p}$ and $\alpha \in \mathfrak{q}$, we get $(\alpha) = \mathfrak{p}\mathfrak{q}$. Hence $[\mathfrak{q}] = [\mathfrak{p}]^{-1}$, so $\text{Cl}_K = \langle [\mathfrak{p}] \rangle$ is cyclic.

Finally take $m = 8$: the equation $u^2 + 23v^2 = 32$ has the solution $u = 3, v = 1$, giving $\beta = \frac{1}{2}(3 + \sqrt{-23})$ with $N(\beta) = 8$. Now (β) has norm $8 = 2^3$, so it is a product of three primes above 2, that is $(\beta) = \mathfrak{p}^a \bar{\mathfrak{p}}^{3-a}$. If both $\mathfrak{p}$ and $\bar{\mathfrak{p}}$ occurred then $(2) = \mathfrak{p}\bar{\mathfrak{p}}$ would divide (β) , forcing $\beta/2 = \frac{1}{4}(3 + \sqrt{-23}) \in \mathcal{O}_K$; but that element has norm 2, which we have seen is impossible. So $(\beta) = \mathfrak{p}^3$ or $\bar{\mathfrak{p}}^3$, and either way $[\mathfrak{p}]^3 = 1$.

Since $[\mathfrak{p}] \neq 1$ and $[\mathfrak{p}]^2 \neq 1$, the class $[\mathfrak{p}]$ has order 3, and

$$\text{Cl}_{\mathbb{Q}(\sqrt{-23})} \cong \mathbb{Z}/3\mathbb{Z}, \quad h_K = 3.$$

Example 35.8.3 ($K = \mathbb{Q}(\sqrt{-17})$)

Here $-17 \equiv 3 \pmod{4}$, so $d_K = -17$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{-17}]$ and $C_K = \frac{2}{\pi}\sqrt{17} \approx 5.25$. We need the primes above 2, 3, 5.

- $p = 2$: $x^2 + 17 \equiv x^2 + 1 = (x+1)^2 \pmod{2}$, so $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, 1 + \sqrt{-17})$, $N(\mathfrak{p}) = 2$.
- $p = 3$: $x^2 + 17 \equiv x^2 - 1 = (x-1)(x+1) \pmod{3}$, so $(3) = \mathfrak{q}\bar{\mathfrak{q}}$ with $\mathfrak{q} = (3, 1 + \sqrt{-17})$, $N(\mathfrak{q}) = 3$.
- $p = 5$: $x^2 + 17 \equiv x^2 + 2 \pmod{5}$, and $-2 \equiv 3$ is not a square mod 5, so 5 is inert and contributes the trivial class.

Now $N(a + b\sqrt{-17}) = a^2 + 17b^2$, which never equals 2 or 3, so neither $\mathfrak{p}$ nor $\mathfrak{q}$ is principal. For a relation, note

$$N(1 + \sqrt{-17}) = 18 = 2 \cdot 3^2.$$

Since $1 + \sqrt{-17}$ lies in both $\mathfrak{p}$ and $\mathfrak{q}$, and the only primes of norm 2 and 3 dividing $(1 + \sqrt{-17})$ are $\mathfrak{p}$ and $\mathfrak{q}, \bar{\mathfrak{q}}$, we must have $(1 + \sqrt{-17}) = \mathfrak{p}\mathfrak{q}^2$ (it cannot involve both $\mathfrak{q}$ and $\bar{\mathfrak{q}}$, since $\mathfrak{q}\bar{\mathfrak{q}} = (3)$ and $3 \nmid 1 + \sqrt{-17}$). Hence in Cl_K ,

$$[\mathfrak{p}][\mathfrak{q}]^2 = 1, \quad [\mathfrak{p}]^2 = 1,$$

so $[\mathfrak{q}]^4 = [\mathfrak{p}]^{-2} = 1$ and $[\mathfrak{q}]^2 = [\mathfrak{p}]^{-1} = [\mathfrak{p}] \neq 1$. Therefore $[\mathfrak{q}]$ has order 4, and since $\text{Cl}_K = \langle [\mathfrak{p}], [\mathfrak{q}] \rangle = \langle [\mathfrak{q}] \rangle$,

$$\text{Cl}_{\mathbb{Q}(\sqrt{-17})} \cong \mathbb{Z}/4\mathbb{Z}, \quad h_K = 4.$$

Example 35.8.4 ($K = \mathbb{Q}(\sqrt{-14})$)

Here $d_K = -56$ and $C_K = \frac{2}{\pi}\sqrt{56} \approx 4.76$, so we need primes above 2 and 3. Modulo 2, $x^2 + 14 \equiv x^2$, so $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, \sqrt{-14})$. Modulo 3, $x^2 + 14 \equiv x^2 + 2 \equiv x^2 - 1$, so $(3) = \mathfrak{q}\bar{\mathfrak{q}}$ with $\mathfrak{q} = (3, 1 + \sqrt{-14})$.

Neither is principal, since $a^2 + 14b^2 \in \{2, 3\}$ is insoluble. But $N(2 + \sqrt{-14}) = 18 = 2 \cdot 3^2$ and $2 + \sqrt{-14} \in \mathfrak{p}$, so exactly as before $(2 + \sqrt{-14}) = \mathfrak{p}\mathfrak{q}^2$ (up to replacing $\mathfrak{q}$ by $\bar{\mathfrak{q}}$). So $[\mathfrak{q}]^4 = 1$ and $[\mathfrak{q}]^2 = [\mathfrak{p}] \neq 1$, giving

$$\text{Cl}_{\mathbb{Q}(\sqrt{-14})} \cong \mathbb{Z}/4\mathbb{Z}, \quad h_K = 4.$$

Real quadratic fields work the same way, but the Minkowski constant is smaller by a factor of $8/\pi$, which often makes life easier.

Example 35.8.5 ($K = \mathbb{Q}(\sqrt{10})$)

Here $10 \equiv 2 \pmod{4}$, so $d_K = 40$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{10}]$, and $C_K = \frac{1}{2}\sqrt{40} \approx 3.16$. We need the primes above 2 and 3.

- $p = 2$: $x^2 - 10 \equiv x^2 \pmod{2}$, so $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, \sqrt{10})$, $N(\mathfrak{p}) = 2$.
- $p = 3$: $x^2 - 10 \equiv x^2 - 1 = (x - 1)(x + 1) \pmod{3}$, so $(3) = \mathfrak{q}\bar{\mathfrak{q}}$ with $\mathfrak{q} = (3, \sqrt{10} - 1)$, $N(\mathfrak{q}) = 3$.

Now $N(a + b\sqrt{10}) = a^2 - 10b^2$. If this is ± 2 then $a^2 \equiv \pm 2 \pmod{5}$; but the squares mod 5 are 0, 1, 4, and $\pm 2 \equiv 2, 3$. So $\mathfrak{p}$ is not principal. Likewise $a^2 - 10b^2 = \pm 3$ forces $a^2 \equiv \pm 3 \equiv 3, 2 \pmod{5}$, impossible; so $\mathfrak{q}$ is not principal.

For a relation, $N(2 + \sqrt{10}) = 4 - 10 = -6$, so $(2 + \sqrt{10}) = \mathfrak{p}\mathfrak{q}$ or $\mathfrak{p}\bar{\mathfrak{q}}$. Either way $[\mathfrak{q}] = [\mathfrak{p}]^{-1} = [\mathfrak{p}]$ (as $[\mathfrak{p}]^2 = 1$), so $\text{Cl}_K = \langle [\mathfrak{p}] \rangle$ has order 2:

$$\text{Cl}_{\mathbb{Q}(\sqrt{10})} \cong \mathbb{Z}/2\mathbb{Z}, \quad h_K = 2.$$

Concretely, the failure of unique factorisation is visible in

$$6 = 2 \cdot 3 = (\sqrt{10} + 2)(\sqrt{10} - 2),$$

where all four factors are irreducible (their norms are 4, 9, 6, 6, and $a^2 - 10b^2 = \pm 2, \pm 3$ are insoluble) and no two are associates. The ideal-theoretic resolution is

$$(6) = \mathfrak{p}^2 \mathfrak{q} \bar{\mathfrak{q}}, \quad (2) = \mathfrak{p}^2, \quad (3) = \mathfrak{q} \bar{\mathfrak{q}}, \quad (\sqrt{10} + 2) = \mathfrak{p} \mathfrak{q}, \quad (\sqrt{10} - 2) = \mathfrak{p} \bar{\mathfrak{q}},$$

exactly as in the $\mathbb{Q}(\sqrt{-5})$ story.

Example 35.8.6 ($K = \mathbb{Q}(\sqrt{-6})$)

Here $d_K = -24$ and $C_K = \frac{2}{\pi} \sqrt{24} \approx 3.12$, so we need primes above 2 and 3. Both ramify: $x^2 + 6 \equiv x^2 \pmod{2}$ and $\pmod{3}$, so

$$(2) = \mathfrak{p}_2^2, \quad \mathfrak{p}_2 = (2, \sqrt{-6}); \quad (3) = \mathfrak{p}_3^2, \quad \mathfrak{p}_3 = (3, \sqrt{-6}).$$

Neither is principal, as $a^2 + 6b^2 \in \{2, 3\}$ is insoluble. But $N(\sqrt{-6}) = 6$, and $\sqrt{-6}$ lies in both, so $(\sqrt{-6}) = \mathfrak{p}_2 \mathfrak{p}_3$ and hence $[\mathfrak{p}_3] = [\mathfrak{p}_2]^{-1} = [\mathfrak{p}_2]$. So $\text{Cl}_K = \langle [\mathfrak{p}_2] \rangle$ has order 2 and $h_K = 2$.

Example 35.8.7 ($K = \mathbb{Q}(\sqrt{15})$)

Here $15 \equiv 3 \pmod{4}$, so $d_K = 60$, $\mathcal{O}_K = \mathbb{Z}[\sqrt{15}]$ and $C_K = \frac{1}{2} \sqrt{60} \approx 3.87$: again the primes above 2 and 3. Modulo 2, $x^2 - 15 \equiv x^2 + 1 = (x+1)^2$, so $(2) = \mathfrak{p}_2^2$ with $\mathfrak{p}_2 = (2, 1 + \sqrt{15})$; modulo 3, $x^2 - 15 \equiv x^2$, so $(3) = \mathfrak{p}_3^2$ with $\mathfrak{p}_3 = (3, \sqrt{15})$.

For principality we need $a^2 - 15b^2 = \pm 2$ or ± 3 . Reducing mod 5 gives $a^2 \equiv \pm 2$ or ± 3 , i.e. $a^2 \in \{2, 3\} \pmod{5}$; but the squares mod 5 are 0, 1, 4. So neither prime is principal. Finally $N(3 + \sqrt{15}) = 9 - 15 = -6$, and $3 + \sqrt{15}$ lies in both $\mathfrak{p}_2$ and $\mathfrak{p}_3$, so $(3 + \sqrt{15}) = \mathfrak{p}_2 \mathfrak{p}_3$ and $[\mathfrak{p}_3] = [\mathfrak{p}_2]$. Hence

$$\text{Cl}_{\mathbb{Q}(\sqrt{15})} \cong \mathbb{Z}/2\mathbb{Z}, \quad h_K = 2.$$

Example 35.8.8 (A Non-Cyclic Class Group: $K = \mathbb{Q}(\sqrt{-30})$)

All our examples so far have had cyclic class group, which is a little misleading. Here $d_K = -120$ and $C_K = \frac{2}{\pi} \sqrt{120} \approx 6.97$, so we need the primes above 2, 3, 5. All three divide d_K , so all three ramify:

$$(2) = \mathfrak{p}_2^2, \quad (3) = \mathfrak{p}_3^2, \quad (5) = \mathfrak{p}_5^2,$$

$$\mathfrak{p}_2 = (2, \sqrt{-30}), \quad \mathfrak{p}_3 = (3, \sqrt{-30}), \quad \mathfrak{p}_5 = (5, \sqrt{-30}).$$

None is principal, since $a^2 + 30b^2 \in \{2, 3, 5\}$ is insoluble; so each class $[\mathfrak{p}_i]$ has order exactly 2. Also $N(\sqrt{-30}) = 30$ and $\sqrt{-30}$ lies in all three primes, so

$$(\sqrt{-30}) = \mathfrak{p}_2 \mathfrak{p}_3 \mathfrak{p}_5, \quad \text{giving} \quad [\mathfrak{p}_2][\mathfrak{p}_3][\mathfrak{p}_5] = 1.$$

So $\text{Cl}_K = \langle [\mathfrak{p}_2], [\mathfrak{p}_3] \rangle$, a quotient of $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. Finally $[\mathfrak{p}_2][\mathfrak{p}_3] \neq 1$, since $\mathfrak{p}_2 \mathfrak{p}_3$ has norm 6 and $a^2 + 30b^2 = 6$ is insoluble. Hence

$$\text{Cl}_{\mathbb{Q}(\sqrt{-30})} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}, \quad h_K = 4.$$

Notice how the ramified primes did all the work: this is typical, and is the beginning of **genus theory**, which computes the 2-torsion of Cl_K for a quadratic field purely in terms of the number of prime divisors of d_K .

Example 35.8.9 (Fields of Class Number One)

If $C_K < 2$ there are no primes to consider at all and $h_K = 1$ automatically. For imaginary quadratic K this reads $\frac{2}{\pi}|d_K|^{1/2} < 2$, i.e. $|d_K| < \pi^2 \approx 9.87$, which happens for $d = -1, -2, -3, -7$. A little more work handles $d = -11, -19, -43, -67, -163$. In fact these nine are all:

Theorem 35.8.10 (Baker–Heegner–Stark)

$\mathcal{O}_{\mathbb{Q}(\sqrt{-d})}$ is a UFD (equivalently a PID) for $d > 0$ squarefree precisely when

$$d \in \{1, 2, 3, 7, 11, 19, 43, 67, 163\}.$$

This is a deep theorem, conjectured by Gauss and settled only in the 1960s. By contrast, it is conjectured – and completely open – that infinitely many *real* quadratic fields have class number one; numerically they seem to be rather common.

Example 35.8.11 (A Cubic Field)

Let $K = \mathbb{Q}(\sqrt[3]{2})$, with $n = 3$, $(r, s) = (1, 1)$ and $d_K = -108$. Then

$$C_K = \frac{4}{\pi} \cdot \frac{3!}{3^3} \cdot \sqrt{108} \approx 2.94,$$

so Cl_K is generated by primes of norm 2. From Example 35.6.8, the only one is $\mathfrak{p} = (2, \sqrt[3]{2}) = (\sqrt[3]{2})$, which is principal. Hence $h_K = 1$ and $\mathbb{Z}[\sqrt[3]{2}]$ is a PID.

Example 35.8.12 (A Cyclotomic Field)

Let $K = \mathbb{Q}(\zeta_5)$, with $n = 4$, $(r, s) = (0, 2)$ and $d_K = 125$. Then

$$C_K = \left(\frac{4}{\pi}\right)^2 \frac{4!}{4^4} \sqrt{125} \approx 1.70 < 2,$$

so there are no prime ideals of norm at most C_K , and $h_K = 1$: the ring $\mathbb{Z}[\zeta_5]$ is a PID.

For $\mathbb{Q}(\zeta_7)$ the bound is larger, $C_K = \left(\frac{4}{\pi}\right)^3 \frac{6!}{6^6} \sqrt{7^5} \approx 4.13$, so we must look at the primes of norm at most 4. But the primes above 2 have residue degree 3 and hence norm 8, the primes above 3 have residue degree 6 and hence norm 729, and the unique prime above 7 has norm 7; so again there is nothing to consider and $h_K = 1$.

This cannot go on forever – for $\mathbb{Q}(\zeta_{11})$ the bound is already about 59 – but in fact $h_{\mathbb{Q}(\zeta_p)} = 1$ for all $p \leq 19$. The first failure is $p = 23$, where $h = 3$; and the first **irregular** prime, meaning one with $p \mid h_{\mathbb{Q}(\zeta_p)}$, is $p = 37$. This is exactly where Kummer’s proof of Fermat’s Last Theorem breaks down (see Section 35.10).

Remark. There is a classical dictionary, going back to Gauss, between ideal classes in

$\mathbb{Q}(\sqrt{D})$ and binary quadratic forms. To a form $q(x, y) = ax^2 + bxy + cy^2$ with $a, b, c \in \mathbb{Z}$ and discriminant $D = b^2 - 4ac$, associate the ideal

$$\mathfrak{a}_q = \left(a, \frac{-b + \sqrt{D}}{2} \right) \trianglelefteq \mathcal{O}_{\mathbb{Q}(\sqrt{D})}.$$

For $D < 0$ and $a > 0$, one can show that $[\mathfrak{a}_q] = [\mathfrak{a}_{q'}]$ if and only if q and q' are equivalent under the action of $SL_2(\mathbb{Z})$ on forms. So h_K is the number of $SL_2(\mathbb{Z})$ -orbits of positive definite binary quadratic forms of discriminant D – Gauss’s *class number*, computed by his reduction theory long before ideals were invented. For example the two classes for $D = -20$ correspond to the reduced forms $x^2 + 5y^2$ and $2x^2 + 2xy + 3y^2$.

§35.9 Dirichlet’s Unit Theorem

We turn to the other end of the exact sequence

$$1 \longrightarrow \mathcal{O}_K^\times \longrightarrow K^\times \longrightarrow I_K \longrightarrow \text{Cl}_K \longrightarrow 1,$$

namely the unit group $\mathcal{O}_K^\times$. We saw in Example 35.4.2 that imaginary quadratic fields have finite unit group, whereas $\mathbb{Z}[\sqrt{2}]^\times$ contains the infinite cyclic group generated by $1 + \sqrt{2}$. Dirichlet’s theorem explains this completely: the answer depends only on the signature.

Definition 35.9.1 (Roots of Unity)

Let $\mu_K = \{\alpha \in K : \alpha^m = 1 \text{ for some } m \geq 1\}$ be the group of roots of unity in K .

Theorem 35.9.2 (Dirichlet’s Unit Theorem)

Let K be a number field of signature (r, s) . Then μ_K is finite and cyclic, and

$$\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r+s-1}.$$

So for K imaginary quadratic, $(r, s) = (0, 1)$ and the rank is 0: the unit group is finite, as we found. For K real quadratic, $(r, s) = (2, 0)$ and the rank is 1, so $\mathcal{O}_K^\times = \{\pm \varepsilon^n : n \in \mathbb{Z}\}$ for a single ε – this is the statement that Pell’s equation has infinitely many solutions, all generated by one. For $K = \mathbb{Q}(\sqrt[3]{2})$, $(r, s) = (1, 1)$ and the rank is again 1.

§35.9.1 The Logarithmic Embedding

The Minkowski embedding turns multiplication into a product of coordinates, and the norm condition $N(u) = \pm 1$ into a multiplicative constraint. Taking logarithms turns this into a linear constraint, which is what we can attack with lattices.

Definition 35.9.3 (Logarithmic Embedding)

Let K have signature (r, s) with embeddings as above. Define

$$\ell : K^\times \rightarrow \mathbb{R}^{r+s},$$

$$\ell(x) = (\log|\sigma_1(x)|, \dots, \log|\sigma_r(x)|, 2\log|\sigma_{r+1}(x)|, \dots, 2\log|\sigma_{r+s}(x)|).$$

The factors of 2 on the complex coordinates are there so that ℓ interacts correctly with the norm; and ℓ does not depend on which member of each conjugate pair we chose, since conjugates have the same absolute value. Since $\log|ab| = \log|a| + \log|b|$, the map ℓ is a group homomorphism from $(K^\times, \times)$ to $(\mathbb{R}^{r+s}, +)$.

Lemma 35.9.4

Let $L = \ell(\mathcal{O}_K^\times) \subseteq \mathbb{R}^{r+s}$. Then

- (i) L is a discrete subgroup of $\mathbb{R}^{r+s}$;
- (ii) $\ker(\ell|_{\mathcal{O}_K^\times}) = \mu_K$, which is a finite cyclic group;
- (iii) L is contained in the hyperplane $H = \{(y_1, \dots, y_{r+s}) : \sum_i y_i = 0\}$.

Proof. (i) and (ii). The map ℓ factors as

$$\mathcal{O}_K^\times \xrightarrow{\sigma} (\mathbb{R}^\times)^r \times (\mathbb{C}^\times)^s \xrightarrow{j} \mathbb{R}^{r+s},$$

where $j(y_1, \dots, y_r, z_1, \dots, z_s) = (\log|y_1|, \dots, 2\log|z_s|)$. For $A > 0$,

$$j^{-1}([-A, A]^{r+s}) = \{(y_i, z_j) : e^{-A} \leq |y_i| \leq e^A, e^{-A/2} \leq |z_j| \leq e^{A/2}\}$$

is a closed and bounded, hence compact, subset of $\mathbb{R}^n$. Since $\sigma(\mathcal{O}_K)$ is a lattice (Lemma 35.7.6), hence discrete, the intersection of this compact set with $\sigma(\mathcal{O}_K^\times) \subseteq \sigma(\mathcal{O}_K)$ is finite. So $L \cap [-A, A]^{r+s}$ is finite for every A , which gives discreteness, and taking $A \rightarrow 0$ shows the kernel of ℓ on $\mathcal{O}_K^\times$ is finite.

A finite subgroup of $K^\times$ consists of elements of finite order, so $\ker \ell \subseteq \mu_K$; conversely every root of unity has all conjugates of absolute value 1, so lies in $\ker \ell$. Hence $\ker \ell = \mu_K$ is finite. Finally, embedding $K \hookrightarrow \mathbb{C}$ realises μ_K as a finite subgroup of $\mathbb{C}^\times$, hence as a group of m th roots of unity for some m ; subgroups of cyclic groups are cyclic.

(iii) If $u \in \mathcal{O}_K^\times$ then $N(u) = \pm 1$ by Lemma 35.4.1, so

$$0 = \log|N(u)| = \sum_{i=1}^r \log|\sigma_i(u)| + \sum_{j=1}^s 2\log|\sigma_{r+j}(u)|,$$

which is exactly the statement that $\ell(u) \in H$. □

Since $H \cong \mathbb{R}^{r+s-1}$, part (i) and Proposition 35.7.2 give $L \cong \mathbb{Z}^a$ for some $a \leq r + s - 1$. The content of Dirichlet's theorem is that a is as large as possible.

Theorem 35.9.5 (Dirichlet, Lattice Form)

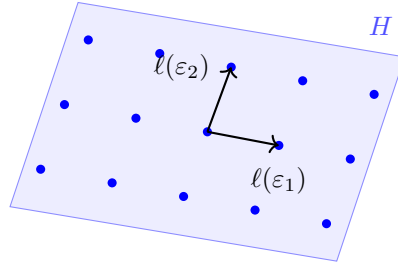
$L = \ell(\mathcal{O}_K^\times)$ is a full lattice in $H \cong \mathbb{R}^{r+s-1}$; that is, $L \cong \mathbb{Z}^{r+s-1}$.

Granting this, Theorem 35.9.2 follows at once: we have a short exact sequence

$$1 \rightarrow \mu_K \rightarrow \mathcal{O}_K^\times \xrightarrow{\ell} L \rightarrow 0$$

with $L \cong \mathbb{Z}^{r+s-1}$ free, so the sequence splits and $\mathcal{O}_K^\times \cong \mu_K \times \mathbb{Z}^{r+s-1}$.

Here is the picture for a field with $r+s=3$ (for instance a totally real cubic field, where $(r,s)=(3,0)$): the image L is a rank-2 lattice sitting inside the plane $y_1 + y_2 + y_3 = 0$.



$$L = \ell(\mathcal{O}_K^\times) \subseteq H = \{y_1 + y_2 + y_3 = 0\}$$

§35.9.2 Proof of Dirichlet's Theorem

The strategy is to manufacture, for each coordinate k , a unit whose logarithmic image is negative in every coordinate except the k th. A linear algebra lemma then shows that such vectors are as independent as they can be. The manufacturing is done by Minkowski's theorem.

Lemma 35.9.6

Fix $1 \leq k \leq r+s$ and let $0 \neq \alpha \in \mathcal{O}_K$ with $\ell(\alpha) = (a_1, \dots, a_{r+s})$. Then there is $0 \neq \beta \in \mathcal{O}_K$ with

$$|\mathbf{N}(\beta)| \leq \left(\frac{2}{\pi}\right)^s |d_K|^{1/2} \quad \text{and} \quad b_i < a_i \text{ for all } i \neq k,$$

where $\ell(\beta) = (b_1, \dots, b_{r+s})$.

Proof. For positive reals $c_1, \dots, c_{r+s}$ let

$$S = \{(y_1, \dots, y_r, z_1, \dots, z_s) \in \mathbb{R}^r \times \mathbb{C}^s : |y_i| \leq c_i \ (i \leq r), |z_j|^2 \leq c_{r+j} \ (j \leq s)\},$$

a closed, bounded, convex, symmetric set of volume $\text{vol}(S) = 2^r \pi^s c_1 \cdots c_{r+s}$. Choose c_i with $0 < c_i < e^{a_i}$ for $i \neq k$, and then define c_k so that

$$c_1 c_2 \cdots c_{r+s} = \left(\frac{2}{\pi}\right)^s |d_K|^{1/2},$$

which makes $\text{vol}(S) = 2^r \pi^s (2/\pi)^s |d_K|^{1/2} = 2^{r+s} |d_K|^{1/2} = 2^n \text{covol}(\sigma(\mathcal{O}_K))$, using $r+2s=n$ and Lemma 35.7.6. By Minkowski's theorem there is $0 \neq \beta \in \mathcal{O}_K$ with $\sigma(\beta) \in S$. Then

$$|\mathbf{N}(\beta)| = \prod_{i \leq r} |y_i| \prod_{j \leq s} |z_j|^2 \leq c_1 \cdots c_{r+s} = \left(\frac{2}{\pi}\right)^s |d_K|^{1/2},$$

and for $i \neq k$ the i th coordinate of $\ell(\beta)$ is at most $\log c_i < a_i$. □

Lemma 35.9.7

Let $A = (a_{ij}) \in \text{Mat}_m(\mathbb{R})$ satisfy $a_{ii} > 0$, $a_{ij} < 0$ for $i \neq j$, and $\sum_j a_{ij} \geq 0$ for all i . Then $\text{rank } A \geq m - 1$.

Proof. We show the first $m - 1$ columns $v_1, \dots, v_{m-1}$ are linearly independent. Suppose $\sum_{i=1}^{m-1} t_i v_i = 0$ with not all t_i zero. Choose $k \leq m - 1$ with $|t_k|$ maximal, and rescale so that $t_k = 1$ and $|t_i| \leq 1$ for all i . Reading off the k th coordinate,

$$0 = \sum_{i=1}^{m-1} t_i a_{ki} = a_{kk} + \sum_{\substack{i \neq k \\ i \leq m-1}} t_i a_{ki}.$$

Now $a_{ki} < 0$ for $i \neq k$ and $t_i \leq 1$, so $t_i a_{ki} \geq a_{ki}$. Hence

$$0 \geq \sum_{i=1}^{m-1} a_{ki} > \sum_{i=1}^m a_{ki} \geq 0,$$

where the strict inequality is because $a_{km} < 0$. This is a contradiction. $\square$

Proof of Theorem 35.9.5. Fix k with $1 \leq k \leq r + s$. Starting from any $\alpha_1 \in \mathcal{O}_K$ nonzero and applying Lemma 35.9.6 repeatedly, we obtain a sequence $\alpha_1, \alpha_2, \dots \in \mathcal{O}_K \setminus \{0\}$ such that

$$|N(\alpha_t)| \leq M := \left(\frac{2}{\pi}\right)^s |d_K|^{1/2} \quad \text{for } t \geq 2,$$

and for each $i \neq k$ the i th coordinate of $\ell(\alpha_t)$ is strictly decreasing in t .

There are only finitely many integers of absolute value at most M , so by pigeonhole some value m occurs as $N(\alpha_t)$ infinitely often. There are also only finitely many residues modulo $m\mathcal{O}_K$ (Lemma 35.4.8), so again by pigeonhole there are $t < t'$ with $N(\alpha_t) = N(\alpha_{t'}) = m$ and $\alpha_t \equiv \alpha_{t'} \pmod{m\mathcal{O}_K}$. Then $\alpha_{t'} = \alpha_t + m\gamma$ for some $\gamma \in \mathcal{O}_K$, and since $m/\alpha_t = \pm \prod_{i \geq 2} \sigma_i(\alpha_t)$ is an algebraic integer lying in K , we get $m/\alpha_t \in \mathcal{O}_K$ and hence

$$\frac{\alpha_{t'}}{\alpha_t} = 1 + \frac{m}{\alpha_t} \gamma \in \mathcal{O}_K.$$

Symmetrically $\alpha_t/\alpha_{t'} \in \mathcal{O}_K$, so $u_k := \alpha_{t'}/\alpha_t \in \mathcal{O}_K^\times$.

By construction $\ell(u_k) = \ell(\alpha_{t'}) - \ell(\alpha_t)$ has i th coordinate strictly negative for every $i \neq k$. Since $\ell(u_k) \in H$ its coordinates sum to zero, so its k th coordinate is strictly positive.

Doing this for each $k = 1, \dots, r + s$ produces units $u_1, \dots, u_{r+s}$, and the matrix A with k th row $\ell(u_k)$ satisfies the hypotheses of Lemma 35.9.7 with $m = r + s$ (the row sums are zero). So $\text{rank } A \geq r + s - 1$, and since the rows lie in the hyperplane H of dimension $r + s - 1$, we get $\text{rank } A = r + s - 1$ exactly. Hence $\ell(u_1), \dots, \ell(u_{r+s})$ span H , so L contains $r + s - 1$ linearly independent vectors and is a discrete subgroup of H : by Proposition 35.7.2, $L \cong \mathbb{Z}^{r+s-1}$ is a full lattice in H . $\square$

Definition 35.9.8 (Fundamental Units, Regulator)

A set of units $\varepsilon_1, \dots, \varepsilon_{r+s-1}$ whose images $\ell(\varepsilon_i)$ form a $\mathbb{Z}$ -basis of L is called a system of **fundamental units**: every unit is then uniquely $\zeta \varepsilon_1^{n_1} \cdots \varepsilon_{r+s-1}^{n_{r+s-1}}$ with $\zeta \in \mu_K$ and $n_i \in \mathbb{Z}$. The **regulator** of K is

$$R_K = \text{covol}(L \subseteq H),$$

computed as the absolute value of the determinant of any $(r+s-1) \times (r+s-1)$ minor of the matrix with rows $\ell(\varepsilon_1), \dots, \ell(\varepsilon_{r+s-1})$.

Example 35.9.9

If $r+s-1 = 0$ (that is, $K = \mathbb{Q}$ or K imaginary quadratic) then $R_K = 1$ by convention. If K is real quadratic with fundamental unit $\varepsilon > 1$, then $\ell(\varepsilon) = (\log \varepsilon, -\log \varepsilon)$ and $R_K = \log \varepsilon$.

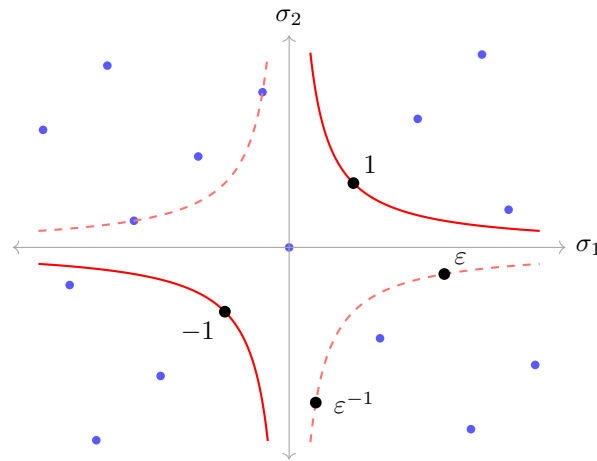
§35.9.3 Real Quadratic Fields and Pell's Equation

The rank-1 case deserves a closer look, since it is by far the most computable and has a beautiful classical shadow. Let $K = \mathbb{Q}(\sqrt{d})$ with $d > 0$ squarefree, so $(r, s) = (2, 0)$ and $\mu_K = \{\pm 1\}$ (as $K \subseteq \mathbb{R}$). Dirichlet's theorem gives

$$\mathcal{O}_K^\times = \{\pm \varepsilon^n : n \in \mathbb{Z}\}$$

for a **fundamental unit** ε , unique if we insist $\varepsilon > 1$ under the embedding $\sqrt{d} > 0$.

Since $N(x + y\sqrt{d}) = x^2 - dy^2$, this is the classical statement about **Pell's equation**: the integer solutions of $x^2 - dy^2 = \pm 1$ (or ± 4 , when $d \equiv 1 \pmod{4}$ and we allow half-integers) are all obtained from a single one by taking powers. Geometrically, $\sigma(\mathcal{O}_K)$ is a lattice in $\mathbb{R}^2$ and the units are precisely its points on the two hyperbolas $y_1 y_2 = \pm 1$.



$\sigma(\mathbb{Z}[\sqrt{2}])$ and the hyperbolas $y_1 y_2 = \pm 1$

Here $\varepsilon = 1 + \sqrt{2}$ has $\sigma(\varepsilon) = (1 + \sqrt{2}, 1 - \sqrt{2}) \approx (2.414, -0.414)$, lying on the dashed branch $y_1 y_2 = -1$ because $N(\varepsilon) = -1$; its inverse $\varepsilon^{-1} = \sqrt{2} - 1$ has $\sigma(\varepsilon^{-1}) \approx (0.414, -2.414)$. The square $\varepsilon^2 = 3 + 2\sqrt{2}$ lands at $(5.83, 0.17)$ on the solid hyperbola $y_1 y_2 = 1$, off the right-hand edge of the picture. As n increases, $\sigma(\varepsilon^n)$ marches off along the hyperbolas, hugging the axes ever more closely – and *every* lattice point on either hyperbola is one of these.

We can prove the rank-1 case of Dirichlet's theorem directly, and the proof is instructive.

Theorem 35.9.10 (Pell)

Let $K = \mathbb{Q}(\sqrt{d})$ with $d > 0$ squarefree. Then $\mathcal{O}_K^\times$ is infinite. Equivalently, $x^2 - dy^2 = 1$ has infinitely many integer solutions.

Proof. Recall $\sigma(x) = (x, \bar{x})$ and $\text{covol}(\sigma(\mathcal{O}_K)) = |d_K|^{1/2}$. For $t > 0$ let

$$S_t = \{(y_1, y_2) \in \mathbb{R}^2 : |y_1| \leq t, |y_2| \leq |d_K|^{1/2}/t\},$$

a closed convex symmetric rectangle whose area is $4|d_K|^{1/2}$, that is, $2^2 \text{covol}(\sigma(\mathcal{O}_K))$. By Minkowski's theorem there is a nonzero $\alpha_t \in \mathcal{O}_K$ with $\sigma(\alpha_t) \in S_t$, and then

$$1 \leq |N(\alpha_t)| = |y_1 y_2| \leq |d_K|^{1/2}.$$

We now produce infinitely many distinct such α . Having chosen $t_1 > t_2 > \dots > t_N > 0$, the set $S_{t_N} \cap \sigma(\mathcal{O}_K)$ is finite (compact meets discrete), and every element of it has nonzero first coordinate (as σ_1 is injective and $\alpha \neq 0$); so we may choose $t_{N+1} > 0$ smaller than all of these first coordinates in absolute value. Then $\alpha_{t_{N+1}}$ is different from all previous α_{t_i} .

So there are infinitely many $\alpha \in \mathcal{O}_K$ with $1 \leq |N(\alpha)| \leq |d_K|^{1/2}$. By pigeonhole some value m is attained infinitely often; and since there are only finitely many ideals of norm $|m|$ (Corollary 35.5.10), some ideal (β) arises as (α) for infinitely many α . For any two such, α/α' is a unit, and distinct α give distinct units. So $\mathcal{O}_K^\times$ is infinite.

Finally, $\mathcal{O}_K^\times$ infinite gives a unit $u \neq \pm 1$ with $N(u) = \pm 1$; then u^2 has norm 1, and writing $u^2 = x + y\sqrt{d}$ (or $\frac{1}{2}(x + y\sqrt{d})$, in which case take u^6) gives infinitely many solutions of $x^2 - dy^2 = 1$ from the powers of u^2 . $\square$

Corollary 35.9.11

$\mathcal{O}_K^\times = \{\pm \varepsilon^n : n \in \mathbb{Z}\}$ for some ε .

Proof. Identify K with $\sigma_1(K) \subseteq \mathbb{R}$. By the theorem there is a unit $u \neq \pm 1$, and $|u| \neq 1$ (since $|u| = 1$ with u real forces $u = \pm 1$); replacing u by u^{-1} if necessary, $E := |u| > 1$. The set

$$\{\alpha \in \mathcal{O}_K : N(\alpha) = \pm 1, 1 < |\alpha| \leq E\}$$

is finite, since $\mathcal{O}_K$ is discrete in $\mathbb{R}^2$ and this set has bounded coordinates ($|\alpha| \leq E$ and $|\bar{\alpha}| = 1/|\alpha| \leq 1$). It is nonempty (it contains u or $-u$), so choose ε in it with $|\varepsilon|$ minimal, and replace ε by $-\varepsilon$ if needed so that $\varepsilon > 1$.

Now let $v \in \mathcal{O}_K^\times$ with $v > 0$; we claim $v = \varepsilon^N$ for some $N \in \mathbb{Z}$. Write $\log v / \log \varepsilon = N + \gamma$ with $N \in \mathbb{Z}$ and $0 \leq \gamma < 1$. Then $w = v\varepsilon^{-N}$ is a unit with $1 \leq w = \varepsilon^\gamma < \varepsilon$, so by minimality $w = 1$, i.e. $\gamma = 0$. Since every unit is $\pm$ a positive one, we are done. $\square$

§35.9.4 Finding Fundamental Units

Locating ε in practice is a matter of searching, and the search can be made short by the following observation: if $\varepsilon = a + b\sqrt{d} > 1$ is a unit with $\bar{\varepsilon} = \pm\varepsilon^{-1}$, then $|\bar{\varepsilon}| < 1$, so

$$2a = \varepsilon + \bar{\varepsilon} > 0 \quad \text{and} \quad 2b\sqrt{d} = \varepsilon - \bar{\varepsilon} > 0,$$

so $a, b > 0$. Hence the fundamental unit is the one with smallest positive b , and we can simply try $b = 1, 2, 3, \dots$ and test whether $db^2 \pm 1$ (or $db^2 \pm 4$) is a square.

Example 35.9.12

Let $d = 2$. Guess $\varepsilon = 1 + \sqrt{2}$, which has $N(\varepsilon) = -1$, so it is a unit. It is fundamental: any smaller unit $u = a + b\sqrt{2} > 1$ would have $a, b \geq 1$ by the above, hence $u \geq 1 + \sqrt{2} = \varepsilon$. So $\mathcal{O}_{\mathbb{Q}(\sqrt{2})}^\times = \{\pm(1 + \sqrt{2})^n\}$ and $R_K = \log(1 + \sqrt{2}) \approx 0.881$.

Example 35.9.13

Let $d = 11$, so $\mathcal{O}_K = \mathbb{Z}[\sqrt{11}]$. Trying $b = 1, 2, 3$: $11 - 1 = 10$, $11 + 1 = 12$, 44 ± 1 , $99 + 1 = 100 = 10^2$. So $\varepsilon^{-1} = 10 - 3\sqrt{11}$ has norm $100 - 99 = 1$ and $\varepsilon = 10 + 3\sqrt{11} \approx 19.95$ is a unit.

Let us verify it is fundamental without appealing to the search. Suppose $u = a + b\sqrt{11}$ is a unit with $1 < u < \varepsilon$. If $N(u) = -1$ then $a^2 - 11b^2 = -1$, so $a^2 \equiv -1 \pmod{11}$; but -1 is not a square mod 11 (the squares are 1, 3, 4, 5, 9). So $N(u) = 1$ and $\bar{u} = u^{-1}$, giving $0 < \bar{u} < 1$. Adding the inequalities $1 < u < 10 + 3\sqrt{11}$ and $-1 < -\bar{u} < 0$:

$$0 < u - \bar{u} = 2b\sqrt{11} < 10 + 3\sqrt{11} < 7\sqrt{11},$$

so $b \in \{1, 2, 3\}$. But $11b^2 + 1 \in \{12, 45, 100\}$ and only 100 is a square, forcing $u = \varepsilon$. So ε is fundamental.

Example 35.9.14

Some fundamental units, to give a sense of how erratically they grow:

d	2	3	5	6	7	11
ε	$1 + \sqrt{2}$	$2 + \sqrt{3}$	$\frac{1}{2}(1 + \sqrt{5})$	$5 + 2\sqrt{6}$	$8 + 3\sqrt{7}$	$10 + 3\sqrt{11}$
d	13	19	31	46		
ε	$\frac{1}{2}(3 + \sqrt{13})$	$170 + 39\sqrt{19}$	$1520 + 273\sqrt{31}$	$24335 + 3588\sqrt{46}$		

The case $d = 5$ is the golden ratio φ , with $\varphi^2 = \varphi + 1$ and $N(\varphi) = -1$. The jump at $d = 19$ and again at $d = 46$ is typical: the size of ε is roughly e^{R_K} , and the regulator behaves very irregularly.

Example 35.9.15 (A Cubic Field)

Let $K = \mathbb{Q}(\sqrt[3]{2})$, with signature $(1, 1)$, so the unit rank is $r+s-1 = 1$ and $\mu_K = \{\pm 1\}$ (as $K \subseteq \mathbb{R}$). Consider

$$\varepsilon = \sqrt[3]{2} - 1.$$

Its minimal polynomial is $(x+1)^3 - 2 = x^3 + 3x^2 + 3x - 1$, whose constant term is

-1 , so $N(\varepsilon) = (-1)^3(-1) = 1$ and ε is a unit. In fact it is fundamental, so

$$\mathcal{O}_{\mathbb{Q}(\sqrt[3]{2})}^\times = \left\{ \pm \left(\sqrt[3]{2} - 1 \right)^n : n \in \mathbb{Z} \right\}, \quad R_K = \log \frac{1}{\sqrt[3]{2} - 1} \approx 1.347.$$

Note that $\varepsilon \approx 0.26 < 1$, so it is $\varepsilon^{-1} = 1 + \sqrt[3]{2} + \sqrt[3]{4} \approx 3.85$ that is the ‘large’ generator; and $\ell(\varepsilon) = (\log|\varepsilon|, -\log|\varepsilon|)$ since the two coordinates must sum to zero.

Remark (Continued Fractions). There is a systematic algorithm. Every real number has a continued fraction expansion

$$t = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}} = [a_0, a_1, a_2, \dots], \quad a_0 = \lfloor t \rfloor,$$

and a classical theorem of Lagrange says that t is a quadratic irrational if and only if this expansion is eventually periodic. For $\sqrt{d}$ the expansion is purely periodic after the first term, say $\sqrt{d} = [a_0, \overline{a_1, \dots, a_m}]$, and then (this is non-examinable, and should not be quoted in exams) if $p/q = [a_0, a_1, \dots, a_{m-1}]$ in lowest terms, the number $p + q\sqrt{d}$ is a unit, and is fundamental when $d \equiv 2, 3 \pmod{4}$.

For instance $\sqrt{7} = [2, \overline{1, 1, 1, 4}]$, and $[2, 1, 1, 1] = 2 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1}}}$ giving $8 + 3\sqrt{7}$ with

$(8 + 3\sqrt{7})(8 - 3\sqrt{7}) = 64 - 63 = 1$. This algorithm runs in time polynomial in R_K , but *not* in $\log|d_K|$; no polynomial-time algorithm is known, and the difficulty is closely related to the erratic growth in the table above.

§35.10 Diophantine Applications

Having built the machinery, let us see it pay for itself. The pattern in each case is the same: an equation over $\mathbb{Z}$ factorises in $\mathcal{O}_K$ for a suitable K , unique factorisation of *ideals* lets us conclude that each factor is (say) a perfect cube as an ideal, and then the class group tells us whether we may promote this to a statement about elements.

The following two observations are what make the pattern work.

Lemma 35.10.1

Let $\mathfrak{a}, \mathfrak{b} \subseteq \mathcal{O}_K$ be coprime nonzero ideals (that is, $\mathfrak{a} + \mathfrak{b} = \mathcal{O}_K$) with $\mathfrak{a}\mathfrak{b} = \mathfrak{c}^m$ for some ideal $\mathfrak{c}$ and $m \geq 1$. Then $\mathfrak{a} = \mathfrak{d}^m$ and $\mathfrak{b} = \mathfrak{e}^m$ for some ideals $\mathfrak{d}, \mathfrak{e}$.

Proof. Write $\mathfrak{a} = \prod_{\mathfrak{p}} \mathfrak{p}^{a_{\mathfrak{p}}}$ and $\mathfrak{b} = \prod_{\mathfrak{p}} \mathfrak{p}^{b_{\mathfrak{p}}}$. Coprimality says no $\mathfrak{p}$ has both $a_{\mathfrak{p}} > 0$ and $b_{\mathfrak{p}} > 0$. The hypothesis says $a_{\mathfrak{p}} + b_{\mathfrak{p}} \equiv 0 \pmod{m}$ for every $\mathfrak{p}$, so for each $\mathfrak{p}$ one of $a_{\mathfrak{p}}, b_{\mathfrak{p}}$ is zero and the other is divisible by m . Take $\mathfrak{d} = \prod_{\mathfrak{p}} \mathfrak{p}^{a_{\mathfrak{p}}/m}$ and $\mathfrak{e} = \prod_{\mathfrak{p}} \mathfrak{p}^{b_{\mathfrak{p}}/m}$. $\square$

Lemma 35.10.2

Suppose $\gcd(m, h_K) = 1$ and $\mathfrak{d} \trianglelefteq \mathcal{O}_K$ is an ideal with $\mathfrak{d}^m$ principal. Then $\mathfrak{d}$ is principal.

Proof. In Cl_K we have $[\mathfrak{d}]^m = 1$, so the order of $[\mathfrak{d}]$ divides m ; it also divides $h_K = |\text{Cl}_K|$ by Lagrange. As $\gcd(m, h_K) = 1$, the order is 1. $\square$

§35.10.1 Sums of Two Squares

The oldest application, and still the prettiest.

Theorem 35.10.3 (Fermat)

An odd prime p is a sum of two integer squares if and only if $p \equiv 1 \pmod{4}$, and then the representation is unique up to signs and order.

Proof. Work in $\mathcal{O}_{\mathbb{Q}(i)} = \mathbb{Z}[i]$, which has $d_K = -4$ and $C_K = \frac{2}{\pi} \cdot 2 < 2$, so $h_K = 1$ and $\mathbb{Z}[i]$ is a PID. Note $N(a + bi) = a^2 + b^2$.

If $p = a^2 + b^2$ then modulo 4 the right-hand side is 0, 1 or 2, and p is odd, so $p \equiv 1 \pmod{4}$.

Conversely suppose $p \equiv 1 \pmod{4}$. Then -1 is a quadratic residue mod p (as the group $\mathbb{F}_p^\times$ is cyclic of order $p - 1 \equiv 0 \pmod{4}$, so contains an element of order 4), so $x^2 + 1$ has a root mod p and factors as $(x - r)(x + r)$. By Dedekind's criterion, p splits: $(p) = \mathfrak{p}\bar{\mathfrak{p}}$ with $N(\mathfrak{p}) = p$. Since $\mathbb{Z}[i]$ is a PID, $\mathfrak{p} = (a + bi)$ for some a, b , and $p = N(\mathfrak{p}) = a^2 + b^2$.

For uniqueness, a representation $p = a^2 + b^2$ gives $(p) = (a + bi)(a - bi)$, a factorisation into primes of norm p ; by unique factorisation these must be $\mathfrak{p}$ and $\bar{\mathfrak{p}}$ in some order, so $a + bi$ is an associate of $a' + b'i$ or of $a' - b'i$, and the units of $\mathbb{Z}[i]$ are $\pm 1, \pm i$, which permute (a, b) up to signs and order. $\square$

Example 35.10.4

$5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $17 = 1^2 + 4^2$, $29 = 2^2 + 5^2$; whereas 3, 7, 11, 19 are not sums of two squares. Note $2 = 1^2 + 1^2$ is the ramified case: $(2) = (1 + i)^2$ up to units.

Corollary 35.10.5

A positive integer n is a sum of two squares if and only if every prime $q \equiv 3 \pmod{4}$ divides n to an even power.

Proof. n is a sum of two squares if and only if $n = N(\alpha)$ for some $\alpha \in \mathbb{Z}[i]$, i.e. $n = N(\mathfrak{a})$ for some ideal $\mathfrak{a}$ (using $h_K = 1$). Now N is multiplicative and, by Proposition 35.6.11, the primes of $\mathbb{Z}[i]$ have norms: 2 (from $(2) = (1 + i)^2$), p and p (for $p \equiv 1 \pmod{4}$, which splits), and q^2 (for $q \equiv 3 \pmod{4}$, which is inert). So the norms of ideals are exactly the integers in which each $q \equiv 3 \pmod{4}$ appears with

even exponent. □

Example 35.10.6 (The Form $x^2 + 2y^2$)

The same argument runs in $\mathbb{Z}[\sqrt{-2}]$, which is also a PID (its Minkowski constant is $\frac{2}{\pi}\sqrt{8} \approx 1.80 < 2$). An odd prime p is of the form $x^2 + 2y^2$ if and only if p splits in $\mathbb{Q}(\sqrt{-2})$, which by Proposition 35.6.11 happens exactly when $\left(\frac{-2}{p}\right) = 1$, that is, when

$$p \equiv 1, 3 \pmod{8}.$$

So $3 = 1 + 2$, $11 = 9 + 2$, $17 = 9 + 8$, $19 = 1 + 18$, whereas $5, 7, 13, 23$ are not of this form. The reader may enjoy checking that the same method with $\mathbb{Z}[\omega]$, $\omega = \frac{1}{2}(-1 + \sqrt{-3})$, gives: p is of the form $x^2 + xy + y^2$ if and only if $p \equiv 1 \pmod{3}$.

Remark. It is essential here that the relevant ring has class number one. For $x^2 + 5y^2$ the class number is 2, and the correct statement splits into two cases according to the ideal class: an odd prime $p \neq 5$ is of the form $x^2 + 5y^2$ if $p \equiv 1, 9 \pmod{20}$, and of the form $2x^2 + 2xy + 3y^2$ – the other reduced quadratic form of discriminant -20 – if $p \equiv 3, 7 \pmod{20}$. This is exactly the phenomenon the class group was invented to describe.

§35.10.2 Mordell Equations

The equation $y^2 = x^3 + k$ is called a **Mordell equation**. Mordell proved that it has only finitely many integer solutions for each $k \neq 0$, but finding them is another matter; in favourable cases number fields do it in a page.

Theorem 35.10.7

The only integer solutions of $y^2 = x^3 - 2$ are $(x, y) = (3, \pm 5)$.

Proof. Write the equation as $y^2 + 2 = x^3$ and work in $K = \mathbb{Q}(\sqrt{-2})$, where $\mathcal{O}_K = \mathbb{Z}[\sqrt{-2}]$, $d_K = -8$, $N(a + b\sqrt{-2}) = a^2 + 2b^2$ and $\mathcal{O}_K^\times = \{\pm 1\}$. Since $C_K = \frac{2}{\pi}\sqrt{8} \approx 1.80 < 2$, we have $h_K = 1$.

Step 1: y is odd. If y were even then $x^3 = y^2 + 2 \equiv 2 \pmod{4}$, so x^3 is even, hence x is even and $8 \mid x^3$ – contradicting $x^3 \equiv 2 \pmod{4}$.

Step 2: the ideals $(y + \sqrt{-2})$ and $(y - \sqrt{-2})$ are coprime. Suppose a prime $\mathfrak{p}$ divides both. Then $\mathfrak{p}$ divides their difference $(2\sqrt{-2}) = (\sqrt{-2})^3$, so $\mathfrak{p} = (\sqrt{-2})$, the unique prime above 2. Then $\mathfrak{p} \mid (y + \sqrt{-2})$ gives $\mathfrak{p} \mid (y)$, so $2 = N(\mathfrak{p})$ divides $N((y)) = y^2$, making y even – contradicting Step 1.

Step 3: cubes. We have $(y + \sqrt{-2})(y - \sqrt{-2}) = (x)^3$ with the factors coprime, so by Lemma 35.10.1 there is an ideal $\mathfrak{d}$ with $(y + \sqrt{-2}) = \mathfrak{d}^3$. As $h_K = 1$, $\mathfrak{d} = (\gamma)$ is principal, so $y + \sqrt{-2} = u\gamma^3$ for a unit $u = \pm 1$; and $\pm 1 = (\pm 1)^3$, so absorbing the unit into γ ,

$$y + \sqrt{-2} = (a + b\sqrt{-2})^3 \quad \text{for some } a, b \in \mathbb{Z}.$$

Step 4: solve. Expanding,

$$(a + b\sqrt{-2})^3 = a^3 - 6ab^2 + (3a^2b - 2b^3)\sqrt{-2},$$

so comparing coefficients of $\sqrt{-2}$,

$$1 = 3a^2b - 2b^3 = b(3a^2 - 2b^2).$$

Hence $b \mid 1$. If $b = 1$ then $3a^2 - 2 = 1$, so $a = \pm 1$; if $b = -1$ then $3a^2 - 2 = -1$, so $3a^2 = 1$, impossible. So $b = 1$, $a = \pm 1$, and

$$y = a^3 - 6ab^2 = a^3 - 6a = \mp 5,$$

giving $y = \pm 5$ and $x^3 = 25 + 2 = 27$, so $x = 3$. $\square$

Now an example where the class group genuinely intervenes.

Theorem 35.10.8

The equation $y^2 = x^3 - 5$ has no integer solutions.

Proof. Write $y^2 + 5 = x^3$ and work in $K = \mathbb{Q}(\sqrt{-5})$, so $\mathcal{O}_K = \mathbb{Z}[\sqrt{-5}]$, $\mathcal{O}_K^\times = \{\pm 1\}$ and $h_K = 2$ by Example 35.8.1.

Step 1: x is odd. If x were even then $8 \mid x^3$, so $y^2 \equiv -5 \equiv 3 \pmod{8}$; but squares mod 8 are 0, 1, 4.

Step 2: coprimality. Suppose a prime $\mathfrak{p}$ divides $(y + \sqrt{-5})$ and $(y - \sqrt{-5})$. Then $\mathfrak{p} \mid (2\sqrt{-5})$, so $\mathfrak{p}$ is the prime $\mathfrak{p}_2 = (2, 1 + \sqrt{-5})$ above 2 or the prime $\mathfrak{p}_5 = (\sqrt{-5})$ above 5.

If $\mathfrak{p} = \mathfrak{p}_2$ then $\mathfrak{p}_2 \mid (y + \sqrt{-5})(y - \sqrt{-5}) = (x)^3$, so $\mathfrak{p}_2 \mid (x)$ and $2 = N(\mathfrak{p}_2)$ divides x^2 , making x even – contradicting Step 1.

If $\mathfrak{p} = \mathfrak{p}_5$ then similarly $5 \mid x^2$, so $5 \mid x$ and $125 \mid x^3$. Then $y^2 = x^3 - 5 \equiv -5 \pmod{25}$, so $5 \mid y^2$, so $5 \mid y$ and $25 \mid y^2$; but then $25 \mid x^3 - y^2 = 5$, absurd.

Step 3: cubes. As before $(y + \sqrt{-5}) = \mathfrak{d}^3$ for some ideal $\mathfrak{d}$. Now $\gcd(3, h_K) = \gcd(3, 2) = 1$, so Lemma 35.10.2 applies: $\mathfrak{d} = (\gamma)$ is principal. Absorbing the unit $\pm 1 = (\pm 1)^3$,

$$y + \sqrt{-5} = (a + b\sqrt{-5})^3 = a^3 - 15ab^2 + (3a^2b - 5b^3)\sqrt{-5}.$$

Step 4: solve. Comparing coefficients, $1 = b(3a^2 - 5b^2)$, so $b = \pm 1$. If $b = 1$ then $3a^2 = 6$, so $a^2 = 2$, impossible; if $b = -1$ then $3a^2 = 4$, impossible. So there are no solutions. $\square$

Remark. Step 3 is exactly where the class group earns its keep, and it is where the argument would break down if 3 divided h_K . If instead we had tried this with $\mathbb{Q}(\sqrt{-26})$, whose class number is 6, the argument would fail – and indeed $y^2 = x^3 - 26$ *does* have solutions, namely $(3, \pm 1)$ and $(35, \pm 207)$. Similarly, one must never forget the unit-absorbing step: it works here only because the unit group is $\{\pm 1\}$ and every unit is a cube. Over a real quadratic field, with its infinite unit group, considerably more care is

needed.

§35.10.3 Cyclotomic Fields and Fermat's Last Theorem

Kummer's original motivation for inventing ideal numbers was Fermat's equation

$$x^p + y^p = z^p, \quad p \text{ an odd prime.}$$

In $\mathbb{Q}(\zeta_p)$ the left-hand side factors completely:

$$x^p + y^p = \prod_{i=0}^{p-1} (x + \zeta_p^i y).$$

Suppose x, y, z are pairwise coprime integers with $p \nmid xyz$ (this is **Case I**). One checks that the ideals $(x + \zeta_p^i y)$ are pairwise coprime, so their product being $(z)^p$ forces

$$(x + \zeta_p y) = \mathfrak{a}^p$$

for some ideal $\mathfrak{a}$, by Lemma 35.10.1. So far this is exactly the Mordell argument. The step that fails in general is the promotion of $\mathfrak{a}^p$ principal to $\mathfrak{a}$ principal: this needs $p \nmid h_{\mathbb{Q}(\zeta_p)}$.

Definition 35.10.9 (Regular Prime)

An odd prime p is **regular** if $p \nmid h_{\mathbb{Q}(\zeta_p)}$.

Theorem 35.10.10 (Kummer)

If p is a regular prime then $x^p + y^p = z^p$ has no solutions in nonzero integers.

The remaining work, which we do not do here, is to control the units of $\mathbb{Z}[\zeta_p]$ (there is a lemma of Kummer saying that every unit is ζ_p^a times a real unit), and to handle Case II where $p \mid xyz$ by a descent.

The smallest irregular prime is 37, and the first few are 37, 59, 67, 101, 103, 131, 149, 157. Kummer gave a beautiful criterion: p is irregular precisely when p divides the numerator of one of the Bernoulli numbers $B_2, B_4, \dots, B_{p-3}$. It is known that infinitely many primes are irregular; whether infinitely many are regular is, remarkably, still open, although heuristics suggest about $e^{-1/2} \approx 61\%$ of primes should be.

Example 35.10.11

For $p \leq 19$ one has $h_{\mathbb{Q}(\zeta_p)} = 1$, so all these primes are regular and Fermat's equation has no solutions for these exponents. The first failure of $h = 1$ is $p = 23$, where $h_{\mathbb{Q}(\zeta_{23})} = 3$; but $23 \nmid 3$, so 23 is still regular.

Remark. Of course, Fermat's Last Theorem is now a theorem of Wiles for all $n \geq 3$, by a route – modular forms and Galois representations – with no visible relation to Kummer's. But Kummer's approach is the origin of algebraic number theory, and the class group of $\mathbb{Q}(\zeta_p)$ remains an object of intense study; the Herbrand–Ribet theorem, for instance, describes exactly which pieces of it are nontrivial in terms of Bernoulli numbers.

§35.11 The Dedekind Zeta Function

This final section is non-examinable. Its purpose is to show that the two invariants we have worked so hard to compute – the class number h_K and the regulator R_K – are not isolated curiosities, but appear together as the residue of a single analytic function; and to use this to prove Dirichlet's theorem on primes in arithmetic progressions, a result about $\mathbb{Z}$ whose only known proofs go through number fields.

§35.11.1 Dirichlet Series

Euler's proof that there are infinitely many primes runs as follows. By unique factorisation in $\mathbb{Z}$,

$$\prod_{p \text{ prime}} \left(1 - \frac{1}{p}\right)^{-1} = \prod_p \left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots\right) = \sum_{n \geq 1} \frac{1}{n},$$

since each n arises exactly once when the product is expanded. The right-hand side diverges, so the product cannot be finite. This is already better than Euclid's proof, because it says something quantitative; but it says nothing about primes in a fixed congruence class, since there is no such product formula for $\prod_{p \equiv a(q)}$. Dirichlet's idea was to introduce a variable.

Definition 35.11.1 (Riemann Zeta Function)

For $s \in \mathbb{C}$ with $\operatorname{Re}(s) > 1$, define $\zeta(s) = \sum_{n \geq 1} n^{-s}$.

More generally, a **Dirichlet series** is a function of the form $\sum_{n \geq 1} a_n n^{-s}$ for a sequence (a_n) of complex numbers. Convergence is controlled by the partial sums of the a_n .

Lemma 35.11.2

Suppose $a_1 + \dots + a_N = O(N^r)$ for some $r \in \mathbb{R}$. Then $\sum_{n \geq 1} a_n n^{-s}$ converges to a holomorphic function on $\operatorname{Re}(s) > r$.

Proof. Write $T(N) = a_1 + \dots + a_N$, so $|T(N)| \leq BN^r$ for some constant B . Abel summation gives

$$\sum_{n=1}^N a_n n^{-s} = \sum_{n=1}^{N-1} T(n) (n^{-s} - (n+1)^{-s}) + \frac{T(N)}{N^s}.$$

If $\operatorname{Re}(s) > r$ then $|T(N)/N^s| \leq BN^{r-\operatorname{Re}(s)} \rightarrow 0$, so it suffices to show that the sum $\sum_n T(n) (n^{-s} - (n+1)^{-s})$ converges absolutely. Now

$$n^{-s} - (n+1)^{-s} = s \int_n^{n+1} \frac{dx}{x^{s+1}},$$

and $n^r \leq x^r$ for $x \in [n, n+1]$ when $r \geq 0$ (and $n^r \leq (x-1)^r$ otherwise, which is

handled the same way), so

$$|T(n)||n^{-s} - (n+1)^{-s}| \leq B|s| \int_n^{n+1} \frac{dx}{x^{\operatorname{Re}(s)+1-r}},$$

and summing over n gives a convergent integral $\int_1^\infty x^{r-\operatorname{Re}(s)-1} dx$. Holomorphy follows since the convergence is locally uniform. $\square$

Proposition 35.11.3 (i) $\zeta(s)$ converges to a holomorphic function on $\operatorname{Re}(s) > 1$;

(ii) on that region, $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, the **Euler product**;

(iii) $\zeta(s) - \frac{1}{s-1}$ extends holomorphically to $\operatorname{Re}(s) > 0$. In particular ζ has a simple pole at $s = 1$ with residue 1.

Proof. (i) is Lemma 35.11.2 with $a_n = 1$, so $T(N) = N = O(N)$.

(ii) Let $p_1, \dots, p_m$ be the first m primes and X the set of positive integers all of whose prime factors lie among them. Expanding geometric series and using unique factorisation, $\prod_{i \leq m} (1 - p_i^{-s})^{-1} = \sum_{n \in X} n^{-s}$, so

$$\left| \zeta(s) - \prod_{i \leq m} (1 - p_i^{-s})^{-1} \right| = \left| \sum_{n \notin X} n^{-s} \right| \leq \sum_{n > m} n^{-\operatorname{Re}(s)} \rightarrow 0,$$

using $1, \dots, m \in X$.

(iii) Since $\frac{1}{s-1} = \sum_{n \geq 1} \int_n^{n+1} t^{-s} dt$, we get

$$\zeta(s) - \frac{1}{s-1} = \sum_{n \geq 1} \int_n^{n+1} (n^{-s} - t^{-s}) dt,$$

and each term is $O(|s|n^{-\operatorname{Re}(s)-1})$ by the mean value theorem, so the sum converges locally uniformly for $\operatorname{Re}(s) > 0$. $\square$

§35.11.2 The Dedekind Zeta Function

The Euler product is exactly the analytic shadow of unique factorisation, so it should have an analogue in any $\mathcal{O}_K$.

Definition 35.11.4 (Dedekind Zeta Function)

Let K be a number field. Its **Dedekind zeta function** is

$$\zeta_K(s) = \sum_{0 \neq \mathfrak{a} \leq \mathcal{O}_K} N(\mathfrak{a})^{-s} = \sum_{n \geq 1} \frac{a_n}{n^s}, \quad a_n = \# \{ \mathfrak{a} \leq \mathcal{O}_K : N(\mathfrak{a}) = n \},$$

which is a finite count by Corollary 35.5.10.

Theorem 35.11.5 (Analytic Class Number Formula)

Let K be a number field of degree n and signature (r, s) . Then

- (i) $\zeta_K(s)$ converges to a holomorphic function on $\operatorname{Re}(s) > 1$;
- (ii) on that region $\zeta_K(s) = \prod_{\mathfrak{p}} (1 - N(\mathfrak{p})^{-s})^{-1}$, the product over nonzero primes of $\mathcal{O}_K$;
- (iii) ζ_K extends meromorphically to $\operatorname{Re}(s) > 1 - \frac{1}{n}$ with a simple pole at $s = 1$ of residue

$$\operatorname{Res}_{s=1} \zeta_K(s) = \frac{2^r (2\pi)^s h_K R_K}{|\mu_K| |d_K|^{1/2}}.$$

Part (ii) is a formal consequence of unique factorisation of ideals and multiplicativity of the norm, exactly as in Proposition 35.11.3(ii). Parts (i) and (iii) follow from Lemma 35.11.2 together with the counting estimate

$$a_1 + \cdots + a_N = \frac{2^r (2\pi)^s h_K R_K}{|\mu_K| |d_K|^{1/2}} \cdot N + O(N^{1-1/n}),$$

which is proved by counting lattice points of bounded norm in each ideal class – a substantial but entirely geometric computation of the same flavour as the Minkowski bound.

It is worth pausing over this formula. On the left is an analytic object built out of counting ideals; on the right sit *every* invariant we have studied: the class number, the regulator, the number of roots of unity, the discriminant and the signature. It is one of the most beautiful formulae in mathematics, and it is the prototype for a vast web of conjectures (Birch–Swinnerton-Dyer, Bloch–Kato) relating special values of L -functions to arithmetic invariants.

Example 35.11.6

For $K = \mathbb{Q}$ we have $r = 1, s = 0, h = 1, R = 1, |\mu| = 2, |d| = 1$, and the formula reads $\operatorname{Res}_{s=1} \zeta(s) = 2/2 = 1$, agreeing with Proposition 35.11.3(iii).

§35.11.3 Factorisation and Dirichlet L -functions

For $K \neq \mathbb{Q}$ the function ζ_K factors, and the factors are interesting. Take $K = \mathbb{Q}(\sqrt{d})$ quadratic with $D = d_K$. Grouping the Euler product by rational primes and using Proposition 35.6.11:

- if p ramifies, $(p) = \mathfrak{p}^2$ with $N(\mathfrak{p}) = p$, contributing $(1 - p^{-s})^{-1}$;
- if p is inert, $N(\mathfrak{p}) = p^2$, contributing $(1 - p^{-2s})^{-1} = (1 - p^{-s})^{-1} (1 + p^{-s})^{-1}$;
- if p splits, two primes of norm p , contributing $(1 - p^{-s})^{-2}$.

In every case the contribution is $(1 - p^{-s})^{-1} (1 - \chi_D(p)p^{-s})^{-1}$ where

$$\chi_D(p) = \begin{cases} 0 & p \text{ ramifies,} \\ -1 & p \text{ inert,} \\ +1 & p \text{ splits} \end{cases} \quad \left(= \left(\frac{D}{p} \right) \text{ for odd } p \right).$$

Extending χ_D multiplicatively to all $n \geq 1$ and setting

$$L(\chi_D, s) = \prod_p (1 - \chi_D(p)p^{-s})^{-1} = \sum_{n \geq 1} \frac{\chi_D(n)}{n^s},$$

we have shown

$$\zeta_{\mathbb{Q}(\sqrt{d})}(s) = \zeta(s) L(\chi_D, s).$$

Definition 35.11.7 (Dirichlet Character)

A **Dirichlet character** of modulus q is a function $\chi : \mathbb{Z} \rightarrow \mathbb{C}$ of the form

$$\chi(n) = \begin{cases} \omega(n \bmod q) & \gcd(n, q) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

for a group homomorphism $\omega : (\mathbb{Z}/q\mathbb{Z})^\times \rightarrow \mathbb{C}^\times$. It is **trivial** if $\omega \equiv 1$.

Such χ are completely multiplicative, so $L(\chi, s) = \prod_p (1 - \chi(p)p^{-s})^{-1} = \sum_n \chi(n)n^{-s}$ makes sense for $\operatorname{Re}(s) > 1$.

Theorem 35.11.8

For d squarefree with $D = d_{\mathbb{Q}(\sqrt{d})}$, the function χ_D is a Dirichlet character of modulus $|D|$.

The proof is a case analysis using quadratic reciprocity; in fact the statement is essentially *equivalent* to quadratic reciprocity, which is a striking way to view that theorem: the law of quadratic reciprocity is the assertion that the splitting behaviour of p in $\mathbb{Q}(\sqrt{d})$ depends only on p modulo d_K .

Example 35.11.9

Let $K = \mathbb{Q}(i)$, so $D = -4$. For odd p , $\chi_{-4}(p) = \left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$, and $\chi_{-4}(2) = 0$ since 2 ramifies. So $\chi_{-4}(n) = (-1)^{(n-1)/2}$ for n odd and 0 for n even, and

$$L(\chi_{-4}, s) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \dots$$

Comparing residues at $s = 1$ in $\zeta_{\mathbb{Q}(i)} = \zeta \cdot L(\chi_{-4}, \cdot)$ and using Theorem 35.11.5 with $r = 0, s = 1, h = 1, R = 1, |\mu| = 4, |d| = 4$:

$$L(\chi_{-4}, 1) = \operatorname{Res}_{s=1} \zeta_{\mathbb{Q}(i)}(s) = \frac{(2\pi) \cdot 1 \cdot 1}{4 \cdot 2} = \frac{\pi}{4},$$

which is Leibniz's formula $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$. The class number formula *proves* Leibniz's series.

Lemma 35.11.10

If χ is a nontrivial Dirichlet character, then $L(\chi, s)$ is holomorphic on $\operatorname{Re}(s) > 0$.

Proof. By orthogonality of characters of the finite abelian group $(\mathbb{Z}/q\mathbb{Z})^\times$, applied to the trivial character and $\omega \neq 1$,

$$\sum_{n=aq+1}^{(a+1)q} \chi(n) = \sum_{u \in (\mathbb{Z}/q\mathbb{Z})^\times} \omega(u) = 0$$

for every a . Hence the partial sums $\sum_{n \leq N} \chi(n)$ are bounded, i.e. $O(N^0)$, and Lemma 35.11.2 applies with $r = 0$. $\square$

Corollary 35.11.11

If $D < 0$ is a fundamental discriminant then

$$L(\chi_D, 1) = \frac{2\pi h_{\mathbb{Q}(\sqrt{D})}}{|\mu_{\mathbb{Q}(\sqrt{D})}| |D|^{1/2}} \neq 0.$$

Proof. Both sides of $\zeta_{\mathbb{Q}(\sqrt{D})} = \zeta \cdot L(\chi_D, \cdot)$ have a simple pole at $s = 1$; comparing residues and using $\text{Res}_{s=1} \zeta = 1$ together with Theorem 35.11.5 (with $r = 0$, $s = 1$, $R = 1$) gives the formula. It is nonzero since $h \geq 1$. $\square$

§35.11.4 Primes in Arithmetic Progressions

To handle general moduli we need the cyclotomic analogue of the factorisation above. Let $q \geq 1$, $L = \mathbb{Q}(\zeta_q)$ and $G = \text{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/q\mathbb{Z})^\times$, of order $\varphi(q)$. By Proposition 35.6.18, for $p \nmid q$ the primes above p number $\varphi(q)/f$ and have norm p^f , where f is the order of p in G . So these primes contribute

$$(1 - p^{-fs})^{-\varphi(q)/f}$$

to ζ_L . Let $\omega_1 = 1, \omega_2, \dots, \omega_{\varphi(q)}$ be the characters of the abelian group G , and χ_i the corresponding Dirichlet characters mod q . Restricted to the cyclic subgroup $\langle p \rangle$ of order f , the ω_i run over the f characters of $\langle p \rangle$, each $\varphi(q)/f$ times; so $\omega_1(p), \dots, \omega_{\varphi(q)}(p)$ are exactly the f th roots of unity, each repeated $\varphi(q)/f$ times. Since $1 - t^f = \prod_{\gamma^f=1} (1 - \gamma t)$, putting $t = p^{-s}$ gives

$$(1 - p^{-fs})^{-\varphi(q)/f} = \prod_{i=1}^{\varphi(q)} (1 - \omega_i(p)p^{-s})^{-1}.$$

Multiplying over all $p \nmid q$ and accounting for the finitely many $p \mid q$ by a correction factor $C(s)$ (a finite product of terms $(1 - p^{-fs})^{-1}$, holomorphic and nonvanishing at $s = 1$):

$$\zeta_{\mathbb{Q}(\zeta_q)}(s) = C(s) \prod_{i=1}^{\varphi(q)} L(\chi_i, s) = C'(s) \zeta(s) \prod_{i=2}^{\varphi(q)} L(\chi_i, s).$$

Theorem 35.11.12

If χ is a nontrivial Dirichlet character then $L(\chi, 1) \neq 0$.

Proof. Every nontrivial Dirichlet character mod q is one of the χ_i above with $i \neq 1$. By Lemma 35.11.10, each such $L(\chi_i, s)$ is holomorphic at $s = 1$. Now compare orders of vanishing at $s = 1$ in the displayed factorisation: the left-hand side has a *simple* pole by Theorem 35.11.5, and on the right ζ contributes a simple pole while C' is holomorphic and nonzero. So $\prod_{i \geq 2} L(\chi_i, 1)$ must be finite and nonzero, whence each factor is nonzero. $\square$

Theorem 35.11.13 (Dirichlet)

Let a, q be coprime positive integers. Then there are infinitely many primes $p \equiv a \pmod{q}$.

Proof. Orthogonality of the columns of the character table of $(\mathbb{Z}/q\mathbb{Z})^\times$ gives, for any n ,

$$\frac{1}{\varphi(q)} \sum_{i=1}^{\varphi(q)} \overline{\chi_i(a)} \chi_i(n) = \begin{cases} 1 & n \equiv a \pmod{q}, \\ 0 & \text{otherwise,} \end{cases}$$

valid even when $\gcd(n, q) > 1$ since then every $\chi_i(n) = 0$. Hence for real $s > 1$,

$$\sum_{p \equiv a \pmod{q}} p^{-s} = \frac{1}{\varphi(q)} \sum_i \overline{\chi_i(a)} \sum_p \chi_i(p) p^{-s}.$$

Now expand the logarithm of the Euler product: for $\operatorname{Re}(s) > 1$,

$$\log L(\chi, s) = - \sum_p \log(1 - \chi(p)p^{-s}) = \sum_p \frac{\chi(p)}{p^s} + \sum_{p, m \geq 2} \frac{\chi(p^m)}{mp^{ms}},$$

and the double sum is bounded as $s \rightarrow 1^+$, since

$$\sum_{p, m \geq 2} p^{-ms} = \sum_p \frac{1}{p^s(p^s - 1)} \leq \sum_{n \geq 2} \frac{1}{n(n-1)} < \infty.$$

So, up to a bounded function,

$$\sum_{p \equiv a \pmod{q}} p^{-s} \sim \frac{1}{\varphi(q)} \sum_i \overline{\chi_i(a)} \log L(\chi_i, s).$$

For $i = 1$ we have $L(\chi_1, s) = \zeta(s) \prod_{p|q} (1 - p^{-s})$, which has a simple pole at $s = 1$, so $\log L(\chi_1, s) \sim \log \frac{1}{s-1} \rightarrow \infty$. For $i \geq 2$, Lemma 35.11.10 and Theorem 35.11.12 say $L(\chi_i, s)$ is holomorphic and nonzero at $s = 1$, so $\log L(\chi_i, s)$ stays bounded. Therefore

$$\sum_{p \equiv a \pmod{q}} p^{-s} \sim \frac{1}{\varphi(q)} \log \frac{1}{s-1} \rightarrow \infty \quad \text{as } s \rightarrow 1^+,$$

which is impossible if there are only finitely many such primes. $\square$

The proof gives more than infinitude: the factor $1/\varphi(q)$ says that the primes are **equidistributed** among the $\varphi(q)$ invertible classes mod q , in the sense of Dirichlet density.

Remark. This is the tip of a very large iceberg. For $L/\mathbb{Q}$ Galois with group G , the zeta function factors as

$$\zeta_L(s) = \prod_{\rho} L(\rho, s)^{\dim \rho},$$

the product over irreducible complex representations of G , where the $L(\rho, s)$ are **Artin L -functions**; the factorisation mirrors the decomposition of the regular representation, and $L(\mathbb{1}, s) = \zeta(s)$. Artin conjectured that $L(\rho, s)$ is entire for $\rho \neq \mathbb{1}$, which is still open. For one-dimensional ρ , matching $L(\rho, s)$ with a Dirichlet L -function is the content of class field theory – the vast generalisation of quadratic reciprocity – and the higher-dimensional case is the Langlands programme. All of it begins with the observation that unique factorisation of ideals gives an Euler product.

§35.12 Further Directions

We close with a brief and impressionistic look at where the subject goes next; none of this is examinable, but all of it is the natural continuation of what we have done.

§35.12.1 Class Field Theory

We saw in Section 35.6.5 that the splitting of a prime q in $\mathbb{Q}(\zeta_m)$ depends only on q modulo m , and that this fact, restricted to the quadratic subfield of $\mathbb{Q}(\zeta_p)$, is quadratic reciprocity. The Kronecker–Weber theorem says that this example is exhaustive:

Theorem 35.12.1 (Kronecker–Weber)

Every finite abelian extension of $\mathbb{Q}$ is contained in some cyclotomic field $\mathbb{Q}(\zeta_m)$.

So the abelian extensions of $\mathbb{Q}$ are completely understood, and splitting in them is always governed by congruence conditions. Class field theory does the same for an arbitrary number field K : it classifies the abelian extensions of K in terms of data intrinsic to K . The simplest instance is the **Hilbert class field** H of K , the maximal unramified abelian extension, which satisfies

$$\mathrm{Gal}(H/K) \cong \mathrm{Cl}_K, \quad [H : K] = h_K,$$

and in which a prime $\mathfrak{p}$ of K splits completely if and only if $\mathfrak{p}$ is principal. So the class group – which we introduced as a measure of the failure of unique factorisation – turns out to be a Galois group.

Example 35.12.2

Take $K = \mathbb{Q}(\sqrt{-5})$, with $h_K = 2$. Its Hilbert class field is $H = K(i) = \mathbb{Q}(i, \sqrt{5})$, of degree 2 over K . This gives a slick explanation of the congruence conditions in the remark of Section 35.10: whether p is $x^2 + 5y^2$ or $2x^2 + 2xy + 3y^2$ is governed by whether p splits completely in H .

For K imaginary quadratic there is a beautiful explicit theory – **complex multiplication** – generating the abelian extensions of K using special values of elliptic modular functions, in exact analogy with the way roots of unity, i.e. special values of $e^{2\pi iz}$, generate the abelian extensions of $\mathbb{Q}$. Doing the same for a general base field is Hilbert’s twelfth problem, and it is still open.

§35.12.2 How Do Class Groups Behave?

We computed a handful of class groups, and they looked rather arbitrary. That impression is correct, and it has been made precise. The **Cohen–Lenstra heuristics** predict, roughly, that the odd part of the class group of a random imaginary quadratic field is a random finite abelian group, weighted inversely by the size of its automorphism group. The predictions are strikingly accurate numerically, and some cases are now theorems; but the basic questions remain hard. It is not known, for instance, whether infinitely many real quadratic fields have class number one, even though numerically they appear to make up more than three quarters of all real quadratic fields.

What *is* known is a strong contrast between the two cases. For imaginary quadratic fields, $h_K \rightarrow \infty$ as $|d_K| \rightarrow \infty$ – indeed $h_K \gg |d_K|^{1/2-\varepsilon}$ by Siegel’s theorem – which is why the list of nine class number one fields in Section 35.8 is finite. For real quadratic fields the class number formula ties h_K to $h_K R_K$, and the regulator can be enormous (recall the fundamental unit for $d = 46$), so a large $|d_K|$ is perfectly compatible with $h_K = 1$.

§35.12.3 Beyond Number Fields

Everything in this course had a geometric shadow. A Dedekind domain is precisely the coordinate ring of a smooth affine curve, and the analogy

$$\mathbb{Z} \longleftrightarrow \mathbb{F}_q[t], \quad \mathbb{Q} \longleftrightarrow \mathbb{F}_q(t), \quad \text{number field} \longleftrightarrow \text{function field of a curve}$$

is close enough to be genuinely useful in both directions. Under it, the class group of $\mathcal{O}_K$ corresponds to the Picard group of a curve, the unit theorem corresponds to a statement about functions with prescribed poles, and the Dedekind zeta function corresponds to the zeta function of a curve over a finite field – where the Riemann hypothesis is a theorem of Weil.

Finally, the pattern of Section 35.10 – factor a Diophantine equation in a suitable ring, use the class group to control the factors – is the ancestor of the modern theory. It is superseded by the theory of elliptic curves for equations like $y^2 = x^3 + k$, where Mordell’s theorem says the rational points form a finitely generated abelian group (a statement whose proof is a descent argument in the same spirit as Dirichlet’s unit theorem), and ultimately by the machinery of modular forms and Galois representations that settled Fermat’s Last Theorem. But all of it starts here, with the observation that $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$.

36 Number Theory

Number theory is the study of the integers, which is to say the study of the two operations $+$ and $\times$ and the friction between them. Addition on its own is dull, and multiplication on its own is not much better; it is the fact that the primes – the atoms of multiplication – are scattered so unpredictably along the additive line $\dots, -1, 0, 1, 2, \dots$ that gives the subject its character. Almost every question we will ask has an embarrassingly simple statement, and almost none of them has an easy answer.

This article constitutes my notes for the ‘Number Theory’ course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§36.1 Introduction

Familiarity with IA Numbers and Sets is assumed throughout, and the first two sections are largely a review of that material – albeit a review conducted with rather different priorities. Where in IA we were content to know that a greatest common divisor exists, here we will care how quickly we can find one; where in IA the primes were a curiosity, here we will want to count them.

Before we start on any of that, it is worth being clear about the sort of questions the subject is trying to answer. Here are three, all of which are still open.

Conjecture 36.1.1 (Twin Prime Conjecture). There are infinitely many primes p for which $p + 2$ is also prime.

Nobody knows. The best that is known, following a spectacular breakthrough of Zhang in 2013 and the subsequent Polymath project, is that there are infinitely many primes p such that at least one of $p + 2, p + 4, \dots, p + 246$ is prime. That is a long way from $246 = 2$, but it is infinitely closer than we were before.

Conjecture 36.1.2 (Riemann Hypothesis). Writing $\pi(x)$ for the number of primes at most x and

$$\text{Li}(x) = \int_2^x \frac{dt}{\log t},$$

we have $|\pi(x) - \text{Li}(x)| \leq \sqrt{x} \log x$ for all $x \geq 3$.

This is one of many equivalent formulations, and it says that the primes are distributed as regularly as one could possibly hope. We will meet the Riemann zeta function in §6, and see where the hypothesis actually lives. The best unconditional result in this direction is of the shape

$$|\pi(x) - \text{Li}(x)| \leq \frac{Cx}{(\log x)^{3/4}} e^{-c\sqrt{\log x}},$$

which is very much weaker.

Conjecture 36.1.3. There is a polynomial-time algorithm which, given $N = pq$ with p and q prime, returns p .

Here the consensus is that the answer is ‘no’, and a great deal of the world’s cryptography is betting on it. We will look at what *can* be done in §8. Notice the change of flavour: this is a question about *algorithms*, and one of the pleasant surprises of the course is how naturally the algorithmic questions sit alongside the classical ones.

Since we will be making claims about algorithms, we should say what we mean.

Definition 36.1.4 (Polynomial Time)

An algorithm taking an integer $N > 0$ as input runs in **polynomial time** if it terminates after at most $C(\log N)^k$ elementary operations, for some constants $C, k > 0$. If the input is a list $N_1, \dots, N_s$, we ask instead for at most $C(\max_i \log N_i)^k$ operations.

The point of taking $\log N$ rather than N is that $\log N$ is (up to a constant) the number of *digits* of N , which is how long it takes merely to write N down. Addition and multiplication of integers run in polynomial time, and we will see shortly that so does Euclid’s algorithm. Trial division does not: to factor $N = pq$ with p, q each of fifty digits by dividing by every integer up to $\sqrt{N}$ would take of the order of 10^{50} divisions, which at a generous 2^{30} divisions per second is around 10^{33} years.

§36.2 Euclid’s Algorithm

We begin at the beginning, with division.

Theorem 36.2.1 (Division Algorithm)

Given $a, b \in \mathbb{Z}$ with $b > 0$, there exist $q, r \in \mathbb{Z}$ with $a = bq + r$ and $0 \leq r < b$. Moreover q and r are unique.

Proof. Let $S = \{a - nb : n \in \mathbb{Z}\}$. Then S contains some nonnegative integer (take n very negative if $a < 0$), so by well-ordering it contains a least nonnegative element r , say $r = a - qb$. We must have $r < b$, as otherwise $r - b \in S$ would be nonnegative and smaller than r . So $a = qb + r$ as required.

For uniqueness, if $a = qb + r = q'b + r'$ with $0 \leq r, r' < b$ then $b(q - q') = r' - r$, and $|r' - r| < b$ forces $q = q'$ and hence $r = r'$. $\square$

Definition 36.2.2

If $r = 0$ above, we write $b \mid a$ and say that b **divides** a ; otherwise we write $b \nmid a$.

Given $a_1, \dots, a_n \in \mathbb{Z}$ not all zero, let

$$I = \{\lambda_1 a_1 + \dots + \lambda_n a_n \mid \lambda_i \in \mathbb{Z}\}$$

be the set of integer combinations of the a_i . Note that for $a, b \in I$ and $l, m \in \mathbb{Z}$ we have $la + mb \in I$; in the language of IB Groups, Rings and Modules, I is an ideal of $\mathbb{Z}$. The following lemma says that $\mathbb{Z}$ is a principal ideal domain, though we will not need to say it that way.

Lemma 36.2.3

$I = d\mathbb{Z} = \{md \mid m \in \mathbb{Z}\}$ for some $d > 0$.

Proof. Let d be the least positive element of I , which exists since I contains a nonzero element and is closed under negation. Certainly $d\mathbb{Z} \subseteq I$. Conversely, if $a \in I$ then we may write $a = qd + r$ with $0 \leq r < d$. If $r = 0$ then $a \in d\mathbb{Z}$. Otherwise $r = a - qd$ is an element of I which is positive and smaller than d , a contradiction. $\square$

In particular $d \mid a_i$ for all i , since each $a_i \in I = d\mathbb{Z}$. Conversely, if $c \mid a_i$ for all i then every integer combination of the a_i is a multiple of c , so $d\mathbb{Z} = I \subseteq c\mathbb{Z}$ and hence $c \mid d$. Thus d deserves its name.

Definition 36.2.4

We write $d = \gcd(a_1, \dots, a_n)$, or more compactly $(a_1, \dots, a_n)$, and call d the **greatest common divisor** of $a_1, \dots, a_n$. If $(a, b) = 1$ we say a and b are **coprime**.

Corollary 36.2.5 (Bézout)

Suppose $a, b, c \in \mathbb{Z}$ with a, b not both 0. Then there exist $x, y \in \mathbb{Z}$ with $ax + by = c$ if and only if $(a, b) \mid c$.

Proof. The set of achievable c is exactly $I = (a, b)\mathbb{Z}$ by Lemma 36.2.3. $\square$

This is all very well, but it is not yet an *algorithm*: Lemma 36.2.3 produces d by minimising over an infinite set. The algorithm is due to Euclid, and consists of iterating the division algorithm.

Theorem 36.2.6 (Euclid's Algorithm)

Given $a, b \in \mathbb{N}$ with $a > b$, set $r_0 = b$ and apply the division algorithm repeatedly:

$$\begin{aligned} a &= q_1 r_0 + r_1, \\ r_0 &= q_2 r_1 + r_2, \\ r_1 &= q_3 r_2 + r_3, \\ &\vdots \\ r_{k-2} &= q_k r_{k-1} + r_k, \\ r_{k-1} &= q_{k+1} r_k + 0, \end{aligned}$$

where $0 < r_i < r_{i-1}$ for $i \leq k$. Then $r_k = (a, b)$.

Proof. The sequence $r_0 > r_1 > \dots \geq 0$ is a strictly decreasing sequence of nonnegative integers, so it does terminate. Reading the last line upwards, $r_k \mid r_{k-1}$, then $r_k \mid r_{k-2}$, and so on all the way up, so $r_k \mid b$ and $r_k \mid a$; hence $r_k \leq (a, b)$. Reading downwards instead, if $m \mid a$ and $m \mid b$ then $m \mid r_1$, then $m \mid r_2$, and so on, so $m \mid r_k$. Taking $m = (a, b)$ gives $(a, b) \leq r_k$, and so $(a, b) = r_k$. $\square$

Unwinding the equations from the bottom expresses r_k as an integer combination of a and b , which is a constructive form of [Corollary 36.2.5](#). Let us do one.

Example 36.2.7

Take $a = 4321$ and $b = 1234$. Euclid's algorithm gives

$$4321 = 3 \times 1234 + 619,$$

$$1234 = 1 \times 619 + 615,$$

$$619 = 1 \times 615 + 4,$$

$$615 = 153 \times 4 + 3,$$

$$4 = 1 \times 3 + 1,$$

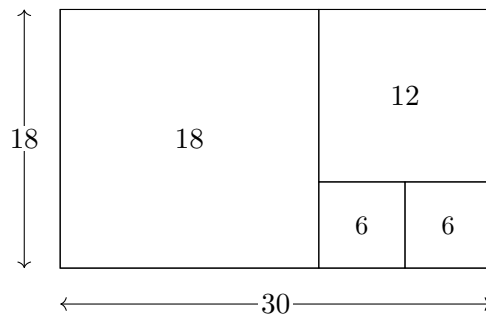
$$3 = 3 \times 1 + 0,$$

so $(4321, 1234) = 1$. Working back up,

$$\begin{aligned} 1 &= 4 - 3 = 4 - (615 - 153 \times 4) = 154 \times 4 - 615 \\ &= 154 \times (619 - 615) - 615 = 154 \times 619 - 155 \times 615 \\ &= 154 \times 619 - 155 \times (1234 - 619) = 309 \times 619 - 155 \times 1234 \\ &= 309 \times (4321 - 3 \times 1234) - 155 \times 1234 = 309 \times 4321 - 1082 \times 1234. \end{aligned}$$

One checks that $309 \times 4321 = 1335189$ and $1082 \times 1234 = 1335188$, so indeed the difference is 1.

There is a pleasant way to see the algorithm. Take an $a \times b$ rectangle and repeatedly cut off the largest square you can; the algorithm terminates exactly when the remaining rectangle is tiled by squares, and the side of the last square is (a, b) .



Here $30 = 1 \times 18 + 12$, then $18 = 1 \times 12 + 6$, then $12 = 2 \times 6$, and $(30, 18) = 6$ is the side of the smallest square.

How fast is this? The key observation is that $r_{i+1} \leq r_{i-1}/2$ always. Indeed, if $r_{i+1} < r_i/2$ then certainly $r_{i+1} < r_{i-1}/2$ since $r_i < r_{i-1}$; and if instead $r_{i+1} \geq r_i/2$, then from $r_{i-1} = q_{i+1}r_i + r_{i+1}$ with $q_{i+1} \geq 1$ we get $r_{i-1} \geq r_i + r_{i+1} \geq 2r_{i+1}$. So the remainders at least halve every two steps, and the algorithm terminates after $O(\log b)$ divisions.

Corollary 36.2.8

Euclid's algorithm runs in polynomial time.

Contrast this with the 'obvious' method of computing (a, b) , namely factorising both and comparing: nobody knows how to factorise in polynomial time, and Euclid's algorithm

cheerfully avoids the need to.

Definition 36.2.9

An integer $n > 1$ is **prime** if its only positive divisors are 1 and n ; otherwise it is **composite**.

Lemma 36.2.10 (Euclid's Lemma)

Let p be prime and $a, b \in \mathbb{Z}$. Then $p \mid ab$ if and only if $p \mid a$ or $p \mid b$.

Proof. The ‘if’ direction is clear. Conversely, suppose $p \mid ab$ but $p \nmid a$. Then (a, p) divides p and is not p , so $(a, p) = 1$ and by Bézout there are $x, y \in \mathbb{Z}$ with $ax + py = 1$. Multiplying by b ,

$$b = b(ax + py) = x(ab) + (by)p,$$

and p divides both terms on the right, so $p \mid b$. $\square$

This fails badly for composite numbers: $6 \mid 4 \times 9$ but $6 \nmid 4$ and $6 \nmid 9$. It is precisely this property that makes the primes the right atoms.

Theorem 36.2.11 (Fundamental Theorem of Arithmetic)

Every integer $n > 1$ can be written as a product of primes, and this factorisation is unique up to reordering.

Proof. Existence follows by strong induction: either n is prime, or $n = ab$ with $1 < a, b < n$ and we apply the inductive hypothesis to each of a, b .

For uniqueness, suppose some $n > 1$ has two genuinely different factorisations and take the least such n , say

$$n = p_1 \cdots p_k = q_1 \cdots q_l.$$

Then $p_1 \mid q_1 \cdots q_l$, so by Euclid's lemma $p_1 \mid q_i$ for some i , and since q_i is prime, $p_1 = q_i$. Relabelling, $p_1 = q_1$, and then $n/p_1 = p_2 \cdots p_k = q_2 \cdots q_l$ is a smaller integer with two different factorisations unless $k = l$ and the remaining primes agree after reordering. This contradicts minimality. $\square$

Remark. Collecting multiplicities, every $n \geq 1$ has a unique expression $n = \prod_p p^{v_p(n)}$ where the product is over primes and all but finitely many exponents vanish. If $m = \prod_p p^{\alpha_p}$ and $n = \prod_p p^{\beta_p}$ then

$$(m, n) = \prod_p p^{\min(\alpha_p, \beta_p)}, \quad [m, n] = \prod_p p^{\max(\alpha_p, \beta_p)},$$

where $[m, n]$ denotes the least common multiple, and in particular $(m, n)[m, n] = mn$. This is a perfectly good *description* of the gcd, and a perfectly awful way to *compute* it.

Definition 36.2.12

For a prime p and $n \geq 1$, the exponent $v_p(n)$ above is the **p -adic valuation** of n .

We record the basic properties: $v_p(mn) = v_p(m) + v_p(n)$, and $p^k \mid n$ if and only if $v_p(n) \geq k$. A slightly less obvious formula, which we will need when we come to Bertrand's postulate, is Legendre's:

$$v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor,$$

the point being that among $1, \dots, n$ exactly $\lfloor n/p^i \rfloor$ are divisible by p^i , and each such contributes one to the count for every i up to its valuation.

Finally, the oldest theorem in the subject.

Theorem 36.2.13 (Euclid)

There are infinitely many primes.

Proof. Suppose there were only finitely many, say $p_1, \dots, p_k$. Then $N = p_1 p_2 \cdots p_k + 1$ is greater than 1, so it has a prime factor; but that factor is some p_i , which then divides $N - p_1 \cdots p_k = 1$, a contradiction. $\square$

Remark. It is worth noticing what this argument does and does not give. It shows that the k -th prime is at most $p_1 \cdots p_{k-1} + 1$, which yields the bound $p_k \leq 2^{2^k}$ and hence $\pi(x) \gg \log \log x$. That is a spectacularly bad bound – the truth is $\pi(x) \sim x / \log x$ – and improving it will occupy us in §6.

§36.3 Congruences

§36.3.1 Modular Inverses

Definition 36.3.1

Let $n \geq 1$ be an integer. We say a is **congruent** to b modulo n , written $a \equiv b \pmod{n}$, if $n \mid a - b$.

This is an equivalence relation on $\mathbb{Z}$, and we write $\mathbb{Z}/n\mathbb{Z}$ for the set of equivalence classes $a + n\mathbb{Z}$. The operations

$$(a + n\mathbb{Z}) + (b + n\mathbb{Z}) = (a + b) + n\mathbb{Z}, \quad (a + n\mathbb{Z})(b + n\mathbb{Z}) = ab + n\mathbb{Z}$$

are well defined, making $\mathbb{Z}/n\mathbb{Z}$ into a ring. The whole point of working modulo n is that it turns an infinite problem into a finite one; the price is that we lose information, and much of the art lies in choosing n so as to lose the right information.

Lemma 36.3.2

Let $a \in \mathbb{Z}$ and $n \geq 1$. The following are equivalent.

- (i) $(a, n) = 1$.
- (ii) There is $b \in \mathbb{Z}$ with $ab \equiv 1 \pmod{n}$.
- (iii) $a + n\mathbb{Z}$ generates the additive group $(\mathbb{Z}/n\mathbb{Z}, +)$.

When these hold we call b a **multiplicative inverse** of a modulo n .

Proof. (i) $\Rightarrow$ (ii). By Bézout there are b, c with $ab + cn = 1$, so $ab \equiv 1 \pmod{n}$.
(ii) $\Rightarrow$ (i). If $ab - 1 = kn$ then $ab - kn = 1$, so any common divisor of a and n divides 1.
(ii) $\Rightarrow$ (iii). The subgroup generated by a contains $ba \equiv 1$, hence contains everything.
(iii) $\Rightarrow$ (ii). If a generates then some multiple ba of a is congruent to 1. $\square$

Note that Euclid's algorithm computes the inverse for us, and does so in polynomial time; this is the reason that modular arithmetic is computationally cheap.

Definition 36.3.3

We write $(\mathbb{Z}/n\mathbb{Z})^\times$ for the group of units of $\mathbb{Z}/n\mathbb{Z}$, that is the set of classes satisfying the conditions of [Lemma 36.3.2](#).

Definition 36.3.4 (Euler's Totient Function)

Let $\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$, so that $\phi(n)$ is the number of $1 \leq a \leq n$ with $(a, n) = 1$. By convention $\phi(1) = 1$.

Theorem 36.3.5 (Euler–Fermat)

If $a, n \in \mathbb{Z}$ with $n \geq 1$ and $(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. By Lagrange's theorem applied to the group $(\mathbb{Z}/n\mathbb{Z})^\times$, the order of a divides the order $\phi(n)$ of the group. $\square$

Theorem 36.3.6 (Fermat's Little Theorem)

If p is prime and $a \in \mathbb{Z}$, then $a^p \equiv a \pmod{p}$. If moreover $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. If $p \mid a$ both sides are 0. Otherwise $(a, p) = 1$ and $\phi(p) = p - 1$, so [Theorem 36.3.5](#) applies. $\square$

Remark. There is a proof of Fermat's little theorem which avoids group theory entirely: by the binomial theorem, $a^p - (a - 1)^p \equiv 1 \pmod{p}$ since $p \mid \binom{p}{j}$ for $0 < j < p$, and then one inducts on a .

§36.3.2 Systems of Linear Congruences

Suppose we want x with

$$x \equiv 4 \pmod{7}, \quad x \equiv 5 \pmod{12}.$$

The trick is to solve the two 'pure' problems first. If we can find u with $u \equiv 1 \pmod{7}$ and $u \equiv 0 \pmod{12}$, and v with $v \equiv 0 \pmod{7}$ and $v \equiv 1 \pmod{12}$, then $x = 4u + 5v$

works. And finding u and v is just Bézout: since $(7, 12) = 1$ we have $7 \times (-5) + 12 \times 3 = 1$, so we may take $u = 36$ and $v = -35$. Then $x = 4 \times 36 + 5 \times (-35) = -31$, and indeed $-31 = -35 + 4 \equiv 4 \pmod{7}$ and $-31 = -36 + 5 \equiv 5 \pmod{12}$. The general solution is $x \equiv -31 \equiv 53 \pmod{84}$.

Generalising is straightforward.

Theorem 36.3.7 (Chinese Remainder Theorem)

Let $m_1, \dots, m_k \in \mathbb{N}$ be pairwise coprime and let $a_1, \dots, a_k \in \mathbb{Z}$. Then the system

$$x \equiv a_i \pmod{m_i}, \quad 1 \leq i \leq k$$

has a solution x , unique modulo $M = m_1 \cdots m_k$.

Proof. Write $M_i = M/m_i$. Since the m_j are pairwise coprime, $(m_i, M_i) = 1$, so there is b_i with $M_i b_i \equiv 1 \pmod{m_i}$. Also $M_i b_i \equiv 0 \pmod{m_j}$ for $j \neq i$, since $m_j \mid M_i$. Hence $x = \sum_i a_i b_i M_i$ satisfies $x \equiv a_i b_i M_i \equiv a_i \pmod{m_i}$ for each i .

For uniqueness, if x and y both satisfy the system then $m_i \mid x - y$ for all i , and since the m_i are pairwise coprime, $M \mid x - y$. $\square$

Algebraically the theorem says that the reduction map

$$\theta : \mathbb{Z}/M\mathbb{Z} \longrightarrow \mathbb{Z}/m_1\mathbb{Z} \times \cdots \times \mathbb{Z}/m_k\mathbb{Z}$$

is an isomorphism of rings. It is worth drawing the picture in a small case, since it makes the statement obvious: below is $\mathbb{Z}/15\mathbb{Z}$ laid out as a 3×5 grid, the row recording the residue mod 3 and the column the residue mod 5. Every cell is filled exactly once, which is precisely the content of the theorem.

		mod 5				
		0	1	2	3	4
mod 3	0	0	6	12	3	9
	1	10	1	7	13	4
	2	5	11	2	8	14

Since θ is a ring isomorphism it restricts to an isomorphism on unit groups,

$$(\mathbb{Z}/M\mathbb{Z})^\times \cong (\mathbb{Z}/m_1\mathbb{Z})^\times \times \cdots \times (\mathbb{Z}/m_k\mathbb{Z})^\times,$$

and comparing sizes gives the following.

Corollary 36.3.8

If $m_1, \dots, m_k$ are pairwise coprime and $M = \prod_i m_i$, then $\phi(M) = \prod_i \phi(m_i)$.

Remark. A warning: one cannot in general split off prime powers further. For example $\mathbb{Z}/4\mathbb{Z}$ is *not* isomorphic to $(\mathbb{Z}/2\mathbb{Z})^2$.

§36.3.3 Multiplicative Functions

The totient function is the first of several arithmetic functions we will meet, all of which share a convenient property.

Definition 36.3.9

A function $f : \mathbb{N} \rightarrow \mathbb{C}$ is **multiplicative** if $f(mn) = f(m)f(n)$ whenever $(m, n) = 1$, and **totally multiplicative** if $f(mn) = f(m)f(n)$ for all m, n .

A multiplicative function with f not identically zero has $f(1) = 1$, and is determined by its values on prime powers. Standard examples are ϕ , the divisor count $\tau(n) = |\{d : d \mid n\}|$, and more generally $\sigma_k(n) = \sum_{d \mid n} d^k$ (so $\sigma_0 = \tau$ and $\sigma_1 = \sigma$, the sum of divisors). None of these is totally multiplicative: $\tau(4) = 3 \neq 4 = \tau(2)^2$.

That σ_k is multiplicative is not quite obvious, and it is worth isolating the reason.

Lemma 36.3.10

If f is multiplicative then so is $g(n) = \sum_{d \mid n} f(d)$.

Proof. If $(m, n) = 1$ then every divisor d of mn factors uniquely as $d = d_1 d_2$ with $d_1 \mid m$, $d_2 \mid n$, and automatically $(d_1, d_2) = 1$. So

$$g(mn) = \sum_{d \mid mn} f(d) = \sum_{d_1 \mid m} \sum_{d_2 \mid n} f(d_1 d_2) = \left(\sum_{d_1 \mid m} f(d_1) \right) \left(\sum_{d_2 \mid n} f(d_2) \right) = g(m)g(n). \quad \square$$

Taking $f(n) = n^k$, which is totally multiplicative, gives that σ_k is multiplicative.

Theorem 36.3.11

Let p be prime and $m, n \geq 1$. Then

- (i) $\phi(p^m) = p^m - p^{m-1} = p^m(1 - p^{-1})$;
- (ii) $\phi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$;
- (iii) $\sum_{d \mid n} \phi(d) = n$.

Proof. (i) Among $1, \dots, p^m$ the integers not coprime to p^m are exactly the multiples of p , of which there are p^{m-1} .

(ii) Follows from (i) and [Corollary 36.3.8](#).

(iii) The function $g(n) = \sum_{d \mid n} \phi(d)$ is multiplicative by [Lemma 36.3.10](#), and $g(p^m) = 1 + (p-1) + \dots + (p^m - p^{m-1}) = p^m$ telescopes. So g and the identity agree on prime powers and are both multiplicative, hence agree everywhere. $\square$

An alternative proof of (iii) which is worth remembering: classify $1 \leq m \leq n$ by the value of (m, n) . There are exactly $\phi(n/d)$ integers m with $(m, n) = d$, namely d times those integers up to n/d coprime to n/d ; summing over $d \mid n$ gives $n = \sum_{d \mid n} \phi(n/d) = \sum_{d \mid n} \phi(d)$.

§36.3.4 Polynomial Congruences

Linear congruences we understand completely. What about higher degree? Here are three warnings.

- Example 36.3.12** (i) $x^2 + 2 \equiv 0 \pmod{5}$ has no solutions, since the squares mod 5 are 0, 1, 4.
 (ii) $x^3 + 1 \equiv 0 \pmod{7}$ has the three solutions $x \equiv 3, 5, 6$.
 (iii) $x^2 - 1 \equiv 0 \pmod{8}$ has the four solutions $x \equiv 1, 3, 5, 7$.

The last of these should be startling: a quadratic with four roots. The explanation is that $\mathbb{Z}/8\mathbb{Z}$ is not an integral domain, and the familiar bound ‘degree n implies at most n roots’ needs one.

Definition 36.3.13

For a commutative ring R , write $R[X]$ for the ring of polynomials $a_0 + a_1X + \cdots + a_nX^n$ with $a_i \in R$.

Remark. Polynomials are equal when their coefficients are equal, which is not the same as the induced functions being equal. Over $R = \mathbb{Z}/p\mathbb{Z}$ the polynomials $X^p - X$ and 0 are distinct elements of $R[X]$ but define the same function $R \rightarrow R$, by Fermat’s little theorem.

Proposition 36.3.14 (Division Algorithm for Polynomials)

Let $f, g \in R[X]$ with $\deg g = m > 0$ and the leading coefficient of g invertible in R . Then there are $q, r \in R[X]$ with $f = qg + r$ and $\deg r < m$.

Proof. Induct on $n = \deg f$. If $n < m$ take $q = 0$, $r = f$. Otherwise write $f = aX^n + \cdots$ and $g = bX^m + \cdots$; then $f' = f - ab^{-1}X^{n-m}g$ has degree strictly less than n , so by induction $f' = q'g + r$ with $\deg r < m$, and $f = (ab^{-1}X^{n-m} + q')g + r$. $\square$

Corollary 36.3.15 (Remainder Theorem)

If $f \in R[X]$ and $\alpha \in R$ then $f(X) = (X - \alpha)q(X) + f(\alpha)$ for some $q \in R[X]$.

Proof. Divide by $X - \alpha$, whose leading coefficient is 1; the remainder has degree 0, so is a constant r , and putting $X = \alpha$ gives $r = f(\alpha)$. $\square$

Theorem 36.3.16

Let R be an integral domain and let $f \in R[X]$ be nonzero of degree n . Then f has at most n distinct roots in R .

Proof. Induct on n . For $n = 0$ the polynomial is a nonzero constant and has no roots. For $n > 0$, if f has no roots we are done; otherwise let α be a root, so $f = (X - \alpha)q$ with $\deg q = n - 1$. Any root β of f satisfies $(\beta - \alpha)q(\beta) = 0$, and

since R is a domain either $\beta = \alpha$ or $q(\beta) = 0$. By induction q has at most $n - 1$ roots, so f has at most n . $\square$

Corollary 36.3.17 (Lagrange)

Let p be prime and let $f(X) = a_0 + a_1X + \cdots + a_nX^n$ be an integer polynomial with $p \nmid a_n$. Then $f(x) \equiv 0 \pmod{p}$ has at most n solutions modulo p .

Proof. $\mathbb{Z}/p\mathbb{Z}$ is a field, hence an integral domain, and the reduction of f has degree n there. $\square$

A pretty application:

Theorem 36.3.18 (Wilson)

If p is prime then $(p - 1)! \equiv -1 \pmod{p}$.

Proof. Consider $f(X) = X^{p-1} - 1 - \prod_{a=1}^{p-1} (X - a)$ over $\mathbb{Z}/p\mathbb{Z}$. Both terms are monic of degree $p - 1$, so $\deg f \leq p - 2$; but by Fermat's little theorem f vanishes at each of $1, \dots, p - 1$, which is $p - 1$ roots. By Theorem 36.3.16 we must have $f = 0$. Comparing constant terms, $-1 - (-1)^{p-1}(p - 1)! \equiv 0$, so $(p - 1)! \equiv -1 \pmod{p}$ for odd p , and directly for $p = 2$. $\square$

Remark. Alternatively, pair each $a \in \{1, \dots, p - 1\}$ with its inverse. The pairing is a genuine involution with no fixed points except where $a^2 \equiv 1$, i.e. $a \equiv \pm 1$, so the product collapses to $1 \times (p - 1) \equiv -1$.

§36.3.5 Primitive Roots

We now come to the real business of this section: understanding $(\mathbb{Z}/n\mathbb{Z})^\times$ as a group.

Definition 36.3.19

For $(a, n) = 1$, the **order** of a modulo n , written $\text{ord}_n(a)$, is the least $d \geq 1$ with $a^d \equiv 1 \pmod{n}$. We call a a **primitive root** modulo n if $\text{ord}_n(a) = \phi(n)$, that is, if a generates $(\mathbb{Z}/n\mathbb{Z})^\times$.

By Lagrange's theorem $\text{ord}_n(a) \mid \phi(n)$ always, and more generally $a^m \equiv 1 \pmod{n}$ if and only if $\text{ord}_n(a) \mid m$.

Theorem 36.3.20

If p is prime then $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic.

Proof. Let $\psi(d)$ be the number of elements of $(\mathbb{Z}/p\mathbb{Z})^\times$ of order exactly d . Every element has some order dividing $p - 1$, so

$$\sum_{d \mid p-1} \psi(d) = p - 1.$$

We claim $\psi(d) \leq \phi(d)$ for each $d \mid p-1$. Indeed, if $\psi(d) = 0$ this is clear. Otherwise pick a of order d ; then $a, a^2, \dots, a^d$ are d distinct solutions of $X^d - 1 \equiv 0 \pmod{p}$, and by [Corollary 36.3.17](#) they are *all* of them. So every element of order d lies in $\langle a \rangle$, and a cyclic group of order d has exactly $\phi(d)$ generators. Hence $\psi(d) \leq \phi(d)$.

But $\sum_{d \mid p-1} \phi(d) = p-1$ by [Theorem 36.3.11](#)(iii), so the inequalities $\psi(d) \leq \phi(d)$ must all be equalities. In particular $\psi(p-1) = \phi(p-1) \geq 1$, so a primitive root exists. $\square$

Example 36.3.21

Take $p = 19$, so $\phi(19) = 18$. The order of 2 divides 18, so it is one of 1, 2, 3, 6, 9, 18. Now $2^9 = 512 = 26 \times 19 + 18 \equiv -1 \pmod{19}$, so the order does not divide 9; and $2^6 = 64 \equiv 7 \not\equiv 1$. So $\text{ord}_{19}(2) = 18$ and 2 is a primitive root modulo 19. There are $\phi(18) = 6$ primitive roots in all, namely 2, 3, 10, 13, 14, 15.

Remark. The proof of [Theorem 36.3.20](#) is thoroughly non-constructive: it tells us that $\phi(p-1)$ of the residues are primitive roots but gives us no way of finding one except by trial. It is a genuinely open problem how small the least primitive root modulo p must be. Unconditionally one can show it is $O(p^{1/4+\varepsilon})$; on the generalised Riemann hypothesis one gets $O(\log^6 p)$. Relatedly:

Conjecture 36.3.22 (Artin). For any integer g which is not a perfect square and not -1 , there are infinitely many primes p for which g is a primitive root.

This is open for every single g , though it is known that at least one of 2, 3, 5 works.

Let us now deal with prime powers. Powers of 2 are badly behaved.

Example 36.3.23

$(\mathbb{Z}/8\mathbb{Z})^\times = \{\pm 1, \pm 3\}$, and every element squares to 1: $3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$. So $(\mathbb{Z}/8\mathbb{Z})^\times$ has exponent 2 and cannot be cyclic (it is a Klein four-group). Since reduction gives a surjection $(\mathbb{Z}/2^k\mathbb{Z})^\times \rightarrow (\mathbb{Z}/8\mathbb{Z})^\times$ for $k \geq 3$, and a quotient of a cyclic group is cyclic, no $(\mathbb{Z}/2^k\mathbb{Z})^\times$ with $k \geq 3$ is cyclic either.

For odd primes the situation is much better, and rests on the following lifting computation.

Lemma 36.3.24

Let $p > 2$ be prime, $k \geq 1$ and $y \in \mathbb{Z}$. Then $(1 + p^k y)^p \equiv 1 + p^{k+1} y \pmod{p^{k+2}}$.

Proof. Expanding by the binomial theorem,

$$(1 + p^k y)^p = 1 + p^{k+1} y + \sum_{j=2}^{p-1} \binom{p}{j} (p^k y)^j + p^{kp} y^p.$$

For $2 \leq j \leq p-1$ we have $p \mid \binom{p}{j}$ and $kj \geq k+2$ (since $k \geq 1$, $j \geq 2$), so those terms are divisible by p^{k+2} comfortably. The last term has exponent $kp \geq 3k \geq k+2$ since $p \geq 3$ and $k \geq 1$. The claim follows. $\square$

Remark. It is exactly here that $p = 2$ fails: for $p = 2, k = 1$ we would need $kp = 2 \geq k + 2 = 3$, which is false, and indeed $(1 + 2)^2 = 9 \equiv 1 \not\equiv 1 + 4 \pmod{8}$. For $p = 2$ and $k \geq 2$ the same proof does work.

Lemma 36.3.25

Let $p > 2$ be prime and $k \geq 1$. If g is a primitive root modulo p with $g^{p-1} \not\equiv 1 \pmod{p^2}$, then g is a primitive root modulo p^k for every $k \geq 1$.

Proof. The case $k = 1$ is given, so assume $k \geq 2$. Let $d = \text{ord}_{p^k}(g)$, so $d \mid \phi(p^k) = p^{k-1}(p-1)$. Reducing modulo p , we see $g^d \equiv 1 \pmod{p}$, so $(p-1) \mid d$. Hence $d = p^j(p-1)$ for some $0 \leq j \leq k-1$, and it suffices to rule out $j \leq k-2$; equivalently to show $g^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$.

Write $g^{p-1} = 1 + py$ with $p \nmid y$, which is exactly the hypothesis. Applying Lemma 36.3.24 repeatedly,

$$g^{p^i(p-1)} = (1 + py)^{p^i} \equiv 1 + p^{i+1}y \pmod{p^{i+2}},$$

by induction on i . Taking $i = k-2$ gives $g^{p^{k-2}(p-1)} \equiv 1 + p^{k-1}y \pmod{p^k}$, and $p \nmid y$ means this is not $1 \pmod{p^k}$. $\square$

Theorem 36.3.26

If $p > 2$ is prime then $(\mathbb{Z}/p^k\mathbb{Z})^\times$ is cyclic for every $k \geq 1$.

Proof. Let g be a primitive root modulo p . If $g^{p-1} \not\equiv 1 \pmod{p^2}$ we are done by Lemma 36.3.25. Otherwise set $h = (1+p)g$, which is still a primitive root modulo p since $h \equiv g \pmod{p}$. Now

$$h^{p-1} = (1+p)^{p-1}g^{p-1} \equiv (1+p)^{p-1} \equiv 1 + (p-1)p \equiv 1 - p \pmod{p^2},$$

which is not 1 modulo p^2 . So h satisfies the hypothesis of Lemma 36.3.25. $\square$

Example 36.3.27

We saw that 3 is a primitive root modulo 7. Now $3^6 - 1 = 728 = 7 \times 104$, and 104 is not divisible by 7, so $3^6 \not\equiv 1 \pmod{49}$. Hence 3 is a primitive root modulo 7^k for every k .

We can now settle the question completely.

Theorem 36.3.28

There is a primitive root modulo n if and only if $n = 1, 2, 4, p^k$ or $2p^k$, where p is an odd prime.

Proof. For $n = 1, 2, 4$ one checks directly (1, 1 and 3 respectively). The case $n = p^k$ is Theorem 36.3.26. For $n = 2p^k$, note $\phi(2p^k) = \phi(p^k)$, and if g is a primitive root

modulo p^k then whichever of g and $g + p^k$ is odd is coprime to $2p^k$ and still has order $\phi(p^k)$ modulo p^k , hence modulo $2p^k$.

Conversely, suppose n is none of these. If $n = 2^k$ with $k \geq 3$ we have seen that $(\mathbb{Z}/n\mathbb{Z})^\times$ is not cyclic. Otherwise n has at least two coprime factors $n = rs$ with $r, s > 2$ and $(r, s) = 1$, so $\phi(r)$ and $\phi(s)$ are both even. Then for any a coprime to n ,

$$a^{\phi(n)/2} = a^{\phi(r)\phi(s)/2} = (a^{\phi(r)})^{\phi(s)/2} \equiv 1 \pmod{r},$$

and symmetrically modulo s , so $a^{\phi(n)/2} \equiv 1 \pmod{n}$ and no element has order $\phi(n)$. $\square$

For completeness, here is the structure in the remaining case.

Proposition 36.3.29

For $k \geq 3$ we have $(\mathbb{Z}/2^k\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$, generated by -1 and 5 .

Proof. The version of Lemma 36.3.24 valid for $p = 2, k \geq 2$ gives $(1 + 4y)^{2^i} \equiv 1 + 2^{i+2}y \pmod{2^{i+3}}$, and applying this with $y = 1$ shows $5^{2^{k-3}} \equiv 1 + 2^{k-1} \not\equiv 1 \pmod{2^k}$, whereas $5^{2^{k-2}} \equiv 1 \pmod{2^k}$. So 5 has order 2^{k-2} . Also $-1 \notin \langle 5 \rangle$, since every power of 5 is $1 \pmod{4}$. Hence $\langle -1 \rangle \times \langle 5 \rangle$ has order $2^{k-1} = \phi(2^k)$ and is everything. $\square$

Combining with the Chinese remainder theorem, we have a complete description of $(\mathbb{Z}/n\mathbb{Z})^\times$ for every n : it is a product of cyclic groups, one for each prime power in the factorisation of n (with the powers of 2 contributing a $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{k-2}\mathbb{Z}$).

§36.4 Quadratic Residues

Having dealt with linear congruences, the next case is quadratic. When is $x^2 \equiv a \pmod{n}$ soluble? By the Chinese remainder theorem it is enough to answer this for prime powers, and by a lifting argument much like Lemma 36.3.24 it is essentially enough to answer it for primes. So we fix an odd prime p and ask which a are squares modulo p . The answer, quadratic reciprocity, is one of the great theorems of the subject, and is genuinely surprising: whether q is a square mod p turns out to be almost the same question as whether p is a square mod q , and there is no obvious reason why the two should have anything to do with one another.

§36.4.1 Euler's Criterion and Gauss's Lemma

Definition 36.4.1

Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. We say a is a **quadratic residue** modulo p if $x^2 \equiv a \pmod{p}$ for some x , and a **quadratic non-residue** otherwise.

Lemma 36.4.2

Let p be an odd prime. There are exactly $(p-1)/2$ quadratic residues modulo p .

Proof. Consider $\sigma : (\mathbb{Z}/p\mathbb{Z})^\times \rightarrow (\mathbb{Z}/p\mathbb{Z})^\times$, $x \mapsto x^2$. This is two-to-one, since $x^2 \equiv y^2$ if and only if $p \mid (x+y)(x-y)$ if and only if $x \equiv \pm y$, and $y \not\equiv -y$ as p is odd. The residues are the image of σ , which therefore has size $(p-1)/2$. $\square$

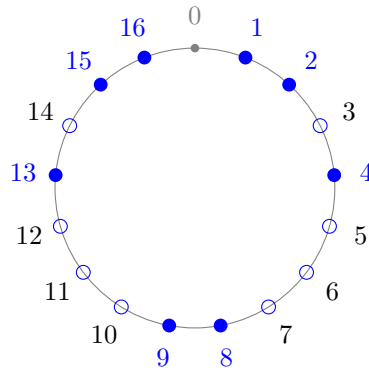
Remark. Equivalently, taking a primitive root g , the residues are exactly the even powers $\{g^{2n}\}$, of which there are $(p-1)/2$. So the squares form the unique index-two subgroup of the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$.

Definition 36.4.3 (Legendre Symbol)

Let p be an odd prime and $a \in \mathbb{Z}$. The **Legendre symbol** is

$$\left(\frac{a}{p}\right) = \begin{cases} 0 & \text{if } p \mid a, \\ 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Here is a picture of the situation modulo 17: the residues 1, 2, 4, 8, 9, 13, 15, 16 are shown as filled dots and the non-residues 3, 5, 6, 7, 10, 11, 12, 14 as open ones.



Notice that the picture is symmetric about the vertical axis, which reflects the fact that -1 is a residue mod 17 (so a and $-a$ are residues together). We will prove that shortly. Notice also that there is no visible pattern in which dots are filled – and indeed finding the least non-residue $n(p)$ quickly is an open problem, the best unconditional bound being $n(p) = O(p^\theta)$ for any $\theta > 1/(4\sqrt{e})$.

The first serious tool is the following.

Theorem 36.4.4 (Euler's Criterion)

Let p be an odd prime and $a \in \mathbb{Z}$. Then

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Proof. If $p \mid a$ both sides are 0, so assume $p \nmid a$. By Fermat's little theorem $(a^{(p-1)/2})^2 = a^{p-1} \equiv 1$, so $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$.

If $a \equiv x^2$ is a residue then $a^{(p-1)/2} \equiv x^{p-1} \equiv 1$. So all $(p-1)/2$ residues are roots of $X^{(p-1)/2} - 1$; but by [Corollary 36.3.17](#) this polynomial has at most $(p-1)/2$ roots, so they are exactly the residues. Hence if a is a non-residue then $a^{(p-1)/2} \equiv -1$. $\square$

Since p is odd, $0, 1, -1$ are distinct modulo p , so this congruence really does determine the symbol. Two immediate consequences:

Corollary 36.4.5

For an odd prime p and $a, b \in \mathbb{Z}$,

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \left(\frac{b}{p}\right).$$

In other words $a \mapsto \left(\frac{a}{p}\right)$ is a homomorphism $(\mathbb{Z}/p\mathbb{Z})^\times \rightarrow \{\pm 1\}$. In particular the product of two non-residues is a residue, which is not obvious from the definition.

Corollary 36.4.6

For an odd prime p ,

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Euler's criterion is already an algorithm: computing $a^{(p-1)/2} \pmod{p}$ by repeated squaring (write $(p-1)/2$ in binary and multiply together the relevant a^{2^i}) takes $O(\log p)$ multiplications, so the Legendre symbol is computable in polynomial time. We will find something better still, but first we need a more refined counting statement.

Definition 36.4.7

The **numerically least residue** of a modulo n is the unique a' with $a' \equiv a \pmod{n}$ and $-n/2 < a' \leq n/2$.

Theorem 36.4.8 (Gauss's Lemma)

Let p be an odd prime and a with $p \nmid a$. For $1 \leq j \leq (p-1)/2$, let a_j be the numerically least residue of aj modulo p . Then

$$\left(\frac{a}{p}\right) = (-1)^\mu, \quad \mu = \left| \left\{ 1 \leq j \leq \frac{p-1}{2} : a_j < 0 \right\} \right|.$$

Proof. Write $a_j = \varepsilon_j c_j$ with $\varepsilon_j \in \{\pm 1\}$ and $1 \leq c_j \leq (p-1)/2$; then $\mu = |\{j : \varepsilon_j = -1\}|$. We claim the c_j are distinct. If $c_j = c_k$ then $aj \equiv \pm ak \pmod{p}$, so $j \equiv \pm k$; but $j, k \in \{1, \dots, (p-1)/2\}$, so $j \equiv -k$ is impossible and $j = k$.

Hence $\{c_1, \dots, c_{(p-1)/2}\} = \{1, \dots, (p-1)/2\}$ and

$$a^{(p-1)/2} \left(\frac{p-1}{2}\right)! \equiv \prod_{j=1}^{(p-1)/2} (a_j) \equiv \prod_{j=1}^{(p-1)/2} \varepsilon_j c_j \equiv (-1)^\mu \left(\frac{p-1}{2}\right)! \pmod{p}.$$

Cancelling the factorial, which is coprime to p , gives $a^{(p-1)/2} \equiv (-1)^\mu$, and Euler's criterion finishes the job. $\square$

Let us use it.

Example 36.4.9

Take $a = 2$. Then $2j$ for $1 \leq j \leq (p-1)/2$ runs over the even numbers $2, 4, \dots, p-1$, and $2j$ has negative least residue exactly when $2j > p/2$, i.e. $j > p/4$. So

$$\mu = \left| \left\{ \frac{p}{4} < j \leq \frac{p-1}{2} \right\} \right| = \frac{p-1}{2} - \left\lfloor \frac{p}{4} \right\rfloor.$$

Checking the four cases $p \equiv 1, 3, 5, 7 \pmod{8}$ gives μ even, odd, odd, even respectively, so

$$\left(\frac{2}{p} \right) = (-1)^{(p^2-1)/8} = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{if } p \equiv \pm 3 \pmod{8}. \end{cases}$$

For instance $\left(\frac{2}{17} \right) = 1$, consistent with $6^2 = 36 \equiv 2 \pmod{17}$.

Example 36.4.10

Taking $a = -1$, every $-j$ has negative least residue, so $\mu = (p-1)/2$ and we recover [Corollary 36.4.6](#).

§36.4.2 Quadratic Reciprocity**Theorem 36.4.11** (Law of Quadratic Reciprocity)

Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q} \right) \left(\frac{q}{p} \right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Equivalently, $\left(\frac{p}{q} \right) = \left(\frac{q}{p} \right)$ unless $p \equiv q \equiv 3 \pmod{4}$, in which case $\left(\frac{p}{q} \right) = -\left(\frac{q}{p} \right)$.

Before the proof, let us set up the counting. Consider the rectangle of lattice points

$$S = \left\{ (j, m) \in \mathbb{Z}^2 : 1 \leq j \leq \frac{p-1}{2}, 1 \leq m \leq \frac{q-1}{2} \right\},$$

so that $|S| = \frac{p-1}{2} \cdot \frac{q-1}{2}$. By Gauss's lemma, $\left(\frac{q}{p} \right) = (-1)^\mu$ where μ counts those j with qj having negative least residue mod p , that is, those j for which there exists m with

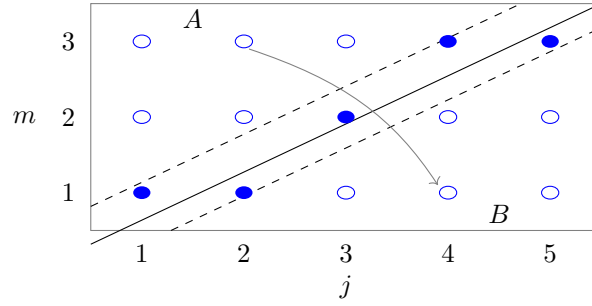
$$(m - \tfrac{1}{2})p < qj < mp, \quad \text{i.e.} \quad 0 < mp - qj < \tfrac{p}{2}.$$

Such an m is unique, and satisfies $1 \leq m \leq (q-1)/2$: indeed $mp < qj + p/2 \leq q(p-1)/2 + p/2 < qp/2$, so $m < q/2$. So $\mu = |\{(j, m) \in S : 0 < mp - qj < p/2\}|$, and symmetrically $\left(\frac{p}{q} \right) = (-1)^\nu$ with $\nu = |\{(j, m) \in S : 0 < qj - mp < q/2\}|$.

Note that $mp - qj$ is never zero on S , since $p \nmid qj$. So S is partitioned into four pieces: the two 'bands' counted by μ and ν , together with

$$A = \left\{ (j, m) \in S : mp - qj > \tfrac{p}{2} \right\}, \quad B = \left\{ (j, m) \in S : qj - mp > \tfrac{q}{2} \right\}.$$

The picture for $p = 11$, $q = 7$ looks as follows, with the fifteen points of S drawn and the three lines $mp = qj$, $mp - qj = p/2$ and $qj - mp = q/2$ marked. The filled points are those counted by μ and ν ; the open points make up A (above) and B (below).



Here $\mu = 3$ (the points $(1, 1), (3, 2), (4, 3)$) and $\nu = 2$ (the points $(2, 1), (5, 3)$), while $|A| = |B| = 5$. The grey arrow indicates the symmetry which pairs A with B ; it takes $(2, 3)$ to $(4, 1)$.

Proof of Theorem 36.4.11. With the notation above, S is the disjoint union of the μ -set, the ν -set, A and B , so

$$\frac{p-1}{2} \cdot \frac{q-1}{2} = |S| = \mu + \nu + |A| + |B|.$$

We claim $|A| = |B|$. Consider the map

$$(j, m) \mapsto (j', m') = \left(\frac{p+1}{2} - j, \frac{q+1}{2} - m \right),$$

which is an involution of S . A short computation gives

$$qj' - m'p = q \left(\frac{p+1}{2} - j \right) - p \left(\frac{q+1}{2} - m \right) = \frac{q-p}{2} - (qj - mp) = (mp - qj) - \frac{p-q}{2}.$$

So if $(j, m) \in A$, meaning $mp - qj > p/2$, then $qj' - m'p > p/2 - (p-q)/2 = q/2$ and $(j', m') \in B$. By symmetry the involution maps B into A , so it restricts to a bijection $A \rightarrow B$ and $|A| = |B|$.

Therefore $\mu + \nu \equiv |S| = \frac{p-1}{2} \cdot \frac{q-1}{2} \pmod{2}$, and

$$\left(\frac{p}{q} \right) \left(\frac{q}{p} \right) = (-1)^\nu (-1)^\mu = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}. \quad \square$$

The practical content is that $\left(\frac{p}{q} \right)$ is determined by $q \pmod{p}$ together with $p, q \pmod{4}$, and that we may flip symbols upside down at the cost of a sign. Combined with [Corollary 36.4.5](#), [Corollary 36.4.6](#) and [Example 36.4.9](#), this gives a fast way to evaluate symbols.

Example 36.4.12 (i) For a prime $p \geq 5$, reciprocity gives $\left(\frac{3}{p} \right) = (-1)^{(p-1)/2} \left(\frac{p}{3} \right)$, and unwinding the four cases modulo 12,

$$\left(\frac{3}{p} \right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{12}, \\ -1 & \text{if } p \equiv \pm 5 \pmod{12}. \end{cases}$$

(ii) Since $73 \equiv 1 \pmod{4}$ we may flip freely:

$$\left(\frac{19}{73}\right) = \left(\frac{73}{19}\right) = \left(\frac{16}{19}\right) = 1,$$

so 19 is a quadratic residue modulo 73. (Indeed $26^2 = 676 = 9 \times 73 + 19$.)

(iii) A longer one:

$$\begin{aligned} \left(\frac{34}{97}\right) &= \left(\frac{2}{97}\right) \left(\frac{17}{97}\right) = \left(\frac{17}{97}\right) = \left(\frac{97}{17}\right) \\ &= \left(\frac{12}{17}\right) = \left(\frac{3}{17}\right) \left(\frac{2}{17}\right)^2 = \left(\frac{3}{17}\right) = -1, \end{aligned}$$

using $97 \equiv 1 \pmod{8}$ for the second symbol and $17 \equiv 5 \pmod{12}$ at the end. So 34 is a non-residue modulo 97.

There is a catch. To apply reciprocity to $\left(\frac{a}{n}\right)$ we must first factorise a into primes, and factorisation is expensive. The fix is to extend the symbol so that it makes sense for composite ‘denominators’ too, and to check that reciprocity survives.

§36.4.3 The Jacobi Symbol

Definition 36.4.13 (Jacobi Symbol)

Let $n \geq 1$ be odd with $n = p_1 \cdots p_k$ a factorisation into (not necessarily distinct) primes, and let $a \in \mathbb{Z}$. The **Jacobi symbol** is

$$\left(\frac{a}{n}\right) = \prod_{i=1}^k \left(\frac{a}{p_i}\right),$$

a product of Legendre symbols, with the convention $\left(\frac{a}{1}\right) = 1$.

Note $\left(\frac{a}{n}\right) = 0$ exactly when $(a, n) \neq 1$.

Proposition 36.4.14

Let $m, n \geq 1$ be odd and $a, b \in \mathbb{Z}$. Then

- (i) $a \equiv b \pmod{n}$ implies $\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right)$;
- (ii) $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right) \left(\frac{b}{n}\right)$ and $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right) \left(\frac{a}{n}\right)$;
- (iii) $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ and $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$.

Proof. (i) and (ii) are immediate from the definition and the corresponding facts for the Legendre symbol. For (iii) one needs the elementary identities

$$\frac{x-1}{2} + \frac{y-1}{2} \equiv \frac{xy-1}{2} \pmod{2}, \quad \frac{x^2-1}{8} + \frac{y^2-1}{8} \equiv \frac{(xy)^2-1}{8} \pmod{2}$$

for odd x, y ; the first because $(x-1)(y-1) \equiv 0 \pmod{4}$, the second because $(x^2-1)(y^2-1) \equiv 0 \pmod{16}$. Writing $n = p_1 \cdots p_k$ and summing the identity over

the factors gives the result. $\square$

Theorem 36.4.15 (Reciprocity for the Jacobi Symbol)

Let $m, n \geq 1$ be odd. Then

$$\left(\frac{m}{n}\right) \left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}.$$

Proof. If $(m, n) > 1$ both sides of $\left(\frac{m}{n}\right) \left(\frac{n}{m}\right)$ vanish, so assume $(m, n) = 1$ and write $m = p_1 \cdots p_k$, $n = q_1 \cdots q_l$ with $\{p_i\} \cap \{q_j\} = \emptyset$. By Theorem 36.4.11,

$$\left(\frac{m}{n}\right) = \prod_{i=1}^k \prod_{j=1}^l \left(\frac{p_i}{q_j}\right) = \prod_{i=1}^k \prod_{j=1}^l (-1)^{\frac{p_i-1}{2} \cdot \frac{q_j-1}{2}} \left(\frac{q_j}{p_i}\right).$$

Using the first identity in the proof of Proposition 36.4.14 repeatedly, $\sum_i \frac{p_i-1}{2} \equiv \frac{m-1}{2}$ and $\sum_j \frac{q_j-1}{2} \equiv \frac{n-1}{2} \pmod{2}$, so the accumulated sign is $(-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}}$, and the remaining product is $\left(\frac{n}{m}\right)$. $\square$

Remark. The Jacobi symbol is *not* a test for quadratic residuacity. If a is a square modulo n then certainly $\left(\frac{a}{n}\right) = 1$, but the converse fails: $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right) \left(\frac{2}{5}\right) = (-1)(-1) = 1$, yet 2 is not a square modulo 15. In general the symbol only records the *parity* of the number of prime factors modulo which a fails to be a square.

What the Jacobi symbol buys us is speed. To compute $\left(\frac{a}{n}\right)$ we may reduce a modulo n , pull out powers of 2 using Proposition 36.4.14(iii), and flip using Theorem 36.4.15 – and at no point do we need to factorise anything. The resulting algorithm is essentially Euclid’s, and runs in polynomial time.

Example 36.4.16

Let us evaluate $\left(\frac{7411}{9283}\right)$. Both are odd; $7411 \equiv 3$ and $9283 \equiv 3 \pmod{4}$, so flipping costs a sign:

$$\left(\frac{7411}{9283}\right) = -\left(\frac{9283}{7411}\right) = -\left(\frac{1872}{7411}\right) = -\left(\frac{2}{7411}\right)^4 \left(\frac{117}{7411}\right) = -\left(\frac{117}{7411}\right),$$

since $1872 = 2^4 \times 117$. Now $117 \equiv 1 \pmod{4}$, so

$$-\left(\frac{117}{7411}\right) = -\left(\frac{7411}{117}\right) = -\left(\frac{40}{117}\right) = -\left(\frac{2}{117}\right)^3 \left(\frac{5}{117}\right) = -\left(\frac{5}{117}\right),$$

as $117 \equiv 5 \pmod{8}$ gives $\left(\frac{2}{117}\right) = -1$ and $(-1)^3 = -1$ combines with $40 = 2^3 \times 5$. Finally $\left(\frac{5}{117}\right) = \left(\frac{117}{5}\right) = \left(\frac{2}{5}\right) = -1$, so

$$\left(\frac{7411}{9283}\right) = -(-1) \cdot (-1) = -1.$$

No factorisation was required at any stage. Note that $9283 = 9283$ is in fact prime, so this really does tell us that 7411 is a non-residue modulo 9283.

Remark. One may ask about higher powers. If $p \equiv 2 \pmod{3}$ then every element of $(\mathbb{Z}/p\mathbb{Z})^\times$ is a cube (the map $x \mapsto x^3$ is injective since $3 \nmid p-1$), so the question is only interesting for $p \equiv 1 \pmod{3}$, where one defines a cubic residue symbol taking values in the cube roots of unity. The ultimate generalisation of all such statements is Artin's reciprocity law, which describes how primes factorise in abelian extensions of number fields; that is a story for Part III.

§36.5 Binary Quadratic Forms

Here is a question which looks like it belongs to the previous section but does not quite fit there: which integers are a sum of two squares? Equivalently, which N are in the image of the function $f(x, y) = x^2 + y^2$? Fermat knew the answer.

Theorem 36.5.1 (Sums of Two Squares)

Let $N \in \mathbb{N}$. Then $N = x^2 + y^2$ for some $x, y \in \mathbb{Z}$ if and only if every prime $p \equiv 3 \pmod{4}$ divides N to an even power.

We can immediately do half of it.

Proof of the 'only if' direction. Suppose $N = x^2 + y^2$ and $p \equiv 3 \pmod{4}$ is a prime dividing N . If $p \nmid y$ then y is invertible mod p and $(xy^{-1})^2 \equiv -1 \pmod{p}$, contradicting [Corollary 36.4.6](#). So $p \mid y$, and then $p \mid x$ too, so $p^2 \mid N$ and $N/p^2 = (x/p)^2 + (y/p)^2$. Repeating, p divides N to an even power. $\square$

For the converse we may reduce a good deal. Writing $N = M^2K$ with K squarefree, we need only represent K ; and by the identity

$$(x^2 + y^2)(w^2 + z^2) = (xw + yz)^2 + (xz - yw)^2,$$

which says that the norm on $\mathbb{Z}[i]$ is multiplicative, it suffices to represent each prime factor of K . Since $2 = 1^2 + 1^2$, everything comes down to the following: *every prime $p \equiv 1 \pmod{4}$ is a sum of two squares*. This turns out to require a genuinely new idea, and the idea is to consider not just $x^2 + y^2$ but all quadratic forms at once.

§36.5.1 Equivalence and Discriminants

Definition 36.5.2

A **binary quadratic form** (BQF) is a polynomial $f(x, y) = ax^2 + bxy + cy^2$ with $a, b, c \in \mathbb{Z}$. We often write $f = (a, b, c)$ for short. We say f **represents** n if $n = f(x, y)$ for some $x, y \in \mathbb{Z}$.

It is helpful to record f as a matrix. Setting

$$M_f = \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}, \quad f(x, y) = \begin{pmatrix} x & y \end{pmatrix} M_f \begin{pmatrix} x \\ y \end{pmatrix}.$$

The entries of M_f need not be integers, but its determinant is a quarter of an integer, which is all we shall need.

Definition 36.5.3

A **unimodular substitution** is a change of variables $(X, Y) = (x, y)A$ with $A \in \text{SL}_2(\mathbb{Z})$. Two BQFs f, g are **equivalent**, written $f \sim g$, if $g(x, y) = f(X, Y)$ for some such substitution.

Since $\text{SL}_2(\mathbb{Z})$ is a group, this is an equivalence relation; concretely, $\text{SL}_2(\mathbb{Z})$ acts on forms by $M_g = AM_fA^\top$. Because A and A^{-1} both have integer entries, equivalent forms represent exactly the same integers – which is the whole point of the definition.

Example 36.5.4

Let $f = (1, 0, 1)$ and $g = (4, 12, 10)$. Then

$$g(x, y) = 4x^2 + 12xy + 10y^2 = (2x + 3y)^2 + y^2 = f(2x + 3y, y),$$

so g is obtained from f by the substitution with matrix $\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix}$. But this matrix has determinant 2, not 1, and indeed f and g are *not* equivalent: f represents 1 and g does not. The inverse substitution does not have integer entries, so the argument above breaks down.

Definition 36.5.5

The **discriminant** of $f = (a, b, c)$ is $\text{disc}(f) = b^2 - 4ac = -4 \det M_f$.

The reason for the name is the completed square: for $a \neq 0$,

$$4af(x, y) = (2ax + by)^2 - dy^2, \quad d = \text{disc}(f),$$

which we shall use repeatedly.

Lemma 36.5.6

Equivalent forms have the same discriminant.

Proof. If $M_g = AM_fA^\top$ with $\det A = 1$ then $\det M_g = \det M_f$, so $\text{disc}(g) = -4 \det M_g = -4 \det M_f = \text{disc}(f)$. $\square$

The converse is false, and this failure is the entire subject: $x^2 + 6y^2$ and $2x^2 + 3y^2$ both have discriminant -24 , but the first represents 1 and the second does not, so they are inequivalent.

Lemma 36.5.7

There is a BQF of discriminant d if and only if $d \equiv 0$ or $1 \pmod{4}$.

Proof. If $d = b^2 - 4ac$ then $d \equiv b^2 \equiv 0, 1 \pmod{4}$. Conversely, if $d \equiv 0 \pmod{4}$ take $(1, 0, -d/4)$, and if $d \equiv 1 \pmod{4}$ take $(1, 1, (1-d)/4)$. $\square$

The form exhibited in each case is called the **principal form** of discriminant d .

Definition 36.5.8

A real quadratic form f in n variables is **positive definite** if $f(\underline{x}) > 0$ for all $\underline{x} \neq 0$, **negative definite** if $f(\underline{x}) < 0$ for all $\underline{x} \neq 0$, and **indefinite** if it takes both signs.

Lemma 36.5.9

Let $f = (a, b, c)$ be a BQF with $d = \text{disc}(f)$.

- (i) If $d < 0$ and $a > 0$, then f is positive definite (and $c > 0$).
- (ii) If $d < 0$ and $a < 0$, then f is negative definite.
- (iii) If $d > 0$, then f is indefinite.
- (iv) If $d = 0$, then $f = l(mx + ny)^2$ for some $l, m, n \in \mathbb{Z}$.

Proof. (i), (ii) From $4af(x, y) = (2ax + by)^2 - dy^2$ and $d < 0$, the right-hand side is positive unless $x = y = 0$. So f has the sign of a .

(iii) If $a = c = 0$ then $f(1, 1) = b$ and $f(1, -1) = -b$ have opposite signs (and $b \neq 0$ as $d > 0$). Otherwise assume $a \neq 0$; then $4af(1, 0) = 4a^2 > 0$ and $4af(-b, 2a) = (-2ab + 2ab)^2 - 4a^2d = -4a^2d < 0$, so $f(1, 0)$ and $f(-b, 2a)$ have opposite signs.

(iv) If $b^2 = 4ac$, write $a = lm^2$ with l squarefree. Then $b^2 = 4lm^2c$, so $l \mid (b/m)^2$. appropriately and one checks $2lm \mid b$; setting $n = b/(2lm)$ gives $f = l(mx + ny)^2$. $\square$

For the rest of this section we restrict to positive definite forms, so $d < 0$ and $a, c > 0$. Our aim is to pick a canonical representative in each equivalence class.

§36.5.2 Reduction

Two particular substitutions do all the work. With $A = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}$, taking $\lambda = \pm 1$ gives

$$T_{\pm} : (a, b, c) \mapsto (a, b \pm 2a, a \pm b + c),$$

which changes b by $2a$ and leaves a alone. With $A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ we get

$$S : (a, b, c) \mapsto (c, -b, a),$$

which swaps the outer coefficients. Between them we can make a small and $|b|$ small.

Definition 36.5.10

A positive definite BQF (a, b, c) is **reduced** if either $-a < b \leq a < c$, or $0 \leq b \leq a = c$.

The awkward second clause is there purely to break ties on the boundary; morally the condition is $|b| \leq a \leq c$.

Proposition 36.5.11

Every positive definite BQF is equivalent to a reduced form.

Proof. If $a > c$, apply S to decrease a while leaving $|b|$ unchanged. If $a \leq c$ and $|b| > a$, apply T_+ or T_- (whichever decreases $|b|$) to replace b by $b \mp 2a$; this leaves a unchanged and strictly decreases $|b|$. Since $a + |b|$ is a nonnegative integer which strictly decreases at every step, the process terminates.

We end with a form satisfying $a \leq c$ and $|b| \leq a$. If $a < c$ then either $b > -a$, in which case the form is reduced, or $b = -a$, in which case T_+ gives (a, a, c) which is reduced. If $a = c$ then either $0 \leq b$, and the form is reduced, or $-a \leq b < 0$, and S gives $(a, -b, a)$ which is reduced. $\square$

Lemma 36.5.12

If $f = (a, b, c)$ is a reduced positive definite BQF of discriminant d , then $b \equiv d \pmod{2}$ and

$$|b| \leq a \leq \sqrt{\frac{|d|}{3}}.$$

Proof. Since $d = b^2 - 4ac$ we have $d \equiv b^2 \equiv b \pmod{2}$. For the inequality, reducedness gives $|b| \leq a \leq c$, so

$$|d| = 4ac - b^2 \geq 4a^2 - a^2 = 3a^2,$$

whence $a^2 \leq |d|/3$. $\square$

In particular there are only finitely many reduced forms of a given discriminant: a is bounded, then b is bounded by a and determined modulo 2, and then $c = (b^2 - d)/(4a)$ is determined. This is an entirely effective search.

The next result says that reduced forms are canonical, and the key is that we can read off a reduced form from the integers it represents.

Lemma 36.5.13

Let $f = (a, b, c)$ be a reduced positive definite BQF. Then the least integers properly represented by f – that is, values $f(x, y)$ with $(x, y) = 1$ – are a , c and $a - |b| + c$, in that order.

Proof. Certainly $f(\pm 1, 0) = a$ and $f(0, \pm 1) = c$, and $0 < a \leq c$. Suppose $(x, y) = 1$ with x, y both nonzero, and say $|x| \geq |y| > 0$. Then

$$f(x, y) \geq a|x|^2 - |b||x||y| + c|y|^2 \geq (a - |b|)|x|^2 + c|y|^2 \geq a - |b| + c,$$

using $|x| \geq |y|$ and $a \geq |b|$. The case $|y| \geq |x| > 0$ is symmetric, using $c \geq a \geq |b|$. Finally $f(1, -\text{sign}(b)) = a - |b| + c$, so the bound is attained. $\square$

Theorem 36.5.14

Every positive definite BQF is equivalent to a *unique* reduced form.

Proof. Suppose (a, b, c) and (a', b', c') are equivalent reduced forms. Equivalent forms properly represent the same integers (a unimodular substitution preserves coprimality of (x, y) , since its inverse is also integral), so by Lemma 36.5.13 we get $a = a'$ and $c = c'$; then equality of discriminants gives $b = \pm b'$.

If $b = 0$ or $b = b'$ we are done, so suppose $b > 0$ and $(a, b, c) \sim (a, -b, c)$ with both reduced. Reducedness of $(a, -b, c)$ forces $-a < -b$, so $|b| < a$, and then reducedness of (a, b, c) with $b > 0$ and $|b| < a$ forces $a < c$. So $a < c < a - |b| + c$ are three strictly increasing values, and by Lemma 36.5.13 each is properly represented in essentially one way: $f(x, y) = a$ only for $(x, y) = (\pm 1, 0)$, and $f(x, y) = c$ only for $(x, y) = (0, \pm 1)$.

So if $A \in \mathrm{SL}_2(\mathbb{Z})$ realises the equivalence, it must send $(\pm 1, 0)$ to $(\pm 1, 0)$ and $(0, \pm 1)$ to $(0, \pm 1)$, forcing $A = \pm I$. But $\pm I$ acts trivially on forms, so $(a, b, c) = (a, -b, c)$ and $b = 0$, a contradiction. $\square$

Definition 36.5.15

For $d < 0$ with $d \equiv 0, 1 \pmod{4}$, the **class number** $h(d)$ is the number of reduced positive definite BQFs of discriminant d ; equivalently, by Theorem 36.5.14, the number of equivalence classes of positive definite BQFs of discriminant d .

- Example 36.5.16** (i) $d = -4$. By Lemma 36.5.12, $a \leq \sqrt{4/3} < 2$, so $a = 1$, hence $|b| \leq 1$ and $b \equiv 0 \pmod{2}$, so $b = 0$ and $c = 1$. Thus $h(-4) = 1$ and $(1, 0, 1)$ is the only class: *every* positive definite form of discriminant -4 is equivalent to $x^2 + y^2$.
- (ii) $d = -24$. Now $a \leq \sqrt{8} < 3$ and b is even. If $a = 1$ then $b = 0$ and $c = 6$. If $a = 2$ then $b \in \{0, \pm 2\}$; $b = \pm 2$ gives $c = (4 + 24)/8 \notin \mathbb{Z}$, so $b = 0$ and $c = 3$. Hence $h(-24) = 2$, with classes $(1, 0, 6)$ and $(2, 0, 3)$.
- (iii) $d = -23$. Here $a \leq \sqrt{23/3} < 3$ and b is odd. If $a = 1$ then $b = 1$ and $c = 6$. If $a = 2$ then $b = \pm 1$ and $c = (1 + 23)/8 = 3$, and both $(2, 1, 3)$ and $(2, -1, 3)$ are reduced since $-2 < \pm 1 \leq 2 < 3$. So $h(-23) = 3$.

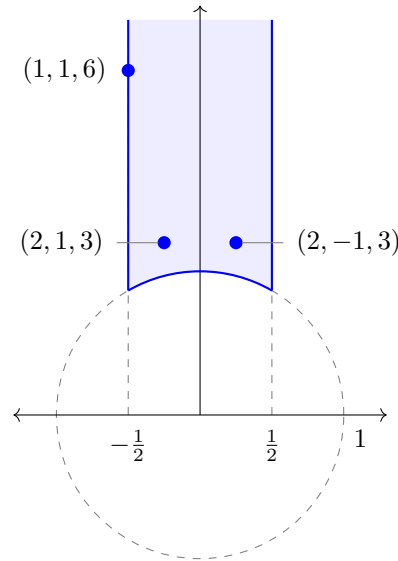
There is a geometric way to see all this which is worth knowing. To a positive definite $f = (a, b, c)$ of discriminant $d < 0$ associate the root

$$z_f = \frac{-b + i\sqrt{|d|}}{2a}$$

of $az^2 + bz + c$ lying in the upper half plane $\mathbb{H}$. Then $\mathrm{Re} z_f = -b/(2a)$ and $|z_f|^2 = c/a$, so the reducedness conditions $|b| \leq a$ and $a \leq c$ say exactly that

$$|\mathrm{Re} z_f| \leq \frac{1}{2} \quad \text{and} \quad |z_f| \geq 1,$$

which is the celebrated fundamental domain for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{H}$. The two operations $T_\pm$ and S correspond to $z \mapsto z \pm 1$ and $z \mapsto -1/z$, and the reduction algorithm is the familiar business of translating into the strip and inverting whenever you fall inside the unit circle. Below is the region, with the three reduced forms of discriminant -23 marked.



Remark. The class number is a delicate object. Heilbronn (1934) showed $h(d) \rightarrow \infty$ as $d \rightarrow -\infty$, and Mertens showed that on average $\frac{1}{N} \sum_{3 \leq -d \leq N} h(d) \sim \frac{\pi}{18} \sqrt{N}$. Determining all d with $h(d) = 1$ was a famous problem finally settled by Baker, Heegner and Stark: the fundamental such discriminants are

$$-3, -4, -7, -8, -11, -19, -43, -67, -163,$$

and there are a handful of non-fundamental ones besides. (A discriminant is **fundamental** if it is not of the form $k^2 d'$ with $k > 1$ and $d' \equiv 0, 1 \pmod{4}$.) For fundamental $d < 0$, Gauss defined a composition law making the classes into a finite abelian group, which turns out to be the ideal class group of $\mathbb{Q}(\sqrt{d})$.

§36.5.3 Representations

We now connect forms back to congruences.

Definition 36.5.17

A BQF f **properly represents** n if $n = f(x, y)$ for some x, y with $(x, y) = 1$.

Lemma 36.5.18

A BQF f properly represents n if and only if $f \sim (n, b, c)$ for some $b, c \in \mathbb{Z}$.

Proof. If $f \sim g = (n, b, c)$ then $g(1, 0) = n$ is properly represented by g , hence by f (proper representation being preserved by unimodular substitutions).

Conversely, suppose $f(\alpha, \beta) = n$ with $(\alpha, \beta) = 1$. By Bézout there are γ, δ with $\alpha\delta - \beta\gamma = 1$, so $A = \begin{pmatrix} \alpha & \gamma \\ \beta & \delta \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$. The form $g(x, y) = f(\alpha x + \gamma y, \beta x + \delta y)$ is equivalent to f and has $g(1, 0) = f(\alpha, \beta) = n$, so its leading coefficient is n . $\square$

Theorem 36.5.19

Let $n \in \mathbb{N}$ and $d \equiv 0, 1 \pmod{4}$. Then n is properly represented by *some* BQF of

discriminant d if and only if $x^2 \equiv d \pmod{4n}$ is soluble.

Proof. If n is properly represented by f with disc $f = d$, then by [Lemma 36.5.18](#) $f \sim (n, b, c)$, and $b^2 - 4nc = d$ gives $b^2 \equiv d \pmod{4n}$.

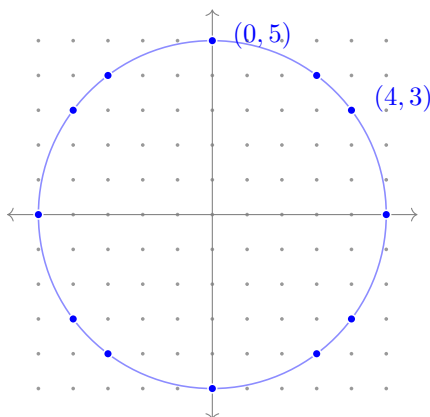
Conversely if $b^2 \equiv d \pmod{4n}$ then $b^2 = d + 4nc$ for some $c \in \mathbb{Z}$, and the form (n, b, c) has discriminant d and represents n properly at $(1, 0)$. $\square$

This is exactly what we need. When $h(d) = 1$ there is only one form to worry about, and the theorem becomes a complete criterion.

Proof of Theorem 36.5.1, completed. It remains to show that every prime $p \equiv 1 \pmod{4}$ is a sum of two squares. By [Corollary 36.4.6](#), $\left(\frac{-1}{p}\right) = 1$, so there is m with $m^2 \equiv -1 \pmod{p}$; replacing m by $m + p$ if necessary we may take m even, and then $(2m)^2 \equiv -4 \pmod{4p}$. So $x^2 \equiv -4 \pmod{4p}$ is soluble, and by [Theorem 36.5.19](#) some form of discriminant -4 properly represents p .

But $h(-4) = 1$ by [Example 36.5.16](#), so that form is equivalent to $(1, 0, 1)$, and equivalent forms represent the same integers. Hence $p = x^2 + y^2$ for some x, y . $\square$

Geometrically, asking whether N is a sum of two squares is asking whether the circle of radius $\sqrt{N}$ passes through a point of the integer lattice. Here is $N = 25$, where it passes through twelve of them, coming from the two essentially different representations $25 = 5^2 + 0^2 = 4^2 + 3^2$.



The method is not special to -4 . Let us run it for another discriminant with class number one.

Example 36.5.20

Which integers are properly represented by $f(x, y) = x^2 + xy + 2y^2$? We have disc $f = 1 - 8 = -7$, and f is positive definite. By [Lemma 36.5.12](#) a reduced form of discriminant -7 has $a \leq \sqrt{7/3} < 2$, so $a = 1$, and b odd with $|b| \leq 1$ gives $b = 1$, $c = 2$. So $h(-7) = 1$ and f is the only class.

By [Theorem 36.5.19](#), f properly represents n if and only if $x^2 \equiv -7 \pmod{4n}$ is soluble. Take $n = p$ an odd prime, $p \neq 7$. Then $x^2 \equiv -7 \pmod{4p}$ is soluble if and only if -7 is a square modulo 4 (always true, as $-7 \equiv 1$) and modulo p ; and by

reciprocity, using $\left(\frac{-1}{p}\right)\left(\frac{7}{p}\right) = \left(\frac{-7}{p}\right)$ and $-7 \equiv 1 \pmod{4}$,

$$\left(\frac{-7}{p}\right) = \left(\frac{p}{7}\right).$$

So f properly represents an odd prime $p \neq 7$ exactly when $p \equiv 1, 2, 4 \pmod{7}$, these being the quadratic residues mod 7. Adding in $f(0, 1) = 2$ and $f(1, -2) = 1 - 2 + 8 = 7$, we conclude that f properly represents a prime p if and only if $p \equiv 0, 1, 2, 4 \pmod{7}$.

To go from proper representations to all representations one notes that $f(kx, ky) = k^2 f(x, y)$, so the values of f are exactly $k^2 n$ with n properly represented; and to go from primes to general n one uses the following lifting lemma together with the Chinese remainder theorem.

Lemma 36.5.21

Let $a \in \mathbb{Z}$.

- (i) If $a \equiv 1 \pmod{8}$ then a is a square modulo 2^k for every $k \geq 1$.
- (ii) If $p > 2$ is prime and $\left(\frac{a}{p}\right) = 1$, then a is a square modulo p^k for every $k \geq 1$.

Proof. (ii) Induct on k . Given $x^2 \equiv a \pmod{p^k}$, say $x^2 = a + mp^k$, we look for a solution of the shape $x + tp^k$:

$$(x + tp^k)^2 = x^2 + 2xtp^k + t^2p^{2k} \equiv a + (m + 2xt)p^k \pmod{p^{k+1}}.$$

Since $p \nmid 2x$, we may choose t with $m + 2xt \equiv 0 \pmod{p}$, and then a is a square modulo p^{k+1} .

(i) The same idea, but $2x$ is now not invertible so one has to work slightly harder. Given $x^2 = a + m2^k$ with $k \geq 3$ and x odd: if m is even then $x^2 \equiv a \pmod{2^{k+1}}$ already. If m is odd, then

$$(x + 2^{k-1})^2 = x^2 + 2^k x + 2^{2k-2} \equiv a + 2^k(m + x) \pmod{2^{k+1}}$$

since $2k - 2 \geq k + 1$, and $m + x$ is even. The base case $k = 3$ is the hypothesis $a \equiv 1 \pmod{8}$. $\square$

Remark. If $h(d) > 1$ this analysis tells us which n are represented by *some* form of discriminant d , but not by which one. Sometimes extra congruence conditions separate the classes – for $d = -24$ one finds that $x^2 + 6y^2$ represents primes $p \equiv 1, 7 \pmod{24}$ while $2x^2 + 3y^2$ represents $p \equiv 5, 11 \pmod{24}$ – but in general no such separation exists, and one needs class field theory. The smallest failure is $d = -92$, where $x^2 + 23y^2$ and $3x^2 \pm 2xy + 8y^2$ cannot be told apart by congruences.

Remark. For indefinite forms everything changes: the number of representations of a given n is typically infinite. For instance $1 = x^2 - 2y^2$ has the solutions $(1, 0)$ and $(3, 2)$, and if (x, y) is a solution so is $(x^2 + 2y^2, 2xy)$, giving infinitely many. We will meet this again in §7 under the name of Pell's equation.

Finally, some context. Lagrange (1770) proved that every positive integer is a sum of

four squares, and Legendre (1797) that n is a sum of three squares if and only if n is not of the form $4^a(8b+7)$. Waring's problem asks for the least $s = g(k)$ such that every positive integer is a sum of s k -th powers; Hilbert (1909) showed $g(k)$ is finite for every k , and we know $g(2) = 4$, $g(3) = 9$, $g(4) = 19$, but the asymptotic behaviour of g remains open.

§36.6 The Distribution of Primes

The sequence $2, 3, 5, 7, 11, \dots$ is one of the most interesting objects in mathematics, and the obvious questions about it are hard. How fast does it grow? How regularly is it distributed? We know from §2 that it is infinite, and Euclid's proof gives the miserable bound $\pi(x) \gg \log \log x$; our first goal is to do considerably better.

Definition 36.6.1

The **prime counting function** is $\pi(x) = |\{p \leq x : p \text{ prime}\}|$.

The most convenient way to construct the primes up to x is the sieve of Eratosthenes: list $2, \dots, x$, and repeatedly take the smallest surviving number, declare it prime, and delete all of its proper multiples. Since any composite $n \leq x$ has a prime factor at most $\sqrt{x}$, we may stop once we pass $\sqrt{x}$. Here it is up to 60, where the primes are shown in blue and the numbers struck out are those deleted as multiples of 2, 3, 5 or 7.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

§36.6.1 What is True

Before working, let us record the theorems we would like to have. The first is the central result of the subject.

Theorem 36.6.2 (Prime Number Theorem)

$$\pi(x) \sim \frac{x}{\log x} \text{ as } x \rightarrow \infty.$$

We will not prove this – the proof needs complex analysis of the Riemann zeta function, and belongs to a Part III course – but we will meet the main character, and we will prove the correct order of magnitude.

Remark. Recall the asymptotic notation. For $f, g : \mathbb{R}_{>0} \rightarrow \mathbb{R}$ we write $f = O(g)$ as $x \rightarrow \infty$ if there are $C > 0$ and x_0 with $|f(x)| \leq C|g(x)|$ for all $x \geq x_0$; and $f = o(g)$ if for every $\varepsilon > 0$ there is x_0 with $|f(x)| \leq \varepsilon|g(x)|$ for $x \geq x_0$. Finally $f \sim g$ means $f = (1 + o(1))g$.

A better approximation to $\pi(x)$ than $x/\log x$ is the logarithmic integral

$$\text{Li}(x) = \int_2^x \frac{dt}{\log t},$$

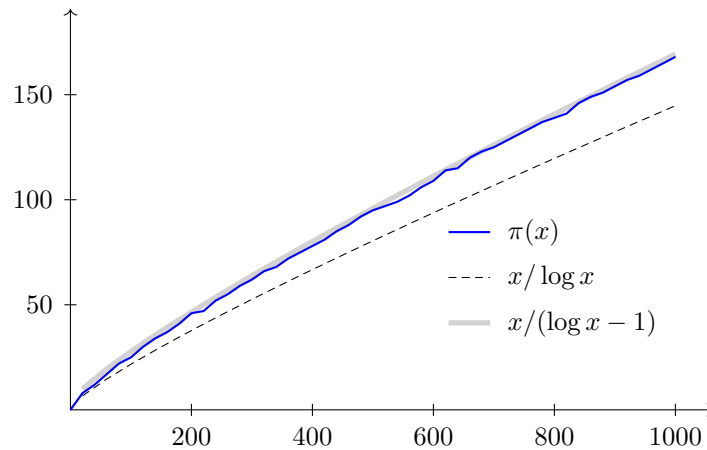
and the strongest known error term is of the following shape.

Theorem 36.6.3

There is a constant $C > 0$ such that

$$\pi(x) = \text{Li}(x) + O\left(x \exp\left(-C\sqrt{\log x}\right)\right).$$

The Riemann hypothesis is the assertion that the error is in fact $O(\sqrt{x} \log x)$, which is very much smaller. The following picture shows $\pi(x)$ against $x/\log x$ and against $x/(\log x - 1)$ for $x \leq 1000$; the last of these is a decent hands-on approximation, and gives a sense of just how good $\text{Li}(x)$ is.



A different flavour of question concerns primes in arithmetic progressions. In IA we saw that there are infinitely many primes congruent to 3 modulo 4; the general statement is much harder.

Theorem 36.6.4 (Dirichlet)

Let a, q be coprime positive integers. Then there are infinitely many primes $p \equiv a \pmod{q}$.

The coprimality is clearly necessary, since $(a, q) > 1$ divides every term of the progression. Quantitatively one expects the primes to be shared out equally among the $\phi(q)$ admissible residues, and this is true in a strong sense.

Theorem 36.6.5 (Siegel–Walfisz)

Let $(a, q) = 1$ and let $\pi(x; q, a)$ be the number of primes $p \leq x$ with $p \equiv a \pmod{q}$. For every $B > 0$ there is $C_B > 0$ such that

$$\pi(x; q, a) = \frac{x}{\phi(q) \log x} + O\left(x \exp\left(-C_B \sqrt{\log x}\right)\right)$$

uniformly for $q \leq (\log x)^B$.

Our own aim, which we will actually achieve, is the following.

Theorem 36.6.6 (Chebyshev)

For all $x \geq 4$,

$$\frac{\log 2}{2} \cdot \frac{x}{\log x} \leq \pi(x) \leq 6 \log 2 \cdot \frac{x}{\log x}.$$

That will take a while, so let us warm up.

§36.6.2 Elementary Bounds

Lemma 36.6.7

For $x \geq 1$ we have $\pi(x) \geq \frac{\log x}{2 \log 2}$.

Proof. Let $p_1, \dots, p_r$ be the primes up to x , so $r = \pi(x)$. Every $n \leq x$ can be written as $n = k^2 m$ with m squarefree, so $m = \prod_i p_i^{\alpha_i}$ with each $\alpha_i \in \{0, 1\}$ and $k \leq \sqrt{x}$. There are at most 2^r choices for m and at most $\sqrt{x}$ for k , so

$$x \leq \sqrt{x} \cdot 2^r = \sqrt{x} \cdot 2^{\pi(x)},$$

giving $\sqrt{x} \leq 2^{\pi(x)}$ and hence the claim. $\square$

This already beats Euclid handsomely: $\log x$ rather than $\log \log x$. It is still far from the truth, but the argument is worth remembering because the ‘squarefree times a square’ decomposition is genuinely useful.

Next we quantify the divergence of $\sum 1/p$. Euler’s observation that this series diverges is a much better proof that there are infinitely many primes than Euclid’s, because it says something about density.

Theorem 36.6.8 (Mertens) (i) For $x \geq 2$ we have $\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} \geq \log x$; in particular the product diverges.

(ii) For $x \geq 5$ we have $\sum_{p \leq x} \frac{1}{p} \geq \log \log x - \frac{1}{2}$; in particular $\sum_p 1/p$ diverges.

Here and below, p ranges over primes.

Proof. (i) Expanding each factor as a geometric series,

$$\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} = \prod_{p \leq x} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \cdots\right) \geq \sum_{n \leq x} \frac{1}{n},$$

since every $n \leq x$ has all its prime factors at most x , so $1/n$ appears exactly once when the product is expanded. Finally $\sum_{n \leq x} 1/n \geq \int_1^{\lfloor x \rfloor + 1} dy/y \geq \log x$.

(ii) Take logarithms in (i) and compare with the sum. Using $-\log(1-t) = t + t^2/2 + t^3/3 + \dots$,

$$\begin{aligned} \log \left(\prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} \right) - \sum_{p \leq x} \frac{1}{p} &= \sum_{p \leq x} \left(\frac{1}{2p^2} + \frac{1}{3p^3} + \dots \right) \\ &\leq \sum_{p \leq x} \frac{1}{2p^2} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \dots \right) \\ &= \sum_{p \leq x} \frac{1}{2p^2} \cdot \frac{1}{1 - p^{-1}} = \sum_{p \leq x} \frac{1}{2p(p-1)} \\ &\leq \sum_{2 \leq n \leq x} \frac{1}{2n(n-1)} = \frac{1}{2} \sum_{2 \leq n \leq x} \left(\frac{1}{n-1} - \frac{1}{n} \right) \leq \frac{1}{2}. \end{aligned}$$

Combining with (i) gives $\sum_{p \leq x} 1/p \geq \log \log x - 1/2$. $\square$

Remark. The truth is sharper: there are constants c and γ (the latter being Euler's constant, $\gamma \approx 0.5772$) with

$$\sum_{p \leq x} \frac{1}{p} = \log \log x + c + O\left(\frac{1}{\log x}\right), \quad \prod_{p \leq x} \left(1 - \frac{1}{p}\right)^{-1} = e^\gamma \log x + O(1).$$

For a sanity check, $\sum_{p \leq 100} 1/p \approx 1.803$ while $\log \log 100 \approx 1.527$.

§36.6.3 The Riemann Zeta Function

The expansion in the proof of [Theorem 36.6.8](#) is worth taking seriously. It is the case $s = 1$ of an identity which holds for all s with $\operatorname{Re} s > 1$, and which is the reason analysis has anything to say about primes at all.

Definition 36.6.9

For $s \in \mathbb{C}$ with $\operatorname{Re} s > 1$, the **Riemann zeta function** is

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Conventionally one writes $s = \sigma + it$.

Lemma 36.6.10

The series converges absolutely for $\sigma > 1$, and uniformly on $\{\sigma \geq 1 + \delta\}$ for any $\delta > 0$. In particular ζ is analytic on $\sigma > 1$.

Proof. $|n^{-s}| = n^{-\sigma}$, and $\sum n^{-\sigma}$ converges for $\sigma > 1$. Uniform convergence on $\sigma \geq 1 + \delta$ follows from the Weierstrass M -test with $M_n = n^{-1-\delta}$, and a uniform limit of analytic functions is analytic. $\square$

Theorem 36.6.11 (Euler Product)

For $\sigma > 1$,

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}.$$

Proof. Fix s with $\sigma > 1$ and let $x \geq 2$. Expanding the finite product,

$$\prod_{p \leq x} (1 - p^{-s})^{-1} = \prod_{p \leq x} (1 + p^{-s} + p^{-2s} + \cdots) = \sum_{n \in \mathcal{N}_x} n^{-s},$$

where $\mathcal{N}_x$ is the set of positive integers all of whose prime factors are at most x ; the rearrangement is legitimate by absolute convergence, and each n occurs exactly once by unique factorisation. Hence

$$\left| \zeta(s) - \prod_{p \leq x} (1 - p^{-s})^{-1} \right| = \left| \sum_{n \notin \mathcal{N}_x} n^{-s} \right| \leq \sum_{n > x} n^{-\sigma} \rightarrow 0$$

as $x \rightarrow \infty$, since every $n \notin \mathcal{N}_x$ has a prime factor exceeding x and so satisfies $n > x$. $\square$

This identity is the analytic form of unique factorisation, and it is the bridge we need: the left-hand side is a smooth analytic object, the right-hand side knows about primes.

Corollary 36.6.12

$\zeta(s) \neq 0$ for $\sigma > 1$.

Proof. For $x \geq 2$ let $P_x = \prod_{p \leq x} (1 - p^{-s})$, which is a finite product of nonzero numbers. Expanding as above,

$$\zeta(s)P_x = 1 + \sum_{n \in \mathcal{N}'_x} n^{-s},$$

where $\mathcal{N}'_x$ consists of those $n > 1$ all of whose prime factors exceed x ; every such n is $> x$, so

$$|\zeta(s)P_x| \geq 1 - \sum_{n > x} n^{-\sigma} \rightarrow 1$$

as $x \rightarrow \infty$. So $\zeta(s)P_x \neq 0$ for large x , whence $\zeta(s) \neq 0$. $\square$

Remark. One can extend ζ meromorphically to all of $\mathbb{C}$. The recipe uses the gamma function $\Gamma(z) = \int_0^\infty e^{-t} t^{z-1} dt$, defined for $\operatorname{Re} z > 0$ and continued meromorphically with simple poles at $0, -1, -2, \dots$; setting

$$\Xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s),$$

one shows the functional equation $\Xi(s) = \Xi(1-s)$, which continues Ξ and hence ζ everywhere. It turns out that ζ has a single simple pole, at $s = 1$ with residue 1, and ‘trivial’ zeros at $s = -2, -4, -6, \dots$ coming from the poles of $\Gamma(s/2)$. Since ζ does not vanish for $\sigma > 1$, the functional equation confines all other zeros to the **critical strip** $0 \leq \sigma \leq 1$. Non-vanishing on the line $\sigma = 1$ is already enough to prove the prime number

theorem; the Riemann hypothesis asserts that all the nontrivial zeros lie on $\sigma = \frac{1}{2}$.

Two more arithmetic functions, both of which have pleasant zeta-flavoured descriptions.

Definition 36.6.13 (Möbius Function)

Define $\mu : \mathbb{N} \rightarrow \{-1, 0, 1\}$ by $\mu(1) = 1$ and

$$\mu(n) = \begin{cases} (-1)^k & \text{if } n \text{ is a product of } k \text{ distinct primes,} \\ 0 & \text{if } n \text{ is not squarefree.} \end{cases}$$

Lemma 36.6.14

μ is multiplicative, and $\sum_{d|n} \mu(d) = \begin{cases} 1 & n = 1, \\ 0 & n > 1. \end{cases}$

Proof. Multiplicativity is immediate from the definition. Hence $\nu(n) = \sum_{d|n} \mu(d)$ is multiplicative by [Lemma 36.3.10](#), and $\nu(p^a) = \mu(1) + \mu(p) = 0$ for $a \geq 1$, while $\nu(1) = 1$. $\square$

This innocuous lemma is what makes inclusion–exclusion work in this setting, and it gives the Möbius inversion formula: if $g(n) = \sum_{d|n} f(d)$ for all n , then $f(n) = \sum_{e|n} \mu(e)g(n/e)$.

Definition 36.6.15

A **Dirichlet series** is a series of the form $\sum_{n \geq 1} a_n n^{-s}$ with $a_n \in \mathbb{C}$.

If $|a_n| \leq Cn^k$ then the series converges absolutely for $\sigma > k + 1$ and uniformly on $\sigma \geq k + 1 + \delta$. Multiplying two Dirichlet series gives a third, whose coefficients are a ‘Dirichlet convolution’:

$$\left(\sum_m \frac{a_m}{m^s} \right) \left(\sum_n \frac{b_n}{n^s} \right) = \sum_{N \geq 1} \frac{c_N}{N^s}, \quad c_N = \sum_{d|N} a_d b_{N/d}.$$

For instance for $\sigma > 2$,

$$\zeta(s-1)\zeta(s) = \left(\sum_m \frac{m}{m^s} \right) \left(\sum_n \frac{1}{n^s} \right) = \sum_{N \geq 1} \frac{1}{N^s} \sum_{d|N} d = \sum_{N \geq 1} \frac{\sigma(N)}{N^s},$$

and by [Lemma 36.6.14](#), $\zeta(s) \sum_n \mu(n)n^{-s} = 1$, i.e. $\sum_n \mu(n)n^{-s} = 1/\zeta(s)$.

Definition 36.6.16 (von Mangoldt Function)

Define $\Lambda : \mathbb{N} \rightarrow \mathbb{R}$ by

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for some prime } p \text{ and } k \geq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Proposition 36.6.17

For $\sigma > 1$,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

Proof. Taking logarithms in [Theorem 36.6.11](#), valid since ζ is nonvanishing on $\sigma > 1$,

$$\log \zeta(s) = - \sum_p \log(1 - p^{-s}).$$

Differentiating term by term (legitimate by local uniform convergence),

$$\frac{\zeta'(s)}{\zeta(s)} = - \sum_p \frac{p^{-s} \log p}{1 - p^{-s}} = - \sum_p (\log p) \sum_{k=1}^{\infty} p^{-ks} = - \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

since the only n contributing are prime powers p^k , each with weight $\log p$. $\square$

So Λ is a weighted indicator function of the prime powers, packaged so as to be visible to ζ . Setting $\psi(x) = \sum_{n \leq x} \Lambda(n)$, one can show that the prime number theorem is equivalent to $\psi(x) \sim x$, and the analytic proof proceeds by writing

$$\psi(x) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \left(-\frac{\zeta'(s)}{\zeta(s)} \right) \frac{x^s}{s} ds$$

for $c > 1$ and moving the contour left, picking up x from the pole of $-\zeta'/\zeta$ at $s = 1$ and a term x^ρ/ρ from each nontrivial zero ρ . One sees very concretely why the location of the zeros controls the error term.

§36.6.4 Counting Primes

We now come back to earth and prove Chebyshev's bounds. First, the sieve of Eratosthenes made quantitative.

Proposition 36.6.18 (Legendre's Formula)

Let $x > 1$ and let $P = \prod_{p \leq \sqrt{x}} p$. Then

$$\pi(x) - \pi(\sqrt{x}) + 1 = |\{1 \leq n \leq x : (n, P) = 1\}| = \sum_{d|P} \mu(d) \left\lfloor \frac{x}{d} \right\rfloor.$$

Proof. For the first equality: an integer $1 \leq n \leq x$ is coprime to P if and only if it has no prime factor at most $\sqrt{x}$, which happens exactly when $n = 1$ or n is a prime in $(\sqrt{x}, x]$. There are $1 + \pi(x) - \pi(\sqrt{x})$ such n .

For the second, use [Lemma 36.6.14](#) to detect coprimality:

$$|\{n \leq x : (n, P) = 1\}| = \sum_{n \leq x} \sum_{d|(n, P)} \mu(d) = \sum_{d|P} \mu(d) |\{n \leq x : d | n\}|,$$

which is $\sum_{d|P} \mu(d) \lfloor x/d \rfloor$. □

Example 36.6.19

Take $x = 100$, so $P = 2 \cdot 3 \cdot 5 \cdot 7 = 210$. The formula reads

$$100 - (50 + 33 + 20 + 14) + (16 + 10 + 7 + 6 + 4 + 2) - (3 + 2 + 1 + 0) + 0 = 22,$$

where the bracketed groups come from d with one, two, three and four prime factors. And indeed $\pi(100) - \pi(10) + 1 = 25 - 4 + 1 = 22$.

This is inclusion–exclusion, and it is exact, but the number of terms is $2^{\pi(\sqrt{x})}$, which grows far too fast to be useful. The successful approach instead extracts information from binomial coefficients, and the following simple observations are the engine.

Lemma 36.6.20

Let $n \geq 5$ and $N = \binom{2n}{n}$.

- (i) If $2n/3 < p \leq n$ then $p \nmid N$.
- (ii) If $p^k \mid N$ for some $k \geq 1$, then $p^k \leq 2n$.

Proof. (i) Write $N = \frac{(2n)(2n-1)\cdots(n+1)}{n(n-1)\cdots 1}$. If $2n/3 < p \leq n$ then $2p \leq 2n < 3p$, so exactly one multiple of p (namely $2p$) lies in $(n, 2n]$ and exactly one (namely p) in $[1, n]$; and $p^2 > 4n^2/9 > 2n$ for $n \geq 5$, so no higher power intervenes. So numerator and denominator each contain exactly one factor of p , which therefore cancels.

(ii) By Legendre's formula for v_p ,

$$v_p(N) = v_p((2n)!) - 2v_p(n!) = \sum_{i \geq 1} \left(\left\lfloor \frac{2n}{p^i} \right\rfloor - 2 \left\lfloor \frac{n}{p^i} \right\rfloor \right).$$

Each summand is $\lfloor 2y \rfloor - 2\lfloor y \rfloor \in \{0, 1\}$, and the summand vanishes once $p^i > 2n$. So if $p^k \mid N$, i.e. $v_p(N) \geq k$, then at least k summands are nonzero and hence $p^k \leq 2n$. □

Proof of Theorem 36.6.6. Upper bound. We first show $\pi(2^k) \leq 3 \cdot 2^k/k$ for all $k \geq 1$ by induction. For $1 \leq k \leq 6$ this is clear, since for even $n \geq 4$ we have $\pi(n) \leq n/2$ and so $\pi(2^k) \leq 2^{k-1} \leq 3 \cdot 2^k/k$ whenever $k \leq 6$.

For the inductive step, note that every prime $p \in (n, 2n]$ divides $\binom{2n}{n}$ exactly once (it divides the numerator and not the denominator), so

$$2^{2n} > \binom{2n}{n} \geq \prod_{n < p \leq 2n} p \geq n^{\pi(2n) - \pi(n)},$$

giving $\pi(2n) - \pi(n) \leq 2n \log 2 / \log n$. Applying this with $n = 2^k$,

$$\pi(2^{k+1}) \leq \pi(2^k) + \frac{2^{k+1} \log 2}{\log 2^k} = \pi(2^k) + \frac{2^{k+1}}{k} \leq \frac{3 \cdot 2^k}{k} + \frac{2^{k+1}}{k} = \frac{5 \cdot 2^{k+1}}{2k} \leq \frac{3 \cdot 2^{k+1}}{k+1},$$

the last step because $5/(2k) \leq 3/(k+1)$ for $k \geq 5$. This completes the induction.

Now given $x \geq 4$, choose k with $2^k < x \leq 2^{k+1}$, so $k \geq 2$. Since $y/\log y$ is increasing for $y \geq e$,

$$\pi(x) \leq \pi(2^{k+1}) \leq \frac{6 \cdot 2^k}{k+1} < \frac{6 \cdot 2^k}{k} = 6 \log 2 \cdot \frac{2^k}{\log 2^k} \leq 6 \log 2 \cdot \frac{x}{\log x}.$$

Lower bound. By [Lemma 36.6.20\(ii\)](#), $p^{v_p(N)} \leq 2n$ for every prime p , where $N = \binom{2n}{n}$; and only primes $p \leq 2n$ divide N . So

$$\log N = \sum_{p \leq 2n} v_p(N) \log p \leq \sum_{p \leq 2n} \frac{\log 2n}{\log p} \log p = \pi(2n) \log 2n.$$

On the other hand N is the largest of the $2n+1$ binomial coefficients $\binom{2n}{j}$, and

$$2^{2n} = \sum_{j=0}^{2n} \binom{2n}{j} = 2 + \sum_{j=1}^{2n-1} \binom{2n}{j} \leq 2 + (2n-1)N \leq 2nN$$

for $n \geq 1$. Hence $\log N \geq 2n \log 2 - \log 2n$, and combining,

$$\pi(2n) \geq \frac{2n \log 2 - \log 2n}{\log 2n} = \log 2 \cdot \frac{2n}{\log 2n} - 1.$$

Finally, for $x \geq 4$ pick n with $2n \leq x \leq 2n+2$; then $2n \geq x-2$ and $\log 2n \leq \log x$, so

$$\pi(x) \geq \pi(2n) \geq \log 2 \cdot \frac{x-2}{\log x} - 1 \geq \frac{\log 2}{2} \cdot \frac{x}{\log x},$$

the last inequality being equivalent to $\frac{\log 2}{2}x \geq 2 \log 2 + \log x$, which holds for $x \geq 11$. For $4 \leq x < 11$ the claimed bound asserts $\pi(x) \geq 1.51$, and $\pi(x) \geq 2$ throughout that range. $\square$

So $\pi(x)$ is of order $x/\log x$, which is the qualitative content of the prime number theorem. It is worth appreciating that the constants $\log 2/2 \approx 0.347$ and $6 \log 2 \approx 4.16$ bracket the truth of 1, so there is still work to be done – but the shape is right, and it was got by hand.

§36.6.5 Bertrand's Postulate

A pleasant application of the same circle of ideas is the following, conjectured by Bertrand in 1845 and proved by Chebyshev.

Theorem 36.6.21 (Bertrand's Postulate)

For every integer $n \geq 2$ there is a prime p with $n \leq p < 2n$.

The key input is a bound on the product of all primes up to x , which is essentially the

statement $\theta(x) = O(x)$ for Chebyshev's θ function.

Lemma 36.6.22

For $x \geq 1$, $P(x) = \prod_{p \leq x} p \leq 4^x$.

Proof. Since P is a step function it suffices to prove this for integers $x = n$, and we induct. The cases $n = 1, 2, 3$ are clear. If n is even and $n \geq 4$ then n is not prime, so $P(n) = P(n-1) \leq 4^{n-1} < 4^n$. So we may assume n is odd, say $n = 2m+1$ with $m \geq 1$.

Every prime p with $m+2 \leq p \leq 2m+1$ divides $\binom{2m+1}{m}$, since it divides the numerator $(2m+1)!$ once and neither $m!$ nor $(m+1)!$. Hence

$$\prod_{m+2 \leq p \leq 2m+1} p \mid \binom{2m+1}{m}, \quad 2 \binom{2m+1}{m} = \binom{2m+1}{m} + \binom{2m+1}{m+1} \leq 2^{2m+1},$$

so $\binom{2m+1}{m} \leq 4^m$. Therefore

$$P(2m+1) = P(m+1) \prod_{m+2 \leq p \leq 2m+1} p \leq 4^{m+1} \cdot 4^m = 4^{2m+1},$$

using the inductive hypothesis for $P(m+1)$. □

Corollary 36.6.23

$\pi(x) \leq \frac{x \log 4}{\log x}$ for $x \geq 2$.

Proof. Every prime in $(\sqrt{x}, x]$ contributes a factor exceeding $\sqrt{x}$ to $P(x)$, so

$$x^{(\pi(x) - \pi(\sqrt{x}))/2} \leq P(x) \leq 4^x.$$

Taking logarithms and absorbing $\pi(\sqrt{x}) \leq \sqrt{x}$ gives the bound. □

Proof of Theorem 36.6.21. Suppose for contradiction that there is no prime in $[n, 2n)$, and set $N = \binom{2n}{n}$. Any prime $p \mid N$ satisfies $p \leq 2n$; and $p \neq 2n$ unless $n = 1$, while our assumption rules out $n \leq p < 2n$. So $p < n$, and then Lemma 36.6.20(i) forces $p \leq 2n/3$.

Split $N = N_1 N_2$ where $N_1 = \prod_{v_p(N)=1} p$ and $N_2 = \prod_{v_p(N) \geq 2} p^{v_p(N)}$. By Lemma 36.6.22,

$$N_1 \leq \prod_{p \leq 2n/3} p = P(2n/3) \leq 4^{2n/3}.$$

For N_2 : if $v_p(N) \geq 2$ then $p^2 \leq p^{v_p(N)} \leq 2n$ by Lemma 36.6.20(ii), so $p \leq \sqrt{2n}$, and

each such prime contributes at most $2n$. Hence

$$N_2 \leq \prod_{p \leq \sqrt{2n}} 2n \leq (2n)^{\sqrt{2n}}.$$

Also $N = \binom{2n}{n} \geq 4^n/(2n)$, as in the proof of [Theorem 36.6.6](#). Combining,

$$\frac{4^n}{2n} \leq N \leq 4^{2n/3}(2n)^{\sqrt{2n}},$$

and taking logarithms, $\frac{n \log 4}{3} \leq (1 + \sqrt{2n}) \log 2n$, i.e.

$$\frac{2 \log 2}{3} n \leq (1 + \sqrt{2n}) \log 2n.$$

The left side grows linearly and the right like $\sqrt{n} \log n$, so this fails for all large n ; a short computation shows it fails for $n \geq 2^9 = 512$.

For $n < 512$ we exhibit primes directly: the list

$$2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631$$

has each term less than twice its predecessor, so for any $n < 512$ some consecutive pair straddles n and provides a prime in $[n, 2n)$. $\square$

§36.7 Continued Fractions

Suppose we want to approximate π by a rational number. Decimals give

$$\left| \pi - \frac{314159}{100000} \right| < 3 \times 10^{-6},$$

which is respectable. But

$$\left| \pi - \frac{355}{113} \right| < 3 \times 10^{-7},$$

which is better by an order of magnitude with a denominator nearly a thousand times smaller. Where did $355/113$ come from, and can we do better still? The answer to both questions is continued fractions.

§36.7.1 Diophantine Approximation

First, how well *can* we approximate? A denominator q divides the line into intervals of length $1/q$, so the best we can hope for from a fixed q is $|\theta - p/q| \leq 1/(2q)$. The following says that some q does very much better than that.

Theorem 36.7.1 (Dirichlet)

Let $\theta \in \mathbb{R}$ and $N \in \mathbb{N}$. Then there are $p, q \in \mathbb{Z}$ with $1 \leq q \leq N$ and

$$\left| \theta - \frac{p}{q} \right| \leq \frac{1}{qN} \leq \frac{1}{q^2}.$$

Proof. Consider the $N+1$ numbers $\{0\}, \{\theta\}, \{2\theta\}, \dots, \{N\theta\}$ where $\{\cdot\}$ denotes fractional part, all lying in $[0, 1)$. Partition $[0, 1)$ into the N intervals $[j/N, (j+1)/N)$; by pigeonhole two of our numbers lie in the same interval, say $\{k\theta\}$ and $\{l\theta\}$ with $k < l$. Then with $q = l - k \in [1, N]$ we have $|q\theta - p| \leq 1/N$ for a suitable integer p , and dividing by q gives the result. $\square$

Corollary 36.7.2

If θ is irrational then $|\theta - p/q| < 1/q^2$ has infinitely many solutions $p/q \in \mathbb{Q}$.

Proof. Given any finite set of rationals p_i/q_i , choose N with $1/N < \min_i |\theta - p_i/q_i|$ (possible as θ is irrational, so the minimum is positive). The p/q produced by Theorem 36.7.1 satisfies $|\theta - p/q| \leq 1/(qN) \leq 1/N$, so it is not on our list. $\square$

§36.7.2 Complete and Partial Quotients

The construction is a greedy algorithm: peel off the integer part, invert what remains, and repeat.

Definition 36.7.3

Let $\theta \in \mathbb{R}$. Set $\theta_0 = \theta$ and $a_0 = \lfloor \theta_0 \rfloor$. If $\theta_0 = a_0$ we stop; otherwise set $\theta_1 = 1/(\theta_0 - a_0)$ and $a_1 = \lfloor \theta_1 \rfloor$, and continue. We call the θ_n the **complete quotients** and the a_n the **partial quotients** of θ , and write

$$\theta = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots}} = [a_0, a_1, a_2, \dots]$$

for the **continued fraction expansion** of θ .

Since $0 < \theta_n - a_n < 1$ whenever the process does not stop, we have $\theta_{n+1} > 1$ and hence $a_n \geq 1$ for all $n \geq 1$; only a_0 may be zero or negative. At every stage we have the exact identity

$$\theta = [a_0, a_1, \dots, a_{n-1}, \theta_n],$$

which is the statement that we have not thrown anything away.

Example 36.7.4

Take $\theta = 59/13$. Then

$$\frac{59}{13} = 4 + \frac{7}{13} = 4 + \frac{1}{\frac{13}{7}} = 4 + \frac{1}{1 + \frac{6}{7}} = 4 + \frac{1}{1 + \frac{1}{1 + \frac{1}{6}}},$$

so $59/13 = [4, 1, 1, 6]$.

Proposition 36.7.5

The continued fraction expansion of θ terminates if and only if $\theta \in \mathbb{Q}$.

Proof. If the expansion terminates then θ is built from integers by finitely many additions and divisions, hence rational. Conversely, if $\theta = a/b$ then the algorithm is Euclid's: writing $a = a_0b + r$ we get $\theta_1 = b/r$, and the numerators $b > r > \dots$ strictly decrease, so the process stops. $\square$

Example 36.7.6

Compare the expansion of $59/13$ above with Euclid's algorithm on $(59, 13)$:

$$59 = 4 \times 13 + 7,$$

$$13 = 1 \times 7 + 6,$$

$$7 = 1 \times 6 + 1,$$

$$6 = 6 \times 1 + 0,$$

whose quotients are $4, 1, 1, 6$ – exactly the partial quotients.

Remark. The expansion of a rational is not quite unique: if $a_n \geq 2$ is the last partial quotient then $[a_0, \dots, a_n] = [a_0, \dots, a_n - 1, 1]$. The algorithm above always produces the version with $a_n \geq 2$ (for $n \geq 1$), and we take that as the canonical one.

§36.7.3 Convergents

Truncating the expansion gives rational approximations, and remarkably they satisfy a simple recursion.

Definition 36.7.7

Given integers $a_0, a_1, a_2, \dots$ with $a_n \geq 1$ for $n \geq 1$, define the **convergents** p_n/q_n by

$$\begin{aligned} p_{-1} &= 1, & q_{-1} &= 0, & p_0 &= a_0, & q_0 &= 1, \\ p_n &= a_n p_{n-1} + p_{n-2}, & q_n &= a_n q_{n-1} + q_{n-2} \quad (n \geq 1). \end{aligned}$$

Note $1 = q_0 \leq q_1 < q_2 < q_3 < \dots$, so the denominators grow at least as fast as the Fibonacci numbers, and in particular $q_n \rightarrow \infty$. It is worth recording the recursion in matrix form,

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix},$$

which one verifies by induction and which makes the next proposition transparent. (A warning about notation: by ‘the convergent p_n/q_n ’ we really mean the *pair* (p_n, q_n) , not merely the rational number; the distinction matters, as (i) below shows the pair is already in lowest terms.)

Proposition 36.7.8

Let (p_n/q_n) be the convergents of $[a_0, a_1, \dots]$. Then

- (i) $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n+1}$; in particular $(p_n, q_n) = 1$;
- (ii) $[a_0, \dots, a_n] = p_n/q_n$;
- (iii) if $\alpha = [a_0, \dots, a_n, \beta]$ for some real $\beta > 0$, then

$$\alpha = \frac{p_n \beta + p_{n-1}}{q_n \beta + q_{n-1}},$$

and α lies strictly between p_n/q_n and p_{n-1}/q_{n-1} .

Proof. (i) Take determinants in the matrix identity: the left side has determinant $p_n q_{n-1} - p_{n-1} q_n$ and the right has determinant $(-1)^{n+1}$.

(iii) Induct on n . For $n = 0$, $[a_0, \beta] = a_0 + 1/\beta = (a_0 \beta + 1)/\beta = (p_0 \beta + p_{-1})/(q_0 \beta + q_{-1})$. For $n \geq 1$, write $\gamma = a_n + 1/\beta$, so $\alpha = [a_0, \dots, a_{n-1}, \gamma]$ and by induction

$$\alpha = \frac{p_{n-1} \gamma + p_{n-2}}{q_{n-1} \gamma + q_{n-2}} = \frac{p_{n-1}(a_n \beta + 1) + \beta p_{n-2}}{q_{n-1}(a_n \beta + 1) + \beta q_{n-2}} = \frac{(a_n p_{n-1} + p_{n-2})\beta + p_{n-1}}{(a_n q_{n-1} + q_{n-2})\beta + q_{n-1}},$$

which is $(p_n \beta + p_{n-1})/(q_n \beta + q_{n-1})$ as claimed. For the last claim, if $x/y < x'/y'$ with $y, y' > 0$ then the *mediant* satisfies $x/y < (x + x')/(y + y') < x'/y'$; applying this with weights gives that $(p_n \beta + p_{n-1})/(q_n \beta + q_{n-1})$ lies strictly between p_n/q_n and p_{n-1}/q_{n-1} .

(ii) Take $\beta = a_n$ in (iii) applied to $n - 1$. □

Theorem 36.7.9

Let θ be irrational with continued fraction $[a_0, a_1, \dots]$. Then for every $n \geq 0$,

$$\left| \theta - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}.$$

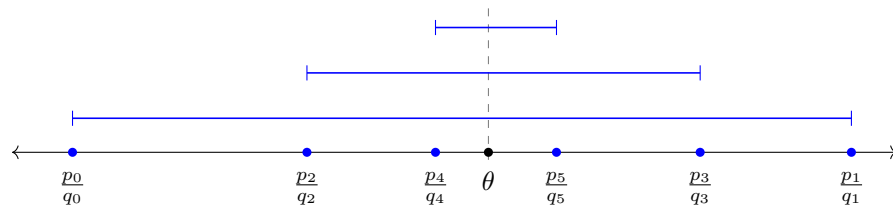
In particular $p_n/q_n \rightarrow \theta$.

Proof. We have $\theta = [a_0, \dots, a_n, \theta_{n+1}]$ exactly, so by [Proposition 36.7.8\(iii\)](#),

$$\theta - \frac{p_n}{q_n} = \frac{p_n \theta_{n+1} + p_{n-1}}{q_n \theta_{n+1} + q_{n-1}} - \frac{p_n}{q_n} = \frac{q_n p_{n-1} - p_n q_{n-1}}{q_n (q_n \theta_{n+1} + q_{n-1})} = \frac{(-1)^n}{q_n (q_n \theta_{n+1} + q_{n-1})}.$$

Since $\theta_{n+1} > a_{n+1}$ we get $q_n \theta_{n+1} + q_{n-1} > a_{n+1} q_n + q_{n-1} = q_{n+1}$, and the bound follows. As $q_n \rightarrow \infty$, the convergents converge to θ . □

Notice that the sign of $\theta - p_n/q_n$ alternates. So the convergents straddle θ , the even ones increasing up to it from below and the odd ones decreasing to it from above, giving a nested sequence of intervals each containing θ :



(The spacing here is schematic: in reality the intervals shrink far faster than the picture suggests.)

Example 36.7.10

The expansion of π begins $\pi = [3, 7, 15, 1, 292, \dots]$, with convergents

$$3, \quad \frac{22}{7}, \quad \frac{333}{106}, \quad \frac{355}{113}, \quad \frac{103993}{33102}, \quad \dots$$

So $355/113$ is the fourth convergent, which explains its quality: the next partial quotient is the enormous 292, and by [Theorem 36.7.9](#) the error is at most $1/(q_3q_4) = 1/(113 \times 33102) \approx 2.7 \times 10^{-7}$. Large partial quotients are exactly the places where continued fractions do unusually well.

§36.7.4 Best Approximation

The convergents are not merely good approximations; they are the best possible.

Theorem 36.7.11

Let θ be irrational with convergents p_n/q_n .

- (i) If $1 \leq q < q_{n+1}$ then $|q\theta - p| \geq |q_n\theta - p_n|$ for every $p \in \mathbb{Z}$.
- (ii) If $|\theta - p/q| < |\theta - p_n/q_n|$ with $q \geq 1$, then $q > q_n$.

Proof. (i) Fix q, p with $1 \leq q < q_{n+1}$ and solve

$$\begin{cases} p_n u + p_{n+1} v = p, \\ q_n u + q_{n+1} v = q. \end{cases}$$

The determinant of the system is $p_n q_{n+1} - p_{n+1} q_n = \pm 1$ by [Proposition 36.7.8\(i\)](#), so u, v are integers, not both zero. Then

$$q\theta - p = u(q_n\theta - p_n) + v(q_{n+1}\theta - p_{n+1}).$$

If $v = 0$ then $u \neq 0$ and $|q\theta - p| = |u||q_n\theta - p_n| \geq |q_n\theta - p_n|$, as required. So suppose $v \neq 0$. If $u = 0$ then $q = q_{n+1}v$, forcing $q \geq q_{n+1}$, contrary to hypothesis. So $u \neq 0$ too, and from $q_n u = q - q_{n+1}v$ with $0 < q < q_{n+1}$ we see u and v have opposite signs. But $\theta - p_n/q_n$ and $\theta - p_{n+1}/q_{n+1}$ also have opposite signs, so $u(q_n\theta - p_n)$ and $v(q_{n+1}\theta - p_{n+1})$ have the *same* sign, and there is no cancellation:

$$|q\theta - p| = |u||q_n\theta - p_n| + |v||q_{n+1}\theta - p_{n+1}| \geq |q_n\theta - p_n|.$$

(ii) If $q \leq q_n < q_{n+1}$ then by (i),

$$\left| \theta - \frac{p}{q} \right| = \frac{1}{q} |q\theta - p| \geq \frac{1}{q} |q_n\theta - p_n| \geq \frac{1}{q_n} |q_n\theta - p_n| = \left| \theta - \frac{p_n}{q_n} \right|,$$

contradicting the hypothesis. □

So among all fractions with denominator at most q_n , the convergent p_n/q_n is the closest

to θ . There is a converse, which is the practical tool: any sufficiently good approximation *must* be a convergent.

Theorem 36.7.12

Let θ be irrational.

- (i) At least one of any two consecutive convergents satisfies $|\theta - p/q| < 1/(2q^2)$.
- (ii) If $p/q \in \mathbb{Q}$ with $q \geq 1$ satisfies $|\theta - p/q| < 1/(2q^2)$, then p/q is a convergent of θ .

Proof. (i) Suppose $|\theta - p_n/q_n| \geq 1/(2q_n^2)$ and $|\theta - p_{n+1}/q_{n+1}| \geq 1/(2q_{n+1}^2)$. Since θ lies between the two convergents,

$$\frac{1}{2} \left(\frac{1}{q_n^2} + \frac{1}{q_{n+1}^2} \right) \leq \left| \theta - \frac{p_n}{q_n} \right| + \left| \theta - \frac{p_{n+1}}{q_{n+1}} \right| = \left| \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}}.$$

This says $(1/q_n - 1/q_{n+1})^2 \leq 0$, so $q_n = q_{n+1}$, which happens only for $n = 0$ with $a_1 = 1$; and one checks directly there.

(ii) Suppose p/q is not a convergent, and choose n with $q_n \leq q < q_{n+1}$. Then $p/q \neq p_n/q_n$, so $|pq_n - p_nq| \geq 1$ and

$$\frac{1}{qq_n} \leq \left| \frac{p}{q} - \frac{p_n}{q_n} \right| \leq \left| \theta - \frac{p}{q} \right| + \left| \theta - \frac{p_n}{q_n} \right| = \frac{1}{q} |q\theta - p| + \frac{1}{q_n} |q_n\theta - p_n|.$$

By Theorem 36.7.11(i), $|q_n\theta - p_n| \leq |q\theta - p| < 1/(2q)$, so

$$\frac{1}{qq_n} < \left(\frac{1}{q} + \frac{1}{q_n} \right) \frac{1}{2q} = \frac{1}{2q^2} + \frac{1}{2qq_n},$$

giving $1/(2qq_n) < 1/(2q^2)$ and hence $q < q_n$, a contradiction. $\square$

Remark. The constant $1/2$ in (ii) cannot be improved to 1 : by Theorem 36.7.1 there are always infinitely many p/q with $|\theta - p/q| < 1/q^2$, and these need not be convergents. In the other direction, Hurwitz showed that $|\theta - p/q| < 1/(\sqrt{5}q^2)$ has infinitely many solutions for every irrational θ , and that $\sqrt{5}$ is optimal – the golden ratio $[1, 1, 1, \dots]$ being the worst-approximable number.

An application in the opposite direction: some numbers cannot be approximated well at all.

Definition 36.7.13

A number $\alpha \in \mathbb{C}$ is **algebraic** if $P(\alpha) = 0$ for some nonzero polynomial P with integer coefficients; the least degree of such a P is the **degree** of α . Numbers which are not algebraic are **transcendental**.

Theorem 36.7.14 (Liouville)

Let $\alpha \in \mathbb{R}$ be algebraic of degree $n > 1$. Then there is $c = c(\alpha) > 0$ with

$$\left| \alpha - \frac{p}{q} \right| > \frac{c}{q^n}$$

for all $p/q \in \mathbb{Q}$ with $q \geq 1$.

Proof. Let P be the minimal polynomial of α , of degree n . We may assume $|\alpha - p/q| < 1$, else take $c \leq 1$. Since P is irreducible of degree $n > 1$ it has no rational roots, so $P(p/q) \neq 0$; and $q^n P(p/q)$ is a nonzero integer, so $|P(p/q)| \geq 1/q^n$.

By the mean value theorem, $P(\alpha) - P(p/q) = (\alpha - p/q)P'(\xi)$ for some ξ between α and p/q , hence with $|\xi - \alpha| < 1$. Let $C = \sup_{|\xi - \alpha| \leq 1} |P'(\xi)|$, which is finite and nonzero. Since $P(\alpha) = 0$,

$$\left| \alpha - \frac{p}{q} \right| = \frac{|P(p/q)|}{|P'(\xi)|} \geq \frac{1}{Cq^n},$$

so $c = \min(1, 1/(2C))$ works, the case $|\alpha - p/q| \geq 1$ being trivial. $\square$

Corollary 36.7.15

The number $\alpha = \sum_{k=1}^{\infty} 10^{-k!}$ is transcendental.

Proof. Truncating at $k = m$ gives a rational p_m/q_m with $q_m = 10^{m!}$ and

$$\left| \alpha - \frac{p_m}{q_m} \right| = \sum_{k>m} 10^{-k!} < 2 \cdot 10^{-(m+1)!} = \frac{2}{q_m^{m+1}}.$$

If α were algebraic of degree n this would contradict Liouville's theorem for $m \geq n$ (and $\alpha \notin \mathbb{Q}$ since its decimal expansion is not eventually periodic). $\square$

§36.7.5 Periodic Continued Fractions

Definition 36.7.16

A continued fraction expansion is **eventually periodic** if it has the form

$$[a_0, \dots, a_{m-1}, \overline{a_m, \dots, a_{m+k-1}}],$$

the bar denoting infinite repetition of the block. It is **purely periodic** if we may take $m = 0$. The **period** is the least such k .

Example 36.7.17

Let us expand $\theta = \sqrt{6}$. Since $2 < \sqrt{6} < 3$ we have $a_0 = 2$, and

$$\theta_1 = \frac{1}{\sqrt{6} - 2} = \frac{\sqrt{6} + 2}{2} = 2 + \frac{\sqrt{6} - 2}{2}, \quad a_1 = 2,$$

$$\begin{aligned}\theta_2 &= \frac{2}{\sqrt{6}-2} = \sqrt{6} + 2 = 4 + (\sqrt{6} - 2), & a_2 &= 4, \\ \theta_3 &= \frac{1}{\sqrt{6}-2} = \theta_1.\end{aligned}$$

So the complete quotients repeat with period 2 and $\sqrt{6} = [2, \overline{2, 4}]$.

Definition 36.7.18

A **quadratic irrational** is an irrational $\theta \in \mathbb{R}$ satisfying $A\theta^2 + B\theta + C = 0$ for some $A, B, C \in \mathbb{Z}$ not all zero. Its **conjugate** θ' is the other root.

Theorem 36.7.19 (Lagrange)

The continued fraction expansion of an irrational θ is eventually periodic if and only if θ is a quadratic irrational.

Proof. ($\Leftarrow$) Suppose $A\theta^2 + B\theta + C = 0$ and consider the BQF $f = (A, B, C)$, so $f(\theta, 1) = 0$. For each $n \geq 0$ define

$$f_n(x, y) = f(p_n x + p_{n-1} y, q_n x + q_{n-1} y) = (A_n, B_n, C_n),$$

which is a unimodular substitution applied to f (up to sign of the determinant), so $\text{disc } f_n = \text{disc } f$ for all n . Since $\theta = (p_n \theta_{n+1} + p_{n-1}) / (q_n \theta_{n+1} + q_{n-1})$ by [Proposition 36.7.8\(iii\)](#), we get

$$f_n(\theta_{n+1}, 1) = (q_n \theta_{n+1} + q_{n-1})^2 f(\theta, 1) = 0,$$

so θ_{n+1} is a root of $A_n x^2 + B_n x + C_n$.

It remains to show the f_n take only finitely many values. Now $A_n = f_n(1, 0) = f(p_n, q_n)$, and writing $g(x) = Ax^2 + Bx + C$ so that $f(p, q) = q^2 g(p/q)$,

$$|A_n| = q_n^2 \left| g\left(\frac{p_n}{q_n}\right) \right| = q_n^2 |A| \left| \frac{p_n}{q_n} - \theta \right| \left| \frac{p_n}{q_n} - \theta' \right| \leq |A| \left| \frac{p_n}{q_n} - \theta' \right|$$

using $|\theta - p_n/q_n| \leq 1/q_n^2$. As $p_n/q_n \rightarrow \theta$, the right-hand side tends to $|A||\theta - \theta'|$, so the A_n are bounded. Similarly $C_n = f_n(0, 1) = f(p_{n-1}, q_{n-1}) = A_{n-1}$ is bounded, and then $B_n^2 = \text{disc } f + 4A_n C_n$ is bounded too. So the f_n take finitely many values, and hence so do the complete quotients θ_{n+1} , each being a root of one of finitely many quadratics. Two must coincide, and once $\theta_i = \theta_j$ the expansion repeats.

($\Rightarrow$) Suppose first that the expansion is purely periodic, say $\phi = [\overline{a_0, \dots, a_{k-1}}]$, so that $\phi = [a_0, \dots, a_{k-1}, \phi]$. By [Proposition 36.7.8\(iii\)](#),

$$\phi = \frac{p_{k-1}\phi + p_{k-2}}{q_{k-1}\phi + q_{k-2}}, \quad \text{i.e.} \quad q_{k-1}\phi^2 + (q_{k-2} - p_{k-1})\phi - p_{k-2} = 0,$$

a quadratic with integer coefficients and $q_{k-1} \neq 0$. In general $\theta = [a_0, \dots, a_{m-1}, \phi]$ for such a ϕ , so $\theta = (p_{m-1}\phi + p_{m-2}) / (q_{m-1}\phi + q_{m-2})$ is a rational function of ϕ with integer coefficients, hence also a quadratic irrational. $\square$

Theorem 36.7.20 (Galois)

Let θ be a quadratic irrational. The continued fraction of θ is purely periodic if and only if $\theta > 1$ and $-1 < \theta' < 0$.

We shall not prove this, but we will use the following consequence, which is what makes Pell's equation tractable.

Corollary 36.7.21

If $d \in \mathbb{N}$ is not a perfect square, then

$$\sqrt{d} = [a_0, \overline{a_1, \dots, a_n}]$$

for some n , with $a_0 = \lfloor \sqrt{d} \rfloor$.

Proof. Let $a_0 = \lfloor \sqrt{d} \rfloor$ and $\theta_1 = 1/(\sqrt{d} - a_0)$. Then $\theta_1 > 1$, and its conjugate is $1/(-\sqrt{d} - a_0) \in (-1, 0)$ since $\sqrt{d} + a_0 > 1$. By [Theorem 36.7.20](#), θ_1 is purely periodic, which is the claim. $\square$

Remark. In fact the periodic part is always a palindrome followed by $2a_0$, so $\sqrt{d} = [a_0, \overline{a_1, a_2, \dots, a_2, a_1, 2a_0}]$. For instance $\sqrt{6} = [2, \overline{2, 4}]$, $\sqrt{17} = [4, \overline{8}]$, $\sqrt{7} = [2, \overline{1, 1, 4}]$ and $\sqrt{23} = [4, \overline{1, 3, 1, 8}]$.

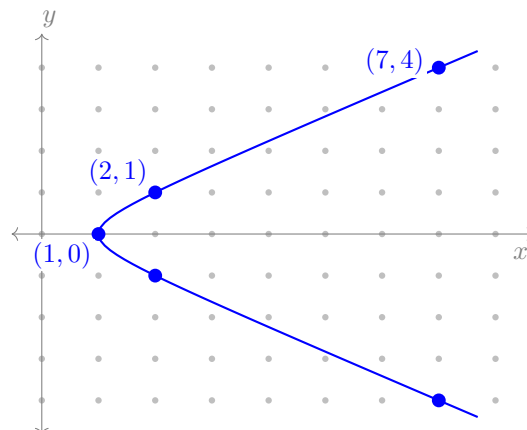
§36.7.6 Pell's Equation**Definition 36.7.22**

Let $d \in \mathbb{N}$ not be a perfect square. The equation

$$x^2 - dy^2 = 1$$

is called **Pell's equation**.

This is an indefinite binary quadratic form set equal to 1, and as we noted in §5 we should expect infinitely many solutions once there is one. Geometrically we are asking for lattice points on a hyperbola; here is $x^2 - 3y^2 = 1$, whose first few solutions are $(1, 0)$, $(2, \pm 1)$ and $(7, \pm 4)$.



The first observation is that solutions can only come from continued fractions.

Proposition 36.7.23

If (x, y) is a solution of $x^2 - dy^2 = 1$ with $x, y > 0$, then x/y is a convergent of $\sqrt{d}$.

Proof. From $(x - y\sqrt{d})(x + y\sqrt{d}) = 1$ we get $x > y\sqrt{d}$ and

$$x - y\sqrt{d} = \frac{1}{x + y\sqrt{d}} < \frac{1}{2y\sqrt{d}},$$

so $|\sqrt{d} - x/y| < 1/(2y^2\sqrt{d}) \leq 1/(2y^2)$. By [Theorem 36.7.12\(ii\)](#), x/y is a convergent of $\sqrt{d}$. $\square$

That there is a solution at all is where periodicity comes in.

Theorem 36.7.24 (Pell)

If $d \in \mathbb{N}$ is not a perfect square then $x^2 - dy^2 = 1$ has a solution with $y \neq 0$.

Proof. By [Corollary 36.7.21](#), $\sqrt{d} = [a_0, \overline{a_1, \dots, a_n}]$; replacing n by $2n$ if necessary we may assume n is even. Then $\theta_1 = \theta_{n+1} = [a_1, \dots, a_n]$, and $\sqrt{d} = a_0 + 1/\theta_1$, so $\theta_{n+1} = 1/(\sqrt{d} - a_0)$. By [Proposition 36.7.8\(iii\)](#),

$$\sqrt{d} = \frac{p_n\theta_{n+1} + p_{n-1}}{q_n\theta_{n+1} + q_{n-1}} = \frac{p_n + p_{n-1}(\sqrt{d} - a_0)}{q_n + q_{n-1}(\sqrt{d} - a_0)}.$$

Clearing denominators,

$$q_{n-1}d + (q_n - q_{n-1}a_0)\sqrt{d} = p_n - a_0p_{n-1} + p_{n-1}\sqrt{d}.$$

Since $\sqrt{d}$ is irrational we may equate ‘rational’ and ‘irrational’ parts: $p_{n-1} = q_n - a_0q_{n-1}$ and $q_{n-1}d = p_n - a_0p_{n-1}$. Hence

$$p_{n-1}^2 - dq_{n-1}^2 = p_{n-1}(q_n - a_0q_{n-1}) - q_{n-1}(p_n - a_0p_{n-1}) = p_{n-1}q_n - p_nq_{n-1},$$

which is $(-1)^n = 1$, using [Proposition 36.7.8\(i\)](#) and that n is even. So (p_{n-1}, q_{n-1}) is a solution. $\square$

Theorem 36.7.25

Let (x_1, y_1) be the solution of $x^2 - dy^2 = 1$ with $x_1, y_1 > 0$ and $\varepsilon = x_1 + y_1\sqrt{d}$ minimal. Then every solution with $x, y > 0$ is given by $x_n + y_n\sqrt{d} = \varepsilon^n$ for some $n \geq 1$.

Proof. Define the norm $N(a + b\sqrt{d}) = a^2 - db^2 = (a + b\sqrt{d})(a - b\sqrt{d})$, which is multiplicative since $\alpha \mapsto \alpha' = a - b\sqrt{d}$ is a ring homomorphism on $\mathbb{Z}[\sqrt{d}]$. So $N(\varepsilon) = 1$ implies $N(\varepsilon^n) = 1$, and every ε^n gives a solution.

Conversely, suppose $\alpha = a + b\sqrt{d}$ with $a, b > 0$ and $N(\alpha) = 1$. Then $\alpha \geq \varepsilon > 1$, so there is $k \geq 1$ with $\varepsilon^k \leq \alpha < \varepsilon^{k+1}$. Set $\beta = \alpha\varepsilon^{-k}$, so $N(\beta) = 1$ and $1 \leq \beta < \varepsilon$.

Writing $\beta = u + v\sqrt{d}$, from $\beta\beta' = 1$ and $\beta \geq 1$ we get $0 < \beta' \leq 1$, whence $2u = \beta + \beta' > 0$ and $2v\sqrt{d} = \beta - \beta' \geq 0$. So (u, v) is a nonnegative solution with $u + v\sqrt{d} < \varepsilon$, and minimality of ε forces $v = 0$, i.e. $\beta = 1$ and $\alpha = \varepsilon^k$. $\square$

Example 36.7.26

Take $d = 6$, so $\sqrt{6} = [2, \overline{2, 4}]$ with period 2. The convergents are

$$\frac{2}{1}, \frac{5}{2}, \frac{22}{9}, \frac{49}{20}, \frac{218}{89}, \frac{485}{198}, \dots$$

and evaluating $p^2 - 6q^2$ along them gives $-2, 1, -2, 1, -2, 1, \dots$. So the fundamental solution is $(5, 2)$: indeed $25 - 24 = 1$. The next is

$$(5 + 2\sqrt{6})^2 = 49 + 20\sqrt{6},$$

giving $(49, 20)$, and $49^2 - 6 \times 20^2 = 2401 - 2400 = 1$. These are precisely the convergents at odd index, as [Proposition 36.7.23](#) predicts.

Example 36.7.27

Take $d = 17$, so $\sqrt{17} = [4, \overline{8}]$ with period 1, which is odd. The convergent $p_0/q_0 = 4/1$ gives $16 - 17 = -1$, so $(4, 1)$ solves the *negative* Pell equation $x^2 - 17y^2 = -1$. Squaring, $(4 + \sqrt{17})^2 = 33 + 8\sqrt{17}$ gives the fundamental solution $(33, 8)$ of $x^2 - 17y^2 = 1$, and indeed $1089 - 1088 = 1$.

Remark. In general, $x^2 - dy^2 = -1$ is soluble if and only if the period of $\sqrt{d}$ is odd, and then the solutions of the two equations interlace along the convergents. For $d = 6$ the period is 2, and correspondingly $x^2 - 6y^2 = -1$ has no solutions – as one can also see modulo 3: $x^2 \equiv -1 \pmod{3}$ is insoluble.

§36.8 Primality Testing and Factorisation

Given a large integer N , we want to decide whether it is prime, and if not, to find a factor. These are different problems, and it is one of the pleasant discoveries of the last fifty years that the first is much easier than the second. Throughout we assume N is odd, since removing factors of 2 is free.

§36.8.1 Fermat Pseudoprimes

The obvious tool is Fermat's little theorem: if N is prime and $(b, N) = 1$ then $b^{N-1} \equiv 1 \pmod{N}$. Computing $b^{N-1} \pmod{N}$ by repeated squaring costs $O(\log N)$ multiplications, so this is a cheap test, and if it fails we have *proved* that N is composite – without finding a single factor.

Example 36.8.1

Take $N = 15$ and $b = 2$. Then $2^{14} = 16384 = 1092 \times 15 + 4$, so $2^{14} \equiv 4 \not\equiv 1 \pmod{15}$ and 15 is composite.

Unfortunately the converse fails.

Definition 36.8.2

Let $b \in \mathbb{N}$. An odd composite $N > 1$ is a **Fermat pseudoprime** to base b if $(b, N) = 1$ and $b^{N-1} \equiv 1 \pmod{N}$.

Example 36.8.3

$N = 91 = 7 \times 13$ is a Fermat pseudoprime to base 3: since $3^6 = 729 = 8 \times 91 + 1$, we have $3^{90} = (3^6)^{15} \equiv 1 \pmod{91}$. It is not one to base 2: $2^{90} \equiv 64 \pmod{91}$.

So a single base can fail us. How many bases can fail us at once?

Proposition 36.8.4

Let $N > 1$ be odd. If N is not a Fermat pseudoprime to some base b_0 coprime to N , then it fails to be one for at least half of the bases in $(\mathbb{Z}/N\mathbb{Z})^\times$.

Proof. The set $B = \{b \in (\mathbb{Z}/N\mathbb{Z})^\times : b^{N-1} \equiv 1\}$ is the kernel of the homomorphism $b \mapsto b^{N-1}$, hence a subgroup of $(\mathbb{Z}/N\mathbb{Z})^\times$. It is proper, since $b_0 \notin B$. So by Lagrange $|B| \leq |(\mathbb{Z}/N\mathbb{Z})^\times|/2$. $\square$

So if there is *any* witness, a handful of random guesses will find one with high probability. Sadly there need not be one.

Definition 36.8.5

An odd composite $N > 1$ is a **Carmichael number** if it is a Fermat pseudoprime to every base b with $(b, N) = 1$.

Example 36.8.6

$N = 561 = 3 \times 11 \times 17$ is the smallest Carmichael number. Indeed $560 = 2^4 \times 5 \times 7$ is divisible by each of $2 = 3 - 1$, $10 = 11 - 1$ and $16 = 17 - 1$, so for $(b, 561) = 1$ we have $b^{560} \equiv 1$ modulo each of 3, 11, 17 and hence modulo 561.

Proposition 36.8.7

An odd composite N is a Carmichael number if and only if N is squarefree and $p - 1 \mid N - 1$ for every prime $p \mid N$.

Proof. $(\Leftarrow)$ is the computation in the example. For $(\Rightarrow)$, suppose $p^2 \mid N$ for some prime p . Then $(\mathbb{Z}/N\mathbb{Z})^\times$ has an element of order p (take a generator of the cyclic factor $(\mathbb{Z}/p^k\mathbb{Z})^\times$ and raise it appropriately), and $p \nmid N - 1$ since $p \mid N$, so that element is not killed by $N - 1$. Hence N is squarefree. Now for $p \mid N$, take b with b a primitive root mod p and $b \equiv 1$ modulo N/p , possible by the Chinese remainder theorem; then $b^{N-1} \equiv 1 \pmod{p}$ forces $(p - 1) \mid N - 1$. $\square$

It was shown by Alford, Granville and Pomerance in 1994 that there are infinitely many

Carmichael numbers, so the Fermat test alone cannot be rescued. We need to ask for more.

§36.8.2 The Solovay–Strassen Test

The refinement is to use Euler’s criterion rather than Fermat’s theorem: if N is an odd prime then $b^{(N-1)/2} \equiv \left(\frac{b}{N}\right) \pmod{N}$, and the right-hand side is a Jacobi symbol which we can compute quickly for composite N too.

Definition 36.8.8

An odd composite $N > 1$ is an **Euler pseudoprime** to base b if $(b, N) = 1$ and

$$b^{(N-1)/2} \equiv \left(\frac{b}{N}\right) \pmod{N},$$

the symbol being Jacobi’s.

Squaring shows that an Euler pseudoprime to base b is also a Fermat pseudoprime to base b , so this is a genuinely stronger requirement. Exactly as in [Proposition 36.8.4](#), the set of bases witnessing Euler pseudoprimality is a subgroup, so if it is proper it has index at least 2. The point is that now it *is* always proper.

Theorem 36.8.9

Let $N > 1$ be odd and composite. Then there is $b \in (\mathbb{Z}/N\mathbb{Z})^\times$ such that N is not an Euler pseudoprime to base b .

Proof. Suppose first N is squarefree, say $N = pm$ with p prime, $p \nmid m$ and $m \geq 3$. Choose u with $\left(\frac{u}{p}\right) = -1$, and by the Chinese remainder theorem choose b with

$$b \equiv u \pmod{p}, \quad b \equiv 1 \pmod{m}.$$

Then $\left(\frac{b}{N}\right) = \left(\frac{b}{p}\right) \left(\frac{b}{m}\right) = (-1) \cdot 1 = -1$. But $b \equiv 1 \pmod{m}$ gives $b^{(N-1)/2} \equiv 1 \pmod{m}$, whereas $-1 \not\equiv 1 \pmod{m}$ as $m \geq 3$. So $b^{(N-1)/2} \not\equiv \left(\frac{b}{N}\right) \pmod{N}$.

Now suppose $p^2 \mid N$ for some prime p . By the Chinese remainder theorem choose b coprime to N with $b \equiv 1 + p \pmod{p^2}$ and $b \equiv 1$ modulo $N/p^{v_p(N)}$. Then

$$b^{N-1} \equiv (1 + p)^{N-1} \equiv 1 + (N-1)p \pmod{p^2},$$

and $p \nmid N-1$ since $p \mid N$, so $b^{N-1} \not\equiv 1 \pmod{p^2}$ and hence $b^{N-1} \not\equiv 1 \pmod{N}$. So $b^{(N-1)/2} \not\equiv \pm 1$, and in particular it is not the Jacobi symbol. $\square$

Corollary 36.8.10

If N is odd and composite, then N is an Euler pseudoprime to at most half of the bases in $(\mathbb{Z}/N\mathbb{Z})^\times$.

This gives the **Solovay–Strassen test**. Given odd $N > 1$, pick $1 < b < N$ at random.

- (1) Compute (b, N) by Euclid’s algorithm. If it is not 1, output COMPOSITE.

- (2) Compute $b^{(N-1)/2} \pmod N$ by repeated squaring, and $(\frac{b}{N})$ by reciprocity. If they differ, output COMPOSITE.
- (3) Otherwise output PROBABLY PRIME.

Every step is polynomial time. If N is prime the test never says COMPOSITE; if N is composite, each run catches it with probability at least $1/2$, so after k independent runs the probability of a composite slipping through is at most 2^{-k} .

§36.8.3 The Miller–Rabin Test

We can squeeze more out of the same idea. If N is an odd prime and $(b, N) = 1$, then $b^{(N-1)/2} \equiv \pm 1$. If it happens to be $+1$ and $(N-1)/2$ is still even, we may take a square root again, and so on. Since the only square roots of 1 modulo a prime are ± 1 , we get a rigid pattern.

Definition 36.8.11

Let $N > 1$ be odd and write $N - 1 = 2^s t$ with t odd and $s \geq 1$. Let $(b, N) = 1$. We say N passes the **strong test** to base b if either

$$b^t \equiv 1 \pmod N \quad \text{or} \quad b^{2^r t} \equiv -1 \pmod N \text{ for some } 0 \leq r < s.$$

An odd composite N passing the strong test to base b is called a **strong pseudo-prime** to base b .

Proposition 36.8.12

Every odd prime N passes the strong test to every base b with $(b, N) = 1$.

Proof. The sequence $b^t, b^{2t}, b^{4t}, \dots, b^{2^{s-1}t} = b^{N-1}$ ends at 1 and each term is the square of the previous one. Let r be least with $b^{2^r t} \equiv 1$. If $r = 0$ we are in the first case. Otherwise $b^{2^{r-1}t}$ is a square root of 1, hence ± 1 , and it is not 1 by minimality, so it is -1 . $\square$

Example 36.8.13

Take $N = 65$, so $N - 1 = 64 = 2^6$ and $s = 6$, $t = 1$.

- Base $b = 8$: we have $8^1 = 8 \not\equiv 1$, but $8^2 = 64 \equiv -1 \pmod{65}$. So 65 passes the strong test to base 8, and is a strong pseudoprime to that base.
- Base $b = 2$: the sequence 2^{2^r} for $r = 0, \dots, 5$ is $2, 4, 16, 61, 16, 61 \pmod{65}$. None of these is $-1 = 64$, and $2^t = 2^1 \not\equiv 1$. So 65 fails the strong test to base 2 and is composite.

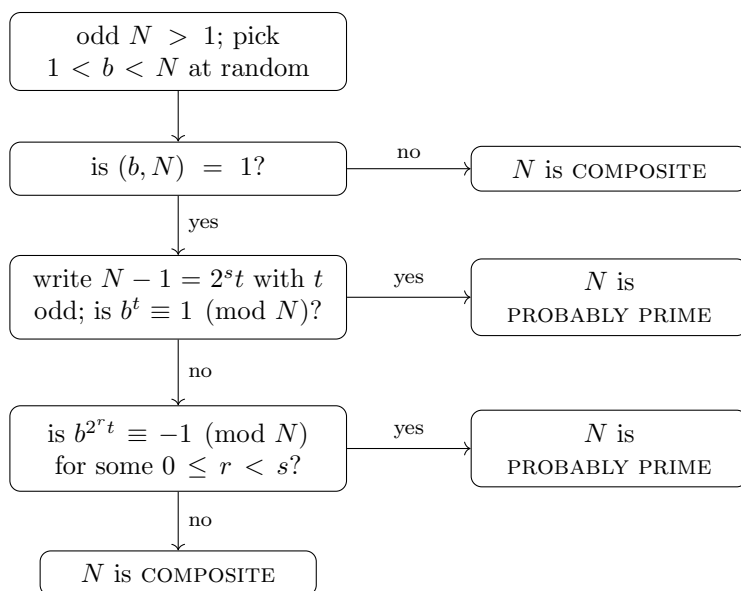
Example 36.8.14

The Carmichael number $N = 561$ has $560 = 2^4 \times 35$, so $s = 4$, $t = 35$. To base 2 we find $2^{35} \equiv 263$, then squaring, $166, 67, 1 \pmod{561}$. Since $2^{35} \not\equiv 1$ and none of $263, 166, 67$ is $-1 = 560$, the number 561 fails the strong test to base 2. So the

strong test detects a Carmichael number that Fermat's test cannot.

Remark. One can show that a strong pseudoprime to base b is an Euler pseudoprime to base b , and hence a Fermat pseudoprime; so the strong test is the sharpest of the three. Rabin proved the quantitative statement we want: an odd composite N is a strong pseudoprime to at most a quarter of the bases in $(\mathbb{Z}/N\mathbb{Z})^\times$.

The **Miller–Rabin test** is then as follows.



Repeating k times, a composite N survives with probability at most 4^{-k} .

Remark. (i) On the generalised Riemann hypothesis, a composite N fails the strong test to some base $b < 2(\log N)^2$. Since the strong test is polynomial time, this would give a *deterministic* polynomial-time primality test.

(ii) Unconditionally, Agrawal, Kayal and Saxena found a deterministic polynomial-time primality test in 2002. In practice Miller–Rabin remains the algorithm of choice, being far faster.

§36.8.4 Fermat Factorisation

Now suppose we know N is composite and want a factor. If $N = ab$ with $a > b$ both odd, then setting

$$r = \frac{a+b}{2}, \quad s = \frac{a-b}{2}$$

we get $N = r^2 - s^2$ with $r > \sqrt{N}$. So we may try $r = \lfloor \sqrt{N} \rfloor + 1, \lfloor \sqrt{N} \rfloor + 2, \dots$ and test whether $r^2 - N$ is a perfect square. This is **Fermat factorisation**, and it is fast exactly when N has two factors of similar size.

Example 36.8.15

Let $N = 200819$. We have $\lfloor \sqrt{N} \rfloor = 448$, so we try $r = 449$, giving $449^2 - N = 201601 - 200819 = 782$, which is not a square; and then $r = 450$, giving

$$450^2 - N = 202500 - 200819 = 1681 = 41^2.$$

Hence $N = 450^2 - 41^2 = (450 + 41)(450 - 41) = 491 \times 409$.

The number of steps is about $(a-b)/2$, so this is hopeless when the factors are far apart. But the underlying idea generalises beautifully.

Proposition 36.8.16

Suppose $x^2 \equiv y^2 \pmod{N}$ with $x \not\equiv \pm y \pmod{N}$. Then $(N, x-y)$ and $(N, x+y)$ are nontrivial factors of N .

Proof. We have $N \mid (x+y)(x-y)$, so $(N, x+y) \neq 1$ or $(N, x-y) \neq 1$; in fact if $(N, x-y) = 1$ then $N \mid x+y$, contradicting $x \not\equiv -y$. Similarly $(N, x+y) \neq 1$. And $N \nmid x \pm y$ by hypothesis, so neither gcd is N . $\square$

So we want congruent squares. Finding them directly is no easier than Fermat factorisation; the trick is to build them out of many small pieces.

§36.8.5 The Factor Base Method

Lemma 36.8.17

Let $p_1, \dots, p_r$ be distinct primes and let $c_1, \dots, c_k$ be nonzero integers all of whose prime factors lie in $\{p_1, \dots, p_r\}$. If $k > r+1$ then there is a nonempty $I \subseteq \{1, \dots, k\}$ with $\prod_{i \in I} c_i$ a perfect square.

Proof. For $J \subseteq \{1, \dots, k\}$ write $c_J = \prod_{i \in J} c_i = (-1)^{\alpha_{J,0}} p_1^{\alpha_{J,1}} \dots p_r^{\alpha_{J,r}} m_J^2$ with each $\alpha_{J,i} \in \{0, 1\}$; that is, record the exponent parities together with the sign. This gives a vector $\underline{\alpha}_J \in \mathbb{F}_2^{r+1}$, and there are only 2^{r+1} possibilities but $2^k > 2^{r+1}$ subsets. So there are $J \neq K$ with $\underline{\alpha}_J = \underline{\alpha}_K$, and then $c_J c_K$ is a square. Taking $I = (J \setminus K) \cup (K \setminus J)$, the elements of $J \cap K$ contribute a square to $c_J c_K$, so c_I is a square too, and $I \neq \emptyset$ as $J \neq K$. $\square$

Definition 36.8.18

A **factor base** is a set $B = \{-1, p_1, \dots, p_r\}$ with the p_i distinct primes. A **B -number** for N is a positive integer x such that the representative $\langle x^2 \rangle$ of x^2 modulo N in $(-N/2, N/2]$ has all its prime factors in B .

The algorithm is now clear. Choose a factor base B , generate more than $|B|$ many B -numbers $x_1, x_2, \dots$, use Lemma 36.8.17 to find I with $\prod_{i \in I} \langle x_i^2 \rangle = y^2$ a square, and set $x = \prod_{i \in I} x_i$. Then $x^2 \equiv y^2 \pmod{N}$, and if we are lucky enough that $x \not\equiv \pm y$, Proposition 36.8.16 hands us a factor. Otherwise, generate more B -numbers and try again.

The obvious way to find B -numbers is to take $x_i = \lceil \sqrt{kN} \rceil$ for small $k \geq 1$, since then $\langle x_i^2 \rangle$ is small in absolute value and so more likely to be smooth.

Example 36.8.19

Let $N = 1829$ and take the factor base $B = \{-1, 2, 3, 5, 7, 11, 13, 19\}$. Now $\lfloor \sqrt{1829k} \rfloor$

equals 42, 60, 74, 85 for $k = 1, 2, 3, 4$, so we try values of x near these:

x	42	43	60	61	74	75	85
$\langle x^2 \rangle$	-65	20	-58	63	-11	138	-91
factorisation	$-5 \cdot 13$	$2^2 \cdot 5$	$-2 \cdot 29$	$3^2 \cdot 7$	-11	$2 \cdot 3 \cdot 23$	$-7 \cdot 13$
B -number?	yes	yes	no	yes	yes	no	yes

Taking I to correspond to $x \in \{42, 43, 61, 85\}$,

$$(-5 \cdot 13)(2^2 \cdot 5)(3^2 \cdot 7)(-7 \cdot 13) = (2 \cdot 3 \cdot 5 \cdot 7 \cdot 13)^2 = 2730^2,$$

while $42 \cdot 43 \cdot 61 \cdot 85 = 9364110 \equiv 1459 \pmod{1829}$ and $2730 \equiv 901 \pmod{1829}$. So

$$1459^2 \equiv 901^2 \equiv 1554 \pmod{1829},$$

and $1459 \not\equiv \pm 901$. Hence $(1829, 1459 - 901) = (1829, 558) = 31$ and $(1829, 1459 + 901) = (1829, 2360) = 59$, giving $1829 = 31 \times 59$.

Remark. Why should the method work? Heuristically, if $N = \prod_{i=1}^t p_i^{e_i}$ with $t \geq 2$ distinct primes, then $x^2 \equiv 1 \pmod{N}$ has 2^t solutions by the Chinese remainder theorem, of which only ± 1 are useless. So a ‘random’ pair (x, y) with $x^2 \equiv y^2 \pmod{N}$ succeeds with probability $1 - 2^{1-t} \geq 1/2$. The case $t = 1$, where N is a prime power, is easy to detect separately.

[Lemma 36.8.17](#) guarantees the subset I exists but the naive search for it is exponential; in practice one finds it by linear algebra over $\mathbb{F}_2$, which is polynomial.

§36.8.6 The Continued Fraction Method

Continued fractions supply an excellent source of B -numbers, because convergents to $\sqrt{N}$ have squares that are unusually close to multiples of N .

Lemma 36.8.20

Let p_n/q_n be a convergent of $\sqrt{N}$. Then $|p_n^2 - Nq_n^2| \leq 2\sqrt{N}$.

Proof. Using [Theorem 36.7.9](#),

$$|p_n^2 - Nq_n^2| = q_n^2 \left| \sqrt{N} - \frac{p_n}{q_n} \right| \left| \sqrt{N} + \frac{p_n}{q_n} \right| \leq \frac{q_n}{q_{n+1}} \left(2\sqrt{N} + \frac{1}{q_n q_{n+1}} \right) \leq 2\sqrt{N},$$

since $q_n \leq q_{n+1}$ and the correction term is tiny. $\square$

So $\langle p_n^2 \rangle = p_n^2 - Nq_n^2$ exactly (it already lies in $(-N/2, N/2]$ for $N > 16$), which makes p_n an excellent B -number candidate. Better still, the recursion $p_n = a_n p_{n-1} + p_{n-2}$ may be run modulo N throughout, since only $p_n \bmod N$ is needed.

Example 36.8.21

Let $N = 12403$. Then $\sqrt{N} = [111, 2, 1, 2, 2, 7, 1, \dots]$, and computing the convergents

modulo N :

n	0	1	2	3	4	5	6
$p_n \bmod N$	111	223	334	891	2116	3300	5416
$\langle p_n^2 \rangle$	-82	117	-71	89	-27	166	-39
factorisation	$-2 \cdot 41$	$3^2 \cdot 13$	-71	89	-3^3	$2 \cdot 83$	$-3 \cdot 13$

The factor base $B = \{-1, 3, 13\}$ picks out $p_1 = 223$, $p_4 = 2116$ and $p_6 = 5416$, and

$$(3^2 \cdot 13)(-3^3)(-3 \cdot 13) = 3^6 \cdot 13^2 = (27 \cdot 13)^2 = 351^2.$$

Meanwhile $223 \times 2116 \times 5416 \equiv 11341 \pmod{12403}$, so $11341^2 \equiv 351^2 \pmod{12403}$, and

$$(12403, 11341 - 351) = (12403, 10990) = 157, \quad (12403, 11341 + 351) = 79.$$

Indeed $12403 = 79 \times 157$.

Remark. A far-reaching refinement of these ideas is the [number field sieve](#), which is the best general-purpose factorisation algorithm known. It runs in time

$$O\left(\exp\left(c(\log N)^{1/3}(\log \log N)^{2/3}\right)\right),$$

which is subexponential but not polynomial. The current record is a 250-digit number, which took thousands of core-years.

§36.8.7 Pollard's $p - 1$ Method

Sometimes we can go faster by looking for factors of a particular shape. Suppose $N = pN_0$ with $(p, N_0) = 1$ and suppose that $p - 1$ happens to be a product of small primes. For any a coprime to N we have $a^{p-1} \equiv 1 \pmod{p}$, so $p \mid (a^{p-1} - 1, N)$; and unless we are unlucky, $N_0 \nmid a^{p-1} - 1$, so the gcd is a nontrivial factor.

We do not know p , so we cannot form a^{p-1} . But we do not need to: any multiple k of $p - 1$ will do, since then $a^k \equiv 1 \pmod{p}$ as well. And if $p - 1$ is a product of prime powers at most m , then

$$k = \text{lcm}(1, 2, \dots, m)$$

is such a multiple.

The algorithm is therefore: choose a bound m , set $k = \text{lcm}(1, \dots, m)$, choose a small $a \geq 2$, check $(a, N) = 1$ (if not we have already won), compute $a^k \bmod N$ by repeated squaring, and evaluate $(a^k - 1, N)$. If this is nontrivial we are done; otherwise increase m and try again.

Example 36.8.22

Let $N = 540143$ and take $m = 8$, so $k = \text{lcm}(1, \dots, 8) = 840$. With $a = 2$,

$$2^{840} \equiv 53047 \pmod{540143},$$

and $(53047 - 1, 540143) = 421$. Indeed $540143 = 421 \times 1283$, and the method worked because

$$421 - 1 = 420 = 2^2 \times 3 \times 5 \times 7$$

is a product of small prime powers, all at most 8. Note that the other factor is no help here: $1283 - 1 = 2 \times 641$, and 641 is far larger than m .

Remark. The running time is $O(m(\log m)(\log^2 N))$, so the method is efficient precisely when some prime factor p of N has $p-1$ smooth. This is why cryptographic applications insist on ‘safe’ primes, for which $(p-1)/2$ is also prime.

For factors of general shape but moderate size (say $N \approx 2^{1000}$ with a factor $p \approx 2^{100}$), the method of choice is Lenstra’s elliptic curve method, which replaces $(\mathbb{Z}/p\mathbb{Z})^\times$ by the group of points of an elliptic curve mod p and thereby gets to vary the group order. And if one is allowed a quantum computer, Shor’s algorithm factors in polynomial time – which is, for now, the only known route to the conjecture with which we began.

37 Principles of Quantum Mechanics

In the IB Quantum Mechanics course, quantum mechanics was a theory of *wavefunctions*: a particle is described by a complex function $\psi(\mathbf{x}, t)$, observables are differential operators acting on it, and everything is computed by doing integrals. That framework is perfectly correct, and it is how one actually solves the hydrogen atom. But it has an obvious defect – there are quantum systems, spin being the first and most important, that have nothing to do with functions on space at all, and for which no wavefunction exists.

The right response is to notice that the wavefunction was never the essential object. What mattered were the *structural* facts: that states form a complex vector space with an inner product, that observables are Hermitian operators on it, and that measurement is expansion in an eigenbasis. In this course we take those facts as the definition. States become abstract vectors $|\psi\rangle$ in a Hilbert space, written in Dirac’s marvellously efficient bra-ket notation; the wavefunction becomes the collection of components $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$ of the state in one particular basis, no more privileged than any other. Once the theory is set up this way, a great deal that looked like a series of miracles in IB – the harmonic oscillator spectrum, the quantisation of angular momentum, the existence of spin – becomes a consequence of a few commutation relations, and can be derived in a page of algebra with no differential equations in sight.

Along the way we will find that almost everything in the subject comes from symmetry. Momentum is what generates translations, angular momentum is what generates rotations, and energy is what generates time evolution; conservation laws and degeneracies are then automatic. We will also develop the approximation methods (perturbation theory, the variational principle, Fermi’s golden rule) without which the theory would be able to describe only a handful of exactly solvable toy systems, and we will end by looking hard at what quantum mechanics says about reality itself, via the EPR paradox and Bell’s inequality.

This article constitutes my notes for the ‘Principles of Quantum Mechanics’ course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§37.1 The Dirac Formalism

We begin by rebuilding the foundations of quantum mechanics in the abstract. Nothing here contradicts IB Quantum Mechanics – the wavefunction picture will drop out as a special case in §37.1.8 – but the emphasis shifts entirely, from the analysis of a particular differential equation to the linear algebra that underlies every quantum system.

§37.1.1 States and the Hilbert Space

The first postulate says what a quantum system *is*.

Axiom 37.1.1 (States)

The states of a quantum system are the non-zero vectors of a complex Hilbert space

$\mathcal{H}$, with two vectors describing the same physical state if and only if they differ by a non-zero complex scalar.

A **Hilbert space** is a complex vector space $\mathcal{H}$ equipped with an inner product $\langle \cdot | \cdot \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{C}$ which is

- (i) *conjugate symmetric*: $\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$;
- (ii) *linear in the second slot*: $\langle \phi | (\lambda_1 |\psi_1\rangle + \lambda_2 |\psi_2\rangle) = \lambda_1 \langle \phi | \psi_1 \rangle + \lambda_2 \langle \phi | \psi_2 \rangle$ (and hence, by (i), *antilinear* in the first slot);
- (iii) *positive definite*: $\langle \psi | \psi \rangle \geq 0$, with equality iff $|\psi\rangle = 0$;

and which is *complete* in the associated norm $\|\psi\| = \langle \psi | \psi \rangle^{1/2}$, meaning every Cauchy sequence converges. We shall also always assume $\mathcal{H}$ is *separable*, i.e. has a countable orthonormal basis.

Two remarks on the postulate. First, since scalar multiples describe the same state, we may and usually will **normalise**, choosing a representative with $\langle \psi | \psi \rangle = 1$. This still leaves a phase ambiguity $|\psi\rangle \mapsto e^{i\alpha} |\psi\rangle$: an *overall* phase is physically meaningless. Second – and this is where all the interesting physics lives – a *relative* phase is not meaningless at all. The states

$$\frac{1}{\sqrt{2}} (|\psi_1\rangle + |\psi_2\rangle) \quad \text{and} \quad \frac{1}{\sqrt{2}} (|\psi_1\rangle - |\psi_2\rangle)$$

are physically different, and it is precisely their difference that produces interference. The technically correct statement of the postulate is that states are *rays* (one-dimensional subspaces) in $\mathcal{H}$; we will not fuss about this, but it is worth remembering when a phase seems to appear from nowhere.

The vector space structure is the **superposition principle**: any linear combination of states is a state. This is not a small assumption, and it is the single most counterintuitive thing about quantum mechanics.

Example 37.1.2 (Some Hilbert Spaces) (i) $\mathcal{H} = \mathbb{C}^n$, with $\langle \phi | \psi \rangle = \sum_i \phi_i^* \psi_i$. Finite-dimensional spaces are always complete, so this is a Hilbert space with nothing to check. The case $n = 2$ is the **qubit**, and is the state space of a spin- $\frac{1}{2}$ particle's spin degree of freedom.

(ii) $\mathcal{H} = L^2(\mathbb{R}^3)$, the square-integrable functions on $\mathbb{R}^3$, with

$$\langle \phi | \psi \rangle = \int_{\mathbb{R}^3} \phi(\mathbf{x})^* \psi(\mathbf{x}) d^3x.$$

This is the state space of a single spinless particle in three dimensions, and is the setting of IB Quantum Mechanics. (Completeness of L^2 is a theorem of measure theory; positive definiteness requires identifying functions that agree almost everywhere.)

(iii) A particle with spin, moving in space, has $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ – a *tensor product*, which we will meet properly in §37.5.1.

Almost all of the structure we need is already familiar from IB Linear Algebra; the only genuinely infinite-dimensional subtleties we shall meet are flagged as they arise, and we will treat them at the level of rigour of a physics course.

§37.1.2 Bras and Kets

Dirac's notation is a piece of notational engineering so good that it does a substantial part of the thinking for you. The idea is to split the inner product bracket $\langle\phi|\psi\rangle$ down the middle.

We write a vector of $\mathcal{H}$ as a **ket** $|\psi\rangle$, with the label inside the ket being whatever is convenient – a Greek letter, a list of quantum numbers $|n, \ell, m\rangle$, an eigenvalue $|\lambda\rangle$, or even an arrow $|\uparrow\rangle$. The labels are just names, and one must remember that (for example) $|\psi_1 + \psi_2\rangle$ is not automatically $|\psi_1\rangle + |\psi_2\rangle$ unless we say so.

Now consider the dual space $\mathcal{H}^*$ of continuous linear maps $\mathcal{H} \rightarrow \mathbb{C}$. Each $|\phi\rangle \in \mathcal{H}$ gives such a map, namely $|\psi\rangle \mapsto \langle\phi|\psi\rangle$, and the Riesz representation theorem (met in IB Analysis and Topology, and proved in Linear Analysis) says that *every* element of $\mathcal{H}^*$ arises this way, exactly once. So $\mathcal{H}^*$ is a faithful copy of $\mathcal{H}$, and we may write its elements as **bras** $\langle\phi|$, defined by

$$\langle\phi| : |\psi\rangle \mapsto \langle\phi|\psi\rangle.$$

The correspondence $|\phi\rangle \leftrightarrow \langle\phi|$ is antilinear:

$$\lambda_1 |\phi_1\rangle + \lambda_2 |\phi_2\rangle \longleftrightarrow \lambda_1^* \langle\phi_1| + \lambda_2^* \langle\phi_2|,$$

which is the notational shadow of conjugate symmetry, and the commonest source of slips. In $\mathbb{C}^n$, if $|\psi\rangle$ is a column vector then $\langle\psi|$ is the conjugate-transposed row vector, and $\langle\phi|\psi\rangle$ is literally the matrix product.

Notation. We will use $\langle\phi|\psi\rangle$ for the inner product throughout, dropping the IB notation $\langle\phi, \psi\rangle$. Both are linear in the second argument and antilinear in the first, matching the physicists' convention; be warned that many pure mathematics texts use the opposite convention.

An **orthonormal basis** of $\mathcal{H}$ is a family $\{|n\rangle\}_{n \in \mathbb{N}}$ with $\langle m|n\rangle = \delta_{mn}$ whose finite linear combinations are dense in $\mathcal{H}$. Given one, every state expands as

$$|\psi\rangle = \sum_n a_n |n\rangle, \quad a_n = \langle n|\psi\rangle,$$

where the coefficients are extracted by taking the inner product with $\langle m|$ and using orthonormality. Substituting the second formula into the first gives an identity worth displaying on its own:

$$|\psi\rangle = \sum_n |n\rangle \langle n|\psi\rangle \quad \text{for all } |\psi\rangle, \quad \text{i.e.} \quad \sum_n |n\rangle \langle n| = I.$$

This is the **completeness relation** (or *resolution of the identity*), and it is the workhorse of the whole formalism: whenever a calculation stalls, the answer is very often to insert a complete set of states in the middle of an expression. Here $|n\rangle \langle n|$ is an **outer product** – an operator, not a number – built by juxtaposing a ket and a bra the ‘wrong’ way round. Dirac's notation makes such objects, and the rule for reading them, completely automatic.

Two standard consequences of the inner product axioms, both proved exactly as in IB:

$$|\langle\phi|\psi\rangle| \leq \|\phi\| \|\psi\| \quad (\text{Cauchy-Schwarz}), \quad \|\phi + \psi\| \leq \|\phi\| + \|\psi\| \quad (\text{triangle inequality}),$$

and Parseval's identity $\|\psi\|^2 = \sum_n |a_n|^2$, which is what makes the Born rule below consistent.

§37.1.3 Operators and Adjoins

An **operator** is a linear map $A : \mathcal{H} \rightarrow \mathcal{H}$. Given an orthonormal basis $\{|n\rangle\}$, an operator is completely determined by its **matrix elements**

$$A_{mn} = \langle m|A|n\rangle,$$

since sandwiching A between two copies of the completeness relation gives

$$A = \left(\sum_m |m\rangle \langle m| \right) A \left(\sum_n |n\rangle \langle n| \right) = \sum_{m,n} A_{mn} |m\rangle \langle n|.$$

So an operator really is a (possibly infinite) matrix once a basis is chosen – and, pleasingly, the composition AB has matrix elements $\sum_k A_{mk} B_{kn}$, again by inserting the identity. All the linear algebra of IB carries over verbatim.

Notation. We will not decorate operators with hats, except occasionally where it aids clarity: whether a symbol denotes an operator or a number will always be clear from context. So the Hamiltonian is H , the position operator is x , and so on.

Definition 37.1.3 (Adjoint)

The **adjoint** (or Hermitian conjugate) of A is the unique operator $A^\dagger$ with

$$\langle \phi|A\psi\rangle = \langle A^\dagger\phi|\psi\rangle \quad \text{for all } |\phi\rangle, |\psi\rangle \in \mathcal{H}.$$

In terms of matrix elements this says $(A^\dagger)_{mn} = A_{nm}^*$ – the conjugate transpose, as expected. The notation is designed so that $A^\dagger$ is ‘ A acting leftwards on a bra’: the ket $A|\psi\rangle$ has as its bra $\langle\psi|A^\dagger$, and so

$$\langle\phi|A|\psi\rangle^* = \langle\psi|A^\dagger|\phi\rangle.$$

The algebraic rules are

$$(\lambda A + \mu B)^\dagger = \lambda^* A^\dagger + \mu^* B^\dagger, \quad (AB)^\dagger = B^\dagger A^\dagger, \quad (A^\dagger)^\dagger = A,$$

and $(|\phi\rangle\langle\psi|)^\dagger = |\psi\rangle\langle\phi|$. In practice one takes the dagger of any expression by reversing the order of everything and daggering each piece; kets become bras, bras become kets, and scalars get conjugated.

Definition 37.1.4 (Hermitian, Unitary, Projector)

An operator A is

- (i) **Hermitian** (or self-adjoint) if $A^\dagger = A$;
- (ii) **unitary** if $U^\dagger U = U U^\dagger = I$, equivalently if U is invertible with $U^{-1} = U^\dagger$;
- (iii) a **projector** if $P^2 = P = P^\dagger$.

Unitaries are exactly the operators preserving the inner product, $\langle U\phi|U\psi\rangle = \langle\phi|\psi\rangle$, so they are the symmetries of the Hilbert space – the quantum analogue of orthogonal transformations. They will describe time evolution, translations and rotations. Projectors are exactly the orthogonal projections onto closed subspaces: if $\{|i\rangle\}_{i \in I}$ is an orthonormal basis of a subspace V , then $P = \sum_{i \in I} |i\rangle\langle i|$ is the projector onto V , and $I - P$ is the projector onto $V^\perp$.

Finally, we will need the trace.

Definition 37.1.5 (Trace)

The **trace** of an operator A is $\text{tr } A = \sum_n \langle n|A|n\rangle$ for any orthonormal basis $\{|n\rangle\}$ (when the sum converges absolutely).

Proposition 37.1.6

The trace is independent of the basis used to compute it, and is cyclic: $\text{tr}(AB) = \text{tr}(BA)$.

Proof. For cyclicity, insert a complete set of states in the middle:

$$\text{tr}(AB) = \sum_n \langle n|AB|n\rangle = \sum_{n,m} \langle n|A|m\rangle \langle m|B|n\rangle = \sum_{m,n} B_{mn} A_{nm} = \sum_m \langle m|BA|m\rangle,$$

which is $\text{tr}(BA)$. Basis independence now follows: any other orthonormal basis is $|n'\rangle = U|n\rangle$ for some unitary U , and

$$\sum_n \langle n'|A|n'\rangle = \sum_n \langle n|U^\dagger A U|n\rangle = \text{tr}(U^\dagger A U) = \text{tr}(A U U^\dagger) = \text{tr } A. \quad \square$$

A useful special case: $\text{tr}(|\phi\rangle\langle\psi|) = \langle\psi|\phi\rangle$.

§37.1.4 Observables and Measurement**Axiom 37.1.7 (Observables)**

Every observable quantity of a quantum system is represented by a Hermitian operator on $\mathcal{H}$.

The reason Hermiticity is the right condition is the following, familiar from IB but worth re-proving in the new notation because the proof is three lines.

Proposition 37.1.8

Let A be Hermitian. Then all eigenvalues of A are real, and eigenvectors with distinct eigenvalues are orthogonal.

Proof. Suppose $A|\psi\rangle = \lambda|\psi\rangle$ with $|\psi\rangle \neq 0$. Then

$$\lambda \langle\psi|\psi\rangle = \langle\psi|A|\psi\rangle = \langle\psi|A^\dagger|\psi\rangle = \langle A\psi|\psi\rangle = \lambda^* \langle\psi|\psi\rangle,$$

and $\langle\psi|\psi\rangle \neq 0$, so $\lambda = \lambda^*$. If also $A|\phi\rangle = \mu|\phi\rangle$ then

$$\mu \langle\psi|\phi\rangle = \langle\psi|A|\phi\rangle = \langle A\psi|\phi\rangle = \lambda^* \langle\psi|\phi\rangle = \lambda \langle\psi|\phi\rangle,$$

so $(\lambda - \mu) \langle\psi|\phi\rangle = 0$, and $\lambda \neq \mu$ forces $\langle\psi|\phi\rangle = 0$. $\square$

We shall assume throughout, as physicists do, that the observables we meet have a complete set of eigenvectors – that is, that $\mathcal{H}$ has an orthonormal basis of eigenvectors of A . In finite dimensions this is the spectral theorem of IB Linear Algebra. In infinite dimensions it is true for a large and well-behaved class of operators, but genuinely fails for

others; the correct general statement is the spectral theorem for unbounded self-adjoint operators, which belongs to a different course. We shall also cheerfully use ‘eigenstates’ like $|x\rangle$ and $|p\rangle$ which are not in $\mathcal{H}$ at all (§37.1.8).

Axiom 37.1.9 (Measurement)

Let A be an observable with eigenvalues $\{\lambda_n\}$ (distinct) and let P_n be the orthogonal projector onto the eigenspace $\mathcal{H}_n = \{|\psi\rangle : A|\psi\rangle = \lambda_n|\psi\rangle\}$. Then a measurement of A in the normalised state $|\psi\rangle$

- (i) returns one of the eigenvalues λ_n ;
- (ii) returns λ_n with probability $\mathbb{P}(\lambda_n) = \|P_n|\psi\rangle\|^2 = \langle\psi|P_n|\psi\rangle$;
- (iii) leaves the system, immediately afterwards, in the state $P_n|\psi\rangle / \|P_n|\psi\rangle\|$.

This is the IB measurement postulate, restated so that degeneracy is handled without extra clauses. If the eigenvalue λ_n is non-degenerate with normalised eigenvector $|n\rangle$, then $P_n = |n\rangle\langle n|$ and the rules reduce to the familiar $\mathbb{P}(\lambda_n) = |\langle n|\psi\rangle|^2$ with collapse to $|n\rangle$. The probabilities sum to one because $\sum_n P_n = I$:

$$\sum_n \mathbb{P}(\lambda_n) = \langle\psi|\sum_n P_n|\psi\rangle = \langle\psi|\psi\rangle = 1.$$

Notice also that multiplying $|\psi\rangle$ by an overall phase changes nothing in (ii) or (iii), as it must.

Definition 37.1.10 (Expectation and Uncertainty)

For a normalised state $|\psi\rangle$ and observable A , the **expectation value** is

$$\langle A \rangle_\psi = \langle\psi|A|\psi\rangle,$$

and the **uncertainty** $(\Delta A)_\psi$ is the standard deviation,

$$(\Delta A)_\psi^2 = \langle (A - \langle A \rangle_\psi)^2 \rangle_\psi = \langle A^2 \rangle_\psi - \langle A \rangle_\psi^2.$$

That $\langle A \rangle_\psi$ is genuinely the mean of the measurement distribution is a one-line check: $\sum_n \lambda_n \langle\psi|P_n|\psi\rangle = \langle\psi|\sum_n \lambda_n P_n|\psi\rangle = \langle\psi|A|\psi\rangle$, using the spectral decomposition we prove next. Note that $(\Delta A)_\psi = 0$ if and only if $|\psi\rangle$ is an eigenstate of A : the uncertainty measures how far the state is from having a definite value of A .

§37.1.5 The Spectral Decomposition

Collecting the projectors of the measurement axiom gives the structural statement that we will use constantly.

Theorem 37.1.11 (Spectral Decomposition)

Let A be an observable with distinct eigenvalues $\{\lambda_n\}$ and eigenprojectors $\{P_n\}$. Then

$$P_n P_m = \delta_{nm} P_n, \quad \sum_n P_n = I, \quad A = \sum_n \lambda_n P_n.$$

Equivalently, in terms of an orthonormal eigenbasis $\{|n\rangle\}$ with $A|n\rangle = \lambda_n|n\rangle$ (repeats allowed),

$$A = \sum_n \lambda_n |n\rangle \langle n|.$$

Proof. Eigenspaces for distinct eigenvalues are orthogonal, giving $P_n P_m = 0$ for $n \neq m$, and $P_n^2 = P_n$ since P_n is a projector. Completeness of the eigenvectors is exactly $\sum_n P_n = I$. Finally, for any $|\psi\rangle$,

$$A|\psi\rangle = A \sum_n P_n |\psi\rangle = \sum_n \lambda_n P_n |\psi\rangle,$$

since $P_n |\psi\rangle$ lies in the λ_n -eigenspace. $\square$

This is more than a restatement of diagonalisability: it lets us *define functions of operators*. Given any function $f: \mathbb{R} \rightarrow \mathbb{C}$, set

$$f(A) = \sum_n f(\lambda_n) P_n.$$

When f is a polynomial this agrees with the obvious definition (check it on $f(\lambda) = \lambda^2$, using $P_n P_m = \delta_{nm} P_n$), and it extends the notion to functions where no power series is available. The most important instances are e^{iA} , which is unitary when A is Hermitian, and A^{-1} , which exists precisely when 0 is not an eigenvalue.

Example 37.1.12 (A Two-State System)

Take $\mathcal{H} = \mathbb{C}^2$ with basis $|\uparrow\rangle, |\downarrow\rangle$, and the observable

$$S_z = \frac{\hbar}{2} (|\uparrow\rangle \langle \uparrow| - |\downarrow\rangle \langle \downarrow|) = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

already written in spectral form: eigenvalues $\pm\hbar/2$ with projectors $|\uparrow\rangle \langle \uparrow|$ and $|\downarrow\rangle \langle \downarrow|$. In the state $|\psi\rangle = \alpha|\uparrow\rangle + \beta|\downarrow\rangle$ (with $|\alpha|^2 + |\beta|^2 = 1$) a measurement gives $+\hbar/2$ with probability $|\alpha|^2$ and $-\hbar/2$ with probability $|\beta|^2$, so

$$\langle S_z \rangle = \frac{\hbar}{2} (|\alpha|^2 - |\beta|^2), \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4},$$

the latter because $S_z^2 = \frac{\hbar^2}{4} I$ – every state is an eigenstate of S_z^2 . Hence

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} (1 - (|\alpha|^2 - |\beta|^2)^2) = \hbar^2 |\alpha|^2 |\beta|^2,$$

which vanishes exactly when α or β is zero, and is maximal for the equal superpositions $|\alpha| = |\beta| = 1/\sqrt{2}$. Also $e^{i\theta S_z/\hbar} = e^{i\theta/2} |\uparrow\rangle \langle \uparrow| + e^{-i\theta/2} |\downarrow\rangle \langle \downarrow|$, which is our first glimpse of the peculiar fact that a spin- $\frac{1}{2}$ state returns to itself only after a 4π rotation.

§37.1.6 Compatible Observables and Complete Sets

Two observables can be measured simultaneously exactly when they commute. This was proved in IB; we restate it here because the language of *complete sets of commuting*

observables organises everything that follows.

Definition 37.1.13 (Commutator)

The **commutator** of A and B is $[A, B] = AB - BA$.

The commutator is bilinear and antisymmetric, satisfies the Leibniz rules

$$[A, BC] = [A, B]C + B[A, C], \quad [AB, C] = A[B, C] + [A, C]B,$$

and the Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$. If A, B are Hermitian then $[A, B]$ is *anti*-Hermitian, so $i[A, B]$ is Hermitian – a fact we use immediately below.

Theorem 37.1.14 (Compatibility)

Two observables A, B admit a complete orthonormal basis of joint eigenvectors if and only if $[A, B] = 0$.

Proof. ($\Rightarrow$) On a joint eigenvector $|i\rangle$ with $A|i\rangle = \lambda_i|i\rangle$, $B|i\rangle = \mu_i|i\rangle$, we have $[A, B]|i\rangle = (\lambda_i\mu_i - \mu_i\lambda_i)|i\rangle = 0$; by completeness $[A, B] = 0$.

($\Leftarrow$) Let $\mathcal{H}_\lambda$ be an eigenspace of A . For $|\psi\rangle \in \mathcal{H}_\lambda$ we have $A(B|\psi\rangle) = B(A|\psi\rangle) = \lambda B|\psi\rangle$, so B maps $\mathcal{H}_\lambda$ to itself. The restriction $B|_{\mathcal{H}_\lambda}$ is Hermitian, hence diagonalisable within $\mathcal{H}_\lambda$; choosing such a basis for each λ and taking the union gives a complete joint eigenbasis. $\square$

Observables with $[A, B] = 0$ are called **compatible**. The proof shows exactly what commuting buys us: B acts *within* each eigenspace of A , so measuring B does not disturb the value of A .

Definition 37.1.15 (Complete Set of Commuting Observables)

A set $\{A_1, \dots, A_k\}$ of pairwise commuting observables is a **complete set of commuting observables** (CSCO) if their joint eigenspaces are all one-dimensional; equivalently, if the list of eigenvalues $(\lambda_1, \dots, \lambda_k)$ determines the state uniquely up to phase.

A CSCO gives an unambiguous labelling $|\lambda_1, \dots, \lambda_k\rangle$ of a basis of $\mathcal{H}$, and the eigenvalues are the **quantum numbers** of the state. This is where the familiar labels come from: for the hydrogen atom $\{H, L^2, L_z\}$ gives $|n, \ell, m\rangle$, and adding spin gives the four quantum numbers behind the periodic table. Conversely, a degeneracy is precisely a failure of some set to be complete, and – as we shall see in §37.6 – degeneracies are almost always caused by a symmetry.

§37.1.7 The Uncertainty Principle

If two operators do not commute, we cannot expect to measure both exactly. This is quantified in an *uncertainty principle*. Let A, B be Hermitian. Then taking $C = A + i\lambda B$ with $\lambda \in \mathbb{R}$,

$$C^\dagger C = A^2 + \lambda^2 B^2 + \lambda i[A, B].$$

The first three are Hermitian, so $i[A, B]$ is also Hermitian. We have

$$\begin{aligned} 0 &\leq \langle C\psi | C\psi \rangle = \langle \psi | C^\dagger C | \psi \rangle \\ &= \langle A^2 \rangle_\psi + \lambda^2 \langle B^2 \rangle_\psi + \lambda \langle i[A, B] \rangle_\psi. \end{aligned}$$

For this to always be nonnegative, the quadratic in λ can have at most one real root, so the discriminant gives

$$\langle A^2 \rangle_\psi \langle B^2 \rangle_\psi \geq \frac{1}{4} (\langle i[A, B] \rangle_\psi)^2.$$

This works for any A, B , so if we apply it to $\tilde{A} = A - \langle A \rangle_\psi$ and $\tilde{B} = B - \langle B \rangle_\psi$, we find $[\tilde{A}, \tilde{B}] = [A, B]$ and hence the following.

Theorem 37.1.16 (Generalised Uncertainty Principle)

For any Hermitian A, B and any normalised state $|\psi\rangle$,

$$(\Delta A)_\psi (\Delta B)_\psi \geq \frac{1}{2} |\langle [A, B] \rangle_\psi|.$$

Most famous is Heisenberg's uncertainty principle, from applying this to the canonical commutation relation $[x, p] = i\hbar$:

$$(\Delta x)(\Delta p) \geq \frac{1}{2}\hbar.$$

It is worth being precise about what this does and does not say. It is *not* a statement that measurement is clumsy, or that our apparatus is imperfect. It says that no state exists in which both spreads are small: prepare a million identical copies, measure x on half of them and p on the other half, and the two sample standard deviations obey the inequality. The bound is also *state-dependent* in general – for spin, $[S_x, S_y] = i\hbar S_z$ gives $\Delta S_x \Delta S_y \geq \frac{\hbar}{2} |\langle S_z \rangle|$, whose right-hand side vanishes for states with $\langle S_z \rangle = 0$.

Tracing through the proof tells us when equality holds: we need $\langle C\psi | C\psi \rangle = 0$ for the optimal λ , that is

$$(\tilde{A} + i\lambda\tilde{B})|\psi\rangle = 0$$

for some real λ . States achieving equality are called **minimum uncertainty states**; for x and p we will find in §37.1.9 that they are exactly the Gaussian wavepackets.

§37.1.8 Position and Momentum Representations

We now recover the wavefunction. The subtlety is that the position operator x (in one dimension, for simplicity) has a *continuous* spectrum, and its ‘eigenstates’ are not normalisable – they do not live in $\mathcal{H}$. Physicists handle this by treating them as a limiting case of an orthonormal basis, replacing sums by integrals and Kronecker deltas by Dirac deltas. Every formula below can be justified rigorously, but doing so properly requires the theory of rigged Hilbert spaces, which we shall not need.

So we postulate a family of states $|x\rangle$, $x \in \mathbb{R}$, with

$$x|x\rangle = x|x\rangle, \quad \langle x|x'\rangle = \delta(x-x'), \quad \int_{-\infty}^{\infty} |x\rangle \langle x| dx = I,$$

the last being the continuum completeness relation. Applying it to a state $|\psi\rangle$,

$$|\psi\rangle = \int |x\rangle \langle x|\psi\rangle dx,$$

so $|\psi\rangle$ is determined by the function

$$\psi(x) := \langle x|\psi\rangle,$$

which is precisely the **wavefunction** of IB. It is the component of the abstract state $|\psi\rangle$ along the ‘basis vector’ $|x\rangle$, exactly as $a_n = \langle n|\psi\rangle$ was in the discrete case. Inner products become the familiar integrals,

$$\langle \phi|\psi\rangle = \int \langle \phi|x\rangle \langle x|\psi\rangle dx = \int \phi(x)^* \psi(x) dx,$$

and $|\psi(x)|^2 dx$ is the probability of finding the particle in $[x, x+dx]$ – the Born rule, now a special case of the measurement axiom.

How do operators act? Position is immediate: $\langle x|X|\psi\rangle = x\psi(x)$, writing X for the operator to distinguish it from the number. For momentum, we *define* the momentum operator by its action in this representation,

$$\langle x|P|\psi\rangle = -i\hbar \frac{d\psi}{dx}(x),$$

and then the canonical commutation relation follows in one line, acting on an arbitrary ψ :

$$[X, P]\psi = x(-i\hbar\psi') - (-i\hbar)(x\psi)' = i\hbar\psi, \quad \text{so} \quad [X, P] = i\hbar.$$

(In three dimensions, $[X_i, P_j] = i\hbar\delta_{ij}$ and $[X_i, X_j] = [P_i, P_j] = 0$.)

There is nothing special about position. The eigenstates $|p\rangle$ of P give an equally good basis, and we can find their wavefunctions immediately: $\langle x|p\rangle$ must satisfy $-i\hbar\partial_x \langle x|p\rangle = p \langle x|p\rangle$, so $\langle x|p\rangle \propto e^{ipx/\hbar}$. Fixing the constant by demanding $\langle p|p'\rangle = \delta(p-p')$ and using $\int e^{i(p-p')x/\hbar} dx = 2\pi\hbar \delta(p-p')$, we get

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \quad \int |p\rangle \langle p| dp = I.$$

The **momentum-space wavefunction** is $\tilde{\psi}(p) = \langle p|\psi\rangle$, and inserting the position completeness relation relates the two:

$$\tilde{\psi}(p) = \int \langle p|x\rangle \langle x|\psi\rangle dx = \frac{1}{\sqrt{2\pi\hbar}} \int \psi(x) e^{-ipx/\hbar} dx.$$

This is the Fourier transform, and it is worth appreciating what has just happened: the deep-looking fact that position and momentum wavefunctions are Fourier transforms of each other is nothing but a change of basis, of exactly the same nature as rotating coordinate axes in $\mathbb{R}^3$. In the momentum representation the roles are swapped,

$$\langle p|P|\psi\rangle = p\tilde{\psi}(p), \quad \langle p|X|\psi\rangle = i\hbar \frac{d\tilde{\psi}}{dp}(p),$$

(the sign change coming from the conjugate in $\langle p|x\rangle$), and the theory is perfectly symmetric between the two.

Remark. This is the real payoff of the abstract formalism. In IB the wavefunction *was* the state, and momentum space felt like a trick. Here neither representation is preferred: $|\psi\rangle$ is the state, and $\psi(x)$, $\tilde{\psi}(p)$, or the coefficients in any other basis are just different ways of writing down the same thing. For systems like spin, no position representation exists at all, and we simply use a different basis.

§37.1.9 Wavepackets

Let us see the uncertainty principle in action, and identify the states that saturate it. Take the **Gaussian wavepacket**

$$\psi(x) = \left(\frac{1}{\pi\sigma^2}\right)^{1/4} \exp\left(-\frac{x^2}{2\sigma^2} + \frac{ip_0x}{\hbar}\right),$$

normalised so that $\int |\psi|^2 = 1$. Since $|\psi(x)|^2$ is an even Gaussian of variance $\sigma^2/2$, we read off

$$\langle X \rangle = 0, \quad (\Delta X)^2 = \frac{\sigma^2}{2}.$$

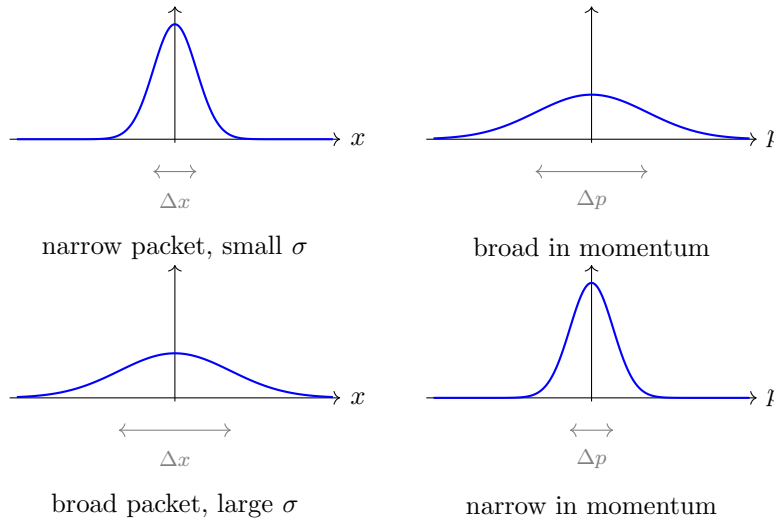
Its Fourier transform is another Gaussian (completing the square in the exponent),

$$\tilde{\psi}(p) = \left(\frac{\sigma^2}{\pi\hbar^2}\right)^{1/4} \exp\left(-\frac{\sigma^2(p-p_0)^2}{2\hbar^2}\right),$$

so $\langle P \rangle = p_0$ and $(\Delta P)^2 = \hbar^2/2\sigma^2$. Multiplying,

$$\Delta X \Delta P = \frac{\sigma}{\sqrt{2}} \cdot \frac{\hbar}{\sqrt{2}\sigma} = \frac{\hbar}{2},$$

exactly the minimum. Indeed the equality condition of the previous section, $(X - \langle X \rangle + i\lambda(P - \langle P \rangle))|\psi\rangle = 0$, becomes the first-order ODE $x\psi + i\lambda(-i\hbar\psi' - p_0\psi) = 0$, whose solutions are precisely the Gaussians above with $\sigma^2 = \lambda\hbar$.



Squeezing the packet in position necessarily broadens it in momentum, and vice versa – which, given that the two are Fourier transforms of each other, is a theorem about Fourier analysis quite as much as about physics. In the limits $\sigma \rightarrow 0$ and $\sigma \rightarrow \infty$ we approach the (non-normalisable) states $|x\rangle$ and $|p\rangle$ respectively, with one uncertainty going to zero and the other diverging.

§37.2 Time Evolution

So far the formalism is entirely kinematic: it describes states at one instant. We now add dynamics, and in doing so we will meet a theme that runs through the rest of the course – that a continuous symmetry is implemented by a family of unitary operators, and is ‘generated’ by a Hermitian observable.

§37.2.1 The Evolution Operator

Axiom 37.2.1 (Time Evolution)

The state of an isolated system evolves according to the **Schrödinger equation**

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle,$$

where the Hamiltonian H is the observable corresponding to the energy of the system.

Since the equation is first order in t and linear, the state at time t is determined by the state at time 0 by a linear operator:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle.$$

$U(t)$ is the **time evolution operator**. Substituting into the Schrödinger equation, and demanding it hold for all initial states, gives the operator equation

$$i\hbar \frac{dU}{dt} = HU(t), \quad U(0) = I,$$

which for a time-independent H has the solution

$$U(t) = \exp\left(-\frac{iHt}{\hbar}\right),$$

where the exponential is defined by the spectral decomposition (or, equivalently here, by its power series). The key property is the following.

Proposition 37.2.2

$U(t)$ is unitary, and $U(s)U(t) = U(s+t)$, $U(t)^{-1} = U(-t)$.

Proof. The group property is immediate from the exponential form (the exponents commute). For unitarity, note that $H^\dagger = H$ gives $U(t)^\dagger = \exp(+iHt/\hbar) = U(-t)$, so $U(t)^\dagger U(t) = U(0) = I$. Alternatively, and more instructively, one can prove it without solving anything: for any $|\psi(t)\rangle$ obeying the Schrödinger equation,

$$\frac{d}{dt} \langle \psi | \psi \rangle = \langle \dot{\psi} | \psi \rangle + \langle \psi | \dot{\psi} \rangle = \frac{i}{\hbar} \langle \psi | H^\dagger | \psi \rangle - \frac{i}{\hbar} \langle \psi | H | \psi \rangle = 0$$

exactly because H is Hermitian. So the norm – total probability – is conserved. $\square$

This is the first instance of a pattern we will see repeatedly. A one-parameter family of unitaries $U(t) = e^{-iHt/\hbar}$ with $U(0) = I$ is determined by its derivative at the identity,

$$H = i\hbar \left. \frac{dU}{dt} \right|_{t=0},$$

and this derivative is Hermitian precisely because the $U(t)$ are unitary. We say H **generates** time translations. In §37.6 momentum will generate spatial translations and angular momentum will generate rotations, by exactly the same argument.

Solving the dynamics is now a matter of diagonalising H . If $H|n\rangle = E_n|n\rangle$ with $\{|n\rangle\}$ an orthonormal basis, then expanding $|\psi(0)\rangle = \sum_n a_n |n\rangle$,

$$|\psi(t)\rangle = \sum_n a_n e^{-iE_n t/\hbar} |n\rangle, \quad a_n = \langle n | \psi(0) \rangle.$$

Time evolution simply attaches a phase to each energy eigenstate. In particular a state which starts as an energy eigenstate stays one, acquiring only an overall phase – these are the **stationary states**, physically unchanging.

Remark (Time-Dependent Hamiltonians). If $H = H(t)$, the formula

$$U(t) = \exp\left(-\frac{i}{\hbar} \int_0^t H(t') dt'\right)$$

is *wrong* in general, because $H(t_1)$ and $H(t_2)$ need not commute. The correct expression is a time-ordered exponential, obtained by iterating the integral equation $U(t) = I - \frac{i}{\hbar} \int_0^t H(t') U(t') dt'$. We will construct the first two terms of this series when we do time-dependent perturbation theory in §37.9. Unitarity of $U(t)$, however, holds regardless, by the norm-conservation argument above.

§37.2.2 The Heisenberg Picture

There is a redundancy in the formalism: all physical predictions are matrix elements $\langle \phi(t) | A | \psi(t) \rangle$, and we can move the time dependence between the states and the operators at will. Writing the evolution out,

$$\langle \phi(t) | A | \psi(t) \rangle = \langle \phi(0) | U(t)^\dagger A U(t) | \psi(0) \rangle,$$

we may either regard the states as evolving with A fixed – the **Schrödinger picture**, which is what we have been doing – or regard the states as fixed and the operators as evolving.

Definition 37.2.3 (Heisenberg Picture)

In the **Heisenberg picture** the states are the fixed vectors $|\psi\rangle_H = |\psi(0)\rangle$, and observables carry the time dependence:

$$A_H(t) = U(t)^\dagger A U(t).$$

By construction the two pictures agree on every matrix element, hence on every physical prediction, and they agree at $t = 0$. Note that $A_H(t)$ is Hermitian if A is, that the transformation preserves all algebraic relations,

$$(AB)_H = U^\dagger AB U = U^\dagger A U U^\dagger B U = A_H B_H, \quad [A, B]_H = [A_H, B_H],$$

and that $H_H = H$ when H is time-independent (it commutes with U). Differentiating the definition gives the dynamical law.

Theorem 37.2.4 (Heisenberg's Equation of Motion)

For a time-independent Hamiltonian and a Schrödinger-picture operator A which may itself depend explicitly on t ,

$$\frac{dA_H}{dt} = \frac{i}{\hbar} [H, A_H] + \left(\frac{\partial A}{\partial t} \right)_H.$$

Proof. Using $dU/dt = -\frac{i}{\hbar}HU$ and $dU^\dagger/dt = \frac{i}{\hbar}U^\dagger H$,

$$\frac{d}{dt}(U^\dagger AU) = \frac{i}{\hbar}U^\dagger HAU - \frac{i}{\hbar}U^\dagger AHU + U^\dagger \frac{\partial A}{\partial t}U = \frac{i}{\hbar}U^\dagger [H, A]U + \left(\frac{\partial A}{\partial t}\right)_H,$$

and $U^\dagger [H, A]U = [H_H, A_H] = [H, A_H]$. $\square$

This is a beautiful result, because it is *formally identical to Hamilton's equations of classical mechanics*, with the Poisson bracket replaced by $\frac{1}{i\hbar}$ times the commutator:

$$\frac{df}{dt} = \{f, H\}_{\text{PB}} + \frac{\partial f}{\partial t} \quad \rightsquigarrow \quad \frac{dA_H}{dt} = \frac{1}{i\hbar}[A_H, H] + \left(\frac{\partial A}{\partial t}\right)_H.$$

Taking expectation values in any fixed state recovers, in the Schrödinger picture, **Ehrenfest's theorem**:

$$\frac{d}{dt}\langle A \rangle_{\psi(t)} = \frac{i}{\hbar}\langle [H, A] \rangle_{\psi(t)} + \left\langle \frac{\partial A}{\partial t} \right\rangle_{\psi(t)}.$$

Corollary 37.2.5 (Conserved Quantities)

If A has no explicit time dependence and $[A, H] = 0$, then A_H is constant, and $\langle A \rangle$ and indeed the whole probability distribution of A are constant in time.

The last claim follows because the eigenprojectors of A also commute with H , so $\langle \psi(t) | P_n | \psi(t) \rangle = \langle \psi(0) | U^\dagger P_n U | \psi(0) \rangle = \langle \psi(0) | P_n | \psi(0) \rangle$. Conservation laws in quantum mechanics are therefore exactly statements that observables commute with the Hamiltonian – and, since the eigenvalues of H label energies, they are also statements that the observable can be simultaneously diagonalised with the energy. This is the tight link between symmetry, conservation and degeneracy that §37.6 makes systematic.

Example 37.2.6 (The Free Particle)

For $H = P^2/2m$ we have $[H, P] = 0$, so $P_H(t) = P$: momentum is conserved, as classically. For position, $[H, X] = \frac{1}{2m}[P^2, X] = \frac{1}{2m}(P[P, X] + [P, X]P) = -\frac{i\hbar}{m}P$, so

$$\frac{dX_H}{dt} = \frac{i}{\hbar}\left(-\frac{i\hbar P}{m}\right) = \frac{P}{m}, \quad \text{hence} \quad X_H(t) = X + \frac{Pt}{m}.$$

This is exactly the classical solution, now as an operator identity. It has genuine content: for instance

$$[X_H(t), X_H(0)] = \frac{t}{m}[P, X] = -\frac{i\hbar t}{m},$$

so positions at different times are *incompatible* observables, and we recover the spreading of a free wavepacket,

$$(\Delta X)_t^2 = \langle X_H(t)^2 \rangle - \langle X_H(t) \rangle^2 = (\Delta X)_0^2 + \frac{t}{m} \langle XP + PX \rangle_c + \frac{t^2}{m^2} (\Delta P)^2,$$

where the middle term is the connected correlator $\langle XP + PX \rangle - 2\langle X \rangle \langle P \rangle$. For large t the packet spreads linearly with rate $\Delta P/m$, which is just the classical statement that a spread of velocities produces a spread of positions.

Remark (Time-Energy Uncertainty). There is no time operator in quantum mechanics – t is a parameter, not an observable – so ‘ $\Delta E \Delta t \geq \hbar/2$ ’ cannot be an instance of the generalised uncertainty principle. It can, however, be given a precise meaning. For any observable A with no explicit time dependence, Ehrenfest plus the uncertainty relation give

$$\Delta E \Delta A \geq \frac{1}{2} |\langle [H, A] \rangle| = \frac{\hbar}{2} \left| \frac{d \langle A \rangle}{dt} \right|,$$

so defining $\tau_A = \Delta A / |d \langle A \rangle / dt|$ – the time needed for $\langle A \rangle$ to shift by one standard deviation, i.e. the timescale on which the change is detectable – we get $\Delta E \tau_A \geq \hbar/2$. A state of sharp energy is stationary and nothing ever happens in it; things can only change on a timescale set by the energy spread.

§37.3 The Harmonic Oscillator

Every quantum system that oscillates is, near equilibrium, a harmonic oscillator, and in IB we solved it by grinding through Hermite’s equation. We now solve it again in three-quarters of a page, using nothing but the commutator $[X, P] = i\hbar$. This is not merely a labour-saving trick: the method – find operators that step between eigenstates, then bound the spectrum below – is exactly the method we will use for angular momentum, and the operators we build are the foundation of quantum field theory.

§37.3.1 Creation and Annihilation Operators

The Hamiltonian is

$$H = \frac{P^2}{2m} + \frac{1}{2} m \omega^2 X^2.$$

If X and P were numbers we would factorise this as a product of $(\cdot + i\cdot)$ and $(\cdot - i\cdot)$. They are not, but let us try anyway.

Definition 37.3.1 (Annihilation and Creation Operators)

Set

$$a = \sqrt{\frac{m\omega}{2\hbar}} \left(X + \frac{iP}{m\omega} \right), \quad a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(X - \frac{iP}{m\omega} \right),$$

the **annihilation** and **creation** operators. (The normalisation is chosen to make them dimensionless.)

These are adjoints of one another, as the notation says, since X and P are Hermitian; they are not themselves Hermitian, and are the first non-Hermitian operators we have found a use for. Inverting,

$$X = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger), \quad P = -i\sqrt{\frac{m\hbar\omega}{2}} (a - a^\dagger),$$

which we will use constantly – almost every harmonic-oscillator computation in the rest of the course is done by substituting these and reading off the answer.

The single fact we need is:

$$[a, a^\dagger] = \frac{m\omega}{2\hbar} \left(\left[X + \frac{iP}{m\omega}, X - \frac{iP}{m\omega} \right] \right) = \frac{m\omega}{2\hbar} \cdot \frac{-2i}{m\omega} [X, P] = \frac{m\omega}{2\hbar} \cdot \frac{-2i}{m\omega} \cdot i\hbar = 1.$$

Now compute the product:

$$a^\dagger a = \frac{m\omega}{2\hbar} \left(X^2 + \frac{P^2}{m^2\omega^2} + \frac{i}{m\omega} [X, P] \right) = \frac{H}{\hbar\omega} - \frac{1}{2}.$$

Definition 37.3.2 (Number Operator)

$N = a^\dagger a$, so that $H = \hbar\omega (N + \frac{1}{2})$.

N is Hermitian, and finding the spectrum of H is now the same as finding the spectrum of N . The Leibniz rule and $[a, a^\dagger] = 1$ give the two commutators that do all the work:

$$[N, a] = [a^\dagger a, a] = [a^\dagger, a]a = -a, \quad [N, a^\dagger] = a^\dagger[a, a^\dagger] = a^\dagger.$$

§37.3.2 The Spectrum

Theorem 37.3.3 (Spectrum of the Harmonic Oscillator)

The eigenvalues of N are exactly $n = 0, 1, 2, \dots$, each non-degenerate, so

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n \in \mathbb{Z}_{\geq 0}.$$

Writing $|n\rangle$ for the normalised eigenstate of N with eigenvalue n ,

$$a|n\rangle = \sqrt{n}|n-1\rangle, \quad a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle, \quad |n\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0\rangle.$$

Proof. Suppose $N|\nu\rangle = \nu|\nu\rangle$ with $|\nu\rangle$ normalised. First,

$$\nu = \langle \nu | a^\dagger a | \nu \rangle = \|a|\nu\rangle\|^2 \geq 0,$$

so all eigenvalues are non-negative, with $\nu = 0$ if and only if $a|\nu\rangle = 0$. Next, using $[N, a] = -a$,

$$N(a|\nu\rangle) = (aN - a)|\nu\rangle = (\nu - 1)a|\nu\rangle,$$

so $a|\nu\rangle$ is either zero or an eigenvector with eigenvalue $\nu - 1$; similarly $N(a^\dagger|\nu\rangle) = (\nu + 1)a^\dagger|\nu\rangle$, with $\|a^\dagger|\nu\rangle\|^2 = \langle \nu | aa^\dagger | \nu \rangle = \nu + 1 > 0$, so $a^\dagger|\nu\rangle$ is never zero.

Now suppose ν is not a non-negative integer. Then applying a repeatedly produces eigenvectors with eigenvalues $\nu - 1, \nu - 2, \dots$, and none of them can ever be zero (a vector is zero only if its eigenvalue was 0, which never occurs in this sequence). But eventually the eigenvalues become negative, contradicting $\nu \geq 0$. Hence $\nu = n \in \mathbb{Z}_{\geq 0}$, and the descending sequence terminates cleanly at $|0\rangle$ with $a|0\rangle = 0$.

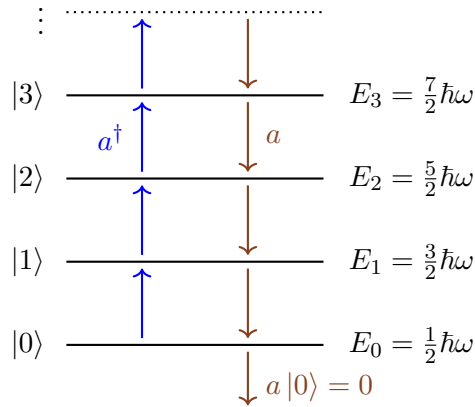
Conversely such states exist: $|0\rangle$ exists (we exhibit it in the next subsection), and $(a^\dagger)^n|0\rangle$ is a non-zero eigenvector with eigenvalue n . The normalisations follow from $\|a|n\rangle\|^2 = n$ and $\|a^\dagger|n\rangle\|^2 = n + 1$, choosing phases so that both constants are positive; iterating $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ gives $|n\rangle = (a^\dagger)^n|0\rangle / \sqrt{n!}$.

Finally, non-degeneracy. In the position representation $a|0\rangle = 0$ reads

$$\left(x + \frac{\hbar}{m\omega} \frac{d}{dx}\right) \psi_0(x) = 0,$$

a first-order linear ODE with a one-dimensional solution space. So the ground state is unique, and since every $|n\rangle$ is obtained from it by $(a^\dagger)^n$ – and conversely any eigenvector of eigenvalue n is mapped by a^n into the $\nu = 0$ eigenspace – every level is non-degenerate. $\square$

So the spectrum is an evenly spaced ladder, with $a^\dagger$ climbing it and a descending it. It is worth drawing.



The operators are named for what the picture shows: $a^\dagger$ creates a quantum of energy $\hbar\omega$ and a annihilates one. The state $|n\rangle$ is naturally thought of not as ‘the n th excited state of one oscillator’ but as ‘ n identical quanta’, with N counting them and $|0\rangle$ the **vacuum**. Applied to the modes of the electromagnetic field, those quanta are photons; this reinterpretation is the beginning of quantum field theory, and it is why this small piece of algebra is so important.

§37.3.3 Wavefunctions and Matrix Elements

We can now recover the IB results with no differential equations beyond the first-order one above. Solving $\psi'_0 = -(m\omega/\hbar)x\psi_0$ and normalising,

$$\psi_0(x) = \langle x|0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar}\right),$$

a Gaussian – and, by §37.1.9, a minimum uncertainty state. The excited states follow by applying

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right)$$

repeatedly, which manufactures a Hermite polynomial times the Gaussian at each step; for instance

$$\psi_1(x) \propto \left(x - \frac{\hbar}{m\omega} \frac{d}{dx}\right) e^{-m\omega x^2/2\hbar} = 2x e^{-m\omega x^2/2\hbar},$$

matching $H_1(\xi) = 2\xi$. Orthonormality of the $|n\rangle$ is automatic (eigenvectors of a Hermitian operator with distinct eigenvalues), which saves proving any Hermite polynomial identities at all.

More useful in practice are matrix elements. From $X = \sqrt{\hbar/2m\omega}(a + a^\dagger)$,

$$\langle m|X|n\rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(\sqrt{n} \delta_{m,n-1} + \sqrt{n+1} \delta_{m,n+1} \right),$$

so X connects only neighbouring levels – a selection rule which we will use in perturbation theory. Similarly

$$\langle m|P|n\rangle = i\sqrt{\frac{m\hbar\omega}{2}} \left(\sqrt{n+1} \delta_{m,n+1} - \sqrt{n} \delta_{m,n-1} \right).$$

Example 37.3.4 (Uncertainties in the Level $|n\rangle$)

Since $\langle n|X|n\rangle = \langle n|P|n\rangle = 0$, we need only the squares. Expanding,

$$X^2 = \frac{\hbar}{2m\omega} \left(a^2 + (a^\dagger)^2 + aa^\dagger + a^\dagger a \right),$$

and $\langle n|a^2|n\rangle = \langle n|(a^\dagger)^2|n\rangle = 0$ (they change the level by two), while $a^\dagger a = N$ and $aa^\dagger = N + 1$. Hence

$$\langle X^2 \rangle_n = \frac{\hbar}{2m\omega} (2n + 1) = \frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right),$$

and identically $\langle P^2 \rangle_n = m\hbar\omega \left(n + \frac{1}{2} \right)$. Therefore

$$\Delta X \Delta P = \left(n + \frac{1}{2} \right) \hbar,$$

saturating the uncertainty bound only in the ground state, as expected. Note also that $\langle P^2/2m \rangle_n = \langle \frac{1}{2}m\omega^2 X^2 \rangle_n = \frac{1}{2}E_n$: kinetic and potential energy contribute equally, which is the quantum virial theorem for a quadratic potential.

Example 37.3.5 (The Oscillator in the Heisenberg Picture)

The Heisenberg equation for a is especially clean. Since $[H, a] = \hbar\omega[N, a] = -\hbar\omega a$,

$$\frac{da_H}{dt} = \frac{i}{\hbar}[H, a_H] = -i\omega a_H \quad \implies \quad a_H(t) = a e^{-i\omega t},$$

and taking adjoints $a_H^\dagger(t) = a^\dagger e^{i\omega t}$. Substituting into $X = \sqrt{\hbar/2m\omega}(a + a^\dagger)$ gives

$$X_H(t) = X \cos \omega t + \frac{P}{m\omega} \sin \omega t, \quad P_H(t) = P \cos \omega t - m\omega X \sin \omega t,$$

which is precisely the classical solution, holding now as an operator identity. This is Ehrenfest's theorem in its most satisfying form: for a quadratic Hamiltonian the expectation values follow the classical trajectory *exactly*, for every state.

Remark (Coherent States). Since a is not Hermitian its eigenvalues need not be real, and in fact a has an eigenstate for *every* complex number α :

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} |n\rangle, \quad a |\alpha\rangle = \alpha |\alpha\rangle,$$

as one checks by applying a term by term. These **coherent states** are minimum-uncertainty Gaussians displaced from the origin, and under time evolution $|\alpha\rangle \mapsto |\alpha e^{-i\omega t}\rangle$ – a wavepacket that oscillates back and forth without ever spreading, tracing out the classical motion. They are the closest thing quantum mechanics has to a classical oscillator, and are the states produced by a laser.

Example 37.3.6 (The Three-Dimensional Isotropic Oscillator)

For $H = \frac{\mathbf{P}^2}{2m} + \frac{1}{2}m\omega^2\mathbf{X}^2$ the three directions decouple: with a_i built from X_i, P_i we have $[a_i, a_j^\dagger] = \delta_{ij}$ and

$$H = \hbar\omega \left(N_1 + N_2 + N_3 + \frac{3}{2} \right),$$

so $E = (n + \frac{3}{2})\hbar\omega$ with $n = n_1 + n_2 + n_3$. The degeneracy of the level n is the number of ways to write n as an ordered sum of three non-negative integers, namely $\binom{n+2}{2} = \frac{1}{2}(n+1)(n+2)$: so 1, 3, 6, 10, ... Some of this is explained by rotational symmetry (§37.6), but not all of it – the isotropic oscillator, like the hydrogen atom, has a ‘hidden’ symmetry larger than rotations.

§37.4 Angular Momentum

In IB we found that the orbital angular momentum operators $L_i = \epsilon_{ijk}X_jP_k$ satisfy

$$[L_i, L_j] = i\hbar \epsilon_{ijk}L_k,$$

and that solving the associated differential equations gives the spherical harmonics, with ℓ a non-negative integer. We now run the harmonic-oscillator strategy on this problem: forget where the operators came from, keep only the commutation relations, and see what the algebra forces. The answer will turn out to be strictly richer than what the differential equations gave us – half-integer values appear, and they are realised in nature as *spin*, an angular momentum with no classical or orbital counterpart at all.

§37.4.1 The Algebra

Definition 37.4.1 (Angular Momentum)

An **angular momentum** is a triple of Hermitian operators J_1, J_2, J_3 on $\mathcal{H}$ satisfying

$$[J_i, J_j] = i\hbar \epsilon_{ijk}J_k.$$

We write $J^2 = J_1^2 + J_2^2 + J_3^2$, and usually J_z for J_3 .

Written out, the relations are $[J_1, J_2] = i\hbar J_3$ and its cyclic images. The first thing to check is that although no two components are compatible, the total is compatible with each one.

Proposition 37.4.2

$[J^2, J_i] = 0$ for each i .

Proof. Take $i = 3$; the rest follow by relabelling. Using $[A, B^2] = B[A, B] + [A, B]B$,

$$[J_1^2, J_3] = J_1[J_1, J_3] + [J_1, J_3]J_1 = -i\hbar(J_1J_2 + J_2J_1),$$

$$[J_2^2, J_3] = J_2[J_2, J_3] + [J_2, J_3]J_2 = i\hbar(J_2J_1 + J_1J_2),$$

and $[J_3^2, J_3] = 0$. The two non-zero pieces cancel. $\square$

So $\{J^2, J_z\}$ is a commuting pair, and we look for their simultaneous eigenstates. This is the best we can do – the length of the angular momentum vector and one of its components – and the choice of z is pure convention.

Definition 37.4.3 (Ladder Operators)

$J_{\pm} = J_1 \pm iJ_2$, so that $J_{\pm}^{\dagger} = J_{\mp}$.

The relations we need are all immediate:

$$[J_z, J_{\pm}] = \pm\hbar J_{\pm}, \quad [J_+, J_-] = 2\hbar J_z, \quad [J^2, J_{\pm}] = 0,$$

and, expanding the products and using $[J_1, J_2] = i\hbar J_3$,

$$J_-J_+ = (J_1 - iJ_2)(J_1 + iJ_2) = J_1^2 + J_2^2 + i[J_1, J_2] = J^2 - J_z^2 - \hbar J_z,$$

and, changing the sign of i throughout, $J_+J_- = J^2 - J_z^2 + \hbar J_z$. The first of these commutators, $[J_z, J_{\pm}] = \pm\hbar J_{\pm}$, is exactly the relation $[N, a^{\dagger}] = a^{\dagger}$ that made the oscillator work, and it does the same job here.

§37.4.2 The Spectrum

Theorem 37.4.4 (Spectrum of Angular Momentum)

Let $|\psi\rangle$ be a simultaneous eigenstate of J^2 and J_z . Then the eigenvalues take the form

$$J^2 |j m\rangle = \hbar^2 j(j+1) |j m\rangle, \quad J_z |j m\rangle = \hbar m |j m\rangle,$$

where

$$j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots\}, \quad m \in \{-j, -j+1, \dots, j-1, j\}.$$

Moreover, with a suitable choice of phases,

$$J_{\pm} |j m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j m \pm 1\rangle.$$

Proof. Write the eigenvalues as $J^2 |\psi\rangle = \hbar^2 \lambda |\psi\rangle$ and $J_z |\psi\rangle = \hbar m |\psi\rangle$ with $|\psi\rangle$ normalised.

Step 1: a bound. Since $J^2 - J_z^2 = J_1^2 + J_2^2$ and each J_i is Hermitian,

$$\hbar^2(\lambda - m^2) = \langle \psi | J_1^2 + J_2^2 | \psi \rangle = \|J_1 |\psi\rangle\|^2 + \|J_2 |\psi\rangle\|^2 \geq 0,$$

so $\lambda \geq m^2$: for fixed λ the possible m are bounded above and below.

Step 2: the ladder. Since $[J^2, J_{\pm}] = 0$ and $[J_z, J_{\pm}] = \pm\hbar J_{\pm}$,

$$J^2 (J_{\pm} |\psi\rangle) = \hbar^2 \lambda (J_{\pm} |\psi\rangle), \quad J_z (J_{\pm} |\psi\rangle) = \hbar(m \pm 1) (J_{\pm} |\psi\rangle),$$

so $J_{\pm}$ moves m by one while preserving λ – unless it gives zero. And its norm is computable:

$$\|J_{\pm}|\psi\rangle\|^2 = \langle\psi|J_{\mp}J_{\pm}|\psi\rangle = \langle\psi|J^2 - J_z^2 \mp \hbar J_z|\psi\rangle = \hbar^2(\lambda - m(m \pm 1)).$$

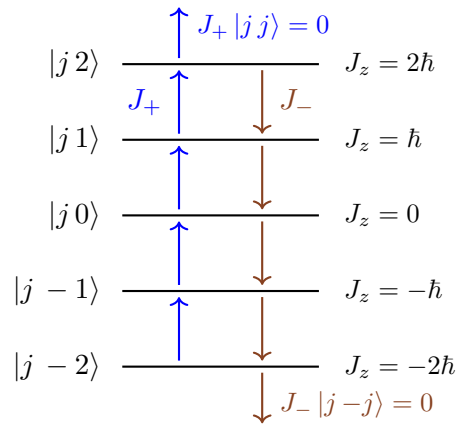
Step 3: termination. By Step 1 the sequence $|\psi\rangle, J_+|\psi\rangle, J_+^2|\psi\rangle, \dots$ cannot continue for ever, so there is a top state with J_z eigenvalue $\hbar j$ (say) killed by J_+ ; by Step 2, $\lambda = j(j+1)$. Likewise there is a bottom state with eigenvalue $\hbar m_{\min}$ killed by J_- , giving $\lambda = m_{\min}(m_{\min} - 1)$. Setting the two equal,

$$j(j+1) = m_{\min}(m_{\min} - 1) \implies m_{\min} = -j$$

(the other root, $m_{\min} = j+1$, is impossible since $m_{\min} \leq j$).

Step 4: quantisation. The bottom state is reached from the top by applying J_- some non-negative integer number of times, so $j - (-j) = 2j \in \mathbb{Z}_{\geq 0}$: that is, j is a non-negative integer or half-integer, and m runs over $-j, -j+1, \dots, j$ in unit steps, giving $2j+1$ values. The normalisation constants come from Step 2, choosing all phases positive. $\square$

Compare this with the IB derivation. There, integrality of ℓ came from demanding that $e^{im\phi}$ be single-valued on the sphere; here, quantisation comes from the algebra alone, and it permits half-integers. Both answers are correct: half-integer j cannot occur for *orbital* angular momentum (§37.4.5), but nothing forbids it in general, and electrons, protons and neutrons all have $j = \frac{1}{2}$.



the multiplet $j = 2$: five states, all with $J^2 = 6\hbar^2$

For each allowed j the $2j+1$ states $|j, m\rangle$ form a **multiplet**, closed under the action of all three J_i (since J_1, J_2 are combinations of $J_{\pm}$). In the language of Representation Theory, a multiplet is an irreducible representation of the Lie algebra $\mathfrak{su}(2)$, and we have just classified them all. Nothing in the argument says which j actually occur in a given physical system, or how many times – that depends on the system.

§37.4.3 Matrix Representations and Spin

Within a single multiplet the operators are finite matrices, which we can write down explicitly.

Example 37.4.5 ($j = 0$)

A single state $|00\rangle$ with $J_i = 0$: no angular momentum at all.

Example 37.4.6 ($j = \frac{1}{2}$)

Two states, $|\frac{1}{2} \frac{1}{2}\rangle =: |\uparrow\rangle$ and $|\frac{1}{2} -\frac{1}{2}\rangle =: |\downarrow\rangle$, spanning $\mathbb{C}^2$. From the theorem, $J_z = \frac{\hbar}{2}\text{diag}(1, -1)$, and

$$J_+ |\downarrow\rangle = \hbar |\uparrow\rangle, \quad J_+ |\uparrow\rangle = 0, \quad J_- |\uparrow\rangle = \hbar |\downarrow\rangle, \quad J_- |\downarrow\rangle = 0,$$

so in the basis $(|\uparrow\rangle, |\downarrow\rangle)$,

$$J_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \text{hence} \quad J_1 = \frac{J_+ + J_-}{2}, \quad J_2 = \frac{J_+ - J_-}{2i}.$$

Writing $\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$ (the traditional notation for spin), this gives the **Pauli matrices**

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Example 37.4.7 ($j = 1$)

Three states; in the basis $|11\rangle, |10\rangle, |1-1\rangle$, the coefficients $\sqrt{j(j+1) - m(m \pm 1)}$ are all $\sqrt{2}$, so

$$J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad J_+ = \sqrt{2}\hbar \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad J_- = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

One can check $J^2 = 2\hbar^2 I$, as it must be.

The Pauli matrices deserve their own list of properties, since spin- $\frac{1}{2}$ is by far the most important case.

Proposition 37.4.8 (Pauli Algebra)

The Pauli matrices are Hermitian and traceless, with $\sigma_i^2 = I$, and

$$\sigma_i \sigma_j = \delta_{ij} I + i\epsilon_{ijk} \sigma_k.$$

Consequently $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk} \sigma_k$ and $\{\sigma_i, \sigma_j\} = 2\delta_{ij} I$, and for any unit vector $\mathbf{n}$,

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I, \quad \exp\left(-\frac{i\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}\right) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}.$$

Proof. The product rule is verified by multiplying the matrices out (or by noting that the symmetric part must be $\delta_{ij} I$ since $\sigma_i^2 = I$, and the antisymmetric part is

$\frac{1}{2}[\sigma_i, \sigma_j]$, which is the angular momentum relation). Then

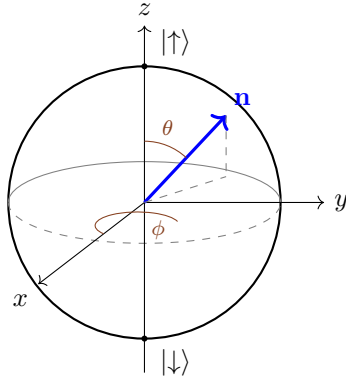
$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = n_i n_j \sigma_i \sigma_j = n_i n_j \delta_{ij} I + i \epsilon_{ijk} n_i n_j \sigma_k = |\mathbf{n}|^2 I = I,$$

the ϵ term dying by symmetry. The exponential then follows by splitting the power series into even and odd parts, exactly as for $e^{i\theta}$. $\square$

Since $(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = I$, the operator $\mathbf{n} \cdot \mathbf{S} = \frac{\hbar}{2} \mathbf{n} \cdot \boldsymbol{\sigma}$ has eigenvalues $\pm \frac{\hbar}{2}$ for *every* direction $\mathbf{n}$: a spin- $\frac{1}{2}$ measurement along any axis gives one of two answers. Writing $\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$, one finds the eigenstates

$$|\mathbf{n} \uparrow\rangle = \cos \frac{\theta}{2} |\uparrow\rangle + e^{i\phi} \sin \frac{\theta}{2} |\downarrow\rangle, \quad |\mathbf{n} \downarrow\rangle = -e^{-i\phi} \sin \frac{\theta}{2} |\uparrow\rangle + \cos \frac{\theta}{2} |\downarrow\rangle.$$

This is a bijection (up to phase) between unit vectors in $\mathbb{R}^3$ and states of a qubit, and gives the standard picture of the spin- $\frac{1}{2}$ state space as a sphere.



This is the **Bloch sphere**. Antipodal points are orthogonal states (note $\theta \mapsto \pi - \theta$, $\phi \mapsto \phi + \pi$ sends $|\mathbf{n} \uparrow\rangle$ to $|\mathbf{n} \downarrow\rangle$ up to phase), so the angle on the sphere is *twice* the angle in Hilbert space – another manifestation of the factor of $\frac{1}{2}$ that pervades spin- $\frac{1}{2}$.

Example 37.4.9 (Measuring Spin in a Rotated Basis)

Prepare a particle in $|\uparrow\rangle$ and measure $\mathbf{n} \cdot \mathbf{S}$. The probability of $+\frac{\hbar}{2}$ is

$$|\langle \mathbf{n} \uparrow | \uparrow \rangle|^2 = \cos^2 \frac{\theta}{2} = \frac{1 + \cos \theta}{2},$$

and of $-\frac{\hbar}{2}$ is $\sin^2 \frac{\theta}{2}$. So $\langle \mathbf{n} \cdot \mathbf{S} \rangle_{\uparrow} = \frac{\hbar}{2} \cos \theta$: the expectation behaves like the projection of a classical vector of length $\frac{\hbar}{2}$ onto $\mathbf{n}$, even though every individual measurement returns $\pm \frac{\hbar}{2}$. For $\theta = \pi/2$ the two outcomes are equally likely – a state with definite S_z is maximally uncertain in S_x .

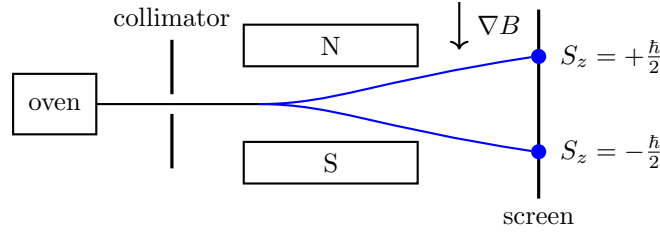
§37.4.4 The Stern–Gerlach Experiment

Spin is not a mathematical curiosity: it was forced on physics by experiment. A neutral particle with spin has a magnetic moment proportional to it,

$$\boldsymbol{\mu} = \gamma \mathbf{S},$$

with γ the **gyromagnetic ratio**, and in a magnetic field it acquires an energy $H = -\boldsymbol{\mu} \cdot \mathbf{B}$. In a field with a gradient the particle also feels a force $\nabla(\boldsymbol{\mu} \cdot \mathbf{B})$, so a beam

passing through an inhomogeneous field is deflected by an amount proportional to μ_z – which is a direct measurement of S_z . Classically, an unpolarised beam has μ_z uniformly distributed and one expects a continuous smear on the screen. What Stern and Gerlach (1922) saw with silver atoms was two discrete spots.



Two spots means two values of S_z , i.e. $2j + 1 = 2$, i.e. $j = \frac{1}{2}$ – a value impossible for orbital angular momentum. The apparatus is also the cleanest illustration of the measurement postulate: block the lower beam, and the surviving atoms are all in $|\uparrow\rangle$; pass them through a second identical magnet and they all go up, but pass them through a magnet oriented along x and they split 50–50, in agreement with the example above. Recombining the two beams coherently, on the other hand, restores the original state entirely – so long as no measurement records which path was taken.

Example 37.4.10 (Larmor Precession)

Put a spin in a uniform field $\mathbf{B} = B\mathbf{e}_3$. Then $H = -\gamma BS_z$, and using $[S_3, S_1] = i\hbar S_2$, $[S_3, S_2] = -i\hbar S_1$, the Heisenberg equations give

$$\frac{dS_1}{dt} = -\gamma B \frac{i}{\hbar} [S_3, S_1] = \omega_L S_2, \quad \frac{dS_2}{dt} = -\omega_L S_1, \quad \frac{dS_3}{dt} = 0,$$

where $\omega_L = \gamma B$. So $\langle \mathbf{S} \rangle$ precesses about the field at the **Larmor frequency** ω_L , exactly like a classical gyroscope. In the Schrödinger picture the same statement is that $|\uparrow\rangle$ and $|\downarrow\rangle$ acquire phases $e^{\pm i\omega_L t/2}$, so a state initially along $\mathbf{e}_1$ becomes

$$\frac{1}{\sqrt{2}} \left(e^{i\omega_L t/2} |\uparrow\rangle + e^{-i\omega_L t/2} |\downarrow\rangle \right),$$

which is the spin pointing along $(\cos \omega_L t, -\sin \omega_L t, 0)$. Note that after a time $2\pi/\omega_L$ the *spin direction* has returned but the *state vector* has acquired a factor of -1 ; only after 4π of rotation does the state itself return.

§37.4.5 Orbital Angular Momentum

Return now to $\mathbf{L} = \mathbf{X} \times \mathbf{P}$. One checks (as in IB, using $[X_i, P_j] = i\hbar \delta_{ij}$) that this satisfies the angular momentum algebra, so all of the above applies with $j = \ell$. The question is which ℓ occur.

In spherical polar coordinates $L_z = -i\hbar \partial_\phi$, and a slightly longer computation gives

$$L_\pm = \hbar e^{\pm i\phi} \left(\pm \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \phi} \right).$$

An eigenfunction of L_z with eigenvalue $\hbar m$ has ϕ -dependence $e^{im\phi}$, and single-valuedness on the sphere forces $m \in \mathbb{Z}$; hence $\ell \in \mathbb{Z}_{\geq 0}$. *Half-integer values are excluded for orbital angular momentum, but only by this topological input, which has no analogue for an intrinsic degree of freedom.* That is the precise sense in which spin is ‘not orbital’.

The ladder operators give a pleasant way to construct the spherical harmonics without touching Legendre's equation. The top state of the multiplet satisfies $L_+ Y_{\ell\ell} = 0$, i.e.

$$\left(\frac{\partial}{\partial \theta} - \ell \cot \theta \right) f(\theta) = 0 \quad \text{for } Y_{\ell\ell} = f(\theta) e^{i\ell\phi},$$

whose solution is $f \propto \sin^\ell \theta$. So

$$Y_{\ell\ell}(\theta, \phi) \propto \sin^\ell \theta e^{i\ell\phi},$$

and every other $Y_{\ell m}$ follows by applying L_- repeatedly, with the normalisations supplied by the theorem. For $\ell = 1$ this gives $Y_{11} \propto \sin \theta e^{i\phi}$, then L_- produces $Y_{10} \propto \cos \theta$ and $Y_{1,-1} \propto \sin \theta e^{-i\phi}$, matching IB.

Remark. Notice how much less work this is than the differential equation route, and how much more it explains. The mysterious constraint $|m| \leq \ell$ and the eigenvalue $\hbar^2 \ell(\ell+1)$ (rather than $\hbar^2 \ell^2$) are now consequences of the commutation relations, and the same algebra will hand us the whole theory of spin and of adding angular momenta.

For a particle with spin moving in space, the total angular momentum is

$$\mathbf{J} = \mathbf{L} + \mathbf{S},$$

where $\mathbf{L}$ acts on the spatial factor of $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ and $\mathbf{S}$ on the spin factor. Since they act on different factors they commute, and one checks immediately that $\mathbf{J}$ satisfies the angular momentum algebra. Understanding the multiplet structure of such a sum is the subject of the next section.

§37.5 Composite Systems and Addition of Angular Momentum

§37.5.1 Tensor Products

If a system is built from two subsystems – two particles, or the position and spin of one particle – how do their Hilbert spaces combine? The answer is not a direct sum but a tensor product, and this innocuous-looking choice is the source of essentially everything strange about quantum mechanics.

Definition 37.5.1 (Tensor Product)

Let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces with orthonormal bases $\{|i\rangle_1\}, \{|a\rangle_2\}$. Their **tensor product** $\mathcal{H}_1 \otimes \mathcal{H}_2$ is the Hilbert space with orthonormal basis $\{|i\rangle_1 \otimes |a\rangle_2\}$, and inner product determined by

$$(\langle \phi |_1 \otimes \langle \chi |_2) (|\psi \rangle_1 \otimes |\omega \rangle_2) = \langle \phi | \psi \rangle_1 \langle \chi | \omega \rangle_2.$$

The map $(|\psi \rangle_1, |\omega \rangle_2) \mapsto |\psi \rangle_1 \otimes |\omega \rangle_2$ is bilinear.

Axiom 37.5.2 (Composite Systems)

The state space of a system composed of two subsystems with state spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ is $\mathcal{H}_1 \otimes \mathcal{H}_2$.

In particular $\dim(\mathcal{H}_1 \otimes \mathcal{H}_2) = \dim \mathcal{H}_1 \cdot \dim \mathcal{H}_2$ – dimensions multiply, they do not add. We will freely abbreviate $|\psi\rangle \otimes |\omega\rangle$ as $|\psi\rangle |\omega\rangle$ or $|\psi\omega\rangle$ when no confusion can arise. An operator A on $\mathcal{H}_1$ acts on the product as $A \otimes I$, and

$$(A \otimes B)(|\psi\rangle \otimes |\omega\rangle) = A|\psi\rangle \otimes B|\omega\rangle, \quad (A \otimes B)(C \otimes D) = AC \otimes BD.$$

Operators belonging to different factors always commute: $[A \otimes I, I \otimes B] = 0$. This is the formal statement that measurements on subsystem 1 are compatible with measurements on subsystem 2.

Example 37.5.3 (Two Spins)

For two spin- $\frac{1}{2}$ particles, $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$ is four-dimensional with basis

$$|\uparrow\uparrow\rangle, \quad |\uparrow\downarrow\rangle, \quad |\downarrow\uparrow\rangle, \quad |\downarrow\downarrow\rangle.$$

A single spin- $\frac{1}{2}$ particle in three dimensions has $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$; a general state is $|\Psi\rangle = |\psi_\uparrow\rangle \otimes |\uparrow\rangle + |\psi_\downarrow\rangle \otimes |\downarrow\rangle$, conveniently written as a two-component **spinor** wavefunction

$$\Psi(\mathbf{x}) = \begin{pmatrix} \psi_\uparrow(\mathbf{x}) \\ \psi_\downarrow(\mathbf{x}) \end{pmatrix},$$

with $|\psi_\uparrow(\mathbf{x})|^2$ the probability density for finding the particle at $\mathbf{x}$ with spin up.

Definition 37.5.4 (Entanglement)

A state of $\mathcal{H}_1 \otimes \mathcal{H}_2$ is a **product state** if it can be written as $|\psi\rangle \otimes |\omega\rangle$, and is **entangled** otherwise.

Most states are entangled: a general state $\sum_{i,a} c_{ia} |i\rangle |a\rangle$ is a product exactly when the matrix (c_{ia}) has rank one. For example,

$$\frac{1}{\sqrt{2}} (|\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle) = |\uparrow\rangle \otimes \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}} \quad \text{is a product,}$$

whereas

$$\frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad \text{is not}$$

(its coefficient matrix is $\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, which is invertible). In an entangled state neither subsystem has a state of its own, yet the pair does – a situation with no classical analogue whatsoever, and the crux of the EPR argument in §37.11.

§37.5.2 Adding Angular Momenta

Suppose subsystems 1 and 2 carry angular momenta $\mathbf{J}^{(1)}$ and $\mathbf{J}^{(2)}$. The total angular momentum of the composite system is

$$\mathbf{J} = \mathbf{J}^{(1)} \otimes I + I \otimes \mathbf{J}^{(2)},$$

which we will write simply as $\mathbf{J} = \mathbf{J}^{(1)} + \mathbf{J}^{(2)}$. Since the two pieces commute, $\mathbf{J}$ satisfies the angular momentum algebra:

$$[J_i, J_j] = [J_i^{(1)}, J_j^{(1)}] + [J_i^{(2)}, J_j^{(2)}] = i\hbar\epsilon_{ijk} (J_k^{(1)} + J_k^{(2)}) = i\hbar\epsilon_{ijk} J_k.$$

So the results of §37.4 apply, and the question is: given multiplets j_1 and j_2 , how does the $(2j_1 + 1)(2j_2 + 1)$ -dimensional product space decompose into multiplets of $\mathbf{J}$?

There are two natural bases. The **uncoupled basis**

$$|j_1 m_1\rangle |j_2 m_2\rangle, \quad m_1 = -j_1, \dots, j_1, \quad m_2 = -j_2, \dots, j_2,$$

diagonalises the CSCO $\{(J^{(1)})^2, J_z^{(1)}, (J^{(2)})^2, J_z^{(2)}\}$; the **coupled basis**

$$|J M\rangle, \quad M = -J, \dots, J,$$

diagonalises $\{(J^{(1)})^2, (J^{(2)})^2, J^2, J_z\}$. Both are legitimate: one checks that J^2 commutes with $(J^{(1)})^2$ and $(J^{(2)})^2$ (each $\mathbf{J}^{(a)}$ commutes with its own square) and with J_z , but that

$$[J^2, J_z^{(1)}] \neq 0,$$

which is why we must give up m_1 and m_2 individually to gain J . Indeed $J^2 = (J^{(1)})^2 + (J^{(2)})^2 + 2\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$, and the cross term does not commute with $J_z^{(1)}$.

Theorem 37.5.5 (Decomposition of a Tensor Product)

The product of multiplets j_1 and j_2 decomposes as

$$j_1 \otimes j_2 = (j_1 + j_2) \oplus (j_1 + j_2 - 1) \oplus \dots \oplus |j_1 - j_2|,$$

each value of J occurring exactly once.

Proof. Since $J_z = J_z^{(1)} + J_z^{(2)}$, the uncoupled basis vectors are already J_z eigenstates with $M = m_1 + m_2$. Let $d(M)$ be the number of uncoupled basis states with that value of M ; assume without loss of generality $j_1 \geq j_2$. Counting lattice points on the line $m_1 + m_2 = M$ inside the rectangle,

$$d(M) = \begin{cases} 0 & |M| > j_1 + j_2, \\ j_1 + j_2 - |M| + 1 & j_1 - j_2 \leq |M| \leq j_1 + j_2, \\ 2j_2 + 1 & |M| \leq j_1 - j_2. \end{cases}$$

On the other hand, a multiplet of total J contributes exactly one state to each M with $|M| \leq J$. So if $n(J)$ is the number of times J occurs,

$$d(M) = \sum_{J \geq |M|} n(J), \quad \text{hence} \quad n(J) = d(J) - d(J+1).$$

Reading off from the formula for d : $n(J) = 1$ for $|j_1 - j_2| \leq J \leq j_1 + j_2$ and $n(J) = 0$ otherwise. $\square$

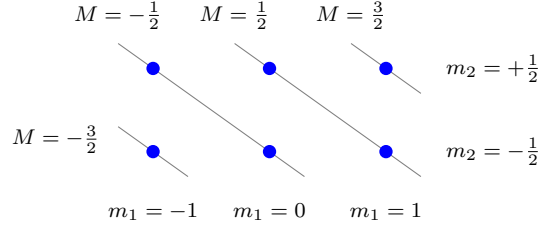
As a check on the arithmetic, the dimensions must agree, and they do:

$$\sum_{J=|j_1-j_2|}^{j_1+j_2} (2J+1) = (2j_1+1)(2j_2+1),$$

which is an easy sum of an arithmetic progression. Note also that the parity of $j_1 + j_2 - J$ organises the answer, and that if j_1 and j_2 are both half-integers then all the resulting

J are integers – so a system of two fermions has integer total angular momentum, and behaves like a boson.

The counting is best seen as a picture. Here is the lattice of uncoupled states for $j_1 = 1$, $j_2 = \frac{1}{2}$, with the lines of constant $M = m_1 + m_2$ drawn in.



The picture makes the counting argument transparent: the multiplicities $d(M)$ along the diagonals are 1, 2, 2, 1 for $M = \frac{3}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$, so $n(\frac{3}{2}) = 1$ and $n(\frac{1}{2}) = 2 - 1 = 1$, giving $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$, of dimension $4 + 2 = 6$.

§37.5.3 Clebsch–Gordan Coefficients

The two bases are related by a change of basis, whose matrix elements have a name.

Definition 37.5.6 (Clebsch–Gordan Coefficients)

Inserting the completeness relation for the uncoupled basis,

$$|J M\rangle = \sum_{m_1, m_2} |j_1 m_1\rangle |j_2 m_2\rangle \langle j_1 m_1; j_2 m_2 | J M \rangle,$$

and the numbers $\langle j_1 m_1; j_2 m_2 | J M \rangle$ are the **Clebsch–Gordan coefficients**.

They vanish unless $M = m_1 + m_2$ and $|j_1 - j_2| \leq J \leq j_1 + j_2$; by convention they are taken real, so the transformation matrix is orthogonal and the inverse expansion uses the same numbers. In practice one computes them by the following recipe, which needs no tables:

- (i) The state of maximal M is unambiguous: $|J=j_1+j_2, M=j_1+j_2\rangle = |j_1 j_1\rangle |j_2 j_2\rangle$.
- (ii) Apply $J_- = J_-^{(1)} + J_-^{(2)}$ repeatedly to descend the top multiplet, using the known action of J_- on each factor.
- (iii) The top state of the next multiplet, $|J=j_1+j_2-1, M=J\rangle$, is the (unique up to phase) normalised state with $M = j_1 + j_2 - 1$ orthogonal to the one found in (ii). Then descend again, and repeat.

§37.5.4 Two Spins: Singlet and Triplet

Let us run the recipe on the most important case, $j_1 = j_2 = \frac{1}{2}$. The theorem says

$$\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0,$$

so the four-dimensional space splits into a three-dimensional $J = 1$ multiplet and a one-dimensional $J = 0$ multiplet. Explicitly, the top state is $|1 1\rangle = |\uparrow\uparrow\rangle$, and applying

$J_- = S_-^{(1)} + S_-^{(2)}$, using $S_- |\uparrow\rangle = \hbar |\downarrow\rangle$ and $S_- |\downarrow\rangle = 0$:

$$J_- |\uparrow\uparrow\rangle = \hbar (|\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle), \quad \text{while} \quad J_- |1\,1\rangle = \hbar\sqrt{2} |1\,0\rangle,$$

so $|1\,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$; one more application gives $|1\,-1\rangle = |\downarrow\downarrow\rangle$. The remaining state is the normalised combination with $M = 0$ orthogonal to $|1\,0\rangle$.

Proposition 37.5.7 (Singlet and Triplet)

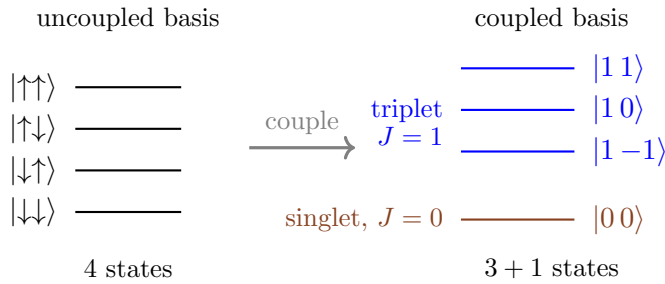
The $J = 1$ **triplet** states are

$$|1\,1\rangle = |\uparrow\uparrow\rangle, \quad |1\,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |1\,-1\rangle = |\downarrow\downarrow\rangle,$$

and the $J = 0$ **singlet** state is

$$|0\,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

The triplet is symmetric under exchange of the two particles, the singlet antisymmetric.



The names come from spectroscopy: in a magnetic field the $J = 1$ states have distinct energies and produce three lines, while the $J = 0$ state gives one. The energy splitting between the two families is already visible without a field, because

$$\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} = \frac{1}{2} (J^2 - (S^{(1)})^2 - (S^{(2)})^2) = \frac{\hbar^2}{2} \left(J(J+1) - \frac{3}{4} - \frac{3}{4} \right) = \begin{cases} +\frac{\hbar^2}{4} & \text{triplet,} \\ -\frac{3\hbar^2}{4} & \text{singlet,} \end{cases}$$

so any interaction of the form $\lambda \mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}$ splits them. (This identity is the standard trick for evaluating dot products of angular momenta: rewrite them in terms of squares, which are diagonal in the coupled basis.) It is worth recording that the singlet is rotationally invariant – being a $J = 0$ state, it is annihilated by every J_i and so looks the same along every axis. This innocuous fact is exactly what makes it the star of the EPR discussion.

Example 37.5.8 (Spin–Orbit Coupling)

An electron in an atomic orbital with $\ell = 1$ has $\mathbf{J} = \mathbf{L} + \mathbf{S}$ with $j_1 = 1$, $j_2 = \frac{1}{2}$, so by the theorem $J = \frac{3}{2}$ or $\frac{1}{2}$: six states in all, splitting $4 + 2$. Relativistic effects generate a term

$$H_{\text{SO}} = \xi(r) \mathbf{L} \cdot \mathbf{S} = \frac{\xi(r)}{2} (J^2 - L^2 - S^2),$$

whose expectation value is $\frac{\hbar^2}{2} \langle \xi \rangle (j(j+1) - \ell(\ell+1) - \frac{3}{4})$, that is $+\frac{\hbar^2}{2} \langle \xi \rangle$ for $j = \frac{3}{2}$ and $-\frac{\hbar^2}{2} \langle \xi \rangle$ for $j = \frac{1}{2}$. This is the origin of the *fine structure* doublet in the sodium spectrum (the famous yellow D lines), and shows why the coupled basis, not the

uncoupled one, is the useful one whenever such a term is present.

§37.6 Transformations and Symmetries

We have already seen the key idea twice: time evolution is a one-parameter family of unitaries generated by the Hermitian operator H , and (in the Larmor example) rotation about the z -axis is generated by S_z . We now make this systematic. The pattern will be that every continuous symmetry of a quantum system is implemented by unitaries $U(\alpha) = \exp(-i\alpha Q/\hbar)$, that the generator Q is an observable, and that Q is conserved precisely when the symmetry is a symmetry of the Hamiltonian. This is Noether's theorem, and in quantum mechanics it is almost a triviality.

§37.6.1 Translations

Consider translating a state of a particle in one dimension by a . In terms of wavefunctions this should send $\psi(x)$ to $\psi(x - a)$ – a bump at the origin moves to a bump at a . Define $U(a)$ by

$$\langle x|U(a)|\psi\rangle = \psi(x - a).$$

For infinitesimal a , Taylor expansion gives $\psi(x - a) = \psi(x) - a\psi'(x) + O(a^2)$, and since $-i\hbar\partial_x$ is the momentum operator,

$$U(a) = I - \frac{iaP}{\hbar} + O(a^2).$$

Composing n translations by a/n and letting $n \rightarrow \infty$ (or simply solving $dU/da = -\frac{i}{\hbar}PU$),

$$U(a) = \exp\left(-\frac{iaP}{\hbar}\right),$$

and in three dimensions $U(\mathbf{a}) = \exp(-i\mathbf{a} \cdot \mathbf{P}/\hbar)$, which makes sense because the components of $\mathbf{P}$ commute with each other. So:

Proposition 37.6.1

Momentum generates translations. The translation operators are unitary and form an abelian group, $U(\mathbf{a})U(\mathbf{b}) = U(\mathbf{a} + \mathbf{b})$.

It is worth checking the effect on operators. Since translating the state and translating the observable should be inverse operations,

$$U(a)^\dagger XU(a) = X + a, \quad U(a)^\dagger PU(a) = P,$$

the first of which one can verify directly from $e^B A e^{-B} = A + [B, A] + \frac{1}{2!}[B, [B, A]] + \dots$ (the *Baker–Campbell–Hausdorff* lemma, proved by differentiating $f(s) = e^{sB} A e^{-sB}$) with $B = iaP/\hbar$: the series terminates after one term since $[P, X] = -i\hbar$ is a constant.

Remark. The canonical commutation relation is thus *equivalent* to the statement that P generates translations of X . Read this way, $[X, P] = i\hbar$ stops looking like an arbitrary postulate.

§37.6.2 Rotations

The same logic applies to rotations, with two differences: the group is non-abelian, and the answer knows about spin. A rotation by angle θ about the axis $\mathbf{n}$ should be implemented by

$$U(\theta, \mathbf{n}) = \exp\left(-\frac{i\theta \mathbf{n} \cdot \mathbf{J}}{\hbar}\right),$$

where $\mathbf{J}$ is the total angular momentum. For a spinless particle we can check this directly: taking $\mathbf{n} = \mathbf{e}_3$ and using $L_z = -i\hbar\partial_\phi$,

$$\exp\left(-\frac{i\theta L_z}{\hbar}\right) = \exp\left(-\theta \frac{\partial}{\partial \phi}\right),$$

which is precisely the Taylor-series operator sending $\psi(r, \theta', \phi)$ to $\psi(r, \theta', \phi - \theta)$: a rotation. The failure of rotations about different axes to commute is encoded in $[J_i, J_j] = i\hbar\epsilon_{ijk}J_k$; indeed expanding both sides of $U(\alpha, \mathbf{e}_1)U(\beta, \mathbf{e}_2)U(\alpha, \mathbf{e}_1)^{-1}U(\beta, \mathbf{e}_2)^{-1}$ to second order reproduces exactly the commutator.

Under a rotation, a **vector operator** $\mathbf{V}$ – one which transforms like $\mathbf{X}$ or $\mathbf{P}$ – satisfies

$$U^\dagger V_i U = R_{ij} V_j, \quad \text{equivalently} \quad [J_i, V_j] = i\hbar\epsilon_{ijk} V_k,$$

where R is the corresponding 3×3 rotation matrix. A **scalar operator** S satisfies $[J_i, S] = 0$; examples are r , P^2 , L^2 and $\mathbf{L} \cdot \mathbf{S}$.

Remark (Spin and the Double Cover). For spin- $\frac{1}{2}$, $U(\theta, \mathbf{n}) = \exp(-\frac{i\theta}{2}\mathbf{n} \cdot \boldsymbol{\sigma}) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \mathbf{n} \cdot \boldsymbol{\sigma}$,

$$U(2\pi, \mathbf{n}) = -I.$$

A rotation by 2π – which is no rotation at all – acts as -1 on the state. This is consistent, because states are rays and an overall sign is unobservable; but it means the map from rotations to unitaries is only defined up to sign. What we really have is a representation of $SU(2)$, which is a double cover of $SO(3)$. Half-integer j gives genuine $SU(2)$ representations that are not representations of $SO(3)$ – another way of seeing why orbital angular momentum, which acts on honest functions on space, must have integer ℓ . And the sign is not entirely unobservable: in an interference experiment where one arm rotates a neutron by 2π and the other does not, the relative sign is real and measurable, and has been measured.

§37.6.3 Symmetries, Conservation and Degeneracy

Definition 37.6.2 (Symmetry)

A unitary operator U is a **symmetry** of a system with Hamiltonian H if

$$U^\dagger H U = H, \quad \text{equivalently} \quad [U, H] = 0.$$

For a continuous family $U(\alpha) = e^{-i\alpha Q/\hbar}$, differentiating at $\alpha = 0$ shows this is equivalent to $[Q, H] = 0$. The consequences are immediate and important.

Theorem 37.6.3 (Symmetry and Conservation)

If Q generates a symmetry of H then Q is a conserved quantity: $\langle Q \rangle$ and the whole

probability distribution of Q are constant in time.

Proof. This is the corollary of §2.2: $[Q, H] = 0$ and Q has no explicit time dependence, so $dQ_H/dt = 0$. $\square$

Theorem 37.6.4 (Symmetry and Degeneracy)

If U is a symmetry and $|\psi\rangle$ is an energy eigenstate with $H|\psi\rangle = E|\psi\rangle$, then $U|\psi\rangle$ is also an energy eigenstate with the same E . Consequently, if the set $\{U|\psi\rangle\}$ over all symmetries U spans a space of dimension $d > 1$, the level E is at least d -fold degenerate.

Proof. $H(U|\psi\rangle) = UH|\psi\rangle = EU|\psi\rangle$. $\square$

This is the deepest structural fact in the subject, and answers a question left dangling in IB: *why* do energy levels come with degeneracies? Because the eigenspaces of H are invariant under the symmetry group, and therefore carry representations of it. For a rotationally invariant Hamiltonian, $[H, J_i] = 0$, so each energy eigenspace is a sum of angular momentum multiplets, and every level automatically has degeneracy $2j + 1$ for some j : the $(2\ell + 1)$ -fold degeneracy of the hydrogen levels in m is a theorem about $SO(3)$, not an accident of the Coulomb potential.

Example 37.6.5 (Accidental Degeneracy)

Hydrogen has a larger degeneracy than rotational symmetry demands: the level E_n has degeneracy n^2 , with all $\ell < n$ appearing, whereas rotational symmetry alone would allow the different ℓ to have different energies (as they do in every other atom). This is not an accident: the Coulomb problem has an extra conserved vector, the quantum *Runge–Lenz vector*

$$\mathbf{A} = \frac{1}{2m} (\mathbf{P} \times \mathbf{L} - \mathbf{L} \times \mathbf{P}) - \frac{e^2}{4\pi\epsilon_0} \frac{\mathbf{X}}{r},$$

which commutes with H and enlarges the symmetry from $SO(3)$ to $SO(4)$. The whole hydrogen spectrum can be derived from this algebra without solving a differential equation, exactly as we did for the oscillator. The three-dimensional isotropic oscillator has a similar story with symmetry group $SU(3)$.

§37.6.4 Parity

Not all symmetries are continuous. The most important discrete one is spatial inversion.

Definition 37.6.6 (Parity)

The **parity operator** Π acts on wavefunctions by $(\Pi\psi)(\mathbf{x}) = \psi(-\mathbf{x})$.

Proposition 37.6.7

Π is Hermitian and unitary with $\Pi^2 = I$, so its eigenvalues are ± 1 . Moreover

$$\Pi \mathbf{X} \Pi = -\mathbf{X}, \quad \Pi \mathbf{P} \Pi = -\mathbf{P}, \quad \Pi \mathbf{L} \Pi = +\mathbf{L}, \quad \Pi \mathbf{S} \Pi = +\mathbf{S}.$$

Proof. Hermiticity is a change of variables in the integral $\int \phi^*(\mathbf{x})\psi(-\mathbf{x})d^3x$, and $\Pi^2 = I$ is clear; together these give unitarity. The action on $\mathbf{X}$ and $\mathbf{P}$ follows from $\Pi\psi(\mathbf{x}) = \psi(-\mathbf{x})$ and the chain rule, and then $\mathbf{L} = \mathbf{X} \times \mathbf{P}$ picks up two signs, so is unchanged. Spin, being an angular momentum, transforms the same way as $\mathbf{L}$. $\square$

Quantities like $\mathbf{L}$ and $\mathbf{S}$ that are *even* under parity are called **axial** (or pseudo-) vectors, while $\mathbf{X}, \mathbf{P}$ are **polar** vectors – the same distinction as in classical mechanics. Since Π commutes with L^2 and $\mathbf{L}$, it acts as a constant on each multiplet, and one finds that constant by evaluating on $Y_{\ell\ell} \propto \sin^\ell \theta e^{i\ell\phi}$: inversion sends $(\theta, \phi) \mapsto (\pi - \theta, \phi + \pi)$, which multiplies this by $(-1)^\ell$. So

$$\Pi Y_{\ell m} = (-1)^\ell Y_{\ell m}.$$

If $[\Pi, H] = 0$ – which holds whenever the potential is even, $U(-\mathbf{x}) = U(\mathbf{x})$ – then parity is conserved and we may choose energy eigenstates of definite parity. The most useful consequence is a *selection rule*.

Proposition 37.6.8 (Parity Selection Rule)

Let $|\psi\rangle$ and $|\phi\rangle$ have definite parities $\eta_\psi, \eta_\phi = \pm 1$, and let V be an operator with $\Pi V \Pi = \eta_V V$. Then

$$\langle \phi | V | \psi \rangle = 0 \quad \text{unless} \quad \eta_\phi \eta_V \eta_\psi = +1.$$

Proof. Insert $\Pi^2 = I$ twice: $\langle \phi | V | \psi \rangle = \langle \phi | \Pi \Pi V \Pi | \psi \rangle = \eta_\phi \eta_V \eta_\psi \langle \phi | V | \psi \rangle$. $\square$

For instance $\mathbf{X}$ is parity-odd, so $\langle \phi | \mathbf{X} | \psi \rangle$ vanishes between states of the same parity. Since the leading interaction of an atom with light is the electric dipole term $\propto \mathbf{E} \cdot \mathbf{X}$, this is exactly the rule that atomic transitions must change parity – so, using $\Pi Y_{\ell m} = (-1)^\ell Y_{\ell m}$, they must change ℓ by an odd number, and a more refined argument using rotational symmetry sharpens this to $\Delta\ell = \pm 1$. Selection rules of this kind are why atomic spectra contain far fewer lines than the number of pairs of levels would suggest.

Remark (Time Reversal). The other classic discrete symmetry is time reversal, which should send $\mathbf{X} \mapsto \mathbf{X}$ and $\mathbf{P} \mapsto -\mathbf{P}$ (and $\mathbf{J} \mapsto -\mathbf{J}$). No unitary can do this: it would preserve $[X, P] = i\hbar$ while flipping the sign of the right-hand side. The resolution, due to Wigner, is that time reversal is implemented by an *antiunitary* operator Θ – one that is antilinear, $\Theta(\lambda|\psi\rangle) = \lambda^* \Theta|\psi\rangle$, and satisfies $\langle \Theta\phi | \Theta\psi \rangle = \langle \phi | \psi \rangle^*$. For a spinless particle Θ is simply complex conjugation of the wavefunction, and indeed if $\psi(\mathbf{x}, t)$ solves the Schrödinger equation with real potential then so does $\psi^*(\mathbf{x}, -t)$. Because Θ is not linear it has no eigenvalues in the usual sense and generates no conservation law; its main structural consequence is *Kramers' degeneracy*, that a time-reversal-invariant system with half-integer total angular momentum has all its energy levels degenerate to even order.

§37.7 Time-Independent Perturbation Theory

The list of quantum systems we can solve exactly is embarrassingly short: the free particle, the square well, the oscillator, the hydrogen atom, and a handful of others. Almost every real problem is one of these with something extra added – an electric field, a magnetic field, a second electron, a slight anharmonicity. Perturbation theory is the systematic way of getting approximate answers by treating the extra piece as small.

The setup throughout is

$$H = H_0 + \lambda V,$$

where the **unperturbed** Hamiltonian H_0 is solved,

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle,$$

with $\{|n^{(0)}\rangle\}$ a complete orthonormal set, and λ is a small dimensionless parameter that we use to organise the expansion (and often set to 1 at the end). We write $V_{mn} = \langle m^{(0)} | V | n^{(0)} \rangle$.

§37.7.1 The Non-Degenerate Case

Assume first that the level $E_n^{(0)}$ we are interested in is non-degenerate, and posit power series

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots, \quad |n\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \lambda^2 |n^{(2)}\rangle + \dots.$$

We may normalise the expansion by choosing $\langle n^{(0)} | n^{(k)} \rangle = 0$ for $k \geq 1$ – the component of the correction along the original state is pure rescaling, and can be removed at the end. Substituting into $H |n\rangle = E_n |n\rangle$ and collecting powers of λ :

$$\begin{aligned} \lambda^0 : \quad & (H_0 - E_n^{(0)}) |n^{(0)}\rangle = 0, \\ \lambda^1 : \quad & (H_0 - E_n^{(0)}) |n^{(1)}\rangle = (E_n^{(1)} - V) |n^{(0)}\rangle, \\ \lambda^2 : \quad & (H_0 - E_n^{(0)}) |n^{(2)}\rangle = (E_n^{(1)} - V) |n^{(1)}\rangle + E_n^{(2)} |n^{(0)}\rangle. \end{aligned}$$

Now project each equation onto the unperturbed basis. Taking $\langle n^{(0)} |$ of the λ^1 equation, the left-hand side vanishes (as H_0 is Hermitian and acts leftwards as $E_n^{(0)}$), giving at once

$$E_n^{(1)} = V_{nn} = \langle n^{(0)} | V | n^{(0)} \rangle.$$

Taking $\langle m^{(0)} |$ with $m \neq n$ gives $(E_m^{(0)} - E_n^{(0)}) \langle m^{(0)} | n^{(1)} \rangle = -V_{mn}$, so

$$|n^{(1)}\rangle = \sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle.$$

Finally, taking $\langle n^{(0)} |$ of the λ^2 equation and using $\langle n^{(0)} | n^{(1)} \rangle = 0$,

$$0 = -\langle n^{(0)} | V | n^{(1)} \rangle + E_n^{(2)}, \quad \text{so} \quad E_n^{(2)} = \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}.$$

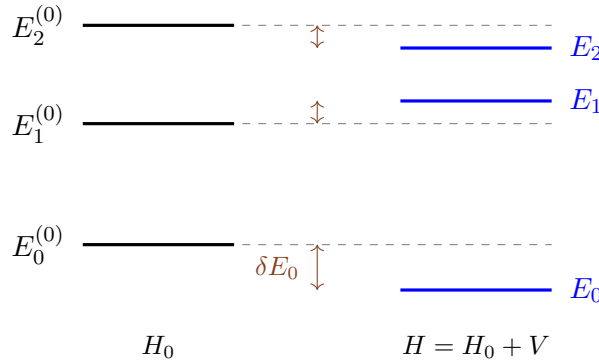
Theorem 37.7.1 (Rayleigh–Schrödinger Perturbation Theory)

To second order, and setting $\lambda = 1$,

$$E_n = E_n^{(0)} + V_{nn} + \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}} + O(V^3),$$

with corresponding state $|n\rangle = |n^{(0)}\rangle + \sum_{m \neq n} \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} |m^{(0)}\rangle + O(V^2)$.

Several things are worth noticing. The first-order shift is just the expectation value of the perturbation in the unperturbed state – entirely intuitive. The second-order shift for the *ground state* is always negative, since every denominator $E_0^{(0)} - E_m^{(0)}$ is negative: mixing in other states can only push the ground state down. More generally levels ‘repel’: a pair of levels contributes to each other’s shift with opposite signs, pushing the higher one up and the lower one down. And the whole thing is an asymptotic expansion in $V/\Delta E$, where ΔE is the typical level spacing; it fails badly if any denominator is small, which is exactly the situation of degeneracy.

**Example 37.7.2** (The Anharmonic Oscillator)

Take $H = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 X^2 + \lambda X^4$. The first-order shift of the level n is $\lambda \langle n|X^4|n\rangle$, which is easy with creation and annihilation operators. Writing $X = \alpha(a + a^\dagger)$ with $\alpha = \sqrt{\hbar/2m\omega}$, we need the terms of $(a + a^\dagger)^4$ with equal numbers of a and $a^\dagger$; keeping track of the orderings,

$$\langle n|X^4|n\rangle = \alpha^4 (3 + 6n + 6n^2) = 3 \left(\frac{\hbar}{2m\omega} \right)^2 (2n^2 + 2n + 1).$$

(A quick check: for $n = 0$ this is $3\alpha^4 = 3 \langle X^2 \rangle_0^2$, as it should be for a Gaussian.) So

$$E_n \approx \left(n + \frac{1}{2} \right) \hbar\omega + \frac{3\lambda\hbar^2}{4m^2\omega^2} (2n^2 + 2n + 1).$$

The correction grows like n^2 , so the expansion is only good for $n \ll m^2\omega^3/\lambda\hbar$: perturbation theory in a fixed λ eventually fails for high levels, which is physically obvious since a highly excited state explores large x where λx^4 dominates $\frac{1}{2}m\omega^2 x^2$.

Example 37.7.3 (A Two-State System, Exactly and Perturbatively)

Let $H = \begin{pmatrix} E_1 & v \\ v & E_2 \end{pmatrix}$ with $E_1 < E_2$ and v real. The exact eigenvalues are

$$E_{\pm} = \frac{E_1 + E_2}{2} \pm \sqrt{\left(\frac{E_2 - E_1}{2}\right)^2 + v^2},$$

and expanding the square root for $v \ll E_2 - E_1$ gives

$$E_- = E_1 - \frac{v^2}{E_2 - E_1} + O(v^4), \quad E_+ = E_2 + \frac{v^2}{E_2 - E_1} + O(v^4),$$

exactly matching the second-order formula (the first-order shifts vanish since the diagonal of V is zero). Note the level repulsion, and note also what happens as $E_2 \rightarrow E_1$: the perturbative answer diverges but the exact one does not – it becomes $E_{\pm} = E_1 \pm v$, a splitting *linear* in v . This is the degenerate case, and it needs different treatment.

§37.7.2 The Degenerate Case

Suppose now that $E^{(0)}$ is an eigenvalue of H_0 with a d -dimensional eigenspace $\mathcal{H}_0$, spanned by orthonormal $|1\rangle, \dots, |d\rangle$. The non-degenerate formulae break down because of the vanishing denominators; but the two-state example above shows what is really going on. *Any* basis of $\mathcal{H}_0$ diagonalises H_0 , so at zeroth order we do not know which states to expand around, and we must let the perturbation tell us.

Write the true eigenstate as $|\psi\rangle = |\psi^{(0)}\rangle + \lambda |\psi^{(1)}\rangle + \dots$ with $|\psi^{(0)}\rangle = \sum_{i=1}^d c_i |i\rangle \in \mathcal{H}_0$ unknown. The order- λ equation is as before,

$$(H_0 - E^{(0)}) |\psi^{(1)}\rangle = (E^{(1)} - V) |\psi^{(0)}\rangle,$$

and projecting onto $\langle j|$ for $j = 1, \dots, d$ kills the left-hand side, leaving

$$\sum_i \langle j|V|i\rangle c_i = E^{(1)} c_j.$$

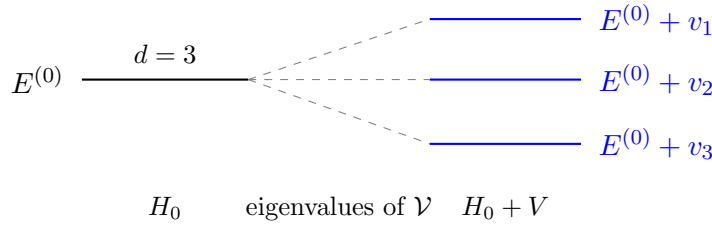
Theorem 37.7.4 (Degenerate Perturbation Theory)

The first-order energy shifts of a degenerate level are the eigenvalues of the $d \times d$ Hermitian matrix

$$\mathcal{V}_{ji} = \langle j|V|i\rangle, \quad i, j = 1, \dots, d,$$

and the correct zeroth-order states are the corresponding eigenvectors.

So the recipe is: restrict the perturbation to the degenerate subspace and diagonalise it. If the eigenvalues of $\mathcal{V}$ are distinct, the degeneracy is completely lifted at first order and one may then continue with non-degenerate perturbation theory in the new basis. If some remain equal, the degeneracy survives to first order and one must go further – often it is protected by a symmetry that V does not break, in which case it survives to all orders.



§37.7.3 Atoms in Fields: Fine Structure, Zeeman and Stark

The classic applications of perturbation theory are all to hydrogen, and they are all degenerate. The spectrum found in IB, $E_n = -\text{Ry}/n^2$ with $\text{Ry} = 13.6 \text{ eV}$, is spectacularly degenerate – the level n has n^2 orbital states, and $2n^2$ once spin is counted – so *everything* we have so far neglected perturbs a degenerate level. Happily, the recipe of the previous subsection is usually easier to apply than it looks, because of the principle of §37.6: if the perturbation commutes with certain observables, we should choose the zeroth-order basis to diagonalise those, and then $\mathcal{V}$ is diagonal (or at worst block diagonal) with no work at all.

Three neglected effects, all of the same relative size $\alpha^2 \approx 5 \times 10^{-5}$, where

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$$

is the **fine structure constant**, make up the **fine structure** of the spectrum.

- (i) *The relativistic correction to the kinetic energy.* Expanding the exact expression,

$$\sqrt{p^2c^2 + m^2c^4} - mc^2 = \frac{p^2}{2m} - \frac{p^4}{8m^3c^2} + \cdots,$$

so the leading correction is $V_{\text{rel}} = -P^4/8m^3c^2$.

- (ii) *Spin-orbit coupling.* In the electron's instantaneous rest frame the proton orbits it, producing a magnetic field which couples to the electron's magnetic moment. Including the relativistic factor of $\frac{1}{2}$ from Thomas precession, this gives

$$V_{\text{SO}} = \frac{1}{2m^2c^2} \frac{1}{r} \frac{dU}{dr} \mathbf{L} \cdot \mathbf{S} = \frac{e^2}{8\pi\epsilon_0 m^2 c^2} \frac{\mathbf{L} \cdot \mathbf{S}}{r^3},$$

the operator we already met in §5.4.

- (iii) *The Darwin term,* $V_D = \frac{\hbar^2}{8m^2c^2} \nabla^2 U = \frac{\hbar^2 e^2}{8\epsilon_0 m^2 c^2} \delta^{(3)}(\mathbf{x})$, a relativistic smearing of the electron over a Compton wavelength which affects only the states with $\ell = 0$, the only ones with $\psi(0) \neq 0$.

Now, which basis? The terms V_{rel} and V_D are scalars: they are built from P^2 and r , and commute with every component of $\mathbf{L}$ and of $\mathbf{S}$. The spin-orbit term does not commute with L_z or S_z separately, but writing it as

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2} (J^2 - L^2 - S^2)$$

shows that it commutes with L^2, S^2, J^2 and J_z . Hence all three perturbations are simultaneously diagonal in the *coupled* basis $|n \ell j m_j\rangle$, and the correct zeroth-order states are those of definite total angular momentum. Not a single matrix has to be diagonalised.

Evaluating the shifts then needs the radial expectation values $\langle r^{-2} \rangle$ and $\langle r^{-3} \rangle$ in hydrogen. That algebra is unedifying and we shall not reproduce it; what is worth knowing is that the three contributions conspire into something startlingly simple.

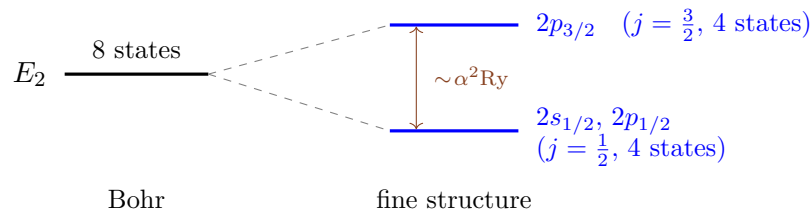
Theorem 37.7.5 (Fine Structure of Hydrogen)

To first order in α^2 , the energy levels of hydrogen are

$$E_{nj} = -\frac{\text{Ry}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right) \right],$$

depending on n and on the total angular momentum j , but not on ℓ .

The shift is always negative, and it lifts the degeneracy in ℓ only partially: states with the same j but different ℓ – such as $2s_{1/2}$ and $2p_{1/2}$, in the spectroscopic notation $n\ell_j$ – remain exactly degenerate. This is a peculiarity of the Coulomb potential, and it survives an exact treatment: solving the Dirac equation for hydrogen gives a spectrum depending only on n and j . The degeneracy is finally broken by quantum electrodynamics, producing the tiny **Lamb shift** of about 4×10^{-6} eV between $2s_{1/2}$ and $2p_{1/2}$ – whose measurement in 1947 was one of the founding experiments of QED.



With the internal structure understood, we can put the atom into an external field.

Example 37.7.6 (The Normal Zeeman Effect)

Put a hydrogen atom in a uniform magnetic field $\mathbf{B} = B\mathbf{e}_3$. The orbital motion of the electron gives a magnetic moment $\mu = -\frac{e}{2m_e}\mathbf{L}$, so (ignoring spin for the moment)

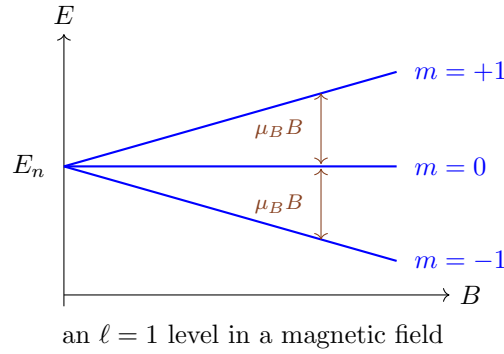
$$V = -\mu \cdot \mathbf{B} = \frac{eB}{2m_e}L_z = \frac{\mu_B B}{\hbar}L_z,$$

where $\mu_B = e\hbar/2m_e$ is the **Bohr magneton**. The unperturbed level E_n is n^2 -fold degenerate, so we should diagonalise V on that subspace – but it is already diagonal in the basis $|n\ell m\rangle$, with eigenvalues $\mu_B B m$. Hence

$$E_{n\ell m} = E_n + \mu_B B m, \quad m = -\ell, \dots, \ell,$$

and each ℓ -multiplet splits into $2\ell + 1$ equally spaced levels. This is exactly the degeneracy-versus-symmetry story of §37.6 run in reverse: the field picks out an axis, breaking $SO(3)$ down to rotations about $\mathbf{e}_3$, and the degeneracy protected by the larger group is destroyed.

The splitting is linear in B , and it is easiest to picture by plotting the levels against the field strength.



Remark (The Anomalous Zeeman Effect). Including spin, the moment is $\mu = -\frac{e}{2m_e}(\mathbf{L} + 2\mathbf{S})$ – the factor of 2 being the electron’s g -factor, a prediction of the Dirac equation – so

$$V = \frac{\mu_B B}{\hbar} (L_z + 2S_z) = \frac{\mu_B B}{\hbar} (J_z + S_z).$$

If the field is weak compared with the spin–orbit coupling, one must work in the coupled basis $|j m_j\rangle$, where S_z is not diagonal but its restriction to fixed j turns out to be proportional to J_z ; the shift is $\mu_B B g_J m_j$ with the **Landé g -factor**

$$g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}.$$

In a strong field the spin–orbit term is instead the perturbation, and one works in the uncoupled basis; the two regimes give different-looking spectra, and the crossover between them (the Paschen–Back effect) is a nice illustration of why choosing the right zeroth-order basis is the whole art of degenerate perturbation theory.

Example 37.7.7 (The Linear Stark Effect)

Put hydrogen in a uniform electric field $\mathcal{E} = \mathcal{E}\mathbf{e}_3$, so $V = e\mathcal{E}Z$. For the ground state $|100\rangle$, the first-order shift vanishes by parity (Z is odd, $|100\rangle$ has definite parity), so the effect is quadratic in $\mathcal{E}$ – the atom has no permanent dipole moment.

The $n = 2$ level is different. It is four-fold degenerate (ignoring spin), spanned by $|200\rangle, |210\rangle, |21\pm 1\rangle$, and we must diagonalise the 4×4 matrix of V . Almost all entries vanish: $\langle 2\ell m | Z | 2\ell' m' \rangle = 0$ unless $m = m'$ (since Z commutes with L_z) and unless ℓ, ℓ' have opposite parity (since Z is parity-odd). The only surviving matrix element is between $|200\rangle$ and $|210\rangle$, and an explicit integral gives $\langle 200 | Z | 210 \rangle = -3a_0$ with a_0 the Bohr radius. So in the basis $(|200\rangle, |210\rangle, |211\rangle, |21-1\rangle)$,

$$\mathcal{V} = -3ea_0\mathcal{E} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

with eigenvalues $\mp 3ea_0\mathcal{E}$ and 0 (twice). The level splits into three, with the outer two shifting *linearly* in the field – so an excited hydrogen atom does behave as though it has a permanent dipole moment of size $3ea_0$. The reason it can is precisely the accidental degeneracy between the $2s$ and $2p$ states, which lets the field mix states of opposite parity at no energy cost.

§37.7.4 The Variational Principle

Perturbation theory needs a nearby solvable problem. When there is none, the following gives a completely different handle, and in particular a rigorous *bound*.

Theorem 37.7.8 (Variational Principle)

Let H have ground state energy E_0 . Then for every normalised $|\psi\rangle \in \mathcal{H}$,

$$\langle\psi|H|\psi\rangle \geq E_0,$$

with equality if and only if $|\psi\rangle$ is a ground state.

Proof. Expand in an orthonormal basis of energy eigenstates, so that $|\psi\rangle = \sum_n a_n |n\rangle$ with coefficients satisfying $\sum_n |a_n|^2 = 1$. Then

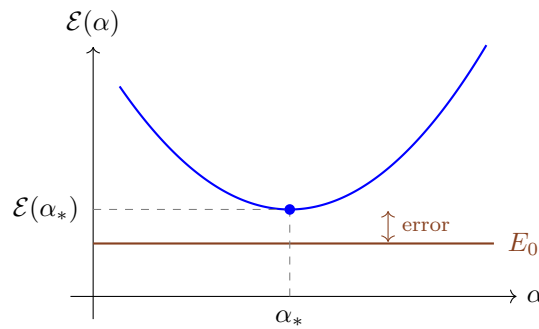
$$\langle\psi|H|\psi\rangle = \sum_n E_n |a_n|^2 \geq E_0 \sum_n |a_n|^2 = E_0,$$

with equality iff $a_n = 0$ whenever $E_n > E_0$. □

The proof is trivial; the power comes from what one does with it. Choose a family of **trial states** $|\psi_\alpha\rangle$ depending on parameters α , compute

$$\mathcal{E}(\alpha) = \frac{\langle\psi_\alpha|H|\psi_\alpha\rangle}{\langle\psi_\alpha|\psi_\alpha\rangle},$$

and minimise over α . Whatever family we pick, the result is a genuine upper bound on E_0 , and if the family happens to contain a good approximation to the true ground state the bound will be tight. Even a crude family usually does well, because $\mathcal{E}$ is *stationary* at the true ground state: an error of size ϵ in the state produces an error of size ϵ^2 in the energy.



Example 37.7.9 (The Quartic Oscillator)

Take $H = \frac{P^2}{2m} + \lambda X^4$, which has no exact solution and no small parameter to perturb in (there is no quadratic term at all). Use the Gaussian family

$$\psi_a(x) = \left(\frac{a}{\pi}\right)^{1/4} e^{-ax^2/2},$$

for which $\langle X^2 \rangle = \frac{1}{2a}$, $\langle X^4 \rangle = 3 \langle X^2 \rangle^2 = \frac{3}{4a^2}$ and $\langle P^2 \rangle = \frac{\hbar^2 a}{2}$. Hence

$$\mathcal{E}(a) = \frac{\hbar^2 a}{4m} + \frac{3\lambda}{4a^2}.$$

Setting $\mathcal{E}'(a) = 0$ gives $a_*^3 = 6m\lambda/\hbar^2$, and substituting back,

$$E_0 \leq \mathcal{E}(a_*) \approx 0.681 \left(\frac{\hbar^2}{m} \right)^{2/3} \lambda^{1/3}.$$

Note first that the $\lambda^{1/3}$ scaling is exact – it follows from dimensional analysis, since $\hbar^2/m$ and λ can be combined into an energy in only one way. Numerical solution of the Schrödinger equation gives the true coefficient as 0.668, so our one-parameter guess is within 2%.

Remark (Excited States). The principle extends: if $|\psi\rangle$ is orthogonal to the exact ground state, then $\langle H \rangle_\psi \geq E_1$, and so on up the spectrum. In practice we rarely know the exact ground state, but symmetry often does the job for free – any odd trial function is automatically orthogonal to an even ground state, so minimising over odd trial functions bounds the first excited state of a symmetric potential.

Remark. A pleasant corollary: *any* attractive potential well in one dimension, however shallow, has at least one bound state. Take a trial Gaussian of width σ ; the kinetic energy costs $O(\hbar^2/m\sigma^2)$ while the potential energy gains $O(-\int U/\sigma)$ for large σ , and the latter wins as $\sigma \rightarrow \infty$, giving $\mathcal{E} < 0$ for some trial state and hence a bound state. The same argument fails in three dimensions, where the kinetic cost scales the same way but the potential gain is weaker – and indeed shallow three-dimensional wells have no bound states.

§37.8 Identical Particles

Classically, two identical billiard balls are still two *different* balls: we can paint one red, or simply follow their trajectories. In quantum mechanics neither option is available – there are no trajectories, and identical particles really are identical. This is not a philosophical nicety; it changes the state space, and its consequences include the structure of the periodic table, the stability of matter, and the existence of lasers.

§37.8.1 Exchange Symmetry

Take two identical particles, each with state space $\mathcal{H}_1$, so the pair lives in $\mathcal{H}_1 \otimes \mathcal{H}_1$.

Definition 37.8.1 (Exchange Operator)

The **exchange** (or swap) operator P is defined on $\mathcal{H}_1 \otimes \mathcal{H}_1$ by

$$P(|\psi\rangle \otimes |\phi\rangle) = |\phi\rangle \otimes |\psi\rangle,$$

extended linearly.

Clearly $P^2 = I$, and P is unitary and Hermitian, so its eigenvalues are ± 1 ; the corresponding eigenspaces are the **symmetric** and **antisymmetric** subspaces. If the par-

ticles are genuinely identical then the Hamiltonian must be unchanged by relabelling them, $[P, H] = 0$, so these subspaces are preserved by time evolution.

Axiom 37.8.2 (Symmetrisation Postulate)

The states of a system of identical particles are either totally symmetric under exchange of any two of them (such particles are **bosons**) or totally antisymmetric (**fermions**). Which of the two holds is a fixed property of the species.

The postulate is stronger than the observation that $[P, H] = 0$: it says the other states simply do not exist in nature. Empirically, particles with integer spin are bosons (photons, pions, helium-4 atoms) and particles with half-integer spin are fermions (electrons, protons, neutrons, helium-3 atoms). That this is forced – the **spin–statistics theorem** – is a genuine theorem, but only within relativistic quantum field theory; here we take it as given.

For N particles, ‘totally symmetric’ means invariant under every transposition (hence under every permutation), and ‘totally antisymmetric’ means picking up $\text{sign}(\sigma)$ under the permutation σ . Given single-particle states $|\chi_1\rangle, \dots, |\chi_N\rangle$ one builds

$$|\Psi_B\rangle \propto \sum_{\sigma \in S_N} |\chi_{\sigma(1)}\rangle \otimes \cdots \otimes |\chi_{\sigma(N)}\rangle, \quad |\Psi_F\rangle \propto \sum_{\sigma \in S_N} \text{sign}(\sigma) |\chi_{\sigma(1)}\rangle \otimes \cdots \otimes |\chi_{\sigma(N)}\rangle.$$

The fermionic expression is a determinant – the **Slater determinant** – of the matrix whose (i, j) entry is the i th state occupied by the j th particle, and this observation gives the fundamental result of the section for free.

Theorem 37.8.3 (Pauli Exclusion Principle)

No two identical fermions can occupy the same single-particle state.

Proof. If $|\chi_i\rangle = |\chi_j\rangle$ for some $i \neq j$, the determinant has two equal rows and vanishes. Directly, for two particles, $|\Psi_F\rangle \propto |\chi\rangle \otimes |\chi\rangle - |\chi\rangle \otimes |\chi\rangle = 0$: there is no antisymmetric state built from two copies of the same state. $\square$

There is no corresponding restriction on bosons – indeed they positively prefer to pile into the same state, which is what a laser and a Bose–Einstein condensate exploit.

§37.8.2 Exchange Forces

Symmetry has consequences even in the absence of any interaction. Suppose two particles occupy orthonormal single-particle states $|\chi_a\rangle, |\chi_b\rangle$, and consider the normalised combinations

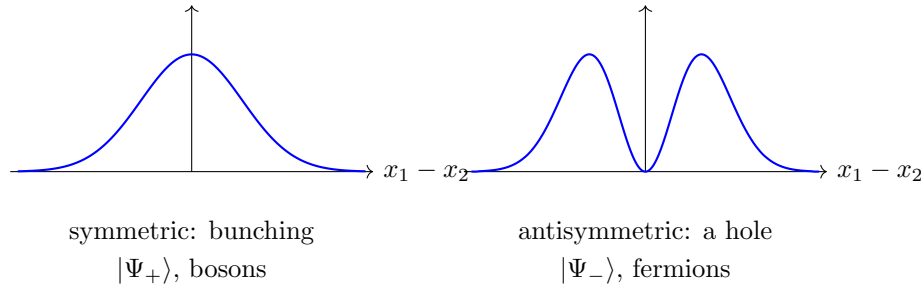
$$|\Psi_{\pm}\rangle = \frac{1}{\sqrt{2}} (|\chi_a\rangle |\chi_b\rangle \pm |\chi_b\rangle |\chi_a\rangle).$$

Working in one dimension for brevity, let us compute the mean square separation of the two particles. A short calculation using orthonormality gives

$$\langle (X_1 - X_2)^2 \rangle_{\pm} = \langle X^2 \rangle_a + \langle X^2 \rangle_b - 2 \langle X \rangle_a \langle X \rangle_b \mp 2 |\langle \chi_a | X | \chi_b \rangle|^2,$$

where the first three terms are exactly what one would get for distinguishable particles. The last term is the **exchange term**. So symmetric (bosonic) states are on average

closer together, and antisymmetric (fermionic) states *further apart*, than distinguishable particles in the same orbitals.



The pictures show the probability density for the separation: bosons show a peak at zero separation ('bunching'), fermions a zero there. It looks like an attractive force between bosons and a repulsive one between fermions, and the effect can be large; but no force is involved at all, only the symmetry of the state. This **exchange force** is what holds covalent bonds together and, in the guise of the Pauli principle, what stops matter collapsing.

For particles with spin, the antisymmetry requirement applies to the *whole* state, space and spin together. For two spin- $\frac{1}{2}$ fermions, $\mathcal{H} = (L^2 \otimes L^2) \otimes (\mathbb{C}^2 \otimes \mathbb{C}^2)$, and using the singlet/triplet decomposition of §5.4 the allowed states are

$$\underbrace{\Psi_{\text{sym}}(\mathbf{x}_1, \mathbf{x}_2)}_{\text{symmetric space}} \otimes \underbrace{|00\rangle}_{\text{singlet}} \quad \text{or} \quad \underbrace{\Psi_{\text{anti}}(\mathbf{x}_1, \mathbf{x}_2)}_{\text{antisymmetric space}} \otimes \underbrace{|1m\rangle}_{\text{triplet}}.$$

So the spins and the spatial wavefunction are correlated even when the Hamiltonian contains no spin-dependent term whatsoever: *the total spin determines the spatial symmetry, and hence the energy*. This is the mechanism behind magnetism, and behind the structure of the helium spectrum, to which we now turn.

§37.8.3 The Helium Atom

Helium is two electrons bound to a nucleus of charge $Z = 2$. Ignoring spin-dependent forces and treating the nucleus as fixed,

$$H = \underbrace{\frac{\mathbf{P}_1^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r_1} + \frac{\mathbf{P}_2^2}{2m} - \frac{Ze^2}{4\pi\epsilon_0 r_2}}_{H_0} + \underbrace{\frac{e^2}{4\pi\epsilon_0 |\mathbf{x}_1 - \mathbf{x}_2|}}_V.$$

Without V the problem separates into two hydrogen-like atoms with $Z = 2$, whose energies are $E_n = -Z^2 \text{Ry}/n^2$ with $\text{Ry} = 13.6 \text{ eV}$. The trouble is that V is not small – it is of the same order as the terms we kept – so we should not expect miracles. Let us see how badly it does.

Example 37.8.4 (The Ground State)

Both electrons in the 1s orbital: the spatial state $\psi_{100}(\mathbf{x}_1)\psi_{100}(\mathbf{x}_2)$ is symmetric, so the spin state must be the singlet, and the Pauli principle is satisfied. The unperturbed energy is

$$E^{(0)} = 2 \times (-Z^2 \text{Ry}) = -8 \text{ Ry} = -108.8 \text{ eV}.$$

The first-order correction is $\langle \Psi | V | \Psi \rangle$, a six-dimensional integral which evaluates to

$$E^{(1)} = \frac{5Z}{8} \cdot \frac{e^2}{4\pi\epsilon_0 a_0} = \frac{5Z}{4} \text{ Ry} = 34.0 \text{ eV},$$

giving $E \approx -74.8 \text{ eV}$ against the measured -79.0 eV . An error of 5% from a perturbation that is not small at all – better than we deserve.

The variational principle does better, and is more illuminating. Take the trial state to be two $1s$ orbitals with an *effective* nuclear charge Z_* treated as a parameter – physically, each electron partially screens the nucleus from the other. Computing $\langle H \rangle$ and minimising gives $Z_* = Z - \frac{5}{16} = 1.6875$ and

$$E_0 \leq -2 \left(Z - \frac{5}{16} \right)^2 \text{ Ry} = -77.5 \text{ eV},$$

within 2% – and with a transparent physical interpretation of the parameter.

Example 37.8.5 (Excited States: Para- and Orthohelium)

Now put one electron in $1s$ and one in some $n\ell$ orbital. The spatial state can be either

$$\Psi_{\pm}(\mathbf{x}_1, \mathbf{x}_2) = \frac{1}{\sqrt{2}} (\psi_{1s}(\mathbf{x}_1)\psi_{n\ell}(\mathbf{x}_2) \pm \psi_{n\ell}(\mathbf{x}_1)\psi_{1s}(\mathbf{x}_2)),$$

with Ψ_+ (symmetric) paired with the spin singlet and Ψ_- (antisymmetric) with the spin triplet. Historically these two families were thought to be different substances and were named **parahelium** ($S = 0$) and **orthohelium** ($S = 1$).

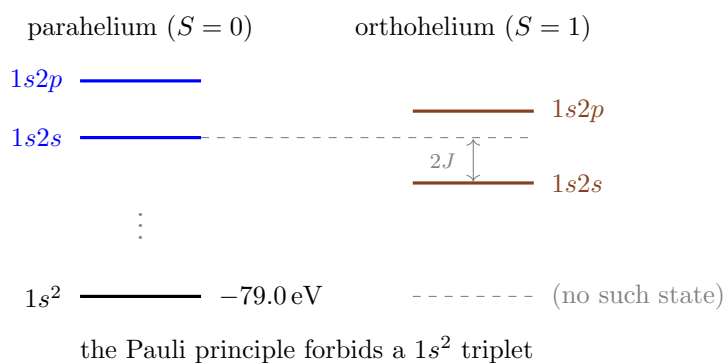
Their energies differ at first order:

$$\langle \Psi_{\pm} | V | \Psi_{\pm} \rangle = I \pm J,$$

where the *direct* integral I is the ordinary electrostatic repulsion of the two charge clouds and the *exchange* integral is

$$J = \frac{e^2}{4\pi\epsilon_0} \int \frac{\psi_{1s}^*(\mathbf{x}_1)\psi_{n\ell}^*(\mathbf{x}_2)\psi_{n\ell}(\mathbf{x}_1)\psi_{1s}(\mathbf{x}_2)}{|\mathbf{x}_1 - \mathbf{x}_2|} d^3x_1 d^3x_2,$$

which turns out to be positive. So the triplet (orthohelium) lies *below* the singlet (parahelium), by $2J$: aligning the spins lowers the energy, even though the Hamiltonian does not mention spin. The reason is precisely the exchange force of the previous subsection – the antisymmetric spatial state keeps the electrons apart, reducing their repulsion.



Note the striking absence at the bottom right: there is no orthohelium ground state, because a $1s^2$ triplet would need an antisymmetric spatial function built from two identical orbitals, which is zero. The Pauli principle removes a state from the spectrum entirely.

§37.8.4 The Periodic Table

The same principle, applied to heavier atoms, produces chemistry. Model an atom of atomic number Z by Z electrons in hydrogen-like orbitals $|n\ell m\rangle \otimes |m_s\rangle$, and fill them from the bottom, respecting exclusion. Each (n, ℓ) **subshell** holds $2(2\ell + 1)$ electrons – the factor of 2 from spin – and the crucial point is that in a many-electron atom the degeneracy in ℓ is broken (the inner electrons screen the nucleus, and low- ℓ orbitals penetrate the screening cloud more effectively), so subshells fill in the order

$$1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, \dots$$

Filling $1s$ and then $2s, 2p$ gives the first two rows of the periodic table, of lengths 2 and 8; the closed-shell configurations $Z = 2, 10, 18$ are the noble gases, chemically inert because promoting an electron across the gap to the next shell is expensive. One electron more (lithium, sodium) or one fewer (fluorine, chlorine) than a closed shell gives the alkali metals and halogens, and the whole familiar periodicity follows. Within a subshell, the residual electron–electron interaction is handled by Hund’s rules, of which the first – *maximise the total spin* – is exactly the orthohelium effect above, and the reason that iron is magnetic.

Remark. It is worth pausing on how much rests on the antisymmetry postulate. Were electrons bosons, every atom would have all its electrons in the $1s$ orbital; there would be no chemistry, no periodic table, and no structure to matter. And white dwarf and neutron stars are held up against gravity by *degeneracy pressure*, the purely quantum-statistical resistance of a fermion gas to compression, which is again just exclusion in disguise.

§37.9 Time-Dependent Perturbation Theory

Time-independent perturbation theory answers ‘how are the energy levels shifted?’. But many of the questions we actually want to ask are about *transitions*: an atom in an excited state emits a photon and drops to the ground state; a beam of particles scatters off a target; a system is shaken by an oscillating field and absorbs energy. For these we need a perturbation that depends on time, and a different technology.

The setup is

$$H(t) = H_0 + V(t),$$

with H_0 time-independent and solved, $H_0 |n\rangle = E_n |n\rangle$, and $V(t)$ small.

§37.9.1 The Interaction Picture

The Schrödinger picture puts all the time dependence in the states, the Heisenberg picture all of it in the operators. When the Hamiltonian splits into a big solvable piece and a small piece, the useful compromise is to let H_0 evolve the operators and V evolve the states.

Definition 37.9.1 (Interaction Picture)

Define

$$|\psi(t)\rangle_I = e^{iH_0 t/\hbar} |\psi(t)\rangle_S, \quad A_I(t) = e^{iH_0 t/\hbar} A e^{-iH_0 t/\hbar}.$$

Matrix elements are unchanged, as always, since we have conjugated by a unitary. The point is the equation of motion:

$$i\hbar \frac{d}{dt} |\psi\rangle_I = -H_0 e^{iH_0 t/\hbar} |\psi\rangle_S + e^{iH_0 t/\hbar} (H_0 + V) |\psi\rangle_S = V_I(t) |\psi(t)\rangle_I,$$

where $V_I(t) = e^{iH_0 t/\hbar} V(t) e^{-iH_0 t/\hbar}$. So in the interaction picture the states evolve *only* under the perturbation – the fast, uninteresting evolution due to H_0 has been transformed away. If $V = 0$ then $|\psi\rangle_I$ is constant, and in general it moves slowly, on a timescale set by the size of V . To summarise the three pictures:

Schrödinger:	states move with H ,	operators fixed;
Heisenberg:	states fixed,	operators move with H ;
Interaction:	states move with V_I ,	operators move with H_0 .

§37.9.2 Transition Amplitudes

Suppose the system starts in the eigenstate $|i\rangle$ of H_0 at $t = 0$, and we ask for the probability of finding it in a different eigenstate $|f\rangle$ at time t . Expanding the interaction-picture state in the unperturbed basis,

$$|\psi(t)\rangle_I = \sum_n c_n(t) |n\rangle, \quad c_n(0) = \delta_{ni},$$

and substituting into $i\hbar \frac{d}{dt} |\psi\rangle_I = V_I |\psi\rangle_I$, then projecting onto $\langle f|$:

$$i\hbar \dot{c}_f(t) = \sum_n \langle f|V_I(t)|n\rangle c_n(t) = \sum_n e^{i\omega_{fn}t} V_{fn}(t) c_n(t), \quad \omega_{fn} := \frac{E_f - E_n}{\hbar}.$$

This is exact. To solve it perturbatively, integrate and iterate, starting from the zeroth-order guess $c_n(t) = \delta_{ni}$:

$$c_f(t) = \delta_{fi} - \frac{i}{\hbar} \int_0^t e^{i\omega_{fi}t'} V_{fi}(t') dt' + O(V^2).$$

(Continuing the iteration produces the **Dyson series**, whose k th term is a k -fold time-ordered integral; we shall only need the first.)

Theorem 37.9.2 (First-Order Transition Probability)

For $f \neq i$, the probability of a transition from $|i\rangle$ to $|f\rangle$ after time t is

$$P_{i \rightarrow f}(t) = |c_f(t)|^2 = \frac{1}{\hbar^2} \left| \int_0^t e^{i\omega_{fi}t'} V_{fi}(t') dt' \right|^2 + O(V^3).$$

The structure of this formula rewards a moment's thought: the amplitude is the Fourier component of the perturbation's matrix element at exactly the Bohr frequency ω_{fi} of the transition. A perturbation which contains no frequency component at ω_{fi} causes no transition; and a perturbation acting for a long time is sharply selective in frequency, while a sudden kick contains all frequencies and can excite anything.

§37.9.3 A Constant Perturbation

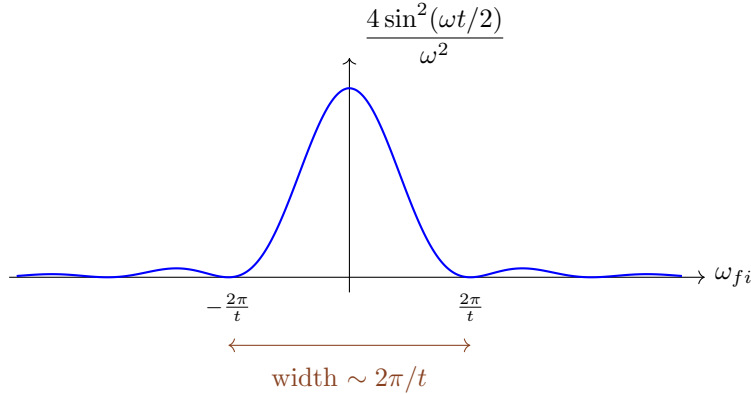
Take the simplest case: V is switched on at $t = 0$, held constant, and we ask for the transition probability at time t . Then

$$\int_0^t e^{i\omega_{fi}t'} dt' = \frac{e^{i\omega_{fi}t} - 1}{i\omega_{fi}},$$

so, writing $\omega = \omega_{fi}$,

$$P_{i \rightarrow f}(t) = \frac{|V_{fi}|^2}{\hbar^2} \cdot \frac{4 \sin^2(\omega t/2)}{\omega^2} = \frac{|V_{fi}|^2 t^2}{\hbar^2} \text{sinc}^2\left(\frac{\omega t}{2}\right),$$

where $\text{sinc } u = \sin u/u$.



The function is sharply peaked at $\omega_{fi} = 0$, that is at $E_f = E_i$: transitions overwhelmingly conserve energy. But the peak has a finite width $\sim 2\pi/t$ and a finite height $\sim t^2$, so at short times energy need not be conserved exactly – the time–energy uncertainty relation again. The two scalings are important:

- (i) the peak height grows like t^2 , so a transition to a single, exactly resonant final state has a probability growing quadratically – which must eventually fail, since probabilities are bounded by 1;
- (ii) the area under the curve grows like t , since height $\times$ width $\sim t^2 \times t^{-1}$.

It is the second scaling that survives when the final state is not a single level but a continuum, and it gives the linear-in-time behaviour we expect of a decay rate.

§37.9.4 Fermi's Golden Rule

Suppose the final states form a continuum – the ejected electron in the photoelectric effect, the emitted photon in atomic decay, the scattered particle in a collision – described by a **density of states** $\rho(E)$, so that the number of final states with energy in $[E, E+dE]$ is $\rho(E)dE$. Assume the matrix element V_{fi} varies slowly over the narrow band that contributes. Then the total transition probability is

$$P(t) = \int \rho(E_f) \frac{|V_{fi}|^2}{\hbar^2} \cdot \frac{4 \sin^2((E_f - E_i)t/2\hbar)}{((E_f - E_i)/\hbar)^2} dE_f.$$

For large t the sharply peaked factor picks out $E_f = E_i$, and using the standard integral $\int_{-\infty}^{\infty} \frac{\sin^2 au}{u^2} du = \pi|a|$ with $a = t/2\hbar$,

$$\int_{-\infty}^{\infty} \frac{4 \sin^2(\Delta t/2\hbar)}{\Delta^2} d\Delta = \frac{2\pi t}{\hbar},$$

so

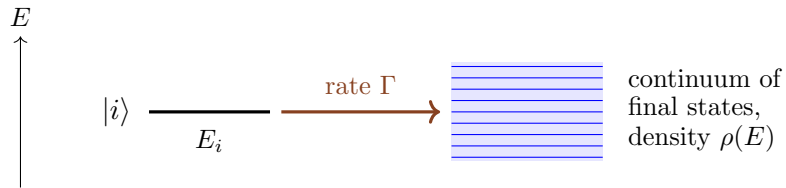
$$P(t) = \frac{2\pi t}{\hbar} |V_{fi}|^2 \rho(E_i).$$

Theorem 37.9.3 (Fermi's Golden Rule)

The transition rate from $|i\rangle$ into a continuum of final states at the same energy is

$$\Gamma = \frac{dP}{dt} = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_f) \Big|_{E_f=E_i}.$$

The probability now grows *linearly* in time, at a constant rate – exactly the behaviour of an exponential decay $P_{\text{survive}} = e^{-\Gamma t}$ in its early stages. Note the conditions hiding behind the derivation: t must be long enough for the peak to be narrower than the scale on which ρ and V_{fi} vary, and short enough that $P \ll 1$ so that first-order perturbation theory is valid. There is usually a wide window in which both hold.



Example 37.9.4 (A Harmonic Perturbation)

Most perturbations of interest oscillate – an atom in a light wave feels $V(t) = V e^{-i\omega t} + V^\dagger e^{i\omega t}$. Repeating the calculation, the time integral becomes

$$\int_0^t \left(e^{i(\omega_{fi}-\omega)t'} V_{fi} + e^{i(\omega_{fi}+\omega)t'} (V^\dagger)_{fi} \right) dt',$$

which is large only when $\omega_{fi} = \pm\omega$, that is when

$$E_f = E_i + \hbar\omega \quad (\text{absorption}) \quad \text{or} \quad E_f = E_i - \hbar\omega \quad (\text{stimulated emission}).$$

The golden rule becomes $\Gamma = \frac{2\pi}{\hbar} |V_{fi}|^2 \rho(E_i \pm \hbar\omega)$. So a system absorbs energy from a driving field only in quanta of $\hbar\omega$, and only when a level exists at the right height

– this is the theoretical content of a spectrum, and of Einstein’s B coefficients. Note that the two terms describe processes whose rates are equal: absorption and stimulated emission are equally likely, which is why a population inversion is needed to make a laser.

Remark (Selection Rules Again). Everything is controlled by $V_{fi} = \langle f|V|i\rangle$, and this is where the symmetry arguments of §37.6 pay off. For light interacting with an atom the leading term is the electric dipole $V \propto \mathcal{E} \cdot \mathbf{X}$, so transitions are forbidden unless $\langle f|\mathbf{X}|i\rangle \neq 0$, which by parity and rotational symmetry requires

$$\Delta\ell = \pm 1, \quad \Delta m \in \{0, \pm 1\}.$$

Transitions violating these are not strictly impossible – higher multipole terms in the expansion can allow them – but they are enormously slower, and the corresponding excited states are ‘metastable’. This is why orthohelium, which cannot decay to the parahelium ground state without flipping a spin, is long-lived enough to have been mistaken for a separate substance.

§37.10 Density Operators and Mixed States

Everything so far has assumed that we know which state the system is in. It may be a superposition, but it is a definite vector $|\psi\rangle$. Yet there are two quite different reasons one might be unable to predict the result of a measurement, and only one of them is described by a vector.

The first is intrinsic. A spin in $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ has no definite value of S_z , and no further information about it exists: this is a complete description, and the state *is* an eigenstate of S_x . The second is ordinary classical ignorance. Suppose the laboratory tossed a fair coin and prepared $|\uparrow\rangle$ or $|\downarrow\rangle$ accordingly, without telling us. Measuring S_z now also gives $\pm\frac{\hbar}{2}$ with equal probabilities; but measuring S_x gives $\pm\frac{\hbar}{2}$ with equal probabilities too, whereas in the superposition it gives $+\frac{\hbar}{2}$ with certainty. The two situations are physically different, and no vector in $\mathcal{H}$ describes the second one.

The fix is to enlarge the notion of state slightly, replacing vectors by operators.

§37.10.1 The Density Operator

Definition 37.10.1 (Density Operator)

Suppose a system is prepared in the normalised state $|\psi_i\rangle$ with probability p_i , where $p_i \geq 0$ and $\sum_i p_i = 1$. Such an **ensemble** is described by the **density operator**

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|.$$

A density operator of the form $\rho = |\psi\rangle \langle \psi|$ is called **pure**; any other is **mixed**.

Note carefully that the $|\psi_i\rangle$ are *not* required to be orthogonal, and that the sum is a sum of *operators*, weighted by classical probabilities – not a superposition of states. A pure state $|\psi\rangle$ is recovered as the ensemble in which one outcome has probability 1, and the projector $|\psi\rangle \langle \psi|$ carries exactly the same information as $|\psi\rangle$ except for the (unphysical) overall phase, which is a small bonus.

Proposition 37.10.2

A density operator satisfies

$$\rho^\dagger = \rho, \quad \rho \geq 0 \text{ (i.e. } \langle \phi | \rho | \phi \rangle \geq 0 \text{ for all } |\phi\rangle), \quad \text{tr } \rho = 1.$$

Conversely, every operator with these three properties is the density operator of some ensemble.

Proof. Each $|\psi_i\rangle \langle \psi_i|$ is Hermitian and positive, with trace $\langle \psi_i | \psi_i \rangle = 1$; the properties follow since the p_i are non-negative reals summing to 1. Conversely, given such a ρ , the spectral decomposition gives $\rho = \sum_n \lambda_n |n\rangle \langle n|$ with $\{|n\rangle\}$ orthonormal, $\lambda_n = \langle n | \rho | n \rangle \geq 0$ and $\sum_n \lambda_n = \text{tr } \rho = 1$: an ensemble. $\square$

So the density operators form a convex set – any probabilistic mixture $t\rho_1 + (1-t)\rho_2$ of density operators is one – whose extreme points, as we are about to see, are exactly the pure states. Everything we can measure is computed from ρ by taking a trace.

Proposition 37.10.3

In the ensemble described by ρ , the expectation value of an observable A is

$$\langle A \rangle_\rho = \text{tr}(\rho A),$$

and a measurement of A returns the eigenvalue λ_n with probability $\text{tr}(\rho P_n)$.

Proof. Using $\text{tr}(|\phi\rangle \langle \psi| A) = \langle \psi | A | \phi \rangle$,

$$\text{tr}(\rho A) = \sum_i p_i \text{tr}(|\psi_i\rangle \langle \psi_i| A) = \sum_i p_i \langle \psi_i | A | \psi_i \rangle,$$

which is the average over the ensemble of the quantum expectation values – exactly what the two layers of uncertainty demand. The second claim is the case $A = P_n$. $\square$

This has an immediate and slightly unsettling consequence: *two ensembles with the same density operator are indistinguishable by any experiment whatsoever*. The decomposition $\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i|$ is therefore not physical data; only ρ is.

Example 37.10.4 (The Same Mixture Twice Over)

Toss a fair coin and prepare $|\uparrow\rangle$ or $|\downarrow\rangle$. Then

$$\rho = \frac{1}{2} |\uparrow\rangle \langle \uparrow| + \frac{1}{2} |\downarrow\rangle \langle \downarrow| = \frac{1}{2} I.$$

Now instead toss a fair coin and prepare $|\rightarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ or $|\leftarrow\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$. Then

$$\rho' = \frac{1}{2} |\rightarrow\rangle \langle \rightarrow| + \frac{1}{2} |\leftarrow\rangle \langle \leftarrow| = \frac{1}{2} I$$

as well, since the cross terms cancel. The two preparation procedures are completely different, and yet no measurement can tell them apart. This $\rho = \frac{1}{2} I$ is the **maxi-**

mally mixed state of a qubit: every observable has expectation $\frac{1}{2} \text{tr } A$, and every measurement is a fair coin toss.

Proposition 37.10.5 (Purity)

For any density operator $\text{tr}(\rho^2) \leq 1$, with equality if and only if ρ is pure.

Proof. Diagonalise, $\rho = \sum_n \lambda_n |n\rangle \langle n|$. Then $\text{tr} \rho^2 = \sum_n \lambda_n^2 \leq (\sum_n \lambda_n)^2 = 1$, since the λ_n are non-negative and the cross terms are non-negative. Equality forces all but one λ_n to vanish, so $\rho = |n\rangle \langle n|$ for a single n . $\square$

The number $\text{tr} \rho^2$ is called the **purity**, and runs from 1 (pure) down to $1/\dim \mathcal{H}$ (maximally mixed). Equivalently, ρ is pure iff $\rho^2 = \rho$, that is iff ρ is a projector.

Finally, dynamics. If each member of the ensemble evolves by the Schrödinger equation, $|\psi_i(t)\rangle = U(t) |\psi_i(0)\rangle$, then $\rho(t) = U(t)\rho(0)U(t)^\dagger$, and differentiating:

Proposition 37.10.6 (Von Neumann Equation)

$$i\hbar \frac{d\rho}{dt} = [H, \rho].$$

Note the sign, which is *opposite* to Heisenberg's equation of motion $i\hbar \dot{A}_H = [A_H, H]$. There is no contradiction: ρ is a state, not an observable, and it evolves like one. A stationary ρ is one commuting with H – for instance any function of H , which is why the thermal state $\rho \propto e^{-\beta H}$ of Statistical Physics is an equilibrium state.

Measurement is described just as before. If A is measured and the outcome λ_n is observed, then

$$\rho \mapsto \frac{P_n \rho P_n}{\text{tr}(\rho P_n)};$$

and if the measurement is made but the result is discarded – a very natural thing to happen when the ‘apparatus’ is the surrounding environment – then averaging over outcomes with their probabilities gives

$$\rho \mapsto \sum_n P_n \rho P_n,$$

which simply deletes all the off-diagonal blocks of ρ in the eigenbasis of A . This is **decoherence**: an unobserved interaction with the environment turns a superposition into a mixture, destroying the relative phases that produce interference, and it is the reason that large objects are never seen in superpositions of macroscopically distinct states.

Example 37.10.7 (The Qubit and the Bloch Ball)

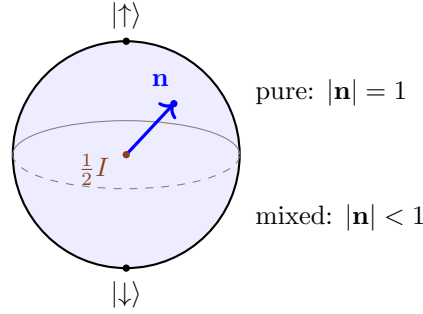
Any Hermitian 2×2 matrix is a real combination of I and the Pauli matrices, and $\text{tr} \sigma_i = 0$, so the trace condition fixes the coefficient of I :

$$\rho = \frac{1}{2} (I + \mathbf{n} \cdot \boldsymbol{\sigma}), \quad \mathbf{n} \in \mathbb{R}^3.$$

The eigenvalues of $\mathbf{n} \cdot \sigma$ are $\pm|\mathbf{n}|$, so those of ρ are $\frac{1}{2}(1 \pm |\mathbf{n}|)$, and positivity holds exactly when $|\mathbf{n}| \leq 1$. Moreover

$$\langle \mathbf{S} \rangle_\rho = \frac{\hbar}{2} \text{tr}(\rho \sigma) = \frac{\hbar}{2} \mathbf{n}, \quad \text{tr} \rho^2 = \frac{1}{2} (1 + |\mathbf{n}|^2),$$

so $\mathbf{n}$ – the **Bloch vector** – is just the mean spin direction, and ρ is pure precisely when $|\mathbf{n}| = 1$. The states of a qubit therefore fill a solid ball of radius 1, with the pure states of §4.3 on its surface (the Bloch sphere) and the maximally mixed state $\frac{1}{2}I$ at its centre.



§37.10.2 Subsystems and the Reduced Density Operator

The second, and more important, reason density operators are unavoidable is that *a subsystem of a system in a pure state need not itself have a pure state*. This is where the tensor product of §37.5.1 takes its revenge.

Suppose the composite system $\mathcal{H}_1 \otimes \mathcal{H}_2$ is in the state ρ , and we have access only to subsystem 1. The observables available to us are those of the form $A \otimes I$, and we would like a ρ_1 on $\mathcal{H}_1$ reproducing all their expectation values.

Definition 37.10.8 (Partial Trace)

Let $\{|a\rangle_2\}$ be an orthonormal basis of $\mathcal{H}_2$. The **reduced density operator** of subsystem 1 is

$$\rho_1 = \text{tr}_2 \rho := \sum_a (I \otimes \langle a|_2) \rho (I \otimes |a\rangle_2),$$

an operator on $\mathcal{H}_1$, and similarly for $\rho_2 = \text{tr}_1 \rho$.

Proposition 37.10.9

ρ_1 is a density operator on $\mathcal{H}_1$, independent of the basis chosen, and

$$\text{tr}(\rho(A \otimes I)) = \text{tr}(\rho_1 A) \quad \text{for every observable } A \text{ on } \mathcal{H}_1.$$

Proof. Hermiticity and positivity are inherited termwise, and $\text{tr} \rho_1 = \text{tr} \rho = 1$ because tracing over $\mathcal{H}_1$ and then $\mathcal{H}_2$ is the same as tracing over the product. For the

displayed identity, evaluate the full trace in a product basis $|i\rangle_1 |a\rangle_2$:

$$\mathrm{tr}(\rho(A \otimes I)) = \sum_{i,a} \langle i, a | \rho(A \otimes I) | i, a \rangle = \sum_i \langle i | \left(\sum_a \langle a | \rho | a \rangle \right) A | i \rangle = \mathrm{tr}(\rho_1 A).$$

Basis independence follows, since the left-hand side does not mention the basis of $\mathcal{H}_2$. $\square$

Example 37.10.10 (The Singlet)

Let the pair of spins be in the pure singlet state $|00\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, so

$$\rho = \frac{1}{2} \left(|\uparrow\downarrow\rangle \langle\uparrow\downarrow| - |\uparrow\downarrow\rangle \langle\downarrow\uparrow| - |\downarrow\uparrow\rangle \langle\uparrow\downarrow| + |\downarrow\uparrow\rangle \langle\downarrow\uparrow| \right).$$

Tracing out the second spin with the basis $|\uparrow\rangle_2, |\downarrow\rangle_2$, the two cross terms die (they have orthogonal states in slot 2), and

$$\rho_1 = \frac{1}{2} |\uparrow\rangle \langle\uparrow| + \frac{1}{2} |\downarrow\rangle \langle\downarrow| = \frac{1}{2} I.$$

The first spin is in the *maximally mixed* state: we know everything there is to know about the pair, and nothing whatsoever about either member of it. Its purity is $\frac{1}{2}$, the smallest possible.

This is the sharpest statement of what entanglement is, and it is worth recording.

Proposition 37.10.11

Let $|\Psi\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$ be a pure state. Then $|\Psi\rangle$ is a product state if and only if $\rho_1 = \mathrm{tr}_2 |\Psi\rangle \langle\Psi|$ is pure.

Proof. If $|\Psi\rangle = |\psi\rangle \otimes |\omega\rangle$ then $\rho = |\psi\rangle \langle\psi| \otimes |\omega\rangle \langle\omega|$ and $\rho_1 = |\psi\rangle \langle\psi|$, which is pure. Conversely, if $\rho_1 = |\psi\rangle \langle\psi|$ is pure then, expanding $|\Psi\rangle = \sum_i |i\rangle_1 |\chi_i\rangle_2$ in an orthonormal basis of $\mathcal{H}_1$ containing $|\psi\rangle$, one computes $\rho_1 = \sum_{ij} \langle\chi_j|\chi_i\rangle |i\rangle \langle j|$; purity forces every $|\chi_i\rangle$ with $|i\rangle \neq |\psi\rangle$ to vanish, leaving $|\Psi\rangle = |\psi\rangle \otimes |\chi_\psi\rangle$. $\square$

So the mixedness of ρ_1 measures how entangled the pair is, and one can turn this into a number (the *entropy of entanglement*, $-\mathrm{tr} \rho_1 \log \rho_1$), which is the starting point of quantum information theory. For us the important corollary is about signalling.

Proposition 37.10.12 (No Signalling)

Suppose Bob, who holds subsystem 2, applies any unitary to it, or measures any observable on it and does not tell Alice the outcome. Then Alice's reduced state ρ_1 is unchanged.

Proof. For a unitary, $\rho \mapsto (I \otimes U)\rho(I \otimes U^\dagger)$, and taking the partial trace over $\mathcal{H}_2$ we may cycle $U^\dagger$ round to meet U within that factor, giving ρ_1 again. For an unrecorded measurement, $\rho \mapsto \sum_n (I \otimes P_n)\rho(I \otimes P_n)$; cycling in the same way turns this into $\mathrm{tr}_2 [(I \otimes \sum_n P_n)\rho] = \mathrm{tr}_2 \rho = \rho_1$. $\square$

This modest-looking result is what reconciles the rest of this course with special relativity: however strongly two separated systems are correlated, nothing Bob does can alter the statistics of *any* measurement Alice makes. Correlation is not communication. Keep it in mind, because in the next section the correlations are going to look very much like communication indeed.

§37.11 The EPR Paradox and Bell's Inequality

We close by asking what quantum mechanics says about the world, rather than about the outcomes of experiments. The formalism is fixed by now and nothing in it will change; what is at stake is whether it can be regarded as a complete description of reality, or merely as a statistical summary of some deeper theory. Remarkably, this apparently metaphysical question turns out to have an experimental answer.

§37.11.1 The EPR Argument

Prepare two spin- $\frac{1}{2}$ particles in the singlet state

$$|00\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

and let them fly apart – to Alice on the left and Bob on the right, arbitrarily far away. Alice measures S_z on her particle. She gets $\pm\frac{\hbar}{2}$ with equal probability; but whatever she gets, she knows with *certainty* that Bob will get the opposite, since the state contains no $|\uparrow\uparrow\rangle$ or $|\downarrow\downarrow\rangle$ term.

Now recall from §5.4 that the singlet is rotationally invariant. Consequently the same argument works along *any* axis: Alice may equally measure S_x , and then predict Bob's S_x with certainty. This is the observation of Einstein, Podolsky and Rosen (1935). Their argument runs as follows.

- (i) *Locality*. Alice's measurement cannot physically affect Bob's distant particle. (The two events can be arranged to be spacelike separated.)
- (ii) *Reality*. 'If, without in any way disturbing a system, we can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to that quantity.'
- (iii) Alice can predict Bob's S_z with certainty, so Bob's S_z has a definite value. She could instead have chosen to predict his S_x , so his S_x has a definite value too. By locality, which choice she makes cannot matter to Bob's particle; so *both* values are real.
- (iv) But no quantum state assigns definite values to both S_x and S_z , since $[S_x, S_z] \neq 0$.

The conclusion EPR drew is that quantum mechanics, while correct as far as it goes, is *incomplete*: there must be additional variables – traditionally written λ and called **hidden variables** – which fix the outcomes of all measurements in advance, and of which the quantum state is only a coarse-grained average. On this view the randomness of quantum mechanics is like the randomness of a coin toss, a statement about our ignorance rather than about the coin.

It is a good argument. Bohr's reply at the time amounted to rejecting (ii), and for thirty years the matter was regarded as a question of taste. What nobody noticed until Bell (1964) was that the two views make *different experimental predictions*.

§37.11.2 Local Hidden Variables and the CHSH Inequality

Let us say exactly what a hidden-variable theory is. Alice chooses a direction $\mathbf{a}$ and measures $\mathbf{a} \cdot \sigma$, getting ± 1 ; Bob chooses $\mathbf{b}$ and gets ± 1 . A **local hidden variable** (LHV) theory posits

- (i) a hidden variable λ , distributed according to some probability density $p(\lambda)$ fixed by the preparation;
- (ii) functions $A(\mathbf{a}, \lambda) = \pm 1$ and $B(\mathbf{b}, \lambda) = \pm 1$ giving the outcomes.

The locality assumption is buried in the notation, and is the whole content of the theorem: A does not depend on $\mathbf{b}$, and B does not depend on $\mathbf{a}$. Alice's outcome is not allowed to know what Bob chose to measure.

The measurable quantity is the **correlation**

$$E(\mathbf{a}, \mathbf{b}) = \int A(\mathbf{a}, \lambda) B(\mathbf{b}, \lambda) p(\lambda) d\lambda,$$

the average of the product of the two outcomes, which is $+1$ for perfect correlation and -1 for perfect anticorrelation.

Theorem 37.11.1 (CHSH Inequality)

In any local hidden variable theory, and for any four directions $\mathbf{a}, \mathbf{a}', \mathbf{b}, \mathbf{b}'$,

$$S := E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}') \quad \text{satisfies} \quad |S| \leq 2.$$

Proof. Fix λ and abbreviate $A = A(\mathbf{a}, \lambda)$, $A' = A(\mathbf{a}', \lambda)$, $B = B(\mathbf{b}, \lambda)$, $B' = B(\mathbf{b}', \lambda)$, each of which is ± 1 . Then

$$AB - AB' + A'B + A'B' = A(B - B') + A'(B + B').$$

Since $B, B' \in \{\pm 1\}$, either $B = B'$, in which case the first bracket vanishes and the second is ± 2 ; or $B = -B'$, in which case the second vanishes and the first is ± 2 . Either way the whole expression is ± 2 . Averaging over λ with the probability density p , which has total mass 1, gives $|S| \leq 2$. $\square$

The proof is three lines and uses nothing about quantum mechanics – indeed nothing about physics. It says only that if each particle carries with it a predetermined answer to every question it might be asked, then four particular correlations cannot all be large at once. It is very hard to see how any theory of the EPR type could evade it.

§37.11.3 The Quantum Prediction

Now compute the same four numbers in quantum mechanics, in the singlet state.

Proposition 37.11.2

In the singlet state of two spin- $\frac{1}{2}$ particles,

$$E(\mathbf{a}, \mathbf{b}) = \langle (\mathbf{a} \cdot \boldsymbol{\sigma}^{(1)}) (\mathbf{b} \cdot \boldsymbol{\sigma}^{(2)}) \rangle = -\mathbf{a} \cdot \mathbf{b} = -\cos \theta_{ab},$$

where θ_{ab} is the angle between the two directions.

Proof. Write $C_{ij} = \langle \sigma_i^{(1)} \sigma_j^{(2)} \rangle$. The singlet is annihilated by every component of $\mathbf{J} = \mathbf{S}^{(1)} + \mathbf{S}^{(2)}$, so it is invariant under simultaneous rotations of both spins, under which C_{ij} transforms as a rank-two tensor; the only rotationally invariant such tensor is a multiple of δ_{ij} , so $C_{ij} = c \delta_{ij}$. To find c , use $J^2 = 0$:

$$0 = (\mathbf{S}^{(1)} + \mathbf{S}^{(2)})^2 = \frac{3\hbar^2}{4} + \frac{3\hbar^2}{4} + 2\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} \implies \mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} = -\frac{3\hbar^2}{4},$$

that is $\langle \sigma^{(1)} \cdot \sigma^{(2)} \rangle = -3$, so $3c = -3$ and $c = -1$. Hence $E(\mathbf{a}, \mathbf{b}) = a_i b_j C_{ij} = -\mathbf{a} \cdot \mathbf{b}$. $\square$

For those who prefer to see the arithmetic, the joint probabilities are

$$\mathbb{P}(\pm, \pm) = \frac{1}{2} \sin^2 \frac{\theta_{ab}}{2}, \quad \mathbb{P}(\pm, \mp) = \frac{1}{2} \cos^2 \frac{\theta_{ab}}{2},$$

whose difference is $\sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2} = -\cos \theta$, as claimed. At $\theta = 0$ the outcomes are perfectly anticorrelated (the EPR observation), and at $\theta = \pi$ perfectly correlated.

Corollary 37.11.3 (Bell's Theorem)

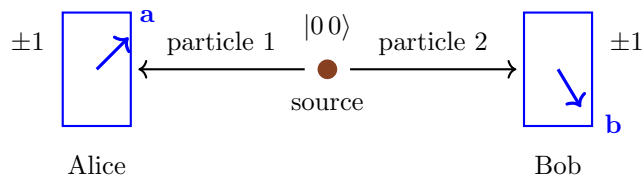
Quantum mechanics violates the CHSH inequality, and therefore no local hidden variable theory can reproduce its predictions.

Proof. Take all four directions in a plane, at angles $0, \frac{\pi}{2}, \frac{\pi}{4}, \frac{3\pi}{4}$ for $\mathbf{a}, \mathbf{a}', \mathbf{b}, \mathbf{b}'$ respectively. Then $\mathbf{a}$ makes an angle $\frac{\pi}{4}$ with $\mathbf{b}$ and $\frac{3\pi}{4}$ with $\mathbf{b}'$, while $\mathbf{a}'$ makes an angle $\frac{\pi}{4}$ with each of $\mathbf{b}, \mathbf{b}'$. So

$$S = -\cos \frac{\pi}{4} + \cos \frac{3\pi}{4} - \cos \frac{\pi}{4} - \cos \frac{\pi}{4} = -\frac{4}{\sqrt{2}} = -2\sqrt{2},$$

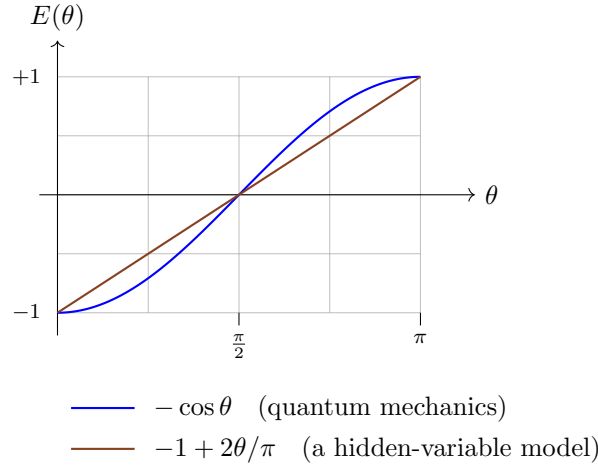
and $|S| = 2\sqrt{2} \approx 2.83 > 2$.

$\square$



The two Stern–Gerlach analysers may be light years apart, and the settings $\mathbf{a}, \mathbf{b}$ chosen at the last moment, so that no signal travelling at or below the speed of light can carry Alice's choice to Bob's apparatus. Yet the correlations between the two columns of ± 1 s are stronger than any pre-agreed strategy could produce. The following picture, of

the correlation as a function of the angle between the settings, shows where the extra strength comes from.



The straight line is the correlation produced by the most obvious hidden-variable model – each pair carries a random direction λ , and each particle answers according to which side of the plane perpendicular to its analyser λ falls on. It agrees with quantum mechanics at $\theta = 0, \frac{\pi}{2}, \pi$ and is wrong in between; and the CHSH combination is precisely designed to probe the region where the cosine bulges away from the line.

§37.11.4 What the Experiments Say

Everything above is theory. The experiment has been done, repeatedly and with increasing care: by Freedman and Clauser (1972), decisively by Aspect and collaborators in the early 1980s using rapidly switched analysers, and in ‘loophole-free’ form in 2015 with the detector settings chosen by fast random number generators at spacelike separation and essentially every photon detected. The answer is unambiguous, and it is $2\sqrt{2}$. Nature violates the CHSH inequality, in quantitative agreement with quantum mechanics; the 2022 Nobel Prize was awarded for this work.

So one of the assumptions of Bell’s theorem is false. Since the derivation used only realism (outcomes determined by λ) and locality (Alice’s outcome does not depend on Bob’s setting), at least one of these must be abandoned. The overwhelmingly standard view is to abandon realism: the outcome of Alice’s measurement simply does not exist before she makes it, and the perfect anticorrelation of the singlet is not explained by a pre-arranged agreement, because there is nothing to arrange. Quantum mechanics is not a statistical approximation to a deeper deterministic theory of the EPR kind.

Three remarks are worth making, to guard against over-interpretation.

Remark (No Signalling, Again). It is tempting to read the violation as instantaneous influence, and thus as a conflict with relativity. It is not, and the reason is the proposition of §37.10: Bob’s reduced density operator is $\frac{1}{2}I$ whatever Alice does, so the statistics he sees are the statistics of a fair coin no matter what she measures or whether she measures at all. The correlations only become visible when the two lists of outcomes are brought together and compared, which requires a classical channel and therefore takes time. Quantum mechanics is *non-local* in the sense that no local hidden variable theory can reproduce it, and yet perfectly *causal*.

Remark (Tsirelson’s Bound). Quantum mechanics does not violate CHSH as much as it logically could: a hypothetical theory with $\mathbb{P}(\text{same}) = 1$ or 0 chosen at will would give $|S| = 4$ while still forbidding signalling. That quantum mechanics stops at $|S| \leq 2\sqrt{2}$ – Tsirelson’s bound, which follows from $(\mathbf{a} \cdot \boldsymbol{\sigma})^2 = I$ and a short calculation – is a genuine and somewhat mysterious feature of the theory.

Remark (Entanglement as a Resource). Modern quantum information theory takes exactly the correlations that troubled Einstein and puts them to work: they are what makes quantum key distribution secure, and what makes teleportation and superdense coding possible. This is the subject of the Part II course Quantum Information and Computation, and it is a fine advertisement for taking the formalism seriously rather than apologising for it.

It is a good place to stop. We began by throwing away the wavefunction in favour of a vector in a Hilbert space, and found that almost everything – the oscillator spectrum, the quantisation of angular momentum, the existence of spin, conservation laws, degeneracies, selection rules – follows from a handful of commutators and the representation theory they generate. We end with the discovery that this abstract structure is not merely a convenient calculus but a statement about the world, and one which the world has been measured to obey.

38 Probability and Measure

Measure theory is the machinery that makes ‘size’ – length, area, volume, probability – into something we can actually do analysis with. The Riemann integral of Analysis I is a perfectly serviceable tool for integrating continuous functions on a closed interval, but it is fragile: it does not interact well with limits, the space of Riemann integrable functions is not complete, and it has nothing whatsoever to say about integration over abstract spaces. Lebesgue’s insight was that if one is willing to work harder at the front end – assigning a measure to sets before integrating functions – then the integral one gets at the back end is enormously more robust.

Probability theory is where this pays off most spectacularly. Kolmogorov’s observation, in 1933, was that a probability is nothing more than a measure of total mass one, a random variable is nothing more than a measurable function, and an expectation is nothing more than an integral. Once one accepts this dictionary, the great limit theorems of probability – the strong law of large numbers, the central limit theorem – become theorems about the convergence of integrals, and the tools of measure theory apply directly. In this course we will build the theory from the ground up and then spend the second half cashing it in.

This article constitutes my notes for the ‘Probability and Measure’ course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§38.1 Introduction

Let us begin by seeing what goes wrong with the integral we already have, because the shape of the fix is visible in the shape of the failure.

Recall from Analysis I that a bounded function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable if the supremum of the lower sums equals the infimum of the upper sums, where the sums are taken over partitions of $[a, b]$ into finitely many intervals. The essential move is to *chop up the domain* into small intervals and, on each one, to pretend that f is constant. That is a good idea when f is continuous, since then f really is nearly constant on small intervals. It is a terrible idea otherwise.

Example 38.1.1 (The indicator of the rationals)

Let $f = \mathbb{1}_{\mathbb{Q} \cap [0, 1]}$, so $f(x) = 1$ if x is rational and 0 otherwise. On every subinterval of $[0, 1]$, however small, f takes both the value 0 and the value 1. So every lower sum is 0 and every upper sum is 1, and f is not Riemann integrable.

But morally the answer ought to be 0: the rationals are a countable set, they are ‘negligible’ in $[0, 1]$, and f is zero off a negligible set. The Riemann integral has no way to express this thought.

The deeper problem is that the Riemann integral behaves badly under limits. Enumerate $\mathbb{Q} \cap [0, 1]$ as $q_1, q_2, \dots$ and let $f_n = \mathbb{1}_{\{q_1, \dots, q_n\}}$. Each f_n is Riemann integrable with integral

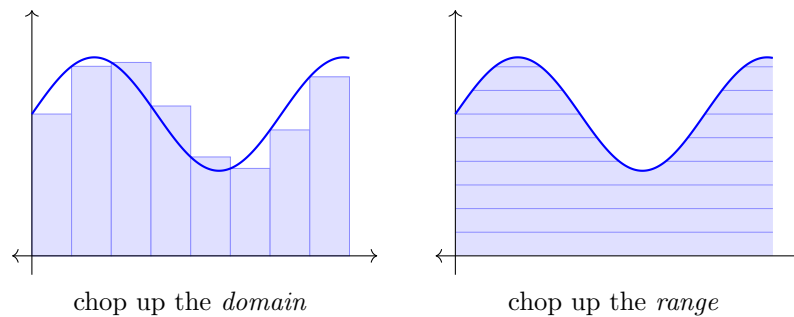
0, and $f_n \uparrow f$ pointwise, and yet the limit is not integrable at all. Any theorem of the form ‘if $f_n \rightarrow f$ then $\int f_n \rightarrow \int f$ ’ must, for the Riemann integral, carry hypotheses strong enough (uniform convergence, say) to rule out most of the situations one actually meets.

Lebesgue’s alternative is to *chop up the range* instead. If $f \geq 0$, then instead of asking ‘how much does f vary on $[x_i, x_{i+1}]$?’ we ask ‘how big is the set on which f exceeds y ?’.

Writing μ for ‘size’, we would like to say something like

$$\int f \approx \sum_j y_j \mu(\{x : y_j \leq f(x) < y_{j+1}\}),$$

approximating f from below by functions taking finitely many values. The point is that the sets $\{y_j \leq f < y_{j+1}\}$ are allowed to be horrible – they need not be intervals, or finite unions of intervals, or anything one can draw – provided only that we can assign them a size.



So the plan for the first third of the course is forced upon us. We must decide which sets get a size (§2), we must decide which functions are compatible with that assignment (§3), and only then can we build the integral (§4). It is worth saying at the outset that the first of these steps is the hard one, and that it cannot be done perfectly: we will prove in §2 that there is no way to assign a length to *every* subset of $\mathbb{R}$ in a translation-invariant, countably additive way. Some sets must be left out. Deciding exactly which ones is the content of the Carathéodory extension theorem, and it will occupy us for a while.

§38.2 Measures

§38.2.1 Set systems

Throughout, E is a non-empty set, and we write $\mathcal{P}(E)$ for its power set. We want to specify a family of subsets of E – the sets to which we are prepared to assign a size – and we want that family to be closed under the operations we expect to perform. The right notion is the following.

Definition 38.2.1 (σ -algebra)

A collection $\mathcal{E} \subseteq \mathcal{P}(E)$ is a **σ -algebra** on E if

- (i) $\emptyset \in \mathcal{E}$;
- (ii) $A \in \mathcal{E}$ implies $A^c = E \setminus A \in \mathcal{E}$;
- (iii) if $(A_n)_{n \in \mathbb{N}}$ is a sequence in $\mathcal{E}$ then $\bigcup_n A_n \in \mathcal{E}$.

The pair $(E, \mathcal{E})$ is then called a **measurable space**, and the elements of $\mathcal{E}$ are called **measurable sets**.

Note that $E = \emptyset^c \in \mathcal{E}$, and that by De Morgan

$$\bigcap_n A_n = \left(\bigcup_n A_n^c \right)^c \in \mathcal{E},$$

so σ -algebras are closed under countable intersections as well as countable unions; and $B \setminus A = B \cap A^c \in \mathcal{E}$, so they are closed under set difference too. The prefix σ is there to record the fact that we allow *countable* unions, and this is exactly the right amount of infinity: finite unions are too weak to say anything about limits, and uncountable unions would be far too strong (every subset of $\mathbb{R}$ is an uncountable union of singletons).

Example 38.2.2 (Two extreme examples)

The collection $\{\emptyset, E\}$ is a σ -algebra, the smallest one. The power set $\mathcal{P}(E)$ is a σ -algebra, the largest one. Neither is very useful on its own: the first is too coarse to distinguish anything, and we will see that the second is too fine to admit a sensible notion of length on $\mathbb{R}$.

In practice one rarely writes down a σ -algebra explicitly. Instead one specifies a handful of sets one certainly wants to be measurable, and takes the smallest σ -algebra containing them.

Definition 38.2.3 (Generated σ -algebra)

Let $\mathcal{A} \subseteq \mathcal{P}(E)$. The **σ -algebra generated by $\mathcal{A}$** is

$$\sigma(\mathcal{A}) = \bigcap \{ \mathcal{E} : \mathcal{E} \text{ is a } \sigma\text{-algebra on } E, \mathcal{A} \subseteq \mathcal{E} \}.$$

This is well defined because $\mathcal{P}(E)$ is a σ -algebra containing $\mathcal{A}$, so the intersection is over a non-empty family; and an arbitrary intersection of σ -algebras is easily checked to be a σ -algebra. By construction $\sigma(\mathcal{A})$ is the smallest σ -algebra containing $\mathcal{A}$.

Definition 38.2.4 (Borel σ -algebra)

Let (E, τ) be a topological space. The **Borel σ -algebra** on E is $\mathcal{B}(E) = \sigma(\tau)$, the σ -algebra generated by the open sets. Its elements are the **Borel sets**. We abbreviate $\mathcal{B} = \mathcal{B}(\mathbb{R})$.

The Borel σ -algebra on $\mathbb{R}$ contains every open set, every closed set, every countable intersection of open sets, every countable union of those, and so on – one really does have to work to write down a subset of $\mathbb{R}$ that is not Borel. (We will manage it, with the help of the axiom of choice, in §2.7.)

Example 38.2.5 (Generators of $\mathcal{B}$)

Each of the following families generates $\mathcal{B}$:

$$\{(a, b) : a < b\}, \quad \{(a, b] : a < b\}, \quad \{(-\infty, y] : y \in \mathbb{R}\}, \quad \{(q, r) : q < r, q, r \in \mathbb{Q}\}.$$

For the last of these: every open subset of $\mathbb{R}$ is a countable union of rational intervals,

so $\sigma(\{(q, r)\}) \supseteq \tau$ and hence contains $\mathcal{B}$; and each (q, r) is open, so the reverse containment holds too. For the third, $(-\infty, y] = \bigcap_n (-\infty, y + 1/n)$ is Borel, and conversely $(a, b) = \bigcup_n ((-\infty, b - 1/n] \setminus (-\infty, a])$.

Working directly with σ -algebras is unpleasant, because the countable operations make it hard to verify anything by hand. It is much more comfortable to work with systems closed only under finitary operations, and then to bootstrap. The following two definitions give the finitary systems that we will use as generators.

Definition 38.2.6 (Ring and algebra)

A collection $\mathcal{A} \subseteq \mathcal{P}(E)$ is a **ring** on E if $\emptyset \in \mathcal{A}$ and, for all $A, B \in \mathcal{A}$, both $B \setminus A \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$.

It is an **algebra** on E if $\emptyset \in \mathcal{A}$ and, for all $A, B \in \mathcal{A}$, both $A^c \in \mathcal{A}$ and $A \cup B \in \mathcal{A}$.

Remark. Every algebra is a ring, since $B \setminus A = (B^c \cup A)^c$. The converse fails: a ring need not contain E itself. For instance the finite unions of bounded intervals of the form $(a, b]$ form a ring on $\mathbb{R}$ but not an algebra. Rings are closed under intersection, because $A \cap B = (A \cup B) \setminus ((A \setminus B) \cup (B \setminus A))$, and hence under symmetric difference as well.

The following triviality gets used so often that it deserves a name.

Proposition 38.2.7 (Disjointification)

Let $\mathcal{A}$ be a ring and let $A_1, A_2, \dots \in \mathcal{A}$. Then there are pairwise disjoint $B_1, B_2, \dots \in \mathcal{A}$ with $B_n \subseteq A_n$ and

$$\bigcup_{n \leq N} A_n = \bigcup_{n \leq N} B_n \quad \text{for every } N,$$

and consequently $\bigcup_n A_n = \bigcup_n B_n$.

Proof. Set $B_1 = A_1$ and $B_{n+1} = A_{n+1} \setminus (A_1 \cup \dots \cup A_n)$, which lies in $\mathcal{A}$ since rings are closed under finite unions and differences. These are disjoint, and an easy induction gives the displayed identity. $\square$

§38.2.2 Measures

We can now say what a size function is.

Definition 38.2.8 (Measure)

Let $(E, \mathcal{E})$ be a measurable space. A **measure** on $(E, \mathcal{E})$ is a function $\mu : \mathcal{E} \rightarrow [0, \infty]$ such that $\mu(\emptyset) = 0$ and, for every sequence $(A_n)_{n \in \mathbb{N}}$ of *pairwise disjoint* sets in $\mathcal{E}$,

$$\mu\left(\bigcup_{n \in \mathbb{N}} A_n\right) = \sum_{n \in \mathbb{N}} \mu(A_n).$$

This last property is **countable additivity**. The triple $(E, \mathcal{E}, \mu)$ is a **measure space**.

We allow the value $+\infty$; the sum on the right is a sum of non-negative terms, so it converges in $[0, \infty]$ regardless of order, and there is no ambiguity. A measure is **finite** if $\mu(E) < \infty$, and **σ -finite** if $E = \bigcup_n E_n$ for some $E_n \in \mathcal{E}$ with $\mu(E_n) < \infty$. It is a **probability measure** if $\mu(E) = 1$.

Example 38.2.9 (Counting measure and Dirac measure)

On $(E, \mathcal{P}(E))$, define $\mu(A) = |A|$ if A is finite and ∞ otherwise. This is the **counting measure**. Fix $x \in E$ and define $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise; this is the **Dirac measure** at x , a probability measure.

Remark. If E is countable and $\mathcal{E} = \mathcal{P}(E)$, then any measure is determined by its values on singletons: $\mu(A) = \sum_{x \in A} \mu(\{x\})$ by countable additivity. This is exactly the notion of a probability mass function from IA Probability. It is the uncountable case that requires all the machinery.

Countable additivity is a strong assumption, and it immediately gives us the properties we would expect of a notion of size.

Proposition 38.2.10 (Basic properties of measures)

Let $(E, \mathcal{E}, \mu)$ be a measure space and let $A, B, A_n \in \mathcal{E}$.

- (i) (Monotonicity) If $A \subseteq B$ then $\mu(A) \leq \mu(B)$.
- (ii) (Countable subadditivity) $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$.
- (iii) (Continuity from below) If $A_n \uparrow A$, meaning $A_1 \subseteq A_2 \subseteq \dots$ and $A = \bigcup_n A_n$, then $\mu(A_n) \rightarrow \mu(A)$.
- (iv) (Continuity from above) If $A_n \downarrow A$ and $\mu(A_1) < \infty$, then $\mu(A_n) \rightarrow \mu(A)$.

Proof. (i) Write $B = A \sqcup (B \setminus A)$; then $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$.

(ii) Disjointify as in Proposition 38.2.7 to get disjoint $B_n \subseteq A_n$ with the same union. Then

$$\mu\left(\bigcup_n A_n\right) = \mu\left(\bigsqcup_n B_n\right) = \sum_n \mu(B_n) \leq \sum_n \mu(A_n)$$

using monotonicity in the last step.

(iii) Set $B_1 = A_1$ and $B_{n+1} = A_{n+1} \setminus A_n$; these are disjoint with $\bigcup_{n \leq N} B_n = A_N$ and $\bigcup_n B_n = A$. So

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \mu(B_n) = \lim_{N \rightarrow \infty} \mu(A_N).$$

(iv) Apply (iii) to $A_1 \setminus A_n \uparrow A_1 \setminus A$. This gives $\mu(A_1) - \mu(A_n) \rightarrow \mu(A_1) - \mu(A)$, and since $\mu(A_1) < \infty$ we may cancel. $\square$

The finiteness hypothesis in (iv) is genuinely needed: with μ the counting measure on $\mathbb{N}$ and $A_n = \{n, n+1, \dots\}$ we have $A_n \downarrow \emptyset$ but $\mu(A_n) = \infty$ for every n .

We will eventually want to build measures by prescribing them on some convenient family of sets and extending. For that we need to talk about functions defined on families that

are not σ -algebras.

Definition 38.2.11 (Set function)

Let $\mathcal{A} \subseteq \mathcal{P}(E)$ with $\emptyset \in \mathcal{A}$. A **set function** on $\mathcal{A}$ is a map $\mu : \mathcal{A} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$. We call μ

- (i) **increasing** if $\mu(A) \leq \mu(B)$ whenever $A \subseteq B$ with $A, B \in \mathcal{A}$;
- (ii) **additive** if $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever $A, B \in \mathcal{A}$ are disjoint and $A \cup B \in \mathcal{A}$;
- (iii) **countably additive** if $\mu(\bigcup_n A_n) = \sum_n \mu(A_n)$ whenever the $A_n \in \mathcal{A}$ are pairwise disjoint and $\bigcup_n A_n \in \mathcal{A}$;
- (iv) **countably subadditive** if $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$ whenever $A_n \in \mathcal{A}$ and $\bigcup_n A_n \in \mathcal{A}$.

On a ring, countable additivity implies all of the others, by exactly the arguments of Proposition 38.2.10. A measure is precisely a countably additive set function on a σ -algebra.

§38.2.3 π -systems, d -systems and Dynkin's lemma

Here is the difficulty we are about to face repeatedly. We will want to prove that two measures agree on a σ -algebra $\mathcal{E}$, knowing only that they agree on some generating family $\mathcal{A}$. The natural strategy is to consider $\mathcal{D} = \{A : \mu_1(A) = \mu_2(A)\}$ and show it is a σ -algebra containing $\mathcal{A}$. Unfortunately $\mathcal{D}$ is *not* obviously a σ -algebra: additivity gives us nothing about $\mu_i(A \cup B)$ when A and B overlap. What $\mathcal{D}$ is closed under is proper differences and increasing limits. This motivates the following.

Definition 38.2.12 (π -system)

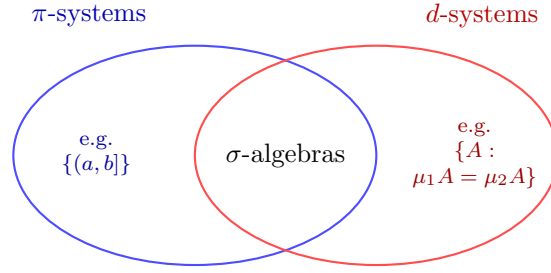
A collection $\mathcal{A} \subseteq \mathcal{P}(E)$ is a **π -system** if $\emptyset \in \mathcal{A}$ and $A, B \in \mathcal{A}$ implies $A \cap B \in \mathcal{A}$.

Definition 38.2.13 (d -system)

A collection $\mathcal{D} \subseteq \mathcal{P}(E)$ is a **d -system** (or Dynkin system) if

- (i) $E \in \mathcal{D}$;
- (ii) if $B_1, B_2 \in \mathcal{D}$ with $B_1 \subseteq B_2$, then $B_2 \setminus B_1 \in \mathcal{D}$;
- (iii) if $A_n \in \mathcal{D}$ with $A_n \uparrow A$, then $A \in \mathcal{D}$.

The point of the pair of definitions is that they are complementary: neither alone is a σ -algebra, but together they are.

**Proposition 38.2.14**

A collection which is both a π -system and a d -system is a σ -algebra.

Proof. Let $\mathcal{D}$ be both. Then $E \in \mathcal{D}$, and taking $B_1 = A$, $B_2 = E$ in (ii) gives $A^c \in \mathcal{D}$; in particular $\emptyset = E^c \in \mathcal{D}$. For countable unions, let $A_n \in \mathcal{D}$. Since $\mathcal{D}$ is a π -system and is closed under complements, it is closed under finite unions:

$$A \cup B = (A^c \cap B^c)^c \in \mathcal{D}.$$

So $\tilde{A}_N = A_1 \cup \dots \cup A_N \in \mathcal{D}$, and $\tilde{A}_N \uparrow \bigcup_n A_n$, whence $\bigcup_n A_n \in \mathcal{D}$ by (iii). $\square$

The engine of the whole subject is now the following lemma, which says that to check a property holds on $\sigma(\mathcal{A})$ it is enough to check that the class of sets with that property is a d -system, provided $\mathcal{A}$ is a π -system.

Lemma 38.2.15 (Dynkin's lemma)

Let $\mathcal{A}$ be a π -system. Then any d -system containing $\mathcal{A}$ contains $\sigma(\mathcal{A})$.

Proof. Let $\mathcal{D}$ be the intersection of all d -systems containing $\mathcal{A}$; one checks immediately that an intersection of d -systems is a d -system, so $\mathcal{D}$ is the smallest d -system containing $\mathcal{A}$, and it suffices to show $\sigma(\mathcal{A}) \subseteq \mathcal{D}$. By Proposition 38.2.14 it is enough to show that $\mathcal{D}$ is a π -system, for then $\mathcal{D}$ is a σ -algebra containing $\mathcal{A}$ and hence contains $\sigma(\mathcal{A})$.

The trick is to do this in two steps. First set

$$\mathcal{D}' = \{B \in \mathcal{D} : B \cap A \in \mathcal{D} \text{ for all } A \in \mathcal{A}\}.$$

We claim $\mathcal{D}'$ is a d -system. Certainly $E \in \mathcal{D}'$, since $E \cap A = A \in \mathcal{A} \subseteq \mathcal{D}$. If $B_1 \subseteq B_2$ both lie in $\mathcal{D}'$ then for $A \in \mathcal{A}$,

$$(B_2 \setminus B_1) \cap A = (B_2 \cap A) \setminus (B_1 \cap A),$$

a proper difference of elements of $\mathcal{D}$, hence in $\mathcal{D}$; so $B_2 \setminus B_1 \in \mathcal{D}'$. If $B_n \uparrow B$ with $B_n \in \mathcal{D}'$ then $B_n \cap A \uparrow B \cap A$, so $B \cap A \in \mathcal{D}$ and $B \in \mathcal{D}'$. Finally $\mathcal{A} \subseteq \mathcal{D}'$, precisely because $\mathcal{A}$ is a π -system. By minimality of $\mathcal{D}$ we get $\mathcal{D} \subseteq \mathcal{D}'$, and the reverse inclusion is by definition, so $\mathcal{D}' = \mathcal{D}$.

Now set

$$\mathcal{D}'' = \{B \in \mathcal{D} : B \cap A \in \mathcal{D} \text{ for all } A \in \mathcal{D}\}.$$

Exactly the same three verifications show $\mathcal{D}''$ is a d -system. And $\mathcal{A} \subseteq \mathcal{D}''$: for $A \in \mathcal{A}$ and $B \in \mathcal{D} = \mathcal{D}'$ we have $B \cap A \in \mathcal{D}$, which is what membership of A in $\mathcal{D}''$ requires. So again $\mathcal{D}'' = \mathcal{D}$ by minimality, and this says exactly that $\mathcal{D}$ is closed under intersections. $\square$

The first payoff is immediate.

Theorem 38.2.16 (Uniqueness of extension)

Let $(E, \mathcal{E})$ be a measurable space and let μ_1, μ_2 be measures on it. Suppose $\mathcal{A} \subseteq \mathcal{E}$ is a π -system with $\sigma(\mathcal{A}) = \mathcal{E}$, that $\mu_1 = \mu_2$ on $\mathcal{A}$, and that $E \in \mathcal{A}$ with $\mu_1(E) = \mu_2(E) < \infty$. Then $\mu_1 = \mu_2$ on $\mathcal{E}$.

Proof. Let $\mathcal{D} = \{A \in \mathcal{E} : \mu_1(A) = \mu_2(A)\}$. By hypothesis $\mathcal{A} \subseteq \mathcal{D}$, so by Dynkin's lemma it suffices to show $\mathcal{D}$ is a d -system.

We have $E \in \mathcal{D}$ by assumption. If $B_1 \subseteq B_2$ lie in $\mathcal{D}$ then, using additivity and the finiteness of $\mu_i(B_1) \leq \mu_i(E) < \infty$ to justify the subtraction,

$$\mu_1(B_2 \setminus B_1) = \mu_1(B_2) - \mu_1(B_1) = \mu_2(B_2) - \mu_2(B_1) = \mu_2(B_2 \setminus B_1),$$

so $B_2 \setminus B_1 \in \mathcal{D}$. Finally if $A_n \uparrow A$ with $A_n \in \mathcal{D}$, then by continuity from below

$$\mu_1(A) = \lim_n \mu_1(A_n) = \lim_n \mu_2(A_n) = \mu_2(A),$$

so $A \in \mathcal{D}$. $\square$

Remark. The finiteness assumption cannot be dropped outright – take μ_1 counting measure and $\mu_2 = 2\mu_1$ on $(\mathbb{R}, \mathcal{B})$, which agree (both being ∞) on the π -system of non-empty open intervals. But it can be relaxed to σ -finiteness: if $E = \bigcup_n E_n$ with $E_n \in \mathcal{A}$ increasing and $\mu_1(E_n) = \mu_2(E_n) < \infty$, apply the theorem to $A \mapsto \mu_i(A \cap E_n)$ for each n and let $n \rightarrow \infty$. We will use it in this form for Lebesgue measure.

§38.2.4 The Carathéodory extension theorem

We now come to the existence half of the story. The situation we have in mind is this: on the ring $\mathcal{A}$ of finite unions of half-open intervals $(a, b] \subseteq \mathbb{R}$, there is an obvious candidate for length, namely the sum of the lengths of the intervals. We would like to promote it to a measure on $\sigma(\mathcal{A}) = \mathcal{B}$. Carathéodory's theorem says that this is always possible, provided the candidate is countably additive on the ring to begin with.

Theorem 38.2.17 (Carathéodory extension theorem)

Let $\mathcal{A}$ be a ring on E and let μ be a countably additive set function on $\mathcal{A}$. Then μ extends to a measure on $\sigma(\mathcal{A})$.

The proof is long but the idea is simple and worth stating first. We define an *outer measure* μ^* on *all* subsets of E by covering them as efficiently as possible with sets from $\mathcal{A}$. This μ^* is defined everywhere but is only subadditive, not additive. We then cut down to the class $\mathcal{M}$ of sets which ‘split every set additively’, and show that on this class μ^* is a genuine measure and that $\mathcal{M} \supseteq \mathcal{A}$ – hence $\mathcal{M} \supseteq \sigma(\mathcal{A})$, and we are done.

Proof. For $B \subseteq E$ define the **outer measure**

$$\mu^*(B) = \inf \left\{ \sum_{n \in \mathbb{N}} \mu(A_n) : A_n \in \mathcal{A}, B \subseteq \bigcup_{n \in \mathbb{N}} A_n \right\},$$

with the convention $\mu^*(B) = \infty$ if no such countable cover exists. Note $\mu^*(\emptyset) = 0$ and μ^* is increasing. Define the **μ^* -measurable sets**

$$\mathcal{M} = \{A \subseteq E : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \cap A^c) \text{ for all } B \subseteq E\}.$$

We proceed in five steps.

Step 1: μ^* is countably subadditive on $\mathcal{P}(E)$. Let $B \subseteq \bigcup_n B_n$; we must show

$$\mu^*(B) \leq \sum_n \mu^*(B_n). \quad (\dagger)$$

We may assume every $\mu^*(B_n) < \infty$. Fix $\varepsilon > 0$. For each n , choose $A_{n,m} \in \mathcal{A}$ with $B_n \subseteq \bigcup_m A_{n,m}$ and

$$\sum_m \mu(A_{n,m}) \leq \mu^*(B_n) + \varepsilon 2^{-n}.$$

Then $\{A_{n,m}\}_{n,m}$ is a countable cover of B by elements of $\mathcal{A}$, so

$$\mu^*(B) \leq \sum_{n,m} \mu(A_{n,m}) \leq \sum_n \mu^*(B_n) + \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, $(\dagger)$ holds.

Step 2: μ^* extends μ . Let $A \in \mathcal{A}$. Covering A by $A, \emptyset, \emptyset, \dots$ gives $\mu^*(A) \leq \mu(A)$. Conversely, suppose $A \subseteq \bigcup_n A_n$ with $A_n \in \mathcal{A}$. Then $A = \bigcup_n (A \cap A_n)$ is a countable union of elements of $\mathcal{A}$; since μ is countably additive on the ring $\mathcal{A}$ it is countably subadditive there, so

$$\mu(A) \leq \sum_n \mu(A \cap A_n) \leq \sum_n \mu(A_n).$$

Taking the infimum over covers, $\mu(A) \leq \mu^*(A)$.

Step 3: $\mathcal{A} \subseteq \mathcal{M}$. Let $A \in \mathcal{A}$ and $B \subseteq E$. The inequality $\mu^*(B) \leq \mu^*(B \cap A) + \mu^*(B \cap A^c)$ is immediate from $(\dagger)$. For the reverse, we may assume $\mu^*(B) < \infty$. Fix $\varepsilon > 0$ and choose $A_n \in \mathcal{A}$ with $B \subseteq \bigcup_n A_n$ and $\sum_n \mu(A_n) \leq \mu^*(B) + \varepsilon$. Now $B \cap A \subseteq \bigcup_n (A_n \cap A)$ and $B \cap A^c \subseteq \bigcup_n (A_n \setminus A)$, and both $A_n \cap A$ and $A_n \setminus A$ lie in the ring $\mathcal{A}$. Hence

$$\begin{aligned} \mu^*(B \cap A) + \mu^*(B \cap A^c) &\leq \sum_n \mu(A_n \cap A) + \sum_n \mu(A_n \setminus A) \\ &= \sum_n [\mu(A_n \cap A) + \mu(A_n \setminus A)] = \sum_n \mu(A_n) \leq \mu^*(B) + \varepsilon, \end{aligned}$$

using additivity of μ on $\mathcal{A}$. Let $\varepsilon \rightarrow 0$.

Step 4: $\mathcal{M}$ is an algebra. That $\emptyset \in \mathcal{M}$ is clear, and the definition of $\mathcal{M}$ is symmetric in A and A^c , so $\mathcal{M}$ is closed under complements. For intersections, let

$A_1, A_2 \in \mathcal{M}$ and $B \subseteq E$. Applying the defining identity for A_1 and then for A_2 (with test set $B \cap A_1$),

$$\mu^*(B) = \mu^*(B \cap A_1 \cap A_2) + \mu^*(B \cap A_1 \cap A_2^c) + \mu^*(B \cap A_1^c).$$

Now observe $A_1 \cap A_2^c = (A_1 \cap A_2)^c \cap A_1$ and $A_1^c = (A_1 \cap A_2)^c \cap A_1^c$, so the last two terms are

$$\mu^*(B \cap (A_1 \cap A_2)^c \cap A_1) + \mu^*(B \cap (A_1 \cap A_2)^c \cap A_1^c) = \mu^*(B \cap (A_1 \cap A_2)^c),$$

using that $A_1 \in \mathcal{M}$ with test set $B \cap (A_1 \cap A_2)^c$. Hence

$$\mu^*(B) = \mu^*(B \cap (A_1 \cap A_2)) + \mu^*(B \cap (A_1 \cap A_2)^c),$$

so $A_1 \cap A_2 \in \mathcal{M}$.

Step 5: $\mathcal{M}$ is a σ -algebra and μ^* is a measure on it. Let $A = \bigcup_n A_n$ with $A_n \in \mathcal{M}$; by Step 4 and disjointification we may assume the A_n are disjoint. Let $B \subseteq E$. Applying the defining identity repeatedly and using disjointness (so that $B \cap A_1^c \cap A_2 = B \cap A_2$, and so on),

$$\begin{aligned} \mu^*(B) &= \mu^*(B \cap A_1) + \mu^*(B \cap A_1^c) \\ &= \mu^*(B \cap A_1) + \mu^*(B \cap A_2) + \mu^*(B \cap A_1^c \cap A_2^c) \\ &= \cdots = \sum_{n \leq N} \mu^*(B \cap A_n) + \mu^*\left(B \cap \bigcap_{n \leq N} A_n^c\right). \end{aligned}$$

Since $\bigcap_{n \leq N} A_n^c \supseteq A^c$ and μ^* is increasing, the last term is at least $\mu^*(B \cap A^c)$. Letting $N \rightarrow \infty$,

$$\mu^*(B) \geq \sum_{n=1}^{\infty} \mu^*(B \cap A_n) + \mu^*(B \cap A^c) \geq \mu^*(B \cap A) + \mu^*(B \cap A^c),$$

the last step by (†). The reverse inequality is (†) again, so $A \in \mathcal{M}$ and $\mathcal{M}$ is a σ -algebra. Moreover, taking $B = A$ in the display above gives

$$\mu^*(A) \geq \sum_{n=1}^{\infty} \mu^*(A_n),$$

and the reverse holds by subadditivity, so μ^* is countably additive on $\mathcal{M}$.

Combining Steps 3 and 5, $\mathcal{M}$ is a σ -algebra containing $\mathcal{A}$, so $\sigma(\mathcal{A}) \subseteq \mathcal{M}$; and by Step 2 the measure $\mu^*|_{\sigma(\mathcal{A})}$ extends μ . $\square$

Remark. The class $\mathcal{M}$ is generally strictly larger than $\sigma(\mathcal{A})$. In the case of Lebesgue measure on $\mathbb{R}$ it is the class of **Lebesgue measurable** sets, and one can show

$$\mathcal{M} = \{A \cup N : A \in \mathcal{B}, N \subseteq B \text{ for some } B \in \mathcal{B} \text{ with } \mu(B) = 0\} \supsetneq \mathcal{B}.$$

In words, $\mathcal{M}$ is obtained from $\mathcal{B}$ by throwing in all subsets of null sets; it is the **completion** of $\mathcal{B}$. There is no harm in working with $\mathcal{B}$ throughout, and it is more convenient, so we shall.

§38.2.5 Lebesgue measure

We can now build the measure that started it all. The construction on $\mathbb{R}$ is the substantial one; higher dimensions will follow painlessly from the theory of product measures in §5.

Theorem 38.2.18 (Lebesgue measure on $\mathbb{R}$)

There is a unique Borel measure μ on $\mathbb{R}$ with

$$\mu((a, b]) = b - a \quad \text{for all } a < b.$$

We call μ **Lebesgue measure**.

Proof. Let $\mathcal{A}$ consist of $\emptyset$ together with all sets of the form

$$A = (a_1, b_1] \cup \cdots \cup (a_n, b_n]$$

with the intervals disjoint and $-\infty < a_i < b_i < \infty$. This $\mathcal{A}$ is a ring: the difference and union of two such sets is again such a set. It is also a π -system, and $\sigma(\mathcal{A}) = \mathcal{B}$, since every open interval $(a, b) = \bigcup_n (a, b - 1/n]$ lies in $\sigma(\mathcal{A})$ and the open intervals generate $\mathcal{B}$.

Define $\mu(A) = \sum_{i=1}^n (b_i - a_i)$ for A as above. This is well defined: if $A = \bigsqcup_j C_j = \bigsqcup_k D_k$ are two such representations then, writing $C_j = \bigsqcup_k (C_j \cap D_k)$ and $D_k = \bigsqcup_j (C_j \cap D_k)$ and using additivity of length on intervals,

$$\sum_j \text{len}(C_j) = \sum_j \sum_k \text{len}(C_j \cap D_k) = \sum_k \sum_j \text{len}(C_j \cap D_k) = \sum_k \text{len}(D_k).$$

Additivity of μ on $\mathcal{A}$ is then clear.

To apply Carathéodory we must upgrade additivity to *countable* additivity on $\mathcal{A}$. For a finitely additive set function on a ring, countable additivity is equivalent to the statement that $\mu(A_n) \rightarrow 0$ whenever $A_n \in \mathcal{A}$ with $A_n \downarrow \emptyset$. (Indeed, if $\bigsqcup_n B_n = B \in \mathcal{A}$ then $A_N = B \setminus \bigsqcup_{n \leq N} B_n \downarrow \emptyset$ and $\mu(B) = \sum_{n \leq N} \mu(B_n) + \mu(A_N)$.) So suppose for contradiction that $B_n \in \mathcal{A}$ with $B_n \downarrow \emptyset$ but $\mu(B_n) \geq 2\varepsilon$ for all n , for some $\varepsilon > 0$.

The idea is to shrink each B_n slightly to a set whose closure is still inside B_n , and then use compactness. Write $B_n = \bigsqcup_{i=1}^{N_n} (a_{ni}, b_{ni}]$ and put

$$C_n = \bigsqcup_{i=1}^{N_n} \left(a_{ni} + \frac{2^{-n}\varepsilon}{N_n}, b_{ni} \right] \in \mathcal{A},$$

so that $\overline{C_n} \subseteq B_n$ and $\mu(B_n \setminus C_n) \leq 2^{-n}\varepsilon$. Since the B_n decrease, $B_N = \bigcap_{n \leq N} B_n$, and so

$$B_N \setminus (C_1 \cap \cdots \cap C_N) = \bigcup_{n \leq N} (B_N \setminus C_n) \subseteq \bigcup_{n \leq N} (B_n \setminus C_n),$$

giving $\mu(B_N \setminus (C_1 \cap \cdots \cap C_N)) \leq \sum_{n \leq N} 2^{-n}\varepsilon \leq \varepsilon$. As $\mu(B_N) \geq 2\varepsilon$, additivity forces

$$\mu(C_1 \cap \cdots \cap C_N) \geq \varepsilon > 0,$$

so in particular $C_1 \cap \cdots \cap C_N \neq \emptyset$ for every N . Setting $K_N = \overline{C_1} \cap \cdots \cap \overline{C_N}$, we obtain a decreasing sequence of non-empty compact subsets of $\mathbb{R}$, so by the finite intersection property there is $x \in \bigcap_N K_N$. But $K_N \subseteq \overline{C_N} \subseteq B_N$, so $x \in \bigcap_N B_N = \emptyset$, a contradiction. Hence μ is countably additive on $\mathcal{A}$, and Carathéodory gives a measure on $\sigma(\mathcal{A}) = \mathcal{B}$ extending it.

For uniqueness, suppose μ, λ are two Borel measures assigning length $b - a$ to $(a, b]$. For each $n \in \mathbb{Z}$ the measures $A \mapsto \mu(A \cap (n, n+1])$ and $A \mapsto \lambda(A \cap (n, n+1])$ are finite measures of total mass 1 agreeing on the π -system $\mathcal{A}$, so they agree on $\mathcal{B}$ by Theorem 38.2.16. Summing over $n \in \mathbb{Z}$ and using countable additivity, $\mu = \lambda$. $\square$

Definition 38.2.19 (Null set)

A set $N \in \mathcal{E}$ with $\mu(N) = 0$ is called **μ -null**. A property is said to hold **μ -almost everywhere** (abbreviated a.e.) if the set where it fails is contained in a null set.

Example 38.2.20 (Countable sets are Lebesgue null)

For $x \in \mathbb{R}$ we have $\{x\} = \bigcap_n (x - 1/n, x]$, a decreasing intersection of sets of finite measure, so by continuity from above $\mu(\{x\}) = \lim_n 1/n = 0$. Hence any countable set is null, being a countable union of null sets. In particular $\mu(\mathbb{Q}) = 0$, and

$$\mu((a, b)) = \mu([a, b]) = \mu((a, b]) = b - a.$$

Not every null set is countable: the Cantor middle-thirds set is uncountable and has measure $\lim_n (2/3)^n = 0$.

Proposition 38.2.21 (Translation invariance)

For $x \in \mathbb{R}$ and $B \in \mathcal{B}$ we have $B + x \in \mathcal{B}$ and $\mu(B + x) = \mu(B)$.

Proof. The map $y \mapsto y - x$ is a homeomorphism of $\mathbb{R}$, so $B \mapsto B + x$ preserves $\mathcal{B}$. Define $\mu_x(B) = \mu(B + x)$. This is a Borel measure and $\mu_x((a, b]) = \mu((a + x, b + x]) = b - a$. By the uniqueness in Theorem 38.2.18, $\mu_x = \mu$. $\square$

Since Lebesgue measure is the unique translation-invariant Borel measure normalised by $\mu((0, 1]) = 1$, translation invariance is really the defining property, and it is what we will exploit in the next subsection to show that not every set can be measured.

§38.2.6 Measures from distribution functions

Before that, it is worth recording that essentially the same construction gives every reasonable measure on $\mathbb{R}$ at once. Recall that a Borel measure μ on a Hausdorff space E is called a **Radon measure** if $\mu(K)$ is finite for every compact set $K \subseteq E$. Note that compact subsets of a Hausdorff space are closed, and hence Borel.

Lemma 38.2.22 (Generalised inverse)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be non-decreasing and right-continuous, and put

$$g(-\infty) = \lim_{y \rightarrow -\infty} g(y), \quad g(+\infty) = \lim_{y \rightarrow \infty} g(y).$$

On the open interval $I = (g(-\infty), g(+\infty))$ define

$$f(x) = \inf\{y \in \mathbb{R} : x \leq g(y)\}.$$

Then f is non-decreasing and left-continuous, and for $x \in I$, $y \in \mathbb{R}$,

$$f(x) \leq y \iff x \leq g(y).$$

Proof. Let $J_x = \{y \in \mathbb{R} : x \leq g(y)\}$. Since $x > g(-\infty)$ the set J_x is bounded below, and since $x < g(+\infty)$ it is non-empty; so $f(x) \in \mathbb{R}$. If $y \in J_x$ and $y' \geq y$ then $y' \in J_x$ as g is non-decreasing, so J_x is an up-set. If $y_n \downarrow y$ with $y_n \in J_x$, then $x \leq g(y_n) \rightarrow g(y)$ by right-continuity, so $y \in J_x$. Hence J_x is closed and $J_x = [f(x), \infty)$, which is exactly the stated equivalence.

If $x \leq x'$ then $J_x \supseteq J_{x'}$, so $f(x) \leq f(x')$. If $x_n \uparrow x$ then $J_x = \bigcap_n J_{x_n}$, so $f(x_n) \uparrow f(x)$, giving left-continuity. $\square$

Theorem 38.2.23 (Lebesgue–Stieltjes measures)

Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be non-decreasing and right-continuous. Then there is a unique Radon measure μ_g on $\mathbb{R}$ with

$$\mu_g((a, b]) = g(b) - g(a) \quad (a < b),$$

and every Radon measure on $\mathbb{R}$ arises this way.

Proof. The construction of Theorem 38.2.18 goes through verbatim with $b - a$ replaced by $g(b) - g(a)$, right-continuity of g being exactly what is needed to make the shrinking argument work. Alternatively – and this is the slicker route – let f be the generalised inverse of Lemma 38.2.22. For $z \in \mathbb{R}$,

$$f^{-1}((-\infty, z]) = \{x \in I : f(x) \leq z\} = \{x \in I : x \leq g(z)\} = I \cap (-\infty, g(z)],$$

which is Borel; since the sets $(-\infty, z]$ generate $\mathcal{B}$, the map f is measurable (see §3.1). Let μ be Lebesgue measure restricted to I and set $\mu_g = \mu \circ f^{-1}$. This is a measure, and

$$\mu_g((a, b]) = \mu(\{x : a < f(x) \leq b\}) = \mu(\{x : g(a) < x \leq g(b)\}) = g(b) - g(a).$$

Uniqueness follows as before. Any compact set lies in some $[a, b]$, whose measure $g(b) - g(a)$ is finite, so μ_g is Radon.

Conversely, given a Radon measure ν on $\mathbb{R}$, define $g(y) = \nu((0, y])$ for $y \geq 0$ and $g(y) = -\nu((y, 0])$ for $y < 0$. Then g is non-decreasing, right-continuous by continuity from above, and $\nu((a, b]) = g(b) - g(a)$ by additivity; uniqueness gives $\nu = \mu_g$. $\square$

We call g the **Stieltjes distribution function** of μ_g . Taking $g(y) = y$ recovers Lebesgue measure; taking $g = \mathbb{1}_{[x, \infty)}$ recovers the Dirac measure δ_x . When μ_g is a probability

measure this g is exactly the distribution function of a random variable, as we will see in §3.

§38.2.7 A non-measurable set

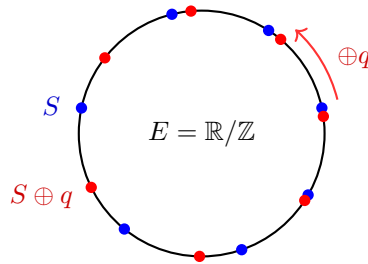
We now show that our caution was justified: one cannot extend Lebesgue measure to all of $\mathcal{P}(\mathbb{R})$. The construction is due to Vitali and uses the axiom of choice in an essential way.

Work on $E = (0, 1]$ with addition modulo 1; concretely, identify E with the circle $\mathbb{R}/\mathbb{Z}$. Lebesgue measure restricted to E is invariant under this addition: for $B \subseteq E$ Borel and $x \in E$, the set $B \oplus x$ is a translate of $B \cap (0, 1 - x]$ union a translate of $B \cap (1 - x, 1]$, and both pieces keep their measure.

Let $Q = E \cap \mathbb{Q}$, a countable subgroup of $(E, \oplus)$. Define $x \sim y$ iff $x - y \in Q$; this is an equivalence relation, whose classes are the cosets $x \oplus Q$. By the axiom of choice, pick one representative from each class and let S be the resulting set. Then

$$E = \bigsqcup_{q \in Q} (S \oplus q),$$

a *disjoint* union: every $x \in E$ lies in the class of exactly one representative, and if $S \oplus q$ met $S \oplus q'$ with $q \neq q'$ then S would contain two representatives of the same class.



Theorem 38.2.24 (Vitali)

The set S is not Lebesgue measurable. Consequently there is no translation-invariant measure defined on all of $\mathcal{P}((0, 1])$ giving the whole space measure 1.

Proof. Suppose S were measurable with $\mu(S) = c$. Each translate $S \oplus q$ is then measurable with the same measure c , and by countable additivity

$$1 = \mu(E) = \sum_{q \in Q} \mu(S \oplus q) = \sum_{q \in Q} c.$$

If $c = 0$ the sum is 0; if $c > 0$ the sum is ∞ (as Q is infinite). Either way we cannot get 1, a contradiction. $\square$

Remark. One can go further. Assuming the continuum hypothesis, a theorem of Banach and Kuratowski says that there is *no* measure at all on $\mathcal{P}((0, 1])$ with $\mu((0, 1]) = 1$ and $\mu(\{x\}) = 0$ for every x – translation invariance is not even needed. So the restriction to a σ -algebra is not an artefact of our construction; it is unavoidable.

§38.2.8 Probability spaces and independence

Everything we have done so far specialises to probability by the simple device of normalising the total mass.

Definition 38.2.25 (Probability space)

A measure space $(E, \mathcal{E}, \mu)$ with $\mu(E) = 1$ is a **probability space**. We change notation to $(\Omega, \mathcal{F}, \mathbb{P})$ and call Ω the **sample space**, elements of $\mathcal{F}$ **events**, and $\mathbb{P}$ the **probability measure**.

Kolmogorov's axioms of probability, as laid down in 1933, say precisely that $\mathbb{P}(\Omega) = 1$, that $0 \leq \mathbb{P}(A) \leq 1$, and that $\mathbb{P}$ is countably additive. So a probability space is nothing more than a measure space of total mass one – there is no new content, only new vocabulary. The reader who recalls the continuity properties of $\mathbb{P}$ from IA Probability will recognise Proposition 38.2.10 as the general form of those results.

Definition 38.2.26 (Independence)

Events $(A_i)_{i \in I}$ in $\mathcal{F}$ are **independent** if for every finite $J \subseteq I$,

$$\mathbb{P}\left(\bigcap_{j \in J} A_j\right) = \prod_{j \in J} \mathbb{P}(A_j).$$

Sub- σ -algebras $(\mathcal{A}_i)_{i \in I}$ of $\mathcal{F}$ are **independent** if, whenever $J \subseteq I$ is finite and $A_j \in \mathcal{A}_j$ for $j \in J$, the events $(A_j)_{j \in J}$ are independent.

Checking independence of σ -algebras directly is hopeless, but Dynkin's lemma reduces it to a check on generators.

Proposition 38.2.27

Let $\mathcal{A}_1, \mathcal{A}_2$ be π -systems contained in $\mathcal{F}$ with $\Omega \in \mathcal{A}_1 \cap \mathcal{A}_2$. Suppose

$$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(A_1)\mathbb{P}(A_2) \quad \text{for all } A_1 \in \mathcal{A}_1, A_2 \in \mathcal{A}_2.$$

Then $\sigma(\mathcal{A}_1)$ and $\sigma(\mathcal{A}_2)$ are independent.

Proof. Fix $A_2 \in \mathcal{A}_2$ and consider the two finite measures $B \mapsto \mathbb{P}(B \cap A_2)$ and $B \mapsto \mathbb{P}(B)\mathbb{P}(A_2)$ on $\sigma(\mathcal{A}_1)$. They agree on the π -system $\mathcal{A}_1$ and have the same total mass $\mathbb{P}(A_2)$, so by Theorem 38.2.16 they agree on $\sigma(\mathcal{A}_1)$. Now fix $B \in \sigma(\mathcal{A}_1)$ and repeat the argument in the second coordinate: the measures $C \mapsto \mathbb{P}(B \cap C)$ and $C \mapsto \mathbb{P}(B)\mathbb{P}(C)$ agree on the π -system $\mathcal{A}_2$, hence on $\sigma(\mathcal{A}_2)$. $\square$

§38.2.9 The Borel–Cantelli lemmas

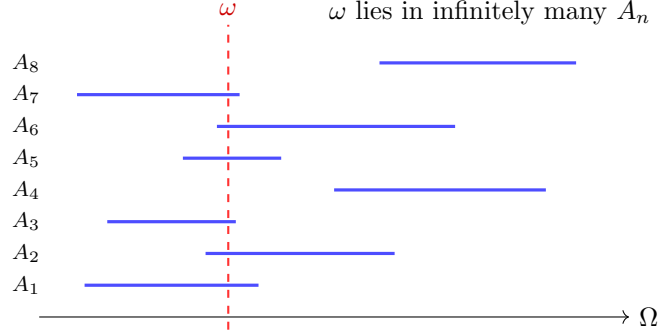
Given a sequence of events (A_n) , two sets naturally record its asymptotic behaviour.

Definition 38.2.28 (Limsup and liminf of events)

For $A_n \in \mathcal{F}$ we define

$$\limsup_n A_n = \bigcap_n \bigcup_{m \geq n} A_m = \{A_n \text{ i.o.}\}, \quad \liminf_n A_n = \bigcup_n \bigcap_{m \geq n} A_m = \{A_n \text{ ev.}\}.$$

Unwinding the quantifiers: $\omega \in \limsup_n A_n$ iff for every n there is some $m \geq n$ with $\omega \in A_m$, that is, iff ω lies in infinitely many of the A_n . Similarly $\omega \in \liminf_n A_n$ iff ω lies in all but finitely many. Both sets are in $\mathcal{F}$.



The picture to keep in mind is the one above: draw Ω horizontally, draw each A_n at height n , and a point ω belongs to $\limsup_n A_n$ exactly when the vertical line through ω meets infinitely many of the segments. The two Borel–Cantelli lemmas give complementary criteria for whether this happens.

Lemma 38.2.29 (First Borel–Cantelli lemma)

Let (A_n) be events with $\sum_n \mathbb{P}(A_n) < \infty$. Then $\mathbb{P}(A_n \text{ i.o.}) = 0$.

Proof. For every n ,

$$\mathbb{P}\left(\bigcap_k \bigcup_{m \geq k} A_m\right) \leq \mathbb{P}\left(\bigcup_{m \geq n} A_m\right) \leq \sum_{m \geq n} \mathbb{P}(A_m),$$

and the right-hand side is the tail of a convergent series, so tends to 0 as $n \rightarrow \infty$. $\square$

Note that this proof never used $\mathbb{P}(\Omega) = 1$; the lemma holds for an arbitrary measure μ in place of $\mathbb{P}$, and we will use it in that generality.

Lemma 38.2.30 (Second Borel–Cantelli lemma)

Let (A_n) be *independent* events with $\sum_n \mathbb{P}(A_n) = \infty$. Then $\mathbb{P}(A_n \text{ i.o.}) = 1$.

Proof. Fix n . By independence and the elementary inequality $1 - a \leq e^{-a}$,

$$\mathbb{P}\left(\bigcap_{m=n}^N A_m^c\right) = \prod_{m=n}^N (1 - \mathbb{P}(A_m)) \leq \exp\left(-\sum_{m=n}^N \mathbb{P}(A_m)\right) \rightarrow 0$$

as $N \rightarrow \infty$, since the series diverges. The sets $\bigcap_{m=n}^N A_m^c$ decrease to $\bigcap_{m \geq n} A_m^c$ as

$N \rightarrow \infty$, so $\mathbb{P}(\bigcap_{m \geq n} A_m^c) = 0$ for every n . Hence

$$\mathbb{P}((A_n \text{ i.o.})^c) = \mathbb{P}\left(\bigcup_n \bigcap_{m \geq n} A_m^c\right) \leq \sum_n \mathbb{P}\left(\bigcap_{m \geq n} A_m^c\right) = 0. \quad \square$$

The two lemmas together are a remarkably sharp tool: for independent events, $\mathbb{P}(A_n \text{ i.o.})$ is 0 or 1 according to whether $\sum_n \mathbb{P}(A_n)$ converges or diverges, with no intermediate case. Here is a typical application, which we can state now and will be able to re-derive later.

Example 38.2.31 (Records of exponential variables)

Let (X_n) be independent with $\mathbb{P}(X_1 > x) = e^{-x}$ for $x \geq 0$, and let $\alpha > 0$. Put $A_n = \{X_n \geq \alpha \log n\}$, so $\mathbb{P}(A_n) = n^{-\alpha}$. Then $\sum_n \mathbb{P}(A_n) < \infty$ iff $\alpha > 1$. By the two lemmas, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\frac{X_n}{\log n} \geq 1 \text{ i.o.}\right) = 1, \quad \mathbb{P}\left(\frac{X_n}{\log n} \geq 1 + \varepsilon \text{ i.o.}\right) = 0.$$

Combining, $\limsup_n X_n / \log n = 1$ almost surely. So the record values of an i.i.d. exponential sequence grow exactly logarithmically.

§38.3 Measurable Functions and Random Variables

§38.3.1 Measurable functions

Having decided which sets we can measure, we must decide which functions respect that structure. As always in mathematics, the right notion of morphism is the one that pulls structure back.

Definition 38.3.1 (Measurable function)

Let $(E, \mathcal{E})$ and $(G, \mathcal{G})$ be measurable spaces. A function $f : E \rightarrow G$ is **$\mathcal{E}$ - $\mathcal{G}$ measurable** (or just **measurable**) if

$$f^{-1}(A) \in \mathcal{E} \quad \text{for every } A \in \mathcal{G}.$$

When $G = \mathbb{R}$ with $\mathcal{G} = \mathcal{B}$, we simply say f is measurable; when in addition E is a topological space and $\mathcal{E} = \mathcal{B}(E)$, we say f is **Borel measurable**. Note the analogy with continuity, which is the statement that preimages of open sets are open.

The saving grace is that measurability can be checked on generators, because preimages commute with all the set operations: $f^{-1}(A^c) = f^{-1}(A)^c$ and $f^{-1}(\bigcup_n A_n) = \bigcup_n f^{-1}(A_n)$.

Lemma 38.3.2 (Measurability on generators)

Let $\mathcal{G} = \sigma(\mathcal{A})$. If $f^{-1}(A) \in \mathcal{E}$ for every $A \in \mathcal{A}$, then f is measurable.

Proof. The collection $\{A \subseteq G : f^{-1}(A) \in \mathcal{E}\}$ is a σ -algebra on G , by the displayed identities. It contains $\mathcal{A}$ by hypothesis, hence contains $\sigma(\mathcal{A}) = \mathcal{G}$. $\square$

Example 38.3.3 (Useful special cases) (i) A function $f : (E, \mathcal{E}) \rightarrow \mathbb{R}$ is measurable if and only if $\{x : f(x) \leq y\} \in \mathcal{E}$ for every $y \in \mathbb{R}$, since the sets $(-\infty, y]$ generate $\mathcal{B}$.

(ii) If E is a topological space with $\mathcal{E} = \mathcal{B}(E)$ and $f : E \rightarrow \mathbb{R}$ is continuous, then f is measurable: preimages of open sets are open, hence Borel, and open sets generate $\mathcal{B}$.

(iii) The indicator $\mathbb{1}_A$ is measurable if and only if $A \in \mathcal{E}$, since $\mathbb{1}_A^{-1}(\{1\}) = A$.

Proposition 38.3.4 (Stability of measurability)

Let $f, g, f_n : (E, \mathcal{E}) \rightarrow \mathbb{R}$ be measurable and $\lambda \in \mathbb{R}$. Then the following are measurable:

$$f+g, \quad fg, \quad \lambda f, \quad \max(f, g), \quad \min(f, g), \quad \inf_n f_n, \quad \sup_n f_n, \quad \liminf_n f_n, \quad \limsup_n f_n,$$

the last four taking values in $[-\infty, \infty]$. Moreover the composition of measurable functions is measurable.

Proof. Composition is immediate: $(g \circ f)^{-1}(A) = f^{-1}(g^{-1}(A))$.

For the supremum, $\{\sup_n f_n \leq y\} = \bigcap_n \{f_n \leq y\} \in \mathcal{E}$; for the infimum, $\{\inf_n f_n < y\} = \bigcup_n \{f_n < y\} \in \mathcal{E}$. Then $\liminf_n f_n = \sup_n \inf_{m \geq n} f_m$ and $\limsup_n f_n = \inf_n \sup_{m \geq n} f_m$ are measurable, and $\max, \min$ of two functions are special cases.

For sums, note $\{f + g > y\} = \bigcup_{q \in \mathbb{Q}} (\{f > q\} \cap \{g > y - q\}) \in \mathcal{E}$. For products, $fg = \frac{1}{4}[(f + g)^2 - (f - g)^2]$, so it suffices to know h^2 is measurable when h is, and this follows from $\{h^2 \leq y\} = \{-\sqrt{y} \leq h \leq \sqrt{y}\}$ for $y \geq 0$. $\square$

That last proposition is worth pausing over. It says that measurability – unlike continuity, or differentiability, or Riemann integrability – is preserved under *pointwise limits*. This is exactly the robustness we were missing, and it is the reason the Lebesgue theory has the convergence theorems it does.

We single out the functions which will serve as building blocks.

Definition 38.3.5 (Simple function)

A function $f : (E, \mathcal{E}) \rightarrow \mathbb{R}$ is **simple** if it can be written as

$$f = \sum_{k=1}^m a_k \mathbb{1}_{A_k}, \quad a_k \geq 0, \quad A_k \in \mathcal{E}, \quad m \in \mathbb{N}.$$

Equivalently, f is measurable, non-negative, and takes only finitely many values.

The following approximation is the single most used construction in the course; we will refer to it as *the standard approximation*.

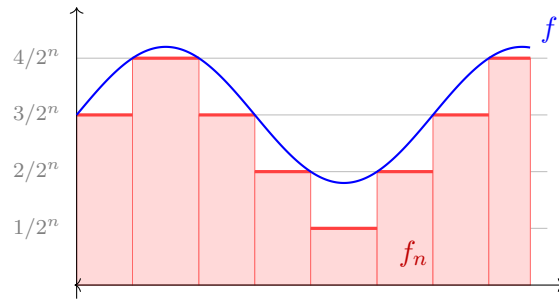
Proposition 38.3.6 (Approximation by simple functions)

Let $f : (E, \mathcal{E}) \rightarrow [0, \infty]$ be measurable. Define

$$f_n = 2^{-n} \lfloor 2^n f \rfloor \wedge n = \sum_{j=0}^{n2^n-1} \frac{j}{2^n} \mathbb{1}_{A_{n,j}} + n \mathbb{1}_{\{f \geq n\}}, \quad A_{n,j} = f^{-1} \left(\left[\frac{j}{2^n}, \frac{j+1}{2^n} \right) \right).$$

Then each f_n is simple, $0 \leq f_n \leq f_{n+1} \leq f$, and $f_n \rightarrow f$ pointwise. If f is bounded the convergence is uniform, with $\|f - f_n\|_\infty \leq 2^{-n}$ once $n > \|f\|_\infty$.

Proof. Each $A_{n,j}$ lies in $\mathcal{E}$ by measurability of f , so f_n is simple. Monotonicity in n holds because refining the mesh from 2^{-n} to $2^{-(n+1)}$ can only increase the value of $\lfloor \cdot \rfloor$ -rounding downwards, and because the truncation level n increases. If $f(x) < \infty$ then for $n > f(x)$ we have $0 \leq f(x) - f_n(x) \leq 2^{-n}$, and if $f(x) = \infty$ then $f_n(x) = n \rightarrow \infty$. $\square$



The picture shows the staircase f_n sitting underneath f . Everything we prove about the integral will be proved first for simple functions, where it is a finite computation, and then transported to general non-negative f along this staircase using the monotone convergence theorem.

§38.3.2 The monotone class theorem

Dynkin's lemma has a functional analogue which will let us prove statements about all bounded measurable functions by checking them only on indicators of a π -system.

Theorem 38.3.7 (Monotone class theorem)

Let $\mathcal{A}$ be a π -system generating $\mathcal{E}$, with $E \in \mathcal{A}$. Let $\mathcal{V}$ be a vector space of bounded functions $E \rightarrow \mathbb{R}$ such that

- (i) $\mathbb{1}_E \in \mathcal{V}$ and $\mathbb{1}_A \in \mathcal{V}$ for all $A \in \mathcal{A}$;
- (ii) if $f_n \in \mathcal{V}$ are non-negative with $f_n \uparrow f$ pointwise and f is bounded, then $f \in \mathcal{V}$.

Then $\mathcal{V}$ contains every bounded measurable function $f : (E, \mathcal{E}) \rightarrow \mathbb{R}$.

Proof. Let $\mathcal{D} = \{A \in \mathcal{E} : \mathbb{1}_A \in \mathcal{V}\}$. Then $E \in \mathcal{D}$ and $\mathcal{A} \subseteq \mathcal{D}$. If $A \subseteq B$ with $A, B \in \mathcal{D}$ then $\mathbb{1}_{B \setminus A} = \mathbb{1}_B - \mathbb{1}_A \in \mathcal{V}$ since $\mathcal{V}$ is a vector space. If $A_n \uparrow A$ with $A_n \in \mathcal{D}$ then $\mathbb{1}_{A_n} \uparrow \mathbb{1}_A$, which is bounded, so $\mathbb{1}_A \in \mathcal{V}$ by (ii). Hence $\mathcal{D}$ is a d -system containing the π -system $\mathcal{A}$, and Dynkin's lemma gives $\mathcal{D} = \mathcal{E}$.

Now let f be bounded, measurable and non-negative, and let f_n be its standard approximation from Proposition 38.3.6. Each f_n is a finite non-negative linear

combination of indicators of sets in $\mathcal{E} = \mathcal{D}$, hence lies in $\mathcal{V}$; and $f_n \uparrow f$ with f bounded, so $f \in \mathcal{V}$ by (ii). For general bounded measurable f , write $f = f^+ - f^-$ with $f^\pm = \max(\pm f, 0)$ bounded, measurable and non-negative; both lie in $\mathcal{V}$, hence so does f . $\square$

§38.3.3 Image measures and random variables

A measurable function transports a measure forwards.

Definition 38.3.8 (Image measure)

Let $f : (E, \mathcal{E}) \rightarrow (G, \mathcal{G})$ be measurable and μ a measure on $(E, \mathcal{E})$. The **image measure** (or pushforward) is

$$(\mu \circ f^{-1})(A) = \mu(f^{-1}(A)), \quad A \in \mathcal{G}.$$

That this is a measure is immediate: preimages preserve disjointness and commute with countable unions.

Definition 38.3.9 (Random variable)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. An **E -valued random variable** is a measurable map $X : (\Omega, \mathcal{F}) \rightarrow (E, \mathcal{E})$. When $E = \mathbb{R}$ (or $\mathbb{R}^d$) with the Borel σ -algebra we simply say **random variable** (or random vector).

The **law** or **distribution** of X is the image measure $\mu_X = \mathbb{P} \circ X^{-1}$, a probability measure on $(E, \mathcal{E})$.

We also write $\sigma(X) = \{X^{-1}(A) : A \in \mathcal{E}\}$, the σ -algebra *generated by* X ; it is the smallest σ -algebra on Ω making X measurable, and morally it encodes exactly the information one learns by observing X .

Definition 38.3.10 (Distribution function)

For a real-valued random variable X , the **distribution function** is

$$F_X(z) = \mu_X((-\infty, z]) = \mathbb{P}(X \leq z).$$

Since the intervals $(-\infty, z]$ form a π -system generating $\mathcal{B}$, and μ_X is a probability measure, Theorem 38.2.16 tells us that F_X determines μ_X completely. From the properties of measures we read off that F_X is non-decreasing, right-continuous (by continuity from above), with $F_X(z) \rightarrow 0$ as $z \rightarrow -\infty$ and $F_X(z) \rightarrow 1$ as $z \rightarrow \infty$. Conversely, given any such F , Lemma 38.2.22 produces a random variable with that distribution function: take $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \mu)$ and $X(\omega) = \inf\{x : \omega \leq F(x)\}$, so that $\mathbb{P}(X \leq z) = \mu(\{\omega : \omega \leq F(z)\}) = F(z)$. This is the *inverse transform* construction familiar from simulation.

Definition 38.3.11 (Independent random variables)

A family $(X_i)_{i \in I}$ of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ is **independent** if the σ -algebras $\sigma(X_i)$ are independent.

For real-valued X_i the sets $\{X_i \leq x\}$ form a π -system generating $\sigma(X_i)$, so by Proposition 38.2.27 (extended to finitely many systems in the obvious way) independence is equivalent to

$$\mathbb{P}(X_1 \leq x_1, \dots, X_n \leq x_n) = \prod_{i=1}^n \mathbb{P}(X_i \leq x_i)$$

for all finite subfamilies and all $x_i \in \mathbb{R}$. This is the criterion one actually uses.

§38.3.4 Constructing independent random variables

It is not obvious that an infinite sequence of independent random variables with prescribed laws exists at all. Happily, one probability space suffices for everything: the unit interval with Lebesgue measure.

Take $(\Omega, \mathcal{F}, \mathbb{P}) = ((0, 1), \mathcal{B}((0, 1)), \mu)$. Every $\omega \in (0, 1)$ has a binary expansion $\omega = \sum_{n \geq 1} \omega_n 2^{-n}$, unique if we forbid expansions ending in all 1s. Define the **Rademacher functions**

$$R_n(\omega) = \omega_n \in \{0, 1\}.$$

Each R_n is a finite sum of indicators of dyadic intervals, hence measurable, and $\mathbb{P}(R_n = 1) = \mathbb{P}(R_n = 0) = 1/2$. For any $x_1, \dots, x_n \in \{0, 1\}$, the event $\{R_1 = x_1, \dots, R_n = x_n\}$ is a single dyadic interval of length 2^{-n} , so

$$\mathbb{P}(R_1 = x_1, \dots, R_n = x_n) = 2^{-n} = \prod_{i=1}^n \mathbb{P}(R_i = x_i),$$

and the R_n are independent Bernoulli(1/2) variables.

Now take a bijection $m : \mathbb{N}^2 \rightarrow \mathbb{N}$ and set $Y_{n,k} = R_{m(n,k)}$; these are again independent. Define

$$Y_n = \sum_{k=1}^{\infty} 2^{-k} Y_{n,k},$$

which converges everywhere. The Y_n are independent, being functions of disjoint subfamilies of the R_m . Moreover each Y_n is uniform on $(0, 1)$: for the dyadic intervals $(i2^{-m}, (i+1)2^{-m}]$, which form a π -system generating $\mathcal{B}((0, 1))$,

$$\mathbb{P}(Y_n \in (i2^{-m}, (i+1)2^{-m}]) = 2^{-m},$$

since this event prescribes the first m binary digits of Y_n . By uniqueness of extension, $Y_n \sim U(0, 1)$.

Finally, given any sequence (F_n) of distribution functions, let g_n be the generalised inverse of F_n from Lemma 38.2.22. Then $X_n = g_n(Y_n)$ is a sequence of independent random variables with $\mathbb{P}(X_n \leq x) = \mathbb{P}(Y_n \leq F_n(x)) = F_n(x)$. So:

Theorem 38.3.12

Given any sequence (F_n) of distribution functions there is a probability space carrying a sequence of independent random variables (X_n) with X_n having distribution function F_n . In particular i.i.d. sequences of any prescribed law exist.

§38.3.5 Convergence of measurable functions

Two notions of convergence make sense at this level of generality.

Definition 38.3.13 (Almost everywhere and in measure)

Let f_n, f be measurable functions on $(E, \mathcal{E}, \mu)$. We say $f_n \rightarrow f$ **almost everywhere** if $\mu(\{x : f_n(x) \not\rightarrow f(x)\}) = 0$, and $f_n \rightarrow f$ **in measure** if for every $\varepsilon > 0$,

$$\mu(\{x : |f_n(x) - f(x)| > \varepsilon\}) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

On a probability space these are called convergence **almost surely** and **in probability**, and we write $X_n \rightarrow X$ a.s. and $X_n \xrightarrow{\mathbb{P}} X$.

Theorem 38.3.14 (Relations between the two)

Let f_n be measurable on $(E, \mathcal{E}, \mu)$.

- (i) If $\mu(E) < \infty$ and $f_n \rightarrow 0$ a.e., then $f_n \rightarrow 0$ in measure.
- (ii) If $f_n \rightarrow 0$ in measure, then $f_{n_k} \rightarrow 0$ a.e. along some subsequence (n_k) .

Proof. (i) Fix $\varepsilon > 0$. The sets $B_n = \bigcap_{m \geq n} \{|f_m| \leq \varepsilon\}$ increase, with

$$\bigcup_n B_n = \{|f_n| \leq \varepsilon \text{ eventually}\} \supseteq \{f_n \rightarrow 0\},$$

a set of full measure. By continuity from below, $\mu(B_n) \rightarrow \mu(E)$. Since $\mu(\{|f_n| \leq \varepsilon\}) \geq \mu(B_n)$ and $\mu(E) < \infty$, we may subtract to get $\mu(\{|f_n| > \varepsilon\}) \leq \mu(E) - \mu(B_n) \rightarrow 0$.

(ii) For each k , convergence in measure with $\varepsilon = 1/k$ lets us choose n_k (increasing) with

$$\mu(|f_{n_k}| > 1/k) \leq 2^{-k}.$$

Then $\sum_k \mu(|f_{n_k}| > 1/k) < \infty$, so by the first Borel–Cantelli lemma (valid for any measure) the set where $|f_{n_k}| > 1/k$ infinitely often is null. Off that set, $|f_{n_k}| \leq 1/k$ eventually, so $f_{n_k} \rightarrow 0$. $\square$

Both hypotheses are needed, as the following show.

Example 38.3.15 (Two counterexamples) (i) On $(\mathbb{R}, \mathcal{B}, \mu)$ let $f_n = \mathbb{1}_{(n, \infty)}$. Then $f_n \rightarrow 0$ pointwise, but $\mu(|f_n| > 1/2) = \infty$ for every n . So (i) genuinely needs $\mu(E) < \infty$.

(ii) Let (A_n) be independent events with $\mathbb{P}(A_n) = 1/n$. Then $\mathbb{1}_{A_n} \rightarrow 0$ in probability, since $\mathbb{P}(\mathbb{1}_{A_n} > \varepsilon) = 1/n \rightarrow 0$. But $\sum_n \mathbb{P}(A_n) = \infty$, so by the second Borel–Cantelli lemma $\mathbb{P}(A_n \text{ i.o.}) = 1$ and $\mathbb{1}_{A_n} \not\rightarrow 0$ almost surely. So (ii) genuinely needs a subsequence.

§38.3.6 Tail events and Kolmogorov's zero-one law

Let $(X_n)_{n \in \mathbb{N}}$ be a sequence of random variables. Define

$$\mathcal{T}_n = \sigma(X_{n+1}, X_{n+2}, \dots), \quad \mathcal{T} = \bigcap_{n \in \mathbb{N}} \mathcal{T}_n.$$

We call $\mathcal{T}$ the **tail σ -algebra**. An event in $\mathcal{T}$ is one whose occurrence is unaffected by changing any finite number of the X_n – for instance $\{X_n \rightarrow 0\}$, or $\{\sum_n X_n \text{ converges}\}$, or $\{\limsup_n \frac{1}{n} \sum_{i \leq n} X_i > c\}$. Kolmogorov's theorem says that for independent X_n such events are deterministic.

Theorem 38.3.16 (Kolmogorov's zero-one law)

Let (X_n) be independent random variables and let $A \in \mathcal{T}$. Then $\mathbb{P}(A) \in \{0, 1\}$. Moreover any $\mathcal{T}$ -measurable random variable Y is almost surely constant.

Proof. Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, which is generated by the π -system of events of the form $\{X_1 \leq x_1, \dots, X_n \leq x_n\}$; and note that $\mathcal{T}_n$ is generated by the π -system of events of the form $\{X_{n+1} \leq x_{n+1}, \dots, X_{n+k} \leq x_{n+k}\}$, for $k \in \mathbb{N}$. By independence of the X_i these two π -systems satisfy the product rule, so by Proposition 38.2.27 the σ -algebras $\mathcal{F}_n$ and $\mathcal{T}_n$ are independent.

Let $\mathcal{F}_\infty = \sigma(X_1, X_2, \dots)$. Then $\bigcup_n \mathcal{F}_n$ is a π -system (the $\mathcal{F}_n$ increase) generating $\mathcal{F}_\infty$. Any $A \in \bigcup_n \mathcal{F}_n$ lies in some $\mathcal{F}_n$, hence is independent of $\mathcal{T}_n \supseteq \mathcal{T}$. So $\bigcup_n \mathcal{F}_n$ and $\mathcal{T}$ satisfy the product rule, and by Proposition 38.2.27 again, $\mathcal{F}_\infty$ is independent of $\mathcal{T}$.

But $\mathcal{T} \subseteq \mathcal{F}_\infty$. So if $A \in \mathcal{T}$ then A is independent of itself:

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)^2,$$

forcing $\mathbb{P}(A) \in \{0, 1\}$.

For the last statement, $F_Y(y) = \mathbb{P}(Y \leq y)$ takes only the values 0 and 1, is non-decreasing and right-continuous with limits 0 and 1 at $\mp\infty$; so $c = \inf\{y : F_Y(y) = 1\}$ is finite and $Y = c$ almost surely. $\square$

Example 38.3.17

If (X_n) are independent then the event that $\sum_n X_n$ converges has probability 0 or 1, and $\limsup_n \frac{1}{n} \sum_{i=1}^n X_i$ is an almost surely constant element of $[-\infty, \infty]$. The zero-one law does not tell us *which* constant – that is the work of the laws of large numbers – but it does tell us that there is no randomness left to speak of.

§38.4 Integration

§38.4.1 Defining the integral

We now carry out the plan of §1. Throughout $(E, \mathcal{E}, \mu)$ is a measure space. We will write the integral of f interchangeably as

$$\mu(f) = \int_E f \, d\mu = \int_E f(x) \, d\mu(x),$$

and when $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X a random variable, we write $\mathbb{E}[X] = \int_{\Omega} X \, d\mathbb{P}$ for its **expectation**. The notation $\mu(f)$ is compact and emphasises that integration is a functional; we will use it freely.

The construction proceeds in three stages: simple functions, non-negative functions, then general functions.

Definition 38.4.1 (Integral of a simple function)

For a simple function $f = \sum_{k=1}^m a_k \mathbb{1}_{A_k}$ with $a_k \geq 0$ and $A_k \in \mathcal{E}$, set

$$\mu(f) = \sum_{k=1}^m a_k \mu(A_k) \in [0, \infty].$$

This requires a check: the representation of f is not unique. Suppose $\sum_k a_k \mathbb{1}_{A_k} = \sum_{\ell} b_{\ell} \mathbb{1}_{B_{\ell}}$. Refining both into the common refinement generated by the A_k and B_{ℓ} (finitely many sets, so this is a finite disjoint partition) and using finite additivity of μ on each piece shows both expressions equal the same number. We omit the routine bookkeeping.

Proposition 38.4.2 (The integral on simple functions)

For simple f, g and $\alpha, \beta \geq 0$:

$$\mu(\alpha f + \beta g) = \alpha \mu(f) + \beta \mu(g), \quad f \leq g \implies \mu(f) \leq \mu(g),$$

$$f = 0 \text{ a.e.} \implies \mu(f) = 0.$$

Proof. Linearity is immediate from the definition once one notes that $\alpha f + \beta g$ is again simple with an obvious representation. Monotonicity: if $f \leq g$ then $g - f$ is a non-negative simple function (take a common refinement), so $\mu(g) = \mu(f) + \mu(g - f) \geq \mu(f)$. For the last part, in a representation with the A_k disjoint and $a_k > 0$, we must have $\mu(A_k) = 0$. $\square$

Definition 38.4.3 (Integral of a non-negative function)

For measurable $f : E \rightarrow [0, \infty]$, define

$$\mu(f) = \sup\{\mu(g) : g \text{ simple, } g \leq f\} \in [0, \infty].$$

By monotonicity on simple functions this is consistent with the previous definition when f is itself simple. It also makes monotonicity for non-negative $f \leq h$ obvious, since the supremum on the left is over a smaller family.

Definition 38.4.4 (Integrable functions)

For measurable $f : E \rightarrow \mathbb{R}$ write $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$, so $f = f^+ - f^-$ and $|f| = f^+ + f^-$. We say f is **μ -integrable** if $\mu(|f|) < \infty$, and then set

$$\mu(f) = \mu(f^+) - \mu(f^-),$$

a well-defined real number. We write $L^1(\mu)$ for the set of integrable functions.

Note the asymmetry with the Riemann theory: a function is Lebesgue integrable if and only if its absolute value is. There is no analogue of a conditionally convergent improper integral here; $\int_0^\infty \frac{\sin x}{x} dx$ exists as an improper Riemann integral but $\frac{\sin x}{x}$ is not Lebesgue integrable on $(0, \infty)$.

§38.4.2 The monotone convergence theorem

Everything now hinges on the following theorem, which is the reason the whole edifice works.

Theorem 38.4.5 (Monotone convergence theorem)

Let f_n, f be measurable and non-negative with $f_n \uparrow f$ pointwise. Then $\mu(f_n) \uparrow \mu(f)$.

Proof. Since $f_n \leq f_{n+1} \leq f$, the sequence $\mu(f_n)$ is non-decreasing and bounded above by $\mu(f)$; let $M = \lim_n \mu(f_n) = \sup_n \mu(f_n) \leq \mu(f)$. It remains to prove $\mu(f) \leq M$, i.e. that $\mu(g) \leq M$ for every simple $g \leq f$.

Fix such a $g = \sum_{k=1}^m a_k \mathbb{1}_{A_k}$ with the $A_k \in \mathcal{E}$ disjoint and $a_k > 0$, and fix $\varepsilon \in (0, 1)$. Put

$$A_k(n) = \{x \in A_k : f_n(x) \geq (1 - \varepsilon)a_k\} \in \mathcal{E}.$$

Since $f_n \uparrow f \geq g = a_k$ on A_k , every $x \in A_k$ eventually satisfies $f_n(x) \geq (1 - \varepsilon)a_k$; so $A_k(n) \uparrow A_k$ and, by continuity from below, $\mu(A_k(n)) \uparrow \mu(A_k)$.

Because the A_k are disjoint,

$$f_n \geq \sum_{k=1}^m f_n \mathbb{1}_{A_k(n)} \geq (1 - \varepsilon) \sum_{k=1}^m a_k \mathbb{1}_{A_k(n)},$$

and the right-hand side is simple, so

$$\mu(f_n) \geq (1 - \varepsilon) \sum_{k=1}^m a_k \mu(A_k(n)).$$

Letting $n \rightarrow \infty$, the right-hand side tends to $(1 - \varepsilon) \sum_k a_k \mu(A_k) = (1 - \varepsilon)\mu(g)$. Hence $M \geq (1 - \varepsilon)\mu(g)$, and letting $\varepsilon \downarrow 0$ gives $M \geq \mu(g)$ as required. $\square$

Remark. Three comments. First, this is a theorem about interchanging limits:

$$\mu\left(\lim_n f_n\right) = \lim_n \mu(f_n),$$

something the Riemann integral cannot deliver without uniform convergence. Second, it suffices that $f_n \uparrow f$ almost everywhere – alter the functions on a null set, which changes no integral. Third, non-negativity is only used to get started; if f_0 is integrable and $f_n \uparrow f$ then applying the theorem to $f_n - f_0$ gives the same conclusion.

Applying the theorem to the standard approximation of Proposition 38.3.6 we get, for measurable $f \geq 0$,

$$\mu(f) = \lim_n \mu(2^{-n} \lfloor 2^n f \rfloor \wedge n),$$

which is a completely explicit recipe: the integral is the limit of the horizontal-slab sums from §1. It also delivers linearity for free.

Theorem 38.4.6 (Properties of the integral, non-negative case)

Let $f, g : E \rightarrow [0, \infty]$ be measurable and $\alpha, \beta \geq 0$. Then

- (i) $\mu(\alpha f + \beta g) = \alpha\mu(f) + \beta\mu(g)$;
- (ii) if $f \leq g$ then $\mu(f) \leq \mu(g)$;
- (iii) $\mu(f) = 0$ if and only if $f = 0$ almost everywhere;
- (iv) if $g_n \geq 0$ are measurable then $\mu(\sum_n g_n) = \sum_n \mu(g_n)$.

Proof. (i) Let $\tilde{f}_n \uparrow f$ and $\tilde{g}_n \uparrow g$ be the standard approximations. Then $\alpha\tilde{f}_n + \beta\tilde{g}_n \uparrow \alpha f + \beta g$, and by linearity on simple functions together with the monotone convergence theorem applied three times,

$$\mu(\alpha f + \beta g) = \lim_n [\alpha\mu(\tilde{f}_n) + \beta\mu(\tilde{g}_n)] = \alpha\mu(f) + \beta\mu(g).$$

(ii) Already noted.

(iii) If $f = 0$ a.e. then every simple $g \leq f$ vanishes a.e., so $\mu(g) = 0$ and hence $\mu(f) = 0$. Conversely suppose $\mu(f) = 0$ and let $A_n = \{f > 1/n\}$. Then $\frac{1}{n}\mathbb{1}_{A_n} \leq f$, so $\mu(A_n) \leq n\mu(f) = 0$; and $\{f > 0\} = \bigcup_n A_n$ is null.

(iv) Apply the monotone convergence theorem to the partial sums, using (i) for finite sums. $\square$

Theorem 38.4.7 (Properties of the integral, general case)

Let $f, g \in L^1(\mu)$ and $\alpha, \beta \in \mathbb{R}$. Then $\alpha f + \beta g \in L^1(\mu)$ with $\mu(\alpha f + \beta g) = \alpha\mu(f) + \beta\mu(g)$; if $f \leq g$ then $\mu(f) \leq \mu(g)$; and $|\mu(f)| \leq \mu(|f|)$.

Proof. For $\alpha \geq 0$, $(\alpha f)^\pm = \alpha f^\pm$ so $\mu(\alpha f) = \alpha\mu(f)$; and $(-f)^\pm = f^\mp$ so $\mu(-f) = -\mu(f)$. For sums, put $h = f + g$. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, so rearranging to avoid subtraction of infinities,

$$h^+ + f^- + g^- = h^- + f^+ + g^+.$$

Both sides are non-negative measurable, so Theorem 38.4.6(i) applies to give

$$\mu(h^+) + \mu(f^-) + \mu(g^-) = \mu(h^-) + \mu(f^+) + \mu(g^+),$$

and all terms are finite, so rearranging gives $\mu(h) = \mu(f) + \mu(g)$. Monotonicity follows since $g - f \geq 0$ has non-negative integral. Finally $|\mu(f)| = |\mu(f^+) - \mu(f^-)| \leq \mu(f^+) + \mu(f^-) = \mu(|f|)$. $\square$

Remark. Part (iii) of Theorem 38.4.6 says that the integral cannot distinguish functions agreeing almost everywhere. So $\mathbb{1}_{\mathbb{Q}}$ is integrable with $\mu(\mathbb{1}_{\mathbb{Q}}) = 0$ – exactly the answer we wanted in §1 and could not get from Riemann. It also means that $\mu(\cdot)$ is not quite a norm on L^1 , a point we return to in §6.

§38.4.3 Fatou's lemma and dominated convergence

Monotone convergence needs monotonicity. Fatou's lemma is what survives without it.

Lemma 38.4.8 (Fatou)

Let $f_n : E \rightarrow [0, \infty]$ be measurable. Then

$$\mu\left(\liminf_n f_n\right) \leq \liminf_n \mu(f_n).$$

Proof. Put $g_n = \inf_{m \geq n} f_m$, so g_n is measurable, non-negative, and $g_n \uparrow \liminf_n f_n$. For every $k \geq n$ we have $g_n \leq f_k$, so $\mu(g_n) \leq \mu(f_k)$ and hence

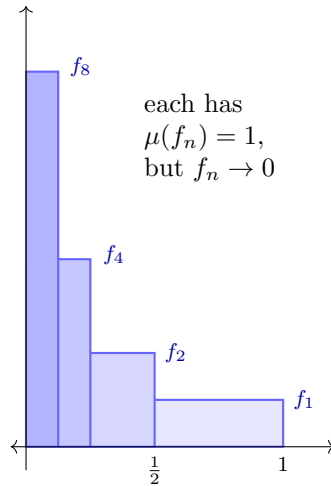
$$\mu(g_n) \leq \inf_{k \geq n} \mu(f_k) \leq \liminf_n \mu(f_n).$$

Now apply the monotone convergence theorem to $g_n \uparrow \liminf_n f_n$. □

The inequality can be strict, and the reason is instructive.

Example 38.4.9 (The escaping bump)

On $([0, 1], \mathcal{B}, \mu)$ let $f_n = n \mathbb{1}_{(0, 1/n)}$. Then $f_n \rightarrow 0$ pointwise, so $\mu(\liminf f_n) = 0$, whereas $\mu(f_n) = 1$ for every n . The mass does not disappear; it escapes to infinity in the vertical direction. On $(\mathbb{R}, \mathcal{B}, \mu)$ the example $f_n = \mathbb{1}_{(n, n+1)}$ does the same thing horizontally.



To rule out escaping mass we need a hypothesis that pins the functions down, and the cheapest one is domination.

Theorem 38.4.10 (Dominated convergence theorem)

Let f_n, f be measurable with $f_n \rightarrow f$ almost everywhere, and suppose there is an integrable g with $|f_n| \leq g$ almost everywhere for every n . Then $f_n, f \in L^1(\mu)$ and

$$\mu(f_n) \rightarrow \mu(f), \quad \text{and indeed } \mu(|f_n - f|) \rightarrow 0.$$

Proof. Since $|f_n| \leq g$ we get $\mu(|f_n|) \leq \mu(g) < \infty$; letting $n \rightarrow \infty$ pointwise gives $|f| \leq g$ a.e., so f is integrable too. Now $|f_n - f| \leq 2g$, so $2g - |f_n - f| \geq 0$, and $2g - |f_n - f| \rightarrow 2g$ a.e. By Fatou,

$$\mu(2g) = \mu\left(\liminf_n (2g - |f_n - f|)\right) \leq \liminf_n \mu(2g - |f_n - f|) = \mu(2g) - \limsup_n \mu(|f_n - f|).$$

As $\mu(2g) < \infty$ we may cancel, giving $\limsup_n \mu(|f_n - f|) \leq 0$. Hence $\mu(|f_n - f|) \rightarrow 0$, and $|\mu(f_n) - \mu(f)| \leq \mu(|f_n - f|) \rightarrow 0$. $\square$

Example 38.4.11 (Bounded convergence)

If $\mu(E) < \infty$, $f_n \rightarrow f$ pointwise, and $\sup_n \|f_n\|_\infty \leq C < \infty$, then the constant function C is integrable and dominates, so $\mu(f_n) \rightarrow \mu(f)$. Note that no uniformity of the convergence is required – contrast with the Riemann theory, where the standard theorem needs uniform convergence.

Example 38.4.12

Compute $\lim_{n \rightarrow \infty} \int_0^\infty (1 + \frac{x}{n})^{-n} \sin(x/n) \, dx$. The integrand tends to 0 pointwise, and for $n \geq 2$ we have $(1 + x/n)^{-n} |\sin(x/n)| \leq (1 + x/2)^{-2}$, which is integrable on $(0, \infty)$. By dominated convergence the limit is 0.

§38.4.4 Comparison with the Riemann integral

It is reassuring to know that we have not thrown anything away.

Theorem 38.4.13

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable. Then f is Lebesgue measurable with respect to the completed Borel σ -algebra, is Lebesgue integrable, and the two integrals agree.

We will not give the full proof, but the mechanism is easy to describe. If f is Riemann integrable, choose partitions P_n refining each other with mesh tending to 0, and let ℓ_n, u_n be the associated lower and upper step functions, so $\ell_n \leq f \leq u_n$ with $\ell_n \uparrow \ell$ and $u_n \downarrow u$ for some measurable $\ell \leq f \leq u$. The Lebesgue integrals of ℓ_n and u_n are exactly the lower and upper Riemann sums, and by monotone (or dominated) convergence they converge to $\mu(\ell)$ and $\mu(u)$. Riemann integrability says these limits agree, so $\mu(u - \ell) = 0$, whence $\ell = f = u$ almost everywhere, and $\mu(f)$ is the common value.

Remark. The same argument gives Lebesgue's criterion: a bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities is Lebesgue null. And for a *continuous* f the fundamental theorem of calculus survives unchanged, since its proof uses nothing about the integral beyond $\int_x^{x+h} dt = h$ and linearity, both of which hold for the Lebesgue integral. So

$$\underbrace{\int_a^x f(t) \, dt}_{\text{Riemann}} = \underbrace{\int_{[a,x]} f \, d\mu}_{\text{Lebesgue}}$$

for continuous f , and all the usual techniques – substitution, integration by parts – extend to bounded measurable functions by the monotone class theorem.

The converse of Theorem 38.4.13 fails badly: $\mathbb{1}_{\mathbb{Q}}$ is Lebesgue integrable but not Riemann integrable. Better still, the Lebesgue theory handles unbounded domains and unbounded functions in a single framework, with no need for the improper-integral gymnastics of Analysis I.

§38.4.5 Integrals depending on a parameter

A standard and constantly useful application of dominated convergence.

Theorem 38.4.14 (Continuity and differentiation under the integral)

Let $U \subseteq \mathbb{R}$ be open, let $(E, \mathcal{E}, \mu)$ be a measure space, and let $f : U \times E \rightarrow \mathbb{R}$ be such that $x \mapsto f(t, x)$ is measurable for each t . Set $F(t) = \int_E f(t, x) \, d\mu(x)$.

- (i) Suppose $t \mapsto f(t, x)$ is continuous for each x , and $|f(t, x)| \leq g(x)$ for all $t \in U$ with g integrable. Then F is continuous on U .
- (ii) Suppose $t \mapsto f(t, x)$ is differentiable for each x , and $|\frac{\partial f}{\partial t}(t, x)| \leq g(x)$ for all $t \in U$ with g integrable. Then F is differentiable on U with

$$F'(t) = \int_E \frac{\partial f}{\partial t}(t, x) \, d\mu(x).$$

Proof. (i) Let $t_n \rightarrow t$ in U . Then $f(t_n, \cdot) \rightarrow f(t, \cdot)$ pointwise, dominated by g , so $F(t_n) \rightarrow F(t)$ by dominated convergence. Since sequential continuity suffices on $\mathbb{R}$, we are done.

(ii) Fix $t \in U$ and let $h_n \rightarrow 0$ with $h_n \neq 0$. Put

$$g_n(x) = \frac{f(t + h_n, x) - f(t, x)}{h_n} - \frac{\partial f}{\partial t}(t, x).$$

By the mean value theorem there is $\tilde{t}$ between t and $t + h_n$ with the difference quotient equal to $\frac{\partial f}{\partial t}(\tilde{t}, x)$, so $|g_n(x)| \leq 2g(x)$, an integrable bound. By differentiability $g_n(x) \rightarrow 0$ for each x , so by dominated convergence $\mu(g_n) \rightarrow 0$. But by linearity

$$\mu(g_n) = \frac{F(t + h_n) - F(t)}{h_n} - \int_E \frac{\partial f}{\partial t}(t, x) \, d\mu(x),$$

which gives the claim. □

§38.4.6 New measures from old

Two constructions come up constantly.

Proposition 38.4.15 (Image measures and the change of variables formula)

Let $f : (E, \mathcal{E}, \mu) \rightarrow (G, \mathcal{G})$ be measurable and $\nu = \mu \circ f^{-1}$. If $g : G \rightarrow \mathbb{R}$ is measurable

and either non-negative or ν -integrable, then

$$\nu(g) = \int_G g \, d\nu = \int_E g \circ f \, d\mu = \mu(g \circ f).$$

Proof. For $g = \mathbb{1}_A$ with $A \in \mathcal{G}$, both sides equal $\mu(f^{-1}(A)) = \nu(A)$. By linearity the identity holds for simple g ; by monotone convergence along the standard approximation it holds for non-negative measurable g ; and by splitting $g = g^+ - g^-$ it holds in the integrable case. $\square$

Specialising to a random variable X on $(\Omega, \mathcal{F}, \mathbb{P})$ with law μ_X gives the formula that we used without proof in IA:

$$\mathbb{E}[g(X)] = \int_{\Omega} g(X(\omega)) \, d\mathbb{P}(\omega) = \int_E g(x) \, d\mu_X(x).$$

In particular, the expectation of $g(X)$ depends only on the law of X , which is why in practice one only ever writes down distributions.

Proposition 38.4.16 (Measures with densities)

Let $f : (E, \mathcal{E}, \mu) \rightarrow [0, \infty]$ be measurable and define

$$\nu(A) = \mu(f \mathbb{1}_A), \quad A \in \mathcal{E}.$$

Then ν is a measure on $(E, \mathcal{E})$, and for measurable $g \geq 0$,

$$\nu(g) = \mu(fg).$$

We call f the **density** of ν with respect to μ and write $d\nu = f \, d\mu$.

Proof. $\nu(\emptyset) = 0$ is clear, and countable additivity follows from Theorem 38.4.6(iv) applied to $\sum_n f \mathbb{1}_{A_n} = f \mathbb{1}_{\bigsqcup A_n}$. The identity $\nu(g) = \mu(fg)$ holds for g an indicator by definition, hence for simple g by linearity, hence in general by monotone convergence. $\square$

If ν is a probability measure we call f a **probability density function**, and this recovers the continuous random variables of IA Probability: the statement ‘ X has density f ’ means precisely that μ_X has density f with respect to Lebesgue measure, and then $\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f(x) \, dx$.

§38.5 Product Measures and Fubini’s Theorem

§38.5.1 Product σ -algebras and sections

If we want to integrate functions of two variables – and we do, since that is how sums of independent random variables work – we need a measure on a product space.

Definition 38.5.1 (Product σ -algebra)

Let $(E_1, \mathcal{E}_1)$ and $(E_2, \mathcal{E}_2)$ be measurable spaces. On $E = E_1 \times E_2$ let

$$\mathcal{A} = \{A_1 \times A_2 : A_1 \in \mathcal{E}_1, A_2 \in \mathcal{E}_2\}$$

be the collection of **measurable rectangles**, a π -system since $(A_1 \times A_2) \cap (B_1 \times B_2) = (A_1 \cap B_1) \times (A_2 \cap B_2)$. The **product σ -algebra** is $\mathcal{E}_1 \otimes \mathcal{E}_2 = \sigma(\mathcal{A})$.

If E_1, E_2 are topological spaces with countable bases then $\mathcal{B}(E_1 \times E_2) = \mathcal{B}(E_1) \otimes \mathcal{B}(E_2)$; in particular $\mathcal{B}(\mathbb{R}^2) = \mathcal{B} \otimes \mathcal{B}$, and inductively $\mathcal{B}(\mathbb{R}^d) = \mathcal{B}^{\otimes d}$.

Lemma 38.5.2 (Sections are measurable)

Let $f : (E_1 \times E_2, \mathcal{E}_1 \otimes \mathcal{E}_2) \rightarrow \mathbb{R}$ be measurable. Then for each fixed $x_1 \in E_1$ the map $x_2 \mapsto f(x_1, x_2)$ is $\mathcal{E}_2$ -measurable.

Proof. Let $\mathcal{V}$ be the set of bounded measurable f for which the conclusion holds. This is a vector space; it contains $\mathbb{1}_E$; and for a rectangle $A = A_1 \times A_2$ we have $\mathbb{1}_A(x_1, x_2) = \mathbb{1}_{A_1}(x_1)\mathbb{1}_{A_2}(x_2)$, which as a function of x_2 is either $\mathbb{1}_{A_2}$ or 0, so $\mathbb{1}_A \in \mathcal{V}$. If $0 \leq f_n \uparrow f$ with $f_n \in \mathcal{V}$ and f bounded, then $x_2 \mapsto f(x_1, x_2)$ is a pointwise limit of measurable functions, hence measurable, so $f \in \mathcal{V}$. By the monotone class theorem (Theorem 38.3.7), $\mathcal{V}$ contains all bounded measurable functions. For general measurable f , apply this to the truncations $(-n) \vee f \wedge n$ and take limits. $\square$

Lemma 38.5.3 (Partial integrals are measurable)

Let $(E_i, \mathcal{E}_i, \mu_i)$ be finite measure spaces and let f on $E_1 \times E_2$ be measurable and either bounded or non-negative. Then

$$x_1 \mapsto \int_{E_2} f(x_1, x_2) \, d\mu_2(x_2)$$

is $\mathcal{E}_1$ -measurable, and is bounded (resp. non-negative) accordingly.

Proof. Again let $\mathcal{V}$ be the collection of bounded measurable f for which the conclusion holds. It is a vector space by linearity of the integral. For a rectangle, $\int_{E_2} \mathbb{1}_{A_1}(x_1)\mathbb{1}_{A_2}(x_2) \, d\mu_2 = \mathbb{1}_{A_1}(x_1)\mu_2(A_2)$, which is measurable and bounded by $\mu_2(E_2) < \infty$; likewise $\mathbb{1}_E \in \mathcal{V}$. If $0 \leq f_n \uparrow f$ with $f_n \in \mathcal{V}$ and f bounded, then by monotone convergence

$$\int_{E_2} f(x_1, x_2) \, d\mu_2(x_2) = \lim_n \int_{E_2} f_n(x_1, x_2) \, d\mu_2(x_2)$$

is an increasing limit of measurable functions, hence measurable, and bounded by $\mu_2(E_2)\|f\|_\infty$. So $\mathcal{V}$ contains all bounded measurable functions. The non-negative case follows by applying this to $f \wedge n$ and using monotone convergence again. $\square$

§38.5.2 The product measure

Theorem 38.5.4 (Existence and uniqueness of the product measure)

Let $(E_1, \mathcal{E}_1, \mu_1)$ and $(E_2, \mathcal{E}_2, \mu_2)$ be finite measure spaces. There is a unique measure

$\mu = \mu_1 \otimes \mu_2$ on $(E_1 \times E_2, \mathcal{E}_1 \otimes \mathcal{E}_2)$ with

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2) \quad \text{for all } A_i \in \mathcal{E}_i.$$

Proof. Uniqueness is Theorem 38.2.16: the rectangles form a π -system generating $\mathcal{E}_1 \otimes \mathcal{E}_2$ and containing E , and the total mass $\mu_1(E_1)\mu_2(E_2)$ is finite.

For existence, define

$$\mu(A) = \int_{E_1} \left(\int_{E_2} \mathbb{1}_A(x_1, x_2) \, d\mu_2(x_2) \right) d\mu_1(x_1),$$

which makes sense by Lemmas 38.5.2 and 38.5.3. On a rectangle,

$$\mu(A_1 \times A_2) = \int_{E_1} \mathbb{1}_{A_1}(x_1) \mu_2(A_2) \, d\mu_1(x_1) = \mu_1(A_1)\mu_2(A_2).$$

Clearly $\mu(\emptyset) = 0$. For countable additivity, let (A_n) be disjoint in $\mathcal{E}_1 \otimes \mathcal{E}_2$. Then $\mathbb{1}_{\bigsqcup_n A_n} = \sum_n \mathbb{1}_{A_n}$, and applying the monotone convergence theorem twice (once in each integral) together with linearity gives

$$\mu\left(\bigsqcup_n A_n\right) = \lim_N \sum_{n=1}^N \int_{E_1} \int_{E_2} \mathbb{1}_{A_n} \, d\mu_2 \, d\mu_1 = \sum_{n=1}^{\infty} \mu(A_n). \quad \square$$

§38.5.3 Fubini's theorem

Theorem 38.5.5 (Fubini–Tonelli)

Let $(E, \mathcal{E}, \mu) = (E_1 \times E_2, \mathcal{E}_1 \otimes \mathcal{E}_2, \mu_1 \otimes \mu_2)$ with μ_1, μ_2 finite.

(i) (Tonelli) If $f : E \rightarrow [0, \infty]$ is measurable then

$$\begin{aligned} \int_E f \, d\mu &= \int_{E_1} \left(\int_{E_2} f(x_1, x_2) \, d\mu_2(x_2) \right) d\mu_1(x_1) \\ &= \int_{E_2} \left(\int_{E_1} f(x_1, x_2) \, d\mu_1(x_1) \right) d\mu_2(x_2). \end{aligned}$$

(ii) (Fubini) If $f \in L^1(\mu)$, let

$$A_1 = \left\{ x_1 \in E_1 : \int_{E_2} |f(x_1, x_2)| \, d\mu_2(x_2) < \infty \right\}$$

and set $f_1(x_1) = \int_{E_2} f(x_1, x_2) \, d\mu_2(x_2)$ for $x_1 \in A_1$ and $f_1 = 0$ elsewhere. Then $\mu_1(A_1^c) = 0$, $f_1 \in L^1(\mu_1)$, and $\mu(f) = \mu_1(f_1)$. Symmetrically in the other coordinate.

Proof. (i) By the construction of μ in Theorem 38.5.4, the first identity holds for $f = \mathbb{1}_A$ with A a measurable rectangle, and the second holds by the symmetric construction; the two constructions agree by uniqueness, so both identities hold for indicators of rectangles, and hence – since the two sides are measures agreeing on a generating π -system – for $f = \mathbb{1}_A$ with $A \in \mathcal{E}$. By linearity they extend to simple f ,

and by monotone convergence along the standard approximation to all measurable $f \geq 0$.

(ii) Let $h(x_1) = \int_{E_2} |f(x_1, x_2)| \, d\mu_2(x_2)$. By part (i) applied to $|f|$,

$$\mu_1(h) = \mu(|f|) < \infty,$$

so h is integrable and hence finite μ_1 -almost everywhere; that is, $\mu_1(A_1^c) = 0$. Applying part (i) to f^+ and f^- separately, the functions $f_1^\pm(x_1) = \int_{E_2} f^\pm(x_1, x_2) \, d\mu_2(x_2)$ satisfy $\mu_1(f_1^\pm) = \mu(f^\pm) < \infty$, and $f_1 = f_1^+ - f_1^-$ on A_1 . Hence

$$\mu(f) = \mu(f^+) - \mu(f^-) = \mu_1(f_1^+) - \mu_1(f_1^-) = \mu_1(f_1). \quad \square$$

Remark. The integrability hypothesis in (ii) matters. On $(0, 1)^2$ with Lebesgue measure, let

$$f(x, y) = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

One computes $\int_0^1 (\int_0^1 f \, dy) \, dx = \pi/4$ and $\int_0^1 (\int_0^1 f \, dx) \, dy = -\pi/4$. The two iterated integrals disagree, and indeed $\int \int |f| = \infty$, so Fubini does not apply. In practice one always runs Tonelli on $|f|$ first to check integrability, and only then applies Fubini to f .

Remark (σ -finite measures and higher dimensions). Everything above extends to σ -finite μ_i by exhausting E_i with sets of finite measure and taking limits. Moreover $(\mathcal{E}_1 \otimes \mathcal{E}_2) \otimes \mathcal{E}_3 = \mathcal{E}_1 \otimes (\mathcal{E}_2 \otimes \mathcal{E}_3)$, by a π -system argument, so the construction iterates: there is a unique measure $\mu_1 \otimes \cdots \otimes \mu_n$ on $\prod_i E_i$ with

$$(\mu_1 \otimes \cdots \otimes \mu_n)(A_1 \times \cdots \times A_n) = \prod_i \mu_i(A_i).$$

In particular, taking each μ_i to be Lebesgue measure on $\mathbb{R}$, we obtain **Lebesgue measure on $\mathbb{R}^d$** : the unique Borel measure with

$$\mu^d\left(\prod_{i=1}^d (a_i, b_i]\right) = \prod_{i=1}^d (b_i - a_i).$$

It is translation invariant, and Fubini says that integration over $\mathbb{R}^d$ may be performed one coordinate at a time in any order, for non-negative or integrable f .

§38.5.4 Product probability spaces and independence

Fubini gives the clean measure-theoretic statement of independence.

Proposition 38.5.6

Let $X = (X_1, \dots, X_n) : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\prod_i E_i, \bigotimes_i \mathcal{E}_i)$ be measurable. The following are equivalent:

- (i) $X_1, \dots, X_n$ are independent;
- (ii) $\mu_X = \bigotimes_{i=1}^n \mu_{X_i}$;
- (iii) for all bounded measurable $f_i : E_i \rightarrow \mathbb{R}$, $\mathbb{E}[\prod_i f_i(X_i)] = \prod_i \mathbb{E}[f_i(X_i)]$.

Proof. (i) $\Rightarrow$ (ii). On the π -system of rectangles,

$$\mu_X\left(\prod_i A_i\right) = \mathbb{P}(X_1 \in A_1, \dots, X_n \in A_n) = \prod_i \mathbb{P}(X_i \in A_i) = \left(\bigotimes_i \mu_{X_i}\right)\left(\prod_i A_i\right),$$

so the two probability measures agree by uniqueness of extension.

(ii) $\Rightarrow$ (iii). By the change of variables formula and Fubini (the integrand is bounded, hence integrable for a probability measure),

$$\mathbb{E}\left[\prod_i f_i(X_i)\right] = \int \prod_i f_i(x_i) d\left(\bigotimes_i \mu_{X_i}\right)(x) = \prod_i \int_{E_i} f_i d\mu_{X_i} = \prod_i \mathbb{E}[f_i(X_i)].$$

(iii) $\Rightarrow$ (i). Take $f_i = \mathbb{1}_{A_i}$. □

Example 38.5.7 (Densities of sums)

Let X, Y be independent real random variables with densities f, g . By Fubini, for $A \in \mathcal{B}$,

$$\mathbb{P}(X + Y \in A) = \int_{\mathbb{R}^2} \mathbb{1}_A(x + y) f(x) g(y) dx dy = \int_{\mathbb{R}} \mathbb{1}_A(v) \left(\int_{\mathbb{R}} f(v - y) g(y) dy \right) dv,$$

substituting $v = x + y$ and using translation invariance. So $X + Y$ has density the convolution $f * g$. This recovers, and now justifies, the formula from IA Probability.

§38.6 Norms and Inequalities

§38.6.1 L^p spaces

Having built the integral we can build function spaces out of it, and this is where measure theory starts to look like analysis again. Recall from IB Analysis and Topology that a **norm** on a real vector space V is a map $\|\cdot\| : V \rightarrow [0, \infty)$ with $\|\lambda v\| = |\lambda| \|v\|$, $\|u + v\| \leq \|u\| + \|v\|$, and $\|v\| = 0$ only for $v = 0$.

Definition 38.6.1 (L^p spaces)

Let $(E, \mathcal{E}, \mu)$ be a measure space and $1 \leq p \leq \infty$. For measurable $f : E \rightarrow \mathbb{R}$ put

$$\|f\|_p = \begin{cases} \left(\int_E |f|^p d\mu \right)^{1/p}, & 1 \leq p < \infty, \\ \text{ess sup } |f| = \inf\{\lambda \geq 0 : |f| \leq \lambda \text{ a.e.}\}, & p = \infty, \end{cases}$$

and let $L^p = L^p(E, \mathcal{E}, \mu)$ be the set of measurable f with $\|f\|_p < \infty$.

There is one wrinkle. If $\|f\|_p = 0$ then $|f|^p = 0$ almost everywhere, so $f = 0$ almost everywhere – but f need not be identically zero. So $\|\cdot\|_p$ is only a seminorm on L^p . The standard fix is to quotient out:

Definition 38.6.2

Write $f \sim g$ if $f = g$ μ -almost everywhere. This is an equivalence relation, and we write $\mathcal{L}^p = L^p / \sim$ for the space of equivalence classes. Then $\|\cdot\|_p$ is a genuine norm on $\mathcal{L}^p$.

In practice everyone abuses notation and talks about ‘the function $f \in \mathcal{L}^p$ ’, meaning a representative of its class; no confusion arises, because every operation we perform is insensitive to modification on a null set. We will do the same.

Homogeneity of $\|\cdot\|_p$ is clear, and the triangle inequality is easy for $p = 1$ (integrate $|f + g| \leq |f| + |g|$) and for $p = \infty$. For $1 < p < \infty$ it is Minkowski’s inequality, which we prove below via Hölder. Before that, two inequalities that need no such machinery.

§38.6.2 Markov, Chebyshev and Jensen

Proposition 38.6.3 (Markov’s inequality)

Let $f : E \rightarrow [0, \infty]$ be measurable. Then for every $\lambda > 0$,

$$\mu(\{f \geq \lambda\}) \leq \frac{\mu(f)}{\lambda}.$$

Proof. Integrate the pointwise inequality $\lambda \mathbb{1}_{\{f \geq \lambda\}} \leq f$. □

That is the whole proof, and yet it is the workhorse of the subject: it converts information about an average into information about a tail.

Corollary 38.6.4 (Chebyshev’s inequality)

If X is a random variable with $\mathbb{E}[X^2] < \infty$ then for $\lambda > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq \lambda) \leq \frac{\text{Var}(X)}{\lambda^2}.$$

More generally, $\mathbb{P}(|X| \geq \lambda) \leq \lambda^{-p} \mathbb{E}[|X|^p]$ for any $p > 0$.

Proof. Apply Markov to $|X - \mathbb{E}[X]|^2$, respectively $|X|^p$, with threshold λ^2 respectively λ^p . □

Next, convexity. Recall that $c : I \rightarrow \mathbb{R}$ on an interval I is **convex** if

$$c(tx + (1 - t)y) \leq tc(x) + (1 - t)c(y) \quad (x, y \in I, t \in [0, 1]),$$

equivalently if the slopes $\frac{c(t) - c(x)}{t - x}$ are non-decreasing in each argument. A convex function is continuous on I° , hence Borel measurable.

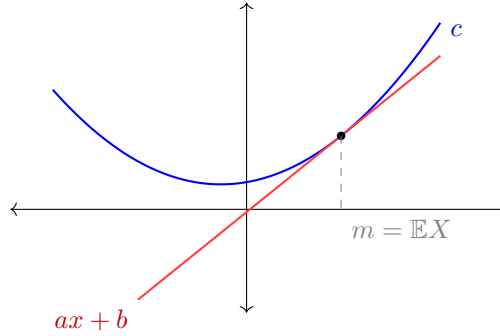
Lemma 38.6.5 (Supporting line)

Let c be convex on I and let $m \in I^\circ$. Then there are $a, b \in \mathbb{R}$ with $c(x) \geq ax + b$ for all $x \in I$ and $c(m) = am + b$.

Proof. Let

$$a = \sup \left\{ \frac{c(m) - c(x)}{m - x} : x \in I, x < m \right\},$$

which is finite by the slope characterisation of convexity (the slopes to the left of m are bounded above by any slope to the right). Set $b = c(m) - am$. For $y > m$ we have $a \leq \frac{c(y) - c(m)}{y - m}$, so $c(y) \geq ay + b$; for $y < m$, by definition of the supremum $\frac{c(m) - c(y)}{m - y} \leq a$, which rearranges to $c(y) \geq ay + b$ again. $\square$



Theorem 38.6.6 (Jensen's inequality)

Let X be a random variable taking values in an interval I with $\mathbb{E}|X| < \infty$, and let $c : I \rightarrow \mathbb{R}$ be convex. Then $\mathbb{E}[c(X)]$ is well defined in $(-\infty, \infty]$ and

$$c(\mathbb{E}[X]) \leq \mathbb{E}[c(X)].$$

Proof. Let $m = \mathbb{E}[X] \in I$. If $m \notin I^\circ$ then m is an endpoint of I and $X = m$ almost surely (a non-degenerate random variable on I has mean in the interior), so both sides are $c(m)$.

Otherwise take a, b as in Lemma 38.6.5, so $c(x) \geq ax + b$ on I . Then $c^-(X) \leq |a||X| + |b|$, so $\mathbb{E}[c^-(X)] < \infty$ and $\mathbb{E}[c(X)] = \mathbb{E}[c^+(X)] - \mathbb{E}[c^-(X)]$ is well defined in $(-\infty, \infty]$. Integrating $c(X) \geq aX + b$,

$$\mathbb{E}[c(X)] \geq a\mathbb{E}[X] + b = am + b = c(m) = c(\mathbb{E}[X]). \quad \square$$

Corollary 38.6.7 (Monotonicity of L^p norms on probability spaces)

Let $1 \leq p \leq q \leq \infty$. Then $\|X\|_p \leq \|X\|_q$ for any random variable X on a probability space, and hence $\mathcal{L}^q(\mathbb{P}) \subseteq \mathcal{L}^p(\mathbb{P})$.

Proof. The case $q = \infty$ is clear. For $q < \infty$, the map $c(x) = |x|^{q/p}$ is convex since $q/p \geq 1$. Applying Jensen to the random variable $|X|^p$ (assume first that X is bounded, so all expectations are finite),

$$(\mathbb{E}|X|^p)^{q/p} = c(\mathbb{E}|X|^p) \leq \mathbb{E}[c(|X|^p)] = \mathbb{E}|X|^q,$$

so $\|X\|_p \leq \|X\|_q$. For general $X \in \mathcal{L}^q$, apply this to $|X| \wedge n$ and let $n \rightarrow \infty$ using monotone convergence. $\square$

This is a genuinely probabilistic phenomenon: it uses $\mathbb{P}(\Omega) = 1$ crucially, and fails for infinite measure spaces, as we will see.

§38.6.3 Hölder and Minkowski

Definition 38.6.8

We call $p, q \in [1, \infty]$ **conjugate exponents** if $\frac{1}{p} + \frac{1}{q} = 1$, with the convention $\frac{1}{\infty} = 0$.

Theorem 38.6.9 (Hölder's inequality)

Let p, q be conjugate and let f, g be measurable on $(E, \mathcal{E}, \mu)$. Then

$$\mu(|fg|) \leq \|f\|_p \|g\|_q.$$

Proof. If $p = 1, q = \infty$: $|fg| \leq |f| \|g\|_\infty$ almost everywhere, and integrate. Symmetrically for $p = \infty$. So assume $1 < p < \infty$.

We may assume $\|f\|_p, \|g\|_q < \infty$, else the right side is infinite. We may also assume $\|f\|_p > 0$ (else $f = 0$ a.e. and both sides vanish), and by homogeneity we may rescale so that $\|f\|_p = 1$.

The point is that $d\mathbb{P} = |f|^p d\mu$ is then a probability measure. Writing

$$\mu(|fg|) = \int_E \frac{|g|}{|f|^{p-1}} \mathbb{1}_{\{|f|>0\}} |f|^p d\mu = \int_E \frac{|g|}{|f|^{p-1}} \mathbb{1}_{\{|f|>0\}} d\mathbb{P},$$

we may apply Corollary 38.6.7 with exponents $1 \leq q$ on the probability space $(E, \mathcal{E}, \mathbb{P})$:

$$\int \frac{|g|}{|f|^{p-1}} \mathbb{1}_{\{|f|>0\}} d\mathbb{P} \leq \left(\int \frac{|g|^q}{|f|^{q(p-1)}} \mathbb{1}_{\{|f|>0\}} d\mathbb{P} \right)^{1/q} = \left(\int_E |g|^q \mathbb{1}_{\{|f|>0\}} d\mu \right)^{1/q} \leq \|g\|_q,$$

where we used $q(p-1) = p$, so that $|f|^{-q(p-1)} |f|^p = 1$ on $\{|f| > 0\}$. This is the claim, since $\|f\|_p = 1$. $\square$

Remark. For $p = q = 2$ this is the Cauchy–Schwarz inequality $\mu(|fg|) \leq \|f\|_2 \|g\|_2$, which the reader has met in several guises already.

Theorem 38.6.10 (Minkowski's inequality)

For $1 \leq p \leq \infty$ and measurable f, g ,

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Proof. The cases $p = 1, \infty$ are immediate. Let $1 < p < \infty$ and assume $f, g \in L^p$. First, $|f + g|^p \leq 2^p(|f|^p + |g|^p)$ pointwise, so $f + g \in L^p$; and we may assume $\|f + g\|_p > 0$. Let q be conjugate to p , so $q(p-1) = p$. Then by the triangle inequality and Hölder,

$$\|f + g\|_p^p = \int_E |f + g|^{p-1} |f + g| d\mu$$

$$\begin{aligned}
&\leq \int_E |f + g|^{p-1} |f| \, d\mu + \int_E |f + g|^{p-1} |g| \, d\mu \\
&\leq \left(\int_E |f + g|^{q(p-1)} \, d\mu \right)^{1/q} (\|f\|_p + \|g\|_p) \\
&= \|f + g\|_p^{p/q} (\|f\|_p + \|g\|_p).
\end{aligned}$$

Dividing by $\|f + g\|_p^{p/q}$ and using $p - p/q = 1$ gives the result. $\square$

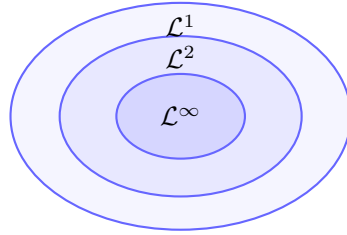
So $(\mathcal{L}^p, \|\cdot\|_p)$ is a normed vector space for every $1 \leq p \leq \infty$.

Example 38.6.11 (Containments and their failure)

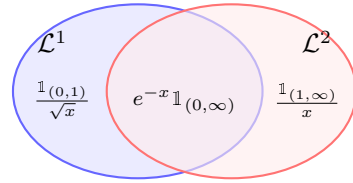
On a probability space we have $\mathcal{L}^\infty \subseteq \mathcal{L}^q \subseteq \mathcal{L}^p \subseteq \mathcal{L}^1$ for $1 \leq p \leq q \leq \infty$, by Corollary 38.6.7. The same holds on any finite measure space, after rescaling. On $(\mathbb{R}, \mathcal{B}, \mu)$ with μ Lebesgue measure, however, *no* containment holds in either direction:

$$f(x) = x^{-1/2} \mathbb{1}_{(0,1)}(x) \in \mathcal{L}^1 \setminus \mathcal{L}^2, \quad g(x) = x^{-1} \mathbb{1}_{(1,\infty)}(x) \in \mathcal{L}^2 \setminus \mathcal{L}^1.$$

The first function is too singular at the origin for $\mathcal{L}^2$; the second has too fat a tail for $\mathcal{L}^1$. On a finite measure space the second failure mode is unavailable, which is exactly why the containments hold there.



$\mu(E) < \infty$



$\mu = \text{Lebesgue on } \mathbb{R}$

§38.6.4 Completeness

The great advantage of $\mathcal{L}^p$ over the space of Riemann integrable functions is that it is complete. Recall that a **Banach space** is a complete normed vector space.

Theorem 38.6.12 (Riesz–Fischer)

For $1 \leq p \leq \infty$, the space $\mathcal{L}^p(\mu)$ is a Banach space.

Proof. The case $p = \infty$ is straightforward (a Cauchy sequence in ess sup is uniformly Cauchy off a null set), so let $1 \leq p < \infty$ and let (f_n) be Cauchy in $\mathcal{L}^p$.

Extract a subsequence (f_{N_k}) with $\|f_{N_{k+1}} - f_{N_k}\|_p \leq 2^{-k}$, so that

$$S = \sum_{k=1}^{\infty} \|f_{N_{k+1}} - f_{N_k}\|_p \leq 1 < \infty.$$

By Minkowski, for every K ,

$$\left\| \sum_{k=1}^K |f_{N_{k+1}} - f_{N_k}| \right\|_p \leq S,$$

and letting $K \rightarrow \infty$ with the monotone convergence theorem (applied to the p -th powers, which increase),

$$\left\| \sum_{k=1}^{\infty} |f_{N_{k+1}} - f_{N_k}| \right\|_p \leq S < \infty.$$

In particular $\sum_k |f_{N_{k+1}} - f_{N_k}|$ is finite almost everywhere. On the set A where it is finite – which has full measure – the telescoping series

$$f_{N_1} + \sum_{k=1}^{\infty} (f_{N_{k+1}} - f_{N_k})$$

converges absolutely, so $f_{N_k}(x)$ converges for $x \in A$ by completeness of $\mathbb{R}$. Define $f = \lim_k f_{N_k}$ on A and $f = 0$ off A ; then f is measurable and $f_{N_k} \rightarrow f$ almost everywhere.

Finally, fix $\varepsilon > 0$ and take N with $\|f_n - f_m\|_p \leq \varepsilon$ for $n, m \geq N$. For $n \geq N$, Fatou's lemma gives

$$\|f_n - f\|_p^p = \mu\left(\lim_k |f_n - f_{N_k}|^p\right) \leq \liminf_k \mu(|f_n - f_{N_k}|^p) \leq \varepsilon^p.$$

So $f_n - f \in \mathcal{L}^p$, hence $f \in \mathcal{L}^p$, and $\|f_n - f\|_p \rightarrow 0$. □

Remark. The proof shows a little more: a sequence converging in $\mathcal{L}^p$ has a subsequence converging almost everywhere. We will use this repeatedly.

Remark (Dense subspaces). Each of

$$C([a, b]), \quad \{\text{simple functions}\}, \quad \{\text{linear combinations of indicators of intervals}\}$$

is dense in $\mathcal{L}^1([a, b])$ for Lebesgue measure. So $\mathcal{L}^1$ is precisely the completion of the continuous functions in the $\|\cdot\|_1$ norm – which is a far more satisfying description of ‘the integrable functions’ than anything the Riemann theory offers, and is the reason completeness matters.

§38.6.5 $\mathcal{L}^2$ as a Hilbert space

When $p = 2$ there is extra structure.

Definition 38.6.13 (Hilbert space)

An **inner product** on a real vector space V is a symmetric bilinear form $\langle \cdot, \cdot \rangle$ with $\langle v, v \rangle \geq 0$ and $\langle v, v \rangle = 0$ only for $v = 0$. It induces a norm $\|v\| = \sqrt{\langle v, v \rangle}$. A **Hilbert space** is an inner product space complete for this norm.

Corollary 38.6.14

$\mathcal{L}^2(\mu)$ is a Hilbert space with $\langle f, g \rangle = \int_E fg \, d\mu$.

Proof. The integral defining $\langle f, g \rangle$ converges by Cauchy–Schwarz, and the form is clearly symmetric and bilinear with $\langle f, f \rangle = \|f\|_2^2$. Completeness is Theorem 38.6.12. $\square$

Two identities follow by expanding, and are worth recording.

Lemma 38.6.15

For $f, g \in \mathcal{L}^2$:

$$\begin{aligned} \|f + g\|_2^2 &= \|f\|_2^2 + 2\langle f, g \rangle + \|g\|_2^2 \quad (\text{Pythagoras}), \\ \|f + g\|_2^2 + \|f - g\|_2^2 &= 2(\|f\|_2^2 + \|g\|_2^2) \quad (\text{parallelogram law}). \end{aligned}$$

We say f and g are **orthogonal**, written $f \perp g$, if $\langle f, g \rangle = 0$; by Pythagoras this happens iff $\|f + g\|_2^2 = \|f\|_2^2 + \|g\|_2^2$.

Example 38.6.16 (The probabilistic dictionary)

On a probability space, for $X, Y \in \mathcal{L}^2(\mathbb{P})$ with means m_X, m_Y ,

$$\text{Var}(X) = \|X - m_X\|_2^2, \quad \text{Cov}(X, Y) = \langle X - m_X, Y - m_Y \rangle.$$

So variance is squared distance from the mean, and uncorrelatedness is orthogonality of the centred variables. The Cauchy–Schwarz inequality becomes $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var } X \text{Var } Y}$, and Pythagoras becomes the additivity of variance for uncorrelated variables.

Definition 38.6.17

For $V \subseteq \mathcal{L}^2(\mu)$ let $V^\perp = \{f \in \mathcal{L}^2 : \langle f, g \rangle = 0 \text{ for all } g \in V\}$. We call a subspace $V \subseteq \mathcal{L}^2$ **closed** if whenever $f_n \in V$ with $f_n \rightarrow f$ in $\mathcal{L}^2$, there is $v \in V$ with $v = f$ almost everywhere.

Theorem 38.6.18 (Orthogonal projection)

Let V be a closed subspace of $\mathcal{L}^2(\mu)$. Then every $f \in \mathcal{L}^2$ has a decomposition

$$f = v + u, \quad v \in V, \, u \in V^\perp,$$

unique up to null sets, and v is the unique closest point of V to f :

$$\|f - v\|_2 \leq \|f - g\|_2 \quad \text{for all } g \in V,$$

with equality only if $g = v$ a.e. We write $v = P_V f$.

Proof. Let $d = d(f, V) = \inf_{g \in V} \|f - g\|_2$ and choose $g_n \in V$ with $\|f - g_n\|_2 \rightarrow d$. We first show (g_n) is Cauchy. By the parallelogram law applied to $f - g_n$ and $f - g_m$,

$$\|g_n - g_m\|_2^2 = 2\|f - g_n\|_2^2 + 2\|f - g_m\|_2^2 - 4\left\|f - \frac{g_n + g_m}{2}\right\|_2^2.$$

Since $\frac{g_n + g_m}{2} \in V$, the last term is at least $4d^2$, so

$$\|g_n - g_m\|_2^2 \leq 2\|f - g_n\|_2^2 + 2\|f - g_m\|_2^2 - 4d^2 \rightarrow 0$$

as $n, m \rightarrow \infty$. By completeness $g_n \rightarrow g$ in $\mathcal{L}^2$ for some g , and since V is closed there is $v \in V$ with $v = g$ a.e. Then $\|f - v\|_2 = \lim_n \|f - g_n\|_2 = d$, so the infimum is attained at v .

Now set $u = f - v$ and let $h \in V$, $t \in \mathbb{R}$. Since $v + th \in V$,

$$d^2 \leq \|f - (v + th)\|_2^2 = d^2 - 2t\langle u, h \rangle + t^2\|h\|_2^2,$$

a quadratic in t minimised at $t = 0$; differentiating at $t = 0$ gives $\langle u, h \rangle = 0$. As $h \in V$ was arbitrary, $u \in V^\perp$.

For uniqueness, if also $f = w + z$ with $w \in V$, $z \in V^\perp$, then $(v - w) + (u - z) = 0$ with $v - w \in V$ and $u - z \in V^\perp$, so by Pythagoras $0 = \|v - w\|_2^2 + \|u - z\|_2^2$, giving $v = w$ and $u = z$ almost everywhere. $\square$

§38.6.6 Conditional expectation

We can now give the definition of conditional expectation that makes it a genuine mathematical object rather than a heuristic. The idea is: $\mathcal{G} \subseteq \mathcal{F}$ is a sub- σ -algebra representing partial information, and $\mathbb{E}[X | \mathcal{G}]$ should be the best prediction of X using only that information. In $\mathcal{L}^2$, ‘best prediction’ means ‘nearest point’, and that is orthogonal projection.

Let us start with the discrete case, which is the one from IA. Suppose $(G_n)_{n \in \mathbb{N}}$ are disjoint events with $\bigcup_n G_n = \Omega$ and $\mathbb{P}(G_n) > 0$, and put $\mathcal{G} = \sigma(G_n : n \in \mathbb{N})$.

Definition 38.6.19 (Conditional expectation, discrete case)

For $X \in \mathcal{L}^1(\mathbb{P})$ define

$$\mathbb{E}[X | \mathcal{G}] = \sum_{n=1}^{\infty} \mathbb{E}[X | G_n] \mathbb{1}_{G_n}, \quad \mathbb{E}[X | G_n] = \frac{\mathbb{E}[X \mathbb{1}_{G_n}]}{\mathbb{P}(G_n)}.$$

So $\mathbb{E}[X | \mathcal{G}]$ is the random variable which, on the event G_n , takes the value of the average of X over G_n . It is $\mathcal{G}$ -measurable by construction: it is constant on each atom of $\mathcal{G}$.

Proposition 38.6.20

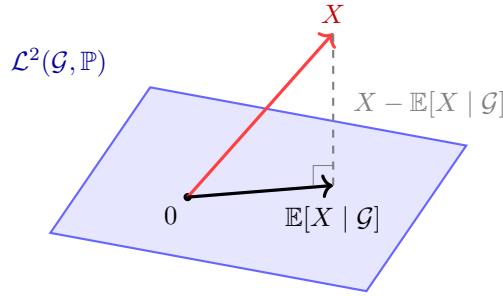
Let $X \in \mathcal{L}^2(\mathbb{P})$ and let $\mathcal{G}$ be as above. Then $Y = \mathbb{E}[X | \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace

$$\mathcal{L}^2(\mathcal{G}, \mathbb{P}) = \{W \in \mathcal{L}^2(\mathbb{P}) : W \text{ is } \mathcal{G}\text{-measurable}\}.$$

Proof. Any $\mathcal{G}$ -measurable W is constant on each G_n , so $W = \sum_n a_n \mathbb{1}_{G_n}$ for some $a_n \in \mathbb{R}$. Then

$$\mathbb{E}[(X-W)^2] = \mathbb{E}\left[\sum_n (X-a_n)^2 \mathbb{1}_{G_n}\right] = \sum_n \mathbb{P}(G_n) \left(\mathbb{E}[X^2 | G_n] - 2a_n \mathbb{E}[X | G_n] + a_n^2 \right).$$

Each summand is a quadratic in a_n minimised at $a_n = \mathbb{E}[X | G_n]$, independently of the others. Hence $\mathbb{E}[(X-W)^2]$ is minimised over $\mathcal{L}^2(\mathcal{G}, \mathbb{P})$ precisely at $W = Y$, and by Theorem 38.6.18 the minimiser is the orthogonal projection. $\square$



The picture is the whole content of the definition: $\mathbb{E}[X | \mathcal{G}]$ is the foot of the perpendicular from X to the subspace of $\mathcal{G}$ -measurable functions. Everything one expects of conditional expectation can be read off from it.

Definition 38.6.21 (Conditional expectation, general case)

Let $\mathcal{G} \subseteq \mathcal{F}$ be a sub- σ -algebra. For $X \in \mathcal{L}^2(\mathbb{P})$ define $\mathbb{E}[X | \mathcal{G}] = P_{\mathcal{L}^2(\mathcal{G}, \mathbb{P})} X$, the orthogonal projection onto the (closed) subspace of $\mathcal{G}$ -measurable square-integrable random variables.

That $\mathcal{L}^2(\mathcal{G}, \mathbb{P})$ is closed is exactly the remark after Theorem 38.6.12: a sequence converging in $\mathcal{L}^2$ has an a.e. convergent subsequence, and an a.e. limit of $\mathcal{G}$ -measurable functions is $\mathcal{G}$ -measurable.

Proposition 38.6.22 (Characterisation and basic properties)

Let $X \in \mathcal{L}^2(\mathbb{P})$ and $Y = \mathbb{E}[X | \mathcal{G}]$. Then Y is the a.s.-unique $\mathcal{G}$ -measurable element of $\mathcal{L}^2$ with

$$\mathbb{E}[Y \mathbb{1}_A] = \mathbb{E}[X \mathbb{1}_A] \quad \text{for all } A \in \mathcal{G}. \quad (*)$$

Consequently $\mathbb{E}[Y] = \mathbb{E}[X]$, the map $X \mapsto \mathbb{E}[X | \mathcal{G}]$ is linear and monotone, and $\|\mathbb{E}[X | \mathcal{G}]\|_1 \leq \|X\|_1$.

Proof. Orthogonality says $\langle X - Y, W \rangle = 0$ for all $\mathcal{G}$ -measurable $W \in \mathcal{L}^2$; taking $W = \mathbb{1}_A$ for $A \in \mathcal{G}$ (which is bounded, hence in $\mathcal{L}^2(\mathbb{P})$) gives $(*)$. Conversely if Y' is $\mathcal{G}$ -measurable and satisfies $(*)$ then $\langle X - Y', \mathbb{1}_A \rangle = 0$ for $A \in \mathcal{G}$, hence by linearity and monotone class $\langle X - Y', W \rangle = 0$ for all bounded $\mathcal{G}$ -measurable W , and by approximation for all $W \in \mathcal{L}^2(\mathcal{G}, \mathbb{P})$; so $Y' = P_{\mathcal{L}^2(\mathcal{G})} X = Y$ a.s.

Taking $A = \Omega$ in $(*)$ gives $\mathbb{E}Y = \mathbb{E}X$. Linearity is linearity of projection. For monotonicity, if $X \geq 0$, take $A = \{Y < 0\} \in \mathcal{G}$: then $\mathbb{E}[Y \mathbb{1}_A] = \mathbb{E}[X \mathbb{1}_A] \geq 0$, and $Y \mathbb{1}_A \leq 0$, so $Y \mathbb{1}_A = 0$ a.s. and $\mathbb{P}(A) = 0$. Finally, applying monotonicity to

$\pm X \leq |X|$ gives $|Y| \leq \mathbb{E}[|X| \mid \mathcal{G}]$ a.s., so $\mathbb{E}|Y| \leq \mathbb{E}[\mathbb{E}[|X| \mid \mathcal{G}]] = \mathbb{E}|X|$. $\square$

The $\mathcal{L}^1$ contraction property in the last line is what lets us extend the definition beyond $\mathcal{L}^2$.

Theorem 38.6.23 (Extension to $\mathcal{L}^1$)

The map $X \mapsto \mathbb{E}[X \mid \mathcal{G}]$ extends uniquely from $\mathcal{L}^2(\mathbb{P})$ to a linear contraction $\mathcal{L}^1(\mathbb{P}) \rightarrow \mathcal{L}^1(\mathbb{P})$, and the extension is still characterised by (*).

Proof. $\mathcal{L}^2(\mathbb{P})$ is dense in $\mathcal{L}^1(\mathbb{P})$: given $X \in \mathcal{L}^1$, the truncations $X_n = (-n) \vee X \wedge n$ are bounded, hence in $\mathcal{L}^2$, and $\|X_n - X\|_1 \rightarrow 0$ by dominated convergence. By Proposition 38.6.22 the map is a contraction for $\|\cdot\|_1$ on the dense subspace $\mathcal{L}^2$, and $\mathcal{L}^1$ is complete, so it extends uniquely by continuity. The identity (*) passes to the limit since $|\mathbb{E}[(X_n - X)\mathbb{1}_A]| \leq \|X_n - X\|_1$. $\square$

Remark. One can also prove existence directly in $\mathcal{L}^1$ via the Radon–Nikodym theorem, and that is the route usually taken in a course on martingales. The $\mathcal{L}^2$ route taken here is more elementary and, I think, more illuminating: it makes it clear that conditional expectation is *geometry*, not a formula.

Example 38.6.24 (Two extremes)

If $\mathcal{G} = \{\emptyset, \Omega\}$ then the only $\mathcal{G}$ -measurable functions are constants, and projecting onto the constants gives $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$. If $\mathcal{G} = \mathcal{F}$ then X itself is $\mathcal{G}$ -measurable and $\mathbb{E}[X \mid \mathcal{F}] = X$. In general $\mathbb{E}[X \mid \mathcal{G}]$ interpolates between knowing nothing and knowing everything.

§38.7 Modes of Convergence

§38.7.1 The four modes

We have met several notions of convergence for random variables, and it is time to organise them.

Definition 38.7.1

Let X_n, X be random variables on $(\Omega, \mathcal{F}, \mathbb{P})$.

- (i) $X_n \rightarrow X$ **almost surely** if $\mathbb{P}(X_n \rightarrow X) = 1$.
- (ii) $X_n \rightarrow X$ **in probability**, written $X_n \xrightarrow{\mathbb{P}} X$, if $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.
- (iii) $X_n \rightarrow X$ **in $\mathcal{L}^p$** if $X_n, X \in \mathcal{L}^p$ and $\|X_n - X\|_p \rightarrow 0$.
- (iv) $X_n \rightarrow X$ **in distribution**, written $X_n \xrightarrow{d} X$, if $F_{X_n}(x) \rightarrow F_X(x)$ at every x at which F_X is continuous.

The last of these is different in kind from the others: it involves only the laws μ_{X_n}, μ_X , and the random variables need not even be defined on the same probability space. The reason for the continuity proviso is that otherwise the definition would be useless: if

$X_n = 1/n$ deterministically then $F_{X_n}(0) = 0$ for all n but $F_0(0) = 1$, and we certainly want $X_n \rightarrow 0$ in distribution.

Theorem 38.7.2 (The implications)

The following implications hold, and no others in general:

$$\begin{aligned} \text{in } \mathcal{L}^q &\implies \text{in } \mathcal{L}^p \ (p \leq q), & \text{in } \mathcal{L}^p &\implies \text{in probability,} \\ & & \text{a.s.} &\implies \text{in probability} \implies \text{in distribution.} \end{aligned}$$

Moreover if $X_n \xrightarrow{\mathbb{P}} X$ then $X_{n_k} \rightarrow X$ almost surely along a subsequence.

Proof. The first implication is Corollary 38.6.7. For the second, Markov's inequality gives

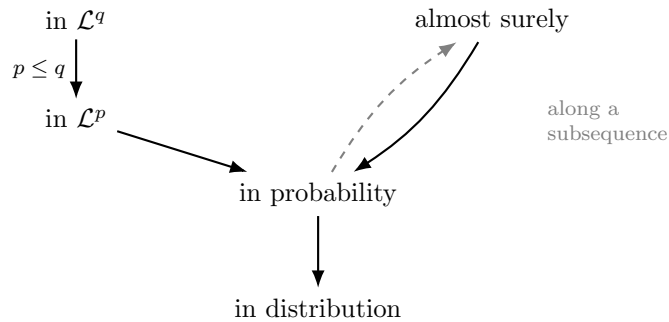
$$\mathbb{P}(|X_n - X| > \varepsilon) \leq \frac{\mathbb{E}|X_n - X|^p}{\varepsilon^p} = \frac{\|X_n - X\|_p^p}{\varepsilon^p} \rightarrow 0.$$

The implication 'a.s. $\implies$ in probability' and the subsequence statement are Theorem 38.3.14, applied to $|X_n - X|$ on the finite measure space $(\Omega, \mathcal{F}, \mathbb{P})$.

Finally suppose $X_n \xrightarrow{\mathbb{P}} X$ and let x be a continuity point of F_X . For $\varepsilon > 0$,

$$\{X_n \leq x\} \subseteq \{X \leq x + \varepsilon\} \cup \{|X_n - X| > \varepsilon\},$$

so $F_{X_n}(x) \leq F_X(x + \varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon)$, and hence $\limsup_n F_{X_n}(x) \leq F_X(x + \varepsilon)$. Symmetrically $\{X \leq x - \varepsilon\} \subseteq \{X_n \leq x\} \cup \{|X_n - X| > \varepsilon\}$ gives $\liminf_n F_{X_n}(x) \geq F_X(x - \varepsilon)$. Letting $\varepsilon \downarrow 0$ and using continuity of F_X at x gives $F_{X_n}(x) \rightarrow F_X(x)$. $\square$



Example 38.7.3 (Counterexamples to the converses)

Work on $((0, 1), \mathcal{B}, \mu)$ unless stated otherwise.

- (i) *In probability but not in $\mathcal{L}^1$.* Let $X_n = n\mathbb{1}_{(0, 1/n)}$. Then $\mathbb{P}(|X_n| > \varepsilon) = 1/n \rightarrow 0$, but $\mathbb{E}[X_n] = 1$ for all n , so $X_n \not\xrightarrow{\mathbb{P}} 0$ in $\mathcal{L}^1$. This is the escaping bump of Example 38.4.9 again.
- (ii) *In probability but not almost surely.* Take the 'typewriter sequence': for $n = 2^k + j$ with $0 \leq j < 2^k$, let $X_n = \mathbb{1}_{(j2^{-k}, (j+1)2^{-k})}$. Then $\mathbb{P}(X_n \neq 0) = 2^{-k} \rightarrow 0$, so $X_n \xrightarrow{\mathbb{P}} 0$; but for every ω the sequence $X_n(\omega)$ takes the value 1 infinitely often, so $X_n \not\xrightarrow{\text{a.s.}} 0$ anywhere.
- (iii) *In $\mathcal{L}^1$ but not in $\mathcal{L}^2$.* Let $X_n = n^{1/2}\mathbb{1}_{(0, 1/n)}$. Then $\|X_n\|_1 = n^{-1/2} \rightarrow 0$ but

$$\|X_n\|_2^2 = 1.$$

- (iv) *In distribution but not in probability.* Let $X \sim N(0, 1)$ and $X_n = -X$ for all n . Since $-X \sim N(0, 1)$, we have $X_n \xrightarrow{d} X$; but $\mathbb{P}(|X_n - X| > \varepsilon) = \mathbb{P}(|2X| > \varepsilon)$ does not tend to zero.
- (v) *Almost surely but not in $\mathcal{L}^1$.* The bump $X_n = n^2 \mathbb{1}_{(0, 1/n)}$ converges to 0 pointwise on $(0, 1)$ but $\mathbb{E}X_n = n \rightarrow \infty$.

The one useful partial converse is that convergence in distribution to a *constant* implies convergence in probability: if $X_n \xrightarrow{d} c$ then $\mathbb{P}(|X_n - c| > \varepsilon) \leq 1 - F_{X_n}(c + \varepsilon/2) + F_{X_n}(c - \varepsilon) \rightarrow 0$.

§38.7.2 Uniform integrability

The examples above suggest the obstruction to upgrading convergence in probability to convergence in $\mathcal{L}^1$ is exactly the escaping-mass phenomenon. Uniform integrability is the hypothesis that rules it out.

We start with a lemma showing that a single integrable random variable never has escaping mass.

Lemma 38.7.4

Let $X \in \mathcal{L}^1(\mathbb{P})$. Then

$$I_X(\delta) = \sup \{ \mathbb{E}[|X| \mathbb{1}_A] : A \in \mathcal{F}, \mathbb{P}(A) \leq \delta \} \rightarrow 0 \quad \text{as } \delta \downarrow 0.$$

Proof. I_X is non-decreasing in δ , so suppose for contradiction that $I_X(\delta) \geq \varepsilon$ for all $\delta > 0$, for some $\varepsilon > 0$. Then we may pick $A_n \in \mathcal{F}$ with $\mathbb{P}(A_n) \leq 2^{-n}$ and $\mathbb{E}[|X| \mathbb{1}_{A_n}] \geq \varepsilon$. Since $\sum_n \mathbb{P}(A_n) < \infty$, the first Borel–Cantelli lemma gives $\mathbb{P}(A) = 0$ where $A = \limsup_n A_n$. Now $\mathbb{1}_{\bigcup_{m \geq n} A_m} \downarrow \mathbb{1}_A$, and $|X| \mathbb{1}_{\bigcup_{m \geq n} A_m} \leq |X| \in \mathcal{L}^1$, so by dominated convergence

$$\varepsilon \leq \mathbb{E}[|X| \mathbb{1}_{A_n}] \leq \mathbb{E}[|X| \mathbb{1}_{\bigcup_{m \geq n} A_m}] \rightarrow \mathbb{E}[|X| \mathbb{1}_A] = 0,$$

a contradiction. □

Definition 38.7.5 (Uniform integrability)

A family $\mathcal{X} \subseteq \mathcal{L}^1(\mathbb{P})$ is **uniformly integrable** (UI) if it is bounded in $\mathcal{L}^1$, i.e. $\sup_{X \in \mathcal{X}} \mathbb{E}|X| < \infty$, and

$$I_{\mathcal{X}}(\delta) = \sup \{ \mathbb{E}[|X| \mathbb{1}_A] : X \in \mathcal{X}, \mathbb{P}(A) \leq \delta \} \rightarrow 0 \quad \text{as } \delta \downarrow 0.$$

By Lemma 38.7.4, any finite family is UI. The following reformulation is usually the easiest to check.

Lemma 38.7.6 (Equivalent form of UI)

A family $\mathcal{X} \subseteq \mathcal{L}^1(\mathbb{P})$ is uniformly integrable if and only if

$$\sup_{X \in \mathcal{X}} \mathbb{E}[|X| \mathbb{1}_{\{|X| > K\}}] \longrightarrow 0 \quad \text{as } K \rightarrow \infty.$$

Proof. ($\Rightarrow$) By Markov's inequality, $\mathbb{P}(|X| > K) \leq \mathbb{E}|X|/K \leq I_{\mathcal{X}}(1)/K$ uniformly in $X \in \mathcal{X}$. So given $\delta > 0$ we may take K large enough that $\mathbb{P}(|X| > K) \leq \delta$ for all $X \in \mathcal{X}$, and then $\mathbb{E}[|X| \mathbb{1}_{\{|X| > K\}}] \leq I_{\mathcal{X}}(\delta)$, which tends to 0.

($\Leftarrow$) Fix $\varepsilon > 0$ and choose K with $\sup_X \mathbb{E}[|X| \mathbb{1}_{\{|X| > K\}}] < \varepsilon/2$. Then

$$\mathbb{E}|X| = \mathbb{E}[|X| \mathbb{1}_{\{|X| \leq K\}}] + \mathbb{E}[|X| \mathbb{1}_{\{|X| > K\}}] \leq K + \frac{\varepsilon}{2},$$

so $\mathcal{X}$ is $\mathcal{L}^1$ -bounded. And for A with $\mathbb{P}(A) \leq \delta$,

$$\mathbb{E}[|X| \mathbb{1}_A] \leq \mathbb{E}[|X| \mathbb{1}_A \mathbb{1}_{\{|X| \leq K\}}] + \mathbb{E}[|X| \mathbb{1}_{\{|X| > K\}}] \leq K\delta + \frac{\varepsilon}{2} < \varepsilon$$

for $\delta < \varepsilon/(2K)$. □

Example 38.7.7 (When UI holds and when it fails) (i) If $\mathcal{X}$ is bounded in $\mathcal{L}^p$ for some $p > 1$ then it is UI: by Hölder with conjugate q ,

$$\mathbb{E}[|X| \mathbb{1}_A] \leq \|X\|_p \mathbb{P}(A)^{1/q} \leq \left(\sup_X \|X\|_p \right) \delta^{1/q} \rightarrow 0.$$

(ii) If $|X| \leq Y$ for all $X \in \mathcal{X}$ with $Y \in \mathcal{L}^1$, then $\mathcal{X}$ is UI, by Lemma 38.7.4 applied to Y . So domination is a special case of uniform integrability.

(iii) The family $\{n \mathbb{1}_{(0, 1/n)} : n \in \mathbb{N}\}$ on $(0, 1)$ is bounded in $\mathcal{L}^1$ (each has norm 1) but not UI: taking $A = (0, 1/n)$ gives $\mathbb{P}(A) \rightarrow 0$ while $\mathbb{E}[X_n \mathbb{1}_A] = 1$.

§38.7.3 The Vitali convergence theorem

We can now state the theorem which makes uniform integrability the right concept: it characterises $\mathcal{L}^1$ convergence exactly. First a bounded case.

Theorem 38.7.8 (Bounded convergence in probability)

Let $X_n \xrightarrow{\mathbb{P}} X$ with $|X_n| \leq C < \infty$ almost surely for all n . Then $X_n \rightarrow X$ in $\mathcal{L}^1$.

Proof. Along a subsequence $X_{n_k} \rightarrow X$ a.s., so $|X| \leq C$ a.s. Given $\varepsilon > 0$, split according to whether $|X_n - X|$ exceeds $\varepsilon/2$:

$$\mathbb{E}|X_n - X| = \mathbb{E}[|X_n - X| \mathbb{1}_{\{|X_n - X| > \varepsilon/2\}}] + \mathbb{E}[|X_n - X| \mathbb{1}_{\{|X_n - X| \leq \varepsilon/2\}}],$$

$$\text{so} \quad \mathbb{E}|X_n - X| \leq 2C \mathbb{P}(|X_n - X| > \varepsilon/2) + \frac{\varepsilon}{2},$$

which is $< \varepsilon$ for n large. □

Theorem 38.7.9 (Vitali convergence theorem)

Let X_n, X be random variables. The following are equivalent:

- (i) $X_n, X \in \mathcal{L}^1(\mathbb{P})$ and $X_n \rightarrow X$ in $\mathcal{L}^1$;
- (ii) $\{X_n : n \in \mathbb{N}\}$ is uniformly integrable and $X_n \xrightarrow{\mathbb{P}} X$.

Proof. (i) $\Rightarrow$ (ii). Markov's inequality gives $\mathbb{P}(|X_n - X| > \varepsilon) \leq \mathbb{E}|X_n - X|/\varepsilon \rightarrow 0$, so $X_n \xrightarrow{\mathbb{P}} X$. For uniform integrability, fix $\varepsilon > 0$ and take N with $\mathbb{E}|X_n - X| < \varepsilon/2$ for $n \geq N$. The finite family $\{X, X_1, \dots, X_N\}$ is UI by Lemma 38.7.4, so there is $\delta > 0$ with $\mathbb{E}[|Y|\mathbb{1}_A] < \varepsilon/2$ for Y in that family and $\mathbb{P}(A) \leq \delta$. For $n > N$ and such A ,

$$\mathbb{E}[|X_n|\mathbb{1}_A] \leq \mathbb{E}[|X_n - X|\mathbb{1}_A] + \mathbb{E}[|X|\mathbb{1}_A] < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So the whole family is UI.

(ii) $\Rightarrow$ (i). Along a subsequence $X_{n_k} \rightarrow X$ a.s., so by Fatou

$$\mathbb{E}|X| = \mathbb{E}\left[\liminf_k |X_{n_k}|\right] \leq \liminf_k \mathbb{E}|X_{n_k}| \leq \sup_n \mathbb{E}|X_n| < \infty,$$

so $X \in \mathcal{L}^1$. Now let $g_K(x) = (-K) \vee x \wedge K$ be the truncation at level K ; it is 1-Lipschitz, so

$$\mathbb{P}(|g_K(X_n) - g_K(X)| > \varepsilon) \leq \mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0,$$

and $|g_K(\cdot)| \leq K$, so by Theorem 38.7.8 $g_K(X_n) \rightarrow g_K(X)$ in $\mathcal{L}^1$. Therefore

$$\begin{aligned} \mathbb{E}|X_n - X| &\leq \mathbb{E}|X_n - g_K(X_n)| + \mathbb{E}|g_K(X_n) - g_K(X)| + \mathbb{E}|g_K(X) - X| \\ &\leq \mathbb{E}[|X_n|\mathbb{1}_{\{|X_n|>K\}}] + \mathbb{E}|g_K(X_n) - g_K(X)| + \mathbb{E}[|X|\mathbb{1}_{\{|X|>K\}}]. \end{aligned}$$

Given $\varepsilon > 0$, choose K so that the first and third terms are each below $\varepsilon/3$ – possible by Lemma 38.7.6 for the first, and by Lemma 38.7.4 for the third – and then n large so the middle term is below $\varepsilon/3$. $\square$

Remark. Vitali's theorem contains the dominated convergence theorem on a probability space as a special case: domination by an integrable Y gives uniform integrability, and a.s. convergence gives convergence in probability. It is strictly stronger, since UI families need not be dominated.

§38.8 Laws of Large Numbers

The empirical content of probability theory is that averages of many independent measurements stabilise. We now prove this, in two forms of increasing strength.

§38.8.1 The weak law

Theorem 38.8.1 (Weak law of large numbers)

Let (X_n) be i.i.d. random variables with $\mathbb{E}[X_1] = \nu$ and $\text{Var}(X_1) = \sigma^2 < \infty$. Then

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\mathbb{P}} \nu \quad \text{as } n \rightarrow \infty.$$

Proof. Replacing X_i by $X_i - \nu$ we may assume $\nu = 0$. Let $S_n = \sum_{i \leq n} X_i$. By independence the variances add, so $\text{Var}(S_n) = n\sigma^2$, and Chebyshev's inequality gives

$$\mathbb{P}\left(\left|\frac{S_n}{n}\right| > \varepsilon\right) \leq \frac{\text{Var}(S_n)}{n^2\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0. \quad \square$$

Notice that the proof also gives convergence in $\mathcal{L}^2$, since $\mathbb{E}[(S_n/n)^2] = \sigma^2/n \rightarrow 0$. The theorem is 'weak' in two respects: it assumes a second moment, and it concludes only convergence in probability. Both can be improved.

§38.8.2 A strong law under fourth moments

Convergence in probability allows the averages to make large excursions infinitely often, provided they do so with vanishing probability – the typewriter sequence of Example 38.7.3 does exactly that. The strong law rules this out. Here is a proof requiring a fourth moment, which is short and instructive; we will remove the hypothesis in §10.

Theorem 38.8.2 (Strong law under a fourth moment)

Let (X_n) be independent with $\mathbb{E}[X_n] = \nu$ and $\mathbb{E}[X_n^4] \leq M < \infty$ for all n . Then

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \nu \quad \text{almost surely.}$$

Proof. Replacing X_n by $Y_n = X_n - \nu$, which satisfies $\mathbb{E}[Y_n^4] \leq 2^4(\mathbb{E}[X_n^4] + \nu^4) < \infty$ by convexity of $t \mapsto t^4$, we may assume $\nu = 0$ and $\mathbb{E}[X_n^4] \leq M$.

Expand $\mathbb{E}[S_n^4]$ where $S_n = \sum_{i \leq n} X_i$. A term $\mathbb{E}[X_i X_j X_k X_\ell]$ with some index appearing exactly once vanishes by independence and $\mathbb{E}[X_i] = 0$; so the only surviving terms are those of the form $\mathbb{E}[X_i^4]$ and $\mathbb{E}[X_i^2 X_j^2]$ with $i \neq j$. There are n of the former and $\binom{n}{2} \binom{4}{2} = 3n(n-1)$ of the latter (choosing which two of the four factors carry index i), and by Cauchy–Schwarz $\mathbb{E}[X_i^2 X_j^2] \leq \sqrt{\mathbb{E}[X_i^4]} \sqrt{\mathbb{E}[X_j^4]} \leq M$. Hence

$$\mathbb{E}[S_n^4] \leq nM + 3n(n-1)M \leq 3n^2M.$$

Therefore, by Theorem 38.4.6(iv),

$$\mathbb{E}\left[\sum_{n=1}^{\infty} \left(\frac{S_n}{n}\right)^4\right] = \sum_{n=1}^{\infty} \frac{\mathbb{E}[S_n^4]}{n^4} \leq \sum_{n=1}^{\infty} \frac{3M}{n^2} < \infty.$$

A non-negative random variable with finite expectation is finite almost surely, so $\sum_n (S_n/n)^4 < \infty$ a.s., whence $(S_n/n)^4 \rightarrow 0$ and $S_n/n \rightarrow 0$ almost surely. $\square$

§38.8.3 The strong law in general

The theorem we are really after is the following, and it is the sharpest possible: no moment beyond the first is required, and the first cannot be dispensed with.

Theorem 38.8.3 (Strong law of large numbers)

Let (X_n) be i.i.d. with $\mathbb{E}|X_1| < \infty$ and $\mathbb{E}[X_1] = \nu$. Then

$$\frac{1}{n} \sum_{i=1}^n X_i \longrightarrow \nu \quad \text{almost surely, and in } \mathcal{L}^1.$$

We will prove this in §10 as a consequence of Birkhoff's ergodic theorem; the ergodic-theoretic route is the one taken in lectures and it is genuinely the cleanest. For now, observe that Kolmogorov's zero-one law already tells us something: the event $\{\frac{1}{n} \sum_{i \leq n} X_i \rightarrow c\}$ is a tail event, so it has probability 0 or 1, and $\limsup_n \frac{1}{n} \sum_{i \leq n} X_i$ is an almost surely constant element of $[-\infty, \infty]$. The strong law identifies that constant as $\mathbb{E}[X_1]$.

Remark. The integrability hypothesis cannot be weakened. If $\mathbb{E}|X_1| = \infty$ then by the second Borel–Cantelli lemma applied to $A_n = \{|X_n| > n\}$ – whose probabilities sum to ∞ precisely when $\mathbb{E}|X_1| = \infty$ – we get $|X_n| > n$ infinitely often almost surely, and hence S_n/n cannot converge (since $\frac{X_n}{n} = \frac{S_n}{n} - \frac{n-1}{n} \frac{S_{n-1}}{n-1}$ would have to tend to zero). For a Cauchy sample, for instance, S_n/n has the same Cauchy distribution for every n and does not converge at all.

Remark (How fast?). The strong law says $S_n/n \rightarrow \nu$; the central limit theorem of §9 will say that the error is of order $n^{-1/2}$ in distribution. Sharper still, the *law of the iterated logarithm* states that if $\mathbb{E}X_1 = 0$ and $\text{Var } X_1 = 1$ then

$$\limsup_n \frac{S_n}{\sqrt{2n \log \log n}} = 1, \quad \liminf_n \frac{S_n}{\sqrt{2n \log \log n}} = -1 \quad \text{a.s.}$$

In particular $S_n/\sqrt{n}$ does *not* converge almost surely – it oscillates forever between $\pm\sqrt{2 \log \log n}$ – so the central limit theorem really is a statement about distributions and nothing more.

§38.9 Fourier Transforms and Characteristic Functions

§38.9.1 The Fourier transform in L^1

We now allow complex-valued functions. For $f = u + iv$ with u, v real and integrable, define $\int f = \int u + i \int v$; this is complex-linear, and the fundamental estimate still holds:

Lemma 38.9.1

For $f \in L^1(\mathbb{R}^d; \mathbb{C})$ we have $|\int f| \leq \int |f|$.

Proof. Choose $\alpha \in \mathbb{C}$ with $|\alpha| = 1$ and $\alpha \int f = |\int f|$. Then $|\int f| = \int \alpha f$ is real, so it equals $\int \text{Re}(\alpha f) \leq \int |\alpha f| = \int |f|$. $\square$

Write $L^p(\mathbb{R}^d)$ for the complex L^p spaces with respect to Lebesgue measure, and $\langle u, x \rangle = \sum_i u_i x_i$.

Definition 38.9.2 (Fourier transform)

For $f \in L^1(\mathbb{R}^d)$ the **Fourier transform** is

$$\hat{f}(u) = \int_{\mathbb{R}^d} f(x) e^{i\langle u, x \rangle} dx, \quad u \in \mathbb{R}^d.$$

For a finite Borel measure μ on $\mathbb{R}^d$, set $\hat{\mu}(u) = \int_{\mathbb{R}^d} e^{i\langle u, x \rangle} d\mu(x)$.

Remark. Immediately $|\hat{f}(u)| \leq \|f\|_1$ and $|\hat{\mu}(u)| \leq \mu(\mathbb{R}^d)$. Moreover if $u_n \rightarrow u$ then $e^{i\langle u_n, x \rangle} \rightarrow e^{i\langle u, x \rangle}$ pointwise, dominated by $|f|$ (respectively by 1, integrable for a finite measure), so by dominated convergence $\hat{f}$ and $\hat{\mu}$ are *continuous and bounded*. If μ has density f then $\hat{\mu} = \hat{f}$.

Different authors put the 2π and the sign in different places; the convention above is the one used in this course, and it is chosen so that the characteristic function below comes out with no constants at all.

Definition 38.9.3 (Characteristic function)

Let X be an $\mathbb{R}^d$ -valued random variable with law μ_X . Its **characteristic function** is

$$\varphi_X(u) = \mathbb{E}[e^{i\langle u, X \rangle}] = \hat{\mu}_X(u).$$

So φ_X is continuous, $\varphi_X(0) = 1$, $|\varphi_X(u)| \leq 1$, and $\varphi_X(-u) = \overline{\varphi_X(u)}$. Our goals are: (a) φ_X determines μ_X ; (b) pointwise convergence of φ_{X_n} implies convergence in distribution of X_n ; and (c) hence the central limit theorem. All three go through Gaussians.

§38.9.2 Convolution**Definition 38.9.4 (Convolution)**

Let $f \in L^1(\mathbb{R}^d)$ and let ν be a probability measure on $\mathbb{R}^d$. Define

$$(f * \nu)(x) = \begin{cases} \int_{\mathbb{R}^d} f(x-y) d\nu(y), & \text{if } y \mapsto f(x-y) \in L^1(\nu), \\ 0, & \text{otherwise.} \end{cases}$$

If ν has density g we write $f * g$. For probability measures μ, ν the convolution $\mu * \nu$ is the law of $X + Y$ with $X \sim \mu$, $Y \sim \nu$ independent, so

$$(\mu * \nu)(A) = \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \mathbb{1}_A(x+y) d\nu(y) d\mu(x).$$

Proposition 38.9.5

For $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^d)$, the map $f \mapsto f * \nu$ is a contraction of $L^p(\mathbb{R}^d)$: $\|f * \nu\|_p \leq \|f\|_p$. In particular $y \mapsto f(x-y)$ lies in $L^p(\nu)$ for almost every x .

Proof. Since ν is a probability measure, Jensen's inequality applied to the convex

function $t \mapsto |t|^p$ gives

$$\left| \int f(x-y) \, d\nu(y) \right|^p \leq \int |f(x-y)|^p \, d\nu(y).$$

Integrating in x and applying Tonelli together with translation invariance of Lebesgue measure,

$$\int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |f(x-y)|^p \, d\nu(y) \, dx = \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |f(x-y)|^p \, dx \right) d\nu(y) = \|f\|_p^p. \quad \square$$

Proposition 38.9.6

$\widehat{f * \nu} = \hat{f} \hat{\nu}$ and $\widehat{\mu * \nu} = \hat{\mu} \hat{\nu}$. If μ has density f then $\mu * \nu$ has density $f * \nu$.

Proof. The second identity is a one-liner: with $X \sim \mu$, $Y \sim \nu$ independent,

$$\widehat{\mu * \nu}(u) = \mathbb{E}[e^{i\langle u, X+Y \rangle}] = \mathbb{E}[e^{i\langle u, X \rangle}] \mathbb{E}[e^{i\langle u, Y \rangle}] = \hat{\mu}(u) \hat{\nu}(u),$$

using Proposition 38.5.6(iii). The first follows by Fubini:

$$\widehat{f * \nu}(u) = \iint f(x-y) e^{i\langle u, x \rangle} \, d\nu(y) \, dx = \int e^{i\langle u, y \rangle} \left(\int f(z) e^{i\langle u, z \rangle} \, dz \right) d\nu(y),$$

which is $\hat{f}(u) \hat{\nu}(u)$, on substituting $z = x - y$. For the density claim, with A Borel,

$$(\mu * \nu)(A) = \iint \mathbb{1}_A(x+y) f(x) \, dx \, d\nu(y) = \int \mathbb{1}_A(v) \left(\int f(v-y) \, d\nu(y) \right) dv. \quad \square$$

In probabilistic language: the characteristic function of a sum of independent random variables is the product of their characteristic functions. That single fact is what makes characteristic functions the tool of choice for limit theorems about sums.

§38.9.3 Gaussians

Definition 38.9.7

For $t > 0$ let g_t denote the density of the $N(0, tI_d)$ distribution on $\mathbb{R}^d$:

$$g_t(x) = (2\pi t)^{-d/2} e^{-|x|^2/(2t)}.$$

Proposition 38.9.8

$\hat{g}_t(u) = e^{-|u|^2 t/2}$, and consequently $\hat{g}_t = (2\pi)^{d/2} t^{d/2} g_{1/t}$. The Fourier inversion formula

$$g_t(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \hat{g}_t(u) \, du$$

holds for every $t > 0$.

Proof. Take $d = 1$ and $t = 1$ first, and let $\varphi(u) = \hat{g}_1(u) = \mathbb{E}[e^{iuZ}]$ where $Z \sim N(0, 1)$. Differentiating under the integral sign – which is legitimate by Theorem 38.4.14,

since $|\partial_u(e^{iux}g_1(x))| \leq |x|g_1(x)$ is integrable – and then integrating by parts,

$$\varphi'(u) = i \int_{\mathbb{R}} x e^{iux} g_1(x) \, dx = \frac{i}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{iux} (x e^{-x^2/2}) \, dx,$$

$$\varphi'(u) = \frac{i}{\sqrt{2\pi}} \left[-e^{iux} e^{-x^2/2} \right]_{-\infty}^{\infty} + \frac{i^2 u}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{iux} e^{-x^2/2} \, dx,$$

so $\varphi'(u) = -u\varphi(u)$. Hence $\frac{d}{du}(e^{u^2/2}\varphi(u)) = 0$, and since $\varphi(0) = 1$ we get $\varphi(u) = e^{-u^2/2}$.

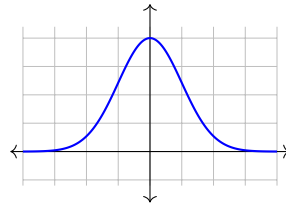
In general, if $Z = (Z_1, \dots, Z_d)$ has i.i.d. $N(0, 1)$ entries then $\sqrt{t}Z$ has density g_t , and by independence

$$\hat{g}_t(u) = \mathbb{E}[e^{i\langle u, \sqrt{t}Z \rangle}] = \prod_{j=1}^d \mathbb{E}[e^{iu_j \sqrt{t}Z_j}] = \prod_{j=1}^d e^{-u_j^2 t/2} = e^{-|u|^2 t/2}.$$

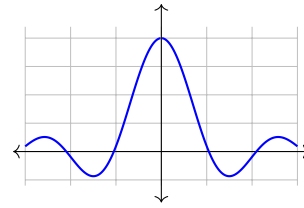
Comparing with the definition of $g_{1/t}$ gives $\hat{g}_t = (2\pi)^{d/2} t^{d/2} g_{1/t}$. Applying the transform twice,

$$\hat{\hat{g}}_t = (2\pi)^{d/2} t^{d/2} \hat{g}_{1/t} = (2\pi)^{d/2} t^{d/2} (2\pi)^{d/2} t^{-d/2} g_t = (2\pi)^d g_t,$$

and since g_t is even, $\hat{\hat{g}}_t(x) = \int e^{-i\langle u, x \rangle} \hat{g}_t(u) \, du$, which is the inversion formula. $\square$



$$X \sim N(0, 1) \\ \varphi(u) = e^{-u^2/2}$$



$$X \sim U[-1, 1] \\ \varphi(u) = \frac{\sin u}{u}$$

Definition 38.9.9 (Gaussian convolution)

A **Gaussian convolution** is a function of the form $f * g_t$ with $f \in L^1(\mathbb{R}^d)$ and $t > 0$.

Gaussian convolutions are the technical device that makes everything work: they are smooth, their Fourier transforms decay rapidly, and they approximate arbitrary L^p functions.

Lemma 38.9.10

Fourier inversion holds for every Gaussian convolution:

$$(f * g_t)(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \widehat{f * g_t}(u) \, du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \hat{f}(u) e^{-|u|^2 t/2} \, du.$$

Proof. Note first that $\widehat{f * g_t}(u) = \hat{f}(u) e^{-|u|^2 t/2}$ is integrable, since $|\hat{f}| \leq \|f\|_1$. Now use inversion for g_t and Fubini (justified since $(x, u) \mapsto f(x - y) \hat{g}_t(u)$ is absolutely

integrable):

$$\begin{aligned}(2\pi)^d(f * g_t)(x) &= \int_{\mathbb{R}^d} f(x-y) \left(\int_{\mathbb{R}^d} e^{-i\langle u, y \rangle} \hat{g}_t(u) \, du \right) dy \\ &= \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \left(\int_{\mathbb{R}^d} f(x-y) e^{i\langle u, x-y \rangle} dy \right) \hat{g}_t(u) \, du \\ &= \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \hat{f}(u) \hat{g}_t(u) \, du,\end{aligned}$$

substituting $z = x - y$ in the inner integral. $\square$

Lemma 38.9.11 (Gaussian convolutions approximate)

Let $1 \leq p < \infty$ and $f \in L^p(\mathbb{R}^d)$. Then $\|f * g_t - f\|_p \rightarrow 0$ as $t \downarrow 0$.

Proof. The space $C_c(\mathbb{R}^d)$ of continuous compactly supported functions is dense in $L^p(\mathbb{R}^d)$. Given $\varepsilon > 0$ pick $h \in C_c$ with $\|f - h\|_p < \varepsilon/3$; by Proposition 38.9.5,

$$\|f * g_t - h * g_t\|_p = \|(f - h) * g_t\|_p \leq \|f - h\|_p < \varepsilon/3,$$

so it suffices to prove the result for $h \in C_c(\mathbb{R}^d)$.

Let $e(y) = \int_{\mathbb{R}^d} |h(x-y) - h(x)|^p \, dx$. Since h is bounded with compact support, dominated convergence shows e is continuous with $e(0) = 0$, and $e(y) \leq 2^{p+1} \|h\|_p^p$ for all y . By Jensen as in Proposition 38.9.5,

$$\begin{aligned}\|h * g_t - h\|_p^p &\leq \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} |h(x-y) - h(x)|^p \, dx g_t(y) \, dy \\ &= \int_{\mathbb{R}^d} e(y) g_t(y) \, dy = \int_{\mathbb{R}^d} e(\sqrt{t}z) g_1(z) \, dz,\end{aligned}$$

substituting $y = \sqrt{t}z$. As $t \downarrow 0$ the integrand tends to $e(0) = 0$ pointwise and is dominated by $2^{p+1} \|h\|_p^p g_1(z)$, so the integral tends to 0. $\square$

§38.9.4 Fourier inversion and Plancherel

Theorem 38.9.12 (Fourier inversion)

Let $f \in L^1(\mathbb{R}^d)$ with $\hat{f} \in L^1(\mathbb{R}^d)$. Then for almost every x ,

$$f(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \hat{f}(u) \, du.$$

Proof. By Lemma 38.9.10, for every $t > 0$,

$$(f * g_t)(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \hat{f}(u) e^{-|u|^2 t/2} \, du.$$

As $t \downarrow 0$ the right-hand side converges, for every x , to $\frac{1}{(2\pi)^d} \int e^{-i\langle u, x \rangle} \hat{f}(u) \, du$ by dominated convergence with dominating function $|\hat{f}| \in L^1$. On the left, Lemma 38.9.11 gives $f * g_t \rightarrow f$ in L^1 , hence in measure by Markov's inequality, hence

almost everywhere along a subsequence $t_k \downarrow 0$. Uniqueness of limits gives the result almost everywhere. $\square$

Corollary 38.9.13

The Fourier transform is injective on $L^1(\mathbb{R}^d)$: if $\hat{f} = \hat{g}$ then $f = g$ almost everywhere.

Proof. Apply the theorem to $f - g$, whose transform is $0 \in L^1$. $\square$

Theorem 38.9.14 (Plancherel identity)

For $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ we have $\|\hat{f}\|_2 = (2\pi)^{d/2} \|f\|_2$.

Proof. Suppose first that $\hat{f} \in L^1$ as well. Then f and $\hat{f}$ are both bounded (being inverse transforms of L^1 functions) and integrable, so Fubini applies to $(x, u) \mapsto f(x)\hat{f}(u)$, and using inversion,

$$\begin{aligned} (2\pi)^d \|f\|_2^2 &= (2\pi)^d \int f(x) \overline{f(x)} \, dx = \int \left(\int e^{-i\langle u, x \rangle} \hat{f}(u) \, du \right) \overline{f(x)} \, dx \\ &= \int \hat{f}(u) \overline{\left(\int e^{i\langle u, x \rangle} f(x) \, dx \right)} \, du = \int |\hat{f}(u)|^2 \, du = \|\hat{f}\|_2^2. \end{aligned}$$

For general $f \in L^1 \cap L^2$, apply the above to $f_t = f * g_t$, whose transform $\hat{f} e^{-|u|^2 t/2}$ is integrable. By Lemma 38.9.11, $\|f_t\|_2 \rightarrow \|f\|_2$; and $|\hat{f}(u)|^2 e^{-|u|^2 t}$ increases to $|\hat{f}(u)|^2$ as $t \downarrow 0$, so $\|\hat{f}_t\|_2^2 \uparrow \|\hat{f}\|_2^2$ by monotone convergence. Passing to the limit in $(2\pi)^d \|f_t\|_2^2 = \|\hat{f}_t\|_2^2$ gives the result. $\square$

Corollary 38.9.15 (The Fourier transform on L^2)

The map $F_0(f) = (2\pi)^{-d/2} \hat{f}$, defined on the dense subspace $L^1 \cap L^2$ of $L^2(\mathbb{R}^d)$, extends uniquely to a linear isometry $F : L^2(\mathbb{R}^d) \rightarrow L^2(\mathbb{R}^d)$, the **Fourier–Plancherel transform**. It is in fact a surjective isometry, with $\langle Ff, Fg \rangle = \langle f, g \rangle$.

Proof. $L^1 \cap L^2$ contains all $f \mathbb{1}_{B(0,n)}$ for $f \in L^2$, hence is dense; F_0 is an isometry there by Theorem 38.9.14; and L^2 is complete, so the extension exists and is unique. Preservation of the inner product follows by polarisation. $\square$

Example 38.9.16

Not every measure has an L^1 transform. For the Dirac mass δ_0 on $\mathbb{R}$ we get $\hat{\delta}_0(u) = \int e^{iux} \, d\delta_0(x) = 1$, which is not integrable, and indeed δ_0 has no density. This is exactly the case the inversion theorem excludes.

§38.9.5 Characteristic functions determine laws

Theorem 38.9.17 (Uniqueness)

Let X be an $\mathbb{R}^d$ -valued random variable. Then φ_X determines μ_X . If moreover $\varphi_X \in L^1(\mathbb{R}^d)$, then μ_X has a density given almost everywhere by

$$f_X(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \varphi_X(u) \, du.$$

Proof. Let Z be a standard Gaussian vector in $\mathbb{R}^d$ independent of X , so $\sqrt{t}Z$ has density g_t and $X + \sqrt{t}Z$ has law $\mu_X * g_t$, which has density $f_t = \mu_X * g_t$. Since $\mu_X * g_t = (\mu_X * g_{t/2}) * g_{t/2}$ and $\mu_X * g_{t/2} \in L^1$, this is a Gaussian convolution, so Lemma 38.9.10 gives

$$f_t(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{-i\langle u, x \rangle} \varphi_X(u) e^{-|u|^2 t/2} \, du,$$

which depends on μ_X only through φ_X .

Now let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and continuous. Then $g(X + \sqrt{t}Z) \rightarrow g(X)$ almost surely as $t \downarrow 0$, and $|g(X + \sqrt{t}Z)| \leq \|g\|_\infty$, so by bounded convergence

$$\int_{\mathbb{R}^d} g(x) f_t(x) \, dx = \mathbb{E}[g(X + \sqrt{t}Z)] \rightarrow \mathbb{E}[g(X)] = \int_{\mathbb{R}^d} g \, d\mu_X.$$

The left-hand side is determined by φ_X , hence so is $\int g \, d\mu_X$ for every bounded continuous g ; and two Borel probability measures agreeing against all such g are equal. So φ_X determines μ_X .

If $\varphi_X \in L^1$, dominated convergence shows $f_t(x) \rightarrow f_X(x)$ for every x , with f_X as displayed, and $f_X \geq 0$ as a limit of the non-negative f_t . For g bounded continuous with compact support, $|g(x)f_t(x)| \leq \|g\|_\infty (2\pi)^{-d} \|\varphi_X\|_1 \mathbb{1}_{\text{supp } g}$, an integrable bound, so

$$\int g f_X = \lim_{t \downarrow 0} \int g f_t = \int g \, d\mu_X,$$

whence μ_X has density f_X . □

§38.9.6 Weak convergence and Lévy's continuity theorem

Definition 38.9.18 (Weak convergence)

Borel probability measures μ_n on $\mathbb{R}^d$ **converge weakly** to μ , written $\mu_n \xrightarrow{w} \mu$, if $\mu_n(g) \rightarrow \mu(g)$ for every bounded continuous $g : \mathbb{R}^d \rightarrow \mathbb{R}$. For random vectors we say $X_n \rightarrow X$ weakly if $\mu_{X_n} \xrightarrow{w} \mu_X$.

Remark. For $d = 1$ weak convergence coincides with convergence in distribution as defined in §7. (One direction: apply the definition to continuous approximations of $\mathbb{1}_{(-\infty, x]}$ from above and below. The other uses the Skorokhod representation or a direct argument with the distribution functions.) It also suffices to test against $g \in C_c^\infty(\mathbb{R}^d)$, the smooth compactly supported functions, by a routine approximation.

Theorem 38.9.19 (Lévy's continuity theorem)

Let X_n, X be $\mathbb{R}^d$ -valued random vectors with $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$ for every $u \in \mathbb{R}^d$. Then $X_n \rightarrow X$ weakly.

Proof. By the remark it suffices to show $\mathbb{E}[g(X_n)] \rightarrow \mathbb{E}[g(X)]$ for $g \in C_c^\infty(\mathbb{R}^d)$. Such a g is bounded, integrable, and Lipschitz with some constant $\|g\|_{\text{Lip}}$.

Let Z be a standard Gaussian vector independent of everything, and let $\varepsilon > 0$. Choose $t > 0$ small enough that $2\|g\|_{\text{Lip}}\sqrt{t}\mathbb{E}|Z| < 2\varepsilon/3$. Then

$$\begin{aligned} |\mathbb{E}[g(X_n)] - \mathbb{E}[g(X)]| &\leq \mathbb{E}|g(X_n) - g(X_n + \sqrt{t}Z)| + \mathbb{E}|g(X) - g(X + \sqrt{t}Z)| \\ &\quad + |\mathbb{E}[g(X_n + \sqrt{t}Z)] - \mathbb{E}[g(X + \sqrt{t}Z)]| \\ &\leq \frac{2\varepsilon}{3} + |\mathbb{E}[g(X_n + \sqrt{t}Z)] - \mathbb{E}[g(X + \sqrt{t}Z)]|, \end{aligned}$$

using the Lipschitz bound on the first two terms. For the remaining term, write $f_{t,n} = \mu_{X_n} * g_t$ for the density of $X_n + \sqrt{t}Z$; by Lemma 38.9.10,

$$\mathbb{E}[g(X_n + \sqrt{t}Z)] = \int_{\mathbb{R}^d} g(x) f_{t,n}(x) \, dx = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} g(x) e^{-i\langle u, x \rangle} \varphi_{X_n}(u) e^{-\frac{|u|^2 t}{2}} \, du \, dx.$$

The integrand converges pointwise as $n \rightarrow \infty$ (since $\varphi_{X_n} \rightarrow \varphi_X$ pointwise) and is dominated by $|g(x)|e^{-|u|^2 t/2}$, which is integrable on $\mathbb{R}^d \times \mathbb{R}^d$ because $|\varphi_{X_n}| \leq 1$. So by dominated convergence the whole expression converges to $\mathbb{E}[g(X + \sqrt{t}Z)]$, and the last term is below $\varepsilon/3$ for n large. $\square$

The converse is trivial: if $X_n \rightarrow X$ weakly, then testing against $x \mapsto \cos\langle u, x \rangle$ and $x \mapsto \sin\langle u, x \rangle$ gives $\varphi_{X_n}(u) \rightarrow \varphi_X(u)$. So convergence in distribution is *equivalent* to pointwise convergence of characteristic functions, and this is what makes the following possible.

§38.9.7 The central limit theorem

Theorem 38.9.20 (Central limit theorem)

Let (X_n) be i.i.d. real random variables with $\mathbb{E}[X_1] = 0$ and $\text{Var}(X_1) = 1$, and let $S_n = \sum_{i=1}^n X_i$. Then

$$\frac{S_n}{\sqrt{n}} \longrightarrow Z \sim N(0, 1) \quad \text{weakly, i.e. } \mathbb{P}\left(\frac{S_n}{\sqrt{n}} \leq x\right) \rightarrow \mathbb{P}(Z \leq x) \text{ for all } x \in \mathbb{R}.$$

Proof. Let $\varphi = \varphi_{X_1}$. Since $\mathbb{E}[X_1^2] < \infty$, Theorem 38.4.14 lets us differentiate twice under the expectation:

$$\varphi'(u) = i\mathbb{E}[X_1 e^{iuX_1}], \quad \varphi''(u) = -\mathbb{E}[X_1^2 e^{iuX_1}],$$

the dominating functions being $|X_1|$ and X_1^2 . So $\varphi(0) = 1$, $\varphi'(0) = i\mathbb{E}[X_1] = 0$ and $\varphi''(0) = -\mathbb{E}[X_1^2] = -1$. Taylor's theorem with Peano remainder gives

$$\varphi(v) = 1 - \frac{v^2}{2} + o(v^2) \quad \text{as } v \rightarrow 0.$$

Now by independence,

$$\varphi_{S_n/\sqrt{n}}(u) = \mathbb{E}\left[e^{i\frac{u}{\sqrt{n}}(X_1 + \dots + X_n)}\right] = \prod_{j=1}^n \mathbb{E}\left[e^{i\frac{u}{\sqrt{n}}X_j}\right] = \left[\varphi\left(\frac{u}{\sqrt{n}}\right)\right]^n = \left[1 - \frac{u^2}{2n} + o\left(\frac{1}{n}\right)\right]^n.$$

Fix u . For n large the bracket is within a neighbourhood of 1 where the principal branch of $\log$ satisfies $\log(1+z) = z + O(|z|^2)$, so

$$\log \varphi_{S_n/\sqrt{n}}(u) = n \log \left(1 - \frac{u^2}{2n} + o(n^{-1}) \right) = n \left(-\frac{u^2}{2n} + o(n^{-1}) \right) \longrightarrow -\frac{u^2}{2}.$$

Hence $\varphi_{S_n/\sqrt{n}}(u) \rightarrow e^{-u^2/2} = \varphi_Z(u)$ for every u , and Lévy's continuity theorem gives $S_n/\sqrt{n} \rightarrow Z$ weakly. Since the distribution function of Z is continuous everywhere, the convergence holds at every x . $\square$

Remark. For general i.i.d. X_i with mean ν and variance $\sigma^2 \in (0, \infty)$, apply the theorem to $(X_i - \nu)/\sigma$ to get

$$\frac{S_n - n\nu}{\sigma\sqrt{n}} \longrightarrow N(0, 1) \quad \text{weakly.}$$

This is the version met in IA Probability, where it was proved for special cases using moment generating functions; the argument here is the general one, and the reason it works is that characteristic functions always exist whereas moment generating functions need not.

§38.9.8 Gaussian random vectors

Definition 38.9.21

A random vector X in $\mathbb{R}^d$ is a **Gaussian vector** if $\langle X, v \rangle$ is a (possibly degenerate) real Gaussian random variable for every $v \in \mathbb{R}^d$.

Proposition 38.9.22

Let X be a Gaussian vector in $\mathbb{R}^d$ with mean $m = \mathbb{E}[X]$ and covariance matrix $V = (\text{Cov}(X_i, X_j))_{ij}$. Then:

- (i) $AX + b$ is a Gaussian vector in $\mathbb{R}^m$ for any $m \times d$ matrix A and $b \in \mathbb{R}^m$;
- (ii) $\varphi_X(u) = \exp(i\langle m, u \rangle - \frac{1}{2}\langle u, Vu \rangle)$, so m and V determine the law of X ;
- (iii) if V is invertible then X has density

$$f_X(x) = (2\pi)^{-d/2} (\det V)^{-1/2} \exp \left(-\frac{1}{2} \langle x - m, V^{-1}(x - m) \rangle \right);$$

- (iv) sub-vectors $X_{(1)}, X_{(2)}$ of X are independent if and only if $\text{Cov}(X_{(1)}, X_{(2)}) = 0$.

Proof sketch. (i) $\langle AX + b, v \rangle = \langle X, A^T v \rangle + \langle b, v \rangle$ is Gaussian.

(ii) $\langle u, X \rangle$ is real Gaussian with mean $\langle m, u \rangle$ and variance $\langle u, Vu \rangle$, and a real $N(\alpha, \sigma^2)$ variable Y has $\mathbb{E}[e^{iY}] = e^{i\alpha - \sigma^2/2}$ by Proposition 38.9.8 and translation. Taking $\varphi_X(u) = \mathbb{E}[e^{i\langle u, X \rangle}]$ gives the formula, and Theorem 38.9.17 gives uniqueness.

(iii) Write $V = \Sigma \Sigma^T$ and check that $m + \Sigma Z$, for Z standard Gaussian, has the stated characteristic function; then change variables in the density of Z .

(iv) If $\text{Cov}(X_{(1)}, X_{(2)}) = 0$ then V is block diagonal and φ_X factorises as $\varphi_{X_{(1)}} \varphi_{X_{(2)}}$, so by uniqueness $\mu_X = \mu_{X_{(1)}} \otimes \mu_{X_{(2)}}$ and Proposition 38.5.6 applies. The converse

is immediate. $\square$

Part (iv) deserves emphasis: for *jointly* Gaussian variables, uncorrelated is the same as independent. This is false in general – if $X \sim N(0, 1)$ and $\epsilon = \pm 1$ with equal probability independently of X , then X and ϵX are each standard Gaussian and uncorrelated, but obviously not independent; the pair $(X, \epsilon X)$ is not a Gaussian vector.

Proposition 38.9.23 (Slutsky-type results)

Let $X_n \rightarrow X$ weakly in $\mathbb{R}^d$. Then

- (i) $h(X_n) \rightarrow h(X)$ weakly for any continuous $h : \mathbb{R}^d \rightarrow \mathbb{R}^k$;
- (ii) if $|X_n - Y_n| \rightarrow 0$ in probability then $Y_n \rightarrow X$ weakly;
- (iii) if $Y_n \rightarrow c$ in probability for a constant c , then $(X_n, Y_n) \rightarrow (X, c)$ weakly.

In particular $X_n + Y_n \rightarrow X + c$ weakly, and for $d = 1$ also $X_n Y_n \rightarrow cX$ weakly.

Proof. (i) If g is bounded continuous then so is $g \circ h$.

(ii) It suffices to test against bounded Lipschitz g . Splitting on whether $|X_n - Y_n|$ exceeds $\eta = \varepsilon/(3\|g\|_{\text{Lip}})$,

$$\mathbb{E}|g(X_n) - g(Y_n)| \leq \|g\|_{\text{Lip}}\eta + 2\|g\|_{\infty}\mathbb{P}(|X_n - Y_n| > \eta) < \frac{2\varepsilon}{3}$$

for n large, and $|\mathbb{E}[g(X_n)] - \mathbb{E}[g(X)]| < \varepsilon/3$ for n large.

(iii) $(X_n, c) \rightarrow (X, c)$ weakly, since $g(\cdot, c)$ is bounded continuous; and $|(X_n, Y_n) - (X_n, c)| = |Y_n - c| \rightarrow 0$ in probability, so (ii) applies. $\square$

These are the results one uses in statistics to justify replacing an unknown variance by a consistent estimator inside a central limit theorem.

§38.10 Ergodic Theory

§38.10.1 Measure-preserving transformations

Ergodic theory studies the long-run behaviour of a system evolving under repeated application of a single map. We will develop just enough of it to prove the strong law of large numbers in full generality, which turns out to be a special case.

Throughout, $(E, \mathcal{E}, \mu)$ is a σ -finite measure space.

Definition 38.10.1 (Measure-preserving)

A measurable map $\Theta : E \rightarrow E$ is **measure-preserving** if $\mu(\Theta^{-1}(A)) = \mu(A)$ for all $A \in \mathcal{E}$.

If Θ is measure-preserving then $\mu \circ \Theta^{-1} = \mu$, so by the change of variables formula (Proposition 38.4.15), for any $f \in L^1(\mu)$ we have $f \circ \Theta \in L^1(\mu)$ with

$$\int_E f \circ \Theta \, d\mu = \int_E f \, d\mu.$$

Definition 38.10.2 (Invariance and ergodicity)

A set $A \in \mathcal{E}$ is **invariant** if $\Theta^{-1}(A) = A$, and a measurable $f : E \rightarrow \mathbb{R}$ is invariant if $f \circ \Theta = f$. We write $\mathcal{E}_\Theta$ for the collection of invariant sets.

We say Θ is **ergodic** if every invariant A satisfies $\mu(A) = 0$ or $\mu(A^c) = 0$.

One checks at once that $\mathcal{E}_\Theta$ is a σ -algebra, and that f is invariant if and only if it is $\mathcal{E}_\Theta$ -measurable. Consequently, if Θ is ergodic then every invariant f is almost everywhere constant: the sets $\{f \leq y\}$ are invariant, so $y \mapsto \mu(\{f \leq y\})$ takes only the values 0 and $\mu(E)$, and f equals $\inf\{y : \mu(f > y) = 0\}$ a.e.

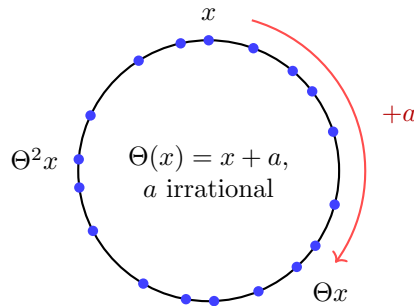
Ergodicity says that the system cannot be split into two non-trivial pieces which do not communicate. It is the dynamical analogue of irreducibility for a Markov chain.

Example 38.10.3 (Circle rotation)

Take $(E, \mathcal{E}, \mu) = ((0, 1], \mathcal{B}, \text{Lebesgue})$ and, for $a \in (0, 1)$,

$$\Theta_a(x) = x + a \bmod 1.$$

This preserves μ by the translation invariance established in §2.5. It is ergodic if and only if a is irrational. If $a = p/q$ is rational then $\Theta_a^q = \text{id}$, so the union of q well-separated small intervals spaced $1/q$ apart is invariant and non-trivial. If a is irrational, the orbit $\{x, x \oplus a, x \oplus 2a, \dots\}$ of any point is dense in the circle, and this can be turned into a proof of ergodicity (one convenient route uses Fourier coefficients: if f is invariant and in L^2 then its Fourier coefficients satisfy $\hat{f}(k)e^{2\pi ika} = \hat{f}(k)$, forcing $\hat{f}(k) = 0$ for $k \neq 0$).



Our object of study is the sequence of **ergodic averages**. For measurable f set

$$S_0(f) = 0, \quad S_n(f) = f + f \circ \Theta + \dots + f \circ \Theta^{n-1}.$$

The question is what $S_n(f)/n$ does as $n \rightarrow \infty$: does the time average of f along an orbit converge, and if so, to what?

§38.10.2 The maximal ergodic lemma

Everything follows from the following deceptively simple inequality.

Lemma 38.10.4 (Maximal ergodic lemma)

Let Θ be measure-preserving on $(E, \mathcal{E}, \mu)$ and let $f \in L^1(\mu)$. Put $S^* = \sup_{n \geq 0} S_n(f)$.

Then

$$\int_{\{S^* > 0\}} f \, d\mu \geq 0.$$

Proof. Put $S_n^* = \max_{0 \leq k \leq n} S_k$, so $S_n^* \geq 0$ (as $S_0 = 0$) and $S_n^* \uparrow S^*$. For $0 \leq k \leq n$,

$$S_{k+1} = f + S_k \circ \Theta \leq f + S_n^* \circ \Theta.$$

Let $A_n = \{S_n^* > 0\}$, so $A_n \uparrow \{S^* > 0\}$. On A_n the maximum defining S_n^* is attained at some $k \geq 1$, so

$$S_n^* = \max_{1 \leq k \leq n} S_k \leq \max_{0 \leq k \leq n-1} (f + S_n^* \circ \Theta) = f + S_n^* \circ \Theta \quad \text{on } A_n.$$

Integrating over A_n ,

$$\int_{A_n} S_n^* \, d\mu \leq \int_{A_n} S_n^* \circ \Theta \, d\mu + \int_{A_n} f \, d\mu.$$

On A_n^c we have $S_n^* = 0 \leq S_n^* \circ \Theta$, so adding $\int_{A_n^c} S_n^* \, d\mu \leq \int_{A_n^c} S_n^* \circ \Theta \, d\mu$ gives

$$\int_E S_n^* \, d\mu \leq \int_E S_n^* \circ \Theta \, d\mu + \int_{A_n} f \, d\mu = \int_E S_n^* \, d\mu + \int_{A_n} f \, d\mu,$$

using that Θ preserves μ . As S_n^* is integrable (being bounded by $\sum_{k \leq n} |f| \circ \Theta^k$), we may cancel and obtain $\int_{A_n} f \, d\mu \geq 0$. Finally $f \mathbb{1}_{A_n} \rightarrow f \mathbb{1}_{\{S^* > 0\}}$ pointwise with $|f \mathbb{1}_{A_n}| \leq |f| \in L^1$, so dominated convergence gives the claim. $\square$

§38.10.3 Birkhoff's ergodic theorem

Theorem 38.10.5 (Birkhoff)

Let Θ be measure-preserving on the σ -finite space $(E, \mathcal{E}, \mu)$ and let $f \in L^1(\mu)$. Then there is an invariant $\bar{f} \in L^1(\mu)$ with $\mu(|\bar{f}|) \leq \mu(|f|)$ and

$$\frac{S_n(f)}{n} \longrightarrow \bar{f} \quad \text{almost everywhere.}$$

If Θ is ergodic and μ is a probability measure, then $\bar{f} = \mu(f)$ almost everywhere.

Proof (non-examinable). Observe that

$$\limsup_n \frac{S_n(f)}{n} = \limsup_n \frac{S_n(f) \circ \Theta}{n},$$

since $S_{n+1}(f) = f + S_n(f) \circ \Theta$ and $f/n \rightarrow 0$; similarly for $\liminf$. So both $\limsup_n S_n/n$ and $\liminf_n S_n/n$ are invariant functions, hence $\mathcal{E}_\Theta$ -measurable. For $a < b$ put

$$D = D_{a,b} = \left\{ \liminf_n \frac{S_n(f)}{n} < a < b < \limsup_n \frac{S_n(f)}{n} \right\},$$

a measurable invariant set. We show $\mu(D) = 0$.

Assume without loss of generality $b > 0$ (otherwise replace f by $-f$ and swap the roles of a and b). Let $B \subseteq D$ be measurable with $\mu(B) < \infty$, and put $g = f - b\mathbb{1}_B \in L^1(\mu)$. Then

$$S_n(g) = S_n(f) - bS_n(\mathbb{1}_B) \geq S_n(f) - bn,$$

which is positive for some n at every point of D , by the definition of $\limsup$. Restrict the measure space to D : since D is invariant, Θ is still measure-preserving for $\mu|_D$, because

$$\mu|_D(\Theta^{-1}A) = \mu(\Theta^{-1}A \cap D) = \mu(\Theta^{-1}(A \cap D)) = \mu(A \cap D) = \mu|_D(A).$$

On this restricted space $\{S^*(g) > 0\} = D$, so the maximal ergodic lemma gives

$$0 \leq \int_D g \, d\mu = \int_D f \, d\mu - b\mu(B),$$

that is, $b\mu(B) \leq \int_D f \, d\mu$. By σ -finiteness we may choose $B_n \uparrow D$ with $\mu(B_n) < \infty$ and let $n \rightarrow \infty$, obtaining $b\mu(D) \leq \int_D f \, d\mu$. Running the same argument with $-f$ and $-a$ in place of f and b gives $-a\mu(D) \leq -\int_D f \, d\mu$, so

$$b\mu(D) \leq \int_D f \, d\mu \leq a\mu(D).$$

Since $a < b$ this forces $\mu(D) = 0$ or $\mu(D) = \infty$; but $\int_D f \, d\mu$ is finite, so $\mu(D) = 0$.

Now let

$$\Delta = \left\{ \liminf_n \frac{S_n(f)}{n} < \limsup_n \frac{S_n(f)}{n} \right\} = \bigcup_{a < b, a, b \in \mathbb{Q}} D_{a,b},$$

a countable union of null sets, so $\mu(\Delta) = 0$. Off Δ the limit $\lim_n S_n(f)/n$ exists in $[-\infty, \infty]$; define $\bar{f}$ to be that limit on Δ^c and 0 on Δ . Then $\bar{f}$ is invariant and $S_n(f)/n \rightarrow \bar{f}$ almost everywhere.

Finally, since Θ is measure-preserving, $\mu(|f \circ \Theta^k|) = \mu(|f|)$, so $\mu(|S_n(f)|) \leq n\mu(|f|)$, and by Fatou

$$\mu(|\bar{f}|) = \mu\left(\liminf_n \left| \frac{S_n(f)}{n} \right| \right) \leq \liminf_n \mu\left(\left| \frac{S_n(f)}{n} \right| \right) \leq \mu(|f|) < \infty.$$

In particular $\bar{f}$ is finite almost everywhere. If Θ is ergodic then $\bar{f}$ is a.e. constant; if additionally $\mu(E) = 1$, the von Neumann theorem below identifies the constant as $\mu(f)$. $\square$

§38.10.4 Von Neumann's L^p ergodic theorem

Theorem 38.10.6 (von Neumann)

Let $(E, \mathcal{E}, \mu)$ be a *finite* measure space, Θ measure-preserving, and $1 \leq p < \infty$. Then for $f \in L^p(\mu)$,

$$\frac{S_n(f)}{n} \longrightarrow \bar{f} \quad \text{in } L^p(\mu),$$

where $\bar{f}$ is the almost-everywhere limit from Birkhoff's theorem.

Proof. Since Θ preserves μ , $\|f \circ \Theta^i\|_p = \|f\|_p$, so by Minkowski

$$\left\| \frac{S_n(f)}{n} \right\|_p \leq \frac{1}{n} \sum_{i=0}^{n-1} \|f \circ \Theta^i\|_p = \|f\|_p. \quad (\#)$$

So $f \mapsto S_n(f)/n$ is a contraction of L^p , uniformly in n ; this is the key.

Fix $\varepsilon > 0$. For $K > 0$ let $f_K = (-K) \vee f \wedge K$. Then

$$\|f - f_K\|_p^p = \int_E |f - f_K|^p \mathbb{1}_{\{|f| > K\}} d\mu \rightarrow 0$$

as $K \rightarrow \infty$ by dominated convergence (the integrand is at most $2^p |f|^p \in L^1$ and tends to 0 pointwise). Choose K with $\|f - f_K\|_p < \varepsilon/3$.

Since μ is finite we have $f_K \in L^1(\mu)$, so Birkhoff gives $S_n(f_K)/n \rightarrow \bar{f}_K$ almost everywhere, and $|S_n(f_K)/n| \leq K$, so that $|\bar{f}_K| \leq K$. By bounded convergence (Theorem 38.4.10, with dominating constant $2K$, integrable as $\mu(E) < \infty$),

$$\left\| \frac{S_n(f_K)}{n} - \bar{f}_K \right\|_p^p \leq (2K)^{p-1} \left\| \frac{S_n(f_K)}{n} - \bar{f}_K \right\|_1 \rightarrow 0,$$

so this term is below $(\varepsilon/3)^p$ for n large.

Also $f \in L^p(\mu) \subseteq L^1(\mu)$ since μ is finite, so Birkhoff applies to f too and $S_n(f)/n \rightarrow \bar{f}$ a.e. By Fatou and the contraction property (#),

$$\begin{aligned} \|\bar{f} - \bar{f}_K\|_p^p &= \int_E \liminf_n \left| \frac{S_n(f - f_K)}{n} \right|^p d\mu \leq \liminf_n \left\| \frac{S_n(f - f_K)}{n} \right\|_p^p \\ &\leq \|f - f_K\|_p^p < (\varepsilon/3)^p. \end{aligned}$$

In particular $\bar{f} \in L^p$. Combining the three estimates via the triangle inequality,

$$\left\| \frac{S_n(f)}{n} - \bar{f} \right\|_p \leq \left\| \frac{S_n(f - f_K)}{n} \right\|_p + \left\| \frac{S_n(f_K)}{n} - \bar{f}_K \right\|_p + \|\bar{f}_K - \bar{f}\|_p < \varepsilon$$

for n large. □

Corollary 38.10.7

If μ is a probability measure and Θ is ergodic then, for $f \in L^1(\mu)$, $\bar{f} = \mu(f)$ almost everywhere.

Proof. Ergodicity makes the invariant function $\bar{f}$ a.e. constant, say $\bar{f} = c$. By von Neumann with $p = 1$, $\mu(S_n(f)/n) \rightarrow \mu(\bar{f}) = c$; but $\mu(S_n(f)/n) = \frac{1}{n} \sum_{k < n} \mu(f \circ \Theta^k) = \mu(f)$ for every n . So $c = \mu(f)$. □

§38.10.5 Bernoulli shifts

To apply the ergodic theorems to i.i.d. sequences we need the right dynamical system. Let $E = \mathbb{R}^{\mathbb{N}}$, the space of real sequences $x = (x_n)$, and let

$$\mathcal{C} = \left\{ \prod_{n=1}^{\infty} A_n : A_n \in \mathcal{B}, A_n = \mathbb{R} \text{ for all large } n \right\}$$

be the collection of **cylinder sets**, a π -system. Put $\mathcal{E} = \sigma(\mathcal{C})$, the **cylindrical σ -algebra**; it coincides with $\sigma(\{f_n : n \in \mathbb{N}\})$ where $f_n(x) = x_n$ are the coordinate maps, and with $\mathcal{B}(\mathbb{R}^{\mathbb{N}})$ for the product topology.

Let m be a probability measure on $\mathbb{R}$, and let (X_n) be i.i.d. with law m – these exist by Theorem 38.3.12. Define $X : \Omega \rightarrow E$ by $X(\omega) = (X_1(\omega), X_2(\omega), \dots)$. This is $\mathcal{F}$ - $\mathcal{E}$ measurable, since for a cylinder set A ,

$$X^{-1}(A) = \bigcap_{n=1}^N X_n^{-1}(A_n) \in \mathcal{F}.$$

Let $\mu = \mathbb{P} \circ X^{-1}$. On cylinder sets,

$$\mu\left(\prod_{n=1}^{\infty} A_n\right) = \lim_{N \rightarrow \infty} \mathbb{P}(X_1 \in A_1, \dots, X_N \in A_N) = \prod_{n=1}^{\infty} m(A_n)$$

(the product being eventually 1). By uniqueness of extension, μ is the unique probability measure on $(E, \mathcal{E})$ with this property; we write $\mu = m^{\otimes \mathbb{N}}$ and call $(E, \mathcal{E}, \mu)$ the **canonical model** for an i.i.d. sequence of law m .

Definition 38.10.8

The **shift map** $\Theta : E \rightarrow E$ is $\Theta(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots)$.

Theorem 38.10.9

The shift map Θ is measure-preserving and ergodic on $(E, \mathcal{E}, \mu)$.

Proof. For a cylinder set $A = \prod_n A_n$ with $A_n = \mathbb{R}$ for $n > N$,

$$\mu(\Theta^{-1}A) = \mathbb{P}(X_2 \in A_1, \dots, X_{N+1} \in A_N) = \prod_{n=1}^N m(A_n) = \mu(A),$$

using that the X_i are i.i.d. As the cylinder sets form a π -system generating $\mathcal{E}$ and both μ and $\mu \circ \Theta^{-1}$ are probability measures, uniqueness of extension gives $\mu \circ \Theta^{-1} = \mu$.

For ergodicity, recall the tail σ -algebra $\mathcal{T} = \bigcap_n \mathcal{T}_n$ with $\mathcal{T}_n = \sigma(f_k : k > n)$. For a cylinder set A ,

$$\Theta^{-n}(A) = \{x : (x_{n+1}, x_{n+2}, \dots) \in A\} \in \mathcal{T}_n,$$

and since $\{A : \Theta^{-n}(A) \in \mathcal{T}_n\}$ is a σ -algebra containing $\mathcal{C}$, this holds for all $A \in \mathcal{E}$. So if A is invariant then $A = \Theta^{-n}(A) \in \mathcal{T}_n$ for every n , whence $A \in \mathcal{T}$. By Kol-

Kolmogorov's zero-one law (Theorem 38.3.16), applied to the independent coordinate variables under μ , we get $\mu(A) \in \{0, 1\}$. $\square$

§38.10.6 The strong law of large numbers

We can now finish the job.

Theorem 38.10.10

Let m be a probability measure on $\mathbb{R}$ with $\int_{\mathbb{R}} |x| \, dm(x) < \infty$ and $\int_{\mathbb{R}} x \, dm(x) = \nu$. Then, on the canonical model $(E, \mathcal{E}, \mu) = (\mathbb{R}^{\mathbb{N}}, \sigma(\mathcal{C}), m^{\otimes \mathbb{N}})$,

$$\mu\left(\left\{x \in \mathbb{R}^{\mathbb{N}} : \frac{x_1 + \cdots + x_n}{n} \rightarrow \nu\right\}\right) = 1.$$

Proof. Take $f(x) = x_1$, the first coordinate map. Then by the change of variables formula,

$$\int_E |f| \, d\mu = \int_{\mathbb{R}} |x| \, dm(x) < \infty,$$

so $f \in L^1(\mu)$, and $\mu(f) = \nu$. Moreover

$$S_n(f)(x) = f(x) + f(\Theta x) + \cdots + f(\Theta^{n-1}x) = x_1 + x_2 + \cdots + x_n.$$

By Theorem 38.10.9 the shift is measure-preserving and ergodic, so Birkhoff's theorem gives $S_n(f)/n \rightarrow \bar{f}$ almost everywhere, and Corollary 38.10.7 identifies $\bar{f} = \mu(f) = \nu$. $\square$

Proof of Theorem 38.8.3. Let (X_n) be i.i.d. on $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}|X_1| < \infty$ and common law m , and let $X : \Omega \rightarrow \mathbb{R}^{\mathbb{N}}$ be the associated map, so $\mathbb{P} \circ X^{-1} = m$. Then

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mathbb{E}[X_1]\right) = \mu\left(\left\{x : \frac{x_1 + \cdots + x_n}{n} \rightarrow \nu\right\}\right) = 1$$

by Theorem 38.10.10, since the event on the left is the X -preimage of the event on the right. Convergence in $\mathcal{L}^1$ follows from von Neumann's theorem with $p = 1$ applied to the same f . $\square$

Remark. It is worth appreciating what has happened. The strong law is a statement about i.i.d. sequences, and its proof consisted of: (a) realising the sequence as a point of a single measure space $\mathbb{R}^{\mathbb{N}}$; (b) noticing that 'shift the sequence along by one' is a measure-preserving ergodic transformation of that space; and (c) applying a completely general theorem about time averages. The independence enters only through Kolmogorov's zero-one law in step (b). Birkhoff's theorem is much more general than the strong law – it applies to stationary sequences that are far from independent – and this is why ergodic theory has become a subject in its own right.

Example 38.10.11 (Normal numbers)

Take m uniform on $\{0, 1, \dots, 9\}$, so that (X_n) are the digits of a uniformly random real in $[0, 1)$. Applying the strong law to $\mathbb{1}_{\{X_n=k\}}$ gives that, almost surely, the

digit k has asymptotic frequency $1/10$ in the decimal expansion, for each k . Since a countable intersection of almost sure events is almost sure, Lebesgue-almost every real number is **normal** to base 10. Exhibiting one explicitly is another matter entirely.

§38.11 Closing Remarks

It is worth stepping back to see the shape of what we have built.

We began with the problem of assigning a size to subsets of a space, and found that we could not do it for all of them – Carathéodory’s theorem told us how far we could get, Dynkin’s lemma told us that the answer was unique, and Vitali told us that we could not do better. From there, the class of measurable functions and the Lebesgue integral came almost for free, and the payoff was immediate: the monotone convergence theorem, Fatou’s lemma and the dominated convergence theorem give us exactly the licence to exchange limits and integrals that the Riemann theory withheld. Fubini’s theorem let us integrate in product spaces, which is to say, handle several random variables at once.

The second half was the harvest. The L^p spaces gave us a complete normed setting for analysis, with L^2 a Hilbert space in which conditional expectation is nothing more exotic than orthogonal projection. Uniform integrability isolated exactly the hypothesis under which convergence in probability upgrades to convergence in L^1 . Characteristic functions turned convolution into multiplication and thereby turned the central limit theorem into a Taylor expansion. And ergodic theory turned the strong law of large numbers into a statement about a shift map on sequence space.

Two threads run through all of it. The first is the π -system argument: almost every uniqueness statement we proved – uniqueness of Lebesgue measure, of product measures, of the law determined by a distribution function or a characteristic function, of conditional expectation – came from Dynkin’s lemma or its functional cousin the monotone class theorem. The second is approximation from below: simple functions approximate non-negative measurable functions, and every theorem about the integral was proved first for indicators, then simple functions, then by monotone convergence in general. If you remember nothing else from this course, remember those two techniques; between them they prove most of it.

39 Quantum Information and Computation

A computer is a physical object, and a computation is something that a physical object does. That sounds like a platitude, but it has teeth: whatever we can and cannot compute is decided not by mathematics alone but by the laws of physics that the computer happens to obey. For most of the twentieth century the physics we implicitly assumed was classical, and the resulting theory of computation – Turing machines, Boolean circuits, the class P – is the one that everybody learns first. Nature, however, is not classical. If we allow our computer to store information in genuinely quantum degrees of freedom, and to process it by genuinely quantum evolutions, we get a strictly richer toolkit: superposition, entanglement, interference, and measurement.

In this course we will see what that toolkit buys us. We will meet tasks that are flatly *impossible* classically but routine quantumly (teleporting a state using only classical communication, distributing a provably secure cryptographic key), tasks that quantum mechanics *forbids* (cloning an unknown state, signalling faster than light), and – the headline act – algorithms that solve computational problems dramatically faster than any classical algorithm we know, culminating in Shor’s polynomial-time factoring algorithm and Grover’s quadratic speedup for unstructured search. Along the way the mathematics stays remarkably elementary: almost everything is linear algebra on finite-dimensional complex vector spaces, done in a notation that makes the bookkeeping painless.

This article constitutes my notes for the ‘Quantum Information and Computation’ course, held in Lent 2023 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§39.1 Quantum States

We begin with the raw material. Everything in this course takes place in a finite-dimensional complex inner product space, and almost everything reduces to manipulating vectors and operators on such a space. The reader will have met all of this before in IB Quantum Mechanics and in IA Vectors and Matrices; the only genuinely new ingredient is a notation, and a habit of thinking of the vectors as *information*.

§39.1.1 Dirac Notation

Let $\mathcal{V}$ be a finite-dimensional complex vector space equipped with a Hermitian inner product¹. Following Dirac, we write a vector of $\mathcal{V}$ as $|v\rangle$, and call it a **ket**. The inner product of $|v\rangle$ and $|w\rangle$ is written $\langle v|w\rangle$, and we adopt the physicists’ convention that it is *antilinear in the first argument and linear in the second*, so that

$$\langle v|\lambda w_1 + \mu w_2\rangle = \lambda \langle v|w_1\rangle + \mu \langle v|w_2\rangle, \quad \langle \lambda v_1 + \mu v_2|w\rangle = \lambda^* \langle v_1|w\rangle + \mu^* \langle v_2|w\rangle,$$

¹Strictly, the state space of a quantum system is a *Hilbert space*: a complex inner product space that is complete with respect to the induced norm. In finite dimensions completeness is automatic, and we will never need infinite-dimensional spaces in this course.

with $\langle v|w\rangle = \langle w|v\rangle^*$ and $\langle v|v\rangle \geq 0$, with equality if and only if $|v\rangle = 0$. The induced norm is $\| |v\rangle \| = \sqrt{\langle v|v\rangle}$.

Once a vector $|v\rangle$ is fixed, the map $|w\rangle \mapsto \langle v|w\rangle$ is a linear functional on $\mathcal{V}$, that is, an element of the dual space $\mathcal{V}^*$. We write it $\langle v|$ and call it a **bra**. Concretely, if we have chosen an orthonormal basis and written $|v\rangle$ as a column vector of components, then $\langle v|$ is the conjugate-transposed row vector,

$$|v\rangle = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \langle v| = (a_1^* \quad \cdots \quad a_n^*),$$

and $\langle v|w\rangle$ is literally the matrix product $\langle v|w\rangle$. This is the whole point of the notation: the ‘bracket’ $\langle v|w\rangle$ splits into a bra and a ket, and every expression we write can be read as a product of compatible matrices.

The other product we can form is the **outer product** $|v\rangle\langle w|$, which is the linear map

$$|v\rangle\langle w| : |x\rangle \longmapsto |v\rangle \langle w|x\rangle.$$

As a matrix this is a column times a row, so it is an $n \times n$ matrix of rank at most one. Outer products are the reason Dirac notation is so pleasant to compute with: operators, states and inner products all live in the same formalism, and juxtaposition always means composition.

Definition 39.1.1 (Complete Orthonormal Basis)

An orthonormal basis $\{|e_1\rangle, \dots, |e_n\rangle\}$ of $\mathcal{V}$ is **complete** if

$$\sum_{i=1}^n |e_i\rangle\langle e_i| = I.$$

In finite dimensions any orthonormal basis is complete – the identity above is just the statement that $|v\rangle = \sum_i |e_i\rangle \langle e_i|v\rangle$, which is the usual expansion of a vector in an orthonormal basis. The completeness relation is nonetheless worth naming, because inserting a resolution of the identity in the middle of an expression is by far the most common computational move we will make.

If $|\psi\rangle = \sum_i c_i |e_i\rangle$ then $c_i = \langle e_i|\psi\rangle$, and

$$\langle \psi|\psi\rangle = \sum_i |c_i|^2.$$

We call $|\psi\rangle$ **normalised** if this is 1, in which case the numbers $|c_i|^2$ form a probability distribution on the basis labels. This is not a coincidence, as we will see when we come to measurement; the c_i are called **probability amplitudes**.

§39.1.2 Qubits

The elementary unit of classical information is the **bit**, an element of $\{0, 1\}$. Its quantum counterpart is a two-dimensional state space.

Definition 39.1.2 (Qubit)

A **qubit** is a quantum system whose state space is $\mathcal{V} = \mathbb{C}^2$, equipped with a distinguished orthonormal basis

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

called the **computational basis** (or standard basis, or Z -basis).

Physically a qubit could be the spin of a spin- $\frac{1}{2}$ particle, with $|0\rangle$ and $|1\rangle$ the spin-up and spin-down states along a chosen axis; or the polarisation of a photon, with $|0\rangle, |1\rangle$ horizontal and vertical. For our purposes the physical realisation is irrelevant: all that matters is that we have a two-dimensional space and can prepare, evolve and measure states in it.

The crucial difference from a classical bit is that a general state of a qubit is a *superposition*

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |a|^2 + |b|^2 = 1.$$

A second basis will be with us constantly.

Definition 39.1.3 (Conjugate Basis)

The **conjugate basis** (or Hadamard basis, or X -basis) of a qubit is $\{|+\rangle, |-\rangle\}$, where

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

One checks immediately that this is orthonormal, and that $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$, so the relationship is symmetric. The pair of bases has the property that a basis vector of one is an equal superposition of the basis vectors of the other; such a pair is called **mutually unbiased**, and this is exactly the property that will make the BB84 protocol of §39.8.2 secure.

There is one further subtlety about states which we should get out of the way immediately. The state of a system is not really a vector, but a *ray*: the vectors $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ represent the same physical state, and no measurement whatsoever can tell them apart. We call $e^{i\theta}$ a **global phase**, and we are free to discard it whenever convenient. What we may *not* discard is a *relative* phase: the states

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad \frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$$

are genuinely different states for $\theta \neq 0$, and are readily distinguished by measuring in the conjugate basis. Keeping this distinction straight is one of the small skills the course teaches.

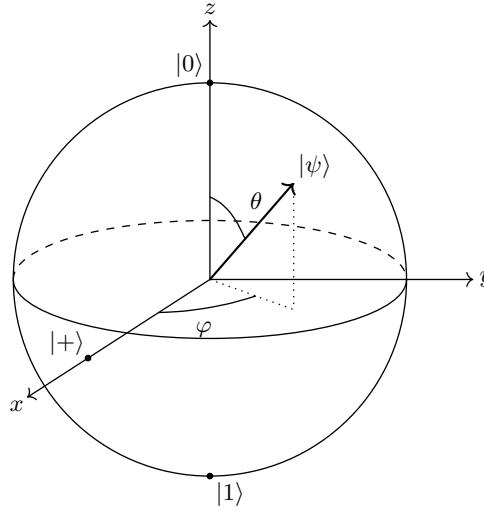
§39.1.3 The Bloch Sphere

Because a qubit state is a normalised vector in $\mathbb{C}^2$ up to global phase, it is described by exactly two real parameters, and there is a very useful picture of the resulting two-parameter family.

Write $|\psi\rangle = a|0\rangle + b|1\rangle$ with $|a|^2 + |b|^2 = 1$. Using the freedom of global phase we may take a real and non-negative, so $a = \cos(\theta/2)$ for a unique $\theta \in [0, \pi]$, and then $|b| = \sin(\theta/2)$, so $b = e^{i\varphi} \sin(\theta/2)$ for some $\varphi \in [0, 2\pi)$. Thus every qubit state can be written

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle,$$

and the pair (θ, φ) is unique except at the poles $\theta = 0, \pi$, where φ is irrelevant. But (θ, φ) are precisely spherical polar coordinates on the unit sphere $S^2 \subseteq \mathbb{R}^3$. So the pure states of a qubit are in bijection with the points of a sphere, called the **Bloch sphere**.



The poles are $|0\rangle$ (north, $\theta = 0$) and $|1\rangle$ (south, $\theta = \pi$); the state $|+\rangle$ where the x -axis meets the equator, with $|-\rangle$ antipodal to it round the back; and $\frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$ sit at the two ends of the y -axis. The picture is a good source of intuition, but it comes with a warning attached: *orthogonal states are antipodal, not perpendicular*. The angle between two states on the Bloch sphere is twice the angle between them in $\mathbb{C}^2$.

Remark. The Bloch sphere is genuinely special to two dimensions. For a d -dimensional system the set of pure states is complex projective space $\mathbb{CP}^{d-1}$, which for $d = 2$ happens to be a 2-sphere but for $d \geq 3$ has no such friendly picture. This is one reason why almost all of our concrete examples will be built from qubits.

§39.1.4 Composite Systems and Tensor Products

A single qubit is not much of a computer. To do anything interesting we need to combine systems, and the mathematical operation that does this is the tensor product.

Definition 39.1.4 (Tensor Product)

Let $\mathcal{V}$ and $\mathcal{W}$ be inner product spaces with orthonormal bases $\{|e_1\rangle, \dots, |e_m\rangle\}$ and $\{|f_1\rangle, \dots, |f_n\rangle\}$. Their **tensor product** $\mathcal{V} \otimes \mathcal{W}$ is the mn -dimensional space with orthonormal basis $\{|e_i\rangle \otimes |f_j\rangle\}_{i,j}$, and the map $(|v\rangle, |w\rangle) \mapsto |v\rangle \otimes |w\rangle$ is extended bilinearly.

Concretely, if $|v\rangle$ has components $(a_1, \dots, a_m)$ and $|w\rangle$ has components $(b_1, \dots, b_n)$, then $|v\rangle \otimes |w\rangle$ has the mn components $a_i b_j$, listed in lexicographic order:

$$|v\rangle \otimes |w\rangle = (a_1 b_1, a_1 b_2, \dots, a_1 b_n, a_2 b_1, \dots, a_m b_n)^\top.$$

The inner product on $\mathcal{V} \otimes \mathcal{W}$ is determined by

$$(\langle v_1| \otimes \langle w_1|)(|v_2\rangle \otimes |w_2\rangle) = \langle v_1|v_2\rangle \langle w_1|w_2\rangle,$$

extended sesquilinearly. In components, if $|A\rangle$ and $|B\rangle$ have coefficient matrices (a_{ij}) and (b_{ij}) with respect to the product basis $|e_i\rangle |f_j\rangle$, then

$$\langle A|B\rangle = \sum_{ij} a_{ij}^* b_{ij},$$

as one would hope.

We will almost always drop the $\otimes$, writing $|v\rangle |w\rangle$, or even $|vw\rangle$, for $|v\rangle \otimes |w\rangle$. This is safe provided we remember that *the order matters*: $|v\rangle |w\rangle$ and $|w\rangle |v\rangle$ live in $\mathcal{V} \otimes \mathcal{W}$ and $\mathcal{W} \otimes \mathcal{V}$ respectively, and even when $\mathcal{V} = \mathcal{W}$ they are different vectors. We also write $\mathcal{V}^{\otimes n}$ for the n -fold tensor power. If $\mathcal{V} = \mathbb{C}^2$ then $\mathcal{V}^{\otimes n}$ has dimension 2^n , with orthonormal basis

$$|i_1 i_2 \cdots i_n\rangle, \quad i_j \in \{0, 1\},$$

indexed by the 2^n bit strings of length n . Note the exponential: n qubits carry 2^n complex amplitudes.

We can now state the first two postulates of quantum mechanics in the form we will use them.

Axiom 39.1.5 (States)

To an isolated quantum system S we associate a finite-dimensional inner product space $\mathcal{V}$. The states of S are the rays of $\mathcal{V}$, represented by normalised vectors $|\psi\rangle$ determined up to a global phase.

Axiom 39.1.6 (Composite Systems)

If S_1 and S_2 are quantum systems with state spaces $\mathcal{V}_1$ and $\mathcal{V}_2$, then the composite system $S_1 S_2$ has state space $\mathcal{V}_1 \otimes \mathcal{V}_2$.

The second postulate is where quantum mechanics diverges most sharply from classical intuition. Classically, the state of a pair of systems is a pair of states; the state space of the composite is the *Cartesian product*, of dimension $m + n$. Quantum mechanically it is the tensor product, of dimension mn , and most of the vectors in it are not of the form $|v\rangle |w\rangle$ at all.

Definition 39.1.7 (Product and Entangled States)

A state $|\Psi\rangle \in \mathcal{V} \otimes \mathcal{W}$ is a **product state** if $|\Psi\rangle = |\psi\rangle \otimes |\varphi\rangle$ for some $|\psi\rangle \in \mathcal{V}$, $|\varphi\rangle \in \mathcal{W}$. Otherwise it is **entangled**.

Example 39.1.8

Let $\mathcal{V} = \mathcal{W} = \mathbb{C}^2$ and consider

$$|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Suppose it were a product state, say $|\phi^+\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$. Expanding,

$$|\phi^+\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle,$$

so we would need $ad = bc = 0$ and $ac = bd = 1/\sqrt{2}$. The latter forces all four of a, b, c, d to be nonzero, contradicting the former. Hence $|\phi^+\rangle$ is entangled.

Counting parameters shows that entanglement is the norm rather than the exception: the product states of two qubits form a set of real dimension $2 + 2 = 4$ inside the 6-real-dimensional set of all two-qubit states. Almost every state of a composite system is entangled, and the whole of §39.4 is devoted to what one can do with that fact.

§39.1.5 Operators

The Dirac notation extends painlessly to linear maps. If A is a linear operator on $\mathcal{V}$ we write $A|\psi\rangle$ for its action, and $\langle\psi|A|\varphi\rangle$ – also written $\langle\psi|A|\varphi\rangle$ – for the inner product of $|\psi\rangle$ with $A|\varphi\rangle$. There is a unique operator $A^\dagger$, the **adjoint**, satisfying

$$\langle v|Aw\rangle = \langle A^\dagger v|w\rangle \quad \text{for all } |v\rangle, |w\rangle,$$

whose matrix is the conjugate transpose of that of A ; one has $(AB)^\dagger = B^\dagger A^\dagger$, and with the convention $|\psi\rangle^\dagger = \langle\psi|$ the rule ‘dagger reverses everything’ holds universally. An operator with $A = A^\dagger$ is **self-adjoint** or **Hermitian**.

In a fixed basis, the outer products $|e_i\rangle\langle e_j|$ are the matrix units, so any operator can be expanded as

$$A = \sum_{i,j} A_{ij} |e_i\rangle\langle e_j|, \quad A_{ij} = \langle e_i|A|e_j\rangle.$$

For a qubit this reads

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a|0\rangle\langle 0| + b|0\rangle\langle 1| + c|1\rangle\langle 0| + d|1\rangle\langle 1|.$$

Another identity worth recording is $\text{Tr}(|v\rangle\langle w|) = \langle w|v\rangle$, and more generally

$$\text{Tr}(A|\psi\rangle\langle\psi|) = \langle\psi|A|\psi\rangle.$$

We will use this repeatedly to convert expectation values into traces, where they are easier to manipulate.

Four operators on $\mathbb{C}^2$ will appear on nearly every page.

Definition 39.1.9 (Pauli Matrices)

The **Pauli matrices** are

$$\sigma_0 = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian, unitary and traceless, they square to the identity, and together with I they form a basis of the space of 2×2 complex matrices. Their action on the computational basis is

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_y|0\rangle = i|1\rangle, \quad \sigma_z|0\rangle = |0\rangle,$$

$$\sigma_x |1\rangle = |0\rangle, \quad \sigma_y |1\rangle = -i|0\rangle, \quad \sigma_z |1\rangle = -|1\rangle,$$

so σ_x is a bit flip, σ_z is a phase flip, and $\sigma_y = i\sigma_x\sigma_z$ is both at once. They satisfy the cyclic relations

$$\sigma_x\sigma_y = i\sigma_z, \quad \sigma_y\sigma_z = i\sigma_x, \quad \sigma_z\sigma_x = i\sigma_y,$$

and any two distinct Pauli matrices anticommute. Geometrically, $\sigma_x, \sigma_y, \sigma_z$ generate rotations about the three axes of the Bloch sphere.

§39.1.6 Projectors

Projection operators will carry all the weight in our discussion of measurement, so we set up the vocabulary now.

Definition 39.1.10 (Projector)

An operator Π on $\mathcal{V}$ is a **(orthogonal) projector** if $\Pi^2 = \Pi$ and $\Pi^\dagger = \Pi$.

If $|v\rangle$ is normalised then $\Pi_v = |v\rangle\langle v|$ is a projector, since

$$\Pi_v\Pi_v = |v\rangle\langle v||v\rangle\langle v| = |v\rangle\langle v|v\rangle\langle v| = |v\rangle\langle v| = \Pi_v,$$

and $\Pi_v|x\rangle = |v\rangle\langle v|x\rangle$ is the component of $|x\rangle$ along $|v\rangle$; in particular Π_v kills everything orthogonal to $|v\rangle$. More generally, if $\mathcal{E} \subseteq \mathcal{V}$ is a subspace with orthonormal basis $|e_1\rangle, \dots, |e_d\rangle$, then

$$\Pi_{\mathcal{E}} = \sum_{k=1}^d |e_k\rangle\langle e_k|$$

is the orthogonal projector onto $\mathcal{E}$, as one checks by extending to an orthonormal basis of $\mathcal{V}$. Its rank is $\dim \mathcal{E}$, and $\text{Tr } \Pi_{\mathcal{E}} = \dim \mathcal{E}$.

If $\mathcal{V} = \mathcal{E}_1 \oplus \dots \oplus \mathcal{E}_d$ is an orthogonal direct sum decomposition, the corresponding projectors satisfy

$$\Pi_i\Pi_j = \delta_{ij}\Pi_i, \quad \sum_{i=1}^d \Pi_i = I.$$

Such a family is exactly the data of a projective measurement, as we will see.

§39.1.7 Tensor Products of Operators

Given linear maps A on $\mathcal{V}$ and B on $\mathcal{W}$, we define $A \otimes B$ on $\mathcal{V} \otimes \mathcal{W}$ by its action on product vectors,

$$(A \otimes B)(|v\rangle \otimes |w\rangle) = (A|v\rangle) \otimes (B|w\rangle),$$

extended linearly. In components this is the familiar Kronecker product: for 2×2 matrices,

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \implies A \otimes B = \begin{pmatrix} aB & bB \\ cB & dB \end{pmatrix} = \begin{pmatrix} ap & aq & bp & bq \\ ar & as & br & bs \\ cp & cq & dp & dq \\ cr & cs & dr & ds \end{pmatrix}.$$

The operators of the form $A \otimes I$ are those acting on the first system only, and $I \otimes B$ those acting on the second; these are the **local** operations, and the constraint that Alice and Bob can only perform local operations on their halves of a shared state is the central restriction in everything we do with entanglement.

Example 39.1.11

Local operations act very differently on an entangled state than one might guess. With $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and A as above,

$$\begin{aligned}(A \otimes I) |\phi^+\rangle &= \frac{1}{\sqrt{2}} [(A|0\rangle)|0\rangle + (A|1\rangle)|1\rangle] = \frac{1}{\sqrt{2}} [a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle], \\ (I \otimes A) |\phi^+\rangle &= \frac{1}{\sqrt{2}} [|0\rangle(A|0\rangle) + |1\rangle(A|1\rangle)] = \frac{1}{\sqrt{2}} [a|00\rangle + c|01\rangle + b|10\rangle + d|11\rangle].\end{aligned}$$

The two differ only by transposing A . In particular, for a symmetric A we have $(A \otimes I) |\phi^+\rangle = (I \otimes A) |\phi^+\rangle$: acting on one half of a maximally entangled pair is the same as acting on the other. This innocuous-looking fact is the engine of superdense coding.

Finally, we record a construction that lets us ‘look at one half’ of a composite state.

Definition 39.1.12 (Partial Inner Product)

Let $|v\rangle \in \mathcal{V}$. The **partial inner product** with $|v\rangle$ is the linear map $\mathcal{V} \otimes \mathcal{W} \rightarrow \mathcal{W}$ determined on the basis by

$$|e_i\rangle |f_j\rangle \mapsto \langle v|e_i\rangle |f_j\rangle.$$

We write it $\langle v|$ again, so that $\langle v|(|\alpha\rangle|\beta\rangle) = \langle v|\alpha\rangle|\beta\rangle$.

When $\mathcal{V}$ and $\mathcal{W}$ are isomorphic one must of course say which factor is meant, and we will do so by labelling the systems, writing $\langle v|_A$ and so on. The notation is heavily overloaded but never actually ambiguous in practice.

§39.2 Measurement

We now come to the part of the formalism with no classical analogue at all: what happens when we look at a quantum system. We state the remaining two postulates, extract the Born rule, and then answer the question that motivates the whole of quantum information theory – *when can two states be told apart?*

§39.2.1 Unitary Evolution

Axiom 39.2.1 (Evolution)

The evolution of a closed quantum system over a finite time interval is given by a unitary operator on its state space: if the system is in state $|\psi(t_1)\rangle$ at time t_1 , then at t_2 it is in state $|\psi(t_2)\rangle = U(t_1, t_2) |\psi(t_1)\rangle$ for some unitary $U(t_1, t_2)$ depending only on t_1 and t_2 .

Recall from IA Vectors and Matrices that for a linear operator U on a finite-dimensional inner product space the following are equivalent: $U^\dagger U = U U^\dagger = I$; U preserves inner products, $\langle Uv|Uw\rangle = \langle v|w\rangle$; U maps some (equivalently, every) orthonormal basis to an orthonormal basis; the columns of the matrix of U form an orthonormal set.

The postulate is a consequence of the Schrödinger equation of IB Quantum Mechanics,

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle,$$

with H the Hermitian Hamiltonian; for time-independent H we get

$$U(t_1, t_2) = \exp\left(-\frac{i}{\hbar} H(t_2 - t_1)\right),$$

which is unitary precisely because H is Hermitian. In this course we will forget about Hamiltonians entirely and simply take ‘the allowed operations are the unitaries’ as a black-box rule.

Two consequences deserve emphasis. First, unitary evolution is *deterministic* and *reversible*: the state at any time determines the state at any other. All the probability in quantum mechanics enters through measurement. Second, since unitaries preserve inner products, *the overlap $\langle\psi|\varphi\rangle$ between two states can never be changed by evolution*. Two states that start out non-orthogonal stay non-orthogonal forever, no matter what we do to them. Almost every impossibility theorem in this course is an application of that one sentence.

§39.2.2 Observables and the Born Rule

Definition 39.2.2 (Observable)

An **observable** of a quantum system with state space $\mathcal{V}$ is a self-adjoint operator A on $\mathcal{V}$.

By the spectral theorem, a self-adjoint A has real eigenvalues $a_1, \dots, a_k$ (which we take to be distinct) and $\mathcal{V}$ decomposes as an orthogonal direct sum of the corresponding eigenspaces. Writing P_j for the projector onto the a_j -eigenspace, we obtain the **spectral decomposition**

$$A = \sum_j a_j P_j, \quad P_i P_j = \delta_{ij} P_i, \quad \sum_j P_j = I.$$

If a_j is a non-degenerate eigenvalue with unit eigenvector $|\varphi_j\rangle$ then $P_j = |\varphi_j\rangle\langle\varphi_j|$ has rank one; if a_j has multiplicity m then $P_j = \sum_{i=1}^m |\varphi_j^i\rangle\langle\varphi_j^i|$ for an orthonormal basis of the eigenspace.

Axiom 39.2.3 (Measurement; the Born Rule)

If the observable $A = \sum_j a_j P_j$ is measured on a system in normalised state $|\psi\rangle$, the outcome is one of the eigenvalues a_j , obtained with probability

$$p(a_j) = \langle\psi|P_j|\psi\rangle = \|P_j|\psi\rangle\|^2,$$

and the post-measurement state of the system is

$$\frac{P_j|\psi\rangle}{\sqrt{p(a_j)}}.$$

Since $\sum_j P_j = I$ and $\langle\psi|\psi\rangle = 1$, the probabilities do sum to one. A measurement of this form is called a **projective** or **von Neumann** measurement. Note that the state after

the measurement is (the normalisation of) the projection of $|\psi\rangle$ onto the eigenspace we found – the notorious ‘collapse of the wavefunction’. Measuring twice in a row gives the same answer both times, since P_j applied to a vector already in its range does nothing.

In practice the eigenvalues themselves rarely matter; what matters is *which* projector fired. So we will usually specify a measurement simply by giving the family of projectors, and speak of ‘the measurement $\{P_j\}$ ’ with outcomes labelled by j .

Definition 39.2.4 (Complete and Incomplete Measurements)

A projective measurement $\{P_j\}$ is **complete** if every P_j has rank 1, and **incomplete** otherwise.

A complete measurement is the same thing as a choice of orthonormal basis $\{|e_j\rangle\}$, with $P_j = |e_j\rangle\langle e_j|$. Then, writing $|\psi\rangle = \sum_j a_j |e_j\rangle$,

$$p(j) = |\langle e_j|\psi\rangle|^2 = |a_j|^2,$$

and the post-measurement state is

$$\frac{P_j |\psi\rangle}{\sqrt{p(j)}} = \frac{|e_j\rangle \langle e_j|\psi\rangle}{\sqrt{p(j)}} = |e_j\rangle$$

up to a global phase. So the amplitudes really are amplitudes for outcomes, exactly as promised in §39.1.1, and the state collapses onto a basis vector. We say we have **measured in the basis $\{|e_j\rangle\}$** .

An incomplete measurement corresponds instead to an orthogonal decomposition $\mathcal{V} = \mathcal{E}_1 \oplus \cdots \oplus \mathcal{E}_d$ into subspaces of arbitrary dimension: the outcome tells us which subspace, and the state collapses into that subspace but is not otherwise disturbed. It is a coarse-graining of a complete measurement, in the sense that if we refine each $\mathcal{E}_i$ into an orthonormal basis and perform the complete measurement instead, then summing the probabilities of the outcomes lying in $\mathcal{E}_i$ recovers $p(i)$.

Example 39.2.5 (Parity Measurement)

Take two qubits and decompose $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathcal{E}_0 \oplus \mathcal{E}_1$, where

$$\mathcal{E}_0 = \text{span}\{|00\rangle, |11\rangle\}, \quad \mathcal{E}_1 = \text{span}\{|01\rangle, |10\rangle\}$$

are the even- and odd-parity subspaces, ‘parity’ meaning $b_1 \oplus b_2$. The corresponding measurement $\{\Pi_0, \Pi_1\}$ with $\Pi_0 = |00\rangle\langle 00| + |11\rangle\langle 11|$, $\Pi_1 = |01\rangle\langle 01| + |10\rangle\langle 10|$ tells us the parity of the two bits and nothing else. Applied to the state $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ it returns 0 with certainty and leaves the state untouched – even though a complete measurement in the computational basis would destroy the entanglement completely. Measuring less can be worth more.

We will constantly want to measure a *subsystem* of a composite system, and there is a useful special case of the Born rule for this.

Proposition 39.2.6 (Extended Born Rule)

Let $S_1 S_2$ be a composite system with state spaces $\mathcal{V}, \mathcal{W}$ having complete orthonormal

bases $\{|e_i\rangle\}_{i=1}^m$, $\{|f_j\rangle\}_{j=1}^n$, and let

$$|\psi\rangle = \sum_{i,j} a_{ij} |e_i\rangle |f_j\rangle.$$

Suppose we measure S_1 in the basis $\{|e_i\rangle\}$, that is, we perform the projective measurement with projectors $P_k \otimes I$, $P_k = |e_k\rangle\langle e_k|$. Then the outcome is k with probability

$$p(k) = \langle\psi|P_k \otimes I|\psi\rangle = \sum_{j=1}^n |a_{kj}|^2,$$

and the post-measurement state is

$$|\psi_{\text{after}}\rangle = \frac{(P_k \otimes I)|\psi\rangle}{\sqrt{p(k)}} = |e_k\rangle \otimes \frac{\sum_j a_{kj} |f_j\rangle}{\sqrt{\sum_j |a_{kj}|^2}}.$$

Proof. The projectors $P_k \otimes I$ project onto the mutually orthogonal subspaces $\mathcal{E}_k = |e_k\rangle \otimes \mathcal{W}$, whose direct sum is $\mathcal{V} \otimes \mathcal{W}$, so this is a legitimate (in general incomplete) projective measurement. Expanding,

$$\langle\psi|P_k \otimes I|\psi\rangle = \sum_{i,j} \sum_{i',j'} a_{ij}^* a_{i'j'} \langle e_i|e_k\rangle \langle e_k|e_{i'}\rangle \langle f_j|f_{j'}\rangle = \sum_j |a_{kj}|^2,$$

using orthonormality. Applying $P_k \otimes I$ to $|\psi\rangle$ retains only the terms with $i = k$, giving $|e_k\rangle \otimes \sum_j a_{kj} |f_j\rangle$, and dividing by $\sqrt{p(k)}$ normalises it. $\square$

Notice what the second half of this says: the state of S_2 after we measure S_1 depends on the outcome k . Measuring here changes the description of the state over there, instantaneously and no matter how far away it is. Whether this permits *signalling* is a question we will settle – in the negative – in §39.3.4.

Example 39.2.7 (Measuring One Qubit of Three)

Let

$$|\psi\rangle = \frac{i}{2} |000\rangle + \frac{1+i}{2\sqrt{2}} |001\rangle - \frac{1}{2} |101\rangle + \frac{3}{10} |110\rangle - \frac{2i}{5} |111\rangle,$$

which one checks is normalised: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{100} + \frac{4}{25} = 1$. Measure the first qubit in the computational basis. By the extended Born rule we sum $|a|^2$ over the basis states beginning with 1:

$$p(1) = \frac{1}{4} + \frac{9}{100} + \frac{4}{25} = \frac{1}{2},$$

so the outcome is 1 with probability $\frac{1}{2}$, and in that case the state becomes

$$|1\rangle \otimes \frac{-\frac{1}{2} |01\rangle + \frac{3}{10} |10\rangle - \frac{2i}{5} |11\rangle}{\sqrt{1/2}} = |1\rangle \otimes \left(-\frac{1}{\sqrt{2}} |01\rangle + \frac{3}{5\sqrt{2}} |10\rangle - \frac{4i}{5\sqrt{2}} |11\rangle \right).$$

Remark. Measuring in an arbitrary basis is no more powerful than measuring in the computational basis, provided we may first apply a unitary. If $\mathcal{B} = \{|e_i\rangle\}$ and $\mathcal{C} = \{|e'_i\rangle\}$ are two orthonormal bases, let U be the unitary with $U|e_i\rangle = |e'_i\rangle$. Measuring $|\psi\rangle$ in $\mathcal{C}$

gives outcome i with probability

$$|\langle e'_i | \psi \rangle|^2 = |\langle e_i | U^\dagger | \psi \rangle|^2,$$

which is exactly the probability of outcome i when $U^\dagger | \psi \rangle$ is measured in $\mathcal{B}$. So ‘apply $U^\dagger$, then measure in the standard basis’ simulates any complete measurement, and we may as well assume that all measurements in a circuit are standard measurements.

§39.2.3 Distinguishing Quantum States

Here is the question that separates quantum information from classical information theory. Someone hands us a system, promising that its state is either $|\alpha_0\rangle$ or $|\alpha_1\rangle$, both known to us. Can we determine which?

Classically the answer is trivially yes: distinct classical states are distinct, and looking tells you which. Quantum mechanically the answer is a sharp and surprising one.

Definition 39.2.8 (Reliably Distinguishable)

Two states $|\psi\rangle, |\varphi\rangle$ are **reliably distinguishable** if there is a measurement with two distinguished outcomes 0, 1 such that measuring $|\psi\rangle$ gives outcome 0 with probability 1, and measuring $|\varphi\rangle$ gives outcome 1 with probability 1.

Theorem 39.2.9

Two states $|\psi\rangle$ and $|\varphi\rangle$ are reliably distinguishable if and only if $\langle \psi | \varphi \rangle = 0$.

Proof. Suppose first that $\langle \psi | \varphi \rangle = 0$. Extend $|\psi\rangle$ to a complete orthonormal basis $\{|\psi\rangle, |f_1\rangle, \dots, |f_{m-1}\rangle\}$ of $\mathcal{V}$ and consider the incomplete measurement $\{\Pi_0, \Pi_1\}$ with $\Pi_0 = |\psi\rangle\langle\psi|$ and $\Pi_1 = I - \Pi_0$. Then measuring $|\psi\rangle$ gives outcome 0 with probability $\langle \psi | \Pi_0 | \psi \rangle = 1$, while measuring $|\varphi\rangle$ gives outcome 0 with probability $|\langle \psi | \varphi \rangle|^2 = 0$, hence outcome 1 with certainty.

Conversely, suppose such a measurement $\{\Pi_0, \Pi_1, \dots\}$ exists. Then $\langle \psi | \Pi_0 | \psi \rangle = 1$ forces $\Pi_0 | \psi \rangle = | \psi \rangle$, since $\|\Pi_0 | \psi \rangle\|^2 = 1 = \| | \psi \rangle \|^2$ and Π_0 is a contraction which fixes a unit vector only if that vector lies in its range. Similarly $\langle \varphi | \Pi_1 | \varphi \rangle = 1$ gives $\Pi_1 | \varphi \rangle = | \varphi \rangle$. But the ranges of Π_0 and Π_1 are orthogonal subspaces, so

$$\langle \psi | \varphi \rangle = \langle \Pi_0 \psi | \Pi_1 \varphi \rangle = \langle \psi | \Pi_0 \Pi_1 | \varphi \rangle = 0. \quad \square$$

So non-orthogonal states cannot be told apart with certainty, ever, by any measurement. Note also the trivial but important corollary that $|\psi\rangle$ and $e^{i\theta} |\psi\rangle$ are indistinguishable, which is what justified our identification of states with rays.

What if we allow ourselves to be wrong sometimes? Suppose the two candidate states are equally likely a priori and we must guess. We could ignore the system entirely and flip a coin, succeeding with probability $\frac{1}{2}$. Can we do better, and how much better?

Before answering, note that there is no point being clever about the *form* of the measurement. If we adjoin an ancilla in a fixed state $|A\rangle$, apply a unitary U to the enlarged system and then measure with $\{\Pi_i\}$, the outcome probabilities are

$$p(i) = \langle \alpha_j A | U^\dagger \Pi_i U | \alpha_j A \rangle,$$

and $\{U^\dagger \Pi_i U\}$ is again a family of orthogonal projectors summing to the identity. So the whole procedure is just *some* projective measurement performed directly on $|\alpha_j\rangle|A\rangle$; and since the ancilla contributes the same factor $\langle A|A\rangle = 1$ in both cases, we may as well restrict attention to two-outcome projective measurements $\{\Pi_0, \Pi_1\}$ on the original system, with $\Pi_0 + \Pi_1 = I$. We guess $|\alpha_0\rangle$ on outcome 0 and $|\alpha_1\rangle$ on outcome 1.

Theorem 39.2.10 (Holevo–Helstrom, Pure States)

Let $|\alpha_0\rangle, |\alpha_1\rangle$ be given with equal prior probabilities $\frac{1}{2}$, and write $|\langle\alpha_0|\alpha_1\rangle| = \cos\theta$ with $\theta \in [0, \pi/2]$. Then the average success probability of any measurement strategy satisfies

$$p_S \leq \frac{1}{2} + \frac{\sin\theta}{2},$$

and there is a measurement attaining this bound.

Proof. With the two-outcome measurement $\{\Pi_0, \Pi_1 = I - \Pi_0\}$ the average success probability is

$$\begin{aligned} p_S &= \frac{1}{2} \langle\alpha_0|\Pi_0|\alpha_0\rangle + \frac{1}{2} \langle\alpha_1|\Pi_1|\alpha_1\rangle \\ &= \frac{1}{2} \langle\alpha_0|\Pi_0|\alpha_0\rangle + \frac{1}{2} (1 - \langle\alpha_1|\Pi_0|\alpha_1\rangle) \\ &= \frac{1}{2} + \frac{1}{2} \text{Tr}[\Pi_0\Delta], \quad \Delta := |\alpha_0\rangle\langle\alpha_0| - |\alpha_1\rangle\langle\alpha_1|, \end{aligned}$$

where we used $\langle\psi|A|\psi\rangle = \text{Tr}(A|\psi\rangle\langle\psi|)$. So we must maximise $\text{Tr}(\Pi_0\Delta)$ over projectors Π_0 .

Now Δ is self-adjoint, so it has real eigenvalues and an orthonormal eigenbasis. It annihilates every vector orthogonal to both $|\alpha_0\rangle$ and $|\alpha_1\rangle$, so it acts nontrivially only on $\text{span}\{|\alpha_0\rangle, |\alpha_1\rangle\}$, and has at most two nonzero eigenvalues. Also $\text{Tr}\Delta = 1 - 1 = 0$, so those eigenvalues are δ and $-\delta$ for some $\delta \geq 0$; write $|p\rangle, |m\rangle$ for corresponding orthonormal eigenvectors, so

$$\Delta = \delta |p\rangle\langle p| - \delta |m\rangle\langle m|.$$

To find δ , pick a normalised $|\alpha_0^\perp\rangle \in \text{span}\{|\alpha_0\rangle, |\alpha_1\rangle\}$ orthogonal to $|\alpha_0\rangle$, so that $\{|\alpha_0\rangle, |\alpha_0^\perp\rangle\}$ is an orthonormal basis of that plane, and write $|\alpha_1\rangle = c_0 |\alpha_0\rangle + c_1 |\alpha_0^\perp\rangle$ with $|c_0|^2 + |c_1|^2 = 1$. In this basis

$$\Delta = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} |c_0|^2 & c_0 c_1^* \\ c_0^* c_1 & |c_1|^2 \end{pmatrix} = \begin{pmatrix} |c_1|^2 & -c_0 c_1^* \\ -c_0^* c_1 & -|c_1|^2 \end{pmatrix},$$

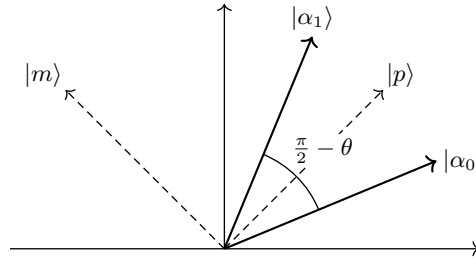
whose determinant is $-|c_1|^4 - |c_0|^2 |c_1|^2 = -|c_1|^2$. Since the eigenvalues are $\pm\delta$, their product is $-\delta^2$, so $\delta = |c_1|$. But $|c_0| = |\langle\alpha_0|\alpha_1\rangle| = \cos\theta$, so $\delta = \sin\theta$.

Therefore

$$p_S = \frac{1}{2} + \frac{\sin\theta}{2} [\langle p|\Pi_0|p\rangle - \langle m|\Pi_0|m\rangle].$$

For any projector and any unit vector, $0 \leq \langle\varphi|\Pi_0|\varphi\rangle \leq 1$, so the bracket is at most 1, giving $p_S \leq \frac{1}{2} + \frac{\sin\theta}{2}$. Equality holds when $\langle p|\Pi_0|p\rangle = 1$ and $\langle m|\Pi_0|m\rangle = 0$, which is achieved by the choice $\Pi_0 = |p\rangle\langle p|$, $\Pi_1 = I - |p\rangle\langle p|$. $\square$

The geometry is worth drawing. In the plane spanned by the two states, $|\alpha_0\rangle$ and $|\alpha_1\rangle$ subtend an angle $\frac{\pi}{2} - \theta$; the optimal measurement basis $\{|p\rangle, |m\rangle\}$ is the pair of orthogonal directions that ‘straddles’ them symmetrically.



Two sanity checks. If the states are orthogonal, $\theta = \pi/2$ and the bound reads $p_S \leq 1$, attained: this is the theorem above. If the states are equal, $\theta = 0$ and $p_S \leq \frac{1}{2}$: we can do no better than guessing, as we must. And in between we interpolate smoothly. There is no sharp line between ‘distinguishable’ and ‘indistinguishable’; there is a continuum, governed entirely by the overlap $|\langle\alpha_0|\alpha_1\rangle|$.

§39.3 No Cloning and No Signalling

Having established what quantum states can do, we now establish two things that they cannot do. Both are one-line consequences of linearity, and both turn out to be indispensable: the first is what makes quantum cryptography possible, and the second is what stops quantum mechanics from contradicting relativity.

§39.3.1 Going to the Church of the Higher Hilbert Space

Before proving impossibility results we should be careful about what we are allowed to try. The most general thing we can do to a system S in state $|\psi\rangle$ is:

- (i) (*ancilla*) adjoin an auxiliary system A in a fixed state $|A\rangle$, so that the joint state of SA is $|\psi\rangle|A\rangle \in \mathcal{V}_S \otimes \mathcal{V}_A$;
- (ii) (*unitary*) apply a unitary U to the whole of SA ;
- (iii) (*measure*) perform a measurement on SA , keep the post-measurement state of S , and discard A .

This is often called *going to the church of the higher Hilbert space*: any messy, noisy, irreversible-looking process on S can be modelled as a clean unitary evolution on a larger closed system, followed by throwing part of it away. The point of the ancilla is not merely bookkeeping – it provides somewhere for S to become entangled *with*, which is exactly how information leaks out of a system in practice.

We will therefore prove our impossibility theorems in this general setting, and they will apply to any physical process whatsoever.

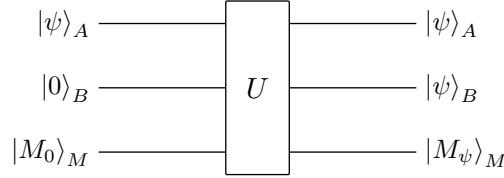
§39.3.2 The No-Cloning Theorem

Classical information can be copied freely: read the bit, write it down again. Can quantum information be copied? A copier would be a device that takes the state $|\psi\rangle$ we

want to duplicate, a blank system B initially in some fixed state $|0\rangle$, and a machine M in some fixed ‘ready’ state $|M_0\rangle$, and produces two copies of $|\psi\rangle$:

$$|\psi\rangle_A |0\rangle_B |M_0\rangle_M \mapsto |\psi\rangle_A |\psi\rangle_B |M_\psi\rangle_M.$$

Note that the machine is allowed to end up in a state $|M_\psi\rangle$ that depends on what was copied – we are not demanding that it be reset.



Theorem 39.3.1 (No-Cloning Theorem)

Let $\mathcal{S}$ be any set of states of a system A containing at least one pair of distinct non-orthogonal states. Then there is no unitary^a U which clones every state in $\mathcal{S}$.

^aAnd hence no process at all: by the discussion of §39.3.1, a general process is a unitary on a larger system followed by a measurement, and we may let M absorb both the ancilla and, by the principle of deferred measurement, any measurement performed.

Proof. Let $|\xi\rangle, |\eta\rangle \in \mathcal{S}$ be distinct and non-orthogonal, so that $0 < |\langle\xi|\eta\rangle| < 1$. Suppose such a U existed. Then

$$U |\xi\rangle_A |0\rangle_B |M_0\rangle_M = |\xi\rangle_A |\xi\rangle_B |M_\xi\rangle_M, \quad U |\eta\rangle_A |0\rangle_B |M_0\rangle_M = |\eta\rangle_A |\eta\rangle_B |M_\eta\rangle_M.$$

Unitaries preserve inner products, so taking the inner product of the two left-hand sides and of the two right-hand sides must give the same answer. The inner product on a tensor product factorises, so

$$\langle\xi|\eta\rangle \langle 0|0\rangle \langle M_0|M_0\rangle = \langle\xi|\eta\rangle \langle\xi|\eta\rangle \langle M_\xi|M_\eta\rangle,$$

that is, $\langle\xi|\eta\rangle = \langle\xi|\eta\rangle^2 \langle M_\xi|M_\eta\rangle$. Taking absolute values and dividing by $|\langle\xi|\eta\rangle| \neq 0$ gives

$$1 = |\langle\xi|\eta\rangle| |\langle M_\xi|M_\eta\rangle|.$$

But $|\langle\xi|\eta\rangle| < 1$ and $|\langle M_\xi|M_\eta\rangle| \leq 1$ by Cauchy–Schwarz, so the right-hand side is strictly less than 1. This is a contradiction. $\square$

The theorem is sharp in both directions. A set of *mutually orthogonal* states *can* be cloned: extend them to an orthonormal basis $\{|e_i\rangle\}$ and let U be any unitary with $U|e_i\rangle|0\rangle = |e_i\rangle|e_i\rangle$, which exists because the images are orthonormal. This is exactly the statement that *classical* information – information encoded in orthogonal states – can be copied. It is the superposition states in between that resist duplication.

Remark. It is worth noticing which feature of quantum mechanics is doing the work. It is linearity, nothing else. If $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$, then the left-hand side is linear in $|\psi\rangle$ and the right-hand side is quadratic, and the two cannot agree.

Why should we care? Suppose we could clone. Then we could take an unknown state, make a million copies, and measure them all in different bases, thereby learning the state to arbitrary precision from a single copy. But the amplitudes of an unknown

state are continuous parameters, so a single qubit would carry an unbounded amount of extractable classical information. That would be strange enough. In fact, it is worse.

§39.3.3 Cloning Would Permit Superluminal Signalling

Suppose Alice and Bob each hold one qubit of the entangled state

$$|\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle),$$

and are separated by a light-year. Alice wants to send Bob one bit *now*. Here is her protocol. To send ‘yes’, she measures her qubit in the basis $\{|0\rangle, |1\rangle\}$; to send ‘no’, she measures it in the basis $\{|+\rangle, |-\rangle\}$.

Consider what happens to Bob’s qubit. Writing $|\phi^+\rangle$ in the conjugate basis,

$$|\phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} (|++\rangle + |--\rangle),$$

as one checks by expanding. So by the extended Born rule, if Alice measures in the standard basis then with probability $\frac{1}{2}$ each Bob’s qubit is left in the state $|0\rangle$ or $|1\rangle$; if she measures in the conjugate basis then with probability $\frac{1}{2}$ each it is left in $|+\rangle$ or $|-\rangle$.

Bob therefore holds a qubit prepared in one of two *ensembles*: an equal mixture of $\{|0\rangle, |1\rangle\}$, or an equal mixture of $\{|+\rangle, |-\rangle\}$. If he could tell these ensembles apart, he would learn Alice’s bit instantly. But he cannot: for any measurement $\{\Pi_i\}$ on his qubit,

$$p_{\text{yes}}(i) = \frac{1}{2} \langle 0|\Pi_i|0\rangle + \frac{1}{2} \langle 1|\Pi_i|1\rangle = \frac{1}{2} \text{Tr}[\Pi_i (|0\rangle\langle 0| + |1\rangle\langle 1|)] = \frac{1}{2} \text{Tr} \Pi_i,$$

$$p_{\text{no}}(i) = \frac{1}{2} \langle +|\Pi_i|+\rangle + \frac{1}{2} \langle -|\Pi_i|-\rangle = \frac{1}{2} \text{Tr}[\Pi_i (|+\rangle\langle +| + |-\rangle\langle -|)] = \frac{1}{2} \text{Tr} \Pi_i,$$

since both $|0\rangle\langle 0| + |1\rangle\langle 1|$ and $|+\rangle\langle +| + |-\rangle\langle -|$ equal the identity. The two probability distributions are *identical*, whatever measurement Bob makes. No signal.

But now suppose Bob could clone. He makes N copies of his qubit and measures each in the standard basis. If Alice sent ‘yes’, his qubit was $|0\rangle$ or $|1\rangle$, so all N outcomes agree. If she sent ‘no’, his qubit was $|\pm\rangle$, and each of the N measurements gives 0 or 1 independently and uniformly, so the probability that they all agree is $2 \cdot 2^{-N} = 2^{-N+1}$. Bob declares ‘yes’ if the outcomes agree and ‘no’ otherwise, and is wrong with probability at most 2^{-N+1} , which he can make as small as he likes.

So cloning would give instantaneous communication across arbitrary distances, in flat contradiction with special relativity. The no-cloning theorem is not a technical curiosity; it is what keeps quantum mechanics consistent with the rest of physics.

§39.3.4 The No-Signalling Principle

The argument above shows that Bob cannot detect Alice’s *choice of basis*. In fact something much stronger is true: Bob cannot detect *whether Alice measured at all*.

Theorem 39.3.2 (No-Signalling Principle)

Let $|\phi_{AB}\rangle \in \mathcal{V}_A \otimes \mathcal{V}_B$ be an arbitrary (possibly entangled) state shared by Alice and Bob, and let Bob perform a complete measurement in a basis $\{|b\rangle\}$ of $\mathcal{V}_B$. The probability distribution of Bob’s outcome is the same whether or not Alice first performs a measurement on her system, and is independent of which measurement

she performs.

Proof. Fix a complete orthonormal basis $\{|a\rangle\}$ of $\mathcal{V}_A$, write $P_a = |a\rangle\langle a|$ and $P_b = |b\rangle\langle b|$, and expand

$$|\phi_{AB}\rangle = \sum_{a,b} c_{ab} |a\rangle |b\rangle.$$

If Alice does nothing, the extended Born rule gives Bob outcome b with probability

$$p(b) = \langle \phi_{AB} | I_A \otimes P_b | \phi_{AB} \rangle = \sum_a |c_{ab}|^2.$$

Suppose instead Alice first measures in the basis $\{|a\rangle\}$. She gets outcome a with probability $p(a) = \sum_b |c_{ab}|^2$, and the joint state becomes

$$|\phi''_{AB}\rangle = \frac{(P_a \otimes I_B) |\phi_{AB}\rangle}{\sqrt{p(a)}}.$$

Bob's conditional probability of outcome b is then

$$\begin{aligned} p(b | a) &= \langle \phi''_{AB} | I_A \otimes P_b | \phi''_{AB} \rangle \\ &= \frac{1}{p(a)} \langle \phi_{AB} | (P_a \otimes I)(I \otimes P_b)(P_a \otimes I) | \phi_{AB} \rangle \\ &= \frac{1}{p(a)} \langle \phi_{AB} | P_a \otimes P_b | \phi_{AB} \rangle, \end{aligned}$$

using $P_a^2 = P_a$ and the fact that operators on different factors commute. Hence the joint distribution is $p(a, b) = p(a)p(b | a) = \langle \phi_{AB} | P_a \otimes P_b | \phi_{AB} \rangle = |c_{ab}|^2$, and Bob's marginal is

$$\sum_a p(a, b) = \sum_a |c_{ab}|^2 = p(b),$$

exactly as before. Since Alice measuring in any other basis is the same as applying a unitary to her half and then measuring in $\{|a\rangle\}$, and unitaries of the form $U \otimes I$ leave Bob's marginal alone by the same computation, the result is independent of her choice. $\square$

So the ‘collapse’ induced at a distance is real as a change in our *description* of Bob's system, but it carries no information whatsoever. It is only when Alice tells Bob, over an ordinary classical channel, what she did and what she got, that the correlations become visible. Every protocol in this course respects this: teleportation, superdense coding and key distribution all require classical communication, and all are therefore limited by the speed of light.

§39.4 Entanglement

Entanglement is the resource that makes quantum information interesting. In this section we meet the canonical entangled states, and then confront the fact – established by Bell, and by now verified experimentally beyond any reasonable doubt – that the correlations they exhibit cannot be reproduced by *any* classical mechanism.

§39.4.1 The Bell Basis

Definition 39.4.1 (Bell States)

The **Bell states** of a pair of qubits are

$$|\phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \quad |\psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle).$$

They form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$, the **Bell basis**.

Orthonormality is immediate from the orthonormality of the computational basis. Each Bell state is entangled, by the same argument we gave for $|\phi^+\rangle$. They are also **maximally** entangled, in a sense we can make precise here informally: measuring either qubit alone in *any* basis gives a uniformly random outcome, so an observer with access to one qubit has no information at all, while the pair together is in a perfectly definite state. All of the information is in the correlations.

These states are also known as **EPR pairs**, after the 1935 paper of Einstein, Podolsky and Rosen, who considered a state of this kind and argued that quantum mechanics must be incomplete. Their reasoning, in modern dress: measuring Alice's qubit of $|\phi^+\rangle$ in the standard basis determines Bob's outcome with certainty, and by no-signalling this cannot be a physical influence; so Bob's outcome must have been predetermined by some *element of reality* which the quantum state fails to record. Call such a hidden parameter λ . EPR concluded that the wavefunction is an incomplete description of reality. It was thirty years before Bell showed that this conclusion is not merely unnecessary but *false*.

Before we get there, a computational fact which we will use constantly.

Proposition 39.4.2

Write $X = \sigma_x$, $Z = \sigma_z$ and $Y = i\sigma_y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Then all four Bell states are obtained from $|\phi^+\rangle$ by a local operation on the first qubit:

$$\begin{aligned} |\phi^+\rangle &= (I \otimes I) |\phi^+\rangle, & |\phi^-\rangle &= (Z \otimes I) |\phi^+\rangle, \\ |\psi^+\rangle &= (X \otimes I) |\phi^+\rangle, & |\psi^-\rangle &= (Y \otimes I) |\phi^+\rangle. \end{aligned}$$

The same holds with the operators acting on the second qubit instead, except that $|\psi^-\rangle = -(I \otimes Y) |\phi^+\rangle$.

Proof. Direct computation. For instance

$$(X \otimes I) |\phi^+\rangle = \frac{1}{\sqrt{2}}((X|0\rangle)|0\rangle + (X|1\rangle)|1\rangle) = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) = |\psi^+\rangle,$$

and

$$(Y \otimes I) |\phi^+\rangle = \frac{1}{\sqrt{2}}((Y|0\rangle)|0\rangle + (Y|1\rangle)|1\rangle) = \frac{1}{\sqrt{2}}(-|10\rangle + |01\rangle) = |\psi^-\rangle,$$

since $Y|0\rangle = -|1\rangle$ and $Y|1\rangle = |0\rangle$. The remaining cases are similar, and the statements for the second qubit follow from the example of §39.1.7, since I, X, Z are symmetric matrices and Y is antisymmetric. $\square$

The moral is striking: Alice, acting only on her own qubit, can steer the joint state to any of four mutually orthogonal – and hence perfectly distinguishable – states. She has

two bits of leverage over a system she only half owns. We will cash this in as superdense coding in §39.6.

§39.4.2 Bell Inequalities and the CHSH Game

Let us take the EPR argument seriously and ask what a *local hidden variable* theory would predict. The setup: Alice and Bob are far apart, sharing some system. Alice chooses one of two binary-outcome measurements, labelled by $s \in \{0, 1\}$, and obtains an outcome $a \in \{-1, +1\}$; Bob chooses $t \in \{0, 1\}$ and gets $b \in \{-1, +1\}$. The experiment is repeated many times and they compare notes, computing the four correlations $\mathbb{E}[ab \mid s, t]$.

A **local hidden variable** (LHV) model asserts that there is a random variable λ , distributed according to some law $p(\lambda)$, and functions

$$A_0(\lambda), A_1(\lambda), B_0(\lambda), B_1(\lambda) \in \{-1, +1\}$$

such that Alice's outcome for setting s is $A_s(\lambda)$ and Bob's for setting t is $B_t(\lambda)$. *Locality* is the assumption that A_s does not depend on t and vice versa – Alice's outcome cannot depend on a choice made outside her light cone. *Realism* is the assumption that all four values exist simultaneously, even though only two of them are measured in any given run.

Theorem 39.4.3 (CHSH Inequality)

In any local hidden variable model, the quantity

$$S = \mathbb{E}[A_0 B_0] + \mathbb{E}[A_0 B_1] + \mathbb{E}[A_1 B_0] - \mathbb{E}[A_1 B_1]$$

satisfies $|S| \leq 2$.

Proof. Fix λ and consider

$$\begin{aligned} & A_0(\lambda)B_0(\lambda) + A_0(\lambda)B_1(\lambda) + A_1(\lambda)B_0(\lambda) - A_1(\lambda)B_1(\lambda) \\ &= A_0(\lambda) [B_0(\lambda) + B_1(\lambda)] + A_1(\lambda) [B_0(\lambda) - B_1(\lambda)]. \end{aligned}$$

Since $B_0(\lambda), B_1(\lambda) \in \{-1, +1\}$, one of $B_0 + B_1$ and $B_0 - B_1$ is 0 and the other is ± 2 . Hence the whole expression is $\pm 2A_s(\lambda)$ for one of $s = 0, 1$, and so lies in $\{-2, +2\}$. Averaging over λ with respect to $p(\lambda)$ and using linearity of expectation gives $|S| \leq 2$. $\square$

The inequality is a genuine constraint on the observable correlations, derived from nothing but locality and realism. And quantum mechanics violates it.

Theorem 39.4.4 (Quantum Violation of CHSH)

Let Alice and Bob share the state $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$, and let each measure the spin observable along a direction in a fixed plane: Alice measures $A(\alpha) = \cos \alpha \sigma_z + \sin \alpha \sigma_x$ and Bob measures $B(\beta) = \cos \beta \sigma_z + \sin \beta \sigma_x$. Then

$$\mathbb{E}[A(\alpha)B(\beta)] = -\cos(\alpha - \beta),$$

and with the choices $\alpha_0 = 0$, $\alpha_1 = \frac{\pi}{2}$, $\beta_0 = \frac{\pi}{4}$, $\beta_1 = -\frac{\pi}{4}$ we obtain

$$S = -2\sqrt{2}, \quad |S| = 2\sqrt{2} > 2.$$

Proof. Each $A(\alpha)$ is Hermitian with $A(\alpha)^2 = I$ (using $\sigma_z^2 = \sigma_x^2 = I$ and $\sigma_z\sigma_x + \sigma_x\sigma_z = 0$), so its eigenvalues are ± 1 and it is a legitimate ± 1 -valued observable. The correlation is the expectation of the observable $A(\alpha) \otimes B(\beta)$ in the state $|\psi^-\rangle$.

We use the fact that $|\psi^-\rangle$ is *rotationally invariant*: for any 2×2 unitary U with $\det U = 1$ we have $(U \otimes U) |\psi^-\rangle = |\psi^-\rangle$, because $|\psi^-\rangle$ spans the antisymmetric part of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and $U \otimes U$ acts on it by $\det U$. Rather than invoke this, though, let us simply compute. Write $|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ and note the useful identities

$$\sigma_z \otimes \sigma_z |\psi^-\rangle = -|\psi^-\rangle, \quad \sigma_x \otimes \sigma_x |\psi^-\rangle = -|\psi^-\rangle,$$

the first because $\sigma_z \otimes \sigma_z |01\rangle = -|01\rangle$ and likewise for $|10\rangle$; the second because $\sigma_x \otimes \sigma_x$ swaps $|01\rangle \leftrightarrow |10\rangle$, which negates the antisymmetric combination. We also need the cross terms:

$$(\sigma_z \otimes \sigma_x) |\psi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\phi^+\rangle,$$

since $\sigma_z \otimes \sigma_x |01\rangle = |00\rangle$ and $\sigma_z \otimes \sigma_x |10\rangle = -|11\rangle$. As $\langle \psi^- | \phi^+ \rangle = 0$, we get $\langle \psi^- | \sigma_z \otimes \sigma_x | \psi^- \rangle = 0$, and symmetrically $\langle \psi^- | \sigma_x \otimes \sigma_z | \psi^- \rangle = 0$. Therefore

$$\begin{aligned} \mathbb{E}[A(\alpha)B(\beta)] &= \langle \psi^- | (\cos \alpha \sigma_z + \sin \alpha \sigma_x) \otimes (\cos \beta \sigma_z + \sin \beta \sigma_x) | \psi^- \rangle \\ &= \cos \alpha \cos \beta (-1) + \sin \alpha \sin \beta (-1) + 0 + 0 \\ &= -\cos(\alpha - \beta). \end{aligned}$$

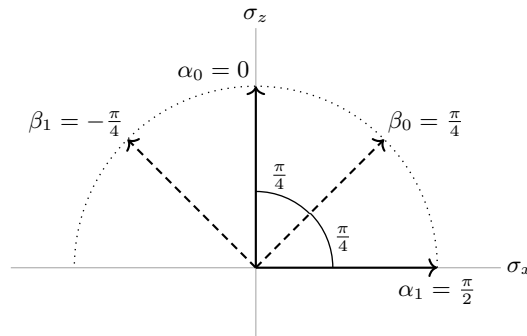
Now we simply evaluate the four correlations. Writing $C_{st} = \mathbb{E}[A(\alpha_s)B(\beta_t)] = -\cos(\alpha_s - \beta_t)$,

$$\begin{aligned} C_{00} &= -\cos(0 - \frac{\pi}{4}) = -\frac{1}{\sqrt{2}}, & C_{01} &= -\cos(0 + \frac{\pi}{4}) = -\frac{1}{\sqrt{2}}, \\ C_{10} &= -\cos(\frac{\pi}{2} - \frac{\pi}{4}) = -\frac{1}{\sqrt{2}}, & C_{11} &= -\cos(\frac{\pi}{2} + \frac{\pi}{4}) = +\frac{1}{\sqrt{2}}, \end{aligned}$$

so that

$$S = C_{00} + C_{01} + C_{10} - C_{11} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} = -\frac{4}{\sqrt{2}} = -2\sqrt{2},$$

and $|S| = 2\sqrt{2} \approx 2.828 > 2$, as claimed. $\square$



Three of the four pairs of directions are separated by $\pi/4$, while α_1 and β_1 are separated by $3\pi/4$; that last pair is the one carrying the minus sign in S , so all four terms contribute $-1/\sqrt{2}$ and we collect $4/\sqrt{2} = 2\sqrt{2}$.

This is a genuinely astonishing state of affairs, and it deserves to be stated plainly. The CHSH inequality is a theorem about *any* theory in which measurement outcomes are determined by local properties carried by the particles. Quantum mechanics predicts, and experiment confirms, that the inequality is violated. Therefore no such theory can be correct. Entanglement is not a matter of pre-arranged correlations, like a pair of gloves posted in separate boxes; it is something with no classical counterpart at all.

Remark. The value $2\sqrt{2}$ is not accidental: **Tsirelson's bound** says that $|S| \leq 2\sqrt{2}$ for *any* quantum states and observables, so quantum mechanics saturates but does not exceed it. There is nothing in the no-signalling principle alone that forbids $S = 4$ – hypothetical ‘PR boxes’ achieving this are perfectly consistent with relativity – so quantum mechanics sits strictly between the classical and the merely-relativistic worlds. Why it sits exactly where it does remains a good question.

§39.4.3 The Schmidt Decomposition

Deciding by hand whether a given bipartite state is entangled, as we did for $|\phi^+\rangle$ in §39.1.7, is not a calculation one wants to repeat often. Happily there is a normal form for bipartite pure states which settles the question at a glance, and tells us rather more besides.

Theorem 39.4.5 (Schmidt Decomposition)

Let $|\Psi\rangle \in \mathcal{V}_A \otimes \mathcal{V}_B$ be normalised, where $\dim \mathcal{V}_A = m$ and $\dim \mathcal{V}_B = n$. Then there are orthonormal sets $\{|a_1\rangle, \dots, |a_k\rangle\} \subseteq \mathcal{V}_A$ and $\{|b_1\rangle, \dots, |b_k\rangle\} \subseteq \mathcal{V}_B$, with $k \leq \min(m, n)$, and reals $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0$ satisfying $\sum_i \lambda_i^2 = 1$, such that

$$|\Psi\rangle = \sum_{i=1}^k \lambda_i |a_i\rangle \otimes |b_i\rangle.$$

The integer k and the numbers λ_i are determined by $|\Psi\rangle$ alone.

Proof. Fix orthonormal bases $\{|e_i\rangle\}$ of $\mathcal{V}_A$ and $\{|f_j\rangle\}$ of $\mathcal{V}_B$ and write

$$|\Psi\rangle = \sum_{i=1}^m \sum_{j=1}^n C_{ij} |e_i\rangle |f_j\rangle,$$

so that the state is nothing but an $m \times n$ complex matrix C with $\sum_{ij} |C_{ij}|^2 = 1$. By the singular value decomposition of IB Linear Algebra we may write $C = UDV$, where U is an $m \times m$ unitary, V is an $n \times n$ unitary, and D is $m \times n$ with non-negative reals $\lambda_1 \geq \dots \geq \lambda_{\min(m,n)}$ on its leading diagonal and zeros elsewhere. Substituting $C_{ij} = \sum_{\ell} U_{i\ell} \lambda_{\ell} V_{\ell j}$ and exchanging the order of summation,

$$|\Psi\rangle = \sum_{\ell} \lambda_{\ell} \left(\sum_i U_{i\ell} |e_i\rangle \right) \otimes \left(\sum_j V_{\ell j} |f_j\rangle \right) = \sum_{\ell} \lambda_{\ell} |a_{\ell}\rangle \otimes |b_{\ell}\rangle,$$

where the bracketed vectors are named $|a_\ell\rangle$ and $|b_\ell\rangle$. The columns of U are orthonormal and the rows of V are orthonormal, so

$$\langle a_\ell | a_{\ell'} \rangle = \sum_i U_{i\ell}^* U_{i\ell'} = \delta_{\ell\ell'}, \quad \langle b_\ell | b_{\ell'} \rangle = \sum_j V_{\ell j} V_{\ell' j}^* = \delta_{\ell\ell'},$$

as required. Deleting the terms with $\lambda_\ell = 0$ and keeping the rest in decreasing order leaves exactly k terms, and $1 = \langle \Psi | \Psi \rangle = \sum_\ell \lambda_\ell^2$.

For uniqueness, observe that the λ_ℓ^2 are precisely the nonzero eigenvalues of $CC^\dagger$, listed with multiplicity. Changing the two bases replaces C by PCQ for unitaries P, Q , which replaces $CC^\dagger$ by $P(CC^\dagger)P^\dagger$ and so leaves the eigenvalues alone. Hence k and the λ_ℓ depend only on $|\Psi\rangle$. $\square$

Definition 39.4.6 (Schmidt Rank and Coefficients)

In the notation of the theorem, k is the **Schmidt rank** of $|\Psi\rangle$ and $\lambda_1, \dots, \lambda_k$ are its **Schmidt coefficients**.

The point of all this is the following one-line criterion.

Corollary 39.4.7

A bipartite pure state is a product state if and only if its Schmidt rank is 1; equivalently, it is entangled if and only if $k \geq 2$.

Proof. If $k = 1$ then $|\Psi\rangle = |a_1\rangle |b_1\rangle$ is a product. Conversely, a product state $|\psi\rangle |\varphi\rangle$ is already in Schmidt form with $k = 1$, and k is unique. $\square$

Example 39.4.8

For $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ we have $C = \frac{1}{\sqrt{2}}I$, whose singular values are $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$. So the Schmidt rank is 2 and the state is entangled – which is the argument of §39.1.7 with the cleverness removed. Similarly $|\psi^-\rangle$ has $C = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, again with both singular values $\frac{1}{\sqrt{2}}$.

The Schmidt coefficients say more than merely ‘entangled or not’: they measure *how* entangled a state is. Notice that they are unchanged by local unitaries, since $(U_A \otimes U_B) |\Psi\rangle = \sum_i \lambda_i (U_A |a_i\rangle) \otimes (U_B |b_i\rangle)$ is again in Schmidt form with the same coefficients. So anything Alice and Bob can compute from the λ_i is a genuine property of their shared resource, invariant under whatever they choose to do locally. A two-qubit state with $\lambda_1 = \lambda_2 = 1/\sqrt{2}$ is called **maximally entangled**, and the four Bell states are exactly the maximally entangled states of two qubits, up to local unitaries. We will meet the λ_i^2 again in §39.15.2 as the eigenvalues of the reduced density matrix, which is where their interpretation as a probability distribution comes from.

Remark. There is no Schmidt decomposition for three or more parties. The three-qubit states

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad |W\rangle = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$$

are both entangled, and neither can be converted into the other by local operations

even with a small probability of success: multipartite entanglement comes in genuinely different kinds, and classifying them is an active research problem.

§39.4.4 Entanglement as a Resource

The modern point of view treats entanglement as a commodity. Alice and Bob are imagined to be restricted to **LOCC**: local operations (unitaries, measurements, ancillas on their own systems) and classical communication. Under this restriction, entanglement cannot be created – a product state stays a product state – so shared entangled pairs, or **ebits**, are a resource which is consumed by protocols.

The two basic conversion rates are established by the protocols of the next section, and they are worth stating in advance:

1 ebit + 1 qubit of communication $\geq$ 2 bits of communication (superdense coding),

1 ebit + 2 bits of communication $\geq$ 1 qubit of communication (teleportation).

Both are optimal. Neither is possible classically, and neither is possible without the shared entanglement.

§39.5 Quantum Gates and Circuits

A quantum computation, in the model we shall use, is a sequence of unitary operators applied to a fixed number of qubits, followed by a measurement. The unitaries are drawn from a fixed repertoire of **gates**, each acting on a small number of qubits, and we picture the whole thing as a **circuit**: time runs left to right, each qubit is a horizontal wire, and a gate is a box straddling the wires it acts on.

§39.5.1 Single-Qubit Gates

The most important gate in the subject is the following.

Definition 39.5.1 (Hadamard Gate)

The **Hadamard gate** is

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

It is Hermitian, unitary, and satisfies $H^2 = I$. Its action is

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle,$$

so H is exactly the change of basis between the computational and conjugate bases. Compactly,

$$H|x\rangle = \frac{1}{\sqrt{2}} (|0\rangle + (-1)^x |1\rangle) = \frac{1}{\sqrt{2}} \sum_{y \in \{0,1\}} (-1)^{xy} |y\rangle, \quad x \in \{0,1\}.$$

The reason H matters so much is that it manufactures superpositions. Applying it to each of n qubits initialised to $|0\rangle$ gives

$$H^{\otimes n} |0\rangle^{\otimes n} = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle,$$

the **equal superposition** over all 2^n bit strings, at a cost of n gates. An exponentially large superposition for a linear price is the starting point of every quantum algorithm we will meet. More generally, one checks by induction that

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle,$$

where $x \cdot y = x_1 y_1 + \cdots + x_n y_n \bmod 2$.

The Pauli operators give three more gates, written $X = \sigma_x$, $Y = i\sigma_y$ and $Z = \sigma_z$ in this context (the factor of i in Y is a global phase convention which makes the entries real). Thus

$$X |k\rangle = |k \oplus 1\rangle, \quad Z |k\rangle = (-1)^k |k\rangle,$$

where $\oplus$ is addition modulo 2: X is the quantum NOT gate, and Z inserts a relative sign. Finally we will need a continuous family.

Definition 39.5.2 (Phase Gate)

For $\theta \in \mathbb{R}$ the **phase gate** is

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}.$$

Note $Z = P_\pi$, and two other special cases have names: $S = P_{\pi/2}$ and $T = P_{\pi/4}$. In circuits, a single-qubit gate is drawn as a labelled box on a wire:



§39.5.2 Two-Qubit Gates

To do anything beyond acting on qubits separately we need gates which couple them.

Definition 39.5.3 (Controlled- X)

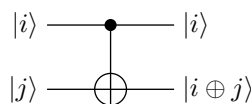
The **controlled-not** or **CNOT** gate CX on two qubits is defined by

$$CX |i\rangle |j\rangle = |i\rangle |i \oplus j\rangle, \quad i, j \in \{0, 1\},$$

so that in the computational basis

$$CX = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}.$$

The first qubit is the **control** and the second the **target**: if the control is $|0\rangle$ nothing happens, and if it is $|1\rangle$ then X is applied to the target. By linearity, $CX |0\rangle |\psi\rangle = |0\rangle |\psi\rangle$ and $CX |1\rangle |\psi\rangle = |1\rangle X |\psi\rangle$ for any $|\psi\rangle$. The gate is drawn with a filled dot on the control wire and a $\oplus$ on the target:



Replacing X by Z gives the **controlled- Z** gate,

$$\text{CZ} = \begin{pmatrix} I & 0 \\ 0 & Z \end{pmatrix}, \quad \text{CZ} |i\rangle |j\rangle = (-1)^{ij} |i\rangle |j\rangle.$$

The second expression makes it obvious that CZ is *symmetric* in its two qubits, so it does not matter which we call the control. That is not at all obvious from the circuit picture, and it is a good illustration of how much easier algebra can be than diagrams.

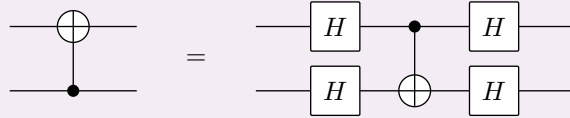
The two controlled gates are conjugate to one another via Hadamards on the target, since $HXH = Z$:

$$\text{CZ} = (I \otimes H) \text{CX} (I \otimes H).$$

A more surprising identity is that conjugating CX by Hadamards on *both* wires reverses the roles of control and target.

Proposition 39.5.4

$(H \otimes H) \text{CX} (H \otimes H)$ is the CNOT gate with control and target interchanged. Diagrammatically,

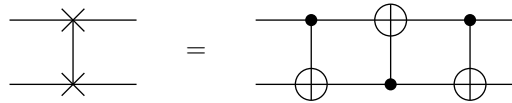


Proof. Since $H^2 = I$ we may split the conjugation into two steps:

$$(H \otimes H) \text{CX} (H \otimes H) = (H \otimes I) \left[(I \otimes H) \text{CX} (I \otimes H) \right] (H \otimes I) = (H \otimes I) \text{CZ} (H \otimes I),$$

using the identity $\text{CZ} = (I \otimes H) \text{CX} (I \otimes H)$ noted above. But CZ is symmetric in its two qubits, so it is equally the controlled- Z gate with the *second* qubit as control; conjugating by H on the first wire therefore turns it into the controlled- X gate with the second qubit as control, which is what we wanted. $\square$

Finally, the **swap** gate $|i\rangle |j\rangle \mapsto |j\rangle |i\rangle$ is drawn with two crosses joined by a wire, and factors as three CNOTs:



§39.5.3 Reversible Computation and the Toffoli Gate

There is an immediate obstacle to running classical computations on a quantum computer: quantum gates are unitary and hence invertible, whereas classical gates typically are not. The AND gate throws away information – from the output 0 one cannot recover the input – and no unitary can do that.

The fix is standard, and it is the same trick we saw with the ancilla. Any function $f : \{0, 1\}^m \rightarrow \{0, 1\}^n$ can be made reversible by remembering its input: define

$$\tilde{f} : \{0, 1\}^{m+n} \rightarrow \{0, 1\}^{m+n}, \quad \tilde{f}(b, c) = (b, c \oplus f(b)).$$

Then $\tilde{f}$ is a bijection – indeed an involution, since

$$\tilde{f}(\tilde{f}(b, c)) = \tilde{f}(b, c \oplus f(b)) = (b, c \oplus f(b) \oplus f(b)) = (b, c),$$

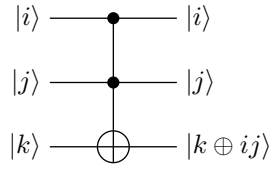
and f is recovered as $f(b) =$ the second component of $\tilde{f}(b, 0)$. So computing f and computing $\tilde{f}$ are equivalent tasks, up to a constant factor in resources.

The gate that makes this work in practice is the following three-qubit gate.

Definition 39.5.5 (Toffoli Gate)

The **Toffoli gate**, or controlled-controlled-NOT, is the map

$$T : |i\rangle |j\rangle |k\rangle \mapsto |i\rangle |j\rangle |k \oplus ij\rangle.$$



Setting $k = 0$ gives $|i\rangle |j\rangle |0\rangle \mapsto |i\rangle |j\rangle |i \wedge j\rangle$, so Toffoli computes AND; setting $j = k = 1$ gives $|i\rangle |1\rangle |1\rangle \mapsto |i\rangle |1\rangle |\neg i\rangle$, so it computes NOT. Since $\{\text{AND}, \text{NOT}\}$ is universal for Boolean logic, the Toffoli gate alone (with the freedom to initialise ancillas and ignore garbage outputs) is universal for classical computation. In particular:

Proposition 39.5.6

Any classical circuit computing f with G gates can be converted into a reversible circuit computing $\tilde{f}$ with $O(G)$ Toffoli gates and $O(G)$ ancillary bits, and hence into a quantum circuit of the same size.

We will not prove this carefully – the only real content is the bookkeeping needed to uncompute the intermediate ‘garbage’ bits so that the ancillas are returned to $|0\rangle$ – but the consequence is important: *a quantum computer can do anything a classical computer can do, with at most polynomial overhead.* Quantum computation is not a different kind of computability; it is a claim about efficiency.

§39.5.4 Universal Gate Sets

Which sets of gates suffice to build everything?

Definition 39.5.7 (Universal Gate Set)

A set $\mathcal{G}$ of gates is **universal** if every unitary on every number of qubits can be written as a finite product of gates from $\mathcal{G}$ (acting on specified qubits).

Theorem 39.5.8

The set consisting of CX together with all single-qubit unitaries is universal.

We shall not give a complete proof, but the shape of the argument is worth knowing, since it is entirely constructive and each of its three steps is elementary.

Proof sketch. Step 1: two-level unitaries suffice. Call a unitary on $\mathbb{C}^{2^n}$ **two-level** if it acts as the identity on all but two of the computational basis vectors, so that it is essentially a 2×2 unitary embedded in a large space. Given any unitary U , one clears the entries of its first column below the diagonal one at a time by multiplying by suitable two-level unitaries – exactly the Givens-rotation procedure used to compute a QR factorisation – then repeats on the second column, and so on. After at most $\frac{1}{2}2^n(2^n - 1)$ such steps we have reduced U to the identity, so U is a product of that many two-level unitaries.

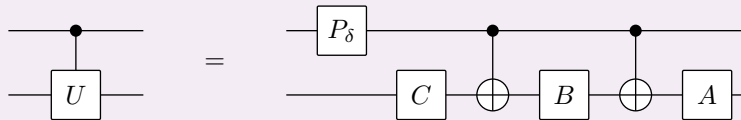
Step 2: a two-level unitary is a multiply-controlled single-qubit gate. Suppose the two-level unitary V acts nontrivially on the span of $|x\rangle$ and $|y\rangle$, where $x, y \in \mathbb{B}_n$. Choose a path $x = z_0, z_1, \dots, z_\ell = y$ in which consecutive strings differ in one bit (a Gray code path), and apply CNOTs to relabel, so that we may assume x and y differ in a single bit, say the last. Then V is a single-qubit gate on the last qubit, controlled on the other $n - 1$ qubits taking the prescribed values – and a control on the value 0 is a control on 1 conjugated by X .

Step 3: multiple controls reduce to single controls. A gate controlled on $n - 1$ qubits can be built from Toffoli gates, which compute the AND of the controls into an ancilla, a singly-controlled gate, and the same Toffolis again to uncompute the ancilla. The proposition below then reduces a singly-controlled single-qubit gate to CNOTs and single-qubit gates, and the Toffoli gate itself is a fixed circuit of CNOTs and single-qubit gates. $\square$

The last step used the following pretty construction, which is worth seeing in full.

Proposition 39.5.9

Let U be a single-qubit unitary. Then there are single-qubit unitaries A, B, C with $ABC = I$ and a phase δ such that $U = e^{i\delta}AXBXC$. Consequently the controlled- U gate factors as



Proof. Writing $R_y(\theta) = \exp(-i\theta\sigma_y/2)$ and $R_z(\theta) = \exp(-i\theta\sigma_z/2)$ for the Bloch-sphere rotations about the y - and z -axes, every single-qubit unitary has an Euler-angle decomposition

$$U = e^{i\delta} R_z(\alpha) R_y(\beta) R_z(\gamma)$$

for some real $\delta, \alpha, \beta, \gamma$ – this is the statement that $SU(2)$ surjects onto $SO(3)$, together with the Euler decomposition of a rotation of $\mathbb{R}^3$. Now put

$$A = R_z(\alpha) R_y\left(\frac{\beta}{2}\right), \quad B = R_y\left(-\frac{\beta}{2}\right) R_z\left(-\frac{\alpha+\gamma}{2}\right), \quad C = R_z\left(\frac{\gamma-\alpha}{2}\right).$$

Then $ABC = R_z(\alpha) R_z\left(-\frac{\alpha+\gamma}{2}\right) R_y\left(\frac{\beta}{2}\right) R_y\left(-\frac{\beta}{2}\right) R_z\left(\frac{\gamma-\alpha}{2}\right) = R_z(\alpha) R_z(-\alpha) = I$. On the other hand $X\sigma_y X = -\sigma_y$ and $X\sigma_z X = -\sigma_z$, so $XR_y(\theta)X = R_y(-\theta)$ and $XR_z(\theta)X = R_z(-\theta)$, whence $AXBX = R_y\left(\frac{\beta}{2}\right) R_z\left(\frac{\alpha+\gamma}{2}\right)$ and

$$AXBXC = R_z(\alpha) R_y\left(\frac{\beta}{2}\right) R_y\left(\frac{\beta}{2}\right) R_z\left(\frac{\alpha+\gamma}{2}\right) R_z\left(\frac{\gamma-\alpha}{2}\right) = R_z(\alpha) R_y(\beta) R_z(\gamma) = e^{-i\delta} U.$$

For the circuit: if the control is $|0\rangle$ the two CNOTs do nothing and the target sees $ABC = I$, while the phase gate P_δ acts trivially on $|0\rangle$. If the control is $|1\rangle$ the target sees $AXBXC = e^{-i\delta}U$ and the phase gate contributes the compensating factor $e^{i\delta}$. So the circuit implements $|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$, which is controlled- U . $\square$

The theorem is very useful but slightly unsatisfying, because it involves a continuum of gates. A physical device will only be able to implement finitely many, so we ask for less.

Definition 39.5.10 (Approximate Universality)

A finite set $\mathcal{G}$ of gates is **approximately universal** if for every n , every unitary U on n qubits and every $\varepsilon > 0$, there is a circuit $\tilde{U}$ built from gates of $\mathcal{G}$ with

$$\|U - \tilde{U}\|_{\text{op}} < \varepsilon,$$

where $\|A\|_{\text{op}} = \sup_{\|\psi\rangle=1} \|A|\psi\rangle\|$ is the operator norm.

Approximation in operator norm is exactly what we want, since if $\|U - \tilde{U}\|_{\text{op}} < \varepsilon$ then all measurement probabilities differ by $O(\varepsilon)$, and errors accumulate at worst additively along a circuit.

Theorem 39.5.11

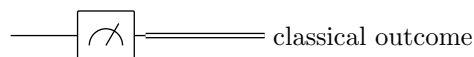
The set $\mathcal{G} = \{H, CX, T\}$, where $T = P_{\pi/4}$, is approximately universal.

Remark. One might worry that approximating to accuracy ε could require a number of gates growing like $1/\varepsilon$, which would destroy any speedup. The **Solovay–Kitaev theorem** says otherwise: for any approximately universal set closed under inverses, accuracy ε on a single qubit is achievable with $O(\log^c(1/\varepsilon))$ gates, for a small constant c . So the overhead is polylogarithmic, and we may cheerfully talk about ‘applying a unitary U ’ without worrying about which gate set is used to build it.

§39.5.5 Circuits and Measurement

A quantum circuit on n qubits consists of: an input state $|b_1\rangle \cdots |b_k\rangle |0\rangle \cdots |0\rangle$, with the input bit string in the first few qubits and the rest as workspace; a sequence of gates from a fixed universal set; and a measurement of some designated qubits in the computational basis at the end. The output is the string of measurement outcomes.

Two conventions make life easier. First, we draw a measurement as a meter symbol, and a wire carrying a classical bit as a doubled line. Second, we use the **principle of deferred measurement**: any circuit in which measurements occur partway through, with later gates conditioned on the outcomes, can be replaced by an equivalent circuit in which all measurements occur at the very end, replacing each classically-controlled gate by a quantum-controlled one. So we lose nothing by assuming all measurements come last; but for exposition it is often clearer to draw them where they physically happen, as we will for teleportation.



Finally, note that we do not need to add randomness to the model by hand, as one

does for classical probabilistic computation. A Hadamard on $|0\rangle$ followed by a standard measurement gives a perfect fair coin. Randomness comes free.

§39.6 Superdense Coding

We can now start spending our entanglement. The first protocol is due to Bennett and Wiesner: Alice sends Bob *two* classical bits by physically transmitting a single qubit, provided they share an ebit in advance.

Recall from §39.4.1 that Alice can convert $|\phi^+\rangle$ into any of the four Bell states by acting on her qubit alone. Since the Bell states are orthonormal, they are reliably distinguishable by a measurement in the Bell basis. That is the entire protocol.

Proposition 39.6.1 (Superdense Coding)

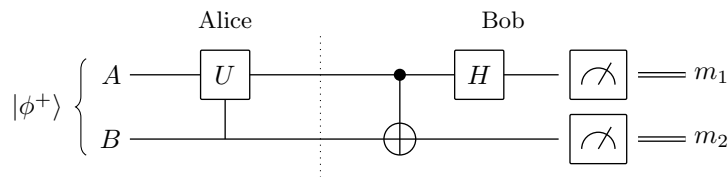
Suppose Alice and Bob share the state $|\phi^+\rangle_{AB}$, with Alice holding qubit A and Bob holding B . Then Alice can transmit any two-bit message to Bob by sending him the single qubit A .

Proof. Alice applies to her qubit the local operation determined by her message:

message	Alice's operation on A	resulting state of AB
00	I	$ \phi^+\rangle$
01	X	$ \psi^+\rangle$
10	Z	$ \phi^-\rangle$
11	Y	$ \psi^-\rangle$

She then sends qubit A to Bob, who now holds both qubits and performs a complete projective measurement in the Bell basis, with projectors $|\phi^+\rangle\langle\phi^+|$, $|\phi^-\rangle\langle\phi^-|$, $|\psi^+\rangle\langle\psi^+|$, $|\psi^-\rangle\langle\psi^-|$. Since the four possible states are exactly these four orthonormal vectors, the outcome identifies Alice's message with probability 1. $\square$

Bob's Bell measurement is performed in practice by rotating the Bell basis to the computational basis and then measuring in the standard way. The circuit that does this is a CNOT followed by a Hadamard, run on the two qubits:



Here $U \in \{I, X, Z, Y\}$ encodes Alice's message, and one checks that the CNOT-then-Hadamard circuit maps

$$|\phi^+\rangle \mapsto |00\rangle, \quad |\psi^+\rangle \mapsto |01\rangle, \quad |\phi^-\rangle \mapsto |10\rangle, \quad |\psi^-\rangle \mapsto |11\rangle,$$

so the pair of measurement outcomes (m_1, m_2) is exactly Alice's message. For instance

$$|\phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \xrightarrow{\text{CX}} \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = |-\rangle|0\rangle \xrightarrow{H \otimes I} |1\rangle|0\rangle,$$

as claimed.

Two remarks. First, the protocol is optimal: *Holevo's theorem* says that a single qubit can never convey more than one bit of classical information on its own, and the shared entanglement is what allows us to double this. Second, the qubit Alice transmits is, by itself, completely uninformative – it is maximally entangled with Bob's, and by the argument of §39.3.3 any measurement on it alone yields uniformly random results. An eavesdropper who intercepts the qubit in transit learns exactly nothing about the message.

§39.7 Quantum Teleportation

Superdense coding sends classical information down a quantum channel. Teleportation does the reverse, and is more startling: Alice transfers an *unknown* qubit state to Bob using only two classical bits and a shared ebit. No quantum system travels between them.

Theorem 39.7.1 (Quantum Teleportation)

Suppose Alice holds a qubit C in an arbitrary unknown state $|\psi\rangle_C$, and that Alice and Bob share the Bell state $|\phi^+\rangle_{AB}$. Then by performing a measurement on CA and sending Bob the two-bit outcome over a classical channel, Alice can arrange for Bob's qubit B to end in the state $|\psi\rangle$.

Proof. Write $|\psi\rangle_C = a|0\rangle + b|1\rangle$. The initial state of the three qubits CAB is

$$|\Psi_0\rangle = |\psi\rangle_C \otimes |\phi^+\rangle_{AB} = \frac{1}{\sqrt{2}} [a|000\rangle + a|011\rangle + b|100\rangle + b|111\rangle].$$

Alice applies CX with C as control and A as target. This flips the second bit exactly when the first is 1:

$$|\Psi_1\rangle = \frac{1}{\sqrt{2}} [a|000\rangle + a|011\rangle + b|110\rangle + b|101\rangle].$$

She then applies H to C , replacing $|0\rangle_C \mapsto |+\rangle_C$ and $|1\rangle_C \mapsto |-\rangle_C$:

$$|\Psi_2\rangle = \frac{1}{\sqrt{2}} [a|+\rangle|00\rangle + a|+\rangle|11\rangle + b|-\rangle|10\rangle + b|-\rangle|01\rangle].$$

Expanding $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ and collecting terms according to the state of CA ,

$$\begin{aligned} |\Psi_2\rangle &= \frac{1}{2} \left[|00\rangle (a|0\rangle + b|1\rangle) + |01\rangle (a|1\rangle + b|0\rangle) \right. \\ &\quad \left. + |10\rangle (a|0\rangle - b|1\rangle) + |11\rangle (a|1\rangle - b|0\rangle) \right] \\ &= \frac{1}{2} \left[|00\rangle |\psi\rangle + |01\rangle X|\psi\rangle + |10\rangle Z|\psi\rangle + |11\rangle XZ|\psi\rangle \right], \end{aligned}$$

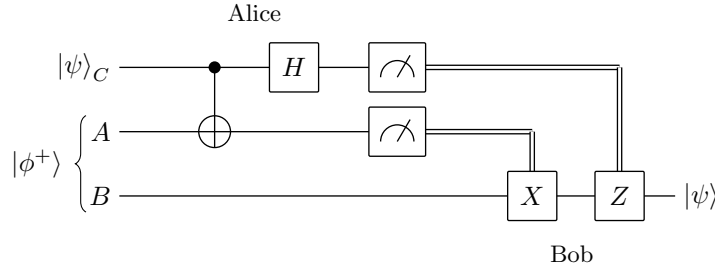
where in the last line we used $X|\psi\rangle = a|1\rangle + b|0\rangle$, $Z|\psi\rangle = a|0\rangle - b|1\rangle$ and $XZ|\psi\rangle = a|1\rangle - b|0\rangle$.

Alice now measures C and A in the computational basis. By the extended Born rule each of the four outcomes ij occurs with probability $\frac{1}{4}$ – *independently of a and b* ,

which is essential, since otherwise the outcome distribution would leak information about $|\psi\rangle$ and hence permit signalling. Given outcome ij , Bob's qubit is left in the state $X^j Z^i |\psi\rangle$.

Alice sends the two bits i, j to Bob classically. Since $X^2 = Z^2 = I$, Bob applies X^j and then Z^i to his qubit, recovering

$$Z^i X^j X^j Z^i |\psi\rangle = |\psi\rangle. \quad \square$$



Several things are worth pausing over.

No cloning is violated. At the end, Bob has $|\psi\rangle$ and Alice does not: her qubit C was measured and collapsed to $|i\rangle$, destroying the original. Teleportation moves a state; it does not copy it.

No signalling is violated. Bob's qubit, before he hears from Alice, is in a uniformly random one of $|\psi\rangle, X|\psi\rangle, Z|\psi\rangle, XZ|\psi\rangle$, and by the computation of §39.3.3 this is indistinguishable from a maximally mixed qubit. He learns nothing until the classical bits arrive, so the protocol runs at the speed of the classical channel.

Alice does not learn $|\psi\rangle$. She acquires two random bits, which by construction carry no information about a and b . This is consistent with the fact that an unknown qubit state cannot be determined from a single copy.

The first half of the protocol is a Bell measurement. Applying CX then H and measuring in the computational basis is precisely a measurement of CA in the Bell basis, as we saw in §39.6 – the same circuit, run in the same direction. So the protocol reads: Alice performs a Bell measurement on the unknown qubit and her half of the entangled pair; the result is two random bits which tell Bob which of four unitaries to undo.

Remark. Teleportation consumes exactly one ebit and two classical bits per qubit sent, and both costs are optimal. One classical bit would not suffice: if it did, then combined with superdense coding one could send two classical bits using one ebit and one classical bit, and iterating would give unlimited communication for free.

§39.8 Quantum Cryptography

Our third application of quantum information is the one that has actually been deployed commercially. The task is **key distribution**: Alice and Bob, who have never met, wish to establish a shared secret bit string, in the presence of an eavesdropper Eve who can listen to everything. Classically this is impossible without computational assumptions. Quantum mechanically it can be done, and the security rests on physics rather than on the presumed hardness of factoring.

§39.8.1 The One-Time Pad

First, why we want a key at all. Suppose Alice and Bob already share a uniformly random secret string $K \in \{0, 1\}^n$, and Alice wishes to send a message $M \in \{0, 1\}^n$ over a public channel.

Definition 39.8.1 (One-Time Pad)

The **one-time pad** protocol is: Alice transmits the ciphertext $C = M \oplus K$; Bob computes $C \oplus K = M \oplus K \oplus K = M$.

Proposition 39.8.2

The one-time pad is perfectly secure: for any message M , the ciphertext C is uniformly distributed on $\{0, 1\}^n$, and hence independent of M .

Proof. For fixed M , the map $K \mapsto M \oplus K$ is a bijection of $\{0, 1\}^n$, so if K is uniform then so is C . Thus $\Pr(C = c \mid M = m) = 2^{-n}$ for every c and m , and Eve's posterior distribution on M equals her prior. $\square$

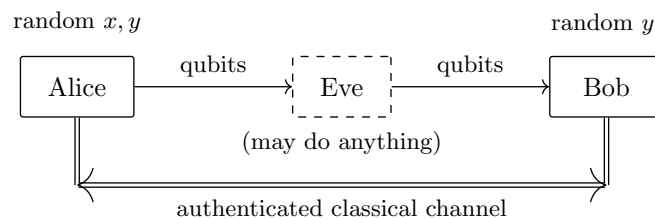
This is unbreakable, not merely hard to break, and there is no cleverness in it whatsoever. Its drawback is in the name: the pad must be used only *once*. If Alice sends $C_1 = M_1 \oplus K$ and $C_2 = M_2 \oplus K$ with the same key, then Eve computes

$$C_1 \oplus C_2 = M_1 \oplus M_2,$$

which typically reveals a great deal about both messages. So Alice and Bob need a fresh key, as long as the message, for every transmission – and they need to establish it privately. That is the problem we now solve.

§39.8.2 The BB84 Protocol

We assume Alice and Bob have two channels: an *authenticated classical channel*, which Eve may listen to but not tamper with or spoof, and a *quantum channel* down which Alice can send qubits, and over which Eve may do absolutely anything.

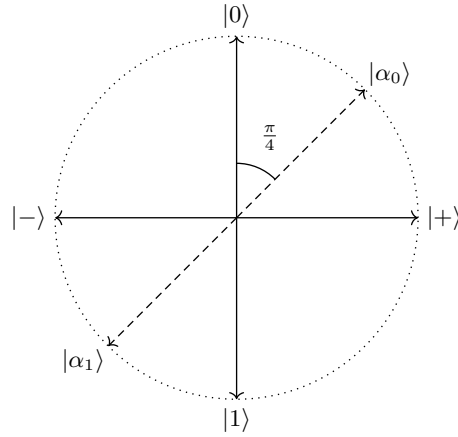


The idea, due to Bennett and Brassard in 1984, exploits two facts we have already proved. Non-orthogonal states cannot be reliably distinguished (§39.2.3), and unknown states cannot be cloned (§39.3.2). So if Alice encodes her bits in randomly chosen states from two mutually unbiased bases, Eve cannot read them without guessing the basis – and guessing wrong disturbs the state, leaving evidence.

Recall the two bases

$$\mathcal{B}_0 = \{|0\rangle, |1\rangle\}, \quad \mathcal{B}_1 = \{|+\rangle, |-\rangle\},$$

which are mutually unbiased: $|\langle a|b\rangle|^2 = \frac{1}{2}$ for every $|a\rangle \in \mathcal{B}_0$, $|b\rangle \in \mathcal{B}_1$. On the Bloch circle in the x - z plane the four states are spaced at right angles:



(The dashed directions are the *Breidbart basis*, which will be Eve's best guess; more on that below. Remember that the Bloch picture doubles angles, so states $\pi/2$ apart on the circle are orthogonal in $\mathbb{C}^2$, and the Breidbart vectors, at $\pi/4$, have overlap $\cos(\pi/8)$ with their neighbours.)

Definition 39.8.3 (BB84)

The **BB84 protocol** runs as follows.

- (i) Alice generates two uniformly random strings $x, y \in \{0, 1\}^m$. The string x holds the raw key bits; the string y records which basis is used for each.
- (ii) Alice prepares the m -qubit state $|\psi_{xy}\rangle = |\psi_{x_1 y_1}\rangle \otimes \cdots \otimes |\psi_{x_m y_m}\rangle$, where

$$|\psi_{00}\rangle = |0\rangle, \quad |\psi_{10}\rangle = |1\rangle, \quad |\psi_{01}\rangle = |+\rangle, \quad |\psi_{11}\rangle = |-\rangle,$$

that is, the bit x_i is encoded in the basis $\mathcal{B}_{y_i}$. She sends the qubits to Bob.

- (iii) Bob generates a uniformly random string $y' \in \{0, 1\}^m$ and measures the i th qubit he receives in the basis $\mathcal{B}_{y'_i}$, recording the outcomes as $x' \in \{0, 1\}^m$.
- (iv) (*Sifting.*) Alice and Bob announce y and y' on the classical channel and discard every position i with $y_i \neq y'_i$. The retained bits form the **sifted key** $\tilde{x}, \tilde{x}'$.
- (v) (*Parameter estimation.*) They publicly compare a random sample of the sifted bits to estimate the **bit error rate**, then discard the sampled positions.
- (vi) (*Reconciliation and privacy amplification.*) They correct residual errors and compress the remaining string to remove any partial information Eve may hold.

Proposition 39.8.4

In the absence of noise or eavesdropping, $x'_i = x_i$ at every retained position, and the expected length of the sifted key is $m/2$.

Proof. If $y'_i = y_i$ then Bob measures in the same basis in which Alice prepared, so $|\psi_{x_i y_i}\rangle$ is a basis vector of $\mathcal{B}_{y'_i}$ and the Born rule gives outcome x_i with probability 1. Since y and y' are independent and uniform, $\Pr(y_i = y'_i) = \frac{1}{2}$ for each i independently, so the expected number of retained positions is $m/2$. $\square$

If Bob happens to choose the wrong basis, the mutual unbiasedness makes his outcome a fair coin, uncorrelated with x_i – which is exactly why those positions are thrown away.

Example 39.8.5

Take $m = 8$, and suppose

$$x = 01110100, \quad y = 11010001,$$

so that Alice sends $|+\rangle, |-\rangle, |1\rangle, |-\rangle, |0\rangle, |1\rangle, |0\rangle, |+\rangle$. Suppose Bob receives these undisturbed and picks $y' = 01110110$.

i	1	2	3	4	5	6	7	8
x_i	0	1	1	1	0	1	0	0
y_i	1	1	0	1	0	0	0	1
sent	$ +\rangle$	$ -\rangle$	$ 1\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ +\rangle$
y'_i	0	1	1	1	0	1	1	0
keep?	$\times$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\times$	$\times$	$\times$

At $i = 1$ Bob measures $|+\rangle$ in $\mathcal{B}_0$ and gets a uniformly random bit, so this position is (rightly) discarded. At $i = 2$ he measures $|-\rangle$ in $\mathcal{B}_1$ and obtains $x'_2 = 1$ with certainty. Sifting keeps positions 2, 4, 5, giving the key $\tilde{x} = 110$.

§39.8.3 Eavesdropping

Now suppose Eve is on the line. Her problem is that she does not know y , so she does not know which basis to measure in, and she cannot make a copy of the qubit to keep for later by the no-cloning theorem. Consider the simplest attack.

Definition 39.8.6 (Intercept-and-Resend)

In the **intercept-and-resend** attack, Eve intercepts each qubit, measures it in some fixed basis, and forwards the post-measurement state to Bob.

Which basis should she use? She is trying to distinguish four equally likely states, but really she wants to guess the *bit* x_i , so she wants a measurement that does well at separating $\{|0\rangle, |+\rangle\}$ (bit 0) from $\{|1\rangle, |-\rangle\}$ (bit 1). The natural candidate is the basis that bisects them.

Definition 39.8.7 (Braidbart Basis)

The **Braidbart basis** is $\{|\alpha_0\rangle, |\alpha_1\rangle\}$, where

$$|\alpha_0\rangle = \cos \frac{\pi}{8} |0\rangle + \sin \frac{\pi}{8} |1\rangle, \quad |\alpha_1\rangle = -\sin \frac{\pi}{8} |0\rangle + \cos \frac{\pi}{8} |1\rangle.$$

These are the dashed vectors in the picture above: on the Bloch circle they sit at $\pi/4$,

halfway between the $|0\rangle$ and $|+\rangle$ directions, and at $5\pi/4$. A short computation gives

$$|\langle\alpha_0|0\rangle|^2 = |\langle\alpha_0|+\rangle|^2 = \cos^2 \frac{\pi}{8} \approx 0.854, \quad |\langle\alpha_1|1\rangle|^2 = |\langle\alpha_1|-\rangle|^2 = \cos^2 \frac{\pi}{8},$$

so whichever of the four states Alice sent, Eve infers the correct bit with probability $\cos^2(\pi/8) \approx 85\%$. (This is precisely the Holevo–Helstrom optimum of §39.2.3 applied to the two equiprobable mixtures.)

That sounds good for Eve, but it is not free.

Proposition 39.8.8

If Eve performs the intercept-and-resend attack in the Breidbart basis, then at every sifted position Bob’s bit disagrees with Alice’s with probability $\frac{1}{4}$.

Proof. Suppose $y_i = y'_i = 0$ and $x_i = 0$, so Alice sends $|0\rangle$ and Bob measures in $\mathcal{B}_0$; the other cases are identical by symmetry. Eve’s measurement returns 0 with probability $|\langle\alpha_0|0\rangle|^2$ and then forwards $|\alpha_0\rangle$; otherwise it returns 1 and she forwards $|\alpha_1\rangle$. Bob then measures in $\mathcal{B}_0$ and gets the wrong answer 1 with probability $|\langle 1|\alpha_j\rangle|^2$. Hence

$$\begin{aligned} \Pr(x'_i \neq x_i) &= |\langle\alpha_0|0\rangle|^2 |\langle 1|\alpha_0\rangle|^2 + |\langle\alpha_1|0\rangle|^2 |\langle 1|\alpha_1\rangle|^2 \\ &= \cos^2 \frac{\pi}{8} \sin^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \\ &= 2 \cos^2 \frac{\pi}{8} \sin^2 \frac{\pi}{8} = \frac{1}{2} \sin^2 \frac{\pi}{4} = \frac{1}{4}. \square \end{aligned}$$

This is the crux of BB84. Eve’s interference is not merely detectable in principle; it produces a bit error rate of 25% on the sifted key, which Alice and Bob will see immediately in step (v) when they compare a random sample. If the observed error rate is above the threshold they expect from channel noise, they abort and try again. If it is below, they may conclude that Eve has only partial information, and proceed to squeeze her out.

Remark. A full security proof must handle arbitrary attacks, not just intercept-and-resend: Eve is entitled to entangle each qubit with an ancilla of her own, forward the qubit, and defer her measurement until after she has heard the basis announcements. Proving security against such *coherent* attacks is genuinely hard and well beyond this course. The conclusion, though, is the one the simple analysis suggests: there is a threshold error rate (around 11% for BB84) below which a secret key can be extracted, and above which no key is possible.

§39.8.4 Reconciliation and Privacy Amplification

Even with no eavesdropper, real channels are noisy, so $\tilde{x}$ and $\tilde{x}'$ will differ in a few positions. Alice and Bob repair this classically.

Example 39.8.9 (Information Reconciliation with a Hamming Code)

Suppose the estimated bit error rate is around $\frac{1}{7}$ and Alice and Bob hold blocks

$a, b \in \{0, 1\}^7$ differing in at most one position. Let

$$\tilde{H} = \begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

be the parity check matrix of the $[7, 4]$ Hamming code, whose columns are the binary representations of $1, \dots, 7$. Alice computes her syndrome $s^A = \tilde{H}a^\top$ and announces it publicly. Bob computes $s^B = \tilde{H}b^\top$ and forms $s = s^B - s^A = \tilde{H}(b - a)^\top$. Since $b - a$ has weight at most one, and the columns of $\tilde{H}$ are distinct and nonzero, s identifies the error position uniquely (or is zero if there is none), and Bob corrects his string to a .

The cost is that Alice has broadcast three bits of information about a . Together with whatever Eve learned from the quantum channel, this leaves her with partial knowledge of the reconciled string. The final step removes it.

Definition 39.8.10 (Privacy Amplification)

Privacy amplification is the compression of a partially-secret string into a shorter, fully-secret one, by applying a publicly agreed function whose output is nearly uniform given Eve's partial information.

Example 39.8.11

Suppose Alice and Bob share $a^* = (a_1, a_2, a_3) \in \{0, 1\}^3$, uniformly distributed, and Eve knows the value of at most one of the three bits. Let

$$c = (a_1 \oplus a_3, a_2 \oplus a_3) \in \{0, 1\}^2.$$

Conditioning on any single bit a_j leaves the other two uniform and independent, and one checks by enumerating the eight possibilities that c is then uniform on $\{0, 1\}^2$. So Eve's knowledge of one bit of a^* tells her nothing at all about c : we have traded three partially-secret bits for two perfectly secret ones.

In practice one uses a randomly chosen function from a two-universal family of hash functions, and a theorem (the **leftover hash lemma**) guarantees that if Eve's information about the n -bit string is at most t bits, then hashing down to roughly $n - t$ bits produces an essentially uniform key. The output of BB84 is then fed to a one-time pad, and Alice and Bob may talk in perfect confidence.

§39.9 Computational Complexity

We turn now from communication to computation. Before we can say that a quantum algorithm is 'faster', we need to be precise about what we are counting, so this section sets up the classical framework and its quantum analogue.

§39.9.1 Computational Tasks and the Circuit Model

Write $\mathbb{B}_n = \{0, 1\}^n$ for the set of bit strings of length n , and $\mathbb{B}^* = \bigcup_{n \geq 1} \mathbb{B}_n$. A **computational task** takes an input string and produces an output string. The special case in which the output has length 1 is a **decision problem**, and is conveniently encoded by a **language** $L \subseteq \mathbb{B}^*$: the task is to decide, given w , whether $w \in L$. For example, the set of primes written in binary is a language, and the associated decision problem is primality testing. Not every task is a decision problem – **FACTOR**(x), which returns a nontrivial factor of x , is not – but the theory is cleanest for decision problems and we will phrase most things that way.

Our model of computation will be the **circuit model**. For each input size n we specify a circuit C_n : the input $x \in \mathbb{B}_n$ is padded with a supply of extra bits set to 0 (the workspace), a prescribed sequence of Boolean gates is applied to prescribed bits, and the answer is read off from designated output bits. The gates are drawn from a fixed finite universal set, say $\{\text{AND}, \text{OR}, \text{NOT}\}$.

Definition 39.9.1 (Time Complexity)

The **size** of the circuit C_n is its number of gates, written $T(n)$; this is the **running time** of the algorithm. If $T(n) = O(n^k)$ for some fixed k , we write $T(n) = O(\text{poly}(n))$ and call the algorithm **poly-time**.

We may also allow the workspace to be seeded with k uniformly random bits, in which case the output is a random variable and we demand that it be correct with probability bounded away from $\frac{1}{2}$.

Definition 39.9.2 (Classical Complexity Classes)

P is the class of languages whose membership problem has a deterministic poly-time algorithm. **BPP** (**bounded-error probabilistic poly-time**) is the class of languages having a randomised poly-time algorithm which answers correctly with probability at least $\frac{2}{3}$ on every input.

The constant $\frac{2}{3}$ is arbitrary: any constant strictly greater than $\frac{1}{2}$ gives the same class, since running the algorithm $O(\log(1/\varepsilon))$ times and taking a majority vote drives the error below ε by a Chernoff bound. Clearly $\text{P} \subseteq \text{BPP}$, and it is widely believed (though not known) that they are equal. **BPP**, not **P**, is the right notion of ‘efficiently solvable in practice’ classically, and it is **BPP** that the quantum class must be compared with.

Definition 39.9.3 (NP)

A **verifier** for a language L is an algorithm V taking two inputs w (the instance) and c (the **certificate**) such that

- (i) if $w \in L$ then $V(w, c)$ accepts for some c ;
- (ii) if $w \notin L$ then $V(w, c)$ rejects for every c .

V is poly-time if it runs in time $O(\text{poly}(|w|))$ (which in particular bounds the length of any useful certificate). The class **NP** consists of the languages admitting a poly-time verifier.

Equivalently, **NP** is what a non-deterministic machine can do in polynomial time: at

each binary choice the machine forks and explores both branches, accepting if any branch accepts. Problems in NP are easy to *check* but may be hard to *solve*.

The archetype is Boolean satisfiability.

Definition 39.9.4 (SAT)

Given a Boolean formula f in variables $x_1, \dots, x_n$, decide whether there is an assignment with $f(x_1, \dots, x_n) = 1$.

Certificates are satisfying assignments, so $\text{SAT} \in \text{NP}$; and by the Cook–Levin theorem every problem in NP reduces to it in polynomial time, so SAT is **NP-complete**. Brute force takes $O(2^n)$ time, and no substantially better general algorithm is known.

§39.9.2 Quantum Circuits and BQP

The quantum circuit model is the direct analogue. For each n we specify a circuit C_n acting on $\text{poly}(n)$ qubits: the input is $|x_1\rangle \cdots |x_n\rangle |0\rangle \cdots |0\rangle$, we apply a sequence of gates drawn from a fixed approximately universal set, and we measure designated qubits in the computational basis at the end. Two remarks, both made earlier: no separate source of randomness is needed, since measurement supplies it; and by deferred measurement we may assume all measurements happen at the end.

Definition 39.9.5 (BQP)

BQP (**bounded-error quantum poly-time**) is the class of languages L for which there is a family of quantum circuits (C_n) , of size $O(\text{poly}(n))$ and generated by a poly-time classical algorithm, whose output is a correct decision of ‘ $x \in L$?’ with probability at least $\frac{2}{3}$ for every input x .

The uniformity condition – that a classical machine can print the description of C_n in time $\text{poly}(n)$ – is needed to prevent us from smuggling uncomputable information into the choice of circuit; the same caveat applies to the classical circuit classes.

What is known about BQP?

Proposition 39.9.6

$\text{P} \subseteq \text{BPP} \subseteq \text{BQP} \subseteq \text{PSPACE}$.

The first inclusion is trivial. The second holds because a quantum computer can simulate a classical one: by §39.5.3 any classical circuit becomes a reversible circuit of Toffoli gates with polynomial overhead, and Hadamards supply the random bits. The third holds because the amplitude of a given final basis state can be written as a sum over computational paths, and a classical machine can evaluate that sum one term at a time, reusing space; this is the quantum analogue of the fact that $\text{NP} \subseteq \text{PSPACE}$.

What is *not* known is anything separating these classes. In particular, nobody has proved $\text{BPP} \neq \text{BQP}$, so we have no proof that quantum computers are more powerful than classical ones; a proof would imply $\text{P} \neq \text{PSPACE}$, which is a famous open problem. What we have instead is strong evidence: problems such as factoring lie in BQP and are not believed to lie in BPP.

Remark. The relationship between BQP and NP is subtle and often misstated. It is *not* believed that $\text{NP} \subseteq \text{BQP}$: quantum computers are not thought to solve NP-complete problems efficiently. Grover’s algorithm gives a quadratic speedup for exhaustive search, taking 2^n to $2^{n/2}$, which is a real improvement but not a polynomial-time algorithm. Nor is $\text{BQP} \subseteq \text{NP}$ believed. Factoring, the flagship quantum success, is in $\text{NP} \cap \text{coNP}$ and is generally thought not to be NP-complete, so Shor’s algorithm is not evidence that $\text{NP} \subseteq \text{BQP}$.

§39.9.3 Query Complexity

Almost all the quantum algorithms we can currently prove things about are stated in a slightly different framework, where the input is not a string but a *function*, presented as a black box.

Definition 39.9.7 (Black-Box Promise Problem)

A **black-box** or **oracle** problem is a computational task whose input is a function $f : \mathbb{B}_m \rightarrow \mathbb{B}_n$, accessible only by evaluating it at points of our choosing. A **promise** is an a priori restriction on which f may occur.

Definition 39.9.8 (Query Complexity)

The **query complexity** of an algorithm for a black-box problem is the number of evaluations of f it performs. Its **total complexity** is the size of the whole circuit, counting each oracle call as one gate.

The virtue of this framework is that we can *prove* separations in it: because the algorithm is only allowed to learn about f through queries, adversary arguments give unconditional lower bounds on the classical query complexity, and these can be beaten provably by quantum algorithms. The vice is that a query separation does not automatically give a separation for explicitly-presented inputs. Still, the two most important quantum algorithms in existence – Shor’s and Grover’s – both come from this circle of ideas.

§39.10 Quantum Oracles and Phase Kickback

§39.10.1 Quantum Oracles

A classical oracle for f is a box that eats x and produces $f(x)$. This will not do quantum mechanically, since it is not reversible. We use instead the reversible form of §39.5.3.

Definition 39.10.1 (Quantum Oracle)

Let $f : \mathbb{B}_m \rightarrow \mathbb{B}_n$. The corresponding **quantum oracle** is the operator U_f on $m+n$ qubits defined on the computational basis by

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle.$$

The first register is the **input register** and the second the **output register**.

Proposition 39.10.2

U_f is unitary.

Proof. U_f acts on the computational basis as the map $\tilde{f}(x, y) = (x, y \oplus f(x))$ of $\mathbb{B}_{m+n}$ to itself, which we showed in §39.5.3 to be an involution and hence a permutation. A linear map permuting an orthonormal basis has a permutation matrix, which has orthonormal columns, so it is unitary. $\square$

Setting $y = 0$ recovers f , so U_f is at least as useful as a classical oracle. But it is strictly more useful, because we may feed it a superposition. If

$$|\varphi_m\rangle = H^{\otimes m} |0\rangle^{\otimes m} = \frac{1}{\sqrt{2^m}} \sum_{x \in \mathbb{B}_m} |x\rangle$$

is the equal superposition, then

$$U_f |\varphi_m\rangle |0\rangle = \frac{1}{\sqrt{2^m}} \sum_{x \in \mathbb{B}_m} |x\rangle |f(x)\rangle.$$

In a *single* query we have produced a state which somehow contains all 2^m values of f at once. This is often called **quantum parallelism**, and it is important to see immediately why it is not by itself useful: if we measure this state we get a single random pair $(x, f(x))$, exactly as if we had queried the classical oracle at a random point. The values are all there, but reading them out destroys all but one.

The art of quantum algorithm design lies in arranging for the unwanted branches to *interfere destructively*, so that the measurement reveals a global property of f rather than one of its values. The first tool for this is the following.

§39.10.2 Phase Kickback**Proposition 39.10.3 (Phase Kickback)**

Let $f : \mathbb{B}_m \rightarrow \mathbb{B}_1$ and set $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then for every $x \in \mathbb{B}_m$,

$$U_f |x\rangle |-\rangle = (-1)^{f(x)} |x\rangle |-\rangle.$$

Proof. We compute directly:

$$\begin{aligned} U_f |x\rangle |-\rangle &= \frac{1}{\sqrt{2}} (|x\rangle |f(x)\rangle - |x\rangle |1 \oplus f(x)\rangle) \\ &= \begin{cases} \frac{1}{\sqrt{2}} |x\rangle (|0\rangle - |1\rangle) = |x\rangle |-\rangle, & f(x) = 0, \\ \frac{1}{\sqrt{2}} |x\rangle (|1\rangle - |0\rangle) = -|x\rangle |-\rangle, & f(x) = 1, \end{cases} \end{aligned}$$

which is $(-1)^{f(x)} |x\rangle |-\rangle$ in both cases. $\square$

So if we prepare the output register in $|-\rangle$ rather than $|0\rangle$, the oracle does not change the state at all except to attach a sign – the value of f has been ‘kicked back’ from the output register into the phase of the input register. Applying this to the equal superposition,

$$U_f (|\varphi_m\rangle \otimes |-\rangle) = \left(\frac{1}{\sqrt{2^m}} \sum_{x \in \mathbb{B}_m} (-1)^{f(x)} |x\rangle \right) \otimes |-\rangle,$$

and the output register factors out and may be discarded. We are left with the m -qubit state

$$|f\rangle := \frac{1}{\sqrt{2^m}} \sum_{x \in \mathbb{B}_m} (-1)^{f(x)} |x\rangle,$$

in which the whole of f is encoded in a pattern of signs. Signs are what interference is made of, and every algorithm in the next section starts here. Note that $|-\rangle$ is prepared from $|0\rangle$ by HX , or equivalently from $|1\rangle$ by H .

§39.11 Quantum Algorithms I: Oracle Problems

§39.11.1 Deutsch's Algorithm

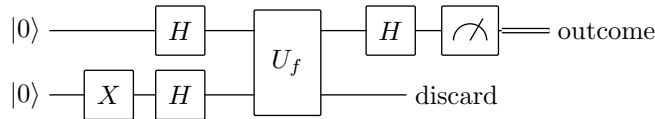
We begin with the smallest possible example, which nonetheless contains the whole idea. Let $f : \mathbb{B}_1 \rightarrow \mathbb{B}_1$ be one of the four functions from one bit to one bit. Two of them ($f(0) = f(1)$) are **constant**, and two ($f(0) \neq f(1)$) are **balanced**.

Problem (Deutsch). *Given an oracle for $f : \mathbb{B}_1 \rightarrow \mathbb{B}_1$, decide whether f is constant or balanced.*

Classically, we must evaluate both $f(0)$ and $f(1)$: knowing one value tells us nothing about the other, so one query is not enough and two queries suffice. Quantum mechanically one query suffices.

Proposition 39.11.1

The following circuit decides Deutsch's problem with certainty, using a single query to U_f . The outcome is 0 if f is constant and 1 if f is balanced.



Proof. After the initial gates the state is $|+\rangle|-\rangle$, that is, $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |-\rangle$. By phase kickback the oracle produces

$$\frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \right) \otimes |-\rangle = \pm \frac{1}{\sqrt{2}} \left(|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle \right) \otimes |-\rangle,$$

where we pulled out the global sign $(-1)^{f(0)}$. The first register is therefore $\pm|+\rangle$ if $f(0) \oplus f(1) = 0$ and $\pm|-\rangle$ if $f(0) \oplus f(1) = 1$. Applying H turns these into $\pm|0\rangle$ and $\pm|1\rangle$ respectively, and the measurement returns $f(0) \oplus f(1)$ with certainty. This is 0 exactly when f is constant. $\square$

The single query has told us a *relational* fact about f – the value of $f(0) \oplus f(1)$ – without telling us either individual value. That trade is the whole of quantum speedup in miniature.

§39.11.2 The Deutsch–Jozsa Algorithm

Now scale it up. Let $f : \mathbb{B}_n \rightarrow \mathbb{B}_1$, and promise that f is either **constant** or **balanced**, meaning that $f(x) = 0$ for exactly half of the 2^n inputs.

Problem (Deutsch–Jozsa). *Given an oracle for such an f , decide with certainty which case holds.*

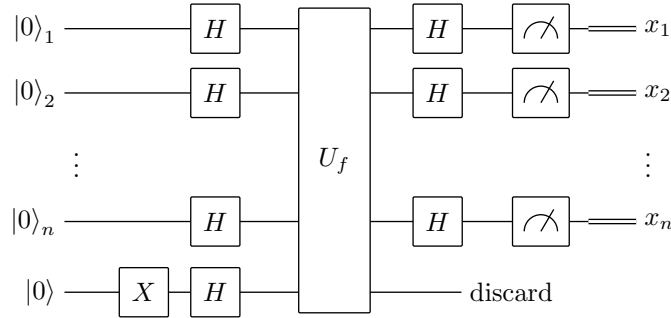
Proposition 39.11.2

Classically, $2^{n-1} + 1$ queries are necessary and sufficient to decide with certainty.

Proof. Sufficient: if we query $2^{n-1} + 1$ distinct points and all the answers agree, f cannot be balanced, since a balanced function takes each value only 2^{n-1} times; and if two answers disagree, f is certainly not constant.

Necessary: suppose an algorithm claims to decide after at most 2^{n-1} queries. Run it against an adversary who answers 0 to every query. After 2^{n-1} queries the algorithm must commit to an answer, but the responses so far are consistent both with the constant function 0 and with a balanced function which is 0 on the queried points and 1 elsewhere. Whatever it answers, one of these two functions defeats it. $\square$

So the classical query complexity is exponential in n . Quantum mechanically it is one.



Theorem 39.11.3 (Deutsch–Jozsa)

The circuit above uses a single query to U_f and $3n + 2$ further operations, and its measurement outcome is $0 \cdots 0$ with probability 1 if f is constant, and with probability 0 if f is balanced.

Proof. The initial gates prepare $|\varphi_n\rangle \otimes |-\rangle$, and by phase kickback (§39.10.2) the oracle leaves the output register in $|-\rangle$ and the input register in

$$|f\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \mathbb{B}_n} (-1)^{f(x)} |x\rangle.$$

We now apply $H^{\otimes n}$ and measure. The amplitude of the outcome $0 \cdots 0$ is

$$\langle 0 \cdots 0 | H^{\otimes n} |f\rangle = \langle \varphi_n | f \rangle = \frac{1}{2^n} \sum_{x \in \mathbb{B}_n} (-1)^{f(x)},$$

using $H^{\otimes n} |0 \cdots 0\rangle = |\varphi_n\rangle$ and $H^{\otimes n} = (H^{\otimes n})^\dagger$.

If f is constant, this sum is $\pm 2^n$, so the amplitude is ± 1 and the outcome is $0 \cdots 0$ with probability 1. (Indeed in that case $|f\rangle = \pm |\varphi_n\rangle$ directly.) If f is balanced, the sum has 2^{n-1} terms equal to $+1$ and 2^{n-1} equal to -1 , so it vanishes and the

outcome $0 \cdots 0$ has probability 0. Hence measuring all zeros certifies ‘constant’, and any other outcome certifies ‘balanced’.

The gate count is $n+1$ Hadamards and one X before the oracle, n Hadamards after, and n measurements, giving $3n+2$ operations besides the single query. $\square$

The interference is visible in the formula: for a balanced function the $+1$ and -1 contributions to the all-zeros amplitude cancel exactly.

Remark. The separation here is exponential but slightly fragile. If we relax ‘with certainty’ to ‘with probability $1 - \varepsilon$ ’, the classical problem becomes easy: query k points at random and answer ‘constant’ if all agree. This errs only when f is balanced but the k samples happened to agree, which has probability $2 \cdot 2^{-k}$, so $k = O(\log(1/\varepsilon))$ queries suffice, independently of n . Deutsch–Jozsa separates *exact* quantum computation from exact classical computation. For a separation which survives bounded error we need Simon’s problem.

§39.11.3 The Bernstein–Vazirani Algorithm

Deutsch–Jozsa extracts a single bit of global information. The next algorithm extracts n bits, and does so in one query rather than n ; it is the cleanest illustration of the fact that a quantum query can be asked ‘in the Fourier basis’.

Problem (Bernstein–Vazirani). Let $a \in \mathbb{B}_n$ be unknown and let $f : \mathbb{B}_n \rightarrow \mathbb{B}_1$ be the linear function

$$f(x) = a \cdot x = a_1 x_1 + \cdots + a_n x_n \bmod 2.$$

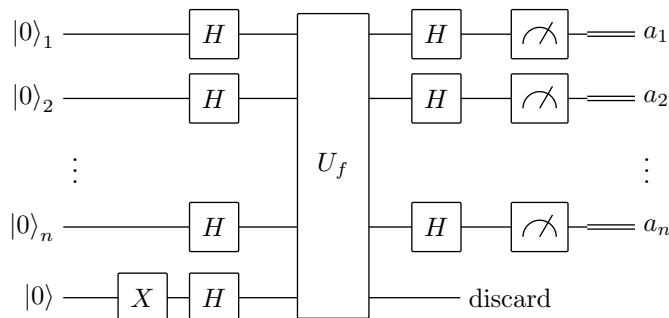
Given an oracle for f , determine a .

Proposition 39.11.4

Classically, n queries are necessary and sufficient.

Proof. Sufficient: query the n standard basis vectors, since $f(e_i) = a_i$. Necessary: each query returns one bit, so k queries have at most 2^k possible transcripts; if $k < n$ then two of the 2^n possible values of a produce the same transcript and cannot be told apart. (One can say this more sharply: the values $f(x)$ for x in a queried set S determine $a \cdot x$ only for $x \in \text{span}(S)$, so they pin a down only to a coset of $\text{span}(S)^\perp$, which is a single point exactly when S spans $\mathbb{B}_n$.) $\square$

The quantum algorithm is the Deutsch–Jozsa circuit, unchanged; only the analysis differs.



Theorem 39.11.5 (Bernstein–Vazirani)

The circuit above outputs a with certainty, using one query to U_f .

Proof. As before the state before the final Hadamards is

$$|f\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \mathbb{B}_n} (-1)^{f(x)} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \mathbb{B}_n} (-1)^{a \cdot x} |x\rangle.$$

But that is exactly the formula $H^{\otimes n} |a\rangle = 2^{-n/2} \sum_x (-1)^{a \cdot x} |x\rangle$ recorded in §39.5. So $|f\rangle = H^{\otimes n} |a\rangle$, and since $H^{\otimes n}$ is its own inverse, applying it once more returns $|a\rangle$ exactly. The measurement reads off the n bits of a with probability 1. $\square$

The algorithm is worth a moment's thought, because the trick is not 'parallelism' in any useful sense. The oracle was applied to a superposition of all 2^n inputs, but what the circuit exploits is that the string a we are looking for is the *Fourier transform* of the sign pattern $(-1)^{a \cdot x}$; the final Hadamards simply undo a transform that the oracle performed for us. Every algorithm in this course has this shape: encode the unknown in a phase pattern, then transform to a basis in which that pattern is a single basis vector.

§39.11.4 Simon's Algorithm

Problem (Simon). Let $f : \mathbb{B}_n \rightarrow \mathbb{B}_n$, promised to be either injective, or two-to-one with a hidden period: there is $\xi \in \mathbb{B}_n$, $\xi \neq 0$, with

$$f(x) = f(y) \iff y = x \text{ or } y = x \oplus \xi.$$

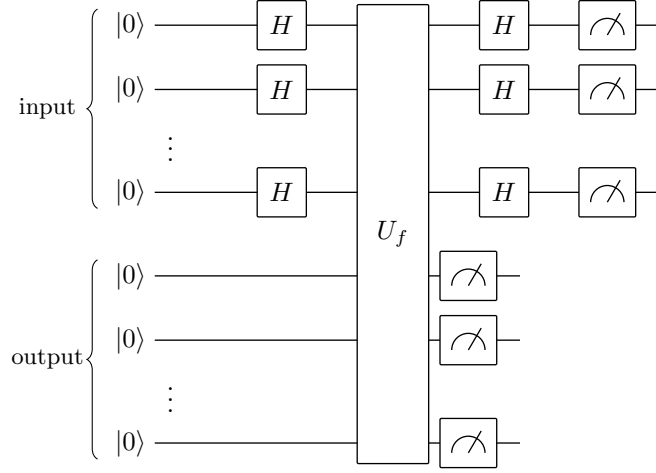
Decide which case holds, and in the second case find ξ .

Proposition 39.11.6

Any classical randomised algorithm solving Simon's problem with bounded error requires $\Omega(2^{n/2})$ queries.

Proof sketch. Classically, the only way to learn anything is to find a collision $f(x) = f(y)$ with $x \neq y$, whereupon $\xi = x \oplus y$. If we query k distinct points, there are $\binom{k}{2}$ pairs, and for a uniformly random ξ the probability that some queried pair differs by ξ is at most $\binom{k}{2} / (2^n - 1)$. For this to be bounded below by a constant we need $k = \Omega(2^{n/2})$. This is the birthday bound, and it is tight. $\square$

The quantum algorithm needs $O(n)$ queries.


Theorem 39.11.7 (Simon)

Simon's problem can be solved with bounded error using $O(n)$ queries to U_f and $O(n^3)$ further classical steps.

Proof. Prepare the input register in the equal superposition and the output register in $|0\rangle$, and query the oracle:

$$U_f |\varphi_n\rangle |0\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \mathbb{B}_n} |x\rangle |f(x)\rangle.$$

Measure the output register. In the two-to-one case the outcome is some value $z = f(x_0) = f(x_0 \oplus \xi)$, and by the extended Born rule the input register collapses to

$$|\text{coset}\rangle = \frac{1}{\sqrt{2}} (|x_0\rangle + |x_0 \oplus \xi\rangle).$$

Measuring this directly would give x_0 or $x_0 \oplus \xi$, each with probability $\frac{1}{2}$, and x_0 is uniformly random, so we would learn nothing. Instead apply $H^{\otimes n}$. Using $H^{\otimes n} |u\rangle = 2^{-n/2} \sum_y (-1)^{u \cdot y} |y\rangle$,

$$\begin{aligned} H^{\otimes n} |\text{coset}\rangle &= \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \mathbb{B}_n} [(-1)^{x_0 \cdot y} + (-1)^{(x_0 \oplus \xi) \cdot y}] |y\rangle \\ &= \frac{1}{\sqrt{2^{n+1}}} \sum_{y \in \mathbb{B}_n} (-1)^{x_0 \cdot y} [1 + (-1)^{\xi \cdot y}] |y\rangle, \end{aligned}$$

using bilinearity of the mod-2 dot product. The bracket is 2 if $\xi \cdot y = 0$ and 0 otherwise, so

$$H^{\otimes n} |\text{coset}\rangle = \frac{1}{\sqrt{2^{n-1}}} \sum_{y: \xi \cdot y = 0} (-1)^{x_0 \cdot y} |y\rangle.$$

Measuring the input register therefore returns a uniformly random y from the 2^{n-1} -element subspace $\xi^\perp = \{y : \xi \cdot y = 0\}$. Note that x_0 has vanished into the phases, exactly as we wanted.

Repeat the whole procedure. Each run yields a uniformly random $y^{(i)} \in \xi^\perp$, that is, a linear equation $\xi \cdot y^{(i)} = 0$ over $\mathbb{F}_2$ satisfied by the unknown ξ . Once we have

$n - 1$ linearly independent equations, the solution space is one-dimensional and its unique nonzero element is ξ ; we find it by Gaussian elimination in $O(n^3)$ steps. Finally we verify by checking $f(0) = f(\xi)$, which also distinguishes the two-to-one case from the injective case (in the injective case, the analysis above is replaced by the observation that the measured y are uniform on all of $\mathbb{B}_n$, and the equations will have only the trivial solution).

It remains to bound the number of repetitions. Each new sample is uniform on the $(n - 1)$ -dimensional space $\xi^\perp$, so if we already have k independent vectors, the probability that the next one lies in their span is $2^k/2^{n-1} \leq \frac{1}{2}$ for $k \leq n - 2$. Hence each run increases the rank with probability at least $\frac{1}{2}$, and by a standard argument $O(n)$ runs suffice to obtain $n - 1$ independent equations with probability at least $\frac{2}{3}$. $\square$

Simon's problem gives an exponential separation between quantum and classical query complexity that is robust to bounded error. Historically it is the crucial precedent: it is a *hidden subgroup problem* for the group $\mathbb{B}_n = (\mathbb{Z}/2\mathbb{Z})^n$, and Shor's algorithm is the same idea for the group $\mathbb{Z}_N$, where the Hadamard transform is replaced by the quantum Fourier transform.

§39.12 The Quantum Fourier Transform

The Hadamard transform $H^{\otimes n}$ is the Fourier transform of the group $(\mathbb{Z}/2\mathbb{Z})^n$, and it was what made Simon's algorithm work. To find periods in $\mathbb{Z}_N$ we need the Fourier transform of $\mathbb{Z}_N$, and remarkably it can be implemented in $O((\log N)^2)$ gates – exponentially faster than the classical fast Fourier transform, which takes $O(N \log N)$ operations on N numbers.

§39.12.1 Definition and Basic Properties

Let $\mathcal{V}_N$ be an N -dimensional state space with orthonormal basis

$$\mathcal{B}_N = \{|0\rangle, |1\rangle, \dots, |N-1\rangle\},$$

whose labels we read as elements of $\mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}$, and put $\omega = e^{2\pi i/N}$, a primitive N th root of unity.

Definition 39.12.1 (Quantum Fourier Transform)

The **quantum Fourier transform** modulo N is the linear map QFT_N on $\mathcal{V}_N$ given on the basis by

$$\text{QFT}_N |k\rangle = \frac{1}{\sqrt{N}} \sum_{\ell=0}^{N-1} \omega^{k\ell} |\ell\rangle.$$

Its matrix has entries $(\text{QFT}_N)_{jk} = \langle j | \text{QFT}_N | k \rangle = \omega^{jk}/\sqrt{N}$, that is,

$$\text{QFT}_N = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots \\ 1 & \omega & \omega^2 & \cdots \\ 1 & \omega^2 & \omega^4 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix},$$

the Vandermonde matrix of the N th roots of unity. For $N = 2$ we have $\omega = -1$ and $\text{QFT}_2 = H$, so the QFT genuinely generalises the Hadamard gate.

Proposition 39.12.2

QFT_N is unitary.

Proof. Everything follows from the geometric sum

$$\sum_{\ell=0}^{N-1} \omega^{j\ell} = \begin{cases} N, & j \equiv 0 \pmod{N}, \\ \frac{1 - \omega^{jN}}{1 - \omega^j} = 0, & \text{otherwise,} \end{cases}$$

valid since $\omega^{jN} = 1$ always, while $\omega^j \neq 1$ when $N \nmid j$. Hence

$$(\text{QFT}_N^\dagger \text{QFT}_N)_{jk} = \sum_{\ell} \overline{(\text{QFT}_N)_{\ell j}} (\text{QFT}_N)_{\ell k} = \frac{1}{N} \sum_{\ell=0}^{N-1} \omega^{\ell(k-j)} = \delta_{jk},$$

so $\text{QFT}_N^\dagger \text{QFT}_N = I$. □

The point of the Fourier transform, here as everywhere, is that it converts periodicity in the label into periodicity in the frequency. That is what we now exploit.

§39.12.2 Periodicity Determination

Problem (Periodicity Determination). *Let $f : \mathbb{Z}_N \rightarrow Y$ be a function, promised to be periodic with some period $r \mid N$ – so $f(x+r) = f(x)$ for all x – and injective within each period. Given an oracle for f , determine r .*

Classically the best possible is $O(\sqrt{N})$ queries, again by a birthday-paradox argument: we must find a collision. We shall get down to $O(\log \log N)$ queries.

Let us first see what goes wrong with the obvious approach. Prepare the equal superposition $|\psi_N\rangle = N^{-1/2} \sum_{x=0}^{N-1} |x\rangle$, set the output register to $|0\rangle$, and query the oracle to obtain

$$|F\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle |f(x)\rangle.$$

Write $A = N/r$, the number of complete periods. Now measure the output register. The possible outcomes are the r values $f(x_0)$ for $x_0 \in \{0, \dots, r-1\}$; the basis states contributing to outcome $f(x_0)$ are those with $x = x_0 + jr$ for $j = 0, \dots, A-1$, so by the extended Born rule each outcome has probability $A/N = 1/r$, and the input register collapses to

$$|\text{per}\rangle = \frac{1}{\sqrt{A}} \sum_{j=0}^{A-1} |x_0 + jr\rangle.$$

This state *is* the period, laid out in front of us. But if we now measure it, we obtain a single uniformly random $x_0 + j_0 r$, and since x_0 itself is uniformly random and unknown, this tells us nothing whatever about r . The information is in the *spacing* of the terms, not in any one of them, and the computational basis is blind to spacing.

The Fourier transform sees nothing but spacing.

Theorem 39.12.3

With $|\text{per}\rangle$ as above and $\omega = e^{2\pi i/N}$,

$$\text{QFT}_N |\text{per}\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} \omega^{x_0 k N/r} |kN/r\rangle.$$

Measuring this state returns $c = k_0 N/r$ for a uniformly random $k_0 \in \{0, \dots, r-1\}$.

Proof. Applying the definition and exchanging the order of summation,

$$\begin{aligned} \text{QFT}_N |\text{per}\rangle &= \frac{1}{\sqrt{A}} \sum_{j=0}^{A-1} \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} \omega^{(x_0+jr)y} |y\rangle \\ &= \frac{1}{\sqrt{NA}} \sum_{y=0}^{N-1} \omega^{x_0 y} \underbrace{\left[\sum_{j=0}^{A-1} (\omega^{ry})^j \right]}_{=: S_y} |y\rangle. \end{aligned}$$

Now S_y is a geometric series with ratio $\omega^{ry} = e^{2\pi i r y/N} = e^{2\pi i y/A}$. This equals 1 exactly when $A \mid y$, in which case $S_y = A$; otherwise

$$S_y = \frac{1 - \omega^{ryA}}{1 - \omega^{ry}} = \frac{1 - e^{2\pi i y}}{1 - \omega^{ry}} = 0.$$

So only the r multiples $y = kA = kN/r$, $k = 0, \dots, r-1$, survive, each with coefficient $A/\sqrt{NA} = \sqrt{A/N} = 1/\sqrt{r}$ times a phase. This gives the stated formula, and since each of the r terms has modulus $1/\sqrt{r}$, the measurement outcomes are uniform. $\square$

Notice what has happened. The unknown offset x_0 , which ruined the naive approach, has been converted into a phase $\omega^{x_0 k N/r}$ and therefore has no effect on measurement probabilities at all. Meanwhile the periodicity r in the label has been inverted into a periodicity N/r in the outcome.

It remains to extract r from a sample. We know $c = k_0 N/r$, hence

$$\frac{c}{N} = \frac{k_0}{r},$$

and the left-hand side is known. If $\gcd(k_0, r) = 1$, then cancelling c/N to lowest terms produces exactly k_0/r , and we read off r as the denominator. If not, we obtain a denominator $\tilde{r}$ which is a proper divisor of r ; we can detect this by testing whether $f(0) = f(\tilde{r})$, and if not, try again.

So the question is how often a random k_0 is coprime to r .

Theorem 39.12.4 (Coprimalty Theorem)

Let φ denote Euler's totient function. There are constants $c > 0$ and r_0 such that for all $r \geq r_0$,

$$\varphi(r) \geq c \frac{r}{\log \log r}.$$

Equivalently $\varphi(r) = \Omega(r/\log \log r)$.

This is a standard result of analytic number theory; it follows from Mertens' theorem on $\prod_{p \leq x} (1 - 1/p)$, proved in the Part II Number Theory course (see the section on the distribution of primes in my notes for that course, `../number-theory/`). Since k_0 is uniform on $\{0, \dots, r-1\}$, the probability that it is coprime to r is $\varphi(r)/r = \Omega(1/\log \log r)$.

Lemma 39.12.5

Suppose a trial succeeds with probability p and is repeated M times independently. Then the probability of at least one success exceeds $1 - \varepsilon$ provided $M \geq \frac{1}{p} \log \frac{1}{\varepsilon}$.

Proof. The probability of no success is $(1 - p)^M \leq e^{-pM}$, which is at most ε when $pM \geq \log(1/\varepsilon)$. $\square$

Corollary 39.12.6

For any fixed $\varepsilon \in (0, 1)$, the period r can be determined with probability at least $1 - \varepsilon$ using $O(\log \log N)$ queries to f and $O(\text{poly}(\log N))$ further steps.

Proof. Each round of the algorithm uses three queries (one to build $|F\rangle$, and two to test whether $f(0) = f(\tilde{r})$), and succeeds with probability $p = \Omega(1/\log \log r)$. By the lemma, $M = O(\log \log r) = O(\log \log N)$ rounds give success probability $1 - \varepsilon$. The remaining work per round is the QFT_N , which by §39.12.3 costs $O((\log N)^2)$ gates when N is a power of two, together with $O(\text{poly}(\log N))$ arithmetic to cancel the fraction c/N . $\square$

§39.12.3 Efficient Implementation of the QFT

Everything above is useless unless QFT_N can actually be built. We show it can, when $N = 2^n$, using $O(n^2)$ gates. (When N is not a power of two one approximates it by one, at the cost of a more delicate analysis; we will meet this in Shor's algorithm.)

Write $x \in \{0, \dots, 2^n - 1\}$ in binary as $x = \sum_{i=0}^{n-1} 2^i x_i$, so that x_{n-1} is the most significant bit, and use the notation

$$0.x_a x_b \cdots x_z = \frac{x_a}{2} + \frac{x_b}{4} + \cdots$$

for binary fractions. The key is the following factorisation.

Proposition 39.12.7

For $N = 2^n$,

$$\begin{aligned} \text{QFT}_N |x\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_0)} |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_1 x_0)} |1\rangle) \\ &\quad \otimes \cdots \otimes \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_{n-1} \cdots x_0)} |1\rangle). \end{aligned}$$

In particular $\text{QFT}_N |x\rangle$ is always a *product* state.

Proof. Writing $y = \sum_{j=0}^{n-1} 2^j y_j$ as well,

$$\text{QFT}_N |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y_0, \dots, y_{n-1} \in \{0,1\}} \exp \left[\frac{2\pi i xy}{2^n} \right] |y_{n-1}\rangle \cdots |y_0\rangle,$$

and

$$\frac{xy}{2^n} = \sum_{j=0}^{n-1} y_j \frac{x}{2^{n-j}}.$$

Only the fractional part of the exponent matters, since $e^{2\pi i m} = 1$ for integers m , and the fractional part of $x/2^{n-j}$ is exactly $0.x_{n-j-1}x_{n-j-2} \cdots x_0$. So the exponent splits as a sum of terms each involving a single y_j , and the sum over all y factorises into a product of n independent sums over $y_j \in \{0,1\}$, the factor associated with y_j being

$$\frac{1}{\sqrt{2}} \sum_{y_j \in \{0,1\}} e^{2\pi i y_j (0.x_{n-j-1} \cdots x_0)} |y_j\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i (0.x_{n-j-1} \cdots x_0)} |1\rangle).$$

Writing the factors in the order $j = n-1, n-2, \dots, 0$, in which the output kets $|y_{n-1}\rangle \cdots |y_0\rangle$ appear, the first is $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (0.x_0)} |1\rangle)$ and the last is $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (0.x_{n-1} \cdots x_0)} |1\rangle)$, as claimed. $\square$

We can now read the circuit straight off. Note that

$$H |x_i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i (0.x_i)} |1\rangle),$$

since the phase is $+1$ if $x_i = 0$ and $e^{i\pi} = -1$ if $x_i = 1$. So a Hadamard creates the leading digit of each fraction, and we need only append the remaining digits, one at a time, with controlled phase gates.

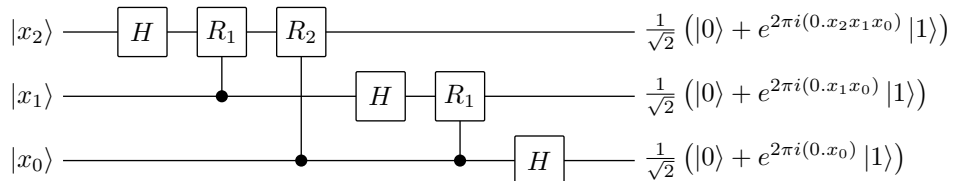
Definition 39.12.8 (Controlled Phase Gate)

For $d \geq 1$ let

$$R_d = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2^d} \end{pmatrix}, \quad \text{where } e^{i\pi/2^d} = e^{2\pi i (0.0 \cdots 01)}$$

with d zeros after the point followed by a 1, and write CR_d for the corresponding two-qubit controlled gate.

Thus $R_1 = \text{diag}(1, i) = S$, and CR_d applied with control $|x_i\rangle$ multiplies the $|1\rangle$ component of the target by $e^{2\pi i (0.0 \cdots 0x_i)}$, appending x_i in the $(d+1)$ st binary place. Here is the circuit for $n = 3$, that is, QFT_8 .



Tracking the top wire, for example:

$$\begin{aligned} |x_2\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_2)} |1\rangle) \\ &\xrightarrow{CR_1} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_2)} e^{2\pi i(0.0x_1)} |1\rangle) \\ &\xrightarrow{CR_2} \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_2)} e^{2\pi i(0.0x_1)} e^{2\pi i(0.00x_0)} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i(0.x_2x_1x_0)} |1\rangle), \end{aligned}$$

which is the factor that the proposition demands. Comparing with the proposition, the outputs appear in the reverse of the natural order, so we finish with $\lfloor n/2 \rfloor$ swap gates.

Proposition 39.12.9

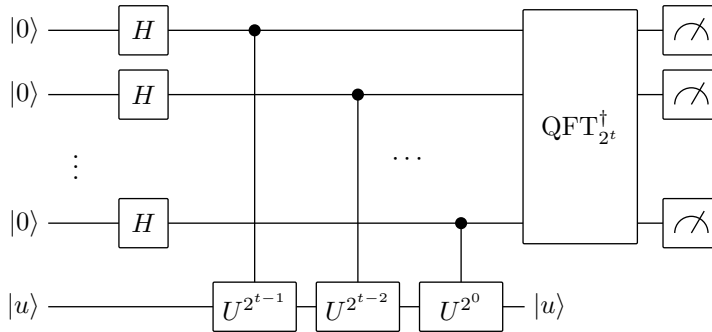
QFT_{2^n} can be implemented exactly using n Hadamard gates, $\frac{1}{2}n(n-1)$ controlled phase gates and $O(n)$ swaps, hence with $O(n^2) = O((\log N)^2)$ gates in total.

Compare the classical fast Fourier transform, which needs $\Theta(N \log N) = \Theta(2^n n)$ operations to transform a vector of 2^n numbers. The quantum circuit is exponentially smaller. Of course we cannot *read out* the transformed amplitudes – a measurement gives us just one sample – so this is not a general-purpose speedup for signal processing. It is only useful when a single sample from the Fourier spectrum is what we want, which, happily, is exactly the situation in period finding.

§39.12.4 Phase Estimation

Period finding is a special case of a more general and very useful primitive.

Problem (Phase Estimation). *Let U be a unitary with an eigenvector $|u\rangle$ and eigenvalue $e^{2\pi i\phi}$, where $\phi \in [0, 1)$. Given the ability to prepare $|u\rangle$ and to apply controlled- U^{2^j} for $j = 0, 1, \dots$, estimate ϕ .*



Proposition 39.12.10

Suppose $\phi = 0.\phi_1\phi_2\cdots\phi_t$ exactly, in t binary digits. Then the circuit above outputs $\phi_1\cdots\phi_t$ with certainty.

Proof. Since $U^{2^j} |u\rangle = e^{2\pi i 2^j \phi} |u\rangle$, applying controlled- U^{2^j} with control qubit in state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ leaves $|u\rangle$ alone and kicks the eigenvalue back into the control:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) |u\rangle \mapsto \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 2^j \phi} |1\rangle) |u\rangle,$$

which is phase kickback again, now with a continuous phase. Doing this on all t

control qubits, with the i th control (counting from the top) applying $U^{2^{t-i}}$, leaves the control register in the product state

$$\bigotimes_{i=1}^t \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 2^{t-i} \phi} |1\rangle \right).$$

Writing $\phi = \sum_{k=1}^t \phi_k 2^{-k}$, the number $2^{t-i} \phi$ has fractional part $0.\phi_{t-i+1} \cdots \phi_t$, so the i th factor is $\frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i (0.\phi_{t-i+1} \cdots \phi_t)} |1\rangle \right)$. Comparing with the previous proposition – taking $x_0 = \phi_t, x_1 = \phi_{t-1}, \dots, x_{t-1} = \phi_1$, that is, $x = 2^t \phi$ – the control register holds precisely $\text{QFT}_{2^t} |2^t \phi\rangle$. Applying $\text{QFT}_{2^t}^\dagger$ therefore returns the basis state $|2^t \phi\rangle = |\phi_1 \phi_2 \cdots \phi_t\rangle$ exactly, and the measurement reads off the digits. $\square$

When ϕ is not an exact t -bit fraction the same circuit returns the best t -bit approximation with probability at least $4/\pi^2 \approx 0.405$, and one can boost this to $1 - \varepsilon$ by using $t + O(\log(1/\varepsilon))$ qubits. We will not need the general estimate in the form stated, but the underlying calculation – bounding a geometric sum whose ratio is close to but not equal to 1 – reappears in the analysis of Shor’s algorithm below, where the truncation of an infinite domain to $\mathbb{Z}_{2^m}$ causes exactly this kind of near-resonance.

Remark. Period finding is phase estimation in disguise. Given $f(x) = a^x \bmod N$ with period r , consider the unitary $U : |y\rangle \mapsto |ay \bmod N\rangle$. Its eigenvectors are the states $|u_k\rangle = r^{-1/2} \sum_{s=0}^{r-1} e^{-2\pi i k s / r} |a^s \bmod N\rangle$ with eigenvalues $e^{2\pi i k / r}$, so estimating a phase of U gives k/r for a random k , which is precisely the output of §39.12.2. This is the modern way to present Shor’s algorithm; we shall follow the original route instead, since the truncation issue is then visible rather than hidden.

§39.13 Shor’s Factoring Algorithm

We come to the most celebrated result in the subject. Given an n -bit integer N , we want a nontrivial factor. The best known classical algorithm, the number field sieve, runs in time $\exp\left(O(n^{1/3}(\log n)^{2/3})\right)$, which is subexponential but far from polynomial; the presumed hardness of this problem is what secures RSA. Shor’s algorithm factors in time $O(n^3)$.

The algorithm has two halves. The first is pure number theory and entirely classical: it reduces factoring to finding the period of a certain function. The second is the quantum period-finding routine of §39.12.2, adapted to cope with the fact that the period does not divide the size of the domain.

§39.13.1 Factoring as a Periodicity Problem

Fix N , and choose $1 < a < N$ at random. If $\gcd(a, N) > 1$ we have already found a factor by Euclid’s algorithm, so assume $\gcd(a, N) = 1$.

Theorem 39.13.1 (Euler)

If $\gcd(a, N) = 1$ then $a^{\varphi(N)} \equiv 1 \pmod{N}$. In particular there is a least $r \geq 1$ with $a^r \equiv 1 \pmod{N}$, called the **order** of a modulo N .

This is proved in IA Numbers and Sets: the units modulo N form a group of order $\varphi(N)$,

and the order of an element divides the order of the group. Consider now the **modular exponentiation function**

$$f : \mathbb{Z} \rightarrow \mathbb{Z}_N, \quad f(k) = a^k \bmod N.$$

Since $f(k_1 + k_2) = f(k_1)f(k_2)$ and $f(r) = 1$, the function f is periodic with period exactly r , and it is injective within each period (if $a^{k_1} \equiv a^{k_2}$ with $0 \leq k_1 < k_2 < r$ then $a^{k_2-k_1} \equiv 1$, contradicting minimality of r). So we have a periodicity problem. Why does its solution factor N ?

Proposition 39.13.2

Suppose r is even and $a^{r/2} \not\equiv -1 \pmod{N}$. Then $\gcd(a^{r/2} - 1, N)$ and $\gcd(a^{r/2} + 1, N)$ are nontrivial factors of N .

Proof. Since r is even we may write

$$a^r - 1 = (a^{r/2} - 1)(a^{r/2} + 1) \equiv 0 \pmod{N},$$

so N divides the product. Now $N \nmid (a^{r/2} - 1)$, since r is the *least* exponent with $a^r \equiv 1$ and $r/2 < r$; and $N \nmid (a^{r/2} + 1)$ by hypothesis. So neither factor is divisible by N , yet their product is: the prime factorisation of N must therefore be split between them, and $\gcd(a^{r/2} \pm 1, N)$ are proper nontrivial divisors, computable by Euclid's algorithm in $O(n^3)$ time. $\square$

Example 39.13.3

Let $N = 15$ and $a = 7$. The powers of 7 modulo 15 are 1, 7, 4, 13, 1, $\dots$, so $r = 4$, which is even, and $a^{r/2} = 49 \equiv 4 \not\equiv -1$. Then

$$\gcd(a^{r/2} - 1, N) = \gcd(48, 15) = 3, \quad \gcd(a^{r/2} + 1, N) = \gcd(50, 15) = 5,$$

and indeed $15 = 3 \times 5$.

Of course the hypotheses might fail. The following says that they usually do not.

Theorem 39.13.4

Let N be odd and not a prime power. If a is chosen uniformly at random from the units modulo N , then with probability at least $\frac{1}{2}$ the order r of a is even and $a^{r/2} \not\equiv -1 \pmod{N}$.

The proof uses the Chinese remainder theorem and the structure of $(\mathbb{Z}/p^k\mathbb{Z})^\times$ as a cyclic group; we take it on trust. Combined with the previous proposition, a random a gives us a factor with probability at least $\frac{1}{2}$, so $O(\log(1/\varepsilon))$ attempts succeed with probability $1 - \varepsilon$.

The excluded cases are easy to handle separately. If N is even, output 2. If $N = c^\ell$ for some $\ell > 1$, we can find c classically in polynomial time: for each $\ell \leq \log_2 N$ compute $\lfloor N^{1/\ell} \rfloor$ to sufficient precision and test.

Method (Shor's Algorithm)

To find a nontrivial factor of N :

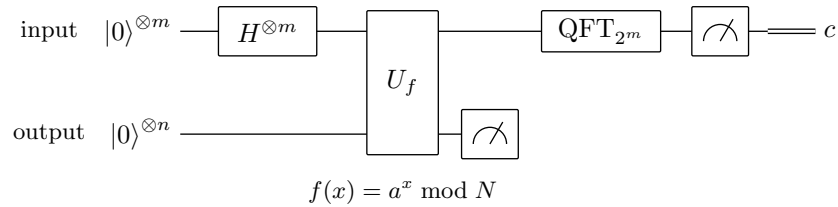
- (i) If N is even, output 2 and stop.
- (ii) If $N = c^\ell$ with $\ell > 1$, output c and stop.
- (iii) Choose $1 < a < N$ uniformly at random and compute $b = \gcd(a, N)$. If $b > 1$, output b and stop.
- (iv) Use the quantum period-finding subroutine to find the order r of a modulo N . If it fails, return to (iii).
- (v) If r is even and $t = \gcd(a^{r/2} + 1, N)$ satisfies $1 < t < N$, output t and stop. Otherwise return to (iii).

Only step (iv) is quantum. Everything else is classical arithmetic that we already know how to do quickly.

§39.13.2 Period Finding on a Truncated Domain

There is a difficulty that we glossed over in §39.12.2. There, the period r divided the size N of the domain, which made the geometric sum collapse exactly. Here f is defined on all of $\mathbb{Z}$, and we must truncate to a finite domain $\mathbb{Z}_M$; but we do not know r , so we cannot choose M to be a multiple of it. The sum will no longer vanish exactly, and we must show that it is nonetheless sharply peaked.

Take $M = 2^m$, where m is minimal with $2^m > N^2$ (the reason for this choice will emerge), and let $D = \{0, 1, \dots, 2^m - 1\}$. Write $2^m = Br + b$ with $0 \leq b < r$, so $B = \lfloor 2^m/r \rfloor$.



Running the circuit, the state after the oracle is

$$\begin{aligned}
 |\Psi\rangle &= \frac{1}{\sqrt{2^m}} \sum_{x \in D} |x\rangle |f(x)\rangle = \frac{1}{\sqrt{2^m}} \sum_{x_0=0}^{b-1} \sum_{j=0}^B |x_0 + jr\rangle |f(x_0)\rangle \\
 &\quad + \frac{1}{\sqrt{2^m}} \sum_{x_0=b}^{r-1} \sum_{j=0}^{B-1} |x_0 + jr\rangle |f(x_0)\rangle,
 \end{aligned}$$

the split reflecting the fact that the residues $x_0 < b$ occur $B + 1$ times in D and the rest occur B times. Measuring the output register gives some $y = f(x_0)$, leaving the input register in

$$|\text{per}\rangle = \frac{1}{\sqrt{A}} \sum_{j=0}^{A-1} |x_0 + jr\rangle, \quad A = \begin{cases} B + 1, & x_0 < b, \\ B, & x_0 \geq b. \end{cases}$$

Applying QFT_{2^m} , with $\omega = e^{2\pi i/2^m}$,

$$\text{QFT}_{2^m} |\text{per}\rangle = \frac{1}{\sqrt{A2^m}} \sum_{c=0}^{2^m-1} \omega^{x_0 c} \left[\sum_{j=0}^{A-1} (\omega^{cr})^j \right] |c\rangle = \sum_{c=0}^{2^m-1} g(c) |c\rangle,$$

say. This is the same expression as before, but now ω^{cr} is in general not a root of unity of order dividing A , so the inner sum does not vanish. It is, however, still a geometric series, and it is large exactly when ω^{cr} is close to 1, that is, when $cr/2^m$ is close to an integer k , that is, when

$$c \approx \frac{k 2^m}{r}.$$

Theorem 39.13.5

For each $k \in \{0, 1, \dots, r-1\}$ let c_k be the unique integer with $|c_k - k2^m/r| < \frac{1}{2}$. Then

$$\Pr(c = c_k) = |g(c_k)|^2 > \frac{\gamma}{r}$$

for an absolute constant γ , which may be taken to be $4/\pi^2$.

Proof sketch. First, c_k is well defined. If $k2^m/r$ were exactly a half-integer $\ell + \frac{1}{2}$, then $k2^{m+1} = r(2\ell + 1)$; writing $r = 2^s u$ with u odd and comparing the power of 2 on each side gives $m+1 \leq s$, so $2^{m+1} \mid r$, contradicting $r < N < 2^m$. Summing the geometric series,

$$\sum_{j=0}^{A-1} (\omega^{cr})^j = \frac{1 - e^{2\pi i A c r / 2^m}}{1 - e^{2\pi i c r / 2^m}},$$

and writing $cr/2^m = k + \delta$ with $|\delta| < r/2^{m+1}$, the numerator and denominator are $|1 - e^{2\pi i A \delta}| = 2|\sin(\pi A \delta)|$ and $2|\sin(\pi \delta)|$. Since $A \approx 2^m/r$ we have $|A\delta| \lesssim \frac{1}{2}$, so we are on the part of the sine curve where $|\sin(\pi A \delta)| \geq 2|A\delta|$, while $|\sin(\pi \delta)| \leq \pi|\delta|$. Hence the ratio is at least $2A/\pi$, and

$$|g(c_k)|^2 \geq \frac{1}{A2^m} \cdot \frac{4A^2}{\pi^2} = \frac{4}{\pi^2} \cdot \frac{A}{2^m} \approx \frac{4}{\pi^2} \cdot \frac{1}{r},$$

as claimed. □

So with constant probability our measured c satisfies $|c - k2^m/r| < \frac{1}{2}$ for some k , and by the coprimality theorem, with probability $\Omega(1/\log \log r)$ this k is coprime to r . Repeating $O(\log \log N)$ times, we may assume both.

§39.13.3 Recovering r by Continued Fractions

Dividing by 2^m , our measurement gives a known rational $c/2^m$ with

$$\left| \frac{c}{2^m} - \frac{k}{r} \right| < \frac{1}{2^{m+1}} < \frac{1}{2N^2},$$

using $2^m > N^2$. Since $r < N$ this is the classic setting of Diophantine approximation, and two facts from the Part II Number Theory course finish the job.

Lemma 39.13.6

There is at most one fraction k/r in lowest terms with $r < N$ and $|c/2^m - k/r| < 1/(2N^2)$.

Proof. Suppose k'/r' and k''/r'' both satisfy this, with $r', r'' < N$, and are distinct. Then

$$\left| \frac{k'}{r'} - \frac{k''}{r''} \right| = \frac{|k'r'' - k''r'|}{r'r''} \geq \frac{1}{r'r''} > \frac{1}{N^2},$$

since the numerator is a nonzero integer. But by the triangle inequality the two fractions are within $2 \cdot \frac{1}{2N^2} = \frac{1}{N^2}$ of each other, a contradiction. $\square$

This is precisely why we chose m minimal with $2^m > N^2$: any smaller and the approximation would not determine k/r ; any larger and we waste qubits. It remains to *find* the fraction, and here we appeal to continued fractions.

Recall from Part II Number Theory² that every rational $x \in (0, 1)$ has a finite continued fraction expansion

$$x = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_\ell}}}} = [a_1, a_2, \dots, a_\ell]$$

with a_i positive integers, obtained by the Euclidean algorithm; the truncations $p_k/q_k = [a_1, \dots, a_k]$ are the **convergents**, and satisfy the recurrences

$$p_k = a_k p_{k-1} + p_{k-2}, \quad q_k = a_k q_{k-1} + q_{k-2},$$

with $p_0 = 0, q_0 = 1, p_1 = 1, q_1 = a_1$, together with $q_k p_{k-1} - p_k q_{k-1} = (-1)^k$ and hence $\gcd(p_k, q_k) = 1$.

Theorem 39.13.7

If c and 2^m are $O(m)$ -bit integers, the continued fraction expansion of $c/2^m$ has length $\ell = O(m)$ and it, together with all its convergents, can be computed in $O(m^3)$ time.

Proof sketch. Each $a_k \geq 1$, so the recurrences give $p_k \geq p_{k-2} + p_{k-1} \geq 2p_{k-2}$ and likewise for q_k ; hence $p_k, q_k \geq 2^{\lfloor k/2 \rfloor}$. Since $q_\ell = 2^m$ (after cancellation, at most 2^m), we need $\ell = O(m)$ steps. Each step is a division of $O(m)$ -bit integers, costing $O(m^2)$, so the total is $O(m^3)$. $\square$

Theorem 39.13.8 (Legendre)

²The continued fraction machinery quoted here – the recurrences for the convergents, the identity $q_k p_{k-1} - p_k q_{k-1} = (-1)^k$, and Legendre's theorem below – is developed in full in the 'Continued Fractions' section of my Number Theory notes, [../number-theory/](#); there Legendre's theorem is proved for irrational x , and the rational case needed here is proved in exactly the same way.

Let $x \in \mathbb{Q}$ with $0 < x < 1$, and let p/q be a rational in lowest terms with

$$\left| x - \frac{p}{q} \right| < \frac{1}{2q^2}.$$

Then p/q is a convergent of the continued fraction expansion of x .

Now apply this with $x = c/2^m$ and $p/q = k/r$: since $r < N$ we have

$$\left| \frac{c}{2^m} - \frac{k}{r} \right| < \frac{1}{2N^2} < \frac{1}{2r^2},$$

so k/r is one of the convergents of $c/2^m$. We compute them all – there are only $O(m)$ – and test each candidate denominator q by checking whether $a^q \equiv 1 \pmod{N}$. The lemma above guarantees that at most one candidate can pass the approximation test, and since $\gcd(k, r) = 1$ the fraction k/r appears in lowest terms, so its denominator is exactly r .

Example 39.13.9

Take $N = 39$ and $a = 7$, which are coprime. Since $39^2 = 1521$ and $2^{10} = 1024 < 1521 < 2048 = 2^{11}$, we set $m = 11$ and $M = 2^{11} = 2048$.

Suppose the measurement returns $c = 853$. We expand

$$\frac{853}{2048} = [2, 2, 2, 42, 4],$$

obtained by the Euclidean algorithm: $2048 = 2 \cdot 853 + 342$, $853 = 2 \cdot 342 + 169$, $342 = 2 \cdot 169 + 4$, $169 = 42 \cdot 4 + 1$, $4 = 4 \cdot 1$. The convergents are

$$\frac{1}{2}, \quad \frac{2}{5}, \quad \frac{5}{12}, \quad \frac{212}{509}, \quad \frac{853}{2048}.$$

We need a denominator $r < 39$, which rules out the last two; and of the remaining three,

$$\left| \frac{853}{2048} - \frac{1}{2} \right| = 0.0835, \quad \left| \frac{853}{2048} - \frac{2}{5} \right| = 0.0165, \quad \left| \frac{853}{2048} - \frac{5}{12} \right| = 0.000163,$$

while $1/(2N^2) = 1/3042 = 0.000329$. Only $5/12$ passes. So we take $r = 12$, and indeed one checks $7^{12} \equiv 1 \pmod{39}$ while no smaller power is 1.

Now $r = 12$ is even, and computing the powers of 7 modulo 39 gives $7^6 \equiv 25$, which is not $-1 \equiv 38$. Since $\gcd(x, N)$ depends only on x modulo N , we compute

$$\gcd(7^6 + 1, 39) = \gcd(26, 39) = 13, \quad \gcd(7^6 - 1, 39) = \gcd(24, 39) = 3,$$

and indeed $39 = 13 \times 3$.

Remark. Without continued fractions, one could search all fractions k/r with $r < N$ directly, but there are $\Theta(N^2)$ of them, and $N^2 = \Theta(\exp(2n))$: the classical post-processing would then dominate and destroy the speedup entirely. The continued fraction algorithm is what makes the last step polynomial, and it is a nice illustration that a quantum algorithm is usually a quantum subroutine wrapped in a great deal of classical cleverness.

§39.13.4 Complexity

Let us total up the cost of one run, with $n = \lceil \log_2 N \rceil$ and $m = O(n)$.

- Preparing the equal superposition: $m = O(n)$ Hadamard gates.
- The oracle U_f for modular exponentiation: computing $a^x \bmod N$ by repeated squaring needs $O(m)$ modular multiplications of n -bit numbers, each $O(n^2)$, so $O(n^3)$ gates in the reversible implementation.
- Measuring the output register: $O(n)$ single-qubit measurements.
- QFT_{2^m} : $O(m^2) = O(n^2)$ gates.
- Measuring the input register: $O(n)$ measurements.
- Continued fractions to recover r : $O(n^3)$ classical steps.
- Euclid's algorithm for the final gcd: $O(n^3)$.

So a single run costs $O(n^3)$, dominated by the modular exponentiation. We need $O(\log \log N) = O(\log n)$ runs to get a good value of c , and $O(\log(1/\varepsilon))$ choices of a to get a useful order r . The total is $O(n^3 \log n)$, and with a little more care $O(n^3)$; in any case, polynomial.

Remark. It is worth being clear about what Shor's algorithm does and does not show. It does not show that factoring is easy in any absolute sense, and it does not show $\text{NP} \subseteq \text{BQP}$ – factoring is not believed to be NP -complete. What it does show is that a problem believed to be outside BPP lies in BQP , which is the strongest evidence we have that quantum computers are more powerful than classical ones. It also, incidentally, breaks RSA.

§39.14 Grover's Search Algorithm

Shor's algorithm exploits a great deal of structure – periodicity – and gets an exponential speedup. What can we do when there is no structure at all?

Problem (Unstructured Search). *We are given a database of N items, exactly one of which is 'good', and the ability to test any single item for goodness. Locate the good item.*

Proposition 39.14.1

Classically, $\Omega(N)$ queries are needed to succeed with any fixed constant probability.

Proof. Testing an item and finding it bad tells us nothing about the location of the good item beyond excluding that one. After M queries we have excluded M possibilities and the good item is uniform on the remaining $N - M$, so the probability of having found it is M/N . For this to be at least $1 - \varepsilon$ we need $M \geq (1 - \varepsilon)N$. $\square$

Grover's algorithm does it in $O(\sqrt{N})$ queries. That is only a quadratic speedup, but it applies to *everything*: unstructured search is the abstract form of brute-force, and in particular of SAT.

§39.14.1 Setup: Reflections

Take $N = 2^n$, so items are labelled by n -bit strings, and let $x_0 \in \mathbb{B}_n$ be the good one. The database is the Boolean function $f : \mathbb{B}_n \rightarrow \mathbb{B}_1$ with $f(x_0) = 1$ and $f(x) = 0$ otherwise, and we are given the oracle U_f .

Definition 39.14.2 (Reflection Operator)

For a normalised $|\alpha\rangle \in \mathcal{V}$, write

$$\Pi_{|\alpha\rangle} = |\alpha\rangle\langle\alpha|, \quad I_{|\alpha\rangle} = I - 2|\alpha\rangle\langle\alpha|.$$

Then $I_{|\alpha\rangle}|\alpha\rangle = -|\alpha\rangle$ and $I_{|\alpha\rangle}|\psi\rangle = |\psi\rangle$ for every $|\psi\rangle \perp |\alpha\rangle$, so $I_{|\alpha\rangle}$ is the reflection of $\mathcal{V}$ in the hyperplane $|\alpha\rangle^\perp$. It is unitary and self-inverse. Note also the conjugation rule

$$UI_{|\alpha\rangle}U^\dagger = U(I - 2|\alpha\rangle\langle\alpha|)U^\dagger = I - 2|U\alpha\rangle\langle U\alpha| = I_{U|\alpha\rangle},$$

valid for any unitary U ; we will use it to build one reflection out of another.

Write $I_{x_0} := I_{|x_0\rangle}$, so that

$$I_{x_0}|x\rangle = \begin{cases} -|x\rangle, & x = x_0, \\ +|x\rangle, & x \neq x_0, \end{cases}$$

and write I_0 for the case $x_0 = 0 \cdots 0$. The oracle gives us I_{x_0} for free.

Proposition 39.14.3

$U_f(|\psi\rangle \otimes |-\rangle) = (I_{x_0}|\psi\rangle) \otimes |-\rangle$ for every $|\psi\rangle$.

Proof. By phase kickback (§39.10.2), $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, which is $-|x\rangle|-\rangle$ if $x = x_0$ and $+|x\rangle|-\rangle$ otherwise. This is exactly the action of $I_{x_0} \otimes I$ on basis states, and both sides are linear. $\square$

So from now on we may forget the output register and simply say that one query implements the reflection I_{x_0} .

§39.14.2 The Grover Iterate

We start, as always, from the equal superposition

$$|\psi_0\rangle = H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{N}} \sum_{x \in \mathbb{B}_n} |x\rangle,$$

which has overlap $\langle x_0|\psi_0\rangle = 1/\sqrt{N}$ with the target – small, but not zero. The idea is to amplify it.

Definition 39.14.4 (Grover Iterate)

Write $H_n = H^{\otimes n}$. The **Grover iterate** is

$$Q = -H_n I_0 H_n I_{x_0}.$$

By the conjugation rule, $H_n I_0 H_n = H_n I_{|0\dots 0\rangle} H_n^\dagger = I_{H_n|0\dots 0\rangle} = I_{|\psi_0\rangle}$, so

$$Q = -I_{|\psi_0\rangle} I_{|x_0\rangle} :$$

a reflection in $|x_0\rangle^\perp$, followed by a reflection in $|\psi_0\rangle^\perp$, followed by an overall sign. Each iterate costs one query (for I_{x_0}) plus $O(n)$ gates (for $H_n I_0 H_n$; the reflection I_0 about the all-zeros state is a multiply-controlled phase, built from $O(n)$ Toffoli gates and ancillas).

Everything now happens in a two-dimensional real plane.

Theorem 39.14.5

Let $\mathcal{P} = \text{span}_{\mathbb{R}}\{|x_0\rangle, |\psi_0\rangle\}$. Then

- (i) $\mathcal{P}$ is invariant under Q , and on $\mathcal{P}$ the operator Q acts as a rotation through 2α , where $\sin \alpha = \langle x_0 | \psi_0 \rangle = 1/\sqrt{N}$;
- (ii) on $\mathcal{P}^\perp$, Q acts as $-I$.

Proof. All the vectors and operators here have real entries in the computational basis, so we may work in real Euclidean space.

For (ii), if $|\xi\rangle$ is orthogonal to both $|x_0\rangle$ and $|\psi_0\rangle$ then both reflections fix it, so $Q|\xi\rangle = -|\xi\rangle$.

For (i), note first that $I_{|x_0\rangle}|v\rangle = |v\rangle - 2\langle x_0|v\rangle|x_0\rangle$ and $I_{|\psi_0\rangle}|v\rangle = |v\rangle - 2\langle \psi_0|v\rangle|\psi_0\rangle$, so each maps $\mathcal{P}$ into $\mathcal{P}$; hence so does Q . Restricted to the plane $\mathcal{P}$, $I_{|x_0\rangle}$ is the reflection in the line $L_1 = \mathcal{P} \cap |x_0\rangle^\perp$ and $I_{|\psi_0\rangle}$ the reflection in $L_2 = \mathcal{P} \cap |\psi_0\rangle^\perp$.

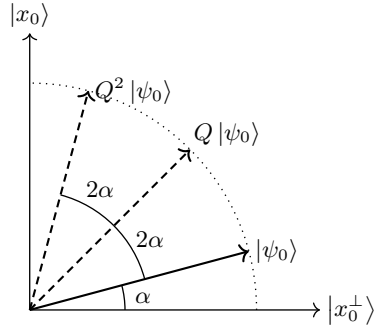
Now recall the elementary fact that the composition of reflections in two lines of the plane meeting at angle θ is a rotation through 2θ . Let β be the angle between $|x_0\rangle$ and $|\psi_0\rangle$, so $\cos \beta = \langle x_0 | \psi_0 \rangle = 1/\sqrt{N}$; then the angle between their perpendiculars L_1 and L_2 is also β , and $I_{|\psi_0\rangle} I_{|x_0\rangle}$ is a rotation through 2β .

Finally we deal with the sign. In a plane, $-R$ where R is the reflection in a line L is the reflection in $L^\perp$; so $-I_{|\psi_0\rangle} = I_{|\psi_0^\perp\rangle}$ on $\mathcal{P}$, where $|\psi_0^\perp\rangle$ is the unit vector in $\mathcal{P}$ perpendicular to $|\psi_0\rangle$. Therefore

$$Q|_{\mathcal{P}} = I_{|\psi_0^\perp\rangle} I_{|x_0\rangle},$$

which is a rotation through twice the angle between the corresponding mirror lines. That angle is now $\frac{\pi}{2} - \beta = \alpha$, where $\sin \alpha = \cos \beta = 1/\sqrt{N}$. Hence Q rotates through 2α . $\square$

The picture, drawn in the plane $\mathcal{P}$ with the orthonormal basis $\{|x_0^\perp\rangle, |x_0\rangle\}$, where $|x_0^\perp\rangle$ is the normalised superposition of all the *bad* items:



The state starts at angle α above the ‘all bad’ axis, and each application of Q swings it a further 2α towards $|x_0\rangle$. When it arrives, a measurement in the computational basis returns x_0 .

Corollary 39.14.6

Let m be the integer nearest to $\frac{\pi/2 - \alpha}{2\alpha}$, so that

$$m \approx \frac{\pi}{4} \sqrt{N}.$$

Then after m iterations a standard measurement returns x_0 with probability $1 - O(1/N)$.

Proof. After m iterations the state makes an angle $(2m + 1)\alpha$ with $|x_0^\perp\rangle$, so the probability of measuring x_0 is $\sin^2((2m + 1)\alpha)$. This is maximised by choosing m so that $(2m + 1)\alpha$ is as close as possible to $\pi/2$. For large N , $\alpha = \arcsin(1/\sqrt{N}) \approx 1/\sqrt{N}$ is small, so

$$m \approx \frac{\pi/2}{2\alpha} \approx \frac{\pi}{4} \sqrt{N},$$

and rounding to the nearest integer costs us at most α in angle, hence at most $O(\alpha^2) = O(1/N)$ in probability. $\square$

Notice that m does not depend on x_0 – only on N . That is essential: we would hardly be able to run the algorithm if the number of steps depended on the answer.

Example 39.14.7

Let $N = 4$, so $n = 2$. Here $\sin \alpha = \frac{1}{2}$, so $\alpha = \pi/6$ and each iterate rotates by $2\alpha = \pi/3$. The initial state $|\psi_0\rangle = |++\rangle$ makes an angle $\alpha = \pi/6$ with $|x_0^\perp\rangle$, so after a single iterate it makes an angle $\pi/6 + \pi/3 = \pi/2$: it is *exactly* $|x_0\rangle$. One query, and the measurement gives x_0 with certainty – whereas classically we would need up to three. Let us verify directly for $x_0 = 11$. We have $|\psi_0\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, so

$$I_{x_0} |\psi_0\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

Now $I_{|\psi_0\rangle} |v\rangle = |v\rangle - 2\langle\psi_0|v\rangle |\psi_0\rangle$, and $\langle\psi_0|I_{x_0}\psi_0\rangle = \frac{1}{4}(1 + 1 + 1 - 1) = \frac{1}{2}$, so

$$-I_{|\psi_0\rangle} I_{x_0} |\psi_0\rangle = |\psi_0\rangle - I_{x_0} |\psi_0\rangle = \frac{1}{2} (0 + 0 + 0 + 2|11\rangle) = |11\rangle,$$

as promised.

§39.14.3 Multiple Targets and Unknown r

Now suppose there are $r \geq 1$ good items $x_1, \dots, x_r$, with r known. Define

$$I_G = I - 2 \sum_{i=1}^r |x_i\rangle\langle x_i|,$$

which flips the sign of every good basis state, and is again implemented by one oracle query. Put $Q_G = -I_{|\psi_0\rangle} I_G$, and let

$$|\psi_G\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^r |x_i\rangle, \quad |\psi_B\rangle = \frac{1}{\sqrt{N-r}} \sum_{x \text{ bad}} |x\rangle$$

be the equal superpositions of good and bad items, which are orthonormal. Then

$$|\psi_0\rangle = \sqrt{\frac{r}{N}} |\psi_G\rangle + \sqrt{\frac{N-r}{N}} |\psi_B\rangle$$

lies in the plane $\mathcal{P}_G = \text{span}\{|\psi_G\rangle, |\psi_B\rangle\}$.

Proposition 39.14.8

On the plane $\mathcal{P}_G$, the operator Q_G acts as a rotation through 2α , where $\sin \alpha = \langle \psi_0 | \psi_G \rangle = \sqrt{r/N}$.

Proof. The only new point is that I_G is not of the form $I_{|v\rangle}$ for a single vector. But restricted to $\mathcal{P}_G$ it is: writing a general element as $a|\psi_G\rangle + b|\psi_B\rangle$, we have

$$I_G (a|\psi_G\rangle + b|\psi_B\rangle) = -a|\psi_G\rangle + b|\psi_B\rangle,$$

since I_G negates each good basis vector and fixes each bad one, so on $\mathcal{P}_G$ it agrees with $I_{|\psi_G\rangle}$. The proof of the previous theorem now goes through verbatim with $|\psi_G\rangle$ in place of $|x_0\rangle$. $\square$

So the number of iterations is now $m \approx \frac{\pi}{4} \sqrt{N/r}$: the more solutions there are, the faster we find one, exactly as one would hope.

The interesting case is when r is *unknown*, which is the realistic one – for SAT we have no idea how many satisfying assignments there are. Here we must be careful: running too many iterations is as bad as running too few, since the state rotates *past* $|\psi_G\rangle$ and back down again. The fix is randomisation.

Method 39.14.9 (Grover with Unknown r)

Choose K uniformly at random in $(0, \frac{\pi}{4} \sqrt{N})$, apply Q_G that many times to $|\psi_0\rangle$, measure, and test the outcome with one further query. Repeat until a good item is found.

Proposition 39.14.10

Assuming $r \ll N$, each round of this procedure finds a good item with probability bounded below by an absolute constant, using $O(\sqrt{N})$ queries.

Proof sketch. Each iterate rotates by $2\alpha \approx 2\sqrt{r/N}$, so applying K of them rotates by a total angle $2K\alpha$, which as K ranges over $(0, \frac{\pi}{4}\sqrt{N})$ ranges over $(0, \frac{\pi}{2}\sqrt{r})$ – many full turns. The final angle is therefore close to uniform on the circle. If the final state makes an angle within $\pm\pi/4$ of $|\psi_G\rangle$, the probability of measuring a good item is at least $\cos^2(\pi/4) = \frac{1}{2}$. Half of all angles have this property, so the overall success probability per round is about $\frac{1}{4}$. Hence $O(1)$ rounds suffice in expectation, each using $O(\sqrt{N})$ queries. $\square$

§39.14.4 Optimality

Is $\sqrt{N}$ the best possible? Yes, and this is one of the few places in the subject where we have a matching lower bound.

Theorem 39.14.11 (BBBV)

Any quantum algorithm that solves unstructured search on N items with bounded error must make $\Omega(\sqrt{N})$ queries. Moreover $\frac{\pi}{4}(1 - \varepsilon)\sqrt{N}$ queries do not suffice for any $\varepsilon > 0$, so Grover’s algorithm is optimal even in its constant.

The proof is a ‘hybrid argument’: one shows that a single query can change the state of the computation by only $O(1/\sqrt{N})$ in norm on average over the possible locations of the target, so that after T queries the algorithm cannot distinguish the N possible oracles unless $T = \Omega(\sqrt{N})$.

The consequence is worth emphasising, because it cuts both ways. On one hand, Grover’s algorithm gives a genuine and completely general speedup: any problem for which we know nothing better than brute-force search over 2^n candidates – which includes every NP-complete problem – can be attacked in time $O(2^{n/2})$ instead of $O(2^n)$. Since SAT is NP-complete, this applies across the whole of NP.

On the other hand, the lower bound says that this is *all* that generic quantum search can do. To do better one must exploit structure, as Shor does. There is no reason to expect NP-complete problems to have the kind of structure that admits an exponential speedup, and most people do not expect quantum computers to solve them efficiently.

Remark. In practice the quadratic speedup is less dramatic than it sounds, since $\sqrt{2^n} = 2^{n/2}$ is still exponential: Grover turns a 128-bit search into a 64-bit search. Its main cryptographic consequence is that symmetric key lengths should be doubled, whereas Shor’s algorithm destroys RSA and elliptic-curve cryptography outright.

§39.15 Mixed States and Density Matrices

Throughout the course we have described states by vectors, and repeatedly run into situations where that is not quite the right language: a qubit which is ‘ $|0\rangle$ or $|1\rangle$ with probability $\frac{1}{2}$ each’, or ‘half of an entangled pair, considered on its own’. In both cases the system has no state vector of its own, yet it certainly has well-defined measurement statistics. The right object is a **density matrix**, and it lets us say cleanly several things we could previously only say by circumlocution.

The general theory of density operators – the ensemble ambiguity, the partial trace, von Neumann entropy, and the way decoherence turns a pure state into a mixture –

is developed at length in the Part II Principles of Quantum Mechanics course, and I have written it up in my notes for that course (see the section ‘Density Operators and Mixed States’ in `../principles-of-quantum-mechanics/`). I shall not repeat that development here. What follows is the minimum needed to make our earlier informal talk of ‘ensembles’ and ‘half of an entangled pair’ precise, together with the one genuinely quantum-information-theoretic use of the formalism: the reduced density matrix as a measure of entanglement.

§39.15.1 Density Matrices

Definition 39.15.1 (Density Matrix)

Suppose a system is prepared in the state $|\psi_i\rangle$ with probability p_i , where $\sum_i p_i = 1$. The **density matrix** of this **ensemble** is

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|.$$

A state described by a rank-one $\rho = |\psi\rangle\langle\psi|$ is called **pure**; otherwise it is **mixed**.

Proposition 39.15.2

An operator ρ is the density matrix of some ensemble if and only if it is self-adjoint, positive semi-definite, and has $\text{Tr } \rho = 1$.

Proof. Each $|\psi_i\rangle\langle\psi_i|$ is self-adjoint and positive semi-definite with trace $\langle\psi_i|\psi_i\rangle = 1$, and these properties are preserved by convex combinations. Conversely, given such a ρ , the spectral theorem gives $\rho = \sum_i \lambda_i |e_i\rangle\langle e_i|$ with $\lambda_i \geq 0$ and $\sum \lambda_i = \text{Tr } \rho = 1$, exhibiting ρ as the density matrix of the ensemble $\{(\lambda_i, |e_i\rangle)\}$. $\square$

The Born rule extends immediately: the probability of outcome j in the measurement $\{P_j\}$ is

$$p(j) = \sum_i p_i \langle\psi_i|P_j|\psi_i\rangle = \text{Tr}(P_j\rho),$$

and unitary evolution acts by $\rho \mapsto U\rho U^\dagger$. Since *all* predictions depend on ρ alone, two ensembles with the same density matrix are physically indistinguishable, however different they look. This is exactly the phenomenon we met in §39.3.3.

Example 39.15.3

The ‘yes’ ensemble $\{(\frac{1}{2}, |0\rangle), (\frac{1}{2}, |1\rangle)\}$ and the ‘no’ ensemble $\{(\frac{1}{2}, |+\rangle), (\frac{1}{2}, |-\rangle)\}$ have density matrices

$$\frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}I, \quad \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = \frac{1}{2}I,$$

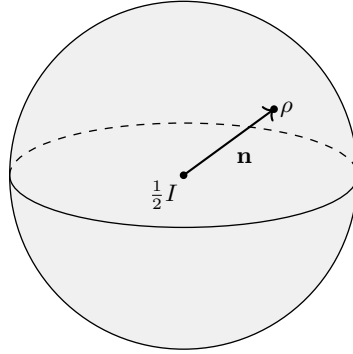
which are equal. This *is* the proof that Alice’s choice of measurement basis is undetectable by Bob, in one line. The state $\frac{1}{2}I$ is called the **maximally mixed** state of a qubit.

A useful invariant is the **purity** $\text{Tr}(\rho^2)$, which satisfies $\frac{1}{d} \leq \text{Tr}(\rho^2) \leq 1$ for a d -dimensional system, with equality on the right exactly for pure states and on the left

exactly for $\frac{1}{d}I$. For a qubit one can write

$$\rho = \frac{1}{2} (I + \mathbf{n} \cdot \vec{\sigma}), \quad \mathbf{n} \in \mathbb{R}^3, \quad |\mathbf{n}| \leq 1,$$

where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$; the pure states are those with $|\mathbf{n}| = 1$. So the density matrices of a qubit fill out the solid *Bloch ball*, with the pure states of §39.1.3 forming its surface and the maximally mixed state at the centre.



surface ($|\mathbf{n}| = 1$): pure states
interior ($|\mathbf{n}| < 1$): mixed states

The picture makes the geometry of mixing transparent: an ensemble $\{(p_i, \rho_i)\}$ has Bloch vector $\sum_i p_i \mathbf{n}_i$, so mixing is literally taking convex combinations inside the ball, and every interior point can be written as a mixture in infinitely many ways – which is precisely the ambiguity of ensembles noted above.

§39.15.2 The Reduced Density Matrix

The other use of density matrices is to describe a subsystem.

Definition 39.15.4 (Partial Trace)

Let ρ_{AB} be a density matrix on $\mathcal{V}_A \otimes \mathcal{V}_B$. Its **partial trace** over B is the operator on $\mathcal{V}_A$ defined by

$$\rho_A = \text{Tr}_B \rho_{AB} = \sum_j (I_A \otimes \langle f_j |) \rho_{AB} (I_A \otimes |f_j\rangle),$$

where $\{|f_j\rangle\}$ is any orthonormal basis of $\mathcal{V}_B$; the result does not depend on the choice.

Proposition 39.15.5

ρ_A is the unique operator on $\mathcal{V}_A$ such that $\text{Tr}(M\rho_A) = \text{Tr}((M \otimes I)\rho_{AB})$ for every operator M on $\mathcal{V}_A$. Consequently ρ_A correctly predicts the statistics of every measurement performed on A alone.

Example 39.15.6

Take $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, so

$$\rho_{AB} = \frac{1}{2} (|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Tracing out B , the cross terms $|00\rangle\langle 11|$ contribute $\langle 1|0\rangle \langle 0|1\rangle = 0$ and similarly for the other, so

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I.$$

Alice's half of a Bell pair is maximally mixed: she has *no* information at all. This is the precise sense in which $|\phi^+\rangle$ is maximally entangled, and it is also a one-line proof of the no-signalling principle for this state.

Notice what this says about the relationship between the whole and its parts: ρ_{AB} is pure (we know the joint state exactly) while ρ_A is maximally mixed (we know nothing about the part). Classically this is impossible – complete knowledge of a pair entails complete knowledge of each member. Entanglement is precisely the failure of that principle.

Proposition 39.15.7

A pure state $|\Psi\rangle_{AB}$ is a product state if and only if ρ_A is pure.

So the purity $\text{Tr}(\rho_A^2)$ is a quantitative measure of how entangled a pure bipartite state is, running from 1 (product) down to $1/\dim \mathcal{V}_A$ (maximally entangled).

§39.16 Quantum Error Correction

The last thing standing between us and a working quantum computer is that quantum systems are appallingly fragile. A qubit interacts with its environment, becomes entangled with it, and – by the discussion above – its reduced state becomes mixed: the information leaks away. This process is called **decoherence**, and without some way of fighting it no quantum computation of any length is possible.

Classically, we fight noise by redundancy: send 000 instead of 0, and take a majority vote. Quantumly this looks hopeless for three reasons.

- (i) We cannot copy an unknown state, by the no-cloning theorem.
- (ii) We cannot measure the encoded qubits to see whether an error has occurred, since measurement destroys superpositions.
- (iii) Errors are continuous, not discrete: instead of a bit flip we may get a rotation by a tiny angle ϵ , and there are uncountably many such.

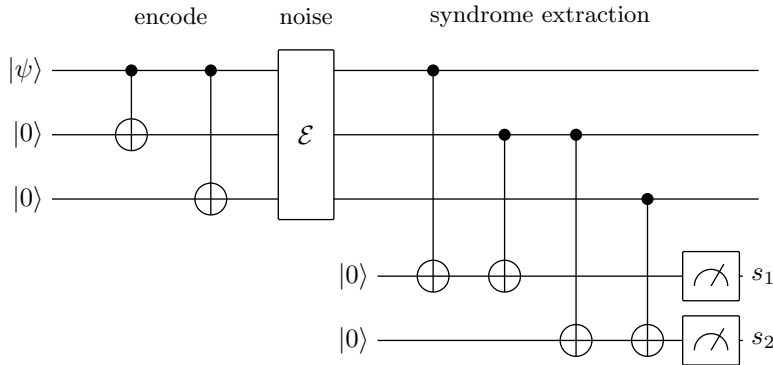
Remarkably, all three objections can be met. We illustrate with the simplest example.

§39.16.1 The Three-Qubit Bit-Flip Code

Suppose the only errors that occur are bit flips X , acting independently on each qubit with probability $p < \frac{1}{2}$, and suppose at most one qubit is affected. Encode

$$|0\rangle \mapsto |0_L\rangle = |000\rangle, \quad |1\rangle \mapsto |1_L\rangle = |111\rangle,$$

so that a general qubit $a|0\rangle + b|1\rangle$ becomes $a|000\rangle + b|111\rangle$. This is *not* cloning: we have not produced three copies of $a|0\rangle + b|1\rangle$, which would be the quite different state $(a|0\rangle + b|1\rangle)^{\otimes 3}$. Encoding is done with two CNOTs:



The two ancillas measure the parities $s_1 = b_1 \oplus b_2$ and $s_2 = b_2 \oplus b_3$ of the three data qubits. This is the key move, and it answers objection (ii): we measure a *comparison* of the qubits, not the qubits themselves. Concretely, the measurement is the incomplete projective measurement with the four rank-two projectors

$$\begin{aligned}\Pi_{00} &= |000\rangle\langle 000| + |111\rangle\langle 111|, & \Pi_{10} &= |100\rangle\langle 100| + |011\rangle\langle 011|, \\ \Pi_{01} &= |001\rangle\langle 001| + |110\rangle\langle 110|, & \Pi_{11} &= |010\rangle\langle 010| + |101\rangle\langle 101|,\end{aligned}$$

projecting onto the four two-dimensional subspaces of $(\mathbb{C}^2)^{\otimes 3}$ on which the parity pattern is constant.

Proposition 39.16.1

If at most one bit flip has occurred, the syndrome (s_1, s_2) identifies which qubit was affected, and the encoded state is undisturbed by the measurement.

Proof. Write $\mathcal{C} = \text{span}\{|000\rangle, |111\rangle\}$ for the code space. The four error operators I, X_1, X_2, X_3 map $\mathcal{C}$ into the four subspaces

$$\mathcal{C}, \quad \text{span}\{|100\rangle, |011\rangle\}, \quad \text{span}\{|010\rangle, |101\rangle\}, \quad \text{span}\{|001\rangle, |110\rangle\},$$

which are mutually orthogonal and are exactly the ranges of $\Pi_{00}, \Pi_{10}, \Pi_{11}, \Pi_{01}$ respectively. So the syndrome measurement determines which of the four occurred; and since the corrupted state $X_j(a|000\rangle + b|111\rangle)$ lies *entirely* inside one of these subspaces, the corresponding projector acts on it as the identity. The state is therefore not disturbed at all, and in particular the superposition coefficients a, b – which we do not know and must not learn – are untouched. Applying X_j to the indicated qubit restores the encoded state. $\square$

That deals with objections (i) and (ii). Objection (iii), the continuum of possible errors, is dealt with by an argument of striking elegance.

Proposition 39.16.2 (Discretisation of Errors)

Suppose the noise acts on the first qubit by an arbitrary operator, which we may expand in the Pauli basis as $E = c_0I + c_1X + c_2Y + c_3Z$. If the code can correct each of I, X, Y, Z on that qubit, then measuring the syndrome projects the corrupted

state onto one of the correctable cases, which we then correct exactly.

Proof idea. Since $\{I, X, Y, Z\}$ spans the space of one-qubit operators, the corrupted state $E_1 |\psi_L\rangle$ is a superposition of the four states $|\psi_L\rangle, X_1 |\psi_L\rangle, Y_1 |\psi_L\rangle, Z_1 |\psi_L\rangle$. If these lie in mutually orthogonal syndrome subspaces, then the syndrome measurement collapses the superposition onto exactly one of them, with the corresponding probability, and we correct it. A continuous error has been *digitised* by the act of measurement. $\square$

The proposition is the reason the whole enterprise is possible. It says that we do not have to fight a continuum: we need only design a code which handles the three Pauli errors on each qubit, and the syndrome measurement will digitise everything else for us. Quantum error correction is therefore, in the end, a discrete combinatorial problem, much like its classical counterpart.

§39.16.2 The Phase-Flip Code

The three-qubit code corrects X errors only. A Z error on any of the three qubits sends $a|000\rangle + b|111\rangle$ to $a|000\rangle - b|111\rangle$, which lies in the code space, produces the syndrome 00, and is therefore invisible; and it is a genuine error, since it takes $|+_L\rangle$ to $|-_L\rangle$.

But a phase flip is a bit flip in disguise. Since $HZH = X$, conjugating everything by $H^{\otimes 3}$ turns Z errors into X errors, and the code

$$|0_L\rangle = |+++\rangle, \quad |1_L\rangle = |--\rangle$$

corrects an arbitrary single Z error, by running the previous analysis in the conjugate basis. Concretely, one encodes with the same two CNOTs followed by a Hadamard on each of the three qubits, and the syndrome measurement extracts the two parities of the $\pm$ labels, which amounts to measuring the observables X_1X_2 and X_2X_3 rather than Z_1Z_2 and Z_2Z_3 . This code, of course, is now blind to X errors: $X|+\rangle = |+\rangle$, so a bit flip on any qubit does nothing detectable, and indeed nothing at all.

Neither code alone is any use against general noise. The way out is to use both at once.

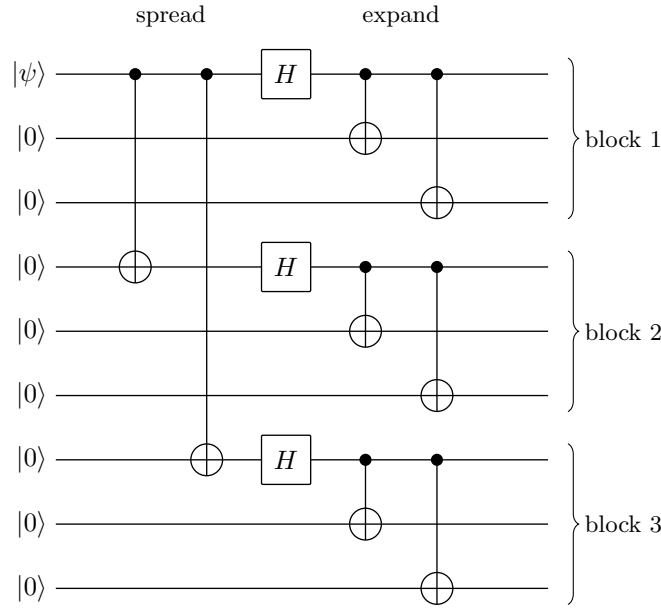
§39.16.3 Shor's Nine-Qubit Code

Definition 39.16.3 (Shor Code)

The **Shor nine-qubit code** encodes one qubit into nine by

$$\begin{aligned} |0_L\rangle &= \frac{1}{2\sqrt{2}} (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle), \\ |1_L\rangle &= \frac{1}{2\sqrt{2}} (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle). \end{aligned}$$

The structure is a **concatenation**: take the phase-flip code $|0\rangle \mapsto |+++\rangle, |1\rangle \mapsto |--\rangle$, and then encode each of its three qubits using the bit-flip code, replacing $|\pm\rangle$ by $\frac{1}{\sqrt{2}}(|000\rangle \pm |111\rangle)$. The encoding circuit is correspondingly two-layered: two CNOTs spread the data qubit across three blocks, a Hadamard on each block leader creates the $|\pm\rangle$ superpositions, and two further CNOTs inside each block expand $|0\rangle, |1\rangle$ to $|000\rangle, |111\rangle$.



Theorem 39.16.4

The Shor code corrects an arbitrary error on any single one of the nine qubits.

Proof. By the discretisation proposition it is enough to correct X , Y and Z on each qubit, and since $Y = iXZ$ up to a phase it is enough to handle X and Z separately.

An X error on some qubit lies inside one block. Measuring the parities Z_1Z_2 and Z_2Z_3 *within each block*, which is six syndrome bits in all, identifies both the affected block and the affected qubit inside it, exactly as in §39.16.1: each block carries a state of the form $\alpha|000\rangle + \beta|111\rangle$. Note that these measurements reveal nothing about α and β , and so nothing about the encoded qubit.

A Z error on any qubit of a block has the same effect on that block, namely to negate the relative sign of $|000\rangle$ and $|111\rangle$; so at the level of the outer code it is a single phase flip on one of the three ‘logical’ qubits $|\pm\rangle$. Measuring the outer syndrome – the two observables comparing the signs of blocks 1,2 and blocks 2,3, that is, $X_1 \cdots X_6$ and $X_4 \cdots X_9$ – identifies which block was affected, and applying Z to any one qubit of that block repairs it.

Since the two syndromes commute and address disjoint information, both can be measured, and an error which is simultaneously a bit flip and a phase flip on one qubit (that is, a Y) is corrected by correcting each part. Finally, an arbitrary single-qubit error is a linear combination of I, X, Y, Z on that qubit, and the syndrome measurement collapses it onto one of the four correctable cases. $\square$

Remark. Nine qubits is not optimal: the five-qubit code achieves the same thing, and one can show that five is the minimum. Modern codes are described in the [stabiliser](#) formalism, where a code space is specified as the joint $+1$ eigenspace of a commuting family of Pauli operators – for the Shor code, the six operators Z_1Z_2, Z_2Z_3 within blocks and the two operators $X_1 \cdots X_6, X_4 \cdots X_9$ across blocks. Syndrome extraction is then just measurement of the stabiliser generators, and the whole theory acquires a pleasantly algebraic flavour, closely parallel to the theory of linear codes in Part II Coding and Cryptography.

§39.16.4 Fault Tolerance

We have been assuming that encoding, syndrome extraction and correction are themselves error-free, which is absurd: the correction circuitry is built from the same faulty components as everything else, and a single faulty gate in a syndrome extraction can spread an error onto several data qubits at once. A **fault-tolerant** construction is one designed so that this does not happen – so that a single fault anywhere produces at most one error per encoded block, which the code can then remove. Given such a construction, one may concatenate: encode each qubit of a code in another copy of the code, and repeat. If a block fails only when two of its constituents fail, then an error rate p at one level becomes something like Cp^2 at the next, and iterating ℓ times gives $(Cp)^{2^\ell}/C$, which tends to zero rapidly provided $p < 1/C$. That threshold behaviour is the content of the following.

Theorem 39.16.5 (Threshold Theorem)

There is a constant $p_{\text{th}} > 0$ such that, if the physical error rate per gate is below p_{th} , then any quantum circuit of size T can be simulated with error at most ε by a fault-tolerant circuit of size $O(T \text{ poly} \log(T/\varepsilon))$.

This is the reason anyone believes large-scale quantum computing to be possible in principle. It says that error correction is not a losing battle – once the hardware is good enough, concatenating codes suppresses errors faster than the extra machinery introduces them, and the overhead is only polylogarithmic.

§39.17 Closing Remarks

Let us take stock of what quantum mechanics has bought us.

Impossibility. Non-orthogonal states cannot be reliably distinguished (§39.2.3); unknown states cannot be cloned (§39.3.2); measurement at a distance cannot signal (§39.3.4). These are not engineering limitations but theorems, and each follows from linearity and unitarity in a few lines.

New protocols. Entanglement lets us send two classical bits down one qubit (§39.6) and one qubit down two classical bits (§39.7), and lets us distribute cryptographic keys whose security rests on physics rather than on computational assumptions (§39.8). None of these is possible classically.

New algorithms. Superposition, phase kickback and interference give exact one-query solutions to Deutsch and Deutsch–Jozsa (§39.11), an exponential bounded-error separation in Simon’s problem (§39.11.4), polynomial-time factoring via period-finding and the quantum Fourier transform (§39.13), and a provably optimal quadratic speedup for unstructured search (§39.14).

And what quantum mechanics does not buy us. A quantum computer computes exactly the same functions as a classical one; the difference is only in efficiency, and even there we have no *proof* of a separation, only very strong evidence. Grover’s bound tells us that generic search cannot do better than quadratic, so NP-complete problems are almost certainly not going to fall. And every protocol we have seen respects relativistic causality: nothing here travels faster than a classical message.

Perhaps the deepest lesson is Landauer's: *information is physical*. The theory of computation is not a branch of pure mathematics floating free of the world, but a description of what physical systems can be made to do. Change the physics and you change the theory. It is a striking thought that the answer to 'what is efficiently computable?' should depend on the correct theory of nature – and a slightly unsettling one that we do not yet know the final answer to either question.

40 Representation Theory

Groups are, in the end, a way of talking about symmetry. But a group presented abstractly – as a set with a multiplication table, or as generators and relations – is a slippery thing to get hold of. Representation theory takes the view that the way to understand a group is to watch it *act*, and the most tractable things a group can act on are vector spaces. A representation is exactly that: a group acting by invertible linear maps. Once we are in the world of linear algebra we can bring the whole toolbox – eigenvalues, traces, direct sums, tensor products – to bear on questions that started out being about groups.

The astonishing thing, and the thing that makes this course so satisfying, is how much information survives when we throw away almost everything. To each representation we attach its **character**, a single complex-valued function on the group recording only the trace of each group element. This looks like a hopeless loss of information, and yet over $\mathbb{C}$ the character determines the representation completely. The characters of the irreducible representations satisfy beautiful orthogonality relations, they can be assembled into a finite square array called the character table, and that little table encodes a remarkable amount about the group. We will use it to prove Burnside's theorem, that every group of order $p^a q^b$ is soluble – a purely group-theoretic statement for which no purely group-theoretic proof was known for decades.

This article constitutes my notes for the 'Representation Theory' course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§40.1 Representations

§40.1.1 Some Preliminaries

Before we begin properly, it is worth being clear about what we are assuming, since this course sits right on top of *Linear Algebra* and *Groups, Rings and Modules*.

Throughout, ‘vector space’ means a *finite dimensional* vector space unless we say otherwise. Our base field will be written F or k , and although a fair amount of what we do works over any field, the theory only becomes really clean when F is algebraically closed of characteristic zero. So the reader should keep $F = \mathbb{C}$ firmly in mind: essentially every theorem with any content in this course is a theorem about complex representations.

Recall from linear algebra that if V is a vector space over F , then

$$\mathrm{GL}(V) = \{\alpha : V \rightarrow V \mid \alpha \text{ is linear and invertible}\}$$

is a group under composition, the **general linear group** of V . If $\dim V = n$ then a choice of basis $e_1, \dots, e_n$ identifies V with F^n and hence identifies $\mathrm{GL}(V)$ with the group $\mathrm{GL}_n(F)$ of invertible $n \times n$ matrices over F . Changing basis changes the identification by conjugation: if α has matrix A in one basis and A' in another, then $A' = P^{-1}AP$ for the change of basis matrix P . This is worth saying out loud now, because *everything* in this course is invariant under conjugation, and that is not a coincidence.

We will need one more piece of linear algebra constantly.

Proposition 40.1.1 (Diagonalisability)

Let $\alpha \in \mathrm{GL}(V)$ with V a finite dimensional $\mathbb{C}$ -vector space, and suppose $\alpha^m = \mathrm{id}$ for some $m \geq 1$. Then α is diagonalisable.

Proof. The polynomial $X^m - 1$ annihilates α , and over $\mathbb{C}$ it factors as $\prod_{j=0}^{m-1} (X - \omega^j)$ where $\omega = e^{2\pi i/m}$, which has no repeated roots. A linear map is diagonalisable if and only if its minimal polynomial splits into distinct linear factors, and the minimal polynomial divides $X^m - 1$, so we are done. $\square$

In particular, if g is an element of a finite group and α is the linear map by which g acts, then α is diagonalisable, with eigenvalues roots of unity. This single observation is responsible for a surprising amount of what follows.

There is also a version for several maps at once, whose proof the reader has essentially seen in *Linear Algebra* when simultaneously diagonalising commuting operators.

Proposition 40.1.2 (Simultaneous diagonalisation)

Let $\alpha, \beta \in \mathrm{GL}(V)$ be diagonalisable, with V a $\mathbb{C}$ -vector space. If $\alpha\beta = \beta\alpha$, then there is a basis of V in which α and β are both diagonal.

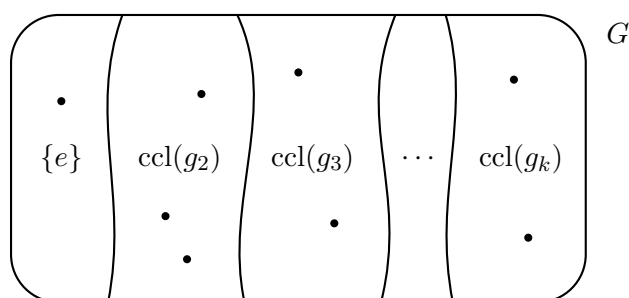
Proof. Write $V = \bigoplus_{\lambda} V_{\lambda}$ as a sum of eigenspaces of α . If $v \in V_{\lambda}$ then $\alpha(\beta v) = \beta(\alpha v) = \lambda\beta v$, so β preserves each V_{λ} . Now $\beta|_{V_{\lambda}}$ is diagonalisable (its minimal polynomial divides that of β , which has distinct linear factors), so pick an eigenbasis of β on each V_{λ} and put them together. $\square$

On the group theory side, recall that an **action** of a group G on a set X is a map $G \times X \rightarrow X$, written $(g, x) \mapsto g \cdot x$, satisfying $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$. Equivalently – and this is the point of view we want – an action of G on X is precisely a group homomorphism $G \rightarrow \text{Sym}(X)$, the group of all bijections $X \rightarrow X$.

Recall also that the **conjugacy class** of $g \in G$ is

$$\text{ccl}_G(g) = \{xgx^{-1} \mid x \in G\},$$

that these classes partition G , and that by the orbit–stabiliser theorem $|\text{ccl}_G(g)| = |G : C_G(g)|$ where $C_G(g) = \{x \in G \mid xg = gx\}$ is the centraliser. Conjugacy classes will play the role for us that ‘points’ play elsewhere, so here is a picture to fix ideas.



The moral of the course, stated far too early, is that the natural ‘coordinates’ on a group are indexed by its conjugacy classes, and the natural ‘functions’ in those coordinates are the irreducible characters. There will be exactly as many of the latter as the former.

§40.1.2 Definitions and First Examples

We now come to the central definition. Since an action of G on a set is a homomorphism into $\text{Sym}(X)$, and since a vector space is a set with extra structure, the right thing to do is to replace Sym by GL .

Definition 40.1.3 (Representation)

Let G be a group and V a vector space over F . A **representation** of G on V is a group homomorphism

$$\rho : G \rightarrow \text{GL}(V).$$

We call V the **representation space**, and we write the representation as the pair (ρ, V) , or often just as ρ or just as V when the rest is clear.

Unwinding the definition of homomorphism, giving a representation is the same as giving, for each $g \in G$, a linear map $\rho(g) : V \rightarrow V$ such that $\rho(e) = \text{id}_V$ and $\rho(gh) = \rho(g)\rho(h)$ for all $g, h \in G$. Note that invertibility comes for free from these two conditions, since $\rho(g)\rho(g^{-1}) = \rho(e) = \text{id}_V$; so we may as well only ask for linear maps.

Definition 40.1.4 (Degree)

The **degree** (or **dimension**) of a representation (ρ, V) is $\dim_F V$.

Definition 40.1.5 (Faithful representation)

A representation ρ is **faithful** if $\ker \rho = \{e\}$, i.e. if ρ is injective.

A faithful representation realises G concretely as a group of matrices, which is often exactly what one wants. Non-faithful representations are not a defect, though; they are how the theory sees the normal subgroups of G .

Everything is much clearer with examples, so let us collect a good supply now. Almost all of these will reappear repeatedly.

Example 40.1.6 (The trivial representation)

For any group G and any vector space V , the map $\rho(g) = \text{id}_V$ for all g is a representation. When $V = F$ is one dimensional this is called **the** trivial representation, and we write it $\mathbf{1}$ or $\mathbf{1}_G$. It is faithful only when G is trivial.

Example 40.1.7 (Representations of a cyclic group)

Let $G = C_m = \langle g \rangle$ be cyclic of order m , and let V be a $\mathbb{C}$ -vector space. A representation of G is determined by $\alpha = \rho(g) \in \text{GL}(V)$, and the only constraint is $\alpha^m = \text{id}$. So representations of C_m on V correspond exactly to elements of $\text{GL}(V)$ of order dividing m .

By our proposition above, such an α is diagonalisable, so there is a basis in which

$$\rho(g) = \begin{pmatrix} \omega^{a_1} & & \\ & \ddots & \\ & & \omega^{a_n} \end{pmatrix}, \quad \omega = e^{2\pi i/m}$$

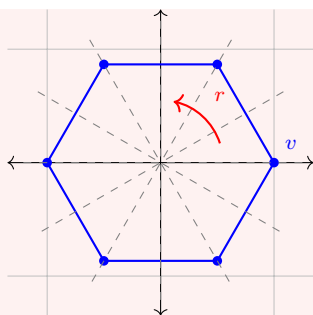
for some $a_1, \dots, a_n$. So every representation of C_m breaks up into one-dimensional pieces $g \mapsto \omega^a$. We will see this is a special case of a general theorem.

Example 40.1.8 (Dihedral groups)

Let $D_{2n} = \langle r, s \mid r^n = s^2 = e, srs^{-1} = r^{-1} \rangle$ be the symmetry group of a regular n -gon. There is an obvious representation on $\mathbb{R}^2$ (and hence on $\mathbb{C}^2$): let r act by rotation through $2\pi/n$ and s by reflection in the x -axis, so

$$\rho(r) = \begin{pmatrix} \cos \frac{2\pi}{n} & -\sin \frac{2\pi}{n} \\ \sin \frac{2\pi}{n} & \cos \frac{2\pi}{n} \end{pmatrix}, \quad \rho(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

One checks the relations hold, and ρ is faithful. This is the representation you should picture whenever D_{2n} comes up.



The dashed lines are the n axes of reflection; r rotates by $2\pi/n$ and each reflection swaps the two half-planes cut out by its axis.

Example 40.1.9 (Symmetric groups on the standard space)

Let $G = S_n$ act on $V = F^n$ with basis $e_1, \dots, e_n$ by permuting coordinates:

$$\rho(\sigma)(e_i) = e_{\sigma(i)}.$$

Extending linearly gives a representation of degree n ; the matrix of $\rho(\sigma)$ is the permutation matrix of σ . It is faithful.

Example 40.1.10 (The sign representation)

Let $G = S_n$ and $V = F$, with $\rho(\sigma) = \text{sign}(\sigma)$. This is a one-dimensional representation, not faithful for $n \geq 3$ (its kernel is A_n).

Two more constructions deserve to be called out because they will be with us for the whole course.

Definition 40.1.11 (Permutation representation)

Let G act on a finite set X . Let FX be the F -vector space with basis $\{e_x \mid x \in X\}$, so $\dim FX = |X|$. Then

$$\rho(g) \left(\sum_{x \in X} \lambda_x e_x \right) = \sum_{x \in X} \lambda_x e_{g \cdot x}$$

defines a representation of G , the **permutation representation** associated to the action.

Definition 40.1.12 (Regular representation)

Taking $X = G$ with G acting on itself by left multiplication gives the **regular representation** of G , on the vector space FG of dimension $|G|$.

The regular representation is always faithful (if $g \neq e$ then $g \cdot e = g \neq e$, so $\rho(g) \neq \text{id}$), which already tells us something: *every* finite group has a faithful representation, so every finite group is isomorphic to a group of matrices. We will see later that FG is much more than a bookkeeping device – it contains every irreducible representation of G , each with multiplicity equal to its degree.

§40.1.3 Equivalence of Representations

The representation of S_3 on $\mathbb{C}^3$ by permutation matrices, and the ‘same’ representation written in a different basis, should not be counted as different. We need to say when two representations are the same.

Definition 40.1.13 (G -homomorphism)

Let (ρ, V) and (ρ', V') be representations of G over F . A linear map $\varphi : V \rightarrow V'$ is a **G -homomorphism** (or is **G -linear**, or **intertwines** ρ and ρ') if

$$\varphi \circ \rho(g) = \rho'(g) \circ \varphi \quad \text{for all } g \in G.$$

We write $\text{Hom}_G(V, V')$ for the set of all such maps.

The condition says exactly that the square

$$\begin{array}{ccc} V & \xrightarrow{\rho(g)} & V \\ \varphi \downarrow & & \downarrow \varphi \\ V' & \xrightarrow{\rho'(g)} & V' \end{array}$$

commutes for every g . Note that $\text{Hom}_G(V, V')$ is an F -subspace of $\text{Hom}_F(V, V')$, since the condition is linear in φ .

Definition 40.1.14 (Isomorphism of representations)

We say (ρ, V) and (ρ', V') are **isomorphic** (or **equivalent**), written $V \cong V'$, if there is a G -homomorphism $\varphi : V \rightarrow V'$ which is a linear isomorphism.

If φ is such an isomorphism then $\rho'(g) = \varphi \rho(g) \varphi^{-1}$, so at the level of matrices, isomorphism of representations of the same degree n means: there is a single $P \in \text{GL}_n(F)$ with $R'(g) = P R(g) P^{-1}$ for all g , where R, R' are the matrix representations. In other words, isomorphism is ‘simultaneous change of basis’. It is straightforward to check that this is an equivalence relation.

Remark. Note the ‘simultaneous’ – one matrix P must work for every g at once. Two representations can have $\rho(g)$ conjugate to $\rho'(g)$ for each individual g and still fail to be isomorphic.

Example 40.1.15

Any two one-dimensional representations $\rho, \rho' : G \rightarrow \text{GL}_1(F) = F^\times$ are isomorphic if and only if they are *equal*, since $\text{GL}_1(F)$ is abelian and conjugation does nothing.

§40.1.4 Representations as Modules

There is a second, slicker way of packaging all this, which the reader will recognise from *Groups, Rings and Modules*. It is worth setting up because it makes several later constructions obvious.

Definition 40.1.16 (Group algebra)

Let G be a finite group and F a field. The **group algebra** FG is the F -vector space with basis $\{g \mid g \in G\}$, that is

$$FG = \left\{ \sum_{g \in G} \lambda_g g \mid \lambda_g \in F \right\},$$

equipped with the multiplication extending the group multiplication bilinearly:

$$\left(\sum_g \lambda_g g \right) \left(\sum_h \mu_h h \right) = \sum_{g,h} \lambda_g \mu_h (gh).$$

This makes FG into a ring (indeed an F -algebra) of dimension $|G|$ over F , with identity the element $1 \cdot e$. It is commutative precisely when G is abelian.

Proposition 40.1.17

Let G be a finite group. There is a bijective correspondence between representations of G over F and FG -modules, under which a representation (ρ, V) corresponds to the module structure

$$\left(\sum_g \lambda_g g \right) \cdot v = \sum_g \lambda_g \rho(g)(v).$$

Under this correspondence, G -homomorphisms correspond to FG -module homomorphisms, and isomorphism of representations to isomorphism of modules.

Proof. Given (ρ, V) , the displayed formula is well defined and one checks the module axioms directly: F -bilinearity is built in, and associativity $(ab)v = a(bv)$ reduces by bilinearity to the case $a = g$, $b = h$ basis elements, where it is $\rho(gh) = \rho(g)\rho(h)$.

Conversely, given an FG -module V , restricting scalars along $F \hookrightarrow FG$ makes V an F -vector space, and $\rho(g) : v \mapsto g \cdot v$ is F -linear with $\rho(g)\rho(g^{-1}) = \text{id}$, so $\rho(g) \in \text{GL}(V)$; and $\rho(gh) = \rho(g)\rho(h)$ by associativity. These two constructions are mutually inverse.

Finally, a linear map $\varphi : V \rightarrow V'$ satisfies $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all g if and only if it satisfies $\varphi(a \cdot v) = a \cdot \varphi(v)$ for all $a \in FG$, by linearity. $\square$

Remark. We insist that our FG -modules are finite dimensional over F , matching our convention on representations. The regular representation is precisely FG regarded as a module over itself.

From now on we will use the two languages interchangeably, saying ‘ V is a G -space’ or ‘ V is a $\mathbb{C}G$ -module’ or ‘ V is a representation of G ’ as convenient, and writing gv for $\rho(g)(v)$.

§40.1.5 Subrepresentations and Irreducibility**Definition 40.1.18** (Subrepresentation)

Let (ρ, V) be a representation of G . A subspace $W \leq V$ is **G -invariant** (or is a **subrepresentation**, or a **G -subspace**) if $\rho(g)(W) \subseteq W$ for all $g \in G$.

If W is G -invariant then in fact $\rho(g)(W) = W$ for each g , since $\rho(g^{-1})(W) \subseteq W$ gives the reverse inclusion. So ρ restricts to a representation $\rho|_W : G \rightarrow \mathrm{GL}(W)$. In module language, subrepresentations are exactly FG -submodules.

Every representation has the **trivial** subrepresentations 0 and V . The interesting question is whether there are any others.

Definition 40.1.19 (Irreducible)

A representation (ρ, V) is **irreducible** (or **simple**) if $V \neq 0$ and the only G -invariant subspaces of V are 0 and V .

Example 40.1.20

Every one-dimensional representation is irreducible, since it has no nonzero proper subspaces at all.

Example 40.1.21 (The standard representation of S_n)

Let $V = \mathbb{C}^n$ carry the permutation action of S_n . This is *not* irreducible for $n \geq 2$: the line

$$U = \langle e_1 + e_2 + \cdots + e_n \rangle$$

is fixed pointwise by every σ , so is a subrepresentation isomorphic to the trivial representation. Its complement

$$W = \left\{ \sum_i \lambda_i e_i \mid \sum_i \lambda_i = 0 \right\}$$

is also G -invariant, of dimension $n - 1$; it is called the **standard representation** of S_n . We have $V = U \oplus W$. We will prove later (using characters, in about two lines) that W is irreducible.

Example 40.1.22 (A reducible but indecomposable representation)

Irreducibility is genuinely sensitive to the ground field. Let $G = C_p = \langle g \rangle$ and let $F = \mathbb{F}_p$. Define a representation on F^2 by

$$\rho(g^k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

This is a homomorphism since the matrices multiply by adding the top-right entries, and $\rho(g^p) = I$ as $p \equiv 0$. The subspace $\langle e_1 \rangle$ is invariant, so ρ is reducible. But it has *no* invariant complement: any invariant line would give an eigenvector of $\rho(g)$ other than a multiple of e_1 , and $\rho(g)$ has 1 as its only eigenvalue with a one-dimensional eigenspace.

So over $\mathbb{F}_p$ things can go wrong. The next section shows that over $\mathbb{C}$ (or any field whose characteristic does not divide $|G|$) they never do.

Definition 40.1.23 (Direct sums)

If (ρ_1, V_1) and (ρ_2, V_2) are representations of G , their **direct sum** is the representation on $V_1 \oplus V_2$ given by $g \cdot (v_1, v_2) = (gv_1, gv_2)$. In matrix terms, choosing bases of V_1 and V_2 and concatenating,

$$\rho(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}.$$

Definition 40.1.24 (Completely reducible)

A representation is **completely reducible** (or **semisimple**) if it is a direct sum of irreducible subrepresentations.

Here is the picture to have in mind. If $V = V_1 \oplus \cdots \oplus V_k$ with each V_i irreducible, then in a suitable basis every matrix $\rho(g)$ is block diagonal with blocks of sizes $\dim V_i$, *simultaneously for all g* :

$$\rho(g) = \begin{array}{|c|c|c|c|} \hline \rho_1 & & & \\ \hline & \rho_2 & & 0 \\ \hline & & \rho_3 & \\ \hline & 0 & & \rho_4 \\ \hline \end{array}$$

block sizes $\dim V_1, \dots, \dim V_k$

Complete reducibility says that we can always reduce to this shape, and so the problem of understanding all representations of G splits into two: find the irreducible ones, and find how a given representation decomposes. The first is the hard and interesting part.

Definition 40.1.25 (Decomposable)

A representation V is **decomposable** if $V = V_1 \oplus V_2$ for nonzero subrepresentations V_1, V_2 , and **indecomposable** otherwise.

Irreducible implies indecomposable, always. The example over $\mathbb{F}_p$ above is indecomposable but not irreducible; the content of Maschke's theorem is that over $\mathbb{C}$ the two notions coincide.

Example 40.1.26 (Quotients)

If $W \leq V$ is a subrepresentation then the quotient space V/W carries a representation, $g(v+W) = gv+W$, which is well defined precisely because W is invariant. This is the **quotient representation**. If $V = W \oplus U$ as representations then $V/W \cong U$.

§40.1.6 Representations of Abelian Groups

Let us do one complete classification straight away, since it is easy and instructive.

Proposition 40.1.27

Let G be a finite abelian group and let $\rho : G \rightarrow \mathrm{GL}(V)$ be a representation over $\mathbb{C}$. Then there is a basis of V in which every $\rho(g)$ is diagonal.

Proof. Each $\rho(g)$ has finite order, hence is diagonalisable. The maps $\{\rho(g)\}_{g \in G}$ pairwise commute since G is abelian. Now induct on $|G|$ using simultaneous diagonalisation: diagonalise $\rho(g_1)$, then $\rho(g_2)$ preserves each of its eigenspaces, and so on. $\square$

Corollary 40.1.28

Every irreducible complex representation of a finite abelian group is one-dimensional.

Proof. In the basis given by the proposition, every coordinate line is G -invariant. So if $\dim V > 1$ then V has a proper nonzero invariant subspace. $\square$

We will see in due course that the converse holds too: if every irreducible complex representation of G is one-dimensional then G is abelian. This is our first hint that the representation theory really does remember the group.

Example 40.1.29 (C_m , completely)

The irreducible complex representations of $C_m = \langle g \rangle$ are exactly the m degree-one representations

$$\rho_j : g^k \mapsto \omega^{jk}, \quad \omega = e^{2\pi i/m}, \quad j = 0, 1, \dots, m-1,$$

and these are pairwise non-isomorphic. Note there are m of them, and C_m has m conjugacy classes.

Example 40.1.30 ($C_2 \times C_2$)

The Klein four-group $V_4 = \{e, a, b, ab\}$ has four one-dimensional representations, determined by the four choices of $(\rho(a), \rho(b)) \in \{\pm 1\}^2$. Again: four irreducibles, four conjugacy classes.

§40.2 Complete Reducibility and Maschke's Theorem

The example over $\mathbb{F}_p$ in the last section was worrying: a representation which is reducible but which cannot be broken up. Maschke's theorem says this cannot happen in the situation we care about. It is the theorem that makes the whole course work.

Theorem 40.2.1 (Maschke's Theorem)

Let G be a finite group and let V be a representation of G over a field F with

$\text{char } F \nmid |G|$ (for instance $F = \mathbb{C}$). If $W \leq V$ is a subrepresentation, then there is a subrepresentation $U \leq V$ with

$$V = W \oplus U.$$

Before proving it, let us see why the hypothesis matters, and what the idea is. The obstruction is that W certainly has *some* vector space complement; the issue is whether we can choose one which is G -invariant. The trick, which we will use again and again, is **averaging over the group**: take any complement, take the projection onto W along it, and average that projection over G to make it G -linear. Averaging requires dividing by $|G|$, and that is exactly where the characteristic hypothesis is used.

Proof. Choose any vector space complement W' of W in V , and let $\pi : V \rightarrow W$ be the corresponding projection: so $\pi(v) = v$ for $v \in W$, and $\ker \pi = W'$. There is no reason for π to be G -linear. Define $\tilde{\pi} : V \rightarrow V$ by

$$\tilde{\pi}(v) = \frac{1}{|G|} \sum_{g \in G} g \pi(g^{-1}v).$$

This makes sense because $|G|$ is invertible in F . We claim $\tilde{\pi}$ is a G -linear projection onto W .

It is linear, being a sum of composites of linear maps.

Its image lies in W : each term $g\pi(g^{-1}v)$ lies in $gW = W$, as W is G -invariant.

It restricts to the identity on W : if $w \in W$ then $g^{-1}w \in W$, so $\pi(g^{-1}w) = g^{-1}w$, so $g\pi(g^{-1}w) = w$. Averaging $|G|$ copies of w gives w .

It is G -linear: for $h \in G$ and $v \in V$,

$$\tilde{\pi}(hv) = \frac{1}{|G|} \sum_{g \in G} g \pi(g^{-1}hv) = \frac{1}{|G|} \sum_{g' \in G} hg' \pi(g'^{-1}v) = h \tilde{\pi}(v),$$

where we substituted $g = hg'$, which is a bijection of G as g ranges over G .

Now set $U = \ker \tilde{\pi}$. This is a subrepresentation, since $\tilde{\pi}$ is G -linear: if $\tilde{\pi}(v) = 0$ then $\tilde{\pi}(gv) = g\tilde{\pi}(v) = 0$. Finally, because $\tilde{\pi}$ is a projection onto W we have $\tilde{\pi}^2 = \tilde{\pi}$ and $V = \text{img } \tilde{\pi} \oplus \ker \tilde{\pi} = W \oplus U$, using $v = \tilde{\pi}(v) + (v - \tilde{\pi}(v))$. $\square$

Over $\mathbb{C}$ there is a second proof which is arguably more memorable, and which will generalise to compact groups in the last section of these notes. It is due to Weyl and is usually called the **unitary trick**. Recall from *Linear Algebra* that a Hermitian inner product on a complex vector space V is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{C}$, linear in the second variable, with $\langle w, v \rangle = \overline{\langle v, w \rangle}$ and $\langle v, v \rangle > 0$ for $v \neq 0$.

Definition 40.2.2 (G -invariant inner product)

An inner product on a representation V of G is **G -invariant** if $\langle gv, gw \rangle = \langle v, w \rangle$ for all $g \in G$ and $v, w \in V$.

Lemma 40.2.3 (Weyl's unitary trick)

Let G be finite and V a complex representation of G . Then V carries a G -invariant

Hermitian inner product. Equivalently, there is a basis of V in which every $\rho(g)$ is a unitary matrix, i.e. $\rho(G) \leq U(V)$.

Proof. Take any Hermitian inner product $(\cdot, \cdot)$ on V – for instance, choose a basis and use the standard one – and average:

$$\langle v, w \rangle = \frac{1}{|G|} \sum_{g \in G} (gv, gw).$$

Sesquilinearity and Hermitian symmetry are inherited termwise. Positive definiteness holds because $\langle v, v \rangle = \frac{1}{|G|} \sum_g (gv, gv)$ is a sum of positive terms when $v \neq 0$. And for $h \in G$,

$$\langle hv, hw \rangle = \frac{1}{|G|} \sum_g (ghv, ghw) = \frac{1}{|G|} \sum_{g'} (g'v, g'w) = \langle v, w \rangle$$

by reindexing $g' = gh$. Choosing an orthonormal basis for $\langle \cdot, \cdot \rangle$ (Gram–Schmidt) makes each $\rho(g)$ unitary. $\square$

Second proof of Maschke’s theorem over $\mathbb{C}$. Equip V with a G -invariant inner product and let $U = W^\perp$. This is a complement to W as a vector space, and it is G -invariant: if $u \in W^\perp$, $w \in W$ and $g \in G$, then

$$\langle w, gu \rangle = \langle g^{-1}w, u \rangle = 0$$

since $g^{-1}w \in W$. So $gu \in W^\perp$. $\square$

The reader should notice that both proofs are the same idea wearing different clothes: average something over G to make it G -equivariant. Almost every construction in this course is an instance of this.

Corollary 40.2.4 (Complete reducibility)

Let G be finite and F a field of characteristic not dividing $|G|$. Then every representation of G over F is a direct sum of irreducible representations.

Proof. Induct on $\dim V$. If $V = 0$ this is the empty direct sum, and if V is irreducible we are done. Otherwise V has a subrepresentation $0 \neq W \leq V$, and by Maschke $V = W \oplus U$ with U a subrepresentation. Both W and U have strictly smaller dimension than V , so by induction each decomposes into irreducibles, and concatenating gives the result. $\square$

Corollary 40.2.5

For G finite and $\text{char } F \nmid |G|$, a representation is irreducible if and only if it is indecomposable.

Remark. Both hypotheses are needed. Finiteness of G matters: the representation of $(\mathbb{Z}, +)$ on $\mathbb{C}^2$ with $n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ is reducible but indecomposable. And the characteristic hypothesis matters, as our C_p over $\mathbb{F}_p$ example shows. The study of the failing case is

modular representation theory and is much harder; we shall stay firmly over $\mathbb{C}$.

Remark (Standing convention). From this point on, unless we explicitly say otherwise, G is a *finite* group and all representations are *finite dimensional over $\mathbb{C}$* . We will relax the finiteness of G only in the final section, on topological groups.

§40.3 Schur's Lemma

Maschke tells us that every representation is built from irreducibles. Schur's lemma tells us how the irreducible pieces interact, and it is astonishingly simple.

Theorem 40.3.1 (Schur's Lemma)

Let V, W be irreducible complex representations of a group G .

- (i) Any G -homomorphism $\varphi : V \rightarrow W$ is either 0 or an isomorphism.
- (ii) Any G -endomorphism $\varphi : V \rightarrow V$ is a scalar multiple of the identity, $\varphi = \lambda \text{id}_V$.

Proof. (i) Let $\varphi \in \text{Hom}_G(V, W)$. Then $\ker \varphi$ is a subrepresentation of V : if $\varphi(v) = 0$ then $\varphi(gv) = g\varphi(v) = 0$. Similarly $\text{img } \varphi$ is a subrepresentation of W . Since V is irreducible, $\ker \varphi$ is 0 or V ; since W is irreducible, $\text{img } \varphi$ is 0 or W . If $\varphi \neq 0$ then $\ker \varphi \neq V$, so $\ker \varphi = 0$, and $\text{img } \varphi \neq 0$, so $\text{img } \varphi = W$. Hence φ is bijective, and its inverse is automatically G -linear.

(ii) Here we use that we are over $\mathbb{C}$: the map φ has an eigenvalue λ , since $\mathbb{C}$ is algebraically closed and $V \neq 0$ is finite dimensional. Then $\varphi - \lambda \text{id}_V$ is a G -homomorphism $V \rightarrow V$ (as id_V certainly is one) with nonzero kernel, so by (i) it is not an isomorphism, hence it is 0. $\square$

Remark. Part (i) works over any field, and indeed for modules over any ring: it says simple modules admit no interesting maps between them. Part (ii) genuinely needs $\mathbb{C}$ algebraically closed. Over $\mathbb{R}$, the rotation representation of C_n on $\mathbb{R}^2$ ($n \geq 3$) is irreducible and its endomorphism algebra is $\mathbb{C}$, not $\mathbb{R}$.

The following restatement is often the most useful form.

Corollary 40.3.2

For irreducible complex representations V, W of G ,

$$\dim_{\mathbb{C}} \text{Hom}_G(V, W) = \begin{cases} 1 & \text{if } V \cong W, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. If $V \not\cong W$ then by (i) every G -homomorphism is zero. If $V \cong W$, fix an isomorphism $\theta : V \rightarrow W$. For any $\varphi \in \text{Hom}_G(V, W)$ the composite $\theta^{-1}\varphi \in \text{End}_G(V)$ is a scalar λ by (ii), so $\varphi = \lambda\theta$. Hence $\text{Hom}_G(V, W) = \mathbb{C}\theta$ is one dimensional. $\square$

Schur's lemma has immediate structural consequences for the group itself.

Corollary 40.3.3

Let G be a finite abelian group. Then every irreducible complex representation of G has degree 1.

Proof. Let V be irreducible and $h \in G$. Then $\rho(h) : V \rightarrow V$ is G -linear, since $\rho(h)\rho(g) = \rho(hg) = \rho(gh) = \rho(g)\rho(h)$. So $\rho(h) = \lambda_h \text{id}_V$ by Schur. But then *every* subspace of V is G -invariant, so irreducibility forces $\dim V = 1$. $\square$

This is our earlier result again, but the proof is better, and it generalises.

Proposition 40.3.4

Let $\rho : G \rightarrow \text{GL}(V)$ be an irreducible complex representation. Then $\rho(z)$ is a scalar for every $z \in Z(G)$. In particular, if ρ is faithful then $Z(G)$ is cyclic.

Proof. The first statement is the argument above applied to $z \in Z(G)$. For the second, ρ restricted to $Z(G)$ lands in the scalars $\mathbb{C}^\times$, so $\rho|_{Z(G)} : Z(G) \rightarrow \mathbb{C}^\times$ is an injective homomorphism (using faithfulness), realising $Z(G)$ as a finite subgroup of $\mathbb{C}^\times$. Every finite subgroup of $\mathbb{C}^\times$ is cyclic (it consists of roots of unity), so $Z(G)$ is cyclic. $\square$

Corollary 40.3.5

The group $G = C_2 \times C_2 \times \cdots$ – or indeed any finite group with non-cyclic centre, such as $D_8 \times C_2$ – has no faithful irreducible complex representation.

Here is a nice application which shows the theory has teeth even at this early stage.

Example 40.3.6

The quaternion group Q_8 has centre $\{\pm 1\} \cong C_2$, which is cyclic, and indeed it does have a faithful irreducible representation – the two-dimensional one we shall meet later. By contrast the Klein four-group $C_2 \times C_2$ is its own centre, which is not cyclic, so the proposition says it has *no* faithful irreducible representation. And indeed all four of its irreducibles have degree 1, and each of them kills a subgroup of order 2.

So already, before we have computed a single character, the theory rules out certain shapes of representation on purely group-theoretic grounds.

§40.3.1 Isotypic Decomposition

Maschke says $V = V_1 \oplus \cdots \oplus V_k$ with V_i irreducible. But is this decomposition unique? Not literally: if V is the trivial representation of any group on $\mathbb{C}^2$, then *every* pair of distinct lines gives a decomposition. What is unique is the multiset of isomorphism classes, and there is a canonical coarser decomposition which is genuinely unique.

Proposition 40.3.7

Let V be a complex representation of the finite group G and let S be an irreducible

representation. If

$$V \cong \bigoplus_{i=1}^k V_i$$

with each V_i irreducible, then the number of i with $V_i \cong S$ equals $\dim \operatorname{Hom}_G(S, V)$, and in particular does not depend on the chosen decomposition.

Proof. The functor $\operatorname{Hom}_G(S, -)$ takes finite direct sums to direct sums: a map into $\bigoplus V_i$ is the same as a tuple of maps into the V_i . So

$$\operatorname{Hom}_G(S, V) \cong \bigoplus_{i=1}^k \operatorname{Hom}_G(S, V_i),$$

which by Schur's lemma has dimension equal to $\#\{i : V_i \cong S\}$. $\square$

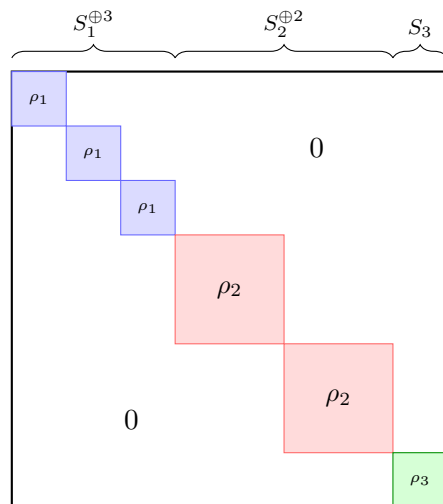
Definition 40.3.8 (Multiplicity)

The number above is the **multiplicity** of S in V . We write $V \cong \bigoplus_S S^{\oplus n_S}$ where n_S is the multiplicity, the sum over isomorphism classes of irreducibles.

Definition 40.3.9 (Isotypic component)

The **S -isotypic component** of V is the sum of all subrepresentations of V isomorphic to S . It is a subrepresentation isomorphic to $S^{\oplus n_S}$, and unlike the individual summands it is canonically determined by V and S .

Picking bases adapted to an isotypic decomposition puts every $\rho(g)$ into the following shape, with the *same* block repeated within each isotypic component.



We can also read off, from Schur's lemma, exactly how big the space of self-maps is.

Proposition 40.3.10

If $V \cong \bigoplus_S S^{\oplus n_S}$ then $\dim \operatorname{End}_G(V) = \sum_S n_S^2$. In particular V is irreducible if and

only if $\dim \text{End}_G(V) = 1$.

Proof. Expand $\text{Hom}_G(\bigoplus_S S^{\oplus n_S}, \bigoplus_T T^{\oplus n_T})$ into a double sum and apply Schur; this gives $\sum_S n_S^2$. If the answer is 1 then exactly one n_S is 1 and the rest vanish, so $V \cong S$ is irreducible; the converse is Schur. $\square$

This is a genuinely useful irreducibility criterion, and it is the shadow of the character-theoretic criterion $\langle \chi, \chi \rangle = 1$ that we will meet shortly.

Finally, let us record what Schur says about the regular representation, since it foreshadows the main counting theorem.

Proposition 40.3.11

Let V be a representation of the finite group G and let S be irreducible. Then the multiplicity of S in the regular representation $\mathbb{C}G$ is $\dim S$.

Proof. By the multiplicity formula it suffices to show $\dim \text{Hom}_G(\mathbb{C}G, S) = \dim S$. Now a $\mathbb{C}G$ -module homomorphism $\varphi : \mathbb{C}G \rightarrow S$ is determined by $\varphi(e) \in S$, since $\varphi(g) = \varphi(g \cdot e) = g\varphi(e)$ and the g span $\mathbb{C}G$; and conversely any choice of $\varphi(e) = s \in S$ gives a well defined homomorphism $\sum \lambda_g g \mapsto \sum \lambda_g gs$. So $\text{Hom}_G(\mathbb{C}G, S) \cong S$ as vector spaces. $\square$

Corollary 40.3.12

If $S_1, \dots, S_r$ are the distinct irreducible complex representations of G , with $d_i = \dim S_i$, then

$$\mathbb{C}G \cong \bigoplus_{i=1}^r S_i^{\oplus d_i}, \quad \text{and hence} \quad |G| = \sum_{i=1}^r d_i^2.$$

Proof. The first statement is the proposition, once we know that every irreducible appears – which it does, with multiplicity $d_i \geq 1$. Comparing dimensions gives $|G| = \sum_i d_i \cdot \dim S_i = \sum d_i^2$. $\square$

We have implicitly used that there are only finitely many irreducibles; this follows since each appears in $\mathbb{C}G$, which is finite dimensional. This identity $|G| = \sum d_i^2$ is one of the two great numerical constraints on the character table – the other being that r is the number of conjugacy classes. It is remarkably effective: for example, if $|G| = 8$ then the only possibilities for the multiset $\{d_i\}$ with all $d_i \geq 1$ and at least one $d_i = 1$ (the trivial representation) are $\{1, 1, 1, 1, 1, 1, 1, 1\}$ and $\{1, 1, 1, 1, 2\}$.

§40.4 Character Theory

We now come to the heart of the course. The idea is almost absurdly reductive: given a representation $\rho : G \rightarrow \text{GL}(V)$, throw away everything about $\rho(g)$ except its trace. What is left is a function $G \rightarrow \mathbb{C}$, and it turns out to remember the representation completely.

Why should the trace be the right thing to keep? Because it is the simplest invariant of

a linear map which does not depend on a choice of basis, and isomorphic representations differ exactly by a choice of basis.

§40.4.1 Characters

Definition 40.4.1 (Character)

Let (ρ, V) be a representation of G over $\mathbb{C}$. The **character** of ρ is the function

$$\chi_V = \chi_\rho : G \rightarrow \mathbb{C}, \quad \chi_\rho(g) = \operatorname{tr} \rho(g),$$

where the trace is that of the matrix of $\rho(g)$ in any basis. The **degree** of χ is $\chi(e) = \dim V$.

Definition 40.4.2 (Irreducible and linear characters)

A character is **irreducible** if it is the character of an irreducible representation. It is **linear** if it has degree 1; equivalently, if $\rho : G \rightarrow \mathbb{C}^\times$ is a homomorphism.

Since we will spend the rest of the course computing with characters, we had better establish their basic properties. Every single one of the following is used constantly.

Proposition 40.4.3 (Basic properties)

Let (ρ, V) be a complex representation of a finite group G , with character χ . Then:

- (i) $\chi(e) = \dim V$.
- (ii) χ is a **class function**: $\chi(hgh^{-1}) = \chi(g)$ for all $g, h \in G$.
- (iii) Isomorphic representations have equal characters.
- (iv) For each g , $\chi(g)$ is a sum of $\dim V$ roots of unity; specifically, if g has order n then $\chi(g)$ is a sum of n -th roots of unity.
- (v) $\chi(g^{-1}) = \overline{\chi(g)}$.
- (vi) $|\chi(g)| \leq \chi(e)$, with equality if and only if $\rho(g)$ is a scalar.
- (vii) $\chi(g) = \chi(e)$ if and only if $g \in \ker \rho$.

Proof. (i) $\rho(e) = \operatorname{id}_V$, whose trace is $\dim V$.

(ii) $\operatorname{tr}(\rho(h)\rho(g)\rho(h)^{-1}) = \operatorname{tr} \rho(g)$, since trace is invariant under conjugation.

(iii) If φ is an isomorphism with $\rho'(g) = \varphi\rho(g)\varphi^{-1}$, take traces.

(iv) The element $\rho(g)$ satisfies $\rho(g)^n = \operatorname{id}$, so it is diagonalisable with eigenvalues $\lambda_1, \dots, \lambda_d$ ($d = \dim V$) satisfying $\lambda_i^n = 1$, and $\chi(g) = \sum \lambda_i$.

(v) With notation as in (iv), $\rho(g^{-1}) = \rho(g)^{-1}$ has eigenvalues $\lambda_i^{-1} = \overline{\lambda_i}$ (as $|\lambda_i| = 1$). So $\chi(g^{-1}) = \sum \overline{\lambda_i} = \overline{\chi(g)}$.

(vi) By the triangle inequality $|\chi(g)| = |\sum \lambda_i| \leq \sum |\lambda_i| = d = \chi(e)$. Equality in the triangle inequality for complex numbers of modulus 1 forces all λ_i to be equal, say to λ ; and then $\rho(g) = \lambda \operatorname{id}_V$ since $\rho(g)$ is diagonalisable.

(vii) If $g \in \ker \rho$ then $\rho(g) = \text{id}$ so $\chi(g) = d$. Conversely if $\chi(g) = d$ then equality holds in (vi), so $\rho(g) = \lambda \text{id}$ with $d\lambda = d$, giving $\lambda = 1$. $\square$

Property (vii) is the reason non-faithful representations are useful: characters detect kernels, so from the character table we can read off all the normal subgroups of G . We will come back to this.

Proposition 40.4.4

If V, W are representations of G then $\chi_{V \oplus W} = \chi_V + \chi_W$.

Proof. In a basis adapted to the decomposition, $\rho(g)$ is block diagonal with blocks $\rho_V(g)$ and $\rho_W(g)$, and the trace of a block diagonal matrix is the sum of the traces of its blocks. $\square$

So characters of representations are exactly the nonnegative integer combinations of irreducible characters. Such things are sometimes called *proper* characters, or just characters; $\mathbb{Z}$ -linear combinations of irreducible characters are called **virtual characters** or **generalised characters**.

Example 40.4.5 (The regular character)

Let π_{reg} be the character of the regular representation $\mathbb{C}G$. In the basis $\{e_g\}$, the matrix of $\rho(h)$ is the permutation matrix of left multiplication by h , and its trace counts the g with $hg = g$. So

$$\pi_{\text{reg}}(g) = \begin{cases} |G| & g = e, \\ 0 & g \neq e. \end{cases}$$

Example 40.4.6 (Permutation characters)

More generally, if G acts on a finite set X and $\mathbb{C}X$ is the associated permutation representation, then the matrix of $\rho(g)$ has (x, x) entry 1 exactly when $g \cdot x = x$, and so

$$\pi_X(g) = |\text{fix}_X(g)| = |\{x \in X : g \cdot x = x\}|.$$

This is an extremely convenient supply of characters, since counting fixed points is easy.

Example 40.4.7

Take $G = S_4$ acting on $X = \{1, 2, 3, 4\}$. Then π_X takes the values 4, 2, 0, 1 and 0 on the classes of e , (12) , $(12)(34)$, (123) and (1234) respectively.

§40.4.2 Class Functions and the Inner Product

Definition 40.4.8 (Class function)

A **class function** on G is a function $f : G \rightarrow \mathbb{C}$ which is constant on conjugacy classes. The set of class functions is written $\mathcal{C}(G)$.

$\mathcal{C}(G)$ is a $\mathbb{C}$ -vector space under pointwise operations, and it has an obvious basis: the indicator functions $\delta_{\mathcal{K}}$ of the conjugacy classes $\mathcal{K}$. So

$$\dim_{\mathbb{C}} \mathcal{C}(G) = \#\{\text{conjugacy classes of } G\} =: r.$$

Characters are class functions, so all the characters of G live in this one r -dimensional space – already a strong constraint, and the source of everything that follows.

We make $\mathcal{C}(G)$ into an inner product space in the natural way.

Definition 40.4.9 (Inner product of class functions)

For $f, f' \in \mathcal{C}(G)$ define

$$\langle f, f' \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} f'(g).$$

This is a Hermitian inner product (linear in the second slot, conjugate-linear in the first, and positive definite since $\langle f, f \rangle = \frac{1}{|G|} \sum |f(g)|^2$). If we group the sum by conjugacy classes with representatives $g_1, \dots, g_r$, it becomes

$$\langle f, f' \rangle = \sum_{j=1}^r \frac{|\text{ccl}_G(g_j)|}{|G|} \overline{f(g_j)} f'(g_j) = \sum_{j=1}^r \frac{\overline{f(g_j)} f'(g_j)}{|\text{C}_G(g_j)|},$$

using $|\text{ccl}_G(g)| \cdot |\text{C}_G(g)| = |G|$. This is the form we use in practice.

Remark. If f, f' are *characters*, then $\overline{f(g)} = f(g^{-1})$, so

$$\langle f, f' \rangle = \frac{1}{|G|} \sum_{g \in G} f(g^{-1}) f'(g),$$

which is visibly symmetric in f and f' . So on characters the inner product is symmetric and we need not fret about which slot is conjugated.

§40.4.3 The Orthogonality Relations

We now prove the central theorem of the course. The strategy is to identify $\langle \chi_V, \chi_W \rangle$ with $\dim \text{Hom}_G(V, W)$, and then apply Schur's lemma. Two lemmas do the work.

Lemma 40.4.10 (Fixed points)

Let V be a representation of G with character χ , and let

$$V^G = \{v \in V \mid gv = v \text{ for all } g \in G\}$$

be the subspace of G -invariant vectors. Then

$$\dim V^G = \frac{1}{|G|} \sum_{g \in G} \chi(g).$$

Proof. Consider

$$\pi = \frac{1}{|G|} \sum_{g \in G} \rho(g) \in \text{End}(V).$$

First, $\text{img } \pi \subseteq V^G$: for $h \in G$,

$$\rho(h)\pi = \frac{1}{|G|} \sum_g \rho(hg) = \frac{1}{|G|} \sum_{g'} \rho(g') = \pi,$$

reindexing. Second, $\pi|_{V^G} = \text{id}$: if $v \in V^G$ then every $\rho(g)v = v$, and we average $|G|$ copies of v .

Together these give $\pi^2 = \pi$, so π is a projection with image V^G . A projection is diagonalisable with eigenvalues 0 and 1, and the multiplicity of the eigenvalue 1 is $\dim \text{img } \pi$. Hence

$$\dim V^G = \text{tr } \pi = \frac{1}{|G|} \sum_g \text{tr } \rho(g) = \frac{1}{|G|} \sum_g \chi(g). \quad \square$$

Lemma 40.4.11 (The Hom representation)

Let V, W be representations of G . Then $\text{Hom}_{\mathbb{C}}(V, W)$ becomes a representation of G via

$$(g \cdot \varphi)(v) = g \varphi(g^{-1}v),$$

and

- (i) $\text{Hom}_{\mathbb{C}}(V, W)^G = \text{Hom}_G(V, W)$;
- (ii) its character is $\chi_{\text{Hom}(V, W)}(g) = \overline{\chi_V(g)} \chi_W(g)$.

Proof. That this is a representation is a direct check: $(e \cdot \varphi) = \varphi$, and

$$(g \cdot (h \cdot \varphi))(v) = g(h \cdot \varphi)(g^{-1}v) = gh \varphi(h^{-1}g^{-1}v) = ((gh) \cdot \varphi)(v).$$

(i) We have $g \cdot \varphi = \varphi$ for all g if and only if $g\varphi(g^{-1}v) = \varphi(v)$ for all g, v , if and only if $\varphi(g^{-1}v) = g^{-1}\varphi(v)$ for all g, v , which is exactly G -linearity.

(ii) Fix $g \in G$. Both $\rho_V(g)$ and $\rho_W(g)$ are diagonalisable, so choose eigenbases $v_1, \dots, v_n$ of V and $w_1, \dots, w_m$ of W with $gv_i = \lambda_i v_i$ and $gw_j = \mu_j w_j$, where all λ_i, μ_j are roots of unity. Let $\varphi_{ij} \in \text{Hom}(V, W)$ be defined by $\varphi_{ij}(v_k) = \delta_{ik} w_j$; these nm maps form a basis. Then

$$(g \cdot \varphi_{ij})(v_k) = g \varphi_{ij}(\lambda_k^{-1} v_k) = \lambda_k^{-1} \delta_{ik} g w_j = \lambda_i^{-1} \mu_j \varphi_{ij}(v_k),$$

so φ_{ij} is an eigenvector with eigenvalue $\lambda_i^{-1} \mu_j = \overline{\lambda_i} \mu_j$. Summing,

$$\chi_{\text{Hom}(V, W)}(g) = \sum_{i,j} \overline{\lambda_i} \mu_j = \left(\sum_i \overline{\lambda_i} \right) \left(\sum_j \mu_j \right) = \overline{\chi_V(g)} \chi_W(g). \quad \square$$

Putting these together gives the key identity.

Theorem 40.4.12

For any representations V, W of the finite group G ,

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}} \operatorname{Hom}_G(V, W).$$

Proof. Apply the fixed point lemma to the representation $\operatorname{Hom}_{\mathbb{C}}(V, W)$:

$$\dim \operatorname{Hom}_G(V, W) = \dim \operatorname{Hom}_{\mathbb{C}}(V, W)^G = \frac{1}{|G|} \sum_{g \in G} \chi_{\operatorname{Hom}(V, W)}(g),$$

and the last expression is $\frac{1}{|G|} \sum_g \overline{\chi_V(g)} \chi_W(g) = \langle \chi_V, \chi_W \rangle$. $\square$

Theorem 40.4.13 (Row orthogonality)

Let $\chi_1, \dots, \chi_r$ be the characters of the distinct (i.e. pairwise non-isomorphic) irreducible complex representations $S_1, \dots, S_r$ of the finite group G . Then

$$\langle \chi_i, \chi_j \rangle = \delta_{ij}.$$

That is, the irreducible characters are orthonormal in $\mathcal{C}(G)$.

Proof. Immediate from the previous theorem and Schur's lemma, which says that $\dim \operatorname{Hom}_G(S_i, S_j)$ is 1 if $i = j$, and is 0 otherwise. $\square$

That is the whole proof, and it is worth appreciating how little it needed: averaging (Maschke's idea), a trace computation, and Schur. Written out in coordinates the statement is

$$\frac{1}{|G|} \sum_{g \in G} \overline{\chi_i(g)} \chi_j(g) = \sum_{k=1}^r \frac{\overline{\chi_i(g_k)} \chi_j(g_k)}{|C_G(g_k)|} = \delta_{ij},$$

a system of $\binom{r+1}{2}$ equations relating the entries of the character table. In practice these equations are so restrictive that they often determine the entire table.

Remark. There is a more computational proof, which is the one Frobenius gave. One takes an arbitrary linear map $\psi : S_i \rightarrow S_j$, averages it to $\tilde{\psi} = \frac{1}{|G|} \sum_g \rho_j(g) \psi \rho_i(g)^{-1}$, applies Schur to $\tilde{\psi}$, and then reads off matrix entries. It gives the same relations, with more index-pushing and less insight; the reader may enjoy carrying it out.

§40.4.4 Consequences of Orthogonality

The corollaries come thick and fast. Throughout, $\chi_1, \dots, \chi_r$ are the irreducible characters of G .

Corollary 40.4.14

The irreducible characters are linearly independent in $\mathcal{C}(G)$. In particular $r \leq \#\{\text{conjugacy classes}\}$, so G has only finitely many irreducible representations up to isomorphism.

Proof. Orthonormal sets are linearly independent, and $\dim \mathcal{C}(G)$ is the number of conjugacy classes. $\square$

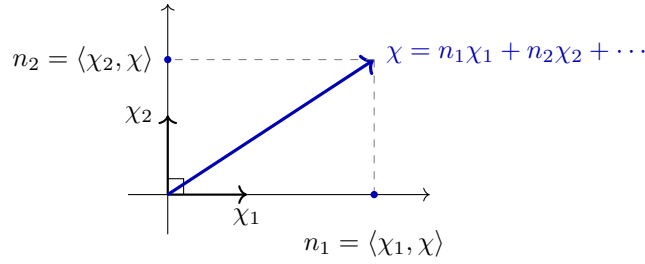
Corollary 40.4.15 (Decomposition)

Let V be a representation with character χ , and write $V \cong \bigoplus_i S_i^{\oplus n_i}$. Then

$$\chi = \sum_i n_i \chi_i, \quad n_i = \langle \chi_i, \chi \rangle = \langle \chi, \chi_i \rangle.$$

Proof. Characters add over direct sums, giving the first formula; then take the inner product with χ_i and use orthonormality. (The two expressions for n_i agree because n_i is real.) $\square$

It is worth taking this geometrically, because that is really all that is going on. The space $\mathcal{C}(G)$ is an r -dimensional inner product space, and the irreducible characters are an orthonormal basis of it. A character χ is then just a vector, its multiplicities are its coordinates, and the coordinates are extracted – as always in an inner product space – by taking inner products with the basis vectors.



$$\langle \chi, \chi \rangle = \sum_i n_i^2, \text{ and } \chi \text{ is irreducible exactly when this is } 1$$

Corollary 40.4.16 (Characters determine representations)

Two complex representations of a finite group are isomorphic if and only if they have the same character.

Proof. One direction we knew. Conversely, if $\chi_V = \chi_W$ then the multiplicities $n_i = \langle \chi_i, \chi_V \rangle = \langle \chi_i, \chi_W \rangle$ agree for every i , and a representation is determined up to isomorphism by its multiplicities. $\square$

This is a genuinely surprising statement. A representation of G of degree d is a huge amount of data – $|G|$ matrices of size $d \times d$ – and it is all encoded in r complex numbers.

Corollary 40.4.17 (Irreducibility criterion)

A representation V with character χ is irreducible if and only if $\langle \chi, \chi \rangle = 1$.

Proof. Writing $\chi = \sum n_i \chi_i$, orthonormality gives $\langle \chi, \chi \rangle = \sum_i n_i^2$. Since the n_i are nonnegative integers, this equals 1 if and only if exactly one n_i is 1 and the rest are 0, i.e. $V \cong S_i$ is irreducible. $\square$

This is the workhorse of the whole subject: to check irreducibility we compute a single number.

Example 40.4.18 (The standard representation of S_n is irreducible)

Let π be the permutation character of S_n on $\{1, \dots, n\}$, so $\pi(\sigma) = |\text{fix}(\sigma)|$. By Burnside's lemma (orbit counting),

$$\langle \pi, \mathbf{1} \rangle = \frac{1}{|G|} \sum_{\sigma} |\text{fix}(\sigma)| = \#\{\text{orbits}\} = 1,$$

and

$$\langle \pi, \pi \rangle = \frac{1}{|G|} \sum_{\sigma} |\text{fix}(\sigma)|^2 = \#\{\text{orbits of } S_n \text{ on } \{1, \dots, n\}^2\} = 2$$

for $n \geq 2$ (the diagonal, and its complement). Since $\pi = \mathbf{1} + \chi$ where χ is the character of the standard representation, we get

$$2 = \langle \pi, \pi \rangle = \langle \mathbf{1} + \chi, \mathbf{1} + \chi \rangle = 1 + 2\langle \mathbf{1}, \chi \rangle + \langle \chi, \chi \rangle = 1 + \langle \chi, \chi \rangle,$$

using $\langle \mathbf{1}, \chi \rangle = \langle \mathbf{1}, \pi \rangle - 1 = 0$. So $\langle \chi, \chi \rangle = 1$ and the standard representation is irreducible, of degree $n - 1$.

Remark. The computation above works verbatim for any **2-transitive** action of a group G on a set X : ‘2-transitive’ means G acts transitively on ordered pairs of distinct elements, which is exactly the statement that there are two orbits on $X \times X$. So for any 2-transitive action, $\pi_X - \mathbf{1}$ is an irreducible character of degree $|X| - 1$. This gives irreducible characters for free for S_n , A_n ($n \geq 4$), $\text{PGL}_2(\mathbb{F}_q)$ acting on the projective line, and many others.

§40.4.5 Counting the Irreducibles

We know $r \leq \#\{\text{conjugacy classes}\}$. We now show equality, which is the second great numerical constraint.

Theorem 40.4.19

The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal *basis* of $\mathcal{C}(G)$. Consequently, the number of irreducible complex representations of G equals the number of conjugacy classes of G .

Proof. We already know they are orthonormal, so it suffices to show that the only class function orthogonal to all of them is 0.

So let $f \in \mathcal{C}(G)$ with $\langle f, \chi_i \rangle = 0$ for all i . For any representation (ρ, V) define

$$\varphi_f = \sum_{g \in G} \overline{f(g)} \rho(g) \in \text{End}(V).$$

We claim φ_f is G -linear. Indeed, for $h \in G$,

$$\varphi_f \rho(h) = \sum_g \overline{f(g)} \rho(gh) = \sum_k \overline{f(kh^{-1})} \rho(k),$$

$$\rho(h) \varphi_f = \sum_g \overline{f(g)} \rho(hg) = \sum_k \overline{f(h^{-1}k)} \rho(k),$$

substituting $k = gh$ and $k = hg$ respectively. But f is a class function, so

$$f(kh^{-1}) = f(h^{-1}(kh^{-1})h) = f(h^{-1}k),$$

and the two expressions agree.

Now take $V = S_i$ irreducible, of degree d_i . By Schur, $\varphi_f = \lambda \text{id}$ for some $\lambda \in \mathbb{C}$, and taking traces,

$$d_i \lambda = \text{tr } \varphi_f = \sum_g \overline{f(g)} \chi_i(g) = |G| \langle f, \chi_i \rangle = 0,$$

so $\lambda = 0$ and $\varphi_f = 0$ on every irreducible representation. By complete reducibility, $\varphi_f = 0$ on *every* representation of G .

Apply this to the regular representation $V = \mathbb{C}G$, and evaluate at the basis vector e :

$$0 = \varphi_f(e) = \sum_{g \in G} \overline{f(g)} g.$$

The elements $g \in G$ are a basis of $\mathbb{C}G$, so $\overline{f(g)} = 0$ for every g , i.e. $f = 0$. $\square$

Corollary 40.4.20

A finite group G is abelian if and only if all its irreducible complex representations have degree 1.

Proof. We proved ‘only if’ earlier. Conversely, if all $d_i = 1$ then $|G| = \sum_i d_i^2 = r$, so G has $|G|$ conjugacy classes, meaning every class is a singleton, meaning every element is central. $\square$

§40.4.6 The Character Table

We can now assemble everything into the object which gives the subject its computational character.

Definition 40.4.21 (Character table)

Let G have conjugacy classes with representatives $g_1 = e, g_2, \dots, g_r$ and irreducible characters $\chi_1 = \mathbf{1}, \chi_2, \dots, \chi_r$. The **character table** of G is the $r \times r$ matrix

$$X = (\chi_i(g_j))_{1 \leq i, j \leq r}.$$

By convention we list the trivial character first and the identity class first, and we record the class sizes $|\text{ccl}_G(g_j)|$ (or the centraliser orders) alongside.

It is a square matrix – that is the content of the counting theorem – and it satisfies two orthogonality relations, one for rows and one for columns. Schematically:

	e	g_2	g_3	$\dots$	g_r	
χ_1						$\left. \begin{array}{l} \text{rows, weighted by } \text{ccl}(g_j) : \\ \langle \chi_i, \chi_k \rangle = \delta_{ik} \end{array} \right\}$
χ_2						
$\vdots$						
χ_r						

columns: $\sum_i \overline{\chi_i(g_j)} \chi_i(g_k) = \delta_{jk} |C_G(g_j)|$

Let us prove the column relation. It is a formal consequence of the row relation, by a small piece of matrix algebra.

Theorem 40.4.22 (Column orthogonality)

With notation as above, for any $1 \leq j, k \leq r$,

$$\sum_{i=1}^r \overline{\chi_i(g_j)} \chi_i(g_k) = \delta_{jk} |C_G(g_j)|.$$

In particular, taking $j = k = 1$ recovers $\sum_i d_i^2 = |G|$.

Proof. Let D be the diagonal matrix whose entries are $D_{jj} = |\text{ccl}_G(g_j)|/|G| = 1/|C_G(g_j)|$. Then row orthogonality says

$$\delta_{ik} = \langle \chi_k, \chi_i \rangle = \sum_j \frac{\overline{\chi_k(g_j)} \chi_i(g_j)}{|C_G(g_j)|} = (XD\overline{X}^T)_{ik},$$

so $XD\overline{X}^T = I$. Thus X is a square invertible matrix, with inverse $X^{-1} = D\overline{X}^T$, and therefore $X^{-1}X = I$ as well:

$$I = D\overline{X}^T X, \quad \text{i.e.} \quad \delta_{jk} = \frac{1}{|C_G(g_j)|} \sum_i \overline{\chi_i(g_j)} \chi_i(g_k),$$

which rearranges to the claim. □

Corollary 40.4.23

Taking $j = k$, for every $g \in G$ we have $\sum_i |\chi_i(g)|^2 = |C_G(g)|$.

Column orthogonality is enormously useful in practice: it lets us fill in a missing row or column of a character table once the rest is known, and it provides an independent check on every table we compute. Whenever the reader writes down a character table, they should verify:

- (i) the table is square, with $r = \text{number of conjugacy classes}$;
- (ii) $\sum_i d_i^2 = |G|$;
- (iii) each row has $\langle \chi_i, \chi_i \rangle = 1$ and distinct rows are orthogonal;

(iv) each column j satisfies $\sum_i |\chi_i(g_j)|^2 = |\mathbf{C}_G(g_j)| = |G|/|\text{ccl}(g_j)|$.

All of the tables in these notes have been checked against all four.

Corollary 40.4.24

Let $g \in G$ with $g \neq e$. Then $\sum_i d_i \chi_i(g) = 0$.

Proof. This is column orthogonality applied to the columns of e and of g , using $\chi_i(e) = d_i$; alternatively, it says $\pi_{\text{reg}}(g) = 0$, which we computed directly. $\square$

§40.4.7 First Character Tables

Let us put the machinery to work.

Example 40.4.25 (Cyclic groups)

Let $G = C_m = \langle g \rangle$. Every element is its own conjugacy class, so $r = m$, and all irreducibles have degree 1 since G is abelian. Indeed $\sum_{i=1}^m d_i^2 = m$ forces every $d_i = 1$. The characters are $\chi_j(g^k) = \omega^{jk}$ with $\omega = e^{2\pi i/m}$. For $m = 4$, with $i = \sqrt{-1}$:

	e	g	g^2	g^3
χ_0	1	1	1	1
χ_1	1	i	-1	$-i$
χ_2	1	-1	1	-1
χ_3	1	$-i$	-1	i

Every centraliser is all of G , of order 4, and indeed every column has $\sum_i |\chi_i(g^k)|^2 = 4$. Rows are orthogonal because $\sum_{k=0}^3 \omega^{(j-j')k} = 0$ for $j \neq j'$ – the character table of a cyclic group is (a scalar multiple of) the discrete Fourier transform matrix.

Example 40.4.26 (S_3)

The group S_3 has order 6 and three conjugacy classes, with representatives e , (12) and (123) , of sizes 1, 3, 2. So there are three irreducible characters with $d_1^2 + d_2^2 + d_3^2 = 6$; since $d_i \geq 1$ and at least one $d_i = 1$, the only solution is $\{1, 1, 2\}$.

We know two of the degree-1 characters already: the trivial character and the sign character. The degree-2 character is that of the standard representation, which we showed is irreducible; its character is $\pi - \mathbf{1}$ where π is the permutation character $\pi = (3, 1, 0)$. So $\chi_3 = (2, 0, -1)$. Alternatively, column orthogonality determines the last row up to a scalar of modulus one.

class size	1	3	2
	e	$(1\ 2)$	$(1\ 2\ 3)$
1	1	1	1
sign	1	-1	1
χ_3	2	0	-1

Let us verify. Degrees: $1 + 1 + 4 = 6 = |S_3|$. Rows: $\langle \chi_3, \chi_3 \rangle = \frac{1}{6}(1 \cdot 4 + 3 \cdot 0 + 2 \cdot 1) = \frac{6}{6} = 1$; $\langle \mathbf{1}, \chi_3 \rangle = \frac{1}{6}(2 + 0 - 2) = 0$; $\langle \mathbf{1}, \text{sign} \rangle = \frac{1}{6}(1 - 3 + 2) = 0$; $\langle \text{sign}, \chi_3 \rangle = \frac{1}{6}(2 - 0 - 2) = 0$. Columns: $1 + 1 + 4 = 6 = |C(e)|$; $1 + 1 + 0 = 2 = |C((1\ 2))| = 6/3$; $1 + 1 + 1 = 3 = |C((1\ 2\ 3))| = 6/2$. Everything checks out.

Example 40.4.27 (Recovering a row)

Suppose we know that a group G of order 12 has four conjugacy classes, of sizes 1, 3, 4, 4, and that three of its irreducible characters are $(1, 1, 1, 1)$, $(1, 1, \omega, \omega^2)$ and $(1, 1, \omega^2, \omega)$, where $\omega = e^{2\pi i/3}$. Then the degrees satisfy $1 + 1 + 1 + d_4^2 = 12$, so $d_4 = 3$; and column orthogonality against the first column gives, for the class of size 3,

$$1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 3\chi_4(g_2) = 0 \implies \chi_4(g_2) = -1,$$

and similarly $1 + \omega^2 + \omega + 3\chi_4(g_3) = 0$ gives $\chi_4(g_3) = 0$, and likewise $\chi_4(g_4) = 0$. The group, of course, is A_4 ; we will meet it again.

§40.5 Permutation Representations and Normal Subgroups

Before building more representations by algebraic means, let us extract everything we can from the ones we get for free – those coming from actions on sets – and see how the character table detects the normal subgroup structure of G .

§40.5.1 Permutation Characters

Recall that an action of G on a finite set X gives the permutation representation $\mathbb{C}X$, with character $\pi_X(g) = |\text{fix}_X(g)|$.

Proposition 40.5.1

Let G act on the finite set X , with permutation character π_X .

- (i) $\langle \pi_X, \mathbf{1} \rangle$ is the number of orbits of G on X .
- (ii) $\langle \pi_X, \pi_X \rangle$ is the number of orbits of G on $X \times X$.
- (iii) If $|X| \geq 2$ and G is transitive on X , then $\pi_X - \mathbf{1}$ is a character of degree $|X| - 1$; it is irreducible if and only if G is 2-transitive on X .

Proof. (i) This is Burnside's orbit-counting lemma: $\langle \pi_X, \mathbf{1} \rangle = \frac{1}{|G|} \sum_g |\text{fix}(g)|$, which is the number of orbits.

(ii) The permutation character of the action on $X \times X$ is $\pi_{X \times X}(g) = |\text{fix}_{X \times X}(g)| =$

$|\text{fix}_X(g)|^2 = \pi_X(g)^2$. Since π_X is real valued, $\langle \pi_X, \pi_X \rangle = \langle \pi_{X \times X}, \mathbf{1} \rangle$, and we apply (i).

(iii) By (i), transitivity gives $\langle \pi_X, \mathbf{1} \rangle = 1$, so $\mathbf{1}$ occurs exactly once in π_X and $\chi := \pi_X - \mathbf{1}$ is a character (of an actual representation, namely the complement of the fixed line $\langle \sum_x e_x \rangle$). Then

$$\langle \chi, \chi \rangle = \langle \pi_X, \pi_X \rangle - 2\langle \pi_X, \mathbf{1} \rangle + 1 = \langle \pi_X, \pi_X \rangle - 1,$$

which equals 1 exactly when G has two orbits on $X \times X$. Since the diagonal is always one orbit, and its complement is a single orbit precisely when G is 2-transitive, we are done. $\square$

Example 40.5.2 (S_n and A_n)

S_n is 2-transitive on $\{1, \dots, n\}$ for $n \geq 2$, and A_n is 2-transitive for $n \geq 4$. So both have an irreducible character of degree $n - 1$, the standard character.

Example 40.5.3 (Two-orbit tricks)

Let $G = S_4$ act on the set X of $\binom{4}{2} = 6$ unordered pairs from $\{1, 2, 3, 4\}$. Then $\pi_X = (6, 2, 2, 0, 0)$ on the classes $e, (1\ 2), (1\ 2)(3\ 4), (1\ 2\ 3), (1\ 2\ 3\ 4)$ – for instance $(1\ 2)$ fixes the pairs $\{1, 2\}$ and $\{3, 4\}$. Then

$$\langle \pi_X, \pi_X \rangle = \frac{1}{24}(36 + 6 \cdot 4 + 3 \cdot 4 + 0 + 0) = \frac{72}{24} = 3,$$

so π_X is a sum of three distinct irreducibles (or has some other decomposition with $\sum n_i^2 = 3$, which forces three distinct constituents). Since $\langle \pi_X, \mathbf{1} \rangle = \frac{1}{24}(6 + 12 + 6) = 1$, one of them is trivial. This is a good source of characters, and we will use it when building the table of S_4 .

§40.5.2 Lifting Characters

Non-faithful representations of G are really representations of quotients of G , and this is a two-way street.

Proposition 40.5.4 (Lifting)

Let $N \trianglelefteq G$ with quotient map $q : G \rightarrow G/N$. Then

$$\tilde{\rho} \mapsto \rho := \tilde{\rho} \circ q$$

is a bijection between representations of G/N and representations of ρ of G with $N \leq \ker \rho$. It preserves irreducibility, direct sums and isomorphism, and on characters it is $\chi(g) = \tilde{\chi}(gN)$.

Proof. If $\tilde{\rho} : G/N \rightarrow \text{GL}(V)$ is a representation then $\rho = \tilde{\rho} \circ q$ is a composite of homomorphisms, hence a representation of G , with $N \leq \ker \rho$. Conversely if ρ is a representation of G with $N \leq \ker \rho$ then ρ factors uniquely through q by the universal property of quotient groups. The two constructions are mutually inverse.

A subspace is $\rho(G)$ -invariant if and only if it is $\tilde{\rho}(G/N)$ -invariant, since $\rho(G) = \tilde{\rho}(G/N)$ as subgroups of $\text{GL}(V)$; so irreducibility is preserved. The character statement is immediate from $\rho(g) = \tilde{\rho}(gN)$. $\square$

$$\begin{array}{ccc}
 G & \xrightarrow{\rho} & \text{GL}(V) \\
 \downarrow q & \nearrow \tilde{\rho} & \\
 G/N & &
 \end{array}
 \quad (\text{exists iff } N \leq \ker \rho)$$

So the character table of G/N sits inside that of G : its rows are exactly those χ_i with $N \leq \ker \chi_i$, and its columns are the classes of G/N , which are the images of the classes of G .

Example 40.5.5 (Lifting from S_3)

S_4 has the normal subgroup $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$ with $S_4/V_4 \cong S_3$. Lifting the three irreducible characters of S_3 gives three irreducible characters of S_4 , of degrees 1, 1, 2: the trivial character, the sign character, and a degree-2 character taking the value 2 on e and on $(12)(34)$, the value 0 on transpositions and 4-cycles, and -1 on 3-cycles. This is exactly how we shall build the character table of S_4 .

More strikingly, the character table sees *all* the normal subgroups.

Proposition 40.5.6

Let χ be a character of G . Then $\ker \chi := \{g \in G : \chi(g) = \chi(e)\}$ is a normal subgroup of G , namely the kernel of any representation affording χ .

Proof. We proved earlier that $\chi(g) = \chi(e)$ if and only if $\rho(g) = \text{id}$. $\square$

Theorem 40.5.7

Every normal subgroup of G is an intersection of kernels of irreducible characters. Precisely, if $N \trianglelefteq G$ then

$$N = \bigcap \{\ker \chi_i : N \leq \ker \chi_i\}.$$

Proof. The inclusion $\subseteq$ is clear. For the reverse, note that the irreducible characters χ_i with $N \leq \ker \chi_i$ are exactly the lifts of the irreducible characters of G/N . Let g lie in the right hand side; then $\tilde{\chi}(gN) = \tilde{\chi}(N)$ for every irreducible character $\tilde{\chi}$ of G/N . Since the regular representation of G/N is faithful and decomposes into irreducibles containing every irreducible, the intersection of the kernels of all irreducible representations of any finite group is trivial. Hence $gN = N$, i.e. $g \in N$. $\square$

Corollary 40.5.8

A finite group G is simple if and only if for every non-trivial irreducible character

χ_i ($i \neq 1$) and every $g \neq e$, we have $\chi_i(g) \neq \chi_i(e)$.

Proof. If some $\chi_i \neq \mathbf{1}$ has $\chi_i(g) = \chi_i(e)$ with $g \neq e$, then $\ker \chi_i$ is a normal subgroup containing g , and it is proper since χ_i is not trivial (if $\ker \chi_i = G$ then ρ_i is trivial so $\chi_i = d_i \mathbf{1}$, forcing $d_i = 1$ and $\chi_i = \mathbf{1}$). Conversely, if N is a proper nontrivial normal subgroup then by the theorem some χ_i with $i \neq 1$ has $N \leq \ker \chi_i$, and any $g \in N \setminus \{e\}$ works. $\square$

This is a genuinely practical test, and we will use it to confirm that A_5 is simple purely by staring at a 5×5 table.

§40.5.3 Linear Characters and the Derived Subgroup

The degree-1 characters are the easiest to find, and they are exactly accounted for by the abelianisation.

Proposition 40.5.9

Let $G' = [G, G]$ be the derived subgroup. The linear characters of G are precisely the lifts of the (linear) characters of the abelian group G/G' , and there are exactly $|G/G'|$ of them. They form a group under pointwise multiplication, isomorphic to G/G' when G/G' is written multiplicatively (non-canonically).

Proof. A homomorphism $\rho : G \rightarrow \mathbb{C}^\times$ has abelian image, so $G' \leq \ker \rho$, so it is a lift from G/G' . Conversely all such lifts are linear characters. Since G/G' is abelian, all of its irreducible characters are linear and there are $|G/G'|$ of them.

Pointwise product of linear characters is a linear character, and the inverse of χ is $\bar{\chi}$; so they form a group, which is the character group of the finite abelian group G/G' , non-canonically isomorphic to it. $\square$

Example 40.5.10

S_n has $S'_n = A_n$ for $n \geq 2$, so S_n has exactly two linear characters: $\mathbf{1}$ and sign . A_5 is perfect ($A'_5 = A_5$), so its only linear character is the trivial one. And Q_8 has $Q'_8 = \{\pm 1\}$ with quotient $C_2 \times C_2$, so Q_8 has four linear characters.

§40.6 New Characters from Old

We have a good supply of representations coming from group actions, but to fill in a character table we usually need to manufacture more. There are three main constructions – duals, tensor products, and (later) induction – and each has a clean effect on characters.

§40.6.1 Dual Representations

Definition 40.6.1 (Dual representation)

Let (ρ, V) be a representation of G and $V^* = \text{Hom}_{\mathbb{C}}(V, \mathbb{C})$ its dual space. The **dual**

(or **contragredient**) representation is

$$\rho^*(g)(f) = f \circ \rho(g^{-1}), \quad \text{i.e.} \quad (g \cdot f)(v) = f(g^{-1}v).$$

The inverse is essential: without it we would get an antihomomorphism. Note this is the special case $W = \mathbb{C}$ (trivial) of the Hom representation from the proof of orthogonality.

Proposition 40.6.2

$$\chi_{V^*}(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)}.$$

Proof. Choose a basis $v_1, \dots, v_n$ of V of eigenvectors for $\rho(g)$, with $gv_i = \lambda_i v_i$, and let $v_1^*, \dots, v_n^*$ be the dual basis. Then $(g \cdot v_i^*)(v_j) = v_i^*(\lambda_j^{-1} v_j) = \lambda_j^{-1} \delta_{ij}$, so $g \cdot v_i^* = \lambda_i^{-1} v_i^*$. Hence $\chi_{V^*}(g) = \sum_i \lambda_i^{-1} = \chi_V(g^{-1})$, which equals $\overline{\chi_V(g)}$ as the λ_i are roots of unity. $\square$

Definition 40.6.3 (Self-dual)

A representation V is **self-dual** if $V \cong V^*$, equivalently if χ_V is real valued.

Proposition 40.6.4

If every element of G is conjugate to its inverse (a **real** group – for example, any S_n , or any D_{2n}), then every character of G is real valued and every representation is self-dual.

Proof. $\overline{\chi(g)} = \chi(g^{-1}) = \chi(g)$ since χ is a class function and g^{-1} is conjugate to g . $\square$

§40.6.2 Tensor Products

The tensor product is the construction that makes the set of characters into a ring, and it is by far the most productive way of building new irreducibles from known ones.

Recall from *Groups, Rings and Modules* the tensor product $V \otimes W$ of vector spaces: if $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ a basis of W , then $V \otimes W$ has basis $\{v_i \otimes w_j\}$, so $\dim(V \otimes W) = \dim V \dim W$, and $\otimes$ is bilinear. Any bilinear map $V \times W \rightarrow U$ factors uniquely through a linear map $V \otimes W \rightarrow U$.

Definition 40.6.5 (Tensor product of representations)

If (ρ, V) and (ρ', W) are representations of G , define $\rho \otimes \rho'$ on $V \otimes W$ by

$$g \cdot (v \otimes w) = (gv) \otimes (gw),$$

extended linearly.

One should check this is well defined: for fixed g , the map $(v, w) \mapsto gv \otimes gw$ is bilinear, so induces a linear map on $V \otimes W$; and $\rho \otimes \rho'(gh) = (\rho \otimes \rho')(g)(\rho \otimes \rho')(h)$ holds on the spanning set of pure tensors.

Proposition 40.6.6

$$\chi_{V \otimes W}(g) = \chi_V(g) \chi_W(g).$$

Proof. Fix g and choose eigenbases (v_i) for $\rho(g)$ with eigenvalues λ_i , and (w_j) for $\rho'(g)$ with eigenvalues μ_j . Then $g \cdot (v_i \otimes w_j) = \lambda_i \mu_j (v_i \otimes w_j)$, so the $v_i \otimes w_j$ form an eigenbasis and

$$\chi_{V \otimes W}(g) = \sum_{i,j} \lambda_i \mu_j = \left(\sum_i \lambda_i \right) \left(\sum_j \mu_j \right) = \chi_V(g) \chi_W(g). \quad \square$$

Remark. Note carefully: this is *not* the same as the representation of $G \times G$ on $V \otimes W$, which we will meet in a moment. Here both factors are acted on by the same g ; this is sometimes called the ‘inner’ tensor product.

Corollary 40.6.7

$\text{Hom}_{\mathbb{C}}(V, W) \cong V^* \otimes W$ as representations of G .

Proof. Both have character $\overline{\chi_V} \chi_W$, and characters determine representations. (One can of course write down the isomorphism $f \otimes w \mapsto (v \mapsto f(v)w)$ directly.) $\square$

Proposition 40.6.8

If χ is a character and λ is a *linear* character, then $\lambda\chi$ is a character, and χ is irreducible if and only if $\lambda\chi$ is.

Proof. $\lambda\chi$ is the character of $L \otimes V$ where L is the one-dimensional space affording λ . Since $|\lambda(g)| = 1$ for all g , we have $\langle \lambda\chi, \lambda\chi \rangle = \frac{1}{|G|} \sum_g |\lambda(g)|^2 |\chi(g)|^2 = \langle \chi, \chi \rangle$, so one is 1 if and only if the other is. $\square$

This is a cheap and very effective way to produce new irreducibles: tensoring the character table of S_n with sign permutes its rows.

The tensor product of two irreducibles is usually *not* irreducible, and decomposing it is a basic computation.

Example 40.6.9

In S_3 , with $\chi_3 = (2, 0, -1)$ the degree-2 character, we have $\chi_3^2 = (4, 0, 1)$. Then

$$\langle \chi_3^2, \mathbf{1} \rangle = \frac{1}{6}(4+0+2) = 1, \quad \langle \chi_3^2, \text{sign} \rangle = \frac{1}{6}(4-0+2) = 1, \quad \langle \chi_3^2, \chi_3 \rangle = \frac{1}{6}(8+0-2) = 1,$$

so $\chi_3 \otimes \chi_3 = \mathbf{1} + \text{sign} + \chi_3$, consistent with degrees $4 = 1 + 1 + 2$.

§40.6.3 Symmetric and Exterior Squares

The decomposition in the last example was not an accident of S_3 : for any V , the space $V \otimes V$ has a canonical decomposition into two pieces, and it is often the fastest way to find new irreducibles.

Let $\tau : V \otimes V \rightarrow V \otimes V$ be the linear map determined by $\tau(v \otimes w) = w \otimes v$ (well defined since $(v, w) \mapsto w \otimes v$ is bilinear). Then $\tau^2 = \text{id}$, and τ is G -linear since g acts diagonally.

Definition 40.6.10 (Symmetric and exterior squares)

Set

$$S^2V = \{x \in V \otimes V : \tau(x) = x\}, \quad \Lambda^2V = \{x \in V \otimes V : \tau(x) = -x\}.$$

These are the **symmetric square** and **exterior square** of V .

Proposition 40.6.11

$V \otimes V = S^2V \oplus \Lambda^2V$ as representations of G . If $v_1, \dots, v_n$ is a basis of V then S^2V has basis $\{v_i v_j := v_i \otimes v_j + v_j \otimes v_i : i \leq j\}$ and Λ^2V has basis $\{v_i \wedge v_j := v_i \otimes v_j - v_j \otimes v_i : i < j\}$, so

$$\dim S^2V = \binom{n+1}{2}, \quad \dim \Lambda^2V = \binom{n}{2}.$$

Proof. Since $\tau^2 = \text{id}$ and we are in characteristic 0, $V \otimes V$ decomposes into the ± 1 eigenspaces of τ via $x = \frac{1}{2}(x + \tau x) + \frac{1}{2}(x - \tau x)$. Both eigenspaces are G -invariant since τ is G -linear. The bases are immediate. $\square$

Proposition 40.6.12

If V has character χ , then

$$\chi_{S^2V}(g) = \frac{1}{2}(\chi(g)^2 + \chi(g^2)), \quad \chi_{\Lambda^2V}(g) = \frac{1}{2}(\chi(g)^2 - \chi(g^2)).$$

Proof. Fix g and take an eigenbasis $v_1, \dots, v_n$ with $gv_i = \lambda_i v_i$. Then $g(v_i v_j) = \lambda_i \lambda_j v_i v_j$ and $g(v_i \wedge v_j) = \lambda_i \lambda_j v_i \wedge v_j$. So

$$\chi_{S^2V}(g) = \sum_{i \leq j} \lambda_i \lambda_j, \quad \chi_{\Lambda^2V}(g) = \sum_{i < j} \lambda_i \lambda_j.$$

Now $\chi(g)^2 = (\sum_i \lambda_i)^2 = \sum_i \lambda_i^2 + 2 \sum_{i < j} \lambda_i \lambda_j$, while $\chi(g^2) = \sum_i \lambda_i^2$ since $g^2 v_i = \lambda_i^2 v_i$. Hence $\sum_{i < j} \lambda_i \lambda_j = \frac{1}{2}(\chi(g)^2 - \chi(g^2))$, and adding $\sum_i \lambda_i^2 = \chi(g^2)$ gives $\sum_{i \leq j} \lambda_i \lambda_j = \frac{1}{2}(\chi(g)^2 + \chi(g^2))$. $\square$

Remark. More generally one can form $S^n V$ and $\Lambda^n V$ for all n , and the reader who has met the tensor algebra will recognise these as the graded pieces of the symmetric and exterior algebras. For this course the squares are what we need, though Λ^n makes an appearance for $\text{SU}(2)$.

Example 40.6.13 (Finding a new irreducible of S_5)

Let $\chi = \pi - 1$ be the standard character of S_5 , of degree 4. We will see later that $\Lambda^2 \chi$ has degree $\binom{4}{2} = 6$ and is irreducible – a character we could not easily have

found by any other elementary means.

§40.6.4 Representations of a Product

Proposition 40.6.14

Let G, H be finite groups. If V is a representation of G and W of H , then $V \otimes W$ is a representation of $G \times H$ via $(g, h) \cdot (v \otimes w) = gv \otimes hw$, with character

$$\chi_{V \otimes W}(g, h) = \chi_V(g)\chi_W(h).$$

Moreover, if V and W are irreducible then so is $V \otimes W$, and every irreducible representation of $G \times H$ arises this way, uniquely.

Proof. The character formula follows exactly as before (diagonalise $\rho_V(g)$ and $\rho_W(h)$ separately). For irreducibility,

$$\langle \chi_V \chi_W, \chi_V \chi_W \rangle_{G \times H} = \frac{1}{|G||H|} \sum_{g, h} |\chi_V(g)|^2 |\chi_W(h)|^2 = \langle \chi_V, \chi_V \rangle_G \langle \chi_W, \chi_W \rangle_H = 1.$$

Distinct pairs give distinct (indeed orthogonal) characters by the same computation. Finally, the conjugacy classes of $G \times H$ are the products of a class of G with a class of H , so there are $r_G r_H$ of them; and we have produced $r_G r_H$ pairwise non-isomorphic irreducibles. Hence there are no others. $\square$

Example 40.6.15

Since $C_6 \cong C_2 \times C_3$, the six characters of C_6 are the products of the two characters of C_2 with the three of C_3 – which is exactly what one sees from $\omega_6 = \omega_2 \omega_3$.

§40.6.5 The Character Ring

We have seen that sums and products of characters are characters. It is convenient to complete this to a ring.

Definition 40.6.16 (Character ring)

The **character ring** (or **representation ring**) of G is

$$R(G) = \left\{ \sum_{i=1}^r n_i \chi_i : n_i \in \mathbb{Z} \right\} \subseteq \mathcal{C}(G),$$

the set of virtual characters, with pointwise addition and multiplication.

Proposition 40.6.17

$R(G)$ is a commutative ring with $1 = \mathbf{1}$, free as a $\mathbb{Z}$ -module with basis $\chi_1, \dots, \chi_r$; and $\chi \in R(G)$ is an irreducible character if and only if $\langle \chi, \chi \rangle = 1$ and $\chi(e) > 0$.

Proof. Closure under multiplication is because $\chi_i\chi_j$ is a genuine character, hence a nonnegative combination of the χ_k . Freeness is linear independence of the χ_i . For the last part, write $\chi = \sum n_i\chi_i$ with $n_i \in \mathbb{Z}$; then $\langle \chi, \chi \rangle = \sum n_i^2 = 1$ forces $\chi = \pm\chi_i$ for a single i , and $\chi(e) = \pm d_i > 0$ picks out the sign. $\square$

This last criterion is used constantly, especially with induced characters: one produces a virtual character by subtracting known constituents, checks its norm is 1, and checks the sign of its degree.

§40.7 Character Tables in Practice

We now have enough theory to compute. This section works out the character tables of a good number of small groups completely; the reader should treat them as templates, since almost every character table computation in this course is one of these arguments in disguise.

The general strategy is always the same.

- (1) Find the conjugacy classes and their sizes. This tells you how many irreducible characters to expect.
- (2) Find the linear characters, by computing G/G' .
- (3) Use $\sum d_i^2 = |G|$ to pin down the remaining degrees, often uniquely.
- (4) Manufacture characters: permutation characters, lifts from quotients, tensor products with linear characters, symmetric and exterior squares, and later induction.
- (5) Test irreducibility with $\langle \chi, \chi \rangle = 1$, and strip off known constituents when it fails.
- (6) If one row is still missing, get it from column orthogonality.
- (7) Check everything.

§40.7.1 Dihedral Groups

Let $G = D_{2n} = \langle r, s \mid r^n = s^2 = e, srs^{-1} = r^{-1} \rangle$, of order $2n$. Note $sr^k s^{-1} = r^{-k}$, so r^k and r^{-k} are always conjugate; and $r^k(sr^j)r^{-k} = sr^{j-2k}$, so the reflections sr^j fall into classes according to j modulo 2.

The case n odd, say $n = 2m + 1$. Then $\gcd(2, n) = 1$, so $j \mapsto j - 2k$ is transitive modulo n and all n reflections are conjugate. The classes are

$$\{e\}, \quad \{r^k, r^{-k}\} \ (1 \leq k \leq m), \quad \{\text{the } n \text{ reflections}\},$$

so there are $m + 2$ of them. Since $srs^{-1}r^{-1} = r^{-2}$ generates $\langle r \rangle$ (as n is odd), $G' = \langle r \rangle$ and $G/G' \cong C_2$: there are exactly two linear characters.

For $1 \leq j \leq m$ define $\rho_j : G \rightarrow \text{GL}_2(\mathbb{C})$ by

$$\rho_j(r) = \begin{pmatrix} \omega^j & 0 \\ 0 & \omega^{-j} \end{pmatrix}, \quad \rho_j(s) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \omega = e^{2\pi i/n}.$$

One checks the defining relations, so this is a representation, with character $\chi_j(r^k) = \omega^{jk} + \omega^{-jk} = 2 \cos(2\pi jk/n)$ and $\chi_j(sr^k) = 0$. It is irreducible:

$$\langle \chi_j, \chi_j \rangle = \frac{1}{2n} \sum_{k=0}^{n-1} |\omega^{jk} + \omega^{-jk}|^2 = \frac{1}{2n} \sum_{k=0}^{n-1} (2 + \omega^{2jk} + \omega^{-2jk}) = \frac{1}{2n} \cdot 2n = 1,$$

since $\sum_k \omega^{2jk} = 0$ when $n \nmid 2j$, which holds as n is odd and $1 \leq j \leq m$. The χ_j are pairwise distinct, and $1 + 1 + 4m = 4m + 2 = 2n$, so we have them all.

class size	1	2	n
	e	$r^k \quad (1 \leq k \leq m)$	s
1	1	1	1
ε	1	1	-1
$\chi_j \quad (1 \leq j \leq m)$	2	$2 \cos \frac{2\pi jk}{n}$	0

The case n even, say $n = 2m$. Now r^m is central, and the reflections split into two classes $\{sr^{\text{even}}\}$ and $\{sr^{\text{odd}}\}$, each of size m . The classes are

$$\{e\}, \quad \{r^m\}, \quad \{r^k, r^{-k}\} \quad (1 \leq k \leq m-1), \quad \{sr^{2i}\}, \quad \{sr^{2i+1}\},$$

giving $m + 3$ classes. Here $G' = \langle r^2 \rangle$ has index 4, so there are four linear characters, determined by $r \mapsto \pm 1$, $s \mapsto \pm 1$. The same ρ_j as before are irreducible for $1 \leq j \leq m-1$ (the computation is identical; $n \nmid 2j$ still holds), and $4 + 4(m-1) = 4m = 2n$, so again we have everything.

class size	1	1	2	m	m
	e	r^m	$r^k \quad (1 \leq k \leq m-1)$	s	sr
1	1	1	1	1	1
ε_1	1	1	1	-1	-1
ε_2	1	$(-1)^m$	$(-1)^k$	1	-1
ε_3	1	$(-1)^m$	$(-1)^k$	-1	1
$\chi_j \quad (1 \leq j \leq m-1)$	2	$2(-1)^j$	$2 \cos \frac{2\pi jk}{n}$	0	0

Geometrically the χ_j are exactly the characters of the rotation-reflection action of D_{2n} on the plane, but with the rotation sped up by a factor of j : ρ_j realises D_{2n} as the symmetries of an n -gon which has been wrapped j times around itself.

Example 40.7.1 (D_8)

Take $n = 4$, so $m = 2$. The classes are $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, r^2s\}$, $\{rs, r^3s\}$ of sizes 1, 1, 2, 2, 2. There is one two-dimensional character, χ_1 , with $\chi_1(e) = 2$, $\chi_1(r^2) = 2(-1)^1 = -2$, $\chi_1(r) = 2 \cos(\pi/2) = 0$, and 0 on both reflection classes.

class size	1	1	2	2	2
	e	r^2	r	s	rs
$\mathbf{1}$	1	1	1	1	1
ε_1	1	1	1	-1	-1
ε_2	1	1	-1	1	-1
ε_3	1	1	-1	-1	1
χ	2	-2	0	0	0

Checks: $1 + 1 + 1 + 1 + 4 = 8 = |D_8|$. $\langle \chi, \chi \rangle = \frac{1}{8}(4 + 4 + 0 + 0 + 0) = 1$. $\langle \mathbf{1}, \chi \rangle = \frac{1}{8}(2 - 2) = 0$, and similarly for the other linear characters. Columns: e gives $1 + 1 + 1 + 1 + 4 = 8 = |C(e)|$; r^2 gives $1 + 1 + 1 + 1 + 4 = 8$, correct since r^2 is central; r gives $1 + 1 + 1 + 1 + 0 = 4 = 8/2$; and likewise for s and rs .

Example 40.7.2 (The quaternion group Q_8)

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$. The classes are $\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}$, of sizes 1, 1, 2, 2, 2. We have $Q'_8 = \{\pm 1\}$ with $Q_8/Q'_8 \cong C_2 \times C_2$, giving four linear characters, and then $8 = 4 \cdot 1 + d^2$ forces one more of degree 2. That degree-2 representation is the obvious one: regard $\mathbb{H} = \mathbb{R}\langle 1, i, j, k \rangle$ inside $M_2(\mathbb{C})$ via

$$1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad i \mapsto \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad k \mapsto \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},$$

whose traces are 2, 0, 0, 0 on $1, i, j, k$ and -2 on -1.

class size	1	1	2	2	2
	1	-1	i	j	k
$\mathbf{1}$	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	1	-1	1	-1
χ_4	1	1	-1	-1	1
χ_5	2	-2	0	0	0

The same four checks go through exactly as for D_8 .

Remark (Character tables do not determine groups). Compare the last two tables: they are *identical*. But $D_8 \not\cong Q_8$ – for instance D_8 has five elements of order dividing 2 and Q_8 has two. So the character table, remarkable as it is, does not determine the group.

What distinguishes them is the *squaring* map, which the character table does not record. A convenient way to package this is the **Frobenius–Schur indicator**

$$\iota(\chi) = \frac{1}{|G|} \sum_{g \in G} \chi(g^2),$$

which turns out to be $+1$ if the representation can be written over $\mathbb{R}$, -1 if χ is real valued but the representation cannot, and 0 if χ is not real valued. For the degree-2 character χ of D_8 , every element squares to e except r, r^3 which square to r^2 , so

$$\iota(\chi) = \frac{1}{8}(1 \cdot 2 + 1 \cdot 2 + 2 \cdot (-2) + 2 \cdot 2 + 2 \cdot 2) = \frac{8}{8} = 1,$$

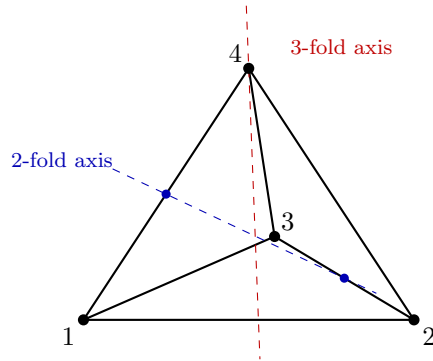
consistent with ρ being the honest rotation-reflection action on $\mathbb{R}^2$. For Q_8 , however, $(\pm i)^2 = (\pm j)^2 = (\pm k)^2 = -1$, so

$$\iota(\chi_5) = \frac{1}{8}(1 \cdot 2 + 1 \cdot 2 + 2 \cdot (-2) + 2 \cdot (-2) + 2 \cdot (-2)) = \frac{-8}{8} = -1,$$

and indeed the two-dimensional representation of Q_8 is quaternionic, not real.

§40.7.2 The Alternating Group A_4

Let $G = A_4$, of order 12, the rotation group of a regular tetrahedron.



The conjugacy classes of S_4 meeting A_4 are those of e , of $(12)(34)$, and of (123) ; the first two remain single A_4 -classes but the class of 3-cycles splits in two, because $C_{S_4}((123)) = \langle (123) \rangle \leq A_4$. So the classes of A_4 are

$$\{e\} \ (1), \quad \{(12)(34), (13)(24), (14)(23)\} \ (3), \quad \text{ccl}((123)) \ (4), \quad \text{ccl}((132)) \ (4),$$

four classes in all. Geometrically the two classes of 3-cycles are the rotations by $+2\pi/3$ and by $-2\pi/3$ about the four 3-fold axes.

Now $A'_4 = V_4$ (the double transpositions together with e), and $A_4/V_4 \cong C_3$. So there are three linear characters, the lifts of the characters of C_3 , and $12 = 1 + 1 + 1 + d_4^2$ gives $d_4 = 3$. The degree-3 character is the standard character $\pi - \mathbf{1}$ of the action on $\{1, 2, 3, 4\}$, with $\pi = (4, 0, 1, 1)$. Writing $\omega = e^{2\pi i/3}$:

class size	1	3	4	4
	e	$(12)(34)$	(123)	(132)
$\mathbf{1}$	1	1	1	1
χ_2	1	1	ω	ω^2
χ_3	1	1	ω^2	ω
χ_4	3	-1	0	0

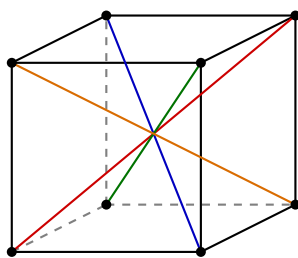
Checks: degrees $1 + 1 + 1 + 9 = 12$. Rows: $\langle \chi_4, \chi_4 \rangle = \frac{1}{12}(9 + 3 \cdot 1 + 0 + 0) = 1$; $\langle \mathbf{1}, \chi_4 \rangle = \frac{1}{12}(3 - 3) = 0$; $\langle \chi_2, \chi_2 \rangle = \frac{1}{12}(1 + 3 + 4 + 4) = 1$; $\langle \chi_2, \chi_3 \rangle = \frac{1}{12}(1 + 3 + 4\bar{\omega}\omega^2 +$

$4\overline{\omega^2\omega} = \frac{1}{12}(4 + 4\omega + 4\omega^2) = 0$ since $1 + \omega + \omega^2 = 0$; $\langle \mathbf{1}, \chi_2 \rangle = \frac{1}{12}(1 + 3 + 4\omega + 4\omega^2) = 0$. Columns: 12, then $1 + 1 + 1 + 1 = 4 = 12/3$, then $1 + 1 + 1 + 0 = 3 = 12/4$ twice. All correct.

Note the character table shows immediately that A_4 is not simple: $\ker \chi_2 = \{e\} \cup \text{ccl}((1\ 2)(3\ 4)) = V_4$ is a proper nontrivial normal subgroup.

§40.7.3 The Symmetric Group S_4

Let $G = S_4$, of order 24, which is the rotation group of the cube: the four space diagonals are permuted, and this gives an isomorphism onto S_4 .



the 4 space diagonals

The classes are indexed by cycle types: e (size 1), transpositions (size 6), double transpositions (size 3), 3-cycles (size 8), 4-cycles (size 6). Five classes, and $1 + 6 + 3 + 8 + 6 = 24$. Since $S'_4 = A_4$, there are two linear characters $\mathbf{1}$ and sign , and

$$24 = 1 + 1 + d_3^2 + d_4^2 + d_5^2$$

with $d_i \geq 2$; the only solution is $\{2, 3, 3\}$.

We get the two degree-3 characters at once: $\chi_4 = \pi - \mathbf{1}$ is the standard character (irreducible since S_4 is 2-transitive), and $\chi_5 = \text{sign} \cdot \chi_4$ is another, distinct from χ_4 since χ_4 is not zero on transpositions. Here $\pi = (4, 2, 0, 1, 0)$, so $\chi_4 = (3, 1, -1, 0, -1)$ and $\chi_5 = (3, -1, -1, 0, 1)$.

The degree-2 character is the lift of the degree-2 character of $S_4/V_4 \cong S_3$. Under the quotient map, e and the double transpositions go to the identity, transpositions and 4-cycles go to transpositions, and 3-cycles go to 3-cycles. Reading off from the table of S_3 , we get $\chi_3 = (2, 0, 2, -1, 0)$.

class size	1	6	3	8	6
	e	$(1\ 2)$	$(1\ 2)(3\ 4)$	$(1\ 2\ 3)$	$(1\ 2\ 3\ 4)$
$\mathbf{1}$	1	1	1	1	1
sign	1	-1	1	1	-1
χ_3	2	0	2	-1	0
χ_4	3	1	-1	0	-1
χ_5	3	-1	-1	0	1

Let us check this properly.

- Degrees: $1 + 1 + 4 + 9 + 9 = 24 = |S_4|$.

- $\langle \chi_3, \chi_3 \rangle = \frac{1}{24}(1 \cdot 4 + 6 \cdot 0 + 3 \cdot 4 + 8 \cdot 1 + 6 \cdot 0) = \frac{24}{24} = 1.$
- $\langle \chi_4, \chi_4 \rangle = \frac{1}{24}(9 + 6 + 3 + 0 + 6) = \frac{24}{24} = 1$, and by symmetry $\langle \chi_5, \chi_5 \rangle = 1.$
- $\langle \mathbf{1}, \text{sign} \rangle = \frac{1}{24}(1 - 6 + 3 + 8 - 6) = 0;$ $\langle \mathbf{1}, \chi_3 \rangle = \frac{1}{24}(2 + 0 + 6 - 8 + 0) = 0;$
 $\langle \mathbf{1}, \chi_4 \rangle = \frac{1}{24}(3 + 6 - 3 + 0 - 6) = 0;$ $\langle \mathbf{1}, \chi_5 \rangle = \frac{1}{24}(3 - 6 - 3 + 0 + 6) = 0.$
- $\langle \text{sign}, \chi_3 \rangle = \frac{1}{24}(2 - 0 + 6 - 8 - 0) = 0;$ $\langle \text{sign}, \chi_4 \rangle = \frac{1}{24}(3 - 6 - 3 + 0 + 6) = 0;$
 $\langle \text{sign}, \chi_5 \rangle = \frac{1}{24}(3 + 6 - 3 + 0 - 6) = 0.$
- $\langle \chi_3, \chi_4 \rangle = \frac{1}{24}(6 + 0 - 6 + 0 + 0) = 0;$ $\langle \chi_3, \chi_5 \rangle = \frac{1}{24}(6 - 0 - 6 + 0 + 0) = 0;$
 $\langle \chi_4, \chi_5 \rangle = \frac{1}{24}(9 - 6 + 3 + 0 - 6) = 0.$
- Columns: $e: 1 + 1 + 4 + 9 + 9 = 24.$ $(12): 1 + 1 + 0 + 1 + 1 = 4 = 24/6.$ $(12)(34):$
 $1 + 1 + 4 + 1 + 1 = 8 = 24/3.$ $(123): 1 + 1 + 1 + 0 + 0 = 3 = 24/8.$ $(1234):$
 $1 + 1 + 0 + 1 + 1 = 4 = 24/6.$

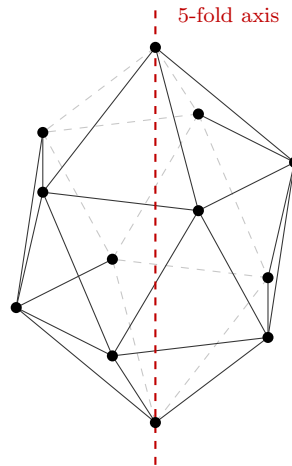
Every relation holds, so the table is correct.

Remark. As a sanity check on χ_3 : earlier we computed the permutation character of S_4 on the six unordered pairs to be $\pi_X = (6, 2, 2, 0, 0)$, and found $\langle \pi_X, \pi_X \rangle = 3$ with $\langle \pi_X, \mathbf{1} \rangle = 1$. Indeed

$$\pi_X = \mathbf{1} + \chi_3 + \chi_4 = (1, 1, 1, 1, 1) + (2, 0, 2, -1, 0) + (3, 1, -1, 0, -1) = (6, 2, 2, 0, 0). \checkmark$$

§40.7.4 The Alternating Group A_5

Now for the most interesting small example. A_5 has order 60 and is the rotation group of the regular icosahedron (equivalently, of the dodecahedron).



The conjugacy classes of A_5 are those of e , of $(12)(34)$, of (123) , and *two* classes of 5-cycles. (The class of 5-cycles in S_5 has 24 elements and splits, since $C_{S_5}((12345)) = \langle (12345) \rangle \leq A_5$; the classes of $(12)(34)$ and (123) do not split, as (12) and (45) respectively centralise a representative while lying outside A_5 .) Sizes:

$$1, \quad 15, \quad 20, \quad 12, \quad 12, \quad \text{summing to } 60.$$

Geometrically these are: the identity; the 15 rotations by π about the 15 edge-midpoint axes; the 20 rotations by $\pm 2\pi/3$ about the 10 face axes; and the 24 rotations by $\pm 2\pi/5$ and $\pm 4\pi/5$ about the 6 vertex axes, splitting $12 + 12$.

A_5 is perfect, so $\mathbf{1}$ is the only linear character, and $60 = 1 + d_2^2 + d_3^2 + d_4^2 + d_5^2$ with $d_i \geq 2$ and (as we shall prove later) each d_i dividing 60. Let us instead just build characters.

Degree 4. A_5 is 2-transitive on $\{1, \dots, 5\}$, so $\chi_4 := \pi - \mathbf{1}$ is irreducible of degree 4, where $\pi = (5, 1, 2, 0, 0)$. Thus $\chi_4 = (4, 0, 1, -1, -1)$.

Degree 5. Compute the symmetric square. We need $\chi_4(g^2)$: for $g = e$ or a double transposition, $g^2 = e$ so $\chi_4(g^2) = 4$; for a 3-cycle, g^2 is a 3-cycle so $\chi_4(g^2) = 1$; and for a 5-cycle g , g^2 is a 5-cycle in the *other* class, but χ_4 takes the value -1 on both, so $\chi_4(g^2) = -1$. Hence

$$\chi_{S^2}(g) = \frac{1}{2}(\chi_4(g)^2 + \chi_4(g^2)) = (10, 2, 1, 0, 0).$$

Now $\langle \chi_{S^2}, \mathbf{1} \rangle = \frac{1}{60}(10 + 15 \cdot 2 + 20 \cdot 1 + 0 + 0) = 1$ and $\langle \chi_{S^2}, \chi_4 \rangle = \frac{1}{60}(40 + 0 + 20 + 0 + 0) = 1$. So

$$\chi_5 := \chi_{S^2} - \mathbf{1} - \chi_4 = (5, 1, -1, 0, 0),$$

and $\langle \chi_5, \chi_5 \rangle = \frac{1}{60}(25 + 15 + 20 + 0 + 0) = 1$: irreducible, of degree 5.

Degree 3, twice. Compute the exterior square:

$$\chi_{\Lambda^2}(g) = \frac{1}{2}(\chi_4(g)^2 - \chi_4(g^2)) = (6, -2, 0, 1, 1).$$

Then $\langle \chi_{\Lambda^2}, \chi_{\Lambda^2} \rangle = \frac{1}{60}(36 + 15 \cdot 4 + 0 + 12 + 12) = \frac{120}{60} = 2$, so it is a sum of two distinct irreducibles; and it is orthogonal to $\mathbf{1}$ and to χ_4 (check: $\frac{1}{60}(6 - 30 + 0 + 12 + 12) = 0$ and $\frac{1}{60}(24 + 0 + 0 - 12 - 12) = 0$) and to χ_5 (check: $\frac{1}{60}(30 - 30 + 0 + 0 + 0) = 0$). So $\chi_{\Lambda^2} = \chi_2 + \chi_3$ for two new irreducibles of degrees summing to 6; since $60 = 1 + 16 + 25 + d_2^2 + d_3^2$ gives $d_2^2 + d_3^2 = 18$ with $d_2 + d_3 = 6$, we get $d_2 = d_3 = 3$.

To find the entries, write $\chi_2 = (3, a, b, c, c')$ and $\chi_3 = (3, \tilde{a}, \tilde{b}, \tilde{c}, \tilde{c}')$ with $\chi_2 + \chi_3 = (6, -2, 0, 1, 1)$. Column orthogonality gives, for the class of $(12)(34)$ (centraliser order $60/15 = 4$),

$$1 + a^2 + \tilde{a}^2 + 0^2 + 1^2 = 4 \implies a^2 + \tilde{a}^2 = 2, \quad a + \tilde{a} = -2 \implies a = \tilde{a} = -1.$$

For the class of (123) (centraliser order 3): $1 + b^2 + \tilde{b}^2 + 1 + 1 = 3$, so $b = \tilde{b} = 0$. For the class of (12345) (centraliser order 5): $1 + c^2 + \tilde{c}^2 + 1 + 0 = 5$, so $c^2 + \tilde{c}^2 = 3$ while $c + \tilde{c} = 1$; hence $c\tilde{c} = \frac{1}{2}((c + \tilde{c})^2 - (c^2 + \tilde{c}^2)) = -1$, so $c, \tilde{c}$ are the roots of $x^2 - x - 1 = 0$, that is

$$\alpha = \frac{1 + \sqrt{5}}{2}, \quad \beta = \frac{1 - \sqrt{5}}{2}.$$

The same holds on the other class of 5-cycles, with the two values swapped (they must be, or $\chi_2 = \chi_3$).

class size	1	15	20	12	12
	e	$(12)(34)$	(123)	(12345)	(13524)
$\mathbf{1}$	1	1	1	1	1
χ_2	3	-1	0	α	β
χ_3	3	-1	0	β	α
χ_4	4	0	1	-1	-1
χ_5	5	1	-1	0	0

where $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$, so that $\alpha + \beta = 1$, $\alpha\beta = -1$ and $\alpha^2 + \beta^2 = 3$.

Let us verify everything, using these three identities.

- Degrees: $1 + 9 + 9 + 16 + 25 = 60 = |A_5|$.
- $\langle \chi_2, \chi_2 \rangle = \frac{1}{60}(9 + 15 + 0 + 12\alpha^2 + 12\beta^2) = \frac{9+15+36}{60} = 1$; same for χ_3 .
- $\langle \chi_4, \chi_4 \rangle = \frac{1}{60}(16 + 0 + 20 + 12 + 12) = 1$; $\langle \chi_5, \chi_5 \rangle = \frac{1}{60}(25 + 15 + 20 + 0 + 0) = 1$.
- $\langle \chi_2, \chi_3 \rangle = \frac{1}{60}(9 + 15 + 0 + 12\alpha\beta + 12\beta\alpha) = \frac{9+15-24}{60} = 0$.
- $\langle \mathbf{1}, \chi_2 \rangle = \frac{1}{60}(3 - 15 + 0 + 12\alpha + 12\beta) = \frac{3-15+12}{60} = 0$; same for χ_3 .
- $\langle \mathbf{1}, \chi_4 \rangle = \frac{1}{60}(4 + 0 + 20 - 12 - 12) = 0$; $\langle \mathbf{1}, \chi_5 \rangle = \frac{1}{60}(5 + 15 - 20 + 0 + 0) = 0$.
- $\langle \chi_2, \chi_4 \rangle = \frac{1}{60}(12 + 0 + 0 - 12\alpha - 12\beta) = \frac{12-12}{60} = 0$; same for χ_3 .
- $\langle \chi_2, \chi_5 \rangle = \frac{1}{60}(15 - 15 + 0 + 0 + 0) = 0$; same for χ_3 ; $\langle \chi_4, \chi_5 \rangle = \frac{1}{60}(20 + 0 - 20 + 0 + 0) = 0$.
- Columns: e : $1+9+9+16+25 = 60$. $(12)(34)$: $1+1+1+0+1 = 4 = 60/15$. (123) : $1+0+0+1+1 = 3 = 60/20$. Each 5-class: $1+\alpha^2+\beta^2+1+0 = 1+3+1 = 5 = 60/12$.

Finally, the table shows that A_5 is simple: for each $i \neq 1$ and each $g \neq e$ we have $\chi_i(g) \neq \chi_i(e)$, since the entries $-1, 0, \alpha, \beta, 1$ are never equal to $3, 4$ or 5 . So no proper nontrivial normal subgroup exists.

Remark. The appearance of the golden ratio is not a coincidence: the two degree-3 representations are the two ways of realising A_5 as the rotation group of the icosahedron (differing by the outer automorphism of A_5 , which swaps the two classes of 5-cycles), and the trace of a rotation by $2\pi/5$ in $\mathbb{R}^3$ is $1 + 2\cos(2\pi/5) = \frac{1+\sqrt{5}}{2} = \alpha$.

§40.7.5 The Symmetric Group S_5

Finally S_5 , of order 120, which has seven conjugacy classes, indexed by cycle type:

cycle type	1^5	21^3	2^21	31^2	32	41	5
class size	1	10	15	20	20	30	24

These sum to 120, as they must. Since $S'_5 = A_5$ there are two linear characters. Let $\pi = (5, 3, 1, 2, 0, 1, 0)$ be the natural permutation character; the standard character is then

$$\chi_3 = \pi - \mathbf{1} = (4, 2, 0, 1, -1, 0, -1),$$

irreducible by 2-transitivity, and $\chi_4 = \text{sign} \cdot \chi_3 = (4, -2, 0, 1, 1, 0, -1)$ is a second one of degree 4.

For more, take squares of χ_3 . We need the class of g^2 for each g : $e, (12), (12)(34)$ all square into 1^5 ; (123) and $(123)(45)$ square to 3-cycles; (1234) squares to a 2^2 -element; and a 5-cycle squares to a 5-cycle. Hence

$$\chi_3(g^2) = (4, 4, 4, 1, 1, 0, -1),$$

and so

$$\begin{aligned}\chi_{S^2}(g) &= \frac{1}{2}(\chi_3(g)^2 + \chi_3(g^2)) = (10, 4, 2, 1, 1, 0, 0), \\ \chi_{\Lambda^2}(g) &= \frac{1}{2}(\chi_3(g)^2 - \chi_3(g^2)) = (6, 0, -2, 0, 0, 0, 1).\end{aligned}$$

Now $\langle \chi_{\Lambda^2}, \chi_{\Lambda^2} \rangle = \frac{1}{120}(36+0+15 \cdot 4+0+0+0+24) = \frac{120}{120} = 1$, so $\chi_5 := \chi_{\Lambda^2}$ is irreducible of degree 6. (Note $\text{sign} \cdot \chi_5 = \chi_5$: this character is ‘self-associate’.)

For the symmetric square, $\langle \chi_{S^2}, \mathbf{1} \rangle = \frac{1}{120}(10+40+30+20+20+0+0) = 1$ and $\langle \chi_{S^2}, \chi_3 \rangle = \frac{1}{120}(40+80+0+20-20+0+0) = 1$, so

$$\chi_6 := \chi_{S^2} - \mathbf{1} - \chi_3 = (5, 1, 1, -1, 1, -1, 0),$$

with $\langle \chi_6, \chi_6 \rangle = \frac{1}{120}(25+10+15+20+20+30+0) = 1$: irreducible of degree 5. And $\chi_7 = \text{sign} \cdot \chi_6 = (5, -1, 1, -1, -1, 1, 0)$ is a seventh, distinct from χ_6 . Degrees: $1+1+16+16+25+25+36=120$, so we have them all.

class size	1	10	15	20	20	30	24
	e	(1 2)	(1 2)(3 4)	(1 2 3)	(1 2 3)(4 5)	(1 2 3 4)	(1 2 3 4 5)
$\mathbf{1}$	1	1	1	1	1	1	1
sign	1	-1	1	1	-1	-1	1
χ_3	4	2	0	1	-1	0	-1
χ_4	4	-2	0	1	1	0	-1
χ_5	6	0	-2	0	0	0	1
χ_6	5	1	1	-1	1	-1	0
χ_7	5	-1	1	-1	-1	1	0

Verification of the column relations, which is the quickest global check:

$$\begin{aligned}
 e : & \quad 1+1+16+16+36+25+25=120=|C(e)|, \\
 (1\ 2) : & \quad 1+1+4+4+0+1+1=12=120/10, \\
 (1\ 2)(3\ 4) : & \quad 1+1+0+0+4+1+1=8=120/15, \\
 (1\ 2\ 3) : & \quad 1+1+1+1+0+1+1=6=120/20, \\
 (1\ 2\ 3)(4\ 5) : & \quad 1+1+1+1+0+1+1=6=120/20, \\
 (1\ 2\ 3\ 4) : & \quad 1+1+0+0+0+1+1=4=120/30, \\
 (1\ 2\ 3\ 4\ 5) : & \quad 1+1+1+1+1+0+0=5=120/24.
 \end{aligned}$$

And a sample of the row relations: $\langle \chi_3, \chi_4 \rangle = \frac{1}{120}(16-40+0+20-20+0+24)=0$; $\langle \chi_6, \chi_7 \rangle = \frac{1}{120}(25-10+15+20-20-30+0)=0$; $\langle \chi_3, \chi_5 \rangle = \frac{1}{120}(24+0+0+0+0+0-24)=0$; $\langle \chi_5, \chi_6 \rangle = \frac{1}{120}(30+0-30+0+0+0+0)=0$; $\langle \mathbf{1}, \chi_5 \rangle = \frac{1}{120}(6+0-30+0+0+0+24)=0$. (All twenty-one orthogonality relations hold; the reader is invited to spot-check a few more.)

Remark. Notice the shape of the answer: seven irreducible characters, in three ‘associate pairs’ $\{\mathbf{1}, \text{sign}\}$, $\{\chi_3, \chi_4\}$, $\{\chi_6, \chi_7\}$ swapped by tensoring with sign, and one self-associate character χ_5 . There is a beautiful combinatorial explanation in terms of partitions of 5 and their transposes, which we come to in the section on the symmetric group.

§40.8 Restriction and Induction

So far all our constructions have taken place inside a fixed group. We now allow ourselves to move between a group and its subgroups, which turns out to be the single most powerful technique for building character tables of larger groups.

Throughout this section $H \leq G$ is a subgroup of index $n = |G : H|$.

§40.8.1 Restriction

This direction is trivial to define and surprisingly informative.

Definition 40.8.1 (Restriction)

Let (ρ, V) be a representation of G and $H \leq G$. The **restriction** $\text{Res}_H^G V$ (also written $V \downarrow_H$) is the representation of H on the same space V given by $\rho|_H$. On characters, $\text{Res}_H^G \chi$ is just $\chi|_H$.

Restricting an irreducible representation of G to H almost never gives an irreducible representation of H , but the failure is controlled.

Proposition 40.8.2

Let V be an irreducible representation of G with character χ , and write $\text{Res}_H^G \chi = \sum_i c_i \psi_i$ over the irreducible characters ψ_i of H . Then

$$\sum_i c_i^2 = \langle \text{Res}_H^G \chi, \text{Res}_H^G \chi \rangle_H \leq |G : H|,$$

with equality if and only if $\chi(g) = 0$ for every $g \in G \setminus H$.

Proof. We have

$$\langle \chi|_H, \chi|_H \rangle_H = \frac{|G|}{|H|} \cdot \frac{1}{|G|} \sum_{h \in H} |\chi(h)|^2 \leq |G : H| \cdot \frac{1}{|G|} \sum_{g \in G} |\chi(g)|^2 = |G : H|,$$

since $\langle \chi, \chi \rangle_G = 1$. The inequality is an equality exactly when the omitted terms $|\chi(g)|^2$, $g \notin H$, all vanish. $\square$

Corollary 40.8.3

If χ is irreducible of degree d and $H \leq G$ has index n , then $\text{Res}_H^G \chi$ has at most n irreducible constituents counted with multiplicity, and in particular $d \leq n \cdot \max_i \psi_i(1)$.

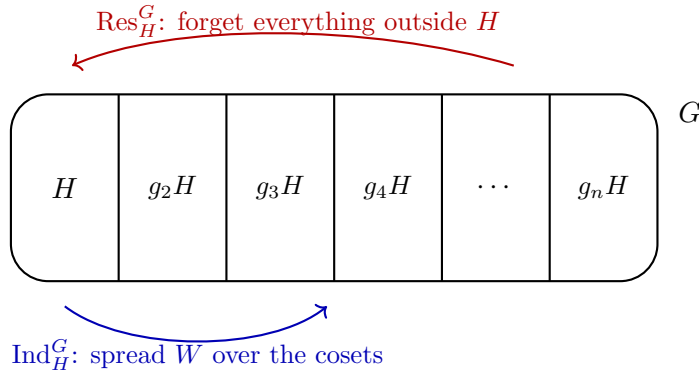
Example 40.8.4

Restrict the degree-2 character $\chi = (2, 0, -1)$ of S_3 to $H = A_3 = C_3$. We get $\chi|_H = (2, -1, -1)$ on $(e, (1\ 2\ 3), (1\ 3\ 2))$, which is $\omega + \omega^2$ -flavoured: indeed $\chi|_H = \chi_1 + \chi_2$, the two nontrivial characters of C_3 . Here $\langle \chi|_H, \chi|_H \rangle = 2 = |G : H|$, consistent with χ vanishing off A_3 .

§40.8.2 Induction

The other direction is the interesting one: from a representation of H we manufacture a representation of G , of degree multiplied by $|G : H|$.

Choose left coset representatives $g_1 = e, g_2, \dots, g_n$ for H in G , so that $G = \bigsqcup_{i=1}^n g_i H$.

**Definition 40.8.5** (Induced representation)

Let W be a representation of H . The **induced representation** is the vector space

$$\text{Ind}_H^G W = \bigoplus_{i=1}^n g_i \otimes W,$$

where $g_i \otimes W$ is a copy of W labelled by the coset $g_i H$, with G acting as follows: given $g \in G$ and i , write $gg_i = g_j h$ with $h \in H$ (which determines j and h uniquely), and set

$$g \cdot (g_i \otimes w) = g_j \otimes hw.$$

Its degree is $|G : H| \cdot \dim W$.

Remark. Invariantly, $\text{Ind}_H^G W = \mathbb{C}G \otimes_{\mathbb{C}H} W$, the extension of scalars along $\mathbb{C}H \hookrightarrow \mathbb{C}G$; this makes the independence of the choice of coset representatives obvious, and identifies induction as the left adjoint of restriction. Readers who found tensor products of modules congenial in *Groups, Rings and Modules* may prefer this definition; the concrete one above is what we compute with.

One should check this is a representation. If $g'g_j = g_k h'$ then $g'g g_i = g'g_j h = g_k h' h$, so

$$g' \cdot (g \cdot (g_i \otimes w)) = g' \cdot (g_j \otimes hw) = g_k \otimes h'hw = (g'g) \cdot (g_i \otimes w),$$

as required, and the identity acts trivially.

Theorem 40.8.6 (Induced character formula)

Let ψ be the character of a representation W of $H \leq G$. Then the character of $\text{Ind}_H^G W$ is

$$\text{Ind}_H^G \psi(g) = \sum_{\substack{1 \leq i \leq n \\ g_i^{-1} g g_i \in H}} \psi(g_i^{-1} g g_i) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1} g x \in H}} \psi(x^{-1} g x).$$

In particular $\text{Ind}_H^G \psi(e) = |G : H| \psi(e)$, and $\text{Ind}_H^G \psi(g) = 0$ if no conjugate of g lies in H .

Proof. Write the matrix of g in a basis adapted to the decomposition $\bigoplus_i g_i \otimes W$, as an $n \times n$ array of blocks of size $\dim W$. The action sends the i -th block into the j -th block, where $gg_i \in g_j H$. So the (i, i) diagonal block is nonzero only when $gg_i \in g_i H$, i.e. $g_i^{-1}gg_i \in H$, and then it is the matrix of $\rho_W(g_i^{-1}gg_i)$. Taking traces of the diagonal blocks gives the first formula.

For the second, every $x \in G$ is uniquely $x = g_i h$ with $h \in H$, and

$$x^{-1}gx = h^{-1}(g_i^{-1}gg_i)h,$$

which lies in H if and only if $g_i^{-1}gg_i$ does, and then has the same ψ -value since ψ is a class function on H . So each term of the first sum is repeated exactly $|H|$ times in the second. $\square$

There is a third form of the formula which is often the most efficient for hand computation.

Corollary 40.8.7

Let $g \in G$ and suppose $\text{ccl}_G(g) \cap H$ is the disjoint union of the H -conjugacy classes of $h_1, \dots, h_m$ (possibly $m = 0$). Then

$$\text{Ind}_H^G \psi(g) = |C_G(g)| \sum_{k=1}^m \frac{\psi(h_k)}{|C_H(h_k)|}.$$

Proof. In the sum $\frac{1}{|H|} \sum_{x: x^{-1}gx \in H} \psi(x^{-1}gx)$, group the x by the value of $x^{-1}gx$. For a fixed $h \in \text{ccl}_G(g) \cap H$, the number of x with $x^{-1}gx = h$ is $|C_G(g)|$. So the sum is $\frac{|C_G(g)|}{|H|} \sum_{h \in \text{ccl}_G(g) \cap H} \psi(h)$, and grouping the h into H -classes of sizes $|H|/|C_H(h_k)|$ gives the result. $\square$

§40.8.3 Frobenius Reciprocity

Restriction and induction are adjoint to one another, and this single fact organises everything.

Theorem 40.8.8 (Frobenius reciprocity)

Let $H \leq G$, let ψ be a class function on H and χ a class function on G . Then

$$\langle \text{Res}_H^G \chi, \psi \rangle_H = \langle \chi, \text{Ind}_H^G \psi \rangle_G.$$

Proof. Extend Ind_H^G to all class functions on H by the formula of the theorem above (which makes sense for any class function). Then

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{\chi(g)} \cdot \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}gx \in H}} \psi(x^{-1}gx)$$

$$= \frac{1}{|G||H|} \sum_{x \in G} \sum_{\substack{g \in G \\ x^{-1}gx \in H}} \overline{\chi(g)} \psi(x^{-1}gx).$$

For fixed x , substitute $h = x^{-1}gx$; as g runs over those elements of G with $x^{-1}gx \in H$, h runs over H , and $\chi(g) = \chi(xhx^{-1}) = \chi(h)$ since χ is a class function on G . So the inner sum is $\sum_{h \in H} \overline{\chi(h)}\psi(h)$, independent of x . Hence

$$\langle \chi, \text{Ind}_H^G \psi \rangle_G = \frac{|G|}{|G||H|} \sum_{h \in H} \overline{\chi(h)}\psi(h) = \langle \text{Res}_H^G \chi, \psi \rangle_H. \quad \square$$

Corollary 40.8.9

$\text{Ind}_H^G \psi$ is a character of G whenever ψ is a character of H , and induction takes nonnegative integer combinations of irreducible characters to nonnegative integer combinations.

Proof. We constructed the induced representation, so this is automatic; but note it also follows formally, since $\langle \chi_i, \text{Ind} \psi \rangle = \langle \text{Res} \chi_i, \psi \rangle \geq 0$ is a nonnegative integer for every irreducible χ_i . $\square$

Corollary 40.8.10

If $H \leq G$ and ψ is an irreducible character of H , then every irreducible constituent of $\text{Ind}_H^G \psi$ has ψ as a constituent of its restriction to H , and vice versa.

Example 40.8.11 (Permutation characters are induced)

Let G act transitively on X and let $H = \text{Stab}_G(x_0)$. Then $\mathbb{C}X \cong \text{Ind}_H^G \mathbf{1}_H$: both have the same character, since

$$\text{Ind}_H^G \mathbf{1}(g) = \frac{1}{|H|} \# \{x \in G : x^{-1}gx \in H\} = \# \{xH : gxH = xH\},$$

and the last quantity is $|\text{fix}_X(g)|$ under the identification $X \leftrightarrow G/H$. Frobenius reciprocity then reproves that $\langle \pi_X, \mathbf{1}_G \rangle = \langle \mathbf{1}_H, \mathbf{1}_H \rangle = 1$ for transitive actions.

Example 40.8.12 (The two-dimensional character of D_8)

Take $G = D_8$ and $H = \langle r \rangle \cong C_4$, of index 2, with coset representatives e, s . Let $\psi(r^k) = i^k$ be a faithful character of H . Since H is normal, every conjugate of r^k lies in H , and

$$\text{Ind}_H^G \psi(r^k) = \psi(r^k) + \psi(s^{-1}r^k s) = i^k + i^{-k},$$

which is $2, 0, -2, 0$ for $k = 0, 1, 2, 3$; while for a reflection g , no conjugate lies in H , so $\text{Ind} \psi(g) = 0$. Thus $\text{Ind}_H^G \psi = (2, -2, 0, 0, 0)$ on the classes e, r^2, r, s, rs – exactly the degree-2 irreducible character of D_8 . This is how one builds the dihedral characters ‘from above’.

Example 40.8.13 (Building characters of A_5 by induction)

Take $G = A_5$ and $H = A_4$, the stabiliser of the point 5, of index 5.

Inducing the trivial character gives the permutation character on 5 points, namely $(5, 1, 2, 0, 0) = \mathbf{1} + \chi_4$.

Now induce the linear character $\psi = \chi_2$ of A_4 , with values $(1, 1, \omega, \omega^2)$ on the classes $e, (12)(34), (123), (132)$ of A_4 . Use the centraliser form of the induced character formula, recalling $|C_{A_5}|$ has values 60, 4, 3, 5, 5 on the five classes of A_5 , and $|C_{A_4}|$ has values 12, 4, 3, 3 on the classes of A_4 .

- $g = e$: $\text{Ind } \psi(e) = 60 \cdot \frac{1}{12} = 5$, as it must be $(5 = |G : H| \cdot 1)$.
- $g = (12)(34)$: the A_5 -class of g meets A_4 in the single A_4 -class of double transpositions, so $\text{Ind } \psi(g) = 4 \cdot \frac{1}{4} = 1$.
- $g = (123)$: the A_5 -class of 3-cycles meets A_4 in *two* A_4 -classes, on which ψ takes the values ω and ω^2 . So $\text{Ind } \psi(g) = 3\left(\frac{\omega}{3} + \frac{\omega^2}{3}\right) = \omega + \omega^2 = -1$.
- g a 5-cycle: A_4 contains no elements of order 5, so $\text{Ind } \psi(g) = 0$.

Hence $\text{Ind}_{A_4}^{A_5} \chi_2 = (5, 1, -1, 0, 0) = \chi_5$, which we found earlier as $S^2 \chi_4 - \mathbf{1} - \chi_4$. Two different routes to the same character is always reassuring.

Remark (Transitivity of induction). If $K \leq H \leq G$ then $\text{Ind}_H^G \text{Ind}_K^H = \text{Ind}_K^G$, and likewise Res is transitive. Both follow from Frobenius reciprocity: for any irreducible χ of G ,

$$\langle \chi, \text{Ind}_H^G \text{Ind}_K^H \psi \rangle = \langle \text{Res}_H \chi, \text{Ind}_K^H \psi \rangle = \langle \text{Res}_K \chi, \psi \rangle = \langle \chi, \text{Ind}_K^G \psi \rangle.$$

§40.8.4 A Character Table Built by Induction

Here is a complete worked example where induction is genuinely the only easy route. Let

$$G = C_7 \rtimes C_3 = \langle a, b \mid a^7 = b^3 = e, bab^{-1} = a^2 \rangle,$$

a nonabelian group of order 21. (Such a group exists because 2 has order 3 modulo 7.)

Conjugacy classes. Conjugating a by b repeatedly gives $a \mapsto a^2 \mapsto a^4 \mapsto a^8 = a$, so $\{a, a^2, a^4\}$ is a class; similarly $\{a^3, a^6, a^5\}$ is a class. The elements $a^i b$ are all conjugate: since $C_G(b) \supseteq \langle b \rangle$ has order at least 3 and $b \notin Z(G)$, the class of b has size dividing 7 and greater than 1, so exactly 7; likewise for b^2 . So the classes are

$$\{e\} (1), \quad \{a, a^2, a^4\} (3), \quad \{a^3, a^6, a^5\} (3), \quad \text{ccl}(b) (7), \quad \text{ccl}(b^2) (7),$$

totalling $1 + 3 + 3 + 7 + 7 = 21$. So there are five irreducible characters.

Degrees. We have $G' = \langle a \rangle$ (as $bab^{-1}a^{-1} = a$), so $G/G' \cong C_3$ and there are exactly three linear characters $\mathbf{1}, \chi_2, \chi_3$, given by $a \mapsto 1, b \mapsto \omega^{\pm 1}$ with $\omega = e^{2\pi i/3}$. Then $21 = 3 + d_4^2 + d_5^2$ gives $d_4 = d_5 = 3$.

The degree-3 characters. Take $H = \langle a \rangle \cong C_7$, of index 3, and let $\zeta = e^{2\pi i/7}$ with $\psi(a^k) = \zeta^k$. Since $H \trianglelefteq G$, every conjugate of an element of H lies in H , and the coset representatives e, b, b^2 give

$$\text{Ind}_H^G \psi(a^k) = \psi(a^k) + \psi(b^{-1}a^kb) + \psi(b^{-2}a^kb^2) = \zeta^k + \zeta^{4k} + \zeta^{2k},$$

using $b^{-1}ab = a^4$ and $b^{-2}ab^2 = a^2$. No conjugate of b or b^2 lies in H , so $\text{Ind}_H^G \psi$ vanishes there. Writing

$$\eta = \zeta + \zeta^2 + \zeta^4, \quad \bar{\eta} = \zeta^3 + \zeta^5 + \zeta^6,$$

(the sums over the quadratic residues and non-residues modulo 7) we get $\text{Ind}_H^G \psi = (3, \eta, \bar{\eta}, 0, 0)$. Note

$$\eta + \bar{\eta} = \sum_{k=1}^6 \zeta^k = -1, \quad \eta \bar{\eta} = 2.$$

The first is the vanishing of the sum of all seventh roots of unity other than 1. For the second, expand into nine terms:

$$(\zeta + \zeta^2 + \zeta^4)(\zeta^3 + \zeta^5 + \zeta^6) = (\zeta^4 + \zeta^6 + 1) + (\zeta^5 + 1 + \zeta) + (1 + \zeta^2 + \zeta^3) = 3 + \sum_{k=1}^6 \zeta^k = 2.$$

So $\eta, \bar{\eta}$ are the roots of $x^2 + x + 2$, that is $\eta = \frac{-1+i\sqrt{7}}{2}$ (which is indeed the complex conjugate of $\bar{\eta}$, justifying the notation). Then

$$\langle \text{Ind} \psi, \text{Ind} \psi \rangle = \frac{1}{21} (9 + 3|\eta|^2 + 3|\bar{\eta}|^2) = \frac{9+6+6}{21} = 1,$$

so $\chi_4 := \text{Ind}_H^G \psi$ is irreducible; and $\chi_5 = \overline{\chi_4} = (3, \bar{\eta}, \eta, 0, 0)$ is a second one, obtained by inducing ψ^3 instead.

class size	1	3	3	7	7
	e	a	a^3	b	b^2
1	1	1	1	1	1
χ_2	1	1	1	ω	ω^2
χ_3	1	1	1	ω^2	ω
χ_4	3	η	$\bar{\eta}$	0	0
χ_5	3	$\bar{\eta}$	η	0	0

where $\eta = \frac{-1+i\sqrt{7}}{2}$, so $\eta + \bar{\eta} = -1$ and $\eta \bar{\eta} = 2$.

Checks: degrees $1 + 1 + 1 + 9 + 9 = 21$. Rows: $\langle \chi_4, \chi_4 \rangle = \frac{1}{21} (9 + 6 + 6) = 1$; $\langle \chi_4, \chi_5 \rangle = \frac{1}{21} (9 + 3\bar{\eta}^2 + 3\eta^2) = \frac{9+3(\eta^2+\bar{\eta}^2)}{21} = \frac{9+3(1-4)}{21} = 0$, using $\eta^2 + \bar{\eta}^2 = (\eta + \bar{\eta})^2 - 2\eta\bar{\eta} = -3$; $\langle \mathbf{1}, \chi_4 \rangle = \frac{1}{21} (3 + 3\eta + 3\bar{\eta}) = 0$; $\langle \chi_2, \chi_4 \rangle = \frac{1}{21} (3 + 3\eta + 3\bar{\eta} + 0 + 0) = 0$; $\langle \mathbf{1}, \chi_2 \rangle = \frac{1}{21} (1 + 3 + 3 + 7\omega + 7\omega^2) = \frac{7-7}{21} = 0$. Columns: e gives 21; a gives $1 + 1 + 1 + |\eta|^2 + |\bar{\eta}|^2 = 3 + 2 + 2 = 7 = 21/3$, and likewise for a^3 ; b gives $1 + 1 + 1 + 0 + 0 = 3 = 21/7$, and likewise for b^2 . All correct.

Remark. This G is the smallest nonabelian group of odd order, and it is a Frobenius group: the complement is $\langle b \rangle \cong C_3$, which satisfies $\langle b \rangle \cap g\langle b \rangle g^{-1} = 1$ for $g \notin \langle b \rangle$, and the Frobenius kernel is $\langle a \rangle \cong C_7$. Frobenius' theorem below explains, from the characters alone, why $\langle a \rangle$ had to be normal.

§40.8.5 Mackey's Formulae

We record, without full proofs, the theorem which describes what happens when induction and restriction are composed the 'wrong' way round. It is not needed in the sequel, but it is the natural completion of the story.

Let $H, K \leq G$. The set $K \backslash G / H$ of **double cosets** KgH partitions G ; each double coset is a union of left cosets of H . For $g \in G$, write $H^g = g^{-1}Hg$ and let gW denote the representation of gHg^{-1} on W given by $(ghg^{-1}) \cdot w = hw$.

Theorem 40.8.14 (Mackey's restriction formula)

With notation as above, for a representation W of H ,

$$\text{Res}_K^G \text{Ind}_H^G W \cong \bigoplus_{KgH \in K \backslash G / H} \text{Ind}_{K \cap {}^gH}^K \text{Res}_{K \cap {}^gH}^{H^g} ({}^gW),$$

where ${}^gH = gHg^{-1}$. The right hand side does not depend on the choice of double coset representatives.

The proof is a bookkeeping exercise: the space $\text{Ind}_H^G W = \bigoplus_{xH} x \otimes W$ is permuted by K according to the action of K on G/H , whose orbits are exactly the double cosets, and one identifies the summand corresponding to each orbit.

Corollary 40.8.15 (Mackey's irreducibility criterion)

Let $H \leq G$ and W an irreducible representation of H . Then $\text{Ind}_H^G W$ is irreducible if and only if, for every $g \in G \setminus H$, the representations

$$\text{Res}_{H \cap {}^gH}^H W \quad \text{and} \quad \text{Res}_{H \cap {}^gH}^{H^g} ({}^gW)$$

of $H \cap {}^gH$ have no irreducible constituent in common.

Proof sketch. By Frobenius reciprocity and Mackey with $K = H$,

$$\langle \text{Ind}_H^G W, \text{Ind}_H^G W \rangle_G = \langle W, \text{Res}_H^G \text{Ind}_H^G W \rangle_H = \sum_{HgH} \langle \text{Res}_{H \cap {}^gH} W, \text{Res}_{H \cap {}^gH} ({}^gW) \rangle.$$

The term for the double coset H itself contributes $\langle W, W \rangle_H = 1$, and all other terms are nonnegative; so the total is 1 if and only if all the others vanish. $\square$

Corollary 40.8.16

If $H \triangleleft G$ and W is irreducible, then $\text{Ind}_H^G W$ is irreducible if and only if ${}^gW \not\cong W$ for all $g \in G \setminus H$.

§40.8.6 An Application: Frobenius' Theorem

Here is a striking theorem which cannot (to this day) be proved without characters. It is the reason induction earns its keep.

Definition 40.8.17 (Frobenius group)

Let G be a finite group with a subgroup $1 < H < G$ such that

$$H \cap gHg^{-1} = 1 \quad \text{for all } g \in G \setminus H.$$

We call H a **Frobenius complement** and G a **Frobenius group**.

Equivalently, G acts transitively on G/H in such a way that no nonidentity element fixes two points, but some nonidentity element fixes a point.

Theorem 40.8.18 (Frobenius, 1901)

Let G be a Frobenius group with complement H . Then

$$K = \left(G \setminus \bigcup_{g \in G} gHg^{-1} \right) \cup \{1\}$$

is a normal subgroup of G with $|K| = |G : H|$ and $G = KH$, $K \cap H = 1$.

K is called the **Frobenius kernel**. We build up to the proof with a lemma which is the whole point.

Lemma 40.8.19

Let G, H be as above and let φ be a class function on H with $\varphi(1) = 0$. Then $\text{Res}_H^G \text{Ind}_H^G \varphi = \varphi$.

Proof. Let $h \in H$ with $h \neq 1$. By the induced character formula,

$$\text{Ind}_H^G \varphi(h) = \frac{1}{|H|} \sum_{\substack{x \in G \\ x^{-1}hx \in H}} \varphi(x^{-1}hx).$$

If $x^{-1}hx \in H$ with $h \neq 1$, then $h \in H \cap xHx^{-1}$, which is trivial unless $x \in H$; so the sum runs over $x \in H$ only, giving $\frac{1}{|H|} \cdot |H| \varphi(h) = \varphi(h)$. And at $h = 1$ we get $\text{Ind}_H^G \varphi(1) = |G : H| \varphi(1) = 0 = \varphi(1)$. $\square$

Proof of Frobenius' theorem. First note $N_G(H) = H$: if g normalises H and $g \notin H$ then $H = H \cap gHg^{-1} = 1$, a contradiction. So H has exactly $|G : H| =: m$ distinct conjugates, and by hypothesis they pairwise intersect trivially. Hence

$$\left| \bigcup_g gHg^{-1} \right| = m(|H| - 1) + 1 = |G| - m + 1,$$

so the set K defined in the statement has exactly m elements. We must show it is a subgroup, and normal.

Let ψ be an irreducible character of H with $\psi \neq \mathbf{1}_H$, and set

$$\varphi = \psi - \psi(1)\mathbf{1}_H, \quad \theta_\psi = \text{Ind}_H^G \varphi + \psi(1)\mathbf{1}_G.$$

This θ_ψ is a virtual character of G (a $\mathbb{Z}$ -combination of irreducible characters), and by the lemma $\text{Res}_H^G \theta_\psi = \varphi + \psi(1)\mathbf{1}_H = \psi$. Now compute its norm using Frobenius reciprocity and the lemma:

$$\langle \text{Ind } \varphi, \text{Ind } \varphi \rangle_G = \langle \varphi, \text{Res Ind } \varphi \rangle_H = \langle \varphi, \varphi \rangle_H = 1 + \psi(1)^2,$$

$$\langle \text{Ind } \varphi, \mathbf{1}_G \rangle_G = \langle \varphi, \mathbf{1}_H \rangle_H = -\psi(1).$$

Therefore

$$\langle \theta_\psi, \theta_\psi \rangle = (1 + \psi(1)^2) + 2\psi(1)(-\psi(1)) + \psi(1)^2 = 1.$$

Since also $\theta_\psi(1) = |G : H| \cdot 0 + \psi(1) > 0$, the virtual character θ_ψ is an *irreducible character* of G .

Now put

$$K' = \bigcap_{\substack{\psi \in \text{Irr}(H) \\ \psi \neq \mathbf{1}}} \ker \theta_\psi,$$

an intersection of kernels of characters, hence a normal subgroup of G .

If $g \in K$ (so $g = 1$ or g lies in no conjugate of H) then no conjugate of g lies in $H \setminus \{1\}$, so the induced character formula gives $\text{Ind } \varphi(g) = \frac{1}{|H|} \cdot |\{x : x^{-1}gx = 1\}| \cdot \varphi(1) = 0$ for $g \neq 1$, and hence $\theta_\psi(g) = \psi(1) = \theta_\psi(1)$. So $K \subseteq K'$.

Conversely, if $g \in K' \cap H$ then $\psi(g) = \theta_\psi(g) = \theta_\psi(1) = \psi(1)$ for every irreducible $\psi \neq \mathbf{1}$ of H , and trivially for $\mathbf{1}$ too; since the irreducible characters of H separate its elements from the identity (their common kernel is trivial), $g = 1$. So $K' \cap H = 1$, whence $|K'| |H| = |K' H| \leq |G|$ and $|K'| \leq m$.

Combining, $K \subseteq K'$ with $|K| = m \geq |K'| \geq |K|$, so $K = K'$ is a normal subgroup of order $m = |G : H|$. Finally $K \cap H = 1$ and $|K| |H| = |G|$ give $G = KH$. $\square$

Example 40.8.20

Take $G = A_4$ and $H = \langle (123) \rangle$. The conjugates of H are the four Sylow 3-subgroups, which pairwise intersect trivially, and $N_{A_4}(H) = H$; so H is a Frobenius complement. The elements of A_4 lying in no conjugate of H are exactly the three double transpositions, so the theorem produces

$$K = \{e, (12)(34), (13)(24), (14)(23)\} = V_4,$$

which is indeed normal, of order $4 = |A_4 : H|$. Of course we knew that already, but the point is that the argument never looked at the multiplication table.

Example 40.8.21

More generally, if $G = \mathbb{F}_q \rtimes \mathbb{F}_q^\times$ is the group of affine maps $x \mapsto ax + b$ on $\mathbb{F}_q$, then $H = \mathbb{F}_q^\times$ (the stabiliser of 0) is a Frobenius complement and $K = \mathbb{F}_q$ (the translations) is the Frobenius kernel. Every sharply 2-transitive group is of this shape.

§40.9 Arithmetic Properties of Characters

Characters are sums of roots of unity, so their values are constrained not just analytically (by orthogonality) but *arithmetically*. Exploiting this gives two beautiful theorems: that the degree of an irreducible representation divides the order of the group, and Burnside's theorem that every group of order $p^a q^b$ is soluble.

§40.9.1 Algebraic Integers

Definition 40.9.1 (Algebraic integer)

A complex number α is an **algebraic integer** if it is a root of some monic polynomial with coefficients in $\mathbb{Z}$. We write $\mathbb{A}$ for the set of algebraic integers.

Example 40.9.2

Every $n \in \mathbb{Z}$ is an algebraic integer (root of $X - n$), as is every root of unity (root of $X^n - 1$), and $\sqrt{2}$, and $\frac{1+\sqrt{5}}{2}$ (root of $X^2 - X - 1$ – so the A_5 table consists of algebraic integers, as it must).

The following criterion is the practical way to recognise algebraic integers, and it is what makes $\mathbb{A}$ a ring.

Proposition 40.9.3

Let $\alpha \in \mathbb{C}$. Then $\alpha \in \mathbb{A}$ if and only if there is a finitely generated nonzero additive subgroup $M \leq (\mathbb{C}, +)$ with $\alpha M \subseteq M$.

Proof. ($\Rightarrow$) If $\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_0 = 0$ with $c_i \in \mathbb{Z}$, take $M = \mathbb{Z}\langle 1, \alpha, \dots, \alpha^{n-1} \rangle$; then $\alpha M \subseteq M$ because α^n is an integer combination of lower powers.

($\Leftarrow$) Let M be generated by $m_1, \dots, m_k$, not all zero. Then $\alpha m_i = \sum_j A_{ij} m_j$ for some integer matrix A . So $(\alpha I - A)\mathbf{m} = 0$ where $\mathbf{m} = (m_1, \dots, m_k)^T \neq 0$, hence $\det(\alpha I - A) = 0$. But $\det(XI - A)$ is a monic polynomial in X with integer coefficients. $\square$

Corollary 40.9.4

$\mathbb{A}$ is a subring of $\mathbb{C}$.

Proof. If $\alpha M \subseteq M$ and $\beta N \subseteq N$ with M, N finitely generated and nonzero, let MN be the additive group generated by all mn . It is finitely generated (by the products of generators), nonzero, and stable under both $\alpha + \beta$ and $\alpha\beta$. $\square$

Proposition 40.9.5

$\mathbb{A} \cap \mathbb{Q} = \mathbb{Z}$.

Proof. Certainly $\mathbb{Z} \subseteq \mathbb{A} \cap \mathbb{Q}$. Conversely let $\alpha = a/b \in \mathbb{Q}$ in lowest terms with $b > 0$, satisfying $\alpha^n + c_{n-1}\alpha^{n-1} + \cdots + c_0 = 0$. Multiplying by b^n ,

$$a^n = -b(c_{n-1}a^{n-1} + c_{n-2}a^{n-2}b + \cdots + c_0b^{n-1}),$$

so $b \mid a^n$. Since $\gcd(a, b) = 1$ this forces $b = 1$. $\square$

Corollary 40.9.6

All character values are algebraic integers.

Proof. $\chi(g)$ is a sum of roots of unity, and $\mathbb{A}$ is a ring containing all roots of unity. $\square$

§40.9.2 Central Characters

To get further we need algebraic integers built out of the class sizes as well.

Definition 40.9.7 (Class sums)

For a conjugacy class C of G , the **class sum** is $\widehat{C} = \sum_{g \in C} g \in \mathbb{C}G$.

Proposition 40.9.8

The class sums $\widehat{C}_1, \dots, \widehat{C}_r$ form a $\mathbb{C}$ -basis of the centre $Z(\mathbb{C}G)$, and

$$\widehat{C}_i \widehat{C}_j = \sum_{k=1}^r a_{ijk} \widehat{C}_k$$

for some *nonnegative integers* a_{ijk} .

Proof. An element $z = \sum_g \lambda_g g$ is central if and only if $hzh^{-1} = z$ for all h , i.e. $\lambda_{hgh^{-1}} = \lambda_g$, i.e. $g \mapsto \lambda_g$ is constant on classes. So $Z(\mathbb{C}G)$ has the class sums as a basis. The product $\widehat{C}_i \widehat{C}_j$ is central, hence a combination of class sums, and its coefficient at $g \in C_k$ is

$$a_{ijk} = |\{(x, y) \in C_i \times C_j : xy = g\}| \in \mathbb{Z}_{\geq 0}. \quad \square$$

Proposition 40.9.9 (Central character)

Let (ρ, V) be an irreducible representation of G with character χ of degree d . Then for each class C with representative g ,

$$\rho(\widehat{C}) = \omega_\chi(C) \operatorname{id}_V, \quad \text{where} \quad \omega_\chi(C) = \frac{|C| \chi(g)}{d}.$$

The function ω_χ is called the **central character**.

Proof. The map $\rho(\widehat{C}) = \sum_{x \in C} \rho(x)$ commutes with every $\rho(h)$, since conjugation by h permutes C . So it is a G -endomorphism of the irreducible V , hence a scalar $\lambda \operatorname{id}_V$ by Schur. Taking traces, $d\lambda = \sum_{x \in C} \chi(x) = |C|\chi(g)$. $\square$

Proposition 40.9.10

Each $\omega_\chi(C)$ is an algebraic integer.

Proof. Applying ρ to $\widehat{C_i C_j} = \sum_k a_{ijk} \widehat{C_k}$ and comparing scalars,

$$\omega_\chi(C_i) \omega_\chi(C_j) = \sum_k a_{ijk} \omega_\chi(C_k), \quad a_{ijk} \in \mathbb{Z}.$$

So $M := \mathbb{Z}\langle\omega_\chi(C_1), \dots, \omega_\chi(C_r)\rangle$ is a finitely generated additive group, nonzero (it contains $\omega_\chi(\{e\}) = 1$), with $\omega_\chi(C_i)M \subseteq M$ for each i . By the criterion, each $\omega_\chi(C_i)$ is an algebraic integer. $\square$

§40.9.3 Degrees Divide the Group Order

Theorem 40.9.11

Let χ be an irreducible character of the finite group G . Then $\chi(1)$ divides $|G|$.

Proof. Write $d = \chi(1)$, and let $C_1, \dots, C_r$ be the conjugacy classes, with representatives $g_1, \dots, g_r$. Then

$$\frac{|G|}{d} = \frac{1}{d} \sum_{j=1}^r |C_j| \chi(g_j) \overline{\chi(g_j)} = \sum_{j=1}^r \frac{|C_j| \chi(g_j)}{d} \cdot \overline{\chi(g_j)} = \sum_{j=1}^r \omega_\chi(C_j) \overline{\chi(g_j)},$$

using $\langle \chi, \chi \rangle = 1$ in the first equality. The right hand side is a sum of products of algebraic integers, so is an algebraic integer; the left hand side is rational. Hence $|G|/d \in \mathbb{A} \cap \mathbb{Q} = \mathbb{Z}$. $\square$

Remark. With a little more work one shows the sharper statement that $\chi(1)$ divides $|G : Z(G)|$; and a theorem of Ito says that if $A \trianglelefteq G$ is abelian then $\chi(1)$ divides $|G : A|$. We shall not need these.

Example 40.9.12

This is a strong constraint. If $|G| = 12$, the degrees of the irreducibles divide 12 and their squares sum to 12, with at least one degree equal to 1. The possible degree multisets are $\{1^{12}\}$, $\{1^4, 2^2\}$ and $\{1^3, 3\}$ – and indeed C_{12} , D_{12} and A_4 realise these.

§40.9.4 Burnside's Lemma on Vanishing

Lemma 40.9.13

Let χ be an irreducible character of G of degree d , and let $C = \text{ccl}_G(g)$ be a class with $\gcd(|C|, d) = 1$. Then either $\chi(g) = 0$ or $|\chi(g)| = d$ (equivalently, $\rho(g)$ is a scalar).

Proof. Choose $a, b \in \mathbb{Z}$ with $a|C| + bd = 1$. Then

$$\frac{\chi(g)}{d} = \frac{a|C|\chi(g) + bd\chi(g)}{d} = a\omega_\chi(C) + b\chi(g)$$

is an algebraic integer; call it α .

Now $\chi(g)$ is a sum of d roots of unity, all lying in $\mathbb{Q}(\zeta)$ where $\zeta = e^{2\pi i/n}$ and n is

the order of g . Let f be the minimal polynomial of α over $\mathbb{Q}$; since α is an algebraic integer, f is monic with integer coefficients. Each root of f is $\sigma(\alpha)$ for some field embedding $\sigma : \mathbb{Q}(\alpha) \rightarrow \mathbb{C}$, and each such σ extends to $\mathbb{Q}(\zeta)$, where it must send $\zeta \mapsto \zeta^k$ for some k coprime to n . Hence

$$\sigma(\alpha) = \frac{1}{d}(\lambda_1^k + \cdots + \lambda_d^k)$$

is again an average of d roots of unity, so $|\sigma(\alpha)| \leq 1$ by the triangle inequality.

The product of all roots of f is $\pm f(0) \in \mathbb{Z}$, and has modulus at most 1 with equality only if every root has modulus exactly 1. So either some root has modulus < 1 , in which case $|f(0)| < 1$ and hence $f(0) = 0$, forcing $\alpha = 0$ (as f is irreducible), or $|\alpha| = 1$.

In the first case $\chi(g) = 0$. In the second $|\chi(g)| = d$, and we showed earlier that this forces $\rho(g)$ to be a scalar matrix. $\square$

Theorem 40.9.14 (Burnside)

Let G be a finite group with a conjugacy class of size p^k for some prime p and some $k \geq 1$. Then G is not simple.

Proof. Let $g \neq e$ with $|\text{ccl}_G(g)| = p^k$, $k \geq 1$, and let $\chi_1 = \mathbf{1}, \chi_2, \dots, \chi_r$ be the irreducible characters, of degrees d_i . Column orthogonality between the columns of e and of g gives

$$0 = \sum_{i=1}^r d_i \chi_i(g) = 1 + \sum_{i=2}^r d_i \chi_i(g).$$

Suppose, for contradiction, that for every $i \geq 2$ either $p \mid d_i$ or $\chi_i(g) = 0$. Then

$$1 = - \sum_{\substack{i \geq 2 \\ p \nmid d_i}} d_i \chi_i(g) = -p \sum_{\substack{i \geq 2 \\ p \nmid d_i}} \frac{d_i}{p} \chi_i(g),$$

so $\frac{1}{p} = - \sum (d_i/p) \chi_i(g)$ is an algebraic integer. But $\frac{1}{p} \in \mathbb{Q} \setminus \mathbb{Z}$, contradicting $\mathbb{A} \cap \mathbb{Q} = \mathbb{Z}$.

So there is some $i \geq 2$ with $p \nmid d_i$ and $\chi_i(g) \neq 0$. Then $\gcd(d_i, p^k) = 1$, so by the lemma $\rho_i(g)$ is a scalar. Let

$$N = \ker \rho_i, \quad Z = \{x \in G : \rho_i(x) \text{ is a scalar}\}.$$

Both are normal subgroups of G (the second is the preimage of the centre of $\rho_i(G)$, which is normal in $\rho_i(G)$). If $N \neq 1$ then, since $\chi_i \neq \mathbf{1}$ gives $N \neq G$, the group G is not simple. So assume $N = 1$, i.e. ρ_i is faithful. Then Z contains $g \neq e$, so $Z \neq 1$. And $Z \neq G$: otherwise $\rho_i(G)$ would consist of scalars, hence be abelian, hence (by faithfulness) G would be abelian and every class would have size 1, contradicting $|\text{ccl}(g)| = p^k > 1$. So Z is a proper nontrivial normal subgroup. $\square$

§40.9.5 Burnside's $p^a q^b$ Theorem

Theorem 40.9.15 (Burnside, 1904)

Let p, q be primes and $a, b \geq 0$. Then every group of order $p^a q^b$ is soluble.

Proof. Induct on $|G|$. The case $|G| = 1$ is trivial.

If $a = 0$ or $b = 0$ then G is a p -group. Every nontrivial finite p -group has nontrivial centre (by the class equation), so $Z(G) \neq 1$; then $Z(G)$ is abelian and $G/Z(G)$ is a smaller p -group, soluble by induction, so G is soluble.

So assume $a, b \geq 1$. Let Q be a Sylow q -subgroup of G , which is nontrivial, and choose $g \in Z(Q)$ with $g \neq e$ (possible as $Z(Q) \neq 1$). Then $Q \leq C_G(g)$, so $|G : C_G(g)|$ divides $|G|/|Q| = p^a$. Hence $|\text{ccl}_G(g)| = p^k$ for some $0 \leq k \leq a$.

If $k = 0$ then $g \in Z(G)$, so $Z(G) \neq 1$; then $Z(G)$ and $G/Z(G)$ both have order of the form $p^{a'} q^{b'}$ and are smaller than G , so both are soluble by induction, hence so is G .

If $k \geq 1$ then by the previous theorem G is not simple, so there is $1 < N \trianglelefteq G$ with $N \neq G$. Both N and G/N have order dividing $p^a q^b$, hence of the form $p^{a'} q^{b'}$, and both are smaller than G ; so both are soluble by induction, and therefore G is soluble. $\square$

Remark. It is worth pausing on what has happened. The statement of Burnside's theorem mentions only groups and their orders; no representation appears anywhere. Yet the only known proofs for over sixty years went through character theory, and the character-free proof eventually found (by Bender, Goldschmidt and Matsuyama in the 1970s) is substantially harder. This is the standard advertisement for representation theory, and it is a fair one.

Remark. The bound is sharp in the sense that three primes are not enough: $|A_5| = 60 = 2^2 \cdot 3 \cdot 5$ and A_5 is not soluble.

§40.10 Representations of the Symmetric Group

The symmetric groups deserve a section of their own. They are the groups for which the general theory works out most completely: there is a purely combinatorial parametrisation of the irreducible representations, an explicit construction of each, and even a closed formula for the degrees. Everything is indexed by *partitions*.

§40.10.1 Partitions and Conjugacy Classes

Definition 40.10.1 (Partition)

A **partition** of n is a sequence $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_k)$ of integers with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k \geq 1$ and $\sum_i \lambda_i = n$. We write $\lambda \vdash n$, and let $p(n)$ be the number of partitions of n .

Proposition 40.10.2

The conjugacy classes of S_n are in bijection with the partitions of n , via cycle type. Consequently S_n has exactly $p(n)$ irreducible complex representations.

Proof. Every permutation is a product of disjoint cycles, and the multiset of cycle lengths (including fixed points as 1-cycles) is a partition of n . Conjugation relabels: if σ has cycle $(a_1 a_2 \cdots a_m)$ then $\tau\sigma\tau^{-1}$ has cycle $(\tau(a_1) \cdots \tau(a_m))$. So conjugate permutations have the same cycle type, and conversely two permutations of the same cycle type are conjugated by any τ matching up their cycles. The last sentence is the theorem that the number of irreducibles equals the number of classes. $\square$

Remark. The class of cycle type $1^{m_1} 2^{m_2} \cdots n^{m_n}$ has size

$$\frac{n!}{\prod_{i=1}^n i^{m_i} m_i!}.$$

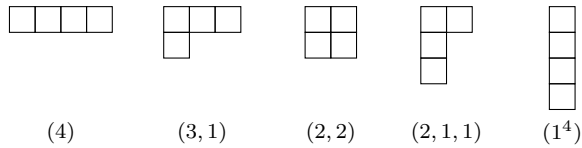
For example in S_5 the type $2^2 1$ has $\frac{120}{2^2 \cdot 2! \cdot 1!} = 15$ elements, agreeing with our table.

So we are looking for a family of irreducible representations indexed by partitions. The combinatorial device that makes this work is the Young diagram.

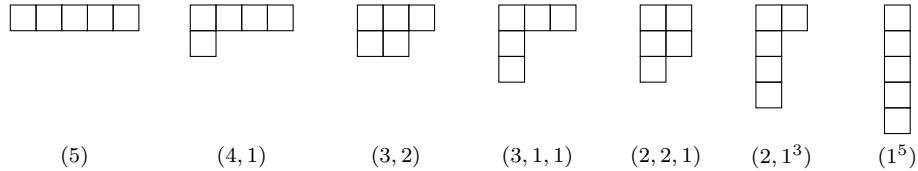
Definition 40.10.3 (Young diagram)

The **Young diagram** of $\lambda = (\lambda_1, \dots, \lambda_k) \vdash n$ is the left-justified array of n boxes with λ_i boxes in row i .

Here are all the partitions of 4:



and all the partitions of 5:



There are $p(4) = 5$ and $p(5) = 7$ of them, matching the five irreducible characters of S_4 and the seven of S_5 that we computed by hand.

Definition 40.10.4 (Conjugate partition)

The **conjugate** (or **transpose**) partition λ' is obtained by reflecting the Young diagram in its main diagonal; that is, $\lambda'_j = \#\{i : \lambda_i \geq j\}$.

For instance $(3, 1)' = (2, 1, 1)$ and $(4, 1)' = (2, 1, 1, 1)$, while $(2, 2)' = (2, 2)$ and $(3, 1, 1)' = (3, 1, 1)$ are **self-conjugate**. Compare this with the tables we computed: the irreducible characters of S_4 and S_5 came in pairs swapped by tensoring with sign, together with one or two fixed characters. That is exactly transposition of diagrams, as we now explain.

§40.10.2 Tableaux, Tabloids and Specht Modules

Definition 40.10.5 (Young tableau)

A **(λ -)tableau** is a filling of the Young diagram of $\lambda \vdash n$ with the numbers $1, \dots, n$, each used once. Two tableaux are **row equivalent** if corresponding rows contain the same sets of entries; an equivalence class is a **tabloid**, written $\{t\}$.

The group S_n acts on tableaux by acting on the entries, and this descends to an action on tabloids.

Definition 40.10.6 (Permutation module)

M^λ is the permutation representation of S_n on the set of λ -tabloids.

Example 40.10.7

If $\lambda = (n)$ there is a single tabloid, and $M^{(n)}$ is the trivial representation. If $\lambda = (1^n)$ a tabloid is just an ordering of $1, \dots, n$, so $M^{(1^n)} = \mathbb{C}S_n$ is the regular representation. If $\lambda = (n-1, 1)$, a tabloid is determined by the entry in the second row, so $M^{(n-1,1)} = \mathbb{C}^n$ is the natural permutation representation.

These modules are far from irreducible, but each contains a distinguished irreducible submodule. Fix a λ -tableau t , and let

$$R_t = \{\sigma : \sigma \text{ preserves each row of } t\}, \quad C_t = \{\sigma : \sigma \text{ preserves each column of } t\}$$

be its row and column groups.

Definition 40.10.8 (Young symmetriser and Specht module)

Define the **signed column sum** $b_t = \sum_{\sigma \in C_t} \text{sign}(\sigma) \sigma \in \mathbb{C}S_n$, and the **polytabloid**

$$e_t = b_t \cdot \{t\} \in M^\lambda.$$

The **Specht module** is

$$S^\lambda = \mathbb{C}S_n\text{-span of } \{e_t : t \text{ a } \lambda\text{-tableau}\} \leq M^\lambda.$$

Since $\sigma e_t = e_{\sigma t}$, the span of the polytabloids is automatically a submodule.

Theorem 40.10.9 (James)

The Specht modules S^λ , for $\lambda \vdash n$, are irreducible, pairwise non-isomorphic, and every irreducible complex representation of S_n is isomorphic to exactly one of them. Moreover

$$S^{\lambda'} \cong S^\lambda \otimes \text{sign}.$$

We do not prove this here – the proof rests on a combinatorial lemma about tableaux (‘if $\lambda \neq \mu$ and $b_t \cdot \{s\} \neq 0$ then λ dominates μ ’) together with an application of Schur’s lemma. What matters for us is that the parametrisation exists and that we can compute with it. In particular, since transposing partitions is an involution on $\{\lambda \vdash n\}$ with fixed points the self-conjugate partitions, the last statement explains the pairing we observed.

Proposition 40.10.10 (Hook length formula, Frame–Robinson–Thrall)

For a box (i, j) of the diagram of λ , define its **hook length** $h(i, j)$ to be 1 plus the number of boxes strictly to its right in row i plus the number strictly below it in column j . Then

$$\dim S^\lambda = \frac{n!}{\prod_{(i,j) \in \lambda} h(i, j)}.$$

Example 40.10.11 (The irreducibles of S_4)

Let us compute the degrees. For $\lambda = (3, 1)$ the hook lengths are

4	2	1
1		

so $\dim S^{(3,1)} = 24/(4 \cdot 2 \cdot 1 \cdot 1) = 3$. Similarly $\dim S^{(2,2)} = 24/(3 \cdot 2 \cdot 2 \cdot 1) = 2$, and $\dim S^{(4)} = \dim S^{(1^4)} = 1$, with $\dim S^{(2,1,1)} = \dim S^{(3,1)} = 3$ by transposition. So the degrees are

$$1, 3, 2, 3, 1 \quad \text{with} \quad 1 + 9 + 4 + 9 + 1 = 24 = |S_4|,$$

exactly the degrees we found. Matching up: $S^{(4)} = \mathbf{1}$, $S^{(1^4)} = \text{sign}$, $S^{(3,1)}$ is the standard representation, $S^{(2,1,1)} = \text{standard} \otimes \text{sign}$, and $S^{(2,2)}$ is the degree-2 representation lifted from $S_4/V_4 \cong S_3$.

Example 40.10.12 (The irreducibles of S_5)

The hook length formula gives

$$\dim S^{(5)} = 1, \quad \dim S^{(4,1)} = \frac{120}{5 \cdot 3 \cdot 2 \cdot 1 \cdot 1} = 4,$$

$$\dim S^{(3,2)} = \frac{120}{4 \cdot 3 \cdot 1 \cdot 2 \cdot 1} = 5, \quad \dim S^{(3,1,1)} = \frac{120}{5 \cdot 2 \cdot 1 \cdot 2 \cdot 1} = 6,$$

and the remaining three follow by transposition: $\dim S^{(2,2,1)} = 5$, $\dim S^{(2,1^3)} = 4$ and $\dim S^{(1^5)} = 1$. The squares sum to

$$1 + 16 + 25 + 36 + 25 + 16 + 1 = 120 = |S_5|. \quad \checkmark$$

Comparing with our computed table: $\lambda = (4, 1)$ gives χ_3 (degree 4), $(2, 1^3)$ gives $\chi_4 = \text{sign} \cdot \chi_3$; $(3, 2)$ gives χ_6 (degree 5) and $(2, 2, 1)$ gives χ_7 ; and $(3, 1, 1)$, which is self-conjugate, gives the degree-6 character $\chi_5 = \Lambda^2 \chi_3$. That is exactly the pattern of associate pairs and one self-associate character that we noticed at the time.

Remark. There is also a combinatorial rule – the Murnaghan–Nakayama rule – computing the value of χ^λ on a permutation of cycle type μ , by removing ‘rim hooks’ from the diagram of λ . It is the natural generalisation of the recursion one finds by hand for small n . The interested reader will find all of this in James’ book *The Representation Theory of the Symmetric Groups*; for this course, the existence of the parametrisation and the hook length formula are the takeaway.

§40.11 Representations of Topological Groups

Everything so far has used the finiteness of G in one essential place: the averaging operator $\frac{1}{|G|} \sum_{g \in G}$. If we can replace that sum by an integral, the entire theory goes through. This is possible exactly when G is a compact topological group, and the resulting theory is one of the most useful things in mathematics – it is, among other things, where the quantum mechanics of angular momentum comes from.

§40.11.1 Topological Groups and Compactness

Definition 40.11.1 (Topological group)

A **topological group** is a group G with a topology such that multiplication $G \times G \rightarrow G$ and inversion $G \rightarrow G$ are continuous.

Example 40.11.2 (i) Any group with the discrete topology; in particular any finite group.

(ii) $(\mathbb{R}, +)$, $(\mathbb{C}^\times, \times)$, and $\mathrm{GL}_n(\mathbb{C})$ with the topology inherited from $\mathbb{C}^{n^2}$.

(iii) The compact groups $\mathrm{O}(n)$, $\mathrm{SO}(n)$, $\mathrm{U}(n)$, $\mathrm{SU}(n)$, and the circle $S^1 = \{z \in \mathbb{C} : |z| = 1\} \cong \mathrm{U}(1)$.

Of these, the ones in (iii) together with the finite groups are compact; $\mathbb{R}$, $\mathbb{C}^\times$ and $\mathrm{GL}_n(\mathbb{C})$ are not.

Definition 40.11.3 (Continuous representation)

A **representation** of a topological group G is a continuous homomorphism $\rho : G \rightarrow \mathrm{GL}(V)$ for a finite dimensional complex vector space V , where $\mathrm{GL}(V)$ has its usual topology. (Equivalently, all the matrix entries $\rho(g)_{ij}$ are continuous functions of g .)

The continuity requirement is not a formality. The circle S^1 has uncountably many one-dimensional representations as an abstract group – it is a divisible abelian group, so has vast numbers of homomorphisms to $\mathbb{C}^\times$ – but only countably many continuous ones, as we shall see.

The replacement for averaging is the following.

Definition 40.11.4 (Haar measure)

Let G be a compact topological group. A **Haar integral** on G is a linear functional $f \mapsto \int_G f$ on the space $C(G, \mathbb{R})$ of continuous real functions such that

- (i) $\int_G 1 = 1$ (normalisation);
- (ii) $\int_G f(xg) \, dg = \int_G f(gx) \, dg = \int_G f(g) \, dg$ for all $x \in G$ (translation invariance);
- (iii) $\int_G f \geq 0$ whenever $f \geq 0$ (positivity).

Theorem 40.11.5

Every compact Hausdorff topological group admits a unique Haar integral.

We shall not prove this – it is a piece of analysis rather than algebra, and for the two groups we care about we can just write the integral down.

Example 40.11.6 (i) If G is finite, $\int_G f = \frac{1}{|G|} \sum_{g \in G} f(g)$.

(ii) If $G = S^1$, $\int_G f = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) d\theta$.

(iii) If $G = \mathrm{SU}(2)$, which we will see is the unit sphere $S^3 \subset \mathbb{R}^4$, the Haar integral is $\frac{1}{2\pi^2}$ times the usual surface integral over S^3 (and $2\pi^2$ is exactly the surface area of the unit 3-sphere).

With the Haar integral in hand, the finite theory transfers essentially verbatim. We record what survives.

Theorem 40.11.7

Let G be a compact Hausdorff topological group and let V, W be (continuous, finite dimensional, complex) representations of G . Then:

- (i) V admits a G -invariant Hermitian inner product; equivalently $\rho(G) \leq \mathrm{U}(V)$ in a suitable basis.
- (ii) (Maschke) Every subrepresentation of V has a G -invariant complement, so V is completely reducible.
- (iii) (Schur) $\dim \mathrm{Hom}_G(V, W) = 1$ if $V \cong W$ are irreducible and 0 if they are non-isomorphic irreducibles.
- (iv) (Orthogonality) With $\langle f, f' \rangle = \int_G \overline{f(g)} f'(g) dg$, the irreducible characters are orthonormal; a representation is irreducible if and only if $\langle \chi, \chi \rangle = 1$; and characters determine representations up to isomorphism.

Proof. For (i), start from any inner product $(\cdot, \cdot)$ and set $\langle v, w \rangle = \int_G (gv, gw) dg$, which converges since $g \mapsto (gv, gw)$ is continuous on a compact space; invariance is property (ii) of the Haar integral, and positive definiteness follows from (iii) together with $\int_G 1 = 1$. Then (ii) follows by taking orthogonal complements, exactly as in the finite case. The proof of (iii) never used finiteness, only that V is a finite dimensional complex vector space.

For (iv), the proof of orthogonality goes through word for word: the map $\pi = \int_G \rho(g) dg$ is a projection onto V^G with $\dim V^G = \int_G \chi_V$, the space $\mathrm{Hom}_{\mathbb{C}}(V, W)$ is a representation with $\mathrm{Hom}_{\mathbb{C}}(V, W)^G = \mathrm{Hom}_G(V, W)$ and character $\overline{\chi_V} \chi_W$, and combining gives $\langle \chi_V, \chi_W \rangle = \dim \mathrm{Hom}_G(V, W)$. Schur finishes it. $\square$

Remark. One thing does *not* transfer: a compact group generally has infinitely many irreducible representations, so there is no finite ‘character table’ and no statement that $\sum d_i^2 = |G|$. The Peter–Weyl theorem is the correct replacement: the matrix coefficients of the irreducible representations span a dense subspace of $C(G)$, and $L^2(G)$ decomposes as $\widehat{\bigoplus_i V_i^{\oplus d_i}}$. For $G = S^1$ this is precisely the theory of Fourier series.

§40.11.2 Representations of the Circle

Theorem 40.11.8

Every irreducible representation of S^1 is one dimensional, of the form $\rho_n : z \mapsto z^n$ for a unique $n \in \mathbb{Z}$.

Proof. Since S^1 is abelian, Schur's lemma gives that every irreducible representation is one dimensional (the argument is identical to the finite case). So let $\rho : S^1 \rightarrow \mathbb{C}^\times$ be a continuous homomorphism. The image $\rho(S^1)$ is a compact subgroup of $\mathbb{C}^\times$, hence bounded, and closed under taking powers; a complex number λ with $|\lambda| \neq 1$ has $|\lambda^n| \rightarrow \infty$ or $\rightarrow 0$, so $\rho(S^1) \subseteq S^1$.

Now consider $\psi : \mathbb{R} \rightarrow S^1$, $\psi(x) = \rho(e^{2\pi i x})$, a continuous homomorphism from $(\mathbb{R}, +)$. Lifting through the covering $\mathbb{R} \rightarrow S^1$ (or, concretely, choosing a continuous branch of $\frac{1}{2\pi i} \log \psi$ near 0 and extending by additivity) gives a continuous homomorphism $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ with $\psi(x) = e^{2\pi i \alpha(x)}$. A continuous additive map $\mathbb{R} \rightarrow \mathbb{R}$ is $x \mapsto \lambda x$: indeed $\alpha(q) = q\alpha(1)$ for $q \in \mathbb{Q}$ by additivity, and continuity extends this to $\mathbb{R}$. Finally $\psi(1) = \rho(1) = 1$ forces $\lambda = \alpha(1) \in \mathbb{Z}$, and then $\rho(z) = z^\lambda$. $\square$

So the characters of S^1 are $\chi_n(e^{i\theta}) = e^{in\theta}$, and orthonormality reads

$$\frac{1}{2\pi} \int_0^{2\pi} e^{-im\theta} e^{in\theta} d\theta = \delta_{mn},$$

which the reader will recognise. Decomposing a representation of S^1 into irreducibles is decomposing a function into its Fourier modes; the Peter–Weyl theorem for S^1 is the completeness of the Fourier basis in $L^2(S^1)$.

§40.11.3 The Group $\mathrm{SU}(2)$

Definition 40.11.9

$\mathrm{SU}(2) = \{A \in \mathrm{GL}_2(\mathbb{C}) : \bar{A}^T A = I, \det A = 1\}$.

Proposition 40.11.10

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix} : a, b \in \mathbb{C}, |a|^2 + |b|^2 = 1 \right\},$$

and hence $\mathrm{SU}(2)$ is homeomorphic to the unit sphere $S^3 \subseteq \mathbb{C}^2 = \mathbb{R}^4$. In particular it is compact and connected.

Proof. If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has determinant 1 then $A^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, and unitarity says $A^{-1} = \bar{A}^T = \begin{pmatrix} \bar{a} & \bar{c} \\ \bar{b} & \bar{d} \end{pmatrix}$. Comparing entries gives $d = \bar{a}$ and $c = -\bar{b}$, and then $\det A = |a|^2 + |b|^2 = 1$. Conversely all such matrices are in $\mathrm{SU}(2)$. $\square$

Remark. Equivalently, $\mathrm{SU}(2)$ is the group of unit quaternions. Identify $\mathbb{H} = \mathbb{R}\langle 1, i, j, k \rangle$ with the set of all matrices $\begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}$, with no norm condition imposed; then the quaternion norm is $\det$, and the unit quaternions are exactly $\mathrm{SU}(2)$. This is why $Q_8 \leq \mathrm{SU}(2)$, and it is the source of the degree-2 representation of Q_8 we met earlier.

The key to everything is the **maximal torus**

$$T = \left\{ t_\theta := \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} : \theta \in \mathbb{R} \right\} \cong S^1 \leq \mathrm{SU}(2).$$

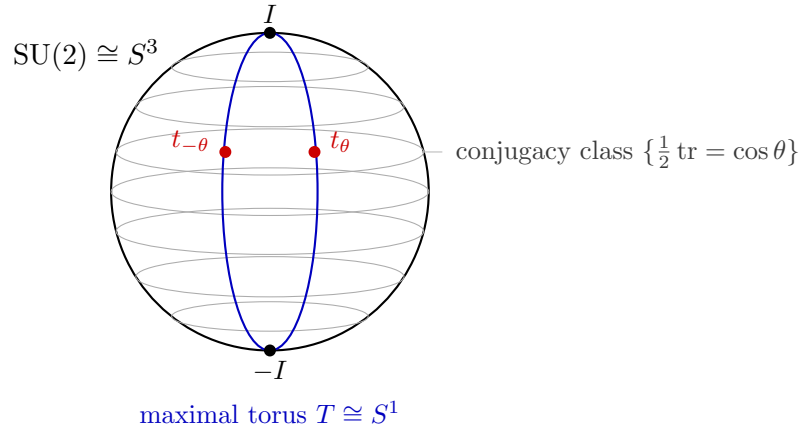
Proposition 40.11.11

Every element of $\mathrm{SU}(2)$ is conjugate to some $t_\theta \in T$, and t_θ is conjugate to $t_{\theta'}$ if and only if $\theta' \equiv \pm\theta$. Consequently the conjugacy classes of $\mathrm{SU}(2)$ are in bijection with $[-1, 1]$, via $A \mapsto \frac{1}{2} \mathrm{tr} A = \cos \theta$.

Proof. A unitary matrix is diagonalisable by a unitary change of basis (the spectral theorem from *Linear Algebra*), so $A = PDP^{-1}$ with $P \in \mathrm{U}(2)$ and D diagonal unitary. Since $\det A = 1$ the diagonal entries are $e^{\pm i\theta}$, so $D \in T$. Replacing P by $\lambda^{-1}P$ where $\lambda^2 = \det P$ gives $P \in \mathrm{SU}(2)$ without changing PDP^{-1} .

Now let $s = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SU}(2)$. A direct computation gives $st_\theta s^{-1} = t_{-\theta}$, so $t_\theta \sim t_{-\theta}$. Conversely conjugate matrices have the same eigenvalues, so $t_\theta \sim t_{\theta'}$ forces $\{e^{i\theta}, e^{-i\theta}\} = \{e^{i\theta'}, e^{-i\theta'}\}$. Finally $\mathrm{tr} t_\theta = 2 \cos \theta$ determines $\{\pm\theta\}$ modulo 2π . $\square$

The picture is worth having. Think of $\mathrm{SU}(2) = S^3$ as a sphere with poles I and $-I$: the conjugacy classes are the ‘latitudes’, which are honest 2-spheres of radius $\sqrt{1-x^2}$ where $x = \frac{1}{2} \mathrm{tr}$, degenerating to points at the poles. The maximal torus is a great circle through both poles, meeting each latitude in exactly the two points t_θ and $t_\theta^{-1} = t_{-\theta}$.



Corollary 40.11.12

A class function on $\mathrm{SU}(2)$ is determined by its restriction to T , and that restriction satisfies $f(t_\theta) = f(t_{-\theta})$.

Integration of class functions can be pushed down to T as well; the resulting formula is the simplest instance of the *Weyl integration formula*.

Proposition 40.11.13 (Weyl integration formula for $\mathrm{SU}(2)$)

For a continuous class function f on $\mathrm{SU}(2)$,

$$\int_{\mathrm{SU}(2)} f = \frac{2}{\pi} \int_0^\pi f(t_\theta) \sin^2 \theta \, d\theta.$$

Sketch. Under $\mathrm{SU}(2) \cong S^3$, the conjugacy class with $\frac{1}{2} \mathrm{tr} = \cos \theta$ is the 2-sphere of radius $\sin \theta$ obtained by intersecting S^3 with the hyperplane at height $\cos \theta$; its surface area is $4\pi \sin^2 \theta$. Slicing S^3 by these spheres and using $\int_{\mathrm{SU}(2)} f = \frac{1}{2\pi^2} \int_{S^3} f$,

$$\int_{\mathrm{SU}(2)} f = \frac{1}{2\pi^2} \int_0^\pi f(t_\theta) \cdot 4\pi \sin^2 \theta \, d\theta = \frac{2}{\pi} \int_0^\pi f(t_\theta) \sin^2 \theta \, d\theta. \quad \square$$

§40.11.4 The Irreducible Representations of $\mathrm{SU}(2)$

Definition 40.11.14

Let V_n be the space of homogeneous polynomials of degree n in two variables X, Y over $\mathbb{C}$:

$$V_n = \mathbb{C}\langle X^n, X^{n-1}Y, \dots, Y^n \rangle, \quad \dim V_n = n + 1.$$

Let $\mathrm{GL}_2(\mathbb{C})$ – and hence $\mathrm{SU}(2)$ – act by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot f(X, Y) = f(aX + cY, bX + dY).$$

One checks this is a (continuous, polynomial in the entries) representation. Note V_0 is the trivial representation, V_1 is the standard representation on $\mathbb{C}^2$, and in general $V_n \cong S^n V_1$.

Proposition 40.11.15

The character of V_n satisfies

$$\chi_n(t_\theta) = e^{in\theta} + e^{i(n-2)\theta} + \dots + e^{-in\theta} = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

Proof. With $z = e^{i\theta}$ we have $t_\theta \cdot X = zX$ and $t_\theta \cdot Y = z^{-1}Y$, so

$$t_\theta \cdot (X^j Y^{n-j}) = z^j z^{-(n-j)} X^j Y^{n-j} = z^{2j-n} X^j Y^{n-j}.$$

So the monomials are an eigenbasis with eigenvalues z^{2j-n} for $j = 0, \dots, n$, and summing the geometric series gives $\sum_{j=0}^n z^{2j-n} = \frac{z^{n+1} - z^{-(n+1)}}{z - z^{-1}}$, which is $\sin((n+1)\theta)/\sin \theta$. $\square$

Theorem 40.11.16

Each V_n is an irreducible representation of $\mathrm{SU}(2)$.

Proof. First, a direct computation:

$$\langle \chi_n, \chi_n \rangle = \frac{2}{\pi} \int_0^\pi \left| \frac{\sin((n+1)\theta)}{\sin \theta} \right|^2 \sin^2 \theta \, d\theta = \frac{2}{\pi} \int_0^\pi \sin^2((n+1)\theta) \, d\theta = \frac{2}{\pi} \cdot \frac{\pi}{2} = 1,$$

so V_n is irreducible.

It is instructive to see this without integration. Let $0 \neq W \leq V_n$ be $\mathrm{SU}(2)$ -invariant. Then W is T -invariant, and the monomials $X^j Y^{n-j}$ span the T -eigenspaces, which are one dimensional with *distinct* eigenvalues z^{2j-n} . Hence W is spanned by a subset of the monomials. Pick $X^j Y^{n-j} \in W$ and apply

$$g = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \in \mathrm{SU}(2),$$

which sends $X \mapsto \frac{1}{\sqrt{2}}(X+Y)$ and $Y \mapsto \frac{1}{\sqrt{2}}(-X+Y)$. Then

$$g \cdot (X^j Y^{n-j}) = 2^{-n/2} (X+Y)^j (-X+Y)^{n-j} \in W$$

has nonzero coefficient of X^n , namely $\pm 2^{-n/2}$; since W is spanned by monomials, $X^n \in W$. Applying g again, $2^{-n/2} (X+Y)^n \in W$ has *every* monomial coefficient nonzero, so every $X^j Y^{n-j}$ lies in W and $W = V_n$. $\square$

Theorem 40.11.17

Every irreducible representation of $\mathrm{SU}(2)$ is isomorphic to V_n for exactly one $n \geq 0$.

Proof. Let V be an irreducible representation with character χ . Restricting V to $T \cong S^1$ and decomposing into the characters $z \mapsto z^m$ of the circle, we get

$$\chi(t_\theta) = \sum_{m \in \mathbb{Z}} a_m z^m, \quad z = e^{i\theta},$$

a Laurent polynomial with $a_m \in \mathbb{Z}_{\geq 0}$ and only finitely many nonzero. Since t_θ and $t_{-\theta}$ are conjugate in $\mathrm{SU}(2)$, we have $a_m = a_{-m}$: the Laurent polynomial is symmetric under $z \mapsto z^{-1}$.

Now the functions $\chi_n|_T = z^n + z^{n-2} + \cdots + z^{-n}$, for $n \geq 0$, form a $\mathbb{Z}$ -basis of the group of symmetric Laurent polynomials with integer coefficients: they are ‘triangular’, in that χ_n has top term z^n with coefficient 1. So we may subtract off successively and write

$$\chi = \sum_{n \geq 0} c_n \chi_n, \quad c_n \in \mathbb{Z},$$

with only finitely many $c_n \neq 0$. But then $1 = \langle \chi, \chi \rangle = \sum_n c_n^2$ by orthonormality of the χ_n , so exactly one c_n is ± 1 and the rest are zero. As $\chi(1) > 0$ the sign is $+$, so $\chi = \chi_n$ and hence $V \cong V_n$. $\square$

§40.11.5 The Clebsch–Gordan Formula

Tensor products of the V_n decompose in a completely explicit way. This is the mathematics behind ‘adding angular momenta’ in quantum mechanics.

Theorem 40.11.18 (Clebsch–Gordan)For $m, n \geq 0$,

$$V_m \otimes V_n \cong V_{m+n} \oplus V_{m+n-2} \oplus \cdots \oplus V_{|m-n|}.$$

Proof. Assume without loss of generality that $m \leq n$. Work on T , writing $z = e^{i\theta}$, and clear the denominator: multiplying by $z - z^{-1}$,

$$\chi_m \chi_n \cdot (z - z^{-1}) = \left(\sum_{j=0}^m z^{m-2j} \right) (z^{n+1} - z^{-(n+1)}) = \sum_{j=0}^m z^{m+n+1-2j} - \sum_{j=0}^m z^{m-n-1-2j}.$$

On the other hand,

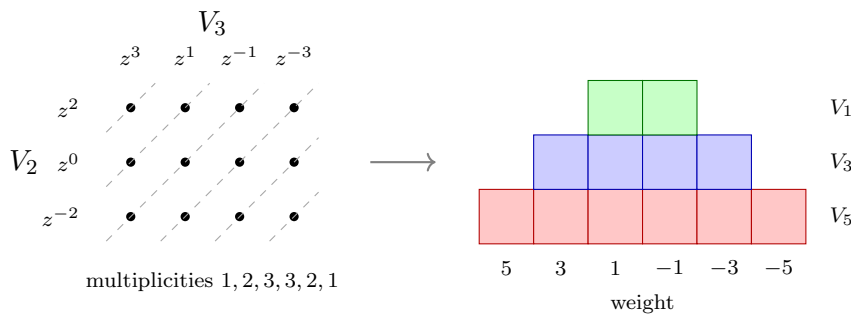
$$\left(\sum_{j=0}^m \chi_{m+n-2j} \right) (z - z^{-1}) = \sum_{j=0}^m z^{m+n+1-2j} - \sum_{j=0}^m z^{-(m+n+1-2j)}.$$

The first sums are literally equal. For the second sums, as j runs from 0 to m the exponents $m - n - 1 - 2j$ run through

$$m - n - 1, m - n - 3, \dots, -m - n - 1,$$

while the exponents $-(m+n+1-2j)$ run through the same list in the opposite order. So the two expressions agree, and since $z - z^{-1}$ is not identically zero we get $\chi_m \chi_n = \sum_{j=0}^m \chi_{m+n-2j}$ on T , hence everywhere. As characters determine representations, the decomposition follows. $\square$

The bookkeeping is clearest as a picture. The T -weights of $V_m \otimes V_n$ are the sums of a weight of V_m and a weight of V_n , so they form a rectangular grid, and the multiplicity of a given weight is the number of lattice points on the corresponding anti-diagonal. Reading off the multiplicity profile and peeling off one layer at a time gives the decomposition. For $V_2 \otimes V_3$:



so $V_2 \otimes V_3 \cong V_5 \oplus V_3 \oplus V_1$, as the theorem predicts.

Example 40.11.19

$V_1 \otimes V_1 \cong V_2 \oplus V_0$: the tensor square of the standard 2-dimensional representation splits as $S^2 V_1 \oplus \Lambda^2 V_1$, and indeed $S^2 V_1 = V_2$ while $\Lambda^2 V_1$ is one dimensional with trivial action (as $\det = 1$ on $\text{SU}(2)$).

§40.11.6 From $SU(2)$ to $SO(3)$

Theorem 40.11.20

There is a surjective continuous homomorphism $\varphi : SU(2) \rightarrow SO(3)$ with kernel $\{\pm I\}$, inducing an isomorphism of topological groups

$$SU(2)/\{\pm I\} \cong SO(3).$$

Proof. Let

$$\mathfrak{su}(2) = \{A \in M_2(\mathbb{C}) : \overline{A}^T = -A, \operatorname{tr} A = 0\} = \mathbb{R}\langle \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \rangle,$$

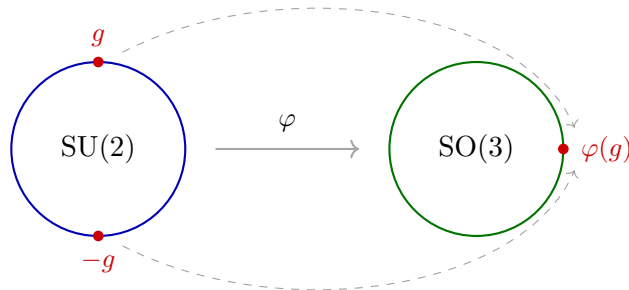
a real vector space of dimension 3. (These are the ‘purely imaginary quaternions’.) Equip it with the inner product $\langle A, B \rangle = \frac{1}{2} \operatorname{tr}(A\overline{B}^T)$, for which the displayed basis is orthonormal, and note $\langle A, A \rangle = \det A$ for $A \in \mathfrak{su}(2)$.

The group $SU(2)$ acts on $\mathfrak{su}(2)$ by conjugation, $\varphi(g)(A) = gAg^{-1}$; this preserves trace and adjoints, so preserves $\mathfrak{su}(2)$, and it preserves $\det$, hence the inner product. So $\varphi : SU(2) \rightarrow O(3)$ is a continuous homomorphism. Since $SU(2) \cong S^3$ is connected and $\varphi(I) = I$, the image lies in the identity component $SO(3)$.

The kernel consists of those g commuting with all of $\mathfrak{su}(2)$, hence with all of $M_2(\mathbb{C})$ (which $\mathfrak{su}(2)$ spans over $\mathbb{C}$ together with I), hence the scalars in $SU(2)$, which are $\pm I$.

For surjectivity, compute: conjugation by t_θ fixes $\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ and rotates the other two basis vectors by 2θ . So the image contains all rotations about one axis by all angles; and conjugating this one-parameter family by other elements of $SU(2)$ – whose image acts transitively on axes, since $SU(2)$ acts transitively on the unit sphere of $\mathfrak{su}(2)$ – gives all rotations. Hence φ is onto. The induced bijection $SU(2)/\{\pm I\} \rightarrow SO(3)$ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism. $\square$

So $SU(2)$ is a *double cover* of $SO(3)$: each rotation has exactly two preimages, $\pm g$. Concretely, the rotation by angle 2θ about a fixed axis is the image of both t_θ and $t_{\theta+\pi} = -t_\theta$; going once round a loop of rotations lifts to a path in $SU(2)$ from g to $-g$.



each rotation has exactly two preimages $\pm g$

Corollary 40.11.21

The irreducible representations of $\mathrm{SO}(3)$ are exactly the V_{2n} for $n \geq 0$, of odd dimensions $1, 3, 5, \dots$

Proof. By the lifting correspondence, representations of $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ are precisely representations of $\mathrm{SU}(2)$ with $-I$ in the kernel, and irreducibility is preserved. Now $-I = t_\pi$ acts on V_n with eigenvalues $(e^{i\pi})^{2j-n} = (-1)^n$, so $-I$ acts as $(-1)^n \mathrm{id}$. This is trivial exactly when n is even. $\square$

Remark. This is why in quantum mechanics the orbital angular momentum states come in multiplets of odd dimension $2\ell+1$ (representations of the rotation group $\mathrm{SO}(3)$), while spin- $\frac{1}{2}$ particles live in the even dimensional representations V_{odd} , which are representations of $\mathrm{SU}(2)$ but only *projective* representations of $\mathrm{SO}(3)$: rotating a spin- $\frac{1}{2}$ state by 2π multiplies it by -1 . The Clebsch–Gordan formula, in the physicists’ labelling $V_n \leftrightarrow \text{spin } j = n/2$, reads

$$j_1 \otimes j_2 = (j_1 + j_2) \oplus (j_1 + j_2 - 1) \oplus \cdots \oplus |j_1 - j_2|,$$

which is the rule for adding angular momenta.

Example 40.11.22 ($\mathrm{SO}(3)$ and $\mathrm{SU}(2)$ inside $\mathrm{U}(2)$)

One can push this further. Every element of $\mathrm{U}(2)$ is λA with $|\lambda| = 1$ and $A \in \mathrm{SU}(2)$, so there is a surjection $S^1 \times \mathrm{SU}(2) \rightarrow \mathrm{U}(2)$ with kernel $\{(1, I), (-1, -I)\}$. Since the irreducible representations of a product are the tensor products of irreducibles, the irreducible representations of $\mathrm{U}(2)$ are the $\rho_m \otimes V_n$ with $m \equiv n \pmod{2}$, i.e. $\det^k \otimes V_n$ for $k \in \mathbb{Z}$ and $n \geq 0$.

Remark (What was left out). We have taken the shortest path to $\mathrm{SU}(2)$. The general theory of compact connected Lie groups – maximal tori, roots and weights, the Weyl group, the Weyl character formula – has $\mathrm{SU}(2)$ as its first and smallest case, and every feature we used (the torus T , the two-element Weyl group generated by $z \mapsto z^{-1}$, the factor $\sin^2 \theta = |z - z^{-1}|^2/4$ in the integration formula, the triangular basis of characters) is an instance of a general phenomenon. That is the subject of a Part III course.

41 Riemann Surfaces

Complex analysis is full of functions that refuse to be functions. The logarithm is the obvious offender: $\exp$ is surjective onto $\mathbb{C}^*$ but wildly non-injective, so ‘the’ logarithm of z is only defined once we have made a choice, and no choice can be made consistently on all of $\mathbb{C}^*$. The same is true of $\sqrt{z}$, of $\sqrt{z^3 - z}$, and of the inverse of essentially any interesting holomorphic map. The traditional response is to slit the plane along a branch cut and work on what is left, but this is a confession of defeat: the cut is an artefact of our bookkeeping, not of the function, and the whole interest of $\log$ lies precisely in what happens when you walk around the origin and come back changed.

Riemann’s idea was to stop mutilating the domain and start enlarging it. If $\log$ wants to take infinitely many values at each point of $\mathbb{C}^*$, then give it a domain with infinitely many points lying over each point of $\mathbb{C}^*$ – an infinite spiral ramp – on which it becomes an honest single-valued holomorphic function. The price is that this new domain is no longer an open subset of $\mathbb{C}$; it is a space which merely *looks like* $\mathbb{C}$ near each of its points. That is a Riemann surface, and once we allow ourselves such objects a great deal falls into place. Multi-valued functions become genuine functions, branch cuts become invisible, and – most strikingly – the analysis starts to control the topology. We will prove the Riemann–Hurwitz formula, which says that a holomorphic map between compact surfaces forces a numerical relation between their genera and its branching, and we will use it to compute the genus of a surface from nothing more than the equation that cuts it out.

This article constitutes my notes for the ‘Riemann Surfaces’ course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§41.1 Holomorphic Functions Revisited

We begin by recalling, mostly without proof, what IB Complex Analysis gives us. Nothing in this section is new, but the results here are the local engine driving everything later on: every theorem we prove about Riemann surfaces will ultimately be a theorem about holomorphic functions on discs, transported by charts.

§41.1.1 Analytic Functions and their Zeros

Definition 41.1.1 (Domain)

A **domain** is a non-empty open connected subset $D \subseteq \mathbb{C}$.

Open discs $D(a, r) = \{z : |z - a| < r\}$, punctured discs $D(a, r) \setminus \{a\}$, annuli, half-planes and slit planes are all domains, and these are essentially the only ones we will ever need to write down explicitly.

Definition 41.1.2 (Holomorphic Function)

Let $D \subseteq \mathbb{C}$ be a domain. A function $f : D \rightarrow \mathbb{C}$ is **holomorphic** (or **analytic**) if it

is complex differentiable at every point of D .

The great theorem of IB Complex Analysis is that this weak-looking hypothesis is enormously strong: a holomorphic function is automatically infinitely differentiable, and is represented by its Taylor series on any disc contained in its domain. In particular ‘complex differentiable everywhere’ and ‘locally the sum of a convergent power series’ mean the same thing, and we shall use the two descriptions interchangeably.¹

The first consequence we need concerns zeros. A holomorphic function cannot vanish on a set with an accumulation point without vanishing identically – its zeros are isolated.

Proposition 41.1.3 (Isolated Zeros)

Let f be holomorphic on a domain D and suppose $f(z_0) = 0$. Then either f is identically zero on D , or there is a punctured disc about z_0 on which f does not vanish.

Proof. Write $f(z) = \sum_{n \geq 0} a_n(z - z_0)^n$ on some disc $U = D(z_0, r) \subseteq D$. If every a_n vanishes then $f \equiv 0$ on U , and a connectedness argument (the set where all derivatives of f vanish is open and closed in D) gives $f \equiv 0$ on all of D .

Otherwise let $m \geq 0$ be minimal with $a_m \neq 0$, so that on U we may write

$$f(z) = (z - z_0)^m g(z), \quad g(z) = \sum_{n \geq 0} a_{m+n}(z - z_0)^n.$$

The series for g converges on U , so g is holomorphic there, and $g(z_0) = a_m \neq 0$. By continuity g is non-vanishing on some disc $D(z_0, \delta)$, and therefore f is non-vanishing on $D(z_0, \delta) \setminus \{z_0\}$. $\square$

The integer m appearing in the proof is the **order** of the zero of f at z_0 , and it is worth noticing now – because the observation is the whole of Section 41.9 in disguise – that near such a point f looks like $z \mapsto z^m$ multiplied by a non-vanishing function.

Corollary 41.1.4 (Identity Theorem)

Let f, g be holomorphic on a domain D . If the set $\{z \in D : f(z) = g(z)\}$ has an accumulation point in D , then $f = g$ on D .

Proof. Apply Proposition 41.1.3 to $f - g$: the agreement set is not discrete, so $f - g$ cannot have isolated zeros, and hence vanishes identically. $\square$

This is the single most important fact in the course. It says a holomorphic function has no local freedom whatsoever: its germ at one point determines it everywhere on a connected domain. All of analytic continuation is an attempt to exploit this rigidity, and all of the trouble with analytic continuation comes from the word ‘connected’.

§41.1.2 Singularities and Meromorphic Functions

¹This is why the words ‘holomorphic’ and ‘analytic’ are used as synonyms throughout. In the real setting they are very much not synonyms.

Definition 41.1.5 (Isolated Singularity)

A function f holomorphic on a punctured disc $D(z_0, r) \setminus \{z_0\}$ is said to have an **isolated singularity** at z_0 .

On such a punctured disc f has a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n,$$

converging locally uniformly, and the singularity is classified by the tail of negative coefficients.

Definition 41.1.6 (Types of Singularity)

With notation as above, we say z_0 is

- (i) a **removable singularity** if $a_n = 0$ for all $n < 0$;
- (ii) a **pole of order** $m > 0$ if $a_{-m} \neq 0$ and $a_n = 0$ for all $n < -m$;
- (iii) an **essential singularity** otherwise.

A pole of order 1 is called **simple**.

Theorem 41.1.7 (Riemann Removable Singularity Theorem)

If f is holomorphic and bounded on a punctured disc about z_0 , then z_0 is a removable singularity, and f extends holomorphically over z_0 .

Theorem 41.1.8 (Casorati–Weierstrass)

The point z_0 is an essential singularity of f if and only if $f(U)$ is dense in $\mathbb{C}$ for every punctured neighbourhood U of z_0 .

Both were proved in IB Complex Analysis and we take them as given. The trichotomy is worth restating in the form we shall actually use: as $z \rightarrow z_0$, either $f(z)$ tends to a limit in $\mathbb{C}$ (removable), or $|f(z)| \rightarrow \infty$ (pole), or f oscillates so violently that it comes arbitrarily close to every complex number (essential). Poles are therefore not really singularities at all if we are willing to let ∞ count as a value, and that willingness is exactly what the Riemann sphere will formalise.

Definition 41.1.9 (Meromorphic Function)

Let D be a domain and $A \subseteq D$ a discrete closed subset. A holomorphic function $f : D \setminus A \rightarrow \mathbb{C}$ with a pole at each point of A is called **meromorphic** on D .

Example 41.1.10

The function

$$f(z) = \frac{1}{e^{1/z} - 1}$$

is meromorphic on the upper half-plane, with simple poles at the points $1/(2\pi in)$ for $n \in \mathbb{N}$. Note what happens if we try to regard f as meromorphic on all of $\mathbb{C}$: the poles accumulate at 0, so the set of poles is no longer discrete as a subset of $\mathbb{C}$, and indeed 0 is an essential singularity of f . The condition that A be discrete *and closed in D* is doing real work.

§41.1.3 The Open Mapping Theorem and the Maximum Modulus Principle

The last two facts we shall want are global in flavour.

Theorem 41.1.11 (Open Mapping Theorem)

A non-constant holomorphic function on a domain $D \subseteq \mathbb{C}$ is an open map.

Theorem 41.1.12 (Maximum Modulus Principle)

Let f be holomorphic on a domain D . If $|f|$ attains a local maximum at some point of D , then f is constant.

The second follows immediately from the first: if $f(D)$ is open then no point of $f(D)$ can be a point of maximum modulus of $f(D)$, since every neighbourhood of a point w contains points of larger modulus.

Finally we record the local statement that will become the definition of multiplicity.

Theorem 41.1.13 (Inverse Function Theorem)

Let f be holomorphic near z_0 with $f'(z_0) \neq 0$. Then there are open neighbourhoods $U \ni z_0$ and $V \ni f(z_0)$ such that $f|_U : U \rightarrow V$ is a biholomorphism, i.e. a holomorphic bijection with holomorphic inverse.

§41.2 Analytic Continuation

We now come to the problem that motivates the whole course. The identity theorem says that a holomorphic function on a domain is determined by its restriction to any small disc. Turning this around: given a holomorphic function on a small disc, can we recover ‘the’ function it came from, on some larger domain? And if we can do it in more than one way, how do the answers differ?

§41.2.1 Function Elements and Direct Continuation

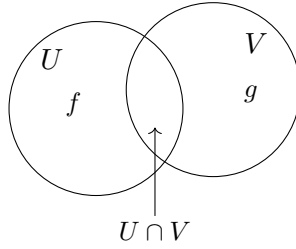
Definition 41.2.1 (Function Element)

Let $D \subseteq \mathbb{C}$ be a domain. A **function element** on D is a pair (f, U) where $U \subseteq D$ is a subdomain and $f : U \rightarrow \mathbb{C}$ is holomorphic.

Definition 41.2.2 (Direct Analytic Continuation)

Two function elements (f, U) and (g, V) on D are **direct analytic continuations**

of one another, written $(f, U) \sim (g, V)$, if $U \cap V \neq \emptyset$ and $f = g$ on $U \cap V$.



By the identity theorem, if $U \cap V$ is connected then g is completely determined by f (and by the choice of V): there is at most one holomorphic g on V agreeing with f on $U \cap V$. If $U \cap V$ is disconnected the definition still makes sense, but we should be careful, since agreeing on one component is then weaker than agreeing on all of them. We will normally arrange for our sets to be discs, so that intersections are convex and this issue evaporates.

The relation $\sim$ is reflexive and symmetric but emphatically *not* transitive – that failure is the entire subject. So we take its transitive closure.

Definition 41.2.3 (Analytic Continuation)

We write $(f, U) \approx (g, V)$ if there is a finite chain of function elements

$$(f, U) = (f_0, U_0) \sim (f_1, U_1) \sim \cdots \sim (f_n, U_n) = (g, V)$$

on D . We then say (g, V) is an **analytic continuation** of (f, U) .

Now $\approx$ is an equivalence relation, being the transitive closure of a reflexive symmetric relation.

Definition 41.2.4 (Complete Analytic Function)

An equivalence class $\mathcal{F}$ of function elements on D under $\approx$ is called a **complete analytic function** on D .

The name is a little grandiose for what is, at this stage, just a set of pairs. But the point of view is the right one: a ‘multi-valued function’ is not a function with several values, it is a *collection of ordinary functions, glued along their overlaps*. Our job over the next few sections is to make the gluing literal.

§41.2.2 The Complex Logarithm

Let us do the fundamental example completely explicitly. Write $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$, and recall that $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ is surjective with $\exp(z) = \exp(w)$ if and only if $z - w \in 2\pi i\mathbb{Z}$. We want to invert it.

For an interval $(\alpha, \beta) \subseteq \mathbb{R}$ with $\beta - \alpha \leq 2\pi$, set

$$U_{(\alpha, \beta)} = \{re^{i\theta} : r > 0, \alpha < \theta < \beta\},$$

an open sector, and define $f_{(\alpha, \beta)} : U_{(\alpha, \beta)} \rightarrow \mathbb{C}$ by

$$f_{(\alpha, \beta)}(re^{i\theta}) = \log r + i\theta, \quad \theta \in (\alpha, \beta).$$

This is well defined precisely because the angle θ of a point of $U_{(\alpha,\beta)}$ is unique in the range (α, β) , and it is holomorphic since it is a local inverse of $\exp$, which has non-vanishing derivative. So

$$F_{(\alpha,\beta)} = (f_{(\alpha,\beta)}, U_{(\alpha,\beta)})$$

is a function element on $\mathbb{C}^*$, and $\exp \circ f_{(\alpha,\beta)} = \text{id}$.

To keep the bookkeeping finite, restrict attention to quarter-turn sectors: for $n \in \mathbb{Z}$ put

$$I(n) = \left(\frac{(n-1)\pi}{2}, \frac{(n+1)\pi}{2} \right),$$

an interval of length π centred on $n\pi/2$, and abbreviate $U_n = U_{I(n)}$, $f_n = f_{I(n)}$, $F_n = F_{I(n)}$. Thus U_0 is the right half-plane, U_1 the upper half-plane, and so on; U_n depends only on n modulo 4, but f_n does not.

Proposition 41.2.5

$F_m \sim F_n$ if and only if $|m - n| \leq 1$.

Proof. First suppose $|m - n| \leq 1$. If $m = n$ there is nothing to do, so take $n = m + 1$. Then $U_m \cap U_{m+1}$ is the open sector of angles in $(m\pi/2, (m+1)\pi/2)$, which is non-empty, and for $z = re^{i\theta}$ in it we may take θ in that common range, whence $f_m(z) = \log r + i\theta = f_{m+1}(z)$. So $F_m \sim F_{m+1}$.

Conversely suppose $|m - n| \geq 2$. If $|m - n| \geq 4$ then $I(m)$ and $I(n)$ are disjoint modulo nothing at all – more precisely the sectors U_m, U_n may still meet, but any z in the intersection has $f_m(z) - f_n(z) \in 2\pi i\mathbb{Z} \setminus \{0\}$, since the two determinations of the argument differ by a non-zero multiple of 2π . If $|m - n| = 2$ then U_m and U_n are opposite half-planes meeting in two opposite quadrants, and $|m - n| = 3$ gives an intersection which is a single quadrant; in every case a point of the intersection has $f_m(z) \neq f_n(z)$, because the arguments used differ by a non-zero multiple of 2π . Hence $F_m \not\sim F_n$. $\square$

Corollary 41.2.6

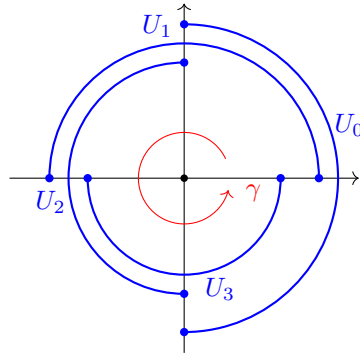
$F_m \approx F_n$ for all $m, n \in \mathbb{Z}$.

Proof. Chain along $F_m \sim F_{m\pm 1} \sim \cdots \sim F_n$. $\square$

So all the F_n lie in a single complete analytic function on $\mathbb{C}^*$, which we may reasonably call *the* complex logarithm. Notice what has happened: $U_0 = U_4$ as subsets of $\mathbb{C}$, but

$$f_0(1) = 0, \quad f_4(1) = 2\pi i.$$

Continuing once round the origin has changed the function. Analytic continuation is therefore *not* path-independent, and the complete analytic function $\log$ is genuinely bigger than any one of its elements.



Each U_n is a half-plane, drawn here as a semicircular arc; the four arcs are given different radii only so that they can be told apart, and the sector boundaries are the two coordinate axes. Together the $U_0, \dots, U_3$ cover $\mathbb{C}^*$, and each overlaps the next in a quadrant. Continuing f_0 around the loop γ takes us through F_1, F_2, F_3 and back to U_0 , but with $f_4 = f_0 + 2\pi i$.

§41.2.3 Complex Roots

Exactly the same analysis handles k -th roots. Let $p_k : \mathbb{C}^* \rightarrow \mathbb{C}^*$ be $z \mapsto z^k$ for $k \geq 2$, and define, on the same sectors U_n as above,

$$g_n(z) = \exp\left(\frac{1}{k} f_n(z)\right), \quad G_n = (g_n, U_n).$$

Then $g_n(z)^k = \exp(f_n(z)) = z$, so each g_n is a holomorphic branch of the k -th root on U_n .

Since $f_{n+4} = f_n + 2\pi i$ we get $g_{n+4} = e^{2\pi i/k} g_n$, and more generally g_n depends only on n modulo $4k$. Repeating the case analysis of Proposition 41.2.5 shows that $G_m \sim G_n$ if and only if $m \equiv n$ or $m \equiv n \pm 1$ modulo $4k$, and hence all the G_n lie in a single complete analytic function. This time, though, going round the origin *four* times returns us to $e^{2\pi i/k} g_0$, and it takes k circuits to get back to where we started. The multi-valuedness of $\sqrt[k]{\cdot}$ is finite, of order k , where that of $\log$ was infinite.

§41.2.4 Natural Boundaries

Before building surfaces, we pause on an obstruction: sometimes a function element admits no continuation at all past the boundary of its natural domain.

Write $\mathbb{D} = D(0, 1)$ and $\mathbb{T} = \partial\mathbb{D}$. Let $f(z) = \sum_{n \geq 0} a_n z^n$ be a power series with radius of convergence exactly 1, so that $(f, \mathbb{D})$ is a function element.

Definition 41.2.7 (Regular and Singular Points)

A point $z_0 \in \mathbb{T}$ is **regular** for f if there is an open neighbourhood U of z_0 and a function element (g, U) with $g = f$ on $U \cap \mathbb{D}$. Otherwise z_0 is **singular** for f .

The set of regular points is visibly open in $\mathbb{T}$, so the singular points form a closed set. It is tempting to think that z_0 is regular exactly when the power series converges at z_0 , and this is false in both directions.

Example 41.2.8

Take $f(z) = 1/(1-z) = \sum_{n \geq 0} z^n$. The only singular point is $z = 1$, so $z = -1$ is regular; but the power series $\sum (-1)^n$ does not converge at -1 . Convergence is not necessary for regularity.

Example 41.2.9

Take $g(z) = \sum_{n \geq 2} \frac{z^n}{n(n-1)}$. This converges absolutely at $z = 1$. But $g''(z) = \sum_{n \geq 2} z^{n-2} = 1/(1-z)$, which is singular at 1; so g cannot be continued past 1 either, and 1 is singular for g . Convergence is not sufficient for regularity.

Proposition 41.2.10

If $f(z) = \sum_{n \geq 0} a_n z^n$ has radius of convergence exactly 1, then some point of $\mathbb{T}$ is singular for f .

Proof. Suppose not. Then every $z_0 \in \mathbb{T}$ has a disc $D(z_0, \varepsilon_{z_0})$ carrying a continuation of f . By compactness finitely many such discs cover $\mathbb{T}$; by the identity theorem (the intersections being convex, hence connected) these continuations agree with each other and with f on overlaps, and so patch together with f to give a holomorphic function on an open set containing $\overline{\mathbb{D}}$. That set contains $D(0, 1 + \delta)$ for some $\delta > 0$, again by compactness, and so the Taylor series of f at 0 would have radius of convergence at least $1 + \delta$, a contradiction. $\square$

Definition 41.2.11 (Natural Boundary)

If every point of $\mathbb{T}$ is singular for f , we call $\mathbb{T}$ the **natural boundary** of f .

Example 41.2.12 (A Lacunary Series)

Let $f(z) = \sum_{n \geq 0} z^{n!}$, which has radius of convergence 1. We claim $\mathbb{T}$ is its natural boundary.

Since the singular set is closed, it suffices to show that the roots of unity are all singular, as these are dense in $\mathbb{T}$. So let $\omega = e^{2\pi i p/q}$ with $p, q \in \mathbb{Z}$, $q \geq 1$. For $n \geq q$ we have $q \mid n!$ and so $\omega^{n!} = 1$. Hence for $r \in (0, 1)$,

$$f(r\omega) = \sum_{n=0}^{q-1} r^{n!} \omega^{n!} + \sum_{n \geq q} r^{n!}.$$

The first sum is a finite sum of terms of modulus at most 1, hence bounded by q . For the second, given any $M > 0$ we have

$$\sum_{n \geq q} r^{n!} \geq \sum_{n=q}^{q+M} r^{n!} \longrightarrow M + 1$$

as $r \rightarrow 1^-$, so $\sum_{n \geq q} r^{n!} > M$ for r close enough to 1. Therefore $f(r\omega) \rightarrow \infty$ as

$r \rightarrow 1^-$. A function element (g, U) agreeing with f near ω would be bounded near ω , so no such element exists and ω is singular.

So continuation can fail entirely. When it does not fail, we would like to understand how the answer depends on the route taken – which requires us to say what a route is.

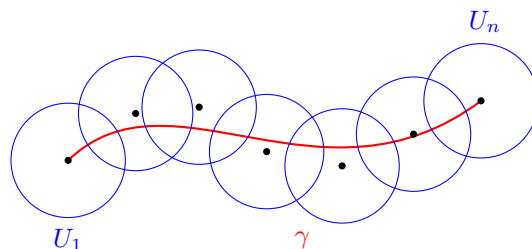
§41.2.5 Continuation Along a Path

Definition 41.2.13 (Continuation Along a Path)

Let $\gamma : [0, 1] \rightarrow D$ be a path and let $(f, U), (g, V)$ be function elements on D with $\gamma(0) \in U$ and $\gamma(1) \in V$. We write $(f, U) \approx_\gamma (g, V)$ if there is a dissection $0 = t_0 < t_1 < \dots < t_n = 1$ and function elements

$$(f, U) = (f_1, U_1), (f_2, U_2), \dots, (f_n, U_n) = (g, V)$$

with $(f_i, U_i) \sim (f_{i+1}, U_{i+1})$ for each i , and $\gamma([t_{i-1}, t_i]) \subseteq U_i$ for each i . We then say (f, U) can be **continued along** γ , with result (g, V) .



The picture is a chain of discs strung along γ , each carrying a holomorphic function agreeing with its neighbours. Definition 41.2.13 is a little unwieldy, and we will replace it in Section 41.7 by something much cleaner: continuation along γ will simply be *lifting* γ to the space of germs. Once we have said that, the key uniqueness statement – the monodromy theorem, that the result of continuing along γ depends only on the homotopy class of γ – becomes a special case of the homotopy lifting property from IB/II Algebraic Topology.

§41.3 The Surfaces of $\log$ and $\sqrt[k]{z}$

Before setting up the general theory, let us build the two surfaces the theory is meant to explain. This is worth doing by hand: everything we do abstractly later is modelled on these two constructions.

§41.3.1 Gluing the Sectors

Recall from Section 41.2.2 the function elements $F_n = (f_n, U_n)$ for $\log$. The trouble was that U_0 and U_4 are the same subset of $\mathbb{C}$ but carry different functions. So let us refuse to identify them. Define

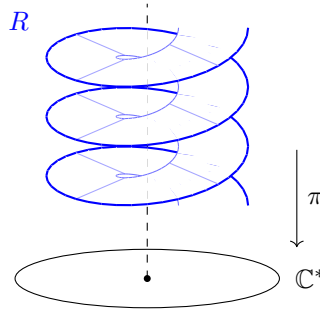
$$R = \left(\coprod_{n \in \mathbb{Z}} U_n \right) / \sim,$$

where a point z_1 in the copy U_m is identified with a point z_2 in the copy U_n if and only if

$$z_1 = z_2 \text{ as elements of } \mathbb{C} \quad \text{and} \quad f_m(z_1) = f_n(z_2),$$

and give R the quotient topology. In words: we lay out infinitely many copies of the sectors, and glue copy m to copy n exactly where the two branches of $\log$ agree. By Proposition 41.2.5 this happens exactly when $|m - n| \leq 1$, so each sector is glued to its two neighbours along a quadrant, and the result spirals.

The picture to have in mind is an infinite multi-storey car park: a helical ramp winding endlessly up and down over the punctured plane, with no floor ever meeting another except its immediate neighbours.



Two maps come along for free. First, the branches assemble into a single global function: define $f : R \rightarrow \mathbb{C}$ by $f([z]) = f_n(z)$ for $z \in U_n$. This is well defined precisely because we built the equivalence relation to make it so. Second, the inclusions $U_n \hookrightarrow \mathbb{C}^*$ assemble into $\pi : R \rightarrow \mathbb{C}^*$, $\pi([z]) = z$, which is well defined because equivalent points are equal in $\mathbb{C}$. These satisfy

$$\exp \circ f = \pi,$$

which says exactly that f deserves the name ‘logarithm’: applying $\exp$ to f recovers the point of $\mathbb{C}^*$ underneath.

Proposition 41.3.1

R is path-connected and Hausdorff.

Proof. Path-connectedness: each U_n is connected, and U_n meets U_{n+1} in R (as $F_n \sim F_{n+1}$), so the images of the U_n in R form a chain of overlapping connected sets covering R .

For Hausdorffness, consider $\Phi : R \rightarrow \mathbb{C}^2$ given by

$$\Phi([z]) = (\pi([z]), f([z])).$$

This is continuous, and it is injective: if $\Phi([z_1]) = \Phi([z_2])$ then $z_1 = z_2$ in $\mathbb{C}$ and the two branch values agree, which is precisely the condition defining $\sim$. A continuous injection into a Hausdorff space separates points, so R is Hausdorff. $\square$

Remark. In fact Φ identifies R with the graph

$$\{(w, z) \in \mathbb{C}^2 : w = \exp z\},$$

and under this identification π and f are the two coordinate projections. So the Riemann surface of $\log$ is nothing more exotic than the graph of $\exp$ with the roles of the axes

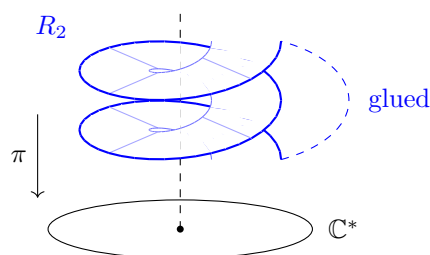
swapped. This is a useful sanity check, and a preview: in general, the Riemann surface of an algebraic function will be (a compactification of) the graph of the equation defining it.

§41.3.2 The Surface of the k -th Root

The same construction with the elements $G_n = (g_n, U_n)$ produces a space R_k . Since g_n depends only on n modulo $4k$, this time only $4k$ copies of the sectors are involved and the spiral closes up after k turns. We again get $g : R_k \rightarrow \mathbb{C}^*$ and $\pi : R_k \rightarrow \mathbb{C}^*$ with

$$p_k \circ g = \pi, \quad p_k(z) = z^k,$$

and, exactly as before, R_k is path-connected and Hausdorff. Here is the picture for $k = 2$: a two-sheeted spiral, closing after two turns.



In both cases g (respectively f) is the honest single-valued function we were after, and π is the map remembering which point of $\mathbb{C}^*$ we are sitting over. The key structural feature of π is that it is a *local homeomorphism*: each little piece of R maps homeomorphically to a piece of $\mathbb{C}^*$. That is what will let us transport the complex structure of $\mathbb{C}^*$ upstairs, and it is where we turn next.

Remark. One thing is unsatisfactory about R_k : the interesting point 0 is missing, and so is ∞ . We would like to add them, obtaining a map $\hat{p}_k : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ which is k -to-1 everywhere except at the two points $0, \infty$, where it is 1-to-1. These exceptional points are the **branch points**, and understanding them is the goal of Section 41.9. Once the points are added, π is no longer a covering map – but it is still holomorphic, and that turns out to be the more useful category.

§41.4 Riemann Surfaces

§41.4.1 Covering Maps

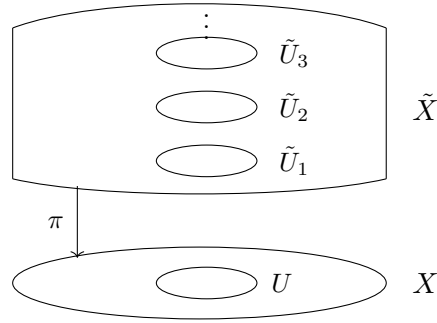
We isolate the topological property of $\pi : R \rightarrow \mathbb{C}^*$ that made the last section work. This material overlaps heavily with II Algebraic Topology; we repeat it because the conventions differ slightly and because we shall need to be careful about exactly which hypotheses are used where.

Definition 41.4.1 (Covering Map)

Let $\tilde{X}, X$ be path-connected Hausdorff spaces. A continuous surjection $\pi : \tilde{X} \rightarrow X$ is a **covering map** if it is a local homeomorphism: every $\tilde{x} \in \tilde{X}$ has an open neighbourhood $\tilde{U}$ such that $\pi|_{\tilde{U}}$ is a homeomorphism onto an open subset of X .

Definition 41.4.2 (Regular Covering Map)

A covering map $\pi : \tilde{X} \rightarrow X$ is **regular** if every $x \in X$ has an open neighbourhood U which is **evenly covered**: there is a discrete set Δ and a homeomorphism $\pi^{-1}(U) \cong U \times \Delta$ under which π becomes the projection $(u, \delta) \mapsto u$. Equivalently, $\pi^{-1}(U)$ is a disjoint union of open sets, the **sheets** over U , each mapped homeomorphically onto U by π .



Regularity is a genuinely stronger condition, and it is the one that matters for lifting paths. The distinction is easy to miss, so here are the examples we shall use.

Example 41.4.3

- (i) The map $\pi : R \rightarrow \mathbb{C}^*$ of Section 41.3 is a regular covering map. Indeed

$$\pi^{-1}(U_n) = \coprod_{m \equiv n \pmod{4}} U_m \cong U_n \times \mathbb{Z},$$

since the copy U_m maps homeomorphically to U_n exactly when $m \equiv n$ modulo 4, and distinct such copies are disjoint in R .

- (ii) $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ is a regular covering map. For an open interval $I \subseteq \mathbb{R}$ of length at most 2π , write $\tilde{V}_I = \mathbb{R} + iI$, a horizontal strip; then $\exp$ restricts to a homeomorphism $\tilde{V}_I \rightarrow U_I$ with inverse a branch of $\log$, and

$$\exp^{-1}(U_{I(n)}) = \coprod_{m \equiv n \pmod{4}} \tilde{V}_{I(m)} \cong U_{I(n)} \times \mathbb{Z}.$$

- (iii) $p_k : \mathbb{C}^* \rightarrow \mathbb{C}^*$, $z \mapsto z^k$, is a regular covering map with k sheets: over the sector U_I with I of length less than $2\pi/k$ sit the k sectors $\zeta^j U_{I/k}$, $\zeta = e^{2\pi i/k}$.
- (iv) (*Non-example.*) The inclusion $\iota : \mathbb{D} \hookrightarrow \mathbb{C}$ is a local homeomorphism, hence a covering map onto its image in the weak sense above, but it is not regular as a map to $\mathbb{C}$: for $z_0 \in \mathbb{T}$ and any neighbourhood U of z_0 , we have $\iota^{-1}(U) = U \cap \mathbb{D}$, whose image is not U . More basically, ι is not surjective. Regularity is what rules out this kind of ‘partial’ cover.

§41.4.2 Charts and Atlases

Now we say what a surface is. The idea is Riemann's: a space that looks locally like an open subset of $\mathbb{C}$, in a way that is compatible from patch to patch.

Definition 41.4.4 (Chart)

Let R be a topological space. A **chart** on R is a pair (ϕ, U) where $U \subseteq R$ is open and $\phi : U \rightarrow \phi(U) \subseteq \mathbb{C}$ is a homeomorphism onto an open subset of $\mathbb{C}$. We call ϕ a **local coordinate** on U .

Definition 41.4.5 (Atlas)

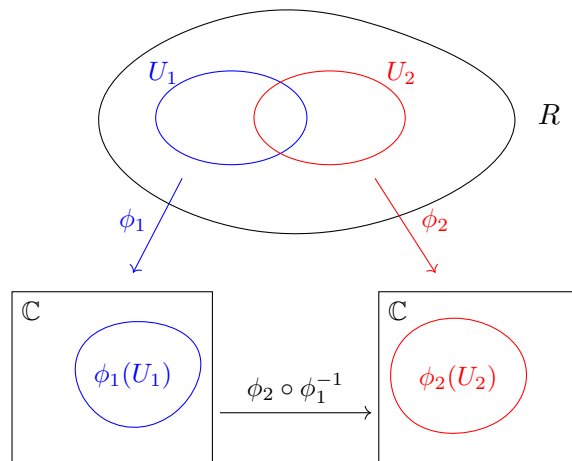
A collection $\mathcal{A}$ of charts on R is an **atlas** if

- (i) $\bigcup_{(\phi, U) \in \mathcal{A}} U = R$;
- (ii) for all $(\phi_1, U_1), (\phi_2, U_2) \in \mathcal{A}$ with $U_1 \cap U_2 \neq \emptyset$, the **transition function**

$$\phi_1 \circ \phi_2^{-1} : \phi_2(U_1 \cap U_2) \rightarrow \phi_1(U_1 \cap U_2)$$

is holomorphic.

Note that the transition function goes between open subsets of $\mathbb{C}$, so ‘holomorphic’ means what it always meant. Note also that its inverse is $\phi_2 \circ \phi_1^{-1}$, which is holomorphic by the same condition with the roles swapped; so transition functions are automatically biholomorphic.



There is an annoying redundancy here, which the next example exposes.

Example 41.4.6

Take $R = \mathbb{C}$. Let $\mathcal{A}$ consist of all pairs (id_U, U) for $U \subseteq \mathbb{C}$ open; this is an atlas. Let $\mathcal{A}'$ consist of all pairs $(z \mapsto z + 1, U)$; this too is an atlas. So is $\mathcal{A} \cup \mathcal{A}'$, since the transition functions are translations. Yet all three describe the same complex analysis on $\mathbb{C}$.

The cure is to take the largest possible atlas, so that no information is hidden in a choice.

Definition 41.4.7 (Conformal Structure)

A **conformal structure** on R is an atlas $\mathcal{A}$ which is **maximal**: whenever a chart (ψ, V) on R has the property that $\phi \circ \psi^{-1}$ is holomorphic (where defined) for every $(\phi, U) \in \mathcal{A}$, we have $(\psi, V) \in \mathcal{A}$.

Definition 41.4.8 (Riemann Surface)

A **Riemann surface** is a pair $(R, \mathcal{A})$ where R is a path-connected Hausdorff topological space and $\mathcal{A}$ is a conformal structure on R .

We will normally suppress $\mathcal{A}$ and just write R . Note that connectedness is part of the definition here – some authors allow disconnected Riemann surfaces, but for us a Riemann surface is connected, which makes the identity theorem available at all times.

Maximal atlases are unwieldy to write down, so in practice we always specify a small atlas and appeal to the following.

Lemma 41.4.9

Every atlas $\mathcal{A}$ on R is contained in a unique conformal structure $\hat{\mathcal{A}}$.

Proof. There is only one possible candidate: let $\hat{\mathcal{A}}$ be the set of all charts (ψ, V) on R such that $\psi \circ \phi^{-1}$ and $\phi \circ \psi^{-1}$ are holomorphic for every $(\phi, U) \in \mathcal{A}$. Any conformal structure containing $\mathcal{A}$ is contained in $\hat{\mathcal{A}}$ by definition of an atlas, and contains $\hat{\mathcal{A}}$ by maximality; so uniqueness is clear once we know $\hat{\mathcal{A}}$ is an atlas.

It covers R since $\mathcal{A} \subseteq \hat{\mathcal{A}}$ does. For compatibility, take $(\psi_1, V_1), (\psi_2, V_2) \in \hat{\mathcal{A}}$ and $p \in V_1 \cap V_2$. Choose $(\phi, U) \in \mathcal{A}$ with $p \in U$. Near $\psi_2(p)$ we may write

$$\psi_1 \circ \psi_2^{-1} = (\psi_1 \circ \phi^{-1}) \circ (\phi \circ \psi_2^{-1}),$$

a composite of holomorphic maps, hence holomorphic at $\psi_2(p)$. As p was arbitrary, $\psi_1 \circ \psi_2^{-1}$ is holomorphic on $\psi_2(V_1 \cap V_2)$. Maximality of $\hat{\mathcal{A}}$ is immediate from its definition. $\square$

So from now on, ‘here is an atlas’ means ‘here is a Riemann surface’.

§41.4.3 First Examples**Example 41.4.10 (The Plane)**

Taking $\mathcal{A} = \{(\text{id}, \mathbb{C})\}$ and extending gives the **canonical conformal structure** on $\mathbb{C}$. Whenever we speak of $\mathbb{C}$ as a Riemann surface, this is what is meant. Note that by Example 41.4.6 the atlas of all translations extends to the same structure.

Example 41.4.11 (The Conjugate Plane)

The single chart $(z \mapsto \bar{z}, \mathbb{C})$ is also an atlas on $\mathbb{C}$, and it does *not* extend to the canonical structure, since $z \mapsto \bar{z}$ is not holomorphic. Write $\bar{\mathbb{C}}$ for $\mathbb{C}$ with this structure. So the same topological space can carry different conformal structures.

(As it happens $\bar{\mathbb{C}} \cong \mathbb{C}$ as Riemann surfaces, via $z \mapsto \bar{z}$ – but they are not *equal*.)

Example 41.4.12 (Open Subsets)

If R is a Riemann surface with conformal structure $\mathcal{A}$ and $S \subseteq R$ is open and connected, then

$$\{(\phi|_{U \cap S}, U \cap S) : (\phi, U) \in \mathcal{A}\}$$

is an atlas on S , making S a Riemann surface. In particular every domain $D \subseteq \mathbb{C}$ – the disc $\mathbb{D}$, the upper half-plane $\mathbb{H}$, the punctured plane $\mathbb{C}^*$ – is a Riemann surface.

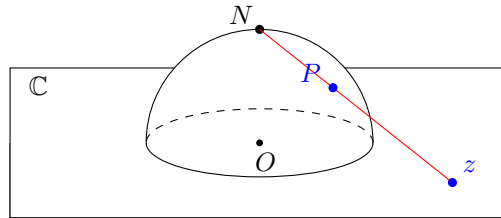
Example 41.4.13 (The Riemann Sphere)

Let $\mathbb{C}_\infty = \mathbb{C} \cup \{\infty\}$ with the topology in which neighbourhoods of ∞ are the complements of compact subsets of $\mathbb{C}$. Take the two charts

$$(\phi_0, U_0) = (\text{id}, \mathbb{C}), \quad (\phi_1, U_1) = (z \mapsto 1/z, \mathbb{C}^* \cup \{\infty\}),$$

where we read $1/\infty = 0$. Each is a homeomorphism onto $\mathbb{C}$, they cover $\mathbb{C}_\infty$, and on the overlap $U_0 \cap U_1 = \mathbb{C}^*$ the transition function is $z \mapsto 1/z$, which is holomorphic there. So this is an atlas, and $\mathbb{C}_\infty$ is a Riemann surface, the **Riemann sphere**.

That $\mathbb{C}_\infty$ deserves to be called a sphere is the content of stereographic projection. Identify $\mathbb{C}$ with the plane $\{x_3 = 0\}$ in $\mathbb{R}^3$ and let S^2 be the unit sphere, with north pole $N = (0, 0, 1)$. Each point $P \in S^2 \setminus \{N\}$ determines a unique point of the plane, namely where the line NP meets it, and this gives a homeomorphism $S^2 \setminus \{N\} \rightarrow \mathbb{C}$; sending $N \mapsto \infty$ extends it to a homeomorphism $S^2 \rightarrow \mathbb{C}_\infty$.



Chasing the geometry, if $P = (x_1, x_2, x_3)$ then the line $N + t(P - N)$ meets $x_3 = 0$ at $t = 1/(1 - x_3)$, giving

$$z = \frac{x_1 + ix_2}{1 - x_3}.$$

Projecting instead from the south pole $S = (0, 0, -1)$ gives $z' = (x_1 - ix_2)/(1 + x_3)$, and one checks $zz' = 1$ using $x_1^2 + x_2^2 + x_3^2 = 1$. So the two stereographic projections differ by $z \mapsto 1/z$ – exactly the transition function of Example 41.4.13. The atlas we wrote down is the geometrically natural one.

Remark. Under this identification, ‘ $z \rightarrow \infty$ ’ means ‘ $P \rightarrow N$ ’, so the point ∞ is in no way special: the sphere has a transitive group of conformal automorphisms (the Möbius transformations) and ∞ can be moved anywhere. That is the real payoff of the Riemann sphere as against the extended plane of a first course.

§41.4.4 Holomorphic Maps

Definition 41.4.14 (Holomorphic Map)

Let R, S be Riemann surfaces. A continuous map $f : R \rightarrow S$ is **holomorphic** (or **analytic**) if for every chart (ϕ, U) on R and every chart (ψ, V) on S , the map

$$\psi \circ f \circ \phi^{-1}$$

is holomorphic on $\phi(U \cap f^{-1}(V))$.

The composite $\psi \circ f \circ \phi^{-1}$ is called the **local expression** of f in the charts ϕ and ψ . Checking the condition for *all* charts is of course impossible in practice; happily it suffices to check one chart at each point.

Lemma 41.4.15

A continuous map $f : R \rightarrow S$ is holomorphic if and only if for each $p \in R$ there exist charts (ϕ_p, U_p) on R with $p \in U_p$ and (ψ_p, V_p) on S with $f(p) \in V_p$ such that $\psi_p \circ f \circ \phi_p^{-1}$ is holomorphic on $\phi_p(U_p \cap f^{-1}(V_p))$.

Proof. Necessity is trivial. For sufficiency, let (ϕ, U) and (ψ, V) be arbitrary charts; we must show $\psi \circ f \circ \phi^{-1}$ is holomorphic at $\phi(p)$ for each $p \in U \cap f^{-1}(V)$. Take $(\phi_p, U_p), (\psi_p, V_p)$ as hypothesised. On a small enough neighbourhood of $\phi(p)$ we may factor

$$\psi \circ f \circ \phi^{-1} = (\psi \circ \psi_p^{-1}) \circ (\psi_p \circ f \circ \phi_p^{-1}) \circ (\phi_p \circ \phi^{-1}).$$

The outer two factors are transition functions, hence holomorphic; the middle factor is holomorphic by hypothesis. So the composite is holomorphic at $\phi(p)$. $\square$

Lemma 41.4.16

If $f : R \rightarrow S$ and $g : S \rightarrow T$ are holomorphic, so is $g \circ f : R \rightarrow T$.

Proof. Immediate from Lemma 41.4.15: locally $g \circ f$ has local expression $(\chi \circ g \circ \psi^{-1}) \circ (\psi \circ f \circ \phi^{-1})$. $\square$

Definition 41.4.17 (Conformal Equivalence)

A **conformal equivalence** (or **biholomorphism**) is a holomorphic bijection $f : R \rightarrow S$ whose inverse is holomorphic. We then write $R \cong S$.

By Lemma 41.4.16 this is an equivalence relation on Riemann surfaces, and it is the notion of sameness we care about. We will see later (Corollary 41.9.12) that the hypothesis on f^{-1} is automatic: a holomorphic bijection of Riemann surfaces is always a conformal equivalence.

Example 41.4.18

The map $\mathbb{C} \rightarrow \bar{\mathbb{C}}, z \mapsto \bar{z}$, is a conformal equivalence: in the chart id upstairs and $z \mapsto \bar{z}$ downstairs, its local expression is $z \mapsto \bar{z} = z$.

Example 41.4.19 (Möbius Maps)

For $a, b, c, d \in \mathbb{C}$ with $ad - bc \neq 0$, the Möbius transformation

$$\mu(z) = \frac{az + b}{cz + d}, \quad \mu(-d/c) = \infty, \quad \mu(\infty) = a/c$$

is a conformal equivalence $\mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$. Indeed μ is holomorphic away from $-d/c$ and ∞ , and at those points one checks holomorphy in the chart $1/z$: for example near $z = \infty$ the local expression is $w \mapsto (cw + d)/(aw + b)$ (in the chart $w = 1/z$ upstairs and $1/z$ downstairs when $a \neq 0$), which is holomorphic at 0. The inverse is again Möbius. In fact *every* conformal automorphism of $\mathbb{C}_\infty$ is of this form, as we shall see.

Example 41.4.20 ($\mathbb{C}^*$ and the Cylinder)

The map $\mathbb{C} \rightarrow \mathbb{C}^*$, $z \mapsto e^z$, is holomorphic and surjective but not injective; it descends to a conformal equivalence $\mathbb{C}/2\pi i\mathbb{Z} \rightarrow \mathbb{C}^*$ once we know how to give the quotient a conformal structure. That is the next subsection.

§41.4.5 Structures Induced by Covering Maps

Here is the tool that lets us construct nearly all our examples: a covering of a Riemann surface is a Riemann surface, in exactly one way.

Lemma 41.4.21

Let R be a Riemann surface and $\pi : \tilde{R} \rightarrow R$ a covering map, with $\tilde{R}$ path-connected and Hausdorff. Then there is a unique conformal structure on $\tilde{R}$ making π holomorphic.

Proof. Existence. Let $p \in \tilde{R}$. Since π is a local homeomorphism, choose an open $\tilde{N} \ni p$ mapped homeomorphically by π onto an open $N \ni \pi(p)$. Choose a chart (ϕ, U) on R with $\pi(p) \in U$, and set

$$\tilde{U} = (\pi|_{\tilde{N}})^{-1}(N \cap U), \quad \tilde{\phi} = \phi \circ \pi|_{\tilde{U}}.$$

Then $\tilde{\phi}$ is a composite of homeomorphisms onto open subsets of $\mathbb{C}$, so $(\tilde{\phi}, \tilde{U})$ is a chart, and these charts cover $\tilde{R}$. For two of them the transition function is

$$\tilde{\phi}_1 \circ \tilde{\phi}_2^{-1} = \phi_1 \circ \pi \circ \pi^{-1} \circ \phi_2^{-1} = \phi_1 \circ \phi_2^{-1},$$

(with the evident restrictions), which is holomorphic since ϕ_1, ϕ_2 came from an atlas on R . So we have an atlas, and by Lemma 41.4.9 a conformal structure $\tilde{\mathcal{A}}$.

In this structure π is holomorphic: with $\tilde{\phi}$ built from ϕ as above, the local expression of π is

$$\phi \circ \pi \circ \tilde{\phi}^{-1} = \phi \circ \pi \circ \pi^{-1} \circ \phi^{-1} = \text{id},$$

and Lemma 41.4.15 applies.

Uniqueness. Let $\tilde{\mathcal{B}}$ be any conformal structure on $\tilde{R}$ making π holomorphic. If $(\psi, V) \in \tilde{\mathcal{B}}$ and $(\tilde{\phi}, \tilde{U})$ is one of the charts constructed above, then on the overlap

$$\tilde{\phi} \circ \psi^{-1} = \phi \circ \pi \circ \psi^{-1}$$

is holomorphic, being the local expression of π in charts ψ and ϕ . Likewise $\psi \circ \tilde{\phi}^{-1}$ is holomorphic, since $\pi|_{\tilde{U}}$ has a holomorphic local inverse (its local expression is the identity). Hence every chart of $\tilde{\mathcal{B}}$ is compatible with our atlas, and by maximality $\tilde{\mathcal{B}} = \tilde{\mathcal{A}}$. $\square$

Example 41.4.22 (The Surface of the Logarithm, Again)

Since $\pi : R \rightarrow \mathbb{C}^*$ is a covering map and R is path-connected and Hausdorff, R becomes a Riemann surface with π holomorphic. Moreover $f : R \rightarrow \mathbb{C}$ is holomorphic, since locally $f|_{U_n} = f_n \circ \pi$ and f_n is holomorphic.

In fact we can identify R outright. Each $f_n : U_n \rightarrow \tilde{V}_{I(n)} = \mathbb{R} + iI(n)$ is a homeomorphism with inverse $\exp$, and these agree on overlaps by construction of $\sim$; so they patch to a bijection $f : R \rightarrow \mathbb{C}$, which is holomorphic with holomorphic inverse. Hence

$$R \cong \mathbb{C}.$$

The Riemann surface of the logarithm is just the plane – but presented as a spiral over $\mathbb{C}^*$, with π the map $\exp$ in disguise. This is our first hint that the interesting data is not the surface alone but the surface *together with its map to the base*.

Example 41.4.23 (The Surface of the k -th Root)

Similarly R_k becomes a Riemann surface with π, g holomorphic, and $g : R_k \rightarrow \mathbb{C}^*$ is a conformal equivalence, with $\pi = p_k \circ g$. So $R_k \cong \mathbb{C}^*$, again realised as a k -sheeted spiral over $\mathbb{C}^*$.

§41.4.6 Complex Tori

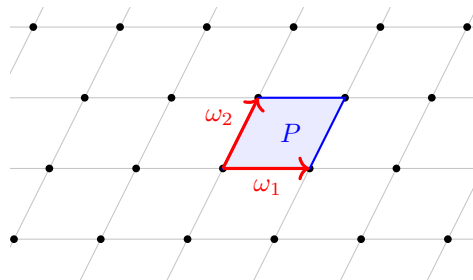
So far our only compact example is $\mathbb{C}_\infty$. Here is a family of others.

Definition 41.4.24 (Lattice)

Let $\omega_1, \omega_2 \in \mathbb{C}^*$ be linearly independent over $\mathbb{R}$. The subgroup

$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2 \leq (\mathbb{C}, +)$$

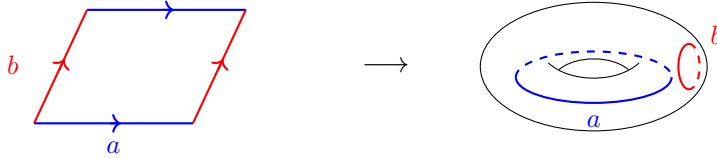
is called a **lattice** in $\mathbb{C}$.



Definition 41.4.25 (Complex Torus)

The quotient group $T = \mathbb{C}/\Lambda$, with the quotient topology, is called a **complex torus**.

The **fundamental parallelogram** $P = \{s\omega_1 + t\omega_2 : s, t \in [0, 1]\}$ maps onto T , and is injective on its interior, so T is compact (being a continuous image of the compact set P) and indeed $T \cong S^1 \times S^1$ topologically, with the two pairs of opposite edges of P glued.



We must give T a conformal structure. Let $\pi : \mathbb{C} \rightarrow T$ be the quotient map.

Proposition 41.4.26

$\pi : \mathbb{C} \rightarrow T$ is a regular covering map, and hence T carries a unique conformal structure making π holomorphic.

Proof. First, Λ is discrete: it is a lattice, so $\mu := \min\{|\lambda| : \lambda \in \Lambda \setminus \{0\}\}$ exists and is positive. (Indeed only finitely many lattice points lie in any bounded set, since $|m\omega_1 + n\omega_2| \geq c(|m| + |n|)$ for a constant $c > 0$ depending on ω_1, ω_2 ; we prove this in Lemma 41.11.9.) Fix $0 < \varepsilon < \mu/2$.

π is an open map, since for $U \subseteq \mathbb{C}$ open, $\pi^{-1}(\pi(U)) = \bigcup_{\lambda \in \Lambda} (U + \lambda)$ is open. Given $z_0 \in \mathbb{C}$, put $V = \pi(D(z_0, \varepsilon))$, which is open, and

$$\pi^{-1}(V) = \bigcup_{\lambda \in \Lambda} (D(z_0, \varepsilon) + \lambda) = \coprod_{\lambda \in \Lambda} (D(z_0, \varepsilon) + \lambda),$$

the union being disjoint because two of these discs meeting would give $|\lambda - \lambda'| < 2\varepsilon < \mu$. Each translate maps bijectively (hence homeomorphically, π being open and continuous) onto V . So π is a regular covering map, Λ with the discrete topology indexing the sheets.

T is Hausdorff: given $\pi(z_0) \neq \pi(z_1)$, the sets $\pi(D(z_0, \delta))$ and $\pi(D(z_1, \delta))$ are disjoint open neighbourhoods once 2δ is less than the distance from z_1 to $z_0 + \Lambda$. It is path-connected as a continuous image of $\mathbb{C}$. Now apply Lemma 41.4.21. $\square$

Explicitly, the charts on T are the maps $(\pi|_{D(z_0, \varepsilon)})^{-1}$ defined on $\pi(D(z_0, \varepsilon))$, and the transition functions are the translations $z \mapsto z + \lambda$, which are certainly holomorphic. This is the cleanest possible atlas, and it is why tori are such a good testing ground.

Remark. All complex tori are homeomorphic, being homeomorphic to $S^1 \times S^1$. They are very far from being all conformally equivalent: one shows on the example sheets that $\mathbb{C}/\Lambda \cong \mathbb{C}/\Lambda'$ if and only if $\Lambda' = a\Lambda$ for some $a \in \mathbb{C}^*$, so that the conformal equivalence classes are parametrised by the modulus $\tau = \omega_2/\omega_1 \in \mathbb{H}$ up to the action of $\text{SL}_2(\mathbb{Z})$. There are uncountably many. So conformal structure is a strictly finer invariant than topology – which is exactly why the subject is interesting.

§41.5 Functions on Riemann Surfaces

We have a supply of surfaces; now we do analysis on them. Every result in this section is proved the same way: take charts, quote the corresponding theorem in $\mathbb{C}$, and check that the conclusion is chart-independent. What makes the section worth reading is the consequences, which are of a kind that has no analogue in the plane.

§41.5.1 The Identity Theorem

Theorem 41.5.1 (Identity Theorem for Riemann Surfaces)

Let $f, g : R \rightarrow S$ be holomorphic maps of Riemann surfaces. If the set

$$A = \{p \in R : f(p) = g(p)\}$$

has an accumulation point in R , then $f = g$.

Proof. Let B be the set of accumulation points of A in R ; by hypothesis $B \neq \emptyset$. We show B is open and closed in R , which by connectedness gives $B = R$ and hence (as $A \supseteq B$ by continuity) $f = g$.

B is open. Let $p \in B$. Then $f(p) = g(p) =: q$ by continuity. Choose a chart (ψ, V) on S about q and a connected chart (ϕ, U) on R about p with $f(U), g(U) \subseteq V$; this is possible since f, g are continuous. The local expressions $\psi f \phi^{-1}$ and $\psi g \phi^{-1}$ are holomorphic on the domain $\phi(U)$ and agree on a set with an accumulation point (namely $\phi(A \cap U)$, which accumulates at $\phi(p)$). By the identity theorem in $\mathbb{C}$ (Corollary 41.1.4) they agree on $\phi(U)$, so $f = g$ on U , and every point of U is an accumulation point of A . Hence $U \subseteq B$.

B is closed. Suppose $p \in \bar{B}$. Take U as above; U meets B , so U contains points of A other than p , whence by the argument above $f = g$ on U and $p \in B$. $\square$

Corollary 41.5.2

A non-constant holomorphic map $f : R \rightarrow S$ is nowhere locally constant, and its fibres $f^{-1}(q)$ are discrete.

Proof. If f were constant $= q$ on an open set, apply Theorem 41.5.1 to f and the constant map q . For the fibres: if $f^{-1}(q)$ had an accumulation point, the same comparison gives f constant. $\square$

§41.5.2 The Open Mapping Theorem

Theorem 41.5.3 (Open Mapping Theorem)

Every non-constant holomorphic map $f : R \rightarrow S$ of Riemann surfaces is an open map.

Proof. Let $W \subseteq R$ be open and $p \in W$; we find a neighbourhood of $f(p)$ inside $f(W)$. Choose charts (ϕ, U) about p and (ψ, V) about $f(p)$, and shrink so that $U \subseteq W$, U is connected and $f(U) \subseteq V$. By Corollary 41.5.2 the local expression

$F = \psi f \phi^{-1}$ is non-constant on the domain $\phi(U)$, so by the open mapping theorem in $\mathbb{C}$ (Theorem 41.1.11) $F(\phi(U))$ is open in $\mathbb{C}$. Hence

$$f(U) = \psi^{-1}(F(\phi(U)))$$

is open in S , being the preimage of an open set under the homeomorphism ψ restricted appropriately. As $f(U) \subseteq f(W)$ contains $f(p)$, we are done. $\square$

Corollary 41.5.4

Let $f : R \rightarrow S$ be a non-constant holomorphic map with R compact. Then f is surjective and S is compact.

Proof. $f(R)$ is open by Theorem 41.5.3 and compact as a continuous image of a compact set, hence closed since S is Hausdorff. As S is connected and $f(R)$ is a non-empty clopen subset, $f(R) = S$. Then $S = f(R)$ is compact. $\square$

Corollary 41.5.5

Every holomorphic function $f : R \rightarrow \mathbb{C}$ on a compact Riemann surface R is constant.

Proof. If f were non-constant then $\mathbb{C}$ would be compact by Corollary 41.5.4. It is not. $\square$

This is a striking statement, and it deserves to be dwelt on: it is a vast generalisation of Liouville's theorem. On a compact surface there is *no* interesting holomorphic function at all. All the analysis on a compact Riemann surface must therefore be carried by maps to other compact targets – to $\mathbb{C}_\infty$, giving meromorphic functions, or to other surfaces. This single fact shapes the rest of the course.

Theorem 41.5.6 (Maximum Modulus Principle)

Let $f : R \rightarrow \mathbb{C}$ be holomorphic on a Riemann surface. If $|f|$ attains a local maximum at some $p \in R$, then f is constant.

Proof. Suppose f is non-constant, so $f(R)$ is open by Theorem 41.5.3. Let W be a neighbourhood of p with $|f(q)| \leq |f(p)|$ for $q \in W$. Then $f(W)$ is an open neighbourhood of $f(p)$ contained in the closed disc $\overline{D}(0, |f(p)|)$; but every neighbourhood of $f(p)$ contains points of modulus greater than $|f(p)|$ (take $f(p)(1 + \epsilon)$ if $f(p) \neq 0$, or any small non-zero point if $f(p) = 0$). This is a contradiction. $\square$

Combining these gives an alternative proof of Corollary 41.5.5: on a compact R the continuous function $|f|$ attains a maximum, so f is constant.

§41.5.3 Harmonic Functions

The open mapping theorem tells us that a non-constant holomorphic function cannot be real-valued, since $\mathbb{R} \subseteq \mathbb{C}$ has empty interior. Real-valued functions on Riemann surfaces are therefore never holomorphic; the right class is the harmonic ones.

Definition 41.5.7 (Harmonic Function)

A smooth function $u : D \rightarrow \mathbb{R}$ on a domain $D \subseteq \mathbb{C}$ is **harmonic** if

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

Lemma 41.5.8

Let $D \subseteq \mathbb{C}$ be a disc. A function $u : D \rightarrow \mathbb{R}$ is harmonic if and only if $u = \operatorname{Re} f$ for some holomorphic f on D .

Proof. If $f = u + iv$ is holomorphic then the Cauchy–Riemann equations $u_x = v_y$, $u_y = -v_x$ give $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$. The converse – constructing a harmonic conjugate v on a disc by integrating $-u_y dx + u_x dy$, which is closed by harmonicity and hence exact on a simply connected domain – was an exercise in IB Complex Analysis. $\square$

Note the appearance of simple connectedness: the converse fails on an annulus, as $u = \log |z|$ shows.

Definition 41.5.9

A function $u : R \rightarrow \mathbb{R}$ on a Riemann surface is **harmonic** if $u \circ \phi^{-1}$ is harmonic for every chart (ϕ, U) .

For this to be a sensible definition we need harmonicity to be preserved by holomorphic changes of coordinate, which it is:

Lemma 41.5.10

A function $u : R \rightarrow \mathbb{R}$ is harmonic if and only if each $p \in R$ has *some* chart (ϕ, U) with $p \in U$ and $u \circ \phi^{-1}$ harmonic.

Proof. One direction is trivial. Conversely, suppose the condition holds and let (ψ, V) be any chart, $p \in V$. Choose (ϕ, U) as hypothesised with $p \in U$; shrinking, we may assume $\phi(U)$ is a disc, so $u \circ \phi^{-1} = \operatorname{Re} f$ for some holomorphic f by Lemma 41.5.8. Then near $\psi(p)$,

$$u \circ \psi^{-1} = (u \circ \phi^{-1}) \circ (\phi \circ \psi^{-1}) = \operatorname{Re} (f \circ (\phi \circ \psi^{-1})),$$

the real part of a holomorphic function, hence harmonic. $\square$

Theorem 41.5.11 (Open Mapping and Identity Theorems for Harmonic Functions)

Let u, v be harmonic on a Riemann surface R .

- (i) If $\{p : u(p) = v(p)\}$ has non-empty interior then $u = v$.
- (ii) If u is non-constant then $u : R \rightarrow \mathbb{R}$ is an open map.

Consequently, every harmonic function on a compact Riemann surface is constant.

Proof. Locally $u = \operatorname{Re} f$ with f holomorphic. For (i), if $u = v$ on an open set then locally $\operatorname{Re}(f - g) = 0$, so $f - g$ is constant (its image lies in $i\mathbb{R}$, which has empty interior, so it cannot be open), and a clopen argument as in Theorem 41.5.1 propagates this. For (ii), openness of u follows from openness of f on any chart where f is non-constant, since $\operatorname{Re} : \mathbb{C} \rightarrow \mathbb{R}$ is open. The last statement follows since $u(R)$ would be an open, compact – hence clopen and non-empty – subset of the connected space $\mathbb{R}$, forcing $u(R) = \mathbb{R}$, which is not compact. $\square$

§41.5.4 Meromorphic Functions

Since holomorphic functions to $\mathbb{C}$ are useless on compact surfaces, we enlarge the target to the compact surface $\mathbb{C}_\infty$.

Definition 41.5.12 (Meromorphic Function)

A **meromorphic function** on a Riemann surface R is a holomorphic map $f : R \rightarrow \mathbb{C}_\infty$ which is not identically ∞ .

Note that $f^{-1}(\infty)$ is then discrete by Corollary 41.5.2, and closed since $\{\infty\}$ is; so a meromorphic function is a holomorphic function on the complement of a discrete closed set. The next result confirms that on a plane domain this agrees with the classical definition.

Proposition 41.5.13

Let $D \subseteq \mathbb{C}$ be a domain. A map $f : D \rightarrow \mathbb{C}_\infty$ is meromorphic if and only if there is a discrete closed $A \subseteq D$ such that $f|_{D \setminus A}$ is holomorphic with values in $\mathbb{C}$, and f has a pole at each point of A .

Proof. ($\Rightarrow$) Let $A = f^{-1}(\infty)$, which is discrete and closed as noted, and certainly f is holomorphic $D \setminus A \rightarrow \mathbb{C}$. Fix $a \in A$ and work in the chart $w \mapsto 1/w$ about ∞ . The local expression $z \mapsto 1/f(z)$ is holomorphic near a and vanishes at a ; it is not identically zero (else $f \equiv \infty$), so by Proposition 41.1.3 we may write $1/f(z) = (z - a)^m g(z)$ with $m \geq 1$ and $g(a) \neq 0$. Shrinking so that g has no zero, $f(z) = (z - a)^{-m}/g(z)$, which has a pole of order m at a .

($\Leftarrow$) f is certainly holomorphic as a map into $\mathbb{C}_\infty$ on $D \setminus A$. At $a \in A$ with a pole of order m we may write $f(z) = (z - a)^{-m}h(z)$ with h holomorphic and $h(a) \neq 0$; then $1/f(z) = (z - a)^m/h(z)$ is holomorphic near a (after shrinking so h does not vanish), and f is continuous into $\mathbb{C}_\infty$ at a since $|f| \rightarrow \infty$. So the local expression in the chart at ∞ is holomorphic, and Lemma 41.4.15 applies. $\square$

Example 41.5.14

The map $\mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ given by a rational function $p(z)/q(z)$ (in lowest terms, with the evident values at the zeros of q and at ∞) is meromorphic. We shall see in Section 41.11 that these are the *only* meromorphic functions on $\mathbb{C}_\infty$.

Example 41.5.15

On a complex torus $T = \mathbb{C}/\Lambda$, a meromorphic function pulls back along $\pi : \mathbb{C} \rightarrow T$ to a meromorphic function on $\mathbb{C}$ invariant under Λ – that is, to a doubly periodic, or **elliptic**, function. Conversely every such function descends. Corollary 41.5.5 then says: a doubly periodic *holomorphic* function on $\mathbb{C}$ is constant, which is Liouville's theorem applied to the (bounded, by compactness of P) function. Section 41.11 is devoted to the non-constant ones.

§41.6 Covering Spaces and Monodromy

To organise analytic continuation we need the topology of covering spaces. Most of this was proved in II Algebraic Topology, and we shall be brisk, but the path-lifting and homotopy-lifting statements are so central that it is worth seeing the proofs again in the form we need.

§41.6.1 Lifting**Definition 41.6.1 (Lift)**

Let $\pi : \tilde{X} \rightarrow X$ be a covering map and $\gamma : [0, 1] \rightarrow X$ a path. A **lift** of γ is a path $\tilde{\gamma} : [0, 1] \rightarrow \tilde{X}$ with $\pi \circ \tilde{\gamma} = \gamma$.

Example 41.6.2

For $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ and $\gamma(t) = e^{2\pi it}$, both $t \mapsto 2\pi it$ and $t \mapsto 2\pi i + 2\pi it$ are lifts. They differ, but they start at different points – and that is the general phenomenon.

Proposition 41.6.3 (Uniqueness of Lifts)

Let $\pi : \tilde{X} \rightarrow X$ be a covering map and $\tilde{\gamma}_1, \tilde{\gamma}_2$ lifts of the same path γ . If $\tilde{\gamma}_1(0) = \tilde{\gamma}_2(0)$ then $\tilde{\gamma}_1 = \tilde{\gamma}_2$.

Proof. Let $I = \{t \in [0, 1] : \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)\}$, which is non-empty and closed (as $\tilde{X}$ is Hausdorff and the $\tilde{\gamma}_i$ continuous). We show I is open, so that $I = [0, 1]$ by connectedness.

Let $t \in I$ and put $\tilde{x} = \tilde{\gamma}_1(t) = \tilde{\gamma}_2(t)$. As π is a local homeomorphism there is an open $\tilde{N} \ni \tilde{x}$ with $\pi|_{\tilde{N}}$ a homeomorphism onto an open N . By continuity there is $\delta > 0$ with $\tilde{\gamma}_i((t - \delta, t + \delta)) \subseteq \tilde{N}$ for $i = 1, 2$. For such s ,

$$\tilde{\gamma}_1(s) = (\pi|_{\tilde{N}})^{-1}(\gamma(s)) = \tilde{\gamma}_2(s),$$

so $(t - \delta, t + \delta) \cap [0, 1] \subseteq I$. □

Existence, however, requires regularity.

Example 41.6.4

Let $\tilde{X} = \mathbb{R} + i(-\pi, 2\pi)$ and $\pi = \exp|_{\tilde{X}} : \tilde{X} \rightarrow \mathbb{C}^*$. This is a covering map (a local

homeomorphism onto $\mathbb{C}^*$), but the loop $t \mapsto e^{2\pi it}$ cannot be lifted starting at 0: the lift would have to be $t \mapsto 2\pi it$, which leaves $\tilde{X}$ at $t = 1$. The map π is not regular.

Proposition 41.6.5 (Path Lifting)

Let $\pi : \tilde{X} \rightarrow X$ be a regular covering map, $\gamma : [0, 1] \rightarrow X$ a path and $\tilde{x} \in \tilde{X}$ with $\pi(\tilde{x}) = \gamma(0)$. Then there is a unique lift $\tilde{\gamma}$ of γ with $\tilde{\gamma}(0) = \tilde{x}$.

Proof. Uniqueness is Proposition 41.6.3. For existence let

$$I = \{t \in [0, 1] : \gamma|_{[0, t]} \text{ lifts to some } \tilde{\gamma} \text{ with } \tilde{\gamma}(0) = \tilde{x}\},$$

which contains 0. Note that by uniqueness, lifts on $[0, t]$ for varying t are compatible, so I is an interval and the lifts patch.

I is open in $[0, 1]$. Let $\tau \in I$ with lift $\tilde{\gamma}$ on $[0, \tau]$. Choose an evenly covered $U \ni \gamma(\tau)$, so $\pi^{-1}(U) = \coprod_{\delta} U_{\delta}$, and let U_{δ_0} be the sheet containing $\tilde{\gamma}(\tau)$. Choose $\varepsilon > 0$ with $\gamma((\tau - \varepsilon, \tau + \varepsilon)) \subseteq U$; then setting $\tilde{\gamma}(t) = (\pi|_{U_{\delta_0}})^{-1}(\gamma(t))$ for $t \in [\tau, \tau + \varepsilon)$ extends the lift.

I is closed. Let $\tau = \sup I$ and choose an evenly covered $U \ni \gamma(\tau)$ with sheets U_{δ} . Pick $\varepsilon > 0$ with $\gamma((\tau - \varepsilon, \tau]) \subseteq U$ and choose $t \in I$ with $t > \tau - \varepsilon$. The connected set $\tilde{\gamma}((\tau - \varepsilon, t])$ lies in $\pi^{-1}(U)$, hence in a single sheet U_{δ_0} , and we may define $\tilde{\gamma}(s) = (\pi|_{U_{\delta_0}})^{-1}(\gamma(s))$ for $s \in (\tau - \varepsilon, \tau]$, which agrees with the existing lift and extends it continuously to τ . So $\tau \in I$.

Hence I is a non-empty clopen subset of $[0, 1]$, so $I = [0, 1]$. $\square$

Theorem 41.6.6 (Homotopy Lifting, or the Monodromy Theorem)

Let $\pi : \tilde{X} \rightarrow X$ be a regular covering map and let $\alpha \simeq \beta$ be homotopic paths in X (rel endpoints). Let $\tilde{\alpha}, \tilde{\beta}$ be their lifts with $\tilde{\alpha}(0) = \tilde{\beta}(0)$. Then $\tilde{\alpha} \simeq \tilde{\beta}$; in particular $\tilde{\alpha}(1) = \tilde{\beta}(1)$.

Proof. This is the homotopy lifting property, proved in II Algebraic Topology: one lifts the homotopy $H : [0, 1]^2 \rightarrow X$ by subdividing the square into small squares each mapping into an evenly covered set, and lifting them one at a time, using uniqueness of lifts to see the pieces agree. The resulting $\tilde{H}$ is a homotopy from $\tilde{\alpha}$ to $\tilde{\beta}$; since H is constant on the two vertical edges, $\tilde{H}$ is too (a lift of a constant path is constant, by uniqueness), so the homotopy is rel endpoints. $\square$

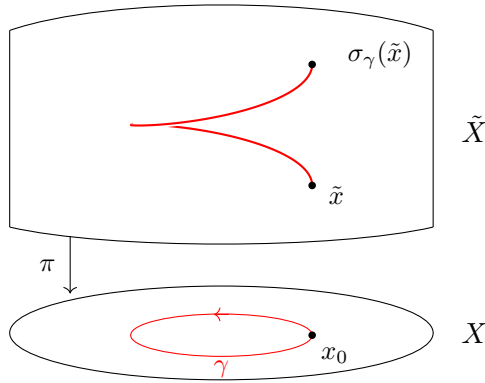
The name ‘monodromy theorem’ attached to this statement is traditional and slightly unfortunate, since we shall also use it for the classical statement about analytic continuation (Theorem 41.7.12). The latter is a corollary of the former, and the two are best regarded as the same theorem stated on either side of the dictionary we are about to build.

§41.6.2 The Monodromy Group

Let $\pi : \tilde{X} \rightarrow X$ be a regular covering map and fix a basepoint $x_0 \in X$. For a loop γ at x_0 and $\tilde{x} \in \pi^{-1}(x_0)$, let $\tilde{\gamma}_{\tilde{x}}$ be the lift starting at $\tilde{x}$; since $\pi(\tilde{\gamma}_{\tilde{x}}(1)) = \gamma(1) = x_0$, the endpoint again lies in the fibre.

Definition 41.6.7 (Monodromy Action)

Define $\sigma_\gamma : \pi^{-1}(x_0) \rightarrow \pi^{-1}(x_0)$ by $\sigma_\gamma(\tilde{x}) = \tilde{\gamma}_{\tilde{x}}(1)$.



Proposition 41.6.8

The assignment $\gamma \mapsto \sigma_\gamma$ satisfies:

- (i) $\sigma_c = \text{id}$ for the constant loop c ;
- (ii) $\sigma_{\bar{\gamma}} = \sigma_\gamma^{-1}$, where $\bar{\gamma}(t) = \gamma(1 - t)$; in particular each σ_γ is a bijection;
- (iii) $\sigma_{\alpha \cdot \beta} = \sigma_\beta \circ \sigma_\alpha$;
- (iv) if $\alpha \simeq \beta$ then $\sigma_\alpha = \sigma_\beta$.

Hence $\{\sigma_\gamma\}$ is a subgroup of $\text{Sym}(\pi^{-1}(x_0))$, and there is a well-defined (right) action of $\pi_1(X, x_0)$ on the fibre.

Proof. (i) is clear. For (iii), if $\tilde{\alpha}_{\tilde{x}}$ is the lift of α at $\tilde{x}$, then the concatenation $\tilde{\alpha}_{\tilde{x}} \cdot \tilde{\beta}_{\tilde{\alpha}_{\tilde{x}}(1)}$ is a lift of $\alpha \cdot \beta$ starting at $\tilde{x}$, so by uniqueness it is *the* lift, and its endpoint is $\sigma_\beta(\sigma_\alpha(\tilde{x}))$. Then (ii) follows from (i) and (iii). Finally (iv) is Theorem 41.6.6. $\square$

Definition 41.6.9 (Monodromy Group)

The subgroup $\{\sigma_\gamma : \gamma \text{ a loop at } x_0\} \leq \text{Sym}(\pi^{-1}(x_0))$ is the **monodromy group** of the covering π .

Up to isomorphism it does not depend on the basepoint, since a path from x_0 to x_1 conjugates the two actions.

Example 41.6.10

Take $p_k : \mathbb{C}^* \rightarrow \mathbb{C}^*$, $z \mapsto z^k$, with basepoint 1. The fibre is the set of k -th roots

of unity ζ^n , $\zeta = e^{2\pi i/k}$. Let $\gamma(t) = e^{2\pi i t}$ be the standard loop; its lift starting at ζ^n is $t \mapsto \zeta^n e^{2\pi i t/k}$, which ends at ζ^{n+1} . So σ_γ is the k -cycle $\zeta^n \mapsto \zeta^{n+1}$. Since $\pi_1(\mathbb{C}^*) \cong \mathbb{Z}$ is generated by $[\gamma]$, the monodromy group is cyclic of order k .

This is the precise sense in which ‘ $\sqrt[k]{z}$ has k branches which are permuted cyclically as z winds once round 0’.

Example 41.6.11

For $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ with basepoint 1, the fibre is $2\pi i\mathbb{Z}$, the loop γ lifts to $t \mapsto 2\pi i t + 2\pi i n$ starting at $2\pi i n$, and σ_γ is the shift $2\pi i n \mapsto 2\pi i(n+1)$. The monodromy group is infinite cyclic, matching the infinitely many branches of $\log$.

§41.7 The Space of Germs

We now build, once and for all, the Riemann surface on which a complete analytic function becomes single-valued. The construction of Section 41.3 was ad hoc: we chose particular sectors and glued them by hand. The trick is to stop choosing. Take *all* function elements at once, remember only their local behaviour, and let the topology sort things out.

§41.7.1 Germs

Fix a domain $D \subseteq \mathbb{C}$.

Definition 41.7.1 (Germ)

Let (f, U) and (g, V) be function elements on D and let $z \in U \cap V$. Write $(f, U) \equiv_z (g, V)$ if $f = g$ on some neighbourhood of z . This is an equivalence relation on the function elements whose domain contains z ; the equivalence class of (f, U) is the **germ** of f at z , written $[f]_z$.

So $[f]_z = [g]_w$ if and only if $z = w$ and f, g agree near z . A germ at z is exactly the same data as a convergent power series centred at z ; but the geometric language will be more useful.

Definition 41.7.2 (The Space of Germs)

The **space of germs** over D is

$$\mathcal{G} = \mathcal{G}_D = \{[f]_z : z \in D, (f, U) \text{ a function element on } D \text{ with } z \in U\}.$$

We topologise $\mathcal{G}$ so that germs of the same function at nearby points are nearby. For a function element (f, U) write

$$[f]_U = \{[f]_z : z \in U\} \subseteq \mathcal{G},$$

a ‘sheet’ of germs.

Lemma 41.7.3

The collection $\{[f]_U\}$, over all function elements (f, U) on D , is a basis for a topology on $\mathcal{G}$.

Proof. The sets cover $\mathcal{G}$ by definition. Suppose $[h]_z \in [f]_U \cap [g]_V$. Then h agrees with f near z and with g near z , so on some open $W \ni z$ with $W \subseteq U \cap V$ we have $h = f = g$, whence $[h]_W \subseteq [f]_U \cap [g]_V$ and $[h]_z \in [h]_W$. $\square$

Lemma 41.7.4

$\mathcal{G}$ is Hausdorff.

Proof. Let $[f]_z \neq [g]_w$. If $z \neq w$, choose disjoint neighbourhoods $U \ni z$, $V \ni w$ inside the domains of f and g ; then $[f]_U$ and $[g]_V$ are disjoint, since germs in them sit over disjoint sets of points.

If $z = w$, choose a *connected* open $U \ni z$ contained in both domains, so that (f, U) and (g, U) represent $[f]_z, [g]_z$. Suppose $[h]_y \in [f]_U \cap [g]_U$. Then h agrees with f near y and with g near y , so $f = g$ on a neighbourhood of y , and hence on all of U by the identity theorem (U being connected). But then $[f]_z = [g]_z$, a contradiction. So $[f]_U \cap [g]_U = \emptyset$. $\square$

Definition 41.7.5

The **forgetful map** $\pi : \mathcal{G} \rightarrow D$ is $\pi([f]_z) = z$.

Lemma 41.7.6

For each connected component G of $\mathcal{G}$, the restriction $\pi : G \rightarrow D$ is a covering map onto its image, which is open in D . In particular π is a local homeomorphism.

Proof. π is continuous: for $U \subseteq D$ open, $\pi^{-1}(U) = \bigcup [f]_V$ over all function elements (f, V) with $V \subseteq U$, which is open. On a basic open set, $\pi|_{[f]_U} : [f]_U \rightarrow U$ is a bijection with inverse $z \mapsto [f]_z$, and this inverse is continuous because the preimage of a basic set $[g]_V$ is the open set $\{z \in U \cap V : f = g \text{ near } z\}$. So $\pi|_{[f]_U}$ is a homeomorphism onto the open set U , and π is a local homeomorphism. Its image is therefore open, and each component G of $\mathcal{G}$ is path-connected (being locally homeomorphic to open subsets of $\mathbb{C}$, hence locally path-connected) and Hausdorff by Lemma 41.7.4. $\square$

Remark. Note that $\pi : G \rightarrow D$ need not be *regular*. For instance, take $D = \mathbb{C}$ and the component containing the germs of $z \mapsto 1/(1-z)$: continuation is blocked at $z = 1$, and the fibre over 1 can be empty while nearby fibres are not. This is exactly the phenomenon of a function element which cannot be continued along every path.

By Lemma 41.4.21, each component G of $\mathcal{G}$ inherits a unique conformal structure making π holomorphic. Explicitly, the charts are the maps $\pi|_{[f]_U}$ on the sets $[f]_U$, and the transition functions are identities. And there is a canonical holomorphic function on it:

Definition 41.7.7 (Evaluation Map)

The **evaluation map** $E : \mathcal{G} \rightarrow \mathbb{C}$ is $E([f]_z) = f(z)$.

This is well defined, since germs which agree near z certainly agree at z . In the chart $\pi|_{[f]_U}$ its local expression is

$$E \circ (\pi|_{[f]_U})^{-1}(z) = E([f]_z) = f(z),$$

which is holomorphic. So E is a holomorphic function on $\mathcal{G}$.

This is the whole point. The multi-valued function has become the pair (π, E) : π remembers which point of D we are over, and E is a genuine single-valued holomorphic function.

§41.7.2 Analytic Continuation as Lifting

Now we cash in.

Theorem 41.7.8

Let $(f, U), (g, V)$ be function elements on a domain D and let $\gamma : [0, 1] \rightarrow D$ be a path with $\gamma(0) \in U$, $\gamma(1) \in V$. Then

$$(f, U) \approx_\gamma (g, V) \iff \gamma \text{ lifts along } \pi \text{ to a path in } \mathcal{G} \text{ from } [f]_{\gamma(0)} \text{ to } [g]_{\gamma(1)}.$$

Proof. ($\Rightarrow$) Suppose we have a chain $(f, U) = (f_1, U_1) \sim \cdots \sim (f_n, U_n) = (g, V)$ and a dissection $0 = t_0 < \cdots < t_n = 1$ with $\gamma([t_{i-1}, t_i]) \subseteq U_i$. Define

$$\tilde{\gamma}(t) = [f_i]_{\gamma(t)}, \quad t \in [t_{i-1}, t_i].$$

This is well defined at the joins: at $t = t_i$ we need $[f_i]_{\gamma(t_i)} = [f_{i+1}]_{\gamma(t_i)}$, which holds since $f_i = f_{i+1}$ on $U_i \cap U_{i+1}$, an open set containing $\gamma(t_i)$. On each $[t_{i-1}, t_i]$ we have $\tilde{\gamma} = (\pi|_{[f_i]_{U_i}})^{-1} \circ \gamma$, which is continuous; by the gluing lemma $\tilde{\gamma}$ is continuous. Clearly $\pi \circ \tilde{\gamma} = \gamma$ and the endpoints are as claimed.

($\Leftarrow$) Suppose $\tilde{\gamma}$ is such a lift. Every point $\tilde{\gamma}(t)$ has a basic neighbourhood $[f_t]_{U_t}$ with U_t a *disc*. By compactness of $[0, 1]$ and the Lebesgue number lemma, there is a dissection $0 = t_0 < \cdots < t_n = 1$ and function elements (f_i, U_i) with U_i a disc and $\tilde{\gamma}([t_{i-1}, t_i]) \subseteq [f_i]_{U_i}$. Applying π gives $\gamma([t_{i-1}, t_i]) \subseteq U_i$. Moreover for each i ,

$$[f_i]_{\gamma(t_i)} = \tilde{\gamma}(t_i) = [f_{i+1}]_{\gamma(t_i)},$$

so $f_i = f_{i+1}$ near $\gamma(t_i) \in U_i \cap U_{i+1}$. Since U_i, U_{i+1} are discs, $U_i \cap U_{i+1}$ is convex hence connected, and the identity theorem gives $f_i = f_{i+1}$ on $U_i \cap U_{i+1}$, i.e. $(f_i, U_i) \sim (f_{i+1}, U_{i+1})$. Finally $[f_1]_{\gamma(0)} = [f]_{\gamma(0)}$ and $[f_n]_{\gamma(1)} = [g]_{\gamma(1)}$, and the identity theorem on the connected sets $U_1 \cap U$ and $U_n \cap V$ upgrades these germ equalities to $(f, U) \sim (f_1, U_1)$ and $(f_n, U_n) \sim (g, V)$. $\square$

This is the theorem that makes the space of germs worth building: *analytic continuation along a path is path lifting*. Every fact about continuation is now a fact about covering spaces.

Corollary 41.7.9

Let $\mathcal{F}$ be a complete analytic function on a domain D . Then

$$G_{\mathcal{F}} = \bigcup_{(f,U) \in \mathcal{F}} [f]_U$$

is a path component of $\mathcal{G}$.

Proof. Two germs lie in the same path component precisely when there is a path in $\mathcal{G}$ between them, which by Theorem 41.7.8 happens precisely when the corresponding function elements are analytic continuations of one another. $\square$

Definition 41.7.10

$G_{\mathcal{F}}$, with the conformal structure of Lemma 41.4.21, is **the Riemann surface of the complete analytic function** $\mathcal{F}$, equipped with $\pi : G_{\mathcal{F}} \rightarrow D$ and $E : G_{\mathcal{F}} \rightarrow \mathbb{C}$.

Example 41.7.11

Take $D = \mathbb{C}^*$ and $\mathcal{F}$ the complete analytic function containing the elements $F_n = (f_n, U_n)$ of Section 41.2.2. Then $G_{\mathcal{F}}$ is precisely the surface R we built by hand, and E is the function f . Indeed $\Phi : R \rightarrow G_{\mathcal{F}}$, $[z] \mapsto [f_n]_z$ for $z \in U_n$, is a well-defined homeomorphism commuting with the projections.

§41.7.3 The Classical Monodromy Theorem**Theorem 41.7.12** (Classical Monodromy Theorem)

Let $D \subseteq \mathbb{C}$ be a domain and (f, U) a function element on D which can be continued along *every* path in D starting in U . Suppose α, β are homotopic paths in D (rel endpoints) from a point of U to a point of V , and

$$(f, U) \approx_{\alpha} (g_1, V), \quad (f, U) \approx_{\beta} (g_2, V).$$

Then $g_1 = g_2$ on V (assuming V connected).

Proof. Let G be the path component of $\mathcal{G}$ containing $[f]_{\alpha(0)}$. The hypothesis says exactly that every path in D starting at $\alpha(0)$ lifts to G starting at $[f]_{\alpha(0)}$, from which one deduces that $\pi : G \rightarrow D$ is a regular covering map: given $z \in D$, take a disc $B \ni z$ in D ; each germ in $\pi^{-1}(z)$ extends over B (continue along radii and use the identity theorem, or note that by Theorem 41.7.8 the radial paths lift), giving a sheet of $\pi^{-1}(B)$ over B , and these sheets are disjoint and exhaust $\pi^{-1}(B)$.

Now let $\tilde{\alpha}, \tilde{\beta}$ be the lifts of α, β starting at $[f]_{\alpha(0)}$, which exist by Theorem 41.7.8 and end at $[g_1]_{\alpha(1)}, [g_2]_{\beta(1)}$ respectively. Since $\alpha \simeq \beta$, Theorem 41.6.6 gives $\tilde{\alpha}(1) = \tilde{\beta}(1)$, i.e.

$$[g_1]_{\alpha(1)} = [g_2]_{\alpha(1)}.$$

So $g_1 = g_2$ near $\alpha(1)$, hence on V by the identity theorem. $\square$

Corollary 41.7.13

Let D be a simply connected domain and (f, U) a function element on D which can be continued along every path in D starting in U . Then there is a (unique) holomorphic $F : D \rightarrow \mathbb{C}$ with $F|_U = f$.

Proof. For $z \in D$ choose a path γ from a point of U to z and define $F(z)$ to be the value at z of the continuation along γ . Any two such paths are homotopic rel endpoints, so Theorem 41.7.12 says F is well defined; and it is holomorphic since near each point it agrees with a function element. $\square$

This explains, at last, the branch cuts of a first course. There is no holomorphic logarithm on $\mathbb{C}^*$, because $\mathbb{C}^*$ is not simply connected and continuation once round the origin changes the branch. Slitting the plane makes it simply connected, so a branch exists; but the slit is a choice, and the honest object is the covering $\pi : R \rightarrow \mathbb{C}^*$ which forgets no branch at all.

§41.8 Gluing Riemann Surfaces

The surfaces we have built so far are all non-compact – and by Corollary 41.5.5 it is compact surfaces where the interesting rigidity lives. The missing operation is compactification: adding the points that continuation cannot reach. The general mechanism is gluing.

§41.8.1 The Gluing Construction**Definition 41.8.1 (Gluing)**

Let X, Y be topological spaces with subspaces $X' \subseteq X$, $Y' \subseteq Y$ and a homeomorphism $\Phi : X' \rightarrow Y'$. The result of **gluing X to Y along Φ** is

$$X \cup_{\Phi} Y = (X \sqcup Y) / \sim,$$

where $\sim$ is generated by $x \sim \Phi(x)$ for $x \in X'$, with the quotient topology.

Proposition 41.8.2

Let R_1, R_2 be Riemann surfaces and $S_j \subseteq R_j$ non-empty open connected subsets, and let $\Phi : S_1 \rightarrow S_2$ be a conformal equivalence. Then $R = R_1 \cup_{\Phi} R_2$ carries a unique conformal structure making the inclusions $i_j : R_j \rightarrow R$ holomorphic (indeed conformal onto their open images). If R is Hausdorff, then R is a Riemann surface.

Proof. Each i_j is an open injection, so we may regard R_1, R_2 as open subsets of R covering it. Take as atlas all charts of the form $(\phi_j \circ i_j^{-1}, i_j(U_j))$ where (ϕ_j, U_j) is a chart of R_j . Two such charts coming from the same R_j have a transition function which is a transition function of R_j . A chart from R_1 and one from R_2 overlap only inside $i_1(S_1) = i_2(S_2)$, where the transition function is

$$\phi_2 \circ i_2^{-1} \circ i_1 \circ \phi_1^{-1} = \phi_2 \circ \Phi \circ \phi_1^{-1},$$

holomorphic since Φ is. So we have an atlas; extend it by Lemma 41.4.9. The local expression of i_j in these charts is the identity, so i_j is holomorphic. Uniqueness follows as in Lemma 41.4.21: any structure making the i_j holomorphic and locally invertible must contain, hence by maximality equal, this one.

Finally R is connected: it is the union of the connected sets $i_1(R_1)$ and $i_2(R_2)$, which meet. $\square$

The Hausdorff hypothesis is not automatic, and one must always check it.

Example 41.8.3 (The Line with Two Origins)

Take $R_1 = R_2 = \mathbb{C}$ and $S_1 = S_2 = \mathbb{C}^*$ with $\Phi = \text{id}$. Then R is the plane with a doubled origin: the two copies of 0 have no disjoint neighbourhoods, since every punctured neighbourhood of either is identified with that of the other. So R is not Hausdorff and is not a Riemann surface.

Example 41.8.4 (The Sphere by Gluing)

Take $R_1 = R_2 = \mathbb{C}$, $S_1 = S_2 = \mathbb{C}^*$, and $\Phi(z) = 1/z$, a conformal equivalence $\mathbb{C}^* \rightarrow \mathbb{C}^*$. The glued space R is Hausdorff: given the two ‘extra’ points $i_1(0)$ and $i_2(0)$, the sets $i_1(D(0, 1))$ and $i_2(D(0, 1))$ are disjoint open neighbourhoods, since Φ maps $D(0, 1) \setminus \{0\}$ to the exterior of the closed unit disc. Other pairs of points are separated inside a single copy of $\mathbb{C}$. It is compact, being the union of the two compact sets $i_j(\overline{D}(0, 1))$. And of course $R \cong \mathbb{C}_\infty$: the map sending $i_1(z) \mapsto z$ and $i_2(w) \mapsto 1/w$ is a conformal equivalence, since the two charts of $\mathbb{C}_\infty$ are related by exactly Φ .

This is the model for everything that follows: to compactify a surface, find a second surface containing the missing points and glue along the common part.

§41.8.2 The Surface of $w^2 = z^3 - z$

Let us carry out a substantial example by hand. Set

$$h(z) = z^3 - z = z(z-1)(z+1),$$

and consider the multi-valued function $w = \sqrt{h(z)}$. Near any point where h does not vanish we can choose a square root holomorphically; the trouble is at $z = 0, \pm 1$, and (as we shall see) at ∞ .

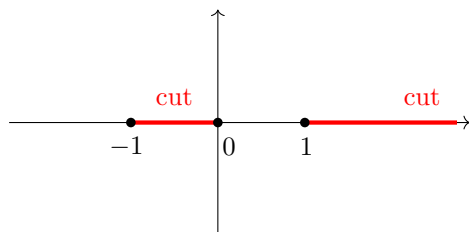
Step 1: a cut domain. Let

$$D = \mathbb{C} \setminus ([-1, 0] \cup [1, \infty)).$$

We construct branches of $\sqrt{h}$ on D . Fix $z_0 \in D$ and w_0 with $w_0^2 = h(z_0)$, and define, for $z \in D$,

$$g(z) = w_0 \exp \left(\frac{1}{2} \int_{\gamma} \frac{h'(\zeta)}{h(\zeta)} d\zeta \right)$$

where γ is any path in D from z_0 to z .



Why this is well defined. If γ is a loop in D , the argument principle gives

$$\frac{1}{2\pi i} \int_{\gamma} \frac{h'}{h} d\zeta = n(\gamma, 0) + n(\gamma, 1) + n(\gamma, -1),$$

the sum of winding numbers about the zeros of h (there are no poles). Now γ lies in D , so it cannot wind around 1 at all – the cut $[1, \infty)$ blocks it – giving $n(\gamma, 1) = 0$; and it must wind the same number of times about 0 and about -1 , since the cut $[-1, 0]$ joins them and γ cannot separate them. Hence the integral is $2\pi i \cdot 2n(\gamma, 0)$, an even multiple of $2\pi i$, and so

$$\exp\left(\frac{1}{2} \int_{\gamma} \frac{h'}{h}\right) = \exp(2\pi i n(\gamma, 0)) = 1.$$

Therefore g does not depend on γ , and it is holomorphic (locally it is $w_0 \exp(\frac{1}{2}L)$ for a local branch L of $\log h$), with

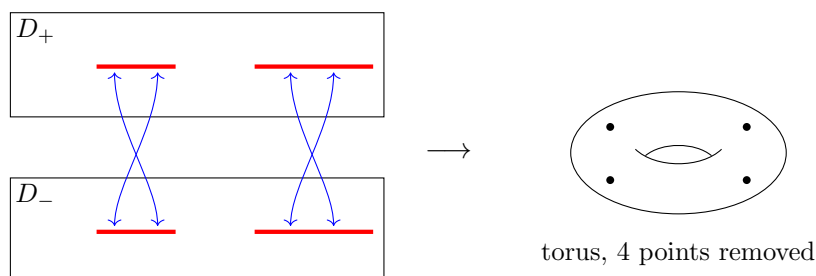
$$g(z)^2 = w_0^2 \exp\left(\int_{\gamma} \frac{h'}{h}\right) = h(z_0) \cdot \frac{h(z)}{h(z_0)} = h(z).$$

The two choices of w_0 give two branches $g_+ = -g_-$. Write D_+, D_- for two copies of D carrying g_+, g_- respectively.

Step 2: gluing along the cuts. As a subset of the sphere, D is $\mathbb{C}_{\infty}$ minus two closed arcs, hence homeomorphic to an open cylinder $S^1 \times \mathbb{R}$. For z_0 in the interior of a cut, and writing $z \rightarrow z_0^{\pm}$ for approach from the upper and lower half planes,

$$\lim_{z \rightarrow z_0^-} g_+(z) = \lim_{z \rightarrow z_0^+} g_-(z), \quad \lim_{z \rightarrow z_0^+} g_+(z) = \lim_{z \rightarrow z_0^-} g_-(z).$$

(Crossing a cut flips the sign of the square root, because the argument of exactly one of the factors of h jumps by 2π .) So we glue the upper side of each cut in D_+ to the lower side of the corresponding cut in D_- , and vice versa.



Each $D_{\pm}$ is a cylinder; gluing two cylinders along two pairs of slits produces a torus with the four points $0, \pm 1, \infty$ removed. (Think of each cylinder as a sphere with two slits; gluing crosswise along two slits is exactly the standard construction of a genus-1 surface from two spheres with two handles' worth of identification.) We shall verify this

by computation in Section 41.10, and the reader should not feel obliged to believe the picture until then.

Call the resulting Riemann surface R . The two branches assemble into a holomorphic $g : R \rightarrow \mathbb{C}$, and the two inclusions $D_{\pm} \hookrightarrow \mathbb{C} \setminus \{0, \pm 1\}$ assemble into a holomorphic covering map $\pi : R \rightarrow \mathbb{C} \setminus \{0, \pm 1\}$, with

$$g(p)^2 = \pi(p)^3 - \pi(p) \quad \text{for all } p \in R.$$

§41.8.3 Compactifying the Cubic

The surface R above is missing four points. To restore them, we use the graph description. Let

$$R_1 = \{(z, w) \in \mathbb{C}^2 : w^2 = z^3 - z\}.$$

One checks (this is an example sheet exercise, and we shall do the general case in Section 41.9.3) that R_1 is a Riemann surface: away from $w = 0$ the projection $(z, w) \mapsto z$ is a chart, and at the three points $(0, 0), (\pm 1, 0)$ the projection $(z, w) \mapsto w$ is a chart. It contains R as the complement of those three points, so R_1 restores 0 and ± 1 . Only ∞ is missing.

To add it, change coordinates so that ∞ becomes finite. Put

$$u = \frac{1}{z}, \quad v = \frac{z}{w},$$

so that $z = 1/u$ and $w = z/v = 1/(uv)$. Substituting into $w^2 = z^3 - z$:

$$\frac{1}{u^2 v^2} = \frac{1}{u^3} - \frac{1}{u} \implies u = v^2(1 - u^2).$$

So set

$$R_2 = \{(u, v) \in \mathbb{C}^2 : u = v^2(1 - u^2)\},$$

again a Riemann surface by the same argument, and note that $(u, v) = (0, 0)$ lies on it – this is our point at infinity. The map

$$\Phi(z, w) = \left(\frac{1}{z}, \frac{z}{w}\right)$$

is a conformal equivalence from $R_1 \setminus \{(0, 0), (\pm 1, 0)\}$ onto $R_2 \setminus \{(0, 0)\}$, with inverse $(u, v) \mapsto (1/u, 1/(uv))$. Define

$$\hat{R} = R_1 \cup_{\Phi} R_2.$$

Proposition 41.8.5

$\hat{R}$ is a compact Riemann surface.

Proof. By Proposition 41.8.2 we need only Hausdorffness and compactness. Define $\hat{\pi} : \hat{R} \rightarrow \mathbb{C}_{\infty}$ by $\hat{\pi}(i_1(z, w)) = z$ and $\hat{\pi}(i_2(u, v)) = 1/u$; these agree on the overlap by construction of Φ , so $\hat{\pi}$ is a well-defined continuous map, and it is holomorphic in the glued structure.

Hausdorff. Points both in $i_1(R_1)$, or both in $i_2(R_2)$, are separated there. The only points not in $i_2(R_2)$ are $i_1(0, 0), i_1(\pm 1, 0)$, and the only point not in $i_1(R_1)$ is $i_2(0, 0)$. These are separated by $\hat{\pi}^{-1}(D(0, 2))$ and $\hat{\pi}^{-1}(\mathbb{C}_{\infty} \setminus \overline{D}(0, 2))$.

Compact. We show sequential compactness (which suffices, $\hat{R}$ being second count-

able and metrisable, or directly by a covering argument). Let (p_n) be a sequence in $\hat{R}$. If some subsequence has $|\hat{\pi}(p_n)| \leq M$, pass to a further subsequence with $\hat{\pi}(p_n) \rightarrow z_0 \in \mathbb{C}$; there are at most two w_0 with $w_0^2 = h(z_0)$, so a further subsequence converges to some $i_1(z_0, w_0)$. Otherwise $|\hat{\pi}(p_n)| \rightarrow \infty$, and since $i_2(0, 0)$ is the unique point with $\hat{\pi} = \infty$, and $\hat{\pi}$ is a proper map near there, $p_n \rightarrow i_2(0, 0)$. $\square$

We have obtained a compact Riemann surface $\hat{R}$ carrying a meromorphic function $\hat{\pi}$ (of degree 2, as we shall see) and a meromorphic $\hat{g}$ extending g , with $\hat{g}^2 = \hat{\pi}^3 - \hat{\pi}$. This is the **elliptic curve** $w^2 = z^3 - z$, and in Section 41.11 we shall discover that it is conformally a complex torus.

§41.9 Branching and the Degree

The map $\hat{p}_k : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$, $z \mapsto z^k$, is k -to-one except over 0 and ∞ , where the k preimages collide. This is the only local phenomenon there is: we now prove that *every* non-constant holomorphic map looks like $z \mapsto z^n$ in suitable coordinates.

§41.9.1 The Local Normal Form

Proposition 41.9.1 (Local Normal Form)

Let $f : R \rightarrow S$ be a non-constant holomorphic map of Riemann surfaces and $p \in R$. Then there is an integer $n \geq 1$ and charts (ϕ, U) about p and (ψ, V) about $f(p)$ with $\phi(p) = 0$, $\psi(f(p)) = 0$ and

$$\psi \circ f \circ \phi^{-1}(z) = z^n$$

on $\phi(U)$.

Proof. Start with any charts (ϕ_0, U_0) about p and (ψ, V) about $f(p)$, normalised so that $\phi_0(p) = 0$ and $\psi(f(p)) = 0$, with $f(U_0) \subseteq V$. The local expression $F = \psi \circ f \circ \phi_0^{-1}$ is holomorphic near 0 with $F(0) = 0$, and is not identically zero by Corollary 41.5.2. So by Proposition 41.1.3 there is $n \geq 1$ and a holomorphic G near 0 with

$$F(z) = z^n G(z), \quad G(0) \neq 0.$$

Choose $\delta > 0$ with $D(0, \delta)$ inside the domain of G and $G(D(0, \delta)) \subseteq D(G(0), |G(0)|)$; the latter disc misses 0 and is simply connected, so there is a holomorphic branch ρ of the n -th root on it. Put

$$h(z) = z \rho(G(z)), \quad z \in D(0, \delta),$$

so that $h(z)^n = z^n G(z) = F(z)$. Now $h(0) = 0$ and $h'(0) = \rho(G(0)) \neq 0$, so by the inverse function theorem (Theorem 41.1.13) h is a biholomorphism from some $D(0, \varepsilon)$ onto an open set.

Set $\phi = h \circ \phi_0$, restricted to $U = \phi_0^{-1}(D(0, \varepsilon))$. This is a chart (a composite of a chart with a biholomorphism) with $\phi(p) = 0$, and it is compatible with the existing

structure since h is biholomorphic. Finally

$$\begin{aligned}\psi \circ f \circ \phi^{-1}(z) &= \psi \circ f \circ \phi_0^{-1} \circ h^{-1}(z) \\ &= F(h^{-1}(z)) = h(h^{-1}(z))^n = z^n. \square\end{aligned}$$

Definition 41.9.2 (Multiplicity)

The integer n of Proposition 41.9.1 is the **multiplicity** (or **ramification index**, or **local degree**) of f at p , written $\text{mult}_f(p)$.

Lemma 41.9.3

$\text{mult}_f(p)$ is well defined, i.e. independent of the charts chosen.

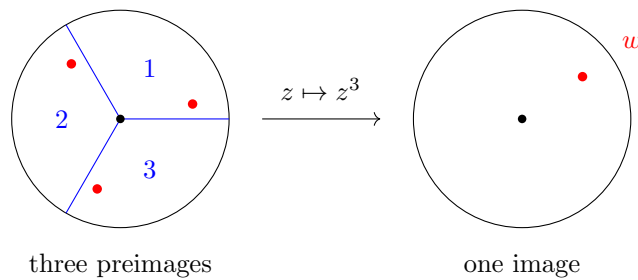
Proof. If z^n and z^m are both local expressions, then they differ by pre-composition and post-composition with biholomorphisms fixing 0. So $\alpha(z)^n = \beta(z^m)$ near 0 with α, β biholomorphic and $\alpha(0) = \beta(0) = 0$. Counting orders of vanishing at 0: α and β have simple zeros (their derivatives are non-zero), so the left side vanishes to order n and the right to order m . Hence $n = m$. $\square$

Equivalently: $\text{mult}_f(p)$ is the order of the zero of $\psi \circ f \circ \phi^{-1} - \psi(f(p))$ at $\phi(p)$, in any charts.

Definition 41.9.4 (Ramification and Branch Points)

If $\text{mult}_f(p) > 1$ we call p a **ramification point** of f , and $f(p)$ a **branch point**. A point $q \in S$ with $f^{-1}(q)$ a single point p of multiplicity $\deg f$ is **totally ramified**.

The local picture is worth having firmly in mind. Under $z \mapsto z^n$, the disc $|z| < \varepsilon$ is wrapped n times around the disc $|w| < \varepsilon^n$: each of the n sectors of angle $2\pi/n$ is opened out to a full disc, and the n preimages of a point $w \neq 0$ merge into the single preimage 0 of 0.



Example 41.9.5

Let $f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ be a polynomial $f(z) = a_d z^d + \cdots + a_0$ with $a_d \neq 0$ and $d \geq 1$, extended by $f(\infty) = \infty$. Away from ∞ , the local expression of f is f itself and $\text{mult}_f(p)$ is the order of the zero of $f(z) - f(p)$ at p ; so the ramification points in $\mathbb{C}$

are exactly the zeros of f' . At ∞ , use the chart $w = 1/z$ upstairs and downstairs:

$$\frac{1}{f(1/w)} = \frac{w^d}{a_d + a_{d-1}w + \cdots + a_0w^d},$$

which vanishes to order d at $w = 0$. So $\text{mult}_f(\infty) = d$, and ∞ is totally ramified whenever $d \geq 2$.

Remark. More generally, if $f : R \rightarrow \mathbb{C}$ is holomorphic and (ϕ, U) is any chart about p with $F = f \circ \phi^{-1}$, then writing $F(z) - F(z_0) = (z - z_0)^m G(z)$ with $G(z_0) \neq 0$ and $m = \text{mult}_f(p)$, we get

$$F'(z) = (z - z_0)^{m-1} (mG(z) + (z - z_0)G'(z)),$$

so $F'(z_0) = 0$ if and only if $m > 1$. The ramification points are the zeros of the derivative in any chart. Since F' is not identically zero (as f is non-constant), its zeros are isolated: *a non-constant holomorphic map has a discrete set of ramification points*. If R is compact, there are only finitely many.

Lemma 41.9.6 (Multiplicativity)

If $f : R \rightarrow S$ and $g : S \rightarrow T$ are non-constant holomorphic, then

$$\text{mult}_{g \circ f}(p) = \text{mult}_g(f(p)) \cdot \text{mult}_f(p).$$

Proof. Choose charts in which f is $z \mapsto z^m$ with $m = \text{mult}_f(p)$, and in which g is $w \mapsto w^n$ with $n = \text{mult}_g(f(p))$; we may take the target chart for f to be the source chart for g , since both are just ‘some chart about $f(p)$ sending it to 0’, and the local normal form allows us to choose the source chart freely at each stage — more carefully, take the normal form charts for g first, then run the proof of Proposition 41.9.1 for f with that target chart. Then $g \circ f$ has local expression $z \mapsto (z^m)^n = z^{mn}$. $\square$

§41.9.2 Proper Maps and the Degree

Definition 41.9.7 (Proper Map)

A continuous map $f : R \rightarrow S$ is **proper** if $f^{-1}(K)$ is compact for every compact $K \subseteq S$.

If R is compact then every continuous $f : R \rightarrow S$ is proper, since a closed subset of a compact space is compact. That is the case we shall use most often, but properness is the right hypothesis in general: it is what makes fibres finite, and what stops points of a fibre escaping to infinity as we move around in S .

Example 41.9.8

- (i) A non-constant polynomial $f : \mathbb{C} \rightarrow \mathbb{C}$ is proper, since $|f(z)| \rightarrow \infty$ as $|z| \rightarrow \infty$ and so the preimage of a bounded set is bounded and closed.
- (ii) $p_k : \mathbb{C}^* \rightarrow \mathbb{C}^*$ is proper: $|z^k| \rightarrow \infty$ as $|z| \rightarrow \infty$ and $|z^k| \rightarrow 0$ as $z \rightarrow 0$, so

preimages of compact subsets of $\mathbb{C}^*$ stay away from both ends.

- (iii) (*Non-examples.*) $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$ is not proper, since $\exp^{-1}(\{1\}) = 2\pi i\mathbb{Z}$ is an infinite discrete set. The inclusion $\mathbb{D} \hookrightarrow \mathbb{C}$ is not proper either, as $\overline{D}(0, 1)$ has non-compact preimage $\mathbb{D}$. Neither map has a sensible degree: $\exp$ is ∞ -to-one, and the inclusion is one-to-one over $\mathbb{D}$ and zero-to-one outside it.

The two properties of proper maps we need are the following.

Lemma 41.9.9

Let $f : R \rightarrow S$ be a proper continuous map of Riemann surfaces. Then f is a closed map. If moreover f is holomorphic and non-constant, every fibre $f^{-1}(q)$ is finite.

Proof. For the fibres, $f^{-1}(q)$ is compact, being the preimage of the compact set $\{q\}$, and discrete by Corollary 41.5.2; a compact discrete space is finite.

For closedness, let $A \subseteq R$ be closed and $q \in \overline{f(A)}$. Pick a chart (ψ, V) about q with $\psi(q) = 0$ and put $N = \psi^{-1}(\overline{D}(0, r))$ for some small $r > 0$ with $\overline{D}(0, r) \subseteq \psi(V)$, a compact neighbourhood of q . Then $f^{-1}(N)$ is compact by properness, so $A \cap f^{-1}(N)$ is compact, and therefore $f(A \cap f^{-1}(N))$ is compact and hence closed in the Hausdorff space S . Every neighbourhood of q contained in the interior of N meets $f(A)$, and the points of $f(A)$ it meets lie in $f(A \cap f^{-1}(N))$; so q is in the closure of that set, which is the set itself. Hence $q \in f(A)$, and $f(A)$ is closed. $\square$

Theorem 41.9.10 (Valency Theorem)

Let $f : R \rightarrow S$ be a non-constant proper holomorphic map of Riemann surfaces. Then the function

$$n(q) = \sum_{p \in f^{-1}(q)} \text{mult}_f(p)$$

is constant on S . In particular this applies to any non-constant holomorphic map of compact Riemann surfaces.

Proof. First, each fibre $f^{-1}(q)$ is finite by Lemma 41.9.9, so n is well defined and $\mathbb{N}$ -valued. Since S is connected it suffices to prove n is locally constant.

Fix $q_0 \in S$ and write $f^{-1}(q_0) = \{p_1, \dots, p_k\}$ with $m_i = \text{mult}_f(p_i)$. Choose a chart (ψ, V) about q_0 with $\psi(q_0) = 0$, and by Proposition 41.9.1 charts (ϕ_i, U_i) about p_i with $\psi \circ \phi_i^{-1}(z) = z^{m_i}$; shrinking, we may take the U_i pairwise disjoint and contained in $f^{-1}(V)$.

Let $U = \bigcup_i U_i$. Then $R \setminus U$ is closed in R , so $K = f(R \setminus U)$ is closed in S by Lemma 41.9.9. Note that $q_0 \notin K$, since the whole fibre over q_0 lies in U . Put

$$V' = V \setminus K,$$

an open neighbourhood of q_0 , and $U'_i = U_i \cap f^{-1}(V')$. Then

$$f^{-1}(V') \subseteq U, \quad \text{so} \quad f^{-1}(V') = \coprod_{i=1}^k U'_i.$$

On U'_i , in the charts ϕ_i and ψ , f is the map $z \mapsto z^{m_i}$ restricted to an open set; such a map takes each value $w \neq 0$ exactly m_i times (counted without multiplicity, all multiplicities being 1) and the value 0 once with multiplicity m_i . Hence for every $q \in V'$,

$$n(q) = \sum_{i=1}^k \sum_{p \in U'_i \cap f^{-1}(q)} \text{mult}_f(p) = \sum_{i=1}^k m_i = n(q_0).$$

So n is constant on V' . □

Definition 41.9.11 (Degree)

The constant value n of Theorem 41.9.10 is the **degree** of f , written $\deg f$.

Thus a non-constant proper holomorphic map is ‘deg f -to-one, when points are counted with multiplicity’; in particular this is true of any non-constant holomorphic map of compact Riemann surfaces, which is the situation we shall almost always be in. Note that $p_k : \mathbb{C}^* \rightarrow \mathbb{C}^*$ has degree k and a polynomial $\mathbb{C} \rightarrow \mathbb{C}$ has degree equal to its degree as a polynomial, so the notion is consistent with the classical one even off compact surfaces.

Corollary 41.9.12

A holomorphic map $f : R \rightarrow S$ of compact Riemann surfaces with $\deg f = 1$ is a conformal equivalence. In particular a holomorphic bijection of compact Riemann surfaces is a conformal equivalence.

Proof. If $\deg f = 1$ then every fibre is a single point of multiplicity 1, so f is a bijection with no ramification points. By Proposition 41.9.1 f is locally biholomorphic, so f^{-1} is holomorphic. □

Example 41.9.13

If f is a polynomial of degree d , regarded as $f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$, then $\deg f = d$: compute $n(\infty)$, which by Example 41.9.5 is $\text{mult}_f(\infty) = d$ since ∞ is the only preimage of ∞ .

Corollary 41.9.14 (Fundamental Theorem of Algebra)

A polynomial $f \in \mathbb{C}[z]$ of degree $d \geq 1$ has exactly d roots in $\mathbb{C}$, counted with multiplicity.

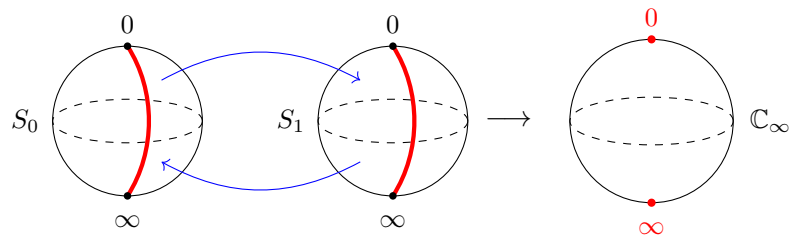
Proof. Extend to $f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$, of degree d . Then $n(0) = d$, and $f^{-1}(0) \subseteq \mathbb{C}$ since $f(\infty) = \infty$. The multiplicity of f at a root is the order of the root. □

Example 41.9.15

A rational function $f = p/q$ in lowest terms with $\deg p = m$, $\deg q = n$ has $\deg f = \max(m, n)$ as a map $\mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$. Indeed if $m > n$ then ∞ is the only preimage of ∞ with multiplicity $m - n$... rather than pursue this, compute $n(w)$ for generic w : the equation $p(z) - wq(z) = 0$ has $\max(m, n)$ roots for all but finitely many w , since the leading coefficient of $p - wq$ vanishes for at most one w .

Example 41.9.16 (The Two-Sheeted Cover of the Sphere)

Consider $\hat{p}_2 : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty, z \mapsto z^2$. It has degree 2, and its branch points are 0 and ∞ , each totally ramified. Topologically: take two copies of the sphere, slit each along an arc from 0 to ∞ , and glue crosswise. The result is again a sphere.



Gluing the upper side of the slit in S_0 to the lower side of the slit in S_1 , and vice versa, returns a sphere; the two red points are the branch points 0 and ∞ , and over every other point there are two preimages. That the total space is again a sphere and not, say, a torus is a genuine constraint, and Riemann–Hurwitz is the tool that makes such constraints computable.

§41.9.3 Riemann Surfaces of Algebraic Functions

The examples $w^2 = z^3 - z$ and $w^k = z$ are instances of a general construction. Let

$$P(z, w) = w^d + c_{d-1}(z)w^{d-1} + \cdots + c_0(z)$$

be a polynomial in w whose coefficients are polynomials in z , and suppose P is irreducible in $\mathbb{C}[z, w]$. The **algebraic function** $w(z)$ defined by $P(z, w) = 0$ is the classical object; the modern one is the surface

$$X' = \{(z, w) \in \mathbb{C}^2 : P(z, w) = 0\}.$$

Proposition 41.9.17

Suppose X' is **smooth**, that is, at every point of X' at least one of $\partial P/\partial z$, $\partial P/\partial w$ is non-zero. Then X' is a Riemann surface, with charts given by the coordinate projections, and $\pi_z : X' \rightarrow \mathbb{C}, (z, w) \mapsto z$, is holomorphic.

Proof sketch. X' is Hausdorff and second countable as a subspace of $\mathbb{C}^2$. Near a point (z_0, w_0) with $\partial P/\partial w \neq 0$, the holomorphic implicit function theorem provides a holomorphic η near z_0 with $\eta(z_0) = w_0$ and $P(z, \eta(z)) = 0$, and the graph of η is a neighbourhood of (z_0, w_0) in X' ; so π_z is a homeomorphism there and serves as a chart. Symmetrically, where $\partial P/\partial z \neq 0$, π_w is a chart. Smoothness says these cover

X' . The transition functions are η and its inverse, which are holomorphic. Connectedness is the one genuinely non-trivial point, and it follows from irreducibility of P ; we verify it by hand in the examples we need. $\square$

Away from the finitely many z for which $P(z, \cdot)$ has a repeated root – the **discriminant locus** – the map π_z is a d -sheeted covering map, since the d roots w vary holomorphically and distinctly. Over the discriminant locus, roots collide and π_z ramifies. Compactifying (by gluing in charts at ∞ , as in Section 41.8.3) gives a compact Riemann surface X with a degree- d meromorphic function $\hat{\pi}_z : X \rightarrow \mathbb{C}_\infty$.

Example 41.9.18 (Fermat Curves)

Let $d \geq 1$ and

$$F'_d = \{(x, y) \in \mathbb{C}^2 : x^d + y^d = 1\}.$$

Here $P = y^d + x^d - 1$, with $\partial P / \partial y = dy^{d-1}$ and $\partial P / \partial x = dx^{d-1}$, which vanish simultaneously only at $(0, 0)$, not a point of F'_d . So F'_d is smooth.

Concretely, π_x has local inverse $x \mapsto (x, \sqrt[d]{1-x^d})$, which exists and is holomorphic near any x_0 with $x_0^d \neq 1$, i.e. away from $y_0 = 0$; and π_y provides charts near the points where $y = 0$, since there $x \neq 0$. These cover F'_d .

For connectedness: let

$$D = \mathbb{C} \setminus \bigcup_{i=0}^{d-1} \{t\zeta_d^i : t \geq 1\}, \quad \zeta_d = e^{2\pi i/d},$$

the plane with d radial slits removed. This is simply connected and $1 - x^d$ does not vanish on it, so there are d holomorphic branches of $(1 - x^d)^{1/d}$ on D , each extending continuously to the slit endpoints ζ_d^i . Given $(x_0, y_0) \in F'_d$ with $x_0 \in D$, choose the branch y with $y(x_0) = y_0$ and a path γ in D from x_0 to 1; then $t \mapsto (\gamma(t), y(\gamma(t)))$ joins (x_0, y_0) to $(1, 0)$. Points with $x_0 \notin D$ can first be joined to nearby points with $x_0 \in D$, as D is dense. So F'_d is path-connected, hence a Riemann surface.

The compactification F_d is obtained by gluing on a second chart at infinity (an example sheet exercise), and $\hat{\pi}_x : F_d \rightarrow \mathbb{C}_\infty$ has degree d . Its ramification points are the d points $(\zeta_d^i, 0)$, each of multiplicity d (there $1 - x^d$ has a simple zero, so $y = (1 - x^d)^{1/d}$ is a d -th root of a simple zero), and $\hat{\pi}_x^{-1}(\infty)$ consists of d distinct unramified points. We compute the genus of F_d in Example 41.10.12.

§41.10 Euler Characteristic and Riemann–Hurwitz

Everything so far has been local or analytic. We now bring in topology, and the payoff is immediate: the degree and ramification of a holomorphic map determine the topology of its source from that of its target.

§41.10.1 Triangulations and the Euler Characteristic

Definition 41.10.1 (Triangulation)

Let S be a compact Riemann surface. A **topological triangle** in S is a continuous embedding $\Delta \hookrightarrow S$ of a closed non-degenerate triangle $\Delta \subseteq \mathbb{R}^2$. A **triangulation**

of S is a finite collection $\{\Delta_i\}$ of topological triangles such that

- (i) $\bigcup_i \Delta_i = S$;
- (ii) for $i \neq j$, $\Delta_i \cap \Delta_j$ is empty, a common vertex, or a common edge;
- (iii) each edge lies in exactly two triangles.

Definition 41.10.2 (Euler Characteristic)

The **Euler characteristic** of a triangulation is $\chi = V - E + F$, where V, E, F count vertices, edges and faces.

Lemma 41.10.3

Every compact Riemann surface admits a triangulation, and χ is independent of the triangulation chosen.

We take this on trust; the first part is Radó's theorem and the second is the topological invariance of simplicial homology, proved in II Algebraic Topology (where $\chi = \sum_i (-1)^i \text{rank } H_i$). We write $\chi(S)$ for the common value.

Example 41.10.4

$\chi(\mathbb{C}_\infty) = 2$. Triangulate the sphere as the boundary of a tetrahedron: $V - E + F = 4 - 6 + 4 = 2$.

Example 41.10.5

$\chi(\mathbb{C}/\Lambda) = 0$. Represent the torus as a square with opposite sides identified and cut it into two triangles by a diagonal. After identification there is $V = 1$ vertex, $E = 3$ edges (two sides and the diagonal), and $F = 2$ faces, giving $1 - 3 + 2 = 0$. (Strictly this is not a triangulation in the sense above, since the two triangles share more than one edge; subdividing repairs this without changing χ .)

We quote from II Algebraic Topology the classification of surfaces: every compact connected orientable surface is homeomorphic to Σ_g , the g -holed torus, for a unique $g \geq 0$, and

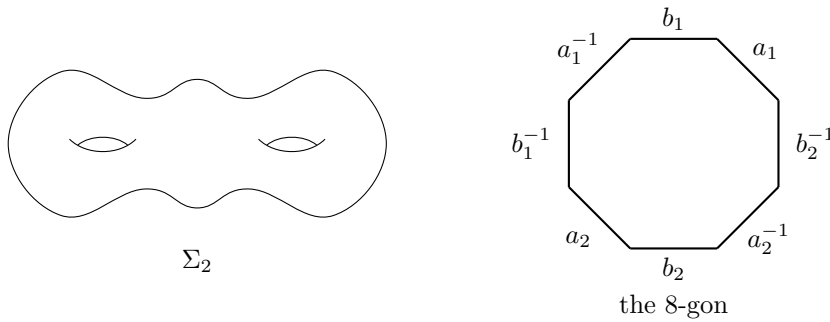
$$\chi(\Sigma_g) = 2 - 2g.$$

A Riemann surface is orientable, since holomorphic transition functions have positive Jacobian determinant $|\phi'|^2 > 0$. So a compact Riemann surface is homeomorphic to some Σ_g , and χ determines it up to homeomorphism.

Definition 41.10.6 (Genus)

The **genus** of a compact Riemann surface S is $g(S) = (2 - \chi(S))/2$.

The standard picture of Σ_g is a sphere with g handles; equivalently it is a $4g$ -gon with its edges identified in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}$. For $g = 2$:



From the octagon picture one reads off $\chi(\Sigma_2)$ directly: all 8 vertices are identified to one, the 8 edges become 4, and there is 1 face, giving $1 - 4 + 1 = -2 = 2 - 2 \cdot 2$.

§41.10.2 The Riemann–Hurwitz Formula

Theorem 41.10.7 (Riemann–Hurwitz)

Let $f : R \rightarrow S$ be a non-constant holomorphic map of compact Riemann surfaces of degree n . Then

$$\chi(R) = n \chi(S) - \sum_{p \in R} (\text{mult}_f(p) - 1).$$

Equivalently, in terms of genera,

$$2g(R) - 2 = n(2g(S) - 2) + \sum_{p \in R} (\text{mult}_f(p) - 1).$$

The sum is finite: as we observed above, the ramification points of f are isolated, and R is compact, so there are finitely many.

Proof. As in the proof of the valency theorem, every $q \in S$ has an open neighbourhood V_q which is **well covered**: $f^{-1}(V_q)$ is a disjoint union of open sets on each of which f is, in suitable charts, a power map $z \mapsto z^m$, and V_q contains no branch point other than possibly q . By compactness of S , finitely many $V_{q_1}, \dots, V_{q_r}$ cover S ; in particular there are finitely many branch points.

Choose a triangulation of S and subdivide it repeatedly (barycentrically, say) until

- (i) every branch point of f is a vertex, and
- (ii) every triangle is contained in some V_{q_i} .

Both are achievable: (i) by first taking a triangulation with the finitely many branch points among the vertices, and (ii) since subdivision makes the triangles arbitrarily small and the V_{q_i} have a Lebesgue number.

Now pull back. If $\Delta \subseteq V_{q_i}$ is a (closed) triangle then $f^{-1}(\Delta)$ is a disjoint union of triangles, each mapped homeomorphically onto Δ by f ; the same for edges. Indeed Δ is simply connected and, since its interior contains no branch point, f restricted over the interior is a covering map of degree n ; and the branch points on $\partial\Delta$ are vertices, where the local picture $z \mapsto z^m$ still sends the m triangle-corners upstairs homeomorphically onto the corner downstairs. So the preimages of the triangles of

S form a triangulation of R , with

$$F_R = nF_S, \quad E_R = nE_S,$$

since each face and each edge downstairs has exactly n preimages.

Vertices are where the count fails. For $q \in S$ a vertex, the valency theorem gives

$$\begin{aligned} |f^{-1}(q)| &= \sum_{p \in f^{-1}(q)} 1 = \sum_{p \in f^{-1}(q)} \text{mult}_f(p) - \sum_{p \in f^{-1}(q)} (\text{mult}_f(p) - 1) \\ &= n - \sum_{p \in f^{-1}(q)} (\text{mult}_f(p) - 1). \end{aligned}$$

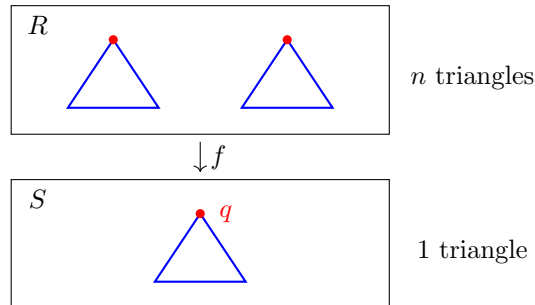
Summing over the vertices q of S , and noting that every ramification point of f lies over a branch point, which is a vertex,

$$V_R = nV_S - \sum_{p \in R} (\text{mult}_f(p) - 1).$$

Therefore

$$\chi(R) = V_R - E_R + F_R = n(V_S - E_S + F_S) - \sum_{p \in R} (\text{mult}_f(p) - 1),$$

which is the first formula. The second follows by substituting $\chi = 2 - 2g$. $\square$



Each triangle and each edge downstairs has exactly n preimages; only the vertices lying under ramification points have fewer than n preimages, and the deficiency is exactly $\sum (\text{mult}_f(p) - 1)$.

Remark. The correction term $\sum_p (\text{mult}_f(p) - 1)$ is always even, being $\chi(R) - n\chi(S) = (2 - 2g(R)) - n(2 - 2g(S))$. This trivial-looking observation is surprisingly useful: it often pins down the ramification at a point we have not analysed.

§41.10.3 First Applications

Corollary 41.10.8

Suppose $f : R \rightarrow S$ is an unramified non-constant holomorphic map of compact Riemann surfaces (equivalently, a covering map) of degree n . Then $g(R) - 1 = n(g(S) - 1)$, and

- (i) if $g(S) = 0$ then $n = 1$ and $R \cong S \cong \mathbb{C}_\infty$;

- (ii) if $g(S) = 1$ then $g(R) = 1$, and n is unrestricted;
- (iii) if $g(S) \geq 2$ then either $n = 1$ and f is a conformal equivalence, or $n \geq 2$ and $g(R) > g(S)$.

Proof. Riemann–Hurwitz with no correction term gives $2g(R) - 2 = n(2g(S) - 2)$. If $g(S) = 0$ then $g(R) - 1 = -n < 0$, forcing $g(R) = 0$ and $n = 1$; then f is a conformal equivalence by Corollary 41.9.12, and a genus 0 compact surface is $\mathbb{C}_\infty$ (which we prove in Corollary 41.13.8). If $g(S) = 1$ then $g(R) = 1$ for any n . If $g(S) \geq 2$ then $g(R) - 1 = n(g(S) - 1) \geq n$, so $n \geq 2$ gives $g(R) \geq 2g(S) - 1 > g(S)$. $\square$

Example 41.10.9

That case (ii) is genuinely unrestricted: take $\Lambda_n = \langle n, i \rangle$ for $n \in \mathbb{Z}_{>0}$. Then $\Lambda_n \leq \Lambda_1 = \langle 1, i \rangle$ with index n , so the identity on $\mathbb{C}$ descends to an unramified degree- n map $\mathbb{C}/\Lambda_n \rightarrow \mathbb{C}/\Lambda_1$ of tori.

Example 41.10.10 (The Cubic Curve)

Let $\hat{R}$ be the compactification of the surface of $w^2 = z^3 - z$ from Section 41.8.3, with $\hat{\pi} : \hat{R} \rightarrow \mathbb{C}_\infty$. Since a generic z has exactly two preimages $(z, \pm\sqrt{h(z)})$, $\deg \hat{\pi} = 2$. The branch points are $0, \pm 1$ (where h has a simple zero, so the two sheets come together) and ∞ ; each is totally ramified, with a single preimage of multiplicity 2. So

$$2g(\hat{R}) - 2 = 2(0 - 2) + 4 \cdot (2 - 1) = -4 + 4 = 0 \implies g(\hat{R}) = 1,$$

confirming that $\hat{R}$ is a torus.

Remark. Suppose we had not analysed the behaviour at ∞ . Writing

$$C = \sum_{p \in \hat{\pi}^{-1}(\infty)} (\text{mult}_{\hat{\pi}}(p) - 1),$$

the correction term is $3 + C$, which must be even by the parity remark above; so C is odd, and since $\deg \hat{\pi} = 2$ we must have $C = 1$, i.e. ∞ is a branch point. The parity argument has done the local analysis for us. This works whenever $\deg f = 2$.

Example 41.10.11 (Hyperelliptic Curves)

More generally, let $h(z)$ be a polynomial of degree d with d distinct roots, and let X be the compactification of $\{w^2 = h(z)\}$, with $\hat{\pi} : X \rightarrow \mathbb{C}_\infty$ of degree 2. The branch points in $\mathbb{C}$ are the d roots of h . What happens at ∞ ? Setting $u = 1/z$, we get $w^2 = u^{-d}(1 + O(u))$, so $w \sim u^{-d/2}$: if d is even there are two points over ∞ , unramified; if d is odd there is one, ramified. Either way the total correction is d or $d + 1$, i.e. $2\lceil d/2 \rceil$, and

$$2g(X) - 2 = 2(-2) + 2\lceil \frac{d}{2} \rceil \implies g(X) = \lceil \frac{d}{2} \rceil - 1 = \left\lfloor \frac{d-1}{2} \right\rfloor.$$

So $d = 3, 4$ give genus 1; $d = 5, 6$ give genus 2; and so on. These are the **hyperelliptic** curves.

Example 41.10.12 (Fermat Curves and the Degree–Genus Formula)

Let F_d be the compactified Fermat curve of Example 41.9.18, with $\hat{\pi}_x : F_d \rightarrow \mathbb{C}_\infty$ of degree d . There are d ramification points $(\zeta_d^i, 0)$, each of multiplicity d , and no others. So

$$2g(F_d) - 2 = d(0 - 2) + d(d - 1) \implies g(F_d) = \frac{(d - 1)(d - 2)}{2}.$$

This is an instance of the **degree–genus formula**: a smooth projective plane curve of degree d has genus $(d - 1)(d - 2)/2$. For $d = 1, 2$ we get $g = 0$ (lines and conics are spheres); for $d = 3$, $g = 1$ (cubics are tori, as we found for $w^2 = z^3 - z$ after homogenising); for $d = 4$, $g = 3$; and g grows quadratically. In particular there exist compact Riemann surfaces of arbitrarily large genus – something not at all obvious from the definitions.

Note that not every genus arises from a plane curve: $(d - 1)(d - 2)/2$ takes the values $0, 0, 1, 3, 6, 10, \dots$, so no smooth plane curve has genus 2. (Genus 2 surfaces exist all the same – they are hyperelliptic, by the previous example with $d = 5$.)

§41.11 Elliptic Functions and Complex Tori

We now describe all the meromorphic functions on the two compact surfaces we understand best: the sphere, and the torus. The sphere is easy and classical. The torus is the heart of the course, and the answer is remarkable – every complex torus is a plane cubic curve, and every elliptic function is a rational expression in one particular function and its derivative.

§41.11.1 Meromorphic Functions on the Sphere**Proposition 41.11.1**

Every meromorphic function $f : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$ is rational:

$$f(z) = c \frac{(z - a_1) \cdots (z - a_m)}{(z - b_1) \cdots (z - b_n)}$$

for some $c \in \mathbb{C}^*$ and $a_i, b_j \in \mathbb{C}$ with $a_i \neq b_j$.

Proof. We may assume f is non-constant. Replacing f by $1/f$ if necessary (which does not affect the conclusion), assume $f(\infty) \neq \infty$. The poles of f form a discrete closed subset of the compact space $\mathbb{C}_\infty$, hence a finite set $\{b_1, \dots, b_n\}$, and none is ∞ . About each b_j write the Laurent expansion

$$f(z) = \sum_{l \geq -k_j} c_{j,l}(z - b_j)^l, \quad c_{j,-k_j} \neq 0,$$

and let $Q_j(z) = \sum_{l=-k_j}^{-1} c_{j,l}(z - b_j)^l$ be the principal part. Then

$$g = f - \sum_{j=1}^{n'} Q_j$$

is holomorphic on all of $\mathbb{C}$ – the principal parts have cancelled every pole – and is bounded near ∞ , since f is (having a finite value there) and each $Q_j \rightarrow 0$. By Liouville, g is constant. So f is a constant plus a sum of principal parts, which is a rational function. $\square$

Remark. If f is written as above in lowest terms, its degree as a holomorphic map is $\max(m, n)$: this counts the preimages of a generic point. The zeros a_i and poles b_j may be prescribed arbitrarily, subject only to being finite in number – there is no constraint on the sphere. On a torus we shall find that there is one.

§41.11.2 Periodic Meromorphic Functions

Definition 41.11.2 (Period)

Let $f : \mathbb{C} \rightarrow \mathbb{C}_\infty$ be meromorphic. A **period** of f is $\omega \in \mathbb{C}$ with $f(z + \omega) = f(z)$ for all z .

The set Ω of periods is a subgroup of $(\mathbb{C}, +)$, and it is closed: if $\omega_n \rightarrow \omega$ with $\omega_n \in \Omega$ then $f(z + \omega) = f(z)$ for all z not a pole, by continuity, and hence everywhere.

Lemma 41.11.3

Let f be a non-constant meromorphic function on $\mathbb{C}$ with period group Ω . Then exactly one of the following holds:

- (i) $\Omega = \{0\}$;
- (ii) $\Omega = \langle \omega \rangle \cong \mathbb{Z}$ for some $\omega \neq 0$;
- (iii) $\Omega = \langle \omega_1, \omega_2 \rangle \cong \mathbb{Z}^2$ is a lattice, with $\omega_1/\omega_2 \notin \mathbb{R}$.

(If f is constant then $\Omega = \mathbb{C}$.)

Proof sketch. Ω is a closed subgroup of $\mathbb{C} \cong \mathbb{R}^2$, and the classification of such subgroups gives $\{0\}$, a discrete rank-1 or rank-2 subgroup, a line, a line plus a discrete complement, or all of $\mathbb{C}$. If Ω contained a line $\mathbb{R}\omega$ then the zeros of $f - f(z_0)$ would be non-discrete for suitable z_0 , contradicting the identity theorem. The remaining possibilities are (i)–(iii). This is on the example sheets. $\square$

Definition 41.11.4

A non-constant meromorphic f on $\mathbb{C}$ is **simply periodic** in case (ii), and **doubly periodic** or **elliptic** in case (iii).

The reason to care is the correspondence with functions on quotients.

Proposition 41.11.5

Let f be meromorphic on $\mathbb{C}$ and suppose $\Lambda \subseteq \Omega$ where $\Lambda \leq \mathbb{C}$ is a lattice (respectively, $\langle \omega \rangle \subseteq \Omega$). Then there is a unique meromorphic function $\bar{f}$ on $\mathbb{C}/\Lambda$ (respectively on $\mathbb{C}^*$) with

$$f = \bar{f} \circ \pi, \quad \pi : \mathbb{C} \rightarrow \mathbb{C}/\Lambda \quad (\text{resp. } z \mapsto e^{2\pi iz/\omega}).$$

Proof. Uniqueness is clear as π is surjective. For existence set $\bar{f}(\pi(z)) = f(z)$, which is well defined by periodicity. It is holomorphic as a map to $\mathbb{C}_\infty$ because π is a covering map with local holomorphic inverses: near any point, $\bar{f} = f \circ (\pi|_U)^{-1}$. $\square$

Thus simply periodic functions are the same thing as meromorphic functions on $\mathbb{C}^* \cong \mathbb{C}/\langle \omega \rangle$, and elliptic functions with $\Lambda \subseteq \Omega$ are the same thing as meromorphic functions on the complex torus $\mathbb{C}/\Lambda$. Everything we know about compact surfaces now applies.

Corollary 41.11.6

A holomorphic doubly periodic function on $\mathbb{C}$ is constant.

Proof. It descends to a holomorphic function on the compact surface $\mathbb{C}/\Lambda$, which is constant by Corollary 41.5.5. $\square$

Corollary 41.11.7

A non-constant elliptic function, viewed as $\bar{f} : \mathbb{C}/\Lambda \rightarrow \mathbb{C}_\infty$, has $\deg \bar{f} \geq 2$.

Proof. If $\deg \bar{f} = 1$ then $\bar{f}$ is a conformal equivalence $\mathbb{C}/\Lambda \rightarrow \mathbb{C}_\infty$ by Corollary 41.9.12, contradicting $\chi(\mathbb{C}/\Lambda) = 0 \neq 2 = \chi(\mathbb{C}_\infty)$. (Equivalently: Riemann–Hurwitz would give $0 = 1 \cdot (-2) + 0$.) $\square$

It is worth also seeing the classical proof, which is where the theory came from.

Second proof of Corollary 41.11.7. Choose z_0 so that no zero or pole of f lies on the boundary of the period parallelogram $P = z_0 + \{s\omega_1 + t\omega_2 : s, t \in [0, 1]\}$; this is possible as there are finitely many such points modulo Λ . The residue theorem gives

$$\sum_{z \text{ pole in } P^\circ} \text{res}_z(f) = \frac{1}{2\pi i} \oint_{\partial P} f(z) dz = 0,$$

because the contributions of opposite sides of P cancel: f takes the same values on them and they are traversed in opposite directions. If f had a single simple pole in P , its residue would be non-zero, contradiction. So f has either no poles (hence is constant) or at least two poles counted with multiplicity. $\square$

§41.11.3 The Weierstrass $\wp$ -Function

We need a non-constant elliptic function to exist at all. By the above the simplest possible has a single double pole per period parallelogram, and this pins down the definition.

Definition 41.11.8 (Weierstrass $\wp$ -function)

Let $\Lambda \subseteq \mathbb{C}$ be a lattice. The **Weierstrass $\wp$ -function** is

$$\wp(z) = \wp_\Lambda(z) = \frac{1}{z^2} + \sum_{\omega \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right).$$

The subtracted terms $1/\omega^2$ are there purely to make the series converge; without them it diverges. To prove convergence we need a lemma on lattice sums.

Lemma 41.11.9

Let $\Lambda = \langle \omega_1, \omega_2 \rangle$ be a lattice and $t \in \mathbb{R}$. Then

$$\sum_{\omega \in \Lambda \setminus \{0\}} \frac{1}{|\omega|^t}$$

converges if and only if $t > 2$.

Proof. Consider $Q = \{(t_1, t_2) \in \mathbb{R}^2 : |t_1| + |t_2| = 1\}$, a compact set, and the continuous function $(t_1, t_2) \mapsto |t_1\omega_1 + t_2\omega_2|$ on it. It attains a maximum M and a minimum m , and $m > 0$ since ω_1, ω_2 are linearly independent over $\mathbb{R}$ so the expression vanishes only at the origin, which is not in Q . Given $(k, l) \in \mathbb{Z}^2 \setminus \{0\}$, apply this with $t_1 = k/(|k| + |l|)$, $t_2 = l/(|k| + |l|)$ to get

$$m(|k| + |l|) \leq |k\omega_1 + l\omega_2| \leq M(|k| + |l|).$$

So the sum in question is bounded above and below by positive multiples of

$$\sum_{(k,l) \neq (0,0)} \frac{1}{(|k| + |l|)^t}.$$

For each $n \geq 1$ there are exactly $4n$ pairs (k, l) with $|k| + |l| = n$, so this equals

$$\sum_{n \geq 1} \frac{4n}{n^t} = 4 \sum_{n \geq 1} \frac{1}{n^{t-1}},$$

which converges if and only if $t - 1 > 1$. □

Notice that the same estimate shows every bounded region contains only finitely many lattice points, which is what we assumed in Proposition 41.4.26.

Theorem 41.11.10

$\wp_\Lambda$ is a well-defined elliptic function whose period group is exactly Λ . It is even, its poles are precisely the points of Λ , each a double pole, and $\deg \wp_\Lambda = 2$ as a map $\mathbb{C}/\Lambda \rightarrow \mathbb{C}_\infty$.

Proof. Convergence. Fix $R > 0$ and work on $|z| \leq R$. For $\omega \neq 0$,

$$\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} = \frac{\omega^2 - (z-\omega)^2}{\omega^2(z-\omega)^2} = \frac{z(2\omega - z)}{\omega^2(z-\omega)^2} = \frac{z}{\omega^2} \left(\frac{2}{z-\omega} + \frac{-z}{(z-\omega)^2} \right) \cdot (-1),$$

so

$$\left| \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right| \leq \frac{|z|}{|\omega|^2} \left(\frac{2}{|z-\omega|} + \frac{|z|}{|z-\omega|^2} \right).$$

For all but finitely many ω we have $|\omega| \geq 2R$, whence $|z-\omega| \geq |\omega| - R \geq |\omega|/2 \geq R$, and the bound becomes

$$\frac{R}{|\omega|^2} \left(\frac{4}{|\omega|} + \frac{R}{(|\omega|/2) \cdot R} \right) = \frac{R}{|\omega|^2} \cdot \frac{6}{|\omega|} = \frac{6R}{|\omega|^3}.$$

By Lemma 41.11.9 with $t = 3$, the series converges absolutely and uniformly on $|z| \leq R$ after discarding finitely many terms. So $\wp$ is a well-defined meromorphic function on $\mathbb{C}$, holomorphic away from Λ , with a double pole at each lattice point (the discarded terms contribute those).

Evenness. Replacing z by $-z$ and ω by $-\omega$ permutes the terms of the (absolutely convergent) sum, so $\wp(-z) = \wp(z)$.

Periodicity. Differentiating term by term (legitimate by local uniform convergence),

$$\wp'(z) = -2 \sum_{\omega \in \Lambda} \frac{1}{(z-\omega)^3},$$

which is manifestly Λ -periodic, since translating z by $\omega_0 \in \Lambda$ just permutes the summands. Fix $\omega_0 \in \Lambda$ and put $F(z) = \wp(z + \omega_0) - \wp(z)$. Then $F' = 0$, so F is constant, say $F \equiv c$. Now take $z = -\omega_0/2$, which is not a pole if $\omega_0/2 \notin \Lambda$ (and if it is, replace ω_0 by a generator and argue for generators, which suffices):

$$c = \wp(\omega_0/2) - \wp(-\omega_0/2) = 0$$

by evenness. So ω_0 is a period, and $\Lambda \subseteq \Omega$.

Conversely the poles of $\wp$ are exactly Λ ; a period must permute the poles, so $\Omega \subseteq \Lambda$ (as 0 is a pole, any $\omega \in \Omega$ satisfies $-\omega \in \Lambda$). Hence $\Omega = \Lambda$.

Degree. On $\mathbb{C}/\Lambda$, $\wp$ has a single pole, at 0, of order 2. So $\deg \wp = n(\infty) = 2$. $\square$

Remark. The properties

- (i) $\wp$ is meromorphic with period lattice Λ ,
- (ii) the poles of $\wp$ are exactly Λ , and
- (iii) $\wp(z) - z^{-2} \rightarrow 0$ as $z \rightarrow 0$

characterise $\wp$ uniquely: if $\tilde{\wp}$ also has them then $\wp - \tilde{\wp}$ is holomorphic and elliptic, hence constant, and the constant is 0 by (iii).

§41.11.4 Ramification of $\wp$

Since $\deg \wp = 2$, Riemann–Hurwitz gives

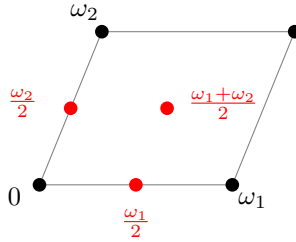
$$0 = 2 \cdot (-2) + \sum_p (\text{mult}_\wp(p) - 1) \implies \sum_p (\text{mult}_\wp(p) - 1) = 4,$$

so $\wp$ has exactly four ramification points, each of multiplicity 2 (multiplicity cannot exceed the degree). Let us find them.

One is $p = 0$: the pole is double, so $\text{mult}_\wp(0) = 2$. The others are the zeros of $\wp'$, since ramification points are the zeros of the derivative in a chart. Now $\wp'$ is odd (differentiate $\wp(-z) = \wp(z)$), elliptic with period lattice Λ , and has a triple pole at each lattice point, so $\deg \wp' = 3$ on $\mathbb{C}/\Lambda$. For any $\omega \in \Lambda$,

$$\wp'(\omega/2) = \wp'(\omega/2 - \omega) = \wp'(-\omega/2) = -\wp'(\omega/2),$$

using periodicity and oddness; hence $\wp'(\omega/2) = 0$ whenever $\omega/2 \notin \Lambda$. Taking $\omega = \omega_1, \omega_2, \omega_1 + \omega_2$ gives three distinct zeros in $\mathbb{C}/\Lambda$, namely the three non-zero **half-periods**. Since $\deg \wp' = 3$ there are no others, and all three are simple.



Proposition 41.11.11

The four ramification points of $\wp : \mathbb{C}/\Lambda \rightarrow \mathbb{C}_\infty$ are $0, \omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$, each of multiplicity 2, and their images

$$\infty, \quad e_1 = \wp(\omega_1/2), \quad e_2 = \wp(\omega_2/2), \quad e_3 = \wp\left(\frac{\omega_1 + \omega_2}{2}\right)$$

are four distinct points of $\mathbb{C}_\infty$.

Proof. We have found the four ramification points and know each has multiplicity 2. They must have distinct images: if two of them mapped to the same q , then $n(q)$ would be at least $4 > 2 = \deg \wp$. $\square$

So $\wp$ realises $\mathbb{C}/\Lambda$ as a two-sheeted cover of the sphere branched over four points – exactly the description of the surface of $w^2 = z^3 - z$ found in Example 41.10.10. This is not a coincidence.

§41.11.5 The Differential Equation

Proposition 41.11.12

There are constants $g_2, g_3 \in \mathbb{C}$, depending only on Λ , such that

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

Proof. Near $z = 0$ we expand. Since $\wp$ is even and $\wp(z) - z^{-2}$ vanishes at 0, its Laurent series has the form

$$\wp(z) = \frac{1}{z^2} + az^2 + bz^4 + O(z^6)$$

(no odd powers, and no constant term). Cubing,

$$\wp(z)^3 = \frac{1}{z^6} + \frac{3a}{z^2} + 3b + O(z^2),$$

while differentiating and squaring,

$$\wp'(z) = \frac{-2}{z^3} + 2az + 4bz^3 + O(z^5), \quad \wp'(z)^2 = \frac{4}{z^6} - \frac{8a}{z^2} - 16b + O(z^2).$$

Hence

$$\wp'(z)^2 - 4\wp(z)^3 = -\frac{20a}{z^2} - 28b + O(z^2).$$

Set $g_2 = 20a$. Then

$$H := (\wp')^2 - 4\wp^3 + g_2\wp$$

is elliptic, and near 0 it equals $-28b + O(z^2) + g_2 \cdot O(z^2)$, so it is holomorphic at 0. Its only possible poles were at Λ , so H is a holomorphic elliptic function, hence constant; call the constant $-g_3$. Rearranging gives the result. $\square$

Remark. The constants are computable: expanding $\wp$ gives $g_2 = 60 \sum_{\omega \neq 0} \omega^{-4}$ and $g_3 = 140 \sum_{\omega \neq 0} \omega^{-6}$, which converge by Lemma 41.11.9. We shall not need this.

Corollary 41.11.13

e_1, e_2, e_3 are the three roots of $4X^3 - g_2X - g_3$; they are distinct, and $e_1 + e_2 + e_3 = 0$. Moreover

$$(\wp')^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3).$$

Proof. At $z = \omega_i/2$ we have $\wp'(z) = 0$ and $\wp(z) = e_i$, so $0 = 4e_i^3 - g_2e_i - g_3$. The e_i are distinct by Proposition 41.11.11, so they are all three roots; the sum of the roots is $-(\text{coefficient of } X^2)/4 = 0$. The factorisation follows. $\square$

§41.11.6 The Torus as a Cubic Curve

Theorem 41.11.14

Let $\Lambda \subseteq \mathbb{C}$ be a lattice with invariants g_2, g_3 as above, and let

$$X' = \{(x, y) \in \mathbb{C}^2 : y^2 = 4x^3 - g_2x - g_3\},$$

with $X = X' \cup \{\infty\}$ its one-point compactification (a compact Riemann surface, as in Section 41.8.3). Then

$$\mathbb{C}/\Lambda \cong X$$

as Riemann surfaces, via $z + \Lambda \mapsto (\wp(z), \wp'(z))$ and $0 \mapsto \infty$.

Proof. Since $4X^3 - g_2X - g_3 = 4(X - e_1)(X - e_2)(X - e_3)$ has distinct roots, X' is smooth and is a Riemann surface with coordinate projections as charts (Proposition 41.9.17), and X is its compactification.

Define $F : \mathbb{C} \setminus \Lambda \rightarrow X'$ by $F(z) = (\wp(z), \wp'(z))$. Its image lies in X' by Proposition 41.11.12. It is holomorphic: composed with the chart π_x it is $\wp$, and composed with π_y it is $\wp'$, both holomorphic. Extending by $F(\omega) = \infty$ for $\omega \in \Lambda$ gives a holomorphic map $\mathbb{C} \rightarrow X$ (the point ∞ of X is exactly where both coordinates blow up, and the local computation there is as in Section 41.8.3). Since F is Λ -periodic it descends, by Proposition 41.11.5, to a holomorphic

$$\Phi : \mathbb{C}/\Lambda \rightarrow X.$$

Φ is non-constant, and $\mathbb{C}/\Lambda$ is compact, so Φ is surjective by Corollary 41.5.4. It remains to prove injectivity, for then $\deg \Phi = 1$ and Corollary 41.9.12 finishes the proof.

Work in the centred period parallelogram P with vertices $\pm(\omega_1 \pm \omega_2)/2$, whose interior P° maps injectively to $\mathbb{C}/\Lambda$ and contains exactly one point of each class except for those on ∂P . Suppose $\Phi(z) = \Phi(z')$ for $z, z' \in P^\circ$. Then $\wp(z) = \wp(z')$; as $\wp$ is even of degree 2, the fibre of $\wp$ over a non-branch value is $\{\pm z\}$, so $z' = \pm z$ (and if z is one of the four ramification points then $z' = z$ directly). Also $\wp'(z) = \wp'(z')$; if $z' = -z$ then, $\wp'$ being odd, $\wp'(z) = -\wp'(z)$, so $\wp'(z) = 0$ and z is a half-period. But the non-zero half-periods lie on ∂P for this centred parallelogram, and $z = 0$ gives $z' = -z = z$. Either way $z = z'$. Hence Φ is injective on P° and so on $\mathbb{C}/\Lambda$. $\square$

So the two families of genus-1 surfaces we have met – complex tori and compactified plane cubics – coincide. The map Φ is what makes the addition law on a cubic curve comprehensible: it is transported from the group $\mathbb{C}/\Lambda$.

§41.11.7 The Field of Elliptic Functions

Theorem 41.11.15

Let f be an elliptic function with period lattice containing Λ , and write $\wp = \wp_\Lambda$. Then there are rational functions Q_1, Q_2 with

$$f(z) = Q_1(\wp(z)) + Q_2(\wp(z)) \wp'(z).$$

If f is even then $Q_2 = 0$.

Proof. Case 1: f even. We first reduce to a convenient normalisation. The function f and $\wp$ each have finitely many branch points on the compact surface $\mathbb{C}/\Lambda$, so we may pick distinct $c, d \in \mathbb{C}$ which are not among the values of f at its ramification points. Replacing f by

$$\frac{f - c}{f - d},$$

which is f composed with a Möbius map, hence still even and elliptic, and from which f can be recovered by the inverse Möbius map, we may assume all the zeros and poles of f are simple and occur away from the ramification points of $\wp$.

Since f is even, its zeros come in pairs $\pm a_i$ and its poles in pairs $\pm b_j$ (modulo Λ), with $a_i \not\equiv \pm a_j$ for $i \neq j$ and similarly for the b_j ; say the zeros are $\{\pm a_1, \dots, \pm a_m\}$ and the poles $\{\pm b_1, \dots, \pm b_n\}$. Consider

$$g(z) = \frac{(\wp(z) - \wp(a_1)) \cdots (\wp(z) - \wp(a_m))}{(\wp(z) - \wp(b_1)) \cdots (\wp(z) - \wp(b_n))}.$$

Each factor $\wp(z) - \wp(a_i)$ is elliptic of degree 2 with zeros exactly $\pm a_i$ (simple, as a_i is not a ramification point of $\wp$) and a double pole at 0. So g has exactly the same zeros and poles as f away from 0, and at 0 the pole orders are $2n - 2m$ in g and, by the valency theorem applied to f , also match. Hence f/g is an elliptic function with no zeros and no poles, so it is holomorphic, hence constant. Therefore $f = \lambda g$ is a rational function of $\wp$.

Case 2: f odd. Then $f/\wp'$ is even (as $\wp'$ is odd) and elliptic, so $f/\wp' = Q_2(\wp)$ by Case 1, giving $f = Q_2(\wp)\wp'$.

General case. Write

$$f(z) = \frac{f(z) + f(-z)}{2} + \frac{f(z) - f(-z)}{2},$$

a sum of an even and an odd elliptic function, and apply the two cases. $\square$

In the language of fields: the field $\mathcal{M}(\mathbb{C}/\Lambda)$ of meromorphic functions on the torus is $\mathbb{C}(\wp)[\wp']$, a degree-2 extension of the rational function field $\mathbb{C}(\wp)$, with $\wp'$ satisfying the quadratic $(\wp')^2 = 4\wp^3 - g_2\wp - g_3$. Compare Proposition 41.11.1, which says $\mathcal{M}(\mathbb{C}_\infty) = \mathbb{C}(z)$. In both cases the meromorphic functions on a compact Riemann surface form a field of transcendence degree 1 over $\mathbb{C}$ – this is true in general, and is the beginning of the dictionary between compact Riemann surfaces and algebraic curves.

§41.12 Differentials on Compact Surfaces

Corollary 41.5.5 told us that a compact Riemann surface carries no non-constant holomorphic function, and we responded by enlarging the target to $\mathbb{C}_\infty$. There is a second, and in the long run more important, response: keep the target $\mathbb{C}$ but change the *kind* of object we are looking at. The observation is that although $f(z)$ is not a well-defined thing to write down on a surface – it depends on the chart – the expression $f(z) dz$ very nearly is, because the chain rule supplies exactly the factor needed to make the two chart-dependencies cancel. Objects of the form $f(z) dz$ are called differentials, and unlike functions they exist in abundance on compact surfaces. Better still, they know the genus.

§41.12.1 Meromorphic Differentials

Definition 41.12.1 (Meromorphic Differential)

Let R be a Riemann surface. A **meromorphic differential** (or **meromorphic 1-form**) ω on R is an assignment to each chart (ϕ, U) of the conformal structure of a meromorphic function f_ϕ on $\phi(U)$, subject to the compatibility condition

$$f_\psi(w) = f_\phi(\tau(w)) \cdot \tau'(w), \quad \tau = \phi \circ \psi^{-1},$$

for all charts $(\phi, U), (\psi, V)$ and all $w \in \psi(U \cap V)$. We write $\omega = f_\phi dz$ for the local expression of ω in the chart ϕ , and call ω **holomorphic** if every f_ϕ is holomorphic.

The compatibility rule is nothing but the chain rule made into a definition: if $z = \tau(w)$ then $dz = \tau'(w) dw$, so

$$f_\phi(z) dz = f_\phi(\tau(w)) \tau'(w) dw = f_\psi(w) dw,$$

and the two local expressions really do describe the same object. As with charts and atlases, it suffices to prescribe the f_ϕ on the charts of some atlas: the compatibility condition then propagates the definition to the whole conformal structure.

Example 41.12.2 (The Sphere)

On $\mathbb{C}$, take $f_{\text{id}} = 1$; the resulting differential is written dz . Now regard it on $\mathbb{C}_\infty$. In the chart $w = 1/z$ at infinity we have $\tau(w) = 1/w$ and $\tau'(w) = -1/w^2$, so

$$dz = -\frac{1}{w^2} dw.$$

So dz extends to a meromorphic differential on the whole sphere, holomorphic except for a double pole at ∞ . Notice that it has no zeros at all – something no meromorphic function on $\mathbb{C}_\infty$ can manage while having only one pole.

Example 41.12.3 (The Torus)

On $T = \mathbb{C}/\Lambda$ the charts of Proposition 41.4.26 are local inverses of π , and the transition functions are the translations $w \mapsto w + \lambda$, whose derivatives are all 1. So the constant assignment $f_\phi = 1$ is compatible, and defines a differential on T , again written dz . It is holomorphic and has no zeros whatsoever.

Example 41.12.4 (The Differential of a Function)

Let f be meromorphic on R . Define df by taking, in the chart (ϕ, U) , the meromorphic function $(f \circ \phi^{-1})'$. Compatibility is the chain rule:

$$(f \circ \psi^{-1})'(w) = (f \circ \phi^{-1} \circ \tau)'(w) = (f \circ \phi^{-1})'(\tau(w)) \cdot \tau'(w).$$

So every meromorphic function has a differential, and $df = 0$ if and only if f is constant.

Lemma 41.12.5

If ω is a meromorphic differential and h a meromorphic function then $h\omega$ (defined chartwise by $hf_\phi \circ \phi^{-1}$) is a meromorphic differential. Conversely, if $\omega \neq 0$ and ω' is any meromorphic differential, then $\omega' = h\omega$ for a unique meromorphic function h on R .

Proof. The first statement is immediate, since multiplying both sides of the compatibility rule by h preserves it. For the second, define h in the chart ϕ to be f'_ϕ/f_ϕ , where f_ϕ, f'_ϕ are the local expressions of ω, ω' . This is meromorphic (the denomina-

tor is not identically zero, by the identity theorem applied on the connected surface R), and the transition factors τ' cancel in the quotient, so the local definitions agree on overlaps and glue to a meromorphic function on R . Uniqueness is clear. $\square$

So the meromorphic differentials form a one-dimensional vector space over the field of meromorphic functions – there are exactly as many differentials as there are functions, once you have found one. What makes them worth introducing is that the *zeros and poles* of a differential are counted differently from those of a function.

Definition 41.12.6 (Order)

Let ω be a meromorphic differential, not identically zero, and $p \in R$. The **order** of ω at p is

$$\text{ord}_p(\omega) = \text{ord}_{\phi(p)}(f_\phi)$$

for any chart (ϕ, U) about p , where ord denotes the order of vanishing (so a pole of order m has $\text{ord} = -m$).

This is well defined: the transition factor τ' is holomorphic and nowhere zero, so multiplying by it changes no order of vanishing. We call p a zero of ω if $\text{ord}_p(\omega) > 0$ and a pole if $\text{ord}_p(\omega) < 0$; by the identity theorem these form a discrete set, hence a finite one when R is compact.

Definition 41.12.7 (Divisor and Degree)

Let R be compact. The **divisor** of a non-zero meromorphic differential ω , or of a non-constant meromorphic function f , is the formal sum

$$\text{div}(\omega) = \sum_{p \in R} \text{ord}_p(\omega) \cdot p, \quad \text{div}(f) = \sum_{p \in R} \text{ord}_p(f) \cdot p,$$

and its **degree** is the integer $\deg \text{div}(\omega) = \sum_p \text{ord}_p(\omega)$.

Lemma 41.12.8

Let f be a non-constant meromorphic function on a compact Riemann surface R . Then $\deg \text{div}(f) = 0$.

Proof. View f as a map $R \rightarrow \mathbb{C}_\infty$. By the valency theorem the sum of the orders of the zeros of f is

$$\sum_{p \in f^{-1}(0)} \text{mult}_f(p) = \deg f,$$

and likewise the sum of the orders of the poles is $\sum_{p \in f^{-1}(\infty)} \text{mult}_f(p) = \deg f$. These cancel. $\square$

In other words, a meromorphic function on a compact surface has exactly as many zeros as poles. The corresponding count for a differential is not zero, and what it is instead is the whole point.

Theorem 41.12.9 (Degree of a Canonical Divisor)

Let R be a compact Riemann surface of genus g carrying a non-constant meromorphic function. Then for every non-zero meromorphic differential ω on R ,

$$\deg \operatorname{div}(\omega) = 2g - 2.$$

Proof. The answer does not depend on ω . If ω' is another non-zero differential then $\omega' = h\omega$ for a meromorphic h by Lemma 41.12.5, and orders add: $\operatorname{ord}_p(\omega') = \operatorname{ord}_p(h) + \operatorname{ord}_p(\omega)$. Summing and applying Lemma 41.12.8 gives $\deg \operatorname{div}(\omega') = \deg \operatorname{div}(\omega)$. (If h is constant this is trivial.)

A convenient choice. Let $f : R \rightarrow \mathbb{C}_\infty$ be a non-constant meromorphic function, of degree n say, and take $\omega = df$, which is not identically zero. We compute $\operatorname{ord}_p(df)$ using the local normal form (Proposition 41.9.1), writing $m = \operatorname{mult}_f(p)$.

If $f(p) \neq \infty$, choose the chart $\psi(w) = w - f(p)$ downstairs and a chart ϕ upstairs putting f in normal form, so that $f \circ \phi^{-1}(z) = f(p) + z^m$. Then df has local expression mz^{m-1} , and

$$\operatorname{ord}_p(df) = m - 1.$$

If $f(p) = \infty$, use the chart $\psi(w) = 1/w$ downstairs, so that normal form reads $1/f \circ \phi^{-1}(z) = z^m$, i.e. $f \circ \phi^{-1}(z) = z^{-m}$. Then df has local expression $-mz^{-m-1}$, and

$$\operatorname{ord}_p(df) = -m - 1.$$

Comparing the two, a pole of f contributes $-m - 1 = (m - 1) - 2m$. Hence

$$\deg \operatorname{div}(df) = \sum_{p \in R} (\operatorname{mult}_f(p) - 1) - 2 \sum_{p \in f^{-1}(\infty)} \operatorname{mult}_f(p) = \sum_{p \in R} (\operatorname{mult}_f(p) - 1) - 2n,$$

the last step by the valency theorem. Finally Riemann–Hurwitz (Theorem 41.10.7) applied to $f : R \rightarrow \mathbb{C}_\infty$, whose target has genus 0, gives

$$2g - 2 = n(0 - 2) + \sum_{p \in R} (\operatorname{mult}_f(p) - 1),$$

so that $\sum_p (\operatorname{mult}_f(p) - 1) = 2g - 2 + 2n$. Substituting, $\deg \operatorname{div}(df) = 2g - 2$. $\square$

Remark. The hypothesis that R carries a non-constant meromorphic function looks like a blemish, and it is: it is a (hard) theorem that every compact Riemann surface does, so the hypothesis is vacuous. All the surfaces we have met come with one for free, and we shall not use the general result.

Divisors of the form $\operatorname{div}(\omega)$ are called **canonical divisors**, and the theorem says they all have the same degree, namely $2g - 2$. This is a genuinely new way to compute the genus: exhibit a single differential and count its zeros and poles.

Example 41.12.10

On $\mathbb{C}_\infty$, the differential dz of Example 41.12.2 has divisor $-2 \cdot \infty$, of degree $-2 = 2 \cdot 0 - 2$. On $\mathbb{C}/\Lambda$, the differential dz of Example 41.12.3 has empty divisor, of degree $0 = 2 \cdot 1 - 2$. Both agree with Theorem 41.12.9, as they had better.

Example 41.12.11 (The Cubic Curve Again)

Let $\hat{R}$ be the compactified curve $w^2 = z^3 - z$ of Section 41.8.3, and consider

$$\omega = \frac{dz}{w}.$$

We claim ω is holomorphic and nowhere vanishing, so that $\deg \operatorname{div}(\omega) = 0$ and hence $g(\hat{R}) = 1$ – recovering Example 41.10.10 without any Riemann–Hurwitz bookkeeping.

Differentiating the equation gives $2w \, dw = (3z^2 - 1)dz$, so that

$$\omega = \frac{dz}{w} = \frac{2 \, dw}{3z^2 - 1}$$

wherever both expressions make sense. Away from the points with $w = 0$ or $z = \infty$, the coordinate z is a chart and $w \neq 0$, so the first expression is holomorphic and non-vanishing. At each of the three points $(z_0, 0)$ with $z_0 \in \{0, \pm 1\}$ the coordinate w is a chart, and $3z_0^2 - 1 \neq 0$ there (indeed $3z^2 - 1$ and $z^3 - z$ have no common root), so the second expression is holomorphic and non-vanishing. Finally at ∞ we use the coordinates $u = 1/z$, $v = z/w$ of Section 41.8.3, in which the curve reads $u = v^2(1 - u^2)$; the implicit function theorem makes v a local coordinate with $u = v^2 + O(v^6)$, so $du = (2v + O(v^5))dv$ and

$$\omega = \frac{dz}{w} = \frac{-du/u^2}{1/(uv)} = -\frac{v \, du}{u} = -\frac{v(2v + O(v^5))}{v^2 + O(v^6)} dv = (-2 + O(v^4)) \, dv,$$

again holomorphic and non-vanishing. So $\operatorname{div}(\omega) = 0$ and $g(\hat{R}) = 1$.

§41.12.2 The Residue Theorem

Differentials, not functions, are the objects one integrates. If γ is a path lying inside a single chart (ϕ, U) , we may define

$$\int_{\gamma} \omega = \int_{\phi \circ \gamma} f_{\phi}(z) \, dz,$$

and the change of variables formula for contour integrals shows this does not depend on the chart: substituting $z = \tau(w)$ turns $f_{\phi}(z) \, dz$ into $f_{\phi}(\tau(w))\tau'(w) \, dw = f_{\psi}(w) \, dw$, which is exactly the compatibility rule. A general path is cut into finitely many such pieces. So $\int_{\gamma} \omega$ makes sense for any path, and this is what a differential is *for*.

Definition 41.12.12 (Residue)

Let ω be a meromorphic differential on R and $p \in R$. The **residue** of ω at p is

$$\operatorname{res}_p(\omega) = \frac{1}{2\pi i} \int_{\gamma} \omega,$$

where γ is a small positively oriented loop about p inside a chart, enclosing no other pole of ω .

Equivalently, $\operatorname{res}_p(\omega)$ is the coefficient of $(z - \phi(p))^{-1}$ in the Laurent expansion of f_{ϕ} ; the

point of the integral formulation is that it makes chart-independence obvious, whereas the coefficient of z^{-1} looks as though it ought to depend on the coordinate. (It is worth pausing on this. For a meromorphic *function* the coefficient of z^{-1} is *not* chart-independent, and ‘the residue of a function at a point of a Riemann surface’ is meaningless. Only differentials have residues.)

Theorem 41.12.13 (Residue Theorem on a Compact Surface)

Let ω be a meromorphic differential on a compact Riemann surface R . Then

$$\sum_{p \in R} \text{res}_p(\omega) = 0,$$

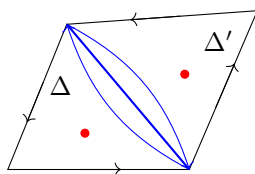
the sum being finite.

Proof. The poles $p_1, \dots, p_k$ of ω are finite in number. Choose a triangulation of R (Lemma 41.10.3) and subdivide until every closed triangle lies inside a single chart; perturbing slightly, we may also assume no p_i lies on an edge or vertex, so each pole is interior to exactly one triangle. Orient every triangle positively, which is possible because R is orientable and the charts are holomorphic.

For a triangle Δ contained in the chart (ϕ, U) , the classical residue theorem applied to f_ϕ on the region $\phi(\Delta)$ gives

$$\int_{\partial\Delta} \omega = 2\pi i \sum_{p_i \in \Delta^\circ} \text{res}_{p_i}(\omega).$$

Summing over all triangles, the right-hand side is $2\pi i \sum_p \text{res}_p(\omega)$, since each pole lies in exactly one triangle. On the left, each edge of the triangulation belongs to exactly two triangles and is traversed once in each direction, so the contributions cancel in pairs and the total is 0. $\square$



The picture is the whole proof: whatever ω does inside the triangles, the boundary integrals cancel along every shared edge, so their total vanishes – and their total is $2\pi i$ times the sum of all the residues.

Example 41.12.14

Take $R = \mathbb{C}_\infty$ and $\omega = f(z) dz$ for a rational f . The theorem says the sum of the residues of f over $\mathbb{C}$, together with the residue at ∞ (which in the chart $w = 1/z$ is the residue at 0 of $-f(1/w)/w^2$), is zero. This is the familiar ‘sum of residues including infinity vanishes’ of IB Complex Analysis, now seen as a statement about the sphere.

Example 41.12.15

Take $R = \mathbb{C}/\Lambda$ and $\omega = f dz$ for an elliptic function f , using the nowhere-vanishing dz of Example 41.12.3; then $\text{res}_p(\omega)$ is the ordinary residue of f . The theorem says the residues of an elliptic function in a period parallelogram sum to zero, which is exactly what the cancellation of opposite sides gave us in the second proof of Corollary 41.11.7. The triangulation argument above is the same cancellation, done invariantly.

§41.12.3 The Degree–Genus Formula

We met the degree–genus formula in Example 41.10.12 as an observation about Fermat curves. It holds for every smooth plane curve, and the proof is a pretty application of Riemann–Hurwitz.

Theorem 41.12.16 (Degree–Genus Formula)

Let $C \subseteq \mathbb{P}^2$ be a smooth projective plane curve of degree $d \geq 1$. Then C is a compact Riemann surface of genus

$$g(C) = \frac{(d-1)(d-2)}{2}.$$

Proof sketch. That C is a compact Riemann surface is Proposition 41.9.17 together with the gluing of charts at infinity, exactly as in Section 41.8.3.

Pick a point $q \in \mathbb{P}^2 \setminus C$ and project from it: each $p \in C$ lies on a unique line through q , and the set of such lines is a copy of $\mathbb{P}^1 \cong \mathbb{C}_\infty$. This gives a holomorphic map

$$\pi_q : C \longrightarrow \mathbb{C}_\infty.$$

A line meets C in d points counted with multiplicity (Bézout’s theorem), so $\deg \pi_q = d$. A point p is a ramification point of π_q precisely when the line qp meets C at p with multiplicity at least 2, that is, when the line is tangent to C at p .

For a generic choice of q – and this is the step we are not proving – every tangent line through q is tangent at exactly one point and to order exactly 2, so each ramification point has multiplicity 2; and the number of such points is $d(d-1)$. (This last count is again Bézout: the points of C whose tangent passes through q are the intersections of C with its *polar curve* with respect to q , which has degree $d-1$.) Riemann–Hurwitz now gives

$$2g(C) - 2 = d(0 - 2) + d(d-1) \cdot (2 - 1) = d^2 - 3d,$$

whence $g(C) = (d^2 - 3d + 2)/2 = (d-1)(d-2)/2$. □

The two computations we did by hand are the special cases we can check independently: the Fermat curve $x^d + y^d = 1$ of Example 41.10.12, where we projected along a coordinate axis rather than from a general point and found the same answer, and the cubic $w^2 = z^3 - z$, which after homogenising is a smooth plane cubic of genus $(3-1)(3-2)/2 = 1$.

Remark. It is worth being clear about what the formula does and does not say. Every smooth plane curve of degree d has genus $(d-1)(d-2)/2$, so smooth plane curves realise only the genera $0, 0, 1, 3, 6, 10, \dots$. Compact Riemann surfaces of the missing genera certainly exist – the hyperelliptic curves supply all of them – but they cannot be

embedded in $\mathbb{P}^2$ without singularities. Genus 2 is the first example. Plane curves are special, and it is exactly the interplay between the abstract theory and such concrete models that makes the subject work.

§41.13 Quotients and Uniformisation

We finish by classifying all Riemann surfaces, at least in principle. The classification is startlingly clean: there are only three simply connected Riemann surfaces, and every other surface is a quotient of one of them.

§41.13.1 Quotients by Group Actions

Definition 41.13.1

Let a group G act on a topological space X by homeomorphisms. The action is

- (i) **properly discontinuous** if for every compact $K \subseteq X$ the set $\{g \in G : g(K) \cap K \neq \emptyset\}$ is finite;
- (ii) **free** if $\text{Stab}_G(x) = \{1\}$ for every $x \in X$.

Example 41.13.2

A lattice Λ acts on $\mathbb{C}$ by translation. This action is free (a non-zero translation has no fixed point) and properly discontinuous (a compact K is bounded, and only finitely many $\lambda \in \Lambda$ have $|\lambda| \leq \text{diam } K$).

Lemma 41.13.3

Let G act freely and properly discontinuously by homeomorphisms on a Riemann surface R . Then $G \backslash R$ is Hausdorff and the quotient map $\pi : R \rightarrow G \backslash R$ is a regular covering map.

Proof. First, π is open, since $\pi^{-1}(\pi(U)) = \bigcup_g g(U)$.

Hausdorff. Let $p, q \in R$ with $\pi(p) \neq \pi(q)$. Choose open $U \ni p$, $V \ni q$ with compact closures, and set $K = \bar{U} \cup \bar{V}$, compact. Then

$$\{g \in G : g(\bar{U}) \cap \bar{V} \neq \emptyset\} \subseteq \{g : g(K) \cap K \neq \emptyset\}$$

is finite, say $\{g_1, \dots, g_k\}$. For each i we have $g_i p \neq q$ (else $\pi(p) = \pi(q)$), so choose disjoint open $U_i \ni p$ and $V_i \ni g_i q$. Shrinking to

$$U' = U \cap \bigcap_i U_i, \quad V' = V \cap \bigcap_i g_i^{-1}(V_i),$$

we get $g(U') \cap V' = \emptyset$ for every $g \in G$. Then $\pi(U')$ and $\pi(V')$ are disjoint open neighbourhoods.

Covering. Let $p \in R$ and choose open $U \ni p$ with $\bar{U}$ compact, so that $\{g : g(\bar{U}) \cap \bar{U} \neq \emptyset\} = \{1, g_1, \dots, g_k\}$ is finite. Freeness gives $g_i p \neq p$, so choose disjoint open $U_i \ni p$,

$V_i \ni g_i p$, and set

$$U' = U \cap \bigcap_i (U_i \cap g_i^{-1}(V_i)).$$

Then $g(U') \cap U' = \emptyset$ for all $g \neq 1$, so $\pi^{-1}(\pi(U')) = \coprod_g g(U')$ and each $g(U')$ maps homeomorphically onto $\pi(U')$ (as π is open, continuous and injective on U'). So $\pi(U')$ is evenly covered. $\square$

Proposition 41.13.4

Let R be a Riemann surface and G a group acting freely and properly discontinuously on R by *conformal equivalences*. Then $S = G \backslash R$ carries a unique conformal structure making $\pi : R \rightarrow S$ holomorphic, and π is then a regular holomorphic covering map.

Proof. S is connected as a continuous image of R , and Hausdorff with π a regular covering map by Lemma 41.13.3. Charts on S are obtained by pushing charts of R forward along local inverses of π : for U' as in the lemma and a chart (ϕ, U'') with $U'' \subseteq U'$, take $(\phi \circ (\pi|_{U'})^{-1}, \pi(U''))$. Two such charts differ by a transition function of R composed with the action of some $g \in G$, which is holomorphic by hypothesis. Uniqueness is as in Lemma 41.4.21. $\square$

This is the converse of Lemma 41.4.21, and together they say: quotients by free properly discontinuous actions of conformal automorphisms are exactly covering spaces, in the holomorphic category. The complex torus is the case $R = \mathbb{C}$, $G = \Lambda$.

Theorem 41.13.5

Let R be a compact Riemann surface with $g(R) \geq 2$ and let G act freely and properly discontinuously on R by conformal equivalences. Then G is finite, with

$$|G| \leq g(R) - 1.$$

Proof. By Proposition 41.13.4, $S = G \backslash R$ is a Riemann surface, compact as a continuous image of R , and $\pi : R \rightarrow S$ is an unramified holomorphic map of degree $|G|$ (the fibres are the G -orbits, which all have size $|G|$ by freeness; in particular G is finite since fibres are finite). Riemann–Hurwitz with no ramification gives

$$g(R) - 1 = |G| (g(S) - 1).$$

Since $g(R) \geq 2$ the left side is positive, so $g(S) \geq 2$ and $g(S) - 1 \geq 1$, whence $|G| \leq g(R) - 1$. $\square$

Remark. This fails for $g = 1$: the torus $\mathbb{C}/\Lambda$ acts on itself freely by translation, and any finite subgroup does too, with no bound on its order. The genus ≥ 2 hypothesis is essential, and it is the first sign that high-genus surfaces are rigid. (The sharp statement, Hurwitz’s automorphism theorem, bounds the full automorphism group by $84(g - 1)$; that is beyond us here.)

§41.13.2 The Uniformisation Theorem

Theorem 41.13.6 (Uniformisation Theorem)

Every simply connected Riemann surface is conformally equivalent to exactly one of

$$\mathbb{C}_\infty, \quad \mathbb{C}, \quad \mathbb{D} = D(0, 1).$$

The proof is well beyond this course. What we can do is check that the three are genuinely distinct, and then extract the consequences.

Proposition 41.13.7

No two of $\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}$ are conformally equivalent.

Proof. $\mathbb{C}_\infty$ is compact and the other two are not. If $\mathbb{D} \cong \mathbb{C}$ then there would be a non-constant holomorphic map $\mathbb{C} \rightarrow \mathbb{D}$, hence a bounded non-constant entire function, contradicting Liouville. $\square$

Corollary 41.13.8

Every conformal structure on S^2 is conformally equivalent to $\mathbb{C}_\infty$; equivalently, a compact Riemann surface of genus 0 is the Riemann sphere.

Proof. S^2 is simply connected, so it is one of the three; it is compact, so it is $\mathbb{C}_\infty$. $\square$

For non-simply-connected surfaces we pass to the universal cover.

Theorem 41.13.9

Every Riemann surface R admits a regular covering map $\pi : \tilde{R} \rightarrow R$ with $\tilde{R}$ simply connected. Moreover $\tilde{R}$ has a unique conformal structure making π holomorphic, and the deck transformation group

$$G = \text{Deck}(\pi) = \{\varphi : \tilde{R} \rightarrow \tilde{R} \text{ conformal} : \pi \circ \varphi = \pi\} \cong \pi_1(R)$$

acts freely and properly discontinuously by conformal equivalences with $G \backslash \tilde{R} \cong R$.

Proof. Existence of the universal cover and the identification $\text{Deck}(\pi) \cong \pi_1(R)$ for a simply connected cover are from II Algebraic Topology (a Riemann surface is locally homeomorphic to a disc, hence locally path-connected and semi-locally simply connected, so the theory applies). The conformal structure on $\tilde{R}$ comes from Lemma 41.4.21, and each deck transformation is then holomorphic (locally it is $(\pi|_{\tilde{U}'})^{-1} \circ \pi$) with holomorphic inverse. Freeness and proper discontinuity are standard properties of deck groups of covering spaces. $\square$

Corollary 41.13.10

Every Riemann surface R is conformally equivalent to $G \backslash \tilde{R}$, where $\tilde{R} \in \{\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}\}$ and $G \cong \pi_1(R)$ acts freely and properly discontinuously by conformal automorphisms of $\tilde{R}$.

We say R is **uniformised by** $\tilde{R}$. To make this useful we need to know the conformal automorphisms of each model.

Proposition 41.13.11 (i) $\text{Aut}(\mathbb{C}_\infty) = \{z \mapsto \frac{az+b}{cz+d} : ad-bc \neq 0\} \cong \text{PGL}_2(\mathbb{C})$, the Möbius group.

(ii) $\text{Aut}(\mathbb{C}) = \{z \mapsto az + b : a \in \mathbb{C}^*, b \in \mathbb{C}\}$.

(iii) $\text{Aut}(\mathbb{D}) = \{z \mapsto e^{i\theta} \frac{z-a}{1-\bar{a}z} : a \in \mathbb{D}, \theta \in \mathbb{R}\}$.

Proof sketch. (i) A conformal automorphism of $\mathbb{C}_\infty$ is a degree-1 meromorphic function, hence rational of degree 1 by Proposition 41.11.1, hence Möbius. (ii) An automorphism of $\mathbb{C}$ extends over ∞ : by Casorati–Weierstrass the singularity at ∞ cannot be essential (an injective function cannot have dense image on every punctured neighbourhood while also being injective near a distant point), so it is a pole or removable; removable would force boundedness and constancy, so it is a pole, and f is a polynomial, which is injective only if it is linear. (iii) This was proved in IB Complex Analysis using the Schwarz lemma. $\square$

Proposition 41.13.12

If R is uniformised by $\mathbb{C}_\infty$ then $R \cong \mathbb{C}_\infty$.

Proof. Say $R = G \backslash \mathbb{C}_\infty$ with G acting freely by Möbius maps. But every Möbius map has a fixed point in $\mathbb{C}_\infty$: the equation $az + b = z(cz + d)$ is a quadratic (or linear) with a root in $\mathbb{C}$, unless $c = 0$ and $a = d$, in which case $z \mapsto z + b/d$ fixes ∞ . So freeness forces $G = \{1\}$. $\square$

Proposition 41.13.13

If R is uniformised by $\mathbb{C}$, then exactly one of the following holds:

(i) $G = \{1\}$ and $R \cong \mathbb{C}$;

(ii) $G \cong \mathbb{Z}$ and $R \cong \mathbb{C}^*$;

(iii) $G \cong \mathbb{Z}^2$ and $R \cong \mathbb{C}/\Lambda$ for a lattice Λ .

Proof. By Proposition 41.13.11(ii), G consists of maps $z \mapsto az + b$. If $a \neq 1$ then $z = b/(1-a)$ is a fixed point, so freeness forces $a = 1$: every element of G is a translation. Identifying G with a subgroup of $(\mathbb{C}, +)$, proper discontinuity forces G to be discrete, and a discrete subgroup of $\mathbb{C}$ is trivial, infinite cyclic, or a lattice (Lemma 41.11.3). These give $\mathbb{C}$, $\mathbb{C}/\langle \omega \rangle \cong \mathbb{C}^*$ (via $z \mapsto e^{2\pi iz/\omega}$), and $\mathbb{C}/\Lambda$. $\square$

Lemma 41.13.14 (Lifting Lemma)

Let $f : R \rightarrow S$ be holomorphic with R simply connected, and let $\pi : \tilde{S} \rightarrow S$ be the universal cover of S . Then there is a holomorphic $F : R \rightarrow \tilde{S}$ with $f = \pi \circ F$.

Proof. The topological lifting criterion from II Algebraic Topology gives a continuous lift F , since $f_*\pi_1(R) = \{1\} \subseteq \pi_*\pi_1(\tilde{S})$. It is holomorphic because locally $F = (\pi|_{\tilde{U}})^{-1} \circ f$, a composite of holomorphic maps. $\square$

Proposition 41.13.15

A Riemann surface is uniformised by at most one of $\mathbb{C}_\infty, \mathbb{C}, \mathbb{D}$.

Proof. The universal cover is unique up to homeomorphism, so this is really the statement that the three models are not conformally equivalent, which we know. Explicitly: suppose R were uniformised both by $\mathbb{D}$ and by $\tilde{R} \in \{\mathbb{C}, \mathbb{C}_\infty\}$, with uniformising maps $\pi : \mathbb{D} \rightarrow R$ and $f : \tilde{R} \rightarrow R$. Since $\tilde{R}$ is simply connected, Lemma 41.13.14 gives holomorphic $F : \tilde{R} \rightarrow \mathbb{D}$ with $f = \pi \circ F$. Restricting to $\mathbb{C} \subseteq \tilde{R}$, F is a bounded entire function, hence constant by Liouville; so f is constant, contradicting surjectivity. $\square$

So every Riemann surface other than $\mathbb{C}_\infty, \mathbb{C}, \mathbb{C}^*$ and the tori is uniformised by the disc. In particular:

Corollary 41.13.16

Every compact Riemann surface of genus ≥ 2 is uniformised by $\mathbb{D}$.

Proof. It is not $\mathbb{C}_\infty$ (genus 0), and by Proposition 41.13.13 the only compact surface uniformised by $\mathbb{C}$ is a torus (genus 1). $\square$

§41.13.3 Fuchsian Groups and the Hyperbolic Plane

The disc case is the generic one, and it connects this course to IB Geometry. Recall the Möbius map

$$z \mapsto \frac{1+iz}{1-iz}$$

carries the upper half-plane $\mathbb{H}$ conformally onto $\mathbb{D}$; under it,

$$\text{Aut}(\mathbb{H}) = \left\{ z \mapsto \frac{az+b}{cz+d} : a, b, c, d \in \mathbb{R}, ad-bc=1 \right\} \cong \text{PSL}_2(\mathbb{R}).$$

These are precisely the orientation-preserving isometries of the hyperbolic metric

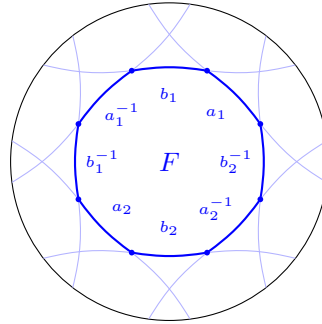
$$\frac{|dz|^2}{(\text{Im } z)^2}$$

on $\mathbb{H}$ (equivalently $4|dz|^2/(1-|z|^2)^2$ on $\mathbb{D}$), as was shown in IB Geometry. So a Riemann surface uniformised by $\mathbb{D}$ inherits a hyperbolic metric, and its conformal geometry is hyperbolic geometry.

Definition 41.13.17 (Fuchsian Group)

A subgroup $\Gamma \leq \text{PSL}_2(\mathbb{R})$ acting properly discontinuously on $\mathbb{H}$ is called a **Fuchsian group**.

By Corollary 41.13.10, the Riemann surfaces uniformised by the disc are exactly the quotients $\Gamma \backslash \mathbb{H}$ for torsion-free Fuchsian Γ . Such a quotient is understood through a **fundamental domain**: a region $F \subseteq \mathbb{D}$ whose Γ -translates tile the disc and cover the quotient exactly once. For a compact genus-2 surface one may take a regular hyperbolic octagon with all angles $2\pi/8$, whose sides are glued in the pattern $a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1}$ – the same octagon as in Section 41.10, but now with a genuine hyperbolic structure.



a fundamental domain in $\mathbb{D}$

§41.13.4 Consequences of Uniformisation

We close with two classical theorems which fall out of the machine.

Theorem 41.13.18 (Riemann Mapping Theorem)

Every simply connected domain $D \subsetneq \mathbb{C}$ is conformally equivalent to $\mathbb{D}$.

Proof. D is a simply connected Riemann surface, so by Theorem 41.13.6 it is $\mathbb{C}_\infty$, $\mathbb{C}$ or $\mathbb{D}$. It is not $\mathbb{C}_\infty$, being non-compact. Suppose $f : \mathbb{C} \rightarrow D$ were a conformal equivalence. Composing with the inclusion $D \hookrightarrow \mathbb{C}$ gives an injective entire function f ; by Casorati–Weierstrass its singularity at ∞ is not essential (an injective function cannot take values densely near ∞ while being open near a fixed finite point), so f extends to a holomorphic $\bar{f} : \mathbb{C}_\infty \rightarrow \mathbb{C}_\infty$. This is non-constant, so it is surjective by Corollary 41.5.4; injectivity of f then forces $\bar{f}(\infty) = \infty$ and $D = f(\mathbb{C}) = \mathbb{C}$, contradicting $D \subsetneq \mathbb{C}$. So $D \cong \mathbb{D}$. $\square$

It is worth noticing how the logic runs: uniformisation is the general theorem and the Riemann mapping theorem is a special case, whereas historically the implication is usually run the other way (the Riemann mapping theorem being an ingredient in proofs of uniformisation).

Theorem 41.13.19 (Little Picard Theorem)

Every holomorphic function $f : \mathbb{C} \rightarrow \mathbb{C} \setminus \{0, 1\}$ is constant.

Proof. The surface $\mathbb{C} \setminus \{0, 1\}$ is uniformised by $\mathbb{D}$: it is not $\mathbb{C}_\infty$ or $\mathbb{C}$ or $\mathbb{C}^*$ or a torus, so by Propositions 41.13.12 and 41.13.13 its universal cover is $\mathbb{D}$. (One can also exhibit the covering explicitly using the modular λ -function; this is on the example sheets.)

Let $\pi : \mathbb{D} \rightarrow \mathbb{C} \setminus \{0, 1\}$ be the uniformising map. As $\mathbb{C}$ is simply connected, Lemma

41.13.14 gives a holomorphic $F : \mathbb{C} \rightarrow \mathbb{D}$ with $f = \pi \circ F$. But F is a bounded entire function, hence constant by Liouville, so f is constant. $\square$

Of course $\{0, 1\}$ may be replaced by any two distinct points of $\mathbb{C}$, by composing with a suitable affine map. The theorem is a dramatic strengthening of Liouville: an entire function which omits merely *two* values is constant. That such a statement should follow from a covering-space argument – with no estimate in sight – is a fair summary of what Riemann surfaces are for.

42 Statistical Physics

Every physics course we have met so far has been microscopic. We wrote down the equation of motion for a particle, or the Schrödinger equation for an electron in a hydrogen atom, and then we worked hard to solve it. Statistical physics asks what happens when we try to do this for 10^{23} particles at once. The honest answer is that we cannot: no computer will ever integrate Avogadro's number of coupled differential equations, and even if one could, the output would be useless to us. What saves the subject – and makes it one of the most powerful in physics – is that we do not *want* to know what all 10^{23} particles are doing. We want to know a handful of macroscopic numbers: the pressure of the gas, its temperature, its heat capacity. The remarkable fact is that these few numbers can be extracted from the microscopic laws by an argument that is essentially probabilistic, and that the extraction is not merely approximate but, for systems this large, absurdly accurate.

The course has three faces. First we build the machinery: microstates, ensembles, entropy, temperature and the partition function, which is the single most useful object in the subject. Then we turn the machinery on real systems – classical gases, and then quantum ones, where we will find that the same formalism produces blackbody radiation, the heat capacity of solids, the physics of white dwarf stars and the strange collective behaviour of Bose–Einstein condensates. Along the way we will meet *classical thermodynamics*, which is the historical route into the subject: a set of laws about heat and work discovered before anyone was sure that atoms existed, and which we can now watch emerge from the statistics. Finally, we look at phase transitions, where a system changes its character discontinuously as we turn a knob, and where – most surprisingly of all – wildly different systems turn out to behave in exactly the same way.

This article constitutes my notes for the ‘Statistical Physics’ course, held in Lent 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered¹.

§42.1 Fundamentals of Statistical Mechanics

We begin by setting up the framework of the subject. The ideas here are not difficult, but they are subtle, and they are the foundation for everything that follows – so it is worth going slowly.

§42.1.1 Microstates and the Microcanonical Ensemble

Our object of study is a system of N particles, where N is a very large number. The canonical picture to have in mind is a box of gas, sealed off from the rest of the universe. We begin by describing the system in as much detail as possible.

¹The course assumes a working knowledge of quantum mechanics at the level of the Part IB course, and it will help enormously to have met the Legendre transform, for instance in the Variational Principles course; I will remind the reader of what is needed as we go.

Definition 42.1.1 (Microstate)

A **microstate** of a system is a complete specification of its state: in the quantum setting, an actual state $|n\rangle$ of the full N -particle system.

A microstate is a hopeless thing to try to write down. But the whole point of statistical physics is that we do not need to: enormously many microstates are macroscopically indistinguishable from one another, and we may as well treat them all on the same footing. So we adopt a probabilistic description. Let $\{|n\rangle\}$ be an orthonormal basis of energy eigenstates,

$$\hat{H} |n\rangle = E_n |n\rangle,$$

and let $p(n)$ be the probability that our system is in the microstate $|n\rangle$. It is important to be clear about what kind of probability this is. It is *not* a quantum amplitude squared; it is a classical probability, encoding our ignorance about which state the system is actually in. (If you have met density matrices, we are describing the mixed state $\rho = \sum_n p(n) |n\rangle \langle n|$.)

Definition 42.1.2 (Ensemble average)

Given probabilities $p(n)$, the **expectation value** of an operator $\mathcal{O}$ is

$$\langle \mathcal{O} \rangle = \sum_n p(n) \langle n | \mathcal{O} | n \rangle.$$

An equivalent and often more intuitive way to think about this: imagine $W \gg 1$ independent copies of our system, with $Wp(n)$ of them in the microstate $|n\rangle$. This collection is called an **ensemble**, and $\langle \mathcal{O} \rangle$ is just the average of $\mathcal{O}$ over the ensemble. Which picture one uses makes no difference to the mathematics.

Throughout, we will assume our systems are in **equilibrium**, meaning that the distribution $p(n)$ does not change with time. This does not mean nothing is happening – the molecules of a gas in equilibrium are hurtling about as fast as ever. It means only that the statistics are stationary. Systems out of equilibrium are enormously harder, and we will not touch them.

Now, what should $p(n)$ be? Suppose all we know about the system is its energy. The microstates consistent with what we know are called **accessible**, and the entire subject rests on the following assumption.

Axiom 42.1.3 (Fundamental Assumption of Statistical Mechanics)

An isolated system in equilibrium is equally likely to be found in any of its accessible microstates.

This is not something we can prove from the microscopic laws (proving it, in the guise of the *ergodic hypothesis*, is genuinely hard and still not fully understood). We take it as an axiom and judge it by its consequences, which are spectacularly successful.

Definition 42.1.4 (Microcanonical ensemble)

In the **microcanonical ensemble** we take our partial knowledge to be that the energy lies in a window $[E, E + \delta E]$, where δE is the resolution of our measuring

apparatus. We write $\Omega(E)$ for the number of microstates with $E \leq E_n \leq E + \delta E$.

The fundamental assumption then reads

$$p(n) = \begin{cases} 1/\Omega(E) & E \leq E_n \leq E + \delta E, \\ 0 & \text{otherwise.} \end{cases}$$

It looks worrying that the theory should depend on the arbitrary parameter δE , and we will see shortly that in fact it does not. Note also that it is essential here that we are working quantum mechanically: energy levels are discrete, so ‘counting states’ makes sense. Doing this classically requires more care, and we will come back to it in Section 42.2.1.

Example 42.1.5

Take $N = 10^{23}$ particles fixed in place, each of which can be in one of two states $|\uparrow\rangle, |\downarrow\rangle$ of the same energy ε . Then every configuration has total energy $N\varepsilon$, and

$$\Omega(N\varepsilon) = 2^{10^{23}}.$$

This is a completely preposterous number – vastly larger than the number of atoms in the observable universe – and it is entirely typical of the numbers in this subject.

Numbers like $2^{10^{23}}$ are unpleasant to work with, and worse, Ω is multiplicative rather than additive when we combine systems. Both problems are cured at a stroke by taking a logarithm.

Definition 42.1.6 (Boltzmann entropy)

The **entropy** of a system is

$$S(E) = k \log \Omega(E),$$

where $k = 1.381 \times 10^{-23} \text{ J K}^{-1}$ is **Boltzmann’s constant**.

The constant k is a historical accident: thermodynamics was developed before anyone knew about microstates, and temperature was given its own unit. Had physics been done in the other order we would simply have set $k = 1$ and measured temperature in joules.

We can now dispose of the worry about δE . If we double the width of our energy window we roughly double Ω , and so we change S by $k \log 2 \approx 10^{-23} \text{ J K}^{-1}$. Meanwhile S itself is of order $Nk \sim 1 \text{ J K}^{-1}$. So the dependence on δE is utterly negligible, and we suppress it. This is the first instance of a pattern that recurs constantly: because N is so large, factors that would be significant anywhere else in physics are invisible here.

The second virtue of the logarithm is additivity. If we have two *non-interacting* systems with energies E_1 and E_2 , then a microstate of the pair is a microstate of each, so $\Omega(E_1, E_2) = \Omega_1(E_1)\Omega_2(E_2)$ and hence

$$S(E_1, E_2) = S_1(E_1) + S_2(E_2).$$

§42.1.2 Entropy and the Second Law

Additivity is not very interesting until we let the systems talk to each other. So suppose we now bring our two systems into contact and allow them to exchange energy, while keeping the total energy $E_{\text{tot}} = E_1 + E_2$ fixed. Now the individual energies can fluctuate, and the number of states available to the combined system is

$$\Omega(E_{\text{tot}}) = \sum_{E_i} \Omega_1(E_i) \Omega_2(E_{\text{tot}} - E_i) = \sum_{E_i} \exp \left(\frac{S_1(E_i) + S_2(E_{\text{tot}} - E_i)}{k} \right),$$

where the sum runs over the energy bins of width δE that we chose earlier.

Now look at the size of the exponents. Each S/k is of order $N \sim 10^{23}$. So if the exponent is even slightly larger for one term than for another, that term dominates the sum by a factor which is the exponential of something like 10^{20} . In other words: *the sum is completely dominated by its largest term*. Writing E_* for the value of E_i that maximises $S_1(E_i) + S_2(E_{\text{tot}} - E_i)$, we have

$$S(E_{\text{tot}}) = k \log \Omega(E_{\text{tot}}) \approx S_1(E_*) + S_2(E_{\text{tot}} - E_*).$$

Of course some neighbouring terms also contribute, but including them multiplies Ω by some modest factor and hence adds something of order k to S , which we cheerfully ignore.

The same estimate tells us something much stronger. The probability that the first system has energy E_1 (writing $E_2 = E_{\text{tot}} - E_1$) is

$$\frac{\Omega_1(E_1)\Omega_2(E_2)}{\Omega(E_{\text{tot}})} = \exp \left(\frac{S_1(E_1) + S_2(E_2) - S(E_{\text{tot}})}{k} \right).$$

Unless $S_1(E_1) + S_2(E_2)$ agrees with its maximum value to fantastic precision, this probability is effectively zero. So although E_1 is in principle free to fluctuate, in practice it is pinned to E_* to a precision far beyond anything we could measure.

Finally, suppose we prepare the two systems separately with $E_1 \neq E_*$ and only then bring them together. The system is not in equilibrium; it will evolve, E_1 will drift to E_* , and in doing so the entropy will increase from $S_1(E_1) + S_2(E_2)$ to its maximum value. This is the second law.

Law 42.1.7 (Second Law of Thermodynamics)

The entropy of an isolated system never decreases, and in equilibrium it attains its maximum value.

It is worth being precise about what we have and have not shown. We have not shown that entropy decrease is *impossible*. What we have shown is that it is overwhelmingly improbable – the odds against it are the exponential of 10^{23} . This is a much more satisfying state of affairs than a bald prohibition, because it explains why the second law has the peculiar status it does: it is a law of statistics rather than of mechanics, and it is the only law of physics that distinguishes the future from the past.

§42.1.3 Temperature

We have a function $S(E)$. The obvious thing to do with it is differentiate it, and the obvious thing to do with the answer is give it a name.

Definition 42.1.8 (Temperature)

The **temperature** of a system is defined by

$$\frac{1}{T} = \frac{dS}{dE}.$$

Why should this deserve to be called temperature? Because it does what temperature is supposed to do. Let us check the two most basic properties we expect.

Proposition 42.1.9

Two systems in thermal contact and in equilibrium have the same temperature.

Proof. We found the equilibrium energy E_* by maximising $S_1(E_1) + S_2(E_{\text{tot}} - E_1)$ over E_1 . Differentiating and setting the result to zero,

$$\left. \frac{dS_1}{dE} \right|_{E_1=E_*} = \left. \frac{dS_2}{dE} \right|_{E_2=E_{\text{tot}}-E_*},$$

that is, $1/T_1 = 1/T_2$, so $T_1 = T_2$. $\square$

Proposition 42.1.10

If two systems at temperatures $T_1 > T_2$ are brought into contact, energy flows from the first to the second.

Proof. The systems are not in equilibrium, so some energy is transferred. Say after a short time $E_1 \mapsto E_1 + \delta E$ and $E_2 \mapsto E_2 - \delta E$. The change in total entropy is

$$\delta S = \frac{dS_1}{dE} \delta E - \frac{dS_2}{dE} \delta E = \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta E.$$

By hypothesis $1/T_1 - 1/T_2 < 0$, and by the second law $\delta S > 0$. Hence $\delta E < 0$: the first system loses energy. $\square$

So heat flows from hot to cold, and bodies in equilibrium have equal temperatures. Notice that these arguments used nothing about T except that $1/T$ is a decreasing function of T ; we could have defined dS/dE to be any monotone decreasing function of ‘temperature’ and got the same results. What singles out this particular choice is that it agrees with the temperature scale everyone already used – the one defined by the ideal gas law – as we will verify in Section 42.2.2.

§42.1.4 Heat Capacity, and a First Example

Differentiating S gave us temperature. If we now invert $T = T(E)$ to get $E = E(T)$ and differentiate again, we get another familiar quantity.

Definition 42.1.11 (Heat capacity)

The **heat capacity** of a system is

$$C = \frac{dE}{dT}.$$

The **specific heat capacity** is C divided by the mass of the system.

Unlike Ω or S , this is something we can go into a laboratory and measure: pump in a known amount of energy, and see how much the temperature rises. And measuring it gives us the entropy for free, because

$$\frac{dS}{dT} = \frac{dS}{dE} \frac{dE}{dT} = \frac{C}{T},$$

and hence, integrating,

$$\Delta S = \int_{T_1}^{T_2} \frac{C(T)}{T} dT.$$

So although entropy was defined by counting microstates – a thoroughly theoretical operation – its changes are experimentally accessible.

The heat capacity also settles a loose end. When we found E_* by setting a derivative to zero, we only checked that it was a stationary point. Differentiating once more,

$$\frac{d^2S}{dE^2} = \frac{d}{dE} \left(\frac{1}{T} \right) = -\frac{1}{T^2} \frac{dT}{dE} = -\frac{1}{T^2 C},$$

so $S_1 + S_2$ has a genuine maximum at E_* provided $C_1, C_2 > 0$. Positivity of heat capacity is thus exactly the condition for thermal equilibrium to be stable.

Enough generalities. Let us compute something.

Example 42.1.12 (The two-state system)

Take N non-interacting particles fixed at lattice sites, each of which has two states, with energies

$$E_{\downarrow} = 0, \quad E_{\uparrow} = \varepsilon.$$

Think of these as spins in a magnetic field. If $N_{\uparrow}$ of them are excited then the total energy is $E = \varepsilon N_{\uparrow}$, and (choosing $\delta E < \varepsilon$, which is legitimate here) the number of microstates is just the number of ways of choosing which particles are excited:

$$\Omega(E) = \binom{N}{N_{\uparrow}} = \frac{N!}{N_{\uparrow}! (N - N_{\uparrow})!}.$$

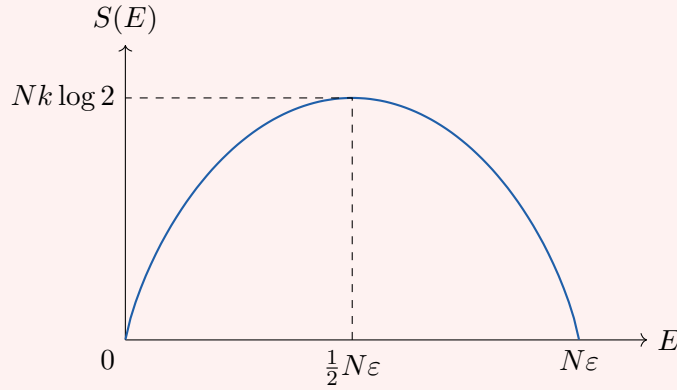
To make progress we need **Stirling's approximation**, which we will use constantly:

$$\log N! = N \log N - N + \frac{1}{2} \log(2\pi N) + O(1/N).$$

Since $N \sim 10^{23}$ we keep only the first two terms. Applying this three times and writing $x = E/(N\varepsilon) = N_{\uparrow}/N$ for the fraction of excited particles,

$$\begin{aligned} S(E) &= k(N \log N - N_{\uparrow} \log N_{\uparrow} - (N - N_{\uparrow}) \log(N - N_{\uparrow})) \\ &= -kN \left(x \log x + (1 - x) \log(1 - x) \right). \end{aligned}$$

This is much more illuminating when plotted.



The entropy vanishes at $E = 0$ (all spins down: one microstate) and at $E = N\epsilon$ (all spins up: one microstate), and is maximal in between at $E = N\epsilon/2$, where it takes the value $Nk \log 2$ – exactly what we expect, since then each spin is free to do as it pleases.

Now differentiate to get the temperature:

$$\frac{1}{T} = \frac{dS}{dE} = \frac{k}{\epsilon} \log \left(\frac{N\epsilon}{E} - 1 \right),$$

which we can invert to give the beautifully simple

$$\frac{N_{\uparrow}}{N} = \frac{E}{N\epsilon} = \frac{1}{e^{\epsilon/kT} + 1}.$$

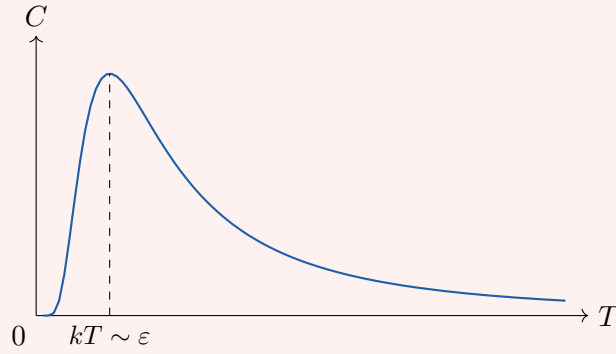
As $T \rightarrow 0$ we get $N_{\uparrow} \rightarrow 0$: everything falls into the ground state, as it should. As $T \rightarrow \infty$ we get $N_{\uparrow}/N \rightarrow \frac{1}{2}$, which is more surprising – one might have guessed that infinite temperature would push everything into the excited state. It does not: at infinite temperature the Boltzmann factors for the two levels become equal and the spins are simply randomised.

This has a strange consequence. The graph of S against E has negative slope for $E > N\epsilon/2$, so such states have *negative temperature*. Moreover, to get from positive T to negative T one must pass through $T = \infty$, so negative temperatures are ‘hotter’ than infinite temperature. This is not a pathology of the definition but a genuine feature of systems whose energy is bounded above; it can only happen because we have frozen the particles in place and thrown away their kinetic energy, which is unbounded. Negative-temperature states have been produced in the laboratory, using nuclear spins in a crystal: align them with a magnetic field, then suddenly reverse the field.

Finally, the heat capacity. Differentiating $E = N\epsilon/(e^{\epsilon/kT} + 1)$,

$$C = \frac{dE}{dT} = \frac{N\epsilon^2}{kT^2} \frac{e^{\epsilon/kT}}{(e^{\epsilon/kT} + 1)^2},$$

which looks like this.



Three things are worth noticing. First, the position of the peak is set by the microscopic energy gap, $kT \sim \varepsilon$; so a macroscopic measurement of C tells us about the microscopic level structure. Secondly, as $T \rightarrow 0$ we have $C \propto T^{-2}e^{-\varepsilon/kT}$, which vanishes faster than any power of T – the classic analysis example of a smooth non-analytic function. This exponential suppression is the signature of an energy gap: at low temperatures there is simply not enough thermal energy to excite anything. Thirdly, $C \rightarrow 0$ at high temperature as well, which is peculiar and again a consequence of the bounded energy spectrum; a realistic system has C increasing with T . In a real solid this spin contribution is swamped by the phonons and electrons, and shows up only as a small bump at low temperature, called a **Schottky anomaly**.

§42.1.5 Pressure, Volume and the First Law

So far our systems have had exactly one macroscopic parameter, the energy. Real systems have more. The most obvious is volume: if we let the walls of our box move, then the energy eigenvalues change, and so Ω and S depend on V as well:

$$S(E, V) = k \log \Omega(E, V).$$

We must therefore be careful about what is held fixed when we differentiate, and we adopt the standard notation of writing the fixed variables as a subscript.

Definition 42.1.13 (Temperature and pressure)

For a system with variable volume,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_V, \quad p = T \left(\frac{\partial S}{\partial V} \right)_E.$$

The definition of p looks unmotivated, but it is forced on us by the same argument as before.

Proposition 42.1.14

If two systems can exchange both energy and volume, with $E_1 + E_2$ and $V_1 + V_2$ fixed, then entropy is maximised when $T_1 = T_2$ and $p_1 = p_2$.

Proof. We must maximise $S_1(E_1, V_1) + S_2(E_{\text{tot}} - E_1, V_{\text{tot}} - V_1)$ over both E_1 and

V_1 . The E derivative gives $T_1 = T_2$ as before, and the V derivative gives

$$\left(\frac{\partial S_1}{\partial V}\right)_E = \left(\frac{\partial S_2}{\partial V}\right)_E,$$

which, on multiplying by the common temperature, is $p_1 = p_2$. $\square$

So p is the quantity that equalises when a piston between two gases is free to move – which is exactly what mechanical pressure does. To see that it really *is* force per unit area, write $S = S(E, V)$ and take the total differential:

$$dS = \left(\frac{\partial S}{\partial E}\right)_V dE + \left(\frac{\partial S}{\partial V}\right)_E dV = \frac{1}{T} dE + \frac{p}{T} dV.$$

Rearranging gives one of the most important equations in the subject.

Proposition 42.1.15 (First Law of Thermodynamics)

$$dE = T dS - p dV.$$

The interpretation is this: $-p dV$ is the mechanical work done on the system when we compress it, exactly as in Newtonian mechanics, which confirms that our p is the ordinary pressure. The remaining term $T dS$ is therefore energy transferred *not* as mechanical work; we call it **heat**. The first law says that energy is conserved provided we remember to count heat as a form of energy – a statement that took most of the nineteenth century to become uncontroversial.

Example 42.1.16 (The ideal gas, done the hard way)

Let us extract the equation of state of an ideal gas from the microcanonical ensemble. We will do this much more slickly later, so treat this as a demonstration that it *can* be done. For N free particles in a box of volume V , the accessible phase space at fixed energy E factorises into a position part and a momentum part. Each particle can be anywhere in the box, giving a factor V^N ; and the momenta are constrained to the sphere $\sum_i \mathbf{p}_i^2 = 2mE$ in $3N$ dimensions, whose area depends only on E and N . So

$$\Omega(E, V) = V^N f(E)$$

for some function f we do not need to know. Then $S = Nk \log V + k \log f(E)$, and

$$p = T \left(\frac{\partial S}{\partial V}\right)_E = \frac{NkT}{V},$$

which is the ideal gas law $pV = NkT$. Notice how little we had to know: only that the positions and momenta decouple.

§42.1.6 The Canonical Ensemble

The microcanonical ensemble is conceptually clean but computationally awful: to use it we must count states at fixed energy, which usually means a nasty combinatorial problem. Worse, it is not a realistic description of most experiments. A beaker of water on a bench is not isolated; it is in contact with the room, and its energy fluctuates.

What is fixed is its *temperature*. This suggests a different setup, and it turns out to be enormously easier to work with.

So: let our system $\mathcal{S}$ be in thermal contact with a **reservoir** $\mathcal{R}$, so much larger than $\mathcal{S}$ that its temperature T is unaffected by anything $\mathcal{S}$ does. The combined system is isolated, with total energy E_{tot} , and we apply the microcanonical ensemble to *it*. If $\mathcal{S}$ is in a definite microstate $|n\rangle$ of energy E_n , then the number of microstates of the combination is just the number available to the reservoir, namely $\Omega_{\mathcal{R}}(E_{\text{tot}} - E_n)$. Hence

$$p(n) \propto \Omega_{\mathcal{R}}(E_{\text{tot}} - E_n) = \exp\left(\frac{S_{\mathcal{R}}(E_{\text{tot}} - E_n)}{k}\right).$$

Now $E_n \ll E_{\text{tot}}$, so we may Taylor expand:

$$S_{\mathcal{R}}(E_{\text{tot}} - E_n) = S_{\mathcal{R}}(E_{\text{tot}}) - E_n \left. \frac{\partial S_{\mathcal{R}}}{\partial E} \right|_{E_{\text{tot}}} + \cdots = S_{\mathcal{R}}(E_{\text{tot}}) - \frac{E_n}{T} + \cdots,$$

and the higher terms are suppressed by further powers of E_n/E_{tot} . The first term is a constant, independent of n , so it is absorbed into the normalisation, and we are left with the single most important formula of the course.

Definition 42.1.17 (Boltzmann distribution and partition function)

A system in equilibrium with a reservoir at temperature T occupies its microstates with probabilities

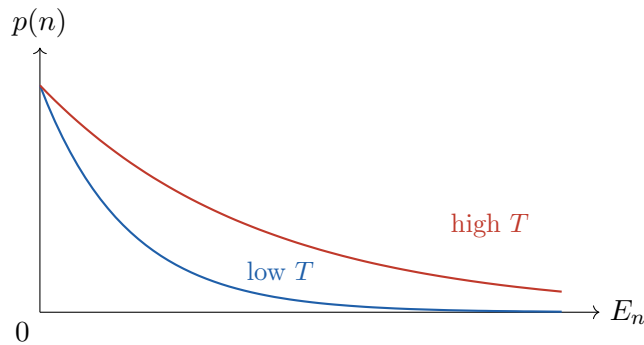
$$p(n) = \frac{e^{-\beta E_n}}{Z}, \quad \beta = \frac{1}{kT},$$

where the normalising constant

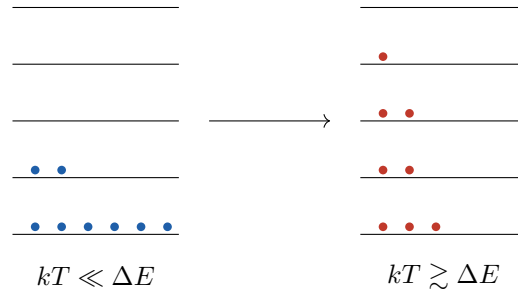
$$Z = \sum_n e^{-\beta E_n}$$

is the **partition function**. This probability distribution is the **Boltzmann distribution**, and the ensemble is the **canonical ensemble**.

The sum defining Z runs over all *states*, not all energy levels; a level of degeneracy g contributes $ge^{-\beta E}$. The factor $e^{-\beta E_n}$ is called the **Boltzmann factor**, and its shape is the whole story of the canonical ensemble: high-energy states are exponentially rare, and the scale on which ‘high’ is measured is kT .



It is worth pausing over what this means for the occupation of energy levels. At low temperature almost all the probability sits in the ground state; as T rises, states higher up the ladder become accessible, and the population spreads out. Schematically:



The partition function was introduced as a normalisation constant, but it is far more useful than that: essentially every thermodynamic quantity can be got out of it by differentiation. The first example is the average energy.

Proposition 42.1.18

The mean energy of a system in the canonical ensemble is

$$\langle E \rangle = \sum_n p(n) E_n = -\frac{\partial}{\partial \beta} \log Z.$$

Proof. Differentiating $Z = \sum_n e^{-\beta E_n}$ with respect to β gives $\partial_\beta Z = -\sum_n E_n e^{-\beta E_n}$, and dividing by Z gives $-\langle E \rangle$. $\square$

Since $\langle E \rangle$ is what we actually mean by ‘the energy’ of a macroscopic system, we will from now on drop the angle brackets and simply write E .

Before computing anything, one more structural remark, which is the reason the canonical ensemble is so much nicer than the microcanonical one. Suppose our system consists of two non-interacting parts, so that the states are pairs and the energies add. Then

$$Z = \sum_{n_1, n_2} e^{-\beta(E_{n_1} + E_{n_2})} = \left(\sum_{n_1} e^{-\beta E_{n_1}} \right) \left(\sum_{n_2} e^{-\beta E_{n_2}} \right) = Z_1 Z_2.$$

Partition functions of independent systems simply multiply. In the microcanonical ensemble the analogous statement was a convolution, which is why everything there was hard.

Example 42.1.19 (The two-state system again)

Each particle has $Z_1 = 1 + e^{-\beta \varepsilon}$, so for N of them $Z = (1 + e^{-\beta \varepsilon})^N$ and

$$E = -\frac{\partial}{\partial \beta} \log Z = -N \frac{\partial}{\partial \beta} \log(1 + e^{-\beta \varepsilon}) = \frac{N \varepsilon}{e^{\beta \varepsilon} + 1},$$

reproducing in three lines what took us a page of Stirling’s formula in the microcanonical ensemble.

§42.1.7 Fluctuations

There is an obvious objection to all this. In the microcanonical ensemble the energy was fixed; in the canonical ensemble it fluctuates. Why should the two give the same answers? The resolution is that the fluctuations are tiny, and we can quantify this precisely.

Proposition 42.1.20

In the canonical ensemble,

$$\Delta E^2 = \langle E^2 \rangle - \langle E \rangle^2 = \frac{\partial^2}{\partial \beta^2} \log Z = kT^2 C_V.$$

Proof. We compute

$$\frac{\partial^2 \log Z}{\partial \beta^2} = \frac{\partial}{\partial \beta} \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right) = \frac{1}{Z} \frac{\partial^2 Z}{\partial \beta^2} - \left(\frac{1}{Z} \frac{\partial Z}{\partial \beta} \right)^2 = \langle E^2 \rangle - \langle E \rangle^2.$$

On the other hand $\langle E \rangle = -\partial_\beta \log Z$, so this equals $-\partial \langle E \rangle / \partial \beta$; and since $\beta = 1/kT$ we have $\partial / \partial \beta = -kT^2 \partial / \partial T$, giving $kT^2 (\partial E / \partial T)_V = kT^2 C_V$. $\square$

Now, both E and C_V are proportional to N . So

$$\frac{\Delta E}{E} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}},$$

which for $N \sim 10^{23}$ is about 10^{-12} . The energy of a macroscopic system in the canonical ensemble is therefore fixed to about twelve significant figures, which is rather better than any experiment can do. This is why the two ensembles agree: in the thermodynamic limit the distinction between ‘fixed energy’ and ‘fixed temperature’ is invisible. We will use whichever is more convenient, which is almost always the canonical one.

The relation $\Delta E^2 = kT^2 C_V$ is also our first **fluctuation–dissipation relation**: it connects the size of spontaneous fluctuations (the left side) to the response of the system to an external prod (the right side). Relations of this type run through the whole of statistical physics.

§42.1.8 The Gibbs Entropy

We defined entropy by counting microstates, which only makes sense when all accessible microstates are equally likely. In the canonical ensemble they are not, so we need a definition that works for a general probability distribution. Here it is.

Definition 42.1.21 (Gibbs entropy)

Given a probability distribution $p(n)$ over microstates, the **Gibbs entropy** is

$$S = -k \sum_n p(n) \log p(n).$$

This agrees with the old definition when it applies: if $p(n) = 1/\Omega$ on Ω accessible states, then

$$S = -k \sum_n \frac{1}{\Omega} \log \frac{1}{\Omega} = k \log \Omega$$

as required. And it is exactly the quantity Shannon later introduced as a measure of the information content of a distribution – entropy is a measure of how much we *don’t* know about the system, which is a rather satisfying way to think about the second law.

Let us check that this definition is consistent with what we have been doing.

Proposition 42.1.22

For the canonical ensemble, the Gibbs entropy satisfies

$$\left(\frac{\partial S}{\partial E}\right)_V = \frac{1}{T}.$$

Proof. Substituting $p(n) = e^{-\beta E_n}/Z$,

$$S = -k \sum_n p(n) (-\beta E_n - \log Z) = k\beta \langle E \rangle + k \log Z,$$

using $\sum_n p(n) = 1$. So, writing everything in terms of β ,

$$\frac{\partial S}{\partial \beta} = k \langle E \rangle + k\beta \frac{\partial \langle E \rangle}{\partial \beta} + k \frac{\partial \log Z}{\partial \beta} = k\beta \frac{\partial \langle E \rangle}{\partial \beta},$$

since $\partial_\beta \log Z = -\langle E \rangle$. Dividing by $\partial \langle E \rangle / \partial \beta$ gives $\partial S / \partial E = k\beta = 1/T$. $\square$

The identity $S = k\beta E + k \log Z$ obtained in that proof is worth remembering; we will use it in a moment.

There is another characterisation of the Boltzmann distribution which is, in a sense, the deepest way of looking at it: it is the distribution which maximises entropy subject to knowing the mean energy.

Proposition 42.1.23 (Maximum entropy principle)

Among all probability distributions $p(n)$ with a given mean energy $\langle E \rangle = \sum_n p(n) E_n$, the Gibbs entropy is maximised by the Boltzmann distribution.

Proof. We extremise S subject to the constraints $\sum_n p(n) = 1$ and $\sum_n p(n) E_n = \langle E \rangle$, using Lagrange multipliers. Consider

$$\Phi = -k \sum_n p(n) \log p(n) + k\alpha \left(\sum_n p(n) - 1 \right) + k\beta \left(\langle E \rangle - \sum_n p(n) E_n \right),$$

where the factors of k and signs are chosen for later convenience. Then

$$\frac{\partial \Phi}{\partial p(n)} = -k (\log p(n) + 1) + k\alpha - k\beta E_n = 0,$$

so $p(n) = e^{\alpha-1} e^{-\beta E_n}$. Imposing $\sum_n p(n) = 1$ fixes $e^{\alpha-1} = 1/Z$, giving the Boltzmann distribution, with β determined by the energy constraint. That this is a maximum rather than a minimum follows since $-p \log p$ is concave. $\square$

So the canonical ensemble is the least prejudiced distribution consistent with what we know. In the same spirit, if we impose no constraint at all beyond normalisation we get the uniform distribution, which is the microcanonical ensemble; the fundamental assumption of statistical mechanics is therefore itself a maximum entropy statement.

§42.1.9 The Helmholtz Free Energy

We have found $S = k\beta E + k \log Z$, which we now rearrange into

$$-kT \log Z = E - TS.$$

The right-hand side is important enough to have a name.

Definition 42.1.24 (Helmholtz free energy)

The **Helmholtz free energy** of a system is

$$F = E - TS = -kT \log Z.$$

Readers who have met the Legendre transform – for instance in the Variational Principles course – will recognise what is going on: F is (minus) the Legendre transform of $E(S, V)$ in the variable S , trading the entropy for its conjugate variable, the temperature. This is exactly the right thing to do, because S is not something one can hold fixed in a laboratory whereas T is. We will return to this systematically in Section 42.4.5.

The point of F is that everything can be computed from it by differentiation. Taking the differential of $F = E - TS$ and using the first law $dE = T dS - p dV$,

$$dF = dE - T dS - S dT = -S dT - p dV.$$

So F is naturally a function of T and V , and

$$S = -\left(\frac{\partial F}{\partial T}\right)_V, \quad p = -\left(\frac{\partial F}{\partial V}\right)_T.$$

Since $F = -kT \log Z$ is directly computable from the partition function, this is our standard route from a microscopic model to an equation of state: compute Z , take a logarithm, differentiate.

There is also a variational characterisation of F , which explains the name. Suppose our system is at temperature T and we ask which of its states it wants to be in. The answer is the one that minimises F : minimising $F = E - TS$ means compromising between minimising energy and maximising entropy, with the temperature setting the exchange rate. At low temperature energy wins and the system falls into its ground state; at high temperature entropy wins and the system disorders. Almost every phenomenon in the second half of this course is a battle between these two terms.

Proposition 42.1.25

Consider a system in contact with a reservoir at temperature T . Then the total entropy of system plus reservoir is maximised when the free energy of the system is minimised.

Proof. If the system has energy E then the reservoir has $E_{\text{tot}} - E$, and

$$S_{\text{tot}} = S(E) + S_{\mathcal{R}}(E_{\text{tot}} - E) \approx S(E) + S_{\mathcal{R}}(E_{\text{tot}}) - \frac{E}{T},$$

expanding as before. Dropping the constant, maximising S_{tot} is the same as max-

imising $S - E/T$, i.e. minimising $E - TS = F$. □

§42.1.10 The Chemical Potential and the Grand Canonical Ensemble

We allowed our systems to exchange energy, and then volume. The remaining thing they might exchange is particles. Repeating the pattern, S now depends on E , V and N , and we define a new derivative.

Definition 42.1.26 (Chemical potential)

The **chemical potential** of a system is

$$\mu = -T \left(\frac{\partial S}{\partial N} \right)_{E,V}.$$

The minus sign is a convention, chosen so that the first law takes a pleasant form. Indeed, taking the differential of $S(E, V, N)$ exactly as before and rearranging,

$$dE = T dS - p dV + \mu dN.$$

So μ is the energy cost of adding one particle at fixed entropy and volume. The same maximisation argument as before shows that two systems which can exchange particles are in equilibrium precisely when their chemical potentials agree, and that particles flow from high μ to low μ .

Now we play the reservoir trick again, but with a reservoir that exchanges particles as well as energy. If our system is in a state with energy E_n and particle number N_n , then

$$p(n) \propto \Omega_{\mathcal{R}}(E_{\text{tot}} - E_n, N_{\text{tot}} - N_n) = \exp \left(\frac{S_{\mathcal{R}}(E_{\text{tot}} - E_n, N_{\text{tot}} - N_n)}{k} \right),$$

and expanding the entropy to first order in both arguments,

$$S_{\mathcal{R}}(E_{\text{tot}} - E_n, N_{\text{tot}} - N_n) \approx \text{const} - \frac{E_n}{T} + \frac{\mu N_n}{T}.$$

Definition 42.1.27 (Grand canonical ensemble)

In the **grand canonical ensemble** the probabilities are

$$p(n) = \frac{e^{-\beta(E_n - \mu N_n)}}{\mathcal{Z}}, \quad \mathcal{Z} = \sum_n e^{-\beta(E_n - \mu N_n)},$$

where $\mathcal{Z}$ is the **grand canonical partition function**.

It is often convenient to organise the sum by particle number. Writing $Z(N)$ for the ordinary canonical partition function of N particles, and introducing the **fugacity**

$$z = e^{\beta\mu},$$

we have

$$\mathcal{Z} = \sum_{N \geq 0} z^N Z(N),$$

so $\mathcal{Z}$ is a generating function for the canonical partition functions. This will be exactly what we need for quantum gases in Section 42.3.

All the same tricks work. Differentiating with respect to μ and β gives

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \log \mathcal{Z}}{\partial \mu}, \quad \langle E \rangle - \mu \langle N \rangle = -\frac{\partial \log \mathcal{Z}}{\partial \beta},$$

and the fluctuations in particle number are

$$\Delta N^2 = \langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta^2} \frac{\partial^2 \log \mathcal{Z}}{\partial \mu^2} = \frac{1}{\beta} \left(\frac{\partial \langle N \rangle}{\partial \mu} \right)_{T,V}.$$

As with the energy, $\Delta N/N \sim 1/\sqrt{N}$, so the particle number is again effectively fixed and the ensemble agrees with the others.

Finally, we have a free energy adapted to this ensemble.

Definition 42.1.28 (Grand canonical potential)

The **grand canonical potential** is

$$\Phi = F - \mu N = -kT \log \mathcal{Z}.$$

Its differential is $d\Phi = -S dT - p dV - N d\mu$, so

$$S = -\left(\frac{\partial \Phi}{\partial T} \right)_{V,\mu}, \quad p = -\left(\frac{\partial \Phi}{\partial V} \right)_{T,\mu}, \quad N = -\left(\frac{\partial \Phi}{\partial \mu} \right)_{T,V}.$$

§42.1.11 Extensive and Intensive Quantities

A macroscopic quantity generally falls into one of two classes.

Definition 42.1.29

A quantity X is **extensive** if it scales linearly when we scale the system,

$$X(\lambda E, \lambda V, \lambda N) = \lambda X(E, V, N),$$

and **intensive** if it is unchanged, $X(\lambda E, \lambda V, \lambda N) = X(E, V, N)$.

Thus E , V , N , S and F are extensive, while T , p and μ are intensive – as they must be, being ratios of extensive quantities. (Extensivity of S requires that interactions be short-ranged, so that the energy of two halves of a system is the sum of their energies up to a negligible surface term. For gravitating systems this fails badly, which is why the thermodynamics of black holes is so strange.)

This simple observation is surprisingly powerful. Consider $\Phi = \Phi(T, V, \mu)$. Of its arguments only V is extensive, and Φ itself is extensive, so we must have

$$\Phi(T, V, \mu) = V \phi(T, \mu)$$

for some intensive function ϕ . But $p = -(\partial \Phi / \partial V)_{T,\mu} = -\phi$, and so

$$\Phi = -pV.$$

Combining with $\Phi = F - \mu N = E - TS - \mu N$ gives the **Euler relation**

$$E = TS - pV + \mu N,$$

and hence, comparing with the first law $dE = T dS - p dV + \mu dN$, the **Gibbs–Duhem relation**

$$S dT - V dp + N d\mu = 0.$$

This says the three intensive quantities T, p, μ are not independent: fixing two determines the third. It is a genuinely non-trivial constraint that we obtained purely from scaling.

§42.2 Classical Gases

We now have a machine: write down the energy levels, compute Z , differentiate. Let us feed it some gases. Everything in this section is classical, in the sense that we will treat the particles as little billiard balls; but we will find, slightly awkwardly, that we cannot entirely forget about quantum mechanics even so.

§42.2.1 The Classical Partition Function

The partition function was defined as a sum over quantum states. Classically there are no discrete states – a particle’s configuration is a point $(\mathbf{q}, \mathbf{p})$ in phase space – so we must replace the sum by an integral. For a single particle in three dimensions the natural guess is

$$Z_1 = \frac{1}{(\text{something})} \int d^3q d^3p e^{-\beta H(\mathbf{q}, \mathbf{p})},$$

and the question is what the normalising constant should be. It cannot be anything: $d^3q d^3p$ has dimensions of (action)³, and Z must be dimensionless. The answer is provided by quantum mechanics. The uncertainty principle says that a quantum state occupies a phase space volume of about $\hbar^3$ per particle – more precisely, of $(2\pi\hbar)^3$, as one can check by counting the momentum states of a particle in a box, which we will do in Section 42.3.1.

Definition 42.2.1 (Classical partition function)

For a single classical particle in three dimensions with Hamiltonian $H(\mathbf{q}, \mathbf{p})$,

$$Z_1 = \frac{1}{(2\pi\hbar)^3} \int d^3q d^3p e^{-\beta H(\mathbf{q}, \mathbf{p})}.$$

This is a slightly uncomfortable state of affairs: a ‘classical’ theory that has $\hbar$ in it. The honest statement is that classical statistical mechanics is not self-consistent, and needs quantum mechanics to fix its normalisation. Happily, for most purposes the constant drops out, since it contributes only an additive constant to $\log Z$ and hence disappears on differentiation. There are two places where it does not drop out – the entropy, and the counting of states – and both will cause us trouble shortly.

§42.2.2 The Monatomic Ideal Gas

Definition 42.2.2 (Ideal gas)

An **ideal gas** is a gas of particles which do not interact with one another.

For a single monatomic particle of mass m in a box of volume V , the Hamiltonian is purely kinetic, $H = \mathbf{p}^2/2m$. So

$$Z_1 = \frac{1}{(2\pi\hbar)^3} \int_V d^3q \int d^3p e^{-\beta \mathbf{p}^2/2m} = \frac{V}{(2\pi\hbar)^3} \left(\int_{-\infty}^{\infty} dp e^{-\beta p^2/2m} \right)^3,$$

the position integral giving V because the integrand does not depend on $\mathbf{q}$. The remaining integral is Gaussian,

$$\int_{-\infty}^{\infty} e^{-\beta p^2/2m} dp = \sqrt{\frac{2\pi m}{\beta}},$$

so

$$Z_1 = V \left(\frac{m}{2\pi\hbar^2\beta} \right)^{3/2} = \frac{V}{\lambda^3},$$

where we have introduced a length that will follow us through the rest of the course.

Definition 42.2.3 (Thermal de Broglie wavelength)

The **thermal de Broglie wavelength** is

$$\lambda = \sqrt{\frac{2\pi\hbar^2}{mkT}}.$$

It is worth understanding what this length means. A particle at temperature T has typical momentum $p \sim \sqrt{mkT}$, and hence de Broglie wavelength $\hbar/p \sim \hbar/\sqrt{mkT}$, which is λ up to a numerical factor. So λ is the size of the quantum-mechanical ‘smear’ of a typical particle. Classical physics should be valid when this is much smaller than the typical interparticle spacing, i.e. when

$$\lambda^3 \ll \frac{V}{N},$$

and we should expect quantum effects when the gas is cold or dense enough that the smears start to overlap. Hold on to this; it is the criterion that will tell us when we need Section 42.3.

Now for N particles. Since they do not interact, one might think $Z = Z_1^N$. This turns out to be wrong, but let us proceed anyway and see what goes wrong – the correction is a factor of $1/N!$ which we will discuss in Section 42.2.5. Taking

$$Z = \frac{1}{N!} Z_1^N = \frac{1}{N!} \frac{V^N}{\lambda^{3N}},$$

we have, using Stirling,

$$\log Z = N \log \frac{V}{\lambda^3} - \log N! \approx N \left(\log \frac{V}{N\lambda^3} + 1 \right).$$

Everything now follows by differentiation. Since $\lambda \propto \beta^{1/2}$, we have $\log Z = -\frac{3}{2}N \log \beta +$ (terms independent of β), and so the energy is

$$E = -\frac{\partial \log Z}{\partial \beta} = \frac{3N}{2\beta} = \frac{3}{2}NkT.$$

The pressure is

$$p = - \left(\frac{\partial F}{\partial V} \right)_T = kT \left(\frac{\partial \log Z}{\partial V} \right)_T = \frac{NkT}{V}.$$

Proposition 42.2.4 (Ideal gas law)

A classical monatomic ideal gas obeys

$$pV = NkT, \quad E = \frac{3}{2}NkT, \quad C_V = \left(\frac{\partial E}{\partial T} \right)_V = \frac{3}{2}Nk.$$

The first of these is the equation everyone meets at school. It is worth appreciating what has just happened: we assumed only the fundamental postulate of statistical mechanics and a free Hamiltonian, and out came a law that was discovered experimentally two centuries earlier. It also justifies, retrospectively, our definition of temperature: the T appearing here is the same T as in the ideal gas thermometer.

Definition 42.2.5 (Equation of state)

An **equation of state** is a relation between the macroscopic variables of a system, such as $p = p(T, V, N)$.

Let us also compute the entropy, since it will be instructive. From $S = -(\partial F / \partial T)_V$ with $F = -kT \log Z$,

$$S = k \log Z + kT \frac{\partial \log Z}{\partial T} = Nk \left(\log \frac{V}{N\lambda^3} + \frac{5}{2} \right),$$

using $\lambda \propto T^{-1/2}$ so that $T \partial_T \log \lambda^{-3} = 3/2$. This is the **Sackur–Tetrode equation**. Notice that $\hbar$ has *not* cancelled: it survives inside λ . So the absolute entropy of a gas is a genuinely quantum quantity, even though we thought we were doing classical physics. This is not merely academic – the Sackur–Tetrode formula can be checked experimentally, and it is right.

§42.2.3 Equipartition

The factor $\frac{3}{2}NkT$ in the energy is an instance of a much more general phenomenon.

Proposition 42.2.6 (Equipartition of energy)

Suppose a classical Hamiltonian depends quadratically on some degree of freedom x (a position or a momentum), say $H = \alpha x^2 + (\text{terms not involving } x)$ with $\alpha > 0$. Then in equilibrium at temperature T that degree of freedom contributes

$$\langle \alpha x^2 \rangle = \frac{1}{2}kT$$

to the mean energy.

Proof. Write $H = \alpha x^2 + \tilde{H}$ where $\tilde{H}$ does not involve x . Then the partition function

factorises as $Z = Z_x \tilde{Z}$ with

$$Z_x = \int_{-\infty}^{\infty} dx e^{-\beta \alpha x^2} = \sqrt{\frac{\pi}{\alpha \beta}} \propto \beta^{-1/2},$$

so $\log Z_x = -\frac{1}{2} \log \beta + \text{const}$, and the contribution to the energy is

$$-\frac{\partial \log Z_x}{\partial \beta} = \frac{1}{2\beta} = \frac{1}{2} kT.$$

□

So each quadratic degree of freedom carries $\frac{1}{2}kT$ of energy. A monatomic gas particle has three (the three components of momentum), giving $\frac{3}{2}kT$ each, as we found. A one-dimensional harmonic oscillator has two, one from the kinetic energy and one from the potential, giving kT – which is why a classical solid, modelled as $3N$ oscillators, has $E = 3NkT$ and $C = 3Nk$, the **Dulong–Petit law**.

Equipartition is a beautiful result, and it is spectacularly wrong at low temperatures. The heat capacity of every real solid tends to zero as $T \rightarrow 0$, not to $3Nk$. The reason is exactly the one we met in the two-state system: when kT drops below the spacing of the energy levels, the degree of freedom is *frozen out* and stops contributing. This failure of equipartition was one of the pieces of evidence that forced quantum mechanics on physics, and we will see how it is repaired in Section 42.3.3.

§42.2.4 The Maxwell Distribution

The partition function tells us about averages, but we can also read off the full distribution of any quantity we like. The most famous example is the distribution of molecular speeds.

The probability that a given particle in an ideal gas has momentum in d^3p about $\mathbf{p}$ and position in d^3q is proportional to the Boltzmann factor $e^{-\beta \mathbf{p}^2/2m}$. Integrating out the position (which contributes a constant V) and changing variables to velocity $\mathbf{v} = \mathbf{p}/m$, the distribution of velocities is

$$f(\mathbf{v}) d^3v \propto e^{-\beta m v^2/2} d^3v.$$

If we only care about the speed $v = |\mathbf{v}|$, we write $d^3v = 4\pi v^2 dv$ and normalise.

Proposition 42.2.7 (Maxwell–Boltzmann distribution)

In a classical ideal gas at temperature T , the probability that a molecule has speed in $[v, v + dv]$ is $f(v) dv$, where

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT}.$$

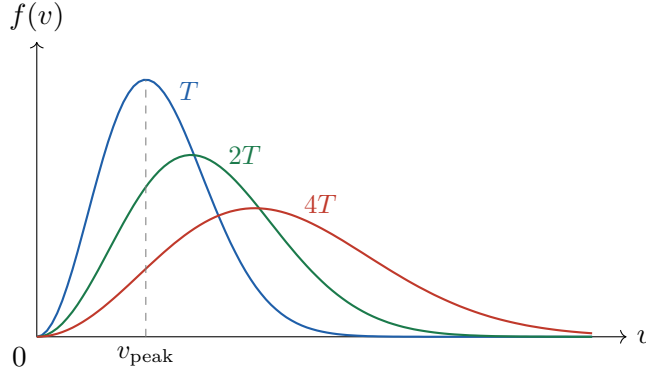
Proof. The functional form is as above; the constant is fixed by $\int_0^\infty f(v) dv = 1$, using the standard integral $\int_0^\infty v^2 e^{-av^2} dv = \frac{1}{4} \sqrt{\pi/a^3}$ with $a = m/2kT$. □

The two factors in f pull in opposite directions: the v^2 comes from the growing area of the sphere of radius v in velocity space and pushes the distribution to large speeds,

while the exponential cuts it off. The competition gives a peak at

$$v_{\text{peak}} = \sqrt{\frac{2kT}{m}},$$

found by differentiating. Here is the distribution at three temperatures T , $2T$ and $4T$; each curve encloses unit area, so as the gas heats up the distribution both broadens and flattens.



Three different ‘average’ speeds are in common use, and it is worth knowing that they differ. Alongside $v_{\text{peak}} = \sqrt{2kT/m}$ we have the mean speed

$$\langle v \rangle = \int_0^\infty v f(v) dv = \sqrt{\frac{8kT}{\pi m}},$$

and the root-mean-square speed, which by equipartition satisfies $\frac{1}{2}m \langle v^2 \rangle = \frac{3}{2}kT$, so

$$v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3kT}{m}}.$$

These are in the ratio $1 : \sqrt{4/\pi} : \sqrt{3/2} \approx 1 : 1.13 : 1.22$, in the order stated – which is what one expects from a distribution with a long tail to the right.

Example 42.2.8

For nitrogen ($m \approx 4.7 \times 10^{-26}$ kg) at room temperature, $v_{\text{rms}} \approx 500 \text{ m s}^{-1}$, comfortably faster than the speed of sound. Molecules in the air around you are moving very quickly indeed; they do not get very far, because they collide constantly.

§42.2.5 Indistinguishability and the Gibbs Paradox

We must now return to the factor of $1/N!$ that we inserted without justification. Suppose we had not: take $Z = Z_1^N$, so that

$$\log Z = N \log \frac{V}{\lambda^3}, \quad S = Nk \left(\log \frac{V}{\lambda^3} + \frac{3}{2} \right).$$

The pressure and energy are unaffected, but the entropy is wrong – and wrong in an instructive way. Under $V \mapsto \lambda V$, $N \mapsto \lambda N$ this S does not scale as λS ; it is not extensive. That is a disaster, since we argued in Section 1.11 that entropy must be extensive.

The physical version of this disaster is the following.

Example 42.2.9 (Gibbs paradox)

Take a box of volume $2V$ containing $2N$ molecules of the same gas at temperature T , and insert a partition down the middle, so we now have two boxes each of volume V containing N molecules. Using the non-extensive formula, the total entropy before was

$$S_{\text{before}} = 2Nk \left(\log \frac{2V}{\lambda^3} + \frac{3}{2} \right),$$

and afterwards

$$S_{\text{after}} = 2 \times Nk \left(\log \frac{V}{\lambda^3} + \frac{3}{2} \right),$$

so $S_{\text{before}} - S_{\text{after}} = 2Nk \log 2 > 0$. Removing the partition would then increase the entropy by $2Nk \log 2$. But nothing has happened! The gas is the same on both sides; inserting and removing a partition is completely reversible, and cannot change the entropy at all.

The resolution, due to Gibbs, is that we have over-counted. Two configurations differing only by swapping particle 1 with particle 2 are not different configurations, because the particles are *identical*: there is no experiment that could distinguish them. When we integrated each particle independently over the box, we counted each genuine configuration $N!$ times, once for each way of labelling the particles. So we should divide by $N!$:

$$Z = \frac{Z_1^N}{N!},$$

which is what we assumed above and which gives the extensive Sackur–Tetrode entropy. Since $\log N! \approx N \log N - N$, the correction changes S by $-Nk(\log N - 1)$, converting $\log(V/\lambda^3)$ into $\log(V/N\lambda^3)$ and curing the paradox: with the correct formula, $S_{\text{before}} = S_{\text{after}}$ exactly.

Two remarks. First, if the two halves contain *different* gases, then removing the partition really does increase the entropy, by exactly $2Nk \log 2$; this is the **entropy of mixing**, and it is real, since unmixing two gases genuinely costs work. Secondly, and more strikingly: indistinguishability of identical particles is a quantum-mechanical fact. Classical particles could in principle be labelled and followed. So this is the second place where classical statistical mechanics needed to be rescued by quantum mechanics – and the $1/N!$ is only an approximation to the correct quantum counting, valid when the chance of two particles occupying the same state is negligible. That is precisely the regime $\lambda^3 \ll V/N$ identified earlier.

§42.2.6 Diatomic Gases

Real gases are usually not monatomic. A diatomic molecule can rotate and vibrate as well as translate, and each of these gives extra degrees of freedom. Since the different motions are (approximately) independent, the partition functions multiply:

$$Z_1 = Z_{\text{trans}} Z_{\text{rot}} Z_{\text{vib}}.$$

The rotational degrees of freedom are the two angles specifying the axis of the molecule. Quantum mechanically, a rigid rotor has energy levels

$$E_j = \frac{\hbar^2}{2I} j(j+1), \quad j = 0, 1, 2, \dots$$

with degeneracy $2j + 1$, where I is the moment of inertia. So

$$Z_{\text{rot}} = \sum_{j=0}^{\infty} (2j + 1) e^{-\beta \hbar^2 j(j+1)/2I}.$$

At high temperature, $kT \gg \hbar^2/2I$, the summand varies slowly and we may replace the sum by an integral; substituting $u = j(j + 1)$, so $du = (2j + 1) dj$,

$$Z_{\text{rot}} \approx \int_0^{\infty} (2j + 1) e^{-\beta \hbar^2 j(j+1)/2I} dj = \int_0^{\infty} e^{-\beta \hbar^2 u/2I} du = \frac{2I}{\beta \hbar^2}.$$

This gives $E_{\text{rot}} = -\partial_{\beta} \log Z_{\text{rot}} = kT$ per molecule, exactly as equipartition predicts for two quadratic degrees of freedom. At low temperature, $kT \ll \hbar^2/2I$, only the first two terms matter,

$$Z_{\text{rot}} \approx 1 + 3e^{-\beta \hbar^2/I},$$

giving $E_{\text{rot}} \approx (3\hbar^2/I)e^{-\beta \hbar^2/I}$, which is exponentially small: the rotational modes are frozen out, and contribute nothing to the heat capacity.

The vibrational mode is a harmonic oscillator of frequency ω , with energy levels $E_n = \hbar\omega(n + \frac{1}{2})$. This we can sum exactly, as a geometric series:

$$Z_{\text{vib}} = \sum_{n=0}^{\infty} e^{-\beta \hbar \omega (n+1/2)} = \frac{e^{-\beta \hbar \omega /2}}{1 - e^{-\beta \hbar \omega}} = \frac{1}{2 \sinh(\beta \hbar \omega /2)},$$

so the mean vibrational energy per molecule is

$$E_{\text{vib}} = -\frac{\partial}{\partial \beta} \log Z_{\text{vib}} = \frac{\hbar \omega}{2} \coth\left(\frac{\beta \hbar \omega}{2}\right) = \hbar \omega \left(\frac{1}{2} + \frac{1}{e^{\beta \hbar \omega} - 1}\right).$$

At high temperature this tends to kT , again as equipartition demands; at low temperature it tends to the zero-point energy $\frac{1}{2}\hbar\omega$, with exponentially small corrections. We will meet this formula again, twice, in the next section.

Putting it together, the heat capacity of a diatomic gas rises in steps as we heat it: $\frac{3}{2}Nk$ when only translations are active, $\frac{5}{2}Nk$ once rotations unfreeze, and $\frac{7}{2}Nk$ once vibrations do. For hydrogen the rotational scale is about 85 K and the vibrational scale about 6000 K, so at room temperature we see the middle plateau – and indeed the measured C_V of hydrogen is close to $\frac{5}{2}Nk$. The steps are unmistakable in the data, and are direct macroscopic evidence for the quantisation of energy levels.

§42.2.7 Interacting Gases and the Virial Expansion

Real gas molecules attract each other at long range and repel at short range, and we would like to know how this modifies the ideal gas law. The strategy is perturbative: for a dilute gas the number density N/V is small, and we expand in it.

Definition 42.2.10 (Virial expansion)

The **virial expansion** of an equation of state is

$$\frac{p}{kT} = \frac{N}{V} + B_2(T) \frac{N^2}{V^2} + B_3(T) \frac{N^3}{V^3} + \dots,$$

whose coefficients $B_k(T)$ are the **virial coefficients**.

The coefficients depend only on T , since p/kT and N/V are both intensive. Our goal is to compute B_2 .

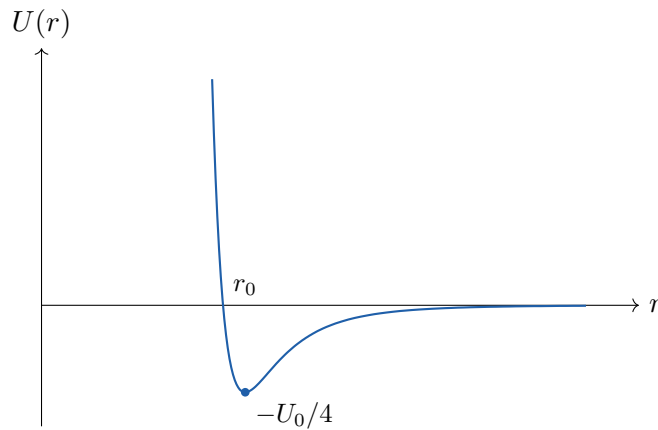
Suppose the particles interact through a pair potential $U(r)$ depending only on the separation. What does U look like? At long range, two neutral atoms attract: although each has zero average electric dipole moment, a fluctuation giving atom 1 a dipole p_1 produces a field $E \sim p_1/r^3$, which induces a dipole $p_2 \propto p_1/r^3$ in atom 2, and the resulting energy is

$$U \sim -p_2 E \sim -\frac{p_1^2}{r^6}.$$

This is the **van der Waals interaction**, and it is attractive. At short range the electron clouds overlap and the Pauli principle produces a strong repulsion. A convenient phenomenological form combining the two is the **Lennard-Jones potential**

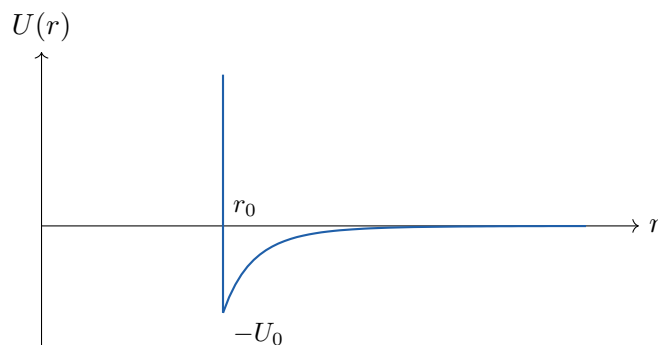
$$U(r) = U_0 \left[\left(\frac{r_0}{r} \right)^{12} - \left(\frac{r_0}{r} \right)^6 \right],$$

where the twelfth power is chosen mostly because it is the square of the sixth and hence easy to compute with.



For calculations it is easier to idealise this into a **hard-core potential**: infinitely repulsive inside r_0 , attractive outside,

$$U(r) = \begin{cases} \infty & r < r_0, \\ -U_0(r_0/r)^6 & r \geq r_0. \end{cases}$$



Now to the calculation. The Hamiltonian of the gas is

$$H = \sum_{i=1}^N \frac{\mathbf{p}_i^2}{2m} + \sum_{i<j} U(r_{ij}), \quad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|,$$

and the partition function no longer factorises. The momentum integrals still do, however, and they give the same λ^{-3N} as before:

$$Z(N, V, T) = \frac{1}{N! \lambda^{3N}} \int \prod_i d^3 r_i e^{-\beta \sum_{j<k} U(r_{jk})}.$$

We would now like to expand the exponential in powers of U , but this is hopeless: U is infinite at short range. The trick is to expand instead in the **Mayer f -function**

$$f(r) = e^{-\beta U(r)} - 1,$$

which is well behaved: it equals -1 inside the hard core and tends to 0 as $r \rightarrow \infty$, so it is supported on a shell of finite size. Writing $f_{jk} = f(r_{jk})$,

$$e^{-\beta \sum_{j<k} U_{jk}} = \prod_{j<k} (1 + f_{jk}) = 1 + \sum_{j<k} f_{jk} + \sum_{j<k} \sum_{\ell<m} f_{jk} f_{\ell m} + \cdots.$$

The first term integrates to V^N , giving the ideal gas. Each term in the second sum gives, say for $j = 1, k = 2$,

$$\int \prod_i d^3 r_i f_{12} = V^{N-2} \int d^3 r_1 d^3 r_2 f(r_{12}) = V^{N-1} I, \quad I = \int d^3 r f(r),$$

where we changed to relative and centre-of-mass coordinates and extended the r integral over all space (legitimate since f decays). There are $\binom{N}{2} \approx N^2/2$ such terms, so

$$Z = \frac{V^N}{N! \lambda^{3N}} \left(1 + \frac{N^2 I}{2V} + \cdots \right) = Z_{\text{ideal}} \left(1 + \frac{NI}{2V} + \cdots \right)^N,$$

where in the last step we pulled an N into the exponent, which is legitimate to this order and has the advantage of making extensivity manifest. Then

$$F = -kT \log Z = F_{\text{ideal}} - NkT \log \left(1 + \frac{NI}{2V} + \cdots \right) \approx F_{\text{ideal}} - \frac{N^2 kT}{2V} I,$$

and differentiating,

$$p = - \left(\frac{\partial F}{\partial V} \right)_T = \frac{NkT}{V} \left(1 - \frac{NI}{2V} + \cdots \right).$$

Proposition 42.2.11

The second virial coefficient is

$$B_2(T) = -\frac{1}{2} I = -\frac{1}{2} \int d^3 r (e^{-\beta U(r)} - 1).$$

The expansion parameter is NI/V , and $I \sim r_0^3$, so we need $N/V \ll r_0^{-3}$: the gas must be much less dense than a liquid, which is exactly the regime in which we expect a dilute-gas expansion to work. Note also the sign: a purely repulsive potential has $U > 0$, hence $f < 0$, hence $B_2 > 0$, and the pressure is increased. A purely attractive potential decreases it. Both are what one would guess.

§42.2.8 The van der Waals Equation

Let us evaluate I for the hard-core potential. Splitting the integral at r_0 ,

$$I = \int_{r < r_0} d^3r (-1) + \int_{r > r_0} d^3r \left(e^{\beta U_0(r_0/r)^6} - 1 \right).$$

The first piece is just $-\frac{4}{3}\pi r_0^3$. For the second we assume high temperature, $\beta U_0 \ll 1$, and expand the exponential to first order:

$$\int_{r_0}^{\infty} 4\pi r^2 dr \beta U_0 \frac{r_0^6}{r^6} = 4\pi \beta U_0 r_0^6 \int_{r_0}^{\infty} \frac{dr}{r^4} = \frac{4\pi r_0^3 U_0}{3kT}.$$

Hence

$$I = \frac{4\pi r_0^3}{3} \left(\frac{U_0}{kT} - 1 \right),$$

and substituting into the virial expansion,

$$\frac{pV}{NkT} = 1 - \frac{N}{V} \left(\frac{a}{kT} - b \right), \quad a = \frac{2\pi r_0^3 U_0}{3}, \quad b = \frac{2\pi r_0^3}{3}.$$

This is already a decent equation of state, but it is traditional (and, as we will see, far more useful) to rearrange it. Inverting,

$$kT = \frac{V}{N} \left(p + \frac{aN^2}{V^2} \right) \left(1 + \frac{bN}{V} \right)^{-1} \approx \left(p + \frac{aN^2}{V^2} \right) \left(\frac{V}{N} - b \right),$$

where we expanded the inverse to first order and discarded higher terms, which are beyond the accuracy we have anyway.

Definition 42.2.12 (van der Waals equation)

The **van der Waals equation of state** is

$$\left(p + \frac{aN^2}{V^2} \right) \left(\frac{V}{N} - b \right) = kT, \quad \text{equivalently} \quad p = \frac{NkT}{V - bN} - \frac{aN^2}{V^2}.$$

The second form makes the interpretation clear. The b term says that the volume available to the gas is reduced, because the molecules take up room; the a term says the pressure is reduced, because the long-range attraction pulls molecules back from the walls.

The factor of bN deserves a comment, since it is not what one would first guess. Two hard spheres of core radius r_0 cannot approach closer than r_0 , so each atom excludes a volume

$$\Omega_{\text{ex}} = \frac{4}{3}\pi r_0^3 = 2b$$

around itself – twice the value of b . Why the factor of two? Because the exclusion is shared between pairs. Placing the atoms in one at a time, the first has volume V available, the second $V - \Omega_{\text{ex}}$, the third $V - 2\Omega_{\text{ex}}$, and so on, so the accessible configuration volume is

$$\frac{1}{N!} \prod_{j=0}^{N-1} (V - j\Omega_{\text{ex}}) \approx \frac{V^N}{N!} \left(1 - \frac{N\Omega_{\text{ex}}}{V} \right) \approx \frac{1}{N!} \left(V - \frac{N\Omega_{\text{ex}}}{2} \right)^N,$$

so the effective excluded volume *per particle* is $\Omega_{\text{ex}}/2 = b$.

We derived the van der Waals equation in a regime where it is a small correction to the ideal gas law. Its real interest, though, is what happens if we take it seriously outside that regime – where, as we will see in Section 42.5.1, it produces a liquid–gas phase transition. That is an entirely illegitimate extrapolation, and it works remarkably well.

Higher virial coefficients can be computed, but not by naively keeping the $\sum \sum f_{jk} f_{\ell m}$ terms; one has to be careful about which diagrams contribute at which order in N/V . The systematic method is the **cluster expansion**, which organises the terms into connected diagrams in a manner very reminiscent of Feynman diagrams. We will not pursue it.

§42.3 Quantum Gases

We saw that the classical treatment of a gas should break down when the thermal wavelength λ becomes comparable with the interparticle spacing, that is, when the gas is cold or dense. In this section we do the calculation properly. This is where statistical physics earns its keep: the same formalism will give us the spectrum of light from a hot object, the heat capacity of a crystal, the pressure that holds up a white dwarf star, and a phase transition with no interactions in sight.

The technical obstacle is that the partition function is a sum over states, and quantum states are discrete. In practice the levels are very finely spaced, so we would like to replace the sum by an integral – but we must keep track of how many states there are per unit energy. That is the density of states.

§42.3.1 Density of States

Consider a single particle in a cubical box of side L , so $V = L^3$. The energy eigenstates are the standing waves

$$\psi(\mathbf{r}) \propto \sin\left(\frac{n_1\pi x}{L}\right) \sin\left(\frac{n_2\pi y}{L}\right) \sin\left(\frac{n_3\pi z}{L}\right), \quad n_i \in \mathbb{N},$$

though for counting purposes it is more convenient to use periodic boundary conditions, which give plane waves $e^{i\mathbf{k}\cdot\mathbf{r}}$ with

$$\mathbf{k} = \frac{2\pi}{L}(n_1, n_2, n_3), \quad n_i \in \mathbb{Z}.$$

Either way the answer is the same: the allowed wavevectors form a lattice in $\mathbf{k}$ -space with one state per volume $(2\pi/L)^3 = (2\pi)^3/V$. So for a large box the number of states with wavevector in d^3k is

$$\frac{V}{(2\pi)^3} d^3k,$$

and equivalently, since $\mathbf{p} = \hbar\mathbf{k}$, one state per phase-space volume $(2\pi\hbar)^3$. This is exactly the normalisation we were forced to guess in Section 42.2.1, and now we have derived it.

Now specialise to a non-relativistic particle, $E = \hbar^2 k^2/2m$. Counting states with energy less than E means counting lattice points inside a sphere of radius $k = \sqrt{2mE}/\hbar$:

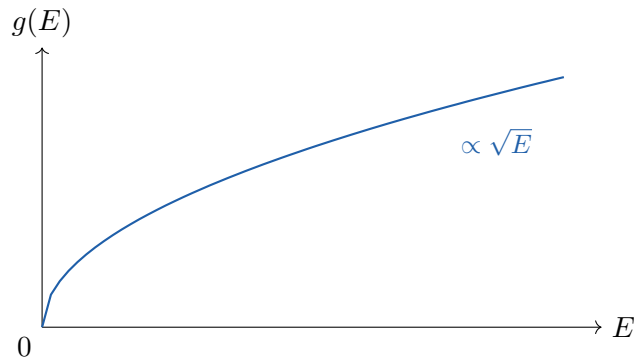
$$g(E) dE = \frac{V}{(2\pi)^3} 4\pi k^2 dk = \frac{V}{2\pi^2} k^2 \frac{dk}{dE} dE.$$

Since $k = \sqrt{2mE}/\hbar$ we have $dk/dE = \sqrt{2m}/(2\hbar\sqrt{E})$, and so we obtain the following.

Definition 42.3.1 (Density of states)

The **density of states** $g(E)$ is defined so that $g(E) dE$ is the number of single-particle states with energy in $[E, E + dE]$. For a non-relativistic particle of mass m in a box of volume V in three dimensions,

$$g(E) = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}.$$



The $\sqrt{E}$ is worth remembering, as is the fact that it depends on the dispersion relation and the dimension. If the particles have spin, or some other internal degree of freedom with g_s states, we simply multiply $g(E)$ by g_s . And for massless particles such as photons, where $E = \hbar\omega = \hbar ck$, the same counting gives

$$g(E) dE = \frac{V}{2\pi^2} k^2 dk = \frac{VE^2}{2\pi^2 \hbar^3 c^3} dE,$$

so $g(E) \propto E^2$ instead.

As a first check, let us recompute the partition function of a classical ideal gas:

$$Z_1 = \int_0^\infty dE g(E) e^{-\beta E} = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty \sqrt{E} e^{-\beta E} dE = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \frac{\sqrt{\pi}}{2\beta^{3/2}},$$

which tidies up to

$$Z_1 = \frac{V}{\lambda^3},$$

agreeing with our earlier result.

§42.3.2 Blackbody Radiation

Our first application is to light. Consider an empty box whose walls are held at temperature T . The walls constantly emit and absorb photons, so the box fills with electromagnetic radiation in thermal equilibrium; this is **blackbody radiation**, and the question is what its spectrum looks like. The problem is a genuinely historic one – getting it wrong is what started quantum mechanics.

The electromagnetic field in a box is equivalent to a collection of harmonic oscillators, one for each mode ($\mathbf{k}$, polarisation), with frequency $\omega = ck$. Quantum mechanically each oscillator has energy levels $E_n = \hbar\omega(n + \frac{1}{2})$, and we interpret n as the number of photons in that mode. Crucially, photon number is not conserved – the walls create and destroy photons freely – so there is no constraint on n , and we may compute the

partition function of each mode separately. Discarding the zero-point energy (which is an infinite constant, and does not affect anything we measure here),

$$Z_{\mathbf{k}} = \sum_{n=0}^{\infty} e^{-\beta n \hbar \omega} = \frac{1}{1 - e^{-\beta \hbar \omega}}.$$

The mean number of photons in the mode is then

$$\langle n \rangle = \frac{1}{Z_{\mathbf{k}}} \sum_{n \geq 0} n e^{-\beta n \hbar \omega} = -\frac{1}{\hbar} \frac{\partial \log Z_{\mathbf{k}}}{\partial \beta} \cdot \frac{1}{\omega},$$

which evaluates to the following.

Definition 42.3.2 (Planck distribution)

The mean occupation number of a photon mode of frequency ω at temperature T is

$$\langle n(\omega) \rangle = \frac{1}{e^{\beta \hbar \omega} - 1}.$$

Now we sum over modes. There are two polarisations for each wavevector, so the number of modes with frequency in $[\omega, \omega + d\omega]$ is

$$2 \cdot \frac{V}{(2\pi)^3} 4\pi k^2 dk = \frac{V \omega^2}{\pi^2 c^3} d\omega,$$

using $\omega = ck$. Multiplying by $\hbar \omega \langle n(\omega) \rangle$ gives the energy.

Proposition 42.3.3 (Planck's law)

The energy density of blackbody radiation per unit frequency is

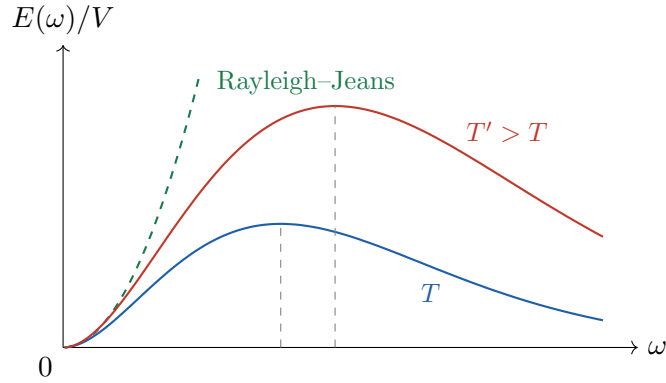
$$\frac{E(\omega)}{V} = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta \hbar \omega} - 1}.$$

Let us look at the two limits. For $\hbar \omega \ll kT$ we may expand $e^{\beta \hbar \omega} - 1 \approx \beta \hbar \omega$, giving

$$\frac{E(\omega)}{V} \approx \frac{\omega^2 kT}{\pi^2 c^3},$$

which is the **Rayleigh–Jeans law**. This is exactly what classical equipartition predicts: each mode is an oscillator with two quadratic degrees of freedom, hence energy kT , and there are $\omega^2 d\omega V / \pi^2 c^3$ of them. It agrees beautifully with experiment at low frequencies, and it is a catastrophe at high ones – the total energy $\int_0^\infty \omega^2 d\omega$ diverges. This is the **ultraviolet catastrophe**. Physically, the trouble is that there are infinitely many modes and classically every one of them must carry kT .

Quantum mechanics saves us. For $\hbar \omega \gg kT$ the Planck distribution is exponentially suppressed, $\langle n \rangle \approx e^{-\beta \hbar \omega}$: a mode whose quantum of energy exceeds kT simply cannot be excited. The two behaviours are stitched together at $\hbar \omega \sim kT$, which is where the spectrum peaks.



Two classical results drop out immediately. Differentiating Planck's law shows that the peak sits at $\hbar\omega_{\max} \approx 2.82 kT$, so

$$\omega_{\max} \propto T,$$

which is **Wien's displacement law** – the reason a heated poker glows first red and then white. And integrating over all frequencies, with $x = \beta\hbar\omega$,

$$\frac{E}{V} = \frac{\hbar}{\pi^2 c^3} \int_0^\infty \frac{\omega^3 d\omega}{e^{\beta\hbar\omega} - 1} = \frac{(kT)^4}{\pi^2 c^3 \hbar^3} \int_0^\infty \frac{x^3 dx}{e^x - 1} = \frac{\pi^2 k^4}{15 c^3 \hbar^3} T^4,$$

where we used $\int_0^\infty x^3/(e^x - 1) dx = \pi^4/15$. This T^4 dependence is the **Stefan–Boltzmann law**. (To evaluate the integral, expand $1/(e^x - 1) = \sum_{n \geq 1} e^{-nx}$ and integrate term by term to get $\Gamma(4)\zeta(4) = 6 \cdot \pi^4/90$.)

It is worth noting the whole thermodynamics of the photon gas. From

$$\log \mathcal{Z} = - \sum_{\mathbf{k}, \text{pol}} \log(1 - e^{-\beta\hbar\omega}) = - \frac{V}{\pi^2 c^3} \int_0^\infty \omega^2 \log(1 - e^{-\beta\hbar\omega}) d\omega,$$

an integration by parts gives $\log \mathcal{Z} = \beta E/3$, and hence, since $F = -kT \log \mathcal{Z}$ and $p = -(\partial F/\partial V)_T$,

$$pV = \frac{1}{3} E.$$

This is the equation of state of *radiation*, and the factor of $\frac{1}{3}$ (as opposed to $\frac{2}{3}$ for a non-relativistic gas, from $pV = NkT$ and $E = \frac{3}{2}NkT$) is a general feature of ultrarelativistic gases. The entropy is $S = \frac{4}{3}(E/T) \propto VT^3$.

§42.3.3 Phonons and the Debye Model

We turn to the heat capacity of a solid. Classically, by equipartition, a crystal of N atoms is $3N$ oscillators and should have $C = 3Nk$ at all temperatures; experimentally $C \rightarrow 0$ as $T \rightarrow 0$, and in insulators it does so like T^3 . The resolution is the same as for blackbody radiation, and the two calculations are almost identical.

The first attempt is due to Einstein: treat the solid as $3N$ independent oscillators, all of the same frequency ω . Then, from our earlier computation of the vibrational partition function,

$$E = 3N\hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right), \quad C = 3Nk (\beta\hbar\omega)^2 \frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}.$$

At high temperature this gives $C \rightarrow 3Nk$, recovering Dulong–Petit. At low temperature it gives $C \sim 3Nk(\beta\hbar\omega)^2 e^{-\beta\hbar\omega}$, which does vanish – but exponentially, whereas experiment says T^3 . Einstein’s model has the right idea and the wrong details.

The flaw is the assumption that the oscillators are independent and identical. In reality the atoms are coupled, and the normal modes of the crystal are sound waves, with a range of frequencies. The quanta of these waves are called **phonons**, and they are treated exactly like photons: a mode of frequency ω has mean occupation $(e^{\beta\hbar\omega} - 1)^{-1}$, and the number of phonons is not conserved.

Definition 42.3.4 (Debye model)

In the **Debye model** we assume a linear dispersion relation $\omega = c_s k$ for the sound waves, with c_s the speed of sound, and we cut the spectrum off at a maximum frequency ω_D chosen so that the total number of modes is $3N$.

The cutoff is not a fudge: a crystal of N atoms has exactly $3N$ degrees of freedom, so there cannot be more than $3N$ modes. There are three polarisations (two transverse, one longitudinal), so

$$3N = 3 \cdot \frac{V}{(2\pi)^3} \int_{k < k_D} d^3k = \frac{V k_D^3}{2\pi^2} \implies \omega_D = c_s \left(\frac{6\pi^2 N}{V} \right)^{1/3}.$$

The energy is then, exactly as for photons but with three polarisations and a cutoff,

$$E = \frac{3V}{2\pi^2 c_s^3} \int_0^{\omega_D} d\omega \frac{\hbar\omega^3}{e^{\beta\hbar\omega} - 1},$$

dropping the zero-point contribution. Substituting $x = \beta\hbar\omega$ and defining the **Debye temperature** T_D by $kT_D = \hbar\omega_D$,

$$E = \frac{3V(kT)^4}{2\pi^2 c_s^3 \hbar^3} \int_0^{T_D/T} \frac{x^3 dx}{e^x - 1} = 9NkT \left(\frac{T}{T_D} \right)^3 \int_0^{T_D/T} \frac{x^3 dx}{e^x - 1}.$$

Now take the two limits.

For $T \gg T_D$ the upper limit is small, so $e^x - 1 \approx x$ and the integral is $\frac{1}{3}(T_D/T)^3$, giving $E = 3NkT$ and $C = 3Nk$: Dulong–Petit, as it must be.

For $T \ll T_D$ the upper limit is effectively infinite, and the integral becomes the constant $\pi^4/15$. Hence

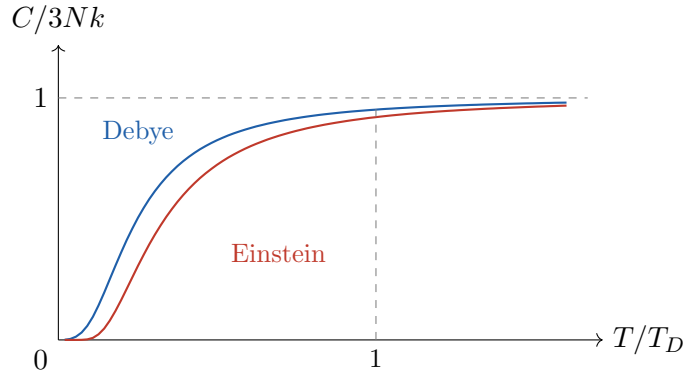
$$E \approx \frac{3\pi^4}{5} NkT \left(\frac{T}{T_D} \right)^3, \quad C \approx \frac{12\pi^4}{5} Nk \left(\frac{T}{T_D} \right)^3.$$

Proposition 42.3.5 (Debye T^3 law)

At low temperatures the phonon contribution to the heat capacity of a solid is

$$C = \frac{12\pi^4}{5} Nk \left(\frac{T}{T_D} \right)^3 \propto T^3.$$

This agrees with experiment. Here are the two models plotted together, in units where $C \rightarrow 3Nk$ at high temperature.



Both curves saturate at the classical value, and both vanish at $T = 0$; the difference is in *how* they vanish, and it is the low-temperature behaviour that distinguishes them. Physically, the Debye model works because there are always arbitrarily low-frequency sound waves available – long-wavelength modes – so however cold the solid, some modes can still be excited. The Einstein model has an energy gap, and so it is exponentially frozen.

§42.3.4 The Quantum Ideal Gas

We now return to gases of massive particles, and do the counting properly. The key fact is quantum statistics: identical particles come in two kinds.

- **Bosons**, which have integer spin, and for which any number of particles may occupy the same single-particle state.
- **Fermions**, which have half-integer spin, and for which the Pauli exclusion principle allows at most one particle per state.

Consider an ideal gas whose single-particle states are labelled by r with energies E_r . A microstate of the whole gas is specified by the **occupation numbers** $\{n_r\}$, with total energy $\sum_r n_r E_r$ and total particle number $\sum_r n_r$. In the canonical ensemble, computing $Z = \sum_{\{n_r\}} e^{-\beta \sum_r n_r E_r}$ subject to $\sum_r n_r = N$ is horrible, because of the constraint. In the grand canonical ensemble the constraint disappears, and everything factorises:

$$\mathcal{Z} = \sum_{\{n_r\}} e^{-\beta \sum_r n_r (E_r - \mu)} = \prod_r \left(\sum_{n_r} e^{-\beta n_r (E_r - \mu)} \right) = \prod_r \mathcal{Z}_r.$$

This is exactly why we built the grand canonical ensemble, and the calculation is now easy.

For bosons the inner sum runs over all $n_r \geq 0$ and is geometric:

$$\mathcal{Z}_r = \sum_{n=0}^{\infty} e^{-\beta n (E_r - \mu)} = \frac{1}{1 - e^{-\beta (E_r - \mu)}},$$

which converges only if $\mu < E_r$ for every r ; in particular μ must lie below the ground state energy, which we take to be 0, so $\mu < 0$ for bosons. For fermions the sum has just two terms:

$$\mathcal{Z}_r = 1 + e^{-\beta (E_r - \mu)}.$$

The mean occupation of a state is $\langle n_r \rangle = -\beta^{-1} \partial \log \mathcal{Z}_r / \partial E_r$, and we obtain the two central distributions of the subject.

Proposition 42.3.6 (Bose–Einstein and Fermi–Dirac distributions)

The mean occupation number of a single-particle state of energy E in an ideal quantum gas at temperature T and chemical potential μ is

$$\langle n(E) \rangle = \frac{1}{e^{\beta(E-\mu)} \mp 1},$$

with the upper sign for bosons (**Bose–Einstein statistics**) and the lower for fermions (**Fermi–Dirac statistics**).

Some sanity checks. For fermions, $\langle n \rangle$ lies between 0 and 1, as the Pauli principle demands. For bosons, $\langle n \rangle$ diverges as $E \rightarrow \mu$, which is the seed of Bose–Einstein condensation. And when $e^{\beta(E-\mu)} \gg 1$ – that is, when occupation numbers are small – both reduce to

$$\langle n(E) \rangle \approx e^{-\beta(E-\mu)},$$

the classical Boltzmann distribution. So classical statistics is the dilute limit of both, as it should be. Setting $\mu = 0$ in the boson formula recovers the Planck distribution: photons are bosons whose number is not conserved, and a non-conserved particle number means exactly that there is no constraint to impose and hence $\mu = 0$.

Everything else follows. Converting sums to integrals with the density of states,

$$N = \int_0^\infty dE \frac{g(E)}{e^{\beta(E-\mu)} \mp 1}, \quad E_{\text{tot}} = \int_0^\infty dE \frac{E g(E)}{e^{\beta(E-\mu)} \mp 1},$$

and the grand potential is

$$\Phi = -kT \log \mathcal{Z} = \pm kT \int_0^\infty dE g(E) \log(1 \mp e^{-\beta(E-\mu)}).$$

Using $g(E) \propto E^{1/2}$ and integrating by parts gives a relation that holds for both statistics and will be useful repeatedly:

$$pV = -\Phi = \frac{2}{3} E_{\text{tot}}.$$

Before specialising, it is worth recording the classical limit as a criterion. In the dilute regime $N = \int g(E) e^{-\beta(E-\mu)} dE = e^{\beta\mu} V / \lambda^3$, so

$$e^{\beta\mu} = \frac{N \lambda^3}{V}.$$

The classical approximation required $e^{\beta\mu} \ll 1$, i.e. $\lambda^3 \ll V/N$, which is exactly the condition we anticipated. Quantum statistics matters when $\lambda^3 N/V \gtrsim 1$.

§42.3.5 Bosons and Bose–Einstein Condensation

Take a gas of N non-relativistic bosons of mass m and spin zero in a box of volume V , and fix N by choosing μ appropriately:

$$\frac{N}{V} = \frac{1}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int_0^\infty dE \frac{\sqrt{E}}{z^{-1} e^{\beta E} - 1}, \quad z = e^{\beta\mu}.$$

Substituting $x = \beta E$ and tidying up, this becomes

$$\frac{N\lambda^3}{V} = g_{3/2}(z), \quad \text{where} \quad g_{3/2}(z) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{x^{1/2} dx}{z^{-1}e^x - 1}.$$

The function $g_{3/2}$ has a nice series expansion: writing $\frac{1}{z^{-1}e^x - 1} = \sum_{n \geq 1} z^n e^{-nx}$ and integrating term by term,

$$g_{3/2}(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^{3/2}}.$$

This is manifestly increasing in z on $0 \leq z \leq 1$, and recall that $\mu < 0$, so $z \leq 1$. Hence $g_{3/2}$ attains its *maximum* at $z = 1$:

$$g_{3/2}(1) = \zeta(3/2) \approx 2.612.$$

And now we have a problem. The left-hand side $N\lambda^3/V$ grows without bound as we cool the gas, since $\lambda \propto T^{-1/2}$. The right-hand side is bounded by 2.612. So below a certain temperature the equation has no solution at all.

Definition 42.3.7 (Critical temperature)

The **critical temperature** for Bose–Einstein condensation is the temperature T_c at which $N\lambda^3/V = \zeta(3/2)$, namely

$$kT_c = \frac{2\pi\hbar^2}{m} \left(\frac{N}{\zeta(3/2)V} \right)^{2/3}.$$

Something has gone wrong with our approximations, and it is easy to see what. In replacing the sum over states by an integral, we weighted the ground state $E = 0$ by $g(0) = 0$ – we threw it away. Usually this is harmless, since a single state out of 10^{23} cannot matter. But as $z \rightarrow 1$ the ground state occupation

$$n_0 = \frac{1}{z^{-1} - 1} = \frac{z}{1 - z}$$

diverges, and it can be macroscopically large. The resolution is to put the ground state back by hand:

$$N = n_0 + \frac{V}{\lambda^3} g_{3/2}(z),$$

where the integral now accounts for the excited states only. For $T > T_c$ we can solve for $z < 1$ with n_0 negligible. For $T < T_c$ the excited states can only hold

$$N_{\text{ex}} = \frac{V}{\lambda^3} \zeta(3/2) = N \left(\frac{T}{T_c} \right)^{3/2}$$

particles, using $\lambda \propto T^{-1/2}$ and the definition of T_c ; and the rest have nowhere to go but the ground state.

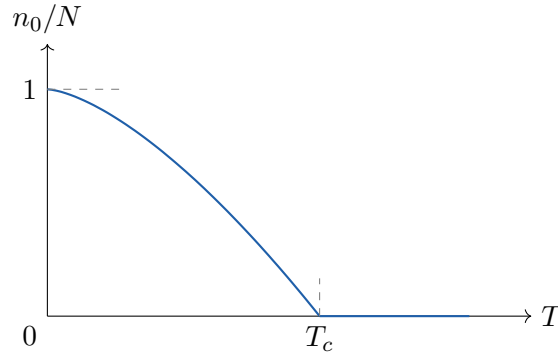
Proposition 42.3.8 (Bose–Einstein condensation)

For $T < T_c$, a macroscopic fraction of the particles occupies the single-particle ground state:

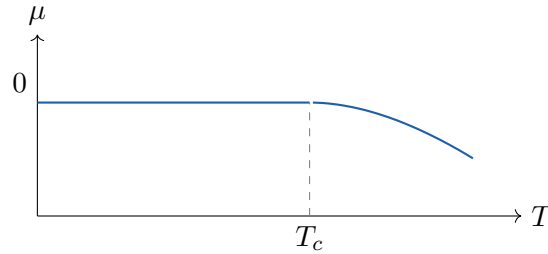
$$\frac{n_0}{N} = 1 - \left(\frac{T}{T_c} \right)^{3/2}.$$

This macroscopically occupied ground state is called the **Bose–Einstein conden-**

sate.



Note the shape. Just below the transition, writing $T = T_c(1 - \delta)$ and expanding gives $n_0/N \approx \frac{3}{2}\delta$, so the condensate fraction grows linearly out of T_c with slope $\frac{3}{2}$; and it reaches 1 at $T = 0$, where every particle is in the ground state. The chemical potential, meanwhile, is negative above T_c and pinned to zero below it:



What is remarkable about this is that it is a genuine phase transition – a discontinuity in the derivative of a thermodynamic function – occurring in a gas of *non-interacting* particles. It is driven purely by quantum statistics. Physically, the condition $N\lambda^3/V \sim 1$ says the transition happens when the thermal wavelengths of the particles begin to overlap, at which point they can no longer be thought of as separate objects.

The thermodynamics below T_c is easy, because the condensate contributes nothing to the energy (its particles have $E = 0$) or the pressure. Setting $z = 1$,

$$E = \frac{V}{\lambda^3} kT \cdot \frac{3}{2} g_{5/2}(1), \quad g_{5/2}(1) = \zeta(5/2) \approx 1.342,$$

so, using $\lambda \propto T^{-1/2}$, we get $E \propto VT^{5/2}$ and hence

$$C_V = \left(\frac{\partial E}{\partial T} \right)_V \propto VT^{3/2}.$$

So the heat capacity vanishes as $T^{3/2}$, in contrast to the classical constant $\frac{3}{2}Nk$. Moreover $p = \frac{2}{3}E/V \propto T^{5/2}$ is independent of V – if we compress a condensate at fixed temperature, the pressure does not rise, because the extra particles simply join the condensate. This is precisely analogous to the behaviour of a vapour in equilibrium with its liquid, and it is one of several respects in which the condensate behaves like a separate phase.

Example 42.3.9 (Superfluid helium)

Liquid ^4He , whose atoms are bosons, undergoes a transition at 2.17 K below which it becomes a **superfluid**: it flows without viscosity, creeps up the walls of its container,

and conducts heat extraordinarily well. Putting the density of liquid helium into our formula gives $T_c \approx 3.1$ K, which is the right order of magnitude – encouraging, given that liquid helium is a strongly interacting system and our calculation assumed no interactions at all. By contrast ^3He , whose atoms are fermions, shows no such transition at these temperatures (it does become superfluid, but at 2.5 mK, and by a quite different mechanism involving pairs of atoms).

§42.3.6 Fermions and the Degenerate Fermi Gas

Now the other case. For fermions the occupation function is

$$\langle n(E) \rangle = \frac{1}{e^{\beta(E-\mu)} + 1},$$

and there is no convergence constraint on μ , which may be positive or negative. The behaviour at low temperature is completely different from the bosonic case, and is dominated by the exclusion principle.

Consider first $T = 0$ exactly. Then $\beta \rightarrow \infty$, and

$$\langle n(E) \rangle \rightarrow \begin{cases} 1 & E < \mu, \\ 0 & E > \mu. \end{cases}$$

So the particles fill up the energy levels from the bottom, one per state, until they run out. This is exactly what one would expect from the Pauli principle.

Definition 42.3.10 (Fermi energy)

The **Fermi energy** E_F is the chemical potential at zero temperature, $E_F = \mu(T = 0)$; equivalently, it is the energy of the highest occupied state at $T = 0$. The corresponding momentum $k_F = \sqrt{2mE_F}/\hbar$ is the **Fermi momentum**, and the sphere $|\mathbf{k}| = k_F$ in momentum space is the **Fermi surface**.

We can compute E_F by counting. For spin- $\frac{1}{2}$ fermions there are two states per wavevector, so

$$N = 2 \cdot \frac{V}{(2\pi)^3} \cdot \frac{4}{3}\pi k_F^3 = \frac{V k_F^3}{3\pi^2} \implies E_F = \frac{\hbar^2}{2m} \left(\frac{3\pi^2 N}{V} \right)^{2/3}.$$

The total energy at $T = 0$ is

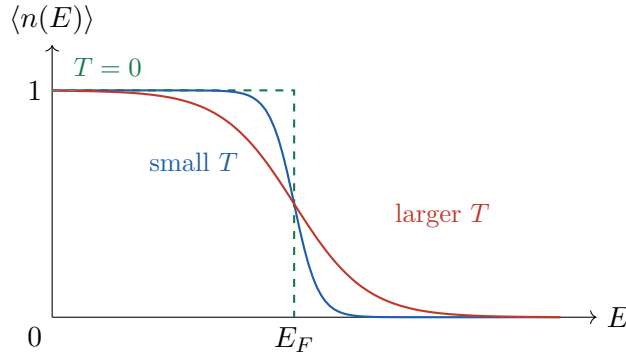
$$E_{\text{tot}} = \int_0^{E_F} E g(E) dE = \frac{3}{5} N E_F,$$

using $g(E) \propto \sqrt{E}$, and hence, by the relation $pV = \frac{2}{3} E_{\text{tot}}$,

$$p = \frac{2N E_F}{5V} \propto \left(\frac{N}{V} \right)^{5/3}.$$

This is **degeneracy pressure**, and it is a purely quantum effect: a Fermi gas at absolute zero exerts an enormous pressure, not because the particles are hot, but because they cannot all sit in the ground state.

At small but non-zero temperature, the step is smoothed over a width of order kT around μ :



This picture is the key to everything. Only the fermions within about kT of the Fermi surface can be excited at all; those deep inside cannot move, because all the nearby states are occupied. So of the N particles, only a fraction $\sim T/T_F$ participate in thermal physics, where $kT_F = E_F$ defines the **Fermi temperature**. Each of those carries $\sim kT$ of thermal energy, so

$$E_{\text{tot}} \approx \frac{3}{5}NE_F + N\frac{T}{T_F}kT \implies C_V \sim Nk\frac{T}{T_F}.$$

A more careful calculation, using the **Sommerfeld expansion** (which exploits the fact that $-\partial \langle n \rangle / \partial E$ is sharply peaked at μ), gives the exact coefficient:

$$C_V = \frac{\pi^2}{2}Nk\frac{T}{T_F} + O(T^3).$$

The essential point is that $C_V \propto T$, linearly, and is suppressed by a factor T/T_F relative to the classical value $\frac{3}{2}Nk$.

Example 42.3.11 (Electrons in metals)

The conduction electrons in a metal form a Fermi gas of density $N/V \sim 10^{29} \text{ m}^{-3}$, giving E_F of a few electronvolts and hence $T_F \sim 10^4\text{--}10^5 \text{ K}$. Room temperature is therefore *extremely* cold for the electrons: $T/T_F \sim 10^{-2}$, and the gas is highly degenerate.

This resolves an old puzzle. In the nineteenth century it was known that metals conduct heat and electricity because they contain free electrons; but classically those electrons should contribute $\frac{3}{2}Nk$ to the heat capacity, and no such contribution was observed. The answer is that the electronic heat capacity is suppressed by $T/T_F \sim 10^{-2}$, and is thus swamped by the phonons at ordinary temperatures. At very low temperature, though, the phonon contribution dies as T^3 while the electronic one dies only as T , so the electrons win. Indeed for a metal at low temperature

$$C \approx \gamma T + \alpha T^3,$$

and plotting C/T against T^2 gives a straight line – which is exactly what is measured.

Example 42.3.12 (White dwarf stars)

A star supports itself against gravity by thermal pressure, generated by nuclear fusion. When the fuel runs out, the star collapses – and for a star of modest mass the collapse is halted by the degeneracy pressure of its electrons. The result is a

white dwarf: an object of roughly the mass of the Sun compressed to the size of the Earth.

We can estimate the equilibrium radius by balancing energies. The gravitational energy of a ball of mass M and radius R is $E_{\text{grav}} \sim -GM^2/R$. The star contains $N \sim M/m_p$ nucleons and a comparable number of electrons, so the electron density is $n \sim N/R^3$ and the degeneracy energy is

$$E_{\text{deg}} \sim NE_F \sim N \frac{\hbar^2}{m_e} \left(\frac{N}{R^3} \right)^{2/3} = \frac{\hbar^2 N^{5/3}}{m_e R^2}.$$

Minimising $E_{\text{grav}} + E_{\text{deg}}$ over R gives

$$R \sim \frac{\hbar^2}{Gm_e m_p^{5/3} M^{1/3}},$$

so, curiously, heavier white dwarfs are *smaller*. Putting in the numbers for $M = M_\odot$ gives R of a few thousand kilometres, which is right.

The relation $R \propto M^{-1/3}$ cannot continue indefinitely. As M increases, R falls, the density rises, and eventually the electrons become relativistic. For an ultrarelativistic gas $E = \hbar ck$, and repeating the computation gives $E_{\text{deg}} \sim \hbar c N^{4/3}/R$ – which scales as $1/R$, exactly like the gravitational energy. The two therefore compete on equal terms, and if M exceeds a critical value gravity wins and there is no equilibrium at all. This is the **Chandrasekhar limit**, about $1.4M_\odot$; heavier remnants collapse to neutron stars or black holes.

§42.3.7 Pauli Paramagnetism

As a final application, consider what happens when we put a Fermi gas in a magnetic field B . Each electron has a magnetic moment, so the energy of an electron of momentum $\mathbf{k}$ is

$$E_{\pm} = \frac{\hbar^2 k^2}{2m} \pm \mu_B B,$$

where $\mu_B = e\hbar/2m_e$ is the Bohr magneton and the sign depends on whether the spin is aligned or anti-aligned. Classically, one would expect the moments to align with the field, giving a susceptibility that diverges as $T \rightarrow 0$ (Curie's law $\chi \propto 1/T$). What actually happens is quite different.

The two spin species have shifted densities of states, $g_{\pm}(E) = \frac{1}{2}g(E \mp \mu_B B)$. The magnetisation is

$$M = \mu_B(N_- - N_+) = \mu_B \int_0^\infty dE \frac{g_-(E) - g_+(E)}{e^{\beta(E-\mu)} + 1}.$$

For small B we expand $g(E \mp \mu_B B) \approx g(E) \mp \mu_B B g'(E)$, so

$$M \approx \mu_B^2 B \int_0^\infty dE \frac{g'(E)}{e^{\beta(E-\mu)} + 1} \approx \mu_B^2 B g(E_F)$$

at low temperature, where the last step integrates by parts and uses the fact that $-\partial \langle n \rangle / \partial E$ is a sharp spike at E_F . So the susceptibility is

$$\chi = \frac{\partial M}{\partial B} \approx \mu_B^2 g(E_F),$$

which is a *constant*, independent of temperature. This is **Pauli paramagnetism**, and the physical reason is by now familiar: only the electrons within kT of the Fermi surface can flip their spins, since the others have nowhere to flip to.

§42.4 Classical Thermodynamics

We now change our point of view entirely. Everything so far has been built up from microscopic foundations: we postulated things about microstates and derived macroscopic consequences. Classical thermodynamics does the opposite. It was developed in the nineteenth century, largely by people trying to build better steam engines, at a time when the atomic hypothesis was still controversial. It makes no assumptions whatsoever about what matter is made of. Instead it starts from a small number of laws, abstracted from experience, and deduces an astonishing amount from them.

This is a strength, not a weakness. Because thermodynamics assumes so little, its conclusions are extraordinarily robust: they apply equally to gases, magnets, black holes and the early universe. Einstein described it as the only physical theory of universal content which he was convinced would never be overthrown. What we have gained from statistical mechanics is an understanding of *why* the laws hold; but the laws themselves stand on their own.

§42.4.1 The Zeroth and First Laws

We begin with some vocabulary. A thermodynamic system is separated from its surroundings by a **wall**, and the properties of the wall determine what can be exchanged.

Definition 42.4.1

A wall is **adiabatic** if it permits no heat to flow through it, and **diathermal** otherwise. A system is in **equilibrium** if its macroscopic variables do not change in time.

The first law we need is so obvious that it was not written down until the others already had numbers.

Law 42.4.2 (Zeroth Law of Thermodynamics)

If system A is in equilibrium with system B , and B is in equilibrium with C , then A is in equilibrium with C .

That is: ‘being in equilibrium with’ is a transitive – indeed an equivalence – relation on systems. This is exactly what is needed to make temperature a meaningful concept: the equivalence classes can be labelled by a number, which we call the temperature, and two systems are in equilibrium precisely when their labels agree. Note that the zeroth law does not tell us *which* labelling to use; any monotone relabelling would do. Fixing the scale requires more, and we will see that the second law does it for us.

Law 42.4.3 (First Law of Thermodynamics)

The change in energy of a system is

$$\Delta E = \mathrm{d}Q + \mathrm{d}W,$$

where $\mathrm{d}Q$ is the heat supplied to the system and $\mathrm{d}W$ is the work done on it.

The content of the first law is that E is a **function of state** – it depends only on the current macroscopic state of the system, not on how the system got there – whereas Q and W separately are not. The notation d is a reminder of this: $\mathrm{d}Q$ is a small amount of heat, not the change in some quantity ‘ Q ’, and it is not an exact differential. Only their sum is.

Definition 42.4.4

A change is **quasi-static** if it is performed slowly enough that the system remains in equilibrium throughout. It is **reversible** if it is quasi-static and can be run backwards, returning both the system and its surroundings to their original states.

For a quasi-static change of a gas, the work done on the system is $\mathrm{d}W = -p \, \mathrm{d}V$, so the first law reads

$$\mathrm{d}E = \mathrm{d}Q - p \, \mathrm{d}V.$$

Comparing with the result $\mathrm{d}E = T \, \mathrm{d}S - p \, \mathrm{d}V$ from statistical mechanics, we see that for a reversible change $\mathrm{d}Q = T \, \mathrm{d}S$. This is the microscopic meaning of heat: heat is the transfer of energy that changes the entropy.

§42.4.2 The Second Law

The second law is the interesting one. Historically it was stated in two apparently quite different ways.

Law 42.4.5 (Kelvin’s Second Law)

There is no thermodynamic process whose sole effect is to extract heat from a reservoir and convert it entirely into work.

Law 42.4.6 (Clausius’s Second Law)

There is no thermodynamic process whose sole effect is to transfer heat from a colder body to a hotter body.

The words ‘sole effect’ are doing a great deal of work in both statements. It is perfectly possible to convert heat entirely into work – an ideal gas expanding isothermally does exactly that – but only at the cost of leaving the gas in an expanded state, which is a change in the world. Likewise a refrigerator moves heat from cold to hot, but only by consuming electricity.

Proposition 42.4.7

Kelvin’s statement and Clausius’s statement are equivalent.

Proof. Suppose Kelvin's law fails. Then we have a device that takes heat Q from a hot reservoir and turns it entirely into work $W = Q$. Feed that work into an ordinary refrigerator, which uses it to move heat from a cold reservoir to the hot one. The net effect is a transfer of heat from cold to hot with no other change, contradicting Clausius.

Conversely, suppose Clausius's law fails, so we can move heat Q from cold to hot for free. Run a heat engine between the two reservoirs, extracting Q_h from the hot one, dumping Q_c to the cold one and doing work $W = Q_h - Q_c$. Choosing $Q = Q_c$, the cold reservoir ends up unchanged, and the sole net effect is the extraction of $Q_h - Q_c$ from the hot reservoir, converted entirely to work. This contradicts Kelvin. $\square$

Both are of course consequences of the statistical statement 'entropy increases', but it is worth appreciating how different they look. The nineteenth-century statements are about what machines cannot do; the statistical statement is about counting.

§42.4.3 Carnot Cycles and Efficiency

A **heat engine** operates in a cycle, extracting heat Q_h from a hot reservoir at T_h , doing work W , and dumping heat Q_c into a cold reservoir at T_c . Since the engine returns to its initial state each cycle, its energy is unchanged, so $W = Q_h - Q_c$.

Definition 42.4.8 (Efficiency)

The **efficiency** of a heat engine is

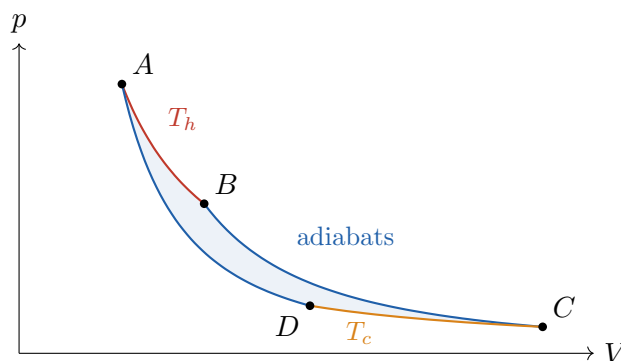
$$\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h}.$$

Kelvin's law says $\eta < 1$: we cannot dispense with the cold reservoir. But we can do much better than that, and the answer is one of the great results of the subject.

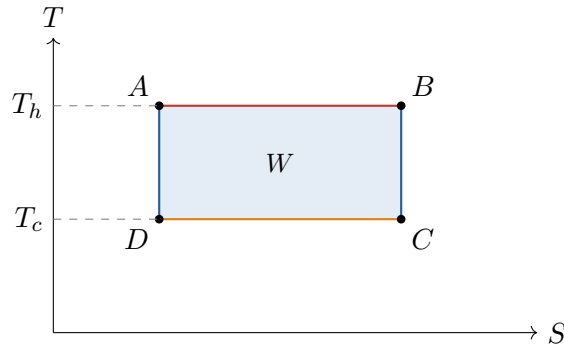
Definition 42.4.9 (Carnot cycle)

A **Carnot cycle** is a reversible cycle consisting of four steps: an isothermal expansion at T_h , an adiabatic expansion from T_h to T_c , an isothermal compression at T_c , and an adiabatic compression back to T_h .

For an ideal gas we can draw this explicitly. On a pV diagram the isotherms are $pV = \text{const}$ and the adiabats are $pV^\gamma = \text{const}$ with $\gamma > 1$, so the adiabats are steeper.



The same cycle is far more revealing on a TS diagram, where the isotherms are horizontal (constant T) and the adiabats are vertical (constant S , since $dQ = T dS = 0$). The Carnot cycle is simply a rectangle.



From this picture the efficiency is immediate. The heat absorbed on $A \rightarrow B$ is $Q_h = T_h \Delta S$, the heat dumped on $C \rightarrow D$ is $Q_c = T_c \Delta S$ with the same ΔS , and the work done is the enclosed area $(T_h - T_c) \Delta S$. So

$$\eta = 1 - \frac{T_c}{T_h}.$$

And this, it turns out, is the best possible.

Theorem 42.4.10 (Carnot's Theorem)

Of all engines operating between two reservoirs at temperatures $T_h > T_c$, a reversible engine is the most efficient; and all reversible engines have the same efficiency,

$$\eta_{\text{Carnot}} = 1 - \frac{T_c}{T_h}.$$

Proof. Let E be any engine and E_R a reversible one, both operating between the same two reservoirs. Since E_R is reversible, we may run it backwards as a refrigerator. Arrange the two so that all the work produced by E is used to drive E_R in reverse; and choose the sizes of the two engines so that this is exactly the work required, that is,

$$W = Q_h - Q_c = Q'_h - Q'_c,$$

where unprimed quantities refer to E and primed to E_R . Combined, the pair extracts $Q_h - Q'_h$ from the hot reservoir and dumps $Q_c - Q'_c$ into the cold one, with no net work. By Clausius's law we cannot be pumping heat from cold to hot, so

$$Q_h - Q'_h \geq 0.$$

But then

$$\eta = \frac{W}{Q_h} \leq \frac{W}{Q'_h} = \eta_R,$$

so E is no more efficient than E_R . If E is itself reversible we may run the argument the other way round to get $\eta_R \leq \eta$, and hence $\eta = \eta_R$: all reversible engines are equally efficient. Since this common efficiency cannot depend on the working substance, we may compute it with an ideal gas, which gives $1 - T_c/T_h$ as above. $\square$

Two things are worth extracting from this. First, the efficiency of a reversible engine

depends *only* on the two temperatures. That means we can use it to define a temperature scale, with no reference to any particular substance: this is the **absolute** or **thermodynamic** temperature scale, and it is what makes the second law able to fix the arbitrariness left by the zeroth law. It agrees, as we have seen, with $1/T = \partial S/\partial E$.

Secondly, real engines fall well short of Carnot. A power station with $T_h \approx 800$ K and $T_c \approx 300$ K has $\eta_{\text{Carnot}} \approx 0.63$; real efficiencies are around 0.4. The gap is the price of running the cycle in finite time, which necessarily makes it irreversible.

Example 42.4.11 (Carnot cycle for an ideal gas)

Let us verify the efficiency directly, which is a good exercise in using the first law. On the isotherm $A \rightarrow B$ at temperature T_h , the energy of an ideal gas does not change, so all the heat absorbed goes into work:

$$Q_h = \int_{V_A}^{V_B} p \, dV = NkT_h \int_{V_A}^{V_B} \frac{dV}{V} = NkT_h \log \frac{V_B}{V_A}.$$

Similarly $Q_c = NkT_c \log(V_C/V_D)$. On an adiabat $dQ = 0$, so $C_V \, dT = -p \, dV = -NkT \, dV/V$, which integrates to $TV^{Nk/C_V} = \text{const}$. Applying this to $B \rightarrow C$ and $D \rightarrow A$,

$$T_h V_B^{Nk/C_V} = T_c V_C^{Nk/C_V}, \quad T_h V_A^{Nk/C_V} = T_c V_D^{Nk/C_V},$$

and dividing gives $V_B/V_A = V_C/V_D$. Hence the two logarithms agree and

$$\eta = 1 - \frac{Q_c}{Q_h} = 1 - \frac{T_c}{T_h}$$

as promised.

§42.4.4 Entropy as a Function of State

We have been using entropy freely, having defined it microscopically. Thermodynamics can define it too, and the argument is elegant. For a Carnot cycle, $Q_h/T_h = Q_c/T_c$; being careful about signs (counting heat absorbed as positive), this reads

$$\frac{Q_h}{T_h} + \frac{Q_c}{T_c} = 0,$$

i.e. the sum of Q/T around the cycle vanishes. This turns out to be true for any reversible cycle, since any cycle can be approximated by a patchwork of small Carnot cycles whose internal edges cancel. In the limit,

$$\oint_{\text{rev}} \frac{dQ}{T} = 0.$$

More generally, for an irreversible cycle one finds **Clausius's inequality**

$$\oint \frac{dQ}{T} \leq 0,$$

which follows by comparing with a reversible engine as in Carnot's theorem.

Now, a quantity whose integral around every closed loop vanishes is the differential of a function of state. So we may define, relative to some fixed reference state O ,

$$S(A) = \int_O^A \frac{dQ}{T},$$

the integral being along any reversible path from O to A , and the result is independent of the path.

Definition 42.4.12 (Entropy, thermodynamically)

The **entropy** of a state A is $S(A) = \int_O^A \mathrm{d}Q/T$ along any reversible path, defined up to an additive constant.

That is a rather remarkable construction: $\mathrm{d}Q$ is not an exact differential, but dividing by T makes it one. ($1/T$ is what a mathematician would call an integrating factor.) With this definition, the first law for a reversible change becomes

$$\mathrm{d}E = T \mathrm{d}S - p \mathrm{d}V,$$

recovering the equation we derived microscopically – but now with no atoms anywhere in sight.

Finally, Clausius’s inequality gives us the second law in its familiar form. For an irreversible change from A to B , close the loop with a reversible path from B back to A ; then

$$\int_A^B \frac{\mathrm{d}Q}{T} + \int_B^A \frac{\mathrm{d}Q_{\text{rev}}}{T} \leq 0 \implies \int_A^B \frac{\mathrm{d}Q}{T} \leq S(B) - S(A).$$

For an isolated system $\mathrm{d}Q = 0$, so $S(B) \geq S(A)$: entropy never decreases.

§42.4.5 Thermodynamic Potentials and Maxwell Relations

The relation $\mathrm{d}E = T \mathrm{d}S - p \mathrm{d}V + \mu \mathrm{d}N$ says that E is naturally a function of S , V and N : these are its **natural variables**, and given $E(S, V, N)$ we can recover everything else by differentiating. The trouble is that S is not a variable we can control in the laboratory. We would much rather work with T .

The tool for changing variables in this way is the **Legendre transform**, which readers will have met in the Variational Principles course (and again in Lagrangian mechanics, where it takes us from the Lagrangian to the Hamiltonian). Recall that if $f = f(x)$ and we set $p = \mathrm{d}f/\mathrm{d}x$, then the transform $g(p) = px - f(x)$ satisfies $\mathrm{d}g = x \mathrm{d}p$, so g is naturally a function of p . Applying this to E in each of its variables generates the family of **thermodynamic potentials**.

Definition 42.4.13 (Thermodynamic potentials)

Define

energy:	$E = E(S, V, N),$	$\mathrm{d}E = T \mathrm{d}S - p \mathrm{d}V + \mu \mathrm{d}N,$
Helmholtz free energy:	$F = E - TS,$	$\mathrm{d}F = -S \mathrm{d}T - p \mathrm{d}V + \mu \mathrm{d}N,$
enthalpy:	$H = E + pV,$	$\mathrm{d}H = T \mathrm{d}S + V \mathrm{d}p + \mu \mathrm{d}N,$
Gibbs free energy:	$G = E + pV - TS,$	$\mathrm{d}G = -S \mathrm{d}T + V \mathrm{d}p + \mu \mathrm{d}N,$
grand potential:	$\Phi = E - TS - \mu N,$	$\mathrm{d}\Phi = -S \mathrm{d}T - p \mathrm{d}V - N \mathrm{d}\mu.$

Each is adapted to a different experimental situation: F for a system at fixed T and V , G for fixed T and p (which is what a chemist working in an open beaker has), H for fixed S and p , and Φ for a system that can exchange particles. And in each case the

relevant potential is minimised in equilibrium, by the same argument we gave for F in Section 1.9.

The Gibbs free energy has a special property worth isolating. It is extensive and its natural variables T and p are both intensive apart from N , so as in Section 1.11 we must have $G = Ng(T, p)$ for some intensive g . But $(\partial G/\partial N)_{T,p} = \mu$, and so

$$\mu(T, p) = \frac{G}{N},$$

the Gibbs free energy per particle. We will use this in the next section.

Now for the Maxwell relations, which are perhaps the most useful pieces of pure book-keeping in physics. They come from the equality of mixed partial derivatives. For instance, from $dE = T dS - p dV$ we read off $T = (\partial E/\partial S)_V$ and $-p = (\partial E/\partial V)_S$, and therefore

$$\frac{\partial^2 E}{\partial V \partial S} = \frac{\partial^2 E}{\partial S \partial V} \implies \left(\frac{\partial T}{\partial V} \right)_S = - \left(\frac{\partial p}{\partial S} \right)_V.$$

Doing the same for each potential in turn gives the following.

Proposition 42.4.14 (Maxwell relations)

At fixed N ,

$$\begin{aligned} \left(\frac{\partial T}{\partial V} \right)_S &= - \left(\frac{\partial p}{\partial S} \right)_V, & \left(\frac{\partial S}{\partial V} \right)_T &= \left(\frac{\partial p}{\partial T} \right)_V, \\ \left(\frac{\partial T}{\partial p} \right)_S &= \left(\frac{\partial V}{\partial S} \right)_p, & \left(\frac{\partial S}{\partial p} \right)_T &= - \left(\frac{\partial V}{\partial T} \right)_p, \end{aligned}$$

coming from E , F , H and G respectively.

These are not obvious statements. The second, for example, relates how the entropy responds to a change in volume – which is hard to measure – to how the pressure responds to a change in temperature – which is easy. The general moral is that if you find yourself needing a derivative of S , a Maxwell relation will usually convert it into something measurable.

Example 42.4.15

Let us use a Maxwell relation to compute how the energy of a gas depends on its volume at fixed temperature. From $dE = T dS - p dV$,

$$\left(\frac{\partial E}{\partial V} \right)_T = T \left(\frac{\partial S}{\partial V} \right)_T - p = T \left(\frac{\partial p}{\partial T} \right)_V - p,$$

using the second Maxwell relation. For an ideal gas, $p = NkT/V$ gives $T(\partial p/\partial T)_V = NkT/V = p$, so $(\partial E/\partial V)_T = 0$: the energy of an ideal gas depends only on temperature, which we knew, but which we have now derived from the equation of state alone. For a van der Waals gas, $p = NkT/(V - bN) - aN^2/V^2$ gives

$$\left(\frac{\partial E}{\partial V} \right)_T = \frac{aN^2}{V^2} > 0,$$

so expanding a real gas at fixed temperature costs energy – as it should, since we

are pulling apart mutually attracting molecules. This is the origin of the cooling one feels when a gas expands freely, which is the principle behind refrigeration.

§42.4.6 The Third Law

For completeness, one more law.

Law 42.4.16 (Third Law of Thermodynamics)

As $T \rightarrow 0$, the entropy of a system tends to a universal constant, which may be taken to be zero:

$$\lim_{T \rightarrow 0} S = 0.$$

Microscopically this is transparent. As $T \rightarrow 0$ the system falls into its ground state, and $S = k \log \Omega$ where Ω is the ground state degeneracy. For a typical system Ω is 1 or some small number, so $S \rightarrow 0$ or at worst to something of order k , which is negligible compared with Nk .

The third law has real consequences. Since $S(T) = \int_0^T C(T') dT'/T'$ must converge, we need $C(T) \rightarrow 0$ as $T \rightarrow 0$ – which rules out equipartition at low temperature and is consistent with everything we found in Section 42.3.3. It also implies that absolute zero cannot be reached in finitely many steps: any cooling process relies on entropy differences between two configurations of the system, and these differences vanish as $T \rightarrow 0$.

§42.5 Phase Transitions

We come to the last part of the course. A **phase transition** is a discontinuous change in the properties of a system as we vary a parameter continuously: water freezes, iron becomes magnetic, helium becomes superfluid. Mathematically, it shows up as a non-analyticity of a thermodynamic potential.

That, at first sight, ought to be impossible. The partition function $Z = \sum_n e^{-\beta E_n}$ is a sum of analytic functions of β , and for a finite system it is a finite sum, hence analytic; so $F = -kT \log Z$ is analytic and nothing dramatic can ever happen. The escape is that phase transitions occur only in the *thermodynamic limit* $N \rightarrow \infty$, where the sum becomes infinite and analyticity can fail. This is a genuine subtlety, and it means that a phase transition is an emergent property of large systems, invisible in any finite one.

We will study two examples in detail – the liquid–gas transition and the Ising model of a magnet – which look as though they have nothing whatever to do with each other. They will turn out to behave identically near their critical points. Explaining why is the last thing we do.

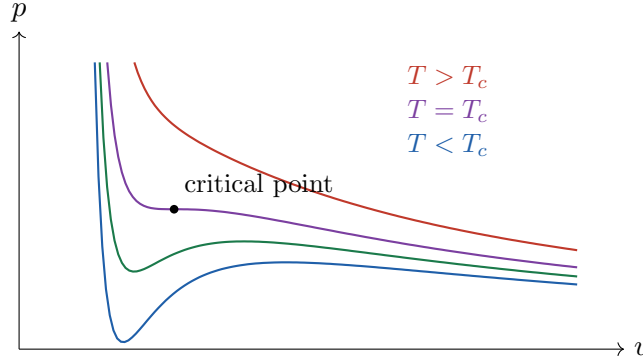
§42.5.1 The Liquid–Gas Transition

We return to the van der Waals equation of Section 42.2.8, written in terms of the volume per particle $v = V/N$:

$$p = \frac{kT}{v - b} - \frac{a}{v^2}.$$

We derived this for a dilute gas, where it is a small correction. We now take it seriously for all $v > b$, which is completely unjustified, and see what happens. The answer is that it produces a liquid–gas transition, which is a good deal more than it deserves.

Here are the isotherms.



There are two regimes, separated by a critical temperature T_c . For $T > T_c$ the isotherms decrease monotonically, as one expects of a gas. For $T < T_c$ they have a wiggle: a local minimum and a local maximum, so that there is a range of pressures at which three different volumes are possible at the same T and p . At $T = T_c$ the two turning points merge into a point of inflection, which locates the critical point:

$$\left(\frac{\partial p}{\partial v}\right)_T = \left(\frac{\partial^2 p}{\partial v^2}\right)_T = 0.$$

Solving these two equations gives

$$v_c = 3b, \quad kT_c = \frac{8a}{27b}, \quad p_c = \frac{a}{27b^2}.$$

Derivation. From $p = kT/(v - b) - a/v^2$ we get

$$\frac{\partial p}{\partial v} = -\frac{kT}{(v - b)^2} + \frac{2a}{v^3}, \quad \frac{\partial^2 p}{\partial v^2} = \frac{2kT}{(v - b)^3} - \frac{6a}{v^4}.$$

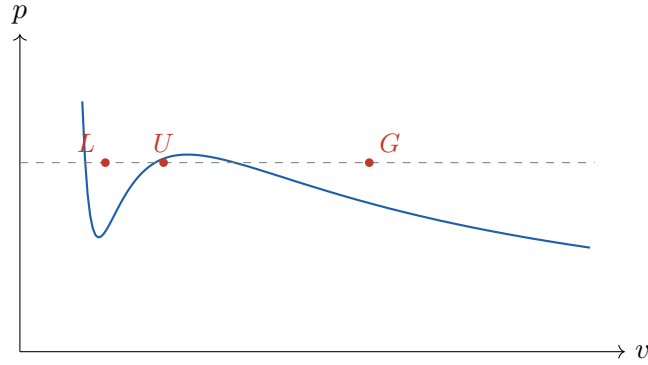
Setting both to zero and dividing one by the other eliminates kT and gives $v - b = \frac{2}{3}v$, so $v_c = 3b$. Substituting back into the first gives $kT_c = 2a(v_c - b)^2/v_c^3 = 8a/27b$, and then $p_c = a/27b^2$. $\square$

It is worth noting the dimensionless combination

$$\frac{p_c v_c}{kT_c} = \frac{3}{8} = 0.375,$$

which is independent of a and b and hence a universal prediction of the model for all substances. Measured values range from about 0.28 to 0.31 – the right ballpark, given how crude the model is.

Let us interpret the three solutions on a sub-critical isotherm, which we label L , U and G in order of increasing v .



At U we have $(\partial p / \partial v)_T > 0$. Such a state is unstable: expand it slightly and the pressure rises, pushing it to expand further; compress it slightly and it collapses. So U is not physical. At L we have v close to b , so the molecules are close-packed, and $|\partial p / \partial v|$ is large, so the substance is hard to compress – this is a **liquid**. At G we have $v \gg b$ and small $|\partial p / \partial v|$: a **gas**. Notice that above T_c there is no distinction between the two at all; one can pass continuously from liquid to gas by going around the critical point, which is why one cannot really say where the boundary between them lies.

§42.5.2 Phase Equilibrium and the Maxwell Construction

So what does the system actually do at a pressure where three solutions exist? There is a fourth possibility we have not considered: part of the substance could be liquid and part gas.

To decide, work at fixed T and p , so the relevant potential is the Gibbs free energy G , and recall that $\mu = G/N$. If $\mu_L > \mu_G$, then the system lowers its free energy by converting liquid into gas, so we end up with pure gas; if $\mu_L < \mu_G$ we get pure liquid. Only if

$$\mu_L = \mu_G$$

can the two coexist, and then they can coexist in any proportion.

We can compute μ along an isotherm. From $dG = -S dT + V dp + \mu dN$ at fixed T, N we have $(\partial \mu / \partial p)_T = v$, so

$$\mu(p) = \mu(p_0) + \int_{p_0}^p v dp'$$

along the isotherm. The condition $\mu_L = \mu_G$ therefore reads

$$\int_L^G v dp = 0,$$

the integral being taken along the van der Waals curve through the unstable region. Integrating by parts, this is equivalent to a statement about areas.

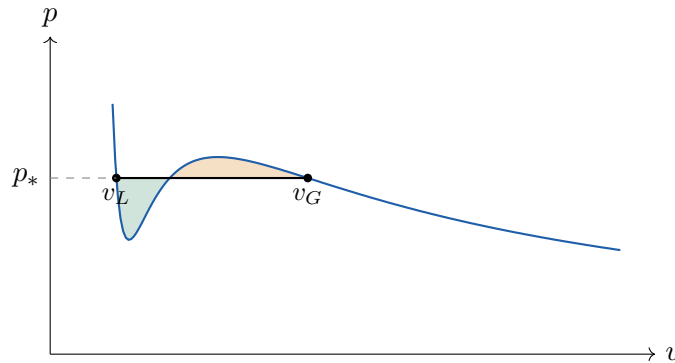
Proposition 42.5.1 (Maxwell construction)

The coexistence pressure p_* at temperature $T < T_c$ is the pressure at which the horizontal line $p = p_*$ cuts the van der Waals isotherm into two lobes of *equal area*.

Proof. Let v_L and v_G be the outer intersections of $p = p_*$ with the isotherm. Then

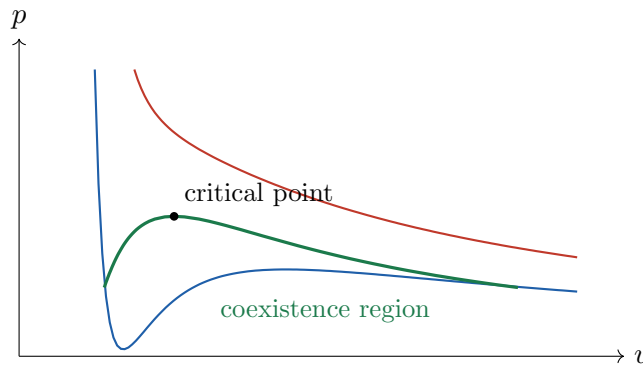
$$0 = \int_L^G v \, dp = [vp]_L^G - \int_{v_L}^{v_G} p \, dv = p_*(v_G - v_L) - \int_{v_L}^{v_G} p \, dv,$$

since p takes the same value p_* at both endpoints. So $\int_{v_L}^{v_G} p \, dv = p_*(v_G - v_L)$: the area under the isotherm equals the area under the horizontal line. Subtracting the common part, the lobe above the line and the lobe below it have equal areas. $\square$



The physical isotherm is therefore not the van der Waals curve at all: it is flat between v_L and v_G . In that region the substance is a mixture of liquid at v_L and gas at v_G , in whatever proportions make the average volume come out right. This is exactly what one sees when boiling a kettle: adding heat at fixed pressure converts liquid to gas at *constant temperature*, and the temperature only rises again once all the liquid is gone. The latent heat $L = T(s_G - s_L)$ per particle is the entropy difference between the two phases.

If we plot the locus of (v_L, p_*) and (v_G, p_*) as T varies we obtain the **coexistence curve**, which meets the critical point from below:



Inside the green curve the substance separates into two phases; outside it there is a single phase.

Finally, the phases are related by the **Clausius–Clapeyron equation**, which we can now derive in three lines. Along the coexistence line in the (p, T) plane we have $\mu_L(T, p) = \mu_G(T, p)$, so moving along it,

$$d\mu_L = d\mu_G \implies -s_L \, dT + v_L \, dp = -s_G \, dT + v_G \, dp,$$

using the Gibbs–Duhem relation per particle, where $s = S/N$. Hence

$$\frac{dp}{dT} = \frac{s_G - s_L}{v_G - v_L} = \frac{L}{T(v_G - v_L)}.$$

Since $v_G > v_L$ and $L > 0$ for boiling, the boiling point rises with pressure – which is why a pressure cooker works. For water freezing, unusually, $v_{\text{ice}} > v_{\text{water}}$, so the melting curve slopes the other way and ice melts under pressure.

§42.5.3 Critical Exponents

Phase transitions come in two flavours.

Definition 42.5.2

A phase transition is **first order** if the first derivatives of the relevant thermodynamic potential are discontinuous, and **second order** (or continuous) if the first derivatives are continuous but some higher derivative is not.

Boiling at $T < T_c$ is first order: the volume $v = (\partial\mu/\partial p)_T$ and the entropy $s = -(\partial\mu/\partial T)_p$ both jump, the latter jump being the latent heat. But if we heat the system along the critical isochore $v = v_c$, the jump shrinks to nothing as $T \rightarrow T_c$, and at the critical point itself the transition is second order. Second-order transitions are the interesting ones, because at such a point the system has no characteristic length scale, and this turns out to make the behaviour universal.

Near the critical point, quantities vanish or diverge as powers of $|T - T_c|$, and the powers are called **critical exponents**. The standard four are defined by

$$c_V \sim |T - T_c|^{-\alpha}, \quad v_G - v_L \sim (T_c - T)^\beta, \quad \left(\frac{\partial p}{\partial v}\right)_{T_c} \sim |v - v_c|^{\delta-1}, \quad \kappa \sim |T - T_c|^{-\gamma},$$

where $\kappa = -v^{-1}(\partial v/\partial p)_T$ is the compressibility. (The clash of notation between the exponent β and $1/kT$ is unfortunate and universal; context will always make clear which is meant.)

Let us compute them for van der Waals. It is convenient to use **reduced variables**

$$\bar{p} = \frac{p}{p_c}, \quad \bar{v} = \frac{v}{v_c}, \quad \bar{T} = \frac{T}{T_c},$$

in terms of which the equation of state becomes

$$\bar{p} = \frac{8\bar{T}}{3\bar{v} - 1} - \frac{3}{\bar{v}^2},$$

with no free parameters at all. This is the **law of corresponding states**: all van der Waals fluids are the same fluid in disguise, and their critical behaviour is identical. (Experimentally this works remarkably well for the noble gases.)

Now expand about the critical point, writing $\bar{T} = 1 + t$ and $\bar{v} = 1 + \epsilon$ with t, ϵ small. A somewhat tedious Taylor expansion gives

$$\bar{p} = 1 + 4t - 6t\epsilon - \frac{3}{2}\epsilon^3 + O(t\epsilon^2, \epsilon^4).$$

From this everything follows.

- *The exponent δ .* On the critical isotherm $t = 0$ we have $\bar{p} - 1 = -\frac{3}{2}\epsilon^3$, so $p - p_c \propto (v - v_c)^3$ and $\delta = 3$.
- *The exponent β .* For $t < 0$, the Maxwell construction requires the coexisting volumes to be symmetric about v_c (which follows from the fact that the above expansion is odd in ϵ at fixed t , so the equal-area condition is met by $\epsilon_G = -\epsilon_L$). Setting $\bar{p}(\epsilon_L) = \bar{p}(\epsilon_G)$ with $\epsilon_G = -\epsilon_L$ gives

$$-6t\epsilon_G - \frac{3}{2}\epsilon_G^3 = 6t\epsilon_G + \frac{3}{2}\epsilon_G^3 \implies \epsilon_G^2 = -4t,$$

so $v_G - v_L \propto (T_c - T)^{1/2}$ and $\beta = \frac{1}{2}$.

- *The exponent γ .* Differentiating, $\partial\bar{p}/\partial\epsilon = -6t - \frac{9}{2}\epsilon^2$. Approaching from above along $\epsilon = 0$ gives $\kappa^{-1} \propto t$; from below along $\epsilon^2 = -4t$ gives $\kappa^{-1} \propto -12t$. Either way $\kappa \propto |t|^{-1}$ and $\gamma = 1$.
- *The exponent α .* One can show that c_V jumps by a finite amount at T_c but does not diverge, so $\alpha = 0$.

Proposition 42.5.3

The van der Waals equation predicts the critical exponents

$$\alpha = 0, \quad \beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3.$$

The measured values, for a wide range of real fluids, are approximately

$$\alpha \approx 0.11, \quad \beta \approx 0.32, \quad \gamma \approx 1.24, \quad \delta \approx 4.8.$$

So the van der Waals prediction is wrong. But notice *how* it is wrong: the experimental values are the same for every fluid. This universality is the deep phenomenon, and it is much more interesting than the failure of any particular model. Hold that thought.

§42.5.4 The Ising Model

We now change subject entirely, to a model of a magnet. Take a d -dimensional lattice of N sites; on each site i put a ‘spin’ variable

$$s_i = \pm 1,$$

and let the energy be

$$E[\{s_i\}] = -J \sum_{\langle ij \rangle} s_i s_j - B \sum_i s_i,$$

where $\sum_{\langle ij \rangle}$ runs over nearest-neighbour pairs, B is an external magnetic field (we set the magnetic moment to 1) and J measures the strength of the interaction. If $J > 0$ neighbouring spins prefer to align, and the material is a **ferromagnet**; if $J < 0$ they prefer to anti-align, an **antiferromagnet**. We take $J > 0$.

In one dimension the lattice and its neighbours are obvious. In two dimensions, on a square lattice, each site has $q = 4$ nearest neighbours:



The physics is a competition. The energy term wants all the spins aligned, since that minimises $-J \sum s_i s_j$. The entropy wants them randomised, since there are vastly more disordered configurations than ordered ones. Which wins depends on the temperature, and the transition between the two regimes is the phase transition we are after.

The quantity to watch is the average spin.

Definition 42.5.4 (Magnetisation)

The **magnetisation** is

$$m = \frac{1}{N} \sum_i \langle s_i \rangle = \frac{1}{N\beta} \left(\frac{\partial \log Z}{\partial B} \right)_T,$$

where $Z = \sum_{\{s_i\}} e^{-\beta E[\{s_i\}]}$.

Before doing any work, it is worth noting that the Ising model is secretly a model of a gas as well. Consider particles on a lattice, with $n_i \in \{0, 1\}$ particles per site and an attractive nearest-neighbour interaction,

$$E[\{n_i\}] = -4J \sum_{\langle ij \rangle} n_i n_j.$$

In the grand canonical ensemble we weight by $e^{-\beta(E - \mu \sum_i n_i)}$. Substituting $n_i = (s_i + 1)/2$ turns this into the Ising energy with

$$B = \frac{\mu}{2} + qJ,$$

so that $\mathcal{Z}_{\text{gas}} = Z_{\text{Ising}}$. The liquid–gas transition and the ferromagnetic transition are, in this ‘lattice gas’ caricature, literally the same problem. That is our first hint of universality.

Can we compute Z ? In $d = 1$, yes, and we will do it in Section 42.5.6. In $d = 2$ with $B = 0$, yes, but the solution (Onsager, 1944) is a tour de force. In $d = 3$, nobody knows. So we need an approximation.

§42.5.5 Mean Field Theory

The difficulty is the interaction term, which couples the spins together. The idea of **mean field theory** is to replace the influence of a spin’s neighbours by their average, turning the problem into N independent single-spin problems.

Concretely, write $s_i = m + (s_i - m)$ and expand:

$$s_i s_j = (s_i - m)(s_j - m) + m(s_i + s_j) - m^2,$$

and then *assume the first term is negligible* – that is, that the fluctuations of neighbouring spins are uncorrelated. This is the whole approximation, and it is a serious one; note that it is not the claim that $\langle (s_i - m)^2 \rangle$ is small, since in fact $\langle (s_i - m)^2 \rangle = 1 - m^2$ which need not be small at all. It is the claim that the *correlations* between different sites are negligible.

Granting it, the energy becomes

$$E \approx -J \sum_{\langle ij \rangle} (m(s_i + s_j) - m^2) - B \sum_i s_i = \frac{1}{2} J N q m^2 - (J q m + B) \sum_i s_i,$$

where q is the number of nearest neighbours and the factor $\frac{1}{2} N q$ counts the bonds. The spins have decoupled! Each now sees an **effective field**

$$B_{\text{eff}} = B + J q m,$$

and we are back to the two-state system of Section 1.4. The partition function is

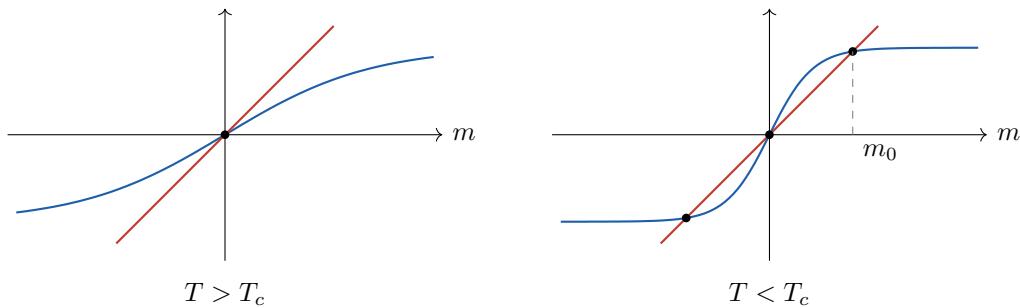
$$Z = e^{-\frac{1}{2} \beta J N q m^2} (e^{\beta B_{\text{eff}}} + e^{-\beta B_{\text{eff}}})^N = e^{-\frac{1}{2} \beta J N q m^2} 2^N \cosh^N(\beta B + \beta J q m).$$

But m was defined as an average, so we cannot simply leave it in the answer; we must impose consistency, $m = (N\beta)^{-1}(\partial \log Z / \partial B)_T$. Doing so gives the central equation of mean field theory.

Proposition 42.5.5 (Mean field equation)

$$m = \tanh(\beta B + \beta J q m).$$

This is a transcendental equation which we cannot solve in closed form, but we can solve it graphically. Set $B = 0$ and plot both sides against m . The left side is the straight line $y = m$; the right is $y = \tanh(\beta J q m)$, an S-shaped curve with slope $\beta J q$ at the origin. Everything hinges on whether that slope exceeds 1.



If $\beta J q < 1$ the $\tanh$ curve starts below the line and stays below it, so the only solution is $m = 0$: the spins are disordered, and entropy has won. If $\beta J q > 1$ the $\tanh$ curve starts above the line, and there are three solutions, $m = 0$ and $m = \pm m_0(T)$. The critical temperature is where the slope is exactly one.

Definition 42.5.6

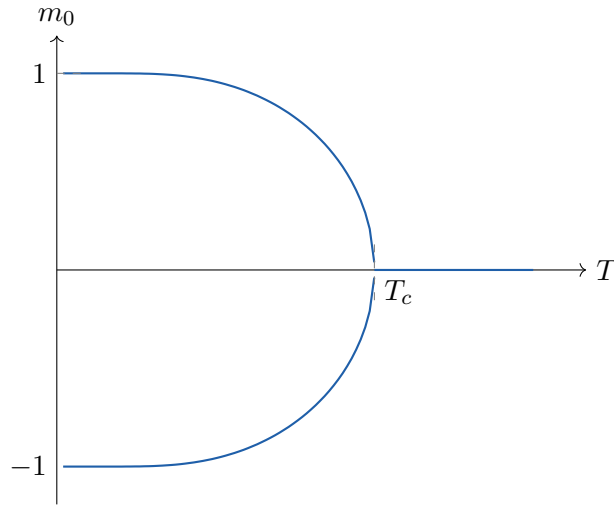
The mean field critical temperature is defined by $\beta_c Jq = 1$, that is,

$$kT_c = Jq.$$

For $T < T_c$ the solution $m = 0$ is unstable – differentiating the mean field equation with respect to B and setting $B = 0$,

$$\left(\frac{\partial m}{\partial B} \right)_T \Big|_{m=0} = \frac{\beta}{1 - \beta Jq} < 0,$$

which is nonsense for a paramagnetic response, so the state is not physical. That leaves $\pm m_0(T)$, the two being related by flipping every spin. So below T_c the system spontaneously magnetises, and it must choose one of two directions; this is **spontaneous symmetry breaking**. Plotting m_0 against T :



At $T = 0$ all spins align and $m_0 = \pm 1$; as T increases m_0 falls, reaching zero at T_c . The manner in which it does so is the critical exponent β , and we can extract it by Taylor expanding. Using $\tanh x \approx x - \frac{1}{3}x^3$ for small x ,

$$m_0 \approx \beta Jq m_0 - \frac{1}{3}(\beta Jq m_0)^3 = \frac{T_c}{T} m_0 - \frac{1}{3} \left(\frac{T_c}{T} \right)^3 m_0^3,$$

so dividing by m_0 and rearranging,

$$m_0^2 = 3 \left(\frac{T}{T_c} \right)^3 \left(\frac{T_c}{T} - 1 \right) \approx 3 \frac{T_c - T}{T_c}$$

near T_c . Hence

$$m_0 \sim (T_c - T)^{1/2}, \quad \beta = \frac{1}{2},$$

which is exactly the exponent we found for van der Waals. That is not a coincidence, and explaining it is the business of the next section but one.

For completeness, the other exponents follow similarly. Turning on a small field B at $T > T_c$ and expanding the mean field equation gives

$$m \approx \beta B + \beta Jq m \implies m = \frac{\beta B}{1 - \beta Jq} \implies \chi = \frac{\partial m}{\partial B} \propto \frac{1}{T - T_c},$$

so $\gamma = 1$. At $T = T_c$ exactly, the linear terms cancel and we must go to cubic order, giving $m \propto B^{1/3}$ and $\delta = 3$. And the heat capacity jumps discontinuously without diverging, so $\alpha = 0$. The full set

$$\alpha = 0, \quad \beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3$$

agrees exactly with van der Waals, and disagrees with experiment in exactly the same way.

§42.5.6 The One-Dimensional Ising Model

Mean field theory predicts a phase transition in every dimension. In one dimension it is simply wrong, and it is instructive to see why, because the exact solution is easy.

Take N spins in a ring, with $s_{N+1} = s_1$, and set $B = 0$ for simplicity. Then

$$Z = \sum_{\{s_i\}} \exp \left(\beta J \sum_{i=1}^N s_i s_{i+1} \right) = \sum_{\{s_i\}} \prod_{i=1}^N e^{\beta J s_i s_{i+1}}.$$

The trick is to write this as a matrix product. Define the 2×2 **transfer matrix**

$$T = \begin{pmatrix} e^{\beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J} \end{pmatrix}, \quad T_{ss'} = e^{\beta J s s'},$$

indexing rows and columns by $s = \pm 1$. Then the sum over configurations is exactly the trace of a product:

$$Z = \sum_{s_1, \dots, s_N} T_{s_1 s_2} T_{s_2 s_3} \cdots T_{s_N s_1} = \text{tr} (T^N).$$

The eigenvalues of T are $\lambda_{\pm} = e^{\beta J} \pm e^{-\beta J}$, i.e. $\lambda_+ = 2 \cosh \beta J$ and $\lambda_- = 2 \sinh \beta J$, so

$$Z = \lambda_+^N + \lambda_-^N = \lambda_+^N \left(1 + \left(\frac{\lambda_-}{\lambda_+} \right)^N \right) \rightarrow \lambda_+^N$$

as $N \rightarrow \infty$, since $\lambda_- < \lambda_+$. Hence

$$F = -kT \log Z = -NkT \log(2 \cosh \beta J),$$

which is a perfectly analytic function of T for all $T > 0$. There is no phase transition in one dimension.

The physical reason is worth understanding, and is due to Peierls. Start from the fully aligned state and consider flipping a whole block of spins. This creates two domain walls, at an energy cost of $4J$ regardless of the size of the block; but the walls can be placed in about N^2 ways, so the entropy gain is $k \log N^2$. The free energy change is

$$\Delta F = 4J - 2kT \log N,$$

which is negative for large N at any $T > 0$. So it is always favourable to break up the ordered state: order is destroyed by fluctuations in one dimension. In two or more dimensions a domain wall costs an energy proportional to its length, and the argument fails – which is why the transition survives.

This should make us suspicious of mean field theory, and rightly so. Mean field theory becomes exact in high dimensions (above $d = 4$, in fact), and becomes worse as the dimension falls, because it is precisely the fluctuations it neglects that matter most in low dimensions.

§42.5.7 Landau Theory

We have now seen two systems – the van der Waals fluid and the mean field Ising magnet – which have nothing physically in common and yet produce identical critical exponents. Landau theory explains this, and does so by throwing away essentially all of the physics.

Look back at what happened in mean field theory. We had a free energy depending on a parameter m ,

$$\tilde{F}(T, B; m) = \frac{1}{2} J N q m^2 - N k T \log(2 \cosh(\beta B + \beta J q m)),$$

and we found the physical value of m by imposing self-consistency. But one can check directly that

$$\left(\frac{\partial \tilde{F}}{\partial m} \right)_{T, B} = 0 \iff m = \tanh(\beta B + \beta J q m),$$

so the self-consistency condition is exactly the statement that m *minimises* $\tilde{F}$. That is the observation Landau theory is built on.

Definition 42.5.7 (Order parameter)

An **order parameter** is a quantity m which vanishes in the disordered phase and is non-zero in the ordered phase.

For a magnet the order parameter is the magnetisation; for a fluid near its critical point it is $\rho - \rho_c$, the deviation of the density from its critical value.

The recipe of **Landau theory** is then:

- (i) identify an order parameter m ;
- (ii) write down a free energy $\tilde{F}(T; m)$, assumed to be an *analytic* function of m and T and to respect the symmetries of the problem;
- (iii) declare that the equilibrium states are the minima of $\tilde{F}$ over m .

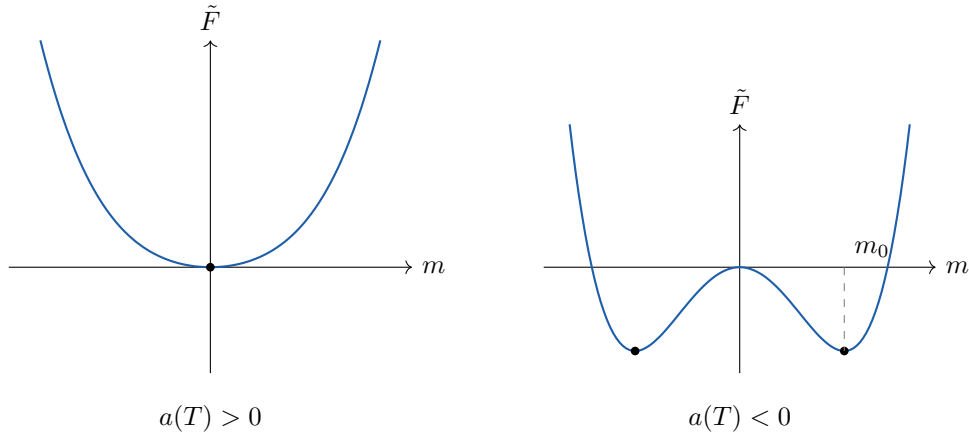
That is all. Notice how little input there is – no Hamiltonian, no lattice, no interactions.

Suppose the system has a symmetry $m \mapsto -m$ (as the Ising model does at $B = 0$). Then $\tilde{F}$ is even in m , and analyticity lets us expand

$$\tilde{F}(T; m) = F_0(T) + a(T)m^2 + b(T)m^4 + \dots$$

The constant F_0 is irrelevant. Assume $b(T) > 0$, so that $\tilde{F}$ is bounded below. Then everything depends on the sign of $a(T)$.

If $a > 0$, the only minimum is at $m = 0$: the disordered phase. If $a < 0$, then $m = 0$ becomes a local maximum and two minima appear at $\pm m_0$: the ordered phase.



So a phase transition occurs at the temperature T_c where $a(T_c) = 0$, and near it we may write

$$a(T) = a_0(T - T_c), \quad b(T) = b_0 > 0,$$

assuming a simple zero. Minimising over m for $T < T_c$ gives

$$m_0 = \sqrt{\frac{-a}{2b}} = \sqrt{\frac{a_0}{2b_0}} (T_c - T)^{1/2},$$

so $\beta = \frac{1}{2}$ once again – and now we can see exactly where it came from. It came from the fact that we truncated an even analytic function at fourth order. Nothing else. The van der Waals fluid, the Ising magnet and any other system whose free energy is a smooth even function of its order parameter must produce $\beta = \frac{1}{2}$, whatever the underlying physics. *That* is the origin of universality.

Substituting back gives the free energy itself,

$$F(T) = \begin{cases} F_0(T) & T > T_c, \\ F_0(T) - \frac{a_0^2(T - T_c)^2}{4b_0} & T < T_c, \end{cases}$$

which is continuous at T_c , as is its first derivative $S = -dF/dT$; but the second derivative, and hence

$$C = T \frac{dS}{dT} = -T \frac{d^2F}{dT^2},$$

jumps by $a_0^2 T_c / 2b_0$. So this is a second-order transition, with $\alpha = 0$. All four exponents come out as before.

Of course, the exponents are wrong. The reason is that we assumed $\tilde{F}$ is analytic in m , which amounts to assuming the system is well described by a single uniform order parameter – exactly the assumption mean field theory made, and exactly the one that fails near a critical point, where fluctuations occur on all length scales. Repairing this is the business of **Landau–Ginzburg theory**, in which m is promoted to a field $m(\mathbf{x})$ and one integrates over its fluctuations, and ultimately of the renormalisation group, which is beyond this course. What Landau theory does give us, cheaply, is a qualitative understanding of *why* phase transitions happen and *why* unrelated systems share critical behaviour.

§42.5.8 First Order Transitions in Landau Theory

Finally, let us see how first-order transitions fit into this picture. There are two ways they arise.

The first is by breaking the symmetry explicitly. In the Ising model with $B \neq 0$, the free energy is no longer even in m ; instead it satisfies $\tilde{F}(T, B; m) = \tilde{F}(T, -B; -m)$, so the expansion acquires odd terms,

$$\tilde{F} = F_0(T, B) + F_1(T, B)m + F_2(T, B)m^2 + \dots,$$

with F_n odd or even in B according as n is odd or even. At low temperature there are two local minima, and a small $B > 0$ tilts the potential so that the right-hand one is the global minimum G , the left-hand one a metastable state M , and the maximum between them the unstable state U . As B passes through zero the two minima swap roles, and the equilibrium m jumps discontinuously from $-m_0$ to $+m_0$:

$$\lim_{B \rightarrow 0^+} m(T, B) = +m_0(T), \quad \lim_{B \rightarrow 0^-} m(T, B) = -m_0(T).$$

That is a first-order transition, occurring as we sweep B at fixed $T < T_c$ – and it is the exact analogue of the liquid–gas transition as we sweep p at fixed T , with the metastable state M corresponding to a supercooled vapour or superheated liquid.

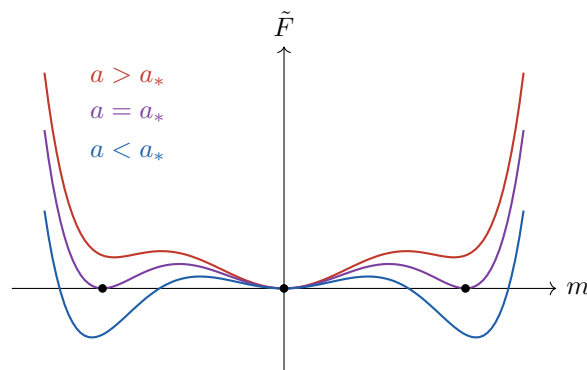
The second mechanism keeps the symmetry but changes the sign of the quartic term. Suppose $b(T) < 0$; then we must keep the sixth-order term to have a free energy bounded below,

$$\tilde{F} = am^2 - bm^4 + cm^6, \quad b, c > 0.$$

Now as we lower a from large positive values, the following happens. For large a there is a single minimum at $m = 0$. For a slightly less than $b^2/3c$, two subsidiary minima appear at non-zero m , but they are higher than the one at the origin, so they are metastable. At

$$a_* = \frac{b^2}{4c}$$

all three minima have exactly the same depth, and below that the outer ones win. So at $a = a_*$ the equilibrium value of m jumps discontinuously from 0 to $\pm\sqrt{b/2c}$: a first-order transition.



The middle (purple) curve is the transition itself: three degenerate minima, which is the Landau-theoretic signature of first-order behaviour, and the analogue of the three-fold coexistence we saw for the van der Waals fluid. Notice that here, unlike the second-order case, the order parameter is already of finite size when the transition occurs; there is no small parameter, no divergent correlation length, and consequently no universality.

First-order transitions are, in this sense, much less interesting than second-order ones – which is a curious inversion of the usual state of affairs in physics, where the discontinuous thing is normally the dramatic one.

§42.5.9 A Last Word

It is worth stepping back. We began with a postulate about counting microstates, and from it we derived temperature, entropy and the laws of thermodynamics; we computed the properties of gases, the spectrum of starlight, the heat capacity of crystals and the structure of white dwarfs; and we ended with a theory of phase transitions in which the microscopic details do not matter at all. That last point deserves emphasis. The reason the van der Waals fluid and the Ising magnet share critical exponents is that near a critical point the correlation length diverges, the system loses all memory of its lattice spacing or molecular size, and only the symmetry and dimensionality survive. Understanding this properly – the renormalisation group – is one of the genuinely great ideas of twentieth-century physics, and this course is the natural place to start.

43 Stochastic Financial Models

Financial markets are, from a mathematician's point of view, a rather remarkable object of study. They are made of people, and people are not easy to axiomatise. And yet if we grant ourselves a single principle – that one cannot make something out of nothing – then a startling amount of structure falls out. Prices of complicated derivative securities are pinned down exactly by the prices of simple ones; the 'right' probability measure to compute with turns out not to be the one describing what will actually happen; and the whole edifice rests on the theory of martingales.

This course is really two courses braided together. One strand is economic: how should an agent with preferences choose a portfolio, and what does market equilibrium look like? The other strand is probabilistic: conditional expectation, martingales, optional stopping, Brownian motion, and a first look at stochastic calculus. The two strands meet in the *fundamental theorem of asset pricing*, which says that a market admits no arbitrage precisely when there is an equivalent measure making discounted prices into martingales. Everything after that theorem is, in one way or another, a computation of a conditional expectation.

This article constitutes my notes for the 'Stochastic Financial Models' course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§43.1 The One-Period Market

Before we can prove anything, we need a model. The models in this course are deliberately, almost offensively, simple; the justification is that even the simplest of them already produces the Black–Scholes formula, and that the qualitative conclusions turn out to be robust.

§43.1.1 Standing Assumptions

We shall work throughout under the following idealisations, which we state once and then quietly forget.

- (i) *No transaction costs.* Buying and selling are free.
- (ii) *Infinite divisibility.* We may hold any real number of units of any asset, not just an integer number.
- (iii) *No bid–ask spread.* There is one price at which an asset may be both bought and sold.
- (iv) *No price impact* (‘linear pricing’). Buying 2θ units costs exactly twice as much as buying θ units, however large θ is. Our trading does not move the market.
- (v) *No short-selling constraints.* We may hold a negative number of units of an asset.
- (vi) *Borrowing and lending at the same rate.* There is a riskless bank account paying interest at rate r , and we may deposit into it or borrow from it freely.

Assumption (iv) is the one doing the most work, and it is also the one most obviously false: if you try to buy every share in a company you will find the price moves against you. But without it there is no hope of a linear theory, and linearity is what makes everything computable.

§43.1.2 Portfolios and Wealth

Our basic market has d **risky assets**, whose prices at time t we write as a random vector

$$S_t = (S_t^1, \dots, S_t^d)^\top.$$

For the moment there are only two times, $t = 0$ and $t = 1$; this is the **one-period model**. The price S_0 is known at time 0, and S_1 is a random vector whose distribution we assume known. In addition there is a **riskless asset** (asset 0, the bank account) which turns one pound at time 0 into $1 + r$ pounds at time 1.

Definition 43.1.1 (Portfolio)

A **portfolio** is a vector $\theta = (\theta^1, \dots, \theta^d)^\top \in \mathbb{R}^d$, where θ^i is the number of units of asset i held between time 0 and time 1, together with a number $\theta^0 \in \mathbb{R}$ of pounds in the bank.

If $\theta^i > 0$ we say the agent is **long** asset i ; if $\theta^i < 0$ the agent is **short** it, meaning that they have borrowed the asset and sold it, and must return it at time 1.

Let X_t denote the agent's wealth at time t , with $X_0 = x$ given. Since the portfolio must be paid for out of the initial wealth, we have the **budget constraint**

$$X_0 = \theta^0 + \theta^\top S_0.$$

At time 1 the bank deposit has grown and the shares are worth $\theta^\top S_1$, so

$$X_1 = \theta^0(1+r) + \theta^\top S_1 = (1+r)(X_0 - \theta^\top S_0) + \theta^\top S_1.$$

Rearranging gives an expression which will be with us for the rest of the course:

$$X_1 = (1+r)X_0 + \theta^\top (S_1 - (1+r)S_0).$$

In other words – the wealth at time 1 is what you would have got by leaving everything in the bank, plus a term measuring how much better the risky assets did than the bank. We give that term a name.

Definition 43.1.2 (Excess return)

The **excess return vector** of the market is

$$\xi = S_1 - (1+r)S_0 \in \mathbb{R}^d.$$

So $X_1 = (1+r)X_0 + \theta^\top \xi$, and the whole of the one-period theory is really the study of the random vector ξ . Note that ξ is genuinely random – it is the only randomness in the model – and that the map $\theta \mapsto \theta^\top \xi$ is linear. Almost every theorem below is a statement about the linear subspace $\{\theta^\top \xi : \theta \in \mathbb{R}^d\}$ of random variables.

We write $\mathcal{X}(x) = \{(1+r)x + \theta^\top \xi : \theta \in \mathbb{R}^d\}$ for the set of **attainable** time-1 wealths starting from initial wealth x . This is an affine subspace of the space of random variables.

Remark. It is worth pausing to note what has *not* been assumed. We have not assumed that the agent knows anything about S_1 beyond its distribution, and we have certainly not assumed that this distribution is correct. Estimating it is a large industry (and something you might recognise from Part IB Statistics); it is not our concern here.

§43.1.3 A First Example

Let us immediately do a computation, so that the abstractions have something to hang on.

Example 43.1.3 (A one-period binomial market)

Take $d = 1$, interest rate $r = 0$, and $S_0 = 5$. Suppose S_1 takes the value 7 or the value 4, each with probability $\frac{1}{2}$. Consider a **call option** with strike $K = 6$: this is the right, but not the obligation, to buy one share at time 1 for the price 6. If $S_1 = 7$ we exercise the right and make 1; if $S_1 = 4$ we do not, and make 0. So the payout is

$$Y = (S_1 - K)^+ = \begin{cases} 1 & \text{if } S_1 = 7, \\ 0 & \text{if } S_1 = 4. \end{cases}$$

What is this option worth at time 0? A first guess might be $\mathbb{E}Y = \frac{1}{2}$. That guess is wrong, and the reason it is wrong is the whole point of the course.

Observe that $Y = \frac{1}{3}(S_1 - 4)$: indeed $\frac{1}{3}(7 - 4) = 1$ and $\frac{1}{3}(4 - 4) = 0$. So if at time

0 we buy $\frac{1}{3}$ of a share and borrow $\frac{4}{3}$ from the bank, our time-1 wealth is exactly $\frac{1}{3}S_1 - \frac{4}{3} = Y$, whatever happens. The cost of setting this up is

$$\frac{1}{3} \cdot 5 - \frac{4}{3} = \frac{5}{3} - \frac{4}{3} = \frac{1}{3}.$$

So a portfolio costing $\frac{1}{3}$ reproduces the option payout exactly. If the option were sold for more than $\frac{1}{3}$, we could sell the option, set up the portfolio, and pocket the difference with no risk whatsoever; if it were sold for less, we would do the reverse. So the only price consistent with the absence of a free lunch is $\frac{1}{3}$.

There is a second route to the same number. Suppose we pretend the probability of the up-move is q rather than $\frac{1}{2}$, chosen so that the share price is ‘fair’, i.e. $\mathbb{E}_{\mathbb{Q}}[S_1] = (1+r)S_0 = 5$. Then $7q + 4(1-q) = 5$, giving $q = \frac{1}{3}$. And under this measure,

$$\frac{\mathbb{E}_{\mathbb{Q}}[Y]}{1+r} = \frac{1}{3} \cdot 1 + \frac{2}{3} \cdot 0 = \frac{1}{3}.$$

The same answer. That this is not a coincidence is the content of the fundamental theorem of asset pricing, which we prove in §4.

Notice that the true probability $p = \frac{1}{2}$ played no role at all in the answer. This is the single most counterintuitive feature of derivative pricing, and it is worth being clear about why it happens: the option payout is a *deterministic function* of the share price, and the share price is already trading at a known price. All the information about probabilities that we need is already baked into $S_0 = 5$.

§43.2 Mean-Variance Analysis

Our agent has to choose a portfolio θ . On what basis? The oldest quantitative answer, due to Markowitz in 1952, is: get the expected return you want, and take as little risk as possible along the way, where ‘risk’ means variance.

§43.2.1 The Markowitz Problem

Definition 43.2.1 (Mean-variance portfolio problem)

Given initial wealth $X_0 = x$ and a target mean $m \in \mathbb{R}$, the **mean-variance portfolio problem** is to

$$\text{minimise } \text{Var}(X_1) \quad \text{subject to} \quad \mathbb{E}[X_1] = m,$$

over portfolios $\theta \in \mathbb{R}^d$.

Throughout this section we assume that S_1 is square integrable, and we write

$$\mu = \mathbb{E}[S_1], \quad V = \text{Cov}(S_1) = \mathbb{E}[(S_1 - \mu)(S_1 - \mu)^\top].$$

The matrix V is always non-negative definite; we assume in addition that it is *positive* definite, so that V^{-1} exists. (If it is not, some portfolio of the risky assets is riskless, and one should first remove the redundancy.) We also assume $\mu \neq (1+r)S_0$, since otherwise the risky assets offer no expected reward over the bank and the problem is uninteresting.

Since $X_1 = (1+r)x + \theta^\top \xi$ with $\xi = S_1 - (1+r)S_0$, we have

$$\mathbb{E}[X_1] = (1+r)x + \theta^\top (\mu - (1+r)S_0), \quad \text{Var}(X_1) = \theta^\top V \theta.$$

So the problem is: minimise the quadratic form $\theta^\top V \theta$ subject to the affine constraint $\theta^\top (\mu - (1+r)S_0) = m - (1+r)x$. This is a problem you could do with Lagrange multipliers, but there is a slicker argument which also delivers uniqueness for free.

Theorem 43.2.2 (Markowitz)

Write $\nu = \mu - (1+r)S_0$. The mean-variance portfolio problem has the unique solution

$$\theta^* = \lambda V^{-1}\nu, \quad \lambda = \frac{m - (1+r)x}{\nu^\top V^{-1}\nu}.$$

Proof. First, θ^* is feasible: $(\theta^*)^\top \nu = \lambda \nu^\top V^{-1}\nu = m - (1+r)x$, as required.

Now let θ be any other feasible portfolio and set $\epsilon = \theta - \theta^* \neq 0$. Expanding,

$$\theta^\top V \theta = (\theta^*)^\top V \theta^* + 2\epsilon^\top V \theta^* + \epsilon^\top V \epsilon.$$

The cross term vanishes: since both θ and θ^* are feasible, $\epsilon^\top \nu = 0$, and hence

$$\epsilon^\top V \theta^* = \lambda \epsilon^\top V V^{-1}\nu = \lambda \epsilon^\top \nu = 0.$$

As V is positive definite and $\epsilon \neq 0$ we get $\epsilon^\top V \epsilon > 0$, so $\theta^\top V \theta > (\theta^*)^\top V \theta^*$. Hence θ^* is the unique minimiser. $\square$

The vector $V^{-1}\nu$ appearing here is important enough to be named.

Definition 43.2.3 (Market portfolio)

The **market portfolio** is $\theta^{\text{mar}} = V^{-1}(\mu - (1+r)S_0)$, and we write $S_t^{\text{mar}} = (\theta^{\text{mar}})^\top S_t$ for its price.

Corollary 43.2.4 (Mutual fund theorem)

A portfolio is mean-variance efficient if and only if it is a scalar multiple of the market portfolio.

Proof. Immediate from Theorem 43.2.2: the optimal θ^* is $\lambda \theta^{\text{mar}}$, and as m ranges over $\mathbb{R}$, λ ranges over $\mathbb{R}$. $\square$

This is a genuinely striking conclusion. Every investor, no matter how risk-hungry or risk-shy, should hold the risky assets *in the same proportions*; the only thing that distinguishes them is how much they put in the bank. In practice this is the theoretical justification for index funds: if the theory holds, you need exactly two products, a bank account and one fund.

§43.2.2 The Efficient Frontier

As θ ranges over $\mathbb{R}^d$, the pair $(\text{Var}(X_1), \mathbb{E}[X_1])$ traces out a subset of the right half plane. Its left-hand boundary is the object of interest.

Definition 43.2.5 (Efficient frontier)

The **mean-variance efficient frontier** is the set of points

$$\{(\min\{\text{Var}(X_1) : \mathbb{E}[X_1] = m\}, m) : m \in \mathbb{R}\}.$$

A portfolio achieving a point of the frontier is called **mean-variance efficient**.

Corollary 43.2.6 (Shape of the frontier)

In the (v, m) plane the efficient frontier is the parabola

$$v = \frac{(m - (1+r)x)^2}{\nu^\top V^{-1}\nu}, \quad \nu = \mu - (1+r)S_0.$$

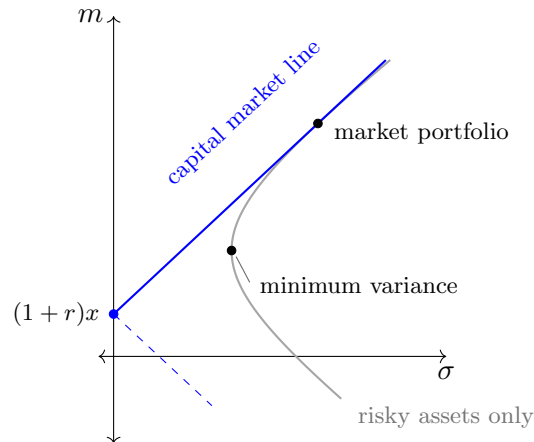
Proof. Substituting $\theta^* = \lambda V^{-1}\nu$ into $v = \theta^\top V \theta$ gives $v = \lambda^2 \nu^\top V^{-1}\nu$, and $\lambda = (m - (1+r)x)/(\nu^\top V^{-1}\nu)$. $\square$

It is more usual, and more informative, to plot the frontier in the (σ, m) plane, where $\sigma = \text{sd}(X_1) = \sqrt{v}$ is the standard deviation. Taking square roots in Corollary 43.2.6, the frontier becomes a pair of *straight lines* through the point $(0, (1+r)x)$, of slopes $\pm\sqrt{\nu^\top V^{-1}\nu}$. Only the upper line is of any interest – given a level of risk one wants the larger mean – and it is called the **capital market line**. Its slope

$$\frac{m - (1+r)x}{\sigma} = \sqrt{\nu^\top V^{-1}\nu}$$

is the **Sharpe ratio**, the amount of excess expected return purchased per unit of risk.

If we forbid the agent from using the bank account, and force them to hold only risky assets, then the corresponding frontier is instead a hyperbola in the (σ, m) plane, with a leftmost point (the **minimum-variance portfolio**). The capital market line is tangent to that hyperbola, and the point of tangency is exactly the market portfolio, normalised to cost x . The picture to hold in your head is the following.



The dashed part of the capital market line corresponds to $\lambda < 0$, i.e. to shorting the market portfolio; those portfolios solve the minimisation problem but no sensible agent would choose them.

Corollary 43.2.7

The problem of minimising $\text{Var}(X_1)$ subject to $\mathbb{E}[X_1] \geq m$ has the same solution as the mean-variance problem whenever $m \geq (1+r)x$. If $m < (1+r)x$, the solution is $\theta = 0$.

Proof. If $m \geq (1+r)x$ then the constraint binds, since variance is increasing in $|\lambda|$ and the mean is increasing in λ . If $m < (1+r)x$ then $\theta = 0$ gives $\text{Var}(X_1) = 0$ and $\mathbb{E}[X_1] = (1+r)x \geq m$, which is clearly optimal. $\square$

Remark. Variance is a strange measure of risk. It punishes unexpectedly *good* outcomes exactly as much as unexpectedly bad ones, and it takes no account of the shape of the tail. We will replace it with something better in §3. On the other hand, mean-variance analysis has the enormous advantage of being solvable in closed form, which is why it is still used.

§43.2.3 The Capital Asset Pricing Model

We now ask an equilibrium question: if *everyone* behaves as in the previous section, what do prices look like? The answer is the capital asset pricing model of Sharpe (1964), and its content is that the expected excess return of any asset is determined by a single number, its exposure to the market as a whole.

We begin with an entirely elementary fact, which is just linear regression.

Proposition 43.2.8

Let X, Y be square integrable random variables with $\text{Var}(X) > 0$. Then there are unique constants α, β such that

$$Y = \alpha + \beta X + Z, \quad \mathbb{E}[Z] = 0, \quad \text{Cov}(Z, X) = 0.$$

Explicitly, $\beta = \text{Cov}(X, Y) / \text{Var}(X)$ and $\alpha = \mathbb{E}[Y] - \beta \mathbb{E}[X]$.

Proof. Setting $Z = Y - \alpha - \beta X$, the two conditions read

$$0 = \mathbb{E}[Y] - \alpha - \beta \mathbb{E}[X], \quad 0 = \text{Cov}(Y, X) - \beta \text{Var}(X),$$

which have the unique solution stated. $\square$

Now let us specialise to the market portfolio. Recall $S_t^{\text{mar}} = (\theta^{\text{mar}})^\top S_t$ and write $\xi = S_1 - (1+r)S_0$, $\xi^{\text{mar}} = S_1^{\text{mar}} - (1+r)S_0^{\text{mar}} = (\theta^{\text{mar}})^\top \xi$.

Theorem 43.2.9

Let $B = \text{Cov}(S_1, S_1^{\text{mar}}) / \text{Var}(S_1^{\text{mar}}) \in \mathbb{R}^d$ and set $Z = \xi - B \xi^{\text{mar}}$. Then $\mathbb{E}[Z] = 0$ and $\text{Cov}(Z, S_1^{\text{mar}}) = 0$.

Proof. Write $\nu = \mu - (1+r)S_0 = \mathbb{E}[\xi]$, so $\theta^{\text{mar}} = V^{-1}\nu$. Then

$$\mathbb{E}[\xi^{\text{mar}}] = (\theta^{\text{mar}})^\top \nu = \nu^\top V^{-1}\nu = (\theta^{\text{mar}})^\top V \theta^{\text{mar}} = \text{Var}(S_1^{\text{mar}}),$$

using $V\theta^{\text{mar}} = \nu$. On the other hand

$$\text{Cov}(S_1, S_1^{\text{mar}}) = \text{Cov}(S_1, S_1)\theta^{\text{mar}} = VV^{-1}\nu = \nu = \mathbb{E}[\xi].$$

Therefore $B = \mathbb{E}[\xi]/\mathbb{E}[\xi^{\text{mar}}]$, whence $\mathbb{E}[Z] = \mathbb{E}[\xi] - B\mathbb{E}[\xi^{\text{mar}}] = 0$. Similarly,

$$\text{Cov}(Z, S_1^{\text{mar}}) = \text{Cov}(S_1, S_1^{\text{mar}}) - B\text{Var}(S_1^{\text{mar}}) = \nu - \nu = 0. \quad \square$$

To turn this into an economic statement we need the market to clear. Suppose asset i exists in total supply n^i , write $n = (n^1, \dots, n^d)^\top$, and define the **index return**

$$R^{\text{index}} = \frac{n^\top S_1}{n^\top S_0} - 1,$$

the return on holding the entire market in proportion to supply. Let $R^i = S_1^i/S_0^i - 1$ be the return on asset i .

Axiom 43.2.10 (CAPM assumptions)

There are K investors, with portfolios $\theta^1, \dots, \theta^K$ satisfying **market clearing** $\sum_k \theta^k = n$. Every investor agrees on the values of $\mu = \mathbb{E}[S_1]$ and $V = \text{Cov}(S_1)$ (*uniform beliefs*) and every investor holds a mean-variance efficient portfolio, though possibly with different target means (*uniform preferences*).

Theorem 43.2.11 (Capital asset pricing model)

Under the above assumptions, write the regression

$$R^i - r = \alpha^i + \beta^i(R^{\text{index}} - r) + \epsilon^i, \quad \mathbb{E}[\epsilon^i] = 0, \quad \text{Cov}(R^{\text{index}}, \epsilon^i) = 0,$$

as in Proposition 43.2.8. Then $\alpha^i = 0$ for every i .

Proof. By the mutual fund theorem each investor's portfolio is $\theta^k = \lambda_k \theta^{\text{mar}}$ for some $\lambda_k \in \mathbb{R}$. Market clearing gives $n = (\sum_k \lambda_k) \theta^{\text{mar}}$, so n is a positive multiple of the market portfolio, and therefore

$$R^{\text{index}} = \frac{n^\top S_1}{n^\top S_0} - 1 = \frac{S_1^{\text{mar}}}{S_0^{\text{mar}}} - 1.$$

Now apply Theorem 43.2.9 coordinatewise. Dividing $\xi^i = B^i \xi^{\text{mar}} + Z^i$ by S_0^i ,

$$R^i - r = \frac{1}{S_0^i} (S_1^i - (1+r)S_0^i) = \frac{B^i S_0^{\text{mar}}}{S_0^i} (R^{\text{index}} - r) + \frac{Z^i}{S_0^i}.$$

Since $\mathbb{E}[Z^i] = 0$ and $\text{Cov}(Z^i, R^{\text{index}}) = 0$, uniqueness in Proposition 43.2.8 identifies $\beta^i = B^i S_0^{\text{mar}}/S_0^i$, $\epsilon^i = Z^i/S_0^i$ and $\alpha^i = 0$. $\square$

So under CAPM the expected excess return on asset i is

$$\mathbb{E}[R^i] - r = \beta^i (\mathbb{E}[R^{\text{index}}] - r),$$

and nothing else about asset i matters. The quantity β^i is a genuinely used piece of market vocabulary. An asset with $\beta = 1.5$ tends to move one and a half times as much

as the market; an asset with $\beta = 0$ is uncorrelated with the market and, however volatile it may be on its own, commands no excess return at all, because its risk can be diversified away.

Remark. It is worth being honest about the assumptions. Uniform beliefs is not plausible: if everyone agreed on μ and V there would be very little trading. Uniform preferences is worse: we have already noted that variance is a poor proxy for risk. And empirically, when one regresses returns against the index, the estimated α 's are not zero – searching for assets with positive α is more or less the definition of active management. Nonetheless CAPM is the standard baseline against which such claims are measured.

§43.3 Utility and Risk Aversion

Mean-variance analysis fixes the mean and minimises variance. This is unsatisfactory in two ways: it does not let an agent trade off return against risk, and variance is the wrong notion of risk. We now set up a framework which does both.

§43.3.1 The Expected Utility Hypothesis

Definition 43.3.1 (Expected utility hypothesis)

Every agent has a **utility function** $U : \mathbb{R} \rightarrow \mathbb{R}$ (or defined on an interval), and prefers a random payout X to a random payout Y – written $X \succ Y$ – if and only if

$$\mathbb{E}[U(X)] > \mathbb{E}[U(Y)].$$

If $\mathbb{E}[U(X)] = \mathbb{E}[U(Y)]$ the agent is **indifferent** between them, written $X \sim Y$.

Two immediate observations. First, if $\tilde{U} = aU + b$ with $a > 0$, then $\tilde{U}$ induces exactly the same preferences as U ; utility functions are only defined up to positive affine transformations, and so ‘utility’ is not a quantity one can meaningfully compare across agents. Second, the criterion depends on X only through its *distribution*, so we may equally think of preferences as an ordering on distribution functions.

Historically the hypothesis was introduced by Daniel Bernoulli to resolve the following puzzle.

Example 43.3.2 (The St. Petersburg paradox)

A fair coin is tossed until it first comes up heads. If this takes n tosses, the casino pays you 2^n pounds. What should you pay to play?

The expected payout is

$$\sum_{n \geq 1} 2^{-n} \cdot 2^n = \sum_{n \geq 1} 1 = \infty,$$

so if one maximises expected wealth one should be willing to pay any finite entry fee. Almost nobody would pay even £100. Bernoulli’s resolution: an agent maximises not expected wealth but expected utility, and if $U(x) = \log x$ then the expected

utility of the game is

$$\sum_{n \geq 1} 2^{-n} \log(2^n) = \log 2 \sum_{n \geq 1} n 2^{-n} = 2 \log 2 = \log 4,$$

so the game is worth exactly £4 to a logarithmic agent. The moral is that a fixed amount of money is worth less to you when you are rich.

A more principled motivation is a representation theorem. Suppose an agent has some preference relation $\succ$ on distribution functions, about which we assume only the following.

Definition 43.3.3 (von Neumann–Morgenstern axioms)

A preference relation $\succ$ on distribution functions satisfies the **von Neumann–Morgenstern axioms** if

- (i) (*Completeness*) for all F, G , exactly one of $F \succ G$, $G \succ F$, $F \sim G$ holds;
- (ii) (*Transitivity*) if $F \succ G$ and $G \succ H$ then $F \succ H$;
- (iii) (*Independence*) if $F \succ G$ and $p \in [0, 1]$ then $pF + (1 - p)H \succ pG + (1 - p)H$ for every H ;
- (iv) (*Continuity*) if $F \succ G \succ H$ then there is $p \in [0, 1]$ with $G \sim pF + (1 - p)H$.

Theorem 43.3.4 (von Neumann–Morgenstern)

If $\succ$ satisfies the above axioms then there exists a function U such that

$$F \succ G \iff \int U \, dF > \int U \, dG.$$

We shall not prove this. The point is that the expected utility hypothesis is not an arbitrary modelling choice: it is forced on us by a short list of conditions each of which looks, at first glance, like a requirement of bare rationality. (The independence axiom is the controversial one, and there is a well-developed literature of experiments in which people violate it.)

§43.3.2 Concavity, Jensen and the Certainty Equivalent

We will almost always assume that U is increasing, and very often that it is concave.

Definition 43.3.5

A utility function U is **increasing** if $X > Y$ almost surely implies $\mathbb{E}[U(X)] > \mathbb{E}[U(Y)]$, and is **concave** if

$$U(px + (1 - p)y) \geq pU(x) + (1 - p)U(y) \quad \text{for all } x, y \in \mathbb{R}, p \in [0, 1].$$

An agent with a concave utility function is called **risk averse**.

If U is twice differentiable then increasing means $U' > 0$ and concave means $U'' \leq 0$. In words: more money is better, but each additional pound is worth slightly less than the

one before.

The connection between concavity and dislike of risk is Jensen's inequality, which you met in Part IA Probability.

Proposition 43.3.6

If U is concave and X is integrable then $\mathbb{E}[U(X)] \leq U(\mathbb{E}[X])$. Hence a risk-averse agent weakly prefers the certain payout $\mathbb{E}[X]$ to the random payout X .

This suggests a way of measuring *how much* an agent dislikes a given gamble.

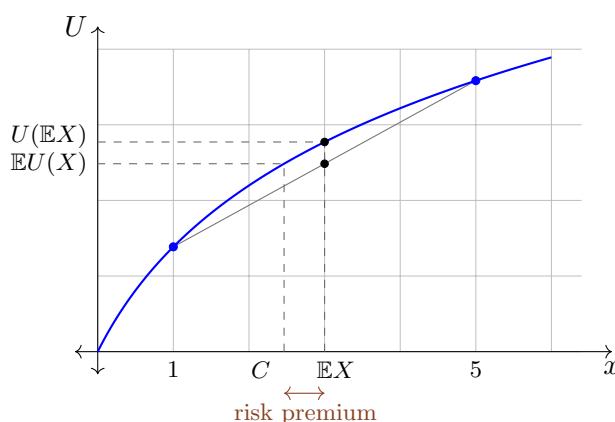
Definition 43.3.7 (Certainty equivalent)

Let U be increasing and continuous and let X be a payout. The **certainty equivalent** of X is the unique $C(X) \in \mathbb{R}$ with

$$U(C(X)) = \mathbb{E}[U(X)],$$

i.e. the sure amount which the agent values exactly as highly as the gamble. The **risk premium** is $\mathbb{E}[X] - C(X)$.

By Jensen, a risk-averse agent has $C(X) \leq \mathbb{E}[X]$, so the risk premium is non-negative: it is the discount the agent demands for bearing the risk. The whole construction is visible in a single picture. Take $U(x) = \log(1+x)$ and let X take the values 1 and 5 with probability $\frac{1}{2}$ each, so $\mathbb{E}[X] = 3$.



The chord joining the two outcomes lies below the graph; its midpoint has height $\mathbb{E}[U(X)]$, and reading horizontally across to the curve and then down gives the certainty equivalent $C \approx 2.46$. The gap between C and $\mathbb{E}X = 3$ is the risk premium, and it is exactly the vertical gap between the chord and the curve, transported through U . A more concave utility bends the curve further away from the chord and so demands a larger premium.

§43.3.3 Arrow–Pratt Risk Aversion

To make ‘more concave’ precise we want a number. The obvious candidate $-U''$ is not good, because U is only defined up to positive affine transformations and $-U''$ is not invariant under $U \mapsto aU + b$. Dividing by U' fixes this.

Definition 43.3.8 (Arrow–Pratt coefficients)

Let U be twice differentiable with $U' > 0$. The **coefficient of absolute risk aversion** at x is

$$A(x) = -\frac{U''(x)}{U'(x)},$$

and the **coefficient of relative risk aversion** at x is

$$R(x) = -\frac{xU''(x)}{U'(x)} = xA(x).$$

Both are invariant under $U \mapsto aU + b$ for $a > 0$, as they should be. The interpretation is the following: if an agent with wealth x is offered a small gamble with mean zero and variance σ^2 , then a second-order Taylor expansion gives

$$\mathbb{E}[U(x + \epsilon)] \approx U(x) + \frac{1}{2}U''(x)\sigma^2, \quad U(x - \pi) \approx U(x) - U'(x)\pi,$$

so the risk premium is $\pi \approx \frac{1}{2}A(x)\sigma^2$. Absolute risk aversion measures the premium demanded for a gamble of a fixed size; relative risk aversion measures the premium for a gamble proportional to wealth.

Two families of utility functions are singled out by asking for these coefficients to be constant.

Example 43.3.9 (CARA utility)

A utility with **constant absolute risk aversion** $\gamma > 0$ satisfies $-U''/U' = \gamma$, so $(\log U')' = -\gamma$ and $U'(x) = Ce^{-\gamma x}$. Integrating,

$$U(x) = -\frac{C}{\gamma}e^{-\gamma x},$$

and normalising, ‘the’ CARA utility is $U(x) = -e^{-\gamma x}$, defined on all of $\mathbb{R}$.

Example 43.3.10 (CRRA utility)

A utility with **constant relative risk aversion** $R > 0$ satisfies $-xU''/U' = R$, so $(\log U')' = -R/x$ and $U'(x) = Cx^{-R}$ for $x > 0$. If $R \neq 1$, integrating gives

$$U(x) = \frac{x^{1-R}}{1-R}, \quad x > 0,$$

up to affine equivalence. If $R = 1$ the integral produces a logarithm, and we obtain the **logarithmic utility** $U(x) = \log x$, $x > 0$.

Remark. CARA utility has the convenient property that the optimal *amount of money* invested in the risky assets does not depend on wealth; CRRA utility has the property that the optimal *proportion* of wealth invested does not. The latter is closer to how people appear to behave, and CRRA utilities are also the ones that combine well with lognormal models, since $\mathbb{E}[e^{(1-R)Z}]$ is computable for Gaussian Z . Note that CRRA utility with $R > 1$ has $U(x) \rightarrow -\infty$ as $x \downarrow 0$, so such an agent will never accept a strategy which can bankrupt them.

§43.3.4 The Utility Maximisation Problem

We can now pose the central optimisation problem of the one-period theory: given a utility U and initial wealth x ,

$$\text{maximise } \mathbb{E}[U(X_1)] \quad \text{over } X_1 \in \mathcal{X}(x),$$

that is, maximise

$$f(\theta) := \mathbb{E}\left[U((1+r)x + \theta^\top \xi)\right] \quad \text{over } \theta \in \mathbb{R}^d.$$

Theorem 43.3.11 (First-order condition)

Suppose U is differentiable and θ^* is optimal, with $X_1^* = (1+r)x + (\theta^*)^\top \xi$. Suppose further that differentiation under the expectation is justified. Then

$$\mathbb{E}[U'(X_1^*) \xi] = 0, \quad \text{i.e.} \quad \mathbb{E}[U'(X_1^*)(S_1 - (1+r)S_0)] = 0.$$

Proof. The function f is differentiable with $\nabla f(\theta) = \mathbb{E}[U'((1+r)x + \theta^\top \xi) \xi]$, provided we may exchange ∇ and $\mathbb{E}$; this is justified by dominated convergence when, for instance, U is concave and differentiable and $U((1+r)x + \theta^\top \xi)$ is integrable for all θ in a neighbourhood of θ^* . Since θ^* is an interior maximum (there are no constraints), $\nabla f(\theta^*) = 0$. $\square$

Note that if U is concave then f is concave, so the first-order condition is sufficient as well as necessary. The equation says that the marginal utility of the optimal wealth is ‘orthogonal’ to every excess return, which is exactly the statement that no small perturbation of the portfolio helps.

§43.3.5 State Price Densities

The first-order condition can be rewritten in a way that removes all mention of the agent, and this turns out to be the key to everything that follows.

Definition 43.3.12 (State price density)

A **state price density** (or **pricing kernel**) is an almost surely positive random variable Z with

$$\mathbb{E}[Z] = \frac{1}{1+r} \quad \text{and} \quad \mathbb{E}[ZS_1] = S_0.$$

The name ‘pricing kernel’ is transparent: multiplying by Z and taking expectations converts time-1 prices into time-0 prices. Note the first condition is just the second applied to the bank account, whose time-1 value is the constant $1+r$.

To see why it is called a *state price* density, consider a market on $\Omega = \{\omega_1, \dots, \omega_d\}$ in which asset i is the **Arrow–Debreu security** paying $S_1^i = \mathbb{1}_{\{\omega_i\}}$, i.e. one pound if state i occurs and nothing otherwise. Then $\mathbb{E}[ZS_1^i] = Z(\omega_i)\mathbb{P}(\{\omega_i\}) = S_0^i$, so

$$Z(\omega_i) = \frac{S_0^i}{\mathbb{P}(\{\omega_i\})}.$$

So Z is the price of a pound delivered in state ω_i , per unit of probability – a density of state prices with respect to $\mathbb{P}$.

Theorem 43.3.13

In the setting of Theorem 43.3.11, suppose additionally that $U' > 0$ and $\mathbb{E}[U'(X_1^*)] < \infty$. Then

$$Z = \frac{U'(X_1^*)}{(1+r) \mathbb{E}[U'(X_1^*)]}$$

is a state price density.

Proof. Certainly $Z > 0$ and $\mathbb{E}[Z] = 1/(1+r)$. For the second condition, Theorem 43.3.11 gives $\mathbb{E}[U'(X_1^*)(S_1 - (1+r)S_0)] = 0$, i.e.

$$\mathbb{E}[U'(X_1^*)S_1] = (1+r)S_0 \mathbb{E}[U'(X_1^*)].$$

Dividing by $(1+r)\mathbb{E}[U'(X_1^*)]$ gives $\mathbb{E}[ZS_1] = S_0$. □

So *if* an agent with an increasing utility can solve their optimisation problem, *then* a state price density exists. This is our first sight of the fundamental theorem: the existence of a solution to some agent's problem forces a consistency condition on the market. In §4 we will strip away the agent entirely and find that the correct hypothesis is simply the absence of arbitrage.

§43.4 Arbitrage and the Fundamental Theorem

We now come to the heart of the course. The idea is to eliminate the agent altogether: rather than asking what a particular investor with a particular utility would pay, we ask what prices are consistent with the impossibility of a free lunch. Remarkably, this is enough to determine prices exactly in a wide class of models.

§43.4.1 Equivalent Measures

We first need a small amount of measure-theoretic machinery for changing probability measures.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let Y be an almost surely positive random variable with $\mathbb{E}_{\mathbb{P}}[Y] = 1$. Define

$$\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}[Y\mathbb{1}_A], \quad A \in \mathcal{F}.$$

Then $\mathbb{Q}(\Omega) = 1$ and countable additivity follows from monotone convergence, so $\mathbb{Q}$ is a probability measure. Moreover, since $Y > 0$ almost surely, $\mathbb{Q}(A) = 0$ if and only if $\mathbb{P}(A) = 0$.

Definition 43.4.1 (Equivalent measures)

Two probability measures $\mathbb{P}, \mathbb{Q}$ on $(\Omega, \mathcal{F})$ are **equivalent**, written $\mathbb{P} \sim \mathbb{Q}$, if they have the same null sets: $\mathbb{P}(A) = 0 \iff \mathbb{Q}(A) = 0$.

Equivalent measures agree on what is possible and what is impossible; they disagree about how likely the possible things are. Every construction above produces an equivalent measure, and the Radon–Nikodym theorem says that these are the only ones.

Theorem 43.4.2 (Radon–Nikodym)

$\mathbb{P}$ and $\mathbb{Q}$ are equivalent if and only if there is an almost surely positive random variable Y , unique up to null sets, with $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}[Y \mathbb{1}_A]$ for all $A \in \mathcal{F}$.

We write $Y = d\mathbb{Q}/d\mathbb{P}$ and call it the **Radon–Nikodym derivative**, or the **likelihood ratio** – a name which should be familiar from Part IB Statistics, where exactly the same object appears in the Neyman–Pearson lemma. The essential computational fact is that for $\mathbb{Q}$ -integrable Z ,

$$\mathbb{E}_{\mathbb{Q}}[Z] = \mathbb{E}_{\mathbb{P}} \left[\frac{d\mathbb{Q}}{d\mathbb{P}} Z \right],$$

which holds for indicators by definition and in general by the usual approximation argument.

Example 43.4.3

If $\Omega = \{\omega_1, \dots, \omega_n\}$ is finite and $\mathbb{P}(\{\omega_i\}) > 0$ for each i , then $\mathbb{P} \sim \mathbb{Q}$ exactly when $\mathbb{Q}(\{\omega_i\}) > 0$ for each i , and

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega_i) = \frac{\mathbb{Q}(\{\omega_i\})}{\mathbb{P}(\{\omega_i\})}.$$

Example 43.4.4 (Changing the mean of a Gaussian)

Let $X \sim N(\mu, \sigma^2)$ under $\mathbb{P}$ and set $Y = e^{X - \mu - \sigma^2/2}$. Then $Y > 0$ and, using the moment generating function of the normal, $\mathbb{E}_{\mathbb{P}}[Y] = e^{-\mu - \sigma^2/2} \mathbb{E}_{\mathbb{P}}[e^X] = 1$. Let $d\mathbb{Q}/d\mathbb{P} = Y$. For bounded measurable f ,

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[f(X)] &= \mathbb{E}_{\mathbb{P}}[e^{X - \mu - \sigma^2/2} f(X)] \\ &= \int_{\mathbb{R}} e^{x - \mu - \sigma^2/2} f(x) \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x - \mu)^2/2\sigma^2} dx \\ &= \int_{\mathbb{R}} f(x) \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x - \mu - \sigma^2)^2/2\sigma^2} dx, \end{aligned}$$

on completing the square. So under $\mathbb{Q}$ we have $X \sim N(\mu + \sigma^2, \sigma^2)$: the change of measure has shifted the mean and left the variance alone. This little computation is the seed of the Cameron–Martin theorem in §7, and hence of the entire Black–Scholes derivation.

§43.4.2 Risk-Neutral Measures**Definition 43.4.5** (Risk-neutral measure)

A **risk-neutral measure** (or **equivalent martingale measure**) for the one-period market is a probability measure $\mathbb{Q} \sim \mathbb{P}$ with

$$\mathbb{E}_{\mathbb{Q}} \left[\frac{S_1}{1+r} \right] = S_0.$$

Under $\mathbb{Q}$, every asset has the same expected return as the bank account, so a $\mathbb{Q}$ -agent is indifferent between them – hence ‘risk neutral’. The connection with state price densities

is immediate.

Proposition 43.4.6

$\mathbb{Q}$ is a risk-neutral measure if and only if $Z = \frac{1}{1+r} \frac{d\mathbb{Q}}{d\mathbb{P}}$ is a state price density. In particular, state price densities and risk-neutral measures are in bijection.

Proof. If $\mathbb{Q} \sim \mathbb{P}$ with density Y , then $Z = Y/(1+r) > 0$ and $\mathbb{E}_{\mathbb{P}}[Z] = 1/(1+r)$ automatically. Furthermore $\mathbb{E}_{\mathbb{P}}[ZS_1] = \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[S_1]$, which equals S_0 precisely when $\mathbb{Q}$ is risk-neutral. Conversely, given a state price density Z , put $Y = (1+r)Z$; then $Y > 0$ and $\mathbb{E}_{\mathbb{P}}[Y] = 1$, so Y defines an equivalent measure $\mathbb{Q}$, which is risk-neutral by the same computation. $\square$

Example 43.4.7 (The one-period binomial model)

Take $d = 1$ and suppose

$$\mathbb{P}(S_1 = (1+b)S_0) = p = 1 - \mathbb{P}(S_1 = (1+a)S_0), \quad a < b, \quad p \in (0, 1).$$

A measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ is determined by $q = \mathbb{Q}(S_1 = (1+b)S_0) \in (0, 1)$, and risk-neutrality demands

$$q(1+b)S_0 + (1-q)(1+a)S_0 = (1+r)S_0 \implies q = \frac{r-a}{b-a}.$$

This lies in $(0, 1)$ if and only if $a < r < b$. So a risk-neutral measure exists exactly when the bank rate lies strictly between the two possible returns on the stock, and then it is unique. Both halves of this statement will be explained by the theorems below.

43.4.3 Arbitrage

Definition 43.4.8 (Arbitrage)

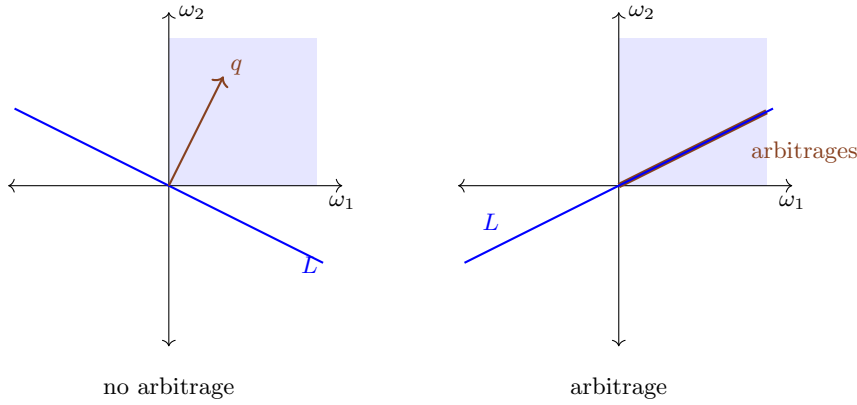
A portfolio $\phi \in \mathbb{R}^d$ is an **arbitrage** if

$$\phi^\top \xi \geq 0 \quad \text{a.s.} \quad \text{and} \quad \mathbb{P}(\phi^\top \xi > 0) > 0,$$

where $\xi = S_1 - (1+r)S_0$. A market with no arbitrage is called **arbitrage-free**.

An arbitrage is a portfolio which costs nothing over and above what the bank would have given you, never loses, and sometimes wins: a free lunch. Note that the definition involves neither a utility function nor an initial wealth, and depends on $\mathbb{P}$ only through its null sets. In particular equivalent measures have exactly the same arbitrages.

Here is the geometry, in the simplest possible case: $d = 1$ and two states ω_1, ω_2 . Identify a random variable with the point of $\mathbb{R}^2$ recording its values in the two states. The attainable excess payoffs form the line $L = \{\theta\xi : \theta \in \mathbb{R}\}$ through the origin, and an arbitrage is a point of L lying in the closed positive quadrant but not at the origin.



On the left the line meets the shaded quadrant only at the origin, and there is a vector q with strictly positive entries orthogonal to L ; normalising q to be a probability vector produces exactly a risk-neutral measure. On the right the line runs into the interior of the quadrant, and every point of the thick segment is an arbitrage; no strictly positive vector can be orthogonal to L . The fundamental theorem is the assertion that this dichotomy is completely general.

Before proving it, let us record the link with the previous section.

Proposition 43.4.9

If the market admits an arbitrage, then no increasing utility function admits an optimal portfolio.

Proof. Let ϕ be an arbitrage and θ any portfolio, with $X_1 = (1+r)x + \theta^\top \xi$ and $\tilde{X}_1 = (1+r)x + (\theta + \phi)^\top \xi$. Then $\tilde{X}_1 \geq X_1$ almost surely, with strict inequality on an event of positive probability. Since U is increasing, $\mathbb{E}[U(\tilde{X}_1)] > \mathbb{E}[U(X_1)]$, so θ was not optimal. $\square$

Combining Proposition 43.4.9 with Theorem 43.3.13: if some agent can solve their optimisation problem then there is no arbitrage, and there is a state price density. The fundamental theorem removes the agent from this chain of reasoning.

§43.4.4 The Fundamental Theorem of Asset Pricing

Theorem 43.4.10 (Fundamental theorem of asset pricing)

The one-period market admits no arbitrage if and only if there exists a risk-neutral measure.

Proof. ($\Leftarrow$) Suppose $\mathbb{Q}$ is risk-neutral and ϕ is an arbitrage. Since $\mathbb{P} \sim \mathbb{Q}$ we have $\phi^\top \xi \geq 0$ $\mathbb{Q}$ -almost surely too. But risk-neutrality says $\mathbb{E}_{\mathbb{Q}}[\xi] = \mathbb{E}_{\mathbb{Q}}[S_1] - (1+r)S_0 = 0$, so

$$\mathbb{E}_{\mathbb{Q}}[\phi^\top \xi] = \phi^\top \mathbb{E}_{\mathbb{Q}}[\xi] = 0.$$

A non-negative random variable with zero expectation vanishes almost surely, so $\phi^\top \xi = 0$ $\mathbb{Q}$ -a.s., hence $\mathbb{P}$ -a.s., contradicting $\mathbb{P}(\phi^\top \xi > 0) > 0$.

($\Rightarrow$) Suppose there is no arbitrage; we construct a risk-neutral measure. We may

make two harmless reductions. First, we may assume the components $\xi^1, \dots, \xi^d$ are linearly independent as random variables: otherwise, discard assets until they are, since a redundant asset contributes nothing new to $\{\theta^\top \xi\}$. Second, we may assume that $\mathbb{E}[e^{-\theta^\top \xi}] < \infty$ for every $\theta \in \mathbb{R}^d$: if not, replace $\mathbb{P}$ by the equivalent measure $\tilde{\mathbb{P}}$ with $d\tilde{\mathbb{P}}/d\mathbb{P} \propto e^{-\|\xi\|^2}$, which has the same arbitrages, and note that $e^{-\theta^\top \xi - \|\xi\|^2}$ is bounded.

Now consider

$$f(\theta) = \mathbb{E}[e^{-\theta^\top \xi}], \quad \theta \in \mathbb{R}^d,$$

which is finite, strictly convex and smooth. Choose (θ_n) with $f(\theta_n) \rightarrow \inf_\theta f(\theta)$.

Case 1: (θ_n) has a bounded subsequence. By Bolzano–Weierstrass we may assume $\theta_n \rightarrow \theta_0$. Continuity of moment generating functions gives $f(\theta_0) = \inf_\theta f$, so θ_0 is a minimiser and

$$\nabla f(\theta_0) = -\mathbb{E}[e^{-\theta_0^\top \xi} \xi] = 0.$$

Setting $d\mathbb{Q}/d\mathbb{P} = e^{-\theta_0^\top \xi} / \mathbb{E}[e^{-\theta_0^\top \xi}]$ defines an equivalent measure (the density is strictly positive) with $\mathbb{E}_{\mathbb{Q}}[\xi] = 0$, i.e. a risk-neutral measure.

Case 2: $\|\theta_n\| \rightarrow \infty$. We derive a contradiction. Put $\phi_n = \theta_n / \|\theta_n\|$; by compactness of the unit sphere we may assume $\phi_n \rightarrow \phi_0$ with $\|\phi_0\| = 1$. We claim $\phi_0^\top \xi \geq 0$ almost surely.

Suppose not, so that $\mathbb{P}(\phi_0^\top \xi < -\epsilon, \|\xi\| < R) > 0$ for some $\epsilon, R > 0$. For n large enough that $\|\phi_n - \phi_0\| \leq \epsilon/(2R)$, on the event $\{\phi_0^\top \xi < -\epsilon, \|\xi\| < R\}$ we have

$$\phi_n^\top \xi = (\phi_n - \phi_0)^\top \xi + \phi_0^\top \xi \leq \|\phi_n - \phi_0\| \|\xi\| + \phi_0^\top \xi \leq \frac{\epsilon}{2} - \epsilon = -\frac{\epsilon}{2}.$$

Since $f(\theta_n) \leq f(0) = 1$ for large n ,

$$1 \geq \mathbb{E}[e^{-\|\theta_n\| \phi_n^\top \xi}] \geq \mathbb{E}[e^{-\|\theta_n\| \phi_n^\top \xi} \mathbb{1}_{\{\phi_0^\top \xi < -\epsilon, \|\xi\| < R\}}] \geq e^{\epsilon \|\theta_n\|/2} \mathbb{P}(\phi_0^\top \xi < -\epsilon, \|\xi\| < R).$$

The right-hand side tends to infinity as $n \rightarrow \infty$, which is absurd. Hence $\phi_0^\top \xi \geq 0$ a.s.

By the absence of arbitrage, $\phi_0^\top \xi \geq 0$ a.s. forces $\phi_0^\top \xi = 0$ a.s. But the ξ^i are linearly independent, so $\phi_0 = 0$, contradicting $\|\phi_0\| = 1$. So Case 2 cannot occur, and we are done. $\square$

Remark. The proof is really a separating hyperplane argument in disguise. The convex function $f(\theta) = \mathbb{E}[e^{-\theta^\top \xi}]$ is a smooth surrogate for the indicator of the ‘no free lunch’ condition, and Case 2 is exactly the statement that if f has no minimiser then the recession direction ϕ_0 is an arbitrage. In finite sample spaces one can give a purely finite-dimensional proof using the separating hyperplane theorem, and the picture above is that proof.

§43.4.5 Contingent Claims and Payoff Diagrams

We now enlarge the market by an extra asset.

Definition 43.4.11 (Contingent claim)

A **contingent claim** with maturity 1 is a random variable Y , interpreted as a

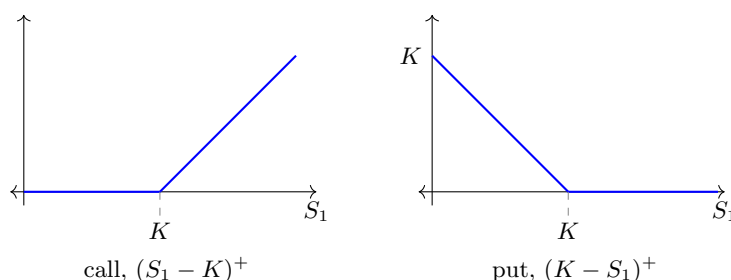
payout at time 1. It is **vanilla** if $Y = g(S_1)$ for some function g .

The two basic examples are the following.

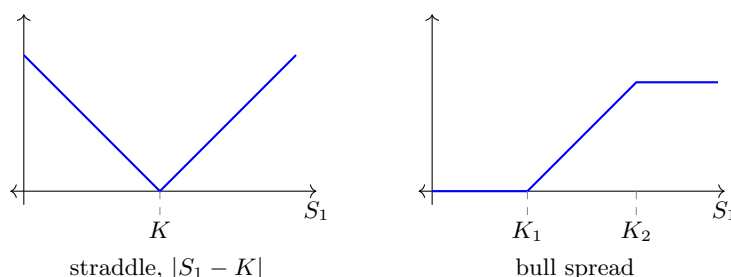
Definition 43.4.12 (Call and put)

A **European call option** with **strike** K is the right, but not the obligation, to buy one unit of the asset at time 1 for the price K ; its payout is $(S_1 - K)^+$. A **European put option** with strike K is the right to sell at price K , with payout $(K - S_1)^+$.

It is customary and extremely useful to draw the payout as a function of S_1 .



Because the space of claims is a vector space and pricing will turn out to be linear, one can build more elaborate payoffs by adding these together. Two standard combinations:



A **straddle** is a call plus a put with the same strike, and pays off when the price moves a long way in either direction: it is a bet on volatility rather than on direction. A **bull spread** is a long call at strike K_1 and a short call at strike $K_2 > K_1$; it is a bet that the price will rise, but with the upside capped in exchange for a lower initial cost. Notice that the bull spread payoff is bounded, which is often what a regulator wants to see.

§43.4.6 No-Arbitrage Pricing

Suppose the fundamental market (the d assets and the bank) is arbitrage-free, and we add a claim with payout Y and initial price π . When is the enlarged market still arbitrage-free?

Theorem 43.4.13 (No-arbitrage pricing)

Let the fundamental market be arbitrage-free. The augmented market containing the claim (Y, π) is arbitrage-free if and only if

$$\pi = \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[Y]$$

for some risk-neutral measure $\mathbb{Q}$ of the fundamental market.

Proof. Apply Theorem 43.4.10 to the augmented market, whose price vectors are $(S_0, \pi)^\top$ and $(S_1, Y)^\top$. The augmented market is arbitrage-free if and only if there is $\mathbb{Q} \sim \mathbb{P}$ with

$$\frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[(S_1, Y)^\top] = (S_0, \pi)^\top.$$

The first d coordinates say exactly that $\mathbb{Q}$ is risk-neutral for the fundamental market, and the last says $\pi = \mathbb{E}_{\mathbb{Q}}[Y]/(1+r)$. $\square$

Proposition 43.4.14

The set of no-arbitrage prices of a claim Y is an interval.

Proof. If $\mathbb{Q}_0, \mathbb{Q}_1$ are risk-neutral, so is $\mathbb{Q}_t = t\mathbb{Q}_1 + (1-t)\mathbb{Q}_0$ for $t \in [0, 1]$: it is a probability measure, it is equivalent to $\mathbb{P}$ (its null sets are the common null sets), and $\mathbb{E}_{\mathbb{Q}_t}[S_1] = t\mathbb{E}_{\mathbb{Q}_1}[S_1] + (1-t)\mathbb{E}_{\mathbb{Q}_0}[S_1] = (1+r)S_0$. The corresponding price is $\pi_t = t\pi_1 + (1-t)\pi_0$, so the set of prices is convex. $\square$

So a claim either has a unique no-arbitrage price, or a whole interval of them, in which case the model simply does not determine the price. The good case has a name.

Definition 43.4.15 (Attainable claim)

A claim Y is **attainable** (or **replicable**) if $Y = a + b^\top S_1$ for some $a \in \mathbb{R}$, $b \in \mathbb{R}^d$; equivalently, if $Y \in \mathcal{X}(y)$ for some y . The portfolio achieving this is called a **replicating portfolio** or a **hedge**.

Theorem 43.4.16

If $Y \in \mathcal{X}(y)$ is attainable, then its unique no-arbitrage price is y , the initial cost of the replicating portfolio.

Proof. Write $Y = a + b^\top S_1$, so $y = a/(1+r) + b^\top S_0$. For any risk-neutral $\mathbb{Q}$,

$$\frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[Y] = \frac{a}{1+r} + \frac{1}{1+r} b^\top \mathbb{E}_{\mathbb{Q}}[S_1] = \frac{a}{1+r} + b^\top S_0 = y,$$

independently of $\mathbb{Q}$. By Theorem 43.4.13 this is the unique no-arbitrage price. $\square$

This is the **law of one price**: two portfolios with the same payout must have the same price. Our very first example was an instance – there the call was attainable, $Y = \frac{1}{3}(S_1 - 4)$, and the price $\frac{1}{3}$ was the cost of the hedge.

The converse of Theorem 43.4.16 is also true, though we shall not prove it here (it is on the example sheet): a claim with a unique no-arbitrage price is attainable.

§43.4.7 Complete Markets

Definition 43.4.17 (Complete market)

A market model is **complete** if every contingent claim is attainable.

Proposition 43.4.18

In a complete market model, S_1 takes at most $d + 1$ values.

Proof. Let $A_1, \dots, A_n$ be disjoint events of positive probability. The indicators $\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_n}$ are linearly independent random variables. By completeness each $\mathbb{1}_{A_i} = a_i + b_i^\top S_1$, so

$$\text{span}\{\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_n}\} \subseteq \text{span}\{1, S_1^1, \dots, S_1^d\},$$

a space of dimension at most $d + 1$. Hence $n \leq d + 1$. As S_1 taking n distinct values would give n such disjoint events, the result follows. $\square$

So completeness is a strong requirement: with one stock ($d = 1$) it forces a binomial model. This is precisely why the binomial model is the workhorse of discrete-time finance.

Theorem 43.4.19 (Second fundamental theorem of asset pricing)

An arbitrage-free market model is complete if and only if the risk-neutral measure is unique.

Proof. ($\Rightarrow$) Suppose the market is complete and $\mathbb{Q}_0, \mathbb{Q}_1$ are both risk-neutral. Set

$$Y = \frac{d\mathbb{Q}_1}{d\mathbb{P}} - \frac{d\mathbb{Q}_0}{d\mathbb{P}},$$

which is a bounded-below claim (in a complete market S_1 takes finitely many values, so Y is a genuine random variable and everything is integrable). Since Y is attainable, write $Y = a + b^\top S_1$; then for $i = 0, 1$,

$$\mathbb{E}_{\mathbb{Q}_i}[Y] = a + b^\top \mathbb{E}_{\mathbb{Q}_i}[S_1] = a + (1 + r)b^\top S_0,$$

the same value for both. But

$$\mathbb{E}_{\mathbb{Q}_1}[Y] - \mathbb{E}_{\mathbb{Q}_0}[Y] = \mathbb{E}_{\mathbb{P}} \left[\left(\frac{d\mathbb{Q}_1}{d\mathbb{P}} - \frac{d\mathbb{Q}_0}{d\mathbb{P}} \right) Y \right] = \mathbb{E}_{\mathbb{P}}[Y^2],$$

so $\mathbb{E}_{\mathbb{P}}[Y^2] = 0$ and hence $Y = 0$ a.s., i.e. $\mathbb{Q}_0 = \mathbb{Q}_1$.

($\Leftarrow$) If the risk-neutral measure is unique then by Theorem 43.4.13 every claim has exactly one no-arbitrage price, and hence (by the converse to Theorem 43.4.16) is attainable. $\square$

Example 43.4.20

Return to Example 43.4.7. If $a < r < b$ the model is arbitrage-free with a unique risk-neutral measure $q = (r - a)/(b - a)$, hence complete: every claim $Y = g(S_1)$ is

replicable, and

$$\pi = \frac{1}{1+r} \left(q g((1+b)S_0) + (1-q) g((1+a)S_0) \right).$$

The replicating portfolio is obtained by solving two linear equations in two unknowns; the number of shares held is

$$\theta = \frac{g((1+b)S_0) - g((1+a)S_0)}{(b-a)S_0},$$

the ratio of the change in payout to the change in stock price. This quantity will reappear as the *delta* of the claim, and it is the single most important number in practical derivative trading.

§43.4.8 Utility Indifference Pricing

Theorem 43.4.13 pins down the price of a claim only when the claim is attainable. When it is not, we must go back to preferences. This subsection is a little more technical and may be skipped on a first reading, but it explains *why* the risk-neutral price is the right answer even in incomplete markets, at least for small trades.

Definition 43.4.21 (Indifference price)

Fix a utility U and initial wealth x . The **utility indifference price** (or **reservation bid price**) of a claim Y is the largest π with

$$\sup\{\mathbb{E}[U(X+Y)] : X \in \mathcal{X}(x-\pi)\} \geq \sup\{\mathbb{E}[U(X)] : X \in \mathcal{X}(x)\}.$$

In words – the most the agent would pay for Y while remaining at least as happy as if they had not bought it. We write $F_Y(\pi)$ for the left-hand side; note that $X \in \mathcal{X}(x-\pi)$ if and only if $X + (1+r)\pi \in \mathcal{X}(x)$, so

$$F_Y(\pi) = \sup\{\mathbb{E}[U(X - (1+r)\pi + Y)] : X \in \mathcal{X}(x)\},$$

which is decreasing and (given mild regularity) continuous in π , with $F_Y(\mp\infty) = U(\pm\infty)$. Thus $\pi(Y)$ is the unique solution of $F_Y(\pi) = F_0(0)$.

Theorem 43.4.22

The map $Y \mapsto \pi(Y)$ is concave.

Proof. Fix claims Y_0, Y_1 and $t \in [0, 1]$, and set $Y_t = tY_1 + (1-t)Y_0$. Let X_i attain the supremum defining $F_{Y_i}(\pi(Y_i))$ for $i = 0, 1$. Then $tX_1 + (1-t)X_0 \in \mathcal{X}(x - t\pi(Y_1) - (1-t)\pi(Y_0))$ by linearity, and so, using concavity of U ,

$$\begin{aligned} F_{Y_t}(\pi(Y_t)) &= F_0(0) = tF_{Y_1}(\pi(Y_1)) + (1-t)F_{Y_0}(\pi(Y_0)) \\ &= t\mathbb{E}[U(X_1 + Y_1)] + (1-t)\mathbb{E}[U(X_0 + Y_0)] \\ &\leq \mathbb{E}[U(tX_1 + (1-t)X_0 + Y_t)] \\ &\leq F_{Y_t}(t\pi(Y_1) + (1-t)\pi(Y_0)). \end{aligned}$$

Since F_{Y_t} is decreasing, $\pi(Y_t) \geq t\pi(Y_1) + (1-t)\pi(Y_0)$. $\square$

Concavity says that an investor values diversification: half of one claim plus half of another is worth at least the average of the two. It also implies that $t \mapsto \pi(tY)/t$ is decreasing, so the limit

$$\pi_0 := \lim_{t \downarrow 0} \frac{\pi(tY)}{t}$$

exists. This is the **marginal** price of the claim – the price per unit for an infinitesimally small trade.

Lemma 43.4.23 (Supporting hyperplane)

If U is concave and differentiable then $U(y) - U(x) \leq U'(x)(y - x)$ for all x, y .

Proof. For $x < x + \epsilon < y$, concavity of U gives that the chord slopes decrease, so

$$\frac{U(y) - U(x)}{y - x} \leq \frac{U(x + \epsilon) - U(x)}{\epsilon}.$$

Letting $\epsilon \downarrow 0$ gives the claim, and the case $y < x$ is symmetric. $\square$

Lemma 43.4.24

If X^* maximises $\mathbb{E}[U(X+Y)]$ over $X \in \mathcal{X}(x)$, then $\mathbb{E}[U'(X^*+Y)\xi] = 0$; consequently $\mathbb{E}[U'(X^*+Y)(X^* - X)] = 0$ for all $X \in \mathcal{X}(x)$.

Proof. Identical to Theorem 43.3.11, differentiating in θ . The second claim follows since $X^* - X$ is of the form $\eta^\top \xi$. $\square$

Theorem 43.4.25 (Marginal pricing)

Let X_0 maximise $\mathbb{E}[U(X)]$ over $X \in \mathcal{X}(x)$. Then

$$\pi_0 = \frac{\mathbb{E}[U'(X_0)Y]}{(1+r)\mathbb{E}[U'(X_0)]} = \mathbb{E}[ZY],$$

where Z is the state price density of Theorem 43.3.13.

Proof. Write $\pi_t = \pi(tY)/t$ and let X_t maximise $\mathbb{E}[U(X+tY)]$ over $X \in \mathcal{X}(x - t\pi_t)$. By definition of the indifference price, $\mathbb{E}[U(X_t + tY)] = \mathbb{E}[U(X_0)]$.

For the upper bound, note that $X_0 - t\pi_t(1+r) \in \mathcal{X}(x - t\pi_t)$, so it is a candidate in the supremum defining X_t , whence

$$\begin{aligned} 0 = \frac{\mathbb{E}[U(X_t + tY) - U(X_0)]}{t} &\geq \frac{\mathbb{E}[U(X_0 - t\pi_t(1+r) + tY) - U(X_0)]}{t} \\ &\xrightarrow{t \downarrow 0} \mathbb{E}[U'(X_0)(Y - (1+r)\pi_0)] \end{aligned}$$

using $\pi_t \downarrow \pi_0$ and dominated convergence. So $\mathbb{E}[U'(X_0)(Y - (1+r)\pi_0)] \leq 0$.

For the lower bound, apply Lemma 43.4.23 and then Lemma 43.4.24:

$$\begin{aligned}
 0 &= \frac{\mathbb{E}[U(X_t + tY) - U(X_0)]}{t} \leq \frac{\mathbb{E}[U'(X_0)(X_t + tY - X_0)]}{t} \\
 &= \mathbb{E}[U'(X_0)(Y - (1+r)\pi_t)] \\
 &\quad + \frac{1}{t}\mathbb{E}[U'(X_0)(X_t + t\pi_t(1+r) - X_0)] \\
 &= \mathbb{E}[U'(X_0)(Y - (1+r)\pi_t)],
 \end{aligned}$$

since $X_t + t\pi_t(1+r) \in \mathcal{X}(x)$ and Lemma 43.4.24 (with $Y = 0$) makes the last expectation vanish. Letting $t \downarrow 0$ gives $\mathbb{E}[U'(X_0)(Y - (1+r)\pi_0)] \geq 0$.

Combining the two bounds gives equality, which rearranges to the stated formula. $\square$

So for small trades the indifference price is linear, and the linear functional is given by a state price density – equivalently, by a risk-neutral expectation. This is the sense in which risk-neutral pricing is the right answer even when replication is impossible: it is the first-order approximation to what an optimising agent would pay.

§43.5 Martingales in Discrete Time

To do finance in more than one period we must be able to say what an agent knows at each time. The mathematical device for this is the σ -algebra, and the resulting theory – conditional expectation and martingales – is the technical backbone of everything that follows.

§43.5.1 Information and σ -Algebras

Suppose we toss a coin twice, so $\Omega = \{HH, HT, TH, TT\}$. After the first toss we know whether the outcome lies in $\{HH, HT\}$ or in $\{TH, TT\}$, but nothing more. So the events we can resolve are

$$\mathcal{G} = \{\emptyset, \{HH, HT\}, \{TH, TT\}, \Omega\}.$$

The general shape of such a collection is clear: if we can resolve A we can resolve A^c ; if we can resolve each of $A_1, A_2, \dots$ we can resolve $\bigcup_n A_n$; and we can always resolve $\emptyset$. That is exactly the definition of a σ -algebra.

Definition 43.5.1 (σ -algebra)

A σ -algebra on a set Ω is a collection $\mathcal{G}$ of subsets of Ω with $\emptyset \in \mathcal{G}$, closed under complements and under countable unions. A random variable X is $\mathcal{G}$ -measurable if $X^{-1}(B) \in \mathcal{G}$ for every Borel set $B \subseteq \mathbb{R}$; equivalently, if $\{X \leq x\} \in \mathcal{G}$ for every $x \in \mathbb{R}$.

The intuition is: $\mathcal{G}$ is ‘the information available’, and X is $\mathcal{G}$ -measurable if X is known once that information is revealed. The two extreme cases are the **trivial σ -algebra** $\{\emptyset, \Omega\}$ (no information – the only measurable random variables are constants) and the power set 2^Ω (everything is known).

Definition 43.5.2

The σ -algebra **generated** by a random variable X is $\sigma(X) = \{X^{-1}(B) : B \text{ Borel}\}$, the smallest σ -algebra making X measurable. More generally $\sigma(X_1, \dots, X_n)$ is the smallest σ -algebra making all of $X_1, \dots, X_n$ measurable.

Fact 43.5.3

A random variable Y is $\sigma(X)$ -measurable if and only if $Y = g(X)$ for some Borel measurable g .

This is the precise sense in which ‘knowing X ’ means ‘knowing every function of X , and nothing else’. We shall use it without further comment.

§43.5.2 Conditional Expectation

In Part IA Probability you defined $\mathbb{E}[X | A] = \mathbb{E}[X \mathbb{1}_A] / \mathbb{P}(A)$ for an event A of positive probability, and $\mathbb{E}[X | Y]$ for a discrete random variable Y as the random variable taking value $\mathbb{E}[X | Y = y]$ on $\{Y = y\}$. We now want to condition on a whole σ -algebra, including ones generated by continuous random variables where every individual event has probability zero. The trick is to characterise the answer rather than construct it.

Note first the key property of the discrete construction: if $f(y) = \mathbb{E}[X | Y = y]$ then for every bounded $\sigma(Y)$ -measurable Z ,

$$\mathbb{E}[XZ] = \mathbb{E}[f(Y)Z],$$

as one checks by decomposing over the values of Y . We turn this into a definition.

Definition 43.5.4 (Conditional expectation)

Let X be an integrable random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{G} \subseteq \mathcal{F}$ a σ -algebra. A random variable $\tilde{X}$ is a **conditional expectation** of X given $\mathcal{G}$ if

- (i) $\tilde{X}$ is $\mathcal{G}$ -measurable and integrable;
- (ii) $\mathbb{E}[XZ] = \mathbb{E}[\tilde{X}Z]$ for every bounded $\mathcal{G}$ -measurable Z .

Proposition 43.5.5

A conditional expectation exists and is almost surely unique. We write it $\mathbb{E}[X | \mathcal{G}]$, and $\mathbb{E}[X | Y] := \mathbb{E}[X | \sigma(Y)]$.

Proof. We prove uniqueness, existence being a genuinely measure-theoretic matter (it follows from the Radon–Nikodym theorem, or from the L^2 projection argument below). Suppose $\tilde{X}_0$ and $\tilde{X}_1$ both work. Take $Z = \mathbb{1}_{\{\tilde{X}_0 < \tilde{X}_1\}}$, which is bounded and $\mathcal{G}$ -measurable. Then

$$\mathbb{E}[(\tilde{X}_1 - \tilde{X}_0) \mathbb{1}_{\{\tilde{X}_0 < \tilde{X}_1\}}] = \mathbb{E}[XZ] - \mathbb{E}[XZ] = 0.$$

The integrand is non-negative and strictly positive on $\{\tilde{X}_0 < \tilde{X}_1\}$, so $\mathbb{P}(\tilde{X}_0 < \tilde{X}_1) = 0$. By symmetry $\mathbb{P}(\tilde{X}_0 > \tilde{X}_1) = 0$. $\square$

The following interpretation is worth recording, since it explains why conditional expectation is ‘the best guess’.

Proposition 43.5.6 (L^2 projection)

Let X be square integrable and $\tilde{X} = \mathbb{E}[X \mid \mathcal{G}]$. Then for every square-integrable $\mathcal{G}$ -measurable Z ,

$$\mathbb{E}[(X - \tilde{X})^2] \leq \mathbb{E}[(X - Z)^2].$$

Proof. Expand:

$$\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - \tilde{X})^2] + 2\mathbb{E}[(X - \tilde{X})(\tilde{X} - Z)] + \mathbb{E}[(\tilde{X} - Z)^2].$$

The middle term vanishes: $\tilde{X} - Z$ is $\mathcal{G}$ -measurable, so the defining property gives $\mathbb{E}[X(\tilde{X} - Z)] = \mathbb{E}[\tilde{X}(\tilde{X} - Z)]$ (first for bounded $\tilde{X} - Z$, then in general by truncation).

The last term is non-negative. $\square$

So $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace of $\mathcal{G}$ -measurable square-integrable random variables. All the familiar properties now follow.

Proposition 43.5.7 (Properties of conditional expectation)

Assume all the conditional expectations below are well defined. Then, almost surely:

- (i) (*Linearity*) $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$;
- (ii) (*Taking out what is known*) if Y is $\mathcal{G}$ -measurable then $\mathbb{E}[XY \mid \mathcal{G}] = Y \mathbb{E}[X \mid \mathcal{G}]$;
- (iii) (*Positivity*) if $X \geq 0$ a.s. then $\mathbb{E}[X \mid \mathcal{G}] \geq 0$ a.s.;
- (iv) (*Independence*) if $\sigma(X)$ is independent of $\mathcal{G}$ then $\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}[X]$;
- (v) (*Jensen*) if f is convex then $\mathbb{E}[f(X) \mid \mathcal{G}] \geq f(\mathbb{E}[X \mid \mathcal{G}])$;
- (vi) (*Tower property*) if $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$ then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{G}]$.

Proof. (i) Both sides are $\mathcal{G}$ -measurable and satisfy the defining identity, so uniqueness applies. (ii) For bounded $\mathcal{G}$ -measurable Z we have $\mathbb{E}[(XY)Z] = \mathbb{E}[X(YZ)] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]YZ]$, since YZ is $\mathcal{G}$ -measurable (first for bounded Y , then by truncation). (iii) Take $Z = \mathbb{1}_{\{\mathbb{E}[X \mid \mathcal{G}] < 0\}}$; then $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]Z] = \mathbb{E}[XZ] \geq 0$, whereas the left-hand side is ≤ 0 and is < 0 unless $\mathbb{P}(\mathbb{E}[X \mid \mathcal{G}] < 0) = 0$. (iv) Immediate from the definition, since $\mathbb{E}[XZ] = \mathbb{E}[X]\mathbb{E}[Z]$ for $\mathcal{G}$ -measurable Z . (v) A convex f satisfies $f(y) \geq f(x) + \lambda(x)(y - x)$ for a suitable $\lambda(x)$; put $x = \mathbb{E}[X \mid \mathcal{G}]$, $y = X$ and condition. (vi) $\mathbb{E}[X \mid \mathcal{G}]$ is $\mathcal{H}$ -measurable so equals its own conditional expectation given $\mathcal{H}$; for the other order, any bounded $\mathcal{G}$ -measurable Z is $\mathcal{H}$ -measurable, so $\mathbb{E}[\mathbb{E}[X \mid \mathcal{H}]Z] = \mathbb{E}[XZ]$. $\square$

Example 43.5.8

Let $(A_n)_{n \geq 1}$ partition Ω with $\mathbb{P}(A_n) > 0$ and let $\mathcal{G} = \sigma(A_1, A_2, \dots)$. A random variable is $\mathcal{G}$ -measurable exactly when it is constant on each A_n , and

$$\mathbb{E}[X \mid \mathcal{G}] = \sum_n \mathbb{E}[X \mid A_n] \mathbb{1}_{A_n},$$

which recovers the Part IA definition.

§43.5.3 Filtrations and Martingales

Definition 43.5.9 (Filtration)

A **filtration** on $(\Omega, \mathcal{F})$ is an increasing sequence $(\mathcal{F}_n)_{n \geq 0}$ of σ -algebras, $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}$. A stochastic process $(X_n)_{n \geq 0}$ is **adapted** to $(\mathcal{F}_n)$ if X_n is $\mathcal{F}_n$ -measurable for every n . The filtration **generated** by a process (X_n) is $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$.

We adopt the convention that $\mathcal{F}_0 = \{\emptyset, \Omega\}$ is trivial, so X_0 is a constant and $\mathbb{E}[Z \mid \mathcal{F}_0] = \mathbb{E}[Z]$ for every integrable Z .

Definition 43.5.10 (Martingale)

An adapted, integrable process $(X_n)_{n \geq 0}$ is a **martingale** with respect to $(\mathcal{F}_n)$ if

$$\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = X_{n-1} \quad \text{for all } n \geq 1.$$

It is a **supermartingale** if $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \leq X_{n-1}$ and a **submartingale** if $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \geq X_{n-1}$.

A martingale is a fair game: given everything you know today, your expected fortune tomorrow is exactly your fortune today. A supermartingale is a game which is unfavourable to you (it goes *down* on average – the terminology is unfortunate but universal), and a submartingale is a favourable one.

Proposition 43.5.11

For an adapted integrable process (X_n) the following are equivalent:

- (i) (X_n) is a martingale;
- (ii) $\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = 0$ for all $n \geq 1$;
- (iii) $\mathbb{E}[X_n \mid \mathcal{F}_m] = X_m$ for all $0 \leq m \leq n$.

Proof. (i) $\iff$ (ii) is immediate from linearity, and (iii) $\implies$ (i) is the case $m = n - 1$. For (i) $\implies$ (iii), induct on $n - m$ using the tower property:

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \mid \mathcal{F}_m] = \mathbb{E}[X_{n-1} \mid \mathcal{F}_m] = X_m. \quad \square$$

Condition (iii) makes no reference to ‘the previous time step’, and so is the definition that generalises to continuous time.

Example 43.5.12 (Martingales built backwards)

Let Z be integrable and $(\mathcal{F}_n)$ a filtration. Then $X_n = \mathbb{E}[Z \mid \mathcal{F}_n]$ is a martingale. Integrability follows from conditional Jensen, $\mathbb{E}|X_n| \leq \mathbb{E}[\mathbb{E}[|Z| \mid \mathcal{F}_n]] = \mathbb{E}|Z| < \infty$, and the martingale property is the tower property. Every martingale on a finite horizon arises this way, with $Z = X_N$.

Example 43.5.13 (Martingales built forwards) (i) Let (Y_n) be independent integrable random variables with $\mathbb{E}[Y_n] = 0$, and let $\mathcal{F}_n = \sigma(Y_1, \dots, Y_n)$. Then $S_0 = 0$, $S_n = Y_1 + \dots + Y_n$ defines a martingale, since

$$\mathbb{E}[S_n - S_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[Y_n] = 0$$

by independence. In particular the simple symmetric random walk is a martingale.

(ii) If instead (Y_n) are independent, positive, with $\mathbb{E}[Y_n] = 1$, then $M_0 = 1$, $M_n = Y_1 \cdots Y_n$ is a martingale by the same computation.

(iii) If (S_n) is a simple symmetric random walk then $Q_n = S_n^2 - n$ is a martingale: indeed $\mathbb{E}[S_n^2 \mid \mathcal{F}_{n-1}] = S_{n-1}^2 + 2S_{n-1}\mathbb{E}[Y_n] + \mathbb{E}[Y_n^2] = S_{n-1}^2 + 1$.

The definition immediately relevant to us is the following, which we shall justify in §6.

Definition 43.5.14 (Risk-neutral measure, multi-period)

For a multi-period market with prices $(S_n)_{n \geq 0}$ adapted to $(\mathcal{F}_n)$ and interest rate r , a probability measure $\mathbb{Q} \sim \mathbb{P}$ is **risk-neutral** if the discounted price process $(S_n/(1+r)^n)_{n \geq 0}$ is a $\mathbb{Q}$ -martingale, i.e. if

$$\mathbb{E}_{\mathbb{Q}}[S_n \mid \mathcal{F}_{n-1}] = (1+r)S_{n-1} \quad \text{for all } n \geq 1.$$

§43.5.4 Stopping Times and Optional Stopping

Definition 43.5.15 (Stopping time)

A random variable T taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ is a **stopping time** for the filtration $(\mathcal{F}_n)$ if $\{T \leq n\} \in \mathcal{F}_n$ for every n .

The condition says that we can decide whether to stop at time n using only information available at time n – no peeking into the future.

Example 43.5.16 (i) If (X_n) is adapted then the first hitting time $T = \inf\{n \geq 0 : X_n > 0\}$ (with $\inf \emptyset = \infty$) is a stopping time, since $\{T \leq n\} = \bigcup_{k \leq n} \{X_k > 0\} \in \mathcal{F}_n$.

(ii) The last time $T = \sup\{n \geq 0 : X_n > 0\}$ is generally *not* a stopping time: $\{T \leq n\} = \bigcap_{k > n} \{X_k \leq 0\}$ requires knowledge of the whole future. ‘Sell at the top’ is not an implementable strategy.

Definition 43.5.17 (Stopped process)

Given a process X and a random time T , the **stopped process** X^T is defined by

$$X_n^T = X_{n \wedge T} = X_0 + \sum_{k=1}^n \mathbb{1}_{\{k \leq T\}} (X_k - X_{k-1}).$$

Proposition 43.5.18

If X is integrable and adapted and T is a stopping time, then X^T is integrable and adapted.

Proof. Integrability follows from the triangle inequality applied to the sum. For adaptedness, note $\{k \leq T\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1} \subseteq \mathcal{F}_n$ for $k \leq n$, and $X_k - X_{k-1}$ is $\mathcal{F}_n$ -measurable. $\square$

Theorem 43.5.19 (Optional sampling)

Let X be a martingale and T a stopping time. Then X^T is a martingale. In particular, if $T \leq N$ almost surely for some constant N , then $\mathbb{E}[X_T] = X_0$.

Proof. We know X^T is adapted and integrable, so it remains to compute, for $n \geq 1$,

$$\begin{aligned} \mathbb{E}[X_n^T - X_{n-1}^T \mid \mathcal{F}_{n-1}] &= \mathbb{E}[\mathbb{1}_{\{n \leq T\}} (X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] \\ &= \mathbb{1}_{\{n \leq T\}} \mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = 0, \end{aligned}$$

where we took out the $\mathcal{F}_{n-1}$ -measurable factor $\mathbb{1}_{\{n \leq T\}}$. If $T \leq N$ a.s. then $X_T = X_{T \wedge N} = X_N^T$, and $\mathbb{E}[X_N^T] = X_0^T = X_0$. $\square$

The boundedness hypothesis is essential.

Example 43.5.20 (Why boundedness matters)

Let (S_n) be a simple symmetric random walk from 0 and let $T = \inf\{n \geq 0 : S_n = b\}$ for $b > 0$. From Part IB Markov Chains we know the walk is recurrent, so $T < \infty$ a.s. and $S_T = b$. Then $\mathbb{E}[S_T] = b \neq 0 = S_0$. Optional sampling fails, because T is unbounded and unboundedly large excursions occur before T .

This is not a mere technicality: it is the martingale-theoretic reason why the ‘double your bet until you win’ strategy does not actually work. It requires an unbounded amount of credit, and $\mathbb{E}[\min_{n \leq T} S_n] = -\infty$.

To handle unbounded stopping times we need an integrability condition.

Theorem 43.5.21 (Optional stopping)

Let X be a martingale and T a stopping time with $T < \infty$ a.s. Suppose there is an integrable random variable Y with $|X_n^T| \leq Y$ for all n . Then $X_n^T \rightarrow X_T$ a.s. and $\mathbb{E}[X_T] = X_0$.

Proof. Since $T < \infty$ a.s., $X_n^T = X_T$ for all $n \geq T$, giving almost sure convergence. By Theorem 43.5.19, $\mathbb{E}[X_n^T] = X_0$ for every n ; the dominated convergence theorem lets us pass to the limit. $\square$

In particular the conclusion holds if T is bounded, or if T is integrable and the increments of X are bounded, or if X^T is bounded. Let us see the last of these in action.

Example 43.5.22 (Gambler's ruin, again)

Let (S_n) be a simple symmetric random walk from 0 and $T = \inf\{n \geq 0 : S_n \in \{-a, b\}\}$ for integers $a, b > 0$. Then $|S_n^T| \leq \max(a, b)$, and $T < \infty$ a.s. (again by recurrence), so Theorem 43.5.21 gives

$$0 = \mathbb{E}[S_T] = -a\mathbb{P}(S_T = -a) + b\mathbb{P}(S_T = b).$$

With $\mathbb{P}(S_T = -a) + \mathbb{P}(S_T = b) = 1$ this yields

$$\mathbb{P}(S_T = -a) = \frac{b}{a+b}, \quad \mathbb{P}(S_T = b) = \frac{a}{a+b},$$

which is the gambler's ruin formula from Part IA. Applying the same argument to the martingale $Q_n = S_n^2 - n$ and the bounded stopping time $T \wedge n$,

$$\mathbb{E}[S_{T \wedge n}^2] = \mathbb{E}[T \wedge n].$$

The right-hand side increases to $\mathbb{E}[T]$ by monotone convergence and the left-hand side converges to $\mathbb{E}[S_T^2]$ by dominated convergence, so

$$\mathbb{E}[T] = \mathbb{E}[S_T^2] = a^2 \frac{a}{a+b} \cdot \frac{b}{a} + b^2 \frac{a}{a+b} = ab,$$

after simplification: the expected duration is ab .

§43.5.5 Martingale Transforms

Definition 43.5.23 (Previsible process)

A process $(H_n)_{n \geq 1}$ is **previsible** (or **predictable**) for $(\mathcal{F}_n)$ if H_n is $\mathcal{F}_{n-1}$ -measurable for every $n \geq 1$.

Definition 43.5.24 (Martingale transform)

The **martingale transform** of X by H is the process

$$(H \cdot X)_n = \sum_{k=1}^n H_k (X_k - X_{k-1}), \quad (H \cdot X)_0 = 0.$$

Think of $X_k - X_{k-1}$ as the winnings per unit stake in game k and H_k as the stake, chosen before the game is played. Then $(H \cdot X)_n$ is the total winnings. The martingale transform is the discrete-time ancestor of the stochastic integral $\int H \, dX$.

Theorem 43.5.25 (You cannot beat the system)

Let H be previsible and bounded.

- (i) If X is a martingale then so is $H \cdot X$.
- (ii) If X is a supermartingale and $H \geq 0$, then $H \cdot X$ is a supermartingale.

Proof. In both cases $H \cdot X$ is adapted (since H_k is $\mathcal{F}_{k-1}$ -measurable and X adapted) and integrable (since H is bounded). Then

$$\mathbb{E}[(H \cdot X)_n - (H \cdot X)_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[H_n(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] = H_n \mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}],$$

which is 0 in case (i) and ≤ 0 in case (ii) as $H_n \geq 0$. $\square$

The name is apt: no gambling strategy, however clever, can turn a fair game into a favourable one, provided your stakes are chosen without knowledge of the future and you are not allowed to bet unboundedly.

Notice that the stopped process is a special case: $X^T - X_0 = H \cdot X$ with $H_k = \mathbb{1}_{\{k \leq T\}}$, which is previsible precisely because T is a stopping time.

§43.5.6 Supermartingales and the Snell Envelope

For optimal stopping problems we need the supermartingale analogue of optional sampling.

Theorem 43.5.26 (Optional sampling for supermartingales)

Let X be a supermartingale and let $S \leq T$ be bounded stopping times. Then $\mathbb{E}[X_S] \geq \mathbb{E}[X_T]$.

Proof. Suppose $T \leq N$. Writing the difference as a sum,

$$X_{N \wedge T} - X_{N \wedge S} = \sum_{k=1}^N \mathbb{1}_{\{S < k \leq T\}} (X_k - X_{k-1}).$$

Now $\{S < k\} = \{S \leq k-1\} \in \mathcal{F}_{k-1}$ and $\{k \leq T\} = \{T \leq k-1\}^c \in \mathcal{F}_{k-1}$, so $\mathbb{1}_{\{S < k \leq T\}}$ is $\mathcal{F}_{k-1}$ -measurable. Hence

$$\mathbb{E}[X_T - X_S] = \sum_{k=1}^N \mathbb{E} \left[\mathbb{1}_{\{S < k \leq T\}} \mathbb{E}[X_k - X_{k-1} \mid \mathcal{F}_{k-1}] \right] \leq 0,$$

since each conditional expectation is ≤ 0 and the indicator is non-negative. $\square$

Now consider the following optimal stopping problem. Fix a horizon N , an adapted process $(Y_n)_{0 \leq n \leq N}$ of payouts, and ask for

$$\sup \{ \mathbb{E}[Y_T] : T \text{ a stopping time with } T \leq N \}.$$

Definition 43.5.27 (Snell envelope)

The **Snell envelope** of $(Y_n)_{n \leq N}$ is the process defined by backward recursion

$$Z_N = Y_N, \quad Z_{n-1} = \max \{Y_{n-1}, \mathbb{E}[Z_n | \mathcal{F}_{n-1}]\}.$$

At each time we compare the payout from stopping now with the expected value of carrying on optimally; the Snell envelope is the value of the problem.

Theorem 43.5.28 (Optimal stopping)

The Snell envelope (Z_n) is the smallest supermartingale dominating (Y_n) . Moreover

$$T^* = \min\{n \geq 0 : Z_n = Y_n\}$$

is an optimal stopping time, and $Z_0 = \mathbb{E}[Y_{T^*}] = \sup_{T \leq N} \mathbb{E}[Y_T]$.

Proof. By construction $Z_n \geq Y_n$ and $Z_{n-1} \geq \mathbb{E}[Z_n | \mathcal{F}_{n-1}]$, so Z is a supermartingale dominating Y . If W is any other such supermartingale then $W_N \geq Y_N = Z_N$, and inductively

$$W_{n-1} \geq \max\{Y_{n-1}, \mathbb{E}[W_n | \mathcal{F}_{n-1}]\} \geq \max\{Y_{n-1}, \mathbb{E}[Z_n | \mathcal{F}_{n-1}]\} = Z_{n-1},$$

so Z is the smallest.

For any stopping time $T \leq N$, Theorem 43.5.26 applied with $S = 0$ gives $\mathbb{E}[Y_T] \leq \mathbb{E}[Z_T] \leq Z_0$.

It remains to show equality for T^* . The definition of T^* gives $Z_n > Y_n$ for $n < T^*$, so on $\{n \leq T^*\}$ we have $Z_{n-1} = \mathbb{E}[Z_n | \mathcal{F}_{n-1}]$ for $n \leq T^*$; hence the stopped process Z^{T^*} satisfies

$$\mathbb{E}[Z_n^{T^*} - Z_{n-1}^{T^*} | \mathcal{F}_{n-1}] = \mathbb{1}_{\{n \leq T^*\}} (\mathbb{E}[Z_n | \mathcal{F}_{n-1}] - Z_{n-1}) = 0,$$

so Z^{T^*} is a martingale. Therefore $Z_0 = \mathbb{E}[Z_N^{T^*}] = \mathbb{E}[Z_{T^*}] = \mathbb{E}[Y_{T^*}]$, using $Z_{T^*} = Y_{T^*}$. $\square$

This is the engine that will price American options. Note the structure of the answer: one solves a backward recursion, and the optimal rule is ‘stop as soon as the payout from stopping equals the value of the problem’.

§43.5.7 Convergence

We record, without proof, the two convergence results that we shall occasionally need. The first is the martingale convergence theorem; you will meet it properly in Part II Probability and Measure.

Theorem 43.5.29 (Martingale convergence)

Let (X_n) be a martingale bounded in L^1 , i.e. $\sup_n \mathbb{E}|X_n| < \infty$. Then $X_n \rightarrow X_\infty$ almost surely for some integrable X_∞ .

In general $\mathbb{E}[X_\infty] \neq X_0$: Example 43.5.20 is a counterexample if one truncates suitably.

What is needed for the limit to be ‘honest’ is uniform integrability, and then $X_n = \mathbb{E}[X_\infty | \mathcal{F}_n]$ for all n .

Theorem 43.5.30 (Monotone and dominated convergence)

If $0 \leq X_n \leq X_{n+1}$ a.s. then $\mathbb{E}[X_n] \rightarrow \mathbb{E}[\lim_n X_n]$. If $X_n \rightarrow X$ a.s. and $|X_n| \leq Y$ for an integrable Y , then $\mathbb{E}[X_n] \rightarrow \mathbb{E}[X]$.

§43.6 Multi-Period Models

We now let the market run for N periods. The state space carries a filtration $(\mathcal{F}_n)_{0 \leq n \leq N}$, prices (S_n) are adapted, and the bank account grows by a factor $1 + r$ each period. Everything from §4 will survive, with ‘random variable’ replaced by ‘adapted process’ and ‘expectation’ by ‘conditional expectation’.

§43.6.1 Self-Financing Portfolios

Let $\theta_n \in \mathbb{R}^d$ be the portfolio of risky assets held over the interval $(n-1, n]$, and θ_n^0 the amount in the bank over the same interval. These must be chosen at time $n-1$, so we require $(\theta_n)_{n \geq 1}$ to be *previsible*. Let X_n be the agent’s wealth at time n .

Definition 43.6.1 (Self-financing)

A previsible portfolio process (θ_n) is **self-financing** if no money is added to or removed from the portfolio after time 0; that is, if

$$X_{n-1} = \theta_n^0 + \theta_n^\top S_{n-1} \quad \text{and} \quad X_n = \theta_n^0(1+r) + \theta_n^\top S_n$$

for all $n \geq 1$.

Eliminating θ_n^0 exactly as in the one-period case gives the wealth recursion

$$X_n = (1+r)X_{n-1} + \theta_n^\top (S_n - (1+r)S_{n-1}).$$

It is neater in discounted terms. Write $\tilde{X}_n = X_n/(1+r)^n$ and $\tilde{S}_n = S_n/(1+r)^n$ for the discounted wealth and prices, i.e. prices measured in units of the bank account rather than in pounds. Dividing the recursion by $(1+r)^n$,

$$\tilde{X}_n - \tilde{X}_{n-1} = \theta_n^\top (\tilde{S}_n - \tilde{S}_{n-1}),$$

and summing,

$$\tilde{X}_n = X_0 + \sum_{k=1}^n \theta_k^\top (\tilde{S}_k - \tilde{S}_{k-1}) = X_0 + (\theta \cdot \tilde{S})_n.$$

In other words – the discounted wealth of a self-financing strategy is the initial wealth plus the martingale transform of the strategy against the discounted prices. Theorem 43.5.25 now does an enormous amount of work for us.

Remark. The change from pounds to units of the bank account is our first **change of numeraire**. A **numeraire** is any asset with strictly positive price which we use as the unit of account; discounting is the statement that in the numeraire of the bank account the riskless asset has constant price 1. Other choices are possible and sometimes much more convenient; we take this up in §6.3.

§43.6.2 Arbitrage in Multi-Period Models

Definition 43.6.2 (Arbitrage)

A previsible process $(\theta_n)_{1 \leq n \leq N}$ is an **arbitrage** if the terminal discounted gain $G = \sum_{k=1}^N \theta_k^\top (\tilde{S}_k - \tilde{S}_{k-1})$ satisfies

$$\mathbb{P}(G \geq 0) = 1 \quad \text{and} \quad \mathbb{P}(G > 0) > 0.$$

Theorem 43.6.3 (Fundamental theorem of asset pricing)

A finite-horizon market model admits no arbitrage if and only if there exists a risk-neutral measure, that is, a $\mathbb{Q} \sim \mathbb{P}$ under which $(\tilde{S}_n)_{0 \leq n \leq N}$ is a martingale.

Proof. ($\Leftarrow$) Let $\mathbb{Q}$ be risk-neutral and let θ be previsible and bounded. By Theorem 43.5.25 the discounted wealth $\tilde{X}_n = X_0 + (\theta \cdot \tilde{S})_n$ is a $\mathbb{Q}$ -martingale, so

$$\mathbb{E}_{\mathbb{Q}}[G] = \mathbb{E}_{\mathbb{Q}}[\tilde{X}_N] - X_0 = 0.$$

If θ were an arbitrage then $G \geq 0$ $\mathbb{P}$ -a.s., hence $\mathbb{Q}$ -a.s. since $\mathbb{P} \sim \mathbb{Q}$; a non-negative random variable of zero mean vanishes a.s., so $G = 0$ $\mathbb{Q}$ -a.s. and therefore $\mathbb{P}$ -a.s., contradicting $\mathbb{P}(G > 0) > 0$. (For unbounded θ one first localises using the stopping times $T_m = \inf\{n : \|\theta_{n+1}\| > m\}$ and then lets $m \rightarrow \infty$.)

($\Rightarrow$) Conversely, suppose there is no arbitrage. Then in particular, for each n there is no one-period arbitrage conditional on $\mathcal{F}_{n-1}$: if η were an $\mathcal{F}_{n-1}$ -measurable random vector with $\eta^\top (\tilde{S}_n - \tilde{S}_{n-1}) \geq 0$ a.s. and positive with positive probability, then the strategy which is η at time n and zero otherwise would be an arbitrage. Applying the one-period Theorem 43.4.10 conditionally on $\mathcal{F}_{n-1}$ produces, for each n , a strictly positive $\mathcal{F}_n$ -measurable Y_n with $\mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] = 1$ and $\mathbb{E}[Y_n \tilde{S}_n \mid \mathcal{F}_{n-1}] = \tilde{S}_{n-1}$. Setting $d\mathbb{Q}/d\mathbb{P} = \prod_{n=1}^N Y_n$ defines an equivalent measure, and a short computation with the tower property shows $\tilde{S}$ is a $\mathbb{Q}$ -martingale. $\square$

Theorem 43.6.4 (Second fundamental theorem)

An arbitrage-free model is complete – every $\mathcal{F}_N$ -measurable claim Y is replicable by a self-financing strategy – if and only if the risk-neutral measure is unique. In that case the unique time- n no-arbitrage price of a claim Y payable at time N is

$$\pi_n = \frac{1}{(1+r)^{N-n}} \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{F}_n].$$

Proof sketch. The equivalence is proved exactly as in Theorem 43.4.19. For the formula, note that $(\pi_n/(1+r)^n)$ must be a $\mathbb{Q}$ -martingale (else the augmented market has an arbitrage, by the argument of Theorem 43.4.13 applied at each time), with terminal value $Y/(1+r)^N$; and a martingale is determined by its terminal value via Proposition 43.5.11(iii). $\square$

§43.6.3 Change of Numeraire

We took it for granted above that the natural thing to divide by is the bank account. It is not the only choice, and being flexible about it is a genuinely useful technique.

Definition 43.6.5 (Numeraire)

A **numeraire** is an adapted process $(M_n)_{0 \leq n \leq N}$ with $M_n > 0$ for all n , which is the price process of a self-financing portfolio. Prices measured in units of the numeraire are S_n/M_n .

The bank account $M_n = (1+r)^n$ is the numeraire we have used so far; another asset, or the price of a bond maturing at a fixed date, will do just as well. The point is that arbitrage is a numeraire-independent notion – multiplying every price by a strictly positive process changes neither the sign of a gain nor its vanishing – so the fundamental theorem must hold for each choice, and we may pick whichever makes a given computation easiest.

Theorem 43.6.6 (Change of numeraire)

Let the market be arbitrage-free with risk-neutral measure $\mathbb{Q}$ (for the bank-account numeraire), and let M be a numeraire with $M_0 = 1$. Define $\mathbb{Q}^M \sim \mathbb{Q}$ by

$$\frac{d\mathbb{Q}^M}{d\mathbb{Q}} = \frac{M_N}{(1+r)^N}.$$

Then $\mathbb{Q}^M$ is a probability measure, and (S_n/M_n) is a $\mathbb{Q}^M$ -martingale. Moreover the time- n price of a claim Y payable at time N is

$$\pi_n = M_n \mathbb{E}_{\mathbb{Q}^M} \left[\frac{Y}{M_N} \mid \mathcal{F}_n \right].$$

Proof. Since M is the value of a self-financing portfolio, $\tilde{M}_n = M_n/(1+r)^n$ is a $\mathbb{Q}$ -martingale with $\tilde{M}_0 = 1$, so the proposed density is positive with $\mathbb{Q}$ -expectation 1 and $\mathbb{Q}^M$ is an equivalent probability measure. For a positive $\mathcal{F}_m$ -measurable random variable Z the density relation together with the tower property gives the conditional Bayes formula

$$\mathbb{E}_{\mathbb{Q}^M} [Z \mid \mathcal{F}_n] = \frac{\mathbb{E}_{\mathbb{Q}} [Z \tilde{M}_m \mid \mathcal{F}_n]}{\tilde{M}_n}, \quad n \leq m,$$

because $\mathbb{E}_{\mathbb{Q}} [d\mathbb{Q}^M / d\mathbb{Q} \mid \mathcal{F}_m] = \tilde{M}_m$. Taking $Z = S_m/M_m$ and $m = N$,

$$\mathbb{E}_{\mathbb{Q}^M} \left[\frac{S_m}{M_m} \mid \mathcal{F}_n \right] = \frac{\mathbb{E}_{\mathbb{Q}} [\tilde{S}_m \mid \mathcal{F}_n]}{\tilde{M}_n} = \frac{\tilde{S}_n}{\tilde{M}_n} = \frac{S_n}{M_n},$$

using that $\tilde{S}$ is a $\mathbb{Q}$ -martingale. The pricing formula is the same computation applied to π_m/M_m , whose $\mathbb{Q}$ -discounted version is a martingale by the second fundamental theorem. $\square$

The formula is worth reading aloud: *the price of anything, measured in units of the numeraire, is a martingale under the measure associated with that numeraire.* We shall

use it in §9.2, where the numeraire is a foreign bank account and the change of measure resolves what looks like a paradox about exchange rates.

§43.6.4 The Binomial Model

We now specialise to the model in which every computation can be carried out explicitly.

Definition 43.6.7 (Cox–Ross–Rubinstein model)

Take $d = 1$ and let $(\xi_n)_{n \geq 1}$ be i.i.d. with

$$\mathbb{P}(\xi = 1 + b) = p = 1 - \mathbb{P}(\xi = 1 + a), \quad a < b, \quad p \in (0, 1),$$

and set $S_n = S_{n-1}\xi_n = S_0\xi_1 \cdots \xi_n$, with $(\mathcal{F}_n)$ the filtration generated by (S_n) . This is the **binomial** or **Cox–Ross–Rubinstein** model.

Proposition 43.6.8

The binomial model is arbitrage-free if and only if $a < r < b$, and in that case the risk-neutral measure is unique: under $\mathbb{Q}$ the ξ_n are i.i.d. with

$$\mathbb{Q}(\xi = 1 + b) = q = \frac{r - a}{b - a}.$$

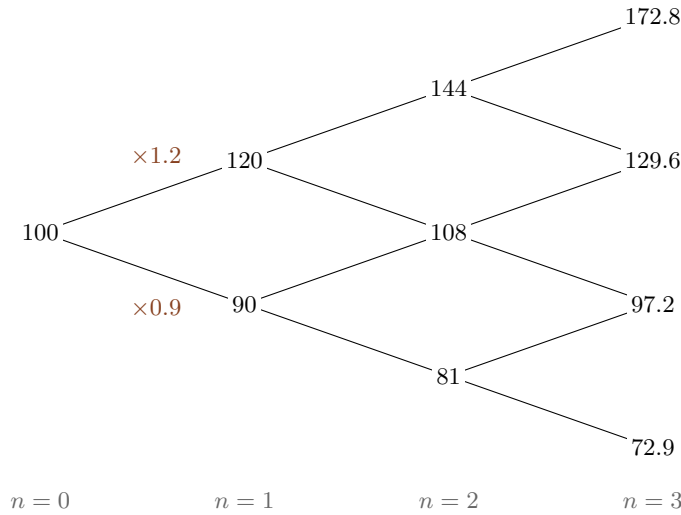
Consequently the model is complete.

Proof. If $r \leq a$ then $\theta_1 = 1$ (buy a share, hold it one period, do nothing thereafter) is an arbitrage, since $S_1 - (1+r)S_0 \geq (1+a-1-r)S_0 \geq 0$ with strict inequality when $\xi_1 = 1 + b$. If $r \geq b$ then $\theta_1 = -1$ is an arbitrage. Conversely, suppose $a < r < b$. A measure $\mathbb{Q}$ under which $\tilde{S}$ is a martingale must satisfy $\mathbb{E}_{\mathbb{Q}}[\xi_n | \mathcal{F}_{n-1}] = 1 + r$; since ξ_n takes two values this determines the conditional law of ξ_n given $\mathcal{F}_{n-1}$ to be the two-point law with mass $q = (r - a)/(b - a)$ at $1 + b$, and $q \in (0, 1)$ exactly when $a < r < b$. This makes the ξ_n i.i.d. under $\mathbb{Q}$, and conversely that choice visibly works. Uniqueness plus the second fundamental theorem gives completeness. $\square$

The price process is **recombining**: an up-move followed by a down-move gives the same price as a down followed by an up, so after n steps there are only $n + 1$ possible prices rather than 2^n . Here is the tree we shall use as a running example, with

$$S_0 = 100, \quad 1 + b = 1.2, \quad 1 + a = 0.9, \quad r = 0.05, \quad N = 3,$$

so that $q = (0.05 + 0.1)/(0.2 + 0.1) = \frac{1}{2}$.



§43.6.5 European Claims: Pricing and Replication

Definition 43.6.9

A **European contingent claim** with expiry N is an $\mathcal{F}_N$ -measurable payout Y , payable at time N . It is **vanilla** if $Y = g(S_N)$.

In the binomial model (S_n) is a Markov chain under $\mathbb{Q}$, so the conditional expectation defining the price depends on $\mathcal{F}_n$ only through S_n . This gives the following.

Theorem 43.6.10

In an arbitrage-free binomial model, the unique time- n price of the vanilla claim $Y = g(S_N)$ is $\pi_n = V(n, S_n)$, where

$$\begin{aligned}
 V(n, s) &= \frac{1}{(1+r)^{N-n}} \mathbb{E}_{\mathbb{Q}}[g(S_N) \mid S_n = s] \\
 &= \frac{1}{(1+r)^{N-n}} \sum_{j=0}^{N-n} \binom{N-n}{j} q^j (1-q)^{N-n-j} g(s(1+b)^j (1+a)^{N-n-j}).
 \end{aligned}$$

Equivalently, V satisfies the backward recursion

$$(1+r)V(n-1, s) = qV(n, s(1+b)) + (1-q)V(n, s(1+a)), \quad V(N, s) = g(s).$$

Proof. Under $\mathbb{Q}$ we have $S_N = S_n \xi_{n+1} \cdots \xi_N$ with the ξ 's i.i.d. and independent of $\mathcal{F}_n$, so $\mathbb{E}_{\mathbb{Q}}[g(S_N) \mid \mathcal{F}_n]$ is the stated function of S_n , and the number of up-moves in $N-n$ steps is binomial with parameters $N-n$ and q . The recursion is the tower property applied one step at a time. $\square$

The recursion is the practical algorithm: fill in the payouts at expiry, then work backwards through the tree, at each node taking the q -weighted average of the two values to the right and discounting by $1+r$. But pricing is only half the story; a bank that sells an option wants to know how to *hedge* it.

Theorem 43.6.11 (Replication)

With V as above, define the previsible process

$$\theta_n = \frac{V(n, S_{n-1}(1+b)) - V(n, S_{n-1}(1+a))}{S_{n-1}(b-a)},$$

and let $X_0 = V(0, S_0)$, $X_n = (1+r)X_{n-1} + \theta_n(S_n - (1+r)S_{n-1})$. Then $X_n = V(n, S_n)$ for all n ; in particular $X_N = g(S_N)$.

Proof. We induct on n , the case $n = 0$ holding by definition. Suppose $X_{n-1} = V(n-1, S_{n-1})$ and write $V_b = V(n, S_{n-1}(1+b))$, $V_a = V(n, S_{n-1}(1+a))$. Using the recursion of Theorem 43.6.10,

$$X_n = qV_b + (1-q)V_a + \frac{V_b - V_a}{b-a}(\xi_n - 1 - r).$$

If $\xi_n = 1+b$ then $\xi_n - 1 - r = b - r$, and since $q = \frac{r-a}{b-a}$, $1-q = \frac{b-r}{b-a}$,

$$\begin{aligned} X_n &= \frac{(r-a)V_b + (b-r)V_a}{b-a} + \frac{(b-r)(V_b - V_a)}{b-a} \\ &= \frac{(r-a+b-r)V_b + (b-r-b+r)V_a}{b-a} = V_b. \end{aligned}$$

If $\xi_n = 1+a$ then $\xi_n - 1 - r = a - r$ and the same computation gives $X_n = V_a$. In both cases $X_n = V(n, S_n)$. $\square$

The quantity θ_n is the **delta** of the claim: the ratio of the change in the option value to the change in the stock price over one step. Holding θ_n shares (funded by the bank) exactly reproduces the option, which is what ‘hedging’ means.

Example 43.6.12 (A European call in the binomial tree)

Take the tree above and a European call with strike $K = 100$ and expiry $N = 3$. At expiry the payouts are

$$(172.8-100)^+ = 72.8, \quad (129.6-100)^+ = 29.6, \quad (97.2-100)^+ = 0, \quad (72.9-100)^+ = 0.$$

Working backwards with $q = \frac{1}{2}$ and $1+r = 1.05$:

$$V(2, 144) = \frac{1}{1.05} \left(\frac{1}{2} \cdot 72.8 + \frac{1}{2} \cdot 29.6 \right) = \frac{51.2}{1.05} = 48.762,$$

$$V(2, 108) = \frac{1}{1.05} \left(\frac{1}{2} \cdot 29.6 + \frac{1}{2} \cdot 0 \right) = \frac{14.8}{1.05} = 14.095,$$

$$V(2, 81) = 0,$$

$$V(1, 120) = \frac{1}{1.05} \cdot \frac{1}{2} (48.762 + 14.095) = 29.932,$$

$$V(1, 90) = \frac{1}{1.05} \cdot \frac{1}{2} (14.095 + 0) = 6.712,$$

$$V(0, 100) = \frac{1}{1.05} \cdot \frac{1}{2} (29.932 + 6.712) = 17.450.$$

As a check, the direct formula of Theorem 43.6.10 gives

$$V(0, 100) = \frac{1}{1.05^3} \left(\frac{1}{8} \cdot 72.8 + \frac{3}{8} \cdot 29.6 \right) = \frac{9.1 + 11.1}{1.157625} = \frac{20.2}{1.157625} = 17.450. \checkmark$$

The initial hedge is

$$\theta_1 = \frac{29.932 - 6.712}{100 \cdot 0.3} = \frac{23.220}{30} = 0.774,$$

so on selling the option for £17.45 one immediately buys 0.774 shares, costing £77.40, borrowing the difference £59.95 from the bank. One period later the position is rebalanced: for instance if the stock rises to 120, the new delta is $(48.762 - 14.095)/(120 \cdot 0.3) = 0.963$, so one buys a further 0.189 shares.

§43.6.6 Put–Call Parity

Calls and puts are not independent objects.

Proposition 43.6.13 (Put–call parity)

Let C_n and P_n be the time- n prices of a European call and put with the same strike K and expiry N . Then

$$C_n - P_n = S_n - \frac{K}{(1+r)^{N-n}}.$$

Proof. At expiry, for every value of S_N ,

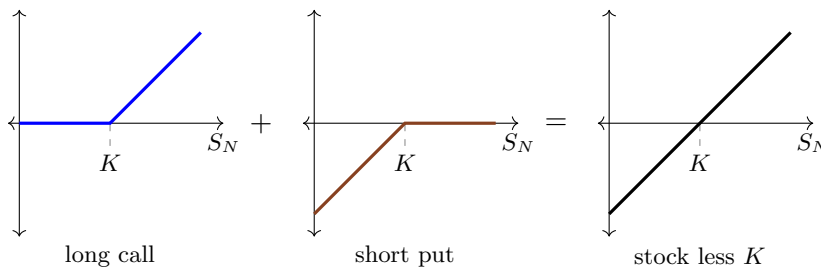
$$(S_N - K)^+ - (K - S_N)^+ = S_N - K,$$

since exactly one of the two options is in the money. Hence a long call and a short put together replicate one share plus a debt of K , and by the law of one price the values must agree. Formally, taking discounted risk-neutral conditional expectations,

$$C_n - P_n = \frac{\mathbb{E}_{\mathbb{Q}}[S_N - K \mid \mathcal{F}_n]}{(1+r)^{N-n}} = S_n - \frac{K}{(1+r)^{N-n}},$$

using that $\tilde{S}$ is a $\mathbb{Q}$ -martingale. $\square$

The identity is best seen as a picture: stacking the call payoff on top of the negated put payoff gives the straight line $S_N - K$.



Parity is used constantly in practice: it means that only one of the call and the put need be modelled, and any violation of it is a pure arbitrage requiring no view on the future at all.

Proposition 43.6.14 (Properties of the call price)

Let $g(s) = (s - K)^+$ and let θ_n be the replicating strategy of Theorem 43.6.11. Then $s \mapsto V(n, s)$ is increasing and convex, and $\theta_n \in [0, 1]$.

Proof. Writing $S_N = S_n Z$ with $Z = \xi_{n+1} \cdots \xi_N$ independent of $\mathcal{F}_n$ and positive,

$$V(n, s) = \frac{1}{(1+r)^{N-n}} \mathbb{E}_{\mathbb{Q}}[(sZ - K)^+].$$

For each fixed value of Z , the map $s \mapsto (sZ - K)^+$ is increasing and convex; both properties are preserved by taking expectations, so $V(n, \cdot)$ is increasing and convex. Since θ_n is a difference quotient of $V(n, \cdot)$, we get $\theta_n \geq 0$.

For the upper bound use parity in the form $(s - K)^+ = s - K + (K - s)^+$:

$$V(n, s) = s - \frac{K}{(1+r)^{N-n}} + \frac{1}{(1+r)^{N-n}} \mathbb{E}_{\mathbb{Q}}[(K - sZ)^+],$$

using $\mathbb{E}_{\mathbb{Q}}[Z] = (1+r)^{N-n}$. The final term is decreasing in s , so $s \mapsto V(n, s) - s$ is decreasing, giving $\theta_n \leq 1$. $\square$

The bounds $0 \leq \theta_n \leq 1$ have a clean interpretation: to hedge one call you never need to short the stock, and never need more than one share.

§43.6.7 American Claims

A European claim may only be exercised at expiry. An American claim may be exercised at any time up to expiry, at the holder's choice.

Definition 43.6.15 (American claim)

An **American contingent claim** with expiry N is an adapted process $(Y_n)_{0 \leq n \leq N}$, where Y_n is the payout if the claim is exercised at time n . The **American call** and **American put** with strike K are $Y_n = (S_n - K)^+$ and $Y_n = (K - S_n)^+$ respectively.

The holder must choose when to exercise using only information available at that time, i.e. must choose a stopping time. So the value is an optimal stopping problem, and §5.6 tells us the answer.

Theorem 43.6.16

In a complete arbitrage-free market with risk-neutral measure $\mathbb{Q}$, the time- n price of the American claim (Y_n) is

$$\pi_n = \max_{\substack{T \text{ stopping time} \\ n \leq T \leq N}} \mathbb{E}_{\mathbb{Q}} \left[\frac{(1+r)^n}{(1+r)^T} Y_T \mid \mathcal{F}_n \right],$$

so that $\pi_n/(1+r)^n$ is the Snell envelope of $Y_n/(1+r)^n$ under $\mathbb{Q}$. Equivalently,

$$\pi_N = Y_N, \quad \pi_{n-1} = \max \left\{ Y_{n-1}, \frac{1}{1+r} \mathbb{E}_{\mathbb{Q}}[\pi_n \mid \mathcal{F}_{n-1}] \right\},$$

and $T^* = \min\{n : \pi_n = Y_n\}$ is an optimal exercise time.

Proof. This is Theorem 43.5.28 applied to the discounted payout process under $\mathbb{Q}$, together with the observation that the seller of the claim must be able to hedge against *every* exercise policy, and the writer's minimal hedging cost is exactly the smallest supermartingale dominating the discounted payouts. $\square$

Example 43.6.17 (An American put)

Take the same tree, and an American put with $K = 100$, $N = 3$. The exercise values are $(100 - S_n)^+$, and we work backwards using $\pi_{n-1} = \max\{Y_{n-1}, \frac{1}{1.05} \cdot \frac{1}{2}(\pi_n^\uparrow + \pi_n^\downarrow)\}$.

At $n = 3$ the payouts are 0, 0, 2.8, 27.1. At $n = 2$,

$$\begin{aligned}\pi(144) &= \max\{0, 0\} = 0, \\ \pi(108) &= \max\{0, 1.4/1.05\} = 1.333, \\ \pi(81) &= \max\{19, 14.95/1.05\} = \max\{19, 14.238\} = 19,\end{aligned}$$

so at the node $S_2 = 81$ it is optimal to exercise. At $n = 1$,

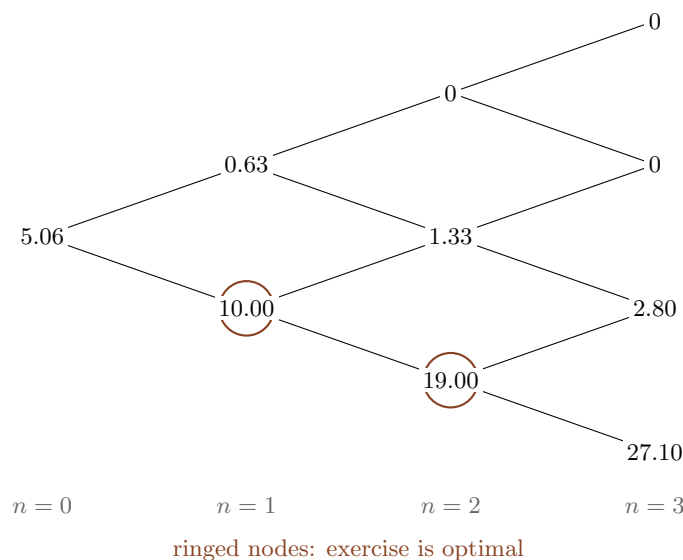
$$\begin{aligned}\pi(120) &= \max\{0, 0.667/1.05\} = 0.635, \\ \pi(90) &= \max\{10, 10.167/1.05\} = \max\{10, 9.683\} = 10,\end{aligned}$$

so we exercise at $S_1 = 90$ too. Finally $\pi_0 = \max\{0, \frac{1}{1.05} \cdot \frac{1}{2}(0.635 + 10)\} = 5.064$.

For comparison, put-call parity gives the *European* put price

$$P_0 = C_0 + \frac{K}{(1+r)^3} - S_0 = 17.450 + 86.384 - 100 = 3.834,$$

so the right of early exercise is worth $5.064 - 3.834 = 1.230$, a premium of over 30%. The tree of American values, with the exercise nodes ringed, is:



The pattern is typical: one exercises the put when the stock has fallen far enough that the interest earned on the strike outweighs the option value of waiting. For calls the situation is quite different.

Proposition 43.6.18

Suppose $(Y_n/(1+r)^n)$ is a $\mathbb{Q}$ -submartingale. Then it is optimal never to exercise the American claim before expiry, and its price coincides with that of the corresponding European claim.

Proof. We show by backward induction that $\pi_n = \mathbb{E}_{\mathbb{Q}}[Y_N/(1+r)^{N-n} \mid \mathcal{F}_n]$. This holds at $n = N$. If it holds at n then

$$\pi_{n-1} = \max \left\{ Y_{n-1}, \frac{1}{(1+r)^{N-n+1}} \mathbb{E}_{\mathbb{Q}}[Y_N \mid \mathcal{F}_{n-1}] \right\},$$

and the submartingale property gives $Y_{n-1}/(1+r)^{n-1} \leq \mathbb{E}_{\mathbb{Q}}[Y_N/(1+r)^N \mid \mathcal{F}_{n-1}]$, so the second term is the larger. Hence the maximum is attained by never stopping. $\square$

Corollary 43.6.19 (Never exercise an American call early)

In an arbitrage-free model with $r \geq 0$ and no dividends, an American call has the same value as the European call, and it is never strictly optimal to exercise it early.

Proof. We check that $(S_n - K)^+/(1+r)^n$ is a $\mathbb{Q}$ -submartingale. The function $x \mapsto x^+$ is convex, so conditional Jensen gives

$$\mathbb{E}_{\mathbb{Q}} \left[\frac{(S_n - K)^+}{(1+r)^n} \mid \mathcal{F}_{n-1} \right] \geq \left(\mathbb{E}_{\mathbb{Q}} \left[\frac{S_n - K}{(1+r)^n} \mid \mathcal{F}_{n-1} \right] \right)^+ = \left(\frac{S_{n-1}}{(1+r)^{n-1}} - \frac{K}{(1+r)^n} \right)^+,$$

using that $\tilde{S}$ is a $\mathbb{Q}$ -martingale. Since $r \geq 0$ this is at least $(S_{n-1} - K)^+/(1+r)^{n-1}$, which is the submartingale property. Now apply Proposition 43.6.18. $\square$

The intuition is worth having: exercising a call early means paying K sooner than necessary and throwing away the insurance that the option provides against the price falling below K . One should sell the option rather than exercise it.

§43.6.8 Dynamic Programming

The Snell envelope is a special case of a much more general technique, which also solves the multi-period utility maximisation problem. We record it briefly.

Let (ξ_n) be i.i.d., let (U_n) be a previsible **control** process taking values in a set $\mathcal{U}$, and let the **controlled Markov process** (X_n) evolve by

$$X_n = G(X_{n-1}, U_n, \xi_n).$$

We wish to maximise $\mathbb{E} \left[\sum_{k=1}^N f(k, U_k) + g(X_N) \mid X_0 = x \right]$ over controls.

Definition 43.6.20

The **value function** is

$$V(n, x) = \sup_{(U_k)} \mathbb{E} \left[\sum_{k=n+1}^N f(k, U_k) + g(X_N) \mid X_n = x \right].$$

Theorem 43.6.21 (Bellman)

Suppose V solves **Bellman's equation**

$$V(n-1, x) = \sup_{u \in \mathcal{U}} \left(f(n, u) + \mathbb{E}[V(n, G(x, u, \xi))] \right), \quad V(N, x) = g(x),$$

and let $u^*(n, x)$ attain the supremum. Then the Markov control $U_n^* = u^*(n, X_{n-1}^*)$ is optimal, and V is the value function.

Proof. Given any control (U_n) with controlled process (X_n) , set $M_n = \sum_{k=1}^n f(k, U_k) + V(n, X_n)$. Then

$$\mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] = f(n, U_n) + \mathbb{E}[V(n, X_n) \mid \mathcal{F}_{n-1}] - V(n-1, X_{n-1}) \leq 0$$

by Bellman's equation, so M is a supermartingale, with equality throughout when $U = U^*$, in which case M is a martingale. Hence

$$\mathbb{E} \left[\sum_{k=1}^N f(k, U_k) + g(X_N) \right] = \mathbb{E}[M_N] \leq M_0 = V(0, x),$$

with equality for $U = U^*$. □

Example 43.6.22 (Merton's problem with CARA utility)

Take $d = 1$, $S_n = S_{n-1}\xi_n$ with (ξ_n) i.i.d., and wealth dynamics $X_n = (1+r)X_{n-1} + \eta_n(\xi_n - (1+r))$, where $\eta_n = \theta_n S_{n-1}$ is the amount of money in the stock – a controlled Markov process. We maximise $\mathbb{E}[U(X_N)]$ for $U(x) = -e^{-\gamma x}$.

Guess $V(n, x) = U(x(1+r)^{N-n})A_n$ with $A_N = 1$. Then

$$\begin{aligned} V(n-1, x) &= \sup_{\eta} \mathbb{E} \left[A_n U((1+r)x + \eta(\xi - (1+r)))(1+r)^{N-n} \right] \\ &= U(x(1+r)^{N-n+1})A_n \inf_t \mathbb{E}[e^{t(\xi - (1+r))}], \end{aligned}$$

on substituting $t = \gamma\eta(1+r)^{N-n}$. So the guess is consistent with $A_{n-1} = \alpha A_n$, where $\alpha = \inf_t \mathbb{E}[e^{t(\xi - (1+r))}]$, whence $A_n = \alpha^{N-n}$, and the optimal holding is

$$\theta_n^* = \frac{t^*}{\gamma(1+r)^{N-n}S_{n-1}},$$

with t^* the minimiser. Note the CARA signature: the optimal *amount of money* η_n^* invested in the stock is a deterministic constant, independent of current wealth.

Example 43.6.23 (Consumption and CRRA utility)

If the agent may also consume an amount C_n during $(n-1, n]$, the dynamics become

$$X_n = (1+r)(X_{n-1} - C_n) + \eta_n(\xi_n - (1+r)),$$

and one maximises $\mathbb{E}[\sum_{k=1}^N U(C_k) + U(X_N)]$. For $U(x) = x^{1-R}/(1-R)$ with $R > 0$, $R \neq 1$, the value function separates as $V(n, x) = U(x)A_n$, and Bellman's equation reduces to a one-dimensional problem in the consumption fraction $s = C/x$:

$$A_{n-1} = \sup_{s \in (0,1)} (s^{1-R} + A_n \alpha (1-s)^{1-R}), \quad \alpha = (1-R) \sup_t \mathbb{E}[U((1+r) + t(\xi - (1+r)))].$$

Solving gives $A_n = ((1 - \alpha^{(N-n+1)/R})/(1 - \alpha^{1/R}))^R$ and the optimal consumption is a fixed *fraction* of wealth at each date. Again the CRRA signature: everything scales with wealth.

§43.7 Brownian Motion

Discrete-time models are perfectly adequate for computation, but they are awkward: the answers depend on the step size, and the formulae are sums of binomial coefficients rather than anything one would want to differentiate. In this section we construct the continuous-time limit.

§43.7.1 From Random Walk to Brownian Motion

Return to the binomial model $S_n = S_0 \xi_1 \cdots \xi_n$ and take logarithms:

$$\log S_n = \log S_0 + \sum_{k=1}^n \log \xi_k,$$

a random walk. Now let each period represent a small time δ , so that $t = n\delta$, and choose the parameters so that the mean and variance accumulate at fixed rates:

$$\mathbb{E}[\log \xi] = \mu\delta, \quad \text{Var}(\log \xi) = \sigma^2\delta.$$

Then $\log S_t$ has mean $\log S_0 + \mu t$ and variance $\sigma^2 t$, independently of δ . Writing

$$W_t = \frac{\log(S_t/S_0) - \mu t}{\sigma},$$

the central limit theorem from Part IA suggests that, as $\delta \downarrow 0$ with t fixed, W_t should converge to a $N(0, t)$ random variable; and since the increments of the walk over disjoint blocks are independent, the same should hold for increments. The limiting object is Brownian motion.

Definition 43.7.1 (Brownian motion)

A **standard Brownian motion** is a stochastic process $(W_t)_{t \geq 0}$ such that

- (i) $W_0 = 0$;
- (ii) for $0 \leq s < t$, the increment $W_t - W_s$ is independent of $(W_u)_{0 \leq u \leq s}$;
- (iii) for $0 \leq s < t$, $W_t - W_s \sim N(0, t - s)$;

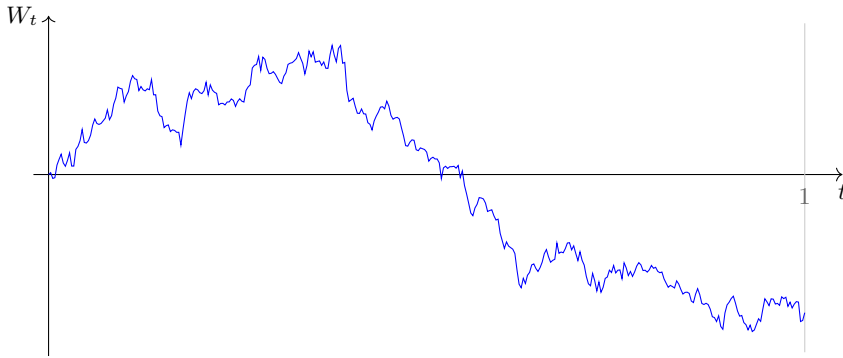
(iv) almost surely, $t \mapsto W_t$ is continuous.

That such a process exists is a genuine theorem, and not an easy one: one is asking for uncountably many random variables with prescribed joint laws whose sample paths are continuous.

Theorem 43.7.2 (Wiener, 1923)

Brownian motion exists.

We shall take this for granted. Here is a simulated path, drawn as an actual random walk with 360 steps on $[0, 1]$ – which, at this resolution, is visually indistinguishable from the real thing.



The path is continuous but visibly rough: it has no intervals of monotonicity, and in fact it is almost surely nowhere differentiable and of infinite variation on every interval. This roughness is not a defect of the model – it is precisely what makes the theory work, since it is the accumulated quadratic variation $\sum (W_{t_{i+1}} - W_{t_i})^2 \rightarrow t$ that produces the second-derivative term in Itô's formula.

§43.7.2 Basic Properties

Definition 43.7.3

A process $(X_t)_{t \geq 0}$ is **Gaussian** if $(X_{t_1}, \dots, X_{t_n})$ is a multivariate normal vector for all $0 \leq t_1 < \dots < t_n$.

Theorem 43.7.4 (Gaussian characterisation)

A continuous process $(W_t)_{t \geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if it is a Gaussian process with

$$\mathbb{E}[W_t] = 0 \quad \text{and} \quad \mathbb{E}[W_s W_t] = s \wedge t \quad \text{for all } s, t \geq 0.$$

Proof. ($\Rightarrow$) If W is a Brownian motion then $W_{t_1}, W_{t_2} - W_{t_1}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent normals, so any linear combination of $W_{t_1}, \dots, W_{t_n}$ is normal, hence W is Gaussian. Clearly $\mathbb{E}[W_t] = 0$, and for $s < t$,

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s(W_t - W_s)] = s + 0 = s.$$

($\Leftarrow$) Conversely, suppose W is Gaussian with the stated moments. For $s < t$, $W_t - W_s$ is normal (being a linear combination of jointly Gaussian variables) with mean 0 and variance

$$\mathbb{E}[W_t^2] - 2\mathbb{E}[W_s W_t] + \mathbb{E}[W_s^2] = t - 2s + s = t - s.$$

For $u < s < t$, $\mathbb{E}[(W_t - W_s)W_u] = u - u = 0$, so $W_t - W_s$ is uncorrelated with $(W_u)_{u \leq s}$; as everything is jointly Gaussian, uncorrelated means independent (a fact from Part IA Probability). Hence W is a Brownian motion. $\square$

This characterisation makes the following symmetries almost trivial to verify.

Theorem 43.7.5 (Invariance properties)

Let $(W_t)_{t \geq 0}$ be a Brownian motion. Then so are:

- (i) (*Scaling*) $\tilde{W}_t = cW_{t/c^2}$, for any $c \neq 0$;
- (ii) (*Simple Markov property*) $\tilde{W}_t = W_{t+T} - W_T$, for any fixed $T \geq 0$;
- (iii) (*Time inversion*) $\tilde{W}_t = tW_{1/t}$ for $t > 0$, $\tilde{W}_0 = 0$;
- (iv) (*Reflection*) $\tilde{W}_t = -W_t$.

Proof. Each is Gaussian with mean zero, so by Theorem 43.7.4 we need only check covariances. For (i), $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = c^2(s/c^2 \wedge t/c^2) = s \wedge t$. For (ii), $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = (s+T) \wedge (t+T) - T = s \wedge t$. For (iii), taking $s < t$, $\mathbb{E}[\tilde{W}_s \tilde{W}_t] = st(1/s \wedge 1/t) = st/t = s$. For (iv) it is immediate. The only subtlety is continuity at 0 in (iii), which amounts to $W_t/t \rightarrow 0$ almost surely as $t \rightarrow \infty$; this is a form of the strong law of large numbers. $\square$

Property (i) is worth dwelling on: Brownian motion is a fractal, statistically identical to itself under the rescaling $(t, w) \mapsto (c^2 t, cw)$. Space scales like the square root of time – which is the CLT again.

Theorem 43.7.6

Brownian motion is a Markov process: for bounded measurable g and $s < t$,

$$\mathbb{E}[g(W_t) \mid \mathcal{F}_s] = \mathbb{E}[g(W_t) \mid W_s],$$

where $\mathcal{F}_s = \sigma(W_u : u \leq s)$.

Proof. Writing $W_t = (W_t - W_s) + W_s$ with $W_t - W_s \sim N(0, t - s)$ independent of $\mathcal{F}_s$,

$$\mathbb{E}[g(W_t) \mid \mathcal{F}_s] = \int_{\mathbb{R}} g(z + W_s) \frac{e^{-z^2/2(t-s)}}{\sqrt{2\pi(t-s)}} dz,$$

which is a function of W_s alone. $\square$

Later we shall want to restart the process not at a fixed time but at a random one, and for that the simple Markov property is not enough. The correct strengthening is the following, which we shall use without proof.

Theorem 43.7.7 (Strong Markov property)

Let W be a Brownian motion and T an almost surely finite stopping time for $(\mathcal{F}_t)$. Then

$$(W_{T+t} - W_T)_{t \geq 0}$$

is a Brownian motion, independent of $\mathcal{F}_T = \{A : A \cap \{T \leq t\} \in \mathcal{F}_t \text{ for all } t\}$.

Remark. The proof is a discretisation argument, and it is worth sketching because it shows exactly where continuity is used. Set $T_n = 2^{-n} \lceil 2^n T \rceil$, a stopping time taking countably many values $k2^{-n}$. On the event $\{T_n = k2^{-n}\} \in \mathcal{F}_{k2^{-n}}$ the simple Markov property applies, and summing over k shows that $(W_{T_n+t} - W_{T_n})_t$ is a Brownian motion independent of $\mathcal{F}_{T_n}$, hence of the smaller σ -algebra $\mathcal{F}_T$. Since $T_n \downarrow T$ and the paths are continuous, $W_{T_n+t} \rightarrow W_{T+t}$ almost surely, and the property passes to the limit. Without path continuity the last step fails, and indeed the strong Markov property is genuinely delicate for processes with jumps.

It is a good moment to make precise the roughness we noticed in the picture. The relevant quantity is the **quadratic variation**, and it is really the whole reason that stochastic calculus differs from ordinary calculus.

Theorem 43.7.8 (Quadratic variation)

Fix $t > 0$ and for each n let $0 = t_0^n < t_1^n < \dots < t_{m_n}^n = t$ be a partition of $[0, t]$ with mesh $\delta_n = \max_i(t_i^n - t_{i-1}^n) \rightarrow 0$. Then

$$Q_n := \sum_{i=1}^{m_n} (W_{t_i^n} - W_{t_{i-1}^n})^2 \longrightarrow t \quad \text{in } L^2.$$

Proof. Write $\Delta_i = W_{t_i^n} - W_{t_{i-1}^n}$ and $h_i = t_i^n - t_{i-1}^n$, so that the Δ_i are independent with $\Delta_i \sim N(0, h_i)$. Then $\mathbb{E}[\Delta_i^2] = h_i$, so $\mathbb{E}[Q_n] = \sum_i h_i = t$ exactly. For the variance, independence gives

$$\text{Var}(Q_n) = \sum_i \text{Var}(\Delta_i^2) = \sum_i (\mathbb{E}[\Delta_i^4] - h_i^2) = \sum_i (3h_i^2 - h_i^2) = 2 \sum_i h_i^2,$$

using $\mathbb{E}[Z^4] = 3$ for standard normal Z . Finally $\sum_i h_i^2 \leq \delta_n \sum_i h_i = \delta_n t \rightarrow 0$, so $\mathbb{E}[(Q_n - t)^2] \rightarrow 0$. $\square$

The contrast with a differentiable function f is stark: there $\sum_i (f(t_i) - f(t_{i-1}))^2 \leq \delta_n \sum_i |f(t_i) - f(t_{i-1})| \rightarrow 0$, since the total variation is finite. So Brownian paths cannot be differentiable, and the following is immediate.

Corollary 43.7.9

Almost surely, the path $s \mapsto W_s$ has infinite total variation on $[0, t]$ for every $t > 0$; in particular it is not monotone on any interval, and is not of bounded variation, so $\int_0^t H_s dW_s$ cannot be defined pathwise as a Stieltjes integral.

Proof. Along a subsequence, $Q_n \rightarrow t$ almost surely. If the path had total variation

$V < \infty$ on $[0, t]$ then, by uniform continuity of the path on the compact interval,

$$Q_n \leq \left(\max_i |W_{t_i^n} - W_{t_{i-1}^n}| \right) V \longrightarrow 0,$$

contradicting $Q_n \rightarrow t > 0$. $\square$

The slogan is that $(dW_t)^2 = dt$: increments of Brownian motion are of size $\sqrt{dt}$, so their squares accumulate at rate dt rather than vanishing. Everything unusual about Itô's formula comes from this one fact.

§43.7.3 Martingale Properties

Definition 43.7.10

A continuous process $(X_t)_{t \geq 0}$ adapted to a filtration $(\mathcal{F}_t)_{t \geq 0}$ is a **martingale** if each X_t is integrable and $\mathbb{E}[X_t | \mathcal{F}_s] = X_s$ for all $0 \leq s \leq t$.

Proposition 43.7.11

Let W be a Brownian motion and $(\mathcal{F}_t)$ its generated filtration. Then the following are martingales:

- (i) W_t ;
- (ii) $W_t^2 - t$;
- (iii) $\mathcal{E}_t^\theta := \exp(\theta W_t - \theta^2 t/2)$, for each fixed $\theta \in \mathbb{R}$.

Proof. In each case we use that $W_t - W_s$ is independent of $\mathcal{F}_s$ with law $N(0, t - s)$.
 (i) $\mathbb{E}[W_t | \mathcal{F}_s] = \mathbb{E}[W_t - W_s | \mathcal{F}_s] + W_s = 0 + W_s$. (ii) $\mathbb{E}[W_t^2 | \mathcal{F}_s] = \mathbb{E}[(W_t - W_s)^2 | \mathcal{F}_s] + 2W_s \mathbb{E}[W_t - W_s | \mathcal{F}_s] + W_s^2 = (t - s) + W_s^2$, and subtracting t gives the claim.
 (iii) $\mathbb{E}[e^{\theta(W_t - W_s)} | \mathcal{F}_s] = e^{\theta^2(t-s)/2}$ by the normal moment generating function, so

$$\mathbb{E}[\mathcal{E}_t^\theta | \mathcal{F}_s] = \mathcal{E}_s^\theta e^{-\theta^2(t-s)/2} \mathbb{E}[e^{\theta(W_t - W_s)} | \mathcal{F}_s] = \mathcal{E}_s^\theta. \quad \square$$

The process $\mathcal{E}^\theta$ is the **exponential martingale**, and it is the single most useful object in this course: it is simultaneously the density of a change of measure, the discounted stock price under the risk-neutral measure, and a generating function for the moments of W .

Remarkably, each of (i)–(iii) characterises Brownian motion.

Theorem 43.7.12

A continuous process $(W_t)_{t \geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if $(e^{\theta W_t - \theta^2 t/2})_{t \geq 0}$ is a martingale for every $\theta \in \mathbb{R}$.

Proof. Necessity is Proposition 43.7.11(iii). Conversely, the martingale property gives

$$\mathbb{E}[e^{\theta(W_t - W_s)} | \mathcal{F}_s] = e^{\theta^2(t-s)/2}$$

for every θ . The right-hand side is deterministic, and it is the moment generating function of $N(0, t-s)$; since moment generating functions determine distributions, $W_t - W_s \sim N(0, t-s)$ and, the conditional law being non-random, $W_t - W_s$ is independent of $\mathcal{F}_s$. $\square$

Theorem 43.7.13 (Lévy's characterisation)

A continuous process $(W_t)_{t \geq 0}$ with $W_0 = 0$ is a Brownian motion if and only if both (W_t) and $(W_t^2 - t)$ are martingales.

We omit the proof, which is a genuine piece of stochastic analysis. It is a strikingly strong statement: two moment conditions plus continuity pin the process down completely.

§43.7.4 Hitting Times and the Reflection Principle

Definition 43.7.14

For $a \in \mathbb{R}$, let $T_a = \inf\{t \geq 0 : W_t = a\}$ be the first **hitting time** of level a , and let $M_t = \max_{0 \leq s \leq t} W_s$ be the running maximum.

Note that $\{T_a \leq t\} = \{M_t \geq a\}$ for $a > 0$, by continuity of paths.

Theorem 43.7.15

For every $a \in \mathbb{R} \setminus \{0\}$, $T_a < \infty$ almost surely.

Proof. By symmetry take $a > 0$; we show $Z := \sup_{t \geq 0} W_t = \infty$ a.s. By the scaling property, for any $c > 0$ the process $\tilde{W}_t = cW_{t/c^2}$ is a Brownian motion, so

$$\mathbb{P}\left(\sup_{t \geq 0} W_t > a\right) = \mathbb{P}\left(\sup_{t \geq 0} \tilde{W}_t > a\right) = \mathbb{P}\left(\sup_{t \geq 0} W_t > a/c\right).$$

Letting $c \downarrow 0$ shows that $\mathbb{P}(Z > a)$ is the same for all $a > 0$, so $Z \in \{0, \infty\}$ almost surely. Now let $\hat{Z} = \sup_{t \geq 1} (W_t - W_1)$, which has the same law as Z and is independent of W_1 . If $p = \mathbb{P}(Z = 0)$ then, since $\{Z = 0\} \subseteq \{W_1 \leq 0\} \cap \{\hat{Z} = 0\}$,

$$p = \mathbb{P}(Z = 0) \leq \mathbb{P}(W_1 \leq 0) \mathbb{P}(\hat{Z} = 0) = \frac{1}{2}p,$$

forcing $p = 0$. Hence $Z = \infty$ a.s. $\square$

The key tool for computing with M_t is the following, which says that if we reflect the path in the level a after it first reaches a , we get another Brownian motion.

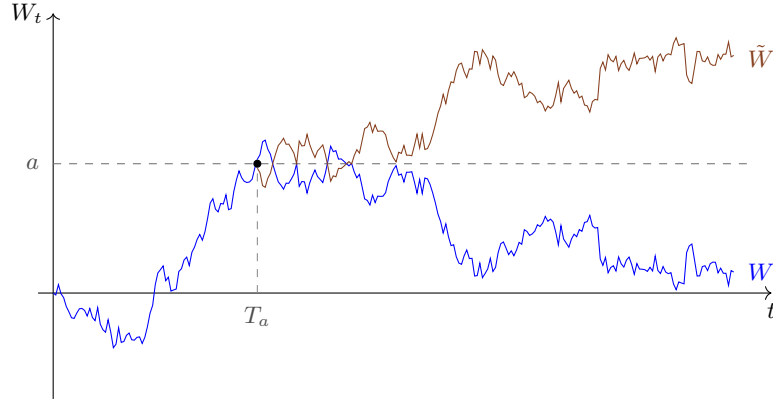
Theorem 43.7.16 (Reflection principle)

Let W be a Brownian motion and $a > 0$. Then

$$\tilde{W}_t = W_t \mathbb{1}_{\{t \leq T_a\}} + (2a - W_t) \mathbb{1}_{\{t > T_a\}}$$

is a Brownian motion.

Proof. By Theorem 43.7.15 the time T_a is an almost surely finite stopping time, so the strong Markov property (Theorem 43.7.7) applies: $(W_{T_a+t} - W_{T_a})_{t \geq 0} = (W_{T_a+t} - a)_{t \geq 0}$ is a Brownian motion independent of $\mathcal{F}_{T_a}$. By the reflection invariance, so is its negative. Since $\tilde{W}_{T_a+t} = a - (W_{T_a+t} - a)$, the process $\tilde{W}$ is obtained from W by replacing the post- T_a increments by their negatives, which does not change the law. $\square$



Lemma 43.7.17

For bounded measurable g and $a > 0$,

$$\mathbb{E}[g(W_t) \mathbb{1}_{\{W_t \leq a, M_t \geq a\}}] = \mathbb{E}[g(2a - W_t) \mathbb{1}_{\{W_t \geq a\}}].$$

Proof. Let $\tilde{W}$ be as in Theorem 43.7.16 and $\tilde{M}_t = \max_{s \leq t} \tilde{W}_s$. Since reflection does not change whether the level a is reached, $\{M_t \geq a\} = \{\tilde{M}_t \geq a\}$; and on that event $W_t = 2a - \tilde{W}_t$. Hence

$$\begin{aligned} \mathbb{E}[g(W_t) \mathbb{1}_{\{W_t \leq a, M_t \geq a\}}] &= \mathbb{E}[g(2a - \tilde{W}_t) \mathbb{1}_{\{\tilde{W}_t \geq a, \tilde{M}_t \geq a\}}] \\ &= \mathbb{E}[g(2a - \tilde{W}_t) \mathbb{1}_{\{\tilde{W}_t \geq a\}}], \end{aligned}$$

because $\{\tilde{W}_t \geq a\} \subseteq \{\tilde{M}_t \geq a\}$. As $\tilde{W}$ has the same law as W , we may drop the tildes. $\square$

Corollary 43.7.18

For $a \geq b$ and $a \geq 0$, $\mathbb{P}(W_t \leq b, M_t \geq a) = \mathbb{P}(W_t \geq 2a - b)$.

Proof. Take $g(w) = \mathbb{1}_{\{w \leq b\}}$ in Lemma 43.7.17: the left side is $\mathbb{P}(W_t \leq b, M_t \geq a)$ (using $b \leq a$), and the right side is $\mathbb{P}(2a - W_t \leq b, W_t \geq a) = \mathbb{P}(W_t \geq 2a - b)$, the constraint $W_t \geq a$ being implied. $\square$

This immediately gives the joint law of (M_t, W_t) and the law of the hitting time.

Theorem 43.7.19

For $t > 0$:

- (i) M_t has the same law as $|W_t|$; explicitly $\mathbb{P}(M_t \geq a) = 2\Phi(-a/\sqrt{t})$ for $a \geq 0$;
(ii) the pair (M_t, W_t) has joint density, for $a \geq b^+$,

$$f_{M_t, W_t}(a, b) = \frac{2(2a - b)}{t^{3/2}\sqrt{2\pi}} \exp\left(-\frac{(2a - b)^2}{2t}\right);$$

- (iii) the hitting time T_a ($a > 0$) has density

$$f_{T_a}(t) = \frac{a}{t^{3/2}\sqrt{2\pi}} e^{-a^2/2t}, \quad t > 0,$$

and $\mathbb{E}[T_a] = \infty$.

Proof. (i) Splitting according to the value of W_t and using Corollary 43.7.18 with $b = a$,

$$\begin{aligned} \mathbb{P}(M_t \geq a) &= \mathbb{P}(M_t \geq a, W_t \geq a) + \mathbb{P}(M_t \geq a, W_t \leq a) \\ &= \mathbb{P}(W_t \geq a) + \mathbb{P}(W_t \geq a) = 2\Phi(-a/\sqrt{t}), \end{aligned}$$

where the first term used $\{W_t \geq a\} \subseteq \{M_t \geq a\}$. This is exactly $\mathbb{P}(|W_t| \geq a)$.

(ii) Differentiate $\mathbb{P}(W_t \leq b, M_t \geq a) = \mathbb{P}(W_t \geq 2a - b) = 1 - \Phi((2a - b)/\sqrt{t})$ once in a and once in b ; the stated formula results.

(iii) By continuity of paths $\{T_a \leq t\} = \{M_t \geq a\}$, so $\mathbb{P}(T_a \leq t) = 2\Phi(-a/\sqrt{t})$; differentiating in t gives the density. Since $f_{T_a}(t) \sim a/(\sqrt{2\pi}t^{3/2})$ as $t \rightarrow \infty$, we get $\int^\infty t f_{T_a}(t) dt = \infty$. $\square$

So Brownian motion hits every level with probability one, but takes infinitely long on average to do so – the same phenomenon we saw for the recurrent random walk in Example 43.5.20, and for exactly the same reason.

§43.7.5 The Cameron–Martin Theorem

Finally we need to be able to change measure so as to add a drift to Brownian motion. The finite-dimensional version is the Gaussian tilt of Example 43.4.4: if $Z \sim N(0, 1)$ and $\alpha \in \mathbb{R}$ then for bounded f ,

$$\mathbb{E}[e^{\alpha Z - \alpha^2/2} f(Z)] = \mathbb{E}[f(Z + \alpha)],$$

and in n dimensions, for $Z \sim N_n(0, I)$ and $\alpha \in \mathbb{R}^n$,

$$\mathbb{E}[e^{\alpha^\top Z - \|\alpha\|^2/2} f(Z)] = \mathbb{E}[f(Z + \alpha)].$$

Theorem 43.7.20 (Cameron–Martin)

Let W be a Brownian motion, $c \in \mathbb{R}$ and $t > 0$. Then for suitably integrable functionals g on $C([0, t])$,

$$\mathbb{E}[g((W_s + cs)_{0 \leq s \leq t})] = \mathbb{E}[e^{cW_t - c^2 t/2} g((W_s)_{0 \leq s \leq t})].$$

Proof. By a monotone class argument it suffices to treat $g(w) = G(w(t_1), \dots, w(t_n))$ for $0 < t_1 < \dots < t_n = t$. Writing $Z_k = (W_{t_k} - W_{t_{k-1}})/\sqrt{t_k - t_{k-1}}$ (with $t_0 = 0$), the Z_k are i.i.d. $N(0, 1)$, and g may be rewritten as $H(Z_1, \dots, Z_n)$ for a suitable H . Adding the drift cs shifts Z_k by $\alpha_k = c\sqrt{t_k - t_{k-1}}$, so

$$\begin{aligned}\mathbb{E}[g((W_s + cs)_{s \leq t})] &= \mathbb{E}[H(Z_1 + \alpha_1, \dots, Z_n + \alpha_n)] \\ &= \mathbb{E}\left[\exp\left(\sum_k \alpha_k Z_k - \frac{1}{2} \sum_k \alpha_k^2\right) H(Z_1, \dots, Z_n)\right] \\ &= \mathbb{E}[e^{cW_t - c^2 t/2} g((W_s)_{s \leq t})],\end{aligned}$$

since $\sum_k \alpha_k Z_k = c \sum_k (W_{t_k} - W_{t_{k-1}}) = cW_t$ and $\sum_k \alpha_k^2 = c^2 t$. $\square$

Corollary 43.7.21 (Girsanov, for constant drift)

Fix $c \in \mathbb{R}$ and a horizon $T > 0$, and define $\mathbb{Q} \sim \mathbb{P}$ by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} = e^{cW_T - c^2 T/2}.$$

Then $\tilde{W}_t = W_t - ct$, $0 \leq t \leq T$, is a $\mathbb{Q}$ -Brownian motion.

Proof. For a bounded functional g ,

$$\mathbb{E}_{\mathbb{Q}}[g((\tilde{W}_t)_{t \leq T})] = \mathbb{E}_{\mathbb{P}}[e^{cW_T - c^2 T/2} g((W_t - ct)_{t \leq T})] = \mathbb{E}_{\mathbb{P}}[g((W_t)_{t \leq T})]$$

by Theorem 43.7.20 applied with drift $-c$. So the law of $\tilde{W}$ under $\mathbb{Q}$ equals the law of W under $\mathbb{P}$. $\square$

Notice what has happened: changing the measure changes the *drift* of the process, but not its volatility. This is the continuous-time avatar of the fact that the risk-neutral measure changes the expected return of a stock but not the size of its fluctuations, and it is the reason why the Black–Scholes price does not depend on μ .

§43.8 The Black–Scholes Model

We are finally in a position to write down the model which made mathematical finance a subject: a single stock whose price is a geometric Brownian motion, and a bank account paying a constant continuously compounded rate.

§43.8.1 The Model

Recall the passage to the limit in §7.1, where we found $\log S_t = \log S_0 + \mu t + \sigma W_t$ in the small-step limit. We take this as our definition.

Definition 43.8.1 (Black–Scholes model)

Fix constants $\mu \in \mathbb{R}$ (the **drift**), $\sigma > 0$ (the **volatility**) and $r \in \mathbb{R}$ (the interest rate), and let W be a Brownian motion. The **Black–Scholes model** has a bank account worth e^{rt} at time t and a stock whose price is the **geometric Brownian**

motion

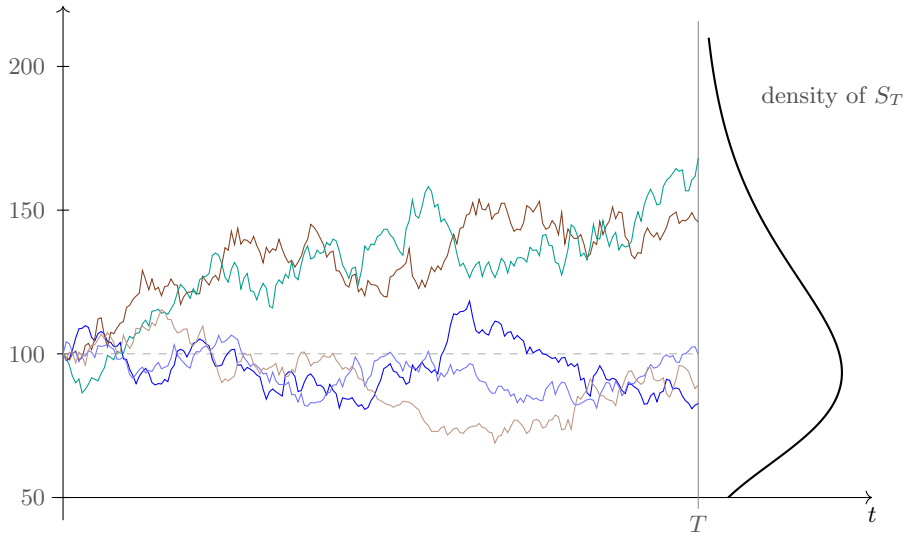
$$S_t = S_0 e^{\mu t + \sigma W_t}.$$

The interest rate convention is the limit of the discrete one: $(1 + r\delta)^{t/\delta} \rightarrow e^{rt}$ as $\delta \downarrow 0$, so a continuously compounded rate r turns one pound into e^{rt} pounds.

Some immediate consequences. Since $W_t \sim N(0, t)$, the price S_t is **lognormal**: $\log S_t \sim N(\log S_0 + \mu t, \sigma^2 t)$. In particular $S_t > 0$ always, which is as it should be for a limited-liability share, and

$$\mathbb{E}[S_t] = S_0 e^{(\mu + \sigma^2/2)t}.$$

Note the $\sigma^2/2$: the expected price grows faster than the typical price, because the lognormal distribution is skewed. Here are five simulated paths with $S_0 = 100$, $\mu = 0.06$, $\sigma = 0.25$ over two years, together with the density of S_T at the terminal time.



The right-hand curve is the lognormal density of S_T , drawn sideways so that heights correspond to prices. Note its asymmetry: a long right tail, and a floor at zero.

§43.8.2 The Risk-Neutral Measure

Proposition 43.8.2

The Black–Scholes model on $[0, T]$ has a risk-neutral measure: there is $\mathbb{Q} \sim \mathbb{P}$ under which $(e^{-rt} S_t)_{0 \leq t \leq T}$ is a martingale. Explicitly, with

$$c = \frac{r - \mu}{\sigma} - \frac{\sigma}{2}, \quad \frac{d\mathbb{Q}}{d\mathbb{P}} = e^{cW_T - c^2 T/2},$$

the process $\tilde{W}_t = W_t - ct$ is a $\mathbb{Q}$ -Brownian motion and

$$S_t = S_0 \exp \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma \tilde{W}_t \right).$$

Proof. By Corollary 43.7.21, $\tilde{W}_t = W_t - ct$ is a $\mathbb{Q}$ -Brownian motion. Substituting $W_t = \tilde{W}_t + ct$ into the definition of S_t ,

$$S_t = S_0 e^{\mu t + \sigma(\tilde{W}_t + ct)} = S_0 e^{(\mu + \sigma c)t + \sigma \tilde{W}_t} = S_0 e^{(r - \sigma^2/2)t + \sigma \tilde{W}_t},$$

since $\mu + \sigma c = r - \sigma^2/2$. Hence $e^{-rt}S_t = S_0 e^{\sigma \tilde{W}_t - \sigma^2 t/2}$, which is the exponential martingale of Proposition 43.7.11 for the $\mathbb{Q}$ -Brownian motion $\tilde{W}$, with $\theta = \sigma$. $\square$

Two things deserve comment. First, the drift μ has vanished from the risk-neutral dynamics: under $\mathbb{Q}$ the stock grows at the riskless rate r , whatever it does in reality. Second, the volatility σ is unchanged – changing measure can shift a Brownian motion by a drift but cannot alter its scale. So of the two parameters of the model, only σ will appear in any option price.

Definition 43.8.3 (Black–Scholes price)

Let Y be the payout of a European claim with maturity T . Its **Black–Scholes price** at time t is

$$\pi_t = e^{-r(T-t)} \mathbb{E}_{\mathbb{Q}}[Y \mid \mathcal{F}_t].$$

By construction $(e^{-rt}\pi_t)$ is a $\mathbb{Q}$ -martingale, so the augmented market has no arbitrage. In fact the Black–Scholes model is complete, so this is the *only* arbitrage-free price; we prove this in §8.5, once we have Itô’s lemma to hand.

§43.8.3 The Black–Scholes Formula

Let $Y = g(S_T)$ be vanilla and write $\tau = T - t$ for the time to maturity. Under $\mathbb{Q}$,

$$S_T = S_t \exp\left(\left(r - \frac{\sigma^2}{2}\right)\tau + \sigma(\tilde{W}_T - \tilde{W}_t)\right),$$

with $\tilde{W}_T - \tilde{W}_t \sim N(0, \tau)$ independent of $\mathcal{F}_t$. So $\pi_t = V(t, S_t)$ where

$$V(t, s) = e^{-r\tau} \mathbb{E}\left[g\left(s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}Z}\right)\right] = e^{-r\tau} \int_{-\infty}^{\infty} g\left(s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}z}\right) \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz$$

for $Z \sim N(0, 1)$. This is a completely explicit one-dimensional integral. For the call it evaluates in closed form.

Theorem 43.8.4 (Black–Scholes formula)

The price of a European call with strike K and maturity T is $V(t, S_t)$ where, with $\tau = T - t$,

$$V(t, s) = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2),$$

$$d_{1,2} = \frac{\log(s/K) + (r \pm \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

and Φ is the standard normal distribution function.

Proof. Write $g(s) = (s - K)^+$ and $S_T = s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}Z}$. The option finishes in the money precisely when

$$s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}z} > K \iff z > \frac{\log(K/s) - (r - \sigma^2/2)\tau}{\sigma\sqrt{\tau}} = -d_2.$$

Hence

$$V(t, s) = e^{-r\tau} \int_{-d_2}^{\infty} \left(s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}z} - K \right) \varphi(z) \, dz,$$

where φ is the standard normal density. The second piece is $-e^{-r\tau}K(1 - \Phi(-d_2)) = -Ke^{-r\tau}\Phi(d_2)$. For the first, note the completing-the-square identity

$$e^{-\sigma^2\tau/2 + \sigma\sqrt{\tau}z} \varphi(z) = \varphi(z - \sigma\sqrt{\tau}),$$

so that

$$\begin{aligned} e^{-r\tau} \int_{-d_2}^{\infty} s e^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}z} \varphi(z) \, dz &= s \int_{-d_2}^{\infty} \varphi(z - \sigma\sqrt{\tau}) \, dz \\ &= s \int_{-d_2 - \sigma\sqrt{\tau}}^{\infty} \varphi(u) \, du = s \Phi(d_2 + \sigma\sqrt{\tau}), \end{aligned}$$

and $d_2 + \sigma\sqrt{\tau} = d_1$. □

Corollary 43.8.5 (Put–call parity)

The price of the corresponding European put is

$$P(t, s) = Ke^{-r\tau}\Phi(-d_2) - s\Phi(-d_1), \quad C(t, s) - P(t, s) = s - Ke^{-r\tau}.$$

Proof. As in Proposition 43.6.13, $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, so

$$C - P = e^{-r\tau} \mathbb{E}_{\mathbb{Q}}[S_T - K \mid \mathcal{F}_t] = s - Ke^{-r\tau}$$

because $e^{-rt}S_t$ is a $\mathbb{Q}$ -martingale. Substituting the formula for C and using $\Phi(x) = 1 - \Phi(-x)$ gives the expression for P . □

Example 43.8.6 (A numerical example)

Take $S_t = 100$, $K = 100$, $r = 0.05$, $\sigma = 0.2$ and $\tau = 1$ year. Then

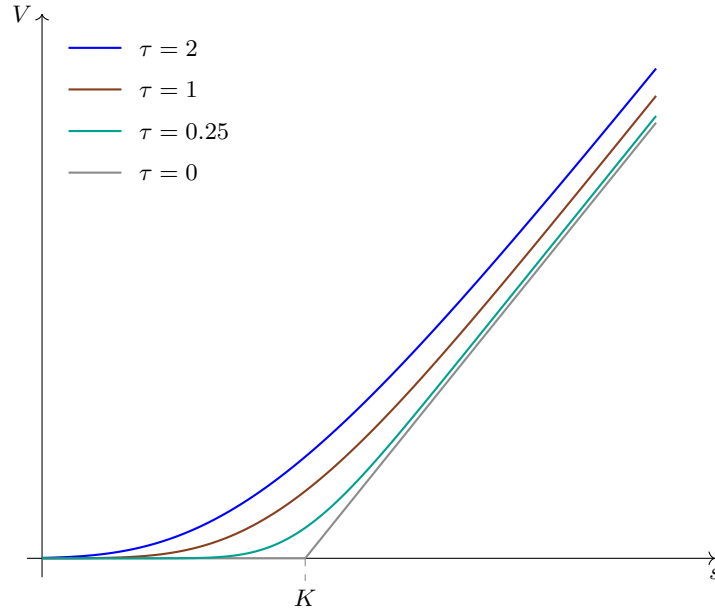
$$d_1 = \frac{0 + (0.05 + 0.02) \cdot 1}{0.2} = 0.35, \quad d_2 = 0.35 - 0.2 = 0.15,$$

and $\Phi(0.35) = 0.6368$, $\Phi(0.15) = 0.5596$, $e^{-0.05} = 0.9512$. So

$$C = 100(0.6368) - 100(0.9512)(0.5596) = 63.68 - 53.23 = 10.45.$$

The corresponding put is $P = C - S + Ke^{-r\tau} = 10.45 - 100 + 95.12 = 5.57$. The hedge ratio is $\Delta = \Phi(d_1) = 0.637$: on selling this call one immediately buys 0.637 shares.

The general shape of V as a function of s is worth internalising. As $\tau \downarrow 0$ the price converges to the payout $(s - K)^+$; for larger τ the curve is a smoothed version of that hockey stick, lying above it, and increasing in τ .



§43.8.4 Itô's Lemma and the Black–Scholes PDE

The derivation above is the modern one. Black and Scholes argued differently, via a partial differential equation, and that route is both historically important and computationally useful. It rests on the fundamental rule of stochastic calculus.

Theorem 43.8.7 (Itô's lemma)

Let X satisfy the stochastic differential equation

$$dX_t = a_t dt + b_t dW_t,$$

and let $f(t, x)$ be twice continuously differentiable. Then

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + a_t \frac{\partial f}{\partial x} + \frac{1}{2} b_t^2 \frac{\partial^2 f}{\partial x^2} \right) dt + b_t \frac{\partial f}{\partial x} dW_t,$$

all partial derivatives being evaluated at (t, X_t) .

We take this as given – it is proved in Part III. The heuristic is that $(dW_t)^2 = dt$: Taylor expanding f to second order, the term $\frac{1}{2} f_{xx} (dX_t)^2 = \frac{1}{2} f_{xx} b_t^2 (dW_t)^2$ is of order dt and must be retained, because by Theorem 43.7.8 the quadratic variation of Brownian motion over $[0, t]$ is t rather than 0. This is precisely the roughness of the Brownian path made useful.

Proposition 43.8.8

The geometric Brownian motion $S_t = S_0 e^{\mu t + \sigma W_t}$ satisfies

$$dS_t = \left(\mu + \frac{\sigma^2}{2} \right) S_t dt + \sigma S_t dW_t.$$

Under $\mathbb{Q}$ this reads $dS_t = r S_t dt + \sigma S_t d\tilde{W}_t$.

Proof. Apply Itô with $X_t = W_t$ (so $a = 0$, $b = 1$) and $f(t, x) = S_0 e^{\mu t + \sigma x}$: the partial derivatives are $f_t = \mu f$, $f_x = \sigma f$, $f_{xx} = \sigma^2 f$, giving $dS_t = (\mu + \sigma^2/2)S_t dt + \sigma S_t dW_t$. For the second statement substitute $W_t = \tilde{W}_t + ct$ and use $\mu + \sigma c + \sigma^2/2 = r$. $\square$

So the parameter $\mu + \sigma^2/2$ is the instantaneous expected rate of return, and σ is the instantaneous standard deviation per unit time.

Theorem 43.8.9 (Black–Scholes PDE)

Let $V(t, s)$ be the Black–Scholes price of a vanilla claim with payout $g(S_T)$. Then V solves

$$\frac{\partial V}{\partial t} + rs \frac{\partial V}{\partial s} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 V}{\partial s^2} = rV, \quad V(T, s) = g(s).$$

Proof. Suppose a self-financing portfolio holds Δ_t shares and the rest in the bank, with value X_t . Then

$$dX_t = r(X_t - \Delta_t S_t) dt + \Delta_t dS_t.$$

On the other hand, applying Itô to $V(t, S_t)$ with $dS_t = (\mu + \sigma^2/2)S_t dt + \sigma S_t dW_t$,

$$dV(t, S_t) = \left(V_t + \left(\mu + \frac{\sigma^2}{2} \right) S_t V_s + \frac{1}{2} \sigma^2 S_t^2 V_{ss} \right) dt + \sigma S_t V_s dW_t.$$

If the portfolio is to replicate the claim, we need $X_t = V(t, S_t)$ for all t , so the two expressions must agree. Matching the dW_t terms forces

$$\Delta_t = V_s(t, S_t),$$

and then matching the dt terms, after substituting dS_t into the first expression, gives

$$rV - rS_t V_s = V_t + \frac{1}{2} \sigma^2 S_t^2 V_{ss},$$

which rearranges to the stated PDE. The terminal condition is $V(T, s) = g(s)$, since at maturity the portfolio must be worth the payout. $\square$

Notice again the disappearance of μ : the drift terms cancelled when we chose $\Delta_t = V_s$. This is the mathematical content of the phrase ‘delta hedging removes the risk’.

Remark (The binomial heuristic). One can also derive the PDE directly as a limit of the binomial recursion, which is illuminating because it shows where each term comes from. Take a time step δ and set

$$1 + b = e^{\mu\delta + \sigma\sqrt{\delta}}, \quad 1 + a = e^{\mu\delta - \sigma\sqrt{\delta}}, \quad r_{\text{bin}} = r\delta,$$

so that $b = \sigma\sqrt{\delta} + (\mu + \frac{\sigma^2}{2})\delta + O(\delta^{3/2})$ and similarly for a . The risk-neutral probability is $q = (r\delta - a)/(b - a)$, and one computes

$$qb + (1 - q)a = r\delta, \quad qb^2 + (1 - q)a^2 = \sigma^2\delta + O(\delta^{3/2}).$$

Now expand the binomial recursion $(1 + r\delta)V(t - \delta, s) = qV(t, s(1 + b)) + (1 - q)V(t, s(1 + a))$ to second order in δ :

$$V + \left(rV - \frac{\partial V}{\partial t} \right) \delta = V + \frac{\partial V}{\partial s} s (qb + (1 - q)a) + \frac{1}{2} \frac{\partial^2 V}{\partial s^2} s^2 (qb^2 + (1 - q)a^2) + o(\delta)$$

$$= V + \frac{\partial V}{\partial s} sr\delta + \frac{1}{2} \frac{\partial^2 V}{\partial s^2} s^2 \sigma^2 \delta + o(\delta).$$

Cancelling V , dividing by δ and letting $\delta \rightarrow 0$ gives the Black–Scholes PDE. Note that the first-order term picks out the interest rate and the second-order term the volatility – exactly as in Itô’s lemma.

Remark (Reduction to the heat equation). The Black–Scholes PDE is the heat equation in disguise. If $u(\tau, x) = \mathbb{E}[f(x + \sqrt{\tau}Z)]$ for $Z \sim N(0, 1)$ then differentiating under the expectation gives $u_\tau = \frac{1}{2}u_{xx}$. Setting

$$u(\tau, x) = e^{r\tau} V\left(T - \frac{\tau}{\sigma^2}, e^{x - (r/\sigma^2 - 1/2)\tau}\right),$$

one checks that V solves the Black–Scholes PDE if and only if u solves $u_\tau = \frac{1}{2}u_{xx}$ with $u(0, x) = g(e^x)$. This is why Black–Scholes prices behave like diffusing heat: the payoff’s kink is smoothed out instantly and increasingly with time to maturity.

§43.8.5 Replication and Completeness

We can now discharge the promise made after Proposition 43.8.2 and show that the Black–Scholes price is not merely *an* arbitrage-free price but the *only* one, because the claim can be replicated exactly.

The continuous-time analogue of the self-financing condition of §6.1 is obtained by letting the rebalancing interval shrink. If the agent holds θ_t shares and θ_t^0 pounds in the bank, with wealth $X_t = \theta_t^0 + \theta_t S_t$, then over $[t, t + dt]$ no money enters or leaves, so the change in wealth is entirely due to the change in the prices of the assets held:

Definition 43.8.10 (Self-financing, continuous time)

An adapted pair (θ_t^0, θ_t) is **self-financing** if the wealth $X_t = \theta_t^0 + \theta_t S_t$ satisfies

$$dX_t = r\theta_t^0 dt + \theta_t dS_t = r(X_t - \theta_t S_t) dt + \theta_t dS_t.$$

Equivalently, writing $\tilde{X}_t = e^{-rt}X_t$ and $\tilde{S}_t = e^{-rt}S_t$, the discounted wealth satisfies $d\tilde{X}_t = \theta_t d\tilde{S}_t$.

The equivalence is a one-line application of Itô to $e^{-rt}X_t$, and it is the exact analogue of the discrete identity $\tilde{X}_n = X_0 + (\theta \cdot \tilde{S})_n$: in the numeraire of the bank account, a self-financing wealth process is a stochastic integral against the discounted prices, with no drift of its own.

Theorem 43.8.11 (Completeness of the Black–Scholes model)

Let g be such that $V(t, s)$, defined by the risk-neutral expectation of §8.3, is $C^{1,2}$ on $[0, T) \times (0, \infty)$ with bounded V_s . Then the self-financing strategy with initial wealth $X_0 = V(0, S_0)$ and

$$\theta_t = V_s(t, S_t), \quad \theta_t^0 = V(t, S_t) - S_t V_s(t, S_t),$$

satisfies $X_t = V(t, S_t)$ for every $t \leq T$; in particular $X_T = g(S_T)$ almost surely.

Hence every such claim is replicable and the Black–Scholes price is the unique arbitrage-free price.

Proof. By Theorem 43.8.9, V solves the Black–Scholes PDE. Apply Itô’s lemma to $V(t, S_t)$ under $\mathbb{Q}$, where $dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$:

$$dV(t, S_t) = \left(V_t + rS_t V_s + \frac{1}{2}\sigma^2 S_t^2 V_{ss} \right) dt + \sigma S_t V_s d\tilde{W}_t = rV(t, S_t) dt + \sigma S_t V_s d\tilde{W}_t,$$

the drift having been replaced using the PDE. On the other hand the self-financing wealth process with $\theta_t = V_s(t, S_t)$ satisfies

$$dX_t = r(X_t - \theta_t S_t) dt + \theta_t(rS_t dt + \sigma S_t d\tilde{W}_t) = rX_t dt + \sigma S_t V_s d\tilde{W}_t.$$

So $D_t = X_t - V(t, S_t)$ satisfies $dD_t = rD_t dt$ with $D_0 = 0$, whence $D_t = D_0 e^{rt} = 0$ for all t . Continuity of V up to $t = T$ gives $X_T = V(T, S_T) = g(S_T)$.

For uniqueness: if π were another arbitrage-free price for the claim, then selling at the higher of π and $V(0, S_0)$, buying at the lower, and running the replicating strategy would produce a riskless profit. $\square$

Remark. The hypotheses are ugly and can be removed. The clean statement is the **martingale representation theorem**: if $(\mathcal{F}_t)$ is the filtration generated by a Brownian motion $\tilde{W}$ and M is a square-integrable $\mathbb{Q}$ -martingale, then there is a previsible H with $M_t = M_0 + \int_0^t H_s d\tilde{W}_s$. Applying this to $M_t = \mathbb{E}_{\mathbb{Q}}[e^{-rT}Y \mid \mathcal{F}_t]$ and setting $\theta_t = H_t/(\sigma \tilde{S}_t)$ replicates *any* integrable $\mathcal{F}_T$ -measurable claim, with no smoothness assumed. This is the exact continuous-time analogue of the second fundamental theorem: the risk-neutral measure is unique, so the model is complete.

§43.8.6 The Greeks and Hedging

The partial derivatives of V have names and are collectively called the **Greeks**. They measure the sensitivity of the price to each input, and a trading desk manages a book by keeping them within limits.

Definition 43.8.12 (The Greeks)

For a claim with price $V(t, s)$, define

$$\Delta = \frac{\partial V}{\partial s}, \quad \Gamma = \frac{\partial^2 V}{\partial s^2}, \quad \Theta = \frac{\partial V}{\partial t}, \quad \mathcal{V} = \frac{\partial V}{\partial \sigma}, \quad \rho = \frac{\partial V}{\partial r},$$

called **delta**, **gamma**, **theta**, **vega** and **rho**. Holding Δ units of the stock is called **delta hedging**.

(The name ‘vega’ is not a Greek letter; the convention is that when one runs out of Greek letters one uses the names of stars, or television celebrities.)

Proposition 43.8.13

For a European call in the Black–Scholes model,

$$\begin{aligned}\Delta &= \Phi(d_1), & \Gamma &= \frac{\varphi(d_1)}{s\sigma\sqrt{\tau}}, & \mathcal{V} &= s\sqrt{\tau}\varphi(d_1), & \rho &= Ke^{-r\tau}\Phi(d_2), \\ \Theta &= -\frac{s\sigma\varphi(d_1)}{2\sqrt{\tau}} - rKe^{-r\tau}\Phi(d_2).\end{aligned}$$

Proof. Differentiating $V = s\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$ in s and using $\partial d_1/\partial s = \partial d_2/\partial s = 1/(s\sigma\sqrt{\tau})$,

$$\Delta = \Phi(d_1) + \frac{s\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)}{s\sigma\sqrt{\tau}}.$$

The bracketed term vanishes, by the identity $s\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, which follows from $d_2 = d_1 - \sigma\sqrt{\tau}$:

$$\frac{\varphi(d_1)}{\varphi(d_2)} = e^{-(d_1^2 - d_2^2)/2} = e^{-\sigma\sqrt{\tau}(d_1 - \sigma\sqrt{\tau}/2)} = e^{-\log(s/K) - r\tau} = \frac{Ke^{-r\tau}}{s}.$$

Hence $\Delta = \Phi(d_1)$, and differentiating once more gives Γ . The remaining Greeks follow similarly, using the same identity to cancel the terms coming from differentiating d_1 and d_2 ; for Θ one uses in addition $\partial d_1/\partial t = -\partial d_1/\partial \tau$ and $\partial(d_1 - d_2)/\partial \tau = \sigma/(2\sqrt{\tau})$. $\square$

Two sanity checks on these formulae are worth making. First, all of $\Delta, \Gamma, \mathcal{V}, \rho$ are positive and Θ is negative: a call is worth more if the stock rises, if volatility rises, if rates rise, and less as time runs out. Second, the Greeks are not independent, because the Black–Scholes PDE of Theorem 43.8.9 is precisely the relation

$$\Theta + rs\Delta + \frac{1}{2}\sigma^2 s^2 \Gamma = rV$$

among them. In practice this is how a desk thinks about the trade-off it faces: a hedged book is delta-neutral, so $\Theta = rV - \frac{1}{2}\sigma^2 s^2 \Gamma$, and one is paid theta exactly in compensation for being short gamma.

Example 43.8.14

In the numerical example above ($s = K = 100$, $r = 0.05$, $\sigma = 0.2$, $\tau = 1$) we have $\varphi(d_1) = \varphi(0.35) = 0.3752$, so

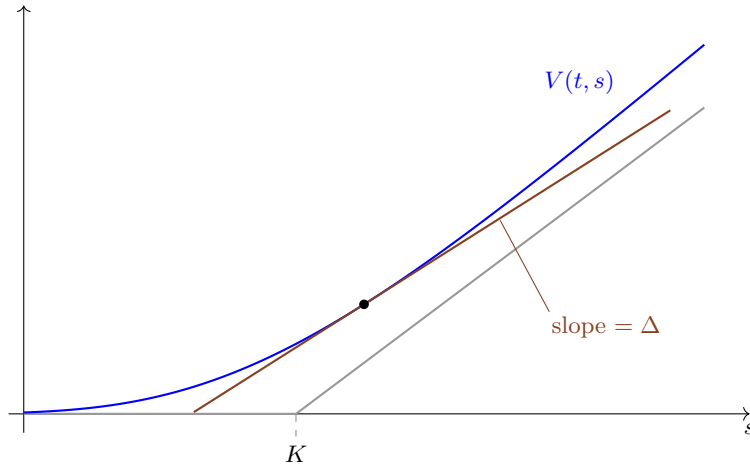
$$\Gamma = \frac{0.3752}{100 \cdot 0.2} = 0.01876, \quad \mathcal{V} = 100 \cdot 0.3752 = 37.52,$$

$$\Theta = -\frac{1}{2}(100)(0.2)(0.3752) - 0.05 \cdot 95.12 \cdot 0.5596 = -6.41.$$

Checking the relation above: $-6.41 + 0.05 \cdot 100 \cdot 0.6368 + \frac{1}{2}(0.04)(10^4)(0.01876) = -6.41 + 3.18 + 3.75 = 0.52 = 0.05 \cdot 10.45$. $\checkmark$

The vega tells us that an extra percentage point of volatility is worth about £0.375 on this option – a useful rule of thumb, since a one-year at-the-money option always has $\mathcal{V} \approx 0.4s$.

Geometrically, Δ is the slope of the price curve, i.e. the number of shares whose value changes at the same rate as the option:



Remark (Why hedging works). Suppose a bank sells the call for $V(0, S_0)$ and continuously holds $\Delta_t = V_s(t, S_t)$ shares, financing the rest at rate r . By the proof of Theorem 43.8.9 the portfolio value satisfies exactly the same stochastic differential equation as $V(t, S_t)$ and starts at the same value, so the two agree for all t ; at maturity the portfolio is worth exactly the payout. The bank has eliminated its risk completely, and its profit is the difference between the price it charged and $V(0, S_0)$.

In reality one rebalances at discrete times, so there is a residual hedging error. Its size is governed by Γ : the larger the gamma, the more the delta moves between rebalancing dates and the worse the approximation. Options near the money and near expiry have very large gamma, which is why they are unpleasant to hedge.

Proposition 43.8.15 (Monotonicity)

If g is increasing then $\Delta \geq 0$; if g is convex then $\Gamma \geq 0$.

Proof. Both are immediate from the representation

$$V(t, s) = e^{-r\tau} \mathbb{E}[g(se^{(r-\sigma^2/2)\tau + \sigma\sqrt{\tau}Z})] :$$

for each fixed value of Z the integrand is an increasing (respectively convex) function of s whenever g is, and both properties survive taking expectations. $\square$

§43.8.7 Implied Volatility

Of the five inputs to the Black–Scholes formula, four – s , K , τ , r – are directly observable. The volatility σ is not. But option prices *are* observable, and the formula can be inverted.

Proposition 43.8.16

For fixed s, K, τ, r , the Black–Scholes call price is a strictly increasing continuous function of σ , mapping $(0, \infty)$ onto $((s - Ke^{-r\tau})^+, s)$.

Proof. Vega is $\mathcal{V} = s\sqrt{\tau}\varphi(d_1) > 0$, giving strict monotonicity. As $\sigma \downarrow 0$ we have $d_1, d_2 \rightarrow \pm\infty$ according to the sign of $\log(s/K) + r\tau$, and the price tends to $(s - Ke^{-r\tau})^+$, the value of the forward. As $\sigma \rightarrow \infty$, $d_1 \rightarrow \infty$ and $d_2 \rightarrow -\infty$, so the price tends to s . $\square$

Definition 43.8.17 (Implied volatility)

Given an observed market price C^{mkt} of a call with strike K and maturity T , the **implied volatility** $\sigma_{\text{imp}}(K, T)$ is the unique σ for which the Black–Scholes formula returns C^{mkt} .

The inversion has no closed form, but the proposition guarantees that a bisection or Newton search converges, and vega – being the derivative one needs for Newton’s method – is available in closed form.

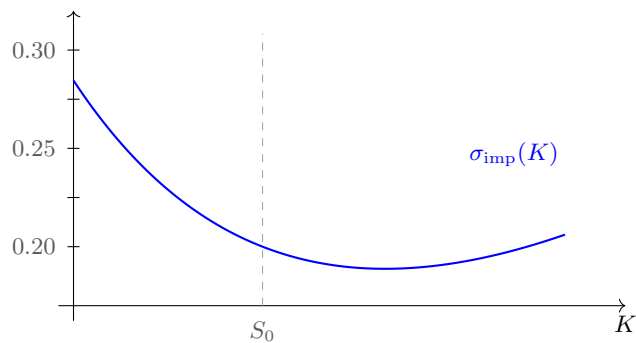
Example 43.8.18

Continue with $s = K = 100$, $r = 0.05$, $\tau = 1$, and suppose the option trades at $C^{\text{mkt}} = 12$ rather than the 10.45 produced by $\sigma = 0.2$. One step of Newton’s method from $\sigma_0 = 0.2$, using $\mathcal{V} = 37.52$ from the previous section, gives

$$\sigma_1 = 0.2 + \frac{12 - 10.45}{37.52} = 0.2413,$$

and the true implied volatility is 0.2411: a single step is already accurate to four decimal places, because the call price is very nearly linear in σ near the money. The market is therefore quoting this option as though the stock will move about 24% over the year rather than 20%.

If the Black–Scholes model were literally correct, then implied volatility would be the same for every strike and maturity, since there is only one σ in the model. It is not. Plotting σ_{imp} against strike for a fixed maturity typically gives a curve like the following, known as the **volatility smile** (or, when it is lopsided as here, the **skew**).



The market charges more than Black–Scholes for low-strike puts, because the real distribution of returns has a fatter left tail than the lognormal: crashes happen. The smile is therefore best read not as a statement about volatility but as a statement about the market’s implied distribution for S_T , which is a nonparametric object recoverable from the prices of calls at all strikes. That the model is systematically wrong and yet universally used is not a paradox: implied volatility is simply a convenient reparametrisation of price, in the same way that a bond’s yield is a reparametrisation of its price.

§43.9 Further Topics

The Black–Scholes machine is flexible, and a great deal can be done by making small modifications to it. We finish with a selection.

§43.9.1 Dividends

Real shares pay dividends. Suppose the stock pays a continuous dividend at rate q , meaning that holding one share over $[t, t + dt]$ produces a cash payment $qS_t dt$ in addition to the capital gain. Then the quantity which must be a martingale after discounting is not the price but the *total return*: reinvesting the dividends, one share at time 0 becomes e^{qt} shares at time t .

Proposition 43.9.1

With a continuous dividend yield q , the risk-neutral dynamics are

$$dS_t = (r - q)S_t dt + \sigma S_t d\tilde{W}_t, \quad \text{so} \quad S_T = S_t e^{(r-q-\sigma^2/2)\tau + \sigma(\tilde{W}_T - \tilde{W}_t)},$$

and the price of a European call is

$$C = se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad d_{1,2} = \frac{\log(s/K) + (r - q \pm \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}.$$

Proof. The traded asset with no intermediate payments is $\hat{S}_t = e^{qt}S_t$ (the reinvested position), so $e^{-rt}\hat{S}_t$ must be a $\mathbb{Q}$ -martingale, i.e. $e^{-(r-q)t}S_t$ is a $\mathbb{Q}$ -martingale. This forces the stated drift. The formula then follows from Theorem 43.8.4 on replacing s by $se^{-q\tau}$ and r by r throughout the derivation; equivalently, on noting that $\mathbb{E}_{\mathbb{Q}}[S_T | \mathcal{F}_t] = se^{(r-q)\tau}$. $\square$

Dividends break the ‘never exercise a call early’ result of §6.7. The reason is transparent: an American call holder receives no dividends, so if q is large enough it can be worth exercising to capture them. Precisely, the submartingale argument required $\mathbb{E}_{\mathbb{Q}}[S_n | \mathcal{F}_{n-1}] \geq (1+r)S_{n-1}$, which fails when $q > 0$.

§43.9.2 Foreign Exchange

Consider two currencies, ‘domestic’ and ‘foreign’, with riskless rates r_d and r_f . Let X_t be the exchange rate, quoted as the number of units of domestic currency per unit of foreign currency.

A domestic investor may hold the foreign bank account. In domestic terms it is worth $X_t e^{r_f t}$, and it is a traded asset with no intermediate payments. Hence, under the domestic risk-neutral measure $\mathbb{Q}_d$, the process $e^{-r_d t} X_t e^{r_f t}$ must be a martingale. So X behaves exactly like a stock paying a dividend at rate r_f :

$$dX_t = (r_d - r_f)X_t dt + \sigma X_t d\tilde{W}_t.$$

Proposition 43.9.2 (Garman–Kohlhagen)

The domestic price of a European call on the exchange rate with strike K and maturity T is

$$C = xe^{-r_f \tau}\Phi(d_1) - Ke^{-r_d \tau}\Phi(d_2), \quad d_{1,2} = \frac{\log(x/K) + (r_d - r_f \pm \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}.$$

There is a pleasant symmetry here which is worth spelling out, because it is a genuine change of numeraire. The foreign investor sees the exchange rate $Y_t = 1/X_t$, and by Itô

$$dY_t = (r_f - r_d + \sigma^2)Y_t dt - \sigma Y_t d\tilde{W}_t$$

under $\mathbb{Q}_d$. The extra σ^2 looks alarming – it appears to say that the two investors disagree about which currency is expected to appreciate, a puzzle known as **Siegel’s paradox**. The resolution is that the foreign investor prices with a different measure $\mathbb{Q}_f$, defined by

$$\frac{d\mathbb{Q}_f}{d\mathbb{Q}_d} = \frac{e^{-r_f t} X_t^{-1}}{X_0^{-1} e^{-r_d t}},$$

the density being (the discounted value of) one unit of domestic currency expressed in foreign terms. Under $\mathbb{Q}_f$ one finds $dY_t = (r_f - r_d)Y_t dt + \sigma Y_t d\hat{W}_t$ for a $\mathbb{Q}_f$ -Brownian motion $\hat{W}$, restoring the symmetry. There is no paradox: expectations are measure-dependent, and prices are not.

§43.9.3 American Options in Continuous Time

In continuous time the price of an American claim with payout $g(S_t)$ is

$$\pi_t = \sup_{t \leq T \leq \mathcal{T}} \mathbb{E}_{\mathbb{Q}}[e^{-r(T-t)} g(S_T) \mid \mathcal{F}_t],$$

the supremum over stopping times. As in discrete time, the value process is the smallest supermartingale dominating the discounted payout. The analytic formulation is a **free boundary problem**: there is a **continuation region** $\mathcal{C}$ in which V solves the Black–Scholes PDE and $V > g$, and a **stopping region** in which $V = g$; the boundary between them is not known in advance and must be found as part of the solution.

At the boundary one imposes **value matching** $V = g$ and **smooth fit** $V' = g'$. The smooth fit condition is not a boundary condition in the usual sense – it is an optimality condition, and it is what selects the correct boundary from the one-parameter family of candidates. The case which can be solved completely is the following.

Example 43.9.3 (The perpetual American put)

Take $g(s) = (K - s)^+$ and no expiry date. By stationarity the value $V(s)$ does not depend on t , and the exercise rule must be of the form ‘exercise when $S_t \leq b$ ’ for some level b to be determined. In the continuation region $s > b$, the Black–Scholes PDE with $V_t = 0$ becomes the Euler equation

$$rsV'(s) + \frac{1}{2}\sigma^2 s^2 V''(s) = rV(s).$$

Trying $V(s) = s^{-\gamma}$ gives the indicial equation $\frac{1}{2}\sigma^2 \gamma(\gamma + 1) - r\gamma - r = 0$, whose positive root is

$$\gamma = \frac{2r}{\sigma^2}.$$

(The other root is $\gamma = -1$, giving the solution $V = s$, which is excluded because V must be bounded as $s \rightarrow \infty$.) So $V(s) = As^{-\gamma}$ for $s > b$. Value matching at $s = b$ gives $Ab^{-\gamma} = K - b$, i.e.

$$V(s) = (K - b) \left(\frac{b}{s}\right)^{\gamma}, \quad s \geq b.$$

Smooth fit requires $V'(b) = -1$, i.e. $-\gamma(K - b)/b = -1$, giving

$$b = \frac{\gamma K}{1 + \gamma} = \frac{2rK}{2r + \sigma^2}.$$

Note the sensible limits: as $\sigma \rightarrow 0$ we get $b \rightarrow K$ (with no volatility there is no reason to wait), and as $\sigma \rightarrow \infty$ we get $b \rightarrow 0$ (with enormous volatility it is always worth waiting). For finite expiry no closed form exists and one must compute numerically.

§43.9.4 Barrier and Path-Dependent Claims

Not every claim is a function of S_T alone. A **barrier option** pays off only if the price has (or has not) crossed a level B before maturity: the **up-and-in** version of a claim Y pays $Y \mathbb{1}_{\{\max_{t \leq T} S_t \geq B\}}$, the **up-and-out** version pays $Y \mathbb{1}_{\{\max_{t \leq T} S_t < B\}}$, and similarly for down-and-in and down-and-out. Adding an in and an out version recovers the original claim, so it suffices to price one of them.

Since $S_t = S_0 e^{\sigma(\tilde{W}_t + ct)}$ with $c = r/\sigma - \sigma/2$ under $\mathbb{Q}$, barrier crossing is a statement about the maximum of a Brownian motion with drift, and the tools of §7 apply.

Proposition 43.9.4

In the Black–Scholes model, the initial price of the up-and-out claim with payout $g(S_T) \mathbb{1}_{\{\max_{t \leq T} S_t < B\}}$ equals the price of the European claim with payout

$$g(S_T) \mathbb{1}_{\{S_T < B\}} - \left(\frac{B}{S_0}\right)^{2r/\sigma^2 - 1} g\left(\frac{B^2 S_T}{S_0^2}\right) \mathbb{1}_{\{S_T < S_0^2/B\}}.$$

Proof sketch. Write $b = \log(B/S_0)/\sigma$ and $S_t = S_0 e^{\sigma(W_t + ct)}$ for a $\mathbb{Q}$ -Brownian motion W . Since $\{\max_t S_t < B\} \subseteq \{S_T < B\}$,

$$g(S_T) \mathbb{1}_{\{\max_t S_t < B\}} = g(S_T) \mathbb{1}_{\{S_T < B\}} - g(S_T) \mathbb{1}_{\{S_T < B, \max_t S_t \geq B\}},$$

so it suffices to compute the expectation of the last term. By the Cameron–Martin theorem (Theorem 43.7.20) we may strip off the drift, then apply the reflection lemma (Lemma 43.7.17) to the driftless maximum, then reinstate the drift:

$$\begin{aligned} \mathbb{E}[g(S_T) \mathbb{1}_{\{S_T < B, \max_t S_t \geq B\}}] &= \mathbb{E}[e^{cW_T - c^2 T/2} g(S_0 e^{\sigma W_T}) \mathbb{1}_{\{W_T < b, \max_t W_t \geq b\}}] \\ &= \mathbb{E}[e^{c(2b - W_T) - c^2 T/2} g(S_0 e^{\sigma(2b - W_T)}) \mathbb{1}_{\{W_T > b\}}] \\ &= e^{2bc} \mathbb{E}[e^{-cW_T - c^2 T/2} g(S_0 e^{2b\sigma} e^{-\sigma W_T}) \mathbb{1}_{\{W_T > b\}}], \end{aligned}$$

and undoing the change of measure gives the stated European payout, with $e^{2bc} = (B/S_0)^{2r/\sigma^2 - 1}$ and $S_0 e^{2b\sigma} = B^2/S_0$. $\square$

Example 43.9.5 (Forward-start call)

A **forward-start** call fixes its strike to be the price at an intermediate date $T_0 < T_1$, so the payout is $(S_{T_1} - S_{T_0})^+$. For $t \geq T_0$ this is an ordinary call with known strike. For $t < T_0$, using that S_{T_1}/S_{T_0} is independent of $\mathcal{F}_{T_0}$ under $\mathbb{Q}$ (the increments of $\tilde{W}$

being independent),

$$\begin{aligned}
 \pi_t &= e^{-r(T_1-t)} \mathbb{E}_{\mathbb{Q}}[(S_{T_1} - S_{T_0})^+ | \mathcal{F}_t] \\
 &= e^{-r(T_1-t)} \mathbb{E}_{\mathbb{Q}} \left[S_{T_0} \left(\frac{S_{T_1}}{S_{T_0}} - 1 \right)^+ \mid \mathcal{F}_t \right] \\
 &= e^{-r(T_1-t)} \mathbb{E}_{\mathbb{Q}}[S_{T_0} | \mathcal{F}_t] \mathbb{E}_{\mathbb{Q}} \left[\left(\frac{S_{T_1}}{S_{T_0}} - 1 \right)^+ \right] \\
 &= e^{-r(T_1-T_0)} S_t \mathbb{E}_{\mathbb{Q}} \left[\left(\frac{S_{T_1}}{S_{T_0}} - 1 \right)^+ \right],
 \end{aligned}$$

and the remaining expectation is the Black–Scholes price of an at-the-money call on a unit stock over the horizon $T_1 - T_0$. So the forward-start call is priced by a homogeneity argument, with no new integration required.

§43.9.5 Numerical Schemes

Beyond a small collection of examples, option prices must be computed numerically. There are three standard approaches: Monte Carlo simulation of the risk-neutral expectation; the binomial tree of §6, which is simply an explicit finite-difference scheme; and direct solution of the Black–Scholes PDE. We say a word about the last.

After the change of variables reducing Black–Scholes to the heat equation $u_t = \frac{1}{2}u_{xx}$, one discretises on a grid with steps Δt and Δx , writing $U_{n,k} \approx u(n\Delta t, k\Delta x)$. The three standard discretisations are

$$\begin{aligned}
 \text{(forward)} \quad & \frac{U_{n+1,k} - U_{n,k}}{\Delta t} = \frac{U_{n,k+1} - 2U_{n,k} + U_{n,k-1}}{2(\Delta x)^2}, \\
 \text{(backward)} \quad & \frac{U_{n+1,k} - U_{n,k}}{\Delta t} = \frac{U_{n+1,k+1} - 2U_{n+1,k} + U_{n+1,k-1}}{2(\Delta x)^2}, \\
 \text{(Crank–Nicolson)} \quad & \frac{U_{n+1,k} - U_{n,k}}{\Delta t} = \frac{U_{n,k+1} - 2U_{n,k} + U_{n,k-1}}{4(\Delta x)^2} \\
 & \quad + \frac{U_{n+1,k+1} - 2U_{n+1,k} + U_{n+1,k-1}}{4(\Delta x)^2},
 \end{aligned}$$

the last being the average of the first two.

Definition 43.9.6 (von Neumann stability)

A scheme is **stable** in the von Neumann sense if $\sup_{\beta} |\varrho(\beta)| \leq 1$, where $\varrho(\beta)$ is defined by requiring that $U_{n,k} = \varrho(\beta)^n e^{i\beta k \Delta x}$ solves the recursion.

Substituting the ansatz and writing $\nu = \Delta t / (\Delta x)^2$ gives

$$\begin{aligned}
 \varrho_{\text{fwd}}(\beta) &= 1 - \nu(1 - \cos(\beta \Delta x)), & \varrho_{\text{bwd}}(\beta) &= \frac{1}{1 + \nu(1 - \cos(\beta \Delta x))}, \\
 \varrho_{\text{CN}}(\beta) &= \frac{2 - \nu(1 - \cos(\beta \Delta x))}{2 + \nu(1 - \cos(\beta \Delta x))}.
 \end{aligned}$$

The forward scheme is stable only if $\nu \leq 1$, which forces very small time steps; the backward and Crank–Nicolson schemes are unconditionally stable, at the cost of solving

a tridiagonal linear system at each step. Crank–Nicolson is additionally second-order accurate in time, and is the usual default.

Remark. It is a pleasing coda that the binomial model, which we introduced as a self-contained discrete market, reappears here as a numerical method for the continuous one; and that the stability condition $\nu \leq 1$ is exactly the requirement that the implied risk-neutral probabilities lie in $[0, 1]$, i.e. that the discretised model be arbitrage-free. Numerical stability and the absence of arbitrage are, in this instance, the same condition.

44 Topics in Analysis

‘Topics in Analysis’ is a course with no real destination. It is a tour of some of the prettier shells lying on the beach of real and complex analysis, picked up because they are pretty rather than because they are needed to build something else. We will prove that you cannot stir a cup of coffee without leaving some point of the surface where it started, that every continuous function on an interval is a uniform limit of polynomials, that there are continuous functions which are differentiable nowhere at all, and that the number $\sum 10^{-n!}$ satisfies no polynomial equation with integer coefficients. Along the way we will meet a theorem which won its author a Nobel prize in economics.

What ties these together is a small number of extremely powerful ideas – compactness, completeness, and the systematic exploitation of approximation. Almost every argument in the course is a variation on one of the following themes: a continuous function on a compact set attains its bounds; a sequence which does not move very much converges; and if you want to prove that an object with strange properties exists, it is often easier to prove that *almost every* object has them than to write one down.

This article constitutes my notes for the ‘Topics in Analysis’ course, held in Lent 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

§44.1 Preliminaries

This first section is mostly a matter of recalling things that we already know, and fixing notation. The course proper begins in the next section; the reader who finds the material below either worryingly familiar or worryingly unfamiliar is asked to suspend judgement until then.

§44.1.1 Metric Spaces and Completeness

Almost all of the spaces we deal with in this course will be $\mathbb{R}^m$, $\mathbb{C}$, or a space of functions or of compact sets. The common framework is that of a metric space.

Definition 44.1.1 (Metric Space)

Let X be a non-empty set. A function $d : X^2 \rightarrow \mathbb{R}$ is a **metric** on X if

- (i) $d(x, y) \geq 0$ for all $x, y \in X$;
- (ii) $d(x, y) = 0$ if and only if $x = y$;
- (iii) $d(x, y) = d(y, x)$ for all $x, y \in X$;
- (iv) $d(x, y) + d(y, z) \geq d(x, z)$ for all $x, y, z \in X$.

We then call (X, d) a **metric space**.

The examples to keep in mind are Euclidean space $\mathbb{R}^m$ with $d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|$, the complex plane with $d(z, w) = |z - w|$, and the space $C([a, b])$ of continuous real-valued

functions on $[a, b]$ with the uniform metric $d(f, g) = \|f - g\|_\infty$. We write

$$B(x, r) = \{y \in X : d(x, y) < r\}, \quad \overline{B}(x, r) = \{y \in X : d(x, y) \leq r\}$$

for the open and closed balls of radius r about x , and we say $E \subseteq X$ is **closed** if $x_n \in E$ and $x_n \rightarrow x$ imply $x \in E$, and $U \subseteq X$ is **open** if for each $u \in U$ there is a $\delta > 0$ with $B(u, \delta) \subseteq U$. These are complementary notions: E is closed if and only if $X \setminus E$ is open.

The single most important property a metric space can have, for our purposes, is completeness.

Definition 44.1.2 (Cauchy Sequence, Completeness)

Let (X, d) be a metric space. A sequence (a_n) in X is a **Cauchy sequence** if, given $\varepsilon > 0$, there is an N with $d(a_n, a_m) < \varepsilon$ for all $n, m \geq N$. We say (X, d) is **complete** if every Cauchy sequence in X converges.

We will need two small observations about Cauchy sequences over and over again, so it is worth isolating them.

Lemma 44.1.3

Let (X, d) be a metric space.

- (i) If a Cauchy sequence (x_n) has a subsequence converging to x , then $x_n \rightarrow x$.
- (ii) Suppose $\varepsilon_n > 0$ with $\sum_{n \geq 1} \varepsilon_n < \infty$. If every sequence (x_n) in X with $d(x_n, x_{n+1}) < \varepsilon_n$ for all n converges, then (X, d) is complete.

Proof. (i) Let $\varepsilon > 0$ and pick N with $d(x_n, x_m) < \varepsilon/2$ for $n, m \geq N$. If $x_{n(k)} \rightarrow x$, choose k with $n(k) \geq N$ and $d(x_{n(k)}, x) < \varepsilon/2$. Then for $n \geq N$ we have $d(x_n, x) \leq d(x_n, x_{n(k)}) + d(x_{n(k)}, x) < \varepsilon$.

(ii) Let (x_n) be Cauchy. Choose $n(1) < n(2) < \dots$ so that $d(x_p, x_q) < \varepsilon_k$ whenever $p, q \geq n(k)$. Then the subsequence $y_k = x_{n(k)}$ satisfies $d(y_k, y_{k+1}) < \varepsilon_k$, so converges by hypothesis, and (i) finishes the job. $\square$

In practice we use (ii) with $\varepsilon_n = 2^{-n}$ or $\varepsilon_n = 100^{-n}$: to check completeness it suffices to deal with sequences whose steps shrink geometrically, which is a considerable convenience. We record without proof the fundamental example.

Theorem 44.1.4

Euclidean space $\mathbb{R}^m$ with the usual metric is complete, as is $\mathbb{C}$.

The proof is the usual one: a Cauchy sequence in $\mathbb{R}^m$ is coordinatewise Cauchy, and $\mathbb{R}$ is complete by the general principle of convergence from Analysis I.

§44.1.2 The Contraction Mapping Theorem

Completeness earns its keep almost immediately. The following theorem was met in IB Analysis and Topology, but it is worth restating here, both because we shall want it and because it is the model against which everything in Section 44.3 should be compared.

Definition 44.1.5 (Contraction)

Let (X, d) be a metric space. A map $f : X \rightarrow X$ is a **contraction mapping** if there is a constant $0 \leq K < 1$ with

$$d(f(x), f(y)) \leq K d(x, y)$$

for all $x, y \in X$.

A contraction is certainly continuous – indeed uniformly so. The point of the definition is that it does not merely fail to increase distances, it shrinks them by a definite factor, and it is that definite factor which makes everything work.

Theorem 44.1.6 (Contraction Mapping Theorem)

Let (X, d) be a non-empty complete metric space and let $f : X \rightarrow X$ be a contraction mapping with constant K . Then f has exactly one fixed point w . Moreover, for any $x_0 \in X$ the iterates $x_{n+1} = f(x_n)$ satisfy $x_n \rightarrow w$, with

$$d(x_n, w) \leq \frac{K^n}{1 - K} d(x_0, x_1).$$

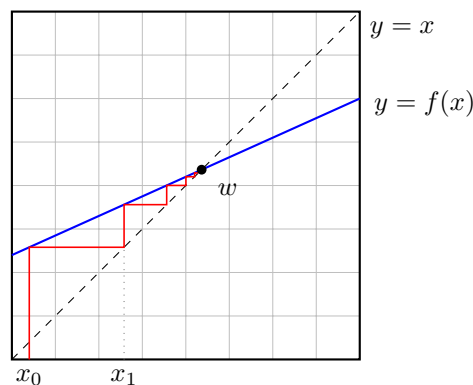
Proof. Uniqueness first, since it is free: if $f(v) = v$ and $f(w) = w$ then $d(v, w) = d(f(v), f(w)) \leq K d(v, w)$, and as $K < 1$ this forces $d(v, w) = 0$.

For existence, fix $x_0 \in X$ and set $x_{n+1} = f(x_n)$. An immediate induction gives $d(x_n, x_{n+1}) \leq K^n d(x_0, x_1)$, so for $m > n$

$$d(x_n, x_m) \leq \sum_{r=n}^{m-1} d(x_r, x_{r+1}) \leq d(x_0, x_1) \sum_{r=n}^{\infty} K^r = \frac{K^n}{1 - K} d(x_0, x_1).$$

Since $K^n \rightarrow 0$, the sequence (x_n) is Cauchy, and by completeness $x_n \rightarrow w$ for some $w \in X$. As f is continuous, $f(w) = \lim f(x_n) = \lim x_{n+1} = w$. Letting $m \rightarrow \infty$ in the displayed inequality gives the error bound. $\square$

The proof is worth a second look, because it is the pattern for a great many existence proofs: we do not find w , we manufacture a sequence which is forced to converge and then let the limit be w . The picture is the familiar staircase, or *cobweb*, of the numerical analyst.



Both hypotheses are needed. Completeness cannot be dropped: on $X = (0, 1]$ with the

usual metric the map $f(x) = x/2$ is a contraction with $K = 1/2$, but it has no fixed point in X . Nor can $K < 1$ be weakened to $d(f(x), f(y)) < d(x, y)$: the map $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = \sqrt{1 + x^2}$ satisfies

$$|f(x) - f(y)| = \frac{|x^2 - y^2|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} = |x - y| \cdot \frac{|x + y|}{\sqrt{1 + x^2} + \sqrt{1 + y^2}} < |x - y|$$

for $x \neq y$, yet $f(x) > x$ everywhere, so there is no fixed point at all.

The contraction mapping theorem is enormously useful – it is what makes Picard’s existence theorem for differential equations and the inverse function theorem work – but the hypothesis is very strong indeed, and it is easy to write down continuous self-maps of a disc which contract nothing. That such maps still have fixed points is the subject of Section 44.3, and the price we pay is that the proof will tell us nothing whatever about how to find one.

§44.1.3 Compactness in Euclidean Space

In this course we deal only with subsets of $\mathbb{R}^m$, and there ‘compact’ may be taken to mean ‘closed and bounded’. (The reader should be warned that this equivalence fails badly in general metric spaces, and that in the world of topological spaces one must further distinguish compactness from sequential compactness. We will not need any of this: if you intend to climb Everest you need your own oxygen supply, but for the Gog Magogs you do not.)

Theorem 44.1.7 (Bolzano–Weierstrass)

If $K > 0$ and $\mathbf{x}_r \in \mathbb{R}^m$ satisfy $\|\mathbf{x}_r\| \leq K$ for all r , then there exist $\mathbf{x} \in \mathbb{R}^m$ and $r(1) < r(2) < \dots$ such that $\mathbf{x}_{r(k)} \rightarrow \mathbf{x}$.

Proof. For $m = 1$ this is Analysis I. For general m , apply the one-dimensional result to the first coordinates to get a subsequence along which they converge, then pass to a further subsequence along which the second coordinates converge, and so on. After m steps every coordinate converges, and convergence in $\mathbb{R}^m$ is coordinatewise. $\square$

Theorem 44.1.8

A subset E of $\mathbb{R}^m$ is closed and bounded if and only if every sequence in E has a subsequence converging to a limit in E .

Proof. If E is closed and bounded, Bolzano–Weierstrass gives a convergent subsequence and closedness puts the limit in E . Conversely, if E is unbounded, choose $\mathbf{x}_r \in E$ with $\|\mathbf{x}_r\| \geq r$; no subsequence can converge. If E is not closed, take $\mathbf{x}_r \in E$ with $\mathbf{x}_r \rightarrow \mathbf{x} \notin E$; every subsequence converges to $\mathbf{x} \notin E$. $\square$

We shall call closed bounded subsets of $\mathbb{R}^m$ **compact**, and refer to the property in Theorem 44.1.8 as the Bolzano–Weierstrass property. Its usefulness comes from the following two results, which will be invoked so frequently that we shall often not bother to mention them.

Theorem 44.1.9

If $E \subseteq \mathbb{R}^m$ is compact and $f : E \rightarrow \mathbb{R}^n$ is continuous, then $f(E)$ is compact.

Proof. Let $\mathbf{y}_r = f(\mathbf{x}_r)$ with $\mathbf{x}_r \in E$. Choose a subsequence $\mathbf{x}_{r(k)} \rightarrow \mathbf{x} \in E$. By continuity $\mathbf{y}_{r(k)} \rightarrow f(\mathbf{x}) \in f(E)$. $\square$

Theorem 44.1.10

If $E \subseteq \mathbb{R}^m$ is non-empty and compact and $f : E \rightarrow \mathbb{R}$ is continuous, then there exist $\mathbf{a}, \mathbf{b} \in E$ with $f(\mathbf{b}) \leq f(\mathbf{x}) \leq f(\mathbf{a})$ for all $\mathbf{x} \in E$.

Proof. $f(E)$ is a non-empty compact subset of $\mathbb{R}$, hence bounded, and contains its supremum and infimum since it is closed. $\square$

We will also want uniform continuity, which is compactness in another guise.

Definition 44.1.11 (Uniform Continuity)

Let (X, d) and (Y, ρ) be metric spaces. A function $f : X \rightarrow Y$ is **uniformly continuous** if, given $\varepsilon > 0$, we can find $\delta > 0$ such that $\rho(f(a), f(b)) < \varepsilon$ whenever $d(a, b) < \delta$.

Theorem 44.1.12

If E is a compact subset of $\mathbb{R}^m$ and $f : E \rightarrow \mathbb{R}^p$ is continuous, then f is uniformly continuous.

Proof. If not, there is an $\varepsilon > 0$ and points $\mathbf{a}_n, \mathbf{b}_n \in E$ with $\|\mathbf{a}_n - \mathbf{b}_n\| < 1/n$ but $\|f(\mathbf{a}_n) - f(\mathbf{b}_n)\| \geq \varepsilon$. Pass to a subsequence with $\mathbf{a}_{n(k)} \rightarrow \mathbf{a} \in E$; then $\mathbf{b}_{n(k)} \rightarrow \mathbf{a}$ too, so both $f(\mathbf{a}_{n(k)})$ and $f(\mathbf{b}_{n(k)})$ tend to $f(\mathbf{a})$, contradicting the inequality $\|f(\mathbf{a}_{n(k)}) - f(\mathbf{b}_{n(k)})\| \geq \varepsilon$. $\square$

Finally, a piece of notation which will be with us all course. If A is a non-empty subset of a metric space and x a point, we write

$$d(x, A) = \inf\{d(x, a) : a \in A\}.$$

Lemma 44.1.13

Let A be a non-empty subset of a metric space (X, d) . Then $x \mapsto d(x, A)$ satisfies $|d(x, A) - d(y, A)| \leq d(x, y)$, and so is (uniformly) continuous. If A is closed, then $d(x, A) = 0$ if and only if $x \in A$.

Proof. For any $a \in A$ we have $d(x, A) \leq d(x, a) \leq d(x, y) + d(y, a)$; taking the infimum over a gives $d(x, A) \leq d(x, y) + d(y, A)$, and symmetry gives the Lipschitz bound. If $d(x, A) = 0$ we can find $a_n \in A$ with $d(x, a_n) \rightarrow 0$, so $a_n \rightarrow x$ and $x \in A$ when A is closed. $\square$

§44.1.4 The Fundamental Theorem of Algebra

Theorem 44.1.10 looks like a piece of abstract nonsense, but it is enough to prove a theorem which – contrary to its name – is a theorem of analysis.

Theorem 44.1.14 (Fundamental Theorem of Algebra)

Every non-constant polynomial P with complex coefficients has a root in $\mathbb{C}$.

Proof. Write $P(z) = \sum_{j=0}^n a_j z^j$ with $n \geq 1$, and (dividing through) assume $a_n = 1$. Then for $z \neq 0$,

$$|P(z)| \geq |z|^n (1 - |a_{n-1}||z|^{-1} - \cdots - |a_0||z|^{-n}),$$

so if we set $R = 2 \left(2 + \sum_{j=0}^{n-1} |a_j| \right)$ then $|z| \geq R$ gives $|P(z)| \geq |z|^n/2 \geq R^n/2 > |a_0| = |P(0)|$.

Now $\overline{B}(0, R)$ is compact and $|P|$ is continuous, so by Theorem 44.1.10 there is a $z_0 \in \overline{B}(0, R)$ at which $|P|$ attains its minimum over $\overline{B}(0, R)$. By the previous paragraph $|P(z_0)| \leq |P(0)| < |P(z)|$ for all $|z| \geq R$, so in fact z_0 is a *global* minimum of $|P|$ on $\mathbb{C}$.

Suppose, for contradiction, that $P(z_0) \neq 0$. Replacing $P(z)$ by $P(z + z_0)$ we may assume $z_0 = 0$, and multiplying P by a suitable unimodular constant we may assume $a_0 = P(0) > 0$. Let $m \geq 1$ be least with $a_m \neq 0$, so that

$$P(z) = a_0 + a_m z^m + \sum_{j=m+1}^n a_j z^j.$$

Choosing ω with $\omega^m = -|a_m|/a_m$ and replacing $P(z)$ by $P(\omega z)$, we may further assume $a_m < 0$. Now for real $\delta > 0$,

$$\frac{P(\delta) - P(0)}{\delta^m} = a_m + \sum_{j=m+1}^n a_j \delta^{j-m} \rightarrow a_m < 0$$

as $\delta \rightarrow 0^+$. So for small positive δ the real part of $P(\delta) - P(0)$ is negative and its imaginary part is $o(\delta^m)$, whence $|P(\delta)| < |P(0)|$. This contradicts the minimality of $|P(0)|$, so $P(z_0) = 0$. $\square$

We shall meet a completely different proof of Theorem 44.1.14, using winding numbers, in the final section of the course. In the meantime we record the standard consequences, which we use freely.

Lemma 44.1.15

Let P be a polynomial of degree n over $\mathbb{C}$.

- (i) If $P(a) = 0$ then $P(z) = (z - a)Q(z)$ for some polynomial Q of degree $n - 1$.
- (ii) $P(z) = A \prod_{j=1}^n (z - a_j)$ for some $A \neq 0$ and $a_j \in \mathbb{C}$.
- (iii) If P has degree at most n and vanishes at $n + 1$ distinct points, then $P = 0$.

Proof. (i) By induction on n , division gives $P(z) = (z-a)Q(z)+r$ with $\deg Q = n-1$ and $r \in \mathbb{C}$; putting $z = a$ gives $r = 0$. (ii) follows by induction using Theorem 44.1.14 and (i). (iii) If $P \neq 0$ has degree $d \leq n$ then by (ii) it has at most $d \leq n$ roots. $\square$

§44.2 Laplace's Equation

Our first genuine topic is a piece of analysis which sits behind a great deal of the theory of partial differential equations. Recall that if ϕ is a real-valued function on (an open subset of) $\mathbb{R}^m$ with enough derivatives, we write

$$\nabla^2 \phi = \sum_{j=1}^m \frac{\partial^2 \phi}{\partial x_j^2},$$

and **Laplace's equation** is the equation $\nabla^2 \phi = 0$. Solutions are called *harmonic*. In Part IA you cheerfully solved boundary value problems for Laplace's equation on discs and rectangles, and you may have been left with the impression that the Dirichlet problem

$$\nabla^2 \phi = 0 \text{ on } \Omega, \quad \phi = f \text{ on } \partial\Omega$$

always has exactly one solution. We shall see that uniqueness holds in great generality, but that existence can fail.

We first need to recall three pieces of point-set vocabulary.

Definition 44.2.1 (Interior, Closure, Boundary)

Let (X, d) be a metric space and $E \subseteq X$.

- (i) The **interior** $\text{Int } E$ is the set of $x \in X$ for which there is a $\delta > 0$ with $B(x, \delta) \subseteq E$.
- (ii) The **closure** $\text{Cl } E$ is the set of $x \in X$ for which there are $e_n \in E$ with $e_n \rightarrow x$.
- (iii) The **boundary** of E is $\partial E = \text{Cl } E \setminus \text{Int } E$.

It is easy to check that $\text{Int } E$ is the largest open set contained in E , that $\text{Cl } E$ is the smallest closed set containing E , and hence that $\partial E = \text{Cl } E \cap \text{Cl}(X \setminus E)$ is closed. If $E \subseteq \mathbb{R}^m$ is bounded then so is $\text{Cl } E$, so $\text{Cl } E$ and ∂E are compact.

§44.2.1 The Maximum Principle

Throughout this subsection Ω is a non-empty bounded open subset of $\mathbb{R}^m$ and $\phi : \text{Cl } \Omega \rightarrow \mathbb{R}$ is continuous and twice differentiable on Ω . Note that $\text{Cl } \Omega$ is compact and non-empty, so ϕ does attain a maximum somewhere; the content of what follows is *where*.

The following lemma is really just the second derivative test, applied one coordinate at a time.

Lemma 44.2.2

If $\nabla^2 \phi > 0$ everywhere on Ω , then ϕ does not attain its maximum at any point of Ω .

Proof. Suppose ϕ attains its maximum over $\text{Cl } \Omega$ at $\mathbf{x}^* \in \Omega$. Since Ω is open there is a $\delta > 0$ with

$$(x_1^* - \delta, x_1^* + \delta) \times \cdots \times (x_m^* - \delta, x_m^* + \delta) \subseteq \Omega.$$

For each j define $g_j(t) = \phi(x_1^*, \dots, x_{j-1}^*, x_j^* + t, x_{j+1}^*, \dots, x_m^*)$ for $|t| < \delta$. Then g_j is twice differentiable with a maximum at $t = 0$, so $g_j'(0) = 0$ and $g_j''(0) \leq 0$. But $g_j''(0) = \partial^2 \phi / \partial x_j^2$ evaluated at $\mathbf{x}^*$, so summing over j gives $\nabla^2 \phi(\mathbf{x}^*) \leq 0$, a contradiction. $\square$

The hypothesis $\nabla^2 \phi > 0$ is of course much stronger than what we want. The trick, which is the same one used to deduce Rolle's theorem from the fact that an interior maximum is stationary, is to add a small strictly subharmonic perturbation and then let it go to zero.

Theorem 44.2.3 (Maximum Principle)

If $\nabla^2 \phi = 0$ on Ω , then ϕ attains its maximum over $\text{Cl } \Omega$ at some point of $\partial \Omega$.

Proof. For $n \geq 1$ set $\phi_n(\mathbf{x}) = \phi(\mathbf{x}) + \frac{1}{n} \sum_{j=1}^m x_j^2$. Then $\nabla^2 \phi_n = 2m/n > 0$ on Ω , so by Lemma 44.2.2 the maximum of ϕ_n over $\text{Cl } \Omega$ is attained at some $\mathbf{x}_n^* \in \partial \Omega$. Since $\partial \Omega$ is compact, we may pass to a subsequence $\mathbf{x}_{n(j)}^* \rightarrow \mathbf{x}^* \in \partial \Omega$.

Let R be such that $\|\mathbf{x}\| \leq R$ for all $\mathbf{x} \in \text{Cl } \Omega$. For every $\mathbf{x} \in \text{Cl } \Omega$ we have

$$\phi(\mathbf{x}_{n(j)}^*) + \frac{R^2}{n(j)} \geq \phi_{n(j)}(\mathbf{x}_{n(j)}^*) \geq \phi_{n(j)}(\mathbf{x}) \geq \phi(\mathbf{x}).$$

Letting $j \rightarrow \infty$ and using continuity of ϕ gives $\phi(\mathbf{x}^*) \geq \phi(\mathbf{x})$, as required. $\square$

Applying the theorem to $-\phi$ shows that the minimum is also attained on the boundary, and uniqueness follows at once.

Theorem 44.2.4 (Uniqueness for the Dirichlet Problem)

Suppose $\phi, \psi : \text{Cl } \Omega \rightarrow \mathbb{R}$ are continuous, twice differentiable on Ω , and satisfy $\nabla^2 \phi = \nabla^2 \psi = 0$ on Ω . If $\phi = \psi$ on $\partial \Omega$, then $\phi = \psi$ on $\text{Cl } \Omega$.

Proof. Set $\chi = \phi - \psi$. Then $\nabla^2 \chi = 0$ on Ω and $\chi = 0$ on $\partial \Omega$, so by Theorem 44.2.3 the maximum of χ is 0, and applying the same to $-\chi$ gives that the minimum is 0. Hence $\chi \equiv 0$. $\square$

Remark. The same argument gives a maximum principle for analytic functions. If $\Omega \subseteq \mathbb{C}$ is bounded and open and $f : \text{Cl } \Omega \rightarrow \mathbb{C}$ is continuous and analytic on Ω , then $\text{Re } f$ and $\text{Im } f$ are harmonic, and one deduces that $|f|$ attains its maximum on $\partial \Omega$: given z_0 , rotate f so that $\text{Re } f(z_0) = |f(z_0)|$ and apply Theorem 44.2.3 to $\text{Re } f$.

§44.2.2 Zaremba's Example

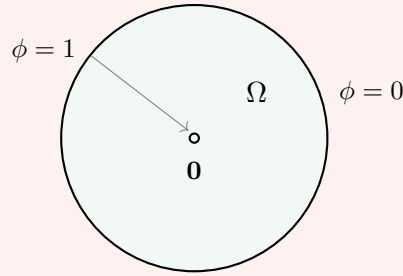
So much for uniqueness. What about existence? The following example, due to Zaremba, shows that we cannot expect a solution for every bounded open Ω and every continuous

boundary datum. It is a nice illustration of how uniqueness theorems can be used to *destroy* existence.

Example 44.2.5 (Zaremba)

Let $D = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| < 1\}$ and let $\Omega = D \setminus \{\mathbf{0}\}$ be the punctured disc. Then Ω is open and bounded, $\text{Cl } \Omega = \text{Cl } D$, and

$$\partial\Omega = \{\mathbf{x} : \|\mathbf{x}\| = 1\} \cup \{\mathbf{0}\}.$$



We claim that there is no continuous $\phi : \text{Cl } \Omega \rightarrow \mathbb{R}$, twice differentiable on Ω , with $\nabla^2 \phi = 0$ on Ω and

$$\phi(\mathbf{x}) = \begin{cases} 0 & \|\mathbf{x}\| = 1, \\ 1 & \mathbf{x} = \mathbf{0}. \end{cases}$$

Suppose such a ϕ existed. Let R_θ be the rotation of $\mathbb{R}^2$ through θ about the origin. Then $\mathbf{x} \mapsto \phi(R_\theta \mathbf{x})$ is also continuous on $\text{Cl } \Omega$, harmonic on Ω , and takes the *same* boundary values (since the boundary data are rotation-invariant). By the uniqueness Theorem 44.2.4 we get $\phi = \phi \circ R_\theta$ for every θ , so ϕ is radially symmetric: $\phi(\mathbf{x}) = f(\|\mathbf{x}\|)$ for some $f : [0, 1] \rightarrow \mathbb{R}$.

Writing $r = \sqrt{x^2 + y^2}$ and computing,

$$\begin{aligned} \frac{\partial \phi}{\partial x} &= \frac{x}{r} f'(r), \\ \frac{\partial^2 \phi}{\partial x^2} &= \left(\frac{1}{r} - \frac{x^2}{r^3} \right) f'(r) + \frac{x^2}{r^2} f''(r), \end{aligned}$$

and similarly for y , so that adding gives the familiar radial form

$$\nabla^2 \phi = f''(r) + \frac{1}{r} f'(r) = \frac{1}{r} \frac{d}{dr} (r f'(r)).$$

Thus $\frac{d}{dr} (r f'(r)) = 0$ for $0 < r < 1$, so $r f'(r) = B$ and $f(r) = A + B \log r$ for constants A, B . But ϕ is continuous at $\mathbf{0}$, and $A + B \log r$ has a limit as $r \rightarrow 0^+$ only when $B = 0$. So f is constant, and ϕ cannot take the value 0 on $\|\mathbf{x}\| = 1$ and 1 at $\mathbf{0}$. This is the contradiction we wanted.

The moral is that the boundary point $\mathbf{0}$ is ‘too thin’ for the boundary condition there to be felt. Lebesgue later produced a three-dimensional example (the *Lebesgue thorn*) in which the offending boundary point is a cusp rather than an isolated point, so that the failure is not simply a matter of the boundary having an isolated point. Deciding exactly which boundary points are ‘regular’ in this sense is a substantial piece of potential theory, and we shall not pursue it.

§44.3 Fixed Point Theorems

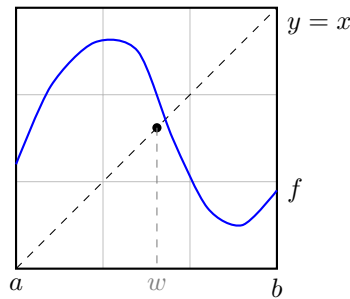
In Analysis I we proved the intermediate value theorem, and used it to prove a very pretty one-dimensional fixed point theorem.

Theorem 44.3.1

If $f : [a, b] \rightarrow [a, b]$ is continuous, then there is a $w \in [a, b]$ with $f(w) = w$.

Proof. Set $g(x) = f(x) - x$. Then g is continuous, $g(a) = f(a) - a \geq 0$ and $g(b) = f(b) - b \leq 0$. By the intermediate value theorem there is a $w \in [a, b]$ with $g(w) = 0$. $\square$

The picture is irresistible: the graph of f lives in the square $[a, b]^2$, it starts on or above the diagonal and ends on or below it, so it must cross.



The implication can be reversed: Theorem 44.3.1 implies the intermediate value theorem, and both are equivalent to the statement that $[0, 1]$ is connected. This is worth bearing in mind, because the object of this section is to prove the two-dimensional analogue, and the proof will consist almost entirely of showing that a chain of superficially unrelated statements are all equivalent to one another.

Theorem 44.3.2 (Brouwer's Fixed Point Theorem)

Let $\overline{D} = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$. If $f : \overline{D} \rightarrow \overline{D}$ is continuous, then there is a $\mathbf{w} \in \overline{D}$ with $f(\mathbf{w}) = \mathbf{w}$.

We shall keep strictly to two dimensions, but the reader should know that the theorem and most of its consequences hold in $\mathbb{R}^n$; the only ingredient which needs real work to generalise is Sperner's lemma, and even there the changes are cosmetic rather than conceptual.

The result is not restricted to discs. The following observation lets us move it around freely, and we shall use it without comment.

Lemma 44.3.3

Suppose $g : \overline{D} \rightarrow A$ is a bijection with g and g^{-1} continuous. If $F : A \rightarrow A$ is continuous, then F has a fixed point.

Proof. The map $g^{-1} \circ F \circ g : \overline{D} \rightarrow \overline{D}$ is continuous, so by Theorem 44.3.2 has a fixed point $\mathbf{w}$. Then $g(\mathbf{w})$ is a fixed point of F . $\square$

In particular Brouwer's theorem holds for closed squares, closed triangles, and any other set homeomorphic to a disc. Throughout we write

$$D = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| < 1\}, \quad \overline{D} = \{\mathbf{x} : \|\mathbf{x}\| \leq 1\}, \quad \partial D = \{\mathbf{x} : \|\mathbf{x}\| = 1\}.$$

§44.3.1 The No Retraction Theorem

The first equivalence is the one which explains why the theorem is true: a fixed-point-free map lets you comb the disc onto its boundary, and that is something you cannot do.

Theorem 44.3.4

The following are equivalent.

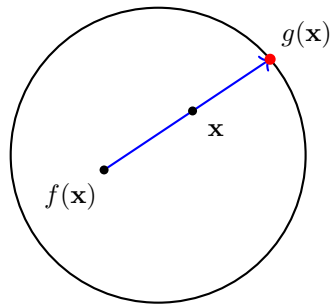
- (i) Every continuous $f : \overline{D} \rightarrow \overline{D}$ has a fixed point.
- (ii) There is no continuous $g : \overline{D} \rightarrow \partial D$ with $g(\mathbf{x}) = \mathbf{x}$ for all $\mathbf{x} \in \partial D$. (Such a g is called a **retraction**.)

Proof. Suppose (i) holds and $g : \overline{D} \rightarrow \partial D$ is a retraction. Let R be rotation through π about $\mathbf{0}$. Then $R \circ g : \overline{D} \rightarrow \overline{D}$ is continuous, so has a fixed point $\mathbf{w}$. But $R(g(\mathbf{w})) = \mathbf{w}$ forces $\mathbf{w} \in R(\partial D) = \partial D$, and then $g(\mathbf{w}) = \mathbf{w}$, so $R\mathbf{w} = \mathbf{w}$, which is false for $\mathbf{w} \neq \mathbf{0}$. Since $\mathbf{0} \notin \partial D$ we have a contradiction, and (ii) holds.

Conversely, suppose (ii) holds and $f : \overline{D} \rightarrow \overline{D}$ is continuous with no fixed point. For $\mathbf{x} \neq \mathbf{y}$ in $\overline{D}$ let $F(\mathbf{x}, \mathbf{y})$ be the point where the ray starting at $\mathbf{y}$ and passing through $\mathbf{x}$ meets ∂D ; explicitly, writing $\mathbf{u} = (\mathbf{x} - \mathbf{y}) / \|\mathbf{x} - \mathbf{y}\|$, we have $F(\mathbf{x}, \mathbf{y}) = \mathbf{x} + t\mathbf{u}$ where $t \geq 0$ is the unique non-negative root of $\|\mathbf{x} + t\mathbf{u}\|^2 = 1$, that is

$$t = -\mathbf{x} \cdot \mathbf{u} + \sqrt{(\mathbf{x} \cdot \mathbf{u})^2 + 1 - \|\mathbf{x}\|^2}.$$

This is a continuous function of $(\mathbf{x}, \mathbf{y})$ on $\{(\mathbf{x}, \mathbf{y}) \in \overline{D}^2 : \mathbf{x} \neq \mathbf{y}\}$. Set $g(\mathbf{x}) = F(\mathbf{x}, f(\mathbf{x}))$, which makes sense since $f(\mathbf{x}) \neq \mathbf{x}$ everywhere. Then g is continuous, $g(\overline{D}) \subseteq \partial D$, and if $\mathbf{x} \in \partial D$ then $\mathbf{x}$ itself is already on the ray and on ∂D , so $g(\mathbf{x}) = \mathbf{x}$. Thus g is a retraction, contradicting (ii). $\square$



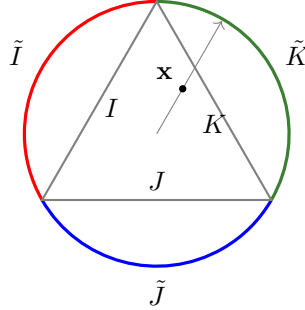
§44.3.2 From Retractions to Colourings

The next step replaces the circle by a triangle and the retraction by a map which preserves three closed pieces of the boundary. Let $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ be unit vectors making angles of $2\pi/3$ with each other, let T be the closed triangle with these vertices, and let

I, J, K be its three closed sides. Correspondingly let

$$\tilde{I} = \{(\cos \theta, \sin \theta) : 0 \leq \theta \leq \frac{2\pi}{3}\}, \quad \tilde{J} = \{(\cos \theta, \sin \theta) : \frac{2\pi}{3} \leq \theta \leq \frac{4\pi}{3}\},$$

and $\tilde{K} = \{(\cos \theta, \sin \theta) : \frac{4\pi}{3} \leq \theta \leq 2\pi\}$ be the three corresponding closed arcs of ∂D .



Lemma 44.3.5

The following three statements are equivalent.

- (i) There is no retraction from $\overline{D}$ to ∂D .
- (ii) There is no continuous $\tilde{k} : \overline{D} \rightarrow \partial D$ with $\tilde{k}(\tilde{I}) \subseteq \tilde{I}$, $\tilde{k}(\tilde{J}) \subseteq \tilde{J}$ and $\tilde{k}(\tilde{K}) \subseteq \tilde{K}$.
- (iii) There is no continuous $k : T \rightarrow \partial T$ with $k(I) \subseteq I$, $k(J) \subseteq J$ and $k(K) \subseteq K$.

Proof. (ii) $\Rightarrow$ (i) is immediate, since a retraction certainly maps each arc into itself. For (i) $\Rightarrow$ (ii), suppose $\tilde{k}$ is as in (ii) and let R be rotation through π . Then $R \circ \tilde{k} : \overline{D} \rightarrow \overline{D}$ has no fixed point: a fixed point $\mathbf{w}$ would have to lie on ∂D , and $\mathbf{w}$ lies in one of the three arcs, say $\tilde{I}$, so $\tilde{k}(\mathbf{w}) \in \tilde{I}$ while $R\tilde{k}(\mathbf{w}) = \mathbf{w} \in \tilde{I}$ means $\tilde{k}(\mathbf{w}) = R\mathbf{w}$, which lies in the arc antipodal to the one containing $\mathbf{w}$. Since an arc of length $2\pi/3$ is disjoint from its antipodal image, this is impossible. By Theorem 44.3.4 this contradicts (i).

For the equivalence of (ii) and (iii), define $\Phi : \overline{D} \rightarrow T$ by $\Phi(\mathbf{0}) = \mathbf{0}$ and, for $\mathbf{x} \neq \mathbf{0}$, $\Phi(\mathbf{x}) = \rho(\mathbf{x})\mathbf{x}$ where $\rho(\mathbf{x}) > 0$ is chosen so that $\rho(\mathbf{x})\mathbf{x} \in \partial T$ when $\|\mathbf{x}\| = 1$; in other words we stretch each radius of the disc onto the corresponding radius of the triangle. Since $\mathbf{0}$ is interior to T and ∂T meets each ray from $\mathbf{0}$ exactly once, ρ is a continuous positive function of the direction of $\mathbf{x}$ alone, and Φ is a homeomorphism carrying ∂D onto ∂T and $\tilde{I}, \tilde{J}, \tilde{K}$ onto I, J, K respectively. So $\tilde{k} \mapsto \Phi \circ \tilde{k} \circ \Phi^{-1}$ is a bijection between maps of the type in (ii) and maps of the type in (iii). $\square$

The next equivalence is the crucial one: it converts a statement about continuous maps into a statement about closed sets, and closed sets are things we can attack combinatorially.

Lemma 44.3.6

The following are equivalent.

- (i) There is no continuous $h : T \rightarrow \partial T$ with $h(I) \subseteq I$, $h(J) \subseteq J$, $h(K) \subseteq K$.
- (ii) If A, B, C are closed subsets of T with $A \supseteq I$, $B \supseteq J$, $C \supseteq K$ and $A \cup B \cup C = T$, then $A \cap B \cap C \neq \emptyset$.

Proof. Suppose (ii) holds and h is as in (i). Put $A = h^{-1}(I)$, $B = h^{-1}(J)$, $C = h^{-1}(K)$. These are closed (preimages of closed sets under a continuous map), they cover T since $I \cup J \cup K = \partial T$, and $A \supseteq I$ since $h(I) \subseteq I$; similarly for B and C . But $I \cap J \cap K = \emptyset$ (the three sides have no common point), so $A \cap B \cap C = h^{-1}(I \cap J \cap K) = \emptyset$, contradicting (ii).

Conversely, suppose (i) holds and A, B, C are closed sets as in (ii) with $A \cap B \cap C = \emptyset$. It is convenient to realise T in barycentric coordinates,^a identifying it with

$$T = \{(x, y, z) \in \mathbb{R}^3 : x, y, z \geq 0, x + y + z = 1\},$$

with $I = \{x = 0\} \cap T$, $J = \{y = 0\} \cap T$ and $K = \{z = 0\} \cap T$. Define

$$h(\mathbf{x}) = \frac{1}{d(\mathbf{x}, A) + d(\mathbf{x}, B) + d(\mathbf{x}, C)} (d(\mathbf{x}, A), d(\mathbf{x}, B), d(\mathbf{x}, C)).$$

The denominator never vanishes: if it did, then $\mathbf{x}$ would be at zero distance from each of the closed sets A, B, C , so $\mathbf{x} \in A \cap B \cap C = \emptyset$. By Lemma 44.1.13 each numerator is continuous, so h is continuous, and clearly $h(\mathbf{x}) \in T$ with at least one coordinate zero (as $A \cup B \cup C = T$ means every $\mathbf{x}$ lies in one of them), so $h(T) \subseteq \partial T$. Finally if $\mathbf{x} \in I$ then $\mathbf{x} \in A$, so $d(\mathbf{x}, A) = 0$ and $h(\mathbf{x}) \in I$; similarly for J and K . This contradicts (i). $\square$

^aBarycentric coordinates were invented by Möbius, and are exactly the right language for everything in this section.

Putting Theorem 44.3.4 and Lemmas 44.3.5 and 44.3.6 together,

Brouwer $\iff$ no retraction $\iff$ three arcs $\iff$ three sides $\iff$ three colours,

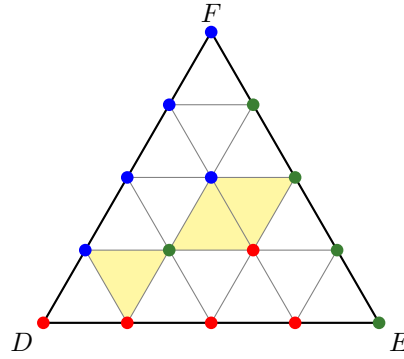
so it is enough to prove the three colours statement. That is what we now do, and the engine is a purely combinatorial lemma of Sperner.

44.3.2 Sperner's Lemma

Lemma 44.3.7 (Sperner)

Consider a triangle DEF divided into a triangular grid by $n - 1$ equally spaced lines parallel to each side. Colour every vertex of the grid red, green or blue in such a way that every vertex on the side DE except E is red, every vertex on EF except F is green, and every vertex on FD except D is blue. Then some small triangle of the grid has all three of its vertices differently coloured.

In fact the proof gives more: the number of such ‘rainbow’ triangles is odd. Here is a grid with $n = 4$; three of the small triangles are rainbow.



Proof. Give each side of each small triangle a value, according to the following rule. Traversing the side in a fixed rotational sense (say clockwise) around its small triangle, set

$$\theta(RG) = \theta(GB) = \theta(BR) = 1, \quad \theta(GR) = \theta(BG) = \theta(RB) = -1,$$

and $\theta = 0$ on a side whose two endpoints have the same colour. Note that reversing the direction of travel changes the sign of θ .

For a small triangle Δ let $\psi(\Delta)$ be the sum of the values of its three sides, traversed clockwise. If the three vertices of Δ are not all different then two of them agree, and a moment's checking shows $\psi(\Delta) = 0$: for example a triangle coloured R, R, G contributes $0 + 1 + (-1) = 0$. If the vertices are all different then going round clockwise we see either R, G, B or R, B, G , giving $\psi(\Delta) = 3$ or $\psi(\Delta) = -3$. In particular $\psi(\Delta) \neq 0$ exactly when Δ is rainbow, and $\psi(\Delta) \equiv 3 \pmod{6}$ in that case.

Now sum $\psi(\Delta)$ over all small triangles Δ . Every interior side of the grid belongs to exactly two small triangles, and is traversed in opposite directions in the two, so its two contributions cancel. Hence

$$\sum_{\Delta} \psi(\Delta) = \sum_{\text{sides on } \partial(DEF)} \theta(\text{side}),$$

the boundary sides being traversed clockwise around the big triangle.

Consider the side DE . All its grid vertices are red except the last, E , which is green. So travelling from D to E we see $R, R, \dots, R, G$ and the total is $\theta(RG) = 1$. Similarly EF reads $G, G, \dots, G, B$ contributing 1, and FD reads $B, B, \dots, B, R$ contributing 1. Hence $\sum_{\Delta} \psi(\Delta) = 3 \neq 0$, so at least one small triangle is rainbow. Indeed, since each rainbow triangle contributes ± 3 and the total is 3, the number of rainbow triangles is odd. $\square$

Proof of Lemma 44.3.6(ii), and of Brouwer's theorem. Let A, B, C be closed subsets of T with $A \supseteq I$, $B \supseteq J$ and $C \supseteq K$ satisfying $A \cup B \cup C = T$, where we label the sides so that $I = DE$, $J = EF$ and $K = FD$.

Fix n and subdivide T into a triangular grid with $n - 1$ lines parallel to each side, so that each small triangle has diameter at most $\text{diam}(T)/n$. Colour each grid vertex $\mathbf{v}$ as follows: if $\mathbf{v} \in A$ colour it red, otherwise if $\mathbf{v} \in B$ colour it green, otherwise colour it blue (which is legitimate as $A \cup B \cup C = T$). Since $I \subseteq A$, every vertex

on I is red; since $J \subseteq B$, every vertex of J not already coloured red is green – and the only vertex of J lying in I is the shared corner E . Rather than fuss over corners, note that we are free to recolour the three corners: D red, E green, F blue is consistent with $D \in I \subseteq A$, $E \in J \subseteq B$, $F \in K \subseteq C$. Thus the hypotheses of Sperner's lemma hold.

So for each n we obtain a small triangle with vertices $\mathbf{a}_n$ (red, so in A), $\mathbf{b}_n$ (green, so in B) and $\mathbf{c}_n$ (blue, so in C), with

$$\|\mathbf{a}_n - \mathbf{b}_n\|, \|\mathbf{b}_n - \mathbf{c}_n\|, \|\mathbf{c}_n - \mathbf{a}_n\| \leq \frac{\text{diam}(T)}{n}.$$

As T is compact we may pass to a subsequence along which $\mathbf{a}_{n(j)} \rightarrow \mathbf{x}$ for some $\mathbf{x} \in T$. Since the three points of each triple are within $\text{diam}(T)/n$ of each other, $\mathbf{b}_{n(j)} \rightarrow \mathbf{x}$ and $\mathbf{c}_{n(j)} \rightarrow \mathbf{x}$ too. As A, B, C are closed, $\mathbf{x} \in A \cap B \cap C$, which is therefore non-empty. $\square$

The reader who is worried about the one-dimensional analogue may like to check that the whole of this section degenerates, when T is replaced by $[0, 1]$ and three colours by two, into the following statement: if we colour $0, 1/n, \dots, 1$ red or blue with 0 red and 1 blue, then some pair of neighbours have different colours. That statement proves the intermediate value theorem, and Sperner's lemma is exactly its two-dimensional cousin.

§44.3.4 Two Applications

Brouwer's theorem is deep, and it is worth seeing that it does real work. Here is a result which the Markov chains course makes one want very much.

Proposition 44.3.8 (Perron–Frobenius, stochastic case)

Let $A = (a_{ij})$ be an $n \times n$ matrix with $a_{ij} \geq 0$ for all i, j and $\sum_{i=1}^n a_{ij} = 1$ for each j . Then A has an eigenvector with eigenvalue 1 lying in

$$\Sigma = \left\{ \mathbf{x} \in \mathbb{R}^n : x_j \geq 0 \text{ for all } j, \sum_j x_j = 1 \right\}.$$

Proof. If $\mathbf{x} \in \Sigma$ then $(A\mathbf{x})_i = \sum_j a_{ij}x_j \geq 0$ and

$$\sum_i (A\mathbf{x})_i = \sum_j x_j \sum_i a_{ij} = \sum_j x_j = 1,$$

so $A\Sigma \subseteq \Sigma$. Now Σ is a closed bounded convex subset of the hyperplane $\sum_j x_j = 1$ with non-empty interior in that hyperplane, so is homeomorphic to a closed disc of dimension $n - 1$ (radially, as in Lemma 44.3.5). Applying Brouwer's theorem in dimension $n - 1$ via Lemma 44.3.3, the continuous map $\mathbf{x} \mapsto A\mathbf{x}$ has a fixed point $\mathbf{x}^* \in \Sigma$, and $A\mathbf{x}^* = \mathbf{x}^*$ is the required eigenvector. $\square$

Reading 'i' and 'j' the other way round, this says that every finite state Markov chain has an invariant distribution – a result which is unpleasant to prove by bare hands, and which becomes almost obvious once we are allowed a fixed point theorem.

The second application is a statement which is 'obvious' in the worst sense of the word.

Proposition 44.3.9

Let $ABCD$ be a square. If γ is a continuous path in the square joining A to C , and τ is a continuous path in the square joining B to D , then γ and τ intersect.

We shall not give the details (they form a rather fiddly exercise), but the idea is worth recording: take the square to be $[-1, 1]^2$ with $\gamma, \tau : [0, 1] \rightarrow [-1, 1]^2$, and suppose $\gamma(s) \neq \tau(t)$ for all s, t . Then

$$F(s, t) = \frac{(\tau_1(t) - \gamma_1(s), \gamma_2(s) - \tau_2(t))}{\|\gamma(s) - \tau(t)\|_\infty}$$

defines a continuous map of $[0, 1]^2$ into itself, and one checks that a fixed point of F would have to lie in a corner, where the boundary conditions on γ and τ make it impossible. This is a baby case of the Jordan curve theorem, which asserts that a simple closed curve in the plane divides it into exactly two components. The Jordan curve theorem is genuinely hard, and is not part of this course; but it is comforting to know that its plausibility is not a delusion.

§44.4 Non-Zero Sum Games

It is said that converting the front garden of a house into a parking space raises the value of that house and lowers the value of every other house on the road. Once everybody has done the conversion, every house is worth less than it was before anybody started. Situations of this kind – where individually rational choices produce a collectively poor outcome – are the meat of game theory, and it turns out that Brouwer's theorem has a great deal to say about them.

Let us set up the simplest possible model. Albert has a choice between actions A_1 and A_2 , and Bertha between B_1 and B_2 . If Albert plays A_i and Bertha plays B_j , then Albert receives a_{ij} and Bertha receives b_{ij} . In IB Optimisation you studied the *zero-sum* case $a_{ij} = -b_{ij}$, where one player's gain is the other's loss; we shall make no such assumption.

Suppose Albert plays A_i with probability p_i and Bertha plays B_j with probability q_j , where $p_1 + p_2 = q_1 + q_2 = 1$ and all entries are non-negative. Writing $\mathbf{p} = (p_1, p_2)$ and $\mathbf{q} = (q_1, q_2)$, the expected payoffs are

$$A(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^2 \sum_{j=1}^2 a_{ij} p_i q_j, \quad B(\mathbf{p}, \mathbf{q}) = \sum_{i=1}^2 \sum_{j=1}^2 b_{ij} p_i q_j.$$

In the zero-sum case you saw that there is a pair of strategies such that neither player would change their choice even knowing the other's. Away from zero-sum we can imagine an endless dance: Albert chooses $\mathbf{p}$, Bertha responds with $\mathbf{q}$, Albert switches to $\mathbf{p}'$, Bertha to $\mathbf{q}'$, and so on. Does the dance ever stop?

Definition 44.4.1 (Nash Equilibrium)

A pair $(\mathbf{p}^*, \mathbf{q}^*)$ of strategies is a **Nash equilibrium** (or **Nash stable point**) if

$$A(\mathbf{p}^*, \mathbf{q}^*) \geq A(\mathbf{p}, \mathbf{q}^*) \quad \text{and} \quad B(\mathbf{p}^*, \mathbf{q}^*) \geq B(\mathbf{p}^*, \mathbf{q})$$

for all strategies $\mathbf{p}, \mathbf{q}$. In other words, neither player can do better by unilaterally

changing their strategy.

Theorem 44.4.2 (Nash)

For any real numbers a_{ij}, b_{ij} there exists a Nash equilibrium.

Proof. Write $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ for the two pure strategies, and let

$$\Gamma = \{(\mathbf{p}, \mathbf{q}) : p_1, p_2, q_1, q_2 \geq 0, p_1 + p_2 = q_1 + q_2 = 1\} \subseteq \mathbb{R}^4,$$

which is (a copy of) the square $[0, 1]^2$, and so closed, bounded and homeomorphic to a disc.

For $(\mathbf{p}, \mathbf{q}) \in \Gamma$ define

$$u_i(\mathbf{p}, \mathbf{q}) = \max\{A(\mathbf{e}_i, \mathbf{q}) - A(\mathbf{p}, \mathbf{q}), 0\}, \quad v_j(\mathbf{p}, \mathbf{q}) = \max\{B(\mathbf{p}, \mathbf{f}_j) - B(\mathbf{p}, \mathbf{q}), 0\},$$

where $\mathbf{f}_j$ are the pure strategies for Bertha. Thus $u_i > 0$ says ‘against $\mathbf{q}$, Albert would do strictly better playing A_i outright’. These are continuous, being maxima of continuous functions. Now set $F(\mathbf{p}, \mathbf{q}) = (\mathbf{p}', \mathbf{q}')$ where

$$p'_i = \frac{p_i + u_i(\mathbf{p}, \mathbf{q})}{1 + u_1(\mathbf{p}, \mathbf{q}) + u_2(\mathbf{p}, \mathbf{q})}, \quad q'_j = \frac{q_j + v_j(\mathbf{p}, \mathbf{q})}{1 + v_1(\mathbf{p}, \mathbf{q}) + v_2(\mathbf{p}, \mathbf{q})}.$$

The denominators are at least 1, the numerators are non-negative, and $p'_1 + p'_2 = 1 = q'_1 + q'_2$, so $F : \Gamma \rightarrow \Gamma$ is a continuous map of Γ into itself. By Brouwer’s theorem (in the form of Lemma 44.3.3) it has a fixed point $(\mathbf{p}^*, \mathbf{q}^*)$.

We claim this is a Nash equilibrium. Write $u_i = u_i(\mathbf{p}^*, \mathbf{q}^*)$. Being a fixed point means $p_i^*(1 + u_1 + u_2) = p_i^* + u_i$, that is

$$p_i^*(u_1 + u_2) = u_i \quad (i = 1, 2).$$

Since $A(\mathbf{p}^*, \mathbf{q}^*) = p_1^*A(\mathbf{e}_1, \mathbf{q}^*) + p_2^*A(\mathbf{e}_2, \mathbf{q}^*)$ is a convex combination, we cannot have $A(\mathbf{e}_i, \mathbf{q}^*) > A(\mathbf{p}^*, \mathbf{q}^*)$ for both i . So at least one u_i vanishes; without loss of generality $u_2 = 0$. Then $p_2^*(u_1 + u_2) = 0$, so $p_2^*u_1 = 0$. If $u_1 > 0$ then $p_2^* = 0$, so $p_1^* = 1$ and $\mathbf{p}^* = \mathbf{e}_1$, whence $u_1 = \max\{A(\mathbf{e}_1, \mathbf{q}^*) - A(\mathbf{e}_1, \mathbf{q}^*), 0\} = 0$, a contradiction. So $u_1 = u_2 = 0$.

Thus $A(\mathbf{e}_i, \mathbf{q}^*) \leq A(\mathbf{p}^*, \mathbf{q}^*)$ for $i = 1, 2$, and so for any strategy $\mathbf{p}$,

$$A(\mathbf{p}, \mathbf{q}^*) = p_1A(\mathbf{e}_1, \mathbf{q}^*) + p_2A(\mathbf{e}_2, \mathbf{q}^*) \leq A(\mathbf{p}^*, \mathbf{q}^*).$$

The same argument applied to v_j gives $B(\mathbf{p}^*, \mathbf{q}) \leq B(\mathbf{p}^*, \mathbf{q}^*)$ for all $\mathbf{q}$. □

Nothing in the argument really used the number 2: with m players having finitely many choices each, exactly the same construction (in a product of simplices) together with Brouwer’s theorem in the appropriate dimension gives a stable point. Note also that the IB theory applied only to two players and only to zero-sum games; Nash’s theorem is a considerable strengthening.

Two warnings. First, equilibria need not be unique. If Albert and Bertha each choose ‘scissors’ or ‘paper’, receiving £1 each if both choose scissors, £2 each if both choose

paper, and nothing if they disagree, then both ‘everyone plays scissors’ and ‘everyone plays paper’ are stable. Second, and more disturbingly, a unique equilibrium may be very bad for everyone concerned. That, however, is not the mathematician’s problem.

Example 44.4.3 (Chicken)

Albert and Bartholomew drive their cars at one another. If both swerve, each loses 1 prestige point. If exactly one swerves, the swerver loses 5 points and the non-swerver gains 10. If neither swerves, both lose 100.

Let p be the probability that Albert swerves and q the probability that Bartholomew does. Then

$$A(p, q) = -pq - 5p(1 - q) + 10(1 - p)q - 100(1 - p)(1 - q) = p(95 - 106q) + 110q - 100,$$

and by symmetry $B(p, q) = q(95 - 106p) + 110p - 100$.

The recipe for problems like this is: first look at the interior of the square, then at the interiors of the edges, then at the vertices. In the interior we need $\partial A/\partial p = 0$ and $\partial B/\partial q = 0$, that is $95 - 106q = 0$ and $95 - 106p = 0$, giving the interior equilibrium

$$(p^*, q^*) = \left(\frac{95}{106}, \frac{95}{106} \right).$$

On the edge $p = 1$ we have $\partial B/\partial q = 95 - 106 = -11 < 0$, so Bartholomew’s best response is $q = 0$; and when $q = 0$ we have $\partial A/\partial p = 95 > 0$, so Albert’s best response is indeed $p = 1$. Hence $(1, 0)$ is an equilibrium (‘Albert always swerves, Bartholomew never does’), and by symmetry so is $(0, 1)$. At $(1, 1)$ we have $\partial A/\partial p = -11 < 0$ and at $(0, 0)$ we have $\partial A/\partial p = 95 > 0$, so neither is stable. So there are exactly three Nash equilibria.

It is genuinely easy to solve toy problems in this way; the difficulty is that with m players we must examine the interior of the cube, then each face, then each edge, then each vertex, and the number of steps explodes. So far as anybody knows, this curse of dimensionality is unavoidable.

§44.5 Dividing the Pot

Faced with the previous section, the tender-hearted ask ‘why not cooperate?’. Even granting that people sometimes do, a question remains: how should the gains from cooperation be divided? Nash gave a striking answer.

Definition 44.5.1 (Convex Set)

A subset E of $\mathbb{R}^m$ is **convex** if $\lambda \mathbf{u} + (1 - \lambda)\mathbf{v} \in E$ whenever $\mathbf{u}, \mathbf{v} \in E$ and $0 \leq \lambda \leq 1$.

The set-up is as follows. There are m participants who must jointly choose a point $\mathbf{x}$ of a closed, bounded, convex set $E \subseteq \mathbb{R}^m$; the value of the outcome to the j th participant is x_j . Convexity is not an arbitrary assumption: if the participants can agree on $\mathbf{u}$ or on $\mathbf{v}$, they can agree to toss a suitable coin, and the expected outcome is then $\lambda \mathbf{u} + (1 - \lambda)\mathbf{v}$. They also know a **status quo** point $\mathbf{s} \in E$, the outcome if they fail to agree.

Nash argues that a ‘best point’ $\mathbf{x}^*$ should have the following properties.

- (1) $x_j^* \geq s_j$ for all j : nobody is worse off than if they had walked away.
- (2) (*Pareto optimality*) If $\mathbf{x} \in E$ and $x_j \geq x_j^*$ for all j , then $\mathbf{x} = \mathbf{x}^*$: there is no available outcome which makes somebody better off and nobody worse off.
- (3) (*Independence of irrelevant alternatives*) If $E' \supseteq E$ is closed, bounded and convex and $\mathbf{x}^{**}$ is the best point for $(E', \mathbf{s})$, then $\mathbf{x}^{**} \in E$ implies that $\mathbf{x}^{**}$ is the best point for $(E, \mathbf{s})$.
- (4) (*Symmetry*) If E is invariant under permutation of coordinates and $\mathbf{s}$ is too, then $x_1^* = x_2^* = \dots = x_m^*$.
- (5) (*Scale invariance*) If $T\mathbf{x} = (a_1x_1 + b_1, \dots, a_mx_m + b_m)$ with $a_j > 0$, and $\mathbf{x}^*$ is the best point for $(E, \mathbf{s})$, then $T\mathbf{x}^*$ is the best point for $(TE, T\mathbf{s})$.

Condition (5) says that we must treat the poor man's penny with the same respect as the rich man's pound: nothing should depend on the units in which each participant measures their satisfaction. It is a genuine assumption, but a reasonable one, and it is the one which does most of the work.

There is no difficulty in remembering these conditions, since each plays a specific role in the proof. We begin with a lemma which is a very concrete instance of the separating hyperplane theorem.

Lemma 44.5.2

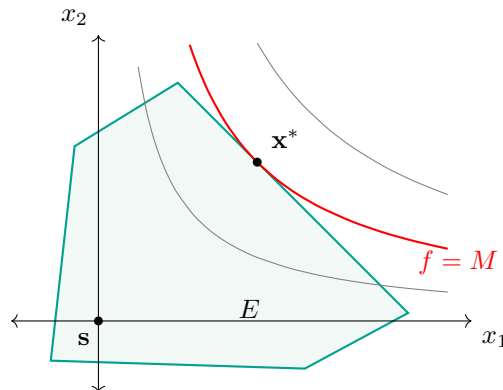
Let K be a convex subset of $\mathbb{R}^n$ with $(1, 1, \dots, 1) \in K$ and $\prod_{j=1}^n x_j \leq 1$ for all $\mathbf{x} \in K$ with every $x_j \geq 0$. Then

$$K \subseteq \left\{ \mathbf{x} : \sum_{j=1}^n x_j \leq n \right\}.$$

Proof. Let $\mathbf{x} \in K$ and let $\mathbf{1} = (1, \dots, 1)$. By convexity $\delta\mathbf{x} + (1 - \delta)\mathbf{1} \in K$ for $0 \leq \delta \leq 1$, and for δ small enough every coordinate $1 + \delta(x_j - 1)$ is positive. So the hypothesis applies and

$$\prod_{j=1}^n (1 + \delta(x_j - 1)) \leq 1.$$

Expanding, the left-hand side equals $1 + \delta \sum_{j=1}^n (x_j - 1) + \delta^2 R(\delta)$ where $R(\delta)$ stays bounded as $\delta \rightarrow 0$. Hence $\sum_{j=1}^n (x_j - 1) + \delta R(\delta) \leq 0$, and letting $\delta \rightarrow 0^+$ gives $\sum_j x_j \leq n$. $\square$



Lemma 44.5.3

Let $E \subseteq \mathbb{R}^m$ be closed, bounded and convex with $\mathbf{s} \in \text{Int } E$. Then the function $f(\mathbf{x}) = \prod_{j=1}^m (x_j - s_j)$ attains a maximum on $E^+ = \{\mathbf{x} \in E : x_j \geq s_j \text{ for all } j\}$ at exactly one point.

Proof. The set E^+ is closed, bounded and non-empty, so f attains a maximum M there. Since $\mathbf{s} \in \text{Int } E$ we can find $\mathbf{x} \in E$ with $x_j > s_j$ for all j , so $M > 0$.

For uniqueness, suppose $f(\mathbf{x}) = f(\mathbf{y}) = M$ with $\mathbf{x}, \mathbf{y} \in E^+$. Let $\mathbf{z} = \frac{1}{2}(\mathbf{x} + \mathbf{y}) \in E^+$ by convexity. By the AM–GM inequality,

$$z_j - s_j = \frac{(x_j - s_j) + (y_j - s_j)}{2} \geq \sqrt{(x_j - s_j)(y_j - s_j)},$$

with equality if and only if $x_j = y_j$. Multiplying over j (all factors are non-negative) gives $f(\mathbf{z}) \geq \sqrt{f(\mathbf{x})f(\mathbf{y})} = M$. As M is the maximum, we have equality throughout, and since $M > 0$ every factor is strictly positive, so equality in AM–GM forces $x_j = y_j$ for every j . $\square$

Theorem 44.5.4 (Nash Bargaining)

Suppose that to each problem $(E, \mathbf{s})$ (with $E \subseteq \mathbb{R}^m$ closed, bounded and convex and $\mathbf{s} \in \text{Int } E$) our conditions assign a best point. Then the best point of $(E, \mathbf{s})$ is the unique maximiser of

$$f(\mathbf{x}) = \prod_{j=1}^m (x_j - s_j)$$

over $\{\mathbf{x} \in E : x_j \geq s_j\}$.

Proof. Let $\mathbf{x}^*$ be the maximiser given by Lemma 44.5.3, so $x_j^* > s_j$ for every j . By condition (5) we may apply the affine change of coordinates $x_j \mapsto (x_j - s_j)/(x_j^* - s_j)$; this replaces $\mathbf{s}$ by $\mathbf{0}$ and $\mathbf{x}^*$ by $\mathbf{1} = (1, 1, \dots, 1)$, and does not change which point is ‘best’. So assume $\mathbf{s} = \mathbf{0}$ and $\mathbf{x}^* = \mathbf{1}$.

Maximality of f now reads: $\prod_j x_j \leq 1$ for every $\mathbf{x} \in E$ with all $x_j \geq 0$. As E is convex and contains $\mathbf{1}$, Lemma 44.5.2 gives $E \subseteq \{\mathbf{x} : \sum_j x_j \leq m\}$. Pick M so large that $E \subseteq [-M, M]^m$ and set

$$\Sigma = \left\{ \mathbf{x} \in [-M, M]^m : \sum_j x_j \leq m \right\},$$

which is closed, bounded, convex, contains E , contains $\mathbf{0}$ in its interior, and is invariant under permutations of the coordinates.

Consider the problem $(\Sigma, \mathbf{0})$. By symmetry (condition (4)) its best point has all coordinates equal, say $\mathbf{y} = (t, t, \dots, t)$ with $mt \leq m$, so $t \leq 1$. By Pareto optimality (condition (2)), since $\mathbf{1} \in \Sigma$ dominates $\mathbf{y}$ coordinatewise when $t < 1$, we must have $t = 1$; that is, the best point of $(\Sigma, \mathbf{0})$ is $\mathbf{1} = \mathbf{x}^*$. Finally $\mathbf{x}^* \in E \subseteq \Sigma$, so by independence of irrelevant alternatives (condition (3)) $\mathbf{x}^*$ is the best point of $(E, \mathbf{0})$. $\square$

So the fair division is the one which maximises the *product* of the gains over the status

quo – a rule with no obvious a priori justification which nonetheless falls out of five innocuous-looking requirements. There is no patent for immortality under the moon, but I suspect Nash's results will be remembered long after the last celluloid copy of *A Beautiful Mind* has crumbled to dust.

§44.6 Approximation by Polynomials

It is a guiding idea of both the calculus and of numerical analysis that well behaved functions look like polynomials. Like most guiding principles, it needs to be used judiciously, and this section is devoted to finding out in what sense it is actually true.

If asked to justify the principle we might mutter something about Taylor's theorem. But Cauchy produced the following example to show that this is not enough.

Example 44.6.1 (Cauchy's Example)

Define $E : \mathbb{R} \rightarrow \mathbb{R}$ by

$$E(t) = \begin{cases} \exp(-1/t^2) & t \neq 0, \\ 0 & t = 0. \end{cases}$$

Then E is infinitely differentiable, and $E^{(n)}(0) = 0$ for every n , yet $E(t) \neq 0$ for $t \neq 0$. So the Taylor series of E about 0 converges everywhere, to the wrong function.

Indeed, for $t \neq 0$ an easy induction gives $E^{(n)}(t) = P_n(1/t)E(t)$ for some polynomial P_n : differentiating $P_n(1/t)E(t)$ gives $(-t^{-2}P'_n(1/t) + 2t^{-3}P_n(1/t))E(t)$, which is of the same shape. At 0 we induct again: if $E^{(n)}(0) = 0$ then

$$\frac{E^{(n)}(h) - E^{(n)}(0)}{h} = \frac{1}{h}P_n(1/h)\exp(-1/h^2) \rightarrow 0$$

as $h \rightarrow 0$, since $x^k e^{-x^2} \rightarrow 0$ as $|x| \rightarrow \infty$ for every k . So $E^{(n+1)}(0)$ exists and is 0.

By Part II this example has been promoted from a counterexample to an invaluable tool: it is what makes partitions of unity and smoothing kernels possible. But it does show that Taylor's theorem is not going to tell us that continuous functions look like polynomials.

§44.6.1 Interpolation and Chebyshev Polynomials

We might instead mutter something about interpolation. Write $\mathcal{P}_n$ for the vector space of real polynomials of degree at most n .

Lemma 44.6.2 (Lagrange Interpolation)

Let $x_0, x_1, \dots, x_n$ be distinct points of $[a, b]$ and let $f : [a, b] \rightarrow \mathbb{R}$.

- (i) There is at most one $P \in \mathcal{P}_n$ with $P(x_j) = f(x_j)$ for $0 \leq j \leq n$.
- (ii) Writing $e_j(t) = \prod_{k \neq j} \frac{t - x_k}{x_j - x_k}$, the polynomial $P = \sum_{j=0}^n f(x_j)e_j$ lies in $\mathcal{P}_n$ and satisfies $P(x_i) = f(x_i)$ for all i .
- (iii) The e_j form a basis of $\mathcal{P}_n$.

Proof. (i) The difference of two such polynomials lies in $\mathcal{P}_n$ and vanishes at $n + 1$ points, so is zero by Lemma 44.1.15. (ii) Each e_j has degree n , and $e_j(x_i) = \delta_{ij}$ by construction, so $P(x_i) = f(x_i)$. (iii) There are $n + 1$ of them and $\dim \mathcal{P}_n = n + 1$; they span by (ii). $\square$

So there is always a unique interpolating polynomial, and if f is smooth we can even say exactly how good it is.

Theorem 44.6.3 (Interpolation Error)

Let $x_0 < x_1 < \cdots < x_n$ be points of $[a, b]$, let $f : [a, b] \rightarrow \mathbb{R}$ be $n + 1$ times differentiable, and let $P \in \mathcal{P}_n$ interpolate f at the x_j . Then for each $t \in [a, b]$ there is a $\xi = \xi(t) \in (a, b)$ with

$$f(t) - P(t) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{j=0}^n (t - x_j).$$

Proof. If $t = x_i$ for some i both sides vanish, so suppose not, and write $\pi(s) = \prod_{j=0}^n (s - x_j)$, which is a monic polynomial of degree $n + 1$ with $\pi(t) \neq 0$. Choose the constant

$$\lambda = \frac{f(t) - P(t)}{\pi(t)}$$

and set $g(s) = f(s) - P(s) - \lambda\pi(s)$. Then g vanishes at the $n + 2$ distinct points $x_0, \dots, x_n, t$. By Rolle's theorem g' vanishes at $n + 1$ points strictly between consecutive zeros of g ; applying Rolle again, g'' vanishes at n points; and after $n + 1$ applications we obtain a $\xi \in (a, b)$ with $g^{(n+1)}(\xi) = 0$. But $P^{(n+1)} = 0$ and $\pi^{(n+1)} = (n+1)!$, so

$$0 = f^{(n+1)}(\xi) - \lambda(n+1)!,$$

which rearranges to the stated formula. $\square$

This is a genuinely informative answer, because it separates the error into two factors of completely different character. The factor $f^{(n+1)}(\xi)/(n+1)!$ depends on f , about which we may know nothing; the factor $\prod_j (t - x_j)$ depends only on *where we chose to interpolate*, and is entirely under our control. So a sensible person picks the x_j to make $\sup_t \left| \prod_j (t - x_j) \right|$ as small as possible. We shall find the optimal choice in Corollary 44.7.4, and it will turn out not to be the obvious one.

Meanwhile, the theorem should not be read as saying that interpolation always works. It has a hypothesis of $n + 1$ derivatives, and even when those exist they may be enormous. Indeed 'interpolates f at $n + 1$ points' is in general a long way from 'is close to f '. To see how badly things can go we introduce a family of polynomials which will be with us for the rest of the course.

Theorem 44.6.4 (Chebyshev Polynomials)

There are polynomials T_n of degree n and U_{n-1} of degree $n - 1$ such that

$$T_n(\cos \theta) = \cos n\theta, \quad U_{n-1}(\cos \theta) = \frac{\sin n\theta}{\sin \theta}$$

for all θ (the second whenever $\sin \theta \neq 0$). The coefficient of t^n in T_n is 2^{n-1} for $n \geq 1$. The roots of T_n are $\cos\left(\frac{(2r+1)\pi}{2n}\right)$ for $0 \leq r \leq n-1$, and the roots of U_{n-1} are $\cos(r\pi/n)$ for $1 \leq r \leq n-1$.

Proof. By de Moivre's theorem and the binomial expansion,

$$\begin{aligned} \cos n\theta + i \sin n\theta &= (\cos \theta + i \sin \theta)^n \\ &= \sum_{0 \leq 2r \leq n} (-1)^r \binom{n}{2r} \cos^{n-2r} \theta \sin^{2r} \theta \\ &\quad + i \sum_{0 \leq 2r+1 \leq n} (-1)^r \binom{n}{2r+1} \cos^{n-2r-1} \theta \sin^{2r} \theta \cdot \sin \theta. \end{aligned}$$

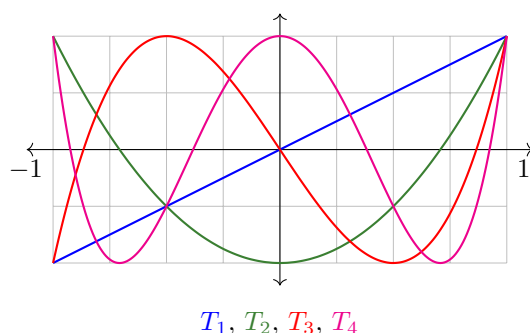
Now $\sin^{2r} \theta = (1 - \cos^2 \theta)^r$, so comparing real and imaginary parts we get $\cos n\theta = T_n(\cos \theta)$ and $\sin n\theta = \sin \theta U_{n-1}(\cos \theta)$ where

$$\begin{aligned} T_n(t) &= \sum_{0 \leq 2r \leq n} (-1)^r \binom{n}{2r} t^{n-2r} (1-t^2)^r, \\ U_{n-1}(t) &= \sum_{0 \leq 2r+1 \leq n} (-1)^r \binom{n}{2r+1} t^{n-2r-1} (1-t^2)^r. \end{aligned}$$

These are polynomials of degrees n and $n-1$. The coefficient of t^n in T_n comes from taking the top power in each term, giving

$$\sum_{0 \leq 2r \leq n} (-1)^r \binom{n}{2r} (-1)^r = \sum_{0 \leq 2r \leq n} \binom{n}{2r} = \frac{(1+1)^n + (1-1)^n}{2} = 2^{n-1}$$

for $n \geq 1$. Finally $T_n(\cos \theta) = 0$ exactly when $n\theta$ is an odd multiple of $\pi/2$, and as θ ranges over $[0, \pi]$ this gives the n distinct roots stated; the argument for U_{n-1} is the same. $\square$

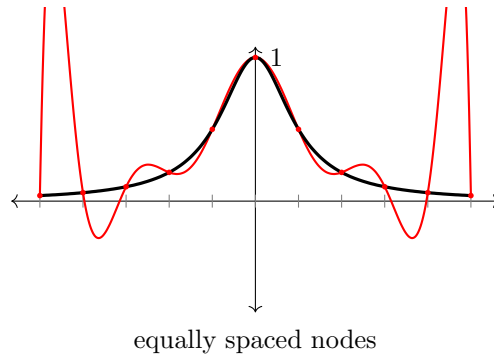


The picture already shows the two features of the T_n which will matter to us: on $[-1, 1]$ they oscillate between $+1$ and -1 , hitting each extreme value alternately and as often as possible, and they are as large as a monic-up-to-scale polynomial can be forced to be while doing so.

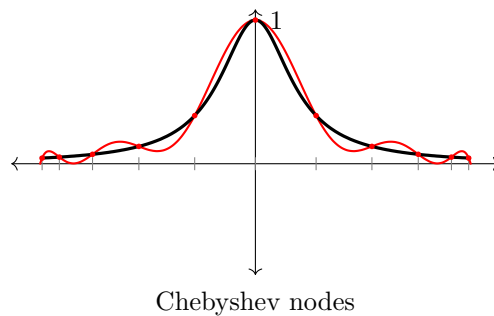
Now look at U_{n-1} . Its roots are the $n-1$ points $\cos(r\pi/n)$, which together with ± 1 are perfectly reasonably spaced points of $[-1, 1]$; but $U_{n-1}(1) = n$, which is large. So the function U_{n-1} is 'well behaved' at $n+1$ well spread points (vanishing at $n-1$ of them)

and yet is very large elsewhere. Interpolation at innocent-looking points can be badly misleading, and it can be shown – though this is harder – that no choice of interpolation points avoids this in general.

The classical illustration is due to Runge. Take $f(x) = (1 + 25x^2)^{-1}$ on $[-1, 1]$, a function about as well behaved as one could wish, and interpolate it at 11 equally spaced points. Below, the true function is drawn in black and the degree-10 interpolating polynomial in red (clipped where it runs off the picture).



The interpolant is excellent in the middle and disastrous near the endpoints, and the disaster gets worse as the degree increases. If instead we interpolate at the roots of the Chebyshev polynomial T_{11} , which cluster near the endpoints, the same degree gives a genuinely good approximation.



This is a warning about interpolation, not about polynomials. The question we actually want to answer is whether *some* polynomial is uniformly close to f , and to that we now turn.

§44.6.2 Which Metric?

Before we can ask whether f looks like g , we must say what we mean. There are several natural metrics on $C([a, b])$, and they are genuinely different.

Example 44.6.5 (Three Norms)

On $C([0, 1])$ the formulae

$$\|f\|_1 = \int_0^1 |f(t)| \, dt, \quad \|f\|_2 = \left(\int_0^1 f(t)^2 \, dt \right)^{1/2}, \quad \|f\|_\infty = \sup_{t \in [0, 1]} |f(t)|$$

all define norms, and hence metrics $d_k(f, g) = \|f - g\|_k$. By the Cauchy–Schwarz

inequality $\|f\|_1 \leq \|f\|_2$, and clearly $\|f\|_2 \leq \|f\|_\infty$, so

$$\|f\|_\infty \geq \|f\|_2 \geq \|f\|_1.$$

These inequalities go only one way. If $f_n(t) = \max\{1 - nt, 0\}$ then $\|f_n\|_\infty = 1$, $\|f_n\|_2 = (3n)^{-1/2}$ and $\|f_n\|_1 = (2n)^{-1}$, so $\|f_n\|_\infty / \|f_n\|_1 = 2n \rightarrow \infty$. A tall thin spike is large in $\|\cdot\|_\infty$ and negligible in $\|\cdot\|_1$.

All three are used in practice: $\|\cdot\|_2$ is the natural one when there is an inner product around (as in least squares fitting, and in Fourier analysis), while $\|\cdot\|_1$ is the robust one favoured by statisticians. We shall work almost exclusively with $\|\cdot\|_\infty$, because it is the one for which ‘looks like’ means what an engineer would want it to mean, and because of the following fact from IB Analysis.

Theorem 44.6.6

The space $C([a, b])$ with the uniform metric $d(f, g) = \|f - g\|_\infty$ is complete.

This is the general principle of uniform convergence: a uniformly Cauchy sequence of continuous functions converges uniformly, and a uniform limit of continuous functions is continuous.

§44.6.3 The Weierstrass Approximation Theorem

We can now state the question precisely, and answer it. Weierstrass proved the following in a paper published when he was seventy.

Theorem 44.6.7 (Weierstrass Approximation Theorem)

Let $f \in C([a, b])$ and $\varepsilon > 0$. Then there is a polynomial P with $\|f - P\|_\infty < \varepsilon$.

Several proofs are known, and it is my belief that one cannot have too many insightful proofs of Weierstrass’s theorem. We give one due to Bernstein, which is based on probability and which has the great virtue of writing down an explicit approximating polynomial. We need one tool from IA Probability.

Theorem 44.6.8 (Chebyshev’s Inequality)

If X is a bounded real random variable with $\sigma^2 = \text{Var } X$, then $\Pr(|X - \mathbb{E}X| \geq a) \leq \sigma^2/a^2$ for all $a > 0$.

Proof. Writing $\mu = \mathbb{E}X$ we have $\sigma^2 = \mathbb{E}(X - \mu)^2 \geq \mathbb{E}((X - \mu)^2 \mathbb{1}_{|X - \mu| \geq a}) \geq a^2 \Pr(|X - \mu| \geq a)$. $\square$

Theorem 44.6.9 (Bernstein)

Let $f \in C([0, 1])$ and define the **Bernstein polynomial**

$$p_n(t) = \sum_{r=0}^n \binom{n}{r} f\left(\frac{r}{n}\right) t^r (1-t)^{n-r}.$$

Then $p_n \in \mathcal{P}_n$ and $\|p_n - f\|_\infty \rightarrow 0$ as $n \rightarrow \infty$.

Proof. The probabilistic idea is this. Fix $t \in [0, 1]$ and let $X_1, \dots, X_n$ be independent with $\Pr(X_r = 1) = t$ and $\Pr(X_r = 0) = 1 - t$ – that is, toss a coin with bias t . Put $Y_n = (X_1 + \dots + X_n)/n$. Then nY_n is binomial, so

$$\mathbb{E}f(Y_n) = \sum_{r=0}^n \Pr\left(Y_n = \frac{r}{n}\right) f\left(\frac{r}{n}\right) = \sum_{r=0}^n \binom{n}{r} t^r (1-t)^{n-r} f\left(\frac{r}{n}\right) = p_n(t).$$

In other words, $p_n(t)$ is the average of f over a distribution concentrated near t . Since $\mathbb{E}Y_n = t$ and $\text{Var } Y_n = t(1-t)/n \leq 1/(4n)$, the concentration improves as n grows, and continuity of f should do the rest.

Let us do this carefully. Let $\varepsilon > 0$. As $[0, 1]$ is compact, f is uniformly continuous (Theorem 44.1.12), so there is a $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Also $\|f\|_\infty < \infty$. Splitting according to whether Y_n is close to t ,

$$\begin{aligned} |p_n(t) - f(t)| &= |\mathbb{E}(f(Y_n) - f(t))| \\ &\leq \mathbb{E}(\mathbb{1}_{|Y_n - t| < \delta} |f(Y_n) - f(t)|) + \mathbb{E}(\mathbb{1}_{|Y_n - t| \geq \delta} |f(Y_n) - f(t)|) \\ &\leq \varepsilon + 2\|f\|_\infty \Pr(|Y_n - t| \geq \delta) \\ &\leq \varepsilon + 2\|f\|_\infty \frac{\text{Var } Y_n}{\delta^2} \leq \varepsilon + \frac{\|f\|_\infty}{2n\delta^2}, \end{aligned}$$

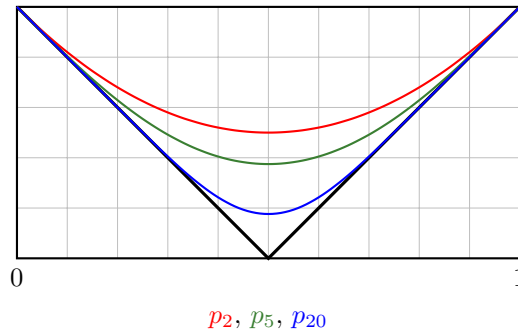
using Chebyshev's inequality. The final bound does not depend on t , so

$$\|p_n - f\|_\infty \leq \varepsilon + \frac{\|f\|_\infty}{2n\delta^2} \leq 2\varepsilon$$

for all n large enough. As $\varepsilon > 0$ was arbitrary, $\|p_n - f\|_\infty \rightarrow 0$. $\square$

Proof of Theorem 44.6.7. Let $f \in C([a, b])$ and set $g(t) = f(a + (b - a)t)$, so $g \in C([0, 1])$. By Theorem 44.6.9 there is a polynomial q with $\|g - q\|_\infty < \varepsilon$, and then $P(x) = q\left(\frac{x-a}{b-a}\right)$ is a polynomial with $\|f - P\|_\infty < \varepsilon$. $\square$

Here is the theorem in action, approximating $f(t) = |t - \frac{1}{2}|$ – a function with a corner, so certainly not analytic – by p_2 , p_5 and p_{20} .



Bernstein's polynomials converge, but not quickly: notice that p_{20} is still visibly rounded at the corner. That is the price of an explicit formula which works for every continuous f . It leads naturally to the question of what the *best* polynomial approximation looks

like, which is the subject of the next section.

§44.7 Best Approximation by Polynomials

Bernstein's theorem gives us an explicit approximating polynomial, but except in very special circumstances it is not the *best* one – indeed we have not yet even shown that a best one exists. Chebyshev was very interested in this question, and gave a beautiful criterion for recognising a best approximation when you see one.

The idea is easy to describe. If P is a good approximation to f then the error $f - P$ should not be lopsided: if it were positive more than negative, we could shift P up a little and do better. Chebyshev's insight is that the error of the *best* approximation must swing between its extreme values as often as it possibly can.

Theorem 44.7.1 (Chebyshev Equiripple Criterion)

Let $f \in C([a, b])$ and $P \in \mathcal{P}_{n-1}$, and write $\sigma = \|f - P\|_\infty$. Suppose we can find points $a \leq a_0 < a_1 < \cdots < a_n \leq b$ and a sign $\zeta \in \{-1, +1\}$ such that

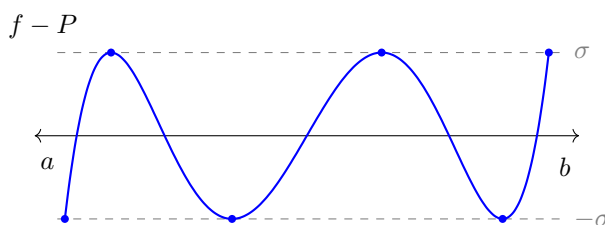
$$f(a_j) - P(a_j) = \zeta(-1)^j \sigma \quad (0 \leq j \leq n).$$

Then $\|f - P\|_\infty \leq \|f - Q\|_\infty$ for every $Q \in \mathcal{P}_{n-1}$.

Proof. Replacing f and P by $-f$ and $-P$ if necessary, we may assume $\zeta = 1$, so $f(a_j) - P(a_j) = (-1)^j \sigma$. Suppose for contradiction that some $Q \in \mathcal{P}_{n-1}$ has $\|f - Q\|_\infty < \sigma$, and set $R = P - Q \in \mathcal{P}_{n-1}$. Then

$$R(a_j) = (f(a_j) - Q(a_j)) - (f(a_j) - P(a_j)) = (f(a_j) - Q(a_j)) - (-1)^j \sigma.$$

Since $|f(a_j) - Q(a_j)| \leq \|f - Q\|_\infty < \sigma$, the sign of $R(a_j)$ is that of $-(-1)^j$, and in particular $R(a_j) \neq 0$. So R takes alternate signs at the $n+1$ points $a_0 < a_1 < \cdots < a_n$, and by the intermediate value theorem R has a root in each of the n intervals (a_j, a_{j+1}) . Thus R is a non-zero element of $\mathcal{P}_{n-1}$ with at least n distinct roots, which is impossible. $\square$



Remark. The equiripple condition is also *necessary*: the best approximation always equioscillates in this way, and in particular it is unique. The proof is a pleasant exercise but is not part of the course.

Chebyshev's criterion is designed to be used in one direction: guess a candidate, check that its error equioscillates, and conclude. Let us see the method in action on a concrete problem.

Example 44.7.2 (The Best Straight Line Through e^t)

We find the best approximation to $f(t) = e^t$ on $[-1, 1]$ from $\mathcal{P}_1$. Write $P(t) = \alpha + \beta t$. Since $n = 2$ here, the criterion asks for three points at which the error takes the values $+\sigma, -\sigma, +\sigma$ (or the negatives).

The error $e^t - \alpha - \beta t$ has derivative $e^t - \beta$, which vanishes exactly once, at $t^* = \log \beta$ (assuming $\beta > 0$). So the error is decreasing then increasing, and its only candidates for extreme values are $t = -1$, $t = t^*$ and $t = 1$. For equioscillation we therefore need

$$e^{-1} - \alpha + \beta = -(e^{t^*} - \alpha - \beta t^*) = e - \alpha - \beta.$$

The outer equality gives $e^{-1} + \beta = e - \beta$, that is

$$\beta = \frac{e - e^{-1}}{2} = \sinh 1 = 1.175201\dots,$$

which is exactly the slope of the chord of f across $[-1, 1]$ – as one would guess from the mean value theorem. Putting $t^* = \log \beta$ into the first equality and rearranging,

$$\alpha = \frac{e^{-1}}{2} + \beta - \frac{\beta \log \beta}{2} = 1.264300\dots$$

The common value of the error is then $\sigma = e^{-1} - \alpha + \beta = 0.278781\dots$, and by Theorem 44.7.1 no other line does better. For comparison, the Taylor polynomial $1 + t$ has error $e - 2 = 0.718\dots$ at $t = 1$: over an interval of this size, the best approximation is better than the Taylor approximation by a factor of nearly three.

Here is the classic application, and it explains where the Chebyshev polynomials really come from.

Theorem 44.7.3

Work on $[-1, 1]$ and let $n \geq 1$. Write $S_n(t) = t^n - 2^{1-n}T_n(t)$, which lies in $\mathcal{P}_{n-1}$ by Theorem 44.6.4. Then

$$\sup_{t \in [-1, 1]} |t^n - Q(t)| \geq \sup_{t \in [-1, 1]} |t^n - S_n(t)| = 2^{1-n}$$

for every $Q \in \mathcal{P}_{n-1}$.

Proof. By construction $t^n - S_n(t) = 2^{1-n}T_n(t)$, and $|T_n(\cos \theta)| = |\cos n\theta| \leq 1$ with $\|T_n\|_\infty = 1$ on $[-1, 1]$, so $\sigma := \|t^n - S_n\|_\infty = 2^{1-n}$. Let $a_j = \cos\left(\frac{(n-j)\pi}{n}\right)$ for $0 \leq j \leq n$, so that $-1 = a_0 < a_1 < \dots < a_n = 1$ and

$$a_j^n - S_n(a_j) = 2^{1-n} \cos((n-j)\pi) = (-1)^{n-j} \sigma = (-1)^n (-1)^j \sigma.$$

So the equiripple criterion of Theorem 44.7.1 applies with $\zeta = (-1)^n$, and S_n is a best approximation to t^n from $\mathcal{P}_{n-1}$. $\square$

So the Chebyshev polynomial $2^{1-n}T_n$ is precisely the monic polynomial of degree n with smallest uniform norm on $[-1, 1]$. This settles the question we left open when we discussed interpolation.

Corollary 44.7.4 (Optimal Interpolation Nodes)

Let $x_0, \dots, x_n$ be distinct points of $[-1, 1]$. Then

$$\sup_{t \in [-1, 1]} \left| \prod_{j=0}^n (t - x_j) \right| \geq 2^{-n},$$

with equality when the x_j are taken to be the $n + 1$ roots

$$x_j = \cos \left(\frac{(2j+1)\pi}{2n+2} \right) \quad (0 \leq j \leq n)$$

of T_{n+1} . Consequently, if f is $n + 1$ times differentiable on $[-1, 1]$ and $P \in \mathcal{P}_n$ interpolates f at those roots, then

$$\|f - P\|_\infty \leq \frac{1}{2^n(n+1)!} \sup_{t \in [-1, 1]} |f^{(n+1)}(t)|.$$

Proof. The polynomial $\pi(t) = \prod_{j=0}^n (t - x_j)$ is monic of degree $n + 1$, so $\pi(t) = t^{n+1} - Q(t)$ with $Q \in \mathcal{P}_n$, and Theorem 44.7.3 (with $n + 1$ in place of n) gives $\|\pi\|_\infty \geq 2^{1-(n+1)} = 2^{-n}$. Taking the x_j to be the roots of T_{n+1} makes $\pi = 2^{-n}T_{n+1}$, since both sides are monic of degree $n + 1$ with the same roots, and $\|T_{n+1}\|_\infty = 1$. The last statement is Theorem 44.6.3 combined with this equality. $\square$

That is precisely why the second of our two pictures was so much better than the first. For equally spaced nodes the product $\prod_j |t - x_j|$ is small in the middle of the interval but exponentially larger near the ends – which is exactly where the interpolant went wrong – whereas the Chebyshev nodes, which crowd towards ± 1 , equalise it. (The reader should not conclude that the corollary *explains* the Runge phenomenon: for $f(x) = (1 + 25x^2)^{-1}$ the derivatives $f^{(n+1)}$ grow so fast that the bound above is useless for either choice of nodes. What is true is that the bound is the best one available which does not look at f , and that in practice Chebyshev interpolation converges for a much wider class of functions than equally spaced interpolation.)

Theorem 44.7.3 also gives useful lower bounds on the size of polynomials with large coefficients.

Corollary 44.7.5

We work on $[-1, 1]$.

- (i) If $P(t) = \sum_{j=0}^n a_j t^j$ with $|a_n| \geq 1$, then $\|P\|_\infty \geq 2^{1-n}$.
- (ii) There is an $\varepsilon(n) > 0$ with the following property: if $P(t) = \sum_{j=0}^n a_j t^j$ and $|a_k| \geq 1$ for *some* $0 \leq k \leq n$, then $\|P\|_\infty \geq \varepsilon(n)$.

Proof. (i) Write $P/a_n = t^n - Q(t)$ with $Q \in \mathcal{P}_{n-1}$. By Theorem 44.7.3, $\|P\|_\infty / |a_n| \geq 2^{1-n}$, and $|a_n| \geq 1$.

(ii) We induct on n . For $n = 0$ we may take $\varepsilon(0) = 1$. Suppose $\varepsilon(n)$ works, and let $P(t) = \sum_{j=0}^{n+1} a_j t^j$ with $|a_k| \geq 1$ for some k . If $|a_{n+1}| \geq \varepsilon(n)/2$ then (i)

applied to P/a_{n+1} gives $\|P\|_\infty \geq |a_{n+1}|2^{-n} \geq \varepsilon(n)2^{-n-1}$. If instead $|a_{n+1}| < \varepsilon(n)/2$ then $k \leq n$, so writing $R(t) = \sum_{j=0}^n a_j t^j$ we have $\|R\|_\infty \geq \varepsilon(n)$ by the inductive hypothesis, and

$$\|P\|_\infty \geq \|R\|_\infty - |a_{n+1}| \|t^{n+1}\|_\infty \geq \varepsilon(n) - \frac{\varepsilon(n)}{2} = \frac{\varepsilon(n)}{2}.$$

So $\varepsilon(n+1) = \varepsilon(n)2^{-n-1}$ will do. $\square$

What part (ii) really says is that on the finite dimensional space $\mathcal{P}_n$ the uniform norm and the norm $\max_j |a_j|$ on coefficients are equivalent. That is exactly what we need to run a compactness argument.

Theorem 44.7.6

If $f \in C([a, b])$ and $n \geq 0$, then there is a $P \in \mathcal{P}_n$ with $\|f - P\|_\infty \leq \|f - Q\|_\infty$ for all $Q \in \mathcal{P}_n$.

Proof. A linear change of variable turns $[a, b]$ into $[-1, 1]$ and carries $\mathcal{P}_n$ bijectively onto itself, so we may assume $[a, b] = [-1, 1]$. Define $\Phi : \mathbb{R}^{n+1} \rightarrow \mathbb{R}$ by

$$\Phi(a_0, \dots, a_n) = \left\| f - \sum_{j=0}^n a_j t^j \right\|_\infty.$$

Then Φ is continuous: indeed $|\Phi(\mathbf{a}) - \Phi(\mathbf{b})| \leq \left\| \sum (a_j - b_j) t^j \right\|_\infty \leq \sum_j |a_j - b_j|$.

Let $\kappa = \inf \Phi$, and note $\kappa \leq \Phi(\mathbf{0}) = \|f\|_\infty$. If $\Phi(\mathbf{a}) \leq \|f\|_\infty$ then $\left\| \sum a_j t^j \right\|_\infty \leq 2\|f\|_\infty$, and so, writing $M = \max_j |a_j|$, Corollary 44.7.5(ii) applied to $M^{-1} \sum a_j t^j$ (which has a coefficient of modulus 1) gives $\left\| \sum a_j t^j \right\|_\infty \geq M\varepsilon(n)$ whenever $M > 0$. Hence

$$M \leq \frac{2\|f\|_\infty}{\varepsilon(n)}.$$

So the set $E = \{\mathbf{a} : \Phi(\mathbf{a}) \leq \|f\|_\infty\}$ is bounded, and it is closed since Φ is continuous. Thus E is compact and non-empty, so Φ attains a minimum on E ; since $\Phi \geq \Phi(\mathbf{0})$ off E , this is a global minimum. $\square$

Remark (Least Squares versus Uniform Approximation). It is worth contrasting this with best approximation in the $\|\cdot\|_2$ norm. There, $\mathcal{P}_n$ is a finite dimensional subspace of an inner product space, and the best approximation is simply the orthogonal projection of f onto $\mathcal{P}_n$: it exists, is unique, is characterised by $\langle f - P, Q \rangle = 0$ for all $Q \in \mathcal{P}_n$, and can be computed by expanding f in an orthonormal basis. All of that is linear algebra. The uniform norm has no inner product behind it, which is why we have had to work for existence, and why the characterisation (equioscillation) is a genuinely analytic statement.

§44.8 Gaussian Quadrature

Suppose we want to estimate $\int_a^b f(x) dx$ knowing only the values of f at finitely many points. The naive approach is to interpolate and integrate the interpolant, which leads at once to the following.

Lemma 44.8.1

Let $x_0, \dots, x_n$ be distinct points of $[a, b]$. Then there are unique real numbers $A_0, \dots, A_n$ with

$$\int_a^b P(x) \, dx = \sum_{j=0}^n A_j P(x_j)$$

for all $P \in \mathcal{P}_n$.

Proof. With e_j as in Lemma 44.6.2, every $P \in \mathcal{P}_n$ satisfies $P = \sum_j P(x_j)e_j$, so setting $A_j = \int_a^b e_j(x) \, dx$ and integrating gives the formula. Conversely, if the formula holds then putting $P = e_i$ gives $\int_a^b e_i = \sum_j A_j e_i(x_j) = A_i$, so the A_j are determined. $\square$

Given the previous section's warnings about interpolation, we should be suspicious of the resulting approximation $\int_a^b f \approx \sum_j A_j f(x_j)$. Our suspicion is justified: when the x_j are equally spaced (the Newton–Cotes rules), computation shows that as n grows the A_j start to vary in sign and $\sum_j |A_j|$ grows rapidly. Since an error of size η in the values $f(x_j)$ produces an error of up to $\eta \sum_j |A_j|$ in the answer, the scheme becomes worse as we use more points.

It is a genuinely surprising fact that a clever choice of points avoids this completely. The clever choice is due to Gauss, and involves the orthogonal polynomials for the inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t)g(t) \, dt$$

on $C([-1, 1])$.

Definition 44.8.2 (Legendre Polynomials)

The **Legendre polynomials** $p_0, p_1, p_2, \dots$ are defined by the conditions that p_n has degree exactly n with positive leading coefficient, and $\langle p_n, p_m \rangle = \delta_{nm}$.

They exist and are unique: apply Gram–Schmidt to $1, t, t^2, \dots$, so that

$$q_n(t) = t^n - \sum_{m=0}^{n-1} \langle t^n, p_m \rangle p_m(t), \quad p_n = \frac{q_n}{\|q_n\|_2}.$$

Here $q_n \neq 0$ because it has leading coefficient 1. By construction $p_0, \dots, p_n$ is an orthonormal basis of $\mathcal{P}_n$, and $p_n \perp \mathcal{P}_{n-1}$. (Other normalisations of the Legendre polynomials are common; they differ from ours only by a positive scalar.)

Lemma 44.8.3

The polynomial p_n has n distinct roots, all lying in $(-1, 1)$.

Proof. Let $y_1, \dots, y_m$ be the distinct points of $(-1, 1)$ at which p_n changes sign, and set $Q(t) = \prod_{j=1}^m (t - y_j)$ (with $Q = 1$ if $m = 0$). Then $p_n Q$ does not change sign

anywhere on $[-1, 1]$, and is not identically zero, so

$$\langle p_n, Q \rangle = \int_{-1}^1 p_n(t)Q(t) dt \neq 0.$$

But $p_n \perp \mathcal{P}_{n-1}$, so we cannot have $\deg Q = m \leq n-1$. Hence $m \geq n$; as p_n has degree n it has at most n roots, so $m = n$, the roots are simple and all lie in $(-1, 1)$. $\square$

Theorem 44.8.4 (Gaussian Quadrature)

Let $\alpha_1, \dots, \alpha_n$ be the roots of p_n and let $A_1, \dots, A_n$ be the weights of Lemma 44.8.1 for these points, so that $\int_{-1}^1 P = \sum_j A_j P(\alpha_j)$ for all $P \in \mathcal{P}_{n-1}$.

- (i) In fact $\int_{-1}^1 Q(x) dx = \sum_{j=1}^n A_j Q(\alpha_j)$ for every $Q \in \mathcal{P}_{2n-1}$.
- (ii) Conversely, if $\beta_j \in [-1, 1]$ and B_j are such that $\int_{-1}^1 Q = \sum_j B_j Q(\beta_j)$ for all $Q \in \mathcal{P}_{2n-1}$, then the β_j are the roots of p_n .

Proof. (i) Let $Q \in \mathcal{P}_{2n-1}$. Dividing by p_n we may write $Q = Gp_n + R$ with $G, R \in \mathcal{P}_{n-1}$. Then

$$\int_{-1}^1 Q = \int_{-1}^1 Gp_n + \int_{-1}^1 R = \langle G, p_n \rangle + \sum_{j=1}^n A_j R(\alpha_j) = 0 + \sum_{j=1}^n A_j R(\alpha_j),$$

using $p_n \perp \mathcal{P}_{n-1}$ and the defining property of the A_j . But $p_n(\alpha_j) = 0$, so $R(\alpha_j) = Q(\alpha_j)$, giving the result.

(ii) Set $W(x) = \prod_{j=1}^n (x - \beta_j)$, a monic polynomial of degree n . For any $R \in \mathcal{P}_{n-1}$ the product WR lies in $\mathcal{P}_{2n-1}$ and vanishes at every β_j , so

$$\langle W, R \rangle = \int_{-1}^1 WR = \sum_{j=1}^n B_j W(\beta_j) R(\beta_j) = 0.$$

Hence $W \in \mathcal{P}_n$ is orthogonal to $\mathcal{P}_{n-1}$. Since $p_0, \dots, p_n$ is an orthonormal basis of $\mathcal{P}_n$ and $p_0, \dots, p_{n-1}$ spans $\mathcal{P}_{n-1}$, the orthogonal complement of $\mathcal{P}_{n-1}$ in $\mathcal{P}_n$ is one-dimensional and spanned by p_n . So W is a scalar multiple of p_n , and the β_j are the roots of p_n . $\square$

Doubling the degree of exactness for free is impressive, but the real payoff is the following, which shows that the objection to Newton–Cotes evaporates.

Theorem 44.8.5

With the notation of Theorem 44.8.4, write $G_n f = \sum_{j=1}^n A_j f(\alpha_j)$.

- (i) $A_j > 0$ for each j .
- (ii) $\sum_{j=1}^n A_j = 2$.
- (iii) If $f \in C([-1, 1])$ and $P \in \mathcal{P}_{2n-1}$, then $\left| \int_{-1}^1 f - G_n f \right| \leq 4 \|f - P\|_\infty$.

(iv) If $f \in C([-1, 1])$ then $G_n f \rightarrow \int_{-1}^1 f(t) dt$ as $n \rightarrow \infty$.

Proof. (i) Fix k and let $w_k(t) = \prod_{j \neq k} (t - \alpha_j)^2$, a polynomial of degree $2n - 2 \leq 2n - 1$. By Theorem 44.8.4(i),

$$0 < \int_{-1}^1 w_k(t) dt = \sum_{j=1}^n A_j w_k(\alpha_j) = A_k w_k(\alpha_k),$$

since w_k vanishes at every α_j with $j \neq k$. As $w_k(\alpha_k) > 0$ we get $A_k > 0$.

(ii) Take $Q = 1$ in Theorem 44.8.4(i).

(iii) Since $\int_{-1}^1 P = G_n P$,

$$\begin{aligned} \left| \int_{-1}^1 f - G_n f \right| &= \left| \int_{-1}^1 (f - P) - \sum_j A_j (f(\alpha_j) - P(\alpha_j)) \right| \\ &\leq \int_{-1}^1 |f - P| + \sum_j |A_j| |f(\alpha_j) - P(\alpha_j)| \\ &\leq 2 \|f - P\|_\infty + \left(\sum_j A_j \right) \|f - P\|_\infty = 4 \|f - P\|_\infty, \end{aligned}$$

where we used (i) to drop the modulus signs on the A_j and (ii) to evaluate their sum.

(iv) Given $\varepsilon > 0$, Weierstrass's Theorem 44.6.7 gives a P with $\|f - P\|_\infty < \varepsilon/4$. For all n with $2n - 1 \geq \deg P$, part (iii) gives

$$\left| \int_{-1}^1 f - G_n f \right| < \varepsilon. \quad \square$$

It is worth carrying the construction out once, if only to see how modest the numbers are.

Example 44.8.6 (The Two-Point Gauss Rule)

Take $n = 2$. Gram-Schmidt gives $p_0(t) = 1/\sqrt{2}$ and, since $\langle t, p_0 \rangle = 0$ and $\int_{-1}^1 t^2 dt = \frac{2}{3}$, also $p_1(t) = \sqrt{3/2}t$. Next

$$q_2(t) = t^2 - \langle t^2, p_0 \rangle p_0(t) - \langle t^2, p_1 \rangle p_1(t) = t^2 - \frac{2/3}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - 0 = t^2 - \frac{1}{3},$$

so the nodes are $\alpha_{1,2} = \pm 1/\sqrt{3}$. By symmetry $A_1 = A_2$, and Theorem 44.8.5(ii) gives $A_1 = A_2 = 1$. Thus

$$\int_{-1}^1 f(t) dt \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right),$$

and Theorem 44.8.4 says that this is exact for every cubic. (One can check this by

hand: $\int_{-1}^1 t^2 dt = \frac{2}{3} = \frac{1}{3} + \frac{1}{3}$ and $\int_{-1}^1 t^3 dt = 0 = -3^{-3/2} + 3^{-3/2}$.)

Let us try it on $f(t) = e^t$, where $\int_{-1}^1 e^t dt = e - e^{-1} = 2.350402\dots$. The rule gives

$$e^{-1/\sqrt{3}} + e^{1/\sqrt{3}} = 2 \cosh\left(\frac{1}{\sqrt{3}}\right) = 2.342760\dots,$$

an error of 0.0076 from two evaluations of f . Simpson's rule, which is also exact on cubics, uses *three* evaluations and gives

$$\frac{1}{3}(e^{-1} + 4e^0 + e) = 2.362054\dots,$$

an error of 0.0117. So Gauss does better with less work, which is the whole point.

It is worth appreciating how the pieces fit together. Positivity of the weights (i) is what stops the errors from amplifying; it is a consequence of the choice of nodes, and it is exactly what fails for equally spaced nodes. Once we have positivity and (ii), the quadrature error is controlled by the best polynomial approximation error, and Weierstrass's theorem then guarantees convergence for *every* continuous f – with no smoothness assumptions at all.

§44.9 Distance and Compact Sets

This section is a digression, but a useful one: we construct a metric on the collection of compact sets. It provides some helpful background for Runge's theorem, and it will supply a striking application of the Baire category theorem later on.

Throughout we work in $\mathbb{R}^m$ with the usual metric, and all sets are non-empty and compact. Recall from Lemma 44.1.13 that $d(\mathbf{x}, A) = \inf_{\mathbf{a} \in A} \|\mathbf{x} - \mathbf{a}\|$ defines a continuous function of $\mathbf{x}$.

Lemma 44.9.1

Let E be a non-empty compact subset of $\mathbb{R}^m$ and $\mathbf{a} \in \mathbb{R}^m$. Then there is a point $\mathbf{e} \in E$ with $\|\mathbf{a} - \mathbf{e}\| = d(\mathbf{a}, E)$.

Proof. The map $\mathbf{x} \mapsto \|\mathbf{a} - \mathbf{x}\|$ is continuous on the compact set E , so attains its infimum by Theorem 44.1.10. $\square$

The nearest point need not be unique – take E to be a circle and $\mathbf{a}$ its centre – but it is unique if E is convex. Indeed if $\mathbf{u}, \mathbf{v} \in E$ both achieve the minimum distance ρ from $\mathbf{a}$ and $\mathbf{u} \neq \mathbf{v}$, then $\frac{1}{2}(\mathbf{u} + \mathbf{v}) \in E$ and, by the parallelogram law,

$$\left\| \mathbf{a} - \frac{\mathbf{u} + \mathbf{v}}{2} \right\|^2 = \rho^2 - \frac{1}{4} \|\mathbf{u} - \mathbf{v}\|^2 < \rho^2,$$

a contradiction.

Lemma 44.9.2

Let $E, F \subseteq \mathbb{R}^m$ be non-empty.

- (i) If E and F are compact, there exist $\mathbf{e} \in E, \mathbf{f} \in F$ with $\|\mathbf{e} - \mathbf{f}\| \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x} \in E, \mathbf{y} \in F$.

(ii) The same holds if E is compact and F is closed.

(iii) It may fail if E and F are merely closed.

Proof. (i) The function $(\mathbf{x}, \mathbf{y}) \mapsto \|\mathbf{x} - \mathbf{y}\|$ is continuous on the compact set $E \times F$.

(ii) Choose $R > 0$ with $E \subseteq \overline{B}(\mathbf{0}, R)$ and $F \cap \overline{B}(\mathbf{0}, R) \neq \emptyset$. Set $F_R = F \cap \overline{B}(\mathbf{0}, 3R)$, a non-empty compact set. By (i) there are $\mathbf{e} \in E$, $\mathbf{f} \in F_R$ minimising the distance over $E \times F_R$, and $\|\mathbf{e} - \mathbf{f}\| \leq 2R$. If $\mathbf{y} \in F \setminus F_R$ then $\|\mathbf{x} - \mathbf{y}\| \geq 3R - R = 2R$ for $\mathbf{x} \in E$, so the minimum over $E \times F$ is achieved too.

(iii) Take $E = \{(x, 1/x) : x > 0\}$ and $F = \{(-x, 1/x) : x > 0\}$. Both are closed, $\|\mathbf{x} - \mathbf{y}\| > 0$ for all $\mathbf{x} \in E$, $\mathbf{y} \in F$, but taking x large shows the infimum is 0. $\square$

§44.9.1 The Hausdorff Metric

We would like a notion of distance between compact sets. The first thing one writes down is

$$\tau(E, F) = \inf_{\mathbf{e} \in E} d(\mathbf{e}, F) = \inf\{\|\mathbf{e} - \mathbf{f}\| : \mathbf{e} \in E, \mathbf{f} \in F\}.$$

This is symmetric and non-negative, and by Lemma 44.9.2 we have $\tau(E, F) = 0$ if and only if $E \cap F \neq \emptyset$. So it fails the axiom $\tau(E, F) = 0 \Rightarrow E = F$ rather badly, and it also fails the triangle inequality: with $A = \{1\}$, $B = \{1, 2\}$, $C = \{2\}$ we have $\tau(A, C) = 1$ but $\tau(A, B) + \tau(B, C) = 0$.

The trouble is that an infimum only notices the closest pair of points. Let us try a supremum instead:

$$\sigma(E, F) = \sup_{\mathbf{e} \in E} d(\mathbf{e}, F).$$

This is non-negative, and $\sigma(E, F) = 0$ if and only if $d(\mathbf{e}, F) = 0$ for every $\mathbf{e} \in E$, that is (as F is closed) if and only if $E \subseteq F$. So σ measures ‘how far E sticks out of F ’, and is emphatically not symmetric. But it does obey the triangle inequality.

Lemma 44.9.3

For non-empty compact E, F, G we have $\sigma(E, G) \leq \sigma(E, F) + \sigma(F, G)$.

Proof. Let $\mathbf{e} \in E$. By Lemma 44.9.1 choose $\mathbf{f} \in F$ with $\|\mathbf{e} - \mathbf{f}\| = d(\mathbf{e}, F)$. For any $\mathbf{g} \in G$,

$$d(\mathbf{e}, G) \leq \|\mathbf{e} - \mathbf{g}\| \leq \|\mathbf{e} - \mathbf{f}\| + \|\mathbf{f} - \mathbf{g}\| = d(\mathbf{e}, F) + \|\mathbf{f} - \mathbf{g}\|.$$

Taking the infimum over $\mathbf{g} \in G$ gives $d(\mathbf{e}, G) \leq d(\mathbf{e}, F) + d(\mathbf{f}, G) \leq \sigma(E, F) + \sigma(F, G)$, and taking the supremum over $\mathbf{e} \in E$ finishes the proof. $\square$

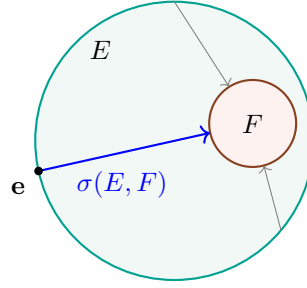
Symmetrising now gives what we want.

Definition 44.9.4 (Hausdorff Metric)

Let $\mathcal{K}$ be the collection of non-empty compact subsets of $\mathbb{R}^m$. For $E, F \in \mathcal{K}$ set

$$\rho(E, F) = \sigma(E, F) + \sigma(F, E) = \sup_{\mathbf{e} \in E} \inf_{\mathbf{f} \in F} \|\mathbf{e} - \mathbf{f}\| + \sup_{\mathbf{f} \in F} \inf_{\mathbf{e} \in E} \|\mathbf{e} - \mathbf{f}\|.$$

We call ρ the **Hausdorff metric**.



here $F \subseteq E$, so $\sigma(F, E) = 0$ while $\sigma(E, F)$ is large

Theorem 44.9.5

ρ is a metric on $\mathcal{K}$.

Proof. Clearly $\rho \geq 0$ and ρ is symmetric. If $\rho(E, F) = 0$ then $\sigma(E, F) = \sigma(F, E) = 0$, so $E \subseteq F$ and $F \subseteq E$, giving $E = F$; and conversely. The triangle inequality follows from Lemma 44.9.3 applied twice:

$$\begin{aligned} \rho(E, G) &= \sigma(E, G) + \sigma(G, E) \\ &\leq (\sigma(E, F) + \sigma(F, G)) + (\sigma(G, F) + \sigma(F, E)) \\ &= \rho(E, F) + \rho(F, G). \square \end{aligned}$$

What makes ρ genuinely useful is that it is complete. We need two preliminary observations.

Lemma 44.9.6

Suppose $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ are non-empty compact subsets of $\mathbb{R}^m$ and let $K = \bigcap_{p \geq 1} K_p$. Then K is non-empty and compact, and $\rho(K_p, K) \rightarrow 0$.

Proof. K is closed (an intersection of closed sets) and bounded, hence compact. To see it is non-empty, pick $\mathbf{k}_p \in K_p$ for each p . All the $\mathbf{k}_p$ lie in the compact set K_1 , so some subsequence $\mathbf{k}_{p(j)} \rightarrow \mathbf{k}$. For each fixed q we have $\mathbf{k}_{p(j)} \in K_q$ for all j with $p(j) \geq q$, and K_q is closed, so $\mathbf{k} \in K_q$. Hence $\mathbf{k} \in K$.

Since $K \subseteq K_p$ we have $\sigma(K, K_p) = 0$, so $\rho(K_p, K) = \sigma(K_p, K)$. Suppose this does not tend to 0. Then there is an $\varepsilon > 0$, a subsequence $p(j)$, and points $\mathbf{x}_{p(j)} \in K_{p(j)}$ with $d(\mathbf{x}_{p(j)}, K) \geq \varepsilon$. Passing to a further subsequence, $\mathbf{x}_{p(j)} \rightarrow \mathbf{x}$, and exactly as above $\mathbf{x} \in K$. But $d(\cdot, K)$ is continuous, so $d(\mathbf{x}, K) \geq \varepsilon > 0$, contradicting $\mathbf{x} \in K$. $\square$

Lemma 44.9.7

If $K \subseteq \mathbb{R}^m$ is compact and $r \geq 0$, then $K + \overline{B}(\mathbf{0}, r) = \{\mathbf{x} + \mathbf{y} : \mathbf{x} \in K, \|\mathbf{y}\| \leq r\}$ is compact.

Proof. It is clearly bounded. If $\mathbf{x}_n + \mathbf{y}_n \rightarrow \mathbf{c}$ with $\mathbf{x}_n \in K$ and $\|\mathbf{y}_n\| \leq r$, pass to a subsequence with $\mathbf{x}_{n(j)} \rightarrow \mathbf{x} \in K$; then $\mathbf{y}_{n(j)} \rightarrow \mathbf{c} - \mathbf{x}$, so $\|\mathbf{c} - \mathbf{x}\| \leq r$ and $\mathbf{c} \in K + \overline{B}(\mathbf{0}, r)$. $\square$

Theorem 44.9.8

The Hausdorff metric ρ is complete on $\mathcal{K}$.

Proof. By Lemma 44.1.3(ii) it suffices to show that any sequence (E_n) in $\mathcal{K}$ with $\rho(E_n, E_{n+1}) \leq 100^{-n}$ converges. Set

$$K_n = E_n + \overline{B}(\mathbf{0}, 10 \cdot 100^{-n}),$$

which is compact by Lemma 44.9.7. Since $\sigma(E_{n+1}, E_n) \leq \rho(E_n, E_{n+1}) \leq 100^{-n}$, every point of E_{n+1} lies within 100^{-n} of E_n , that is $E_{n+1} \subseteq E_n + \overline{B}(\mathbf{0}, 100^{-n})$. Hence

$$K_{n+1} \subseteq E_n + \overline{B}(\mathbf{0}, 100^{-n}) + \overline{B}(\mathbf{0}, 10 \cdot 100^{-n-1}) \subseteq E_n + \overline{B}(\mathbf{0}, 2 \cdot 100^{-n}) \subseteq K_n.$$

By Lemma 44.9.6 the set $K = \bigcap_n K_n$ is a non-empty compact set with $\rho(K_n, K) \rightarrow 0$. Finally $E_n \subseteq K_n$ and every point of K_n is within $10 \cdot 100^{-n}$ of E_n , so $\rho(E_n, K_n) \leq 10 \cdot 100^{-n} \rightarrow 0$, and therefore

$$\rho(E_n, K) \leq \rho(E_n, K_n) + \rho(K_n, K) \rightarrow 0. \quad \square$$

§44.10 Runge's Theorem

Weierstrass's theorem tells us that every continuous real function on $[a, b]$ is a uniform limit of polynomials. Is there a complex analogue? Cauchy's theorem answers with a resounding no.

Example 44.10.1

Let $\overline{D} = \{z : |z| \leq 1\}$ and $f(z) = \bar{z}$. Then $\sup_{z \in \overline{D}} |f(z) - P(z)| \geq 1$ for every polynomial P .

Indeed, let C be the contour $\theta \mapsto e^{i\theta}$, $0 \leq \theta \leq 2\pi$. By Cauchy's theorem $\int_C P(z) dz = 0$, so

$$2\pi \|f - P\|_\infty \geq \left| \int_C (\bar{z} - P(z)) dz \right| = \left| \int_C \bar{z} dz \right| = \left| \int_0^{2\pi} e^{-i\theta} \cdot ie^{i\theta} d\theta \right| = 2\pi.$$

That example is unfair: $\bar{z}$ is nowhere analytic, and we know that a uniform limit of analytic functions is analytic. So let us ask for analytic f . Cauchy's theorem still causes trouble.

Example 44.10.2

Let $T = \{z : \frac{1}{2} \leq |z| \leq 2\}$ and $f(z) = 1/z$, which is analytic on the open set $\Omega = \{z : 0 < |z| < 3\} \supseteq T$. Then $\sup_{z \in T} |f(z) - P(z)| \geq 1$ for every polynomial P ,

since with C as above,

$$2\pi \|f - P\|_\infty \geq \left| \int_C \left(\frac{1}{z} - P(z) \right) dz \right| = |2\pi i| = 2\pi.$$

The obstruction here is that T has a hole, and f misbehaves inside it. So the best we can hope for is a theorem about sets without holes, and the following turns out to be the right notion.

Definition 44.10.3 (Path-Connectedness)

A set $U \subseteq \mathbb{C}$ is **path-connected** if, given $z_0, z_1 \in U$, we can find a continuous $\gamma : [0, 1] \rightarrow U$ with $\gamma(0) = z_0$ and $\gamma(1) = z_1$.

Theorem 44.10.4 (Runge)

Let $\Omega \subseteq \mathbb{C}$ be open and $f : \Omega \rightarrow \mathbb{C}$ analytic. Suppose $K \subseteq \Omega$ is compact with $\mathbb{C} \setminus K$ path-connected. Then given $\varepsilon > 0$ there is a polynomial P with

$$\sup_{z \in K} |f(z) - P(z)| < \varepsilon.$$

Note that this is genuinely stronger than Taylor's theorem, which says only that f is uniformly approximable by polynomials on any *disc* contained in Ω ; Runge's theorem is a global statement. (And there is no hope of simply expanding in a Taylor series: on $\Omega = \{z : 10^{-2} < |z| < 1\} \setminus (-\infty, 0]$ the function $1/z$ is analytic and bounded but is not given by any single power series.)

The proof falls into three steps. First we express f on K as a sum of contour integrals along segments lying outside K ; then we approximate each such integral by a finite sum of simple poles; then we show that each simple pole can be pushed off to infinity and replaced by a polynomial. The last step is where path-connectedness enters.

§44.10.1 Step One: A Contour Representation

Lemma 44.10.5

Let $\Omega \subseteq \mathbb{C}$ be open and $K \subseteq \Omega$ compact and non-empty. Then we can find finitely many line segments $l_1, \dots, l_M$, each contained in $\Omega \setminus K$, such that

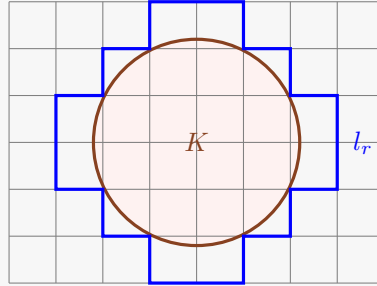
$$f(z) = \sum_{r=1}^M \frac{1}{2\pi i} \int_{l_r} \frac{f(w)}{w - z} dw$$

for every $z \in K$ and every analytic $f : \Omega \rightarrow \mathbb{C}$. Moreover $d(K, l_1 \cup \dots \cup l_M) > 0$.

Proof. Since K is compact and $\mathbb{C} \setminus \Omega$ is closed and disjoint from K , Lemma 44.9.2(ii) gives a $\delta > 0$ with $|k - u| \geq 100\delta$ for all $k \in K$ and $u \notin \Omega$ (if $\Omega = \mathbb{C}$, take any $\delta > 0$).

Tile the plane by the closed squares of side δ with corners in $\delta\mathbb{Z} + i\delta\mathbb{Z}$, and let Γ be the (finite) collection of those squares which meet K . Every $S \in \Gamma$ satisfies

$S \subseteq K + \overline{B}(0, 2\delta) \subseteq \Omega$, so f is analytic on a neighbourhood of each $S \in \Gamma$. Write $G = \bigcup_{S \in \Gamma} S$. If $z \in K$ then every square containing z meets K and so lies in Γ ; as the squares containing z cover a neighbourhood of z , we get $K \subseteq \text{Int } G$.



Now fix $z \in \text{Int } G$ which does not lie on the boundary of any square of the tiling. Then z lies in the interior of exactly one square S_0 , and $S_0 \in \Gamma$ since $z \in G$. Orienting each ∂S anticlockwise, Cauchy's integral formula and Cauchy's theorem give

$$\frac{1}{2\pi i} \int_{\partial S} \frac{f(w)}{w-z} dw = \begin{cases} f(z) & S = S_0, \\ 0 & S \in \Gamma, S \neq S_0, \end{cases}$$

so summing over $S \in \Gamma$ yields $f(z) = \sum_{S \in \Gamma} \frac{1}{2\pi i} \int_{\partial S} \frac{f(w)}{w-z} dw$.

Each edge of the tiling which is shared by two squares of Γ is traversed once in each direction, so those contributions cancel. Let $l_1, \dots, l_M$ be the remaining edges – the edges of squares of Γ whose neighbouring square is not in Γ – with their induced orientations. Then

$$f(z) = \sum_{r=1}^M \frac{1}{2\pi i} \int_{l_r} \frac{f(w)}{w-z} dw$$

for all such z . Each l_r lies in Ω , and $l_r \cap K = \emptyset$: if l_r met K then the square on its far side would meet K too and would belong to Γ , so l_r would have cancelled. Since $\bigcup_r l_r$ and K are disjoint compact sets, Lemma 44.9.2(i) gives $d(K, \bigcup_r l_r) > 0$.

Finally, both sides of the displayed identity are continuous functions of z on the open set $\text{Int } G$ (for the right-hand side, because the integrands have denominators bounded away from zero there). They agree on $\text{Int } G$ minus the union of the grid lines, which is dense in the open set $\text{Int } G$. Hence they agree on all of $\text{Int } G \supseteq K$. $\square$

§44.10.2 Step Two: Approximation by Simple Poles

Lemma 44.10.6

Let Ω be open, $K \subseteq \Omega$ compact, and $f : \Omega \rightarrow \mathbb{C}$ analytic. Given $\varepsilon > 0$ we can find N , complex numbers $A_1, \dots, A_N$ and points $\alpha_1, \dots, \alpha_N \in \Omega \setminus K$ such that

$$\left| f(z) - \sum_{n=1}^N \frac{A_n}{z - \alpha_n} \right| \leq \varepsilon$$

for all $z \in K$.

Proof. By Lemma 44.10.5 it suffices to approximate a single term $\frac{1}{2\pi i} \int_l \frac{f(w)}{w-z} dw$, where l is a segment disjoint from K and contained in Ω . Parametrise l by $\gamma(t) = w_0 + t(w_1 - w_0)$ for $t \in [0, 1]$, so that

$$\frac{1}{2\pi i} \int_l \frac{f(w)}{w-z} dw = \int_0^1 F(t, z) dt, \quad F(t, z) = \frac{1}{2\pi i} \cdot \frac{f(\gamma(t))(w_1 - w_0)}{\gamma(t) - z}.$$

Since $d(K, l) > 0$, the function F is continuous on the compact set $[0, 1] \times K$, hence uniformly continuous there by Theorem 44.1.12. So given $\eta > 0$ there is an m with $|F(t, z) - F(r/m, z)| < \eta$ whenever $t \in [r/m, (r+1)/m]$ and $z \in K$. Therefore

$$\left| \int_0^1 F(t, z) dt - \frac{1}{m} \sum_{r=0}^{m-1} F(r/m, z) \right| \leq \eta$$

for all $z \in K$, and the Riemann sum is exactly of the form $\sum_r \frac{A_r}{z - \alpha_r}$ with $\alpha_r = \gamma(r/m) \in l \subseteq \Omega \setminus K$ and suitable constants A_r . Doing this for each of the M segments with $\eta = \varepsilon/M$ completes the proof. $\square$

§44.10.3 Step Three: Pushing Poles to Infinity

Everything now reduces to the following statement, which is where the topology of K finally matters.

Lemma 44.10.7

Let $K \subseteq \mathbb{C}$ be compact with $\mathbb{C} \setminus K$ path-connected. Then for every $\alpha \notin K$ and every $\varepsilon > 0$ there is a polynomial P with $|P(z) - \frac{1}{z-\alpha}| < \varepsilon$ for all $z \in K$.

Write $\Lambda(K)$ for the set of $\alpha \notin K$ with this property. We must show $\Lambda(K) = \mathbb{C} \setminus K$, and we do it in three moves.

Proof. (a) $\Lambda(K)$ contains all sufficiently large α . Choose R with $K \subseteq B(0, R/2)$. If $|\alpha| \geq R$ then for $z \in K$,

$$\frac{1}{z-\alpha} = \frac{-1}{\alpha(1-z/\alpha)} = -\frac{1}{\alpha} \sum_{n=0}^{\infty} \left(\frac{z}{\alpha}\right)^n,$$

and since $|z/\alpha| \leq \frac{1}{2}$ on K the series converges uniformly on K by the Weierstrass M -test. Its partial sums are polynomials, so $\alpha \in \Lambda(K)$.

(b) $\Lambda(K)$ is ‘locally closed under small moves’. Suppose $\alpha \in \Lambda(K)$ and $|\alpha - \beta| < \frac{1}{2}d(\alpha, K)$. Put $\delta = d(\alpha, K)$, so $|z - \alpha| \geq \delta$ for $z \in K$. Then for $z \in K$,

$$\frac{1}{z-\beta} = \frac{1}{(z-\alpha) - (\beta-\alpha)} = \frac{1}{z-\alpha} \cdot \frac{1}{1 - \frac{\beta-\alpha}{z-\alpha}} = \sum_{n=0}^{\infty} \frac{(\beta-\alpha)^n}{(z-\alpha)^{n+1}},$$

and since $\left|\frac{\beta-\alpha}{z-\alpha}\right| \leq \frac{1}{2}$ on K the series converges uniformly on K . So given $\varepsilon > 0$ we

can find N with

$$\left| \frac{1}{z - \beta} - \sum_{n=0}^N \frac{(\beta - \alpha)^n}{(z - \alpha)^{n+1}} \right| < \frac{\varepsilon}{2} \quad (z \in K).$$

Now $\alpha \in \Lambda(K)$ means $(z - \alpha)^{-1}$ is uniformly approximable on K by polynomials; hence so is each power $(z - \alpha)^{-(n+1)}$ (a product of uniformly bounded uniformly approximable functions is uniformly approximable), and hence so is the finite sum. Choosing a polynomial within $\varepsilon/2$ of it gives $\beta \in \Lambda(K)$.

(c) Path-connectedness. Let $\beta \notin K$. By (a) choose α_0 with $|\alpha_0|$ large enough that $\alpha_0 \in \Lambda(K)$. Since $\mathbb{C} \setminus K$ is path-connected there is a continuous $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus K$ with $\gamma(0) = \alpha_0$, $\gamma(1) = \beta$. The image $\gamma([0, 1])$ is compact and disjoint from the compact set K , so by Lemma 44.9.2(i) there is a $\delta > 0$ with $d(\gamma(t), K) \geq 4\delta$ for all t . Also γ is uniformly continuous, so there is an N with $|\gamma(s) - \gamma(t)| < \delta$ whenever $|s - t| \leq 1/N$. Setting $w_j = \gamma(j/N)$ we have $|w_j - w_{j+1}| < \delta \leq \frac{1}{2}d(w_j, K)$, so by (b) and induction $w_0 \in \Lambda(K)$ gives $w_N = \beta \in \Lambda(K)$. $\square$

Proof of Runge's theorem. Given $\varepsilon > 0$, Lemma 44.10.6 provides A_n and $\alpha_n \notin K$ with $\left| f(z) - \sum_{n=1}^N \frac{A_n}{z - \alpha_n} \right| \leq \varepsilon/2$ on K . By Lemma 44.10.7 each $\frac{1}{z - \alpha_n}$ is within $\frac{\varepsilon}{2N(1+|A_n|)}$ of a polynomial P_n uniformly on K . Then $P = \sum_n A_n P_n$ satisfies $\sup_K |f - P| < \varepsilon$. $\square$

§44.10.4 An Application: Pointwise Limits Need Not Be Continuous

If analytic functions converge *uniformly*, the limit is analytic. It is natural to ask whether uniformity is really needed, and Runge's theorem lets us answer emphatically.

Example 44.10.8

Let $D = \{z : |z| < 1\}$ and define $f : \mathbb{C} \rightarrow \mathbb{C}$ by $f(0) = 0$ and

$$f(re^{i\theta}) = r^{3/2}e^{3i\theta/2} \quad (r > 0, 0 < \theta \leq 2\pi).$$

Then f is not continuous on D (approaching a positive real x from above gives $x^{3/2}$, from below gives $-x^{3/2}$), yet there are polynomials P_n with $P_n(z) \rightarrow f(z)$ for every $z \in D$.

Proof. The function f is analytic on $\Omega = \mathbb{C} \setminus [0, \infty)$, being a branch of $z^{3/2}$ there. For $n \geq 1$ let $f_n(z) = f(z - i/n)$, which is analytic on $\Omega_n = \mathbb{C} \setminus (\frac{i}{n} + [0, \infty))$, and let

$$K_n = \{z : |z| \leq 1, d(z, \frac{i}{n} + [0, \infty)) \geq \frac{1}{n^2}\}.$$

Then K_n is compact, $K_n \subseteq \Omega_n$, and $\mathbb{C} \setminus K_n$ is path-connected: K_n is a closed disc with a slit-shaped neighbourhood removed, and any point of the complement can be joined to infinity by going along the slit and out. By Runge's theorem there is a polynomial P_n with $\sup_{z \in K_n} |f_n(z) - P_n(z)| \leq 2^{-n}$.

Now fix z with $|z| < 1$. If z is a non-negative real then $d(z, \frac{i}{n} + [0, \infty)) = \frac{1}{n} \geq \frac{1}{n^2}$, so $z \in K_n$ for every n . Otherwise $d(z, [0, \infty)) = c > 0$ and so $d(z, \frac{i}{n} + [0, \infty)) \geq$

$c - \frac{1}{n} \geq \frac{1}{n^2}$ for all large n . Either way $z \in K_n$ eventually, and then

$$|P_n(z) - f(z)| \leq 2^{-n} + |f(z - i/n) - f(z)| \rightarrow 0,$$

since $z - i/n \rightarrow z$ from below, and f is continuous from below across the positive real axis (as $\theta \rightarrow 2\pi^-$ we get $r^{3/2}e^{3i\pi} = -r^{3/2}$, which is the assigned value at $\theta = 2\pi$); at $z = 0$ we simply have $f(-i/n) = n^{-3/2}e^{9i\pi/4} \rightarrow 0 = f(0)$. $\square$

§44.11 Odd Numbers

According to Von Neumann, ‘in mathematics you don’t understand things, you just get used to them’. The real line is a case in point. A single innocuous-looking axiom – every bounded increasing sequence converges – calls into being an object of indescribable complexity. We know that $\mathbb{R}$ is uncountable; on the other hand, if our alphabet has n symbols then there are at most n^m phrases of length exactly m , so the collection of *describable* real numbers is a countable union of finite sets, and hence countable. Almost every real number cannot be described at all.

In this section we look at a few of the prettier shells: numbers we can describe, and about which we can say something.

§44.11.1 Irrationality

Theorem 44.11.1

The number e is irrational.

Proof. Suppose $e = p/q$ with $p, q \in \mathbb{N}$. Recall $e = \sum_{r=0}^{\infty} \frac{1}{r!}$. Consider

$$N = q! \left(e - \sum_{r=0}^q \frac{1}{r!} \right) = q! e - \sum_{r=0}^q \frac{q!}{r!}.$$

Now $q! e = q! p/q = (q-1)! p$ is an integer, and each $q!/r!$ with $r \leq q$ is an integer, so $N \in \mathbb{Z}$. On the other hand

$$0 < N = q! \sum_{r=q+1}^{\infty} \frac{1}{r!} = \sum_{k=1}^{\infty} \frac{q!}{(q+k)!} \leq \sum_{k=1}^{\infty} \frac{1}{(q+1)^k} = \frac{1}{q} \leq 1,$$

and the inequality $q!/(q+k)! \leq (q+1)^{-k}$ is strict for $k \geq 2$, so $0 < N < 1$. There is no such integer. $\square$

The proof of the corresponding statement for π is harder, and rests on the following computation.

Lemma 44.11.2

Write $f_n(x) = x^n(\pi - x)^n$. Then

$$\int_0^{\pi} f_n(x) \sin x \, dx = n! \sum_{j=0}^n a_j \pi^j$$

for some integers a_j (depending on n).

Proof. Expanding, $f_n(x) = \sum_{r=0}^n \binom{n}{r} (-1)^r \pi^{n-r} x^{n+r}$, a polynomial with terms of degrees between n and $2n$. So $f_n^{(k)}(0) = 0$ for $k < n$ and for $k > 2n$, while for $n \leq k \leq 2n$ only the term with $n+r = k$ survives and

$$f_n^{(k)}(0) = k! \binom{n}{k-n} (-1)^{k-n} \pi^{2n-k}.$$

Since $n \leq k$, the number $k!/n!$ is an integer, so $f_n^{(k)}(0) = n! m_k \pi^{2n-k}$ for some $m_k \in \mathbb{Z}$, and the exponent $2n-k$ lies between 0 and n . Moreover $f_n(\pi-x) = f_n(x)$, so $f_n^{(k)}(\pi) = (-1)^k f_n^{(k)}(0)$ is of the same shape.

Now integrate by parts twice. Writing $I_k = \int_0^\pi f_n^{(k)}(x) \sin x \, dx$,

$$\begin{aligned} I_k &= \left[-f_n^{(k)}(x) \cos x \right]_0^\pi + \int_0^\pi f_n^{(k+1)}(x) \cos x \, dx \\ &= f_n^{(k)}(\pi) + f_n^{(k)}(0) + \int_0^\pi f_n^{(k+1)}(x) \cos x \, dx, \end{aligned}$$

and $\int_0^\pi f_n^{(k+1)} \cos x \, dx = \left[f_n^{(k+1)} \sin x \right]_0^\pi - I_{k+2} = -I_{k+2}$. Hence

$$I_k = f_n^{(k)}(\pi) + f_n^{(k)}(0) - I_{k+2}.$$

Since $f_n^{(k)} \equiv 0$ for $k > 2n$, we have $I_k = 0$ for $k > 2n$, and iterating the recursion downwards from $k = 0$ expresses $I_0 = \int_0^\pi f_n \sin x \, dx$ as an integer combination of the numbers $f_n^{(k)}(0)$ and $f_n^{(k)}(\pi)$. Each of these is $n!$ times an integer times a power π^j with $0 \leq j \leq n$, which is the claim. $\square$

Theorem 44.11.3

The number π is irrational.

Proof. Write $P_n(X) = \sum_{j=0}^n a_j X^j \in \mathbb{Z}[X]$ as in Lemma 44.11.2, so that the integral $\int_0^\pi f_n(x) \sin x \, dx$ equals $n! P_n(\pi)$. On $(0, \pi)$ we have $x^n(\pi-x)^n \sin x > 0$, so this integral is strictly positive. On the other hand, by AM–GM we have $x(\pi-x) \leq (\pi/2)^2$, so

$$0 < n! P_n(\pi) = \int_0^\pi f_n(x) \sin x \, dx \leq \int_0^\pi \left(\frac{\pi}{2}\right)^{2n} \, dx = \pi \left(\frac{\pi^2}{4}\right)^n.$$

Suppose now $\pi = p/q$ with $p, q \in \mathbb{N}$. Since $\deg P_n \leq n$, the number $q^n P_n(p/q)$ is an integer, and it is strictly positive, so $q^n P_n(\pi) \geq 1$. Hence

$$n! \leq q^n P_n(\pi) \leq \pi \left(\frac{q\pi^2}{4}\right)^n.$$

But $C^n/n! \rightarrow 0$ for every constant C , so this fails for large n . $\square$

Faced with a proof like that, the reader may cry ‘how did you think of looking at f_n ?’ The honest answer is ‘I did not, I learnt it from somebody else’; but even granting that we could not have invented it in a thousand years, once presented with it we can see a path which might have led there. We are all familiar with the fact that $\int_0^\pi x^n \sin x \, dx$ takes the form $U(\pi)$ for a polynomial U of degree at most n with integer coefficients, and hence so does $\int_0^\pi Q(x) \sin x \, dx$ for any $Q \in \mathbb{Z}[X]$. If $\pi = p/q$ then $q^n U(\pi)$ is an integer. So we experiment, trying to squeeze $\int_0^\pi Q(x) \sin x \, dx$ between 0 and 1 in the manner of our proof for e ; the factor $n!$ appears because we need the integral to be very small, and the symmetric choice $x^n(\pi - x)^n$ is the natural one.

It is worth remembering that after three hundred years we still do not know whether Euler’s constant

$$\gamma = \lim_{N \rightarrow \infty} \left(\sum_{n=1}^N \frac{1}{n} - \log N \right)$$

is irrational.

§44.11.2 Transcendence

If the rationals are the best understood numbers, the algebraic numbers are the next best understood.

Definition 44.11.4 (Algebraic and Transcendental)

A real number α is **algebraic** if it is a root of some non-zero polynomial with integer coefficients. Otherwise α is **transcendental**.

Multiplying up denominators, α is algebraic if and only if it is a root of a non-zero polynomial with rational coefficients.

Lemma 44.11.5

The set of algebraic numbers is countable. Consequently transcendental numbers exist, and indeed the transcendental numbers are uncountable.

Proof. For each n and each bound B there are finitely many integer polynomials of degree at most n with all coefficients at most B in modulus, and each has at most n roots. So the algebraic numbers are a countable union of finite sets, hence countable. Since $\mathbb{R}$ is uncountable, the complement is uncountable. $\square$

This argument is Cantor’s, and it is beautiful but entirely non-constructive: it tells us that transcendentals abound while showing us none. Historically, the first proof came earlier, and is due to Liouville. It is longer, but it produces actual examples. The key point is that algebraic numbers are *badly approximable* by rationals.

Theorem 44.11.6 (Liouville)

Let α be an irrational root of a polynomial $P(X) = a_n X^n + \cdots + a_0$ with $a_j \in \mathbb{Z}$, $n \geq 1$ and $a_n \neq 0$. Then there is a constant $c = c(\alpha) > 0$ such that

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{c}{q^n}$$

for all $p, q \in \mathbb{Z}$ with $q \geq 1$.

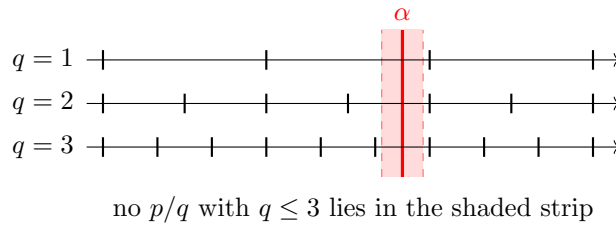
Proof. Since P has finitely many roots, we can find $\delta > 0$ such that P has no root other than α in $J = [\alpha - \delta, \alpha + \delta]$. As P' is continuous on the compact set J , it is bounded there, say $|P'| \leq M$ with $M > 0$; by the mean value theorem $|P(t) - P(s)| \leq M|t - s|$ for $s, t \in J$.

Let $p, q \in \mathbb{Z}$ with $q \geq 1$. Suppose first that $p/q \in J$. Then $P(p/q) \neq 0$, since α is the only root of P in J and α is irrational. But $q^n P(p/q) = \sum_j a_j p^j q^{n-j}$ is an integer, so $|q^n P(p/q)| \geq 1$ and hence $|P(p/q)| \geq q^{-n}$. Therefore

$$\left| \alpha - \frac{p}{q} \right| \geq \frac{1}{M} |P(\alpha) - P(p/q)| = \frac{|P(p/q)|}{M} \geq \frac{1}{Mq^n}.$$

If instead $p/q \notin J$ then $\left| \alpha - \frac{p}{q} \right| > \delta \geq \delta/q^n$. So $c = \min\{1/M, \delta\}$ works. $\square$

The content of the theorem is best seen geometrically. Rationals with denominator q are spaced $1/q$ apart, so we should expect to be able to find one within $1/(2q)$ of α , and the theory of continued fractions in the next section will do much better, giving one within $1/q^2$. Liouville's theorem says that for an algebraic α of degree n we can never do better than c/q^n : around each algebraic number there is a forbidden zone which the good rational approximations may not enter.



Turning this on its head: if a number *can* be approximated absurdly well by rationals, it cannot be algebraic.

Theorem 44.11.7

The number $\gamma = \sum_{j=0}^{\infty} \frac{1}{10^{j!}} = 1.110001000000000000000001\dots$ is transcendental.

Proof. First, γ is irrational: its decimal expansion consists of 1's in the places $j!$ and 0's elsewhere, and so is neither terminating nor eventually periodic (the gaps between consecutive 1's grow without bound).

Let $N \geq 1$ and put $\frac{p_N}{q_N} = \sum_{j=0}^N 10^{-j!}$, where $q_N = 10^{N!}$ and $p_N \in \mathbb{N}$. Then

$$0 < \gamma - \frac{p_N}{q_N} = \sum_{j=N+1}^{\infty} 10^{-j!} \leq \sum_{k=(N+1)!}^{\infty} 10^{-k} = \frac{10}{9} \cdot 10^{-(N+1)!} \leq \frac{2}{q_N^{N+1}},$$

using $(N+1)! = (N+1) \cdot N!$, so that $10^{-(N+1)!} = q_N^{-(N+1)}$.

Suppose γ were algebraic, of degree n say, and let $c > 0$ be as in Theorem 44.11.6.

Then for every N ,

$$\frac{c}{q_N^n} \leq \left| \gamma - \frac{p_N}{q_N} \right| \leq \frac{2}{q_N^{N+1}}, \quad \text{so} \quad c \leq \frac{2}{q_N^{N+1-n}}.$$

Taking $N \geq n+1$ and letting $N \rightarrow \infty$, the right-hand side tends to 0 (as $q_N \rightarrow \infty$), contradicting $c > 0$. $\square$

Remark. Replacing γ by $\sum_{j \geq 0} \varepsilon_j 10^{-j!}$ with $\varepsilon_j \in \{1, 2\}$ gives an uncountable family of such numbers, all transcendental by the same argument. So Liouville's method also proves that there are uncountably many transcendentals, and this time we can point at them.

Numbers which admit approximations of this quality are called **Liouville numbers**. They are, in a precise sense, very rare – the set of them has Lebesgue measure zero – yet they are the only transcendentals we can construct with bare hands. As one might expect, proving that a *naturally occurring* number is transcendental is much harder. Hermite proved that e is transcendental, and Lindemann adapted his method to show that π is transcendental, thereby settling the two-thousand-year-old problem of squaring the circle in the negative. These proofs are not part of this course; Alan Baker's *Transcendental Number Theory* contains accessible (but miraculous) versions of both.

§44.12 The Baire Category Theorem

The theorem of this section is much more useful than its somewhat abstract formulation makes it appear. The informal version is this. We work in a non-empty complete metric space (X, d) , and we have a list of properties $P_1, P_2, \dots$ which points might have. Suppose each P_j is *stable* (if x has P_j then so does everything sufficiently close to x) while $\neg P_j$ is *unstable* (however close we look to any point, we can find something with P_j). Then there is a single point which has *all* of the properties at once.

That is a remarkable thing to be able to say. It means that to prove the existence of an object with infinitely many strange properties, it is enough to show that each property individually is achievable by a small perturbation. We shall use it to construct nowhere differentiable functions, and to construct compact sets which are simultaneously very thin and very rich.

Theorem 44.12.1 (Baire Category Theorem)

Let (X, d) be a non-empty complete metric space and let $U_1, U_2, \dots$ be open sets whose complements have empty interior. Then $\bigcap_{j \geq 1} U_j \neq \emptyset$.

Note that ' U_j is open and U_j^c has empty interior' says exactly that U_j is open and dense.

Proof. Since U_1^c has empty interior, $U_1 \neq \emptyset$; choose $x_1 \in U_1$, and since U_1 is open choose $\delta_1 \in (0, 1)$ with $\overline{B}(x_1, 4\delta_1) \subseteq U_1$.

Now proceed inductively. Given x_n and $\delta_n > 0$, the ball $B(x_n, \delta_n)$ is not contained in U_{n+1}^c (which has empty interior), so we may choose $x_{n+1} \in U_{n+1}$ with $d(x_n, x_{n+1}) < \delta_n$. Since U_{n+1} is open we may then choose δ_{n+1} with $0 < \delta_{n+1} \leq \delta_n/4$ and $\overline{B}(x_{n+1}, 4\delta_{n+1}) \subseteq U_{n+1}$.

Then $d(x_n, x_{n+1}) < \delta_n \leq 4^{1-n}\delta_1$, so $\sum_n d(x_n, x_{n+1}) < \infty$ and (x_n) is Cauchy; by completeness $x_n \rightarrow x$ for some $x \in X$. Moreover, for $m > n$,

$$d(x_n, x_m) \leq \sum_{r=n}^{m-1} d(x_r, x_{r+1}) \leq \delta_n \left(1 + \frac{1}{4} + \frac{1}{16} + \cdots\right) \leq 2\delta_n,$$

so letting $m \rightarrow \infty$ gives $d(x_n, x) \leq 2\delta_n$, whence $x \in \overline{B}(x_n, 4\delta_n) \subseteq U_n$. As n was arbitrary, $x \in \bigcap_n U_n$. $\square$

Taking complements gives an equivalent statement which is often the more convenient one.

Theorem 44.12.2

Let (X, d) be a non-empty complete metric space and let $F_1, F_2, \dots$ be closed sets with empty interior. Then $\bigcup_{j \geq 1} F_j \neq X$.

And the informal version we began with is the following.

Theorem 44.12.3

Let (X, d) be a non-empty complete metric space and let $P_1, P_2, \dots$ be properties such that for each j :

- (i) if x has property P_j then there is an $\varepsilon > 0$ such that every y with $d(x, y) < \varepsilon$ has property P_j ;
- (ii) given $x \in X$ and $\varepsilon > 0$, we can find y with $d(x, y) < \varepsilon$ having property P_j .

Then there is an $x_0 \in X$ having every property $P_1, P_2, \dots$

Proof. Let $U_j = \{x : x \text{ has } P_j\}$. Condition (i) says U_j is open and (ii) says U_j is dense, so U_j^c is closed with empty interior. Apply Theorem 44.12.1. $\square$

Baire's theorem has given rise to some standard, and rather unfortunate, terminology.

Definition 44.12.4 (Category)

Let (X, d) be a metric space.

- (i) A set $E \subseteq X$ is **nowhere dense** if $\text{Cl } E$ has empty interior.
- (ii) A set E is of **first category** if it is contained in a countable union of closed sets with empty interior.
- (iii) A set which is not of first category is said to be of **second category**.

The nomenclature is poor – there is nothing categorical about it, and ‘second category’ tells one nothing – so I shall try to avoid saying ‘second category’ and speak of ‘not of first category’ instead. If everything outside a set of first category has some property, I shall say that **quasi-all** points have that property. Sets of first category are the topological analogue of null sets: small, but small in a sense which has nothing to do with measure.

Lemma 44.12.5

Let (X, d) be a metric space.

- (i) If X is non-empty and complete and $E \subseteq X$ is of first category, then $E \neq X$; indeed $E^c \neq \emptyset$.
- (ii) A countable union of sets of first category is of first category.

Proof. (i) is Theorem 44.12.2. (ii) is because a countable union of countable unions is a countable union. $\square$

§44.12.1 Uncountability, Again**Definition 44.12.6**

Let (X, d) be a metric space and $E \subseteq X$. A point $x \in E$ is **isolated** in E if there is a $\delta > 0$ with $B(x, \delta) \cap E = \{x\}$. We say E has **no isolated points** if no point of E is isolated in E .

Theorem 44.12.7

A non-empty complete metric space with no isolated points is uncountable.

Proof. Suppose $X = \{x_1, x_2, \dots\}$ is countable. Each singleton $F_n = \{x_n\}$ is closed, and has empty interior precisely because x_n is not isolated (every ball about x_n contains another point of X). By Theorem 44.12.2, $\bigcup_n F_n \neq X$, contradicting the enumeration. $\square$

Corollary 44.12.8

$\mathbb{R}$ is uncountable. More generally, a non-empty closed subset of $\mathbb{R}$ with no isolated points is uncountable.

Proof. $\mathbb{R}$ is complete with no isolated points. A closed subset of a complete space is complete. $\square$

This proof is much closer to Cantor's original argument than the diagonal argument of Part IA, and it has the advantage of avoiding extraneous notions like decimal expansions.

§44.12.2 Nowhere Differentiable Functions

Banach was a master of the Baire category theorem, and the following is one of his results. Recall from Theorem 44.6.6 that $C([0, 1])$ with the uniform norm is a non-empty complete metric space.

Theorem 44.12.9 (Banach)

The set of functions in $C([0, 1])$ which are differentiable at at least one point is of first category. In particular, continuous nowhere differentiable functions exist.

The key is to replace ‘differentiable somewhere’ by something closed.

Lemma 44.12.10

If $f : [0, 1] \rightarrow \mathbb{R}$ is continuous and differentiable at some $x \in [0, 1]$, then there is an $M > 0$ with $|f(x) - f(s)| \leq M|x - s|$ for all $s \in [0, 1]$.

Proof. Since $(f(s) - f(x))/(s - x) \rightarrow f'(x)$ as $s \rightarrow x$, there is a $\delta > 0$ with

$$\left| \frac{f(s) - f(x)}{s - x} - f'(x) \right| \leq 1 \quad \text{whenever } 0 < |s - x| < \delta,$$

and hence $|f(s) - f(x)| \leq (1 + |f'(x)|)|s - x|$ for such s . If $|s - x| \geq \delta$ then $|f(s) - f(x)| \leq 2\|f\|_\infty \leq \frac{2\|f\|_\infty}{\delta}|s - x|$. Take M to be the larger of the two constants. $\square$

So let

$$E_m = \{f \in C([0, 1]) : \exists x \in [0, 1] \text{ with } |f(x) - f(s)| \leq m|x - s| \text{ for all } s \in [0, 1]\}.$$

By Lemma 44.12.10, every function differentiable somewhere lies in some E_m . So it suffices to show each E_m is closed with empty interior.

Lemma 44.12.11

E_m is closed in $C([0, 1])$.

Proof. Suppose $f_r \in E_m$ and $\|f_r - f\|_\infty \rightarrow 0$. For each r pick $x_r \in [0, 1]$ with $|f_r(x_r) - f_r(s)| \leq m|x_r - s|$ for all s . By compactness of $[0, 1]$, pass to a subsequence with $x_{r(k)} \rightarrow x$. Then for any $s \in [0, 1]$, writing r for $r(k)$,

$$\begin{aligned} |f(x) - f(s)| &\leq |f(x) - f(x_r)| + |f(x_r) - f_r(x_r)| + |f_r(x_r) - f_r(s)| + |f_r(s) - f(s)| \\ &\leq |f(x) - f(x_r)| + 2\|f - f_r\|_\infty + m|x_r - s|. \end{aligned}$$

Letting $k \rightarrow \infty$, the first term tends to 0 by continuity of f , the second by uniform convergence, and the third to $m|x - s|$. Hence $|f(x) - f(s)| \leq m|x - s|$ for all s , so $f \in E_m$. $\square$

Lemma 44.12.12

E_m has empty interior: given $f \in C([0, 1])$ and $\varepsilon > 0$, there is an $h \notin E_m$ with $\|h - f\|_\infty < \varepsilon$.

Proof. The idea is to move first from f to a very nice function, and then from the nice function to a very bad one.

By the Weierstrass approximation theorem there is a polynomial g with $\|g - f\|_\infty < \varepsilon/3$. Being a polynomial, g is continuously differentiable, so $|g'| \leq M$ on $[0, 1]$ for some M , and hence $|g(s) - g(t)| \leq M|s - t|$ for all s, t .

Now set

$$h(t) = g(t) + \frac{\varepsilon}{3} \cos(2\pi Nt)$$

for an integer N to be chosen. Certainly $\|h - f\|_\infty < \varepsilon$. Let $t_k = k/(2N)$ for $0 \leq k \leq 2N$, so that $\cos(2\pi N t_k) = (-1)^k$ and

$$|h(t_{k+1}) - h(t_k)| = \left| g(t_{k+1}) - g(t_k) \mp \frac{2\varepsilon}{3} \right| \geq \frac{2\varepsilon}{3} - \frac{M}{2N}.$$

Choose N so large that $M/(2N) \leq \varepsilon/3$ and $N > 3m/\varepsilon$. Then $|h(t_{k+1}) - h(t_k)| \geq \varepsilon/3$ for every k .

Now let $x \in [0, 1]$ be arbitrary, and pick k with $t_k \leq x \leq t_{k+1}$. By the triangle inequality

$$\max \{ |h(t_k) - h(x)|, |h(t_{k+1}) - h(x)| \} \geq \frac{1}{2} |h(t_{k+1}) - h(t_k)| \geq \frac{\varepsilon}{6},$$

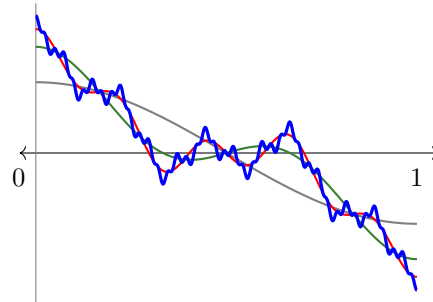
while $|t_k - x|, |t_{k+1} - x| \leq \frac{1}{2N} < \frac{\varepsilon}{6m}$. So for some $s \in \{t_k, t_{k+1}\}$ we have $|h(x) - h(s)| \geq \varepsilon/6 > m|x - s|$. Since x was arbitrary, $h \notin E_m$. $\square$

Proof of Theorem 44.12.9. By Lemmas 44.12.11 and 44.12.12 each E_m is closed with empty interior, so $\bigcup_m E_m$ is of first category and, by Lemma 44.12.5, is not all of $C([0, 1])$. Any $f \notin \bigcup_m E_m$ is, by Lemma 44.12.10, differentiable nowhere. $\square$

Notice what a strange proof this is. We have shown that quasi-all continuous functions are nowhere differentiable – that the differentiable functions, which are the only ones anybody ever draws, form a vanishingly small part of $C([0, 1])$ – without exhibiting a single nowhere differentiable function. One can of course write one down: Weierstrass's original example was

$$W(t) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi t)$$

for suitable $0 < a < 1$ and odd integers b with $ab > 1$. Here are the partial sums for $a = 1/2$, $b = 3$ up to $k = 0, 1, 2$ and 4. Each new term adds wiggles which are smaller in amplitude but far steeper, and in the limit the graph has a corner at every point.



partial sums to $k = 0$, $k = 1$, $k = 2$, $k = 4$

§44.12.3 Perfect Nowhere Dense Sets

Here is a second, and rather different, application. A closed subset of $\mathbb{R}$ with no isolated points is uncountable, by the corollary above. Are there such sets which are nevertheless nowhere dense – containing no interval at all? The Cantor set is one; but we can prove existence without constructing anything, by applying Baire's theorem in a space of sets.

Let $\mathcal{K}$ denote the collection of non-empty compact subsets of $[0, 1]$, with the Hausdorff

metric ρ of the previous section. Since the compact subsets of $[0, 1]$ form a closed subset of the space of compact subsets of $\mathbb{R}$, Theorem 44.9.8 tells us that $(\mathcal{K}, \rho)$ is a non-empty complete metric space.

Lemma 44.12.13

For $k \geq 1$ let $\mathcal{E}_k = \{E \in \mathcal{K} : \exists x \in E \text{ with } (x - \frac{1}{k}, x + \frac{1}{k}) \cap E = \{x\}\}$.

- (i) $\mathcal{E}_k$ is closed in $(\mathcal{K}, \rho)$.
- (ii) $\mathcal{E}_k$ has empty interior.
- (iii) The collection of compact sets possessing an isolated point is of first category in $\mathcal{K}$.

Proof. (i) Let $E_r \in \mathcal{E}_k$ with $\rho(E_r, E) \rightarrow 0$, and pick $x_r \in E_r$ with $(x_r - \frac{1}{k}, x_r + \frac{1}{k}) \cap E_r = \{x_r\}$. Passing to a subsequence, $x_r \rightarrow x$; since $d(x_r, E) \leq \sigma(E_r, E) \rightarrow 0$ and E is closed, $x \in E$.

Suppose $y \in E$ with $y \neq x$ and $|x - y| < \frac{1}{k}$. Put $4\eta = \frac{1}{k} - |x - y| > 0$. Since $\sigma(E, E_r) \rightarrow 0$ we can find $y_r \in E_r$ with $|y_r - y| \rightarrow 0$. For r large we have $|x_r - y_r| < |x - y| + 2\eta < \frac{1}{k}$ and $y_r \neq x_r$ (as $|x - y| > 0$), so y_r lies in $(x_r - \frac{1}{k}, x_r + \frac{1}{k}) \cap E_r$ and differs from x_r . This contradiction shows $(x - \frac{1}{k}, x + \frac{1}{k}) \cap E = \{x\}$, so $E \in \mathcal{E}_k$.

(ii) Let $E \in \mathcal{K}$ and $\varepsilon > 0$. Take $0 < \eta < \varepsilon$ and set $F = \{x \in [0, 1] : d(x, E) \leq \eta\}$, which is a non-empty compact subset of $[0, 1]$ with $E \subseteq F$, so $\sigma(E, F) = 0$ and $\sigma(F, E) \leq \eta$, giving $\rho(E, F) \leq \eta < \varepsilon$. But F has no isolated points at all: if $x \in F$, take $e \in E$ with $|x - e| \leq \eta$; the whole segment between e and x lies in F , and if $x = e$ then one of $[x, x + \eta] \cap [0, 1]$, $[x - \eta, x] \cap [0, 1]$ is a non-degenerate interval contained in F . Either way every neighbourhood of x meets $F \setminus \{x\}$. So $F \notin \mathcal{E}_k$.

(iii) A compact set has an isolated point if and only if it lies in $\mathcal{E}_k$ for some k , so the collection in question is $\bigcup_k \mathcal{E}_k$, a countable union of closed sets with empty interior. $\square$

Lemma 44.12.14

For integers $k \geq 1$ and $0 \leq j < k$, let $\mathcal{F}_{j,k} = \{E \in \mathcal{K} : E \supseteq [\frac{j}{k}, \frac{j+1}{k}]\}$. Then $\mathcal{F}_{j,k}$ is closed with empty interior, and the collection of compact subsets of $[0, 1]$ with non-empty interior is of first category.

Proof. Write $I = [\frac{j}{k}, \frac{j+1}{k}]$. If $E_r \in \mathcal{F}_{j,k}$ and $\rho(E_r, E) \rightarrow 0$, then for $y \in I$ we have $y \in E_r$, so $d(y, E) \leq \sigma(E_r, E) \rightarrow 0$ and hence $y \in E$; thus $E \in \mathcal{F}_{j,k}$ and $\mathcal{F}_{j,k}$ is closed.

For empty interior, let $E \in \mathcal{K}$ and $\varepsilon > 0$. If $E \notin \mathcal{F}_{j,k}$ we are done. Otherwise $E \supseteq I$; let a be the midpoint of I , choose $0 < \eta < \min\{\varepsilon, \frac{1}{2k}\}$ and set $F = E \setminus (a - \eta, a + \eta)$. Then F is compact and non-empty (it still contains the endpoints of I), $\sigma(F, E) = 0$, and every removed point is within η of $a \pm \eta \in F$, so $\sigma(E, F) \leq \eta$ and $\rho(E, F) \leq \eta < \varepsilon$. But $a \notin F$, so $F \notin \mathcal{F}_{j,k}$.

Finally, a compact $E \subseteq [0, 1]$ has non-empty interior if and only if it contains some non-degenerate interval, hence some $[\frac{j}{k}, \frac{j+1}{k}]$; so the collection of such E is

$$\bigcup_{j,k} \mathcal{F}_{j,k}.$$

□

Theorem 44.12.15

Quasi-all non-empty compact subsets of $[0, 1]$ have empty interior and no isolated points. In particular such sets exist, and by Theorem 44.12.7 they are uncountable.

Proof. By Lemmas 44.12.13 and 44.12.14 the sets we wish to exclude form a union of two sets of first category, hence a set of first category, which by Lemma 44.12.5 is not all of $\mathcal{K}$. □

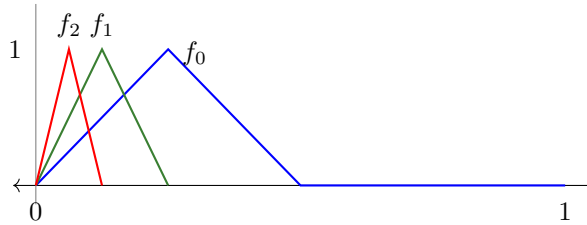
Sets which are closed with no isolated points are called *perfect*; if they make pretty pictures they are called *fractals*. The reader is not obliged to remember either name.

§44.12.4 Pointwise Convergence and Boundedness

Our last application concerns the gap between pointwise and uniform convergence. Recall the ‘witch’s hat’ functions $f_n : [0, 1] \rightarrow \mathbb{R}$,

$$f_n(x) = \begin{cases} 2^{n+1}x & 0 \leq x \leq 2^{-(n+1)}, \\ 2 - 2^{n+1}x & 2^{-(n+1)} < x \leq 2^{-n}, \\ 0 & \text{otherwise,} \end{cases}$$

which satisfy $f_n(x) \rightarrow 0$ for every x while $\|f_n\|_\infty = 1$ for every n : the hat becomes narrower but never shorter, and slides off to the left.



The trouble is confined to a shrinking interval, and one might hope that pointwise convergence forces uniform convergence somewhere. Alas, it does not.

Lemma 44.12.16

There exist $g_n \in C([0, 1])$ with $\|g_n\|_\infty \leq 1$ and $g_n(x) \rightarrow 0$ for every $x \in [0, 1]$, such that (g_n) fails to converge uniformly on every non-degenerate interval $I \subseteq [0, 1]$.

Proof. Enumerate the subintervals of $[0, 1]$ with distinct rational endpoints as $[a_1, b_1]$, $[a_2, b_2]$, $\dots$, and let $p_1, p_2, \dots$ be distinct primes. Define $g_m = 0$ unless $m = p_k^r$ for some $k, r \geq 1$ (which determines k and r uniquely), and in that case

$$g_{p_k^r}(x) = \begin{cases} 2^{-k} f_r\left(\frac{x - a_k}{b_k - a_k}\right) & x \in [a_k, b_k], \\ 0 & \text{otherwise.} \end{cases}$$

Since $f_r(0) = f_r(1) = 0$, each g_m is continuous, and clearly $\|g_m\|_\infty \leq 2^{-k} \leq 1$.

Pointwise convergence: fix x and $\varepsilon > 0$, and choose K with $2^{-K} < \varepsilon$. For $k > K$ we have $|g_{p_k^r}(x)| \leq 2^{-k} < \varepsilon$, whatever r may be. For each of the finitely many $k \leq K$ we have $f_r(y) \rightarrow 0$ as $r \rightarrow \infty$ for each fixed y , so there is an R_k with $|g_{p_k^r}(x)| < \varepsilon$ whenever $r \geq R_k$. Hence $|g_m(x)| \geq \varepsilon$ for at most finitely many m , and $g_m(x) \rightarrow 0$.

Non-uniformity: any non-degenerate interval I contains some $[a_k, b_k]$, and

$$\sup_{x \in [a_k, b_k]} |g_{p_k^r}(x)| = 2^{-k}$$

for every r , so g_m does not tend to 0 uniformly on I . $\square$

In spite of this, Baire's theorem gives a remarkable positive result: the failure can never be one of *size*.

Theorem 44.12.17 (Osgood)

Let $f_n \in C([0, 1])$ with $f_n(x) \rightarrow 0$ for every $x \in [0, 1]$. Then given $\varepsilon > 0$ there is a non-degenerate interval $(a, b) \subseteq [0, 1]$ and an N such that $|f_n(x)| \leq \varepsilon$ for all $n \geq N$ and all $x \in (a, b)$.

Proof. For $n \geq 1$ set

$$E_n = \{x \in [0, 1] : |f_m(x)| \leq \varepsilon \text{ for all } m \geq n\} = \bigcap_{m \geq n} f_m^{-1}([- \varepsilon, \varepsilon]),$$

an intersection of closed sets and hence closed. Since $f_m(x) \rightarrow 0$ for each x , every x lies in some E_n , so $\bigcup_n E_n = [0, 1]$. Now $[0, 1]$ is a non-empty complete metric space, so by Theorem 44.12.2 not every E_n can have empty interior. Take N with E_N containing an interval (a, b) . $\square$

Corollary 44.12.18 (Uniform Boundedness)

Let $f_n \in C([0, 1])$ with $f_n(x) \rightarrow 0$ for every x . Then there is a non-degenerate interval (a, b) and an $M > 0$ with $|f_n(x)| \leq M$ for all $n \geq 1$ and all $x \in (a, b)$.

Proof. Apply Theorem 44.12.17 with $\varepsilon = 1$ to get (a, b) and N with $|f_n| \leq 1$ on (a, b) for $n \geq N$. The finitely many remaining functions $f_1, \dots, f_{N-1}$ are each bounded on $[0, 1]$, so $M = \max\{1, \|f_1\|_\infty, \dots, \|f_{N-1}\|_\infty\}$ works. $\square$

Comparing this with Lemma 44.12.16, we see that the functions g_n constructed there are as bad as pointwise-convergent sequences are permitted to be: they refuse to converge uniformly on any interval, but they cannot be allowed to blow up on every interval as well. This is the elementary ancestor of the principle of uniform boundedness in Part II Linear Analysis, which is proved by exactly the same Baire category argument in a Banach space.

§44.13 Continued Fractions

We are used to writing real numbers as decimals, but there are other ways of specifying a real number which are often more convenient and always more interesting. The oldest

is the method of continued fractions.

Suppose x is irrational with $0 < x < 1$. Then $1/x > 1$ is irrational, so there is a unique integer $N(x) \geq 1$ with

$$\frac{1}{N(x)+1} < x < \frac{1}{N(x)}, \quad \text{that is} \quad N(x) = \left\lfloor \frac{1}{x} \right\rfloor,$$

and setting $T(x) = \frac{1}{x} - \lfloor \frac{1}{x} \rfloor$ we have $0 < T(x) < 1$, $T(x)$ irrational, and

$$x = \frac{1}{N(x) + T(x)}.$$

We may now do the same thing to $T(x)$, obtaining $T(x) = \frac{1}{NT(x) + T^2(x)}$, and so

$$x = \frac{1}{N(x) + \frac{1}{NT(x) + T^2(x)}} = \frac{1}{N(x) + \frac{1}{NT(x) + \frac{1}{NT^2(x) + T^3(x)}}} = \dots,$$

and so on indefinitely. We call

$$\cfrac{1}{N(x) + \cfrac{1}{NT(x) + \cfrac{1}{NT^2(x) + \cfrac{1}{NT^3(x) + \ddots}}}}$$

the **continued fraction expansion** of x . For the moment this is nothing more than a pretty way of writing down the sequence of positive integers $N(x), NT(x), NT^2(x), \dots$; in the next subsection we shall show that it can be assigned a numerical value, and that the value is x . For a general irrational y we prefix the integer part, writing the expansion of y as $\lfloor y \rfloor$ followed by the expansion of $y - \lfloor y \rfloor$.

The same procedure works for rationals, except that it may stop. Let us see why.

Example 44.13.1

Take $x = 100/37$. Then

$$\begin{aligned} \frac{100}{37} &= 2 + \frac{26}{37} = 2 + \frac{1}{\frac{37}{26}} = 2 + \frac{1}{1 + \frac{11}{26}} = 2 + \frac{1}{1 + \frac{1}{\frac{26}{11}}} = 2 + \frac{1}{1 + \frac{1}{2 + \frac{4}{11}}} \\ &= 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2 + \frac{3}{4}}}} = 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{1 + \frac{1}{3}}}}} \end{aligned}$$

$$= 2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{1 + \frac{1}{3}}}}}.$$

Lemma 44.13.2

The continued fraction algorithm applied to a rational number is Euclid's algorithm, and in particular it terminates.

Proof. Suppose at some stage we are looking at r_k/s_k with $0 < r_k < s_k$ coprime positive integers, and we write

$$\frac{r_k}{s_k} = \frac{1}{a_k + \frac{r_{k+1}}{s_{k+1}}}, \quad a_k = \left\lfloor \frac{s_k}{r_k} \right\rfloor.$$

Then $s_k/r_k = a_k + r_{k+1}/s_{k+1}$, so $s_{k+1} = r_k$ and $s_k = a_k s_{k+1} + r_{k+1}$ with $0 \leq r_{k+1} < s_{k+1}$. That is exactly one step of the Euclidean algorithm applied to the pair (s_k, r_k) , and r_{k+1} is the remainder. Since $s_1 > s_2 > \dots$ is a strictly decreasing sequence of positive integers, the process stops. (Coprimality is preserved, again as in Euclid's algorithm.) $\square$

Consequently a number with a non-terminating continued fraction expansion is irrational, and this gives a rather satisfying proof of an old fact.

Theorem 44.13.3

$\sqrt{2}$ is irrational.

Proof. We have $\sqrt{2} = 1 + (\sqrt{2} - 1)$ and

$$\sqrt{2} - 1 = \frac{(\sqrt{2} - 1)(\sqrt{2} + 1)}{\sqrt{2} + 1} = \frac{1}{1 + \sqrt{2}} = \frac{1}{2 + (\sqrt{2} - 1)}.$$

So writing $\beta = \sqrt{2} - 1 \in (0, 1)$ we have $\beta = \frac{1}{2+\beta}$, and the continued fraction algorithm applied to β produces $N(\beta) = 2$ and $T(\beta) = \beta$ again. It therefore never terminates, giving

$$\sqrt{2} = 1 + \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \ddots}}},$$

and by Lemma 44.13.2 the number $\sqrt{2}$ cannot be rational. $\square$

§44.13.1 Gauss's Observation

Before we make sense of the value of a continued fraction, here is a beautiful aside. The decimal expansion has a map behind it too: if $x \in [0, 1)$ and $Dx = 10x - \lfloor 10x \rfloor$, then the digits of x are read off by iterating D , and if X is uniformly distributed on $[0, 1]$ then so is DX – which is why the digits of a ‘random’ number are independent and uniform.

What is the analogue for continued fractions? The map T does *not* preserve the uniform distribution. Gauss found the distribution it does preserve.

Lemma 44.13.4 (Gauss)

Let X be a random variable on $[0, 1)$ with density

$$f(x) = \frac{1}{\log 2} \cdot \frac{1}{1+x}.$$

Then TX has the same density.

Proof. First, f is a density: $\int_0^1 \frac{dx}{\log 2(1+x)} = \frac{\log 2}{\log 2} = 1$. Now for $0 \leq a \leq 1$,

$$\begin{aligned} \Pr(TX \leq a) &= \Pr\left(\frac{1}{X} - \left\lfloor \frac{1}{X} \right\rfloor \leq a\right) = \sum_{n=1}^{\infty} \Pr\left(n \leq \frac{1}{X} \leq n+a\right) \\ &= \sum_{n=1}^{\infty} \Pr\left(\frac{1}{n+a} \leq X \leq \frac{1}{n}\right) = \frac{1}{\log 2} \sum_{n=1}^{\infty} \int_{1/(n+a)}^{1/n} \frac{dx}{1+x} \\ &= \frac{1}{\log 2} \sum_{n=1}^{\infty} \left[\log\left(1 + \frac{1}{n}\right) - \log\left(1 + \frac{1}{n+a}\right) \right]. \end{aligned}$$

Both parts telescope. Summing to N ,

$$\begin{aligned} \sum_{n=1}^N (\log(n+1) - \log n) &= \log(N+1), \\ \sum_{n=1}^N (\log(n+1+a) - \log(n+a)) &= \log(N+1+a) - \log(1+a), \end{aligned}$$

so the partial sum is $\log(N+1) - \log(N+1+a) + \log(1+a) \rightarrow \log(1+a)$ as $N \rightarrow \infty$. Hence

$$\Pr(TX \leq a) = \frac{\log(1+a)}{\log 2} = \int_0^a f(x) dx. \quad \square$$

Corollary 44.13.5

With X as above, for each $m \geq 0$ and each $j \geq 1$,

$$\Pr(NT^m X = j) = \frac{1}{\log 2} \int_{1/(j+1)}^{1/j} \frac{dx}{1+x} = \frac{1}{\log 2} \log\left(\frac{(j+1)^2}{j(j+2)}\right) \sim \frac{1}{j^2 \log 2}$$

as $j \rightarrow \infty$.

Proof. By Lemma 44.13.4, $T^m X$ has the same density as X , and $NY = j$ if and only if $\frac{1}{j+1} < Y \leq \frac{1}{j}$. Evaluating the integral gives $\frac{1}{\log 2} \left(\log \frac{j+1}{j} - \log \frac{j+2}{j+1} \right)$, which is the stated logarithm; and $\log \frac{(j+1)^2}{j(j+2)} = \log \left(1 + \frac{1}{j^2+2j} \right) \sim j^{-2}$. $\square$

With a little more work (which we shall not do) one can show that if X has Gauss's density then $NX, NTX, NT^2X, \dots$ are independent and identically distributed; and it is true, though harder, that if Y is merely *uniform* on $[0, 1]$ then the proportion of the first M partial quotients of Y which equal j still tends to the above value. So in a 'typical' continued fraction the digit 1 appears about 41% of the time, and large partial quotients are rare.

§44.13.2 Convergents

We now return to the question of what a continued fraction *means*.

Definition 44.13.6

Let a_0 be an integer and $a_1, a_2, \dots$ strictly positive integers. The associated **convergents** are the rationals

$$\frac{p_n}{q_n} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\ddots + \frac{1}{a_n}}}}.$$

Evaluating from the bottom up is the natural thing to try. Write

$$\frac{r_k}{s_k} = a_k + \frac{1}{a_{k+1} + \frac{1}{\ddots + \frac{1}{a_n}}},$$

so that $r_n = a_n$, $s_n = 1$, and

$$\frac{r_k}{s_k} = a_k + \frac{1}{r_{k+1}/s_{k+1}} = \frac{a_k r_{k+1} + s_{k+1}}{r_{k+1}}, \quad \text{i.e.} \quad \begin{pmatrix} r_k \\ s_k \end{pmatrix} = \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} r_{k+1} \\ s_{k+1} \end{pmatrix}.$$

Hence

$$\begin{pmatrix} p_n \\ q_n \end{pmatrix} = \begin{pmatrix} r_0 \\ s_0 \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_{n-1} & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_n \\ 1 \end{pmatrix}.$$

Similarly $\begin{pmatrix} p_{n-1} \\ q_{n-1} \end{pmatrix}$ is the same product with the last factor replaced by $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, so putting the two columns side by side,

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}.$$

This is the pay-off for recasting things in matrix form: every fact we want now falls out.

Theorem 44.13.7

With p_n, q_n as above (and the conventions $p_{-1} = 1, q_{-1} = 0$):

- (i) $p_{n+1} = a_{n+1}p_n + p_{n-1}$ and $q_{n+1} = a_{n+1}q_n + q_{n-1}$;
- (ii) $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n+1}$, and in particular p_n and q_n are coprime;
- (iii) $\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} = \frac{(-1)^{n+1}}{q_n q_{n-1}}$;
- (iv) $q_n \geq F_{n+1} \rightarrow \infty$, where F_m is the m th Fibonacci number;
- (v) $\frac{p_0}{q_0} < \frac{p_2}{q_2} < \frac{p_4}{q_4} < \dots < \dots < \frac{p_5}{q_5} < \frac{p_3}{q_3} < \frac{p_1}{q_1}$, and $p_n/q_n \rightarrow \alpha$ for some real α with $\left| \alpha - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}$.

Proof. (i) Multiply the matrix identity for $n+1$ as

$$\begin{pmatrix} p_{n+1} & p_n \\ q_{n+1} & q_n \end{pmatrix} = \begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} \begin{pmatrix} a_{n+1} & 1 \\ 1 & 0 \end{pmatrix}$$

and read off the first column.

(ii) Take determinants: each factor $\begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix}$ has determinant -1 , and there are $n+1$ of them, so $p_n q_{n-1} - q_n p_{n-1} = (-1)^{n+1}$. Any common factor of p_n and q_n would divide this.

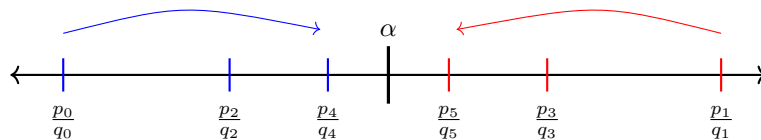
(iii) Divide (ii) by $q_n q_{n-1}$.

(iv) By (i), $q_{n+1} = a_{n+1}q_n + q_{n-1} \geq q_n + q_{n-1}$, with $q_0 = 1$ and $q_1 = a_1 \geq 1$, so induction gives $q_n \geq F_{n+1}$.

(v) From (i) and (iii),

$$\begin{aligned} \frac{p_{n+1}}{q_{n+1}} - \frac{p_{n-1}}{q_{n-1}} &= \left(\frac{p_{n+1}}{q_{n+1}} - \frac{p_n}{q_n} \right) + \left(\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right) \\ &= \frac{(-1)^{n+2}}{q_{n+1}q_n} + \frac{(-1)^{n+1}}{q_n q_{n-1}} = \frac{(-1)^{n+1}}{q_n} \left(\frac{1}{q_{n-1}} - \frac{1}{q_{n+1}} \right), \end{aligned}$$

which has the sign of $(-1)^{n+1}$ since $q_{n+1} > q_{n-1}$. So the even convergents increase and the odd ones decrease. By (iii) every even convergent is below every odd one, so both are bounded monotone sequences and converge; and since $\left| \frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right| = \frac{1}{q_n q_{n-1}} \rightarrow 0$ by (iv), the two limits agree, say to α . As α lies between p_n/q_n and p_{n+1}/q_{n+1} , we get $\left| \alpha - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}}$. $\square$



So a continued fraction does converge, and it converges in the pleasantest possible way: the convergents alternate around the limit, so each one gives an error bound for the previous one. It remains to check that when we start with a number x we get back to x .

Lemma 44.13.8

If $x \in (0, 1)$ is irrational and $a_1 = N(x)$, $a_2 = NT(x)$, $\dots$ are the partial quotients produced by the continued fraction algorithm (with $a_0 = 0$), then $p_n/q_n \rightarrow x$.

Proof. Truncating the algorithm at stage n gives x exactly as the continued fraction with entries $a_0, a_1, \dots, a_n + T^n(x)$ in place of $a_0, \dots, a_n$, where $0 < T^n(x) < 1$. Increasing a partial quotient a_n increases p_n/q_n when n is even and decreases it when n is odd (this is immediate from the fact, used repeatedly, that $t \mapsto \frac{1}{a+t}$ is decreasing on $t \geq 0$, and that these substitutions are nested). It follows that

$$\frac{p_{2k}}{q_{2k}} < x < \frac{p_{2k-1}}{q_{2k-1}}$$

for every k . As both sides tend to the limit α of Theorem 44.13.7(v), we get $x = \alpha$. $\square$

§44.13.3 Convergents Are the Best Approximations

The convergents are not merely good approximations: they are the best possible ones for their size of denominator.

Lemma 44.13.9

Let $u, v \in \mathbb{Z}$ with $1 \leq v \leq q_n$, and suppose $\left| \frac{u}{v} - \frac{p_{n+1}}{q_{n+1}} \right| \leq \frac{1}{q_n q_{n+1}}$. Then $v = q_n$ and (provided $q_{n+1} > 2$) $u = p_n$.

Proof. First, $\frac{u}{v} \neq \frac{p_{n+1}}{q_{n+1}}$: since p_{n+1}, q_{n+1} are coprime this would force $q_{n+1} \mid v$, impossible as $1 \leq v \leq q_n < q_{n+1}$. Hence $|uq_{n+1} - vp_{n+1}| \geq 1$ and

$$\left| \frac{u}{v} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{|uq_{n+1} - vp_{n+1}|}{vq_{n+1}} \geq \frac{1}{vq_{n+1}} \geq \frac{1}{q_n q_{n+1}}.$$

So both inequalities are equalities: $v = q_n$ and $|uq_{n+1} - q_n p_{n+1}| = 1$. But also $|p_n q_{n+1} - q_n p_{n+1}| = 1$ by Theorem 44.13.7(ii), so subtracting gives $(u - p_n)q_{n+1} \in \{0, \pm 2\}$, and if $q_{n+1} > 2$ this forces $u = p_n$. $\square$

Theorem 44.13.10

Let α be the value of the continued fraction with convergents p_n/q_n , and suppose $q_{n+1} > 2$. Then

$$\left| \alpha - \frac{p_n}{q_n} \right| \leq \left| \alpha - \frac{p}{q} \right|$$

for all integers p, q with $1 \leq q \leq q_n$.

Proof. Since α lies between $\frac{p_n}{q_n}$ and $\frac{p_{n+1}}{q_{n+1}}$, we have

$$\left| \alpha - \frac{p_n}{q_n} \right| + \left| \alpha - \frac{p_{n+1}}{q_{n+1}} \right| = \left| \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}}.$$

Suppose $1 \leq q \leq q_n$ and $\left| \alpha - \frac{p}{q} \right| < \left| \alpha - \frac{p_n}{q_n} \right|$. Then

$$\left| \frac{p}{q} - \frac{p_{n+1}}{q_{n+1}} \right| \leq \left| \frac{p}{q} - \alpha \right| + \left| \alpha - \frac{p_{n+1}}{q_{n+1}} \right| < \left| \alpha - \frac{p_n}{q_n} \right| + \left| \alpha - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}},$$

so by Lemma 44.13.9 we get $q = q_n$ and $p = p_n$, contradicting the strict inequality. $\square$

Corollary 44.13.11

If x is irrational, there are infinitely many pairs of integers (u, v) with $v \geq 1$ and

$$\left| x - \frac{u}{v} \right| < \frac{1}{v^2}.$$

Proof. Take $u = p_n$, $v = q_n$. By Theorem 44.13.7(v) and $q_{n+1} > q_n$,

$$\left| x - \frac{p_n}{q_n} \right| \leq \frac{1}{q_n q_{n+1}} < \frac{1}{q_n^2},$$

and $q_n \rightarrow \infty$ gives infinitely many distinct pairs. $\square$

This should be compared with Liouville's Theorem 44.11.6, which says that an algebraic irrational of degree n satisfies $|x - p/q| \geq cq^{-n}$. For $n = 2$ the two statements are in direct competition, and neither can be improved to the point of contradicting the other: quadratic irrationals are approximable to order q^{-2} and no better.

§44.13.4 Badly Approximable Numbers

Large partial quotients a_j produce large jumps in $q_{n+1} = a_{n+1}q_n + q_{n-1}$, and hence unusually good approximations. So the most *resistant* number to rational approximation should be the one whose partial quotients are all as small as possible.

Example 44.13.12 (The Golden Ratio)

Let

$$\sigma = \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \ddots}}}.$$

Then $\sigma = \frac{1}{1+\sigma}$, so $\sigma^2 + \sigma - 1 = 0$ and (taking the positive root) $\sigma = \frac{\sqrt{5}-1}{2} = 1/\varphi$, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio.

Here every $a_j = 1$, so by Theorem 44.13.7(i) we have $p_{n+1} = p_n + p_{n-1}$ and $q_{n+1} = q_n + q_{n-1}$. With $a_0 = 0$ we get $p_0 = 0$, $q_0 = 1$, $p_1 = q_1 = 1$, and induction gives

$$p_n = F_n, \quad q_n = F_{n+1},$$

where $F_0 = 0$, $F_1 = 1$, $F_{m+1} = F_m + F_{m-1}$ are the Fibonacci numbers. Theorem 44.13.7(ii) becomes the identity $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$.

Solving the recurrence, $F_m = \frac{1}{\sqrt{5}} (\varphi^m - (-\varphi)^{-m})$. Since $\sigma\varphi = 1$, a short computation gives

$$\sigma - \frac{F_n}{F_{n+1}} = \frac{\sigma F_{n+1} - F_n}{F_{n+1}} = \frac{(-1)^n \varphi^{-n} (1 + \varphi^{-2})}{\sqrt{5} F_{n+1}},$$

and as $F_{n+1} \sim \varphi^{n+1}/\sqrt{5}$ this gives

$$\left| \sigma - \frac{p_n}{q_n} \right| \sim \frac{1}{\sqrt{5} q_n^2}.$$

So the convergents to σ approximate it to order q^{-2} and to no better order – the constant $1/\sqrt{5}$ is exactly what the general theory guarantees, and not a whit more. This lets us prove that σ is *badly approximable*.

Proposition 44.13.13

There is an $m > 0$ such that $\left| \sigma - \frac{p}{q} \right| > \frac{m}{q^2}$ for all integers p and all $q \geq 1$.

Proof. Fix p, q with $q \geq 1$, and let n be least with $q \leq q_n$ and $q_{n+1} > 2$. By Theorem 44.13.10,

$$\left| \sigma - \frac{p}{q} \right| \geq \left| \sigma - \frac{p_n}{q_n} \right| \geq \frac{c}{q_n^2}$$

for some constant $c > 0$, by Example 44.13.12. By minimality of n (and $q_n = q_{n-1} + q_{n-2} \leq 2q_{n-1}$) we have $q_n \leq 2q_{n-1} \leq 2q$ for all n beyond the first couple, so $\left| \sigma - \frac{p}{q} \right| \geq \frac{c}{4q^2}$. Adjusting the constant to deal with the finitely many small cases gives the result. $\square$

Hardy and Wright's *An Introduction to the Theory of Numbers* contains a chapter on approximation by rationals in which, among many other things, it is shown that σ really is the most resistant of all irrationals in this sense. If I were asked to nominate a single book to be taken away by somebody leaving professional mathematics but wishing to keep up an interest in the subject, that book would be my first choice.

A Nice, But Starred, Formula

This subsection is not examinable, but it is too pretty to leave out. Continued fractions generalise in many directions; here is one of them. We are used to approximating a function by polynomials; sometimes it is better to approximate by a *rational* function P/Q , and if approximation by polynomials leads to Taylor series, approximation by rational functions leads to something like continued fractions. Lambert found the following.

Theorem 44.13.14 (Lambert)

Let

$$R_n(x) = \frac{x}{1 - \frac{x^2}{3 - \frac{x^2}{5 - \frac{x^2}{\ddots - \frac{x^2}{2n-1}}}}}.$$

Then $R_n(x) \rightarrow \tan x$ as $n \rightarrow \infty$ for all real x with $|x| \leq 1$.

The proof runs along lines which will now be familiar. One first generalises the matrix identity: if p_n, q_n are defined by

$$\begin{pmatrix} p_n & b_n p_{n-1} \\ q_n & b_n q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & b_0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & b_1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & b_n \\ 1 & 0 \end{pmatrix},$$

then p_n/q_n is the generalised continued fraction with numerators b_j and denominators a_j , and

$$p_n = a_n p_{n-1} + b_{n-1} p_{n-2}, \quad q_n = a_n q_{n-1} + b_{n-1} q_{n-2}.$$

One then writes

$$S_n(x) = \frac{1}{2^n n!} \int_0^x (x^2 - t^2)^n \cos t \, dt$$

and checks, by integrating by parts twice in the manner of Lemma 44.11.2, that $S_n(x) = q_n(x) \sin x - p_n(x) \cos x$ where p_n, q_n satisfy exactly the recurrences

$$p_n(x) = (2n-1)p_{n-1}(x) - x^2 p_{n-2}(x), \quad q_n(x) = (2n-1)q_{n-1}(x) - x^2 q_{n-2}(x)$$

with $p_0 = 0, q_0 = 1, p_1(x) = x, q_1 = 1$. These are the recurrences for the continued fraction above, and the estimate $|S_n(x)| \leq \frac{|x|^{2n+1}}{2^n n!} \rightarrow 0$ then gives $p_n(x)/q_n(x) \rightarrow \tan x$.

§44.14 Winding Numbers

We all know that complex analysis has a great deal to say about the number of times a curve goes round a point. In this final section we make the notion precise without using any complex analysis at all, and then use it to give a second proof of two theorems from earlier in the course. The moral will be that parts of complex analysis are really topology in disguise.

§44.14.1 Lifting

Write $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. The basic fact is that a continuous path on the circle can be followed continuously by an angle.

Theorem 44.14.1 (Path Lifting)

Let $g : [0, 1] \rightarrow \mathbb{T}$ be continuous with $g(0) = e^{i\theta_0}$. Then there is a unique continuous $\theta : [0, 1] \rightarrow \mathbb{R}$ with $\theta(0) = \theta_0$ and $g(t) = e^{i\theta(t)}$ for all t .

Proof. For *existence*, note first that on the half-plane $H = \{z : \operatorname{Re} z > 0\}$ we have a continuous argument function

$$\arg(u + iv) = \arctan(v/u) \in (-\frac{\pi}{2}, \frac{\pi}{2}),$$

satisfying $w = |w|e^{i\arg w}$ and $\arg 1 = 0$.

Now g is uniformly continuous by Theorem 44.1.12, so we may choose N with $|g(s) - g(t)| < 1$ whenever $|s - t| \leq 1/N$. Fix r and $t \in [\frac{r}{N}, \frac{r+1}{N}]$. Then $\left| \frac{g(t)}{g(r/N)} - 1 \right| = |g(t) - g(r/N)| < 1$, and a point of $\mathbb{T}$ within distance 1 of 1 has positive real part, so $g(t)/g(r/N) \in H$. We may therefore define θ successively on the intervals $[\frac{r}{N}, \frac{r+1}{N}]$ by

$$\theta(t) = \theta\left(\frac{r}{N}\right) + \arg\left(\frac{g(t)}{g(r/N)}\right),$$

starting from $\theta(0) = \theta_0$. This is continuous on each interval, agrees at the shared endpoints, and satisfies $e^{i\theta(t)} = e^{i\theta(r/N)} \cdot \frac{g(t)}{g(r/N)} = g(t)$ by induction on r .

For *uniqueness*, suppose $\psi, \phi : [0, 1] \rightarrow \mathbb{R}$ are continuous with $e^{i\psi(t)} = e^{i\phi(t)}$ for all t . Then $\frac{\psi - \phi}{2\pi}$ is a continuous function on $[0, 1]$ taking values in $\mathbb{Z}$; since $[0, 1]$ is connected and $\mathbb{Z}$ is discrete, it is constant. If in addition $\psi(0) = \phi(0)$, the constant is 0. $\square$

Corollary 44.14.2

If $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ is continuous with $\gamma(0) = |\gamma(0)|e^{i\theta_0}$, then there is a unique continuous $\theta : [0, 1] \rightarrow \mathbb{R}$ with $\theta(0) = \theta_0$ and $\gamma(t) = |\gamma(t)|e^{i\theta(t)}$.

Proof. Apply Theorem 44.14.1 to $g(t) = \gamma(t)/|\gamma(t)|$, which is continuous since γ never vanishes. $\square$

Definition 44.14.3 (Winding Number)

Let $\gamma : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ be continuous and let θ be as in Corollary 44.14.2. We define the **winding number** of γ about 0 to be

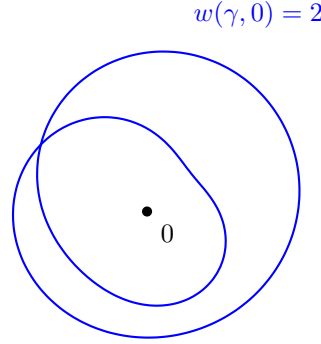
$$w(\gamma, 0) = \frac{\theta(1) - \theta(0)}{2\pi}.$$

By the uniqueness in Theorem 44.14.1, any two choices of θ differ by a constant, so $w(\gamma, 0)$ is well defined. From now on we shall only be interested in **closed** curves, that is those with $\gamma(0) = \gamma(1)$.

Lemma 44.14.4

If γ is closed then $w(\gamma, 0) \in \mathbb{Z}$, and every integer arises.

Proof. $\gamma(0) = \gamma(1)$ gives $e^{i\theta(0)} = e^{i\theta(1)}$, so $\theta(1) - \theta(0) \in 2\pi\mathbb{Z}$. For the converse take $\gamma(t) = e^{2\pi i n t}$, for which $\theta(t) = 2\pi n t$ and $w(\gamma, 0) = n$. $\square$



Winding numbers behave exactly as one would hope under multiplication.

Lemma 44.14.5

If $\gamma_1, \gamma_2 : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ are continuous, then $w(\gamma_1\gamma_2, 0) = w(\gamma_1, 0) + w(\gamma_2, 0)$.

Proof. Suppose $\gamma_j(t) = |\gamma_j(t)|e^{i\theta_j(t)}$ with θ_j continuous. Then

$$\gamma_1(t)\gamma_2(t) = |\gamma_1(t)\gamma_2(t)|e^{i(\theta_1(t)+\theta_2(t))},$$

and $\theta_1 + \theta_2$ is continuous, so it is a valid choice of angle function for the product $\gamma_1\gamma_2$. $\square$

§44.14.2 Walking the Dog

Lemma 44.14.6 (Dog Walking Lemma)

Let $\gamma_1, \gamma_2 : [0, 1] \rightarrow \mathbb{C}$ be continuous closed curves with $\gamma_1(t) \neq 0$ and $|\gamma_2(t)| < |\gamma_1(t)|$ for all t . Then $\gamma_1 + \gamma_2$ never vanishes and

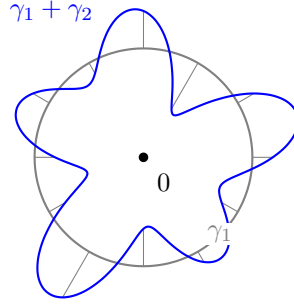
$$w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0).$$

Proof. Write $\gamma_1 + \gamma_2 = \gamma_1 \left(1 + \frac{\gamma_2}{\gamma_1}\right)$. Since $\left|\frac{\gamma_2(t)}{\gamma_1(t)}\right| < 1$, the curve $\eta = 1 + \frac{\gamma_2}{\gamma_1}$ takes values in $\{z : \operatorname{Re} z > 0\}$, and in particular never vanishes; hence neither does $\gamma_1 + \gamma_2$. By Lemma 44.14.5,

$$w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0) + w(\eta, 0),$$

so it suffices to show $w(\eta, 0) = 0$. But on $\{\operatorname{Re} z > 0\}$ we may take $\theta(t) = \arg \eta(t) \in (-\frac{\pi}{2}, \frac{\pi}{2})$ as our continuous angle function, so $|\theta(1) - \theta(0)| < \pi$; as η is closed, $\theta(1) - \theta(0) \in 2\pi\mathbb{Z}$, forcing $\theta(1) = \theta(0)$. $\square$

The name comes from the following picture. A man walks round a closed path γ_1 which avoids a lamp-post at the origin, taking his dog with him on a lead of length always shorter than his own distance from the lamp-post. The dog's path is $\gamma_1 + \gamma_2$. Then, however much the dog dashes about, man and dog go round the lamp-post the same number of times.



The lemma is really a statement about small perturbations. The right way to say ‘small perturbation’ in general is the following.

Definition 44.14.7 (Homotopy)

Let $\gamma_0, \gamma_1 : [0, 1] \rightarrow \mathbb{C} \setminus \{0\}$ be closed curves. We say γ_0 and γ_1 are **homotopic** through closed curves avoiding 0 if there is a continuous $\Gamma : [0, 1]^2 \rightarrow \mathbb{C} \setminus \{0\}$ with

$$\Gamma(s, 0) = \Gamma(s, 1) \quad \forall s, \quad \Gamma(0, t) = \gamma_0(t) \quad \forall t, \quad \Gamma(1, t) = \gamma_1(t) \quad \forall t.$$

We write $\gamma_0 \simeq \gamma_1$; it is easy to check that $\simeq$ is an equivalence relation.

Theorem 44.14.8 (Dog Walking Along a Canal)

If $\gamma_0 \simeq \gamma_1$ then $w(\gamma_0, 0) = w(\gamma_1, 0)$.

Proof. Let Γ be a homotopy. Since $|\Gamma|$ is continuous and strictly positive on the compact set $[0, 1]^2$, it attains a strictly positive minimum, so there is a $\delta > 0$ with $|\Gamma(s, t)| \geq 8\delta$ everywhere. Also Γ is uniformly continuous, so we may find N with

$$|\Gamma(s, t) - \Gamma(s', t')| < \delta \quad \text{whenever } |s - s'|, |t - t'| \leq \frac{1}{N}.$$

Set $\tau_r(t) = \Gamma(r/N, t)$ for $0 \leq r \leq N$, a closed curve avoiding 0. Then for every t ,

$$|\tau_{r+1}(t) - \tau_r(t)| < \delta < 8\delta \leq |\tau_r(t)|,$$

so by the dog walking Lemma 44.14.6 (with $\gamma_1 = \tau_r$ and $\gamma_2 = \tau_{r+1} - \tau_r$) we get $w(\tau_{r+1}, 0) = w(\tau_r, 0)$. Chaining these,

$$w(\gamma_0, 0) = w(\tau_0, 0) = w(\tau_1, 0) = \cdots = w(\tau_N, 0) = w(\gamma_1, 0). \quad \square$$

§44.14.3 Two Old Theorems Again

The following is the workhorse: a non-zero winding number on the boundary forces a zero inside.

Corollary 44.14.9

Let $f : \overline{B}(0, R) \rightarrow \mathbb{C}$ be continuous with $f(z) \neq 0$ for $|z| = R$, and define $\gamma(t) = f(Re^{2\pi it})$. If $w(\gamma, 0) \neq 0$ then $f(z) = 0$ for some z with $|z| < R$.

Proof. Suppose f has no zero in $\overline{B}(0, R)$. Then $\Gamma(s, t) = f(sRe^{2\pi it})$ is a continuous map $[0, 1]^2 \rightarrow \mathbb{C} \setminus \{0\}$ with $\Gamma(s, 0) = \Gamma(s, 1)$, so it is a homotopy from the constant curve $t \mapsto f(0)$ to γ . The constant curve has winding number 0, so by Theorem 44.14.8 $w(\gamma, 0) = 0$, a contradiction. $\square$

Theorem 44.14.10 (Fundamental Theorem of Algebra, again)

Every non-constant complex polynomial has a root.

Proof. We may assume $P(z) = z^n + \sum_{j=0}^{n-1} a_j z^j$ with $n \geq 1$. Choose any $R > 1$ which also exceeds $\sum_{j=0}^{n-1} |a_j|$, and set

$$\gamma_1(t) = R^n e^{2\pi i n t}, \quad \gamma_2(t) = \sum_{j=0}^{n-1} a_j R^j e^{2\pi i j t},$$

so that $\gamma_1 + \gamma_2$ is the curve $t \mapsto P(Re^{2\pi it})$. For every t ,

$$|\gamma_2(t)| \leq \sum_{j=0}^{n-1} |a_j| R^j \leq R^{n-1} \sum_{j=0}^{n-1} |a_j| < R^n = |\gamma_1(t)|,$$

so by Lemma 44.14.6 we get $w(\gamma_1 + \gamma_2, 0) = w(\gamma_1, 0) = n \neq 0$. Now Corollary 44.14.9 gives a root in $B(0, R)$. $\square$

This is the same idea as our very first proof in Theorem 44.1.14, but the bookkeeping is now done by the winding number rather than by hand. Finally, we recover the no retraction theorem, and with it Brouwer's fixed point theorem in two dimensions.

Theorem 44.14.11

There is no continuous $f : \overline{D} \rightarrow \partial D$ with $f(z) = z$ for all $z \in \partial D$.

Proof. Suppose such an f existed and set $\Gamma(s, t) = f(se^{2\pi it})$ for $s, t \in [0, 1]$. This is continuous, satisfies $\Gamma(s, 0) = \Gamma(s, 1)$, and never vanishes since $|f| \equiv 1$ on ∂D . So Γ is a homotopy through closed curves avoiding 0 between $t \mapsto \Gamma(0, t) = f(0)$, a constant curve of winding number 0, and $t \mapsto \Gamma(1, t) = f(e^{2\pi it}) = e^{2\pi it}$, which has winding number 1. This contradicts Theorem 44.14.8. $\square$

It is worth comparing the two proofs of the no retraction theorem. The one via Sperner's lemma is combinatorial, elementary, and generalises to $\mathbb{R}^n$ with only cosmetic changes; the one just given is slicker, but is tied to the plane, because the whole notion of a winding number is.

More generally, the contents of this section show that some of what one learns in a complex analysis course – the argument principle, Rouché's theorem, the fundamental theorem of algebra – is really a special case of general topological facts about continuous maps, and has nothing to do with differentiability. Other parts of the subject, such as Taylor's theorem and Cauchy's theorem itself, depend crucially on the fact that we are dealing with the very restricted class of functions satisfying the Cauchy–Riemann equations. In traditional courses this distinction appears late, if at all; Beardon's *Complex*

Analysis shows that it is possible to do things differently, and is well worth a look.